# 人类数学猜想有多少？

规模评估与可追溯资料汇编

17,614

条已发布记录：编号、题名、完整问题表述与来源入口

17 个学科类别 / 1,117 条题名标记索引 / 1,417 条归档索引

中文评估 + 英文及 LaTeX 原始表述 + 可点击导航

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调度 AI 生成

严格的全球总数未知。本报告不把数据库条目数当作猜想总数。

# 01 结论：能数清什么，不能数清什么

目前无法给出“人类全部数学猜想”的可信精确总数，也无法证明一次搜集已经穷尽全部。可以严谨完成的是：明确计数口径，完整提取某个有边界的公开语料库，并把所有记录交给读者复核。

## 本次评估

| 问题 | 可支持的回答 |
| --- | --- |
| 人类历史上所有猜想，一共有多少个？ | 未知。未发表、已遗失、口头提出的猜想无法被完整观测；等价命题和猜想族也没有统一计数约定。 |
| 公开可检索的猜想与开放问题，大致多大？ | 数据库记录已达万至十万量级。这里说的是宽口径记录规模，不能解释为严格去重的未证猜想数量。 |
| 本报告实际收集了多少？ | 完整下载 19,031 条原始记录；逐条刊入其中 17,614 条 published=true 记录；另列 1,417 条归档记录索引。 |
| 标题明确含猜想类词语的有多少？ | 1,117 条标题命中 conjecture(s) 或 hypothesis/hypotheses。只是可重复的检索指标，不是严格猜想总数。 |
| 这些记录已经彻底去重了吗？ | 没有。ID 唯一；规范化表述后仍有 4 条冗余记录，得到 17,610 种文本。语义等价与变体尚未全面归并。 |

如果必须给一个数量级判断：广义、公开记录中的“猜想 + 开放问题 + 历史问题”已经是数万乃至接近十万条目的研究对象；对严格意义的“目前未解决、语义去重后的数学猜想”，本次证据不足以给出可靠区间。不要把 17,614、90,717 或二者之和称作人类猜想总数。[S1-S5]

这里没有用少数样本外推一个“精确全球数”，也没有将开放问题、形式化命题、定理和数据库记录混为一谈。

# 02 范围、定义与计数规则

## 猜想与开放问题

猜想通常是被提出并被认为可能成立、但尚未确立的数学断言。开放问题可以只是一个问题，甚至要求分类、构造、最优常数或算法；它未必包含被猜测为真的明确命题。两者有交集，不能互换。

## 历史猜想与当前猜想

已经被证明或被反驳的历史猜想仍可能保留名称。若“人类所有猜想”按历史口径统计，它们应保留；若问“尚待解决”，则必须区分已解决、部分解决、独立性、争议性声称等状态。

## 一个名称不一定是一个命题

广义黎曼猜想、标准猜想、Langlands 纲领等可以包含多个版本或命题族。一个猜想也可能有多个名称和等价形式。按名称计数、按论文中的编号计数、按逻辑等价类计数，会得到不同答案。

## 有限历史记录与潜在无限命题

人类在某个日期前实际写下的有限记录，不等于所有可以形式化表达的数学命题。参数族的无限多个实例也不应自动计为人类提出了无限多个独立猜想。

## 本报告的基本单位

采用数据集中的 problem_number 作为记录主键。published=true 的每条记录均保留，已解决记录也保留。published=false 记录仅进入归档索引，不计入 17,614。

## “完整”的限定

本报告完整刊入所下载版本的 17,614 条已发布记录的 statement 字段；它是“该语料版本的完整汇编”，不是“数学史的完整全集”。没有对所有原论文逐页复核，也没有声称所有表述均正确、自足或仍未解决。

# 03 来源地图：不同数据库不能直接相加

| 来源 | 本次观察 | 计数对象与处理 |
| --- | --- | --- |
| UnsolvedMath v1.8.0 [S1-S2] | 19,031 / 17,614 | 已下载原始 JSON 并自行计数；前者为全部记录，后者为 published=true。本报告的主体。 |
| MathDB [S3] | 90,717 | 网页显示的 problems 数量。涵盖多种知识条目和状态；全库下载需 API 订阅，本次未获取全库。 |
| Erdős 社区库 [S4] | 1,221 | 带固定编号的问题记录；已包含在 UnsolvedMath 主体中，不能再加一次。 |
| Formal Conjectures [S5] | 4,411 | 首页 Open 1,599 + Solved 2,812 的正式命题计数；页面另列教学、测试、API 类目，不能按总页面元素数量计猜想。 |
| AIM / AMR [S6-S7] | 目录型来源 | 原始研讨会问题及问题集入口；一个链接可包含很多问题。不是单条猜想登记册。 |
| Kourovka Notebook [S8] | 第 21 期 | 编者提供群论问题、历史答案与持续更新；语料库中的该集合有 1,934 条记录。 |
| Clay 千禧年问题 [S9-S10] | 7 个历史题目 | 一个经过筛选的著名问题集，不能代表整个数学界。三维庞加莱猜想已解决。 |

MathDB 的“90,717”是网页快照，不是本报告逐条验算的数。Formal Conjectures 中同一问题可能对应多个 formal statement；形式化表述存在，也不意味着该问题已有形式化证明。[S3,S5]

**来源间高度重叠，抽取粒度和状态标准也不同，因此不能使用“各库数量相加”的方式估算全球总量。未观察到的部分也没有随机抽样概率，无法给出有统计意义的覆盖率或置信区间。**

# 04 语料审计与可重复统计

| 指标 | 本次程序复核结果 |
| --- | --- |
| 下载文件 / 版本 | problems.json / UnsolvedMath v1.8.0 |
| 原始记录总数 | 19,031 |
| 刊入正文的已发布记录 | 17,614 |
| 归档记录 | 1,417（重复记录和非问题摘录等，不全是重复猜想） |
| 已发布记录的主键数 | 17,614 个唯一 problem_number；17,614 个唯一 id |
| 空 statement 字段 | 0 |
| 标题完全相同的额外记录 | 121 条；同名不能直接认定数学内容重复 |
| 规范化后不同 statement 文本 | 17,610 种；仅做 Unicode NFKC 与空白合并 |
| 标题猜想关键词命中 | 1,117 条 |
| 直接 source_url 字段非空 | 10,381 条 |
| source_url 缺失 / 空 | 7,233 条；尽量使用背景链接或问题集入口 |
| 已发布表述总长度 | 5,729,542 个字符（原始 statement 字段） |

## 四组完全相同的表述

GEO-016 / GEOM-016；NT-041 / NUM-003；NT-047 / NUM-007；NT-049 / NUM-005。为保留与上游数据的一一对应关系，正文仍保留全部记录并加重复提示。

这只是文本去重，不会发现“表述不同但数学等价”的记录。例如不同语境下的同一问题、一个猜想的特殊情形、独立分问，都需要领域专家决定是否合并。

# 05 来源状态：15,332 条未完全解决标记

下列状态是 UnsolvedMath v1.8.0 的数据标签。open 与 partially_solved 合计 15,332 条，仅表示该数据快照未标为 solved，不能当作截至检索日仍未解决的独立猜想数。

open / 开放 8,283

partially_solved / 部分解决 7,049

solved / 已解决 2,282

| 题名命中子集 | 条数 |
| --- | --- |
| open | 574 |
| partially_solved | 462 |
| solved | 81 |
| 合计 | 1117 |

## 如何理解“已解决”

在问题汇编中，solved 可能包括被证明、被反驳或得到其他完整解答。它不保证原猜想为真。partially_solved 也可能覆盖部分情形、已报告但未经审定的结果等不同情况，不能从标签反推确定的数学结论。

## 状态证据的限制

17,614 条记录中，13,734 条没有专门的 status_review.evidence 字段；这不等于没有任何参考文献，但说明不能把一个统一的自动标签视作逐条专家认证。另有 538 条记录标记 unrefereed_claim。上游说明指出部分研究摘要和状态整理使用机器辅助；本报告不把这些摘要当作新定理。[S1]

Kourovka 编者明确说明：带有尚未得到数学家确认的 AI 解答的问题，仍应视为未解决。遇到重要结果，应回到原论文、编者更新和正式发表记录核验。[S8]

# 06 学科分布：本语料的覆盖形状

| 类别 | 记录数 | 占主体 |
|---|---|---|
| 数论 / Number Theory | 1,657 | 9.4% |
| 代数 / Algebra | 1,003 | 5.7% |
| 群论 / Group Theory | 2,655 | 15.1% |
| 代数几何 / Algebraic Geometry | 1,159 | 6.6% |
| 几何 / Geometry | 1,854 | 10.5% |
| 拓扑 / Topology | 1,974 | 11.2% |
| 分析 / Analysis | 1,379 | 7.8% |
| 偏微分方程 / Partial Differential Equations | 438 | 2.5% |
| 动力系统 / Dynamical Systems | 888 | 5.0% |
| 组合数学 / Combinatorics | 1,032 | 5.9% |
| 图论 / Graph Theory | 1,181 | 6.7% |
| 概率 / Probability | 733 | 4.2% |
| 集合论 / Set Theory | 221 | 1.3% |
| 逻辑 / Logic | 374 | 2.1% |
| 理论计算机科学 / Computer Science | 554 | 3.1% |
| 数学物理 / Mathematical Physics | 363 | 2.1% |
| 其他 / Miscellaneous | 149 | 0.8% |

学科类别沿用数据集的主分类，每条记录只在本表出现一次。现实中的问题可能跨学科。群论、拓扑与几何占比高，反映所选问题集的来源结构，不代表人类数学猜想的真实全球分布。

# 07 来源集合分布与别名恢复

| 语料所属集合 | 刊入条数 |
| --- | --- |
| Oberwolfach 报告问题汇编 | 5,903 |
| AMR 问题清单汇编 | 4,387 |
| AIM 研讨会问题汇编 | 3,026 |
| Kourovka 群论问题集 | 1,934 |
| Erdős 问题 | 1,221 |
| Open Problem Garden | 377 |
| Kirby 低维拓扑问题 | 366 |
| 独立 / 未分配集合 | 257 |
| Ben Green 问题 | 81 |
| DARPA 挑战 | 17 |
| Hilbert 问题 | 14 |
| Richard Guy 数论问题 | 10 |
| Smale 问题 | 8 |
| 千禧年问题 | 6 |
| Landau 问题 | 4 |
| Hardy-Littlewood 猜想 | 3 |

这些是去归档后保留在各集合下的记录数。例如“Hilbert 的 23 个问题”在本语料中的所属集合只有 14 条，不能据此把历史上的 23 改写成 14；其他条目可能以别名或不同归属出现，也可能未覆盖。

## 别名案例：Collatz 猜想

NT-002 “Collatz Conjecture”在原数据中被归档，duplicate_of 指向 EP-1135。正文的 EP-1135 因而补充该别名。若只按标题搜 conjecture，会漏掉这个著名猜想。这直接说明 1,117 只是题名筛选结果，不是已收集猜想的完整数量。

# 08 阅读方式与重要边界

使用 PDF 书签可直接进入 17 个学科、题名索引或归档索引；全文可搜索英文名称、原始编号和 LaTeX 表达式。附录中的原始题名保留英文，避免自行翻译人名和专业术语造成新别名。

## 每条记录包含什么

依次为：本报告流水号与原始编号、题名、来源状态和集合、完整 statement 字段、可点击来源入口。可以恢复的归档别名也列在正文。编号用于定位，来源链接用于回查，二者都不代表本报告已验证题目的真伪。

## 公式为什么保留 LaTeX

各条表述中的 $...$、\frac、\forall 等以原有 LaTeX 文本显示，而非重新解释或自动改写为公式图片。这样保留括号、量词和上下标结构，便于复制检索。原始文本的断行会合并；排版使用的长短横线和少量兼容字形作显示规范化。PDF 内嵌数据附件保留原始字段。

## 来源入口的三种含义

“来源字段”来自原数据 source_url；“背景链接”是 background 字段中提取的首个网址，未逐条鉴别其优先级；“问题集入口 / 汇编入口”表示没有可直接采用的单条原始网址。链接可能失效，也可能只指向原始文献的一整页或一本问题集。

## 机器可读附件

本 PDF 内嵌 math_catalogue_data.zip，可由支持附件功能的 PDF 阅读器提取。它包含所有 19,031 条记录的选定原始字段、统计清单、17,614 条正文页码映射、归档关系和来源信息。部分浏览器预览不显示附件，但不影响阅读正文。

## 没有被穷尽的部分

付费文献、未数字化资料、未发表猜想、语言覆盖不足的资料、新产生的猜想，以及大型数据库里尚未取得的记录均可能遗漏。历史上已证明或反驳的猜想也没有得到系统全集覆盖。即使把一个语料库完整印出，仍不能声称收齐人类所有猜想。

# 09 著名问题导航与口径示例

以下是阅读导航，不另外增加主体记录数。概述仅用于区分计数类型；严谨条件与完整表述以原始来源为准。[S9-S11]

| 常见名称 | 正文主键 |
| --- | --- |
| 黎曼猜想 | MPP-002 |
| 霍奇猜想 | MPP-006 |
| Birch-Swinnerton-Dyer 猜想 | MPP-005 |
| P 与 NP 问题 | MPP-001 |
| Navier-Stokes 存在性与光滑性 | MPP-004 |
| Yang-Mills 存在性与质量间隙 | MPP-003 |
| Collatz / 3n+1 猜想 | EP-1135 |
| 强哥德巴赫猜想 | LAN-001 |
| 孪生素数猜想 | LAN-002 |
| abc 猜想 | NT-005 |
| 光滑四维庞加莱猜想 | TOP-001 |

黎曼猜想：黎曼 ζ 函数的非平凡零点实部是否均为 1/2。

霍奇猜想：研究特定上同调类与代数循环之间的关系。

Birch-Swinnerton-Dyer 猜想：研究椭圆曲线的有理点秩与 L 函数在特定点的消失阶。

P 与 NP 问题：问 P 是否等于 NP；是开放问题，不能不加说明就称为一个取定方向的猜想。

Navier-Stokes 存在性与光滑性：具体方程与初值条件下的存在性、光滑性问题。

Yang-Mills 存在性与质量间隙：要求构造满足相应条件的量子理论并确立质量间隙。

Collatz / 3n+1 猜想：正整数按奇数 3n+1、偶数除以 2 迭代，猜测最终到达 1；原名在归档别名中。

历史对照：三维庞加莱猜想已经由 Perelman 解决，Clay 官方页有确认。[S10] 这份语料的 6 条 MPP 记录并不是 7 个历史千禧年问题的完整历史登记。

# 10 复现方法、来源与许可

## 统计步骤

1. 下载 problems.json 并保存校验值。2. 以 published=true 筛选正文。3. 检查 id 与 problem_number 的唯一性。4. 对 statement 作 NFKC 规范化并合并空白，比较文本重复。5. 在原始 title 上进行不区分大小写的词边界匹配：conjecture、conjectures、hypothesis、hypotheses。6. 按来源 category、set_id 和 status 汇总。7. 将归档记录的 duplicate_of 映射到已发布主键并生成导航。

## 输入文件 SHA-256

7fcc68b82cb84e39561511a839a7de9893a6c1267cef35602151a73d3bb1e6c9

检索快照为 2026-10-25；主体是下载时标为 v1.8.0 的语料，不是对 2026-10-25 以前每篇论文的实时审查。网页会变化，以上校验值用于识别本次处理的具体输入。

[S1] ulamai. UnsolvedMath 数据说明，v1.8.0，2026-10-16。

https://huggingface.co/datasets/ulamai/UnsolvedMath/blob/main/README.md

[S2] ulamai. problems.json：本次完整下载并程序复核的原始数据。

https://huggingface.co/datasets/ulamai/UnsolvedMath/resolve/main/problems.json

[S3] MathDB. Database；本次网页显示 90,717 problems。API 页说明全库下载属于订阅功能。

https://mathdb.com/database

[S3a] MathDB. API access / dataset downloads。

https://mathdb.com/datasets

[S4] teorth/erdosproblems. 社区库自动生成的统计页。

https://github.com/teorth/erdosproblems/blob/main/STATISTICS.md

[S5] The Formal Conjectures Authors. Formal Conjectures 首页及类别计数。

https://google-deepmind.github.io/formal-conjectures/

[S6] American Institute of Mathematics. Workshop Problem Lists。

https://aimath.org/problemlists/

[S7] Association for Mathematical Research. Problem Lists。

https://amathr.org/problems/

[S8] E. Khukhro 与 V. Mazurov 编. Kourovka Notebook 官方更新页，21 期。

https://kourovkanotebookorg.wordpress.com/

[S9] Clay Mathematics Institute. Millennium Prize Problems。

https://www.claymath.org/millennium-problems/

[S10] Clay Mathematics Institute. Poincaré Conjecture。

https://www.claymath.org/millennium/poincare-conjecture/

[S11] Terence Tao. Almost all orbits of the Collatz map attain almost bounded values。

https://arxiv.org/abs/1909.03562

[S12] Ralf Stephan. Prove or Disprove. 100 Conjectures from the OEIS。

https://arxiv.org/abs/math/0409509

[S13] Creative Commons Attribution 4.0 International。

https://creativecommons.org/licenses/by/4.0/

# 资料使用、来源核验与归属说明

## 数据归属与本报告新增内容

基础语料由 ulamai / UnsolvedMath 发布，数据卡标示 CC BY 4.0。各题原始数学内容归其作者与来源；逐条出处随记录提供。本报告新增中文评估、统计、导航、分组和排版。

原始字段与本报告的分析应当区分：记录标题、问题表述、状态、类别、归档关系来自上游；重新计数、重复文本检测、中文说明与页码导航由本报告生成。

## 使用某条问题开展研究前

先用编号与标题定位条目，再打开链接核对原始文献。尤其需要检查全部假设、量词、维数、底域、范畴、边界条件，以及原文是否将一个命题拆成多个分问。语料里的简写或抽取表述可能省略上下文。

查看状态时，应核查论文版本、正式发表情况、编者更新与后续纠错。一个数据库标签、一份预印本或一个 AI 证明声称，都不能单独代替对数学证明的验证。

## 检索日与数据日不相同

2026-10-25 是本次检索和整理日期。主体采用 v1.8.0 数据快照；不同记录的状态审查可能发生在更早的日期。网页上的最新动态没有被自动覆盖写入全部历史记录。

## 完整性与去重的实际含义

本报告逐条刊入所有 published=true 记录，并保留归档索引和数据附件。完整性针对这一明确的输入版本。17,614 是记录数，17,610 是简单规范化后的表述文本数，两者都不是经专家完成逻辑等价去重的猜想数。

## 引用本报告时建议写法

“该报告基于 UnsolvedMath v1.8.0 整理了 17,614 条已发布数学问题记录，并以 2026-10-25 的公开来源评估可见数据库规模；它不提供人类所有数学猜想的精确总数。”

引用具体数学事实时，请优先引用条目对应的原论文或问题集。本报告是导航与资料汇编，不是对全部原始数学结论的重新审定。

# 11 主体目录：按学科阅读

点击页码可跳转阅读。　正文共有 17,614 条，按学科分组；条目英文表述与公式源码可复制搜索。

数论 / Number Theory 1,657 条 14

代数 / Algebra 1,003 条 200

群论 / Group Theory 2,655 条 319

代数几何 / Algebraic Geometry 1,159 条 620

几何 / Geometry 1,854 条 757

拓扑 / Topology 1,974 条 1003

分析 / Analysis 1,379 条 1265

偏微分方程 / Partial Differential Equations 438 条 1452

动力系统 / Dynamical Systems 888 条 1508

组合数学 / Combinatorics 1,032 条 1620

图论 / Graph Theory 1,181 条 1760

概率 / Probability 733 条 1877

集合论 / Set Theory 221 条 1971

逻辑 / Logic 374 条 1998

理论计算机科学 / Computer Science 554 条 2049

数学物理 / Mathematical Physics 363 条 2113

其他 / Miscellaneous 149 条 2161

附录 A：1,117 条猜想类题名索引 2188

附录 B：1,417 条归档记录索引 2214

# 资料汇编 01 | 数论 / Number Theory

1,657 条记录。O=开放；P=部分解决；S=已解决。状态与英文/LaTeX 表述均沿用来源。

00001 | OWR-1586-003

### $K(\pi,1)$ Theorems without Cyclotomic Class-Number Conditions

S / solved | Oberwolfach 报告问题汇编

Let $k$ be a number field, let $p$ be a prime, let $S$ be a finite set of primes disjoint from $S_p$, and let $T$ have Dirichlet density one. Without assuming $\zeta_p\notin k$ or $p\nmid|\operatorname{Cl}(k)|$, does there exist a finite subset $T_1\subset T$ such that $\operatorname{Spec}(\mathcal O_k)\setminus(S\cup T_1)$ is a $K(\pi,1)$ for $p$?

来源字段：https://doi.org/10.4171/owr/2007/30

00002 | OWR-15170-014

### $p$-Adic Definability Rings with Finite Additive Descriptions

P / partially_solved | Oberwolfach 报告问题汇编

For a prime $p$, with $\gamma_p(x)=p^{-1}(x^p-x)/((x^p-x)^2-1)$ and $\Gamma_p(\mathbb Q)=\{a(1+pb)^{-1}:a,b\in\mathbb Z_{(p)}[\gamma_p(\mathbb Q)]\}$, does there exist $n_p$ such that $\Gamma_p(\mathbb Q)=\{a(1+pb)^{-1}:a,b\in\sum_{i=1}^{n_p}\gamma_p(\mathbb Q)\}$?

来源字段：https://doi.org/10.4171/owr/2016/49

00003 | OWR-1703867-005

### $p$-Adic Regulators for Triple-Product $L$-Functions

P / partially_solved | Oberwolfach 报告问题汇编

In the indefinite triple-product setting, can one formulate and prove a Bloch-Kato-style $p$-adic regulator law governing the behavior of $L_p(f,g,h)$ near $(2,1,1)$?

来源字段：https://doi.org/10.4171/owr/2020/20

00004 | OWR-4416-007

### $p$-Adic Representations and $L$-Functions in Knot Theory

O / open | Oberwolfach 报告问题汇编

Can analogues of the $p$-adic representation $\rho_p$ and the algebraic $p$-adic $L$-function $L_p(T)$ be constructed in knot theory?

来源字段：https://doi.org/10.4171/owr/2010/35

00005 | OWR-1318-004

### $p$-Adic Special-Value Formulas over Totally Real Fields

P / partially_solved | Oberwolfach 报告问题汇编

For a totally real number field $k$, an odd integer $n\geq2$, and any prime $p$, is $$L_p(n,\omega_p^{1-n},k)D_k^{1/2,p}=q_p(n,k)\operatorname{Eul}_p(n,k)R_{n,p}(k)$$ for some $q_p(n,k)\in\mathbb Q^*$, with $q_p(n,k)=q(n,k)$ from Borel's complex formula? Are $L_p(n,\omega_p^{1-n},k)$ and $R_{n,p}(k)$ nonzero?

来源字段：https://doi.org/10.4171/owr/2006/32

00006 | OPG-37396

### 3 is a primitive root modulo primes of the form 16 q^4 + 1, where q>3 is prime

P / partially_solved | Open Problem Garden

Conjecture $3\sim$ is a primitive root modulo $\sim p$ for all primes $\sim p=16\cdot q^4+1$, where $\sim q>3$ is prime.

背景链接：http://www.openproblemgarden.org/op/primes_p_such_that_3_is_a_primitive_root_modulo_p

00007 | AMR-108-0049

### 7.1 (Long) - Principal and Euclidean rings of integers from totally real polynomials

O / open | AMR 问题清单汇编

Let $f(x)\in\mathbb{Z}[x]$ be irreducible over $\mathbb{Q}$ with all roots real, let $f(\alpha)=0$, let $k=\mathbb{Q}(\alpha)$, and let $\mathcal{O}_k$ be its ring of integers. How often is $\mathcal{O}_k$ a principal ideal domain; does this happen infinitely often? How often is $\mathcal{O}_k$ Euclidean with respect to the standard norm?

背景链接：https://arxiv.org/pdf/1512.04620#page=5

00008 | AMR-108-0050

### 7.2 (Long) - Clique numbers in unit- and prime-difference graphs

S / solved | AMR 问题清单汇编

For $\mathcal{O}_k$ as in item 7.1, let $\Gamma_{\mathrm{unit}}$ have vertex set $\mathcal{O}_k$, joining two elements when their difference is a unit, and let $\Gamma_{\mathrm{prime}}$ also join elements whose difference is a prime. The source records $$|\operatorname{clique}(\Gamma_{\mathrm{prime}})|\leq|\operatorname{clique}(\Gamma_{\mathrm{unit}})|\min_{I\ \mathrm{prime}}|\mathcal{O}_k/I|.$$ Is this inequality an equality?

背景链接：https://arxiv.org/pdf/1512.04620#page=6

00009 | AMR-108-0051

### 7.3 (Manning) - Number fields as trace fields

P / partially_solved | AMR 问题清单汇编

If $k$ is a number field that is not totally real, is there a hyperbolic $3$-manifold with trace field $k$?

背景链接：https://arxiv.org/pdf/1512.04620#page=6

00010 | AIM-ANALYTIC_NUMBER_THEORY-0099

### A Bailey-Chebyshev-Bloch dictionary and terminal-choice blindness

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.4.** Relate Bailey pair techniques with Bloch group techniques. In particular, develop a dictionary. Perhaps this might might help with the computation of torsion. Similarly, connect orthogonal polynomials to Bloch group techniques.

来源字段：https://aimath.org/WWN/mockmodular/mockmodular.pdf

00011 | AIM-ANALYTIC_NUMBER_THEORY-0028

### A binary strongly stationary compatibility model for the unipotent torus

P / partially_solved | AIM 研讨会问题汇编

Can a Liouville system be isomorphic to $(\mathbb T^2,m_{\mathbb T^2},T)$, where \[ T(x,y)=(x+y,y)\pmod 1? \]

The record is internally coherent. The transpose convention $T(s,t)=(s,t+s)$ occurs in Frantzikinakis--Host; exchanging the two coordinates conjugates it to the AIM convention.

There is one terminological ambiguity worth preserving. A **Liouville system** is a Furstenberg system obtained from the Liouville sequence $\lambda\in\{-1,1\}^{\mathbb Z}$ by a subsequential weak-star limit of orbit averages. The averages may be Cesàro or logarithmic, and the two resulting systems need not agree. Frantzikinakis's original unqualified formulation is Cesàro, while much of the strongest unconditional structure theory is logarithmic. The AIM record does not specify the averaging convention. The construction below is a statement about the proposed target and is relevant to either convention; comparisons with proved Liouville correlations are explicitly labeled logarithmic.

来源字段：http://aimpl.org/sarnakconjecture/3/

00012 | OWR-12866-010

### A BMO Approach to the Lindelöf Hypothesis

O / open | Oberwolfach 报告问题汇编

Can a BMO theory or another functional-analytic method be used to prove the Lindelöf hypothesis?

来源字段：https://doi.org/10.4171/owr/2014/06

00013 | AIM-ALGEBRAIC_NUMBER_THEORY-0086

### A Brauer-free finite local sieve for Fano complete intersections

P / partially_solved | AIM 研讨会问题汇编

**Recovered Question 2.** If \(X\) is a smooth projective geometrically rationally connected variety over a number field \(k\), is \[ \overline{X(k)}=X(\mathbf A_k)^{\operatorname{Br}X}? \] Here the closure is taken in the adelic topology.

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

00014 | AIM-ALGEBRAIC_NUMBER_THEORY-0002

### A Brauer-transfer sieve and reduction of the cubic route

P / partially_solved | AIM 研讨会问题汇编

Is there a \(\mathbb Q\)-irrational del Pezzo surface \(X\) which is \(\mathbb Q\)-unirational for which \(\operatorname{Sym}^dX\) is \(\mathbb Q\)-rational for some \(d>1\)?

来源字段：http://aimpl.org/degreedsurface/1/

00015 | AIM-ALGEBRAIC_NUMBER_THEORY-0069

### A Catalan correspondence criterion for middle Tate tensors

P / partially_solved | AIM 研讨会问题汇编

**6. (J. Getz)** Take a product of modular curves \(X=\prod_i X_i\). Many smaller products embed diagonally into \(X\). How much of the middle cohomology can be accounted for by these constructions?

**Comments.** 1) Ribet considered the product of two modular curves \(X=X_1\times X_2\); here \(NS(X)\) is just the quotient of \(\operatorname{Hom}(J(X_1),J(X_2))\) by the pullbacks of divisors on the two factors. He shows it is necessary to consider twisting correspondences, in addition to the Hecke correspondences, to complete the Tate classes. Are there similarly interesting, yet explicit, correspondences for \(X\) when we consider higher dimensional factors \(X_i\)? (J. Getz, D. Ramakrishnan, D. Ulmer)

2) For a product of four Shimura curves, there are interesting Tate classes, which are also Hodge classes, which we don't know how to represent by cycles. (D. Ramakrishnan)

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

00016 | AIM-ALGEBRAIC_NUMBER_THEORY-0106

### A certified orbit-lattice algorithm for M_3(Z)

S / solved | AIM 研讨会问题汇编

**Question 21.** Let \(A/\mathbf Z\) be an algebra with \({\rm rk}_{\mathbf Z}A=9\) given by its multiplication table. Suppose you know that \[ A\simeq M_3(\mathbf Z); \] find an algorithm which gives an explicit isomorphism. (Stoll)

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

00017 | AIM-ALGEBRAIC_NUMBER_THEORY-0080

### A classwise CM implication and a one-product coniveau certificate

P / partially_solved | AIM 研讨会问题汇编

7. (W. Raskind) In the situations where the Hodge Conjecture implies the Tate Conjecture, is it also true that the generalized Hodge Conjecture implies the generalized Tate Conjecture, e.g., for CM abelian varieties?

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

00018 | AIM-ANALYTIC_NUMBER_THEORY-0076

### A compensated-Laplace window reduction for fractional zeta moments

P / partially_solved | AIM 研讨会问题汇编

**Problem.** Evaluate fractional moments of zeta.

**Remark.** There are conjectures on lower order terms, but there is no analogue yet of Conrey--Keating.

来源字段：http://aimpl.org/zetamoments/8/

00019 | AIM-ANALYTIC_NUMBER_THEORY-0012

### A conductor-aware four-stratum reduction for the double level moment

P / partially_solved | AIM 研讨会问题汇编

**Asymptotic for second moment of $\mathrm{GL}_2\times\mathrm{GL}_2$ in level aspect.** Let $\mathcal B_k(p)$ be a basis for holomorphic or Hecke--Maass forms for $\Gamma_0(p)\subset SL_2(\mathbb Z)$ of weight $k$. Obtain an asymptotic with a power-saving error term for \[ \sum_{f\in\mathcal B_k(p)}\sum_{g\in\mathcal B_\ell(q)} |L(1/2,f\times g)|^2, \] as the levels $p$ and $q$ become larger, the weights $k$ and $\ell$ being fixed.

来源字段：http://aimpl.org/deltasubconvex2/1/

00020 | AIM-ANALYTIC_NUMBER_THEORY-0017

### A conductor-normalized hybrid phase diagram

P / partially_solved | AIM 研讨会问题汇编

**Strong hybrid bounds for \(L(1/2,f\times\chi)\).** Let \(f\) be a holomorphic or Hecke--Maass cusp form of level \(p\), and \(\chi\) be a primitive Dirchlet character of level \(q\). Conrey--Iwaniec obtained a Weyl type subconvexity bound for \(L(1/2,f\times\chi)\) when \(p=q\), while Khan has obtained a Burgess-type bound when \(p\asymp q\), and Yang has proved a Burgess-type bound over number fields in a similar range. Prove strong hybrid bounds for other ranges of \(p\) and \(q\).

来源字段：http://aimpl.org/deltasubconvex2/1/

00021 | AIM-ANALYTIC_NUMBER_THEORY-0016

### A conductor-safe audit for the number-field delta method

S / solved | AIM 研讨会问题汇编

**Delta method over Number fields.** Design a delta symbol method over number fields in order to generalize the classical techniques to the number fields setting.

来源字段：http://aimpl.org/deltasubconvex2/1/

00022 | AIM-ALGEBRAIC_NUMBER_THEORY-0042

### A CRT lower bound for fixed-q higher divisor correlations

P / partially_solved | AIM 研讨会问题汇编

Inspection of the typeset AIM PDF confirms that the displayed expression is only \[ \sum_f d_k(f)d_k(f+h), \tag{1.1} \] followed by “where \(f\) ranges over degree \(n\) polynomials” [AIM14]. There is no \(\mathcal M_n\) under the summation sign. The PDF also does not define \(d_k\), say that \(f\) is monic, or exclude \(h=0\).

The most natural recovered formulation is nevertheless the one used by the question's author immediately afterward [ABSR15]: \[ S_k(n;h):=\sum_{f\in\mathcal M_n}d_k(f)d_k(f+h), \tag{1.2} \] where \(\mathcal M_n\) is the set of monic degree-\(n\) polynomials and \(d_k(f)\) counts ordered factorizations of the monic polynomial \(f\) into \(k\) monic factors, factors equal to \(1\) being allowed. One takes \(0\ne h\in\mathbb F_q[T]\) fixed. For all sufficiently large \(n\), \(\deg h<n\), so \(f+h\) is again monic of degree \(n\).

This reconstruction is labeled rather than silently imposed. Proposition 5.3 below gives the exact conversion if “degree \(n\) polynomials” literally means every leading coefficient. Proposition 5.2 treats \(h=0\), which is a genuinely different asymptotic regime.

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

00023 | AIM-ALGEBRAIC_NUMBER_THEORY-0051

### A cubefree local model for trigonal discriminants

P / partially_solved | AIM 研讨会问题汇编

**Question 21 (Jordan Ellenberg).** In what ways are discriminants of random trigonal curves like random polynomials?

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

00024 | AIM-ALGEBRAIC_NUMBER_THEORY-0066

### A d=9 exception and a CM-synchronization reduction

P / partially_solved | AIM 研讨会问题汇编

**Problem 5 (D. Ulmer).** Consider the curve \[ C_{d,a}: y^d=x(x-1)(x-a) \] defined over \(\mathbf C\) (though \(\overline{\mathbf Q}\) or \(\mathbf F_q\) will also work). For which values of \(a\) does it happen that this curve has CM? (More precisely, for which values of \(a\) does the Jacobian of \(C_{d,a}\) have endomorphism algebra of dimension \(2g\) over \(\mathbf Q\), where \(g=g(C_{a,d})=d-1\) is the genus?) It happens (for dull reasons) when \[ a=-1,\quad \frac12,\quad 2,\quad \zeta _6,\quad \zeta _6^{-1}. \] For a fixed \(d>7\), there are only finitely many values of \(a\) for which it happens. Are there infinitely many \(d\) such that there is another value of \(a\) (besides the trivial ones listed above) for which \(C_{d,a}\) has CM?

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

00025 | AIM-ANALYTIC_NUMBER_THEORY-0082

### A differential certificate and first-normal-jet obstruction for elliptic reduction

P / partially_solved | AIM 研讨会问题汇编

别名：A normal-symbol criterion for lower-dimensional modular reduction [AIM-PHYSICS-0015]

When can modular forms on Calabi Yau 3-folds moduli spaces be reduced to lower dimensional forms?

来源字段：http://aimpl.org/gromwitnumthry/3/

00026 | AIM-ALGEBRAIC_NUMBER_THEORY-0063

### A dihedral projector test for explicit exotic divisor classes

P / partially_solved | AIM 研讨会问题汇编

2. (D. Ramakrishnan) Let X be a Hilbert modular surface over Q. All of the Tate classes that arise from modular curves on X are defined over Qab; however, there also exist Tate classes which are algebraic, but not defined over Qab. (In fact, they are defined over dihedral extensions of Q.) We know the algebraicity of these classes, but do not know specific representatives. Comments: 1) One possible suggestion for such representatives could be non-congruence curves on X. (D. Ramakrishnan) 2) If we can find representatives, we can intersect them with the modular curves: this may be a way to find interesting non-Heegner points on the modular curves. (D. Ramakrishnan)

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

00027 | AIM-ALGEBRAIC_NUMBER_THEORY-0075

### A dimension-two counterexample with an explicit Galois projector

S / solved | AIM 研讨会问题汇编

**12. (W. Raskind)** Take an abelian variety over a finite extension \(K\) of \(\mathbf Q_p\) with multiplicative reduction. Show the isomorphism

\[ \operatorname{End}(A)\otimes \mathbf Q_p \xrightarrow{\ \sim\ } \operatorname{End}(V_p(A))^{G_K}. \]

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

00028 | OWR-12866-017

### A Dirichlet Series with Exactly One Zero

O / open | Oberwolfach 报告问题汇编

Does there exist an ordinary Dirichlet series that converges in an open half-plane and has exactly one zero there?

来源字段：https://doi.org/10.4171/owr/2014/06

00029 | OPG-819

### A discrete iteration related to Pierce expansions

P / partially_solved | Open Problem Garden

Conjecture Let $a > b > 0$ be integers. Set $b_1 = b$ and $b_{i+1} = {a \bmod {b_i}}$ for $i \geq 0$. Eventually we have $b_{n+1} = 0$; put $P(a,b) = n$.

Example: $P(35, 22) = 7$, since $b_1 = 22$, $b_2 = 13$, $b_3 = 9$, $b_4 = 8$, $b_5 = 3$, $b_6 = 2$, $b_7 = 1$, $b_8 = 0$.

Prove or disprove: $P(a,b) = O((\log a)^2)$.

背景链接：http://www.openproblemgarden.org/op/a_discrete_iteration_related_to_pierce_expansions

00030 | AIM-ALGEBRAIC_NUMBER_THEORY-0011

### A fibered transfer from density constructions to maps

P / partially_solved | AIM 研讨会问题汇编

Given some specific geometric construction showing that $d\in\delta(X/k)$, can one use it to construct a dominant rational map $X\dashrightarrow\mathbb P^2$ of degree $nd$, for some small $n$?

来源字段：http://aimpl.org/degreedsurface/1/

00031 | AIM-ALGEBRAIC_NUMBER_THEORY-0116

### A fiberwise harmonic bridge on the quintic del Pezzo torsor

P / partially_solved | AIM 研讨会问题汇编

**Question 31.** Harmonic analysis and nonabelian torsors: Takloo-Bighash: Do these give you methods to find rational points? Hassett: Is there a harmonic analysis argument to count points on the quintic del Pezzo surface? Wooley: no examples known where you can combine harmonic analysis with torsors.

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

00032 | AIM-ALGEBRAIC_NUMBER_THEORY-0092

### A finite binary criterion for Brauer-compatible lifts to an Enriques K3 cover

P / partially_solved | AIM 研讨会问题汇编

Let \(X\) [be] an Enriques surface over a number field \(K\), and let \(p:Y\to X\) be the double cover, where \(Y\) is a K3 surface. Let \(X(\mathbb A_K)^{\operatorname{Br}}\) be the Brauer--Manin set. Take \(m=(m_v)\in X(\mathbb A_K)^{\operatorname{Br}}\), and lift it to an adelic point \(P=(P_v)\) on \(Y\). Under what assumptions on \(m\) will it be liftable to a point in \(Y(\mathbb A_K)^{\operatorname{Br}}\), or at least in \(Y(\mathbb A_K)^{\operatorname{Br}_1}\)?

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

00033 | AIM-ANALYTIC_NUMBER_THEORY-0064

### A finite local prime model for the Bogomolny--Keating Type-II input

P / partially_solved | AIM 研讨会问题汇编

Is there a random model on the primes that leads to the Type II calculations of Bogomolny and Keating (e.g. use Hardy-Littlewood for the probability that both $n$ and $n+h$ are prime)?

来源字段：http://aimpl.org/zetamoments/6/

00034 | AIM-ANALYTIC_NUMBER_THEORY-0005

### A finite-field dictionary between the Nelson and Sharma kernels

P / partially_solved | AIM 研讨会问题汇编

**Character sums appearing in the works of Nelson, and Sharma.** Relate or compare the exponential sums appearing in the work of Paul Nelson, and the work of Prahlad Sharma in the proof for level aspect subconvexity bound for \(\mathrm{GL}(1)\) twists of \(\mathrm{GL}(3)\times \mathrm{GL}(2)\) \(L\)-functions.

来源字段：http://aimpl.org/deltasubconvex2/1/

00035 | AIM-ALGEBRAIC_NUMBER_THEORY-0038

### A finite-order marked-moduli certificate for the fixed-field Poisson conjecture

P / partially_solved | AIM 研讨会问题汇编

**Question 8 (Kiran Kedlaya).** Fix \(q\). Describe (conjecturally) the distribution of the number of \(\mathbb F_q\) points on curves of genus \(g\) as \(g\) tends to \(\infty\).

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

00036 | AIM-ALGEBRAIC_NUMBER_THEORY-0055

### A finite-period Levinson-Speiser identity for function-field numerators

P / partially_solved | AIM 研讨会问题汇编

**Question 25 (Brian Conrey).** Carry out (the other) Levinson's method for function fields.

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

00037 | AIM-ANALYTIC_NUMBER_THEORY-0093

### A finite-theta certificate for sum-side modularity

P / partially_solved | AIM 研讨会问题汇编

Find a method to prove modularity directly from the sum (q-hypergeometric) expansion. For motivation, consider \[ G(q)=\sum_{n=0}^{\infty}\frac{q^{n^2}}{(q;q)_n},\qquad H(q)=\sum_{n=0}^{\infty}\frac{q^{n^2+n}}{(q;q)_n}. \] The Rogers--Ramanujan product identities make the modularity of \(q^{-1/60}G(q)\) and \(q^{11/60}H(q)\) apparent. How can it be seen from the left-hand sides?

来源字段：https://aimath.org/WWN/mockmodular/mockmodular.pdf

00038 | AIM-ALGEBRAIC_NUMBER_THEORY-0107

### A fixed-real-subfield criterion for a complex archimedean ball

P / partially_solved | AIM 研讨会问题汇编

Let \(k\) be a number field, \(k\not\subset\mathbb R\). Construct an algebraic set \(S\subset k^n\) for some \(n\) such that the projection onto one of the coordinates is exactly the set of elements of \(k\) with \(|k|\leq 1\) for some (archimedean) absolute value \(|\,\cdot\,|\) of \(k\).

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

00039 | AIM-ANALYTIC_NUMBER_THEORY-0095

### A forced mixing invariant for the Rogers-Ramanujan modular matrices

P / partially_solved | AIM 研讨会问题汇编

What intrinsic information is carried by diagonalizing the matrices $\rho(T)$ and $\rho(S)$ of a vector-valued modular form, especially for the Rogers--Ramanujan vector, where $T:\tau\mapsto\tau+1$ is diagonal in the $q$-expansion basis but $S:\tau\mapsto-1/\tau$ is not?

来源字段：https://aimath.org/WWN/mockmodular/mockmodular.pdf

00040 | AIM-ANALYTIC_NUMBER_THEORY-0069

### A four-cycle benchmark and a Type-II reduction for zeta four-correlation

P / partially_solved | AIM 研讨会问题汇编

Compute the $4$-point correlation function rigorously under suitable hypotheses (here, Type II terms come in).

来源字段：http://aimpl.org/zetamoments/7/

00041 | AIM-ALGEBRAIC_NUMBER_THEORY-0037

### A genus-ordering obstruction for cyclic quartic covers

P / partially_solved | AIM 研讨会问题汇编

Question 7. Does (A) fail for Z/4Z function field extensions, counted by genus?

It is Question 7 in the AIM problem-session document for the January 2014 workshop *Arithmetic statistics over finite fields and function fields*. The surrounding questions ask which ways of counting field extensions satisfy Principles (A) and (B), whether cyclic triple covers recover them when counted by genus, and whether conditioning on ramification helps.

There is a source defect that matters mathematically: neither the three-page AIM problem document nor the workshop report defines the labels (A) and (B). I therefore do **not** assign the letter (A) to a particular assertion. The strongest source-supported reconstruction comes from Melanie Matchett Wood's paper cited by the workshop report. It foregrounds exactly two properties:

- the **conditional Chebotarev property**: conditional on a place being unramified, its Frobenius/splitting distribution has the usual Chebotarev proportions; - the **local-independence property**: prescribed local behaviors at distinct fixed places are asymptotically independent.

Wood proves that both behave well for abelian number-field extensions ordered by conductor and can fail under discriminant ordering. Her non-independence result for cyclic groups of order $p^2$ makes local independence the especially plausible intended reading of Question 7, but the letter assignment remains unverified. This report uses neutral terminology throughout.

The main arithmetic scope below is the tame Kummer regime

\[ q\equiv1\pmod4,\qquad K=\mathbb F_q(x), \]

with $q$ fixed and geometrically connected cyclic $4$-covers of $\mathbb P^1$ ordered by source genus $g$. The geometric theorem remains valid over an algebraically closed field of characteristic different from $2$. Characteristic $2$, $q\equiv3\pmod4$, constant-field extensions, and alternative weights are explicitly excluded from the arithmetic model.

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

00042 | AIM-ALGEBRAIC_NUMBER_THEORY-0056

### A Hardy-space cyclicity criterion for curve RH

P / partially_solved | AIM 研讨会问题汇编

**Question 26 (Brian Conrey).** Do the analogue of Nymon-Barling criterion for RH over function fields (leading to a different proof of RH).

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

00043 | AIM-ANALYTIC_NUMBER_THEORY-0014

### A harmonic family bound and the exact individual-extraction barrier

P / partially_solved | AIM 研讨会问题汇编

Shifted convolution sum for $\mathrm{GL}_3\times\mathrm{GL}_2$ in level aspect. Let $\pi$ be a fixed Hecke--Maass cusp form for $SL_3(\mathbb Z)$ with Fourier coefficients $\{\lambda_\pi(r,n)\}$ and $f$ be a holomorphic for Hecke Maass cusp form for $\Gamma_0(M)$ with Fourier coefficients $\{\lambda_f(n)\}$. Obtain a non-trivial bound for \[ \sum_{n<N}\lambda_\pi(r,n)\lambda_f(n+1) \] in the convexity range $N\sim M^{3/2}$.

来源字段：http://aimpl.org/deltasubconvex2/1/

00044 | AIM-ALGEBRAIC_NUMBER_THEORY-0118

### A height-index transfer for a conic bundle over a rank-one elliptic curve

P / partially_solved | AIM 研讨会问题汇编

**Problem/Question 33 (Poonen).** Instead of counting points of bounded height on a variety, count points of bounded height in a Diophantine set. (I.e., counting points in a base the fibre above which has a rational point.) What rates of growth can you get? Example: conic bundle over an elliptic curve. Heuristically, it appears rate of growth can be \(\log\log B\), whereas for varieties it always turns out to be \(cB^\alpha(\log B)^\beta\).

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

00045 | AIM-ANALYTIC_NUMBER_THEORY-0026

### A linear-complexity obstruction and a rank-energy coloring criterion

S / solved | AIM 研讨会问题汇编

Let $f$ be a multiplicative function with countable range. Can we find a coloring of that range with (say) two colors, such that the number of “sign patterns” of $f$ (in terms of the colors) grows super-linearly?

来源字段：http://aimpl.org/sarnakconjecture/2/

00046 | AIM-ANALYTIC_NUMBER_THEORY-0081

### A local-curve criterion for noncompact elliptic genera

P / partially_solved | AIM 研讨会问题汇编

别名：Compact-fixed-locus definitions and pole obstructions for noncompact elliptic genera [AIM-PHYSICS-0014]

**When can the elliptic genus be defined for noncompact Calabi--Yau manifolds?**

来源字段：http://aimpl.org/gromwitnumthry/3/

00047 | AIM-ALGEBRAIC_NUMBER_THEORY-0113

### A local-defect lower bound for the 3-Selmer group over cyclic cubic fields

P / partially_solved | AIM 研讨会问题汇编

Let \(E/\mathbb Q\) be an elliptic curve, let \(P\in E(\mathbb Q)[3]\) with \(P\ne O\), and suppose that \({\rm rk}\,E(\mathbb Q)=0\). Let \(K/\mathbb Q\) be cyclic of degree \(3\), of conductor \(f\), and let \(\nu=\nu(f)\) be the number of distinct prime divisors of \(f\). Is \[ 3^{\nu-1}\mid \#\operatorname{Sel}_{3}(E/K)? \]

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

00048 | AIM-ANALYTIC_NUMBER_THEORY-0116

### A moment ladder and tuple-error barrier for a converse from bounded gaps

P / partially_solved | AIM 研讨会问题汇编

**17. Elliott--Halberstam from bounded gaps.**

During the workshop, much attention focused on how to prove \(\mathrm{EH}(\theta)\) with \(\theta>1/2\), so we can deduce the existence of bounded gaps between primes. V. Blomer asked whether it is possible to go the opposite way and derive an improvement on the Bombieri--Vinogradov theorem from a “strong” quantitative result on bounded gaps between primes.

R. C. Vaughan commented that such an improvement would encode so much information about \(L\)-functions that the hypothesis would have to be extremely strong.

K. Soundararajan added that the “strong quantitative result” could be as strong as the Hardy--Littlewood \(k\)-tuple conjecture with a sharp error term, say \(O(N^{1/2+\epsilon})\).

来源字段：https://aimath.org/WWN/primegaps/WorkshopProblems.pdf

00049 | AIM-ANALYTIC_NUMBER_THEORY-0042

### A Möbius--Jewett--Krieger formulation

P / partially_solved | AIM 研讨会问题汇编

What is the M\"obius disjointness variant of the Jewett-Krieger theorem?

来源字段：http://aimpl.org/sarnakconjecture/7/

00050 | AIM-ANALYTIC_NUMBER_THEORY-0107

### A multiplier-repetition obstruction for long prime chains

P / partially_solved | AIM 研讨会问题汇编

Let \(x\) be large and let \(p_1,\ldots,p_k\leq x\) be primes such that, for every \(j=2,\ldots,k\), \[ p_j=m_jp_{j-1}+1\qquad (m_j\in\mathbb N). \] Obviously \(k=O(\log x)\). Prove or disprove that \(k=o(\log x)\).

Show that there are only finitely many \(n\) such that \(n-2^k\) is prime for every positive integer \(k\) with \(2^k<n\).

The recovered main problem is:

The same item records Montgomery's comparison with an Erdős problem:

It then says that Vaughan recalled Erdős believing that \(105\) is the largest such \(n\).

- Plain-text strings `p1`, `pk`, `pj`, `mj`, and `pj-1` were restored as \(p_1,p_k,p_j,m_j,p_{j-1}\). - In the PDF, the exponent \(k\) is set on a separate extraction line after each `2`. Thus the two corrupted strings `n - 2k` and `2 k < n` are \(n-2^k\) and \(2^k<n\), respectively. - The broken name `Erd ¨ os` was restored to Erdős. - The PDF literally says “integers \(k\),” but negative \(k\) would make \(n-2^k\) nonintegral. The standard positive-\(k\) reading is confirmed by Vaughan's 1973 formulation \(1\leq k\leq\log_2 n\) [Vaughan1973]. This is a scope clarification, not a change to the prime-chain problem.

For example, the displayed exponents make the recollection about \(105\) coherent: \(105-2^k\) for \(1\leq k\leq6\) is \(103,101,97,89,73,41\), all prime.

来源字段：https://aimath.org/WWN/primegaps/WorkshopProblems.pdf

00051 | AIM-ALGEBRAIC_NUMBER_THEORY-0083

### A necessary Bockstein fingerprint for a bounded Weil-étale topology on Spec Z

P / partially_solved | AIM 研讨会问题汇编

**20. (M. Flach)** Find a natural definition of the Weil-\'etale topology so that the cohomology of \(Z\) vanishes in degrees greater than 3.

**Comments:** (1) The current definition only vanishes for odd degrees and gives infinitely generated groups in even degrees. (M. Flach) (2) It it not clear how hard this problem is. A better definition might be just around the corner or the only solution might be to truncate the cohomology complex. (M. Flach)

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

00052 | AIM-ALGEBRAIC_NUMBER_THEORY-0072

### A new-Lefschetz defect reduction for rational Hodge pairings

P / partially_solved | AIM 研讨会问题汇编

**9. (J. Milne)** Let \(A\) be a CM abelian variety over \(\overline{\mathbf Q}\) and let \(c\) be an absolute Hodge class. Reduce \(A\) to \(A_0\) over \(\mathbf F_p\) and take a Lefschetz class \(x\) on \(A_0\) of complementary dimension. Then \(x\cdot\bar c\in\mathbf Q_\ell\); is it in \(\mathbf Q\) and independent of \(\ell\)?

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

00053 | AIM-ALGEBRAIC_NUMBER_THEORY-0114

### A PAC counterexample and the projectivity boundary

S / solved | AIM 研讨会问题汇编

**Recovered Question 29.** Suppose every smooth projective geometrically connected $K$-variety $X$ satisfying $H^i(X,\mathcal O_X)=0$ for all $i>0$ has a $K$-rational point. Must $G_K$ be topologically generated by one element?

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

00054 | AIM-ANALYTIC_NUMBER_THEORY-0051

### A parity-projector certificate for fixed-field hyperelliptic moments

P / partially_solved | AIM 研讨会问题汇编

Develop a heuristic for moments of L-functions over function fields. Can one carry out the analogous Conrey-Keating calculation for high moments (fixed $q$, large degree $g$)?

来源字段：http://aimpl.org/zetamoments/2/

00055 | AIM-ANALYTIC_NUMBER_THEORY-0019

### A power-saving GL(4) short second moment

P / partially_solved | AIM 研讨会问题汇编

**Bound for short second moment of \(\mathrm{GL}_4\) \(L\)-functions.** Let \(f\) be a fixed Hecke--Maass cusp form for \(\mathrm{GL}_4(\mathbb Z)\), and let \(T\) and \(\Delta\) be parameters. Prove a non-trivial bound for \[ \int_T^{T+\Delta}\left|L\!\left(\frac12+it,f\right)\right|^2\,dt \] for \(\Delta=T^\delta\) with some \(\delta<1\).

来源字段：http://aimpl.org/deltasubconvex2/1/

00056 | AIM-ANALYTIC_NUMBER_THEORY-0063

### A quantitative Möbius-ratios-prime bridge

P / partially_solved | AIM 研讨会问题汇编

Is there some version of Möbius randomness that would lead to similar calculations? And vice versa, can one go back to primes e.g. from analogs for ratios conjectures?

来源字段：http://aimpl.org/zetamoments/6/

00057 | AIM-ANALYTIC_NUMBER_THEORY-0022

### A quantitative negative-rectangle obstruction to bounded fixed-shift Liouville correlations

P / partially_solved | AIM 研讨会问题汇编

\[ \left|\sum_{n\le X_k}\lambda(n)\lambda(n+h)\right|\longrightarrow\infty. \]

It also suggests variants for multiplicative \(f:\mathbb N\to\{\pm1\}\), notes conditional arguments from four-point Chowla or arbitrarily long constant-sign strings, asks whether a Sudoku-type argument is possible, and observes that allowing the shift to vary should be easier. Here

\[ \lambda(n)=(-1)^{\Omega(n)} \]

is the Liouville function. The repository record is internally unambiguous.

\[ b_h(n):=\lambda(n)\lambda(n+h),\qquad C_h(X):=\sum_{n\le X}b_h(n). \]

The question is exactly whether \(\sup_X|C_h(X)|=\infty\). If the supremum is infinite, selecting successive record values gives the requested sequence \(X_k\).

来源字段：http://aimpl.org/sarnakconjecture/1/

00058 | AIM-ANALYTIC_NUMBER_THEORY-0113

### A quantitative short-interval transfer for small values of ap-q

P / partially_solved | AIM 研讨会问题汇编

14. Small values of linear forms at prime arguments. A. Balog proposed the following problem: Show that the difference \(|ap-q|\), where \(p,q\) are primes and \(a>1\) is a fixed integer, can be small. More generally, consider the case when \(a\) is real, \(a,1\).”

“More generally, consider the case when \(a\) is real, \(a\ne1\).

来源字段：https://aimath.org/WWN/primegaps/WorkshopProblems.pdf

00059 | AIM-ANALYTIC_NUMBER_THEORY-0034

### A quantized gap for the Liouville Goldbach correlation

P / partially_solved | AIM 研讨会问题汇编

Show that \[ \left|\sum_{1\le n<N}\lambda(n)\lambda(N-n)\right|<N-1. \]

Here \(\lambda(n)=(-1)^{\Omega(n)}\) is the Liouville function. Write \[ \mathcal L_\lambda(N):=\sum_{1\le n<N}\lambda(n)\lambda(N-n). \]

Mangerel's published paper identifies it as AIM Problem 5.1 and records the intended reading as “for every sufficiently large \(N\).” The natural domain used here is integers \(N\ge2\). This convention matters: at \(N=1\) the sum is empty and the asserted strict inequality would read \(0<0\).

来源字段：http://aimpl.org/sarnakconjecture/5/

00060 | AIM-ALGEBRAIC_NUMBER_THEORY-0050

### A ramification-bias criterion for predicted mean point counts

P / partially_solved | AIM 研讨会问题汇编

**Question 20 (Melanie Matchett Wood). When do (A) and (B) predict an interesting average value of \(\#C(\mathbb F_q)\)?**

Thus the verified typeset question is

The three-page problem sheet does **not** define Principles (A) and (B). The workshop report says that Wood's opening talk asked when point counts in a family can be modeled by a sum of independent identically distributed fiber variables. Questions 3--7 also connect (A), (B), automorphism weights, local behavior of field extensions, the counting invariant, and independence. Wood's contemporaneous papers make the following a natural contextual reconstruction, but it is not textually verified as the workshop's exact labeling:

- **Reconstructed (A), local marginal/mass principle:** at a rational place, a local degree-\(n\) algebra \(L\) has relative mass \(1/(|\operatorname{Aut}L|\,|D_L|)\), or the corresponding structured centralizer-weighted version when a monodromy group is fixed. - **Reconstructed (B), independence principle:** local types at distinct rational places become asymptotically independent (and are identically distributed when the geometry has no distinguished rational places).

Everything proved below is therefore a rigorous answer to this explicitly labeled reconstruction. Recovering Wood's workshop slides or notes could change which label is attached to which principle, and possibly the exact normalization. No claim below depends on calling the two principles A or B; it depends only on the stated mass and independence hypotheses.

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

00061 | AIM-ANALYTIC_NUMBER_THEORY-0015

### A rank-compatible GL(3) Kuznetsov delta reduction

P / partially_solved | AIM 研讨会问题汇编

**\(\mathrm{GL}_3\) Kuznetsov trace formula as a \(\delta\)-method.** Obtain \(t\)-aspect subconvexity for a \(\mathrm{GL}_2\) \(L\)-function by using the \(\mathrm{GL}_3\) Kuznetsov trace formula as a \(\delta\)-method.

来源字段：http://aimpl.org/deltasubconvex2/1/

00062 | AIM-ANALYTIC_NUMBER_THEORY-0090

### A rational j-equation for a mixed mock Kaneko-Zagier solution

P / partially_solved | AIM 研讨会问题汇编

别名：Hauptmodul differential equations for an even-weight mixed-mock class [AIM-PHYSICS-0023]

Do mixed mock modular forms satisfy differential equations with respect to a modular function?

来源字段：http://aimpl.org/gromwitnumthry/5/

00063 | AIM-ALGEBRAIC_NUMBER_THEORY-0110

### A representation-sensitive decidability dichotomy for integer rotation constraints

P / partially_solved | AIM 研讨会问题汇编

Is there an algorithm to decide solubility of a system of linear equations \[ a_{i1}x_1+\cdots+a_{in}x_n=b_i \] for integer coefficients and integer unknowns, together with conditions of the form \[ \{x_i\alpha_j\}<r_j \] for \(\alpha_j\in\mathbb R\setminus\mathbb Q\) and rational \(r_j\)?

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

00064 | AIM-ANALYTIC_NUMBER_THEORY-0049

### A safe-half-plane Euler product for ratios in the hyperelliptic ensemble

P / partially_solved | AIM 研讨会问题汇编

Evaluate moments of ratios of \(L\)-functions over function fields.

来源字段：http://aimpl.org/zetamoments/2/

00065 | AIM-ANALYTIC_NUMBER_THEORY-0060

### A safe-range bridge between character and t-aspect orthogonality

P / partially_solved | AIM 研讨会问题汇编

The mechanism for computing moments in families such as Dirichlet characters \(\chi\pmod q\), \(q\leq Q\), uses congruence orthogonality, unlike the \(t\)-aspect mechanism that selects pairs with \(|m-n|\) small. Yet the predicted moment formulae have the same unitary-family structure. Why?

来源字段：http://aimpl.org/zetamoments/5/

00066 | AIM-ALGEBRAIC_NUMBER_THEORY-0077

### A semisimple-unipotent reduction for the torsion field of J(X(ell))

P / partially_solved | AIM 研讨会问题汇编

**14. (C. Hall)** Let \(X(\ell)\) be a modular curve. What can we say about the splitting field of the \(\ell\)-torsion of the Jacobian \(J(X(\ell))\)?

**Comment.** This is needed to speak about the \(L\)-function of \(E\to X(\ell)\) (mod \(\ell\)) in \(\mathbf F_\ell[T]\).

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

00067 | AMR-092-0003

### A seventeenth-century proof of Fermat's Last Theorem

O / open | AMR 问题清单汇编

Can Fermat's Last Theorem be proved using only mathematical techniques that were available in the seventeenth century?

背景链接：https://unsolvedproblems.org/index_files/FLT.htm

00068 | OPG-508

### A sextic counterexample to Euler's sum of powers conjecture

O / open | Open Problem Garden

Problem Find six positive integers $x_1, x_2, \dots, x_6$ such that $$x_1^6 + x_2^6 + x_3^6 + x_4^6 + x_5^6 = x_6^6$$ or prove that such integers do not exist.

背景链接：http://www.openproblemgarden.org/op/a_sextic_counterexample_to_eulers_sum_of_powers_conjecture

00069 | AIM-ANALYTIC_NUMBER_THEORY-0115

### A smooth-modulus 21/40 small-residue maximum and a weight-quantifier audit

P / partially_solved | AIM 研讨会问题汇编

The record is Problem 16 in the AIM workshop list *Gaps between primes* (2005). With \[ \psi(x;q,a)=\sum_{\substack{n\le x\\n\equiv a\pmod q}}\Lambda(n), \] the three questions attributed to J. B. Friedlander are:

1. Show, for some fixed \(\theta>1/2\) and \(D=x^\theta\), that \[ \sum_{\substack{d\le D\\(d,a)=1}}\mu(d) \left(\psi(x;d,a)-\frac{x}{\phi(d)}\right) \ll x(\log x)^{-A}. \] 2. Prove anything beyond the Bombieri--Vinogradov theorem for \[ \sum_{d\le D}\max_{\substack{(a,d)=1\\|a|<(\log x)^{2005}}} \left|\psi(x;d,a)-\frac{x}{\phi(d)}\right|. \] 3. For any “reasonable” weights \(\lambda_d\), show, for some fixed \(\theta>1/2\) and \(D=x^\theta\), that \[ \sum_{\substack{d\le D\\(d,a)=1}}\lambda_d \left(\psi(x;d,\bar a)-\frac{x}{\phi(d)}\right) \ll x(\log x)^{-A}, \qquad a\bar a\equiv1\pmod d. \]

The item continues with two broader prompts: obtain Bombieri--Vinogradov-type theorems for arithmetic functions other than \(\Lambda\), and formulate a reasonable zeros-of-\(L\)-functions conjecture implying \(\mathrm{EH}(\theta)\) for some \(\theta>1/2\).

The PDF displays \(A\) without a quantifier. The standard and useful reading, adopted below, is “for every fixed \(A>0\),” with constants allowed to depend on fixed parameters. This is a reconstruction, not a literal quantifier in the source.

来源字段：https://aimath.org/WWN/primegaps/WorkshopProblems.pdf

00070 | AIM-ANALYTIC_NUMBER_THEORY-0054

### A split-dependent diagonal obstruction in the four-factor approximate functional equation

P / partially_solved | AIM 研讨会问题汇编

Can one prove the fourth moment formula starting from \[ \int \prod_{j=1}^4\left(\sum_{n\leq x_j}\frac1{n^{1/2+it}} +\chi(s)\sum_{n\leq t/x_j}\frac1{n^{1/2-it}}\right)\,dt\ ? \]

The preceding AIM record (3.1) explicitly asks for a robust recipe based on *asymmetric approximate functional equations*. This context, the workshop report, and the standard fourth-moment notation determine the intended reading, but the display itself suppresses important information.

Put \[ s=\tfrac12+it,\qquad q(t)=\frac{t}{2\pi},\qquad \chi(s)=\pi^{s-1/2}\frac{\Gamma((1-s)/2)}{\Gamma(s/2)}, \] so that \(\zeta(s)=\chi(s)\zeta(1-s)\) and \(\chi(s)\chi(1-s)=1\). The two asymmetric approximants that are actually needed are \[ F_{+,x}(t)=\sum_{n\leq x}n^{-s} +\chi(s)\sum_{n\leq q(t)/x}n^{s-1}, \tag{1} \] and \[ F_{-,x}(t)=\sum_{n\leq x}n^{s-1} +\chi(1-s)\sum_{n\leq q(t)/x}n^{-s}. \tag{2} \] Thus the intended object is \[ \int w(t/T)F_{+,x_1}(t)F_{+,x_2}(t) F_{-,x_3}(t)F_{-,x_4}(t)\,dt, \tag{3} \] with \(w\) smooth and supported, for example, in \([1,2]\). Formula (3) models \(\zeta(s)^2\zeta(1-s)^2=|\zeta(s)|^4\). Taken literally, the AIM display has four copies of (1) and therefore models \(\zeta(s)^4\), not the fourth absolute moment. Moreover, under the displayed standard definition of \(\chi\), complementary cutoffs satisfy \(xy=q(t)=t/(2\pi)\), not \(xy=t\). The missing \(2\pi\) can only be harmlessly suppressed if it has been absorbed into the convention for the \(x_j\). Hard cutoffs also create moving boundary terms; a proof should first smooth them or track them explicitly.

The recovered target, at minimum, is Ingham's leading formula \[ \int_0^T|\zeta(\tfrac12+it)|^4\,dt \sim \frac{1}{2\pi^2}T(\log T)^4. \tag{4} \] The stronger established target is \(T P_4(\log T)+E_2(T)\), where \(P_4\) is an explicit polynomial of degree four with leading coefficient \(1/(2\pi^2)\).

来源字段：http://aimpl.org/zetamoments/3/

00071 | AIM-ANALYTIC_NUMBER_THEORY-0062

### A square-root shifted-divisor hypothesis for the Conrey--Keating one-swap block

P / partially_solved | AIM 研讨会问题汇编

Can one formulate a conjecture (e.g. Kloosterman sum estimates) that lead to the calculations of Conrey and Keating in a rigorous way? We need a better understanding (even mechanically) of these calculations.

来源字段：http://aimpl.org/zetamoments/6/

00072 | AIM-ALGEBRAIC_NUMBER_THEORY-0031

### A three-state correction to the local quadratic-twist rank ratio

P / partially_solved | AIM 研讨会问题汇编

**Question 1 (Brian Conrey).** Let $E/\mathbb F_q(T)$. Let $f$ vary over square free, large degree $n$ and let $P\in\mathbb P^1$; does $$ \lim_{n\to\infty} \frac{\#\{f\in\mathbb F_q[T]_n:\operatorname{rank}E_f\ge 2 \text{ and } f(P)\in(\mathbb F_q^*)^2-\{0\}\}} {\#\{f\in\mathbb F_q[T]_n:\operatorname{rank}E_f\ge 2 \text{ and } f(P)\notin(\mathbb F_q^*)^2\}} \stackrel{?}{=} \sqrt{\frac{\#E_P(\mathbb F_q)}{\#\widetilde E_P(\mathbb F_q)}}\ ? $$

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

00073 | AIM-ANALYTIC_NUMBER_THEORY-0008

### A weighted large-sieve reduction and an exponent obstruction

P / partially_solved | AIM 研讨会问题汇编

It asks for a bound

$$ \sum_{f,g\in \mathcal B_k} L(1/2,\operatorname{sym}^2 f)L(1/2,\operatorname{sym}^2 g) \left|\sum_{n\leq k}\lambda_f(n)\lambda_g(n+h)\right|^2 \ll k^{4-\delta} \tag{1.1} $$

for $|h|\ll1$ and some $\delta>0$, by a delta method. It says that (1.1), conditionally on nonnegativity of the central values, would imply the holomorphic QUE result of Holowinsky and Soundararajan.

The record does not define $\mathcal B_k$, the normalization of $\lambda_f(n)$, or whether the outer sum is harmonic. The standard reconstruction used here is:

* $k$ is a large even integer; * $\mathcal B_k=H_k$ is the Hecke-normalized orthogonal basis of level-one, weight-$k$ cusp forms, with $f(z)=\sum_{n\geq1}\lambda_f(n)n^{(k-1)/2}e(nz)$ and $\lambda_f(1)=1$; * hence $|H_k|=k/12+O(1)$, and Deligne gives $|\lambda_f(n)|\leq d(n)$; * the outer sum in (1.1) is a natural, not a Petersson-harmonic, family sum.

This is an explicit reconstruction, not text verified on the unavailable AIM page. If $\mathcal B_k$ instead means an orthonormal Fourier basis, or if harmonic weights are intended, both the scale and the applicable trace formula change.

There is a second ambiguity: $|h|\ll1$ includes $h=0$, whereas the QUE shifted-convolution criterion has a Rankin--Selberg main term at zero shift. Cancellation should be requested for fixed $h\ne0$; $h=0$ must be separated and its main term retained. The algebra below remains valid whenever all $n+h$ are positive.

来源字段：http://aimpl.org/deltasubconvex2/1/

00074 | NT-005

### ABC Conjecture

P / partially_solved | 独立 / 未分配集合

别名：The abc Conjecture [NT-030]

For any $\epsilon > 0$, there exist only finitely many triples $(a, b, c)$ of coprime positive integers with $a + b = c$ such that $c > \text{rad}(abc)^{1+\epsilon}$, where $\text{rad}(n)$ is the product of distinct prime factors of $n$.

背景链接：https://ems.press/journals/prims/issues/1507

00075 | OWR-1053-005

### Abelian Covering Conjectures for Curves over Number Fields

O / open | Oberwolfach 报告问题汇编

Let $C$ be a curve over a number field $k$, and let $X\in C(\mathbb A_k)$ be disjoint from $C(k)$. Does there exist an unramified covering $D\to C$ such that every twist $D_j$ has image disjoint from $X$? Can $D$ always be chosen abelian?

来源字段：https://doi.org/10.4171/owr/2005/32

00076 | OWR-12583-005

### Abelian Galois Groups of Imaginary Quadratic Fields

O / open | Oberwolfach 报告问题汇编

For density one of the imaginary quadratic fields of prime class number, is the absolute abelian Galois group isomorphic to $A_0=\widehat{\mathbb Z}^{\times}\times\prod_{n\geq1}\mathbb Z/n\mathbb Z$?

来源字段：https://doi.org/10.4171/owr/2013/35

00077 | OWR-4127-007

### Abelianizations of Arithmetic Orthogonal Groups

P / partially_solved | Oberwolfach 报告问题汇编

What can be said about the abelianisation of arithmetic groups, particularly arithmetic groups of orthogonal type?

来源字段：https://doi.org/10.4171/owr/2009/44

00078 | OWR-16158-001

### Absolute Galois Symmetries of Moduli Stacks

P / partially_solved | Oberwolfach 报告问题汇编

How is the absolute Galois group $G_{\mathbb Q}$ encoded in the geometric symmetries of the moduli stacks $\mathcal M_{g,[m]}$?

来源字段：https://doi.org/10.4171/owr/2018/17

00079 | OWR-15174-015

### Abundance of Zeros in Families of Quadratic Forms

O / open | Oberwolfach 报告问题汇编

Fix integral quadratic polynomials $Q_1,\ldots,Q_h$ in $t$ variables. For coefficient vectors $\mathbf x=(x_1,\ldots,x_h)$ with $\lvert x_i\rvert\leq B^\theta$, is it true for a positive proportion of $\mathbf x$ that

$$\#\{\mathbf y\in\mathbb Z^t:\lvert y_j\rvert\leq B,\ \textstyle\sum_{i=1}^h x_iQ_i(\mathbf y)=0\}\gg B^{t-2-\theta}$$

for some $(h-7)/(h-1)\leq\theta<1$?

来源字段：https://doi.org/10.4171/owr/2016/53

00080 | AIM-ALGEBRAIC_NUMBER_THEORY-0043

### Additive gaps and an affine-coordinate audit

P / partially_solved | AIM 研讨会问题汇编

Question 13 (Pollack). What can we say (resp. what do we know) about gaps between irreducible polynomials over Fq?

The typeset question ends after \(\mathbb F_q\).

There are several inequivalent notions that the word “gap” could denote in \(\mathbb F_q[T]\). The intended one can nevertheless be recovered with high confidence. Question 12 immediately before this record concerns additive shifts \(f\) and \(f+h\). More decisively, the paper of Castillo, Hall, Lemke Oliver, Pollack, and Thompson states that its work began at this same AIM workshop and develops “bounded gaps” over \(\mathbb F_q(t)\). In its fixed-degree application, a gap is the polynomial difference between two monic irreducibles of the same degree.

Accordingly, let \(\mathcal P_{q,n}\) be the set of monic irreducible polynomials of degree \(n\) in \(A=\mathbb F_q[T]\).

1. For which nonzero \(h\in A\) are there infinitely many \(n\) and \(P,Q\in\mathcal P_{q,n}\) with \(P-Q=h\)? 2. For a fixed \(h\), what is the asymptotic count of such pairs as \(n\to\infty\) with \(q\) fixed? 3. Can one at least prove the existence of pairs for which \(\deg(P-Q)\), equivalently \(|P-Q|=q^{\deg(P-Q)}\), stays bounded as \(n\to\infty\)? 4. How large is the gap set \[ \mathcal D_{q,n}=\{P-Q:P,Q\in\mathcal P_{q,n},\ P\ne Q\}\subseteq\{h:\deg h<n\}? \]

This is also the short-interval interpretation: \(\deg(P-Q)\le d\) precisely when \(P\) and \(Q\) lie in the same radius-\(q^d\) ball for the norm at the chosen place at infinity.

Two other readings are rejected. A “degree gap” between available irreducible degrees is trivial, since irreducibles exist in every positive degree.

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

00081 | OWR-13952-007

### Adelic Harmonic Metrics on Line Bundles

P / partially_solved | Oberwolfach 报告问题汇编

Does every line bundle $M$ on $X$ have an admissible adelic metrization $\overline{M}$ whose metric at each place has harmonic curvature and satisfies $\overline{M}\cdot\omega^n=\lambda(M)\omega^{n+1}$, unique up to multiplication by a degree-zero element of $\operatorname{Div}(K)$?

来源字段：https://doi.org/10.4171/owr/2015/57

00082 | OWR-14746-007

### Adelic Harmonic Metrizations over Global Fields

O / open | Oberwolfach 报告问题汇编

Let $X$ be a projective variety of dimension $n$ over a global field $K$, let $\overline L$ be a positive adelic metrized line bundle, and put $\omega=c_1(\overline L)/(\deg L)^{1/n}$. Does every line bundle $M$ on $X$ admit an admissible metrization $\overline M$ whose curvature is harmonic at every place and which satisfies

$$\overline M\cdot\omega^n=\lambda(M)\omega^{n+1},$$

uniquely up to multiplication by an adelic divisor from $K$ of degree zero?

来源字段：https://doi.org/10.4171/owr/2016/38

00083 | OWR-3388-003

### Admissible Groups over Global Function Fields

O / open | Oberwolfach 报告问题汇编

For the function field $F$ of a curve over a finite field, characterize the finite groups $G$ that are admissible over $F$, meaning that $G$ occurs as the Galois group of a maximal subfield of an $F$-division algebra.

来源字段：https://doi.org/10.4171/owr/2009/05

00084 | NT-087

### Agoh-Giuga Conjecture

P / partially_solved | 独立 / 未分配集合

Is $p$ prime if and only if $pB_{p-1} \equiv -1 \pmod{p}$ for the Bernoulli number $B_{p-1}$?

背景链接：https://doi.org/10.2307/2975190

00085 | AMR-093-0114

### Agrawal's conjecture

O / open | AMR 问题清单汇编

Agrawal's conjecture that given coprime positive integers $n$ and $r$, if $(X - 1)^n \equiv X^n - 1 \pmod{n, X^r - 1}$, then either $n$ is prime or $n^{2} \equiv 1 \pmod{r}$

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00086 | AIM-ALGEBRAIC_NUMBER_THEORY-0004

### Albanese sums of symmetric powers: thresholds, arithmetic reduction, and explicit fibers

P / partially_solved | AIM 研讨会问题汇编

With some hypotheses on \(X\) and \(X\to\operatorname{Alb}(X)\), study the image of \(\operatorname{Sym}^d(X)\to\operatorname{Alb}(X)\) for some small \(d\).

What are examples of \(X\) for which the fibers of \(\operatorname{Sym}^dX\to\operatorname{Alb}(X)\) are “well-understood”? For example, when are they finite or rational?

来源字段：http://aimpl.org/degreedsurface/1/

00087 | OWR-1393-004

### Albert's Cyclicity Conjecture

O / open | Oberwolfach 报告问题汇编

Is every central simple algebra of prime degree cyclic?

来源字段：https://doi.org/10.4171/owr/2006/55

00088 | OPG-37413

### Alexa's Conjecture on Primality

P / partially_solved | Open Problem Garden

Definition Let $r_i$ be the unique integer (with respect to a fixed $p\in\mathbb{N}$ ) such that

$$(2i+1)^{p-1} \equiv r_i \pmod p ~~\text{ and } ~ 0 \le r_i < p.$$

Conjecture A natural number $p \ge 8$ is a prime iff $$\displaystyle \sum_{i=1}^{\left \lfloor \frac{\sqrt[3]p}{2} \right \rfloor} r_i = \left \lfloor \frac{\sqrt[3]p}{2} \right \rfloor$$

背景链接：http://www.openproblemgarden.org/op/alexas_conjecture_on_primality

00089 | OWR-4076-010

### Algebraic Cycles Realizing Modular Differential Forms

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist $\Pi^*\in CH^{r+1}(W_r\times A^{r+1})(K)\otimes\mathbb{Q}$ such that $\Pi^*_{\mathrm{dR}}([\omega_A^{r+1}])\sim[\omega_\psi^{r+1}]$and $\Pi^*_{\mathrm{dR}}([\omega_A^j\overline{\omega}_A^{r+1-j}])=0$ for every $1\leq j\leq r$, where $\sim$ denotes equality up to a nonzero rational scalar?

来源字段：https://doi.org/10.4171/owr/2009/33

00090 | OPG-1786

### Algebraic independence of pi and e

O / open | Open Problem Garden

Conjecture $\pi$ and $e$ are algebraically independent

背景链接：http://www.openproblemgarden.org/op/algebraic_independence_of_pi_and_e

00091 | NUM-009

### Algebraic Number Normality

O / open | 独立 / 未分配集合

Are all irrational algebraic numbers normal?

背景链接：https://doi.org/10.1017/CBO9781139019672

00092 | OWR-4076-012

### Algebraicity of Complex ATR Points

P / partially_solved | Oberwolfach 报告问题汇编

Let $K/F$ be an ATR extension embedded as an $F$-algebra in $M_2(F)$. Is the associated complex ATR point $AJ_f(\Delta_\Psi)$ algebraic? More precisely, after choosing the isogeny in its definition, does it lie in $E(H_{c_\Psi})$ for every $\Psi$?

来源字段：https://doi.org/10.4171/owr/2009/33

00093 | OWR-17300-010

### Algebraicity of Rigid Cocycle Values at Real Multiplication Points

O / open | Oberwolfach 报告问题汇编

If $J$ is a rigid meromorphic cocycle, then there is a finite extension $H_J$ of $\mathbb{Q}$ such that for every real-multiplication point $\tau$, the value $J[\tau]$ lies in the compositum $H_JH_\tau$, where $H_\tau$ is the narrow ring class field attached to $\tau$.

来源字段：https://doi.org/10.4171/owr/2019/57

00094 | OWR-14299909-012

### Algebraicity Under Central-Binomial Untwisting

O / open | Oberwolfach 报告问题汇编

Let $f(x)=\sum_{n\geq 0}a_nx^n\in\mathbb Z[[x]]$ and $F(x)=\sum_{n\geq 0}\binom{2n}{n}a_nx^n$. If $F$ satisfies a second-order homogeneous linear differential equation with polynomial coefficients and regular singularities, must $f$ be algebraic?

来源字段：https://doi.org/10.4171/owr/2026/7

00095 | AIM-ALGEBRAIC_NUMBER_THEORY-0021

### All-valuation isotropy of a quadratic pencil and the limits of Boolean local tests

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.** Browning: Can we develop a version of the circle method over $\mathbf Q(t)$? Wooley: The major arcs are difficult to understand. Browning: For example, consider the diagonal quadratic form $$ A_1x_1^2+\cdots+A_sx_s^2=0 \tag{1} $$ with $A_j\in\mathbf Q(t)$. There are “obvious” local conditions, e.g. arising from discrete valuation rings for $\mathbf Q(t)$. Does the Hasse principle hold? Wooley: If $A_j$ are linear, say $A_j=c_j+td_j$ with $c_j,d_j\in\mathbf Q$, then (1) is equivalent to the system of equations $$ \sum_{j=1}^s c_jx_j^2=\sum_{j=1}^s d_jx_j^2=0 \tag{2} $$ by Amer--Brumer, which defines a quartic del Pezzo surface over $\mathbf Q$. Do the “obvious” local obstructions over $\mathbf Q(t)$ for (1) capture the Brauer--Manin obstruction over $\mathbf Q$ for (2)? Harari: We may ask these questions over $\mathbf Q_p(t)$; see work of Harari--Szamuely.

来源字段：https://aimath.org/pastworkshops/ratlhigherdimvarproblems.pdf

00096 | NT-044

### Almost Perfect Numbers Beyond Powers of 2

P / partially_solved | 独立 / 未分配集合

Do any almost perfect numbers exist that are not powers of 2?

背景链接：https://arxiv.org/abs/1602.04248

00097 | OWR-3388-008

### Almost-Rational Automorphisms of Higher Prime-Power Order

O / open | Oberwolfach 报告问题汇编

Classify the almost-rational automorphisms $\sigma$ of $k[[t]]$ of order $p^n>p$ without assuming that every power $\sigma^j$, for $j\neq1$, is almost rational with respect to the same Artin-Schreier root.

来源字段：https://doi.org/10.4171/owr/2009/05

00098 | AIM-ANALYTIC_NUMBER_THEORY-0114

### Alternative GPY weights: exact perturbation geometry and a radial escape

P / partially_solved | AIM 研讨会问题汇编

This is Problem 15 in the AIM workshop list *Gaps between primes*. The official PDF is the authority for the reconstruction below.

Let $$ P_{\mathcal H}(n)=\prod_{i=1}^k(n+h_i),\qquad \log^+x=\max(0,\log x). $$ The PDF defines the truncated divisor sum $$ \Lambda_R(n;\mathcal H)=\sum_{d\mid P_{\mathcal H}(n)}\lambda_d, \qquad \lambda_d=\mu(d)\frac{\bigl(\log^+(R/d)\bigr)^{k+\ell}}{(k+\ell)!}. \tag{1.1} $$ Conrey asked whether a quantitatively stronger method results from other divisor coefficients. One proposed family was $$ \lambda_d=\mu(d) \left\{\prod_{p\mid d}f\!\left(\frac{\log p}{\log R}\right)\right\} Q\!\left(\log^+(R/d)\right), \tag{1.2} $$ with smooth functions $f,Q$. Vaughan suggested testing other traditional sieve weights. A third suggestion was to combine upper and lower sieves in the style of Chen; Soundararajan warned that the negative contribution would have to be calculated.

There is a crucial notational distinction. The object in (1.1) is a possibly signed **divisor sum**. In the GPY positivity argument, the actual test weight is $$ w(n)=\Lambda_R(n;\mathcal H)^2\geq 0. \tag{1.3} $$ More generally, one can square a linear combination of divisor sums. Negative divisor coefficients or negative cross terms inside this square do not make $w$ signed. A genuinely signed upper-minus-lower weight is a different object and loses the automatic positivity inference unless its negative contribution is controlled.

来源字段：https://aimath.org/WWN/primegaps/WorkshopProblems.pdf

00099 | NT-046

### Amicable Numbers of Opposite Parity

P / partially_solved | 独立 / 未分配集合

Do any pairs of amicable numbers exist where one is odd and one is even?

背景链接：https://arxiv.org/abs/math/0501402

00100 | AIM-ALGEBRAIC_NUMBER_THEORY-0010

### An Albanese threshold obstruction for Hilbert schemes of points

P / partially_solved | AIM 研讨会问题汇编

Is $\operatorname{Hilb}^2(C\times D)(k)$ Zariski dense in $\operatorname{Hilb}^2(C\times D)$, where $C,D/k$ are hyperelliptic curves over a number field. More generally, for which varieties $X$ over a number field is there a $d$ such that $\operatorname{Hilb}^d(X)(k)$ is Zariski dense?

Is $\operatorname{Hilb}^2(C\times D)(k)$ Zariski dense in $\operatorname{Hilb}^2(C\times D)$, where $C,D/k$ are hyperelliptic curves over a number field?

来源字段：http://aimpl.org/degreedsurface/1/

00101 | AIM-ANALYTIC_NUMBER_THEORY-0021

### An averaged short-additive-twist criterion for a GL(3) Hardy theorem

P / partially_solved | AIM 研讨会问题汇编

**Hardy-type theorem for \(\mathrm{GL}_3\) \(L\)-functions.** Use a \(\delta\)-method to prove that there are infinitely many critical zeros for a \(\mathrm{GL}_3(\mathbb Z)\) \(L\)-functions.

来源字段：http://aimpl.org/deltasubconvex2/1/

00102 | AIM-ALGEBRAIC_NUMBER_THEORY-0019

### An exact nonrational double-cover source for dense quadratic points

P / partially_solved | AIM 研讨会问题汇编

The following is asked in analogy with a result of Harris and Silverman showing that, for a curve $C/k$, $2\in\delta(C/k)\iff C$ is a double cover of $\mathbb P^1$ or a positive rank elliptic curve (see https://www.ams.org/journals/proc/1991-112-02/S0002-9939-1991-1055774-0/S0002-9939-1991-1055774-0.pdf).

Classify surfaces $X/k$ of general type with $2\in\delta(X/k)$, possibly using geometric and/or dynamical constructions.

Vojta has prove that for a hyperelliptic curve $C/k$ of genus $\ge4$, all but finitely many degree $2$ points are contracted by its hyperelliptic map.

If $\operatorname{irr}(X)=2$, there exists a dominant map $X\dashrightarrow\mathbb P^2$ of degree $2$. What conditions on $X$ guarantee that the set of quadratic points on $X$ not arising from this map is not Zariski dense?

来源字段：http://aimpl.org/degreedsurface/1/

00103 | AIM-ANALYTIC_NUMBER_THEORY-0101

### An exact q-dilogarithmic action and finite-q Nahm equation

P / partially_solved | AIM 研讨会问题汇编

Problem 2.6. How do q-analogues of the dilogarithm figure into the theory?

**Recovered question.** Explain whether q-dilogarithms provide a finite-\(q\) or “quantized” version of the classical dilogarithm/Bloch-group mechanism governing Nahm sums, and how that mechanism interacts with q-hypergeometric identities and modular or mock-modular behavior.

来源字段：https://aimath.org/WWN/mockmodular/mockmodular.pdf

00104 | AIM-ALGEBRAIC_NUMBER_THEORY-0099

### An exact residue-matching reduction for least points on diagonal cubic surfaces

P / partially_solved | AIM 研讨会问题汇编

Consider \[ a_0X_0^3+a_1X_1^3+a_2X_2^3+a_3X_3^3=0. \] (a) If it is soluble, give an upper bound for the smallest solution in terms of the \(a_i\). (b) For \(|a_i|<A\), tabulate the smallest solution and conjecture its growth in \(A\).

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

00105 | AIM-ALGEBRAIC_NUMBER_THEORY-0079

### An explicit CM projector and Kummer correspondence

P / partially_solved | AIM 研讨会问题汇编

Let \(E\) be an elliptic curve over \(\mathbb Q\) with CM by \(K\) and with associated Hecke character \(\chi\). Then \(\chi^2\) is associated to a modular form of weight 3 appearing in \(H^2(\mathcal E)\), \(\mathcal E\to X\) for a modular curve \(X\) (whose level is the norm of the conductor of \(\chi^2\) times the discriminant of \(K\)). The Tate and Hodge Conjectures predict the existence of a correspondence \(E\times E\to\mathcal E\) because the same motive appears in \(H^2\) of both sides. Can we exhibit this correspondence explicitly?

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

00106 | AIM-ALGEBRAIC_NUMBER_THEORY-0105

### An explicit cubic point and its descent certificate

S / solved | AIM 研讨会问题汇编

Put \[ g(T)=3(T^4-54T^2-117T-243),\qquad p(X)=X^2+1,\qquad q(X)=X^2+2. \] The official AIM workshop PDF, *Rational and integral points on higher dimensional varieties*, Question 20, asks whether the surface with affine open \[ S_{\mathrm{AIM}}:\qquad y^2=g(t)p(x),\qquad z^2=g(t)q(x) \tag{1} \] has a point over some odd-degree number field. The question attributes to Skorobogatov the facts that a smooth projective model \(S/\mathbb Q\) has \(S(\mathbb Q)=\varnothing\) and \(S(\mathbb A_{\mathbb Q})^{\mathrm{Br}}\ne\varnothing\).

There is an important coordinate convention. Skorobogatov's example and the opening display in Creutz's paper use \[ S_0:\qquad p(x)\widetilde y^2=q(x)\widetilde z^2=g(t). \tag{2} \] On the common open where \(p(x)q(x)\ne0\), the two presentations are related by \[ y=p(x)\widetilde y,\qquad z=q(x)\widetilde z, \tag{3} \] with inverse \(\widetilde y=y/p(x)\), \(\widetilde z=z/q(x)\). Creutz explicitly describes (1) as a change of variables from (2). All point coordinates below are labeled by convention. Since the point obtained lies in this common affine open, it gives a point on the intended smooth projective surface under either presentation.

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

00107 | AIM-ALGEBRAIC_NUMBER_THEORY-0098

### An explicit geometric Picard-rank-one quintic over F_2

S / solved | AIM 研讨会问题汇编

**Problem/Question 13.** Find a smooth quintic hypersurface in \(\mathbb P^3\) with Picard number 1 over \(\mathbb F_2\). (Voloch)

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

00108 | AIM-ANALYTIC_NUMBER_THEORY-0067

### An orthogonalized Type-II lower-bound certificate and rank obstruction

P / partially_solved | AIM 研讨会问题汇编

We can get lower bounds for the Type I off-diagonal terms using sifted sequences. Is there any analog of this for Type II terms? Can one improve lower bounds for $F(\alpha,T)$ in this way (as in Goldston-Gonek-Ozluk-Snyder)?

Can one get lower bounds for moments in general?

来源字段：http://aimpl.org/zetamoments/7/

00109 | OWR-14297740-016

### Anabelian Interpretation of Homomorphisms between Fundamental Groups

O / open | Oberwolfach 报告问题汇编

Is there an anabelian interpretation of $\operatorname{Hom}_{\mathrm{op}}(\Pi_{X_1}^{\bullet},\Pi_{X_2}^{\bullet})$?

来源字段：https://doi.org/10.4171/owr/2023/42

00110 | OWR-4990381-016

### Anabelian Reconstruction of Genus-Zero Configuration Spaces

P / partially_solved | Oberwolfach 报告问题汇编

In genus zero, does an isomorphism of étale fundamental groups between a hyperbolic polycurve and a configuration space of a hyperbolic curve imply that the varieties are isomorphic?

来源字段：https://doi.org/10.4171/owr/2021/12

00111 | OWR-8415342-010

### Analytic Continuation and Functional Equations of Automorphic L-Functions

P / partially_solved | Oberwolfach 报告问题汇编

For a number field $k$, a split reductive group $G$, a cuspidal automorphic representation $\sigma$ of $G(\mathbb A)$, and a finite-dimensional dual-group representation $\rho$, prove that the automorphic $L$-function $L(s,\sigma,\rho)$ has meromorphic continuation and satisfies $L(s,\sigma,\rho)=\epsilon(s,\sigma,\rho)L(1-s,\widetilde\sigma,\rho)$.

来源字段：https://doi.org/10.4171/owr/2021/39

00112 | AIM-ANALYTIC_NUMBER_THEORY-0030

### Antipodal lifts in every nonzero Möbius-square fiber

P / partially_solved | AIM 研讨会问题汇编

Let $X_{\mu^2}=\overline{\{T^k(\mu^2):k\in\mathbb N\}}$ and $X_\mu=\overline{\{T^k(\mu):k\in\mathbb N\}}$, for $T$ the shift map. What are the pre-images of $f:X_\mu\to X_{\mu^2}$ given by $(x_i)\mapsto(x_i^2)$ in $X_{\mu^2}$?

来源字段：http://aimpl.org/sarnakconjecture/3/

00113 | AIM-ALGEBRAIC_NUMBER_THEORY-0119

### Arakelov replacement and a surface exponent-transfer lemma

P / partially_solved | AIM 研讨会问题汇编

**Problem/Question 34. (Bogomolov)** Replace Heath-Brown by Arakelov (et al). \[ \#\{(x,y,z)\in\mathbb P^2: |x|,|y|,|z|\le B,\ P(x,y,z)=0\} \ll B^{\alpha+\epsilon} \] with \(\alpha\le 2/d\), \(d=\deg P\). Analogue for surfaces, etc.

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

00114 | OWR-1104-004

### Arakelov Self-Intersection Formula for Arithmetic Curves

O / open | Oberwolfach 报告问题汇编

For a regular arithmetic model $\mathcal X$ of a smooth projective curve $X/K$, with dualizing sheaf $\omega$ carrying the Arakelov metric, is $\widehat{\deg}(\omega^2)$ equal to $a_1(2g-2)\log|\Delta_{K/\mathbb Q}|+a_2\sum_{x\in\mathcal X_{\rm sing}}\log\#k(x)+\sum_{\sigma:K\hookrightarrow\mathbb C}\delta_{\rm Fal}(X_\sigma(\mathbb C))+a_3$ for constants $a_1,a_2,a_3\in\mathbb R$?

来源字段：https://doi.org/10.4171/owr/2005/43

00115 | AMR-093-0138

### Are all Euclid numbers square-free

O / open | AMR 问题清单汇编

Are all Euclid numbers square-free?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00116 | OPG-55810

### Are all Fermat Numbers square-free?

O / open | Open Problem Garden

Conjecture Are all Fermat Numbers $$ F_n = 2^{2^{n } } + 1 $$ Square-Free?

背景链接：http://www.openproblemgarden.org/op/are_all_fermat_numbers_square_free

00117 | OPG-59976

### Are all Mersenne Numbers with prime exponent square-free?

P / partially_solved | Open Problem Garden

Conjecture Are all Mersenne Numbers with prime exponent ${2^p-1}$ Square free?

背景链接：http://www.openproblemgarden.org/op/are_all_mersenne_numbers_with_prime_exponent_square_free

00118 | AMR-030-0051

### Are the positive integer powers of 3/2 mod 1 uniformly distributed in the unit interval

O / open | AMR 问题清单汇编

Are the positive integer powers of 3/2 mod 1 uniformly distributed in the unit interval? One would think so, but apparently this is a hard question. See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

00119 | OPG-37192

### Are there an infinite number of lucky primes?

O / open | Open Problem Garden

Conjecture If every second positive integer except 2 is remaining, then every third remaining integer except 3, then every fourth remaining integer etc., an infinite number of the remaining integers are prime.

背景链接：http://www.openproblemgarden.org/op/are_there_an_infinite_number_of_lucky_primes

00120 | AMR-093-0141

### Are there any composite c satisfying 2^{c - 1} ≡ 1 (mod c^{2})

O / open | AMR 问题清单汇编

Are there any composite c satisfying 2^{c - 1} ≡ 1 (mod c^{2})?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00121 | AMR-093-0020

### Are there any pairs of betrothed numbers which have same parity

O / open | AMR 问题清单汇编

Are there any pairs of betrothed numbers which have same parity?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00122 | AMR-093-0021

### Are there any pairs of relatively prime amicable numbers

O / open | AMR 问题清单汇编

Are there any pairs of relatively prime amicable numbers?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00123 | AMR-093-0143

### Are there any Wieferich primes in base 47

O / open | AMR 问题清单汇编

Are there any Wieferich primes in base 47?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00124 | OPG-59977

### Are there infinite number of Mersenne Primes?

O / open | Open Problem Garden

Conjecture A Mersenne prime is a Mersenne number $$ M_n = 2^p - 1 $$ that is prime.

Are there infinite number of Mersenne Primes?

背景链接：http://www.openproblemgarden.org/op/are_there_infinite_number_of_mersenne_primes

00125 | AMR-093-0144

### Are there infinitely many balanced primes

O / open | AMR 问题清单汇编

Are there infinitely many balanced primes?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00126 | AMR-093-0023

### Are there infinitely many betrothed numbers

O / open | AMR 问题清单汇编

Are there infinitely many betrothed numbers?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00127 | AMR-093-0145

### Are there infinitely many cluster primes

O / open | AMR 问题清单汇编

Are there infinitely many cluster primes?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00128 | AMR-093-0146

### Are there infinitely many cousin primes

O / open | AMR 问题清单汇编

Are there infinitely many cousin primes?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00129 | AMR-093-0147

### Are there infinitely many Cullen primes

O / open | AMR 问题清单汇编

Are there infinitely many Cullen primes?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00130 | AMR-093-0148

### Are there infinitely many Euclid primes

O / open | AMR 问题清单汇编

Are there infinitely many Euclid primes?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00131 | AMR-093-0150

### Are there infinitely many Kummer primes

O / open | AMR 问题清单汇编

Are there infinitely many Kummer primes?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00132 | AMR-093-0151

### Are there infinitely many Kynea primes

O / open | AMR 问题清单汇编

Are there infinitely many Kynea primes?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00133 | AMR-093-0152

### Are there infinitely many Lucas primes

O / open | AMR 问题清单汇编

Are there infinitely many Lucas primes?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00134 | NT-007

### Are there infinitely many Mersenne primes?

O / open | 独立 / 未分配集合

别名：Are There Infinitely Many Mersenne Primes? [NT-017]

Are there infinitely many prime numbers of the form $M_p = 2^p - 1$ where $p$ is prime?

背景链接：https://www.mersenne.org/

00135 | AMR-093-0154

### Are there infinitely many Newman-Shanks-Williams primes

O / open | AMR 问题清单汇编

Are there infinitely many Newman-Shanks-Williams primes?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00136 | AMR-093-0155

### Are there infinitely many palindromic primes to every base

P / partially_solved | AMR 问题清单汇编

Are there infinitely many palindromic primes to every base?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00137 | AMR-093-0156

### Are there infinitely many Pell primes

O / open | AMR 问题清单汇编

Are there infinitely many Pell primes?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00138 | NT-008

### Are there infinitely many perfect powers in the Fibonacci sequence?

P / partially_solved | 独立 / 未分配集合

Besides 1, 8, and 144, are there any other perfect powers (numbers of the form $a^b$ where $a, b > 1$) in the Fibonacci sequence?

背景链接：https://annals.math.princeton.edu/2006/163-3/p05

00139 | AMR-093-0157

### Are there infinitely many Pierpont primes

O / open | AMR 问题清单汇编

Are there infinitely many Pierpont primes?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00140 | AMR-093-0158

### Are there infinitely many prime quadruplets

O / open | AMR 问题清单汇编

Are there infinitely many prime quadruplets?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00141 | AMR-093-0159

### Are there infinitely many prime triplets

O / open | AMR 问题清单汇编

Are there infinitely many prime triplets?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00142 | AMR-093-0161

### Are there infinitely many sexy primes

O / open | AMR 问题清单汇编

Are there infinitely many sexy primes?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00143 | NT-019

### Are There Infinitely Many Sophie Germain Primes?

O / open | 独立 / 未分配集合

Are there infinitely many primes $p$ such that $2p + 1$ is also prime?

背景链接：https://fme.upc.edu/ca/la-facultat/activitats-fme-personalitats-del-curs/personalitat-del-curs/2025-2026-germain/nombresprimers

00144 | AMR-093-0163

### Are there infinitely many Wagstaff primes

O / open | AMR 问题清单汇编

Are there infinitely many Wagstaff primes?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00145 | AMR-093-0166

### Are there infinitely many Wolstenholme primes

O / open | AMR 问题清单汇编

Are there infinitely many Wolstenholme primes?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00146 | AMR-093-0167

### Are there infinitely many Woodall primes

O / open | AMR 问题清单汇编

Are there infinitely many Woodall primes?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00147 | OPG-55812

### Are there only finite Fermat Primes?

O / open | Open Problem Garden

Conjecture A Fermat prime is a Fermat number $$ F_n = 2^{2^n } + 1 $$ that is prime. The only known Fermat primes are F_0 =3,F_1=5,F_2=17,F_3 =257,F_4=65537 It is unknown if other fermat primes exist.

背景链接：http://www.openproblemgarden.org/op/are_only_finite_fermat_primes

00148 | AIM-ALGEBRAIC_NUMBER_THEORY-0074

### Arithmetic descent of a rank-two curve factorization

P / partially_solved | AIM 研讨会问题汇编

**11. (J. Getz)** C. Simpson proved that if \(X\) is a variety and if \[ \rho_X:\pi_1^{\mathrm{geom}}(X)\longrightarrow \mathrm{GL}_2(\mathbb Q_\ell) \] is a representation whose algebraic enveloping contains \(\mathrm{SL}_2\), then \(\rho_X\) factors through \[ \rho_Y:\pi_1^{\mathrm{geom}}(Y)\longrightarrow \mathrm{GL}_2(\mathbb Q_\ell), \] where \(Y\) is either a curve or a Shimura variety; the factorization is induced from a map from \(X\) to some cover of \(Y\). Is there a version of this result using just \(\pi_1\) in place of \(\pi_1^{\mathrm{geom}}\)?

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

00149 | OWR-12010-016

### Arithmetic Fundamental Lemma in All Dimensions

P / partially_solved | Oberwolfach 报告问题汇编

For an unramified quadratic extension $E/F$ of $p$-adic fields, let $Y=\mathcal N_n$ embed diagonally in $\mathcal N_n\times\mathcal N_{n+1}$. For $\delta\in U(n+1)$ and $\gamma\in\operatorname{GL}_{n+1}(E)$ matching under $\operatorname{GL}_n(E)$, does the arithmetic fundamental lemma identity $\langle Y,(1\times\delta)^*Y\rangle=\operatorname{orb}^{\prime}(\gamma,\mathbf 1_{S_{n+1}(\mathcal O_F)})$ hold in every dimension $n$?

来源字段：https://doi.org/10.4171/owr/2012/38

00150 | AIM-ALGEBRAIC_NUMBER_THEORY-0012

### Arithmetic gonality in a covering family

P / partially_solved | AIM 研讨会问题汇编

Given a covering family \(T\leftarrow\mathcal C\dashrightarrow X\) of curves, where \(T(k)\) is Zariski dense, how does \(\operatorname{gon}(C)\) (for \(C\) a generic member of the family) compare to \(\min\delta(X/k)\)?

来源字段：http://aimpl.org/degreedsurface/1/

00151 | OWR-14299289-019

### Arithmetic Identities from Geometric Trace Formulas

O / open | Oberwolfach 报告问题汇编

别名：Geometric Numerical Conjecture Lead-In [OWR-14299289-018]

For a cuspidal Hecke eigensheaf with trace function $f_\phi$, does the geometric trace formula give the predicted arithmetic identity equating the normalized global period of $f_\phi$ with the normalized sheared $L$-value attached to the dual parameter $\phi$?

来源字段：https://doi.org/10.4171/owr/2025/18

00152 | OWR-3480-003

### Arithmetic Intersection Formula for Special Cycles

P / partially_solved | Oberwolfach 报告问题汇编

For $f\in H_L^{1-n/2,\bar\rho}$ with vanishing constant term $c^+(0,0)$, does $\langle\widehat{Z}(f),Z(U)\rangle_{\mathrm{Fal}}=\frac{\deg Z(U)}{2}L'(\xi(f),U,0)$ hold?

来源字段：https://doi.org/10.4171/owr/2009/30

00153 | OWR-14299523-004

### Arithmetic Meaning of Euler-Kronecker Constants

O / open | Oberwolfach 报告问题汇编

For a number field $F$, determine what arithmetic information is encoded by its Euler-Kronecker constant $\gamma_F$, and characterize this information through primes, zeros, class-field data, or other intrinsic invariants of $F$.

来源字段：https://doi.org/10.4171/owr/2025/38

00154 | AIM-ALGEBRAIC_NUMBER_THEORY-0026

### Arithmetic purity: solved quadric cases, a sieve criterion, and diagonal topology

P / partially_solved | AIM 研讨会问题汇编

Let \(X\) be a smooth variety over a number field \(K\), let \(S\) be a finite set of places, and assume that \(X\) satisfies strong approximation outside \(S\). If \(Z\subset X\) is closed of codimension two, must \(X\setminus Z\) satisfy strong approximation outside \(S\)? Is the set of \(S\)-integral points Zariski dense?

来源字段：https://aimath.org/pastworkshops/ratlhigherdimvarproblems.pdf

00155 | AIM-ALGEBRAIC_NUMBER_THEORY-0024

### Arithmetic surjectivity forces recurrence of locally soluble fibers

P / partially_solved | AIM 研讨会问题汇编

**Problem 4. Viray:** Let \(\phi:X\to E\) be a fibration over an elliptic curve of positive rank over \(\mathbb Q\) whose generic fiber is smooth and geometrically irreducible. Let \[ Z=\{p\in E(\mathbb Q)\mid X_p=\phi^{-1}(p)\text{ has points everywhere locally}\}. \] What can we say about \(Z\)? Is \(|Z|<\infty\) with \(Z\ne\varnothing\) possible?

The motivation is that work of Poonen, Skorobogatov-Harpaz and Colliot-Thélène-Pál-Skorobogatov constructs \(X\) failing the Hasse principle such that none of the known obstructions apply. All of these use a map \(X\to C\) to a curve with \(0<|C(\mathbb Q)|<\infty\).

**Browning:** The case where \(\phi\) is a conic bundle may already be interesting.

来源字段：https://aimath.org/pastworkshops/ratlhigherdimvarproblems.pdf

00156 | OWR-14299289-003

### Arithmetic Topology over Rings of S-Integers

O / open | Oberwolfach 报告问题汇编

How does the arithmetic-topology correspondence change when the base ring is replaced by the ring of $S$-integers of a global field $F$, where $S$ contains all archimedean places when $F$ is a number field?

来源字段：https://doi.org/10.4171/owr/2025/18

00157 | AIM-ALGEBRAIC_NUMBER_THEORY-0049

### Arithmetic, geometric, and Frobenius monodromy over global function fields

P / partially_solved | AIM 研讨会问题汇编

Question 19 (Kiran Kedlaya). Fix g, fix C/K a curve of genus g, where K = Fq(X) and X is not necessarily rational. Can we compute the monodromy group of C? I.e., can we compute {im(Gal(K) to Aut(H1(C,Qp))): C in Mg(K)}

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

00158 | OWR-4804-009

### Arthur Parameters from Automorphic Descent Integrals

P / partially_solved | Oberwolfach 报告问题汇编

Let $\sigma$ and $\pi$ be irreducible cuspidal automorphic representations of $\operatorname{SO}_{2k}(\mathbb A)$ and $\operatorname{SO}_{2ab+2k}(\mathbb A)$, respectively. Suppose the automorphic descent integral $I_{a,k,l}^{\psi}(\tau,\sigma;\pi)$ is nonzero. If $b$ is even when $\tau$ is of symplectic type and odd when $\tau$ is of orthogonal type, does $\sigma$ have global Arthur parameter $\psi_{\operatorname{SO}_{2k}}$ if and only if $\pi$ has parameter $\psi_{\operatorname{SO}_{2k}}\oplus(\tau,b)$?

来源字段：https://doi.org/10.4171/owr/2011/14

00159 | OWR-14298805-012

### Arthur-Langlands Reciprocity

O / open | Oberwolfach 报告问题汇编

Establish the general Arthur-Langlands reciprocity correspondence between constituents of the discrete automorphic spectrum of a connected reductive group $G$ and the appropriate global Galois or Arthur parameters, compatibly with localization and automorphic $L$-functions.

来源字段：https://doi.org/10.4171/owr/2024/54

00160 | NT-029

### Artin's Conjecture on Primitive Roots

P / partially_solved | 独立 / 未分配集合

For how many prime numbers $p$ is a given integer $a$ (not $\pm 1$ or a perfect square) a primitive root modulo $p$?

背景链接：https://eudml.org/doc/150785

00161 | OWR-1586-006

### Artin-Twist Parity for Elliptic Curves

O / open | Oberwolfach 报告问题汇编

Let $p\geq5$, let $E/F$ be an elliptic curve, put $F_\infty=F(E[p^\infty])$ and $G=\operatorname{Gal}(F_\infty/F)$, and let $\rho:G\to GL_{d_\rho}(\overline{\mathbb Q}_p)$ be an irreducible self-dual Artin representation. If $s(E,\rho)$ is the multiplicity of $\rho$ in the Pontryagin dual of the Selmer group over a finite field through which $\rho$ factors, is $$w(E,\rho)=(-1)^{s(E,\rho)}$$?

来源字段：https://doi.org/10.4171/owr/2007/30

00162 | OWR-1590-004

### Artin’s Primitive-Root Density for Two

O / open | Oberwolfach 报告问题汇编

Does the set $S=\{p\text{ prime}:2\text{ is a primitive root modulo }p\}$ have natural density $C=0.3739\ldots$ among the primes?

来源字段：https://doi.org/10.4171/owr/2007/34

00163 | AIM-ANALYTIC_NUMBER_THEORY-0053

### Asymmetric lengths: a safe-range obstruction and conductor-matched boundary cancellation

P / partially_solved | AIM 研讨会问题汇编

Can one find a version of the recipe using asymmetric approximate functional equations, giving a “robust” version of the calculations? Similarly, can one apply the Conrey-Keating method using two asymmetric Dirichlet polynomials \[ \sum_{m\leq X}\frac{\tau_A(m)}{m^s} \qquad\text{and}\qquad \sum_{n\leq Y}\frac{\tau_B(n)}{n^{1-s}} \] to get the same answer as before, i.e. when $X=Y$?

来源字段：http://aimpl.org/zetamoments/3/

00164 | OWR-4410-006

### Asymptotic Counting in General Circle Packings

P / partially_solved | Oberwolfach 报告问题汇编

For a locally finite circle packing $P$ in the complex plane, do there exist a Borel measure $\omega_P$ and an exponent $\alpha$ such that, for every suitable bounded Borel set $E$, the number of circles meeting $E$ and having curvature less than $T$ is asymptotic to $\omega_P(E)T^\alpha$ as $T\to\infty$? What analogous results hold for circle packings beyond Apollonian packings?

来源字段：https://doi.org/10.4171/owr/2010/29

00165 | OWR-13750333-003

### Asymptotic Counting Near Curved Submanifolds

P / partially_solved | Oberwolfach 报告问题汇编

Let $M\subset\mathbb R^m$ be a bounded immersed submanifold with boundary and codimension $R$, satisfying appropriate curvature conditions. Prove $N(M;Q,\delta)\sim c_M\delta^RQ^{\dim M+1}$ whenever $\delta\geq Q^{-1/R+\epsilon}$ and $Q\to\infty$.

来源字段：https://doi.org/10.4171/owr/2023/22

00166 | OWR-14299578-009

### Asymptotic Density for Quaternary Quadratic Forms

O / open | Oberwolfach 报告问题汇编

For a quaternary quadratic form $Q$, choose the appropriate main term $M_T(r)$ and determine whether the normalized error in the count $\#\{x\in\mathbb{Z}^4\cap[-r,r]^4:Q(x)=0\}-M_T(r)$ has an asymptotic density as $r\to\infty$.

来源字段：https://doi.org/10.4171/owr/2025/52

00167 | OWR-11135-002

### Asymptotic Norms of Saito-Kurokawa Lifts

O / open | Oberwolfach 报告问题汇编

For a normalized Hecke eigenform $f\in S_{2k}(\operatorname{SL}_2(\mathbb Z))$ and its Saito-Kurokawa lift $F_f$, can one prove unconditionally that $N(F_f)\sim2$ as $k\to\infty$?

来源字段：https://doi.org/10.4171/owr/2011/43

00168 | OWR-12866-008

### Attainable Zero Counts of Nonnegative Bivariate Polynomials

O / open | Oberwolfach 报告问题汇编

Let $P(x,y)\geq0$ be a real polynomial of degree $n$ whose real zeros are isolated. Its number of real zeros is at most $n^2$, while examples give at least $\lfloor n^2/4\rfloor$. Can every integer between these two bounds occur as the number of real zeros of such a polynomial?

来源字段：https://doi.org/10.4171/owr/2014/06

00169 | OWR-8415342-021

### Automorphic Functions as Sections on Arithmetic Local Systems

O / open | Oberwolfach 报告问题汇编

Prove that there is an isomorphism $\operatorname{Func}(\operatorname{Bun}_G(\mathbb{F}_q))\simeq\Gamma(\operatorname{LocSys}^{\operatorname{arithm}}_{\widehat G},\omega)$ compatible with Hecke operators; equivalently, prove that Frobenius traces of nilpotent $\operatorname{IndCoh}$ recover the dualizing global sections.

来源字段：https://doi.org/10.4171/owr/2021/39

00170 | OWR-8415342-014

### Automorphic L-Functions from Sigma-Rho Poisson Summation

O / open | Oberwolfach 报告问题汇编

Prove that the conjectural $(\sigma,\rho)$-Poisson summation formula implies meromorphic continuation and the functional equation of the automorphic $L$-function $L(s,\sigma,\rho)$.

来源字段：https://doi.org/10.4171/owr/2021/39

00171 | OWR-1734-007

### Automorphic Realization of Motivic L-Functions

O / open | Oberwolfach 报告问题汇编

Given number fields $E$ and $F$ and a motive $M$ over $F$ with coefficients in $E$, find a connected reductive algebraic group $G/F$, an automorphic representation $\pi$ of $G(\mathbb{A}_F)$, and a representation $r$ of the L-group of $G$ such that $L(M,s)=L(s,\pi,r)$. Find $G$ as small as possible.

来源字段：https://doi.org/10.4171/owr/2007/51

00172 | AIM-ALGEBRAIC_NUMBER_THEORY-0034

### Automorphism weights forced by groupoid mass

P / partially_solved | AIM 研讨会问题汇编

For finite families of field extensions, covers, or curves over finite fields, which automorphism group should appear in an inverse-automorphism weight so that quotient, local-mass, marking, and moduli-stack formulas are simplest?

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

00173 | OWR-1734-009

### Automorphy Conjecture for Motives

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a quasi-split connected reductive group over a number field $F$, let $r$ be a representation of its L-group, and let $\varphi_v$ be the local Langlands parameters attached to a motive $M$. Does there exist an irreducible automorphic representation $\pi=\bigotimes_v\pi_v$ of $G(\mathbb{A}_F)$ such that $L(s,\pi,r)=L(M,s)$, each $\pi_v$ lies in $\Pi_{\varphi_v}$, $\pi$ is cuspidal when $M$ is absolutely irreducible, and $\pi$ is tempered and essentially unitary?

来源字段：https://doi.org/10.4171/owr/2007/51

00174 | OWR-15169-008

### Badly Approximable Points on Planar Curves

P / partially_solved | Oberwolfach 报告问题汇编

For weights $i, j \geq 0$ with $i + j = 1$, let $\operatorname{Bad}(i, j) \subset \mathbb{R}^2$ denote the set of $(i, j)$-badly approximable points in the plane, i.e., pairs $(x, y)$ for which there exists $c > 0$ such that $\max(q^i \|qx\|, q^j \|qy\|) \geq c$ for all integers $q \geq 1$. Given a nondegenerate planar curve $C \subset \mathbb{R}^2$ and weights $i, j \geq 0$ with $i + j = 1$, is the intersection $C \cap \operatorname{Bad}(i, j)$ always uncountable?

来源字段：https://doi.org/10.4171/owr/2016/48

00175 | OWR-11695860-003

### Bateman-Horn Asymptotics for Prime Polynomial Values

O / open | Oberwolfach 报告问题汇编

For an irreducible integer polynomial $f$ of degree at least two, establish the Bateman-Horn asymptotic for $\sum_{n\leq x}\Lambda(f(n))$, including fixed polynomials such as $f(t)=t^2+1$.

来源字段：https://doi.org/10.4171/owr/2022/50

00176 | NT-078

### Beal's Conjecture

O / open | 独立 / 未分配集合

For $A^x + B^y = C^z$ with $x, y, z > 2$, must $A$, $B$, and $C$ share a common prime factor?

背景链接：https://www.jointmathematicsmeetings.org/prizes-awards/paview.cgi?parent_id=41

00177 | AMR-093-0060

### Beilinson conjectures on special values of motivic L-functions

P / partially_solved | AMR 问题清单汇编

For a motive (or the cohomology of a smooth projective variety) and an appropriate integer argument, is the order of vanishing of its L-function the predicted motivic-cohomology rank, and is its first nonzero Taylor coefficient, up to the prescribed rational factor, the determinant of the Beilinson regulator?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00178 | OWR-16171-007

### Beilinson's Regulator Conjecture

O / open | Oberwolfach 报告问题汇编

Let $X$ be a variety over $\mathbb Q$ and let $m<2i$. Does the regulator map carry the appropriate integral subspace of motivic cohomology to a $\mathbb Q$-structure on Deligne cohomology, with determinant equal to the special value predicted by Beilinson's conjecture?

来源字段：https://doi.org/10.4171/owr/2018/30

00179 | AIM-ANALYTIC_NUMBER_THEORY-0074

### Bessel endpoint averages and a Chebyshev moment reduction

P / partially_solved | AIM 研讨会问题汇编

Evaluate \[ \int_0^T f\!\left(\left|\zeta\!\left(\tfrac12+it\right)\right|\right)\,dt, \] where \(f=J_{2k}\), the Bessel function, or \(f=T_{2k}\), the Chebyshev polynomial.

来源字段：http://aimpl.org/zetamoments/8/

00180 | OWR-4990381-012

### Birational Sections and Rational Points of Hyperbolic Curves

P / partially_solved | Oberwolfach 报告问题汇编

Let $K$ be the function field of a proper hyperbolic curve $X/k$. Is every section of $G_K\to G_k$ induced by a unique point of $X(k)$?

来源字段：https://doi.org/10.4171/owr/2021/12

00181 | MPP-005

### Birch and Swinnerton-Dyer Conjecture

O / open | 千禧年问题

别名：The Birch and Swinnerton-Dyer Conjecture [AG-003]; Birch & Swinnerton-Dyer conjecture [OPG-37423]; Birch-Swinnerton-Dyer Conjecture [OWR-1053-008]

The conjecture relates the rank of the abelian group of rational points of an elliptic curve to the order of zero of the associated L-function at $s=1$.

背景链接：https://www.claymath.org/millennium-problems/

00182 | OWR-16171-009

### Birch-Swinnerton-Dyer Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

The Birch-Swinnerton-Dyer conjecture asserts for every elliptic curve $E/\mathbb{Q}$ that $\operatorname{ord}_{s=1}L(E,s)=\operatorname{rank}E(\mathbb{Q})$, together with a formula for the leading Taylor coefficient in terms of arithmetic invariants; the rank part remains largely open when the rank is at least 2.

来源字段：https://doi.org/10.4171/owr/2018/30

00183 | AMR-093-0081

### Bombieri-Lang conjecture

P / partially_solved | AMR 问题清单汇编

Bombieri-Lang conjecture: $K$-rational points on a variety of general type over a number field $K$ are not a dense set in Zariski topology.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00184 | OWR-1188-012

### Bounded Local Solutions of Global Differential Modules

O / open | Oberwolfach 报告问题汇编

别名：Duplicate: Bounded Local Solutions of Global Differential Modules [OWR-1188-013]

Let $(M,\partial_M)$ be a global differential module over the global differential ring $(\mathcal O_{F'},\partial_F)$, and suppose its $p$-curvature vanishes for almost all primes $p$. Near every nonsingular prime of degree one, are the solutions of $M$ locally bounded power series over the constant field?

来源字段：https://doi.org/10.4171/owr/2006/06

00185 | OWR-17293-015

### Bounded Weighted Integral Points on a Rank-One Elliptic Curve

O / open | Oberwolfach 报告问题汇编

For integer solutions $(x,y,z)$ with $|x|,|y|,|z|\leq B$ of $zy^2+yz^2=x^3-xz^2$, is the counting quantity $\sum r_0(z)$, where $r_0(z)=1$ if $z$ is a sum of two squares and $0$ otherwise, bounded as $B\to\infty$?

来源字段：https://doi.org/10.4171/owr/2019/50

00186 | AIM-ALGEBRAIC_NUMBER_THEORY-0030

### Bounded-degree decisions and local-density support for rational curves on quartic K3 surfaces

P / partially_solved | AIM 研讨会问题汇编

Let \(U\) be the complement of all rational curves over \(\mathbf Q\).

The relevant opening assertion is:

For a smooth quartic surface \(X\subset \mathbf P^3_{\mathbf Q}\), assuming \(U(\mathbf Q)\ne\varnothing\), the source reports numerical evidence for \[ N_U(B):=\#\{u\in U(\mathbf Q):H(u)\le B\} \sim c(\log B)^{\rho(X)}, \tag{1} \] where \(\rho(X)\) is the rank of the Picard group and \(c\) is described as a product of local densities. It then asks the following related questions.

1. Can (1) hold when weak approximation fails? 2. Given a quartic K3 surface \(X/\mathbf Q\) and \(x\in X(\mathbf Q)\), is there a procedure deciding whether \(x\) lies on a rational curve over \(\mathbf Q\)? 3. What changes if elliptic curves are also removed? 4. What numerical evidence exists, and what is the “Peyre freedom” of low-degree rational curves? 5. If \(X/\mathbf Q\) is a K3 surface with a rational point, must it contain a rational curve defined over \(\mathbf Q\)? Must one pass through a prescribed \(x\in X(\mathbf Q)\)? 6. For \(X/\mathbf F_q\) and \(x\in X(\mathbf F_q)\), is there an \(\mathbf F_q\)-defined rational curve through \(x\)? Does \(X\) contain any \(\mathbf F_q\)-defined rational curve? 7. The source notes a Bogomolov--Tschinkel result for Kummer surfaces. 8. Can a K3 surface \(X/\mathbf Q\) with \(\operatorname{Pic}(X_{\mathbf C})\simeq\mathbf Z\) have infinitely many rational curves defined over \(\mathbf Q\)?

These ranks must not be silently identified.

The source does not normalize \(H\). For definiteness, this report uses the standard multiplicative height induced by \(\mathcal O_X(1)\), obtained from primitive integral projective coordinates by \(H([x_0:\cdots:x_3])=\max_i|x_i|\). Changing the adelic metric changes the constant \(c\), not the logical conclusions below. The complement \(U\) is a set-theoretic complement of a countable union and need not be Zariski open.

来源字段：https://aimath.org/pastworkshops/ratlhigherdimvarproblems.pdf

00187 | AIM-ALGEBRAIC_NUMBER_THEORY-0020

### Bounded-degree quotient obstructions to solvable points

P / partially_solved | AIM 研讨会问题汇编

1.) Find effective obstructions to the existence of solvable points on varieties (in any characteristic).

2.) Do there exist curves (over surfaces) over $\mathbb Q$ without solvable points?

来源字段：http://aimpl.org/degreedsurface/1/

00188 | OWR-15170-007

### Bounded-House Finiteness for Totally Real Algebraic Integers

O / open | Oberwolfach 报告问题汇编

For the ring $\mathcal O_{\mathbb Q_{\mathrm{tr}}^{(d)}}$ of totally real algebraic integers, is $\{\alpha:|\alpha|<X\}$ finite for every $X\geq1$, where $|\alpha|$ is the maximum modulus of the conjugates of $\alpha$?

来源字段：https://doi.org/10.4171/owr/2016/49

00189 | GREEN-062

### Bounds for Birch's Theorem

P / partially_solved | Ben Green 问题

别名：Bounds for Homogeneous Polynomial Zeros [GREEN-098]

Let $d \geq 3$ be odd. Give bounds on $\nu(d)$ such that if $n > \nu(d)$ then any homogeneous polynomial $F(\mathbf{x}) \in \mathbb{Z}[x_1, \dots, x_n]$ of degree $d$ has a nontrivial integer zero.

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

00190 | OWR-14211-007

### Bounds for Quaternary Local-Global Representation Exceptions

O / open | Oberwolfach 报告问题汇编

If Q is a positive-definite quaternary quadratic form with discriminant $D(Q)$, is the largest positive integer n that is locally represented by Q but not represented by Q always $O_\varepsilon(D(Q)^{1+\varepsilon})$ for every $\varepsilon>0$?

来源字段：https://doi.org/10.4171/owr/2016/3

00191 | OWR-4076-003

### Brauer-Manin Conjecture for K3 Surfaces

O / open | Oberwolfach 报告问题汇编

For every K3 surface $X$ over a number field $k$, is $X(k)$ dense in its Brauer-Manin set of adelic points? Equivalently, is the Brauer-Manin obstruction the only obstruction to the Hasse principle and weak approximation on $X$?

来源字段：https://doi.org/10.4171/owr/2009/33

00192 | OWR-16166-001

### Brauer-Manin Obstruction for Rationally Connected Varieties

O / open | Oberwolfach 报告问题汇编

Let $X$ be a smooth, proper, geometrically irreducible, rationally connected variety over a number field $k$. Is $X(k)$ dense in the Brauer-Manin set $X(\mathbb A_k)^{\operatorname{Br}}$?

来源字段：https://doi.org/10.4171/owr/2018/25

00193 | OWR-17293-003

### Brauer-Manin Obstruction for Rationally Connected Varieties

O / open | Oberwolfach 报告问题汇编

For every smooth, proper, geometrically integral, rationally connected variety, the Brauer-Manin obstruction is the only obstruction to the Hasse principle.

来源字段：https://doi.org/10.4171/owr/2019/50

00194 | OWR-15170-001

### Brauer-Manin Obstructions for Rationally Connected Fibrations

P / partially_solved | Oberwolfach 报告问题汇编

Let $k$ be a number field and $f:X\to\mathbb P^1_k$ a projective surjective morphism with geometrically rationally connected generic fibre. If the Brauer-Manin obstruction is the only obstruction to the Hasse principle for every smooth fibre of $f$, is it also the only obstruction for $X$?

来源字段：https://doi.org/10.4171/owr/2016/49

00195 | OWR-12010-012

### Breuil-Mézard Multiplicity Formula

P / partially_solved | Oberwolfach 报告问题汇编

For a residual representation $\bar\rho:G_K\to\operatorname{GL}_2(\mathbb F)$, do there exist nonnegative integers $\mu_a(\bar\rho)$, independent of the regular $p$-adic Hodge type $\lambda$ and inertial type $\tau$, such that $e(R_{\lambda,\tau}/\pi_E R_{\lambda,\tau})=\sum_a n(a)\mu_a(\bar\rho)$ for every $\lambda$ and $\tau$?

来源字段：https://doi.org/10.4171/owr/2012/38

00196 | OWR-4537-002

### Brumer-Stark Class Group Annihilation Conjecture

S / solved | Oberwolfach 报告问题汇编

Let $K$ be a finite abelian extension of $\mathbb{Q}$ and let $S$ contain the infinite and ramified places. Define $\theta_S(s)=\prod_{P\notin S}(1-\sigma_P^{-1}(NP)^{-s})^{-1}$. Does $e\theta_S(0)$ annihilate $\operatorname{Cl}_K$, where $e=|\mu_K|$?

来源字段：https://doi.org/10.4171/owr/2010/55

00197 | AMR-093-0182

### Brumer-Stark conjecture

S / solved | AMR 问题清单汇编

For a finite abelian extension of number fields, does the Brumer-Stark element have the predicted ideal-class annihilation and principalization property, with anti-units whose roots generate the prescribed abelian Kummer extensions, including the remaining 2-primary cases?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00198 | NT-084

### Bunyakovsky Conjecture

O / open | 独立 / 未分配集合

Does an irreducible integer polynomial with no fixed prime divisor produce infinitely many primes?

背景链接：https://encyclopediaofmath.org/wiki/Bunyakovskii_conjecture

00199 | AMR-093-0001

### Büchi's problem

P / partially_solved | AMR 问题清单汇编

Büchi's problem on sufficiently large sequences of square numbers with constant second difference.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00200 | OWR-15170-019

### Büchi's Problem for Sequences of Integer Squares

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a positive integer $M$ such that every sequence of $M$ integer squares with second differences equal to $2$ is necessarily of the form $(x+i)^2$ for some $x\in\mathbb Z$ and $1\leq i\leq M$?

来源字段：https://doi.org/10.4171/owr/2016/49

00201 | AMR-093-0094

### Can a discrete logarithm on a elliptic curve be computed in sub-exponential time

O / open | AMR 问题清单汇编

Can a discrete logarithm on a elliptic curve be computed in sub-exponential time?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00202 | AMR-093-0168

### Can a prime p satisfy $2^{p-1}\equiv 1\pmod{p^2}$ and $3^{p-1}\equiv 1\pmod{p^2}$ simultaneously

O / open | AMR 问题清单汇编

Can a prime p satisfy $2^{p-1}\equiv 1\pmod{p^2}$ and $3^{p-1}\equiv 1\pmod{p^2}$ simultaneously?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00203 | AMR-093-0112

### Can every integer be written as a sum of four perfect cubes

S / solved | AMR 问题清单汇编

Can every integer be written as a sum of four perfect cubes?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00204 | OWR-11695860-011

### Cancellation in Twisted-Multiplicative Exponential Sums

O / open | Oberwolfach 报告问题汇编

For $W_f(a;q)=q^{-1/2}\sum_{x\bmod q}e(af(x)/q)$, develop methods for estimating $\sum_{q\leq x}W_f(a;q)$ without replacing each summand by its absolute value.

来源字段：https://doi.org/10.4171/owr/2022/50

00205 | OWR-13102-005

### Canonical Plectic Lifts of Shimura-Variety Cohomology

P / partially_solved | Oberwolfach 报告问题汇编

Does the étale cohomology complex of a Shimura variety admit a canonical plectic lift with the indicated $\operatorname{Aut}(F\otimes\overline{\mathbb{Q}}/F)[\mu]$-action, compatible with products and the action of $G(\mathbb{Q})$, so that plectic $\ell$-adic étale cohomology can be defined by taking the corresponding group cohomology?

来源字段：https://doi.org/10.4171/owr/2014/32

00206 | OWR-1703867-006

### Canonical Triple-Product $p$-Adic $L$-Values and Rational Points

P / partially_solved | Oberwolfach 报告问题汇编

Is there a canonical multiple $L_p^{\mathrm{can}}(f,g,h)$ of $L_p(f,g,h)$ whose value at $(2,1,1)$ equals the formal-group logarithm of a rational point $P_\rho\in E(\overline{\mathbb Q})_\rho$, with $P_\rho$ nontrivial if and only if $L(E,\rho,s)$ has a simple zero at $s=1$?

来源字段：https://doi.org/10.4171/owr/2020/20

00207 | AMR-093-0002

### Carmichael's totient function conjecture

O / open | AMR 问题清单汇编

Carmichael's totient function conjecture: do all values of Euler's totient function have multiplicity greater than $1$?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00208 | NT-016

### Catalan-Mersenne Conjecture

O / open | 独立 / 未分配集合

Are all Catalan-Mersenne numbers $C_n$ composite for $n > 4$? Here $C_0 = 2$ and $C_{n+1} = 2^{C_n} - 1$.

背景链接：https://en.wikipedia.org/wiki/Double_Mersenne_number

00209 | AMR-093-0003

### Catalan-Dickson conjecture on aliquot sequences

O / open | AMR 问题清单汇编

Catalan-Dickson conjecture on aliquot sequences: no aliquot sequences are infinite but non-repeating.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00210 | AIM-ANALYTIC_NUMBER_THEORY-0020

### Certified Petersson resolution for finite point counts

P / partially_solved | AIM 研讨会问题汇编

Various circle methods and $\delta$-methods have been successfully used to count points on varieties. Could one use Petersson or Kuznetsov trace formula as a $\delta$-method to count point on varieties?

来源字段：http://aimpl.org/deltasubconvex2/1/

00211 | OWR-17293-004

### CFKRS Formula for Shifted Zeta Function Moments

O / open | Oberwolfach 报告问题汇编

The CFKRS conjecture predicts an asymptotic formula for smoothed averages of products of zeta-functions with small complex shifts: the main term is the sum of all equal-cardinality swap terms, with Dirichlet-series factor $B(C,D)=\sum_{n\geq 1}\tau_C(n)\tau_D(n)/n$, and an error $O(T^{1-\delta})$ as $T\to\infty$.

来源字段：https://doi.org/10.4171/owr/2019/50

00212 | OWR-14299578-003

### Character Sums for General Polynomial Families

O / open | Oberwolfach 报告问题汇编

Extend the available stratification and cancellation estimates for multiplicative character sums from the treated polynomial family to broad, explicitly characterized families of multivariable polynomials.

来源字段：https://doi.org/10.4171/owr/2025/52

00213 | AIM-ANALYTIC_NUMBER_THEORY-0103

### Character-phase evidence for Elliott--Halberstam and a sparse twisted-psi diagonal

P / partially_solved | AIM 研讨会问题汇编

What evidence exists in support of the Elliott--Halberstam conjecture (EH)? Vaughan suggested that progress would most likely exploit cancellation among the values of \(\overline\chi(a)\) in \[ \psi(x;q,a)=\frac{1}{\phi(q)}\sum_{\chi\bmod q}\overline\chi(a)\psi(x,\chi). \] Conrey asked whether known asymptotics for \[ \sum_{\chi\bmod q}^{*}\chi(a)|L(1/2,\chi)|^2 \] suggest that the desired cancellation is nonexistent. Soundararajan replied that the latter behavior comes from the Euler product and has no analogue for \(\psi(x,\chi)\).

With only those typographical defects repaired, the key passage is:

Here \(^*\) means primitive characters. These corrections are confirmed by page 1 of the PDF.

For a precise modern target, put \[ \Delta(x;q,a)=\psi(x;q,a)-\frac{x}{\phi(q)},\qquad \psi(x;q,a)=\sum_{\substack{n\le x\\n\equiv a\pmod q}}\Lambda(n). \] The standard Elliott--Halberstam assertion is that, for every \(A>0\) and every fixed \(\theta<1\), \[ \sum_{q\le x^\theta}\max_{(a,q)=1}|\Delta(x;q,a)| \ll_{A,\theta}\frac{x}{(\log x)^A}. \tag{EH} \] Thus the record contains three connected questions: what theorem-level evidence supports (EH), what character-phase cancellation would explain it, and whether a twisted second moment of central \(L\)-values obstructs that mechanism.

来源字段：https://aimath.org/WWN/primegaps/WorkshopProblems.pdf

00214 | OWR-12493-003

### Characteristic-Zero Lifts of Local Galois Extensions

O / open | Oberwolfach 报告问题汇编

Let $k$ be an algebraically closed field of characteristic $p$, let $L/k((t))$ be a Galois extension with group $G$, and let $A$ be the integral closure of $k[[t]]$ in $L$. Does there exist a complete discrete valuation ring $R$ of characteristic zero with residue field $k$ and a $G$-Galois lift whose reduction is $k[[t]]$-isomorphic to $A$?

来源字段：https://doi.org/10.4171/owr/2013/30

00215 | OWR-13351-009

### Characteristic-Zero Lifts of Local Galois Extensions

O / open | Oberwolfach 报告问题汇编

Given an algebraically closed field $k$ of characteristic $p$ and a $G$-Galois extension $k[[z]]/k[[t]]$, does there exist a complete characteristic-zero discrete valuation ring $(R,\pi)$ with residue field $k$ and a $G$-Galois extension $R[[Z]]/R[[T]]$ reducing to $k[[z]]/k[[t]]$ modulo $\pi$?

来源字段：https://doi.org/10.4171/owr/2014/49

00216 | AMR-093-0059

### Characterize all algebraic number fields that have some power basis

O / open | AMR 问题清单汇编

Characterize all algebraic number fields that have some power basis.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00217 | NT-064

### Class Number Problem

O / open | 独立 / 未分配集合

别名：Class Number Problem [NUM-012]

Are there infinitely many real quadratic number fields with unique factorization?

背景链接：https://en.wikipedia.org/wiki/List_of_number_fields_with_class_number_one

00218 | OWR-12010-010

### Classification and Specialization of p-Divisible Groups

O / open | Oberwolfach 报告问题汇编

Classify all $p$-divisible groups $D$ with fixed dimension and codimension, including their specialization relations, finite-level $\operatorname{Hom}$ groups, and useful specialization-monotone invariants.

来源字段：https://doi.org/10.4171/owr/2012/38

00219 | OWR-14298584-013

### Classification of Abelian-Surface L-Functions of Small Conductor

O / open | Oberwolfach 报告问题汇编

Prove the proposed classification of the $456$ $L$-functions of abelian surfaces over $\mathbb{Q}$ of conductor at most $1000$, including completeness of the list and the asserted division into products, quadratic-coefficient newforms, Weil restrictions, and generic abelian surfaces.

来源字段：https://doi.org/10.4171/owr/2024/40

00220 | OWR-13351-011

### Classification of Local Oort Groups

O / open | Oberwolfach 报告问题汇编

What is the complete list of local Oort groups, namely groups for which every local G-extension lifts to characteristic zero?

来源字段：https://doi.org/10.4171/owr/2014/49

00221 | OWR-1586-007

### CM Hida Families and p-Adic L-Function Ideals

P / partially_solved | Oberwolfach 报告问题汇编

For a CM Hida deformation, is $(L_p^{\mathrm{an}}(\operatorname{GL}(1)/K))=(L_p^{\mathrm{an}}(\operatorname{GL}(2)/\mathbb{Q}))$ as ideals?

来源字段：https://doi.org/10.4171/owr/2007/30

00222 | OWR-11127-010

### CM Rational Points on Nonsplit Cartan Modular Curves

O / open | Oberwolfach 报告问题汇编

Does there exist a constant $c$ such that, for every prime $p>c$, the only rational points on the modular curve $X_{\mathrm{ns}}(p)$ over $\mathbb Q$ are CM points?

来源字段：https://doi.org/10.4171/owr/2011/35

00223 | AIM-ANALYTIC_NUMBER_THEORY-0083

### Coefficientwise indices and a Calabi-Yau threefold level-one test

P / partially_solved | AIM 研讨会问题汇编

别名：Coefficientwise indices and a level-two Calabi--Yau/CFT test [AIM-PHYSICS-0016]

Is there an index theoretic interpretation of elliptic genus? When it can, how do we compute it and what is the geometric interpretation of the coefficients? Check that this matches CFT calculations.

来源字段：http://aimpl.org/gromwitnumthry/3/

00224 | GUY-A19b

### Cohen-Selfridge Problem on $\pm p^a \pm 2^b$

P / partially_solved | Richard Guy 数论问题

What is the least positive odd number not of the form $\pm p^a \pm 2^b$, where $p$ is an odd prime?

背景链接：https://www.ams.org/journals/mcom/1975-29-129/S0025-5718-1975-0376583-0/S0025-5718-1975-0376583-0.pdf

00225 | OWR-4076-009

### Cohen-Lenstra Densities for Class-Group $\ell$-Ranks

P / partially_solved | Oberwolfach 报告问题汇编

Let $\ell$ be prime and $r\geq0$. Is the density of fundamental discriminants $D$ for which $\operatorname{rk}_\ell(C_D)=r$ equal to $\ell^{-r}\eta_\infty(\ell)\eta_r(\ell)^{-2}$ for $D<0$ and to $\ell^{-r(r+1)}\eta_\infty(\ell)\eta_r(\ell)^{-1}\eta_{r+1}(\ell)$ for $D>0$, where $\eta_k(t)=\prod_{j=1}^k(1-t^{-j})$?

来源字段：https://doi.org/10.4171/owr/2009/33

00226 | OWR-1542-005

### Cohen-Lenstra Moments for Narrow Class Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $D$ range over fundamental discriminants, let $C_D$ be the narrow class group of $\mathbb{Q}(\sqrt D)$, and put $\operatorname{rk}_\ell(A)=\dim_{\mathbb{F}_\ell}(A/A^\ell)$. If $M^-$ and $M^+$ denote means over negative and positive fundamental discriminants, respectively, is it true for every prime $\ell$ and integer $k>0$ that $M^-(\prod_{0\leq i<k}(\ell^{\operatorname{rk}_\ell(C_D^2)}-\ell^i))=1$ and $M^+(\prod_{0\leq i<k}(\ell^{\operatorname{rk}_\ell(C_D^2)}-\ell^i))=\ell^{-k}$?

来源字段：https://doi.org/10.4171/owr/2007/26

00227 | AIM-ALGEBRAIC_NUMBER_THEORY-0095

### Cohomological dimension one is not a rational-point principle

S / solved | AIM 研讨会问题汇编

**Question 11 (Colliot-Thélène).** Let \(X/k\) be a smooth projective rationally connected variety and suppose that \(\operatorname{cd}(k)\leq 1\). Must \(X(k)\) be nonempty?

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

00228 | AIM-ALGEBRAIC_NUMBER_THEORY-0023

### Common twists, sparse families, and the marked-torsor bottleneck

P / partially_solved | AIM 研讨会问题汇编

Does Bhargava's machinery have implications for the Hasse principle for special surfaces? For example, let $F,G\in\mathbb Q[x,y]$ be homogeneous polynomials of degree $3$. Consider the cubic surface \[ S=\{F(x,y)=G(z,w)\}\subset\mathbb P^3_{\mathbb Q}. \] The defining equation is equivalent to the system of equations \[ u^3=tF(x,y),\qquad v^3=tG(z,w). \] This defines a family of cubic twists of curves of genus $1$ over the $t$-line. Swinnerton-Dyer has discussed how to search for $t$ such that this system is solvable over $\mathbb Q$ in the diagonal case [Ann. Sci. ENS]. Can we extend his work beyond the diagonal case? Similarly, consider Kummer K3 surfaces defined by \[ z^2=f(x)g(y), \] where $f,g$ are quartic separable polynomials. This is equivalent to the family of quadratic twists of curves of genus $1$ defined by \[ u^2=tf(x),\qquad v^2=tg(y). \] The goal is to eliminate the condition in Swinnerton-Dyer's work that (the $2$-primary part of) $\Sha$ has finite order for quadratic twists, using the recent work presented in Bhargava's talk.

来源字段：https://aimath.org/pastworkshops/ratlhigherdimvarproblems.pdf

00229 | AMR-116-0001

### Comparative Prime Number Theory - Problem 1

O / open | AMR 问题清单汇编

- (Greg Martin and Lior Silberman, University of British Columbia) An *ordinate* of an $L$-function is a positive imaginary part of a nontrivial zero of that $L$-function. LI implies that at most one positive ordinate can be rational, and presumably even that is not possible (for classical analytic $L$-functions).

1}: Prove that $\zeta(s)$ has at least one(!) irrational ordinate.

背景链接：https://arxiv.org/abs/2407.03530

00230 | AMR-116-0012

### Comparative Prime Number Theory - Problem 1

O / open | AMR 问题清单汇编

The best improvement to this lower bound I am aware of is by Morrill, Platt, and Trudgian [MPT] (improving work of Kaczorowski [K2]), who proved that \[ \liminf \frac{W^\psi(T)}{\log T} \ge \frac{\gamma_1}{\pi} + 1.867\cdot10^{-30}. \]

Improve this lower bound, even assuming RH. Here are some suggested target lower bounds for $\liminf {W^\psi(T)}/{\log T}$: (a) $1.01 \frac{\gamma_1}\pi$; (b) $2 \frac{\gamma_1}\pi$; (c) $\infty$.

What we expect to be true is that the large-scale intervals of positive and negative values of $\psi(x)-x$ do indeed alternate at about the speed given by the repeated-averaging argument, but that each such sign change detected is actually a flurry of many sign changes on a small scale (this is how random walks behave, for example). In particular, we expect $W^\psi(T)$ and related sign-change counting functions to grow more like $\sqrt T$ (up to logarithmic factors) than like $\log T$.

背景链接：https://arxiv.org/abs/2407.03530

00231 | AMR-116-0015

### Comparative Prime Number Theory - Problem 1

O / open | AMR 问题清单汇编

- (Nathan Ng, University of Lethbridge) Feuerverger and Martin [FeM] numerically computed the densities for some three-way races, as well as for the four-way races for the moduli $5$, $8$, and $12$. Their paper contains a general formula for densities in races with any number of competitors; but there have been few (if any) subsequent computations of densities for races with three or more competitors.

Numerically compute some further three-way (or higher) races.

Fiorilli and Martin [FiM] determined the 117 most biased two-way races.

背景链接：https://arxiv.org/abs/2407.03530

00232 | AMR-116-0009

## Comparative Prime Number Theory - Problem 10

O / open | AMR 问题清单汇编

- (Stanley Yao Xiao, University of Northern British Columbia) Let $f \in \mathbb{Z}[x,y]$ be a primitive, positive definite binary quadratic form such that $f(x,1) \not\equiv x(x+1) \pmod{2}$. Consider the counting function for the generalized Friedlander--Iwaniec primes \[ \pi_f(x) = \# \{p \leq x \colon p = f(a,b^2) \text{ for some integers $a$ and $b$}\}. \] By the main theorem in my paper [X], we have $\pi_f(x) \sim C(\operatorname{Disc}(f)) x^{3/4}/\log x$ for some positive constant $C(\operatorname{Disc}(f))$ depending only on the discriminant of $f$.

If $f$ and $g$ are two distinct such quadratic forms with the same discriminant, how do the quantities $\pi_f(x)$ and $\pi_g(x)$ compare as $x \rightarrow \infty$?

A similar question can be asked for indefinite binary quadratic forms, where the analogous counting function must restrict $a$ and $b$ to a suitable ``box'' depending on $x$.

Heath-Brown and Moroz [HBM] gave a similar asymptotic formula for primitive binary cubic forms $F$ such that $F(x,y) \not\equiv xy(x+y) \pmod{2}$; one could again look for pairs $F,G$ of distinct such cubic forms such that $\pi_F(x) \sim \pi_G(x)$ and compare those counting functions to each other.

- (Shin-ya Koyama, Toyo University) A new formulation of Chebyshev's bias was given in [AK] in terms of the difference of the weighted counting function of prime numbers $p$ as \begin{equation*} \sum_{\genfrac{}{}{0pt}{1}{p\le x}{p\equiv 3\mod4}} \frac 1{\sqrt p} -\sum_{\genfrac{}{}{0pt}{1}{p\le x}{p\equiv 1\mod4}} \frac 1{\sqrt p} \sim\frac12\log\log x\quad(x\to\infty). \end{equation*} **Problem**: What is the logical inclusion or relation of this new definition to the conventional definiton of ``Chebyshev's bias'' in terms of the logarithmic density of the set of $x$ satisfying $\pi(x,4,3)>\pi(x,4,1)$ being very close to 1?

- (Greg Martin, University of British Columbia) This question was asked by Carl Pomerance and appears in [Mar, Section 3]. Let $a$ and $b$ be distinct reduced residue classes modulo $q$.

背景链接：https://arxiv.org/abs/2407.03530

00233 | AMR-116-0010

## Comparative Prime Number Theory - Problem 11

O / open | AMR 问题清单汇编

- (Greg Martin, University of British Columbia) This question was asked by Carl Pomerance and appears in [Mar, Section 3]. Let $a$ and $b$ be distinct reduced residue classes modulo $q$.

Prove unconditionally that the set of $x$ for which $\pi(x;q,a)=\pi(x;q,b)$ (the set of ``ties'' in the prime number race) has (natural or logarithmic) density $0$.

If we assume GRH and a very mild version of LI (see [Devin, Theorem 2.2]), the limiting logarithmic distribution of $\bigl( \pi(x;q,a)=\pi(x;q,b) \bigr) (\log x)/\sqrt x$ exists and is absolutely continuous with respect to Lebesgue measure. Consequently, for any function $f(x)$ that is $o(\sqrt x/\log x)$, the set of $x$ for which $\bigl| \pi(x;q,a)=\pi(x;q,b) \bigr| < f(x)$ has logarithmic density $0$; this conclusion is far stronger than ``ties have density $0$''. But ``ties have density $0$'' is such a weak assertion that we might hope to prove it unconditionally.

背景链接：https://arxiv.org/abs/2407.03530

00234 | AMR-116-0011

## Comparative Prime Number Theory - Problem 12

O / open | AMR 问题清单汇编

- (Daniel R. Johnston, University of New South Wales--Canberra at ADFA) Define \[ S(x)=\sum_{n\leq x}(-2)^{\Omega(n)} \qquad\text{and}\qquad \alpha = \limsup_{x\to\infty}\frac{|S(x)|}{x}. \] In a soon-to-be-released preprint by students at UNSW--Canberra, it is shown that $\alpha$ exists, and a (large) upper bound is given. Note that $S(x)$ has jump discontinuities of size $x$ when $x$ is a power of $2$, which implies the lower bound $\alpha\ge\tfrac{1}{2}$. A conjecture [sun2016pair, Conjecture 1.1], supported by extensive computations [mossinghoff2021oscillations], asks whether $\alpha\leq 1$.

What is the exact value of $\alpha$?

- (Greg Martin, University of British Columbia) The standard way to investigate the frequency of sign changes of error terms such as $\psi(x)-x$ is to use repeated averaging so that the largest term in the explicit formula eventually dominates the sum of all other terms. In this way, one shows that the number of sign changes is essentially at least as large as the number of sign changes of that largest term. For example, Kaczorowski [K1] showed unconditionally that \[ \liminf \frac{W^\psi(T)}{\log T} \ge \frac{\gamma_1}{\pi}. \] This result can be generalized to many other error terms (although more generally one must assme that the relevant $L$-functions have no real zeros, and GRH tends to be necessary for error terms with biases).

The best improvement to this lower bound I am aware of is by Morrill, Platt, and Trudgian [MPT] (improving work of Kaczorowski [K2]), who proved that \[ \liminf \frac{W^\psi(T)}{\log T} \ge \frac{\gamma_1}{\pi} + 1.867\cdot10^{-30}. \]

背景链接：https://arxiv.org/abs/2407.03530

00235 | AMR-116-0014

## Comparative Prime Number Theory - Problem 15

O / open | AMR 问题清单汇编

- (Alexandre Bailleul, Université Paris--Saclay) This question was raised by Nilotpal Sinha [Sinha]. Let $P^+(n)$ denote the largest prime factor of $n$. Numerical investigation suggests that there is a bias towards integers $n$ for which $P^+(n)\equiv3\mod4$ over integers $n$ for which $P^+(n)\equiv1\mod4$, but the asymptotic behaviour of $\sum_{n\le x} \chi_{-4}(P^+(n))$ is still unknown.

Prove or disprove that $\sum_{n\le x} \chi_{-4}(P^+(n)) < 0$ for all $n \geq 2$. Possibly even find an asymptotic formula for the sum.

背景链接：https://arxiv.org/abs/2407.03530

00236 | AMR-116-0017

## Comparative Prime Number Theory - Problem 18

O / open | AMR 问题清单汇编

It could be more challenging to bound the densities in three-way races sufficiently well to verify that such a list is exhaustive: as Lamzouri [Lam] showed, the rate of convergence of three-way race densities to $\frac16$ is only $O(1/\log q)$, as opposed to a $q^{-1/2+o(1)}$ rate of convergence to $\frac12$ for densities in the two-way case.

- (Greg Martin, University of British Columbia)

Compute (assuming GRH and LI) the densities in the five-way race among the quadratic residues and among the quadratic nonresidues modulo $11$.

There are $120$ densities of the form $\delta_{11;a_1,a_2,a_3,a_4,a_5}$ where $\{a_1,a_2,a_3,a_4,a_5\} = \{1,3,4,5,9\}$ (the quadratic residues modulo $11$). There is a known formula [FeM, Theorem 4] for $\delta_{q;a_1,a_2,a_3,a_4,a_5}$ that can in principle be computed to any desired accuracy; however, the formula involves a four-dimensional integral that was beyond the authors' computational ability at the time.

In addition to being the simplest natural five-way prime number race, Bays and Hudson [BH] noticed that this particular race exhibited an interesting cyclic phenonemon, which the computation of these densities might shed some light on.

Because of the symmetries in [FeM, Theorem 2], there are only $8$ (presumably) distinct values among these $120$ densities, so all that would need to be computed are $\delta_{11;1, 3, 4, 5, 9}$, $\delta_{11;1, 3, 4, 9, 5}$, $\delta_{11;1, 3, 5, 4, 9}$, $\delta_{11;1, 3, 5, 9, 4}$, $\delta_{11;1, 3, 9, 5, 4}$, $\delta_{11;1, 5, 3, 4, 9}$, $\delta_{11;1, 5, 3, 9, 4}$, and $\delta_{11;1, 5, 9, 3, 4}$. By the same theorem, these values also represent all possible densities in the five-way race among the quadratic nonresidues $\{2,6,7,8,10\}$ modulo $11$.

- (Nathan Ng, University of Lethbridge) My Ph.D. thesis [NgPhd] contains some computations of numerical examples of prime number races in Chebotarev's density theorem. There is now a large database of zeros of Artin $L$-functions at the LMFDB [LMFDB], as well as Rubinstein's lcalc program which has been integrated into SageMath.

背景链接：https://arxiv.org/abs/2407.03530

00237 | AMR-116-0018

## Comparative Prime Number Theory - Problem 19

O / open | AMR 问题清单汇编

Investigate further examples. Determine the most biased two-way Chebotarev races.

- (Greg Martin, University of British Columbia) In 1959 Shanks [Shanks] made an observation that seems to have been overlooked since then: some combinatorial reasoning led him to assert that for each integer $k\ge1$, the odd numbers $n$ with $\Omega(n)=k$ should be biased towards the residue class $(-1)^k\mod4$, where the case $k=1$ is Chebyshev's original observation. In other words, $(-1)^{\Omega(n)}\chi_{-4}(n)$ should be biased towards $1$ over $-1$.

背景链接：https://arxiv.org/abs/2407.03530

00238 | AMR-116-0013

## Comparative Prime Number Theory - Problem 2

O / open | AMR 问题清单汇编

What we expect to be true is that the large-scale intervals of positive and negative values of $\psi(x)-x$ do indeed alternate at about the speed given by the repeated-averaging argument, but that each such sign change detected is actually a flurry of many sign changes on a small scale (this is how random walks behave, for example). In particular, we expect $W^\psi(T)$ and related sign-change counting functions to grow more like $\sqrt T$ (up to logarithmic factors) than like $\log T$.

Make any progress whatsoever towards advancing our knowledge of sign changes beyond the repeated-averaging techniques.

- (Alexandre Bailleul, Université Paris--Saclay) This question was raised by Nilotpal Sinha [Sinha]. Let $P^+(n)$ denote the largest prime factor of $n$. Numerical investigation suggests that there is a bias towards integers $n$ for which $P^+(n)\equiv3\mod4$ over integers $n$ for which $P^+(n)\equiv1\mod4$, but the asymptotic behaviour of $\sum_{n\le x} \chi_{-4}(P^+(n))$ is still unknown.

背景链接：https://arxiv.org/abs/2407.03530

00239 | AMR-116-0016

### Comparative Prime Number Theory - Problem 2

O / open | AMR 问题清单汇编

**Problem 1**: Numerically compute some further three-way (or higher) races.

Fiorilli and Martin [FiM] determined the 117 most biased two-way races.

Determine the $100$ (for example) most biased three-way races.

It could be more challenging to bound the densities in three-way races sufficiently well to verify that such a list is exhaustive: as Lamzouri [Lam] showed, the rate of convergence of three-way race densities to $\frac16$ is only $O(1/\log q)$, as opposed to a $q^{-1/2+o(1)}$ rate of convergence to $\frac12$ for densities in the two-way case.

- (Greg Martin, University of British Columbia)

背景链接：https://arxiv.org/abs/2407.03530

00240 | AMR-116-0019

### Comparative Prime Number Theory - Problem 20

O / open | AMR 问题清单汇编

- (Greg Martin, University of British Columbia) In 1959 Shanks [Shanks] made an observation that seems to have been overlooked since then: some combinatorial reasoning led him to assert that for each integer $k\ge1$, the odd numbers $n$ with $\Omega(n)=k$ should be biased towards the residue class $(-1)^k\mod4$, where the case $k=1$ is Chebyshev's original observation. In other words, $(-1)^{\Omega(n)}\chi_{-4}(n)$ should be biased towards $1$ over $-1$.

Assuming GRH and LI, calculate the logarithmic density of the sets \[ \{n\ge1\colon (-1)^{\Omega(n)}\chi_{-4}(n) = 1\} \quad\text{and}\quad \{n\ge1\colon (-1)^{\Omega(n)}\chi_{-4}(n) = -1\}. \] Presumably a similar phenomenon should hold for $(-1)^{\Omega(n)}\chi(n)$ for any quadratic character $\chi$; the case $\chi=\chi_{-3}$ (the nonprincipal Dirichlet character modulo $3$) might be particularly pronounced.

By the way, [Shanks] also contains the timeless quip ``Since $\log\log\log n\to\infty$ (with great dignity)\dots.''

- (Daniel R. Johnston, University of New South Wales--Canberra at ADFA). It has been shown [johnston2023average, Theorem 1.1] that RH is equivalent to \[ {\mathfrak{A}}_1^\pi(x) = \int_0^x \bigl( \pi(t)-\mathop\mathrm{li}(t) \bigr) dt < 0 \quad\text{for all $x>2$.} \] Roughly, the reason this inequality holds under RH is that the constant $\sum_{\rho} 1/\rho(\rho+1)$ occuring in the explicit formula for ${\mathfrak{A}}_1^\pi(x)$ is too small to overcome the bias. However, Martin pointed out (private communication) that when looking at the analogous \[ {\mathfrak{A}}_1^\pi(x;q,a) = \int_0^x \bigl( \phi(q)\pi(t;q,a)-\mathop\mathrm{li}(t) \bigr) dt, \] the corresponding sum over zeros of Dirichlet $L$-functions grows faster than the bias as $q$ increases, and thus at some point the inequality ${\mathfrak{A}}_1^\pi(x;q,a) < 0$ should fail when $q$ is large.

- (Daniel R. Johnston, University of New South Wales--Canberra at ADFA). It has been shown [johnston2023average, Theorem 1.1] that RH is equivalent to \[ {\mathfrak{A}}_1^\pi(x) = \int_0^x \bigl( \pi(t)-\mathop\mathrm{li}(t) \bigr) dt < 0 \quad\text{for all $x>2$.} \] Roughly, the reason this inequality holds under RH is that the constant $\sum_{\rho} 1/\rho(\rho+1)$ occuring in the explicit formula for ${\mathfrak{A}}_1^\pi(x)$ is too small to overcome the bias. However, Martin pointed out (private communication) that when looking at the analogous \[ {\mathfrak{A}}_1^\pi(x;q,a) = \int_0^x \bigl( \phi(q)\pi(t;q,a)-\mathop\mathrm{li}(t) \bigr) dt, \] the corresponding sum over zeros of Dirichlet $L$-functions grows faster than the bias as $q$ increases, and thus at some point the inequality ${\mathfrak{A}}_1^\pi(x;q,a) < 0$ should fail when $q$ is large.

背景链接：https://arxiv.org/abs/2407.03530

00241 | AMR-116-0020

### Comparative Prime Number Theory - Problem 21

O / open | AMR 问题清单汇编

- (Daniel R. Johnston, University of New South Wales--Canberra at ADFA). It has been shown [johnston2023average, Theorem 1.1] that RH is equivalent to \[ {\mathfrak{A}}_1^\pi(x) = \int_0^x \bigl( \pi(t)-\mathop\mathrm{li}(t) \bigr) dt < 0 \quad\text{for all $x>2$.} \] Roughly, the reason this inequality holds under RH is that the constant $\sum_{\rho} 1/\rho(\rho+1)$ occuring in the explicit formula for ${\mathfrak{A}}_1^\pi(x)$ is too small to overcome the bias. However, Martin pointed out (private communication) that when looking at the analogous \[ {\mathfrak{A}}_1^\pi(x;q,a) = \int_0^x \bigl( \phi(q)\pi(t;q,a)-\mathop\mathrm{li}(t) \bigr) dt, \] the corresponding sum over zeros of Dirichlet $L$-functions grows faster than the bias as $q$ increases, and thus at some point the inequality ${\mathfrak{A}}_1^\pi(x;q,a) < 0$ should fail when $q$ is large.

Assuming GRH, classify those values of $q$ and $a$ such that ${\mathfrak{A}}_1^\pi(x;q,a)<0$ for all sufficiently large $x$.

背景链接：https://arxiv.org/abs/2407.03530

00242 | AMR-116-0002

### Comparative Prime Number Theory - Problem 3

O / open | AMR 问题清单汇编

One needs to be careful with higher-degree $L$-functions since there are known barriers to LI, such as: the factorization of Dedekind $\zeta$-functions and Artin $L$-functions into other $L$-functions; the vanishing of elliptic curve $L$-functions at the central point; and the ability to shift an entire $L$-function vertically if we don't restrict to the central character.

- (Alexandre Bailleul, Université Paris--Saclay) As a concrete instance of the above problem:

Show that $\gamma_1$ is irrational.

- (Nicol Leong, University of New South Wales--Canberra at ADFA)

背景链接：https://arxiv.org/abs/2407.03530

00243 | AMR-116-0003

### Comparative Prime Number Theory - Problem 4

O / open | AMR 问题清单汇编

Can we obtain upper bounds for $1/{|\zeta'(\rho)|}$ as $\rho$ runs over the nontrivial zeros of $\zeta(s)$?

Assuming SZ for $\zeta(s)$, the Mertens sum has the explicit formula \begin{equation*} M_0(x) = \sum_\rho \frac{x^\rho}{\rho \zeta'(\rho)} -2 + \sum_{n= 1}^\infty \frac{(-1)^{n-1}(2\pi)^{2n}}{(2n)!n\zeta(2n+1)x^{2n}}. \end{equation*} The order of the quantity $1/|\zeta'(\rho)|$ is a huge open problem that is difficult to approach. We have an exact formula (see [Tao]) \begin{equation}

\frac{1}{\zeta'(\rho)} = \lim_{s\to \rho}\frac{s-\rho}{\zeta(s)}= -\frac{2e^{1-B\rho}\rho(\rho-1)\Gamma(1+\rho/2)\pi^{-\rho/2}}{\prod_{\rho'\neq \rho}(1-\tfrac{\rho}{\rho'})e^{\rho/\rho'}}, \end{equation} where $B= -0.0230957\ldots$ is the constant appearing in the Weierstrass product for $\xi(s)$. Thus the main issue here is bounding the term $1- \tfrac{\rho}{\rho'}$, in other words, obtaining a lower bound for the distance $|\rho -\rho'|$ from a zero to its nearest neighbour. Even assuming SZ, we cannot yet rule out the existence of two extremely close zeros.

- (Nathan Ng, University of Lethbridge)

背景链接：https://arxiv.org/abs/2407.03530

00244 | AMR-116-0004

### Comparative Prime Number Theory - Problem 5

O / open | AMR 问题清单汇编

- (Nathan Ng, University of Lethbridge) My Ph.D. thesis [NgPhd] contains the conjecture \[ \sum_{0 < \gamma < T} \biggl| \frac{\zeta(2 \rho)}{\zeta'(\rho)} \biggr|^2 \sim \frac{1}{2 \pi} T. \] **Problem 1**: Use the argument in Milinovich and Ng [MN] to establish the lower bound \[ \sum_{0 < \gamma < T} \biggl| \frac{\zeta(2 \rho)}{\zeta'(\rho)} \biggr|^2 \ge \biggl( \frac{1}{4 \pi} + o(1) \biggr) T. \] **Problem 2**: Prove lower bounds for the discrete moments of the shape \[ \sum_{0 < \gamma < T} \biggl| \frac{\zeta(2 \rho)}{\zeta'(\rho)} \biggr|^{2k} \gg T (\log T)^{(k+1)^2}. \] See the articles of Heap, Li, and Zhao [HLZ] and Gao and Zhao [GZ], where the authors prove lower bounds of the type \[ \sum_{0 < \gamma < T} \frac{ 1}{|\zeta'(\rho)|^{2k}} \gg T (\log T)^{(k+1)^2} \] for $k >0$.

- (Nicol Leong, University of New South Wales--Canberra at ADFA)

Can we improve upper bounds for the multiplicities of nontrivial zeros of $\zeta(s)$?

Let $m(\rho)$ denote the multiplicity of a nontrivial zero of $\zeta(s)$. We know unconditionally that \begin{equation*} m(\beta +i\gamma) \ll \log \gamma, \end{equation*} while under the assumption of the Lindel\"of hypothesis or the Riemann hypothesis, the bound can be improved to \begin{equation*} o(\log \gamma) \qquad \text{or} \qquad \ll\frac{\log \gamma}{\log\log \gamma}, \end{equation*} respectively. Ivi\'c has proved a couple of unconditional results of order $\ll \log\gamma$, but this is still far from Karatsuba's conjectured size of $\ll 1$. (Of course, under the LI conjecture, $m(\rho)$ would equal $1$ for all $\rho$.) For more details on the problem and Ivi\'c's approach, see for instance [Ivic1,Ivic2].

- (Alexandre Bailleul, Université Paris--Saclay) Skewes's number is defined as the smallest $x$ such that $\pi(x) > \mathop\mathrm{li}(x)$. It was shown to exist by Littlewood, and a first (enormous) bound was found by Skewes. Its value is still unknown, but it must be quite large since it is known to lie in the interval $(10^{19}, 1.3971653 \times 10^{316})$ (the lower bound is due to Büthe [Buthe0], while the upper bound is due to Saouter, Trudgian, and Demichel [STD]); experts suggest that it is quite close to this upper bound. The corresponding logarithmic density for the race between $\pi(x)$ and $\mathop\mathrm{li}(x)$ was determined to be extremely small by Rubinstein and Sarnak [RS], namely $\delta(\pi,\mathop\mathrm{li}) \approx 2.6 \times 10^{-7}$.

Another example of a race with a large bias is the race between primes congruent to $2\mod3$ versus primes congruent to $1\mod3$. Rubinstein and Sarnak [RS] calculated the corresponding logarithmic density $\delta_{q;1,2} \approx 1.0\times10^{-3}$, while Bays and Hudson shows that the first sign change of $\pi(x;3,2) - \pi(x;3,1)$ occurs at $x=608{,}981{,}813{,}029$. Therefore, it seems natural to think that the size of the ``Skewes's number'' of a prime number race is correlated with how biased that race is. In particular, classical prime number races have their logarithmic densities tending to $\frac12$ as $q \to \infty$, and so we might expect the corresponding Skewes's numbers to be less and less extreme (but of course bounded below by the modulus $q$).

背景链接：https://arxiv.org/abs/2407.03530

00245 | AMR-116-0005

## Comparative Prime Number Theory - Problem 6

O / open | AMR 问题清单汇编

Another example of a race with a large bias is the race between primes congruent to $2\mod3$ versus primes congruent to $1\mod3$. Rubinstein and Sarnak [RS] calculated the corresponding logarithmic density $\delta_{q;1,2} \approx 1.0\times10^{-3}$, while Bays and Hudson shows that the first sign change of $\pi(x;3,2) - \pi(x;3,1)$ occurs at $x=608{,}981{,}813{,}029$. Therefore, it seems natural to think that the size of the ``Skewes's number'' of a prime number race is correlated with how biased that race is. In particular, classical prime number races have their logarithmic densities tending to $\frac12$ as $q \to \infty$, and so we might expect the corresponding Skewes's numbers to be less and less extreme (but of course bounded below by the modulus $q$).

Obtain an upper bound for the first sign change of $\pi(x;q,a)-\pi(x;q,b)$ in terms of $q$, preferably one that increases as slowly as possible.

- (Daniel R. Johnston, University of New South Wales--Canberra at ADFA) This question was posed during Ethan Lee's talk at the CPNTS.

Given a number field $\mathbb{K}$ with ring of integers $\mathcal{O}_{\mathbb{K}}$, let $\kappa_{\mathbb{K}}$ be the residue at the pole $s=1$ for the Dedekind zeta-function $\zeta_{\mathbb{K}}(s)$ associated to $\mathbb{K}$. Consider the difference \[ \Delta_{\mathbb{K}}(x)=\prod_{N(\mathfrak{p})\leq x}\left(1-\frac{1}{N(\mathfrak{p})}\right)^{-1}-e^{\gamma}\kappa_{\mathbb{K}}\log x, \] where the product runs over the prime ideals $\mathfrak{p}$ of $\mathcal{O}_{\mathbb{K}}$ with norm $N(\mathfrak{p})$ up to $x$; this difference $\Delta_K(x)$, which tends to be positive, is connected to Mertens's third theorem for number fields. B\"uthe [buthe2015first] showed that the first negative value of $\Delta_{\mathbb{Q}}(x)$ occurs before $2\cdot 10^{215}$. On the other hand, as shown in [hathi2021mertens, Table 1], the tendency for $\Delta_{\mathbb{K}}(x)$ to be positive is weaker (in terms of logarithmic density) for the simple number fields $\mathbb{K}=\mathbb{Q}(\sqrt{5})$ and $\mathbb{K}=\mathbb{Q}(\sqrt{13})$ than for $\mathbb{K}=\mathbb{Q}$. It's unclear whether this should mean that the first negative values of $\Delta_{\mathbb{K}}(x)$ in these cases should occur sooner than those of $\Delta_{\mathbb{Q}}(x)$.

背景链接：https://arxiv.org/abs/2407.03530

00246 | AMR-116-0006

### Comparative Prime Number Theory - Problem 7

O / open | AMR 问题清单汇编

Localize the first negative value for $\Delta_{\mathbb{K}}(x)$ when $\mathbb{K}$ is $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\sqrt{13})$, or some other simple number field. In general, obtain an upper bound for the first sign change of $\Delta_{\mathbb{K}}(x)$ for a family of number fields $\mathbb{K}$.

- (Alexandre Bailleul, Université Paris--Saclay) There exist highly biased races, such as the (appropriately normalized) race between quadratic residues and nonresidues for composite moduli $q$. Fiorilli [Fio] formulated a conjecture for the Skewes's number of this race between residues and nonresidues (the smallest real number for which the counting function for primes that are quadratic residues overcomes the bias): it is roughly of the form $\exp\bigl( \exp\bigl( {\rho(q)}/\log{\mathrm{rad}(q)} \bigr) \bigr)$, where $\rho(q)$ is the index of the subgroup of squares in $(\mathbb{Z}/q\mathbb{Z})^{\times}$ and $\mathrm{rad}(q)$ is the radical (largest squarefree divisor) of $q$.

More recently, Fiorilli and Jouve [FJ] and Bailleul [Bai] have investigated similar highly-biased races in number fields.

背景链接：https://arxiv.org/abs/2407.03530

00247 | AMR-116-0007

### Comparative Prime Number Theory - Problem 8

O / open | AMR 问题清单汇编

More recently, Fiorilli and Jouve [FJ] and Bailleul [Bai] have investigated similar highly-biased races in number fields.

Bound Skewes's numbers of other highly biased races in terms of the parameters defining the race, either conjecturally or provably.

- (Alexandre Bailleul, Université Paris--Saclay) Consider Chebyshev's bias in the function field setting. As is customary since the work of Cha [Cha], in addition to assuming the linear independence (over ${\mathbb{Q}}$) of the corresponding arguments of zeros of $L$-functions, one typically assumes that those arguments are linearly independent with $\pi$ as well. While there are no known rational linear relations among the arguments of these zeros themselves, there are known instances where such an argument is a rational multiple of $\pi$. These instances usually come from special geometric conditions, such as supersingularity of elliptic curves or jacobians that are not absolutely simple.

背景链接：https://arxiv.org/abs/2407.03530

00248 | AMR-116-0008

### Comparative Prime Number Theory - Problem 9

O / open | AMR 问题清单汇编

- (Alexandre Bailleul, Université Paris--Saclay) Consider Chebyshev's bias in the function field setting. As is customary since the work of Cha [Cha], in addition to assuming the linear independence (over ${\mathbb{Q}}$) of the corresponding arguments of zeros of $L$-functions, one typically assumes that those arguments are linearly independent with $\pi$ as well. While there are no known rational linear relations among the arguments of these zeros themselves, there are known instances where such an argument is a rational multiple of $\pi$. These instances usually come from special geometric conditions, such as supersingularity of elliptic curves or jacobians that are not absolutely simple.

Show that the failure of linear independence with $\pi$ always has a geometric explanation.

- (Stanley Yao Xiao, University of Northern British Columbia) Let $f \in \mathbb{Z}[x,y]$ be a primitive, positive definite binary quadratic form such that $f(x,1) \not\equiv x(x+1) \pmod{2}$. Consider the counting function for the generalized Friedlander--Iwaniec primes \[ \pi_f(x) = \# \{p \leq x \colon p = f(a,b^2) \text{ for some integers $a$ and $b$}\}. \] By the main theorem in my paper [X], we have $\pi_f(x) \sim C(\operatorname{Disc}(f)) x^{3/4}/\log x$ for some positive constant $C(\operatorname{Disc}(f))$ depending only on the discriminant of $f$.

背景链接：https://arxiv.org/abs/2407.03530

00249 | OWR-17300-013

### Comparing Eisenstein and Twisted Doubling L Functions

O / open | Oberwolfach 报告问题汇编

For Brylinski-Deligne covering groups, how do the L-functions obtained from constant terms of Eisenstein series compare with the L-functions obtained from twisted doubling integrals?

来源字段：https://doi.org/10.4171/owr/2019/57

00250 | OWR-13952-002

### Comparison of Algebraic and Analytic Modular Forms

O / open | Oberwolfach 报告问题汇编

For a holomorphic modular form viewed both as an algebraic global section of $\omega^{\otimes k}$ on a modular curve and as an analytic function on the upper half-plane, how can these two points of view be identified or related?

来源字段：https://doi.org/10.4171/owr/2015/57

00251 | OWR-1318-005

### Complex and $p$-Adic Artin $L$-Value Formulas

P / partially_solved | Oberwolfach 报告问题汇编

In the totally-real/odd and CM/even parity cases for the Artin representation $\rho$ on $E[G]\pi$, prove $$L(n,\rho\otimes\operatorname{id},\mathbb Q)D(M_\pi^E)^{1/2,\infty}=e(n,M_\pi^E)R_{n,\infty}(M_\pi^E)$$ and $$L_p(n,\rho\otimes\omega_p^{1-n},\mathbb Q)D(M_\pi^E)^{1/2,p}=e_p(n,M_\pi^E)\operatorname{Eul}_p(n,\rho\otimes\operatorname{id},\mathbb Q)R_{n,p}(M_\pi^E),$$ with $e_p(n,M_\pi^E)=e(n,M_\pi^E)$, and prove that the $p$-adic $L$-value and regulator are nonzero.

来源字段：https://doi.org/10.4171/owr/2006/32

00252 | OWR-8415342-007

### Component-Group Factorization on Dualizing Cohomology

O / open | Oberwolfach 报告问题汇编

For every algebraically closed field $K$ and every $\gamma\in C^{\bullet}(K)$, does the action of the stabilizer group $P_\gamma$ on $R\Gamma(f^{-1}(\gamma),\omega_{f^{-1}(\gamma)})$ factor through the component group $\pi_0(P_\gamma)$?

来源字段：https://doi.org/10.4171/owr/2021/39

00253 | OWR-782-001

### Congruences for Extremal Modular Forms

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Extremal Modular-Form Congruence [OWR-782-002]

Let $f(\tau)=1+\sum_{i\geq1}a_iq^i$ be the extremal modular form of weight $12\mu$, and let $p$ be a prime such that $p$ divides $a_i$ whenever $p$ does not divide $i$, while not every $a_i$ is divisible by $p$. Must there be a modular form $g$ of smaller weight such that $f(\tau)\equiv g(p\tau)\pmod p$? Moreover, can $g$ be chosen extremal and can one obtain $f(\tau)\equiv g(p^r\tau)\pmod p$ for some $r\in\mathbb{N}$?

来源字段：https://doi.org/10.4171/owr/2005/01

00254 | OWR-12010-006

### Congruent Numbers in Three Residue Classes

O / open | Oberwolfach 报告问题汇编

Is every squarefree positive integer congruent to $5$, $6$, or $7$ modulo $8$ a congruent number?

来源字段：https://doi.org/10.4171/owr/2012/38

00255 | OWR-12725-013

### Consecutive Integers with Restricted Prime Divisors

O / open | Oberwolfach 报告问题汇编

Let $T$ be the set of positive integers all of whose prime divisors are not congruent to $1\pmod 4$. Does $T$ contain infinitely many pairs of consecutive integers?

来源字段：https://doi.org/10.4171/owr/2013/51

00256 | AIM-ANALYTIC_NUMBER_THEORY-0027

### Constant Liouville runs and the diagonal-progression barrier

P / partially_solved | AIM 研讨会问题汇编

Can we show arbitrarily large patterns $++\ldots+$ or $--\ldots -$ in $(\lambda(n))_{n \in \mathbb N}$? An approach through the entropy decrement argument (roughly) reduces this to showing the existence of arbitrarily large progressions $n,n+p,\dots,n+kp$ with $\lambda(n)=\lambda(n+p)=\dots=\lambda(n+kp)$.

来源字段：http://aimpl.org/sarnakconjecture/2/

00257 | OWR-12866-018

### Continuity and Strict Monotonicity of Nyman-Beurling Distances

O / open | Oberwolfach 报告问题汇编

In $L^2((0,\infty),t^{-2}dt)$, let $\chi(t)=\mathbf 1_{(1,\infty)}(t)$, let $e_\alpha(t)=\{t/\alpha\}$, and define $$D(\lambda)=\operatorname{dist}\!\left(\chi,\operatorname{span}\{e_\alpha:1\leq\alpha\leq\lambda\}\right),\qquad\lambda>1.$$ Is $D(\lambda)$ continuous? Is it strictly decreasing?

来源字段：https://doi.org/10.4171/owr/2014/06

00258 | OWR-14297746-002

### Convergence of Multiplicative Polynomial Averages

O / open | Oberwolfach 报告问题汇编

Let $f:\mathbb N\to\mathbb R$ be bounded and completely multiplicative, and let $P\in\mathbb Z[x,y]$ be homogeneous and positive on $\mathbb N^2$. Does the limit $\lim_{N\to\infty}N^{-2}\sum_{m,n=1}^N f(P(m,n))$ exist?

来源字段：https://doi.org/10.4171/owr/2023/37

00259 | OWR-11127-016

### Correct Modularity Criteria for Nahm Series

O / open | Oberwolfach 报告问题汇编

What is the correct necessary-and-sufficient criterion on rational $A$, $B$, and $C$ for the Nahm series $F_{A,B,C}(q)$ to be modular, and how should Nahm's conjectural criterion be repaired?

来源字段：https://doi.org/10.4171/owr/2011/35

00260 | AIM-ALGEBRAIC_NUMBER_THEORY-0029

### Correcting the threshold and computing the singular locus

P / partially_solved | AIM 研讨会问题汇编

For a positive integer \(k\), consider \[ \mathcal Q_k(s)= \left\{ Q(y_1^k,\ldots,y_s^k)\in \mathbb Q[y_1,\ldots,y_s]: Q\in\mathbb Q[x_1,\ldots,x_s]\text{ is a quadratic form} \right\}. \] These are forms of degree \(2k\). The source defines \[ h(k)=\inf_{s\in\mathbb N} \left\{ s:\text{ the Hasse principle holds for every member of }\mathcal Q_k(s) \right\} \] and asks for \[ \log h(k)=o(k). \] It records the estimate \[ h(k)\leq 2k\,2^{2k} \] from Birch's theorem and compares it with the diagonal degree-\(k\) bound \[ h_{\mathrm{diag}}(k)\ll 2k\log k. \] The symbol in the PDF is \(\ll\), not an asymptotic equivalence. Browning also suggests replacing the \(k\)-th powers by norm forms from fixed degree-\(k\) extensions.

来源字段：https://aimath.org/pastworkshops/ratlhigherdimvarproblems.pdf

00261 | AIM-ANALYTIC_NUMBER_THEORY-0045

### Correlation with a uniquely ergodic system

P / partially_solved | AIM 研讨会问题汇编

The Chowla conjecture implies that $\mu$ cannot be orthogonal to all uniquely ergodic systems (allowing non-zero entropy). Can we find a concrete example of a uniquely ergodic system that correlates with $\mu$? A way to find a uniquely ergodic system (of positive entropy) that does not correlate with $\lambda$ is known.

Since the system of Problem 3.1 is disjoint from all ergodic systems, an example of an ergodic sequence (or uniquely ergodic system) that correlates with the Liouville function would give a negative answer to Problem 3.1.

来源字段：http://aimpl.org/sarnakconjecture/8/

00262 | OPG-511

### Counterexamples to the Baillie-PSW primality test

O / open | Open Problem Garden

Problem (1) Find a counterexample to Baillie-PSW primality test or prove that there is no one.

Problem (2) Find a composite $n\equiv 3$ or $7\pmod{10}$ which divides both $2^{n-1} - 1$ (see Fermat pseudoprime) and the Fibonacci number $F_{n+1}$ (see Lucas pseudoprime), or prove that there is no such $n$.

背景链接：http://www.openproblemgarden.org/op/counterexamples_to_the_baillie_psw_primality_test

00263 | OWR-10252929-006

### Counting Integer Matrices with Fixed Characteristic Polynomial

O / open | Oberwolfach 报告问题汇编

Let $R_n(H;f)$ count the integer $n\times n$ matrices of height at most $H$ with characteristic polynomial $f$. Uniformly over $f$, is $R_n(H;f)\leq H^{n(n-1)/2+o(1)}$ as $H\to\infty$?

来源字段：https://doi.org/10.4171/owr/2022/21

00264 | OWR-17293-014

### Counting Integer Matrices with Prescribed Minor Scales

O / open | Oberwolfach 报告问题汇编

Let $f(n)$ be one of $\sqrt n,n,n^{3/2},n^2,n^3$, let $M=[f(m_{ij})]$, let $\Delta_k$ be the maximum absolute determinant of a $k\times k$ minor, and count the tuples $(m_{ij})$ with prescribed scales $\Delta_k\asymp D_k$; what upper bounds can be obtained for this counting function?

来源字段：https://doi.org/10.4171/owr/2019/50

00265 | AIM-ALGEBRAIC_NUMBER_THEORY-0033

### Counting orders, local laws, and dominant-residue factorization

P / partially_solved | AIM 研讨会问题汇编

For a specified family of extensions, weight, and height/counting invariant, determine whether its limiting local laws satisfy (Ch) and (Ind), after explicitly accounting for globally inviable local specifications.

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

00266 | OWR-14299577-020

### Counting Palindromic and Digit-Reversible Squares

P / partially_solved | Oberwolfach 报告问题汇编

Determine the asymptotic number of palindromic squares in $[1,X]$, and decide whether infinitely many exist outside the four known infinite families. Determine the number of nonpalindromic squares whose digit reversal is also a square, and decide whether there are infinitely many.

来源字段：https://doi.org/10.4171/owr/2025/51

00267 | AIM-ANALYTIC_NUMBER_THEORY-0098

### Coupled rank-two Nahm sums as modular units and partition series

P / partially_solved | AIM 研讨会问题汇编

Problem 2.3. Look at explicit examples where r ≥ 2 and fA,B,C (τ) is modular. Can we write these in terms of natural objects? Similarly, can we related them to combinatorial identities; in the case r = 1, the two triples given relate to the Rogers-Ramanujan identities.

The record reads, exactly:

The preceding paragraph of the original PDF supplies the notation. In standard typesetting, it is \[ f_{A,B,C}(\tau) :=\sum_{n_1,\ldots,n_r\geq 0} \frac{q^{\frac12\mathbf n^{T}A\mathbf n+B^{T}\mathbf n+C}} {(q;q)_{n_1}\cdots(q;q)_{n_r}}, \qquad q=e^{2\pi i\tau}, \] where \(A\in M_{r\times r}(\mathbb Q)\) is symmetric positive definite, \(B\in\mathbb Q^r\), \(C\in\mathbb Q\), and \[ (q;q)_m=\prod_{k=1}^{m}(1-q^k),\qquad (q;q)_0=1. \] The PDF prints the quadratic term using vector arrows and a transpose; comparison with the clean definition in the later literature confirms the displayed reconstruction.

There are two separate source issues:

The task is an open-ended request for examples and interpretations, rather than a single proposition. This report gives a proved, genuinely coupled rank-two pair, identifies it simultaneously as a product of modular units and as a partition generating function, and supplies an explicit partition bijection for the key rank-two-to-rank-one collapse.

来源字段：https://aimath.org/WWN/mockmodular/mockmodular.pdf

00268 | AIM-ANALYTIC_NUMBER_THEORY-0087

### Coupling-weight criteria and formal obstructions for an all-genus series

P / partially_solved | AIM 研讨会问题汇编

别名：Formal, Jacobi, and wave-function transformations of the string coupling [AIM-PHYSICS-0020]

Does \(F=\sum f_g(\tau)\lambda^{2g-2}\) have transformation properties with respect to \(\lambda\)?

来源字段：http://aimpl.org/gromwitnumthry/4/

00269 | GREEN-085

### Covering by Residue Classes

O / open | Ben Green 问题

Let $N$ be large. For each prime $p$ with $N^{0.51} \leq p < 2N^{0.51}$, pick a residue $a(p) \in \mathbb{Z}/p\mathbb{Z}$. Is $\#\{n \in [N] : n \equiv a(p) \pmod p \text{ for some } p\} \gg N^{1-o(1)}$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

00270 | GREEN-034

### Covering Squares with Sumsets

P / partially_solved | Ben Green 问题

Suppose $A + A$ contains the first $n$ squares. Is $|A| \geq n^{1-o(1)}$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

00271 | OPG-493

### Covering systems with big moduli

S / solved | Open Problem Garden

Problem Does for every integer $N$ exist a covering system with all moduli distinct and at least equal to~ $N$?

背景链接：http://www.openproblemgarden.org/op/covering_systems_with_big_moduli

00272 | OWR-15956-008

### Critical Mean Values of Dirichlet Polynomials

O / open | Oberwolfach 报告问题汇编

Let $D(s)=\sum_{n\leq N}a_nn^{-s}$ with $|a_n|\leq1$. In the critical range $T^{1/2}<N<T$ with $N^\kappa=T$, is $\int_0^T|D(it)|^{2\kappa}\,dt\ll T^{2+\varepsilon}$?

来源字段：https://doi.org/10.4171/owr/2017/51

00273 | AIM-ALGEBRAIC_NUMBER_THEORY-0001

### Cubic points in the plane and the two exceptional sets

P / partially_solved | AIM 研讨会问题汇编

It asks for a Manin-type conjecture for degree-\(d\) points on a Fano variety \(X/k\). Its displayed counting function is \[ N_{X,\mathscr L}(U,B;d) = \#\{x\in U(\overline k):h_{\mathscr L}(x)\leq B,\ [\kappa(x):k]=d\}, \] and the proposed form is \(cB^a(\log B)^b\). It asks for \(a,b,c\), for the correct shape of \(U\), and particularly for \(X=\mathbb P^2_{\mathbb Q}\), \(d=3\).

There is an essential convention to repair. In much of arithmetic geometry \(h_{\mathscr L}\) denotes a **logarithmic** Weil height. A bound \(h_{\mathscr L}\leq B\) then produces exponential, not polynomial, growth in \(B\). Throughout this report, \[ H_{\mathscr L}:=\exp(h_{\mathscr L}) \] is the multiplicative height and the bound is \(H_{\mathscr L}\leq B\), equivalently \(h_{\mathscr L}\leq\log B\). If the source's inequality is read literally with logarithmic \(h_{\mathscr L}\), every occurrence of \(B^\alpha\) below becomes \(\exp(\alpha B)\).

Replacing a metric by an equivalent one preserves power and logarithmic exponents but can change the leading constant. Thus no numerical \(c\) is meaningful until the metric and the exceptional set are specified.

来源字段：http://aimpl.org/degreedsurface/1/

00274 | OWR-14299578-010

### Curves with Unbounded Rational-Point-to-Genus Ratio

O / open | Oberwolfach 报告问题汇编

Construct, or prove impossible, a sequence of smooth projective curves $C_n/\mathbb{Q}$ with genus $g(C_n)\geq2$ such that $\#C_n(\mathbb{Q})/g(C_n)\to\infty$.

来源字段：https://doi.org/10.4171/owr/2025/52

00275 | AIM-ALGEBRAIC_NUMBER_THEORY-0035

### Cyclic triple covers ordered by genus

P / partially_solved | AIM 研讨会问题汇编

Question 5. For cyclic triple covers of P1 can we recover (A) and (B) while counting by genus?

It is Question 5 in the AIM problem-session document for the January 2014 workshop *Arithmetic statistics over finite fields and function fields*. The immediately surrounding questions ask which ways of counting field extensions satisfy Principles (A) and (B), which automorphism groups give the cleanest formulas, whether conditioning on ramification helps, and whether an analogous assertion fails for \(\mathbb Z/4\mathbb Z\)-extensions counted by genus.

There is a source defect that matters mathematically: neither the three-page problem document nor the workshop report defines the labels (A) and (B). I therefore do **not** assign a letter to either assertion below. The strongest source-supported reconstruction comes from two facts:

1. the workshop report says that the group working on Question 5 aimed to use Wood's results on local behavior of abelian extensions to count \(\mathbb Z/3\mathbb Z\)-covers by the whole genus; and 2. Wood's cited paper foregrounds exactly two properties: conditional Chebotarev proportions for unramified splitting and independence of local specifications at distinct places.

Accordingly, this report uses the neutral names:

- **Chebotarev property:** conditional on a place being unramified, its splitting behavior has the usual Chebotarev proportions; - **independence property:** prescribed local behaviors at any fixed finite set of places become independent in the genus limit.

This is a high-confidence reconstruction of the intended mathematical pair, but the assignment of these properties to the letters (A) and (B) remains unverified. A third closely related feature, the explicit local mass formula, is recorded as part of the same theorem rather than silently identified with either letter.

The published work arising from the workshop makes the relevant ensemble precise: fix \(\mathbb F_q\) with \(q\equiv 1\pmod 3\), let \(g\to\infty\), and count the full moduli space \(\mathcal H_{g,3}\) of cyclic degree-three Galois covers of \(\mathbb P^1\) using the stack weights \(1/|\operatorname{Aut}(C)|\) specified in that paper.

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

00276 | OWR-3480-005

### Cyclotomic-Anticyclotomic Iwasawa Main Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $E/\mathbb{Q}$ be an elliptic curve, let $p$ be a prime of good ordinary or multiplicative reduction, let $k$ be an imaginary quadratic field, and let $k_\infty$ be the compositum of the cyclotomic and anticyclotomic $\mathbb{Z}_p$-extensions of $k$, with $G=\operatorname{Gal}(k_\infty/k)$. Is the dual Selmer group $X(E/k_\infty)$ torsion over $\Lambda(G)$, and does $(L_p(f,k_\infty))=(\operatorname{char}_{\Lambda(G)}X(E/k_\infty))$ hold?

来源字段：https://doi.org/10.4171/owr/2009/30

00277 | OPG-563

### Davenport's constant

P / partially_solved | Open Problem Garden

For a finite (additive) abelian group $G$, the Davenport constant of $G$, denoted $s(G)$, is the smallest integer $t$ so that every sequence of elements of $G$ with length $\ge t$ has a nontrivial subsequence which sums to zero.

Conjecture $s( {\mathbb Z}_n^d) = d(n-1) + 1$

背景链接：http://www.openproblemgarden.org/op/davenports_constant

00278 | AIM-ALGEBRAIC_NUMBER_THEORY-0006

### Degree sets on Skorobogatov's bielliptic surface

P / partially_solved | AIM 研讨会问题汇编

For a nice variety \(X/k\),

$$ \mathcal D(X/k)=\{d>0:X\text{ has a closed point of degree }d\}, $$

\(\delta(X/k)\) is the set of \(d>0\) for which degree-\(d\) closed points are Zariski dense, and

$$ \mathcal P(X/k)=\bigcup_{[k':k]<\infty}\delta(X/k'). $$

The question concerns Skorobogatov's bielliptic surface \(S/\mathbb Q\), for which

$$ S(\mathbb A_{\mathbb Q})^{\operatorname{Br}}\ne\varnothing, \qquad S(\mathbb Q)=\varnothing. $$

$$ p(x)=x^2+1,\qquad q(x)=x^2+2,\qquad g(t)=3(t^4-54t^2-117t-243). $$

The intended \(S\) is the smooth projective model of

$$ p(x)y^2=q(x)z^2=g(t). \tag{1} $$

Equivalently, after setting \(\widetilde y=p(x)y\) and \(\widetilde z=q(x)z\), an affine chart is

$$ \widetilde y^2=g(t)p(x),\qquad \widetilde z^2=g(t)q(x). \tag{2} $$

来源字段：http://aimpl.org/degreedsurface/1/

00279 | OWR-8415337-003

### Degree-$d$ Points on Algebraic Curves

O / open | Oberwolfach 报告问题汇编

Given an integer $d$ and a smooth projective curve $X$ over a number field, is the set of degree-$d$ points on $X$ finite or infinite?

来源字段：https://doi.org/10.4171/owr/2021/34

00280 | AIM-ALGEBRAIC_NUMBER_THEORY-0018

### Degree-d points on a generic surface are line sections

S / solved | AIM 研讨会问题汇编

Given a general smooth degree \(d\) hypersurface \(X_d\subset\mathbb P^3_k\), for \(d\) sufficiently large, must a general degree \(d\) point arise from intersecting \(X\) with a line?

来源字段：http://aimpl.org/degreedsurface/1/

00281 | OWR-11695860-018

### Degree-Five Two-Superirreducible Polynomials

O / open | Oberwolfach 报告问题汇编

Does there exist $f\in\mathbb Z[x]$ of degree five such that $f(g(x))$ is irreducible for every $g\in\mathbb Z[x]$ of degree at most two?

来源字段：https://doi.org/10.4171/owr/2022/50

00282 | AIM-ALGEBRAIC_NUMBER_THEORY-0013

### Del Pezzo irrationality values and a global-field quartic characteristic dichotomy

P / partially_solved | AIM 研讨会问题汇编

For a variety \(X/F\), define its degree of irrationality \(\operatorname{irr}(X)\) to be the minimum degree of a rational map \(X\dashrightarrow\mathbb P^2\) (defined over \(F\)).

Compute \(\operatorname{irr}(X/F)\) for a del Pezzo surface \(X\) over a general field \(F\).

来源字段：http://aimpl.org/degreedsurface/1/

00283 | OWR-12980-010

### Deligne's Conjecture for Critical Motive Values

P / partially_solved | Oberwolfach 报告问题汇编

For a motive $M$ over $\mathbb Q$ with coefficients in $E$, if $n\in\mathbb Z$ is critical, does Deligne's conjecture give $\frac{L(M,n)}{(2\pi i)^{d_{\pm}n}c_{\pm}(M)}\in E$, where the sign $\pm$ equals $(-1)^n$?

来源字段：https://doi.org/10.4171/owr/2014/22

00284 | AIM-ANALYTIC_NUMBER_THEORY-0018

### Delta-method status and a conductor-drop phase diagram for GL(2)

P / partially_solved | AIM 研讨会问题汇编

**Subconvexity bound \(\mathrm{GL}_2\) in the “conductor dropping” case.** Let \(f\) be a holomorphic or Hecke--Maass cusp form for \(\mathrm{GL}_2\). Use a \(\delta\)-method to prove:

1. a level-aspect subconvexity bound for \(L(1/2,f)\), where \(f\) has prime level \(p\) and may have nontrivial nebentypus; 2. a subconvexity bound for \(L(1/2+it_f,f)\), where \(f\) is a level-one Maass form with spectral parameter \(t_f\).

来源字段：http://aimpl.org/deltasubconvex2/1/

00285 | OWR-14746-008

### Dense Frobenius Eigenvectors in Completed Enveloping Algebras

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Dense Frobenius-Eigenvector Conjecture [OWR-14746-009]

Let $X$ be a variety over a finite field $k$, let $\ell$ be a prime, and let $I$ be the augmentation ideal in $\mathbb Z_\ell[[\pi_1^{\mathrm{\acute et},\ell}(X_{\overline k},\overline x)]]$. Assume $I^n/I^{n+1}$ is torsion-free for every $n$. Does there exist $r>0$ such that the convergent group algebra

$$\mathbb Q_\ell[[\pi_1^{\mathrm{\acute et},\ell}(X_{\overline k},\overline x)]]^{\leq\ell^{-r}}$$

admits Frobenius eigenvectors whose span is dense?

来源字段：https://doi.org/10.4171/owr/2016/38

00286 | OWR-14603-014

### Dense Multiplicative Orbits on the Two-Torus

O / open | Oberwolfach 报告问题汇编

If $1,x,y$ are multiplicatively independent over $\mathbb Q$ and $a,b$ are relatively prime positive integers, is $\{a^n b^k(x,y)\bmod 1:n,k\in\mathbb N\}$ dense in $[0,1]^2$?

来源字段：https://doi.org/10.4171/owr/2016/21

00287 | OWR-1274-009

### Dense Settling and Arboreal Galois Representations

P / partially_solved | Oberwolfach 报告问题汇编

For a quadratic polynomial $f\in\mathbb Z[x]$, is the image of its arboreal Galois representation densely settled? For $f=(x+1)^2-2$, determine the kernel of the quotient from the maximal $2$-extension unramified outside $\{2,\infty\}$ onto its arboreal Galois group. Also construct a natural zeta function or $L$-function for the rooted tree of iterates.

来源字段：https://doi.org/10.4171/owr/2006/25

00288 | AIM-ALGEBRAIC_NUMBER_THEORY-0009

### Density degrees under finite covers

P / partially_solved | AIM 研讨会问题汇编

Given a finite map \(f:X\to Y\) (over a number field), how are \(\delta(X/k)\) and \(\delta(Y/k)\) related? Are there examples showing that the obvious relations are sharp?

It could be interesting to consider the special case where \(f\) is a Galois cover. If \(X,Y\) are curves, you can also ask about containment relations between there \(\delta_{\mathbb P^1}\)'s and their \(\delta_{\operatorname{AV}}\)'s.

来源字段：http://aimpl.org/degreedsurface/1/

00289 | AIM-ALGEBRAIC_NUMBER_THEORY-0015

### Density degrees under finite extension

P / partially_solved | AIM 研讨会问题汇编

Is there a surface \(X\) over a number field \(k\) and a finite extension \(k'/k\) such that \(\delta(X/k)\not\subset\delta(X/k')\)?

来源字段：http://aimpl.org/degreedsurface/1/

00290 | OWR-11695860-016

### Density Hypothesis for Riemann-Zeta Zeros

P / partially_solved | Oberwolfach 报告问题汇编

Let $N(\sigma,T)$ count zeros $\rho$ of $\zeta$ with $\operatorname{Re}\rho\geq\sigma$ and $0\leq\operatorname{Im}\rho\leq T$. Prove $N(\sigma,T)<T^{2-2\sigma+o(1)}$.

来源字段：https://doi.org/10.4171/owr/2022/50

00291 | OWR-4990381-001

### Density of Specializations of High-Genus Galois Covers

P / partially_solved | Oberwolfach 报告问题汇编

Let $E/\mathbb Q(T)$ be a $\mathbb Q$-regular Galois extension with group $G$ and function-field genus $g\geq2$. Does $$ \frac{|\operatorname{Sp}(E)\cap S(G,x)|}{|S(G,x)|}\longrightarrow0 $$ as $x\to\infty$?

来源字段：https://doi.org/10.4171/owr/2021/12

00292 | OWR-14298584-011

### Density of Sums of Two Rational Cubes

P / partially_solved | Oberwolfach 报告问题汇编

Determine the asymptotic proportion of positive integers that can be expressed as a sum of two rational cubes. In particular, is the density equal to $1/2$?

来源字段：https://doi.org/10.4171/owr/2024/40

00293 | GUY-A20

### Density of Symmetric Primes

P / partially_solved | Richard Guy 数论问题

Given pairs of odd primes $p, q$, define $S(q,p)$ as the number of lattice points $(m, n)$ in the rectangle $0 < m < p/2$, $0 < n < q/2$ below the diagonal. A pair is symmetric if $S(p, q) = S(q,p)$. Is the number of symmetric primes less than $x$ equal to $x/(\ln x)^{\sigma+o(1)}$, where $\sigma = 2 - (1+\ln \ln 2)/\ln 2 \approx 1.08607$?

背景链接：https://arxiv.org/abs/1908.06161

00294 | NT-063

### Density of Ulam Numbers

O / open | 独立 / 未分配集合

Do the Ulam numbers have a positive density?

背景链接：https://arxiv.org/abs/2007.02697

00295 | OWR-14299909-002

### Derivative Formula for Multiple Eisenstein Series

O / open | Oberwolfach 报告问题汇编

For every word $w\in\mathbb Q\langle z_2,z_3,\ldots\rangle$, prove the conjectural derivative identity $2\pi i\,\frac{d}{d\tau}G(w)=G(\theta(w))$, where $\theta=-D\circ\varphi$ is defined using the Drop1 operator.

来源字段：https://doi.org/10.4171/owr/2026/7

00296 | OWR-10252932-005

### Determinant Formula for Leading Zeta Values

P / partially_solved | Oberwolfach 报告问题汇编

For a proper regular arithmetic scheme $X/\mathbb{Z}$ and $n\in\mathbb{Z}$, does the determinant isomorphism satisfy the conjectural leading-value formula $\det(\lambda_{X;n})=\pm\zeta^*(X;n)$?

来源字段：https://doi.org/10.4171/owr/2022/24

00297 | NT-085

### Dickson's Conjecture

P / partially_solved | 独立 / 未分配集合

Do finitely many linear forms simultaneously take prime values infinitely often, barring congruence obstructions?

背景链接：https://annals.math.princeton.edu/2015/181-1/p07

00298 | GREEN-038

### Difference Sets Containing Squares

P / partially_solved | Ben Green 问题

Is there $c > 0$ such that whenever $A \subset [N]$ has size $N^{1-c}$, the difference set $A - A$ contains a nonzero square?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

00299 | AIM-ANALYTIC_NUMBER_THEORY-0079

### Differential closure and linear-ODE obstructions for eta-products

P / partially_solved | AIM 研讨会问题汇编

别名：Differential fingerprints of eta-products [AIM-PHYSICS-0012]

Eta-products and root systems and holomorphic anomaly equation

As it turns out mock modular forms (such as eta-products) do not satisfy the holomorphic anaomaly equation.

Do they satisfy other differential equations?

来源字段：http://aimpl.org/gromwitnumthry/1/

00300 | OWR-12493-004

### Dihedral Groups as Oort Groups

O / open | Oberwolfach 报告问题汇编

For an odd prime $p$ and $n\geq1$, is the dihedral group $D_{p^n}$ of order $2p^n$ an Oort group; that is, does every local Galois extension with this group lift to characteristic zero?

来源字段：https://doi.org/10.4171/owr/2013/30

00301 | OWR-1274-023

### Diophantine Obstructions in the Regular Inverse Galois Problem

O / open | Oberwolfach 报告问题汇编

Determine the source of the difficulty of the regular inverse Galois problem: is it fundamentally diophantine rather than group-theoretic?

来源字段：https://doi.org/10.4171/owr/2006/25

00302 | OPG-16555

### Diophantine quintuple conjecture

P / partially_solved | Open Problem Garden

Definition A set of m positive integers $\{a_1, a_2, \dots, a_m\}$ is called a Diophantine $m$-tuple if $a_i\cdot a_j + 1$ is a perfect square for all $1 \leq i < j \leq m$.

Conjecture (1) Diophantine quintuple does not exist.

It would follow from the following stronger conjecture [Da]:

Conjecture (2) If $\{a, b, c, d\}$ is a Diophantine quadruple and $d > \max \{a, b, c\}$, then $d = a + b + c + 2bc + 2\sqrt{(ab+1)(ac+1)(bc+1)}.$

背景链接：http://www.openproblemgarden.org/op/diophantine_quintuple_does_not_exist

00303 | OWR-1734-004

### Direct Modularity from the Hypergeometric Partition Expansion

O / open | Oberwolfach 报告问题汇编

Let $$P(q)=\sum_{n\geq0}p(n)q^n=1+\sum_{n=1}^\infty\frac{q^{n^2}}{(1-q)^2(1-q^2)^2\cdots(1-q^n)^2}=\frac{q^{1/24}}{\eta(z)}.$$ Prove directly from the hypergeometric expansion that $P(q)$ is modular.

来源字段：https://doi.org/10.4171/owr/2007/51

00304 | OWR-8415342-013

### Direct Poisson Summation for Split Classical Groups

P / partially_solved | Oberwolfach 报告问题汇编

Prove the $(\sigma,\rho)$-Poisson summation formula directly for split classical groups $G$, with $\rho$ the standard representation of the complex dual group.

来源字段：https://doi.org/10.4171/owr/2021/39

00305 | OWR-12010-021

### Direct-Summand Property for Selmer Restriction Images

O / open | Oberwolfach 报告问题汇编

Let $\beta:H^1(k,E[n])\to H^1(A,E[n])$ be the restriction map in the Selmer model. Is $\operatorname{im}\beta$ a direct summand for $100\%$ of elliptic curves $E$ over $k$?

来源字段：https://doi.org/10.4171/owr/2012/38

00306 | NT-071

### Dirichlet's Divisor Problem

P / partially_solved | 独立 / 未分配集合

What is the optimal exponent in the error term for the divisor summatory function?

背景链接：https://dlmf.nist.gov/27.11

00307 | OWR-13750333-006

### Discrete Spectrum for Nonlattice Quotients

P / partially_solved | Oberwolfach 报告问题汇编

For a nonlattice subgroup $\Gamma<G$, determine whether $L^2(\Gamma\backslash X)$ can contain a Laplace eigenfunction and whether $L^2(\Gamma\backslash G)$ can be non-tempered.

来源字段：https://doi.org/10.4171/owr/2023/22

00308 | OWR-14603-022

### Distinct Number Fields from Fibers of Rational Functions

P / partially_solved | Oberwolfach 报告问题汇编

Let $X/\mathbb Q$ be a curve of genus $g$ and $t\in\mathbb Q(X)$ a nonconstant rational function of degree $\nu\geq2$. For $t(P_n)=n$, are at least $c(g,\nu)N$ of the fields $\mathbb Q(P_1),\ldots,\mathbb Q(P_N)$ distinct for all sufficiently large $N$?

来源字段：https://doi.org/10.4171/owr/2016/21

00309 | OWR-2491-008

### Distinguished Arthur Parameters and Euler Products of Periods

P / partially_solved | Oberwolfach 报告问题汇编

For an automorphic representation on a symmetric variety, under suitable multiplicity-one hypotheses, can a nonzero global $H$-invariant functional be expressed as an Euler product of the local $H_v$-invariant functionals in the Plancherel formula? Is the support of Plancherel measure for $L^2(X_v)$ contained in the representations with $X$-distinguished Arthur parameters?

来源字段：https://doi.org/10.4171/owr/2008/53

00310 | OWR-4804-004

### Distinguished Representations of Symmetric Pairs

O / open | Oberwolfach 报告问题汇编

For a symmetric pair $H=G^\theta$, which irreducible representations $\pi$ of $G$ are $H$-distinguished, meaning that $\operatorname{Hom}_H(\pi,\mathbb C)\ne0$?

来源字段：https://doi.org/10.4171/owr/2011/14

00311 | OPG-36961

### Distribution and upper bound of mimic numbers

P / partially_solved | Open Problem Garden

Problem

Let the notation $a|b$ denote " $a$ divides $b$ ". The mimic function in number theory is defined as follows [1].

Definition For any positive integer $\mathcal{N} = \sum_{i=0}^{n}\mathcal{X}_{i}\mathcal{M}^{i}$ divisible by $\mathcal{D}$, the mimic function, $f(\mathcal{D} | \mathcal{N})$, is given by,

$$f(\mathcal{D} | \mathcal{N}) = \sum_{i=0}^{n}\mathcal{X}_{i}(\mathcal{M}-\mathcal{D})^{i}$$

By using this definition of mimic function, the mimic number of any non-prime integer is defined as follows [1].

Definition The number $m$ is defined to be the mimic number of any positive integer $\mathcal{N} = \sum_{i=0}^{n}\mathcal{X}_{i}\mathcal{M}^{i}$, with respect to $\mathcal{D}$, for the minimum value of which $f^{m}(\mathcal{D} | \mathcal{N}) = \mathcal{D}$.

Given these two definitions and a positive integer $\mathcal{D}$, find the distribution of mimic numbers of those numbers divisible by $\mathcal{D}$.

Again, find whether there is an upper bound of mimic numbers for a set of numbers divisible by any fixed positive integer $\mathcal{D}$.

背景链接：http://www.openproblemgarden.org/op/distribution_and_upper_bound_of_mimic_numbers

00312 | OWR-12010-017

### Distribution of Elliptic-Curve Selmer Exact Sequences

O / open | Oberwolfach 报告问题汇编

Given an abstract short exact sequence $S$ of $\mathbb Z_p$-modules, what is the density of elliptic curves $E$ over $k$ for which $0\to E(k)\otimes\mathbb Q_p/\mathbb Z_p\to\operatorname{Sel}_{p^\infty}(E)\to\operatorname{Sha}(E)[p^\infty]\to0$ is isomorphic to $S$?

来源字段：https://doi.org/10.4171/owr/2012/38

00313 | OWR-4076-005

### Distribution of Rational Points on Varieties

O / open | Oberwolfach 报告问题汇编

For a suitably regular variety $X$ over a number field $k$ with $X(k)\neq\varnothing$, how are its rational points distributed, particularly with respect to finiteness, infinitude, and Zariski density?

来源字段：https://doi.org/10.4171/owr/2009/33

00314 | OWR-1590-001

### Distributions of Class-Group Ranks and $S_5$ Extensions

O / open | Oberwolfach 报告问题汇编

Let $r_p(D)$ be the $p$-rank of the ideal class group of $\mathbb Q(\sqrt{-D})$. As $0<D<X$, determine the average of $p^{r_p(D)}$, the distribution of $r_p(D)$, and upper bounds for $r_p(D)$ as functions of $D$. Let $n(D)$ be the number of $S_5$-extensions of $\mathbb Q$ with squarefree discriminant $D$. As $-X<D<X$, determine the average and distribution of $n(D)$ and upper bounds for $n(D)$ as functions of $D$.

来源字段：https://doi.org/10.4171/owr/2007/34

00315 | OPG-506

### Divisibility of central binomial coefficients

O / open | Open Problem Garden

Problem (1) Prove that there exist infinitely many positive integers $n$ such that $$\gcd({2n\choose n}, 3\cdot 5\cdot 7) = 1.$$

Problem (2) Prove that there exists only a finite number of positive integers $n$ such that $$\gcd({2n\choose n}, 3\cdot 5\cdot 7\cdot 11) = 1.$$

背景链接：http://www.openproblemgarden.org/op/divisibility_of_central_binomial_coefficients

00316 | AMR-093-0028

### Do any (2, 5)-perfect numbers exist

O / open | AMR 问题清单汇编

Do any (2, 5)-perfect numbers exist?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00317 | AMR-093-0026

### Do any odd noncototients exist

O / open | AMR 问题清单汇编

Do any odd noncototients exist?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00318 | AMR-093-0029

### Do any Taxicab(5, 2, n) exist for n > 1

O / open | AMR 问题清单汇编

Do any Taxicab(5, 2, n) exist for n > 1?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00319 | AMR-093-0169

### Does every prime number appear in the Euclid-Mullin sequence

O / open | AMR 问题清单汇编

Does every prime number appear in the Euclid-Mullin sequence?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00320 | AMR-093-0095

### Does every rational number with an odd denominator have an odd greedy expansion

O / open | AMR 问题清单汇编

Does every rational number with an odd denominator have an odd greedy expansion?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00321 | AMR-093-0137

### Does the converse of Wolstenholme's theorem hold for all natural numbers

O / open | AMR 问题清单汇编

Does the converse of Wolstenholme's theorem hold for all natural numbers?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00322 | AMR-093-0120

### Dubner's conjecture

O / open | AMR 问题清单汇编

Dubner's conjecture: every even number greater than $4208$ is the sum of two primes which both have a twin.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00323 | OWR-1703867-009

### Effective Enumeration of Rational Points on Curves

O / open | Oberwolfach 报告问题汇编

Given equations for a smooth geometrically irreducible projective curve $C$ of genus at least $2$ over a number field $F$, is there an algorithm that produces the complete list of points in $C(F)$?

来源字段：https://doi.org/10.4171/owr/2020/20

00324 | OWR-14299908-005

### Effective G-Function Bounds for Unlikely Intersections

P / partially_solved | Oberwolfach 报告问题汇编

Determine the strongest fully explicit G-function height bounds for unlikely intersections in algebraic tori and abelian varieties. In particular, extend the current explicit bounds from line sections in $\mathbb G_m^3$ to arbitrary algebraic curves in $\mathbb G_m^n$, and develop analogous explicit bounds for curves in powers $A^n$ of a simple abelian variety $A$.

来源字段：https://doi.org/10.4171/owr/2026/3

00325 | OWR-14299578-006

### Effective Mordell Bounds for Rational Points

O / open | Oberwolfach 报告问题汇编

Give an effective form of the Mordell conjecture: for every smooth projective curve $C$ of genus at least two over a number field $K$, compute an explicit bound for the heights and number of points in $C(K)$ from effective arithmetic data of $C$ and $K$.

来源字段：https://doi.org/10.4171/owr/2025/52

00326 | OWR-1104-003

### Effective Mordell-Vojta Conjecture

O / open | Oberwolfach 报告问题汇编

Let $K$ be a number field and $X/K$ a smooth projective curve. For every $\varepsilon>0$, does there exist $\kappa(\varepsilon)$ such that $h_\omega(P)\leq(1+\varepsilon)d(P)+\kappa(\varepsilon)$ for every algebraic point $P\in X(\overline K)$, where $h_\omega$ is the normalized dualizing-sheaf height and $d(P)$ is the logarithmic discriminant of $K(P)$?

来源字段：https://doi.org/10.4171/owr/2005/43

00327 | OWR-14603-031

### Effective Siegel Theorems for Genus-Two Curves

O / open | Oberwolfach 报告问题汇编

For a genus-two curve $C$ over a number field $k$ and a nonconstant $\phi\in k(C)$, can Siegel's theorem on $S$-integral points with respect to $\phi$ be made effective in general, i.e. can one give an algorithm or explicit bound for all such points?

来源字段：https://doi.org/10.4171/owr/2016/21

00328 | OWR-11784-002

### Effective Thue-Mahler Equations over Finitely Generated Domains

P / partially_solved | Oberwolfach 报告问题汇编

Let $F\in A[X,Y]$ be a binary form of degree at least $3$ without multiple factors over a finitely generated domain $A$. Given that $F(x,y)\in A^*$ has only finitely many solution classes under simultaneous multiplication by units, can representatives $(x_i,y_i)$ of every class be determined effectively?

来源字段：https://doi.org/10.4171/owr/2012/22

00329 | AIM-ANALYTIC_NUMBER_THEORY-0023

### Efficient correlation with Liouville and entropy

P / partially_solved | AIM 研讨会问题汇编

Is there a bounded sequence $a(n)$ such that each $a(n)$ is computable in time $O(\log^A n)$ for some fixed $A > 0$ and such that \[ \left| \sum_{n \leq X} \lambda(n) a(n) \right| > \epsilon X \] for all $X$ and some fixed $\epsilon > 0$? Can we say something about the Furstenberg systems of such an $a(n)$? In particular, do they have positive entropy?

来源字段：http://aimpl.org/sarnakconjecture/1/

00330 | OWR-12583-004

### Efficient Relation Generation for Curve Class Groups

O / open | Oberwolfach 报告问题汇编

Can the heuristic expected running time $$\exp\!\left(O\!\left(\max\{\log q,\,n(\log q)^{1/2}\}\right)\right)$$ for relation generation in the discrete-logarithm algorithm be achieved for broader families of elliptic curves, or for curves of any fixed genus, by weakening or removing the hypothesis $n\leq C\log q$?

来源字段：https://doi.org/10.4171/owr/2013/35

00331 | OWR-833-005

### Eichler-Shimura Theory for Families of Modular Forms

P / partially_solved | Oberwolfach 报告问题汇编

Does an extension of the Eichler-Shimura theorem directly relate the full family of modular forms to the corresponding family of modular symbols?

来源字段：https://doi.org/10.4171/owr/2005/28

00332 | OWR-13750341-005

### Eisenstein Maps and Modular Symbols in Cyclotomic K-Theory

P / partially_solved | Oberwolfach 报告问题汇编

Let $I$ be the Eisenstein ideal and let $\widetilde\Pi$ and $\Pi$ be the modular-symbol maps in the cyclotomic K-theory construction. Is $\widetilde\Pi$ Eisenstein, meaning $\widetilde\Pi\circ T=0$ for every $T\in I$? Is $\Pi\bmod I$ an isomorphism?

来源字段：https://doi.org/10.4171/owr/2023/28

00333 | OWR-13750341-006

### Eisenstein Maps from Bianchi Homology to K2

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist an Eisenstein map $\Pi:H_1(\widetilde X_1(N),\mathbb Z')\to K_2(\mathcal O(N))\otimes\mathbb Z'$ that extends from homology relative to the non-infinity cusps and sends each Cremona symbol $[u:v]$ to the corresponding Steinberg symbol?

来源字段：https://doi.org/10.4171/owr/2023/28

00334 | OWR-11695860-013

### Elementary Fourth-Moment Bounds for Cubic Exponential Sums

O / open | Oberwolfach 报告问题汇编

For $f(X)=X^3+X$, prove by methods no deeper than Chebotarev density that the normalized fourth moment of $W_f(a;p)$ either tends to $2$ or is at least $3+O(p^{-1/2})$ on a positive-density set of primes.

来源字段：https://doi.org/10.4171/owr/2022/50

00335 | NT-088

### Elliott-Halberstam Conjecture

P / partially_solved | 独立 / 未分配集合

Do primes distribute uniformly in arithmetic progressions up to nearly $x$ (instead of $x^{1/2}$)?

背景链接：https://michaelnielsen.org/polymath/index.php?title=Bounded_gaps_between_primes

00336 | OWR-17300-012

### Elliptic Curve Values of Rigid Cocycles

O / open | Oberwolfach 报告问题汇编

For an elliptic rigid cocycle attached to a weight-two newform $f$ and every real-multiplication point $\tau$, the values $J_f^+[\tau]$ and $J_f^-[\tau]$ should belong to the elliptic curve $E(H_\tau)$.

来源字段：https://doi.org/10.4171/owr/2019/57

00337 | AIM-ALGEBRAIC_NUMBER_THEORY-0027

### Elliptic scrolls with Hasse failure and ratio tending to one half

P / partially_solved | AIM 研讨会问题汇编

Can one construct smooth projective varieties \(X_k\subset \mathbf P^k_{\mathbf Q}\) such that \(X_k(\mathbf Q_v)\ne\varnothing\) for every place \(v\), \(X_k(\mathbf Q)=\varnothing\), and \[ \frac{\dim X_k}{\deg X_k} \] is unbounded?

来源字段：https://aimath.org/pastworkshops/ratlhigherdimvarproblems.pdf

00338 | AIM-ALGEBRAIC_NUMBER_THEORY-0045

### Elliptic surfaces, orthogonal monodromy, and BKLPR

P / partially_solved | AIM 研讨会问题汇编

**Question 15 (Jordan Ellenberg). What algebro-geometric facts about elliptic surfaces explain Poonen-Rains/BKLPR?**

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

00339 | OWR-15174-011

### Elliptic-Curve Bijections and Additive Bases from Restricted Primes

O / open | Oberwolfach 报告问题汇编

Let $p>3$, let $t\not\equiv0\pmod p$, and put

$$E_p(t)=\{(x,y)\in(\mathbb F_p^\times)^2:x+x^{-1}+y+y^{-1}=t\}.$$

Is there an explicit bijection $E_p(t)\to E_p(16/t)$? Also, if $S$ is the set of positive integers all of whose prime divisors are congruent to $3$ modulo $4$, is every sufficiently large even integer a sum of two elements of $S$?

来源字段：https://doi.org/10.4171/owr/2016/53

00340 | OWR-8415342-031

### Endoscopic Identities and Archimedean Independence of Local Packets

P / partially_solved | Oberwolfach 报告问题汇编

For the local packet $\Pi^{\theta(\psi)}_v(G_v)$ attached to an Arthur parameter, prove the endoscopic character identities. At Archimedean places $v$, is the packet independent of the auxiliary group, and does it agree with Arthur's packet in the quasi-split case?

来源字段：https://doi.org/10.4171/owr/2021/39

00341 | AIM-ANALYTIC_NUMBER_THEORY-0024

### Endpoint variation and a correlation-budget criterion for Liouville short sums

P / partially_solved | AIM 研讨会问题汇编

Can we obtain lower bounds for \[ \int_X^{2X}\left|\sum_{x\le n\le x+H}\lambda(n)\right|\,dx? \] In particular, can we get a lower bound of the form \(XH^{1/2-\epsilon}\)?

来源字段：http://aimpl.org/sarnakconjecture/1/

00342 | AMR-042-0005

### Entropy values and Lehmer's problem

P / partially_solved | AMR 问题清单汇编

Given $\varepsilon>0$, does there exist a polynomial $f(x)=\prod_{i=1}^d(x-\alpha_i)\in\mathbb{Z}[x]$ whose logarithmic Mahler measure $$m(f)=\sum_{i:\,|\alpha_i|>1}\log|\alpha_i|$$ satisfies $0<m(f)<\varepsilon$?

背景链接：https://www.imath.kiev.ua/~skolyada/kevin.pdf

00343 | OWR-17293-018

### Equal Sum Counts for Increasing Ratio Sequences

O / open | Oberwolfach 报告问题汇编

Given real numbers $1<a_1<a_2<\cdots$ whose successive ratios $a_{n+1}/a_n$ are increasing, what can be said about the number of solutions of $a_{i_1}+\cdots+a_{i_k}=a_{j_1}+\cdots+a_{j_k}$?

来源字段：https://doi.org/10.4171/owr/2019/50

00344 | OWR-16764-005

### Equality of Automorphic Weight Cones

P / partially_solved | Oberwolfach 报告问题汇编

Let $C_{\mathrm{zip}}$ be the cone of automorphic weights defined by global sections on the $G$-zip stack, and let $C_K(\mathbb F_p)$ be the corresponding cone for the special fiber of the Shimura variety. Is $$ \langle C_{\mathrm{zip}}\rangle=\langle C_K(\mathbb F_p)\rangle $$ ?

来源字段：https://doi.org/10.4171/owr/2019/2

00345 | OWR-8415342-030

### Equality of Discrete and Automorphic Multiplicities

P / partially_solved | Oberwolfach 报告问题汇编

Let $F$ be a number field, let $G$ be an even orthogonal or unitary group over $F$, let $\psi$ be an elliptic Arthur parameter of $G$, and let $\pi$ be an automorphic representation in the near-equivalence class represented by $\psi$. Prove $m_{\mathrm{disc}}(\pi)=m_{\mathrm{aut}}(\pi)$.

来源字段：https://doi.org/10.4171/owr/2021/39

00346 | OWR-15583-007

### Equidistribution of Automorphic L Functions over Function Fields

O / open | Oberwolfach 报告问题汇编

Establish equidistribution of local factors, or of complete $L$-functions, in families of automorphic representations over function fields with specified local conditions, preferably by a method that also proves the Ramanujan conjecture for those families.

来源字段：https://doi.org/10.4171/owr/2017/40

00347 | GREEN-052

### Equidistribution of Integer Multiples

P / partially_solved | Ben Green 问题

Let $c > 0$ and let $A$ be a set of $n$ distinct integers. Does there exist $\theta$ such that no interval of length $\frac{1}{n}$ in $\mathbb{R}/\mathbb{Z}$ contains more than $n^c$ of the numbers $\theta a \pmod 1$, for $a \in A$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

00348 | NT-055

### Erdős Conjecture on Arithmetic Progressions

P / partially_solved | 独立 / 未分配集合

If the sum of reciprocals of a set of positive integers diverges, does the set contain arbitrarily long arithmetic progressions?

背景链接：https://doi.org/10.4007/annals.2025.202.1.1

00349 | EP-1

### Erdős Problem #1

S / solved | Erdős 问题

If $A\subseteq \{1,\ldots,N\}$ with $\lvert A\rvert=n$ is such that the subset sums $\sum_{a\in S}a$ are distinct for all $S\subseteq A$ then\[N \gg 2^{n}.\]

来源字段：https://www.erdosproblems.com/1

00350 | EP-10

### Erdős Problem #10

O / open | Erdős 问题

Is there some $k$ such that every large integer is the sum of a prime and at most $k$ powers of 2?

来源字段：https://www.erdosproblems.com/10

00351 | EP-1000

### Erdős Problem #1000: Number Theory, Diophantine Approximation

S / solved | Erdős 问题

Let $A=\{n_1<n_2<\cdots\}$ be an infinite sequence of integers, and let $\phi_A(k)$ count the number of $1\leq m\leq n_k$ such that the fraction $\frac{m}{n_k}$ does not have denominator $n_j$ for $j<k$ when written in lowest form; equivalently,\[\frac{n_k}{(m,n_k)}\neq n_j\]for all $1\leq j<k$.

Is there a sequence $A$ such that\[\lim_{N\to \infty}\frac{1}{N}\sum_{k\leq N}\frac{\phi_A(k)}{n_k}=0?\]

来源字段：https://www.erdosproblems.com/1000

00352 | EP-1001

### Erdős Problem #1001: Number Theory, Diophantine Approximation

S / solved | Erdős 问题

Let $S(N,A,c)$ be the measure of the set of those $\alpha\in (0,1)$ such that\[\left\lvert \alpha-\frac{x}{y}\right\rvert< \frac{A}{y^2}\]for some $N\leq y\leq cN$ and $(x,y)=1$. Does\[\lim_{N\to \infty}S(N,A,c)=f(A,c)\]exist? What is its explicit form?

来源字段：https://www.erdosproblems.com/1001

00353 | EP-1003

### Erdős Problem #1003

P / partially_solved | Erdős 问题

Are there infinitely many solutions to $\phi(n)=\phi(n+1)$, where $\phi$ is the Euler totient function?

来源字段：https://www.erdosproblems.com/1003

00354 | EP-1004

### Erdős Problem #1004

P / partially_solved | Erdős 问题

Let $c>0$. If $x$ is sufficiently large then does there exist $n\leq x$ such that the values of $\phi(n+k)$ are all distinct for $1\leq k\leq (\log x)^c$, where $\phi$ is the Euler totient function?

来源字段：https://www.erdosproblems.com/1004

00355 | EP-1005

### Erdős Problem #1005

S / solved | Erdős 问题

Let $\frac{a_1}{b_1},\frac{a_2}{b_2},\ldots$ be the Farey fractions of order $n\geq 4$. Let $f(n)$ be the largest integer such that if $1\leq k<l\leq k+f(n)$ then $\frac{a_k}{b_k}$ and $\frac{a_l}{b_l}$ are similarly ordered - in other words,\[(a_k-a_l)(b_k-b_l)\geq 0.\]Estimate $f(n)$ - in particular, is there a constant $c>0$ such that $f(n)=(c+o(1))n$ for all large $n$?

来源字段：https://www.erdosproblems.com/1005

00356 | EP-1017

### Erdős Problem #1017

P / partially_solved | Erdős 问题

Let $f(n,k)$ be such that every graph on $n$ vertices and $k$ edges can be partitioned into at most $f(n,k)$ edge-disjoint complete graphs. Estimate $f(n,k)$ for $k>n^2/4$.

来源字段：https://www.erdosproblems.com/1017

00357 | EP-1049

### Erdős Problem #1049

P / partially_solved | Erdős 问题

Let $t>1$ be a rational number. Is\[\sum_{n=1}^\infty\frac{1}{t^n-1}=\sum_{n=1}^\infty \frac{\tau(n)}{t^n}\]irrational, where $\tau(n)$ counts the divisors of $n$?

来源字段：https://www.erdosproblems.com/1049

00358 | EP-1050

### Erdős Problem #1050: Is\[\sum_{n=1}^\infty \frac{1}{2^n-3}\]irrational?

S / solved | Erdős 问题

Is\[\sum_{n=1}^\infty \frac{1}{2^n-3}\]irrational?

来源字段：https://www.erdosproblems.com/1050

00359 | EP-1051

### Erdős Problem #1051

S / solved | Erdős 问题

Is it true that if $1\leq a_1<a_2<\cdots$ is a sequence of integers with\[\liminf a_n^{1/2^n}>1\]then\[\sum_{n=1}^\infty \frac{1}{a_na_{n+1}}\]is irrational?

来源字段：https://www.erdosproblems.com/1051

00360 | EP-1052

### Erdős Problem #1052

P / partially_solved | Erdős 问题

A unitary divisor of $n$ is $d\mid n$ such that $(d,n/d)=1$. A number $n\geq 1$ is a unitary perfect number if it is the sum of its unitary divisors (aside from $n$ itself).

Are there only finitely many unitary perfect numbers?

来源字段：https://www.erdosproblems.com/1052

00361 | EP-1053

### Erdős Problem #1053

P / partially_solved | Erdős 问题

Call a number $k$-perfect if $\sigma(n)=kn$, where $\sigma(n)$ is the sum of the divisors of $n$. Must $k=o(\log\log n)$?

来源字段：https://www.erdosproblems.com/1053

00362 | EP-1054

### Erdős Problem #1054

P / partially_solved | Erdős 问题

Let $f(n)$ be the minimal integer $m$ such that $n$ is the sum of the $k$ smallest divisors of $m$ for some $k\geq 1$.

Is it true that $f(n)=o(n)$? Or is this true only for almost all $n$, and $\limsup f(n)/n=\infty$?

来源字段：https://www.erdosproblems.com/1054

00363 | EP-1055

### Erdős Problem #1055

O / open | Erdős 问题

A prime $p$ is in class $1$ if the only prime divisors of $p+1$ are $2$ or $3$. In general, a prime $p$ is in class $r$ if every prime factor of $p+1$ is in some class $\leq r-1$, with equality for at least one prime factor.

Are there infinitely many primes in each class? If $p_r$ is the least prime in class $r$, then how does $p_r^{1/r}$ behave?

来源字段：https://www.erdosproblems.com/1055

00364 | EP-1056

### Erdős Problem #1056

P / partially_solved | Erdős 问题

Let $k\geq 2$. Does there exist a prime $p$ and consecutive intervals $I_1,\ldots,I_k$ such that\[\prod_{n\in I_i}n \equiv 1\pmod{p}\]for all $1\leq i\leq k$?

来源字段：https://www.erdosproblems.com/1056

00365 | EP-1057

### Erdős Problem #1057

P / partially_solved | Erdős 问题

别名：Erdős Conjecture on Carmichael Numbers [GUY-A13]

Let $C(x)$ count the number of Carmichael numbers in the interval $[1,x]$. Is it true that $C(x)=x^{1-o(1)}$?

来源字段：https://www.erdosproblems.com/1057

00366 | EP-1058

### Erdős Problem #1058: Number Theory

S / solved | Erdős 问题

Let $2=p_1<p_2<\cdots$ be the sequence of prime numbers. Are there only finitely many $n$ such that $n\in [p_{k-1},p_k)$ and the only primes dividing $n!+1$ are $p_{k}$ and $p_{k+1}$?

来源字段：https://www.erdosproblems.com/1058

00367 | EP-1059

### Erdős Problem #1059

O / open | Erdős 问题

Are there infinitely many primes $p$ such that $p-k!$ is composite for each $k$ such that $1\leq k!<p$?

来源字段：https://www.erdosproblems.com/1059

00368 | EP-1060

### Erdős Problem #1060

O / open | Erdős 问题

Let $f(n)$ count the number of solutions to $k\sigma(k)=n$, where $\sigma(k)$ is the sum of divisors of $k$. Is it true that $f(n)\leq n^{o(\frac{1}{\log\log n})}$? Perhaps even $\leq (\log n)^{O(1)}$?

来源字段：https://www.erdosproblems.com/1060

00369 | EP-1061

### Erdős Problem #1061

O / open | Erdős 问题

How many solutions are there to\[\sigma(a)+\sigma(b)=\sigma(a+b)\]with $a+b\leq x$, where $\sigma$ is the sum of divisors function? Is it $\sim cx$ for some constant $c>0$?

来源字段：https://www.erdosproblems.com/1061

00370 | EP-1064

### Erdős Problem #1064: Number Theory

S / solved | Erdős 问题

Prove that $\phi(n)>\phi(n-\phi(n))$ for almost all $n$, but that $\phi(n)<\phi(n-\phi(n))$ for infinitely many $n$, where $\phi$ is Euler's totient function.

来源字段 : https://www.erdosproblems.com/1064

00371 | EP-1065

### Erdős Problem #1065

O / open | Erdős 问题

Are there infinitely many primes $p$ such that $p=2^kq+1$ for some prime $q$ and $k\geq 0$? Or $p=2^k3^lq+1$?

来源字段 : https://www.erdosproblems.com/1065

00372 | EP-1072

### Erdős Problem #1072

O / open | Erdős 问题

For any prime $p$, let $f(p)$ be the least integer such that $f(p)!+1\equiv 0\pmod{p}$.

Is it true that there are infinitely many $p$ for which $f(p)=p-1$? Is it true that $f(p)/p\to 0$ for almost all $p$?

来源字段 : https://www.erdosproblems.com/1072

00373 | EP-1073

### Erdős Problem #1073

O / open | Erdős 问题

Let $A(x)$ count the number of composite $u<x$ such that $n!+1\equiv 0\pmod{u}$ for some $n$. Is it true that $A(x)\leq x^{o(1)}$?

来源字段 : https://www.erdosproblems.com/1073

00374 | EP-1074

### Erdős Problem #1074

P / partially_solved | Erdős 问题

Let $S$ be the set of all $m\geq 1$ such that there exists a prime $p\not\equiv 1\pmod{m}$ such that $m!+1\equiv 0\pmod{p}$. Does\[\lim \frac{\lvert S\cap [1,x]\rvert}{x}\]exist? What is it?

Similarly, if $P$ is the set of all primes $p$ such that there exists an $m$ with $p\not\equiv 1\pmod{m}$ such that $m!+1\equiv 0\pmod{p}$, then does\[\lim \frac{\lvert P\cap [1,x]\rvert}{\pi(x)}\]exist? What is it?

来源字段 : https://www.erdosproblems.com/1074

00375 | EP-1081

### Erdős Problem #1081: Number Theory, Powerful

S / solved | Erdős 问题

Let $A(x)$ count the number of $n\leq x$ which are the sum of two squarefull numbers (a number $m$ is squarefull if $p\mid m$ implies $p^2\mid m$). Is it true that\[A(x) \sim c \frac{x}{\sqrt{\log x}}\]for some $c>0$?

来源字段 : https://www.erdosproblems.com/1081

00376 | EP-1084

### Erdős Problem #1084

P / partially_solved | Erdős 问题

Let $f_d(n)$ be minimal such that in any collection of $n$ points in $\mathbb{R}^d$, all of distance at least $1$ apart, there are at most $f_d(n)$ many pairs of points which are distance $1$ apart. Estimate $f_d(n)$.

来源字段 : https://www.erdosproblems.com/1084

00377 | EP-1093

### Erdős Problem #1093

P / partially_solved | Erdős 问题

For $n\geq 2k$ we define the deficiency of $\binom{n}{k}$ as follows. If $\binom{n}{k}$ is divisible by a prime $p\leq k$ then the deficiency is undefined. Otherwise, the deficiency is the number of $0\leq i<k$ such that $n-i$ is $k$-smooth, that is, divisible only by primes $\leq k$.

Are there infinitely many binomial coefficients with deficiency $1$? Are there only finitely many with deficiency $>1$?

来源字段 : https://www.erdosproblems.com/1093

00378 | EP-1094

### Erdős Problem #1094

P / partially_solved | Erdős 问题

For all $n\geq 2k$ the least prime factor of $\binom{n}{k}$ is $\leq \max(n/k,k)$, with only finitely many exceptions.

来源字段 : https://www.erdosproblems.com/1094

00379 | EP-1095

### Erdős Problem #1095

P / partially_solved | Erdős 问题

Let $g(k)>k+1$ be the smallest $n$ such that all prime factors of $\binom{n}{k}$ are $>k$. Estimate $g(k)$.

来源字段 : https://www.erdosproblems.com/1095

00380 | EP-1097

### Erdős Problem #1097

P / partially_solved | Erdős 问题

Let $A$ be a set of $n$ integers. How many distinct $d$ can occur as the common difference of a three-term arithmetic progression in $A$?

In particular, are there always $O(n^{3/2})$ many such $d$?

来源字段 : https://www.erdosproblems.com/1097

00381 | EP-1098

### Erdős Problem #1098: Group Theory

S / solved | Erdős 问题

Let $G$ be a group and $\Gamma=\Gamma(G)$ be the non-commuting graph, with vertices the elements of $G$ and an edge between $g$ and $h$ if and only if $g$ and $h$ do not commute, $gh\neq hg$.

If $\Gamma$ contains no infinite complete subgraph, then is there a finite bound on the size of complete subgraphs of $\Gamma$?

来源字段：https://www.erdosproblems.com/1098

00382 | EP-1099

### Erdős Problem #1099: Number Theory, Divisors

S / solved | Erdős 问题

Let $1=d_1<\cdots<d_{\tau(n)}=n$ be the divisors of $n$, and for $\alpha>1$ let\[h_\alpha(n) = \sum_i \left( \frac{d_{i+1}}{d_i}-1\right)^\alpha.\]Is it true that\[\liminf_{n\to \infty}h_\alpha(n) \ll_\alpha 1?\]

来源字段：https://www.erdosproblems.com/1099

00383 | EP-1100

### Erdős Problem #1100

P / partially_solved | Erdős 问题

If $1=d_1<\cdots<d_{\tau(n)}=n$ are the divisors of $n$, then let $\tau_\perp(n)$ count the number of $i$ for which $(d_i,d_{i+1})=1$.

Is it true that $\tau_\perp(n)/\omega(n)\to \infty$ for almost all $n$? Is it true that\[\tau_\perp(n)< \exp((\log n)^{o(1)})\]for all $n$?

Let\[g(k) = \max_{\omega(n)=k}\tau_\perp(n),\]where $\omega(n)$ counts the number of distinct prime divisors of $n$, and $n$ is restricted to squarefree integers. Determine the growth of $g(k)$.

来源字段：https://www.erdosproblems.com/1100

00384 | EP-1101

### Erdős Problem #1101

P / partially_solved | Erdős 问题

If $u=\{u_1<u_2<\cdots\}$ is a sequence of integers such that $(u_i,u_j)=1$ for all $i\neq j$ and $\sum \frac{1}{u_i}<\infty$ then let $\{a_1<a_2<\cdots\}$ be the sequence of integers which are not divisible by any of the $u_i$. For any $x$ define $t_x$ by\[u_1\cdots u_{t_x}\leq x< u_1\cdots u_{t_x}u_{t_x+1}.\]We call such a sequence $u_i$ good if, for all $\epsilon>0$, if $x$ is sufficiently large then\[\max_{a_k<x} (a_{k+1}-a_k) < (1+\epsilon)t_x \prod_{i}\left(1-\frac{1}{u_i}\right)^{-1}.\]Is there a good sequence such that $u_n< n^{O(1)}$? Is there a good sequence such that $u_n\leq e^{o(n)}$?

来源字段：https://www.erdosproblems.com/1101

00385 | EP-1102

### Erdős Problem #1102: Number Theory

S / solved | Erdős 问题

We say that $A\subseteq \mathbb{N}$ has property $P$ if, for all $n\geq 1$, there are only finitely many $a\in A$ such that $n+a$ is squarefree.

We say that $A$ has property $Q$ if there are infinitely many $n$ such that $n+a$ is squarefree for all $a<n$.

How fast must sequences $A=\{a_1<a_2<\cdots\}$ with properties $P$ or $Q$ increase?

来源字段：https://www.erdosproblems.com/1102

00386 | EP-1103

### Erdős Problem #1103

P / partially_solved | Erdős 问题

Let $A$ be an infinite sequence of integers such that every $n\in A+A$ is squarefree. How fast must $A$ grow?

来源字段：https://www.erdosproblems.com/1103

00387 | EP-1106

### Erdős Problem #1106

P / partially_solved | Erdős 问题

Let $p(n)$ denote the partition function of $n$ and let $F(n)$ count the number of distinct prime factors of\[\prod_{1\leq k\leq n}p(k).\]Does $F(n)\to \infty$ with $n$? Is $F(n)>n$ for all sufficiently large $n$?

来源字段：https://www.erdosproblems.com/1106

00388 | EP-1107

### Erdős Problem #1107

P / partially_solved | Erdős 问题

Let $r\geq 2$. A number $n$ is $r$-powerful if for every prime $p$ which divides $n$ we have $p^r\mid n$. Is every large integer the sum of at most $r+1$ many $r$-powerful numbers?

来源字段：https://www.erdosproblems.com/1107

00389 | EP-1109

### Erdős Problem #1109

P / partially_solved | Erdős 问题

Let $f(N)$ be the size of the largest subset $A\subseteq \{1,\ldots,N\}$ such that every $n\in A+A$ is squarefree. Estimate $f(N)$. In particular, is it true that $f(N)\leq N^{o(1)}$, or even $f(N) \leq (\log N)^{O(1)}$?

来源字段：https://www.erdosproblems.com/1109

00390 | EP-1110

### Erdős Problem #1110

P / partially_solved | Erdős 问题

Let $p>q\geq 2$ be two coprime integers. We call $n$ representable if it is the sum of integers of the form $p^kq^l$, none of which divide each other.

If $\{p,q\}\neq \{2,3\}$ then what can be said about the density of non-representable numbers? Are there infinitely many coprime non-representable numbers?

来源字段：https://www.erdosproblems.com/1110

00391 | EP-1112

### Erdős Problem #1112

P / partially_solved | Erdős 问题

Let $1\leq d_1<d_2$ and $k\geq 3$. Does there exist an integer $r$ such that if $B=\{b_1<\cdots\}$ is a lacunary sequence of positive integers with $b_{i+1}\geq rb_i$ then there exists a sequence of positive integers $A=\{a_1<\cdots\}$ such that\[d_1\leq a_{i+1}-a_i\leq d_2\]for all $i\geq 1$ and $(kA)\cap B=\emptyset$, where $kA$ is the $k$-fold sumset?

来源字段：https://www.erdosproblems.com/1112

00392 | EP-1113

### Erdős Problem #1113

P / partially_solved | Erdős 问题

A positive odd integer $m$ such that none of $2^km+1$ are prime for $k\geq 0$ is called a Sierpinski number. We say that a set of primes $P$ is a covering set for $m$ if every $2^km+1$ is divisible by some $p\in P$.

Are there Sierpinski numbers with no finite covering set of primes?

来源字段：https://www.erdosproblems.com/1113

00393 | EP-112

### Erdős Problem #112

O / open | Erdős 问题

Let $k=k(n,m)$ be minimal such that any directed graph on $k$ vertices must contain either an independent set of size $n$ or a transitive tournament of size $m$. Determine $k(n,m)$.

来源字段：https://www.erdosproblems.com/112

00394 | EP-1123

### Erdős Problem #1123: Algebra

S / solved | Erdős 问题

Let $B_1$ be the Boolean algebra of sets of integers modulo sets of density $0$ (that is, in which two sets are equivalent if and only if they differ by a set of density $0$) and let $B_2$ be the Boolean algebra of sets modulo sets of logarithmic density $0$.

Prove that $B_1$ and $B_2$ are not isomorphic.

来源字段：https://www.erdosproblems.com/1123

00395 | EP-1134

### Erdős Problem #1134: \]\[x\mapsto 3x+1,\]and\[x\mapsto 6x+1.\]Does $A$ have positive lower density?

S / solved | Erdős 问题

Let $A\subseteq \mathbb{N}$ be the smallest set which contains $1$ and is closed under the operations\[x\mapsto 2x+1,\]\[x\mapsto 3x+1,\]and\[x\mapsto 6x+1.\]Does $A$ have positive lower density?

来源字段：https://www.erdosproblems.com/1134

00396 | EP-1135

### Erdős Problem #1135

P / partially_solved | Erdős 问题

别名：Collatz Conjecture [NT-002]; The 3n+1 conjecture [OPG-432]

Define $f:\mathbb{N}\to \mathbb{N}$ by $f(n)=n/2$ if $n$ is even and $f(n)=\frac{3n+1}{2}$ if $n$ is odd.

Given any integer $m\geq 1$ does there exist $k\geq 1$ such that $f^{(k)}(m)=1$?

来源字段：https://www.erdosproblems.com/1135

00397 | EP-1136

### Erdős Problem #1136: Number Theory

S / solved | Erdős 问题

Does there exist $A\subset \mathbb{N}$ with lower density $>1/3$ such that $a+b\neq 2^k$ for any $a,b\in A$ and $k\geq 0$?

来源字段：https://www.erdosproblems.com/1136

00398 | EP-1137

### Erdős Problem #1137: Number Theory, Primes

O / open | Erdős 问题

Let $d_n=p_{n+1}-p_n$, where $p_n$ denotes the $n$th prime. Is it true that\[\frac{\max_{n<x}d_{n}d_{n-1}}{(\max_{n<x}d_n)^2}\to 0\]as $x\to \infty$?

来源字段：https://www.erdosproblems.com/1137

00399 | EP-1138

### Erdős Problem #1138: Number Theory, Primes

S / solved | Erdős 问题

Let $x/2<y<x$ and $C>1$. If $d=\max_{p_n<x}(p_{n+1}-p_n)$, where $p_n$ denotes the $n$th prime, then is it true that\[\pi(y+Cd)-\pi(y)\sim\frac{Cd}{\log y}?\]

来源字段：https://www.erdosproblems.com/1138

00400 | EP-1139

### Erdős Problem #1139: Is it true that\[\limsup \frac{u_{k+1}-u_k}{\log k}=\infty?\]

O / open | Erdős 问题

Let $1\leq u_1<u_2<\cdots$ be the sequence of integers with at most $2$ prime factors. Is it true that\[\limsup \frac{u_{k+1}-u_k}{\log k}=\infty?\]

来源字段：https://www.erdosproblems.com/1139

00401 | EP-1140

### Erdős Problem #1140: Number Theory

S / solved | Erdős 问题

Do there exist infinitely many $n$ such that $n-2x^2$ is prime for all $x$ with $2x^2<n$?

来源字段：https://www.erdosproblems.com/1140

00402 | EP-1141

### Erdős Problem #1141: Number Theory, Primes

S / solved | Erdős 问题

Are there infinitely many $n$ such that $n-k^2$ is prime for all $k$ with $(n,k)=1$ and $k^2<n$?

来源字段：https://www.erdosproblems.com/1141

00403 | EP-1142

### Erdős Problem #1142: Number Theory, Primes

O / open | Erdős 问题

别名：Erdős Conjecture on $n - 2^k$ Prime [GUY-A19a]

Are there infinitely many $n$ (or any $n>105$) such that $n-2^k$ is prime for all $1<2^k<n$?

来源字段：https://www.erdosproblems.com/1142

00404 | EP-1143

### Erdős Problem #1143: Number Theory, Primes

O / open | Erdős 问题

Let $p_1<\cdots<p_u$ be primes and let $k\geq 1$. Let $F_k(p_1,\ldots,p_u)$ be such that every interval of $k$ positive integers contains at least $F_k(p_1,\ldots,p_u)$ multiples of at least one of the $p_i$.

Estimate $F_k(p_1,\ldots,p_u)$, particularly in the range $k=\alpha p_u$ for constant $\alpha>2$.

来源字段：https://www.erdosproblems.com/1143

00405 | EP-1144

### Erdős Problem #1144: Number Theory, Probability

O / open | Erdős 问题

Let $f$ be a random completely multiplicative function, where for each prime $p$ we independently choose $f(p)\in \{-1,1\}$ uniformly at random. Is it true that\[\limsup_{N\to \infty}\frac{\sum_{m\leq N}f(m)}{\sqrt{N}}=\infty\]with probability $1$?

来源字段：https://www.erdosproblems.com/1144

00406 | EP-1146

### Erdős Problem #1146: Number Theory

O / open | Erdős 问题

We say that $A\subset \mathbb{N}$ is an essential component if $d_s(A+B)>d_s(B)$ for every $B\subset \mathbb{N}$ with $0<d_s(B)<1$ where $d_s$ is the Schnirelmann density.

Is $B=\{2^m3^n: m,n\geq 0\}$ an essential component?

来源字段：https://www.erdosproblems.com/1146

00407 | EP-1147

### Erdős Problem #1147: Irrational, Additive Basis

S / solved | Erdős 问题

Let $\alpha>0$ be an irrational number. Is the set\[A=\left\{ n\geq 1: \| \alpha n^2\| < \frac{1}{\log n}\right\},\]where $\|\cdot\|$ denotes the distance to the nearest integer, an additive basis of order $2$?

来源字段：https://www.erdosproblems.com/1147

00408 | EP-1148

### Erdős Problem #1148: Number Theory

S / solved | Erdős 问题

Can every large integer $n$ be written as $n=x^2+y^2-z^2$ with $\max(x^2,y^2,z^2)\leq n$?

来源字段：https://www.erdosproblems.com/1148

00409 | EP-1149

### Erdős Problem #1149: Number Theory

S / solved | Erdős 问题

Let $\alpha>0$ be a real number, not an integer. The density of integers $n\geq 1$ for which $(n,\lfloor n^\alpha\rfloor)=1$ is $6/\pi^2$.

来源字段：https://www.erdosproblems.com/1149

00410 | EP-1160

### Erdős Problem #1160: denote the number of groups of order $n$. If $n\leq 2^m$ then $g(n)\leq g(2^m)$

O / open | Erdős 问题

Let $g(n)$ denote the number of groups of order $n$. If $n\leq 2^m$ then $g(n)\leq g(2^m)$.

来源字段：https://www.erdosproblems.com/1160

00411 | EP-1161

### Erdős Problem #1161: Group Theory

S / solved | Erdős 问题

Let $f_k(n)$ count the number of elements of $S_n$ of order $k$. For which values of $k$ will $f_k(n)$ be maximal?

来源字段：https://www.erdosproblems.com/1161

00412 | EP-1162

### Erdős Problem #1162: Group Theory

O / open | Erdős 问题

Give an asymptotic formula for the number of subgroups of $S_n$. Is there a statistical theorem on their order?

来源字段：https://www.erdosproblems.com/1162

00413 | EP-1163

### Erdős Problem #1163: Group Theory

O / open | Erdős 问题

Describe (by statistical means) the arithmetic structure of the orders of subgroups of $S_n$.

来源字段：https://www.erdosproblems.com/1163

00414 | EP-1164

### Erdős Problem #1164:

S / solved | Erdős 问题

Let $R_n$ be the maximal integer such that almost every random walk from the origin in $\mathbb{Z}^2$ visits every $x\in\mathbb{Z}^2$ with $\| x\|\leq R_n$ in at most $n$ steps.

Is it true that\[\log R_n \asymp \sqrt{\log n}?\]

来源字段：https://www.erdosproblems.com/1164

00415 | EP-1165

### Erdős Problem #1165: Probability

S / solved | Erdős 问题

Given a random walk $s_0,s_1,\ldots$ in $\mathbb{Z}^2$, starting at the origin, let $f_n(x)$ count the number of $0\leq k\leq n$ such that $s_k=x$.

Let\[F(n)=\{ x: f_n(x) = \max_y f_n(y)\}\]be the set of 'favourite values'. Find\[\mathbb{P}(\lvert F(n)\rvert=r\textrm{ infinitely often})\]for $r\geq 3$.

来源字段：https://www.erdosproblems.com/1165

00416 | EP-1166

### Erdős Problem #1166: Probability

S / solved | Erdős 问题

Given a random walk $s_0,s_1,\ldots$ in $\mathbb{Z}^2$, starting at the origin, let $f_k(x)$ count the number of $0\leq l\leq k$ such that $s_l=x$.

Let\[F(k)=\{ x: f_k(x) = \max_y f_k(y)\}\]be the set of 'favourite values'. Is it true that\[\left\lvert \bigcup_{k\leq n}F(k)\right\rvert \leq (\log n)^{O(1)}\]almost surely, for all but finitely many $n$?

来源字段：https://www.erdosproblems.com/1166

00417 | EP-1180

### Erdős Problem #1180: Number Theory

S / solved | Erdős 问题

Let $\epsilon>0$. Does there exist a constant $C_\epsilon$ such that, for all primes $p$, every residue modulo $p$ is the sum of at most $C_\epsilon$ many elements of\[\{ n^{-1}: 1\leq n\leq p^\epsilon\}\]where $n^{-1}$ denotes the inverse of $n$ modulo $p$?

来源字段：https://www.erdosproblems.com/1180

00418 | EP-1181

### Erdős Problem #1181: Number Theory

O / open | Erdős 问题

Let $q(n,k)$ denote the least prime which does not divide $\prod_{1\leq i\leq k}(n+i)$. Is it true that there exists some $c>0$ such that, for all large $n$,\[q(n,\log n)<(1-c)(\log n)^2?\]

来源字段：https://www.erdosproblems.com/1181

00419 | EP-1184

### Erdős Problem #1184: Number Theory, Primes

O / open | Erdős 问题

Let $f(n,k)$ count the number of $1\leq i\leq k$ such that $P(n+i)>k$ (where $P(m)$ is the largest prime divisor of $m$). Is it true that, if $\alpha>1$ is such that $n=k^{\alpha+o(1)}$, then\[f(n,k)=(1-\rho(\alpha)+o(1))k,\]where $\rho$ is the Dickman function?

来源字段：https://www.erdosproblems.com/1184

00420 | EP-1187

### Erdős Problem #1187: Number Theory, Primes

S / solved | Erdős 问题

Let $k\geq 3$. Is it true that, in any finite colouring of the integers, there are monochromatic arithmetic progressions of primes of length $k$?

Are there monochromatic arithmetic progressions of length $k$ whose common difference is a prime?

来源字段：https://www.erdosproblems.com/1187

00421 | EP-1188

### Erdős Problem #1188: Number Theory, Covering Systems

O / open | Erdős 问题

Call a set of distinct integers $1<n_1<\cdots<n_k$ with associated congruence classes $a_i\pmod{n_i}$ a distinct covering system if every integer satisfies at least one of these congruences. A minimal distinct covering system is one such that no proper subset forms a covering system.

Let $F(x)$ count the number of minimal distinct covering systems with all moduli in $[1,x]$. Estimate $F(x)$.

来源字段：https://www.erdosproblems.com/1188

00422 | EP-1189

### Erdős Problem #1189: Number Theory, Covering Systems

O / open | Erdős 问题

Call a set of distinct integers $1<n_1<\cdots<n_k$ a covering set if there is a choice of $a_i\pmod{n_i}$ for $1\leq i\leq k$ such that every integer satisfies at least one of these congruences. A set is an irreducible covering set if no proper subset is a covering set.

How many irreducible covering sets of size $k$ are there?

What is the minimum and maximum that $n_k$ can be?

Determine or estimate $\max \sum\frac{1}{n_i}$, where the maximum ranges over all irreducible covering sets of size $k$.

Are there infinitely many $n$ such that the divisors of $n$ (which are $>1$) form an irreducible covering set?

来源字段：https://www.erdosproblems.com/1189

00423 | EP-1190

### Erdős Problem #1190: Number Theory, Covering Systems

S / solved | Erdős 问题

Let\[\epsilon_m=\max \sum \frac{1}{n_i}\]where the maximum is taken over all finite sequences $m<n_1<\cdots<n_k$ for which there exist congruences $a_i\pmod{n_i}$ such that no integer satisfies two such congruences.

Estimate $\epsilon_m$.

来源字段：https://www.erdosproblems.com/1190

00424 | EP-1195

### Erdős Problem #1195: y\in S$

S / solved | Erdős 问题

Let $S\subset \mathbb{R}$ be a set of infinite measure such that $x/y$ is never an integer for all distinct $x,y\in S$.

How fast can $\lvert S\cap (0,x)\rvert$ tend to infinity?

来源字段：https://www.erdosproblems.com/1195

00425 | EP-1196

### Erdős Problem #1196: Number Theory, Primitive Sets

S / solved | Erdős 问题

Is it true that, for any $x$, if $A\subset [x,\infty)$ is a primitive set of integers (so that no distinct elements of $A$ divide each other) then\[\sum_{a\in A}\frac{1}{a\log a}<1+o(1),\]where the $o(1)$ term $\to 0$ as $x\to \infty$?

来源字段：https://www.erdosproblems.com/1196

00426 | EP-11

### Erdős Problem #11: Is every large odd integer $n$ the sum of a squarefree number and a power of 2?

O / open | Erdős 问题

Is every large odd integer $n$ the sum of a squarefree number and a power of 2?

来源字段：https://www.erdosproblems.com/11

00427 | EP-12

### Erdős Problem #12

P / partially_solved | Erdős 问题

Let $A$ be an infinite set such that there are no distinct $a,b,c\in A$ such that $a\mid (b+c)$ and $b,c>a$. Is there such an $A$ with\[\liminf \frac{\lvert A\cap\{1,\ldots,N\}\rvert}{N^{1/2}}>0?\]Does there exist some absolute constant $c>0$ such that there are always infinitely many $N$ with\[\lvert A\cap\{1,\ldots,N\}\rvert<N^{1-c}?\]Is it true that\[\sum_{n\in A}\frac{1}{n}<\infty?\]

来源字段：https://www.erdosproblems.com/12

00428 | EP-1200

### Erdős Problem #1200: Number Theory, Primes

O / open | Erdős 问题

There exists a constant $C$ such that for all large $x$ there is a collection of primes $p_1<\ldots<p_k<x$ with $\sum\frac{1}{p_i}<C$ together with a system of congruences $a_i\pmod{p_i}$ such that every integer $n<x$ satisfies at least one of these congruences.

来源字段：https://www.erdosproblems.com/1200

00429 | EP-1201

### Erdős Problem #1201: Number Theory, Primes

O / open | Erdős 问题

Is it true that for every $\epsilon,\eta>0$ there exists a $k$ such that the density of $n$ for which\[P(n(n+1)\cdots(n+k))>n^{1-\epsilon}\]is at least $1-\eta$ (where $P(m)$ is the greatest prime divisor of $m$)?

来源字段：https://www.erdosproblems.com/1201

00430 | EP-1202

### Erdős Problem #1202: Number Theory, Primes

S / solved | Erdős 问题

Let $\epsilon,\eta>0$. Does there exist a $k$ such that, given any set of $k$ primes $p_1<\cdots<p_k<n^{1-\epsilon}$, each with a set $A_i$ of $\frac{p_i-1}{2}$ many congruence classes modulo $p_i$, the number of $m\leq n$ such that $m\not\in A_i\pmod{p_i}$ for all $i$ is at most $\epsilon n$?

来源字段：https://www.erdosproblems.com/1202

00431 | EP-1203

### Erdős Problem #1203: Number Theory

O / open | Erdős 问题

If $\omega(n)$ counts the number of distinct prime divisors of $n$ then let\[F(n)=\max_k \omega(n+k)\frac{\log\log k}{\log k}.\]Prove that $F(n)\to \infty$ as $n\to \infty$.

来源字段 : https://www.erdosproblems.com/1203

00432 | EP-1204

### Erdős Problem #1204: Number Theory

O / open | Erdős 问题

We call a sequence of integers $0\leq a_1<\cdots <a_k$ admissible if it is missing at least one congruence class modulo every prime $p$. Let $A(k)=\min a_k$. Estimate $A(k)$ - in particular, is it true that\[A(k)\sim k\log k?\]Estimate\[B(k)=\min \frac{a_1+\cdots+a_k}{k}.\]

来源字段 : https://www.erdosproblems.com/1204

00433 | EP-1205

### Erdős Problem #1205: Number Theory

S / solved | Erdős 问题

Let $F(x)$ be maximal such that there exists, for all $n\leq x$, a congruence class $a_n\pmod{n}$ such that every $m\leq x$ satisfies at least $F(x)$ of these congruences.

Estimate $F(x)$.

来源字段 : https://www.erdosproblems.com/1205

00434 | EP-1206

### Erdős Problem #1206: Does $\{1,2^3,\ldots,N^3\}$ contain a Sidon set of size $\gg N$?

O / open | Erdős 问题

Does $\{1,2^3,\ldots,N^3\}$ contain a Sidon set of size $\gg N$?

Is there an infinite set $A\subset \mathbb{N}$ of positive density such that $\{a^3: a\in A\}$ is a Sidon set?

来源字段 : https://www.erdosproblems.com/1206

00435 | EP-1209

### Erdős Problem #1209: Number Theory

O / open | Erdős 问题

Let $A=\{a_1<a_2<\cdots\}$ be a sequence of integers which tends to infinity sufficiently fast. If there is an $n$ such that all $n+a_k$ are primes then must there exist infinitely many such $n$?

What if we ask for $n+a_k$ to be squarefree instead of prime?

Are there $n$ such that $n+2^{2^k}$ is always a prime (or always squarefree, or infinitely often a prime, or infinitely often squarefree)?

来源字段 : https://www.erdosproblems.com/1209

00436 | EP-1210

### Erdős Problem #1210: Number Theory

O / open | Erdős 问题

Let $A\subseteq [1,n)$ be a set of integers such that $(a,b)=1$ for all distinct $a,b\in A$. Is it true that\[\sum_{a\in A}\frac{1}{n-a}\leq \sum_{p<n}\frac{1}{p}+O(1)?\]

来源字段 : https://www.erdosproblems.com/1210

00437 | EP-1211

### Erdős Problem #1211: Number Theory, Ramsey Theory

S / solved | Erdős 问题

Let $\mathbb{N}=A\cup B$ where $A$ and $B$ are disjoint. Let $S(A)$ be those integers which are the sum of finitely many distinct elements of $A$, and similarly for $S(B)$.

If\[\overline{\delta}(X)=\limsup_{x\to \infty}\frac{1}{\log x}\sum_{\substack{n\in X\\ n<x}}\frac{1}{n}\]is the upper logarithmic density of $X$ then how large must\[\max(\overline{\delta}(S(A)),\overline{\delta}(S(B)))\]be?

来源字段 : https://www.erdosproblems.com/1211

00438 | EP-1212

### Erdős Problem #1212: Number Theory, Primes

O / open | Erdős 问题

Let $G$ be the graph with vertex set those pairs $(x,y)\in \mathbb{N}^2$ with $\mathrm{gcd}(x,y)=1$, in which we join two vertices if the differ in only one coordinate, and there by $\pm 1$.

Is there a path going to infinity on $G$, say $P$, such that for all $(x,y)\in P$ both $\min(x,y)>1$ and at least one of $x$ or $y$ is composite?

来源字段 : https://www.erdosproblems.com/1212

00439 | EP-1214

### Erdős Problem #1214: Number Theory

S / solved | Erdős 问题

Let $x,y\geq 1$ be integers such that, for all $n\geq 1$, the set of primes dividing $x^{n}-1$ is equal to set of primes dividing $y^n-1$. Must $x=y$?

来源字段 : https://www.erdosproblems.com/1214

00440 | EP-1217

### Erdős Problem #1217:

S / solved | Erdős 问题

Let $A=\{a_1<a_2<\cdots\}$ be an infinite sequence of positive integers with positive lower logarithmic density.

Must there exist a sequence $n_1<\cdots$ such that\[a_{n_i}\mid a_{n_{i+1}}\]for all $i\geq 1$ and\[\limsup_{x\to \infty}\frac{1}{\log\log x}\sum_{a_{n_i}<x}1 \geq \limsup_{x\to \infty} \frac{1}{\log\log x}\sum_{a_n<x}\frac{1}{a_n\log a_n}?\]

来源字段 : https://www.erdosproblems.com/1217

00441 | EP-121

### Erdős Problem #121: Number Theory, Squares

S / solved | Erdős 问题

Let $F_{k}(N)$ be the size of the largest $A\subseteq \{1,\ldots,N\}$ such that the product of no $k$ many distinct elements of $A$ is a square. Is $F_5(N)=(1-o(1))N$? More generally, is $F_{2k+1}(N)=(1-o(1))N$?

来源字段 : https://www.erdosproblems.com/121

00442 | EP-122

### Erdős Problem #122

P / partially_solved | Erdős 问题

For which number theoretic functions $f$ is it true that, for any $F(n)$ such that $F(n)/f(n)\to 0$ for almost all $n$, there are infinitely many $x$ such that\[\frac{\#\{ n\in \mathbb{N}: n+f(n)\in (x,x+F(x))\}}{F(x)}\to \infty?\]

来源字段 : https://www.erdosproblems.com/122

00443 | EP-123

### Erdős Problem #123

S / solved | Erdős 问题

Let $a,b,c> 1$ be three integers which are pairwise coprime. Is every large integer the sum of distinct integers of the form $a^kb^lc^m$ ($k,l,m\geq 0$), none of which divide any other?

来源字段 : https://www.erdosproblems.com/123

00444 | EP-124

### Erdős Problem #124

P / partially_solved | Erdős 问题

For any $d\geq 1$ and $k\geq 0$ let $P(d,k)$ be the set of integers which are the sum of distinct powers $d^i$ with $i\geq k$. Let $3\leq d_1<d_2<\cdots <d_r$ be integers such that\[\sum_{1\leq i\leq r}\frac{1}{d_r-1}\geq 1.\]Can all sufficiently large integers be written as a sum of the shape $\sum_i c_ia_i$ where $c_i\in \{0,1\}$ and $a_i\in P(d_i,0)$?

If we further have $\mathrm{gcd}(d_1,\ldots,d_r)=1$ then, for any $k\geq 1$, can all sufficiently large integers be written as a sum of the shape $\sum_i c_ia_i$ where $c_i\in \{0,1\}$ and $a_i\in P(d_i,k)$?

来源字段 : https://www.erdosproblems.com/124

00445 | EP-125

### Erdős Problem #125

S / solved | Erdős 问题

Let $A = \{ \sum\epsilon_k3^k: \epsilon_k\in \{0,1\}\}$ be the set of integers which have only the digits $0,1$ when written base $3$, and $B=\{ \sum\epsilon_k4^k: \epsilon_k\in \{0,1\}\}$ be the set of integers which have only the digits $0,1$ when written base $4$.

Does $A+B$ have positive lower density?

来源字段 : https://www.erdosproblems.com/125

00446 | EP-126

### Erdős Problem #126

S / solved | Erdős 问题

Let $f(n)$ be maximal such that if $A\subseteq\mathbb{N}$ has $\lvert A\rvert=n$ then $\prod_{a\neq b\in A}(a+b)$ has at least $f(n)$ distinct prime factors. Is it true that $f(n)/\log n\to\infty$?

来源字段 : https://www.erdosproblems.com/126

00447 | EP-130

### Erdős Problem #130

O / open | Erdős 问题

Let $A\subset\mathbb{R}^2$ be an infinite set which contains no three points on a line and no four points on a circle. Consider the graph with vertices the points in $A$, where two vertices are joined by an edge if and only if they are an integer distance apart.

How large can the chromatic number and clique number of this graph be? In particular, can the chromatic number be infinite?

来源字段 : https://www.erdosproblems.com/130

00448 | EP-137

### Erdős Problem #137

O / open | Erdős 问题

We say that $N$ is powerful if whenever $p\mid N$ we also have $p^2\mid N$. Let $k\geq 3$. Can the product of any $k$ consecutive positive integers ever be powerful?

来源字段 : https://www.erdosproblems.com/137

00449 | EP-138

### Erdős Problem #138

O / open | Erdős 问题

Let the van der Waerden number $W(k)$ be such that whenever $N\geq W(k)$ and $\{1,\ldots,N\}$ is $2$-coloured there must exist a monochromatic $k$-term arithmetic progression. Improve the bounds for $W(k)$ - for example, prove that $W(k)^{1/k}\to \infty$.

来源字段 : https://www.erdosproblems.com/138

00450 | EP-13

### Erdős Problem #13: Number Theory

S / solved | Erdős 问题

Let $A\subseteq \{1,\ldots,N\}$ be such that there are no $a,b,c\in A$ such that $a\mid(b+c)$ and $a<\min(b,c)$. Is it true that $\lvert A\rvert\leq N/3+O(1)$?

来源字段 : https://www.erdosproblems.com/13

00451 | EP-14

### Erdős Problem #14

O / open | Erdős 问题

Let $A\subseteq \mathbb{N}$. Let $B\subseteq \mathbb{N}$ be the set of integers which are representable in exactly one way as the sum of two elements from $A$.

Is it true that for all $\epsilon>0$ and large $N$\[\lvert \{1,\ldots,N\}\backslash B\rvert \gg_\epsilon N^{1/2-\epsilon}?\]Is it possible that\[\lvert \{1,\ldots,N\}\backslash B\rvert =o(N^{1/2})?\]

来源字段：https://www.erdosproblems.com/14

00452 | EP-141

### Erdős Problem #141

O / open | Erdős 问题

别名：Consecutive Primes in Arithmetic Progression [GUY-A6]

Let $k\geq 3$. Are there $k$ consecutive primes in arithmetic progression?

来源字段：https://www.erdosproblems.com/141

00453 | EP-143

### Erdős Problem #143

P / partially_solved | Erdős 问题

Let $A\subset (1,\infty)$ be a countably infinite set such that for all $x\neq y\in A$ and integers $k\geq 1$ we have\[ \lvert kx -y\rvert \geq 1.\]Does this imply that $A$ is sparse? In particular, does this imply that\[\sum_{x\in A}\frac{1}{x\log x}<\infty\]or\[\sum_{\substack{x <n\\ x\in A}}\frac{1}{x}=o(\log n)?\]

来源字段：https://www.erdosproblems.com/143

00454 | EP-144

### Erdős Problem #144: Number Theory, Divisors

S / solved | Erdős 问题

The density of integers which have two divisors $d_1,d_2$ such that $d_1<d_2<2d_1$ exists and is equal to $1$.

来源字段：https://www.erdosproblems.com/144

00455 | EP-146

### Erdős Problem #146

S / solved | Erdős 问题

If $H$ is bipartite and is $r$-degenerate, that is, every induced subgraph of $H$ has minimum degree $\leq r$, then\[\mathrm{ex}(n;H) \ll n^{2-1/r}.\]

来源字段：https://www.erdosproblems.com/146

00456 | EP-148

### Erdős Problem #148

O / open | Erdős 问题

Let $F(k)$ be the number of solutions to\[ 1= \frac{1}{n_1}+\cdots+\frac{1}{n_k},\]where $1\leq n_1<\cdots<n_k$ are distinct integers. Find good estimates for $F(k)$.

来源字段：https://www.erdosproblems.com/148

00457 | EP-15

### Erdős Problem #15

O / open | Erdős 问题

Is it true that\[\sum_{n=1}^\infty(-1)^n\frac{n}{p_n}\]converges, where $p_n$ is the sequence of primes?

来源字段：https://www.erdosproblems.com/15

00458 | EP-164

### Erdős Problem #164: Number Theory, Primitive Sets

S / solved | Erdős 问题

A set $A\subset \mathbb{N}$ is primitive if no member of $A$ divides another. Is the sum\[\sum_{n\in A}\frac{1}{n\log n}\]maximised over all primitive sets when $A$ is the set of primes?

来源字段：https://www.erdosproblems.com/164

00459 | EP-169

### Erdős Problem #169

O / open | Erdős 问题

Let $k\geq 3$ and $f(k)$ be the supremum of $\sum_{n\in A}\frac{1}{n}$ as $A$ ranges over all sets of positive integers which do not contain a $k$-term arithmetic progression. Estimate $f(k)$.

Is\[\lim_{k\to \infty}\frac{f(k)}{\log W(k)}=\infty\]where $W(k)$ is the van der Waerden number?

来源字段：https://www.erdosproblems.com/169

00460 | EP-16

### Erdős Problem #16: Number Theory, Additive Basis

S / solved | Erdős 问题

Is the set of odd integers not of the form $2^k+p$ the union of an infinite arithmetic progression and a set of density $0$?

来源字段：https://www.erdosproblems.com/16

00461 | EP-17

### Erdős Problem #17

O / open | Erdős 问题

Are there infinitely many primes $p$ such that every even number $n\leq p-3$ can be written as a difference of primes $n=q_1-q_2$ where $q_1,q_2\leq p$?

来源字段：https://www.erdosproblems.com/17

00462 | EP-172

### Erdős Problem #172

O / open | Erdős 问题

Is it true that in any finite colouring of $\mathbb{N}$ there exist arbitrarily large finite $A$ such that all sums and products of distinct elements in $A$ are the same colour?

来源字段：https://www.erdosproblems.com/172

00463 | EP-174

### Erdős Problem #174

O / open | Erdős 问题

A finite set $A\subset \mathbb{R}^n$ is called Ramsey if, for any $k\geq 1$, there exists some $d=d(A,k)$ such that in any $k$-colouring of $\mathbb{R}^d$ there exists a monochromatic copy of $A$. Characterise the Ramsey sets in $\mathbb{R}^n$.

来源字段：https://www.erdosproblems.com/174

00464 | EP-175

### Erdős Problem #175: Number Theory, Binomial Coefficients

S / solved | Erdős 问题

Show that, for any $n\geq 5$, the binomial coefficient $\binom{2n}{n}$ is not squarefree.

来源字段：https://www.erdosproblems.com/175

00465 | EP-177

### Erdős Problem #177

O / open | Erdős 问题

Find the smallest $h(d)$ such that the following holds. There exists a function $f:\mathbb{N}\to\{-1,1\}$ such that, for every $d\geq 1$,\[\max_{P_d}\left\lvert \sum_{n\in P_d}f(n)\right\rvert\leq h(d),\]where $P_d$ ranges over all finite arithmetic progressions with common difference $d$.

来源字段：https://www.erdosproblems.com/177

00466 | EP-18

### Erdős Problem #18

O / open | Erdős 问题

We call $m$ practical if every integer $1\leq n<m$ is the sum of distinct divisors of $m$. If $m$ is practical then let $h(m)$ be such that $h(m)$ many divisors always suffice.

Are there infinitely many practical $m$ such that\[h(m) < (\log\log m)^{O(1)}?\]Is it true that $h(n!)<n^{o(1)}$? Or perhaps even $h(n!)<(\log n)^{O(1)}$?

来源字段：https://www.erdosproblems.com/18

00467 | EP-183

### Erdős Problem #183

S / solved | Erdős 问题

Let $R(3;k)$ be the minimal $n$ such that if the edges of $K_n$ are coloured with $k$ colours then there must exist a monochromatic triangle. Determine\[\lim_{k\to \infty}R(3;k)^{1/k}.\]

来源字段：https://www.erdosproblems.com/183

00468 | EP-187

### Erdős Problem #187

O / open | Erdős 问题

Find the best function $f(d)$ such that, in any 2-colouring of the integers, at least one colour class contains an arithmetic progression with common difference $d$ of length $f(d)$ for infinitely many $d$.

来源字段：https://www.erdosproblems.com/187

00469 | EP-188

### Erdős Problem #188

O / open | Erdős 问题

What is the smallest $k$ such that $\mathbb{R}^2$ can be red/blue coloured with no pair of red points unit distance apart, and no $k$-term arithmetic progression of blue points with distance $1$?

来源字段：https://www.erdosproblems.com/188

00470 | EP-195

### Erdős Problem #195

O / open | Erdős 问题

What is the largest $k$ such that in any permutation of $\mathbb{Z}$ there must exist a monotone $k$-term arithmetic progression $x_1<\cdots<x_k$?

来源字段：https://www.erdosproblems.com/195

00471 | EP-197

### Erdős Problem #197

O / open | Erdős 问题

Can $\mathbb{N}$ be partitioned into two sets, each of which can be permuted to avoid monotone 3-term arithmetic progressions?

来源字段：https://www.erdosproblems.com/197

00472 | EP-200

### Erdős Problem #200

O / open | Erdős 问题

Does the longest arithmetic progression of primes in $\{1,\ldots,N\}$ have length $o(\log N)$?

来源字段：https://www.erdosproblems.com/200

00473 | EP-201

### Erdős Problem #201

O / open | Erdős 问题

Let $G_k(N)$ be such that any set of $N$ integers contains a subset of size at least $G_k(N)$ which does not contain a $k$-term arithmetic progression. Determine the size of $G_k(N)$. How does it relate to $R_k(N)$, the size of the largest subset of $\{1,\ldots,N\}$ without a $k$-term arithmetic progression? Is it true that\[\lim_{N\to \infty}\frac{R_3(N)}{G_3(N)}=1?\]

来源字段：https://www.erdosproblems.com/201

00474 | EP-202

### Erdős Problem #202

S / solved | Erdős 问题

Let $n_1<\cdots < n_r\leq N$ with associated $a_i\pmod{n_i}$ such that the congruence classes are disjoint (that is, every integer is $\equiv a_i\pmod{n_i}$ for at most one $1\leq i\leq r$). How large can $r$ be in terms of $N$?

来源字段：https://www.erdosproblems.com/202

00475 | EP-203

### Erdős Problem #203

O / open | Erdős 问题

Is there an integer $m\geq 1$ with $(m,6)=1$ such that none of $2^k3^\ell m+1$ are prime, for any $k,\ell\geq 0$?

来源字段：https://www.erdosproblems.com/203

00476 | EP-204

### Erdős Problem #204: Covering Systems, Divisors

S / solved | Erdős 问题

Are there $n$ such that there is a covering system with moduli the divisors of $n$ which is 'as disjoint as possible'?

That is, for all $d\mid n$ with $d>1$ there is an associated $a_d$ such that every integer is congruent to some $a_d\pmod{d}$, and if there is some integer $x$ with\[x\equiv a_d\pmod{d}\textrm{ and }x\equiv a_{d'}\pmod{d'}\]then $(d,d')=1$.

来源字段：https://www.erdosproblems.com/204

00477 | EP-205

### Erdős Problem #205: Number Theory

S / solved | Erdős 问题

Is it true that all sufficiently large $n$ can be written as $2^k+m$ for some $k\geq 0$, where $\Omega(m)<\log\log m$? (Here $\Omega(m)$ is the number of prime divisors of $m$ counted with multiplicity.) What about $<\epsilon \log\log m$? Or some more slowly growing function?

来源字段：https://www.erdosproblems.com/205

00478 | EP-206

### Erdős Problem #206: Number Theory, Unit Fractions

S / solved | Erdős 问题

Let $x>0$ be a real number. For any $n\geq 1$ let\[R_n(x) = \sum_{i=1}^n\frac{1}{m_i}<x\]be the maximal sum of $n$ distinct unit fractions which is $<x$.

Is it true that, for almost all $x$, for sufficiently large $n$, we have\[R_{n+1}(x)=R_n(x)+\frac{1}{m},\]where $m$ is minimal such that $m$ does not appear in $R_n(x)$ and the right-hand side is $<x$? (That is, are the best underapproximations eventually always constructed in a 'greedy' fashion?)

来源字段：https://www.erdosproblems.com/206

00479 | EP-213

### Erdős Problem #213

O / open | Erdős 问题

Let $n\geq 4$. Are there $n$ points in $\mathbb{R}^2$, no three on a line and no four on a circle, such that all pairwise distances are integers?

来源字段：https://www.erdosproblems.com/213

00480 | EP-218

### Erdős Problem #218

O / open | Erdős 问题

Let $d_n=p_{n+1}-p_n$. The set of $n$ such that $d_{n+1}\geq d_n$ has density $1/2$, and similarly for $d_{n+1}\leq d_n$. Furthermore, there are infinitely many $n$ such that $d_{n+1}=d_n$.

来源字段：https://www.erdosproblems.com/218

00481 | EP-219

### Erdős Problem #219: Are there arbitrarily long arithmetic progressions of primes?

S / solved | Erdős 问题

Are there arbitrarily long arithmetic progressions of primes?

来源字段：https://www.erdosproblems.com/219

00482 | EP-220

### Erdős Problem #220: Number Theory

S / solved | Erdős 问题

Let $n\geq 1$ and\[A=\{a_1<\cdots <a_{\phi(n)}\}=\{ 1\leq m<n: (m,n)=1\}.\]Is it true that\[ \sum_{1\leq k<\phi(n)}(a_{k+1}-a_k)^2 \ll \frac{n^2}{\phi(n)}?\]

来源字段：https://www.erdosproblems.com/220

00483 | EP-221

### Erdős Problem #221: Number Theory, Additive Basis

S / solved | Erdős 问题

Is there a set $A\subset\mathbb{N}$ such that, for all large $N$,\[\lvert A\cap\{1,\ldots,N\}\rvert \ll N/\log N\]and such that every large integer can be written as $2^k+a$ for some $k\geq 0$ and $a\in A$?

来源字段：https://www.erdosproblems.com/221

00484 | EP-222

### Erdős Problem #222

O / open | Erdős 问题

Let $n_1<n_2<\cdots$ be the sequence of integers which are the sum of two squares. Explore the behaviour of (i.e. find good upper and lower bounds for) the consecutive differences $n_{k+1}-n_k$.

来源字段：https://www.erdosproblems.com/222

00485 | EP-233

### Erdős Problem #233

O / open | Erdős 问题

Let $d_n=p_{n+1}-p_n$, where $p_n$ is the $n$th prime. Prove that\[\sum_{1\leq n\leq N}d_n^2 \ll N(\log N)^2.\]

来源字段：https://www.erdosproblems.com/233

00486 | EP-234

### Erdős Problem #234

O / open | Erdős 问题

For every $c\geq 0$ the density $f(c)$ of integers for which\[\frac{p_{n+1}-p_n}{\log n}< c\]exists and is a continuous function of $c$.

来源字段：https://www.erdosproblems.com/234

00487 | EP-235

### Erdős Problem #235: Number Theory

S / solved | Erdős 问题

Let $N_k=2\cdot 3\cdots p_k$ and $\{a_1<a_2<\cdots <a_{\phi(N_k)}\}$ be the integers $<N_k$ which are relatively prime to $N_k$. Then, for any $c\geq 0$, the limit\[\frac{\#\{ a_i-a_{i-1}\leq c \frac{N_k}{\phi(N_k)}: 2\leq i\leq \phi(N_k)\}}{\phi(N_k)}\]exists and is a continuous function of $c$.

来源字段：https://www.erdosproblems.com/235

00488 | EP-236

### Erdős Problem #236

O / open | Erdős 问题

Let $f(n)$ count the number of solutions to $n=p+2^k$ for prime $p$ and $k\geq 0$. Is it true that $f(n)=o(\log n)$?

来源字段：https://www.erdosproblems.com/236

00489 | EP-237

### Erdős Problem #237: Number Theory, Primes

S / solved | Erdős 问题

Let $A\subseteq \mathbb{N}$ be a set such that $\lvert A\cap \{1,\ldots,N\}\rvert \gg \log N$ for all large $N$. Let $f(n)$ count the number of solutions to $n=p+a$ for $p$ prime and $a\in A$. Is it true that $\limsup f(n)=\infty$?

来源字段：https://www.erdosproblems.com/237

00490 | EP-238

### Erdős Problem #238

O / open | Erdős 问题

Let $c_1,c_2>0$. Is it true that, for any sufficiently large $x$, there exist more than $c_1\log x$ many consecutive primes $\leq x$ such that the difference between any two is $>c_2$?

来源字段：https://www.erdosproblems.com/238

00491 | EP-239

### Erdős Problem #239: Is it true that\[ \lim_{N\to \infty}\frac{1}{N}\sum_{n\leq N}f(n)\]always exists?

S / solved | Erdős 问题

Let $f:\mathbb{N}\to \{-1,1\}$ be a multiplicative function. Is it true that\[ \lim_{N\to \infty}\frac{1}{N}\sum_{n\leq N}f(n)\]always exists?

来源字段：https://www.erdosproblems.com/239

00492 | EP-240

### Erdős Problem #240: Number Theory, Primes

S / solved | Erdős 问题

Is there an infinite set of primes $P$ such that if $\{a_1<a_2<\cdots\}$ is the set of integers divisible only by primes in $P$ then $\lim a_{i+1}-a_i=\infty$?

来源字段：https://www.erdosproblems.com/240

00493 | EP-242

### Erdős Problem #242: Number Theory, Unit Fractions

O / open | Erdős 问题

别名：The Erdős-Straus Conjecture [NT-012]; Erdős-Straus Conjecture [NT-024]; Erdős-Straus conjecture [OPG-37397]

For every $n>2$ there exist distinct integers $1\leq x<y<z$ such that\[\frac{4}{n} = \frac{1}{x}+\frac{1}{y}+\frac{1}{z}.\]

来源字段：https://www.erdosproblems.com/242

00494 | EP-243

### Erdős Problem #243

O / open | Erdős 问题

Let $1\leq a_1<a_2<\cdots$ be a sequence of integers such that\[\lim_{n\to \infty}\frac{a_n}{a_{n-1}^2}=1\]and $\sum\frac{1}{a_n}\in \mathbb{Q}$. Then, for all sufficiently large $n\geq 1$,\[ a_n = a_{n-1}^2-a_{n-1}+1.\]

来源字段：https://www.erdosproblems.com/243

00495 | EP-244

### Erdős Problem #244

P / partially_solved | Erdős 问题

Let $C>1$. Does the set of integers of the form $p+\lfloor C^k\rfloor$, for some prime $p$ and $k\geq 0$, have density $>0$?

来源字段：https://www.erdosproblems.com/244

00496 | EP-246

### Erdős Problem #246: Number Theory

S / solved | Erdős 问题

Let $(a,b)=1$. The set $\{a^kb^l: k,l\geq 0\}$ is complete - that is, every large integer is the sum of distinct integers of the form $a^kb^l$ with $k,l\geq 0$.

来源字段：https://www.erdosproblems.com/246

00497 | EP-247

### Erdős Problem #247

O / open | Erdős 问题

Let $1\leq a_1<a_2<\cdots$ be a sequence of integers such that\[\limsup \frac{a_n}{n}=\infty.\]Is\[\sum_{n=1}^\infty \frac{1}{2^{a_n}}\]transcendental?

来源字段：https://www.erdosproblems.com/247

00498 | EP-248

### Erdős Problem #248: Number Theory

S / solved | Erdős 问题

Are there infinitely many $n$ such that, for all $k\geq 1$,\[ \omega(n+k) \ll k?\](Here $\omega(n)$ is the number of distinct prime divisors of $n$.)

来源字段：https://www.erdosproblems.com/248

00499 | EP-249

### Erdős Problem #249

O / open | Erdős 问题

Is\[\sum_n \frac{\phi(n)}{2^n}\]irrational? Here $\phi$ is the Euler totient function.

来源字段：https://www.erdosproblems.com/249

00500 | EP-25

### Erdős Problem #25

O / open | Erdős 问题

Let $1\leq n_1<n_2<\cdots$ be an arbitrary sequence of integers, each with an associated residue class $a_i\pmod{n_i}$. Let $A$ be the set of integers $n$ such that for every $i$ either $n<n_i$ or $n\not\equiv a_i\pmod{n_i}$. Must the logarithmic density of $A$ exist?

来源字段：https://www.erdosproblems.com/25

00501 | EP-250

### Erdős Problem #250: Number Theory, Irrationality

S / solved | Erdős 问题

Is\[\sum \frac{\sigma(n)}{2^n}\]irrational? (Here $\sigma(n)$ is the sum of divisors function.)

来源字段：https://www.erdosproblems.com/250

00502 | EP-251

### Erdős Problem #251

O / open | Erdős 问题

Is\[\sum \frac{p_n}{2^n}\]irrational? (Here $p_n$ is the $n$th prime.)

来源字段：https://www.erdosproblems.com/251

00503 | EP-252

### Erdős Problem #252

P / partially_solved | Erdős 问题

Let $k\geq 1$ and $\sigma_k(n)=\sum_{d\mid n}d^k$. Is\[\sum \frac{\sigma_k(n)}{n!}\]irrational?

来源字段：https://www.erdosproblems.com/252

00504 | EP-253

### Erdős Problem #253: Number Theory

S / solved | Erdős 问题

Let $1\leq a_1<a_2<\cdots $ be an infinite sequence of integers such that $a_{i+1}/a_i\to 1$. If every infinite arithmetic progression contains infinitely many integers which are the sum of distinct $a_i$ then every sufficiently large integer is the sum of distinct $a_i$.

来源字段：https://www.erdosproblems.com/253

00505 | EP-254

### Erdős Problem #254

O / open | Erdős 问题

Let $A\subseteq \mathbb{N}$ be such that\[\lvert A\cap [1,2x]\rvert -\lvert A\cap [1,x]\rvert \to \infty\textrm{ as }x\to \infty\]and\[\sum_{n\in A} \{ \theta n\}=\infty\]for every $\theta\in (0,1)$, where $\{x\}$ is the distance of $x$ from the nearest integer. Then every sufficiently large integer is the sum of distinct elements of $A$.

来源字段：https://www.erdosproblems.com/254

00506 | EP-256

### Erdős Problem #256

P / partially_solved | Erdős 问题

Let $n\geq 1$ and $f(n)$ be maximal such that for any integers $1\leq a_1\leq \cdots \leq a_n$ we have\[\max_{\lvert z\rvert=1}\left\lvert \prod_{i}(1-z^{a_i})\right\rvert\geq f(n).\]Estimate $f(n)$ - in particular, is it true that there exists some constant $c>0$ such that\[\log f(n) \gg n^c?\]

来源字段：https://www.erdosproblems.com/256

00507 | EP-257

### Erdős Problem #257

O / open | Erdős 问题

Let $A\subseteq \mathbb{N}$ be an infinite set. Is\[\sum_{n\in A}\frac{1}{2^n-1}\]irrational?

来源字段：https://www.erdosproblems.com/257

00508 | EP-258

### Erdős Problem #258

S / solved | Erdős 问题

Let $a_1,a_2,\ldots$ be a sequence of positive integers with $a_n\to \infty$. Is\[\sum_{n} \frac{\tau(n)}{a_1\cdots a_n}\]irrational, where $\tau(n)$ is the number of divisors of $n$?

来源字段：https://www.erdosproblems.com/258

00509 | EP-259

### Erdős Problem #259: Is the sum\[\sum_{n} \mu(n)^2\frac{n}{2^n}\]irrational?

S / solved | Erdős 问题

Is the sum\[\sum_{n} \mu(n)^2\frac{n}{2^n}\]irrational?

来源字段：https://www.erdosproblems.com/259

00510 | EP-261

### Erdős Problem #261

P / partially_solved | Erdős 问题

Are there infinitely many $n$ such that there exists some $t\geq 2$ and distinct integers $a_1,\ldots,a_t\geq 1$ such that\[\frac{n}{2^n}=\sum_{1\leq k\leq t}\frac{a_k}{2^{a_k}}?\]Is this true for all $n$? Is there a rational $x$ such that\[x = \sum_{k=1}^\infty \frac{a_k}{2^{a_k}}\]has at least $2^{\aleph_0}$ solutions?

来源字段：https://www.erdosproblems.com/261

00511 | EP-262

### Erdős Problem #262: Irrationality

S / solved | Erdős 问题

Suppose $a_1<a_2<\cdots$ is a sequence of integers such that for all integer sequences $t_n$ with $t_n\geq 1$ the sum\[\sum_{n=1}^\infty \frac{1}{t_na_n}\]is irrational. How slowly can $a_n$ grow?

来源字段：https://www.erdosproblems.com/262

00512 | EP-263

### Erdős Problem #263

P / partially_solved | Erdős 问题

Let $a_n$ be an increasing sequence of positive integers such that for every sequence of positive integers $b_n$ with $b_n/a_n\to 1$ the sum\[\sum\frac{1}{b_n}\]is irrational. Is $a_n=2^{2^n}$ such a sequence? Must such a sequence satisfy $a_n^{1/n}\to \infty$?

来源字段：https://www.erdosproblems.com/263

00513 | EP-264

### Erdős Problem #264

P / partially_solved | Erdős 问题

Let $a_n$ be a sequence of positive integers such that for every bounded sequence of integers $b_n$ (with $a_n+b_n\neq 0$ and $b_n\neq 0$ for all $n$) the sum\[\sum \frac{1}{a_n+b_n}\]is irrational. Are $a_n=2^n$ or $a_n=n!$ examples of such a sequence?

来源字段：https://www.erdosproblems.com/264

00514 | EP-265

### Erdős Problem #265

P / partially_solved | Erdős 问题

Let $1\leq a_1<a_2<\cdots$ be an increasing sequence of integers. How fast can $a_n\to \infty$ grow if\[\sum\frac{1}{a_n}\quad\textrm{and}\quad\sum\frac{1}{a_n-1}\]are both rational?

来源字段：https://www.erdosproblems.com/265

00515 | EP-266

### Erdős Problem #266: There exists some integer $t\geq 1$ such that\[\sum \frac{1}{a_n+t}\]is irrational

S / solved | Erdős 问题

Let $a_n$ be an infinite sequence of positive integers such that $\sum \frac{1}{a_n}$ converges. There exists some integer $t\geq 1$ such that\[\sum \frac{1}{a_n+t}\]is irrational.

来源字段：https://www.erdosproblems.com/266

00516 | EP-267

### Erdős Problem #267

O / open | Erdős 问题

Let $F_1=F_2=1$ and $F_{n+1}=F_n+F_{n-1}$ be the Fibonacci sequence. Let $n_1<n_2<\cdots $ be an infinite sequence with $n_{k+1}/n_k \geq c>1$. Must\[\sum_k\frac{1}{F_{n_k}}\]be irrational?

来源字段：https://www.erdosproblems.com/267

00517 | EP-268

### Erdős Problem #268: Number Theory

S / solved | Erdős 问题

Let $X\subseteq \mathbb{R}^3$ be the set of all points of the shape\[\left( \sum_{n\in A} \frac{1}{n},\sum_{n\in A}\frac{1}{n+1},\sum_{n\in A} \frac{1}{n+2}\right) \]as $A\subseteq\mathbb{N}$ ranges over all infinite sets with $\sum_{n\in A}\frac{1}{n}<\infty$. Does $X$ contain an open set?

来源字段：https://www.erdosproblems.com/268

00518 | EP-269

### Erdős Problem #269

O / open | Erdős 问题

Let $P$ be a finite set of primes with $\lvert P\rvert \geq 2$ and let $\{a_1<a_2<\cdots\}=\{ n\in \mathbb{N}: \textrm{if }p\mid n\textrm{ then }p\in P\}$. Is the sum\[\sum_{n=1}^\infty \frac{1}{[a_1,\ldots,a_n]},\]where $[a_1,\ldots,a_n]$ is the lowest common multiple of $a_1,\ldots,a_n$, irrational?

来源字段：https://www.erdosproblems.com/269

00519 | EP-26

### Erdős Problem #26: Number Theory, Divisors

S / solved | Erdős 问题

Let $A\subset\mathbb{N}$ be infinite. Must there exist some $k\geq 1$ such that almost all integers have a divisor of the form $a+k$ for some $a\in A$?

来源字段：https://www.erdosproblems.com/26

00520 | EP-270

### Erdős Problem #270: Irrationality

S / solved | Erdős 问题

Let $f(n)\to \infty$ as $n\to \infty$. Is it true that\[\sum_{n\geq 1} \frac{1}{(n+1)\cdots (n+f(n))}\]is irrational?

来源字段：https://www.erdosproblems.com/270

00521 | EP-271

### Erdős Problem #271

P / partially_solved | Erdős 问题

Let $A(n)=\{a_0<a_1<\cdots\}$ be the sequence defined by $a_0=0$ and $a_1=n$, and for $k\geq 1$ define $a_{k+1}$ as the least positive integer such that there is no three-term arithmetic progression in $\{a_0,\ldots,a_{k+1}\}$.

Can the $a_k$ be explicitly determined? How fast do they grow?

来源字段：https://www.erdosproblems.com/271

00522 | EP-272

### Erdős Problem #272

P / partially_solved | Erdős 问题

Let $N\geq 1$. What is the largest $t$ such that there are $A_1,\ldots,A_t\subseteq \{1,\ldots,N\}$ with $A_i\cap A_j$ a non-empty arithmetic progression for all $i\neq j$?

来源字段：https://www.erdosproblems.com/272

00523 | EP-273

### Erdős Problem #273

O / open | Erdős 问题

Is there a covering system all of whose moduli are of the form $p-1$ for some primes $p\geq 5$?

来源字段：https://www.erdosproblems.com/273

00524 | EP-274

### Erdős Problem #274

P / partially_solved | Erdős 问题

If $G$ is a group then can there exist an exact covering of $G$ by more than one cosets of different sizes? (i.e. each element is contained in exactly one of the cosets)

来源字段：https://www.erdosproblems.com/274

00525 | EP-275

### Erdős Problem #275: Number Theory, Covering Systems

S / solved | Erdős 问题

If a finite system of $r$ congruences $\{ a_i\pmod{n_i}: 1\leq i\leq r\}$ (the $n_i$ are not necessarily distinct) covers $2^r$ consecutive integers then it covers all integers.

来源字段：https://www.erdosproblems.com/275

00526 | EP-276

### Erdős Problem #276

P / partially_solved | Erdős 问题

Is there an infinite Lucas sequence $a_0,a_1,\ldots$ where $a_{n+2}=a_{n+1}+a_n$ for $n\geq 0$ such that all $a_k$ are composite, and yet no integer has a common factor with every term of the sequence?

来源字段：https://www.erdosproblems.com/276

00527 | EP-277

### Erdős Problem #277: Number Theory, Covering Systems

S / solved | Erdős 问题

Is it true that, for every $c$, there exists an $n$ such that $\sigma(n)>cn$ but there is no covering system whose moduli all distinct divisors of $n$ (which are $>1$)?

来源字段：https://www.erdosproblems.com/277

00528 | EP-278

### Erdős Problem #278

P / partially_solved | Erdős 问题

Let $A=\{n_1<\cdots<n_r\}$ be a finite set of positive integers. What is the maximum density of integers covered by a suitable choice of congruences $a_i\pmod{n_i}$?

Is the minimum density achieved when all the $a_i$ are equal?

来源字段：https://www.erdosproblems.com/278

00529 | EP-279

### Erdős Problem #279

O / open | Erdős 问题

Let $k\geq 3$. Is there a choice of congruence classes $a_p\pmod{p}$ for every prime $p$ such that all sufficiently large integers can be written as $a_p+tp$ for some prime $p$ and integer $t\geq k$?

来源字段：https://www.erdosproblems.com/279

00530 | EP-27

### Erdős Problem #27: Number Theory, Covering Systems

S / solved | Erdős 问题

An $\epsilon$-almost covering system is a set of congruences $a_i\pmod{n_i}$ for distinct moduli $n_1<\cdots<n_k$ such that the density of those integers which satisfy none of them is $\leq \epsilon$.

Is there a constant $C>1$ such that for every $\epsilon>0$ and $N\geq 1$ there is an $\epsilon$-almost covering system with $N\leq n_1<\cdots <n_k\leq CN$?

来源字段：https://www.erdosproblems.com/27

00531 | EP-28

### Erdős Problem #28

O / open | Erdős 问题

If $A\subseteq \mathbb{N}$ is such that $A+A$ contains all but finitely many integers then $\limsup 1_A\ast 1_A(n)=\infty$.

来源字段：https://www.erdosproblems.com/28

00532 | EP-280

### Erdős Problem #280: Number Theory, Covering Systems

S / solved | Erdős 问题

Let $n_1<n_2<\cdots $ be an infinite sequence of integers with associated $a_k\pmod{n_k}$, such that for some $\epsilon>0$ we have $n_k>(1+\epsilon)k\log k$ for all $k$. Then\[\#\{ m<n_k: m\not\equiv a_i\pmod{n_i} \textrm{ for }1\leq i\leq k\}\neq o(k).\]

来源字段：https://www.erdosproblems.com/280

00533 | EP-281

### Erdős Problem #281

S / solved | Erdős 问题

Let $n_1<n_2<\cdots$ be an infinite sequence such that, for any choice of congruence classes $a_i\pmod{n_i}$, the set of integers not satisfying any of the congruences $a_i\pmod{n_i}$ has density $0$.

Is it true that for every $\epsilon>0$ there exists some $k$ such that, for every choice of congruence classes $a_i$, the density of integers not satisfying any of the congruences $a_i\pmod{n_i}$ for $1\leq i\leq k$ is less than $\epsilon$?

来源字段：https://www.erdosproblems.com/281

00534 | EP-282

### Erdős Problem #282

O / open | Erdős 问题

Let $A\subseteq \mathbb{N}$ be an infinite set and consider the following greedy algorithm for a rational $x\in (0,1)$: choose the minimal $n\in A$ such that $n\geq 1/x$ and repeat with $x$ replaced by $x-\frac{1}{n}$. If this terminates after finitely many steps then this produces a representation of $x$ as the sum of distinct unit fractions with denominators from $A$.

Does this process always terminate if $x$ has odd denominator and $A$ is the set of odd numbers? More generally, for which pairs $x$ and $A$ does this process terminate?

来源字段：https://www.erdosproblems.com/282

00535 | EP-283

### Erdős Problem #283

S / solved | Erdős 问题

Let $p:\mathbb{Z}\to \mathbb{Z}$ be a polynomial whose leading coefficient is positive and such that there exists no $d\geq 2$ with $d\mid p(n)$ for all $n\geq 1$. Is it true that, for all sufficiently large $m$, there exist integers $1\leq n_1<\cdots <n_k$ such that\[1=\frac{1}{n_1}+\cdots+\frac{1}{n_k}\]and\[m=p(n_1)+\cdots+p(n_k)?\]

来源字段：https://www.erdosproblems.com/283

00536 | EP-284

### Erdős Problem #284: \]Is it true that\[f(k)=(1+o(1))\frac{k}{e-1}?\]

S / solved | Erdős 问题

Let $f(k)$ be the maximal value of $n_1$ such that there exist $n_1<n_2<\cdots <n_k$ with\[1=\frac{1}{n_1}+\cdots+\frac{1}{n_k}.\]Is it true that\[f(k)=(1+o(1))\frac{k}{e-1}?\]

来源字段：https://www.erdosproblems.com/284

00537 | EP-285

### Erdős Problem #285: \]Is it true that\[f(k)=(1+o(1))\frac{e}{e-1}k?\]

S / solved | Erdős 问题

Let $f(k)$ be the minimal value of $n_k$ such that there exist $n_1<n_2<\cdots <n_k$ with\[1=\frac{1}{n_1}+\cdots+\frac{1}{n_k}.\]Is it true that\[f(k)=(1+o(1))\frac{e}{e-1}k?\]

来源字段：https://www.erdosproblems.com/285

00538 | EP-286

### Erdős Problem #286: Number Theory, Unit Fractions

S / solved | Erdős 问题

Let $k\geq 2$. Is it true that there exists an interval $I$ of width $(e-1+o(1))k$ and integers $n_1<\cdots<n_k\in I$ such that\[1=\frac{1}{n_1}+\cdots+\frac{1}{n_k}?\]

来源字段：https://www.erdosproblems.com/286

00539 | EP-287

### Erdős Problem #287: Number Theory, Unit Fractions

O / open | Erdős 问题

Let $k\geq 2$. Is it true that, for any distinct integers $1<n_1<\cdots <n_k$ such that\[1=\frac{1}{n_1}+\cdots+\frac{1}{n_k}\]we must have $\max(n_{i+1}-n_i)\geq 3$?

来源字段：https://www.erdosproblems.com/287

00540 | EP-290

### Erdős Problem #290: Number Theory, Unit Fractions

S / solved | Erdős 问题

Let $a\geq 1$. Must there exist some $b>a$ such that\[\sum_{a\leq n\leq b}\frac{1}{n}=\frac{r_1}{s_1}\textrm{ and }\sum_{a\leq n\leq b+1}\frac{1}{n}=\frac{r_2}{s_2},\]with $(r_i,s_i)=1$ and $s_2<s_1$? If so, how does this $b(a)$ grow with $a$?

来源字段：https://www.erdosproblems.com/290

00541 | EP-291

### Erdős Problem #291

P / partially_solved | Erdős 问题

Let $n\geq 1$ and define $L_n$ to be the least common multiple of $\{1,\ldots,n\}$ and $a_n$ by\[\sum_{1\leq k\leq n}\frac{1}{k}=\frac{a_n}{L_n}.\]Is it true that $(a_n,L_n)=1$ and $(a_n,L_n)>1$ both occur for infinitely many $n$?

来源字段：https://www.erdosproblems.com/291

00542 | EP-292

### Erdős Problem #292: Explore $A$. In particular, does $A$ have density $1$?

S / solved | Erdős 问题

Let $A$ be the set of $n\in \mathbb{N}$ such that there exist $1\leq m_1<\cdots <m_k=n$ with $\sum\tfrac{1}{m_i}=1$. Explore $A$. In particular, does $A$ have density $1$?

来源字段：https://www.erdosproblems.com/292

00543 | EP-293

### Erdős Problem #293

O / open | Erdős 问题

Let $k\geq 1$ and let $v(k)$ be the minimal integer which does not appear as some $n_i$ in a solution to\[1=\frac{1}{n_1}+\cdots+\frac{1}{n_k}\]with $1\leq n_1<\cdots <n_k$. Estimate the growth of $v(k)$.

来源字段：https://www.erdosproblems.com/293

00544 | EP-294

### Erdős Problem #294: Number Theory, Unit Fractions

S / solved | Erdős 问题

Let $N\geq 1$ and let $t(N)$ be the least integer $t$ such that there is no solution to\[1=\frac{1}{n_1}+\cdots+\frac{1}{n_k}\]with $t=n_1<\cdots <n_k\leq N$. Estimate $t(N)$.

来源字段：https://www.erdosproblems.com/294

00545 | EP-296

### Erdős Problem #296: Number Theory, Unit Fractions

S / solved | Erdős 问题

Let $N\geq 1$ and let $k(N)$ be maximal such that there are $k$ disjoint $A_1,\ldots,A_k\subseteq \{1,\ldots,N\}$ with $\sum_{n\in A_i}\frac{1}{n}=1$ for all $i$. Estimate $k(N)$. Is it true that $k(N)=o(\log N)$?

来源字段：https://www.erdosproblems.com/296

00546 | EP-297

### Erdős Problem #297: Number Theory, Unit Fractions

S / solved | Erdős 问题

Let $N\geq 1$. How many $A\subseteq \{1,\ldots,N\}$ are there such that $\sum_{n\in A}\frac{1}{n}=1$?

来源字段：https://www.erdosproblems.com/297

00547 | EP-298

### Erdős Problem #298: Number Theory, Unit Fractions

S / solved | Erdős 问题

Does every set $A\subseteq \mathbb{N}$ of positive density contain some finite $S\subset A$ such that $\sum_{n\in S}\frac{1}{n}=1$?

来源字段：https://www.erdosproblems.com/298

00548 | EP-299

### Erdős Problem #299: Number Theory, Unit Fractions

S / solved | Erdős 问题

Is there an infinite sequence $a_1<a_2<\cdots $ such that $a_{i+1}-a_i=O(1)$ and no finite sum of $\frac{1}{a_i}$ is equal to $1$?

来源字段：https://www.erdosproblems.com/299

00549 | EP-29

### Erdős Problem #29: Number Theory, Additive Basis

S / solved | Erdős 问题

Is there an explicit construction of a set $A\subseteq \mathbb{N}$ such that $A+A=\mathbb{N}$ but $1_A\ast 1_A(n)=o(n^\epsilon)$ for every $\epsilon>0$?

来源字段：https://www.erdosproblems.com/29

00550 | EP-2

### Erdős Problem #2: Can the smallest modulus of a covering system be arbitrarily large?

S / solved | Erdős 问题

Can the smallest modulus of a covering system be arbitrarily large?

来源字段：https://www.erdosproblems.com/2

00551 | EP-3

### Erdős Problem #3

O / open | Erdős 问题

别名：Erdős $3000 Conjecture on Arithmetic Progressions [GUY-A5b]

If $A\subseteq \mathbb{N}$ has $\sum_{n\in A}\frac{1}{n}=\infty$ then must $A$ contain arbitrarily long arithmetic progressions?

来源字段：https://www.erdosproblems.com/3

00552 | EP-30

### Erdős Problem #30

O / open | Erdős 问题

Let $h(N)$ be the maximum size of a Sidon set in $\{1,\ldots,N\}$. Is it true that, for every $\epsilon>0$,\[h(N) = N^{1/2}+O_\epsilon(N^\epsilon)?\]

来源字段：https://www.erdosproblems.com/30

00553 | EP-300

### Erdős Problem #300: Number Theory, Unit Fractions

S / solved | Erdős 问题

Let $A(N)$ denote the maximal cardinality of $A\subseteq \{1,\ldots,N\}$ such that $\sum_{n\in S}\frac{1}{n}\neq 1$ for all $S\subseteq A$. Estimate $A(N)$.

来源字段：https://www.erdosproblems.com/300

00554 | EP-301

### Erdős Problem #301

O / open | Erdős 问题

Let $f(N)$ be the size of the largest $A\subseteq \{1,\ldots,N\}$ such that there are no solutions to\[\frac{1}{a}= \frac{1}{b_1}+\cdots+\frac{1}{b_k}\]with distinct $a,b_1,\ldots,b_k\in A$?

Estimate $f(N)$. In particular, is it true that $f(N)=(\tfrac{1}{2}+o(1))N$?

来源字段：https://www.erdosproblems.com/301

00555 | EP-302

### Erdős Problem #302

O / open | Erdős 问题

Let $f(N)$ be the size of the largest $A\subseteq \{1,\ldots,N\}$ such that there are no solutions to\[\frac{1}{a}= \frac{1}{b}+\frac{1}{c}\]with distinct $a,b,c\in A$?

Estimate $f(N)$. In particular, is $f(N)=(\tfrac{1}{2}+o(1))N$?

来源字段：https://www.erdosproblems.com/302

00556 | EP-303

### Erdős Problem #303: Number Theory, Unit Fractions

S / solved | Erdős 问题

Is it true that in any finite colouring of the integers there exists a monochromatic solution to\[\frac{1}{a}=\frac{1}{b}+\frac{1}{c}\]with distinct $a,b,c$?

来源字段：https://www.erdosproblems.com/303

00557 | EP-304

### Erdős Problem #304

O / open | Erdős 问题

For integers $1\leq a<b$ let $N(a,b)$ denote the minimal $k$ such that there exist integers $1<n_1<\cdots<n_k$ with\[\frac{a}{b}=\frac{1}{n_1}+\cdots+\frac{1}{n_k}.\]Estimate $N(b)=\max_{1\leq a<b}N(a,b)$. Is it true that $N(b) \ll \log\log b$?

来源字段：https://www.erdosproblems.com/304

00558 | EP-305

### Erdős Problem #305: Number Theory, Unit Fractions

S / solved | Erdős 问题

For integers $1\leq a<b$ let $D(a,b)$ be the minimal value of $n_k$ such that there exist integers $1\leq n_1<\cdots<n_k$ with\[\frac{a}{b}=\frac{1}{n_1}+\cdots+\frac{1}{n_k}.\]Estimate $D(b)=\max_{1\leq a<b}D(a,b)$. Is it true that\[D(b) \ll b(\log b)^{1+o(1)}?\]

来源字段：https://www.erdosproblems.com/305

00559 | EP-306

### Erdős Problem #306

O / open | Erdős 问题

Let $a/b\in \mathbb{Q}_{>0}$ with $b$ squarefree. Are there integers $1<n_1<\cdots<n_k$, each the product of two distinct primes, such that\[\frac{a}{b}=\frac{1}{n_1}+\cdots+\frac{1}{n_k}?\]

来源字段：https://www.erdosproblems.com/306

00560 | EP-307

### Erdős Problem #307: Number Theory, Unit Fractions

O / open | Erdős 问题

Are there two finite sets of primes $P,Q$ such that\[1=\left(\sum_{p\in P}\frac{1}{p}\right)\left(\sum_{q\in Q}\frac{1}{q}\right)?\]

来源字段：https://www.erdosproblems.com/307

00561 | EP-308

### Erdős Problem #308: Number Theory, Unit Fractions

S / solved | Erdős 问题

Let $N\geq 1$. What is the smallest integer not representable as the sum of distinct unit fractions with denominators from $\{1,\ldots,N\}$? Is it true that the set of integers representable as such has the shape $\{1,\ldots,m\}$ for some $m$?

来源字段：https://www.erdosproblems.com/308

00562 | EP-309

### Erdős Problem #309: Number Theory, Unit Fractions

S / solved | Erdős 问题

Let $N\geq 1$. How many integers can be written as the sum of distinct unit fractions with denominators from $\{1,\ldots,N\}$? Are there $o(\log N)$ such integers?

来源字段：https://www.erdosproblems.com/309

00563 | EP-310

### Erdős Problem #310: Number Theory, Unit Fractions

S / solved | Erdős 问题

Let $\alpha >0$ and $N\geq 1$. Is it true that for any $A\subseteq \{1,\ldots,N\}$ with $\lvert A\rvert \geq \alpha N$ there exists some $S\subseteq A$ such that\[\frac{a}{b}=\sum_{n\in S}\frac{1}{n}\]with $a\leq b=O_\alpha(1)$?

来源字段：https://www.erdosproblems.com/310

00564 | EP-312

### Erdős Problem #312

O / open | Erdős 问题

Does there exist some $c>0$ such that, for any $K>1$, whenever $A$ is a sufficiently large finite multiset of positive integers with $\sum_{n\in A}\frac{1}{n}>K$ there exists some $S\subseteq A$ such that\[1-e^{-cK} < \sum_{n\in S}\frac{1}{n}\leq 1?\]

来源字段：https://www.erdosproblems.com/312

00565 | EP-313

### Erdős Problem #313

O / open | Erdős 问题

Are there infinitely many solutions to\[\frac{1}{p_1}+\cdots+\frac{1}{p_k}=1-\frac{1}{m},\]where $m\geq 2$ is an integer and $p_1<\cdots<p_k$ are distinct primes?

来源字段：https://www.erdosproblems.com/313

00566 | EP-314

### Erdős Problem #314: Number Theory, Unit Fractions

S / solved | Erdős 问题

Let $n\geq 1$ and let $m$ be minimal such that $\sum_{n\leq k\leq m}\frac{1}{k}\geq 1$. We define\[\epsilon(n) = \sum_{n\leq k\leq m}\frac{1}{k}-1.\]How small can $\epsilon(n)$ be? Is it true that\[\liminf n^2\epsilon(n)=0?\]

来源字段：https://www.erdosproblems.com/314

00567 | EP-315

### Erdős Problem #315: Number Theory, Unit Fractions

S / solved | Erdős 问题

Let $u_1=1$ and $u_{n+1}=u_n(u_n+1)$, so that $\sum_{k\geq 1}\frac{1}{u_k+1}$ and $u_k=\lfloor c_0^{2^k}+1\rfloor$ for $k\geq 1$, where\[c_0=\lim u_n^{1/2^n}=1.264085\cdots.\]Let $a_1<a_2<\cdots $ be any other sequence with $\sum \frac{1}{a_k}=1$. Is it true that\[\liminf a_n^{1/2^n}<c_0=1.264085\cdots?\]

来源字段：https://www.erdosproblems.com/315

00568 | EP-316

### Erdős Problem #316: Number Theory, Unit Fractions

S / solved | Erdős 问题

Is it true that if $A\subset \mathbb{N}\backslash\{1\}$ is a finite set with $\sum_{n\in A}\frac{1}{n}<2$ then there is a partition $A=A_1\sqcup A_2$ such that\[\sum_{n\in A_i}\frac{1}{n}<1\]for $i=1,2$?

来源字段：https://www.erdosproblems.com/316

00569 | EP-319

### Erdős Problem #319

O / open | Erdős 问题

What is the size of the largest $A\subseteq \{1,\ldots,N\}$ such that there is a function $\delta:A\to \{-1,1\}$ such that\[\sum_{n\in A}\frac{\delta_n}{n}=0\]and\[\sum_{n\in A'}\frac{\delta_n}{n}\neq 0\]for all non-empty $A'\subsetneq A$?

来源字段：https://www.erdosproblems.com/319

00570 | EP-31

### Erdős Problem #31: Number Theory, Additive Basis

S / solved | Erdős 问题

Given any infinite set $A\subset \mathbb{N}$ there is a set $B$ of density $0$ such that $A+B$ contains all except finitely many integers.

来源字段：https://www.erdosproblems.com/31

00571 | EP-32

### Erdős Problem #32

P / partially_solved | Erdős 问题

Is there a set $A\subset\mathbb{N}$ such that\[\lvert A\cap\{1,\ldots,N\}\rvert = o((\log N)^2)\]and such that every large integer can be written as $p+a$ for some prime $p$ and $a\in A$?

Can the bound $O(\log N)$ be achieved? Must such an $A$ satisfy\[\liminf \frac{\lvert A\cap\{1,\ldots,N\}\rvert}{\log N}> 1?\]

来源字段：https://www.erdosproblems.com/32

00572 | EP-322

### Erdős Problem #322

P / partially_solved | Erdős 问题

Let $k\geq 3$ and $A\subset \mathbb{N}$ be the set of $k$th powers. What is the order of growth of $1_A^{(k)}(n)$, i.e. the number of representations of $n$ as the sum of $k$ many $k$th powers? Does there exist some $c>0$ and infinitely many $n$ such that\[1_A^{(k)}(n) >n^c?\]

来源字段：https://www.erdosproblems.com/322

00573 | EP-323

### Erdős Problem #323

O / open | Erdős 问题

Let $1\leq m\leq k$ and $f_{k,m}(x)$ denote the number of integers $\leq x$ which are the sum of $m$ many nonnegative $k$th powers. Is it true that\[f_{k,k}(x) \gg_\epsilon x^{1-\epsilon}\]for all $\epsilon>0$? Is it true that if $m<k$ then\[f_{k,m}(x) \gg x^{m/k}\]for sufficiently large $x$?

来源字段：https://www.erdosproblems.com/323

00574 | EP-324

### Erdős Problem #324

P / partially_solved | Erdős 问题

Does there exist a polynomial $f(x)\in\mathbb{Z}[x]$ such that all the sums $f(a)+f(b)$ with $a<b$ nonnegative integers are distinct?

来源字段：https://www.erdosproblems.com/324

00575 | EP-325

### Erdős Problem #325

P / partially_solved | Erdős 问题

Let $k\geq 3$ and $f_{k,3}(x)$ denote the number of integers $\leq x$ which are the sum of three nonnegative $k$th powers. Is it true that\[f_{k,3}(x) \gg x^{3/k}\]or even $\gg_\epsilon x^{3/k-\epsilon}$?

来源字段：https://www.erdosproblems.com/325

00576 | EP-328

### Erdős Problem #328: Number Theory, Additive Combinatorics

S / solved | Erdős 问题

Suppose $A\subseteq\mathbb{N}$ and $C>0$ is such that $1_A\ast 1_A(n)\leq C$ for all $n\in\mathbb{N}$. Can $A$ be partitioned into $t$ many subsets $A_1,\ldots,A_t$ (where $t=t(C)$ depends only on $C$) such that $1_{A_i}\ast 1_{A_i}(n)<C$ for all $1\leq i\leq t$ and $n\in \mathbb{N}$?

来源字段：https://www.erdosproblems.com/328

00577 | EP-329

### Erdős Problem #329

O / open | Erdős 问题

Suppose $A\subseteq \mathbb{N}$ is a Sidon set. How large can\[\limsup_{N\to \infty}\frac{\lvert A\cap \{1,\ldots,N\}\rvert}{N^{1/2}}\]be?

来源字段：https://www.erdosproblems.com/329

00578 | EP-33

### Erdős Problem #33

P / partially_solved | Erdős 问题

Let $A\subset\mathbb{N}$ be such that every large integer can be written as $n^2+a$ for some $a\in A$ and $n\geq 0$. What is the smallest possible value of\[\limsup \frac{\lvert A\cap\{1,\ldots,N\}\rvert}{N^{1/2}}?\]Is\[\liminf \frac{\lvert A\cap\{1,\ldots,N\}\rvert}{N^{1/2}}>1?\]

来源字段：https://www.erdosproblems.com/33

00579 | EP-330

### Erdős Problem #330

S / solved | Erdős 问题

Does there exist a minimal basis with positive density, say $A\subset\mathbb{N}$, such that for any $n\in A$ the (upper) density of integers which cannot be represented without using $n$ is positive?

来源字段：https://www.erdosproblems.com/330

00580 | EP-331

### Erdős Problem #331: Number Theory, Additive Combinatorics

S / solved | Erdős 问题

Let $A,B\subseteq \mathbb{N}$ such that for all large $N$\[\lvert A\cap \{1,\ldots,N\}\rvert \gg N^{1/2}\]and\[\lvert B\cap \{1,\ldots,N\}\rvert \gg N^{1/2}.\]Is it true that there are infinitely many solutions to $a_1-a_2=b_1-b_2\neq 0$ with $a_1,a_2\in A$ and $b_1,b_2\in B$?

来源字段：https://www.erdosproblems.com/331

00581 | EP-332

### Erdős Problem #332

O / open | Erdős 问题

Let $A\subseteq \mathbb{N}$ and $D(A)$ be the set of those numbers which occur infinitely often as $a_1-a_2$ with $a_1,a_2\in A$. What conditions on $A$ are sufficient to ensure $D(A)$ has bounded gaps?

来源字段：https://www.erdosproblems.com/332

00582 | EP-333

### Erdős Problem #333: Number Theory, Additive Basis

S / solved | Erdős 问题

Let $A\subseteq \mathbb{N}$ be a set of density zero. Does there exist a $B$ such that $A\subseteq B+B$ and\[\lvert B\cap \{1,\ldots,N\}\rvert =o(N^{1/2})\]for all large $N$?

来源字段：https://www.erdosproblems.com/333

00583 | EP-334

### Erdős Problem #334

O / open | Erdős 问题

Find the best function $f(n)$ such that every $n$ can be written as $n=a+b$ where both $a,b$ are $f(n)$-smooth (that is, are not divisible by any prime $p>f(n)$.)

来源字段：https://www.erdosproblems.com/334

00584 | EP-336

### Erdős Problem #336

O / open | Erdős 问题

For $r\geq 2$ let $h(r)$ be the maximal finite $k$ such that there exists a basis $A\subseteq \mathbb{N}$ of order $r$ (so every large integer is the sum of at most $r$ integers from $A$) and exact order $k$ (so every large integer is the sum of exactly $k$ integers from $A$).

Find the value of\[\lim_r \frac{h(r)}{r^2}.\]

来源字段：https://www.erdosproblems.com/336

00585 | EP-337

### Erdős Problem #337: Number Theory, Additive Combinatorics

S / solved | Erdős 问题

Let $A\subseteq \mathbb{N}$ be an additive basis (of any finite order) such that $\lvert A\cap \{1,\ldots,N\}\rvert=o(N)$. Is it true that\[\lim_{N\to \infty}\frac{\lvert (A+A)\cap \{1,\ldots,N\}\rvert}{\lvert A\cap \{1,\ldots,N\}\rvert}=\infty?\]

来源字段：https://www.erdosproblems.com/337

00586 | EP-338

### Erdős Problem #338

O / open | Erdős 问题

The restricted order of a basis is the least integer $t$ (if it exists) such that every large integer is the sum of at most $t$ distinct summands from $A$. What are necessary and sufficient conditions that this exists? Can it be bounded (when it exists) in terms of the order of the basis? What are necessary and sufficient conditions that this is equal to the order of the basis?

来源字段：https://www.erdosproblems.com/338

00587 | EP-339

### Erdős Problem #339: Number Theory, Additive Basis

S / solved | Erdős 问题

Let $A\subseteq \mathbb{N}$ be a basis of order $r$. Must the set of integers representable as the sum of exactly $r$ distinct elements from $A$ have positive lower density?

来源字段：https://www.erdosproblems.com/339

00588 | EP-340

### Erdős Problem #340

O / open | Erdős 问题

Let $A=\{1,2,4,8,13,21,31,45,66,81,97,\ldots\}$ be the greedy Sidon sequence: we begin with $1$ and iteratively include the next smallest integer that preserves the Sidon property (i.e. there are no non-trivial solutions to $a+b=c+d$). What is the order of growth of $A$? Is it true that\[\lvert A\cap \{1,\ldots,N\}\rvert \gg N^{1/2-\epsilon}\]for all $\epsilon>0$ and large $N$?

来源字段：https://www.erdosproblems.com/340

00589 | EP-341

### Erdős Problem #341

O / open | Erdős 问题

Let $A=\{a_1<\cdots<a_k\}$ be a finite set of positive integers and extend it to an infinite sequence $\overline{A}=\{a_1<a_2<\cdots \}$ by defining $a_{n+1}$ for $n\geq k$ to be the least integer exceeding $a_n$ which is not of the form $a_i+a_j$ with $i,j\leq n$. Is it true that the sequence of differences $a_{m+1}-a_m$ is eventually periodic?

来源字段：https://www.erdosproblems.com/341

00590 | EP-342

### Erdős Problem #342

O / open | Erdős 问题

With $a_1=1$ and $a_2=2$ let $a_{n+1}$ for $n\geq 2$ be the least integer $>a_n$ which can be expressed uniquely as $a_i+a_j$ for $i<j\leq n$.

What can be said about this sequence? Do infinitely many pairs $a,a+2$ occur? Does this sequence eventually have periodic differences? Is the density $0$?

来源字段：https://www.erdosproblems.com/342

00591 | EP-343

### Erdős Problem #343: Number Theory, Complete Sequences

S / solved | Erdős 问题

If $A\subseteq \mathbb{N}$ is a multiset of integers such that\[\lvert A\cap \{1,\ldots,N\}\rvert\gg N\]for all $N$ then must $A$ be subcomplete? That is, must\[P(A) = \left\{\sum_{n\in B}n: B\subseteq A\textrm{ finite }\right\}\]contain an infinite arithmetic progression?

来源字段：https://www.erdosproblems.com/343

00592 | EP-344

### Erdős Problem #344: Number Theory, Complete Sequences

S / solved | Erdős 问题

If $A\subseteq \mathbb{N}$ is a set of integers such that\[\lvert A\cap \{1,\ldots,N\}\rvert\gg N^{1/2}\]for all $N$ then must $A$ be subcomplete? That is, must\[P(A) = \left\{\sum_{n\in B}n: B\subseteq A\textrm{ finite }\right\}\]contain an infinite arithmetic progression?

来源字段：https://www.erdosproblems.com/344

00593 | EP-345

### Erdős Problem #345

O / open | Erdős 问题

Let $A\subseteq \mathbb{N}$ be a complete sequence, and define the threshold of completeness $T(A)$ to be the least integer $m$ such that all $n\geq m$ are in\[P(A) = \left\{\sum_{n\in B}n: B\subseteq A\textrm{ finite }\right\}\](the existence of $T(A)$ is guaranteed by completeness).

Is it true that there are infinitely many $k$ such that $T(n^k)>T(n^{k+1})$?

来源字段：https://www.erdosproblems.com/345

00594 | EP-346

### Erdős Problem #346

S / solved | Erdős 问题

Let $A=\{1\leq a_1< a_2<\cdots\}$ be a set of integers such that

- $A\backslash B$ is complete for any finite subset $B$ and

- $A\backslash B$ is not complete for any infinite subset $B$. (Here 'complete' means all sufficiently large integers can be written as a sum of distinct members of the sequence.)

Is it true that if $a_{n+1}/a_n \geq 1+\epsilon$ for some $\epsilon>0$ and all $n$ then\[\lim_n \frac{a_{n+1}}{a_n}=\frac{1+\sqrt{5}}{2}?\]

来源字段：https://www.erdosproblems.com/346

00595 | EP-347

### Erdős Problem #347: Number Theory, Complete Sequences

S / solved | Erdős 问题

Is there a sequence $A=\{a_1\leq a_2\leq \cdots\}$ of integers with\[\lim \frac{a_{n+1}}{a_n}=2\]such that\[P(A')= \left\{\sum_{n\in B}n: B\subseteq A'\textrm{ finite }\right\}\]has density $1$ for every cofinite subsequence $A'$ of $A$?

来源字段：https://www.erdosproblems.com/347

00596 | EP-348

### Erdős Problem #348

O / open | Erdős 问题

For what values of $0\leq m<n$ is there a complete sequence $A=\{a_1\leq a_2\leq \cdots\}$ of integers such that

- $A$ remains complete after removing any $m$ elements, but

- $A$ is not complete after removing any $n$ elements?

来源字段：https://www.erdosproblems.com/348

00597 | EP-349

### Erdős Problem #349

O / open | Erdős 问题

For what values of $t,\alpha \in (0,\infty)$ is the sequence $\lfloor t\alpha^n\rfloor$ complete (that is, all sufficiently large integers are the sum of distinct integers of the form $\lfloor t\alpha^n\rfloor$)?

来源字段：https://www.erdosproblems.com/349

00598 | EP-34

### Erdős Problem #34: Number Theory

S / solved | Erdős 问题

For any permutation $\pi\in S_n$ of $\{1,\ldots,n\}$ let $S(\pi)$ count the number of distinct consecutive sums, that is, sums of the shape $\sum_{u\leq i\leq v}\pi(i)$. Is it true that\[S(\pi) = o(n^2)\]for all $\pi\in S_n$?

来源字段：https://www.erdosproblems.com/34

00599 | EP-350

### Erdős Problem #350: Number Theory, Additive Combinatorics

S / solved | Erdős 问题

If $A\subset\mathbb{N}$ is a finite set of integers which is dissociated (that is, all of the subset sums are distinct) then\[\sum_{n\in A}\frac{1}{n}<2.\]

来源字段：https://www.erdosproblems.com/350

00600 | EP-351

### Erdős Problem #351

S / solved | Erdős 问题

Let $p(x)\in \mathbb{Q}[x]$ with positive leading coefficient. Is it true that\[A=\{ p(n)+1/n: n\in \mathbb{N}\}\]is strongly complete, in the sense that, for any finite set $B$,\[\left\{\sum_{n\in X}n: X\subseteq A\backslash B\textrm{ finite }\right\}\]contains all sufficiently large integers?

来源字段：https://www.erdosproblems.com/351

00601 | EP-354

### Erdős Problem #354

O / open | Erdős 问题

Let $\alpha,\beta\in \mathbb{R}_{>0}$ such that $\alpha/\beta$ is irrational. Is the multiset\[\{ \lfloor \alpha\rfloor,\lfloor 2\alpha\rfloor,\lfloor 4\alpha\rfloor,\ldots\}\cup \{ \lfloor \beta\rfloor,\lfloor 2\beta\rfloor,\lfloor 4\beta\rfloor,\ldots\}\]complete? That is, can all sufficiently large natural numbers $n$ be written as\[n=\sum_{s\in S}\lfloor 2^s\alpha\rfloor+\sum_{t\in T}\lfloor 2^t\beta\rfloor\]for some finite $S,T\subset \mathbb{N}$?

What if $2$ is replaced by some $\gamma\in(1,2)$?

来源字段：https://www.erdosproblems.com/354

00602 | EP-355

### Erdős Problem #355: Number Theory, Unit Fractions

S / solved | Erdős 问题

Is there a lacunary sequence $A\subseteq \mathbb{N}$ (so that $A=\{a_1<a_2<\cdots\}$ and there exists some $\lambda>1$ such that $a_{n+1}/a_n\geq \lambda$ for all $n\geq 1$) such that\[\left\{ \sum_{a\in A'}\frac{1}{a}: A'\subseteq A\textrm{ finite}\right\}\]contains all rationals in some open interval?

来源字段：https://www.erdosproblems.com/355

00603 | EP-356

### Erdős Problem #356: Number Theory

S / solved | Erdős 问题

Is there some $c>0$ such that, for all sufficiently large $n$, there exist integers $a_1<\cdots<a_k\leq n$ such that there are at least $cn^2$ distinct integers of the form $\sum_{u\leq i\leq v}a_i$?

来源字段：https://www.erdosproblems.com/356

00604 | EP-357

### Erdős Problem #357

O / open | Erdős 问题

Let $1\leq a_1<\cdots <a_k\leq n$ be integers such that all sums of the shape $\sum_{u\leq i\leq v}a_i$ are distinct. Let $f(n)$ be the maximal such $k$.

How does $f(n)$ grow? Is $f(n)=o(n)$?

来源字段：https://www.erdosproblems.com/357

00605 | EP-358

### Erdős Problem #358

S / solved | Erdős 问题

Let $A=\{a_1<\cdots\}$ be an infinite sequence of integers. Let $f(n)$ count the number of solutions to\[n=\sum_{u\leq i\leq v}a_i.\]Is there such an $A$ for which $f(n)\to \infty$ as $n\to \infty$? Or even where $f(n)\geq 2$ for all large $n$?

来源字段：https://www.erdosproblems.com/358

00606 | EP-359

### Erdős Problem #359

O / open | Erdős 问题

Let $a_1<a_2<\cdots$ be an infinite sequence of integers such that $a_1=n$ and $a_{i+1}$ is the least integer which is not a sum of consecutive earlier $a_j$s. What can be said about the density of this sequence?

In particular, in the case $n=1$, can one prove that $a_k/k\to \infty$ and $a_k/k^{1+c}\to 0$ for any $c>0$?

来源字段：https://www.erdosproblems.com/359

00607 | EP-35

### Erdős Problem #35: Number Theory, Additive Basis

S / solved | Erdős 问题

Let $B\subseteq\mathbb{N}$ be an additive basis of order $k$ with $0\in B$. Is it true that for every $A\subseteq\mathbb{N}$ we have\[d_s(A+B)\geq \alpha+\frac{\alpha(1-\alpha)}{k},\]where $\alpha=d_s(A)$ and\[d_s(A) = \inf \frac{\lvert A\cap\{1,\ldots,N\}\rvert}{N}\]is the Schnirelmann density?

来源字段：https://www.erdosproblems.com/35

00608 | EP-36

### Erdős Problem #36

O / open | Erdős 问题

Find the optimal constant $c>0$ such that the following holds.

For all sufficiently large $N$, if $A\sqcup B=\{1,\ldots,2N\}$ is a partition into two equal parts, so that $\lvert A\rvert=\lvert B\rvert=N$, then there is some $x$ such that the number of solutions to $a-b=x$ with $a\in A$ and $b\in B$ is at least $cN$.

来源字段：https://www.erdosproblems.com/36

00609 | EP-360

### Erdős Problem #360: Number Theory

S / solved | Erdős 问题

Let $f(n)$ be minimal such that $\{1,\ldots,n-1\}$ can be partitioned into $f(n)$ classes so that $n$ cannot be expressed as a sum of distinct elements from the same class. How fast does $f(n)$ grow?

来源字段：https://www.erdosproblems.com/360

00610 | EP-361

### Erdős Problem #361

O / open | Erdős 问题

Let $c>0$ and $n$ be some large integer. What is the size of the largest $A\subseteq \{1,\ldots,\lfloor cn\rfloor\}$ such that $n$ is not a sum of a subset of $A$? Does this depend on $n$ in an irregular way?

来源字段：https://www.erdosproblems.com/361

00611 | EP-362

### Erdős Problem #362: Number Theory

S / solved | Erdős 问题

Let $A\subseteq \mathbb{N}$ be a finite set of size $N$. Is it true that, for any fixed $t$, there are\[\ll \frac{2^N}{N^{3/2}}\]many $S\subseteq A$ such that $\sum_{n\in S}n=t$?

If we further ask that $\lvert S\rvert=l$ (for any fixed $l$) then is the number of solutions\[\ll \frac{2^N}{N^2},\]with the implied constant independent of $l$ and $t$?

来源字段：https://www.erdosproblems.com/362

00612 | EP-363

### Erdős Problem #363: Number Theory

S / solved | Erdős 问题

Is it true that there are only finitely many collections of disjoint intervals $I_1,\ldots,I_n$ of size $\lvert I_i\rvert \geq 4$ for $1\leq i\leq n$ such that\[\prod_{1\leq i\leq n}\prod_{m\in I_i}m\]is a square?

来源字段：https://www.erdosproblems.com/363

00613 | EP-364

### Erdős Problem #364: Number Theory, Powerful

O / open | Erdős 问题

Are there any triples of consecutive positive integers all of which are powerful (i.e. if $p\mid n$ then $p^2\mid n$)?

来源字段：https://www.erdosproblems.com/364

00614 | EP-365

### Erdős Problem #365

P / partially_solved | Erdős 问题

Do all pairs of consecutive powerful numbers $n$ and $n+1$ come from solutions to Pell equations? In other words, must either $n$ or $n+1$ be a square?

Is the number of such $n\leq x$ bounded by $(\log x)^{O(1)}$?

来源字段：https://www.erdosproblems.com/365

00615 | EP-366

### Erdős Problem #366: Number Theory, Powerful

O / open | Erdős 问题

Are there any $2$-full $n$ such that $n+1$ is $3$-full? That is, if $p\mid n$ then $p^2\mid n$ and if $p\mid n+1$ then $p^3\mid n+1$.

来源字段：https://www.erdosproblems.com/366

00616 | EP-367

### Erdős Problem #367

P / partially_solved | Erdős 问题

Let $B_2(n)$ be the $2$-full part of $n$ (that is, $B_2(n)=n/n'$ where $n'$ is the product of all primes that divide $n$ exactly once). Is it true that, for every fixed $k\geq 1$,\[\prod_{n\leq m<n+k}B_2(m) \ll n^{2+o(1)}?\]Or perhaps even $\ll_k n^2$?

来源字段：https://www.erdosproblems.com/367

00617 | EP-368

### Erdős Problem #368

O / open | Erdős 问题

How large is the largest prime factor of $n(n+1)$?

来源字段：https://www.erdosproblems.com/368

00618 | EP-369

### Erdős Problem #369

S / solved | Erdős 问题

Let $\epsilon>0$ and $k\geq 2$. Is it true that, for all sufficiently large $n$, there is a sequence of $k$ consecutive integers in $\{1,\ldots,n\}$ all of which are $n^\epsilon$-smooth?

来源字段：https://www.erdosproblems.com/369

00619 | EP-370

### Erdős Problem #370: Number Theory

S / solved | Erdős 问题

Are there infinitely many $n$ such that the largest prime factor of $n$ is $<n^{1/2}$ and the largest prime factor of $n+1$ is $<(n+1)^{1/2}$?

来源字段：https://www.erdosproblems.com/370

00620 | EP-371

### Erdős Problem #371

O / open | Erdős 问题

Let $P(n)$ denote the largest prime factor of $n$. Show that the set of $n$ with $P(n)<P(n+1)$ has density $1/2$.

来源字段：https://www.erdosproblems.com/371

00621 | EP-372

### Erdős Problem #372: Number Theory

S / solved | Erdős 问题

Let $P(n)$ denote the largest prime factor of $n$. There are infinitely many $n$ such that $P(n)>P(n+1)>P(n+2)$.

来源字段：https://www.erdosproblems.com/372

00622 | EP-373

### Erdős Problem #373

O / open | Erdős 问题

Show that the equation\[n! = a_1!a_2!\cdots a_k!,\]with $n-1>a_1\geq a_2\geq \cdots \geq a_k\geq 2$, has only finitely many solutions.

来源字段：https://www.erdosproblems.com/373

00623 | EP-374

### Erdős Problem #374

O / open | Erdős 问题

For any $m\in \mathbb{N}$, let $F(m)$ be the minimal $k\geq 2$ (if it exists) such that there are $a_1<\cdots <a_k=m$ with $a_1!\cdots a_k!$ a square. Let $D_k=\{ m: F(m)=k\}$. What is the order of growth of $\lvert D_k\cap\{1,\ldots,n\}\rvert$ for $3\leq k\leq 6$? For example, is it true that $\lvert D_6\cap \{1,\ldots,n\}\rvert \gg n$?

来源字段：https://www.erdosproblems.com/374

00624 | EP-375

### Erdős Problem #375: Number Theory, Primes

O / open | Erdős 问题

Is it true that for any $n,k\geq 1$, if $n+1,\ldots,n+k$ are all composite then there are distinct primes $p_1,\ldots,p_k$ such that $p_i\mid n+i$ for $1\leq i\leq k$?

来源字段：https://www.erdosproblems.com/375

00625 | EP-376

### Erdős Problem #376

O / open | Erdős 问题

Are there infinitely many $n$ such that $\binom{2n}{n}$ is coprime to $105$?

来源字段：https://www.erdosproblems.com/376

00626 | EP-377

### Erdős Problem #377

O / open | Erdős 问题

Is there some absolute constant $C>0$ such that\[\sum_{p\leq n}1_{p\nmid \binom{2n}{n}}\frac{1}{p}\leq C\]for all $n$ (where the summation is restricted to primes $p\leq n$)?

来源字段：https://www.erdosproblems.com/377

00627 | EP-378

### Erdős Problem #378: Number Theory, Binomial Coefficients

S / solved | Erdős 问题

Let $r\geq 0$. Does the density of integers $n$ for which $\binom{n}{k}$ is squarefree for at least $r$ values of $1\leq k<n$ exist? Is this density $>0$?

来源字段：https://www.erdosproblems.com/378

00628 | EP-379

### Erdős Problem #379: Number Theory, Binomial Coefficients

S / solved | Erdős 问题

Let $S(n)$ denote the largest integer such that, for all $1\leq k<n$, the binomial coefficient $\binom{n}{k}$ is divisible by $p^{S(n)}$ for some prime $p$ (depending on $k$). Is it true that\[\limsup S(n)=\infty?\]

来源字段：https://www.erdosproblems.com/379

00629 | EP-37

### Erdős Problem #37: Number Theory, Additive Combinatorics

S / solved | Erdős 问题

We say that $A\subset \mathbb{N}$ is an essential component if $d_s(A+B)>d_s(B)$ for every $B\subset \mathbb{N}$ with $0<d_s(B)<1$ where $d_s$ is the Schnirelmann density.

Can a lacunary set $A\subset\mathbb{N}$ be an essential component?

来源字段：https://www.erdosproblems.com/37

00630 | EP-38

### Erdős Problem #38

S / solved | Erdős 问题

Does there exist $B\subset\mathbb{N}$ which is not an additive basis, but is such that for every set $A\subseteq\mathbb{N}$ of Schnirelmann density $\alpha$ and every $N$ there exists $b\in B$ such that\[\lvert (A\cup (A+b))\cap \{1,\ldots,N\}\rvert\geq (\alpha+f(\alpha))N\]where $f(\alpha)>0$ for $0<\alpha <1$?

The Schnirelmann density is defined by\[d_s(A) = \inf_{N\geq 1}\frac{\lvert A\cap\{1,\ldots,N\}\rvert}{N}.\]

来源字段：https://www.erdosproblems.com/38

00631 | EP-380

### Erdős Problem #380

S / solved | Erdős 问题

We call an interval $[u,v]$ 'bad' if the greatest prime factor of $\prod_{u\leq m\leq v}m$ occurs with an exponent greater than $1$. Let $B(x)$ count the number of $n\leq x$ which are contained in at least one bad interval. Is it true that\[B(x)\sim \#\{ n\leq x: P(n)^2\mid n\},\]where $P(n)$ is the largest prime factor of $n$?

来源字段：https://www.erdosproblems.com/380

00632 | EP-381

### Erdős Problem #381: Number Theory, Divisors

S / solved | Erdős 问题

A number $n$ is highly composite if $\tau(m)<\tau(n)$ for all $m<n$, where $\tau(m)$ counts the number of divisors of $m$. Let $Q(x)$ count the number of highly composite numbers in $[1,x]$.

Is it true that\[Q(x)\gg_k (\log x)^k\]for every $k\geq 1$?

来源字段：https://www.erdosproblems.com/381

00633 | EP-382

### Erdős Problem #382

O / open | Erdős 问题

Let $u\leq v$ be such that the largest prime dividing $\prod_{u\leq m\leq v}m$ appears with exponent at least $2$. Is it true that $v-u=v^{o(1)}$? Can $v-u$ be arbitrarily large?

来源字段：https://www.erdosproblems.com/382

00634 | EP-383

### Erdős Problem #383

O / open | Erdős 问题

Is it true that for every $k$ there are infinitely many primes $p$ such that the largest prime divisor of\[\prod_{0\leq i\leq k}(p^2+i)\]is $p$?

来源字段：https://www.erdosproblems.com/383

00635 | EP-384

### Erdős Problem #384: Number Theory, Binomial Coefficients

S / solved | Erdős 问题

If $1<k<n-1$ then $\binom{n}{k}$ is divisible by a prime $p<n/2$ (except $\binom{7}{3}=5\cdot 7$).

来源字段：https://www.erdosproblems.com/384

00636 | EP-385

### Erdős Problem #385

O / open | Erdős 问题

Let\[F(n) = \max_{\substack{m<n\\ m\textrm{ composite}}} m+p(m),\]where $p(m)$ is the least prime divisor of $m$. Is it true that $F(n)>n$ for all sufficiently large $n$? Does $F(n)-n\to \infty$ as $n\to\infty$?

来源字段：https://www.erdosproblems.com/385

00637 | EP-386

### Erdős Problem #386

O / open | Erdős 问题

Let $2\leq k\leq n-2$. Can $\binom{n}{k}$ be the product of consecutive primes infinitely often? For example\[\binom{21}{2}=2\cdot 3\cdot 5\cdot 7.\]

来源字段：https://www.erdosproblems.com/386

00638 | EP-387

### Erdős Problem #387

S / solved | Erdős 问题

Is there an absolute constant $c>0$ such that, for all $1\leq k< n$, the binomial coefficient $\binom{n}{k}$ has a divisor in $(cn,n]$?

来源字段：https://www.erdosproblems.com/387

00639 | EP-391

### Erdős Problem #391: Number Theory, Factorials

S / solved | Erdős 问题

Let $t(n)$ be maximal such that there is a representation\[n!=a_1\cdots a_n\]with $t(n)=a_1\leq \cdots \leq a_n$. Obtain good bounds for $t(n)/n$. In particular, is it true that\[\lim \frac{t(n)}{n}=\frac{1}{e}?\]Furthermore, does there exist some constant $c>0$ such that\[\frac{t(n)}{n} \leq \frac{1}{e}-\frac{c}{\log n}\]for infinitely many $n$?

来源字段：https://www.erdosproblems.com/391

00640 | EP-392

### Erdős Problem #392: Number Theory, Factorials

S / solved | Erdős 问题

Let $A(n)$ denote the least value of $t$ such that\[n!=a_1\cdots a_t\]with $a_1\leq \cdots \leq a_t\leq n^2$. Is it true that\[A(n)=\frac{n}{2}-\frac{n}{2\log n}+o\left(\frac{n}{\log n}\right)?\]

来源字段：https://www.erdosproblems.com/392

00641 | EP-393

### Erdős Problem #393

O / open | Erdős 问题

Let $f(n)$ denote the minimal $m\geq 1$ such that\[n! = a_1\cdots a_t\]with $a_1<\cdots <a_t=a_1+m$. What is the behaviour of $f(n)$?

来源字段：https://www.erdosproblems.com/393

00642 | EP-394

### Erdős Problem #394

O / open | Erdős 问题

Let $t_k(n)$ denote the least $m$ such that\[n\mid m(m+1)(m+2)\cdots (m+k-1).\]Is it true that\[\sum_{n\leq x}t_2(n)\ll \frac{x^2}{(\log x)^c}\]for some $c>0$?

Is it true that, for $k\geq 2$,\[\sum_{n\leq x}t_{k+1}(n) =o\left(\sum_{n\leq x}t_k(n)\right)?\]

来源字段：https://www.erdosproblems.com/394

00643 | EP-396

### Erdős Problem #396

O / open | Erdős 问题

Is it true that for every $k$ there exists $n$ such that\[\prod_{0\leq i\leq k}(n-i) \mid \binom{2n}{n}?\]

来源字段：https://www.erdosproblems.com/396

00644 | EP-397

### Erdős Problem #397: Number Theory, Binomial Coefficients

S / solved | Erdős 问题

Are there only finitely many solutions to\[\prod_i \binom{2m_i}{m_i}=\prod_j \binom{2n_j}{n_j}\]with the $m_i,n_j$ distinct?

来源字段：https://www.erdosproblems.com/397

00645 | EP-398

### Erdős Problem #398: Are the only solutions to\[n!=x^2-1\]when $n=4,5,7$?

O / open | Erdős 问题

别名：Brocard's Problem [NT-010]; Brocard's Conjecture (Prime Gaps) [NT-086]

Are the only solutions to\[n!=x^2-1\]when $n=4,5,7$?

来源字段：https://www.erdosproblems.com/398

00646 | EP-399

### Erdős Problem #399: Number Theory, Factorials

S / solved | Erdős 问题

Is it true that there are no solutions to\[n! = x^k\pm y^k\]with $x,y,n\in \mathbb{N}$, with $xy>1$ and $k>2$?

来源字段：https://www.erdosproblems.com/399

00647 | EP-400

### Erdős Problem #400

O / open | Erdős 问题

For any $k\geq 2$ let $g_k(n)$ denote the maximum value of\[(a_1+\cdots+a_k)-n\]where $a_1,\ldots,a_k$ are integers such that $a_1!\cdots a_k! \mid n!$. Can one show that\[\sum_{n\leq x}g_k(n) \sim c_k x\log x\]for some constant $c_k$? Is it true that there is a constant $c_k$ such that for almost all $n<x$ we have\[g_k(n)=c_k\log x+o(\log x)?\]

来源字段：https://www.erdosproblems.com/400

00648 | EP-401

### Erdős Problem #401: Number Theory, Factorials

S / solved | Erdős 问题

Is there some function $f(r)$ such that $f(r)\to \infty$ as $r\to\infty$, such that, for infinitely many $n$, there exist $a_1,a_2$ with\[a_1+a_2> n+f(r)\log n\]such that $a_1!a_2! \mid n!2^n3^n\cdots p_r^n$?

来源字段：https://www.erdosproblems.com/401

00649 | EP-402

### Erdős Problem #402: Number Theory

S / solved | Erdős 问题

Prove that, for any finite set $A\subset\mathbb{N}$, there exist $a,b\in A$ such that\[\mathrm{gcd}(a,b)\leq a/\lvert A\rvert.\]

来源字段：https://www.erdosproblems.com/402

00650 | EP-403

### Erdős Problem #403: Number Theory, Factorials

S / solved | Erdős 问题

Does the equation\[2^m=a_1!+\cdots+a_k!\]with $a_1<a_2<\cdots <a_k$ have only finitely many solutions?

来源字段：https://www.erdosproblems.com/403

00651 | EP-404

### Erdős Problem #404

O / open | Erdős 问题

For which integers $a\geq 1$ and primes $p$ is there a finite upper bound on those $k$ such that there are $a=a_1<\cdots<a_n$ with\[p^k \mid (a_1!+\cdots+a_n!)?\]If $f(a,p)$ is the greatest such $k$, how does this function behave?

Is there a prime $p$ and an infinite sequence $a_1<a_2<\cdots$ such that if $p^{m_k}$ is the highest power of $p$ dividing $\sum_{i\leq k}a_i!$ then $m_k\to \infty$?

来源字段：https://www.erdosproblems.com/404

00652 | EP-405

### Erdős Problem #405: Is it true that the equation\[(p-1)!+a^{p-1}=p^k\]has only finitely many solutions?

S / solved | Erdős 问题

Let $p$ be an odd prime. Is it true that the equation\[(p-1)!+a^{p-1}=p^k\]has only finitely many solutions?

来源字段：https://www.erdosproblems.com/405

00653 | EP-406

### Erdős Problem #406

O / open | Erdős 问题

Is it true that there are only finitely many powers of $2$ which have only the digits $0$ and $1$ when written in base $3$?

来源字段：https://www.erdosproblems.com/406

00654 | EP-407

### Erdős Problem #407: Number Theory

S / solved | Erdős 问题

Let $w(n)$ count the number of solutions to\[n=2^a+3^b+2^c3^d\]with $a,b,c,d\geq 0$ integers. Is it true that $w(n)$ is bounded by some absolute constant?

来源字段：https://www.erdosproblems.com/407

00655 | EP-408

### Erdős Problem #408

O / open | Erdős 问题

Let $\phi(n)$ be the Euler totient function and $\phi_k(n)$ be the iterated $\phi$ function, so that $\phi_1(n)=\phi(n)$ and $\phi_k(n)=\phi(\phi_{k-1}(n))$. Let\[f(n) = \min \{ k: \phi_k(n)=1\}.\]Does $f(n)/\log n$ have a distribution function? Is $f(n)/\log n$ almost always constant? What can be said about the largest prime factor of $\phi_k(n)$ when, say, $k=\log\log n$?

来源字段：https://www.erdosproblems.com/408

00656 | EP-409

### Erdős Problem #409

O / open | Erdős 问题

How many iterations of $n\mapsto \phi(n)+1$ are needed before a prime is reached? Can infinitely many $n$ reach the same prime? What is the density of $n$ which reach any fixed prime?

来源字段：https://www.erdosproblems.com/409

00657 | EP-410

### Erdős Problem #410

O / open | Erdős 问题

Let $\sigma_1(n)=\sigma(n)$, the sum of divisors function, and $\sigma_k(n)=\sigma(\sigma_{k-1}(n))$. Is it true that for all $n\geq 2$\[\lim_{k\to \infty} \sigma_k(n)^{1/k}=\infty?\]

来源字段：https://www.erdosproblems.com/410

00658 | EP-411

### Erdős Problem #411

O / open | Erdős 问题

Let $g_1=g(n)=n+\phi(n)$ and $g_k(n)=g(g_{k-1}(n))$. For which $n$ and $r$ is it true that $g_{k+r}(n)=2g_k(n)$ for all large $k$?

来源字段：https://www.erdosproblems.com/411

00659 | EP-412

### Erdős Problem #412

O / open | Erdős 问题

Let $\sigma_1(n)=\sigma(n)$, the sum of divisors function, and $\sigma_k(n)=\sigma(\sigma_{k-1}(n))$.

Is it true that, for every $m,n\geq 2$, there exist some $i,j$ such that $\sigma_i(m)=\sigma_j(n)$?

来源字段：https://www.erdosproblems.com/412

00660 | EP-413

### Erdős Problem #413

P / partially_solved | Erdős 问题

Let $\omega(n)$ count the number of distinct primes dividing $n$. Are there infinitely many $n$ such that, for all $m<n$, we have $m+\omega(m) \leq n$?

Can one show that there exists an $\epsilon>0$ such that there are infinitely many $n$ where $m+\epsilon \omega(m)\leq n$ for all $m<n$?

来源字段：https://www.erdosproblems.com/413

00661 | EP-414

### Erdős Problem #414

O / open | Erdős 问题

Let $h_1(n)=h(n)=n+\tau(n)$ (where $\tau(n)$ counts the number of divisors of $n$) and $h_k(n)=h(h_{k-1}(n))$. Is it true, for any $m,n$, there exist $i$ and $j$ such that $h_i(m)=h_j(n)$?

来源字段：https://www.erdosproblems.com/414

00662 | EP-415

### Erdős Problem #415

P / partially_solved | Erdős 问题

For any $n$ let $F(n)$ be the largest $k$ such that any of the $k!$ possible ordering patterns appears in some sequence of $\phi(m+1),\ldots,\phi(m+k)$ with $m+k\leq n$. Is it true that\[F(n)=(c+o(1))\log\log\log n\]for some constant $c$? Is the first pattern which fails to appear always\[\phi(m+1)>\phi(m+2)>\cdots >\phi(m+k)?\]Is it true that the 'natural' ordering which mimics what happens to $\phi(1),\ldots,\phi(k)$ is the most likely to appear?

来源字段：https://www.erdosproblems.com/415

00663 | EP-416

### Erdős Problem #416

O / open | Erdős 问题

Let $V(x)$ count the number of $n\leq x$ such that $\phi(m)=n$ is solvable. Does $V(2x)/V(x)\to 2$? Is there an asymptotic formula for $V(x)$?

来源字段：https://www.erdosproblems.com/416

00664 | EP-418

### Erdős Problem #418: Are there infinitely many positive integers not of the form $n-\phi(n)$?

S / solved | Erdős 问题

Are there infinitely many positive integers not of the form $n-\phi(n)$?

来源字段：https://www.erdosproblems.com/418

00665 | EP-419

### Erdős Problem #419: Number Theory, Factorials

S / solved | Erdős 问题

If $\tau(n)$ counts the number of divisors of $n$, then what is the set of limit points of\[\frac{\tau((n+1)!)}{\tau(n!)}?\]

来源字段：https://www.erdosproblems.com/419

00666 | EP-420

### Erdős Problem #420

O / open | Erdős 问题

If $\tau(n)$ counts the number of divisors of $n$ then let\[F(f,n)=\frac{\tau((n+\lfloor f(n)\rfloor)!)}{\tau(n!)}.\]Is it true that\[\lim_{n\to \infty}F((\log n)^C,n)=\infty\]for large $C$?

Is it true that $F(\log n,n)$ is everywhere dense in $(1,\infty)$?

More generally, if $f(n)\leq \log n$ is a monotonic function such that $f(n)\to \infty$ as $n\to \infty$, then is $F(f,n)$ everywhere dense?

来源字段：https://www.erdosproblems.com/420

00667 | EP-422

### Erdős Problem #422

O / open | Erdős 问题

Let $f(1)=f(2)=1$ and for $n>2$\[f(n) = f(n-f(n-1))+f(n-f(n-2)).\]Does $f(n)$ miss infinitely many integers? What is its behaviour?

来源字段：https://www.erdosproblems.com/422

00668 | EP-423

### Erdős Problem #423

P / partially_solved | Erdős 问题

Let $a_1=1$ and $a_2=2$ and for $k\geq 3$ choose $a_k$ to be the least integer $>a_{k-1}$ which is the sum of at least two consecutive terms of the sequence. What is the asymptotic behaviour of this sequence?

来源字段：https://www.erdosproblems.com/423

00669 | EP-424

### Erdős Problem #424

O / open | Erdős 问题

Let $a_1=2$ and $a_2=3$ and continue the sequence by appending to $a_1,\ldots,a_n$ all possible values of $a_ia_j-1$ with $i\neq j$. Is it true that the set of integers which eventually appear has positive density?

来源字段：https://www.erdosproblems.com/424

00670 | EP-425

### Erdős Problem #425

P / partially_solved | Erdős 问题

Let $F(n)$ be the maximum possible size of a subset $A\subseteq\{1,\ldots,N\}$ such that the products $ab$ are distinct for all $a<b$. Is there a constant $c$ such that\[F(n)=\pi(n)+(c+o(1))n^{3/4}(\log n)^{-3/2}?\]If $A\subseteq \{1,\ldots,n\}$ is such that all products $a_1\cdots a_r$ are distinct for $a_1<\cdots <a_r$ then is it true that\[\lvert A\rvert \leq \pi(n)+O(n^{\frac{r+1}{2r}})?\]

来源字段：https://www.erdosproblems.com/425

00671 | EP-427

### Erdős Problem #427: Number Theory, Primes

S / solved | Erdős 问题

Is it true that, for every $n$ and $d$, there exists $k$ such that\[d \mid p_{n+1}+\cdots+p_{n+k},\]where $p_r$ denotes the $r$th prime?

来源字段：https://www.erdosproblems.com/427

00672 | EP-428

### Erdős Problem #428

O / open | Erdős 问题

Is there a set $A\subseteq \mathbb{N}$ such that, for infinitely many $n$, all of $n-a$ are prime for all $a\in A$ with $0<a<n$ and\[\liminf\frac{\lvert A\cap [1,x]\rvert}{\pi(x)}>0?\]

来源字段：https://www.erdosproblems.com/428

00673 | EP-429

### Erdős Problem #429: Number Theory

S / solved | Erdős 问题

Is it true that, if $A\subseteq \mathbb{N}$ is sparse enough and does not cover all residue classes modulo $p$ for any prime $p$, then there exists some $n$ such that $n+a$ is prime for all $a\in A$?

来源字段：https://www.erdosproblems.com/429

00674 | EP-430

### Erdős Problem #430

O / open | Erdős 问题

Fix some integer $n$ and define a decreasing sequence in $[1,n)$ by $a_1=n-1$ and, for $k\geq 2$, letting $a_k$ be the greatest integer in $[1,a_{k-1})$ such that all of the prime factors of $a_k$ are $>n-a_k$.

Is it true that, for sufficiently large $n$, not all of this sequence can be prime?

来源字段：https://www.erdosproblems.com/430

00675 | EP-431

### Erdős Problem #431

O / open | Erdős 问题

Are there two infinite sets $A$ and $B$ such that $A+B$ agrees with the set of prime numbers up to finitely many exceptions?

来源字段：https://www.erdosproblems.com/431

00676 | EP-432

### Erdős Problem #432

P / partially_solved | Erdős 问题

Let $A,B\subseteq \mathbb{N}$ be two infinite sets. How dense can $A+B$ be if all elements of $A+B$ are pairwise relatively prime?

来源字段：https://www.erdosproblems.com/432

00677 | EP-433

### Erdős Problem #433: Number Theory

S / solved | Erdős 问题

If $A\subset \mathbb{N}$ is a finite set then let $G(A)$ denote the greatest integer which is not expressible as a finite sum of elements from $A$ (with repetitions allowed). Let\[g(k,n)=\max G(A)\]where the maximum is taken over all $A\subseteq \{1,\ldots,n\}$ of size $\lvert A\rvert=k$ which has no common divisor. Is it true that\[g(k,n)\sim \frac{n^2}{k-1}?\]

来源字段：https://www.erdosproblems.com/433

00678 | EP-434

### Erdős Problem #434: Number Theory

S / solved | Erdős 问题

Let $k\leq n$. What choice of $A\subseteq \{1,\ldots,n\}$ (with $\mathrm{gcd}(A)=1$) of size $\lvert A\rvert=k$ maximises the number of integers not representable as the sum of finitely many elements from $A$ (with repetitions allowed)? Is it $\{n,n-1,\ldots,n-k+1\}$?

来源字段：https://www.erdosproblems.com/434

00679 | EP-435

### Erdős Problem #435: Number Theory

S / solved | Erdős 问题

Let $n\in\mathbb{N}$ with $n\neq p^k$ for any prime $p$ and $k\geq 0$. What is the largest integer not of the form\[\sum_{1\leq i<n}c_i\binom{n}{i}\]where the $c_i\geq 0$ are integers?

来源字段：https://www.erdosproblems.com/435

00680 | EP-436

### Erdős Problem #436

P / partially_solved | Erdős 问题

If $p$ is a prime and $k,m\geq 2$ then let $r(k,m,p)$ be the minimal $r$ such that $r,r+1,\ldots,r+m-1$ are all $k$th power residues modulo $p$. Let\[\Lambda(k,m)=\limsup_{p\to \infty} r(k,m,p).\]Is it true that $\Lambda(k,2)$ is finite for all $k$? Is $\Lambda(k,3)$ finite for all odd $k$? How large are they?

来源字段：https://www.erdosproblems.com/436

00681 | EP-437

### Erdős Problem #437: Number Theory

S / solved | Erdős 问题

Let $1\leq a_1<\cdots<a_k\leq x$. How many of the partial products $a_1,a_1a_2,\ldots,a_1\cdots a_k$ can be squares? Is it true that, for any $\epsilon>0$, there can be more than $x^{1-\epsilon}$ squares?

来源字段：https://www.erdosproblems.com/437

00682 | EP-438

### Erdős Problem #438: How large can $A\subseteq \{1,\ldots,N\}$ be if $A+A$ contains no square numbers?

S / solved | Erdős 问题

How large can $A\subseteq \{1,\ldots,N\}$ be if $A+A$ contains no square numbers?

来源字段：https://www.erdosproblems.com/438

00683 | EP-439

### Erdős Problem #439: Number Theory, Ramsey Theory

S / solved | Erdős 问题

Is it true that, in any finite colouring of the integers, there must be two integers $x\neq y$ of the same colour such that $x+y$ is a square? What about a $k$th power?

来源字段：https://www.erdosproblems.com/439

00684 | EP-440

### Erdős Problem #440: a_{i+1})\leq x$. Is it true that $A(x) \ll x^{1/2}$?

S / solved | Erdős 问题

Let $A=\{a_1<a_2<\cdots\}\subseteq \mathbb{N}$ be infinite and let $A(x)$ count the number of indices for which $\mathrm{lcm}(a_i,a_{i+1})\leq x$. Is it true that $A(x) \ll x^{1/2}$?

How large can\[\liminf \frac{A(x)}{x^{1/2}}\]be?

来源字段：https://www.erdosproblems.com/440

00685 | EP-441

### Erdős Problem #441: Number Theory

S / solved | Erdős 问题

Let $N\geq 1$. What is the size of the largest $A\subset \{1,\ldots,N\}$ such that $[a,b]\leq N$ for all $a,b\in A$, where $[a,b]$ is the least common multiple of $a$ and $b$?

Is it attained by choosing all integers in $[1,(N/2)^{1/2}]$ together with all even integers in $[(N/2)^{1/2},(2N)^{1/2}]$?

来源字段：https://www.erdosproblems.com/441

00686 | EP-442

### Erdős Problem #442: Number Theory

S / solved | Erdős 问题

Is it true that if $A\subseteq\mathbb{N}$ is such that\[\frac{1}{\log\log x}\sum_{n\in A\cap [1,x)}\frac{1}{n}\to \infty\]then\[\left(\sum_{n\in A\cap [1,x)}\frac{1}{n}\right)^{-2} \sum_{\substack{a,b\in A\cap (1,x]\\ a<b}}\frac{1}{\mathrm{lcm}(a,b)}\to \infty?\]

来源字段：https://www.erdosproblems.com/442

00687 | EP-443

### Erdős Problem #443: Number Theory

S / solved | Erdős 问题

Let $m,n\geq 1$. What is\[\# \{ k(m-k): 1\leq k\leq m/2\} \cap \{ l(n-l): 1\leq l\leq n/2\}?\]Can it be arbitrarily large? Is it $\leq (mn)^{o(1)}$ for all sufficiently large $m,n$?

来源字段：https://www.erdosproblems.com/443

00688 | EP-444

### Erdős Problem #444: Number Theory, Divisors

S / solved | Erdős 问题

Let $A\subseteq\mathbb{N}$ be infinite and $d_A(n)$ count the number of $a\in A$ which divide $n$. Is it true that, for every $k$,\[\limsup_{x\to \infty} \frac{\max_{n<x}d_A(n)}{\left(\sum_{n\in A\cap[1,x)}\frac{1}{n}\right)^k}=\infty?\]

来源字段：https://www.erdosproblems.com/444

00689 | EP-445

### Erdős Problem #445

O / open | Erdős 问题

Is it true that, for any $c>1/2$, if $p$ is a sufficiently large prime then, for any $n\geq 0$, there exist $a,b\in(n,n+p^c)$ such that $ab\equiv 1\pmod{p}$?

来源字段：https://www.erdosproblems.com/445

00690 | EP-446

### Erdős Problem #446: Number Theory, Divisors

S / solved | Erdős 问题

Let $\delta(n)$ denote the density of integers which are divisible by some integer in $(n,2n)$. What is the growth rate of $\delta(n)$?

If $\delta_1(n)$ is the density of integers which have exactly one divisor in $(n,2n)$ then is it true that $\delta_1(n)=o(\delta(n))$?

来源字段：https://www.erdosproblems.com/446

00691 | EP-448

### Erdős Problem #448: Number Theory, Divisors

S / solved | Erdős 问题

Let $\tau(n)$ count the divisors of $n$ and $\tau^+(n)$ count the number of $k$ such that $n$ has a divisor in $[2^k,2^{k+1})$. Is it true that, for all $\epsilon>0$,\[\tau^+(n) < \epsilon \tau(n)\]for almost all $n$?

来源字段：https://www.erdosproblems.com/448

00692 | EP-449

### Erdős Problem #449: Number Theory, Divisors

S / solved | Erdős 问题

Let $r(n)$ count the number of $d_1,d_2$ such that $d_1\mid n$ and $d_2\mid n$ and $d_1<d_2<2d_1$. Is it true that, for every $\epsilon>0$,\[r(n) < \epsilon \tau(n)\]for almost all $n$, where $\tau(n)$ is the number of divisors of $n$?

来源字段：https://www.erdosproblems.com/449

00693 | EP-450

### Erdős Problem #450

P / partially_solved | Erdős 问题

How large must $y=y(\epsilon,n)$ be such that the number of integers in $(x,x+y)$ with a divisor in $(n,2n)$ is at most $\epsilon y$?

来源字段：https://www.erdosproblems.com/450

00694 | EP-451

### Erdős Problem #451

P / partially_solved | Erdős 问题

Estimate $n_k$, the smallest integer $>2k$ such that $\prod_{1\leq i\leq k}(n_k-i)$ has no prime factor in $(k,2k)$.

来源字段：https://www.erdosproblems.com/451

00695 | EP-452

### Erdős Problem #452

O / open | Erdős 问题

Let $\omega(n)$ count the number of distinct prime factors of $n$. What is the size of the largest interval $I\subseteq [x,2x]$ such that $\omega(n)>\log\log n$ for all $n\in I$?

来源字段：https://www.erdosproblems.com/452

00696 | EP-453

### Erdős Problem #453: Number Theory

S / solved | Erdős 问题

别名：Pomerance's Questions on Good Primes [GUY-A14a]

Is it true that, for all sufficiently large $n$, there exists some $i<n$ such that\[p_n^2 < p_{n+i}p_{n-i},\]where $p_k$ is the $k$th prime?

来源字段：https://www.erdosproblems.com/453

00697 | EP-454

### Erdős Problem #454

O / open | Erdős 问题

Let\[f(n) = \min_{i<n} (p_{n+i}+p_{n-i}),\]where $p_k$ is the $k$th prime. Is it true that\[\limsup_n (f(n)-2p_n)=\infty?\]

来源字段：https://www.erdosproblems.com/454

00698 | EP-455

### Erdős Problem #455

O / open | Erdős 问题

Let $q_1<q_2<\cdots$ be a sequence of primes such that\[q_{n+1}-q_n\geq q_n-q_{n-1}.\]Must\[\lim_n \frac{q_n}{n^2}=\infty?\]

来源字段：https://www.erdosproblems.com/455

00699 | EP-456

### Erdős Problem #456

P / partially_solved | Erdős 问题

Let $p_n$ be the smallest prime $\equiv 1\pmod{n}$ and let $m_n$ be the smallest integer such that $n\mid \phi(m_n)$.

Is it true that $m_n<p_n$ for almost all $n$? Does $p_n/m_n\to \infty$ for almost all $n$? Are there infinitely many primes $p$ such that $p-1$ is the only $n$ for which $m_n=p$?

来源字段：https://www.erdosproblems.com/456

00700 | EP-457

### Erdős Problem #457

S / solved | Erdős 问题

Is there some $\epsilon>0$ such that there are infinitely many $n$ where all primes $p\leq (2+\epsilon)\log n$ divide\[\prod_{1\leq i\leq \log n}(n+i)?\]

来源字段：https://www.erdosproblems.com/457

00701 | EP-458

### Erdős Problem #458: Number Theory, Primes

O / open | Erdős 问题

Let $[1,\ldots,n]$ denote the least common multiple of $\{1,\ldots,n\}$. Is it true that, for all $k\geq 1$,\[[1,\ldots,p_{k+1}-1]< p_k[1,\ldots,p_k]?\]

来源字段：https://www.erdosproblems.com/458

00702 | EP-459

### Erdős Problem #459: v)$ is composed entirely of primes dividing $uv$. Estimate $f(u)$

S / solved | Erdős 问题

Let $f(u)$ be the largest $v$ such that no $m\in (u,v)$ is composed entirely of primes dividing $uv$. Estimate $f(u)$.

来源字段：https://www.erdosproblems.com/459

00703 | EP-45

### Erdős Problem #45: Number Theory, Unit Fractions

S / solved | Erdős 问题

Let $k\geq 2$. Is there an integer $n_k$ such that, if $D=\{1<d<n_k: d\mid n_k\}$, then for any $k$-colouring of $D$ there is a monochromatic subset $D'\subseteq D$ such that $\sum_{d\in D'}\frac{1}{d}=1$?

来源字段：https://www.erdosproblems.com/45

00704 | EP-460

### Erdős Problem #460

P / partially_solved | Erdős 问题

Let $a_0=0$ and $a_1=1$, and in general define $a_k$ to be the least integer $>a_{k-1}$ for which $(n-a_k,n-a_i)=1$ for all $0\leq i<k$. Does\[\sum_{0<a_i< n}\frac{1}{a_i}\to \infty\]as $n\to \infty$? What about if we restrict the sum to those $i$ such that $n-a_j$ is divisible by some prime $\leq a_j$, or the complement of such $i$?

来源字段：https://www.erdosproblems.com/460

00705 | EP-461

### Erdős Problem #461

O / open | Erdős 问题

Let $s_t(n)$ be the $t$-smooth component of $n$ - that is, the product of all primes $p$ (with multiplicity) dividing $n$ such that $p<t$. Let $f(n,t)$ count the number of distinct possible values for $s_t(m)$ for $m\in [n+1,n+t]$. Is it true that\[f(n,t)\gg t\](uniformly, for all $t$ and $n$)?

来源字段：https://www.erdosproblems.com/461

00706 | EP-462

### Erdős Problem #462

O / open | Erdős 问题

Let $p(n)$ denote the least prime factor of $n$. There is a constant $c>0$ such that\[\sum_{\substack{n<x\\ n\textrm{ not prime}}}\frac{p(n)}{n}\sim c\frac{x^{1/2}}{(\log x)^2}.\]Is it true that there exists a constant $C>0$ such that\[\sum_{x\leq n\leq x+Cx^{1/2}(\log x)^2}\frac{p(n)}{n} \gg 1\]for all large $x$?

来源字段：https://www.erdosproblems.com/462

00707 | EP-463

### Erdős Problem #463

O / open | Erdős 问题

Is there a function $f$ with $f(n)\to \infty$ as $n\to \infty$ such that, for all large $n$, there is a composite number $m$ such that\[n+f(n)<m<n+p(m)?\](Here $p(m)$ is the least prime factor of $m$.)

来源字段：https://www.erdosproblems.com/463

00708 | EP-464

### Erdős Problem #464: Number Theory

S / solved | Erdős 问题

Let $A=\{n_1<n_2<\cdots\}\subset \mathbb{N}$ be a lacunary sequence (so there exists some $\epsilon>0$ with $n_{k+1}\geq (1+\epsilon)n_k$ for all $k$). Must there exist an irrational $\theta$ such that\[\{ \|\theta n_k\|: k\geq 1\}\]is not dense in $[0,1]$ (where $\| x\|$ is the distance to the nearest integer)?

来源字段：https://www.erdosproblems.com/464

00709 | EP-465

### Erdős Problem #465: Number Theory

S / solved | Erdős 问题

Let $N(X,\delta)$ denote the maximum number of points $P_1,\ldots,P_n$ which can be chosen in a circle of radius $X$ such that\[\| \lvert P_i-P_j\rvert \| \geq \delta\]for all $1\leq i<j\leq n$. (Here $\|x\|$ is the distance from $x$ to the nearest integer.)

Is it true that, for any $0<\delta<1/2$, we have\[N(X,\delta)=o(X)?\]In fact, is it true that (for any fixed $\delta>0$)\[N(X,\delta)<X^{1/2+o(1)}?\]

来源字段：https://www.erdosproblems.com/465

00710 | EP-466

### Erdős Problem #466: Number Theory

S / solved | Erdős 问题

Let $N(X,\delta)$ denote the maximum number of points $P_1,\ldots,P_n$ which can be chosen in a circle of radius $X$ such that\[\| \lvert P_i-P_j\rvert \| \geq \delta\]for all $1\leq i<j\leq n$. (Here $\|x\|$ is the distance from $x$ to the nearest integer.)

Is there some $\delta>0$ such that\[\lim_{x\to \infty}N(X,\delta)=\infty?\]

来源字段：https://www.erdosproblems.com/466

00711 | EP-467

### Erdős Problem #467

P / partially_solved | Erdős 问题

Prove the following for all large $x$: there is a choice of congruence classes $a_p$ for all primes $p\leq x$ and a decomposition $\{p\leq x\}=A\sqcup B$ into two non-empty sets such that, for all $n<x$, there exist some $p\in A$ and $q\in B$ such that $n\equiv a_p\pmod{p}$ and $n\equiv a_q\pmod{q}$.

来源字段：https://www.erdosproblems.com/467

00712 | EP-468

### Erdős Problem #468

O / open | Erdős 问题

For any $n$ let $D_n$ be the set of sums of the shape $d_1,d_1+d_2,d_1+d_2+d_3,\ldots$ where $1<d_1<d_2<\cdots$ are the divisors of $n$.

What is the size of $D_n\backslash \cup_{m<n}D_m$?

If $f(N)$ is the minimal $n$ such that $N\in D_n$ then is it true that $f(N)=o(N)$? Perhaps just for almost all $N$?

来源字段：https://www.erdosproblems.com/468

00713 | EP-469

### Erdős Problem #469

S / solved | Erdős 问题

Let $A$ be the set of all $n$ such that $n=d_1+\cdots+d_k$ with $d_i$ distinct proper divisors of $n$, but this is not true for any $m\mid n$ with $m<n$. Does\[\sum_{n\in A}\frac{1}{n}\]converge?

来源字段：https://www.erdosproblems.com/469

00714 | EP-46

### Erdős Problem #46: Number Theory, Unit Fractions

S / solved | Erdős 问题

Does every finite colouring of the integers have a monochromatic solution to $1=\sum \frac{1}{n_i}$ with $2\leq n_1<\cdots <n_k$?

来源字段：https://www.erdosproblems.com/46

00715 | EP-470

### Erdős Problem #470

O / open | Erdős 问题

Call $n$ weird if $\sigma(n)\geq 2n$ and $n$ is not pseudoperfect, that is, it is not the sum of any set of its divisors.

Are there any odd weird numbers? Are there infinitely many primitive weird numbers, i.e. those such that no proper divisor of $n$ is weird?

来源字段：https://www.erdosproblems.com/470

00716 | EP-471

### Erdős Problem #471: Number Theory

S / solved | Erdős 问题

Given a finite set of primes $Q=Q_0$, define a sequence of sets $Q_i$ by letting $Q_{i+1}$ be $Q_i$ together with all primes formed by adding three distinct elements of $Q_i$. Is there some initial choice of $Q$ such that the $Q_i$ become arbitrarily large?

来源字段：https://www.erdosproblems.com/471

00717 | EP-472

### Erdős Problem #472

O / open | Erdős 问题

Given some initial finite sequence of primes $q_1<\cdots<q_m$ extend it so that $q_{n+1}$ is the smallest prime of the form $q_n+q_i-1$ for $n\geq m$. Is there an initial starting sequence so that the resulting sequence is infinite?

来源字段：https://www.erdosproblems.com/472

00718 | EP-473

### Erdős Problem #473: Number Theory

S / solved | Erdős 问题

Is there a permutation $a_1,a_2,\ldots$ of the positive integers such that $a_k+a_{k+1}$ is always prime?

来源字段：https://www.erdosproblems.com/473

00719 | EP-475

### Erdős Problem #475: Number Theory, Additive Combinatorics

O / open | Erdős 问题

Let $p$ be a prime. Given any finite set $A\subseteq \mathbb{F}_p\backslash \{0\}$, is there always a rearrangement $A=\{a_1,\ldots,a_t\}$ such that all partial sums $\sum_{1\leq k\leq m}a_{k}$ are distinct, for all $1\leq m\leq t$?

来源字段：https://www.erdosproblems.com/475

00720 | EP-476

### Erdős Problem #476: Number Theory, Additive Combinatorics

S / solved | Erdős 问题

Let $A\subseteq \mathbb{F}_p$. Let\[A\hat{+}A = \{ a+b: a\neq b \in A\}.\]Is it true that\[\lvert A\hat{+}A\rvert \geq \min(2\lvert A\rvert-3,p)?\]

来源字段：https://www.erdosproblems.com/476

00721 | EP-478

### Erdős Problem #478

O / open | Erdős 问题

Let $p$ be a prime and\[A_p = \{ k! \pmod{p}: 1\leq k<p\}.\]Is it true that\[\lvert A_p\rvert \sim (1-\tfrac{1}{e})p?\]

来源字段：https://www.erdosproblems.com/478

00722 | EP-47

### Erdős Problem #47: Number Theory, Unit Fractions

S / solved | Erdős 问题

If $\delta>0$ and $N$ is sufficiently large in terms of $\delta$, and $A\subseteq\{1,\ldots,N\}$ is such that $\sum_{a\in A}\frac{1}{a}>\delta \log N$ then must there exist $S\subseteq A$ such that $\sum_{n\in S}\frac{1}{n}=1$?

来源字段：https://www.erdosproblems.com/47

00723 | EP-480

### Erdős Problem #480: Number Theory

S / solved | Erdős 问题

Let $x_1,x_2,\ldots\in [0,1]$ be an infinite sequence. Is it true that\[\inf_n \liminf_{m\to \infty} n \lvert x_{m+n}-x_m\rvert\leq 5^{-1/2}\approx 0.447?\]

来源字段：https://www.erdosproblems.com/480

00724 | EP-481

### Erdős Problem #481: Number Theory

S / solved | Erdős 问题

Let $a_1,\ldots,a_r,b_1,\ldots,b_r\in \mathbb{N}$ such that $\sum_{i}\frac{1}{a_i}>1$. For any finite sequence of $n$ (not necessarily distinct) integers $A=(x_1,\ldots,x_n)$ let $T(A)$ denote the sequence of length $rn$ given by\[(a_ix_j+b_i)_{1\leq j\leq n, 1\leq i\leq r}.\]Prove that, if $A_1=(1)$ and $A_{i+1}=T(A_i)$, then there must be some $A_k$ with repeated elements.

来源字段：https://www.erdosproblems.com/481

00725 | EP-482

### Erdős Problem #482: Number Theory

S / solved | Erdős 问题

Define a sequence by $a_1=1$ and\[a_{n+1}=\lfloor\sqrt{2}(a_n+1/2)\rfloor\]for $n\geq 1$. The difference $a_{2n+1}-2a_{2n-1}$ is the $n$th digit in the binary expansion of $\sqrt{2}$.

Find similar results for $\theta=\sqrt{m}$, and other algebraic numbers.

来源字段：https://www.erdosproblems.com/482

00726 | EP-483

### Erdős Problem #483

O / open | Erdős 问题

Let $f(k)$ be the minimal $N$ such that if $\{1,\ldots,N\}$ is $k$-coloured then there is a monochromatic solution to $a+b=c$. Estimate $f(k)$. In particular, is it true that $f(k) < c^k$ for some constant $c>0$?

来源字段：https://www.erdosproblems.com/483

00727 | EP-484

### Erdős Problem #484: Number Theory, Additive Combinatorics

S / solved | Erdős 问题

Prove that there exists an absolute constant $c>0$ such that, whenever $\{1,\ldots,N\}$ is $k$-coloured (and $N$ is large enough depending on $k$) then there are at least $cN$ many integers in $\{1,\ldots,N\}$ which are representable as a monochromatic sum (that is, $a+b$ where $a,b\in \{1,\ldots,N\}$ are in the same colour class and $a\neq b$).

来源字段：https://www.erdosproblems.com/484

00728 | EP-486

### Erdős Problem #486

P / partially_solved | Erdős 问题

Let $A\subseteq \mathbb{N}$, and for each $n\in A$ choose some $X_n\subseteq \mathbb{Z}/n\mathbb{Z}$. Let\[B = \{ m\in \mathbb{N}: m\not\in X_n\pmod{n}\textrm{ for all }n\in A\textrm{ with }m>n\}.\]Must $B$ have a logarithmic density, i.e. is it true that\[\lim_{x\to \infty} \frac{1}{\log x}\sum_{\substack{m\in B\\ m<x}}\frac{1}{m}\]exists?

来源字段：https://www.erdosproblems.com/486

00729 | EP-487

### Erdős Problem #487: Number Theory

S / solved | Erdős 问题

Let $A\subseteq \mathbb{N}$ have positive density. Must there exist distinct $a,b,c\in A$ such that $[a,b]=c$ (where $[a,b]$ is the least common multiple of $a$ and $b$)?

来源字段：https://www.erdosproblems.com/487

00730 | EP-488

### Erdős Problem #488

O / open | Erdős 问题

Let $A$ be a finite set and\[B=\{ n \geq 1: a\mid n\textrm{ for some }a\in A\}.\]Is it true that, for every $m>n\geq \max(A)$,\[\frac{\lvert B\cap [1,m]\rvert }{m}< 2\frac{\lvert B\cap [1,n]\rvert}{n}?\]

来源字段：https://www.erdosproblems.com/488

00731 | EP-489

### Erdős Problem #489

O / open | Erdős 问题

Let $A\subseteq \mathbb{N}$ be a set such that $\lvert A\cap [1,x]\rvert=o(x^{1/2})$. Let\[B=\{ n\geq 1: a\nmid n\textrm{ for all }a\in A\}.\]If $B=\{b_1<b_2<\cdots\}$ then is it true that\[\lim \frac{1}{x}\sum_{b_i<x}(b_{i+1}-b_i)^2\]exists (and is finite)?

来源字段：https://www.erdosproblems.com/489

00732 | EP-48

### Erdős Problem #48: Are there infinitely many integers $n,m$ such that $\phi(n)=\sigma(m)$?

S / solved | Erdős 问题

Are there infinitely many integers $n,m$ such that $\phi(n)=\sigma(m)$?

来源字段：https://www.erdosproblems.com/48

00733 | EP-490

### Erdős Problem #490: Is it true that\[\lvert A\rvert \lvert B\rvert \ll \frac{N^2}{\log N}?\]

S / solved | Erdős 问题

Let $A,B\subseteq \{1,\ldots,N\}$ be such that all the products $ab$ with $a\in A$ and $b\in B$ are distinct. Is it true that\[\lvert A\rvert \lvert B\rvert \ll \frac{N^2}{\log N}?\]

来源字段：https://www.erdosproblems.com/490

00734 | EP-491

### Erdős Problem #491: Number Theory

S / solved | Erdős 问题

Let $f:\mathbb{N}\to \mathbb{R}$ be an additive function (i.e. $f(ab)=f(a)+f(b)$ whenever $(a,b)=1$). If there is a constant $c$ such that $\lvert f(n+1)-f(n)\rvert <c$ for all $n$ then must there exist some $c'$ such that\[f(n)=c'\log n+O(1)?\]

来源字段：https://www.erdosproblems.com/491

00735 | EP-492

### Erdős Problem #492: Number Theory

S / solved | Erdős 问题

Let $A=\{a_1<a_2<\cdots\}\subseteq \mathbb{N}$ be infinite such that $a_{i+1}/a_i\to 1$. For any $x\geq a_1$ let\[f(x) = \frac{x-a_i}{a_{i+1}-a_i}\in [0,1),\]where $x\in [a_i,a_{i+1})$. Is it true that, for almost all $\alpha$, the sequence $f(\alpha n)$ is uniformly distributed in $[0,1)$?

来源字段：https://www.erdosproblems.com/492

00736 | EP-493

### Erdős Problem #493: Number Theory

S / solved | Erdős 问题

Does there exist a $k$ such that every sufficiently large integer can be written in the form\[\prod_{i=1}^k a_i - \sum_{i=1}^k a_i\]for some integers $a_i\geq 2$?

来源字段：https://www.erdosproblems.com/493

00737 | EP-495

### Erdős Problem #495

O / open | Erdős 问题

别名：Littlewood Conjecture [NT-076]; Littlewood Conjecture [NUM-018]

Let $\alpha,\beta \in \mathbb{R}$. Is it true that\[\liminf_{n\to \infty} n \| n\alpha \| \| n\beta\| =0\]where $\|x\|$ is the distance from $x$ to the nearest integer?

来源字段：https://www.erdosproblems.com/495

00738 | EP-496

### Erdős Problem #496: Number Theory, Diophantine Approximation

S / solved | Erdős 问题

Let $\alpha \in \mathbb{R}$ be irrational and $\epsilon>0$. Are there positive integers $x,y,z$ such that\[\lvert x^2+y^2-z^2\alpha\rvert <\epsilon?\]

来源字段：https://www.erdosproblems.com/496

00739 | EP-49

### Erdős Problem #49: Number Theory, Primes

S / solved | Erdős 问题

Let $A=\{a_1<\cdots<a_t\}\subseteq \{1,\ldots,N\}$ be such that $\phi(a_1)<\cdots<\phi(a_t)$. The primes are such an example. Are they the largest possible? Can one show that $\lvert A\rvert<(1+o(1))\pi(N)$ or even $\lvert A\rvert=o(N)$?

来源字段：https://www.erdosproblems.com/49

00740 | EP-4

### Erdős Problem #4: Number Theory, Primes

S / solved | Erdős 问题

别名：Erdős $5000 Problem on Prime Gaps [GUY-A8a]

Is it true that, for any $C>0$, there are infinitely many $n$ such that\[p_{n+1}-p_n> C\frac{\log\log n\log\log\log\log n}{(\log\log \log n)^2}\log n?\]

来源字段：https://www.erdosproblems.com/4

00741 | EP-5

### Erdős Problem #5

O / open | Erdős 问题

Let $C\geq 0$. Is there an infinite sequence of $n_i$ such that\[\lim_{i\to \infty}\frac{p_{n_i+1}-p_{n_i}}{\log n_i}=C?\]

来源字段：https://www.erdosproblems.com/5

00742 | EP-51

### Erdős Problem #51

O / open | Erdős 问题

Is there an infinite set $A\subset \mathbb{N}$ such that for every $a\in A$ there is an integer $n$ such that $\phi(n)=a$, and yet if $n_a$ is the smallest such integer then $n_a/a\to \infty$ as $a\to\infty$?

来源字段：https://www.erdosproblems.com/51

00743 | EP-52

### Erdős Problem #52

O / open | Erdős 问题

Let $A$ be a finite set of integers. Is it true that for every $\epsilon>0$\[\max( \lvert A+A\rvert,\lvert AA\rvert)\gg_\epsilon \lvert A\rvert^{2-\epsilon}?\]

来源字段：https://www.erdosproblems.com/52

00744 | EP-520

### Erdős Problem #520

O / open | Erdős 问题

Let $f$ be a Rademacher multiplicative function: a random $\{-1,0,1\}$-valued multiplicative function, where for each prime $p$ we independently choose $f(p)\in \{-1,1\}$ uniformly at random, and for square-free integers $n$ we extend $f(p_1\cdots p_r)=f(p_1)\cdots f(p_r)$ (and $f(n)=0$ if $n$ is not squarefree). Does there exist some constant $c>0$ such that, almost surely,\[\limsup_{N\to \infty}\frac{\sum_{m\leq N}f(m)}{\sqrt{N\log\log N}}=c?\]

来源字段：https://www.erdosproblems.com/520

00745 | EP-530

### Erdős Problem #530

P / partially_solved | Erdős 问题

Let $\ell(N)$ be maximal such that in any finite set $A\subset \mathbb{R}$ of size $N$ there exists a Sidon subset $S$ of size $\ell(N)$ (i.e. the only solutions to $a+b=c+d$ in $S$ are the trivial ones). Determine the order of $\ell(N)$.

In particular, is it true that $\ell(N)\sim N^{1/2}$?

来源字段：https://www.erdosproblems.com/530

00746 | EP-532

### Erdős Problem #532: Number Theory, Ramsey Theory

S / solved | Erdős 问题

If $\mathbb{N}$ is 2-coloured then is there some infinite set $A\subseteq \mathbb{N}$ such that all finite subset sums\[\sum_{n\in S}n\](as $S$ ranges over all non-empty finite subsets of $A$) are monochromatic?

来源字段：https://www.erdosproblems.com/532

00747 | EP-534

### Erdős Problem #534: Number Theory, Intersecting Family

S / solved | Erdős 问题

What is the largest possible subset $A\subseteq\{1,\ldots,N\}$ which contains $N$ such that $\mathrm{gcd}(a,b)>1$ for all $a\neq b\in A$?

来源字段：https://www.erdosproblems.com/534

00748 | EP-535

### Erdős Problem #535

O / open | Erdős 问题

Let $r\geq 3$, and let $f_r(N)$ denote the size of the largest subset of $\{1,\ldots,N\}$ such that no subset of size $r$ has the same pairwise greatest common divisor between all elements. Estimate $f_r(N)$.

来源字段：https://www.erdosproblems.com/535

00749 | EP-537

### Erdős Problem #537: Number Theory

S / solved | Erdős 问题

Let $\epsilon>0$ and $N$ be sufficiently large. If $A\subseteq \{1,\ldots,N\}$ has $\lvert A\rvert \geq \epsilon N$ then must there exist $a_1,a_2,a_3\in A$ and distinct primes $p_1,p_2,p_3$ such that\[a_1p_1=a_2p_2=a_3p_3?\]

来源字段：https://www.erdosproblems.com/537

00750 | EP-538

### Erdős Problem #538

O / open | Erdős 问题

Let $r\geq 2$ and suppose that $A\subseteq\{1,\ldots,N\}$ is such that, for any $m$, there are at most $r$ solutions to $m=pa$ where $p$ is prime and $a\in A$. Give the best possible upper bound for\[\sum_{n\in A}\frac{1}{n}.\]

来源字段：https://www.erdosproblems.com/538

00751 | EP-53

### Erdős Problem #53: Number Theory, Additive Combinatorics

S / solved | Erdős 问题

Let $A$ be a finite set of integers. Is it true that, for every $k$, if $\lvert A\rvert$ is sufficiently large depending on $k$, then there are least $\lvert A\rvert^k$ many integers which are either the sum or product of distinct elements of $A$?

来源字段：https://www.erdosproblems.com/53

00752 | EP-540

### Erdős Problem #540: Number Theory

S / solved | Erdős 问题

Is it true that if $A\subseteq \mathbb{Z}/N\mathbb{Z}$ has size $\gg N^{1/2}$ then there exists some non-empty $S\subseteq A$ such that $\sum_{n\in S}n\equiv 0\pmod{N}$?

来源字段：https://www.erdosproblems.com/540

00753 | EP-541

### Erdős Problem #541: Number Theory

S / solved | Erdős 问题

Let $a_1,\ldots,a_p$ be (not necessarily distinct) residues modulo $p$, such that there exists some $r$ so that if $S\subseteq [p]$ is non-empty and\[\sum_{i\in S}a_i\equiv 0\pmod{p}\]then $\lvert S\rvert=r$. Must there be at most two distinct residues amongst the $a_i$?

来源字段：https://www.erdosproblems.com/541

00754 | EP-542

### Erdős Problem #542: Number Theory

S / solved | Erdős 问题

Is it true that if $A\subseteq\{1,\ldots,n\}$ is a set such that $[a,b]>n$ for all $a\neq b$, where $[a,b]$ is the least common multiple, then\[\sum_{a\in A}\frac{1}{a}\leq \frac{31}{30}?\]Is it true that there must be $\gg n$ many $m\leq n$ which do not divide any $a\in A$?

来源字段：https://www.erdosproblems.com/542

00755 | EP-54

### Erdős Problem #54: Number Theory, Ramsey Theory

S / solved | Erdős 问题

A set of integers $A$ is Ramsey $2$-complete if, whenever $A$ is $2$-coloured, all sufficiently large integers can be written as a monochromatic sum of elements of $A$.

Burr and Erdős [BuEr85] showed that there exists a constant $c>0$ such that it cannot be true that\[\lvert A\cap \{1,\ldots,N\}\rvert \leq c(\log N)^2\]for all large $N$ and that there exists a Ramsey $2$-complete $A$ such that for all large $N$\[\lvert A\cap \{1,\ldots,N\}\rvert < (2\log_2N)^3.\]Improve either of these bounds.

来源字段：https://www.erdosproblems.com/54

00756 | EP-552

### Erdős Problem #552

P / partially_solved | Erdős 问题

Determine the Ramsey number\[R(C_4,S_n),\]where $S_n=K_{1,n}$ is the star on $n+1$ vertices.

In particular, is it true that, for any $c>0$, there are infinitely many $n$ such that\[R(C_4,S_n)\leq n+\sqrt{n}-c?\]

来源字段：https://www.erdosproblems.com/552

00757 | EP-555

### Erdős Problem #555

P / partially_solved | Erdős 问题

Let $R_k(G)$ denote the minimal $m$ such that if the edges of $K_m$ are $k$-coloured then there is a monochromatic copy of $G$. Determine the value of\[R_k(C_{2n}).\]

来源字段：https://www.erdosproblems.com/555

00758 | EP-55

### Erdős Problem #55: Number Theory, Ramsey Theory

S / solved | Erdős 问题

A set of integers $A$ is Ramsey $r$-complete if, whenever $A$ is $r$-coloured, all sufficiently large integers can be written as a monochromatic sum of elements of $A$. Prove any non-trivial bounds about the growth rate of such an $A$ for $r>2$.

来源字段：https://www.erdosproblems.com/55

00759 | EP-562

### Erdős Problem #562

O / open | Erdős 问题

Let $R_r(n)$ denote the $r$-uniform hypergraph Ramsey number: the minimal $m$ such that if we $2$-colour all edges of the complete $r$-uniform hypergraph on $m$ vertices then there must be some monochromatic copy of the complete $r$-uniform hypergraph on $n$ vertices.

Prove that, for $r\geq 3$,\[\log_{r-1} R_r(n) \asymp_r n,\]where $\log_{r-1}$ denotes the $(r-1)$-fold iterated logarithm. That is, does $R_r(n)$ grow like\[2^{2^{\cdots n}}\]where the tower of exponentials has height $r-1$?

来源字段 : https://www.erdosproblems.com/562

00760 | EP-564

### Erdős Problem #564

O / open | Erdős 问题

Let $R_3(n)$ be the minimal $m$ such that if the edges of the $3$-uniform hypergraph on $m$ vertices are $2$-coloured then there is a monochromatic copy of the complete $3$-uniform hypergraph on $n$ vertices.

Is there some constant $c>0$ such that\[R_3(n) \geq 2^{2^{cn}}?\]

来源字段 : https://www.erdosproblems.com/564

00761 | EP-56

### Erdős Problem #56: Number Theory, Intersecting Family

S / solved | Erdős 问题

Let $N\geq p_k$ where $p_k$ is the $k$th prime. Suppose $A\subseteq \{1,\ldots,N\}$ is such that there are no $k+1$ elements of $A$ which are relatively prime. An example is the set of all multiples of the first $k$ primes. Is this the largest such set?

来源字段 : https://www.erdosproblems.com/56

00762 | EP-572

### Erdős Problem #572

P / partially_solved | Erdős 问题

Show that for $k\geq 3$\[\mathrm{ex}(n;C_{2k})\gg n^{1+\frac{1}{k}}.\]

来源字段 : https://www.erdosproblems.com/572

00763 | EP-586

### Erdős Problem #586: Is there a covering system such that no two of the moduli divide each other?

S / solved | Erdős 问题

Is there a covering system such that no two of the moduli divide each other?

来源字段 : https://www.erdosproblems.com/586

00764 | EP-587

### Erdős Problem #587: Number Theory

S / solved | Erdős 问题

What is the size of the largest $A\subseteq \{1,\ldots,N\}$ such that, for all $\emptyset\neq S\subseteq A$, $\sum_{n\in S}n$ is not a square?

来源字段 : https://www.erdosproblems.com/587

00765 | EP-591

### Erdős Problem #591

S / solved | Erdős 问题

Let $\alpha$ be the infinite ordinal $\omega^{\omega^2}$. Is it true that in any red/blue colouring of the edges of $K_\alpha$ there is either a red $K_\alpha$ or a blue $K_3$?

来源字段 : https://www.erdosproblems.com/591

00766 | EP-592

### Erdős Problem #592

P / partially_solved | Erdős 问题

Determine which countable ordinals $\beta$ have the property that, if $\alpha=\omega^{^\beta}$, then in any red/blue colouring of the edges of $K_\alpha$ there is either a red $K_\alpha$ or a blue $K_3$.

来源字段 : https://www.erdosproblems.com/592

00767 | EP-595

### Erdős Problem #595

P / partially_solved | Erdős 问题

Is there an infinite graph $G$ which contains no $K_4$ and is not the union of countably many triangle-free graphs?

来源字段 : https://www.erdosproblems.com/595

00768 | EP-60

### Erdős Problem #60

O / open | Erdős 问题

Does every graph on $n$ vertices with $>\mathrm{ex}(n;C_4)$ edges contain $\gg n^{1/2}$ many copies of $C_4$?

来源字段 : https://www.erdosproblems.com/60

00769 | EP-601

### Erdős Problem #601

P / partially_solved | Erdős 问题

For which limit ordinals $\alpha$ is it true that if $G$ is a graph with vertex set $\alpha$ then $G$ must have either an infinite path or independent set on a set of vertices with order type $\alpha$?

来源字段 : https://www.erdosproblems.com/601

00770 | EP-604

### Erdős Problem #604

P / partially_solved | Erdős 问题

Given $n$ distinct points $A\subset\mathbb{R}^2$ must there be a point $x\in A$ such that\[\#\{ d(x,y): y \in A\} \gg n^{1-o(1)}?\]Or even $\gg n/\sqrt{\log n}$?

来源字段：https://www.erdosproblems.com/604

00771 | EP-633

### Erdős Problem #633

S / solved | Erdős 问题

Classify those triangles which can only be cut into a square number of congruent triangles.

来源字段：https://www.erdosproblems.com/633

00772 | EP-634

### Erdős Problem #634

P / partially_solved | Erdős 问题

Find all $n$ such that there is at least one triangle which can be cut into $n$ congruent triangles.

来源字段：https://www.erdosproblems.com/634

00773 | EP-645

### Erdős Problem #645: Number Theory, Additive Combinatorics

S / solved | Erdős 问题

If $\mathbb{N}$ is 2-coloured then must there exist a monochromatic three-term arithmetic progression $x,x+d,x+2d$ such that $d>x$?

来源字段：https://www.erdosproblems.com/645

00774 | EP-646

### Erdős Problem #646: Number Theory, Factorials

S / solved | Erdős 问题

Let $p_1,\ldots,p_k$ be distinct primes. Are there infinitely many $n$ such that $n!$ is divisible by an even power of each of the $p_i$?

来源字段：https://www.erdosproblems.com/646

00775 | EP-647

### Erdős Problem #647: Number Theory

O / open | Erdős 问题

Let $\tau(n)$ count the number of divisors of $n$. Is there some $n>24$ such that\[\max_{m<n}(m+\tau(m))\leq n+2?\]

来源字段：https://www.erdosproblems.com/647

00776 | EP-648

### Erdős Problem #648: Number Theory

S / solved | Erdős 问题

Let $g(n)$ denote the largest $t$ such that there exist integers $2\leq a_1<a_2<\cdots <a_t <n$ such that\[P(a_1)>P(a_2)>\cdots >P(a_t)\]where $P(m)$ is the greatest prime factor of $m$. Estimate $g(n)$.

来源字段：https://www.erdosproblems.com/648

00777 | EP-649

### Erdős Problem #649: Number Theory

S / solved | Erdős 问题

Let $P(m)$ denote the greatest prime factor of $m$. Is it true that, for any two primes $p,q$, there exists some integer $n$ such that $P(n)=p$ and $P(n+1)=q$?

来源字段：https://www.erdosproblems.com/649

00778 | EP-650

### Erdős Problem #650

S / solved | Erdős 问题

Let $f(m)$ be such that if $A\subseteq \{1,\ldots,N\}$ has $\lvert A\rvert=m$ then every interval in $[1,\infty)$ of length $2N$ contains $\geq f(m)$ many distinct integers $b_1,\ldots,b_r$ where each $b_i$ is divisible by some $a_i\in A$, where $a_1,\ldots,a_r$ are distinct.

Estimate $f(m)$. In particular is it true that $f(m)\leq \sqrt{m}$?

来源字段：https://www.erdosproblems.com/650

00779 | EP-656

### Erdős Problem #656: Number Theory, Additive Combinatorics

S / solved | Erdős 问题

Let $A\subset \mathbb{N}$ be a set with positive upper density. Must there exist an infinite set $B\subseteq A$ and integer $t$ such that\[\{b_1+b_2: b_1\neq b_2\in B\}+t\subseteq A?\]

来源字段：https://www.erdosproblems.com/656

00780 | EP-663

### Erdős Problem #663

P / partially_solved | Erdős 问题

Let $k\geq 2$ and $q(n,k)$ denote the least prime which does not divide $\prod_{1\leq i\leq k}(n+i)$. Is it true that, if $k$ is fixed and $n$ is sufficiently large, we have\[q(n,k)<(1+o(1))\log n?\]

来源字段：https://www.erdosproblems.com/663

00781 | EP-665

### Erdős Problem #665

P / partially_solved | Erdős 问题

A pairwise balanced design for $\{1,\ldots,n\}$ is a collection of sets $A_1,\ldots,A_m\subseteq \{1,\ldots,n\}$ such that $2\leq \lvert A_i\rvert <n$ and every pair of distinct elements $x,y\in \{1,\ldots,n\}$ is contained in exactly one $A_i$.

Is there a constant $C>0$ and, for all large $n$, a pairwise balanced design such that\[\lvert A_i\rvert > n^{1/2}-C\]for all $1\leq i\leq m$?

来源字段：https://www.erdosproblems.com/665

00782 | EP-667

### Erdős Problem #667

P / partially_solved | Erdős 问题

Let $p,q\geq 1$ be fixed integers. We define $H(n)=H(N;p,q)$ to be the largest $m$ such that any graph on $n$ vertices where every set of $p$ vertices spans at least $q$ edges must contain a complete graph on $m$ vertices. Is\[c(p,q)=\liminf \frac{\log H(n)}{\log n}\]a strictly increasing function of $q$ for $1\leq q\leq \binom{p-1}{2}+1$?

来源字段：https://www.erdosproblems.com/667

00783 | EP-672

### Erdős Problem #672: Number Theory

O / open | Erdős 问题

Can the product of an arithmetic progression of positive integers $n,n+d,\ldots,n+(k-1)d$ of length $k\geq 4$ (with $(n,d)=1$) be a perfect power?

来源字段：https://www.erdosproblems.com/672

00784 | EP-673

### Erdős Problem #673: Number Theory, Divisors

S / solved | Erdős 问题

Let $1=d_1<\cdots <d_{\tau(n)}=n$ be the divisors of $n$ and\[G(n) = \sum_{1\leq i<\tau(n)}\frac{d_i}{d_{i+1}}.\]Is it true that $G(n)\to \infty$ for almost all $n$? Can one prove an asymptotic formula for $\sum_{n\leq X}G(n)$?

来源字段：https://www.erdosproblems.com/673

00785 | EP-674

### Erdős Problem #674: Are there any integer solutions to $x^xy^y=z^z$ with $x,y,z>1$?

S / solved | Erdős 问题

Are there any integer solutions to $x^xy^y=z^z$ with $x,y,z>1$?

来源字段：https://www.erdosproblems.com/674

00786 | EP-675

### Erdős Problem #675

P / partially_solved | Erdős 问题

We say that $A\subset \mathbb{N}$ has the translation property if, for every $n$, there exists some integer $t_n\geq 1$ such that, for all $1\leq a\leq n$,\[a\in A\quad\textrm{ if and only if }\quad a+t_n\in A.\]

- Does the set of the sums of two squares have the translation property?

- If we partition all primes into $P\sqcup Q$, such that each set contains $\gg x/\log x$ many primes $\leq x$ for all large $x$, then can the set of integers only divisible by primes from $P$ have the translation property?

- If $A$ is the set of squarefree numbers then how fast does the minimal such $t_n$ grow? Is it true that $t_n>\exp(n^c)$ for some constant $c>0$?

来源字段：https://www.erdosproblems.com/675

00787 | EP-676

### Erdős Problem #676

O / open | Erdős 问题

Is every sufficiently large integer of the form\[ap^2+b\]for some prime $p$ and integer $a\geq 1$ and $0\leq b<p$?

来源字段：https://www.erdosproblems.com/676

00788 | EP-677

### Erdős Problem #677

O / open | Erdős 问题

别名：Congruent Products of Consecutive Numbers [GUY-A15]

Let $M(n,k)=[n+1,\ldots,n+k]$ be the least common multiple of $\{n+1,\ldots,n+k\}$.

Is it true that for all $m\geq n+k$\[M(n,k) \neq M(m,k)?\]

来源字段：https://www.erdosproblems.com/677

00789 | EP-678

### Erdős Problem #678: \ldots,n+k\}$

S / solved | Erdős 问题

Let $M(n,k)=[n+1,\ldots,n+k]$ be the least common multiple of $\{n+1,\ldots,n+k\}$.

Are there infinitely many $m,n$ and $k\geq 3$ with $m\geq n+k$ such that\[M(n,k)>M(m,k+1)?\]

来源字段：https://www.erdosproblems.com/678

00790 | EP-679

### Erdős Problem #679

P / partially_solved | Erdős 问题

Let $\epsilon>0$ and $\omega(n)$ count the number of distinct prime factors of $n$. Are there infinitely many values of $n$ such that\[\omega(n-k) < (1+\epsilon)\frac{\log k}{\log\log k}\]for all $k<n$ which are sufficiently large depending on $\epsilon$ only?

Can one show the stronger version with\[\omega(n-k) < \frac{\log k}{\log\log k}+O(1)\]is false?

来源字段：https://www.erdosproblems.com/679

00791 | EP-68

### Erdős Problem #68

O / open | Erdős 问题

Is\[\sum_{n\geq 2}\frac{1}{n!-1}\]irrational?

来源字段：https://www.erdosproblems.com/68

00792 | EP-680

### Erdős Problem #680

O / open | Erdős 问题

Is it true that, for all sufficiently large $n$, there exists some $k$ such that\[p(n+k)>k^2+1,\]where $p(m)$ denotes the least prime factor of $m$?

Can one prove this is false if we replace $k^2+1$ by $e^{(1+\epsilon)\sqrt{k}}+C_\epsilon$, for all $\epsilon>0$, where $C_\epsilon>0$ is some constant?

来源字段：https://www.erdosproblems.com/680

00793 | EP-681

### Erdős Problem #681

O / open | Erdős 问题

Is it true that for all large $n$ there exists $k$ such that $n+k$ is composite and\[p(n+k)>k^2,\]where $p(m)$ is the least prime factor of $m$?

来源字段：https://www.erdosproblems.com/681

00794 | EP-682

### Erdős Problem #682: Number Theory, Primes

S / solved | Erdős 问题

Is it true that for almost all $n$ there exists some $m\in (p_n,p_{n+1})$ such that\[p(m) \geq p_{n+1}-p_n,\]where $p(m)$ denotes the least prime factor of $m$?

来源字段：https://www.erdosproblems.com/682

00795 | EP-683

### Erdős Problem #683

O / open | Erdős 问题

Is it true that for every $1\leq k\leq n$ the largest prime divisor of $\binom{n}{k}$, say $P(\binom{n}{k})$, satisfies\[P\left(\binom{n}{k}\right)\geq \min(n-k+1, k^{1+c})\]for some constant $c>0$?

来源字段：https://www.erdosproblems.com/683

00796 | EP-684

### Erdős Problem #684

P / partially_solved | Erdős 问题

For $0\leq k\leq n$ write\[\binom{n}{k} = uv\]where the only primes dividing $u$ are in $[2,k]$ and the only primes dividing $v$ are in $(k,n]$.

Let $f(n)$ be the smallest $k$ such that $u>n^2$. Give bounds for $f(n)$.

来源字段：https://www.erdosproblems.com/684

00797 | EP-685

### Erdős Problem #685

P / partially_solved | Erdős 问题

Let $\epsilon>0$ and $n$ be large depending on $\epsilon$. Is it true that for all $n^\epsilon<k\leq n^{1-\epsilon}$ the number of distinct prime divisors of $\binom{n}{k}$ is\[(1+o(1))k\sum_{k<p<n}\frac{1}{p}?\]Or perhaps even when $k \geq (\log n)^c$?

来源字段：https://www.erdosproblems.com/685

00798 | EP-686

### Erdős Problem #686

P / partially_solved | Erdős 问题

Can every integer $N\geq 2$ be written as\[N=\frac{\prod_{1\leq i\leq k}(m+i)}{\prod_{1\leq i\leq k}(n+i)}\]for some $k\geq 2$ and $m\geq n+k$?

来源字段：https://www.erdosproblems.com/686

00799 | EP-687

### Erdős Problem #687

P / partially_solved | Erdős 问题

Let $Y(x)$ be the maximal $y$ such that there exists a choice of congruence classes $a_p$ for all primes $p\leq x$ such that every integer in $[1,y]$ is congruent to at least one of the $a_p\pmod{p}$.

Give good estimates for $Y(x)$. In particular, can one prove that $Y(x)=o(x^2)$ or even $Y(x)\ll x^{1+o(1)}$?

来源字段：https://www.erdosproblems.com/687

00800 | EP-688

### Erdős Problem #688

P / partially_solved | Erdős 问题

Define $\epsilon_n$ to be maximal such that there exists some choice of congruence class $a_p$ for all primes $n^{\epsilon_n}<p\leq n$ such that every integer in $[1,n]$ satisfies at least one of the congruences $\equiv a_p\pmod{p}$.

Estimate $\epsilon_n$ - in particular is it true that $\epsilon_n=o(1)$?

来源字段：https://www.erdosproblems.com/688

00801 | EP-689

### Erdős Problem #689

P / partially_solved | Erdős 问题

Let $n$ be sufficiently large. Is there some choice of congruence class $a_p$ for all primes $2\leq p\leq n$ such that every integer in $[1,n]$ satisfies at least two of the congruences $\equiv a_p\pmod{p}$?

来源字段：https://www.erdosproblems.com/689

00802 | EP-690

### Erdős Problem #690

S / solved | Erdős 问题

Let $d_k(p)$ be the density of those integers whose $k$th smallest prime factor is $p$ (i.e. if $p_1<p_2<\cdots$ are the primes dividing $n$ then $p_k=p$).

For fixed $k\geq 1$ is $d_k(p)$ unimodular in $p$? That is, it first increases in $p$ until its maximum then decreases.

来源字段：https://www.erdosproblems.com/690

00803 | EP-691

### Erdős Problem #691

P / partially_solved | Erdős 问题

Given $A\subseteq \mathbb{N}$ let $M_A=\{ n \geq 1: a\mid n\textrm{ for some }a\in A\}$ be the set of multiples of $A$. Find a necessary and sufficient condition on $A$ for $M_A$ to have density $1$.

来源字段：https://www.erdosproblems.com/691

00804 | EP-692

### Erdős Problem #692: Number Theory, Divisors

S / solved | Erdős 问题

Let $\delta_1(n,m)$ be the density of the set of integers with exactly one divisor in $(n,m)$. Is $\delta_1(n,m)$ unimodular for $m>n+1$ (i.e. increases until some $m$ then decreases thereafter)? For fixed $n$, where does $\delta_1(n,m)$ achieve its maximum?

来源字段：https://www.erdosproblems.com/692

00805 | EP-693

### Erdős Problem #693

O / open | Erdős 问题

Let $k\geq 2$ and $n$ be sufficiently large depending on $k$. Let $A=\{a_1<a_2<\cdots \}$ be the set of those integers in $[n,n^k]$ which have a divisor in $(n,2n)$. Estimate\[\max_{i} a_{i+1}-a_i.\]Is this $\leq (\log n)^{O(1)}$?

来源字段：https://www.erdosproblems.com/693

00806 | EP-694

### Erdős Problem #694

S / solved | Erdős 问题

Let $f_{\max}(n)$ be the largest $m$ such that $\phi(m)=n$, and $f_{\min}(n)$ be the smallest such $m$, where $\phi$ is Euler's totient function. Investigate\[\max_{n\leq x}\frac{f_{\max}(n)}{f_{\min}(n)}\](where the maximum is restricted to those $n$ of the form $n=\phi(m)$ for some $m$.)

来源字段：https://www.erdosproblems.com/694

00807 | EP-695

### Erdős Problem #695

P / partially_solved | Erdős 问题

Let $p_1<p_2<\cdots$ be a sequence of primes such that $p_{i+1}\equiv 1\pmod{p_i}$. Is it true that\[\lim_k p_k^{1/k}=\infty?\]Does there exist such a sequence with\[p_k \leq \exp(k(\log k)^{1+o(1)})?\]

来源字段：https://www.erdosproblems.com/695

00808 | EP-696

### Erdős Problem #696

S / solved | Erdős 问题

Let $h(n)$ be the largest $\ell$ such that there is a sequence of primes $p_1<\cdots < p_\ell$ all dividing $n$ with $p_{i+1}\equiv 1\pmod{p_i}$. Let $H(n)$ be the largest $u$ such that there is a sequence of integers $d_1<\cdots < d_u$ all dividing $n$ with $d_{i+1}\equiv 1\pmod{d_i}$.

Estimate $h(n)$ and $H(n)$. Is it true that $H(n)/h(n)\to \infty$ for almost all $n$?

来源字段：https://www.erdosproblems.com/696

00809 | EP-697

### Erdős Problem #697: Number Theory, Divisors

S / solved | Erdős 问题

Let $\delta(m,\alpha)$ denote the density of the set of integers which are divisible by some $d\equiv 1\pmod{m}$ with $1<d<\exp(m^\alpha)$. Does there exist some $\beta\in (1,\infty)$ such that\[\lim_{m\to \infty}\delta(m,\alpha)\]is $0$ if $\alpha<\beta$ and $1$ if $\alpha>\beta$?

来源字段：https://www.erdosproblems.com/697

00810 | EP-698

### Erdős Problem #698: Number Theory, Binomial Coefficients

S / solved | Erdős 问题

Is there some $h(n)\to \infty$ such that for all $2\leq i<j\leq n/2$\[\textrm{gcd}\left( \binom{n}{i},\binom{n}{j}\right) \geq h(n)?\]

来源字段：https://www.erdosproblems.com/698

00811 | EP-699

### Erdős Problem #699: Number Theory, Binomial Coefficients

O / open | Erdős 问题

Is it true that for every $1\leq i<j\leq n/2$ there exists some prime $p\geq i$ such that\[p\mid \textrm{gcd}\left(\binom{n}{i}, \binom{n}{j}\right)?\]

来源字段：https://www.erdosproblems.com/699

00812 | EP-69

### Erdős Problem #69: Number Theory, Irrationality

S / solved | Erdős 问题

Is\[\sum_{n\geq 2}\frac{\omega(n)}{2^n}\]irrational? (Here $\omega(n)$ counts the number of distinct prime divisors of $n$.)

来源字段：https://www.erdosproblems.com/69

00813 | EP-6

### Erdős Problem #6: Let $d_n=p_{n+1}-p_n$. Are there infinitely many $n$ such that $d_n<d_{n+1}<d_{n+2}$?

S / solved | Erdős 问题

别名：Erdős $100 Problem on Increasing and Decreasing Gaps [GUY-A11]

Let $d_n=p_{n+1}-p_n$. Are there infinitely many $n$ such that $d_n<d_{n+1}<d_{n+2}$?

来源字段：https://www.erdosproblems.com/6

00814 | EP-700

### Erdős Problem #700

P / partially_solved | Erdős 问题

Let\[f(n)=\min_{1<k\leq n/2}\textrm{gcd}\left(n,\binom{n}{k}\right).\]

- Characterise those composite $n$ such that $f(n)=n/P(n)$, where $P(n)$ is the largest prime dividing $n$.

- Are there infinitely many composite $n$ such that $f(n)>n^{1/2}$?

- Is it true that, for every composite $n$,\[f(n) \ll_A \frac{n}{(\log n)^A}\]for every $A>0$?

来源字段：https://www.erdosproblems.com/700

00815 | EP-708

### Erdős Problem #708

O / open | Erdős 问题

Let $g(n)$ be minimal such that for any $A\subseteq [2,\infty)\cap \mathbb{N}$ with $\lvert A\rvert =n$ and any set $I$ of $\max(A)$ consecutive integers there exists some $B\subseteq I$ with $\lvert B\rvert=g(n)$ such that\[\prod_{a\in A} a \mid \prod_{b\in B}b.\]Is it true that\[g(n) \leq (2+o(1))n?\]Or perhaps even $g(n)\leq 2n$?

来源字段：https://www.erdosproblems.com/708

00816 | EP-709

### Erdős Problem #709

P / partially_solved | Erdős 问题

Let $f(n)$ be minimal such that, for any $A=\{a_1,\ldots,a_n\}\subseteq [2,\infty)\cap\mathbb{N}$ of size $n$, in any interval $I$ of $f(n)\max(A)$ consecutive integers there exist distinct $x_1,\ldots,x_n\in I$ such that $a_i\mid x_i$.

Obtain good bounds for $f(n)$, or even an asymptotic formula.

来源字段：https://www.erdosproblems.com/709

00817 | EP-710

### Erdős Problem #710

O / open | Erdős 问题

Let $f(n)$ be minimal such that in $(n,n+f(n))$ there exist distinct integers $a_1,\ldots,a_n$ such that $k\mid a_k$ for all $1\leq k\leq n$. Obtain an asymptotic formula for $f(n)$.

来源字段：https://www.erdosproblems.com/710

00818 | EP-711

### Erdős Problem #711

P / partially_solved | Erdős 问题

Let $f(n,m)$ be minimal such that in $(m,m+f(n,m))$ there exist distinct integers $a_1,\ldots,a_n$ such that $k\mid a_k$ for all $1\leq k\leq n$. Prove that\[\max_m f(n,m) \leq n^{1+o(1)}\]and that\[\max_m (f(n,m)-f(n,n))\to \infty.\]

来源字段：https://www.erdosproblems.com/711

00819 | EP-713

### Erdős Problem #713

O / open | Erdős 问题

Is it true that, for every bipartite graph $G$, there exists some $\alpha\in [1,2)$ and $c>0$ such that\[\mathrm{ex}(n;G)\sim cn^\alpha?\]Must $\alpha$ be rational?

来源字段：https://www.erdosproblems.com/713

00820 | EP-721

### Erdős Problem #721: Number Theory, Additive Combinatorics

S / solved | Erdős 问题

Let $W(3,k)$ be the van der Waerden number defined as the minimum $n$ such that in any red/blue colouring of $\{1,\ldots,n\}$ there exists either a red $3$-term arithmetic progression or a blue $k$-term arithmetic progression.

Give reasonable bounds for $W(3,k)$. In particular, give any non-trivial lower bounds for $W(3,k)$ and prove that $W(3,k) < \exp(k^c)$ for some constant $c<1$.

来源字段：https://www.erdosproblems.com/721

00821 | EP-724

### Erdős Problem #724

P / partially_solved | Erdős 问题

Let $f(n)$ be the maximum number of mutually orthogonal Latin squares of order $n$. Is it true that\[f(n) \gg n^{1/2}?\]

来源字段 : https://www.erdosproblems.com/724

00822 | EP-726

### Erdős Problem #726

O / open | Erdős 问题

As $n\to \infty$ ranges over integers\[\sum_{p\leq n}1_{n\in (p/2,p)\pmod{p}}\frac{1}{p}\sim \frac{\log\log n}{2}.\]

来源字段 : https://www.erdosproblems.com/726

00823 | EP-727

### Erdős Problem #727

P / partially_solved | Erdős 问题

Let $k\geq 2$. Does\[(n+k)!^2 \mid (2n)!\]for infinitely many $n$?

来源字段 : https://www.erdosproblems.com/727

00824 | EP-728

### Erdős Problem #728: Number Theory, Factorials

S / solved | Erdős 问题

Let $C>0$ and $\epsilon>0$ be sufficiently small. Are there infinitely many integers $a,b,n$ with $a\geq \epsilon n$ and $b\geq \epsilon n$ such that\[a! b! \mid n!(a+b-n)!\]and $a+b>n+C\log n$?

来源字段 : https://www.erdosproblems.com/728

00825 | EP-729

### Erdős Problem #729: Number Theory, Factorials

S / solved | Erdős 问题

Let $C>0$ be a constant. Are there infinitely many integers $a,b,n$ with $a+b> n+C\log n$ such that the denominator of\[\frac{n!}{a!b!}\]contains only primes $\ll_C 1$?

来源字段 : https://www.erdosproblems.com/729

00826 | EP-730

### Erdős Problem #730

S / solved | Erdős 问题

Are there infinitely many pairs of integers $n\neq m$ such that $\binom{2n}{n}$ and $\binom{2m}{m}$ have the same set of prime divisors?

来源字段 : https://www.erdosproblems.com/730

00827 | EP-731

### Erdős Problem #731

P / partially_solved | Erdős 问题

Find some reasonable function $f(n)$ such that, for almost all integers $n$, the least integer $m$ such that $m\nmid \binom{2n}{n}$ satisfies\[m\sim f(n).\]

来源字段 : https://www.erdosproblems.com/731

00828 | EP-740

### Erdős Problem #740

P / partially_solved | Erdős 问题

Let $\mathfrak{m}$ be an infinite cardinal and $G$ be a graph with chromatic number $\mathfrak{m}$. Let $r\geq 1$. Must $G$ contain a subgraph of chromatic number $\mathfrak{m}$ which does not contain any odd cycle of length $\leq r$?

来源字段 : https://www.erdosproblems.com/740

00829 | EP-748

### Erdős Problem #748: Number Theory

S / solved | Erdős 问题

Let $f(n)$ count the number of sum-free $A\subseteq \{1,\ldots,n\}$, i.e. $A$ contains no solutions to $a=b+c$ with $a,b,c\in A$. Is it true that\[f(n)=2^{(1+o(1))\frac{n}{2}}?\]

来源字段 : https://www.erdosproblems.com/748

00830 | EP-757

### Erdős Problem #757

P / partially_solved | Erdős 问题

Let $A\subset \mathbb{R}$ be a set of size $n$ such that every subset $B\subseteq A$ with $\lvert B\rvert =4$ has $\lvert B-B\rvert\geq 11$. Find the best constant $c>0$ such that $A$ must always contain a Sidon set of size $\geq cn$.

来源字段 : https://www.erdosproblems.com/757

00831 | EP-763

### Erdős Problem #763: Number Theory, Additive Combinatorics

S / solved | Erdős 问题

Let $A\subseteq \mathbb{N}$. Can there exist some constant $c>0$ such that\[\sum_{n\leq N} 1_A\ast 1_A(n) = cN+O(1)?\]

来源字段 : https://www.erdosproblems.com/763

00832 | EP-764

### Erdős Problem #764: Number Theory, Additive Combinatorics

S / solved | Erdős 问题

Let $A\subseteq \mathbb{N}$. Can there exist some constant $c>0$ such that\[\sum_{n\leq N} 1_A\ast 1_A\ast 1_A(n) = cN+O(1)?\]

来源字段 : https://www.erdosproblems.com/764

00833 | EP-768

### Erdős Problem #768

P / partially_solved | Erdős 问题

Let $A\subset\mathbb{N}$ be the set of $n$ such that for every prime $p\mid n$ there exists some $d\mid n$ with $d>1$ such that $d\equiv 1\pmod{p}$. Is it true that there exists some constant $c>0$ such that for all large $N$\[\frac{\lvert A\cap [1,N]\rvert}{N}=\exp(-(c+o(1))\sqrt{\log N}\log\log N).\]

来源字段：https://www.erdosproblems.com/768

00834 | EP-769

### Erdős Problem #769

P / partially_solved | Erdős 问题

Let $c(n)$ be minimal such that if $k\geq c(n)$ then the $n$-dimensional unit cube can be decomposed into $k$ homothetic $n$-dimensional cubes. Give good bounds for $c(n)$ - in particular, is it true that $c(n) \gg n^n$?

来源字段：https://www.erdosproblems.com/769

00835 | EP-770

### Erdős Problem #770

O / open | Erdős 问题

Let $h(n)$ be minimal such that $2^n-1,3^n-1,\ldots,h(n)^n-1$ are mutually coprime.

Does, for every prime $p$, the density $\delta_p$ of integers with $h(n)=p$ exist? Does $\liminf h(n)=\infty$? Is it true that if $p$ is the greatest prime such that $p-1\mid n$ and $p>n^\epsilon$ then $h(n)=p$?

来源字段：https://www.erdosproblems.com/770

00836 | EP-771

### Erdős Problem #771: Number Theory

S / solved | Erdős 问题

Let $f(n)$ be maximal such that, for every $m\geq 1$, there exists some $S\subseteq \{1,\ldots,n\}$ with $\lvert S\rvert=f(n)$ such that $m\neq \sum_{a\in A}a$ for all $A\subseteq S$.

Is it true that\[f(n) = \left(\frac{1}{2}+o(1)\right)\frac{n}{\log n}?\]

来源字段：https://www.erdosproblems.com/771

00837 | EP-772

### Erdős Problem #772: Number Theory, Sidon Sets

S / solved | Erdős 问题

Let $k\geq 1$ and $H_k(n)$ be the maximal $r$ such that if $A\subset\mathbb{N}$ has $\lvert A\rvert=n$ and $\| 1_A\ast 1_A\|_\infty \leq k$ then $A$ contains a Sidon set of size at least $r$.

Is it true that $H_k(n)/n^{1/2}\to \infty$? Or even $H_k(n) > n^{1/2+c}$ for some constant $c>0$?

来源字段：https://www.erdosproblems.com/772

00838 | EP-773

### Erdős Problem #773

P / partially_solved | Erdős 问题

What is the size of the largest Sidon subset $A\subseteq\{1,2^2,\ldots,N^2\}$? Is it $N^{1-o(1)}$?

来源字段：https://www.erdosproblems.com/773

00839 | EP-774

### Erdős Problem #774

P / partially_solved | Erdős 问题

We call $A\subset \mathbb{N}$ dissociated if $\sum_{n\in X}n\neq \sum_{m\in Y}m$ for all finite $X,Y\subset A$ with $X\neq Y$.

Let $A\subset \mathbb{N}$ be an infinite set. We call $A$ proportionately dissociated if every finite $B\subset A$ contains a dissociated set of size $\gg \lvert B\rvert$.

Is every proportionately dissociated set the union of a finite number of dissociated sets?

来源字段：https://www.erdosproblems.com/774

00840 | EP-779

### Erdős Problem #779: Number Theory, Primes

O / open | Erdős 问题

Let $n> 1$ and $p_1<\cdots<p_n$ denote the first $n$ primes. Let $P=\prod_{1\leq i\leq n}p_i$. Does there always exist some prime $p$ with $p_n<p<P$ such that $P+p$ is prime?

来源字段：https://www.erdosproblems.com/779

00841 | EP-783

### Erdős Problem #783

S / solved | Erdős 问题

Fix some constant $C>0$ and let $N$ be large. Let $A\subseteq \{2,\ldots,N\}$ be such that $(a,b)=1$ for all $a\neq b\in A$ and $\sum_{n\in A}\frac{1}{n}\leq C$.

What choice of such an $A$ minimises the number of integers $m\leq N$ not divisible by any $a\in A$?

来源字段：https://www.erdosproblems.com/783

00842 | EP-784

### Erdős Problem #784: Number Theory

S / solved | Erdős 问题

Let $C>0$. Does there exist a $c>0$ (depending on $C$) such that, for all sufficiently large $x$, if $A\subseteq [1,x]$ has $\sum_{n\in A}\frac{1}{n}\leq C$ then\[\#\{ m\leq x: a\nmid m\textrm{ for all }a\in A\}\gg\frac{x}{(\log x)^c}?\]

来源字段：https://www.erdosproblems.com/784

00843 | EP-786

### Erdős Problem #786

P / partially_solved | Erdős 问题

Let $\epsilon>0$. Is there some set $A\subset \mathbb{N}$ of density $>1-\epsilon$ such that $a_1\cdots a_r=b_1\cdots b_s$ with $a_i,b_j\in A$ can only hold when $r=s$?

Similarly, can one always find a set $A\subset\{1,\ldots,N\}$ with this property of size $\geq (1-o(1))N$?

来源字段：https://www.erdosproblems.com/786

00844 | EP-792

### Erdős Problem #792

P / partially_solved | Erdős 问题

Let $f(n)$ be maximal such that in any $A\subset \mathbb{Z}$ with $\lvert A\rvert=n$ there exists some sum-free subset $B\subseteq A$ with $\lvert B\rvert \geq f(n)$, so that there are no solutions to\[a+b=c\]with $a,b,c\in B$. Estimate $f(n)$.

来源字段：https://www.erdosproblems.com/792

00845 | EP-793

### Erdős Problem #793

S / solved | Erdős 问题

Let $F(n)$ be the maximum possible size of a subset $A\subseteq\{1,\ldots,n\}$ such that $a\nmid bc$ whenever $a,b,c\in A$ with $a\neq b$ and $a\neq c$. Is there a constant $C$ such that\[F(n)=\pi(n)+(C+o(1))n^{2/3}(\log n)^{-2}?\]

来源字段：https://www.erdosproblems.com/793

00846 | EP-795

### Erdős Problem #795: Number Theory

S / solved | Erdős 问题

Let $g(n)$ be the maximal size of $A\subseteq \{1,\ldots,n\}$ such that the products $\prod_{n\in S}n$ are distinct for all $S\subseteq A$. Is it true that\[g(n) \leq \pi(n)+\pi(n^{1/2})+o\left(\frac{x^{1/2}}{\log n}\right)?\]

来源字段：https://www.erdosproblems.com/795

00847 | EP-796

### Erdős Problem #796

P / partially_solved | Erdős 问题

Let $k\geq 2$ and let $g_k(n)$ be the largest possible size of $A\subseteq \{1,\ldots,n\}$ such that every $m$ has $<k$ solutions to $m=a_1a_2$ with $a_1<a_2\in A$.

Is it true that\[g_3(n)=\frac{\log\log n}{\log n}n+(c+o(1))\frac{n}{\log n}\]for some constant $c$?

来源字段：https://www.erdosproblems.com/796

00848 | EP-7

### Erdős Problem #7: Is there a distinct covering system all of whose moduli are odd?

O / open | Erdős 问题

Is there a distinct covering system all of whose moduli are odd?

来源字段：https://www.erdosproblems.com/7

00849 | EP-81

### Erdős Problem #81

O / open | Erdős 问题

Let $G$ be a chordal graph on $n$ vertices - that is, $G$ has no induced cycles of length greater than $3$. Can the edges of $G$ be partitioned into $n^2/6+O(n)$ many cliques?

来源字段：https://www.erdosproblems.com/81

00850 | EP-820

### Erdős Problem #820

P / partially_solved | Erdős 问题

Let $H(n)$ be the smallest integer $l$ such that there exist $k<l$ with $(k^n-1,l^n-1)=1$.

Is it true that $H(n)=3$ infinitely often? (That is, $(2^n-1,3^n-1)=1$ infinitely often?)

Estimate $H(n)$. Is it true that there exists some constant $c>0$ such that, for all $\epsilon>0$,\[H(n) > \exp(n^{(c-\epsilon)/\log\log n})\]for infinitely many $n$ and\[H(n) < \exp(n^{(c+\epsilon)/\log\log n})\]for all large enough $n$?

Does a similar upper bound hold for the smallest $k$ such that $(k^n-1,2^n-1)=1$?

来源字段：https://www.erdosproblems.com/820

00851 | EP-821

### Erdős Problem #821

P / partially_solved | Erdős 问题

Let $g(n)$ count the number of $m$ such that $\phi(m)=n$. Is it true that, for every $\epsilon>0$, there exist infinitely many $n$ such that\[g(n) > n^{1-\epsilon}?\]

来源字段：https://www.erdosproblems.com/821

00852 | EP-822

### Erdős Problem #822: Does the set of integers of the form $n+\phi(n)$ have positive (lower) density?

S / solved | Erdős 问题

Does the set of integers of the form $n+\phi(n)$ have positive (lower) density?

来源字段：https://www.erdosproblems.com/822

00853 | EP-823

### Erdős Problem #823: Number Theory

S / solved | Erdős 问题

Let $\alpha\geq 1$. Is there a sequence of integers $n_k,m_k$ such that $n_k/m_k\to \alpha$ and $\sigma(n_k)=\sigma(m_k)$ for all $k\geq 1$, where $\sigma$ is the sum of divisors function?

来源字段：https://www.erdosproblems.com/823

00854 | EP-824

### Erdős Problem #824

O / open | Erdős 问题

Let $h(x)$ count the number of integers $1\leq a<b<x$ such that $(a,b)=1$ and $\sigma(a)=\sigma(b)$, where $\sigma$ is the sum of divisors function.

Is it true that $h(x)>x^{2-o(1)}$?

来源字段：https://www.erdosproblems.com/824

00855 | EP-825

### Erdős Problem #825

S / solved | Erdős 问题

Is there an absolute constant $C>0$ such that every integer $n$ with $\sigma(n)>Cn$ is the distinct sum of proper divisors of $n$?

来源字段：https://www.erdosproblems.com/825

00856 | EP-828

### Erdős Problem #828

O / open | Erdős 问题

Is it true that, for any $a\in\mathbb{Z}$, there are infinitely many $n$ such that\[\phi(n) \mid n+a?\]

来源字段：https://www.erdosproblems.com/828

00857 | EP-836

### Erdős Problem #836

P / partially_solved | Erdős 问题

Let $r\geq 2$ and $G$ be a $r$-uniform hypergraph with chromatic number $3$ (that is, there is a $3$-colouring of the vertices of $G$ such that no edge is monochromatic).

Suppose any two edges of $G$ have a non-empty intersection. Must $G$ contain $O(r^2)$ many vertices? Must there be two edges which meet in $\gg r$ many vertices?

来源字段：https://www.erdosproblems.com/836

00858 | EP-839

### Erdős Problem #839

P / partially_solved | Erdős 问题

Let $1\leq a_1<a_2<\cdots$ be a sequence of integers such that no $a_i$ is the sum of consecutive $a_j$ for $j<i$. Is it true that\[\limsup \frac{a_n}{n}=\infty?\]Or even\[\lim \frac{1}{\log x}\sum_{a_n<x}\frac{1}{a_n}=0?\]

来源字段：https://www.erdosproblems.com/839

00859 | EP-840

### Erdős Problem #840

P / partially_solved | Erdős 问题

Let $f(N)$ be the size of the largest quasi-Sidon subset $A\subset\{1,\ldots,N\}$, where we say that $A$ is quasi-Sidon if\[\lvert A+A\rvert=(1+o(1))\binom{\lvert A\rvert}{2}.\]How does $f(N)$ grow?

来源字段：https://www.erdosproblems.com/840

00860 | EP-841

### Erdős Problem #841: Number Theory

S / solved | Erdős 问题

Let $t_n$ be minimal such that $\{n+1,\ldots,n+t_n\}$ contains a subset whose product with $n$ is a square number (and let $t_n=0$ if $n$ is itself square). Estimate $t_n$.

来源字段：https://www.erdosproblems.com/841

00861 | EP-843

### Erdős Problem #843: Are the squares Ramsey $2$-complete?

S / solved | Erdős 问题

Are the squares Ramsey $2$-complete?

That is, is it true that, in any 2-colouring of the square numbers, every sufficiently large $n\in \mathbb{N}$ can be written as a monochromatic sum of distinct squares?

来源字段：https://www.erdosproblems.com/843

00862 | EP-844

### Erdős Problem #844: for all $a,b\in A$, the product $ab$ is not squarefree

S / solved | Erdős 问题

Let $A\subseteq \{1,\ldots,N\}$ be such that, for all $a,b\in A$, the product $ab$ is not squarefree.

Is the maximum size of such an $A$ achieved by taking $A$ to be the set of even numbers and odd non-squarefree numbers?

来源字段：https://www.erdosproblems.com/844

00863 | EP-845

### Erdős Problem #845: Number Theory

S / solved | Erdős 问题

Let $C>0$. Is it true that the set of integers of the form\[n=b_1+\cdots+b_t\textrm{ with }b_1<\cdots<b_t\]where $b_i=2^{k_i}3^{l_i}$ for $1\leq i\leq t$ and $b_t\leq Cb_1$ has density $0$?

来源字段：https://www.erdosproblems.com/845

00864 | EP-848

### Erdős Problem #848: Number Theory

O / open | Erdős 问题

Is the maximum size of a set $A\subseteq \{1,\ldots,N\}$ such that $ab+1$ is never squarefree (for all $a,b\in A$) achieved by taking those $n\equiv 7\pmod{25}$?

来源字段：https://www.erdosproblems.com/848

00865 | EP-849

### Erdős Problem #849

P / partially_solved | Erdős 问题

别名：Singmaster's Conjecture [COMB-014]; Singmaster's Conjecture [NUM-001]; Singmaster's conjecture [OPG-60034]

Is it true that, for every integer $t\geq 1$, there is some integer $a$ such that\[\binom{n}{k}=a\](with $1\leq k\leq n/2$) has exactly $t$ solutions?

来源字段：https://www.erdosproblems.com/849

00866 | EP-850

### Erdős Problem #850

P / partially_solved | Erdős 问题

Can there exist two distinct integers $x$ and $y$ such that $x,y$ have the same prime factors, $x+1,y+1$ have the same prime factors, and $x+2,y+2$ also have the same prime factors?

来源字段：https://www.erdosproblems.com/850

00867 | EP-851

### Erdős Problem #851

S / solved | Erdős 问题

Let $\epsilon>0$. Is there some $r\ll_\epsilon 1$ such that the density of integers of the form $2^k+n$, where $k\geq 0$ and $n$ has at most $r$ prime divisors, is at least $1-\epsilon$?

来源字段：https://www.erdosproblems.com/851

00868 | EP-852

### Erdős Problem #852

P / partially_solved | Erdős 问题

Let $d_n=p_{n+1}-p_n$, where $p_n$ is the $n$th prime. Let $h(x)$ be maximal such that for some $n<x$ the numbers $d_n,d_{n+1},\ldots,d_{n+h(x)-1}$ are all distinct. Estimate $h(x)$. In particular, is it true that\[h(x)>(\log x)^c\]for some constant $c>0$, and\[h(x)=o(\log x)?\]

来源字段：https://www.erdosproblems.com/852

00869 | EP-853

### Erdős Problem #853

O / open | Erdős 问题

Let $d_n=p_{n+1}-p_n$, where $p_n$ is the $n$th prime. Let $r(x)$ be the smallest even integer $t$ such that $d_n=t$ has no solutions for $n\leq x$.

Is it true that $r(x)\to \infty$? Or even $r(x)/\log x \to \infty$?

来源字段：https://www.erdosproblems.com/853

00870 | EP-854

### Erdős Problem #854

P / partially_solved | Erdős 问题

Let $n_k$ denote the $k$th primorial, i.e. the product of the first $k$ primes.

If $1=a_1<a_2<\cdots a_{\phi(n_k)}=n_k-1$ is the sequence of integers coprime to $n_k$, then estimate the smallest even integer not of the form $a_{i+1}-a_i$. Are there\[\gg \max_i (a_{i+1}-a_i)\]many even integers of the form $a_{j+1}-a_j$?

来源字段：https://www.erdosproblems.com/854

00871 | EP-855

### Erdős Problem #855: Number Theory, Primes

O / open | Erdős 问题

If $\pi(x)$ counts the number of primes in $[1,x]$ then is it true that (for large $x$ and $y$)\[\pi(x+y) \leq \pi(x)+\pi(y)?\]

来源字段：https://www.erdosproblems.com/855

00872 | EP-856

### Erdős Problem #856

P / partially_solved | Erdős 问题

Let $k\geq 3$ and $f_k(N)$ be the maximum value of $\sum_{n\in A}\frac{1}{n}$, where $A$ ranges over all subsets of $\{1,\ldots,N\}$ which contain no subset of size $k$ with the same pairwise least common multiple.

Estimate $f_k(N)$.

来源字段：https://www.erdosproblems.com/856

00873 | EP-858

### Erdős Problem #858

S / solved | Erdős 问题

Let $A\subseteq \{1,\ldots,N\}$ be such that there is no solution to $at=b$ with $a,b\in A$ and the smallest prime factor of $t$ is $>a$. Estimate the maximum of\[\frac{1}{\log N}\sum_{n\in A}\frac{1}{n}.\]

来源字段：https://www.erdosproblems.com/858

00874 | EP-859

### Erdős Problem #859

P / partially_solved | Erdős 问题

Let $t\geq 1$ and let $d_t$ be the density of the set of integers $n\in\mathbb{N}$ for which $t$ can be represented as the sum of distinct divisors of $n$.

Do there exist constants $c_1,c_2>0$ such that\[d_t \sim \frac{c_1}{(\log t)^{c_2}}\]as $t\to \infty$?

来源字段：https://www.erdosproblems.com/859

00875 | EP-860

### Erdős Problem #860

P / partially_solved | Erdős 问题

Let $h(n)$ be such that, for any $m\geq 1$, in the interval $(m,m+h(n))$ there exist distinct integers $a_i$ for $1\leq i\leq \pi(n)$ such that $p_i\mid a_i$, where $p_i$ denotes the $i$th prime.

Estimate $h(n)$.

来源字段：https://www.erdosproblems.com/860

00876 | EP-861

### Erdős Problem #861: Number Theory, Sidon Sets

S / solved | Erdős 问题

Let $f(N)$ be the size of the largest Sidon subset of $\{1,\ldots,N\}$ and $A(N)$ be the number of Sidon subsets of $\{1,\ldots,N\}$. Is it true that\[A(N)/2^{f(N)}\to \infty?\]Is it true that\[A(N) = 2^{(1+o(1))f(N)}?\]

来源字段：https://www.erdosproblems.com/861

00877 | EP-862

### Erdős Problem #862: Number Theory, Sidon Sets

S / solved | Erdős 问题

Let $A_1(N)$ be the number of maximal Sidon subsets of $\{1,\ldots,N\}$. Is it true that\[A_1(N) < 2^{o(N^{1/2})}?\]Is it true that\[A_1(N) > 2^{N^c}\]for some constant $c>0$?

来源字段：https://www.erdosproblems.com/862

00878 | EP-865

### Erdős Problem #865

S / solved | Erdős 问题

There exists a constant $C>0$ such that, for all large $N$, if $A\subseteq \{1,\ldots,N\}$ has size at least $\frac{5}{8}N+C$ then there are distinct $a,b,c\in A$ such that $a+b,a+c,b+c\in A$.

来源字段：https://www.erdosproblems.com/865

00879 | EP-866

### Erdős Problem #866

O / open | Erdős 问题

Let $k\geq 3$ and $g_k(N)$ be minimal such that if $A\subseteq \{1,\ldots,2N\}$ has $\lvert A\rvert \geq N+g_k(N)$ then there exist integers $b_1,\ldots,b_k$ such that all $\binom{k}{2}$ pairwise sums are in $A$ (but the $b_i$ themselves need not be in $A$).

Estimate $g_k(N)$.

来源字段：https://www.erdosproblems.com/866

00880 | EP-868

### Erdős Problem #868: Number Theory, Additive Basis

S / solved | Erdős 问题

If $A$ is an additive basis of order $2$, and $1_A\ast 1_A(n)\to \infty$ as $n\to \infty$, then must $A$ contain a minimal additive basis of order $2$? (i.e. such that deleting any element creates infinitely many $n\not\in A+A$)

What if $1_A\ast 1_A(n) >\epsilon \log n$ (for all large $n$, for arbitrary fixed $\epsilon>0$)?

来源字段：https://www.erdosproblems.com/868

00881 | EP-869

### Erdős Problem #869

S / solved | Erdős 问题

If $A_1,A_2$ are disjoint additive bases of order $2$ (i.e. $A_i+A_i$ contains all large integers) then must $A=A_1\cup A_2$ contain a minimal additive basis of order $2$ (one such that deleting any element creates infinitely many $n\not\in A+A$)?

来源字段：https://www.erdosproblems.com/869

00882 | EP-870

### Erdős Problem #870

P / partially_solved | Erdős 问题

Let $k\geq 3$ and $A$ be an additive basis of order $k$. Does there exist a constant $c=c(k)>0$ such that if $r(n)\geq c\log n$ for all large $n$ then $A$ must contain a minimal basis of order $k$? (Here $r(n)$ counts the number of representations of $n$ as the sum of at most $k$ elements from $A$.)

来源字段：https://www.erdosproblems.com/870

00883 | EP-871

### Erdős Problem #871: Number Theory, Additive Basis

S / solved | Erdős 问题

Let $A$ be an additive basis of order $2$, and suppose $1_A\ast 1_A(n)\to \infty$ as $n\to \infty$. Can $A$ be partitioned into two disjoint additive bases of order $2$?

来源字段：https://www.erdosproblems.com/871

00884 | EP-872

### Erdős Problem #872

P / partially_solved | Erdős 问题

Consider the two-player game in which players alternately choose integers from $\{2,3,\ldots,n\}$ to be included in some set $A$ (the same set for both players) such that no $a\mid b$ for $a\neq b\in A$.

The game ends when no legal move is possible. One player wants the game to last as long as possible, the other wants the game to end quickly. How long can the game be guaranteed to last for?

At least $\epsilon n$ moves? (For $\epsilon>0$ and $n$ sufficiently large.) At least $(1-\epsilon)\frac{n}{2}$ moves?

来源字段：https://www.erdosproblems.com/872

00885 | EP-874

### Erdős Problem #874: Number Theory, Additive Combinatorics

S / solved | Erdős 问题

Let $k(N)$ denote the size of the largest set $A\subseteq \{1,\ldots,N\}$ such that the sets\[S_r = \{ a_1+\cdots +a_r: a_1<\cdots<a_r\in A\}\]are disjoint for distinct $r\geq 1$. Estimate $k(N)$ - in particular, is it true that $k(N)\sim 2N^{1/2}$?

来源字段：https://www.erdosproblems.com/874

00886 | EP-878

### Erdős Problem #878

P / partially_solved | Erdős 问题

If $n=\prod_{1\leq i\leq t} p_i^{k_i}$ is the factorisation of $n$ into distinct primes then let\[f(n)=\sum p_i^{\ell_i},\]where $\ell_i$ is chosen such that $n\in [p_i^{\ell_i},p_i^{\ell_i+1})$. Furthermore, let\[F(n)=\max \sum_{i} a_i\]where the maximum is taken over all distinct $a_1,\ldots,a_k\leq n$ such that $(a_i,a_j)=1$ for $i\neq j$ and all prime factors of each $a_i$ are prime factors of $n$.

Is it true that, for almost all $n$,\[f(n)=o(n\log\log n)\]and\[F(n) \gg n\log\log n?\]Is it true that\[\max_{n\leq x}f(n)\sim \frac{x\log x}{\log\log x}?\]Is it true that (for all $x$, or perhaps just for all large $x$)\[\max_{n\leq x}f(n)=\max_{n\leq x}F(n)?\]Find an asymptotic formula for the number of $n<x$ such that $f(n)=F(n)$. Find an asymptotic formula for\[H(x)=\sum_{n<x}\frac{f(n)}{n}.\]Is it true that\[H(x) \ll x\log\log\log\log x?\]

来源字段：https://www.erdosproblems.com/878

00887 | EP-879

### Erdős Problem #879

P / partially_solved | Erdős 问题

Call a set $S\subseteq \{1,\ldots,n\}$ admissible if $(a,b)=1$ for all $a\neq b\in S$. Let\[G(n) = \max_{S\subseteq \{1,\ldots,n\}} \sum_{a\in S}a\]and\[H(n)=\sum_{p<n}p+ n\pi(n^{1/2}).\]Is it true that\[G(n) >H(n)-n^{1+o(1)}?\]Is it true that, for every $k\geq 2$, if $n$ is sufficiently large then the admissible set which maximises $G(n)$ contains at least one integer with at least $k$ prime factors?

来源字段：https://www.erdosproblems.com/879

00888 | EP-880

### Erdős Problem #880: Number Theory, Additive Basis

S / solved | Erdős 问题

Let $A\subset\mathbb{N}$ be an additive basis of order $k$. Let $B=\{b_1<b_2<\cdots\}$ be the set of integers which are the sum of $k$ or fewer distinct $a\in A$. Is it true that $b_{n+1}-b_n=O(1)$? (Where the implied constant may depend on both $A$ and $k$.)

来源字段：https://www.erdosproblems.com/880

00889 | EP-882

### Erdős Problem #882: Number Theory, Primitive Sets

S / solved | Erdős 问题

What is the size of the largest $A\subseteq \{1,\ldots,n\}$ such that in the set\[\left\{ \sum_{a\in S} a: \emptyset\neq S\subseteq A\right\}\]no two distinct elements divide each other?

来源字段：https://www.erdosproblems.com/882

00890 | EP-883

### Erdős Problem #883

P / partially_solved | Erdős 问题

For $A\subseteq \{1,\ldots,n\}$ let $G(A)$ be the graph with vertex set $A$, where two integers are joined by an edge if they are coprime.

Is it true that if\[\lvert A\rvert >\lfloor\tfrac{n}{2}\rfloor+\lfloor\tfrac{n}{3}\rfloor-\lfloor\tfrac{n}{6}\rfloor\]then $G(A)$ contains all odd cycles of length $\leq \frac{n}{3}+1$?

Is it true that, for every $\ell\geq 1$, if $n$ is sufficiently large and\[\lvert A\rvert >\lfloor\tfrac{n}{2}\rfloor+\lfloor\tfrac{n}{3}\rfloor-\lfloor\tfrac{n}{6}\rfloor\]then $G(A)$ must contain a complete $(1,\ell,\ell)$ triparite graph on $2\ell+1$ vertices?

来源字段：https://www.erdosproblems.com/883

00891 | EP-884

### Erdős Problem #884

S / solved | Erdős 问题

Is it true that, for any $n$, if $d_1<\cdots <d_t$ are the divisors of $n$, then\[\sum_{1\leq i<j\leq t}\frac{1}{d_j-d_i} \ll 1+\sum_{1\leq i<t}\frac{1}{d_{i+1}-d_i},\]where the implied constant is absolute?

来源字段：https://www.erdosproblems.com/884

00892 | EP-885

### Erdős Problem #885

P / partially_solved | Erdős 问题

For integer $n\geq 1$ we define the factor difference set of $n$ by\[D(n) = \{\lvert a-b\rvert: n=ab\}.\]Is it true that, for every $k\geq 1$, there exist integers $N_1<\cdots<N_k$ such that\[\lvert \cap_i D(N_i)\rvert \geq k?\]

来源字段：https://www.erdosproblems.com/885

00893 | EP-886

### Erdős Problem #886

P / partially_solved | Erdős 问题

Let $\epsilon>0$. Is it true that, for all large $n$, the number of divisors of $n$ in $(n^{1/2},n^{1/2}+n^{1/2-\epsilon})$ is $O_\epsilon(1)$?

来源字段：https://www.erdosproblems.com/886

00894 | EP-887

### Erdős Problem #887

P / partially_solved | Erdős 问题

Is there an absolute constant $K$ such that, for every $C>0$, if $n$ is sufficiently large then $n$ has at most $K$ divisors in $(n^{1/2},n^{1/2}+C n^{1/4})$.

来源字段：https://www.erdosproblems.com/887

00895 | EP-888

### Erdős Problem #888

S / solved | Erdős 问题

What is the size of the largest $A\subseteq \{1,\ldots,n\}$ such that if $a\leq b\leq c\leq d\in A$ are such that $abcd$ is a square then $ad=bc$?

来源字段：https://www.erdosproblems.com/888

00896 | EP-889

### Erdős Problem #889

P / partially_solved | Erdős 问题

For $k\geq 0$ and $n\geq 1$ let $v(n,k)$ count the prime factors of $n+k$ which do not divide $n+i$ for $0\leq i<k$. Equivalently, $v(n,k)$ counts the number of prime factors of $n+k$ which are $>k$.

Is it true that\[v_0(n)=\max_{k\geq 0}v(n,k)\to \infty\]as $n\to \infty$?

来源字段：https://www.erdosproblems.com/889

00897 | EP-89

### Erdős Problem #89

O / open | Erdős 问题

Does every set of $n$ distinct points in $\mathbb{R}^2$ determine $\gg n/\sqrt{\log n}$ many distinct distances?

来源字段：https://www.erdosproblems.com/89

00898 | EP-890

### Erdős Problem #890

P / partially_solved | Erdős 问题

If $\omega_k(n)$ counts the number of distinct prime factors of $n$ which are $>k$, then is it true that, for every $k\geq 1$,\[\liminf_{n\to \infty}\sum_{0\leq i<k}\omega_k(n+i)\leq k?\]Is it true that\[\limsup_{n\to \infty}\left(\sum_{0\leq i<k}\omega(n+i)\right) \frac{\log\log n}{\log n}=1,\]where $\omega$ counts the number of distinct prime factors without restriction?

来源字段：https://www.erdosproblems.com/890

00899 | EP-891

### Erdős Problem #891

P / partially_solved | Erdős 问题

Let $2=p_1<p_2<\cdots$ be the primes and $k\geq 2$. Is it true that, for all sufficiently large $n$, there must exist an integer in $[n,n+p_1\cdots p_k)$ with $>k$ many prime factors?

来源字段：https://www.erdosproblems.com/891

00900 | EP-892

### Erdős Problem #892

P / partially_solved | Erdős 问题

Is there a necessary and sufficient condition for a sequence of integers $b_1<b_2<\cdots$ that ensures there exists a primitive sequence $a_1<a_2<\cdots$ (i.e. no element divides another) with $a_n \ll b_n$ for all $n$?

In particular, is this always possible if there are no non-trivial solutions to $(b_i,b_j)=b_k$?

Similarly, find necessary and sufficient conditions on a sequence $n_1<n_2<\cdots$ that ensure there exists a primitive set $A$ such that\[\lvert A\cap [1,2^{n_i}]\rvert \gg 2^{n_i}\]for every $i$.

来源字段：https://www.erdosproblems.com/892

00901 | EP-893

### Erdős Problem #893

P / partially_solved | Erdős 问题

If $\tau(n)$ counts the divisors of $n$ then let\[f(n)=\sum_{1\leq k\leq n}\tau(2^k-1).\]Does $f(2n)/f(n)$ tend to a limit?

来源字段：https://www.erdosproblems.com/893

00902 | EP-894

### Erdős Problem #894:

S / solved | Erdős 问题

Let $A=\{n_1<n_2<\cdots\}\subset \mathbb{N}$ be a lacunary sequence (so there exists some $\epsilon>0$ with $n_{k+1}\geq (1+\epsilon)n_k$ for all $k$).

Is it true that there must exist a finite colouring of $\mathbb{N}$ with no monochromatic solutions to $a-b\in A$?

来源字段：https://www.erdosproblems.com/894

00903 | EP-897

### Erdős Problem #897: Number Theory

S / solved | Erdős 问题

Let $f(n)$ be an additive function (so that $f(ab)=f(a)+f(b)$ if $(a,b)=1$) such that\[\limsup_{p,k}\frac{f(p^k)}{\log p^k}=\infty.\]Is it true that\[\limsup_n \frac{f(n+1)-f(n)}{\log n}=\infty?\]Or perhaps even\[\limsup_n \frac{f(n+1)}{f(n)}=\infty?\]

来源字段：https://www.erdosproblems.com/897

00904 | EP-8

### Erdős Problem #8: Number Theory, Covering Systems

S / solved | Erdős 问题

For any finite colouring of the integers is there a covering system all of whose moduli are monochromatic?

来源字段：https://www.erdosproblems.com/8

00905 | EP-9

### Erdős Problem #9

O / open | Erdős 问题

Let $A$ be the set of all odd integers $\geq 1$ not of the form $p+2^{k}+2^l$ (where $k,l\geq 0$ and $p$ is prime). Is the upper density of $A$ positive?

来源字段：https://www.erdosproblems.com/9

00906 | EP-91

### Erdős Problem #91

O / open | Erdős 问题

Let $n$ be a sufficiently large integer. Suppose $A\subset \mathbb{R}^2$ has $\lvert A\rvert=n$ and minimises the number of distinct distances between points in $A$. Prove that there are at least two (and probably many) such $A$ which are non-similar.

来源字段：https://www.erdosproblems.com/91

00907 | EP-910

### Erdős Problem #910: Topology

S / solved | Erdős 问题

Does every connected set in $\mathbb{R}^n$ contain a connected subset which is not a point and not homeomorphic to the original set?

If $n\geq 2$ does every connected set in $\mathbb{R}^n$ contain more than $2^{\aleph_0}$ many connected subsets?

来源字段：https://www.erdosproblems.com/910

00908 | EP-912

### Erdős Problem #912

P / partially_solved | Erdős 问题

If\[n! = \prod_i p_i^{k_i}\]is the factorisation into distinct primes then let $h(n)$ count the number of distinct exponents $k_i$.

Prove that there exists some $c>0$ such that\[h(n) \sim c \left(\frac{n}{\log n}\right)^{1/2}\]as $n\to \infty$.

来源字段：https://www.erdosproblems.com/912

00909 | EP-913

### Erdős Problem #913

O / open | Erdős 问题

Are there infinitely many $n$ such that if\[n(n+1) = \prod_i p_i^{k_i}\]is the factorisation into distinct primes then all exponents $k_i$ are distinct?

来源字段：https://www.erdosproblems.com/913

00910 | EP-928

### Erdős Problem #928

P / partially_solved | Erdős 问题

Let $\alpha,\beta\in (0,1)$ and let $P(n)$ denote the largest prime divisor of $n$. Does the density of integers $n$ such that $P(n)<n^{\alpha}$ and $P(n+1)<(n+1)^\beta$ exist?

来源字段：https://www.erdosproblems.com/928

00911 | EP-929

### Erdős Problem #929

P / partially_solved | Erdős 问题

Let $k\geq 2$ be large and let $S(k)$ be the minimal $x$ such that there is a positive density set of $n$ where\[n+1,n+2,\ldots,n+k\]are all divisible by primes $\leq x$.

Estimate $S(k)$ - in particular, is it true that $S(k)\geq k^{1-o(1)}$?

来源字段：https://www.erdosproblems.com/929

00912 | EP-930

### Erdős Problem #930

P / partially_solved | Erdős 问题

Is it true that, for every $r$, there is a $k$ such that if $I_1,\ldots,I_r$ are disjoint intervals of consecutive integers, all of length at least $k$, then\[\prod_{1\leq i\leq r}\prod_{m\in I_i}m\]is not a perfect power?

来源字段：https://www.erdosproblems.com/930

00913 | EP-931

### Erdős Problem #931

O / open | Erdős 问题

Let $k_1\geq k_2\geq 3$. Are there only finitely many $n_2\geq n_1+k_1$ such that\[\prod_{1\leq i\leq k_1}(n_1+i)\textrm{ and }\prod_{1\leq j\leq k_2}(n_2+j)\]have the same prime factors?

来源字段：https://www.erdosproblems.com/931

00914 | EP-932

### Erdős Problem #932

O / open | Erdős 问题

Let $p_k$ denote the $k$th prime. For infinitely many $r$ there are at least two integers $p_r<n<p_{r+1}$ all of whose prime factors are $<p_{r+1}-p_r$.

来源字段：https://www.erdosproblems.com/932

00915 | EP-933

### Erdős Problem #933

P / partially_solved | Erdős 问题

If $n(n+1)=2^k3^l+m$, where $(m,6)=1$, then is it true that\[\limsup_{n\to \infty} \frac{2^k3^l}{n\log n}=\infty?\]

来源字段：https://www.erdosproblems.com/933

00916 | EP-934

### Erdős Problem #934

P / partially_solved | Erdős 问题

Let $h_t(d)$ be minimal such that every graph $G$ with $h_t(d)$ edges and maximal degree $\leq d$ contains two edges whose shortest path between them has length $\geq t$.

Estimate $h_t(d)$.

来源字段：https://www.erdosproblems.com/934

00917 | EP-935

### Erdős Problem #935

P / partially_solved | Erdős 问题

For any integer $n=\prod p^{k_p}$ let $Q_2(n)$ be the powerful part of $n$, so that\[Q_2(n) = \prod_{\substack{p\\ k_p\geq 2}}p^{k_p}.\]Is it true that, for every $\epsilon>0$ and $\ell\geq 1$, if $n$ is sufficiently large then\[Q_2(n(n+1)\cdots(n+\ell))<n^{2+\epsilon}?\]If $\ell\geq 2$ then is\[\limsup_{n\to \infty}\frac{Q_2(n(n+1)\cdots(n+\ell))}{n^2}\]infinite?

If $\ell\geq 2$ then is\[\lim_{n\to \infty}\frac{Q_2(n(n+1)\cdots(n+\ell))}{n^{\ell+1}}=0?\]

来源字段：https://www.erdosproblems.com/935

00918 | EP-937

### Erdős Problem #937: Number Theory, Powerful

S / solved | Erdős 问题

Are there infinitely many four-term arithmetic progressions of coprime powerful numbers (i.e. if $p\mid n$ then $p^2\mid n$)?

来源字段：https://www.erdosproblems.com/937

00919 | EP-939

### Erdős Problem #939

P / partially_solved | Erdős 问题

Let $r\geq 2$. An $r$-powerful number $n$ is one such that if $p\mid n$ then $p^r\mid n$.

If $r\geq 4$ then can the sum of $r-2$ coprime $r$-powerful numbers ever be itself $r$-powerful? Are there at most finitely many such solutions?

Are there infinitely many triples of coprime $3$-powerful numbers $a,b,c$ such that $a+b=c$?

来源字段：https://www.erdosproblems.com/939

00920 | EP-940

### Erdős Problem #940

P / partially_solved | Erdős 问题

Let $r\geq 3$. A number $n$ is $r$-powerful if for every prime $p$ which divides $n$ we have $p^r\mid n$.

Are there infinitely many integers which are not the sum of at most $r$ many $r$-powerful numbers? Does the set of integers which are the sum of at most $r$ $r$-powerful numbers have density $0$?

来源字段：https://www.erdosproblems.com/940

00921 | EP-941

### Erdős Problem #941: Number Theory, Powerful

S / solved | Erdős 问题

Are all large integers the sum of at most three powerful numbers (i.e. if $p\mid n$ then $p^2\mid n$)?

来源字段：https://www.erdosproblems.com/941

00922 | EP-942

### Erdős Problem #942

P / partially_solved | Erdős 问题

Let $h(n)$ count the number of powerful (if $p\mid m$ then $p^2\mid m$) integers in $[n^2,(n+1)^2)$. Estimate $h(n)$. In particular is there some constant $c>0$ such that\[h(n) < (\log n)^{c+o(1)}\]and, for infinitely many $n$,\[h(n) >(\log n)^{c-o(1)}?\]

来源字段：https://www.erdosproblems.com/942

00923 | EP-944

### Erdős Problem #944

P / partially_solved | Erdős 问题

A critical vertex, edge, or set of edges, is one whose deletion lowers the chromatic number.

Let $k\geq 4$ and $r\geq 1$. Must there exist a graph $G$ with chromatic number $k$ such that every vertex is critical, yet every critical set of edges has size $>r$?

来源字段：https://www.erdosproblems.com/944

00924 | EP-945

### Erdős Problem #945

P / partially_solved | Erdős 问题

Let $F(x)$ be the maximal $k$ such that there exist $n+1,\ldots,n+k\leq x$ with $\tau(n+1),\ldots,\tau(n+k)$ all distinct (where $\tau(m)$ counts the divisors of $m$). Estimate $F(x)$. In particular, is it true that\[F(x) \leq (\log x)^{O(1)}?\]In other words, is there a constant $C>0$ such that, for all large $x$, every interval $[x,x+(\log x)^C]$ contains two integers with the same number of divisors?

来源字段：https://www.erdosproblems.com/945

00925 | EP-946

### Erdős Problem #946: Number Theory, Divisors

S / solved | Erdős 问题

Are there infinitely many $n$ such that $\tau(n)=\tau(n+1)$, where $\tau$ is the divisor function?

来源字段：https://www.erdosproblems.com/946

00926 | EP-947

### Erdős Problem #947: Number Theory, Covering Systems

S / solved | Erdős 问题

There is no exact covering system - that is, a finite collection of congruence classes $a_i\pmod{n_i}$ with distinct $n_i$ such that every integer satisfies exactly one of these congruence classes.

来源字段：https://www.erdosproblems.com/947

00927 | EP-948

### Erdős Problem #948

S / solved | Erdős 问题

Is there a function $f(n)$ and a $k$ such that in any $k$-colouring of the integers there exists a sequence $a_1<\cdots$ such that $a_n<f(n)$ for infinitely many $n$ and the set\[\left\{ \sum_{i\in S}a_i: \textrm{finite }S\right\}\]does not contain all colours?

来源字段：https://www.erdosproblems.com/948

00928 | EP-950

### Erdős Problem #950

O / open | Erdős 问题

Let\[f(n) = \sum_{p<n}\frac{1}{n-p}.\]Is it true that\[\liminf f(n)=1\]and\[\limsup f(n)=\infty?\]Is it true that $f(n)=o(\log\log n)$ for all $n$?

来源字段：https://www.erdosproblems.com/950

00929 | EP-951

### Erdős Problem #951

P / partially_solved | Erdős 问题

Let $1<a_1<\cdots$ be a sequence of real numbers such that\[\left\lvert \prod_i a_i^{k_i}-\prod_j a_j^{\ell_j}\right\rvert \geq 1\]for every distinct pair of non-negative finitely supported integer tuples $k_i,\ell_j\geq 0$. Is it true that\[\#\{ a_i \leq x\} \leq \pi(x)?\]

来源字段：https://www.erdosproblems.com/951

00930 | EP-952

### Erdős Problem #952

O / open | Erdős 问题

别名：Walking to Infinity on Gaussian Primes [GUY-A16]

Is there an infinite sequence of distinct Gaussian primes $x_1,x_2,\ldots$ such that\[\lvert x_{n+1}-x_n\rvert \ll 1?\]

来源字段：https://www.erdosproblems.com/952

00931 | EP-953

### Erdős Problem #953

P / partially_solved | Erdős 问题

Let $A\subset \{ x\in \mathbb{R}^2: \lvert x\rvert <r\}$ be a measurable set with no integer distances, that is, such that $\lvert a-b\rvert \not\in \mathbb{Z}$ for any distinct $a,b\in A$. How large can the measure of $A$ be?

来源字段：https://www.erdosproblems.com/953

00932 | EP-954

### Erdős Problem #954

P / partially_solved | Erdős 问题

Let $0=a_0<a_1<a_2<\cdots$ be the sequence of integers defined by $a_0=0$ and $a_1=1$, and $a_{k+1}$ is the smallest integer $n$ for which the number of solutions to $a_i+a_j \leq n$ (with $0\leq i\leq j\leq k$ and $j\geq 1$) is $<n$.

Is the number of solutions to $a_i+a_j \leq x$ equal to $x+O(x^{1/4+o(1)})$?

来源字段：https://www.erdosproblems.com/954

00933 | EP-955

### Erdős Problem #955

P / partially_solved | Erdős 问题

Let\[s(n)=\sigma(n)-n=\sum_{\substack{d\mid n\\ d<n}}d\]be the sum of proper divisors function.

If $A\subset \mathbb{N}$ has density $0$ then $s^{-1}(A)$ must also have density $0$.

来源字段：https://www.erdosproblems.com/955

00934 | EP-961

### Erdős Problem #961

P / partially_solved | Erdős 问题

Let $f(k)$ be the minimal $n$ such that every set of $n$ consecutive integers $>k$ contains an integer divisible by a prime $>k$. Estimate $f(k)$.

来源字段：https://www.erdosproblems.com/961

00935 | EP-962

### Erdős Problem #962

P / partially_solved | Erdős 问题

Let $k(n)$ be the maximal $k$ such that there exists $m\leq n$ such that each of the integers\[m+1,\ldots,m+k\]are divisible by at least one prime $>k$. Estimate $k(n)$ - in particular, is it true that\[\log k(n) \leq (\log n)^{1/2+o(1)}?\]

来源字段：https://www.erdosproblems.com/962

00936 | EP-964

### Erdős Problem #964: Number Theory, Divisors

S / solved | Erdős 问题

Let $\tau(n)$ count the number of divisors of $n$. Is the sequence\[\frac{\tau(n+1)}{\tau(n)}\]everywhere dense in $(0,\infty)$?

来源字段：https://www.erdosproblems.com/964

00937 | EP-966

### Erdős Problem #966: Number Theory, Additive Combinatorics

S / solved | Erdős 问题

Let $k,r\geq 2$. Does there exist a set $A\subseteq \mathbb{N}$ that contains no non-trivial arithmetic progression of length $k+1$, yet in any $r$-colouring of $A$ there must exist a monochromatic non-trivial arithmetic progression of length $k$?

来源字段：https://www.erdosproblems.com/966

00938 | EP-967

### Erdős Problem #967: Number Theory, Analysis

S / solved | Erdős 问题

Let $1<a_1<\cdots $ be a sequence of integers such that $\sum\frac{1}{a_i}<\infty$. Is it true that, for every $t\in \mathbb{R}$,\[1+\sum_{k}\frac{1}{a_k^{1+it}}\neq 0?\]

来源字段：https://www.erdosproblems.com/967

00939 | EP-968

### Erdős Problem #968

P / partially_solved | Erdős 问题

Let $u_n=p_n/n$, where $p_n$ is the $n$th prime. Does the set of $n$ such that $u_n<u_{n+1}$ have positive density?

来源字段：https://www.erdosproblems.com/968

00940 | EP-969

### Erdős Problem #969

P / partially_solved | Erdős 问题

Let $Q(x)$ count the number of squarefree integers in $[1,x]$. Determine the order of magnitude in the error term in the asymptotic\[Q(x)=\frac{6}{\pi^2}x+E(x).\]

来源字段：https://www.erdosproblems.com/969

00941 | EP-970

### Erdős Problem #970

P / partially_solved | Erdős 问题

Let $h(k)$ be Jacobsthal's function, defined to as the minimal $m$ such that, if $n$ has at most $k$ prime factors, then in any set of $m$ consecutive integers there exists an integer coprime to $n$. Determine the order of magnitude of $h(k)$. In particular, is it true that\[h(k) \ll k^2?\]

来源字段：https://www.erdosproblems.com/970

00942 | EP-971

### Erdős Problem #971

P / partially_solved | Erdős 问题

Let $p(a,d)$ be the least prime congruent to $a\pmod{d}$. Does there exist a constant $c>0$ such that, for all large $d$,\[p(a,d) > (1+c)\phi(d)\log d\]for $\gg \phi(d)$ many values of $a$?

来源字段：https://www.erdosproblems.com/971

00943 | EP-972

### Erdős Problem #972

O / open | Erdős 问题

Let $\alpha>1$ be irrational. Are there infinitely many primes $p$ such that $\lfloor p\alpha\rfloor$ is also prime?

来源字段：https://www.erdosproblems.com/972

00944 | EP-975

### Erdős Problem #975

P / partially_solved | Erdős 问题

Let $f\in \mathbb{Z}[x]$ be an irreducible non-constant polynomial such that $f(n)\geq 1$ for all large $n\in\mathbb{N}$. Does there exist a constant $c=c(f)>0$ such that\[\sum_{n\leq X} \tau(f(n))\sim cX\log X,\]where $\tau$ is the divisor function?

来源字段：https://www.erdosproblems.com/975

00945 | EP-976

### Erdős Problem #976

P / partially_solved | Erdős 问题

Let $f\in \mathbb{Z}[x]$ be an irreducible polynomial of degree $d\geq 2$. Let $F_f(n)$ be maximal such that there exists $1\leq m\leq n$ with $f(m)$ is divisible by a prime $\geq F_f(n)$. Equivalently, $F_f(n)$ is the greatest prime divisor of\[\prod_{1\leq m\leq n}f(m).\]Estimate $F_f(n)$. In particular, is it true that $F_f(n)\gg n^{1+c}$ for some constant $c>0$? Or even $\gg n^d$?

来源字段：https://www.erdosproblems.com/976

00946 | EP-977

### Erdős Problem #977: Number Theory

S / solved | Erdős 问题

If $P(m)$ is the greatest prime divisor of $m$, then is it true that\[\frac{P(2^n-1)}{n}\to \infty\]as $n\to \infty$?

来源字段：https://www.erdosproblems.com/977

00947 | EP-978

### Erdős Problem #978

P / partially_solved | Erdős 问题

Let $f\in \mathbb{Z}[x]$ be an irreducible polynomial of degree $k>2$ (and suppose that $k\neq 2^l$ for any $l\geq 1$) such that the leading coefficient of $f$ is positive.

Does the set of integers $n\geq 1$ for which $f(n)$ is $(k-1)$-power-free have positive density?

If $k>3$, and for all primes $p$ there exists $n$ such that $p^{k-2}\nmid f(n)$, then are there infinitely many $n$ for which $f(n)$ is $(k-2)$-power-free?

In particular, does\[n^4+2\]represent infinitely many squarefree numbers?

来源字段：https://www.erdosproblems.com/978

00948 | EP-979

### Erdős Problem #979

P / partially_solved | Erdős 问题

Let $k\geq 2$, and let $f_k(n)$ count the number of solutions to\[n=p_1^k+\cdots+p_k^k,\]where the $p_i$ are prime numbers. Is it true that $\limsup f_k(n)=\infty$?

来源字段：https://www.erdosproblems.com/979

00949 | EP-980

### Erdős Problem #980: Number Theory

S / solved | Erdős 问题

Let $k\geq 2$ and $n_k(p)$ denote the least $k$th power nonresidue of $p$. Is it true that\[\sum_{p<x} n_k(p)\sim c_k \frac{x}{\log x}\]for some constant $c_k>0$?

来源字段：https://www.erdosproblems.com/980

00950 | EP-981

### Erdős Problem #981: and $f_\epsilon(p)$ be the smallest integer $m$ such that $\sum_{n\leq N}

S / solved | Erdős 问题

Let $\epsilon>0$ and $f_\epsilon(p)$ be the smallest integer $m$ such that $\sum_{n\leq N} \left(\frac{n}{p}\right)<\epsilon N$ for all $N\geq m$. Prove that\[\sum_{p<x}f_\epsilon(p)\sim c_\epsilon \frac{x}{\log x}\]for some $c_\epsilon>0$.

来源字段：https://www.erdosproblems.com/981

00951 | EP-983

### Erdős Problem #983

P / partially_solved | Erdős 问题

Let $n\geq 2$ and $\pi(n)<k\leq n$. Let $f(k,n)$ be the smallest integer $r$ such that in any $A\subseteq \{1,\ldots,n\}$ of size $\lvert A\rvert=k$ there exist primes $p_1,\ldots,p_r$ such that $>r$ many $a\in A$ are only divisible by primes from $\{p_1,\ldots,p_r\}$.

Is it true that\[2\pi(n^{1/2})-f(\pi(n)+1,n)\to \infty\]as $n\to \infty$?

In general, estimate $f(k,n)$, particularly when $\pi(n)+1<k=o(n)$.

来源字段：https://www.erdosproblems.com/983

00952 | EP-985

### Erdős Problem #985

P / partially_solved | Erdős 问题

Is it true that, for every prime $p$, there is a prime $q<p$ which is a primitive root modulo $p$?

来源字段：https://www.erdosproblems.com/985

00953 | EP-987

### Erdős Problem #987

S / solved | Erdős 问题

Let $x_1,x_2,\ldots \in (0,1)$ be an infinite sequence and let\[A_k=\limsup_{n\to \infty}\left\lvert \sum_{j\leq n} e(kx_j)\right\rvert,\]where $e(x)=e^{2\pi ix}$.

Is it true that\[\limsup_{k\to \infty} A_k=\infty?\]Is it possible for $A_k=o(k)$?

来源字段：https://www.erdosproblems.com/987

00954 | EP-992

### Erdős Problem #992

S / solved | Erdős 问题

Let $x_1<x_2<\cdots$ be an infinite sequence of integers. Is it true that, for almost all $\alpha \in [0,1]$, the discrepancy\[D(N)=\max_{I\subseteq [0,1]} \lvert \#\{ n\leq N: \{ \alpha x_n\}\in I\} - \lvert I\rvert N\rvert\]satisfies\[D(N) \ll N^{1/2}(\log N)^{o(1)}?\]Or even\[D(N)\ll N^{1/2}(\log\log N)^{O(1)}?\]

来源字段：https://www.erdosproblems.com/992

00955 | EP-995

### Erdős Problem #995

P / partially_solved | Erdős 问题

Let $n_1<n_2<\cdots$ be a lacunary sequence of integers and $f\in L^2([0,1])$. Estimate the growth of, for almost all $\alpha$,\[\sum_{1\leq k\leq N}f(\{ \alpha n_k\}).\]For example, is it true that, for almost all $\alpha$,\[\sum_{1\leq k\leq N}f(\{ \alpha n_k\})=o(N\sqrt{\log\log N})?\]

来源字段：https://www.erdosproblems.com/995

00956 | EP-996

### Erdős Problem #996

O / open | Erdős 问题

Let $n_1<n_2<\cdots$ be a lacunary sequence of integers, and let $f\in L^2([0,1])$. Let $f_n$ be the $n$th partial sum of the Fourier series of $f(x)$. Is there an absolute constant $C>0$ such that, if\[\| f-f_n\|_2 \ll \frac{1}{(\log\log\log n)^{C}}\]then\[\lim_{N\to\infty}\frac{1}{N}\sum_{k\leq N}f(\{\alpha n_k\})=\int_0^1 f(x)\mathrm{d}x\]for almost every $\alpha$?

来源字段：https://www.erdosproblems.com/996

00957 | EP-997

### Erdős Problem #997

S / solved | Erdős 问题

Call $x_1,x_2,\ldots \in (0,1)$ well-distributed if, for every $\epsilon>0$, if $k$ is sufficiently large then, for all $n>0$ and intervals $I\subseteq [0,1]$,\[\lvert \# \{ n<m\leq n+k: x_m\in I\} - \lvert I\rvert k\rvert < \epsilon k.\]Is it true that, for every $\alpha$, the sequence $\{ \alpha p_n\}$ is not well-distributed, if $p_n$ is the sequence of primes?

来源字段：https://www.erdosproblems.com/997

00958 | EP-999

### Erdős Problem #999: Number Theory, Diophantine Approximation

S / solved | Erdős 问题

For any function $f:\mathbb{N}\to \mathbb{N}$ the property that, for almost all $\alpha$\[\left\lvert \alpha-\frac{p}{q}\right\rvert < \frac{f(q)}{q}\]has infinitely many solutions with $(p,q)=1$, is equivalent to\[\sum_{q\geq 1}\phi(q)\frac{f(q)}{q}=\infty.\]

来源字段：https://www.erdosproblems.com/999

00959 | GUY-A18

### Erdős-Selfridge Classification: Infinitely Many Primes in Each Class

O / open | Richard Guy 数论问题

In the Erdős-Selfridge classification of primes, are there infinitely many primes in each class? Prime $p$ is in class 1 if the only prime divisors of $p+1$ are 2 or 3; and $p$ is in class $r$ if every prime factor of $p+1$ is in some class $\le r-1$, with equality for at least one prime factor.

背景链接：https://oeis.org/wiki/Erd%C5%91s%E2%80%93Selfridge_classification_of_primes

00960 | NT-056

### Erdős-Turán Conjecture on Additive Bases

P / partially_solved | 独立 / 未分配集合

If $B$ is an additive basis of order 2, must the representation function tend to infinity?

背景链接：https://doi.org/10.1016/j.jcta.2026.106166

00961 | AMR-093-0122

### Erdős-Mollin-Walsh conjecture

O / open | AMR 问题清单汇编

Erdős-Mollin-Walsh conjecture: no three consecutive numbers are all powerful.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00962 | AMR-093-0105

### Erdős-Moser problem

P / partially_solved | AMR 问题清单汇编

Erdős-Moser problem: is $1^1 + 2^1 = 3^1$ the only solution to the Erdős-Moser equation?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00963 | OWR-13351-007

### Essential Finite Generation of Valuation Rings

S / solved | Oberwolfach 报告问题汇编

For a finite valued-field extension $(F|L,v)$, do there exist $a_1,\ldots,a_r\in O_F$ such that $O_F=O_L[a_1,\ldots,a_r]_{O_L[a_1,\ldots,a_r]\cap\mathfrak{m}_F}$?

来源字段：https://doi.org/10.4171/owr/2014/49

00964 | OWR-15174-014

### Euclid-Type Expressions for Primes Congruent to Three Modulo Four

O / open | Oberwolfach 报告问题汇编

Let $p_1<p_2<\cdots$ be all primes and $q_1<q_2<\cdots$ the primes congruent to $3$ modulo $4$. Writing $P_k=\prod_{i=1}^kp_i$ and $Q_k=\prod_{i=1}^kq_i$, can one construct a Euclid-type expression in $Q_k$ whose least prime divisor is $q_{k+1}$?

来源字段：https://doi.org/10.4171/owr/2016/53

00965 | OPG-37329

### Euler-Mascheroni constant

P / partially_solved | Open Problem Garden

Question Is Euler-Mascheroni constant an transcendental number?

背景链接：http://www.openproblemgarden.org/op/euler_mascheroni_constant

00966 | NUM-017

### Euler-Mascheroni Constant Irrationality

O / open | 独立 / 未分配集合

Is the Euler-Mascheroni constant γ irrational? Transcendental?

背景链接：https://mathworld.wolfram.com/Euler-MascheroniConstant.html

00967 | GUY-A12c

### Even Fibonacci Pseudoprimes

S / solved | Richard Guy 数论问题

Does there exist an even Fibonacci pseudoprime?

背景链接：https://www.fq.math.ca/Scanned/31-2/diporto.pdf

00968 | OWR-11127-002

### Even Mod-p Galois Representations in Langlands Reciprocity

O / open | Oberwolfach 报告问题汇编

What does the arithmetic Langlands reciprocity program predict or establish for absolutely irreducible even mod-$p$ Galois representations $\rho:G_{\mathbb Q}\to\operatorname{GL}_2(\mathbb F_p)$?

来源字段：https://doi.org/10.4171/owr/2011/35

00969 | AMR-030-0056

### Every sequence of 2n-1 elements from a group of order n (written multiplicatively) has an n element subsequence with pro

P / partially_solved | AMR 问题清单汇编

Olson: Every sequence of 2n-1 elements from a group of order n (written multiplicatively) has an n element subsequence with product 1 (in the given order). This is the Erdős-Ginzburg-Ziv Theorem for nonabelian groups. See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

00970 | AIM-ALGEBRAIC_NUMBER_THEORY-0041

### Exact boundary corrections for stack quotients and branch collisions

P / partially_solved | AIM 研讨会问题汇编

**Question 11 (John Voight). When do counts get nicer if we are willing to partially compactify?**

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

00971 | AIM-ALGEBRAIC_NUMBER_THEORY-0005

### Exact density spectra for projective bundles and large Severi-Brauer gaps

P / partially_solved | AIM 研讨会问题汇编

For a nice variety $X/k$, it asks how far

$$\min \delta(X/k),\qquad \operatorname{uni.irr}(X),\qquad \operatorname{cov.gon}(X),\qquad \operatorname{conn.cov.gon}(X)$$

can differ, where

$$\delta(X/k)=\{d\geq 1:\{x\in X\text{ closed}:[k(x):k]=d\}\text{ is Zariski dense}\}.$$

The record specifically suggests $X=Y\times\mathbb P^1$, asserting that the covering gonality is $1$ while the minimum density degree is unchanged.

There is a transcription error in the two displayed gonality definitions in the record. They minimize $d$ subject to

$$\operatorname{gon}(\mathcal C_b)\geq d.$$

Read literally, $d=1$ works for every family of integral curves, so this would make both minima vacuous whenever the relevant family exists. The standard definition of Bastianelli--De Poi--Ein--Lazarsfeld--Ullery is the **least gonality of a covering family** (respectively a family connecting two general points): the general fiber has gonality $d$, equivalently at most $d$. All gonality assertions below use this standard reconstructed definition. This reconstruction is also consistent with the source remark about $Y\times\mathbb P^1$.

Unless explicitly stated otherwise, $k$ is a number field and a nice variety is smooth, projective, and geometrically integral over $k$. Covering and connecting gonality are taken geometrically, as in the standard literature. When a $k$-defined version of unirational degree is used, it is denoted $\operatorname{uni.irr}_k$.

来源字段：http://aimpl.org/degreedsurface/1/

00972 | AIM-ANALYTIC_NUMBER_THEORY-0104

### Exact indicator extremizers for complete-grid large sieve inequalities

P / partially_solved | AIM 研讨会问题汇编

H. Helfgott asked whether the standard large sieve inequality for the exponential sum \[ S(\alpha)=\sum_{n=M+1}^{M+N}a_n e(\alpha n) \] is best possible for coefficient sequences with \(a_n\in\{0,1\}\).

**Recovered analytic formulation.** If \(\alpha_1,\ldots,\alpha_J\in\mathbb R/\mathbb Z\) satisfy \[ \|\alpha_r-\alpha_s\|\geq \delta\qquad(r\ne s), \] then \[ \sum_{r=1}^{J}|S(\alpha_r)|^2 \leq (N-1+\delta^{-1}) \sum_{n=M+1}^{M+N}|a_n|^2. \tag{LS} \] The historically common slightly weaker factor is \(N+\delta^{-1}\). The question is whether restricting every \(a_n\) to \(0\) or \(1\) permits a smaller uniform factor.

来源字段：https://aimath.org/WWN/primegaps/WorkshopProblems.pdf

00973 | AIM-ANALYTIC_NUMBER_THEORY-0037

### Exact level sets inherit bounded-multiplicative logarithmic disjointness

S / solved | AIM 研讨会问题汇编

Let \((X,\mu,T)\) be a totally uniquely ergodic \(0\)-entropy system. We know from recent work of Frantzikinakis--Host that if \(g:\mathbb N\to\mathbb C\) is a bounded multiplicative function, then for all \(f\in C(X)\) with \(\int_X f\,d\mu=0\) and \(x\in X\) we have \[ \mathbb E^{\log}_{n\in\mathbb N}g(n)f(T^n x)=0. \] Now, if \(g:\mathbb N\to\mathbb C\) is a (possibly unbounded) multiplicative function, show that for all \(f\in C(X)\), \(x\in X\), and fixed \(c\in\mathbb C\), \[ \mathbb E^{\log}_{n\in\mathbb N}{\bf1}_{g(n)=c}f(T^n x) =\mathbb E^{\log}_{n\in\mathbb N}{\bf1}_{g(n)=c}\cdot\int_X f\,d\mu. \] This can be done for totally ergodic nilsystems.

来源字段：http://aimpl.org/sarnakconjecture/6/

00974 | AIM-ANALYTIC_NUMBER_THEORY-0059

### Exact Petersson--Kloosterman reduction of the fixed-weight symmetric-square second moment

P / partially_solved | AIM 研讨会问题汇编

Find an exact formula for $\sum_{f\in H_k}^h L(\frac12,\operatorname{Sym}^2 f)^2$.

来源字段：http://aimpl.org/zetamoments/5/

00975 | AIM-ANALYTIC_NUMBER_THEORY-0004

### Exact-conductor projectors as the finite bridge from trivial delta to automorphic kernels

P / partially_solved | AIM 研讨会问题汇编

**Relationship between Trivial delta method and Nelson's Kernel method.** Let \(f\) be a holomorphic or Hecke--Maass cusp form of level \(1\) and \(\chi\) be a Dirichlet character of level \(p\). For fixed \(f\), a level aspect Burgess-type bound for \(L(f\times\chi,1/2)\) can be obtained by (among many others) these two methods: (1) Trivial delta method, (2) Using the Kernel \[ K(z,w)=\sum_{\gamma\in\Gamma_0(p)}\omega(z^{-1}\gamma w) \] for \(z,x\in\mathbb H\) and \(\omega\) a smooth bump function, defined appropriately. Find a precise relationship between the two methods.

来源字段：http://aimpl.org/deltasubconvex2/1/

00976 | AIM-ALGEBRAIC_NUMBER_THEORY-0003

### Exact-degree points from automorphism orbits

P / partially_solved | AIM 研讨会问题汇编

Let \(X\) be a variety. Suppose one has a diagram \[ X\dashleftarrow Y\dashrightarrow \mathbb P^n, \] where \(Y\dashrightarrow\mathbb P^n\) is finite of degree \(d\). Then, as a consequence of Hilbert irreducibility, \(X\) will have a Zariski dense set of degree \(d\) points.

Can we give a method to construct a Zariski dense set of degree \(d\) points on a variety different from the one just mentioned?

* Maybe one can replace \(\mathbb P^n\) with another sort of variety? * Maybe one can do something totally different and give a construction not coming from a correspondence.

来源字段：http://aimpl.org/degreedsurface/1/

00977 | OWR-4333-011

### Exceptional Diophantine Approximation by Matrix Systems

O / open | Oberwolfach 报告问题汇编

For the Diophantine-approximation function $$\psi_L(k)=\inf_{\substack{\|q\|_\infty\leq k,\ q\in\mathbb Z^m,\ p\in\mathbb Z^\ell}}\|Lq-p\|_\infty$$ of a matrix $L\in M_{\ell,m}(\mathbb R)$, what exceptional behaviors relative to the generic case are possible, and how large are the corresponding exceptional sets?

来源字段：https://doi.org/10.4171/owr/2010/20

00978 | OWR-8415337-006

### Exceptional Galois Images of Non-CM Elliptic Curves

O / open | Oberwolfach 报告问题汇编

For non-CM elliptic curves over $\mathbb Q$, is the known list of $22$ exceptional $\ell$-adic Galois images complete? Equivalently, must every counterexample arise from an unexpected rational point on $X_{\mathrm{ns}}^+(N)$ for $N\in\{33,5^2,7^2,11^2\}$ or a prime $\ell\geq17$, or on one of the two level-$49$ genus-$9$ modular curves?

来源字段：https://doi.org/10.4171/owr/2021/34

00979 | OWR-14298584-001

### Exceptional Galois Points on Modular Curves

O / open | Oberwolfach 报告问题汇编

Extend the Khawaja-Siksek finiteness theorem for primitive algebraic points from symmetric and alternating Galois groups to other families of simple and almost simple groups. More specifically, let $N$ be large and let $j:X_0(N)\to X(1)$ be the usual map. Hilbert irreducibility gives a thin set $S\subset X(1)(\mathbb{Q})$ such that, for $\alpha\in X(1)(\mathbb{Q})\setminus S$, the fiber $j^{-1}(\alpha)$ has Galois group $\mathrm{PGL}_2(\mathbb{Z}/N\mathbb{Z})$. Is $S$ finite or infinite? If it is infinite, which Galois groups occur infinitely often? Is there an infinite family of algebraic points on $X_0(N)$ with Galois group $\mathrm{PGL}_2(\mathbb{Z}/N\mathbb{Z})$ that does not arise from $j$?

来源字段：https://doi.org/10.4171/owr/2024/40

00980 | OWR-4990381-015

### Exceptional Reducibility Sets in Polynomial Specialization

P / partially_solved | Oberwolfach 报告问题汇编

For an irreducible polynomial $F(t,X)\in\mathbb Q[t,X]$, determine, up to a finite set, $$ R_F=\{t_0\in\mathbb Q:F(t_0,X)\text{ is reducible}\}, $$ and find analogous descriptions over number fields and for integral parameters.

来源字段：https://doi.org/10.4171/owr/2021/12

00981 | OWR-11784-007

### Exceptional-Unit Bounds by Algebraic Degree

O / open | Oberwolfach 报告问题汇编

Let $E_0(\alpha)$ be the largest $n$ such that $\alpha^j-1$ is a unit for every $1\leq j\leq n$, or $0$ if $\alpha-1$ is not a unit, and let $e(d)=\max E_0(\alpha)$ over algebraic integers $\alpha$ of degree $d$. Is $e(d)<d$ for every $d\geq7$?

来源字段：https://doi.org/10.4171/owr/2012/22

00982 | AMR-092-0001

### Existence of a four-dimensional Euler brick

O / open | AMR 问题清单汇编

Do there exist positive integers $a,b,c,d$ such that all six pairwise face diagonals $\sqrt{a^2+b^2}$, $\sqrt{a^2+c^2}$, $\sqrt{a^2+d^2}$, $\sqrt{b^2+c^2}$, $\sqrt{b^2+d^2}$, and $\sqrt{c^2+d^2}$ are integers?

背景链接：https://unsolvedproblems.org/index_files/4DEulerBrick.htm

00983 | AIM-ALGEBRAIC_NUMBER_THEORY-0048

### Explicit bounds for distinct curve zeta functions over a fixed finite field

P / partially_solved | AIM 研讨会问题汇编

**Question 18.** Fix \(q\). How many different zeta functions are attached to curves of genus \(g\gg q\)?

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

00984 | AIM-ANALYTIC_NUMBER_THEORY-0080

### Explicit integration and a shifted-canonical obstruction for elliptic Calabi--Yau holomorphic anomalies

P / partially_solved | AIM 研讨会问题汇编

别名：Holomorphic anomaly, fibrations, and the BPS multiple-cover transform [AIM-PHYSICS-0013]

Investigate the Holomorphic anomaly equation (HAE) in fibered Calabi-Yau 3-folds and BPS counting. `\label{hae-cy}`

For example, does the HAE ``integrate'' over elliptic fibrations or K3 fibers?

来源字段：http://aimpl.org/gromwitnumthry/1/

00985 | AMR-093-0004

### Exponent pair conjecture

P / partially_solved | AMR 问题清单汇编

Exponent pair conjecture: for all $\varepsilon > 0$, is the pair $(\varepsilon, 1/2 + \varepsilon)$ an exponent pair?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00986 | OWR-14603-023

### Exponential Degree Growth in Composita of Fibers

P / partially_solved | Oberwolfach 报告问题汇编

Let $X/\mathbb Q$ be a curve of genus $g$, let $t\in\mathbb Q(X)$ have degree $\nu\geq2$, and choose points $P_n$ with $t(P_n)=n$. If $t$ has a finite critical value outside $\mathbb Q$, or if $\overline{\mathbb Q}(X)/\overline{\mathbb Q}(t)$ is nonabelian, does there exist $c(g,\nu)>0$ such that

$$[\mathbb Q(P_1,\ldots,P_N):\mathbb Q]\geq e^{c(g,\nu)N}$$

for all sufficiently large $N$?

来源字段：https://doi.org/10.4171/owr/2016/21

00987 | OWR-1795-001

### Exponential Distribution of Normalized Prime Gaps

O / open | Oberwolfach 报告问题汇编

If $p_n$ is the $n$th prime, then the normalized prime gaps $(p_{n+1}-p_n)/\log p_n$ should be asymptotically distributed like an exponential random variable with density $e^{-x}$ for $0\leq x<\infty$.

来源字段：https://doi.org/10.4171/owr/2008/14

00988 | OWR-15952-004

### Exponential Separation of Roots of Bounded Coefficient Polynomials

O / open | Oberwolfach 报告问题汇编

Let $\mathcal P_d$ be the set of polynomials of degree at most $d$ whose coefficients all lie in $\{-1,0,1\}$. Does there exist a constant $s>0$, independent of $d$, such that any distinct roots $\alpha$ and $\beta$ of polynomials in $\mathcal P_d$ satisfy $|\alpha-\beta|>s^d$?

来源字段：https://doi.org/10.4171/owr/2017/47

00989 | OWR-14603-013

### Exponential Small-Value Bounds for Littlewood Polynomials

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Small-Value Sparsity Question for Littlewood Polynomials [OWR-14603-011]

Let $\mathcal P_d=\{a_dx^d+\cdots+a_0:a_i\in\{-1,0,1\}\}$. For every rational $p/q\in[9/10,19/20]$, does there exist an absolute $C>0$ such that $\#\{P\in\mathcal P_d:|P(p/q)|<c_{p,q}e^{-Cd}\}<e^{d/100}$ for some constant $c_{p,q}$?

来源字段：https://doi.org/10.4171/owr/2016/21

00990 | OWR-9790364-004

### Extreme Normalized Gaps between Riemann-Zeta Zeros

O / open | Oberwolfach 报告问题汇编

If $0<\gamma_1\leq\gamma_2\leq\cdots$ are the ordinates of the nontrivial zeros of the Riemann zeta function, are $$ \liminf_{n\to\infty}\frac{(\gamma_{n+1}-\gamma_n)\log\gamma_n}{2\pi}=0 \quad\text{and}\quad \limsup_{n\to\infty}\frac{(\gamma_{n+1}-\gamma_n)\log\gamma_n}{2\pi}=\infty $$ ?

来源字段：https://doi.org/10.4171/owr/2022/13

00991 | AMR-092-0005

### Factor RSA-1024

O / open | AMR 问题清单汇编

Find the two prime factors of the RSA-1024 challenge integer $135066410865995223349603216278805969938881475605667027524485143851526510604859533833940287150571909441798207282164471551373680419703964191743046496589274256239341020864383202110372958725762358509643110564073501508187510676594629205563685529475213500852879416377328533906109750544334999811150056977236890927563$.

背景链接：https://unsolvedproblems.org/index_files/RSA.htm

00992 | AIM-ANALYTIC_NUMBER_THEORY-0052

### Families and moment heuristics for automorphic L-functions over function fields

P / partially_solved | AIM 研讨会问题汇编

Develop a heuristic for moments of L-functions of automorphic forms over function fields. What are families in this context?

来源字段：http://aimpl.org/zetamoments/2/

00993 | AMR-093-0123

### Feit-Thompson conjecture

O / open | AMR 问题清单汇编

Feit-Thompson conjecture: for all distinct prime numbers $p$ and $q$, $(p^q - 1)/(p - 1)$ does not divide $(q^p - 1)/(q - 1)$

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

00994 | NT-081

### Fermat-Catalan Conjecture

P / partially_solved | 独立 / 未分配集合

Are there finitely many solutions to $a^m + b^n = c^k$ with coprime $a,b,c$ and $1/m + 1/n + 1/k < 1$?

背景链接：https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/blms/27.6.513

00995 | OPG-156

### Few subsequence sums in Z_n x Z_n

P / partially_solved | Open Problem Garden

Conjecture For every $0 \le t \le n-1$, the sequence in ${\mathbb Z}_n^2$ consisting of $n-1$ copes of $(1,0)$ and $t$ copies of $(0,1)$ has the fewest number of distinct subsequence sums over all zero-free sequences from ${\mathbb Z}_n^2$ of length $n-1+t$.

背景链接：http://www.openproblemgarden.org/op/limiting_subsequence_sums_in_z_n_x_z_n

00996 | OWR-14298584-007

### Field Extensions and Ordinarity in Arithmetic Specialization

P / partially_solved | Oberwolfach 报告问题汇编

In the arithmetic specialization result of Bright and Newton, when is the finite extension $k'/k$ necessary? Can the ordinarity hypothesis be removed?

来源字段：https://doi.org/10.4171/owr/2024/40

00997 | OWR-14297740-022

### Field Reconstruction from Rationalized Milnor K-Theory

P / partially_solved | Oberwolfach 报告问题汇编

Let $F$ be a field of Kronecker dimension at least $2$. Is the isomorphism type of its inseparable closure $F^i$ determined by the graded $\mathbb{Q}$-algebra $K_*^M(F)\otimes\mathbb{Q}$?

来源字段：https://doi.org/10.4171/owr/2023/42

00998 | OWR-15173-006

### Fields of Definition of Eutactic Forms

O / open | Oberwolfach 报告问题汇编

Can one predict the field of definition of a eutactic form over $\mathbb Q$ or another number field?

来源字段：https://doi.org/10.4171/owr/2016/52

00999 | OWR-14297740-007

### Fields of Definition under Gluing of Hurwitz Components

P / partially_solved | Oberwolfach 报告问题汇编

In the arithmetic geometry of Hurwitz spaces parametrizing branched covers of curves with fixed monodromy group and conjugacy classes, let $K$ be a number field over which two irreducible components of Hurwitz spaces are defined. Are the irreducible components of higher-degree Hurwitz spaces obtained by topological gluing of components defined over $K$ themselves defined over $K$ (Seguin)?

来源字段：https://doi.org/10.4171/owr/2023/42

01000 | OWR-16764-003

### Filtration along Modular Theta Cycles

P / partially_solved | Oberwolfach 报告问题汇编

If $\tau(f),\ldots,\tau^p(f)$ is the theta cycle of a modular form $f$, how does the filtration of $f$ change along this cycle?

来源字段：https://doi.org/10.4171/owr/2019/2

01001 | AMR-093-0062

### Find the value of the De Bruijn-Newman constant

P / partially_solved | AMR 问题清单汇编

Find the value of the De Bruijn-Newman constant.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01002 | AIM-ANALYTIC_NUMBER_THEORY-0096

### Finite asymptotic certificates and a cusp-width span bound for fixed-matrix Nahm sums

P / partially_solved | AIM 研讨会问题汇编

Problem 2.1. Given a particular A, can we bound the number of possible B? Similarly, given a particular A, can we bound the dimension of the vector space determined by the B?

The bounds may or may not be effective.

The statement is not self-contained.

* \(r\geq 1\), \(A\in M_r(\mathbb Q)\) is symmetric positive definite; * \(B\in\mathbb Q^r\), \(C\in\mathbb Q\); * \(q=e^{2\pi i\tau}\), \((q)_m=\prod_{j=1}^m(1-q^j)\), and \((q)_0=1\); * the associated Nahm sum is \[ f_{A,B,C}(\tau)=F_{A,B,C}(q) :=\sum_{n\in\mathbb Z_{\geq0}^r} \frac{q^{\frac12n^tAn+B^tn+C}} {(q)_{n_1}\cdots(q)_{n_r}}. \]

The preceding paragraph calls \((A,B,C)\) a choice for which this function is modular. In current terminology it is a **modular triple**. Thus “possible \(B\)” is most naturally read as \[ \mathcal B_A:=\{B\in\mathbb Q^r:\text{there is }C\in\mathbb Q \text{ such that }(A,B,C)\text{ is a scalar modular triple}\}. \] The 2024 work of Wang--Zhang confirms the natural interpretation of “the vector space determined by the \(B\)” as the span of the Nahm sums with the same \(A\), often organized as a vector-valued modular function.

There remains an ambiguity that matters: the workshop does not prescribe one modular group, multiplier, or level for all \(B\). A dimension bound in a fixed space of modular functions therefore needs a **common finite-index group** hypothesis. The result below states that hypothesis explicitly. It does not silently assume a uniform level.

来源字段：https://aimath.org/WWN/mockmodular/mockmodular.pdf

01003 | AMR-030-0054

### Finite field Sylvester-Gallai: Suppose S is a tranversal of Z_(p)^(2), i

P / partially_solved | AMR 问题清单汇编

**(Conjecture)** Every transversal $S$ of $AG(2,p)$ that is **not contained in a line** has an *ordinary line*: a line meeting $S$ in exactly two points.

背景链接：https://people.math.sc.edu/cooper/combprob.html

01004 | OWR-11127-015

### Finite Secant-Tangent Generation on Cubic Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

For a smooth cubic surface $S$ over a field $K$, does there always exist a finite subset $B\subseteq S(K)$ such that all $K$-points of $S$ can be generated from $B$ by successive secant and tangent constructions?

来源字段：https://doi.org/10.4171/owr/2011/35

01005 | AIM-ALGEBRAIC_NUMBER_THEORY-0109

### Finite-level criteria and an exact finite dense-prime example

P / partially_solved | AIM 研讨会问题汇编

**Question 24.** Given an elliptic curve \(E/\mathbf Q\), \({\rm rk}\,E(\mathbf Q)>0\), describe the set of primes \(p\) such that \(E(\mathbf Q)\) is dense in \(E(\mathbf Q_p)\). Is this set nonempty? Is there a modular interpretation of this problem? (Takloo-Bighash)

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01006 | AIM-ANALYTIC_NUMBER_THEORY-0035

### Finite-prime parity laws and a covariance reduction

P / partially_solved | AIM 研讨会问题汇编

Show that \[ \left| \mathbb{E}_{n \in \mathbb{N}}^{\log} \lambda(n^2+1) \right| < 1-\epsilon, \] and that \[ \left| \mathbb{E}_{p \in \mathbb{P}} \lambda(p-1) \right| < 1-\epsilon \] for some \(\epsilon>0\). It would even be interesting to show that both sets \(\{p:\lambda(p-1)=1\}\) and \(\{p:\lambda(p-1)=-1\}\) are infinite. For the first bound, using the entropy decrement argument, wouldn't it be enough to show that \[ \left|\sum_{p\approx P}\sum_{n\approx X}\lambda(n^2+p^2)\right| =o\!\left(X\frac{P}{\log P}\right) \] with \(P=\log\log\log X\)?

来源字段：http://aimpl.org/sarnakconjecture/5/

01007 | OWR-12985-003

### Finiteness of Common Zeros of Coprime Exponential Sums

O / open | Oberwolfach 报告问题汇编

Let $f$ and $g$ be one-variable exponential sums with no nonconstant common divisor. Is the common zero set $\{z:f(z)=g(z)=0\}$ finite?

来源字段：https://doi.org/10.4171/owr/2014/27

01008 | OWR-1537-012

### Finiteness of Nonsquare Diophantine Quadruples

O / open | Oberwolfach 报告问题汇编

If $n$ is not a perfect square, then there should be only finitely many $D(n)$-quadruples, where a $D(n)$-quadruple is a set of four positive integers such that $xy+n$ is a perfect square for every pair of distinct elements $x,y$.

来源字段：https://doi.org/10.4171/owr/2007/21

01009 | NT-027

### Firoozbakht's Conjecture

O / open | 独立 / 未分配集合

Is the sequence $p_n^{1/n}$ strictly decreasing, where $p_n$ is the $n$-th prime?

背景链接：https://arxiv.org/abs/1904.00499

01010 | AMR-093-0064

### First Hardy-Littlewood zeta-function conjecture

S / solved | AMR 问题清单汇编

For every $\varepsilon>0$, is there a $T_0(\varepsilon)$ such that, whenever $T\geq T_0$ and $H=T^{1/4+\varepsilon}$, the interval $(T,T+H]$ contains an odd-order zero of $\zeta(1/2+it)$?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01011 | AIM-ANALYTIC_NUMBER_THEORY-0072

### Fixed rational twists of zeta moments and a prime-exponent response law

S / solved | AIM 研讨会问题汇编

Develop a heuristic for twisted moments, e.g. \[ \int_0^T \left(\frac{m}{n}\right)^{it} \left|\zeta\!\left(\tfrac12+it\right)\right|^{2k}\,dt. \]

来源字段：http://aimpl.org/zetamoments/8/

01012 | AIM-ANALYTIC_NUMBER_THEORY-0056

### Fixed-form automorphic diagonals and the missing Type-I mechanism

P / partially_solved | AIM 研讨会问题汇编

**Problem.** Develop a heuristic for moments of \(L\)-functions of automorphic forms.

**Remark.** One of the problems here is that \(\sum a_m a_{m+h}=0\), so there are no Type I terms.

Fix a primitive cuspidal automorphic form \(f\) (initially on \(\mathrm{GL}_2/\mathbb Q\)) and develop a Conrey--Keating/CFKRS heuristic for \[ M_{f,k}(T;w)=\int_{\mathbb R}w(t/T) |L(\tfrac12+it,f)|^{2k}\,dt, \qquad k\in\mathbb N, \] as \(T\to\infty\).

来源字段：http://aimpl.org/zetamoments/5/

01013 | AIM-ALGEBRAIC_NUMBER_THEORY-0100

### Fixed-length sums of 2/(x^2-2)

P / partially_solved | AIM 研讨会问题汇编

**Problem/Question 15.** Characterize the rational numbers \(\alpha\) that can be written as \[ \alpha=\frac{2}{x_1^2-2}+\frac{2}{x_2^2-2}+\cdots+\frac{2}{x_n^2-2} \] for \(x_i\in\mathbb Q\) and fixed \(n\). (Poonen)

It is necessary that \(|\alpha|_p\leq1\) for \((2/p)=-1\) and \(p=2\). Is it sufficient?

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01014 | OWR-11127-020

### Fixed-Rank Cohen-Lenstra Distributions

O / open | Oberwolfach 报告问题汇编

Fix an odd prime $p$ and a positive integer $g$. Among imaginary quadratic fields $K$ with $p$-rank $g$, ordered by discriminant, is the probability that $A_K\cong G$ equal to

$$\frac{1}{|\operatorname{Aut}(G)|}\,p^{-g^2}\prod_{k=1}^g(p^g-p^{g-k})^2?$$

Equivalently, does the displayed expression give the limiting frequency?

来源字段：https://doi.org/10.4171/owr/2011/35

01015 | AIM-ANALYTIC_NUMBER_THEORY-0002

### Fixed-shift cancellation and a four-detector residual ledger

P / partially_solved | AIM 研讨会问题汇编

For a holomorphic or a Hecke--Maass cusp form \(f\), with \(n\)-th Fourier coefficient \(\lambda_f(n)\), obtain cancellation in

\[ \sum_{n\leq N}\lambda_f(n)\lambda_f(n+1) \]

using (i) the Duke--Friedlander--Iwaniec delta method, (ii) a trivial delta method, (iii) the \({\rm GL}_2\) Petersson trace formula as a delta method, and (iv) the \({\rm GL}_3\) Kuznetsov trace formula as a delta method. The record adds that an explicit \({\rm GL}_3\) Kuznetsov formula with arbitrary level and nebentypus was not yet available.

来源字段：http://aimpl.org/deltasubconvex2/1/

01016 | OWR-14298588-016

### Fontaine-Mazur Conjecture for Unramified Galois Representations

O / open | Oberwolfach 报告问题汇编

Prove the Fontaine-Mazur conjecture for the indicated unramified Galois representations, extending the known special unramified Tate results to the general setting.

来源字段：https://doi.org/10.4171/owr/2024/33

01017 | AMR-093-0048

### Fontaine-Mazur geometric Galois-representation conjecture

P / partially_solved | AMR 问题清单汇编

Let $K$ be a number field and let $\rho$ be an irreducible $p$-adic representation of $\operatorname{Gal}(\overline K/K)$ that is unramified outside finitely many places and potentially semistable at the places above $p$. Must $\rho$, up to the standard finite-image and Tate-twist qualifications, arise from the étale cohomology of an algebraic variety over $K$?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01018 | AMR-093-0172

### For any given integer a > 0, are there infinitely many primes p such that a^{p - 1} ≡ 1 (mod p^{2})

O / open | AMR 问题清单汇编

For any given integer a > 0, are there infinitely many primes p such that a^{p - 1} ≡ 1 (mod p^{2})?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01019 | AMR-093-0124

### Fortune's conjecture

O / open | AMR 问题清单汇编

Fortune's conjecture that no Fortunate number is composite.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01020 | NT-073

### Four Exponentials Conjecture

O / open | 独立 / 未分配集合

别名：Four Exponentials Conjecture [NUM-019]

If $x_1, x_2$ are linearly independent over $\mathbb{Q}$ and $y_1, y_2$ are linearly independent over $\mathbb{Q}$, is at least one of $e^{x_1 y_1}, e^{x_1 y_2}, e^{x_2 y_1}, e^{x_2 y_2}$ transcendental?

背景链接：https://webusers.imj-prg.fr/~michel.waldschmidt/texts.html

01021 | AIM-ALGEBRAIC_NUMBER_THEORY-0091

### Four-part status and a finite canonical-cover descent audit

P / partially_solved | AIM 研讨会问题汇编

I checked the record against the AIM PDF and the AIM HTML rendering [AIM].

Let \(k\) be a number field and let \(X/k\) be an Enriques surface.

1. If \(X(k)\ne\varnothing\), must \(X(k)\) be Zariski dense in \(X\)? 2. Does an Enriques surface exist that violates the Hasse principle? 3. Does one exist for which \[ X(\mathbf A_k)^{\operatorname{Br}}=\varnothing \quad\text{and}\quad X(\mathbf A_k)\ne\varnothing? \] 4. Does one exist for which \[ X(k)=\varnothing \quad\text{and}\quad X(\mathbf A_k)^{\operatorname{Br}}\ne\varnothing? \]

The neighboring AIM Question 8 makes the intended descent connection explicit.

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01022 | AIM-ANALYTIC_NUMBER_THEORY-0043

### Fourier uniformity, the entropy-scale gap, and a fixed-shift obstruction

S / solved | AIM 研讨会问题汇编

Is is possible to show that \[ \int_X^{2X}\left|\sum_{x\le n\le x+H}\lambda(n)e(n\alpha)\right|\,dx=o(XH) \] with \(H>X^\epsilon\) implies \[ \sum_{n\le X}\frac{\lambda(n)\lambda(n+1)}n=o(\log X)? \]

来源字段：http://aimpl.org/sarnakconjecture/8/

01023 | OWR-11695860-007

### Fractional Moments of Möbius Character and Archimedean Twists

P / partially_solved | Oberwolfach 报告问题汇编

For $0\leq q\leq1$ and fixed $A>0$, prove $\frac1{r-1}\sum_{\chi\bmod r}|\sum_{n\leq x}\mu(n)\chi(n)|^{2q}\ll(\frac{x}{1+(1-q)\sqrt{\log\log x}})^q$ for $x\leq r^A$, and $\frac1{2T}\int_{-T}^T|\sum_{n\leq x}\mu(n)n^{it}|^{2q}\,dt\ll(\frac{x}{1+(1-q)\sqrt{\log\log x}})^q$ for $x\leq T^A$.

来源字段：https://doi.org/10.4171/owr/2022/50

01024 | OWR-1188-014

### Frey-Jarden Infinite-Rank Conjecture

O / open | Oberwolfach 报告问题汇编

For every nonzero abelian variety $A$ over a number field $K$, is $\operatorname{rk}A(K^{\mathrm{ab}})=\infty$?

来源字段：https://doi.org/10.4171/owr/2006/06

01025 | OPG-17958

### Frobenius number of four or more integers

P / partially_solved | Open Problem Garden

Problem Find an explicit formula for Frobenius number $g(a_1, a_2, \dots, a_n)$ of co-prime positive integers $a_1, a_2, \dots, a_n$ for $n\geq 4$.

背景链接：http://www.openproblemgarden.org/op/frobenius_number_of_four_or_more_integers

01026 | OWR-4076-002

### Frobenius Polynomials of Degeneration Pencils

O / open | Oberwolfach 报告问题汇编

Let $f_0\in\mathbb Q[x_0,\ldots,x_{n+1}]$ be homogeneous of degree $d$, put $f_1=x_0^d+\cdots+x_{n+1}^d$, and consider the pencil of degree-$d$, dimension-$n$ hypersurfaces $$(1-t)f_0+tf_1=0,$$ which is generically smooth and may degenerate at $t=0$. For a suitable prime $p$, let $F$ and $N$ be the Frobenius and nilpotent monodromy matrices in the associated limiting cohomology, with entries in $\mathbb Q_p$ and $\mathbb Q$, respectively, and satisfying $NF=pFN$. Does $\det(1-TF|_{\ker N})$ have integer coefficients, with each reciprocal root having complex absolute value $p^{i/2}$ for an integer $i$ satisfying $0\leq i\leq n$?

来源字段：https://doi.org/10.4171/owr/2009/33

01027 | OWR-11127-012

### Functional Equations for Higher Bloch Groups

O / open | Oberwolfach 报告问题汇编

For higher polylogarithms, especially $n>3$, can one construct functional equations that generate the universal relations needed to define the higher Bloch groups $B_n(F)$ and prove the corresponding form of Zagier's conjecture?

来源字段：https://doi.org/10.4171/owr/2011/35

01028 | OWR-1734-003

### Functorial Lifts of Siegel Modular Forms

O / open | Oberwolfach 报告问题汇编

Let $f$ and $g$ be Siegel modular forms of genus $2m$ and weights $k>2m$ and $l=k-2m$, with Satake parameters $(\alpha_i)$ and $(\beta_i)$. Does there exist a Siegel modular form $F$ of genus $4m$ and weight $k$ whose Satake parameters satisfy $\gamma_0=\alpha_0\beta_0$, $\gamma_i=\alpha_i$ for $1\leq i\leq2m$, and $\gamma_{2m+j}=\beta_j$ for $1\leq j\leq2m$?

来源字段：https://doi.org/10.4171/owr/2007/51

01029 | OWR-4076-011

### Galois Equivariance of Generalized Heegner-Cycle Images

P / partially_solved | Oberwolfach 报告问题汇编

If $H\subseteq K^{\mathrm{ab}}$ and $\Delta_\varphi\in CH^{r+1}(X_r)_0(H)$ is a generalized Heegner cycle, is $\Phi_C(\Delta_\varphi)\in A(H)\otimes\mathbb{Q}$ and does $\Phi_C(\Delta_\varphi^\sigma)=\Phi_C(\Delta_\varphi)^\sigma$ for every $\sigma\in\operatorname{Gal}(H/K)$?

来源字段：https://doi.org/10.4171/owr/2009/33

01030 | OWR-12867-007

### Galois Orbits of Finite Planar Trees

P / partially_solved | Oberwolfach 报告问题汇编

Describe the orbits of $\operatorname{Gal}(\overline{\mathbb Q}/\mathbb Q)$ on isotopy classes of finite planar trees, equivalently on the associated Shabat polynomials whose only critical values are $0$ and $1$.

来源字段：https://doi.org/10.4171/owr/2014/07

01031 | OWR-7155439-003

### Galois Representations from Hecke Maximal Ideals

P / partially_solved | Oberwolfach 报告问题汇编

For every maximal ideal $\mathfrak m$ of the Hecke algebra $\mathbb T_S(K)$, does there exist a continuous semisimple Galois representation $\bar\rho_{\mathfrak m}$ with the prescribed Hecke characteristic polynomials, maximally odd complex conjugation, and Fontaine-Laffaille Hodge-Tate weights? If $\mathfrak m$ is non-Eisenstein, does this representation have a lift $\rho_{\mathfrak m}$ with analogous properties, and is the Taylor-Wiles deformation map $R_S\to\mathbb T_S(K)_{\mathfrak m}$ an isomorphism?

来源字段：https://doi.org/10.4171/owr/2021/18

01032 | OWR-15588-010

### Galois Semisimplicity of Cohomology over Finitely Generated Fields

P / partially_solved | Oberwolfach 报告问题汇编

For every smooth projective $X$ over a finitely generated field $K$ of characteristic zero, every $i,n$, and every prime $\ell$, is $H^i(X)(n)$ semisimple as a representation of the absolute Galois group $\Gamma_K$?

来源字段：https://doi.org/10.4171/owr/2017/45

01033 | AIM-ALGEBRAIC_NUMBER_THEORY-0097

### Galois-stable cyclotomic interlacing pairs

S / solved | AIM 研讨会问题汇编

Classify finite sets \(A,B\subset\mu_\infty\) such that \(\#A=\#B\), each set is stable under \(\operatorname{Gal}(\overline{\mathbb Q}/\mathbb Q)\), and the points of \(A\) and \(B\) strictly alternate around the unit circle.

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01034 | OPG-337

### Gao's theorem for nonabelian groups

O / open | Open Problem Garden

For every finite multiplicative group $G$, let $s(G)$ ( $s'(G)$ ) denote the smallest integer $m$ so that every sequence of $m$ elements of $G$ has a subsequence of length $>0$ (length $|G|$ ) which has product equal to 1 in some order.

Conjecture $s'(G) = s(G) + |G| - 1$ for every finite group $G$.

背景链接：http://www.openproblemgarden.org/op/gaos_theorem_for_nonabelian_groups

01035 | OWR-13750333-012

### Gap Distribution of Prime Multiples Modulo One

P / partially_solved | Oberwolfach 报告问题汇编

For irrational $\alpha$, determine the limiting distribution and possible sizes of the gaps between the points $p\alpha\bmod 1$ as $p$ ranges over primes.

来源字段：https://doi.org/10.4171/owr/2023/22

01036 | GREEN-039

### Gaps Between Sums of Two Squares

P / partially_solved | Ben Green 问题

Is there always a sum of two squares between $X - \frac{1}{10}X^{1/4}$ and $X$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

01037 | OWR-12725-003

### Gaussian Limits for Arithmetic Progressions Under Weaker Growth

O / open | Oberwolfach 报告问题汇编

Does the Gaussian limit theorem for normalized cusp-form and divisor-function values in arithmetic progressions remain valid when its auxiliary function is assumed only to satisfy $\Phi(x)=O_\varepsilon(x^{1-\varepsilon})$ for some $\varepsilon>0$?

来源字段：https://doi.org/10.4171/owr/2013/51

01038 | OWR-11695860-001

### Gaussian-Prime Paths with Bounded Steps

O / open | Oberwolfach 报告问题汇编

Does there exist an infinite path through the Gaussian primes whose successive Euclidean step lengths are bounded by an absolute constant?

来源字段：https://doi.org/10.4171/owr/2022/50

01039 | OWR-14298805-013

### General Langlands Functoriality

O / open | Oberwolfach 报告问题汇编

Given a homomorphism of dual groups $\xi:\widehat H\to\widehat G$, construct the predicted transfer $\xi_*:\operatorname{AutRep}(H)\to\operatorname{AutRep}(G)$ satisfying $L(s,\xi_*(\pi),\tau)=L(s,\pi,\tau\circ\xi)$ for every finite-dimensional representation $\tau$ of $\widehat G$.

来源字段：https://doi.org/10.4171/owr/2024/54

01040 | OWR-4076-001

### Generalized Fermat Equation $a^3+b^3=c^n$

P / partially_solved | Oberwolfach 报告问题汇编

For every integer $n\geq3$, does the equation $a^3+b^3=c^n$ have no nontrivial primitive integer solutions?

来源字段：https://doi.org/10.4171/owr/2009/33

01041 | OWR-12697688-003

### Generalized Ihara Lemma for Locally Symmetric Spaces

P / partially_solved | Oberwolfach 报告问题汇编

If $\mathfrak m$ is non-Eisenstein, prove that $\pi_{1*}+\pi_{2*}:H^d(X_K,\mathbb F_\ell)_{\mathfrak m}^{\oplus 2}\to H^d(X_{K_0(\mathfrak p)},\mathbb F_\ell)_{\mathfrak m}$ is injective.

来源字段：https://doi.org/10.4171/owr/2023/5

01042 | OWR-13102-002

### Generalized Kato Classes at Vanishing Artin-Twisted L-Values

O / open | Oberwolfach 报告问题汇编

For an elliptic curve $E$ over $\mathbb{Q}$ and an Artin representation $\rho$, assuming $L(E,\rho,1)=0$, does the generalized Kato class lie in $E(H)_\rho\otimes\mathbb{Q}_p$ and become trivial when the equivariant analytic rank is greater than $2$, with the rank-$2$ class given by the proposed $p$-adic-period formula?

来源字段：https://doi.org/10.4171/owr/2014/32

01043 | AMR-093-0078

### Generalized Ramanujan conjecture for automorphic representations

P / partially_solved | AMR 问题清单汇编

Let $K$ be a number field and let $\pi$ be a cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb A_K)$ with unitary central character. Is every local component $\pi_v$ tempered; equivalently, at unramified places do all normalized Satake parameters have absolute value one?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01044 | OWR-15583-003

### Generalized Riemann Hypothesis for Automorphic L-Functions

O / open | Oberwolfach 报告问题汇编

For every automorphic $L$-function having an Euler product, the Ramanujan bounds, analytic continuation, and a functional equation, do all nontrivial zeros lie on the critical line $\operatorname{Re}(s)=1/2$?

来源字段：https://doi.org/10.4171/owr/2017/40

01045 | AMR-093-0071

### Generalized Riemann hypothesis for Selberg class

O / open | AMR 问题清单汇编

Generalized Riemann hypothesis for Selberg class: do the nontrivial zeros of all functions in Selberg class lie on the critical line $1/2 + it$ with real $t$?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01046 | AIM-ALGEBRAIC_NUMBER_THEORY-0016

### Generic minimum density degree and a quartic counterexample

P / partially_solved | AIM 研讨会问题汇编

The following is asked as an analogue of a result due to Debarre and Klassen (arXiv:alg-geom/9210004).

Let $X_d\subset\mathbb P^3_k$ ($k$ a number field) be a smooth surface of degree $d$. Are the degree $\le d-3$ points not Zariski dense? What is $\min\delta(X_d/k)$?

You probably want to assume that $d$ is sufficiently large.

One can also ask this question for the generic degree $d$ hypersurface, defined over the field of rational functions in the appropriate number of variables.

For $d$ small, can we construct surfaces such that the degree $\le d-3$ points *are* Zariski dense?

来源字段：http://aimpl.org/degreedsurface/1/

01047 | OWR-12697688-004

### Generic Vanishing of Localized Shimura-Variety Cohomology

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathfrak m_p$ be a generic maximal ideal of the local Hecke algebra. Prove that $H^i(X_K,R)_{\mathfrak m_p}\neq 0$ implies $i\geq q_0$, and that $H_c^i(X_K,R)_{\mathfrak m_p}\neq 0$ implies $i\leq q_0+l_0$.

来源字段：https://doi.org/10.4171/owr/2023/5

01048 | AIM-ALGEBRAIC_NUMBER_THEORY-0017

### Generic-line gonality and the pencil-descent obstruction

P / partially_solved | AIM 研讨会问题汇编

Given a dominant map $\psi:X\dashrightarrow\mathbb P^{\dim X}$ of minimal degree $d$, is the gonality of the pullback of a general line equal to $d$?

来源字段：http://aimpl.org/degreedsurface/1/

01049 | OWR-4197-005

### Genus Three Hecke Trace Formula

P / partially_solved | Oberwolfach 报告问题汇编

For $a\geq b\geq c\geq0$, prove or disprove the genus-3 conjectural formula for the trace of the Hecke operator $T(p)$ on $S_{a-b,b-c,c+4}$: $\operatorname{tr}T(p)=e_c(A_3\otimes\mathbb{F}_p,V_{a,b,c})-e_{3,\mathrm{extra}}(a,b,c)$, where $e_{3,\mathrm{extra}}=-e_c(A_2,V_{a+1,b+1})-e_{2,\mathrm{extra}}(a+1,b+1)\otimes S[c+2]+e_c(A_2,V_{a+1,c})+e_{2,\mathrm{extra}}(a+1,c)\otimes S[b+3]-e_c(A_2,V_{b,c})-e_{2,\mathrm{extra}}(b,c)\otimes S[a+4]$.

来源字段：https://doi.org/10.4171/owr/2010/02

01050 | OWR-14298805-006

### Geometric and Local Arthur Packets for Classical Groups

P / partially_solved | Oberwolfach 报告问题汇编

For all local Arthur parameters of a $p$-adic classical group, prove the conjectural comparison between local Arthur packets and the corresponding geometric ABV packets, including the predicted pairing with equivariant perverse sheaves.

来源字段：https://doi.org/10.4171/owr/2024/54

01051 | OWR-14605-005

### Geometric Interpretations of Lattice Modular Forms

O / open | Oberwolfach 报告问题汇编

Let $M$ be a primitive $2$-elementary Lorentzian sublattice of the K3 lattice $L_{K3}$, put $\Lambda=M^\perp\subset L_{K3}$, and let $F_\Lambda$ be the associated vector-valued modular form. What is the geometric meaning of $F_\Lambda$? In particular, does it arise as the elliptic genus of a bundle over a K3 surface beyond the cases where this is already known?

来源字段：https://doi.org/10.4171/owr/2016/23

01052 | AIM-ANALYTIC_NUMBER_THEORY-0047

### Geometric rigidity and a Hecke obstruction for the apparent Ramanujan coefficients

P / partially_solved | AIM 研讨会问题汇编

Some coefficients of the Ramanujan tau function appear as coefficients of the piecewise polynomial in the work of Keating, Rodgers, Roditty-Gershon, and Rudnick. Is this a coincidence or not?

来源字段：http://aimpl.org/zetamoments/1/

01053 | AIM-ALGEBRAIC_NUMBER_THEORY-0014

### Geometric specialization versus arithmetic oscillation

P / partially_solved | AIM 研讨会问题汇编

How do all of these invariants (e.g. $\min\delta(X/k),\operatorname{irr}(X/k),\operatorname{cov.gon}(X/k)$, etc.) behave under specialization/deformation in a smooth family? This includes the arithmetic case, where the base is something like $\operatorname{Spec}\mathscr O_{k,S}$.

来源字段：http://aimpl.org/degreedsurface/1/

01054 | OWR-8415342-022

### Geometric Supercuspidals, Base Change, and Ramified Temperedness

P / partially_solved | Oberwolfach 报告问题汇编

Determine which Yu representations are monomial geometric supercuspidals; extend the proof beyond monomial representations; remove or replace the cyclic-base-change assumption; and determine when ramified Hecke operators establish temperedness at ramified places.

来源字段：https://doi.org/10.4171/owr/2021/39

01055 | OWR-3480-008

### Geometry of Galois Representation Spaces

O / open | Oberwolfach 报告问题汇编

For the rigid analytic space $X$ of Galois representations and its geometric locus $X_g$, determine the structure of $X_g$ and describe its Zariski closure in $X$.

来源字段：https://doi.org/10.4171/owr/2009/30

01056 | OWR-14211-005

### Geometry of Lattices from Finite Abelian Groups

P / partially_solved | Oberwolfach 报告问题汇编

Given a finite Abelian group G and $S=\{0=g_0,g_1,\ldots,g_{n-1}\}\subseteq G$, determine the geometric properties of $L_G(S)=\{x\in A_{n-1}:\sum_{j=1}^{n-1}x_jg_j=0\}$, including its minimum norm and determinant, number of minimal vectors, well-roundedness, minimal-vector bases, and automorphism group.

来源字段：https://doi.org/10.4171/owr/2016/3

01057 | OWR-17293-011

### Gilbreath Conjecture for Iterated Prime Differences

O / open | Oberwolfach 报告问题汇编

Does current knowledge about the primes suffice to prove Gilbreath's conjecture that every row after the first in the iterated unsigned-difference table of consecutive primes begins with 1?

来源字段：https://doi.org/10.4171/owr/2019/50

01058 | NT-009

### Gilbreath's Conjecture

O / open | 独立 / 未分配集合

别名：Gilbreath's Conjecture [NUM-010]; Gilbreath's Conjecture [GUY-A10]

Starting with the sequence of primes and repeatedly taking absolute differences of consecutive terms, the first term of each row is always 1.

背景链接：https://link.springer.com/article/10.1007/s00208-023-02579-w

01059 | AMR-093-0126

### Gillies' conjecture

O / open | AMR 问题清单汇编

Gillies' conjecture on the distribution of prime divisors of Mersenne numbers.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01060 | OPG-37411

### Giuga's Conjecture on Primality

O / open | Open Problem Garden

Conjecture $p$ is a prime iff $~\displaystyle \sum_{i=1}^{p-1} i^{p-1} \equiv -1 \pmod p$

背景链接：http://www.openproblemgarden.org/op/giugas_conjecture_on_primality

01061 | GUY-A17

### Giuga's Conjecture on Prime Characterization

P / partially_solved | Richard Guy 数论问题

Is it true that if $n$ divides $1^{n-1} + 2^{n-1} + \dots + (n-1)^{n-1} + 1$, then $n$ is prime?

背景链接：https://carmamaths.org/jon/giuga2013.pdf

01062 | AMR-030-0052

### Given any subset S of the integers modulo a prime p, what is the least K=K(p) for which there always exists an m so that

P / partially_solved | AMR 问题清单汇编

(Alon-Peres) Given any subset $S$ of the integers modulo a prime $p$, what is the least $K = K(p)$ for which there always exists an $m$ so that $mS$ has no gap of length greater than $K$? This question is particularly interesting if $S$ contains about half of the elements of $\mathbb{Z}_p$, since then it bears on questions concerning quadratic residues.

背景链接：https://people.math.sc.edu/cooper/combprob.html

01063 | OWR-12593-006

### Global L-Functions from Adelic Cyclic Homology

P / partially_solved | Oberwolfach 报告问题汇编

Can Connes-Consani cyclic homology with adelic coefficients be extended to give a global definition of the completed $L$-function $\Lambda(s,M)$, including analytic continuation, local epsilon factors, and a global duality that yields the functional equation?

来源字段：https://doi.org/10.4171/owr/2013/45

01064 | OWR-16158-015

### Global Realization of Prescribed Local Galois Extensions

O / open | Oberwolfach 报告问题汇编

Let $k$ be a number field, $G$ a finite group, $S$ a finite set of primes, and $F_{\mathfrak p}/k_{\mathfrak p}$ prescribed local Galois extensions whose Galois groups embed in $G$. Does there exist a global Galois extension $F/k$ with group $G$ whose completion at every $\mathfrak p\in S$ is $F_{\mathfrak p}/k_{\mathfrak p}$?

来源字段：https://doi.org/10.4171/owr/2018/17

01065 | LAN-001

### Goldbach's Conjecture

P / partially_solved | Landau 问题

别名：Goldbach conjecture [OPG-706]

Every even integer greater than 2 can be expressed as the sum of two primes.

背景链接：https://www.ams.org/bookstore/pspdf/gsm-134-prev.pdf

01066 | OWR-11695860-006

### Gonek’s Conjecture on Shifted Negative Zeta Moments

P / partially_solved | Oberwolfach 报告问题汇编

For fixed $k>0$, determine the uniform asymptotic order of $I_{-k}(\alpha,T)=T^{-1}\int_T^{2T}|\zeta(\tfrac12+\alpha+it)|^{-2k}\,dt$ throughout $0<\alpha\leq1$, including every transition regime near $\alpha=1/\log T$.

来源字段：https://doi.org/10.4171/owr/2022/50

01067 | AMR-093-0108

### Goormaghtigh conjecture

O / open | AMR 问题清单汇编

Goormaghtigh conjecture on solutions to $(x^m - 1)/(x - 1) = (y^n - 1)/(y - 1)$ where $x > y > 1$ and $m, n > 2$.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01068 | NT-069

### Grand Riemann Hypothesis

P / partially_solved | 独立 / 未分配集合

Do all automorphic L-functions have their nontrivial zeros on the critical line?

背景链接：https://arxiv.org/abs/2004.14410

01069 | AMR-093-0050

### Greenberg's Iwasawa-invariants conjecture

P / partially_solved | AMR 问题清单汇编

For every totally real number field $F$ and prime $p$, do the Iwasawa invariants $\lambda(F_\infty/F)$ and $\mu(F_\infty/F)$ of the cyclotomic $\mathbb Z_p$-extension both vanish; equivalently, is the $p$-part of the class numbers in the tower bounded?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01070 | AMR-093-0181

### Greenberg's p-rationality conjecture

P / partially_solved | AMR 问题清单汇编

For every odd prime $p$ and every positive integer $t$, does there exist a $p$-rational number field $K$ with $\operatorname{Gal}(K/\mathbb Q)\cong(\mathbb Z/2\mathbb Z)^t$?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01071 | AMR-093-0179

### Greenberg's pseudo-null conjecture

O / open | AMR 问题清单汇编

Let $F$ be totally real, let $\widetilde F$ be the compositum of all $\mathbb Z_p$-extensions of $F$, let $\widetilde L$ be its maximal unramified abelian pro-$p$ extension, and set $\widetilde X=\operatorname{Gal}(\widetilde L/\widetilde F)$. Is $\widetilde X$ pseudo-null over $\widetilde\Lambda=\mathbb Z_p[[\operatorname{Gal}(\widetilde F/F)]]$?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01072 | OWR-1274-005

### Greenberg's Pseudo-Nullity Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For every number field $F$, is the Galois group $X_{\widetilde F}$ pseudo-null as a module over the Iwasawa algebra $\widetilde\Lambda$?

来源字段：https://doi.org/10.4171/owr/2006/25

01073 | NT-033

### Grimm's Conjecture

P / partially_solved | 独立 / 未分配集合

Can each element of a set of consecutive composite numbers be assigned a distinct prime divisor?

背景链接：https://doi.org/10.1142/S1793042106000498

01074 | OWR-4804-003

### Gross-Prasad Labels for Multiplicity-One Pairs

P / partially_solved | Oberwolfach 报告问题汇编

Under the Langlands parametrizations of the two orthogonal-group $L$-packets, is the unique multiplicity-one pair $$\bigl(\pi(\phi,\epsilon_{\phi,\phi'}),\pi(\phi',\epsilon'_{\phi,\phi'})\bigr),$$ where the Gross-Prasad characters $\epsilon_{\phi,\phi'}$ and $\epsilon'_{\phi,\phi'}$ are defined by epsilon factors?

来源字段：https://doi.org/10.4171/owr/2011/14

01075 | OWR-3388-009

### Grothendieck's Section Conjecture over Number Fields

O / open | Oberwolfach 报告问题汇编

Let $X$ be a smooth proper geometrically integral curve of genus at least $2$ over a number field $k$. Is the natural map from $X(k)$ to conjugacy classes of continuous sections of $$\pi_1(X/k)\longrightarrow\operatorname{Gal}(\overline k/k)$$ a bijection?

来源字段：https://doi.org/10.4171/owr/2009/05

01076 | OWR-1188-010

### Grothendieck-Katz $p$-Curvature Conjecture

O / open | Oberwolfach 报告问题汇编

P-curvature Conjecture: let $(F,\partial_F)$ be a one-variable differential field over a number field $K$, and let $(M,\partial_M)$ be a differential module over $F$. Is it true that $M$ has a full system of algebraic solutions if and only if its $p$-curvature is trivial for almost all primes $p$?

来源字段：https://doi.org/10.4171/owr/2006/06

01077 | OWR-13952-005

### Grothendieck-Katz p-Curvature Conjecture

O / open | Oberwolfach 报告问题汇编

If $X$ is a smooth variety over $\mathbb C$ and an integrable connection $(E,\nabla)$ has trivial $p$-curvature at every closed point of a nonempty open of definition, must $(E,\nabla)$ have finite monodromy, equivalently be trivialized by a finite étale cover?

来源字段：https://doi.org/10.4171/owr/2015/57

01078 | NT-034

### Hall's Conjecture

P / partially_solved | 独立 / 未分配集合

For any $\varepsilon > 0$, is there a constant $c(\varepsilon)$ such that either $y^2 = x^3$ or $|y^2 - x^3| > c(\varepsilon) x^{1/2-\varepsilon}$?

背景链接：https://doi.org/10.1007/BF01140190

01079 | OWR-1786-003

### Harder Congruences and Yoshida-Type Siegel Modular Liftings

P / partially_solved | Oberwolfach 报告问题汇编

Let $f=\sum_{n\geq1}a(n)q^n$ be a normalized eigenform of weight $l+m+4$ for $\operatorname{SL}_2(\mathbb Z)$, and let $\ell$ be a sufficiently large prime in its coefficient field dividing $\Lambda(f,l+3)$. Must there exist a Siegel modular eigenform $F$ of weight $(l-m,m+3)$ whose Hecke eigenvalues satisfy $\lambda(p)\equiv p^{m+1}+a(p)+p^{l+2}\pmod{\ell^{\prime}}$ for a prime $\ell^{\prime}$ above $\ell$? If $f\in S_{l+m+4}(\Gamma_0(2))^{\mathrm{new}}$ and $g\in S_{l-m+2}(\Gamma_0(2))^{\mathrm{new}}$, must there exist a Siegel eigenform $F\in S_{l-m,m+3}(\Gamma_2[2])$ with $L(F,s)=L(f,s)L(g,s-m-1)$, with multiplicity five when the two newforms have the same $w_2$-eigenvalue and multiplicity one otherwise? Does the corresponding level-four lifting occur with multiplicity five in $S_{l-m,m+3}(\Gamma_2[2])$?

来源字段：https://doi.org/10.4171/owr/2008/05

01080 | OWR-17300-006

### Harder Congruences for Holomorphic Jacobi Forms

P / partially_solved | Oberwolfach 报告问题汇编

别名：Conditional Harder-Congruence Theorem [OWR-17300-007]

Assume $k\geq 3$, $j\geq 0$, and $j\equiv 2\pmod 4$. For a Hecke-eigen cusp form $f$ in $S_{2k+j-2}(\Gamma_1)$, let $\phi_0$ be the corresponding degree-one Jacobi form and $E(\phi_0)$ its degree-two Klingen lift. The holomorphic Jacobi-form version of Harder's conjecture asserts that there is a degree-two holomorphic Jacobi form $\Phi$ outside the lift and a prime ideal $\mathfrak l$ dividing $L^{\mathrm{alg}}(k+j,f)$ such that the Hecke eigenvalues of $\Phi$ and $E(\phi_0)$ are congruent modulo $\mathfrak l$ for every integral Hecke operator.

来源字段：https://doi.org/10.4171/owr/2019/57

01081 | HL-A

### Hardy-Littlewood Conjecture A (Prime k-tuples)

O / open | Hardy-Littlewood 猜想

别名：General Patterns of Consecutive Primes [GUY-A9]

Let $a_1, \ldots, a_k$ be given integers. Then there exist infinitely many positive integers $n$ such that $n + a_1, \ldots, n + a_k$ are all prime, provided that for every prime $p$, there exists an integer $m$ such that $(m + a_i, p) = 1$ for all $i$.

背景链接：https://arxiv.org/abs/1311.4600

01082 | HL-B

### Hardy-Littlewood Conjecture B (Second Conjecture)

O / open | Hardy-Littlewood 猜想

For all integers $x, y \geq 2$, we have $\pi(x+y) \leq \pi(x) + \pi(y)$, where $\pi(n)$ denotes the prime counting function (the number of primes less than or equal to $n$).

背景链接：https://doi.org/10.4064/aa-25-4-375-391

01083 | HL-F

### Hardy-Littlewood Conjecture F (Primes in Quadratic Polynomials)

P / partially_solved | Hardy-Littlewood 猜想

For a polynomial $f(x) = ax^2 + bx + c$ with $a > 0$, $\gcd(a,b,c) = 1$, and discriminant $\Delta = b^2 - 4ac$ not a perfect square, the polynomial takes infinitely many prime values. Furthermore, the number $P(n)$ of primes of the form $f(x) \leq n$ satisfies an asymptotic formula $P(n) \sim A \cdot \frac{\sqrt{n}}{\log n}$ where $A$ depends on $a$, $b$, $c$ but not on $n$.

背景链接：https://arxiv.org/abs/math/0703284

01084 | OWR-8415337-001

### Hardy-Littlewood Prime Tuple Conjecture

O / open | Oberwolfach 报告问题汇编

For every admissible finite set of distinct integer shifts $a_1,\ldots,a_k$, are there infinitely many integers $n$ for which $n+a_1,\ldots,n+a_k$ are all prime?

来源字段：https://doi.org/10.4171/owr/2021/34

01085 | OWR-13952-006

### Harmonic Metrizations of Line Bundles

P / partially_solved | Oberwolfach 报告问题汇编

For any line bundle $M$ on $X$, does there exist a metrization $\overline{M}$, unique up to scale, whose curvature $c_1(\overline{M})$ is $\omega$-harmonic, meaning it is $\omega$-bounded and satisfies $c_1(\overline{M})\omega^{n-1}=\lambda(M)\omega^n$?

来源字段：https://doi.org/10.4171/owr/2015/57

01086 | OWR-15169-006

### Hausdorff Dimension of Intersections of Badly Approximable Sets

O / open | Oberwolfach 报告问题汇编

For any specified system of Diophantine inequalities defining a generalized badly approximable set $\operatorname{Bad}$, does $\operatorname{Bad}$ have full Hausdorff dimension? More generally, for a countable family $\operatorname{Bad}_1,\operatorname{Bad}_2,\ldots$, determine the size and Hausdorff dimension of $\bigcap_{i=1}^{\infty}\operatorname{Bad}_i$.

来源字段：https://doi.org/10.4171/owr/2016/48

01087 | AIM-ALGEBRAIC_NUMBER_THEORY-0061

### Heegner transfer on a split genus-two locus

P / partially_solved | AIM 研讨会问题汇编

**1. (C. Hall)** We know how to construct points on an elliptic curve $E$ over $\mathbb F_q(T)$ when the analytic rank is 1 by using (Drinfel'd) Heegner points. What if $X$ is a hyperelliptic curve over $\mathbb F_q(T)$ of higher genus? Is there any way of constructing the points on the Jacobian of $X$ that ought to be there?

**Comments.** 1) We have a lot of explicit examples when the analytic rank is 1. (C. Hall)

2) If we start with any surface $S$ over $\mathbb F_q$ we can, via Lefschetz fibration, make it into a curve over $\mathbb F_q(T)$. Typically (since the rank of $H^2$ is even), one has two Tate classes. One of these is given by hyperplane sections; can we “see” or construct the other one? This question is like the one above, but without requiring $C$ to be hyperelliptic. So, in some sense, the problem is almost as hard as the problem of treating general surfaces. (C. Schoen)

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

01088 | OWR-12725-016

### Height Counts for Closed Projective Immersions

O / open | Oberwolfach 报告问题汇编

Let $f:\mathbb P^n\to\mathbb P^m$ be a degree-$d$ morphism induced by degree-$d$ forms with no common zero, and suppose that $f$ is a closed immersion. Does

$$n_f(B)=O_{d,n,\varepsilon}\!\left(B^{(n+1)/d+\varepsilon}\right),$$

where $n_f(B)=\#\{x\in\mathbb P^n(\mathbb Q):H(f(x))\leq B\}$?

来源字段：https://doi.org/10.4171/owr/2013/51

01089 | OWR-4076-007

### Height Heuristics for Bombieri-Lang in Higher Dimensions

O / open | Oberwolfach 报告问题汇编

What is the higher-dimensional analogue of the curve condition $g>1$, and can height-counting heuristics be generalized to predict the Bombieri-Lang conjecture for every variety of general type?

来源字段：https://doi.org/10.4171/owr/2009/33

01090 | AIM-ALGEBRAIC_NUMBER_THEORY-0085

### Height-window bounds and modern status for random hypersurface solubility

P / partially_solved | AIM 研讨会问题汇编

$$\mathcal F_{d,n}(H)=\{f\in\mathbb Z[x_0,\ldots,x_n]_d:\max_I |a_I|\le H\},\qquad m=\binom{n+d}{d},$$

where $f=\sum_Ia_Ix^I$ is homogeneous of degree $d$. Then

$$N_{\rm tot}(H)=\#\mathcal F_{d,n}(H)=(2\lfloor H\rfloor+1)^m,$$

$N(H)$ counts the members having a nonzero integral zero (equivalently a rational point on $f=0\subset\mathbb P^n$), and $N_{\rm loc}(H)$ counts those having a point over every completion of $\mathbb Q$.

1. If $d>n+1$, does $N(H)/N_{\rm tot}(H)\to0$, and does this follow from Lang's conjectures? 2. Is $N_{\rm loc}(H)/N_{\rm tot}(H)$ bounded below by a positive constant? 3. For a random plane cubic, how likely is a rational point? 4. For cubic surfaces, and more generally in the Fano range $d\le n$, is $$\frac{N(H)}{N_{\rm tot}(H)}\sim \frac{N_{\rm loc}(H)}{N_{\rm tot}(H)}?$$

The PDF literally says "max coefficient". This reconstruction is verified by the definitions in Poonen--Voloch, the update linked from the AIM record. Their paper also makes precise that counting primitive equations or projective hypersurfaces instead of all integral equations does not alter the limiting ratios.

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01091 | AIM-ALGEBRAIC_NUMBER_THEORY-0052

### Heilbronn sums as exact Fourier quadrature

P / partially_solved | AIM 研讨会问题汇编

The PDF literally prints

\[ \sum_{x\bmod p} \exp\!\left(\frac{2\pi i x^p(1+pt)}{p^2}\right) = \sum_{x\bmod p} \exp\!\left(\frac{2\pi i x^p}{p^2}\right) \cdot \exp\!\left(\frac{2\pi i pt}{p^2}\right), \]

and then asks why these sums, as $t$ varies, "seem to approximate ... the (push forward of) Haar measure better than expected."

There is a source-level error in the displayed equality. The last factor on the printed right-hand side is independent of $x$, so that side is merely $e_p(t)$ times the sum at $t=0$. Direct algebra instead gives

\[ e_{p^2}\!\left(x^p(1+pt)\right) =e_{p^2}(x^p)e_{p^2}(ptx^p) =e_{p^2}(x^p)e_p(tx), \tag{1.1} \]

where $e_N(u)=\exp(2\pi i u/N)$ and $x^p\equiv x\pmod p$. Thus the algebraically forced correction is

\[ S_p(t):=\sum_{x\bmod p}e_{p^2}\!\left(x^p(1+pt)\right) =\sum_{x\bmod p}e_{p^2}(x^p)e_p(tx). \tag{1.2} \]

The PDF does **not** specify a range for $t$, a normalization, the compact group carrying Haar measure, or the pushforward map. The left side is periodic in $t\pmod p$, so the effective range is $t\in\mathbb F_p$. The following is the minimal natural reconstruction compatible with (1.2): the right side samples one real Laurent polynomial at all $p$-th roots of unity, and the comparison measure is the pushforward of normalized Haar measure on the unit circle by that same polynomial. A different random-matrix Haar target cannot be recovered from the PDF alone.

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

01092 | OWR-10252929-010

### Hermite and GL2 Equivalence Classes of Number Fields

P / partially_solved | Oberwolfach 报告问题汇编

Does $\operatorname{PI}(K)$ contain only finitely many Hermite-equivalence classes that split into more than one $\operatorname{GL}_2(\mathbb{Z})$-equivalence class?

来源字段：https://doi.org/10.4171/owr/2022/21

01093 | AMR-093-0051

### Hermite's problem

P / partially_solved | AMR 问题清单汇编

Hermite's problem: is it possible, for any natural number $n$, to assign a sequence of natural numbers to each real number such that the sequence for $x$ is eventually periodic if and only if $x$ is algebraic of degree $n$?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01094 | AIM-ANALYTIC_NUMBER_THEORY-0073

### High moments of a prime Dirichlet polynomial: collision corrections and a finite-height cutoff

P / partially_solved | AIM 研讨会问题汇编

Develop a heuristic for \[ \int_T^{2T}\left|\sum_{p\leq X}\frac{1}{p^{1/2+it}}\right|^{2k}\,dt. \] Find connections with the work of Bogomolny--Keating and with the Ratios Conjectures.

来源字段：http://aimpl.org/zetamoments/8/

01095 | OWR-15426-001

### Higher Central-Derivative Formulas for L-Functions

P / partially_solved | Oberwolfach 报告问题汇编

Develop formulas for higher-order central derivatives of automorphic $L$-functions that extend the Waldspurger and Gross-Zagier formulas over number fields and function fields.

来源字段：https://doi.org/10.4171/owr/2017/18

01096 | OWR-14299577-016

### Higher-Rank Shifted Convolution Sums

O / open | Oberwolfach 报告问题汇编

For Fourier coefficients $a(n)$ and $b(n)$ of higher-rank automorphic forms, obtain nontrivial estimates and asymptotic formulas for shifted convolution sums such as $\sum_{n\leq x}a(n)b(n+1)$ beyond the known low-rank cases.

来源字段：https://doi.org/10.4171/owr/2025/51

01097 | HIL-011

### Hilbert's 11th Problem: Quadratic Forms over Algebraic Number Fields

P / partially_solved | Hilbert 问题

Extend the theory of quadratic forms with algebraic numerical coefficients.

背景链接：https://encyclopediaofmath.org/wiki/Hilbert_problems

01098 | HIL-012

### Hilbert's 12th Problem: Extension of Kronecker-Weber Theorem

P / partially_solved | Hilbert 问题

别名：Hilbert's Twelfth Problem [NT-065]; Hilbert's 12th Problem [NUM-013]

Extend the Kronecker-Weber theorem on abelian extensions of the rationals to any base number field.

背景链接：https://encyclopediaofmath.org/wiki/Hilbert_problems

01099 | HIL-007

### Hilbert's 7th Problem: Transcendence of Certain Numbers

S / solved | Hilbert 问题

If $\alpha$ is algebraic and irrational, and $\beta$ is algebraic and irrational, is $\alpha^\beta$ transcendental?

背景链接：https://doi.org/10.1515/crll.1935.172.65

01100 | HIL-009

### Hilbert's 9th Problem: Reciprocity Laws

P / partially_solved | Hilbert 问题

Generalize the reciprocity law of number theory to arbitrary number fields.

背景链接：https://encyclopediaofmath.org/wiki/Hilbert_problems

01101 | HIL-010

### Hilbert's Tenth Problem for Number Fields

P / partially_solved | Hilbert 问题

别名：Hilbert's Tenth Problem for Number Fields [MOD-003]

For which number fields is there an algorithm to determine solvability of Diophantine equations?

背景链接：https://doi.org/10.1007/s00222-025-01392-3

01102 | NT-068

### Hilbert-Pólya Conjecture

O / open | 独立 / 未分配集合

Do the nontrivial zeros of the Riemann zeta function correspond to eigenvalues of a self-adjoint operator?

背景链接：https://en.wikipedia.org/wiki/Hilbert%E2%80%93P%C3%B3lya_conjecture

01103 | OWR-8415342-023

### Hiraga-Ichino-Ikeda Degrees for Essentially Discrete Series

O / open | Oberwolfach 报告问题汇编

For an essentially discrete-series representation $\pi$ of $G$ with extended parameter $(\phi,\rho)$, prove the Hiraga-Ichino-Ikeda formula $d(\pi,\nu_G)=\frac{\dim\rho}{|S_\phi|}\lvert\gamma(0,\phi,\operatorname{Ad}_G)\rvert$.

来源字段：https://doi.org/10.4171/owr/2021/39

01104 | OWR-8415342-003

### Hiraga-Ichino-Ikeda Formal Degree Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For every discrete-series representation $\sigma$ of $H$, prove the Hiraga-Ichino-Ikeda formula $d_H(\sigma)=\frac{\dim(\rho_\sigma)}{|S_\sigma|}\lvert\gamma(0,\sigma,\operatorname{Ad}_H)\rvert$.

来源字段：https://doi.org/10.4171/owr/2021/39

01105 | OWR-8415342-002

### Hiraga-Ichino-Ikeda-Langlands Plancherel Density Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Hiraga-Ichino-Ikeda-Langlands Conjecture [OWR-8415342-004]

For almost every tempered representation $\sigma$ of a reductive group $H$, prove the Hiraga-Ichino-Ikeda-Langlands formula $d_H(\sigma)=\frac{\dim(\rho_\sigma)}{|S_\sigma|}\lvert\gamma^*(0,\sigma,\operatorname{Ad}_H)\rvert$, where $\gamma^*$ is the suitably normalized first nonzero Laurent coefficient at $s=0$.

来源字段：https://doi.org/10.4171/owr/2021/39

01106 | OWR-14297740-017

### Homeomorphism Conjecture for Moduli of Fundamental Groups

P / partially_solved | Oberwolfach 报告问题汇编

For pointed stable curves of type $(g,n)$ over algebraically closed fields of characteristic $p>0$, is the natural continuous map $\pi_{g,n}^{\mathrm{adm}}:\mathcal M_{g,n}^{\mathrm{adm}}\twoheadrightarrow\Pi_{g,n}$ from Frobenius-equivalence classes of curves to isomorphism classes of their admissible fundamental groups a homeomorphism?

来源字段：https://doi.org/10.4171/owr/2023/42

01107 | AMR-030-0067

### How quickly do the gaps between successive primes grow

P / partially_solved | AMR 问题清单汇编

How quickly do the gaps between successive primes grow? Is it slower than n^(l') for every l' > 0? See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

01108 | AIM-ANALYTIC_NUMBER_THEORY-0086

### Hurwitz class numbers: missing-context audit, all-index formula, and CM groupoid count

P / partially_solved | AIM 研讨会问题汇编

别名：Hurwitz class numbers, CM enumeration, and local conductor layers [AIM-PHYSICS-0019]

Here $h_m$ is the $m$-th Hurwitz class number, i.e. the number of equivalence classes of positive definite binary quadratic forms of discriminant $-m$ with the class containing $x^2+y^2$ weighted by 1/2 and the class containing $x^2 + xy + y^2$ weighted by 1/3. Moreover by convention $h_0=-1/12.$

For more information in regards to the connection between physics and number theory, see the following work of Kathrin Bringmann and Ben Kane (arXiv:1305.0112), Katrin Bringmann and Sameer Murthy (arXiv:1208.3476), and Katrin Bringmann and Jan Manschot (arXiv:1304.7208).

Do the coefficients of $h_m$ have enumerative significance? Can we find $h_m$ for $m \geq 9$?

来源字段：http://aimpl.org/gromwitnumthry/4/

01109 | AMR-030-0060

### If F is a finite field with at least 4 elements and A is an invertible n by n matrix over F, then there are vectors x, y

P / partially_solved | AMR 问题清单汇编

Jaeger: If F is a finite field with at least 4 elements and A is an invertible n by n matrix over F, then there are vectors x, y in F^(n) which haveall nonzero coordinates so that Ax = y. See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

01110 | OWR-8415350-006

### Igusa Bounds for Finite-Ring Exponential Sums

P / partially_solved | Oberwolfach 报告问题汇编

Let $f\in\mathbb Z[x_1,\ldots,x_n]$ be nonconstant of degree $d>1$, let $f_d$ be its highest-degree homogeneous part, and let $s$ be the dimension of the critical locus of $f_d$. For every $\varepsilon>0$ and prime $p$, is there a $C_{p,\varepsilon}>0$ such that $$ |E_f(p,m)|\leq C_{p,\varepsilon}p^{m(-(n-s)/d+\varepsilon)} $$ for every $m\geq1$, with $C_{p,\varepsilon}=1$ for all sufficiently large $p$?

来源字段：https://doi.org/10.4171/owr/2021/47

01111 | OWR-16171-012

### Images of Infinite Galois Representations

P / partially_solved | Oberwolfach 报告问题汇编

Let $\rho:G_{\mathbb Q}\to\operatorname{GL}_2(A)$ be a continuous representation that is irreducible, is not induced from a character of an index-two subgroup, and has infinite image. What is the structure of $\operatorname{Im}\rho$?

来源字段：https://doi.org/10.4171/owr/2018/30

01112 | AMR-030-0050

### In the binary expansion of sqrt(2), are there arbitrarily long sequences of 0's

P / partially_solved | AMR 问题清单汇编

Erdős: In the binary expansion of sqrt(2), are there arbitrarily long sequences of 0's? Can you find a single algebraic number with this property?

背景链接：https://people.math.sc.edu/cooper/combprob.html

01113 | OWR-14211-008

### Indefinite Theta Series and Real Jacobi Representations

O / open | Oberwolfach 报告问题汇编

Can indefinite theta series formed from lattice sums over cones, together with modular completions annihilated by the real-Jacobi Casimir operator, be realized through principal-series representations of the real Jacobi group?

来源字段：https://doi.org/10.4171/owr/2016/3

01114 | OWR-14297740-020

### Inertia Actions on Torsion Points under Supersingular Reduction

P / partially_solved | Oberwolfach 报告问题汇编

Let $X/K$ be a geometrically connected curve of genus greater than one with Jacobian $J$, and embed $X$ in $J$ using a point $x_0\in X(K)$. If $J$ has supersingular good reduction over $K$, is the inertia action on $X_{\mathrm{tors}}=X(K)\cap J(K)_{\mathrm{tors}}$ trivial?

来源字段：https://doi.org/10.4171/owr/2023/42

01115 | OWR-3388-005

### Infinite Rank of Abelian Varieties over $\mathbb Q^{\mathrm{ab}}$

O / open | Oberwolfach 报告问题汇编

Does every nonzero abelian variety $A/\mathbb Q$ acquire infinite rank over the maximal abelian extension $\mathbb Q^{\mathrm{ab}}$?

来源字段：https://doi.org/10.4171/owr/2009/05

01116 | NT-047

### Infinitely Many Amicable Pairs

P / partially_solved | 独立 / 未分配集合

文本重复：NUM-007

Are there infinitely many pairs of amicable numbers?

背景链接：https://www.erdosproblems.com/830

01117 | NT-048

### Infinitely Many Giuga Numbers

P / partially_solved | 独立 / 未分配集合

Are there infinitely many Giuga numbers?

背景链接：https://gauss.dartmouth.edu/~carlp/giugafinal.pdf

01118 | NT-041

### Infinitely Many Perfect Numbers

P / partially_solved | 独立 / 未分配集合

文本重复：NUM-003

Are there infinitely many perfect numbers?

背景链接：https://www.mersenne.org/primes/

01119 | OWR-11695860-004

### Infinitely Many Prime Values of Integer Polynomials

O / open | Oberwolfach 报告问题汇编

Let $P\in\mathbb Z[X]$ be irreducible of degree at least two with no fixed prime divisor. Prove that $P(n)$ is prime for infinitely many integers $n$.

来源字段：https://doi.org/10.4171/owr/2022/50

01120 | NUM-007

### Infinitude of Amicable Pairs

O / open | 独立 / 未分配集合

文本重复：NT-047

Are there infinitely many pairs of amicable numbers?

背景链接：https://en.wikipedia.org/wiki/Amicable_numbers

01121 | NUM-003

### Infinitude of Perfect Numbers

O / open | 独立 / 未分配集合

文本重复：NT-041

Are there infinitely many perfect numbers?

背景链接：https://www.mersenne.org/primes/

01122 | GUY-A7a

### Infinitude of Sophie Germain Primes

O / open | Richard Guy 数论问题

Are there infinitely many Sophie Germain primes? A prime $p$ is called a Sophie Germain prime if $2p + 1$ is also prime.

背景链接：https://fme.upc.edu/ca/la-facultat/activitats-fme-personalitats-del-curs/personalitat-del-curs/2025-2026-germain/nombresprimers

01123 | OWR-15170-018

### Infinitude of Twin Primes

O / open | Oberwolfach 报告问题汇编

Are there infinitely many twin-prime pairs; equivalently, for every $n$, are there at least $n$ pairs of primes differing by $2$?

来源字段：https://doi.org/10.4171/owr/2016/49

01124 | NT-077

### Integer Factorization in Polynomial Time

O / open | 独立 / 未分配集合

别名：Integer Factorization Polynomial Time [NUM-020]

Can integer factorization be solved in polynomial time on a classical computer?

背景链接：https://arxiv.org/abs/quant-ph/9508027

01125 | AIM-ALGEBRAIC_NUMBER_THEORY-0032

### Integral BKLPR matrices and the rank-2-to-rank-3 exponent transfer

S / solved | AIM 研讨会问题汇编

**Question 2 (Melanie Matchett Wood).** Develop a version of the BKLPR heuristics with \(\mathbb Z_p\) replaced by \(\mathbb Z\), counting up to height \(B\). Use the rank 2 case to calibrate \(B\) with the height used when counting elliptic curves. Then use that calibration to make a rank 3 prediction and compare this with predictions coming from random matrix theory.

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

01126 | AIM-ALGEBRAIC_NUMBER_THEORY-0081

### Integral cycle classes on the theta line

P / partially_solved | AIM 研讨会问题汇编

**18. (E. Izadi)** Take a principally polarized abelian variety \(A\) and assume \(\operatorname{rank}(\operatorname{NS}(A))=1\). What is the index of \(C^i(A)\) in \(H^{2i}(A,\mathbb Z(i))\)? What is the image? What if we just restrict ourselves to the line spanned by \([\theta]^i\), where \(\theta\) is the polarization?

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

01127 | OWR-16764-001

### Integral Models of Parahoric Shimura Varieties

O / open | Oberwolfach 报告问题汇编

Determine the existence and characterize the uniqueness of integral models of Shimura varieties with parahoric level structure.

来源字段：https://doi.org/10.4171/owr/2019/2

01128 | OWR-1537-016

### Integral Points on Finite Covers of Semiabelian Varieties

O / open | Oberwolfach 报告问题汇编

Let $G$ be a semiabelian variety over a number field $k$, let $V$ be a smooth $k$-variety, let $\pi:V\to G$ be a finite $k$-morphism, and let $S$ be a finite set of places of $k$. If the $S$-integral points $V(\mathcal{O}_S)$ are Zariski-dense in $V$, then $\pi$ should be an unramified covering; in particular, $V$ should have a semiabelian variety structure for which $\pi$ is an isogeny.

来源字段：https://doi.org/10.4171/owr/2007/21

01129 | OWR-8415337-002

### Integral Points over Almost All Primitive Number Fields

O / open | Oberwolfach 报告问题汇编

Let $X/\mathbb Q$ be a hyperbolic curve with a model over $\mathbb Z$, and let $n\geq1$. When primitive degree-$n$ number fields $K$ are ordered by discriminant, does $X(\mathcal O_K)=X(\mathbb Z)$ for $100$ percent of them?

来源字段：https://doi.org/10.4171/owr/2021/34

01130 | OWR-14613-003

### Integral Reduced Norms in Division-Algebra K-Theory

O / open | Oberwolfach 报告问题汇编

Let $K$ be a complete discretely valued field with finite residue field of characteristic $p$ and $D$ a central division algebra over $K$. Is the reduced norm isomorphism $\operatorname{Nrd}_{D/K}:K_j(D,\mathbb Z_p)\to K_j(K,\mathbb Z_p)$ induced by a map of K-theory spectra, equivalently yielding an integral reduced-norm isomorphism on K-groups? In particular, are the $p$-primary torsion subgroups of the integral K-groups of bounded exponent?

来源字段：https://doi.org/10.4171/owr/2016/31

01131 | OWR-1053-002

### Integral Sequences and Algebraic Generating Functions

O / open | Oberwolfach 报告问题汇编

Characterize the integer sequences $(u_n)_{n\geq0}$ whose ordinary generating function $\sum_{n\geq0}u_nz^n$ is algebraic.

来源字段：https://doi.org/10.4171/owr/2005/32

01132 | OWR-1795-007

### Integral Zeros and Counting for Polynomial Forms

P / partially_solved | Oberwolfach 报告问题汇编

For a form $F\in\mathbb{Z}[x_1,\ldots,x_n]$ of degree $d$, when does $F$ have a nontrivial zero in $\mathbb{Z}^n$? How does $N(B)=|\{x\in\mathbb{Z}^n:F(x)=0,\ \max_i|x_i|\leq B\}|$ behave as $B\to\infty$, especially for nonsingular cubic and quartic forms beyond diagonal or high-dimensional cases?

来源字段：https://doi.org/10.4171/owr/2008/14

01133 | OWR-1542-009

### Integral-Point Non-Density on Plane-Divisor Complements

O / open | Oberwolfach 报告问题汇编

别名：Duplicate: Integral-Point Non-Density on Plane-Divisor Complements [OWR-1542-010]

Let $D\subset\mathbb P^2_{\mathbb Q}$ be a normal-crossing divisor of degree at least four, set $X=\mathbb P^2_{\mathbb Q}\setminus D$, choose an affine embedding $X\hookrightarrow\mathbb A^n_{\mathbb Q}$, and let $S$ be a finite set of primes. Are the $S$-integral points on $X$ non-Zariski-dense?

来源字段：https://doi.org/10.4171/owr/2007/26

01134 | AIM-ANALYTIC_NUMBER_THEORY-0078

### Integrating holomorphic anomaly equations: known cases and a channel-rank obstruction

P / partially_solved | AIM 研讨会问题汇编

别名：When holomorphic anomaly equations survive fiber integration [AIM-PHYSICS-0011]

Does the holomorphic anomaly equation "integrate" over elliptic fibrations?

来源字段：http://aimpl.org/gromwitnumthry/1/

01135 | AIM-ANALYTIC_NUMBER_THEORY-0068

### Interior-frequency isolation of the genuine zeta triple term

P / partially_solved | AIM 研讨会问题汇编

Compute the $3$-point correlation function $R_3(x_1,x_2,x_3)$ for zeta rigorously under suitable hypotheses (e.g. Hardy-Littlewood conjecture for twin primes).

来源字段：http://aimpl.org/zetamoments/7/

01136 | OWR-11137-006

### Internal Isomorphism Invariants of Number Fields

O / open | Oberwolfach 报告问题汇编

Can an invariant internal to a number field classify it up to isomorphism, yielding an internal analogue of the known external reconstruction results?

来源字段：https://doi.org/10.4171/owr/2011/45

01137 | OWR-12593-005

### Intrinsic Criteria for Expected L-Function Properties

O / open | Oberwolfach 报告问题汇编

Which intrinsic properties of an arithmetic or geometric object $M$ ensure that its $L$-function has the expected local-to-global properties: an Euler product, an archimedean factor, meromorphic continuation, and a functional equation?

来源字段：https://doi.org/10.4171/owr/2013/45

01138 | OWR-15169-010

### Intrinsic Diophantine Exponents of Submanifolds

O / open | Oberwolfach 报告问题汇编

For a nondegenerate $k$-dimensional submanifold $M\subset\mathbb R^d$, define $$ \omega(M)=\inf_{x\in M}\limsup_{p/q\in M}\frac{-\log|x-p/q|}{\log q}, $$ and let $\omega(k,d)$ be the supremum over such $M$. Can the available upper bound be improved, can $\omega(k,d)$ be computed exactly for more pairs $(k,d)$, and does the essential-infimum variant $\omega'(k,d)$ equal $\omega(k,d)$?

来源字段：https://doi.org/10.4171/owr/2016/48

01139 | AIM-ANALYTIC_NUMBER_THEORY-0013

### Intrinsic level and the prime-power breakthrough

P / partially_solved | AIM 研讨会问题汇编

**Subconvexity bound for \(\mathrm{GL}_3\times\mathrm{GL}_2\) in level aspect.** Let \(\pi\) be a fixed Hecke--Maass cusp form for \(SL_3(\mathbb Z)\) and \(f\) be a holomorphic for Hecke Maass cusp form for \(\Gamma_0(M)\). Obtain a subconvexity bound estimate for \(L(1/2,\pi\times f)\) as \(M\) grows.

\(f\) is a holomorphic or Hecke--Maass cusp form for \(\Gamma_0(M)\).

来源字段：http://aimpl.org/deltasubconvex2/1/

01140 | OWR-12866-005

### Invariant Dirichlet-Series Subspace Characterized by Zeta Divisibility

O / open | Oberwolfach 报告问题汇编

Let $E^2$ be the Hilbert space of Dirichlet series $D(s)$ for which $D(s)/s$ extends to a function in $H^2(\operatorname{Re}s>1/2)$. Let $V$ be the closure in $E^2$ of the functions $$\zeta(s)\sum_{k=1}^n c_k k^{-s}\qquad\text{with}\qquad \sum_{k=1}^n\frac{c_k}{k}=0.$$ Is $$V=\{D\in E^2:D(s)/\zeta(s)\text{ is holomorphic for }\operatorname{Re}s>1/2\}$$?

来源字段：https://doi.org/10.4171/owr/2014/06

01141 | OWR-8415342-011

### Invariant Functionals and Automorphic Fourier Transforms

O / open | Oberwolfach 报告问题汇编

For a pair $(G,\rho)$ and a cuspidal automorphic representation $\sigma$, do there exist nonzero $k^\times$-invariant linear functionals $E_{\sigma,\rho}$ and $E_{\widetilde{\sigma},\rho}$ such that $E_{\sigma,\rho}(\phi)=E_{\widetilde{\sigma},\rho}(F_{\sigma,\rho,\psi}\phi)$ for every $\phi\in S_{\sigma,\rho}(\mathbb{A}^\times)$?

来源字段：https://doi.org/10.4171/owr/2021/39

01142 | OWR-12866-004

### Irrational Local Maxima of the Fractional-Part Autocorrelation

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Composite: Nyman Criterion and Autocorrelation Maximum [OWR-12866-003]

For $$A(\lambda)=\int_0^\infty \{t\}\{\lambda t\}\,\frac{dt}{t^2},$$ does there exist a positive irrational $\lambda$ at which $A$ has a strict local maximum?

来源字段：https://doi.org/10.4171/owr/2014/06

01143 | OWR-10252929-004

### Irrationality of Catalan’s Constant

O / open | Oberwolfach 报告问题汇编

Is Catalan's constant $G=\sum_{n\geq0}(-1)^n/(2n+1)^2$ irrational?

来源字段：https://doi.org/10.4171/owr/2022/21

01144 | NT-074

### Irrationality of Euler's Constant

O / open | 独立 / 未分配集合

Is the Euler-Mascheroni constant $\gamma$ irrational?

背景链接：https://doi.org/10.1073/pnas.2321440121

01145 | GREEN-058

### Irreducibility of Random {0,1} Polynomials

P / partially_solved | Ben Green 问题

Is a random polynomial with coefficients in $\{0, 1\}$ and nonzero constant term almost surely irreducible?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

01146 | NT-053

### Is 10 a Solitary Number?

P / partially_solved | 独立 / 未分配集合

Is 10 a solitary number (no other number shares its abundancy index)?

背景链接：https://arxiv.org/abs/2310.15900

01147 | AMR-093-0176

### Is 509,203 the lowest Riesel number

O / open | AMR 问题清单汇编

Is 509,203 the lowest Riesel number?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01148 | AMR-093-0175

### Is every Fermat number $2^{2^n} + 1$ composite for $n > 4$

O / open | AMR 问题清单汇编

Is every Fermat number $2^{2^n} + 1$ composite for $n > 4$?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01149 | AMR-030-0059

### Is it possible to choose 2n points in an n by n grid in the plane so that no three are collinear

O / open | AMR 问题清单汇编

Dudeney: Is it possible to choose 2n points in an n by n grid in the plane so that no three are collinear? Conjecture: no. In fact, it is conjectured that the answer is still "no" unless 2 is changed to something less than Ŕ/3 = 1.813799.... However, this problem dates back to 1917 and little is known about it. See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

01150 | AMR-030-0063

### Is it possible, for each positive integer n, to find positive integers a, b, and c so that 4/n = 1/a + 1/b + 1/c

P / partially_solved | AMR 问题清单汇编

Erdős-Strauss : Is it possible, for each positive integer n, to find positive integers a, b, and c so that 4/n = 1/a + 1/b + 1/c ? See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

01151 | AMR-030-0062

### Is it true that, for every n, there is an integer M(n), so that whenever a linear homogeneous equation in n variables is

O / open | AMR 问题清单汇编

Rado: Is it true that, for every n, there is an integer M(n), so that whenever a linear homogeneous equation in n variables is Ramsey (in the positive integers) for M(n)-colors, it is Ramsey for any number of colors?

背景链接：https://people.math.sc.edu/cooper/combprob.html

01152 | AMR-093-0063

### Is Selberg class of Dirichlet series equal to class of automorphic L-functions

O / open | AMR 问题清单汇编

Is Selberg class of Dirichlet series equal to class of automorphic L-functions?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01153 | OPG-37366

### Is Skewes' number e^e^e^79 an integer?

O / open | Open Problem Garden

Conjecture

Skewes' number $e^{e^{e^{79}}}$ is not an integer.

背景链接：http://www.openproblemgarden.org/op/is_skewes_number_e_e_e_79_an_integer

01154 | AMR-030-0069

### Is the least quadratic residue modulo p at most p^(l')^( )for any l' > 0

P / partially_solved | AMR 问题清单汇编

Is the least quadratic residue modulo p at most p^(l')^( )for any l' > 0?

背景链接：https://people.math.sc.edu/cooper/combprob.html

01155 | AMR-030-0066

### Is there a dense of points in the real plane so that every two points are at a rational distance

P / partially_solved | AMR 问题清单汇编

Ulam : Is there a dense of points in the real plane so that every two points are at a rational distance? See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

01156 | AMR-030-0068

### Is there a prime between n^(2)and (n+1)^(2 )for every n > 0

P / partially_solved | AMR 问题清单汇编

Erdős: Is there a prime between n^(2)and (n+1)^(2 )for every n > 0?

背景链接：https://people.math.sc.edu/cooper/combprob.html

01157 | AMR-030-0061

### Is x^(2)+y^(2)=z^(2) partition regular

P / partially_solved | AMR 问题清单汇编

Graham: Is x^(2)+y^(2)=z^(2) partition regular? That is, is it true that every coloring of the positive integers by a finite number of colors contains a monochromatic Pythagorean triple? (Update 2017: Heule+Kullman+Marek showed that this cannot be done with two colors, and that the smallest n for which it becomes impossible to 2-color [n] is exactly 7825!)

背景链接：https://people.math.sc.edu/cooper/combprob.html

01158 | OWR-3480-006

### Iwasawa Main Conjecture for Dihedral Ring Class Towers

P / partially_solved | Oberwolfach 报告问题汇编

Let $K$ be a finite extension of the imaginary quadratic field $k$ inside its cyclotomic extension, and let $F$ be the maximal totally real subfield of $K$. Assume that the root number of $L(E/k,s)$ is $+1$, that $p\geq5$, that the representation on $E[p]$ has image $\operatorname{GL}_2(\mathbb{F}_p)$, that $p$ does not divide the minimal modular degree, and that $E[p]$ is an irreducible $I_v$-module whenever $v^2\mid N\mathcal{O}_F$ and $p\mid\operatorname{N}(v)+1$. For the ring-class tower $K[p^\infty]$, does $(L_p(\phi_F,K[p^\infty]))=(\operatorname{char}_{\Lambda(J_K(\infty))}X(E/K[p^\infty]))$ hold? Torsion and the inclusion from the $p$-adic $L$-function ideal into the characteristic ideal are known; the open content is the reverse divisibility.

来源字段：https://doi.org/10.4171/owr/2009/30

01159 | AMR-093-0180

### Iwasawa mu-invariant conjecture for number fields

P / partially_solved | AMR 问题清单汇编

For every number field $K$ and every prime $\ell$, does the Iwasawa invariant $\mu_\ell(K)$ vanish?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01160 | OWR-1586-002

### Iwasawa Reciprocity in Sharifi’s Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Construct an isomorphism of Iwasawa $\Lambda$-modules $\varphi:X_K^-\to Y^-/IY^-$ such that, for $\Xi=\varphi\otimes\psi^-$, $$\Xi\bigl(\Psi_K(1-\zeta)^-\bigr)=\mathcal L^-\pmod{IY^-\otimes\Lambda^-}.$$ Does the candidate $\sigma\mapsto\overline b(\sigma)(\alpha)$ obtained from Galois action and twisted Poincaré duality have this property?

来源字段：https://doi.org/10.4171/owr/2007/30

01161 | OWR-3480-010

### Kato-Suslin Homology of Smooth Connected Varieties

O / open | Oberwolfach 报告问题汇编

If $X$ is smooth and connected over a perfect field, is $H_i^{KS}(X,\mathbb{Z})=0$ for $i>0$ and $H_0^{KS}(X,\mathbb{Z})=\mathbb{Z}$?

来源字段：https://doi.org/10.4171/owr/2009/30

01162 | OWR-5159-009

### Kato’s Main Conjecture for Modular Galois Representations

P / partially_solved | Oberwolfach 报告问题汇编

For the Galois representation $T$ attached to a normalized eigen-newform and the cyclotomic Iwasawa algebra $\Lambda$, does Kato's main-conjecture equality

$$\operatorname{Char}(H^2(T)^\eta)=\operatorname{Char}((H^1(T)/\Lambda z_f)^\eta)$$

hold for every character $\eta$ of the torsion subgroup of $\Gamma=\operatorname{Gal}(\mathbb Q(\mu_{p^\infty})/\mathbb Q)$, where $z_f$ is Kato's Euler-system element?

来源字段：https://doi.org/10.4171/owr/2011/31

01163 | OWR-14603-004

### Kawaguchi-Silverman Arithmetic Degree Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $f$ be a dominant self-map of a smooth projective variety $X$ over a number field, and let $P\in X(\overline{\mathbb Q})$. If the forward orbit of $P$ is Zariski dense in $X$, must the arithmetic degree $\alpha(f,P)$ equal the dynamical degree $\delta(f)$?

来源字段：https://doi.org/10.4171/owr/2016/21

01164 | OWR-15583-008

### Keating-Snaith Distribution of Central L-Values

P / partially_solved | Oberwolfach 报告问题汇编

In each standard family of central $L$-values, is the logarithm of the central value asymptotically Gaussian with the Keating-Snaith mean and variance? More precisely, for primitive characters modulo $q$, are the mean and variance asymptotic to $0$ and $\tfrac12\log\log q$; for quadratic characters $\chi_d$, to $\tfrac12\log\log X$ and $\log\log X$; and for root-number-$1$ quadratic twists $f\otimes\chi_d$, to $-\tfrac12\log\log X$ and $\log\log X$?

来源字段：https://doi.org/10.4171/owr/2017/40

01165 | AMR-093-0065

### Keating-Snaith moment conjecture for the Riemann zeta function

P / partially_solved | AMR 问题清单汇编

For fixed admissible $k$, does $T^{-1}\int_0^T|\zeta(1/2+it)|^{2k}\,dt$ have the Keating-Snaith asymptotic $a(k)G(k+1)^2G(2k+1)^{-1}(\log T)^{k^2}$, with $a(k)$ the predicted arithmetic Euler product and $G$ the Barnes $G$-function?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01166 | OWR-15956-005

### Keating-Snaith Moment Conjecture for the Zeta Function

O / open | Oberwolfach 报告问题汇编

For fixed $k$, is $\frac{1}{T}\int_0^T|\zeta(1/2+it)|^{2k}\,dt\sim a(k)\frac{G(k+1)^2}{G(2k+1)}\left(\log\frac{T}{2\pi}\right)^{k^2}$, where $G$ is the Barnes G-function and $a(k)$ is the arithmetic factor?

来源字段：https://doi.org/10.4171/owr/2017/51

01167 | OWR-14299909-004

### Kernel Presentation for Multiple Eisenstein Relations

O / open | Oberwolfach 报告问题汇编

Let $G$ map words in $H^{\geq 2}=\mathbb Q\langle z_2,z_3,\ldots\rangle$ to multiple Eisenstein series, and set $\operatorname{DR}^*=\operatorname{span}_{\mathbb Q}\{\theta^n(R(u,v))*w:n\geq 0,\ u,v,w\in H^{\geq 2}\}$. Prove that $\operatorname{DR}^*=\ker G$.

来源字段：https://doi.org/10.4171/owr/2026/7

01168 | OWR-8415342-015

### Kernels and Involutivity of Exotic Fourier Transforms

O / open | Oberwolfach 报告问题汇编

Compute explicitly the kernel $\phi^G_\rho$ defining the exotic Fourier transform $F^G_\rho$ on functions on $G^F$, and extend $F^G_\rho$ to an involutive operator.

来源字段：https://doi.org/10.4171/owr/2021/39

01169 | OWR-15169-011

### Khintchine Theory for Symbolic Self-Similar Approximation

O / open | Oberwolfach 报告问题汇编

Let $\phi_1,\ldots,\phi_N$ be contractions of $\mathbb R^n$ with attractor $F$, choose $x\in F$ and an approximating function $\psi$ on finite words, and let $W(x,\psi)$ be the set of points of $F$ lying in $$ B\bigl(\phi_{s_1}\circ\cdots\circ\phi_{s_k}(x),\psi(s_1,\ldots,s_k)\bigr) $$ for infinitely many words. Classify the functions $\psi$ for which $W(x,\psi)$ is large, and establish a Khintchine theory when $\dim_{\mathrm H}F$ equals the similarity dimension.

来源字段：https://doi.org/10.4171/owr/2016/48

01170 | OWR-1542-008

### Kim's Nonabelian Selmer-Dimension Conjecture

O / open | Oberwolfach 报告问题汇编

Let $X/\mathbb{Q}$ be a smooth absolutely irreducible projective curve of genus at least two with rational basepoint $x_0$, and let $p$ be a prime of good reduction. For the étale and de Rham realizations $\Pi_{\mathrm{et},n}$ and $\Pi_{\mathrm{dR},n}$ of the truncated unipotent fundamental group, is $\dim H_f^1(\mathbb{Q},\Pi_{\mathrm{et},n})<\dim(\Pi_{\mathrm{dR},n}/F^0)$ for all sufficiently large $n$?

来源字段：https://doi.org/10.4171/owr/2007/26

01171 | OWR-13351-004

### Kontsevich-Zagier Equivalence of Real Period Integrals

O / open | Oberwolfach 报告问题汇编

Let $X_1,X_2\subset\mathbb R^n$ be semialgebraic sets and let $f_1,f_2$ be semialgebraic functions for which the integrals converge. If $$\int_{X_1}f_1=\int_{X_2}f_2,$$ can one pass from one integral to the other using additivity, semialgebraic changes of variables, and Stokes' formula?

来源字段：https://doi.org/10.4171/owr/2014/49

01172 | AMR-093-0055

### Kummer-Vandiver conjecture

O / open | AMR 问题清单汇编

Kummer-Vandiver conjecture: primes $p$ do not divide the class number of the maximal real subfield of the $p$-th cyclotomic field.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01173 | OWR-14211-001

### Kummer-Vandiver Conjecture

O / open | Oberwolfach 报告问题汇编

Kummer-Vandiver conjecture: for every odd prime $p$, if $C$ is the $p$-Sylow subgroup of the ideal class group of $\mathbb{Q}(\zeta_p)$, then the plus part $C^+$, equivalently each even character component $C^{(i)}$ for $0\leq i\leq p-3$, is zero.

来源字段：https://doi.org/10.4171/owr/2016/3

01174 | LAN-004

### Landau's Fourth Problem: Primes of the Form n² + 1

P / partially_solved | Landau 问题

Are there infinitely many primes of the form $n^2 + 1$?

背景链接：https://eudml.org/doc/142575

01175 | NT-058

### Lander-Parkin-Selfridge Conjecture

O / open | 独立 / 未分配集合

别名：Lander-Parkin-Selfridge Conjecture [NUM-011]

If the sum of $m$ $k$-th powers equals the sum of $n$ $k$-th powers, must $m + n \geq k$?

背景链接：https://en.wikipedia.org/wiki/Lander%2C_Parkin%2C_and_Selfridge_conjecture

01176 | AMR-093-0056

### Lang and Trotter's conjecture

P / partially_solved | AMR 问题清单汇编

Lang and Trotter's conjecture on supersingular primes that the number of supersingular primes less than a constant $X$ is within a constant multiple of $\sqrt{X}/\ln{X}$

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01177 | OWR-14603-006

### Lang-Silverman Height Bounds for Simple Abelian Varieties

P / partially_solved | Oberwolfach 报告问题汇编

For every integer $g\geq1$ and number field $K$, does there exist $c_2(K,g)>0$ such that every simple abelian variety $A/K$ of dimension $g$, every ample symmetric line bundle $L$ on $A$, and every nontorsion point $P\in A(K)$ satisfy

$$\widehat h_{A,L}(P)\geq c_2(K,g)\max\{h_F(A/K),1\}$$?

来源字段：https://doi.org/10.4171/owr/2016/21

01178 | OWR-12725-002

### Lang-Trotter Asymptotics for Fixed Frobenius Fields

O / open | Oberwolfach 报告问题汇编

Let $E/\mathbb Q$ be a non-CM elliptic curve and let $K$ be an imaginary quadratic field. Is

$$\Pi_E(K;x)\sim c(E,K)\frac{\sqrt{x}}{\log x}$$

for some $c(E,K)>0$, where $\Pi_E(K;x)$ counts ordinary primes of good reduction $p\leq x$ for which $\mathbb Q(\pi_p(E))\cong K$?

来源字段：https://doi.org/10.4171/owr/2013/51

01179 | GREEN-021

### Large Sieve and Quadratic Sets

P / partially_solved | Ben Green 问题

Suppose that a large sieve process leaves a set of quadratic size. Is that set quadratic?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

01180 | AIM-ANALYTIC_NUMBER_THEORY-0033

### Large-field local Fourier uniformity and a higher-order obstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.3.** Can we prove the local Fourier uniformity conjecture in function fields? Would that suffice to obtain the full logarithmic Chowla conjecture?

来源字段：http://aimpl.org/sarnakconjecture/4/

01181 | OWR-11127-004

### Large-Image Even Galois Representations

O / open | Oberwolfach 报告问题汇编

Does there exist an absolutely irreducible continuous even representation $\rho:G_{\mathbb Q}\to\operatorname{GL}_2(\mathbb F_p)$ that is unramified outside $p$ and whose image contains $\operatorname{SL}_2(\mathbb F_p)$? In particular, is there at most one prime $p<10^6$ for which such a representation exists?

来源字段：https://doi.org/10.4171/owr/2011/35

01182 | LAN-003

### Legendre's Conjecture

O / open | Landau 问题

For every positive integer $n$, there exists a prime number between $n^2$ and $(n+1)^2$.

背景链接：https://pure.royalholloway.ac.uk/en/publications/on-the-difference-between-consecutive-primes-ii/

01183 | OWR-4416-009

### Lehmer Conjecture on Mahler Measure

O / open | Oberwolfach 报告问题汇编

Lehmer's conjecture states that there exists a constant $m>1$ such that the Mahler measure $M(P)\geq m$ for every non-cyclotomic integer polynomial $P$.

来源字段：https://doi.org/10.4171/owr/2010/35

01184 | NT-035

### Lehmer's Totient Problem

P / partially_solved | 独立 / 未分配集合

If Euler's totient function $\phi(n)$ divides $n-1$, must $n$ be prime?

背景链接：https://doi.org/10.1090/S0002-9904-1932-05518-3

01185 | OWR-15581-002

### Lehmer’s Mahler Measure Problem

O / open | Oberwolfach 报告问题汇编

Given $\varepsilon>0$, does there exist an integral polynomial $f(x)$ with $1<M(f)<1+\varepsilon$?

来源字段：https://doi.org/10.4171/owr/2017/38

01186 | NT-059

### Lemoine's Conjecture

P / partially_solved | 独立 / 未分配集合

Can every odd integer greater than 5 be expressed as the sum of an odd prime and an even semiprime?

背景链接：https://cs.uwaterloo.ca/journals/JIS/VOL27/Juhasz/juhasz3.html

01187 | NT-066

### Leopoldt's Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Leopoldt's Conjecture [NUM-014]; Are there infinitely many Wilson primes [AMR-093-0165]

Does the $p$-adic regulator of an algebraic number field not vanish?

背景链接：https://encyclopediaofmath.org/wiki/Leopoldt_conjecture

01188 | AMR-030-0072

### Let f be the formal power series over Z/2Z whose nth coefficient is independently chosen to be 1 with probability Ü(n^(

O / open | AMR 问题清单汇编

: Let f be the formal power series over Z/2Z whose nth coefficient is independently chosen to be 1 with probability Ü(n^(-2)) (and probability 1 for n=0). Is it true that the density of 1's in the power series 1/f is 1/2?

背景链接：https://people.math.sc.edu/cooper/combprob.html

01189 | AMR-030-0071

### Let f be the formal power series over Z/2Z whose nth coefficient is the parity of the divisor function Ă_(0)(n)

O / open | AMR 问题清单汇编

/Riasanovsky: Let f be the formal power series over Z/2Z whose nth coefficient is the parity of the divisor function Ă_(0)(n). Is it true that the density of 1's in the power series 1/f is 1/32?

背景链接：https://people.math.sc.edu/cooper/combprob.html

01190 | OWR-11127-003

### Level-One Serre Modularity over Imaginary Quadratic Fields

O / open | Oberwolfach 报告问题汇编

Is Serre's conjecture true for representations $\rho:G_E\to\operatorname{GL}_2(\mathbb F_p)$ of level $N(\rho)=1$ when $E$ is an imaginary quadratic field of class number one?

来源字段：https://doi.org/10.4171/owr/2011/35

01191 | OWR-14299909-001

### Lie Structure of Formal $q$-Zeta Algebras

O / open | Oberwolfach 报告问题汇编

Prove that the balanced quasi-shuffle relations and $\tau$-invariance generate all algebraic relations among balanced multiple $q$-zeta values, giving $\mathcal G^{\mathrm f}\cong\mathcal G=\mathcal Z_q$. Prove that the primitive space $\mathfrak{bm}_0$ is a Lie algebra under its proposed bracket and establish $\mathcal G^{\mathrm f}\cong\widetilde M_{\mathbb Q}(\operatorname{SL}_2(\mathbb Z))\otimes U(\mathfrak{bm}_0)^\vee$.

来源字段：https://doi.org/10.4171/owr/2026/7

01192 | NT-067

### Lindelöf Hypothesis

P / partially_solved | 独立 / 未分配集合

别名：Lindelöf hypothesis [OPG-37255]; Are there infinitely many Wieferich primes [AMR-093-0164]

For all $\varepsilon > 0$, does $\zeta(1/2 + it) = o(t^\varepsilon)$ as $t \to \infty$?

背景链接：https://arxiv.org/abs/1408.5794

01193 | OWR-1537-011

### Linear Independence Across Tschakaloff Parameters

O / open | Oberwolfach 报告问题汇编

别名：Linear Independence of Tschakaloff Function Values [OWR-1537-001]

For the Tschakaloff function $T_q(z)=\sum_{\nu\geq0}q^{-\nu(\nu+1)/2}z^\nu$, where $|q|>1$ and $z$ is rational, prove that the values $T_q(z)$ corresponding to distinct admissible parameters $q$ are linearly independent over $\mathbb{Q}$.

来源字段：https://doi.org/10.4171/owr/2007/21

01194 | OWR-1537-013

### Liouville Classification of the Gelfond Constant

O / open | Oberwolfach 报告问题汇编

Is $e^\pi$ a Liouville number? The conjecture is that it is not.

来源字段：https://doi.org/10.4171/owr/2007/21

01195 | OWR-11784-005

### Littlewood Conjecture

O / open | Oberwolfach 报告问题汇编

For every $(\alpha,\beta)\in\mathbb R^2$, is $\liminf_{q\to\infty}q\lVert q\alpha\rVert\,\lVert q\beta\rVert=0$, where $\lVert\cdot\rVert$ denotes distance to the nearest integer?

来源字段：https://doi.org/10.4171/owr/2012/22

01196 | OWR-15169-001

### Littlewood's Simultaneous Approximation Conjecture

O / open | Oberwolfach 报告问题汇编

For every $(x,y)\in\mathbb R^2$, does $\liminf_{n\to\infty}n\|nx\|\,\|ny\|=0$?

来源字段：https://doi.org/10.4171/owr/2016/48

01197 | OWR-3388-002

### Local $p$-Adic Section Conjecture

O / open | Oberwolfach 报告问题汇编

Let $k$ be a local $p$-adic field and $X/k$ a smooth projective curve of genus at least $2$. Is the natural map from $X(k)$ to conjugacy classes of continuous sections of $$\pi_1(X/k)\longrightarrow\operatorname{Gal}(\overline k/k)$$ a bijection?

来源字段：https://doi.org/10.4171/owr/2009/05

01198 | OWR-13860-005

### Local Determination of Serre Weights for Unitary Groups

O / open | Oberwolfach 报告问题汇编

In the weight part of Serre's conjecture for unitary groups of semisimple rank greater than one, is the set of modular Serre weights W(r) determined by the local restrictions $r|_{G_{F_w}}$ through the predicted local sets $W^?(r|_{G_{F_w}})$?

来源字段：https://doi.org/10.4171/owr/2015/39

01199 | AIM-ANALYTIC_NUMBER_THEORY-0031

### Local Fourier uniformity down to exponent 2/5 and spectral-moment transfer

P / partially_solved | AIM 研讨会问题汇编

How much can we lower \(H\) in the local Fourier uniformity conjecture \[ \int_X^{2X} \sup_\alpha \left| \sum_{x\leq n\leq x+H} \lambda(n)e(n\alpha) \right|\,dx=o(HX)? \]

来源字段：http://aimpl.org/sarnakconjecture/4/

01200 | OWR-4804-002

### Local Gross-Prasad Multiplicity for Orthogonal Groups

P / partially_solved | Oberwolfach 报告问题汇编

For generic $\phi\in\Phi^{\mathrm{orth}}(d)$ and $\phi'\in\Phi^{\mathrm{orth}}(d-1,\delta)$, is there exactly one pair $(\pi,\sigma)$ in the corresponding orthogonal-group $L$-packets with multiplicity $m(\pi,\sigma)=1$, with every other pair having multiplicity zero?

来源字段：https://doi.org/10.4171/owr/2011/14

01201 | OWR-1703872-005

### Local Maxima of Special Eigenfunctions at Spherical Points

O / open | Oberwolfach 报告问题汇编

Assume there are no nonzero functions on

$$D^\times(\mathbb Q)\backslash D^\times(\mathbb A_\mathbb Q)/\bigl((K_\infty\cap D^\times(\mathbb A_\mathbb Q))Z(D^\times(\mathbb A_\mathbb Q))\bigr)$$

that transform according to the $3$-dimensional spherical representation of $\operatorname{SO}(3)$. For all but finitely many special eigenfunctions $f$, is every $x\in D^\times(\mathbb A_\mathbb Q)$ a local maximum of $f$?

来源字段：https://doi.org/10.4171/owr/2020/26

01202 | OWR-4076-016

### Local Period Factors for Quaternionic Transfers

P / partially_solved | Oberwolfach 报告问题汇编

Let $\pi$ be the automorphic representation attached to $f$. For every place $v$ where $\pi_v$ is discrete series, does there exist $c_v\in\mathbb{C}$ such that $\langle f,f\rangle/\langle f_B,f_B\rangle=\prod_{v\mid\operatorname{disc}(B)}c_v$ up to a $p$-adic unit?

来源字段：https://doi.org/10.4171/owr/2009/33

01203 | AIM-ALGEBRAIC_NUMBER_THEORY-0094

### Local R-equivalence and a geometric Brauer obstruction to eventual universal R-triviality

P / partially_solved | AIM 研讨会问题汇编

**Question 10 (Colliot-Thélène).** Are there results at the level of number fields arising from the techniques of rationally connected varieties? For example, recently Kollár got nice results over local fields: if \(X/k\) is a projective variety, we say \(P,Q\in X(k)\) are \(R\)-equivalent if one can link them by a chain of curves of genus zero over \(k\); if \(X\) is smooth with \(k\hookrightarrow\mathbb C\), then \(X\) is rationally connected if \(X(\mathbb C)/R\) is a single point. Kollár proved that if \(k\) is a local field and \(X/k\) is rationally connected, then \(X(k)/R\) is finite. Kollár and Szabó proved that if \(k\) is a number field and \(X/k\) is rationally connected, then \(X(k_v)/R\) is trivial for almost all \(v\).

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01204 | AIM-ANALYTIC_NUMBER_THEORY-0106

### Local singular series and multiplicative-order correlations for Elliott's shifted-prime ratio

P / partially_solved | AIM 研讨会问题汇编

`ak = p + 1` `q + 1 (2)`

In particular, it contains the lines

and renders the proposed bound with missing comparison and epsilon symbols. Direct inspection of the typeset AIM problem PDF verifies that equation (2) is

\[ a^k=\frac{p+1}{q+1}, \tag{1} \]

where \(a>1\) is fixed, \(p,q\) are primes, and \(k\) is an integer. The PDF states Elliott's range as \(2\leq k\leq \log q\), and the proposed improvement as

\[ k\ll (\log q)^{1/2+\varepsilon}. \tag{2} \]

Thus the mathematical task is to improve the size of an exponent that is allowed to vary from solution to solution; it is **not** the much stronger assertion that a prescribed fixed \(k\) gives infinitely many prime pairs.

Put \(A=a^k\). Equation (1) is exactly the simultaneous-primality problem

\[ L_1(n)=n,\qquad L_2(n)=An+A-1, \tag{3} \]

with \(n=q\) and \(p=L_2(q)\). Equivalently, on writing \(m=q+1\), the two forms are \(m-1\) and \(a^k m-1\).

There is a minor source discrepancy worth recording. The local calculations below are made from (1), using (3). The same local root counts apply to the all-plus family because multiplying every forbidden residue by \(-1\) does not change their number.

来源字段：https://aimath.org/WWN/primegaps/WorkshopProblems.pdf

01205 | OWR-833-004

### Local Systems from Frobenius-Invariant Sections

O / open | Oberwolfach 报告问题汇编

Let $M$ be a $(\varphi, \Gamma)$-module (or p-adic Hodge module) over the Robba/Fontaine ring associated to a $p$-adic family over a rigid analytic base $F^a$. Do the Frobenius invariants $M^{\varphi = 1}$ form a local system of $\mathbb{Q}_p$-vector spaces on $F^a$?

来源字段：https://doi.org/10.4171/owr/2005/28

01206 | OWR-4990381-002

### Local-Global Density of Galois Cover Specializations

P / partially_solved | Oberwolfach 报告问题汇编

Let $E/\mathbb Q(T)$ be a $\mathbb Q$-regular Galois extension with group $G$ and genus $g\geq2$. Does $$ \frac{|\operatorname{Sp}(E)\cap S(G,x)|}{|\operatorname{Sp}(E)^{\mathrm{loc}}\cap S(G,x)|}\longrightarrow0 $$ as $x\to\infty$?

来源字段：https://doi.org/10.4171/owr/2021/12

01207 | OWR-4990381-017

### Local-Global Principles for Torsors over Semi-Global Fields

O / open | Oberwolfach 报告问题汇编

For a semi-global field $F$ and a linear algebraic group $G$, when does the local-global principle for $G$-torsors hold, and what is an explicit description of its obstruction?

来源字段：https://doi.org/10.4171/owr/2021/12

01208 | OWR-12004-003

### Local-Model Diagrams for Parahoric Shimura Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Does every Shimura variety with parahoric level admit a local-model diagram

$$S_K\longleftarrow\widetilde S_K\longrightarrow M^{\mathrm{loc}}$$

in which the left map is a $K$-torsor and the right map is $K$-equivariant and smooth?

来源字段：https://doi.org/10.4171/owr/2012/32

01209 | OWR-3388-006

### Local-Global Principles for Homogeneous Spaces over Semi-Global Fields

O / open | Oberwolfach 报告问题汇编

Let $K$ be a complete discretely valued field with residue field $k$, let $F$ be the function field of a curve over $K$, and let $F_v$ be the completion at a discrete valuation $v$ of $F$. If an $F$-variety $Y$ is a homogeneous space of a connected linear $F$-group and $Y(F_v)\neq\varnothing$ for every $v$, must $Y(F)\neq\varnothing$?

来源字段：https://doi.org/10.4171/owr/2009/05

01210 | AIM-ANALYTIC_NUMBER_THEORY-0077

### Location-sensitive asymptotics for sparse fourth moments

P / partially_solved | AIM 研讨会问题汇编

别名：A one-coupling J-Schur certificate for projected spectral enclosures [AIM-PHYSICS-0010]

Evaluate \[ \sum_j\int_{t_j}^{t_j+T^{7/8}} \left|\zeta\!\left(\tfrac12+it\right)\right|^4\,dt, \] where the sum is over a set of $t_j$'s with modulus at most $T$, spaced more than $T^{7/8}$ apart (related to work of Heath-Brown and Zavorotnyi).

来源字段：http://aimpl.org/zetamoments/8/

01211 | OWR-17293-005

### Long Dirichlet-Polynomial Recipe for Quadratic L-Functions

P / partially_solved | Oberwolfach 报告问题汇编

For fundamental discriminants $d$ and quadratic characters $\chi_d$, the long-polynomial recipe conjectures the stated smoothed average of $\sum_{n\leq X}\tau_A(n)\chi_d(n)/\sqrt n$ is given by the corresponding integral of the quadratic $L$-function swap terms, up to $O(D^{1-\delta})$ as $D\to\infty$.

来源字段：https://doi.org/10.4171/owr/2019/50

01212 | AIM-ALGEBRAIC_NUMBER_THEORY-0111

### Low-degree integral curves on the Fermat fourfold

P / partially_solved | AIM 研讨会问题汇编

Describe the variety of curves of low degree on the Fermat variety \[ F:x_1^d+x_2^d+\cdots+x_6^d=0. \] (Heath-Brown)

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01213 | OWR-1703872-006

### Lower Bounds for Function-Field Quaternionic Periods

O / open | Oberwolfach 报告问题汇编

Fix a quaternion algebra $D$ over $\mathbb F_q(t)$. Let $F/\mathbb F_q(t)$ vary over quadratic extensions of genus $g$, let $K\subset D^\times(\mathbb A_F)$ be compact, and let $m$ be the index of $K\cap D^\times(\mathbb A_{\mathbb F_q(t)})$ in a maximal compact subgroup. For every normalized Hecke eigenfunction $f$ on

$$D^\times(F)\backslash D^\times(\mathbb A_F)/(KZ(\mathbb A_F)),$$

is its period over $D^\times(\mathbb F_q(t))\backslash D^\times(\mathbb A_{\mathbb F_q(t)})$ bounded below by $\Omega(m^{-C}q^{g/2})$ for a constant $C$ depending only on $q$ and $D$?

来源字段：https://doi.org/10.4171/owr/2020/26

01214 | OWR-12014-013

### Lower Bounds for Sums of Roots of Unity

O / open | Oberwolfach 报告问题汇编

Let $C_d=\{e^{2\pi i k/d}:k\in\mathbb Z\}$, and let $mC_d$ denote its $m$-fold Minkowski sum. For a set $A\subset\mathbb C$, write $\min_0(A)=\min\{|x|:x\in A\setminus\{0\}\}$. For every fixed positive integer $m$, does there exist $E=E(m)>0$ such that $$\min_0(mC_d)>d^{-E}$$ for all $d$?

来源字段：https://doi.org/10.4171/owr/2012/42

01215 | OWR-11695860-012

### Lower Bounds for Twisted-Multiplicative Exponential Sums

O / open | Oberwolfach 报告问题汇编

For $W_f(a;q)=q^{-1/2}\sum_{x\bmod q}e(af(x)/q)$, obtain lower bounds of the same order as the best available upper estimates for $\sum_{q\leq x}|W_f(a;q)|$.

来源字段：https://doi.org/10.4171/owr/2022/50

01216 | OPG-37402

### Lucas Numbers Modulo m

P / partially_solved | Open Problem Garden

Conjecture The sequence {L(n) mod m}, where L(n) are the Lucas numbers, contains a complete residue system modulo m if and only if m is one of the following: 2, 4, 6, 7, 14, 3^k, k >=1.

背景链接：http://www.openproblemgarden.org/op/lucas_numbers_modulo_m

01217 | NUM-005

### Lychrel Numbers

O / open | 独立 / 未分配集合

文本重复：NT-049

Do Lychrel numbers exist in base 10?

背景链接：https://en.wikipedia.org/wiki/Lychrel_number

01218 | NT-049

### Lychrel Numbers in Base 10

O / open | 独立 / 未分配集合

文本重复：NUM-005

Do Lychrel numbers exist in base 10?

背景链接：https://mathworld.wolfram.com/LychrelNumber.html

01219 | OPG-671

### MacEachen Conjecture

P / partially_solved | Open Problem Garden

Conjecture Every odd prime number must either be adjacent to, or a prime distance away from a primorial or primorial product.

背景链接：http://www.openproblemgarden.org/op/maceachen_conjecture

01220 | OWR-12865-004

### Maeda Conjecture for Level-One Cusp Forms

O / open | Oberwolfach 报告问题汇编

For every normalized Hecke eigenform $f\in S_k(1)$, is $[\mathbb Q_f:\mathbb Q]=\dim S_k(1)=m_k$, with the Galois group of the normal closure of $\mathbb Q_f/\mathbb Q$ isomorphic to the symmetric group $S_{m_k}$?

来源字段：https://doi.org/10.4171/owr/2014/05

01221 | NT-036

### Magic Square of Squares

O / open | 独立 / 未分配集合

别名：Magic square of squares [OPG-16570]

Does there exist a 3×3 magic square composed entirely of distinct perfect squares?

背景链接：https://eudml.org/doc/207247

01222 | NT-037

### Mahler's 3/2 Problem

P / partially_solved | 独立 / 未分配集合

Is there a real number $x$ such that the fractional parts of $x(3/2)^n$ are all less than $1/2$ for every positive integer $n$?

背景链接：https://doi.org/10.1017/S144678870000613X

01223 | OWR-1274-025

### Main Conjecture for Modular Towers

P / partially_solved | Oberwolfach 报告问题汇编

For every Modular Tower component branch and every number field $K$, prove that the corresponding component has no $K$-rational points at all sufficiently high levels.

来源字段：https://doi.org/10.4171/owr/2006/25

01224 | OWR-16409-008

### Malle Conjecture for Number Field Counts

O / open | Oberwolfach 报告问题汇编

For a number field $k$ and a transitive permutation group $G\subseteq S_n$, determine the asymptotic behavior of $$N_k(G,X)=\#\{K/k:\operatorname{Gal}(K/k)\simeq G,\ \operatorname{Disc}(K/k)<X\}.$$ In the cases where Malle's original logarithmic exponent is valid, is $$N_k(G,X)\sim C(G,k)X^{1/a(G)}(\log X)^{b(G,k)-1}$$?

来源字段：https://doi.org/10.4171/owr/2018/34

01225 | OWR-14297740-014

### Malle Lower Bounds over Rational Function Fields

P / partially_solved | Oberwolfach 报告问题汇编

For a transitive group $G\leq S_m$, prove the conjectural lower bound in Malle's asymptotic for the number $N_{G,\mathbb F_q(t)}(X)$ of degree-$m$ extensions of $\mathbb F_q(t)$ with Galois group $G$ and discriminant norm at most $X$.

来源字段：https://doi.org/10.4171/owr/2023/42

01226 | OWR-14743-006

### Malle-Type Asymptotics for Refined Cover Counts

P / partially_solved | Oberwolfach 报告问题汇编

For a countable transitive subgroup $G\leq S_n$ and a conjugation-invariant subset $c\subseteq G$, does the refined function $N^c_{G,q}(X)$, counting geometrically connected $G$-covers with ramification monodromy in $c$, satisfy $N^c_{G,q}(X)\sim C X^a(\log X)^{b-1}$ for constants $a,b,C$ depending on $G$ and $c$?

来源字段：https://doi.org/10.4171/owr/2016/35

01227 | AMR-093-0083

### Manin conjecture

P / partially_solved | AMR 问题清单汇编

Manin conjecture: if K-rational points on Fano variety are Zariski-dense subset, then the distribution of points of height: $H(x)\leq B$ in any Zariski-open subset $U$ is proportional to $B \log (B)^{r-1}$, where $r$ is rank of Picard group of that variety.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01228 | GREEN-088

### Maximal Covering Interval

P / partially_solved | Ben Green 问题

What is the largest $y$ for which one may cover the interval $[y]$ by residue classes $a_p \pmod p$, one for each prime $p \leq x$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

01229 | OWR-1274-010

### Maximal Pro-$p$ Galois Groups of Higher Local Fields

P / partially_solved | Oberwolfach 报告问题汇编

Let $K$ be a complete discrete valuation field with residue field of characteristic $p$, and let $\Gamma_K(p)$ be the Galois group of its maximal $p$-extension. Determine the structure of $\Gamma_K(p)$, particularly when $K$ is an $N$-dimensional local field.

来源字段：https://doi.org/10.4171/owr/2006/25

01230 | OWR-14299577-003

### Maximum Normalized Mahler Measure of Littlewood Polynomials

P / partially_solved | Oberwolfach 报告问题汇编

Let $P_n(x)=\sum_{k=0}^n a_kx^k$ range over Littlewood polynomials with $a_k\in\{-1,1\}$. Determine the extremal limit superior $b_m=\limsup_{n\to\infty}M_0(P_n)/\sqrt{n}$, where $M_0$ denotes Mahler measure.

来源字段：https://doi.org/10.4171/owr/2025/51

01231 | OWR-15956-006

### Maximum Size of the Zeta Function

O / open | Oberwolfach 报告问题汇编

Is $\max_{0\leq t\leq T}|\zeta(1/2+it)|=\exp\!\left((1+o(1))\sqrt{\tfrac12\log T\log\log T}\right)$ as $T\to\infty$?

来源字段：https://doi.org/10.4171/owr/2017/51

01232 | AIM-ANALYTIC_NUMBER_THEORY-0006

### Mellin-density obstruction and a degree-four hyper-Kloosterman reduction

P / partially_solved | AIM 研讨会问题汇编

Subconvexity bounds for \(\mathrm{GL}(1)\) twists of a fixed \(\mathrm{GL}(2)\times\mathrm{GL}(2)\) \(L\)-function has been proved using a delta symbol by C. Raju in the archimedean aspect, and by A. Ghosh in the level aspect. Can we generalize these techniques to replace the \(\mathrm{GL}(1)\) twist by a trace function, or an appropriate archimedean analog?

来源字段：http://aimpl.org/deltasubconvex2/1/

01233 | AIM-ANALYTIC_NUMBER_THEORY-0112

### Merged GPY and Gaussian-prime questions with a finite-direction occupancy reduction

O / open | AIM 研讨会问题汇编

12. Limits of the GPY method.

• H. Helfgott noted that it seems that the GPY method is by nature a lower bound method and will most likely never yield an asymptotic formula for the number of primes with a given spacing. He asked those who were well-versed in the GPY method whether this is indeed so and also whether the GPY method is capable of producing a right-order lower bound.

• J. Pintz replied that an asymptotic formula of any kind appears to be out of the reach of the method. He also noted that the method could potentially yield a "right-order lower bound", but that that will involve considerable technical di ffi culties. Some ideas in that direction were sketched in C. Yildirim's talk. It should be noted that in this context, a "right-order lower bound" means that one can show that a positive proportion of the k-tuples we consider contain at least two primes. 313. Distances between Gaussian primes. Several members of the audience asked whether the GPY method can be adapted to study small distances between Gaussian primes. T.D. Wooley remarked that such an adaptation should be straightforward.

来源字段：https://aimath.org/WWN/primegaps/WorkshopProblems.pdf

01234 | OWR-12493-005

### Merkurjev-Suslin Theory for Higher Massey Products

O / open | Oberwolfach 报告问题汇编

Can the Merkurjev-Suslin theorem be generalized from the twofold cup product to $n$-fold Massey products?

来源字段：https://doi.org/10.4171/owr/2013/30

01235 | AIM-ANALYTIC_NUMBER_THEORY-0046

### Microscopic even moments and a diagonal crossover mechanism for short divisor sums

P / partially_solved | AIM 研讨会问题汇编

**Moments of divisor functions in short intervals.** Evaluate \[ \int_X^{2X}\left|\sum_{x\le n\le x+H}d_k(n)-\mbox{main term}\right|^{2\ell}\,dx \] for different ranges of \(H\).

来源字段：http://aimpl.org/zetamoments/1/

01236 | OWR-12697688-002

### Middle-Degree Concentration of Non-Eisenstein Cohomology

P / partially_solved | Oberwolfach 报告问题汇编

If $\mathfrak m$ is non-Eisenstein, meaning $\overline\rho_{\mathfrak m}$ is absolutely irreducible, prove that $H^*(X_K(\mathbb C),\mathbb F_\ell)_{\mathfrak m}$ is concentrated in the middle degree $d$.

来源字段：https://doi.org/10.4171/owr/2023/5

01237 | OWR-13351-010

### Minimal Coefficient Rings for Cyclic Local Lifts

O / open | Oberwolfach 报告问题汇编

Let $k$ be an algebraically closed field of characteristic $p > 0$, and let $k[[z]] / k[[t]]$ be a cyclic $G$-Galois extension of power-series rings. In the local lifting problem for cyclic groups $G$ (which asks for a complete discrete valuation ring $R$ of characteristic $0$ with residue field $k$ and a $G$-Galois extension $R[[Z]] / R[[T]]$ reducing to $k[[z]] / k[[t]]$), can the lifting ring always be chosen as $R = W(k)[\zeta_{|G|}]$ (the ring of Witt vectors over $k$ adjoined a primitive $|G|$-th root of unity) for all cyclic groups $G$?

来源字段：https://doi.org/10.4171/owr/2014/49

01238 | OWR-4990381-010

### Minimal Ramification over Rational Function Fields

P / partially_solved | Oberwolfach 报告问题汇编

For $q=p^\nu$ and a finite group $G$, is $$ r_{\mathbb F_q(T)}(G)=\max\{1,d((G/p(G))^{\mathrm{ab}})\}, $$ where $r_{\mathbb F_q(T)}(G)$ is the minimum number of ramified primes and $G/p(G)$ is the maximal quotient of order prime to $p$?

来源字段：https://doi.org/10.4171/owr/2021/12

01239 | OWR-15174-012

### Minimal Separations Among Lattice Points on Circles

P / partially_solved | Oberwolfach 报告问题汇编

Let $n$ be a product of distinct primes congruent to $1$ modulo $4$, and consider the lattice points on $x^2+y^2=n$. Can three such points attain separation of order $R^{1/3}$, where $R=\sqrt n$, and how small can the separation be for configurations of $k$ such points?

来源字段：https://doi.org/10.4171/owr/2016/53

01240 | OWR-1053-003

### Minimal-Coefficient Polynomial Selection for the Number Field Sieve

P / partially_solved | Oberwolfach 报告问题汇编

Given degrees $d_f$ and $d_g$, construct irreducible coprime polynomials $f,g\in\mathbb Z[x]$ of those degrees with a common root modulo $N$ and coefficients of minimal possible absolute size.

来源字段：https://doi.org/10.4171/owr/2005/32

01241 | OWR-4416-005

### Minimally Ramified Extensions with Prescribed Galois Group

O / open | Oberwolfach 报告问题汇编

Given a finite nontrivial group $G$ and a positive integer $g$, does there exist a $G$-extension of $\mathbb{Q}$ ramified at $g$ primes, including the infinite prime, and if so how many; conjecturally, existence is equivalent to $g$ being at least the minimal number of generators of $G^{\mathrm{ab}}$?

来源字段：https://doi.org/10.4171/owr/2010/35

01242 | AIM-ANALYTIC_NUMBER_THEORY-0088

### Mirror-quintic periods and a base-change-stable obstruction to classical modularity

P / partially_solved | AIM 研讨会问题汇编

别名：Mirror-quintic periods, modular-type structures, and a conifold obstruction [AIM-PHYSICS-0021]

Is $\sum \frac{(5n)!}{(n!)^5}z^n$ related to (non-holomorphic) forms? Is it related to automorphic objects? Do other solutions to the Picard-Fuchs equation have modular properties?

来源字段：http://aimpl.org/gromwitnumthry/4/

01243 | AIM-ANALYTIC_NUMBER_THEORY-0041

### Missing unit-disc hypothesis and a universal unbounded counterexample

P / partially_solved | AIM 研讨会问题汇编

Suppose that for all fixed \(\chi\) mod \(q\) and \(t\in\mathbb R\), \(f:\mathbb N\to\mathbb C\) is multiplicative with \[ \sum_p\frac{1-\operatorname{Re}f(p)\overline{\chi(p)p^{it}}}{p}=\infty. \] Does there exists a subsequence \(X_k\to\infty\) such that \[ \sum_{n\leq X_j}f(n)\overline{f(n+1)}=o(X_j)? \] Alternatively, is \[ \sum_{n\leq X}f(n)f(n+h_1)\cdots f(n+h_k)=o(X) \] for distinct \(h_1,\ldots,h_k\)?

来源字段：http://aimpl.org/sarnakconjecture/7/

01244 | OWR-11784-010

### Mixed Littlewood Conjecture for Pseudo-Absolute Values

O / open | Oberwolfach 报告问题汇编

For a pseudo-absolute-value sequence $D=(n_k)$ with $n_0=1$ and $n_k\mid n_{k+1}$, define $|q|_D=\min\{n_k^{-1}:q\in n_k\mathbb Z\}$. Is $\inf_{q\geq1}|q|_D\,\lVert qx\rVert=0$ for every such $D$ and every $x\in\mathbb R$?

来源字段：https://doi.org/10.4171/owr/2012/22

01245 | AIM-ALGEBRAIC_NUMBER_THEORY-0090

### Mixed resolution of the quartic K3 questions and an exact 2-adic separation lemma

P / partially_solved | AIM 研讨会问题汇编

(a) Is there a K3 surface \(X/\mathbb Q\) such that \(0<\#X(\mathbb Q)<\infty\)?

(b) Is there a nonsingular quartic surface with this property?

(c) Find a third rational point on \[ X_1^4+2X_2^4=X_3^4+4X_4^4. \] (The source says that there are only two points of height at most \(2^{15}\).)

(d) Find a smooth quartic \(X_4\subset\mathbb P^3\) with \(\operatorname{Pic}(\overline X_4)\cong\mathbb Z\) and \(X_4(K)\) infinite.

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01246 | AIM-ANALYTIC_NUMBER_THEORY-0100

### Mock forms as boundary pieces of twisted Nahm sums

P / partially_solved | AIM 研讨会问题汇编

Extend the Nahm-sum, Bloch-group, and asymptotic framework so that it can recognize mock modular \(q\)-hypergeometric series, their shadows, and their controlled failure of holomorphic modularity.

The pronoun “this” is resolved by the immediately preceding source context. Section 2 of the 2010 AIM problem list introduces

\[ f_{A,B,C}(\tau)= \sum_{\boldsymbol n\in\mathbb Z_{\geq0}^r} \frac{q^{\frac12\boldsymbol n^tA\boldsymbol n+B^t\boldsymbol n+C}}{(q;q)_{n_1}\cdots(q;q)_{n_r}}, \qquad q=e^{2\pi i\tau}, \]

where \(A\) is rational, symmetric, and positive definite, \(B\in\mathbb Q^r\), and \(C\in\mathbb Q\). The section discusses Nahm's proposed connection between modularity and the Bloch-group element obtained from the Nahm equations. Problems 2.1--2.4 concern the possible \(B\)'s, asymptotic jumps, higher-rank modular examples, and a Bailey-pair/Bloch-group dictionary; Problem 2.6 asks about quantum dilogarithms.

Thus the recovered task is:

来源字段：https://aimath.org/WWN/mockmodular/mockmodular.pdf

01247 | AIM-ANALYTIC_NUMBER_THEORY-0091

### Mock modularity in geometric counting: polar data, wall crossing, and holomorphic ambiguity

P / partially_solved | AIM 研讨会问题汇编

别名：Mock modularity, geometric invariants, and finite pole tomography [AIM-PHYSICS-0024]

Partition the known connections by mechanism, distinguish proved enumerative statements from physical conjectures, and give a rigorous criterion that tests when two proposed geometric/black-hole realizations have the same mock anomaly and when that information is insufficient to identify their coefficients.

来源字段：http://aimpl.org/gromwitnumthry/5/

01248 | OWR-13351-017

### Mod-$\ell$ Bogomolov-Pop Reconstruction Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Statement of the Mod-$\ell$ Bogomolov-Pop Conjecture [OWR-13351-016]

Let $K/k$ and $L/l$ be function fields of transcendence degree at least $2$ over algebraically closed fields whose characteristics differ from $\ell$. Is the canonical map from field isomorphisms modulo Frobenius to Milnor-$K$-theory-preserving isomorphisms $K^\times/\ell\to L^\times/\ell$ bijective?

来源字段：https://doi.org/10.4171/owr/2014/49

01249 | OWR-14298584-003

### Mod-p Automorphic Induction and p-Adic Regulators

O / open | Oberwolfach 报告问题汇编

Can one construct and characterize mod-$p$ automorphic induction without passing through mod-$p$ Galois representations? Can one predict the $p$-adic valuation of the relevant ratios of regulators or, when nonzero, of $\#H^i(M(\mathcal{O}_1),\mathbb{Z})/\#H^i(M(\mathcal{O}_2),\mathbb{Z})$? Can one define $p$-adic regulators whose ratios equal the corresponding rational ratios of real regulators?

来源字段：https://doi.org/10.4171/owr/2024/40

01250 | OWR-14298588-001

### Mod-p Reductions of Crystalline Galois Representations

P / partially_solved | Oberwolfach 报告问题汇编

Let $\rho:\operatorname{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)\to\mathrm{GL}_n(\overline{\mathbb{Q}}_p)$ be crystalline with Hodge-Tate weights $h_1\geq\cdots\geq h_n$. Classify the possible semisimplified reductions $\overline{\rho}:\operatorname{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)\to\mathrm{GL}_n(\overline{\mathbb{F}}_p)$. In particular, are the known inertial-weight constraints best possible for $n>2$?

来源字段：https://doi.org/10.4171/owr/2024/33

01251 | OWR-14299909-011

### Modular $q$-Analogues of Ramanujan’s 9801 Series

O / open | Oberwolfach 报告问题汇编

Construct a modular or mock-modular $q$-analogue of Ramanujan’s 9801 series for $1/\pi$.

来源字段：https://doi.org/10.4171/owr/2026/7

01252 | OWR-12010-005

### Modular Construction of Prescribed $\mathbb Z_p$-Extensions

O / open | Oberwolfach 报告问题汇编

Let $F_\infty/F$ be the cyclotomic $\mathbb Z_p$-extension. Can modular methods construct a $\mathbb Z_p$-extension $L_Q/F_\infty$ that is Galois over $F$, on which $\operatorname{Gal}(F_\infty/F)$ acts cyclotomically, and that is unramified outside prescribed auxiliary primes $Q$ but ramified at $Q$?

来源字段：https://doi.org/10.4171/owr/2012/38

01253 | OWR-707-002

### Modularity of Calabi-Yau Cohomological $L$-Series

P / partially_solved | Oberwolfach 报告问题汇编

Are the cohomological $L$-series of Calabi-Yau varieties completely determined by suitable modular cusp forms?

来源字段：https://doi.org/10.4171/owr/2004/24

01254 | OWR-14299909-010

### Modularity of Higher-$t$ Generating-Function Coefficients

O / open | Oberwolfach 报告问题汇编

Determine whether the coefficients of higher powers of $t$ in the hypergeometric generating function for the integrals $R_{k,1}(X)$ are modular forms or higher-order modular antiderivatives, and rigorously identify the corresponding structure for $R_{k,\ell}(X)$ when $\ell>1$.

来源字段：https://doi.org/10.4171/owr/2026/7

01255 | AMR-117-0001

### Moments of L-functions - Moments of the Riemann zeta function - Problem 1

O / open | AMR 问题清单汇编

Give an explicit version of Harper's [Ha] upper bound for $I_k(T)$, assuming the Riemann hypothesis. That is, for $k>0$ determine $C_k$ such that \begin{equation}

I_k(T) \le C_k T (\log T)^{k^2} \end{equation} for $T$ sufficiently large. Note that this is related to the following conjecture of Soundararajan [S2] for $k \ge 2$ and $T \ge 10$, \[ I_k(T) \le T (\log T)^{k^2}. \] (Nathan Ng, Lethbridge) \\

背景链接：https://arxiv.org/abs/2405.17800

01256 | AMR-117-0008

### Moments of L-functions - Moments of the Riemann zeta function - Problem 10

O / open | AMR 问题清单汇编

We have been informed a graduate student has worked extensively on this problem and nearly has a result.

背景链接：https://arxiv.org/abs/2405.17800

01257 | AMR-117-0009

### Moments of L-functions - Moments of the Riemann zeta function - Problem 11

O / open | AMR 问题清单汇编

Is it possible to prove an omega theorem for $I_3(T)$ assuming the ternary additive divisor conjecture? Prove that there exists $\Theta >0$, such that \[ I_3(T) = T \mathcal{P}_9(\log T) + \Omega(T^{\Theta}) \] where $\mathcal{P}_9$ is a certain degree nine polynomial. \\ (Nathan Ng, Lethbridge)

背景链接：https://arxiv.org/abs/2405.17800

01258 | AMR-117-0010

### Moments of L-functions - Moments of the Riemann zeta function - Problem 12

O / open | AMR 问题清单汇编

Ng, Shen, and Wong [NSW] showed that the Riemann hypothesis and a smoothed quaternary additive divisor conjecture imply the conjectured asymptotic \[ I_4(T) \sim C_4 T (\log T)^{16} \] for a certain positive constant $C_4$. Under the same assumptions, obtain the full main term, up to an error term of size $o(T)$ or a power savings error term. Obtain a shifted version of this result. Under the same assumptions, asymptotically evaluate \[ \int_{-\infty}^{\infty} \omega(t) \Big( \prod_{j=1}^{4} \zeta(\tfrac{1}{2}+a_j+it) \zeta(\tfrac{1}{2}+b_j-it) \Big) dt \] where $a_j, b_j$, $j=1, \ldots, 4$ are complex numbers satisfying $|a_j|, |b_j| \ll \frac{1}{\log T}$ and $\omega(t)$ is a smooth, compactly supported function in $[c_1T, c_2T]$ with $0 < c_1 < c_2$.

背景链接：https://arxiv.org/abs/2405.17800

01259 | AMR-117-0011

### Moments of L-functions - Moments of the Riemann zeta function - Problem 13

O / open | AMR 问题清单汇编

In [CFKRS] a conjectural method known as the `recipe' is introduced. The `recipe' was used to produce (conjecturally) a full asymptotic formula for $I_k(T)$. Can the `recipe' be made rigorous in the case of the fourth moment? That is, can we give a new proof of the asymptotic formula with power savings error term for $I_2(T)$ following the `recipe' method. \\ (Nathan Ng, Lethbridge) \\

背景链接：https://arxiv.org/abs/2405.17800

01260 | AMR-117-0012

### Moments of L-functions - Moments of the Riemann zeta function - Problem 14

O / open | AMR 问题清单汇编

Can the conjectural lower order terms for $I_k(T)$ (due to CFKRS [CFKRS] and Diaconu-Goldfeld-Hoffstein [DGH]) be recovered by refining the splitting conjecture in the Gonek-Hughes-Keating [GHK] hybrid Euler-Hadamard product? \\ (Anurag Sahay, Rochester)

背景链接：https://arxiv.org/abs/2405.17800

01261 | AMR-117-0013

### Moments of L-functions - Moments of the Riemann zeta function - Problem 15

O / open | AMR 问题清单汇编

Prove third and fourth moment asymptotics with best possible error terms for $L(1/2,f)$ as $f$ runs over holomorphic Hecke eigenforms of weight $k_f \leq T$. It should be possible to show, using current technology, that \[\sum_{k_f \leq T} \frac{L\left(\frac{1}{2},f\right)^3}{L(1,\operatorname{ad}f)} = T^2 P_3(\log T) + O_{\varepsilon}(T^{1+\varepsilon})\] and \[\sum_{k_f \leq T} \frac{L\left(\frac{1}{2},f\right)^4}{L(1,\operatorname{ad}f)} = T^2 P_6(\log T) + O_{\varepsilon}(T^{4/3+\varepsilon}),\] where $P_j(x)$ denotes some polynomial of degree $j$ in $x$. For Hecke--Maass cusp forms, this is known for the fourth moment via work of Ivic, while Zhi Qi proved earlier this year the analogous result for the third moment.\\ (Peter Humphries, Virginia)

背景链接：https://arxiv.org/abs/2405.17800

01262 | AMR-117-0014

### Moments of L-functions - Moments of the Riemann zeta function - Problem 16

O / open | AMR 问题清单汇编

Heath-Brown proved the famous Weyl-strength twelfth moment bound \[\int_{0}^{T} \left|\zeta\left(\frac{1}{2}+it\right)\right|^{12} dt \ll_{\varepsilon} T^{2 + \varepsilon}\] for the Riemann zeta function. Jutila generalised this to prove an analogous result for the sixth moment in the $t$-aspect of $L(1/2 + it,f)$, where $f$ is a Hecke--Maass cusp form. Recently, Milicevic-White and Nunes have independently proven complementary results on the twelfth moment of Dirichlet $L$-functions in the level aspect. Can one also prove analogous bounds for the sixth moment of $L(1/2,f \otimes \chi)$?\\ (Peter Humphries, Virginia)

背景链接：https://arxiv.org/abs/2405.17800

01263 | AMR-117-0015

### Moments of L-functions - Moments of the Riemann zeta function - Problem 17

O / open | AMR 问题清单汇编

Jutila proved the Weyl-strength twelfth moment bound \[\sum_{t_f \leq T} \frac{L\left(\frac{1}{2},f\right)^{12}}{L(1,\operatorname{ad}f)} \ll_{\varepsilon} T^{4 + \varepsilon},\] where the sum is over Hecke--Maass cusp forms of level 1 and spectral parameter at most $T$. Can one prove a level-aspect analogue of this, where one sums over Hecke--Maass cusp forms of level $q$ and bounded spectral parameter?\\ (Peter Humphries, Virginia)

背景链接：https://arxiv.org/abs/2405.17800

01264 | AMR-117-0016

### Moments of L-functions - Moments of the Riemann zeta function - Problem 18

O / open | AMR 问题清单汇编

Good showed that given a Hecke--Maass cusp form of level 1 and fixed spectral parameter $t_f$, we have that \[\int_{0}^{T} \left|L\left(\frac{1}{2}+it,f\right)\right|^2 dt = T P_1(\log T) + O_{t_f,\varepsilon}(T^{2/3 + \varepsilon}),\] where $P_1(x)$ denotes some polynomial of degree 1 in x. Crucially, the dependence on $t_f$ in the error term is not made explicit. Can one make this dependence explicit? In particular, does this error term not depend on $t_f$ if, say, $T \geq t_f^{3/2 + \delta}$ for some $\delta > 0$? How about for holomorphic Hecke eigenforms of weight $k_f$?\\ (Peter Humphries, Virginia)

背景链接：https://arxiv.org/abs/2405.17800

01265 | AMR-117-0017

### Moments of L-functions - Moments of the Riemann zeta function - Problem 19

O / open | AMR 问题清单汇编

Use the recent work of Humphries-Khan [HK] to compute weight aspect upper bounds for the 6th moment of $L(\frac12,f\otimes g)$ where $f$ and $g$ vary over the set of primitive forms of level 1 and weights $k$ and $\ell$ respectively, with $|k-\ell|=O(1)$ and $K<k+\ell\leq2K$.\\ (Alia Hamieh, UNBC)

背景链接：https://arxiv.org/abs/2405.17800

01266 | AMR-117-0018

### Moments of L-functions - Moments of the Riemann zeta function - Problem 20

O / open | AMR 问题清单汇编

Remove the integral in $t$ from Chandee-Li's work and give an asymptotic formula for $$ \frac{2}{\phi(q)} \sum_{\substack{\chi \mod{q}\\ \chi(-1)=(-1)^k}} \sum_{f\in\mathcal{H}_\chi}^h \left|L\left({\textstyle \frac{1}{2}},f\right)\right|^6 $$ See work of Chandee-Li-Matomaki-Radziwill on the sixth moment of Dirichlet $L$-functions.\\ (Joshua Stucky, University of Georgia)

背景链接：https://arxiv.org/abs/2405.17800

01267 | AMR-117-0019

### Moments of L-functions - Moments of the Riemann zeta function - Problem 21

O / open | AMR 问题清单汇编

Consider the family of Rankin-Selberg $L$-functions $L(\frac12,f\otimes g)$ where $g$ is fixed and $f$ varies over the set of primitive forms of weight $k$ and level $p^{\nu}$ as $\nu\to\infty$. Establish upper bounds for third and fourth moments of this family following the work in [ChL1] and [ChL2]. Note that the 4th moment of the family $L(\frac12,f)$ with $f$ varying as above was studied by Balkanova [balkanova]. \\ (Alia Hamieh, UNBC)

背景链接：https://arxiv.org/abs/2405.17800

01268 | AMR-117-0020

### Moments of L-functions - Moments of the Riemann zeta function - Problem 23

O / open | AMR 问题清单汇编

Determine a polynomial $P$ of degree 10 such that \begin{equation}

\sum_{0 < d < X} L(\tfrac{1}{2}, \chi_{8d})^4 = X P(\log X) + o(X) \end{equation} where $d$ ranges through odd square-free numbers and $\chi_{8d}$ denotes the quadratic character associated to the fundamental discriminant $8d$. Can the error term $o(X)$ be replaced by a power savings error term? Can the Multiple Dirichlet series method be used to obtain the first coefficients of $P$ (for instance see [DPP]). \\ (Nathan Ng, Lethbridge)

背景链接：https://arxiv.org/abs/2405.17800

01269 | AMR-117-0021

### Moments of L-functions - Moments of the Riemann zeta function - Problem 24

O / open | AMR 问题清单汇编

Shen and Stucky [ShSt] have determined the first four coefficients of this polynomial.

背景链接：https://arxiv.org/abs/2405.17800

01270 | AMR-117-0022

### Moments of L-functions - Moments of the Riemann zeta function - Problem 25

O / open | AMR 问题清单汇编

Compute moments for families of Artin-Schreier $L$-functions and improve results on 1-level zero density.\\ Notation: $p>2$ prime, $q$ power of $p$, $\mathcal{L}(u,f,\psi)$ is the Artin-Schreier $L$-function associated with rational function $f\in\mathbb{F}_q(x)$ and additive character $\psi$, $$\mathcal{AS}^0_d=\{f\in\mathbb{F}_q[x]:\deg f=d\}, (d,p)=1\}$$ $$\mathcal{AS}^{0,\mathrm{odd}}_d=\{f\in\mathbb{F}_q[x]:\deg f=d, f(x)=-f(-x)\}, (d,2p)=1\}$$ \begin{multline*}\mathcal{AS}^{\mathrm{ord}}_d=\left\{f=h/g: h,g\in\mathbb{F}_q[x],(g,h)=1,g\mbox{ squarefree},\right.\\ \left. \deg f=\max(\deg h,\deg g)=d, \deg g\in\{d,d-1\}\right\}.\end{multline*} $M_k=\langle \mathcal{L}(q^{1/2},f,\psi)^k\rangle$ is the $k$-th moment. All problems below concern the regime where $q$ is fixed and $d\to\infty$. 1-level density of zeros is defined w.r.t a Schwartz test function $\phi$.

Estimated difficulty scheme: {green = easy}, {blue = challenging}, {red = hard}, black = $$. \\ \\ $\mathcal{AS}^{\mathrm{ord}}$: compute 1-level density for $\mathrm{supp} \hat\phi\subset{(-1-\delta,1+\delta)\mbox{ for some }\delta>0}$ or $(-2-\delta,2+\delta),\mbox{ for some }\delta>0$.\\ Compute ${M_1},{M_2},{M_3,M_4},M_5,M_6$. \\ \\ $\mathcal{AS}^0$: compute 1-level density for $\mathrm{supp} \hat\phi\subset(-(2-2/p)-\delta,2-2/p+\delta)$.\\ Compute ${M_1,M_2,M_3},M_4$. \\ \\ $\mathcal{AS}^{0,\mathrm{odd}}$: compute 1-level density for $\mathrm{supp} \hat\phi\subset(-(1-1/p)-\delta,1-1/p+\delta)$.\\ Compute ${M_1},M_2$.

(Alexei Entin, Tel Aviv University)

背景链接：https://arxiv.org/abs/2405.17800

01271 | AMR-117-0023

### Moments of L-functions - Moments of the Riemann zeta function - Problem 26

O / open | AMR 问题清单汇编

Investigate the dependence of zero-density on moments of the zeta-function? More specifically, Ingham’s zero-density estimate makes use of the fourth moment $\int_{T}^{2T} |\zeta(1/2 + it)|^{4} dt$. Can we use other non-trivial moments (like, say, Heath-Brown’s estimate of the 12th moment) in such zero-density results, and can we generalise this to other L-functions?

(Timothy Trudgian, UNSW Canberra)

\section*{Bounds for $L$-functions}

背景链接：https://arxiv.org/abs/2405.17800

01272 | AMR-117-0024

### Moments of L-functions - Moments of the Riemann zeta function - Problem 27

O / open | AMR 问题清单汇编

Chandee and Soundararajan [CS] proved that the Riemann Hypothesis implies \[ |\zeta(\tfrac{1}{2}+it)| \ll \exp \left( (C_0 +o(1)) \frac{\log t}{\log \log t} \right) \] with $C_0 = \frac{\log 2}{2}= 0.1505 \ldots $. Reduce the value of $C_0$. \\ (N. Ng, Lethbridge)

背景链接：https://arxiv.org/abs/2405.17800

01273 | AMR-117-0025

### Moments of L-functions - Moments of the Riemann zeta function - Problem 28

O / open | AMR 问题清单汇编

Let $S(t)=\frac{1}{\pi} \arg \zeta(\tfrac{1}{2}+it)$. Carneiro, Chandee, and Milinovich [CCM] proved that the Riemann hypothesis implies \[ |S(t)| \le (C_0+o(1)) \cdot \frac{ \log t}{\log \log t} \] with $C_0 =\frac{1}{4}$. Reduce the value of $C_0$. \\ (Nathan Ng, Lethbridge)

背景链接：https://arxiv.org/abs/2405.17800

01274 | AMR-117-0026

### Moments of L-functions - Moments of the Riemann zeta function - Problem 29

O / open | AMR 问题清单汇编

Let $f$ be a Hecke--Maass cusp form of level 1. Prove the Weyl-strength subconvex bound $L(1/2+it,f) \ll_{\varepsilon} ((1 + |t_f - t|)(1 + |t_f + t|))^{1/6 + \varepsilon}$. This is known, via work of Jutila and Motohashi, for all $t \in \mathbb{R}$ away from the conductor-dropping regimes, namely when either $t_f - t$ or $t_f + t$ is $o(\max\{t_f,|t|\})$. Michel-Venkatesh prove a subconvex bound in the conductor-dropping regime, but the exponent is not explicit (and certainly not Weyl-strength).\\ (Peter Humphries, Virginia)

\section*{Additive divisor sums/shifted convolution sums}

背景链接：https://arxiv.org/abs/2405.17800

01275 | AMR-117-0002

### Moments of L-functions - Moments of the Riemann zeta function - Problem 3

O / open | AMR 问题清单汇编

We have also been informed a graduate student has worked extensively on this problem and nearly has a result.

背景链接：https://arxiv.org/abs/2405.17800

01276 | AMR-117-0027

### Moments of L-functions - Moments of the Riemann zeta function - Problem 30

O / open | AMR 问题清单汇编

Topacogullari [T1], [T3] and Drappeau [Dr] have proven asymptotic formulae for $D_{k,2}(x,h)$. Prove an omega theorem for $D_{k,2}(x,h)-M_{k,2}(x,h)$ for $k \ge 3$ where $M_{k,2}(x,h)$ is the main term in the asymptotic formula. \\ (N. Ng, Lethbridge)

背景链接：https://arxiv.org/abs/2405.17800

01277 | AMR-117-0028

### Moments of L-functions - Moments of the Riemann zeta function - Problem 31

O / open | AMR 问题清单汇编

Topacogullari [T1] proved that for $1 \le h \ll x^{\frac{2}{3}}$ \[ D_{3,2}(x,h) = x P_3(\log x) + O(x^{\frac{8}{9}+\varepsilon}) \] for a certain degree 3 polynomial where the coefficients of $P_3$ depend on $h$. A result was also obtained for $D_{2,3}(x,h)$.

背景链接：https://arxiv.org/abs/2405.17800

01278 | AMR-117-0029

### Moments of L-functions - Moments of the Riemann zeta function - Problem 32

O / open | AMR 问题清单汇编

Improve the range of $h$ and the size of the error term.

背景链接：https://arxiv.org/abs/2405.17800

01279 | AMR-117-0030

### Moments of L-functions - Moments of the Riemann zeta function - Problem 33

O / open | AMR 问题清单汇编

Compute exact simple formulae for the coefficients of $P_3$. Numerically check the size of $| D_{3,2}(x,h) - x P_3(\log x)| $. Is the error term of size $\sqrt{x}$? See the recent related article of Nguyen [Nguyen].

(Nathan Ng, Lethbridge)

背景链接：https://arxiv.org/abs/2405.17800

01280 | AMR-117-0031

### Moments of L-functions - Moments of the Riemann zeta function - Problem 35

O / open | AMR 问题清单汇编

Asymptotically evaluate \[ \sum_{n \le x} d_3(n) d_3(n+1). \] (folklore)

背景链接：https://arxiv.org/abs/2405.17800

01281 | AMR-117-0032

### Moments of L-functions - Moments of the Riemann zeta function - Problem 36

O / open | AMR 问题清单汇编

Asymptotically evaluate \[ \sum_{n \le x} d(n) d(n+1) d(n+2). \] (folklore)

背景链接：https://arxiv.org/abs/2405.17800

01282 | AMR-117-0033

### Moments of L-functions - Moments of the Riemann zeta function - Problem 37

O / open | AMR 问题清单汇编

Let $\kappa \in (0,1)$. Asymptotically evaluate \[ \sum_{n \le x} d_{\kappa}(n) d(n+1) d(n+2). \]

背景链接：https://arxiv.org/abs/2405.17800

01283 | AMR-117-0034

### Moments of L-functions - Moments of the Riemann zeta function - Problem 38

O / open | AMR 问题清单汇编

Let $\kappa \in (2,3)$. Asymptotically evaluate \[ \sum_{n \le x} d_{\kappa}(n) d_3(n+1). \]

背景链接：https://arxiv.org/abs/2405.17800

01284 | AMR-117-0035

### Moments of L-functions - Moments of the Riemann zeta function - Problem 39

O / open | AMR 问题清单汇编

Find a (conjectural) asymptotic formula for \[ \sum_{n \le x} d_{k_1}(n+h_1) d_{k_2}(n+h_2) \cdots d_{k_r}(n+h_r) \] where $(k_1, \ldots, k_r) \in \mathbb{N}^r$ and $h_1, \ldots, h_r$ are distinct positive integers with $r \ge 3$. This would be analogous to the. conjectural asymptotic formula for \[ \sum_{n \le x} \Lambda(n+h_1) \cdots \Lambda(n+h_r) \] that arises from the Hardy-Littlewood conjectures. Note that Ng-Thom and Tao have used different probabilistic arguments to derive conjectures in the case $r=2$.

背景链接：https://arxiv.org/abs/2405.17800

01285 | AMR-117-0003

### Moments of L-functions - Moments of the Riemann zeta function - Problem 4

O / open | AMR 问题清单汇编

Smoothed fourth moment. Let $\omega(t)$ be a smooth compactly supported function supported in $[c_1T, c_2T]$ where $0 < c_1 < c_2$. Show that \[ \int_{-\infty}^{\infty} \omega(t) |\zeta(\tfrac{1}{2}+i t)|^{4} dt = O(T^{\frac{1}{2} + \theta+\varepsilon }) \] where $\theta$ is upper bound in Ramanujan's conjecture. See [Ber, Corollary 2] where an analogous result was proven for modular $L$-functions.

背景链接：https://arxiv.org/abs/2405.17800

01286 | AMR-117-0036

### Moments of L-functions - Moments of the Riemann zeta function - Problem 40

O / open | AMR 问题清单汇编

Let $k \in \mathbb{N}$ and $k \ge 3$. For some small $\epsilon_0\geq0$ asymptotically evaluate \[ \int_{0}^{T} \sum_{m \le T^{2+\epsilon_0}} \frac{d_k(m)}{m^{\frac{1}{2}+it}} \sum_{n \le T^{2+\epsilon_0}} \frac{d(n)}{n^{\frac{1}{2}-it}} dt. \] Note that for mean values of long Dirichlet polynomials we know how to evaluate these sums (conjecturally on the additive divisor conjecture) when the length of the polynomials is $N=T^{2-\epsilon}$ (see [HN]).

背景链接：https://arxiv.org/abs/2405.17800

01287 | AMR-117-0037

### Moments of L-functions - Moments of the Riemann zeta function - Problem 41

O / open | AMR 问题清单汇编

Let $g$ be a primitive cusp form of weight $k$ for $\Gamma_{0}(N)$. Assume that $g$ is arithmetically normalized so that its first Fourier coefficient $a_g(1)=1$ and write $g(z)=\sum_{n=0}^{\infty}\lambda_{g}(n)n^{\frac{k-1}{2}}e(nz)$. Let $X, Y, P\geq 1$, and let $F(x,y)$ be a smooth weight function supported on $(X/4,4X]\times[Y/4,4Y]$ satisfying \[x^iy^j\frac{\partial^{i}}{\partial{x^i}}\frac{\partial^{j}}{\partial v^j}f(u,v)\ll P^{i+j}\] for all $i,j\geq0$. Consider the shifted convolution sum \[D_{g}(\ell_1,\ell_2;h)=\sum_{\ell_1m\pm \ell_2n=h}\lambda_g(n)\lambda_g(m)F(\ell_1m,\ell_2n),\] where $\ell_1$ and $\ell_2$ are coprime. Harcos [Harcos] proved an asymptotic upper bound for $D_{g}(\ell_1,\ell_2,h)$ where the implicit constants depend on $g$. Blomer and Harcos [BH] established an asymptotic bound for an averaged version of this shifted convolution sum in which the implicit constants also depend on $g$.

背景链接：https://arxiv.org/abs/2405.17800

01288 | AMR-117-0038

### Moments of L-functions - Moments of the Riemann zeta function - Problem 42

O / open | AMR 问题清单汇编

Determine the dependence of the explicit constant on $k$ and $N$.

背景链接：https://arxiv.org/abs/2405.17800

01289 | AMR-117-0039

### Moments of L-functions - Moments of the Riemann zeta function - Problem 43

O / open | AMR 问题清单汇编

For a primitive cusp form $g$ of weight $k$ for the full modular group, establish an asymptotic upper bound that is uniform in the weight aspect. Such bounds are useful in moment computations for Rankin-Selberg $L$-functions associated with primitive forms of varying large weights.

(A. Hamieh, UNBC)

背景链接：https://arxiv.org/abs/2405.17800

01290 | AMR-117-0004

### Moments of L-functions - Moments of the Riemann zeta function - Problem 5

O / open | AMR 问题清单汇编

In 1989 Zavorotnyi proved that \[ I_2(T) = T P_4 (\log T) + O( T^{\frac{2}{3}+\varepsilon}), \] where $P_4$ is a specific degree $4$ polynomial. Improve the bound in the error term. \\ (Nathan Ng, Lethbridge)

背景链接：https://arxiv.org/abs/2405.17800

01291 | AMR-117-0005

### Moments of L-functions - Moments of the Riemann zeta function - Problem 6

O / open | AMR 问题清单汇编

Conrey *el al* made the conjecture \[ I_k(T) = TP_{k^2} (\log T) + O(T^{\frac{1}{2} + \varepsilon}) \] where $P_{k^2}$ is a certain degree $k^2$ polynomial. Let \[ E_k(T) = I_k(T) -TP_{k^2} (\log T) \] denote the error term. Since the publication of [CFKRS] there has been some debate on what should be the correct conjectural size of $E_k(T) $. Note that it is widely believed that $E_1(T) = O(T^{\frac{1}{4}+\varepsilon})$ and $E_2(T) = O(T^{\frac{1}{2}+\varepsilon})$ are best possible. There are omega results for both of these cases. It would be desirable to come up with a conjecture for optimal constants $\theta_k$ such that \[ E_k(T) = O(T^{\theta_k+\varepsilon}) \] holds for $k \in\mathbb{N}$. N. Ng believes that $\theta_1=\frac{1}{4}$ and $\theta_2=\frac{1}{2}$ are the correct answers for $k=1,2$. What about $k \ge 3$? Both Ivic and Motohashi suggested $\theta_3 = \frac{3}{4}$. Can the Multiple Dirichlet Series method be used to derive an appropriate conjecture? (Work in progress of Baluyot and Cech is studying this.) \\ (Nathan Ng, Lethbridge)

背景链接：https://arxiv.org/abs/2405.17800

01292 | AMR-117-0006

### Moments of L-functions - Moments of the Riemann zeta function - Problem 7

O / open | AMR 问题清单汇编

Improve Chandee's [Ch, Theorem 1.2]. \begin{theorem} Let $k_i$ be positive real numbers. Let $\alpha_i=\alpha_i(T)$ be real-valued functions of $T$ such that $\alpha_i =o(T)$. Assume that $\lim_{T\rightarrow \infty} \alpha_i\log T$ and $\lim_{T\rightarrow \infty} (\alpha_i-\alpha_j)\log T$ exist or equal $\pm \infty$. Assume that for $i\neq j$, $\alpha_i\neq \alpha_j$ and $\alpha_i-\alpha_j =O(1)$. Then the Riemann Hypothesis implies that for $T$ sufficiently large, one has $$ M_{**k**}(T, **\alpha**)\ll_{**k**,\varepsilon} T(\log T)^{k_1^2 + \cdots k_m^2 +\varepsilon} \prod_{i<j} \left( \min\left\{\frac{1}{|\alpha_i-\alpha_j|}, \log T \right\}\right)^{2k_ik_j} $$ where \[ M_{**k**}(T, **\alpha**) = \int_{0}^{T} |\zeta(\tfrac{1}{2} + (t+\alpha_1) )|^{2 k_1} \cdots |\zeta( \tfrac{1}{2} + (t+\alpha_m)|^{2 k_m} dt. \] \end{theorem} Use Harper's method [Ha] to remove the $\varepsilon$. Note that the case of two shifts ($m=2$ and $k_1=k_2$) has been dealt with in [NSW]. \\ (N. Ng, Lethbridge)

背景链接：https://arxiv.org/abs/2405.17800

01293 | AMR-117-0007

### Moments of L-functions - Moments of the Riemann zeta function - Problem 9

O / open | AMR 问题清单汇编

Soundararajan [S] proved the following lower bound for $I_3(T)$: \[ I_3(T) \ge \frac{20.26 a_3}{9!} (1+o(1)) T (\log T)^9, \] for $T$ sufficiently large. Improve the value of 20.26. The best possible result would be 42. (Likewise improve the lower bounds for $I_k(T)$ for $k >3$.) Note that the works [BBLR], [BCR], and [HY] are now available and could be useful. \\ (Nathan Ng, Lethbridge)

背景链接：https://arxiv.org/abs/2405.17800

01294 | OWR-14299577-006

### Moments of Quadratic Dirichlet L-Functions

P / partially_solved | Oberwolfach 报告问题汇编

For every integer $k\geq1$, prove that there is an explicit polynomial $Q_k$ of degree $k(k+1)/2$ such that $\frac1N\sum_{|d|\leq N,\ d\text{ squarefree}}L(1/2,\chi_d)^k=Q_k(\log N)+o(1)$.

来源字段：https://doi.org/10.4171/owr/2025/51

01295 | OWR-15174-004

### Moments of Quadratic Dirichlet L-Functions over Function Fields

P / partially_solved | Oberwolfach 报告问题汇编

Fix $q\equiv1\pmod 4$, and let $\mathcal H_{2g+1}$ be the set of monic square-free polynomials of degree $2g+1$ over $\mathbb F_q$. For fixed $k$, does

$$\sum_{D\in\mathcal H_{2g+1}}L(1/2,\chi_D)^k=q^{2g+1}P_k(2g+1)+o(q^{2g+1})$$

hold as $g\to\infty$, where $P_k$ has degree $k(k+1)/2$?

来源字段：https://doi.org/10.4171/owr/2016/53

01296 | OWR-14299577-007

### Moments of Quadratic L-Functions over Function Fields

P / partially_solved | Oberwolfach 报告问题汇编

Let $q$ be an odd prime power. For every integer $k\geq1$, prove that there is an explicit polynomial $R_{k,q}$ of degree $k(k+1)/2$ such that $q^{-(2g+1)}\sum_{d}L(1/2,\chi_d)^k=R_{k,q}(g)+o(1)$ as $g\to\infty$, where the sum is over monic squarefree $d\in\mathbb{F}_q[t]$ of degree $2g+1$.

来源字段：https://doi.org/10.4171/owr/2025/51

01297 | OWR-14299577-005

### Moments of the Riemann Zeta Function

O / open | Oberwolfach 报告问题汇编

For every integer $k\geq1$, prove that there is an explicit polynomial $P_k$ of degree $k^2$ such that $\frac{1}{T}\int_0^T|\zeta(1/2+it)|^{2k}\,dt=P_k(\log T)+o(1)$ as $T\to\infty$.

来源字段：https://doi.org/10.4171/owr/2025/51

01298 | OWR-13102-007

### Monodromy Exactness for Isocrystals on Semistable Curves

O / open | Oberwolfach 报告问题汇编

For an $F$-convergent isocrystal $E$ on a semistable curve, for which coefficient objects is the sequence $H^1_{\mathrm{rig}}(X_k,E)\to H^1_{\mathrm{log-cris}}(X_k,E)\xrightarrow{N_E}H^1_{\mathrm{log-cris}}(X_k,E)$ exact?

来源字段：https://doi.org/10.4171/owr/2014/32

01299 | AIM-ALGEBRAIC_NUMBER_THEORY-0040

### Monotone means and a Poisson moving-target limit for complete-intersection curves

P / partially_solved | AIM 研讨会问题汇编

**Question 10 (Melanie Matchett Wood).** Look at complete intersections, and some examples with a moving target space.

The record is exactly as terse as this quotation: it does not specify an ensemble, statistic, asymptotic variable, or formal conjecture. The nearby questions ask for the distribution of the number of $\mathbb F_q$-points on curves as genus grows and for “principle (A)” for tri-canonically embedded curves. The AIM workshop report supplies the missing common theme: Wood's talk asked when the number of points on curves in a family has a distribution given by a sum of independent identically distributed random variables. The source file itself never defines its labels “Principles (A) and (B),” and no definition was recovered from the AIM report. Consequently, this report does **not** assign those labels to a reconstructed statement.

The closest precise complete-intersection model is the one of Bucur--Kedlaya. Let $X\subseteq \mathbb P^r_{\mathbb F_q}$ be smooth of dimension $m$, let $$ S_d=H^0(\mathbb P^r,\mathcal O(d)),\qquad \mathbf f=(f_1,\ldots,f_k)\in S_{d_1}\times\cdots\times S_{d_k}, $$ choose the **ordered tuple of equations** uniformly, and condition $X\cap H_{f_1}\cap\cdots\cap H_{f_k}$ to be smooth of dimension $m-k$. The statistic is its number of projective $\mathbb F_q$-points. For fixed $X,q,m,k$, and large degrees, the precise local-independence principle is that this point count converges to a sum of $\#X(\mathbb F_q)$ independent Bernoulli variables.

At a rational point, put $$ L(q,m,k)=\prod_{j=0}^{k-1}(1-q^{-(m-j)}). $$ The probability that the first jets of the equations are locally compatible with smoothness is $$ a(q,m,k)=1-q^{-k}+q^{-k}L(q,m,k), $$ and, conditional on that local event, the point lies on the intersection with probability $$ \rho(q,m,k)=\frac{q^{-k}L(q,m,k)}{1-q^{-k}+q^{-k}L(q,m,k)}. \tag{1.1} $$ Thus the recovered fixed-target principle is a rigorously proved **fiberwise/local asymptotic independence statement**, not an assertion about uniformly sampled isomorphism classes of curves.

“Moving target space” is naturally read as a triangular array in which the ambient $X_i$, hence $N_i=\#X_i(\mathbb F_q)$, also varies. This reading is confirmed by the paper of Kurlberg--Wigman cited by Bucur--Kedlaya: they vary projective surfaces having increasingly many rational points and obtain Gaussian point-count statistics for high-degree curve sections. The present report also studies the explicit moving targets $X_n=\mathbb P^n$, with the codimension moving simultaneously to $k_n=n-1$. This second shape is in a different probabilistic regime: the number of ambient points grows, but the local hit probability decays at the compensating rate, leading to a Poisson rather than a Gaussian limit.

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

01300 | NT-070

### Montgomery's Pair Correlation Conjecture

P / partially_solved | 独立 / 未分配集合

Does the pair correlation function of Riemann zeta zeros match that of random Hermitian matrices?

背景链接：https://doi.org/10.1090/pspum/024/0338455

01301 | OWR-12985-004

### Mordell-Lang Finiteness Outside Anomalous Loci

O / open | Oberwolfach 报告问题汇编

Let $A\subset(\mathbb C^*)^n$ have codimension greater than $k$, let $A'$ be a subvariety of its anomalous points, and let $L\subset\mathbb C^n$ have dimension $k$ with $\exp(L)\subset G$ and $G/\exp(L)$ of finite rank. Is $(A\setminus A')\cap G$ finite?

来源字段：https://doi.org/10.4171/owr/2014/27

01302 | OWR-13351-005

### Motivic Kontsevich-Zagier Equivalence

P / partially_solved | Oberwolfach 报告问题汇编

Does equality of two Cluckers-Loeser motivic integrals imply that they are related by the motivic analogues of additivity, semialgebraic change of variables, and Stokes' formula?

来源字段：https://doi.org/10.4171/owr/2014/49

01303 | AIM-ANALYTIC_NUMBER_THEORY-0036

### Moving-cutoff smooth-number correlations and a weighted mean-value consequence

P / partially_solved | AIM 研讨会问题汇编

What can we say about correlations with itself of $$ \mathbb{1}_{p^+(n)\leq n^\delta}, $$ with $\delta\to0$? Here $p^+(n):=\max\{p:p\mid n\}$. Even upper bounds would be welcome. This kind of result would imply sharp mean values for Dirichlet polynomials supported on smooth numbers.

来源字段：http://aimpl.org/sarnakconjecture/5/

01304 | OWR-14298805-002

### Moy-Prasad Data and Langlands Parameters

O / open | Oberwolfach 报告问题汇编

Let $\pi\in\operatorname{Irr}_{\mathrm{sm}}(G(F))$, let $(X,r)$ be its associated Moy-Prasad datum with $r>0$, and let $\phi_\pi$ be its Langlands parameter. Prove that $\phi_\pi|_{W_F^{>r}}=1$ and $\phi_\pi|_{W_F^{=r}}=\phi_X$ in general.

来源字段：https://doi.org/10.4171/owr/2024/54

01305 | OWR-15169-007

### Multiplicatively Badly Approximable Pairs

O / open | Oberwolfach 报告问题汇编

What is the correct multiplicative analogue of badly approximable points? For $M_\lambda=\{(\alpha,\beta)\in\mathbb R^2:\liminf_{q\to\infty}q(\log q)^\lambda\|q\alpha\|\,\|q\beta\|>0\}$, is $M_\lambda$ empty for $\lambda<1$ and of full Hausdorff dimension for $\lambda\geq1$?

来源字段：https://doi.org/10.4171/owr/2016/48

01306 | GREEN-036

### Multiplicatively Closed Set Density

O / open | Ben Green 问题

Let $A$ be the smallest set containing 2 and 3, and closed under the operation $a_1a_2 - 1$ (if $a_1, a_2 \in A$, then $a_1a_2 - 1 \in A$). Does $A$ have positive density?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

01307 | OWR-4804-006

### Multiplicity Formulas for Galois-Invariant Induced Representations

O / open | Oberwolfach 报告问题汇编

Let $E/F$ be a quadratic extension of local fields, let $\tau$ be its nontrivial Galois involution, set $G=\operatorname{GL}_n(E)$, and let $X$ be the variety of nonsingular Hermitian matrices over $E/F$ with its natural right $G$-action. If $\pi=\delta_1\times\cdots\times\delta_k$ is $\tau$-invariant and each $\delta_i$ is essentially square-integrable, is $\dim\operatorname{Hom}_G(X,\pi^*)=2^k$?

来源字段：https://doi.org/10.4171/owr/2011/14

01308 | OWR-4804-007

### Multiplicity of Galois-Invariant Imprimitive Products

O / open | Oberwolfach 报告问题汇编

Let $E/F$ be a quadratic extension of local fields, let $\tau$ be its nontrivial Galois involution, set $G=\operatorname{GL}_n(E)$, and let $X$ be the variety of nonsingular Hermitian matrices over $E/F$ with its natural right $G$-action. If $\pi_1,\ldots,\pi_k$ are $\tau$-invariant imprimitive representations and $\pi=\pi_1\times\cdots\times\pi_k$ is irreducible, is $\dim\operatorname{Hom}_G(X,\pi^*)=2^k$?

来源字段：https://doi.org/10.4171/owr/2011/14

01309 | AIM-ANALYTIC_NUMBER_THEORY-0057

### Multiplicity-sensitive moments of products of Dirichlet L-functions

P / partially_solved | AIM 研讨会问题汇编

Develop a heuristic for moments of imprimitive $L$-functions, or if your $L$-function factors into degree $1$ factors, e.g. $$\int |L(\tfrac{1}{2}+it,\chi_1)\cdots L(\tfrac{1}{2}+it,\chi_k)|^2\,dt$$ (or different powers).

来源字段：http://aimpl.org/zetamoments/5/

01310 | OWR-1973-010

### Natural Boundaries of Polynomial Euler Products

P / partially_solved | Oberwolfach 报告问题汇编

Let $W(x,y)=\sum_{n,m}a_{n,m}x^ny^m$ be integral with $W(x,0)=1$, and define $D(s)=\prod_pW(p,p^{-s})$. Is $D$ meromorphically continuable to the whole plane iff it is a finite product of Riemann zeta-functions? Conjecturally, otherwise $\operatorname{Re}(s)=\beta$, where $\beta=\max\{(n+1)/m:a_{n,m}\neq0,\ m>0\}$, is its natural boundary.

来源字段：https://doi.org/10.4171/owr/2008/28

01311 | AIM-ANALYTIC_NUMBER_THEORY-0001

### Near-rational improvements and a linear-phase delta-method bottleneck

P / partially_solved | AIM 研讨会问题汇编

Obtain cancellations in additive twists of GL(3) Fourier coefficients.

Let $\pi$ be a GL(3) Hecke--Maass cusp form with the n-th Fourier coefficient $\lambda_\pi(1,n)$. For $\alpha\in\mathbb R$, prove a bound of the form $$\sum_{n\ll N}\lambda_\pi(1,n)e(n\alpha)\ll_\pi N^{1-\delta},$$ for some $\delta>0$ using a delta method. Aim to improve upon Miller's bound.

来源字段：http://aimpl.org/deltasubconvex2/1/

01312 | AMR-093-0133

### New Mersenne conjecture

O / open | AMR 问题清单汇编

New Mersenne conjecture: for any odd natural number $p$, if any two of the three conditions $p = 2^k \pm 1$ or $p = 4^k \pm 3$, $2^p - 1$ is prime, and $(2^{p} + 1)/3$ is prime are true, then the third condition is also true.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01313 | NT-038

### Newman's Conjecture

P / partially_solved | 独立 / 未分配集合

Does the partition function satisfy any arbitrary congruence infinitely often?

背景链接：https://doi.org/10.1016/j.aim.2025.110367

01314 | AIM-ALGEBRAIC_NUMBER_THEORY-0070

### Newton divisors versus Hirzebruch--Zagier cycles on Hilbert modular surfaces

P / partially_solved | AIM 研讨会问题汇编

**Problem 7 (K. Murty).** A Hilbert modular surface in characteristic \(p\) has interesting cycles parametrizing Hilbert--Blumenthal abelian varieties with specified Newton polygon behavior. Are these contained in the space of special cycles made up of (Hecke translates of) Hirzebruch--Zagier curves? How can we check this?

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

01315 | OWR-16764-006

### Newton Strata Meeting Weakly Admissible Loci

P / partially_solved | Oberwolfach 报告问题汇编

For which $[b']\in B(G)$ does the Newton stratum $\mathcal{F}(G,b,\mu)^{[b']}$ intersect the weakly admissible locus $\mathcal{F}(G,b,\mu)^{wa}$ nontrivially?

来源字段：https://doi.org/10.4171/owr/2019/2

01316 | OWR-14297740-019

### Next-Most-Probable Galois Groups of Random Polynomials

O / open | Oberwolfach 报告问题汇编

In the large-box model for random integer polynomials of fixed degree, determine the next-most-probable Galois group after the full symmetric group, including whether the alternating group or a point stabilizer dominates and with what asymptotic probability.

来源字段：https://doi.org/10.4171/owr/2023/42

01317 | OWR-14297740-012

### Non-Density of Bounded-Rank Galois Loci

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a smooth geometrically connected variety of dimension $\geq 2$ over a number field $k$, let $\rho_\ell : \pi_1(X) \to G_\ell \subset \mathrm{GL}_r(\mathbb{Z}_\ell)$ be an $\ell$-adic Galois representation whose image $G_\ell$ is a $\mathbb{Z}_\ell$-Lie group with finite abelianization, let $|X|_{\leq d}$ be the set of closed points of degree $\leq d$ over $k$, and let $|X|_{V_\ell}$ be the Cadoret--Tamagawa locus of closed points $x \in |X|$ where the image of the decomposition group $D_x$ fails to be open in $G_\ell$ (or lies in a proper subgroup scheme $V_\ell$). Is the intersection $|X|_{V_\ell} \cap |X|_{\leq d}$ non-Zariski-dense in $X$?

来源字段：https://doi.org/10.4171/owr/2023/42

01318 | OWR-14297740-013

### Non-Density of Toric Galois Loci

P / partially_solved | Oberwolfach 报告问题汇编

Under the hypotheses defining the geometric $\ell$-adic local system $V_\ell$ on $X$, is the toric locus $|X|^{\mathrm{tor}}_{V_\ell}\cap|X|$ non-Zariski-dense in $X$? Is this toric locus independent of $\ell$?

来源字段：https://doi.org/10.4171/owr/2023/42

01319 | OWR-3388-004

### Non-Integrality Conjecture for Rational Hypersurfaces

O / open | Oberwolfach 报告问题汇编

For $n\geq2$, let $V\subset\mathbb{A}^n$ be a hypersurface over $\mathbb{Q}$ with $V(\mathbb{Q})$ Zariski dense. Must $V(\mathbb{Q})\cap((\mathbb{Q}\setminus\mathbb{Z})\times\mathbb{Q}^{n-1})$ also be Zariski dense?

来源字段：https://doi.org/10.4171/owr/2009/05

01320 | OWR-3480-014

### Noncommutative $p$-adic Zeta Functions

S / solved | Oberwolfach 报告问题汇编

For a totally real $p$-adic Lie extension $F_\infty/F$, construct the unique $\zeta\in K_1(\Lambda(G)_S)$ satisfying $\partial(\zeta)=-[C^\bullet]$ and $\zeta(\rho\kappa_F^k)=L_\Sigma(\rho,1-k)$ for Artin $\rho$ and $k>0$ divisible by $p-1$.

来源字段：https://doi.org/10.4171/owr/2009/30

01321 | OWR-14299577-011

### Nonconstancy of the Multiplication-Table Limit Profile

O / open | Oberwolfach 报告问题汇编

For the limiting profile $f_{\mathrm{Mult}}$ governing the multiplication-table problem, prove that $\max_x f_{\mathrm{Mult}}(x)/\min_x f_{\mathrm{Mult}}(x)>1$; equivalently, prove that the profile is not constant.

来源字段：https://doi.org/10.4171/owr/2025/51

01322 | AIM-ANALYTIC_NUMBER_THEORY-0048

### Nonsquare dual polynomials in Florea's Poisson formula

S / solved | AIM 研讨会问题汇编

Analyze Alexandra Florea's formula for $V$ not equal to a square.

来源字段：http://aimpl.org/zetamoments/2/

01323 | OWR-13750341-007

### Nontriviality of Heegner-Point Kolyvagin Systems

O / open | Oberwolfach 报告问题汇编

For an elliptic curve $E/\mathbb Q$, an odd prime $p$ of good ordinary reduction, and an admissible imaginary quadratic field $K$, is the Kolyvagin system $\{\kappa_n\}_{n\in\mathcal N}$ derived from Heegner points nontrivial; equivalently, does $\kappa_n\neq0$ for some square-free product $n$ of admissible inert primes?

来源字段：https://doi.org/10.4171/owr/2023/28

01324 | OWR-13750341-004

### Nonvanishing Statistics of Twisted Central L-Values

O / open | Oberwolfach 报告问题汇编

Fix a modular form $f$ of weight $k\geq2$ and an integer $d\geq1$. If $K_d(X)$ is the set of primitive Dirichlet characters of order $d$ and conductor at most $X$, what is the asymptotic behavior of $\#\{\chi\in K_d(X):L(f,\chi,k/2)\neq0\}$ as $X\to\infty$?

来源字段：https://doi.org/10.4171/owr/2023/28

01325 | NT-052

### Normality of Irrational Algebraic Numbers

O / open | 独立 / 未分配集合

Are all irrational algebraic numbers normal in every base?

背景链接：https://doi.org/10.1017/CBO9781139019672

01326 | NT-051

### Normality of Pi

O / open | 独立 / 未分配集合

Is $\pi$ a normal number in base 10?

背景链接：https://doi.org/10.1080/10586458.2002.10504475

01327 | AIM-ALGEBRAIC_NUMBER_THEORY-0046

### Normalized discriminants and a trace-square reduction

P / partially_solved | AIM 研讨会问题汇编

Question 16 (Brian Conrey). Fix q, let g → ∞, and describe the distribution of discrimi-nants of L-polynomials of curves (or hyperelliptic curves).

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

01328 | OWR-4428-014

### Normalized Mod-p Cohomology in Arithmetic Lattice Towers

O / open | Oberwolfach 报告问题汇编

Let $\Gamma$ be an arithmetic lattice in $\operatorname{SL}_2(\mathbb C)$, let $p$ be prime, and let $(\Gamma_i)$ be any normal chain of $p$-power-index subgroups with trivial intersection. Does

$$\frac{\dim H^1(\Gamma_i,\mathbb F_p)}{[\Gamma:\Gamma_i]}$$

tend to $0$ as $i\to\infty$?

来源字段：https://doi.org/10.4171/owr/2010/40

01329 | OWR-15170-004

### Northcott Property for Bounded-Degree Composita

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Northcott-Property Question [OWR-15170-005]

Let $\mathbb Q^{(d)}$ be the compositum of all extensions of $\mathbb Q$ of degree at most $d$; does it have Property (N), meaning that for every $X\geq1$ only finitely many elements have absolute multiplicative Weil height $H(\alpha)<X$?

来源字段：https://doi.org/10.4171/owr/2016/49

01330 | OWR-15170-006

### Northcott Property for Totally Real Integer Rings

O / open | Oberwolfach 报告问题汇编

For the ring of integers $\mathcal O_{\mathbb Q_{\mathrm{tr}}^{(d)}}$ of the maximal totally real subfield of $\mathbb Q^{(d)}$, does Property (N) hold?

来源字段：https://doi.org/10.4171/owr/2016/49

01331 | OWR-1537-006

### Number Fields Whose Integers Are Sums of Units

O / open | Oberwolfach 报告问题汇编

Determine all number fields $K$ such that every algebraic integer in $K$ can be represented as a finite sum of units of the ring of integers of $K$.

来源字段：https://doi.org/10.4171/owr/2007/21

01332 | AIM-ALGEBRAIC_NUMBER_THEORY-0073

### Numerical surface classes on the square of a general abelian surface

P / partially_solved | AIM 研讨会问题汇编

`0. (C. Schoen) Let J be a general abelian surface (over C, say). Describe the surfaces in J ×J.3\n1`

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

01333 | AIM-ALGEBRAIC_NUMBER_THEORY-0028

### Odd Brauer torsion on K3 surfaces: status and a Picard-rank filter

P / partially_solved | AIM 研讨会问题汇编

The record is Problem 8 in the AIM workshop list *Rational and integral points on higher-dimensional varieties*, dated May 28, 2014. I interpret \(\operatorname{Br}(X)[u]\) as the subgroup killed by the odd integer \(u\).

1. Can a K3 surface \(X/\mathbb Q\) have odd-order torsion in \(\operatorname{Br}(X)\) that obstructs the Hasse principle? Can this happen already in \[ \operatorname{Br}_1(X)=\ker\bigl(\operatorname{Br}(X)\to \operatorname{Br}(\overline X)\bigr)? \] 2. For a smooth quartic K3 surface \(X\subset \mathbb P^3\), does \[ \bigl(\operatorname{Br}(X)/\operatorname{Br}(\mathbb Q)\bigr)[2]=0 \] imply the Hasse principle? What happens to weak approximation? 3. What is the corresponding situation for a degree-six K3 surface \(X\subset \mathbb P^4\)? In the K3 context this is the genus-four polarization; for a smooth nondegenerate model it is the familiar quadric--cubic complete intersection.

The source also sketches a zero-cycle argument: an odd class on a quartic can be made orthogonal to local zero-cycles of degree one, and the degree-one zero-cycle conjecture invoked by Colliot-Thélène would then produce a global zero-cycle of degree one. The remaining issue is that index one does not by itself produce a rational point.

来源字段：https://aimath.org/pastworkshops/ratlhigherdimvarproblems.pdf

01334 | OWR-13351-012

### Odd Dihedral Groups as Oort Groups

P / partially_solved | Oberwolfach 报告问题汇编

For odd primes p, are the dihedral groups $D_{p^n}$ of order $2p^n$ Oort groups?

来源字段：https://doi.org/10.4171/owr/2014/49

01335 | OPG-491

### Odd incongruent covering systems

O / open | Open Problem Garden

Conjecture There is no covering system whose moduli are odd, distinct, and greater than 1.

背景链接：http://www.openproblemgarden.org/op/odd_incongruent_covering_systems

01336 | NT-001

### Odd Perfect Numbers

O / open | 独立 / 未分配集合

别名：Odd Perfect Numbers [NUM-002]; Odd perfect numbers [OPG-2147]

Does there exist an odd perfect number? A perfect number is a positive integer that is equal to the sum of its proper divisors (excluding itself). For example, $6 = 1 + 2 + 3$ is perfect.

背景链接：https://www.ams.org/journals/mcom/2012-81-279/S0025-5718-2012-02563-4/S0025-5718-2012-02563-4.pdf

01337 | NT-050

### Odd Weird Numbers

P / partially_solved | 独立 / 未分配集合

别名：Odd Weird Numbers [NUM-006]

Do any odd weird numbers exist?

背景链接：https://www.erdosproblems.com/470

01338 | AIM-ALGEBRAIC_NUMBER_THEORY-0047

### Odd-degree Jacobian mean and its arithmetic correction

P / partially_solved | AIM 研讨会问题汇编

Question 17 (John Voight). What is the average number of points on the Jacobian of a hyperelliptic curve?

It is Question 17 in the AIM workshop list *Arithmetic statistics over finite fields and function fields*. The statement does not specify the ground field, genus, family, limiting regime, or weighting. “Points” must also be interpreted: over an algebraic closure a positive-dimensional Jacobian has infinitely many points, whereas over a finite field its set of rational points is finite.

Question 16 immediately before this record begins “Fix \(q\), let \(g\to\infty\),” and Question 18 immediately after it again begins “Fix \(q\).” This makes the following a well-motivated, but not textually forced, reconstruction:

- \(q\) is an odd prime power and \(A=\mathbb F_q[T]\); - \(\mathcal H_{2g+1,q}\) is the set of monic square-free \(D\in A\) of degree \(2g+1\); - \(C_D\) is the smooth projective curve with affine model \(y^2=D(T)\), and \(J_D=\operatorname{Jac}(C_D)\); - the desired average is \[ M_g(q):=\frac{1}{\#\mathcal H_{2g+1,q}} \sum_{D\in\mathcal H_{2g+1,q}}\#J_D(\mathbb F_q), \qquad g\longrightarrow\infty \] with \(q\) fixed.

Here \(\#\mathcal H_{2g+1,q}=(q-1)q^{2g}\). Odd degree marks the unique rational point at infinity and is the standard “imaginary hyperelliptic” equation ensemble. It is not interchangeable with any of the following:

- uniform isomorphism classes of curves; - stack weighting by \(1/\#\operatorname{Aut}(C)\); - even-degree equations; - characteristic-two Artin--Schreier models; - fixed genus with \(q\to\infty\).

The exact source wording is therefore preserved, and all results below state their weighting and limiting regime explicitly.

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

01339 | OPG-155

### Olson's Conjecture

P / partially_solved | Open Problem Garden

Conjecture If $a_1,a_2,\ldots,a_{2n-1}$ is a sequence of elements from a multiplicative group of order $n$, then there exist $1 \le j_1 < j_2 \ldots < j_n \le 2n-1$ so that $\prod_{i=1}^n a_{j_i} = 1$.

背景链接：http://www.openproblemgarden.org/op/olsons_conjecture

01340 | OWR-1188-004

### Optimal Descent Degrees in Reduced Hurwitz Spaces

O / open | Oberwolfach 报告问题汇编

Let $\Pi:H(C)\to H^{\mathrm{rd}}(C)$ be the quotient from the Hurwitz space of $G$-covers with inertia data $C$ to the reduced Hurwitz space. Given $p^{\mathrm{rd}}\in H^{\mathrm{rd}}(C)(k)$, find an optimal upper bound for $m_k(p^{\mathrm{rd}})=\min_{p\in\Pi^{-1}(p^{\mathrm{rd}})}[k(p):k]$.

来源字段：https://doi.org/10.4171/owr/2006/06

01341 | OWR-17293-002

### Optimal Exponent for Counting Monogenic Number Fields

O / open | Oberwolfach 报告问题汇编

The exponent $1/2+1/n$ in the lower bound for the number of degree-$n$ monogenic number fields with Galois group $S_n$ and discriminant less than $X$ is optimal.

来源字段：https://doi.org/10.4171/owr/2019/50

01342 | OWR-11695860-005

### Optimal Gaps Between Binary Quadratic-Form Values

P / partially_solved | Oberwolfach 报告问题汇编

Determine the optimal order of the largest gaps between consecutive positive integers represented by a fixed primitive positive-definite binary quadratic form.

来源字段：https://doi.org/10.4171/owr/2022/50

01343 | OWR-17293-012

### Optimal Pólya-Vinogradov Constants Across Dirichlet Characters

O / open | Oberwolfach 报告问题汇编

For a prime $p$ and nonprincipal Dirichlet characters $\chi\pmod p$, determine the best constant $C$ such that $\min_{\chi}\max_{0\leq x\leq p}\left|\sum_{0<n\leq x}\chi(n)\right|<C\sqrt p$.

来源字段：https://doi.org/10.4171/owr/2019/50

01344 | AIM-ANALYTIC_NUMBER_THEORY-0085

### Orbifold expansions of the mirror-quintic free energies

P / partially_solved | AIM 研讨会问题汇编

别名：Normalization-aware leading quintic amplitudes at the Gepner point [AIM-PHYSICS-0018]

Can we calculate the leading terms of $F_g^B(q)$ at the orbifold point $z=0$ of the quintic \[ (\sum x_i^5+z\prod x_i=0)/\mathbb Z_5^3? \]

来源字段：http://aimpl.org/gromwitnumthry/4/

01345 | AIM-ALGEBRAIC_NUMBER_THEORY-0059

### Orthogonal low-lying zeros and forced-zero deflation

P / partially_solved | AIM 研讨会问题汇编

For an orthogonal family of function-field $L$-functions---naturally, quadratic twists $L(E\otimes\chi_D,u)$ of a fixed elliptic curve $E/\mathbb F_q(t)$---compute low-lying one-level or higher-level statistics after mean-spacing-one unfolding; condition on the functional-equation sign when needed; distinguish $O$, $SO(\mathrm{even})$, and $SO(\mathrm{odd})$; and state whether forced central zeros are retained or removed. In the fixed-$q$ degree aspect, let $\deg D\to\infty$.

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

01346 | OWR-14298795-012

### Outer Automorphisms of p-Adic Absolute Galois Groups

O / open | Oberwolfach 报告问题汇编

Determine the structure of the outer automorphism group $\operatorname{Out}(G_K)$ of the absolute Galois group of a $p$-adic field $K$.

来源字段：https://doi.org/10.4171/owr/2024/45

01347 | OWR-12697688-008

### p-Adic Arithmetic Gan-Gross-Prasad Formula

P / partially_solved | Oberwolfach 报告问题汇编

Assume $\pi$ is tempered and ordinary at every $p$-adic place. For $\phi\in\pi_f$ and $\phi^\vee\in\pi_f^\vee$, prove, up to a nonzero explicit factor, $(\int_{\operatorname{Sh}(H)}\phi,\int_{\operatorname{Sh}(H)}\phi^\vee)_p=e_p(\pi)^{-1}L_p'(\tfrac12,\pi,R)\prod_v\alpha_v(\phi_v,\phi_v^\vee)$.

来源字段：https://doi.org/10.4171/owr/2023/5

01348 | OWR-17300-009

### p-Adic Gross-Kudla-Zagier Formula for Diagonal Cycles

O / open | Oberwolfach 报告问题汇编

For the canonical $p$-adic height pairing on the modified diagonal cycles of three modular elliptic curves, the conjectural $p$-adic Gross-Kudla-Zagier formula asserts that $\langle h^*\Delta^\dagger_\xi,h'^*\Delta^\dagger_\xi\rangle_{\mathrm{Nek}}(\sqrt{3}-1)E_p(\operatorname{Fil}^+V_pE)=L'_p(E,1)I^\natural(h\otimes h')\cdot(\text{a power of }2)(\zeta(2)/\pi^2)^2$ for all admissible $h$ and $h'$.

来源字段：https://doi.org/10.4171/owr/2019/57

01349 | OWR-760-002

### Pair Correlation of Zeta Zeros

O / open | Oberwolfach 报告问题汇编

Assume the Riemann hypothesis, and write the nontrivial zeros of $\zeta$ as $\frac12+i\gamma$. For every suitable even test function $f$, does the following asymptotic hold? $$\sum_{0<\gamma,\gamma'\leq T} f\!\left(\frac{(\gamma-\gamma')\log T}{2\pi}\right)\sim\frac{T\log T}{2\pi}\left[f(0)+\int_{-\infty}^{\infty}f(u)\left(1-\left(\frac{\sin(\pi u)}{\pi u}\right)^2\right)\,du\right]$$.

来源字段：https://doi.org/10.4171/owr/2004/46

01350 | OWR-8415337-005

### Pairs of $p$-Congruent Elliptic Curves

O / open | Oberwolfach 报告问题汇编

Are the two known nontrivial $17$-congruent pairs, together with their simultaneous quadratic twists, the only nontrivial $17$-congruent pairs over $\mathbb Q$, and are there no nontrivial $p$-congruent pairs for primes $p>17$?

来源字段：https://doi.org/10.4171/owr/2021/34

01351 | OWR-1734-006

### Paramodular Conjecture for Abelian Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Let $N$ be odd and squarefree, and let $K$ be a totally real field of degree $d$ with ring of integers $\mathcal O$. Is there a bijection between Hecke newforms $f$ in the weight-two, degree-two paramodular space for $K(N)$ whose eigenvalues generate $K$, and isogeny classes of simple abelian varieties $A/\mathbb Q$ of dimension $2d$ and conductor $N^d$ with endomorphisms by $\mathcal O$, such that the product of the spinor $L$-functions of the Galois conjugates of $f$ equals the Hasse-Weil $L$-function of $A$?

来源字段：https://doi.org/10.4171/owr/2007/51

01352 | OWR-4990375-007

### Paramodular Correspondence for Abelian Surfaces and Fourfolds

O / open | Oberwolfach 报告问题汇编

别名：Paramodular Correspondence for Abelian Surfaces and Fourfolds [OWR-4990375-008]

For $N\in\mathbb N$, let $A_N$ be the isogeny classes of abelian surfaces over $\mathbb Q$ of conductor $N$ with $\operatorname{End}_{\mathbb Q}(A)=\mathbb Z$, let $B_N$ be the isogeny classes of quaternionic-multiplication abelian fourfolds over $\mathbb Q$ of conductor $N^2$, and let $P_N$ be the cuspidal nonlift Siegel paramodular newforms of level $N$ with rational Hecke eigenvalues, modulo nonzero scaling. Is there a bijection $A_N\cup B_N\leftrightarrow P_N$ matching Hasse-Weil and spin $L$-functions, with the spin $L$-function squared for $B_N$?

来源字段：https://doi.org/10.4171/owr/2021/6

01353 | OWR-4990375-001

### Paramodular Forms and Isogeny Classes of Abelian Surfaces

O / open | Oberwolfach 报告问题汇编

Does the Brumer-Kramer paramodular correspondence give a bijection between isogeny classes of abelian surfaces over $\mathbb Q$ of conductor $N$ and the appropriate paramodular forms of level $N$?

来源字段：https://doi.org/10.4171/owr/2021/6

01354 | OWR-14605-002

### Paramodularity of Antisymmetric Borcherds Products

O / open | Oberwolfach 报告问题汇编

Are the holomorphic antisymmetric paramodular Borcherds products of weight $2$ and levels $N=713$ and $N=893$ modular with respect to the known rank-one abelian surfaces over $\mathbb Q$ of conductor $N$, as predicted by the Paramodular Conjecture?

来源字段：https://doi.org/10.4171/owr/2016/23

01355 | OWR-1590-009

### Parity Conjectures for Elliptic Curves over Number Fields

P / partially_solved | Oberwolfach 报告问题汇编

For an elliptic curve $E$ over a number field $K$, do the parity conjectures hold: assuming finiteness of the Tate-Shafarevich group, is $(-1)^{\operatorname{rank}E(K)}=w(E/K)$, and, for every prime $p$, is the corresponding $p$-Selmer-rank parity equal to $w(E/K)$? The stronger version asks for the rank-parity equality without assuming finiteness of the Tate-Shafarevich group.

来源字段：https://doi.org/10.4171/owr/2007/34

01356 | OWR-3480-004

### Parity of 2-Primary Tate-Shafarevich Groups

O / open | Oberwolfach 报告问题汇编

For every elliptic curve $E/K$ over a number field, the $\mathbb{F}_2$-dimension of $\operatorname{Sha}(E/K)[2]$ is even.

来源字段：https://doi.org/10.4171/owr/2009/30

01357 | AIM-ANALYTIC_NUMBER_THEORY-0029

### Parity-twisted Liouville correlations and a synchronization obstruction

P / partially_solved | AIM 研讨会问题汇编

Is $-1$ in the spectrum of a Liouville system? Equivalently, can we show that \[ \sum_{n\leq X}(-1)^{n-1}\frac{\lambda(n+h_1)\cdots\lambda(n+h_k)}{n} =o(\log X)? \]

来源字段：http://aimpl.org/sarnakconjecture/3/

01358 | OWR-15956-007

### Partition Regularity of Pythagorean Triples

O / open | Oberwolfach 报告问题汇编

Is the Pythagorean equation $x^2+y^2=z^2$ partition regular on $\mathbb{N}$, meaning that every finite coloring of $\mathbb{N}$ has a nontrivial monochromatic solution?

来源字段：https://doi.org/10.4171/owr/2017/51

01359 | AIM-ANALYTIC_NUMBER_THEORY-0108

### PDF-splice repair and a gap-scale reduction for two independent prime questions

O / open | AIM 研讨会问题汇编

**7. Small gaps between primes in thin sequences.** A. Kantorovich asked whether it is possible to prove the existence of small gaps between primes from the sequence \[ \{p\leq x: p=\lfloor n\log n\rfloor\text{ for some }n\in\mathbb N\}. \] This is a thin set of primes: \[ \#\{p\leq x: p=\lfloor n\log n\rfloor\text{ for some }n\in\mathbb N\} \sim \frac{x}{(\log x)^2}. \] Several participants suggested the analogous question for Piatetski--Shapiro primes.

**8. Mersenne composites.** A. Granville asked whether it is possible to use the GPY method, assuming the Elliott--Halberstam conjecture if necessary, to prove that there are infinitely many composites of the form \[ 2^p-1. \] C. Elsholtz noted possible related work of M. R. Murty under Artin's primitive-root conjecture.

来源字段：https://aimath.org/WWN/primegaps/WorkshopProblems.pdf

01360 | OWR-16941-002

### Perfectness of Grothendieck's Component-Group Pairing

O / open | Oberwolfach 报告问题汇编

Let $K$ be a complete discrete valuation field with perfect residue field of characteristic $p>0$, and let $A$ and $B$ be dual abelian varieties with Néron special fibers $A_x$ and $B_x$. Is the canonical pairing $$ \pi_0(A_x)\times\pi_0(B_x)\longrightarrow\mathbb Q/\mathbb Z $$ perfect?

来源字段：https://doi.org/10.4171/owr/2019/29

01361 | OWR-1732-002

### Period Formula for Special Orthogonal Automorphic Representations

P / partially_solved | Oberwolfach 报告问题汇编

Let $k$ be a global field of characteristic different from $2$, let $(V_0,Q_0)\subset(V_1,Q_1)$ be quadratic spaces of ranks $n$ and $n+1$, and set $G_i=\operatorname{SO}_{Q_i}$. For tempered cuspidal automorphic representations $\pi_i$ of $G_i$ and cusp forms $\varphi_i\in\pi_i$, does there exist an integer $\beta$ such that $\frac{|\langle\varphi_1|_{G_0},\varphi_0\rangle|^2}{\langle\varphi_1,\varphi_1\rangle\langle\varphi_0,\varphi_0\rangle}=2^\beta C_0\Delta_{G_1}^S P_{\pi_1,\pi_0}^S(1/2)\prod_{v\in S}\frac{I(\varphi_{1,v},\varphi_{0,v})}{\langle\varphi_{1,v},\varphi_{1,v}\rangle_v\langle\varphi_{0,v},\varphi_{0,v}\rangle_v}$, with the component-group normalization predicted for $\pi_1$ and $\pi_0$?

来源字段：https://doi.org/10.4171/owr/2007/49

01362 | OWR-12004-001

### Perverse Intermediate Extensions at Parahoric Level

P / partially_solved | Oberwolfach 报告问题汇编

If $\pi_v$ is an irreducible cuspidal representation of $GL_g(F_v)$ whose mod-$\ell$ reduction is supercuspidal, does the conjectured isomorphism

$${}^pj_{!*}\mathcal F_{\overline{\mathbb F}_\ell}(\pi_v,t)[d-tg]\simeq{}^{p+}j_{!*}\mathcal F_{\overline{\mathbb F}_\ell}(\pi_v,t)[d-tg]$$

hold for every $1\leq t\leq\lfloor d/g\rfloor$?

来源字段：https://doi.org/10.4171/owr/2012/32

01363 | NUM-008

### Pi Normality

O / open | 独立 / 未分配集合

Is π a normal number (all digits equally frequent in all bases)?

背景链接：https://doi.org/10.1080/10586458.2002.10504475

01364 | NT-023

### Pillai's Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Pillai's conjecture [AMR-093-0110]

For each positive integer $k$, does the equation $|2^m - 3^n| = k$ have only finitely many solutions in positive integers $m$ and $n$?

背景链接：https://doi.org/10.1007/BF01388636

01365 | AMR-093-0076

### Piltz divisor problem

P / partially_solved | AMR 问题清单汇编

Piltz divisor problem on bounding $\Delta_k(x) = D_k(x) - xP_k(\log(x))$

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01366 | OWR-8415342-028

### Plancherel Densities on Yu-Induced Tempered Components

O / open | Oberwolfach 报告问题汇编

Verify the Hiraga-Ichino-Ikeda Plancherel-measure conjecture on the nondiscrete tempered-dual components arising by parabolic induction from nonsingular Yu supercuspidal representations.

来源字段：https://doi.org/10.4171/owr/2021/39

01367 | OWR-4804-008

### Plancherel Support of Spherical Homogeneous Spaces

O / open | Oberwolfach 报告问题汇编

For a spherical homogeneous space $X=H\backslash G$, which irreducible representations $\pi$ occur in $L^2(X)$ according to the Plancherel decomposition, equivalently in its Fell-topology support?

来源字段：https://doi.org/10.4171/owr/2011/14

01368 | OWR-1703872-011

### Plancherel Volumes of Rankin-Selberg Metric Balls

O / open | Oberwolfach 报告问题汇编

For the local Rankin-Selberg pseudometric on irreducible supercuspidal representations of $\operatorname{GL}_n(F)$, what is the Plancherel measure of the ball $B_r(\pi)$ of radius $r$ centered at $\pi$?

来源字段：https://doi.org/10.4171/owr/2020/26

01369 | OWR-13750341-011

### Poincaré Duality for Pro-Étale Cohomology of Stein Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a smooth geometrically irreducible Stein variety over $K$. Are $H^i_{\mathrm{pro\acute et}}(X,\mathbb{Q}_p(j))$ nuclear Fréchet spaces and $H^i_{\mathrm{pro\acute et},c}(X,\mathbb{Q}_p(j))$ spaces of compact type? Does the cup-product pairing give perfect duality between ordinary and compactly supported pro-étale cohomology?

来源字段：https://doi.org/10.4171/owr/2023/28

01370 | OWR-1703872-004

### Pointwise Asymptotics of Special Quaternionic Eigenfunctions

O / open | Oberwolfach 报告问题汇编

For $y\in D^\times(\mathbb A_\mathbb Q)$ and every special eigenfunction $f$, is

$$f(y)=(1+o(1))\int_{D^\times(\mathbb Q)\backslash D^\times(\mathbb A_\mathbb Q)}f(x)\,d\mu(x)$$?

来源字段：https://doi.org/10.4171/owr/2020/26

01371 | OWR-1703872-007

### Pointwise Function-Field Equidistribution from Period Integrals

O / open | Oberwolfach 报告问题汇编

Fix a quaternion algebra $D$ over $\mathbb F_q(t)$. Let $F/\mathbb F_q(t)$ vary over quadratic extensions of genus $g$, let $K\subset D^\times(\mathbb A_F)$ be compact, and let $m$ be the index of $K\cap D^\times(\mathbb A_{\mathbb F_q(t)})$ in a maximal compact subgroup. For a normalized Hecke eigenfunction $f$ on

$$D^\times(F)\backslash D^\times(\mathbb A_F)/(KZ(\mathbb A_F)),$$

do constants $C,\delta>0$, depending only on $q$ and $D$, exist such that for every $y\in D^\times(\mathbb A_{\mathbb F_q(t)})$,

$$f(y)=\int_{D^\times(\mathbb F_q(t))\backslash D^\times(\mathbb A_{\mathbb F_q(t)})}f(x)\,d\mu(x)+O\!\left(m^Cq^{(1-\delta)g/2}\right),$$

with the implicit constant independent of $F$ and $K$?

来源字段：https://doi.org/10.4171/owr/2020/26

01372 | OWR-1590-002

### Poisson Statistics for S5 Number-Field Extensions

O / open | Oberwolfach 报告问题汇编

Let $K$ be the function field of a curve over $\mathbb F_q$, and let $n(D)$ count $S_5$-extensions of $K$ with squarefree discriminant $D$. Does $n(D)$ follow a Poisson distribution with mean $1$, and can this prediction be tested experimentally? Does the analogous count over $\mathbb Q$ follow a Poisson distribution with expected mean $\frac{13}{120}\zeta(2)^{-1}$?

来源字段：https://doi.org/10.4171/owr/2007/34

01373 | OWR-12725-006

### Polignac Conjecture for Consecutive Prime Gaps

O / open | Oberwolfach 报告问题汇编

Does every positive even integer occur infinitely often as a gap $p_{n+1}-p_n$ between consecutive primes?

来源字段：https://doi.org/10.4171/owr/2013/51

01374 | NT-022

### Polignac's Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Polignac's Conjecture [OPG-37289]

For every even number $n$, are there infinitely many pairs of consecutive primes differing by $n$?

背景链接：https://en.wikipedia.org/wiki/Polignac%27s_conjecture

01375 | AMR-093-0177

### Pollock's octahedral-number conjecture

O / open | AMR 问题清单汇编

Is every positive integer expressible as a sum of at most seven octahedral numbers?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01376 | AMR-093-0041

### Pollock's tetrahedral-number conjecture

O / open | AMR 问题清单汇编

Is every positive integer expressible as a sum of at most five tetrahedral numbers?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01377 | OWR-1542-013

### Polygonal Period Relations and Formal Multiple Zeta Values

O / open | Oberwolfach 报告问题汇编

Let $FC$ be the formal algebra generated by polygon pairs for periods on $M_{0,n}$, modulo symmetric-group invariance, one-point shuffle relations, and product-map relations. Is $FC$ isomorphic to the formal multiple-zeta-value algebra; equivalently, do these three families generate all relations among the corresponding periods?

来源字段：https://doi.org/10.4171/owr/2007/26

01378 | OWR-12725-009

### Polynomial Asymptotics for Shifted Divisor Correlations

O / open | Oberwolfach 报告问题汇编

For $r\geq 3$ and fixed $h\neq 0$, is

$$D_r(X;h)=\sum_{n\leq X}d_r(n)d_r(n+h)\sim X P_{2(r-1)}(\log X;h),$$

where $P_{2(r-1)}(u;h)$ is a polynomial of degree $2(r-1)$ in $u$?

来源字段：https://doi.org/10.4171/owr/2013/51

01379 | OWR-12725-010

### Polynomial Images of Subgroups and Function-Field Curve Points

O / open | Oberwolfach 报告问题汇编

For polynomials $f,g\in\mathbb F_p[X]$ and multiplicative subgroups $G,H\subseteq\mathbb F_p^*$, estimate $\#(f(G)\cap g(H))$. Separately, let $K=\mathbb F_q(T)$ and let $F\in K[X,Y]$ be absolutely irreducible of degree $d$. Bound the number of solutions to $F(x,y)=0$ with $x,y\in\mathbb F_q[T]$ of degree at most $n$ as $n\to\infty$.

来源字段：https://doi.org/10.4171/owr/2013/51

01380 | AIM-ALGEBRAIC_NUMBER_THEORY-0057

### Polynomial Ramanujan sums and finite von Mangoldt expansions

S / solved | AIM 研讨会问题汇编

Define the polynomial/function-field analogue of \(c_q(x)\). Determine its basic identities and explain how Ramanujan-sum formulas for the von Mangoldt function and other arithmetic functions appear over a function field.

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

01381 | OWR-12014-012

### Polynomial Separation at Roots of Unity

O / open | Oberwolfach 报告问题汇编

For every $f\in\mathbb Z[t_1,\ldots,t_n]$, does there exist $E>0$ such that for all roots of unity $z_1,\ldots,z_n$ of order at most $d$, either $f(z_1,\ldots,z_n)=0$ or $|f(z_1,\ldots,z_n)|>d^{-E}$?

来源字段：https://doi.org/10.4171/owr/2012/42

01382 | OWR-13853-006

### Polynomial Time Zeta Functions for Hyperelliptic Curves

O / open | Oberwolfach 报告问题汇编

Can Zp(T), the zeta function of a genus-g hyperelliptic curve over a finite field of characteristic p, be computed in time polynomial in both g and log p?

来源字段：https://doi.org/10.4171/owr/2015/32

01383 | OWR-782-007

### Positivity of Saito's Eta Products

S / solved | Oberwolfach 报告问题汇编

别名：Duplicate Saito Positivity Conjecture [OWR-782-008]

For every natural number $h$, does the eta product $\eta(h\tau)^{\varphi(h)}\prod_{d\mid h}\eta(d\tau)^{-\mu(d)}$ have only nonnegative Fourier coefficients? In particular, prove or disprove this conjecture when $h$ is divisible by at least two distinct primes.

来源字段：https://doi.org/10.4171/owr/2005/01

01384 | OWR-15583-005

### Power-Rate Arithmetic Quantum Unique Ergodicity

O / open | Oberwolfach 报告问题汇编

For Hecke-Maass cusp forms $\phi$ on the modular surface with Laplace eigenvalue $\lambda\to\infty$, prove quantitative equidistribution of the probability measures $\mu_\phi$ toward hyperbolic area measure $\mu$ at a power rate $O(\lambda^{-\delta})$ for some $\delta>0$.

来源字段：https://doi.org/10.4171/owr/2017/40

01385 | OWR-2043-006

### Prime-Entry Matrices of Fixed Determinant

O / open | Oberwolfach 报告问题汇编

Let $B_T$ be the norm ball in the variety of $2\times2$ real matrices of determinant $2$. Are there infinitely many integral matrices in $B_T$ whose four entries are prime? More quantitatively, is their number asymptotic to $CT^2/(\log T)^4$ for some $C>0$ as $T\to\infty$?

来源字段：https://doi.org/10.4171/owr/2008/32

01386 | AIM-ANALYTIC_NUMBER_THEORY-0003

### Prime-level direct delta solution and an exponent-transfer lemma

P / partially_solved | AIM 研讨会问题汇编

**Recovered standard reading.** Let \(f\) be a primitive holomorphic or Maass \(GL(2)\) newform of level \(q\to\infty\), with fixed archimedean parameter. Prove, by a direct delta-symbol argument rather than a spectral moment average, that

\[ L(1/2,f)\ll_{\varepsilon,f_\infty} C(f)^{1/4-\delta+\varepsilon} \]

for some \(\delta>0\), where \(C(f)\) is the analytic conductor.

来源字段：http://aimpl.org/deltasubconvex2/1/

01387 | AIM-ANALYTIC_NUMBER_THEORY-0038

### Prime-power obstructions and divisor-bounded positive results

P / partially_solved | AIM 研讨会问题汇编

Can anything be done for correlations of unbounded functions? What is the behavior in almost all intervals of \[ \sum_{x\leq n\leq x+H}g(n), \] with \(g\) an unbounded multiplicative function such that \(g(p)=O(1)\), and \(H=\log^A x\) for \(A>0\) fixed?

来源字段：http://aimpl.org/sarnakconjecture/6/

01388 | AIM-ALGEBRAIC_NUMBER_THEORY-0071

### Prime-specific failures and determinant certificates for integral Tate saturation

P / partially_solved | AIM 研讨会问题汇编

In cases where we know all Tate classes are in the image of \(C^i\otimes\mathbb Q_\ell\), what can we say about the image of \(C^i\otimes\mathbb Z_\ell\)? For instance, for which primes \(\ell\) can we have non-surjectivity?

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

01389 | GREEN-037

### Primes with p-2 Having Odd Omega

O / open | Ben Green 问题

Do there exist infinitely many primes $p$ for which $p-2$ has an odd number of prime factors (counting multiplicity)?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

01390 | AIM-ANALYTIC_NUMBER_THEORY-0118

### Primitive balanced proper triples of sums of two squares

P / partially_solved | AIM 研讨会问题汇编

**19. Triples in prime-like sequences.**

• C. Elsholtz asked whether it is possible to find infinitely many triples \(s_n,s_{n+1},s_{n+2}\), with \(s_{n+2}-s_n\) bounded, in sequences that are commonly considered to be similar to but more manageable than the primes (e.g., integers that are sums of two squares).

• K. B. Ford noted that for sums of two squares the problem is trivial, because of triples of the form \(n^2,n^2+1,n^2+4\), for example.

• J. B. Friedlander suggested that such trivialities can be avoided by requiring that the two squares be of comparable sizes. S. W. Graham and T. D. Wooley proposed another option: produce “many” gaps.

• R. C. Vaughan proposed considering the more general question of triples of values of norm-forms.

来源字段：https://aimath.org/WWN/primegaps/WorkshopProblems.pdf

01391 | AIM-ANALYTIC_NUMBER_THEORY-0007

### Primitive character moments as centered parity variance

P / partially_solved | AIM 研讨会问题汇编

**Subconvexity for \(\mathrm{GL}(3)\times\mathrm{GL}(1)\) via averaging over a family.** Let \(\pi\) be a fixed Hecke--Maass cusp form for \(\mathrm{SL}_3(\mathbb Z)\), and \(\chi\) be a character of level \(M\). Munshi used Petersson trace formula as a delta method to obtain a subconvexity bound for \(L(1/2,\pi\times\chi)\) as \(M\) varies. Can we obtain a subconvexity bound by averaging over a family, e.g. by bounding \[ \sum_{\chi\bmod M}|L(1/2,\chi)|^2? \]

来源字段：http://aimpl.org/deltasubconvex2/1/

01392 | OPG-37318

### Primitive pythagorean n-tuple tree

P / partially_solved | Open Problem Garden

Conjecture Find linear transformation construction of primitive pythagorean n-tuple tree!

背景链接：http://www.openproblemgarden.org/op/primitive_pythagorean_n_tuple_tree

01393 | AIM-ANALYTIC_NUMBER_THEORY-0070

### Primitive reciprocal resonance at the Type-II T-squared threshold

P / partially_solved | AIM 研讨会问题汇编

**Problem 8.8.** Extend the work of Goldston and Gonek on mean value theorems for long Dirichlet polynomials by using Type II sums.

来源字段：http://aimpl.org/zetamoments/8/

01394 | OWR-1590-006

### Primitive-Root Density Formulas for Rank-One Tori

O / open | Oberwolfach 报告问题汇编

Let $K$ be a number field, let $T$ be a rank-one torus over $K$, and let $A\in T(K)$ be nontorsion. For each prime $p$, put $B_p=\langle T(K),\sqrt[p]{A},Z_p\rangle\subset T(\overline K)$, where $Z_p$ is a nontrivial $p$-torsion point, let $B_\infty$ be generated by all $B_p$, let $E$ be the finite abelian cokernel of $\operatorname{Gal}(K(B_\infty)/K)\to\operatorname{Aut}_{T(K)}(B_\infty)$, and put $A_p=\operatorname{Aut}_{T(K)}(B_p)$. Is the density of primes $\mathfrak q$ of $K$ for which the reduction of $A$ generates $T(\mathbb F_{\mathfrak q})$ equal to $C_{T,A}\prod_p(1-1/\#A_p)$, where $C_{T,A}=\sum_{\chi\in E^\vee}\prod_{p:\,\chi(A_p)\neq1}-1/(\#A_p-1)$?

来源字段：https://doi.org/10.4171/owr/2007/34

01395 | OWR-14297740-006

### Pro-l Congruence Subgroup Problem for Mapping Class Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $\Gamma_{g,r}$ be the mapping class group of a surface of genus $g$ with $r$ punctures, and for a prime $\ell$ let $\rho^{\ell\text{-csp}}_{g,r}$ denote the pro-$\ell$ congruence-subgroup representation induced by the action of $\Gamma_{g,r}$ on the pro-$\ell$ completion of the surface fundamental group. For which $(g, r, \ell)$ is $\rho^{\ell\text{-csp}}_{g,r}$ injective?

来源字段：https://doi.org/10.4171/owr/2023/42

01396 | GREEN-035

### Products of Primes Modulo p

P / partially_solved | Ben Green 问题

Let $p$ be a large prime, and let $A$ be the set of all primes less than $p$. Is every $x \in \{1, \dots, p-1\}$ congruent to some product $a_1a_2$ modulo $p$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

01397 | OWR-14297740-005

### Profinite Congruence Subgroup Problem for Mapping Class Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $\Pi_{g,r}$ be the fundamental group of a hyperbolic topological surface of genus $g$ with $r$ punctures, let $\Gamma_{g,r}$ be its mapping class group, and for a class of finite groups $P$ let $\rho^P_{g,r} : \widehat{\Gamma}_{g,r} \to \operatorname{Out}(\Pi_{g,r}^P)$ be the universal pro-$P$ outer monodromy representation (the pro-$P$ congruence subgroup problem for mapping class groups). Is $\rho^P_{g,r}$ injective?

来源字段：https://doi.org/10.4171/owr/2023/42

01398 | OWR-12493-009

### Pythagoras Numbers of Real Function Fields

O / open | Oberwolfach 报告问题汇编

For the function field $F$ of a variety over a real closed field, determine its Pythagoras number $p(F)$, or obtain good lower bounds, where $p(F)$ is the least $n$ such that every sum of squares in $F$ is a sum of $n$ squares.

来源字段：https://doi.org/10.4171/owr/2013/30

01399 | AMR-093-0132

### Quadratic bound in Linnik's least-prime problem

P / partially_solved | AMR 问题清单汇编

For coprime integers $1\leq a<d$, is the least prime $p(a,d)$ congruent to $a\pmod d$ always less than $d^2$?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01400 | AIM-ALGEBRAIC_NUMBER_THEORY-0008

### Quadratic density and potential gaps on symmetric squares of curves

P / partially_solved | AIM 研讨会问题汇编

Let $C$ be a curve of genus $\ge 2$ over a number field. Must $2\in\delta(\operatorname{Sym}^2_C/k)$ or in $\mathcal P(\operatorname{Sym}^2_C/k)$? What do these sets look like?

来源字段：http://aimpl.org/degreedsurface/1/

01401 | AIM-ALGEBRAIC_NUMBER_THEORY-0007

### Quadratic density, diagonal quotients, and correspondence components

P / partially_solved | AIM 研讨会问题汇编

Let \(C,D\) be hyperelliptic curves of genus \(\ge2\) over a number field \(k\). Must \(2\in\delta(C\times D/k)\)? Must it be in \(\mathcal P(C\times D/k)\)?

More generally, let \(Z\subset C\times D\) be the closure of the degree \(d\) points. When is there a component of \(Z\) which dominates both \(C\) and \(D\)?

来源字段：http://aimpl.org/degreedsurface/1/

01402 | AIM-ALGEBRAIC_NUMBER_THEORY-0064

### Quadratic descent and character-valued Tate multiplicities

P / partially_solved | AIM 研讨会问题汇编

**3. (J. Getz)** Let X be a Hilbert modular variety. At some level (say, that of cohomology), these look like a product of modular curves. Under what circumstances does X have Tate classes not coming from modular subvarieties? Thinking of these classes as submotives defined by automorphic forms (and thus as Galois representations), do these exotic classes come from CM automorphic forms as in Question 2?

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

01403 | AIM-ANALYTIC_NUMBER_THEORY-0055

### Quadratic moments, Poisson squares, and swap terms

P / partially_solved | AIM 研讨会问题汇编

Using the ideas of Conrey and Keating, develop a heuristic for conjecturing asymptotic formulas for high moments of quadratic Dirichlet \(L\)-functions. What new terms contribute to the main term when \(k\geq 3\)?

来源字段：http://aimpl.org/zetamoments/5/

01404 | AIM-ANALYTIC_NUMBER_THEORY-0025

### Quantitative discrepancy for completely multiplicative signs

S / solved | AIM 研讨会问题汇编

Can we get a reasonable omega result for \[ \left|\sum_{n\leq X} f(n)\right|, \] where \(f:\mathbb N\to\{\pm1\}\) is totally multiplicative? Anything better than \(\log\log\log\log\log X\) would be interesting.

来源字段：http://aimpl.org/sarnakconjecture/1/

01405 | OWR-14603-015

### Quantitative Entropy of Rational Multiplicative Orbits

O / open | Oberwolfach 报告问题汇编

For multiplicatively independent integers $a,b$, do there exist $N,M,K$ depending only on $a,b$ such that, whenever $(s,ab)=(s,r)=1$, the set $A=\{a^n b^k r/s\bmod 1:n,k\leq M\log s\}$ satisfies $\log N(A,(\log s)^{-N})\geq K\log\log s$?

来源字段：https://doi.org/10.4171/owr/2016/21

01406 | OWR-16158-008

### Quantitative Thin-Set Descriptions of Reducible Specializations

O / open | Oberwolfach 报告问题汇编

Let $f(t,x)\in\mathbb Q[t,x]$ be irreducible, and let $\operatorname{Red}_f$ be the set of parameters $t$ for which $f(t,x)$ is reducible. In a thin-set presentation of $\operatorname{Red}_f$ by maps $\varphi_i$, how many maps are needed, what degrees can they have, and how do these quantities control the growth of $\operatorname{Red}_f\cap\mathbb Z$?

来源字段：https://doi.org/10.4171/owr/2018/17

01407 | AIM-ANALYTIC_NUMBER_THEORY-0032

### Quantitative toral transference for local polynomial uniformity

P / partially_solved | AIM 研讨会问题汇编

Can we extend the local Fourier uniformity conjecture to nilsequences? Warm-up: do it for polynomial phases.

来源字段：http://aimpl.org/sarnakconjecture/4/

01408 | OPG-791

### Quartic rationally derived polynomials

P / partially_solved | Open Problem Garden

Call a polynomial $p \in {\mathbb Q}[x]$ rationally derived if all roots of $p$ and the nonzero derivatives of $p$ are rational.

Conjecture There does not exist a quartic rationally derived polynomial $p \in {\mathbb Q}[x]$ with four distinct roots.

背景链接：http://www.openproblemgarden.org/op/quartic_rationally_derived_polynomials

01409 | NT-043

### Quasiperfect Numbers

P / partially_solved | 独立 / 未分配集合

别名：Quasiperfect Numbers [NUM-004]

Do quasiperfect numbers exist?

背景链接：https://arxiv.org/abs/2608.02066

01410 | AIM-ANALYTIC_NUMBER_THEORY-0097

### Radial Laplace obstruction and an exact C-law for Stokes jumps

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.2.** Consider the Laplace transform of the q-hypergeometric series above. Note that the Mellin transform will not converge. Look at the properties, in particular the “jumps;” what can we say about the dependence on \(A,B,\) and \(C\)?

The exact AIM record asks:

\[ f_{A,B,C}(\tau)=F_{A,B,C}(q) :=\sum_{n=(n_1,\ldots,n_r)^t\in\mathbb Z_{\geq0}^r} \frac{q^{Q(n)}}{(q)_{n_1}\cdots(q)_{n_r}}, \qquad Q(n)=\tfrac12n^tAn+B^tn+C, \]

where \(q=e^{2\pi i\tau}\), \((q)_m=\prod_{j=1}^m(1-q^j)\), and \((q)_0=1\).

The PDF does **not** specify a transform variable, a kernel, a contour, a subtraction of the dominant cusp exponential, or what “jumps” means. At least three readings are therefore possible:

1. the ordinary transform of the radial function \(R(t)=F(e^{-t})\); 2. a transform after the reciprocal change \(x=1/t\) and a subtraction or tail cutoff; 3. the Laplace step in Borel--Laplace summation of the formal cusp expansion, with “jumps” meaning lateral Stokes discontinuities.

The third reading is historically plausible: Costin--Garoufalidis used exactly this language for the related Kontsevich--Zagier series in work published immediately after the 2010 workshop. It is nevertheless not stated in the AIM PDF. The results below therefore separate the literal reading (an unconditional obstruction) from the Borel--Laplace reading (an exact parameter law conditional only on the relevant lateral sums existing).

来源字段：https://aimath.org/WWN/mockmodular/mockmodular.pdf

01411 | OWR-13750333-011

### Ramanujan Bounds for Siegel Cusp-Form Coefficients

P / partially_solved | Oberwolfach 报告问题汇编

Let $F$ be a holomorphic Siegel Hecke cusp form of weight $k$ for $\operatorname{Sp}_4(\mathbb Z)$. Prove that some fixed $\delta>0$ satisfies $a(F,S)\ll_F\det(S)^{(k-1)/2-\delta}$ for every fundamental-discriminant form $S$, and ultimately prove $a(F,S)\ll_{F,\epsilon}\det(S)^{k/2-3/4+\epsilon}$.

来源字段：https://doi.org/10.4171/owr/2023/22

01412 | OWR-1060-008

### Ramanujan Conjecture for Cuspidal Representations

O / open | Oberwolfach 报告问题汇编

Every irreducible unitary representation occurring in $L^2_{\mathrm{cusp}}(\Gamma(N)\backslash SL(2,\mathbb{R}))$ should be tempered.

来源字段：https://doi.org/10.4171/owr/2005/39

01413 | OWR-14299909-008

### Ramanujan Tau Nonvanishing from Double Eisenstein Series

O / open | Oberwolfach 报告问题汇编

Use double-Eisenstein-series representations of the discriminant modular form to prove that Ramanujan’s $\tau(n)$ never vanishes, or at least that $\tau(p)=0$ for only finitely many primes $p$.

来源字段：https://doi.org/10.4171/owr/2026/7

01414 | AIM-ALGEBRAIC_NUMBER_THEORY-0036

### Ramification conditioning: preserved marks, induced dependence, and fixed-conductor limits

P / partially_solved | AIM 研讨会问题汇编

**Question 6 (Alina Bucur, Kiran Kedlaya).** Are principles (A) and (B) easier/harder/true after conditioning on ramification information?

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

01415 | AIM-ALGEBRAIC_NUMBER_THEORY-0117

## Random hypersurfaces, local conditioning, and height transfer

P / partially_solved | AIM 研讨会问题汇编

In cleaned mathematical notation, the record asks three related but distinct questions.

1. For fixed degree $d$, projective dimension $n$, and number field $K$, order all degree-$d$ hypersurfaces in $\mathbf P^n_K$ by coefficient height. What proportion has a $K$-point, and what proportion of the everywhere locally soluble members satisfies the Hasse principle? 2. For the diagonal family

\[ a_1x_1^d+\cdots+a_sx_s^d=0,\qquad |a_i|\leq A, \]

determine the asymptotic of

\[ A^{-s}\#\{\mathbf a:\ |a_i|\leq A, \ a_1x_1^d+\cdots+a_sx_s^d=0\text{ satisfies HP}\}. \]

Wooley and Venkatesh tentatively suggested a product of local densities, at least for $s\geq 3d+1$. 3. For plane cubics, relate such proportions to $3$-Selmer groups of elliptic curves. Stoll predicted that, conditional on local solubility, the rational-point proportion should be $0$, and Mazur asked for a decay exponent.

The official source confirms all three clauses and the threshold $s\geq 3d+1$ [AIM]. They must not be conflated: the diagonal locus is a zero-density subfamily of the full coefficient space, while plane cubics are a borderline genus-one family rather than a Fano family.

There is also a semantic trap in the phrase “satisfies HP.” For a single variety with no adelic point, the implication “local points imply a global point” is vacuously true. That cannot be the intended count, since the displayed local product measures locally soluble members. Throughout this report the quantitative HP assertion means

\[ \#\{\text{globally soluble members}\} =\#\{\text{everywhere locally soluble members}\}+o(\#\Omega_H), \]

or, conditionally, that the locally soluble failure proportion tends to zero. Nonlocally soluble members are not reclassified as globally soluble by vacuous implication.

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01416 | OWR-14299578-004

## Rank Distribution in Quadratic-Twist Families

O / open | Oberwolfach 报告问题汇编

Fix an elliptic curve $E$ over a global field $K$, a norm $\operatorname{Nm}$ on $K$, and $j\geq0$. If $\mathcal{F}(X)$ is the set of quadratic extensions $K(\sqrt{d})/K$ with squarefree $d$ and $\operatorname{Nm}(d)\leq X$, determine the limit as $X\to\infty$ of the proportion for which $\operatorname{rk} E(K(\sqrt{d}))-\operatorname{rk}E(K)=j$.

来源字段：https://doi.org/10.4171/owr/2025/52

01417 | OWR-1053-007

## Rank Growth over Cubic Radical Extensions

O / open | Oberwolfach 报告问题汇编

Given an elliptic curve $E$ over $\mathbb{Q}$, for which cube-free integers $m$ does $E$ acquire new rational points of infinite order (i.e., strictly larger Mordell--Weil rank $\operatorname{rank} E(\mathbb{Q}(\sqrt[3]{m})) > \operatorname{rank} E(\mathbb{Q})$) over the cubic radical extension $\mathbb{Q}(\sqrt[3]{m})$?

来源字段：https://doi.org/10.4171/owr/2005/32

01418 | OWR-16171-008

## Rank-One Equivalences for Elliptic Curves

P / partially_solved | Oberwolfach 报告问题汇编

For an elliptic curve $E/\mathbb{Q}$ and $r\in\{0,1\}$, the following are conjectured equivalent: $\operatorname{rank}E(\mathbb{Q})=r$ and the Tate-Shafarevich group $\operatorname{Sha}(E/\mathbb{Q})$ is finite; $\operatorname{corank}_{\mathbb{Z}_p}\operatorname{Sel}_{p^\infty}(E/\mathbb{Q})=r$; and $\operatorname{ord}_{s=1}L(s,E/\mathbb{Q})=r$.

来源字段：https://doi.org/10.4171/owr/2018/30

01419 | AIM-ALGEBRAIC_NUMBER_THEORY-0103

## Rational descent obstructions for geometric elliptic powers in hyperelliptic Jacobians

P / partially_solved | AIM 研讨会问题汇编

**Question 18.** (a) Find a separable polynomial \(g(t)\in \mathbf Q[t]\) such that the Jacobian of the hyperelliptic curve \[ s^2=g(t) \] is isogenous over \(\mathbf Q\) to \(E^r\times B\), with \(E\) an elliptic curve, and either \(r\geq 4\), or \(r=3\) and \(\dim B\leq 1\). (Silverberg)

(b) Give an example of a map \[ \mathbf P^1\longrightarrow E^4/\{\pm1\}, \] where \(\{\pm1\}\) acts diagonally, whose image does not lie in \(H/\{\pm1\}\) for any proper subgroup \(H\subsetneq E^4\). (Ellenberg)

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01420 | OWR-14299909-006

## Rational Dimensions of Critical Modular $L$-Values

O / open | Oberwolfach 报告问题汇编

Let $f\in S_k(\operatorname{SL}_2(\mathbb Z))$ be a normalized Hecke eigenform. If $L_f^+$ and $L_f^-$ are the $\mathbb Q$-spans of its odd and even critical values $\Lambda(f;s)$, respectively, prove that $\dim_{\mathbb Q}L_f^\pm=\dim_{\mathbb C}S_k(\operatorname{SL}_2(\mathbb Z))$.

来源字段：https://doi.org/10.4171/owr/2026/7

01421 | AIM-ALGEBRAIC_NUMBER_THEORY-0101

### Rational lines on cubic hypersurfaces: a three-layer local obstruction and an explicit pointed example

P / partially_solved | AIM 研讨会问题汇编

**Question 16 (Wooley).** Solve the local-global solubility problem for finding lines on a cubic hypersurface.

(a) Find an example of a cubic hypersurface over \(\mathbb Q_p\) (in as many variables as possible) with no rational line.

(b) Find an example of a cubic hypersurface over \(\mathbb Q\) (in as many variables as possible) with no rational line.

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01422 | OWR-13750336-010

### Rational Point Counts near Hypersurfaces

P / partially_solved | Oberwolfach 报告问题汇编

For a compact $C^1$ hypersurface $M\subset\mathbb R^n$, characterize the local obstructions to the uniform estimate $N_M(\delta,Q)\asymp\delta Q^n$ for $Q\geq1$ and $Q^{\varepsilon-1}<\delta<1/2$, where $N_M(\delta,Q)$ counts pairs $(q,p)\in\mathbb Z\times\mathbb Z^n$ with $Q\leq q<2Q$ and $\operatorname{dist}(p/q,M)\leq\delta/q$.

来源字段：https://doi.org/10.4171/owr/2023/25

01423 | OWR-14299577-021

### Rational Points on a Singular Twisted Cayley Cubic

O / open | Oberwolfach 报告问题汇编

Let $N(B)$ count primitive integer solutions $(x,y,z,w)\in[-B,B]^4$ with $x\neq0$ to $x^3+2xyz+xw^2-y^2w+wz^2=0$. Prove an asymptotic $N(B)\sim(c_1+c_2)B(\log B)^3$ with explicit positive constants, where $c_1$ is the predicted product of local densities and $c_2$ accounts for the Zariski-dense thin set.

来源字段：https://doi.org/10.4171/owr/2025/51

01424 | OWR-12725-001

### Rational Points on Cartan Modular Curves

P / partially_solved | Oberwolfach 报告问题汇编

Let $X_G$ be the modular curve of level $N$ attached to a subgroup $G\subseteq\operatorname{GL}_2(\mathbb Z/N\mathbb Z)$ with full determinant. Describe $X_G(\mathbb Q)$, especially when $N=p$ is prime and $G$ is a Borel subgroup or the normalizer of a split or nonsplit Cartan subgroup.

来源字段：https://doi.org/10.4171/owr/2013/51

01425 | OWR-8415337-007

### Rational Points on Nonsplit Cartan Modular Curves

O / open | Oberwolfach 报告问题汇编

Determine the rational points $X(\Gamma_{\mathrm{ns}}^+(p))(\mathbb Q)$ on the modular curve associated with the normalizer of a nonsplit Cartan subgroup.

来源字段：https://doi.org/10.4171/owr/2021/34

01426 | OWR-12583-002

### Rational Points on Prime-Level Modular Curves

P / partially_solved | Oberwolfach 报告问题汇编

For every prime $p>37$, are the rational points on $X_0^+(p)$, $X_{\mathrm{sp}}^+(p)$, and $X_{\mathrm{ns}}^+(p)$ only cusps and points arising from elliptic curves with complex multiplication?

来源字段：https://doi.org/10.4171/owr/2013/35

01427 | OWR-13750333-002

### Rational-Point Counting Near Immersed Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Determine the asymptotic behavior of the rational-point counting function $N(M;Q,\delta)$ as $Q\to\infty$ and $\delta\to 0$ for compact immersed manifolds $M$.

来源字段：https://doi.org/10.4171/owr/2023/22

01428 | OWR-4990381-005

### Rationality of Quasi-Monomial Fixed Fields

O / open | Oberwolfach 报告问题汇编

Let $K/k$ be a finite field extension and let $G$ act on $K(x_1,\ldots,x_n)$ by quasi-monomial $k$-automorphisms. Under what conditions is $K(x_1,\ldots,x_n)^G$ rational over $k$?

来源字段：https://doi.org/10.4171/owr/2021/12

01429 | AIM-ALGEBRAIC_NUMBER_THEORY-0025

### Real approximation and odd-kernel reduction for finite stabilizers

P / partially_solved | AIM 研讨会问题汇编

**Problem 5.** Harari: The following question is due to Borovoi: Consider weak approximation for $X=\operatorname{SL}_n/G$ over $\mathbf Q$, where $G$ is a finite group scheme that is not necessarily constant. For example, is $X(\mathbf Q)$ dense in $X(\mathbf R)$? If $G$ is constant, this is known to be true.

A variant is the following. Given $X=\operatorname{SL}_n/G$ with a constant finite group scheme $G$ over a number field $K$ with $r\geq 2$ real places $v_1,\ldots,v_r$, is $X(K)$ dense in $\prod_{j=1}^{r}X(K_{v_j})$? A formulation via non-abelian Galois cohomology is given in the case $K=\mathbf Q$ as follows: Is $H^1\!\left(\operatorname{Gal}(\overline{\mathbf Q}/\mathbf Q),G(\overline{\mathbf Q})\right) \longrightarrow H^1\!\left(\operatorname{Gal}(\mathbf C/\mathbf R),G(\mathbf C)\right)$ surjective?

Lucchini Arteche: The algebraic Brauer--Manin obstruction says nothing for this problem. Skorobogatov: The result is known for abelian $G$, due to Borovoi.

来源字段：https://aimath.org/pastworkshops/ratlhigherdimvarproblems.pdf

01430 | OWR-11784-003

### Real Fixed Points of the Minkowski Question-Mark Function

O / open | Oberwolfach 报告问题汇编

How many real fixed points does Minkowski’s question-mark function have?

来源字段：https://doi.org/10.4171/owr/2012/22

01431 | OWR-12866-002

### Real-Variable Proof of the Nyman Criterion

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Composite: Nyman Criterion and Autocorrelation Maximum [OWR-12866-003]

Give a real-variable proof of the equivalence between the following assertions: for every $\varepsilon>0$, $$\pi(x)=\operatorname{li}(x)+O(x^{1/2+\varepsilon});$$ and, for every $\delta>0$, there are coefficients $c_\alpha$, with $\alpha>1$, such that $$\int_0^\infty\left|\chi(t)-\sum_\alpha c_\alpha e_\alpha(t)\right|^2\frac{dt}{t^2}<\delta,$$ where $\chi(t)=1$ for $t>1$ and $0$ otherwise, and $e_\alpha(t)=\{t/\alpha\}$.

来源字段：https://doi.org/10.4171/owr/2014/06

01432 | NT-060

### Recamán's Sequence Completeness

O / open | 独立 / 未分配集合

Does every nonnegative integer appear in Recamán's sequence?

背景链接：https://oeis.org/wiki/Recam%C3%A1n%27s_sequence

01433 | AIM-ANALYTIC_NUMBER_THEORY-0084

### Reciprocal fibre widths and Fricke symmetry in Calabi--Yau curve counts

P / partially_solved | AIM 研讨会问题汇编

别名：Paramodular consistency tests for Calabi-Yau counting series [AIM-PHYSICS-0017]

Investigate the role of paramodular groups in counting problems in Calabi Yau 3-folds

来源字段：http://aimpl.org/gromwitnumthry/3/

01434 | OWR-11695860-014

### Recovering Polynomials from Fourier-Magnitude Data

P / partially_solved | Oberwolfach 报告问题汇编

Let $f$ and $g$ be polynomials satisfying $|W_f(a;p)|=|W_g(a;p)|$ for every $a$ coprime to one large prime $p$, or for all sufficiently large primes. Determine the algebraic relationship between $f$ and $g$.

来源字段：https://doi.org/10.4171/owr/2022/50

01435 | OWR-1537-002

### Reduced Length and Mahler-Measure Inequality

O / open | Oberwolfach 报告问题汇编

Let $L(R)$ be the sum of the absolute values of the coefficients of a real polynomial $R$, let $\ell(P)=\inf_{Q\in\mathbb{R}[x]\text{ monic}}L(PQ)$ be the reduced length, and let $M(P)$ be the Mahler measure. Is $\ell((x^2+tx+1)P(x))\geq2M(P)$ for every $t\in[-2,2]$ and every $P\in\mathbb{R}[x]$?

来源字段：https://doi.org/10.4171/owr/2007/21

01436 | OWR-14299523-003

### Regularity of Dedekind Zeta Functions in Zp-Extensions

O / open | Oberwolfach 报告问题汇编

Identify and prove further regularity phenomena for Dedekind zeta functions and their invariants along arbitrary $\mathbb{Z}_p$-extensions, beyond the known Iwasawa-type asymptotics.

来源字段：https://doi.org/10.4171/owr/2025/38

01437 | OWR-14603-008

### Regulator Finiteness for Elliptic Curves from ABC

O / open | Oberwolfach 报告问题汇编

Assume the ABC conjecture. Over a fixed number field $K$, are there only finitely many $\mathbb Q$-isomorphism classes of elliptic curves with bounded positive Mordell-Weil rank and bounded regulator?

来源字段：https://doi.org/10.4171/owr/2016/21

01438 | OWR-14603-007

### Regulator Finiteness from Lang-Silverman Heights

O / open | Oberwolfach 报告问题汇编

Assume the Lang-Silverman height conjecture. For a fixed number field $K$ and integer $g\geq1$, are there only finitely many $\mathbb Q$-isomorphism classes of simple abelian varieties over $K$ of dimension $g$, bounded positive Mordell-Weil rank, and bounded regulator?

来源字段：https://doi.org/10.4171/owr/2016/21

01439 | OWR-14299909-009

### Relations Among Multiple Weierstrass Function Spaces

O / open | Oberwolfach 报告问题汇编

For the spaces $\mathcal P_k$ of multiple Weierstrass functions and Eisenstein-series spaces $\mathcal E_k$, prove that $C_n(\mathcal P_{k+n})=\mathcal E_k$ for all $k\geq 0$ and $n\geq 3$, and that $\frac{d}{dz}(\mathcal E\cdot\mathcal P)\subseteq\mathcal P$. Also prove $\mathcal E_k\cap\mathcal P_k=\Delta\mathcal E_{k-12}$ and $\mathcal E_k\cap(\mathcal P_{k-2}\wp_2)=\Delta\mathcal E_{k-14}\wp_2$.

来源字段：https://doi.org/10.4171/owr/2026/7

01440 | OWR-15581-008

### Relations Between Arithmetic Milnor Invariants

P / partially_solved | Oberwolfach 报告问题汇编

For $\ell=2,3$, is there a relation between the mod-$\ell$ Milnor invariants $\mu_\ell(l)$ of primes and the $\ell$-adic Milnor invariants $\mu^{(\ell)}(g;l)$ of Galois elements?

来源字段：https://doi.org/10.4171/owr/2017/38

01441 | OWR-15169-012

### Removing Logarithmic Losses in Multiplicative Approximation

O / open | Oberwolfach 报告问题汇编

The sets $\{(\alpha,\beta):\liminf_{q\to\infty}(q\log q\log\log q)\lVert q\alpha\rVert\lVert q\beta\rVert>0\}$ and $\{\alpha:\liminf_{q\to\infty}(q\log q\log\log q)|q|_p\lVert q\alpha\rVert>0\}$ have full Hausdorff dimension. Can the factor $\log\log q$ be removed? Are these sets winning? What happens when they are intersected with a planar curve?

来源字段：https://doi.org/10.4171/owr/2016/48

01442 | OWR-14211-006

### Representation of Quadratic Forms by Lattices and Genera

P / partially_solved | Oberwolfach 报告问题汇编

For a lattice $\Lambda$ and $T_d=\{T:\det(T)=d\}$, study which forms in subsets $S_d\subseteq T_d$ are represented by $\Lambda$ or by every $\Lambda'\in\operatorname{gen}(\Lambda)$, determine how many $T\in T_d$ a given $\Lambda$ represents, and determine the average and asymptotics of $r(\Lambda,T)$ over $T\in T_d$.

来源字段：https://doi.org/10.4171/owr/2016/3

01443 | OWR-4428-008

### Residually Finite Torsion Groups with Restricted Subgroups

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist an infinite finitely generated residually finite torsion group in which every subgroup is finite or of finite index? For a number field $K$, a prime $p$, and a finite set $S$ of primes, if $G=G_{K,p,S}$ has positive weighted deficiency, can one choose a valuation $W$ naturally described by the arithmetic of $K$ such that $\operatorname{def}_W(G)>0$?

来源字段：https://doi.org/10.4171/owr/2010/40

01444 | GREEN-087

### Residue Class Multiple Coverage

O / open | Ben Green 问题

Can we pick residue classes $a_p \pmod p$, one for each prime $p \leq N$, such that every integer $\leq N$ lies in at least 10 of them?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

01445 | OWR-8415342-016

### Restriction of Unramified GL Representations to Iwahori Blocks

P / partially_solved | Oberwolfach 报告问题汇编

For an unramified generic representation $\pi$ of $\operatorname{GL}_{n+1}(F)$, determine and model its restriction to the Iwahori block of $\operatorname{GL}_n(F)$, including whether the isomorphism class is independent of $\pi$ and how products of Steinberg representations behave.

来源字段：https://doi.org/10.4171/owr/2021/39

01446 | OWR-11135-003

### Restriction Spectra of $\operatorname{GL}_3$ Maass Forms

O / open | Oberwolfach 报告问题汇编

For a $\operatorname{GL}_3$ Maass form restricted to the codimension-$2$ submanifold arising from the upper-triangular embedding of $\operatorname{GL}_2$, what is its inner product with a given $\operatorname{SL}_2(\mathbb Z)$ Maass form, or more generally its spectral decomposition?

来源字段 : https://doi.org/10.4171/owr/2011/43

01447 | OWR-14299577-001

### Reversible Primes and Prime Palindromes

O / open | Oberwolfach 报告问题汇编

Are there infinitely many primes whose base-ten digit reversals are also prime? Are there infinitely many palindromic primes?

来源字段 : https://doi.org/10.4171/owr/2025/51

01448 | AIM-ALGEBRAIC_NUMBER_THEORY-0093

### Rigid Calabi--Yau intermediate Jacobians: periods, cycle-theoretic separation, and symmetry certificates

P / partially_solved | AIM 研讨会问题汇编

Use the intermediate Jacobian (when it is an abelian variety) in arithmetic? To fix ideas, \(\dim X=3\), look at algebraic 1-cycles modulo rational equivalence. Compute this for a rigid Calabi--Yau 3-fold \(X\) (over \(\mathbf C\)), i.e. \(K_X\sim0\), \(X\) simply connected, \(h^{1,0}=h^{2,0}=h^{2,1}=0\), and \(h^{3,0}=1\). In this case the intermediate Jacobian \(J^2(X)\cong E\) is of dimension 1. Determine \(E\) given \(X\), i.e. give its \(j\)-invariant.

来源字段 : https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01449 | OWR-15956-011

### Rigidity of a Multiplicative Functional Equation

O / open | Oberwolfach 报告问题汇编

If $f:\mathbb R_{>0}\to\mathbb R_{>0}$ is nondecreasing and $$ f(x)+f(x/2)+f(x/3)+f(x/5)=\left(\frac{61}{30}+o(1)\right)x $$ as $x\to\infty$, must $f(x)=(1+o(1))x$?

来源字段 : https://doi.org/10.4171/owr/2017/51

01450 | OWR-8415342-024

### Root Numbers and Central Characters of Langlands Parameters

P / partially_solved | Oberwolfach 报告问题汇编

Let $k$ be a non-Archimedean local field, let $G$ be a reductive $k$-group, and let $\phi$ be the Langlands parameter of an essentially discrete-series representation of $G(k)$. Prove that $\frac{\omega(\operatorname{Ad}_G\circ\phi)}{\omega(\operatorname{Ad}_G\circ\phi_{\operatorname{prin}})}=\chi_\phi(z_{\operatorname{Ad}_G})$, where $\phi_{\operatorname{prin}}$ is the principal parameter, $z_{\operatorname{Ad}_G}$ is the associated central involution, and $\chi_\phi$ is the central character attached to $\phi$.

来源字段 : https://doi.org/10.4171/owr/2021/39

01451 | OWR-14603-010

### Root Separation and Rational Values of Littlewood Polynomials

O / open | Oberwolfach 报告问题汇编

Let

$$\mathcal P_d=\{a_dx^d+\cdots+a_0:a_i\in\{-1,0,1\}\}.$$

Is the minimum nonzero distance between roots of two polynomials in $\mathcal P_d$ bounded below by $\exp(-Cd)$ for some absolute $C>0$? For a fixed rational $p/q\in[9/10,19/20]$, can the trivial lower bound on $\min_{P\in\mathcal P_d}\lvert P(p/q)\rvert$ be improved to $c_{p,q}\exp(-Cd)$?

来源字段 : https://doi.org/10.4171/owr/2016/21

01452 | AIM-ANALYTIC_NUMBER_THEORY-0058

### Root-number conditioning in quadratic-twist moments

P / partially_solved | AIM 研讨会问题汇编

**Problem.** Develop a heuristic for quadratic twists of \(L\)-functions.

**Remark.** What role does the root number play?

来源字段 : http://aimpl.org/zetamoments/5/

01453 | OWR-11127-009

### Sato-Tate Equidistribution for Exceptional Abelian Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Which exceptional abelian varieties admit a proof of the numerical Sato-Tate equidistribution conjecture, and how can all such exceptional cases be classified?

来源字段 : https://doi.org/10.4171/owr/2011/35

01454 | NUM-016

### Schanuel's Conjecture

O / open | 独立 / 未分配集合

别名：Schanuel's Conjecture [OPG-1788]

For e and π: are they algebraically independent? Is e+π, eπ, π^e, etc. transcendental?

背景链接：https://doi.org/10.1007/s00029-023-00853-y

01455 | NT-028

### Schinzel's Hypothesis H

O / open | 独立 / 未分配集合

If polynomials satisfy certain necessary divisibility conditions, do they simultaneously produce infinitely many primes for integer inputs?

背景链接：https://eudml.org/doc/206115

01456 | NT-039

### Scholz Conjecture

P / partially_solved | 独立 / 未分配集合

Is the shortest addition chain for $2^n - 1$ at most $n - 1$ plus the length of the shortest addition chain for $n$?

背景链接：https://projecteuclid.org/journals/bulletin-of-the-american-mathematical-society/volume-45/issue-10/On-addition-chains/bams/1183502523.full

01457 | OWR-11127-021

### Schur $\sigma$-Group Frequency Heuristics

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Schur $\sigma$-Group Frequency Heuristic [OWR-11127-022]

Let $G$ be a finite $p$-group that is a Schur $\sigma$-group of rank $g$, or a maximal $p$-class-$c$ quotient of one. Among imaginary quadratic fields $K$ with $A_K$ of rank $g$, is the probability that the corresponding $G_K$, or its maximal $p$-class-$c$ quotient, is isomorphic to $G$ equal to

$$\frac{z(G)^g}{|\operatorname{Aut}(G)|}\,p^{-gh}\prod_{k=1}^g(p^g-p^{g-k})\prod_{k=1}^h(p^g-p^{h-k})?$$

Equivalently, does the displayed expression give the limiting frequency?

来源字段：https://doi.org/10.4171/owr/2011/35

01458 | AIM-ANALYTIC_NUMBER_THEORY-0075

### Second moments on periodic comb sets

P / partially_solved | AIM 研讨会问题汇编

Evaluate moments of zeta over sets other than $[0,T]$ or $[T,2T]$.

来源字段：http://aimpl.org/zetamoments/8/

01459 | OWR-3480-017

### Section Conjecture for Hyperbolic Curves Without Boundary Points

O / open | Oberwolfach 报告问题汇编

别名：Section Conjecture for Hyperbolic Curves [OWR-3480-018]

If $k$ is a number field or a finite extension of $\mathbb{Q}_p$ and $X$ is a smooth geometrically connected hyperbolic curve over $k$ whose boundary has no $k$-rational point, does $a\mapsto s_a$ biject $X(k)$ with conjugacy classes of sections of $\pi_1(X/k)$?

来源字段：https://doi.org/10.4171/owr/2009/30

01460 | AIM-ALGEBRAIC_NUMBER_THEORY-0084

### Section retractions obstruct the fibre-product line topoi

P / partially_solved | AIM 研讨会问题汇编

**21. (M. Flach)** Using the current definition of the Weil-étale topology, define the Weil-étale topos for an arbitrary arithmetic scheme (i.e., a scheme of finite type over \(\operatorname{Spec}(\mathbb Z)\)) as a fibred product. Can one compute the cohomology of the sheaf \(\mathbb Z\) on, say, the affine or projective line?

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

01461 | OWR-4990381-011

### Sections and Rational Points of Hyperbolic Curves

P / partially_solved | Oberwolfach 报告问题汇编

Let $k$ be a finitely generated extension of $\mathbb Q$ and let $X/k$ be a proper hyperbolic curve. Is every section of $\pi_1(X)\to G_k$ induced by a unique point of $X(k)$? For a smooth nonproper hyperbolic curve $U$ with compactification $X$, is every section of $\pi_1(U)\to G_k$ likewise induced by a unique point of $X(k)$?

来源字段：https://doi.org/10.4171/owr/2021/12

01462 | AIM-ALGEBRAIC_NUMBER_THEORY-0104

### Sections on every open and the Brauer--Manin set

P / partially_solved | AIM 研讨会问题汇编

**Problem/Question 19.** Let \(X\) be a variety over a number field \(k\), and suppose that for every Zariski-dense open \(U\subset X\), the map

\[ \pi_1(U)\longrightarrow \operatorname{Gal}(\bar k/k) \]

has a splitting \(s\) (for example, this holds if \(X(k)\) is Zariski dense). If \(X(\mathbf A_k)\ne\varnothing\), is \[ X(\mathbf A_k)^{\operatorname{Br}}\ne\varnothing? \] (Ellenberg)

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01463 | AIM-ALGEBRAIC_NUMBER_THEORY-0062

### Seeing a primitive surface class in the generic Jacobian

P / partially_solved | AIM 研讨会问题汇编

If we start with any surface \(S\) over \(\mathbb F_q\) we can, via Lefschetz fibration, make it into a curve over \(\mathbb F_q(T)\). Typically (since the rank of \(H^2\) is even), one has two Tate classes. One of these is given by hyperplane sections; can we “see” or construct the other one? This question is like the one above, but without requiring \(C\) to be hyperelliptic. So, in some sense, the problem is almost as hard as the problem of treating general surfaces. (C. Schoen)

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

01464 | OWR-14299577-014

### Selberg Eigenvalue Conjecture for Congruence Groups

O / open | Oberwolfach 报告问题汇编

Prove that every nonzero Laplace eigenvalue of a Maass cusp form for a congruence subgroup is at least $1/4$, equivalently that exceptional Maass cusp forms with eigenvalue below $1/4$ do not exist.

来源字段：https://doi.org/10.4171/owr/2025/51

01465 | AMR-093-0079

### Selberg's 1/4 conjecture

P / partially_solved | AMR 问题清单汇编

Selberg's 1/4 conjecture: the eigenvalues of the Laplace operator on Maass wave forms of congruence subgroups are at least $1/4$.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01466 | AMR-093-0080

### Selberg's orthogonality conjecture

O / open | AMR 问题清单汇编

Selberg's orthogonality conjecture: generalization of Mertens' theorem for functions in Selberg class.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01467 | OWR-14297740-021

### Self-Duality of Quasi-Supersingular Group Schemes

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a $p$-torsion finite flat commutative group scheme over the Witt vectors $W$ of rank $p^{2n}$ whose special fiber $\overline G$ is quasi-supersingular. For $n\geq3$, must $G$ and $\overline G$ be self-dual?

来源字段：https://doi.org/10.4171/owr/2023/42

01468 | AMR-093-0136

### Selfridge's conjecture

O / open | AMR 问题清单汇编

Selfridge's conjecture: is 78,557 the lowest Sierpiński number?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01469 | GUY-A12b

### Selfridge-Wagstaff-Pomerance Prize Problem

P / partially_solved | Richard Guy 数论问题

Does there exist a composite number $n \equiv 3$ or $7 \pmod{10}$ which divides both $2^n - 2$ and the Fibonacci number $u_{n+1}$?

背景链接：https://doi.org/10.2140/obs.2019.2.411

01470 | AMR-092-0006

### Semi-magic square of distinct positive cubes

O / open | AMR 问题清单汇编

Does there exist a $3\times3$ semi-magic square whose nine entries are distinct positive integer cubes and whose three row sums and three column sums are all equal?

背景链接：https://unsolvedproblems.org/index_files/SquareofCubes.htm

01471 | OWR-12583-003

### Serre's Uniformity Problem for Elliptic Curves

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a constant $p_0$ such that, for every elliptic curve $E/\mathbb Q$ without complex multiplication and every prime $p>p_0$, the representation $$\rho_{E,p}:\operatorname{Gal}(\overline{\mathbb Q}/\mathbb Q)\longrightarrow\operatorname{GL}_2(\mathbb F_p)$$ is surjective?

来源字段：https://doi.org/10.4171/owr/2013/35

01472 | OPG-414

### Sets with distinct subset sums

S / solved | Open Problem Garden

Say that a set $S \subseteq {\mathbb Z}$ has distinct subset sums if distinct subsets of $S$ have distinct sums.

Conjecture There exists a fixed constant $c$ so that $|S| \le \log_2(n) + c$ whenever $S \subseteq \{1,2,\ldots,n\}$ has distinct subset sums.

背景链接：http://www.openproblemgarden.org/op/sets_with_distinct_subset_sums

01473 | OWR-14297746-010

### Settled Frobenius Elements in Arboreal Representations

P / partially_solved | Oberwolfach 报告问题汇编

Let $f$ be a polynomial of degree $d\geq 2$ over a number field $K$, let $\alpha\in K$, and let $\operatorname{Gal}_{f,\alpha}\subseteq\operatorname{Aut}(T_d)$ be the image of the associated arboreal representation. Prove that the images of Frobenius elements are settled, and hence that settled elements are dense in $\operatorname{Gal}_{f,\alpha}$.

来源字段：https://doi.org/10.4171/owr/2023/37

01474 | OWR-1274-008

### Settled Quadratic Polynomials and Markov Factorization Models

O / open | Oberwolfach 报告问题汇编

Is every irreducible quadratic polynomial $f\in\mathbb F_p[x]$ settled, meaning that the proportion of the degree of $f^n$ occupied by $f$-stable irreducible factors tends to $1$? More generally, does the large-$n$ distribution of critical-orbit labels of the irreducible factors of $f^n$ follow the long-term behavior of the appropriate Markov process?

来源字段：https://doi.org/10.4171/owr/2006/25

01475 | AIM-ANALYTIC_NUMBER_THEORY-0089

### Shadow-controlled functional-equation defects for mixed mock coefficient series

P / partially_solved | AIM 研讨会问题汇编

别名：Functional equations and an Euler-product obstruction for mixed mock L-series [AIM-PHYSICS-0022]

**Problem.** Do the L-series of mixed mock modular forms have interesting properties?

来源字段：http://aimpl.org/gromwitnumthry/5/

01476 | GUY-A7b

### Shanks Chains of Length 7

S / solved | Richard Guy 数论问题

Are there any Shanks chains of length 7 with $p_{i+1} = 4p_i^2 - 17$?

背景链接：https://mathworld.wolfram.com/ShanksChain.html

01477 | OWR-1586-001

### Sharifi’s Modular-Symbol Isomorphism

P / partially_solved | Oberwolfach 报告问题汇编

For each level $r$, is there an isomorphism of $\mathbb Z_p[\operatorname{Gal}(F_r/\mathbb Q)]$-modules $$\nu_r:H^2_{\mathrm{et}}(\mathbb Z[\zeta_{p^r},1/p],\mathbb Z_p(2))^+\longrightarrow H_1(X_1(p^r);\mathbb Z_p)^+/I_r$$ sending the cup product $(1-\zeta_{p^r}^u,1-\zeta_{p^r}^v)_r^+$ to $\xi_r(u:v)^+$ modulo $I_r$?

来源字段：https://doi.org/10.4171/owr/2007/30

01478 | OWR-16158-003

### Sharp Density of Parametric Regular Quadratic Extensions

O / open | Oberwolfach 报告问题汇编

The proportion $f(H)$ of parametric regular quadratic extensions is known to satisfy

$$f(H)=1-O\!\left(\frac{\log H}{\sqrt H}\right)$$

Can the error term be removed unconditionally?

来源字段：https://doi.org/10.4171/owr/2018/17

01479 | OWR-1703872-010

### Sharp Moments of Zeta and Character Averages

O / open | Oberwolfach 报告问题汇编

Are both estimates below sharp for every $0\leq q\leq1$?

$$\frac1T\int_T^{2T}\left(\int_{-1/2}^{1/2}|\zeta(1/2+it+ih)|^2\,dh\right)^qdt \ll\left(\frac{\log T}{1+(1-q)\sqrt{\log\log T}}\right)^q,$$

and, for a prime $r$, $1\leq x\leq r$, and $L=\min\{x,r/x\}$,

$$\frac1{r-2}\sum_{\substack{\chi\ ({\rm mod}\ r)\\\chi\neq\chi_0}}\left|\sum_{n\leq x}\chi(n)\right|^{2q} \ll\left(\frac{x}{1+(1-q)\sqrt{\log\log(10L)}}\right)^q$$?

来源字段：https://doi.org/10.4171/owr/2020/26

01480 | OWR-3480-009

### Sharper Ramification Bounds for Local Galois Representations

O / open | Oberwolfach 报告问题汇编

Let $p>2$, let $T_n$ be the reduction modulo $p^n$ of a crystalline representation with Hodge-Tate weights in $[0,r]$, and let $L_n$ be the fixed field of its kernel. Write $r/(p-1)=p^{\alpha'}\beta'$ with $\alpha'\in\mathbb{N}$ and $p^{-1}<\beta'\leq1$. Does $v_K(\mathcal{D}_{L_n/K})\leq1+e(n+\alpha'+\beta')-p^{-(n+\alpha')}$ hold?

来源字段：https://doi.org/10.4171/owr/2009/30

01481 | OWR-14298805-004

### Sheaves on Bundles and Isocrystals

P / partially_solved | Oberwolfach 报告问题汇编

Construct the conjectural canonical $\operatorname{Perf}(\operatorname{Par}_G)$-linear equivalence $\operatorname{Shv}(\operatorname{Bun}_G,\Lambda)\simeq\operatorname{Shv}(\operatorname{Isoc}_G,\Lambda)$ in full generality.

来源字段：https://doi.org/10.4171/owr/2024/54

01482 | OWR-4076-015

### Shimura Period Relations for Jacquet-Langlands Transfers

P / partially_solved | Oberwolfach 报告问题汇编

Let $F$ be totally real, let $f$ be a Hilbert newform, and let $f_B$ be its Jacquet-Langlands transfer. Do constants $c_v\in\mathbb C$, for $v\in\Sigma_\infty$, exist such that, as $B$ varies, $$\langle f_B,f_B\rangle\sim_{\overline{\mathbb Q}^\times}\prod_{\substack{v\in\Sigma_\infty\\v\nmid\operatorname{disc}(B)}} c_v$$?

来源字段：https://doi.org/10.4171/owr/2009/33

01483 | OWR-14299577-017

### Short Coefficient Detection of Cusp-Form Norms

O / open | Oberwolfach 报告问题汇编

For every $\varepsilon>0$, prove that there is a constant $C=C(k,\varepsilon)>0$ such that every cusp form $f\in S_k(N)$ satisfies $\|f\|^2\ll_{k,\varepsilon}\sum_{n\leq C N^{1+\varepsilon}}|a_f(n)|^2$, without assuming squarefree level or Atkin-Lehner eigenform conditions.

来源字段：https://doi.org/10.4171/owr/2025/51

01484 | OWR-1703872-013

### Short-Interval Möbius Orthogonality to Nilsequences

O / open | Oberwolfach 报告问题汇编

Let $G/\Gamma$ be a nilmanifold of degree $k\geq1$ and $H=H(X)\to\infty$. For every fixed Lipschitz map $F:G/\Gamma\to\mathbb{C}$, is it true that for almost all $x\in[X,2X]$, $\sup_{g\in\operatorname{Poly}(\mathbb{Z}\to G)}\left|\sum_{x\leq n\leq x+H}\mu(n)F(g(n)\Gamma)\right|=o(H)$?

来源字段：https://doi.org/10.4171/owr/2020/26

01485 | OWR-1703872-012

### Short-Interval Möbius Orthogonality to Polynomial Phases

O / open | Oberwolfach 报告问题汇编

For every $k\geq1$ and every $H=H(X)\to\infty$ as $X\to\infty$, is it true that for almost all $x\in[X,2X]$, $\sup_{P\in\mathbb{R}[t],\ \deg P=k}\left|\sum_{x\leq n\leq x+H}\mu(n)e(P(n))\right|=o(H)$?

来源字段：https://doi.org/10.4171/owr/2020/26

01486 | AMR-030-0053

### Show that there exists a B so that, for every n > 0, there exists a k relatively prime to n whose continued fraction has

O / open | AMR 问题清单汇编

Niederreiter: Show that there exists a B so that, for every n > 0, there exists a k relatively prime to n whose continued fraction has partial quotients all bounded in average by B. (The sequence a, b, c, d is bounded in average by B if a <= B, a+b <= 2B, a+b+c <= 3B, etc.) Is B=3 already sufficient? If such a B exists, it means that there are "good multipliers" for quasirandom permutations.

背景链接：https://people.math.sc.edu/cooper/combprob.html

01487 | AMR-030-0064

### Show that there is some B so that no integer appears more than B times among the binomial coefficients

P / partially_solved | AMR 问题清单汇编

Singmaster : Show that there is some B so that no integer appears more than B times among the binomial coefficients. See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

01488 | OWR-17300-018

### Siegel Type Fourier Expansions Without Complex Structures

P / partially_solved | Oberwolfach 报告问题汇编

If a reductive $\mathbb{Q}$-group $G$ has an associated symmetric space without complex structure, is there a notion of modular forms for $G$ possessing a Fourier expansion analogous to the Fourier expansion of Siegel modular forms?

来源字段：https://doi.org/10.4171/owr/2019/57

01489 | NUM-015

### Siegel Zeros

P / partially_solved | 独立 / 未分配集合

Do Siegel zeros (real zeros of Dirichlet L-functions near s=1) exist?

背景链接：https://en.wikipedia.org/wiki/Siegel_zero

01490 | AMR-093-0160

### Siegel's conjecture

O / open | AMR 问题清单汇编

Siegel's conjecture: are there infinitely many regular primes, and if so is their natural density as a subset of all primes $e^{-1/2}$?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01491 | GREEN-086

### Sieving by Many Small Primes

P / partially_solved | Ben Green 问题

Sieve $[N]$ by removing half the residue classes mod $p_i$, for primes $2 \leq p_1 < p_2 < \dots < p_{1000} < N^{9/10}$. Does the remaining set have size at most $\frac{1}{10}N$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

01492 | AIM-ANALYTIC_NUMBER_THEORY-0066

### Signed height weights that extract one moment coefficient

P / partially_solved | AIM 研讨会问题汇编

Can one formulate a weighted version of moments with potentially simpler main terms?

来源字段：http://aimpl.org/zetamoments/6/

01493 | AIM-ANALYTIC_NUMBER_THEORY-0109

### Signed large and recurrent fixed second differences of consecutive primes

P / partially_solved | AIM 研讨会问题汇编

T. D. Wooley proposed the following problem: How often are the second differences \[ p_{n+2}-2p_{n+1}+p_n \] small (large)? He noted that Erdős has proved that \[ |p_{n+2}-2p_{n+1}+p_n|\ge (1+\delta)\log p_n \tag{3} \] infinitely often. A. Balog added that he had thought about the problem in the past and had convinced himself that the left side of (3) is infinitely often as large as the largest known gap between consecutive primes.

来源字段：https://aimath.org/WWN/primegaps/WorkshopProblems.pdf

01494 | OWR-15583-004

### Simplicity of Nonreal Zeros of Cusp-Form L-Functions

O / open | Oberwolfach 报告问题汇编

For a primitive holomorphic cusp form $f$, are all nonreal zeros of $L(s,f)$ simple?

来源字段：https://doi.org/10.4171/owr/2017/40

01495 | OWR-1537-015

### Simultaneous Real and $p$-Adic Algebraicity

O / open | Oberwolfach 报告问题汇编

Let $p$ be prime and let $(a_n)_{n\geq1}$ satisfy $a_n\in\{0,1,\ldots,p-1\}$. Define the real number $\alpha=\sum_{n\geq1}a_n/p^n$ and the $p$-adic number $\alpha_p=\sum_{n\geq1}a_np^n$. Prove that $\alpha$ and $\alpha_p$ are algebraic if and only if both are rational.

来源字段：https://doi.org/10.4171/owr/2007/21

01496 | OWR-1537-014

### Single-Solution Bounds for Diophantine Equations

O / open | Oberwolfach 报告问题汇编

Can Matiyasevich’s theorem for exponential Diophantine equations-an equation with at most one solution in the variables for each parameter, while some parameter has a solution exceeding every prescribed total effectively computable bound-be improved to genuine Diophantine equations without exponentiation, possibly allowing finitely many rather than at most one solution?

来源字段：https://doi.org/10.4171/owr/2007/21

01497 | OWR-12725-004

### Singular Locus Dimension of Nonsingular Form Systems

S / solved | Oberwolfach 报告问题汇编

For a nonsingular system of $R$ homogeneous forms of the same degree, is the dimension $B$ of its singular locus always equal to $R-1$?

来源字段：https://doi.org/10.4171/owr/2013/51

01498 | AIM-ALGEBRAIC_NUMBER_THEORY-0022

### Singularities, Fano type, and anticanonical height zeta functions

P / partially_solved | AIM 研讨会问题汇编

What is the “right” assumption on a variety \(V\) for considering height zeta functions? Definitely smooth \(V\) with ample anticanonical sheaf \(\omega_V^{-1}\) should be allowed. How about klt (i.e. Kawamata log terminal) \(V\)? Or \(V\) of Fano type (i.e. there is an effective \(\mathbb Q\)-divisor \(\Delta\) such that \((V,\Delta)\) is ample and \(-(K_V+\Delta)\) is ample)? An example for the latter: a quasi-smooth hypersurface \(V\subset \mathbb P(a_0,\ldots,a_n)\) in weighted projective space of degree \(\deg(V)<a_0+\cdots+a_n\).

来源字段：https://aimath.org/pastworkshops/ratlhigherdimvarproblems.pdf

01499 | OWR-11695860-019

### Six Problems from an Analytic Number Theory Session

O / open | Oberwolfach 报告问题汇编

Resolve the following six problems: (1) construct a degree-five $2$-superirreducible polynomial; (2) for each finite abelian group $G$, optimize $n$ and $d$ in a representation $G\cong\{z^d:z\in(\mathbb Z/n\mathbb Z)^\times\}$; (3) find homogeneous forms $f$ and $g$ of degrees $3$ and $2$ in five variables with $f^2+g^3=N_{K/\mathbb Q}(x_1\omega_1+\cdots+x_5\omega_5)$; (4) cover $\{1,\ldots,N\}$ at least twice by one residue class modulo each prime, controlling both the primes and $\sum_p1/p$; (5) produce arbitrary prescribed patterns of consecutive prime residues; and (6) develop circle-method, Weyl-sum, or Poisson-summation techniques for quaternions.

来源字段：https://doi.org/10.4171/owr/2022/50

01500 | NT-061

### Skolem Problem

P / partially_solved | 独立 / 未分配集合

Can an algorithm determine if a constant-recursive sequence contains a zero?

背景链接：https://theoretics.episciences.org/14219

01501 | AIM-ANALYTIC_NUMBER_THEORY-0061

### Small moments from defect-stable mollified correlations

P / partially_solved | AIM 研讨会问题汇编

Is there any way to get lower bounds for small moments (e.g. first moment) in families, even assuming GRH?

来源字段：http://aimpl.org/zetamoments/5/

01502 | OWR-15956-003

### Small Möbius Sums in Generalized Integer Systems

O / open | Oberwolfach 报告问题汇编

For a Beurling generalized integer system whose counting function satisfies $N(x)=x^{1+o(1)}$, how small can the generalized Möbius summatory function $$ M_{\mathcal P}(x)=\sum_{\substack{n\leq x\\ n\in\mathcal N}}\mu_{\mathcal P}(n) $$ be?

来源字段：https://doi.org/10.4171/owr/2017/51

01503 | GREEN-022

### Small Sieve Maximal Sets

O / open | Ben Green 问题

Suppose that a small sieve process leaves a set of maximal size. What is the structure of that set?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

01504 | AIM-ANALYTIC_NUMBER_THEORY-0071

### Smoothed second moments and the sigma=1 boundary term

P / partially_solved | AIM 研讨会问题汇编

Can one calculate moments of zeta closer to the \(\sigma=1\) line instead of on the \(\sigma=\frac12\) line? Can one also identify the lower order terms in the \(\sigma=1\) case?

来源字段：http://aimpl.org/zetamoments/8/

01505 | OPG-655

### Snevily's conjecture

P / partially_solved | Open Problem Garden

Conjecture Let $G$ be an abelian group of odd order and let $A,B \subseteq G$ satisfy $|A| = |B| = k$. Then the elements of $A$ and $B$ may be ordered $A = \{a_1,\ldots,a_k\}$ and $B = \{b_1,\ldots,b_k\}$ so that the sums $a_1+b_1, a_2+b_2 \ldots, a_k + b_k$ are pairwise distinct.

背景链接：http://www.openproblemgarden.org/op/snevilys_conjecture

01506 | GREEN-063

### Solutions to Polynomial Equations in Dense Sets

P / partially_solved | Ben Green 问题

别名：Polynomial Solutions in Dense Sets [GREEN-099]

Finding a single solution to $F(x_1, \dots, x_n) = C$ can be very difficult. What conditions on $A$ ensure that the number of solutions in $A$ is roughly $\alpha^n$ times the number in $[X]$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

01507 | AIM-ANALYTIC_NUMBER_THEORY-0040

### Solved divisibility-parity means and exact correlations in the complementary convergent regime

S / solved | AIM 研讨会问题汇编

Let \(\mathcal{A}=\{a_1,a_2,\dots\}\) be a sequence of integers with \((a_i,a_j)=1\) for \(i\ne j\). Let \[ \pi(n):=(-1)^{\#\{a_j:a_j\mid n\}}. \] Is it true that \[ \sum_{n\leq X}\pi(n)=o(X)? \] Can we find a set \(\mathcal A\) such that \[ \sum_{n\leq X}\pi(n+h_1)\dots\pi(n+h_k)=o(X) \] for all distinct \(h_1,\dots,h_k\)?

Remark: Let \[ \lambda_y(p)= \begin{cases} -1,&\text{if }p\leq y,\\ +1,&\text{if }p>y. \end{cases} \] If \(y=X^{o(1)}\), then \[ \sum_{n\leq X}\lambda_y(n+h_1)\cdots\lambda_y(n+h_k)=o(X). \] For \(k=2\) this was done by Daboussi--Sárközy.

来源字段：http://aimpl.org/sarnakconjecture/7/

01508 | AIM-ALGEBRAIC_NUMBER_THEORY-0087

### Solved intended local pencil problem and dyadic rank-seven annuli

P / partially_solved | AIM 研讨会问题汇编

Let \(f_1(x_0,\ldots,x_n)=f_2(x_0,\ldots,x_n)=0\) define a smooth complete intersection \(X_{2,2}\subset \mathbf P^n\) over a number field \(k\), for simplicity \(k=\mathbf Q\). If \(n\geq 8\), then \(X(\mathbf R)\ne\varnothing\) implies \(X(\mathbf Q)\ne\varnothing\). For \(n\geq 9\) (at least ten variables), this can be proved by the circle method. For \(n=7\), let \(k\) be a \(p\)-adic field and let \(f_1,f_2\) be quadratic forms in eight variables whose common projective zero locus is smooth, so that \[ P(\lambda,\mu)=\det(\lambda f_1+\mu f_2) \] is separable. Does there exist \([\lambda:\mu]\in\mathbf P^1(k)\) for which \(\lambda f_1+\mu f_2\) contains three hyperbolic planes? A solution would give a local--global principle for \(n=7\).

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01509 | OWR-1795-015

## Soundararajan Conjecture on Zeta Zero Spacing

P / partially_solved | Oberwolfach 报告问题汇编

Let $\beta'+i\gamma'$ be a nontrivial zero of $\zeta'(s)$, let $\gamma_c$ be the ordinate of the zero of $\zeta(s)$ nearest to $\gamma'$, and let $\gamma^+$ be the next zero ordinate after $\gamma$. Assuming the Riemann Hypothesis, is $\liminf_{\gamma'\to\infty}(\beta'-1/2)\log\gamma'=0$ equivalent to $\liminf_{\gamma\to\infty}(\gamma^+-\gamma)\log\gamma=0$?

来源字段：https://doi.org/10.4171/owr/2008/14

01510 | AIM-ANALYTIC_NUMBER_THEORY-0110

## Sparse-gap data do not determine Montgomery pair correlation

P / partially_solved | AIM 研讨会问题汇编

The record is Problem 10 in Angel Kumchev's notes from the December 2005 ARCC workshop *Gaps Between Primes*. The mathematically unambiguous repaired statement is as follows. Write

\[ g_n=p_{n+1}-p_n. \]

Heath-Brown proved, under the Riemann hypothesis (RH) and the relevant local forms of Montgomery's pair-correlation conjecture, that

\[ \liminf_{n\to\infty}\frac{g_n}{\log p_n}=0 \tag{1} \]

\[ g_n\ll \sqrt{p_n\log p_n}. \tag{2} \]

The workshop asks (i) whether there is a converse to the implication leading to (1), and (ii) whether random-matrix theory can improve (2).

* “Di ff erences” is merely broken PDF font extraction and means “Differences.” * The control character U+001C in the last display is the Vinogradov symbol \(\ll\). * Most importantly, in the official PDF the radical overbar spans the entire product \(p_n\log p_n\). Thus (2) is \(g_n\ll\sqrt{p_n\log p_n}\), not \(g_n\ll\sqrt{p_n}\log p_n\). This is also Corollary 2 on p. 87 of Heath-Brown's paper. * The word “converse” is not expanded in the workshop notes. Its only reading consistent with “which ... would be unconditional post-GPY” is

\[ \liminf g_n/\log p_n=0\quad\Longrightarrow\quad \text{Montgomery pair correlation} \tag{3} \]

(with whatever additional baseline hypothesis, such as RH, is stated). Indeed, GPY made the antecedent of (3) unconditional. The reverse reading would just repeat Heath-Brown's theorem and would not become a new unconditional pair-correlation result after GPY.

The workshop formulation says “RH and Montgomery's Pair Correlation Conjecture,” which is sufficient but stronger than Heath-Brown needs locally. For (1), his displayed conclusion on p. 88 follows when the pair-correlation asymptotic holds uniformly in a neighborhood of the frequency parameter \(\alpha=1\). For (2), bounded pair correlation in a neighborhood of \(\alpha=2\) suffices.

来源字段：https://aimath.org/WWN/primegaps/WorkshopProblems.pdf

01511 | AIM-ANALYTIC_NUMBER_THEORY-0039

## Sparse-orbit counterexamples and the exact Polynomial Chowla implication

P / partially_solved | AIM 研讨会问题汇编

1. **Polynomial Sarnak 1 (PS1).** If \((X,T)\) is minimal and has zero entropy, then \[ \sum_{n\leq N}\lambda(n)f(T^{p(n)}x)=o(N). \] 2. **Polynomial Sarnak 2 (PS2).** If \(p\in\mathbb Z[t]\) is not a constant multiple of the square of a polynomial, then \[ \sum_{n\leq N}\lambda(p(n))f(T^n x)=o(N). \] 3. **Polynomial Chowla (PC).** Under the same nonsquare condition, \[ \sum_{n\leq N}\lambda(p(n))=o(N). \]

The question is whether the two Polynomial Sarnak statements are true and whether either follows from Polynomial Chowla. The source uses \(X\) both for the phase space and the summation cutoff; this report uses \(N\) for the cutoff. We extend \(\lambda\) to \(\mathbb Z\) by \(\lambda(-m)=\lambda(m)\) and \(\lambda(0)=0\), as is standard in the polynomial Chowla literature. Polynomial zeros affect only finitely many summands.

The archived source contains two later remarks. The name “Z. Liang” in one remark is a typo: the cited author is **Zhengxing Lian**, with Ruxi Shi. The source also states that PS2 follows from PC by the usual Chowla-to-Sarnak argument. Section 7 below supplies the algebraic and ergodic details needed to justify that assertion from the displayed, one-polynomial formulation of PC.

来源字段：http://aimpl.org/sarnakconjecture/7/

01512 | OPG-37300

## Special Primes

P / partially_solved | Open Problem Garden

Conjecture Let $p$ be a prime natural number. Find all primes $q\equiv1\left(\mathrm{mod}\: p\right)$, such that $2^{\frac{\left(q-1\right)}{p}}\equiv1\left(\mathrm{mod}\: q\right)$.

背景链接：http://www.openproblemgarden.org/op/special_primes

01513 | OWR-16941-001

## Special Zeta Values via Weil-Étale Cohomology

O / open | Oberwolfach 报告问题汇编

For a regular scheme $X$ that is projective and flat over the integers, can its special zeta values be expressed conjecturally in terms of Weil-étale motivic, singular, and de Rham cohomology, compatibly with Serre's functional equation?

来源字段：https://doi.org/10.4171/owr/2019/29

01514 | OWR-14298805-003

### Spectral Categorical Langlands for Quasisplit Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a quasisplit group $G$ with Whittaker datum $(U,\psi)$, construct the conjectural canonical $\operatorname{Perf}(\operatorname{Par}_G)$-linear equivalence $\operatorname{Coh}_{\operatorname{Nilp}}(\operatorname{Par}_G)\simeq\operatorname{Shv}(\operatorname{Isoc}_G,\Lambda)^\omega$.

来源字段：https://doi.org/10.4171/owr/2024/54

01515 | OWR-14746-003

### Spectral Halo Breaking Points for Hilbert Modular Forms

O / open | Oberwolfach 报告问题汇编

In the Hilbert modular setting, what should replace the following spectral-halo assertion for classical overconvergent modular forms: if $(n_i,m_i)$ are the breaking points of the Newton polygon of the characteristic series modulo $p$, then there exists a rational $0<r<1$ such that, for every weight $\kappa$ in the annulus $r<\lvert T\rvert<1$, the breaking points after specialization are

$$\bigl(n_i,\operatorname{val}_p(\kappa(1+p)-1)m_i\bigr)$$?

来源字段：https://doi.org/10.4171/owr/2016/38

01516 | AIM-ANALYTIC_NUMBER_THEORY-0011

### Spectral large-sieve bounds for GL(3) and a GL(4) conductor window

P / partially_solved | AIM 研讨会问题汇编

Let \(\alpha\in\mathbb R\) and \(f_j\) be a Maass form for \({\rm SL}(4,\mathbb Z)\) with eigenvalue \(t_j\) and Fourier coefficients \(\{\lambda_j(1,1,n)\}_{n\geq1}\). For parameters \(T\) and \(N\), prove a non-trivial bound for the averaged sum \[ \sum_{t_j\sim T}\left|\sum_{n\leq N}\lambda_j(1,1,n)e(n\alpha)\right|^2. \] If \(\{f_j\}\) are Maass forms for \({\rm SL}(3,\mathbb Z)\), improve Miller's bound on average over the spectral parameter \(t_j\).

来源字段：http://aimpl.org/deltasubconvex2/1/

01517 | OWR-9790364-005

### Spectral-Deformation Methods for the Riemann Hypothesis

O / open | Oberwolfach 报告问题汇编

Can the spectral-deformation method based on $H_\lambda(t)$ prove the Riemann hypothesis despite the failure of the analogous assertion for $H_0(t)$ in general?

来源字段：https://doi.org/10.4171/owr/2022/13

01518 | OWR-1703872-016

### Sphere-Packing Bounds and Fourier Eigenfunction Constructions

O / open | Oberwolfach 报告问题汇编

The linear-programming upper bound for sphere-packing density in $\mathbb R^n$ is conjectured to be $2^{-(\lambda+o(1))n}$ with $\lambda\approx0.6044$, and there is conjectured to exist a nonzero Schwartz function $f:\mathbb R^2\times\mathbb R^2\to\mathbb R$ with $f(0,0)=0$, $\widehat f=-f$, and $f(x,y)\geq0$ whenever $|x|^2-|y|^2\in2\mathbb Z$ and $|x|^2+|y|^2\geq4/3$.

来源字段：https://doi.org/10.4171/owr/2020/26

01519 | OWR-12487-003

### Spherical Hecke Modules on Homogeneous Schemes

P / partially_solved | Oberwolfach 报告问题汇编

For a homogeneous spherical $G$-scheme $X$ over the integers of a nonarchimedean local field, describe the module structure of the $K$-invariant compactly supported functions $C_c^\infty(X)^K$ over the spherical Hecke algebra $\mathcal H(G,K)$, including a description of $\mathcal H(G,K)$.

来源字段：https://doi.org/10.4171/owr/2013/24

01520 | OWR-16166-009

### Square Discriminants and Minimal Ramification of Alternating Group Powers

O / open | Oberwolfach 报告问题汇编

Let $$\mathcal M_n=\left\{X^n+\sum_{i=0}^{n-1}\varepsilon_iX^i:\varepsilon_i\in\{\pm1\}\right\}.$$ Does the proportion of polynomials in $\mathcal M_n$ with square discriminant tend to zero as $n\to\infty$, and can the number of such polynomials be bounded by $O(1)$? Separately, for $G=A_5^n$, is the Boston-Markin prediction $$m(G)=\max\{d(G/[G,G]),1\}=1$$ valid?

来源字段：https://doi.org/10.4171/owr/2018/25

01521 | GUY-A12a

### Square Pseudoprimes

P / partially_solved | Richard Guy 数论问题

Are there any square pseudoprimes (base 2) other than multiples of $1194649 = 1093^2$ or $12327121 = 3511^2$?

背景链接：https://cs.uwaterloo.ca/journals/JIS/VOL14/Klyve/klyve3.html

01522 | OWR-1590-008

### Square Values of Primitive Binary Cubic Forms

O / open | Oberwolfach 报告问题汇编

Let $F$ be any primitive binary cubic form. Does $F$ primitively represent a square; that is, do there always exist coprime integers $x,z$ and an integer $y$ such that $F(x,z)=y^2$?

来源字段：https://doi.org/10.4171/owr/2007/34

01523 | OWR-11695860-008

### Square-Root-Scale Cancellation in Short Möbius Sums

P / partially_solved | Oberwolfach 报告问题汇编

Prove cancellation in $\sum_{x<n\leq x+y}\mu(n)$ for intervals of length $y$ at the square-root scale or below. A concrete range arising from the proposed fractional-moment bound is $\sqrt{x}/(\log\log x)^{1/4}\ll y\leq x^{1-\epsilon}$.

来源字段：https://doi.org/10.4171/owr/2022/50

01524 | OWR-15174-002

### Squarefree Group Orders in Elliptic-Curve Reductions

O / open | Oberwolfach 报告问题汇编

Let $E/\mathbb Q$ be an elliptic curve with $\operatorname{End}_{\mathbb Q}(E)\cong\mathbb Z$. Does there exist an explicit constant $\delta(E)$ such that

$$\#\{p\leq x:\#E(\mathbb F_p)\text{ is square-free}\}\sim\delta(E)\pi(x)$$

as $x\to\infty$?

来源字段：https://doi.org/10.4171/owr/2016/53

01525 | OWR-12725-007

### Squarefree Kernel Counts Outside the Established Range

O / open | Oberwolfach 报告问题汇编

Let $k(n)$ be the squarefree kernel of $n$ and

$$N(x,y)=\#\{n\leq x:k(n)\leq y\}.$$

Determine the asymptotic behavior of $N(x,y)$ outside

$$H_\varepsilon=\{(x,y):\exp((\log\log x)^\varepsilon)\leq y\leq x\},$$

apart from the known small-$y$ range.

来源字段：https://doi.org/10.4171/owr/2013/51

01526 | AMR-093-0058

### Stark conjectures on leading terms of Artin L-functions

P / partially_solved | AMR 问题清单汇编

For an Artin $L$-function attached to a Galois extension of number fields, is its leading Taylor coefficient at $s=0$ the product of the corresponding Stark regulator of $S$-units with the predicted algebraic factor, including the refined rank-one prediction of Stark units?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01527 | OWR-16171-013

### Stark Units in Abelian Towers

P / partially_solved | Oberwolfach 报告问题汇编

Prove the Gross-Popescu conjecture describing the behavior of Stark units in towers of abelian extensions of number fields.

来源字段：https://doi.org/10.4171/owr/2018/30

01528 | OWR-13750341-010

### Steinness of Local Shimura Varieties

O / open | Oberwolfach 报告问题汇编

Is every local Shimura variety a Stein space?

来源字段：https://doi.org/10.4171/owr/2023/28

01529 | OWR-14297740-018

### Strict Higher Étale-Homotopy Obstructions over Global Fields

O / open | Oberwolfach 报告问题汇编

Find varieties over global fields $K$ for which the full étale-homotopy obstruction is strictly stronger than its two-stage truncation; equivalently, find examples with $X(\mathbb{A}_K)^h\subsetneq X(\mathbb{A}_K)^{h,2}$.

来源字段：https://doi.org/10.4171/owr/2023/42

01530 | OWR-1188-007

### Strong Uniform Torsion Conjecture for Abelian Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Strong Torsion Conjecture: for every $g,d\geq1$, does there exist $n(d,g)\geq1$ such that no abelian variety of dimension at most $g$, defined over a number field of degree at most $d$, has a rational torsion point of order at least $n(d,g)$?

来源字段：https://doi.org/10.4171/owr/2006/06

01531 | OWR-14603-027

### Structure of Counterexamples to the p-Adic Littlewood Conjecture

O / open | Oberwolfach 报告问题汇编

How should a real number $x$ behave if it is a counterexample to the $p$-adic Littlewood conjecture $\liminf_{q\to\infty}q|q|_p\|qx\|=0$?

来源字段：https://doi.org/10.4171/owr/2016/21

01532 | OWR-1188-009

### Structure of Finitely Generated Maximal Pro-$p$ Galois Groups

O / open | Oberwolfach 报告问题汇编

Let $p$ be prime and let $F$ have characteristic different from $p$ and contain the $p$-th roots of unity. If the maximal pro-$p$ Galois group $G_F(p)$ is finitely generated, is it a free pro-$p$ product of groups isomorphic to $\mathbb{Z}_p$, to $\mathbb{Z}/2\mathbb{Z}$ when $p=2$, or to decomposition groups of valuations with nontrivial inertia?

来源字段：https://doi.org/10.4171/owr/2006/06

01533 | OWR-15956-004

### Sub-Square-Root Möbius Bounds for Generalized Primes

O / open | Oberwolfach 报告问题汇编

Does there exist a Beurling generalized prime system for which $M_{\mathcal P}(x)=O(x^\alpha)$ for some $\alpha<1/2$? Can constructions based on periodic errors $N(x)-x$ be adapted to nonperiodic approximations of $\lfloor x\rfloor$ whose associated Mellin transforms have zeros with real parts approaching $1/2$?

来源字段：https://doi.org/10.4171/owr/2017/51

01534 | OWR-1703872-003

### Subconvex Pointwise Equidistribution of Hecke-Laplace Eigenfunctions

O / open | Oberwolfach 报告问题汇编

Does there exist $\delta>0$ such that, for $y\in D^\times(\mathbb A_\mathbb Q)$ and every Hecke-Laplace eigenfunction $f$ of eigenvalue $\lambda$,

$$f(y)=\int_{D^\times(\mathbb Q)\backslash D^\times(\mathbb A_\mathbb Q)}f(x)\,d\mu(x)+O(\lambda^{1/4-\delta})$$?

来源字段：https://doi.org/10.4171/owr/2020/26

01535 | AIM-ANALYTIC_NUMBER_THEORY-0010

### Subconvex prime-twist families for GL4xGL2 and GL3xGL3

P / partially_solved | AIM 研讨会问题汇编

**Subconvexity bound for Rankin-Selberg \(L\)-functions.** Obtain a subconvexity bound for a \({\rm GL}_n\times {\rm GL}_m\) Rankin--Selberg \(L\)-function when \(n\) and \(m\) are not coprime.

The simplest first cases would be obtaining a subconvexity bound (in any aspect) for a \({\rm GL}_4\times {\rm GL}_2\) \(L\)-function or a \({\rm GL}_3\times {\rm GL}_3\) \(L\)-function.

来源字段：http://aimpl.org/deltasubconvex2/1/

01536 | OWR-794-004

### Subconvexity Coefficient Bounds for Triple $L$-Functions

O / open | Oberwolfach 报告问题汇编

For fixed Maass forms $\phi$ and $\phi'$, let $c_i=\int_Y\phi\phi'\phi_i\,dv$ be their triple periods and set $d_i=\gamma(\lambda_i)|c_i|^2$ using the standard gamma-factor normalization. Is it true that, for every $\varepsilon>0$, $d_i\ll|\lambda_i|^\varepsilon$?

来源字段：https://doi.org/10.4171/owr/2005/13

01537 | OWR-1703872-017

### Subconvexity for Symmetric-Square L-Functions

P / partially_solved | Oberwolfach 报告问题汇编

Let $u_j$ be a Hecke--Maass cusp form on $\mathrm{SL}_2(\mathbb{Z})$ with Laplace eigenvalue $1/4 + t_j^2$, and let $L(\operatorname{sym}^2 u_j, s)$ be its symmetric-square $L$-function. Prove a subconvexity bound in the spectral $t_j$-aspect at the central point $s = 1/2$, i.e., $L(\operatorname{sym}^2 u_j, 1/2) \ll t_j^{1/2 - \delta}$ for some absolute constant $\delta > 0$.

来源字段：https://doi.org/10.4171/owr/2020/26

01538 | OWR-12725-014

### Subpolynomial Bounds for an Additive Multiplicative Equation

O / open | Oberwolfach 报告问题汇编

Let $SP(n)$ be the number of positive-integer solutions to $abc+a+b+c=n$. For every $\varepsilon>0$, does there exist $C(\varepsilon)$ such that $SP(n)<C(\varepsilon)n^\varepsilon$ for all $n$?

来源字段：https://doi.org/10.4171/owr/2013/51

01539 | OPG-37404

### Sum of prime and semiprime conjecture

P / partially_solved | Open Problem Garden

Conjecture Every even number greater than $10$ can be represented as the sum of an odd prime number and an odd semiprime.

背景链接：http://www.openproblemgarden.org/op/sum_of_prime_and_semiprime_conjecture

01540 | GREEN-032

### Sums of Smooth Numbers

P / partially_solved | Ben Green 问题

Is every $n \leq N$ the sum of two integers, all of whose prime factors are at most $N^\varepsilon$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

01541 | GREEN-031

### Sumsets Containing Composites

O / open | Ben Green 问题

Suppose $A, B \subset \{1, \dots, N\}$ both have size $N^{0.49}$. Does $A + B$ contain a composite number?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

01542 | GREEN-033

### Sumsets of Perfect Squares

P / partially_solved | Ben Green 问题

Is there an absolute constant $c > 0$ such that if $A \subset \mathbb{N}$ is a set of squares of size at least 2, then $|A + A| \geq |A|^{1+c}$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

01543 | AIM-ANALYTIC_NUMBER_THEORY-0111

### Support-gap obstructions and a local-discrepancy converse for zeta

P / partially_solved | AIM 研讨会问题汇编

Traditionally, analytic information about the zeta-function has been used to derive upper bounds for differences between consecutive primes. Y. Motohashi asked whether this process can be reversed. For example, what (if anything) can be said about \(\zeta(s)\) in the critical strip under (H)?

来源字段：https://aimath.org/WWN/primegaps/WorkshopProblems.pdf

01544 | AMR-030-0048

### Suppose I have a sequence of positive integers whose reciprocals sum to infinity

P / partially_solved | AMR 问题清单汇编

Erdős: Suppose I have a sequence of positive integers whose reciprocals sum to infinity. Must that sequence contain arbitrarily long arithmetic progressions? Apparently very hard, though obnoxiously simple.

背景链接：https://people.math.sc.edu/cooper/combprob.html

01545 | AMR-030-0057

### Suppose k runners having distinct constant speeds start at a common point and run laps on a unit length circular track

P / partially_solved | AMR 问题清单汇编

Wills, Cusick: Suppose k runners having distinct constant speeds start at a common point and run laps on a unit length circular track. Then for any given runner x, there is a time at which runner x is distance at least 1/k away from every other runner. See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

01546 | AMR-030-0058

### Suppose that S is a set of positive integers with the property that no element is the sum of a nonempty set of other ele

P / partially_solved | AMR 问题清单汇编

Erdős: Suppose that S is a set of positive integers with the property that no element is the sum of a nonempty set of other elements. Such a set is called "sum-free." Let R(S) be the sum of the reciprocals of S. How large can R(S) be? Denote by R the supremum of R(S) over all sum-free S. It is known that R is less than 4, and R > 2.064 (Abbott and Levine-O'Sullivan, respectively).

背景链接：https://people.math.sc.edu/cooper/combprob.html

01547 | AMR-030-0055

### Suppose that S is a set of positive integers with the property that S+S -- that is, all sums of the form s_(1)+s_(2) for

P / partially_solved | AMR 问题清单汇编

Erdős-Turán: Suppose that S is a set of positive integers with the property that S+S -- that is, all sums of the form s_(1)+s_(2) for s_(1), s_(2) in S -- contains all sufficiently large integers. (Then S is called an "additive basis".) Is it possible for the number of representations of n as one of these sums to be bounded, for all n? (Conjectural answer: of course not!)

背景链接：https://people.math.sc.edu/cooper/combprob.html

01548 | OWR-1786-004

### Symmetric-Power Functoriality and Twisted Period Relations

O / open | Oberwolfach 报告问题汇编

For a holomorphic cuspidal representation $\pi$ of $\operatorname{GL}_2$ over a number field $K$ and every $m$, does the $m$th symmetric-power $L$-function arise from an automorphic representation $\operatorname{Sym}^m(\pi)$ on $\operatorname{GL}_{m+1}$? After twisting by algebraic Hecke characters, do the resulting $L$-functions satisfy the predicted period relations and $\operatorname{Aut}(\mathbb C)$-equivariance?

来源字段：https://doi.org/10.4171/owr/2008/05

01549 | AMR-093-0090

### Szpiro's conjecture

P / partially_solved | AMR 问题清单汇编

Szpiro's conjecture: for any $\varepsilon > 0$, there is some constant $C(\varepsilon)$ such that, for any elliptic curve $E$ defined over $\mathbb{Q}$ with minimal discriminant $\Delta$ and conductor $f$, we have $|\Delta| \leq C(\varepsilon) \cdot f^{6+\varepsilon}$.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01550 | AMR-030-0049

### Take any positive integer, and apply the following process: (1) divide it by two if it's even, multiply by three and add

O / open | AMR 问题清单汇编

3n+1 ("Collatz" or "Ulam") problem: Take any positive integer, and apply the following process: (1) divide it by two if it's even, multiply by three and add one if it's odd, (2) repeat until you reach one. Must this process terminate? This one is famously difficult, and is a spectacular way to waste days, weeks, or years of your life. See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

01551 | OWR-15429-006

### Tamagawa Formula for Connected Linear Algebraic Groups

P / partially_solved | Oberwolfach 报告问题汇编

For every connected linear algebraic group $G$ over a global field $k$, is the Tamagawa-number identity $$\tau(G)=\frac{\#\operatorname{Ext}^1(G,G_m)}{\#X(G)}$$ valid?

来源字段：https://doi.org/10.4171/owr/2017/21

01552 | OWR-1274-020

### Tame Fontaine-Mazur Conjecture

O / open | Oberwolfach 报告问题汇编

If $S$ contains no primes above $p$, does every continuous homomorphism $G_{K,S}\to GL_n(\mathbb Z_p)$, with $n\geq1$, have finite image? Equivalently, does $G_{K,S}$ have no infinite $p$-adic analytic quotient?

来源字段：https://doi.org/10.4171/owr/2006/25

01553 | OWR-14297740-011

### Tate-Shafarevich Finiteness for Tori over Function Fields

P / partially_solved | Oberwolfach 报告问题汇编

Let $K$ be the function field of a smooth curve $C$ over a number field $k$. For every $K$-torus, or more generally every group of multiplicative type, is the Tate-Shafarevich group $\operatorname{Sha}_C^1(T)$ finite?

来源字段：https://doi.org/10.4171/owr/2023/42

01554 | OWR-7155439-004

### Taylor-Wiles Deformation Maps to Augmented Hecke Algebras

O / open | Oberwolfach 报告问题汇编

Does there exist an $\mathcal O[\Delta_Q]$-algebra homomorphism from $R_{S\cup Q}$ to the $Q$-augmented Hecke algebra $\mathbb T_{S\cup Q,Q\text{-aug}}(K_0(Q)/K_1(Q))$ localized at the maximal ideal $\mathfrak m_{\bar\rho}$ determined by the residual representation $\bar\rho$, carrying the characteristic polynomial of $\rho_{S\cup Q}(\operatorname{Fr}_v)$ to $P_v(X)$ for every $v\notin S\cup Q$?

来源字段：https://doi.org/10.4171/owr/2021/18

01555 | OWR-11127-017

### The Abelian Rank-One Stark Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $K/k$ be an abelian extension of number fields with Galois group $G$, and let $S$ contain at least two places, including a place totally split in $K/k$. Does there exist an $S$-unit $\varepsilon$ in $K$ satisfying the Stark relations for all characters, such that $K(\varepsilon^{1/w_K})/k$ is abelian and $\varepsilon$ is a unit when $|S|\geq3$?

来源字段：https://doi.org/10.4171/owr/2011/35

01556 | OWR-5159-003

### The Arithmetic Fundamental Lemma Beyond Minuscule Cases

P / partially_solved | Oberwolfach 报告问题汇编

In the arithmetic fundamental lemma setting, let $F$ be a $p$-adic field, let $E/F$ be an unramified quadratic extension, and let a regular semisimple $\gamma$ in the unitary symmetric space match $g$ in the corresponding unitary group. Does the AFL identity

$$O'_\gamma(1_S)=-\omega(\gamma)\left\langle\Delta(N_{n-1}),(1,g)\Delta(N_{n-1})\right\rangle$$

hold, where the right-hand side is the arithmetic intersection number of the indicated cycles on the unitary Rapoport-Zink space?

来源字段：https://doi.org/10.4171/owr/2011/31

01557 | OWR-4804-005

### The Converse Unitary Distinction Criterion

P / partially_solved | Oberwolfach 报告问题汇编

Let $\pi$ be a Langlands quotient of characters of $\mathbb C^\times$. If $\pi$ is $U_{p,q}$-distinguished, then it is invariant under complex conjugation and the number of conjugation orbits of size two is at most $\min(p,q)$. Is the converse true for all such $\pi$, not only generic ones?

来源字段：https://doi.org/10.4171/owr/2011/14

01558 | AMR-093-0068

### The density hypothesis for zeroes of the Riemann zeta function

P / partially_solved | AMR 问题清单汇编

The density hypothesis for zeroes of the Riemann zeta function.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01559 | OWR-14603-019

### The Dynamical Mordell-Lang Conjecture for Endomorphisms

P / partially_solved | Oberwolfach 报告问题汇编

For an endomorphism $f:A\to A$ of a complex variety with $\deg(f)>1$, a subvariety $V\subseteq A$, and $P\in A(\mathbb C)$, is $\{n\in\mathbb N:f^n(P)\in V(\mathbb C)\}$ always a finite union of one-sided arithmetic progressions?

来源字段：https://doi.org/10.4171/owr/2016/21

01560 | OPG-367

### The Erdos-Turan conjecture on additive bases

O / open | Open Problem Garden

Let $B \subseteq {\mathbb N}$. The representation function $r_B: {\mathbb N} \rightarrow {\mathbb N}$ for $B$ is given by the rule $r_B(k) = \#\{ (i,j) \in B \times B: i + j = k \}$. We call $B$ an additive basis if $r_B$ is never $0$.

Conjecture If $B$ is an additive basis, then $r_B$ is unbounded.

背景链接：http://www.openproblemgarden.org/op/the_erdos_turan_conjecture_on_additive_bases

01561 | NT-025

### The Gauss Circle Problem

P / partially_solved | 独立 / 未分配集合

别名：Gauss Circle Problem [NT-032]

What is the optimal error term in the formula for the number of lattice points inside a circle of radius $r$?

背景链接：https://mathoverflow.net/questions/355861/what-is-the-most-general-formulation-of-gausss-circle-problem

01562 | OWR-14298365-002

### The Hasse Principle for Unirational Varieties

O / open | Oberwolfach 报告问题汇编

Does every unirational variety over $\mathbb{Q}$ satisfy the Hasse principle?

来源字段：https://doi.org/10.4171/owr/2024/17

01563 | OWR-14299289-005

### The Ichino-Ikeda Period Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $F$ be a global field and let $\pi=\pi_V\boxtimes\pi_{V\oplus F}$ be a multiplicity-one tempered cuspidal automorphic representation of $SO(V)\times SO(V\oplus F)$. Does the square of its global period equal the central Rankin-Selberg value divided by the adjoint value, multiplied by the normalized product of local periods and the factor determined by the component group of the global Arthur parameter?

来源字段：https://doi.org/10.4171/owr/2025/18

01564 | AIM-ALGEBRAIC_NUMBER_THEORY-0102

### The infinity 5-torsion line and Kummer quotient of the pentagonal quintic

P / partially_solved | AIM 研讨会问题汇编

Draw a regular pentagon \(P\), construct the circle through the 5 vertices \(C\), and consider the curve \(E:P+\lambda C^2=0\). This is a quintic curve with 5 double points, and therefore has geometric genus 1. This curve has five points at \(\infty\) (given by the slopes of the lines). Compute the 5-torsion of \(E\). (McCallum)

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01565 | AIM-ANALYTIC_NUMBER_THEORY-0050

### The local inverse-zeta fingerprint of a forced rational point

P / partially_solved | AIM 研讨会问题汇编

Consider a family of elliptic curves with a known rational point. Evaluate moments of $L$-functions in this family at certain points. How does the fixed rational point analytically affect calculations? E.g. $y^2=x^3+ax+b^2$ with $a,b$ integers having $|a|,|b|\leq X$ (or in $\mathbb F_q[t]$).

来源字段：http://aimpl.org/zetamoments/2/

01566 | OWR-8415342-009

### The Local Langlands Correspondence for Reductive Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a connected reductive group $G$ over a non-Archimedean local field $F$, construct the canonical local Langlands map $L_{G/F}:\Pi(G/F)\to\Phi(G/F)$ and prove that every parameter has a finite, nonempty $L$-packet.

来源字段：https://doi.org/10.4171/owr/2021/39

01567 | OWR-14603-028

### The Mixed Littlewood Conjecture for Arbitrary Pseudo-Absolute Values

O / open | Oberwolfach 报告问题汇编

For every sequence $D=(d_i)_{i\in\mathbb N}$ with $d_i\geq2$ and every $x\in\mathbb R$, is $\liminf_{q\to\infty}q|q|_D\|qx\|=0$, where $|a|_D=\min\{D_n^{-1}:a\in D_n\mathbb Z\}$ and $D_n=\prod_{i=1}^n d_i$?

来源字段：https://doi.org/10.4171/owr/2016/21

01568 | AMR-093-0088

### The n-conjecture

P / partially_solved | AMR 问题清单汇编

Fix $n\geq3$. If coprime nonzero integers $a_1,\ldots,a_n$ have sum zero and no proper subsum zero, is it true that for every $\varepsilon>0$ there is $C_{n,\varepsilon}$ such that $\max_i|a_i|<C_{n,\varepsilon}\operatorname{rad}(|a_1\cdots a_n|)^{2n-5+\varepsilon}$?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01569 | NT-045

### The Number of Idoneal Numbers

P / partially_solved | 独立 / 未分配集合

Are there exactly 65 idoneal numbers, or could there be 66 or 67?

背景链接：https://eudml.org/doc/205250

01570 | NT-026

### The Odd Perfect Number Conjecture

P / partially_solved | 独立 / 未分配集合

Do there exist any odd perfect numbers? (A perfect number equals the sum of its proper divisors.)

背景链接：https://mathworld.wolfram.com/OddPerfectNumber.html

01571 | GUY-A4

### The Prime Number Race

P / partially_solved | Richard Guy 数论问题

Let $\pi(n; a, b)$ be the number of primes $p \le n$ with $p \equiv a \pmod b$. For every $a$ and $b$ with $a \perp b$, are there infinitely many values of $n$ for which $\pi(n; a, b) > \pi(n; a_1, b)$ for every $a_1 \not\equiv a \pmod b$?

背景链接：https://doi.org/10.1080/10586458.1994.10504289

01572 | AIM-ALGEBRAIC_NUMBER_THEORY-0112

### The quadratic degree scale and an irreducibility diagnostic

S / solved | AIM 研讨会问题汇编

Given a plane curve of degree \(d\), the number of points on \(X\) of height at most \(d\) is \(\ll d^3\). There are curves which have the number of rational points \(\gg d^2\). Can you do better than \(\ll d^3\)? (Heath-Brown)

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01573 | OWR-8415342-025

### The Refined Formal-Degree Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Refined formal-degree conjecture: if the split center $A_G$ is trivial, then $(-1)^{r(G)}\deg(\pi,\mu_G)=\pm\frac{\dim\rho}{|S_\phi|}\frac{\gamma(0,\operatorname{Ad}_G\circ\phi)}{\gamma(0,\operatorname{Ad}_G\circ\phi_{\operatorname{prin}})}$, with sign given by the quotient of the corresponding root numbers.

来源字段：https://doi.org/10.4171/owr/2021/39

01574 | MPP-002

### The Riemann Hypothesis

O / open | 千禧年问题

别名：The Riemann Hypothesis [OPG-573]; The Riemann Hypothesis [OWR-1703864-001]

Do all non-trivial zeros of the Riemann zeta function $\zeta(s)$ have real part equal to $\frac{1}{2}$?

背景链接：https://www.claymath.org/millennium-problems/

01575 | OWR-8415342-012

### The Sigma-Rho Poisson Summation Conjecture

O / open | Oberwolfach 报告问题汇编

The $(\sigma,\rho)$-Poisson summation conjecture asserts that the nonzero invariant functionals on the Schwartz spaces attached to $\sigma$ and its contragredient satisfy $E_{\sigma,\rho}(\phi)=E_{\widetilde{\sigma},\rho}(F_{\sigma,\rho,\psi}\phi)$ for every test function $\phi$.

来源字段：https://doi.org/10.4171/owr/2021/39

01576 | AIM-ALGEBRAIC_NUMBER_THEORY-0089

### The solved D4 cubic and an exact deleted-line count

S / solved | AIM 研讨会问题汇编

Consider

\[ xyz=t(x+y+z)^2, \]

with height \(H(x,y,z,t)=\max(|x|,|y|,|z|,|t|)\), and restrict to \(\gcd(x,y,z,t)=1\). Excluding trivial solutions (on lines), can one prove a counting function which is

\[ \sim cB(\log B)^6 < B^{1+\varepsilon}? \]

This is a singular cubic surface, so an explicit constant is also expected.

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01577 | AIM-ANALYTIC_NUMBER_THEORY-0009

### The solved maximal-parabolic formula and an all-level Borel support reduction

P / partially_solved | AIM 研讨会问题汇编

**$GL(3)$ Kuznetsov trace formula in the level aspect.**

Develop a Kuznetsov trace formula for automorphic forms for the group \[ \Gamma_0(N)=\left\{\begin{pmatrix} \star&\star&\star\\ \star&\star&\star\\ 0&0&\star \end{pmatrix}\bmod N\right\} \] for any integer \(N\geq 1\).

For the purpose of applications, develop a Kuznetsov trace formula for automorphic forms for the group \[ \Gamma_B(N)=\left\{\begin{pmatrix} \star&\star&\star\\ 0&\star&\star\\ 0&0&\star \end{pmatrix}\bmod N\right\}. \]

来源字段：http://aimpl.org/deltasubconvex2/1/

01578 | OWR-14299289-020

### The Tempered Numerical Period Conjecture

O / open | Oberwolfach 报告问题汇编

For an everywhere-unramified tempered automorphic representation $\pi$ and a hyperspherical dual pair, does the normalized automorphic period of $\pi$ equal the normalized $L$-value prescribed by the extended Langlands parameter?

来源字段：https://doi.org/10.4171/owr/2025/18

01579 | OWR-8415342-008

### The Twisted Fourier-Jacobi Gan-Gross-Prasad Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For étale quadratic algebras $E$ and $K$ over a local or global field $F$, a skew-Hermitian $E$-space $V$, an irreducible representation $\Pi$ of $U(V_K)$, and the Weil representation $\omega_{V,\psi,\mu}$ of $U(V)$, determine the local branching multiplicity $\operatorname{Hom}_{U(V)}(\Pi,\omega_{V,\psi,\mu})$ and the corresponding global period in the twisted Fourier-Jacobi Gan-Gross-Prasad problem.

来源字段：https://doi.org/10.4171/owr/2021/39

01580 | AMR-030-0065

### There is no n so that the only integer m with phi(n) = phi(m) is m=n

P / partially_solved | AMR 问题清单汇编

Carmichael : There is no n so that the only integer m with phi(n) = phi(m) is m=n. ("phi" is the Euler phi/totient function). See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

01581 | OWR-16764-002

### Theta Operators for Unitary Shimura Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Develop, for unitary Shimura varieties, analogues of the characteristic-$p$ modular-form structures involving the Hasse invariant, the theta derivation, and theta cycles.

来源字段：https://doi.org/10.4171/owr/2019/2

01582 | AIM-ALGEBRAIC_NUMBER_THEORY-0088

### Thin-deleted asymptotic and an explicit split-fibre surplus

P / partially_solved | AIM 研讨会问题汇编

Let \[ X=\left\{([\mathbf x],[\mathbf y])\in \mathbf P^3\times\mathbf P^3: x_1y_1^2+x_2y_2^2+x_3y_3^2+x_4y_4^2=0\right\}. \] For primitive integral representatives, take \[ H([\mathbf x],[\mathbf y])=\|\mathbf x\|_\infty^3\|\mathbf y\|_\infty^2, \] delete the locus on which some \(x_i\) or some \(y_i\) is zero, and estimate the counting function.

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01583 | OWR-15173-001

### Third Homology and Pre-Bloch Groups of $S$-Arithmetic $\operatorname{SL}_2$

O / open | Oberwolfach 报告问题汇编

For a global field $F$ and every sufficiently large finite set $S$ of places, is the natural map

$$\pi_S:\ker\!\left(H_3(\operatorname{SL}_2(\mathcal O_S),\mathbb Z[1/2])\to H_3(\operatorname{SL}_2(F),\mathbb Z[1/2])\right)\longrightarrow\bigoplus_{v\in S\setminus S_\infty}\mathcal P(k(v))\otimes\mathbb Z[1/2]$$

an isomorphism?

来源字段：https://doi.org/10.4171/owr/2016/52

01584 | OWR-15174-010

### Three Problems from an Analytic Number Theory Session

O / open | Oberwolfach 报告问题汇编

Does there exist a Dirichlet series $D(s)$ that converges in a half-plane $\operatorname{Re}s>\alpha$ and has exactly one zero there? For the ordered Laplace eigenvalues $0<\lambda_1\leq\lambda_2\leq\cdots$ of Maass forms on $\operatorname{SL}_2(\mathbb Z)\backslash\mathbb H$, are

$$\liminf_{j\to\infty}(\lambda_{j+1}-\lambda_j)=0, \qquad \limsup_{j\to\infty}(\lambda_{j+1}-\lambda_j)=\infty?$$

If $x^2\equiv a\pmod m$ has many roots for highly composite $m$, can one uniformly bound how many of those roots lie in an interval of length $m^{1-\varepsilon}$?

来源字段：https://doi.org/10.4171/owr/2016/53

01585 | OWR-14299577-019

### Three-Full Sums and the Erdős-Ingham Dirichlet Zero Problem

O / open | Oberwolfach 报告问题汇编

Let $A$ be the set of integers representable as a sum of three $3$-full numbers. Does $A$ have positive density, and does it contain every sufficiently large integer? Separately, for $f(s)=1+2^{-s}+3^{-s}+5^{-s}$, does there exist $t\in\mathbb{R}$ with $f(1+it)=0$?

来源字段：https://doi.org/10.4171/owr/2025/51

01586 | AIM-ANALYTIC_NUMBER_THEORY-0094

### Torsion continuous q-Hermite expansions as Siegel modular units

P / partially_solved | AIM 研讨会问题汇编

Is there any kind of relations between the modularity of a series and the expansion in terms of q-orthogonal polynomials?

来源字段：https://aimath.org/WWN/mockmodular/mockmodular.pdf

01587 | NT-075

### Transcendence of Apéry's Constant

O / open | 独立 / 未分配集合

Is $\zeta(3) = 1 + 1/8 + 1/27 + 1/64 + \cdots$ transcendental?

背景链接：https://doi.org/10.1070/RM2001v056n04ABEH000427

01588 | AIM-ANALYTIC_NUMBER_THEORY-0117

### Tuple extension requires shifted correlation

P / partially_solved | AIM 研讨会问题汇编

**18. From \(k\)-tuples to \((k+1)\)-tuples.** J. B. Conrey asked whether it is possible to deduce the existence of prime \((k+1)\)-tuples from a version of the Elliott--Halberstam conjecture for prime \(k\)-tuples.

J. Pintz proposed to try to use EH for twin primes to obtain two pairs of twin primes that are close to each other.

The heading and first question are:

The two displayed estimates in the typeset PDF are, with their absolute values restored from the page layout,

\[ \sum_{q\leq Q}\max_{(a,q)=1} \sum_{\substack{1\leq b\leq q\\(b,q)=1}} \left| \sum_{n\leq x/q}\varpi(qn+a)\varpi(qn+b) -MT(x,q;a,b) \right| \ll x^2(\log x)^{-A}, \tag{W1} \]

and Vaughan's suggested alternative

\[ \sum_{q\leq Q}\max_{(a,q)=1} \left| \sum_{\substack{1\leq b\leq B\\(b,q)=1}} \left( \sum_{n\leq x/q}\varpi(qn+a)\varpi(qn+b) -MT(x,q;a,b) \right) \right| \ll Bx(\log x)^{-A}. \tag{W2} \]

Thus (W1)--(W2) are unweighted prime counts. The distinction matters because the cited papers use von Mangoldt weights.

The position of the absolute values is also essential. In (W1) one sums \(|E_{q,a,b}|\) over \(b\); in (W2) the sum over \(b\) is taken first and only then is its absolute value taken. Parentheses around the inner error in (W2) confirm this reading.

The PDF does **not** state:

- whether \(A>0\) is arbitrary; - how \(Q\) may depend on \(x\), and in particular no exponent of distribution; - whether \(B\) is fixed, grows, or is related to \(q,Q,x\); - the permitted dependence of the implied constant; or - a definition of \(MT(x,q;a,b)\).

Consequently (W1) and (W2) are templates, not complete quantified conjectures. The displayed range is exactly \(n\leq x/q\); floors and the lower endpoint are not specified.

For \(a\ne b\), the natural Hardy--Littlewood reconstruction is

\[ MT(x,q;a,b)\ \approx\ \mathfrak S(q;a,b) \int_{1}^{x/q} \frac{dt}{\log(qt+a)\log(qt+b)}, \tag{1.1} \]

\[ \mathfrak S(q;a,b)= \prod_p \frac{1-\nu_p(q;a,b)/p}{(1-1/p)^2},\qquad \nu_p(q;a,b)= \#\{r\pmod p:(qr+a)(qr+b)\equiv0\pmod p\}. \tag{1.2} \]

This is a reconstruction, not a definition found in the workshop notes. It vanishes when there is a local obstruction. The displayed \(b\)-range includes \(b=a\), where the two forms coincide; that diagonal has a one-prime main term of order \(x/(q\log x)\), not the nondegenerate two-form main term (1.1). A complete version of (W1)--(W2) must therefore separate or explicitly normalize the diagonal.

The final bullet, also verified in the PDF, is:

来源字段：https://aimath.org/WWN/primegaps/WorkshopProblems.pdf

01589 | OWR-760-003

### Turán-Type Reformulations for Shifted Primes

O / open | Oberwolfach 报告问题汇编

What is an analogue of Turán's reformulation for the monotonicity problem for additive arithmetic functions on shifted primes?

来源字段：https://doi.org/10.4171/owr/2004/46

01590 | LAN-002

### Twin Prime Conjecture

O / open | Landau 问题

别名：Twin Prime Conjecture [GUY-A8b]; Twin prime conjecture [OPG-36952]; Twin Prime Conjecture [OWR-1795-011]

Are there infinitely many twin primes? Twin primes are pairs of primes that differ by 2, such as (3, 5), (5, 7), (11, 13), (17, 19), (29, 31).

背景链接：https://arxiv.org/abs/1407.4897

01591 | AIM-ALGEBRAIC_NUMBER_THEORY-0078

### Two explicit new divisor classes on the level-4 surface modulo 3

P / partially_solved | AIM 研讨会问题汇编

(J. Ellenberg, D. Ulmer) Let $X$ be a modular curve, which parametrizes elliptic curves with some level structure, and let $E$ be the universal family of elliptic curves over $X$. This defines an elliptic surface, typically over $\mathbb Q$. It often happens that, for many primes $p$, the reduction modulo $p$ of this surface acquires new Tate classes, not coming from characteristic zero. Can we find algebraic cycles (living in characteristic $p$), which account for such Tate classes?

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

01592 | AIM-ALGEBRAIC_NUMBER_THEORY-0065

### Two Tate lines and the missing adjoint correspondence

P / partially_solved | AIM 研讨会问题汇编

Let $F$ be a real quadratic field, let $\pi$ be an automorphic form over $\mathbf Q$, and let $\Pi$ be its base change to $F$. Then $\Pi$ contributes a subspace $H^2(\Pi)$ to $H^2(S_F)$ for a Hilbert modular surface $S_F$ for $\mathrm{GL}_2(F)$, while $H^1(\pi)$ occurs in $H^1(M)$ for a modular curve $M$. As $\operatorname{Gal}(\overline{\mathbf Q}/F)$-modules,

$$ H^2(\Pi)\simeq H^1(\pi)\otimes H^1(\pi). $$

Take two real quadratic fields $F_1,F_2$, the base changes $\Pi_1,\Pi_2$ of $\pi$, and associated Hilbert modular surfaces $S_{F_1},S_{F_2}$. The summand

$$ H^2(\Pi_1)\otimes H^2(\Pi_2)\subset H^4(S_{F_1}\times S_{F_2}) $$

becomes $H^1(\pi)^{\otimes4}$ over $\operatorname{Gal}(\overline{\mathbf Q}/F_1F_2)$. One therefore expects codimension-two cycles on $S_{F_1}\times S_{F_2}$, defined over $F_1F_2$. What are they?

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

01593 | OWR-14605-007

### Two-Reflective Automorphic Forms for Hyperbolic Lattices

P / partially_solved | Oberwolfach 报告问题汇编

别名：Expository Automorphic-Correction Description of Lorentzian Kac-Moody Algebras [OWR-14605-006]

For each of the $59$ even hyperbolic lattices $S$ whose $2$-reflection Weyl group has finite index and whose fundamental chamber admits a lattice Weyl vector, does there exist an integer $m$ and a positive-weight $2$-reflective holomorphic automorphic form $\Phi$ on $\Omega(U(m)\oplus S)$, with integral Fourier coefficients and divisor equal with multiplicity one to the rational quadratic divisors orthogonal to square-$2$ vectors?

来源字段：https://doi.org/10.4171/owr/2016/23

01594 | AIM-ALGEBRAIC_NUMBER_THEORY-0108

### Two-square coordinates and curve-source images over a rank-one elliptic curve

P / partially_solved | AIM 研讨会问题汇编

**Question 23 (Poonen).** Let $E\subset \mathbf P^2$ be an elliptic curve over $\mathbf Q$, and suppose $E(\mathbf Q)\simeq \mathbf Z$.

(a) Describe $S\subset E(\mathbf Q)$, where $$ S=\{(x,y):y=a^2+b^2,\ a,b\in\mathbf Q\}. $$ Is $S$ finite?

(b) More generally, $\pi:X\to E$ gives a subset $\pi(X(\mathbf Q))\subset E(\mathbf Q)$; what others can you build?

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01595 | AMR-042-0004

### Typical group automorphisms

P / partially_solved | AMR 问题清单汇编

Choose a random subset $Q$ of the primes by independently retaining each prime with probability $1/2$. Is it almost surely true that $$\limsup_{n\to\infty}\frac1n\log(2^n-1)\prod_{p\in Q}|2^n-1|_p=\log 2?$$

背景链接：https://www.imath.kiev.ua/~skolyada/kevin.pdf

01596 | OWR-17293-006

### Typical Size of the Erdős-Hooley Function

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Erdős-Hooley Typical-Size Extraction [OWR-17293-007]; Duplicate Erdős-Hooley Optimal-Exponent Extraction [OWR-17293-008]

For a typical integer $n$, determine the asymptotic size of the Erdős-Hooley divisor-concentration function

$$\Delta(n)=\max_i\#\{d\mid n:e^i\leq d\leq e^{i+1}\}.$$

In particular, is the exponent $\eta\approx0.353$ in the almost-all lower bound $\Delta(n)\gg(\log\log n)^{\eta+o(1)}$ optimal?

来源字段：https://doi.org/10.4171/owr/2019/50

01597 | GREEN-007

### Ulam's Sequence

O / open | Ben Green 问题

Define Ulam's sequence $1, 2, 3, 4, 6, 8, 11, 13, 16, 18, 26, 28, 36, \ldots$ where $u_1 = 1, u_2 = 2$, and $u_{n+1}$ is the smallest number uniquely expressible as $u_i + u_j$ for $i < j \leq n$. Does this sequence have positive density? Can one explain its curious Fourier properties?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

01598 | OWR-13853-004

### Unbounded Degrees of Full Belyi Maps

O / open | Oberwolfach 报告问题汇编

For a finite set of primes $P$, let $N(P)$ be the set of degrees $n\geq4$ for which a full Belyi map over $\mathbb{Q}$ has bad reduction only at primes in $P$. If $P$ contains all prime divisors of the order of some nonabelian finite simple group, is $N(P)$ infinite?

来源字段：https://doi.org/10.4171/owr/2015/32

01599 | OWR-12583-010

### Unbounded Symmetric and Alternating Extensions with Restricted Ramification

P / partially_solved | Oberwolfach 报告问题汇编

Let $P$ be a finite set of primes containing every prime dividing the order of some finite nonabelian simple group, and let $F_P(m)$ count degree-$m$ number fields with Galois group $A_m$ or $S_m$ that ramify only at primes in $P$. Is the sequence $F_P(m)$ unbounded?

来源字段：https://doi.org/10.4171/owr/2013/35

01600 | OWR-12583-011

### Unbounded Torsion and Selmer Groups in Quadratic Extensions

P / partially_solved | Oberwolfach 报告问题汇编

Let $\ell$ be a prime, $K$ a number field, and $A/K$ an abelian variety. As quadratic extensions $L/K$ vary, are $\#\operatorname{Pic}(\mathcal O_L)[\ell]$ and $\#\operatorname{Sel}_\ell(A_L)$ unbounded?

来源字段：https://doi.org/10.4171/owr/2013/35

01601 | OWR-8415337-004

### Unified Selmer and Cohen-Lenstra-Martinet Heuristics

O / open | Oberwolfach 报告问题汇编

Can one formulate unified heuristics for $n$-Selmer groups of possibly nonproper algebraic groups that also imply the Cohen-Lenstra-Martinet heuristics for class groups of number fields?

来源字段：https://doi.org/10.4171/owr/2021/34

01602 | OWR-14299574-001

### Uniform Bounds for Transcendental Rational Points

O / open | Oberwolfach 报告问题汇编

Let $X\subseteq\mathbb{C}_p^n$ be definable in a Hensel-minimal structure. Can $\#X^{\mathrm{trans}}(\mathbb{Q},H)$ be bounded uniformly for all $H\geq1$, with bounds analogous to those known in dimension one?

来源字段：https://doi.org/10.4171/owr/2025/45

01603 | OWR-14603-020

### Uniform Composition Powers of Rational Functions

O / open | Oberwolfach 报告问题汇编

For rational functions $u,f\in K(X)$, if $u\circ v=f^n$ for some $n>0$ and $v\in K(X)$, must there be $w\in K(X)$ with $u\circ w=f^N$ for a positive $N$ depending only on $\deg(u)$? More generally, when does the reduction to irreducible genus-at-most-one fibered products produce multiple towers?

来源字段：https://doi.org/10.4171/owr/2016/21

01604 | AIM-ANALYTIC_NUMBER_THEORY-0065

### Uniform CUE moment asymptotics and the k ~ N^(1/3) crossover

P / partially_solved | AIM 研讨会问题汇编

**Can we make the calculations uniform in \(k\)?**

来源字段：http://aimpl.org/zetamoments/6/

01605 | AIM-ALGEBRAIC_NUMBER_THEORY-0076

### Uniform edge-pole bounds under base extension

P / partially_solved | AIM 研讨会问题汇编

**13. (K. Murty)** According to the Tate Conjecture, if one takes the L-function of a variety \(X\) over \(K\) at the edge of the critical strip, and then increases \(K\), the order of the pole should stay bounded. Can we prove this same statement without assuming the Tate Conjecture?

**Recovered mathematical question.** Let \(X/K\) be a smooth projective geometrically connected variety over a global field, fix \(m\geq 0\), and put

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

01606 | AIM-ALGEBRAIC_NUMBER_THEORY-0082

### Uniform Frobenius exponents and all-good-fiber algebraicity cases

P / partially_solved | AIM 研讨会问题汇编

(D. Ramakrishnan) The following question is due to P. Deligne. Let \(X\) be a variety defined over a (finitely-generated) field of characteristic 0. Given a Tate class on \(c\) on \(X\), is there a prime \(p\) such that, upon reducing \(X\) mod \(p\), the reduction of \(c\) is algebraic?

**Recovered Problem 19.** Let \(X\) be a variety over a finitely generated field of characteristic zero. Given a Tate class \(c\) on \(X\), is there a prime \(p\) such that, after reducing \(X\) modulo \(p\), the specialized class \(c_p\) is algebraic?

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

01607 | OWR-12725-015

### Uniform Lattice-Point Bounds on Short Circle Arcs

P / partially_solved | Oberwolfach 报告问题汇编

Let $A$ be an arc of length $R^\theta$ on a circle of radius $R$. If $\theta<1$, is the number of lattice points on $A$ bounded by a constant $C(\theta)$ independent of $R$ and $A$?

来源字段：https://doi.org/10.4171/owr/2013/51

01608 | OWR-14603-021

### Uniform Lower Bounds for Specialized Canonical Heights

P / partially_solved | Oberwolfach 报告问题汇编

Let $K$ be a number field and $f_t$ a non-isotrivial family of rational functions of generic degree $d\geq2$ over a curve $U$. Do there exist $\varepsilon>0$ and $N\geq0$ such that for every $t\in U(K)$, $\hat h_{f_t}(z)\geq\varepsilon h(t)$ for all but at most $N$ points $z\in\mathbb P^1(K)$?

来源字段：https://doi.org/10.4171/owr/2016/21

01609 | AIM-ANALYTIC_NUMBER_THEORY-0044

### Uniform Mobius disjointness for horocycles: an AOP obstruction and a moving-basepoint reduction

P / partially_solved | AIM 研讨会问题汇编

Does the Sarnak conjecture hold uniformly in \(x\in X\) in the case of horocycle flows? It is known that the uniform Sarnak conjecture follows from the Sarnak conjecture, but we don't know that it follows case by case.

来源字段：http://aimpl.org/sarnakconjecture/8/

01610 | OWR-13952-010

### Uniform Mordell Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For every integer $g\geq2$, does there exist $N(g)$ such that every smooth, proper, geometrically connected curve $C/\mathbb Q$ of genus $g$ satisfies $\lvert C(\mathbb Q)\rvert\leq N(g)$?

来源字段：https://doi.org/10.4171/owr/2015/57

01611 | OWR-1590-005

### Uniform Rank Bounds for Elliptic Function Fields

O / open | Oberwolfach 报告问题汇编

For a nonconstant elliptic curve $E$ over $F(T)$, with $F$ an arbitrary field of characteristic zero and in particular $F\subseteq\mathbb C$, is the rank of $E(F(T))$ bounded uniformly as $E$ and/or $F$ vary? More generally, what can be said about the rank of elliptic curves over $F(T)$ for general fields $F$?

来源字段：https://doi.org/10.4171/owr/2007/34

01612 | OWR-11101913-014

### Uniform Torsion Bounds for Chow Groups in Families

O / open | Oberwolfach 报告问题汇编

For a family $\pi:X\to S$ and a degree bound $d$, does there exist a positive integer $C$, depending only on $\pi$ and $d$, such that $C\,\operatorname{CH}^i(X_s)_{\mathrm{tors}}=0$ for every number field $k/\mathbb{Q}$ of degree at most $d$, every $i>0$, and every $k$-point $s\in S$?

来源字段：https://doi.org/10.4171/owr/2022/32

01613 | OWR-760-004

### Uniform Zero-Distribution Limits for Characteristic Test Functions

O / open | Oberwolfach 报告问题汇编

The zero-distribution limit formula for zeta or Dirichlet $L$-function zeros should continue to hold when the test function $h$ is the characteristic function of $[0,y]$, uniformly in $y$.

来源字段：https://doi.org/10.4171/owr/2004/46

01614 | OWR-1274-024

### Uniformly Bounded Ramification in $p$-Frattini-Tower Realizations

O / open | Oberwolfach 报告问题汇编

For a $p$-perfect finite group $G$ and its $p$-Frattini quotients $G_k$, does there exist $r_0$ such that every $G_k$ satisfies the regular inverse Galois problem using covers with at most $r_0$ ramification classes?

来源字段：https://doi.org/10.4171/owr/2006/25

01615 | OWR-12865-002

### Unifying Constructions of Exceptional Monodromy Local Systems

O / open | Oberwolfach 报告问题汇编

Can the automorphic and constructive approaches to producing local systems with prescribed exceptional algebraic monodromy be unified?

来源字段：https://doi.org/10.4171/owr/2014/05

01616 | AIM-ALGEBRAIC_NUMBER_THEORY-0044

### Universal capacity bounds and an inseparable coordinate obstruction

P / partially_solved | AIM 研讨会问题汇编

**Question 14 (Pollack).** Over a function field, how many coefficients of a polynomial (and in which positions) can be specified without eliminating the possibility of irreducibility?

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

01617 | OWR-1274-004

### Unramified Pro-$p$ Galois Groups over Infinite Extensions

P / partially_solved | Oberwolfach 报告问题汇编

Let $F$ be a number field and $L/F$ a pro-$p$ Galois extension unramified outside a finite set $S$ containing the primes above $p$. Determine the Galois group $G_L$ of the maximal unramified pro-$p$ extension of $L$ and its maximal abelian quotient $X_L$.

来源字段：https://doi.org/10.4171/owr/2006/25

01618 | OWR-12493-007

### Valuative Form of Grothendieck's Section Conjecture

O / open | Oberwolfach 报告问题汇编

别名：Duplicate: Valuative Form of Grothendieck's Section Conjecture [OWR-12493-006]

Let $k$ be a number field and $X/k$ a hyperbolic curve. Is every Galois section $s:G_k\to\pi_1(X)$ induced, up to conjugacy, by a suitable $k$-valuation of the function field $k(X)$?

来源字段：https://doi.org/10.4171/owr/2013/30

01619 | AIM-ALGEBRAIC_NUMBER_THEORY-0053

### Variance coefficients as hook multiplicities

P / partially_solved | AIM 研讨会问题汇编

**Question 23 (Jordan Ellenberg).** Describe the right hand side of variance of \(\Lambda(n)\sim q^{?}\cdot n\) (as \(q\to\infty\)) geometrically.

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

01620 | OWR-15583-006

### Variance of Gaussian Prime Angles in Short Arcs

O / open | Oberwolfach 报告问题汇编

Let $N$ be the number of Gaussian-prime angles with norm at most $X$, and let $N_{K,X}(\theta)$ count those in an arc of angular scale $1/K$. Is $$\frac{\operatorname{Var}(N_{K,X})}{N/K}\sim\min\!\left\{1,\frac{2\log K}{\log N}\right\}$$ as $X\to\infty$?

来源字段：https://doi.org/10.4171/owr/2017/40

01621 | AIM-ALGEBRAIC_NUMBER_THEORY-0054

### Variance of polynomial von Mangoldt pair correlations

P / partially_solved | AIM 研讨会问题汇编

Question 24 (Brian Conrey). What is the variance of $\sum_f \Lambda(f)\Lambda(f+h)$ (considered as a function of $h$).

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

01622 | OWR-11695860-002

### Vinogradov-Type Bounds for Large-Order Dirichlet Characters

O / open | Oberwolfach 报告问题汇编

For primitive characters $\chi$ modulo primes $q$ whose orders tend to infinity, find general conditions ensuring that the least $n$ with $\chi(n)\notin\{0,1\}$ is $O_\epsilon(q^\epsilon)$ and that $\chi$ takes varied values below $q^\epsilon$.

来源字段：https://doi.org/10.4171/owr/2022/50

01623 | OWR-14298805-007

### Vogan Conjecture for Quasisplit Classical Groups

P / partially_solved | Oberwolfach 报告问题汇编

For $G=\operatorname{GL}_n$, $\operatorname{SO}_{2n+1}$, or $\operatorname{Sp}_{2n}$, prove that local Arthur packets equal the corresponding ABV packets and that Arthur's stable distributions and their endoscopic transfers are computed by the normalized vanishing-cycle functors predicted by Vogan's conjecture.

来源字段：https://doi.org/10.4171/owr/2024/54

01624 | AMR-093-0087

### Vojta's conjecture

P / partially_solved | AMR 问题清单汇编

Vojta's conjecture: points on non-singular algebraic variety over algebraic number field that not satisfy certain height inequality are contained in some Zariski-closed set.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01625 | OWR-10252929-011

### Vojta's Diophantine Approximation Conjecture

O / open | Oberwolfach 报告问题汇编

Let $X$ be a smooth projective variety over a number field $k$, let $D$ be a reduced normal-crossings divisor, let $K_X$ be a canonical divisor, and let $A$ be ample. For every $\varepsilon>0$, does there exist a proper Zariski-closed set $Z_\varepsilon\subsetneq X$ such that $\sum_{v\in S}\lambda_v(D,P)+h_{K_X}(P)<\varepsilon h_A(P)+O(1)$ for every $P\in X(k)\setminus Z_\varepsilon$?

来源字段：https://doi.org/10.4171/owr/2022/21

01626 | OPG-822

### Wall-Sun-Sun primes and Fibonacci divisibility

O / open | Open Problem Garden

Conjecture For any prime $p$, there exists a Fibonacci number divisible by $p$ exactly once.

Equivalently:

Conjecture For any prime $p>5$, $p^2$ does not divide $F_{p-\left(\frac p5\right)}$ where $\left(\frac mn\right)$ is the Legendre symbol.

背景链接：http://www.openproblemgarden.org/op/fibonacci_divisibility

01627 | GREEN-040

### Waring's Problem Over Finite Fields

P / partially_solved | Ben Green 问题

Determine bounds for Waring's problem over finite fields.

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

01628 | NT-062

### Waring's Problem: Exact Values

P / partially_solved | 独立 / 未分配集合

What are the exact values of $g(k)$ and $G(k)$ for all $k$ in Waring's problem?

背景链接：https://dlmf.nist.gov/27.13

01629 | AIM-ALGEBRAIC_NUMBER_THEORY-0120

### Weak approximation for finite quotients with a controlled auxiliary branch point

P / partially_solved | AIM 研讨会问题汇编

**Question 35 (recovered).** Study weak approximation over \(L\). It was known for a connected linear algebraic group. For a homogeneous space \(G/H\), it was suggested that the same method should work when \(H\) is connected. In particular, decide whether weak approximation holds for \(\mathrm{GL}_n/G\) when \(G\) is a finite subgroup of \(\mathrm{GL}_n\). A further claim attributed to de Jong was that every smooth projective geometrically rationally connected \(L\)-variety satisfies weak approximation, although this was not in the de Jong--Starr paper. Finally, an Enriques surface over \(\mathbb C(t)\) with a rational point but without weak approximation was mentioned.

来源字段：https://aimath.org/WWN/qptsurface2/qptsurface2.pdf

01630 | OWR-1274-016

### Weak Greenberg Conjecture for CM Fields

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate: Weak Greenberg Conjecture for CM Fields [OWR-1274-017]

For an odd prime $p$ and every CM field $K$ containing $\mu_p$, is $\lambda^-\to\lambda^+$ unless $\lambda^-=\lambda^+=0$?

来源字段：https://doi.org/10.4171/owr/2006/25

01631 | OWR-3388-010

### Weak Section Conjecture over Number Fields

O / open | Oberwolfach 报告问题汇编

For a smooth projective curve $X$ of genus at least $2$ over a number field $k$, is $X(k)\neq\varnothing$ equivalent to the existence of a continuous section of $$\pi_1(X/k)\longrightarrow\operatorname{Gal}(\overline k/k)$$?

来源字段：https://doi.org/10.4171/owr/2009/05

01632 | OWR-4990375-003

### Weight-Dimension Coincidences for Modular Forms and Representations

O / open | Oberwolfach 报告问题汇编

For $3\leq g\leq10$, is the equality $\operatorname{weight}(F^{(g)})-19=\dim(V_g)$ between the modular-form weight and the associated representation dimension merely accidental?

来源字段：https://doi.org/10.4171/owr/2021/6

01633 | AIM-ANALYTIC_NUMBER_THEORY-0092

### Weight-only modularity recognition has a finite-data obstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.1.** Given a q-series, determine fast methods to find (heuristically) if it is modular. The method should require knowledge of the weight, but not require knowledge of the level or group.

来源字段：https://aimath.org/WWN/mockmodular/mockmodular.pdf

01634 | OWR-15169-009

### Weighted Bad Approximation on Affine Lines

O / open | Oberwolfach 报告问题汇编

For $L_{a,b}=\{(x,ax+b):x\in\mathbb R\}$, weights $i,j\geq0$ with $i+j=1$, and $\sigma=\min\{i,j\}$, can the known hypothesis $\liminf_{q\to\infty}q^{1/\sigma-\varepsilon}\max\{\|qa\|,\|qb\|\}>0$ for some $\varepsilon>0$ be relaxed to $\varepsilon=0$ while retaining $\dim_H(L_{a,b}\cap\operatorname{Bad}(i,j))=1$?

来源字段：https://doi.org/10.4171/owr/2016/48

01635 | OWR-14298805-014

### Weighted Fundamental Lemma for Classical Groups

P / partially_solved | Oberwolfach 报告问题汇编

Prove the weighted fundamental lemma for nonsplit classical groups, thereby making the endoscopic classification of representations of classical groups unconditional.

来源字段：https://doi.org/10.4171/owr/2024/54

01636 | OWR-14613-007

### Weil-Arakelov and Weil-Étale Cohomology for Arithmetic Schemes

O / open | Oberwolfach 报告问题汇编

For every arithmetic scheme $X$, meaning a separated scheme of finite type over $\mathbb Z$, do there exist compactly supported complexes $R\Gamma_{W,c}(X,\mathbb Z(n))$ and $R\Gamma_{\mathrm{ar},c}(X,A(n))$ with the following properties: $R\Gamma_{W,c}(X,\mathbb Z(n))$ is a perfect complex of abelian groups; $R\Gamma_{\mathrm{ar},c}(X,\mathbb R(n))$ is a perfect complex of real vector spaces; and cup product with the fundamental class $\theta\in H^1(\operatorname{Spec}\mathbb Z,\mathbb R(0))$ makes the cohomology of the latter complex acyclic?

来源字段：https://doi.org/10.4171/owr/2016/31

01637 | OWR-1318-007

### Weil-Étale Determinant Formulas for L-Values

O / open | Oberwolfach 报告问题汇编

Are there finite-dimensional Weil-étale cohomology spaces and cup-product isomorphisms $\beta$ and $\gamma$ such that $L_M(0)=\det(\beta)/\det(\gamma)$ up to a rational factor?

来源字段：https://doi.org/10.4171/owr/2006/32

01638 | AIM-ALGEBRAIC_NUMBER_THEORY-0068

### What Belyi reparametrization can and cannot detect

P / partially_solved | AIM 研讨会问题汇编

3. (Y. Zarhin) 4) People like Wolfart have studied the problem of determining which Jacobians of curves with Belyi parametrizations are CM. Does it help here to reparametrize this equation in the Belyi form, i.e., to express this equation as a cover of P1 ramified only at 0, 1, infinity? (D. Ramakrishnan)

来源字段：https://aimath.org/WWN/tateconjecture/tateconjecture.pdf

01639 | AMR-093-0170

### What is the smallest Skewes's number

O / open | AMR 问题清单汇编

What is the smallest Skewes's number?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01640 | AMR-030-0070

### What is Σ_(n≥1 )φ(n)/2^(n), where φ(n) is the Euler phi (totient) function, counting the number of integers less than n

O / open | AMR 问题清单汇编

Erdős: What is Σ_(n≥1 )φ(n)/2^(n), where φ(n) is the Euler phi (totient) function, counting the number of integers less than n which are relatively prime to n? (Note: a closed form answer is desired.) Is it irrational?

背景链接：https://people.math.sc.edu/cooper/combprob.html

01641 | AMR-093-0111

### Which integers can be written as the sum of three perfect cubes

P / partially_solved | AMR 问题清单汇编

Which integers can be written as the sum of three perfect cubes?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01642 | AMR-093-0099

### Which transcendental numbers are (exponential) periods

O / open | AMR 问题清单汇编

Which transcendental numbers are (exponential) periods?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01643 | OWR-14299289-017

### Whittaker Special Cases of Global Geometric Duality

O / open | Oberwolfach 报告问题汇编

Establish the Whittaker case of global geometric duality by identifying the Whittaker period sheaf with the corresponding spectral object under geometric Langlands.

来源字段：https://doi.org/10.4171/owr/2025/18

01644 | AMR-093-0100

### Wikipedia number-theory item 100: How well can non-quadratic irrational numbers be approximated? What is the irrationality measure of…

O / open | AMR 问题清单汇编

How well can non-quadratic irrational numbers be approximated? What is the irrationality measure of specific (suspected) transcendental numbers such as $\pi$ and $\gamma$?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01645 | AMR-093-0104

### Wikipedia number-theory item 104: Congruent number problem (a corollary to Birch and Swinnerton-Dyer conjecture, per Tunnell's theorem…

P / partially_solved | AMR 问题清单汇编

Congruent number problem (a corollary to Birch and Swinnerton-Dyer conjecture, per Tunnell's theorem): determine precisely what rational numbers are congruent numbers.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01646 | AMR-093-0109

### Wikipedia number-theory item 109: The uniqueness conjecture for Markov numbers that every Markov number is the largest number in exact…

O / open | AMR 问题清单汇编

The uniqueness conjecture for Markov numbers that every Markov number is the largest number in exactly one normalized solution to the Markov Diophantine equation.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01647 | AMR-093-0171

### Wikipedia number-theory item 171: For any given integer a > 0, are there infinitely many Lucas-Wieferich primes associated with the pa…

O / open | AMR 问题清单汇编

For any given integer a > 0, are there infinitely many Lucas-Wieferich primes associated with the pair (a, -1)? (Specially, when a = 1, this is the Fibonacci-Wieferich primes, and when a = 2, this is the Pell-Wieferich primes)

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01648 | AMR-093-0173

### Wikipedia number-theory item 173: For any given integer b which is not a perfect power and not of the form -4k^{4} for integer k, are…

O / open | AMR 问题清单汇编

For any given integer b which is not a perfect power and not of the form -4k^{4} for integer k, are there infinitely many repunit primes to base b?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01649 | AMR-093-0174

### Wikipedia number-theory item 174: For any given integers $k\geq 1, b\geq 2, c\neq 0$, with gcd(k, c) = 1 and gcd(b, c) = 1, are there…

O / open | AMR 问题清单汇编

For any given integers $k\geq 1, b\geq 2, c\neq 0$, with gcd(k, c) = 1 and gcd(b, c) = 1, are there infinitely many primes of the form $(k\times b^n+c)/\gcd(k+c,b-1)$ with integer n ≥ 1?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

01650 | AIM-ALGEBRAIC_NUMBER_THEORY-0058

### Wild ramification strata and an exact characteristic-two point-count law

P / partially_solved | AIM 研讨会问题汇编

How does wild ramification alter the geometry, weighting, and independent-fibre mechanisms used to study point-count distributions of covers over a fixed finite field, especially hyperelliptic double covers in characteristic $2$?

来源字段：https://aimath.org/pastworkshops/arithstatffieldproblems.pdf

01651 | OWR-8415353-003

### Zaremba Conjecture for Bounded Continued Fraction Digits

O / open | Oberwolfach 报告问题汇编

别名：Zaremba Conjecture for Bounded Continued Fraction Digits [OWR-8415353-004]

Does there exist a constant $A$ such that every natural number is the denominator of a reduced fraction whose continued-fraction partial quotients are all at most $A$?

来源字段：https://doi.org/10.4171/owr/2021/50

01652 | OWR-11135-005

### Zaremba’s Bounded Continued-Fraction Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a fixed $A$ such that every positive integer is the denominator of a finite continued fraction $[a_1,\ldots,a_k]$ with $1\leq a_i\leq A$?

来源字段：https://doi.org/10.4171/owr/2011/43

01653 | OWR-15956-010

### Zero-Free Dirichlet Polynomials on the Unit Line

O / open | Oberwolfach 报告问题汇编

For $$ D(s)=1+2^{-s}+3^{-s}+5^{-s}, $$ is $D(s)\neq0$ whenever $\operatorname{Re}(s)=1$?

来源字段：https://doi.org/10.4171/owr/2017/51

01654 | OWR-11695860-017

### Zero-One Laws in Simultaneous Diophantine Approximation

O / open | Oberwolfach 报告问题汇编

Let $f:\mathbb R^d\to\mathbb R^m$, put $n=d+m$ and $\alpha_{d+j}=f_j(\alpha_1,\ldots,\alpha_d)$, and let $A(h)$ consist of points admitting denominators $q_j\leq h$ with $\|q_j\alpha_j\|\leq\psi(h)$. Under suitable nondegeneracy assumptions on $f$, prove that the limsup $A^*$ has measure zero or full measure according as $\sum_{h=1}^\infty h^{n-1}\psi(h)$ converges or diverges.

来源字段：https://doi.org/10.4171/owr/2022/50

01655 | OWR-15174-003

### Zeta Moments from Shifted Divisor Correlations

O / open | Oberwolfach 报告问题汇编

How can the conjectured $2k$th moments of the Riemann zeta function be derived arithmetically from shifted-divisor problems, including the required $k$-fold convolutions of shifted divisor sums?

来源字段：https://doi.org/10.4171/owr/2016/53

01656 | OWR-11784-008

### Zilber Conjecture for Torsion-Anomalous Intersections

O / open | Oberwolfach 报告问题汇编

For every algebraic subset $W\subseteq\mathbb G_m^N$, is there a finite collection $\mathcal U_W$ of codimension-one torsion cosets such that, for every torsion coset $T_0$ and every irreducible component $Y$ of $W\cap T_0$ with $\dim Y>\dim W-\operatorname{codim}T_0$, some $T\in\mathcal U_W$ satisfies $Y\subseteq W\cap T$?

来源字段：https://doi.org/10.4171/owr/2012/22

01657 | AMR-093-0091

### Zilber-Pink conjecture

O / open | AMR 问题清单汇编

Zilber-Pink conjecture that if $X$ is a mixed Shimura variety or semiabelian variety defined over $\mathbb{C}$, and $V \subseteq X$ is a subvariety, then $V$ contains only finitely many maximal atypical subvarieties.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

# 资料汇编 02 | 代数 / Algebra

1,003 条记录。O=开放；P=部分解决；S=已解决。状态与英文/LaTeX 表述均沿用来源。

01658 | OWR-12175-003

### $\mathcal Z$-Stability and Regularity of Nuclear $C^*$-Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Does $\mathcal Z$-stability imply finite nuclear dimension for all unital simple separable nuclear $C^*$-algebras? More broadly, does the equivalence between $\mathcal Z$-stability and strict comparison hold without the additional hypotheses currently known?

来源字段：https://doi.org/10.4171/owr/2012/52

01659 | OWR-12175-018

### $K$-Theory of Easy Quantum Group $C^*$-Algebras

O / open | Oberwolfach 报告问题汇编

What is the $K$-theory of the $C^*$-algebras associated with easy quantum groups?

来源字段：https://doi.org/10.4171/owr/2012/52

01660 | OWR-11580-004

### $q$-Whittaker Eigenfunctions of Lattice Quantum Toda Operators

O / open | Oberwolfach 报告问题汇编

For a semisimple simply connected group $G$, prove existence and uniqueness of the $q$-Whittaker polynomials $\Psi_{\check{\lambda}}(q,z)$ satisfying vanishing outside dominant weights, $\Psi_0=1$, and $M_f^{\mathrm{lat}}(\Psi(q,z))=f(z)\Psi(q,z)$ for every Weyl-invariant $f$.

来源字段：https://doi.org/10.4171/owr/2012/13

01661 | AMR-108-0007

### 2.2 (Cooper) - An invariant polynomial on a tensor product

S / solved | AMR 问题清单汇编

Does there exist a nonzero polynomial on $U\otimes V\otimes W$ invariant under $SL(U)\times SL(V)\times SL(W)$ when $\dim U=\dim V=4$ and $\dim W=8$?

背景链接：https://arxiv.org/pdf/1512.04620#page=2

01662 | AIM-REPRESENTATION_THEORY-0096

### A categorical trilemma for the proposed category of supercharacters

P / partially_solved | AIM 研讨会问题汇编

Problem 1.12. Is superclass theory for Un tame?. That is, is there a tame algebra whose category of modules is equivalent to the category of supercharacters of Un?.

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

01663 | AIM-REPRESENTATION_THEORY-0020

### A characteristic-two obstruction to degree-two tca noetherianity

S / solved | AIM 研讨会问题汇编

**Noetherianity of degree two tca's.**

1. [T. Church] Is \(\operatorname{Sym}(\operatorname{Sym}^2)\) noetherian over \(\mathbb Z\) (or over a general noetherian ring)? 2. [J. Wilson] Same question for \(\bigwedge(\operatorname{Sym}^2)\).

来源字段：http://aimpl.org/repnstability/1/

01664 | AIM-REPRESENTATION_THEORY-0017

### A componentwise affine non-noetherian sub-FI-algebra

P / partially_solved | AIM 研讨会问题汇编

**Noetherianity of \(\mathrm{FI}\)-algebras.** Let \(V_n=\mathbf k[x_1,x_2,\ldots,x_n]\). Find a concrete example of a sub-\(\mathrm{FI}\)-algebra which is not noetherian. Is there a version for finitely presented modules? Does it depend on characteristic?

来源字段：http://aimpl.org/repnstability/1/

01665 | AIM-REPRESENTATION_THEORY-0067

### A cubic genuineness and central-overlap audit for exceptional theta lifting

P / partially_solved | AIM 研讨会问题汇编

What is theta correspondence for $\widetilde{G}$?

e.g.) $SL_2 \times \widetilde{SL}^{(3)}_2 \hookrightarrow \widetilde{G_2}^{(3)}$ (G-R-S)

$SL_3 \times \widetilde{SL}^{(2)}_3 \hookrightarrow \widetilde{F_4}^{(2)}$

$\widetilde{Spin}^{(2)}_{2a+1} \times \widetilde{Spin}^{(2)}_{2b} \hookrightarrow \widetilde{Spin}^{(2)}_{2(a+b)+1}$ (Loke-Savin)

$\widetilde{Spin}^{(2)}_{5} \times \widetilde{Spin}^{(2)}_{4} \hookrightarrow \widetilde{Spin}^{(2)}_{9}$

来源字段：http://aimpl.org/autoformcovergp/1/

01666 | AIM-REPRESENTATION_THEORY-0023

### A cubic homological obstruction for the doubled standard q-tca

P / partially_solved | AIM 研讨会问题汇编

Berenstein--Zwicknagl define quantum analog of \(\operatorname{Sym}(\mathbf S_\lambda)\). For small \(\lambda\), this behaves like the classical case, but for large \(\lambda\) (and \(q\) generic) its isotypic decomposition has smaller multiplicities than the classical case. Are modules over this algebra (as \(q\)-tca's) equivalent to modules in the classical case (as tca's)?

来源字段：http://aimpl.org/repnstability/1/

01667 | AIM-LINEAR_ALGEBRA-0011

### A degree-independent ternary rhombus bound and a seven-variable obstruction

P / partially_solved | AIM 研讨会问题汇编

Given positive scalars $a_0, \dots, a_d$ such that \[ \frac{a_k^2}{a_{k-1}a_{k+1}}\geq 4 \qquad (0<k<d), \] Hutchinson's theorem says that the generating polynomial $p(x)=\sum_{j=0}^d a_jx^j$ is real-rooted. One can homogenize this to $p(x,y)$.

Now consider the analogue for homogeneous polynomials of three variables. It is known that for each degree $d\geq1$, there exists $\lambda_d>0$ such that if $\lambda\geq\lambda_d$, then the “$\lambda$-rhombus log-concavity” of the Maclaurin coefficients of a homogeneous polynomial $p(x,y,z)$ implies that $p$ is real-stable.

Can one remove the dependence of $\lambda_d$ on $d$? If yes, then what about for $n>3$ variables?

来源字段：http://aimpl.org/totalpos/1/

01668 | AIM-REPRESENTATION_THEORY-0054

### A degree-two criterion for highest-weight perverse hearts

P / partially_solved | AIM 研讨会问题汇编

More generally, when is a category of perverse sheaves constructible with respect to a fixed stratification highest-weight?

来源字段：http://aimpl.org/sheavemodular/9/

01669 | AIM-REPRESENTATION_THEORY-0045

### A degreewise torsion detector for Iwahori IC stalks

P / partially_solved | AIM 研讨会问题汇编

Fix the affine Grassmannian \(Gr\) and its stratification by orbits of an Iwahori subgroup \(I\). For which primes \(p\) do all integral intersection cohomology objects \[ \mathrm{IC}_y(\mathcal O)\in\operatorname{Perv}_I(Gr,\mathcal O), \] one for each \(I\)-orbit closure \(\overline{Gr_y}\), have torsion-free stalk cohomology?

来源字段：http://aimpl.org/sheavemodular/5/

01670 | AIM-REPRESENTATION_THEORY-0027

### A direct algebraic S-infinity model for configuration cohomology

P / partially_solved | AIM 研讨会问题汇编

Let \(S_\infty=\bigcup_nS_n\) be the group of finitary permutations of \(\mathbb N\). For fixed \(i\), form the canonical FI-colimit \[ \mathcal H^i_\infty=\varinjlim_{S\subset\mathbb N,\ |S|<\infty} H^i(\operatorname{PConf}_S(\mathbb C);\mathbb Q), \] compute it as an algebraic \(S_\infty\)-representation directly from the Arnol'd algebra, and compare it with stable finite \(S_n\)-decompositions.

来源字段：http://aimpl.org/repnstability/2/

01671 | AIM-REPRESENTATION_THEORY-0021

### A fixed quadratic tca presentation with square-root syzygy excess

P / partially_solved | AIM 研讨会问题汇编

**Bounds on syzygies of modules over degree two tca's.** Let \(A=\operatorname{Sym}(\operatorname{Sym}^2)\) or \(\bigwedge(\operatorname{Sym}^2)\), and let \(M\) be an \(A\)-module generated in degree \(d\) and related in degree \(r\). Are there any good bounds on the syzygies? In other words, bound generators of the terms of the free resolutions. Can start with characteristic \(0\).

来源字段：http://aimpl.org/repnstability/1/

01672 | AIM-REPRESENTATION_THEORY-0030

### A fixed-d Toeplitz obstruction to degree-wise coherence for FI_d

P / partially_solved | AIM 研讨会问题汇编

**Coherence.** By [Theorem B, Ra], the category of \(\mathrm{FI}\)-modules presented in finite degree is abelian. Is this true for other categories?

A. Snowden: Probably false for \(\mathrm{FI}_d\) (\(d>1\)). [Ra] Eric Ramos, *On the degree-wise coherence of \(\mathrm{FI}_G\)-modules*.

来源字段：http://aimpl.org/repnstability/3/

01673 | AIM-REPRESENTATION_THEORY-0041

### A fixed-degree obstruction and finite Morita test for the quasimap model

P / partially_solved | AIM 研讨会问题汇编

Is $\mathrm{Rep}_0(G_1T)$ equivalent to a category of perverse sheaves on a space of quasi-maps from $\mathbb{P}^1$ to $G/B$?

来源字段：http://aimpl.org/sheavemodular/2/

01674 | AIM-REPRESENTATION_THEORY-0057

### A Frobenius-dependent modular character-sheaf model for split tori

P / partially_solved | AIM 研讨会问题汇编

Can one develop a theory of modular character sheaves on \(G\) that relates to the representation theory of \(G(\mathbb F_p)\) in non-defining characteristic?

来源字段：http://aimpl.org/sheavemodular/12/

01675 | AIM-REPRESENTATION_THEORY-0077

### A genuine-character obstruction and packet test for lifts to covered inner forms

P / partially_solved | AIM 研讨会问题汇编

Is there a natural lifting from representations of $G_F$ to $\widetilde{G'}_F$?

The exact AIM record asks:

The record itself does not define $G'$, the cover, the class of representations, or the sense of “natural.” The immediately preceding AIM problem (3.3) asks for transfer from $\widetilde G_F$ to $\widetilde{G'}_F$ “where two linear groups $G_F$ and $G'_F$ are inner forms each other,” and warns that the centers of the covers may differ. I therefore use the following conservative reconstruction:

* $F$ is a local field (the surrounding base-change problem specifies a characteristic-zero $p$-adic field); * $G$ and $G'$ are inner forms of a common quasi-split reductive group $G^*$; * $1\to A\to \widetilde{G'}\to G'(F)\to1$ is a finite central cover, normally of Brylinski--Deligne or étale-metaplectic origin; * a target representation is required to be $\epsilon$-genuine for a fixed injective character $\epsilon:A\hookrightarrow\mathbb C^\times$.

This reconstruction is supported by the neighboring repository records, not by the unavailable web page: on 12 August 2026, `http://aimpl.org/autoformcovergp/3/` returned HTTP 502. No claim below depends on silently adding further structure.

来源字段：http://aimpl.org/autoformcovergp/3/

01676 | AIM-REPRESENTATION_THEORY-0052

### A geometric highest-weight proof and its degree-two obstruction

S / solved | AIM 研讨会问题汇编

**Is there a geometric proof (i.e. avoiding Geometric Satake) that \(\operatorname{Perv}_{\mathrm{sph}}(Gr,k)\) is highest-weight?**

来源字段：http://aimpl.org/sheavemodular/9/

01677 | AIM-REPRESENTATION_THEORY-0084

### A Levi-defect criterion for Brylinski-Deligne covers of GL(r)

P / partially_solved | AIM 研讨会问题汇编

Study the residual spectrum for $\widetilde{GL}(n)$.

-- Can the lifting of Levi subgroups be done over $p$-adic? If so, is it one of the Brylinski-Deligne extensions?

-- Describe covers of $GL(n)$ not arising from Kazhdan-Petterson.

来源字段：http://aimpl.org/autoformcovergp/4/

01678 | AIM-REPRESENTATION_THEORY-0029

### A local-system Schur transform for configuration-space FI-modules

P / partially_solved | AIM 研讨会问题汇编

**Schur weyl duals.** Are there direct constructions for the Schur--Weyl dual of $\mathrm{FI}$-modules like $\mathrm{H}^{\bullet}(\mathrm{PConf}^n{X}, \mathbb{Q})$? E.g., can one construct a $\mathbf{GL}_{\infty}$-space and a $\mathbf{GL}_{\infty}$-equivariant sheaf whose cohomology is the Schur--Weyl dual?

来源字段：http://aimpl.org/repnstability/2/

01679 | AMR-046-0026

### A moments problem I

O / open | AMR 问题清单汇编

Let $f(x_1,\ldots,x_n)\in\mathbb{C}[x_1,\ldots,x_n]$ satisfy $$M_m:=\int_0^1\cdots\int_0^1 f(x_1,\ldots,x_n)^m\,dx_1\cdots dx_n=0\qquad(m\geq1).$$ Must $f=0$?

背景链接：https://arxiv.org/abs/2012.02524

01680 | AMR-046-0027

### A moments problem II

O / open | AMR 问题清单汇编

Let $f(x)\in\mathbb{C}[x]$ have $k$ monomials. Does there exist $N(k)$ such that if $$M_n:=\int_0^1f(x)^n\,dx=0\qquad(1\leq n\leq N(k)),$$ then $f=0$? If so, determine $N(k)$ or give a good upper bound.

背景链接：https://arxiv.org/abs/2012.02524

01681 | AIM-REPRESENTATION_THEORY-0113

### A normalization audit of the higher-rank Serre weight conjecture

P / partially_solved | AIM 研讨会问题汇编

If $\sigma$ is any mod-$p$ representation of $\mathrm{GL}_n(\mathbf F_p)$, then there is a Hecke eigenclass $x\in H^d(\Gamma_1(N),\sigma)$, for some $d$, satisfying $$ T_\ell x=\operatorname{Tr}(\bar\rho(\operatorname{Frob}_\ell))x \quad(\ell\nmid Np), $$ with $p\nmid N$, if and only if $$ N(\bar\rho)\mid N, \qquad \operatorname{JH}(\sigma)\cap \operatorname{Wt}(\bar\rho|_{I_p})\ne\varnothing. $$

Let $$ \bar\rho:G_{\mathbf Q}\longrightarrow \mathrm{GL}_n(\overline{\mathbf F}_p) $$ be continuous and irreducible. The preceding paragraph imposes the higher-rank oddness condition $$ \left|\dim \bar\rho^{c=1}-\dim \bar\rho^{c=-1}\right|\leq 1 $$ for complex conjugation $c$. Let $N(\bar\rho)$ be its prime-to-$p$ Artin conductor and $$ \Gamma_1(N)=\{g\in \mathrm{SL}_n(\mathbf Z): \text{the bottom row of }g\text{ is }(0,\ldots,0,1)\bmod N\}. $$

The source's **Conjecture 6 (approx.)** is:

It asks whether a specific degree $d$ should occur, expresses the hope $$ \operatorname{Wt}(\bar\rho|_{I_p})\subset \operatorname{Wt}((\bar\rho|_{I_p})^{\mathrm{ss}}), $$ sketches an analogue for a unitary group over an imaginary quadratic field, recalls highest-weight modules and Hodge--Tate numbers $$ a_1+n-1,a_2+n-2,\ldots,a_{n-1}+1,a_n, $$ and gives a tame-inertia/Deligne--Lusztig recipe. The next item in the PDF is Conjecture 7 and is not part of this record.

If $q=p^r$ and $S_r=(p^r-1)/(p-1)$, the displayed map is $$ R(a_1,\ldots,a_n)= (a_n-(n-1)S_r,a_{n-1}-(n-2)S_r,\ldots,a_2-S_r,a_1) \pmod{q-1}. $$ Also, the niveau-$s$ characters in the tame recipe are $\chi,\chi^p,\ldots,\chi^{p^{s-1}}$, not the malformed OCR exponents.

来源字段：https://aimath.org/WWN/padicmodularity/padicmodularity.pdf

01682 | AIM-REPRESENTATION_THEORY-0003

### A normalization-safe sign and family obstruction for unitary central derivatives

P / partially_solved | AIM 研讨会问题汇编

**Non-vanishing of the central \(L\)-derivative for \(\mathrm{U}(n)\times\mathrm{U}(n-1)\).**

Given an automorphic representation \(\pi\) of \(\mathrm{U}(n)\), show that there exists an automorphic representation \(\sigma\) of \(\mathrm{U}(n-1)\) such that \[ L'(\pi\otimes\sigma,\tfrac12)\ne 0. \]

来源字段：http://aimpl.org/aagaautomorphic/1/

01683 | AIM-REPRESENTATION_THEORY-0086

### A prefix-cut obstruction for an NCQSym supercharacter realization

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.2.** Can one construct a nested supercharacter theory that would realize the Hopf algebra NCQSym (indexed by ordered partitions)?

**Remark.** Nested means a projective system of groups. See *Algebraic structures on Grothendieck groups of a tower of algebras*, by H. Li.

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

01684 | AIM-REPRESENTATION_THEORY-0032

### A projective scalar quotient of VI and its module theory

P / partially_solved | AIM 研讨会问题汇编

**VI like categories**

VI is obtained from FI by replacing finite sets with finite vector spaces. In combinatorics, often better to replace finite sets by finite projective spaces.

Does this lead to an interesting analogue of VI-modules?

来源字段：http://aimpl.org/repnstability/3/

01685 | AIM-REPRESENTATION_THEORY-0038

### A projective Steinberg cone and a sharp rank-one boundary

P / partially_solved | AIM 研讨会问题汇编

What representations of $G(\mathbb F_q)$ arise by restricting a tilting module of $G$?

来源字段：http://aimpl.org/sheavemodular/1/

01686 | AIM-REPRESENTATION_THEORY-0098

### A q-power-sum bridge to supercharacters and a block-interval-order model

S / solved | AIM 研讨会问题汇编

Problem 1.14. To get a better understanding of the supercharacter basis, give a basis in NC-Sym that is nicely related to supercharacters. For example, see remark about p-basis pp.20 in *Supercharacters, symmetric functions in non commuting variables, and related Hopf algebras*.

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

01687 | AIM-REPRESENTATION_THEORY-0073

### A quadratic self-opposition criterion for double covers

P / partially_solved | AIM 研讨会问题汇编

**Problem.** What is special about $2$-fold covers?

`Maybe, $\pm \in \mathbb Q$?`

来源字段：http://aimpl.org/autoformcovergp/2/

01688 | AIM-REPRESENTATION_THEORY-0053

### A quasihereditary criterion and an extension obstruction

P / partially_solved | AIM 研讨会问题汇编

Given a symplectic resolution \(\pi\colon Y\to X\), with \(G=\operatorname{Aut}(X,\omega)\), consider the subcategory of \(\operatorname{Perv}_G(X,k)\) generated by subquotients of \(\pi_*\underline{k}_Y\). When is this highest-weight?

来源字段：http://aimpl.org/sheavemodular/9/

01689 | AIM-REPRESENTATION_THEORY-0059

### A rank-one boundary-Mellin-Poisson audit for basic functions

P / partially_solved | AIM 研讨会问题汇编

(1) What does one need to know about "basic functions" to put them in the trace formula? What are asymptotics of the "basic functions"? (2) Define and explain the Fourier transform for Vinberg monoids. Find their relations to the Langlands-Shahidi method and the Rankin-Selberg method.

Endoscopy and Beyond:

来源字段：https://aimath.org/pastworkshops/automorphkernelproblems.pdf

01690 | AIM-LINEAR_ALGEBRA-0012

### A Schur-positive order-two case for affine elementary-symmetric Toeplitz matrices

P / partially_solved | AIM 研讨会问题汇编

There are many classical theorems about operations that preserve negative-real-rootedness of polynomials: for instance, \(d/dx+a\) or \(x\,d/dx+a\) with \(a\geq 0\); or Hadamard product; or Hadamard product with an extra \(n!\) (sometimes known as Schur composition); or Brändén's (2011) log-concavity operation. These can be reinterpreted as statements about *pointwise* total positivity for certain Toeplitz matrices involving elementary symmetric functions (e.g. \((n+a)e_n\)). Can these results be upgraded to total positivity in the *monomial* basis? Computational tests suggest that the answer is yes. But we have been unable, thus far, to come up with a plausible strategy for proving any of these conjectures.

来源字段：http://aimpl.org/totalpos/1/

01691 | AIM-REPRESENTATION_THEORY-0070

### A sector-and-twist test for covering-group ABV

P / partially_solved | AIM 研讨会问题汇编

**ABV for covering groups.** There is a canonical isomorphism as $K$-groups $$ KRep\,\widetilde G_{\mathbb R}\cong\bigl(KPer_H(X)\bigr)^*, $$ where $X$ is a $\mathbb C$-algebraic variety and $H$ is a $\mathbb C$-algebraic group (for the trivial cover, $H={}^\vee G$).

The exact AIM record (2013 workshop *Automorphic forms and harmonic analysis on covering groups*, “Fundamental properties,” Conjecture 2.3) is:

The original AIMPL URL returned a 502 error when checked on 12 August 2026. The formula is a compressed version of the Adams--Barbasch--Vogan (ABV) perfect pairing: $KRep$ is a Grothendieck group of finite-length representations, $KPer$ is a Grothendieck group of equivariant perverse sheaves on a geometric parameter space, and ${}^*$ is an algebraic dual.

Two qualifications suppressed in the AIM slogan are mathematically essential.

1. ABV works most cleanly after fixing an infinitesimal character, and its global notation includes strong real forms and canonical projective representations. A finite regular block is the right test object. 2. If $1\to A\to\widetilde G_{\mathbb R}\to G(\mathbb R)\to1$ is a finite central cover, “genuine” means that a specified character $\epsilon:A\to\mathbb C^\times$ acts on every representation. Modern covering-group $L$-groups also depend on this $\epsilon$.

Accordingly, the recovered target is a blockwise, fixed-$\epsilon$ pairing

$$ K_0\bigl(\mathcal B_\epsilon(\widetilde G_{\mathbb R})\bigr) \simeq K_0\bigl(\operatorname{Per}_{H_\epsilon}(X_{\epsilon,\mathcal O},\beta_\epsilon)\bigr)^*, \tag{1.1} $$

where a possible twist $\beta_\epsilon$ may instead be absorbed into the definitions of $H_\epsilon$ and $X_{\epsilon,\mathcal O}$. This is a reconstruction, not a claim that these objects have been constructed in general.

来源字段：http://aimpl.org/autoformcovergp/2/

01692 | AIM-LINEAR_ALGEBRA-0017

### A sharp codimension-one compound threshold

P / partially_solved | AIM 研讨会问题汇编

In the strict convention used below, \(TP_r\) means that every minor of order at most \(r\) is positive, and \(TP\) means that every minor is positive.

The record begins with the dimension-free Katkova--Vishnyakova phenomenon: a positive square matrix satisfying \[ ad-\lambda bc>0,\qquad \lambda\geq4, \] on its \(2\times2\) submatrices is TP. It asks for higher-order quantitative conditions that upgrade \(TP_{k-1}\) to TP and for optimal constants.

There is an important formulation distinction. The AIM problem literally says “every \(2\times2\) submatrix,” which ordinarily means every choice of two rows and two columns. The primary theorem of Katkova--Vishnyakova needs only the **local adjacent-entry** inequalities \[ a_{ij}a_{i+1,j+1}>c\,a_{i,j+1}a_{i+1,j}. \tag{1.1} \] Thus the literal AIM hypothesis is stronger.

来源字段：http://aimpl.org/totalpos/1/

01693 | AIM-REPRESENTATION_THEORY-0060

### A shifted beyond-endoscopic detector for sharp Jiang--Liu Ramanujan bounds

P / partially_solved | AIM 研讨会问题汇编

For a totally imaginary number field \(F\) and odd \(n\geq5\), can a beyond-endoscopic trace formula prove that some cuspidal automorphic representation of \(\operatorname{Sp}_{2n}(\mathbb A_F)\) has an unramified exponent \((n-1)/2\)?

来源字段：https://aimath.org/pastworkshops/automorphkernelproblems.pdf

01694 | AIM-REPRESENTATION_THEORY-0049

### A stabilized-center criterion for distributions

P / partially_solved | AIM 研讨会问题汇编

Is there a geometric description of the center of \(\mathrm{Rep}_0G\)? Possibly related to affine Springer fibers?

**If one had a geometric description of the categorical center of the principal block \(\mathrm{Rep}_0(G)\), would it describe the algebraic center \(Z(\operatorname{Dist}(G))\) of the distribution algebra?**

The pronoun “this” is not meaningful in isolation.

Thus the conservative reconstruction of 6.2 is:

This is a reconstruction from the adjacent record, not a claim that the proposed geometric description in 6.1 exists. The AIM page is terse and specifies neither the ground field nor hypotheses on \(G\). In this report, \(k\) is algebraically closed of characteristic \(p>0\), and \(G\) is a smooth connected affine algebraic group; reductivity is needed only when discussing the workshop's geometric context.

来源字段：http://aimpl.org/sheavemodular/6/

01695 | AIM-REPRESENTATION_THEORY-0082

### A trace-and-type normalization ledger for formal degrees

P / partially_solved | AIM 研讨会问题汇编

Interpret formal degrees of representations of $\widetilde{G}_{F}$ on level of Hecke algebra. ($\grave{\mathrm a}$ la Opdam, Reeder, et al.)

Here $\widetilde G_F$ is read as the group of $F$-points of a covering group. The broader workshop summary makes the motivating example more precise: the Bernstein blocks containing the even and odd Weil representations of the two-fold metaplectic group should be compared with Iwahori-spherical blocks of equal-rank odd orthogonal groups. The summary singles out preservation of the natural $L^2$ norm under the Hecke-algebra isomorphism, because that is exactly what transports Plancherel measure and hence formal degrees.

Thus the problem has two levels:

1. identify formal degree intrinsically on a Hecke algebra equipped with its positive trace; and 2. keep track of the normalization needed to recover the group formal degree from the type Hecke algebra.

来源字段：http://aimpl.org/autoformcovergp/4/

01696 | AIM-REPRESENTATION_THEORY-0012

### A two-prime Hecke-Fourier transfer criterion for vertical mod-p nonvanishing

P / partially_solved | AIM 研讨会问题汇编

**Applications of the Hecke orbit conjecture.** Are there applications of the recent progress on the Hecke orbit conjecture to vertical mod-\(p\) non-vanishing (as in Iwasawa theory)?

来源字段：http://aimpl.org/aagaautomorphic/6/

01697 | AIM-REPRESENTATION_THEORY-0013

### A two-stage Whittaker and cohomological normalization for Bianchi eigenforms

P / partially_solved | AIM 研讨会问题汇编

**Normalization of Bianchi modular forms.** Consider a Bianchi modular form, is there a correct way to normalize its coefficients in its Whittaker expansion (in \(H^2\))?

来源字段：http://aimpl.org/aagaautomorphic/6/

01698 | AIM-REPRESENTATION_THEORY-0065

### A type-theoretic category for depth-zero genuine unipotent representations

P / partially_solved | AIM 研讨会问题汇编

What is the category of depth-zero unipotent representations for $\widetilde{G}_{F}$ for a $p$-adic field $F$?

来源字段：http://aimpl.org/autoformcovergp/1/

01699 | OWR-14299511-006

### A-Infinity Realization of Finite-Depth Subfactor Indices

P / partially_solved | Oberwolfach 报告问题汇编

Prove that every index realized by a finite-depth irreducible hyperfinite subfactor is also realized by an $A_\infty$ subfactor.

来源字段：https://doi.org/10.4171/owr/2025/34

01700 | OWR-3472-014

### a-Invariants of Quasi-Gorenstein Projective Cones

O / open | Oberwolfach 报告问题汇编

For a normal projective variety $X$, let $A(X)=\{a(R):R\text{ is quasi-Gorenstein graded and }\operatorname{Proj}R=X\}$. Classify the possible sets $A(X)$. In particular, if $-K_X$ is big, must $A(X)$ contain a negative integer? If $X$ is Fano of index $1$, is $A(X)$ the set of positive odd integers?

来源字段：https://doi.org/10.4171/owr/2009/22

01701 | OWR-17294-008

### Abelian Derived Racks and Their Structure Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a left nondegenerate Yang-Baxter solution $(X,r)$ and its derived rack, what property of $(X,r)$ corresponds to the derived rack having an abelian structure group, and what property corresponds to the rack itself being abelian? Does splitting of the associated central extension, with an additional condition, characterize rack structure groups, and are there group-theoretic characterizations of them?

来源字段：https://doi.org/10.4171/owr/2019/51

01702 | OWR-4413-002

### Absolute Galois Groups among Profinite Groups

O / open | Oberwolfach 报告问题汇编

Which profinite groups occur as absolute Galois groups of fields?

来源字段：https://doi.org/10.4171/owr/2010/32

01703 | OWR-1465-008

### Abstract Pierce-Birkhoff Conjecture

O / open | Oberwolfach 报告问题汇编

Let $A$ be a polynomial ring over a real closed field, let $f$ be piecewise polynomial, and let $\alpha,\beta\in\operatorname{Sper}A$. If $f_\alpha,f_\beta\in A$ are local representatives of $f$ at $\alpha$ and $\beta$, must $f_\alpha-f_\beta$ belong to the separating ideal $\langle\alpha,\beta\rangle$?

来源字段：https://doi.org/10.4171/owr/2007/14

01704 | OWR-14299281-013

### Acting Groups and Identity-Component Automorphisms

O / open | Oberwolfach 报告问题汇编

Let $X=G/P$, where $G$ is simple adjoint and $P$ is a nonreduced parabolic subgroup scheme. When is the natural homomorphism $G\hookrightarrow\operatorname{Aut}^0(X)$ an isomorphism, and what is the structure of $\operatorname{Aut}^0(X)$ when the inclusion is strict?

来源字段：https://doi.org/10.4171/owr/2025/20

01705 | OWR-14299088-004

### Additive Orthogonal Bases for Invariant Polynomials

P / partially_solved | Oberwolfach 报告问题汇编

For invariant-polynomial rings under general group actions, characterize approximately orthogonal bases and determine when such a basis can also be chosen to satisfy an additive property analogous to that of tensor cumulants.

来源字段：https://doi.org/10.4171/owr/2025/9

01706 | OWR-13494-005

### Additivity of Waring Rank in Independent Variables

P / partially_solved | Oberwolfach 报告问题汇编

If $F_1,\ldots,F_r$ are forms in independent sets of variables, is their Waring rank additive, namely $\operatorname{rank}(F_1+\cdots+F_r)=\operatorname{rank}(F_1)+\cdots+\operatorname{rank}(F_r)$?

来源字段：https://doi.org/10.4171/owr/2015/9

01707 | OWR-11583-017

### Affine Algebra Models for Principal Representation Blocks

O / open | Oberwolfach 报告问题汇编

Let $A$ be the integral algebra whose complex module category is equivalent to the principal block $\mathcal O_0$ of category $\mathcal O$ and whose reduction over a field $k$ models the corresponding modular category. Fiebig's affine analogue controls the full Lusztig conjecture. How is the module category of this affine algebra related to the entire principal block $\operatorname{Rep}(G_k)$?

来源字段：https://doi.org/10.4171/owr/2012/16

01708 | OWR-1393-001

### Affine Crystal Structures on Quantum Demazure Modules

O / open | Oberwolfach 报告问题汇编

For a quantum Demazure module $D_q(m,\lambda)$, does adding appropriate arrows labelled $0$ to its crystal graph produce the crystal graph of an irreducible $U_q(\widehat{\mathfrak{g}})$-representation?

来源字段：https://doi.org/10.4171/owr/2006/55

01709 | OWR-8415355-008

### Affine Springer Cohomology and Quantum-Group Centers

P / partially_solved | Oberwolfach 报告问题汇编

For an affine Springer fiber $\mathcal F\ell^J_\gamma$, a small quantum group $u^J_\zeta$, the coweight lattice $\check\Lambda$, and the affine Weyl group $W_{\mathrm{af}}$, are there canonical isomorphisms $$H^\bullet(\mathcal F\ell^J_\gamma)^{\check\Lambda}\cong Z(u^J_\zeta)^{\check T},\qquad H^\bullet(\mathcal F\ell^J_\gamma)^{W_{\mathrm{af}}}\cong Z(u^J_\zeta)^{\check G}$$?

来源字段：https://doi.org/10.4171/owr/2021/52

01710 | AIM-REPRESENTATION_THEORY-0100

### Algebra-group supercharacter refinements and a class-two formula

S / solved | AIM 研讨会问题汇编

Le and Magaard have a way of refining supercharacters of \(U_n(q)\). Is there an analogous refinement for arbitrary finite algebra groups \(G=1+\mathfrak n\)?

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

01711 | OWR-14298797-002

### Algebraic Generators and Equivalence of Loop Invariants

O / open | Oberwolfach 报告问题汇编

Prove that every loop invariant is a shuffle combination of trivial loop invariants, signed areas, and conjugation invariants. Also characterize the equivalence relation $X\sim X'$ defined by equality of all loop-invariant signature coordinates.

来源字段：https://doi.org/10.4171/owr/2024/49

01712 | OWR-17294-006

### Algebraic Invariants of Yang-Baxter Map Order

O / open | Oberwolfach 报告问题汇编

Is there an algebraic invariant that determines the order of the Yang-Baxter map $r$?

来源字段：https://doi.org/10.4171/owr/2019/51

01713 | AMR-014-0007

### Algebraic Stories - Degree of the Waring map

P / partially_solved | AMR 问题清单汇编

Calculate the degree of $\widetilde{W}_{k,d}$ for perfect pairs $(k,d)$.

背景链接：https://arxiv.org/abs/1801.01692

01714 | AMR-014-0023

### Algebraic Stories - Exterior algebras

P / partially_solved | AMR 问题清单汇编

Let $f$ be a form of odd degree $d$ in $E$. Is it true that $({\rm Ann}(f))_i = (f)_i$, for $i < (n-d)/2$?

背景链接：https://arxiv.org/abs/1801.01692

01715 | AMR-014-0024

### Algebraic Stories - Exterior algebras

P / partially_solved | AMR 问题清单汇编

Let $f$ and $g$ be generic quadratic forms in $E$ and let $\ell_1$ and $\ell_2$ be two generic linear forms in $S$. Then the Hilbert series of $E/(f,g)$ is equal to the Hilbert series of $S/(x_1^2,\ldots,x_n^2,\ell_1^2,\ell_2^2)$ and is given by $1 + a(n,1)\, t + a(n,2)\, t^2 + \cdots + a(n,s)t^s + \cdots$, where $a(n,s)$ is the number of lattice paths inside the rectangle $(n+2-2s)\times (n+2)$ starting from the bottom-left corner and ending at the top-right corner by using only moves of two types: either $(x,y)\rightarrow (x+1,y+1) \textrm{\ or\ } (x-1,y+1)$.

背景链接：https://arxiv.org/abs/1801.01692

01716 | AMR-014-0001

### Algebraic Stories - Generic $k$-rank

O / open | AMR 问题清单汇编

Given a triple of positive integers $(k,d,n)$, calculate $\operatorname{rk}_k^\circ(kd,n).$

背景链接：https://arxiv.org/abs/1801.01692

01717 | AMR-014-0002

### Algebraic Stories - Generic $k$-rank

O / open | AMR 问题清单汇编

The $k$-rank of a general form of degree $kd$ in $n$ variables is given by $$ \operatorname{rk}_k^\circ(kd,n)=\begin{cases} \min \left\{s\ge 1 \mid s\binom{n+d-1}{n-1}-\binom {s}{2}\ge \binom{n+2d-1}{n-1}\right\}, & \text{ for } k=2; \min \left\{s\ge 1 \mid s\binom{n+d-1}{n-1}\ge \binom{n+kd-1}{n-1}\right\}, & \text{ for } k\ge 3. \end{cases} $$

背景链接：https://arxiv.org/abs/1801.01692

01718 | AMR-014-0015

### Algebraic Stories - Hilbert functions of fat points.

P / partially_solved | AMR 问题清单汇编

Given a scheme of generic fat points $X \subset \mathbb{P}^{n_1-1}\times\ldots\times \mathbb{P}^{n_t-1}$, what is the multi-graded Hilbert function $\operatorname{Hilb}_{S/I_X}(l)$, for $l \in \mathbb{N}^t$?

背景链接：https://arxiv.org/abs/1801.01692

01719 | AMR-014-0009

### Algebraic Stories - Hilbert series of generic power ideals.

P / partially_solved | AMR 问题清单汇编

[Fr\"oberg-Iarrobino Conjecture] Given generic linear forms $\ell_1, \ldots, \ell_r$ and a positive integer $d$, let $I$ be the power ideal generated by $\ell_1^d,\ldots,\ell_r^d$. Then the Hilbert function of $R = S/I$ is as in [source label: eq:RALF], except for the cases $(n,r) = (3,7), (3,8), (4,9), (5,14)$ and possibly for $r = n+2$ and $r = n+3$.

背景链接 : https://arxiv.org/abs/1801.01692

01720 | AMR-014-0017

### Algebraic Stories - Hilbert series of numerical semigroup rings

P / partially_solved | AMR 问题清单汇编

$\mathcal{S}$ is cyclotomic if and only if $k[\mathcal{S}]$ is a complete intersection.

背景链接 : https://arxiv.org/abs/1801.01692

01721 | AMR-014-0010

### Algebraic Stories - Hilbert series of other classes of ideals

O / open | AMR 问题清单汇编

For $\mu \neq (d)$, does a generic $\mu$-power ideal have the same Hilbert function as in [source label: eq:RALF]?

背景链接 : https://arxiv.org/abs/1801.01692

01722 | AMR-014-0011

### Algebraic Stories - Hilbert series of other classes of ideals

P / partially_solved | AMR 问题清单汇编

[] For generic forms $g_1,\ldots,g_r$ of degree $d>1$, the ideal $(g_1^k,\ldots,g_r^k)$ has the same Hilbert series as the one generated by $r$ generic forms of degree $dk$.

背景链接 : https://arxiv.org/abs/1801.01692

01723 | AMR-014-0008

### Algebraic Stories - Ideals of generic forms

P / partially_solved | AMR 问题清单汇编

[Fr\"oberg's Conjecture, 1985] Let $f_1, \dots, f_r$ be generic forms of degrees $d_1, \dots, d_r$, respectively. Then the Hilbert series of the quotient algebra $R = S/(f_1,\ldots,f_r)$ is given by $$ \operatorname{Hilb}_R(t)=\left[\frac{\prod_{i=1}^r(1-t^{d_i})}{(1-t)^n}\right]_+. $$ Here $[\sum_{i\ge0}a_iz^i]_+:=\sum_{i\ge0}b_iz^i$, with $b_i=a_i$ if $a_j\ge0$ for all $j\le i$ and $b_i=0$. In other words, $[\sum_{i\ge0}a_iz^i]_+$ is the truncation of a power series at its first non-positive coefficient.

背景链接 : https://arxiv.org/abs/1801.01692

01724 | AMR-014-0012

### Algebraic Stories - Lefschetz properties of graded algebras

O / open | AMR 问题清单汇编

It has been conjectured that each complete intersection $R=S/(f_1,\ldots,f_n)$ satisfies the WLP and also the SLP, see . Does the same hold for $R=S/(f_1,\ldots,f_r)$, with $f_1,\ldots,f_r$ being generic forms, and $r > n$?

背景链接 : https://arxiv.org/abs/1801.01692

01725 | AMR-014-0013

### Algebraic Stories - Lefschetz properties of graded algebras

P / partially_solved | AMR 问题清单汇编

When are the WLP and the SLP true for $T_{n,d,k}$?

背景链接 : https://arxiv.org/abs/1801.01692

01726 | AMR-014-0014

### Algebraic Stories - Lefschetz properties of graded algebras

P / partially_solved | AMR 问题清单汇编

For $R= S/(f_1,\ldots,f_r)$, where $f_1,\ldots,f_r$ are generic forms, does $R$ satisfy the $\mu$-Lefschetz property for all partitions $\mu$?

背景链接 : https://arxiv.org/abs/1801.01692

01727 | AMR-014-0003

### Algebraic Stories - Maximal $k$-rank

O / open | AMR 问题清单汇编

Given a triple of positive integers $(k,d,n)$, calculate $\operatorname{rk}_k^{\max}(kd,n).$

背景链接 : https://arxiv.org/abs/1801.01692

01728 | AMR-014-0004

### Algebraic Stories - Maximal $k$-rank

P / partially_solved | AMR 问题清单汇编

For any positive integers $k,d$, the maximal $k$-rank $\operatorname{rk}^{\max}_k(kd,2)$ of binary forms equals $k$. Additionally, in the above notation, binary forms representable by $\ell_1 \ell_2^{kd-1}$, where $\ell_1$ and $\ell_2$ are non-proportional linear forms, have the latter maximal $k$-rank.

背景链接 : https://arxiv.org/abs/1801.01692

01729 | AMR-014-0018

### Algebraic Stories - Non-negative forms

P / partially_solved | AMR 问题清单汇编

Are $B_{n,m}$ and $B'_{n,m}$ finite for any pair $(n,m)$ with even $n$?

背景链接 : https://arxiv.org/abs/1801.01692

01730 | AMR-014-0019

### Algebraic Stories - Non-negative forms

O / open | AMR 问题清单汇编

For any given pair $(n,m)$ with even $n$, $B'_{n,m}=\left(\frac{n}{2}\right)^{m-1}$.

背景链接：https://arxiv.org/abs/1801.01692

01731 | AMR-014-0020

### Algebraic Stories - Non-negative forms

O / open | AMR 问题清单汇编

Determine $\lim_{n\to\infty}\frac{B_{n,3}}{n^2}$.

背景链接：https://arxiv.org/abs/1801.01692

01732 | AMR-014-0021

### Algebraic Stories - Polynomial generation

P / partially_solved | AMR 问题清单汇编

For $n=1$ and given $p$, what are the (lengths of the) possible periods of $\phi$?

背景链接：https://arxiv.org/abs/1801.01692

01733 | AMR-014-0022

### Algebraic Stories - Polynomial generation

P / partially_solved | AMR 问题清单汇编

For $n = 1$ and given $p$, find the minimal positive integer $i$ such that $\psi^i$ is the identity map on the space of polynomials of degree at most $p-1$.

背景链接：https://arxiv.org/abs/1801.01692

01734 | AMR-014-0016

### Algebraic Stories - Symbolic powers vs. ordinary powers.

O / open | AMR 问题清单汇编

For the ideal $I$ of $s$ general points in $\mathbb P^{n-1}$, what is the difference between the Hilbert series of the $m$-th symbolic power and the $m$-th ordinary power?

背景链接：https://arxiv.org/abs/1801.01692

01735 | AMR-014-0005

### Algebraic Stories - The $k$-rank of monomials

P / partially_solved | AMR 问题清单汇编

Given $k \geq 3$ and a monomial $m$ of degree $kd$, determine the monomial $k$-rank $\operatorname{rk}_k(m)$.

背景链接：https://arxiv.org/abs/1801.01692

01736 | AMR-014-0006

### Algebraic Stories - The $k$-rank of monomials

P / partially_solved | AMR 问题清单汇编

Given $k \geq 3$ and a monomial $x^a y^b$ of degree $a+b=kd$, it is known that $\operatorname{rk}_k(x^{a}y^{b}) \leq \max(s,t)+1$, where $s$ and $t$ are the remainders of the division of $a$ and $b$ by $k$, see . Is it true that the latter inequality is, in fact, an equality?

背景链接：https://arxiv.org/abs/1801.01692

01737 | OWR-14298013-003

### Algebraic Structures for Transposed Bi-Hamiltonian Systems

O / open | Oberwolfach 报告问题汇编

Develop the theory of transposed bi-Hamiltonian algebras. More generally, identify the algebraic structures obtained by replacing the Poisson bracket with compatible Lie brackets in Poisson and transposed Poisson algebras.

来源字段：https://doi.org/10.4171/owr/2023/46

01738 | OWR-12872-019

### Algebraicity and D-Finiteness of Group-Algebra Series

O / open | Oberwolfach 报告问题汇编

Let $G$ be a group, let $u\in\mathbb ZG$, define $f_u(n)=[1]u^n$ to be the coefficient of the identity in $u^n$, and set $F_u(x)=\sum_{n\geq 1}f_u(n)x^n$. Is $F_u(x)$ D-finite for every $u$ when $G=\operatorname{GL}(d,\mathbb Z)$? More generally, for which groups $G$ is $F_u(x)$ algebraic for every $u\in\mathbb ZG$, and for which groups is it always D-finite?

来源字段：https://doi.org/10.4171/owr/2014/12

01739 | OWR-1188-011

### Algebraicity of Global Iterative Differential Modules

O / open | Oberwolfach 报告问题汇编

Is every global iterative differential module $(M,\partial_M)$ over a global iterative differential ring $(\mathcal O_{F'},\partial_F)$ algebraic?

来源字段：https://doi.org/10.4171/owr/2006/06

01740 | OWR-12732-001

### Algebras with Natural Cluster Structures

O / open | Oberwolfach 报告问题汇编

Which algebras carry natural cluster algebra structures?

来源字段：https://doi.org/10.4171/owr/2013/58

01741 | OWR-3400-008

### Alperin Weight Conjecture and Perfect Localities

P / partially_solved | Oberwolfach 报告问题汇编

In modular representation theory, understand and prove Alperin's Weight Conjecture, including the more general form suggested by Frobenius categories; also determine whether a perfect locality (centric linking system) associated with a Frobenius category exists and is unique.

来源字段：https://doi.org/10.4171/owr/2009/17

01742 | OWR-11101917-004

### Amenable Actions on Unital Simple Stably Finite Algebras

O / open | Oberwolfach 报告问题汇编

Does a nonamenable group admit an amenable action on a unital, simple, stably finite $C^*$-algebra?

来源字段：https://doi.org/10.4171/owr/2022/36

01743 | AIM-REPRESENTATION_THEORY-0092

### An abelianization supercharacter tower for Sylow subgroups of symmetric groups

P / partially_solved | AIM 研讨会问题汇编

**Recovered Problem 1.8.** Find a natural supercharacter theory for Sylow \(p\)-subgroups of \(\mathfrak S_{p^n}\). Are these theories nested? Does the restriction of a supercharacter for a Sylow \(p\)-subgroup of \(\mathfrak S_{p^n}\) to a Sylow \(p\)-subgroup of \(\mathfrak S_{p^{n-1}}\) split into supercharacters?

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

01744 | AIM-REPRESENTATION_THEORY-0031

### An affine-scheme VIC category and its rational representations

P / partially_solved | AIM 研讨会问题汇编

**Enrichments of $\mathrm{VIC}(\mathbb F)$.** Look at enrichments of $\mathrm{VIC}(\mathbb F)$ ($\mathbb F$ a field) over schemes (i.e., use algebraic representations of $\mathbf{GL}$). What can you say about their representations?

来源字段：http://aimpl.org/repnstability/3/

01745 | AIM-REPRESENTATION_THEORY-0051

### An effective-root-datum and dg-Morita reduction for small-order and singular ABG equivalences

P / partially_solved | AIM 研讨会问题汇编

1. Is there a version of the derived equivalences of Arkhipov--Bezrukavnikov--Ginzburg that applies when \(\ell\) (the order of the root of unity) is less than or equal to \(h\) (the Coxeter number)? 2. Same question for singular blocks rather than the principal block.

来源字段：http://aimpl.org/sheavemodular/8/

01746 | AIM-LINEAR_ALGEBRA-0006

### An elementary boundary and finite-atom compactness reduction for AESW

P / partially_solved | AIM 研讨会问题汇编

The Aissen--Edrei--Schoenberg--Whitney (AESW) theorem states that a real sequence $(a_0,a_1,a_2,\ldots)$ with $a_0 \neq 0$ is a one-sided PF sequence $\Leftrightarrow F(z)=a_0 e^{\delta z} \prod\limits_{j=0}^{\infty}\frac{1+\alpha_j z}{1-\beta_jz}$ for $\alpha_j, \beta_j, \delta, a_0 \geq 0$ and $\sum_j (\alpha_j + \beta_j) < \infty$. Find a proof that does not use Nevanlinna theory: in other words, the fact that if $F(z)$ is entire of genus at most one, and has no zeros, then it is an exponential function $ae^{\delta z}$ for some $\delta \geq 0$.

来源字段：http://aimpl.org/totalpos/1/

01747 | AIM-REPRESENTATION_THEORY-0088

### An elementary-generator characterization of the André-Yan theory

P / partially_solved | AIM 研讨会问题汇编

Characterize the André-Yan theory of \(U_n(q)\) intrinsically in terms of its elementary, single-box irreducible characters, and determine whether it is the coarsest supercharacter theory in which those characters remain individual supercharacters.

The original PDF was checked directly. It gives the same wording (with André printed without the accent) and no additional remark.

The dependence on the preceding item is essential. Problem 1.3 reads:

The following Problem 1.6 also records the key structural fact that, for \(U_n\), every supercharacter is a product of supercharacters that happen to be irreducible. Thus the most literal precise reconstruction of Problem 1.4 is:

This report proves such a characterization. Since the source request is open-ended and never defines what class of “characterizations” is intended, the result is conservatively classified as a rigorous partial result rather than as an unconditional full solution to every possible reading.

The list came from the AIM workshop *Supercharacters and combinatorial Hopf algebras*, held May 17-21, 2010.

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

01748 | AIM-LINEAR_ALGEBRA-0014

### An independent-cycle obstruction to Pólya frequency

S / solved | AIM 研讨会问题汇编

The record is Problem 1.5 in the AIM workshop list *Theory and applications of total positivity* and is attributed there to Pavlo Pylyavskyy. The source display is a nested sum: for each \(k\), its inner sum contributes one copy of \(z^k\) for every matching with \(f(\mu')=k\). Equivalently,

\[ F_{G,\mu}(z):=\sum_{\substack{\mu'\text{ perfect}\\ \mu'\ne\mu}} z^{f(\mu')}=\sum_{k\geq0}a_kz^k, \qquad a_k=\#\{\mu'\ne\mu:f(\mu')=k\}. \]

Here \(f(\mu')\) is the number of noncontractible cycles obtained by superimposing \(\mu\) and \(\mu'\). Equivalently, one may discard the doubled common edges and use the alternating cycles of \(\mu\mathbin\triangle\mu'\); doubled common edges are contractible and do not affect \(f\).

The questions are whether \((a_k)\) is always a Pólya-frequency (PF) sequence and, after putting indeterminate weights on the edges, whether the associated infinite triangular Toeplitz matrix is coefficientwise totally nonnegative (TN). The source notes that coefficientwise \(\mathrm{TN}_2\) is known.

来源字段：http://aimpl.org/totalpos/1/

01749 | AIM-REPRESENTATION_THEORY-0033

### An obstruction to double-covering FI and a framed spin-FI supercategory

P / partially_solved | AIM 研讨会问题汇编

**FI like categories.** Modify $\mathrm{FI}$ to use double cover of symmetric groups.

The exact AIM record, from the workshop *Representation stability*, section "Representation theory," problem 3.4, is:

We therefore distinguish three plausible readings.

1. Construct a category over FI whose automorphism group at $[n]$ is a Schur double cover \(1\to\{1,z\}\to\widehat S_n\to S_n\to1\), and whose every Hom-set is a two-sheeted lift of the corresponding FI Hom-set. 2. Construct a linear category whose endomorphism algebras are the genuine spin quotients \(k[\widehat S_n]/(z+1)\), so its modules are sequences of genuine spin representations. 3. Use the Sergeev/Hecke--Clifford superalgebras, which are super-Morita equivalent to spin symmetric-group algebras, and formulate stability in the super setting.

The first reading is the most literal but is impossible. The obstruction is already visible on the map from the empty set. We prove it below. We then give a coherent correction for readings (2)--(3): retain an ordered frame of the complement of an injection, lift those framed morphisms to compatible Schur covers, and pass to the genuine superlinear sector. The correction has explicit composition, stabilizers, free modules, and a classification of its modules as compatible sequences.

来源字段：http://aimpl.org/repnstability/3/

01750 | ALG-009

### Andrews-Curtis Conjecture

P / partially_solved | 独立 / 未分配集合

别名：The Andrews-Curtis Conjecture [ALG-023]; Andrews-Curtis Conjecture [ALG-053]; Exact escape of exponent rectangles in Thompson's group F [AIM-GEOMETRIC_GROUP_THEORY-0087]

Can every balanced presentation of the trivial group be transformed into a trivial presentation by a sequence of Nielsen transformations and conjugations of relators?

背景链接：https://doi.org/10.1007/s10817-026-09759-8

01751 | OWR-3475-005

### Anisotropic Bilinear Forms in Characteristic Two

P / partially_solved | Oberwolfach 报告问题汇编

Over a field $F$ of characteristic $2$, for integers $h>0$ and $d\geq0$, which anisotropic regular symmetric bilinear forms have height $h$ and degree $d$?

来源字段：https://doi.org/10.4171/owr/2009/25

01752 | AIM-REPRESENTATION_THEORY-0081

### Applications and a positive nonvanishing sieve for covering-group trace formulas

P / partially_solved | AIM 研讨会问题汇编

Are there applications of the invariant trace formula for $\widetilde{G}_{F}$?

来源字段：http://aimpl.org/autoformcovergp/4/

01753 | OWR-12697689-012

### Arc Hilbert-Poincaré Series of ADE Surface Singularities

O / open | Oberwolfach 报告问题汇编

Does the arc Hilbert-Poincaré series characterize ADE surface singularities?

来源字段：https://doi.org/10.4171/owr/2023/6

01754 | ALG-015

### Are All Groups Surjunctive?

O / open | 独立 / 未分配集合

Is every group surjunctive? That is, for any group $G$, if $\phi: A^G \to A^G$ is a cellular automaton that is injective, must it also be surjective?

背景链接：https://en.wikipedia.org/wiki/Surjunctive_group

01755 | AIM-REPRESENTATION_THEORY-0008

### Arithmetic consequences and height gaps from horizontal CM packet equidistribution

S / solved | AIM 研讨会问题汇编

Fix an imaginary quadratic field \(K\). For the order \[ \mathcal O_C=\mathbf Z+C\mathcal O_K \] of squarefree conductor \(C\), let \(P_C\) be a full ring-class/Galois packet of CM points of endomorphism order \(\mathcal O_C\) on a fixed modular or quaternionic Shimura curve, subject to the fixed level and local embedding conditions. As the admissible squarefree integers \(C\) tend to infinity, the uniform measures on \(P_C\) converge to the canonical probability measure. What arithmetic consequences can be extracted?

来源字段：http://aimpl.org/aagaautomorphic/4/

01756 | AMR-046-0028

### Around Kouchnirenko's conjecture I

P / partially_solved | AMR 问题清单汇编

Find a reasonable, or sharp, upper bound in terms of $m_1,m_2$ for the maximum number of simple positive-coordinate solutions of a real polynomial system $f_1(x,y)=f_2(x,y)=0$, where $m_i$ is the number of monomials of $f_i$.

背景链接：https://arxiv.org/abs/2012.02524

01757 | AMR-046-0029

### Around Kouchnirenko's conjecture II

P / partially_solved | AMR 问题清单汇编

Is $(2m_1-1)(2m_2-1)$ the maximum number of simple solutions of a real polynomial system $f_1(x,y)=f_2(x,y)=0$, where $m_i$ is the number of monomials of $f_i$?

背景链接：https://arxiv.org/abs/2012.02524

01758 | ALG-034

### Arthur's Conjectures

P / partially_solved | 独立 / 未分配集合

What is the structure of the discrete spectrum of automorphic forms on reductive groups?

背景链接：https://bookstore.ams.org/COLL/61

01759 | OWR-16773-002

### Artin-Schelter Gorenstein Property of Noetherian Hopf Algebras

O / open | Oberwolfach 报告问题汇编

Is every noetherian Hopf algebra Artin-Schelter Gorenstein?

来源字段：https://doi.org/10.4171/owr/2019/11

01760 | OWR-1272-004

### AS-Gorenstein Properties of Noetherian Hopf Algebras

O / open | Oberwolfach 报告问题汇编

Is every Noetherian Hopf algebra AS-Gorenstein?

来源字段：https://doi.org/10.4171/owr/2006/23

01761 | OWR-15178-020

### Asymptotic Lengths of Local Cohomology Modules

P / partially_solved | Oberwolfach 报告问题汇编

Let $(R,\mathfrak m)$ be a d-dimensional local ring, I an ideal, and $(I_n)$ a decreasing sequence cofinal with the powers of I. If $\operatorname{length}H^i_{\mathfrak m}(R/I_n)<\infty$ for all sufficiently large n, what is the asymptotic behavior of $I_n^i=\operatorname{length}H^i_{\mathfrak m}(R/I_n)$?

来源字段：https://doi.org/10.4171/owr/2016/57

01762 | OWR-2654827-015

### Asymptotic Minimal Quadratic Growth of Gorenstein Algebras

O / open | Oberwolfach 报告问题汇编

Let

$$\nu(r)=\min_{A\in QG(r)}\dim A_2,$$

where $QG(r)$ consists of codimension-$r$ Artinian Gorenstein algebras presented by quadrics and having socle degree $4$. Is

$$\lim_{r\to\infty}\frac{\nu(r)}{r^{2/3}}=6$$?

来源字段：https://doi.org/10.4171/owr/2020/31

01763 | OWR-3393-007

### Atypicality and Support Varieties of Simple Supermodules

P / partially_solved | Oberwolfach 报告问题汇编

For a simple $\mathfrak g$-module $L(\lambda)$ of highest weight $\lambda$, is its atypicality equal to the dimension of the associated support/rank variety $V_{(\mathfrak e,\mathfrak e_{\bar 0})}(L(\lambda))$?

来源字段：https://doi.org/10.4171/owr/2009/10

01764 | OWR-4216-010

### Atypicality Detection by Ambidextrous Modified Dimensions

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathfrak g$ be a basic classical Lie superalgebra, let $J$ be a simple $\mathfrak g$-supermodule admitting an ambidextrous trace, and let $L\in\mathcal I_J$ be simple. Is

$$\operatorname{atyp}(L)=\operatorname{atyp}(J)$$

if and only if $d_J(L)\neq0$?

来源字段：https://doi.org/10.4171/owr/2010/15

01765 | AIM-REPRESENTATION_THEORY-0089

### Automorphism coarsenings and the reversal-quotient theory of U_n(q)

P / partially_solved | AIM 研讨会问题汇编

Problem 1.5. Consider all the automorphisms of the unitriangular group. Do these automor-phisms give additional nice cumpling (coarsification) to the theory?

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

01766 | OWR-1452-021

### Automorphisms of Weyl-Algebra Automorphism Groups

O / open | Oberwolfach 报告问题汇编

For $n>1$, are the groups $\operatorname{Aut}(\operatorname{Aut}(W_n))$ and $\operatorname{Aut}(\operatorname{Sympl}(\mathbb{C}^{2n}))$ generated by conjugations and automorphisms of the ground field?

来源字段：https://doi.org/10.4171/owr/2007/01

01767 | OWR-1542-012

### Ax-Schanuel Theory for Non-Isoconstant D-Groups

O / open | Oberwolfach 报告问题汇编

Extend Ax’s generalized Schanuel transcendence theorem, including its logarithmic and exponential differential-Galois formulations, from constant commutative algebraic groups over $\mathbb C$ to non-isoconstant algebraic $D$-groups over a function field.

来源字段：https://doi.org/10.4171/owr/2007/26

01768 | OWR-1452-006

### Bad Hyperplanes and Residual Variables in Polynomial Rings

O / open | Oberwolfach 报告问题汇编

For $d\geq3$ and a characteristic-zero ring $R$, can constructions of bad lines and bad residual variables be generalized to produce bad hyperplanes and bad residual variables in $R[x_1,\ldots,x_d]$?

来源字段：https://doi.org/10.4171/owr/2007/01

01769 | OWR-14298370-004

### Balmer Spectra of Stable Finite Tensor Categories

P / partially_solved | Oberwolfach 报告问题汇编

Describe the Balmer spectrum of the stable category of a finite tensor category in terms of its cohomology ring.

来源字段：https://doi.org/10.4171/owr/2024/20

01770 | OWR-14299915-003

### Balmer Spectra of Stable Finite Tensor Categories

P / partially_solved | Oberwolfach 报告问题汇编

For every finite tensor category $\mathcal C$, identify the Balmer spectrum of $\operatorname{stab}\mathcal C$ with the projective spectrum of the categorical center $Z_{\mathcal C}(H^*(\mathcal C))$.

来源字段：https://doi.org/10.4171/owr/2026/9

01771 | AIM-REPRESENTATION_THEORY-0074

### Base change for tame covering groups and a degree-normalization obstruction

P / partially_solved | AIM 研讨会问题汇编

**Base change.** Let $F$ be a $p$-adic field of characteristic zero, $G_F$ an unramified linear group (quasi-split over $F$ and split over a finite unramified extension $L/F$), and $\widetilde G_F$ a covering group of degree coprime to $p$. Given a genuine irreducible representation $\widetilde\pi$ of $\widetilde G_F$, describe $\operatorname{Lift}_{L/F}(\widetilde\pi)$, supposed to be a virtual representation of $\widetilde G_L$.

来源字段：http://aimpl.org/autoformcovergp/3/

01772 | OWR-5155-006

### Beilinson-Soulé Vanishing Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For every regular ring $R$, is $K_m(R)=F^\gamma_nK_m(R)$ for $m\geq\max\{2n-1,1\}$, equivalently, does the Adams eigenspace $K_m(R)^{(n)}$ vanish in the excluded range except for $K_0(R)^{(0)}$ (the Beilinson-Soulé vanishing conjecture)?

来源字段：https://doi.org/10.4171/owr/2011/27

01773 | OWR-14298157-002

### Belavin-Drinfeld Cluster Correspondence

P / partially_solved | Oberwolfach 报告问题汇编

Do Poisson-Lie and Poisson-homogeneous structures arising from Belavin-Drinfeld quasitriangular solutions of the classical Yang-Baxter equation admit compatible cluster structures?

来源字段：https://doi.org/10.4171/owr/2024/2

01774 | OWR-734-002

### Betti-Preserving Initial Ideals in Fixed Coordinates

P / partially_solved | Oberwolfach 报告问题汇编

Given a classical ideal $I$, does it have an initial ideal $H$ in the given coordinates with identical graded Betti numbers in every bidegree?

来源字段：https://doi.org/10.4171/owr/2004/32

01775 | OWR-15219-022

### Bianalytic Free Spectrahedron Maps Without Genericity

O / open | Oberwolfach 报告问题汇编

In the partial converse classifying bianalytic maps between bounded free spectrahedra, can the genericity hypothesis on the coefficient tuples $A$ and $B$ be removed?

来源字段：https://doi.org/10.4171/owr/2017/14

01776 | OWR-11101916-011

### Bicategorical Limits of Nonproper Product Systems

P / partially_solved | Oberwolfach 报告问题汇编

Does a bicategorical limit exist for a nonproper product system over a monoid, and if so how can it be described?

来源字段：https://doi.org/10.4171/owr/2022/35

01777 | OWR-16768-012

### Big Test Ideals from Plus Closure

O / open | Oberwolfach 报告问题汇编

Let $R$ be a localization of a numerically $\mathbb Q$-Gorenstein normal domain of finite type over an $F$-finite field, and set

$$P(R)=\bigcap_{I\subseteq R}(I:I^+),\qquad I^+=IR^+\cap R.$$

Is the big test ideal $\tau_b(R)$ equal to $P(R)$?

来源字段：https://doi.org/10.4171/owr/2019/6

01778 | OWR-1272-003

### Bijectivity of Antipodes in Noetherian Hopf Algebras

O / open | Oberwolfach 报告问题汇编

Is the antipode $S$ bijective for every Noetherian Hopf algebra $A$?

来源字段：https://doi.org/10.4171/owr/2006/23

01779 | OWR-17294-017

### Bijectivity of the Quantum Symmetrizer

O / open | Oberwolfach 报告问题汇编

Is the quantum symmetrizer $\operatorname{QS}$ bijective for general set-theoretic Yang-Baxter solutions?

来源字段：https://doi.org/10.4171/owr/2019/51

01780 | OWR-13681-014

### Bilax Monoidal Structures for Stable Morita Equivalences

P / partially_solved | Oberwolfach 报告问题汇编

For a stable equivalence of Morita type $(M,N,\varphi,\psi)$ between algebras $A$ and $B$, can the functor $M\otimes_B(-)\otimes_B N:\operatorname{Mod}(B^{\mathrm{ev}})\to\operatorname{Mod}(A^{\mathrm{ev}})$ be made bilax monoidal with split-surjective unit map $M\otimes_B N\twoheadrightarrow A\oplus P\to A$?

来源字段：https://doi.org/10.4171/owr/2015/24

01781 | ALG-017

### Birch-Tate Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Birch-Tate Conjecture [ALG-051]; Birch-Tate Conjecture [ALG-055]

Does the order of the center of the Steinberg group of the ring of integers of a number field relate to the value of the Dedekind zeta function at $s=-1$?

背景链接：https://en.wikipedia.org/wiki/Birch%E2%80%93Tate_conjecture

01782 | OWR-12697690-005

### Birkhoff Classification over $\mathbb Z/p^6\mathbb Z$

O / open | Oberwolfach 报告问题汇编

Classify monomorphisms of finite abelian groups over $\mathbb{Z}/p^6\mathbb{Z}$ up to Birkhoff equivalence.

来源字段：https://doi.org/10.4171/owr/2023/7

01783 | OWR-11580-013

### Block Decompositions of Critical-Level Affine Modules

O / open | Oberwolfach 报告问题汇编

Using the center of the critical-level affine enveloping algebra, determine whether its abelian category of modules admits a canonical direct-product decomposition into blocks.

来源字段：https://doi.org/10.4171/owr/2012/13

01784 | OWR-5149-003

### Blocks of Symmetric-Group Centralizer Algebras

O / open | Oberwolfach 报告问题汇编

For the centralizer algebra $kS_n^{S_\ell}$, are all blocks represented by products $ef$ of a block idempotent $e$ of $kS_n$ and a block idempotent $f$ of $kS_\ell$?

来源字段：https://doi.org/10.4171/owr/2011/21

01785 | AIM-REPRESENTATION_THEORY-0010

### Boecherer central values determine autocorrelation, not Fourier phases

P / partially_solved | AIM 研讨会问题汇编

Let \(d\) be a positive integer and let \(F\) be a Siegel modular form with Fourier coefficients \(a(F)\). In many cases, we know the (generalized) Böcherer conjecture \[ \left|\sum_{S\in\mathrm{Cl}_d}a(F,S)\Lambda(S)\right|^2 =L(\pi_F\times\theta_\Lambda,1/2) =L(\mathrm{BC}_{K/\mathbb Q}(\pi_F)\times\Lambda,1/2). \] Knowing this equality and the values for many \(\Lambda\), can we compute the Fourier coefficients \(a(F,S)\)?

来源字段：http://aimpl.org/aagaautomorphic/5/

01786 | AIM-REPRESENTATION_THEORY-0039

### Borel restrictions of tilting modules: a rank-one classification and Frobenius obstruction

P / partially_solved | AIM 研讨会问题汇编

What rational representations of $B$ arise by restricting a tilting module of $G$?

来源字段：http://aimpl.org/sheavemodular/1/

01787 | AIM-REPRESENTATION_THEORY-0094

### Borel supercharacters and a split torus-degree Hopf structure

S / solved | AIM 研讨会问题汇编

Try to carry over all of the \(U_n\) work to the Borel subgroup \(B_n\) of \(GL_n\). Is there a similar (module to a) Hopf algebra?

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

01788 | OWR-2048-002

### Bost Assembly Conjecture with $C^*$-Algebra Coefficients

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a locally compact second-countable Hausdorff group and $A$ a $G$-$C^*$-algebra. Is the Bost assembly map $$\mu^{G,A}_{L^1}:K_*^{G,\mathrm{top}}(EG;A)\longrightarrow K_*(L^1(G,A))$$ an isomorphism for every coefficient algebra $A$?

来源字段：https://doi.org/10.4171/owr/2008/37

01789 | ALG-010

### Bounded Burnside Problem

P / partially_solved | 独立 / 未分配集合

别名：The Bounded Burnside Problem [ALG-024]

For which positive integers $m$ and $n$ is the free Burnside group $B(m,n)$ finite? In particular, is $B(2,5)$ finite?

背景链接：https://mathshistory.st-andrews.ac.uk/HistTopics/Burnside_problem/

01790 | OWR-12591-002

### Bounded Maps Between Close Operator Algebras

O / open | Oberwolfach 报告问题汇编

What methods produce bounded linear maps between close, or completely close, operator algebras?

来源字段：https://doi.org/10.4171/owr/2013/43

01791 | GREEN-090

### Bounds for Approximate Group Classification

P / partially_solved | Ben Green 问题

Find bounds in the classification theorem for approximate groups.

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

01792 | OWR-17466-001

### Braid-Group Extensions of Zamolodchikov Actions

O / open | Oberwolfach 报告问题汇编

Let $\Delta$ and $\Delta'$ be simply laced Dynkin diagrams with Coxeter numbers $k$ and $n-k$. Does the cyclic action generated by the Zamolodchikov transformation on the cluster algebra $\mathcal A(\Delta\square\Delta')$ extend to an action of the extended braid group $\widetilde{\operatorname{Br}}_d$, where $d=\gcd(k,n)$?

来源字段：https://doi.org/10.4171/owr/2020/3

01793 | OWR-14298013-004

### BV Structures with Nonsemisimple Nakayama Automorphisms

O / open | Oberwolfach 报告问题汇编

Let $A_h$ be a twisted Calabi-Yau algebra whose Nakayama automorphism is not semisimple. Determine whether $HH^\bullet(A_h)$ admits a Batalin-Vilkovisky structure.

来源字段：https://doi.org/10.4171/owr/2023/46

01794 | OWR-17466-014

### Caldero-Chapoton Maps for General Hereditary Algebras

O / open | Oberwolfach 报告问题汇编

For a general hereditary algebra, what is the appropriate analogue of the Euler characteristic of a submodule quiver Grassmannian for defining the Caldero-Chapoton map in all relevant cases?

来源字段：https://doi.org/10.4171/owr/2020/3

01795 | OWR-17289-006

### Canonical Bases from Quiver Grassmannian Orbit Closures

O / open | Oberwolfach 报告问题汇编

How is the canonical basis formed by the intersection-cohomology functions $f_{\mathcal O}$ on quiver-Grassmannian orbit closures related to variants of Lusztig's canonical basis in quantized enveloping algebras?

来源字段：https://doi.org/10.4171/owr/2019/46

01796 | OWR-14298157-005

### Canonical Decomposition Cones and TF Equivalence

P / partially_solved | Oberwolfach 报告问题汇编

For every finite-dimensional algebra $A$ and $\theta\in K_0(\operatorname{proj}A)$, is the TF-equivalence class $[\theta]_{\mathrm{TF}}$ equal to the relative interior of the cone generated by $\operatorname{ind}(\mathbb{N}\theta)$?

来源字段：https://doi.org/10.4171/owr/2024/2

01797 | OWR-12480-001

### Canonical Dimension of Prime-Power-Split Anisotropic Tori

O / open | Oberwolfach 报告问题汇编

Is $\operatorname{cdim}(T)=\dim T$ for every anisotropic algebraic torus $T$ split by a Galois extension of prime-power degree?

来源字段：https://doi.org/10.4171/owr/2013/17

01798 | AIM-REPRESENTATION_THEORY-0085

### Canonical supercharacter products: an exact divisibility criterion

P / partially_solved | AIM 研讨会问题汇编

Let \(G\) be a finite group. Let \(\mathcal X\) be a partition of \(\operatorname{Irr}(G)\) defining a supercharacter theory, and put \(\sigma_X=\sum_{\chi\in X}\chi(1)\chi\). When is the product of two such \(\sigma\)'s a positive integer combination of the \(\sigma\)'s?

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

01799 | OWR-15219-021

### Carathéodory Bounds and Independent Evaluation Vectors

O / open | Oberwolfach 报告问题汇编

For $d\geq3$, are the Carathéodory number $C_{2d}$ and the maximal finite projective-zero count $\beta(2d)$ related by $$\beta(2d)\leq C_{2d}\leq\beta(2d)+1?$$ If $f\in B_{n,2d}$ is nonnegative on $\mathbb R^n$ and has projective zero set $\{z_1,\ldots,z_k\}$, must the evaluation vectors $\{s_{B_{n,2d}}(z_i):1\leq i\leq k\}$ be linearly independent?

来源字段：https://doi.org/10.4171/owr/2017/14

01800 | OWR-14218-006

### Cartan Determinant Conjecture

O / open | Oberwolfach 报告问题汇编

If $A$ is an Artin algebra of finite global dimension, must its Cartan determinant satisfy $\det C(A)=1$?

来源字段：https://doi.org/10.4171/owr/2016/10

01801 | OWR-12170-002

### Cartan Schemes and Positive-Characteristic Lie Algebras

O / open | Oberwolfach 报告问题汇编

Are the objects in the classification of finite Cartan schemes that do not occur in the classification of arithmetic root systems related to Lie algebras in positive characteristic?

来源字段：https://doi.org/10.4171/owr/2012/47

01802 | OWR-11101917-006

### Cartan Subalgebras from Twisted Groupoid Subgroupoids

P / partially_solved | Oberwolfach 报告问题汇编

Characterize the clopen normal subgroupoids $S$ of a twisted étale groupoid $(G,\Sigma)$ for which $C_r^*(S;\Sigma|_S)$ is a Cartan subalgebra of $C_r^*(G;\Sigma)$.

来源字段：https://doi.org/10.4171/owr/2022/36

01803 | OWR-15585-008

### Cartan Subalgebras in Classifiable $C^*$-Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Does the Razak-Jacelon algebra $\mathcal W$ have a Cartan subalgebra? Does the Jiang-Su algebra $\mathcal Z$ have a distinguished one-dimensional Cartan subalgebra? Do strongly self-absorbing Cartan pairs admit useful existence and uniqueness theorems? What $K$-theory map is induced by the inclusion of a Cartan subalgebra, what equivalence relation models Popa intertwining, and how do Cartan subalgebras interact with completely positive approximation? Does the CAR algebra have a Cartan subalgebra with connected spectrum? When does an equivariant embedding into $M_{2^\infty}$ have maximal-abelian image? Is every order-two automorphism of $\mathcal O_2$ strongly approximately inner?

来源字段：https://doi.org/10.4171/owr/2017/42

01804 | ALG-008

### Casas-Alvero Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Casas-Alvero Conjecture [ALG-056]; Existence of Nonsofic Groups [OWR-5154-008]; Existence of Nonsofic Groups [OWR-14299906-006]

If a univariate polynomial $f$ of degree $d$ over a field of characteristic 0 shares a common factor with each of its first $d-1$ derivatives, must $f$ be a power of a linear polynomial?

背景链接：https://arxiv.org/abs/2501.09272

01805 | OWR-14298807-001

### Casimir Substitutes for Kac-Moody Enveloping Algebras

O / open | Oberwolfach 报告问题汇编

Does a substitute for the Casimir element exist in the universal enveloping algebra of $\mathfrak{k}(A)$, or in a suitable completion?

来源字段：https://doi.org/10.4171/owr/2024/53

01806 | OWR-8415347-010

### Categorical Determination of KMS-State Structure

O / open | Oberwolfach 报告问题汇编

Can further information about the structure of KMS states be derived from purely categorical considerations?

来源字段：https://doi.org/10.4171/owr/2021/44

01807 | OWR-13682-002

### Categorical Diagonalization of Coxeter Full Twists

P / partially_solved | Oberwolfach 报告问题汇编

For any finite Coxeter group $W$, is the Rouquier complex $F_T$ of the full twist categorically diagonalizable, with spectrum indexed by the two-sided cells of $W$ and with the predicted eigenshifts?

来源字段：https://doi.org/10.4171/owr/2015/25

01808 | OWR-14754-009

### Categorical Functors from Soergel Bimodules to Flag Hilbert Schemes

O / open | Oberwolfach 报告问题汇编

Do there exist adjoint functors between $K_b(\operatorname{SBim}_n)$ and $D_b(\operatorname{Coh}^T(\operatorname{FHilb}_n(\mathbb{C})))$ such that the forward functor is monoidal and sends $L_1\otimes\cdots\otimes L_{k-1}$ to $C(FT_k)$?

来源字段：https://doi.org/10.4171/owr/2016/46

01809 | OWR-3400-004

### Cellular Models for Unequal Parameter Hecke Algebras

P / partially_solved | Oberwolfach 报告问题汇编

For the type-$B$ Hecke algebra with unequal parameters $Q$ and $q$ and a positive irrational weight parameter $\xi$, does Kazhdan-Lusztig theory provide the conjectured cellular datum indexed by bipartitions and standard bitableaux, yielding Specht modules whose specialized simple quotients realize all of Ariki's parametrizations and whose decomposition numbers are given by the canonical-basis transition matrix?

来源字段：https://doi.org/10.4171/owr/2009/17

01810 | OWR-11783-008

### Cellular Resolutions for Consistent Toric Algebras

O / open | Oberwolfach 报告问题汇编

Let $X$ be an $n$-dimensional Gorenstein affine toric variety, let $\mathcal E=(E_0,\ldots,E_r)$ be rank-one reflexive sheaves, and set $A=\operatorname{End}_{\mathcal O_X}(\bigoplus_i E_i)$. If $A$ is a consistent toric algebra of global dimension $n$, does its higher quiver exist, and is the minimal projective $(A,A)$-bimodule resolution the cellular resolution encoded by that higher quiver?

来源字段：https://doi.org/10.4171/owr/2012/21

01811 | OWR-14299281-010

### Centers of Small Quantum Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $W_P\subseteq W$ be the stabilizer of a weight $\lambda$, and let $DR_W^{W_P}$ be the canonical quotient of the diagonal coinvariant algebra. Is the $G$-invariant part $z_\lambda^G$ of the corresponding block of the small-quantum-group center isomorphic to $DR_W^{W_P}$ as a bigraded vector space?

来源字段：https://doi.org/10.4171/owr/2025/20

01812 | OWR-12175-022

### Central Sequence Invariants for $C^*$-Algebra Classification

O / open | Oberwolfach 报告问题汇编

Does the invariant $F(A)$ contain all technical information needed to decide whether $A$ is classifiable, together with information about its class? Do the $K$-groups of $F_{\mathrm{cont}}(A)$ determine the $KK$-equivalence class of $A$?

来源字段：https://doi.org/10.4171/owr/2012/52

01813 | AIM-REPRESENTATION_THEORY-0028

## Central stability of the standard VIC module and the complement-orbit correction

P / partially_solved | AIM 研讨会问题汇编

It starts over a field \(\mathbf k\) with

\[ \mathrm{PBC}_{p}^{n} =\{(v_1,\ldots,v_p,C):v_i\text{ are linearly independent and } C\text{ complements their span}\}. \]

It says that these sets form a semisimplicial \(\mathbf{VIC}(\mathbf k)\)-set, that the resulting space is \((n-3)/2\)-connected, and that the augmented chain complex

\[ \mathbb Z[\mathrm{PBC}_\bullet]\longrightarrow\mathbb Z\longrightarrow0 \]

is “acyclic (exact).” It describes a face as removing one vector and adding that vector to the complement. It then claims the same over a general ring \(R\), including the identification

\[ \mathbb Z[\mathrm{PBC}_p] =\mathbb Z[\mathrm{GL}_n(R)] \otimes_{\mathbb Z[\mathrm{GL}_{n-(p+1)}(R)]}\mathbb Z, \]

and asks whether

\[ R[\mathrm{GL}_n(R)]\otimes_{R[\mathrm{GL}_{n-(\bullet+1)}(R)]} R^{n-(\bullet+1)}\longrightarrow R^n\longrightarrow0 \tag{1} \]

is highly acyclic. The record says that an affirmative answer for \(R=\mathbb Z\) would imply that \(H_2(\mathrm{IA}_n)\) is presented in finite degree.

The display requires five repairs.

1. **Semisimplicial indexing.** A \(p\)-simplex has \(p+1\) vectors, conventionally written \((v_0,\ldots,v_p,C)\). The exponent \(n-(p+1)\) in (1) confirms this convention. The source definition with only \(p\) vectors is shifted. 2. **Connectivity is not total exactness.** An \(r\)-connected realization has vanishing reduced homology only through degree \(r\). It need not have an exact augmented chain complex in every degree. A concrete counterexample appears in Theorem C below. 3. **Ring versus coefficient ring.** The same symbol \(R\) is being used for the ring defining \(\mathrm{GL}_n(R)\) and for chain coefficients. This is legitimate for a commutative ring, but it must be stated; over a noncommutative ring one must also fix left/right conventions. 4. **The denominator needs a block embedding.** We use \(\mathrm{GL}_{n-p-1}(R)\hookrightarrow\mathrm{GL}_n(R)\) acting on the standard complement and fixing the last \(p+1\) standard vectors. There is no omitted unipotent group when the complement itself is distinguished: preserving it kills the off-diagonal block. 5. **General rings have an orbit problem.** If “complement” permits a nonfree stably free summand, the \(\mathrm{GL}_n(R)\)-action is not transitive and the single induced module in the source sees only the free-complement orbit. Theorem B gives the correct orbit decomposition and an explicit example.

With these repairs, (1) is exactly the central stability complex of the standard \(\mathrm{VIC}(R)\)-module \(A_n=R^n\).

来源字段：http://aimpl.org/repnstability/2/

01814 | OWR-4533-004

## Centralizer Realization of Cartan Subalgebras in $L^*$-Algebras

S / solved | Oberwolfach 报告问题汇编

Is every Cartan subalgebra of a simple $L^*$-algebra the centralizer of one of its elements?

来源字段：https://doi.org/10.4171/owr/2010/51

01815 | OWR-16635-001

## Character Determination from Jantzen’s Sum Formula

P / partially_solved | Oberwolfach 报告问题汇编

How far can Jantzen's sum formula determine the characters of simple representations, beyond the groups where it is already complete?

来源字段：https://doi.org/10.4171/owr/2018/52

01816 | OWR-4800-006

## Character Multiplicities and Kac-Moody Roots

P / partially_solved | Oberwolfach 报告问题汇编

For a generic tuple of irreducible characters $X_1,\ldots,X_k$ of $\operatorname{GL}_n(\mathbb F_q)$ and its associated quiver dimension vector $v$, is $\langle X_1\otimes\cdots\otimes X_k,1\rangle$ an integer polynomial in $q$ with positive coefficients when the characters are split and leading coefficient $1$, nonzero exactly when $v$ is a Kac-Moody root, and equal to $1$ exactly for a real root?

来源字段：https://doi.org/10.4171/owr/2011/10

01817 | OWR-14299911-028

## Characterization of Graphical Apiculate and Generalized Median Algebras

O / open | Oberwolfach 报告问题汇编

A ternary algebra is called apiculate if it is the apex algebra of an apiculate graph, and graphical if it and all of its subalgebras are apiculate algebras. Characterize the apiculate graphs whose apex algebras are graphical. Likewise, characterize the generalized median graphs whose generalized median algebras are graphical.

来源字段：https://doi.org/10.4171/owr/2026/8

01818 | OWR-13678-005

## Characterization of Hermite Ideal Projectors

O / open | Oberwolfach 报告问题汇编

Which finite-dimensional ideal projectors are Hermite projectors?

来源字段：https://doi.org/10.4171/owr/2015/21

01819 | OWR-11101917-009

### Characterization of Isometric Shift Absorption

P / partially_solved | Oberwolfach 报告问题汇编

Let $\alpha:G\curvearrowright A$ be an amenable action on a Kirchberg algebra such that $A\rtimes H$ is a Kirchberg algebra for every closed subgroup $H\subseteq G$. Must $\alpha$ be isometrically shift-absorbing?

来源字段：https://doi.org/10.4171/owr/2022/36

01820 | AIM-REPRESENTATION_THEORY-0001

### Characterwise Jochnowitz congruences via an integral class-group comparison

P / partially_solved | AIM 研讨会问题汇编

Let $f$ be a modular form of weight $2$ and let $\chi$ be a character of the class group of an imaginary quadratic number field $K$. Formulate a direct connection \[ L'(f\otimes\theta_\chi,\tfrac12) \longleftrightarrow L(g\otimes\theta_\chi,\tfrac12), \] for a modular form $g$ on a definite quaternion algebra with some congruence to $f$.

The AIM record (workshop *Analytic, arithmetic, and geometric aspects of automorphic forms*, Group Problems 1.1) asks:

Two clarifications are essential.

1. The displayed center $s=1/2$ is the unitary automorphic normalization. For a weight-$2$ classical newform, \[ L_{\mathrm u}(s,\pi_f\times\operatorname{AI}_K^{\mathbf Q}\chi) =L_{\mathrm{cl}}(s+\tfrac12,f/K,\chi). \] Thus the same center is $s=1$ in the classical notation used by Bertolini--Darmon and Vigni. 2. A weight-$2$ form on a definite quaternion algebra is an algebraic automorphic form on a finite double-coset set. Its Jacquet--Langlands transfer is the corresponding classical $\mathrm{GL}_2$ newform. The definite form itself does not have a holomorphic $q$-expansion.

Here $\operatorname{AI}_K^{\mathbf Q}\chi$ is represented by the weight-$1$ theta series $\theta_\chi$ when the induction is cuspidal. For exceptional characters for which the induction decomposes, the equivalent base-change $L$-function formulation remains the unambiguous one.

来源字段：http://aimpl.org/aagaautomorphic/1/

01821 | OWR-734-006

### Charalambous-Reeves Standard-Matching Conjecture

O / open | Oberwolfach 报告问题汇编

Let $S=k[x_1,\ldots,x_n]$, let $\mathfrak a\subset S$ be a monomial ideal with minimal generating set $G(\mathfrak a)$, and put $A=S/\mathfrak a$. If $M$ is a standard matching on the Taylor resolution, is the multigraded Poincaré series $P_A(\mathbf x,t)$ equal to $\prod_{i=1}^n(1+tx_i)$ divided by $1+\sum_I(-1)^{\operatorname{cl}(I)}m_I t^{\operatorname{cl}(I)+|I|}$, where the sum runs over subsets $I\subseteq G(\mathfrak a)$ left unmatched by $M$, $m_I=\operatorname{lcm}(I)$, and $\operatorname{cl}(I)$ is the number of connected components under the relation generated by non-coprimality?

来源字段：https://doi.org/10.4171/owr/2004/32

01822 | OWR-14211-009

### Chern Class Comparison for Arithmetic General Linear Groups

P / partially_solved | Oberwolfach 报告问题汇编

For which integers n is the natural morphism $\gamma_{n,\mathbb{Z}[1/2]}$ from Grothendieck's mod-2 Chern-class algebra to $H^*(GL_n(\mathbb{Z}[1/2]);\mathbb{F}_2)$ an isomorphism?

来源字段：https://doi.org/10.4171/owr/2016/3

01823 | OWR-12862-004

### Classifiable Conjugacy in Automorphism Groups of Simple $C^*$-Algebras

O / open | Oberwolfach 报告问题汇编

Does there exist an infinite-dimensional simple $C^*$-algebra $A$ such that conjugacy in $\operatorname{Aut}(A)$ is classifiable?

来源字段：https://doi.org/10.4171/owr/2014/02

01824 | OWR-14299580-016

### Classifiable Subcategories of Braid-Category Monoidal Functors

O / open | Oberwolfach 报告问题汇编

Which subcategories of $\operatorname{MonFun}(\mathcal B,\operatorname{Mat})$ admit a tractable classification?

来源字段：https://doi.org/10.4171/owr/2025/49

01825 | OWR-8415347-003

### Classification and K-Theory of Compact Quantum Groups

O / open | Oberwolfach 报告问题汇编

Determine when the reduced $C^*$-algebra of a compact quantum group is simple and compute its K-theory, even for the basic examples for which these questions remain open.

来源字段：https://doi.org/10.4171/owr/2021/44

01826 | OWR-12175-010

### Classification Complexity of Unitary Equivalence

O / open | Oberwolfach 报告问题汇编

For a separable $C^*$-algebra $A$ with nonsmooth unitary-equivalence relation $E_A^{\mathrm{u.e.}}$, is $E_A^{\mathrm{u.e.}}$ nonclassifiable by countable structures?

来源字段：https://doi.org/10.4171/owr/2012/52

01827 | OWR-8415347-007

### Classification of Finite-Growth Pre-Nichols Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a finitely generated group and let $V\in{}^{\mathbb{C}G}_{\mathbb{C}G}\mathcal{YD}$ satisfy $\operatorname{GKdim}(\mathcal{B}(V)\#\mathbb{C}G)<\infty$. Classify the pre-Nichols algebras $R$ of $V$ for which $\operatorname{GKdim}(R\#\mathbb{C}G)<\infty$. In addition, if $\mathcal{O}$ is a finite conjugacy class of type C, prove that $\operatorname{GKdim}\mathcal{B}(V)=\infty$ whenever $\operatorname{supp}V=\mathcal{O}$.

来源字段：https://doi.org/10.4171/owr/2021/44

01828 | OWR-17466-011

### Classification of Generalized Quaternion-Type Algebras

O / open | Oberwolfach 报告问题汇编

别名：Weighted-Surface Classification of Quaternion-Type Algebras [OWR-17466-016]; Galactic Conjecture for Quaternion-Type Algebras [OWR-17466-017]

Is every algebra of generalized quaternion type Morita equivalent to a weighted surface algebra or to one of the nine exotic algebras?

来源字段：https://doi.org/10.4171/owr/2020/3

01829 | OWR-13943-005

### Classification of Laminations and Higher Tropical Frieze Patterns

O / open | Oberwolfach 报告问题汇编

Is there a useful description of all A-laminations, and an explicit description of all higher tropical frieze patterns?

来源字段：https://doi.org/10.4171/owr/2015/48

01830 | OWR-1703864-003

### Classification of Liquid Condensed Modules

O / open | Oberwolfach 报告问题汇编

How can liquid condensed $R$-modules be characterized? Equivalently, determine the liquidifications of the compact projective generators $R[\beta S]$.

来源字段：https://doi.org/10.4171/owr/2020/17

01831 | OWR-14299580-015

### Classification of Monoidal Functors from the Braid Category

O / open | Oberwolfach 报告问题汇编

Fix a monoidal subcategory $\mathcal Y$ of the matrix category. Classify the strict monoidal functors in $\operatorname{MonFun}(\mathcal B,\mathcal Y)$, up to the appropriate equivalences.

来源字段：https://doi.org/10.4171/owr/2025/49

01832 | OWR-1062-006

### Classification of Simple Real-Rank-Zero $C^*$-Algebras

O / open | Oberwolfach 报告问题汇编

For simple, separable, nuclear, stable $C^*$-algebras of real rank zero, is every algebra stably finite or purely infinite? Is $K_0$ weakly unperforated? Is $(K_0,K_0^+,K_1)$ a complete invariant?

来源字段：https://doi.org/10.4171/owr/2005/41

01833 | OWR-14298169-005

### Classification of Stein Groups

P / partially_solved | Oberwolfach 报告问题汇编

Classify Stein's groups as topological full groups.

来源字段：https://doi.org/10.4171/owr/2024/12

01834 | OWR-11137-005

### Classification of Unitary Drinfeld Twists

O / open | Oberwolfach 报告问题汇编

Let $G$ be a compact semisimple Lie group and let $G_q$ be its Drinfeld--Jimbo $q$-deformation ($0 < q < 1$). How can unitary Drinfeld $2$-cocycle twists (up to coboundary equivalence) for the compact quantum group $G_q$ be classified or explicitly described?

来源字段：https://doi.org/10.4171/owr/2011/45

01835 | OWR-12170-001

### Classification Problems for Finite Weyl Groupoids

P / partially_solved | Oberwolfach 报告问题汇编

Are all finite Weyl groupoids invariants of Nichols algebras? Can simplicial arrangements be classified? How far does the connection between reflection groupoids and cluster algebras extend? What is the connection with toric varieties? How can sporadic finite Weyl groupoids be explained? Can the classification of finite-dimensional Nichols algebras of diagonal type be derived from finite Weyl groupoids?

来源字段：https://doi.org/10.4171/owr/2012/47

01836 | OWR-12697689-006

### Classifying Rings Up to Singularity Equivalence

O / open | Oberwolfach 报告问题汇编

How fine is the invariant $D_{\mathrm{sg}}(R)$; equivalently, when do two rings $R$ and $S$ satisfy $D_{\mathrm{sg}}(R)\simeq D_{\mathrm{sg}}(S)$?

来源字段：https://doi.org/10.4171/owr/2023/6

01837 | OWR-12868-004

### Cluster Algebras of Open Richardson Varieties

P / partially_solved | Oberwolfach 报告问题汇编

For $v\leq w$ in the Bruhat order, is the cluster algebra $R_{v,w}$ equal to the coordinate ring $\mathbb C[\mathcal R_{v,w}]$ of the open Richardson variety $\mathcal R_{v,w}$?

来源字段：https://doi.org/10.4171/owr/2014/08

01838 | OWR-15429-008

### Cluster Duality and Classical Canonical Bases

O / open | Oberwolfach 报告问题汇编

What is the relationship between the canonical basis $B_{\mathrm{can}}$ constructed by cluster duality and the classical canonical basis $B$ of $\mathbb C[G/U]$?

来源字段：https://doi.org/10.4171/owr/2017/21

01839 | OWR-3398-004

### Cluster Monomials and Simple Quantum Affine Modules

P / partially_solved | Oberwolfach 报告问题汇编

For a simple Lie algebra $\mathfrak{g}$ of type $A$, $D$, or $E$, let $\mathcal{C}_\ell$ be the monoidal subcategory of finite-dimensional quantum-affine $\mathfrak{g}$-modules with Grothendieck ring $R_\ell$. Is every cluster monomial of $R_\ell$ represented by a simple object? Conversely, is a simple object's Grothendieck class a cluster monomial exactly when its tensor square is simple?

来源字段：https://doi.org/10.4171/owr/2009/15

01840 | OWR-14299283-010

### Cluster Monomials as Simple Shifted Quantum Affine Modules

P / partially_solved | Oberwolfach 报告问题汇编

For the cluster algebra $\mathcal{A}_{\Gamma'_\infty}$ and category $\mathcal{O}$ of the associated shifted quantum affine algebra, prove that every cluster monomial is represented by a simple object under the cluster-algebra isomorphism relating these structures.

来源字段：https://doi.org/10.4171/owr/2025/25

01841 | OWR-13682-004

### Cluster Monomials in Dual Canonical Bases

P / partially_solved | Oberwolfach 报告问题汇编

Do all cluster monomials in quantum cluster algebras belong to the dual canonical basis? In Grothendieck-ring cluster algebras of quantum affine representations, do all cluster monomials belong to the basis of simple modules?

来源字段：https://doi.org/10.4171/owr/2015/25

01842 | OWR-12732-003

### Cluster Structures for Belavin-Drinfeld Data

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a simple complex Lie group and let $T=(\Gamma_1,\Gamma_2,\gamma)$ be a Belavin-Drinfeld triple, with associated class $R_T$ of classical $R$-matrices. Does there exist a cluster structure $\mathcal C_T$ on $G$ that is compatible with the Poisson bracket for every $r\in R_T$ and whose upper cluster algebra is naturally isomorphic to $\mathcal O(G)$?

来源字段：https://doi.org/10.4171/owr/2013/58

01843 | OWR-14298159-001

### Cocycle-Twist Classification of Pointed Hopf Algebras

O / open | Oberwolfach 报告问题汇编

Let $H$ be a finite-dimensional pointed Hopf algebra with group of group-like elements $G$ and infinitesimal braiding $V$. Prove that $H$ is a cocycle twist of $\mathcal B(V)\#\mathbb CG$.

来源字段：https://doi.org/10.4171/owr/2024/5

01844 | OWR-14299580-009

### Cocycle-Weighted Representations of Welded Braid Groups

O / open | Oberwolfach 报告问题汇编

For a welded pair $(X,r,s)$, do cocycle weights make the second solution $s$ representation-theoretically relevant? Are the resulting representations of the welded braid group isomorphic to representations arising from a welded pair $(X,r',q)$ built from the derived solution $r'$?

来源字段：https://doi.org/10.4171/owr/2025/49

01845 | OWR-8415355-021

### Coefficient Criterion for Irredundant String-Cone Inequalities

O / open | Oberwolfach 报告问题汇编

For a string cone presented by linear inequalities, is the presentation irredundant if and only if every coefficient in every defining linear form belongs to $\{-1,0,1\}$?

来源字段：https://doi.org/10.4171/owr/2021/52

01846 | OWR-8415355-022

### Coefficient Criterion for Redundant Tropical String Inequalities

O / open | Oberwolfach 报告问题汇编

Does a term of the potential $W|_{T_i^\vee}$ yield a redundant inequality after tropicalization if and only if its coefficient is at most $1$?

来源字段：https://doi.org/10.4171/owr/2021/52

01847 | OWR-822-003

### Coefficient Positivity in Stretched Tensor Multiplicities

O / open | Oberwolfach 报告问题汇编

For highest weights $\lambda,\mu,\nu$ of a classical Lie algebra, write the stretched Clebsch-Gordan multiplicity $C_{n\lambda,n\mu}^{n\nu}$ as its constituent polynomials $f_i(n)$. Must every coefficient of every $f_i$ be nonnegative?

来源字段：https://doi.org/10.4171/owr/2005/17

01848 | OWR-14298370-008

### Cohomological Finite Generation for Finite Tensor Categories

P / partially_solved | Oberwolfach 报告问题汇编

Does every finite tensor category have finitely generated cohomology, and what homological lower bounds analogous to Carlsson's conjecture hold for its perfect complexes?

来源字段：https://doi.org/10.4171/owr/2024/20

01849 | OWR-12494-002

### Cohomological Invariants of Simple Algebraic Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a simple algebraic group $G$ and a Galois module $M$, compute the group of normalized cohomological invariants $\operatorname{Inv}^n(G,M)^{\mathrm{norm}}$, including the basic case $M=\mathbb Z/2\mathbb Z$.

来源字段：https://doi.org/10.4171/owr/2013/31

01850 | AIM-REPRESENTATION_THEORY-0110

### Cohomological tools for pseudorepresentations and a two-character visibility-order lemma

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.** Classically, if one is looking at deformation rings of Galois representations one has various cohomological tools. One would like analogues of these in the situation of pseudo-representations.

This is a request for a tangent-and-obstruction theory, not a yes/no conjecture. In the PDF the next line is the heading “9.

The immediate source context matters. Sections 8.1-8.2 discuss two-dimensional pseudocharacters, their characteristic polynomials and Cayley-Hamilton identities, the relation between representation and pseudorepresentation moduli, the absolutely irreducible theorem of Nyssen-Rouquier, and the geometry near a sum of two distinct characters. Thus “cohomological tools” naturally means at least:

1. a description of first-order pseudodeformations; 2. obstruction classes for lifting through small extensions; 3. a way to impose local or global deformation conditions; and 4. a replacement, in the reducible case, for the usual \(H^1(G,\operatorname{ad}\bar\rho)\) and \(H^2(G,\operatorname{ad}\bar\rho)\).

来源字段：https://aimath.org/WWN/padicmodularity/padicmodularity.pdf

01851 | OWR-14298159-007

### Cohomology and Balmer Spectra of Hopf Algebras

O / open | Oberwolfach 报告问题汇编

For a finite-dimensional, not necessarily cocommutative Hopf algebra $A$, determine when $H^*(A,k)$ is Noetherian, when the natural map $\Psi:\operatorname{Spec}(\operatorname{stab}A)\to\operatorname{Proj}H^*(A,k)$ is a homeomorphism, and how to compute $\operatorname{Proj}H^*(A,k)$.

来源字段：https://doi.org/10.4171/owr/2024/5

01852 | OWR-14213-012

### Cohomology Kernels for Inseparable Pfister Composita

O / open | Oberwolfach 报告问题汇编

Let $L/F$ be a finite purely inseparable extension, let $B$ be an anisotropic bilinear Pfister form over $F$, and let $K/F$ be a separable quadratic extension, with $B$ remaining anisotropic over $L$. Is $H_2^{n+1}(L\mathbin{\cdot}K(B)/F)=H_2^{n+1}(L/F)+H_2^{n+1}(K/F)+H_2^{n+1}(F(B)/F)$?

来源字段：https://doi.org/10.4171/owr/2016/5

01853 | OWR-14299523-001

### Cohomology Theory for Subalgebra Zeta Functions

O / open | Oberwolfach 报告问题汇编

Construct a cohomology theory for subalgebra zeta functions that unifies the arithmetic zeta-function theory with the combinatorial theory of flag Hilbert-Poincaré series.

来源字段：https://doi.org/10.4171/owr/2025/38

01854 | OWR-10252931-004

### Coincidence of Degeneration Orders for Representation-Directed Algebras

O / open | Oberwolfach 报告问题汇编

For representation-directed algebras, do the geometric and homological degeneration orders coincide?

来源字段：https://doi.org/10.4171/owr/2022/23

01855 | OWR-14299523-007

### Coincidence of Reduced and Topological Zeta Invariants

O / open | Oberwolfach 报告问题汇编

For a $d$-dimensional algebra $A$, define $m_{\mathrm{top}}(A)$ from the leading behavior of $Z_A^{\mathrm{top}}(s)$ at $s=0$ and $m_{\mathrm{red}}(A)$ from that of $Z_A^{\mathrm{red}}(T)$ at $T=1$. Prove that $m_{\mathrm{red}}(A)=m_{\mathrm{top}}(A)$ and that their common value is a positive rational number.

来源字段：https://doi.org/10.4171/owr/2025/38

01856 | AIM-REPRESENTATION_THEORY-0025

### Color-shear Ext classes and the annulus filtration correction

P / partially_solved | AIM 研讨会问题汇编

It asks for naturally occurring nontrivial $\mathrm{FI}_2$-modules. For a manifold $X$ with two specified boundary components it claims that

$$ V_n=H_i(\operatorname{PConf}^n(X)) $$

has an $\mathrm{FI}_2$-module structure. For the cylinder $X=S^1\times I$ and $i=1$, it records

$$ 0\longrightarrow \widetilde M(1)\longrightarrow V \longrightarrow\widetilde M(2)\longrightarrow0, $$

where the tildes mean pullback from $\mathrm{FI}$, says that $V$ has an ordinary $\mathrm{FI}$-structure but is not such a pullback, and asks:

1. what generally happens for $H_i(\operatorname{PConf}(X))$; and 2. whether $\operatorname{Ext}(M,N)$ can be calculated in $\mathrm{FI}_2$-modules.

The literature note attributes to Andrew Snowden the observation that Koszul resolutions calculate these Ext groups and cites Church--Ellenberg--Farb [CEF].

Two corrections are necessary before doing mathematics.

**Notation correction.** CEF define $M(a)=k[\operatorname{Inj}([a],-)]$, the free module generated by the regular $S_a$-representation, and more generally $M(W)$. Literal $M(2)$ has dimension $n(n-1)$. But

$$ \operatorname{rank}H_1(\operatorname{PConf}_n(S^1\times I)) =n+\binom n2. $$

Thus the cylinder statement can only use the module induced from the trivial $S_2$-representation, of dimension $\binom n2$. We denote this unambiguously by

$$ M_{(2)}(S)=k\binom S2, \qquad M_{(1)}(S)=kS. $$

Presumably the partition $(2)$, or its Young diagram, was flattened in the source.

**Filtration correction.** Under the standard homology stabilization by inserting points at boundary collars, pair-collision meridians are preserved. They therefore form the $M_{(2)}$ submodule.

$$ \boxed{0\longrightarrow\widetilde M_{(2)}\longrightarrow H_1 \longrightarrow\widetilde M_{(1)}\longrightarrow0,} $$

the reverse of the recorded display. This is proved in §7. If a different variance or dual convention was intended, it must be stated; it is not the covariant homology stabilization in the record.

Here $\operatorname{PConf}_n(X)$ means ordered configurations of $n$ distinct interior points. Replacing the compact cylinder by its interior does not change the homotopy calculation and identifies it with $\mathbf C^*$.

来源字段：http://aimpl.org/repnstability/2/

01857 | OWR-9790358-002

### Combinatorial Approaches to the Gasca-Maeztu Conjecture

O / open | Oberwolfach 报告问题汇编

Can combinatorial properties of maximal lines be used to prove or generalize the Gasca-Maeztu conjecture?

来源字段：https://doi.org/10.4171/owr/2022/7

01858 | OWR-11101920-013

### Combinatorial Description of Plethysm Coefficients

O / open | Oberwolfach 报告问题汇编

Find a combinatorial description of the plethysm coefficients $a^\nu_{\lambda,\mu}$ in general.

来源字段：https://doi.org/10.4171/owr/2022/39

01859 | OWR-12172-005

### Combinatorial Freeness of Piecewise-Polynomial Fan Modules

O / open | Oberwolfach 报告问题汇编

For the piecewise-polynomial module $C^0(\Sigma)$ associated with a polyhedral fan $\Sigma$, does there exist a simple combinatorial condition equivalent to its freeness?

来源字段：https://doi.org/10.4171/owr/2012/49

01860 | OWR-8415355-014

### Combinatorial Invariance of Kazhdan-Lusztig Polynomials

P / partially_solved | Oberwolfach 报告问题汇编

For a Coxeter group $(W,S)$ and elements $x,y\in W$, does the Kazhdan-Lusztig polynomial $P_{x,y}$ depend only on the isomorphism class of the Bruhat interval $[x,y]$?

来源字段：https://doi.org/10.4171/owr/2021/52

01861 | OWR-4216-005

### Combinatorics and Cauchy Identities for Parabolic KL Elements

O / open | Oberwolfach 报告问题汇编

Can one find a combinatorial interpretation for evaluations of Kazhdan-Lusztig elements for non-maximal parabolic modules, and a Cauchy summation formula for the Kazhdan-Lusztig elements of a maximally parabolic module?

来源字段：https://doi.org/10.4171/owr/2010/15

01862 | OWR-13682-005

### Common Triangular Bases for Quantum Cluster Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Does every injective-reachable quantum cluster algebra admit a unique common triangular basis parametrized by tropical points, as predicted by the Fock-Goncharov conjecture?

来源字段：https://doi.org/10.4171/owr/2015/25

01863 | OWR-1452-017

### Commuting Locally Nilpotent Derivations Conjecture

O / open | Oberwolfach 报告问题汇编

If $D_1,\ldots,D_{n-1}$ are pairwise commuting locally nilpotent derivations of $\mathbb{C}[x_1,\ldots,x_n]$ that are linearly independent over that ring, is their common kernel $\mathbb{C}[f]$ for some coordinate $f$?

来源字段：https://doi.org/10.4171/owr/2007/01

01864 | OWR-4800-004

### Compactified Jacobians and Kac Polynomials of Quivers

O / open | Oberwolfach 报告问题汇编

For every quiver $\Gamma$ and dimension vector $v$, construct the projective curve $C_{\Gamma,v}$ with one rational component of multiplicity $v_i$ per vertex and nodal intersections along edges, form its compactified Jacobian $J_v$, and seek a cohomological interpretation of $A_\Gamma(v,q)$ via $J_v$ and the associated character variety.

来源字段：https://doi.org/10.4171/owr/2011/10

01865 | OWR-12697710-008

### Compactly Supported Representing Measures for Linear Functionals

P / partially_solved | Oberwolfach 报告问题汇编

Let $A$ be a unital commutative real algebra with nonempty character space $X(A)$, and let $L:A\to\mathbb R$ be linear with $L(1)=1$. Characterize when there is a compactly supported Radon measure $\nu$ on $X(A)$ such that $$L(a)=\int_{X(A)}\alpha(a)\,d\nu(\alpha)$$ for every $a\in A$.

来源字段：https://doi.org/10.4171/owr/2023/14

01866 | OWR-1703864-002

### Comparison of Analytic and Periodic Cyclic Homology

O / open | Oberwolfach 报告问题汇编

For which bornological algebras is analytic cyclic homology isomorphic to periodic cyclic homology? In particular, can the known comparison results for analytically quasi-free algebras be extended to a broader class?

来源字段：https://doi.org/10.4171/owr/2020/17

01867 | OWR-12697690-006

### Comparison of Integral and Truncated-Polynomial Birkhoff Categories

O / open | Oberwolfach 报告问题汇编

Compare the representation theory of $\operatorname{mono}(1\to2,\operatorname{mod}\mathbb{Z}/p^6\mathbb{Z})$ with that of $\operatorname{mono}(1\to2,\operatorname{mod}k[x]/(x^6))$. Are the two monomorphism categories equivalent in an appropriate representation-theoretic sense?

来源字段：https://doi.org/10.4171/owr/2023/7

01868 | OWR-12175-025

### Comparison Properties and Jiang-Su Absorption

P / partially_solved | Oberwolfach 报告问题汇编

For every nonelementary, stably finite, simple, separable, unital, nuclear $C^*$-algebra, are $\mathcal Z$-absorption, local weak comparison, and strict comparison equivalent? In particular, are local weak comparison and strict comparison equivalent?

来源字段：https://doi.org/10.4171/owr/2012/52

01869 | OWR-9790358-009

### Compatible Polynomial Interpolation on Algebraic Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Given algebraic varieties $V_1,\ldots,V_m\subseteq\mathbb C^d$ and polynomials $p_1,\ldots,p_m$, when does a polynomial $f\in\mathbb C[x_1,\ldots,x_d]$ exist such that $f|_{V_j}=p_j|_{V_j}$ for every $j$?

来源字段：https://doi.org/10.4171/owr/2022/7

01870 | OWR-13351-013

### Complete Integral Closures of Total Blow-Up Rings

O / open | Oberwolfach 报告问题汇编

Let $(R, v)$ be a valuation domain (or local Noetherian domain dominated by a valuation $v$), and let $S = B_\infty(R, v)$ be the total blow-up ring obtained as the union of the local blow-ups of $R$ along $v$. Is the complete integral closure of $S = B_\infty(R, v)$ always a valuation ring?

来源字段：https://doi.org/10.4171/owr/2014/49

01871 | OWR-13678-006

### Complete-Intersection Kernels of Ideal Projectors

S / solved | Oberwolfach 报告问题汇编

If an ideal projector $P$ has a zero-dimensional kernel generated by exactly $d$ polynomials in $\mathbb{C}[x_1,\ldots,x_d]$, must $P$ be a Hermite projector?

来源字段：https://doi.org/10.4171/owr/2015/21

01872 | OWR-17466-007

### Completion of Presilting Complexes

S / solved | Oberwolfach 报告问题汇编

Is every presilting complex with $n$ indecomposable summands already silting?

来源字段：https://doi.org/10.4171/owr/2020/3

01873 | OWR-3393-004

### Complexity Drops for p-by-p Specht Modules

P / partially_solved | Oberwolfach 报告问题汇编

If $\lambda$ is a $p\times p$ partition of block weight $w$, is the complexity of $S^\lambda$ equal to $w-1$, or can it be smaller? Can these cases be described further?

来源字段：https://doi.org/10.4171/owr/2009/10

01874 | AIM-REPRESENTATION_THEORY-0111

### Component extraction in completed-cohomology factorizations

P / partially_solved | AIM 研讨会问题汇编

The document itself warns that it was typeset during the talks and may contain transcription errors.

Let \(\mathbb F\) be a finite field, let \(\mathcal O=\mathcal O_K\) be a complete discrete valuation ring with residue field \(\mathbb F\), and let \[ \bar\rho:G_{\mathbb Q}\longrightarrow \mathrm{GL}_2(\mathbb F) \] be absolutely irreducible and modular. Let \(R\) be its universal deformation ring for deformations unramified outside a finite set \(\Sigma\), and let \(R^{\mathrm{mod}}\) be the image of \(R\) in the localized Hecke algebra. The notes describe \(\operatorname{Spec}R^{\mathrm{mod}}\) as the Zariski closure of the classical modular points. Put \[ \widehat H^1=\widehat H^1_{\Sigma,\bar\rho}, \] the localized completed cohomology of modular curves, and let \(I\) define a component of \(R^{\mathrm{mod}}\).

After restoring notation from the primary PDF, Conjecture 4 asks for an equivariant isomorphism \[ \tag{3} \widehat H^1[I]\stackrel{??}{\cong} \rho_{R^{\mathrm{mod}}/I} \widehat\otimes_{R^{\mathrm{mod}}/I} \widehat\pi_p\!\left( \rho_{R^{\mathrm{mod}}/I}|_{G_{\mathbb Q_p}}\right) \widehat\otimes \left( \widehat\bigotimes_{\substack{\ell\neq \ell\in\Sigma}} \widehat\pi^{\mathrm{mod}}_\ell\!\left( \rho_{R^{\mathrm{mod}}/I}|_{G_{\mathbb Q_\ell}}\right) \right). The factors carry, respectively, the actions of \(G_{\mathbb Q}\), \(\mathrm{GL}_2(\mathbb Q_p)\), and \(\mathrm{GL}_2(\mathbb Q_\ell)\). The source calls the \(p\)-adic factor an orthonormalizable \(R^{\mathrm{mod}}/I\)-Banach module supplied by \(p\)-adic local Langlands and calls the factors away from \(p\) the \(\mathfrak m\)-adic completions of modified local Langlands in families.

The notes impose: (1) a scalar-endomorphism/genericity condition on the residual restriction at \(p\), excluding in particular a twist of \(\left(\begin{smallmatrix}\omega^{-1}&*\\0&1\end{smallmatrix}\right)\); and (2) existence and classical-point interpolation of \(\widehat\pi_p\). The punctuation in (1) is defective: the most coherent reading is “irreducible, or a nonsplit extension \(\left(\begin{smallmatrix}\chi&*\\0&\psi\end{smallmatrix}\right)\) with \(\chi\ne\psi\),” together with the stated exceptional exclusion. Under these two hypotheses, Theorem 14 in the same source states that (3) holds.

The first source remark includes Corollary 15, which gives classicality for potentially semistable trianguline specializations with distinct Hodge--Tate weights and finite-slope overconvergent modularity for trianguline specializations. The genuine second remark ends with anticipated Fontaine--Mazur, eigencurve \(p\)-adic \(L\)-function, and mod-\(p\) consequences. Schien” belongs to the next section and is unrelated boundary contamination.

来源字段：https://aimath.org/WWN/padicmodularity/padicmodularity.pdf

01875 | OWR-16166-008

### Computing Galois Closures over Rational Function Fields

O / open | Oberwolfach 报告问题汇编

Let $k$ be a field, let $a_1,\ldots,a_n$ be algebraically independent over $k$, set $K=k(a_1,\ldots,a_n)$, and let $L/K$ be an explicitly given finite separable extension. Construct an algorithm that computes the Galois group of the Galois closure of $L/K$.

来源字段：https://doi.org/10.4171/owr/2018/25

01876 | OWR-1465-009

### Connectedness in the Real Spectrum

O / open | Oberwolfach 报告问题汇编

Let $A = R[x_1, \ldots, x_n]$ be a polynomial ring over a real closed field $R$, let $\operatorname{Sper} A$ be its real spectrum, and for $\alpha, \beta \in \operatorname{Sper} A$ let $\langle \alpha, \beta \rangle \subseteq A$ denote the separating ideal in the sense of the Pierce--Birkhoff conjecture. If $g_1, \ldots, g_s \in A$ lie outside $\langle \alpha, \beta \rangle$, is there a connected subset $C \subseteq \operatorname{Sper} A$ containing $\alpha$ and $\beta$ and disjoint from every zero set $\{g_i = 0\}$? Equivalently, do $\alpha$ and $\beta$ lie in the same connected component of $\operatorname{Sper} A \setminus \bigcup_{i=1}^s \{g_i = 0\}$?

来源字段：https://doi.org/10.4171/owr/2007/14

01877 | OWR-13943-001

### Connections Among Weyl Groupoids, Nichols Algebras, and Cluster Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Is there a deeper connection between Weyl groupoids or Nichols algebras and cluster algebras?

来源字段：https://doi.org/10.4171/owr/2015/48

01878 | OWR-14299910-005

### Connectivity of Cluster-Tilting Exchange Graphs

O / open | Oberwolfach 报告问题汇编

For a stably $2$-Calabi-Yau Frobenius category $\mathcal C$, is the exchange graph whose vertices are cluster-tilting objects and whose edges are mutations always connected?

来源字段：https://doi.org/10.4171/owr/2026/6

01879 | ALG-005

### Connes Embedding Problem

S / solved | 独立 / 未分配集合

别名：Connes Embedding Problem [ALG-057]

Can every finite von Neumann algebra be embedded into an ultrapower of the hyperfinite $\mathrm{II}_1$ factor?

背景链接：https://arxiv.org/abs/2001.04383

01880 | OWR-4133-001

### Constructive Homological Algebra

O / open | Oberwolfach 报告问题汇编

When can homology be computed constructively, what algebraic conditions are required, and which homological-algebra results and techniques can be made constructive and implemented in software or proof assistants?

来源字段：https://doi.org/10.4171/owr/2009/50

01881 | AIM-LINEAR_ALGEBRA-0001

### Convention audit and connected linear-multiplicity obstructions

P / partially_solved | AIM 研讨会问题汇编

Let $G$ be a graph with maximum degree bounded by $\Delta$. For the following choices of Hermitian matrix $M_G$ associated with $G$, what is the maximum multiplicity of the second largest eigenvalue of $M_G$?

The four choices are (1) the adjacency matrix, (2) the Laplacian matrix, (3) a Schrödinger operator $D+A_G$ with arbitrary diagonal $D$, and (4) a Hermitian weighted adjacency matrix satisfying $M_G[i,j]=0$ whenever $\{i,j\}\notin E(G)$.

1. The order $n=|V(G)|$ and connectedness are not stated in the first sentence. The cited adjacency literature formulates the intended problem for connected $n$-vertex graphs, with $\Delta$ fixed and $n\to\infty$. 2. For the cited adjacency problem, eigenvalues are listed with multiplicity, $$\lambda_1\geq \lambda_2\geq\cdots\geq\lambda_n,$$ and the multiplicity of $\lambda_2$ means the multiplicity of its numerical value. For a connected unweighted graph Perron--Frobenius makes $\lambda_1$ simple, so this agrees with the second *distinct* eigenvalue. That equivalence fails for the other matrix classes.

This report keeps the ordered-with-multiplicity convention used by Haiman--Schildkraut--Zhang--Zhao and treats “second-largest” literally. Results for a second-distinct convention are called out separately. “Laplacian” means the combinatorial Laplacian $L=\operatorname{diag}(d)-A$, and “second” means second-largest as written in the prompt, not the usual second-smallest algebraic-connectivity eigenvalue.

来源字段：http://aimpl.org/spectralhypergraph/1/

01882 | OWR-16768-009

### Coordinate Realization of Characteristic-Polyhedron Multiplicity

O / open | Oberwolfach 报告问题汇编

Let $R$ be an excellent regular local ring with regular parameters $(u,y)$, and let $f\in R\setminus\langle u\rangle$. Write $m=m(f)$ and let $\operatorname{in}_0(f)\in k[Y_1,\ldots,Y_r]$ be its degree-$m$ initial form. Assume there is no proper $k$-subspace $T\subset k[Y_1,\ldots,Y_r]_1$ with $\operatorname{in}_0(f)\in k[T]$. If $\widehat y_j=y_j+\varphi_j$, with $\varphi_j\in\langle u\rangle\widehat R$, and $f\in\langle\widehat y\rangle^m$, do elements $z_1,\ldots,z_r\in R$ exist such that $(u,z)$ is a regular system of parameters,

$$\langle z\rangle\widehat R=\langle\widehat y\rangle,$$

and $f\in\langle z\rangle^m$?

来源字段：https://doi.org/10.4171/owr/2019/6

01883 | OWR-4338-003

### Correspondence between Infinite-Dimensional Lie Groups and Lie 2-Groups

O / open | Oberwolfach 报告问题汇编

What can be learned from infinite-dimensional Lie groups about Lie 2-groups, and conversely?

来源字段：https://doi.org/10.4171/owr/2010/25

01884 | AIM-REPRESENTATION_THEORY-0043

### Costandard-probe Hom formulas and a rank-one Ext obstruction

S / solved | AIM 研讨会问题汇编

Is there a homological interpretation of the \(p\)-Kazhdan--Lusztig polynomials?

来源字段：http://aimpl.org/sheavemodular/4/

01885 | AIM-REPRESENTATION_THEORY-0037

### Cotangent-bundle vanishing for projective-space parabolics

P / partially_solved | AIM 研讨会问题汇编

Let $P$ be a parabolic subgroup of $G$. Is it true in characteristic $p$ that \[ R^i\operatorname{Ind}_P^G\operatorname{Sym}(\mathfrak n_P^\bullet)=0 \qquad(i>0)? \] (Known for $P=B$ and for certain other parabolics. Possible connection with normality of nilpotent orbit closures.)

来源字段：http://aimpl.org/sheavemodular/1/

01886 | OWR-1453-012

### Cotorsion Modules and Pure-Injective Envelopes

O / open | Oberwolfach 报告问题汇编

Are there noncoherent rings $R$ for which every cotorsion left $R$-module is pure-injective? Which ring-theoretic properties are equivalent to every cokernel of a pure-injective envelope being pseudoflat?

来源字段：https://doi.org/10.4171/owr/2007/02

01887 | AIM-REPRESENTATION_THEORY-0046

### Criteria for normalized parity extensions to equal simple modular IC sheaves

P / partially_solved | AIM 研讨会问题汇编

Give nontrivial sufficient conditions for indecomposable parity sheaves on $(G/B)$ or $(Gr)$ or $(Fl)$ to be simple perverse sheaves.

来源字段：http://aimpl.org/sheavemodular/5/

01888 | ALG-041

### Crouzeix's Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Crouzeix's Conjecture [ALG-058]

Is $\|f(A)\| \leq 2\sup_{z \in W(A)} |f(z)|$ for all matrices $A$ and functions $f$ analytic on the numerical range?

背景链接：https://arxiv.org/abs/2608.03841

01889 | OWR-11782-002

### Cuntz Splices and Morita Equivalence of Leavitt Path Algebras

O / open | Oberwolfach 报告问题汇编

Is the determinant a Morita-equivalence invariant for Leavitt path algebras; in particular, does the Cuntz splice preserve Morita equivalence of Leavitt path algebras?

来源字段：https://doi.org/10.4171/owr/2012/20

01890 | AIM-REPRESENTATION_THEORY-0024

### Cyclotomic bounds and even-dimensional transfer for stable multiplicities

P / partially_solved | AIM 研讨会问题汇编

The AIM record is Problem 2.1, “Quasi-polynomial behavior,” in the Topology section of the Representation Stability workshop list. Its first example concerns

$$ V(1,1)_n=\bigwedge^2\mathbb Q^{n-1}\cong S^{(n-2,1,1)} $$

and the multiplicity of this stable irreducible in $(H^i(\operatorname{PConf}_n(\mathbb C);\mathbb Q))$. The displayed stable answer is

$$ d_i(1,1)= \begin{cases} 2i-2,&i\equiv0\pmod4,\\ 2i-3,&i\equiv1\pmod4,\\ 2i-2,&i\equiv2\pmod4,\\ 2i-1,&i\equiv3\pmod4, \end{cases} \qquad (i\ge2). $$

The record then says that, for fixed $(\lambda)$, the stable multiplicity is quasi-polynomial in $(i)$, and asks:

1. For which manifolds $(X)$ does analogous behavior hold?
2. What additional hypothesis on a finitely generated FI-algebra forces analogous behavior?

- Both inner products in the source are missing the closing delimiter before the subscript $(S_n)$. They are read as $(\langle H^i(\operatorname{PConf}_n(X);\mathbb Q),V(\lambda)_n\rangle_{S_n})$. - The source calls the first number $(a_n)$, although after stabilization it is a function of $(i)$. We write $$ d_i^X(\lambda)=\lim_{n\to\infty} \left\langle H^i(\operatorname{PConf}_n(X);\mathbb Q),V(\lambda)_n\right\rangle_{S_n}. $$ - For a partition $(\lambda)$, $(V(\lambda)_n)$ means the irreducible $(S_n)$-representation $(S^{(n-|\lambda|,\lambda)})$, defined once $(n-|\lambda|\ge\lambda_1)$. - The order of operations is essential: fix $(i)$, stabilize in $(n)$, and only then study the resulting sequence as $(i)$ varies. No simultaneous two-variable limit is asserted. - The FI-algebra question suppresses the grading. Without graded pieces there is no variable $(i)$. We interpret it as asking about $(\lim_n\langle A_n^i,V(\lambda)_n\rangle_{S_n})$ for a cohomologically graded FI-algebra $(A^\bullet)$.

The original AIM URL did not return usable content in the literature check. The reconstruction above is independently confirmed by Chen’s notation, transfer identity, stable generating function, and the exact $(V(1,1))$ example. Chen’s coefficient table also confirms that the displayed four-case formula begins at $(i=2)$.

来源字段：http://aimpl.org/repnstability/2/

01891 | OWR-4216-008

### Cyclotomic Quiver-Hecke Categorification of Highest-Weight Modules

O / open | Oberwolfach 报告问题汇编

For a symmetrizable Kac-Moody algebra $\mathfrak g$ and dominant integral weight $\Lambda$, is there an isomorphism of $U_q(\mathfrak g)$-modules

$$\mathbb Q(q)\otimes K^\Lambda(\Gamma)\cong V(\Lambda),$$

where $K^\Lambda(\Gamma)$ is the Grothendieck group of graded finite-dimensional modules over the cyclotomic quiver Hecke algebra and simple modules correspond to the dual canonical basis?

来源字段：https://doi.org/10.4171/owr/2010/15

01892 | ALG-035

### Dade's Conjecture

P / partially_solved | 独立 / 未分配集合

Is there a relationship between the numbers of irreducible characters in blocks of a finite group and its local subgroups?

背景链接：https://ems.press/journals/jems/articles/14667

01893 | OWR-8415352-001

### Deciding Positivity in Star Algebras

P / partially_solved | Oberwolfach 报告问题汇编

How can one decide whether an element of a $*$-algebra is positive, meaning that it maps to a positive operator in every $*$-representation on a Hilbert space?

来源字段：https://doi.org/10.4171/owr/2021/49

01894 | OWR-5149-007

### Decomposition Numbers of Symmetric Groups

O / open | Oberwolfach 报告问题汇编

Determine all decomposition numbers of the symmetric group $S_n$ over a field of prime characteristic.

来源字段：https://doi.org/10.4171/owr/2011/21

01895 | OWR-1790-005

### Degenerations of Elliptic Yang-Baxter Solutions

P / partially_solved | Oberwolfach 报告问题汇编

For nondegenerate solutions of the classical Yang-Baxter equation, construct and understand the degenerations of elliptic solutions into trigonometric solutions and then into rational solutions, including the corresponding degenerations of the associated vector-bundle constructions on singular cubic curves.

来源字段：https://doi.org/10.4171/owr/2008/09

01896 | OWR-14299523-005

### Degree of Topological Subalgebra Zeta Functions

O / open | Oberwolfach 报告问题汇编

Let $A$ be an algebra of dimension $d$ in the subalgebra-zeta setting. Prove the degree conjecture $\deg Z_A^{\mathrm{top}}(s)=-d$.

来源字段：https://doi.org/10.4171/owr/2025/38

01897 | OWR-13681-010

### Degree Spectra and Index of Brauer-Equivalent Azumaya Algebras

O / open | Oberwolfach 报告问题汇编

Given an Azumaya algebra $A$ on a connected ringed space $X$, what are the possible degrees of Azumaya algebras $A'$ that are Brauer-equivalent to $A$, and how can their greatest common divisor, the index $\operatorname{ind}([A])$, be characterized?

来源字段：https://doi.org/10.4171/owr/2015/24

01898 | OWR-1790-003

### Denominators of Cluster Variables from Nonacyclic Seeds

P / partially_solved | Oberwolfach 报告问题汇编

For a cluster algebra $A(\mathbf x,Q)$ whose initial seed is not acyclic, how should the denominator of a cluster variable be interpreted? In particular, can every cluster variable be written with exponent vector $d_y=\dim\operatorname{Hom}_{\mathcal C_Q}(T_y,M)$ for $T=\bigoplus_y T_y$?

来源字段：https://doi.org/10.4171/owr/2008/09

01899 | OWR-12863-003

### Density of Whittaker Distributions on Spherical Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be quasisplit with Borel subgroup $B$ and nilradical $U$, let $\psi$ be a nondegenerate character of $U$, let $H\subset G$ be reductive and spherical, and put $X=G/H$. If $O$ is the union of the open $B$-orbits in $G$, is $$\sum_{x\in O}S^*(Ux)^{U,\psi}$$ dense in $S^*(X)^{U,\psi}$?

来源字段：https://doi.org/10.4171/owr/2014/03

01900 | OWR-3398-006

### Derived Equivalence from Intersection Cohomology Extensions

P / partially_solved | Oberwolfach 报告问题汇编

If a complex reductive group $G$ acts on a projective variety $X$ with finitely many orbits, is there an equivalence of triangulated categories $D^b_{G,c}(X)\simeq\operatorname{Perf}(\operatorname{Ext}(IC))$?

来源字段：https://doi.org/10.4171/owr/2009/15

01901 | OWR-787-005

### Derived Equivalence of Maximal Orthogonal Subcategories

P / partially_solved | Oberwolfach 报告问题汇编

Let $\Lambda$ be an isolated-singularity $R$-order. Are the endomorphism rings $\operatorname{End}_{\Lambda}(M)$ derived equivalent for all maximal $(n-1)$-orthogonal subcategories $\operatorname{add}M$ of $\operatorname{CM}\Lambda$?

来源字段：https://doi.org/10.4171/owr/2005/06

01902 | OWR-16635-002

### Derived Equivalence of Unipotent Unitary and Linear Representations

O / open | Oberwolfach 报告问题汇编

Are the categories of unipotent representations of $\operatorname{GU}_n(q)$ and $\operatorname{GL}_n(-q)$ derived equivalent through an equivalence compatible with dominance order, Lusztig's $a$-function, and the Deligne-Lusztig geometric realizations on both sides?

来源字段：https://doi.org/10.4171/owr/2018/52

01903 | OWR-17466-004

### Derived Invariants from Gentle Quivers with Relations

O / open | Oberwolfach 报告问题汇编

Given only the quiver with relations $(Q(\Delta),I(\Delta))$ defining a gentle algebra $A(\Delta)$, find an algorithm that computes its full derived-equivalence invariant, including the genus, puncture and boundary data, winding-number invariants, and the remaining parity or greatest-common-divisor invariants.

来源字段：https://doi.org/10.4171/owr/2020/3

01904 | OWR-4990375-005

### Descent of Freeness for Orthogonal Modular Form Algebras

O / open | Oberwolfach 报告问题汇编

Let $\Gamma<O^+(M)$ be a finite-index subgroup generated by reflections, and let $\Gamma'$ be a finite-index subgroup of $\Gamma$. If $M_*(\Gamma')$ is a free graded algebra of modular forms, must $M_*(\Gamma)$ also be free?

来源字段：https://doi.org/10.4171/owr/2021/6

01905 | ALG-043

### Determinantal Conjecture

P / partially_solved | 独立 / 未分配集合

Characterize the determinant of the sum of two normal matrices.

背景链接：https://doi.org/10.1016/0024-3795(85

01906 | AIM-REPRESENTATION_THEORY-0108

### Determinants over arbitrary coefficient rings

S / solved | AIM 研讨会问题汇编

**Problem 1.** Recast the theory so that $(d!^{-1})$ is not necessarily in $(A)$. [[This is not simply a problem about divided powers]].

来源字段：https://aimath.org/WWN/padicmodularity/padicmodularity.pdf

01907 | OWR-15217-003

### DG Models for Higher Cluster Categories

O / open | Oberwolfach 报告问题汇编

Is there a natural differential graded category $B_m^{(d)}$ with one object, endomorphism algebra $B_m$, and perfect derived category equal to the higher cluster category $C_{\infty,m}^{(d)}$, preferably with a topological interpretation?

来源字段：https://doi.org/10.4171/owr/2017/12

01908 | OWR-12978-004

### Diagram Folding and Structural Equivalence of Hopf Algebras

O / open | Oberwolfach 报告问题汇编

What does diagram folding of Nichols algebras represent under the associated representation-theoretic correspondence? If $H'$ is the maximal pointed Hopf subalgebra of a Hopf algebra $H$, does there exist a Hopf ideal $I$ such that $H=H'\oplus I$, and which algebraic and homological relations connect $H$ and $H'$? If two Hopf algebras have the same set of Frobenius-Schur indicator values, are their module categories gauge equivalent?

来源字段：https://doi.org/10.4171/owr/2014/20

01909 | OWR-8415355-016

### Diamond-Complete Hypercube Formulas for Kazhdan-Lusztig Polynomials

P / partially_solved | Oberwolfach 报告问题汇编

Does the proposed hypercube-cluster formula hold for every $c'\in[x,y]$ such that $[x,c']$ is diamond complete and the edges leaving it span a hypercube cluster?

来源字段：https://doi.org/10.4171/owr/2021/52

01910 | OWR-12170-003

### Dimension and Quadraticity of Transposition Nichols Algebras

O / open | Oberwolfach 报告问题汇编

For $n\geq6$, determine the dimension of the Nichols algebra $\mathfrak B(\mathcal O_2^n,-1)$ associated with the conjugacy class of transpositions in $S_n$ and the constant cocycle $-1$. Is this algebra quadratic?

来源字段：https://doi.org/10.4171/owr/2012/47

01911 | OWR-16162-004

### Dimension Constraints for Quadratic Forms over Quadric Fields

P / partially_solved | Oberwolfach 报告问题汇编

Let $p$ and $q$ anisotropic quadratic forms over a field $F$, let $F(p)$ be the function field of the projective quadric $p=0$, let $s$ be the unique integer such that $2^s<\dim(p)\leq 2^{s+1}$, and put $k=\dim(q)-2i_0(q_{F(p)})$. Must $k\geq0$, and must there be $a\in\mathbb N$ and $\epsilon\in\mathbb Z$ with $-k\leq\epsilon\leq k$ such that $\dim(q)=a2^{s+1}+\epsilon$?

来源字段：https://doi.org/10.4171/owr/2018/21

01912 | OWR-8415347-011

### Dimension Rigidity of Quantum Automorphism-Group Factors

O / open | Oberwolfach 报告问题汇编

For a finite-dimensional $C^*$-algebra $B$ equipped with its canonical trace, let $G^+(B)$ denote the compact quantum automorphism group of $B$, and let $L^\infty(G^+(B))$ be its associated von Neumann algebra. For $n = \dim B \geq 5$, does the von Neumann algebra isomorphism class of $L^\infty(G^+(B))$ depend only on $n$?

来源字段：https://doi.org/10.4171/owr/2021/44

01913 | OWR-9790358-012

### Dimensions of Planar Spline Spaces in Intermediate Degrees

P / partially_solved | Oberwolfach 报告问题汇编

For a triangulated planar polygonal subdivision $\Delta \subset \mathbb{R}^2$, let $S_d^r(\Delta)$ denote the vector space of piecewise polynomial splines on $\Delta$ of degree at most $d$ and global smoothness $C^r$. Determine the exact dimension $\dim S_d^r(\Delta)$ throughout the intermediate degree range $r + 1 \leq d \leq 3r$.

来源字段：https://doi.org/10.4171/owr/2022/7

01914 | OWR-12591-005

### Dini Spaces as Primitive Spectra of Amenable $C^*$-Algebras

O / open | Oberwolfach 报告问题汇编

Is every Dini space homeomorphic to the primitive spectrum of a separable amenable $C^*$-algebra?

来源字段：https://doi.org/10.4171/owr/2013/43

01915 | OWR-12863-004

### Dirac Cohomology of Tempered Modules

O / open | Oberwolfach 报告问题汇编

Does every tempered module with a solvable central character have nonzero Dirac cohomology?

来源字段：https://doi.org/10.4171/owr/2014/03

01916 | AIM-REPRESENTATION_THEORY-0061

### Directionality in rank-one transfer

P / partially_solved | AIM 研讨会问题汇编

**(2)** Develop relative endoscopy in conjunction with the relative trace formula. **(3)** Referring to MR3117742, understand transfer in the direction from \(\mathrm{GL}_2\) to an elliptic torus, as opposed to the torus-to-\(\mathrm{GL}_2\) direction. **(4)** Explain the relation between intertwining operators and character relations, with Labesse--Langlands as the rank-one model.

来源字段：https://aimath.org/pastworkshops/automorphkernelproblems.pdf

01917 | OWR-15181-013

### Discrete Initial Subgroups of the Surreal Numbers

O / open | Oberwolfach 报告问题汇编

Is every discrete initial subgroup of the surreal numbers $\mathbf{No}$ isomorphic to an initial subgroup of the omnific integers $\mathbf{Oz}$?

来源字段：https://doi.org/10.4171/owr/2016/60

01918 | OWR-7155441-004

### Discrete-Valuation Local-Global Principle for Homogeneous Varieties and Torsors

P / partially_solved | Oberwolfach 报告问题汇编

For a projective homogeneous variety or torsor $V$ over $F$, is $V(F)$ nonempty if and only if $V(M_x)$ is nonempty for every $x$ in the set $S$ of points corresponding to discrete valuations of $F$?

来源字段：https://doi.org/10.4171/owr/2021/20

01919 | OWR-15181-008

### Distinct Extensions of Surreal Derivations

O / open | Oberwolfach 报告问题汇编

Can there be distinct surreal derivations on $\mathbf{No}$ whose restrictions to $\mathbb{R}\langle L\rangle$ agree?

来源字段：https://doi.org/10.4171/owr/2016/60

01920 | OWR-14298590-007

### Distributivity of Compact Weight Scales

O / open | Oberwolfach 报告问题汇编

Do the $\omega$-compact weight-scale constructions in enriched higher category theory satisfy the required distributivity law?

来源字段：https://doi.org/10.4171/owr/2024/39

01921 | OWR-12868-002

### Divisor Class Groups of Cluster Algebras

P / partially_solved | Oberwolfach 报告问题汇编

What is the divisor class group $\operatorname{Cl}(\mathcal A(\mathbf x,Q))$ of the cluster algebra associated with a quiver $Q$?

来源字段：https://doi.org/10.4171/owr/2014/08

01922 | OWR-1452-013

### Dixmier Conjecture for Weyl-Algebra Endomorphisms

P / partially_solved | Oberwolfach 报告问题汇编

Is every $\mathbb{C}$-endomorphism of the Weyl algebra $A_n=\mathbb{C}[X_1,\ldots,X_n,D_1,\ldots,D_n]$, with $[D_i,X_j]=\delta_{ij}$ and all $X_i$ and all $D_i$ commuting, an automorphism?

来源字段：https://doi.org/10.4171/owr/2007/01

01923 | OWR-1265-023

### Domino-Insertion Description of Hecke Left Cells

P / partially_solved | Oberwolfach 报告问题汇编

For the type-$B_n$ unequal-parameter Hecke setting, the partition into left cells should be given by domino insertion relative to an $r$-sized $2$-core and hence should depend only on $r$.

来源字段：https://doi.org/10.4171/owr/2006/15

01924 | OWR-13681-004

### Donkin’s Good Filtration Conjecture

O / open | Oberwolfach 报告问题汇编

Let $k$ be algebraically closed of characteristic $p>0$, let $G$ be simple and simply connected, and let $\mathrm{St}_r=L((p^r-1)\rho)$ be the $r$th Steinberg module. Donkin's conjecture asserts that a finite-dimensional rational $G$-module $M$ has a good $(p,r)$-filtration if and only if $\mathrm{St}_r\otimes M$ has a good filtration.

来源字段：https://doi.org/10.4171/owr/2015/24

01925 | OWR-14218-001

### Donovan’s Finiteness Conjecture for Fixed Defect Groups

O / open | Oberwolfach 报告问题汇编

For a fixed finite $p$-group $P$, are there only finitely many Morita-equivalence classes of block algebras of finite groups whose defect groups are isomorphic to $P$?

来源字段：https://doi.org/10.4171/owr/2016/10

01926 | OWR-2654827-001

### Double Jordan Stratifications of Artinian Gorenstein Algebras

O / open | Oberwolfach 报告问题汇编

For an Artinian Gorenstein algebra and principal ideals $J,K$, can the double $Q_{J,K}$-stratification be redefined to have the expected dimensions while retaining its relation to the single-ideal $Q_J(a)$- and $Q_K(a)$-stratifications? Can the $Q_{\ell,m}$-decomposition define a Jordan order type for nongraded Artinian Gorenstein algebras? Do combinatorial invariants of the poset determine the $Q_{x,y}$-decomposition for a binomial, including for variables of different degrees or for three variables?

来源字段：https://doi.org/10.4171/owr/2020/31

01927 | OWR-4216-012

### Doubly Singular Irreducible Partitions in Type A

O / open | Oberwolfach 报告问题汇编

Classify the doubly singular $(2,0)$-irreducible partitions for the type-A Iwahori-Hecke algebra.

来源字段：https://doi.org/10.4171/owr/2010/15

01928 | OWR-14299580-006

### Drinfeld-Twist Classification of Set-Theoretic Solutions

O / open | Oberwolfach 报告问题汇编

Classify set-theoretic Yang-Baxter solutions up to Drinfeld twists. In particular, if two rack solutions on the same set are related by a Drinfeld twist, must they coincide?

来源字段：https://doi.org/10.4171/owr/2025/49

01929 | OWR-14298157-003

### Dual Canonical Basis Analogues for General Cluster Algebras

P / partially_solved | Oberwolfach 报告问题汇编

What are analogues of the dual canonical basis $B^*$ for other cluster algebras $k[A]$, and can the Fomin-Zelevinsky conjecture be generalized when $A$ is not contained in $N$?

来源字段：https://doi.org/10.4171/owr/2024/2

01930 | OWR-16773-004

### Duality Invariance of Finite Type Tensor Categories

P / partially_solved | Oberwolfach 报告问题汇编

Is the finite-type property of finite tensor categories preserved under duality?

来源字段：https://doi.org/10.4171/owr/2019/11

01931 | OWR-17289-010

### Dyck-Path Inequalities for Newton-Okounkov Polytopes

P / partially_solved | Oberwolfach 报告问题汇编

Let $\Lambda_i=(\lambda_i^i,\ldots,\lambda_i^n)$ and put $\lambda_j^i=\lambda_j^j$ for $j>i$. Is the Newton-Okounkov polytope $\Delta_{v_0}(X_{w_0},L(\Lambda_1,\ldots,\Lambda_{n-1}))$ defined by $u_{lm}\geq0$ and $$ \sum_{(l,m)\in D}u_{lm}\leq\sum_{s=1}^j(\lambda_k^s-\lambda_{i+j}^s) $$ for every indicated Dyck path $D$ from $\lambda_k$ to $u_{ij}$, where $1<k\leq j<n$ and $1\leq i\leq n-j$?

来源字段：https://doi.org/10.4171/owr/2019/46

01932 | OWR-14214-001

### Dynamical Reconstruction from Bare Crossed Product Banach Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Can the dynamics $I_K$ acting on $X_K$ be reconstructed from the non-involutive Banach algebra obtained as the closure of $C(X_K)\rtimes_{\mathrm{alg}} I_K$ in its natural bounded-operator representation; can this be done without preserving the MASA $C(X_K)$; and can K be reconstructed from $A_K$ without using $\sigma_K$?

来源字段：https://doi.org/10.4171/owr/2016/6

01933 | OWR-824-004

### Effective Factorization in Monoid Rings

P / partially_solved | Oberwolfach 报告问题汇编

Let $R$ be a Dedekind domain, let $M$ be a submonoid of a rational vector space, and fix $n\geq3$, $c\geq2$, and $A\in SL_n(R[M])$. It is known that some $t\in\mathbb{N}$ satisfies $A\in E_n(R[M^{1/c^t}])$. Can one always take $t=1$? Can the effective factorization of $A$ into elementary matrices over a suitable root extension of $R[M]$ be accelerated?

来源字段：https://doi.org/10.4171/owr/2005/19

01934 | OWR-17466-005

### Efficient Extraction of Interval Summands

P / partially_solved | Oberwolfach 报告问题汇编

Given a pointwise finite-dimensional representation $M$ of $\mathbb R^d$, determine efficiently whether $M$ has interval summands and, when it does, extract them, preferably by local and readily checkable conditions rather than by decomposing $M$ completely.

来源字段：https://doi.org/10.4171/owr/2020/3

01935 | OWR-8415341-004

### Efficient Normalizers of Non-Chevalley Derivation Lie Algebras

O / open | Oberwolfach 报告问题汇编

Can normalizers of the non-Chevalley simple Lie algebras that arise as derivation algebras be constructed efficiently?

来源字段：https://doi.org/10.4171/owr/2021/38

01936 | ALG-044

### Eilenberg-Ganea Conjecture

P / partially_solved | 独立 / 未分配集合

Does every group with cohomological dimension 2 have a 2-dimensional Eilenberg-MacLane space K(G,1)?

背景链接：https://doi.org/10.1007/s002220050168

01937 | OWR-824-009

### Eisenbud-Mazur Symbolic-Square Conjecture

O / open | Oberwolfach 报告问题汇编

Let $(R,\mathfrak{m})$ be a regular local ring of equal characteristic zero and $P\subset R$ a prime ideal. Is $P^{(2)}\subseteq\mathfrak{m}P$?

来源字段：https://doi.org/10.4171/owr/2005/19

01938 | AIM-REPRESENTATION_THEORY-0087

### Elementary characters generate Andre's supercharacter theory

S / solved | AIM 研讨会问题汇编

Problem 1.3.

In the original supercharacter theory of Andre, the supercharacters/superclasses associated to single boxes above the identity are irreducible. Is Andre's supercharacter theory the coarsest containing this?. If not, what is the coarsest supercharacter theory containing the irreducibles?

A supercharacter theory \(\mathsf T\) contains the elementary irreducibles if each elementary irreducible character is a singleton block of the character partition of \(\mathsf T\).

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

01939 | OWR-7155441-010

### Elementary Proofs and Computations of Equivariant Multiplicities

O / open | Oberwolfach 报告问题汇编

Can the equivariant-multiplicity formula involving the Duistermaat-Heckman measure be proved without using such measures, and can concrete examples be computed?

来源字段：https://doi.org/10.4171/owr/2021/20

01940 | OPG-23298

### Elementary symmetric of a sum of matrices

P / partially_solved | Open Problem Garden

Problem

Given a Matrix $A$, the $k$-th elementary symmetric function of $A$, namely $S_k(A)$, is defined as the sum of all $k$-by- $k$ principal minors.

Find a closed expression for the $k$-th elementary symmetric function of a sum of N $n$-by- $n$ matrices, with $0\le N\le k\le n$ by using partitions.

背景链接：http://www.openproblemgarden.org/op/elementary_symmetric_of_a_sum_of_matrices

01941 | AIM-REPRESENTATION_THEORY-0105

### Emerton's mod-p local-global factorization and a finite-level one-prime ratio law

P / partially_solved | AIM 研讨会问题汇编

The record is Conjecture 3 in Michael Volpato's notes from the 2006 AIM workshop *(p)-adic representations, modularity, and beyond*, followed by questions of Calegari and Diamond. The PDF warns that the notes may contain transcription errors.

Let \[ \bar\rho:G_{\mathbb Q}\longrightarrow \mathrm{GL}_2(\overline{\mathbb F}_p) \] be absolutely irreducible and odd, and let \(\mathfrak m_N\) be the maximal ideal of the level- \(N\) Hecke algebra determined away from \(N\) by \(T_\ell-\operatorname{tr}\bar\rho(\operatorname{Frob}_\ell)\). Then Emerton predicted an isomorphism of \(G_{\mathbb Q}\times\mathrm{GL}_2(\mathbb A_f)\)-representations of the form \[ \varinjlim_N H^1\!\left(X(N)_{\overline{\mathbb Q}}, \overline{\mathbb F}_p\right)[\mathfrak m] \;\cong\; \left(\bigotimes_\ell'\pi^{\mathrm{modified}}_\ell (\bar\rho|_{G_{\mathbb Q_\ell}})\right)\otimes\bar\rho. \tag{1} \] Each local factor is to be a finite-length admissible smooth representation of \(\mathrm{GL}_2(\mathbb Q_\ell)\). The later formulation in Emerton's paper uses etale cohomology of the full, geometrically disconnected modular curve \(Y(N)\), a finite coefficient field \(k\), and the Galois multiplicity space \(\operatorname{Hom}_{k[G_{\mathbb Q}]}(\bar\rho,H^1_k)\). That formulation is safer in characteristic \(p\), where an isotypic summand need not be semisimple.

Calegari then asks for explicit finite-level multiplicities, noting that the factors away from \(p\) should be easy and that the factor at \(p\) was a thesis-sized problem. Diamond asks for reducible residual representations and for an analogue on Shimura curves attached to quaternion algebras unramified at \(p\).

来源字段：https://aimath.org/WWN/padicmodularity/padicmodularity.pdf

01942 | OWR-11580-007

### Equality of $p$-Canonical and Kazhdan-Lusztig Bases

O / open | Oberwolfach 报告问题汇编

Determine for which primes $p$ and affine Weyl-group elements $x$ the $p$-canonical basis element $\!{}^pH_x$ equals the Kazhdan-Lusztig basis element $H_x$.

来源字段：https://doi.org/10.4171/owr/2012/13

01943 | OWR-16768-011

### Equality of Finitistic and Big Test Ideals

O / open | Oberwolfach 报告问题汇编

For an $F$-finite Noetherian domain $R$ of characteristic $p>0$, a nonzero ideal $\mathfrak a$, and $t>0$, is

$$\tau_{\mathrm{fg}}(\mathfrak a^t)=\tau_b(\mathfrak a^t).$$

Does this equality hold?

来源字段：https://doi.org/10.4171/owr/2019/6

01944 | OWR-17133-001

### Equivalence of Nuclear Dimension, Z-Stability, and Strict Comparison

O / open | Oberwolfach 报告问题汇编

For simple, separable, nonelementary, nuclear $C^*$-algebras, are finite nuclear dimension, $\mathcal Z$-stability, and strict comparison equivalent?

来源字段：https://doi.org/10.4171/owr/2019/37

01945 | OWR-16941-006

### Equivariant Haar Measures for Algebra Actions

O / open | Oberwolfach 报告问题汇编

What is an appropriate notion of an equivariant Haar measure for locally compact abelian groups with a right action by a finite-dimensional semisimple algebra?

来源字段：https://doi.org/10.4171/owr/2019/29

01946 | OWR-14299915-002

### Etingof-Ostrik Cohomological Finite-Generation Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathcal C$ be a finite tensor category. Prove that $H^*(\mathcal C)=\operatorname{Ext}^*_{\mathcal C}(\mathbf 1,\mathbf 1)$ is Noetherian and that $\operatorname{Ext}^*_{\mathcal C}(M,N)$ is a finite $H^*(\mathcal C)$-module for all objects $M,N\in\mathcal C$.

来源字段：https://doi.org/10.4171/owr/2026/9

01947 | AIM-REPRESENTATION_THEORY-0055

### Even morphisms characterize even polynomial quasi-hereditary geometric extension algebras

P / partially_solved | AIM 研讨会问题汇编

Given an algebraic group \(G\) acting on a variety \(X\) with finitely many orbits and a \(G\)-equivariant morphism \(\pi\colon \widetilde X\to X\), where \(\widetilde X\) is nonsingular (not necessarily connected), consider \[ A=\operatorname{End}^{\bullet}_{D^b_G(X)} (\pi_*\underline{k}_{\widetilde X}), \] an algebra over \(H_G^\bullet(\mathrm{pt};k)\). What geometric condition on \(\pi\) is equivalent to \(A\) being quasi-hereditary over \(H_G^\bullet(\mathrm{pt};k)\)?

来源字段：http://aimpl.org/sheavemodular/10/

01948 | OWR-2654827-016

### Eventual Lefschetz Dichotomies for Symmetric Ideal Chains

O / open | Oberwolfach 报告问题汇编

Let $(I_n)$ be a $\operatorname{Sym}$-invariant chain of Artinian ideals. Is each Lefschetz property eventually constant along the chain: either $I_n$ has the strong, respectively weak, Lefschetz property for every sufficiently large $n$, or it fails that property for every sufficiently large $n$?

来源字段：https://doi.org/10.4171/owr/2020/31

01949 | OWR-8415346-003

## Eventual Linearity of Equivariant Projective Dimensions

P / partially_solved | Oberwolfach 报告问题汇编

Let $I=(I_n)_{n\geq 1}$ be an $\operatorname{Inc}(\mathbb{N})$-invariant filtration of nonzero proper homogeneous ideals. Is the sequence $(\operatorname{pd}(R_n/I_n))_{n\geq 1}$ eventually linear?

来源字段：https://doi.org/10.4171/owr/2021/43

01950 | OWR-2654827-017

## Eventual Linearity of Homological Invariants in Symmetric Ideal Chains

P / partially_solved | Oberwolfach 报告问题汇编

Let $(I_n)$ be a $\operatorname{Sym}$-invariant chain of ideals in polynomial rings $R_n$. Are $\operatorname{pd}_{R_n}(R_n/I_n)$ and $\operatorname{reg}(I_n)$ eventually linear in $n$? Equivalently, do integers $A,B,C,D$ exist such that

$$\operatorname{pd}_{R_n}(R_n/I_n)=An+B,\qquad \operatorname{reg}(I_n)=Cn+D$$

for all sufficiently large $n$?

来源字段：https://doi.org/10.4171/owr/2020/31

01951 | AIM-LINEAR_ALGEBRA-0015

## Exact block witnesses for the Barrett-Johnson converse

S / solved | AIM 研讨会问题汇编

If \(F_\lambda(A)\geq F_\mu(A)\) for every \(n\times n\) totally nonnegative matrix \(A\), must \(\lambda\preceq\mu\)?

This is Problem 1.55 in the AIM list from the workshop *Theory and applications of total positivity*. Its literature field says, “This has now been answered affirmatively.”

The official AIM workshop report and the published solution identify the intended reading: the sums are **over all ordered set partitions** of the specified types. The listed AIM Problem Lists URL returned HTTP 502 during this attempt, so it was not used to resolve the ambiguity.

For a partition \(\alpha=(\alpha_1,\ldots,\alpha_t)\vdash n\), let \(\mathcal P(\alpha)\) be the set of ordered tuples \((I_1,\ldots,I_t)\) of pairwise disjoint subsets whose union is \([n]\) and for which \(|I_k|=\alpha_k\). Define \[ S_\alpha(A)=\sum_{(I_1,\ldots,I_t)\in\mathcal P(\alpha)}\prod_{k=1}^t\det A_{I_k,I_k}, \qquad F_\alpha(A)=\left(\prod_{k=1}^t\alpha_k!\right)S_\alpha(A). \] The empty determinant convention is \(\det A_{\varnothing,\varnothing}=1\) if trailing zero parts are used.

For partitions \(\lambda,\mu\vdash n\), write \[ \lambda\preceq\mu \quad\Longleftrightarrow\quad \sum_{i=1}^k\lambda_i\leq\sum_{i=1}^k\mu_i \quad\text{for every }k\geq1, \] after padding with zeros. Thus “\(\mu\) majorizes \(\lambda\)” means \(\lambda\preceq\mu\). The precise question is:

The source takes the same number of positive parts for the two partitions. The published result allows different lengths, and hence strictly contains the source formulation.

The factorials are exactly the normalization for an average. Since \(|\mathcal P(\alpha)|=n!/(\alpha_1!\cdots\alpha_t!)\), \[ F_\alpha(A)=n!\,\frac{1}{|\mathcal P(\alpha)|}\sum_{I\in\mathcal P(\alpha)}\prod_k\det A_{I_k,I_k}. \]

来源字段：http://aimpl.org/totalpos/1/

01952 | AIM-REPRESENTATION_THEORY-0026

## Exact character formulas and minimal FI-generators via full label support

P / partially_solved | AIM 研讨会问题汇编

**Cohomology of \(\mathrm{PConf}^n\).** Find the Specht module decomposition for \(\mathrm H^i(\mathrm{PConf}^n\mathbb C)\), and its \(\mathrm{FI}\)-module generators. It might involve a derangement recurrence. Recurrence is going to involve both \(i\) and \(n\). (This question might relate to W. Chen's question.)

来源字段：http://aimpl.org/repnstability/2/

01953 | AIM-REPRESENTATION_THEORY-0106

### Exact classification of Berger's inverse-psi representations of the mirabolic group

S / solved | AIM 研讨会问题汇编

**Recovered Question 1 (Berger).** Let $V$ be an irreducible finite-dimensional $\overline{\mathbb F}_p$-representation of $G_{\mathbb Q_p}=\operatorname{Gal}(\overline{\mathbb Q}_p/\mathbb Q_p)$, of arbitrary dimension. If $D(V)$ is its étale $(\varphi,\Gamma)$-module, let

来源字段：https://aimath.org/WWN/padicmodularity/padicmodularity.pdf

01954 | OWR-14213-009

### Exact Indices of Orthogonal Grassmannians

O / open | Oberwolfach 报告问题汇编

For every even $m\geq 18$, the index of the $\lfloor m/4\rfloor$-th orthogonal Grassmannian of any generic m-dimensional quadratic form $\varphi/F$ in $I^3_q$ is $2^{\lfloor m/4\rfloor}$.

来源字段：https://doi.org/10.4171/owr/2016/5

01955 | AIM-LINEAR_ALGEBRA-0008

### Exact orthant certificates and Newton obstructions for polynomials in minors

P / partially_solved | AIM 研讨会问题汇编

An algorithmic process is known to identify inequalities that compare the products of two minors for all \(TN\) matrices.

Classify real linear combinations of products of minors that are nonnegative for all \(TN\) matrices. More concretely, given a polynomial \(p(\{x_{I\times J}\mid I,J\subset[n], |I|=|J|\})\), when can one certify, via some constructive algorithmic process, that evaluating \(p\) at \(x_{I\times J}=\det A_{I\times J}\) yields a non-negative output for all \(TN\) matrices \(A\).

来源字段：http://aimpl.org/totalpos/1/

01956 | OWR-16635-009

### Exceptional Isomorphism of Two W-Algebras

O / open | Oberwolfach 报告问题汇编

Is

$$W_{-7+7/3}(\mathfrak{sl}_7,f)\cong W_{-4+4/3}(\mathfrak{sl}_4,f'),$$

where $f$ has orbit partition $(3,2^2)$ in $\mathfrak{sl}_7$ and $f'$ has orbit partition $(2^2)$ in $\mathfrak{sl}_4$?

来源字段：https://doi.org/10.4171/owr/2018/52

01957 | ALG-030

### Existence of Generalized Moonshine

S / solved | 独立 / 未分配集合

Does generalized moonshine exist for all elements of the Monster group?

背景链接：https://arxiv.org/abs/1804.04161

01958 | OWR-14221-001

### Existence of Global Monomial Orders for PBW Bases

P / partially_solved | Oberwolfach 报告问题汇编

For a simple finite-dimensional Lie algebra $\mathfrak g$, is the set $S_{\mathrm{gm}}$ of global monomial degrees nonempty? For every simply laced simple $\mathfrak g$, is $S_{\mathrm{gm}}\cap D^q_{w_0}$ nonempty for some reduced decomposition $w_0$?

来源字段：https://doi.org/10.4171/owr/2016/13

01959 | ALG-012

### Existence of Perfect Cuboids

O / open | 独立 / 未分配集合

Does there exist a rectangular cuboid where all edges, face diagonals, and space diagonals have integer lengths?

背景链接：https://en.wikipedia.org/wiki/Perfect_cuboid

01960 | OWR-11129-005

### Exotic Units in a Finite Group Ring

S / solved | Oberwolfach 报告问题汇编

Does the group $C_2\times C_2\times C_2\times C_6$ have exotic units in its integral group ring $\mathbb{Z}G$?

来源字段：https://doi.org/10.4171/owr/2011/37

01961 | AIM-REPRESENTATION_THEORY-0072

### Explicit BD pushouts, coprime split-field cocycles, and a descent obstruction for p-adic tori

P / partially_solved | AIM 研讨会问题汇编

**Covering group of tori.** Let \(T\) be a (not necessary split) torus over a \(p\)-adic field. Construct \(\widetilde T\) \[ 1\longrightarrow\mu_F\longrightarrow\widetilde T \longrightarrow T\longrightarrow1 \] explicitly within Brylinski--Deligne framework.

Nice subcases: when \(T\) is anisotropic or unramified.

来源字段：http://aimpl.org/autoformcovergp/2/

01962 | OWR-14298370-011

### Explicit Deformation Algebras for Operated Structures

P / partially_solved | Oberwolfach 报告问题汇编

Construct explicitly the dg Lie or $L_\infty$ algebra governing deformations of operated algebraic structures, and develop their homotopy versions.

来源字段：https://doi.org/10.4171/owr/2024/20

01963 | OWR-1538-004

### Explicit Formulas for the Affine Satake Isomorphism

O / open | Oberwolfach 报告问题汇编

Let $G$ be reductive, let $G_{\mathrm{aff}}$ be the corresponding affine Kac-Moody group, and let $G_{\mathrm{aff}}^\vee$ be its affine Langlands dual. The affine Satake isomorphism identifies the affine Hecke algebra with a completion of $K(\operatorname{Rep}(G_{\mathrm{aff}}^\vee))\otimes\mathbb{C}$. Is this isomorphism given by a formula analogous to Lusztig's explicit $q$-analogue formula?

来源字段：https://doi.org/10.4171/owr/2007/22

01964 | AIM-REPRESENTATION_THEORY-0064

### Explicit genuine depth-zero supercuspidals for the metaplectic cover of SL2

P / partially_solved | AIM 研讨会问题汇编

Construct supercuspidal representations of \(\widetilde{G}_{F}\).

来源字段：http://aimpl.org/autoformcovergp/1/

01965 | OWR-7155441-008

### Exponent Criterion for Nonredundant String-Cone Inequalities

P / partially_solved | Oberwolfach 报告问题汇编

Are the inequalities defining the string cone $S_{\mathbf i}$ nonredundant if and only if the absolute value of the exponent of every variable in $W|_{T_{\mathbf i}^{\vee}}$ is at most $1$?

来源字段：https://doi.org/10.4171/owr/2021/20

01966 | OWR-1790-007

### Ext-Algebra Models for Crossingless Matchings

P / partially_solved | Oberwolfach 报告问题汇编

For the coherent-sheaf model on the Slodowy-slice variety $S_{k,k}$ and crossingless matchings $A,B$, is the direct-sum Ext algebra $\bigoplus_{A,B}\operatorname{Ext}^{\bullet}_{\operatorname{Coh}(S_{k,k})}(i_*\Omega(A)^{1/2},j_*\Omega(B)^{1/2})$ isomorphic, as an algebra, to the corresponding convolution algebra $H^{\bullet}$?

来源字段：https://doi.org/10.4171/owr/2008/09

01967 | OWR-1703864-005

### Extension of Group Cyclic Cocycles to Smooth Subalgebras

P / partially_solved | Oberwolfach 报告问题汇编

Given a group $\Gamma$ and a suitable smooth dense subalgebra $A\subset C_r^*(\Gamma)$, when does a cyclic cocycle on $\mathbb C\Gamma$ extend continuously to a cyclic cocycle on $A$?

来源字段：https://doi.org/10.4171/owr/2020/17

01968 | OWR-782-009

### Extension of Representations along $\tau$

O / open | Oberwolfach 报告问题汇编

Given representations $\sigma:K\to M$ and a map $\tau:K\to N$, under what conditions does there exist a global representation $\rho:N\to M$ such that $\rho\circ\tau=\sigma$ on $K$?

来源字段：https://doi.org/10.4171/owr/2005/01

01969 | OWR-12175-015

### Extensions of Graph $C^*$-Algebras

P / partially_solved | Oberwolfach 报告问题汇编

If $I$ and $A/I$ are graph $C^*$-algebras, when is $A$ a graph $C^*$-algebra, and can the obstructions be described using $K$-theoretic data?

来源字段：https://doi.org/10.4171/owr/2012/52

01970 | OWR-4332-004

### Facet Connectivity in the Fan of $D_h$

O / open | Oberwolfach 报告问题汇编

For each cone $\sigma$ in the fan defining $D_h$, is there a sequence of cones $\sigma_1=\sigma,\ldots,\sigma_r$ such that consecutive cones meet in a common facet and one facet of $\sigma_r$ is defined by $d_i=0$ for some $i$?

来源字段：https://doi.org/10.4171/owr/2010/19

01971 | AIM-REPRESENTATION_THEORY-0090

### Factorization into irreducible algebra-group supercharacters

P / partially_solved | AIM 研讨会问题汇编

For Un every supercharacter is a product of supercharacters that happen to be irreducible. This is not true for every pattern groups. When is it true (even for algebra groups)?

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

01972 | OWR-14214-002

### Factorization of Equivariant Dirac Operators and Spectral Triples

O / open | Oberwolfach 报告问题汇编

What should it mean for an equivariant Dirac operator on a principal G-bundle $M\to M/G$ to factor into Dirac operators on G and M/G, and how can this factorization be generalized to noncommutative equivariant spectral triples?

来源字段：https://doi.org/10.4171/owr/2016/6

01973 | ALG-045

### Farrell-Jones Conjecture

P / partially_solved | 独立 / 未分配集合

Are the assembly maps in algebraic K-theory and L-theory isomorphisms?

背景链接：https://arxiv.org/abs/2507.11337

01974 | OWR-11580-003

### Feigin-Frenkel-Lusztig Character Formula

O / open | Oberwolfach 报告问题汇编

Prove the Feigin-Frenkel-Lusztig character formula expressing the character of a critical-level simple module $L(\lambda)$ in terms of periodic polynomials.

来源字段：https://doi.org/10.4171/owr/2012/13

01975 | OWR-10252931-001

### Fermionized Yangians from Quivers with Potential

P / partially_solved | Oberwolfach 报告问题汇编

别名：Partial Fermionization of Cohomological Hall Algebras [OWR-10252931-002]

Can the cohomological Hall-algebra construction for preprojective algebras be partially fermionized to recover Yangians of classical Lie superalgebras such as $\mathfrak{gl}(m|n)$? Equivalently, can one choose a quiver $Q'$ and potential $W'$ so that $A_{Q',W'}$ partially fermionizes $A_{\widetilde Q,\widetilde W}$?

来源字段：https://doi.org/10.4171/owr/2022/23

01976 | OWR-1316-003

### Finite Generation and Separation in Invariant Rings

O / open | Oberwolfach 报告问题汇编

For a linear algebraic group $G$ acting on an affine variety $X$, is $K[X]^G$ finitely generated? If so, find generators; determine structural properties of $K[X]^G$; and determine whether invariant functions separate distinct $G$-orbits.

来源字段：https://doi.org/10.4171/owr/2006/30

01977 | OWR-12868-008

### Finite Generation for Gorenstein Algebras

P / partially_solved | Oberwolfach 报告问题汇编

For which finite-dimensional Gorenstein algebras does condition $\mathrm{Fg}$ hold; that is, when is the Hochschild cohomology ring noetherian and the Yoneda algebra of the direct sum of the simple modules finitely generated over it?

来源字段：https://doi.org/10.4171/owr/2014/08

01978 | OWR-12170-004

### Finite Generation of Cohomology for Finite Tensor Categories

P / partially_solved | Oberwolfach 报告问题汇编

Is the cohomology of every finite tensor category finitely generated; equivalently, can the Etingof-Ostrik conjecture be proved or disproved, first for further classes of finite-dimensional Hopf algebras?

来源字段：https://doi.org/10.4171/owr/2012/47

01979 | OWR-16773-005

### Finite Generation of Noncocommutative Hopf Algebra Cohomology

P / partially_solved | Oberwolfach 报告问题汇编

Is the cohomology of every finite-dimensional noncocommutative Hopf algebra finitely generated? In particular, for odd characteristic $p$, is finite generation valid for the remaining algebra $A_5(0)$ among the connected Hopf algebras of dimension $p^3$?

来源字段：https://doi.org/10.4171/owr/2019/11

01980 | ALG-046

### Finite Lattice Representation Problem

P / partially_solved | 独立 / 未分配集合

别名：Finite Lattice Representation Problem [OPG-37283]

Is every finite lattice isomorphic to the congruence lattice of some finite algebra?

背景链接：https://en.wikipedia.org/wiki/Finite_lattice_representation_problem

01981 | OWR-14299587-002

### Finite Length of Permutation Modules over Arbitrary Fields

P / partially_solved | Oberwolfach 报告问题汇编

Let $M$ be an $\omega$-categorical structure, let $G=\operatorname{Aut}(M)$, and let $W$ be an oligomorphic $G$-set. Is the permutation module $F[W]$ of finite length over $FG$ for every field $F$?

来源字段：https://doi.org/10.4171/owr/2025/58

01982 | AIM-REPRESENTATION_THEORY-0015

### Finite residual certificates for adjoint Stark-unit reconstruction

P / partially_solved | AIM 研讨会问题汇编

**Local-global description of Stark units.** Let \(f\) be a weight-\(1\) modular form with associated Galois representation \(\rho_f\). The Harris--Venkatesh conjecture gives a description of Stark units for \(\mathrm{Ad}^0(\rho_f)\) modulo \(p\) for almost all \(p\). In explicit exotic cases, can this be used to describe the Stark units for \(\mathrm{Ad}^0(\rho_f)\) themselves?

来源字段：http://aimpl.org/aagaautomorphic/6/

01983 | OWR-12480-003

### Finite Subgroups and Torsion Primes of Cremona Groups

P / partially_solved | Oberwolfach 报告问题汇编

Write $\operatorname{Aff}_n$ for the affine group, $\operatorname{Cr}_n$ for the Cremona group, and $\operatorname{Tors}(G)$ for the set of torsion primes of a group $G$. Do there exist isomorphic finite subgroups of $\operatorname{Aff}_n$ that are not conjugate in $\operatorname{Cr}_n$? Are the sets $\operatorname{Tors}(\operatorname{Cr}_n)$, $\operatorname{Tors}(\operatorname{Aut}(\mathbb A^n))$, and $\operatorname{Tors}(\operatorname{Aut}^*(\mathbb A^n))$ finite, and what is the least $n$ for which one of them contains $7$? Finally, which analogues of the standard Jordan-decomposition properties for connected reductive algebraic groups hold for these three groups?

来源字段：https://doi.org/10.4171/owr/2013/17

01984 | OWR-12975-010

### Finite-Dimensional Detection of Positivity in Surface Groups

O / open | Oberwolfach 报告问题汇编

For the fundamental group of every closed surface, is positivity of a Hermitian element in all unitary representations equivalent to positivity in all finite-dimensional unitary representations, and hence to membership in the cone of sums of Hermitian squares?

来源字段：https://doi.org/10.4171/owr/2014/17

01985 | OWR-14298159-003

### Finite-Dimensional Nichols Algebras over Finite Groups

O / open | Oberwolfach 报告问题汇编

Classify the pairs $(G,V)$ for which $G$ is finite, $V\in{}_{\mathbb CG}^{\mathbb CG}\mathcal{YD}$, and $\dim\mathcal B(V)<\infty$, and compute defining relations for every such Nichols algebra.

来源字段：https://doi.org/10.4171/owr/2024/5

01986 | OWR-8415347-005

### Finite-Growth Hopf Algebras in Yetter-Drinfeld Categories

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a finitely generated group. Classify, or at least characterize, all coradically graded connected Hopf algebras $E=\bigoplus_{n\in\mathbb{N}_0}E_n$ in the Yetter-Drinfeld category ${}^{\mathbb{C}G}_{\mathbb{C}G}\mathcal{YD}$ such that $\operatorname{GKdim}(E\#\mathbb{C}G)<\infty$.

来源字段：https://doi.org/10.4171/owr/2021/44

01987 | OWR-8415347-006

### Finite-Growth Nichols Algebras over Finitely Generated Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a finitely generated group. Classify all Yetter-Drinfeld modules $V\in{}^{\mathbb{C}G}_{\mathbb{C}G}\mathcal{YD}$ such that the bosonized Nichols algebra satisfies $\operatorname{GKdim}(\mathcal{B}(V)\#\mathbb{C}G)<\infty$.

来源字段：https://doi.org/10.4171/owr/2021/44

01988 | OWR-13750328-014

### Finitely Generated Cohomology in Stable Module Categories

O / open | Oberwolfach 报告问题汇编

For a finite group $G$ and a field $k$ of characteristic $p$, prove that the thick subcategory of $\operatorname{StMod}(kG)$ consisting of modules with finitely generated cohomology is generated by the thick subcategories of finitely generated modules and modules with no cohomology.

来源字段：https://doi.org/10.4171/owr/2023/19

01989 | OWR-12697693-004

### Finiteness and Goodness for Property-S Skew Braces

O / open | Oberwolfach 报告问题汇编

Let $B=(B,+,\circ)$ be a skew brace, write $a*b=-a+a\circ b-b$, and define $$\operatorname{Fix}^r(x)=\{b\in B:x*b=0\},\qquad \operatorname{Fix}^l(x)=\{b\in B:b*x=0\}.$$ Say that $B$ has property $(S)$ when, for every $x\in B$, both $$[(B,+):\operatorname{Fix}^r(x)\cap C_x^+]\quad\text{and}\quad[(B,\circ):\operatorname{Fix}^l(x)\cap C_x^\circ]$$ are finite, where $C_x^+$ and $C_x^\circ$ are the centralizers of $x$ in the two groups. Is every finitely generated ideal of such a $B$ finite? If an ideal $I$ is called good when every finitely generated ideal $J\subseteq I$ is finitely generated as a skew brace, is every skew brace with property $(S)$ good?

来源字段：https://doi.org/10.4171/owr/2023/9

01990 | OWR-13673-012

### Finiteness of Automorphism Groups of Rational Vertex Operator Algebras

O / open | Oberwolfach 报告问题汇编

If $V$ is a rational, $C_2$-cofinite VOA of CFT type with $V_1=0$, then conjecturally $\operatorname{Aut}(V)$ is finite.

来源字段：https://doi.org/10.4171/owr/2015/16

01991 | ALG-031

### Finiteness of Finitely Presented Periodic Groups

O / open | 独立 / 未分配集合

Is every finitely presented periodic group finite?

背景链接：https://www.mathnet.ru/links/63460d5ff426dd8772ad636b68cd111d/im8978_eng.pdf

01992 | OWR-3472-011

### Finiteness of Nonzero Bockstein Maps in Local Cohomology

O / open | Oberwolfach 报告问题汇编

For a polynomial ring $R$ over $\mathbb Z$ and a fixed ideal $\mathfrak a\subseteq R$, are the Bockstein homomorphisms $\beta_k^p:H_{\mathfrak a}^k(R/pR)\to H_{\mathfrak a}^{k+1}(R/pR)$ zero for all but finitely many primes $p$?

来源字段：https://doi.org/10.4171/owr/2009/22

01993 | OWR-14299578-008

### Finiteness of R-Equivalence Classes in Nonabelian Cohomology

O / open | Oberwolfach 报告问题汇编

Let $K$ be a number field and $G$ a finite solvable group. Is the quotient $H^1(K,G)/R$ by $R$-equivalence finite? If so, relate its cardinality to the minimum number of torsors over a purely transcendental extension of $K$ needed to parametrize $H^1(K,G)$ densely.

来源字段：https://doi.org/10.4171/owr/2025/52

01994 | OWR-10252926-004

### Finkelberg-Mirković Equivalence in Modular Geometric Satake

O / open | Oberwolfach 报告问题汇编

Let $k$ have characteristic $p$, let $G$ be reductive with Weyl group $W$, and let $\operatorname{Rep}^{\mathrm{ext}}_0(G_k^\vee)$ be the Serre subcategory generated by the modules $L(\lambda)$ for $\lambda\in W\mathbin{\cdot}0\cap X^+$. Does an equivalence $$ Q:\operatorname{Perv}_{\mathrm{Iw}}(\operatorname{Gr}_G,k)\longrightarrow\operatorname{Rep}^{\mathrm{ext}}_0(G_k^\vee) $$ exist that geometrically realizes Frobenius twist?

来源字段：https://doi.org/10.4171/owr/2022/18

01995 | OWR-13750328-003

### First Hochschild Cohomology of Nontrivial-Defect Blocks

O / open | Oberwolfach 报告问题汇编

For every block $B$ of a finite group with nontrivial defect group, is $HH^1(B)$ nonzero?

来源字段：https://doi.org/10.4171/owr/2023/19

01996 | AIM-LINEAR_ALGEBRA-0016

### Fisk's higher Toeplitz-minor transform in degree at most three

P / partially_solved | AIM 研讨会问题汇编

Is Fisk's conjecture true for $3\times3$ minors / $n\times n$ minors? If $\sum_{k=0}^d a_kz^k$ is real-rooted with all $a_k>0$, Br\"and\'en proved that \[ \sum_{k\geq0} \left|\begin{matrix}a_k&a_{k-1}\\a_{k+1}&a_k\end{matrix}\right|z^k \] is real-rooted. Is the analogous polynomial \[ \sum_{k\geq0} \left|\begin{matrix} a_k&a_{k-1}&a_{k-2}\\ a_{k+1}&a_k&a_{k-1}\\ a_{k+2}&a_{k+1}&a_k \end{matrix}\right|z^k \] also real-rooted, and likewise for higher-order minors?

来源字段：http://aimpl.org/totalpos/1/

01997 | AIM-LINEAR_ALGEBRA-0003

### Fixed second adjacency eigenvalue multiplicity: formulation audit and exact boundary cases

P / partially_solved | AIM 研讨会问题汇编

For a fixed choice of parameters $\rho$ and $\Delta$, what is the largest possible multiplicity of the second eigenvalue of the adjacency matrix $A_G$ with the constraint that $\lambda_2(A_G) = \rho$.

“Let $G$ be a graph with maximum degree bounded by $\Delta$.”

The surrounding working-group heading is “Multiplicity of $\lambda_2$ for Cayley graphs of bounded degree,” but item (1) and item (2) themselves say “graph,” and the report separately says that the group investigated Cayley graphs. Thus restricting the formal question to Cayley graphs would be an additional assumption, not a safe source reconstruction.

Several quantifiers remain implicit even in the PDF:

- The graph is naturally read as a finite, undirected, simple graph, as in the cited adjacency-spectrum literature, but the PDF does not say this in item (2). - Connectedness is essential and is not stated. The cited theorem of Jiang-Tidor-Yao-Zhang-Zhao and the later papers formulating this fixed-eigenvalue question use connected graphs. - No order $n=|V(G)|$ appears. This may be deliberate: fixing $\rho$ and $\Delta$ suggests asking for a uniform bound over all orders. If that supremum is infinite, an order-dependent extremal function is the useful replacement. - The literature orders adjacency eigenvalues with multiplicity, \[ \lambda_1(G)\ge \lambda_2(G)\ge\cdots\ge\lambda_n(G). \] “Second distinct eigenvalue” is a different convention. For connected graphs the Perron root is simple, so the value $\lambda_2$ is also the second distinct value. For disconnected graphs the conventions can differ. - A maximum-degree bound is integral. If the source parameter $\Delta$ is real, replace it by $D=\lfloor\Delta\rfloor$.

Three readings must therefore be separated.

1. **Canonical sentence literally:** $\Delta$ never constrains $G$. 2. **PDF-context reading:** $G$ has maximum degree at most $D$, but need not be connected. 3. **Repaired connected reading:** $G$ is a finite connected simple graph of maximum degree at most $D$, and eigenvalues are ordered with multiplicity.

For the repaired reading define

\[ F_{\mathrm{conn}}(\rho,D) =\sup_G m_G(\rho), \]

where the supremum runs over all finite connected simple graphs with $\Delta(G)\le D$ and $\lambda_2(G)=\rho$, and $m_G(\rho)=\dim\ker(A_G-\rho I)$. An empty admissible family is called **infeasible**, not assigned the misleading value zero. The order-specific repair is

\[ M_{\mathrm{conn}}(n;\rho,D) =\max_{\substack{|V(G)|=n,\ G\ {\rm connected,\ simple}\\ \Delta(G)\le D,\ \lambda_2(G)=\rho}}m_G(\rho), \]

with the understanding that some orders have no admissible graph.

来源字段：http://aimpl.org/spectralhypergraph/1/

01998 | AIM-REPRESENTATION_THEORY-0007

### Fixed vectors as compact-torus distinction

P / partially_solved | AIM 研讨会问题汇编

**Fixed vectors for $\mathrm{GSp}_{2n}$.** Let $\pi$ be a irreducible admissible infinite-dimensional representation of $\mathrm{GSp}_{2n}(\mathbb Q_p)$. Consider the subgroup \[ R(n):=\{g\equiv(\text{matrix with $1$ on the diagonal for first $n$ entries, $a$ on the diagonal for the last $n$ entries})\pmod{p^n}\text{ for some }a\in\mathbb Z_p^\times\}. \] Does $\pi$ have a $R(n)$-fixed vector for some $n$?

来源字段：http://aimpl.org/aagaautomorphic/3/

01999 | OWR-14298159-004

### Fomin-Kirillov Algebras as Nichols Algebras

O / open | Oberwolfach 报告问题汇编

Determine whether the quadratic Fomin-Kirillov algebra associated with the symmetric group is a Nichols algebra in every rank, extending the known small-rank cases.

来源字段：https://doi.org/10.4171/owr/2024/5

02000 | OWR-16770-010

### Formal Group Models for Commuting p-Adic Power Series

P / partially_solved | Oberwolfach 报告问题汇编

Let $(f,u)$ be a pair of commuting stable power series over a $p$-adic field $K$, with $f$ noninvertible, $u$ invertible, and $f\not\equiv0\pmod{\mathfrak m_K}$. Does there exist a formal group $S$ such that $f$ and $u$ are semiconjugate to endomorphisms of $S$?

来源字段：https://doi.org/10.4171/owr/2019/8

02001 | OWR-4533-002

### Forms and Multiloop Algebras in Nullity Two

P / partially_solved | Oberwolfach 报告问题汇编

Let $$R_n=\mathbb C[t_1^{\pm1},\ldots,t_n^{\pm1}]$$ and let $\mathfrak g$ be a finite-dimensional complex semisimple Lie algebra. Classify the $R_n$-forms of $\mathfrak g\otimes_\mathbb C R_n$ and characterize which are multiloop algebras. In nullity two, are the only non-multiloop examples of Margaux type?

来源字段：https://doi.org/10.4171/owr/2010/51

02002 | AIM-REPRESENTATION_THEORY-0101

### Four symmetric-group supercharacter theories and a canonical restriction dichotomy

P / partially_solved | AIM 研讨会问题汇编

Problem 1.17. Is it true that there are four infinite families of supercharacter theories for symmetric groups and 6 or 7 exceptions? Are those four families nested? See *Supercharacter theories of cyclic p-groups* by A. Hendrickson.

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

02003 | AIM-REPRESENTATION_THEORY-0014

### Fourier, slope, and weight gates for special lifts in differential-operator images

P / partially_solved | AIM 研讨会问题汇编

**\(p\)-adic differential operators for higher-rank groups.** For a higher-rank group (e.g. \(\mathrm{GSp}_4\)), there are some standard \(p\)-adic differential operators. Do the images of these differential operators contain special automorphic forms coming from lower-rank groups (e.g. Yoshida lifts, Saito--Kurokawa lifts)?

来源字段：http://aimpl.org/aagaautomorphic/6/

02004 | OWR-14298370-006

### Fractional Calabi-Yau Properties of Higher Representation-Finite Algebras

O / open | Oberwolfach 报告问题汇编

Is every higher representation-finite algebra, or more generally every twisted fractionally Calabi-Yau algebra, fractionally Calabi-Yau?

来源字段：https://doi.org/10.4171/owr/2024/20

02005 | OWR-8415347-004

### Free Actions of Rigid Tensor Categories on the Hyperfinite Factor

P / partially_solved | Oberwolfach 报告问题汇编

Can every rigid $C^*$-tensor category act freely on the hyperfinite $\mathrm{II}_1$ factor? In the nonamenable case, are there infinitely many such actions up to cocycle conjugacy? What happens for the Temperley-Lieb-Jones categories at index values greater than $4$?

来源字段：https://doi.org/10.4171/owr/2021/44

02006 | OWR-17288-003

### Free Decompositions of Compact Quantum Groups

O / open | Oberwolfach 报告问题汇编

When does a compact matrix quantum group $(G,u)$ admit a free decomposition of the sequence $(\operatorname{Fix}(u^{\otimes n}))_{n\geq1}$? Is this equivalent to $G$ having a free fusion semiring, and how can the notion be generalized to allow negative free cumulants?

来源字段：https://doi.org/10.4171/owr/2019/45

02007 | OWR-14298003-005

### Free Structures and Operadic Symmetries for Rough Paths

O / open | Oberwolfach 报告问题汇编

Find new combinatorial models and free structures for rough paths, develop an operadic account of them, and describe chain-rule and Itô-isometry symmetries through maps from free Novikov structures.

来源字段：https://doi.org/10.4171/owr/2023/54

02008 | OWR-17133-009

### Freely Complemented and Maximal Amenable MASAs

O / open | Oberwolfach 报告问题汇编

Let $p=(p_1,\ldots,p_d)$ be evaluated on a semicircular $d$-tuple $S$. Under what conditions does $p_1(S)$ generate a freely complemented masa? Is this generic? Are most masas in $L(F_d)$ maximal amenable, and is every maximal amenable masa freely complemented?

来源字段：https://doi.org/10.4171/owr/2019/37

02009 | AIM-REPRESENTATION_THEORY-0075

### Galois descent for a covering group

P / partially_solved | AIM 研讨会问题汇编

Let $F$ be a characteristic-zero $p$-adic local field, let $L/F$ be a finite unramified extension, let $G_F$ be an unramified reductive group over $F$ (quasi-split over $F$ and split over $L$), and let $\widetilde{G}_F$ be a central topological covering group of $G_F(F)$ of degree coprime to $p$, with corresponding base-change covering group $\widetilde{G}_L$ over $L$. Does the Galois group $\operatorname{Gal}(L/F)$ act naturally on $\widetilde{G}_L$?

来源字段：http://aimpl.org/autoformcovergp/3/

02010 | OWR-9790358-001

### Gasca-Maeztu Maximal-Line Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

别名：Gasca-Maeztu Maximal Line Conjecture [OWR-9790358-006]; Structure of n-Independent and GCn Interpolation Sets [OWR-9790358-007]

Does every planar $GC_n$ set contain a maximal line?

来源字段：https://doi.org/10.4171/owr/2022/7

02011 | OWR-8415347-017

### Gaussian Functionals on q-Deformed Compact Lie Groups

P / partially_solved | Oberwolfach 报告问题汇编

Does the Lévy-Khintchine decomposition method for $U_q(n)$ extend to the $q$-deformations $G_q$ of other simple compact Lie groups?

来源字段：https://doi.org/10.4171/owr/2021/44

02012 | OWR-12732-004

### Generalized Cluster Structures on Poisson-Lie Doubles

P / partially_solved | Oberwolfach 报告问题汇编

For every Belavin-Drinfeld triple $T$, does the double $D(G)$ admit a generalized cluster structure compatible with the corresponding Poisson-Lie structure?

来源字段：https://doi.org/10.4171/owr/2013/58

02013 | OWR-12591-003

### Generalized Pimsner-Voiculescu Embeddings

O / open | Oberwolfach 报告问题汇编

Can the Pimsner-Voiculescu embedding of an irrational rotation algebra be generalized to higher-dimensional noncommutative tori or more general discrete abelian groups; can spectra of unitary generators be broken in more general crossed products by locally compact abelian groups; and which properties such as maximality or extendibility of automorphisms characterize this embedding?

来源字段：https://doi.org/10.4171/owr/2013/43

02014 | OWR-13685-006

### Generation of Tail Algebras by Diagonal Subalgebras

O / open | Oberwolfach 报告问题汇编

In free probability theory, let $(A, \phi)$ be a $W^*$-probability space generated by a sequence of random variables $(x_n)_{n \geq 1}$ with tail algebra $T_\phi = \bigcap_{n \geq 1} W^*(x_k : k \geq n)$ and exchangeable/distributional symmetry algebra $D_\phi$. Is the tail von Neumann algebra $T_\phi$ generated as a von Neumann algebra by $D_\phi$?

来源字段：https://doi.org/10.4171/owr/2015/28

02015 | OWR-12175-009

### Generator Rank of $\mathcal Z$-Stable $C^*$-Algebras

P / partially_solved | Oberwolfach 报告问题汇编

If $A$ is unital and separable and absorbs the Jiang-Su algebra $\mathcal Z$, or is unital, separable, nuclear, of real rank zero, and of stable rank one, must $\operatorname{gr}(A)\leq1$?

来源字段：https://doi.org/10.4171/owr/2012/52

02016 | OWR-12175-008

### Generator Rank of Direct Sums

P / partially_solved | Oberwolfach 报告问题汇编

Given separable $C^*$-algebras $A$ and $B$, is $\operatorname{gr}(A\oplus B)=\max\{\operatorname{gr}(A),\operatorname{gr}(B)\}$?

来源字段：https://doi.org/10.4171/owr/2012/52

02017 | OWR-14298007-008

### Generic Anisotropy of Gorenstein Graded Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Identify classes of Gorenstein standard graded algebras that have the generic anisotropy property.

来源字段：https://doi.org/10.4171/owr/2023/58

02018 | OWR-4800-002

### Generic Bases for Cluster Algebras

P / partially_solved | Oberwolfach 报告问题汇编

For the cluster algebra $A(B,x)$, is the set $G=\{x^a C_Z\mid Z\in\operatorname{Irr}_{sr}(\operatorname{Rep}_{\mathbf d}(J)),\ a,\mathbf d\in\mathbb N^m,\ a\cdot\mathbf d=0\}$ a basis?

来源字段：https://doi.org/10.4171/owr/2011/10

02019 | OWR-8415347-020

### Genuine Finite-Growth Nichols Algebras over Nonabelian Groups

O / open | Oberwolfach 报告问题汇编

Find a genuine Nichols algebra over a nonabelian group whose support is nonabelian, whose dimension is infinite, and whose Gelfand-Kirillov dimension is finite. In particular, for the rack of transpositions in $S_3$, what is the Nichols algebra corresponding to the constant cocycle $1$?

来源字段：https://doi.org/10.4171/owr/2021/44

02020 | OWR-8415347-013

### Geometric and Sheaf-Theoretic Models of Nichols Algebras

O / open | Oberwolfach 报告问题汇编

Can bosonization and the reflection functors be interpreted geometrically? Can the associated polynomials be interpreted in geometric terms? Can extensions of local systems, quiver descriptions of perverse sheaves, or asymptotic calculations yield a combinatorial description or finite-presentation results for Nichols algebras?

来源字段：https://doi.org/10.4171/owr/2021/44

02021 | OWR-14298590-004

### Geometric Fixed Points for Poincaré Infinity-Categories

O / open | Oberwolfach 报告问题汇编

How should geometric fixed-point constructions such as $\Phi^{C_2}\mathrm{THR}(R)$ extend to Poincaré $\infty$-categories?

来源字段：https://doi.org/10.4171/owr/2024/39

02022 | OWR-734-001

### Geometric Interpretations of Coxeter-Group R-Polynomials

O / open | Oberwolfach 报告问题汇编

What geometric interpretation or geometric connections do R-polynomials of Coxeter groups have?

来源字段：https://doi.org/10.4171/owr/2004/32

02023 | AIM-REPRESENTATION_THEORY-0042

### Geometric models for Frobenius-kernel representations and an all-level grading test

P / partially_solved | AIM 研讨会问题汇编

What geometric category is equivalent to $\mathrm{Rep}_0(G_rT)$ or $\mathrm{Rep}_{\mathrm{res}}(\mathfrak g)$?

来源字段：http://aimpl.org/sheavemodular/2/

02024 | OWR-9790358-013

### Geometry and Combinatorics of Planar Spline Dimensions

O / open | Oberwolfach 报告问题汇编

Can the dimension of a planar spline space be determined from the geometry and combinatorics of its underlying subdivision?

来源字段：https://doi.org/10.4171/owr/2022/7

02025 | OWR-8415347-014

### Geometry and Local Systems Behind Finite Nichols Algebras

O / open | Oberwolfach 报告问题汇编

Can one describe more explicitly the geometry behind $F(K_n)\simeq T_{!*}(V_n)$ for $n=3,4,5$? What can be said about the local systems and sheaves whose Nichols algebras have finite dimension or finite Gelfand-Kirillov dimension?

来源字段：https://doi.org/10.4171/owr/2021/44

02026 | OWR-17472-020

### Geometry behind a Plethystic Representation Isomorphism

O / open | Oberwolfach 报告问题汇编

Find a geometric or invariant-theoretic interpretation of the isomorphism $\nabla_{(3\ell-3,2^{\ell-1})}\operatorname{Sym}^{\ell}E\cong\nabla_{(\ell+1,1^{\ell-2})}\operatorname{Sym}^{3\ell-4}E$ for all $\ell\ge2$.

来源字段：https://doi.org/10.4171/owr/2020/9

02027 | AIM-REPRESENTATION_THEORY-0083

### Global applications and the ordinary-twist quotient for covering tori

P / partially_solved | AIM 研讨会问题汇编

Are there global applications of the trace formula, automorphic forms, Whittaker models, the Langlands-Shahidi method for $\widetilde{G}$ ?

-- Classify automorphic representations of covers of split tori.

来源字段：http://aimpl.org/autoformcovergp/4/

02028 | AIM-REPRESENTATION_THEORY-0080

### Global covering groups with arithmetic applications and a reciprocity-coherence criterion

P / partially_solved | AIM 研讨会问题汇编

**What global covering groups have arithmetic applications?**

来源字段：http://aimpl.org/autoformcovergp/4/

02029 | AMR-093-0178

### Global Gan-Gross-Prasad conjecture

P / partially_solved | AMR 问题清单汇编

In the global Gan-Gross-Prasad setting, is nonvanishing of the relevant $H$-period on an irreducible cuspidal automorphic representation equivalent to nonvanishing of every corresponding local Hom space together with the central Rankin-Selberg value $L_E(1/2,\pi_1\times\pi_2)$?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

02030 | OWR-14299523-006

### Gorenstein Hilbert Series for Reduced Subalgebra Zeta Functions

O / open | Oberwolfach 报告问题汇编

For every $d$-dimensional algebra $A$ in the subalgebra-zeta setting, construct a natural $\mathbb{N}_0$-graded Gorenstein algebra of dimension $d$ whose Hilbert series is the reduced subalgebra zeta function $Z_A^{\mathrm{red}}(T)$.

来源字段：https://doi.org/10.4171/owr/2025/38

02031 | OWR-734-003

### Gorenstein Initial Ideals with Minimal Generators

P / partially_solved | Oberwolfach 报告问题汇编

Given a Gorenstein classical ideal $I$, does it have a Gorenstein initial ideal $H$ in the given coordinates with the same number of generators?

来源字段：https://doi.org/10.4171/owr/2004/32

02032 | OWR-14298169-006

### Graded Classification Conjecture for Leavitt Path Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Is the pointed graded $K_0$-group a complete invariant of naturally $\mathbb{Z}$-graded Leavitt path algebras of finite graphs?

来源字段：https://doi.org/10.4171/owr/2024/12

02033 | OWR-1465-011

### Gradient-Tentacle Certificates for Polynomial Nonnegativity

O / open | Oberwolfach 报告问题汇编

Let $f\in\mathbb R[X_1,\ldots,X_n]$, let $f^*=\inf f(\mathbb R^n)$, and let $J$ be generated by $X_j(\partial f/\partial X_i)-X_i(\partial f/\partial X_j)$ for $1\leq i<j\leq n$. Without assuming that $f^*$ is attained, are the following equivalent: $f\geq0$ on $\mathbb R^n$; $f\geq0$ on $V(J)\cap\mathbb R^n$; $f\in\sum\mathbb R[\bar X]^2+\sqrt J$; and $f+\varepsilon\in\sum\mathbb R[\bar X]^2+J$ for every $\varepsilon>0$?

来源字段：https://doi.org/10.4171/owr/2007/14

02034 | OWR-4800-005

### Graphical Non-Abelian Hodge Correspondence

O / open | Oberwolfach 报告问题汇编

For a quiver $\Gamma$ and dimension vector $v$, let $C_{\Gamma,v}$ be the projective curve with one rational component of multiplicity $v_i$ per vertex and nodal intersections along the edges, let $J_v$ be its compactified Jacobian, and let $\mathcal M_v$ be the Crawley-Boevey-Shaw character variety. Do graphical non-Abelian Hodge isomorphisms $H^*(J_v)\cong H^*(\mathcal M_v)$ identify $W_{2k}H^*(\mathcal M_v)$ with $P_kH^*(J_v)$ and yield $$P(H;J_v;t)=P(P;\mathcal M_v;t)=t^{2d_v}A_\Gamma(v,t^{-2})$$?

来源字段：https://doi.org/10.4171/owr/2011/10

02035 | AMR-015-0011

### Green's conjecture

P / partially_solved | AMR 问题清单汇编

For a non-hyperelliptic algebraic curve, is its Clifford index determined by the extent to which its canonical embedding has linear syzygies?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767

02036 | AIM-LINEAR_ALGEBRA-0018

### Grone--Merris majorization and a sharp complete-bipartite Laplacian-energy gap

S / solved | AIM 研讨会问题汇编

**Question 6 (Grone--Merris conjecture).** Is \(\lambda\) majorized by \(d^*\)?

It was shown at the workshop that, if the Grone--Merris conjecture is true, then \[ E_L(G)\leq \sum_{i=1}^n \left|d_i^*-\frac{2m}{n}\right|. \]

来源字段：https://aimath.org/WWN/matrixspectrum/matrixspectrum.pdf

02037 | AMR-015-0012

### Grothendieck-Katz p-curvature conjecture

O / open | AMR 问题清单汇编

Prove the conjectured local-to-global principle for linear ordinary differential equations known as the Grothendieck-Katz $p$-curvature conjecture.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767

02038 | OWR-12494-006

### Grothendieck-Serre Conjecture for Regular Local Rings

P / partially_solved | Oberwolfach 报告问题汇编

Let $R$ be a regular local ring with fraction field $K$, and let $G$ be a reductive group scheme over $R$. If a principal $G$-bundle over $R$ becomes trivial over $K$, must it already be trivial over $R$? Equivalently, does $H^1_{\mathrm{\acute et}}(R,G)\to H^1_{\mathrm{\acute et}}(K,G)$ have trivial kernel?

来源字段：https://doi.org/10.4171/owr/2013/31

02039 | OWR-17294-018

### Groups Arising as Quandle Structure Groups

O / open | Oberwolfach 报告问题汇编

Characterize and further analyze the groups $G\cong\operatorname{Grp}(Q)$ that arise as structure groups of quandles $Q$.

来源字段：https://doi.org/10.4171/owr/2019/51

02040 | OWR-14299281-005

### Groups Realizable as Variety Automorphism Groups

P / partially_solved | Oberwolfach 报告问题汇编

Which groups occur as automorphism groups $\operatorname{Aut}(X)$ of algebraic varieties $X$?

来源字段：https://doi.org/10.4171/owr/2025/20

02041 | ALG-004

### Hadamard Conjecture

O / open | 独立 / 未分配集合

For every positive integer $k$, does there exist a Hadamard matrix of order $4k$?

背景链接：https://ajc.maths.uq.edu.au/pdf/93/ajc_v93_p422.pdf

02042 | ALG-047

### Hadamard Matrix Conjecture

O / open | 独立 / 未分配集合

Does a Hadamard matrix of order 4k exist for every positive integer k?

背景链接：https://en.wikipedia.org/wiki/Hadamard_matrix#Hadamard_conjecture

02043 | AMR-015-0016

### Hadamard's maximal determinant problem

P / partially_solved | AMR 问题清单汇编

For each order $n$, determine the largest possible absolute determinant of an $n\times n$ matrix whose entries are all $1$ or $-1$.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767

02044 | OWR-17289-005

### Hall Multiplication on Principal Quiver Grassmannians

O / open | Oberwolfach 报告问题汇编

For principal quiver Grassmannians and the invariant-function space $A=\bigoplus_{[P],[I],e}\mathbb{Q}^{\operatorname{Aut}_{Q}(P\oplus I)}[\operatorname{Gr}_{e}(P\oplus I)]$, can one define a Hall-algebra-type multiplication making $A$ graded associative and inducing an isomorphism $U_q(\mathfrak b^+)\otimes U_q(\mathfrak b^+)\to A$?

来源字段：https://doi.org/10.4171/owr/2019/46

02045 | OWR-14298169-003

### Hazrat Conjecture for Leavitt Path Algebras

P / partially_solved | Oberwolfach 报告问题汇编

For essential square matrices $A$ and $B$ over $\mathbb{N}$, are $A$ and $B$ shift equivalent if and only if the Leavitt path algebras $L_A$ and $L_B$ are graded Morita equivalent?

来源字段：https://doi.org/10.4171/owr/2024/12

02046 | OWR-14298370-009

### Heisenberg Actions on Hochschild-Serre Cohomology

O / open | Oberwolfach 报告问题汇编

Is there a Heisenberg algebra controlling the Hochschild-Serre cohomology of symmetric quotient stacks?

来源字段：https://doi.org/10.4171/owr/2024/20

02047 | OWR-13686-006

### Helton's Sum-of-Squares Theorem for Low-Dimensional Group Rings

O / open | Oberwolfach 报告问题汇编

Does Helton's sum-of-squares theorem hold for group rings of groups of cohomological dimension at most two?

来源字段：https://doi.org/10.4171/owr/2015/29

02048 | MOD-007

### Henson Graphs Finite Model Property

P / partially_solved | 独立 / 未分配集合

Do Henson graphs have the finite model property?

背景链接：https://www.ma.imperial.ac.uk/~dmevans/hattingen.pdf

02049 | AIM-REPRESENTATION_THEORY-0114

### Herzig's semisimple weight recipe and extension sensitivity

P / partially_solved | AIM 研讨会问题汇编

The document warns that it was typeset during the talks and may contain transcription errors.

The immediately preceding discussion is essential. For $q=p$, write the semisimplified inertial representation as $$\tau:=\rho|_{I_p}^{\mathrm{ss}} =\bigoplus_i\left(\chi_i\oplus\chi_i^p\oplus\cdots \oplus\chi_i^{p^{s_i-1}}\right), \qquad \sum_i s_i=n, $$ where $\chi_i$ has minimal niveau $s_i$ and factors through $\mathbb F_{p^{s_i}}^\times$. This determines a maximal torus $$ T=\prod_i\mathbb F_{p^{s_i}}^\times\subset\mathrm{GL}_n(\mathbb F_p) $$ and a Teichmüller lift $\widetilde\chi$ of the product character. The notes attach a characteristic-zero representation $$ V(\tau)\overset{\mathrm{def}}=V(T,\widetilde\chi) $$ of $\mathrm{GL}_n(\mathbb F_p)$, described there as induction from a product of cuspidal representations. In Herzig’s later formalization this is the genuine Deligne--Lusztig representation attached to the corresponding maximally split dual pair.

For a Serre weight $F(a_1,\ldots,a_n)$, the reflection/regularization operator in the $q=p$ case is $$ R\bigl(F(a_1,\ldots,a_n)\bigr) =F(a_n-(n-1),a_{n-1}-(n-2),\ldots,a_2-1,a_1)_{\mathrm{reg}}. $$ The subscript means the regular Serre weight corresponding to the resulting class modulo $p-1$. On regular weights, $$ R^2(F)=F\otimes\det^{\,1-n}. $$ The workshop puts quotation marks around “map” and says that it does not always associate a regular weight; this records a genuine 2006 ambiguity at irregular boundaries and for the contemplated $q=p^r$ extension. The single-valued $q=p$ formula above and the later multivalued $n=2,q=p^r$ extension are taken from Herzig’s precise 2009 formulation, not silently attributed to the rough notes.

With this context, the exact recovered conjecture is $$ \tag{C7} \operatorname{Wt}\!\left(\rho|_{I_p}^{\mathrm{ss}}\right)_{\mathrm{reg}} =R\!\left( \operatorname{JH}\!\left( \overline{V\!\left(\rho|_{I_p}^{\mathrm{ss}}\right)} \right)\right). $$ Here the overline is the reduction modulo $p$ of a stable lattice, and $\operatorname{JH}$ is the set of Jordan--Hölder constituents, with multiplicities forgotten. The PDF prints “Conjecture 7 (Herzig(?)).”

The two examples following (C7) say:

- for $n=3,q=p$, $R$ can be extended as a one-to-many map and yields all predicted weights; - for $n=2,q=p^r$, such an extension agrees with “Fred’s conjecture” (Fred Diamond’s part in the emerging Buzzard--Diamond--Jarvis recipe), although the notes say that several extensions were then possible.

The global eigenclass and cohomological-degree assertions in Conjectures 5 and 6 are not taken over here.

来源字段：https://aimath.org/WWN/padicmodularity/padicmodularity.pdf

02050 | ALG-011

### Herzog-Schönheim Conjecture

P / partially_solved | 独立 / 未分配集合

别名：The Herzog-Schönheim Conjecture [ALG-026]; Herzog-Schönheim Conjecture [ALG-054]

If a finite system of left cosets of subgroups of a group $G$ partitions $G$, then must at least two of the subgroups have the same index in $G$?

背景链接：https://en.wikipedia.org/wiki/Herzog%E2%80%93Sch%C3%B6nheim_conjecture

02051 | OWR-16635-008

### Higgs Branches as Associated Varieties of Vertex Algebras

O / open | Oberwolfach 报告问题汇编

For a four-dimensional $\mathcal N=2$ Argyres-Douglas theory, is its Higgs branch isomorphic to the associated variety of the corresponding vertex algebra under the four-dimensional/two-dimensional duality? This includes algebras of the forms $L_{-h^\vee+h^\vee/q}(\mathfrak g)$ and $W_{-h^\vee+h^\vee/q}(\mathfrak g,f)$ for $\mathfrak g$ of type $A$, $D$, or $E$.

来源字段：https://doi.org/10.4171/owr/2018/52

02052 | OWR-14298590-008

### Higher Algebra Structures on Graph-Cooperad Deformations

O / open | Oberwolfach 报告问题汇编

Does the deformation complex of the chain cooperad $\mathrm{EGraphs}_n^c$ carry a natural $E_{n+2}$-algebra structure, with the action of the $L_\infty$-algebra $\mathrm{EGC}_{2n}$ inducing a quasi-isomorphism

$$U_{E_{n+2}}(\mathrm{EGC}_{2n})\xrightarrow{\sim}\operatorname{Def}(\mathrm{EGraphs}_n^c,\mathrm{EGraphs}_n^c)?$$

Equivalently, establish the proposed compatibility of these higher-algebraic structures.

来源字段：https://doi.org/10.4171/owr/2024/39

02053 | OWR-15429-002

### Higher Extensions Between Additive and Multiplicative Algebraic Groups

O / open | Oberwolfach 报告问题汇编

In the proposed analogue of the Milnor conjecture, describe $\operatorname{Ext}^2_{\mathcal C}(G_a,G_m)$ by generators and relations. Does $\operatorname{Ext}^i_{\mathcal C}(G_a,G_m)=0$ for all sufficiently large $i$?

来源字段：https://doi.org/10.4171/owr/2017/21

02054 | OWR-7155441-003

### Higher Product Rigidity for Wound Unipotent Targets

O / open | Oberwolfach 报告问题汇编

Let $U$ be a wound unipotent group over $k$, let $X_1,\ldots,X_n$ be nonempty open subschemes of $\mathbb P_k^1$ with chosen points $x_i\in X_i(k)$, and let $D$ be the union of their complementary divisors. If $n$ exceeds the degree of primitivity of $D$, is every $k$-morphism $\prod_iX_i\to U$ that is trivial whenever one coordinate is fixed at $x_i$ the constant morphism $1$?

来源字段：https://doi.org/10.4171/owr/2021/20

02055 | OWR-12697689-008

### Higher Structure Map Criterion for LICCI Ideals

P / partially_solved | Oberwolfach 报告问题汇编

Let $I$ be a perfect codimension-$3$ ideal over a local Gorenstein ring $R$. Is $I$ LICCI if and only if a higher structure map in $W(d_1)$ has a unit entry; and do unit entries in $W(d_2)$ and $W(d_3)$ characterize linkage to ideals with smaller resolution formats?

来源字段：https://doi.org/10.4171/owr/2023/6

02056 | OWR-12697690-008

### Higher-Auslander Koszul Duals of Auslander Algebras

O / open | Oberwolfach 报告问题汇编

For which representation-finite algebras is the Koszul dual of their Auslander algebra a higher Auslander algebra?

来源字段：https://doi.org/10.4171/owr/2023/7

02057 | OWR-14299523-008

### Higher-Depth Mock Modular Forms and Polyharmonic Maass Forms

O / open | Oberwolfach 报告问题汇编

Relate higher-depth mixed mock modular forms and iterated integrals to polyharmonic weak Maass forms. Use this connection to explain the passage from finite-length representation theory for the modular group to a well-behaved representation theory for the ambient real Lie group. Determine first whether iterated Eisenstein integrals correspond directly to polyharmonic Eisenstein series or require cusp-form or weak-form corrections.

来源字段：https://doi.org/10.4171/owr/2025/38

02058 | OWR-14298590-002

### Higher-Dimensional Categorical Nerves

O / open | Oberwolfach 报告问题汇编

How can categorical nerve constructions and their universal properties be extended to higher dimensions?

来源字段：https://doi.org/10.4171/owr/2024/39

02059 | OWR-14298807-002

### Higher-Level Generalization of Conformal-Block Operators

O / open | Oberwolfach 报告问题汇编

Do the conformal-block operators constructed at level $\ell=3$ admit corresponding generalizations at every level $\ell>3$?

来源字段：https://doi.org/10.4171/owr/2024/53

02060 | OWR-9790358-008

### Higher-Order Vanishing Ideals in Polynomial Interpolation

O / open | Oberwolfach 报告问题汇编

Given a finite point set $X$, determine the least degree of a nonzero polynomial in the symbolic power $I_X^{(m)}$, determine the dimension of each graded component $(I_X^{(m)})_\delta$, and characterize when polynomials vanishing to order $m$ on $X$ factor into products of polynomials vanishing on $X$.

来源字段：https://doi.org/10.4171/owr/2022/7

02061 | OWR-4340-008

### Hilbert Polynomials of Homogeneous Ideals

O / open | Oberwolfach 报告问题汇编

Does there exist a homogeneous ideal $I$ in a polynomial ring $S$ whose quotient has Hilbert polynomial $p_{S/I}(i)=i^2$?

来源字段：https://doi.org/10.4171/owr/2010/27

02062 | HIL-013

### Hilbert's 13th Problem: Seventh Degree Equations

P / partially_solved | Hilbert 问题

Prove that the general equation of the seventh degree cannot be solved using functions of only two variables.

背景链接：https://encyclopediaofmath.org/wiki/Hilbert_problems

02063 | HIL-014

### Hilbert's 14th Problem: Finite Generation of Rings

S / solved | Hilbert 问题

Is the ring of invariants of a linear algebraic group acting on a polynomial ring always finitely generated?

背景链接：https://doi.org/10.2307/2372927

02064 | HIL-017

### Hilbert's 17th Problem: Expression of Definite Forms

S / solved | Hilbert 问题

Can every non-negative rational function be expressed as a sum of squares of rational functions?

背景链接：https://doi.org/10.1007/BF02952513

02065 | OWR-12170-005

### Hilbert-Series Factorization for Finite-Dimensional Nichols Algebras

O / open | Oberwolfach 报告问题汇编

Does every finite-dimensional Nichols algebra have a Hilbert series of the form $H_{\mathfrak B}(t)=(\alpha_1)_t\cdots(\alpha_n)_t(\beta_1)_{t^2}\cdots(\beta_m)_{t^2}$, where $(r)_t=1+t+\cdots+t^{r-1}$?

来源字段：https://doi.org/10.4171/owr/2012/47

02066 | AIM-REPRESENTATION_THEORY-0102

### Hilbert-series obstructions to supercharacter Hopf quotients of Sym

P / partially_solved | AIM 研讨会问题汇编

Problem 1.18. Can the four infinite families of supercharacter theories of S n be realized as isomorphic to a Hopf quotient of S ym?. Can we define multiplication and comultiplication in one of the families to obtain Hopf algebra structure?.

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

02067 | OWR-14298370-005

### Hochschild and Categorical Entropy in dg Dynamics

P / partially_solved | Oberwolfach 报告问题汇编

When does Hochschild (co)homology entropy of a dg categorical dynamical system coincide with the entropy of the corresponding categorical dynamical system?

来源字段：https://doi.org/10.4171/owr/2024/20

02068 | OWR-13750328-004

### Hochschild Cohomology of Symmetric-Group Algebras

O / open | Oberwolfach 报告问题汇编

Prove that $HH^1(\mathbb F_pS_n)$ is a solvable Lie algebra for every $n\geq 1$, and extend the known Hochschild-cohomology generating functions to blocks of alternating groups, finite Coxeter groups, Hecke algebras, and diagram algebras.

来源字段：https://doi.org/10.4171/owr/2023/19

02069 | OWR-14218-002

### Hochschild Regularity Under Separable Equivalence

O / open | Oberwolfach 报告问题汇编

Do separable equivalences between symmetric $k$-algebras preserve the Castelnuovo-Mumford regularity of Hochschild cohomology?

来源字段：https://doi.org/10.4171/owr/2016/10

02070 | OWR-13681-013

### Hochschild-Cohomology Lie Structure of a Stratified Algebra

S / solved | Oberwolfach 报告问题汇编

Let $K$ have characteristic $2$, let $B=\begin{pmatrix}K(\mathbb Z_2\times\mathbb Z_2)&K\mathbb Z_2\\0&K\end{pmatrix}$ with stratifying idempotent $e=\begin{pmatrix}0&0\\0&1\end{pmatrix}$. Determine the precise Lie algebra structure of $\operatorname{HH}^*(B)$, and decide whether the long exact sequence from $0\to BeB\to B\to B/BeB\to0$ determines it.

来源字段：https://doi.org/10.4171/owr/2015/24

02071 | OWR-5149-013

### Homological and Filtration Problems for Symmetric-Group Modules

O / open | Oberwolfach 报告问题汇编

Is the Young dimension of an $S_n$-module $M$ equal to the injective dimension of its associated Schur-algebra module $G(M)$? For $\lambda$ maximal in its block, is the algebra $\operatorname{Ext}_{S_d}^{\bullet}(Y^\lambda, Y^\lambda)$ graded-commutative, and can it act naturally on the modules in that block? Does $\operatorname{Lie}(n)$, and likewise the Foulkes module, admit a Specht or dual Specht filtration?

来源字段：https://doi.org/10.4171/owr/2011/21

02072 | OWR-12983-001

### Homological Determinant of the Nakayama Automorphism

P / partially_solved | Oberwolfach 报告问题汇编

For every Artin-Schelter Gorenstein algebra $A$, is the homological determinant of its Nakayama automorphism equal to $1$; that is, does $\operatorname{hdet}(\mu_A)=1$?

来源字段：https://doi.org/10.4171/owr/2014/25

02073 | OWR-12863-006

### Homology Comparison for Real Spherical Subalgebras

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathfrak h$ be a real spherical subalgebra, let $V$ be a Harish-Chandra module, and let $V^\infty$ be its smooth moderate-growth completion. Is the natural map $$\Phi_p:H_p(\mathfrak h,V)\longrightarrow H_p(\mathfrak h,V^\infty)$$ an isomorphism for every $p$?

来源字段：https://doi.org/10.4171/owr/2014/03

02074 | OWR-15987-026

### Homology of Free-Lie Current Algebras over Square-Zero Extensions

O / open | Oberwolfach 报告问题汇编

Let $E$ and $V$ be finite-dimensional complex vector spaces, let $A=\mathbb C\oplus E$ with $E^2=0$, and let $\mathfrak g=\operatorname{Lie}(V)$ be the free Lie algebra. Determine the Lie algebra homology $H_*(\mathfrak g\otimes A)$, preferably as a sum of polynomial functors in $E$ and $V$.

来源字段：https://doi.org/10.4171/owr/2018/2

02075 | OWR-11583-005

### Homomorphisms between Specht Modules

O / open | Oberwolfach 报告问题汇编

Compute the homomorphism spaces $\operatorname{Hom}_{F S_n}(S^{\lambda},S^{\mu})$ between Specht modules over arbitrary fields, thereby determining the associated decomposition-number problem.

来源字段：https://doi.org/10.4171/owr/2012/16

02076 | OWR-1062-005

### Hopf $C^*$-Algebras from Multiplicative Unitaries

P / partially_solved | Oberwolfach 报告问题汇编

Explain how a multiplicative unitary gives rise to a pair of Hopf $C^*$-algebras.

来源字段：https://doi.org/10.4171/owr/2005/41

02077 | OWR-13495-001

### Hopf Algebra Extensions and Their Conformal Field Theories

O / open | Oberwolfach 报告问题汇编

Classify all extensions $u\longrightarrow H\longrightarrow U(\mathfrak g'')$ of finite-dimensional pointed Hopf algebras $u$ by universal enveloping algebras, and determine the corresponding conformal field theories.

来源字段：https://doi.org/10.4171/owr/2015/10

02078 | OWR-12871-002

### Hopf Graphical Calculus for Deligne-Type Operations

O / open | Oberwolfach 报告问题汇编

What is the analogue, for the Hopf case, of the graphical calculus for operations in Deligne's conjecture?

来源字段：https://doi.org/10.4171/owr/2014/11

02079 | OWR-2048-006

### Horn Spectral Data in Finite $\mathrm{II}_1$ Factors

P / partially_solved | Oberwolfach 报告问题汇编

Let $M$ be a finite $\mathrm{II}_1$ factor with normalized trace $\tau$, and let $a,b\in M$ be self-adjoint with prescribed spectral distributions. Characterize the possible spectral distribution of $a+b$. Equivalently, characterize the spectral data of self-adjoint $a,b,c\in M$ satisfying $a+b+c=0$.

来源字段：https://doi.org/10.4171/owr/2008/37

02080 | OWR-1738-008

### Hypercenter and Prime-Graph Problems for Integral Group Rings

O / open | Oberwolfach 报告问题汇编

Let $U=U(\mathbb{Z}G)$, let $Z_\infty(U)$ be its hypercenter, let $\Delta(U)$ be its finite-conjugacy center, and let $N_U(G)$ be the normalizer of $G$ in $U$. Are $Z_\infty(U)\subseteq\Delta(U)$ and $\Delta(U)\subseteq N_U(G)$? For a finite group $G$, is every simple section of a finite subgroup of $V(\mathbb{Z}G)$ isomorphic to a section of $G$, and does $\Pi(V(\mathbb{Z}G))=\Pi(G)$ hold for the prime graphs?

来源字段：https://doi.org/10.4171/owr/2007/55

02081 | OWR-8415355-015

### Hypercube Decomposition of Kazhdan-Lusztig Polynomial Derivatives

P / partially_solved | Oberwolfach 报告问题汇编

For permutations $x\leq y$ in Bruhat order and the canonical element $c$, does the proposed decomposition $$\partial P_{x,y}=I_{x,y,c}+H_{x,y,c}$$ hold, with the inductive and hypercube-cluster terms determined by the corresponding Bruhat-interval data?

来源字段：https://doi.org/10.4171/owr/2021/52

02082 | OWR-16165-007

### Ideal Theory of Witt and Virasoro Enveloping Algebras

O / open | Oberwolfach 报告问题汇编

For the full Witt algebra $W$ and the Virasoro algebra $V$, do their enveloping algebras have the following properties: every nonzero ideal has a quotient of finite Gelfand-Kirillov dimension, two-sided ideals satisfy the ascending chain condition, and an ideal containing a quadratic expression in the generators has a quotient of finite Gelfand-Kirillov dimension?

来源字段：https://doi.org/10.4171/owr/2018/24

02083 | OWR-10252931-003

### Ideals and Central Reductions of the Virasoro Enveloping Algebra

P / partially_solved | Oberwolfach 报告问题汇编

Determine the containment relations among the Poisson primitive ideals of $S(\operatorname{Vir})$. If $\lambda\neq0$, is $U(\operatorname{Vir})/(z-\lambda)$ simple? Does $U(\operatorname{Vir})$ satisfy the ascending chain condition on two-sided ideals?

来源字段：https://doi.org/10.4171/owr/2022/23

02084 | OWR-1274-022

### Ideals in Iwasawa Algebras from Inner Derivations

O / open | Oberwolfach 报告问题汇编

What information about ideals in Iwasawa algebras can be obtained from the Lie algebra of inner derivations and its induced Poisson-bracket structure on the associated graded algebra?

来源字段：https://doi.org/10.4171/owr/2006/25

02085 | OWR-16165-006

### Ideals in the Positive Witt Algebra

P / partially_solved | Oberwolfach 报告问题汇编

For the positive Witt algebra $W_+$ with $[e_i,e_j]=(j-i)e_{i+j}$, prove or disprove that every nonzero ideal $I$ of $U(W_+)$ gives a quotient of finite Gelfand-Kirillov dimension, and that the two-sided ideals of $U(W_+)$ satisfy the ascending chain condition.

来源字段：https://doi.org/10.4171/owr/2018/24

02086 | AIM-REPRESENTATION_THEORY-0018

### Indecomposable injective FI-modules in characteristic zero

P / partially_solved | AIM 研讨会问题汇编

**Classify torsion free injectives.** What's the injective hull of the trivial \(\mathrm{FI}\)-modules (\(V_n=\mathrm{triv}_n\) for each \(n\))? In general, classify all indecomposable injectives. Is it true that these indecomposable injectives are degree-wise finitely generated (same question over a more general combinatorial category)?

来源字段：http://aimpl.org/repnstability/1/

02087 | OWR-8415354-001

### Indecomposable Representations of Quantum Loop Algebras

O / open | Oberwolfach 报告问题汇编

Classify the finite-dimensional indecomposable representations of quantum loop algebras; equivalently, split their Grothendieck groups.

来源字段：https://doi.org/10.4171/owr/2021/51

02088 | OWR-12697689-007

### Independent Defects and Rough Deep Ramification

O / open | Oberwolfach 报告问题汇编

Is every valued field that admits only independent defects roughly deeply ramified?

来源字段：https://doi.org/10.4171/owr/2023/6

02089 | AIM-REPRESENTATION_THEORY-0099

### Induction from supercharacters to unipotent class functions of GL_n(q)

P / partially_solved | AIM 研讨会问题汇编

Problem 1.15. Is there any relation between supercharacter theory of Un and properties of GLn?. Is there an analogous problem where this has been worked out?.

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

02090 | MOD-009

### Infinite Minimal Field Algebraic Closure

P / partially_solved | 独立 / 未分配集合

Is every infinite minimal field of characteristic zero algebraically closed?

背景链接：https://arxiv.org/abs/1201.5709

02091 | ALG-029

### Infinitude of Leinster Groups

P / partially_solved | 独立 / 未分配集合

Are there infinitely many Leinster groups?

背景链接：https://arxiv.org/abs/math/0104012

02092 | OWR-14299283-002

### Information outside the Thin Affine Grassmannian

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Thin-Affine-Grassmannian Question Continuation [OWR-14299283-006]

What representation-theoretic or geometric information is carried by the components of the fixed-point affine Grassmannian $\operatorname{Gr}^{\varpi}$ lying outside its thin part? Also determine how equivariant and Iwahori-Whittaker Smith-Treumann localization are related and compare the Leslie-Lonergan and Riche-Williamson constructions.

来源字段：https://doi.org/10.4171/owr/2025/25

02093 | OWR-824-007

### Injective Ext Systems for Cofinal Ideal Chains

P / partially_solved | Oberwolfach 报告问题汇编

Let $R$ be a polynomial ring over a field. For which ideals $\mathfrak{a}\subset R$ does there exist a descending chain $\{\mathfrak{a}_t\}_{t\in\mathbb{N}}$, cofinal with $\{\mathfrak{a}^t\}_{t\in\mathbb{N}}$, such that the natural maps $$\operatorname{Ext}_R^i(R/\mathfrak{a}_t, R)\longrightarrow\operatorname{Ext}_R^i(R/\mathfrak{a}_{t+1},R)$$ are injective for every $i$ and $t$?

来源字段：https://doi.org/10.4171/owr/2005/19

02094 | OWR-7155441-011

### Injectivity of the Quantum K-Theoretic Hall Homomorphism

O / open | Oberwolfach 报告问题汇编

Let $(Q, W)$ be the quiver with potential associated to a symmetric quiver (such as the tripled quiver with canonical cubic potential), let $K(Q, W)$ be its critical $K$-theoretic Hall algebra (KHA), and let $U^+$ be the positive half of the corresponding quantum affine algebra. Is the canonical graded algebra homomorphism $\psi : U^+ \to K(Q, W)$ injective?

来源字段：https://doi.org/10.4171/owr/2021/20

02095 | OWR-16773-003

### Inner Automorphisms in the Fourth Antipode Formula

O / open | Oberwolfach 报告问题汇编

Let $H$ be an Artin-Schelter Gorenstein Hopf algebra. Determine the inner automorphism $\gamma$ in the fourth-antipode identity $$S^4=\gamma\circ\tau_{\chi_\ell}\circ\tau_{\chi_r}^{-1}$$.

来源字段：https://doi.org/10.4171/owr/2019/11

02096 | OWR-1272-006

### Inner Automorphisms in the Fourth-Power Antipode Formula

O / open | Oberwolfach 报告问题汇编

For a Noetherian AS-Gorenstein Hopf algebra, determine the inner automorphism $\gamma$ in $S^4=\gamma\circ\tau_\pi^r\circ(\tau_\pi^\ell)^{-1}$.

来源字段：https://doi.org/10.4171/owr/2006/23

02097 | AIM-REPRESENTATION_THEORY-0076

### Inner-form transfer for covering groups: a center-radical and central-sector audit

P / partially_solved | AIM 研讨会问题汇编

Make sense of transferring representation from $\widetilde G_F$ to $\widetilde {G'}_F$, where the two linear groups $G_F$ and $G'_F$ are inner forms of each other.

Possible trouble: centres of $\widetilde G_F$ and $\widetilde {G'}_F$ can be different.

来源字段：http://aimpl.org/autoformcovergp/3/

02098 | OWR-14218-007

### Intrinsic Gerstenhaber Brackets from Projective Resolutions

P / partially_solved | Oberwolfach 报告问题汇编

Given a projective resolution $P\to A\to0$ of $A$ over $A^e$, how can the Gerstenhaber Lie bracket on $\operatorname{HH}^*(A)$ be expressed in terms of $P$, and how can one intrinsically define a bimap on $D(A^e)(A,\Sigma^*A)$ that corresponds to this bracket under $\operatorname{HH}^*(A)\cong D(A^e)(A,\Sigma^*A)$?

来源字段：https://doi.org/10.4171/owr/2016/10

02099 | OWR-14218-005

### Invariant and Smash-Product Hochschild Cohomology

P / partially_solved | Oberwolfach 报告问题汇编

When $\operatorname{char}(k)$ divides $|G|$, how are the graded algebras $\operatorname{HH}^*(A,M)^G$ and $\operatorname{HH}^*(A\# kG,M)$ related?

来源字段：https://doi.org/10.4171/owr/2016/10

02100 | OWR-14299281-011

### Invariant Centers of Small Quantum Groups

O / open | Oberwolfach 报告问题汇编

For a small quantum group of type $A$, is the invariant center $z(u_q(\mathfrak g))^G$ isomorphic to the full center $z(u_q(\mathfrak g))$? More generally, determine the bigraded and multiplicative structure of the center and construct a natural Weyl-group action compatible with its diagonal-coinvariant model.

来源字段：https://doi.org/10.4171/owr/2025/20

02101 | OWR-8415354-005

### Invariant Involutions for Quantum Affine Evaluation Modules

P / partially_solved | Oberwolfach 报告问题汇编

For every quantum affine evaluation module $V(z)$, does there exist an invariant automorphism $\zeta$ of the second kind such that $\zeta_q^*V=V$?

来源字段：https://doi.org/10.4171/owr/2021/51

02102 | OWR-11129-003

### Invariant Ring Structure and Finite Linear Actions

O / open | Oberwolfach 报告问题汇编

How do structural properties of the invariant ring $K[V]^G$, such as being polynomial, a complete intersection, Gorenstein, or Cohen-Macaulay, relate to properties of the finite linear group action of $G$ on $V$?

来源字段：https://doi.org/10.4171/owr/2011/37

02103 | OWR-1790-006

### Invariant Rings of Exceptional-Mutation Actions

P / partially_solved | Oberwolfach 报告问题汇编

Let $R$ be the polynomial ring on the entries of an upper-triangular Cartan matrix, with $\Gamma$ acting through exceptional mutations and allowed sign changes. Do the algebraically independent $\Gamma$-invariant polynomials $F_i$ generate $R^\Gamma$?

来源字段：https://doi.org/10.4171/owr/2008/09

02104 | OWR-12175-028

### Invariant Translation Approximation Property

O / open | Oberwolfach 报告问题汇编

Does every discrete group $\Gamma$ have the invariant translation approximation property, i.e., $C_r^*(\Gamma)=UC_r^*(\Gamma)^\Gamma$?

来源字段：https://doi.org/10.4171/owr/2012/52

02105 | ALG-001

### Inverse Galois Problem

O / open | 独立 / 未分配集合

别名：The Inverse Galois Problem [ALG-027]; Inverse Galois Problem [OPG-3572]

Is every finite group the Galois group of some Galois extension of the rational numbers $\mathbb{Q}$?

背景链接：https://www-fourier.univ-grenoble-alpes.fr/~mbrion/InverseGaloisProblem.pdf

02106 | OWR-17294-002

### Involutivity and Cancellativity of Yang-Baxter Solutions

P / partially_solved | Oberwolfach 报告问题汇编

For a set-theoretic Yang-Baxter solution $(X,r)$ with structure algebra $kM(X,r)$, characterize in ring-theoretic or homological terms when $(X,r)$ is involutive and when the structure monoid $M$ is cancellative.

来源字段：https://doi.org/10.4171/owr/2019/51

02107 | OWR-2654827-005

### Irreducibility of Gorenstein-Algebra Families with Fixed Hilbert Function

P / partially_solved | Oberwolfach 报告问题汇编

Given a Gorenstein sequence $T$, is the family $\operatorname{Gor}(T)$ of local Artinian Gorenstein algebras with Hilbert function $T$ irreducible? If not, how many irreducible components does it have?

来源字段：https://doi.org/10.4171/owr/2020/31

02108 | OWR-1593-007

### Irreducibility of Restricted Classical-Group Tensor Representations

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathcal{G}=\operatorname{Cl}(W)$ be a simple classical group, and let $G\leq\mathcal{G}$ be a Zariski-closed subgroup. If $V$ is the largest composition factor of the $\mathcal{G}$-module $W\otimes W^*$, $\operatorname{Sym}^k(W)$, or $\bigwedge^k(W)$, for small $k$, when is the restriction $V\mathbin{\downarrow}_G$ irreducible?

来源字段：https://doi.org/10.4171/owr/2007/37

02109 | OWR-4426-001

### Irreducible Coherent-Configuration Representations Without Polynomial Splitting

O / open | Oberwolfach 报告问题汇编

For representations of a coherent-configuration matrix algebra, can the desired irreducible action be computed without assuming that a chosen self-adjoint element has a characteristic polynomial splitting into distinct linear factors over the relevant splitting field?

来源字段：https://doi.org/10.4171/owr/2010/38

02110 | OWR-1593-006

### Irreducible Restrictions of Almost-Simple Group Modules

P / partially_solved | Oberwolfach 报告问题汇编

Let $F$ be an algebraically closed field of characteristic $\ell$. Classify all triples $(K,V,H)$ such that $K/Z(K)$ is almost simple, $V$ is an $FK$-module of dimension greater than one, $H<K$ is proper, and the restriction $V\!\downarrow_H$ is irreducible.

来源字段：https://doi.org/10.4171/owr/2007/37

02111 | OWR-4216-011

### Irreducible Specht Modules for Type-A Hecke Algebras

P / partially_solved | Oberwolfach 报告问题汇编

For an Iwahori-Hecke algebra $H_n$ of type $A$ and a field $F$ with parameter $q$, classify the triples $(\lambda,F,q)$ for which the Specht module $S^\lambda$ is irreducible, equivalently the $(e,p)$-irreducible partitions.

来源字段：https://doi.org/10.4171/owr/2010/15

02112 | OWR-1738-006

### Isomorphism and Torsion-Unit Problems for Group Algebras

P / partially_solved | Oberwolfach 报告问题汇编

If $G$ is a finite $p$-group whose group algebra $kG$ has $G$ as a complement factor in its unit group and $kG\cong kH$ as $k$-algebras, must $G\cong H$? For $G=\operatorname{GL}_n(k)$ over a finite field, is the partial augmentation $\varepsilon_{\pi(u)}(u)$ nonzero for every torsion unit $u\in V(\mathbb{Z}G)$? Finally, if the torsion units $u_p$ constructed from central primitive idempotents of localized cyclotomic group algebras are mutually conjugate by units of $\mathbb{C}G$, must one, and hence every, $u_p$ be conjugate to an element of $G$ by a unit of $\mathbb{C}G$?

来源字段：https://doi.org/10.4171/owr/2007/55

02113 | OWR-4084-008

### Isomorphism Data in Correspondence Two-Categories

O / open | Oberwolfach 报告问题汇编

Why is it useful in the correspondence 2-category of $C^*$-algebras to distinguish isomorphic correspondences rather than identify them?

来源字段：https://doi.org/10.4171/owr/2009/41

02114 | OWR-1315-002

### Isomorphism of Trace Forms for G-Galois Algebras

P / partially_solved | Oberwolfach 报告问题汇编

For two $G$-Galois algebras $L$ and $L'$ over a field $k$ of characteristic not $2$, when are their trace $G$-forms $q_L$ and $q_{L'}$ isomorphic? More weakly, for which quadratic forms $\varphi$ over $k$ are $\varphi\otimes q_L$ and $\varphi\otimes q_{L'}$ isomorphic?

来源字段：https://doi.org/10.4171/owr/2006/29

02115 | AMR-124-0001

### Iwahori-Hecke Algebras - Problem 1

O / open | AMR 问题清单汇编

Let $I$ be a finite set and let $(m_{ij})_{(i,j)\in I\times I}$ be a symmetric matrix whose diagonal entries are $1$ and whose nondiagonal entries are integers $\ge2$ or $\infty$. Let $W$ be the group with generators $\{s_i;i\in I\}$ and relations $(s_is_j)^{m_{ij}}=1$ for any $i,j$ such that $m_{ij}<\infty$; this is a Coxeter group. (Examples of Coxeter groups are the Weyl groups of simple Lie algebras; these are finite groups. Other examples are the affine Weyl groups which are almost finite.) For $w\in W$ let $|w|$ be the smallest integer $n\ge0$ such that $w$ is a product of $n$ generators $s_i,i\in I$. We assume that we are given a weight function $L:W@>>>\mathbb{N}$ that is, a function such that $L(w)>0$ for all $w\in W-\{1\}$ and $L(ww')=L(w)+L(w')$ for any $w,w'$ in $W$ such that $|ww'|=|w|+|w'|$. (For example, $w\mapsto|w|$ is a weight function.) Let $A=\mathbb{Z}[v,v - ]$ where $v$ is an indeterminate. Let $H$ be the free $A$-module with basis $\{T_w;w\in W\}$. There is a unique structure of associative $A$-algebra on $H$ for which $(T_{s_i}+v^{-L(s_i)})(T_{s_i}-v^{L(s_i)})=0$ for $i\in I$ and $T_wT_{w'}=T_{ww'}$ for any $w,w'$ in $W$ such that $|ww'|=|w|+|w'|$; this is the Iwahori-Hecke algebra associated to $W,L$.

For $c\in\mathbb{C}-\{0\}$ let $H_c=\mathbb{C}\otimes AH$ where $\mathbb{C}$ is viewed as an $A$-algebra via the ring homomorphism $A@>>>\mathbb{C},v\mapsto c$. Now $H_c$ is also referred to as an Iwahori-Hecke algebra.

(a) {\it Show that the algebras associated in [S20] to a supercuspidal representation of a parabolic subgroup of a $p$-adic reductive group are (up to extension by a group algebra of a small finite group) of the form $H_q$ where $q$ is a power of $p$, with $H$ associated to an affine Weyl group $W$ and with $L$ in the collection $\Sigma_W$ of weight functions on $W$ described in [L91, \mathbb{S]17}, [L95], [L02].} \newline For example, if $W$ is of affine type $F_4$, then $\Sigma_W$ consists of all $L$ whose values on $\{s_i;i\in I\}$ are $(1,1,1,1,1)$ or $(1,1,1,2,2)$ or $(2,2,2,1,1)$ or $(1,1,1,4,4)$; if $W$ is of affine type $G_2$, then $\Sigma_W$ consists of all $L$ whose values on $\{s_i;i\in I\}$ are $(1,1,1)$ or $(1,1,3)$ or $(3,3,1)$ or $(1,1,9)$.

The statement analogous to (a) for groups with connected centre over a finite field $F_q$ instead of $p$-adic groups is known to hold, without the words in parenthesis; in that case, $W$ is a Weyl group and $\Sigma_W$ consists of the weight functions on $W$ described in [L78, p.35].

背景链接：https://arxiv.org/abs/2006.08535

02116 | AMR-124-0002

### Iwahori-Hecke Algebras - Problem 2

O / open | AMR 问题清单汇编

There is a unique group homomorphism $\bar{}:H@>>>H$ such that $\overline{v^nT_w}T_{w\i}=v^{-n}$ for $n\in\mathbb{Z},w\in W$; it is a ring isomorphism. Let $H_{\le0}=\sum_{w\in W}\mathbb{Z}[v - ]T_w\subset H$. For any $w\in W$ there is a unique element $c_w\in H_{\le0}$ such that $\bar c_w=c_w$ and $c_w-T_w\in v - H_{\le0}$ (see [KL],[L03]). Then $\{c_w;w\in W\}$ is an $A$-basis of $H$. For $x,y,z$ in $W$ we define $f_{x,y,z}\in A$, $h_{x,y,z}\in A$ by $T_xT_y=\sum_{z\in W}f_{x,y,z}T_z$, $c_xc_y=\sum_{z\in W}h_{x,y,z}c_z$.

(b) {\it Show that there exists an integer $N\ge0$ such that for any $x,y,z$ in $W$ we have $v^{-N}f_{x,y,z}\in\mathbb{Z}[v - ]$.} \newline (See [L03, 13.4].) If $W$ is finite this is obvious. If $W$ is an affine Weyl group, this is known.

We will now assume that (b) holds. With $N$ as in (b), we see that $v^{-N}h_{x,y,z}\in\mathbb{Z}[v - ]$ for any $x,y,z$ in $W$. It follows that for any $z\in W$ there is a unique integer $a(z)\ge0$ such that $h_{x,y,z}\in v^{a(z)}\mathbb{Z}[v - ]$ for all $x,y$ in $W$ and $h_{x,y,z}\notin v^{a(z)-1}\mathbb{Z}[v - ]$ for some $x,y$ in $W$. Hence for $x,y,z$ in $W$ there is a well defined integer $\gamma_{x,y,z\i}$ such that $h_{x,y,z}=\gamma_{x,y,z\i}v^{a(z)}\mod v^{a(z)-1}\mathbb{Z}[v - ]$. Let $J$ be the free abelian group with basis $\{t_w;w\in W\}$. For $x,y$ in $W$ we set $t_xt_y=\sum_{z\in W}\gamma_{x,y,z\i}t_z$. (This is a finite sum.)

(c) *Show that this defines an (associative) ring structure on $J$ (without $1$ in general).* \newline Assume now that $W$ is a Weyl group or an affine Weyl group and $L=|$. In this case, (c) is known to be true and the ring $J$ does have a unit element.

More generally, assume that $W$ is an affine Weyl group and $L\in\Sigma_W$ (see (a)); in this case there is a (conjectural) geometric description [L16, 3.11] of the elements $c_w$. From this one should be able to deduce (c) as well as the well-definedness of the $\mathbb{C}$-algebra homomorphism $H_q@>>>\underline J$ in [L03, 18.9], where $H_q$ is as in (a) and $\underline J=\mathbb{C}\otimes J$ is independent of $q$. One should expect that the irreducible (finite dimensional) $\underline J$-modules, when viewed as $H_q$-modules, form a basis of the Grothendieck group of $H_q$-modules. (This is indeed so if $L=|$.) This should provide a construction of the ``standard modules'' of $H_q$ which, unlike the construction in [L95],[L02], does not involve the geometry of the dual group.

背景链接：https://arxiv.org/abs/2006.08535

02117 | AMR-124-0003

### Iwahori-Hecke Algebras - Problem 3

O / open | AMR 问题清单汇编

Assume that $W$ is finite and that $L=||$. Let $C$ be a conjugacy class in $W$; let $C_{min}$ be the set of all $w\in C$ such that $|w|$ is minimal. For $w\in C$ let $N^w\in A$ be the trace of the $A$-linear map $H@>>>H$, $h\mapsto v^{2|w|}T_whT_{w\i}$. We have $N^w\in\mathbb{Z}[v^2]$. (Note that $N^w|_{v=1}$ is the order of the centralizer of $w$ in $W$.) From [GP] one can deduce that for $w\in C_{min}$, $N^w$ depends only on $C$, not on $w$. We say that $C$ is *positive* if $N^w\in\mathbb{N}[v^2]$. For example, if $C$ is an elliptic regular conjugacy classes (in the sense of [S74]) then $C$ is positive (see [L18]). If $W$ is of type $A_n$, the positive conjugacy classes are $\{1\}$ and the class of the Coxeter element. In the case where $W$ is a Weyl group of exceptional type a complete list of positive conjugacy classes in $W$ is given in [L18].

(d) *Make a list of all positive conjugacy classes in $W$ assuming that $W$ is a Weyl group of type $B_n$ or $D_n$.*

\widestnumber\key{L78}

背景链接：https://arxiv.org/abs/2006.08535

02118 | OWR-17466-010

### Jacobi Algebras as Contraction Algebras

O / open | Oberwolfach 报告问题汇编

Does every finite-dimensional Jacobi algebra arise as the contraction algebra of a flopping curve?

来源字段：https://doi.org/10.4171/owr/2020/3

02119 | ALG-006

### Jacobson's Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Problem 6.2 - Is the Burau representation of B4 faithful? [AMR-103-0110]

For a left-and-right Noetherian ring $R$, is the intersection of all powers of the Jacobson radical $J(R)$ equal to zero?

背景链接：https://doi.org/10.1016/0021-8693(74

02120 | OWR-4086-008

### Jet-Closure Equality in Regular Local Algebras

P / partially_solved | Oberwolfach 报告问题汇编

For every ideal $\mathfrak a$ in a noetherian regular local $k$-algebra $A$, does the associated closure $\mathfrak a^\ast:=\bigcap_{m\in\mathbb N}\{f\in A:X_m(0)\subset H_m(0)\}$ equal $\mathfrak a$?

来源字段：https://doi.org/10.4171/owr/2009/43

02121 | OWR-12175-023

### Jiang-Su Embeddings in Character-Free Central Sequence Algebras

O / open | Oberwolfach 报告问题汇编

If $F(A)$ has no characters, must it contain a unital copy of $\mathcal Z$? Does $\mathcal Z$ embed unitally into $\bigotimes_{n=1}^{\infty}D$ for every unital $C^*$-algebra $D$ without characters?

来源字段：https://doi.org/10.4171/owr/2012/52

02122 | OWR-4535-010

### Johnson-Homomorphism Images as General Linear Modules

P / partially_solved | Oberwolfach 报告问题汇编

Determine the $GL(n,\mathbb Z)$-module structure of the image, equivalently the cokernel, of the Johnson homomorphism $\tau_{k,\mathbb Q}$.

来源字段：https://doi.org/10.4171/owr/2010/53

02123 | AIM-REPRESENTATION_THEORY-0009

### Joint CM packets are joinings, not products of marginals

P / partially_solved | AIM 研讨会问题汇编

**Joint equidistribution of CM points.** Determine the joint equidistribution of CM points over \(K\) of products of modular curves over \(F\) as \(\operatorname{disc}(K)\to\infty\).

来源字段：http://aimpl.org/aagaautomorphic/4/

02124 | OWR-14298013-002

### Jordan Brackets and Simple Transposed Poisson Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Characterize the Jordan brackets that arise from transposed Poisson algebras and determine whether their associated Jordan algebras are special. Also determine whether a finite-dimensional simple transposed Poisson algebra can have both multiplications simple.

来源字段：https://doi.org/10.4171/owr/2023/46

02125 | OWR-17466-003

### Jordan-Recoverable Quiver Subcategories

O / open | Oberwolfach 报告问题汇编

Which additive subcategories $\mathcal C$ of the representation category $\operatorname{rep}Q$ of a quiver $Q$ are Jordan recoverable, meaning that a representation can be recovered up to isomorphism from its generic nilpotent Jordan data?

来源字段：https://doi.org/10.4171/owr/2020/3

02126 | OWR-16941-003

### K-Regularity Criterion for Ring Regularity

O / open | Oberwolfach 报告问题汇编

If $A$ is an algebra essentially of finite type over a field $k$, of Krull dimension $d$, and $A$ is $K_{d+1}$-regular, must $A$ be regular?

来源字段：https://doi.org/10.4171/owr/2019/29

02127 | OWR-13682-006

### K-Theoretic Geometric Satake Equivalence

P / partially_solved | Oberwolfach 报告问题汇编

For a semisimple complex group $G$ with Langlands dual $G^\vee$, let $\mathrm{Gr}$ be its affine Grassmannian. Is the convolution category $K\mathrm{Conv}(\mathrm{Gr})$ equivalent to a full subcategory of $U_q(\mathfrak g)$-equivariant $\mathcal O_q(G)$-modules?

来源字段：https://doi.org/10.4171/owr/2015/25

02128 | OWR-14299283-012

### K-Theory Equivalence of Hall and Quasi-BPS Categories

O / open | Oberwolfach 报告问题汇编

Prove that the algebraic or topological $K$-theory of $D\text{-}\operatorname{mod}_{\mathrm{coh}}(\mathcal{X}(d))_{w/d}$ is isomorphic, under the specialization map, to the $K$-theory of the quasi-BPS category $\mathbb{T}(d)_w$. Establish the corresponding slope-zero Hall-category equivalence.

来源字段：https://doi.org/10.4171/owr/2025/25

02129 | OWR-4800-003

### Kac Polynomials and Character-Variety Cohomology

P / partially_solved | Oberwolfach 报告问题汇编

For the associated character variety $M_\mu$, is $P_c(M_\mu;t)=t^{2d_\mu}A_\Gamma(\mu,t^2)$, and consequently is $A_\Gamma(\mu,q)\in\mathbb N[q]$?

来源字段：https://doi.org/10.4171/owr/2011/10

02130 | OWR-12175-005

### Kadison-Kastler Conjecture for Separable $C^*$-Algebras

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate: Kadison-Kastler Conjecture for $C^*$-Algebras [OWR-12175-006]

For a separable Hilbert space $H$, does there exist $\delta>0$ such that every pair of separable $C^*$-algebras $A,B\subset B(H)$ with $d(A,B)<\delta$ are spatially isomorphic?

来源字段：https://doi.org/10.4171/owr/2012/52

02131 | ALG-002

### Kaplansky's Conjectures

O / open | 独立 / 未分配集合

A set of conjectures about group rings: (1) Zero divisor conjecture: If $G$ is a torsion-free group and $K$ is a field, then $K[G]$ has no zero divisors. (2) Idempotent conjecture: The only idempotents in $K[G]$ are 0 and 1. (3) Unit conjecture: The only units in $\mathbb{Z}[G]$ are of the form $\pm g$ for $g \in G$.

背景链接：https://annals.math.princeton.edu/2021/194-3/p09

02132 | OWR-14213-011

### Kato Cohomology Kernels of Quadratic Pfister Forms

O / open | Oberwolfach 报告问题汇编

Let $Q=\langle 1,a_1\rangle\otimes\cdots\otimes\langle 1,a_m\rangle\otimes[1,b]$ be an anisotropic quadratic Pfister form over $F$. Is the Kato-cohomology kernel $H_2^{n+1}(F(Q)/F)$ equal to $b\,\frac{d a_1}{a_1}\wedge\cdots\wedge\frac{d a_m}{a_m}\wedge\nu_F(n-m)$ when $n\geq m$, and zero when $n<m$?

来源字段：https://doi.org/10.4171/owr/2016/5

02133 | OWR-17294-012

### Kegel-Wielandt Theory for Skew Braces

P / partially_solved | Oberwolfach 报告问题汇编

For skew left braces, is the sum of a trivial skew left brace and a right-nilpotent skew left brace right-nilpotent? Can the Kegel-Wielandt theorem be adapted to skew left braces? Can Huppert's result that finite sums of finite cyclic groups are supersolvable be adapted to skew left braces, and what are the corresponding notions' impacts on Yang-Baxter solutions?

来源字段：https://doi.org/10.4171/owr/2019/51

02134 | MOD-010

### Keisler's Order

P / partially_solved | 独立 / 未分配集合

Determine the structure of Keisler's order on first-order theories.

背景链接：https://arxiv.org/abs/1503.08341

02135 | OWR-1452-028

### Kernels of Completed Locally Nilpotent Derivations

O / open | Oberwolfach 报告问题汇编

Let $B$ be a factorial affine domain over $\mathbb{C}$, let $\delta$ be a locally nilpotent derivation of $B$, set $A=\ker(\delta)$, let $M$ be a maximal ideal of $B$, and set $m=M\cap A$. If $\delta$ extends to a derivation $\widehat{\delta}_M$ of the completion $\widehat{B}_M$ and $\ker(\delta_M)=A_m$ for the corresponding local derivation, is $\ker(\widehat{\delta}_M)=\widehat{A}_m$?

来源字段：https://doi.org/10.4171/owr/2007/01

02136 | AIM-REPRESENTATION_THEORY-0104

### Kisin's framed Breuil-Mezard multiplicity conjecture

S / solved | AIM 研讨会问题汇编

**Conjecture 2 (Kisin), recovered.** The Hilbert--Samuel multiplicity of \[ R^{\square,\psi}(k,\tau,\bar\rho)/(\varpi) \] is equal to \(\mu_{\mathrm{Aut}}(k,\tau,\bar\rho)\).

Colmez's functor and an expectation.' The actual record ends there. It is not part of Conjecture 2 and is not treated as an assigned problem here.

The source first fixes an inertial Galois type \(\tau:I_{\mathbf Q_p}\to\mathrm{GL}_2(E)\), a residual representation \(\bar\rho\), a determinant datum \(\psi\), and the potentially semistable framed deformation quotient. Its statement is:

Here \(k\geq 2\) specifies Hodge--Tate weights \((0,k-1)\), \(\varpi\) is a uniformizer of the coefficient ring, and \[ \mu_{\mathrm{Aut}}(k,\tau,\bar\rho) =\sum_{0\leq n\leq p-1\atop 0\leq m\leq p-2} a(n,m)\,\mu_{n,m}(\bar\rho), \] where \(a(n,m)\) is the multiplicity of \(\sigma_{n,m}=\operatorname{Sym}^n\mathbb F^2\otimes\det^m\) in the semisimplified reduction of a stable lattice \(L_{k,\tau}\).

- the control character U+0003 after \(R\) is the lost framed symbol \(\square\); - the final argument is the residual representation \(\bar\rho\), whose bar is lost in the extraction; - the source's \(\pi\) is written \(\varpi\) to avoid confusion with a representation; - \(\mu_{\mathrm{Aut}}\) is restored with its arguments; - the source PDF itself displays the framed symbol imperfectly in text extraction, but its definition as the universal **framed** deformation ring makes the reconstruction unambiguous.

The PDF also warns that these notes were typeset on the fly and may contain transcription errors. This recovery uses only the paragraph ending immediately before section 1.2.

来源字段：https://aimath.org/WWN/padicmodularity/padicmodularity.pdf

02137 | OWR-16171-014

### Kontsevich's Period Conjecture

O / open | Oberwolfach 报告问题汇编

Kontsevich's period conjecture asserts that the evaluation map from the algebra of formal periods over $\mathbb{Q}$ to the algebra of actual periods is injective.

来源字段：https://doi.org/10.4171/owr/2018/30

02138 | OWR-14221-007

### Kostka-Shoji Formula for Graded Modules

O / open | Oberwolfach 报告问题汇编

Does the conjectural Kostka-Shoji formula yield the graded $GL_N$-module identity $\Gamma(T_r^*B_N^r,O(\mu))=\Gamma(B_N^r,\operatorname{Sym}^{\bullet}T_rB_N^r\otimes O(\mu))=\bigoplus_{\lambda\geq\mu}K_{\lambda\mu}^{-,\bullet}\otimes V_\lambda$?

来源字段：https://doi.org/10.4171/owr/2016/13

02139 | OWR-12697689-014

### Koszulness from the Kähler Package

O / open | Oberwolfach 报告问题汇编

Is every Artinian quadratic Gorenstein algebra with the Kähler package Koszul?

来源字段：https://doi.org/10.4171/owr/2023/6

02140 | OWR-14218-004

### Koszulness from Vanishing Higher Koszul Homology

O / open | Oberwolfach 报告问题汇编

For a quadratic algebra $A$, is $A$ Koszul if and only if its higher Koszul homology vanishes in positive degrees?

来源字段：https://doi.org/10.4171/owr/2016/10

02141 | OWR-14299283-009

### Lagrangian p-Supports of Holonomic Quantization Modules

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a conical symplectic singularity, let $A_\lambda$ be a quantization, and let $M$ be a holonomic finite $A_\lambda$-module. Prove that for all sufficiently large primes $p$, the support of the reduction $M_p$ over the $p$-center is Lagrangian.

来源字段：https://doi.org/10.4171/owr/2025/25

02142 | OWR-1452-020

### Lagrangian Varieties from Weyl-Algebra Annihilators

O / open | Oberwolfach 报告问题汇编

For the algebraic subvariety of $\operatorname{Spec}Z(W_n)$ defined by the central annihilator arising after reduction in the Weyl-algebra construction, is it Lagrangian? Conversely, can every Lagrangian variety arise in this way?

来源字段：https://doi.org/10.4171/owr/2007/01

02143 | OWR-13943-004

### Lamination Realization of Higher Tropical Friezes

O / open | Oberwolfach 报告问题汇编

Can every higher tropical frieze pattern be realized by an A-lamination?

来源字段：https://doi.org/10.4171/owr/2015/48

02144 | OWR-8415354-002

### Langlands-Dual Parametrization of Non-Simply-Laced Truncation Modules

P / partially_solved | Oberwolfach 报告问题汇编

Are simple modules of a non-simply-laced truncation parametrized by data from the Langlands-dual quantum affine Lie algebra?

来源字段：https://doi.org/10.4171/owr/2021/51

02145 | OWR-7155441-001

### Langlands-Dual Parametrization of Shifted Quantum Affine Modules

P / partially_solved | Oberwolfach 报告问题汇编

Are the simple modules of a truncated shifted quantum affine algebra of non-simply-laced type parametrized by monomials in the Langlands-dual q-character of a finite-dimensional representation of the Langlands-dual quantum affine algebra?

来源字段：https://doi.org/10.4171/owr/2021/20

02146 | OWR-12481-010

### Large Nonnegative Rank at Bounded Ordinary Rank

P / partially_solved | Oberwolfach 报告问题汇编

Do there exist nonnegative $n\times n$ matrices of bounded ordinary rank whose nonnegative rank is $\Omega(n)$? In particular, do matrices of rank $3$ with nonnegative rank $n$ exist?

来源字段：https://doi.org/10.4171/owr/2013/18

02147 | OWR-15178-021

### Large Regularity Quadratic Ideals with Linear First Syzygies

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a family of quadratic ideals $I_m\subset K[x_1,\ldots,x_m]$ with linearly generated first syzygies and $\lim_{m\to\infty}\operatorname{reg}(S/I_m)/m>0$? Also, is there a constant $\lambda$ such that $\operatorname{reg}K[\Delta]\leq\lambda$ whenever $K[\Delta]$ is Cohen-Macaulay and satisfies $N_2$?

来源字段：https://doi.org/10.4171/owr/2016/57

02148 | GREEN-072

### Large Subsets of Approximate Groups

O / open | Ben Green 问题

Suppose $A$ is a $K$-approximate group (not necessarily abelian). Is there $S \subset A$ with $|S| \gg K^{-O(1)}|A|$ and $S^8 \subset A^4$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

02149 | OWR-11782-001

### Leavitt Path Algebra Isomorphisms and Graph $C^*$-Algebras

O / open | Oberwolfach 报告问题汇编

For arbitrary graphs $E$ and $F$, does a ring isomorphism $L_{\mathbb C}(E)\cong L_{\mathbb C}(F)$ imply $C^*(E)\cong C^*(F)$, and conversely?

来源字段：https://doi.org/10.4171/owr/2012/20

02150 | OWR-15178-003

### Lech Multiplicity Conjecture for Flat Local Extensions

O / open | Oberwolfach 报告问题汇编

Lech's conjecture: if $(R,\mathfrak m)\to(S,\mathfrak n)$ is a flat local extension of Noetherian local rings, then $e(R)\leq e(S)$.

来源字段：https://doi.org/10.4171/owr/2016/57

02151 | OWR-13494-016

### Lefschetz Elements for Powers of General Linear Forms

O / open | Oberwolfach 报告问题汇编

For the Cohen-Macaulay codimension-two algebras arising from tangent-space ideals to the variety $X_{n,\lambda}$ of reducible degree-$d$ forms of factorization type $\lambda$, do the Lefschetz elements needed to determine improper intersections always exist?

来源字段：https://doi.org/10.4171/owr/2015/9

02152 | OWR-13490-011

### Lefschetz Injectivity for Doubly Cohen-Macaulay Complexes

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Lefschetz-Injectivity Extraction [OWR-13490-010]

Does the Stanley-Reisner ring $R$ of every doubly Cohen-Macaulay simplicial complex admit a linear system of parameters $\Theta$ and a linear form $\omega$ such that $\omega^{s-2i}:(R/\Theta R)_i\to(R/\Theta R)_{s-i}$ is injective for every $i\leq s/2$, where $s$ is the rank of the complex?

来源字段：https://doi.org/10.4171/owr/2015/5

02153 | OWR-16933-001

### Lefschetz Properties of PL Sphere Face Rings

O / open | Oberwolfach 报告问题汇编

Let $\Sigma$ be a piecewise-linear $(d-1)$-sphere with graded face ring $R[\Sigma]$. Are there open dense sets of Artinian reductions $A(\Sigma)$ and degree-one elements $\ell$ such that, for every $k\leq d/2$, multiplication by $\ell^{d-2k}$ gives an isomorphism $A^k(\Sigma)\to A^{d-k}(\Sigma)$ and the form $Q_{\ell,k}(a,b)=\deg(ab\ell^{d-2k})$ is nondegenerate on every squarefree monomial ideal and its annihilator?

来源字段：https://doi.org/10.4171/owr/2019/21

02154 | OWR-14298590-006

### Left-Exact Characterization of Enriched Topoi

O / open | Oberwolfach 报告问题汇编

Is a presentable enriched category $E$ a $V$-topos if and only if the colimit functor $\mathcal{P}(E)\to E$ is left exact?

来源字段：https://doi.org/10.4171/owr/2024/39

02155 | OWR-1790-008

### Levels and Thick Subcategories in Triangulated Categories

O / open | Oberwolfach 报告问题汇编

Given a class $\mathcal C$ of objects in a triangulated category $\mathcal T$ and an object $M$, how can one determine the level of $M$ with respect to $\mathcal C$? More modestly, how can one decide whether that level is finite, equivalently whether $M$ belongs to the thick subcategory generated by $\mathcal C$?

来源字段：https://doi.org/10.4171/owr/2008/09

02156 | OWR-8415347-016

### Lévy-Khintchine Decomposition for Star-Bialgebras

P / partially_solved | Oberwolfach 报告问题汇编

Characterize the $*$-bialgebras with the Lévy-Khintchine property (LK): every generating functional decomposes into a Gaussian part and a purely non-Gaussian part.

来源字段：https://doi.org/10.4171/owr/2021/44

02157 | OWR-14299281-004

### Lie Algebras and Reconstruction from Variety Automorphisms

P / partially_solved | Oberwolfach 报告问题汇编

For an algebraic variety $X$, does $operatorname{Aut}(X)$ admit a natural Lie algebra that embeds into the algebraic vector fields on $X$? More generally, to what extent is $X$ determined by $operatorname{Aut}(X)$?

来源字段：https://doi.org/10.4171/owr/2025/20

02158 | OWR-1452-007

### Lifting Polynomial Automorphisms from Dual Numbers

O / open | Oberwolfach 报告问题汇编

Let $d\geq3$ and let $R$ be a reduced ring containing $\mathbb{Q}$; put $Y=(y_1,\ldots,y_{d-1})$. Does there exist $\bar\gamma\in\operatorname{Aut}_x(R[t]/(t^2)[x,Y])\cap\operatorname{Aut}_{R[t]}(R[t][x,Y])$ with $\bar\gamma\notin\operatorname{Aut}_x(R[t][x,Y])$?

来源字段：https://doi.org/10.4171/owr/2007/01

02159 | OWR-13494-006

### Linear Computability Under Addition of Forms

P / partially_solved | Oberwolfach 报告问题汇编

Is the sum of two linearly computable forms always linearly computable?

来源字段：https://doi.org/10.4171/owr/2015/9

02160 | OWR-12732-007

### Linear Independence of Cluster Monomials

P / partially_solved | Oberwolfach 报告问题汇编

For a cluster algebra $\mathcal A$ over a coefficient semifield $\mathbb P$, are the cluster monomials linearly independent over $\mathbb Z\mathbb P$?

来源字段：https://doi.org/10.4171/owr/2013/58

02161 | AIM-LINEAR_ALGEBRA-0002

### Linear Jordan-block bounds and a counterexample to the literal nonbacktracking Alon--Boppana claim

P / partially_solved | AIM 研讨会问题汇编

The exact statement is:

The AIM page does **not** specify four material conventions.

1. **Simple versus multi-graph.** The later papers used below explicitly work with finite simple undirected graphs. All proved claims and the counterexample in this report use that convention. 2. **Connectedness.** The page says “any graph” and imposes no connectedness condition. This omission is decisive for part (2). 3. **No leaves.** Literally, a leaf is a degree-one vertex, so “no leaves” permits isolates. Heysse, Lorenzen, and Reinhart (2025) instead describe the intended defectiveness problem as one for graphs of minimum degree at least two. Results below distinguish these readings. 4. **Second modulus and the error term.** For the nonnormal matrix $B_G$, define $$ |\lambda|_1(B_G)\ge |\lambda|_2(B_G)\ge\cdots $$ by listing complex eigenvalues with algebraic multiplicity and sorting their moduli. The counterexample below has a unique eigenvalue outside a fixed circle, so it also works if the second *distinct* modulus is intended. The expression $o_n(1)$ is interpreted in the only nonvacuous all-graph sense: a single error sequence $\varepsilon_n\to0$ must work uniformly for every graph in the stated $n$-vertex class.

来源字段：http://aimpl.org/spectralhypergraph/1/

02162 | AMR-130-0001

### Linear Systems and Eigenvalue Problems - Problem 1

O / open | AMR 问题清单汇编

We propose focusing the benchmark problems on linear systems that are discretizations of partial differential equations (PDEs) due to both their wide application in science and engineering and to their connection to graph Laplacians that have been a focus of TCS research in linear system solvers. Furthermore, matrix collections such as the SuiteSparse collection [DavisHu2011] provide a set of matrices, but they do not provide a pipeline that defines parametrized discretizations of model problems from which a large set of such sample matrices could be generated. Parameterized classes of problems are important to understand the range of characteristics and behaviors that a solver might encounter for a class of problem.

We therefore propose the following two problems.

Develop software to generate benchmark parameterized linear systems that are generally accepted as relevant in practice and suitable for theoretical analysis.

背景链接：https://arxiv.org/abs/2602.05394

02163 | AMR-130-0010

### Linear Systems and Eigenvalue Problems - Problem 10

O / open | AMR 问题清单汇编

What is the bit complexity of solving an $n\times n$ linear system with $\text{nnz}(A)$ non-zeros? In particular, does the $\tilde O(n\cdot \text{nnz}(A))$ runtime extend to the Word-RAM model.

背景链接：https://arxiv.org/abs/2602.05394

02164 | AMR-130-0011

### Linear Systems and Eigenvalue Problems - Problem 11

O / open | AMR 问题清单汇编

Another reason for focusing on condition number reduction rather than ``improving'' the entire eigenvalue distribution by preconditioning is that the eigenvalue distribution is usually unknown in practical applications. Yet, the recent line of work in [GerMadNieStr19,GerNieStr20,GerNieStr22,NieStr24] has shown that for certain elliptic PDE model problems, for example $-\nabla\cdot (k(x)\nabla u)=f$ with $k(x)$ being a measurable, bounded, and uniformly positive function from ${\mathbb{R}}^d$ to ${\mathbb{R}}$, one can obtain the exact localization of the eigenvalues of the discretized operators (i.e., the matrices), and of the left-preconditioned matrices under Laplacian preconditioning. This approach yields new insights into the convergence of CG for the preconditioned systems.

Several specific open problems that exist in this context are mentioned in [NieStr24, Section 9]. More generally, try to extend the approach in [GerMadNieStr19,GerNieStr20,GerNieStr22,NieStr24] to other types of PDEs or preconditioners.

背景链接：https://arxiv.org/abs/2602.05394

02165 | AMR-130-0012

### Linear Systems and Eigenvalue Problems - Problem 12

O / open | AMR 问题清单汇编

The behavior of iterative methods such as GMRES for solving nonsymmetric (more precisely, nonnormal) linear algebraic systems is much less understood than the behavior of CG. In general, the performance in this case may depend not only on the eigenvalue distribution, but also on further quantities including the conditioning of the eigenvectors of $A$, which motivates considering $\epsilon$-pseudospectra or the field of values of $A$ in the convergence analysis; see, e.g., [LieStr13, Section 5.7] for a survey of results. Despite numerous existing approaches, the following problem is largely open (cf. also [Emb22]):

Develop a descriptive convergence bound for GMRES applied to nonnormal linear algebraic systems, that particularly describes the (transient) behavior early in the iteration.

背景链接：https://arxiv.org/abs/2602.05394

02166 | AMR-130-0013

### Linear Systems and Eigenvalue Problems - Problem 13

O / open | AMR 问题清单汇编

Find an interesting (nonselfdajoint resp. nonsymmetric) model problem, where the theoretical result on the eigenvalue distribution of the preconditioned matrix indeed corresponds to the actual convergence behavior of GMRES (e.g., $M^{-1}A$ has $m$ eigenvalue clusters and an acceptable approximate solution is obtained by GMRES in ${\mathcal{O}}(m)$ steps). Determine which particular properties of the problem implied this correspondence.

背景链接：https://arxiv.org/abs/2602.05394

02167 | AMR-130-0014

### Linear Systems and Eigenvalue Problems - Problem 14

O / open | AMR 问题清单汇编

Explain the convergence acceleration of GMRES after the initial (transient) phase that is observed in some convection-diffusion model problems such as the one in [LieStr05]. In particular, explain whether or not the speed of convergence can be related to eigeninformation (i.e., eigenvalue location and eigenvector conditioning) of the discretized convection-diffusion operators.

背景链接：https://arxiv.org/abs/2602.05394

02168 | AMR-130-0015

### Linear Systems and Eigenvalue Problems - Problem 15

O / open | AMR 问题清单汇编

A number of methods for solving linear systems (with or without a preconditioner) can be implemented in the following form: Starting with an initial guess $x_0$ for the solution to $Ax=b$, for $k=1,2, \ldots$, set \begin{eqnarray*} x_k & = & x_{k-1} + a_{k-1} p_{k-1} \\ r_k & = & r_{k-1} - a_{k-1} A p_{k-1} . \end{eqnarray*} The updated vectors $r_k$ are not the true residuals $b-A x_k$ when these formulas are implemented in finite precision arithmetic. It is pointed out in [Greenbaum97] that these vectors often shrink well below the machine precision, and this is used to prove results about the size of the true residuals. But this is not proved.

Taking the above recurrences to represent the CG or steepest descent algorithm, determine conditions on $A$ that will insure that the norms of the vectors $r_k$ drop below the machine precision in finite precision arithmetic or show some examples where they do not.

背景链接：https://arxiv.org/abs/2602.05394

02169 | AMR-130-0016

### Linear Systems and Eigenvalue Problems - Problem 16

O / open | AMR 问题清单汇编

Show that the CG and Lanczos algorithms do or do not behave similarly in finite precision arithmetic when used to solve a symmetric positive definite linear system. This will, of course, depend on the implementation of each method, so use standard good implementations of both.

背景链接：https://arxiv.org/abs/2602.05394

02170 | AMR-130-0017

### Linear Systems and Eigenvalue Problems - Problem 17

O / open | AMR 问题清单汇编

How many bits of precision are necessary to guarantee that CG applied to an $n$ by $n$ SPD linear system $Ax=b$ obtains an approximate solution with normwise backward error at most $\epsilon$ in $n$ (or fewer) steps?

背景链接：https://arxiv.org/abs/2602.05394

02171 | AMR-130-0018

### Linear Systems and Eigenvalue Problems - Problem 18

O / open | AMR 问题清单汇编

Analyze the Lanczos algorithm under general models of inexact matrix-vector multiplication and/or the use of a preconditioner. Progress is made if these bounds generalize Paige's analysis and/or provide new, easier to understand, insight into the behavior of the algorithm.

背景链接：https://arxiv.org/abs/2602.05394

02172 | AMR-130-0019

### Linear Systems and Eigenvalue Problems - Problem 19

O / open | AMR 问题清单汇编

Extend Paige's analysis of the Lanczos algorithm [paige_76,paige_80] to the block Lanczos algorithm. The exact analysis will depend on the use of an orthogonalization scheme used within blocks, so either specify certain assumptions on the accuracy of the scheme, or analyze a standard ``good'' algorithm.

背景链接：https://arxiv.org/abs/2602.05394

02173 | AMR-130-0002

### Linear Systems and Eigenvalue Problems - Problem 2

O / open | AMR 问题清单汇编

For each class of parameterized benchmark linear systems, develop theoretically and practically fast algorithms.

背景链接：https://arxiv.org/abs/2602.05394

02174 | AMR-130-0020

### Linear Systems and Eigenvalue Problems - Problem 20 (Deterministic Pseudospectral Shattering)

O / open | AMR 问题清单汇编

Give a provably correct and efficient *deterministic* algorithm for the following task: on input $A\in\mathbb{C}^{n\times n}$ and $\delta>0$, output a perturbation $E$ with $\|E\|\le \delta$ such that $\kappa_{eig}(A+E)\le C(\delta/n)^c$, where $C,c$ are universal constants. In this context, efficient means $O(n^3\log^c(n/\delta))$ arithmetic operations, either in exact arithmetic or, more ambitiously, in finite arithmetic with $O(\log^c(n/\delta))$ bits of precision.

背景链接：https://arxiv.org/abs/2602.05394

02175 | AMR-130-0021

### Linear Systems and Eigenvalue Problems - Problem 21 (Deterministic Minami Perturbation)

O / open | AMR 问题清单汇编

Minami [minami1996local] showed that if $A$ is $n\times n$ Hermitian tridiagonal and $E$ is a random diagonal matrix with independent absolutely continuous entries, then $gap(A+\delta E)\ge C(\delta/n)^c$ for some constants $C,c$ depending on the distribution of the entries. Give an efficient deterministic algorithm for finding such a perturbation. Here, efficient means faster than the parent algorithm in which it is used, so $O(n^3\log^c(n/\delta))$ for diagonalization and, ambitiously, $O(nnz(A)\log^c(n/\delta))$ or $O(nnz(A)\log^c(n)/\delta^c)$ for Krylov methods.

背景链接：https://arxiv.org/abs/2602.05394

02176 | AMR-130-0022

### Linear Systems and Eigenvalue Problems - Problem 22

O / open | AMR 问题清单汇编

We propose the following problem, which will improve the result in [Banks-et-al-23] (for diagonalization) and in [banks2022global] (for the Schur form). The motivation is that the amount of precision required by the best currently known provable algorithms for diagonalization/Schur form is much larger than the gold standard of $\log(1/\delta)+c\log(n)$ bits preferred in numerical analysis (which has been achieved for the Hermitian diagonalization problem [shah2025hermitian]).

Give a provable randomized algorithm for backward error diagonalization or Schur form with runtime $O(n^3\log^c(n/\delta))$ that only uses $O(\log(n/\delta))$ bits of precision.

背景链接：https://arxiv.org/abs/2602.05394

02177 | AMR-130-0023

### Linear Systems and Eigenvalue Problems - Problem 23

O / open | AMR 问题清单汇编

For a given $A\in{\mathbb{C}}^{n\times n}$, let $\widehat{\mathcal{V}}$ denote an approximate invariant subspace of dimension $d$ of $A$, in the sense that there exists some $d$-dimensional invariant subspace ${\mathcal{V}}$ of $A$ such that $\angle({\mathcal{V}},\widehat{\mathcal{V}})$ is small. Let ${\mathcal{W}}$ be some other subspace of dimension $r<n-d$. Suppose the columns of $Q$ form an orthonormal basis for the (generically) $d+r$ dimensional space ${\mathcal{Q}} = \widehat{\mathcal{V}} + {\mathcal{W}}$.

Provide conditions that will ensure that $d$ eigenvalues of $Q^*AQ$ approximate the $d$ eigenvalues of $A$ associated with the invariant subspace ${\mathcal{V}}$. Impose assumptions about the eigenvalues of $A$ corresponding to $\mathcal{V}$, as needed.

背景链接：https://arxiv.org/abs/2602.05394

02178 | AMR-130-0024

### Linear Systems and Eigenvalue Problems - Problem 24

O / open | AMR 问题清单汇编

Let $Q$ be an orthonormal basis for a Krylov space for arbitrary $A\in{\mathbb{C}}^{n\times n}$ with random starting vector $b$. Is $\kappa_V(Q^*AQ)$ bounded (by a polynomial in $n$) with high probability? Or is ${\mathbb{E}}[\operatorname{area}\Lambda_\varepsilon(Q^*AQ)]$ (the expected value of the area in ${\mathbb{C}}$ of the $\varepsilon$-pseudospectrum) bounded by $\mathrm{poly}(n)\varepsilon^\beta$ for $\beta$ close to 2? (This problem is open even when $A$ is a circulant shift matrix.)

背景链接：https://arxiv.org/abs/2602.05394

02179 | AMR-130-0025

### Linear Systems and Eigenvalue Problems - Problem 25

O / open | AMR 问题清单汇编

Consider a non-Hermitian $A\in{\mathbb{C}}^{n\times n}$ and suppose $Q\in{\mathbb{C}}^{n\times k}$ has orthonormal columns. We know that the Ritz values (eigenvalues of $Q^*AQ$) are located in the numerical range, $W(A) = \{v^*Av: \|v\|_2=1\}$, a closed, convex subset of ${\mathbb{C}}$ that contains the eigenvalues of $A$ (and, potentially, points far from these eigenvalues).

- What can be said (deterministically or probabalistically) about the distribution of the Ritz values across $W(A)$ for $1<k<n$?

- What if $Q$ is restricted to be a basis for a Krylov subspace? What if $Q$ is the basis for a random subspace?

背景链接：https://arxiv.org/abs/2602.05394

02180 | AMR-130-0026

### Linear Systems and Eigenvalue Problems - Problem 26

O / open | AMR 问题清单汇编

There is currently no rigorous proof (in the vein of theoretical computer science) of success for a floating-point implementation of MR$^3$ when applied to an arbitrary symmetric tridiagonal matrix. Willems and Lang [Willems_framework] have given a proof of correctness which is conditional on several underlying assumptions, some of which cannot be verified until the algorithm is executed on a given input. At the same time, it is known that MR$^3$ can fail in the presence of roundoff when eigenvalues are too tightly clustered, if implemented without care (see [Parlett:CSD-04-1367]).

Give a ``TCS-style'' analysis of MR$^3$ which proves its correctness and convergence under easily verifiable conditions on the input.

背景链接：https://arxiv.org/abs/2602.05394

02181 | AMR-130-0027

### Linear Systems and Eigenvalue Problems - Problem 27

O / open | AMR 问题清单汇编

Let $B$ be an $n \times n$ (upper) bidiagonal matrix. Is there an algorithm that, in $O(kn)$ operations, can produce $k$ triples $(\sigma_i,v_i,u_i)$ such that $\sigma_i \geq 0$, $v_i, u_i \in {\mathbb{C}}^n$ are unit vectors, and, for machine precision $\epsilon$,

- $\|Bv_i-\sigma_iu_i\|_2 = O(\epsilon n\|B\|_2)$;

- $|v_i^Hv_j|$, $|u_i^Hu_j| = O(\epsilon n)$ if $i \neq j$.

背景链接：https://arxiv.org/abs/2602.05394

02182 | AMR-130-0028

### Linear Systems and Eigenvalue Problems - Problem 28

O / open | AMR 问题清单汇编

Consider a general nonnormal matrix $A = V \Lambda V^{-1}$ with largest eigenvalue magnitude $\lambda_1$ and finite eigenvector condition number $\kappa_V = \|V\| \|V^{-1}\|$. Is there an algorithm that uses $\mathrm{poly}(\log(n\kappa_V),1/\epsilon)$ matrix-vector products with $A$ and returns $\lambda$ so that $|\lambda - \lambda'| \le \epsilon |\lambda_1|$, where $\lambda'$ is any eigenvalue of $A$ with $|\lambda' | \ge (1-\epsilon)\mleft|\lambda_1\mright|$?

背景链接：https://arxiv.org/abs/2602.05394

02183 | AMR-130-0029

### Linear Systems and Eigenvalue Problems - Problem 29

O / open | AMR 问题清单汇编

Prove stronger theoretical bounds for GECP applied to $K$, by exploiting the structure of the kernel $K$.

背景链接：https://arxiv.org/abs/2602.05394

02184 | AMR-130-0003

### Linear Systems and Eigenvalue Problems - Problem 3 (Algebraic Multigrid Analysis for Theoretical Computer Scientists)

O / open | AMR 问题清单汇编

Show that for polynomially bounded SWCDDM matrices $\mathbf{M}$, Algebraic Multigrid runs in time $\tilde{O}(\operatorname{nnz}(\mathbf{M}))$ to compute an (implicitly represented) symmetric matrix $\mathbf{Z}$, which can be applied in $O(\operatorname{nnz}(\mathbf{M}))$ time, and satisfies the condition that $ \Omega(1) \mathbf{M}^{-1} \preceq \mathbf{Z} \preceq O(1) \mathbf{M}^{-1}$. In particular, the constants are independent of the problem size $n$.

背景链接：https://arxiv.org/abs/2602.05394

02185 | AMR-130-0030

### Linear Systems and Eigenvalue Problems - Problem 30

O / open | AMR 问题清单汇编

It is known that there exists $I$ such that $\|Q(I,:)^{-1}\|_2 \le \sqrt{k(n-k+1)}$ [hong1992rank] and there is a fairly efficient $O(nk^2)$ algorithm that satisfies this bound [Osinsky2025close].

Does the QRCP algorithm typically used in practice satisfy the above, or a similar, bound.

背景链接：https://arxiv.org/abs/2602.05394

02186 | AMR-130-0031

### Linear Systems and Eigenvalue Problems - Problem 31

O / open | AMR 问题清单汇编

Consider selecting $r = \mathcal{O}(k)$ uniformly random row indices $I$ and then choosing column indices $\tilde{J}$ so that $A(I,:) \approx A(I,\tilde{J}) \tilde{T}$. What are sufficient conditions on $r$ and on the kernel $f$ which guarantee that \begin{equation} \|\Pi_{\tilde{J}} A - A\|_F \leq c(n, m, k) \|\Pi_J A - A\|_F \end{equation} with high probability, where $c(n, m, k)$ grows at most logarithmically in its arguments? In other words, when does column selection on a random subset of the rows perform similarly to column selection on the full matrix?

背景链接：https://arxiv.org/abs/2602.05394

02187 | AMR-130-0032

### Linear Systems and Eigenvalue Problems - Problem 32

O / open | AMR 问题清单汇编

Consider $n$ positive real numbers $\lambda$, and consider the set of all $n \times n$ orthonormal matrices $S$. Prove a tightness bound(s) between the (1) trace error of the optimal column subset selection of the worst case matrix $K$ possessing eigenvalues $\lambda$: \begin{equation} x_k (\lambda) := \max_{V \in S} \min_{\mathcal{I} \subset [n], |\mathcal{I}| = k} \mathrm{Tr}\left[K - K_{:,\mathcal{I}} K_{\mathcal{I}, \mathcal{I}}^{-1} K_{\mathcal{I},:} \biggr\rvert_{K=V^\top \mathrm{diag}(\lambda) V}\right] \end{equation} and (2) the objective yielded by volume sampling any such $K$: \begin{equation} y_k (\lambda) := \left(\mathbb{E}_{\mathcal{I} \sim k-\mathrm{DPP}(K)} \mathrm{Tr}\left[K- K_{:,\mathcal{I}} K_{\mathcal{I}, \mathcal{I}}^{-1} K_{\mathcal{I}, :}\right]\right) \bigr\rvert_{K=\mathrm{diag}(\lambda)} = (k+1) \frac{e_{k+1} (\lambda)}{e_k(\lambda)} \end{equation} which is dependent on $\lambda$ alone in terms of the elementary symmetric polynomials $e_k$[Guruswami2012-optimal].

背景链接：https://arxiv.org/abs/2602.05394

02188 | AMR-130-0033

### Linear Systems and Eigenvalue Problems - Problem 33 (Sketch-and-solve)

O / open | AMR 问题清单汇编

Let $A \in \mathbb{R}^{n \times d}$ be a full-rank matrix, $B\in\mathbb{R}^{n \times p}$ be a matrix, and let $\Omega \in \mathbb{R}^{n \times k}$ be an oblivious subspace injection with dimension $d$ and injectivity $1-\varepsilon$. Let $\tilde X = (\Omega^\top A)^+(\Omega^\top B) \in \mathbb{R}^{d \times p}$ be the sketch-and-solve approximation to the ordinary least-squares solution $X = A^+B$. (Here, ${}^+$ denotes the Moore--Penrose pseudoinverse.) Prove or disprove: Does it necessarily hold that $$ \|A \tilde X - B\|_\mathrm{F} \leq (1+\mathcal{O}(\varepsilon)) \min_{X \in \mathbb{R}^{d \times p}}\|AX - B\|_\mathrm{F}? $$

背景链接：https://arxiv.org/abs/2602.05394

02189 | AMR-130-0034

### Linear Systems and Eigenvalue Problems - Problem 34 (Randomized SVD)

O / open | AMR 问题清单汇编

Let \(A \in \mathbb{R}^{n \times d}\) be a matrix, fix $r > 0$, and let \(\Omega \in \mathbb{R}^{n \times k}\) be an oblivious subspace injection with dimension $r$ and injectivity \(1-\varepsilon\). Let $\tilde{A} = (A\Omega)(A\Omega)^+A$ denote the randomized SVD approximation to $A$, and let $A_r$ be the best rank-$r$ approximation to $A$ with respect to the Frobenius norm. Prove or disprove: Does it necessarily hold that \[ \|A - \tilde{A}\|_\mathrm{F} \leq (1+\mathcal{O}(\varepsilon)) \|A-A_r\|_\mathrm{F}? \]

背景链接：https://arxiv.org/abs/2602.05394

02190 | AMR-130-0035

### Linear Systems and Eigenvalue Problems - Problem 35 (Regression in the $\ell_p$ norm)

O / open | AMR 问题清单汇编

Consider the $\ell_p$ linear regression problem

\begin{equation*} \operatorname{minimize}_{x \in\mathbb{R}^d} \|Ax - b\|_{\ell_p} \end{equation*}

for a matrix \(A \in \mathbb{R}^{n \times d}\) and a vector \(b\in\mathbb{R}^{n}\). For a sketching matrix \(\Omega \in \mathbb{R}^{n \times k}\), the sketch-and-solve solution is \[\tilde x \in \operatorname{argmin}_{x \in\mathbb{R}^d} \|(\Omega^\top Ax)-(\Omega^\top b)\|_{\ell_p}.\] Formulate a generalization of the OSI property to the $\ell_p$ regression problem which ensures that this procedure succeeds to constant-factor accuracy \[ \|A \tilde x - b \|_{\ell_p} \leq \mathrm{const}\cdot \min_{x \in \mathbb{R}^d}\|Ax - b\|_{\ell_p}. \]

背景链接：https://arxiv.org/abs/2602.05394

02191 | AMR-130-0036

### Linear Systems and Eigenvalue Problems - Problem 36

O / open | AMR 问题清单汇编

Prove or disprove the Nelson--Nguyen conjecture: A SparseStack embedding with parameters $\zeta = \mathcal{O}((\log r) / \varepsilon)$ and $k = \mathcal{O}(r / \varepsilon^2)$ is an oblivious subspace embedding with parameters $\alpha = 1 - \varepsilon$ and $\beta = 1+\varepsilon$.

背景链接：https://arxiv.org/abs/2602.05394

02192 | AMR-130-0037

### Linear Systems and Eigenvalue Problems - Problem 37 (Subspace injection properties of sparse random matrices)

O / open | AMR 问题清单汇编

Assume the condition $r\le \mathcal{O}(n/\log n)$ relating the subspace dimension $r$ and the ambient dimension $n$. Prove or disprove: A SparseStack embedding with parameters $\zeta = \mathcal{O}(1)$ and $k = \mathcal{O}(r)$ is an oblivious subspace injection with parameter $\alpha = \mathrm{const}$. More specifically, do parameters $\zeta = 4$ and $k = 2r$ suffice to achieve the OSI property (for all sufficiently large $r$)?

背景链接：https://arxiv.org/abs/2602.05394

02193 | AMR-130-0038

### Linear Systems and Eigenvalue Problems - Problem 38 (Rerandomized subsampled trigonometric transforms)

O / open | AMR 问题清单汇编

Assume that $n = \Omega(\log r)$. Prove or disprove: The rerandomized SRHT of dimension $k = \mathcal{O}(r/\varepsilon^2)$ is an oblivious subspace embedding with $\alpha = 1-\varepsilon$ and $\beta = 1+\varepsilon$.

背景链接：https://arxiv.org/abs/2602.05394

02194 | AMR-130-0039

### Linear Systems and Eigenvalue Problems - Problem 39

O / open | AMR 问题清单汇编

Given any choice of $m$-node tree tensor network (e.g., tensor train or Tucker) and choice of ranks (dimensions of edges in the tensor network), let the set of representable tensors in this tensor network be $S\subset \mathbb{R}^{d_1\times \cdots \times d_n}$. Find a polynomial-time algorithm that takes as input any tensor $\mathcal{A}\in\mathbb{R}^{d_1\times \cdots \times d_n}$ and yields an approximation $\mathcal{X}\in\mathbb{S}$ such that \[\|\mathcal{X}-\mathcal{A}\|_F^2 < (m-1)\min_{\mathcal{Y}\in S}\|\mathcal{Y}-\mathcal{A}\|_F^2,\] or show that an $(m-1)$-approximation is optimal under complexity-theoretic assumptions.

背景链接：https://arxiv.org/abs/2602.05394

02195 | AMR-130-0004

### Linear Systems and Eigenvalue Problems - Problem 4 (Polynomially decaying eigenvalues)

O / open | AMR 问题清单汇编

Fix constants $\varepsilon > 0$ and $p > 0$. Let $\mathbf{A}_n \in \mathbb{C}^{n \times n}$, $n\geq 1$, be a sequence of random positive definite matrices constructed via \begin{equation*} \mathbf{A}_n = \mathbf{U} \mathbf{\Lambda}_n \mathbf{U}^*, \quad \text{where } \begin{cases} \mathbf{U} \in \mathbb{C}^{n \times n} \text{ is Haar unitary}, \\ \mathbf{\Lambda}_n = \operatorname{diag}\{1^{-p}, 2^{-p}, \ldots, n^{-p}\}. \end{cases} \end{equation*} Let $\mathbf{b}_n = \mathbf{A}_n \mathbf{z}_n$ be a sequence of random vectors where $\mathbf{z}_n \in \mathbb{C}^n$ is distributed uniformly on the sphere. What is the asymptotic behavior of $T_\mathrm{CG}(\mathbf{A}_n,\mathbf{b}_n,\epsilon)$ compared with $T_\mathrm{RCD}(\mathbf{A}_n,\mathbf{b}_n,\epsilon)$ as $n \rightarrow \infty$?

背景链接：https://arxiv.org/abs/2602.05394

02196 | AMR-130-0040

### Linear Systems and Eigenvalue Problems - Problem 40

O / open | AMR 问题清单汇编

Let $\Pi_{2^m}^*$ be the set of univariate polynomials whose corresponding matrix function can be computed with $m$ matrix-matrix multiplications and an arbitrary number of matrix additions and scalings. We consider the problem of determining the best such polynomial that approximates the sign function in $I:=[-1,-\delta]\cup [\delta,1],$ \begin{equation} \epsilon_{m}^*=\min_{p\in\Pi_{2^m}^*} \max_{x\in I} |p(x)-\mathrm{sign}(x)|. \end{equation}

What is the asymptotic error $\epsilon_m^*$ as a function of $m$ and $\delta$?

背景链接：https://arxiv.org/abs/2602.05394

02197 | AMR-130-0005

### Linear Systems and Eigenvalue Problems - Problem 5

O / open | AMR 问题清单汇编

*Complexity for sparse matrices.* Importantly, the matrix-vector product model is most relevant for sparse matrices, in which case the extra cost paid by the Krylov solver is only $\tilde O(\mathrm{nnz}(A)\cdot k)$, where $\mathrm{nnz}(A)$ is the number of non-zero entries in $A$ (although, importantly, additional costs arise in finite precision arithmetic). Many randomized linear algebra methods can also leverage input sparsity for faster running time, so it is natural to expect that a similar improvement may be attainable in the direct access model.

Can we solve an $(n,k,0,\kappa)$ linear system $Ax=b$ to high accuracy using $\tilde O(k^\omega + \mathrm{nnz}(A)\kappa)$ arithmetic operations?

背景链接：https://arxiv.org/abs/2602.05394

02198 | AMR-130-0006

### Linear Systems and Eigenvalue Problems - Problem 6

O / open | AMR 问题清单汇编

Find the smallest $p>0$ such that we can solve an $n\times n$ linear system $Ax=b$ to high accuracy using $\tilde O(\mathcal{T}_A\cdot\bar\kappa_p)$ arithmetic operations, where $\bar\kappa_p$ is the $p$-averaged condition number of the singular values of $A$, whereas $\mathcal{T}_A$ is either $n^2$ or $\mathrm{nnz}(A)$.

背景链接：https://arxiv.org/abs/2602.05394

02199 | AMR-130-0007

### Linear Systems and Eigenvalue Problems - Problem 7

O / open | AMR 问题清单汇编

What is the complexity of solving $(n,0,l,\kappa)$ linear systems, and how is this affected by the model of computation (direct access vs matrix-vector product)?

背景链接：https://arxiv.org/abs/2602.05394

02200 | AMR-130-0008

### Linear Systems and Eigenvalue Problems - Problem 8

O / open | AMR 问题清单汇编

Let $A$ be an $n\times n$ random matrix with i.i.d. Gaussian (or Rademacher, or Bernoulli) entries, and take any $n$-dimensional vector $b$. Is it possible (conditioned on a high-probability event) to solve the linear system $Ax=b$ to high accuracy faster than $O(n^\omega)$?

背景链接：https://arxiv.org/abs/2602.05394

02201 | AMR-130-0009

### Linear Systems and Eigenvalue Problems - Problem 9

O / open | AMR 问题清单汇编

What is the bit complexity of solving $(n,k,l,\kappa)$ linear systems? In particular, does the $\tilde O(k^\omega + n^2\kappa)$ running time for $(n,k,0,\kappa)$ systems extend to the Word-RAM model?

背景链接：https://arxiv.org/abs/2602.05394

02202 | OWR-16162-003

### Linkage and Vanishing in Kato-Milne Cohomology

P / partially_solved | Oberwolfach 报告问题汇编

Let $F$ be a field of characteristic $p$. If the Kato-Milne cohomology group $H_p^n(F)$ is linked, must $H_p^{n+2}(F)$ vanish? If $H_p^n(F)$ is triple linked, must $H_p^{n+1}(F)$ vanish?

来源字段：https://doi.org/10.4171/owr/2018/21

02203 | OWR-3397-002

### Local Closedness of Symplectic Cores

S / solved | Oberwolfach 报告问题汇编

For a complex affine Poisson algebra $R$, are its symplectic cores-the equivalence classes of maximal ideals with the same Poisson core-locally closed in the maximal spectrum?

来源字段：https://doi.org/10.4171/owr/2009/14

02204 | AMR-093-0049

### Local Gan-Gross-Prasad conjecture

S / solved | AMR 问题清单汇编

For the classical-group pairs and generic local $L$-parameters in the Gan-Gross-Prasad setting, is there exactly one relevant representation $\pi$ in the associated $L$-packet for which $\operatorname{Hom}_H(\pi\otimes\overline\nu,\mathbb C)$ is nonzero, namely the representation indexed by the distinguished Gan-Gross-Prasad character?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

02205 | OWR-14299522-008

### Local Lifting Properties of Free-Group Product $C^*$-Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Does the full group $C^*$-algebra $C^*(F_2\times F_2)$ have the local lifting property or the lifting property? More generally, does a $C^*$-algebra $A$ have the local lifting property if and only if $\operatorname{Ext}(A)$ is a group?

来源字段：https://doi.org/10.4171/owr/2025/35

02206 | OWR-16768-003

### Localization of Finitistic Test Ideals

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Localization Conjecture for Finitistic Test Ideals [OWR-16768-010]

Let $R$ be an $F$-finite Noetherian domain of characteristic $p>0$, let $\mathfrak a\neq0$ be an ideal, let $t>0$, and let $S\subset R$ be multiplicatively closed. Is

$$S^{-1}\tau_{\mathrm{fg}}(\mathfrak a^t)=\tau_{\mathrm{fg}}\bigl((S^{-1}\mathfrak a)^t\bigr).$$

Does this equality hold?

来源字段：https://doi.org/10.4171/owr/2019/6

02207 | AIM-REPRESENTATION_THEORY-0107

### Locally analytic vectors determine the Hodge filtration through topology and extension data

S / solved | AIM 研讨会问题汇编

**Question 2.** Let $V$ be a two-dimensional irreducible potentially crystalline representation of $G_{\mathbb Q_p}$, and assume that it is of supercuspidal type (i.e. the associated Weil--Deligne representation is irreducible). Let $B(V)$ be the associated (conjecturally) irreducible admissible Banach-space representation. Can one prove that the locally analytic vectors in $B(V)$ determine the Hodge filtration on $D_{\mathrm{pcris}}(V)$?

**3.4.1 (Emerton).** More generally, relate $B(V)^{\mathrm{an}}$ to $D_{\mathrm{rig}}(V)$. Can one relate $(B(V)^{\mathrm{an}})'$ to the de Rham cohomology of the coverings of the $p$-adic upper half-plane? (Here $\{\}'$ is the dual.)

来源字段：https://aimath.org/WWN/padicmodularity/padicmodularity.pdf

02208 | OWR-12494-001

### Local-Global Isotropy over Function Fields

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be an integral scheme on which $2$ is invertible, let $K$ be its function field, and let $\Omega$ be the set of rank-one discrete valuations of $K$. If a nondegenerate quadratic form $q$ over $K$ is isotropic over the completion $K_v$ for every $v\in\Omega$, must $q$ be isotropic over $K$? More specifically, does this local-global principle hold for quadratic forms of dimension at least $5$ over function fields of curves over totally imaginary number fields?

来源字段：https://doi.org/10.4171/owr/2013/31

02209 | OWR-3475-003

### Local-Global Principles for Homogeneous Spaces

O / open | Oberwolfach 报告问题汇编

Let $K$ be a complete discretely valued field and $F$ a one-variable function field over $K$. If $Y$ is a homogeneous space of a connected linear algebraic group over $F$ and $Y(F_v)\neq\varnothing$ for every rank-one discrete valuation $v$ of $F$, must $Y(F)\neq\varnothing$?

来源字段：https://doi.org/10.4171/owr/2009/25

02210 | AIM-LINEAR_ALGEBRA-0007

### Loop-group factorization and normalization obstructions for 2 by 2 totally nonnegative block Toeplitz matrices

P / partially_solved | AIM 研讨会问题汇编

Let $$ M_{\mathbf A}= \begin{pmatrix} A_0&0&0&\cdots\\ A_1&A_0&0&\cdots\\ A_2&A_1&A_0&\cdots\\ \vdots&\vdots&\vdots&\ddots \end{pmatrix}\in TN, $$ and let $F(t)=A_0+tA_1+t^2A_2+\cdots\in M_2[[t]]$, where $A_0$ is invertible. Is there an AESW-type factorization for $F(t)$?

来源字段：http://aimpl.org/totalpos/1/

02211 | AIM-REPRESENTATION_THEORY-0019

### Lower-finite highest weight structure on the characteristic-zero generic FI category

P / partially_solved | AIM 研讨会问题汇编

**Highest weight category.** Working over a field. Is the Serre quotient \[ \operatorname{Mod}_{\mathrm{FI}}/\operatorname{Mod}_{\mathrm{FI}}^{\mathrm{tors}} \] a highest weight category?

来源字段：http://aimpl.org/repnstability/1/

02212 | OWR-792-003

### MacWilliams Duality for Convolutional Codes

S / solved | Oberwolfach 报告问题汇编

Does a MacWilliams duality theorem hold for general classes of convolutional codes?

来源字段：https://doi.org/10.4171/owr/2005/11

02213 | OWR-13678-008

### Matrix Characterization of Complete Intersections

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Complete-Intersection Extraction [OWR-13678-009]

Let $P$ be an ideal projector and let $M_j$ be multiplication by $x_j$ on $\mathbb{C}[x_1,\ldots,x_d]/\ker P$. Characterize when $\ker P$ is a complete intersection in terms of the commuting matrices $M_1,\ldots,M_d$.

来源字段：https://doi.org/10.4171/owr/2015/21

02214 | OWR-3475-002

### Matrix Factorization over Curve Patching Fields

P / partially_solved | Oberwolfach 报告问题汇编

Let $F=F_U\cap F_P\subset F_{\wp}$ be the patching fields associated with a curve. For which linear algebraic groups $G/F$ does every $g\in G(F_{\wp})$ factor as $g=g_Ug_P$ with $g_U\in G(F_U)$ and $g_P\in G(F_P)$?

来源字段：https://doi.org/10.4171/owr/2009/25

02215 | OWR-3393-002

### Maximal Complexity of Non-p-by-p Specht Modules

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Non-$p\times p$ Specht-Complexity Conjecture [OWR-3393-003]

For a Specht module $S^\lambda$ of a symmetric group in a block of weight $w$, prove that every remaining case with $\lambda$ not $p\times p$ has complexity $w$, the maximal complexity in its block; equivalently, resolve the remaining direction of the VIGRE conjecture.

来源字段：https://doi.org/10.4171/owr/2009/10

02216 | OWR-1538-001

### Maximal Poisson-Commutative Subalgebras and Flatness

O / open | Oberwolfach 报告问题汇编

Let $\mathfrak{q}$ be an algebraic Lie algebra over an algebraically closed field of characteristic zero, let $\ell=\operatorname{ind}\mathfrak{q}$, and set $b(\mathfrak{q})=(\dim\mathfrak{q}+\ell)/2$. Is every maximal Poisson-commutative subalgebra of $S(\mathfrak{q})$ of Krull dimension $b(\mathfrak{q})$? Now suppose the coadjoint representation has the codimension-$2$ property and $Z(\mathfrak{q})$ contains algebraically independent homogeneous elements $f_1,\ldots,f_\ell$ with $\sum_i\deg f_i=b(\mathfrak{q})$. For regular $\xi\in\mathfrak{q}^*$, let $F_\xi(Z(\mathfrak{q}))$ be generated by the $\xi$-shifts of the Poisson center. Is the natural morphism $\pi:\mathfrak{q}^*\to\operatorname{Spec}(F_\xi(Z(\mathfrak{q})))$ flat?

来源字段：https://doi.org/10.4171/owr/2007/22

02217 | OWR-1274-006

### Maximal Strongly Free Quadratic Relator Sequences

S / solved | Oberwolfach 报告问题汇编

For strongly free sequences of quadratic relators on $d$ generators, is the largest possible length $t(d)=d^2/4$ for even $d$ and $t(d)=(d-1)^2/4$ for odd $d$?

来源字段：https://doi.org/10.4171/owr/2006/25

02218 | AIM-REPRESENTATION_THEORY-0093

### Maximal-degree characters from rank vectors

P / partially_solved | AIM 研讨会问题汇编

The AIM problem list for the workshop *Supercharacters and combinatorial Hopf algebras* gives:

The PDF places a line break after “super-”.

The phrase “as supercharacters” is potentially stronger than merely asking for large constituents of supercharacters. This report distinguishes:

1. ordinary irreducible characters of maximal degree; 2. standard Diaconis--Isaacs supercharacters which happen to be irreducible; 3. the degree of a reducible supercharacter.

They need not agree for a general pattern group. The main theorem below gives a class in which they do agree completely.

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

02219 | ALG-014

### McKay Conjecture

S / solved | 独立 / 未分配集合

别名：McKay Conjecture for Finite Groups [OWR-1265-004]

For a finite group $G$ and prime $p$, is the number of irreducible complex characters of $G$ whose degree is not divisible by $p$ equal to the corresponding number for the normalizer of a Sylow $p$-subgroup?

背景链接：https://arxiv.org/abs/2410.20392

02220 | AIM-REPRESENTATION_THEORY-0066

### Metaplectic Steinberg via the sign type, its formal degree, and the even-Weil trivial analogue

P / partially_solved | AIM 研讨会问题汇编

In the even-Weil block of the twofold metaplectic group, transport the trivial and Steinberg objects from split $\operatorname{SO}(2r+1)$ and compute the latter's degree using the Hilbert-algebra normalization.

来源字段：http://aimpl.org/autoformcovergp/1/

02221 | OWR-10252925-001

### Minimal Resolutions from Virtual Multigraded Resolutions

O / open | Oberwolfach 报告问题汇编

Prove that the Fourier-Mukai complex $G^\bullet$ associated with a multigraded module $M$ is the minimal free resolution of its truncation $M_{\geq d}$.

来源字段：https://doi.org/10.4171/owr/2022/17

02222 | OWR-12868-005

### Mixed Hodge Polynomials of Character Varieties

P / partially_solved | Oberwolfach 报告问题汇编

For a $k$-tuple of partitions $\mu$, let $\mathcal M_\mu$ be the corresponding generic $\operatorname{GL}_n(\mathbb C)$ character variety. Define $\mathbb H_\mu(z,w)$ by $$ (z^2-1)(1-w^2)\operatorname{Log}\!\left(\sum_\lambda \mathcal H_\lambda(z,w)\prod_{i=1}^k\widetilde H_\lambda(x_i;z^2,w^2)\right) =\sum_\mu\mathbb H_\mu(z,w)m_\mu(x), $$ where $\widetilde H_\lambda$ are Macdonald polynomials and $$ \mathcal H_\lambda(z,w)=\prod_{s\in\lambda} \frac{(z^{2a(s)+1}-w^{2l(s)+1})^{2g}} {(z^{2a(s)+2}-w^{2l(s)})(z^{2a(s)}-w^{2l(s)+2})}. $$ Are the functions $\mathbb H_\mu(z,w)$ polynomials that encode the mixed Hodge numbers of $\mathcal M_\mu$?

来源字段：https://doi.org/10.4171/owr/2014/08

02223 | OWR-12697710-007

### Mixed Odd Powers of Sum-of-Squares Polynomials

S / solved | Oberwolfach 报告问题汇编

If $p^{2i+1}$ and $q^{2j+1}$ are sums of squares, must $(p+q)^{2i+2j+1}$ also be a sum of squares? In addition, define $c_k$ with $0\leq c_k<3$ by $$\left\{c:\left(x^4y^2+x^2y^4+1-cx^2y^2\right)^{2k+1}\text{ is a sum of squares}\right\}=\{c:c\leq c_k\}.$$ What is $\lim_{k\to\infty}c_k$?

来源字段：https://doi.org/10.4171/owr/2023/14

02224 | OWR-14299511-004

### Modular S-Transform Equivalence for Strongly Finite Vertex Algebras

P / partially_solved | Oberwolfach 报告问题汇编

For a strongly finite $C_2$-cofinite vertex operator algebra, prove that the categorical $S$-transformation defined by Hopf links agrees with the modular $S$-transformation induced by $\tau\mapsto-1/\tau$.

来源字段：https://doi.org/10.4171/owr/2025/34

02225 | OWR-9790367-001

### Module Categories of Affine Vertex Algebras

O / open | Oberwolfach 报告问题汇编

For affine vertex algebras at admissible level, classify simple modules and their projective covers and injective hulls, prove projectives have finite length, and construct rigid braided tensor-category structures with explicit fusion rules.

来源字段：https://doi.org/10.4171/owr/2022/16

02226 | OWR-12697690-009

### Module Parametrization by $F$-Polynomials

O / open | Oberwolfach 报告问题汇编

Is the parametrization map $\Phi$ associated with the $F$-polynomials of modules over $\Lambda\cong kQ/I$ an isomorphism?

来源字段：https://doi.org/10.4171/owr/2023/7

02227 | OWR-716-001

### Moduli Spaces of Global Lie-Algebra Deformations

P / partially_solved | Oberwolfach 报告问题汇编

Can the equivalence classes of global deformations of a Lie algebra be endowed naturally with the structure of a topological or geometric moduli space?

来源字段：https://doi.org/10.4171/owr/2004/06

02228 | OWR-1185-001

### Moduli Spaces of Lie Algebra Structures

O / open | Oberwolfach 报告问题汇编

Does the set of Lie algebra structures, considered up to equivalence, admit a moduli space whose local structure at a fixed Lie algebra is reflected by its deformations?

来源字段：https://doi.org/10.4171/owr/2006/03

02229 | OWR-12697693-008

### Monoid and Group Invariants of qL-Algebras

O / open | Oberwolfach 报告问题汇编

Develop a theory of $qL$-algebras and their monoid-valued and group-valued invariants. Do these invariants carry Garside-like structures?

来源字段：https://doi.org/10.4171/owr/2023/9

02230 | OWR-8415347-009

### Monoidal Invariance of Hopf-Algebra Cohomological Dimension

P / partially_solved | Oberwolfach 报告问题汇编

If Hopf algebras $A$ and $B$ have equivalent linear tensor categories of comodules, must their Hochschild cohomological dimensions agree, $\operatorname{cd}(A)=\operatorname{cd}(B)$?

来源字段：https://doi.org/10.4171/owr/2021/44

02231 | OWR-8415347-018

### Morita Classification of Quantum SU-Two Projective Representations

P / partially_solved | Oberwolfach 报告问题汇编

If $(M_F,\alpha_F)$ is an ergodic coaction of $L^\infty(FO_F)$, does there exist $G\in F_q$ such that $M_G$ is equivariantly $W^*$-Morita equivalent to a coideal von Neumann subalgebra of $L^\infty(FO_G)$? If not, can such coactions be characterized? If $(M,\alpha)$ is an ergodic coaction of $L^\infty(SU_q(2))$ on a type-I von Neumann algebra, is it equivariantly $W^*$-Morita equivalent to a coideal von Neumann subalgebra?

来源字段：https://doi.org/10.4171/owr/2021/44

02232 | OWR-12980-009

### Multiplicity Formula for Elliptic Arthur Parameters

O / open | Oberwolfach 报告问题汇编

For every elliptic Arthur parameter $\phi$ of Ramanujan type, do we have $c_{\pi^{\psi_N}(\phi)}^{\psi_N}=|S_\phi|$?

来源字段：https://doi.org/10.4171/owr/2014/22

02233 | OWR-15178-012

### Multiplicity Inequality for Quotients by Perfect Ideals

O / open | Oberwolfach 报告问题汇编

If $(T,\mathfrak n)$ is a local ring and $J\subset T$ is a perfect ideal, must the multiplicities satisfy $e(T)\leq e(T/J)$?

来源字段：https://doi.org/10.4171/owr/2016/57

02234 | OWR-5149-010

### Murphy-Element Description of Centralizer-Algebra Centers

O / open | Oberwolfach 报告问题汇编

Is the center of the centralizer algebra $kS_n^{S_\ell}$ the algebra of polynomials in the Murphy elements that are symmetric under $S_\ell\times S_{n-1}$?

来源字段：https://doi.org/10.4171/owr/2011/21

02235 | OWR-4335-005

### Mutation Generation of E-Invariant-Zero Representations

O / open | Oberwolfach 报告问题汇编

For an indecomposable $(Q,S)$-representation $M$ with $E(M)=0$, must $M$ be obtainable from a negative simple representation by a sequence of mutations?

来源字段：https://doi.org/10.4171/owr/2010/22

02236 | OWR-12697690-002

### Mutation Theories from Extriangulated Categories

O / open | Oberwolfach 报告问题汇编

Which established mutation theories can be recovered from the mutation theorem for $0$-Auslander extriangulated categories, and which new mutation operations arise from it?

来源字段：https://doi.org/10.4171/owr/2023/7

02237 | OWR-11101917-008

### Necessity of Z-Stability in KK-Uniqueness

O / open | Oberwolfach 报告问题汇编

Let $C,D,E$ be separable $C^*$-algebras, with $D$ a closed two-sided ideal of $E$, and let $\Phi,\Psi:C\to E$ be relatively absorbing homomorphisms agreeing modulo $D$. Can the $\mathcal Z$-stability assumption on $E$ be removed from the equivalence between $[\Phi,\Psi]_{KK}=0$ and asymptotic unitary equivalence through unitaries in $\widetilde D$?

来源字段：https://doi.org/10.4171/owr/2022/36

02238 | OWR-14298159-008

### Nichols Algebras over Nonabelian Simple Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a finite nonabelian simple group and let $0\neq V\in{}_{\mathbb CG}^{\mathbb CG}\mathcal{YD}$. Must the Nichols algebra $\mathcal B(V)$ be infinite-dimensional?

来源字段：https://doi.org/10.4171/owr/2024/5

02239 | OWR-14299580-017

### Nichols Algebras over Simple Groups

O / open | Oberwolfach 报告问题汇编

Let $G$ be a finite simple group and let $V$ be a nonzero Yetter-Drinfeld module over $\mathbb C G$. Must the Nichols algebra $\mathcal B(V)$ be infinite-dimensional?

来源字段：https://doi.org/10.4171/owr/2025/49

02240 | OWR-14299580-005

### Noetherian Enveloping Algebras of Lie Algebras

O / open | Oberwolfach 报告问题汇编

If the universal enveloping algebra $U(\mathfrak g)$ of a Lie algebra $\mathfrak g$ is Noetherian, must $\mathfrak g$ be finite-dimensional? If a group algebra $kG$ is Noetherian, must $G$ be polycyclic-by-finite?

来源字段：https://doi.org/10.4171/owr/2025/49

02241 | OWR-14299915-006

### Noetherian Twisted Cohomology of Finite $p$-Groups

O / open | Oberwolfach 报告问题汇编

For every finite $p$-group $G$, prove that its permutation-twisted cohomology ring $H^{\bullet\bullet}(G)$ is Noetherian.

来源字段：https://doi.org/10.4171/owr/2026/9

02242 | OWR-16162-017

### Non-Unital Composition Algebras over Rings

O / open | Oberwolfach 报告问题汇编

How can non-unital composition algebras over rings, including octonion isotopes, be classified and described?

来源字段：https://doi.org/10.4171/owr/2018/21

02243 | OWR-1272-008

### Noncrossing Partitions in Wild Representation Type

S / solved | Oberwolfach 报告问题汇编

Does the bijection between noncrossing partitions and finitely generated wide subcategories extend to wild representation type?

来源字段：https://doi.org/10.4171/owr/2006/23

02244 | OWR-11101917-007

### Nonembeddable Tracial Algebras with Wasserstein Lifting

O / open | Oberwolfach 报告问题汇编

Does there exist a non-Connes-embeddable tracial von Neumann algebra whose noncommutative law has coincident weak-star and Wasserstein topologies and for which every ultraproduct embedding lifts to factorizable completely positive maps?

来源字段：https://doi.org/10.4171/owr/2022/36

02245 | OWR-741-002

### Nonnegative Stretched Clebsch-Gordan Coefficients

O / open | Oberwolfach 报告问题汇编

For highest weights $\lambda,\mu,\nu$ of a Lie algebra of type $A_r$, $B_r$, $C_r$, or $D_r$, must every constituent polynomial of the Ehrhart quasipolynomial $C_{N\lambda,N\mu}^{N\nu}$ have nonnegative coefficients?

来源字段：https://doi.org/10.4171/owr/2004/39

02246 | OWR-9790358-010

### Nonsingularity of Bernstein-Basis Collocation Matrices

P / partially_solved | Oberwolfach 报告问题汇编

Let $T$ be a triangle, let $D_{d,T}$ be the degree-$d$ domain points, and let $\Gamma\subseteq D_{d,T}$. Is the collocation matrix $$ \bigl(B_\eta^d(\xi)\bigr)_{\xi,\eta\in\Gamma} $$ always nonsingular?

来源字段：https://doi.org/10.4171/owr/2022/7

02247 | OWR-2654827-011

### Nonunimodal Codimension-Four Gorenstein h-Vectors

O / open | Oberwolfach 报告问题汇编

Do there exist non-unimodal Gorenstein h-vectors of codimension 4 for standard graded Artinian Gorenstein algebras?

来源字段：https://doi.org/10.4171/owr/2020/31

02248 | OWR-12983-002

### Nonvanishing of Derived Categories for Gorenstein Algebras

O / open | Oberwolfach 报告问题汇编

Let $A$ be a connected graded Artin--Schelter Gorenstein algebra with Nakayama automorphism $\mu$, and for a graded algebra automorphism $\varepsilon$ of $A$ let $D_\varepsilon(A)$ be the triangulated category of $\varepsilon$-skew Calabi--Yau / twisted Cohen--Macaulay modules introduced by Reyes, Rogalski, and Zhang. For every Artin--Schelter Gorenstein algebra $A$, is $D_\varepsilon(A)$ nonzero for some (or every) twisting automorphism $\varepsilon$?

来源字段：https://doi.org/10.4171/owr/2014/25

02249 | OWR-12868-001

### Nonzerodivisors and Stabilization of Cohomology Annihilators

P / partially_solved | Oberwolfach 报告问题汇编

Let $\Lambda$ be a noetherian algebra finite over its center, let $\operatorname{ca}(\Lambda)$ be its cohomology-annihilator ideal, and let $\operatorname{ca}_s(\Lambda)$ denote its degree-$s$ approximation. Does $\operatorname{ca}(\Lambda)$ contain a nonzerodivisor? What is the least $s$ such that $\operatorname{ca}(\Lambda)=\operatorname{ca}_s(\Lambda)$?

来源字段：https://doi.org/10.4171/owr/2014/08

02250 | AIM-REPRESENTATION_THEORY-0005

### Norm descent reduces a quartic CM subfamily to two imaginary-quadratic Hecke motives

P / partially_solved | AIM 研讨会问题汇编

**Beilinson conjecture for Hecke characters of quartic CM fields**

Prove the Beilinson conjecture for Hecke characters of quartic CM fields.

Literature note: *Subproblem: how to generalize the Eisenstein symbol?*

来源字段：http://aimpl.org/aagaautomorphic/1/

02251 | OWR-5149-011

### Normal Nodes and Specht Restriction Filtrations

S / solved | Oberwolfach 报告问题汇编

For the restriction filtration of a $p$-regular Specht module $S^\lambda$, is $J_i^\lambda\ne J_{i+1}^\lambda$ if and only if the $i$-th removable node of $\lambda$ is normal in Kleshchev's sense, and what is the minimal $n$ for which this equivalence holds?

来源字段：https://doi.org/10.4171/owr/2011/21

02252 | OWR-7155441-009

### Normal Schubert Varieties in Full Affine Flag Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be absolutely simple over a Laurent-series field of characteristic $p>0$, with $p\mid|\pi_1(G)|$. Classify all normal Schubert varieties in the neutral component $\operatorname{Fl}_{G,a}^{\circ}$ of the full affine flag variety; equivalently, classify the corresponding affine Weyl-group elements or alcoves.

来源字段：https://doi.org/10.4171/owr/2021/20

02253 | OWR-12697689-003

### Normalized Ext Limits and Epsilon Multiplicity

O / open | Oberwolfach 报告问题汇编

Does the limit $\lim_{d\to\infty}\ell(\operatorname{Ext}^3_S(S/I^d,S))/d^6$ exist, as predicted by higher $\varepsilon$-multiplicity theory?

来源字段：https://doi.org/10.4171/owr/2023/6

02254 | OWR-14603-017

### Northcott Property for Rings of Integers in Bounded-Degree Composita

O / open | Oberwolfach 报告问题汇编

For $d>2$, let $\mathbb Q^{(d)}$ be the compositum of all extensions of $\mathbb Q$ of degree at most $d$ and let $\mathcal O_{\mathbb Q^{(d)}}$ be its ring of integers. Does $\mathcal O_{\mathbb Q^{(d)}}$ have Property (N), meaning that it contains only finitely many elements of bounded absolute Weil height?

来源字段：https://doi.org/10.4171/owr/2016/21

02255 | OWR-12175-027

### Nuclear Dimension after Jiang-Su Stabilization

O / open | Oberwolfach 报告问题汇编

Is $\operatorname{dim}_{\mathrm{nuc}}(A\otimes\mathcal Z)<\infty$ for every nuclear $C^*$-algebra $A$? What is the decomposition rank of $C(X)\subset C(X,M_n)$? Is it less than $\dim X$, or does dimension drop only after tensoring with a UHF algebra?

来源字段：https://doi.org/10.4171/owr/2012/52

02256 | OWR-12591-011

### Nuclear Dimension and Z-Stability Beyond Simple Algebras

O / open | Oberwolfach 报告问题汇编

For $C^*$-algebras in which no ideal has a type-I representation, are finite nuclear dimension and amenability together with $\mathcal Z$-stability equivalent even without assuming simplicity and unitality?

来源字段：https://doi.org/10.4171/owr/2013/43

02257 | OWR-15585-002

### Nuclear Dimension of Transformation-Group Crossed Products

P / partially_solved | Oberwolfach 报告问题汇编

For a continuous action of a locally compact group $G$ on a locally compact Hausdorff space $X$, when does the crossed product $C_0(X)\rtimes G$ have finite nuclear dimension?

来源字段：https://doi.org/10.4171/owr/2017/42

02258 | MOD-008

### O-Minimal Theory with Trans-Exponential Growth

O / open | 独立 / 未分配集合

Does there exist an o-minimal first-order theory with a trans-exponential (rapid growth) function?

背景链接：https://arxiv.org/abs/2604.03477

02259 | AIM-REPRESENTATION_THEORY-0056

### Odd cohomology vanishes by a pointed-support filtration

S / solved | AIM 研讨会问题汇编

Let \(X\) be the moduli stack of torsion coherent sheaves on \(\mathbb P^1\) over \(\mathbb C\). Is it true that \(H^{\mathrm{odd}}(X,k)=0\) for all fields \(k\)?

(Motivated by KLR algebras for affine \(\mathfrak{sl}_2\).)

来源字段：http://aimpl.org/sheavemodular/11/

02260 | OWR-1452-010

### Odd-Dimensional Projective-Module Splitting

O / open | Oberwolfach 报告问题汇编

Let $X=\operatorname{Spec}A$ be a smooth affine variety of odd dimension $n\geq2$ over a characteristic-zero field, and let $P$ be a rank-$n$ projective $A$-module with $c_n(P)=0$ in $CH_0(X)$. Does there exist a rank-$(n-1)$ projective $A$-module $Q$ such that $P\simeq A\oplus Q$?

来源字段：https://doi.org/10.4171/owr/2007/01

02261 | AIM-REPRESENTATION_THEORY-0050

### Odd-prime Frobenius-twist geometry and formal Satake diagnostics

P / partially_solved | AIM 研讨会问题汇编

Under geometric Satake, what operation on perverse sheaves corresponds to Frobenius twist?

来源字段：http://aimpl.org/sheavemodular/7/

02262 | AMR-139-0002

### Open Problems in the Theory of Ample Fields - Conjecture 2 (Shafarevich)

O / open | AMR 问题清单汇编

The importance of this question lies in the fact that, via Pop's results on ample fields, a positive answer would give a proof of the following conjecture of Shafarevich, cf. [DD]:

The absolute Galois group $G_{\mathbb{Q}_\mathrm{ab}}$ of $\mathbb{Q}_\mathrm{ab}$ is a free profinite group.

背景链接：https://arxiv.org/abs/1106.1310

02263 | AMR-139-0007

### Open Problems in the Theory of Ample Fields - Conjecture 7 (Podewski)

O / open | AMR 问题清单汇编

An infinite field $K$ is called **minimal** if every subset $A\subseteq K$ which is definable by a first-order formula in the language of rings is finite or cofinite. For example, algebraically closed fields have this property, and an old conjecture of Podewski says that they should be the only ones:

Every minimal field is algebraically closed.

背景链接：https://arxiv.org/abs/1106.1310

02264 | AMR-139-0008

### Open Problems in the Theory of Ample Fields - Conjecture 8 (D\`ebes-Deschamps)

O / open | AMR 问题清单汇编

D\`ebes and Deschamps combined these two conjectures into one. A **regular solution** to an embedding problem $G\rightarrow\mathrm{Gal}(L/K)$ over $K$ is a solution $\gamma:\mathrm{Gal}(F/K(t))\rightarrow G$ to the induced embedding problem $G\rightarrow\mathrm{Gal}(L/K)\cong\mathrm{Gal}(L(t)/K(t))$ over $K(t)$ such that $F/L$ is regular.

Every finite split embedding problem over any field $K$ has a regular solution.

背景链接：https://arxiv.org/abs/1106.1310

02265 | AMR-139-0001

### Open Problems in the Theory of Ample Fields - Question 1

O / open | AMR 问题清单汇编

Our first question can be simply stated as follows.

Is $\mathbb{Q}_\mathrm{ab}$ ample?

背景链接：https://arxiv.org/abs/1106.1310

02266 | AMR-139-0003

### Open Problems in the Theory of Ample Fields - Question 3

O / open | AMR 问题清单汇编

Colliot-Th\'el\`ene observed that a field $K$ is ample whenever $G_K$ is a pro-$p$ group for some prime number $p$, [ColliotThelene,JardenAmple]. This covers again in particular the archimedean local fields $\mathbb{R}$ and $\mathbb{C}$. Since pro-$p$ groups are prosolvable, and all local fields have prosolvable absolute Galois groups we want to ask for the following generalization of Colliot-Th\'el\`ene's result:

Is every infinite field $K$ with prosolvable absolute Galois group $G_K$ ample?

背景链接：https://arxiv.org/abs/1106.1310

02267 | AMR-139-0004

### Open Problems in the Theory of Ample Fields - Question 4

O / open | AMR 问题清单汇编

Another Galois theoretic property that the local fields ${\mathbb{Q}}_p$, $\mathbb{R}$, and $\mathbb{C}$ share, apart from having a prosovable absolute Galois group, is that their absolute Galois groups are all finitely generated (as topological groups). The following question occurs as a conjecture in [JunkerKoenigsmann]:

Is every infinite field $K$ with finitely generated absolute Galois group $G_K$ ample?

背景链接：https://arxiv.org/abs/1106.1310

02268 | AMR-139-0005

### Open Problems in the Theory of Ample Fields - Question 5

O / open | AMR 问题清单汇编

Trivially, every ample field is virtually ample, and every virtually ample field is weakly ample. One could think that every virtually ample field should be ample, but no proof is known. As evidence one usually refers to the fact that similar descent properties hold for Henselian valued fields. We want to ask more generally:

What is the relation between ample, virtually ample, and weakly ample fields? Is every weakly ample field virtually ample? Is every virtually ample field ample?

背景链接：https://arxiv.org/abs/1106.1310

02269 | AMR-139-0006

### Open Problems in the Theory of Ample Fields - Question 6

O / open | AMR 问题清单汇编

A field $K$ is called **radically closed** if the multiplicative group $K^\times$ is divisible. Any radically closed field contains all roots of unity and hence, by Kummer theory, has no solvable finite Galois extensions. One would like to know the following:

Is every radically closed field ample?

背景链接：https://arxiv.org/abs/1106.1310

02270 | AMR-136-0022

### Open Problems on Central Simple Algebras - Conjecture 24 (Tate's Conjecture T)

O / open | AMR 问题清单汇编

Let $\ell\neq p$ be a prime. Let $P_2(X,t)$ be the characteristic polynomial of the action of the (geometric) Frobenius morphism on the $\ell$-adic cohomology ${\text\mathrm{H}}^2(\overline{X},{\mathbb{Q}}_{\ell}(1))$, where $\overline{X} = X \times_{k} k_{{\mathrm{sep}}}$. Let $\operatorname{NS}(X) = \mathrm{Pic}(X)/\mathrm{Pic}^0(X)$ denote the *N\'eron--Severi group* of $X$ and $\rho(X) = \operatorname{rank}_{{\mathbb{Z}}} \operatorname{NS}(X)$ its *Picard number*. Tate [Ta65, \mathbb{S}3] has earlier conjectured a relationship between the Picard number and this characteristic polynomial for surfaces.

The Picard number of $X$ is equal to the multiplicity of the root $t=q^{-1}$ in the polynomial $P_2(X,t)$.

背景链接：https://arxiv.org/abs/1006.3304

02271 | AMR-136-0023

### Open Problems on Central Simple Algebras - Conjecture 27 (BSD Conjecture)

O / open | AMR 问题清单汇编

Let $K$ be a global field of characteristic $p$, $A$ an abelian $K$-variety, and $L^*(A,s)$ the $L$-function of $A$ with respect to a suitably normalized Tamagawa measure. Then $L^*(A,s)$ has an analytic continuation to the complex plane and $$ \lim_{s\to 1} \frac{L^*(A,s)}{(s-1)^{r}} = \frac{|\mathrm{Ш}(A)| \det(h_A)}{|A(K)_{\mathrm{tors}}| |\hat{A}(K)_{\mathrm{tors}}|} $$ where $h_A$ is the N\'eron--Tate height pairing on $A(K)$ modulo torsion, $\hat{A}$ is the dual abelian variety, and $r = \operatorname{rank}_{{\mathbb{Z}}}A(K)$ is the rank of the Mordell--Weil group.

背景链接：https://arxiv.org/abs/1006.3304

02272 | AMR-136-0028

### Open Problems on Central Simple Algebras - Conjecture 32 (Amitsur's Conjecture)

O / open | AMR 问题清单汇编

He then posed the following converse of (2):

If $A$ and $B$ have the same degree and generate the same subgroup of the Brauer group, then $F(A)$ and $F(B)$ are isomorphic.

背景链接：https://arxiv.org/abs/1006.3304

02273 | AMR-136-0034

### Open Problems on Central Simple Algebras - Conjecture 39

O / open | AMR 问题清单汇编

The ``if'' direction can be found in [Vos, p. 186]. The full theorem amounts to combining the results from ibid.\ with [Gille:KT, Th. 5.9]. Gille's theorem furthermore shows that the conditions are equivalent to the abstract group $\operatorname{SL}_r(A)(L)$ being ``projectively simple'' for every extension $L/F$ and every $r \ge 2$.

The main problem from this point of view is to prove the following converse to Wang's result:

[Suslin \protect{[Suslin:SK1, p. 75]}] If $\operatorname{SK}_1(A \otimes L) = 0$ for every extension $L/F$, then the index of $A$ is square-free.

背景链接：https://arxiv.org/abs/1006.3304

02274 | AMR-136-0036

### Open Problems on Central Simple Algebras - Conjecture 41

O / open | AMR 问题清单汇编

[Brumer--Rosen \protect{[BrumerRosen]}] For a given field $F$ and each prime $p$, one of the following holds:

\renewcommand{\alph{enumi}}{\alph{enumi}}

- $(\operatorname{Br} F)_p = 0$.

- $(\operatorname{Br} F)_p$ contains a nonzero divisible subgroup.

- $p = 2$ and $(\operatorname{Br} F)_2$ is an elementary abelian $2$-group.

背景链接：https://arxiv.org/abs/1006.3304

02275 | AMR-136-0001

### Open Problems on Central Simple Algebras - Problem 1

O / open | AMR 问题清单汇编

Perhaps the most important open problem in the theory of central simple algebras is:

Given a prime $p$, construct a non-cyclic division algebra of degree $p$ over some field $F$.

背景链接：https://arxiv.org/abs/1006.3304

02276 | AMR-136-0010

### Open Problems on Central Simple Algebras - Problem 10

O / open | AMR 问题清单汇编

And one can ask the weaker question:

Is $\operatorname{Br}_p[n](F)$ generated by algebras of degree $n$?

背景链接：https://arxiv.org/abs/1006.3304

02277 | AMR-136-0011

### Open Problems on Central Simple Algebras - Problem 11

O / open | AMR 问题清单汇编

Suppose $m$ and $n$ have the same prime factors, $m\mid n$, and $\operatorname{Br}_p[m](F)$ is generated by cyclic algebras. Let $\ell_F(m,n)$ denote the minimal number such that every central simple $F$-algebra of degree $n$ and period (dividing) $m$ is similar to a tensor product of $\ell_F(m,n)$ cyclic algebras of degree $m$, unless no such bound exists, in which case set $\ell_F(m,n)=\infty$. If $F$ contains the $m$-th roots of unity, the generic division algebra can be used to show that $\ell_F(m,n)$ is finite, by the Merkurjev--Suslin theorem. In this case, we may assume $m$ is a power of a prime $p$ by [Tig84, Th. 2.2], and $\ell_F(m,n)$ is known as the *symbol length*.

Suppose $p$ is a prime, $r\leq s$, and $\operatorname{Br}_p[p^r](F)$ is generated by cyclic algebras. Compute $\ell_F(p^r,p^s)$.

背景链接：https://arxiv.org/abs/1006.3304

02278 | AMR-136-0012

### Open Problems on Central Simple Algebras - Problem 12

O / open | AMR 问题清单汇编

If $F$ is finitely generated over a field $F_0$ with $\operatorname{Br.\!dim}(F_0)$ finite, is $\operatorname{Br.\!dim}(F)$ necessarily finite?

It is natural to start by considering only certain fields $F_0$. For example, suppose that $F$ has prime characteristic $p$ and we take $F_0$ to be its prime field, so $\operatorname{Br.\!dim}(F_0) = 0$. If $F$ has transcendence degree 1 over $F_0$, then $F$ is global and $\operatorname{Br.\!dim}(F) = 1$. If $F$ has transcendence degree 2 over $F_0$, then $\operatorname{Br.\!dim}(F) \le 3$ by [Lieblich:PI].

背景链接：https://arxiv.org/abs/1006.3304

02279 | AMR-136-0013

### Open Problems on Central Simple Algebras - Problem 13

O / open | AMR 问题清单汇编

Define a notion of ``dimension'' for some class of fields $F$ such that \[ \operatorname{Br.\!dim}(F) \le \dim F - 1 \quad \text{and} \quad \dim(F(t)) = \dim F + 1. \] One possibility would be to set a $C_i$ field to have dimension $i$. The notion of cohomological dimension is obviously not the right one in view of Merkurjev's example of a field $F$ with cohomological dimension 2 and $\operatorname{Br.\!dim}(F) = \infty$ from [M:simple].

背景链接：https://arxiv.org/abs/1006.3304

02280 | AMR-136-0014

### Open Problems on Central Simple Algebras - Problem 14

O / open | AMR 问题清单汇编

To simplify the discussion note the result of Katsylo [Katsylo] and Schofield [Schofield] that if $n = ab$ for $a$ and $b$ relatively prime, then $Z(F,n)$ is stably isomorphic to the field compositum $Z(F,a) Z(F,b)$. Similar statements are possible for the $Z_m(F,n)$ and $Z_H(F,n)$. This often (but not always) allows reduction to the case where $n$ is a prime power.

*[procesi:67, p. 240]* Is $Z(F,n)/F$ rational, stably rational, or retract rational? The same question for $Z_m(F,n)$ and $Z_H(F,n)$.

背景链接：https://arxiv.org/abs/1006.3304

02281 | AMR-136-0015

### Open Problems on Central Simple Algebras - Problem 15

O / open | AMR 问题清单汇编

There are a few results for the $Z_m(F,n)$. The field $Z_2(F,4)$ is stably rational by Saltman [Salt:symp], and Saltman--Tignol [SaltTig] showed that $Z_2(F,8)$ is retract rational. Beneish [Beneish:28] showed that $Z_2(F,8)$ is stably rational. Finally, we leave to the well-read reader to prove that when $H \subset S_n$ is cyclic and transitive and $F$ contains a primitive $n$-th root of unity, then $Z_H(F,n)$ is rational (and results *are* available for general $F$, $n$ not of the form $8m$, and retract rationality). Thus the first interesting specific question along these lines is:

Determine if $Z_H(F,9)$ is rational, stably rational, or retract rational, where

$H = C_3 \times C_3 \subset S_9$ is transitive.

背景链接：https://arxiv.org/abs/1006.3304

02282 | AMR-136-0016

### Open Problems on Central Simple Algebras - Problem 16

O / open | AMR 问题清单汇编

Compute the essential dimension and the essential $p$-dimension of $\operatorname{PGL}_{n}$. In particular:

- Compute $\mathrm{ed}_F(\operatorname{PGL}_4)$ when $\operatorname{char} F=2$.

- Compute $\mathrm{ed}_F(\operatorname{PGL}_5)$.

背景链接：https://arxiv.org/abs/1006.3304

02283 | AMR-136-0017

### Open Problems on Central Simple Algebras - Problem 17

O / open | AMR 问题清单汇编

Determine structural conclusions about central simple algebras from essential ($p$-)dimension bounds.

- Calculate bounds for $\mathrm{ed}(\operatorname{CPAlg}_n)$ and $\mathrm{ed}(\mathrm{Alg}_G)$.

- Compute the difference $\mathrm{ed}(\mathrm{Alg}_G)-\mathrm{ed}(G)$.

- If $\mathrm{ed}(\operatorname{CPAlg}_n)=\mathrm{ed}(\operatorname{PGL}_n)$, determine whether every algebra of degree $n$ over every extension of $F$ is a crossed product.

背景链接：https://arxiv.org/abs/1006.3304

02284 | AMR-136-0018

### Open Problems on Central Simple Algebras - Problem 18

O / open | AMR 问题清单汇编

Compute $\mathrm{ed}_F(\operatorname{GL}_{8}/\mu_{2})$ when $\operatorname{char} F = 2$; $\mathrm{ed}_F(\operatorname{GL}_{16}/\mu_{2})$ when $\operatorname{char} F \ne 2$; and $\mathrm{ed}_F(\operatorname{GL}_{p^2}/\mu_{p})$ for odd primes $p$ and all $F$.

背景链接：https://arxiv.org/abs/1006.3304

02285 | AMR-136-0019

### Open Problems on Central Simple Algebras - Problem 19

O / open | AMR 问题清单汇编

Suppose $\sigma$ is an anisotropic orthogonal or symplectic involution on a central simple algebra $A$ over a field $F$. Does $\sigma$ remain anisotropic after scalar extension to any odd-degree field extension of $F$?

背景链接：https://arxiv.org/abs/1006.3304

02286 | AMR-136-0002

### Open Problems on Central Simple Algebras - Problem 2

O / open | AMR 问题清单汇编

As mentioned above, the smallest non-crossed product $p$-algebras known have degree $p^3$. Therefore even the following is open:

Determine whether $p$-algebras of degree $p^2$ are crossed products.

背景链接：https://arxiv.org/abs/1006.3304

02287 | AMR-136-0020

### Open Problems on Central Simple Algebras - Problem 20

O / open | AMR 问题清单汇编

Suppose $\sigma$ is an anisotropic orthogonal involution on a central simple algebra $A$ over a field $F$. Does $\sigma$ remain anisotropic after scalar extension to the function field $F_A$ of the Severi--Brauer variety of $A$?

背景链接：https://arxiv.org/abs/1006.3304

02288 | AMR-136-0021

### Open Problems on Central Simple Algebras - Problem 21

O / open | AMR 问题清单汇编

Let $G$ be an absolutely almost simple linear algebraic group that is anisotropic over $F$. Does $G$ remain anisotropic over every finite extension $K/F$ of dimension not divisible by any prime in $S(G)$?

背景链接：https://arxiv.org/abs/1006.3304

02289 | AMR-136-0024

### Open Problems on Central Simple Algebras - Problem 28

O / open | AMR 问题清单汇编

Given a geometrically connected, smooth, projective surface $X$ over a finite field $k$, is there a non-degenerate alternating pairing on $\operatorname{Br}(X)(2)$ with values in ${\mathbb{Q}}/{\mathbb{Z}}$?

背景链接：https://arxiv.org/abs/1006.3304

02290 | AMR-136-0025

### Open Problems on Central Simple Algebras - Problem 29

O / open | AMR 问题清单汇编

Is there a structural criterion for the tensor product $A\otimes_F B$ of two $F$-division algebras $A$ and $B$ to have a zero divisor?

背景链接：https://arxiv.org/abs/1006.3304

02291 | AMR-136-0003

### Open Problems on Central Simple Algebras - Problem 3

O / open | AMR 问题清单汇编

In view of the fact that all known examples of non-crossed product algebras occur over fields of cohomological dimension at least 3, it is natural to ask:

If $F$ has cohomological dimension at most 2, is every central simple $F$-algebra a crossed product?

背景链接：https://arxiv.org/abs/1006.3304

02292 | AMR-136-0026

### Open Problems on Central Simple Algebras - Problem 30

O / open | AMR 问题清单汇编

Suppose $(a,b)_n$ and $(c,d)_n$ are symbol algebras of degree $n$ over $F$, and $(a,b)_n \otimes_F (c,d)_n$ has index $< n^2$. Are there necessarily symbol algebras $(a',b')_n$ and $(c',d')_n$ that share a common maximal subfield and such that $(a, b)_n \otimes_F (c, d)_n \cong (a',b')_n \otimes_F (c', d')_n$?

背景链接：https://arxiv.org/abs/1006.3304

02293 | AMR-136-0027

### Open Problems on Central Simple Algebras - Problem 31

O / open | AMR 问题清单汇编

One can also consider splitting fields instead of maximal subfields:

If $\operatorname{ind}(A\otimes_F B) < \operatorname{ind} (A) \operatorname{ind}(B)$, what are the common splitting fields for the $F$-division algebras $A$ and $B$?

背景链接：https://arxiv.org/abs/1006.3304

02294 | AMR-136-0029

### Open Problems on Central Simple Algebras - Problem 33

O / open | AMR 问题清单汇编

Determine whether two division algebras that have the same splitting fields in a given class (say finite, or of transcendence degree at most one), necessarily generate the same subgroup in the Brauer group.

背景链接：https://arxiv.org/abs/1006.3304

02295 | AMR-136-0030

### Open Problems on Central Simple Algebras - Problem 34

O / open | AMR 问题清单汇编

The smallest open case appears to be:

Determine whether two division algebras of degree $3$ that have the same splitting fields of transcendence degree $\leq 1$ generate the same subgroup of the Brauer group.

背景链接：https://arxiv.org/abs/1006.3304

02296 | AMR-136-0031

### Open Problems on Central Simple Algebras - Problem 35

O / open | AMR 问题清单汇编

Show that if $F$ is a field which is finitely generated over a prime field or over an algebraically closed field, then any two quaternion division algebras having the same finite splitting fields (or even just the same maximal subfields) must be isomorphic.

背景链接：https://arxiv.org/abs/1006.3304

02297 | AMR-136-0032

### Open Problems on Central Simple Algebras - Problem 37

O / open | AMR 问题清单汇编

Let $F$ be a field and $G$ a finite group. We say $G$ is *$F$-admissible* if there exists a $G$-crossed product $F$-division algebra $A$.

Determine which groups are $F$-admissible, for a field $F$.

背景链接：https://arxiv.org/abs/1006.3304

02298 | AMR-136-0033

### Open Problems on Central Simple Algebras - Problem 38

O / open | AMR 问题清单汇编

This problem is an extension of the inverse Galois problem, and is especially interesting for $F={\mathbb{Q}}$. It appears as part of Problem 5 in [Am:surv].

The next problem, also known as the \emph{${\mathbb{Q}}$-admissibility conjecture}, was first recognized in [Schacher:th1].

Prove every Sylow-metacyclic group is ${\mathbb{Q}}$-admissible.

背景链接：https://arxiv.org/abs/1006.3304

02299 | AMR-136-0004

### Open Problems on Central Simple Algebras - Problem 4

O / open | AMR 问题清单汇编

Determine the smallest splitting group(s) of an $F$-division algebra $A$ arising from the Galois closure of a maximal subfield of $A$. Determine in particular whether every $A$ of degree $n$ has a separable maximal subfield whose Galois closure has group the alternating group $A_n$.

背景链接：https://arxiv.org/abs/1006.3304

02300 | AMR-136-0035

### Open Problems on Central Simple Algebras - Problem 40

O / open | AMR 问题清单汇编

Amongst the first examples one sees, probably the fields ${\mathbb{R}}$ or iterated Laurent series ${\mathbb{R}}((x_1))((x_2))\cdots((x_n))$ are the only ones whose Brauer group is nonzero and finite. This leads to the following naive question:

For which natural numbers $n$ is there a field $F$ such that $|\!\operatorname{Br}(F)| = n$?

背景链接：https://arxiv.org/abs/1006.3304

02301 | AMR-136-0005

### Open Problems on Central Simple Algebras - Problem 5

O / open | AMR 问题清单汇编

Determine whether every division algebra is split by an abelian group, and find degree bounds over specific fields.

背景链接：https://arxiv.org/abs/1006.3304

02302 | AMR-136-0006

### Open Problems on Central Simple Algebras - Problem 6

O / open | AMR 问题清单汇编

Determine groups $G$ and $G'$ and conditions on $F$ under which $G\Rightarrow_F G'$. In particular:

- Determine, for $m$ prime-to-$p$, whether $C_p\rtimes C_m\Rightarrow_F C_p\times C_m$.

- Determine whether all quasi-symbols are cyclic.

背景链接：https://arxiv.org/abs/1006.3304

02303 | AMR-136-0007

### Open Problems on Central Simple Algebras - Problem 7

O / open | AMR 问题清单汇编

{\textrm{[Albert:cyclics, p. 126]}} Is the $n$-torsion subgroup $\operatorname{Br}_p[n](F)$ generated by the classes of cyclic algebras of degree (dividing) $n$?

背景链接：https://arxiv.org/abs/1006.3304

02304 | AMR-136-0008

### Open Problems on Central Simple Algebras - Problem 8

O / open | AMR 问题清单汇编

- *$F$ contains a primitive $n$-th root of unity.* This was pointed out above. The Merkurjev-Suslin theorem, which establishes an isomorphism between $K^M_2(F) / nK^M_2(F)$ and ${\text\mathrm{H}}^2(F,\mu_n^{\otimes 2})$, is now a special case of the recently-proved Bloch--Kato Conjecture.

The smallest open case for $n$ prime is therefore:

Is $\operatorname{Br}_p[7](F)$ generated by cyclic algebras of degree $7$ when $\operatorname{char} F\neq 7$ and $F$ does not contain a primitive $7$-th root of unity?

背景链接：https://arxiv.org/abs/1006.3304

02305 | AMR-136-0009

### Open Problems on Central Simple Algebras - Problem 9

O / open | AMR 问题清单汇编

(These problems are related to the following very special question stated by M. Mahdavi-Hezavehi: \emph{If the multiplicative group $F^\times$ is divisible, does it follow that $\operatorname{Br}(F)$ is zero?} The answer is ``yes'' if $\operatorname{Br}(F)$ is generated by cyclic algebras, because cyclic algebras are split over such a field. Therefore the answer is also ``yes'' if $F$ has characteristic zero, because in that case $F$ has all the roots of unity so the Merkurjev--Suslin theorem applies.)

For composite $n$, even the following is unclear:

Is $\operatorname{Br}_p[4](F)$ generated by cyclic algebras of degree $4$ when $\operatorname{char} F \neq 2$ and $\sqrt{-1} \not\in F$?

背景链接：https://arxiv.org/abs/1006.3304

02306 | OWR-14299281-008

### Open Problems on Ind-Groups of Automorphisms

O / open | Oberwolfach 报告问题汇编

Let $X$ be an affine variety. Determine whether the following assertions hold. First, if $G\subseteq\operatorname{Aut}(X)$ is connected and closed and $N\triangleleft G$ is proper and closed, then $\operatorname{Lie}(N)\subsetneq\operatorname{Lie}(G)$. Second, if closed subgroups $H,G\subseteq\operatorname{Aut}(X)$ satisfy that $G$ is connected, $\operatorname{Lie}(G)\subseteq\operatorname{Lie}(H)$, and every element of $G$ is algebraic, then $G\subseteq H$. Third, a closed subgroup consisting of algebraic elements, and more generally the closure of any such subgroup, is nested. Fourth, if $\operatorname{Aut}(X)^\circ$ is nontrivial, it contains a positive-dimensional algebraic subgroup. Fifth, the normalizer of the tame subgroup $\langle\operatorname{SAff}_n,\operatorname{SJonq}_n\rangle$ is dense in $\operatorname{SAut}(\mathbb A^n)$.

来源字段：https://doi.org/10.4171/owr/2025/20

02307 | OWR-11580-015

### Oper Support of Induced Critical-Level Modules

P / partially_solved | Oberwolfach 报告问题汇编

If $Z(\mathfrak{g}_n)$ acts on a $\mathfrak{g}_n$-module $V$ through $\Lambda\in(\mathfrak{h}^*/W)_n$, prove that the induced critical-level module is scheme-theoretically supported on the oper subscheme $\operatorname{Op}^{\operatorname{ord}_n}_{\check G}(D)_{-\Lambda}$.

来源字段：https://doi.org/10.4171/owr/2012/13

02308 | OWR-1062-003

### Operator Models for Universal Hopf $C^*$-Algebras

O / open | Oberwolfach 报告问题汇编

For universal Hopf $C^*$-algebras beyond $A_u(n)$, $A_o(n)$, and $A_s(n)$, in particular quantum automorphism algebras $A(X)$ of finite graphs, find concrete operator models and analogues of the spectral-measure faithfulness criterion used for the basic examples.

来源字段：https://doi.org/10.4171/owr/2005/41

02309 | OWR-8415355-010

### Operator Relations on Yified Khovanov-Rozansky Homology

P / partially_solved | Oberwolfach 报告问题汇编

What are the relations in the algebra acting on y-ified Khovanov-Rozansky homology that is generated by $x_i$, $y_i$, $F_k$, and the $\mathfrak{sl}_2$ operators $f$ and $h$?

来源字段：https://doi.org/10.4171/owr/2021/52

02310 | OWR-15219-001

### Optimal Degree Bounds for Sum-of-Squares Representations

P / partially_solved | Oberwolfach 报告问题汇编

Determine the best possible degree bounds for the polynomials in sum-of-squares representations of nonnegative polynomials, or equivalently in the associated rational-function representations.

来源字段：https://doi.org/10.4171/owr/2017/14

02311 | OWR-17289-004

### Orbit Closures in Degenerate Flag Varieties

P / partially_solved | Oberwolfach 报告问题汇编

For closures of $G_a$-orbits in the degenerate flag variety $\operatorname{Fl}^a(V)$, parametrized by tuples of coordinate subsets, what is the closure relation, what are the minimal singularities, and how can the intersection cohomology complexes of the orbit closures be described?

来源字段：https://doi.org/10.4171/owr/2019/46

02312 | OWR-17289-008

### Orbit Structure of Maximal Flat Flag Degenerations

O / open | Oberwolfach 报告问题汇编

For the natural action of $\operatorname{Aut}_Q(M)$ on $\operatorname{Fl}^{mf}(V)$, are there finitely many orbits, how can they be parametrized, and what is their closure relation?

来源字段：https://doi.org/10.4171/owr/2019/46

02313 | OWR-13494-020

### Ordinary and Symbolic Powers of Point-Set Ideals

O / open | Oberwolfach 报告问题汇编

Which Cohen-Macaulay point sets $X$ in $\mathbb P^1\times\mathbb P^1$, equivalently line unions in $\mathbb P^3$, satisfy $I(X)^3=I(X)^{(3)}$; and for every positive integer $t$, does there exist an ideal $I_t$ with $I_t^m=I_t^{(m)}$ for $m\leq\operatorname{bight}(I_t)+t-1$ but $I_t^m\ne I_t^{(m)}$ when $m=\operatorname{bight}(I_t)+t$?

来源字段：https://doi.org/10.4171/owr/2015/9

02314 | OWR-8415354-003

### Origins of Yangian R-Matrix Factors

P / partially_solved | Oberwolfach 报告问题汇编

Can the diagonal factor $R_0(s)$ of a Yangian R-matrix be derived from four-dimensional Chern-Simons theory, and what is the interpretation of its lower-triangular factor $R_-(s)$ in Maulik-Okounkov geometry?

来源字段：https://doi.org/10.4171/owr/2021/51

02315 | OWR-14298593-005

### Ozone Groups of Calabi-Yau and PI AS-Regular Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Classify the Calabi-Yau algebras whose ozone group $\operatorname{Oz}(A)=\operatorname{Gal}(A/Z(A))$ is trivial. Also classify the PI Artin-Schelter regular algebras whose ozone groups have maximal rank.

来源字段：https://doi.org/10.4171/owr/2024/44

02316 | AIM-REPRESENTATION_THEORY-0016

### Pairwise independence of integral FI-models away from finitely many primes

S / solved | AIM 研讨会问题汇编

Given a map $\phi_{\mathbb Q}:V_{\mathbb Q}\to W_{\mathbb Q}$ of finitely generated FI-modules over $\mathbb Q$ (or perhaps over an algebraically closed field), and choices $V,W$ of lattices in the two rational FI-modules, is the reduced map $$ V\otimes_{\mathbb Z}\mathbb F_p\longrightarrow W\otimes_{\mathbb Z}\mathbb F_p $$ independent of $V,W$, up to isomorphism, for sufficiently large $p$?

来源字段：http://aimpl.org/repnstability/1/

02317 | OWR-16171-011

### Parametrization of Supersingular Cohomology Modules

P / partially_solved | Oberwolfach 报告问题汇编

Let $G = \mathrm{GL}_n(\mathbb{Q}_p)$ (or a reductive $p$-adic group), let $B_s^b$ be a Deligne--Lusztig-type or Bruhat--Tits stratum in the $p$-adic period domain, and let $H_s^{(1)}$ be the pro-$p$ Iwahori--Hecke algebra at a supersingular parameter $s$. Can the supersingular $H_s^{(1)}$-modules appearing in the $\bmod\, p$ cohomology of $B_s^b$ be canonically parametrized by the central parameter $t$?

来源字段：https://doi.org/10.4171/owr/2018/30

02318 | ALG-042

### Perfect Cuboid

P / partially_solved | 独立 / 未分配集合

别名：Perfect Cuboid [ALG-060]; Perfect cuboid [OPG-37221]

Does there exist a rectangular cuboid with integer edges, face diagonals, and space diagonal?

背景链接：https://arxiv.org/abs/2401.06784

02319 | OWR-11580-019

### Perfect Weyl-Group Complexes of Flag Varieties

O / open | Oberwolfach 报告问题汇编

For a complex reductive group $G$ with maximal torus $T$ and Weyl group $W$, describe the perfect complex $R\Gamma(G/T,k)$ of $k[W]$-modules for an arbitrary field $k$.

来源字段：https://doi.org/10.4171/owr/2012/13

02320 | OWR-12868-006

### Periodic Simple Modules and Periodic Algebras

O / open | Oberwolfach 报告问题汇编

Let $A$ be a finite-dimensional algebra. If every simple module in $\operatorname{mod}A$ is periodic, must $A$ be a periodic algebra? Classify periodic algebras up to Morita and derived equivalence.

来源字段：https://doi.org/10.4171/owr/2014/08

02321 | OWR-12868-007

### Periodicity of Stable Auslander Algebras

O / open | Oberwolfach 报告问题汇编

Is the stable Auslander algebra of every hypersurface singularity of finite Cohen-Macaulay type periodic?

来源字段：https://doi.org/10.4171/owr/2014/08

02322 | OWR-17466-015

### Periodicity of Symmetric Tame Algebras

O / open | Oberwolfach 报告问题汇编

If $A$ is an indecomposable symmetric tame algebra of non-polynomial growth for which every simple module is periodic, must $A$ be a periodic algebra of period $4$?

来源字段：https://doi.org/10.4171/owr/2020/3

02323 | OWR-12175-007

### Perturbation Stability of Jiang-Su Absorption for $C^*$-Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist $\delta>0$ such that, if a separable $C^*$-algebra $A$ absorbs the Jiang-Su algebra tensorially, $A\cong A\otimes\mathcal Z$, and $d(A,B)<\delta$, then $B\cong B\otimes\mathcal Z$?

来源字段：https://doi.org/10.4171/owr/2012/52

02324 | AIM-REPRESENTATION_THEORY-0044

### Perversity of normalized parity sheaves on flags and affine-Grassmannian truncations

P / partially_solved | AIM 研讨会问题汇编

1. For which primes \(p\) is it true that all indecomposable \(B\)-constructible parity \(\mathbb F_p\)-sheaves on \(G/B\) are perverse? 2. Same question for \(I\)-constructible parity \(\mathbb F_p\)-sheaves on \(\mathrm{Gr}\).

(Expect this to be true for large \(p\).)

来源字段：http://aimpl.org/sheavemodular/5/

02325 | OWR-1465-007

### Pierce-Birkhoff Supremum-Infimum Representation Conjecture

O / open | Oberwolfach 报告问题汇编

Let $A=\mathbb R[x_1,\ldots,x_n]$ and let $f$ be a piecewise-polynomial function on $\mathbb R^n$. Must there be a finite family of polynomials $g_{ij}\in A$ such that $f(x)=\sup_i\inf_j g_{ij}(x)$ for every $x\in\mathbb R^n$?

来源字段：https://doi.org/10.4171/owr/2007/14

02326 | OWR-3397-003

### Poisson Dixmier-Moeglin Equivalence

S / solved | Oberwolfach 报告问题汇编

Does the Dixmier-Moeglin equivalence hold for Poisson-primitive ideals of affine Poisson algebras? Equivalently, when do the Poisson-primitive, Poisson-rational, and Poisson-locally-closed prime ideals coincide?

来源字段：https://doi.org/10.4171/owr/2009/14

02327 | OWR-14298013-005

### Poisson Nilpotency and Solvability in Small Characteristics

O / open | Oberwolfach 报告问题汇编

Let $L$ be a Lie algebra over a field of characteristic $p>0$. If $s(L)$ is Lie nilpotent, do its Lie nilpotency class and strong Lie nilpotency class coincide for $p=2$ or $3$? If $p>2$ and $s(L)$ is solvable, do its derived length and strong derived length coincide?

来源字段：https://doi.org/10.4171/owr/2023/46

02328 | OWR-8415354-004

### Pole Orders and Extensions of Kirillov-Reshetikhin Modules

O / open | Oberwolfach 报告问题汇编

For Kirillov-Reshetikhin modules $V$ and $W$, is the order of the pole at $z=1$ of the normalized R-matrix between $V$ and $W$ equal to the dimension of the first extension group between their generic-kernel modules over the associated generalized preprojective algebra?

来源字段：https://doi.org/10.4171/owr/2021/51

02329 | OWR-15429-011

### Polynomial Identities and Symbolic Calculus for Noncommutative Algebras

O / open | Oberwolfach 报告问题汇编

For a noncommutative algebra $A$, particularly a matrix algebra, characterize its polynomial identities and construct the symbolic calculus determined by those identities.

来源字段：https://doi.org/10.4171/owr/2017/21

02330 | OWR-8415346-007

### Polynomial Specht Multiplicities in Symmetric-Variety Cohomology

P / partially_solved | Oberwolfach 报告问题汇编

Let a symmetric ideal in the inverse-limit ring of symmetric polynomials define a sequence of symmetric varieties $V_n$, and fix a partition $\lambda$ and an integer $p\geq0$. Is the multiplicity of the Specht module indexed by $\{\lambda\}_n$ in $H^p(V_n;\mathbb Q)$ eventually a polynomial in $n$?

来源字段：https://doi.org/10.4171/owr/2021/43

02331 | OWR-11133-002

### Polynomial Subalgebras Admitting SAGBI Bases

O / open | Oberwolfach 报告问题汇编

Which subalgebras of a polynomial ring have a SAGBI basis?

来源字段：https://doi.org/10.4171/owr/2011/41

02332 | OWR-13108-013

### Positive Bases for Surface Skein Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Let $S(\Sigma)$ be the Kauffman bracket skein algebra of a surface $\Sigma$ over $\mathbb Z[t^{\pm1}]$. For an integer lamination $\mu=\{(a_i,C_i)\}$, set $b_T(\mu)=\prod_i T_{a_i}(C_i)$ using the Chebyshev polynomials $T_j$. Is $b_T=\{b_T(\mu)\}$ a positive basis of $S(\Sigma)$, meaning that all multiplication structure constants lie in $\mathbb Z_{\geq0}[t^{\pm1}]$?

来源字段：https://doi.org/10.4171/owr/2014/38

02333 | OWR-12697689-002

### Positive Characteristic Cohomology of Flag Variety Line Bundles

P / partially_solved | Oberwolfach 报告问题汇编

Determine the cohomology groups of line bundles on a flag variety in characteristic $p>0$, including their general vanishing and nonvanishing behavior.

来源字段：https://doi.org/10.4171/owr/2023/6

02334 | OWR-14298157-001

### Positivity Criterion for Quantum Greedy Bases

P / partially_solved | Oberwolfach 报告问题汇编

For the quantum greedy basis, is positivity preserved whenever $\delta_1\mid\delta_2$ or $\delta_2\mid\delta_1$?

来源字段：https://doi.org/10.4171/owr/2024/2

02335 | OWR-11101920-002

### Positivity in Affine Kazhdan-Lusztig Theory

P / partially_solved | Oberwolfach 报告问题汇编

In affine type $A$, is the decomposition of $N_{\lambda_{i+1}}$ in terms of the $N_{\mu_i}$ positive?

来源字段：https://doi.org/10.4171/owr/2022/39

02336 | OWR-4216-002

### Positivity in Specialized Macdonald-Polynomial Expansions

O / open | Oberwolfach 报告问题汇编

Let $\lambda$ be a partition of $n$ and let $k\in\mathbb N$. Is the coefficient of $s_\lambda$ in $J_\lambda(X;q,q^k)/(1-q)^n$ always a polynomial in $\mathbb N[q]$?

来源字段：https://doi.org/10.4171/owr/2010/15

02337 | OWR-4800-001

### Positivity of Cohomological and K-Theoretic Quiver Coefficients

P / partially_solved | Oberwolfach 报告问题汇编

For any quiver cycle $\Omega\subset V$, are the cohomological quiver coefficients $c_{\mu}(\Omega)$ nonnegative when $|\mu|=\operatorname{codim}(\Omega;V)$, are only finitely many $K$-theoretic coefficients $c_{\mu}(\Omega)$ nonzero, and, if $\Omega$ has rational singularities, do they satisfy $(-1)^{|\mu|-\operatorname{codim}(\Omega;V)}c_{\mu}(\Omega)\geq0$?

来源字段：https://doi.org/10.4171/owr/2011/10

02338 | OWR-15217-008

### Positivity of Kac-Type Polynomials for Concealed-Canonical Algebras

O / open | Oberwolfach 报告问题汇编

If $C$ is an almost concealed-canonical algebra, are all coefficients of the polynomial $P_{C,d}(T)$ nonnegative for every dimension vector $d$?

来源字段：https://doi.org/10.4171/owr/2017/12

02339 | OWR-4800-008

### Positivity of Quiver Laurent Polynomials $\Omega_\alpha$

P / partially_solved | Oberwolfach 报告问题汇编

For every quiver $Q$ and $\alpha\in\mathbb N^{Q_0}$, is $\Omega_\alpha(-q^{1/2})$ a Laurent polynomial in $q^{1/2}$ with nonnegative integer coefficients?

来源字段：https://doi.org/10.4171/owr/2011/10

02340 | OWR-12494-004

### Possible Values of Characteristic-Two Quadratic-Form Invariants

P / partially_solved | Oberwolfach 报告问题汇编

Let $F$ be a field of characteristic $2$, let $Q'(F)$ be the set of anisotropic quadratic forms over $F$ that are not totally singular, and let $\operatorname{rdim}(\varphi)$ and $\operatorname{ql}(\varphi)$ denote the regular dimension and quasilinear part of $\varphi$. Determine the possible values of

$$\widetilde u(F)=\sup_{\varphi\in Q'(F)}\dim\varphi,$$

$$u_r(F)=\sup\{\dim\varphi:\varphi\in Q'(F),\ \operatorname{rdim}(\varphi)=2r\},$$

and $\widetilde u_s(F)=\sup\{\dim\varphi:\varphi\in Q'(F),\ \dim\operatorname{ql}(\varphi)=s\}$, for integers $r,s\geq 1$.

来源字段：https://doi.org/10.4171/owr/2013/31

02341 | OWR-14298013-001

### Post-Lie Structures with Semisimple Target

O / open | Oberwolfach 报告问题汇编

Let $\mathfrak g$ be a perfect nonsemisimple complex Lie algebra and let $\mathfrak n$ be semisimple. Prove that there is no post-Lie algebra structure on $(\mathfrak g,\mathfrak n)$.

来源字段：https://doi.org/10.4171/owr/2023/46

02342 | OWR-12732-002

### Potentials for Quivers from Twice-Punctured Surfaces

O / open | Oberwolfach 报告问题汇编

For quivers arising from triangulations of closed positive-genus surfaces with exactly two punctures, does there exist a unique non-degenerate potential up to weak right equivalence?

来源字段：https://doi.org/10.4171/owr/2013/58

02343 | OWR-734-005

### Powers of Linearly Presented Primary Ideals

O / open | Oberwolfach 报告问题汇编

If $I$ is a homogeneous $\mathfrak m$-primary ideal in $S=K[x_1,\ldots,x_n]$, generated in one degree $d$ and linearly presented, must $I^{n-1}=\mathfrak m^{(n-1)d}$?

来源字段：https://doi.org/10.4171/owr/2004/32

02344 | OWR-17294-010

### Prevalence of Multipermutation Yang-Baxter Solutions

O / open | Oberwolfach 报告问题汇编

Does the ratio of the number of multipermutation Yang-Baxter solutions of size $n$ to the total number of solutions of size $n$ tend to $1$ as $n\to\infty$?

来源字段：https://doi.org/10.4171/owr/2019/51

02345 | OWR-1272-005

### Prime Affine Hopf Algebras of GK Dimension One

P / partially_solved | Oberwolfach 报告问题汇编

Describe all prime affine Hopf algebras of Gelfand-Kirillov dimension 1.

来源字段：https://doi.org/10.4171/owr/2006/23

02346 | OWR-4335-003

### Prime Ideals and Coordinate-Ring Maps of $R(\alpha,\beta)$

O / open | Oberwolfach 报告问题汇编

Let $k$ be an uncountable algebraically closed field, let $\alpha,\beta$ be algebraically independent over its prime subfield, and let $$R=R(\alpha,\beta)=k\langle x_1,x_2,x_3,x_4\rangle/(f_1,\ldots,f_6),$$ where $$\begin{aligned}f_1&=x_1(\alpha x_1-x_3)+x_3(x_1-\alpha x_3),&f_2&=x_1(\alpha x_2-x_4)+x_3(x_2-\alpha x_4),\\f_3&=x_2(\alpha x_1-x_3)+x_4(x_1-\alpha x_3),&f_4&=x_2(\alpha x_2-x_4)+x_4(x_2-\alpha x_4),\\f_5&=x_1(\beta x_1-x_2)+x_4(x_1-\beta x_2),&f_6&=x_1(\beta x_3-x_4)+x_4(x_3-\beta x_4).\end{aligned}$$ Is the augmentation ideal $R_+$ the only nontrivial graded prime ideal of $R$, and does $R$ admit no nontrivial map to a twisted homogeneous coordinate ring?

来源字段：https://doi.org/10.4171/owr/2010/22

02347 | OWR-1274-003

### Prime Ideals in Iwasawa Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a uniform, torsion-free, $p$-adic analytic pro-$p$ group whose $\mathbb{Q}_p$-Lie algebra is simple (for example, a congruence subgroup of $\mathrm{SL}_n(\mathbb{Z}_p)$), and let $\Omega G = \mathbb{F}_p[[G]]$ be its mod-$p$ Iwasawa algebra. Are $\{0\}$ and the Jacobson radical $J(\Omega G)$ the only two-sided prime ideals of $\Omega G$?

来源字段：https://doi.org/10.4171/owr/2006/25

02348 | OWR-17294-004

### Prime Quotients of Yang-Baxter Structure Algebras

P / partially_solved | Oberwolfach 报告问题汇编

For a prime ideal $P$ of $kM(X,r)$, determine the structure of $kM(X,r)/P$, including when it is a domain, a maximal order, Koszul, Auslander-regular, Cohen-Macaulay, or otherwise homologically well behaved. When does the quotient come from a related Yang-Baxter solution, and how do its properties affect $(X,r)$?

来源字段：https://doi.org/10.4171/owr/2019/51

02349 | OWR-1265-011

### Prime-Reduced Standard and Costandard Modules

O / open | Oberwolfach 报告问题汇编

Assume $p>h$. For each regular dominant weight $\lambda=\lambda_0+p\lambda_1$, with $\lambda_0\in X_1^+$ and $\lambda_1\in X^+$, set $\Delta'_{\mathrm{red}}(\lambda)=L(\lambda_0)\otimes\Delta(\lambda_1)^{(1)}$ and $\nabla'_{\mathrm{red}}(\lambda)=L(\lambda_0)\otimes\nabla(\lambda_1)^{(1)}$. Must $\Delta'_{\mathrm{red}}(\lambda)[-l(\lambda)]\in E^L$ and $\nabla'_{\mathrm{red}}(\lambda)[-l(\lambda)]\in E^R$?

来源字段：https://doi.org/10.4171/owr/2006/15

02350 | OWR-8415355-003

### Primitive Generators of Preprojective Cohomological Hall Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Let $Q$ be a quiver with preprojective algebra $\Pi_Q$, let $\mathcal{A}_{\Pi_Q}$ be its cohomological Hall algebra (CoHA), and for a dimension vector $d$ let $\operatorname{IC}(X_d)$ denote the intersection cohomology of the coarse moduli space $X_d$ of semisimple $\Pi_Q$-modules of dimension vector $d$. Does $\operatorname{IC}(X_d)$ form a complete space of primitive degree-$d$ generators for the preprojective cohomological Hall algebra $\mathcal{A}_{\Pi_Q}$?

来源字段：https://doi.org/10.4171/owr/2021/52

02351 | OWR-1274-015

### Principal Ideals and Characteristic Elements for Iwasawa Modules

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a compact $p$-valued group with no elements of order $p$, and let $\Lambda_G$ be its Iwasawa algebra. For a finitely generated torsion $\Lambda_G$-module, can the reflexive right ideals $L_i$ in the structure theorem be replaced by principal ideals? How should a characteristic element be defined in general?

来源字段：https://doi.org/10.4171/owr/2006/25

02352 | AMR-021-0001

### Problems Around Polynomials - Conjecture 1

P / partially_solved | AMR 问题清单汇编

[ Maxwell, seems bad, no tools] For any system of $N$ isolated fixed point charges in $\mathbb{R}^3$, the number of points of equilibrium (assumed finite) of the created electrostatic field $$E(\bar x)=\sum_{i=1}^N\frac{\xi_i(\bar x-\bar x_i)}{|\bar x-\bar x_i|^3},$$ is at most $(N-1)^2$. Here $\xi_i$ is a charge placed at $\bar x_i\in \mathbb{R}^3$ and $\bar x\in \mathbb{R}^3$ is a variable vector.

背景链接：https://arxiv.org/abs/1503.05295

02353 | AMR-021-0015

### Problems Around Polynomials - Conjecture 10

O / open | AMR 问题清单汇编

[J. Forsg\aa rd, B. Sh., seems good, see ]] Let $f(z) = \sum_{k=0}^n a_k z^k$ be a polynomial with positive coefficients. Consider the differences \[ c_k = a_k^2 - a_{k-1}a_{k+1}, \] where $a_{-1} = a_{n+1} = 0$. Let $0=k_1 < k_2 < \dots < k_m = n$ be the sequence of indices such that $c_{k_i}$ is non-negative, and let $v(f)$ be the number of changes in the sequence $\{k_i \mod 2\}_{i=0}^m$. Then the number of real zeros of $f(z)$ does not exceed $v(f)$.

背景链接：https://arxiv.org/abs/1503.05295

02354 | AMR-021-0018

### Problems Around Polynomials - Conjecture 11

P / partially_solved | AMR 问题清单汇编

[B. Sh., seems good] For any real polynomial $p(x)$ of degree $k$ with simple real zeros, $$ \#_{r}\left[(k-1)(p'(x))^2-kp(x)p''(x)\right] \le \#_{nr}p(x), $$ i.e. "Hawaiian`` conjecture holds for $G_1(x)$ as well.

背景链接：https://arxiv.org/abs/1503.05295

02355 | AMR-021-0019

### Problems Around Polynomials - Conjecture 12

P / partially_solved | AMR 问题清单汇编

[B. Sh] For any real polynomial $p(x)$ of even degree, $$ \#_{r}\left[(k-1)(p'(x))^2-kp(x)p''(x)\right] + \#_{r}p(x)>0, $$

背景链接：https://arxiv.org/abs/1503.05295

02356 | AMR-021-0020

### Problems Around Polynomials - Conjecture 13

O / open | AMR 问题清单汇编

[B. Sh] For any degree $k$ polynomial $p(x)$ with real coefficients, $$ \#_{r}P_{i}(x) \le \min\{{\deg{P_i(x)},k}\}. $$

背景链接：https://arxiv.org/abs/1503.05295

02357 | AMR-021-0002

### Problems Around Polynomials - Conjecture 2

O / open | AMR 问题清单汇编

[folklore, very irritating] For any set of charges of the same sign in $\mathbb{R}^n$, the set of its points of equilibrium is finite.

背景链接：https://arxiv.org/abs/1503.05295

02358 | AMR-021-0003

### Problems Around Polynomials - Conjecture 3

O / open | AMR 问题清单汇编

[A. Gabrielov, D. Novikov, B. Sh., seems good, but no progress] Let $(x_1,y_1),(x_2,y_2),\dots, (x_N,y_N)$ be a collection of points in $\mathbb{R}^2$, $\xi_1, \xi_2, \dots , \xi_N$ be arbitrary real charges and $\alpha\ge 1/2$. Then the rational univariate function $$\Psi(x)=\sum_{i=1}^N\frac{\xi_i}{((x-x_i)^2+y_i^2)^{\alpha}},\; x\in \mathbb{R},$$ has at most $N$ local maxima on the whole real line.

背景链接：https://arxiv.org/abs/1503.05295

02359 | AMR-021-0007

### Problems Around Polynomials - Conjecture 4

O / open | AMR 问题清单汇编

[G. Ottaviani, B. Sh., seems good] For any number of variables, $\widetilde\#(2k,l)=k^l$.

背景链接：https://arxiv.org/abs/1503.05295

02360 | AMR-021-0009

### Problems Around Polynomials - Conjecture 5

O / open | AMR 问题清单汇编

[P. Br\"anden, I. Krasikov, B. Sh., hopefully good, see ] A difference operator $T(p(x))=a_0p(x)+a_1p(x-1)+\cdots+a_kp(x-k)$ with constant coefficients preserves the set of real-rooted polynomials of degree at most $m$ whose mesh is at least $1$ if and only if the polynomial $T((x)_m)$ is real-rooted and has mesh at least one. Here $(x)_m=x(x-1)(x-2)\dots (x-m+1)$ is the $m$-th Pochhammer polynomial.

背景链接：https://arxiv.org/abs/1503.05295

02361 | AMR-021-0010

### Problems Around Polynomials - Conjecture 6

O / open | AMR 问题清单汇编

If $p$ and $q$ are real-rooted polynomials of degree at most $d$ and of mesh $\geq 1$, then so is $p \bullet q$.

背景链接：https://arxiv.org/abs/1503.05295

02362 | AMR-021-0012

### Problems Around Polynomials - Conjecture 7

S / solved | AMR 问题清单汇编

[J. Forsg\aa rd, V. Kostov, B. Sh, hopefully good, see ] For an arbitrary sign pattern $\sigma$, the only type of pairs $(pos,neg)$ which can be non-realizable has either $pos$ or $neg$ vanishing. In other words, for any sign pattern $\sigma$, each pair $(pos,neg)$ satisfying [source label: stand] with positive $pos$ and $neg$ is realizable.

背景链接：https://arxiv.org/abs/1503.05295

02363 | AMR-021-0013

### Problems Around Polynomials - Conjecture 8

O / open | AMR 问题清单汇编

[J. Forsg\aa rd, B. Sh., seems good, see ] Let $f(z) = \sum_{k=0}^n a_k z^k$ be a polynomial with positive coefficients, and consider the related (weighted) tropical polynomial \[ f_{trop}(x) =\max_{k}\left(\Log(a_k)+k x + \Log{n\choose k}\right). \] Then the number of real zeros of $f(z)$ does not exceed the number of points in the tropical variety defined by $f_{trop}$, i.e. the number of corners of the continuous piecewise-linear function $ f_{trop}(x),\; x\in \mathbb{R}$.

背景链接：https://arxiv.org/abs/1503.05295

02364 | AMR-021-0014

### Problems Around Polynomials - Conjecture 9

O / open | AMR 问题清单汇编

[J. Forsg\aa rd, B. Sh., seems good, see ] Let $f(z) = \sum_{k=0}^n a_k z^k$ be a polynomial with positive coefficients. Consider the differences \[ \tilde c_k = (k+1)a_k^2 - k a_{k-1}a_{k+1}, \] where $a_{-1} = a_{n+1} = 0$. Let $0=k_1 < k_2 < \dots < k_m = n$ be the sequence of indices such that $\tilde c_{k_i}$ is positive, and let $v(f)$ be the number of changes in the sequence $\{k_i \mod 2\}_{i=0}^m$. Then the number of real zeros of $f(z)$ does not exceed $v(f)$.

背景链接：https://arxiv.org/abs/1503.05295

02365 | AMR-021-0004

### Problems Around Polynomials - Problem 1

O / open | AMR 问题清单汇编

[B. Sh., looks bad, but very important] Does there exist an upper bound for the number of real roots valid for all non-trivial solutions of all equations [source label: eq:triv] in $\Omega_k$?

背景链接：https://arxiv.org/abs/1503.05295

02366 | AMR-021-0005

### Problems Around Polynomials - Problem 2

O / open | AMR 问题清单汇编

[D. Khavinson, I. Itenberg, B. Sh., apparently bad] Find the maximal possible number $\#(2k, l)$ of isolated zeros for real non-negative polynomials of degree $2k$ in $l$ variables.

背景链接：https://arxiv.org/abs/1503.05295

02367 | AMR-021-0006

### Problems Around Polynomials - Problem 3

O / open | AMR 问题清单汇编

[G. Ottaviani, B. Sh., seems good] Find the maximal possible number $\widetilde\#(2k, l)$ of isolated zeros for real non-negative polynomials of degree $2k$ in $l$ variables which are representable as the sums of squares of real polynomials of degree at most $k$.

背景链接：https://arxiv.org/abs/1503.05295

02368 | AMR-021-0008

### Problems Around Polynomials - Problem 4

O / open | AMR 问题清单汇编

[S. Fisk, seems bad, see , p. 575] Given a pair of real polynomials $(p,q),$ give restrictions on the location of the roots of $p+iq$ in terms of the location of the roots of $p$ and $q$.

背景链接：https://arxiv.org/abs/1503.05295

02369 | AMR-021-0011

### Problems Around Polynomials - Problem 5

P / partially_solved | AMR 问题清单汇编

[seems bad, but might be ugly] For a given sign pattern $\sigma,$ which admissible pairs $(pos,neg)$ are realizable by polynomials whose signs of coefficients are given by $\sigma$?

背景链接：https://arxiv.org/abs/1503.05295

02370 | AMR-021-0016

### Problems Around Polynomials - Problem 6

O / open | AMR 问题清单汇编

[V. Kostov, B. Sh., looks ugly, see ] What additional restrictions besides [source label: eq:1] exist on configurations $\mathcal A_{f}=\{x^{(i)}_{l}\}$ coming from real-rooted polynomial-like functions of a given degree $n$? Or, more ambitiously, given a configuration $\mathcal A=\{x^{(i)}_{l}\vert\; i=0,\ldots,n-1;\,l=1,\ldots n-i\}$ of $\binom {n+1} 2$ real numbers satisfying standard Rolle's restrictions, is it possible to determine if there exists a real-rooted polynomial-like $f$ of degree $n$ such that $\mathcal A_{f}=\mathcal A$?

背景链接：https://arxiv.org/abs/1503.05295

02371 | AMR-021-0017

### Problems Around Polynomials - Problem 7

O / open | AMR 问题清单汇编

[looks ugly] What symbolic sequences can occur for strictly real-rooted polynomials of degree $n$?

背景链接：https://arxiv.org/abs/1503.05295

02372 | GREEN-005

### Product-Free Sets in Finite Groups

P / partially_solved | Ben Green 问题

Which finite groups have the smallest largest product-free sets?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

02373 | AIM-REPRESENTATION_THEORY-0091

### Projected superinduction and the transitivity defect

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.7.** Make sense out of superinduction and restriction more generally than for algebra groups.

**Remark.** To start with, trying the supercharacter theory induced by automorphisms and character-istic subgroups. Look at the article Induction for association schemes. Johnson & Smith (1986).

K. W. Johnson and J. D. H. Smith, “A Note on Character Induction in Association Schemes,” *European Journal of Combinatorics* **7**(2) (1986), p. 139, DOI 10.1016/S0195-6698(86)80039-7.

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

02374 | OWR-5149-002

### Projective-Free Lie Modules of Symmetric Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $p$ divide $n$ and let $\operatorname{Lie}(n)$ be the Lie module of $S_n$, with projective-free part $\operatorname{Lie}^{\mathrm{pf}}(n)$. Determine the size of $\operatorname{Lie}^{\mathrm{pf}}(n)$ and how far $\operatorname{Lie}(n)$ is from being projective. When $n=p^k$, is $\operatorname{Lie}^{\mathrm{pf}}(n)$ indecomposable with regular elementary abelian vertex $E_n$, and what are the isomorphism types of its endo-permutation sources? What changes when $n$ is not a power of $p$?

来源字段：https://doi.org/10.4171/owr/2011/21

02375 | OWR-12697689-015

### Proxy-Small Modules and Cohomological Support Varieties

O / open | Oberwolfach 报告问题汇编

Let $R$ be a Noetherian local ring. Is $R$ a complete intersection whenever every finitely generated $R$-module $M$ is proxy small, meaning that there is an object $P\in\langle M\rangle\cap\langle R\rangle$ with the same ordinary support as $M$? More generally, how small can the cohomological support variety $V_R(M)$ be, and which conical subvarieties occur as $V_R(M)$ for $M\in D^b(R)$?

来源字段：https://doi.org/10.4171/owr/2023/6

02376 | AIM-REPRESENTATION_THEORY-0079

### Published Rankin-Selberg integrals for higher covers and a two-place Eulerianity test

S / solved | AIM 研讨会问题汇编

Are there instances of Rankin-Selberg methods for higher covers?

来源字段：http://aimpl.org/autoformcovergp/4/

02377 | OWR-12494-003

### Pythagoras Numbers of Real Function Fields

O / open | Oberwolfach 报告问题汇编

For a real function field $F$, determine its Pythagoras number $p(F)$, or obtain effective lower bounds for it, where $p(F)$ is the least integer such that every sum of squares in $F$ is a sum of at most $p(F)$ squares.

来源字段：https://doi.org/10.4171/owr/2013/31

02378 | OWR-1272-011

### Quadratic Nil Algebras and Algebraic Division Algebras

O / open | Oberwolfach 报告问题汇编

Does there exist an infinite-dimensional quadratic nil algebra? More specifically, let $F$ be a field of characteristic $p>0$, let $R=F[t_0,t_1,\ldots\mid t_i^p=0]$, and put $\partial_i=\partial/\partial t_i$. Is the associative algebra generated by $V_1=\sum_{i\geq1}(\prod_{j=0}^{i-2}t_j^{p-1})\partial_i$ and $V_2=\sum_{i\geq2}(\prod_{j=1}^{i-2}t_j^{p-1})\partial_i$ nil? If $L$ is the Lie algebra generated by $V_1,V_2$, $U=U(L)$ is its universal enveloping algebra, $Z$ is the center of $U$, and $D=(Z\setminus\{0\})^{-1}U$, is $D$ an algebraic division algebra?

来源字段：https://doi.org/10.4171/owr/2006/23

02379 | OWR-14299511-005

### Quadrilaterals of Subfactors with Two Intermediate Factors

O / open | Oberwolfach 报告问题汇编

Develop the planar-algebra theory of finite-index subfactor quadrilaterals $N\subset P,Q\subset M$ with two intermediate subfactors, accounting for the relative position of $P$ and $Q$ beyond the known commuting-square and restricted cocommuting cases.

来源字段：https://doi.org/10.4171/owr/2025/34

02380 | OWR-14299906-003

### Quantum Automorphisms in Limiting Cuntz-Krieger Isometry Groups

O / open | Oberwolfach 报告问题汇编

Let $A$ be a finite zero-one matrix. If $\operatorname{QAut}(A)\neq\operatorname{Aut}(A)$, does the limiting quantum isometry group $G_A^\infty$ contain $\operatorname{QAut}(A)$ as a quantum subgroup?

来源字段：https://doi.org/10.4171/owr/2026/4

02381 | OWR-8415355-018

### Quantum Grothendieck Rings Beyond Simply-Laced Types

P / partially_solved | Oberwolfach 报告问题汇编

For a category $\mathcal C$ of finite-dimensional representations of a quantum affine algebra, does the bilinear product $*$ constructed from the monoidal Jantzen filtration make $(K(\mathcal C)_t,*)$ an algebra with anti-involution, isomorphic to the quantum Grothendieck ring $K_t(\mathcal C)$ with its standard bases identified?

来源字段：https://doi.org/10.4171/owr/2021/52

02382 | OWR-8415347-015

### Quantum-Group Interpretations of Quasi-Macdonald Polynomials

O / open | Oberwolfach 报告问题汇编

Do the quasi-polynomial generalizations of symmetric Macdonald polynomials admit natural quantum-group interpretations?

来源字段：https://doi.org/10.4171/owr/2021/44

02383 | OWR-16635-004

### Quasi-Cuspidal Representations and Generators of Equivariant D-Modules

O / open | Oberwolfach 报告问题汇编

Let $G$ be connected and reductive, and put $P(V)=D_{\mathfrak g}\otimes_{U\mathfrak g}V$. Characterize the quasi-cuspidal representations $V$ for which $P(V)$ has a cuspidal summand, and find a minimal set of irreducible $V$ whose modules $P(V)$ generate $D\!\operatorname{-mod}(\mathfrak g)^G$.

来源字段：https://doi.org/10.4171/owr/2018/52

02384 | OWR-4800-011

### Quasi-Hereditary Categorifications of $U_q(\mathfrak{sl}_2)$ Tensor Powers

P / partially_solved | Oberwolfach 报告问题汇编

Can the $U_q(\mathfrak{sl}_2)$-representations $V_1^{\otimes n}(j)$ be categorified by finite-dimensional quasi-hereditary graded algebras $A_{j,n}$ so that $K_0(A_{j,n}\text{-gmod})\cong V_1^{\otimes n}(j)$, with the simple modules indexed by binary strings of weight $j$?

来源字段：https://doi.org/10.4171/owr/2011/10

02385 | OWR-787-004

### Quiver Discriminants from Generic Maximal Minors

O / open | Oberwolfach 报告问题汇编

For the free divisor $f$ formed by the maximal minors of a generic $(n-2)\times(n-1)$ matrix, determine the quivers $Q$ and dimension vectors $d$ for which $f$ occurs as a discriminant in a quiver representation variety.

来源字段：https://doi.org/10.4171/owr/2005/06

02386 | OWR-12978-005

### Quotient Rings and Homological Properties of Hopf-Module Algebras

O / open | Oberwolfach 报告问题汇编

For Hopf-module algebras, does every $H$-semiprime right-noetherian algebra have an artinian classical right quotient ring? Can an $H$-simple semiprime noetherian algebra finite over its center have infinite global dimension, or fail to be Gorenstein? If a cosemisimple Hopf algebra acts inner faithfully on a polynomial-identity domain of degree $d$, is $\operatorname{PIdeg}(H^*)\leq d^2$? Is every noetherian Artin-Schelter regular algebra a braided Hopf algebra?

来源字段：https://doi.org/10.4171/owr/2014/20

02387 | GREEN-089

### Random Walk Mixing on Alternating Groups

P / partially_solved | Ben Green 问题

Pick $x_1, \dots, x_k \in A_n$ at random. Is it true that, almost surely as $n \to \infty$, the random walk on this set of generators and their inverses equidistributes in time $O(n \log n)$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

02388 | OWR-1274-012

### Rank Bounds for Modular-Representation Grothendieck Groups

O / open | Oberwolfach 报告问题汇编

Let $k$ be a finite field of characteristic $p$, let $G$ be a compact $p$-adic analytic group with no elements of order $p$, and let $\mathcal F_t$ be the category of finitely generated $kG$-modules of canonical dimension $\dim_G(M)\leq t$. For the inclusion-induced map $\theta_t:K_0(\mathcal F_t)\to K_0(\mathcal F_d)$, improve the bound on $\operatorname{rank}(\operatorname{im}\theta_t)$ given by Galois orbits of conjugacy classes of finite-order elements whose centralizers have dimension at most $t$. What is the analogue with $k$ replaced by the $p$-adic integers?

来源字段：https://doi.org/10.4171/owr/2006/25

02389 | AIM-REPRESENTATION_THEORY-0022

### Rank four gives a nonalgebraic determinantal tca Hilbert series

S / solved | AIM 研讨会问题汇编

**Hilbert series of modules over degree two tca's.** What can we say about Hilbert series of modules over \(A=\operatorname{Sym}(\operatorname{Sym}^2)\)? More precisely, take \(M\) to be \(A\) modulo a determinantal ideal. Is it true that its (non-exponential) Hilbert series is algebraic? Assume characteristic \(0\).

来源字段：http://aimpl.org/repnstability/1/

02390 | OWR-17289-009

### Rank Stratifications and KLR Actions on Flag Degenerations

O / open | Oberwolfach 报告问题汇编

For the flat family $F_U\to U$ of linear degenerations, study the stratification by the rank conditions defining $S(A)$. Describe $H_*^{\mathrm{BM}}(F_U\times_U F_U)$ as a quotient of a quiver Hecke, or KLR, algebra, together with its standard representations indexed by Motzkin paths.

来源字段：https://doi.org/10.4171/owr/2019/46

02391 | OWR-3393-006

### Rank Varieties of Specht Modules

P / partially_solved | Oberwolfach 报告问题汇编

For partitions whose parts are divisible by $p$, characterize the associated Specht modules through their rank varieties. Is the multiplicity integer in the rank-variety formula always $1$, and what is the sharp upper bound for the degree of the projectivized rank variety?

来源字段：https://doi.org/10.4171/owr/2009/10

02392 | OWR-15219-019

### Rank-One Kernel Boundary Points of Free Spectrahedra

O / open | Oberwolfach 报告问题汇编

Let $L$ be a Hermitian monic linear pencil and let $D(L)$ be its free spectrahedron. Must there exist a boundary point $X$ satisfying $\dim\ker L(X)=1$; equivalently, is $\partial^1D(L)$ nonempty?

来源字段：https://doi.org/10.4171/owr/2017/14

02393 | OWR-12697710-015

### Rational Certificates for Truncated Moment Nonrepresentability

O / open | Oberwolfach 报告问题汇编

Let $K=\{x\in\mathbb R^n:g_1(x),\ldots,g_r(x)\geq0\}$ and let $Q(g)$ be the quadratic module generated by $g_1,\ldots,g_r$. For a rational truncated moment vector that is not representable by a measure on $K$, can there be a rational separating polynomial $p\in1+Q(g)$ for which every sum-of-squares certificate of membership in $1+Q(g)$ requires irrational coefficients?

来源字段：https://doi.org/10.4171/owr/2023/14

02394 | OWR-12697690-010

### Realizability of Complete Sign-Coherent Fans

O / open | Oberwolfach 报告问题汇编

Characterize the complete sign-coherent fans in $\mathbb R^d$ that can be realized as the $g$-fans of finite-dimensional algebras.

来源字段：https://doi.org/10.4171/owr/2023/7

02395 | AIM-REPRESENTATION_THEORY-0069

### Realization fibers for real Brylinski--Deligne covers and a split-torus parity theorem

P / partially_solved | AIM 研讨会问题汇编

Define the real realization functor from BD triples to topological double covers of $G(\mathbb R)$; determine its image and fibers, or at least give computable necessary conditions and explicit inverses in tractable cases.

来源字段：http://aimpl.org/autoformcovergp/2/

02396 | OWR-17466-006

### Realization Functors as Derived Equivalences

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathcal A$ be an abelian category contained in the essential image of the realization functor $G$ from the bounded derived category of its heart $\mathcal H$ to the ambient triangulated category. Is $G$ a derived equivalence?

来源字段：https://doi.org/10.4171/owr/2020/3

02397 | OWR-2048-009

### Realizing Compact Spectra of Pathological Masas

O / open | Oberwolfach 报告问题汇编

Can every compact Hausdorff space occur as the spectrum of a maximal abelian self-adjoint subalgebra $C$ of a $C^*$-algebra $A$ for which tensoring creates a non-maximal-abelian inclusion?

来源字段：https://doi.org/10.4171/owr/2008/37

02398 | OWR-17133-011

### Realizing Lower Semicontinuous Trace Functions by Positive Elements

O / open | Oberwolfach 报告问题汇编

Let $A$ be a separable, unital, simple, exact $C^*$-algebra, and let $f\in\operatorname{LAff}(T(A))_{++}$. Does there exist $a\in(A\otimes\mathcal K)^+$ such that

$$d_\tau([a])=f(\tau)$$

for every $\tau\in T(A)$?

来源字段：https://doi.org/10.4171/owr/2019/37

02399 | OWR-14299580-003

### Recomposition and Cabling of Set-Theoretic Yang-Baxter Solutions

O / open | Oberwolfach 报告问题汇编

Can indecomposable set-theoretic Yang-Baxter solutions be systematically recomposed into new solutions? Can these solutions be classified up to cabling, or distinguished by computable invariants of cabling equivalence?

来源字段：https://doi.org/10.4171/owr/2025/49

02400 | OWR-12732-005

### Recovery of Support $\tau$-Tilting Orders from $g$-Vectors

P / partially_solved | Oberwolfach 报告问题汇编

For a basic finite-dimensional $k$-algebra $\Lambda$, do the $g$-vectors determine the partial order on $\mathrm{s}\tau\text{-}\mathrm{tilt}\,\Lambda$?

来源字段：https://doi.org/10.4171/owr/2013/58

02401 | OWR-17288-001

### Reduced $C^*$-Algebras of Free Orthogonal Quantum Groups

O / open | Oberwolfach 报告问题汇编

For distinct positive integers $M$ and $N$, are the reduced $C^*$-algebras $C_r^*(\mathrm{FO}_M)$ and $C_r^*(\mathrm{FO}_N)$ isomorphic?

来源字段：https://doi.org/10.4171/owr/2019/45

02402 | OWR-1265-009

### Reduced Standard Modules for Regular Dominant Weights

O / open | Oberwolfach 报告问题汇编

Assume $p>h$. For every regular dominant weight $\lambda$, do $\Delta_{\mathrm{red}}(\lambda)[-l(\lambda)]\in E^L$ and $\nabla_{\mathrm{red}}(\lambda)[-l(\lambda)]\in E^R$?

来源字段：https://doi.org/10.4171/owr/2006/15

02403 | OWR-7155441-007

### Redundant Inequalities in String Cones

O / open | Oberwolfach 报告问题汇编

For an arbitrary reduced word $\mathbf i$, how many inequalities defining its string cone are redundant, and can the redundancies be classified?

来源字段：https://doi.org/10.4171/owr/2021/20

02404 | OWR-1265-012

### Regular Filtrations of Standard and Costandard Modules

O / open | Oberwolfach 报告问题汇编

For every regular dominant weight $\lambda$, does $\Delta(\lambda)$, respectively $\nabla(\lambda)$, have a filtration with sections $\Delta_{\mathrm{red}}(\nu)$, respectively $\nabla_{\mathrm{red}}(\nu)$, for regular dominant $\nu$?

来源字段：https://doi.org/10.4171/owr/2006/15

02405 | OWR-4133-002

### Regular Sequences without the Regular Element Theorem

O / open | Oberwolfach 报告问题汇编

How can one avoid reliance on the Regular Element Theorem when developing a theory of regular sequences over arbitrary commutative rings without Noetherian hypotheses?

来源字段：https://doi.org/10.4171/owr/2009/50

02406 | AIM-REPRESENTATION_THEORY-0095

### Regular-locus excision for unitriangular conjugacy

P / partially_solved | AIM 研讨会问题汇编

In what precise sense is the uniform description of the conjugacy classes of \(U_n(q)\) difficult, as \(n\) and the finite field vary? Is even the class number \(k(U_n(q))\) difficult to determine?

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

02407 | OWR-3472-010

### Regularity Bounds for Ext Modules Under Serre Conditions

O / open | Oberwolfach 报告问题汇编

Let $S=k[x_1,\ldots,x_n]$, let $I$ be graded, let $R=S/I$ have dimension $d$, and let $r\geq1$. Which graded ideals satisfy $\operatorname{reg}\operatorname{Ext}_S^{n-i}(R,\omega_S)\leq\dim\operatorname{Ext}_S^{n-i}(R,\omega_S)$, and, when $R$ satisfies $(S_r)$, do they satisfy the corresponding bound by $i-r$ for every $0\leq i\leq d-1$, even for monomial ideals?

来源字段：https://doi.org/10.4171/owr/2009/22

02408 | OWR-734-004

### Regularity of Cycles for Linearly Resolved Ideals

O / open | Oberwolfach 报告问题汇编

Let $I\subset S=K[x_1,\ldots,x_n]$ be generated in degree $d$, be of linear type on the punctured spectrum, and have a linear free resolution. Is the regularity of the $j$-th module $B_j$ equal to $jd$?

来源字段：https://doi.org/10.4171/owr/2004/32

02409 | OWR-8415346-002

### Regularity of Equivariant Monomial-Ideal Filtrations

O / open | Oberwolfach 报告问题汇编

Let $I=(I_n)_{n\geq1}$ be an $\operatorname{Inc}(\mathbb N)$-invariant filtration of nonzero proper homogeneous ideals with stability index $r$. Is $\operatorname{reg}(I_n)$ eventually linear in $n$? If the filtration is monomial, does there exist a constant $D$ such that $$\operatorname{reg}(I_n)=w(I_r)-1)n+D$$ for all sufficiently large $n$? The question remains open even for nonsaturated filtrations of squarefree monomial ideals.

来源字段：https://doi.org/10.4171/owr/2021/43

02410 | OWR-12591-009

### Regularity Properties of Nuclear $C^*$-Algebras

P / partially_solved | Oberwolfach 报告问题汇编

For a separable unital simple nuclear stably finite infinite-dimensional $C^*$-algebra $A$, are the following equivalent: $\mathcal Z$-stability ($A\cong A\otimes\mathcal Z$), strict comparison, and finite decomposition rank?

来源字段：https://doi.org/10.4171/owr/2013/43

02411 | OWR-17294-005

### Related Solutions from Prime Structure-Algebra Quotients

P / partially_solved | Oberwolfach 报告问题汇编

For prime quotients $kM(X,r)/P$ that arise from related Yang-Baxter solutions, how do the properties of those solutions, as $P$ varies, reflect the properties of $(X,r)$?

来源字段：https://doi.org/10.4171/owr/2019/51

02412 | OWR-3472-003

### Relative Betti Inequalities for Lex-Plus-Powers Ideals

O / open | Oberwolfach 报告问题汇编

Let $S = k[x_1, \ldots, x_n]$ be a polynomial ring over a field $k$, let $F = (x_1^{a_1}, \ldots, x_n^{a_n})$ be a regular sequence of pure powers, let $P = (f_1, \ldots, f_n)$ be a homogeneous regular sequence with $\deg f_i = a_i$, and let $I \supset P$ be a homogeneous ideal with corresponding lex-plus-powers (LPP) ideal $L \supset F$ having the same Hilbert function. In the setting of the Lex-Plus-Powers Conjecture, does the graded Betti-number inequality $b_{i,j}^{S/P}(L) \geq b_{i,j}^{S/F}(I)$ hold for all $i, j$?

来源字段：https://doi.org/10.4171/owr/2009/22

02413 | OWR-2048-007

### Relative Commutants in Maximal Tensor Products

O / open | Oberwolfach 报告问题汇编

Let $H\cong\ell^2(\mathbb{N})$, let $C\subset B(H)$ be a maximal abelian self-adjoint algebra isomorphic to $L^\infty(0,1)$, and let $(C\otimes1)^c$ be its relative commutant in $B(H)\otimes_{\max}C_r^*(F_2)$. Is $(C\otimes1)^c=C\otimes C_r^*(F_2)$?

来源字段：https://doi.org/10.4171/owr/2008/37

02414 | OWR-14299289-013

### Removing Dual-Group Invariants from Plancherel Geometry

O / open | Oberwolfach 报告问题汇编

Does local geometric duality admit an equivariant upgrade that removes the $G^\vee$-invariance from the Borel-Moore-homology description and recovers the full function algebra $\mathcal O(M^\vee)$ as an algebra object over $\operatorname{QCoh}(\mathfrak g^{\vee *})^{G^\vee}$?

来源字段：https://doi.org/10.4171/owr/2025/18

02415 | OWR-4332-003

### Representation Theory of Classical Finite W-Algebras

O / open | Oberwolfach 报告问题汇编

Determine the structure and representation theory of finite W-algebras associated with classical Lie algebras.

来源字段：https://doi.org/10.4171/owr/2010/19

02416 | OWR-1790-004

### Representation-Theoretic Meaning of Cluster Denominator Vectors

P / partially_solved | Oberwolfach 报告问题汇编

For a cluster variable represented as a Laurent rational function with denominator $\prod_y y^{d_y}$, how can the exponents $d_y$ be interpreted representation-theoretically?

来源字段：https://doi.org/10.4171/owr/2008/09

02417 | OWR-14299283-001

### Representation-Theoretic Meaning of Steinberg Quotient Characters

P / partially_solved | Oberwolfach 报告问题汇编

For each dominant weight $\lambda$, identify a natural representation-theoretic or geometric object whose character is the Steinberg quotient $t(\lambda)$. Determine in particular whether it is the character of a finite-dimensional module.

来源字段：https://doi.org/10.4171/owr/2025/25

02418 | AIM-REPRESENTATION_THEORY-0004

### Residual blocks for Dirichlet L-values at zero

P / partially_solved | AIM 研讨会问题汇编

**$p$-indivisibility of $L(\chi,0)$.** Fix a prime $p$. Let $q$ be a prime and let $\chi$ be an odd character of $(\mathbb Z/q\mathbb Z)^\times$. How often is $L(\chi,0)$ indivisible by $p$?

来源字段：http://aimpl.org/aagaautomorphic/1/

02419 | GREEN-064

### Residually Finite Groups

P / partially_solved | Ben Green 问题

别名：Sofic Groups [GREEN-100]

Is every group well-approximated by finite groups?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

02420 | AIM-REPRESENTATION_THEORY-0062

### Residue isolation and an exact finite-field hyper-Kloosterman transform

P / partially_solved | AIM 研讨会问题汇编

(1) Isolate contributions of the continuous "special" representations on the geo-metric side of the trace formula. Here, special refers to larger poles of rela-tive L-functions. Some special cases may be related to the Rankin-Selberg integral method, for instance, the doubling integral method of Piatetski-Shapiro and Rallis on GL n or GL 2.(2) Take the current literature on beyond endoscopy, and make a list of the precise local and global statements required. For example: local matching and global matching. (3) Arthur has introduced the so-called r-trace formula. Describe (and name!) the r-"beyond endoscopy" groups and discuss the meaning of geometric matching. (4) How does the functional equation give an expression for the r-trace formula? In other words, how does the r-Fourier transform play roles here? (5) Isolate the contribution of hyperkloosterman sums to the Kuznetsov trace formula and explain it spectrally.

来源字段：https://aimath.org/pastworkshops/automorphkernelproblems.pdf

02421 | OWR-12172-007

### Resolutions and Higher-Dimensional Structure of Orlik-Terao Algebras

P / partially_solved | Oberwolfach 报告问题汇编

For the Orlik-Terao algebra $R/I_A$ of a hyperplane arrangement $A$, what phenomena occur in higher dimensions and for reflection arrangements, and what is its minimal free resolution, even when $A\subseteq\mathbb P^2$?

来源字段：https://doi.org/10.4171/owr/2012/49

02422 | OWR-14299915-004

### Revised Donkin Tilting-Module Conjectures

O / open | Oberwolfach 报告问题汇编

Let $G$ be a simple simply connected algebraic group over an algebraically closed field of characteristic $p>0$, with root system $\Phi$ and Coxeter number $h$. Prove that Donkin’s Tilting Module Conjecture holds for every prime $p$ exactly when $\Phi$ has type $A_n$ or $B_2$, and prove that it holds whenever $p>h$.

来源字段：https://doi.org/10.4171/owr/2026/9

02423 | OWR-12591-001

### Rigidity of Completely Close $C^*$-Algebras

O / open | Oberwolfach 报告问题汇编

For $C^*$-algebras $A,B\subseteq B(H)$, what structural properties and invariants are forced by small completely bounded Kadison-Kastler distance $d_{\mathrm{cb}}(A,B)$, and must sufficiently close algebras be isomorphic?

来源字段：https://doi.org/10.4171/owr/2013/43

02424 | OWR-7155441-002

### Rigidity of Morphisms to Wound Unipotent Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $U$ be a wound unipotent group over $k$, let $x$ be a closed point of $\mathbb P_k^1$, put $X=\mathbb P_k^1\setminus\{x\}$, and choose $y_1,y_2\in X(k)$. Is every $k$-morphism $f:X\times X\to U$ that is trivial on $\{y_1\}\times X$ and $X\times\{y_2\}$ the constant morphism $1$?

来源字段：https://doi.org/10.4171/owr/2021/20

02425 | OWR-3400-006

### Rigidity Under Stable Equivalence of Quantum Complete Intersections

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a quantum complete intersection over a field $k$ and $Y$ a finite-dimensional symmetric local $k$-algebra stably equivalent to $X$ by an equivalence of Morita type, with $\dim_kY=\dim_kX$ and $Z(X)\cong Z(Y)$ as $k$-algebras. Must $X$ and $Y$ be isomorphic as $k$-algebras?

来源字段：https://doi.org/10.4171/owr/2009/17

02426 | OWR-4084-012

### Ring $C^*$-Algebras with Zero Divisors

O / open | Oberwolfach 报告问题汇编

For ring $C^*$-algebras of rings with zero divisors, how can the support projections of partial isometries implementing ring multiplication be organized while retaining all relevant multiplicative information?

来源字段：https://doi.org/10.4171/owr/2009/41

02427 | OWR-14299580-012

### Root Systems and Classification of Generalized Nichols Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Develop a root-system and Weyl-groupoid theory for Nichols algebra objects in arbitrary braided tensor categories, and classify the resulting generalized Nichols algebras.

来源字段：https://doi.org/10.4171/owr/2025/49

02428 | OWR-15219-013

### Root Vector Majorization for Doubly Stochastic Stable Polynomials

O / open | Oberwolfach 报告问题汇编

Let $D_{m,d}$, $P_\infty$, and $\lambda(P)=(\lambda_1(P),\ldots,\lambda_d(P))$ be as in the doubly stochastic stable-polynomial setting. Is $\lambda(P)$ majorized by $\lambda(P_\infty)$ for every $P\in D_{m,d}$? Moreover, if $P\in D_{m,d}\cap\operatorname{Sym}_m$, $0\leq r\leq s$, and $\Psi_s=\exp(s(I-T))$, is $\lambda(\Psi_r(P))$ majorized by $\lambda(\Psi_s(P))$?

来源字段：https://doi.org/10.4171/owr/2017/14

02429 | ALG-048

### Rota's Basis Conjecture

P / partially_solved | 独立 / 未分配集合

Given n bases of an n-dimensional matroid, can we find n disjoint rainbow bases?

背景链接：https://arxiv.org/abs/2008.06045

02430 | AIM-REPRESENTATION_THEORY-0011

### Saito--Kurokawa content obstruction and the period-phase barrier

S / solved | AIM 研讨会问题汇编

**Siegel modular forms.** Let \(F\) be a Siegel cusp form of full level and weight \(k\) that is a Hecke eigenform. Assume any standard conjecture on \(L\)-values (e.g. GRH) and all conjectural period formulae. Prove that for some \(\delta>0\), \[ |a(F,S)|\ll_F \det(S)^{K/2-1/2-\delta}. \]

来源字段：http://aimpl.org/aagaautomorphic/5/

02431 | OWR-734-015

### Scarf Resolutions for Weighted Digraph Ideals

O / open | Oberwolfach 报告问题汇编

Does the ideal $J_G$ associated with a saturated weighted digraph $G$ have a Scarf-type minimal free resolution?

来源字段：https://doi.org/10.4171/owr/2004/32

02432 | OWR-1265-022

### Schur-Algebra Models for Symmetric-Group Blocks

O / open | Oberwolfach 报告问题汇编

Every block of a symmetric group over $\mathcal{O}$ should be equivalent to a quotient of the deformation of the double of the Schur algebra of $\mathcal{O}Q$ and to the analogous quotient for its Koszul dual.

来源字段：https://doi.org/10.4171/owr/2006/15

02433 | AIM-LINEAR_ALGEBRA-0013

### Schur-ratio minima on bounded interval boxes

P / partially_solved | AIM 研讨会问题汇编

It was recently shown that if $\lambda,\mu$ are $N$-tuples of positive integers, then the ratio of Schur polynomials $s_\lambda/s_\mu$, when evaluated at $N$ real variables with values in $(0,1]^N$ (the “log-negative orthant”), is minimized at $(1,\dots,1)$ if and only if $-\lambda$ weakly majorizes $-\mu$.

The analogous statement, in which the minimum over the entire positive orthant $(0,\infty)^N$ is achieved at $(1,\dots,1)$, is equivalent to $\lambda$ majorizing $\mu$.

What can one say these domains are replaced by $I^N$ for other intervals $I\subset(0,\infty)$?

What can one say **when** these domains are replaced by $I^N$ for other intervals $I\subset(0,\infty)$?

来源字段：http://aimpl.org/totalpos/1/

02434 | OWR-17294-007

### Second-Cohomology Torsion of Finite Abelian Quandles

P / partially_solved | Oberwolfach 报告问题汇编

Compute the torsion subgroup of the second cohomology groups $H^2$ of finite abelian quandles.

来源字段：https://doi.org/10.4171/owr/2019/51

02435 | OWR-14299522-010

### Selected Lifting and Perturbation Problems in Operator Algebras

O / open | Oberwolfach 报告问题汇编

Is there an automorphism $\alpha$ of $\mathcal{O}_\infty\otimes\mathcal{K}$ that fixes the diagonal and induces $-\mathrm{id}$ on $K_0$, and can such an automorphism have the Rokhlin property? Does there exist a relatively free group $G$ for which $C_r^*(G)$ is selfless? Does there exist a simple $C^*$-algebra that is not $K_1$-injective? Is every approximate $C^*$-algebra close to an actual $C^*$-algebra?

来源字段：https://doi.org/10.4171/owr/2025/35

02436 | OWR-11580-010

### Self-Extensions of Simple Quantum-Affine Modules

O / open | Oberwolfach 报告问题汇编

Characterize the simple finite-dimensional quantum-affine modules $V$ satisfying $\dim\operatorname{Ext}^1(V,V)=1$.

来源字段：https://doi.org/10.4171/owr/2012/13

02437 | OWR-14299522-006

### Selflessness and Strict Comparison for Reduced Group $C^*$-Algebras

O / open | Oberwolfach 报告问题汇编

For a discrete group $G$, is strict comparison for $C_r^*(G)$ equivalent to $C^*$-simplicity? Is selflessness equivalent to strict comparison together with $C^*$-simplicity? Find a dynamical characterization of selflessness, and more generally characterize selflessness of reduced crossed products through dynamics or stabilizer subgroups.

来源字段：https://doi.org/10.4171/owr/2025/35

02438 | OWR-14299522-007

### Selflessness of the Reduced $C^*$-Algebra of $\mathrm{SL}_3(\mathbb{Z})$

O / open | Oberwolfach 报告问题汇编

Is the reduced group $C^*$-algebra $C_r^*(\mathrm{SL}_3(\mathbb{Z}))$ selfless?

来源字段：https://doi.org/10.4171/owr/2025/35

02439 | OWR-1393-002

### Semi-Infinite Fusion for Arbitrary Dominant Weights

O / open | Oberwolfach 报告问题汇编

Does the semi-infinite fusion construction for an arbitrary simple Lie algebra work for every dominant weight, not only for multiples of $\Lambda_0$?

来源字段：https://doi.org/10.4171/owr/2006/55

02440 | OWR-3397-001

### Semiclassical Correspondence of Primitive and Poisson Spectra

P / partially_solved | Oberwolfach 报告问题汇编

If an algebra $A$ is a suitably generic member of a flat family with semiclassical limit a Poisson algebra $R$, is the primitive ideal space of $A$ homeomorphic to the Poisson-primitive spectrum of $R$, and is the prime spectrum of $A$ homeomorphic to the Poisson-prime spectrum of $R$, with the homeomorphisms preserving the appropriate dimensions and group actions?

来源字段：https://doi.org/10.4171/owr/2009/14

02441 | OWR-8415355-020

### Semicuspidal and Schur Algebras of Affine Quivers

O / open | Oberwolfach 报告问题汇编

For any affine-type quiver $Q$ over $k=\mathbb{F}_q$, is $C(n\delta)\simeq S_n(\mathbb{Z}^{Q_0})\otimes_{\mathbb{Z}}\mathbb{F}_q$?

来源字段：https://doi.org/10.4171/owr/2021/52

02442 | OWR-12697693-005

### Semilattice Realizations by Wedge Semibraces

O / open | Oberwolfach 报告问题汇编

Which semilattices occur as the underlying semilattice of a $\wedge$-semibrace, and which occur when the $\wedge$-semibrace is also an $L$-algebra?

来源字段：https://doi.org/10.4171/owr/2023/9

02443 | OWR-17294-003

### Semiprime Yang-Baxter Structure Algebras

P / partially_solved | Oberwolfach 报告问题汇编

When is the structure algebra $kM(X,r)$ semiprime, and how does its semiprimeness affect the Yang-Baxter solution $(X,r)$?

来源字段：https://doi.org/10.4171/owr/2019/51

02444 | OWR-8415347-019

### Separated Augmentation Filtrations and Quantum Group Generation

P / partially_solved | Oberwolfach 报告问题汇编

Let $I$ be the augmentation ideal of a Hopf algebra $A$ and set $I^\infty=\bigcap_{n\in\mathbb{N}}I^n$. For which $A$ does $I^\infty=\{0\}$, and more generally how can $A/I^\infty$ be described in terms of $A$? In particular, what is $I^\infty$ for $\mathcal{O}(O_N^+)$? Does there exist a residually torsion-free nilpotent discrete group $\Gamma$ such that $O_N^+$ is topologically generated by $SO_N$ and $\widehat{\Gamma}$?

来源字段：https://doi.org/10.4171/owr/2021/44

02445 | AIM-REPRESENTATION_THEORY-0112

### Serre weights and complete Hecke packets

P / partially_solved | AIM 研讨会问题汇编

The section declares \(p>2\) and uses arithmetic Frobenius. Its goal is to generalize the weight part of Serre's conjecture from \(\mathrm{GL}_2/\mathbf Q\) to \(\mathrm{GL}_n/\mathbf Q\).

The mathematical content of Conjecture 5 is recovered as follows. Let \[ \bar\rho:G_{\mathbf Q}\longrightarrow \mathrm{GL}_2(\overline{\mathbf F}_p) \] be continuous, odd, and irreducible, let \(p\nmid N\), and let \(\sigma\) be a finite-dimensional characteristic-\(p\) representation of \(\mathrm{GL}_2(\mathbf F_p)\). Then \[ H^1(\Gamma_1(N),\sigma) \] should contain a Hecke eigenclass on which \[ T_\ell\quad\text{acts by}\quad \operatorname{tr}\bar\rho(\operatorname{Frob}_\ell) \qquad(\ell\nmid Np) \] if and only if \[ N(\bar\rho)\mid N, \qquad \operatorname{JH}(\sigma)\cap \operatorname{Wt}(\bar\rho|_{I_p})\ne\varnothing. \tag{1.1} \] The notes then ask how to generalize this and begin the Ash--Sinnott/Ash--Doud--Pollack setup: take an irreducible \(\bar\rho:G_{\mathbf Q}\to\mathrm{GL}_n(\overline{\mathbf F}_p)\) satisfying \[ \left|\dim \bar\rho(c)^{+1}-\dim \bar\rho(c)^{-1}\right|\le 1, \tag{1.2} \] define its prime-to-\(p\) conductor \(N(\bar\rho)\) and a set of Serre weights, and set \[ \Gamma_1(N)=\left\{g\in\mathrm{SL}_n(\mathbf Z): g\equiv \begin{pmatrix} *&*\\0&1\end{pmatrix}\pmod N\right\}. \tag{1.3} \] Here the lower-left block is a zero row of length \(n-1\), so equivalently the last row is congruent to \((0,\ldots,0,1)\). For \(\ell\nmid Np\), the record ends with the complete displayed definition \[ T_\ell= \left[\Gamma_1(N)\operatorname{diag}(\ell,1,\ldots,1)\Gamma_1(N)\right]. \tag{1.4} \]

The next sentence of the PDF begins the separately numbered Conjecture 6, which asks for an eigenclass in \(H^d\) for some degree \(d\). Thus record 0112 really does end at (1.4). The nearby continuation is needed to understand what the setup was preparing, but it is not part of this record and is not reproduced as its claimed statement.

来源字段：https://aimath.org/WWN/padicmodularity/padicmodularity.pdf

02446 | ALG-013

### Serre's Conjecture II

P / partially_solved | 独立 / 未分配集合

For simply connected semisimple algebraic groups over fields of cohomological dimension ≤2, is $H^1(F,G) = 0$?

背景链接：https://en.wikipedia.org/wiki/Serre%27s_conjecture_II

02447 | ALG-022

### Serre's Positivity Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Serre's Positivity Conjecture [ALG-061]

If $R$ is a regular local ring and $P, Q$ are prime ideals with intersecting dimensions satisfying a certain condition, is the intersection multiplicity positive?

背景链接：https://arxiv.org/abs/1510.05146

02448 | OWR-824-008

### Set-Theoretic Cohen-Macaulay Ideals in Characteristic Zero

O / open | Oberwolfach 报告问题汇编

Let $K$ be a field of characteristic zero, let $R=K[w,x,y,z]$, and let $$\varphi:R\longrightarrow K[s^4,s^3t,st^3,t^4]$$ send $w,x,y,z$ to $s^4,s^3t,st^3,t^4$, respectively. Writing $\mathfrak{a}=\ker\varphi$, does there exist an ideal $\mathfrak{b}\subset R$ such that $\sqrt{\mathfrak{b}}=\mathfrak{a}$ and $R/\mathfrak{b}$ is Cohen-Macaulay?

来源字段：https://doi.org/10.4171/owr/2005/19

02449 | AIM-LINEAR_ALGEBRA-0005

### Sharp degree-two coefficientwise-preserver criteria and local Turan-power families

P / partially_solved | AIM 研讨会问题汇编

The source asks five related questions. In exact mathematical terms:

1. For $$ \mathcal P_d=\left\{\sum_{j=0}^d a_jz^j:a_j\ge0, \text{ and every zero is real}\right\}, $$ classify functions $F:[0,\infty)\to[0,\infty)$ with $F(0)=0$ for which $$ F[f](z)=\sum_{j=0}^d F(a_j)z^j $$ remains in $\mathcal P_d$. Equivalently, the source phrases this as entrywise preservation of total nonnegativity of the lower-triangular Toeplitz matrix $T_{\mathbf a}$. 2. Ask the same question for fixed $d$. The natural reading is therefore that item (i) asks for preservation uniformly over all finite degrees. 3. Replace either the input or output ordinary generating polynomial, or both, by the exponential normalization $$ \sum_{j=0}^d \frac{a_jz^j}{j!}, $$ giving four ordinary/exponential cases. 4. Allow a coefficient weight $g(j)$: $$ F_g[f](z)=\sum_{j=0}^d g(j)F(a_j)z^j. $$ 5. Classify local rules $$ F_h[f](z)=\sum_{j=0}^d h(a_{j-1},a_j,a_{j+1})z^j, \qquad a_{-1}=a_{d+1}=0, $$ beyond the Turan rule $h(x,y,z)=y^2-xz$, and ask about higher-order determinants.

- Since the coefficients are nonnegative, every real zero of a nonzero polynomial is automatically nonpositive. - The logical definition in the source excludes the zero polynomial: it vanishes at nonreal numbers. In contrast, the zero Toeplitz matrix is totally nonnegative. Thus the stated equivalence implicitly uses the common PF/TN convention that the zero sequence is allowed. Below, zero-map statements are labeled by convention; all main constructive results use a positive scale and produce a nonzero polynomial. - The polynomial formulation permits zero coefficients, while the displayed Toeplitz formulation takes $\mathbf a\in(0,\infty)^{d+1}$. For arbitrary discontinuous $F$, these are not automatically interchangeable as test domains. The degree-two results below use the literal nonnegative-coefficient polynomial formulation and handle vanishing coefficients directly.

来源字段：http://aimpl.org/totalpos/1/

02450 | OWR-4084-006

### Similarity of Homotopic Group-Ring Projections

O / open | Oberwolfach 报告问题汇编

Are homotopic projections in matrices over a complex group ring similar?

来源字段：https://doi.org/10.4171/owr/2009/41

02451 | OWR-17294-011

### Simple Braces as Asymmetric Products

O / open | Oberwolfach 报告问题汇编

Let $B$ be a nontrivial finite simple left brace whose multiplicative group is metabelian with abelian Sylow subgroups. Is $B$ the asymmetric product of two trivial left braces?

来源字段：https://doi.org/10.4171/owr/2019/51

02452 | OWR-8415355-004

### Simple Modules of Nonsimply-Laced Quantum-Group Truncations

P / partially_solved | Oberwolfach 报告问题汇编

别名：Langlands Dual Parametrization from Baxter Polynomiality [OWR-8415355-005]

Can the simple modules of a nonsimply laced truncation be parametrized in terms of the Langlands dual quantum affine Lie algebra?

来源字段：https://doi.org/10.4171/owr/2021/52

02453 | OWR-5149-001

### Simple Specht Modules for Hecke Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Classify simple Specht modules for Hecke algebras over fields of prime characteristic.

来源字段：https://doi.org/10.4171/owr/2011/21

02454 | OWR-3400-007

### Simple Tops of Young Modules

O / open | Oberwolfach 报告问题汇编

For a partition $\lambda$ of $r$, is the top of the Young module $Y^\lambda$ simple if and only if there exists $n\geq\ell(\lambda)$ such that the indecomposable injective polynomial module $I_n(\lambda)$ is projective; equivalently, if and only if $\lambda$ belongs to the De Visscher-Donkin index set $\Lambda^{\mathrm{in}}$?

来源字段：https://doi.org/10.4171/owr/2009/17

02455 | OWR-16768-023

### Simplicity of Differential-Operator Rings on Invariant Rings

O / open | Oberwolfach 报告问题汇编

Let $G$ be a linearly reductive group acting linearly on a polynomial ring $R$ over $\mathbb C$. Is the differential-operator ring $D_{\mathbb C}(R^G)$ simple?

来源字段：https://doi.org/10.4171/owr/2019/6

02456 | OWR-14299281-007

### Simplicity of Special Automorphism Groups of Affine Space

P / partially_solved | Oberwolfach 报告问题汇编

For $n\geq3$, is the special automorphism group $\operatorname{SAut}(\mathbb A^n)$ simple as an abstract group?

来源字段：https://doi.org/10.4171/owr/2025/20

02457 | OWR-14298159-005

### Simplicity of Twisted Hopf Algebras

O / open | Oberwolfach 报告问题汇编

Let $G=\prod_{i=1}^n(D_i\rtimes_{\varphi_i}M_i)$, where every $D_i\rtimes_{\varphi_i}M_i$ is a Frobenius group, every $M_i$ is an abelian group of prime-power order, $M=\prod_iM_i$ has square order, and $J$ arises from a $2$-cocycle on $\widehat M$. For $n\geq3$, determine whether the twisted Hopf algebra $(KG)^J$ is simple.

来源字段：https://doi.org/10.4171/owr/2024/5

02458 | OWR-14298590-003

### Simpson’s Weak-Units Conjecture for Higher Categories

O / open | Oberwolfach 报告问题汇编

Are fair $n$-categories equivalent to weakly globular $n$-fold categories, as predicted by Simpson's weak-units conjecture?

来源字段：https://doi.org/10.4171/owr/2024/39

02459 | OWR-14299522-012

### Single Generation and Free-Group $C^*$-Superrigidity

P / partially_solved | Oberwolfach 报告问题汇编

Is every unital simple separable $C^*$-algebra singly generated? Is the free group $F_2$ determined, among hyperbolic groups, by its reduced $C^*$-algebra, possibly in a superrigid sense?

来源字段：https://doi.org/10.4171/owr/2025/35

02460 | OWR-15585-001

### Single-Prime Characterizations of the UCT Problem

O / open | Oberwolfach 报告问题汇编

Does the characterization of the UCT problem remain valid if the quantifier over primes is restricted exclusively to $p=2$, or to another single prime?

来源字段：https://doi.org/10.4171/owr/2017/42

02461 | OWR-17289-007

### Singular Loci of Maximal Flat Flag Degenerations

P / partially_solved | Oberwolfach 报告问题汇编

For an irreducible component $C(A)$ of the maximal flat linear degeneration $\operatorname{Fl}^{mf}(V)$ of the flag variety, what is its singular locus?

来源字段：https://doi.org/10.4171/owr/2019/46

02462 | OWR-12697689-005

### Singularity Categories as Invariants of Singularities

O / open | Oberwolfach 报告问题汇编

What information about a singularity is determined by its singularity category $D_{\mathrm{sg}}(R)$?

来源字段：https://doi.org/10.4171/owr/2023/6

02463 | OWR-13750328-015

### Singularity-Category Support and the Group-Theoretic Nucleus

O / open | Oberwolfach 报告问题汇编

For every finite group $G$, prove that the support of the singularity category $D_{\mathrm{sg}}(C^*BG)$ is exactly the group-theoretic nucleus $\Theta_G$.

来源字段：https://doi.org/10.4171/owr/2023/19

02464 | OWR-1191-002

### Skew Fields between $\mathbb{C}G$ and $U(G)$

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a torsion-free group, let $L(G)$ be its group von Neumann algebra, and let $U(G)$ be the algebra of densely defined closed operators affiliated to $L(G)$. Does there exist a skew field (division ring) $D$ satisfying $\mathbb{C}G \subseteq D \subseteq U(G)$ (the Linnell skew-field question)?

来源字段：https://doi.org/10.4171/owr/2006/09

02465 | OWR-12697693-001

### Smallest Solvable Group Excluded from Abelian-Type Skew Braces

O / open | Oberwolfach 报告问题汇编

Among solvable finite groups that are not the multiplicative group of any skew brace of abelian type, determine one of minimum order.

来源字段：https://doi.org/10.4171/owr/2023/9

02466 | OWR-15217-004

### Smooth Moduli Spaces of Tame Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Let $A$ be a tame finite-dimensional algebra, let $C\subseteq\operatorname{rep}(A,d)$ be a closed irreducible invariant subvariety with $\theta$-stable decomposition $C=m_1C_1\dotplus\cdots\dotplus m_rC_r$, and suppose no $C_i$ is an orbit closure. Is the natural finite birational surjection $$ \Psi:\prod_{i=1}^r S^{m_i}(\mathcal M(C_i)_{\theta}^{ss})\longrightarrow\mathcal M(C)_{\theta}^{ss} $$ always an isomorphism? More generally, are moduli spaces of semistable representations of tame algebras always smooth?

来源字段：https://doi.org/10.4171/owr/2017/12

02467 | OWR-14299281-001

### Smooth Projective Realization of Connected Algebraic Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a smooth connected algebraic group over a field $k$. Does there exist a smooth projective $k$-variety $X$ whose automorphism group is isomorphic to $G$?

来源字段：https://doi.org/10.4171/owr/2025/20

02468 | OWR-16768-006

### Smooth Varieties with Squarefree Gröbner Degenerations

O / open | Oberwolfach 报告问题汇编

Let $I$ be a homogeneous ideal in a standard graded polynomial ring $S$ over a field $K$. Suppose $I$ defines a nonsingular projective variety and has a square-free initial ideal for some monomial order. Must $S/I$ be Cohen-Macaulay with negative $a$-invariant? For curves, must every such nonsingular projective curve have genus zero?

来源字段：https://doi.org/10.4171/owr/2019/6

02469 | OWR-13943-002

### Snake-Graph Models for Generalized Cluster Variables

O / open | Oberwolfach 报告问题汇编

For a generalized cluster algebra of type $C_n$ with generalized exchange polynomials, does every cluster variable correspond to a connected subgraph of its generalized snake graph? Do analogous snake-graph descriptions hold for the affine types covered by the same construction?

来源字段：https://doi.org/10.4171/owr/2015/48

02470 | OWR-16635-003

### Socle Filtrations from Duflo-Serganova Kernels

P / partially_solved | Oberwolfach 报告问题汇编

For $k\leq\min(m,n)$, let $K^k_{m\mid n}$ be generated by the objects annihilated by every Duflo-Serganova functor $\operatorname{DS}_x$ with $x\in X_k$. Is the equality

$$\operatorname{soc}_kK_{m\mid n}=K^k_{m\mid n}$$

valid?

来源字段：https://doi.org/10.4171/owr/2018/52

02471 | OWR-14299910-001

### Socles of Products from Lusztig Components

O / open | Oberwolfach 报告问题汇编

Let $C$ and $D$ be irreducible components of Lusztig nilpotent varieties of Dynkin type $A$, with associated simple modules $\pi(C)$ and $\pi(D)$. If $C$ is rigid, prove that $\operatorname{soc}(\pi(C)\times\pi(D))=\pi(C*D)$.

来源字段：https://doi.org/10.4171/owr/2026/6

02472 | OWR-13750328-006

### Source-Algebra Classification of Cyclic Blocks

P / partially_solved | Oberwolfach 报告问题汇编

For a cyclic block $B$ of a finite group $G$, determine whether $W(B)$ is recoverable from the character table of $G$; determine $W(B)$ for every nonuniserial cyclic block and every cyclic block of a finite quasisimple group; and reduce the general classification to the quasisimple case.

来源字段：https://doi.org/10.4171/owr/2023/19

02473 | OWR-5149-014

### Specht Modules with Abelian Vertices

P / partially_solved | Oberwolfach 报告问题汇编

Classify all indecomposable Specht modules over a field of characteristic $p$ whose vertices are abelian.

来源字段：https://doi.org/10.4171/owr/2011/21

02474 | OWR-12697689-013

### Specialness of Local Koszul Homomorphisms

O / open | Oberwolfach 报告问题汇编

Is every local Koszul homomorphism special Koszul?

来源字段：https://doi.org/10.4171/owr/2023/6

02475 | OWR-8415346-004

### Spectrahedrality Under the Eigenvalue-Set Transformation

O / open | Oberwolfach 报告问题汇编

Let $M\subseteq\mathbb R^n$ be invariant under coordinate permutations, let $\mathcal S^n$ be the space of real symmetric $n\times n$ matrices, and define $$\Lambda(M)=\{A\in\mathcal S^n:\lambda(A)\in M\},$$ where $\lambda(A)$ is the ordered eigenvalue vector of $A$. Is $M$ a spectrahedron if and only if $\Lambda(M)$ is a spectrahedron?

来源字段：https://doi.org/10.4171/owr/2021/43

02476 | OWR-12732-006

### Spectral Coordinates for Grassmannian Cluster Structures

O / open | Oberwolfach 报告问题汇编

For the affine cone over $\operatorname{Gr}(k,n)$, viewed through decorated $\operatorname{GL}(k)$-connections, do spectral coordinates recover the cluster structure on its homogeneous coordinate ring, with a spectral network corresponding to every cluster?

来源字段：https://doi.org/10.4171/owr/2013/58

02477 | OWR-1062-004

### Spectral Measures of Arbitrary Corepresentation Coefficients

O / open | Oberwolfach 报告问题汇编

Determine spectral measures for arbitrary coefficients of a corepresentation $u$ of a universal Hopf algebra, extending the known formulas for its character $\chi(u)$.

来源字段：https://doi.org/10.4171/owr/2005/41

02478 | OWR-1265-008

### Spectral Sequences for Extensions of Categories

P / partially_solved | Oberwolfach 报告问题汇编

For extensions of categories, do the five-term cohomology sequences arise from Lyndon-Hochschild-Serre spectral sequences, as they do for group extensions?

来源字段：https://doi.org/10.4171/owr/2006/15

02479 | OWR-1062-002

### Spectral-Measure Algebras of Free Quantum Groups

O / open | Oberwolfach 报告问题汇编

Classify the algebras arising in the spectral-measure theory of free quantum groups.

来源字段：https://doi.org/10.4171/owr/2005/41

02480 | OWR-14297741-013

### Spectrum Comparison for Finite Tensor Categories

O / open | Oberwolfach 报告问题汇编

Let $\mathcal T$ be a finite tensor category, let $\underline{\mathcal T}$ be its stable category, and let $C_{\mathcal T}^{\bullet}$ be the categorical center of its cohomology ring. Is the continuous central-support comparison map $$\rho:\operatorname{Spc}(\underline{\mathcal T})\longrightarrow\operatorname{Proj} C_{\mathcal T}^{\bullet}$$ from the Balmer spectrum of the stable category to the projective spectrum of the categorical center a homeomorphism for every $\mathcal T$?

来源字段：https://doi.org/10.4171/owr/2023/40

02481 | OWR-794-002

### Splitting Bundles for Hilbert-Scheme Equivalences

P / partially_solved | Oberwolfach 报告问题汇编

Choose a splitting bundle $V_0$ on the Frobenius-twisted punctual Hilbert scheme $\operatorname{Hilb}^{(1)}_0$ such that $\Gamma(\operatorname{Hilb}^{(1)}_0,V_0(-1)) =L_{\mathrm{sign}}$. Prove that the resulting composite derived equivalence preserves the natural $t$-structures and sends each simple $S_n$-module object $L_\tau$ to $\tau_H$.

来源字段：https://doi.org/10.4171/owr/2005/13

02482 | OWR-1452-009

### Splitting Projective Modules with Vanishing Top Chern Class

O / open | Oberwolfach 报告问题汇编

Let $X=\operatorname{Spec}A$ be a smooth affine variety of dimension $n\geq2$ over a field, and let $P$ be a projective $A$-module of rank $n$. If its top Chern class $c_n(P)\in CH_0(X)$ vanishes, must $P$ split off a free rank-one summand?

来源字段：https://doi.org/10.4171/owr/2007/01

02483 | OWR-8415355-007

### Splitting Subgroups of General Linear Supergroups

P / partially_solved | Oberwolfach 报告问题汇编

In the representation theory of the general linear supergroup $G = \mathrm{GL}(m|n)$ (with $m \geq n$), a subgroup $K \subseteq G$ is called a *splitting subgroup* if the restriction of every projective $G$-module to $K$ remains projective, and let $H = \mathrm{SL}(1|1)^n \subseteq \mathrm{GL}(m|n)$ be the minimal standard splitting subgroup of Serganova and Sherman. Does every splitting subgroup $K$ of $\mathrm{GL}(m|n)$ contain a subgroup conjugate to $H = \mathrm{SL}(1|1)^n$?

来源字段：https://doi.org/10.4171/owr/2021/52

02484 | OWR-824-003

### Square-Free Initial Ideals of Linear Determinantal Ideals

P / partially_solved | Oberwolfach 报告问题汇编

For every choice of the coefficients $a_{ijk}$ and every term order $<$ on $K[t_{ij}]$, the initial ideal of $I_2(L)$ should be square-free in the $\mathbb{Z}^m$-graded sense; that is, it should be generated by monomials $t_{i_1j_1}\cdots t_{i_kj_k}$ with $i_1<i_2<\cdots<i_k$. Does the same statement hold for every distributive lattice?

来源字段：https://doi.org/10.4171/owr/2005/19

02485 | OWR-1381-008

### Stable Cohomology of Orthogonal Groups over Binary Fields

O / open | Oberwolfach 报告问题汇编

Does the inverse system of cohomology groups for orthogonal groups over $\mathbb{F}_2$, analogous to the stabilizing system for general linear groups, stabilize? Is its stable value described by an analogue of the Betley-Suslin theorem in the functor category $\mathcal{F}_{\mathrm{quad}}$?

来源字段：https://doi.org/10.4171/owr/2006/43

02486 | MOD-006

### Stable Field Conjecture

P / partially_solved | 独立 / 未分配集合

Is every infinite field with a stable first-order theory separably closed?

背景链接：https://doi.org/10.1017/jsl.2022.59

02487 | OWR-11101920-014

### Stable Limits of Plethysm Multiplicities

P / partially_solved | Oberwolfach 报告问题汇编

What is the limit of $A_{\ell,m}(\nu,\lambda)$, and can the known stability results for plethysm multiplicities be generalized from $(m-1,1)$ to arbitrary partitions of $m$?

来源字段：https://doi.org/10.4171/owr/2022/39

02488 | AIM-REPRESENTATION_THEORY-0068

### Stable matching through commutator fibers

P / partially_solved | AIM 研讨会问题汇编

Determine notion of stable conjugacy when stable conjugacy classes in $\widetilde G_F$ and $\widetilde H_L$ naturally are related.

来源字段：http://aimpl.org/autoformcovergp/2/

02489 | AIM-REPRESENTATION_THEORY-0035

### Stable modular factors: finite periodic reduction and a sharp period correction

P / partially_solved | AIM 研讨会问题汇编

Hard problem in modular \(S_n\) representation theory: determining simple factors of Specht modules. Are stable values (for example if the representations are coming from a finitely generated \(\mathrm{FI}\)-module) easier?

来源字段：http://aimpl.org/repnstability/3/

02490 | OWR-17133-010

### Stable Rank One for Simple Reduced Group $C^*$-Algebras

O / open | Oberwolfach 报告问题汇编

For every discrete group $G$ such that $C^*_{\lambda}(G)$ is simple, does $C^*_{\lambda}(G)$ have stable rank one?

来源字段：https://doi.org/10.4171/owr/2019/37

02491 | OWR-16628-004

### Stable Symbolic-Power Containments from a Single Containment

O / open | Oberwolfach 报告问题汇编

Let $R$ be a regular ring and let $I$ be a radical ideal of big height $h$. If $I^{(hk-C)}\subseteq I^k$ for some $k$, must $I^{(hn-C)}\subseteq I^n$ for all sufficiently large $n$?

来源字段：https://doi.org/10.4171/owr/2018/45

02492 | AIM-REPRESENTATION_THEORY-0058

### Steenrod stability, parity summands, and the Satake fiber

P / partially_solved | AIM 研讨会问题汇编

1. What is the representation-theoretic meaning of the Steenrod algebra action on mod-\(p\) cohomology? For example, on the cohomology of \(Gr\)? 2. Does the Steenrod algebra act on the cohomology of any parity sheaf? (Note: examples of Goresky show that this fails for mod-\(p\) IC sheaves.)

来源字段：http://aimpl.org/sheavemodular/13/

02493 | OWR-14299915-023

### Steenrod-Closed Parameter Ideals at Every Prime

O / open | Oberwolfach 报告问题汇编

For every prime $p$, classify the Steenrod-closed parameter ideals in $\mathbb F_p[a,b]^{\operatorname{SL}_2(p)}$ and prove that no ideals occur beyond the fibered, twisted, and mixed families identified for $p=3$.

来源字段：https://doi.org/10.4171/owr/2026/9

02494 | OWR-13681-006

### Steinberg Tensor Products with Good Filtrations

O / open | Oberwolfach 报告问题汇编

Let $G$ be a simple algebraic group over an algebraically closed field of characteristic $p > 0$, let $X^+$ be the set of dominant weights, let $L(\lambda)$ be the simple rational $G$-module of highest weight $\lambda \in X^+$, and let $\mathrm{St}_r = L((p^r - 1)\rho)$ be the $r$-th Steinberg representation. For which dominant weights $\lambda \in X^+$ does the tensor product $\mathrm{St}_r \otimes L(\lambda)$ admit a good filtration (i.e., a filtration with dual-Weyl / costandard quotients $\nabla(\mu)$) as a rational $G$-module?

来源字段：https://doi.org/10.4171/owr/2015/24

02495 | OWR-14298370-003

### Strict Convergent Quantization of Poisson Structures

P / partially_solved | Oberwolfach 报告问题汇编

Which Poisson structures admit strict convergent quantizations?

来源字段：https://doi.org/10.4171/owr/2024/20

02496 | OWR-15585-006

### Strong Approximate Innerness of Quasi-Free Actions

P / partially_solved | Oberwolfach 报告问题汇编

Is every finite abelian group action on a simple purely infinite Cuntz-Krieger algebra by diagonal quasi-free automorphisms strongly approximately inner in Izumi's sense?

来源字段：https://doi.org/10.4171/owr/2017/42

02497 | OWR-2654827-002

### Strong Lefschetz Property of the Zeroth Jordan Component

O / open | Oberwolfach 报告问题汇编

For an Artinian Gorenstein algebra $A$ and the component $Q(0)$ of its Jordan decomposition, can $A$ have the strong Lefschetz property while $Q(0)$ does not?

来源字段：https://doi.org/10.4171/owr/2020/31

02498 | OWR-13673-004

### Strong Locality and Reconstruction of Vertex Operator Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Is every unitary vertex operator algebra strongly local? Does every irreducible conformal net arise from a unitary vertex operator algebra through the conformal-net construction?

来源字段：https://doi.org/10.4171/owr/2015/16

02499 | OWR-787-006

### Strong Nakayama Conjecture

O / open | Oberwolfach 报告问题汇编

Does every Artin algebra $A$ have finite finitistic dimension? For every nonzero finitely generated $A$-module $M$, must there be an integer $n\geq0$ such that $\operatorname{Ext}_A^n(M,A)\neq0$?

来源字段：https://doi.org/10.4171/owr/2005/06

02500 | OWR-14299522-011

### Structural Problems for $C^*$- and von Neumann Algebras

O / open | Oberwolfach 报告问题汇编

Do the normalized random matrices $X_k$ converge strongly to a free circular distribution? Are $q$-Gaussian $C^*$-algebras selfless for $-1<q<1$ and $q\neq0$? Are the indicated tracial ultrapowers mutually isomorphic? Does there exist a group that has a mixed identity and whose reduced $C^*$-algebra is selfless? Does $C^*(F_2\times F_2)$ have the local lifting property? Does a nonamenable group without free subgroups exist whose reduced $C^*$-algebra has rapid decay? For compact convex $K$, does equality of its minimal and maximal operator-system structures imply that $K$ is a Choquet simplex? Are the free group factors $L(F_2)$ and $L(F_3)$ nonisomorphic? Are the free-product factors $Z(r,\tau)^{*n}$ and $Z(r,\tau)^{*m}$ isomorphic, and is there a dichotomy? Is every unital simple separable $C^*$-algebra singly generated? Do classifiable $C^*$-algebras admit filtrations with rapid decay? Is $F_2$ superrigid among hyperbolic groups at the level of reduced $C^*$-algebras? When is a von Neumann algebra with a faithful state selfless?

来源字段：https://doi.org/10.4171/owr/2025/35

02501 | OWR-17294-001

### Structure Algebras of Yang-Baxter Solutions

P / partially_solved | Oberwolfach 报告问题汇编

For a set-theoretic Yang-Baxter solution $(X,r)$ with structure algebra $kM(X,r)$ and associated injective $1$-cocycle $\pi$, when is $\pi$ injective, and what is the algebraic structure of $kM(X,r)$ in the injective and general cases?

来源字段：https://doi.org/10.4171/owr/2019/51

02502 | OWR-12978-003

### Structure and Classification Problems for Noetherian Hopf Algebras

O / open | Oberwolfach 报告问题汇编

Let $H$ be a noetherian Hopf algebra. Are the quotients $H/P$ by minimal prime ideals Morita equivalent, and if not, which properties do they share, including polynomial-identity status and Gelfand-Kirillov dimension? Is every noetherian Hopf algebra strongly noetherian? For extensions of enveloping algebras by finite-dimensional pointed Hopf algebras, classify the extensions with fixed $H$ or fixed Gelfand-Kirillov dimension and describe their root systems. Can every known finite-dimensional Nichols algebra be obtained from a diagonal one by diagram folding, and do folded liftings exhaust all liftings?

来源字段：https://doi.org/10.4171/owr/2014/20

02503 | OWR-12978-001

### Structure and Finiteness of General Nichols Algebras

O / open | Oberwolfach 报告问题汇编

For a braided vector space $V$, describe the structure of its Nichols algebra $\mathcal B(V)$. Characterize when $\mathcal B(V)$ is finite-dimensional, noetherian, or of finite Gelfand-Kirillov dimension.

来源字段：https://doi.org/10.4171/owr/2014/20

02504 | OWR-12697693-007

### Structure Groups of Anisotropic Bilinear Forms

O / open | Oberwolfach 报告问题汇编

Let $b(v,w)=v^{\mathsf T}Aw$ be an anisotropic, not necessarily symmetric, bilinear form on $K^n$. Find a representation of the structure group $G(X(b))$ that generalizes the pure paraunitary model $$G(X(b))\cong\{\varphi(t)\in\operatorname{GL}_n(K[t,t^{-1}]):\varphi(t^{-1})^{\mathsf T}A\varphi(t)=A,\ \varphi(1)=I\}$$ available in the symmetric case.

来源字段：https://doi.org/10.4171/owr/2023/9

02505 | OWR-14299580-010

### Structure Objects and Deformations of Welded Pairs

O / open | Oberwolfach 报告问题汇编

What object plays for a welded pair the role that the structure skew brace plays for a solution? Is there a common framework for the two families of deformations?

来源字段：https://doi.org/10.4171/owr/2025/49

02506 | OWR-8415355-002

### Structure of Cohomological Hall Algebras of Preprojective Algebras

P / partially_solved | Oberwolfach 报告问题汇编

What is the structure of the entire cohomological Hall algebra $A_{\Pi_Q}$ of the preprojective algebra $\Pi_Q$?

来源字段：https://doi.org/10.4171/owr/2021/52

02507 | OWR-1538-003

### Structure of Generalized Verma Modules

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathfrak{g}$ be a semisimple complex Lie algebra, let $\mathfrak{p}$ be a parabolic subalgebra containing a Borel subalgebra, let $\mathfrak{a}$ be the reductive part of $\mathfrak{p}$, and let $V$ be a simple $\mathfrak{a}$-module. Determine the structure of the generalized Verma module $\Delta(\mathfrak{p},V)=U(\mathfrak{g})\otimes_{U(\mathfrak{p})}V$, including its simple composition factors and their multiplicities.

来源字段：https://doi.org/10.4171/owr/2007/22

02508 | OPG-48715

### Sub-atomic product of funcoids is a categorical product

P / partially_solved | Open Problem Garden

Conjecture In the category of continuous funcoids (defined similarly to the category of topological spaces) the following is a direct categorical product:

- Product morphism is defined similarly to the category of topological spaces. - Product object is the sub-atomic product. - Projections are sub-atomic projections.

See details, exact definitions, and attempted proofs here.

背景链接：http://www.openproblemgarden.org/op/sub_atomic_product_of_funcoids_is_a_categorical_product

02509 | OWR-12175-024

### Subhomogeneous Character-Free Subalgebras of Infinite Tensor Powers

O / open | Oberwolfach 报告问题汇编

Let $B$ be a separable unital $C^*$-algebra without characters and with separating tracial states. Does $B\otimes_{\min}B\otimes_{\min}\cdots$ contain a subhomogeneous $C^*$-algebra without characters?

来源字段：https://doi.org/10.4171/owr/2012/52

02510 | OWR-15217-001

### Support Ideals in Constructible-Function Convolution Algebras

O / open | Oberwolfach 报告问题汇编

For the convolution algebra of constructible functions associated with quiver data $\Pi$, is the direct sum of the homogeneous subspaces consisting of functions $f$ such that $$ \dim\operatorname{supp}(f)<\dim G(d)-q_D^C(d/D) $$ an ideal?

来源字段：https://doi.org/10.4171/owr/2017/12

02511 | OWR-1452-008

### Surjectivity of Reduction Maps for Special Polynomial Automorphisms

O / open | Oberwolfach 报告问题汇编

For a commutative ring $R$ containing $\mathbb{Q}$, $m\geq1$, and $Y=(y_1,\ldots,y_n)$, is the reduction map $\operatorname{SAut}_{R[t]}(R[t][Y])\to\operatorname{SAut}_{R[t]/(t^m)}(R[t]/(t^m)[Y])$ surjective, where $\operatorname{SAut}$ denotes polynomial automorphisms with Jacobian determinant $1$?

来源字段：https://doi.org/10.4171/owr/2007/01

02512 | OWR-3472-012

### Symbolic Rees Rings and Nagata’s Conjecture

O / open | Oberwolfach 报告问题汇编

For pairwise coprime positive integers $a,b,c$, determine the relationship between finite generation of $R_s(P(a,b,c))=\bigoplus_{m\geq0}P(a,b,c)^{(m)}$ for the monomial curve $(t^a,t^b,t^c)$ and Nagata's Conjecture for general points in $\mathbb P^2$; in particular, characterize when this symbolic Rees ring is Noetherian.

来源字段：https://doi.org/10.4171/owr/2009/22

02513 | OWR-12863-001

### Symplectic Periods of Essentially Speh Representations

O / open | Oberwolfach 报告问题汇编

Let $\mathcal{G}'$ consist of representations $Q(\Delta_1,\ldots,\Delta_{2r})$ built from ladder representations such that every $Q(\Delta_{2i+1},\Delta_{2i+2})$ is essentially Speh. Is every $\theta\in\mathcal{G}'$ endowed with a symplectic period?

来源字段：https://doi.org/10.4171/owr/2014/03

02514 | OWR-14754-006

### Symplectic Reduction of Visible Polar Representations

O / open | Oberwolfach 报告问题汇编

For a visible polar representation $(G,V)$, is the natural Poisson morphism $(\mathfrak c\oplus\mathfrak c^*)/W\to(V\oplus V^*)\mathbin{///}G$ an isomorphism?

来源字段：https://doi.org/10.4171/owr/2016/46

02515 | AIM-REPRESENTATION_THEORY-0063

### Tame covering-group Iwahori--Matsumoto presentation and a rank-one parity diagnostic

P / partially_solved | AIM 研讨会问题汇编

For a split connected reductive group $G/F$ and a finite Brylinski--Deligne central cover $\widetilde G$, describe the Hecke algebra attached to a split Iwahori and a fixed genuine central character, first in the tame/unramified case.

来源字段：http://aimpl.org/autoformcovergp/1/

02516 | OWR-12871-003

### Tannakian Structures in Feynman Categories

O / open | Oberwolfach 报告问题汇编

Can fiber functors and their associated Tannakian categories be seen directly in the Feynman-category picture, what is the role of Hopf cyclic cohomology, and is there a categorification connecting this story to Hopf algebroids?

来源字段：https://doi.org/10.4171/owr/2014/11

02517 | AIM-REPRESENTATION_THEORY-0047

### Target-local maximal-minor certificates for spherical parity sheaves

P / partially_solved | AIM 研讨会问题汇编

Given a dominant coweight $\lambda$ and a prime $p$, compute the normalized indecomposable spherical parity complex $\mathcal E_\lambda^{(p)}$ on $Gr$, and, when it is perverse, regard it as an object of $\operatorname{Perv}_{\mathrm{sph}}(Gr,\mathbb F_p)$. Can its graded stalk/costalk character be computed more efficiently than by constructing a large portion of the spherical $p$-canonical basis?

来源字段：http://aimpl.org/sheavemodular/5/

02518 | MOD-005

### Tarski's Exponential Function Problem

O / open | 独立 / 未分配集合

别名：Tarski's exponential function problem [OPG-1790]

Is the theory of the real numbers with addition, multiplication, and exponentiation decidable?

背景链接：https://www.maths.ox.ac.uk/node/6577

02519 | OWR-1265-019

### Tensor Products and Extensions of Specht-Filtered Modules

O / open | Oberwolfach 报告问题汇编

Assume $p>3$, and let $M$ and $N$ be indecomposable self-dual $k\Sigma_d$-modules having both Specht and dual Specht filtrations. Must $M\otimes N$ be a direct sum of signed Young modules, and must $\operatorname{Ext}^i_{k\Sigma_d}(M,N)=0$ for $1\leq i\leq p-2$?

来源字段：https://doi.org/10.4171/owr/2006/15

02520 | OWR-14299522-001

### Tensor-Factorization Uniqueness for SAW$^*$-Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Let $A_i$ for $i<m$ and $B_j$ for $j<n$ be SAW$^*$-algebras such that $\bigotimes_{i<m}A_i\cong\bigotimes_{j<n}B_j$. Must $m=n$, and, after permuting the factors, must $A_i$ be stably isomorphic to $B_i$ for every $i$? In particular, for the Calkin algebra $\mathcal{Q}(H)$, is $\mathcal{Q}(H)\otimes\mathcal{Q}(H)$ isomorphic to $\mathcal{Q}(H)\otimes\mathcal{Q}(H)\otimes\mathcal{Q}(H)$?

来源字段：https://doi.org/10.4171/owr/2025/35

02521 | OWR-2048-008

### Tensor-Product Masas and the Extension Property

P / partially_solved | Oberwolfach 报告问题汇编

Let $C\subseteq B(H)$ be a maximal abelian self-adjoint algebra. Is $(C\otimes1)^c=C\otimes B(H)$ inside $B(H)\otimes_{\max}B(H)$? More generally, is $C_1\otimes C_2$ maximal abelian in $B(H)\otimes_{\max}B(H)$ for arbitrary maximal abelian self-adjoint algebras $C_1,C_2$? If $C\cong\ell^\infty(\mathbb{N})$, does $C$ have the extension property, and is $C\otimes C$ maximal abelian in $B(H)\otimes_{\max}B(H)$?

来源字段：https://doi.org/10.4171/owr/2008/37

02522 | OWR-8415345-006

### The Alekseev-Torossian Symmetry-Group Conjecture

O / open | Oberwolfach 报告问题汇编

Prove the Alekseev-Torossian symmetry-group identities $KV\cong GT_1\times K$ and $KRV\cong GRT_1\times K$ for Kashiwara-Vergne solutions.

来源字段：https://doi.org/10.4171/owr/2021/42

02523 | ALG-016

### The Babai Conjecture on Graph Isomorphism

S / solved | 独立 / 未分配集合

Can graph isomorphism be decided in quasi-polynomial time for all graphs?

背景链接：https://arxiv.org/abs/1512.03547

02524 | AIM-REPRESENTATION_THEORY-0048

### The center of the modular principal block as a pro-center of geometric truncations

P / partially_solved | AIM 研讨会问题汇编

Is there a geometric description of the center of $\mathrm{Rep}_0G$? Possibly related to affine Springer fibers?

来源字段：http://aimpl.org/sheavemodular/6/

02525 | ALG-039

### The Cherlin-Zilber Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Cherlin-Zilber Conjecture [LOGIC-002]; Cherlin-Zilber Conjecture [MOD-001]

Is every simple group with a stable first-order theory an algebraic group over an algebraically closed field?

背景链接：https://arxiv.org/abs/2606.18207

02526 | AIM-REPRESENTATION_THEORY-0002

### The compact obstruction to a relative conductor

P / partially_solved | AIM 研讨会问题汇编

**Relative conductor of $p$-adic groups.** Let $H\leq G$ be groups over $\mathbb Q_p$, $\pi$ be an admissible representation of $G$, and $K_G$ be a maximal compact of $G$. Define $$ K_H(p^n):=\{g\in K_G\mid (g\bmod p^n)\in H\}. $$ Does $\pi$ have a non-zero vector fixed by $K_H(p^n)$ for some $n$? If so, what is the smallest such $n$?

来源字段：http://aimpl.org/aagaautomorphic/1/

02527 | ALG-036

### The Demazure Conjecture

P / partially_solved | 独立 / 未分配集合

Can representations of semisimple algebraic groups be characterized over the integers?

背景链接：https://www.numdam.org/article/PMIHES_1987__65__61_0.pdf

02528 | OWR-17133-004

### The Dixmier Unitarisability Problem

O / open | Oberwolfach 报告问题汇编

Is every unitarisable discrete group amenable?

来源字段：https://doi.org/10.4171/owr/2019/37

02529 | AIM-REPRESENTATION_THEORY-0097

### The dual NCSym realization and its internal coproduct

P / partially_solved | AIM 研讨会问题汇编

Describe explicitly the Hopf isomorphism $$ SC^{(2)*}\longrightarrow \Pi^*\cong\Pi QSym $$ in these two presentations, and make it compatible with an internal comultiplication.

Its key fragments are

Comparing the display with the later polynomial realization in Aguiar et al.

Let $\Pi=\mathrm{NCSym}$ be the Hopf algebra of symmetric functions in noncommuting variables. For a set partition $\mu\vdash[n]$, its monomial basis is

$$ \mathbf m_\mu=M_\mu=\sum_{\nabla\omega=\mu}\omega, \tag{1.1} $$

where $\omega\in A^*$, $A=\{a_1,a_2,\ldots\}$ is a noncommuting alphabet, and $\nabla\omega$ is the partition of the positions of $\omega$ into equal-letter classes.

For the dual polynomial realization, let the $x_{ij}$ be commuting variables with the rook relations

$$ x_{ij}x_{ik}=0\quad(j\ne k), \qquad x_{ik}x_{jk}=0\quad(i\ne j). \tag{1.2} $$

For $\sigma\in\mathfrak S_n$, put

$$ M_\sigma=\sum_{i_1<\cdots<i_n} x_{i_1,i_{\sigma(1)}}\cdots x_{i_n,i_{\sigma(n)}} \tag{1.3} $$

$$ U_\mu=\sum_{\operatorname{csupp}(\sigma)=\mu}M_\sigma, \tag{1.4} $$

where $\operatorname{csupp}(\sigma)$ is the partition of $[n]$ whose blocks are the supports of the disjoint cycles of $\sigma$. The span of the $U_\mu$ is denoted $\Pi QSym$ and realizes $\Pi^*$.

Thus the recovered question is:

The infinite-index sum (1.3) is the rigorous published version of the abbreviated monomial in the workshop display.

There remains one genuine mathematical ambiguity: the workshop sheet does not define “internal comultiplication.” The natural operation on $SC^{(2)*}_n$ is the dual of pointwise multiplication of superclass functions in the fixed group $UT_n(2)$. This report solves that reading. The familiar internal coproduct on the primal algebra $\Pi$, if that was intended instead, dualizes to an **internal product** on $\Pi^*$, not to a coproduct; identifying the two requires extra self-duality data not specified in the question.

来源字段：https://aimath.org/WWN/supercharacters/supercharacters.pdf

02530 | AIM-REPRESENTATION_THEORY-0036

### The finite-point shadow of derived Borel induction

P / partially_solved | AIM 研讨会问题汇编

What does the induction theorem of Achar--Riche/Hodge--Karuppuchamy--Scott imply about the relationship between representations of \(G(\mathbb F_q)\) and \(B(\mathbb F_q)\) in defining characteristic?

来源字段：http://aimpl.org/sheavemodular/1/

02531 | ALG-040

### The Generalized Star Height Problem

O / open | 独立 / 未分配集合

别名：Generalized Star Height Problem [MOD-002]

Can all regular languages be expressed with generalized regular expressions of bounded star height?

背景链接：https://www.labri.fr/perso/tplace/Files/TOCS18.pdf

02532 | ALG-025

### The Guralnick-Thompson Conjecture

S / solved | 独立 / 未分配集合

What are the composition factors of finite groups appearing in genus-0 systems?

背景链接：https://doi.org/10.2307/3062099

02533 | ALG-028

### The Isomorphism Problem for Coxeter Groups

P / partially_solved | 独立 / 未分配集合

Is there an algorithm to determine whether two Coxeter groups given by presentations are isomorphic?

背景链接：https://arxiv.org/abs/math/0506572

02534 | SMA-016

### The Jacobian Conjecture

P / partially_solved | Smale 问题

别名：Jacobian Conjecture [OPG-1768]; Jacobian Conjecture for Polynomial Maps [OWR-1452-011]

If $F: \mathbb{C}^n \to \mathbb{C}^n$ is a polynomial map with constant non-zero Jacobian determinant, then $F$ is invertible.

背景链接：https://terrytao.wordpress.com/2026/07/21/a-digestion-of-the-jacobian-conjecture-counterexample/

02535 | ALG-003

### The Köthe Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Köthe Conjecture [ALG-050]; Köthe Conjecture [ALG-059]

A ring has no non-zero nil ideal (an ideal all of whose elements are nilpotent) if and only if it has no non-zero nil one-sided ideal.

背景链接：https://arxiv.org/abs/2609.07996

02536 | OWR-3472-002

### The Lex-Plus-Powers Betti Number Conjecture

O / open | Oberwolfach 报告问题汇编

Let $S=k[x_1,\ldots,x_n]$, let $F=(f_1,\ldots,f_r)$ be a homogeneous regular sequence with $2\leq e_1\leq\cdots\leq e_r$, and let $P=(x_1^{e_1},\ldots,x_r^{e_r})$. For every homogeneous ideal $I\supseteq F$, does there exist a unique lex-plus-$P$ ideal $L$ with the same Hilbert function, with $b_{i,j}(S/L)\geq b_{i,j}(S/I)$ for all $i,j$?

来源字段：https://doi.org/10.4171/owr/2009/22

02537 | AMR-113-0001

### The Lyapin's Notebook - Problem 1

O / open | AMR 问题清单汇编

(S. Kublanovsky, M. Sapir) Describe the types of equations or systems of equations for which there exists an algorithm that decides for any values of the constants whether the given system is solvable in some finite extension of the original semigroup.

背景链接：https://arxiv.org/abs/2604.04763

02538 | AMR-113-0010

### The Lyapin's Notebook - Problem 10

O / open | AMR 问题清单汇编

(E.S. Lyapin) Under what conditions is a system of identical inequalities (identical relations of precedence) equivalent to a system of identities?\\ This problem is solved for systems consisting of a single identical inequality in [Mak1].

背景链接：https://arxiv.org/abs/2604.04763

02539 | AMR-113-0011

### The Lyapin's Notebook - Problem 11

O / open | AMR 问题清单汇编

(E.S. Lyapin) Describe all varieties of ordered semigroups that have covers in the lattice of varieties of ordered semigroups. \\ Some such varieties are described in works [Mak2], [Mak3]. In the lattice of varieties of ordered semigroups, not every variety has a cover (see [Mak2]).

背景链接：https://arxiv.org/abs/2604.04763

02540 | AMR-113-0012

### The Lyapin's Notebook - Problem 12

O / open | AMR 问题清单汇编

(V. Makaridina) Describe all varieties of ordered semigroups in which every semigroup has only the trivial order.\\ Some examples are given in [Mak4].

背景链接：https://arxiv.org/abs/2604.04763

02541 | AMR-113-0013

### The Lyapin's Notebook - Problem 13

O / open | AMR 问题清单汇编

Let $K$ be a class of semigroups. For each natural number $n$, denote by $f_K(n)$ the minimal number of endomorphisms among all finite semigroups from $K$ having at least $n$ elements. The function $f_K(n)$ is called the *endomorphic function* of the class $K$. A class $K$ is called *endomorphic* if its endomorphic function is unbounded. In particular, we are interested in endomorphic varieties of semigroups. The following problems arise concerning endomorphic functions.

(S. Kublanovsky) Describe endomorphic varieties of semigroups.\\ Some sufficient conditions were obtained in [kubl].

背景链接：https://arxiv.org/abs/2604.04763

02542 | AMR-113-0014

### The Lyapin's Notebook - Problem 14

O / open | AMR 问题清单汇编

(S. Kublanovsky) Is it true that for every variety of commutative semigroups its endomorphic function is identical? \\ The variety of commutative semigroups is endomorphic (see [kubl]).

背景链接：https://arxiv.org/abs/2604.04763

02543 | AMR-113-0015

### The Lyapin's Notebook - Problem 15

O / open | AMR 问题清单汇编

(S. Kublanovsky) Is it true that for any endomorphic variety of semigroups its endomorphic function is identical?

背景链接：https://arxiv.org/abs/2604.04763

02544 | AMR-113-0016

### The Lyapin's Notebook - Problem 16

O / open | AMR 问题清单汇编

(S. Kublanovsky) Obtain estimates for the values of the endomorphic function for various basic classes of semigroups.

背景链接：https://arxiv.org/abs/2604.04763

02545 | AMR-113-0017

### The Lyapin's Notebook - Problem 17

O / open | AMR 问题清单汇编

(A.A. Borisov) Describe finitely generated Lesokhin semigroups without idempotents.\\ It is proved in [Borisov4] that finitely generated separative Lesokhin semigroups without idempotents should be isomorphic to subsemigroups of an infinite monogenic semigroup.

背景链接：https://arxiv.org/abs/2604.04763

02546 | AMR-113-0018

### The Lyapin's Notebook - Problem 18

O / open | AMR 问题清单汇编

(A.A. Borisov) Describe commutative semigroups all of whose endomorphisms are injective. \\ It is proved in [Borisov4] that finitely generated commutative semigroups all of whose endomorphisms are injective are precisely subsemigroups of an infinite monogenic semigroups or finite ordinal sums of such semigroups.

背景链接：https://arxiv.org/abs/2604.04763

02547 | AMR-113-0019

### The Lyapin's Notebook - Problem 19

O / open | AMR 问题清单汇编

(A.A. Borisov) Describe finitely generated commutative separative E-semigroups. \\ It is proved in [Borisov4] that for a finitely defined commutative semigroup the problem of deciding whether the semigroup of all its endomorphisms (under pointwise multiplication) is Archimedean is algorithmically solvable.

背景链接：https://arxiv.org/abs/2604.04763

02548 | AMR-113-0002

### The Lyapin's Notebook - Problem 2

O / open | AMR 问题清单汇编

(S. Kublanovsky) Describe the divisors of the following subclasses of finite 0-simple semigroups:

- the class of finite Brandt semigroups;

- the class of finite 0-simple semigroups presented in the Rees representation by a matrix that consists only of zeros and ones.

Divisors of the class of all finite 0-simple semigroups are described by positive universal formulas (see [Kubl]).

背景链接：https://arxiv.org/abs/2604.04763

02549 | AMR-113-0020

### The Lyapin's Notebook - Problem 20

O / open | AMR 问题清单汇编

(A.A. Borisov) Describe commutative semigroups whose semigroups of endomorphisms (with pointwise multiplication) is cancelative.

背景链接：https://arxiv.org/abs/2604.04763

02550 | AMR-113-0021

### The Lyapin's Notebook - Problem 21

O / open | AMR 问题清单汇编

(A.A. Borisov) Describe commutative semigroups whose semigroups of endomorphisms (with pointwise multiplication) are isomorphic to one of the following semigroups:

- the additive semigroup of all positive rational numbers;

- the additive semigroup of all non negative rational numbers.

背景链接：https://arxiv.org/abs/2604.04763

02551 | AMR-113-0022

### The Lyapin's Notebook - Problem 23

O / open | AMR 问题清单汇编

(A.A. Borisov) Is it true that the chain rank of any finitely generated (or finitely presented) semigroup without idempotents is finite?\\ It is proved in [Borisov4] that finitely generated commutative semigroups without idempotents have finite chain rank.

背景链接：https://arxiv.org/abs/2604.04763

02552 | AMR-113-0023

### The Lyapin's Notebook - Problem 24

O / open | AMR 问题清单汇编

(A.A. Borisov) Describe commutative semigroups that have exactly four endomorphisms. \\ It is proved in [Borisov3] that every finite commutative semigroup with more than four elements has at least five endomorphisms.

背景链接：https://arxiv.org/abs/2604.04763

02553 | AMR-113-0024

### The Lyapin's Notebook - Problem 25

O / open | AMR 问题清单汇编

In his book *Semigroups*, E.S. Lyapin emphasized that establishing connections between the properties of endomorphism semigroups and the properties of the underlying structured sets is one of the most important directions in semigroup theory. This direction may be viewed as a natural generalization of the ideas of Galois theory.

(E.S. Lyapin) For natural classes of mathematical structures - such as ordered sets, relatively free algebras, and for subsemigroups of endomorphism semigroups of these structures - determine:

- Subclasses of structures defined up to isomorphism by these semigroups;

- Subsemigroups of endomorphism semigroups that determine these structures up to isomorphism.

For various classes of partially ordered sets E.S. Lyapin found endomorphism semigroups that determine these classes up to isomorphism (see [Lyapin4], [L5], [L6]).

背景链接：https://arxiv.org/abs/2604.04763

02554 | AMR-113-0025

### The Lyapin's Notebook - Problem 26

O / open | AMR 问题清单汇编

(G. Mashevitzky) Describe the endomorphism semigroup and the tower of endomorphisms (automorphisms)

- for endomorphism semigroups of semigroups free in basic classes of semigroups, in particular in the classes of idempotent, commutative Clifford, and Clifford semigroups;

- for endomorphism semigroups of algebras free in basic classes of mathematical structures.

For the class of all semigroups and inverse semigroups, the description of semigroups of endo- morphisms were obtained in [MS] and [MSZ]. Towers of automorphisms of free inverse semigroups are described in [Mash3].

背景链接：https://arxiv.org/abs/2604.04763

02555 | AMR-113-0026

### The Lyapin's Notebook - Problem 27

O / open | AMR 问题清单汇编

(G. Mashevitzky)

- Describe transformation semigroups that are closed in the pointwise topology.

- Describe endomorphism semigroups that are closed in a topology of Zarisky type.

Motivation and definitions can be found in [Cameron], [Mash1], [MPP], [Mash2]. Transformation semigroups closed in the pointwise topology are precisely the semigroups of endomorphisms of mathematical structures (see [Mash2]). It was shown in [Mash1] that the Zariski-type topology is weaker than the pointwise topology and conditions were obtained under which these topologies coincide. Several special cases of the problem were solved in [Mash2].

背景链接：https://arxiv.org/abs/2604.04763

02556 | AMR-113-0027

### The Lyapin's Notebook - Problem 28

O / open | AMR 问题清单汇编

(S.I. Kublanovsky) Let $C_0$ be a class of finite $0$-simple semigroups. Is the class $ \mathbb{K}(C_0)$ decidable, where $\mathbb{K}$ is one of the operators $\mathbb{H}\mathbb{P}, \mathbb{H}\mathbb{P}\mathbb{S}, \mathbb{S}\mathbb{H}\mathbb{P}\mathbb{S}$?\\ For all other Birkhoff operators, the answer has been obtained in [Kubl6]).

背景链接：https://arxiv.org/abs/2604.04763

02557 | AMR-113-0028

### The Lyapin's Notebook - Problem 29

O / open | AMR 问题清单汇编

(S.I. Kublanovsky)

- Is the class of finite $0$-simple semigroups over groups, belonging to a pseudovariety of groups with decidable universal theory, completely solvable?

- The same question for the class of finite Brandt semigroups, under a similar restriction on maximal subgroups.

背景链接：https://arxiv.org/abs/2604.04763

02558 | AMR-113-0003

### The Lyapin's Notebook - Problem 3

O / open | AMR 问题清单汇编

(S. Bershadsky, S. Kublanovsky) Is there an algorithm that decides whether a given finite semigroup is embeddable into a power semigroup over some finite group (periodic group) (see Chapter 5 of this notebook)?

Not every finite semigroup has this property, but every semigroup is a divisor of the power semigroup over a periodic group (see [BersKubl]). For divisors of power semigroups over finite groups the answer is positive (see [Henckell]).

背景链接：https://arxiv.org/abs/2604.04763

02559 | AMR-113-0029

### The Lyapin's Notebook - Problem 30

O / open | AMR 问题清单汇编

(S.I. Kublanovsky) Is the class of finite regular aperiodic semigroups completely solvable?

Recall that a semigroup is aperiodic if all its subgroups are trivial.

背景链接：https://arxiv.org/abs/2604.04763

02560 | AMR-113-0030

### The Lyapin's Notebook - Problem 31

O / open | AMR 问题清单汇编

(S.I. Kublanovsky)

- Is the class of finite regular orthodox semigroups completely solvable?

- The same question for the class of finite regular semigroups, whose idempotents belong to a given variety of idempotent semigroups.

背景链接：https://arxiv.org/abs/2604.04763

02561 | AMR-113-0031

### The Lyapin's Notebook - Problem 33

O / open | AMR 问题清单汇编

(S.I. Kublanovsky) Let $\mathbb{K}$ be one of the operators $\mathbb{H}\mathbb{P}$, $\mathbb{H}\mathbb{P}\mathbb{S}$, $\mathbb{S}\mathbb{H}\mathbb{P}$, $\mathbb{S}\mathbb{H}\mathbb{P}\mathbb{S}$, and let $\Phi$ be an arbitrary finite class of finite semigroups. Is the class $Cor(\mathbb{K}(\Phi))$ decidable?

For all other Birkhoff operators the answer is positive (see [Kubl6]).

背景链接：https://arxiv.org/abs/2604.04763

02562 | AMR-113-0032

### The Lyapin's Notebook - Problem 34

O / open | AMR 问题清单汇编

(E. M. Mogiljanskaja) Does the isomorphism of power semigroups implies the isomorphism of the original semigroups in the class of finite semigroups?

背景链接：https://arxiv.org/abs/2604.04763

02563 | AMR-113-0033

### The Lyapin's Notebook - Problem 35

O / open | AMR 问题清单汇编

(S.G. Bershadsky, S.I. Kublanovsky) Find necessary and sufficient conditions for the embeddability of a regular semigroup into the power one over a periodic group.

For completely regular semigroups such conditions are inversion and periodicity (see [Bersh+Kubl]).

背景链接：https://arxiv.org/abs/2604.04763

02564 | AMR-113-0034

### The Lyapin's Notebook - Problem 36

O / open | AMR 问题清单汇编

(S.G. Bershadsky, S.I. Kublanovsky) Is every periodic commutative semigroup embeddable into the power semigroup over a periodic Abelian group?

This is true for all regular commutative semigroups and for some finite non-regular commutative semigroups (see [Bersh+Kubl]).

背景链接：https://arxiv.org/abs/2604.04763

02565 | AMR-113-0035

### The Lyapin's Notebook - Problem 37

O / open | AMR 问题清单汇编

(S.G. Bershadsky, S.I. Kublanovsky) Is it true that if every finitely generated subsemigroup of a semigroup $S$ is embeddable into the power semigroup over some periodic group then $S$ is also embeddable into the power semigroup over a periodic group?

背景链接：https://arxiv.org/abs/2604.04763

02566 | AMR-113-0036

### The Lyapin's Notebook - Problem 38

O / open | AMR 问题清单汇编

(S.G. Bershadsky, S.I. Kublanovsky) Find general embeddability criteria (necessary and/or sufficient conditions) for a semigroup to be embeddable into the power semigroup over a periodic group.

Some partial conditions were obtained in [Donald], [Bersh+Kubl]

背景链接：https://arxiv.org/abs/2604.04763

02567 | AMR-113-0037

### The Lyapin's Notebook - Problem 39

O / open | AMR 问题清单汇编

(G. Mashevitzky) Find a theorem for inclusive varieties analogous to the Birkhoff theorem for varieties; that is, determine a (minimal) set of Birkhoff type operators such that a class of semigroups forms an inclusive variety if and only if it is closed under these operators.

Inclusive varieties are known to be closed under subsemigroups and homomorphic images.

背景链接：https://arxiv.org/abs/2604.04763

02568 | AMR-113-0004

### The Lyapin's Notebook - Problem 4

O / open | AMR 问题清单汇编

(E.S. Lyapin) Describe the identities of the full transformation semigroup $T(n)$ on an $n$-element set. \\ A description of some classes of identities was obtained in the works [PSSSV], [Ma3].

背景链接：https://arxiv.org/abs/2604.04763

02569 | AMR-113-0038

### The Lyapin's Notebook - Problem 40

O / open | AMR 问题清单汇编

(G. Mashevitzky) Determine all inclusive varieties with finite or countable lattices of inclusive subvarieties.

A partial solution can be found in [M2].

背景链接：https://arxiv.org/abs/2604.04763

02570 | AMR-113-0039

### The Lyapin's Notebook - Problem 41

O / open | AMR 问题清单汇编

(G. Mashevitzky)

- Find minimal (maximal) inclusive varieties of semigroups that cannot be defined by elementary identical inclusions.

- Describe elementary inclusive varieties of semigroups all of whose inclusive subvarieties are elementary.

- Describe nonelementary inclusive varieties of periodic semigroups whose inclusive supervarieties in the class of periodic semigroups are nonelementary.

All minimal inclusive varieties of Abelian groups that cannot be defined by elementary identical inclusions are described in [M3].

背景链接：https://arxiv.org/abs/2604.04763

02571 | AMR-113-0040

### The Lyapin's Notebook - Problem 42

O / open | AMR 问题清单汇编

We call two semigroups inclusively equivalent if they satisfy the same identical inclusions; equivalently if they generate the same inclusive variety.

(G. Mashevitzky) Classify semigroups from the basic classes of semigroups up to inclusive equivalence.

背景链接：https://arxiv.org/abs/2604.04763

02572 | AMR-113-0041

### The Lyapin's Notebook - Problem 43

O / open | AMR 问题清单汇编

(G. Mashevitzky) Find semigroups from the basic classes of semigroups determined by their identical inclusions up to isomorphism.

Two-element semilattice ([L1]), quasicyclic groups ([BM]) and any finite group ([M2]) are determined by identical inclusions up to isomorphism.

背景链接：https://arxiv.org/abs/2604.04763

02573 | AMR-113-0042

### The Lyapin's Notebook - Problem 44

O / open | AMR 问题清单汇编

(G. Mashevitzky) Describe ordinary semigroup varieties such that their finitely generated free semigroups can be separated from each other by identical inclusions.

The variety of semilattices satisfies this condition ([M1, M01]).

背景链接：https://arxiv.org/abs/2604.04763

02574 | AMR-113-0043

### The Lyapin's Notebook - Problem 45

O / open | AMR 问题清单汇编

(G. Mashevitzky) Does there exist a nonelementary inclusive variety of idempotent semigroups?

背景链接：https://arxiv.org/abs/2604.04763

02575 | AMR-113-0044

### The Lyapin's Notebook - Problem 46

O / open | AMR 问题清单汇编

(G. Mashevitzky) Does there exist a variety of semigroups such that the class of its finite (or finitely generated) semigroups forms an elementary inclusive variety?

背景链接：https://arxiv.org/abs/2604.04763

02576 | AMR-113-0045

### The Lyapin's Notebook - Problem 47

O / open | AMR 问题清单汇编

(G. Mashevitzky) Does there exist a finite semigroup that generates an inclusive variety with an infinite lattice of inclusive subvarieties?

The lattice of inclusive subvarieties of an inclusive variety, generated by a finite group or a finite nilsemigroup is finite (see [M3]).

背景链接：https://arxiv.org/abs/2604.04763

02577 | AMR-113-0046

### The Lyapin's Notebook - Problem 48

O / open | AMR 问题清单汇编

(G. Mashevitzky) Find an algorithm that decides whether the inclusive variety generated by a given finite set of finite semigroups, can be defined by a finite system of (elementary) identical inclusions or prove that such an algorithm does not exist. (This problem is similar to Tarski's problem for identities).

In [M4], a finite semigroup was constructed that does not have a finite basis for its elementary identical inclusions.

背景链接：https://arxiv.org/abs/2604.04763

02578 | AMR-113-0047

### The Lyapin's Notebook - Problem 49

O / open | AMR 问题清单汇编

(G. Mashevitzky)

- Describe classes of semigroups in which elementary inclusive varieties form a sublattice of the lattice of inclusive varieties.

- Describe nonelementary inclusive varieties that form an upper subsemilattice in the lattice of inclusive varieties within the class of periodic semigroups.

Elementary inclusive varieties form a sublattice, and nonelementary inclusive varieties form an upper subsemilattice in the lattice of inclusive varieties of finite Abelian groups ([M3]).

背景链接：https://arxiv.org/abs/2604.04763

02579 | AMR-113-0005

### The Lyapin's Notebook - Problem 5

O / open | AMR 问题清单汇编

(S. Kublanovsky) Describe locally residually finite varieties of semigroups for known predicates.\\ In the global case, such description is obtained in [Kubl1], [Kubl2]. For the equality predicate in the local case, the description is obtained in [Sapir]. Locally residually finite varieties of rings were described in [Kubl3].

背景链接：https://arxiv.org/abs/2604.04763

02580 | AMR-113-0048

### The Lyapin's Notebook - Problem 50

O / open | AMR 问题清单汇编

(E.S. Lyapin)

- Characterize pargoids of projections (idempotents in the semigroup of all transformations).

- Characterize the pargoids of matrix units.

- Characterize the pargoids of idempotents of the basic classes of semigroups.

Some classes of pargoids were characterized in the paper [ESL5].

背景链接：https://arxiv.org/abs/2604.04763

02581 | AMR-113-0049

### The Lyapin's Notebook - Problem 51

O / open | AMR 问题清单汇编

(E.S. Lyapin)

- For the main classes of mathematical structures and natural pargoids of transformations, determine the classes of structures that are defined by these pargoids up to isomorphism.

- Find pargoids of transformations that are determined up to isomorphism by their semigroup of endomorphisms (or group of automorphisms).

E.S. Lyapin proved that partially ordered sets are determined up to isomorphism by the pargoids of their closure operators on these sets (see [ESL2], [ESL3], [ESL4]).

背景链接：https://arxiv.org/abs/2604.04763

02582 | AMR-113-0050

### The Lyapin's Notebook - Problem 52

O / open | AMR 问题清单汇编

(E.S. Lyapin) Characterize pargoids of transformations that admit an internal semigroup extension.

Some sufficient conditions are given in [ESL6].

背景链接：https://arxiv.org/abs/2604.04763

02583 | AMR-113-0051

### The Lyapin's Notebook - Problem 53

O / open | AMR 问题清单汇编

(O.M. Gornostaev, S.I. Kublanovsky) Describe varieties of involuted semigroups for which the quasivarieties of models representable in these varieties are finitely axiomatizable.

For classical varieties such a description has been obtained in [Gorn1].

背景链接：https://arxiv.org/abs/2604.04763

02584 | AMR-113-0006

### The Lyapin's Notebook - Problem 6

O / open | AMR 问题清单汇编

(S. Kublanovsky) Describe the covers of the semigroup variety $RS_n$ in the lattice of varieties of all semigroups and in the lattice of varieties satisfying the identity $x^2 = x^{n+2}$. \\ It follows from the results of [Kubl5] that in the latter lattice the number of covers is finite. It is also known that every subvariety of $RS_1$ has only finitely many covers (see [Kubl4]).

背景链接：https://arxiv.org/abs/2604.04763

02585 | AMR-113-0007

### The Lyapin's Notebook - Problem 7

O / open | AMR 问题清单汇编

(S. Kublanovsky) Describe periodic varieties generated by regular semigroups. \\ For periodic varieties generated by completely 0-simple semigroups, such a description was obtained in [Kubl4].

背景链接：https://arxiv.org/abs/2604.04763

02586 | AMR-113-0008

### The Lyapin's Notebook - Problem 8

O / open | AMR 问题清单汇编

(I. Melnichuk) Let $c(u)$ denote the number of distinct letters occuring in the word $u$. Let $m(u)$ denote the minimal number of letters in a finite alphabet $A$ for which there exists an infinite word over $A$ that does not contain images of $u$ under substitutions. Determine the number $m(u)$ for a non-blocking word $u$ containing $c(u)$ letters. \\ Some estimates were obtained in works [[Melnich1], [Melnich2], [McNulty], [Baker]].

背景链接：https://arxiv.org/abs/2604.04763

02587 | AMR-113-0009

### The Lyapin's Notebook - Problem 9

O / open | AMR 问题清单汇编

(G. Mashevitzky) Is the set of finite finitely based completely 0-simple semigroups over Abelian groups, in particular, over a two-element group, recursive?\\ An infinite chain of varieties, generated by completely 0-simple semigroups over the two-element group, in which finitely based and non-finitely based varieties alternate, was constructed in [M4].

背景链接：https://arxiv.org/abs/2604.04763

02588 | ALG-021

### The Pierce-Birkhoff Conjecture

P / partially_solved | 独立 / 未分配集合

Is every piecewise-polynomial function $f: \mathbb{R}^n \to \mathbb{R}$ the maximum of finitely many minimums of finite collections of polynomials?

背景链接：https://arxiv.org/abs/1003.1188

02589 | OWR-12697693-003

### The Property-s Elements of a Skew Brace

O / open | Oberwolfach 报告问题汇编

Let $B=(B,+,\circ)$ be a skew brace, write $a*b=-a+a\circ b-b$, and for $x\in B$ set $$\operatorname{Fix}^r(x)=\{b\in B:x*b=0\},\qquad \operatorname{Fix}^l(x)=\{b\in B:b*x=0\}.$$ If $C_x^+$ and $C_x^\circ$ are the centralizers of $x$ in $(B,+)$ and $(B,\circ)$, respectively, call $x$ an element with property $(s)$ when both indices $$[(B,+):\operatorname{Fix}^r(x)\cap C_x^+]\quad\text{and}\quad[(B,\circ):\operatorname{Fix}^l(x)\cap C_x^\circ]$$ are finite. Do the elements with property $(s)$ form an ideal of $B$?

来源字段：https://doi.org/10.4171/owr/2023/9

02590 | AIM-REPRESENTATION_THEORY-0103

### The proved Breuil-Mézard equality and a weight-support certificate

S / solved | AIM 研讨会问题汇编

Conjecture 1 (Breuil-M´ ezard). $\mu\mathrm{Gal}=\mu\mathrm{Aut}$.

Its core text is

The rest of the record says that global arguments generally give the inequality from the Galois side to the automorphic side and then introduces a framed potentially semistable deformation ring.

Let $p$ be a prime, let $E/\mathbb Q_p$ be a sufficiently large finite extension with ring of integers $\mathcal O$, uniformizer $\varpi$, and residue field $\mathbb F$, and let $\bar\rho:G_{\mathbb Q_p}\to\mathrm{GL}_2(\mathbb F)$ be a residual representation. For an inertial type $\tau$ and Hodge--Tate weights $(0,k-1)$, the Galois multiplicity $\mu_{\mathrm{Gal}}(k,\tau,\bar\rho)$ is the Hilbert--Samuel multiplicity of the special fibre of the relevant potentially semistable local deformation ring. Every irreducible mod-$p$ representation of $\mathrm{GL}_2(\mathbb Z_p)$ is inflated from a Serre weight

\[ \sigma_{n,m}=\operatorname{Sym}^n\mathbb F^2\otimes\det^m, \qquad 0\le n\le p-1,\quad 0\le m\le p-2. \]

If $L_{k,\tau}$ is a stable lattice in the characteristic-zero type, write

\[ (L_{k,\tau}/\varpi L_{k,\tau})^{\mathrm{ss}} \cong \bigoplus_{n,m}\sigma_{n,m}^{\,a_{k,\tau}(n,m)}. \]

Breuil and Mézard define integers $\mu_{n,m}(\bar\rho)\ge0$ and

\[ \mu_{\mathrm{Aut}}(k,\tau,\bar\rho) =\sum_{n,m}a_{k,\tau}(n,m)\mu_{n,m}(\bar\rho). \]

The recovered conjecture is therefore the numerical equality

\[ \boxed{\ \mu_{\mathrm{Gal}}(k,\tau,\bar\rho) =\sum_{n,m}a_{k,\tau}(n,m)\mu_{n,m}(\bar\rho)\ }. \tag{BM} \]

来源字段：https://aimath.org/WWN/padicmodularity/padicmodularity.pdf

02591 | AIM-REPRESENTATION_THEORY-0109

### The reducible fiber of the representation-to-pseudorepresentation map

P / partially_solved | AIM 研讨会问题汇编

\[ V_{\mathbb F}=\omega _1\oplus\omega _2\longmapsto T_{\mathbb F}, \]

and the completion appearing later is \(\widehat R_x\). The final token “8.3.” belongs to the next subsection, whose separate problem asks for cohomological tools for pseudorepresentations; it is not part of Problem 2.

With those corrections, but without supplying definitions absent from the notes, the problem is the following. Let \(\mathbb F\) be a finite field, let \(\omega _1\ne\omega _2\) be characters, and let \(T_{\mathbb F}\) be the pseudorepresentation attached to the semisimple representation \(\omega _1\oplus\omega _2\). Study representations over \(W(\mathbb F)\)-algebras whose residual representation is a non-split extension of \(\omega _1\) by \(\omega _2\). The notes propose a morphism

\[ X_{\omega _2}\longrightarrow \operatorname{Spec}R(T_{\mathbb F}) \]

whose special fiber involves

\[ \bigl(\operatorname{Ext}^1(\omega _1,\omega _2)\setminus\{0\}\bigr) /\mathbb F^\times. \]

A point \(x\) of that fiber should determine an extension \(V_x\), and the completion \(\widehat R_x\) is suggested to be the universal deformation ring of \(V_x\). A universal rank-two bundle \(V_{\omega _2}\) on \(X_{\omega _2}\) is proposed, as is a bundle \(\pi^*V_{\omega _2}\) on the pseudodeformation space. The notes then put \(X_{\omega _2}\) and the opposite-extension space \(X_{\omega _1}\) over the same base and ask whether they can be glued “along the trivial extension,” perhaps into an algebraic stack. They also assert, without giving the construction of \(X_{\omega_i}\), that \(X_{\omega_2}\) is proper by the valuative criterion.

来源字段：https://aimath.org/WWN/padicmodularity/padicmodularity.pdf

02592 | ALG-033

### The Sofic Groups Conjecture

P / partially_solved | 独立 / 未分配集合

Is every discrete countable group sofic?

背景链接：https://cdn.openai.com/pdf/ten-proofs-oai.pdf

02593 | AIM-LINEAR_ALGEBRA-0004

### The square-energy bound is proved, while the proposed edge-addition monotonicities fail

P / partially_solved | AIM 研讨会问题汇编

Consider a connected graph $G$ with adjacency eigenvalues $$ \lambda_1\geq\cdots\geq\lambda_s\geq0>\lambda_{s+1}\geq\cdots\geq\lambda_n $$ and $$ S^+(G)=\sum_{i=1}^s\lambda_i^2, \qquad S^-(G)=\sum_{i=s+1}^n\lambda_i^2. $$ It contains four items:

1. prove $\min\{S^+(G),S^-(G)\}\geq n-1$; 2. after ordering the complement edges as $e_1,\ldots,e_\ell$ and setting $G_i=G+\{e_1,\ldots,e_i\}$, prove that $S^+(G_i)$ is nondecreasing and $S^-(G_i)$ is unimodal; 3. ask whether $\sum_{t=1}^r\lambda_t(G_i)^2$ is nondecreasing for every $r\geq1$; 4. ask whether $\sum_{t:\lambda_t(G_i)\geq\tau}\lambda_t(G_i)^2$ is nondecreasing for a fixed threshold $\tau$.

It defines the positive and negative square energies using strictly positive and strictly negative eigenvalues and records the $n-1$ conjecture and the one-edge $S^+$ conjecture.

Following the AIM report and all primary papers checked, graphs are taken to be finite, simple, undirected, and unweighted.

The natural and useful reading of item 2 is universal over complement-edge orderings. We set $G_0=G$. “Unimodal” means the standard peak convention: for some $j$, $$ a_0\leq\cdots\leq a_j\geq\cdots\geq a_\ell. $$ This is also the convention required by the source's claimed implication to item 1. The counterexamples below violate these statements in a prefix and therefore remain counterexamples after the remaining complement edges are appended in any order.

来源字段：http://aimpl.org/spectralhypergraph/1/

02594 | AIM-REPRESENTATION_THEORY-0006

### The surface target fails, but an exact inverse-paramodularity gate remains

S / solved | AIM 研讨会问题汇编

**Construction of abelian surfaces for paramodular newforms.** Given a paramodular newform \(F\) of weight \(2\) with rational coefficients, find a construction of the abelian surface to which it corresponds.

In the setting of Problem 1.4 (Relative conductor of \(p\)-adic groups): when is the space of \(K_H(p^n)\)-fixed vectors one-dimensional? Is there always an \(H\) for which this happens? Is there a commonality between such \(H\)?

Given a rational weight-two **nonlift** paramodular eigenform \(F\) of level \(N\), first decide whether its four-dimensional compatible system is realized by a typical abelian surface or twice inside a QM fourfold. In the surface case, construct the \(\mathbb Q\)-isogeny class of a conductor-\(N\) abelian surface \(A\) whose good spin Euler factors equal those of \(F\); if a principal polarization exists over \(\mathbb Q\), seek an explicit genus-two curve \(C/\mathbb Q\) with \(A\sim_{\mathbb Q}\operatorname{Jac}(C)\).

来源字段：http://aimpl.org/aagaautomorphic/3/

02595 | ALG-032

### The Surjunctivity Conjecture

O / open | 独立 / 未分配集合

Is every group surjunctive?

背景链接：https://arxiv.org/abs/2511.06586

02596 | AIM-REPRESENTATION_THEORY-0078

### The tempered endoscopic character identity and finite packet inversion for Mp(2n)

S / solved | AIM 研讨会问题汇编

What is the local character identity for \(Mp(2n)\)?

**Recovered problem.** Let \(F\) be a local field of characteristic zero, fix a nontrivial additive character \(\psi:F\to\mathbb C^\times\), and let \(Mp(W)\) be the metaplectic double cover of \(Sp(W)\), \(\dim W=2n\). For a tempered Langlands parameter \(\phi:WD_F\to Sp(2n,\mathbb C)\), determine the endoscopic transfer identity relating the stable characters on the elliptic endoscopic groups \[ H_{n',n''}=SO(2n'+1)\times SO(2n''+1),\qquad n'+n''=n, \] to the characters in the theta-correspondence packet of \(Mp(W)\).

来源字段：http://aimpl.org/autoformcovergp/3/

02597 | ALG-020

### The Uniform Boundedness Conjecture

P / partially_solved | 独立 / 未分配集合

Is there a bound $B(g, d)$ such that every curve of genus $g$ over a number field of degree $d$ has at most $B(g, d)$ rational points?

背景链接：https://arxiv.org/abs/alg-geom/9601004

02598 | OWR-12494-005

### The Value Nine and Universality for Quadratic-Form Invariants

O / open | Oberwolfach 报告问题汇编

Does there exist a field $F$ of characteristic $2$ with $\widetilde u(F)=9$? If $u(F)<\infty$, is every anisotropic nonsingular quadratic form over $F$ of dimension $u(F)$ universal?

来源字段：https://doi.org/10.4171/owr/2013/31

02599 | OWR-3393-005

### Thick Subcategories of Noetherian Derived Categories

P / partially_solved | Oberwolfach 报告问题汇编

For a noetherian ring $A$, classify the thick subcategories of $D_f(A)$, the derived category of $A$-modules with finitely generated cohomology. Equivalently, via compact objects in $K(\operatorname{Inj}A)$, classify the localizing subcategories of the homotopy category of complexes of injective $A$-modules.

来源字段：https://doi.org/10.4171/owr/2009/10

02600 | AIM-REPRESENTATION_THEORY-0040

### Tilting objects under the modular induction equivalence

P / partially_solved | AIM 研讨会问题汇编

What objects of the derived category of representations of $B$ correspond to tilting modules of $G$?

来源字段：http://aimpl.org/sheavemodular/1/

02601 | AIM-LINEAR_ALGEBRA-0009

### Toeplitz corner tests and a PF-to-positroid support rule

P / partially_solved | AIM 研讨会问题汇编

Consider $TN$ Toeplitz matrices. What are the allowed patterns of zero and nonzero minors (i.e., which positroid cells contain Toeplitz matrices)? How does this relate to P\'olya frequency sequences? (Check work by Rietsch for the lower triangular Toeplitz case.) Is it enough to check all the corner minors to identify $TN$ Toeplitz matrices?

来源字段：http://aimpl.org/totalpos/1/

02602 | OWR-12175-001

### Toms-Winter Regularity Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate: Toms-Winter Regularity Conjecture [OWR-12175-002]

For every unital, simple, separable, nuclear $C^*$-algebra $A$, are the following conditions equivalent: $A$ has finite nuclear dimension; $A\otimes\mathcal Z\cong A$; and $A$ has strict comparison?

来源字段：https://doi.org/10.4171/owr/2012/52

02603 | OWR-15585-003

### Toms-Winter Regularity Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For every nonelementary simple separable unital nuclear $C^*$-algebra $A$, are finite nuclear dimension, tensorial absorption of the Jiang-Su algebra $\mathcal Z$, and strict comparison of positive elements equivalent? When $A$ is stably finite, is finite decomposition rank equivalent to these conditions as well?

来源字段：https://doi.org/10.4171/owr/2017/42

02604 | OWR-12495-007

### Topological Nilpotence Degree of Finite-Group Chow Rings

O / open | Oberwolfach 报告问题汇编

For a finite group $G$, is the topological nilpotence degree $d_0(CH_G^*)$ of its mod-$p$ Chow ring equal to the largest integer $d$ such that $CH_G^*$, as a module over the Steenrod algebra, contains the $d$th suspension of a nonzero unstable module?

来源字段：https://doi.org/10.4171/owr/2013/32

02605 | OWR-734-016

### Toric Analogues of Property $N_1$

O / open | Oberwolfach 报告问题汇编

What are the toric analogues of property $(N_1)$, and does every ample line bundle on a smooth toric variety satisfy property $(N_1)$?

来源字段：https://doi.org/10.4171/owr/2004/32

02606 | OWR-12868-003

### Torsion-Freeness of Cluster Algebra Class Groups

S / solved | Oberwolfach 报告问题汇编

Is the divisor class group $\operatorname{Cl}(\mathcal A(\mathbf x,B))$ of every cluster algebra torsion-free?

来源字段：https://doi.org/10.4171/owr/2014/08

02607 | OWR-1738-009

### Torsion-Unit Conjugacy and Local Endomorphism Orders

P / partially_solved | Oberwolfach 报告问题汇编

Given a finite group $H$, does there exist a finite group $G\supset H$ such that every torsion unit in $V(\mathbb{Z}H)$ is conjugate to an element of $H$ by a unit of $\mathbb{Q}G$? Can the conjugating units be chosen in $\mathbb{Q}H$? For a finite group $G$, a prime $p$, a simple $\mathbb{Q}G$-module $X$, and a $\mathbb{Z}_{(p)}G$-lattice $L\subset X$, determine $E=\operatorname{End}_{\mathbb{Z}_{(p)}G}(L)$ and its local-order building blocks arising from the central primitive idempotents of $\mathbb{Q}G$. Which such local orders can occur?

来源字段：https://doi.org/10.4171/owr/2007/55

02608 | OPG-3031

### trace inequality

S / solved | Open Problem Garden

Let $A,B$ be positive semidefinite, by Jensen's inequality, it is easy to see $[tr(A^s+B^s)]^{\frac{1}{s}}\leq [tr(A^r+B^r)]^{\frac{1}{r}}$, whenever $s>r>0$.

What about the $tr(A^s+B^s)^{\frac{1}{s}}\leq tr(A^r+B^r)^{\frac{1}{r}}$, is it still valid?

背景链接：http://www.openproblemgarden.org/op/trace_inequality

02609 | OWR-12175-020

### Trace-Cone Classification after Razak-Jacelon Stabilization

O / open | Oberwolfach 报告问题汇编

For separable nuclear $C^*$-algebras $A$ and $B$, does $T(A)\cong T(B)$ if and only if $A\otimes\mathcal W\otimes\mathcal K\cong B\otimes\mathcal W\otimes\mathcal K$, assuming the trace-cone isomorphism is a linear homeomorphism?

来源字段：https://doi.org/10.4171/owr/2012/52

02610 | OWR-15429-010

### Transfer Matrix Eigenvalue Polynomials for Quantum Affine Algebras

O / open | Oberwolfach 报告问题汇编

For the joint eigenvalues of transfer matrices acting on a common generalized eigenspace $W$ in a tensor product of finite-dimensional quantum-affine-algebra modules, characterize the Baxter polynomials $Q_{i,k}(z)$ that occur in the eigenvalue formulas.

来源字段：https://doi.org/10.4171/owr/2017/21

02611 | OWR-4800-012

### Tree Modules for Roots of Acyclic Quivers

P / partially_solved | Oberwolfach 报告问题汇编

Does every root $\mathbf d\in\mathbb N^{Q_0}$ of an acyclic quiver admit an indecomposable representation having a basis whose coefficient quiver is a tree?

来源字段：https://doi.org/10.4171/owr/2011/10

02612 | OWR-1738-004

### Trivial Restrictions of Twists in Nilpotent Group Algebras

O / open | Oberwolfach 报告问题汇编

Let $G$ be a torsion-free nilpotent group of finite cohomological dimension $n$, let $\alpha\in H^2(G,\mathbb{C}^\times)$, and suppose $1\leq r\leq n$ with $\operatorname{gl.dim}(\mathbb{C}^\alpha G)=r$. Must there exist a subgroup $H\leq G$ such that $\alpha$ restricts trivially to $H$ and $\operatorname{gl.dim}(\mathbb{C}H)=r$?

来源字段：https://doi.org/10.4171/owr/2007/55

02613 | OWR-12697693-009

### Trivialization Structures for Nondegenerate Cycle Sets

O / open | Oberwolfach 报告问题汇编

For a nondegenerate cycle set $X$, develop a trivialization theory relating $X$ to $X_{\mathrm{tr}}$. Determine the relations between $S(X)$ and $S(X_{\mathrm{tr}})$ and between $G(X)$ and $G(X_{\mathrm{tr}})$, decide whether they admit a brace-like description, and study the $L$-algebra trivialization defined by $$x\mathbin{\cdot_{\mathrm{tr}}}y=\begin{cases}e,&x\leq y,\\y,&x\nleq y.\end{cases}$$ Determine the resulting structure.

来源字段：https://doi.org/10.4171/owr/2023/9

02614 | OWR-11101917-003

### Twisted Groupoid Reconstruction of $C^*$-Algebra Pairs

O / open | Oberwolfach 报告问题汇编

Characterize the pairs of $C^*$-algebras arising from quasi-Cartan groupoid reconstruction, extend quasi-Cartan groupoid reconstruction to twisted $C^*$-algebras, and determine whether its isotropy hypotheses can be weakened.

来源字段：https://doi.org/10.4171/owr/2022/36

02615 | OWR-1391-002

### Unbounded Representatives of the Chern-Connes Character

O / open | Oberwolfach 报告问题汇编

Let $(A,H,D)$ be a $p$-summable odd spectral triple and let $F=D|D|^{-1}$. Can the Chern-Connes character of its Fredholm module be constructed using only $D$ and commutators $[D,a]$ with elements $a\in A$?

来源字段：https://doi.org/10.4171/owr/2006/53

02616 | OWR-9790367-002

### Unified Construction of Deformed W-Algebras

O / open | Oberwolfach 报告问题汇编

Determine the structure of the deformed $W$-algebras of exceptional and higher-rank types and decide whether they arise from one uniform construction.

来源字段：https://doi.org/10.4171/owr/2022/16

02617 | ALG-049

### Uniform Boundedness Conjecture for Rational Points

O / open | 独立 / 未分配集合

Is there a bound N(g,d) such that all curves of genus g≥2 over degree d number fields have at most N(g,d) rational points?

背景链接：https://en.wikipedia.org/wiki/Uniform_boundedness_conjecture

02618 | OWR-12697690-001

### Uniform Expansion Stability for Wild Quivers

O / open | Oberwolfach 报告问题汇编

Does every wild quiver $Q$ admit a stability condition $\mu$ such that $(Q,\mu)$ has uniform expansion?

来源字段：https://doi.org/10.4171/owr/2023/7

02619 | OWR-3472-013

### Uniform Generic Degree Bounds for Ideal Membership

O / open | Oberwolfach 报告问题汇编

For a standard graded $k$-algebra $R$ of dimension $d$ and generic homogeneous generators $f_1,\ldots,f_n$ of prescribed degrees $a_1,\ldots,a_n$, does there exist a generic degree bound $m_0$, depending only on $d$ and the degrees $a_i$, such that $R_{m_0}\subseteq(f_1,\ldots,f_n)$?

来源字段：https://doi.org/10.4171/owr/2009/22

02620 | OWR-12697689-011

### Uniform Local Cohomology Annihilators for Symbolic Powers

O / open | Oberwolfach 报告问题汇编

Let $(R,\mathfrak m,k)$ be a complete Cohen-Macaulay normal domain of dimension $d$, and let $I$ have pure height $h$. For each $1\leq j\leq d-h-1$, does there exist an ideal $\mathfrak a_j$ of height $d-j+1$ that uniformly annihilates $H^j_{\mathfrak m}(R/I^{(n)})$ for every $n\in\mathbb{N}$?

来源字段：https://doi.org/10.4171/owr/2023/6

02621 | OWR-15585-004

### Uniform Property Gamma for Nuclear $C^*$-Algebras

O / open | Oberwolfach 报告问题汇编

Does every simple separable unital nuclear $C^*$-algebra have uniform property $\Gamma$; that is, for every $k\geq2$, does its central sequence algebra contain $k$ pairwise orthogonal positive contractions whose trace is $1/k$ for every limit trace?

来源字段：https://doi.org/10.4171/owr/2017/42

02622 | OWR-15181-011

### Unique Factorization of Germ-Like Generalized Series

P / partially_solved | Oberwolfach 报告问题汇编

Does every nonzero germ-like series in $K((\mathbb{R}^{\leq 0}))$ admit a unique factorization into irreducibles?

来源字段：https://doi.org/10.4171/owr/2016/60

02623 | OWR-14298159-002

### Uniqueness of Finite-Dimensional Post-Nichols Algebras

O / open | Oberwolfach 报告问题汇编

Let $G$ be a group and let $V\in{}_{\mathbb CG}^{\mathbb CG}\mathcal{YD}$ satisfy $\dim\mathcal B(V)<\infty$. Is $\mathcal B(V)$ the only finite-dimensional post-Nichols algebra of $V$?

来源字段：https://doi.org/10.4171/owr/2024/5

02624 | OWR-15994-003

### Uniqueness of Homological Primes over Triangular Primes

P / partially_solved | Oberwolfach 报告问题汇编

Given a triangular prime $P$ in a tensor-triangulated category, is there a unique maximal Serre tensor-ideal homological prime $\mathcal B$ whose Yoneda preimage is $P$?

来源字段：https://doi.org/10.4171/owr/2018/9

02625 | OWR-12697690-007

### Uniqueness of Surface Models for Locally Gentle Algebras

O / open | Oberwolfach 报告问题汇编

Does every finite-dimensional locally gentle algebra admit a unique tiled-surface model up to rotations of the tiles?

来源字段：https://doi.org/10.4171/owr/2023/7

02626 | OWR-14299910-006

### Universal $F$-Polynomial Configuration-Space Identities

O / open | Oberwolfach 报告问题汇编

For every finite-dimensional algebra $\Lambda$, prove that the configuration spaces defined by its universal $F$-polynomials and its $u$-equations satisfy $\widetilde{\mathcal M}_\Lambda=\widetilde{\mathcal U}_\Lambda$ and $\mathcal M_\Lambda=\mathcal U_\Lambda$.

来源字段：https://doi.org/10.4171/owr/2026/6

02627 | OWR-9790358-014

### Universal Four-Point Polynomial Interpolation Subspaces

O / open | Oberwolfach 报告问题汇编

What is the minimum size of a family of four-dimensional subspaces of $K[x,y]$ such that every set of four planar points admits arbitrary-data interpolation in at least one member of the family?

来源字段：https://doi.org/10.4171/owr/2022/7

02628 | OWR-17292-007

### Universal Grading of Hyperfinite Factor Bimodules

O / open | Oberwolfach 报告问题汇编

Determine the universal grading group $U_R$ of the tensor category of dualizable $R$-$R$ bimodules, where $R$ is the hyperfinite $\mathrm{II}_1$ factor.

来源字段：https://doi.org/10.4171/owr/2019/49

02629 | OWR-16162-015

### Unramified Cohomology of Mixed-Type Classifying Spaces

O / open | Oberwolfach 报告问题汇编

If $G$ is a semisimple algebraic group of mixed Dynkin types over an algebraically closed field, is the unramified degree-three cohomology $H_{\mathrm{nr}}^3(F(BG))$ trivial?

来源字段：https://doi.org/10.4171/owr/2018/21

02630 | AIM-LINEAR_ALGEBRA-0010

### Vanishing criterion for minors of one-sided totally nonnegative Toeplitz matrices

S / solved | AIM 研讨会问题汇编

Consider semi-infinite lower triangular $TN$ Toeplitz matrices. Under what conditions can a non-trivial $k\times k$ minor be zero?

来源字段：http://aimpl.org/totalpos/1/

02631 | OWR-14298370-002

### Vanishing Lie Brackets for Non-Quasitriangular Hopf Algebras

S / solved | Oberwolfach 报告问题汇编

For a Hopf algebra whose module category is not braided, is the graded Lie bracket on its positive-degree cohomology always zero?

来源字段：https://doi.org/10.4171/owr/2024/20

02632 | OWR-12697689-016

### Vasconcelos Conjecture for Kähler Differentials

O / open | Oberwolfach 报告问题汇编

Let $k$ be a field of characteristic zero and let $S$ be a commutative Noetherian $k$-algebra essentially of finite type. If the module of Kähler differentials $\Omega^1_{S/k}$ has finite projective dimension over $S$, must $S$ be a reduced complete intersection?

来源字段：https://doi.org/10.4171/owr/2023/6

02633 | LOGIC-001

### Vaught Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Vaught Conjecture [MOD-004]

Is the number of countable models of a complete first-order theory finite, $\aleph_0$, or $2^{\aleph_0}$?

背景链接：https://encyclopediaofmath.org/wiki/Vaught_conjecture

02634 | OWR-2048-004

### Vector-Bundle Invariants of Cuntz-Pimsner Fields

P / partially_solved | Oberwolfach 报告问题汇编

Let $E$ be a complex rank-$n$ vector bundle over a compact Hausdorff space $X$, and let $O_E$ be its associated locally trivial Cuntz-Pimsner continuous field with fiber $O_n$. Which invariants of $E$ are captured by the isomorphism class of $O_E$?

来源字段：https://doi.org/10.4171/owr/2008/37

02635 | OWR-15581-005

### Verlinde Formulas and Lusztig Characters for New Weight Families

P / partially_solved | Oberwolfach 报告问题汇编

Can one find Verlinde-like dimension formulas for other families of dominant weights, and do the resulting character formulas coincide with those predicted by Lusztig’s conjecture?

来源字段：https://doi.org/10.4171/owr/2017/38

02636 | OWR-11580-014

### Verma-to-Wakimoto Isomorphisms for Jet-Level Characters

P / partially_solved | Oberwolfach 报告问题汇编

For a jet-level character $\theta\in\mathfrak h_n^*$, construct a Wakimoto module $W(\chi_\theta)$ with a natural nonzero map $V(\theta)\to W(\chi_\theta)$, and characterize the conditions on $\theta$ under which this map is an isomorphism.

来源字段：https://doi.org/10.4171/owr/2012/13

02637 | OWR-4990375-004

### Vertex Algebra Partition Functions and Siegel Modular Forms

P / partially_solved | Oberwolfach 报告问题汇编

For vertex algebras not arising from quadratic forms, is it true that for sufficiently large genus their partition functions are not pullbacks of Siegel modular forms, and is a vertex algebra uniquely determined by its collection of partition functions?

来源字段：https://doi.org/10.4171/owr/2021/6

02638 | OWR-5149-008

### Vertices of Modular Symmetric-Group Representations

O / open | Oberwolfach 报告问题汇编

Determine the vertices of naturally occurring modular representations of symmetric groups, in particular the Specht modules, beyond the currently known special cases.

来源字段：https://doi.org/10.4171/owr/2011/21

02639 | OWR-5149-012

### Vertices, Sources, and Filtrations of Symmetric-Group Modules

O / open | Oberwolfach 报告问题汇编

Let $F$ be a field, let $G$ be a finite group, and let $H\leq G$. For $i\geq1$, let $$T_i=\underbrace{FG\otimes_{FH}\cdots\otimes_{FH}FG}_{i\text{ copies of }FG},$$ viewed as an $(FH,FH)$-bimodule. Determine the minimal $n$ for which $T_n$ and $T_{n+1}$ have the same indecomposable direct summands up to multiplicity, including the alternating-group case. Determine the vertices and sources of irreducible $S_n$-modules $D^\lambda$, and decide whether a nontrivial source in characteristic $2$ must have even dimension. Is $\operatorname{Lie}(n)\otimes\operatorname{Lie}(n)^*$ a permutation module? Does every nonsimple indecomposable Specht module in characteristic $2$ restrict indecomposably to $A_n$? Determine the graded multiplicities of $D^\lambda$ in the coinvariant algebras of symmetric groups and $\operatorname{GL}_n(q)$, or in their socles.

来源字段：https://doi.org/10.4171/owr/2011/21

02640 | OWR-12697685-005

### Very Ample Internal Strongly Minimal Sets

O / open | Oberwolfach 报告问题汇编

Is there a very ample strongly minimal set internal to $M$; more specifically, is there a definable equivalence relation $E$ on $M$ such that $M/E$ is very ample?

来源字段：https://doi.org/10.4171/owr/2023/2

02641 | OWR-14299580-014

### Virtual Abelianity of Unitary Yang-Baxter Braid Images

O / open | Oberwolfach 报告问题汇编

Let $(R,V)$ be a unitary solution of the Yang-Baxter equation. Is the braid-group image $\rho_R(B_n)$ virtually abelian for every $n$?

来源字段：https://doi.org/10.4171/owr/2025/49

02642 | OPG-60031

### Waring rank of determinant

P / partially_solved | Open Problem Garden

Question What is the Waring rank of the determinant of a $d \times d$ generic matrix?

For simplicity say we work over the complex numbers. The $d \times d$ generic matrix is the matrix with entries $x_{i,j}$ for $1 \leq i,j \leq d$. Its determinant is a homogeneous form of degree $d$, in $d^2$ variables. If $F$ is a homogeneous form of degree $d$, a power sum expression for $F$ is an expression of the form $F = \ell_1^d+\dotsb+\ell_r^d$, the $\ell_i$ (homogeneous) linear forms. The Waring rank of $F$ is the least number of terms $r$ in any power sum expression for $F$. For example, the expression $xy = \frac{1}{4}(x+y)^2 - \frac{1}{4}(x-y)^2$ means that $xy$ has Waring rank $2$ (it can't be less than $2$, as $xy \neq \ell_1^2$ ).

The $2 \times 2$ generic determinant $x_{1,1}x_{2,2}-x_{1,2}x_{2,1}$ (or $ad-bc$ ) has Waring rank $4$. The Waring rank of the $3 \times 3$ generic determinant is at least $14$ and no more than $20$, see for instance Lower bound for ranks of invariant forms, Example 4.1. The Waring rank of the permanent is also of interest. The comparison between the determinant and permanent is potentially relevant to Valiant's "VP versus VNP" problem.

背景链接：http://www.openproblemgarden.org/op/waring_rank_of_determinant

02643 | OWR-12697689-010

### Weak and Strong F-Regularity in Excellent Rings

P / partially_solved | Oberwolfach 报告问题汇编

别名：Introduction to the Weak-Implies-Strong Conjecture [OWR-12697689-009]

If $R$ is an excellent weakly $F$-regular ring of prime characteristic $p>0$, is $R$ strongly $F$-regular?

来源字段：https://doi.org/10.4171/owr/2023/6

02644 | OWR-13494-014

### Weak Lefschetz Properties of Complete Intersections

O / open | Oberwolfach 报告问题汇编

Does every complete-intersection quotient of $k[x_1,\ldots,x_r]$ have the weak Lefschetz property when $r\geq4$?

来源字段：https://doi.org/10.4171/owr/2015/9

02645 | OWR-12984-011

### Weighted $q$-Boundaries for Presimplicial Modules

O / open | Oberwolfach 报告问题汇编

Given a presimplicial module with boundary $\partial_n=\sum_{i=0}^n(-1)^i d_i$, what useful homological structure can be extracted from the weighted operator $\partial_n^q=\sum_{i=0}^n q^i d_i$ on $C_n^q=C_n\otimes_k\mathbb Z[q]$, despite the fact that these operators do not generally form a chain complex?

来源字段：https://doi.org/10.4171/owr/2014/26

02646 | OWR-14299580-008

### Welding Set-Theoretic Yang-Baxter Solutions

P / partially_solved | Oberwolfach 报告问题汇编

When does a set-theoretic Yang-Baxter solution admit an extension of its braid-group actions to welded braid groups?

来源字段：https://doi.org/10.4171/owr/2025/49

02647 | OWR-1452-019

### Weyl-Algebra Automorphisms and Polynomial Symplectomorphisms

O / open | Oberwolfach 报告问题汇编

Is the automorphism group of the $n$th Weyl algebra $W_n$ over $\mathbb{C}$ isomorphic to the group of polynomial symplectomorphisms of $\mathbb{C}^{2n}$?

来源字段：https://doi.org/10.4171/owr/2007/01

02648 | AIM-REPRESENTATION_THEORY-0034

### Whitehouse modules: explicit rational splitting and the FI obstruction

S / solved | AIM 研讨会问题汇编

V. Reiner described Whitehouse modules as virtual representations. Is it clear why the negative part is a submodule of the positive part, i.e., can we fine explicit embedding? Is there a way to think about this in terms of \(\mathrm{FI}\)-modules?

来源字段：http://aimpl.org/repnstability/3/

02649 | AMR-015-0014

### Williamson matrix existence problem

P / partially_solved | AMR 问题清单汇编

Determine for which orders Williamson matrices exist; such matrices give a construction of Hadamard matrices.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767

02650 | OWR-1315-001

### Witt Groups of Stable Module Categories

O / open | Oberwolfach 报告问题汇编

Do the Witt groups of the stable module category $kG\text{-}\mathrm{stab}$ vanish for every finite group $G$ in characteristic $p$?

来源字段：https://doi.org/10.4171/owr/2006/29

02651 | OWR-1315-006

### Witt Indices along Bilinear-Form Splitting Towers

O / open | Oberwolfach 报告问题汇编

Let $B$ be an anisotropic bilinear form with standard splitting tower $(F_i,B_i)_{0\leq i\leq h}$. If $B_L$ becomes isotropic over $L/F$, must $i_W(B_L)=i_W(B_{F_k})$ for some $1\leq k\leq h$?

来源字段：https://doi.org/10.4171/owr/2006/29

02652 | OWR-3398-002

### Yangian Realization of Universal Affine Verma Modules

P / partially_solved | Oberwolfach 报告问题汇编

Let $M$ be the localized equivariant cohomology representation of the moduli spaces of parabolic sheaves on $\mathbb{P}^1\times\mathbb{P}^1$, equipped with the affine Yangian action. Is $M$ isomorphic, as an affine Yangian module, to the universal Verma module over $U(\widehat{\mathfrak{gl}}_n)$, with the affine Gelfand-Tsetlin fixed-point basis specializing to bases of integrable modules?

来源字段：https://doi.org/10.4171/owr/2009/15

02653 | OWR-17294-014

### Yang-Baxter Solutions from Generalized Left Semi-Braces

O / open | Oberwolfach 报告问题汇编

Which generalized left semi-braces $S$ have the property that $$ r(a,b)=\bigl(a\circ(a^{-}+b),(a^{-}+b)^{-}\circ b\bigr) $$ is a set-theoretic solution of the Yang-Baxter equation?

来源字段：https://doi.org/10.4171/owr/2019/51

02654 | OWR-17294-016

### Yang-Baxter Solutions from Noncommutative Lattices

O / open | Oberwolfach 报告问题汇编

Do the Yang-Baxter solutions obtained from skew lattices remain solutions for other noncommutative lattices, and do noncommutative lattices yield new solutions?

来源字段：https://doi.org/10.4171/owr/2019/51

02655 | OWR-17294-015

### Yang-Baxter Solutions from Skew Lattices

O / open | Oberwolfach 报告问题汇编

Can skew lattices be used to describe more set-theoretic solutions of the Yang-Baxter equation, and what kinds of solutions result?

来源字段：https://doi.org/10.4171/owr/2019/51

02656 | OWR-12175-021

### Z-Stability of Elliott Classification Counterexamples

O / open | Oberwolfach 报告问题汇编

Are all nonelementary counterexamples to the Elliott classification conjecture $C^*$-algebras that do not absorb the Jiang-Su algebra $\mathcal Z$? Equivalently, for every such unital algebra $A$, does $\mathcal Z$ fail to embed unitally into $A_\omega\cap A'$?

来源字段：https://doi.org/10.4171/owr/2012/52

02657 | AMR-015-0031

### Zariski-Lipman conjecture

O / open | AMR 问题清单汇编

Let $V$ be a complex algebraic variety with coordinate ring $R$. If the module of derivations of $R$ is free over $R$, must $V$ be smooth?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767

02658 | ALG-007

### Zauner's Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Zauner's Conjecture (SIC-POVM) [ALG-052]

Do SIC-POVMs (Symmetric Informationally Complete Positive Operator-Valued Measures) exist in all finite dimensions?

背景链接：https://doi.org/10.3842/SIGMA.2024.044

02659 | OWR-4216-003

### Zeros and Poles of Quantum-Group R-Matrices

O / open | Oberwolfach 报告问题汇编

For finite-dimensional $U_q(\mathfrak g)$-modules $V,V'$, determine the zeros and poles of the meromorphic universal $R$-matrix intertwiners

$$V(z)\otimes V'\longrightarrow V'\otimes V(z),$$

and their relation to reducibility of tensor products.

来源字段：https://doi.org/10.4171/owr/2010/15

02660 | GREEN-030

### Φ(G) and Φ'(G) Coincidence

P / partially_solved | Ben Green 问题

Do $\Phi(G)$ and $\Phi'(G)$ coincide?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

# 资料汇编 03 | 群论 / Group Theory

2,655 条记录。O=开放；P=部分解决；S=已解决。状态与英文/LaTeX 表述均沿用来源。

02661 | OWR-1274-007

### $2$-Adic Analyticity of the Groups $G(A)$ and $G(B)$

O / open | Oberwolfach 报告问题汇编

The closures $G(A)$ and $G(B)$ are $\mathbb Z_2$-analytic groups; more precisely, $G(B)$ is a pro-$2$ Sylow subgroup of $SL_2(\mathbb Z_2)$.

来源字段：https://doi.org/10.4171/owr/2006/25

02662 | OWR-14749-007

### $C^*$-Simplicity from Nonfree Boundary Actions

P / partially_solved | Oberwolfach 报告问题汇编

If a countable group $G$ acts on a $G$-boundary that is not topologically free, under what conditions is $G$ $C^*$-simple?

来源字段：https://doi.org/10.4171/owr/2016/41

02663 | OWR-16167-005

### $C^1$ Smoothability of Monod and Lodha-Moore Groups

O / open | Oberwolfach 报告问题汇编

Are the Monod groups $H(A)$ and the Lodha-Moore group $G_0$ smoothable by $C^1$ diffeomorphisms, or do they admit faithful $C^1$ actions on the real line or a closed interval?

来源字段：https://doi.org/10.4171/owr/2018/26

02664 | OWR-3477-011

### $FP_{1}$ Elementary Amenable Groups and Virtual Cyclicity

P / partially_solved | Oberwolfach 报告问题汇编

Are elementary amenable groups of type $FP_1$ necessarily virtually cyclic?

来源字段：https://doi.org/10.4171/owr/2009/27

02665 | OWR-1703863-005

### $L^2$ Euler Characteristics and HNN Splittings

O / open | Oberwolfach 报告问题汇编

Let $G=F_n\rtimes_\alpha\mathbb Z$ be free-by-cyclic and let $\phi:G\to\mathbb Z$ be a character. Is

$$-\chi^{(2)}(\ker\phi)=\min\{-\chi(A):G=A*_{B}\text{ is an HNN splitting inducing }\phi,\ A,B\text{ finitely generated}\}$$?

来源字段：https://doi.org/10.4171/owr/2020/16

02666 | OWR-14299086-003

### $L^p$ Cohomology at the Semisimple Rank Threshold

P / partially_solved | Oberwolfach 报告问题汇编

Let $F$ be a local field, let $G$ be the $F$-points of a semisimple algebraic group over $F$, and let $r=\operatorname{rk}_F(G)$. Is $H^k_{\mathrm{ct}}(G,L^p(G))=0$ for every $1<p<\infty$ and $k<r$? Does there exist $1<p<\infty$ such that $H^r_{\mathrm{ct}}(G,L^p(G))\neq 0$?

来源字段：https://doi.org/10.4171/owr/2025/6

02667 | OWR-1062-008

### $L^{2}$-Betti$ Numbers of Groups and Group Factors

O / open | Oberwolfach 报告问题汇编

For a group $\Gamma$, is the $k$-th $L^2$-Betti number $\beta_k(L\Gamma,\tau)$ equal to $\beta_k(\Gamma)$?

来源字段：https://doi.org/10.4171/owr/2005/41

02668 | OWR-1319-006

### $R\infty$ Property for Relatively Hyperbolic Groups

P / partially_solved | Oberwolfach 报告问题汇编

Does every relatively hyperbolic group, and hence every Kleinian group, have property $R_\infty$?

来源字段：https://doi.org/10.4171/owr/2006/33

02669 | AMR-108-0008

### 2.3 (Cooper) - Hausdorff limits of conjugates of an isometry group

O / open | AMR 问题清单汇编

Let $\beta$ be a nondegenerate bilinear form on a finite-dimensional real vector space $V$, and let $G=\operatorname{Isom}(\beta)\subset GL(V)$. Which subgroups $H\subset GL(V)$ are Hausdorff limits of sequences of conjugates of $G$?

背景链接：https://arxiv.org/pdf/1512.04620#page=2

02670 | AMR-108-0019

### 3.11 (Walsh) - Unbounded CAT(0) cubical dimension

P / partially_solved | AMR 问题清单汇编

The CAT(0) cubical dimension of a group $G$ is the least dimension of a CAT(0) cubical space on which $G$ acts geometrically. Is there a sequence of closed hyperbolic $3$-manifold groups whose CAT(0) cubical dimensions tend to infinity? More generally, is there a sequence of CAT(0) groups for which the difference between CAT(0) dimension and CAT(0) cubical dimension tends to infinity?

背景链接：https://arxiv.org/pdf/1512.04620#page=3

02671 | AMR-108-0015

### 3.7 (Agol) - Kleinian groups with closed quasi-Fuchsian surface subgroups

P / partially_solved | AMR 问题清单汇编

Which Kleinian groups admit closed quasi-Fuchsian surface subgroups?

背景链接：https://arxiv.org/pdf/1512.04620#page=2

02672 | AMR-108-0020

### 4.1 (Agol) - Virtual embeddings in hyperbolic reflection groups

O / open | AMR 问题清单汇编

Do closed hyperbolic $3$-manifold groups have finite-index subgroups that embed in a word-hyperbolic reflection group?

背景链接：https://arxiv.org/pdf/1512.04620#page=3

02673 | AMR-108-0021

### 4.2 (Futer) - The surface subgroup conjecture for cubulated hyperbolic groups

O / open | AMR 问题清单汇编

Does every freely indecomposable cubulated hyperbolic group contain the fundamental group of a closed hyperbolic surface?

背景链接：https://arxiv.org/pdf/1512.04620#page=3

02674 | AMR-108-0022

### 4.3 (Cooper) - 3-manifold groups acting on the affine building for $SL(4,\mathbb{R})$

O / open | AMR 问题清单汇编

If $M$ is a closed $3$-manifold, when does $\pi_1M$ act on the affine building for $SL(4,\mathbb{R})$ so that the quotient retracts to $M$?

背景链接：https://arxiv.org/pdf/1512.04620#page=3

02675 | AMR-108-0023

### 4.4 (Kassel, Mann) - Proper affine actions on $\mathbb{R}^5$

P / partially_solved | AMR 问题清单汇编

Let $\Gamma$ be a discrete group acting properly discontinuously by affine transformations on $\mathbb{R}^5$. Is $\Gamma$ virtually an extension of a free group by a solvable group?

背景链接：https://arxiv.org/pdf/1512.04620#page=3

02676 | AMR-108-0024

### 4.5 (Kassel, Mann) - Proper affine surface-group actions on $\mathbb{R}^6$

P / partially_solved | AMR 问题清单汇编

Classify all properly discontinuous affine actions of a given closed surface group on $\mathbb{R}^6$.

背景链接：https://arxiv.org/pdf/1512.04620#page=3

02677 | AMR-108-0025

### 4.6 (Kassel, Mann) - Minimal dimension of a proper affine Coxeter-group action

P / partially_solved | AMR 问题清单汇编

For a given right-angled Coxeter group, what is the least $n$ for which it admits a proper affine action on $\mathbb{R}^n$?

背景链接：https://arxiv.org/pdf/1512.04620#page=3

02678 | AMR-108-0037

### 5.12 (Reid) - Finite quotients of finite-covolume Kleinian groups

P / partially_solved | AMR 问题清单汇编

Let $\Gamma$ be a Kleinian group of finite covolume, and let $\mathcal{C}(\Gamma)$ be the set of isomorphism classes of its finite quotient groups. Does $\mathcal{C}(\Gamma)$ determine $\Gamma$ up to isomorphism?

背景链接：https://arxiv.org/pdf/1512.04620#page=4

02679 | AMR-108-0038

### 5.13 (Reid) - Finite quotients of free groups

P / partially_solved | AMR 问题清单汇编

For $\Gamma=F_r$ with $r\geq2$, does $\mathcal{C}(F_r)$ determine $F_r$ up to isomorphism?

背景链接：https://arxiv.org/pdf/1512.04620#page=4

02680 | AMR-108-0039

### 5.14 (Reid) - Finite-quotient rigidity among 3-manifold groups

P / partially_solved | AMR 问题清单汇编

Let $\Gamma$ be a Kleinian group of finite covolume. Does its set $\mathcal{C}(\Gamma)$ of finite quotient isomorphism classes determine $\Gamma$ among Kleinian groups, or among fundamental groups of compact $3$-manifolds?

背景链接：https://arxiv.org/pdf/1512.04620#page=4

02681 | AMR-108-0041

### 5.16 (Walsh) - Hyperbolic groups with Kleinian-type boundaries

P / partially_solved | AMR 问题清单汇编

If $G$ is a Gromov-hyperbolic group whose boundary is homeomorphic to the limit set of a convex-cocompact Kleinian group, is $G$ virtually a convex-cocompact Kleinian group?

背景链接：https://arxiv.org/pdf/1512.04620#page=5

02682 | AMR-108-0042

### 5.17 (Walsh) - Limit sets of convex-cocompact Kleinian groups

O / open | AMR 问题清单汇编

Which subsets of $S^2$ can occur as limit sets of convex-cocompact Kleinian groups?

背景链接：https://arxiv.org/pdf/1512.04620#page=5

02683 | AMR-108-0043

### 5.18 (Walsh) - Sierpiński carpets and continua in Kleinian limit sets

P / partially_solved | AMR 问题清单汇编

For which Kleinian groups does the limit set contain a Sierpiński carpet? For which Kleinian groups does the limit set contain a continuum?

背景链接：https://arxiv.org/pdf/1512.04620#page=5

02684 | AMR-108-0044

### 5.19 (Walsh) - Limit sets of graph Kleinian groups

O / open | AMR 问题清单汇编

Characterize the limit sets of graph Kleinian groups and iterated graph-Kleinian groups. A graph Kleinian group is a convex-cocompact Kleinian group for which the double of the convex core is a graph manifold. An iterated graph-Kleinian group is one whose Bowditch decomposition contains only hanging Fuchsian and graph Kleinian pieces.

背景链接：https://arxiv.org/pdf/1512.04620#page=5

02685 | AMR-108-0052

### 7.4 (Agol) - Algebraic trace fields of degenerate Kleinian groups

O / open | AMR 问题清单汇编

Can there be a degenerate Kleinian group that is not the fiber of a fibration and has algebraic trace field?

背景链接：https://arxiv.org/pdf/1512.04620#page=6

02686 | AMR-108-0053

### 7.5 (Schleimer) - Singly degenerate Kleinian groups over a number field

O / open | AMR 问题清单汇编

Is there a singly degenerate Kleinian group for which all matrix entries of all group elements lie in one fixed number field?

背景链接：https://arxiv.org/pdf/1512.04620#page=6

02687 | AIM-GEOMETRIC_GROUP_THEORY-0010

### A canonical congruence-kernel reduction for Out(F_n)

P / partially_solved | AIM 研讨会问题汇编

**Congruence subgroup property**

Let \(Q\) be a finite quotient of \(\mathrm{Out}(F_n)\) for \(n > 2\). Does there exist a finite index characteristic subgroup \(K\) of \(F_n\) such that the quotient \(\mathrm{Out}(F_n) \twoheadrightarrow Q\) factors the quotient \(\mathrm{Out}(F_n) \twoheadrightarrow \mathrm{Out}(F_n /K)\)?

For every epimorphism \(q:\Gamma\twoheadrightarrow Q\) to a finite group, does there exist a characteristic finite-index subgroup \(K\leq F\) such that \[ \ker\rho_K\leq\ker q? \] Equivalently, is there a homomorphism \(\bar q:\operatorname{im}\rho_K\to Q\) such that \[ q=\bar q\circ\rho_K? \]

来源字段：http://aimpl.org/freebycyclic/2/

02688 | AIM-GEOMETRIC_GROUP_THEORY-0082

### A central-quotient reduction for Wise's power alternative

P / partially_solved | AIM 研讨会问题汇编

Given \(a,b \in A\), is there an \(n=n(a,b)\) such that \(\langle a^n,b^n\rangle\) is free or free abelian?

来源字段：http://aimpl.org/geomartingp/8/

02689 | AIM-GEOMETRIC_GROUP_THEORY-0055

### A character-twist obstruction on the extended Deligne Helly graph

P / partially_solved | AIM 研讨会问题汇编

Does the $\widetilde A_2$ Artin group act properly and cocompactly on a Helly graph?

来源字段：http://aimpl.org/geomartingp/4/

02690 | AIM-GEOMETRIC_GROUP_THEORY-0166

### A coefficient-sampling reduction for lattice descent of rapid decay

P / partially_solved | AIM 研讨会问题汇编

The database record has accidentally concatenated all of Sections 1 and 2, page numbers, and the heading of Section 3.

- **Section 1, RD for cocompact lattices**, consists of items 1--7: the generalized Valette conjecture; invariance between two uniform lattices in one ambient group (and a measurable-topological-coupling version); the obstruction for many nonuniform lattices; a request for a relative version of RD; descent to a cocompact normal subgroup; permanence in a short exact sequence; and the proposed `(**)`-relative-hyperbolicity approach. - **Section 2, RD for unitary representations**, consists of Definition 2.1 and items 1--5, including Cowling's quasi-regular-representation remark. - The final words "3 RD for length functions" are only the next section heading. Section 3 is the next canonical corpus record (source index 166), and is not part of the mathematics considered here.

The principal question, quoted with only punctuation normalized, is:

The source calls this the generalized Valette conjecture. The original Valette conjecture asks for RD of uniform lattices in semisimple groups over local fields. The short-exact-sequence question is:

\[ 1\longrightarrow H\longrightarrow \Gamma\longrightarrow P \longrightarrow 1, \qquad P\text{ has RD}; \]

is \(\Gamma\) RD if and only if \(H\) is RD for the restriction of a word length of \(\Gamma\)?

1. "locally compact compactly generated" means "locally compact, compactly generated"; the word *compact* is not duplicated. 2. `S0(2,3)` is \(SO(2,3)\). The displayed equality with \(Sp(4,\mathbb R)\) is not literal: one should use the identity component and central isogeny, for example \(PSp(4,\mathbb R)\cong SO_0(2,3)\). 3. The notation \(SL(3,\mathbb O)\) is informal. The exceptional real case in Chatterji's theorem is the symmetric space associated with \(E_{6(-26)}\), alongside \(SL_3(\mathbb H)\). 4. The isolated `1`, `2`, and `22.` are page-number contamination, not mathematical exponents or item numbers. 5. The extracted Definition 2.1 loses the squared modulus and even retains the impossible phrase "and every \(g\)" after integrating over \(g\). The standard intended coefficient formulation, confirmed in Boyer's later paper, is: a unitary representation \(\pi\) has RD with respect to \(L\) if, for some \(s,C>0\), \[ \int_G \frac{|\langle \pi(g)v,w\rangle|^2} {(1+L(g))^{2s}}\,d\mu(g)\le C \] for all unit vectors \(v,w\). Replacing \(2s\) by \(s\) is the same convention after renaming the exponent. The literal unsquared complex integral is not a usable definition of RD.

来源字段：https://aimath.org/WWN/rapiddecay/rapiddecay.pdf

02691 | AIM-GEOMETRIC_GROUP_THEORY-0119

### A countable Ozawa Liouville action with abelian stabilizers

S / solved | AIM 研讨会问题汇编

Does a countable non-amenable discrete group admit a transitive action \(G\curvearrowright X\) with amenable point stabilizers and a non-degenerate probability measure \(\mu\) on \(G\) for which every bounded \(P_\mu\)-harmonic function on \(X\) is constant?

来源字段：http://aimpl.org/amenablediscrete/4/

02692 | AIM-GEOMETRIC_GROUP_THEORY-0127

### A counterexample sieve for infinite amenable groups

P / partially_solved | AIM 研讨会问题汇编

**Question.** Is it true that every infinite amenable group contains an infinite abelian subgroup?

来源字段：http://aimpl.org/amenablediscrete/5/

02693 | AIM-GEOMETRIC_GROUP_THEORY-0076

### A current boundary and rank-two within-class rigidity for Artin groups

P / partially_solved | AIM 研讨会问题汇编

**Problem.** Which Artin groups are profinitely rigid?

来源字段：http://aimpl.org/geomartingp/8/

02694 | AIM-GEOMETRIC_GROUP_THEORY-0062

### A curvature sieve for four Artin-group complexes

P / partially_solved | AIM 研讨会问题汇编

What kind of non-positive curvature might these complexes have? (systolic, Helly, CUB, $\mathrm{CAT}(0)$?)

Let $A$ be an Artin group with Deligne complex $D$, Salvetti complex $S$, ``pure'' Salvetti complex $P$, and complex of irreducible parabolics $X$.

来源字段：http://aimpl.org/geomartingp/5/

02695 | AIM-GEOMETRIC_GROUP_THEORY-0084

### A doubled-Coxeter integral representation for right-angled Artin groups

P / partially_solved | AIM 研讨会问题汇编

Find a “good”/“natural” (e.g., finite dimensional) linear representation for Artin groups.

来源字段：http://aimpl.org/geomartingp/8/

02696 | AIM-GEOMETRIC_GROUP_THEORY-0100

### A dynamical residual-finiteness obstruction to finite presentation

P / partially_solved | AIM 研讨会问题汇编

Find a minimal faithful $\mathbf Z^2$-action on a Cantor set for which the commutator subgroup of the topological full group is finitely presented.

来源字段：http://aimpl.org/amenablediscrete/2/

02697 | AIM-GEOMETRIC_GROUP_THEORY-0005

### A finite fibre-intersection criterion for Cannon--Thurston maps

P / partially_solved | AIM 研讨会问题汇编

Let $G$ be a hyperbolic free-by-cyclic group. Which subgroups of $G$ admit Cannon--Thurston maps?

来源字段：http://aimpl.org/freebycyclic/1/

02698 | AIM-GEOMETRIC_GROUP_THEORY-0001

### A finite height-kernel case of the quasiconvex-subgroup conjecture

P / partially_solved | AIM 研讨会问题汇编

A subgroup \(F\) of a group \(G\) is a *semi-fibre* if there exists an injective non-surjective homomorphism \(\varphi:F\to F\) such that \(G\) is the HNN extension \(G=F*_{\varphi}\). A subgroup \(F\) of \(G\) is a *virtual semi-fibre* if there exists a finite-index subgroup \(G'\) of \(G\) such that \(F\) is a semi-fibre of \(G'\).

Let \(G\) be a hyperbolic free-by-cyclic group and \(H\) a non-quasiconvex subgroup of \(G\). Then \(H\) contains a subgroup \(F\) which is a virtual fibre or a virtual semi-fibre of a free-by-cyclic subgroup of \(G\).

Its statement is:

There are two source issues that matter mathematically.

The AIM workshop report instead formulates the motivating problem for a **distorted** subgroup, and the later precise formulation in Linton's Conjecture 1.3 explicitly assumes that \(H\) is finitely generated. In a hyperbolic group, quasiconvex subgroups are finitely generated, so every infinitely generated subgroup is automatically non-quasiconvex. Thus omitting finite generation makes the literal statement substantially stronger, not merely stylistically different. All results below are stated for the intended finitely generated reading; no claim is made that the literal unrestricted statement is false.

The partial theorem below uses the still narrower atoroidal fully irreducible case exactly where an existing quasiconvexity theorem is needed. The height-kernel lemma itself has no irreducibility hypothesis.

Write \[ G=F_n\rtimes_{\phi}\langle t\rangle, \qquad p:G\longrightarrow \mathbb Z, \] where \(p(t)=1\) and \(p(F_n)=0\). For \(H\leq G\), its **height kernel** is \[ K_H=H\cap F_n=\ker(p|_H). \]

来源字段：http://aimpl.org/freebycyclic/1/

02699 | AIM-GEOMETRIC_GROUP_THEORY-0045

### A finite twist canonicalizer for the large-type frontier and correction of the Coxeter status

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.2.** Solve the isomorphism problem for 2-dimensional Artin groups.

来源字段：http://aimpl.org/geomartingp/3/

02700 | AIM-GEOMETRIC_GROUP_THEORY-0022

### A finite-cover fixed-vector obstruction for virtual residual torsion-free nilpotence

P / partially_solved | AIM 研讨会问题汇编

Are all free-by-cyclic groups virtually (residually torsion-free nilpotent)?

来源字段：http://aimpl.org/freebycyclic/4/

02701 | AIM-GEOMETRIC_GROUP_THEORY-0104

### A finite-diagonal Følner criterion for the universal Grigorchuk group

P / partially_solved | AIM 研讨会问题汇编

Is the universal Grigorchuk group amenable?

If this group is amenable, then the Folner function of this group is universal bound on Folner function of $G_\omega$

来源字段：http://aimpl.org/amenablediscrete/3/

02702 | AIM-GEOMETRIC_GROUP_THEORY-0096

### A finite-gluing obstruction for amenable clopen restrictions

P / partially_solved | AIM 研讨会问题汇编

Is non-amenability of a topological full group preserved under taking restrictions?

To be more precise: let \(G\) act on a Cantor set \(C\) minimally. Assume \(G\) is equal to its full group and is non-amenable.

Take a clopen subset \(U\) of the Cantor set \(C\), let \(G_U\) be the restriction of \(G\) to \(U\). Is \(G_U\) also non-amenable?

whenever \(h\in\operatorname{Homeo}(C)\) has a finite clopen partition \(C=\bigsqcup_i C_i\) and elements \(g_i\in G\) such that \(h|_{C_i}=g_i|_{C_i}\), then \(h\in G\).

来源字段：http://aimpl.org/amenablediscrete/2/

02703 | AIM-GEOMETRIC_GROUP_THEORY-0018

### A finite-index virtual-geometric nonvanishing criterion

P / partially_solved | AIM 研讨会问题汇编

Clay shows that \(\ell^2\)-torsion vanishes for free-by-cyclic groups with polynomially-growing monodromies (Clay 2017, MR3667215).

If \(G\) is a free-by-cyclic group with exponentially growing monodromy then it has non-vanishing \(\ell^2\)-torsion.

Thus the second sentence is the conjecture to be studied. Write \[ G_\Phi=F_n\rtimes_\Phi\langle t\rangle =\langle F_n,t\mid t^{-1}xt=\Phi(x)\ (x\in F_n)\rangle, \] where \(\Phi\in\operatorname{Aut}(F_n)\), and put \(\phi=[\Phi]\in\operatorname{Out}(F_n)\). Following Clay's sign convention, it is useful to set \[ \tau^{(2)}(G):=-\rho^{(2)}(G). \] For free-by-cyclic groups \(\tau^{(2)}(G)\geq 0\), so the conjecture asks whether exponential growth forces \(\tau^{(2)}(G)>0\).

来源字段：http://aimpl.org/freebycyclic/4/

02704 | AIM-GEOMETRIC_GROUP_THEORY-0098

### A finite-patch exterior criterion for the Penrose full group

P / partially_solved | AIM 研讨会问题汇编

Is the group of polygon rearrangements (PRG) amenable? Specifically, is the full group of the Penrose tiling amenable?

来源字段：http://aimpl.org/amenablediscrete/2/

02705 | AIM-GEOMETRIC_GROUP_THEORY-0112

### A finite-product counterexample would force an exponential-growth supramenable subgroup

P / partially_solved | AIM 研讨会问题汇编

**Definition:** A group $G$ is *supramenable* if for every triplet $(G,X,E)$ (where $E$ is a nonempty subset of $X$ and $G \def\actson{\curvearrowright} X$), there is an invariant measure normalized on $E$.

**Fact:** Subexponential growth implies supramenability.

1. **Q:** (Rosenblatt 74') Is there a supramenable group of exponential growth? 2. Is the direct product of $2$ supramenable groups also supramenable?

来源字段：http://aimpl.org/amenablediscrete/4/

02706 | AIM-GEOMETRIC_GROUP_THEORY-0035

### A four-generator theorem and a rank-five frontier sieve

P / partially_solved | AIM 研讨会问题汇编

**Does the $K(\pi,1)$ conjecture hold for all Artin groups?**

来源字段：http://aimpl.org/geomartingp/1/

02707 | AIM-GEOMETRIC_GROUP_THEORY-0025

### A four-way CAT(0)--virtual-RFRS equivalence for one-vertex tubular mapping tori

P / partially_solved | AIM 研讨会问题汇编

What's the relationship between being CAT(0) and RFRS for free-by-cyclic groups?

来源字段：http://aimpl.org/freebycyclic/5/

02708 | AIM-GEOMETRIC_GROUP_THEORY-0159

### A homological gate at the first unresolved target rank

P / partially_solved | AIM 研讨会问题汇编

The second paragraph begins “In the other direction” and is the lead-in to the separately displayed Problem 4.2. This report therefore studies

\[ \operatorname{Hom}(\operatorname{Mod}(S_g),\operatorname{Out}(F_n)) \]

and injectivity, while auditing the reverse-direction paragraph only as source context.

The PDF does not state the surface convention. We use the standard one in this literature: \(S_g\) is a connected, closed, orientable surface of genus \(g\), and

\[ \operatorname{Mod}(S_g)=\pi_0\operatorname{Homeo}^+(S_g). \]

Extended or nonorientable mapping class groups are different questions. We take \(n\geq 1\).

来源字段：https://aimath.org/WWN/outerauto/outerauto.pdf

02709 | AIM-GEOMETRIC_GROUP_THEORY-0063

### A mapping-class-free unicorn proof with a uniform 17-centered bound

P / partially_solved | AIM 研讨会问题汇编

Find a direct combinatorial proof (i.e., without reference to mapping class groups) that \(X\) is \(\delta\)-hyperbolic for the braid groups.

Let \(A\) be an Artin group with Deligne complex \(D\), Salvetti complex \(S\), “pure” Salvetti complex \(P\), and complex of irreducible parabolics \(X\).

来源字段：http://aimpl.org/geomartingp/5/

02710 | AIM-GEOMETRIC_GROUP_THEORY-0077

### A maximal-clique criterion for Serre goodness of even Artin groups

P / partially_solved | AIM 研讨会问题汇编

We say a group is *good in the sense of Serre* if it has the same cohomology as its profinite completion.

Are all Artin groups good in the sense of Serre?

来源字段：http://aimpl.org/geomartingp/8/

02711 | AIM-GEOMETRIC_GROUP_THEORY-0072

### A maximal-envelope reduction for affine parabolic intersections

P / partially_solved | AIM 研讨会问题汇编

Are intersections of parabolic subgroups themselves parabolic? More specifically: is this true in the Euclidean case?

来源字段：http://aimpl.org/geomartingp/7/

02712 | AIM-GEOMETRIC_GROUP_THEORY-0016

### A measure-equivalence phase boundary for free-by-cyclic groups

P / partially_solved | AIM 研讨会问题汇编

What are the measure equivalence classes of free-by-cyclic groups?

来源字段：http://aimpl.org/freebycyclic/3/

02713 | AIM-GEOMETRIC_GROUP_THEORY-0081

### A minimal-counterexample obstruction and acylindrical hierarchy for Artin Tits alternatives

P / partially_solved | AIM 研讨会问题汇编

Does the Tits alternative hold for all Artin groups?

来源字段：http://aimpl.org/geomartingp/8/

02714 | OWR-16167-023

### A Nonamenable Extension Beyond Thompson's Group

O / open | Oberwolfach 报告问题汇编

Let $\alpha(t)=t+1/2$ and let $\beta$ be the piecewise-projective homeomorphism of $\mathbb R$ defined by $\beta(t)=t$ for $t\leq0$, $\beta(t)=t/(1-t)$ for $0\leq t\leq1/2$, $\beta(t)=3-1/t$ for $1/2\leq t\leq1$, and $\beta(t)=t+1$ for $t\geq1$. Is the subgroup $\langle\alpha,\beta\rangle$ of Monod's group nonamenable, and what structural role does the coefficient $1/2$ play in its relation to Thompson's group $F$?

来源字段：https://doi.org/10.4171/owr/2018/26

02715 | AIM-GEOMETRIC_GROUP_THEORY-0023

### A pairwise CAT(0) certificate for triangular free-by-cyclic groups

P / partially_solved | AIM 研讨会问题汇编

Characterise the free-by-cyclic groups which act geometrically on CAT(0) spaces.

来源字段：http://aimpl.org/freebycyclic/5/

02716 | AIM-GEOMETRIC_GROUP_THEORY-0020

### A periodic non-geometric counterexample and a transfer obstruction

S / solved | AIM 研讨会问题汇编

Let $G$ be a non-geometric free-by-cyclic group and $A$ a finitely generated abelian group. Does there exist a finite index subgroup $G'$ of $G$ such that $A$ is isomorphic to a direct summand of the abelianisation of $G'$?

It attributes the problem to Tam Cheetham-West and states, verbatim:

The page gives no remark, status, literature, definition of “non-geometric,” rank convention for the free kernel, or growth hypothesis on the monodromy.

There are two standard readings of “non-geometric”:

1. a specified splitting $G=F_n\rtimes_\phi\mathbb Z$ has a monodromy class $[\phi]\in\operatorname{Out}(F_n)$ that is not induced by a homeomorphism of a compact surface with boundary; 2. the abstract free-by-cyclic group is not the fundamental group of a compact surface mapping torus, equivalently in this setting is not a $3$-manifold group.

The counterexample below satisfies both readings. It has a noncyclic free kernel $F_3$, so it is not a rank-one edge case. Its monodromy is periodic, however. If the workshop intended “exponentially growing non-geometric monodromy,” that restriction is absent from the literal source and the counterexample does not settle the strengthened version. This distinction matters because the two immediately preceding AIM problems explicitly say “exponentially growing monodromy” when that is intended.

The quantifier over finitely generated abelian $A$ includes finite groups. This is also consistent with the surrounding literature on prescribed homological torsion. If “abelian” were intended to mean “free abelian,” the counterexample would not apply; indeed Proposition 5.5 below shows that every finite-rank free abelian group does occur in the counterexample's finite covers.

来源字段：http://aimpl.org/freebycyclic/4/

02717 | AIM-GEOMETRIC_GROUP_THEORY-0129

### A presentation-and-structure sieve for finitely presented torsion groups

P / partially_solved | AIM 研讨会问题汇编

Is there a f.p. infinite torsion group? **Conjecture:** (Grigorchuk) no.

Does there exist an infinite group \(G\) admitting an ordinary finite presentation such that every element of \(G\) has finite order?

来源字段：http://aimpl.org/amenablediscrete/5/

02718 | AIM-GEOMETRIC_GROUP_THEORY-0158

### A quantitative Dehn-twist growth criterion

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.7.** Prove (uniform) uniform exponential growth for two-generator subgroups of \(\operatorname{Out}(F_n)\), or in particular two Dehn twists.

Matt Clay in his talk had proposed a “baby question” towards proving the Dehn twist case: bounding a certain ratio (which ought to be recalled here) appearing in his joint work with Alexandra Pettet. Martin Bridson suggested using linear quotients, as it is potentially easier to prove uniform exponential growth by mapping to well-understood subgroups.

来源字段：https://aimath.org/WWN/outerauto/outerauto.pdf

02719 | AIM-GEOMETRIC_GROUP_THEORY-0135

### A quantitative small-spectrum criterion for analytic L2-torsion approximation

P / partially_solved | AIM 研讨会问题汇编

Let \(M\) be a closed Riemannian manifold and let \[ \Gamma=\pi_1(M)>\Gamma_1>\Gamma_2>\cdots \] be a decreasing sequence of finite-index normal subgroups of \(\Gamma\) with \(\bigcap_n\Gamma_n=1\). Let \(\widetilde M\) be the universal cover of \(M\) and let \(M_n=\widetilde M/\Gamma_n\). Is \[ \rho^{(2)}(\widetilde M)= \lim_n\frac{\rho(M_n)}{|\Gamma:\Gamma_n|}? \]

来源字段：http://aimpl.org/l2invariantsgroups/1/

02720 | AIM-GEOMETRIC_GROUP_THEORY-0162

### A rank transition for finite-index homomorphisms into mapping class groups

P / partially_solved | AIM 研讨会问题汇编

It is known, by Martin and Juan's results, that all the Poison groups in \(\operatorname{Out}(F_n)\) would give virtually abelian image under a homomorphism to \(\operatorname{MCG}(S)\). However, subgroups of finite index change the story.

Let \(\Gamma<\operatorname{Out}(F_n)\) have finite index. Determine \(\operatorname{Hom}(\Gamma,\operatorname{Mod}(S))\). Can such a map have infinite image, especially for \(n\geq4\)? Must its image be virtually abelian? What can be said about injectivity?

来源字段：https://aimath.org/WWN/outerauto/outerauto.pdf

02721 | AIM-GEOMETRIC_GROUP_THEORY-0060

### A recursive 2/infinity-assembly criterion for HHG Artin groups

P / partially_solved | AIM 研讨会问题汇编

**Which Artin groups are HHGs?**

来源字段：http://aimpl.org/geomartingp/4/

02722 | AIM-GEOMETRIC_GROUP_THEORY-0161

### A rose-symmetry obstruction and finite-image genus range

P / partially_solved | AIM 研讨会问题汇编

Does every homomorphism \(\phi:\operatorname{Out}(F_n)\to \operatorname{MCG}(S)\) have virtually abelian image?

Its main question is:

\[ W_n=(\mathbb Z/2)^n\rtimes S_n=(\mathbb Z/2)\wr S_n, \qquad |W_n|=2^n n!. \]

Thus `Z2 o S n` is the wreath product and `Zn · n!` is \(2^n\cdot n!\), not a product involving \(\mathbb Z^n\).

There is, however, a missing logical qualification. The trivial homomorphism exists for every surface, so a genus lower bound cannot apply to *every* homomorphism. The statement must concern an injective map, a map faithful on the finite rose stabilizer, or (as proved below) a map with infinite image. The PDF does not say whether \(S\) is closed, orientable, punctured, or has boundary, nor whether `MCG` includes orientation-reversing classes. The main result below uses the standard group \(\operatorname{Mod}(S_g)\) of orientation-preserving mapping classes of a closed connected orientable surface of genus \(g\ge2\).

来源字段：https://aimath.org/WWN/outerauto/outerauto.pdf

02723 | AIM-GEOMETRIC_GROUP_THEORY-0167

### A row-column Sobolev operator space with a sharp amenable cb norm

P / partially_solved | AIM 研讨会问题汇编

Make $H_L^s(\Gamma)$ an operator space in a natural way. For instance, prove that if $\Gamma$ has polynomial growth, then it has "completely bounded RD" for the operator norms $\|\cdot\|_n$.

来源字段：https://aimath.org/WWN/rapiddecay/rapiddecay.pdf

02724 | AIM-GEOMETRIC_GROUP_THEORY-0009

### A separability reduction for hyperbolic free-by-cyclic groups

P / partially_solved | AIM 研讨会问题汇编

**Construct a hyperbolic free-by-cyclic group which is LERF.**

来源字段：http://aimpl.org/freebycyclic/2/

02725 | AIM-GEOMETRIC_GROUP_THEORY-0145

### A sharp free-product case of the Osin-Thom normal-rank inequality

P / partially_solved | AIM 研讨会问题汇编

It is known that for every finitely generated group we have \[ \beta_1^{(2)}(\Gamma)\leq d(\Gamma)-1, \] where \(d\) stands for the minimal number of generators of \(\Gamma\). A group \(\Gamma\) is normally generated by \(S\subseteq\Gamma\) if the only normal subgroup of \(\Gamma\) containing \(S\) is \(\Gamma\) itself. Let \(nrk(\Gamma)\) be the normal rank of \(\Gamma\), i.e. the minimal number of normal generators.

Do we have \[ \beta_1^{(2)}(\Gamma)\leq nrk(\Gamma)-1 \] for a torsion free group \(\Gamma\)?

来源字段：http://aimpl.org/l2invariantsgroups/3/

02726 | AIM-GEOMETRIC_GROUP_THEORY-0163

### A simplexwise Patterson-Sullivan inequality and entropy-metric Hessian

P / partially_solved | AIM 研讨会问题汇编

For all T, T' in Xn as above, is it true i(T, mu of T') i(T', mu of T) >= 1? At least, where is the inequality true?

The immediately preceding paragraph in the source PDF supplies the omitted meaning of “as above”: \(T\) is a tree in Culler--Vogtmann Outer space \(X_n\), \(\mu_T\) is its Patterson--Sullivan current, and the representative of that projective current is normalized by

\[ \langle T,\mu_T\rangle=1. \]

Thus the precise question is whether

\[ I(T,S):=\langle S,\mu_T\rangle\langle T,\mu_S\rangle\geq 1 \tag{1.1} \]

for every \(T,S\in X_n\), and, failing that, on which pairs or subspaces it holds. I use the modern brackets \(\langle T,\nu\rangle\) for the continuous tree--current intersection form. This avoids confusing it with Guirardel's tree--tree intersection number, which is also often denoted \(i\).

The PDF was checked directly. The mathematical formulas above are present there. Most importantly, the paragraph beginning “The next question comes from Martin Lustig” and mentioning “Huber[?]” and Besson is the introduction to Problem 5.2. It is not a hypothesis or continuation of Problem 5.1. The bracketed uncertainty “Huber[?]” really occurs in the PDF and is preserved rather than silently reconstructed.

There are two common normalizations of Outer space: unit covolume and projectivized metrics. They cause no ambiguity here. If \(c>0\), then the projective Patterson--Sullivan current is unchanged on replacing \(T\) by \(cT\), while self-intersection normalization gives \(\mu_{cT}=c^{-1}\mu_T\). Consequently \(I(cT,dS)=I(T,S)\).

来源字段：https://aimath.org/WWN/outerauto/outerauto.pdf

02727 | AIM-GEOMETRIC_GROUP_THEORY-0056

### A spherical-rank obstruction and finite-label triangle audit for systolic Artin groups

P / partially_solved | AIM 研讨会问题汇编

**Problem.** Further classify the systolic Artin groups.

来源字段：http://aimpl.org/geomartingp/4/

02728 | AIM-GEOMETRIC_GROUP_THEORY-0057

### A status-corrected residual reduction for acylindrical hyperbolicity of Artin groups

P / partially_solved | AIM 研讨会问题汇编

Acylindrical hyperbolicity

All non-spherical irreducible Artin groups are acylindrically hyperbolic.

来源字段：http://aimpl.org/geomartingp/4/

02729 | AIM-GEOMETRIC_GROUP_THEORY-0088

### A structure map and explicit wreath closure for subgroups of Thompson's group F

P / partially_solved | AIM 研讨会问题汇编

What can be said about the structure of subgroups of \(F\)?

来源字段：http://aimpl.org/amenablediscrete/1/

02730 | AIM-GEOMETRIC_GROUP_THEORY-0003

### A surface-shift criterion inside the free fiber

P / partially_solved | AIM 研讨会问题汇编

If $G$ is a hyperbolic free-by-cyclic group then does $G$ contain a closed surface subgroup?

来源字段：http://aimpl.org/freebycyclic/1/

02731 | AIM-GEOMETRIC_GROUP_THEORY-0109

### A torsion-image obstruction for the finitely presented amenable Grigorchuk HNN overgroup

P / partially_solved | AIM 研讨会问题汇编

Is there a finitely presented *amenable* group that surjects onto a group of intermediate growth?

来源字段：http://aimpl.org/amenablediscrete/3/

02732 | AIM-GEOMETRIC_GROUP_THEORY-0069

### A two-ended radial wonderful compactification in spherical type

P / partially_solved | AIM 研讨会问题汇编

Is there a ``natural'' (cellular?) compactification of the hyperplane complement associated to an Artin group?

来源字段：http://aimpl.org/geomartingp/5/

02733 | OWR-13750328-017

### Abelian Fields Generated by Height-Zero Characters

P / partially_solved | Oberwolfach 报告问题汇编

Let $p$ be prime and let $F$ be an abelian number field of conductor $n=p^am$ with $(m,p)=1$. Prove that $F=\mathbb Q(\chi)$ for some height-zero irreducible character $\chi$ in a $p$-block of a finite group if and only if $[\mathbb Q_n:\langle F,\mathbb Q_m\rangle]$ is coprime to $p$.

来源字段：https://doi.org/10.4171/owr/2023/19

02734 | OWR-793-004

### Abelian Root Groups in Special Moufang Sets

P / partially_solved | Oberwolfach 报告问题汇编

Let $M=M(U,\tau)$ be a Moufang set whose little projective group $G^\dagger$ is not sharply two-transitive on $X$. Is $M$ special if and only if $U$ is abelian? If $M$ is special with $U$ abelian, must it arise as $M(J,U,1)$ from a quadratic Jordan division algebra over a commutative field?

来源字段：https://doi.org/10.4171/owr/2005/12

02735 | OWR-1593-008

### Abelian Root Groups in Special Moufang Sets

O / open | Oberwolfach 报告问题汇编

If $M(U,\tau)$ is a special Moufang set, must its root group $U$ be abelian?

来源字段：https://doi.org/10.4171/owr/2007/37

02736 | OWR-16167-025

### Abelianizations of Finitely Presented Thompson $F$ Subgroups

O / open | Oberwolfach 报告问题汇编

Does every finitely presented subgroup of Thompson's group $F$ have torsion-free abelianization?

来源字段：https://doi.org/10.4171/owr/2018/26

02737 | OWR-14299915-005

### Abstract Cohomology of Pro-$p$ Groups in Cross Characteristic

O / open | Oberwolfach 报告问题汇编

Let $G$ be a pro-$p$ group and let $q\neq p$ be a prime. Must the abstract-group cohomology $H^n_{\mathrm{abs}}(G;\mathbb F_q)$ vanish for every $n>0$, as continuous cohomology does?

来源字段：https://doi.org/10.4171/owr/2026/9

02738 | OWR-13488-006

### Action Dimension versus $L^{2}$ Cohomological Dimension

O / open | Oberwolfach 报告问题汇编

Does every group $G$ satisfy the action-dimension inequality $\operatorname{actdim}(G)\geq2\dim_{(2)}(G)$, where $\dim_{(2)}(G)$ is its $L^2$-cohomological dimension?

来源字段：https://doi.org/10.4171/owr/2015/3

02739 | OWR-16770-002

### Admissibility and Finite Length on the Drinfeld Tower

P / partially_solved | Oberwolfach 报告问题汇编

别名：Admissibility and Finite Length on the Drinfeld Tower [OWR-16770-003]

Let $F$ be a $p$-adic field, let $G=\operatorname{GL}_2(F)$, and let $\Sigma_n$ be the level-$n$ space in the Drinfeld tower. For every $n\geq0$, is $\mathcal O(\Sigma_n)^*$ an admissible locally analytic representation of $G$, and does it have finite length?

来源字段：https://doi.org/10.4171/owr/2019/8

02740 | OWR-14748-013

### AF Full Group $C^*$-Algebras and Local Finiteness

P / partially_solved | Oberwolfach 报告问题汇编

For a countable group $G$, does $C^*(G)$ being an AF algebra imply that $G$ is locally finite?

来源字段：https://doi.org/10.4171/owr/2016/40

02741 | OWR-1452-032

### Algebraic Groups with Large Cayley Degree

O / open | Oberwolfach 报告问题汇编

For a connected reductive algebraic group $G$, let $\operatorname{Cay}(G)$ be the minimum degree of a dominant $G$-equivariant rational map $G\dashrightarrow\operatorname{Lie}(G)$. Does there exist an algebraic group $G$ with $\operatorname{Cay}(G)>2$? Does there exist a simple algebraic group with this property?

来源字段：https://doi.org/10.4171/owr/2007/01

02742 | OWR-5155-002

### Algebraic K-Theory and Rigidity Conjectures for Discrete Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a torsion-free group $G$ and an integral domain $R$, are $0$ and $1$ the only idempotents of $RG$? Do the negative K-groups of $\mathbb{Z}G$, the reduced group $K_0(\mathbb{Z}G)$, and $\operatorname{Wh}(G)$ vanish? Are higher signatures homotopy invariant, closed aspherical manifolds topologically rigid, and groups of type FP of type FF? Is every finitely presented Poincaré-duality group of dimension at least five the fundamental group of an aspherical homology ANR-manifold? Is every hyperbolic group with boundary $S^n$ the fundamental group of a closed aspherical manifold? Are the Farrell-Jones assembly maps in algebraic K- and L-theory bijective?

来源字段：https://doi.org/10.4171/owr/2011/27

02743 | OWR-14745-009

### Algorithms and Exponent Bounds for Dimension Subgroups

P / partially_solved | Oberwolfach 报告问题汇编

For a finitely presented group $G$ and a ring $k$, let $\Delta(kG)$ be the augmentation ideal and define

$$\gamma_{n,k}=\{g\in G:g-1\in\Delta(kG)^n\}.$$

Writing $\gamma_m$ for the lower central series, one has

$$\gamma_{n,\mathbb F_p}=\prod_{mp^j\geq n}\gamma_m^{p^j},\qquad \gamma_{n,\mathbb Q}=\sqrt{\gamma_n}.$$

How can $G/\gamma_{n,\mathbb F_p}$ and $\gamma_{n,\mathbb Z}$ be computed efficiently? Is $\gamma_{n,\mathbb Z}=\bigcap_p\gamma_{n,\mathbb F_p}$? If $c_n$ bounds the exponent of $\gamma_{n,\mathbb Z}/\gamma_n$, is $c_n=2$?

来源字段：https://doi.org/10.4171/owr/2016/37

02744 | AIM-GEOMETRIC_GROUP_THEORY-0140

### Almost treeability of finite-volume hyperbolic 3-manifold groups

S / solved | AIM 研讨会问题汇编

**Definition.** \(\Gamma\) is almost treable if it admits a free action on probability space such that the equivalence relation associated to this action is almost treeable, i.e., it is an increasing union of treeable subequivalence relations.

The property is stable under measure equivalence and taking subgroups. From the property it follows that \(\Gamma\) has the Haagerup property and is sofic. Moreover, \(\lambda_{cb}(\Gamma)=1\) if \(\Gamma\) is almost treeable.

Examples include: \(\mathbb F_2\times H\), where \(H\) is amenable.

Are the fundamental groups of hyperbolic \(3\)-manifolds almost treeable?

来源字段：http://aimpl.org/l2invariantsgroups/2/

02745 | OWR-1063-002

### Alperin-Goldschmidt Chains in Ordinary Weight Formulas

O / open | Oberwolfach 报告问题汇编

For a block $B$, do only chains of $B$-subpairs beginning with $(U,b_U)$ such that $Z(U)$ is a defect group of $b_U$ and $O_p(N_G(U,b_U)/(U C_G(U)))=1$ contribute to the alternating sums in the Ordinary Weight Conjecture?

来源字段：https://doi.org/10.4171/owr/2005/42

02746 | OWR-13750328-009

### Alperin-McKay Conjecture

O / open | Oberwolfach 报告问题汇编

Let $b$ be an $\ell$-block of a finite group $G$ with defect group $D$, and let $B$ be its Brauer correspondent in $N_G(D)$. Prove $|\operatorname{Irr}_0(b)|=|\operatorname{Irr}_0(B)|$.

来源字段：https://doi.org/10.4171/owr/2023/19

02747 | OWR-17295-003

### Alternativity of Parameters for $A_2$-Graded Groups

O / open | Oberwolfach 报告问题汇编

Is every ring $R$ that parametrizes an $A_2$-graded group alternative?

来源字段：https://doi.org/10.4171/owr/2019/52

02748 | AIM-GEOMETRIC_GROUP_THEORY-0110

### Amenability across the polynomial-activity hierarchy

P / partially_solved | AIM 研讨会问题汇编

Are polynomial activity automata groups amenable?

Is every group generated by a finite invertible automaton whose states have polynomially growing activity amenable?

来源字段：http://aimpl.org/amenablediscrete/3/

02749 | OWR-4411-002

### Amenability of Interval-Exchange Transformation Groups

O / open | Oberwolfach 报告问题汇编

If $G$ is a subgroup of the group of interval-exchange transformations, must $G$ be amenable?

来源字段：https://doi.org/10.4171/owr/2010/30

02750 | OWR-15216-021

### Amenability of Thompson Group and Interval Homeomorphism Groups

O / open | Oberwolfach 报告问题汇编

Is Thompson's group $F$ amenable? Is the group $\mathrm{PL}_+(I)$ of orientation-preserving piecewise-linear homeomorphisms of the interval amenable?

来源字段：https://doi.org/10.4171/owr/2017/11

02751 | OWR-4792-011

### Amenability of Thompson's Group $F$

O / open | Oberwolfach 报告问题汇编

Is Thompson's group $F$ amenable; equivalently, does it support a finitely additive translation-invariant probability measure?

来源字段：https://doi.org/10.4171/owr/2011/02

02752 | AIM-GEOMETRIC_GROUP_THEORY-0026

### An analytic-convexity reduction for forward and backward stretch factors

P / partially_solved | AIM 研讨会问题汇编

Let \(G\) be a free-by-cyclic group and \(F\) a component of the BNS invariant of \(G\). When is it the case that the stretch factor functions \(\lambda_{+}, \lambda_{-} \colon F \to \mathbb{R}\) agree on a convex subset of \(F\)?

来源字段：http://aimpl.org/freebycyclic/6/

02753 | AIM-GEOMETRIC_GROUP_THEORY-0071

### An equivariant curve complex for affine type C and a correction to the orbifold premise

P / partially_solved | AIM 研讨会问题汇编

In (Allcock 2002, MR1911508), several Artin groups are shown to be mapping class groups of certain orbifolds.

For an Artin group $A$ and orbifold $\mathcal O$ with $A \cong MCG(\mathcal O)$, is there a (reasonably defined) curve complex for $\mathcal O$ which agrees with the complex of irreducible parabolics for $A$?

Daniel Allcock, “Braid pictures for Artin groups,” *Transactions of the American Mathematical Society* **354** (2002), no. 9, 3455--3474, MR1911508, DOI 10.1090/S0002-9947-02-02944-6.

来源字段：http://aimpl.org/geomartingp/6/

02754 | AIM-GEOMETRIC_GROUP_THEORY-0134

### An exact lower-degree defect for homological torsion growth

P / partially_solved | AIM 研讨会问题汇编

Let $M$ be a closed Riemannian manifold and let \[ \Gamma=\pi_1(M)>\Gamma_1>\Gamma_2>\cdots \] be a decreasing sequence of finite-index normal subgroups with $\bigcap_n\Gamma_n=1$. Let $\widetilde M$ be the universal cover and $M_n=\widetilde M/\Gamma_n$. Suppose in addition that $M$ is aspherical, $\dim M=2k+1$, and $\beta_j^{(2)}=0$ for every $j$. Is \[ \rho^{(2)}(\widetilde M)=(-1)^k\lim_{n\to\infty} \frac{\log|\operatorname{tors}H_k(M_n;\mathbb Z)|} {|\Gamma:\Gamma_n|}? \]

来源字段：http://aimpl.org/l2invariantsgroups/1/

02755 | AIM-GEOMETRIC_GROUP_THEORY-0095

### An explicit abstract-embeddability antichain in Monod's group

P / partially_solved | AIM 研讨会问题汇编

Let \(H\) be the group of all piecewise-projective homeomorphisms of \(S^1\) that fix \(\infty\).

What is the subgroup structure of \(H\)?

Is there an infinite anti-chain of subgroups of \(H\)?

Are there only finitely many obstructions to amenability? (i.e., is there a finite collection of subgroups of \(H\) such that every non amenable subgroup of \(H\) contains a copy of one of them?) This question is also interesting for subgroups of Thompson's group \(F\) or more generally, for subgroups of \(\mathrm{PL}_o(I)\) (the group of piecewise linear orientation preserving homeomorphisms of the interval \([0,1]\) with finitely many breakpoints).

来源字段：http://aimpl.org/amenablediscrete/1/

02756 | AIM-GEOMETRIC_GROUP_THEORY-0105

### An explicit Følner window and quantitative good-ball radii for the first Grigorchuk group

P / partially_solved | AIM 研讨会问题汇编

Find bounds on Folner functions for the Grigorchuk group \(G\).

来源字段：http://aimpl.org/amenablediscrete/3/

02757 | AIM-GEOMETRIC_GROUP_THEORY-0015

### An explicit lamination-depth gap for the AIM free-by-cyclic pair

P / partially_solved | AIM 研讨会问题汇编

Let \(F = F(a,b,c,d,e)\) be a free group of rank five. Define an automorphism \(\phi \in \mathrm{Aut}(F)\) such that \[ \begin{split} \phi \colon F(a,b,c,d,e) &\to F(a,b,c,d,e) \\ a &\mapsto b\\ b &\mapsto c\\ c &\mapsto ab\\ d &\mapsto ea \\ e &\mapsto ed \end{split} \] Let \(F'\) be the free factor of \(F\) generated by \(\{a,b,c\}\). Let \(G = F \rtimes_{\phi} \mathbb{Z}\) be the mapping torus of \(\phi\), and let \(\Gamma\) be the mapping torus of the restriction of \(\phi\) to \(F'\). Then \(\Gamma\) and \(G\) are not quasi-isometric.

来源字段：http://aimpl.org/freebycyclic/3/

02758 | AIM-GEOMETRIC_GROUP_THEORY-0051

### An index-ratio obstruction for the open F4/H4 commensurability case

P / partially_solved | AIM 研讨会问题汇编

Find the commensurability and Q.I. classification for \(\infty\)-free Artin groups.

A specific case: are \(F_4\) and \(H_4\) commensurable?

来源字段：http://aimpl.org/geomartingp/3/

02759 | AIM-GEOMETRIC_GROUP_THEORY-0103

### An infinite-antichain obstruction for weakly branch groups

P / partially_solved | AIM 研讨会问题汇编

Is there a f.p. branch group? Is there a f.p. amenable branch group?

来源字段：http://aimpl.org/amenablediscrete/3/

02760 | AIM-GEOMETRIC_GROUP_THEORY-0117

### An orbitwise dyadic-volume criterion for extensive amenability

P / partially_solved | AIM 研讨会问题汇编

Are group actions with polynomial-growth Schreier graphs extensively amenable?

Extensive amenability is known for recurrent actions.

来源字段：http://aimpl.org/amenablediscrete/4/

02761 | AIM-GEOMETRIC_GROUP_THEORY-0027

### An unbounded family of BNS component orbits for free-by-cyclic groups

P / partially_solved | AIM 研讨会问题汇编

Find families of examples of free-by-cyclic groups $G$ such that $b_1(G)>2$ and the BNS invariant of $G$ has many components (up to the action of $\operatorname{Out}(G)$).

来源字段：http://aimpl.org/freebycyclic/6/

02762 | OWR-1967-005

### Andersen-Masbaum-Ueno Conjecture

O / open | Oberwolfach 报告问题汇编

Let $\Sigma$ be a compact orientable surface with negative Euler characteristic, and let $\rho_k$ be a Reshetikhin-Turaev TQFT representation of its mapping class group at level $k$. Does a mapping class $\phi$ have a pseudo-Anosov piece if and only if there is $k_0(\phi)$ such that $\rho_k(\phi)$ has infinite order for every $k\geq k_0(\phi)$?

来源字段：https://doi.org/10.4171/owr/2008/22

02763 | OWR-1106-002

### Approximately Free Finitely Generated Groups

O / open | Oberwolfach 报告问题汇编

What natural class of finitely generated groups should be regarded as approximately free groups?

来源字段：https://doi.org/10.4171/owr/2005/45

02764 | OWR-16929-006

### Arithmeticity Problems for Hyperbolic Lattices and Submanifolds

O / open | Oberwolfach 报告问题汇编

Does there exist a hyperbolic $n$-manifold, with $n\geq4$, containing no arithmetic totally geodesic $m$-submanifold for any $1<m<n$? Is the trace field of every lattice in $\operatorname{PO}(n,1)$, for $n\geq4$, totally real?

来源字段：https://doi.org/10.4171/owr/2019/17

02765 | OWR-4990377-003

### Artin Groups in Helly and Injective Geometry

O / open | Oberwolfach 报告问题汇编

Which two-dimensional Artin groups of hyperbolic type are virtually Helly; which Artin groups act properly on CAT(0) cube complexes; are parabolic subgroups stable under intersection; is the complex of proper parabolic two-dimensional Artin subgroups NPC; is the three-generator all-label-3 Artin group Helly; are all two-dimensional Artin groups systolic; is reducibility equivalent to direct-product decomposition; are Artin groups and dihedral Artin groups hierarchically hyperbolic; do non-Helly groups act geometrically on coarse Helly graphs; are uniform lattices in $SL_n(K)$ injective or in $GL_n(K)$ Helly; and what are the injective hulls of $GL_n(R)/O(n)$ and the hyperbolic plane, and the Coxeter groups realizable as reflection groups in injective metric spaces?

来源字段：https://doi.org/10.4171/owr/2021/8

02766 | OWR-1453-007

### Asymptotic Dynamics of Group Automorphism Orbits

O / open | Oberwolfach 报告问题汇编

Let $G$ be a finitely generated group, let $\alpha$ be an automorphism of $G$, and let $g\in G$. What can be said about $\alpha^n(g)$ as $n\to\infty$? More specifically, for periodic $g$, what bounds or descriptions exist for its period; for nonperiodic $g$, how does its length grow; and does the sequence converge in a suitable sense, possibly after replacing $\alpha$ by a power?

来源字段：https://doi.org/10.4171/owr/2007/02

02767 | OWR-2053-009

### Asymptotic Growth Under Group Automorphisms

O / open | Oberwolfach 报告问题汇编

Given a finitely generated group $G$, an automorphism $\alpha$, and an element $g\in G$, how does $\alpha^n(g)$ grow as $n\to\infty$, and does the sequence have a limit in a suitable sense?

来源字段：https://doi.org/10.4171/owr/2008/42

02768 | OWR-11129-006

### Asymptotic Proportion of $Q$ in Classical Groups

O / open | Oberwolfach 报告问题汇编

Let $G=\operatorname{Class}(n,q)$ be a finite classical group of dimension $n$ over a field of odd order $q$, with natural module $V$. Let $Q$ be the set of even-order elements $g\in G$ for which the fixed space of $g^{|g|/2}$ has dimension at least $n/3$ and less than $2n/3$. Determine the asymptotic behavior of $|Q|/|G|$. In particular, is it of order $1/\log n$?

来源字段：https://doi.org/10.4171/owr/2011/37

02769 | OWR-14298004-015

### Atiyah Conjectures and Irrational-Covolume Lattices for TDLC Groups

O / open | Oberwolfach 报告问题汇编

Does an Atiyah conjecture hold for totally disconnected locally compact groups, or do such groups have irrational $L^2$-Betti numbers? Do compactly generated totally disconnected locally compact groups admit lattices of irrational covolume?

来源字段：https://doi.org/10.4171/owr/2023/55

02770 | OWR-4428-009

### Attainable $\ell^2$-Betti Numbers of a Fixed Group

O / open | Oberwolfach 报告问题汇编

For a countable discrete group $G$, what is the set $C(G)$ of $\ell^2$-Betti numbers arising as von Neumann dimensions of kernels of matrices over $\mathbb QG$?

来源字段：https://doi.org/10.4171/owr/2010/40

02771 | AIM-GEOMETRIC_GROUP_THEORY-0149

### Audit of candidate Out(F_N) complexes and abelian edge-rank collapse

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.2.** Do various candidate complexes satisfy the conditions of Problem 1.1-namely, constructing a $\delta$-hyperbolic graph with an $\operatorname{Out}(F_n)$-action such that: (1) every irreducible-with-irreducible-powers (iwip) outer automorphism $\gamma$ has positive translation distance ($|\gamma| > 0$); (2) polynomially growing $\gamma$ are elliptic ($|\gamma| = 0$); (3) the construction is natural and useful; (4) the local structure permits inductive arguments; and (5) the graph is highly connected with highly connected links?

Specifically, investigate these properties for the following candidate complexes: 1. the free-factor complex; 2. the free-splitting complex (with trivial edge groups), quasi-isometric to the sphere complex; 3. variants of splitting complexes allowing non-trivial edge groups, such as splittings over $\mathbb{Z}^r$ or splittings of rank less than $k$; 4. a poset of commensurability classes of abelian subgroups consisting of linearly growing automorphisms; and 5. the Kapovich--Lustig graph.

来源字段：https://aimath.org/WWN/outerauto/outerauto.pdf

02772 | OWR-16167-007

### Automaticity of Thompson's Group $F$

O / open | Oberwolfach 报告问题汇编

Is Thompson's group $F$ automatic?

来源字段：https://doi.org/10.4171/owr/2018/26

02773 | OWR-4428-011

### Automaton Kernel Series and $\ell^2$-Betti Values

P / partially_solved | Oberwolfach 报告问题汇编

Fix integers $n$ and $C$, and let $A$ be an automaton whose letters are $n\times n$ integer matrices with entries in $\{-C,-C+1,\ldots,C\}$. Say that a matrix $M$ of dimension at least $n$ is recognized by $A$ if its entries vanish outside the $n$-neighborhood of the diagonal and $A$ recognizes the word of consecutive $n\times n$ minors along the diagonal. Enumerate the recognized matrices as $M_1,M_2,\ldots$, and let $d(M)$ be the dimension of $M$. What values can $$F_A(x)=\sum_{i\geq1}\dim\ker(M_i)x^{d(M_i)}$$ take at rational $x$? In particular, can $F_A(q)$ be an irrational algebraic number for some automaton $A$ and rational $q$?

来源字段：https://doi.org/10.4171/owr/2010/40

02774 | OWR-14299097-009

### Babai Polylogarithmic Diameter Conjecture for Finite Simple Groups

O / open | Oberwolfach 报告问题汇编

Prove that there is an absolute constant $C$ such that, for every finite simple group $G$ and every generating subset $A\subseteq G$, one has $A^\ell=G$ for some $\ell\leq(\log|G|)^C$.

来源字段：https://doi.org/10.4171/owr/2025/16

02775 | OWR-1965-006

### Babai's Diameter Conjecture for Finite Simple Groups

O / open | Oberwolfach 报告问题汇编

For every nonabelian finite simple group $G$ and generating set $A$, is $$\operatorname{diam}(\operatorname{Cay}(G,A))\ll(\log|G|)^c$$ for some absolute constant $c$?

来源字段：https://doi.org/10.4171/owr/2008/20

02776 | OWR-14298365-007

### Babai’s Polylogarithmic Diameter Conjecture

O / open | Oberwolfach 报告问题汇编

If finite simple groups satisfy a diameter bound with exponent $c$, does there exist $c'(c)$ such that $\operatorname{diam}_{\mathrm{worst}}(G)\leq(\log|G|)^{c'(c)}$?

来源字段：https://doi.org/10.4171/owr/2024/17

02777 | OWR-1738-005

### Basis Numbers and Torsion Units in Skew Group Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Let $t:G\to\operatorname{Aut}(K)$ be a group action on a field $K$. Does the skew group algebra $K^tG$ have invariant basis number? In the recursive construction of torsion units $w_i$ from a torsion element $a$ and an element $b\in G$, what can be said about $S(a,b)=\{w_i\}$ and the group it generates? Finally, if $G$ is a finite nonabelian $p$-group over a field $K$ of characteristic $p>2$, does $KG$ lack a filtered multiplicative basis?

来源字段：https://doi.org/10.4171/owr/2007/55

02778 | OWR-832-005

### Baum-Connes Conjecture for Locally Compact Groups

O / open | Oberwolfach 报告问题汇编

For every locally compact group $H$, the Baum-Connes assembly map from the equivariant $K$-homology of $EH$ to $K_*(C_r^*(H))$ should be an isomorphism.

来源字段：https://doi.org/10.4171/owr/2005/27

02779 | OWR-13488-011

### Bestvina-Davis Candidates for the Eilenberg-Ganea Conjecture

O / open | Oberwolfach 报告问题汇编

Let $L$ be a finite acyclic two-dimensional flag complex whose fundamental group has a nontrivial unitary representation, let $W_L$ be the associated right-angled Coxeter group, and let $H\leq W_L$ be torsion-free of finite index. Although $\operatorname{cd}(H)=2$, must $\operatorname{gd}(H)=3$, making $H$ a counterexample to the Eilenberg-Ganea conjecture?

来源字段：https://doi.org/10.4171/owr/2015/3

02780 | OWR-2053-024

### Bilipschitz Group Equivalences and 3-Manifold Hyperbolicity

O / open | Oberwolfach 报告问题汇编

For every finitely generated group $G$, is there a bi-Lipschitz bijection $G\to G\times\mathbb{Z}^2$? If $G_1\subset G$ has finite index, is there a bi-Lipschitz bijection $G_1\to G$? For two uniform lattices $L_1,L_2$ in the same Lie group, is there a bi-Lipschitz bijection $L_1\to L_2$? Finally, if a compact three-manifold $M$ admits a partially hyperbolic diffeomorphism with one-dimensional stable, center, and unstable bundles, what restrictions does this impose on the topology of $M$?

来源字段：https://doi.org/10.4171/owr/2008/42

02781 | OWR-2091-006

### Bimonster Quotients of Complex Hyperbolic Reflection Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be the complex-hyperbolic reflection group acting on the complex $13$-ball, and let $X_0$ be the complement of its mirror arrangement modulo $G$. If order-$2$ rather than order-$3$ relations are imposed on loops around the discriminant, the resulting quotient satisfies all known bimonster relations except possibly $w^{10}=1$. Is this quotient the bimonster? Do the $16$ corresponding loops generate $\pi_1(X_0)$, and what other relations does $\pi_1(X_0)$ have?

来源字段：https://doi.org/10.4171/owr/2008/45

02782 | OWR-1452-029

### Birational Asymmetry Between Representations and Their Duals

O / open | Oberwolfach 报告问题汇编

Over an algebraically closed field $k$ of characteristic zero, do there exist a connected semisimple algebraic group $G$ and a finite-dimensional algebraic $G$-module $V$ with trivial stabilizers in general position such that $V$ and its dual $V^*$ are not birationally $G$-isomorphic?

来源字段：https://doi.org/10.4171/owr/2007/01

02783 | OWR-1452-030

### Birational Equivalence of Twisted Algebraic-Group Representations

O / open | Oberwolfach 报告问题汇编

Let $G$ be a connected semisimple algebraic group, let $V$ be a finite-dimensional algebraic $G$-module with trivial stabilizers in general position, and let $\sigma$ be an automorphism of $G$. Must $V$ and its twist $V^{(\sigma)}$ be birationally $G$-isomorphic? In the explicit special case $G=\operatorname{SL}_d/\mu_d$, $V=\operatorname{Sym}^d(k^d)$ with its natural $G$-action, and $\sigma$ induced by $g\mapsto(g^T)^{-1}$, are $V$ and $V^{(\sigma)}$ birationally $G$-isomorphic? For $d>3$, this special case has trivial stabilizers in general position.

来源字段：https://doi.org/10.4171/owr/2007/01

02784 | OWR-1703870-003

### Biset-Functor Structure of Dade-Type Groups

O / open | Oberwolfach 报告问题汇编

For a finite group $G$ and an odd prime $p$, is $C_{ba+}(G,p)=\ker\Psi_G$? Does $D_\Omega(G)$ admit a well-defined biset-functor structure?

来源字段：https://doi.org/10.4171/owr/2020/23

02785 | OWR-16776-006

### Blockwise Chain Cancellation for Finite Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a finite group $G$, a prime $p$, and a positive-defect block idempotent $b$ of $\mathbb{Z}_pG$, does the alternating sum over conjugacy classes of chains of nontrivial $p$-subgroups, $\sum_{\sigma\in[\mathcal{P}(G)/G]}(-1)^{|\sigma|}\operatorname{Mat}_{|G:G_\sigma|}(\mathbb{Q}G_\sigma b_\sigma)$, equal zero?

来源字段：https://doi.org/10.4171/owr/2019/14

02786 | OWR-1596-010

### BNS Invariants and First Resonance Varieties

O / open | Oberwolfach 报告问题汇编

For a finitely generated group $G$, what is the relationship between its Bieri-Neumann-Strebel invariant $\Sigma^1(G)$ and its first resonance variety $\mathcal R_1^1(G)$? In particular, when does $\Sigma^1(G)\cup-\Sigma^1(G)=H^1(G;\mathbb R)\setminus\mathcal R_1^1(G)$?

来源字段：https://doi.org/10.4171/owr/2007/40

02787 | AIM-GEOMETRIC_GROUP_THEORY-0008

### BNS symmetry, separability, and a subgroup-tameness reduction

P / partially_solved | AIM 研讨会问题汇编

**Hyperbolic case.** If $G$ is a hyperbolic free-by-cyclic group with irreducible monodromy then $G$ is LERF if and only if the BNS invariant of every finite index subgroup of $G$ is symmetric.

The adjacent AIM background page <http://aimpl.org/freebycyclic/6/> gives the Cayley-graph definition of the Bieri--Neumann--Strebel invariant. We write

\[ S(V)=\bigl(\operatorname{Hom}(V,\mathbb R)\setminus\{0\}\bigr)/\mathbb R_{>0} \]

and interpret “symmetric” as antipodal symmetry \(\Sigma^1(V)=-\Sigma^1(V)\) in $S(V)$.

There is one important scope distinction. In current usage, an outer automorphism $\Phi\in\operatorname{Out}(F_n)$ is **irreducible** if it preserves no nontrivial free-factor system (equivalently, no free splitting of the form described by Hughes--Kudlinska). It is **fully irreducible** if every positive power is irreducible. The AIM hypothesis says irreducible, not fully irreducible. Hyperbolicity separately forces the monodromy to be atoroidal. The conditional theorem below is stated first in the fully irreducible subclass because the available subgroup-tameness conjecture has exactly that hypothesis; an explicitly analogous tameness assumption gives the same implication in the merely irreducible case.

来源字段：http://aimpl.org/freebycyclic/2/

02788 | OWR-14298004-009

### BNSR Invariants of Algebraic Bieri-Strebel Groups

O / open | Oberwolfach 报告问题汇编

Determine the Bieri-Neumann-Strebel-Renz invariant of every algebraic Bieri-Strebel group, extending the known computation for $F_\tau$.

来源字段：https://doi.org/10.4171/owr/2023/55

02789 | OWR-5158-007

### Borel-Serre-Type Compactifications for General Lattices

P / partially_solved | Oberwolfach 报告问题汇编

For a general lattice in a semisimple Lie group, does there exist an analogue of the Borel-Serre compactification that is a cofinite model for its proper classifying space?

来源字段：https://doi.org/10.4171/owr/2011/30

02790 | OWR-16415-012

### Boundary Maps for Distorted Hyperbolic Subgroups

O / open | Oberwolfach 报告问题汇编

Let $H$ be a Gromov-hyperbolic subgroup of a Lie group $G$, without assuming that $H$ is undistorted. Is there an asymptotically continuous map from the Gromov boundary of $H$ to $G/P$? In rank one, is $H$ asymptotically embedded even when it is not convex cocompact?

来源字段：https://doi.org/10.4171/owr/2018/40

02791 | OWR-9790362-009

### Bounded Conjugation-Invariant Metrics on Higher-Rank Lattices

O / open | Oberwolfach 报告问题汇编

Let $G$ be a lattice in a higher-rank simple Lie group with finite center, and let $d_S$ be the conjugation-invariant word metric induced by the union of all conjugates of a finite generating set. Is $d_S$ bounded?

来源字段：https://doi.org/10.4171/owr/2022/11

02792 | OWR-1319-003

### Bounded Unipotent Generation of Noncocompact $SL_{3}$ Lattices

P / partially_solved | Oberwolfach 报告问题汇编

Does every noncocompact lattice in $SL_3(\mathbb R)$ or $SL_3(\mathbb C)$ have a finite-index subgroup that is a product of finitely many unipotent subgroups?

来源字段：https://doi.org/10.4171/owr/2006/33

02793 | OWR-15433-005

### Bounded-Degree Property for Reductive Groups

P / partially_solved | Oberwolfach 报告问题汇编

Does every reductive group $G$ over a number field satisfy property (BD) for $S=S_{\mathrm{fin}}$, the set of all non-Archimedean places?

来源字段：https://doi.org/10.4171/owr/2017/25

02794 | OWR-1593-003

### Bounded-Generator Expander Families of Finite Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $(G_i)$ be an infinite family of finite groups. Is it possible to choose generating sets of uniformly bounded size so that the corresponding Cayley graphs form an expander family?

来源字段：https://doi.org/10.4171/owr/2007/37

02795 | OWR-1319-009

### Boundedness of Volume-Preserving Diffeomorphism Groups of Spheres

O / open | Oberwolfach 报告问题汇编

Is the group of volume-preserving diffeomorphisms of $S^n$ isotopic to the identity bounded or unbounded for $n\geq3$?

来源字段：https://doi.org/10.4171/owr/2006/33

02796 | OWR-4413-007

### Bounds on Cohomological Exponents of Finite $p$-Groups

O / open | Oberwolfach 报告问题汇编

For a finite $p$-group $P$, is $e(P)\leq2n(P)-c(P)$, where $c(P)$ is the rank of its maximal central elementary abelian $p$-subgroup and $n(P)$ is the minimal dimension of a faithful complex representation of $P$?

来源字段：https://doi.org/10.4171/owr/2010/32

02797 | AIM-GEOMETRIC_GROUP_THEORY-0014

### Bourdon-building no-go and an infinite-rank near-pair

P / partially_solved | AIM 研讨会问题汇编

Construct two hyperbolic irreducible free-by-cyclic groups which are not quasi-isometric.

来源字段：http://aimpl.org/freebycyclic/3/

02798 | OWR-1265-017

### Broué Abelian Defect-Group Conjecture

O / open | Oberwolfach 报告问题汇编

A block with abelian defect group should be derived equivalent to its Brauer correspondent in the normalizer of a defect group.

来源字段：https://doi.org/10.4171/owr/2006/15

02799 | OWR-11583-002

### Broué's Abelian Defect Group Conjecture

O / open | Oberwolfach 报告问题汇编

Is every block of a finite group with abelian defect group derived equivalent to its Brauer correspondent?

来源字段：https://doi.org/10.4171/owr/2012/16

02800 | OWR-13675-004

### Broué’s Abelian Defect Group Conjecture

O / open | Oberwolfach 报告问题汇编

If $B$ is a block of $\mathcal{O}G$ with abelian defect group $P$, then $B$ and its Brauer correspondent in $N_G(P)$ are derived equivalent.

来源字段：https://doi.org/10.4171/owr/2015/18

02801 | OPG-760

### Burnside problem

P / partially_solved | Open Problem Garden

Conjecture If a group has $r$ generators and exponent $n$, is it necessarily finite?

背景链接：http://www.openproblemgarden.org/op/burnside_problem

02802 | OWR-14297746-001

### CAM Actions of Residually Finite Groups

O / open | Oberwolfach 报告问题汇编

Does every residually finite group admit a chaotic almost minimal (CAM) action?

来源字段：https://doi.org/10.4171/owr/2023/37

02803 | OWR-14299097-005

### Cameron Greedy Base-Size Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Prove that there is an absolute constant $c$ such that every finite primitive permutation group $G$ has greedy base size $G(G)\leq c\,b(G)$, where $b(G)$ is its minimum base size.

来源字段：https://doi.org/10.4171/owr/2025/16

02804 | OWR-16406-003

### Cannon’s Hyperbolic Three-Space Conjecture

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Cannon-Conjecture Extraction [OWR-16406-004]

If a hyperbolic group $G$ has boundary homeomorphic to $S^2$, must $G$ act properly, cocompactly, and isometrically on $\mathbb H^3$?

来源字段：https://doi.org/10.4171/owr/2018/31

02805 | AIM-GEOMETRIC_GROUP_THEORY-0133

### Canonical nesting and the amenability problem for full Hanoi Towers groups

P / partially_solved | AIM 研讨会问题汇编

Are the Hanoi tower groups \(H_n\), \(n\geq4\), amenable?

来源字段：http://aimpl.org/amenablediscrete/5/

02806 | AIM-GEOMETRIC_GROUP_THEORY-0017

### Canonical peripheral reduction and a geometric rigidity case

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 3.6 (Leininger).** Let $G$ and $H$ be relatively hyperbolic, non-hyperbolic free-by-cyclic groups with a trivial JSJ decomposition. If $G$ and $H$ are quasi-isometric, then they are commensurable.

来源字段：http://aimpl.org/freebycyclic/3/

02807 | OWR-17472-007

### Canonical Subsets in McKay Bijections

O / open | Oberwolfach 报告问题汇编

Which subset of $\operatorname{Irr}_{p^{\prime}}(G)$, if it exists, corresponds under a McKay bijection to $\operatorname{Irr}(N_G(P)/P)$, and what is the analogous answer for groups of Lie type?

来源字段：https://doi.org/10.4171/owr/2020/9

02808 | AIM-GEOMETRIC_GROUP_THEORY-0124

### Cantor Cayley spectrum: normalization audit and quotient-torus obstruction

P / partially_solved | AIM 研讨会问题汇编

Is there an amenable group whose spectrum is a Cantor set?

**Remark (Dudko).** For Grigorchuk's group $G$, the spectrum is a union of two intervals: $[-1,0]\cup[1/2,1]$; the question is whether there can be something a tad crazier.

**Recovered isotropic question.** Does there exist an infinite finitely generated amenable group $G$ and a finite symmetric generating set $S$ such that $\sigma(M_S)$ is homeomorphic to the Cantor set?

来源字段：http://aimpl.org/amenablediscrete/4/

02809 | OWR-11780-007

### Cartan and Polar Decompositions for Kac-Moody Groups

O / open | Oberwolfach 报告问题汇编

For a Kac-Moody group $G$ with unitary form $K$, when do Cartan or polar decompositions exist? Does $G=KAK$ imply that $G$ is spherical? Does $\operatorname{SL}_n(F[t,t^{-1}])$ admit a polar decomposition? If $U$ is the unipotent radical of a Borel subgroup, when does $G=KUK$ hold in the nonspherical case, particularly for $\operatorname{SL}_n(F[t,t^{-1}])$?

来源字段：https://doi.org/10.4171/owr/2012/18

02810 | OWR-12871-001

### CAT(0) Actions of Artin Groups

O / open | Oberwolfach 报告问题汇编

Does every Artin group act properly and cocompactly by isometries on a CAT(0) space?

来源字段：https://doi.org/10.4171/owr/2014/11

02811 | OWR-1452-031

### Cayley Degrees of Simple Algebraic Groups

O / open | Oberwolfach 报告问题汇编

For a connected reductive algebraic group $G$, define $\operatorname{Cay}(G)$ as the minimum degree of a dominant $G$-equivariant rational map $G\dashrightarrow\operatorname{Lie}(G)$, where $G$ acts on itself by conjugation and on $\operatorname{Lie}(G)$ by the adjoint action. Determine $\operatorname{Cay}(G)$ for every connected simple algebraic group, in particular $\operatorname{Cay}(\operatorname{SL}_5)$.

来源字段：https://doi.org/10.4171/owr/2007/01

02812 | AIM-GEOMETRIC_GROUP_THEORY-0037

### Centers of Artin groups in rank at most four

P / partially_solved | AIM 研讨会问题汇编

Are the irreducible non-spherical Artin groups centerless? If not, describe the centers.

来源字段：http://aimpl.org/geomartingp/1/

02813 | OWR-1738-007

### Centers, Normalizers, and Units of Integral Group Rings

P / partially_solved | Oberwolfach 报告问题汇编

If the centers of the integral group rings $\mathbb{Z}G$ and $\mathbb{Z}H$ of finite groups $G$ and $H$ are isomorphic, must $G$ and $H$ have equivalent character tables, with class sums corresponding? For a torsion-free group $G$, is the unit group $U(KG)$ trivial? If the normalizer problem holds for every finite normal subgroup of $G$, must it hold for $G$, and can the groups satisfying it be classified? For the finite-dimensional algebra $A$ arising from a noncentral finite-conjugacy center of the unit group of a $\mathbb{Z}$-order, what can be said about $A$ and its Wedderburn-Malcev decomposition?

来源字段：https://doi.org/10.4171/owr/2007/55

02814 | AIM-GEOMETRIC_GROUP_THEORY-0053

### Central-kernel and free-by-free reductions for proper cubulation of B4

P / partially_solved | AIM 研讨会问题汇编

Prove $Br_4$ does not act properly on a $\mathrm{CAT}(0)$ cube complex.

来源字段：http://aimpl.org/geomartingp/4/

02815 | OWR-12697693-012

### Centrally Nilpotent Skew Braces of Maximal Class

P / partially_solved | Oberwolfach 报告问题汇编

Study centrally nilpotent skew braces of maximal nilpotency class, using the lower central series $$\Gamma_{n+1}(B)=\sum_{i=1}^{n}\bigl(\Gamma_i(B)*\Gamma_{n+1-i}(B)+[\Gamma_i(B),\Gamma_{n+1-i}(B)]_+\bigr).$$ Classify the braces that attain maximal class.

来源字段：https://doi.org/10.4171/owr/2023/9

02816 | OWR-4428-004

### Chain Independence of Rank Gradient

O / open | Oberwolfach 报告问题汇编

If $\Gamma$ is finitely generated and residually finite and $(\Gamma_n)$ is a normal chain with trivial intersection, is the rank gradient $$\operatorname{RG}(\Gamma,(\Gamma_n))=\lim_{n\to\infty}\frac{d(\Gamma_n)-1}{[\Gamma:\Gamma_n]}$$ independent of the chosen chain?

来源字段：https://doi.org/10.4171/owr/2010/40

02817 | OWR-11454-003

### Chain-Independent Homology Growth over Arbitrary Fields

P / partially_solved | Oberwolfach 报告问题汇编

For which groups $G$ does

$$\lim_i\frac{b_1(G_i;K)}{[G:G_i]}$$

exist for every inverse system of finite-index normal subgroups $G_i$ with trivial intersection, independently of the system and the field $K$?

来源字段：https://doi.org/10.4171/owr/2011/56

02818 | OWR-16776-011

### Character Bounds beyond Split Levi Subgroups

O / open | Oberwolfach 报告问题汇编

Can character bounds currently available for split Levi subgroups be extended to non-split Levi subgroups? Can useful bounds be obtained for every $g\in G^F$ and every $\chi\in\operatorname{Irr}(G^F)$ when $G$ is simple and is not of type $A$?

来源字段：https://doi.org/10.4171/owr/2019/14

02819 | OWR-1265-005

### Character Bounds for Blocks with One Brauer Character

O / open | Oberwolfach 报告问题汇编

In modular representation theory of finite groups, let $B$ be a $p$-block of a finite group with defect group $D$, let $k(B)$ be the number of ordinary irreducible characters in $B$, and let $l(B)$ be the number of irreducible Brauer characters in $B$. If $l(B) = 1$, must $k(B) \leq k(D)$ (a special case of Brauer's $k(B)$ conjecture)?

来源字段：https://doi.org/10.4171/owr/2006/15

02820 | OWR-8415341-012

### Character Degrees of Graph-Defined Exponent-$p$ Groups

O / open | Oberwolfach 报告问题汇编

For a finite simple graph $\Gamma$ and an odd prime $p$, determine the ordinary representation theory of the exponent-$p$, nilpotency-class-two group $G_\Gamma(\mathbb F_p)$. For each $i\geq1$, how does the number of its irreducible characters of degree $p^i$ depend on $p$?

来源字段：https://doi.org/10.4171/owr/2021/38

02821 | OWR-11583-010

### Character Degrees, Group Algebras, and Nilpotent Blocks

O / open | Oberwolfach 报告问题汇编

Determine whether the set of irreducible character degrees determines solvability of a finite group, whether its complex group algebra determines the existence of a normal Sylow $p$-subgroup, and whether nilpotent blocks are exactly those whose height-zero characters all have the same degree.

来源字段：https://doi.org/10.4171/owr/2012/16

02822 | OWR-16776-012

### Character Determination by Jantzen's Sum Formula

P / partially_solved | Oberwolfach 报告问题汇编

How far does Jantzen's sum formula determine the characters of simple highest-weight modules for reductive groups?

来源字段：https://doi.org/10.4171/owr/2019/14

02823 | OWR-17472-006

### Character Fields of Odd-Degree Representations

S / solved | Oberwolfach 报告问题汇编

Which fields occur as $\mathbb Q(\chi)$ for odd-degree irreducible characters $\chi\in\operatorname{Irr}_{2'}(G)$ of finite groups $G$? Equivalently, determine $$ F_2=\{\mathbb Q(\chi)/\mathbb Q\mid G\text{ finite},\ \chi\in\operatorname{Irr}_{2'}(G)\} $$ .

来源字段：https://doi.org/10.4171/owr/2020/9

02824 | OWR-16776-007

### Character-Counting Chain Cancellation in Blocks

P / partially_solved | Oberwolfach 报告问题汇编

For a positive-defect block idempotent $b$ of $\mathcal{O}G$ and every admissible parameter tuple $(d,r,L,h)$, does the alternating sum over conjugacy classes of chains of nontrivial $p$-subgroups, $\sum_{\sigma\in[\mathcal{P}(G)/G]}(-1)^{|\sigma|}|\operatorname{Irr}_K(G_\sigma,b_\sigma,d,r,L,h)|$, equal zero? Here $\operatorname{Irr}_K$ denotes the irreducible $K$-characters in the corresponding Brauer-correspondent blocks with the specified defect, residue, character field, and Hasse invariant.

来源字段：https://doi.org/10.4171/owr/2019/14

02825 | OWR-4216-006

### Character-Degree Criteria for Nilpotent Blocks

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a finite group and $B$ a $p$-block of $G$. Is $B$ nilpotent if and only if all height-zero character degrees in $B$ are equal?

来源字段：https://doi.org/10.4171/owr/2010/15

02826 | OWR-13675-014

### Character-Degree Rigidity of Nonabelian Simple Groups

P / partially_solved | Oberwolfach 报告问题汇编

If $H$ is a nonabelian simple group and $\operatorname{cd}(G)=\operatorname{cd}(H)$, then $G$ is isomorphic to $H\times A$ for some abelian group $A$.

来源字段：https://doi.org/10.4171/owr/2015/18

02827 | AIM-GEOMETRIC_GROUP_THEORY-0138

### Characteristic and generator defects in a residual p-tower

P / partially_solved | AIM 研讨会问题汇编

Let $\Gamma$ be a finitely presented residually $p$-group. Let $\Gamma_n$ be a normal $p$-chain with $\bigcap \Gamma_n=1$, then $$\lim \frac{b_{\mathbb Q}(\Gamma_n)}{|\Gamma:\Gamma_n|}\leq \lim \frac{b_{F_p}}{|\Gamma:\Gamma_n|}\leq \lim \frac{rk(\Gamma_n)}{|\Gamma:\Gamma_n|}.$$ Can these inequalities be strict?

The notation is repaired as follows, with every repair made explicit:

1. $b_{F_p}$ in the middle numerator is $b_1(\Gamma_n;\mathbb F_p)$; the argument $\Gamma_n$ was visibly dropped. 2. $b_{\mathbb Q}(\Gamma_n)$ is $b_1(\Gamma_n;\mathbb Q)$. 3. $rk(\Gamma_n)$ is the minimal number $d(\Gamma_n)$ of abstract generators. 4. A normal $p$-chain is a descending chain $\Gamma=\Gamma_0\geq \Gamma_1\geq\cdots$ of finite-index subgroups normal in $\Gamma$, with every $[\Gamma:\Gamma_n]$ a power of $p$ (equivalently, $\Gamma/\Gamma_n$ a finite $p$-group).

This reading is confirmed by the immediately preceding AIM problem 11.4, which asks about the mod-$p$ first Betti gradient and the generator gradient, and by Ershov--Lück's definitions arising from the same workshop.

来源字段：http://aimpl.org/l2invariantsgroups/1/

02828 | OWR-4428-002

### Characteristic-p Approximation of Normalized Betti Numbers

O / open | Oberwolfach 报告问题汇编

Let $k$ be a field of positive characteristic, let $\Gamma$ be a residually finite group with residual chain $(\Gamma_i)$, and let $X$ be a finite free $\Gamma$-CW complex. For every $n$, does $$\lim_{i\to\infty}\frac{\dim_k H_n(\Gamma_i\backslash X;k)}{[\Gamma:\Gamma_i]}$$ exist for arbitrary $\Gamma$, is it independent of the residual chain, and what invariant equals this limit?

来源字段：https://doi.org/10.4171/owr/2010/40

02829 | OWR-17135-013

### Clasp Formula for Local Intersection Numbers

O / open | Oberwolfach 报告问题汇编

Let $V=\mathbb C^n$, let $\lambda$ be dominant, and let $\mu$ be a weight of $\Lambda^kV$ such that $\lambda+\mu$ is dominant. If $\kappa_{\lambda,\mu}$ is the resulting one-dimensional local intersection form, is $$\kappa_{\lambda,\mu}=\prod_\alpha\frac{\langle\lambda+\rho,\alpha\rangle}{\langle\lambda+\rho+\mu,\alpha\rangle},$$ where the product is over positive roots $\alpha$ for which the denominator is exactly one less than the numerator?

来源字段：https://doi.org/10.4171/owr/2019/39

02830 | OWR-14299292-004

### Class-Breadth Inequalities for Odd-Prime Groups

O / open | Oberwolfach 报告问题汇编

For a finite $p$-group $G$, write $p^{\operatorname{br}(g)}=|g^G|$ and $\operatorname{br}(G)=\max_{g\in G}\operatorname{br}(g)$. If $p$ is odd, must the nilpotency class satisfy $\operatorname{cl}(G)\leq\operatorname{br}(G)+1$?

来源字段：https://doi.org/10.4171/owr/2025/27

02831 | OWR-9790362-011

### Classification of Coarsifications of the Integers

O / open | Oberwolfach 报告问题汇编

Classify the coarsifications of $\mathbb{Z}$; in particular, determine when the word metric $d_S$ induced by an infinite generating set $S$ is bounded and whether two such sets induce the same coarse structure.

来源字段：https://doi.org/10.4171/owr/2022/11

02832 | OWR-1593-010

### Classification of Extremely Primitive Affine Groups

S / solved | Oberwolfach 报告问题汇编

Classify the insoluble, simply primitive, extremely primitive affine permutation groups. Are all such groups among the known groups $\mathbb Z_2^d\rtimes H$ whose socle is sporadic, alternating, or of Lie type, with no further exceptions?

来源字段：https://doi.org/10.4171/owr/2007/37

02833 | OWR-17295-001

### Classification of Rank-One Groups

O / open | Oberwolfach 报告问题汇编

Is the known list of rank-one groups, consisting of examples of algebraic origin such as groups related to algebraic or classical groups and small-characteristic variants, complete?

来源字段：https://doi.org/10.4171/owr/2019/52

02834 | OWR-1453-008

### Classifying and Ubiquitous Infinite Group Actions

O / open | Oberwolfach 报告问题汇编

For an infinite finitely presented group $G$, call $G$ classifying if free $G$-actions with different numbers of regular orbits cannot be transformed into one another, and call $G$ ubiquitous if every semifree $G$-action can be transformed into a free $G$-action. Does there exist a finitely presented, or at least finitely generated, group that is not classifying? Is every finitely presented amenable group ubiquitous?

来源字段：https://doi.org/10.4171/owr/2007/02

02835 | OWR-14298365-012

### Classifying Thin Two-Generator Matrix Groups

O / open | Oberwolfach 报告问题汇编

Classify pairs $A,B\in\operatorname{GL}_n(\mathbb{Z})$, for $n\geq3$, such that $\langle A,B\rangle$ is thin in its Zariski closure.

来源字段：https://doi.org/10.4171/owr/2024/17

02836 | AIM-GEOMETRIC_GROUP_THEORY-0090

### Closed Cantor strata in the perfect kernel of Sub(F)

P / partially_solved | AIM 研讨会问题汇编

The derivative of a subset \(S\) of a topological space \(X\), denoted \(S'\), is the set of limit points of \(S\). By transfinite induction, one defines the \(\alpha\)-derivative of \(S\) for every ordinal \(\alpha\). The Cantor-Bendixon rank of \(S\) is the smallest ordinal \(\beta\) such that the \(\beta\)-derivative of \(S\) and the \((\beta+1)\)-derivative of \(S\) coincide. What is the Cantor-Bendixon rank of the space of Subgroups of \(F\)?

来源字段：http://aimpl.org/amenablediscrete/1/

02837 | OWR-1784-006

### Closed Contraction Groups of Totally Disconnected Groups

O / open | Oberwolfach 报告问题汇编

For an automorphism $\alpha$ of a totally disconnected locally compact group $G$, define $U_\alpha=\{x\in G:\alpha^n(x)\to1\text{ as }n\to\infty\}$. Under what conditions is $U_\alpha$ closed?

来源字段：https://doi.org/10.4171/owr/2008/03

02838 | OWR-16167-011

### Closed Spraiges in the Braided Thompson Group

O / open | Oberwolfach 报告问题汇编

A closed spraige is a closed split-braid-merge band diagram representing conjugacy data in the braided Thompson group $V_{\mathrm{br}}$. Beyond its use in studying conjugacy, what structural or dynamical information about $V_{\mathrm{br}}$ can be recovered from closed spraiges?

来源字段：https://doi.org/10.4171/owr/2018/26

02839 | OWR-9790362-010

### Coarse Automorphism Groups of Free Groups

O / open | Oberwolfach 报告问题汇编

For $n\geq 2$, what is the structure of the coarse automorphism group $\operatorname{Aut}_{\mathrm{Crs}}(F_n,d_S)$ and its subgroups?

来源字段：https://doi.org/10.4171/owr/2022/11

02840 | OWR-4333-015

### Coarse-Manifold Fillings in S-Arithmetic Lattices

O / open | Oberwolfach 报告问题汇编

Let $K$ be a global field, $S$ a finite set of inequivalent valuations, $G$ a noncommutative absolutely almost simple $K$-isotropic group, and $G(\mathcal O_S)\subset G_S$ its arithmetic lattice. Given a scale $r_1$, does there exist a linear polynomial $f$ and a scale $r_2$ such that every coarse $n$-manifold $\Sigma\subset G_S$ of scale $r_1$ with $\partial\Sigma\subset G(\mathcal O_S)$ and $n<k(G,S)$ can be replaced by a coarse $n$-manifold $\Sigma'\subset G(\mathcal O_S)$ of scale $r_2$ and the same topological type, with $\partial\Sigma'=\partial\Sigma$ and $\operatorname{vol}(\Sigma')\leq f(\operatorname{vol}(\Sigma))$?

来源字段：https://doi.org/10.4171/owr/2010/20

02841 | OWR-9790354-003

### Cocompact Amenable Subgroups of Type-I Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a second-countable locally compact group. If $G$ is type I, must $G$ have a cocompact amenable subgroup?

来源字段：https://doi.org/10.4171/owr/2022/3

02842 | AIM-GEOMETRIC_GROUP_THEORY-0054

### Cocompact and one-dimensional obstructions for hyperbolic triangle Artin groups

P / partially_solved | AIM 研讨会问题汇编

Prove that hyperbolic triangle Artin groups do not act properly on a \(\mathrm{CAT}(0)\) cube complex.

来源字段：http://aimpl.org/geomartingp/4/

02843 | OWR-15177-018

### Coherence and Free-Product Subgroups of $\operatorname{SL}_3(\mathbb Z)$

O / open | Oberwolfach 报告问题汇编

Is $\operatorname{SL}_3(\mathbb Z)$ coherent, or does it contain a subgroup isomorphic to $(\mathbb Z\times\mathbb Z)*\mathbb Z$?

来源字段：https://doi.org/10.4171/owr/2016/56

02844 | OWR-4333-009

### Cohomological Dimensions of Elementary Amenable Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be an elementary amenable group and $k$ a commutative ring. Is $\operatorname{cd}_k(G)<\infty$ equivalent to finite Hirsch length, $|G|<\aleph_\omega$, and invertibility in $k$ of every finite-subgroup order? For an arbitrary group $G$, is $\operatorname{hd}_k(G)\leq1$ equivalent to $G$ being a filtered colimit of groups of $k$-cohomological dimension at most one? Finally, for elementary amenable $G$, is $\operatorname{cd}_k(G)=2$ equivalent both to an action on a contractible $2$-complex with finite stabilizers whose orders are invertible in $k$, and to one of the following: $G$ has a finite-index subgroup $\langle x,y\mid yxy^{-1}=x^n\rangle$ for some nonzero integer $n$; $G$ has a countably infinite locally finite normal subgroup $T$ with $G/T$ infinite cyclic or infinite dihedral; or $G$ is locally finite of cardinality $\aleph_1$?

来源字段：https://doi.org/10.4171/owr/2010/20

02845 | OWR-11583-008

### Cohomological Disjointness for Parabolic Deligne-Lusztig Varieties

P / partially_solved | Oberwolfach 报告问题汇编

For a parabolic Deligne-Lusztig variety $X_P$, prove that the cohomology modules $H_c^i(X_P,\mathbb{Q}_\ell)$ are pairwise disjoint and that $\operatorname{End}_{G^F}\!\left(\bigoplus_i H_c^i(X_P,\mathbb{Q}_\ell)\right)\cong\mathbb{Q}_\ell W_S$.

来源字段：https://doi.org/10.4171/owr/2012/16

02846 | OWR-1973-005

### Cohomology Bounds for Irreducible Finite-Group Modules

O / open | Oberwolfach 报告问题汇编

Let $G$ be finite, $k$ a field, and $M$ an absolutely irreducible $kG$-module. For every $j>0$, is there $C_j$ such that $\dim H^j(G,M)\leq C_j(\dim M)^{j-1}$?

来源字段：https://doi.org/10.4171/owr/2008/28

02847 | OWR-3477-013

### Cohomology Classes of Undistorted Central Extensions

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate: Cohomology Classes of Undistorted Central Extensions [OWR-3477-014]

If a central $\mathbb Z$-extension $0\to\mathbb Z\to E\to G\to1$ is undistorted, what restrictions does this impose on the cohomology class of its defining two-cocycle?

来源字段：https://doi.org/10.4171/owr/2009/27

02848 | OWR-5149-009

### Cohomology Vanishing for Principal-Block Simple Modules

O / open | Oberwolfach 报告问题汇编

For a finite group $G$, its principal-block simple module $S$, and $i>0$, can $H^i(G,S)=0$ for every $i$? Can this occur, or be verified, for symmetric groups?

来源字段：https://doi.org/10.4171/owr/2011/21

02849 | OWR-16415-013

### Collected Problems in Higher Teichmüller and Representation Theory

O / open | Oberwolfach 报告问题汇编

Resolve the following problems in higher Teichmüller and representation theory.

1. Define a cluster superalgebra incorporating spin structures and a Laurent phenomenon, and determine whether spectral networks have a super analogue in higher super-Teichmüller theory.

2. Is the direct sum of a quasi-Fuchsian representation and a geometrically infinite representation stable under small deformations in $\operatorname{SO}(6,2)$, $\operatorname{SO}(4,4)$, or $\operatorname{SL}(8,\mathbb R)$? More generally, construct stable discrete faithful representations that are not Anosov.

3. For a finite graph $\Gamma$, determine the least $n$ for which the right-angled Artin group $A(\Gamma)$ embeds in $\operatorname{SL}_n(\mathbb R)$, including the unipotent, axial, and pentagon cases.

来源字段：https://doi.org/10.4171/owr/2018/40

02850 | OWR-17135-015

### Combinatorial Explanation of the Clasp Conjecture

O / open | Oberwolfach 报告问题汇编

Give a combinatorial explanation of the product formula for the one-dimensional local intersection forms in the clasp conjecture for tensor products $L_\lambda\otimes\Lambda^k\mathbb C^n$.

来源字段：https://doi.org/10.4171/owr/2019/39

02851 | OWR-16776-003

### Combinatorial Formula for Kronecker Coefficients

O / open | Oberwolfach 报告问题汇编

Give an efficient combinatorial description of the Kronecker coefficients $g(\lambda,\mu,\nu)$ analogous to the Littlewood-Richardson rule.

来源字段：https://doi.org/10.4171/owr/2019/14

02852 | OWR-14299292-003

### Common-Neighbor Conjecture for Generalized Saxl Graphs

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a finite primitive permutation group on $\Omega$ with base size $b(G)\geq2$. Form the generalized Saxl graph $\Sigma(G)$ on $\Omega$ by joining two vertices when they lie in a base of size $b(G)$. Must every two vertices of $\Sigma(G)$ have a common neighbor?

来源字段：https://doi.org/10.4171/owr/2025/27

02853 | OWR-1319-027

### Commutator Diameters in Hamiltonian Groups

O / open | Oberwolfach 报告问题汇编

For a closed symplectic manifold $M$, is $A(M)=\sup_{f,g\in \operatorname{Ham}(M)}\rho_H(\mathrm{id},[f,g])$ finite or infinite?

来源字段：https://doi.org/10.4171/owr/2006/33

02854 | OWR-8415341-011

### Compact Graph Representations of Permutation Groups

O / open | Oberwolfach 报告问题汇编

For permutation groups represented by graphs, determine which classes admit representations with $|V|+|E|$ of order $|\Omega|$, $|\Omega|\log|\Omega|$, or a low-order polynomial in $|\Omega|$; develop construction methods and fast algorithms; and determine whether low-index supergroups can be represented when the group itself cannot be compactly represented.

来源字段：https://doi.org/10.4171/owr/2021/38

02855 | OWR-12975-011

### Completely Positive Separation for the Integer Lattice

O / open | Oberwolfach 报告问题汇编

Can the finite-dimensional completely-positive-functional and separation argument used for free groups be extended to $\mathbb Z^2$?

来源字段：https://doi.org/10.4171/owr/2014/17

02856 | OWR-14743-003

### Completion Isomorphisms in Real Group L-Theory

O / open | Oberwolfach 报告问题汇编

For every countable discrete group $G$, is the completion map $L_*(\mathbb RG)[1/2]\to L_*(C_r^*(G;\mathbb R))[1/2]$ an isomorphism?

来源字段：https://doi.org/10.4171/owr/2016/35

02857 | OWR-1319-015

### Complex-Hyperbolic Lattices in Hamiltonian Diffeomorphism Groups

O / open | Oberwolfach 报告问题汇编

If $G$ is a lattice in $PU(n,1)$, with $n\geq2$, does $G$ admit a monomorphism into $\operatorname{Ham}(M,\omega)$ for some symplectic manifold, possibly with interesting dynamical properties?

来源字段：https://doi.org/10.4171/owr/2006/33

02858 | OWR-3477-016

### Compression, Amenability, and Uniform Embeddability of Groups

P / partially_solved | Oberwolfach 报告问题汇编

What is the Hilbert-space compression of Grigorchuk's group of intermediate growth? Can an amenable group have Hilbert-space compression less than $1/2$? Is every uniformly embeddable group weakly amenable? Does a finitely generated group exist with no uniform embedding into any uniformly convex Banach space?

来源字段：https://doi.org/10.4171/owr/2009/27

02859 | OWR-14298004-008

### Condensed-Set Equivalence of CH and Regular Model Structures

O / open | Oberwolfach 报告问题汇编

After replacing the ambient category by condensed sets, do the weak equivalences in the compactly generated Hausdorff and regular model structures coincide?

来源字段：https://doi.org/10.4171/owr/2023/55

02860 | OWR-4428-013

### Congruence Kernels of Arithmetic Lattices in $\operatorname{SL}_2(\mathbb C)$

O / open | Oberwolfach 报告问题汇编

Is the congruence kernel of an arithmetic lattice in $\operatorname{SL}_2(\mathbb C)$ isomorphic to the free profinite group $F_\omega$? More generally, what is the structure of this congruence kernel?

来源字段：https://doi.org/10.4171/owr/2010/40

02861 | OWR-8415341-003

### Conjectural Schur-Multiplier Exponents

O / open | Oberwolfach 报告问题汇编

Is the exponent of the Schur multiplier of $H(8,12)$ equal to $32$, and is the exponent of the Schur multiplier of $H(7,13)$ equal to $49$?

来源字段：https://doi.org/10.4171/owr/2021/38

02862 | OWR-1973-007

### Conjugacy Classes and Characters of Finite p-Groups

O / open | Oberwolfach 报告问题汇编

For a finite $p$-group $P$, determine general methods to parametrize its conjugacy classes, compute $h_P$, describe $\operatorname{Irr}(P)$, and compute the number of irreducible characters of degree $p^k$ for each $k$.

来源字段：https://doi.org/10.4171/owr/2008/28

02863 | OWR-11127-006

### Conjugacy Classes Distinguishing Finite Groups from Subgroups

P / partially_solved | Oberwolfach 报告问题汇编

How many conjugacy classes of a finite group $G$ are needed to distinguish $G$ from each of its proper subgroups?

来源字段：https://doi.org/10.4171/owr/2011/35

02864 | OWR-16167-010

### Conjugacy in Braided Thompson Groups

O / open | Oberwolfach 报告问题汇编

Is the conjugacy problem decidable for the braided Thompson groups $F_{\mathrm{br}}$ and $T_{\mathrm{br}}$?

来源字段：https://doi.org/10.4171/owr/2018/26

02865 | OWR-1319-023

### Conjugacy-Class Growth in Locally Symmetric Lattices

O / open | Oberwolfach 报告问题汇编

For a symmetric space $X$ and a uniform or nonuniform lattice $\Gamma$, what is the growth rate of the number $P(T)$ of conjugacy classes in $\Gamma$ of length less than $T$, especially for $SL(n,\mathbb Z)<SL(n,\mathbb R)$?

来源字段：https://doi.org/10.4171/owr/2006/33

02866 | OWR-14298365-006

### Conjugacy-Class Intersections with Generating Sets

O / open | Oberwolfach 报告问题汇编

For a rank-$r$ linear algebraic group $G$ over a finite field, a regular semisimple $g\in G(F)$, and a generating set $A\subset G(F)$, prove $|A\cap\operatorname{Cl}(g)|\leq C|A^k|^{1-r/\dim G}$ with $C$ and $k$ as small as possible, and determine whether character theory can help.

来源字段：https://doi.org/10.4171/owr/2024/17

02867 | OWR-14299097-003

### Constant-Dimension Framings of Simple Locally Finite Localities

O / open | Oberwolfach 报告问题汇编

Let $\mathcal{L}$ be a simple locally finite locality at a prime $p$. Prove that $\mathcal{L}$ admits a framing by injective homomorphisms of finite localities $\mathcal{L}_n\to\mathcal{L}_{n+1}$ such that every $\mathcal{L}_n$ is simple and $\dim_p(\mathcal{L}_n)$ is independent of $n$.

来源字段：https://doi.org/10.4171/owr/2025/16

02868 | OWR-14299292-010

### Constructive Product Criteria for Alternating-Group Conjugacy Classes

O / open | Oberwolfach 报告问题汇编

Given three conjugacy classes $C_1,C_2,C_3$ in $A_n$, decide constructively whether $C_1C_2\supseteq C_3$. Prove effective large-class criteria, such as the proposed bounds in terms of $|C_i|=|A_n|^{\eta_i}$, or give explicit counterexample families. Also determine whether, for every normal subset $S\subseteq A_n$ and every conjugacy class $C$, the product $SC$ contains at least as many conjugacy classes as $S$.

来源字段：https://doi.org/10.4171/owr/2025/27

02869 | OWR-1316-001

### Constructive Recognition of Matrix-Group Representations

O / open | Oberwolfach 报告问题汇编

In constructive recognition of a matrix group $G$, determine whether $G$ is classical in its natural representation, another natural-characteristic matrix representation, or a black-box representation, and determine whether its natural characteristic is odd or even.

来源字段：https://doi.org/10.4171/owr/2006/30

02870 | OWR-17298-007

### Cost and First $L^2$ Betti Numbers

O / open | Oberwolfach 报告问题汇编

For every finitely generated group $\Gamma$, is $\operatorname{cost}(\Gamma)=1+\beta_1^{(2)}(\Gamma)$, where $\beta_1^{(2)}$ is the first $L^2$-Betti number?

来源字段：https://doi.org/10.4171/owr/2019/55

02871 | OWR-13488-010

### Countable Groups of Prescribed Transitivity Degree

O / open | Oberwolfach 报告问题汇编

Does there exist an infinite countable group of transitivity degree $k$ for every $k\in\mathbb N$? More generally, is there a reasonable classification of highly transitive groups or of groups of high transitivity degree?

来源字段：https://doi.org/10.4171/owr/2015/3

02872 | AIM-GEOMETRIC_GROUP_THEORY-0137

### Counterexamples to rank versus mod-p homology gradient equality

S / solved | AIM 研讨会问题汇编

Is \[ \lim \frac{b_{F_p}(\Gamma_n)}{|\Gamma:\Gamma_n|} = \lim \frac{\operatorname{rk}(\Gamma_n)}{|\Gamma:\Gamma_n|} \] for any (not necessarily finitely generated) group?

来源字段：http://aimpl.org/l2invariantsgroups/1/

02873 | OWR-2487-010

### Cowling-Haagerup Constant and the Haagerup Property

O / open | Oberwolfach 报告问题汇编

Does every countable group $G$ with Cowling-Haagerup constant $\Lambda(G)=1$ have the Haagerup property; equivalently, is every such group a-T-menable?

来源字段：https://doi.org/10.4171/owr/2008/49

02874 | OWR-1703863-003

### Coxeter Quotients of Automorphism Groups of Coxeter Groups

O / open | Oberwolfach 报告问题汇编

For every infinite Coxeter group $W_\Gamma$, does $\operatorname{Aut}(W_\Gamma)$ virtually map onto an infinite Coxeter group?

来源字段：https://doi.org/10.4171/owr/2020/16

02875 | OWR-14299911-025

### Cubulability and Residual Finiteness of Hyperbolic Groups

O / open | Oberwolfach 报告问题汇编

Let $G$ be a hyperbolic group acting properly and cocompactly on a finite-rank median space. Must $G$ be cocompactly cubulable? Must $G$ be residually finite, or even linear?

来源字段：https://doi.org/10.4171/owr/2026/8

02876 | OWR-16415-007

### Cubulation of Hyperbolic Surface-by-Free Groups

O / open | Oberwolfach 报告问题汇编

If $1\to\pi_1(S)\to G\to Q\to1$ is exact, $S$ is a closed surface of genus greater than one, $G$ is hyperbolic, and $Q$ is finitely generated free, is $G$ cubulable, meaning that it acts freely, properly, and cocompactly on a $\operatorname{CAT}(0)$ cube complex?

来源字段：https://doi.org/10.4171/owr/2018/40

02877 | AIM-GEOMETRIC_GROUP_THEORY-0064

### Curvature does not replace stabilizer membership

P / partially_solved | AIM 研讨会问题汇编

Let $A$ be an Artin group with Deligne complex $D$, Salvetti complex $S$, “pure” Salvetti complex $P$, and complex of irreducible parabolics $X$.

If one of the complexes has non-positive curvature and all cell stabilizers have solvable word problem, does this mean the whole group has solvable word problem? Can this be generalized to arbitrary cocompact actions on NPC complexes? $\mathrm{CAT}(0)$ cube complexes?

In the general case, most likely need additional assumptions, namely the membership problem for cell stabilizers into larger cell stabilizers.

来源字段：http://aimpl.org/geomartingp/5/

02878 | AIM-GEOMETRIC_GROUP_THEORY-0073

### Curvature, stabilizer recognition, and parabolic intersections

P / partially_solved | AIM 研讨会问题汇编

Are intersections of parabolic again parabolic when the Deligne complex has some notion of non-positive curvature? (i.e., systolic, injective, CUB, $\mathrm{CAT}(0)$)

来源字段：http://aimpl.org/geomartingp/7/

02879 | OWR-12697693-010

### Cycle Set Deformations into Cycloids

O / open | Oberwolfach 报告问题汇编

Find further deformations of cycle sets into cycloids, and determine the commutative semigroup operations on the structure monoid $M(X)$ induced by these deformations.

来源字段：https://doi.org/10.4171/owr/2023/9

02880 | OWR-14299904-010

### Cyclic Labelings and Full-Twist Braid Words

O / open | Oberwolfach 报告问题汇编

Do cyclic labelings correspond exactly to reduced words for the full twist in the squares of the Birman-Ko-Lee generators of the braid group?

来源字段：https://doi.org/10.4171/owr/2026/2

02881 | OWR-1452-033

### Cyclicity of Surface Quotients by Connected Groups

O / open | Oberwolfach 报告问题汇编

Suppose a connected reductive algebraic group $G$ acts algebraically on $\mathbb{C}^n$ and its two-dimensional categorical quotient is isomorphic to $\mathbb{C}^2/\Gamma$ for a finite group $\Gamma$ acting algebraically on $\mathbb{C}^2$. If $G$ is connected, then $\Gamma$ should be cyclic.

来源字段：https://doi.org/10.4171/owr/2007/01

02882 | OWR-14745-007

### Decidability of Subgroup Membership in SL3 of the Integers

O / open | Oberwolfach 报告问题汇编

Is the subgroup membership problem for subgroups of $\operatorname{SL}(3,\mathbb Z)$ decidable?

来源字段：https://doi.org/10.4171/owr/2016/37

02883 | AIM-GEOMETRIC_GROUP_THEORY-0146

### Deep fall: automatic regimes and a closed-range analytic counterexample

P / partially_solved | AIM 研讨会问题汇编

It asks for examples and nonexamples of the deep-fall property, its relation to the integral Atiyah conjecture, and analogues for nonfree ordered actions and matrices over \(\mathbb C\Gamma\).

The record needs four small but mathematically relevant corrections.

1. The paper cited as “Sumultiplicativity” and later as “Sumbmultiplicativity” is Igor Mineyev's *Submultiplicativity and the Hanna Neumann Conjecture* [Min12a]. 2. A free action and a left order on \(\Gamma\) do not by themselves produce a unique order on the cells. Mineyev chooses a fundamental domain \(\overline\Sigma_i\), orders it, identifies \(\Sigma_i=\Gamma\times\overline\Sigma_i\), and uses \[ (g,\bar\sigma)<(h,\bar\tau) \quad\Longleftrightarrow\quad g<h\ \text{ or }\ (g=h\text{ and }\bar\sigma<\bar\tau). \] 3. The sets \(\mathbb E_i\) and \(\mathbb I_i\) consist of \(i\)-**cells**, not necessarily edges. “Edges” at this point of the published paper is itself a harmless dimension-one slip. 4. Mineyev introduces deep fall for a \(\Gamma\)-ordered complex of type F, meaning that \(\Gamma\backslash\widehat Y\) is finite. More generally, the formula makes analytic sense when the cellular boundary extends to a bounded map on \(\ell^2\); uniform local finiteness is a sufficient hypothesis. The AIM transcription omits this scope condition.

Let \[ T=\partial_i:\ell^2(\Sigma_i)\longrightarrow \ell^2(\Sigma_{i-1}) \] be bounded, and put \(v_\sigma=T\delta_\sigma\). For \(S_{<\sigma}=\{\tau:\tau<\sigma\}\), call \(\sigma\) **order-essential** when \[ v_\sigma\in M_{<\sigma}:=\overline{\operatorname{span}} \{v_\tau:\tau<\sigma\}; \] it is **order-inessential** otherwise. Write \(\mathbb E\) and \(\mathbb I\) for the two sets. Deep fall means \[ v_\sigma\in \overline{\operatorname{span}} \{v_\tau:\tau<\sigma,\ \tau\in\mathbb I\} \qquad(\sigma\in\mathbb E). \tag{DF} \] The closure is taken in the codomain. The apparently counterintuitive word “essential” is faithful to Mineyev: an essential cell is the one whose boundary is already in the closed span of earlier boundaries.

来源字段：http://aimpl.org/l2invariantsgroups/4/

02884 | OWR-17140-012

### Depth-Controlled Relative Local Character Expansions

P / partially_solved | Oberwolfach 报告问题汇编

Let a reductive group $H$ over $F$ act on $V$, let $\mathcal N_V$ be its nilpotent cone, and let $V_{r+}$ be the Moy-Prasad neighborhood. For an $H$-distinguished representation $\pi$ of depth $r$, is the relative local character expansion valid on $V_{r+}$? Equivalently, does $$\operatorname{res}_{D(V)_{r+}}\widetilde J(V)_{r+}=\operatorname{res}_{D(V)_{r+}}J(\mathcal N_V)$$ hold for every real $r$?

来源字段：https://doi.org/10.4171/owr/2019/44

02885 | OWR-17140-008

### Derived Equivalence for Spherical Hamiltonian Duality

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathfrak o=k[[t]]$, where $k$ is finite or $k=\mathbb C$. Is the derived category of $G(\mathfrak o)$-equivariant constructible sheaves on $X(F)$ equivalent to the differential graded category of $\check G$-equivariant coherent sheaves on $$\check M\simeq\check V_X\times_{\check G_X}\check G$$?

来源字段：https://doi.org/10.4171/owr/2019/44

02886 | OWR-1738-012

### Derived Subgroups, Automorphisms, and Solvable Group-Algebra Units

P / partially_solved | Oberwolfach 报告问题汇编

For a free group $G$, is $G\cap(1+\delta_n(\mathbb{Z}G))$ equal to the $n$-th derived subgroup $G^{(n)}$, and what is the analogous intersection over a field of positive characteristic or for an arbitrary group? For a finite $p$-group $G$ over the field $k$ with $p$ elements, can every automorphism of $kG$ be written as an automorphism induced by $G$ followed by a unipotent automorphism? For a Lie-solvable group algebra $KG$ over a field of odd characteristic, does there exist $f:\mathbb{N}\to\mathbb{N}$ such that $\operatorname{dl}(U(KG))\leq f(\operatorname{dl}^L(KG))$?

来源字段：https://doi.org/10.4171/owr/2007/55

02887 | OWR-17295-004

### Descent of Rational Points from Laurent Series Fields

O / open | Oberwolfach 报告问题汇编

Let $k$ be a field, let $G$ be an affine algebraic $k$-group, and let $X$ be a homogeneous space of $G$. If $X(k((t)))\neq\varnothing$, must $X(k)\neq\varnothing$?

来源字段：https://doi.org/10.4171/owr/2019/52

02888 | OWR-17472-004

### Detecting Abelian Sylow Subgroups from Character Tables

O / open | Oberwolfach 报告问题汇编

For a finite group $G$ and a Sylow $p$-subgroup $P$, does the $p$-element submatrix $X_p(G)$ of the character table determine whether $P$ is abelian in all cases, as it does when $p\notin\{3,5\}$?

来源字段：https://doi.org/10.4171/owr/2020/9

02889 | OWR-2053-021

### Detection and Uniqueness of Lie Quasi States

P / partially_solved | Oberwolfach 报告问题汇编

Given a Lie group $G$, its Lie algebra $\mathfrak{g}$, and an adjoint-invariant cone $K\subset\mathfrak{g}$, characterize all Lie quasi-states $\zeta:\mathfrak{g}\to\mathbb{R}$ that are linear on commutative subspaces and nonnegative on $K$. Is the Maslov Lie quasi-state for $\operatorname{Sp}(2n)$ unique up to isomorphism? Does $(T^4,\omega_{\mathrm{std}})$ admit a nonlinear Lie quasi-state? How many Lie quasi-states do $(\mathbb{CP}^2,\omega_{\mathrm{std}})$ and $(S^2\times S^2,\omega_{\mathrm{std}}\oplus\omega_{\mathrm{std}})$ admit?

来源字段：https://doi.org/10.4171/owr/2008/42

02890 | AIM-GEOMETRIC_GROUP_THEORY-0120

### Diagonal Liouville stability, periodicity, and finite-fibre obstructions

P / partially_solved | AIM 研讨会问题汇编

**Diagonal/convolution reading.** Is there a transitive $\mu$-Liouville action $G\curvearrowright X$ for which the diagonal action of $G$ on some orbit $\mathcal O\subseteq X\times X$ is not $\mu^{*2}$-Liouville?

来源字段：http://aimpl.org/amenablediscrete/4/

02891 | OWR-13675-013

### Diagrammatic Actions on Principal Tilting Categories

P / partially_solved | Oberwolfach 报告问题汇编

For a semisimple simply connected group $G$ in characteristic $p$ greater than its Coxeter number, can the principal tilting category $\mathrm{Tilt}_0$ be given a right $\mathcal{D}$-module structure with $B_s$ acting as translation, together with an equivalence $\mathrm{AS}_{\mathrm{deg}}\simeq \mathrm{Tilt}_0$ of right $\mathcal{D}$-module categories?

来源字段：https://doi.org/10.4171/owr/2015/18

02892 | OWR-14298365-014

### Diameter and Product Structure in High-Rank Linear Groups

O / open | Oberwolfach 报告问题汇编

Can one prove improved diameter bounds for $\operatorname{SL}_n(\mathbb{F}_q)$, and what size and structure can be guaranteed for high powers of subsets with permutation-rich restrictions?

来源字段：https://doi.org/10.4171/owr/2024/17

02893 | OWR-14298365-005

### Diameter Bounds with Permutation-Rich Generating Sets

O / open | Oberwolfach 报告问题汇编

Let $A\subset\operatorname{SL}_n(F)$ generate the group and contain all permutation matrices. Prove a strong, ideally polylogarithmic, upper bound on $\operatorname{diam}\Gamma(\operatorname{SL}_n(F),A)$.

来源字段：https://doi.org/10.4171/owr/2024/17

02894 | AIM-GEOMETRIC_GROUP_THEORY-0048

### Dihedral embedding spectra in two-dimensional Artin groups

P / partially_solved | AIM 研讨会问题汇编

**Problem.** For fixed \(A\), which spherical and/or dihedral Artin groups embed in \(A\)?

来源字段：http://aimpl.org/geomartingp/3/

02895 | OWR-13675-002

### Direct McKay Reduction for Self-Normalizing Sylow Two-Subgroups

O / open | Oberwolfach 报告问题汇编

For self-normalizing Sylow 2-subgroups, can the McKay conjecture be directly reduced to almost simple groups?

来源字段：https://doi.org/10.4171/owr/2015/18

02896 | AIM-GEOMETRIC_GROUP_THEORY-0151

### Directed axes, power minsets, and the shortcut defect in Lipschitz Outer space

P / partially_solved | AIM 研讨会问题汇编

This is Problem 2.1 in the AIM workshop notes *The geometry of the outer automorphism group of a free group* (25--29 October 2010, edited by Johanna Mangahas). In normalized notation, it asks for analogies between Teichmüller/Weil--Petersson geodesics and geodesics in Culler--Vogtmann Outer space with the Lipschitz metric:

1. Which geodesics or lines have strongly contracting projection maps? 2. Is there a useful “thick part”? 3. How does geodesic behavior relate to boundary theory? 4. For an iwip \(\phi\), describe \[ \operatorname{Min}(\phi)= \left\{X:d(X,\phi X)=\inf_Y d(Y,\phi Y)\right\}. \] Is it CAT(0), contained in a CAT(0) space, an axis bundle, or quasi-isometric to a line? 5. How close are \(\operatorname{Min}(\phi)\) and \(\operatorname{Min}(\phi^{-1})\)?

The discussion asks in addition whether thick geodesics are contracting, whether iwip axes are quantitatively thick, whether taking powers changes the minset, and how train-track points and axis bundles enter.

The all-caps line `THE GEOMETRY ... 3` is a running page header. The final `3. GROUP PROPERTIES` is the heading of the next section and is not part of Problem 2.1.

The metric belongs to Outer space \(CV_n\), not to \(\operatorname{Out}(F_n)\) itself. The group acts on \(CV_n\) by isometries. Crucially, the Lipschitz metric is asymmetric, so every claim below records its direction.

来源字段：https://aimath.org/WWN/outerauto/outerauto.pdf

02897 | OWR-1973-003

### Discrete and Profinite Relation Ranks of Finite Groups

O / open | Oberwolfach 报告问题汇编

For a finite group $G$, let $r(G)$ be the minimum number of relations in a discrete presentation and $\widehat r(G)$ the corresponding minimum for a profinite presentation. Is $r(G)=\widehat r(G)$ for every finite group?

来源字段：https://doi.org/10.4171/owr/2008/28

02898 | OWR-14298004-001

### Discrete First Cohomology of Totally Disconnected Groups

O / open | Oberwolfach 报告问题汇编

Let $G=\operatorname{Aut}_F(E)$, where $F$ has characteristic zero and $E/F$ has finite transcendence degree. Compute the discrete cohomology group $dH^1(G,E^\times)$.

来源字段：https://doi.org/10.4171/owr/2023/55

02899 | OWR-1319-025

### Displacement-Energy Bounds for Bi-Invariant Hamiltonian Metrics

O / open | Oberwolfach 报告问题汇编

For every bi-invariant distance $\rho$ on a Hamiltonian group $G$, does there exist $C>0$ such that the displacement energy $e(\rho,B)$ of every symplectic ball $B$ of capacity $a$ satisfies $e(\rho,B)\geq Ca$?

来源字段：https://doi.org/10.4171/owr/2006/33

02900 | OWR-17298-002

### Dixmier's Amenability Problem for Unitarizable Groups

O / open | Oberwolfach 报告问题汇编

Dixmier's problem asks whether every unitarizable discrete group is amenable.

来源字段：https://doi.org/10.4171/owr/2019/55

02901 | OWR-1062-011

### Dixmier's Unitarizability Problem

O / open | Oberwolfach 报告问题汇编

Does every unitarizable group have to be amenable?

来源字段：https://doi.org/10.4171/owr/2005/41

02902 | AIM-GEOMETRIC_GROUP_THEORY-0038

### Dual Artin isomorphisms for cograph and complete multipartite RAAGs

P / partially_solved | AIM 研讨会问题汇编

**Is each Artin group isomorphic to its dual?**

Recorded literature note: “Known for spherical and Euclidean. Open for RAAGs.”

来源字段：http://aimpl.org/geomartingp/1/

02903 | OWR-14299289-010

### Dual Hamiltonian Geometry of Equivariant Sheaf Categories

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Local Geometric Duality Formulation [OWR-14299289-011]

For a hyperspherical pair with dual Hamiltonian space $M^\vee$, is there an equivalence $\operatorname{Shv}_{G(\mathcal D)}(X(\mathring{\mathcal D}))\simeq\operatorname{QCoh}(M^\vee)^{G^\vee}$ that is linear over the spherical Hecke category?

来源字段：https://doi.org/10.4171/owr/2025/18

02904 | OWR-4413-008

### Dyer-Lashof Finite Generation of Tate Cohomology

O / open | Oberwolfach 报告问题汇编

For a finite group $G$, is its Tate cohomology ring finitely generated as an algebra over the Dyer-Lashof algebra?

来源字段：https://doi.org/10.4171/owr/2010/32

02905 | AIM-GEOMETRIC_GROUP_THEORY-0155

### Effective characteristic refinement and exact Aut-Out transfer for free-group congruence kernels

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.4.** Does every finite index subgroup contain one of these \(\Gamma\)?

Mladen Bestvina asked if there is an effective way to construct these subgroups. Verbal subgroups, or pass to Burnside group, suggested Martin Bridson. Karen Vogtmann commented that you would classify characteristic subgroups of \(\operatorname{Out}(F_n)\).

Bridson offered a baby question in that direction: if \(n\geq 4\), is \(\operatorname{Out}(F_n)\) large? Meaning, does there exist a finite-index subgroup which acts on a tree, thus maps onto a free group?

来源字段：https://aimath.org/WWN/outerauto/outerauto.pdf

02906 | AIM-GEOMETRIC_GROUP_THEORY-0004

### Effective coherence in the periodic-monodromy and finite-height-kernel regimes

P / partially_solved | AIM 研讨会问题汇编

Given a finite set \(S\) of elements of a free-by-cyclic group \(G\), algorithmically find a presentation of the sugroup of \(G\) generated by \(S\).

来源字段：http://aimpl.org/freebycyclic/1/

02907 | OWR-14298004-014

### Embedding $\sigma$-Compact TDLC Groups

O / open | Oberwolfach 报告问题汇编

Does every $\sigma$-compact totally disconnected locally compact group embed into a compactly generated totally disconnected locally compact group?

来源字段：https://doi.org/10.4171/owr/2023/55

02908 | OWR-1703861-004

### Embedding Classification of Piecewise-Linear Interval Groups

O / open | Oberwolfach 报告问题汇编

For finitely generated subgroups of Thompson's group $F$ and of $\mathrm{PL}_o(I)$ ordered by homomorphic embedding, is there a substantial classifiable initial segment and where does the quasiorder become intractable? Which finitely generated subgroups are comparable with every finitely generated subgroup, and does $F$ have this property; must a subgroup of $\mathrm{PL}_o(I)$ not containing $F$ be elementary amenable; and are the finitely generated subgroups of $F$ well quasiordered?

来源字段：https://doi.org/10.4171/owr/2020/14

02909 | OWR-12869-005

### Embedding Finitely Presented Groups into Kähler Groups

O / open | Oberwolfach 报告问题汇编

Does every finitely presented group occur as a subgroup of some Kähler group?

来源字段：https://doi.org/10.4171/owr/2014/09

02910 | OWR-16160-016

### Embedding Independence of Congruence Subgroups

S / solved | Oberwolfach 报告问题汇编

Let $G$ be a finitely generated group embedded in $\operatorname{SL}_N(\mathbb Z)$, and define its congruence subgroups by intersecting $G$ with congruence subgroups of $\operatorname{SL}_N(\mathbb Z)$. Does the resulting family depend on the chosen embedding?

来源字段：https://doi.org/10.4171/owr/2018/19

02911 | OWR-15216-022

### Embedding Interval Groups Without Thompson F into the G-Hierarchy

O / open | Oberwolfach 报告问题汇编

Does every finitely generated subgroup $G\leq\mathrm{PL}_+(I)$ that contains no copy of Thompson's group $F$ embed into $G_\xi$ for some $\xi<\varepsilon_0$? In particular, does every finitely generated elementary amenable subgroup of $F$ embed into some $G_\xi$?

来源字段：https://doi.org/10.4171/owr/2017/11

02912 | OWR-16167-013

### Empty Regions in the Biembeddability Poset

O / open | Oberwolfach 报告问题汇编

In the poset of biembeddability classes of finitely generated subgroups of $\operatorname{PL}_o(I)$, are all three of the following regions empty: elementary-amenable classes incomparable with $F$, non-elementary-amenable classes incomparable with $F$, and non-elementary-amenable subgroups of $F$ into which $F$ does not embed?

来源字段：https://doi.org/10.4171/owr/2018/26

02913 | OWR-11125-001

### Entropy Rigidity for Hyperbolic Group Actions

O / open | Oberwolfach 报告问题汇编

Let $X$ be a proper geodesic hyperbolic space with a geometric group action by $\Gamma$ and boundary a topological $n$-sphere, and let $e(\Gamma)$ be the critical exponent of the action. Is $e(\Gamma)\geq n$, with equality if and only if $X$ is rough-isometric to real hyperbolic space $\mathbb H^{n+1}$?

来源字段：https://doi.org/10.4171/owr/2011/33

02914 | OWR-15433-011

### Epsilon Dichotomy for Generic General Linear Representations

P / partially_solved | Oberwolfach 报告问题汇编

Let $\pi$ be a generic representation of $\operatorname{GL}_6(F)$ with central character $\chi^2$. Is $$m(\pi)=1\iff\epsilon\!\left(\frac12,(\wedge^3\pi)\otimes\chi^{-1}\right)=1,$$ and equivalently $m(\pi)=0$ precisely when this epsilon factor is $-1$?

来源字段：https://doi.org/10.4171/owr/2017/25

02915 | OWR-12697684-034

### Equal Products from Disjoint Subsets of Finite Groups

O / open | Oberwolfach 报告问题汇编

For every $k\geq2$, does there exist $c(k)$ such that every subset $S$ of a finite group $G$ of order $n$, with $|S|\geq c(k)\log n$, contains pairwise disjoint subsets $T_1,\ldots,T_k$ that can each be ordered to give the same product?

来源字段：https://doi.org/10.4171/owr/2023/1

02916 | OWR-15987-013

### Equality of the Andreadakis and Representation-Ring Filtrations

O / open | Oberwolfach 报告问题汇编

Let $A_n(k)$ be the Andreadakis-Johnson filtration of $\operatorname{Aut}(F_n)$, and let $D_n(k)$ be the kernel of its action on $J/J^{k+1}$, where $J$ is the augmentation ideal of the universal $\operatorname{SL}_2(\mathbb C)$-representation ring of $F_n$. Is

$$A_n(k)=D_n(k)$$

for every $k\geq1$?

来源字段：https://doi.org/10.4171/owr/2018/2

02917 | OWR-9790362-005

### Equivalence Classes of Homological Filling Functions

O / open | Oberwolfach 报告问题汇编

Are there uncountably many equivalence classes of homological filling functions for groups; in particular, are the homological filling functions for $\operatorname{GL}(S)$ and $\operatorname{GL}(T)$ equivalent?

来源字段：https://doi.org/10.4171/owr/2022/11

02918 | OWR-14299097-015

### Equivalence of Exotic and Block-Exotic Fusion Systems

P / partially_solved | Oberwolfach 报告问题汇编

Is a fusion system exotic if and only if it is block-exotic?

来源字段：https://doi.org/10.4171/owr/2025/16

02919 | OWR-1319-017

### Equivariant Real-Hyperbolic Embeddings for Deligne-Mostow Lattices

O / open | Oberwolfach 报告问题汇编

For a uniform arithmetic lattice $\Gamma<PU(2,1)$ generated by order-five complex reflections, is there a $\Gamma$-equivariant totally geodesic real embedding $\mathbb H^2_{\mathbb R}\times\pi_1(\Sigma_g)\to\mathbb H^2_{\mathbb C}$ whose image avoids the reflection fixed-point sets?

来源字段：https://doi.org/10.4171/owr/2006/33

02920 | OWR-1703863-002

### Euler Characteristic and Coherence of Random Few-Relator Groups

P / partially_solved | Oberwolfach 报告问题汇编

In the few-relators random-group model $G(n,m,\ell)$, is a random group of Euler characteristic $\chi$ coherent when $\chi\leq0$ and incoherent when $\chi>0$, in each case with probability tending to $1$ as $\ell\to\infty$?

来源字段：https://doi.org/10.4171/owr/2020/16

02921 | OWR-14298004-004

### Euler Characteristics from Formal Dirichlet Series of Merohoric Pairs

O / open | Oberwolfach 报告问题汇编

For which meromorphic pairs $(G,O)$, with $G$ unimodular, rationally of type $\mathrm{FP}_\infty$, and of finite rational cohomological dimension, does $\chi_G=\widehat\zeta_{G,O}(-1)\,\mu_O$ hold?

来源字段：https://doi.org/10.4171/owr/2023/55

02922 | AIM-GEOMETRIC_GROUP_THEORY-0079

### Exact commutator and Newton-polytope reductions for Artin group rings

P / partially_solved | AIM 研讨会问题汇编

Does the group ring of an Artin group have zero divisors?

来源字段：http://aimpl.org/geomartingp/8/

02923 | AIM-GEOMETRIC_GROUP_THEORY-0123

### Exact Liouville stability under aperiodic randomization of convolution time

P / partially_solved | AIM 研讨会问题汇编

Is the Liouville property stable under the choice of symmetric, finitely supported, generating measure?

来源字段：http://aimpl.org/amenablediscrete/4/

02924 | AIM-GEOMETRIC_GROUP_THEORY-0093

### Exact local routes and quantitative non-quotient constraints for sofic models of Thompson's group F

P / partially_solved | AIM 研讨会问题汇编

Is $F$ sofic?

来源字段：http://aimpl.org/amenablediscrete/1/

02925 | AIM-GEOMETRIC_GROUP_THEORY-0033

### Exact product cones and a group-data obstruction

P / partially_solved | AIM 研讨会问题汇编

Develop a fibred face theory for four-manifolds which fibre over the circle with fibers handlebodies or connect sums of $\mathbb S^2\times\mathbb S^1$.

来源字段：http://aimpl.org/freebycyclic/7/

02926 | AIM-GEOMETRIC_GROUP_THEORY-0021

### Exact virtual-RFRS criterion for a non-hyperbolic triangular family

P / partially_solved | AIM 研讨会问题汇编

**RFRS.** A group $G$ is *residually finite rationally solvable (RFRS)* if there exists a non-ascending residual chain $(H_i)_{i\in\mathbb N}$ of finite index normal subgroups $N_i\trianglelefteq G$ with $H_0=G$, such that $\ker\alpha_i\leq H_{i+1}$ for every $i$, where $\alpha_i:H_i\to H_i^{\mathrm{fab}}$ is the free abelianisation map.

Characterise the non-hyperbolic free-by-cyclic groups which are virtually RFRS.

来源字段：http://aimpl.org/freebycyclic/4/

02927 | OWR-1703863-001

### Exotic Group $C^*$-Algebras for Nonamenable Locally Compact Groups

O / open | Oberwolfach 报告问题汇编

Does every nonamenable locally compact group admit an exotic group $C^*$-algebra?

来源字段：https://doi.org/10.4171/owr/2020/16

02928 | OWR-11101917-005

### Explicit Connes-Embedding Counterexamples and Nonhyperlinear Groups

P / partially_solved | Oberwolfach 报告问题汇编

Find a more algebraic proof of $\operatorname{MIP}^*=\operatorname{RE}$, construct an explicit Connes-embedding counterexample including its tracial state, identify analogous consequences, and construct a nonhyperlinear group.

来源字段：https://doi.org/10.4171/owr/2022/36

02929 | AIM-GEOMETRIC_GROUP_THEORY-0047

### Explicit outer automorphisms for a RAAG join family and a free-product obstruction

P / partially_solved | AIM 研讨会问题汇编

For an Artin group \(A\), when is \(\operatorname{Out}(A)\) finite/finitely generated? Find explicit generators.

The record itself contains no qualifications on the Coxeter matrix. Thus it is a programmatic question about all finite-rank Artin groups, not a single yes/no conjecture. No corruption or ambiguity is visible in the source record. Throughout, an Artin group is assumed to have a finite standard generating set.

This report separates three logically different requests:

1. decide when \(\operatorname{Out}(A)\) is finite; 2. decide when it is finitely generated (or finitely presented); 3. give abstract group generators, which may be strictly more numerous than graph symmetries of a chosen Artin presentation.

The strongest new work in this attempt is a complete, elementary outer-automorphism calculation for a two-parameter family of right-angled Artin groups and a free-product obstruction to finiteness.

来源字段：http://aimpl.org/geomartingp/3/

02930 | OWR-15173-007

### Explicit Reduction Theory for Rank-One Unitary Arithmetic Groups

O / open | Oberwolfach 报告问题汇编

Let $L=\mathbb Q(\sqrt{-d})$ with $d>0$ square-free, let

$$G(\mathbb Q)=\{g\in\operatorname{SL}_3(L):g^*Jg=J\}$$

be a $\mathbb Q$-rank-one unitary group, and let $\Gamma\leq G(\mathbb Q)$ be an arithmetic subgroup. Can one carry out explicit reduction theory for $\Gamma$?

来源字段：https://doi.org/10.4171/owr/2016/52

02931 | OWR-11101920-005

### Extending Character-Field Bounds Beyond p-Solvable Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $P$ be a Sylow $p$-subgroup of a finite group $G$, and let $\chi\in\operatorname{Irr}(G)$ have conductor $p^am$ with $p\nmid m$. Is

$$[\mathbb Q_{p^a}:\mathbb Q_p(\chi_P)]\leq\chi(1)_p$$

for arbitrary finite $G$, without a $p$-solvability assumption?

来源字段：https://doi.org/10.4171/owr/2022/39

02932 | OWR-8415341-009

### Extending Generator Maps to Finite Group Homomorphisms

O / open | Oberwolfach 报告问题汇编

Given $G=\langle g_1,\ldots,g_k\rangle\leq\operatorname{SL}(n,\mathbb{Z})$ and a map on the generators sending them to a generating set of a finite group $H$, determine whether the map extends to a homomorphism $G\to H$, including the case where it does not factor through a congruence homomorphism.

来源字段：https://doi.org/10.4171/owr/2021/38

02933 | OWR-1265-021

### Extension Vanishing in Abelian-Defect Blocks

P / partially_solved | Oberwolfach 报告问题汇编

If $D_\lambda,D_\mu$ are simple modules in a symmetric-group block with abelian defect group and $\delta_e(\lambda)=\delta_e(\mu)$, then $\operatorname{Ext}^1(D_\lambda,D_\mu)=0$.

来源字段：https://doi.org/10.4171/owr/2006/15

02934 | OWR-14298365-017

### Extremal Product-Free Sets in Quasirandom Groups

P / partially_solved | Oberwolfach 报告问题汇编

Determine the maximum density and structure of product-free subsets in finite quasirandom groups.

来源字段：https://doi.org/10.4171/owr/2024/17

02935 | OWR-1973-008

### Fake Degrees and Class Numbers of Unitriangular Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $P$ be a finite $p$-group with a generalized Cayley map onto a Lie algebra, let $\mathcal O$ be its coadjoint orbits, and let $\operatorname{Irr}(P)$ be its irreducible characters. When does $\{\chi\in\operatorname{Irr}(P):\chi(1)=p^k\}=\{O\in\mathcal O:|O|=p^{2k}\}$? A related question asks whether there is $t\in\mathbb Z[T]$ such that $h_{U_n(q)}=t(q)$.

来源字段：https://doi.org/10.4171/owr/2008/28

02936 | AIM-GEOMETRIC_GROUP_THEORY-0058

### Family-dependent hyperbolic models and a quantitative RAAG centralizer electrification

P / partially_solved | AIM 研讨会问题汇编

**Problem.** Describe the hyperbolic space on which the acylindrically hyperbolic Artin groups act.

来源字段：http://aimpl.org/geomartingp/4/

02937 | OWR-3477-012

### Farrell-Jones Assembly Maps for Torsion-Free Groups

O / open | Oberwolfach 报告问题汇编

Let $G$ be a torsion-free discrete group and $R$ a regular ring. Are the algebraic $K$- and $L$-theory assembly maps $$H_n(BG;\mathbf K_R)\longrightarrow K_n(RG),\qquad H_n(BG;\mathbf L_R^{\langle-\infty\rangle})\longrightarrow L_n^{\langle-\infty\rangle}(RG)$$ bijective for every $n\in\mathbb Z$?

来源字段：https://doi.org/10.4171/owr/2009/27

02938 | OWR-4333-004

### Farrell-Jones Conjecture for Torsion-Free Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a torsion-free group $G$ and a regular ring $R$, are the algebraic $K$- and $L$-theory assembly maps $H_n(BG;\mathbf K_R)\to K_n(RG)$ and $H_n(BG;\mathbf L_R^{\langle-\infty\rangle})\to L_n^{\langle-\infty\rangle}(RG)$ bijective for every $n\in\mathbb Z$ (the Farrell-Jones conjecture)?

来源字段：https://doi.org/10.4171/owr/2010/20

02939 | OWR-16167-014

### Fast Generators for F-Less Homeomorphism Groups

O / open | Oberwolfach 报告问题汇编

Does every finitely generated subgroup $H<\operatorname{PL}_o(I)$ that contains no copy of Thompson's group $F$ have a finite fast generating set $A$ with the same biembeddability class? Can fast subgroups of $\operatorname{PL}_o(I)$ that contain no copy of $F$ be characterized?

来源字段：https://doi.org/10.4171/owr/2018/26

02940 | AIM-GEOMETRIC_GROUP_THEORY-0024

### Fast monodromy and the cocompact cubulation bottleneck

P / partially_solved | AIM 研讨会问题汇编

Let $F=F(a,b,c)$ be a free group of rank $3$. Define $\phi\in\operatorname{Aut}(F)$ by \[ \phi(a)=a,\qquad \phi(b)=ab,\qquad \phi(c)=bcb. \] Does $G=F\rtimes_\phi\mathbb Z$ act geometrically on a CAT(0) cube complex?

来源字段：http://aimpl.org/freebycyclic/5/

02941 | OWR-11101920-003

### Fields of Values of Irreducible Characters

O / open | Oberwolfach 报告问题汇编

What restrictions hold on the field of values $\mathbb{Q}(\chi)$ for $\chi\in\operatorname{Irr}(G)$?

来源字段：https://doi.org/10.4171/owr/2022/39

02942 | OWR-8415341-007

### Finding Free Subgroups Containing a Given Matrix

O / open | Oberwolfach 报告问题汇编

Given a linear group $G$ and a matrix $M\in G$, determine whether $M$ belongs to a nontrivial free subgroup of $G$ and, if so, find a basis for that free subgroup.

来源字段：https://doi.org/10.4171/owr/2021/38

02943 | OWR-1381-005

### Finite Generation of Group-Ring Cohomology

O / open | Oberwolfach 报告问题汇编

If a group $\Gamma$ is virtually of type FP, is $H^*(\Gamma;\mathbb{Z}\Gamma)$ always a finitely generated $\Gamma$-module?

来源字段：https://doi.org/10.4171/owr/2006/43

02944 | OWR-1274-014

### Finite Groups Admitting Beauville Structures

O / open | Oberwolfach 报告问题汇编

Classify the finite groups $G$ that admit Beauville structures and therefore can be used to construct Beauville surfaces.

来源字段：https://doi.org/10.4171/owr/2006/25

02945 | AMR-099-0002

### Finite groups in the property-T collapse window

O / open | AMR 问题清单汇编

In Gromov's density model with generators $a,a^{-1},b,b^{-1}$, choose $3^{nd}$ relators independently and uniformly from the reduced words of length $n$, for a density parameter $d$, and add the relators sequentially. The resulting quotient groups eventually collapse to the trivial group. Give an upper bound on the number of pairwise nonisomorphic finite nontrivial groups that can occur after the last infinite group and before the trivial group. In particular, prove the conjecture that this number is almost surely bounded uniformly in $n$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/hplast.pdf

02946 | OWR-14298160-009

### Finite Moves for Coxeter Presentations

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a finite list of vertical and horizontal moves such that any two vertices in one connected component of the Coxeter galaxy are joined by a finite sequence of those moves? Can an algorithm also specify the kinds of moves needed along such a path?

来源字段：https://doi.org/10.4171/owr/2024/4

02947 | AIM-GEOMETRIC_GROUP_THEORY-0002

### Finite palette closure and explicit distortion families in free-by-cyclic groups

P / partially_solved | AIM 研讨会问题汇编

What are the possible distortion functions for subgroups of free-by-cyclic groups?

来源字段：http://aimpl.org/freebycyclic/1/

02948 | AIM-GEOMETRIC_GROUP_THEORY-0115

### Finite support is the remaining boundary

P / partially_solved | AIM 研讨会问题汇编

Is there an infinite finitely generated amenable simple non-Liouville group (with respect to symmetric, finitely supported, generating measures)?

来源字段：http://aimpl.org/amenablediscrete/4/

02949 | OWR-2053-013

### Finite Width of the 14-Triangle Complex Group

O / open | Oberwolfach 报告问题汇编

Let $G=\langle x,y\mid r_1,r_2,r_3\rangle$, where $r_1=yxyxyxy^{-3}x^{-3}$, $r_2=yx^{-1}y^{-1}x^{-3}y^2x^{-1}yxy$, and $r_3=y^3x^{-1}yxyx^2y^2xyx$, and let $\gamma_i(G)$ be its lower central series. For every $i\geq2$, is $[\gamma_i(G):\gamma_{i+1}(G)]=8$ when $i\equiv0$ or $1\pmod3$, and $[\gamma_i(G):\gamma_{i+1}(G)]=4$ when $i\equiv2\pmod3$? Equivalently, does $G$ have finite width $3$?

来源字段：https://doi.org/10.4171/owr/2008/42

02950 | AIM-GEOMETRIC_GROUP_THEORY-0092

### Finite-index maximal subgroups and an endpoint-core obstruction to elementary amenability

P / partially_solved | AIM 研讨会问题汇编

What are the maximal subgroups of \(F\), and specifically, are there elementary amenable maximal ones?

来源字段：http://aimpl.org/amenablediscrete/1/

02951 | OWR-1965-001

### Finite-Lattice Representations as Subgroup Intervals

P / partially_solved | Oberwolfach 报告问题汇编

Is every nonempty finite lattice isomorphic to the lattice $O_G(H)$ of overgroups of a subgroup $H$ in a finite group $G$?

来源字段：https://doi.org/10.4171/owr/2008/20

02952 | AIM-GEOMETRIC_GROUP_THEORY-0130

### Finite-power rigidity at small Tarski number

P / partially_solved | AIM 研讨会问题汇编

Is the Tarski number of \(G\times G\) equal to the Tarski number of \(G\)?

来源字段：http://aimpl.org/amenablediscrete/5/

02953 | AIM-GEOMETRIC_GROUP_THEORY-0144

### Finite-quotient descent for central double covers of sofic groups

P / partially_solved | AIM 研讨会问题汇编

Let $\Gamma/\mathbb Z_2$ be a sofic group. Is $\Gamma$ sofic?

来源字段：http://aimpl.org/l2invariantsgroups/3/

02954 | AIM-GEOMETRIC_GROUP_THEORY-0013

### Finite-rank RFRS reduction with exact Fisher theorem indices

P / partially_solved | AIM 研讨会问题汇编

If a group $G$ is quasi-isometric to a free-by-cyclic group then is it true that $G$ is virtually free-by-cyclic?

Suppose that $H$ is virtually RFRS and quasi-isometric to a free-by-cyclic group $G$. Then H is virtually free-by-cyclic.

来源字段：http://aimpl.org/freebycyclic/3/

02955 | AIM-GEOMETRIC_GROUP_THEORY-0086

### Finite-relation separation and an abelianization obstruction for property R-infinity

P / partially_solved | AIM 研讨会问题汇编

**Problem 8.6.** Which Artin groups have property \(R_\infty\)?

来源字段：http://aimpl.org/geomartingp/8/

02956 | AIM-GEOMETRIC_GROUP_THEORY-0036

### Finite-subgroup localization for Artin groups

P / partially_solved | AIM 研讨会问题汇编

Are all Artin groups torsion-free?

来源字段：http://aimpl.org/geomartingp/1/

02957 | OWR-15987-017

### Finitely Generated Categories for Slowly Growing Braid Representations

O / open | Oberwolfach 报告问题汇编

别名：Duplicate FI-Type Stability Problem for Braid Groups [OWR-15987-018]

Is there a category playing the role for braid groups $B_n$ that FI plays for symmetric groups $S_n$, such that every slowly growing sequence of $B_n$-representations arises from a finitely generated module over that category?

来源字段：https://doi.org/10.4171/owr/2018/2

02958 | OWR-13488-005

### Finitely Presented Counterexamples for Conjugation-Invariant Norms

O / open | Oberwolfach 报告问题汇编

Does there exist a finitely presented group $G$, generated by a finite union of conjugacy classes and equipped with the associated conjugation-invariant word norm, such that either some cyclic subgroup is neither bounded nor undistorted, or some cyclic subgroup is neither bounded nor detected by a homogeneous quasimorphism?

来源字段：https://doi.org/10.4171/owr/2015/3

02959 | OWR-16167-016

### Finiteness and Rigidity of Fast Bump Groups

O / open | Oberwolfach 报告问题汇编

For every $n>1$, are all groups generated by $n$ fast one-bump functions, apart from nontrivial direct sums and wreath products, of type $F_\infty$? Are all such groups isomorphic?

来源字段：https://doi.org/10.4171/owr/2018/26

02960 | OWR-16162-008

### Finiteness of Algebraic Forms with Good Reduction

P / partially_solved | Oberwolfach 报告问题汇编

For a field $K$ and an absolutely almost simple simply connected or adjoint algebraic $K$-group $G$, when can $K$ be equipped with a natural set $V$ of discrete valuations such that, for every finite $S\subset V$, only finitely many $K$-isomorphism classes of inner $K$-forms of $G$ have good reduction at all $v\in V\setminus S$?

来源字段：https://doi.org/10.4171/owr/2018/21

02961 | OWR-16162-011

### Finiteness of Tate-Shafarevich Sets for Algebraic Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be an absolutely almost simple group over a finitely generated field $K$ of good characteristic, and let $S$ be a finite set of places. Is the Tate-Shafarevich set given by the fiber over the distinguished point of $$ H^1(K,G)\longrightarrow\prod_{v\notin S}H^1(K_v,G) $$ finite?

来源字段：https://doi.org/10.4171/owr/2018/21

02962 | OWR-16167-003

### Finiteness Properties of Hyperbolic Tessellation Groups

O / open | Oberwolfach 报告问题汇编

Let $\Delta$ be a tessellation of the hyperbolic plane, let $S_\infty$ be the finitary symmetric group, and suppose that $\phi(\Delta)/S_\infty$ is isomorphic to Thompson's group $V$. Do $\phi(\Delta)$ and $\phi(\Delta)/S_\infty$ have the same finiteness properties?

来源字段：https://doi.org/10.4171/owr/2018/26

02963 | OWR-4990377-005

### Fixed Points for Infinite-Dimensional Locally Elliptic Actions

O / open | Oberwolfach 报告问题汇编

When does a locally elliptic action on an infinite-dimensional CAT(0) cube complex have a global fixed point?

来源字段：https://doi.org/10.4171/owr/2021/8

02964 | OWR-4990377-006

### Fixed Points, Expanding Combings, and Lattices in Metric Geometry

O / open | Oberwolfach 报告问题汇编

Does every locally elliptic action of a finitely generated group on a finite-dimensional CAT(0) space, in particular on a Euclidean building of arbitrary dimension, have a global fixed point; does the analogous statement hold for finite-dimensional Helly complexes; do central extensions of hyperbolic groups admit coherent expanding combings; how can one construct an automatic group without an expanding combing; are the listed automatic structures expanding; and are irreducible lattices in products of two trees and non-uniform lattices in two-dimensional right-angled hyperbolic buildings finitely generated?

来源字段：https://doi.org/10.4171/owr/2021/8

02965 | AIM-GEOMETRIC_GROUP_THEORY-0139

### Fixed price one for products of infinite countable groups

P / partially_solved | AIM 研讨会问题汇编

Let $\Gamma_1$ and $\Gamma_2$ be infinite countable groups. Does $\Gamma_1\times\Gamma_2$ have fixed price $1$?

By Abert-Weiss, it is enough to show that Bernoulli action of $\Gamma_1\times\Gamma_2$ have cost equals to $1$.

来源字段：http://aimpl.org/l2invariantsgroups/2/

02966 | OWR-4990377-009

### Fixed Subgroups of Right-Angled Artin Group Automorphisms

O / open | Oberwolfach 报告问题汇编

Given a right-angled Artin group $A_\Gamma$ and an automorphism $\phi\in\operatorname{Aut}(A_\Gamma)$, what can be said about the fixed subgroup $\operatorname{Fix}(\phi)\leq A_\Gamma$?

来源字段：https://doi.org/10.4171/owr/2021/8

02967 | OWR-14299292-011

### Fixed-Point-Free Representations of Alternating Groups

O / open | Oberwolfach 报告问题汇编

Let $p$ and $r$ be primes. Does the alternating group $A_p$ admit a faithful irreducible representation over a field of characteristic $r>0$ in which every element of order $p$ and every element of order $p-2$ acts without nonzero fixed vectors?

来源字段：https://doi.org/10.4171/owr/2025/27

02968 | OWR-4428-015

### Flat Closing and Rigidity for Intermediate-Rank Groups

O / open | Oberwolfach 报告问题汇编

For the random intermediate-rank groups $G$ acting freely and cocompactly on $\operatorname{CAT}(0)$ spaces $X$, if $X$ contains an isometric copy of $\mathbb R^2$, must $G$ contain a copy of $\mathbb Z^2$? What rigidity properties do these groups have, and are they linear or residually finite?

来源字段：https://doi.org/10.4171/owr/2010/40

02969 | OWR-13750328-022

### Four Modular Isomorphism Problems

O / open | Oberwolfach 报告问题汇编

Determine whether the Modular Isomorphism Problem has a positive answer for groups of odd order and for groups of nilpotency class $2$; whether there are $p$-groups $G,H$ and a characteristic-$p$ field $k$ with $\mathbb F_pG\not\cong\mathbb F_pH$ but $kG\cong kH$; and which invariants of $G$ are determined by $kG$.

来源字段：https://doi.org/10.4171/owr/2023/19

02970 | OWR-1319-008

### Fragmentation Norms on the Torus and Open Möbius Strip

P / partially_solved | Oberwolfach 报告问题汇编

Is there a uniform constant $C$ such that every diffeomorphism of $T^2$ isotopic to the identity is a product of at most $C$ disk-supported diffeomorphisms? Is the analogous fragmentation norm bounded on the compactly supported identity component of the open Möbius strip?

来源字段：https://doi.org/10.4171/owr/2006/33

02971 | OWR-2053-020

### Free and Orthogonal Subgroups of Interval Exchange Groups

P / partially_solved | Oberwolfach 报告问题汇编

For the group $G$ of interval exchange transformations, do free groups, in particular $F_2$, embed into $G$? Does $\mathrm{SO}(n)$ embed into $G$?

来源字段：https://doi.org/10.4171/owr/2008/42

02972 | AIM-GEOMETRIC_GROUP_THEORY-0141

### Free groups of ranks two and three under topological orbit equivalence

S / solved | AIM 研讨会问题汇编

Are $2$- and $3$-generated groups topologically orbit equivalent?

来源字段：http://aimpl.org/l2invariantsgroups/2/

02973 | OWR-17298-004

### Free Quasi-Actions and Almost-Ramanujan Subgroups

O / open | Oberwolfach 报告问题汇编

For a standard probability space $(X,\mu)$, how many essentially free quasi-actions of $\operatorname{Aut}(X,\mu)$ exist up to conjugacy, and are all factors of the canonical product quasi-action $\rho_\infty$? Separately, for $k\geq 2$ and $\varepsilon>0$, does there exist $C$ such that every $k$-generated subgroup of a free group with no nontrivial reduced-word relations has normalized adjacency operator norm at most the Ramanujan bound $\sqrt{2k-1}/k+\varepsilon$?

来源字段：https://doi.org/10.4171/owr/2019/55

02974 | OWR-14299086-010

### Free-Group Word-Measure Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $F$ be a nonabelian free group of finite rank, and let $w,w'\in F$. If $w$ and $w'$ induce the same word measure on every finite group, prove that some $\varphi\in\operatorname{Aut}(F)$ satisfies $\varphi(w)=w'$.

来源字段：https://doi.org/10.4171/owr/2025/6

02975 | OWR-11127-007

### Frobenius Certificates for Finite Galois-Group Isomorphisms

P / partially_solved | Oberwolfach 报告问题汇编

Given an injective homomorphism from a finite Galois group into a fixed finite group $G$, what finite collection of conjugacy classes or Frobenius data suffices to certify that the homomorphism is an isomorphism?

来源字段：https://doi.org/10.4171/owr/2011/35

02976 | OWR-13750328-012

### Frobenius-Schur Indicators in Real Nilpotent Blocks

P / partially_solved | Oberwolfach 报告问题汇编

Let $B$ be a real nilpotent nonprincipal $2$-block with defect pair $(D,E)$. Prove that there is a height-preserving bijection $\Gamma:\operatorname{Irr}(D)\to\operatorname{Irr}(B)$ such that $\epsilon(\Gamma(\lambda))=|D|^{-1}\sum_{e\in E\setminus D}\lambda(e^2)$ for every $\lambda\in\operatorname{Irr}(D)$.

来源字段：https://doi.org/10.4171/owr/2023/19

02977 | OWR-4792-013

### Functional Characterization of Thompson Group Amenability

O / open | Oberwolfach 报告问题汇编

Let $\mathcal{T}$ be the variable-free terms generated by $1$ under a binary operation $\star$. Is the amenability of Thompson's group $F$ equivalent to the existence of strictly increasing nonidentity functions $f_T:\mathbb{N}\to\mathbb{N}$, indexed by $T\in\mathcal{T}$, such that $f_S=f_T$ whenever $S$ and $T$ are related by $A\star(B\star C)=(A\star B)\star(A\star C)$, $\operatorname{crit}(f_{A\star B})=f_A(\operatorname{crit}(f_B))$, and the set of critical points is $\mathbb{N}$?

来源字段：https://doi.org/10.4171/owr/2011/02

02978 | OWR-14299097-008

### Fusion Systems of Sectional Rank Three

P / partially_solved | Oberwolfach 报告问题汇编

Let $p$ be odd, let $S$ be a $p$-group of sectional rank three, and let $\mathcal{F}$ be a saturated fusion system on $S$ with $O_p(\mathcal{F})=1$ and $\mathcal{F}=O^p(\mathcal{F})$. Prove that either $S$ has maximal nilpotency class or $S$ has an abelian maximal subgroup.

来源字段：https://doi.org/10.4171/owr/2025/16

02979 | OWR-4792-012

### Følner Sets from Variable-Free Terms

O / open | Oberwolfach 报告问题汇编

Let $\mathcal{T}$ be the variable-free terms generated by $1$ under a binary operation $\star$. Associate to each $T\in\mathcal{T}$ a strictly increasing nonidentity function $f_T:\mathbb{N}\to\mathbb{N}$ such that $f_S=f_T$ whenever $S$ and $T$ are related by $A\star(B\star C)=(A\star B)\star(A\star C)$, $\operatorname{crit}(f_{A\star B})=f_A(\operatorname{crit}(f_B))$, and the set of critical points is $\mathbb{N}$. Define $\phi(1)=0$, $\phi(A\star B)=\max\{\phi(A),f_A(\phi(B))\}$, and let $A_{n,p}$ consist of the right divisors of the $(p+1)$st right-associated power of $1$ for which $\phi(T)\leq n$. For every $\varepsilon>0$, is there $p_0$ such that, whenever $p>p_0$, the set $A_{n,p}$ is $\varepsilon$-Følner for all but finitely many $n$?

来源字段：https://doi.org/10.4171/owr/2011/02

02980 | OWR-16776-009

### Geck's Unitriangularity Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a finite reductive group and let $\ell$ be a very good prime. Can the unipotent decomposition matrix be ordered by unipotent classes so that it is block lower triangular, with identity diagonal blocks indexed by the relevant families of unipotent characters?

来源字段：https://doi.org/10.4171/owr/2019/14

02981 | OWR-14299086-001

### Generating Sets Realizing Minimal Growth

O / open | Oberwolfach 报告问题汇编

Let $G=F_n\rtimes_\phi\mathbb Z$ be a free-by-cyclic group, and let $\xi(G)=\{e(G,X):X\text{ is a finite generating set of }G\}$. Characterize the finite generating sets $X$ for which $e(G,X)=\min\xi(G)$.

来源字段：https://doi.org/10.4171/owr/2025/6

02982 | OWR-1319-016

### Genus Dependence of Hamiltonian Surface Subgroups

O / open | Oberwolfach 报告问题汇编

Does some finitely generated group embed in $\operatorname{Ham}(\Sigma_g)$ but not in $\operatorname{Ham}(\Sigma_h)$ for $g>h$? Conversely, does every finitely generated subgroup of $\operatorname{Ham}(\Sigma_h)$ embed in $\operatorname{Ham}(\Sigma_g)$?

来源字段：https://doi.org/10.4171/owr/2006/33

02983 | OWR-16162-013

### Genus Finiteness for Algebraic Groups

O / open | Oberwolfach 报告问题汇编

别名：Finiteness of Genera of Simply Connected Groups [OWR-16162-012]

Let $G$ be an absolutely almost simple simply connected algebraic group over a field $K$. When is the genus $\operatorname{gen}_K(G)$, consisting of the $K$-isomorphism classes of $K$-forms having the same maximal $K$-tori as $G$, finite? In particular, is it finite when $K$ is finitely generated and has good characteristic?

来源字段：https://doi.org/10.4171/owr/2018/21

02984 | OWR-15433-009

### Geometric Description of Pure Local Arthur Packets

P / partially_solved | Oberwolfach 报告问题汇编

别名：Geometric Description of Pure Local Arthur Packets [OWR-15433-010]

For every Arthur parameter $\psi$ of $G$, is the pure local Arthur packet $$\Pi^{\mathrm{pure},\psi}(G/F)=\{[\pi,\delta]\in\Pi^{\mathrm{pure}}(G/F):\operatorname{Ev}^0_\psi(P(\pi,\delta))\neq0\},$$ with pairing $$\langle a,[\pi,\delta]\rangle_\psi=\operatorname{Trace}\!\left(\operatorname{Ev}^0_\psi(P(\pi,\delta))(a)\right)\qquad(a\in A_\psi)$$?

来源字段：https://doi.org/10.4171/owr/2017/25

02985 | OWR-17140-009

### Geometric Langlands Duals of Quantized Theta Sheaves

P / partially_solved | Oberwolfach 报告问题汇编

Let $M$ be a quantization whose geometric theta series is a distinguished $D$-module on $\operatorname{Bun}_G$. Is its geometric-Langlands dual a sheaf on the moduli space of $\check G$-local systems arising from a coherent quantization of $\check M$?

来源字段：https://doi.org/10.4171/owr/2019/44

02986 | OWR-14299292-006

### Geometric Realizability of Algebraic Two-Complexes over Group Rings

O / open | Oberwolfach 报告问题汇编

Let $G$ be a finitely presented group. Characterize which algebraic two-complexes of $\mathbb{Z}[G]$-modules are geometrically realizable as the cellular chain complexes of finite two-dimensional CW-complexes with fundamental group $G$.

来源字段：https://doi.org/10.4171/owr/2025/27

02987 | OWR-1106-003

### Geometry and Measure Equivalence of Limit Groups

P / partially_solved | Oberwolfach 报告问题汇编

Do all hyperbolic limit groups admit compact negatively curved classifying spaces? More generally, which countable groups are measure equivalent to finitely generated free groups, and in particular are all limit groups measure equivalent to free groups?

来源字段：https://doi.org/10.4171/owr/2005/45

02988 | OWR-14299511-011

### Goldman-Type Theorem for Sharply Transitive Permutation Groups

O / open | Oberwolfach 报告问题汇编

Let $G$ be a sharply $k$-transitive finite permutation group with point stabilizer $H$, and let $\Gamma_{G>H}$ be the principal graph of the group-subgroup subfactor $R\rtimes G\supset R\rtimes H$. Prove that every subfactor $P\subset Q$ with principal graph $\Gamma_{G>H}$ is itself of the form $R\rtimes_\alpha G\supset R\rtimes_\alpha H$ for an outer action $\alpha$ of $G$ on a factor $R$.

来源字段：https://doi.org/10.4171/owr/2025/34

02989 | OWR-16162-009

### Good-Reduction Finiteness over Finitely Generated Fields

P / partially_solved | Oberwolfach 报告问题汇编

For every finitely generated field $K$ of good characteristic and every absolutely almost simple $K$-group $G$, is there a natural set $V$ of discrete valuations such that, for each finite $S\subset V$, only finitely many $K$-isomorphism classes of inner $K$-forms of $G$ have good reduction at every $v\in V\setminus S$?

来源字段：https://doi.org/10.4171/owr/2018/21

02990 | OWR-14298004-005

### Gorenstein Equivalences under One-Sided Finiteness

O / open | Oberwolfach 报告问题汇编

Can the equivalence among the complete-cohomology, singularity, totally acyclic complex, and stable Gorenstein-projective or Gorenstein-injective categories for $A_G$ be proved assuming finiteness of only one of $\operatorname{silp}(A_G)$ and $\operatorname{spli}(A_G)$?

来源字段：https://doi.org/10.4171/owr/2023/55

02991 | OWR-14299518-006

### Graham’s Rearrangement Conjecture over Prime Fields

P / partially_solved | Oberwolfach 报告问题汇编

For every prime $p$, does every subset $S\subseteq\mathbb{F}_p\setminus\{0\}$ admit an ordering whose successive partial sums are all distinct?

来源字段：https://doi.org/10.4171/owr/2025/42

02992 | OWR-16167-008

### Graph-Automatic Structures for Thompson's Group $F$

O / open | Oberwolfach 报告问题汇编

Why should Thompson's group $F$ fail to be automatic, and does $F$ admit a graph-automatic structure?

来源字段：https://doi.org/10.4171/owr/2018/26

02993 | OWR-14299097-006

### Greedy Base Sizes for Uniform-Subset Actions

O / open | Oberwolfach 报告问题汇编

Determine the greedy base size for the standard action of a primitive permutation group on $r$-element subsets when the ground-set size is less than $4r^2$, completing the remaining uniform-subset cases.

来源字段：https://doi.org/10.4171/owr/2025/16

02994 | OWR-14297743-002

### Group Actions with Differentiability Obstructions

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a countable or finitely generated group $\Gamma_{r,M}\leq\operatorname{Diff}^r(M)$ such that, for every $s>r$, every action of $\Gamma_{r,M}$ on $M$ by $C^s$ diffeomorphisms is trivial or has a large kernel?

来源字段：https://doi.org/10.4171/owr/2023/33

02995 | OWR-17140-011

### Group-Level Realization of the c-Operation

O / open | Oberwolfach 报告问题汇编

Construct a group-level realization of the $c$-operation for a reductive $p$-adic group $G$ that recovers its known realization on the Iwahori-Hecke algebra.

来源字段：https://doi.org/10.4171/owr/2019/44

02996 | OWR-1268-003

### Groups Acting Minimally on Real Trees

O / open | Oberwolfach 报告问题汇编

Classify or otherwise describe finitely generated groups that act minimally and without a global fixed point on an $\mathbb R$-tree.

来源字段：https://doi.org/10.4171/owr/2006/19

02997 | OWR-1965-004

### Growth Functions of Finitely Generated Groups

O / open | Oberwolfach 报告问题汇编

Let $A$ be a finite generating set of an infinite group $G$, and let $B(t)$ count the elements expressible as products of at most $t$ elements of $A$. How does $B(t)$ grow as $t\to\infty$?

来源字段：https://doi.org/10.4171/owr/2008/20

02998 | OWR-12491-001

### Growth Functions of Finitely Generated Groups

O / open | Oberwolfach 报告问题汇编

Which functions occur, up to growth equivalence, as the growth function $v_G$ of a finitely generated group $G$? How far can $\inf_S v_{G,S}$ differ from $v_G$?

来源字段：https://doi.org/10.4171/owr/2013/28

02999 | OPG-702

### Growth of finitely presented groups

O / open | Open Problem Garden

Problem Does there exist a finitely presented group of intermediate growth?

背景链接：http://www.openproblemgarden.org/op/growth_of_finitely_presented_groups

03000 | AIM-GEOMETRIC_GROUP_THEORY-0065

### Halfspace pullbacks in the pure Salvetti complex

P / partially_solved | AIM 研讨会问题汇编

Let $\Sigma$ be the Davis complex of the associated Artin group. Take intersections of halfspaces in $\Sigma$, then pull back via the natural surjection from $P$. What can we say about these subspaces of $P$? Specifically, when are they $\pi_1$-injective?

来源字段：http://aimpl.org/geomartingp/5/

03001 | OWR-14298004-010

### Higher Finiteness Embeddings and TDLC Analogues

O / open | Oberwolfach 报告问题汇编

Does every finitely presented group embed in a group of type $F_3$? Does every group of type $\mathrm{FP}_2$ embed in a group of type $\mathrm{FP}_3$? Formulate and prove, or disprove, a totally disconnected locally compact analogue of the intended embedding theorem.

来源字段：https://doi.org/10.4171/owr/2023/55

03002 | OWR-1319-026

### Hofer-Geodesic Representatives of Hamiltonian Fundamental Groups

O / open | Oberwolfach 报告问题汇编

Is every element of the fundamental group of a Hamiltonian diffeomorphism group represented by a closed geodesic for the Hofer metric?

来源字段：https://doi.org/10.4171/owr/2006/33

03003 | OWR-15987-019

### Homological Stability for Indefinite Orthogonal Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a maximal ordered field $k$ and the inclusion $O_n(k)\times O_i(k)\hookrightarrow O_{n,i}(k)$, does the induced map on discrete group homology $H_d(-,\mathbb Z)$ become an isomorphism for $n$ sufficiently large relative to $d$ and $i$?

来源字段：https://doi.org/10.4171/owr/2018/2

03004 | OWR-14299904-008

### Homotopy Between Salvetti and Bessis-Brady-Watt Complexes

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Homotopy Question in Permutahedron-Triangulation Exposition [OWR-14299904-007]

Construct an explicit homotopy between the Salvetti complex and the Bessis-Brady-Watt complex.

来源字段：https://doi.org/10.4171/owr/2026/2

03005 | AIM-GEOMETRIC_GROUP_THEORY-0078

### Hopficity through finite quotients and large-type geometry

P / partially_solved | AIM 研讨会问题汇编

Which Artin groups are Hopfian?

来源字段：http://aimpl.org/geomartingp/8/

03006 | OWR-4333-017

### Hyperbolic Quotients of Large Coxeter Groups

O / open | Oberwolfach 报告问题汇编

Does every large Coxeter group surject onto a hyperbolic group?

来源字段：https://doi.org/10.4171/owr/2010/20

03007 | OWR-1319-019

### Hyperbolic Surface-by-Surface Group Extensions

O / open | Oberwolfach 报告问题汇编

Does there exist a hyperbolic group $\Gamma$ fitting into $1\to\pi_1(\Sigma_g)\to\Gamma\to\pi_1(\Sigma_h)\to1$? If so, are there only finitely many such groups for each $g$ and $h$?

来源字段：https://doi.org/10.4171/owr/2006/33

03008 | AIM-GEOMETRIC_GROUP_THEORY-0097

### IET amenability remains open: rank localization and a finite-permutation certificate

P / partially_solved | AIM 研讨会问题汇编

Is the discrete group \(\operatorname{IET}\) amenable?

来源字段：http://aimpl.org/amenablediscrete/2/

03009 | OWR-14299292-015

### Immediate Descendants of Elliptic-Curve p-Groups

O / open | Oberwolfach 报告问题汇编

For which pairs $(E,P)$ does the associated class-two group $G_{E,P}(\mathbb{F}_p)$ admit a central extension $1\to\mathbb{Z}/p\mathbb{Z}\to G\to G_{E,P}(\mathbb{F}_p)\to1$ with $|G|=p^{10}$ and nilpotency class three? Determine whether the total number of immediate descendants, summed over the relevant isomorphism classes, is a PORC function of $p$.

来源字段：https://doi.org/10.4171/owr/2025/27

03010 | OWR-1265-020

### Induced Character Models for Hyperoctahedral Groups

S / solved | Oberwolfach 报告问题汇编

Let $W_n$ be the hyperoctahedral group, let $\operatorname{Bip}(n)$ be the set of bipartitions of $n$, and let $C_\lambda$ be the conjugacy class associated with $\lambda\in\operatorname{Bip}(n)$. Suppose $(E_\lambda)_{\lambda\in\operatorname{Bip}(n)}$ is a family of orthogonal primitive idempotents in the rational Mantaci-Reutenauer algebra of $W_n$ satisfying $\sum_\lambda E_\lambda=1$ and $\dim_{\mathbb{Q}}(\mathbb{Q}W_nE_\lambda)=|C_\lambda|$. For $w_\lambda\in C_\lambda$, does there exist a linear character $\zeta_\lambda$ of $C_{W_n}(w_\lambda)$ such that $\mathbb{Q}W_nE_\lambda$ affords $\operatorname{Ind}_{C_{W_n}(w_\lambda)}^{W_n}\zeta_\lambda$?

来源字段：https://doi.org/10.4171/owr/2006/15

03011 | OWR-13750328-010

### Inductive Alperin-McKay Condition for Isolated Blocks

P / partially_solved | Oberwolfach 报告问题汇编

For finite simple groups of Lie type $B$ or $C$ and primes $\ell\geq 5$ outside the defining characteristic, verify the inductive Alperin-McKay condition for every isolated $\ell$-block by constructing the required equivariant height-zero-character bijection.

来源字段：https://doi.org/10.4171/owr/2023/19

03012 | OWR-11583-013

### Inductive Blockwise Alperin Weight Condition

P / partially_solved | Oberwolfach 报告问题汇编

Does the inductive blockwise Alperin weight condition hold for every finite simple group and every prime $p$, and hence imply the blockwise Alperin weight conjecture?

来源字段：https://doi.org/10.4171/owr/2012/16

03013 | OWR-11583-012

### Inductive Proofs of Alperin Conjectures

P / partially_solved | Oberwolfach 报告问题汇编

Prove the Alperin-McKay and Alperin weight conjectures, including their blockwise versions, by verifying the corresponding inductive conditions on finite simple groups.

来源字段：https://doi.org/10.4171/owr/2012/16

03014 | OWR-16929-001

### Infinite Families of Nonarithmetic Complex-Hyperbolic Lattices

O / open | Oberwolfach 报告问题汇编

For every $n\geq2$, are there infinitely many commensurability classes of nonarithmetic lattices in $\operatorname{PU}(n,1)$?

来源字段：https://doi.org/10.4171/owr/2019/17

03015 | OWR-1973-009

### Infinite-Index Maximal Subgroups of Just-Infinite Groups

P / partially_solved | Oberwolfach 报告问题汇编

Is every maximal subgroup of a finitely generated branch group of finite index? Conversely, does every finitely generated hereditary just-infinite group have a maximal subgroup of infinite index?

来源字段：https://doi.org/10.4171/owr/2008/28

03016 | OWR-14298004-011

### Integral Finiteness from Rational or Modular Finiteness

O / open | Oberwolfach 报告问题汇编

Let $G$ be of type $\mathrm{FP}_n(F)$ for $F\in\{\mathbb Q,\mathbb F_p:p\text{ prime}\}$. For which classes of groups must $G$ be of type $\mathrm{FP}_n(\mathbb Z)$? Does this hold when there is an exact sequence $G\to H\to\mathbb Z$ with $H$ of type $F$?

来源字段：https://doi.org/10.4171/owr/2023/55

03017 | OWR-1265-015

### Integral Local Grothendieck-Module Isomorphism

O / open | Oberwolfach 报告问题汇编

别名：Repeated Local Grothendieck-Group Conjecture [OWR-1265-014]; Repeated Local Grothendieck-Group Conjecture [OWR-1265-016]

For every block $b$ of a finite group $G$, let $\operatorname{Out}(G)_b$ be the stabilizer of $b$ in the outer automorphism group. Is there an $\mathcal{O}\operatorname{Out}(G)_b$-module isomorphism $G_k(G,b)\cong LG_k(G,b)$?

来源字段：https://doi.org/10.4171/owr/2006/15

03018 | OWR-9790362-012

### Interval Structure of Cohomology-Vanishing Exponents

O / open | Oberwolfach 报告问题汇编

Given a continuous homomorphism $$\sigma:G\longrightarrow\operatorname{Aut}(X,[\mu])\ltimes L^0(X,\mu;\{-1,1\}),$$ let $\pi_{p,\sigma}$ be the induced isometric representation on $L^p(X,\mu)$. Is the set $$\{p>0:H^1(G,\pi_{p,\sigma})=0\}$$ always an interval?

来源字段：https://doi.org/10.4171/owr/2022/11

03019 | OWR-2487-007

### Invariance of $L^2$-Torsion Under Large-Scale Equivalence

P / partially_solved | Oberwolfach 报告问题汇编

Let $G_0$ and $G_1$ have finite classifying-space models and vanishing $L^2$-Betti numbers. If $G_0$ and $G_1$ are quasi-isometric or orbit equivalent, must their $L^2$-torsion vanish simultaneously; that is, does $\rho^{(2)}(G_0)=0$ if and only if $\rho^{(2)}(G_1)=0$?

来源字段：https://doi.org/10.4171/owr/2008/49

03020 | OWR-1732-005

### Invariant Distribution Sheaves for Rank-One Lie Groups

O / open | Oberwolfach 报告问题汇编

Let $G$ be a connected simple real-rank-one Lie group, let $P=MAN$ be a minimal parabolic subgroup, and let $U\subset N$ be connected. For a generic principal-series induction parameter, the space of $U$-invariant distribution vectors is the space of global sections of a sheaf on an auxiliary manifold $Z_U$. Is this sheaf canonically isomorphic to the sheaf of distributional sections of a suitable vector bundle over $Z_U$?

来源字段：https://doi.org/10.4171/owr/2007/49

03021 | OWR-5158-015

### Invariant Orders on Finitely Generated Infinite Groups

O / open | Oberwolfach 报告问题汇编

For a finitely generated infinite group $\Gamma$, does there exist an invariant order relation on $\Gamma$? Consider separately left-invariant and bi-invariant orders, and total and partial orders.

来源字段：https://doi.org/10.4171/owr/2011/30

03022 | OWR-16166-010

### Inverse-Galois and Decidability Problems for Profinite Groups

O / open | Oberwolfach 报告问题汇编

Let $G$ be a finite group and let $C_1,\ldots,C_{m(G)+r}$ be conjugacy classes of cyclic subgroups that generate $G$. Is there a tamely ramified extension $L/\mathbb Q$ with Galois group $G$ and inertia groups $C_1,\ldots,C_{m(G)+r}$? Can every finite $p$-group occur as a regular Galois group over $\mathbb Q(t)$? For $g\geq2$, is the profinite or pro-$p$ completion of the surface group $\Sigma_g$ an absolute Galois group in the corresponding characteristic? Does every decidable projective profinite group have only decidable open subgroups?

来源字段：https://doi.org/10.4171/owr/2018/25

03023 | OWR-4333-002

### Isomorphism and Commensurability of Panel-Regular Building Lattices

O / open | Oberwolfach 报告问题汇编

Given the combinatorial data used to construct panel-regular lattices in buildings of type $\widetilde{A}_2$ and $\widetilde{C}_2$, can two different data sets produce isomorphic buildings or commensurable lattices? It is also open whether the corresponding buildings are classical or exotic.

来源字段：https://doi.org/10.4171/owr/2010/20

03024 | OWR-11780-005

### Iwasawa Decompositions Without Borel Exchange

O / open | Oberwolfach 报告问题汇编

For a group $G$ with a twin $BN$-pair and involution $\theta$, how does the Iwasawa decomposition change when the hypothesis $\theta(B^+)=B^-$ is removed?

来源字段：https://doi.org/10.4171/owr/2012/18

03025 | OWR-9790362-007

### Kaplansky's Conjectures for Group Rings

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a group and $K$ a field. If $G$ is torsion-free, must $K[G]$ have no nontrivial units, no nonzero zero divisors, and no idempotents other than $0$ and $1$; and for arbitrary $G$, must $K[G]$ be directly finite?

来源字段：https://doi.org/10.4171/owr/2022/11

03026 | KOU-1.1

### Kourovka Notebook Problem 1.1

S / solved | Kourovka 群论问题集

Do there exist non-trivial finitely-generated divisible groups? Equivalently, do there exist non-trivial finitely-generated divisible simple groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03027 | KOU-1.10

### Kourovka Notebook Problem 1.10

S / solved | Kourovka 群论问题集

An automorphism $\varphi$ of a group $G$ is called *splitting* if $gg^{\varphi}\cdots g^{\varphi^{n-1}}=1$ for any element $g\in G$, where $n$ is the order of $\varphi$. Is a soluble group admitting a regular splitting automorphism of prime order necessarily nilpotent?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03028 | KOU-1.11

### Kourovka Notebook Problem 1.11

S / solved | Kourovka 群论问题集

(E. Artin). The conjugacy problem for the braid groups ${\mathcal{B}}_n$, $n>4$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03029 | KOU-1.12

### Kourovka Notebook Problem 1.12

O / open | Kourovka 群论问题集

(W. Magnus). The problem of the isomorphism to the trivial group for all groups with $n$ generators and $n$ defining relations, where $n > 2$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03030 | KOU-1.13

### Kourovka Notebook Problem 1.13

S / solved | Kourovka 群论问题集

(J. Stallings). If a finitely presented group is trivial, is it always possible to replace one of the defining words by a primitive element without changing the group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03031 | KOU-1.14

### Kourovka Notebook Problem 1.14

S / solved | Kourovka 群论问题集

(B. H. Neumann). Does there exist an infinite simple finitely presented group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03032 | KOU-1.17

### Kourovka Notebook Problem 1.17

S / solved | Kourovka 群论问题集

Write an explicit set of generators and defining relations for a universal finitely presented group.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03033 | KOU-1.18

### Kourovka Notebook Problem 1.18

S / solved | Kourovka 群论问题集

(A. Tarski). a) Does there exist an algorithm for determining the solubility of equations in a free group?

b) Describe the structure of all solutions of an equation when it has at least one solution.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03034 | KOU-1.19

### Kourovka Notebook Problem 1.19

S / solved | Kourovka 群论问题集

(A. I. Mal'cev). Which subgroups (subsets) are first order definable in a free group? Which subgroups are relatively elementarily definable in a free group? In particular, is the derived subgroup first order definable (relatively elementarily definable) in a free group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03035 | KOU-1.2

### Kourovka Notebook Problem 1.2

S / solved | Kourovka 群论问题集

Let $G$ be a group, $F$ a free group with free generators $x_1,\ldots , x_n$, and $R$ the free product of $G$ and $F$. An *equation $($in the unknowns $x_1,\ldots ,x_n)$ over* $G$ is an expression of the form $v(x_1,\ldots x_n)=1$, where on the left is an element of $R$ not conjugate in $R$ to any element of $G$. We call $G$ *algebraically closed* if every equation over $G$ has a solution in $G$. Do there exist algebraically closed groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03036 | KOU-1.20

### Kourovka Notebook Problem 1.20

O / open | Kourovka 群论问题集

For which groups (classes of groups) is the lattice of normal subgroups first order definable in the lattice of all subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03037 | KOU-1.21

### Kourovka Notebook Problem 1.21

S / solved | Kourovka 群论问题集

Are there only finitely many finite simple groups of a given exponent $n$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03038 | KOU-1.23

### Kourovka Notebook Problem 1.23

S / solved | Kourovka 群论问题集

Does there exist an infinite simple locally finite group of finite rank?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03039 | KOU-1.24

### Kourovka Notebook Problem 1.24

S / solved | Kourovka 群论问题集

Does every infinite group possess an infinite abelian subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03040 | KOU-1.25

### Kourovka Notebook Problem 1.25

S / solved | Kourovka 群论问题集

a) Is the universal theory of the class of finite groups decidable?

b) Is the universal theory of the class of finite nilpotent groups decidable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03041 | KOU-1.26

### Kourovka Notebook Problem 1.26

S / solved | Kourovka 群论问题集

Does elementary equivalence of two finitely generated nilpotent groups imply that they are isomorphic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03042 | KOU-1.27

### Kourovka Notebook Problem 1.27

O / open | Kourovka 群论问题集

Describe the universal theory of free groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03043 | KOU-1.28

### Kourovka Notebook Problem 1.28

O / open | Kourovka 群论问题集

Describe the universal theory of a free nilpotent group.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03044 | KOU-1.29

### Kourovka Notebook Problem 1.29

S / solved | Kourovka 群论问题集

(A. Tarski). Is the elementary theory of a free group decidable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03045 | KOU-1.3

### Kourovka Notebook Problem 1.3

O / open | Kourovka 群论问题集

(Well-known problem). Can the group ring $\mathbb{Z}[G]$ of a torsion-free group $G$ contain zero divisors?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03046 | KOU-1.30

### Kourovka Notebook Problem 1.30

S / solved | Kourovka 群论问题集

Is the universal theory of the class of soluble groups decidable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03047 | KOU-1.31

### Kourovka Notebook Problem 1.31

O / open | Kourovka 群论问题集

Is a residually finite group with the maximum condition for subgroups almost polycyclic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03048 | KOU-1.32

### Kourovka Notebook Problem 1.32

S / solved | Kourovka 群论问题集

Is the Frattini subgroup of a finitely generated matrix group over a field nilpotent?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03049 | KOU-1.33

### Kourovka Notebook Problem 1.33

O / open | Kourovka 群论问题集

(A. I. Mal'cev). Describe the automorphism group of a free solvable group.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03050 | KOU-1.34

### Kourovka Notebook Problem 1.34

S / solved | Kourovka 群论问题集

Has every orderable polycyclic group a faithful representation by matrices over the integers?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03051 | KOU-1.35

### Kourovka Notebook Problem 1.35

P / partially_solved | Kourovka 群论问题集

c) (A. I. Mal'cev, L. Fuchs). Do there exist simple pro-orderable groups? A group is said to be *pro-orderable* if each of its partial orderings can be extended to a linear ordering.

[Archived solved part(s) of 1.35:] A group is called *pro-orderable* if every partial ordering of the group extends to a linear ordering.

a) Is the wreath product of two arbitrary pro-orderable groups again pro-orderable?

b) (A. I. Mal'cev). Is every subgroup of a pro-orderable group again pro-orderable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03052 | KOU-1.36

### Kourovka Notebook Problem 1.36

S / solved | Kourovka 群论问题集

If a group $G$ is factorizable by $p $-subgroups, that is, $ G = AB$, where $A$ and $B$ are $p $-subgroups, does it follows that $G$ is itself a $p $-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03053 | KOU-1.37

### Kourovka Notebook Problem 1.37

S / solved | Kourovka 群论问题集

Is it true that every subgroup of a locally nilpotent group is quasi-invariant?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03054 | KOU-1.38

### Kourovka Notebook Problem 1.38

S / solved | Kourovka 群论问题集

An *$N^0$-group* is a group in which every cyclic subgroup is a term of some normal system of the group. Is every $N^0$-group an $\widetilde N$-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03055 | KOU-1.39

### Kourovka Notebook Problem 1.39

S / solved | Kourovka 群论问题集

Is a group binary nilpotent if it is the product of two normal binary nilpotent subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03056 | KOU-1.4

### Kourovka Notebook Problem 1.4

S / solved | Kourovka 群论问题集

(A. I. Mal'cev). Does there exist a ring without zero divisors which is not embeddable in a skew field, while the multiplicative semigroup of its non-zero elements is embeddable in a group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03057 | KOU-1.40

### Kourovka Notebook Problem 1.40

O / open | Kourovka 群论问题集

Is a group a nilgroup if it is the product of two normal nilsubgroups?

By definition, a nilgroup is a group consisting of nilelements, in other words, of (not necessarily boundedly) Engel elements.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03058 | KOU-1.41

### Kourovka Notebook Problem 1.41

S / solved | Kourovka 群论问题集

A subgroup of a linearly orderable group is called *relatively convex* if it is convex with respect to some linear ordering of the group. Under what conditions is a subgroup of an orderable group relatively convex?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03059 | KOU-1.42

### Kourovka Notebook Problem 1.42

S / solved | Kourovka 群论问题集

Is the centre of a relatively convex subgroup relatively convex?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03060 | KOU-1.43

### Kourovka Notebook Problem 1.43

S / solved | Kourovka 群论问题集

Is the centralizer of a relatively convex subgroup relatively convex?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03061 | KOU-1.44

### Kourovka Notebook Problem 1.44

S / solved | Kourovka 群论问题集

Is a maximal abelian normal subgroup relatively convex?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03062 | KOU-1.45

### Kourovka Notebook Problem 1.45

S / solved | Kourovka 群论问题集

Is the largest locally nilpotent normal subgroup relatively convex?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03063 | KOU-1.46

### Kourovka Notebook Problem 1.46

O / open | Kourovka 群论问题集

What conditions ensure the normalizer of a relatively convex subgroup to be relatively convex?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03064 | KOU-1.47

### Kourovka Notebook Problem 1.47

S / solved | Kourovka 群论问题集

A subgroup $H$ of a group $G$ is said to be *strictly isolated* if, whenever $xg_1^{-1}xg_1\cdots g_n^{-1}xg_n$ belongs to $H$, so do $x$ and each $g_i^{-1}xg_i$. A group in which the identity subgroup is strictly isolated is called an *$S$-group*. Do there exist $S$-groups that are not orderable groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03065 | KOU-1.48

### Kourovka Notebook Problem 1.48

S / solved | Kourovka 群论问题集

Is a free product of two orderable groups with an amalgamated subgroup that is relatively convex in each of the factors again an orderable group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03066 | KOU-1.49

### Kourovka Notebook Problem 1.49

S / solved | Kourovka 群论问题集

An automorphism $\varphi$ of a linearly ordered group is called *order preserving* if $ x < y$ implies $x^{\varphi} <y^{\varphi}$. Is it possible to order an abelian strictly isolated normal subgroup of an $S$-group (see Archive 1.47) in such a way that the order is preserved under the action of the inner automorphisms of the whole group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03067 | KOU-1.5

### Kourovka Notebook Problem 1.5

O / open | Kourovka 群论问题集

(Well-known problem). Does there exist a group whose group ring does not contain zero divisors and is not embeddable into a skew field?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03068 | KOU-1.50

### Kourovka Notebook Problem 1.50

S / solved | Kourovka 群论问题集

Do the order-preserving automorphisms of a linearly ordered group form an orderable group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03069 | KOU-1.51

### Kourovka Notebook Problem 1.51

O / open | Kourovka 群论问题集

What conditions ensure a matrix group over a field (of complex numbers) to be orderable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03070 | KOU-1.52

### Kourovka Notebook Problem 1.52

S / solved | Kourovka 群论问题集

Describe the groups that can be ordered linearly in a unique way (reversed orderings are not regarded as different).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03071 | KOU-1.53

### Kourovka Notebook Problem 1.53

S / solved | Kourovka 群论问题集

Describe all possible linear orderings of a free nilpotent group with finitely many generators.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03072 | KOU-1.54

### Kourovka Notebook Problem 1.54

O / open | Kourovka 群论问题集

Describe all linear orderings of a free metabelian group with a finite number of generators.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03073 | KOU-1.55

### Kourovka Notebook Problem 1.55

O / open | Kourovka 群论问题集

Give an elementary classification of linearly ordered free groups with a fixed number of generators.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03074 | KOU-1.56

### Kourovka Notebook Problem 1.56

S / solved | Kourovka 群论问题集

Is a torsion-free group pro-orderable (see Archive 1.35) if the factor-group by its centre is pro-orderable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03075 | KOU-1.6

### Kourovka Notebook Problem 1.6

O / open | Kourovka 群论问题集

(A. I. Mal'cev). Is the group ring of a right-ordered group embeddable into a skew field?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03076 | KOU-1.60

### Kourovka Notebook Problem 1.60

S / solved | Kourovka 群论问题集

Can an orderable metabelian group be embedded in a radicable orderable group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03077 | KOU-1.61

### Kourovka Notebook Problem 1.61

S / solved | Kourovka 群论问题集

Can any orderable group be embedded into an orderable group

a) with a radicable maximal locally nilpotent normal subgroup?

b) with a radicable maximal abelian normal subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03078 | KOU-1.63

### Kourovka Notebook Problem 1.63

S / solved | Kourovka 群论问题集

A group $G$ is called *dense* if it has no proper isolated subgroups other than its trivial subgroup.

a) Do there exist dense torsion-free groups that are not locally cyclic?

b) Suppose that any two non-trivial elements $x$ and $y$ of a torsion-free group $G$ satisfy the relation $ x^k = y^l$, where $k$ and $l$ are non-zero integers depending on $x$ and $y$. Does it follow that $G$ is abelian?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03079 | KOU-1.64

### Kourovka Notebook Problem 1.64

S / solved | Kourovka 群论问题集

A torsion-free group is said to be *separable* if it can be represented as the set-theoretic union of two of its proper subsemigroups. Is every $R$-group separable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03080 | KOU-1.65

### Kourovka Notebook Problem 1.65

O / open | Kourovka 群论问题集

Is the class of groups of abelian extensions of abelian groups closed under taking direct sums $(A,B) \mapsto A \oplus B$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03081 | KOU-1.66

### Kourovka Notebook Problem 1.66

S / solved | Kourovka 群论问题集

Suppose that $T$ is a periodic abelian group, and ${\mathfrak{m}}$ an uncountable cardinal number. Does there always exist an abelian torsion-free group $U(T,{\mathfrak{m}})$ of cardinality ${\mathfrak{m}}$ with the following property: for any abelian torsion-free group $A$ of cardinality $\leq {\mathfrak{m}}$, the equality $\mathrm{Ext} (A,T)=0$ holds if and only if $A$ is embeddable in $U(T,{\mathfrak{m}})$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03082 | KOU-1.67

### Kourovka Notebook Problem 1.67

O / open | Kourovka 群论问题集

Suppose that $G$ is a finitely presented group, $F$ a free group whose rank is equal to the minimal number of generators of $G$, with a fixed homomorphism of $F$ onto $G$ with kernel $N$. Find a complete system of invariants of the factor-group of $N$ by the commutator subgroup $[F,N]$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03083 | KOU-1.68

### Kourovka Notebook Problem 1.68

S / solved | Kourovka 群论问题集

(A. Tarski). Let ${\mathfrak{K}}$ be a class of groups and $Q{\mathfrak{K}}$ the class of all homomorphic images of groups from ${\mathfrak{K}}$. If ${\mathfrak{K}}$ is axiomatizable, does it follow that $Q{\mathfrak{K}}$ is?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03084 | KOU-1.69

### Kourovka Notebook Problem 1.69

S / solved | Kourovka 群论问题集

(B. I. Plotkin). Do there exist locally nilpotent torsion-free groups without the property $RN^*$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03085 | KOU-1.70

### Kourovka Notebook Problem 1.70

S / solved | Kourovka 群论问题集

Let $p$ be a prime number and let $G$ be the group of all matrices of the form {\small $\left( \begin{array}{cc} 1+p\alpha & p\beta\\ p\gamma & 1+p\delta \end{array}\right)$}, where $\alpha$, $\beta$, $\gamma$, $\delta$ are rational numbers with denominators coprime to $p$. Does $G$ have the property $ \overline{\! RN}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03086 | KOU-1.71

### Kourovka Notebook Problem 1.71

S / solved | Kourovka 群论问题集

Let $G$ be a connected algebraic group over an algebraically closed field. Is the number of conjugacy classes of maximal soluble subgroups of $G$ finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03087 | KOU-1.72

### Kourovka Notebook Problem 1.72

S / solved | Kourovka 群论问题集

D. Hertzig has shown that a connected algebraic group over an algebraically closed field is soluble if it has a rational regular automorphism. Is this result true for an arbitrary field?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03088 | KOU-1.73

### Kourovka Notebook Problem 1.73

S / solved | Kourovka 群论问题集

Are there only finitely many conjugacy classes of maximal periodic subgroups in a finitely generated linear group over the integers?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03089 | KOU-1.74

### Kourovka Notebook Problem 1.74

O / open | Kourovka 群论问题集

Describe all minimal topological groups, that is, non-discrete groups all of whose closed subgroups are discrete. The minimal locally compact groups can be described without much effort. At the same time, the problem is probably complicated in the general case.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03090 | KOU-1.75

### Kourovka Notebook Problem 1.75

S / solved | Kourovka 群论问题集

Classify the infinite simple periodic linear groups over a field of characteristic $p > 0$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03091 | KOU-1.76

### Kourovka Notebook Problem 1.76

S / solved | Kourovka 群论问题集

Does there exist a simple, locally nilpotent, locally compact, topological group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03092 | KOU-1.78

### Kourovka Notebook Problem 1.78

S / solved | Kourovka 群论问题集

Let a group $G$ be the product of two divisible abelian $p$-groups of finite rank. Is then $G$ itself a divisible abelian $p$-group of finite rank?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03093 | KOU-1.80

### Kourovka Notebook Problem 1.80

S / solved | Kourovka 群论问题集

Does there exist a finite simple group whose Sylow 2-subgroup is a direct product of quaternion groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03094 | KOU-1.81

### Kourovka Notebook Problem 1.81

S / solved | Kourovka 群论问题集

The *width* of a group $G$ is, by definition, the smallest cardinal $ m = m(G)$ with the property that any subgroup of $G$ generated by a finite set $S\subseteq G$ is generated by a subset of $S$ of cardinality at most $m$.

a) Does a group of finite width satisfy the minimum condition for subgroups?

b) Does a group with the minimum condition for subgroups have finite width?

c) The same questions under the additional condition of local finiteness. In particular, is a locally finite group of finite width a Chernikov group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03095 | KOU-1.82

### Kourovka Notebook Problem 1.82

S / solved | Kourovka 群论问题集

Two sets of identities are said to be *equivalent* if they determine the same variety of groups. Construct an infinite set of identities which is not equivalent to any finite one.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03096 | KOU-1.83

### Kourovka Notebook Problem 1.83

S / solved | Kourovka 群论问题集

Does there exist a simple group, the orders of whose elements are unbounded, in which a non-trivial identity relation holds?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03097 | KOU-1.84

### Kourovka Notebook Problem 1.84

S / solved | Kourovka 群论问题集

Is it true that a polycyclic group $G$ is residually a finite $p$-group if and only if $G$ has a nilpotent normal torsion-free subgroup of $p$-power index?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03098 | KOU-1.85

### Kourovka Notebook Problem 1.85

S / solved | Kourovka 群论问题集

Is it true that the identity relations of a metabelian group have a finite basis?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03099 | KOU-1.86

### Kourovka Notebook Problem 1.86

O / open | Kourovka 群论问题集

Is it true that the identical relations of a polycyclic group have a finite basis?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03100 | KOU-1.87

### Kourovka Notebook Problem 1.87

O / open | Kourovka 群论问题集

The same question for matrix groups (at least over a field of characteristic 0).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03101 | KOU-1.88

### Kourovka Notebook Problem 1.88

S / solved | Kourovka 群论问题集

Is it true that if a matrix group over a field of characteristic 0 does not satisfy any non-trivial identity relation, then it contains a non-abelian free subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03102 | KOU-1.89

### Kourovka Notebook Problem 1.89

S / solved | Kourovka 群论问题集

Is the following assertion true? Let $G$ be a free soluble group, and $a$ and $b$ elements of $G$ whose normal closures coincide. Then there is an element $ x\in G$ such that $ b^{\pm 1} = x^{-1}ax$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03103 | KOU-1.9

### Kourovka Notebook Problem 1.9

S / solved | Kourovka 群论问题集

Can the factor-group of a locally normal group by the second term of its upper central series be embedded (isomorphically) in a direct product of finite groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03104 | KOU-1.90

### Kourovka Notebook Problem 1.90

S / solved | Kourovka 群论问题集

A subgroup $H$ of a group $G$ is called *$2$-infinitely isolated* in $G$ if, whenever the centralizer $C_G(h)$ in $G$ of some element $h\ne 1$ of $H$ contains at least one involution and intersects $H$ in an infinite subgroup, it follows that $C_G(h)\leq H$. Let $G$ be an infinite simple locally finite group whose Sylow 2-subgroups are Chernikov groups, and suppose $G$ has a proper 2-infinitely isolated subgroup $H$ containing some Sylow 2-subgroup of $G$. Does it follow that $G$ is isomorphic to a group of the type $PSL_2(k)$, where $k$ is a field of odd characteristic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03105 | KOU-10.1

### Kourovka Notebook Problem 10.1

S / solved | Kourovka 群论问题集

Let $p$ be a prime number. Describe the groups of order $p^9$ of nilpotency class 2 which contain subgroups $X$ and $Y$ such that $|X|=|Y|=p^3$ and any non-identity elements $x \in X$, $y \in Y$ do not commute. An answer to this question would yield a description of the semifields of order $p^3$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03106 | KOU-10.10

### Kourovka Notebook Problem 10.10

O / open | Kourovka 群论问题集

Is it true that a quasivariety generated by a finitely generated torsion-free soluble group and containing a non-abelian free metabelian group, can be defined in the class of torsion-free groups by an independent system of quasiidentities?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03107 | KOU-10.11

### Kourovka Notebook Problem 10.11

O / open | Kourovka 群论问题集

Is it true that every finitely presented group contains either a free subsemigroup on two generators or a nilpotent subgroup of finite index?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03108 | KOU-10.12

### Kourovka Notebook Problem 10.12

O / open | Kourovka 群论问题集

Does there exist a finitely generated semigroup $S$ with cancellation having non-exponential growth and such that its group of left quotients $G=S^{-1}S$ (which exists) is a group of exponential growth? An affirmative answer would give a positive solution to Problem 12 in (S. Wagon, *The Banach--Tarski Paradox*, Cambridge Univ. Press, 1985).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03109 | KOU-10.13

### Kourovka Notebook Problem 10.13

S / solved | Kourovka 群论问题集

Does there exist a *massive set* of independent elements in a free group $F_2$ on free generators $a, b$, that is, a set $ E$ of irreducible words on the alphabet $a, b, a^{-1}, b^{-1}$ such that

1) no element $w \in E$ belongs to the normal closure of $E\setminus \{ w\}$, and

2) the massiveness condition is satisfied: $\mathop{\overline{\mathrm{lim}}}\limits_{n\rightarrow \infty} \sqrt[n]{|E_n|}=3,$ where $E_n$ is the set of all words of length $n$ in $E$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03110 | KOU-10.14

### Kourovka Notebook Problem 10.14

S / solved | Kourovka 群论问题集

a) Does every group satisfying the minimum condition on subgroups satisfy the weak maximum condition on subgroups?

b) Does every group satisfying the maximum condition on subgroups satisfy the weak minimum condition on subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03111 | KOU-10.15

### Kourovka Notebook Problem 10.15

O / open | Kourovka 群论问题集

For every (known) finite quasisimple group and every prime $p$, find the faithful $ p $-modular absolutely irreducible linear representations of minimal degree.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03112 | KOU-10.16

### Kourovka Notebook Problem 10.16

O / open | Kourovka 群论问题集

A class of groups is called a *direct variety* if it is closed under taking subgroups, factor-groups, and direct products (Yu. M. Gorchakov, *Groups with Finite Classes of Conjugate Elements*, Moscow, Nauka, 1978 (Russian)). It is obvious that the class of $ FC $-groups is a direct variety. P. Hall (*J. London Math. Soc.*, **34**, no. 3 (1959), 289--304) showed that the class of finite groups and the class of abelian groups taken together do not generate the class of $ FC $-groups as a direct variety, and it was shown in (L. A. Kurdachenko, *Ukrain. Math. J.*, **39**, no. 3 (1987), 255--259) that the direct variety of $FC $-groups is also not generated by the class of groups with finite derived subgroups. Is the direct variety of $ FC $-groups generated by the class of groups with finite derived subgroups together with the class of $ FC $-groups having quasicyclic derived subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03113 | KOU-10.17

### Kourovka Notebook Problem 10.17

O / open | Kourovka 群论问题集

(M. J. Tomkinson). Let $G$ be an $FC $-group whose derived subgroup is embeddable in a direct product of finite groups. Must $G/Z(G)$ be embeddable in a direct product of finite groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03114 | KOU-10.18

### Kourovka Notebook Problem 10.18

O / open | Kourovka 群论问题集

(M. J. Tomkinson). Let $G$ be an $FC $-group which is residually in the class of groups with finite derived subgroups. Must $G/Z(G)$ be embeddable in a direct product of finite groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03115 | KOU-10.19

### Kourovka Notebook Problem 10.19

O / open | Kourovka 群论问题集

Characterize the radical associative rings such that the set of all normal subgroups of the adjoint group coincides with the set of all ideals of the associated Lie ring.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03116 | KOU-10.2

### Kourovka Notebook Problem 10.2

O / open | Kourovka 群论问题集

A *mixed identity* of a group $G$ is, by definition, an identity of an algebraic system obtained from $G$ by supplementing its signature by some set of 0-ary operations. One can develop a theory of mixed varieties of groups on the basis of this notion (see, for example, V. S. Anashin, *Math. USSR Sbornik*, **57** (1987), 171--182). Construct an example of a class of groups which is not a mixed variety, but which is closed under taking factor-groups, Cartesian products, and those subgroups of Cartesian powers that contain the diagonal subgroup.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03117 | KOU-10.20

### Kourovka Notebook Problem 10.20

O / open | Kourovka 群论问题集

In a Chevalley group of rank $\leq 6$ over a finite field of order $\leq 9$, describe all subgroups of the form $H=\left< H \cap U, H \cap V\right>$ that are not contained in any proper parabolic subgroup, where $U$ and $V$ are opposite unipotent subgroups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03118 | KOU-10.23

### Kourovka Notebook Problem 10.23

S / solved | Kourovka 群论问题集

Is it true that extraction of roots in braid groups is unique up to conjugation?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03119 | KOU-10.24

### Kourovka Notebook Problem 10.24

S / solved | Kourovka 群论问题集

A braid is said to be *coloured* if its strings represent the identity permutation. Is it true that links obtained by closing coloured braids are equivalent if and only if the original braids are conjugate in the braid group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03120 | KOU-10.25

### Kourovka Notebook Problem 10.25

S / solved | Kourovka 群论问题集

(Well-known problem). Does there exist an algorithm which decides for a given automorphism of a free group whether this automorphism has a non-trivial fixed point?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03121 | KOU-10.26

### Kourovka Notebook Problem 10.26

P / partially_solved | Kourovka 群论问题集

b) Does there exist an algorithm which decides whether the equation of the form $w(x_{i_1}^{\varphi _1},\ldots ,x_{i_n}^{\varphi _n})=1$ is soluble in a free group where $\varphi _1,\ldots ,\varphi _n$ are automorphisms of this group?

[Archived solved part(s) of 10.26:] a) Does there exist an algorithm which decides, for given elements $a, b$ and an automorphism $ \varphi$ of a free group, whether the equation $ax^{\varphi}=xb$ is soluble in this group? This question seems to be useful for solving the problem of equivalence of two knots.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03122 | KOU-10.27

### Kourovka Notebook Problem 10.27

O / open | Kourovka 群论问题集

a) Let $t$ be an involution of a finite group $G$ and suppose that the set $D=t^G \cup \{ t^xt^y\mid x,y \in G, \; \; |t^xt^y|=2\} $ does not intersect $O_2(G)$. Prove that if $t \in O_2(C(d))$ for any involution $d$ from $C_D(t)$, then $ D=t^G$ (and in this case the structure of the group $\left< D\right> $ is known).

b) A significantly more general question. Let $ t$ be an involution of a finite group $G$ and suppose that $t \in Z^{\displaystyle{*}}(N(X))$ for every non-trivial subgroup $X$ of odd order which is normalized, but not centralized, by $t$. What is the group $\left< t^G\right>$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03123 | KOU-10.28

### Kourovka Notebook Problem 10.28

S / solved | Kourovka 群论问题集

Is it true that finite strongly regular graphs with $\lambda =1$ have rank 3?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03124 | KOU-10.29

### Kourovka Notebook Problem 10.29

O / open | Kourovka 群论问题集

Describe the finite groups which contain a set of involutions $D$ such that, for any subset $D_0$ of $ D$ generating a 2-subgroup, the normalizer $N_D(D_0)$ also generates a 2-subgroup.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03125 | KOU-10.3

### Kourovka Notebook Problem 10.3

O / open | Kourovka 群论问题集

Characterize (in terms of bases of mixed identities or in terms of generating groups) minimal mixed varieties of groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03126 | KOU-10.30

### Kourovka Notebook Problem 10.30

S / solved | Kourovka 群论问题集

Does there exist a non-right-orderable group which is residually finite $p $-group for a finite set of prime numbers $p$ containing at least two different primes? If a group is residually finite $p $-group for an infinite set of primes $p$, then it admits a linear order (A. H. Rhemtulla, *Proc. Amer. Math. Soc.*, **41**, no. 1 (1973), 31--33).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03127 | KOU-10.31

### Kourovka Notebook Problem 10.31

O / open | Kourovka 群论问题集

Suppose that $G$ is an algebraic group, $H$ is a closed normal subgroup of $G$ and $f\! : G \rightarrow G/H$ is the canonical homomorphism. What conditions ensure that there exists a rational section $s\! : G/H \rightarrow G$ such that $ sf=1$? For partial results, see (Yu. I. Merzlyakov, *Rational Groups*, Nauka, Moscow, 1987, $\S$ 37 (Russian)).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03128 | KOU-10.32

## Kourovka Notebook Problem 10.32

O / open | Kourovka 群论问题集

A group word is said to be *universal on a group* $G$ if its values on $G$ run over the whole of $G$. For an arbitrary constant $c > 8/5$, J. L. Brenner, R. J. Evans, and D. M. Silberger (*Proc. Amer. Math. Soc.*, **96**, no. 1 (1986), 23--28) proved that there exists a number $N_0=N_0(c)$ such that the word $x^ry^s$, $rs \ne 0$, is universal on the alternating group ${\mathbb{A}_n$ for any $ n \geq \max \{ N_0, \; c\cdot \log m(r,s)\} $, where $m(r,s)$ is the product of all primes dividing $rs$ if $r,s \not\in \{ -1, 1\} $, and $m(r,s)=1$ if $r,s \in \{ -1, 1\} $. For example, one can take $ N_0(5/2)=5$ and $ N_0(2)=29$. Find an analogous bound for the degree of the symmetric group ${\mathbb{S}_n$ under hypothesis that at least one of the $r, s$ is odd: up to now, here only a more crude bound $ n \geq \max \{ 6, \; 4m(r,s) - 4\} $ is known (M. Droste, *Proc. Amer. Math. Soc.*, **96**, no. 1 (1986), 18--22).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03129 | KOU-10.33

## Kourovka Notebook Problem 10.33

S / solved | Kourovka 群论问题集

For $HNN $-extensions of the form $ G=\left< t, A\mid t^{-1}Bt=C, \; \varphi \right> $, where $A$ is a finitely generated abelian group and $\varphi \! : B \rightarrow C$ is an isomorphism of two of its subgroups, find

a) a criterion to be residually finite;

b) a criterion to be Hopfian.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03130 | KOU-10.34

## Kourovka Notebook Problem 10.34

S / solved | Kourovka 群论问题集

Does there exist a non-soluble finite group which coincides with the product of any two of its non-conjugate maximal subgroups?

*Editors' comment:* the answer is negative for almost simple groups (T. V. Tikhonenko, V. N. Tyutyanov, *Siberian Math. J.*, **51**, no. 1 (2010), 174--177).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03131 | KOU-10.35

## Kourovka Notebook Problem 10.35

O / open | Kourovka 群论问题集

Is it true that every finitely generated torsion-free subgroup of $GL_n({\mathbb{C}})$ is residually in the class of torsion-free subgroups of $GL_n(\overline{{\mathbb{Q}}})$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03132 | KOU-10.36

## Kourovka Notebook Problem 10.36

O / open | Kourovka 群论问题集

Is it true that $SL_n({\mathbb{Z}[x_1,\ldots ,x_r])$, for $n$ sufficiently large, is a group of type $(FP)_m$ (which is defined in the same way as $(FP)_{\infty}$ was defined in 6.3, but with that weakening that the condition of being finitely generated is not imposed on the terms of the resolution with numbers $ > m$). An affirmative answer is known for $m=0$, $n \geq 3$ (A. A. Suslin, *Math. USSR Izvestiya*, **11** (1977), 221--238) and for $m=1$, $n \geq 5$ (M. S. Tulenbayev, *Math. USSR Sbornik*, **45** (1983), 139--154; \ U. Rehmann, C. Soul\'e, in: *Algebraic $K$-theory, Proc. Conf., Northwestern Univ., Evanston, Ill., 1976, $($Lect. Notes Math.*, **551**), Springer, Berlin, 1976, 164--169).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03133 | KOU-10.37

## Kourovka Notebook Problem 10.37

S / solved | Kourovka 群论问题集

Suppose that $G$ is a finitely generated metabelian group all of whose integral homology groups are finitely generated. Is it true that $G$ is a group of finite rank? The answer is affirmative if $G$ splits over the derived subgroup (J. R. J. Groves, *Quart. J. Math.*, **33**, no. 132 (1982), 405--420).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03134 | KOU-10.38

## Kourovka Notebook Problem 10.38

O / open | Kourovka 群论问题集

(W. van der Kallen). Does $E_{n+3}({\mathbb{Z}[x_1,\ldots ,x_n])$, $n \geq 1$, have finite breadth with respect to the set of transvections?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03135 | KOU-10.39

## Kourovka Notebook Problem 10.39

O / open | Kourovka 群论问题集

a) Is the membership problem soluble for the subgroup $E_n(R)$ of $SL_n(R)$ where $R$ is a commutative ring?

b) Is the word problem soluble for the groups $K_i(R)$ where $K_i$ are the Quillen $K $-functors and $R$ is a commutative ring?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03136 | KOU-10.4

## Kourovka Notebook Problem 10.4

S / solved | Kourovka 群论问题集

Let $p$ be a prime number. Is it true that a mixed variety of groups generated by an arbitrary finite $p $-group of sufficiently large nilpotency class is a variety of groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03137 | KOU-10.40

## Kourovka Notebook Problem 10.40

O / open | Kourovka 群论问题集

(H. Bass). Let $G$ be a group, $e$ an idempotent matrix over ${\mathbb{Z}G$ and let $\mathrm{tr} e=\sum\limits_{g\in G}e_gg, e_g \in {\mathbb{Z}.$ *Strong conjecture:* for any non-trivial $x \in G$, the equation $\sum\limits_{g\sim x} e_g=0$ holds where $\sim $ denotes conjugacy in $G$. *Weak conjecture:* $\sum\limits_{g\in G} e_g=0$. The strong conjecture has been proved for finite, abelian and linear groups and the weak conjecture has been proved for residually finite groups.

*H. Bass' Comment of 2005:* Significant progress has been made; a good up-to-date account is in the paper (A. J. Berrick, I. Chatterji, G. Mislin, *Math. Ann.*, **329** (2004), 597--621).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03138 | KOU-10.41

## Kourovka Notebook Problem 10.41

S / solved | Kourovka 群论问题集

(Well-known problem). Let $\Gamma$ be an almost polycyclic group with no non-trivial finite normal subgroups, and let $k$ be a field. The complete ring of quotients $Q(k\Gamma )$ is a matrix ring $M_n(D)$ over a skew field. Conjecture: $n$ is the least common multiple of the orders of the finite subgroups of $\Gamma$. An equivalent formulation (M. Lorenz) is as follows: $\rho (G_0(k\Gamma ))=\rho (G_0(k\Gamma )_{{\mathcal{F}}})$, where $G_0(k\Gamma )$ is the Grothendieck group of the category of finitely-generated $k\Gamma$-modules, $G_0(k\Gamma )_{{\mathcal{F}}}$ is the subgroup generated by classes of modules induced from finite subgroups of $\Gamma$, and $\rho$ is the Goldie rank. There is a stronger conjecture: $G_0(k\Gamma )=G_0(k\Gamma )_{{\mathcal{F}}}$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03139 | KOU-10.42

## Kourovka Notebook Problem 10.42

O / open | Kourovka 群论问题集

(J. T. Stafford). Let $G$ be a poly-${\mathbb{Z} $-group and let $k$ be a field. Is it true that every finitely generated projective $kG $-module is either a free module or an ideal?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03140 | KOU-10.43

## Kourovka Notebook Problem 10.43

O / open | Kourovka 群论问题集

Let $R$ and $S$ be associative rings with identity such that 2 is invertible in $ S$. Let $\Lambda _I \! : GL_n(R) \rightarrow GL_n(R/I)$ be the homomorphism corresponding to an ideal $I$ of $ R$ and let $ E_n(R)$ be the subgroup of $ GL_n(R)$ generated by elementary transvections $t_{ij}(x)$. Let $a_{ij}=t_{ij}(1)t_{ji}(-1)t_{ij}(1)$, let the bar denote images in the factor-group of $ GL_n(R)$ by the centre and let $PG= \overline{\! G}$ for $G \leq GL_n(R)$. A homomorphism $$ \Lambda \! : E_n(R) \rightarrow GL(W)=GL_m(S)$$

is called *standard* if $S^m=P \oplus \cdots \oplus P \oplus Q$ (a direct sum of $S $-modules in which there are $n$ summands $P$) and $$\Lambda x=g^{-1}\tau (\delta ^{\displaystyle{*}} (x)f + (^t\delta ^{\displaystyle{*}}(x)^{\nu })^{-1}(1 - f))g,\quad \quad x \in E_n(R),$$ where $\delta ^{\displaystyle{*}} \! : GL_n(R) \rightarrow GL_n(\mathrm{End} P)$ is the homomorphism induced by a ring homomorphism $\delta \! : R \rightarrow \mathrm{End} P$ taking identity to identity, $g$ is an isomorphism of the module $W$ onto $ S^m$, $\tau \! : GL_n(\mathrm{End} P) \rightarrow GL(gW)$ is an embedding, $f$ is a central idempotent of $\delta R$, $t$ denotes transposition and $\nu$ is an antiisomorphism of $\delta R$. Let $n \geq 3$, $m \geq 2$. One can show that the homomorphism

$$\Lambda _0\! : PE_n(R) \rightarrow GL(W)=GL_m(S)$$

is induced by a standard homomorphism $ \Lambda$ if $\Lambda _0 \overline{\! a}_{ij}=g^{-1}\tau (a^{\displaystyle{*}}_{ij})g$ for some $g$ and $\tau$ and for any $i \ne j$ where $a^{\displaystyle{*}}_{ij}$ denotes the matrix obtained from $a_{ij}$ by replacing 0 and 1 from $R$ by 0 and 1 from $\mathrm{End} P$. Find (at least in particular cases) the form of a homomorphism $\Lambda _0$ that does not satisfy the last condition.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03141 | KOU-10.44

## Kourovka Notebook Problem 10.44

O / open | Kourovka 群论问题集

Let $R$ and $S$ be associative rings with identity such that 2 is invertible in $S$. Prove that every standard homomorphism $\Lambda$ of $ E_n(R) \subset GL_n(R)=GL(V)$ into $E_m(S) \subset GL_m(S)=GL(W)$ originates from a collineation and a correlation, that is, has the form $\Lambda x=(g^{-1}xg)f + (h^{-1}xh)(1 - f),$ where $g$ is a semilinear isomorphism $V_R \rightarrow W_S$ (collineation) and $h$ is a semilinear isomorphism $V_R \rightarrow _S\! W'_{S^0}$ (correlation).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03142 | KOU-10.45

### Kourovka Notebook Problem 10.45

O / open | Kourovka 群论问题集

Let $R$ be an associative ring with identity. Let $n \geq 3$ and suppose that $N$ is a subgroup of $ GL_n(R)$ which is normalized by $E_n(R)$. Prove that either $ N$ contains $E_n(R)$ or $\Lambda _I[N,E_n(R)]=1$ for a suitable ideal $I \ne R$ of $R$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03143 | KOU-10.46

### Kourovka Notebook Problem 10.46

S / solved | Kourovka 群论问题集

Let $R$ be an associative ring with identity. Prove that for every element $\sigma \in GL_n(R)$, $n \geq 3$, there exist transvections $\tau _1, \tau _2$ such that $[[\sigma ,\tau _1], \tau _2]$ is a unipotent element.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03144 | KOU-10.47

### Kourovka Notebook Problem 10.47

O / open | Kourovka 群论问题集

Describe the automorphisms of $PE_2(R)$ in the case where $R$ is an associative commutative ring and 2 and 3 have inverses in it.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03145 | KOU-10.48

### Kourovka Notebook Problem 10.48

S / solved | Kourovka 群论问题集

Let $V$ be a vector space of finite dimension over a field of prime order. A subset $R$ of $GL(V) \cup \{ 0\}$ is called *regular* if $|R|=|V|$, $0,1 \in R$ and $vx \ne vy$ for any non-trivial vector $v \in V$ and any distinct elements $x,y \in R$. It is obvious that $\tau ,\varepsilon ,\mu _g$ transform a regular set into a regular one, where $x^{\tau}=x^{-1}$ for $x \ne 0$ and $0^{\tau}=0$, $x^{\varepsilon}=1 - x$, \ $x^{\mu _g}=xg^{-1}$ and $ g$ is a non-zero element of the set being transformed. We say that two regular subsets are *equivalent* if one can be obtained from the other by a sequence of such transformations. \ \ \

a) Study the equivalence classes of regular subsets. \ \ \

b) Is every regular subset equivalent to a subgroup of $GL(V)$ together with $0$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03146 | KOU-10.49

### Kourovka Notebook Problem 10.49

O / open | Kourovka 群论问题集

Does there exist a group $G$ satisfying the following four conditions:

1) $G$ is simple, moreover, there is an integer $n$ such that $G=C^n$ for any conjugacy class $C$,

2) all maximal abelian subgroups of $G$ are conjugate in $ G$,

3) every maximal abelian subgroup of $G$ is self-normalizing and it is the centralizer of any of its nontrivial elements,

4) there is an integer $m$ such that if $H$ is a maximal abelian subgroup of $G$ and $a \in G\setminus H$ then every element in $G$ is a product of $m$ elements in $aH$?

*Comment of 2005:* There is a partial solution in (E. Jaligot, A. Ould Houcine, *J. Algebra*, **280** (2004), 772--796).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03147 | KOU-10.5

### Kourovka Notebook Problem 10.5

O / open | Kourovka 群论问题集

Construct an example of a finite group whose mixed identities do not have a finite basis. Is the group constructed in (R. Bryant, *Bull. London Math. Soc.*, **14**, no. 2 (1982), 119--123) such an example?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03148 | KOU-10.50

### Kourovka Notebook Problem 10.50

O / open | Kourovka 群论问题集

Complex characters of a finite group $G$ induced by linear characters of cyclic subgroups are called *induced cyclic characters* of $G$. Computer-aided computations (A. V. Rukolaine, *Abstracts of the 10th All--Union Sympos. on Group Theory*, Gomel', 1986, p. 199 (Russian)) show that there exist groups (for example, ${\mathbb{S}}_5$, $ SL_2(13)$, $M_{11}$) all of whose irreducible complex characters are integral linear combinations of induced cyclic characters and the principal character of the group. Describe all finite groups with this property.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03149 | KOU-10.51

### Kourovka Notebook Problem 10.51

O / open | Kourovka 群论问题集

Describe the structure of thin abelian $p $-groups. Such a description is known in the class of separable $ p $-groups (C. Megibben, *Mich. Math. J.*, **13**, no. 2 (1966), 153--160).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03150 | KOU-10.52

### Kourovka Notebook Problem 10.52

O / open | Kourovka 群论问题集

(R. Mines). It is well known that the topology on the completion of an abelian group under the $ p $-adic topology is the $p $-adic topology. R. Warfield has shown that this is also true in the category of nilpotent groups. Find a categorical setting for this theorem which includes the case of nilpotent groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03151 | KOU-10.53

### Kourovka Notebook Problem 10.53

O / open | Kourovka 群论问题集

(M. Dugas). Let ${\mathfrak{R}}$ be the *Reid class*, i. e. the smallest containing ${\mathbb{Z}$ and closed under direct sums and direct products. Is ${\mathfrak{R}}$ closed under direct summands?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03152 | KOU-10.54

### Kourovka Notebook Problem 10.54

O / open | Kourovka 群论问题集

(R. G\"obel). For a cardinal number $\mu$, let $${\mathbb{Z}^{<\mu}=\{ f \in {\mathbb{Z}^{\mu}\mid |\mathrm{supp} (f)| < \mu \} \; \; \; \; \mbox{ and} \; \; \; \; G_{\mu}={\mathbb{Z}^{\mu}/{\mathbb{Z}^{<\mu}.$$ a) Find a non-zero direct summand $D$ of $G_{\omega _1}$ such that $D\not\cong G_{\omega _1}$.

b) Investigate the structure of $G_{\mu}$ (the structure of $G_{\omega _0}$ is well known).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03153 | KOU-10.55

### Kourovka Notebook Problem 10.55

O / open | Kourovka 群论问题集

(A. Mader). ``Standard $B$'', that is, $B={\mathbb{Z}(p) \oplus {\mathbb{Z}(p^2 ) \oplus \cdots$, is slender as a module over its endomorphism ring (A. Mader, in: *Abelian Groups and Modules, Proc., Udine, 1984*, Springer, 1984, 315--327). Which abelian $p $-groups are slender as modules over their endomorphism rings?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03154 | KOU-10.56

### Kourovka Notebook Problem 10.56

S / solved | Kourovka 群论问题集

Is the lattice of formations of finite nilpotent groups of class $\leq 4$ distributive?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03155 | KOU-10.57

### Kourovka Notebook Problem 10.57

O / open | Kourovka 群论问题集

What are the minimal non-$A $-formations? An *$A $-formation* is, by definition, the formation of the finite groups all of whose Sylow subgroups are abelian.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03156 | KOU-10.58

### Kourovka Notebook Problem 10.58

O / open | Kourovka 群论问题集

Is the subsemigroup generated by the undecomposable formations in the semigroup of formations of finite groups free?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03157 | KOU-10.59

### Kourovka Notebook Problem 10.59

O / open | Kourovka 群论问题集

Is a $p' $-group $G$ locally nilpotent if it admits a splitting automorphism $\varphi$ of prime order $p$ such that all subgroups of the form $\langle g, g^{\varphi}, \ldots , g^{\varphi ^{p-1}}\rangle $ are nilpotent? An automorphism $\varphi$ of order $p$ is called *splitting* if $gg^{\varphi }g^{\varphi ^{2}}\cdots g^{\varphi ^{p-1}}= 1$ for all $g \in G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03158 | KOU-10.6

### Kourovka Notebook Problem 10.6

S / solved | Kourovka 群论问题集

Is it true that in an abelian group every non-discrete group topology can be strengthened up to a non-discrete group topology such that the group becomes a complete topological group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03159 | KOU-10.60

### Kourovka Notebook Problem 10.60

O / open | Kourovka 群论问题集

Does every periodic group ($p $-group) $A$ of regular automorphisms of an abelian group have non-trivial centre?

*Editors' comment (2001):* The answer is affirmative if $A$ contains an element of order 2 or 3 (A. Kh. Zhurtov, *Siberian Math. J.*, **41**, no. 2 (2000), 268--275).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03160 | KOU-10.61

### Kourovka Notebook Problem 10.61

O / open | Kourovka 群论问题集

Suppose that $H$ is a proper subgroup of a group $G$, $a \in H$, $a^2 \ne 1$ and for every $g \in G\setminus H$ the subgroup $\left< a, a^g\right>$ is a Frobenius group whose complement contains $a$. Does the set-theoretic union of the kernels of all Frobenius subgroups of $G$ with complement $\left< a\right>$ constitute a subgroup? For definitions see 6.55; see also (A. I. Sozutov, *Algebra and Logic*, **34**, no. 5 (1995), 295--305).

*Editors' comment (2005):* The answer is affirmative if the order of $a$ is even (A. M. Popov, A. I. Sozutov, *Algebra and Logic*, **44**, no. 1 (2005), 40--45) or if the order of $a$ is not 3 or 5 and the group $\left< a, a^g\right>$ is finite for any $g\not\in H$ (A. M. Popov, *Algebra and Logic*, **43**, no. 2 (2004), 123--127).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03161 | KOU-10.62

### Kourovka Notebook Problem 10.62

S / solved | Kourovka 群论问题集

Construct an example of a (periodic) group without subgroups of index 2 which is generated by a conjugacy class of involutions $X$ such that the order of the product of any two involutions from $X$ is odd.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03162 | KOU-10.63

### Kourovka Notebook Problem 10.63

S / solved | Kourovka 群论问题集

Is there a doubly transitive permutation group in which the stabilizer of a point is infinite cyclic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03163 | KOU-10.64

### Kourovka Notebook Problem 10.64

S / solved | Kourovka 群论问题集

Does there exist a non-periodic doubly transitive permutation group with a periodic stabilizer of a point?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03164 | KOU-10.65

### Kourovka Notebook Problem 10.65

O / open | Kourovka 群论问题集

Determine the structure of infinite 2-transitive permutation groups $(G,\Omega )$ in which the stabilizer of a point $\alpha \in \Omega$ has the form $G_{\alpha}=A\cdot G_{\alpha \beta}$ where $G _{\alpha \beta}$ is the stabilizer of two points $\alpha , \beta $, $\alpha \ne \beta $, such that $G _{\alpha \beta}$ contains an element inverting the subgroup $A$. Suppose, in particular, that $A\setminus \{ 1\}$ contains at most two conjugacy classes of $G_{\alpha}$; does $G$ then possess a normal subgroup isomorphic to $PSL_2$ over a field?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03165 | KOU-10.66

### Kourovka Notebook Problem 10.66

S / solved | Kourovka 群论问题集

Is a group $G$ non-simple if it contains two non-trivial subgroups $A$ and $B$ such that $AB \ne G$ and $AB^g=B^gA$ for any $g \in G$? This is true if $G$ is finite (O. H. Kegel, *Arch. Math.*, **12**, no. 2 (1961), 90--93).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03166 | KOU-10.67

### Kourovka Notebook Problem 10.67

O / open | Kourovka 群论问题集

The class $LN{\mathfrak{M}}_p$ of locally nilpotent groups admitting a splitting automorphism of prime order $p$ (for definition see 10.59) is a variety of groups with operators (E. I. Khukhro, *Math. USSR Sbornik*, **58** (1987), 119--126). Is it true that $$LN{\mathfrak{M}}_p=({\mathfrak{N}}_{c(p)} \cap LN{\mathfrak{M}}_p) \vee ({\mathfrak{B}}_p \cap LN{\mathfrak{M}}_p)$$ where ${\mathfrak{N}}_{c(p)}$ is the variety of nilpotent groups of some $p $-bounded class $c(p)$ and ${\mathfrak{B}}_p$ is the variety of groups of exponent $p$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03167 | KOU-10.68

### Kourovka Notebook Problem 10.68

S / solved | Kourovka 群论问题集

Suppose that a finite $ p $-group $G$ admits an automorphism of order $p$ with exactly $p^m$ fixed points. Is it true that $G$ has a subgroup whose index is bounded by a function of $p$ and $m$ and whose nilpotency class is bounded by a function of $m$ only? The results on $p $-groups of maximal class give an affirmative answer in the case $m=1$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03168 | KOU-10.69

### Kourovka Notebook Problem 10.69

S / solved | Kourovka 群论问题集

Suppose that $S$ is a closed oriented surface of genus $g > 1$ and $G=\pi _1S=G_1*_A G_2$, where $G_1 \ne A \ne G_2$ and the subgroup $A$ is finitely generated (and hence free). Is this decomposition geometrical, that is, do there exist connected surfaces $S_1, S_2, T$ such that $S=S_1 \cup S_2$, $S_1 \cap S_2=T$, $G_i=\pi _1S_i$, $A=\pi _1T$ and embeddings $S_i \subset S$, $T \subset S$ induce, in a natural way, embeddings $G_i \subset G$, $A \subset G$? This is true for $A={\mathbb{Z}}$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03169 | KOU-10.7

### Kourovka Notebook Problem 10.7

S / solved | Kourovka 群论问题集

Is it true that in a countable group $G$, any non-discrete group topology satisfying the first axiom of countability can be strengthened up to a non-discrete group topology such that $G$ becomes a complete topological group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03170 | KOU-10.70

### Kourovka Notebook Problem 10.70

O / open | Kourovka 群论问题集

Find a geometrical justification for the Whitehead method for free products similar to the substantiation given for free groups by Whitehead himself and more visual than given in (D. J. Collins, H. Zieschang, *Math. Z.*, **185**, no. 4 (1984), 487--504; **186**, no. 3 (1984), 335--361).

*D. J. Collins' comment:* there is a partial solution in (D. McCullough, A. Miller, *Symmetric automorphisms of free products*, Mem. Amer. Math. Soc. **582** (1996)).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03171 | KOU-10.71

### Kourovka Notebook Problem 10.71

O / open | Kourovka 群论问题集

Is it true that the centralizer of any automorphism (any finite set of automorphisms) in the automorphism group of a free group of finite rank is a finitely presented group? This is true in the case of rank 2 and in the case of inner automorphisms for any finite rank.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03172 | KOU-10.72

### Kourovka Notebook Problem 10.72

S / solved | Kourovka 群论问题集

Prove that the formation of all finite $p $-groups does not decompose into a product of two non-trivial subformations.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03173 | KOU-10.73

### Kourovka Notebook Problem 10.73

O / open | Kourovka 群论问题集

Enumerate all formations of finite groups all of whose subformations are $S_n $-closed.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03174 | KOU-10.74

### Kourovka Notebook Problem 10.74

O / open | Kourovka 群论问题集

Suppose that a group $G$ contains an element $a$ of prime order such that its centralizer $ C_G(a)$ is finite and all subgroups $\left< a, a^g\right> $, $g \in G$, are finite and almost all of them are soluble. Is $G$ locally finite? This problem is closely connected with 6.56. The question was solved in the positive for a number of very important partial cases by the author (*Abstracts on Group Theory of the Mal'cev Int. Conf. on Algebra*, Novosibirsk, 1989, p. 145 (Russian)).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03175 | KOU-10.75

### Kourovka Notebook Problem 10.75

O / open | Kourovka 群论问题集

Suppose that a group $G$ contains an element $a$ of prime order $p$ such that the normalizer of every finite subgroup containing $a$ has finite periodic part and all subgroups $\left< a, a^g\right> $, $g \in G$, are finite and almost all of them are soluble. Does $G$ possess a periodic part if $p > 2$? It was proved in (V. P. Shunkov, *Groups with involutions*, Preprints no. 4, 5, 12 of the Comput. centre of SO AN SSSR, Krasnoyarsk, 1986 (Russian)) that if $a$ is a point, then the answer is affirmative; on the other hand, a group with a point $a$ of order 2 satisfying the given hypothesis, which has no periodic part, was exhibited in the same works. For the definition of a *point* see (V. I. Senashov, V. P. Shunkov, *Algebra and Logic*, **22**, no. 1 (1983), 66--81).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03176 | KOU-10.76

### Kourovka Notebook Problem 10.76

S / solved | Kourovka 群论问题集

Suppose that $G$ is a periodic group having an infinite Sylow 2-subgroup $S$ which is either elementary abelian or a Suzuki 2-group, and suppose that the normalizer $ N_G(S)$ is strongly embedded in $G$ and is a Frobenius group (see 6.55) with locally cyclic complement. Must $G$ be locally finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03177 | KOU-10.77

### Kourovka Notebook Problem 10.77

O / open | Kourovka 群论问题集

Suppose that $G$ is a periodic group containing an elementary abelian subgroup $R$ of order 4. Must $G$ be locally finite

a) if $C_G(R)$ is finite?

b) if the centralizer of every involution of $R$ in $G$ is a Chernikov group?

*Remark of 1999:* P. V. Shumyatsky (*Quart. J. Math. Oxford (2)*, **49**, no. 196 (1998), 491--499) gave a positive answer to the question a) in the case where $G$ is residually finite.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03178 | KOU-10.78

### Kourovka Notebook Problem 10.78

O / open | Kourovka 群论问题集

Does there exist a non-Chernikov group which is a product of two Chernikov subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03179 | KOU-10.8

### Kourovka Notebook Problem 10.8

O / open | Kourovka 群论问题集

Does there exist a topological group which cannot be embedded in the multiplicative semigroup of a topological ring?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03180 | KOU-10.9

### Kourovka Notebook Problem 10.9

S / solved | Kourovka 群论问题集

Let $p$ be a prime number and let $L_p$ denote the set of all quasivarieties each of which is generated by a finite $ p $-group. Is $ L_p$ a sublattice of the lattice of all quasivarieties of groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03181 | KOU-11.1

### Kourovka Notebook Problem 11.1

O / open | Kourovka 群论问题集

(Well-known problem). Describe the structure of the centralizers of unipotent elements in almost simple groups of Lie type.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03182 | KOU-11.10

### Kourovka Notebook Problem 11.10

P / partially_solved | Kourovka 群论问题集

(R. C. Lyndon). a) Does there exist an algorithm that, given a group word $ w(a,x)$, recognizes whether $a$ is equal to the identity element in the group $\left< a,x\mid a^n=1,\right.$ $\left. w(a,x)=1\right>$?

[Archived solved part(s) of 11.10:] (R. C. Lyndon). b) Is it true that $a \ne 1$ in $G=\langle a,x\mid a^5=1,\; a^{x^2}=[a,a^x]\rangle $?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03183 | KOU-11.100

### Kourovka Notebook Problem 11.100

O / open | Kourovka 群论问题集

Is it true that every periodic conjugacy biprimitively finite group (see 6.59) can be obtained from 2-groups and binary finite groups by taking extensions? This question is of independent interest for $p $-groups, $p$ an odd prime.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03184 | KOU-11.101

### Kourovka Notebook Problem 11.101

S / solved | Kourovka 群论问题集

Does there exist a Golod group (see 9.76) with finite centre?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03185 | KOU-11.102

### Kourovka Notebook Problem 11.102

O / open | Kourovka 群论问题集

Does there exist a residually soluble but insoluble group satisfying the maximum condition on subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03186 | KOU-11.103

### Kourovka Notebook Problem 11.103

S / solved | Kourovka 群论问题集

Is a 2-group satisfying the minimum condition for centralizers necessarily locally finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03187 | KOU-11.104

### Kourovka Notebook Problem 11.104

S / solved | Kourovka 群论问题集

Let $G$ be a finite group of order $p^a\cdot q^b\cdots $, where $p, q, \ldots $ are distinct primes. Introduce distinct variables $x_p, x_q, \ldots $ corresponding to $p, q, \ldots $ . Define functions $f, \varphi$ from the lattice of subgroups of $G$ to the polynomial ring ${\mathbb{Z}[x_p,x_q,\ldots ]$ as follows:

(1) if $H$ has order $p^{\alpha}\cdot q^{\beta }\cdots $, then $f(H)=x_p^{\alpha}\cdot x_q^{\beta}\cdots $;

(2) for all $H \leq G$, we have $\sum\limits_{K\leq H}\varphi (K)=f(H)$.

Then $f(G), \varphi (G)$ may be called the *order* and *Eulerian polynomials* of $G$. Substituting $p^m, q^m, \ldots $ for $x_p, x_q, \ldots $ in these polynomials we get the $m$th power of the order of $G$ and the number of ordered $m$-tuples of elements that generate $G$ respectively.

It is known that if $G$ is $p $-solvable, then $\varphi (G)$ is a product of a polynomial in $x_p$ and a polynomial in the remaining variables. Consequently, if $G$ is solvable, $\varphi (G)$ is the product of a polynomial in $x_p$ by a polynomial in $x_q$ by $\ldots $ . Are the converses of these statements true \ a) for solvable groups? \ b) for $p $-solvable groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03188 | KOU-11.105

### Kourovka Notebook Problem 11.105

P / partially_solved | Kourovka 群论问题集

b) Let ${\mathfrak{V}}$ be a variety of groups. Its relatively free group of given rank has a presentation $F/N$, where $F$ is absolutely free of the same rank and $N$ fully invariant in $F$. The associated Lie ring ${\mathcal{L}}(F/N)$ has a presentation ${ L}/J$, where ${ L}$ is the free Lie ring of the same rank and $J$ an ideal of ${ L}$. Is $J$ fully invariant in ${ L}$ if ${\mathfrak{V}}$ is the Burnside variety of all groups of given exponent $q$, where $q$ is a prime-power, $q \geq 4$?

[Archived solved part(s) of 11.105:] a) Let ${\mathfrak{V}}$ be a variety of groups. Its relatively free group of given rank has a presentation $F/N$, where $F$ is absolutely free of the same rank and $N$ fully invariant in $F$. The associated Lie ring ${\mathcal{L}}(F/N)$ has a presentation ${ L}/J$, where ${ L}$ is the free Lie ring of the same rank and $J$ an ideal of ${ L}$. Is $J$ always fully invariant in ${ L}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03189 | KOU-11.106

### Kourovka Notebook Problem 11.106

S / solved | Kourovka 群论问题集

Can every periodic group be embedded in a simple periodic group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03190 | KOU-11.107

### Kourovka Notebook Problem 11.107

O / open | Kourovka 群论问题集

Is there a non-linear locally finite simple group each of whose proper subgroups is residually finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03191 | KOU-11.108

### Kourovka Notebook Problem 11.108

S / solved | Kourovka 群论问题集

Is every locally finite simple group absolutely simple? A group $G$ is said to be *absolutely simple* if the only composition series of $G$ is $\{ 1, G\} $. For equivalent formulations see (R. E. Phillips, *Rend. Sem. Mat. Univ. Padova*, **79** (1988), 213--220).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03192 | KOU-11.109

### Kourovka Notebook Problem 11.109

O / open | Kourovka 群论问题集

Is it true that the union of an ascending series of groups of Lie type of equal ranks over some fields should be also a group of Lie type of the same rank over a field? (For locally finite fields the answer is ``yes'', the theorem in this case is due to V. Belyaev, A. Borovik, B. Hartley \& G. Shute, and S. Thomas.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03193 | KOU-11.11

### Kourovka Notebook Problem 11.11

S / solved | Kourovka 群论问题集

The well-known Baer--Suzuki theorem states that if every two conjugates of an element $a$ of a finite group $G$ generate a finite $p $-subgroup, then $a$ is contained in a normal $p $-subgroup.

a) Does such a theorem hold in the class of periodic groups? The case $p=2$ is of particular interest.

b) Does such a theorem hold in the class of binary finite groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03194 | KOU-11.110

### Kourovka Notebook Problem 11.110

S / solved | Kourovka 群论问题集

Is it possible to embed $SL(2,{\mathbb{Q}})$ in the multiplicative group of some division ring?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03195 | KOU-11.111

### Kourovka Notebook Problem 11.111

O / open | Kourovka 群论问题集

Does every infinite locally finite simple group $G$ contain a nonabelian finite simple subgroup? Is there an infinite tower of such subgroups in $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03196 | KOU-11.112

### Kourovka Notebook Problem 11.112

P / partially_solved | Kourovka 群论问题集

Let $L=L(K(p))$ be the associated Lie ring of a free countably generated group $K(p)$ of the Kostrikin variety of locally finite groups of a given prime exponent $p$. Is it true that

a) $L$ is a relatively free Lie ring?

c) all identities of $L$ follow from a finite number of identities of $L$?

[Archived solved part(s) of 11.112:] b) Let $L=L(K(p))$ be the associated Lie ring of a free countably generated group $K(p)$ of the Kostrikin variety of locally finite groups of a given prime exponent $p$. Is it true that all identities of $L$ follow from multilinear identities of $L$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03197 | KOU-11.113

### Kourovka Notebook Problem 11.113

O / open | Kourovka 群论问题集

(B. Hartley). Is it true that the derived length of a nilpotent periodic group admitting a regular automorphism of prime-power order $p^n$ is bounded by a function of $p$ and $n$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03198 | KOU-11.114

### Kourovka Notebook Problem 11.114

O / open | Kourovka 群论问题集

Is every locally graded group of finite special rank almost hyperabelian? This is true in the class of periodic locally graded groups (*Ukrain. Math. J.*, **42**, no. 7 (1990), 855--861).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03199 | KOU-11.115

### Kourovka Notebook Problem 11.115

O / open | Kourovka 群论问题集

Suppose that $X$ is a free non-cyclic group, $N$ is a non-trivial normal subgroup of $X$ and $T$ is a proper subgroup of $ X$ containing $N$. Is it true that $[T,N] < [X,N]$ if $N$ is not maximal in $X$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03200 | KOU-11.116

### Kourovka Notebook Problem 11.116

S / solved | Kourovka 群论问题集

The *dimension* of a partially ordered set $\left< P, \leq \right> $ is, by definition, the least cardinal number $\delta$ such that the relation $\leq$ is an intersection of $\delta$ relations of linear order on $P$. Is it true that, for any Chernikov group which does not contain a direct product of two quasicyclic groups over the same prime number, the subgroup lattice has finite dimension? *Expected answer:* yes.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03201 | KOU-11.117

### Kourovka Notebook Problem 11.117

O / open | Kourovka 群论问题集

Let ${\mathfrak{X}}$ be a soluble non-empty Fitting class. Is it true that every finite non-soluble group possesses an ${\mathfrak{X}}$-injector?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03202 | KOU-11.118

### Kourovka Notebook Problem 11.118

O / open | Kourovka 群论问题集

What are the hereditary soluble local formations ${\mathfrak{F}}$ of finite groups such that every finite group has an ${\mathfrak{F}}$-covering subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03203 | KOU-11.119

### Kourovka Notebook Problem 11.119

O / open | Kourovka 群论问题集

Is it true that, for any non-empty set of primes $\pi$, the $\pi $-length of any finite $\pi $-soluble group does not exceed the derived length of its Hall $\pi $-subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03204 | KOU-11.12

### Kourovka Notebook Problem 11.12

O / open | Kourovka 群论问题集

a) Suppose that $G$ is a simple locally finite group in which the centralizer of some element is a linear group, that is, a group admitting a faithful matrix representation over a field. Is $G$ itself a linear group?

b) The same question with replacement of the word ``linear'' by ``finitary linear''. A group $H$ is *finitary linear* if it admits a faithful representation on an infinite-dimensional vector space $V$ such that the residue subspaces $V(1-h)$ have finite dimensions for all $h\in H$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03205 | KOU-11.120

### Kourovka Notebook Problem 11.120

O / open | Kourovka 群论问题集

Let ${\mathfrak{F}}$ be a soluble saturated Fitting formation. Is it true that $l_{\mathfrak{F}}(G) \leq f(c_{\mathfrak{F}}(G))$, where $c_{\mathfrak{F}}(G)$ is the length of a composition series of some ${\mathfrak{F}}$-covering subgroup of a finite soluble group $G$? This is Problem 17 in (L. A. Shemetkov, *Formations of Finite Groups*, Moscow, Nauka, 1978 (Russian)). For the definition of the {\it ${\mathfrak{F}}$-length} $l_{{\mathfrak{F}}}(G)$, see *ibid*.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03206 | KOU-11.121

### Kourovka Notebook Problem 11.121

O / open | Kourovka 群论问题集

Does there exist a local formation of finite groups which has a non-trivial decomposition into a product of two formations and in any such a decomposition both factors are non-local? Local products of non-local formations do exist.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03207 | KOU-11.122

### Kourovka Notebook Problem 11.122

O / open | Kourovka 群论问题集

Does every non-zero submodule of a free module over the group ring of a torsion-free group contain a free cyclic submodule?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03208 | KOU-11.123

### Kourovka Notebook Problem 11.123

O / open | Kourovka 群论问题集

(Well-known problem). For a given group $G$, define the following sequence of groups: $A_1(G)=G$, $A_{i+1}(G)=\mathrm{Aut} (A_i(G))$. Does there exist a finite group $G$ for which this sequence contains infinitely many non-isomorphic groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03209 | KOU-11.124

### Kourovka Notebook Problem 11.124

S / solved | Kourovka 群论问题集

Let $F$ be a non-cyclic free group and $R$ a non-cyclic subgroup of $F$. Is it true that if $[R,R]$ is a normal subgroup of $F$ then $R$ is also a normal subgroup of $F$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03210 | KOU-11.125

### Kourovka Notebook Problem 11.125

O / open | Kourovka 群论问题集

Let $G$ be a finite group admitting a regular elementary abelian group of automorphisms $V$ of order $p^n$. Is it true that the subgroup $H=\bigcap\limits_{v\in V\setminus \{ 1\}} [G,v]$ is nilpotent? In the case of an affirmative answer, does there exist a function depending only on $p$, $n$, and the derived length of $G$ which bounds the nilpotency class of $H$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03211 | KOU-11.126

### Kourovka Notebook Problem 11.126

S / solved | Kourovka 群论问题集

Do there exist a constant $h$ and a function $f$ with the following property: if a finite soluble group $G$ admits an automorphism $\varphi$ of order 4 such that $|C_G(\varphi)| \leq m$, then $G$ has a normal series $1\leq M\leq N\leq G$ such that the index $|G:N|$ does not exceed $f(m)$, the group $N/M$ is nilpotent of class $\leq 2$, and the group $M$ is nilpotent of class $\leq h$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03212 | KOU-11.127

### Kourovka Notebook Problem 11.127

O / open | Kourovka 群论问题集

Is every group of exponent 12 locally finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03213 | KOU-11.128

### Kourovka Notebook Problem 11.128

S / solved | Kourovka 群论问题集

A group $G$ is said to be a *$K$-group* if for every subgroup $A \leq G$ there exists a subgroup $B \leq G$ such that $A \cap B=1$ and $\left< A,B\right> =G$. Is it true that normal subgroups of $K$-groups are also $K$-groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03214 | KOU-11.13

### Kourovka Notebook Problem 11.13

S / solved | Kourovka 群论问题集

Suppose that $G$ is a periodic group with an involution $ i$ such that $i^g \cdot i$ has odd order for any $g \in G$. Is it true that the image of $i$ in $G/O(G)$ belongs to the centre of $G/O(G)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03215 | KOU-11.14

### Kourovka Notebook Problem 11.14

O / open | Kourovka 群论问题集

Is every finite simple group characterized by its Cartan matrix over an algebraically closed field of characteristic 2? In (R. Brandl, *Arch. Math.*, **38** (1982), 322--323) it is shown that for any given finite group there exist only finitely many finite groups with the same Cartan matrix.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03216 | KOU-11.15

### Kourovka Notebook Problem 11.15

O / open | Kourovka 群论问题集

It is known that for each prime number $p$ there exists a series $a_1, a_2, \ldots $ of words in two variables such that the finite group $G$ has abelian Sylow $p $-subgroups if and only if $ a_k(G)=1$ for almost all $k$. For $p=2$ such a series is known explicitly (R. Brandl, *J. Austral. Math. Soc.*, **31** (1981), 464--469). What about $p > 2$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03217 | KOU-11.16

### Kourovka Notebook Problem 11.16

O / open | Kourovka 群论问题集

Let $V_r$ be the class of all finite groups $G$ satisfying a law $[x, _r y]=[x, _s y]$ for some $s=s(G) > r$. Here $[x, _1 y] = [x,y]$ and $[x, _{i+1} y]=[[x, _i y],y]$.

a) Is there a function $f$ such that every soluble group in $V_r$ has Fitting length $< f(r)$? For $r < 3$ see (R. Brandl, *Bull. Austral. Math. Soc.*, **28** (1983), 101--110).

b) Is it true that $V_r$ contains only finitely many nonabelian simple groups? This is true for $r < 4$.

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03218 | KOU-11.17

### Kourovka Notebook Problem 11.17

O / open | Kourovka 群论问题集

Let $G$ be a finite group and let $d=d(G)$ be the least positive integer such that $G$ satisfies a law $[x, _ry]=[x, _{r+d} y]$ for some nonnegative integer $r=r(G)$.

a) Let $e=1$ if $d(G)$ is even and $e=2$ otherwise. Is it true that the exponent of $G/F(G)$ divides $e\cdot d(G)$?

b) If $G$ is a nonabelian simple group, does the exponent of $G$ divide $d(G)$?

Part a) is true for soluble groups (N. D. Gupta, H. Heineken, *Math. Z.*, **95** (1967), 276--287). I have checked part b) for ${\mathbb{A}_n$, $PSL(2,q)$, and a number of sporadic groups.

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03219 | KOU-11.18

### Kourovka Notebook Problem 11.18

O / open | Kourovka 群论问题集

Let $G(a,b)=\left< x,y\mid x=[x, _a y], \; y=[y, _b x]\right> $. Is $G(a,b)$ finite?

It is easy to show that $G(1,b)=1$ and one can show that $G(2,2)=1$. Nothing is known about $G(2,3)$. If one could show that every minimal simple group is a quotient of some $ G(a,b)$, then this would yield a very nice sequence of words in two variables to characterize soluble groups, see (R. Brandl, J. S. Wilson, *J. Algebra*, **116** (1988), 334--341.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03220 | KOU-11.19

### Kourovka Notebook Problem 11.19

O / open | Kourovka 群论问题集

(C. Sims). Is the $n$th term of the lower central series of an absolutely free group the normal closure of the set of basic commutators (in some fixed free generators) of weight exactly $n$?

D. Jackson announced a positive answer for $n\leq 5$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03221 | KOU-11.2

### Kourovka Notebook Problem 11.2

S / solved | Kourovka 群论问题集

Classify the simple groups that are isomorphic to the multiplicative groups of finite rings, in particular, of the group rings of finite groups over finite fields and over ${\mathbb{Z}/n{\mathbb{Z}$, $n \in {\mathbb{Z}$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03222 | KOU-11.20

### Kourovka Notebook Problem 11.20

S / solved | Kourovka 群论问题集

Suppose we have $[a,b]=[c,d]$ in an absolutely free group, where $a, b, [a,b]$ are basic commutators (in some fixed free generators). If $c$ and $d$ are arbitrary (proper) commutators, does it follow that $a=c$ and $b=d$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03223 | KOU-11.21

### Kourovka Notebook Problem 11.21

S / solved | Kourovka 群论问题集

Let ${\mathfrak{N}}_p$ denote the formation of all finite $p $-groups, for a given prime number $p$. Is it true that, for every subformation ${\mathfrak{F}}$ of ${\mathfrak{N}}_p$, there exists a variety ${\mathfrak{M}}$ such that ${\mathfrak{F}}={\mathfrak{N}}_p \cap {\mathfrak{M}}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03224 | KOU-11.22

### Kourovka Notebook Problem 11.22

O / open | Kourovka 群论问题集

Characterize all $p $-groups, $p$ a prime, that can be faithfully represented as $n\times n$ triangular matrices over a division ring of characteristic $p$.

If $p \geq n$ the solution is given in (B. A. F. Wehrfritz, *Bull. London Math. Soc.*, **19** (1987), 320--324). The corresponding question with $p=0$ was solved by B. Hartley and P. Menal (*Bull. London Math. Soc.*, **15** (1983), 378--383), but see above reference for a second proof.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03225 | KOU-11.23

### Kourovka Notebook Problem 11.23

S / solved | Kourovka 群论问题集

An automorphism $\varphi$ of a group $G$ is called a *nil-automorphism* if, for every $a \in G$, there exists $n$ such that $[a, _n \varphi ] = 1$. Here $[x, _1 y]=[x,y]$ and $[x, _{i+1} y]=[[x, _i y],y]$. An automorphism $\varphi$ is called an *$e$-automorphism* if, for any two $\varphi $-invariant subgroups $A$ and $B$ such that $A \not\subseteq B$, there exists $a \in A\setminus B$ such that $[a,\varphi ] \in B$. Is every $e$-automorphism of a group a nil-automorphism?

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03226 | KOU-11.24

### Kourovka Notebook Problem 11.24

S / solved | Kourovka 群论问题集

A Fitting class ${\mathfrak{F}}$ is said to be *local* if there exists a group function $f$ (for definition see (L. A. Shemetkov, *Formations of Finite Groups*, Moscow, Nauka, 1978 (Russian)) such that $f(p)$ is a Fitting class for every prime number $p$ and ${\mathfrak{F}}={\mathfrak{G}}_{\pi ( {\mathfrak{F}})} \bigcap \left(\bigcap_{p\in \pi ({\mathfrak{F}})} f(p){\mathfrak{N}}_p{\mathfrak{G}}_{p'}\right) $. Is every hereditary Fitting class of finite groups local?

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03227 | KOU-11.25

### Kourovka Notebook Problem 11.25

P / partially_solved | Kourovka 群论问题集

b) Do there exist local Fitting classes which are decomposable into a non-trivial product of Fitting classes and in every such a decomposition all factors are non-local? \ For the definition of the *product* of Fitting classes see (N. T. Vorob'\"ev, *Math. Notes*, **43**, no. 2 (1988), 91--94).

[Archived solved part(s) of 11.25:] a) Does there exist a local product (different from the class of all finite groups and from the class of all finite soluble groups) of Fitting classes each of which is not local and is not a formation? See the definition of the *product* of Fitting classes in (N. T. Vorob'ev, *Math. Notes*, **43**, No 1--2 (1988), 91--94).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03228 | KOU-11.26

### Kourovka Notebook Problem 11.26

S / solved | Kourovka 群论问题集

Does there exist a group which is not isomorphic to outer automorphism group of a metabelian group with trivial center?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03229 | KOU-11.27

### Kourovka Notebook Problem 11.27

S / solved | Kourovka 群论问题集

What are the minimum numbers of generators for groups $G$ satisfying $S \leq G \leq \mathrm{Aut} S$ where $S$ is a finite simple non-abelian group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03230 | KOU-11.28

### Kourovka Notebook Problem 11.28

O / open | Kourovka 群论问题集

Suppose the prime graph of the finite group $G$ is disconnected. (This means that the set of prime divisors of the order of $G$ is the disjoint union of non-empty subsets $\pi$ and $\pi '$ such that $G$ contains no element of order $pq$ where $p \in \pi$, $q \in \pi '$.) Then P. A. Linnell (*Proc. London Math. Soc.*, **47**, no. 1 (1983), 83--127) has proved mod CFSG that there is a decomposition of ${\mathbb{Z}G$-modules ${\mathbb{Z} \oplus {\mathbb{Z}G=A \oplus B$ with $A$ and $B$ non-projective. Find a proof independent of CFSG.

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03231 | KOU-11.29

## Kourovka Notebook Problem 11.29

P / partially_solved | Kourovka 群论问题集

Let $F$ be a free group and ${\mathfrak{f}}={\mathbb{Z}F(F - 1)$ the augmentation ideal of the integral group ring ${\mathbb{Z}F$. For any normal subgroup $R$ of $F$ define the corresponding ideal ${\mathfrak{r}}={\mathbb{Z}F(R - 1)= _{\mathrm{id}}} (r - 1\mid r \in R)$. One may identify, for instance, $F \cap (1 + {\mathfrak{r}} {\mathfrak{f}})=R'$, where $F$ is naturally imbedded into ${\mathbb{Z}F$ and $1 + {\mathfrak{r}} {\mathfrak{f}}=\{ 1 + a\mid a \in {\mathfrak{r}}{\mathfrak{f}}\} $.

Identify in an analogous way in terms of corresponding subgroups of $F$

a) $F \cap (1 + {\mathfrak{r}}_1 {\mathfrak{r}_2}\cdot \cdot \cdot {\mathfrak{r}}_n)$, where $R_i$ are normal subgroups of $F$, $i=1, 2,\ldots ,n$;

b) $F \cap (1 + {\mathfrak{r}}_1{\mathfrak{r}}_2{\mathfrak{r}}_3)$;

c) $F \cap (1 + {\mathfrak{f}} {\mathfrak{s}} + {\mathfrak{f}}^n)$, where $F/S$ is finitely generated nilpotent;

d) $F \cap (1 + {\mathfrak{f}}{\mathfrak{s}}{\mathfrak{f}} + {\mathfrak{f}}^n)$;

e) $F \cap (1 + {\mathfrak{r}}(k) + {\mathfrak{f}}^n)$, $n > k \geq 2$, where ${\mathfrak{r}}(k)={\mathfrak{r}}{\mathfrak{f}}^{k-1} + {\mathfrak{f}}{\mathfrak{r}} {\mathfrak{f}}^{k-2} + \cdots + {\mathfrak{f}}^{k-1}{\mathfrak{r}}$.

[Archived solved part(s) of 11.29:] f) Let $F$ be a free group and ${\mathfrak{f}}={\mathbb{Z}F(F - 1)$ the augmentation ideal of the integral group ring ${\mathbb{Z}F$. For any normal subgroup $R$ of $F$ define the corresponding ideal ${\mathfrak{r}}={\mathbb{Z}F(R - 1)= _{\mathrm{id}}} (r - 1\mid r \in R)$. One may identify, for instance, $F \cap (1 + {\mathfrak{r}} {\mathfrak{f}})=R'$, where $F$ is naturally imbedded into ${\mathbb{Z}F$ and $1 + {\mathfrak{r}} {\mathfrak{f}}=\{ 1 + a\mid a \in {\mathfrak{r}}{\mathfrak{f}}\} $. Is the quotient group $(F \cap (1 + {\mathfrak{r}} + {\mathfrak{f}}^n))/R\cdot \gamma _n(F)$ always abelian?

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03232 | KOU-11.3

## Kourovka Notebook Problem 11.3

O / open | Kourovka 群论问题集

(M. Aschbacher). A $p$-local subgroup $H$ in a group $G$ is said to be a *superlocal* if $H= N_G(O_p(H))$. Describe the superlocals in alternating groups and in groups of Lie type.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03233 | KOU-11.30

## Kourovka Notebook Problem 11.30

O / open | Kourovka 群论问题集

Is it true that the rank of a torsion-free soluble group is equal to the rank of any of its subgroups of finite index? The answer is affirmative for groups having a rational series (D. I. Zaitsev, in: *Groups with restrictions on subgroups*, Naukova dumka, Kiev, 1971, 115--130 (Russian)). We note also that every torsion-free soluble group of finite rank contains a subgroup of finite index which has a rational series.

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03234 | KOU-11.31

## Kourovka Notebook Problem 11.31

O / open | Kourovka 群论问题集

Is a radical group polycyclic if it is a product of two polycyclic subgroups? The answer is affirmative for soluble and for hyperabelian groups (D. I. Zaitsev, *Math. Notes*, **29**, no. 4 (1981), 247--252; \ J. C. Lennox, J. E. Roseblade, *Math. Z.*, **170** (1980), 153--154).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03235 | KOU-11.32

## Kourovka Notebook Problem 11.32

S / solved | Kourovka 群论问题集

Describe the primitive finite linear groups that contain a matrix with *simple spectrum*, that is, a matrix all of whose eigenvalues are of multiplicity 1. See partial results in (A. Zalesskii, I. D. Suprunenko, *Commun. Algebra*, **26**, no. 3 (1998), 863--888; **28**, no. 4 (2000), 1789--1833; \ Ch. Rudloff, A. Zalesski, *J. Group Theory*, **10** (2007), 585--612; \ L. Di Martino, A. E. Zalesski, *J. Algebra Appl.*, **11**, no. 2 (2012), 1250038; \ L. Di Martino, M. Pellegrini, A. Zalesski, {Commun. Algebra}, **42** (2014), 880--908, where the problem is considered in a more general context, when all but one eigenvalue have multiplicity 1.).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03236 | KOU-11.33

## Kourovka Notebook Problem 11.33

P / partially_solved | Kourovka 群论问题集

b) Let $G(q)$ be a simple Chevalley group over a field of order $q$. Prove that there exists $m$ such that the restriction of every non-one-dimensional representation of $G(q^m)$ over a field of prime characteristic not dividing $q$ to $G(q)$ contains all irreducible representations of $G(q)$ as composition factors.

[Archived solved part(s) of 11.33:] a) Let $G(q)$ be a simple Chevalley group over a field of order $q$. Prove that there exists $m$ such that the restriction of every non-one-dimensional complex representation of $G(q^m)$ to $G(q)$ contains all irreducible representations of $G(q)$ as composition factors.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03237 | KOU-11.34

## Kourovka Notebook Problem 11.34

O / open | Kourovka 群论问题集

Describe the complex representations of quasisimple finite groups which remain irreducible after reduction modulo any prime number $q$. An important example: representations of degree $(p^k - 1)/2$ of the symplectic group $Sp(2k,p)$ where $k \in {\mathbb{N}}$ and $p$ is an odd prime. See partial results in (P. H. Tiep, A. E. Zalesski, *Proc. London Math. Soc.*, **84** (2002), 439--472; \ P. H. Tiep, A. E. Zalesski, *Proc. Amer. Math. Soc.*, **130** (2002), 3177--3184); \ C. Parker, M. van Beek, *Preprint*, 2024, https://arxiv.org/abs/2411.16379).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03238 | KOU-11.35

## Kourovka Notebook Problem 11.35

S / solved | Kourovka 群论问题集

Suppose that $H$ is a finite linear group over ${\mathbb{C}}$ and $h$ is an element of $H$ of prime order $p$ which is not contained in any abelian normal subgroup. Is it true that $h$ has at least $(p - 1)/2$ different eigenvalues?

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03239 | KOU-11.36

## Kourovka Notebook Problem 11.36

P / partially_solved | Kourovka 群论问题集

Let $G=B(m,n)$ be the free Burnside group of rank $m$ and of odd exponent $n \gg 1$. Are the following statements true?

a) Every 2-generated subgroup of $G$ is isomorphic to the Burnside $n$-product of two cyclic groups.

b) Every automorphism $\varphi$ of $G$ such that $\varphi ^n=1$ and $b^{\varphi}\cdot b^{\varphi ^2} \cdots b^{ \varphi ^n}=1$ for all $b \in G$ is an inner automorphism (here $m > 1$).

*Editors' comment:* for large prime $n$ this follows from (E. A. Cherepanov, *Int. J. Algebra Comput.*, **16** (2006), 839--847), for prime $n\geq 1009$ from (V. S. Atabekyan, *Izv. Math.*, **75**, no. 2 (2011), 223--237); this is also true for odd $n\geq 1003$ if in addition the order of $\varphi$ is a prime power (V. S. Atabekyan, *Math. Notes*, **95**, no. 5 (2014), 586--589).

c) The group $G$ is Hopfian if $m < \infty$.

d) All retracts of $G$ are free.

[Archived solved part(s) of 11.36:] e) Let $G=B(m,n)$ be the free Burnside group of rank $m$ and of odd exponent $n \gg 1$. Is it true that all zero divisors in the group ring ${\mathbb{Z}G$ are trivial? which means that if $ab=0$ then $a=a_1c$, $b=db_1$ where $a_1, c, b_1, d \in {\mathbb{Z}G$, \ $cd=0$, and the set \ $\mathrm{supp} c \cup \mathrm{supp} d$ \ is contained in a cyclic subgroup of $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03240 | KOU-11.37

## Kourovka Notebook Problem 11.37

P / partially_solved | Kourovka 群论问题集

a) Can the free Burnside group $ B(m,n)$, for any $m$ and $ n$, be given by defining relations of the form $v^n=1$ such that for any natural divisor $d$ of $n$ distinct from $n$ the element $v^d$ is not trivial in $B(m,n)$? This is true for odd $n \geq 665$, and for all $n\geq 2^{48}$ divisible by $2^9$.

[Archived solved part(s) of 11.37:] b) Can the free Burnside group $ B(m,n)$, for any $m$ and $ n=2^l \gg 1$, be given by defining relations of the form $v^n=1$ such that for any natural divisor $d$ of $n$ distinct from $n$ the element $v^d$ is not trivial in $B(m,n)$?

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03241 | KOU-11.38

## Kourovka Notebook Problem 11.38

O / open | Kourovka 群论问题集

Does there exist a finitely presented Noetherian group which is not almost polycyclic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03242 | KOU-11.39

## Kourovka Notebook Problem 11.39

O / open | Kourovka 群论问题集

(Well-known problem). Does there exist a group which is not almost polycyclic and whose integral group ring is Noetherian?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03243 | KOU-11.4

## Kourovka Notebook Problem 11.4

S / solved | Kourovka 群论问题集

Is it true that the lattice of centralizers in a group is modular if it is a sublattice of the lattice of all subgroups? This is true for finite groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03244 | KOU-11.40

## Kourovka Notebook Problem 11.40

O / open | Kourovka 群论问题集

Prove or disprove that a torsion-free group $G$ with the small cancellation condition $C'(\lambda )$ where $\lambda \ll 1$ necessarily has the ${\mathcal{U}P}$-property (and therefore $KG$ has no zero divisors).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03245 | KOU-11.42

## Kourovka Notebook Problem 11.42

S / solved | Kourovka 群论问题集

Does there exist a torsion-free group having exactly three conjugacy classes and containing a subgroup of index 2?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03246 | KOU-11.43

## Kourovka Notebook Problem 11.43

S / solved | Kourovka 群论问题集

For a finite group $X$, we denote by $k(X)$ the number of its conjugacy classes. Is it true that $k(AB) \leq k(A)k(B)$?

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03247 | KOU-11.44

## Kourovka Notebook Problem 11.44

O / open | Kourovka 群论问题集

For a finite group $X$, we denote by $r(X)$ its sectional rank. Is it true that the sectional rank of a finite $p $-group, which is a product $AB$ of its subgroups $A$ and $ B$, is bounded by some linear function of $r(A)$ and $r(B)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03248 | KOU-11.45

## Kourovka Notebook Problem 11.45

O / open | Kourovka 群论问题集

A $t$-$(v,k,\lambda )$ *design* ${\mathcal{D}}=(X,{\mathcal{B}})$ contains a set $X$ of $v$ points and a set ${\mathcal{B}}$ of $k$-element subsets of $X$ called *blocks* such that each $t$-element subset of $X$ is contained in $\lambda$ blocks. Prove that there are no nontrivial block-transitive 6-designs. (We have shown that there are no nontrivial block-transitive 8-designs and there are certainly some block-transitive, even flag-transitive, 5-designs.)

*Comment of 2009:* In (*Finite Geometry and Combinatorics (Deinze 1992)*, Cambridge Univ. Press, 1993, 103--119) we showed that a block-transitive group $G$ on a nontrivial 6-design is either an affine group $AGL(d,2)$ or is between $PSL(2,q)$ and $P\Gamma L(2,q)$; in (M. Huber, *J. Combin. Theory Ser. A*, **117**, no. 2 (2010), 196--203) it is shown that for the case $\lambda=1$ the group $G$ may only be $P\Gamma L(2,p^e)$, where $p$ is $2$ or $3$ and $e$ is an odd prime power. *Comment of 2013:* In the case $\lambda=1$ there are no block-transitive $7$-designs (M. Huber, *Discrete Math. Theor. Comput. Sci.*, **12**, no. 1 (2010), 123--132). *Comment of 2021:* There are no nontrivial $6$-designs with $k\leq 10^4$ in the case where the automorphism group is almost simple (Q. Tan, W. Liu, J. Chen, *Algebra Colloq.*, **21**, no. 2 (2014), 231--234).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03249 | KOU-11.46

## Kourovka Notebook Problem 11.46

P / partially_solved | Kourovka 群论问题集

a) Does there exist a finite 3-group $G$ of nilpotency class 3 with the property $[a, a^{\varphi}]=1$ for all $a \in G$ and all endomorphisms $\varphi$ of $G$? (See A. Caranti, *J. Algebra*, **97**, no. 1 (1985), 1--13.)

[Archived solved part(s) of 11.46:] b) Does there exist a finite $p $-group of nilpotency class greater than 2, with $\mathrm{Aut} G=\mathrm{Aut}_c G\cdot \mathrm{Inn} G$, where $\mathrm{Aut}_c G$ is the group of central automorphisms of $G$?

c) Does there exist a 2-Engel finite $p $-group $G$ of nilpotency class greater than 2 such that $\mathrm{Aut} G=\mathrm{Aut}_c G\cdot \mathrm{Inn} G$, where $\mathrm{Aut}_c G$ is the group of central automorphisms of $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03250 | KOU-11.47

## Kourovka Notebook Problem 11.47

S / solved | Kourovka 群论问题集

Let ${\mathcal{L}}_d$ be the homogeneous component of degree $d$ in a free Lie algebra ${\mathcal{L}}$ of rank 2 over the field of order 2. What is the dimension of the fixed point space in ${\mathcal{L}}_d$ for the automorphism of ${\mathcal{L}}$ which interchanges two elements of a free generating set of ${\mathcal{L}}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03251 | KOU-11.48

## Kourovka Notebook Problem 11.48

O / open | Kourovka 群论问题集

Is the commutator $[x, y, y, y, y, y, y]$ a product of fifth powers in the free group $\left< x,y\right>$? If not, then the Burnside group $B(2,5)$ is infinite.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03252 | KOU-11.49

## Kourovka Notebook Problem 11.49

O / open | Kourovka 群论问题集

B. Hartley (*Proc. London Math. Soc.*, **35**, no. 1 (1977), 55--75) constructed an example of a non-countable Artinian ${\mathbb{Z}G$-module where $G$ is a metabelian group with the minimum condition for normal subgroups. It follows that there exists a non-countable soluble group (of derived length 3) satisfying Min-$n$. The following question arises in connection with this result and with the study of some classes of soluble groups with the weak minimum condition for normal subgroups. Is an Artinian ${\mathbb{Z}G$-module countable if $G$ is a soluble group of finite rank (in particular, a minimax group)?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03253 | KOU-11.5

### Kourovka Notebook Problem 11.5

S / solved | Kourovka 群论问题集

Let $k$ be a commutative ring and let $G$ be a torsion-free almost polycyclic group. Suppose that $ P$ is a finitely generated projective module over the group ring $kG$ and $P$ contains two elements independent over $kG$. Is $P$ a free module?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03254 | KOU-11.50

### Kourovka Notebook Problem 11.50

O / open | Kourovka 群论问题集

Let $A, C$ be abelian groups. If $A[n]= 0$, i. e. for $a \in A$, $na=0$ implies $a=0$, then the sequence \ $\displaystyle{ \frac{\mathrm{Hom} (C,A)}{n\mathrm{Hom} (C,A)}\; \; \rightarrowtail \mathrm{Hom} \left(\frac{C}{nC},\; \frac{A}{nA}\right) \twoheadrightarrow \mathrm{Ext} (C,A)[n]} $ \ is exact. Given \ $ \displaystyle{f \in \mathrm{Hom}\left(C,\frac{A}{nA}\right)=\mathrm{Hom} \left(\frac{C}{nC},\; \frac{A}{nA}\right) } $ \ the corresponding extension $X_f$ is obtained as a pull-back

$$\displaystyle{ \begin{array}{ccccc}A& \rightarrowtail &X_f& \twoheadrightarrow & C \cr \downarrow &\vphantom{_{_j}}&\downarrow &&\downarrow \cr A& \rightarrowtail &A& \twoheadrightarrow &\displaystyle{\frac{A}{nA}} \end{array}}. $$ Use this scheme to classify certain extensions of $A$ by $C$. The case $nC=0$, $A$ being torsion-free is interesting. Here $\mathrm{Ext} (C,A)[n] = \mathrm{Ext} (C,A)$. (See E. L. Lady, A. Mader, *J. Algebra*, **140** (1991), 36--64.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03255 | KOU-11.51

### Kourovka Notebook Problem 11.51

O / open | Kourovka 群论问题集

Are there (large, non-trivial) classes ${\mathfrak{X}}$ of torsion-free abelian groups such that for $A, C \in {\mathfrak{X}}$ the group $\mathrm{Ext} (C,A)$ is torsion-free? It is a fact (E. L. Lady, A. Mader, *J. Algebra*, **140** (1991), 36--64) that two groups of such a class are nearly isomorphic if and only if they have equal $p $-ranks for all $p$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03256 | KOU-11.52

### Kourovka Notebook Problem 11.52

S / solved | Kourovka 群论问题集

(Well-known problem). A permutation group on a set $\Omega $ is called *sharply doubly transitive* if for any two pairs $(\alpha , \beta )$ and $(\gamma , \delta )$ of elements of $\Omega$ such that $\alpha \ne \beta$ and $\gamma \ne \delta $, there is exactly one element of the group taking $\alpha$ to $\gamma$ and $\beta$ to $\delta$. Does every sharply doubly transitive group possess a non-trivial abelian normal subgroup? A positive answer is well known for finite groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03257 | KOU-11.53

### Kourovka Notebook Problem 11.53

S / solved | Kourovka 群论问题集

(P. Kleidman). Do the sporadic simple groups of Rudvalis $Ru$, Mathieu $M_{22}$, and Higman--Sims $HS$ embed into the simple group $E_7(5)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03258 | KOU-11.54

### Kourovka Notebook Problem 11.54

S / solved | Kourovka 群论问题集

Is it true that in the group of coloured braids only the identity braid is a conjugate to its inverse? (For definition see Archive, 10.24.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03259 | KOU-11.55

### Kourovka Notebook Problem 11.55

S / solved | Kourovka 群论问题集

Is it true that extraction of roots in the group of coloured braids is uniquely determined?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03260 | KOU-11.56

### Kourovka Notebook Problem 11.56

O / open | Kourovka 群论问题集

a) Does every infinite residually finite group contain an infinite abelian subgroup? This is equivalent to the following: does every infinite residually finite group contain a non-identity element with an infinite centralizer?

By a famous theorem of Shunkov a torsion group with an involution having a finite centralizer is a virtually soluble group. Therefore we may assume that in our group all elements have odd order. One should start, perhaps, with the following:

b) Does every infinite residually $p $-group contain an infinite abelian subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03261 | KOU-11.57

### Kourovka Notebook Problem 11.57

S / solved | Kourovka 群论问题集

An *upper composition factor* of a group $G$ is a composition factor of some finite quotient of $ G$. Is there any restriction on the set of non-abelian upper composition factors of a finitely generated group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03262 | KOU-11.58

### Kourovka Notebook Problem 11.58

O / open | Kourovka 群论问题集

Describe the finite groups which contain a tightly embedded subgroup $H$ such that a Sylow 2-subgroup of $H$ is a direct product of a quaternion group of order 8 and a non-trivial elementary group.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03263 | KOU-11.59

### Kourovka Notebook Problem 11.59

O / open | Kourovka 群论问题集

A $TI$-subgroup $A$ of a group $G$ is called a *subgroup of root type* if $[A, A^g]=1$ whenever $N_A(A^g) \ne 1$. Describe the finite groups containing a cyclic subgroup of order 4 as a subgroup of root type.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03264 | KOU-11.6

### Kourovka Notebook Problem 11.6

S / solved | Kourovka 群论问题集

Let $p$ be an odd prime. Is it true that every finite $p $-group possesses a set of generators of equal orders?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03265 | KOU-11.60

### Kourovka Notebook Problem 11.60

O / open | Kourovka 群论问题集

Is it true that the hypothetical Moore graph with 3250 vertices of valence 57 has no automorphisms of order 2? M. Aschbacher (*J. Algebra*, **19**, no. 4 (1971), 538--540) proved that this graph is not a graph of rank 3.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03266 | KOU-11.61

### Kourovka Notebook Problem 11.61

O / open | Kourovka 群论问题集

Let $F$ be a non-abelian free pro-$p $-group. Is it true that the subset $\{ r \in F\mid \mathrm{cd} (F/(r)) \leq 2\} $ is dense in $F$? Here $ (r)$ denotes the closed normal subgroup of $F$ generated by $r$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03267 | KOU-11.62

### Kourovka Notebook Problem 11.62

O / open | Kourovka 群论问题集

Describe the groups over which any equation is soluble. In particular, is it true that this class of groups coincides with the class of torsion-free groups? It is easy to see that a group, over which every equation is soluble, is torsion-free. On the other hand, S. D. Brodskiĭ (*Siberian Math. J.*, **25**, no. 2 (1984), 235--251) showed that any equation is soluble over a locally indicable group.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03268 | KOU-11.63

### Kourovka Notebook Problem 11.63

O / open | Kourovka 群论问题集

Suppose that $G$ is a one-relator group containing non-trivial elements of finite order and $N$ is a subgroup of $G$ generated by all elements of finite order. Is it true that any subgroup of $G$ that intersects $N$ trivially is a free group? One can show that the answer is affirmative in the cases where $ G/N$ has non-trivial centre or satisfies a non-trivial identity.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03269 | KOU-11.64

### Kourovka Notebook Problem 11.64

S / solved | Kourovka 群论问题集

Let $\pi (G)$ denote the set of prime divisors of the order of a finite group $G$. Are there only finitely many finite simple groups $G$, different from alternating groups, which have a proper subgroup $H$ such that $\pi (H)=\pi (G)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03270 | KOU-11.65

### Kourovka Notebook Problem 11.65

O / open | Kourovka 群论问题集

*Conjecture:* any finitely generated soluble torsion-free pro-$p $-group with decidable elementary theory is an analytic pro-$p $-group.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03271 | KOU-11.66

### Kourovka Notebook Problem 11.66

O / open | Kourovka 群论问题集

(Yu. L. Ershov). Is the elementary theory of a free pro-$p $-group decidable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03272 | KOU-11.67

### Kourovka Notebook Problem 11.67

S / solved | Kourovka 群论问题集

Does there exist a torsion-free group with exactly 3 classes of conjugate elements such that no non-trivial conjugate class contains a pair of inverse elements?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03273 | KOU-11.68

### Kourovka Notebook Problem 11.68

S / solved | Kourovka 群论问题集

Can every fully ordered group be embedded in a fully ordered group (continuing the given order) with only 3 classes of conjugate elements?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03274 | KOU-11.69

### Kourovka Notebook Problem 11.69

O / open | Kourovka 群论问题集

A group $G$ acting on a set $\Omega$ will be said to be {\it \mathrm{1}-$\{ 2\}$-transitive} if it acts transitively on the set $ \Omega ^{1,\{ 2\}}=\{ (\alpha , \{ \beta ,\gamma \} )\mid \alpha ,\beta ,\gamma \mbox{ distinct}\} .$ Thus $G$ is 1-$\{ 2\}$-transitive if and only if it is transitive and a stabilizer $G_{\alpha}$ is 2-homogeneous on $\Omega \setminus \{ \alpha \} $. The problem is to classify all (infinite) permutation groups that are 1-$\{ 2\}$-transitive but not 3-transitive.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03275 | KOU-11.7

### Kourovka Notebook Problem 11.7

O / open | Kourovka 群论问题集

a) Find conditions for a group $ G$, given by its presentation, under which the property *some term of the lower central series of $G$ is a free group* $(*)$ implies the residual finiteness of $G$.

b) Find conditions on the structure of a residually finite group $G$ which ensure property $(*)$ for $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03276 | KOU-11.70

### Kourovka Notebook Problem 11.70

P / partially_solved | Kourovka 群论问题集

Let $F$ be an infinite field or a skew-field.

a) Find all transitive subgroups of $ PGL(2,F)$ acting on the projective line $F \cup \{ \infty \} $.

c) What are the flag-transitive subgroups of $PGL(d + 1, F)$?

d) What subgroups of $PGL(d + 1, F)$ are 2-transitive on the points of $PG(d,F)$?

[Archived solved part(s) of 11.70:] b) Let $F$ be an infinite field or a skew-field. What conditions on the subring $R$ of $F$ will ensure that $PGL(d + 1, R)$ is flag-transitive on the projective $d$-space $PG(d,F)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03277 | KOU-11.71

### Kourovka Notebook Problem 11.71

O / open | Kourovka 群论问题集

Let A be a finite group with a normal subgroup $H$. A subgroup $U$ of $H$ is called an *$A$-covering subgroup* of $H$ if $\bigcup\limits_{a\in A} U^a=H$. Is there a function $f\!: {\mathbb{N} \rightarrow {\mathbb{N}$ such that whenever $U< H < A$, where $A$ is a finite group, $H$ is a normal subgroup of $ A$ of index $n$, and $U$ is an $A$-covering subgroup of $H$, the index $|H:U| \leq f(n)$? (We have shown that the answer is ``yes'' if $U$ is a maximal subgroup of $H$.)

*Comments of 2025*: The conjecture is proved in the case where $H$ acts innately transitively on the coset space $[H:U]$ (M. Fusari, A. Previtali, P. Spiga, *J. Group Theory*, **27** (2024), 929--965); \ for $n=3$ with $|H:U|\leq 10$ (L. Gogniat, P. Spiga, *Preprint*, 2025, https://arxiv.org/abs/2502.01287); and if $H=UL$ where $L$ is a minimal $A$-invariant subgroup of $H$ (M. Fusari, S. Harper, P. Spiga, *Bull. Austral. Math. Soc.*, 2025, https://doi.org/10.1017/S0004972725000176).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03278 | KOU-11.72

### Kourovka Notebook Problem 11.72

O / open | Kourovka 群论问题集

Suppose that a variety of groups ${\mathfrak{V}}$ is *non-regular*, that is, the free group $F_{n+1}({\mathfrak{V}})$ is embeddable in $F_n({\mathfrak{V}})$ for some $ n$. Is it true that then every countable group in ${\mathfrak{V}}$ is embeddable in an $n$-generated group in ${\mathfrak{V}}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03279 | KOU-11.73

### Kourovka Notebook Problem 11.73

O / open | Kourovka 群论问题集

If a relatively free group is finitely presented, is it virtually nilpotent?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03280 | KOU-11.74

### Kourovka Notebook Problem 11.74

S / solved | Kourovka 群论问题集

Let $G$ be a non-elementary hyperbolic group and let $G^n$ be the subgroup generated by the $n$th powers of the elements of $G$.

a) (M. Gromov). Is it true that $G/G^n$ is infinite for some $n = n(G)$?

b) Is it true that $\bigcap\limits_{n=1}^{\infty} G^n=\{ 1\} $?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03281 | KOU-11.75

### Kourovka Notebook Problem 11.75

S / solved | Kourovka 群论问题集

Let us consider the class of groups with $n$ generators and $ m$ relators. A subclass of this class is called *dense* if the ratio of the number of presentations of the form $\left< a_1,\ldots ,a_n\mid R_1,\ldots ,R_m\right>$ (where $|R_i|=d_i$) for groups from this subclass to the number of all such presentations converges to 1 when $d_1 + \cdots + d_m$ tends to infinity. Prove that for every $k < m$ and for any $n$ the subclass of groups all of whose $k$-generator subgroups are free is dense.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03282 | KOU-11.76

### Kourovka Notebook Problem 11.76

O / open | Kourovka 群论问题集

(Well-known problem). Is the group of collineations of a finite non-Desarguasian projective plane defined over a semi-field soluble? (The hypothesis on the plane means that the corresponding regular set, see Archive, 10.48, is closed under addition.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03283 | KOU-11.77

### Kourovka Notebook Problem 11.77

O / open | Kourovka 群论问题集

(Well-known problem). Describe the finite translation planes whose collineation groups act doubly transitively on the set of points of the line at infinity.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03284 | KOU-11.78

### Kourovka Notebook Problem 11.78

O / open | Kourovka 群论问题集

An isomorphism of groups of points of algebraic groups is called *semialgebraic* if it can be represented as a composition of an isomorphism of translation of the field of definition and a rational morphism.

a) Is it true that the existence of an isomorphism of groups of points of two directly undecomposable algebraic groups with trivial centres over an algebraically closed field implies the existence of a semialgebraic isomorphism of the groups of points?

b) Is it true that every isomorphism of groups of points of directly undecomposable algebraic groups with trivial centres defined over algebraic fields is semialgebraic? A field is called *algebraic* if all its elements are algebraic over the prime subfield.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03285 | KOU-11.79

### Kourovka Notebook Problem 11.79

S / solved | Kourovka 群论问题集

Let $G$ be a finite group of automorphisms of an infinite field $F$ of characteristic $p$. Taking integral powers of the elements of $F$ and the action of $G$ define the action of the group ring ${\mathbb{Z}G$ of $G$ on the multiplicative group of $F$. Is it true that any subfield of $F$ that contains the images of all elements of $F$ under the action of some fixed element of ${\mathbb{Z}G\setminus p{\mathbb{Z}G$ contains infinitely many $ G$-invariant elements of $F$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03286 | KOU-11.8

### Kourovka Notebook Problem 11.8

P / partially_solved | Kourovka 群论问题集

b) For a finite group $X$, let $\chi _1(X)$ denote the totality of the degrees of all irreducible complex characters of $X$ with allowance for their multiplicities. Suppose that $\chi _1(G)=\chi _1(H)$ for groups $G$ and $H$. Clearly, then $|G|=|H|$. Is it true that $H$ is soluble if $G$ is soluble?

It is known that $H$ is a Frobenius group if $G$ is a Frobenius group.

[Archived solved part(s) of 11.8:] a) For a finite group $X$, let $\chi _1(X)$ denote the totality of the degrees of all irreducible complex characters of $X$ with allowance for their multiplicities. Suppose that $\chi _1(G)=\chi _1(H)$ for groups $G$ and $H$. Clearly, then $|G|=|H|$. Is it true that $H$ is simple if $G$ is simple?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03287 | KOU-11.80

### Kourovka Notebook Problem 11.80

O / open | Kourovka 群论问题集

Let $G$ be a primitive permutation group on a finite set $\Omega$ and suppose that, for $\alpha \in \Omega $, $G_{\alpha}$ acts 2-transitively on one of its orbits in $\Omega \setminus \{ \alpha \} $. By (C. E. Praeger, *J. Austral. Math. Soc. (A)*, **45**, 1988, 66--77) either

(a) $T \leq G \leq \mathrm{Aut} T$ for some nonabelian simple group $T$, or

(b) $G$ has a unique minimal normal subgroup which is regular on $\Omega$.

For what classes of simple groups in (a) is a classification feasible? Describe the examples in as explicit a manner as possible. Classify all groups in (b).

*Remark of 1999:* Two papers by X. G. Fang and C. E. Praeger (*Commun. Algebra*, **27** (1999), 3727--3754 and 3755--3769) show that this is feasible for $T$ a Suzuki and Ree simple group, and a paper of J. Wang and C. E. Praeger (*J. Algebra*, **180** (1996), 808--833) suggests that this may not be the case for $T$ an alternating group. *Remark of 2001:* In (X. G. Fang, G. Havas, J. Wang, *European J. Combin.*, **20**, no. 6 (1999), 551--557) new examples are constructed with $G=PSU(3,q)$. *Remark of 2009:* It was shown in (D. Leemans, *J. Algebra*, **322**, no. 3 (2009), 882--892) that for part (a) classification is feasible for self-paired 2-transitive suborbits with $T$ a sporadic simple group; in this case these suborbits correspond to 2-arc-transitive actions on undirected graphs.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03288 | KOU-11.81

### Kourovka Notebook Problem 11.81

O / open | Kourovka 群论问题集

A topological group is said to be *$F$-balanced* if for any subset $X$ and any neighborhood of the identity $U$ there is a neighborhood of the identity $V$ such that $VX \subseteq XU$. Is every $ F$-balanced group *balanced*, that is, does it have a basis of neighborhoods of the identity consisting of invariant sets?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03289 | KOU-11.82

### Kourovka Notebook Problem 11.82

S / solved | Kourovka 群论问题集

Let $R$ be the normal closure of an element $r$ in a free group $F$ with the natural length function and suppose that $s$ is an element of minimal length in $R$. Is it true that $s$ is conjugate to one of the following elements: $r$, $r^{-1}$, $[r,f]$, $[r^{-1},f]$ for some $f \in F$?

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03290 | KOU-11.83

### Kourovka Notebook Problem 11.83

O / open | Kourovka 群论问题集

Is the conjugacy problem soluble for finitely generated abelian-by-polycyclic groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03291 | KOU-11.84

### Kourovka Notebook Problem 11.84

O / open | Kourovka 群论问题集

Is the isomorphism problem soluble

a) for finitely generated metabelian groups?

b) for finitely generated soluble groups of finite rank?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03292 | KOU-11.85

### Kourovka Notebook Problem 11.85

O / open | Kourovka 群论问题集

Let $F$ be a free pro-$p $-group with a basis $X$ and let $ R=r^F$ be the closed normal subgroup generated by an element $ r$. We say that an element $s$ of $F$ is *associated to $r$* if $s^F=R$.

a) Suppose that none of the elements associated to $r$ is a $p$th power of an element in $F$. Is it true that $F/R$ is torsion-free?

b) Among the elements associated to $r$ there is one that depends on the minimal subset $X'$ of the basis $X$. Let $x \in X'$. Is it true that the images of the elements of $X\setminus \{ x\}$ in $F/R$ freely generate a free pro-$p $-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03293 | KOU-11.86

### Kourovka Notebook Problem 11.86

O / open | Kourovka 群论问题集

Does every group $G=\left< x_1,\ldots ,x_n\mid r_1=\cdots =r_m=1\right> $ possess, in a natural way, a homomorphic image $H=\left< x_1,\ldots ,x_n\mid s_1=\cdots =s_m=1\right> $

a) such that $H$ is a torsion-free group?

b) such that the integral group ring of $H$ is embeddable in a skew field?

If the stronger assertion ``b'' is true, then this will give an explicit method of finding elements $x_{i_1},\ldots ,x_{i_{n-m}}$ which generate a free group in $G$. Such elements exist by N. S. Romanovskiĭ's theorem (*Algebra and Logic*, **16**, no. 1 (1977), 62--67).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03294 | KOU-11.87

### Kourovka Notebook Problem 11.87

O / open | Kourovka 群论问题集

(Well-known problem). Is the automorphism group of a free metabelian group of rank $n \geq 4$ finitely presented?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03295 | KOU-11.88

### Kourovka Notebook Problem 11.88

S / solved | Kourovka 群论问题集

We define the *length* $l(g)$ of an Engel element $g$ of a group $G$ to be the smallest number $l$ such that $[h,g; l]=1$ for all $h \in G$. Here $[h,g; 1]=[h,g]$ and $[h,g; i + 1]=[[h,g; i],g]$. Does there exist a polynomial function $\varphi (x,y)$ such that $l(uv) \leq \varphi (l(u),l(v))$? Up to now, it is unknown whether a product of Engel elements is again an Engel element.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03296 | KOU-11.89

### Kourovka Notebook Problem 11.89

O / open | Kourovka 群论问题集

Let $k$ be an infinite cardinal number. Describe the epimorphic images of the Cartesian power $\prod\limits_{k}{\mathbb{Z}}$ of the group ${\mathbb{Z}}$ of integers. Such a description is known for $k=\omega _0$ (R. Nunke, *Acta Sci. Math. (Szeged)*, **23** (1963), 67--73).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03297 | KOU-11.9

### Kourovka Notebook Problem 11.9

S / solved | Kourovka 群论问题集

(I. I. Pyatetskiĭ-Shapiro). Does there exist a finite non-soluble group $G$ such that the set of characters induced by the trivial characters of representatives of all conjugacy classes of subgroups of $G$ is linearly independent?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03298 | KOU-11.90

### Kourovka Notebook Problem 11.90

O / open | Kourovka 群论问题集

Let ${\mathfrak{V}}$ be a variety of groups and let $k$ be an infinite cardinal number. A group $G \in {\mathfrak{V}}$ of rank $k$ is called {\it almost free in ${\mathfrak{V}}$} if each of its subgroups of rank less than $k$ is contained in a subgroup of $G$ which is free in ${\mathfrak{V}}$. For what $k$ do there exist almost free but not free in ${\mathfrak{B}}$ groups of rank $k$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03299 | KOU-11.91

### Kourovka Notebook Problem 11.91

S / solved | Kourovka 群论问题集

Prove that a hereditary formation ${\mathfrak{F}}$ of finite soluble groups is local if every finite soluble non-simple minimal non-${\mathfrak{F}}$-group is a Shmidt group (that is, a non-nilpotent finite group all of whose proper subgroups are nilpotent).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03300 | KOU-11.92

### Kourovka Notebook Problem 11.92

O / open | Kourovka 群论问题集

What are the soluble hereditary non-one-generator formations of finite groups all of whose proper hereditary subformations are one-generated?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03301 | KOU-11.93

### Kourovka Notebook Problem 11.93

S / solved | Kourovka 群论问题集

Is the variety of all lattices generated by the block lattices (see Archive, 9.62) of finite groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03302 | KOU-11.94

### Kourovka Notebook Problem 11.94

S / solved | Kourovka 群论问题集

Describe all *simply reducible* groups, that is, groups such that all their characters are real and the tensor product of any two irreducible representations contains no multiple components. This question is interesting for physicists. Is every finite simply reducible group soluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03303 | KOU-11.95

### Kourovka Notebook Problem 11.95

O / open | Kourovka 群论问题集

Suppose that $G$ is a $p $-group $G$ containing an element $a$ of order $p$ such that the subgroup $\left< a, a^g\right> $ is finite for any $g$ and the set $C_G(a) \cap a^G$ is finite. Is it true that $G$ has non-trivial centre? This is true for 2-groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03304 | KOU-11.96

### Kourovka Notebook Problem 11.96

S / solved | Kourovka 群论问题集

(a) Is it true that, for a given number $n$, there exist only finitely many finite simple groups each of which contains an involution which commutes with at most $n$ involutions of the group?

(b) Is it true that there are no infinite simple groups satisfying this condition?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03305 | KOU-11.97

### Kourovka Notebook Problem 11.97

S / solved | Kourovka 群论问题集

Are there only finitely many finite simple groups with a given set of all different values of irreducible characters on a single element?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03306 | KOU-11.98

### Kourovka Notebook Problem 11.98

P / partially_solved | Kourovka 群论问题集

a) (R. Brauer). Find the best-possible estimate of the form $|G| \leq f(r)$ where $r$ is the number of conjugacy classes of elements in a finite (simple) group $G$.

[Archived solved part(s) of 11.98:] b) Let $r$ be the number of conjugacy classes of elements in a finite (simple) group $G$. Is it true that $|G|\leq \mathrm{exp}(r)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03307 | KOU-11.99

### Kourovka Notebook Problem 11.99

O / open | Kourovka 群论问题集

Find (in group-theoretic terms) necessary and sufficient conditions for a finite group to have complex irreducible characters having defect 0 for more than one prime number dividing the order of the group. Express the number of such characters in the same terms.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03308 | KOU-12.1

### Kourovka Notebook Problem 12.1

P / partially_solved | Kourovka 群论问题集

a) H. Bass (*Topology*, **4**, no. 4 (1966), 391--400) has constructed explicitly a proper subgroup of finite index in the group of units of the integer group ring of a finite cyclic group. Calculate the index of Bass' subgroup.

[Archived solved part(s) of 12.1:] b) (A. A. Bovdi). H. Bass (*Topology*, **4**, no. 4 (1966), 391--400) has constructed explicitly a proper subgroup of finite index in the group of units of the integer group ring of a finite cyclic group. Construct analogous subgroups for finite abelian non-cyclic groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03309 | KOU-12.10

### Kourovka Notebook Problem 12.10

S / solved | Kourovka 群论问题集

(P. Neumann). Can the free group on two generators be embedded in $\mathrm{Sym} ({\mathbb{N}})$ so that the image of every non-identity element has only a finite number of orbits?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03310 | KOU-12.100

### Kourovka Notebook Problem 12.100

O / open | Kourovka 群论问题集

Is every periodic group with a regular automorphism of order 4 locally finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03311 | KOU-12.101

### Kourovka Notebook Problem 12.101

O / open | Kourovka 群论问题集

We call a group $G$ containing an involution $i$ a *$T_0 $-group* if

1) the order of the product of any two involutions conjugate to $i$ is finite;

2) all 2-subgroups of $G$ are either cyclic or generalized quaternion;

3) the centralizer $C$ of the involution $i$ in $G$ is infinite, distinct from $G$, and has finite periodic part;

4) the normalizer of any non-trivial $i $-invariant finite subgroup in $G$ either is contained in $C$ or has periodic part which is a Frobenius group (see 6.55) with abelian kernel and finite complement of even order;

5) for every element $c$ not contained in $C$ for which $ci$ is an involution there is an element $s$ of $C$ such that $\left< c,c^s\right>$ is an infinite subgroup.

Does there exist a simple $T_0 $-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03312 | KOU-12.102

### Kourovka Notebook Problem 12.102

S / solved | Kourovka 群论问题集

Is every proper factor-group of a group of Golod (see 9.76) residually finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03313 | KOU-12.11

### Kourovka Notebook Problem 12.11

O / open | Kourovka 群论问题集

Suppose that $G, H$ are countable or finite groups and $A$ is a proper subgroup of $G$ and $H$ containing no non-trivial subgroup normal in both $G$ and $H$. Can $G*_A H$ be embedded in $\mathrm{Sym} ({\mathbb{N}})$ so that the image is highly transitive?

*Comment of 2013:* for partial results, see (S. V. Gunhouse, *Highly transitive representations of free products on the natural numbers*, Ph. D. Thesis Bowling Green State Univ., 1993; \ K. K. Hickin, *J. London Math. Soc. (2)*, **46**, no. 1 (1992), 81--91).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03314 | KOU-12.12

### Kourovka Notebook Problem 12.12

O / open | Kourovka 群论问题集

Is the conjugacy problem for nilpotent finitely generated lattice-ordered groups soluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03315 | KOU-12.13

### Kourovka Notebook Problem 12.13

O / open | Kourovka 群论问题集

If $\left< \Omega , \leq \right>$ is the countable universal poset, then $G=\mathrm{Aut} (\left< \Omega , \leq \right> )$ is simple (A. M. W. Glass, S. H. McCleary, M. Rubin, *Math. Z.*, **214**, no. 1 (1993), 55--66). If $H$ is a subgroup of $G$ such that $|G:H| < 2^{\aleph _0}$ and $H$ is transitive on $\Omega$, does $H=G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03316 | KOU-12.14

### Kourovka Notebook Problem 12.14

S / solved | Kourovka 群论问题集

If $T$ is a countable theory, does there exist a model ${\mathfrak{A}}$ of $T$ such that the theory of $\mathrm{Aut} ({\mathfrak{A}})$ is undecidable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03317 | KOU-12.15

### Kourovka Notebook Problem 12.15

S / solved | Kourovka 群论问题集

Suppose that, in a finite 2-group $G$, any two elements are conjugate whenever their normal closures coincide. Is it true that the derived subgroup of $G$ is abelian?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03318 | KOU-12.16

### Kourovka Notebook Problem 12.16

O / open | Kourovka 群论问题集

Is the class of groups of recursive automorphisms of arbitrary models closed with respect to taking free products?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03319 | KOU-12.17

### Kourovka Notebook Problem 12.17

O / open | Kourovka 群论问题集

Find a description of autostable periodic abelian groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03320 | KOU-12.19

### Kourovka Notebook Problem 12.19

O / open | Kourovka 群论问题集

Is it true that, for every $n \geq 2$ and every two epimorphisms $\varphi$ and $\psi$ of a free group $F_{2n}$ of rank $2n$ onto $F_n \times F_n$, there exists an automorphism $\alpha$ of $F_{2n}$ such that $\alpha \varphi =\psi $?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03321 | KOU-12.2

### Kourovka Notebook Problem 12.2

S / solved | Kourovka 群论问题集

Is it true that every non-discrete group topology (in an abelian group) can be strengthened up to a maximal complete group topology?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03322 | KOU-12.20

### Kourovka Notebook Problem 12.20

O / open | Kourovka 群论问题集

(Well-known problem). Is R. Thompson's group $$F=\left< x_0, x_1,\ldots \mid x_n^{x_i}= x_{n+1}, \; \; i < n,\;\;n=1, 2, \ldots \right> =$$ $$= \left< x_0,\ldots ,x_4\mid x_1^{x_0}=x_2, \; \; x_2^{x_0}=x_3, \; \; x_2^{x_1}=x_3, \; \; x_3^{x_1}=x_4, \; \; x_3^{x_2}=x_4\right> $$ amenable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03323 | KOU-12.21

### Kourovka Notebook Problem 12.21

O / open | Kourovka 群论问题集

Let $R$ be a commutative ring with identity and let $G$ be a finite group. Prove that the ring $a(RG)$ of $R $-representations of $G$ has no non-trivial idempotents.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03324 | KOU-12.22

### Kourovka Notebook Problem 12.22

S / solved | Kourovka 群论问题集

Let $\Delta (G)$ be the augmentation ideal of the integer group ring of an arbitrary group $G$. Then $D_n(G)=G \cap (1 + \Delta ^n(G))$ contains the $n $th lower central subgroup $\gamma _n(G)$ of $G$.

a) Is it true that $D_n(G)/\gamma_n(G)$ is central in $G/\gamma _n(G)$?

b) Is it true that $D_n(G)/\gamma _n(G)$ has exponent dividing 2?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03325 | KOU-12.23

### Kourovka Notebook Problem 12.23

O / open | Kourovka 群论问题集

The *index of permutability* of a group $G$ is defined to be the minimal integer $k \geq 2$ such that for each $k $-tuple $x_1,\ldots ,x_k$ of elements in $G$ there is a non-identity permutation $\sigma$ on $k$ symbols such that $x_1\cdots x_k=x_{\sigma (1)}\cdots x_{\sigma (k)}$. Determine the index of permutability of the symmetric group ${\mathbb{S}_n$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03326 | KOU-12.24

### Kourovka Notebook Problem 12.24

S / solved | Kourovka 群论问题集

Given a ring $R$ with identity, the automorphisms of $R[[x]]$ sending $x$ to $x\left( 1 + \sum_{i=1}^{\infty} a_ix^i\right) $, $a_i\in R$, form a group $N(R)$. We know that $N({\mathbb{Z})$ contains a copy of the free group $F_2$ of rank 2 and, from work of A. Weiss, that $N({\mathbb{Z}/p{\mathbb{Z})$ contains a copy of every finite $p $-group (but not of ${\mathbb{Z}_{p^{\infty}})$, $p$ a prime. Does $N({\mathbb{Z}/p{\mathbb{Z})$ contain a copy of $F_2$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03327 | KOU-12.25

### Kourovka Notebook Problem 12.25

S / solved | Kourovka 群论问题集

Let $G$ be a finite group acting irreducibly on a vector space $V$. An orbit $\alpha ^G$ for $\alpha \in V$ is said to be *$p$-regular* if the stabilizer of $\alpha$ in $G$ is a $p' $-subgroup. Does $G$ have a regular orbit on $V$ if it has a $p $-regular orbit for every prime $p$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03328 | KOU-12.26

### Kourovka Notebook Problem 12.26

S / solved | Kourovka 群论问题集

(Shi Shengming). Is it true that a finite $p $-soluble group $G$ has a $p $-block of defect zero if and only if there exists an element $x \in O_{p'}(G)$ such that $C_G(x)$ is a $p' $-subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03329 | KOU-12.27

### Kourovka Notebook Problem 12.27

O / open | Kourovka 群论问题集

Let $G$ be a simple locally finite group. We say that $G$ is a *group of finite type* if there is a non-trivial permutational representation of $G$ such that some finite subgroup of $G$ has no regular orbits. Investigate and, perhaps, classify the simple locally finite groups of finite type. Partial results see in (B. Hartley, A. Zalesski, *Isr. J. Math.*, **82** (1993), 299--327, *J. London Math. Soc.*, **55** (1997), 210--230; \ F. Leinen, O. Puglisi, *Illinois J. Math.*, **47** (2003), 345--360).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03330 | KOU-12.28

### Kourovka Notebook Problem 12.28

O / open | Kourovka 群论问题集

Let $G$ be a group. A function $f\! : G \rightarrow {\mathbb{C}$ is called \begin{enumerate}

\item[1)] *normed* if $f(1)=1$;

\item[2)] *central* if $f(gh)=f(hg)$ for all $g,h \in G$;

\item[3)] *positive-definite* if $\sum\limits_{k, l}f(g_k^{-1}g_l) \overline{\! c}_kc_l \geq 0$ for any $g_1,\ldots ,g_n \in G$ and any $c_1,\ldots ,c_n \in {\mathbb{C}$. \end{enumerate}

Classify the infinite simple locally finite groups $G$ which possess functions satisfying 1)--3). The simple Chevalley groups are known to have no such functions, while such functions exist on locally matrix (or stable) classical groups over finite fields. The question is motivated by the theory of $C^{\displaystyle{*}} $-algebras, see $\S$ 9 in (A. M. Vershik, S. V. Kerov, *J. Sov. Math.*, **38** (1987), 1701--1733). Partial results see in (F. Leinen, O. Puglisi, *J. Pure Appl. Algebra*, **208** (2007), 1003--1021, *J. London Math. Soc.*, **70** (2004), 678--690).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03331 | KOU-12.29

### Kourovka Notebook Problem 12.29

O / open | Kourovka 群论问题集

Classify the locally finite groups for which the augmentation ideal of the complex group algebra is a simple ring. The problem goes back to I. Kaplansky (1965). For partial results, see (K. Bonvallet, B. Hartley, D. S. Passman, M. K. Smith, *Proc. Amer. Math. Soc.*, **56**, no. 1 (1976), 79--82; \ A. E. Zalesskiĭ, *Algebra i Analiz*, **2**, no. 6 (1990), 132--149 (Russian); \ Ch. Praeger, A. E. Zalesskii, *Proc. London Math. Soc.*, **70**, no. 2 (1995), 313--335; \ B. Hartley, A. Zalesski, *J. London Math. Soc.*, **55** (1997), 210--230).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03332 | KOU-12.3

### Kourovka Notebook Problem 12.3

O / open | Kourovka 群论问题集

A $p $-group is called *thin* if every set of pairwise incomparable (by inclusion) normal subgroups contains $\leq p + 1$ elements. Is the number of thin pro-$p $-groups finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03333 | KOU-12.30

### Kourovka Notebook Problem 12.30

O / open | Kourovka 群论问题集

(O. N. Golovin). On the class of all groups, do there exist associative operations which satisfy the postulates of MacLane and Mal'cev (that is, which are free functorial and hereditary) and which are different from taking free and direct products?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03334 | KOU-12.31

### Kourovka Notebook Problem 12.31

S / solved | Kourovka 群论问题集

For relatively free groups $G$, prove or disprove the following conjecture of P. Hall: if a word $v$ takes only finitely many values on $G$ then the verbal subgroup $vG$ is finite.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03335 | KOU-12.32

### Kourovka Notebook Problem 12.32

S / solved | Kourovka 群论问题集

Prove an analogue of Higman's theorem for the Burnside variety ${\mathfrak{B}}_n$ of groups of odd exponent $n \gg 1$, that is, prove that every recursively presented group of exponent $n$ can be embedded in a finitely presented (in ${\mathfrak{B}}_n$) group of exponent $n$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03336 | KOU-12.33

### Kourovka Notebook Problem 12.33

O / open | Kourovka 群论问题集

Suppose that $G$ is a finite group and $x$ is an element of $G$ such that the subgroup $\left< x,y\right>$ has odd order for any $y$ conjugate to $x$ in $G$. Prove, without using CFSG, that the normal closure of $x$ in $G$ is a group of odd order.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03337 | KOU-12.34

### Kourovka Notebook Problem 12.34

O / open | Kourovka 群论问题集

Describe the finite groups $G$ such that the sum of the cubes of the degrees of all irreducible complex characters is at most $|G|\cdot \log _2|G|$. The question is interesting for applications in the theory of signal processing.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03338 | KOU-12.35

### Kourovka Notebook Problem 12.35

O / open | Kourovka 群论问题集

Suppose that ${\mathfrak{F}}$ is a radical composition formation of finite groups. Prove that $\left< H,K\right> ^{{\mathfrak{F}}} =\left< H^{{\mathfrak{F}}}, K^{{\mathfrak{F}}}\right>$ for every finite group $G$ and any subnormal subgroups $H$ and $K$ of $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03339 | KOU-12.36

### Kourovka Notebook Problem 12.36

S / solved | Kourovka 群论问题集

Let $p$ be a prime, $V$ an $n $-dimensional vector space over the field of $p$ elements, and let $G$ be a subgroup of $GL(V)$. Let $S=S[V ^{\displaystyle{*}} ]$ be the symmetric algebra on $V ^{\displaystyle{*}} $, the dual of $V$. Let $T=S^G$ be the ring of invariants and let $b_m$ be the dimension of the homogeneous component of degree $m$. Then the Poincar\'e series $\sum\limits_{m\geq 0}b_mt^m$ is a rational function with a Laurent power series expansion $\sum\limits_{i\geq - n}a_i(1 - t)^i$ about $t=1$ where $a_{- n}=\frac{\displaystyle{1}}{\displaystyle{|G|}}$.

*Conjecture:* $a_{- n+1}= \frac{\displaystyle{r}}{\displaystyle{2|G|}}$ where $r=\sum\limits_{W} ((p - 1)\alpha _W + s_W - 1)$, the sum is taken over all maximal subspaces $W$ of $V$, and $\alpha _W$, $s_W$ are defined by $|G_W|=p^{\alpha _W}\cdot s_W$ where $p\nmid s_W$ and $G_W$ denotes the pointwise stabilizer of $W$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03340 | KOU-12.37

### Kourovka Notebook Problem 12.37

S / solved | Kourovka 群论问题集

(J. G. Thompson). For a finite group $G$ and natural number $n$, set $G(n)=\{ x \in G\mid x^n=1\} $ and define the *type* of $G$ to be the function whose value at $n$ is the order of $G(n)$. Is it true that a group is soluble if its type is the same as that of a soluble one?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03341 | KOU-12.38

### Kourovka Notebook Problem 12.38

S / solved | Kourovka 群论问题集

(J. G. Thompson). For a finite group $G$, we denote by $N(G)$ the set of all orders of the conjugacy classes of $G$. Is it true that if $G$ is a finite non-abelian simple group, $H$ a finite group with trivial centre and $N(G)=N(H)$, then $G$ and $H$ are isomorphic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03342 | KOU-12.39

### Kourovka Notebook Problem 12.39

S / solved | Kourovka 群论问题集

(W. J. Shi). Must a finite group and a finite simple group be isomorphic if they have equal orders and the same set of orders of elements?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03343 | KOU-12.4

### Kourovka Notebook Problem 12.4

O / open | Kourovka 群论问题集

Let $G$ be a group and assume that we have $[x,y]^3=1$ for all $x,y \in G$. Is $G'$ of finite exponent? Is $G$ soluble? By a result of N. D. Gupta and N. S. Mendelssohn, 1967, we know that $G'$ is a 3-group.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03344 | KOU-12.40

### Kourovka Notebook Problem 12.40

P / partially_solved | Kourovka 群论问题集

Let $\varphi$ be an irreducible $p $-modular character of a finite group $G$. Find the best-possible estimate of the form $\varphi (1)_p \leq f(|G|_p)$, where $n_p$ is the $p $-part of a positive integer $n$,

a) for $p=2$;

b) for $p\ne 2$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03345 | KOU-12.41

### Kourovka Notebook Problem 12.41

S / solved | Kourovka 群论问题集

Let $F$ be a free group on two generators $x,y$ and let $\varphi$ be the automorphism of $F$ defined by $x \rightarrow y$, $y \rightarrow xy$. Let $G$ be a semidirect product of $F/(F''(F')^2)$ and $\left< \varphi \right>$. Then $G$ is just-non-polycyclic. What is the cohomological dimension of $G$ over ${\mathbb{Q}}$? (It is either 3 or 4.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03346 | KOU-12.42

### Kourovka Notebook Problem 12.42

S / solved | Kourovka 群论问题集

Describe the automorphisms of the Sylow $p $-subgroup of a Chevalley group of normal type over ${\mathbb{Z}/p^m{\mathbb{Z}} $, $m \geq 2$, where $p$ is a prime.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03347 | KOU-12.43

### Kourovka Notebook Problem 12.43

O / open | Kourovka 群论问题集

(Well-known problem). Does there exist an infinite finitely-generated residually finite $p $-group such that each subgroup is either finite or of finite index?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03348 | KOU-12.44

### Kourovka Notebook Problem 12.44

S / solved | Kourovka 群论问题集

(P. Hall). Is there a non-trivial group which is isomorphic with every proper extension of itself by itself?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03349 | KOU-12.45

### Kourovka Notebook Problem 12.45

S / solved | Kourovka 群论问题集

(P. Hall). Must a non-trivial group, which is isomorphic to each of its non-trivial normal subgroups, be either free of infinite rank, simple, or infinite cyclic? (Lennox, Smith and Wiegold, 1992, have shown that a finitely generated group of this kind which has a proper normal subgroup of finite index is infinite cyclic.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03350 | KOU-12.46

### Kourovka Notebook Problem 12.46

S / solved | Kourovka 群论问题集

Let $F$ be the nonabelian free group on two generators $x,y$. For $a,b \in {\mathbb{C}$, $|a|=|b|=1$, let $\vartheta _{a,b}$ be the automorphism of ${\mathbb{C}F$ defined by $\vartheta _{a,b}(x)=ax$, $\vartheta _{a,b}(y)=by$. Given $0 \ne \alpha \in {\mathbb{C}F$, can we always find $a,b \in {\mathbb{C}\setminus \{ 1\} $ with $|a|=|b|=1$ such that $\alpha {\mathbb{C}F \cap \vartheta _{a,b}(\alpha ){\mathbb{C}F \ne 0$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03351 | KOU-12.47

### Kourovka Notebook Problem 12.47

S / solved | Kourovka 群论问题集

Let $k$ be a field, let $p$ be a prime, and let $G$ be the Wreath product ${\mathbb{Z}_p\wr {\mathbb{Z}$ (so the base group has exponent $p$). Does $kG$ have a classical quotient ring? (i. e. do the non-zero-divisors of $kG$ form an Ore set?)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03352 | KOU-12.48

### Kourovka Notebook Problem 12.48

O / open | Kourovka 群论问题集

Let $G$ be a sharply doubly transitive permutation group on a set $\Omega$ (see Archive 11.52 for a definition).

(a) Does $G$ possess a regular normal subgroup if a point stabilizer is locally finite?

(b) Does $G$ possess a regular normal subgroup if a point stabilizer has an abelian subgroup of finite index?

*Comments of 2022*: an affirmative answer to part (b) is obtained in permutation characteristic 0 (F. O. Wagner, *Preprint*, 2022, https://hal.archives-ouvertes.fr/hal-03590818).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03353 | KOU-12.49

### Kourovka Notebook Problem 12.49

S / solved | Kourovka 群论问题集

Construct all non-split extensions of elementary abelian 2-groups $V$ by $H=PSL_2(q)$ for which $H$ acts irreducibly on $V$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03354 | KOU-12.5

### Kourovka Notebook Problem 12.5

S / solved | Kourovka 群论问题集

Does there exist a countable non-trivial filter in the lattice of quasivarieties of metabelian torsion-free groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03355 | KOU-12.50

### Kourovka Notebook Problem 12.50

O / open | Kourovka 群论问题集

(Well-known problem). Find an algorithm which decides, by a given finite set of matrices in $SL_3({\mathbb{Z}})$, whether the first matrix of this set is contained in the subgroup generated by the remaining matrices. An analogous problem for $SL_4({\mathbb{Z}})$ is insoluble since a direct product of two free groups of rank 2 embeds into $SL_4({\mathbb{Z}})$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03356 | KOU-12.51

### Kourovka Notebook Problem 12.51

S / solved | Kourovka 群论问题集

Does the free group $F_{\eta}$ ($1 < \eta < \infty$) have a finite subset $S$ for which there is a unique total order of $F_{\eta}$ making all elements of $S$ positive? Equivalently, does the free representable $l $-group of rank $\eta$ have a basic element? (The analogs for right orders of $F_{\eta}$ and free $l $-groups have negative answers.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03357 | KOU-12.52

### Kourovka Notebook Problem 12.52

O / open | Kourovka 群论问题集

Is the free $l $-group ${\mathcal{F}}_{\eta}$ of rank $\eta$ ($1 < \eta < \infty$) Hopfian? That is, are $l $-homomorphisms from ${\mathcal{F}}_{\eta}$ onto itself necessarily one-to-one?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03358 | KOU-12.53

### Kourovka Notebook Problem 12.53

O / open | Kourovka 群论问题集

Is it decidable whether or not two elements of a free $l $-group are conjugate?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03359 | KOU-12.54

### Kourovka Notebook Problem 12.54

O / open | Kourovka 群论问题集

Is there a normal valued $l $-group $G$ for which there is no abelian $l $-group $A$ with $C(A) \cong C(G)$? (Here $C(G)$ denotes the lattice of convex $l $-subgroups of $G$. If $G$ is not required to be normal valued, this question has an affirmative answer.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03360 | KOU-12.55

### Kourovka Notebook Problem 12.55

O / open | Kourovka 群论问题集

Let $f(n,p)$ be the number of groups of order $p^n$. Is $f(n,p)$ an increasing function of $p$ for any fixed $n \geq 5$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03361 | KOU-12.56

### Kourovka Notebook Problem 12.56

O / open | Kourovka 群论问题集

Let $\left< X\mid R\right>$ be a finite presentation (the words in $R$ are assumed cyclically reduced). Define the *length* of the presentation to be the sum of the number of generators and the lengths of the relators. Let $F(n)$ be the number of (isomorphism types of) groups that have a presentation of length at most $n$. What can one say about the function $F(n)$? It can be shown that $F(n)$ is not recursive (D. Segal) and it is at least exponential (L. Pyber).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03362 | KOU-12.57

### Kourovka Notebook Problem 12.57

O / open | Kourovka 群论问题集

Suppose that a cyclic group $A$ of order 4 is a $TI $-subgroup of a finite group $G$.

a) Does $A$ centralize a component $L$ of $G$ if $A$ intersects $L\cdot C(L)$ trivially?

b) What is the structure of the normal closure of $A$ in the case where $A$ centralizes each component of $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03363 | KOU-12.58

### Kourovka Notebook Problem 12.58

O / open | Kourovka 群论问题集

A *generalized quadrangle* $GQ(s,t)$ with parameters $s, t$ is by definition an incidence system consisting of points and lines in which every line consists of $s + 1$ points, every two different lines have at most one common point, each point belongs to $t + 1$ lines, and for any point $a$ not lying on a line $L$ there is a unique line containing $a$ and intersecting $L$.

a) Does $GQ(4,11)$ exist?

b) Can the automorphism group of a hypothetical $GQ(5,7)$ contain involutions?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03364 | KOU-12.59

### Kourovka Notebook Problem 12.59

O / open | Kourovka 群论问题集

Does there exist a strongly regular graph with parameters (85, 14, 3, 2) and a non-connected neighborhood of some vertex?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03365 | KOU-12.6

### Kourovka Notebook Problem 12.6

O / open | Kourovka 群论问题集

Let $G$ be a finitely generated group and suppose that $H$ is a $p $-subgroup of $G$ such that $H$ contains no non-trivial normal subgroups of $G$ and $HX=XH$ for any subgroup $ X$ of $G$. Is then $G/C_G(H^G)$ a $p $-group where $H^G$ is the normal closure of $H$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03366 | KOU-12.60

### Kourovka Notebook Problem 12.60

O / open | Kourovka 群论问题集

Describe the strongly regular graphs in which the neighborhoods of vertices are generalized quadrangles (see 12.58).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03367 | KOU-12.62

### Kourovka Notebook Problem 12.62

O / open | Kourovka 群论问题集

A *Frobenius group* is a transitive permutation group in which the stabilizer of any two points is trivial. Does there exist a Frobenius group of infinite degree which is primitive as permutation group, and in which the stabilizer of a point is cyclic and has only finitely many orbits?

*Conjecture:* No. Note that such a group cannot be 2-transitive (Gy. K\'arolyi, S. J. Kov\'acs, P. P. P\'alfy, *Aequationes Math.*, **39** (1990), 161--166; \ V. D. Mazurov, *Siberian Math. J.*, **31** (1990), 615--617; \ P. M. Neumann, P. J. Rowley, in: *Geometry and cohomology in group theory $($London Math. Soc. Lect. Note Ser.*, **252**), Cambridge Univ. Press, 1998, 291--295).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03368 | KOU-12.63

### Kourovka Notebook Problem 12.63

O / open | Kourovka 群论问题集

Does there exist a soluble permutation group of infinite degree that has only finitely many orbits on triples? *Conjecture:* No.

*Comment of 2001:* It is proved (D. Macpherson, *in: Advances in algebra and model theory $($Algebra Logic Appl.*, **9**), Gordon \& Breach, Amsterdam, 1997, 87--92) that every infinite soluble permutation group has infinitely many orbits on quadruples.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03369 | KOU-12.64

### Kourovka Notebook Problem 12.64

O / open | Kourovka 群论问题集

Is it true that for a given number $k \geq 2$ and for any (prime) number $n$, there exists a number $N=N(k,n)$ such that every finite group with generators $A=\{ a_1,\ldots ,a_k\} $ has exponent $\leq n$ if $(x_1\cdots x_N)^n=1$ for any $x_1,\ldots ,x_N \in A \cup \{ 1\}$?

For given $k$ and $n$, a negative answer implies, for example, the infiniteness of the free Burnside group $B(k,n)$, and a positive answer, in the case of sufficiently large $n$, gives, for example, an opportunity to find a hyperbolic group which is not residually finite (and in this group a hyperbolic subgroup of finite index which has no proper subgroups of finite index).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03370 | KOU-12.65

### Kourovka Notebook Problem 12.65

S / solved | Kourovka 群论问题集

Let ${\mathcal{P}}=(P_0,P_1,P_2)$ be a parabolic system in a finite group $G$, belonging to the $C_3$ Coxeter diagram \ $\parbox[t]{2ex}{ $ \circ $ \\ \mathstrut $ {}^1$} \rule[0.5ex]{8ex}{0.1ex} \; \parbox[t]{2ex}{ $ \circ $ \\ \mathstrut $ {}^2$} \underline{\rule[0.5ex]{8ex{0.1ex}}} \; \parbox[t]{2ex}{ $ \circ $ \\ \mathstrut $ {}^3$} $ \ and let the Borel subgroup have index at least 3 in $P_0$ and $P_1$. It is known that, if we furthermore assume that the chamber system of ${\mathcal{P}}$ is geometric and that the projective planes arising as $\{ 0,1\} $-residues from ${\mathcal{P}}$ are desarguesian, then either $G^{(\infty )}$ is a Chevalley group of type $C_3$ or $B_3$ or $G=A_7$. Can we obtain the same conclusion in general, without assuming the previous two hypothesis?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03371 | KOU-12.66

### Kourovka Notebook Problem 12.66

O / open | Kourovka 群论问题集

Describe the finite translation planes whose collineation groups act doubly transitively on the affine points.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03372 | KOU-12.67

### Kourovka Notebook Problem 12.67

O / open | Kourovka 群论问题集

Describe the structure of the locally compact soluble groups with the maximum (minimum) condition for closed non-compact subgroups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03373 | KOU-12.68

### Kourovka Notebook Problem 12.68

O / open | Kourovka 群论问题集

Describe the locally compact abelian groups in which any two closed subgroups of finite rank generate a subgroup of finite rank.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03374 | KOU-12.69

### Kourovka Notebook Problem 12.69

O / open | Kourovka 群论问题集

Let $F$ be a countably infinite field and $G$ a finite group of automorphisms of $F$. The group ring ${\mathbb{Z}G$ acts on $F$ in a natural way. Suppose that $S$ is a subfield of $F$ satisfying the following property: for any $x \in F$ there is a non-zero element $f \in {\mathbb{Z}G$ such that $x^f \in S$. Is it true that either $F=S$ or the field extension $F/S$ is purely inseparable? One can show that for an uncountable field an analogous question has an affirmative answer.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03375 | KOU-12.7

### Kourovka Notebook Problem 12.7

S / solved | Kourovka 群论问题集

Is it true that every radical hereditary formation of finite groups is a composition one?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03376 | KOU-12.70

### Kourovka Notebook Problem 12.70

S / solved | Kourovka 群论问题集

Let $p$ be a prime number, $F$ a free pro-$p $-group of finite rank, and $\Delta \ne 1$ an automorphism of $F$ whose order is a power of $p$. Is the rank of $\mathrm{Fix}_F(\Delta )=\{ x \in F\mid \Delta (x)=x\}$ finite? If the order of $\Delta$ is prime to $p$, then $\mathrm{Fix}_F(\Delta )$ has infinite rank (W. Herfort, L. Ribes, *Proc. Amer. Math. Soc.*, **108** (1990), 287--295).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03377 | KOU-12.71

### Kourovka Notebook Problem 12.71

S / solved | Kourovka 群论问题集

Let $d(G)$ denote the smallest cardinality of a generating set of the group $G$. Let $A$ and $B$ be finite groups. Is there a finite group $G$ such that $A,B \leq G$, $G=\left< A,B\right>$, and $d(G)=d(A) + d(B)$? The corresponding question has negative answer in the class of solvable groups (L. G. Kov\'acs, H.-S. Sim, *Indag. Math.*, **2**, no. 2 (1991), 229--232).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03378 | KOU-12.72

### Kourovka Notebook Problem 12.72

O / open | Kourovka 群论问题集

Let ${\mathfrak{F}}$ be a soluble local hereditary formation of finite groups. Prove that ${\mathfrak{F}}$ is radical if every finite soluble minimal non-${\mathfrak{F}}$-group $G$ is a minimal non-${\mathfrak{N}}^{l(G)-1}$-group. Here ${\mathfrak{N}}$ is the formation of all finite nilpotent groups, and $l(G)$ is the nilpotent length of $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03379 | KOU-12.73

### Kourovka Notebook Problem 12.73

O / open | Kourovka 群论问题集

A formation ${\mathfrak{H}}$ of finite groups is said to have *length* $t$ if there is a chain of formations $\varnothing ={\mathfrak{H}}_0 \subset {\mathfrak{H}}_1 \subset \cdots \subset {\mathfrak{H}}_t={\mathfrak{H}}$ in which ${\mathfrak{H}}_{i-1}$ is a maximal subformation of ${\mathfrak{H}}_i$. Is the lattice of soluble formations of length $\leq 4$ distributive?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03380 | KOU-12.74

### Kourovka Notebook Problem 12.74

S / solved | Kourovka 群论问题集

Let ${\mathfrak{F}}$ be a non-primary one-generator composition formation of finite groups. Is it true that if ${\mathfrak{F}}={\mathfrak{M}}{\mathfrak{H}}$ and the formations ${\mathfrak{M}}$ and ${\mathfrak{H}}$ are non-trivial, then ${\mathfrak{M}}$ is a composition formation?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03381 | KOU-12.75

### Kourovka Notebook Problem 12.75

O / open | Kourovka 群论问题集

(B. Jonsson). Is the class $N$ of the lattices of normal subgroups of groups a variety? It is proved (C. Herrman, W. Poguntke, *Algebra Univers.*, **4**, no. 3 (1974), 280--286) that $N$ is an infinitely based quasivariety.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03382 | KOU-12.76

### Kourovka Notebook Problem 12.76

S / solved | Kourovka 群论问题集

Is every group generated by a set of 3-transpositions locally finite? A *set of $3$-transpositions* is, by definition, a normal set of involutions such that the orders of their pairwise products are at most 3.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03383 | KOU-12.77

### Kourovka Notebook Problem 12.77

S / solved | Kourovka 群论问题集

(Well-known problem). Does the order (if it is greater than $p^2$) of a finite non-cyclic $p$-group divide the order of its automorphism group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03384 | KOU-12.78

### Kourovka Notebook Problem 12.78

S / solved | Kourovka 群论问题集

(M. J. Curran). a) Does there exist a group of order $p^6$ (here $p$ is a prime number), whose automorphism group has also order $p^6$?

b) Is it true that for $p \equiv 1 (\mathrm{mod} 3)$, the smallest order of a $p$-group that is the automorphism group of a $p$-group is $p^7$?

c) The same question for $p=3$ with $3^9$ replacing $p^7$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03385 | KOU-12.79

### Kourovka Notebook Problem 12.79

S / solved | Kourovka 群论问题集

Suppose that $a$ and $b$ are two elements of a finite group $G$ such that the function $\varphi (g)=1^G(g) - 1_{\left< a\right>}^G(g) - 1_{\left< b\right>}^G(g) - 1_{\left< ab\right>}^G(g) + 2$ is a character of $G$. Is it true that $G=\left< a,b\right>$? The converse statement is true.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03386 | KOU-12.8

### Kourovka Notebook Problem 12.8

O / open | Kourovka 群论问题集

Let ${\mathcal{V}}$ be a non-trivial variety of groups and let $a_1,\ldots ,a_r$ freely generate a free group $F_r(V)$ in ${\mathcal{V}}$. We say $F_r({\mathcal{V}})$ *strongly discriminates* ${\mathcal{V}}$ just in case every finite system of inequalities $w_i(a_1,\ldots ,a_r,x_1,\ldots ,x_k) \ne 1$ for $1 \leq i \leq n$ having a solution in some $F_s({\mathcal{V}})$ containing $F_r({\mathcal{V}})$ as a varietally free factor in the sense of ${\mathcal{V}}$, already has a solution in $F_r({\mathcal{V}})$. Does there exist ${\mathcal{V}}$ such that for some integer $r > 0$, $F_r({\mathcal{V}})$ discriminates but does not strongly discriminate ${\mathcal{V}}$? What about the analogous question for general algebras in the context of universal algebra?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03387 | KOU-12.80

### Kourovka Notebook Problem 12.80

S / solved | Kourovka 群论问题集

(K. W. Roggenkamp). a) Is it true that the number of $p$-blocks of defect 0 of a finite group $G$ is equal to the number of the conjugacy classes of elements $g \in G$ such that the number of solutions of the equation $[x,y]=g$ in $G$ is not divisible by $p$?

b) The same question in the case of $G$ being a simple group.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03388 | KOU-12.81

### Kourovka Notebook Problem 12.81

O / open | Kourovka 群论问题集

What is the cardinality of the set of subvarieties of the group variety ${\mathfrak{A}}_p^3$ (where ${\mathfrak{A}}_p$ is the variety of abelian groups of prime exponent $p$)?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03389 | KOU-12.82

### Kourovka Notebook Problem 12.82

S / solved | Kourovka 群论问题集

Find all pairs $(n,r)$ such that the symmetric group ${\mathbb{S}}_n$ contains a maximal subgroup isomorphic to ${\mathbb{S}}_r$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03390 | KOU-12.84

### Kourovka Notebook Problem 12.84

S / solved | Kourovka 群论问题集

(Well-known problem). Is it true that if there exist two non-isomorphic groups with the given set of orders of the elements, then there are infinitely many groups with this set of orders of the elements?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03391 | KOU-12.85

### Kourovka Notebook Problem 12.85

O / open | Kourovka 群论问题集

Does every variety which is generated by a (known) finite simple group containing a soluble subgroup of derived length $d$ contain a 2-generator soluble group of derived length $d$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03392 | KOU-12.86

### Kourovka Notebook Problem 12.86

O / open | Kourovka 群论问题集

For each known finite simple group, find its maximal 2-generator direct power.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03393 | KOU-12.87

### Kourovka Notebook Problem 12.87

O / open | Kourovka 群论问题集

Let $\Gamma$ be a connected undirected graph without loops or multiple edges and suppose that the automorphism group $\mathrm{Aut} (\Gamma )$ acts transitively on the vertex set of $\Gamma$. Is it true that at least one of the following assertions holds?

1.

The stabilizer of a vertex of $\Gamma$ in $\mathrm{Aut} (\Gamma )$ is finite.

2.

The group $\mathrm{Aut} (\Gamma )$ as a permutation group on the vertex set of $\Gamma$ admits an imprimitivity system $\sigma$ with finite blocks for which the stabilizer of a vertex of the factor-graph $\Gamma /\sigma$ in $\mathrm{Aut} (\Gamma /\sigma )$ is finite.

3.

There exists a natural number $n$ such that the graph obtained from $\Gamma$ by adding edges joining distinct vertices the distance between which in $\Gamma$ is at most $n$ contains a regular tree of valency 3.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03394 | KOU-12.88

### Kourovka Notebook Problem 12.88

O / open | Kourovka 群论问题集

An undirected graph is called a *locally finite Cayley graph of a group* $G$ if its vertex set can be identified with the set of elements of $G$ in such a way that, for some finite generating set $X=X^{-1}$ of $G$ not containing $1$, two vertices $g$ and $h$ are adjacent if and only if $g^{-1}h \in X$. Do there exist two finitely generated groups with the same locally finite Cayley graph, one of which is periodic and the other is not periodic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03395 | KOU-12.89

### Kourovka Notebook Problem 12.89

O / open | Kourovka 群论问题集

Describe the infinite connected graphs to which the sequences of finite connected graphs with primitive automorphism groups converge. For definitions, see (V. I. Trofimov, *Algebra and Logic*, **28**, no. 3 (1989), 220--237).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03396 | KOU-12.9

## Kourovka Notebook Problem 12.9

O / open | Kourovka 群论问题集

Following Bass, call a group *tree-free* if there is an ordered Abelian group $\Lambda$ and a $\Lambda $-tree $X$ such that $G$ acts freely without inversion on $X$.

a) Must every finitely generated tree-free group satisfy the maximal condition for Abelian subgroups?

b) The same question for finitely presented tree-free groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03397 | KOU-12.90

## Kourovka Notebook Problem 12.90

S / solved | Kourovka 群论问题集

Let $G$ be a finitely generated soluble minimax group and let $H$ be a finitely generated residually finite group which has precisely the same finite images as $G$. Must $H$ be a minimax group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03398 | KOU-12.91

## Kourovka Notebook Problem 12.91

S / solved | Kourovka 群论问题集

Every metabelian group belonging to a Fitting class of (finite) supersoluble groups is nilpotent. Does the following generalization also hold: Every group belonging to a Fitting class of supersoluble groups has only central minimal normal subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03399 | KOU-12.92

## Kourovka Notebook Problem 12.92

O / open | Kourovka 群论问题集

For a field $K$ of characteristic 2, each finite group $G=\{ g_1,\ldots ,g_n\}$ of odd order $n$ is determined up to isomorphism by its group determinant (Formanek--Sibley, 1990) and even by its *reduced norm* which is defined as the last coefficient $s_m(x)$ of the minimal polynomial $\varphi (\lambda ;x)=\lambda ^m - s_1(x)\lambda ^{m-1} + \cdots + (- 1)^{m}s_m(x)$ for the generic element $x=x_1g_1 + \cdots + x_ng_n$ of the group ring $KG$ (Hoehnke, 1991). Is it possible in this theorem to replace $s_m(x)$ by some other coefficients $s_i(x)$, $i < m$? For notation see (G. Frobenius, *Sitzungsber. Preuss. Akad. Wiss. Berlin*, 1896, 1343--1382) and (K. W. Johnson, *Math. Proc. Cambridge Phil. Soc.*, **109** (1991), 299--311).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03400 | KOU-12.93

## Kourovka Notebook Problem 12.93

S / solved | Kourovka 群论问题集

Let $N \rightarrowtail G \twoheadrightarrow Q$ be an extension of nilpotent groups, with $Q$ finitely generated, which splits at every prime. Does the extension split? This is known to be true if $N$ is finite or commutative.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03401 | KOU-12.94

## Kourovka Notebook Problem 12.94

S / solved | Kourovka 群论问题集

Let $G$ be a finitely generated pro-$p $-group not involving the wreath product $C_p\wr {\mathbb{Z}_p$ as a closed section (where $C_p$ is a cyclic group of order $p$ and ${\mathbb{Z}_p$ is the group of $p $-adic integers). Does it follow that $G$ is $p $-adic analytic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03402 | KOU-12.95

## Kourovka Notebook Problem 12.95

O / open | Kourovka 群论问题集

Let $G$ be a finitely generated pro-$p $-group and let $g_1,\ldots ,g_n \in G$. Let $H$ be an open subgroup of $G$, and suppose there exists a non-trivial word $w=w(X_1,\ldots ,X_n)$ such that $w(a_1,\ldots ,a_n)=1$ whenever $a_1 \in g_1H,\ldots ,a_n \in g_nH$ (that is, $G$ satisfies a *coset identity*). Does it follow that $G$ satisfies some non-trivial identity?

A positive answer to this question would imply that an analogue of the Tits Alternative holds for finitely generated pro-$p $-groups. Note that J. S. Wilson and E. I. Zelmanov (*J. Pure Appl. Algebra*, **81** (1992), 103--109) showed that the graded Lie algebra $L_p(G)$ with respect to the dimension subgroups of $G$ in characteristic $p$ satisfies a polynomial identity.

*Comment of 2017*: This has been shown to be true for finitely generated linear groups (M. Larsen, A. Shalev, *Algebra Number Theory*, **10**, no. 6 (2016), 1359--1371).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03403 | KOU-12.96

## Kourovka Notebook Problem 12.96

S / solved | Kourovka 群论问题集

Find a non-empty Fitting class ${\mathfrak{F}}$ and a non-soluble finite group $G$ such that $G$ has no ${\mathfrak{F}} $-injectors.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03404 | KOU-12.97

### Kourovka Notebook Problem 12.97

S / solved | Kourovka 群论问题集

Let ${\mathfrak{F}}$ be the formation of all finite groups all of whose composition factors are isomorphic to some fixed simple non-abelian group $T$. Prove that ${\mathfrak{F}}$ is indecomposable into a product of two non-trivial subformations.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03405 | KOU-12.98

### Kourovka Notebook Problem 12.98

S / solved | Kourovka 群论问题集

Let $F$ be a free group of finite rank, $R$ its recursively defined normal subgroup. Is it true that

a) the word problem for $F/R$ is soluble if and only if it is soluble for $F/[R,R]$?

b) the conjugacy problem for $F/R$ is soluble if and only if it is soluble for $F/[R,R]$?

c) the conjugacy problem for $F/[R,R]$ is soluble if the word problem is soluble for $F/[R,R]$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03406 | KOU-13.1

### Kourovka Notebook Problem 13.1

O / open | Kourovka 群论问题集

Let $U(K)$ denote the group of units of a ring $K$. Let $G$ be a finite group, ${\mathbb{Z}} G$ the integral group ring of $G$, and ${\mathbb{Z}} _pG$ the group ring of $G$ over the residues modulo a prime number $p$. Describe the homomorphism from $U({\mathbb{Z}} G)$ into $U({\mathbb{Z}} _pG)$ induced by reducing the coefficients modulo $p$. More precisely, find the kernel and the image of this homomorphism and an explicit transversal over the kernel.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03407 | KOU-13.10

### Kourovka Notebook Problem 13.10

S / solved | Kourovka 群论问题集

Is there a function $f\! : {\mathbb{N}} \rightarrow {\mathbb{N}}$ such that, for every soluble group $G$ of derived length $k$ generated by a set $A$, the validity of the identity $x^4=1$ on each subgroup generated by at most $f(k)$ elements of $A$ implies that $G$ is a group of exponent 4?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03408 | KOU-13.11

### Kourovka Notebook Problem 13.11

S / solved | Kourovka 群论问题集

Is a torsion-free group almost polycyclic if it has a finite set of generators $a_1,\ldots , a_n$ such that every element of the group has a unique presentation in the form $a_1^{k_1}\cdots a_n^{k_n}$, where $k_1,\ldots , k_n\in {\mathbb{Z}}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03409 | KOU-13.12

### Kourovka Notebook Problem 13.12

O / open | Kourovka 群论问题集

Is the group of all automorphisms of an arbitrary hyperbolic group finitely presented?

*Comment of 2001:* This is proved for the group of automorphisms of a torsion-free hyperbolic group that cannot be decomposed into a free product (Z. Sela, *Geom. Funct. Anal.*, **7**, no. 3 (1997) 561--593).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03410 | KOU-13.13

### Kourovka Notebook Problem 13.13

O / open | Kourovka 群论问题集

Let $G$ and $H$ be finite $p$-groups with isomorphic Burnside rings. Is the nilpotency class of $H$ bounded by some function of the class of $G$? There is an example where $G$ is of class 2 and $H$ is of class 3.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03411 | KOU-13.14

### Kourovka Notebook Problem 13.14

O / open | Kourovka 群论问题集

Is the lattice of quasivarieties of nilpotent torsion-free groups of nilpotency class $\leq 2$ distributive?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03412 | KOU-13.15

### Kourovka Notebook Problem 13.15

O / open | Kourovka 群论问题集

Does a free non-abelian nilpotent group of class 3 possess an independent basis of quasiidentities in the class of torsion-free groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03413 | KOU-13.16

### Kourovka Notebook Problem 13.16

S / solved | Kourovka 群论问题集

Is every locally nilpotent group with minimum condition on centralizers hypercentral?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03414 | KOU-13.17

### Kourovka Notebook Problem 13.17

O / open | Kourovka 群论问题集

A representation of a group $G$ on a vector space $V$ is called a *nil-representation* if for every $v$ in $V$ and $g$ in $G$ there exists $n=n(v,g)$ such that $v (g-1)^n = 0$. Is it true that in zero characteristic every irreducible nil-representation is trivial? (In prime characteristic it is not true.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03415 | KOU-13.18

### Kourovka Notebook Problem 13.18

S / solved | Kourovka 群论问题集

Let $F$ be a finitely generated non-Abelian free group and let $G$ be the Cartesian (unrestricted) product of countable infinity of copies of $F$. Must the Abelianization $G/G'$ of $G$ be torsion-free?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03416 | KOU-13.19

### Kourovka Notebook Problem 13.19

O / open | Kourovka 群论问题集

Suppose that both a group $Q$ and its normal subgroup $H$ are subgroups of the direct product $G_1\times \cdots \times G_n$ such that for each $i$ the projections of both $Q$ and $H$ onto $G_i$ coincide with $G_i$. If $Q/H$ is a $p$-group, is $Q/H$ a regular $p$-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03417 | KOU-13.2

### Kourovka Notebook Problem 13.2

O / open | Kourovka 群论问题集

Does there exist a finitely based variety of groups ${\mathfrak{V}}$ such that the word problem is solvable in $F_n({\mathfrak{V}})$ for every positive integer $n$, but is unsolvable in $F_{\infty}({\mathfrak{V}})$ (with respect to a free generating system)?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03418 | KOU-13.20

### Kourovka Notebook Problem 13.20

O / open | Kourovka 群论问题集

Is it true that the generating series of the growth function of every finitely generated group with one defining relation represents an algebraic (or even a rational) function?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03419 | KOU-13.21

### Kourovka Notebook Problem 13.21

P / partially_solved | Kourovka 群论问题集

b) What is the minimal possible rate of growth of the function $ \pi (n)=\max\limits_{\delta (g)\leq n} |g|$ for the class of groups indicated in part ``a'' of this problem (see Archive)? It is known (R. I. Grigorchuk, *Math. USSR--Izv.*, **25** (1985), 259--300) that there exist $p$-groups with $\pi (n)\leq n^{\lambda}$ for some $\lambda >0$. At the same time, it follows from a result of E. I. Zel'manov that $\pi (n)$ is not bounded if $G$ is infinite.

[Archived solved part(s) of 13.21:] a) Is there an infinite finitely generated residually finite $p$-group, in which the order $|g|$ of an arbitrary element $g$ does not exceed $f(\delta (g))$, where $\delta (g)$ is the length of $g$ with respect to a fixed set of generators and $f(n)$ is a function growing at $n\rightarrow \infty$ slower than any power function $n^{\lambda}$, $\lambda >0$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03420 | KOU-13.22

### Kourovka Notebook Problem 13.22

O / open | Kourovka 群论问题集

Let the group $G=AB$ be the product of two polycyclic subgroups $A$ and $B$, and assume that $G$ has an ascending normal series with locally nilpotent factors (i. e. $G$ is radical). Is it true that $G$ is polycyclic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03421 | KOU-13.23

### Kourovka Notebook Problem 13.23

S / solved | Kourovka 群论问题集

Let the group $G$ have a finite normal series with infinite cyclic factors (containing of course $G$ and $\{1\}$). Is it true that $G$ has a non-trivial outer automorphism?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03422 | KOU-13.24

### Kourovka Notebook Problem 13.24

O / open | Kourovka 群论问题集

Let $G$ be a non-discrete topological group with only finitely many ultrafilters that converge to the identity. Is it true that $G$ contains a countable open subgroup of exponent 2?

*Comment of 2005:* This is proved for $G$ countable (E. G. Zelenyuk, *Mat. Studii*, **7** (1997), 139--144).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03423 | KOU-13.25

### Kourovka Notebook Problem 13.25

O / open | Kourovka 群论问题集

Let $(G,\tau )$ be a topological group with finite semigroup $\tau (G)$ of ultrafilters converging to the identity (I. V. Protasov, *Siberian Math. J.*, **34**, no. 5 (1993), 938--951). Is it true that $\tau (G)$ is a semigroup of idempotents?

*Comment of 2005:* This is proved for $G$ countable (E. G. Zelenyuk, *Mat. Studii*, **14** (2000), 121--140).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03424 | KOU-13.26

### Kourovka Notebook Problem 13.26

S / solved | Kourovka 群论问题集

Is it true that a countable topological group of exponent 2 with unique free ultrafilter converging to the identity has a basis of neighborhoods of the identity consisting of subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03425 | KOU-13.27

### Kourovka Notebook Problem 13.27

O / open | Kourovka 群论问题集

(B. Amberg). Suppose that $G=AB=AC=BC$ for a group $G$ and its subgroups $A,B,C$.

a) Is $G$ a Chernikov group if $ A,B,C$ are Chernikov groups?

b) Is $G$ almost polycyclic if $A,B,C$ are almost polycyclic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03426 | KOU-13.28

### Kourovka Notebook Problem 13.28

S / solved | Kourovka 群论问题集

(D. M. Evans). A permutation group on an infinite set is *cofinitary* if its non-identity elements fix only finitely many points. Is it true that a closed cofinitary permutation group is locally compact (in the topology of pointwise convergence)?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03427 | KOU-13.29

### Kourovka Notebook Problem 13.29

S / solved | Kourovka 群论问题集

Given an infinite set $\Omega$, define an algebra $A$ (the *reduced incidence algebra of finite subsets*) as follows. Let $V_n$ be the set of functions from the set of $n $-element subsets of $\Omega$ to the rationals $\mathbb{Q}$. Now let $A=\bigoplus V_n$, with multiplication as follows: for $f\in V_n$, $g\in V_m$, and $|X|=m+n$, let $(fg)(X)=\sum f(Y)g(X\setminus Y),$ where the sum is over the $n $-element subsets $Y$ of $X$. If $G$ is a permutation group on $\Omega$, let $A^G$ be the algebra of $G $-invariants in $A$.

*Conjecture:* If $G$ has no finite orbits on $\Omega$, then $A^G$ is an integral domain.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03428 | KOU-13.3

### Kourovka Notebook Problem 13.3

O / open | Kourovka 群论问题集

Let ${\mathfrak{M}}$ be an arbitrary variety of groups. Is it true that every infinitely generated projective group in ${\mathfrak{M}}$ is an ${\mathfrak{M}} $-free product of countably generated projective groups in ${\mathfrak{M}}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03429 | KOU-13.30

### Kourovka Notebook Problem 13.30

O / open | Kourovka 群论问题集

A group $G$ is called a *$B$-group* if every primitive permutation group which contains the regular representation of $G$ is doubly transitive. Are there any countable $B $-groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03430 | KOU-13.31

### Kourovka Notebook Problem 13.31

O / open | Kourovka 群论问题集

Let $G$ be a permutation group on a set $\Omega$. A sequence of points of $\Omega$ is a *base* for $G$ if its pointwise stabilizer in $G$ is the identity. *The greedy algorithm* for a base chooses each point in the sequence from an orbit of maximum size of the stabilizer of its predecessors. Is it true that there is a universal constant $c$ with the property that, for any finite primitive permutation group, the greedy algorithm produces a base whose size is at most $c$ times the minimal base size?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03431 | KOU-13.32

### Kourovka Notebook Problem 13.32

O / open | Kourovka 群论问题集

(Well-known problem). On a group, a partial order that is directed upwards and has linearly ordered cone of positive elements is said to be *semilinear* if this order is invariant under right multiplication by the group elements. Can every semilinear order of a group be extended to a right order of the group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03432 | KOU-13.33

### Kourovka Notebook Problem 13.33

S / solved | Kourovka 群论问题集

(F. Gross). Is a normal subgroup of a finite $D_{\pi} $-group (see Archive, 3.62) always a $D_{\pi} $-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03433 | KOU-13.34

### Kourovka Notebook Problem 13.34

S / solved | Kourovka 群论问题集

(I. D. Macdonald). If the identity $[x, y]^n=1$ holds on a group, is the derived subgroup of the group periodic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03434 | KOU-13.35

### Kourovka Notebook Problem 13.35

O / open | Kourovka 群论问题集

Does every non-soluble pro-$p $-group of cohomological dimension 2 contain a free non-abelian pro-$p $-subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03435 | KOU-13.36

### Kourovka Notebook Problem 13.36

P / partially_solved | Kourovka 群论问题集

For a finitely generated pro-$p $-group $G$ set $a_n(G)=\mathrm{dim}_{ {\mathbb{F}_p}I^n/I^{n+1}$, where $I$ is the augmentation ideal of the group ring ${\mathbb{F}_p[[G]]$. We define the *growth* of $G$ to be the growth of the sequence $\{ a_n(G)\} _{n\in {\mathbb{N}}$.

a) If the growth of $G$ is exponential, does it follow that $G$ contains a free pro-$p $-subgroup of rank 2?

c) Do there exist pro-$p $-groups of finite cohomological dimension which are not $p $-adic analytic, and whose growth is slower than an exponential one?

[Archived solved part(s) of 13.36:] b) (A. Lubotzky, A. Shalev). For a finitely generated pro-$p $-group $G$ set $a_n(G)=\mathrm{dim}_{ {\mathbb{F}_p}I^n/I^{n+1}$, where $I$ is the augmentation ideal of the group ring ${\mathbb{F}_p[[G]]$. We define the *growth* of $G$ to be the growth of the sequence $\{ a_n(G)\} _{n\in {\mathbb{N}}$. Is the growth of $G$ exponential if $G$ contains a finitely generated closed subgroup of exponential growth?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03436 | KOU-13.37

### Kourovka Notebook Problem 13.37

O / open | Kourovka 群论问题集

Let $G$ be a torsion-free pro-$p $-group, $U$ an open subgroup of $G$. Suppose that $U$ is a pro-$p $-group with a single defining relation. Is it true that then $G$ is also a pro-$p $-group with a single defining relation?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03437 | KOU-13.38

### Kourovka Notebook Problem 13.38

S / solved | Kourovka 群论问题集

Let $G=F/R$ be a pro-$p $-group with one defining relation, where $R$ is the normal subgroup of a free pro-$p $-group $F$ generated by a single element $r\in F^p[F, F]$.

a) Suppose that $r=t^p$ for some $t\in F$; can $G$ contain a Demushkin group as a subgroup?

b) Do there exist two pro-$p $-groups $G_1\supset G_2$ with one defining relation, where $G_1$ has elements of finite order, while the subgroup $G_2$ is torsion-free?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03438 | KOU-13.39

## Kourovka Notebook Problem 13.39

P / partially_solved | Kourovka 群论问题集

Let $A$ be an associative ring with unity and with torsion-free additive group, and let $F^A$ be the tensor product of a free group $F$ by $A$ (A. G. Myasnikov, V. N. Remeslennikov, *Siberian Math. J.*, **35**, no. 5 (1994), 986--996); then $F^A$ is a *free exponential group over $A$*; in (A. G. Myasnikov, V. N. Remeslennikov, *Int. J. Algebra Comput.*, **6** (1996), 687--711), it is shown how to construct $F^A$ in terms of free products with amalgamation.

b) Is $F^{A}$ a linear group?

In the case where $\left< 1\right>$ is a pure subgroup of the additive group of $A$, there is an affirmative answer to ``b'' (A. M. Gaglione, A. G. Myasnikov, V. N. Remeslennikov, D. Spellman, *Commun. Algebra*, **25** (1997), 631--648). It is also known that the Magnus homomorphism is one-to-one on any subgroup of $F^{\mathbb{Q}}$ of the type $\left< F,t \mid u = t^n\right>$ (G. Baumslag, *Commun. Pure Appl. Math.*, **21** (1968), 491--506).

d) Is the universal theory of $F^{A}$ decidable?

e) (G. Baumslag). Can free $A $-groups be characterized by a length function?

f) (G. Baumslag). Does a free $\mathbb{Q} $-group admit a free action on some $\Lambda $-tree? See definition in (R. Alperin, H. Bass, *in: Combinatorial group theory and topology, Alta, Utah, 1984 $($Ann. Math. Stud.*, **111**), Princeton Univ. Press, 1987, 265--378).

[Archived solved part(s) of 13.39:] Let $A$ be an associative ring with unity and with torsion-free additive group, and let $F^A$ be the tensor product of a free group $F$ by $A$ (A. G. Myasnikov, V. N. Remeslennikov, *Siberian Math. J.*, **35**, no. 5 (1994), 986--996); then $F^A$ is a *free exponential group over $A$*; in (A. G. Myasnikov, V. N. Remeslennikov, *Int. J. Algebra Comput.*, **6** (1996), 687--711), it is shown how to construct $F^A$ in terms of free products with amalgamation.

a) (G. Baumslag). Is $F^{A}$ residually nilpotent torsion-free?

c) (G. Baumslag). Is the Magnus homomorphism of $F^{\mathbb{Q}}$ into the group of power series over the rational number field $\mathbb{Q}$ faithful or not?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03439 | KOU-13.4

## Kourovka Notebook Problem 13.4

O / open | Kourovka 群论问题集

Let $G$ be a group with a normal (pro-) 2-subgroup $N$ such that $G/N$ is isomorphic to $GL_n(2)$, inducing its natural module on $N/\Phi (N)$, the Frattini factor-group of $N$. For $n=3$ determine $G$ such that $N$ is as large as possible. (For $n>3$ it can be proved that already the Frattini subgroup $\Phi (N)$ of $N$ is trivial; for $n=2$ there exists $G$ such that $N$ is the free pro-2-group generated by two elements).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03440 | KOU-13.40

## Kourovka Notebook Problem 13.40

S / solved | Kourovka 群论问题集

A group $G$ is said to be $\omega$-*residually free* if, for every finite set of non-trivial elements of $G$, there is a homomorphism of $G$ into a free group such that the images of all these elements are non-trivial. Is every finitely-generated $\omega$-residually free group embeddable in a free ${\mathbb{Z}} [x]$-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03441 | KOU-13.41

## Kourovka Notebook Problem 13.41

O / open | Kourovka 群论问题集

Is the elementary theory of the class of all groups acting freely on $\Lambda $-trees decidable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03442 | KOU-13.42

## Kourovka Notebook Problem 13.42

O / open | Kourovka 群论问题集

Prove that the tensor $A $-completion (see A. G. Myasnikov, V. N. Remeslennikov, *Siberian Math. J.*, **35**, no. 5 (1994), 986--996) of a free nilpotent group can be non-nilpotent.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03443 | KOU-13.43

## Kourovka Notebook Problem 13.43

S / solved | Kourovka 群论问题集

(G. R. Robinson). Let $G$ be a finite group and $B$ be a $p $-block of characters of $G$. *Conjecture:* If the defect group $D=D(B)$ of the block $B$ is non-abelian, and if $|D:Z(D)| = p^a$, then each character in $B$ has height strictly less than $a$.

*G. R. Robinson's comment of 2020*: the conjecture is proved mod CFSG for $p\ne 2$ in (Z. Feng, C. Li, Y. Liu, G. Malle, J. Zhang, *Compos. Math.*, **155**, no. 6 (2019), 1098--1117).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03444 | KOU-13.44

## Kourovka Notebook Problem 13.44

O / open | Kourovka 群论问题集

For any partition of an arbitrary group $G$ into finitely many subsets $G=A_1\cup \cdots \cup A_n$, there exists a subset of the partition $A_i$ and a finite subset $F\subseteq G$, such that $G=A_i^{-1}A_iF$ (I. V. Protasov, *Siberian Math. J.*, **34**, no. 5 (1993), 938--952). Can the subset $F$ always be chosen so that $|F|\leq n$? This is true for amenable groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03445 | KOU-13.45

### Kourovka Notebook Problem 13.45

S / solved | Kourovka 群论问题集

Every infinite group $G$ of regular cardinality ${\mathfrak{m}}$ can be partitioned into two subsets $G=A_1\cup A_2$ so that $A_1F\ne G$ and $A_2F\ne G$ for every subset $F\subset G$ of cardinality less than ${\mathfrak{m}}$. Is this statement true for groups of singular cardinality?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03446 | KOU-13.46

### Kourovka Notebook Problem 13.46

S / solved | Kourovka 群论问题集

Can every uncountable abelian group of finite odd exponent be partitioned into two subsets so that neither of them contains cosets of infinite subgroups? Among countable abelian groups, such partitions exist for groups with finitely many involutions.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03447 | KOU-13.47

### Kourovka Notebook Problem 13.47

S / solved | Kourovka 群论问题集

Can every countable abelian group with finitely many involutions be partitioned into two subsets that are dense in every group topology?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03448 | KOU-13.48

### Kourovka Notebook Problem 13.48

O / open | Kourovka 群论问题集

(W. W. Comfort, J. van Mill). A topological group is said to be *irresolvable* if every two of its dense subsets intersect non-trivially. Does every non-discrete irresolvable group contain an infinite subgroup of exponent 2?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03449 | KOU-13.49

### Kourovka Notebook Problem 13.49

O / open | Kourovka 群论问题集

(V. I. Malykhin). Can a topological group be partitioned into two dense subsets, if there are infinitely many free ultrafilters on the group converging to the identity?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03450 | KOU-13.5

### Kourovka Notebook Problem 13.5

O / open | Kourovka 群论问题集

Groups $A$ and $B$ are said to be *locally equivalent* if for every finitely generated subgroup $X\leq A$ there is a subgroup $Y\leq B$ isomorphic to $X$, and conversely, for every finitely generated subgroup $Y\leq B$ there is a subgroup $X\leq A$, isomorphic to $Y$. We call a group $G$ *categorical* if $G$ is isomorphic to any group that is locally equivalent to $G$. Is it true that a periodic locally soluble group $G$ is categorical if and only if $G$ is hyperfinite with Chernikov Sylow subgroups? (A group is *hyperfinite* if it has a well ordered ascending normal series with finite factors.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03451 | KOU-13.50

### Kourovka Notebook Problem 13.50

S / solved | Kourovka 群论问题集

Let ${\mathfrak{F}}$ be a local Fitting class. Is it true that there are no maximal elements in the partially ordered by inclusion set of the Fitting classes contained in ${\mathfrak{F}}$ and distinct from ${\mathfrak{F}}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03452 | KOU-13.51

### Kourovka Notebook Problem 13.51

S / solved | Kourovka 群论问题集

Is every finite modular lattice embeddable in the lattice of formations of finite groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03453 | KOU-13.52

### Kourovka Notebook Problem 13.52

O / open | Kourovka 群论问题集

The *dimension* of a finitely based variety of algebras ${\mathcal{V}}$ is defined to be the maximal length of a *basis* (that is, an independent generating set) of the $SC $-theory $SC({\mathcal{V}})$, which consists of the strong Mal'cev conditions satisfied on ${\mathcal{V}}$. The dimension is defined to be infinite if the lengths of bases in $SC({\mathcal{V}})$ are not bounded. Does every finite abelian group generate a variety of finite dimension?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03454 | KOU-13.53

### Kourovka Notebook Problem 13.53

O / open | Kourovka 群论问题集

Let $a, b$ be elements of finite order of the infinite group $G=\left< a, b\right>$. Is it true that there are infinitely many elements $g\in G$ such that the subgroup $\left< a, b^g\right>$ is infinite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03455 | KOU-13.54

### Kourovka Notebook Problem 13.54

P / partially_solved | Kourovka 群论问题集

a) Is it true that, for $ p$ sufficiently large, every (finite) $p $-group can be a complement in some Frobenius group (see 6.55)?

[Archived solved part(s) of 13.54:] b) Is it true that every group is embeddable in the kernel of some Frobenius group (see Archive, 6.53)?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03456 | KOU-13.55

### Kourovka Notebook Problem 13.55

O / open | Kourovka 群论问题集

Does there exist a Golod group (see 9.76), which is isomorphic to an $AT $-group? For a definition of an *$AT $-group* see (A. V. Rozhkov, *Math. Notes*, **40**, no. 5 (1986), 827--836).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03457 | KOU-13.56

### Kourovka Notebook Problem 13.56

S / solved | Kourovka 群论问题集

(A. Shalev). Let $G$ be a finite $p $-group of sectional rank $r$, and $\varphi$ an automorphism of $G$ having exactly $m$ fixed points. Is the derived length of $G$ bounded by a function depending on $r$ and $m$ only?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03458 | KOU-13.57

### Kourovka Notebook Problem 13.57

O / open | Kourovka 群论问题集

Let $\varphi$ be an automorphism of prime order $p$ of a finite group $G$ such that $C_G(\varphi )\leq Z(G)$.

a) Is $G$ soluble if $p=3$? V. D. Mazurov and T. L. Nedorezov proved in (*Algebra and Logic*, **35**, no. 6 (1996), 392--397) that the group $G$ is soluble for $p=2$, and there are examples of unsoluble $G$ for all $p>3$.

b) If $G$ is soluble, is the derived length of $G$ bounded in terms of $p$?

c) (V. K. Kharchenko). If $G$ is a $p $-group, is the derived length of $G$ bounded in terms of $p$? (V. V. Bludov produced a simple example showing that the nilpotency class cannot be bounded.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03459 | KOU-13.58

### Kourovka Notebook Problem 13.58

S / solved | Kourovka 群论问题集

Let $\varphi$ be an automorphism of prime order $p$ of a nilpotent (periodic) group $G$ such that $C_G(\varphi )$ is a group of finite sectional rank $r$. Does $G$ possess a normal subgroup $N$ which is nilpotent of class bounded by a function of $p$ only and is such that $G/N$ is a group of finite sectional rank bounded in terms of $r$ and $p$?

This was proved for $p=2$ in (P. Shumyatsky, *Arch. Math.*, **71** (1998), 425--432).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03460 | KOU-13.59

### Kourovka Notebook Problem 13.59

O / open | Kourovka 群论问题集

One can show that any extension of shape ${\mathbb{Z}^d. {\mathcal{S}L}_d({\mathbb{Z}})$ is residually finite, unless possibly $d=3$ or $d=5$. Are there in fact any extensions of this shape that fail to be residually finite when $d=5$? As shown in (P. R. Hewitt, *Groups/St. Andrews'93 in Galway, Vol. 2 $($London Math. Soc. Lecture Note Ser.*, **212**), Cambridge Univ. Press, 1995, 305--313), there is an extension of ${\mathbb{Z} ^3$ by ${\mathcal{S}L}_3({\mathbb{Z} )$ that is not residually finite. Usually it is true that an extension of an arithmetic subgroup of a Chevalley group over a rational module is residually finite. Is it ever false, apart from the examples of shape ${\mathbb{Z}^3. {\mathcal{S}L}_3({\mathbb{Z})$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03461 | KOU-13.6

### Kourovka Notebook Problem 13.6

O / open | Kourovka 群论问题集

(B. Hartley). Is it true that a locally finite group containing an element with Chernikov centralizer is almost soluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03462 | KOU-13.60

### Kourovka Notebook Problem 13.60

O / open | Kourovka 群论问题集

If a locally graded group $G$ is a product of two subgroups of finite special rank, is $G$ of finite special rank itself? The answer is affirmative if both factors are periodic groups. (N. S. Chernikov, *Ukrain. Math. J.*, **42**, no. 7 (1990), 855--861).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03463 | KOU-13.61

### Kourovka Notebook Problem 13.61

S / solved | Kourovka 群论问题集

We call a metric space *narrow* if it is quasiisometric to a subset of the real line, and *wide* otherwise. Let $G$ be a group with the finite set of generators $A$, and let $\Gamma =\Gamma (G,A)$ be the Cayley graph of $G$ with the natural metric. Suppose that, after deleting any narrow subset $L$ from $\Gamma$, at most two connected components of the graph $\Gamma \setminus L$ can be wide, and there exists at least one such a subset $L$ yielding exactly two wide components in $\Gamma \setminus L$. Is it true that $\Gamma$ is quasiisometric to an Euclidean or a hyperbolic plane?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03464 | KOU-13.62

### Kourovka Notebook Problem 13.62

S / solved | Kourovka 群论问题集

Let $U$ and $V$ be non-cyclic subgroups of a free group. Does the inclusion $[U, U]\leq [V, V]$ imply that $U\leq V$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03465 | KOU-13.63

### Kourovka Notebook Problem 13.63

S / solved | Kourovka 群论问题集

Let $\pi _e(G)$ denote the set of orders of elements of a group $G$. For $\Gamma \subseteq {\mathbb{N}}$ let $h(\Gamma )$ denote the number of non-isomorphic finite groups $G$ with $\pi _e(G)=\Gamma$. Is there a number $k$ such that, for every $\Gamma$, either $h(\Gamma )\leq k$, or $h(\Gamma )=\infty$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03466 | KOU-13.64

### Kourovka Notebook Problem 13.64

O / open | Kourovka 群论问题集

Let $\pi _e(G)$ denote the set of orders of elements of a group $G$. A group $G$ is said to be an *$OC_n$-group* if $\pi _e(G)=\{ 1, 2, \ldots , n\}$. Is every $OC_n$-group locally finite? Do there exist infinite $OC_n$-groups for $n\geq 7$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03467 | KOU-13.65

### Kourovka Notebook Problem 13.65

O / open | Kourovka 群论问题集

A finite simple group is called a *$K_n$-group* if its order is divisible by exactly $n$ different primes. The number of $K_3$-groups is known to be 8. The $K_4$-groups are classified mod CFSG (W. J. Shi, *in: Group Theory in China ($Math. Appl.*, **365**), Kluwer, 1996, 163--181) and some significant further results are obtained in (Yann Bugeaud, Zhenfu Cao, M. Mignotte, *J. Algebra*, **241** (2001), 658--668). But the question remains: is the number of $K_4$-groups finite or infinite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03468 | KOU-13.66

### Kourovka Notebook Problem 13.66

S / solved | Kourovka 群论问题集

Let $F$ be a \

a) free; \

b) free metabelian \

group of finite rank. Let $M$ denote the set of all endomorphisms of $F$ with non-cyclic images. Can one choose two elements $g, h\in F$ such that, for every $\varphi , \psi \in M$, equalities $\varphi (g)=\psi (g)$ and $\varphi (h)=\psi (h)$ imply that $\varphi =\psi$, that is, the endomorphisms in $M$ are uniquely determined by their values at $g$ and $h$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03469 | KOU-13.67

### Kourovka Notebook Problem 13.67

O / open | Kourovka 群论问题集

Let $G$ be a $T_0$-group (see 12.101), $i$ an involution in $G$ and $G=\left< i^g\mid g\in G\right>$. Is the centralizer $C_G(i)$ residually periodic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03470 | KOU-13.7

### Kourovka Notebook Problem 13.7

O / open | Kourovka 群论问题集

(B. Hartley). Is it true that a simple locally finite group containing a finite subgroup with finite centralizer is linear?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03471 | KOU-13.8

### Kourovka Notebook Problem 13.8

P / partially_solved | Kourovka 群论问题集

(B. Hartley). Is it true that a locally soluble periodic group has a finite normal series with locally nilpotent factors if it contains an element

a) with finite centralizer?

b) with Chernikov centralizer?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03472 | KOU-13.9

### Kourovka Notebook Problem 13.9

O / open | Kourovka 群论问题集

Is it true that a locally soluble periodic group containing a finite nilpotent subgroup with Chernikov centralizer has a finite normal series with locally nilpotent factors?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03473 | KOU-14.1

### Kourovka Notebook Problem 14.1

S / solved | Kourovka 群论问题集

Suppose that $G$ is a finite group with no non-trivial normal subgroups of odd order, and $\varphi$ is its $2$-automorphism centralizing a Sylow $2$-subgroup of $G$. Is it true that $\varphi ^2$ is an inner automorphism of $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03474 | KOU-14.10

### Kourovka Notebook Problem 14.10

P / partially_solved | Kourovka 群论问题集

c) Find an explicit and *" natural"* \ finitely presented group $\Gamma_n$ and an embedding of $GL_n({\mathbb{Q}})$ in $\Gamma _n$.

Another phrasing of the same problems is: find a simplicial complex $X$ which covers a finite complex such that the fundamental group of $X$ is ${\mathbb{Q}}$ or, respectively, $GL_n({\mathbb{Q}})$.

[Archived solved part(s) of 14.10:] a) (Well-known problem). It is known that any recursively presented group embeds in a finitely presented group (G. Higman, *Proc. Royal Soc. London Ser. A*, **262** (1961), 455--475). Find an explicit and *" natural"* \ finitely presented group $\Gamma$ and an embedding of the additive group of the rationals ${\mathbb{Q}}$ in $\Gamma$.

b) Find an explicit embedding of $\mathbb{Q}$ in a finitely generated group; such a group exists by Theorem IV in (G. Higman, B. H. Neumann, H. Neumann, *J. London Math. Soc.*, **24** (1949), 247--254).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03475 | KOU-14.100

### Kourovka Notebook Problem 14.100

O / open | Kourovka 群论问题集

Is it true that in a Shunkov group (i. e. conjugately biprimitively finite group, see 6.59) having infinitely many elements of finite order every element of prime order is contained in some infinite locally finite subgroup? This is true under the additional condition that any two conjugates of this element generate a soluble subgroup (V. P. Shunkov, *$M_p$-groups*, Moscow, Nauka, 1990 (Russian)).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03476 | KOU-14.101

### Kourovka Notebook Problem 14.101

O / open | Kourovka 群论问题集

A group $G$ is *saturated* with groups from a class $X$ if every finite subgroup $K\leq G$ is contained in a subgroup $L\leq G$ isomorphic to some group from $X$. Is it true that a periodic group saturated with finite simple groups of Lie type of uniformly bounded ranks is itself a simple group of Lie type of finite rank?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03477 | KOU-14.102

### Kourovka Notebook Problem 14.102

S / solved | Kourovka 群论问题集

(V. Lin). Let $B_n$ be the braid group on $n$ strings, and let $n>4$.

a) Does $B_n$ have any non-trivial non-injective endomorphisms with non-cyclic images?

b) Is it true that every non-trivial endomorphism of the derived subgroup $[B_n, B_n]$ is an automorphism?

[Archived solved part(s) of 14.102:] c) (V. Lin). Let $B_n$ be the braid group on $n$ strings, and let $n>4$. Does $B_n$ have proper non-abelian torsion-free factor-groups?

*Comment of 2001:* S. P. Humphries, (*Int. J. Algebra Comput.*, **11**, no. 3 (2001), 363--373) has constructed a representation of $B_n$ which is shown to provide torsion-free non-abelian factor groups of $B_n$ as well as of $[B_n, B_n]$ for $n < 7$. It is likely that the same representation should work for other values of $n$ as well.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03478 | KOU-14.103

### Kourovka Notebook Problem 14.103

S / solved | Kourovka 群论问题集

Let $H$ be a proper subgroup of a group $G$ and let elements $a,b\in H$ have distinct prime orders $p, q$. Suppose that, for every $g\in G\setminus H$, the subgroup $\langle a,b^g\rangle$ is a finite Frobenius group with complement of order $pq$. Does the subgroup generated by the union of the kernels of all Frobenius subgroups of $G$ with complement $\langle a\rangle$ intersect $\langle a\rangle$ trivially? The case where all groups $\langle a,b^g\rangle$, $g\in G$, are finite is of special interest.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03479 | KOU-14.104

### Kourovka Notebook Problem 14.104

S / solved | Kourovka 群论问题集

An infinite group $G$ is called a *monster of the first kind* if it has elements of order $> 2$ and for any such an element $a$ and for any proper subgroup $H$ of $G$, there is an element $g$ in $G\setminus H$, such that $\left< a, a^g\right> =G$. A. Yu. Olshanskii showed that there are continuously many monsters of the first kind (see Archive, 6.63). Does there exist, for any such a monster, a torsion-free group which is a central extension of a cyclic group by the given monster?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03480 | KOU-14.11

### Kourovka Notebook Problem 14.11

O / open | Kourovka 群论问题集

(Yu. I. Merzlyakov). It is a well-known fact that for the ring $R={\mathbb{Q} [x,y]$ the elementary group $E_2(R)$ is distinct from $SL_2(R)$. Find a minimal subset $A\subseteq SL_2(R)$ such that $\left< E_2(R), A\right> =SL_2(R)$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03481 | KOU-14.12

### Kourovka Notebook Problem 14.12

O / open | Kourovka 群论问题集

(Yu. I. Merzlyakov, J. S. Birman). Is it true that all braid groups $B_n$, $n\geq 3$, are conjugacy separable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03482 | KOU-14.13

### Kourovka Notebook Problem 14.13

S / solved | Kourovka 群论问题集

a) By definition the *commutator length* of an element $z$ of the derived subgroup of a group $G$ is the least possible number of commutators from $G$ whose product is equal to $z$. Does there exist a simple group on which the commutator length is not bounded?

b) Does there exist a finitely presented simple group on which the commutator length is not bounded?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03483 | KOU-14.14

### Kourovka Notebook Problem 14.14

O / open | Kourovka 群论问题集

(C. C. Edmunds, G. Rosenberger). We call a pair of natural numbers $(k,m)$ *admissible* if in the derived subgroup $F_2'$ of a free group $F_2$ there is an element $w$ such that the commutator length of the element $w^m$ is equal to $k$. Find all admissible pairs.

It is known that every pair $(k, 2)$, $k \geq 2$, is admissible (V. G. Bardakov, *Algebra and Logic*, **39** (2000), 224--251).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03484 | KOU-14.15

### Kourovka Notebook Problem 14.15

S / solved | Kourovka 群论问题集

For the automorphism group $A_n=\mathrm{Aut }F_n$ of a free group $F_n$ of rank $n\geq 3$ find the supremum $k_n$ of the commutator lengths of the elements of $A_n'$.

It is easy to show that $k_2=\infty$. On the other hand, the commutator length of any element of the derived subgroup of $\lim\limits_{\longrightarrow}A_n$ is at most $2$ (R. K. Dennis, L. N. Vaserstein, *$K$-Theory*, **2**, N 6 (1989), 761--767).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03485 | KOU-14.16

### Kourovka Notebook Problem 14.16

O / open | Kourovka 群论问题集

Following Yu. I. Merzlyakov we define the *width* of a verbal subgroup $V(G)$ of a group $G$ with respect to the set of words $V$ as the smallest $m\in {\mathbb{N}\cup \{ \infty\}$ such that every element of $V(G)$ can be written as a product of $\leq m$ values of words from $V\cup V^{-1}$. It is known that the width of any verbal subgroup of a finitely generated group of polynomial growth is finite. Is this statement true for finitely generated groups of intermediate growth?

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03486 | KOU-14.17

### Kourovka Notebook Problem 14.17

O / open | Kourovka 群论问题集

Let $\mathrm{IMA}(G)$ denote the subgroup of the automorphism group $\mathrm{Aut }G$ consisting of all automorphisms that act trivially on the second derived quotient $G/G''$. Find generators and defining relations for $\mathrm{IMA}(F_n)$, where $F_n$ is a free group of rank $n\geq 3$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03487 | KOU-14.18

### Kourovka Notebook Problem 14.18

O / open | Kourovka 群论问题集

We say that a family of groups ${\mathcal{D}}$ *discriminates* a group $G$ if for any finite subset $\{a_1,\ldots ,a_n\}\subseteq G\setminus\{ 1\}$ there exists a group $D\in {\mathcal{D}}$ and a homomorphism $\varphi :G\rightarrow D$ such that $a_j\varphi \ne 1$ for all $j=1,\ldots ,n$. Is every finitely generated group acting freely on some $\Lambda$-tree discriminated by torsion-free hyperbolic groups?

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03488 | KOU-14.19

### Kourovka Notebook Problem 14.19

S / solved | Kourovka 群论问题集

We say that a group $G$ has the *Noetherian Equation Property* if every system of equations over $G$ in finitely many variables is equivalent to some finite part of it. Does an arbitrary hyperbolic group have the Noetherian Equation Property?

*Comment of 2013:* the conjecture holds for torsion-free hyperbolic groups (Z. Sela, *Proc. London Math. Soc.*, **99**, no. 1 (2009), 217--273) and for the larger class of toral relatively hyperbolic groups (D. Groves, *J. Geom. Topol.*, **9** (2005), 2319--2358).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03489 | KOU-14.2

### Kourovka Notebook Problem 14.2

O / open | Kourovka 群论问题集

(S. D. Berman). Prove that every automorphism of the centre of the integral group ring of a finite group induces a monomial permutation on the set of the class sums.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03490 | KOU-14.20

### Kourovka Notebook Problem 14.20

S / solved | Kourovka 群论问题集

Does a free product of two groups have the Noetherian Equation Property if this property is enjoyed by the factors?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03491 | KOU-14.21

### Kourovka Notebook Problem 14.21

O / open | Kourovka 群论问题集

Does a free pro-$p $-group have the Noetherian Equation Property?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03492 | KOU-14.22

### Kourovka Notebook Problem 14.22

O / open | Kourovka 群论问题集

Prove that any irreducible system of equations $S(x_1, \ldots,x_n) = 1$ with coefficients in a torsion-free linear group $G$ is equivalent over $G$ to a finite system $T(x_1, \ldots,x_n) = 1$ satisfying an analogue of Hilbert's Nullstellensatz, i. e. $Rad_G(T) = \sqrt T$. This is true if $G$ is a free group (O. Kharlampovich, A. Myasnikov, *J. Algebra*, **200** (1998), 472--570).

Here both $S$ and $T$ are regarded as subsets of $G[X] = G \ast F(X)$, a free product of $G$ and a free group on $X = \{x_1, \ldots, x_n \}$. By definition, $ Rad_G(T) = \{ w(x_1, \ldots,x_n) \in G[X] \mid w(g_1, \ldots, g_n) = 1$ for any solution $g_1, \ldots, g_n \in G$ of the system $T(X) = 1\}$, and $\sqrt T$ is the minimal normal isolated subgroup of $G[X]$ containing $T$.

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03493 | KOU-14.23

### Kourovka Notebook Problem 14.23

S / solved | Kourovka 群论问题集

Let $F_n$ be a free group with basis $\{ x_1,\ldots ,x_n \}$, and let $|\cdot |$ be the length function with respect to this basis. For $\alpha\in \mathrm{Aut }F_n$ we put $|\! |\alpha |\! |=\max \{ |\alpha (x_1)|,\ldots ,|\alpha (x_n)| \}$. Is it true that there is a recursive function $f: {\mathbb{N} \rightarrow {\mathbb{N}}$ with the following property: for any $\alpha\in \mathrm{Aut} F_n$ there is a basis $\{ y_1,\ldots ,y_k \}$ of $\mathrm{Fix}(\alpha )=\{ x\mid \alpha (x)=x\}$ such that $|y_i|\leq f(|\! |\alpha |\! |)$ for all $i=1,\ldots ,k$?

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03494 | KOU-14.24

### Kourovka Notebook Problem 14.24

O / open | Kourovka 群论问题集

Let $\mathrm{Aut }F_n$ be the automorphism group of a free group of rank $n$ with norm $|\! |\cdot |\! |$ as in 14.23. Does there exist a recursive function $f: {\mathbb{N} \times {\mathbb{N} \rightarrow {\mathbb{N}}$ with the following property: for any two conjugate elements $\alpha,\beta\in \mathrm{Aut }F_n$ there is an element $\gamma \in \mathrm{Aut }F_n$ such that $\gamma ^{-1}\alpha \gamma =\beta$ and $|\! |\gamma |\! |\leq f(|\! |\alpha |\! |,|\! |\beta |\! |)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03495 | KOU-14.25

### Kourovka Notebook Problem 14.25

S / solved | Kourovka 群论问题集

Let $qG$ denote the quasivariety generated by a group $G$. Is it true that there exists a finitely generated group $G$ such that the set of proper maximal subquasivarieties in $qG$ is infinite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03496 | KOU-14.26

### Kourovka Notebook Problem 14.26

O / open | Kourovka 群论问题集

A quasivariety ${\mathcal{M}}$ is {\it closed under direct ${\mathbb{Z}}$-wreath products} if the direct wreath product $G\wr {\mathbb{Z}}$ belongs to ${\mathcal{M}}$ for every $G\in {\mathcal{M}}$ (here ${\mathbb{Z}}$ is an infinite cyclic group). Is the quasivariety generated by the class of all nilpotent torsion-free groups closed under direct ${\mathbb{Z}}$-wreath products?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03497 | KOU-14.27

### Kourovka Notebook Problem 14.27

S / solved | Kourovka 群论问题集

Let $\Gamma$ be a group generated by a finite set $S$. Assume that there exists a nested sequence $F_1\subset F_2\subset\cdots$ of finite subsets of $\Gamma$ such that \ (i) $F_k\ne F_{k+1}$ for all $k\geq 1$, \ \ (ii) $\displaystyle{\Gamma=\bigcup_{k \geq 1}F_k}$, \ \ (iii) $\displaystyle{\lim_{k\to\infty}\frac{\vert\partial F_k\vert}{\vert F_k\vert}=0}$, where, by definition, $\partial F_k = \{ \gamma\in\Gamma \setminus F_k \mid \text{there exists}\; s\in S \text{ such that}\;\gamma s\in F_k \}$, \ and \ (iv) there exist constants $c\geq 0, d\geq 1$ such that $\vert F_k\vert\le c k^d$ for all $k\geq 1$. Does it follow that $\Gamma$ has polynomial growth?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03498 | KOU-14.28

### Kourovka Notebook Problem 14.28

O / open | Kourovka 群论问题集

Let ${\mathfrak{F}}$ be a soluble Fitting formation of finite groups with Kegel's property, that is, ${\mathfrak{F}}$ contains every finite group of the form $G=AB=BC=CA$ if $A,B,C$ are in ${\mathfrak{F}}$. Is ${\mathfrak{F}}$ a saturated formation?

*Comment of 2013:* Some progress was made in (A. Ballester-Bolinches, L. M. Ezquerro, *J. Group Theory*, **8**, no. 5 (2005), 605--611).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03499 | KOU-14.29

### Kourovka Notebook Problem 14.29

O / open | Kourovka 群论问题集

Is there a soluble Fitting class of finite groups ${\mathfrak{F}}$ such that ${\mathfrak{F}}$ is not a formation and $ A_{{\mathfrak{F}}}\cap B_{{\mathfrak{F}}}\subseteq G_{{\mathfrak{F}}}$ for every finite soluble group of the form $G=AB$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03500 | KOU-14.3

### Kourovka Notebook Problem 14.3

O / open | Kourovka 群论问题集

Is it true that every central unit of the integral group ring of a finite group is a product of a central element of the group and a symmetric central unit? (A unit is *symmetric* if it is fixed by the canonical antiinvolution that transposes the coefficients at the mutually inverse elements.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03501 | KOU-14.30

### Kourovka Notebook Problem 14.30

O / open | Kourovka 群论问题集

Let $\mathrm{lFit}{\mathfrak{X}}$ be the local Fitting class generated by a set of groups ${\mathfrak{X}}$ and let $\Psi (G)$ be the smallest normal subgroup of a finite group $G$ such that $\mathrm{lFit}(\Psi (G)\cap M)=\mathrm{lFit}M$ for every $M\mbox{$\lhd\lhd$} G$ (K. Doerk, P. Hauck, *Arch. Math.*, **35**, no. 3 (1980), 218--227). We say that a Fitting class ${\mathfrak{F}}$ is *saturated* if $\Psi (G)\in {\mathfrak{F}}$ implies that $G\in {\mathfrak{F}}$. Is it true that every non-empty soluble saturated Fitting class is local?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03502 | KOU-14.31

### Kourovka Notebook Problem 14.31

O / open | Kourovka 群论问题集

Is the lattice of Fitting subclasses of the Fitting class generated by a finite soluble group finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03503 | KOU-14.32

### Kourovka Notebook Problem 14.32

S / solved | Kourovka 群论问题集

Extending the classical definition of formations, let us define a *formation* of (not necessarily finite) groups as a nonempty class of groups closed under taking homomorphic images and subdirect products with finitely many factors. Must every first order axiomatizable formation of groups be a variety?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03504 | KOU-14.33

### Kourovka Notebook Problem 14.33

S / solved | Kourovka 群论问题集

Does there exist a finitely presented pro-$p $-group ($p$ being a prime) which contains an isomorphic copy of every countably based pro-$p $-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03505 | KOU-14.34

### Kourovka Notebook Problem 14.34

S / solved | Kourovka 群论问题集

By definition, a locally compact group has the *Kazhdan $T$-property* if the trivial representation is an isolated point in the natural topological space of unitary representations of the group. Does there exist a profinite group with two dense discrete subgroups one of which is amenable, and the other has the Kazhdan $T$-property?

See (A. Lubotzky, *Discrete groups, expanding graphs and invariant measures* (*Progress in Mathematics, Boston, Mass.*, **125**), Birkh\"{a}user, Basel, 1994) for further motivation.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03506 | KOU-14.35

### Kourovka Notebook Problem 14.35

O / open | Kourovka 群论问题集

Is every finitely presented group of prime exponent finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03507 | KOU-14.36

### Kourovka Notebook Problem 14.36

O / open | Kourovka 群论问题集

A group $G$ is called a *$T$-group* if every subnormal subgroup of $G$ is normal, while $G$ is said to be a {\it $\overline{\! T}$-group} if all of its subgroups are $T$-groups. Is it true that every non-periodic locally graded $ \overline{\! T}$-group must be abelian?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03508 | KOU-14.37

### Kourovka Notebook Problem 14.37

O / open | Kourovka 群论问题集

Let $G(n)$ be one of the classical groups (special, orthogonal, or symplectic) of $(n\times n)$-matrices over an infinite field $K$ of non-zero characteristic, and $M(n)$ the space of all $(n\times n)$-matrices over $K$. The group $G(n)$ acts diagonally by conjugation on the space $M(n)^m=\underbrace{M(n)\oplus\cdots\oplus M(n)}_{m}$. Find generators of the algebra of invariants $K[M(n)^m]^{G(n)}$.

In characteristic $0$ they were found in (C. Procesi, *Adv. Math.*, **19** (1976), 306--381). *Comment of 2001:* in positive characteristic the problem is solved for all cases excepting the orthogonal groups in characteristic 2 and special orthogonal groups of even degree (A. N. Zubkov, *Algebra and Logic*, **38**, no. 5 (1999), 299--318). *Comment of 2009:* ...and for special orthogonal groups of even degree over (infinite) fields of odd characteristic (A. A. Lopatin, *J. Algebra*, **321** (2009), 1079--1106).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03509 | KOU-14.38

### Kourovka Notebook Problem 14.38

O / open | Kourovka 群论问题集

For every pro-$p $-group $G$ of $(2\times 2)$-matrices for $p\ne 2$ an analogue of the Tits Alternative holds: either $G$ is soluble, or the variety of pro-$p $-groups generated by $G$ contains the group {\small $\overline{\left< \left( \!\!\begin{array}{cc}1&t\cr 0&1 \end{array}\!\!\right), \left(\!\! \begin{array}{cc}1&0\cr t&1\end{array}\!\!\right) \right> }$} $\leq SL_2({\mathbb{F} _p[[t]])$ (A. N. Zubkov, *Algebra and Logic*, **29**, no. 4 (1990), 287--301). Is the same result true for matrices of size $\geq 3$ for $p\ne 2$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03510 | KOU-14.39

### Kourovka Notebook Problem 14.39

O / open | Kourovka 群论问题集

Let $F(V)$ be a free group of some variety of pro-$p $-groups $V$. Is there a uniform bound for the exponents of periodic elements in $F(V)$? This is true if the variety $V$ is metabelian (A. N. Zubkov, *Dr. Sci. Diss.*, Omsk, 1997 (Russian)).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03511 | KOU-14.4

### Kourovka Notebook Problem 14.4

S / solved | Kourovka 群论问题集

a) Is it true that there exists a nilpotent group $G$ for which the lattice ${\mathcal{L}}(G)$ of all group topologies is not modular? (It is known that for abelian groups the lattice ${\mathcal{L}}(G)$ is modular and that there are groups for which this lattice is not modular: V. I. Arnautov, A. G. Topale, *Izv. Akad. Nauk Moldova Mat.*, **1997**, no. 1, 84--92 (Russian).)

b) Is it true that for every countable nilpotent non-abelian group $G$ the lattice ${\mathcal{L}}(G)$ of all group topologies is not modular?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03512 | KOU-14.40

### Kourovka Notebook Problem 14.40

O / open | Kourovka 群论问题集

Is a free pro-$p $-group representable as an abstract group by matrices over a commutative-associative ring with $1$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03513 | KOU-14.41

### Kourovka Notebook Problem 14.41

O / open | Kourovka 群论问题集

A group $G$ is said to be *para-free* if all factors $\gamma _i(G)/\gamma _{i+1}(G)$ of its lower central series are isomorphic to the corresponding lower central factors of some free group, and $\bigcap_{i=1}^{\infty}\gamma _i(G) =1$. Is an arbitrary para-free group representable by matrices over a commutative-associative ring with $1$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03514 | KOU-14.42

### Kourovka Notebook Problem 14.42

O / open | Kourovka 群论问题集

Is a free pro-$p$-group representable by matrices over an associative-commutative profinite ring with 1?

A negative answer is equivalent to the fact that every linear pro-$p$-group satisfies a non-trivial pro-$p$-identity. This is known to be true in dimension $2$ for $p\ne 2$ (A. N. Zubkov, *Siberian Math. J.*, **28**, no. 5 (1987), 742--747). It is also proved that a 2-dimensional linear pro-$2$-group in characteristic 2 satisfies a non-trivial pro-$2$-identity (D. E.-C. Ben-Ezra, E. Zelmanov, *Trans. Amer. Math. Soc.*, **374**, no. 6 (2021), 4093--4128).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03515 | KOU-14.43

### Kourovka Notebook Problem 14.43

O / open | Kourovka 群论问题集

Suppose that a finite group $G$ has the form $G=AB $, where $A$ and $B$ are nilpotent subgroups of classes $\alpha$ and $\beta$ respectively; then $G$ is soluble (O. H. Kegel, H. Wielandt, 1961). Although the derived length $\mathrm{dl}(G)$ of $G$ need not be bounded by $\alpha +\beta$ (see Archive, 5.17), can one bound $\mathrm{dl}(G)$ by a (linear) function of $\alpha$ and $\beta$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03516 | KOU-14.44

### Kourovka Notebook Problem 14.44

O / open | Kourovka 群论问题集

Let $k(X)$ denote the number of conjugacy classes of a finite group $X$. Suppose that a finite group $G=AB$ is a product of two subgroups $A,B$ of coprime orders. Is it true that $k(AB) \leq k(A)k(B)$?

Note that one cannot drop the coprimeness condition and the answer is positive if one of the subgroups is normal, see Archive, 11.43.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03517 | KOU-14.45

### Kourovka Notebook Problem 14.45

O / open | Kourovka 群论问题集

Does there exist a (non-abelian simple) linearly right-orderable group all of whose proper subgroups are cyclic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03518 | KOU-14.46

### Kourovka Notebook Problem 14.46

O / open | Kourovka 群论问题集

A finite group $G$ is said to be *almost simple* if $T\leq G\leq \mathrm{Aut}(T)$ for some nonabelian simple group $T$. By definition, a *finite linear space* consists of a set $V$ of points, together with a collection of $k$-element subsets of $V$, called *lines* ($k\geq 3$), such that every pair of points is contained in exactly one line. Classify the finite linear spaces which admit a line-transitive almost simple subgroup $G$ of automorphisms which acts transitively on points.

A. Camina, P. Neumann and C. E. Praeger have solved this problem in the case where $T$ is an alternating group. In (A. Camina, C. E. Praeger, *Aequat. Math.*, **61** (2001), 221--232) it is shown that a line-transitive group of automorphisms of a finite linear space which is point-*quasiprimitive* (i. e. all of whose non-trivial normal subgroups are point-transitive) is almost simple or affine. \

*Remarks of 2005:* \ The cases of the following groups, but not almost simple groups with this socle, were dealt with: $PSU(3,q)$ (W. Liu, *Linear Algebra Appl.*, **374** (2003), 291--305); $PSL(2,q)$ (W. Liu, *J. Combin. Theory (A)*, **103** (2003), 209--222); $Sz(q)$ (W. Liu, *Discrete Math.*, **269** (2003), 181--190); $Ree(q)$ (W. Liu, *Europ. J. Combin.*, **25** (2004), 311--325); sporadic simple groups were done in (A. R. Camina, F. Spiezia, *J. Combin. Des.*, **8** (2000), 353--362). \ *Remark of 2009:* It was shown (N. Gill, *Trans. Amer. Math. Soc.*, **368** (2016), 3017--3057) that for non-desarguesian projective planes a point-quasiprimitive group would be affine. \ *Remark of 2013:* The problem is solved for large-dimensional classical groups (A. Camina, N. Gill, A. E. Zalesski, *Bull. Belg. Math. Soc. Simon Stevin*, **15**, no. 4 (2008), 705--731).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03519 | KOU-14.47

### Kourovka Notebook Problem 14.47

O / open | Kourovka 群论问题集

Is the lattice of all soluble Fitting classes of finite groups modular?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03520 | KOU-14.48

### Kourovka Notebook Problem 14.48

S / solved | Kourovka 群论问题集

If an equation over a free group $F$ has no solution in $F$, is there a finite quotient of $F$ in which the equation has no solution?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03521 | KOU-14.49

## Kourovka Notebook Problem 14.49

S / solved | Kourovka 群论问题集

Is $SL_3({\mathbb{Z} )$ a factor group of the modular group $ PSL_2({\mathbb{Z} )$? Since the latter is isomorphic to the free product of two cyclic groups of orders 2 and 3, the question asks if $SL_3({\mathbb{Z} )$ can be generated by two elements of orders $2$ and $3$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03522 | KOU-14.5

## Kourovka Notebook Problem 14.5

P / partially_solved | Kourovka 群论问题集

Let $G$ be an infinite group admitting non-discrete Hausdorff group topologies, and ${\mathcal{L}}(G)$ the lattice of all group topologies on $G$.

a) Is it true that for any natural number $k$ there exists a non-refinable chain $\tau_0 < \tau_1<\cdots < \tau_k$ of length $k$ of Hausdorff topologies in ${\mathcal{L}}(G)$? (For countable nilpotent groups this is true (A. G. Topale, *Deposited in VINITI*, 25.12.98, no. 3849--V 98 (Russian)).)

b) Let $k,m,n$ be natural numbers and let $G$ be a nilpotent group of class $k$. Suppose that $\tau_0 <\tau_1 <\cdots < \tau_m$ and $\tau'_0 <\tau'_1 <\cdots < \tau'_n$ are non-refinable chains of Hausdorff topologies in ${\mathcal{L}}(G)$ such that $\tau_0 =\tau'_0$ and $\tau_m =\tau'_n$. Is it true that $m\leq n\cdot k$ and this inequality is best-possible? (This is true if $k=1$, since for $G$ abelian the lattice ${\mathcal{L}}(G)$ is modular.)

c) Is it true that there exists a countable $G$ such that in the lattice ${\mathcal{L}}(G)$ there are a finite non-refinable chain $\tau_0 <\tau_1 <\cdots < \tau_k$ of Hausdorff topologies and an infinite chain $\{\tau'_{\gamma}\mid \gamma\in\Gamma\}$ of topologies such that $\tau_0<\tau'_{\gamma} < \tau_k$ for any $\gamma\in\Gamma$?

d) Let $G$ be an abelian group, $k$ a natural number. Let $A_k$ be the set of all those Hausdorff group topologies on $G$ that, for every topology $\tau\in A_k$, any non-refinable chain of topologies starting from $\tau$ and terminating at the discrete topology has length $k$. Is it true that $A_k\bigcap \{\tau'_{\gamma}\mid \gamma\in\Gamma \}\neq\varnothing$ for any infinite non-refinable chain $\{\tau'_{\gamma}\mid \gamma\in\Gamma \}$ of Hausdorff topologies containing the discrete topology? (This is true for $k=1$.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03523 | KOU-14.50

## Kourovka Notebook Problem 14.50

S / solved | Kourovka 群论问题集

(Z. I. Borevich). A subgroup $A$ of a group $G$ is said to be *paranormal* (respectively, *polynormal*) if $A^x\leq \left< A^u \mid u\in \left< A, A^x\right> \right> $ (respectively, $A^x\leq \left< A^u \mid u\in A^{\left< x \right>}\right> $) for any $x\in G$. Is every polynormal subgroup of a finite group paranormal?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03524 | KOU-14.51

## Kourovka Notebook Problem 14.51

O / open | Kourovka 群论问题集

(Well-known problems). Is there a finite basis for the identities of any

a) abelian-by-nilpotent group?

b) abelian-by-finite group?

c) abelian-by-(finite nilpotent) group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03525 | KOU-14.52

## Kourovka Notebook Problem 14.52

S / solved | Kourovka 群论问题集

It is known that if a finitely generated group is residually torsion-free nilpotent, then the group is residually finite $p$-group, for every prime $p$. Is the converse true?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03526 | KOU-14.53

## Kourovka Notebook Problem 14.53

O / open | Kourovka 群论问题集

*Conjecture:* Let $G$ be a profinite group such that the set of solutions of the equation $x^n=1$ has positive Haar measure. Then $G$ has an open subgroup $H$ and an element $t$ such that all elements of the coset $tH$ have order dividing $n$.

This is true in the case $n=2$. It would be interesting to see whether similar results hold for profinite groups in which the set of solutions of some equation has positive measure.

*Comment of 2021*: This is also proved for $n=3$ (A. Abdollahi, M. S. Malekan, *Adv. Group Theory Appl.*, **13** (2022), 71--81). *Comment of 2025*: Significant progress was obtained in (S. Kionke, N. Otmen, T. Toti, M. Vannacci, T. Weigel, *Preprint*, 2025, https://arxiv.org/pdf/2507.19086).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03527 | KOU-14.54

### Kourovka Notebook Problem 14.54

S / solved | Kourovka 群论问题集

Let $k(G)$ denote the number of conjugacy classes of a finite group $G$. Is it true that $k(G)\leq |N|$ for some nilpotent subgroup $N$ of $G$? This is true if $G$ is simple; besides, always $k(G)\leq |S|$ for some soluble subgroup $S\leq G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03528 | KOU-14.55

### Kourovka Notebook Problem 14.55

P / partially_solved | Kourovka 群论问题集

b) Prove that the Nottingham group $J=N({\mathbb{Z}/p{\mathbb{Z}})$ (as defined in Archive, 12.24) is finitely presented for $p=2$.

[Archived solved part(s) of 14.55:] a) Prove that the Nottingham group $J=N({\mathbb{Z}/p{\mathbb{Z}})$ (as defined in Archive, 12.24) is finitely presented for $p>2$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03529 | KOU-14.56

### Kourovka Notebook Problem 14.56

O / open | Kourovka 群论问题集

Prove that if $G$ is an infinite pro-$p $-group with $G/\gamma_{2p+1}(G)$ isomorphic to $J/\gamma_{2p+1}(J)$ then $G$ is isomorphic to $J$, where $J$ is the Nottingham group.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03530 | KOU-14.57

### Kourovka Notebook Problem 14.57

O / open | Kourovka 群论问题集

Describe the hereditarily just infinite pro-$p $-groups of finite width.

Definitions: A pro-$p $-group $G$ has finite *width* $p^d$ if $|\gamma _i(G)/\gamma _{i+1}(G)|\leq p^d$ for all $i$, and $G$ is *hereditarily just infinite* if $G$ is infinite and each of its open subgroups has no closed normal subgroups of infinite index.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03531 | KOU-14.58

### Kourovka Notebook Problem 14.58

P / partially_solved | Kourovka 群论问题集

a) Suppose that $A$ is a periodic group of regular automorphisms of an abelian group. Is $A$ cyclic if $A$ has prime exponent?

[Archived solved part(s) of 14.58:] b) Suppose that $A$ is a periodic group of regular automorphisms of an abelian group. Is $A$ finite if $A$ is generated by elements of order $3$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03532 | KOU-14.59

### Kourovka Notebook Problem 14.59

O / open | Kourovka 群论问题集

Suppose that $G$ is a triply transitive group in which a stabilizer of two points contains no involutions, and a stabilizer of three points is trivial. Is it true that $G$ is similar to $PGL_2(P)$ in its natural action on the projective line $P\cup \{ \infty \}$, for some field $P$ of characteristic $2$? This is true under the condition that the stabilizer of two points is periodic.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03533 | KOU-14.6

### Kourovka Notebook Problem 14.6

O / open | Kourovka 群论问题集

A group $\Gamma$ is said to have {\it Property $P_{\rm{nai}}$} if, for any finite subset $F$ of $\Gamma \setminus \{1\},$ there exists an element $y_0 \in \Gamma$ of infinite order such that, for each $x \in F$, the canonical epimorphism from the free product $\langle x \rangle * \langle y_0\rangle$ onto the subgroup $\langle x , y_0 \rangle$ of $\Gamma$ generated by $x$ and $y_0$ is an isomorphism. For $n \in \{2, 3,\ldots\},$ does $PSL_n({\mathbb{Z} )$ have Property $P_{\rm{nai}}$? More generally, if $\Gamma$ is a lattice in a connected real Lie group $G$ which is simple and with centre reduced to $\{ 1\},$ does $\Gamma$ have Property $P_{\rm{nai}}$?

Answers are known to be " yes" \ if $n = 2$, and more generally if $G$ has real rank $1$ (M. Bekka, M. Cowling, P. de la Harpe, *Publ. Math. IHES*, **80** (1994), 117--134).

*Comments of 2018*: Property $P_{\rm{nai}}$ has been established for relatively hyperbolic groups (G. Arzhantseva, A. Minasyan, *J. Funct. Anal.*, **243**, no. 1 (2007), 345--351) and for groups acting on CAT(0) cube complexes with appropriate conditions (A. Kar, M. Sageev, *Comment. Math. Helv.*, **91** (2016), 543--561). Some progress has been made in the preprint (T. Poznansky, http://arxiv.org/abs/0812.2486).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03534 | KOU-14.60

### Kourovka Notebook Problem 14.60

S / solved | Kourovka 群论问题集

Suppose that $H$ is a non-trivial normal subgroup of a finite group $G$ such that the factor-group $G/H$ is isomorphic to one of the simple groups $L_n(q)$, $n\geq 3$. Is it true that $G$ has an element whose order is distinct from the order of any element in $G/H$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03535 | KOU-14.61

### Kourovka Notebook Problem 14.61

O / open | Kourovka 群论问题集

Determine all pairs $({\mathcal{S}},G)$, where $\mathcal{S}$ is a semipartial geometry and $G$ is an almost simple flag-transitive group of automorphisms of $\mathcal{S}$. A system of points and lines $(P,B)$ is a *semipartial geometry* with parameters $(\alpha,s,t,\mu)$ if every point belongs to exactly $t+1$ lines (two different points belong to at most one line); every line contains exactly $s+1$ points; for any anti-flag $(a,l)\in (P,B)$ the number of lines containing $a$ and intersecting $l$ is either $0$ or $\alpha$; and for any non-collinear points $a,b$ there are exactly $\mu$ points collinear with $a$ and with $b$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03536 | KOU-14.62

### Kourovka Notebook Problem 14.62

S / solved | Kourovka 群论问题集

Suppose that $H$ is a non-soluble normal subgroup of a finite group $G$. Does there always exist a maximal soluble subgroup $S$ of $H$ such that $G=H\cdot N_G(S)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03537 | KOU-14.63

### Kourovka Notebook Problem 14.63

S / solved | Kourovka 群论问题集

What are the composition factors of non-soluble finite groups all of whose normalizers of Sylow subgroups are $2$-nilpotent, in particular, supersoluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03538 | KOU-14.64

### Kourovka Notebook Problem 14.64

O / open | Kourovka 群论问题集

(M. F. Newman). Classify the finite $5$-groups of maximal class; computer calculations suggest some conjectures (M. F. Newman, in: *Groups--Canberra, 1989 $($Lecture Notes Math.*, **1456**), Springer, Berlin, 1990, 49--62).

*M. F. Newman's comment of 2025*: Recent work on related conjectures can be found in (A. Cant, H. Dietrich, B. Eick, T. Moede, *J. Algebra*, **604** (2022), 429--450).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03539 | KOU-14.65

### Kourovka Notebook Problem 14.65

O / open | Kourovka 群论问题集

(Well-known problem). For a finite group $G$ let $\rho(G)$ denote the set of prime numbers dividing the order of some conjugacy class, and $\sigma(G)$ the maximum number of primes dividing the order of some conjugacy class. Is it true that $|\rho(G)|\leq3\sigma(G)$?

A possible linear bound for $|\rho(G)|$ in terms of $\sigma(G)$ cannot be better than $3\sigma(G)$, since there is a family of groups $\{G_n\}$ such that $\lim_{n\to\infty}|\rho(G_n)|/\sigma(G_n)=3$ (C. Casolo, S. Dolfi, *Rend. Sem. Mat. Univ. Padova*, **96** (1996), 121--130).

*Comment of 2009:* A quadratic bound was obtained in (A. Moret\'o, *Int. Math. Res. Not.*, **2005**, no. 54, 3375--3383), and a linear bound in (C. Casolo, S. Dolfi, *J. Group Theory*, **10**, no. 5 (2007), 571--583).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03540 | KOU-14.66

### Kourovka Notebook Problem 14.66

S / solved | Kourovka 群论问题集

(Well-known problem). Let $G$ be a finite soluble group, $\pi(G)$ the set of primes dividing the order of $G$, and $\nu(G)$ the maximum number of primes dividing the order of some element. Does there exist a linear bound for $|\pi(G)|$ in terms of $\nu(G)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03541 | KOU-14.67

### Kourovka Notebook Problem 14.67

S / solved | Kourovka 群论问题集

Suppose that $a$ is a non-trivial element of a finite group $G$ such that $|C_{G}(a)| \geq |C_{G}(x)|$ for every non-trivial element $x\in G$, and let $H$ be a nilpotent subgroup of $G$ which is normalized by $C_{G}(a)$. Is it true that $H \leq C_{G}(a)$? This is true if $H$ is abelian, or if $H$ is a $p'$-group for some prime number $p\in \pi(Z(C_G(a)))$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03542 | KOU-14.68

### Kourovka Notebook Problem 14.68

O / open | Kourovka 群论问题集

(Well-known problem). Suppose that $F$ is an automorphism of order $2$ of the polynomial ring $R_n={\mathbb{C}}[x_1, \ldots ,x_n]$, $n> 2$. Does there exist an automorphism $G$ of $R_n$ such that $G^{-1}FG$ is a linear automorphism?

This is true for $n=2$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03543 | KOU-14.69

## Kourovka Notebook Problem 14.69

P / partially_solved | Kourovka 群论问题集

For every finite simple group find the minimum of the number of generating involutions satisfying an additional condition, in each of the following cases.

a) The product of the generating involutions equals $1$.

b) (Malle--Saxl--Weigel). All generating involutions are conjugate.

c) (Malle--Saxl--Weigel). The conditions a) and b) are simultaneously satisfied.

*Editors' comment:* partial progress was obtained in (J. Ward, PhD Thesis, QMW, 2009, http://www.maths.qmul.ac.uk/ raw/JWardPhD.pdf).

d) All generating involutions are conjugate and two of them commute.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03544 | KOU-14.7

## Kourovka Notebook Problem 14.7

O / open | Kourovka 群论问题集

Let $\Gamma_g$ be the fundamental group of a closed surface of genus $g \geq 2.$ For each finite system $S$ of generators of $\Gamma_g,$ let $\beta_S(n)$ denote the number of elements of $\Gamma _g$ which can be written as products of at most $n$ elements of $S \cup S^{-1},$ and let $\omega (\Gamma _g,S) = \limsup_{n \to \infty}\root n \of {\beta _S(n)}$ be the growth rate of the sequence $\left( \beta _S(n) \right)_{n \geq 0}.$ Compute the infimum $\omega(\Gamma _g)$ of the $\omega (\Gamma _g,S)$ over all finite sets of generators of $\Gamma _g.$

It is easy to see that, for a free group $F_k$ of rank $k \geq 2,$ the corresponding infimum is $\omega(F_k) = 2k-1$ (M. Gromov, *Structures m\'etriques pour les vari\'et\'es riemanniennes,* Cedic/F. Nathan, Paris, 1981, Ex. 5.13). As any generating set of $\Gamma_g$ contains a subset of $2g-1$ elements generating a subgroup of infinite index in $\Gamma_g$ with abelianization ${\mathbb{Z} ^{2g-1},$ hence a subgroup which is free of rank $2g-1,$ it follows that $\omega(\Gamma_g) \geq 4g-3.$

问题集入口：https://kourovkanotebookorg.wordpress.com/

03545 | KOU-14.70

## Kourovka Notebook Problem 14.70

O / open | Kourovka 群论问题集

A group $G$ is called *$n$-Engel* if it satisfies the identity $[x,y,\ldots ,y]=1$, where $y$ is taken $n$ times. Are there non-nilpotent finitely generated $n$-Engel groups?

*Comment of 2017*: For $n=2,3,4$, finitely generated $n$-Engel groups are nilpotent. A course of lectures devoted to constructing non-nilpotent finitely generated $n$-Engel groups for sufficiently large $n$ is given by E. Rips and is available on the Internet.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03546 | KOU-14.71

## Kourovka Notebook Problem 14.71

S / solved | Kourovka 群论问题集

Consider a free group $F$ of finite rank and an arbitrary group $G$. Define the *$G$-closure* $\mathrm{cl}_G(T)$ of any subset $T\subseteq F$ as the intersection of the kernels of all those homomorphisms $\mu:F \rightarrow G$ of $F$ into $G$ that vanish on $T$: \ $ \mathrm{cl}_G(T) = \bigcap \{ \mathrm{Ker} \mu \mid \mu :F \rightarrow G;\; T\subseteq \mathrm{Ker} \mu \}$. \ Groups $G$ and $H$ are called *geometrically equivalent* if for every free group $F$ and every subset $T\subset F$ the $G$- and $H$-closures of $T$ coincide: $\mathrm{cl}_G(T)=\mathrm{cl}_H(T)$. It is easy to see that if $G$ and $H$ are geometrically equivalent then they have the same quasiidentities. Is it true that if two groups have the same quasiidentities then they are geometrically equivalent? This is true for nilpotent groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03547 | KOU-14.72

## Kourovka Notebook Problem 14.72

S / solved | Kourovka 群论问题集

Let $X$ be a regular algebraic variety over a field of arbitrary characteristic, and $G$ a finite cyclic group of automorphisms of $X$. Suppose that the fixed point variety $X^G$ of $G$ is a regular hypersurface of $X$ (of codimension $1$). Is the quotient variety $X/G$ regular?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03548 | KOU-14.73

## Kourovka Notebook Problem 14.73

O / open | Kourovka 群论问题集

*Conjecture:* There is a function $f$ on the natural numbers such that, if $\Gamma$ is a finite, vertex-transitive, locally-quasiprimitive graph of valency $v$, then the number of automorphisms fixing a given vertex is at most $f(v)$. (By definition, a vertex-transitive graph $\Gamma$ is *locally-quasiprimitive* if the stabilizer in $Aut(\Gamma)$ of a vertex $\alpha$ is quasiprimitive (see 14.46) in its action on the set of vertices adjacent to $\alpha$.)

To prove the conjecture above one need only consider the case where $Aut(\Gamma)$ has the property that every non-trivial normal subgroup has at most two orbits on vertices (C. E. Praeger, *Ars Combin.*, **19 A** (1985), 149--163). The analogous conjecture for finite, vertex-transitive, locally-primitive graphs was made by R. Weiss in 1978 and is still open. For non-bipartite graphs, there is a ``reduction'' of Weiss' conjecture to the case where the automorphism group is almost simple (see 14.46 for definition) (M. Conder, C. H. Li, C. E. Praeger, *Proc. Edinburgh Math. Soc. (2)*, **43**, no. 1 (2000), 129--138).

*Comment of 2013:* These conjectures have been proved in the case where there is an upper bound on the degree of any alternating group occurring as a quotient of a subgroup (C. E. Praeger, L. Pyber, P. Spiga, E. Szab\'{o}, *Proc. Amer. Math. Soc.*, **140**, no. 7 (2012), 2307--2318). *Comment of 2021:* This has been proved in the case where the group induced on the neighbourhood of a vertex has an abelian regular normal subgroup (P. Spiga, *Bull. London Math. Soc.*, **48**, no. 1 (2016), 12--18.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03549 | KOU-14.74

## Kourovka Notebook Problem 14.74

O / open | Kourovka 群论问题集

Let $k(G)$ denote the number of conjugacy classes of a finite group $G$. Is it true that $k(G)\leq k(P_1)\cdots k(P_s)$, where $P_1,\ldots ,P_s$ are Sylow subgroups of $G$ such that $|G|=|P_1|\cdots |P_s|$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03550 | KOU-14.75

## Kourovka Notebook Problem 14.75

O / open | Kourovka 群论问题集

Suppose that ${\mathfrak{G}}=\{ G_1, G_2,\ldots \}$ is a family of finite 2-generated groups which generates the variety of all groups. Is it true that a free group of rank 2 is residually in ${\mathfrak{G}}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03551 | KOU-14.76

## Kourovka Notebook Problem 14.76

O / open | Kourovka 群论问题集

Does there exist an absolute constant $c$ such that any finite $p $-group $P$ has an abelian section $A$ satisfying $|A|^c>|P|$?

By a result of A. Yu. Olshanskii (*Math. Notes*, **23** (1978), 183--185) we cannot require $A$ to be a subgroup. By a result of J. G. Thompson (*J. Algebra*, **13** (1969), 149--151) the existence of such a section $A$ would imply the existence of a class $2$ subgroup $H$ of $P$ with $|H|^c >|P|$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03552 | KOU-14.77

## Kourovka Notebook Problem 14.77

S / solved | Kourovka 群论问题集

Let $p$ be a prime number and $X$ a finite set of powers of $p$ containing $1$. Is it true that $X$ is the set of all lengths of the conjugacy classes of some finite $p $-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03553 | KOU-14.78

## Kourovka Notebook Problem 14.78

O / open | Kourovka 群论问题集

Suppose that ${\mathfrak{H}}\subseteq {\mathfrak{F}}_{1}\subseteq {\mathfrak{M}}$ \ and \ ${\mathfrak \ H} \subseteq {\mathfrak{F}}_{2}\subseteq {\mathfrak{M}}$ where ${\mathfrak{H}}$ and ${\mathfrak{M}}$ are local formations of finite groups and ${\mathfrak{F}}_{1}$ is a complement for ${\mathfrak{F}}_{2}$ in the lattice of all formations between ${\mathfrak{H}}$ and $ {\mathfrak{M}}$. Is it true that ${\mathfrak{F}}_{1}$ and ${\mathfrak{F}}_{2}$ are local formations? This is true in the case ${\mathfrak{H}}=(1)$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03554 | KOU-14.79

## Kourovka Notebook Problem 14.79

O / open | Kourovka 群论问题集

Suppose that $ {\mathfrak{F}}={\mathfrak{M}H}=\mathrm{lFit}( G)$ is a soluble one-generated local Fitting class of finite groups where ${\mathfrak{H}}$ and ${\mathfrak{M}}$ are Fitting classes and ${\mathfrak{M}}\neq {\mathfrak{F}}$. Is ${\mathfrak{H}}$ a local Fitting class?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03555 | KOU-14.8

### Kourovka Notebook Problem 14.8

O / open | Kourovka 群论问题集

Let $G$ denote the group of germs at $+\infty$ of orientation-preserving homeomorphisms of the real line ${\mathbb{R}}$. Let $\alpha \in G$ be the germ of $x \mapsto x+1.$ What are the germs $\beta \in G$ for which the subgroup $\langle \alpha , \beta \rangle$ of $G$ generated by $\alpha$ and $\beta$ is free of rank $2$?

If $\beta$ is the germ of $x \mapsto x^k$ for an odd integer $k \geq 3$, it is known that $\langle \alpha , \beta \rangle$ is free of rank 2. The proofs of this rely on Galois theory (for $k$ an odd prime: S. White, *J. Algebra*, **118** (1988), 408--422; \ for any odd $k\geq 3$: S. A. Adeleke, A. M. W. Glass, L. Morley, *J. London Math. Soc.*, **43** (1991), 255--268, and for any odd $k\ne \pm 1$ and any even $k>0$ in several papers by S. D. Cohen and A. M. W. Glass).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03556 | KOU-14.80

### Kourovka Notebook Problem 14.80

S / solved | Kourovka 群论问题集

Is the lattice of all totally local formations of finite groups modular? The definition of a *totally local* formation see in (L. A. Shemetkov, A. N. Skiba, *Formatsii algebraicheskikh system*, Moscow, Nauka, 1989 (Russian)).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03557 | KOU-14.81

### Kourovka Notebook Problem 14.81

O / open | Kourovka 群论问题集

Prove that the formation generated by a finite group has only finitely many $S_{n}$-closed subformations.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03558 | KOU-14.82

### Kourovka Notebook Problem 14.82

S / solved | Kourovka 群论问题集

(Well-known problem). Describe the finite simple groups in which every element is a product of two involutions.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03559 | KOU-14.83

### Kourovka Notebook Problem 14.83

O / open | Kourovka 群论问题集

We say that an infinite simple group $G$ is a *monster of the third kind* if for every non-trivial elements $a,b$, of which at least one is not an involution, there are infinitely many elements $g\in G$ such that $\left< a,b^g\right> =G$. (Compare with V. P. Shunkov's definitions in Archive, 6.63, 6.64.) Is it true that every simple quasi-Chernikov group is a monster of the third kind? This is true for quasi-finite groups (A. I. Sozutov, *Algebra and Logic*, **36**, no. 5 (1997), 336--348). (We say that a non-$\sigma$ group is *quasi-$\sigma$* if all of its proper subgroups have the property $\sigma$.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03560 | KOU-14.84

### Kourovka Notebook Problem 14.84

O / open | Kourovka 群论问题集

An element $g$ of a (relatively) free group $F_r({\mathfrak{M}})$ of rank $r$ of a variety ${\mathfrak{M}}$ is said to be *primitive* if it can be included in a basis of $F_r({\mathfrak{M}})$.

a) Do there exist a group $F_r({\mathfrak{M}})$ and a non-primitive element $h\in F_r({\mathfrak{M}})$ such that for some monomorphism $\alpha$ of $F_r({\mathfrak{M}})$ the element $\alpha (h)$ is primitive?

b) Do there exist a group $F_r({\mathfrak{M}})$ and a non-primitive element $h\in F_r({\mathfrak{M}})$ such that for some $n>r$ the element $h$ is primitive in $F_n({\mathfrak{M}})$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03561 | KOU-14.85

### Kourovka Notebook Problem 14.85

S / solved | Kourovka 群论问题集

Suppose that an endomorphism $\varphi$ of a free metabelian group of rank $r$ takes every primitive element to a primitive one. Is $\varphi$ necessarily an automorphism? This is true for $r\leq 2$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03562 | KOU-14.86

### Kourovka Notebook Problem 14.86

S / solved | Kourovka 群论问题集

Does there exist an infinite locally nilpotent $p $-group that is equal to its commutator subgroup and in which every proper subgroup is nilpotent?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03563 | KOU-14.88

### Kourovka Notebook Problem 14.88

S / solved | Kourovka 群论问题集

We say that an element $u$ of a group $G$ is a *test element* if for any endomorphism $\varphi$ of $G$ the equality $\varphi (u)=u$ implies that $\varphi$ is an automorphism of $G$. Does a free soluble group of rank $2$ and derived length $d>2$ have any test elements?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03564 | KOU-14.89

### Kourovka Notebook Problem 14.89

O / open | Kourovka 群论问题集

(E. A. O'Brien, A. Shalev). Let $P$ be a finite $p$-group of order $p^m$ and let $m=2n+e$ with $e=0$ or $1$. By a theorem of P. Hall the number of conjugacy classes of $P$ has the form $n(p^2-1) + p^e + a(p^2-1)(p-1)$ for some integer $a\geq 0$, which is called the *abundance* of $P$.

a) Is there a bound for the coclass of $P$ which depends only on $a$? Note that $a=0$ implies coclass $1$ and that all known examples with $a=1$ have coclass $\le 3$. (The group $P$ has *coclass* $r$ if $|P|=p^{c+r}$ where $c$ is the nilpotency class of $P$.)

b) Is there an element $s\in P$ such that $|C_P(s)|\le p^{f(a)}$ for some $f(a)$ depending only on $a$? We already know that we can take $f(0)=2$ and it seems that $f(1)=3$.

Note that A. Jaikin-Zapirain (*J. Group Theory*, **3**, no. 3 (2000), 225--231) has proved that $|P|\le p^{f(p,a)}$ for some function $f$ of $p$ and $a$ only.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03565 | KOU-14.9

### Kourovka Notebook Problem 14.9

O / open | Kourovka 群论问题集

(Well-known problem). Let $W^*(F_k)$ denote the von Neumann algebra of the free group of rank $k \in \{ 2, 3, \ldots ,\aleph_0 \}.$ Is $W^*(F_k)$ isomorphic to $W^*(F_l)$ for $k \ne l$?

For a group $G$, recall that $W^*(G)$ is an appropriate completion of the group algebra ${\mathbb{C} G} (see e. g. S. Sakai, *$C^*$-algebras and $W^*$-algebras,* Springer, 1971, in particular Problem 4.4.44). It is known that either $W^*(F_k) \cong W^*(F_l)$ for all $k,l \in \{ 2, 3, \ldots ,\aleph_0 \},$ or that the $W^*(F_k)$ are pairwise non-isomorphic (F. Radulescu, *Invent. Math.*, **115** (1994), 347--389, Corollary 4.7).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03566 | KOU-14.90

### Kourovka Notebook Problem 14.90

O / open | Kourovka 群论问题集

Let $P$ be a finite $p$-group of abundance $a$ and nilpotency class $c$. Does there exist an integer $t=t(a)$ such that $\gamma_i(G)=\zeta _{c-i+1}(G)$ for $i\geq t$? This holds for $a=0$ with $t=1$, since $P$ has maximal class; it can be proved that for $a=1$ one can take $t=3$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03567 | KOU-14.91

### Kourovka Notebook Problem 14.91

O / open | Kourovka 群论问题集

Let $p$ be a fixed prime. Do there exist finite $p$-groups of abundance $a$ for any $a\geq 0$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03568 | KOU-14.92

### Kourovka Notebook Problem 14.92

S / solved | Kourovka 群论问题集

(I. D. Macdonald). Every finite $p$-group has at least $p-1$ conjugacy classes of maximum size. I. D. Macdonald (*Proc. Edinburgh Math. Soc.*, **26** (1983), 233--239) constructed groups of order $2^n$ for any $n\geq 7$ with just one conjugacy class of maximum size. Are there any examples with exactly $p-1$ conjugacy classes of maximum size for odd $p$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03569 | KOU-14.93

### Kourovka Notebook Problem 14.93

O / open | Kourovka 群论问题集

Let $N({\mathbb{Z}/p{\mathbb{Z}})$ be the group defined in Archive, 12.24 (the so-called ``Nottingham group'', or the ``Wild group''). Find relations of $N({\mathbb{Z}/p{\mathbb{Z}})$ as a pro-$p$-group (it has two generators, e. g., $x+x^2$ and $x/(1-x)$). *Comment of 2009:* for $p>2$ see progress in (M. V. Ershov, *J. London Math. Soc.*, **71** (2005), 362--378).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03570 | KOU-14.94

### Kourovka Notebook Problem 14.94

O / open | Kourovka 群论问题集

For each positive integer $r$ find the $p$-cohomological dimension $\mathrm{cd}_p(H_r)$ where $H_r$ is the closed subgroup of $N({\mathbb{Z}/p{\mathbb{Z}})$ consisting of the series \ $x\left( 1 + \sum\limits_{i=1}^{\infty} a_ix^{p^ri}\right) $, $a_i\in {\mathbb{Z}/p{\mathbb{Z}}$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03571 | KOU-14.95

### Kourovka Notebook Problem 14.95

O / open | Kourovka 群论问题集

(C. R. Leedham-Green, P. M. Neumann, J. Wiegold). For a finite $p $-group $P$, denote by $c=c(P)$ its nilpotency class and by $b=b(P)$ its *breadth*, that is, $p^b$ is the maximum size of a conjugacy class in $P$. *Class-Breadth Problem:* Is it true that $c\le b+1$ if $p\ne 2$?

So far, the best known bound is $c < \frac{p}{p-1}b+1$ (C. R. Leedham-Green, P. M. Neumann, J. Wiegold, *J. London Math. Soc. (2)*, **1** (1969), 409--420). For $p=2$ for every $n\in{\mathbb{N}}$ there exists a $2$-group $T_n$ such that $c(T_n)\geq b(T_n)+n$ (W. Felsch, J. Neub\"user, W. Plesken, *J. London Math. Soc. (2)*, **24** (1981), 113--122).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03572 | KOU-14.96

### Kourovka Notebook Problem 14.96

S / solved | Kourovka 群论问题集

Suppose that a finite $p $-group $P$ admits an automorphism of order $p^n$ having exactly $p^m$ fixed points. By (E. I. Khukhro, *Russ. Acad. Sci. Sbornik Math.*, **80** (1995), 435--444) then $P$ has a subgroup of index bounded in terms of $p$, $n$ and $m$ which is soluble of derived length bounded in terms of $p^n$. Is it true that $P$ has also a subgroup of index bounded in terms of $p$, $n$ and $m$ which is soluble of derived length bounded in terms of $m$? There are positive answers in the cases of $m=1$ (S. McKay, *Quart. J. Math. Oxford, Ser. (2)*, **38** (1987), 489--502; \ I. Kiming, *Math. Scand.*, **62** (1988), 153--172) and $n=1$ (Yu. A. Medvedev, see Archive, 10.68).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03573 | KOU-14.97

### Kourovka Notebook Problem 14.97

O / open | Kourovka 群论问题集

Is it true that for any two different prime numbers $p$ and $q$ there exists a non-primary periodic locally soluble $\{ p,q\}$-group that can be represented as the product of two of its $p $-subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03574 | KOU-14.98

### Kourovka Notebook Problem 14.98

S / solved | Kourovka 群论问题集

We say that a metric space is a *$2$-end* one (a *narrow* one), if it is quasiisometric to the real line ${\mathbb{R}}$ (respectively, to a subset of ${\mathbb{R}}$). All other spaces are said to be *wide*. Suppose that the Caley graph $\Gamma = \Gamma (G, A)$ of a group $G$ with a finite set of generators $A$ in the natural metric contains a 2-end subset, and suppose that there is $\varepsilon > 0$ such that the complement in $\Gamma$ to the $\varepsilon$-neighbourhood of any connected 2-end subset contains exactly two wide connected components. Is it true that the group $G$ in the word metric is quasiisometric to the Euclidean or hyperbolic plane?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03575 | KOU-14.99

### Kourovka Notebook Problem 14.99

P / partially_solved | Kourovka 群论问题集

a) A formation ${\mathfrak{F}}$ of finite groups is called *superradical* if it is $S_{n}$-closed and contains every finite group of the form $G=AB$ where $A$ and $B$ are ${\mathfrak{F}}$-subnormal ${\mathfrak \ F}$-subgroups. Find all superradical local formations.

[Archived solved part(s) of 14.99:] b) A formation ${\mathfrak{F}}$ of finite groups is called *superradical* if it is $S_{n}$-closed and contains every finite group of the form $G=AB$ where $A$ and $B$ are ${\mathfrak{F}}$-subnormal ${\mathfrak \ F}$-subgroups. Prove that every $S$-closed superradical formation is a solubly saturated formation.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03576 | KOU-15.1

### Kourovka Notebook Problem 15.1

O / open | Kourovka 群论问题集

(P. Longobardi, M. Maj, A. H. Rhemtulla). Let $w = w(x_1, \ldots , x_n)$ be a group word in $n$ variables $x_1, \dots , x_n$, and $V(w)$ the variety of groups defined by the law $w= 1$. Let $V(w^*)$ (respectively, $V(w^{\# })$) be the class of all groups $G$ in which for every $n$ infinite subsets $S_1,\ldots , S_n$ there exist $s_i \in S_i$ such that $w(s_1,\ldots ,s_n)=1$ (respectively, $\left< s_1,\ldots ,s_n \right> \in V(w)$).

a) Is there some word $w$ and an infinite group $G$ such that $G \in V(w^\#)$ but $G \notin V(w)$?

b) Is there some word $w$ and an infinite group $G$ such that $G \in V(w^*)$ but $G\notin V(w^{\# })$?

The answer to both of these questions is likely to be ``yes''. It is known that in a) \ $w$ cannot be any of several words such as $x_1^n$, \ $[x_1,\ldots ,x_n]$, \ $[x_1,x_2]^2$, \ $(x_1x_2)^3x_2^{-3}x_1^{-3}$, and $x_1^{a_1}\cdots x_n^{a_n}$ for any non-zero integers $a_1,\ldots ,a_n$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03577 | KOU-15.10

### Kourovka Notebook Problem 15.10

S / solved | Kourovka 群论问题集

(Yu. I. Merzlyakov). Is the group of all automorphisms of the free group $F_n$ that act trivially on $F_n/[F_n,F_n]$ linear for $n\geq 3$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03578 | KOU-15.100

### Kourovka Notebook Problem 15.100

O / open | Kourovka 群论问题集

Is a periodic group locally finite if it has a non-cyclic subgroup of order 4 that coincides with its centralizer?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03579 | KOU-15.101

### Kourovka Notebook Problem 15.101

O / open | Kourovka 群论问题集

Is a periodic group locally finite if it has an involution whose centralizer is a locally finite group with Sylow 2-subgroup of order 2?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03580 | KOU-15.102

### Kourovka Notebook Problem 15.102

O / open | Kourovka 群论问题集

(W. Magnus). An element $r$ of a free group $F_n$ is called a *normal root* of an element $u\in F_n$ if $u$ belongs to the normal closure of $r$ in $F_n$. Can an element $u$ that lies outside the commutator subgroup $[F_n, F_n]$ have infinitely many non-conjugate normal roots?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03581 | KOU-15.103

### Kourovka Notebook Problem 15.103

O / open | Kourovka 群论问题集

(Well-known problem). Is the group $\mathrm{Out}(F_3)$ of outer automorphisms of a free group of rank $3$ linear?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03582 | KOU-15.104

### Kourovka Notebook Problem 15.104

O / open | Kourovka 群论问题集

Let $n$ be a positive integer and let $w$ be a group word in the variables $x_1, x_2,\ldots $. Suppose that a residually finite group $G$ satisfies the identity $ w^n=1$. Does it follow that the verbal subgroup $w(G)$ is locally finite?

This is the Restricted Burnside Problem if $w=x_1$. A positive answer was obtained also in a number of other particular cases (P. V. Shumyatsky, *Quart. J. Math.*, **51**, no. 4 (2000), 523--528).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03583 | KOU-15.11

### Kourovka Notebook Problem 15.11

O / open | Kourovka 群论问题集

(M. Morigi). An automorphism of a group is called a *power automorphism* if it leaves every subgroup invariant. Is every finite abelian $p $-group the group of all power automorphisms of some group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03584 | KOU-15.12

### Kourovka Notebook Problem 15.12

O / open | Kourovka 群论问题集

Let $G$ be a group acting faithfully and level-transitively by automorphisms on a rooted tree $\mathcal{T}$. For a vertex $v$ of $\mathcal{T}$, the *rigid vertex stabilizer* at $v$ consists of those elements of $G$ whose support in $\mathcal{T}$ lies entirely in the subtree ${\mathcal{T}}_v$ rooted at $v$. For a non-negative integer $n$, the *$n$-th rigid level stabilizer* is the subgroup of $G$ generated by all rigid vertex stabilizers corresponding to the vertices at the level $n$ of the tree $\mathcal{T}$. The group $G$ is a *branch group* if all rigid level stabilizers have finite index in $G$. For motivation, examples and known results see (R. I. Grigorchuk, *in: New horizons in pro-$p$ groups*, Birkh\"auser, Boston, 2000, 121--179).

Do there exist branch groups with Kazhdan's $T$-property? (See A14.34.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03585 | KOU-15.13

### Kourovka Notebook Problem 15.13

O / open | Kourovka 群论问题集

Do there exist finitely presented branch groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03586 | KOU-15.14

### Kourovka Notebook Problem 15.14

S / solved | Kourovka 群论问题集

b) Do there exist finitely generated branch groups that are non-amenable and do not contain the free group $F_2$ on two generators?

[Archived solved part(s) of 15.14:] Do there exist finitely generated branch groups (see 15.12)

a) that are non-amenable?

c) that contain $F_2$?

d) that have exponential growth?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03587 | KOU-15.15

### Kourovka Notebook Problem 15.15

S / solved | Kourovka 群论问题集

Is every maximal subgroup of a finitely generated branch group necessarily of finite index?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03588 | KOU-15.16

### Kourovka Notebook Problem 15.16

O / open | Kourovka 群论问题集

Do there exist groups whose rate of growth is $e^{\sqrt{n}}$

a) in the class of finitely generated branch groups?

b) in the whole class of finitely generated groups? This question is related to 9.9.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03589 | KOU-15.17

### Kourovka Notebook Problem 15.17

S / solved | Kourovka 群论问题集

An infinite group is *just infinite* if all of its proper quotients are finite. Is every finitely generated just infinite group of intermediate growth necessarily a branch group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03590 | KOU-15.18

### Kourovka Notebook Problem 15.18

S / solved | Kourovka 群论问题集

A group is *hereditarily just infinite* if it is residually finite and all of its non-trivial normal subgroups are just infinite.

a) Do there exist finitely generated hereditarily just infinite torsion groups? (It is believed there are none.)

b) Is every finitely generated hereditarily just infinite group necessarily linear?

A positive answer to the question b) would imply a negative answer to a).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03591 | KOU-15.19

### Kourovka Notebook Problem 15.19

P / partially_solved | Kourovka 群论问题集

Let $p$ be a prime, and ${\mathcal{F}}_p$ the class of finitely generated groups acting faithfully on a $p$-regular rooted tree by finite automata. Any group in ${\mathcal{F}}_p$ is residually-$p$ (residually in the class of finite $p$-groups) and has word problem that is solvable in (at worst) exponential time. There exist therefore groups that are residually-$p$, have a solvable word problem, and do not belong to ${\mathcal{F}}_p$; though no concrete example is known. For instance:

a) Is it true that some (or even all) the groups given in (R. I. Grigorchuk, *Math. USSR--Sb.*, **54** (1986), 185--205) do not belong to ${\mathcal{F}}_p$ when the sequence $\omega$ is computable, but not periodic?

b) Does ${\mathbb{Z}\wr({\mathbb{Z}\wr {\mathbb{Z}})$ belong to ${\mathcal{F}}_2$? (Here the wreath products are restricted.)

See (A. M. Brunner, S. Sidki, *J. Algebra*, **257** (2002), 51--64) and (R. I. Grigorchuk, V. V. Nekrashevich, V. I. Sushchanskiĭ, *in: Dynamical systems, automata, and infinite groups. Proc. Steklov Inst. Math.*, **231** (2000), 128--203).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03592 | KOU-15.2

### Kourovka Notebook Problem 15.2

O / open | Kourovka 群论问题集

By a theorem of W. Burnside, if $\chi \in \mathrm{Irr}(G)$ and $\chi(1) > 1$, then there exists $x \in G$ such that $\chi(x) = 0$, that is, only the linear characters are ``nonvanishing''. It is interesting to consider the dual notion of *nonvanishing elements* of a finite group $G$, that is, the elements $x \in G$ such that $\chi(x) \ne 0$ for all $\chi \in \mathrm{Irr}(G)$.

a) It is proved (I. M. Isaacs, G. Navarro, T. R. Wolf, *J. Algebra*, **222**, no. 2 (1999), 413--423) that if $G$ is solvable and $x \in G$ is a nonvanishing element of odd order, then $x$ belongs to the Fitting subgroup $**F**( G)$. Is this true for elements of even order too? (M. Miyamoto showed in 2008 that every nontrivial abelian normal subgroup of a finite group contains a nonvanishing element.)

b) Which nonabelian simple groups have nonidentity nonvanishing elements? (For example, ${\mathbb{A}_7$ has.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03593 | KOU-15.20

### Kourovka Notebook Problem 15.20

O / open | Kourovka 群论问题集

(B. Hartley). An infinite transitive permutation group is said to be *barely transitive* if each of its proper subgroups has only finite orbits. Can a locally finite barely transitive group coincide with its derived subgroup?

Note that there are no simple locally finite barely transitive groups (B. Hartley, M. Kuzucuo\v{g}lu, *Proc. Edinburgh Math. Soc.*, **40** (1997), 483--490), any locally finite barely transitive group is a $p $-group for some prime $p$, and if the stabilizer of a point in a locally finite barely transitive group $G$ is soluble of derived length $d$, then $G$ is soluble of derived length bounded by a function of $d$ (V. V. Belyaev, M. Kuzucuo\v{g}lu, *Algebra and Logic*, **42** (2003), 147--152).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03594 | KOU-15.21

### Kourovka Notebook Problem 15.21

O / open | Kourovka 群论问题集

(B. Hartley). Do there exist torsion-free barely transitive groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03595 | KOU-15.22

### Kourovka Notebook Problem 15.22

O / open | Kourovka 群论问题集

A permutation group is said to be *finitary* if each of its elements moves only finitely many points. Do there exist finitary barely transitive groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03596 | KOU-15.23

### Kourovka Notebook Problem 15.23

O / open | Kourovka 群论问题集

A transitive permutation group is said to be *totally imprimitive* if every finite set of points is contained in some finite block of the group. Do there exist totally imprimitive barely transitive groups that are not locally finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03597 | KOU-15.24

### Kourovka Notebook Problem 15.24

S / solved | Kourovka 群论问题集

Suppose that a finite $p $-group $G$ has a subgroup of exponent $p$ and order $p^n$. Is it true that if $p$ is sufficiently large relative to $n$, then $G$ contains a normal subgroup of exponent $p$ and order $p^n$?

J. L. Alperin and G. Glauberman (*J. Algebra*, **203**, no. 2 (1998), 533--566) proved that if a finite $p $-group contains an elementary abelian subgroup of order $p^n$, then it contains a normal elementary abelian subgroup of the same order provided $p>4n-7$, and the analogue for arbitrary abelian subgroups is proved in (G. Glauberman, *J. Algebra*, **272** (2004), 128--153).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03598 | KOU-15.25

### Kourovka Notebook Problem 15.25

S / solved | Kourovka 群论问题集

A finite group $G$ is said to be *rational* if every irreducible character of $G$ takes only rational values. Are the Sylow $2$-subgroups of the symmetric groups ${\mathbb{S}}_{2^n}$ rational?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03599 | KOU-15.26

### Kourovka Notebook Problem 15.26

S / solved | Kourovka 群论问题集

A *partition* of a group is a representation of it as a set-theoretic union of some of its proper subgroups (*components*) that intersect pairwise trivially. Is it true that every nontrivial partition of a finite $p$-group has an abelian component?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03600 | KOU-15.27

### Kourovka Notebook Problem 15.27

S / solved | Kourovka 群论问题集

Is it possible that $\text{Aut} G\cong\text{Aut} H$ for a finite $p $-group $G$ of order $>2$ and a proper subgroup $H<G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03601 | KOU-15.28

### Kourovka Notebook Problem 15.28

O / open | Kourovka 群论问题集

Suppose that a finite $p $-group $G$ is the product of two subgroups: $G=AB$.

a) Is the exponent of $G$ bounded in terms of the exponents of $A$ and $B$?

b) Is the exponent of $G$ bounded if $A$ and $B$ are groups of exponent $p$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03602 | KOU-15.29

### Kourovka Notebook Problem 15.29

O / open | Kourovka 群论问题集

(A. Mann). The dihedral group of order $8$ is isomorphic to its own automorphism group. Are there other non-trivial finite $p $-groups with this property?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03603 | KOU-15.3

### Kourovka Notebook Problem 15.3

O / open | Kourovka 群论问题集

Let $\alpha$ and $\beta$ be faithful non-linear irreducible characters of a finite group $G$. There are non-solvable groups $G$ giving examples when the product $\alpha\beta$ is again an irreducible character (for some of such $\alpha,\beta$). One example is $G=SL_2(5)$ with two irreducible characters of degree 2. In (I. Zisser, *Israel J. Math.*, **84**, no. 1--2 (1993), 147--151) it is proved that such an example exists in an alternating group ${\mathbb{A}_n$ if and only if $n$ is a square exceeding $4$. But do solvable examples exist? Evidence (but no proof) that they do not is given in (I. M. Isaacs, *J. Algebra*, **223**, no. 2 (2000), 630--646).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03604 | KOU-15.30

### Kourovka Notebook Problem 15.30

O / open | Kourovka 群论问题集

Is it true that every finite abelian $p $-group is isomorphic to the Schur multiplier of some nonabelian finite $p $-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03605 | KOU-15.31

### Kourovka Notebook Problem 15.31

O / open | Kourovka 群论问题集

M. R. Vaughan-Lee and J. Wiegold (*Proc. R. Soc. Edinburgh Sect. A*, **95** (1983), 215--221) proved that if a finite $p $-group $G$ is generated by elements of breadth $\leq n$ (that is, having at most $p^n$ conjugates), then $G$ is nilpotent of class $\leq n^2+1$; the bound for the class was later improved by A. Mann (*J. Group Theory*, **4**, no. 3 (2001), 241--246) to $\leq n^2-n+1$. Is there a linear bound for the class of $G$ in terms of $n$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03606 | KOU-15.32

### Kourovka Notebook Problem 15.32

S / solved | Kourovka 群论问题集

Does there exist a function $f(k)$ (possibly depending also on $p$) such that if a finite $p $-group $G$ of order $p^m$ with $m\geq f(k)$ has an automorphism of order $p^{m-k}$, then $G$ possesses a cyclic subgroup of index $p^k$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03607 | KOU-15.33

### Kourovka Notebook Problem 15.33

S / solved | Kourovka 群论问题集

Suppose that all $2$-generator subgroups of a finite $2$-group $G$ are metacyclic. Is the derived length of $G$ bounded? This is true for finite $p $-groups if $p\ne 2$, see 1.1.8 in (M. Suzuki, *Structure of a group and the structure of its lattice of subgroups*, Springer, Berlin, 1956).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03608 | KOU-15.34

### Kourovka Notebook Problem 15.34

S / solved | Kourovka 群论问题集

Is any free product of linearly ordered groups with an amalgamated subgroup right-orderable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03609 | KOU-15.35

### Kourovka Notebook Problem 15.35

S / solved | Kourovka 群论问题集

Let $F$ be the free group of finite rank $ r$ with basis $\{ x_1,\ldots, x_r\}$. Is it true that there exists a number $C=C(r)$ such that any reduced word of length $n>1$ in the $x_i$ lies outside some subgroup of $F$ of index at most $C \log n$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03610 | KOU-15.36

### Kourovka Notebook Problem 15.36

O / open | Kourovka 群论问题集

For a class $M$ of groups let $L(M)$ be the class of groups $G$ such that the normal subgroup $\left< a^G\right>$ generated by an element $a$ belongs to $M$ for any $a\in G$. Is it true that the class $L(M)$ is finitely axiomatizable if $M$ is the quasivariety generated by a finite group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03611 | KOU-15.37

### Kourovka Notebook Problem 15.37

O / open | Kourovka 群论问题集

Let $G$ be a group satisfying the minimum condition for centralizers. Suppose that $X$ is a normal subset of $G$ such that $[x,y,\ldots ,y]=1$ for any $x,y\in X$ with $y$ repeated $f(x,y)$ times. Does $X$ generate a locally nilpotent (whence hypercentral) subgroup?

If the numbers $f(x,y)$ can be bounded, the answer is affirmative (F. O. Wagner, *J. Algebra*, **217**, no. 2 (1999), 448--460).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03612 | KOU-15.38

### Kourovka Notebook Problem 15.38

S / solved | Kourovka 群论问题集

Does there exist a non-local hereditary composition formation $\mathfrak {F} $ of finite groups such that the set of all $\mathfrak {F} $-subnormal subgroups is a sublattice of the subgroup lattice in any finite group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03613 | KOU-15.39

### Kourovka Notebook Problem 15.39

S / solved | Kourovka 群论问题集

Axiomatizing the basic properties of subnormal subgroups, we say that a functor $\tau$ associating with every finite group $G$ some non-empty set $\tau (G) $ of its subgroups is an *$ETP$-functor* if

1) $ \tau (A) ^ {\varphi} \subseteq \tau (B ) $ and $\tau (B) ^ {\varphi^ {-1}} \subseteq\tau (A) $ for any epimorphism $\varphi : A\longmapsto B$, as well as $\{ H\cap R\mid R\in\tau (G) \} \subseteq\tau (H) $ for any subgroup $H\leq G$;

2) $\tau (H) \subseteq\tau (G) $ for any subgroup $H\in\tau (G) $;

3) $\tau (G) $ is a sublattice of the lattice of all subgroups of $G$.

Let $\tau$ be an $ETP$-functor. Does there exist a hereditary formation $\mathfrak {F} $ such that $\tau (G) $ coincides with the set of all $\mathfrak {F} $-subnormal subgroups in any finite group $G$? This is true for finite soluble groups (A. F. Vasil'ev, S. F. Kamornikov, *Siberian Math. J.*, **42**, no. 1 (2001), 25--33).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03614 | KOU-15.4

### Kourovka Notebook Problem 15.4

S / solved | Kourovka 群论问题集

Is it true that large growth implies non-amenability? More precisely, consider a number $\epsilon > 0$, an integer $k \geq 2$, a group $\Gamma$ generated by a set $S$ of $k$ elements, and the corresponding exponential growth rate $\omega(\Gamma,S)$ defined as in 14.7. For $\epsilon$ small enough, does the inequality \ $\omega(\Gamma,S) \geq 2k-1-\epsilon$ \ imply that $\Gamma$ is non-amenable?

If $\omega(\Gamma,S) = 2k-1$, it is easy to show that $\Gamma$ is free on $S$, and in particular non-amenable; see Section 2 in (R. I. Grigorchuk, P. de la Harpe, *J. Dynam. Control Syst.*, **3**, no. 1 (1997), 51--89).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03615 | KOU-15.40

### Kourovka Notebook Problem 15.40

S / solved | Kourovka 群论问题集

Let $N$ be a nilpotent subgroup of a finite simple group $G$. Is it true that there exists a subgroup $N_1$ conjugate to $N$ such that $N\cap N_1=1$?

The answer is known to be affirmative if $N$ is a $p$-group. *Comment of 2013:* an affirmative answer for alternating groups is obtained in (R. K. Kurmazov, *Siberian Math. J.*, **54**, no. 1 (2013), 73--77).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03616 | KOU-15.41

### Kourovka Notebook Problem 15.41

O / open | Kourovka 群论问题集

Let $R(m,p)$ denote the largest finite $m$-generator group of prime exponent $p$.

a) Can the nilpotency class of $R(m,p)$ be bounded by a polynomial in $m$? (This is true for $p=2, 3, 5, 7$.)

b) Can the nilpotency class of $R(m,p)$ be bounded by a linear function in $m$? (This is true for $p=2, 3, 5$.)

c) In particular, can the nilpotency class of $R(m,7)$ be bounded by a linear function in $m$?

My guess is ``no'' to the first two questions for general $p$, but ``yes'' to the third. By contrast, a beautiful and simple argument of Mike Newman shows that if $m\geq 2$ and $k\geq 2$ ($k\geq 3$ for $p=2$), then the order of $R(m,p^k)$ is at least $p^{p^{.^{.^{.^{p^m}}}}}$, with $p$ appearing $k$ times in the tower; see (M. Vaughan-Lee, E. I. Zelmanov, *J. Austral. Math. Soc. (A)*, **67**, no. 2 (1999), 261--271).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03617 | KOU-15.42

### Kourovka Notebook Problem 15.42

O / open | Kourovka 群论问题集

Is it true that the group algebra $k[F]$ of R. Thompson's group $F$ (see 12.20) over a field $k$ satisfies the Ore condition, that is, for any $a,b \in k[F]$ there exist $u,v \in k[F]$ such that $au=bv$ and either $u$ or $v$ is nonzero? If the answer is negative, then $F$ is not amenable.

*Comment of 2026*: If the answer is positive, then $F$ is amenable (so problem 15.42 is equivalent to 12.20), because the ``if and only if'' statement holds for groups satisfying Kaplansky's zero-divisor conjecture (D. Kielak, *Prepint*, 2026, https://arxiv.org/pdf/1605.09133v2), and $F$ does satisfy it.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03618 | KOU-15.43

## Kourovka Notebook Problem 15.43

S / solved | Kourovka 群论问题集

Let $G$ be a finite group of order $n$.

a) Is it true that $|\mathrm{Aut} G|\ge\varphi(n)$ where $\varphi$ is Euler's function?

b) Is it true that $G$ is cyclic if $|\mathrm{Aut} G|=\varphi(n)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03619 | KOU-15.44

## Kourovka Notebook Problem 15.44

P / partially_solved | Kourovka 群论问题集

a) Let $G$ be a reductive group over an algebraically closed field $K$ of arbitrary characteristic. Let $X$ be an affine $G$-variety such that, for a fixed Borel subgroup $B\leq G$, the coordinate algebra $K[X]$ as a $G$-module is the union of an ascending chain of submodules each of whose factors is an induced module $\mathrm{Ind}^G_BV$ of some one-dimensional $B$-module $V$. (See S. Donkin, *Rational representations of algebraic groups. Tensor products and filtration $($Lect. Notes Math.*, **1140**), Springer, Berlin, 1985). Suppose in addition that $K[X]$ is a Cohen--Macaulay ring, that is, a free module over the subalgebra generated by any homogeneous system of parameters. Is then the ring of invariants $K[X]^G$ Cohen--Macaulay? \ \ *Comment of 2005:* This is proved in the case where $X$ is a rational $G$-module (M. Hashimoto, *Math. Z.*, **236** (2001), 605--623).

[Archived solved part(s) of 15.44:] b) Is the ring of invariants $K[M(n)^m]^{GL(n)}$ Cohen--Macaulay in all characteristics? (Here $M(n)^m$ is the direct sum of $m$ copies of the space of $n\times n$ matrices.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03620 | KOU-15.45

## Kourovka Notebook Problem 15.45

P / partially_solved | Kourovka 群论问题集

We define the class of *hierarchically decomposable* groups in the following way. First, if ${\mathfrak{X}}$ is any class of groups, then let $**H**_1{\mathfrak{X}}$ denote the class of groups which admit an admissible action on a finite-dimensional contractible complex in such a way that every cell stabilizer belongs to ${\mathfrak{X}}$. Then the ``big'' class $**H**{\mathfrak{X}}$ is defined to be the smallest $**H**_1$-closed class containing ${\mathfrak{X}}$.

a) Let ${\mathfrak{F}}$ be the class of all finite groups. Find an example of an $**H**{\mathfrak{F}}$ group which is not in $**H**_3{\mathfrak{F}}$ $(=**H**_1**H**_1**H**_1F)$.

b) Prove or disprove that there is an ordinal $\alpha$ such that $**H**_{\alpha}{\mathfrak{F}}=**H**{\mathfrak{F}}$, where $**H**_{\alpha}$ is the operator on classes of groups defined by transfinite induction in the obvious way starting from $**H**_1$. \ *Editor's comment*: It is proved that such an ordinal cannot be countable (T. Januszkiewicz, P. H. Kropholler, I. J. Leary, *Bull. London Math. Soc.*, **42**, No. 5 (2010), 896--904).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03621 | KOU-15.46

## Kourovka Notebook Problem 15.46

O / open | Kourovka 群论问题集

Can the question 7.28 on conditions for admissibility of an elementary carpet ${\mathfrak{A}} = \{ {\mathfrak{A}}_r\mid r \in \Phi \}$ be reduced to Lie rank 1 if $K$ is a field? A carpet ${\mathfrak{A}}$ of type $\Phi$ of additive subgroups of $K$ is called *addmissible* if in the Chevalley group over $K$ associated with the root system $\Phi$ the subgroup $\left< x_r({\mathfrak{A}}_r)\mid r \in \Phi \right>$ intersects $x_r(K)$ in $x_r({\mathfrak{A}}_r)$. More precisely, is it true that the carpet ${\mathfrak{A}}$ is admissible if and only if the subcarpets $\{ {\mathfrak{A}}_r, {\mathfrak{A}}_{-r} \}$, $r \in \Phi$, of rank 1 are admissible? The answer is known to be affirmative if the field $K$ is locally finite (V. M. Levchuk, *Algebra and Logic*, **22**, no. 5 (1983), 362--371).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03622 | KOU-15.47

### Kourovka Notebook Problem 15.47

O / open | Kourovka 群论问题集

Let $M<G\leq \mathrm{Sym}(\Omega)$, where $\Omega$ is finite, be such that $M$ is transitive on $\Omega$ and there is a $G$-invariant partition $\mathcal{P}$ of $\Omega\times\Omega\setminus\{(\alpha,\alpha)\mid \alpha\in\Omega\}$ such that $G$ is transitive on the set of parts of $\mathcal{P}$ and $M$ fixes each part of $\mathcal{P}$ setwise. (Here $\mathcal{P}$ can be identified with a decomposition of the complete directed graph with vertex set $\Omega$ into edge-disjoint isomorphic directed graphs.) If $G$ induces a cyclic permutation group on $\mathcal{P}$, then we showed (\!*Trans. Amer. Math. Soc.*, **355**, no. 2 (2003), 637--653) that the numbers $n=|\Omega|$ and $k=|\mathcal{P}|$ are such that the $r$-part $n_r$ of $n$ satisfies $n_r\equiv 1\pmod{k}$ for each prime $r$. Are there examples with $G$ inducing a non-cyclic permutation group on $\mathcal{P}$ for any $n, k$ not satisfying this congruence condition?

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03623 | KOU-15.48

### Kourovka Notebook Problem 15.48

O / open | Kourovka 群论问题集

Let $G$ be any non-trivial finite group, and let $X$ be any generating set for $G$. Is it true that every element of $G$ can be obtained from $X$ using fewer than $2\log_2|G|$ multiplications? (When counting the number of mutiplications on a path from the generators to a given element, at each step one can use the elements obtained at previous steps.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03624 | KOU-15.49

### Kourovka Notebook Problem 15.49

S / solved | Kourovka 群论问题集

A group $G$ is a *unique product group* if, for any finite nonempty subsets $X,Y$ of $G$, there is an element of $G$ which can be written in exactly one way in the form $xy$ with $x \in X$ and $y \in Y$. Does there exist a unique product group which is not left-orderable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03625 | KOU-15.5

### Kourovka Notebook Problem 15.5

S / solved | Kourovka 群论问题集

(Well-known problem). Does there exist an infinite finitely generated group which is simple and amenable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03626 | KOU-15.50

### Kourovka Notebook Problem 15.50

O / open | Kourovka 群论问题集

Let $G$ be a group of automorphisms of an abelian group of prime exponent. Suppose that there exists $x\in G$ such that $x$ is regular of order 3 and the order of $[x,g]$ is finite for every $g\in G$. Is it true that $\langle x^G\rangle$ is locally finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03627 | KOU-15.51

### Kourovka Notebook Problem 15.51

O / open | Kourovka 群论问题集

Suppose that $G$ is a periodic group satisfying the identity $[x,y]^5=1$. Is then the derived subgroup $[G,G]$ a $5$-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03628 | KOU-15.52

### Kourovka Notebook Problem 15.52

O / open | Kourovka 群论问题集

(Well-known problem). By a famous theorem of Wielandt the sequence $G_0=G, G_1,\ldots $, where $ G_{i+1}=\mathrm{Aut }G_i$, stabilizes for any finite group $G$ with trivial centre. Does there exist a function $f$ of natural argument such that $|G_i|\leq f(|G|)$ for all $i=0, 1,\ldots $ for an arbitrary finite group $G$ and the same kind of sequence?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03629 | KOU-15.53

### Kourovka Notebook Problem 15.53

S / solved | Kourovka 群论问题集

Let $S$ be the set of all prime numbers $p$ for which there exists a finite simple group $G$ and an absolutely irreducible $G$-module $V$ over a field of characteristic $p$ such that the order of any element in the natural semidirect product $VG$ coincides with the order of some element in $G$. Is $S$ finite or infinite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03630 | KOU-15.54

### Kourovka Notebook Problem 15.54

O / open | Kourovka 群论问题集

Suppose that $G$ is a periodic group containing an involution $i$ such that the centralizer $C_G(i)$ is a locally cyclic $2$-group. Does the set of all elements of odd order in $G$ that are inverted by $i$ form a subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03631 | KOU-15.55

### Kourovka Notebook Problem 15.55

O / open | Kourovka 群论问题集

a) The Monster, $M$, is a $6$-transposition group. Pairs of Fischer transpositions generate 9 \ $M$-classes of dihedral groups. The order of the product of a pair is the coefficient of the highest root of affine type $E_8$. Similar properties hold for Baby $B$, and $F_{24}'$ with respect to $E_7$ and $E_6$ when the product is read modulo centres ($2.B$, $3.F_{24}'$). Explain this. \ *Editors' Comment of 2005:* Some progress was made in (C. H. Lam, H. Yamada, H. Yamauchi, *Trans. Amer. Math. Soc.*, **359**, no. 9 (2007), 4107--4123).

b) Note that the Schur multiplier of the sporadic simple groups $M$, $B$, $F_{24}'$ is the fundamental group of type $E_8$, $E_7$, $E_6$, respectively. Why?

See (J. McKay, in: {\it Finite groups, Santa Cruz Conf. 1979 $($Proc. Symp. Pure Math., **37**}), Amer. Math. Soc., Providence, RI, 1980, 183--186) and (R. E. Borcherds, *Doc. Math., J. DMV Extra Vol. ICM Berlin 1998*, Vol. I (1998), 607--616).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03632 | KOU-15.56

### Kourovka Notebook Problem 15.56

O / open | Kourovka 群论问题集

Is there a spin manifold such that the Monster acts on its loop space? Perhaps 24-dimensional? hyper-K\"{a}hler, non-compact? See (F. Hirzebruch, T. Berger, R. Jung, *Manifolds and modular forms*, Vieweg, Braunschweig, 1992).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03633 | KOU-15.57

### Kourovka Notebook Problem 15.57

O / open | Kourovka 群论问题集

Suppose that $H$ is a subgroup of $SL_2({\mathbb{Q}} )$ that is dense in the Zariski topology and has no nontrivial finite quotients. Is then $H=SL_2({\mathbb{Q}} )$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03634 | KOU-15.58

### Kourovka Notebook Problem 15.58

O / open | Kourovka 群论问题集

Suppose that a free profinite product $G*H$ is a free profinite group of finite rank. Must $G$ and $H$ be free profinite groups?

By (J. Neukirch, *Arch. Math.*, **22**, no. 4 (1971), 337--357) this may not be true if the rank of $G*H$ is infinite.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03635 | KOU-15.59

### Kourovka Notebook Problem 15.59

O / open | Kourovka 群论问题集

Does there exist a profinite group $G$ that is not free but can be represented as a projective limit $G=\lim\limits_{\longleftarrow}(G/N_{\alpha})$, where all the $G/N_{\alpha}$ are free profinite groups of finite ranks?

The finiteness condition on the ranks of the $G/N_{\alpha}$ is essential. Such a group $G$ cannot satisfy the first axiom of countability (O. V. Mel'nikov, *Dokl. AN BSSR*, **24**, no. 11 (1980), 968--970 (Russian)).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03636 | KOU-15.6

### Kourovka Notebook Problem 15.6

S / solved | Kourovka 群论问题集

(Well-known problem). Is it true that Golod $p $-groups are non-amenable?

These infinite finitely generated torsion groups are defined in (E. S. Golod, *Amer. Math. Soc. Transl. (2)*, **48** (1965), 103--106).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03637 | KOU-15.60

### Kourovka Notebook Problem 15.60

S / solved | Kourovka 群论问题集

Is it true that any finitely generated $p^\prime$-isolated subgroup of a free group is separable in the class of finite $p $-groups? It is easy to see that this is true for cyclic subgroups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03638 | KOU-15.61

### Kourovka Notebook Problem 15.61

O / open | Kourovka 群论问题集

Is it true that \ $l_ {\pi} ^ {n} (G) \le n (G_ {\pi} ) -1+\max_ {p\in \pi} l_p (G) $ \ for any $\pi$-soluble group $G$? Here $n (G_ {\pi} ) $ is the nilpotent length of a Hall $\pi$-subgroup $G_ {\pi} $ of the group $G$ and $l_ {\pi} ^ {n} (G) $ is the nilpotent $\pi$-length of $G$, that is, the minimum number of $\pi$-factors in those normal series of $G$ whose factors are either $\pi '$-groups, or nilpotent $\pi$-groups. The answer is known to be affirmative in the case when all proper subgroups of $G_ {\pi} $ are supersoluble.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03639 | KOU-15.62

## Kourovka Notebook Problem 15.62

S / solved | Kourovka 群论问题集

Given an ordinary irreducible character $\chi$ of a finite group $G$ write $p^{e_p(\chi)}$ to denote the $p $-part of $\chi(1)$ and put $e_p(G)=\max\{e_p(\chi)\mid\chi\in\mathrm{Irr}(G)\}$. Suppose that $P$ is a Sylow $p $-subgroup of a group $G$. Is it true that $e_p(P)$ is bounded above by a function of $e_p(G)$?

*Comment of 2005:* An affirmative answer in the case of solvable groups has been given in (A. Moret\'o, T. R. Wolf, *Adv. Math.*, **184** (2004), 18--36).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03640 | KOU-15.63

## Kourovka Notebook Problem 15.63

P / partially_solved | Kourovka 群论问题集

a) Let $F_n$ be the free group of finite rank $n$ on the free generators $x_1,\ldots ,x_n$. An element $u \in F_n$ is called *positive* if $u$ belongs to the semigroup generated by the $x_i$. An element $u \in F_n$ is called *potentially positive* if $\alpha (u)$ is positive for some automorphism $\alpha$ of $F_n$. Is the property of an element to be potentially positive algorithmically recognizable?

*Comment of 2013:* In (R. Goldstein, *Contemp. Math.*, Amer. Math. Soc., **421** (2006), 157--168) the problem was solved in the affirmative in the special case $n=2$.

[Archived solved part(s) of 15.63:] b) Let $F_n$ be the free group of finite rank $n$ on the free generators $x_1,\ldots ,x_n$. An element $u \in F_n$ is called *positive* if $u$ belongs to the semigroup generated by the $x_i$. An element $u \in F_n$ is called *potentially positive* if $\alpha (u)$ is positive for some automorphism $\alpha$ of $F_n$. Finally, $u \in F_n$ is called *stably potentially positive* if it is potentially positive as an element of $F_m$ for some $m \geq n$. Are there stably potentially positive elements that are not potentially positive?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03641 | KOU-15.64

## Kourovka Notebook Problem 15.64

O / open | Kourovka 群论问题集

For finite groups $G$, $X$ define $r(G;X)$ to be the number of inequivalent actions of $G$ on $X$, that is, the number of equivalence classes of homomorphisms $G \to \hbox\mathrm{Aut} X$, where equivalence is defined by conjugation by an element of $\hbox\mathrm{Aut} X$. Now define $ r_G(n) := \max\{r(G;X) : |X| = n\}. $

Is it true that $r_G(n)$ may be bounded as a function of $\lambda(n)$, the total number (counting multiplicities) of prime factors of $n$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03642 | KOU-15.65

## Kourovka Notebook Problem 15.65

O / open | Kourovka 群论问题集

A square matrix is said to be *separable* if its minimal polynomial has no repeated roots, and *cyclic* if its minimal and characteristic polynomials are equal. For a matrix group $G$ over a finite field define $s(G)$ and $c(G)$ to be the proportion of separable and of cyclic elements respectively in $G$. For a classical group $\mathrm{X}(d,q)$ of dimension $d$ defined over the field with $q$ elements let \ $S(\hbox\mathrm{X}; q) := \lim_{d \to \infty}s(\hbox\mathrm{X}(d,q))$ \ and \ $ C(\hbox\mathrm{X}; q) := \lim_{d \to \infty}c(\hbox\mathrm{X}(d,q)).$ Independently G. E. Wall (*Bull. Austral. Math. Soc.*, **60**, no. 2 (1999), 253--284) and J. Fulman (*J. Group Theory*, **2**, no. 3 (1999), 251--289) have evaluated $S(\hbox\mathrm{GL}; q)$ and $C(\hbox\mathrm{GL}; q)$, and have found them to be rational functions of $q$. Are $S(\hbox\mathrm{X}; q)$ and $C(\hbox\mathrm{X}; q)$ rational functions of $q$ also for the unitary, symplectic, and orthogonal groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03643 | KOU-15.66

## Kourovka Notebook Problem 15.66

O / open | Kourovka 群论问题集

For a class $\mathfrak{X}$ of groups let $g_{\mathfrak{X}}(n)$ be the number of groups of order $n$ in the class $\mathfrak{X}$ (up to isomorphism). Many years ago I formulated the following problem: find good upper bounds for the quotient $g_{\mathfrak{V}}(n)/g_{\mathfrak{U}}(n)$, where $\mathfrak{V}$ is a variety that is defined by its finite groups and $\mathfrak{U}$ is a subvariety of $\mathfrak{V}$. (This quotient is not defined for all $n$ but only for those for which there are groups of order $n$ in $\mathfrak{U}$.) Some progress has been made by G. Venkataraman (*Quart. J. Math. Oxford (2)*, **48**, no. 189 (1997), 107--125) when $\mathfrak{V}$ is a variety generated by finite groups all of whose Sylow subgroups are abelian. *Conjecture:* if $\mathfrak{V}$ is a locally finite variety of $p $-groups and $\mathfrak{U}$ is a non-abelian subvariety of ${\mathfrak{V}}$, then $g_{\mathfrak{V}}(p^m)/g_{\mathfrak{U}}(p^m) < p^{O(m^2)}$.

Moreover, this seems a possible way to attack the Sims Conjecture that when we write the number of groups of order $p^m$ as $p^{{2\over27}m^3 + \varepsilon(m)}$ the error term $\varepsilon(m)$ is $O(m^2)$.

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03644 | KOU-15.67

### Kourovka Notebook Problem 15.67

O / open | Kourovka 群论问题集

Which adjoint Chevalley groups (of normal type) over the integers are generated by three involutions two of which commute?

The groups $SL_n({\mathbb{Z}})$, $n>13$, satisfy this condition (M. C. Tamburini, P. Zucca, *J. Algebra*, **195**, no. 2 (1997), 650--661). The groups $PSL_n({\mathbb{Z}})$ satisfy it if and only if $n>4$ (N. Ya. Nuzhin, *Vladikavkaz. Mat. Zh.*, **10**, no. 1 (2008), 68--74 (Russian)). The group $PSp_4({\mathbb{Z}})$ does not satisfy it, which follows from the corresponding fact for $PSp_4(3)$, see Archive, 7.30. The group $G_2$ satisfies it (I. A. Timofeenko, *J. Siberian Fed. Univ. Ser. Math. Phys.*, **8**, no. 1 (2015), 104--108. The group $E_l$ satisfies it (I. A. Timofeenko, *Sibrsk. \`Electron. Mat. Izv.*, **14** (2017), 807--820 (Russian)).

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03645 | KOU-15.68

### Kourovka Notebook Problem 15.68

O / open | Kourovka 群论问题集

Does there exist an infinite finitely generated 2-group (of finite exponent) all of whose proper subgroups are locally finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03646 | KOU-15.69

### Kourovka Notebook Problem 15.69

O / open | Kourovka 群论问题集

Is it true that every hyperbolic group has a free normal subgroup with the factor-group of finite exponent?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03647 | KOU-15.7

### Kourovka Notebook Problem 15.7

S / solved | Kourovka 群论问题集

(Well-known problem). Is it true that the reduced $C^*$-algebra of a countable group without amenable normal subgroups distinct from $\{ 1\}$ is always a simple $C^*$-algebra with unique trace? See (M. B. Bekka, P. de la Harpe, *Expos. Math.*, **18**, no. 3 (2000), 215--230).

*Comment of 2009*: This was proved for linear groups (T. Poznansky, https://arxiv.org/pdf/0812.2486.pdf).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03648 | KOU-15.70

### Kourovka Notebook Problem 15.70

S / solved | Kourovka 群论问题集

Do there exist groups of arbitrarily large cardinality that satisfy the minimum condition for subgroups?

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03649 | KOU-15.71

### Kourovka Notebook Problem 15.71

O / open | Kourovka 群论问题集

(B. Huppert). Let $G$ be a finite solvable group, and let $\rho(G)$ denote the set of prime divisors of the degrees of irreducible characters of $G$. Is it true that there always exists an irreducible character of $G$ whose degree is divisible by at least $|\rho(G)|/2$ different primes?

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03650 | KOU-15.72

### Kourovka Notebook Problem 15.72

O / open | Kourovka 群论问题集

For a fixed prime $p$ does there exist a sequence of groups $P_n$ of order $p^n$ such that the number of conjugacy classes $k(P_n)$ satisfies $\lim_{n\to\infty} \log k(P_n) / \sqrt{n} = 0$?

Note that J. M. Riedl (*J. Algebra*, **218** (1999), 190--215) constructed $p $-groups for which the above limit is $2 \log p$.

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03651 | KOU-15.73

### Kourovka Notebook Problem 15.73

O / open | Kourovka 群论问题集

Is it true that for every finite lattice $L$ there exists a finite group $G$ and a subgroup $H \le G$ such that the interval $Int(H;G)$ in the subgroup lattice of $G$ is isomorphic to $L$? (Probably not.)

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03652 | KOU-15.74

### Kourovka Notebook Problem 15.74

O / open | Kourovka 群论问题集

For every prime $p$ find a finite $p $-group of nilpotence class $p$ such that its lattice of normal subgroups cannot be embedded into the subgroup lattice of any abelian group. (Solved for $p=2$, $3$.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03653 | KOU-15.75

### Kourovka Notebook Problem 15.75

P / partially_solved | Kourovka 群论问题集

b) Consider the sequence $u_1 = [x,y],\ldots$, $ u_{n+1} = [[u_n, x],[u_n,y]]$. Is it true that an arbitrary finite group is soluble if and only if it satisfies one of these identities $u_n = 1$?

[Archived solved part(s) of 15.75:] a) Does there exist a sequence of identities in two variables $u_1=1$, $u_2=1,\ldots $ with the following properties: \ 1) each of these identities implies the next one, and \ 2) an arbitrary finite group is soluble if and only if it satisfies one of the identities $u_n = 1$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03654 | KOU-15.76

### Kourovka Notebook Problem 15.76

P / partially_solved | Kourovka 群论问题集

b) If $\Theta$ is a variety of groups, then let $\Theta ^0$ denote the category of all free groups of finite rank in $\Theta$. It is proved (G. Mashevitzky, B. Plotkin, E. Plotkin *J. Algebra*, **282** (2004), 490--512) that if $\Theta$ is the variety of all groups, then every automorphism of the category $\Theta ^0$ is an inner one. The same is true if $\Theta$ is the variety of all nilpotent groups of class $\leq d$, for $d\geq 2$ (A. Tsurkov, *Int. J. Algebra Comput.*, **17** (2007), 1273--1281). Is this true for the varieties of solvable groups? of metabelian groups?

An automorphism $\varphi$ of a category is called *inner* if it is isomorphic to the identity automorphism. Let $s:1\to\varphi$ be a function defining this isomorphism. Then for every object $A$ we have an isomorphism $s_A: A\to \varphi(A)$ and for any morphism of objects $\mu: A\to B$ we have $\varphi(\mu)=s_B\mu s_A^{-1}$.

[Archived solved part(s) of 15.76:] a) If $\Theta$ is a variety of groups, then let $\Theta ^0$ denote the category of all free groups of finite rank in $\Theta$. It is proved (G. Mashevitzky, B. Plotkin, E. Plotkin *J. Algebra*, **282** (2004), 490--512) that if $\Theta$ is the variety of all groups, then every automorphism of the category $\Theta ^0$ is an inner one. The same is true if $\Theta$ is the variety of all abelian groups. Is this true for the variety of nilpotent groups of class 2?

An automorphism $\varphi$ of a category is called *inner* if it is isomorphic to the identity automorphism. Let $s:1\to\varphi$ be a function defining this isomorphism. Then for every object $A$ we have an isomorphism $s_A: A\to \varphi(A)$ and for any morphism of objects $\mu: A\to B$ we have $\varphi(\mu)=s_B\mu s_A^{-1}$.

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03655 | KOU-15.77

### Kourovka Notebook Problem 15.77

S / solved | Kourovka 群论问题集

Elements $a,b$ of a group $G$ are said to be *symmetric with respect to an element* $ g\in G$ if $a=gb^{-1}g$. Let $G$ be an infinite abelian group, $\alpha$ a cardinal, $\alpha <|G|$. Is it true that for any $n$-colouring $\chi : G\rightarrow\{ 0, 1,\ldots ,n-1\}$ there exists a monochrome subset of cardinality $\alpha$ that is symmetric with respect to some element of $G$? This is known to be true for $n\leq 3$.

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03656 | KOU-15.78

### Kourovka Notebook Problem 15.78

O / open | Kourovka 群论问题集

(R. I. Grigorchuk). Is it true that for any $n$-colouring of a free group of any rank there exists a monochrome subset that is symmetric with respect to some element of the group? This is true for $n=2$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03657 | KOU-15.79

### Kourovka Notebook Problem 15.79

S / solved | Kourovka 群论问题集

Does there exist a Hausdorff group topology on ${\mathbb{Z}}$ such that the sequence $ \{ 2^n+3^n \}$ converges to zero?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03658 | KOU-15.8

### Kourovka Notebook Problem 15.8

P / partially_solved | Kourovka 群论问题集

b) Let $G$ be any Lie group (indeed, any separable continuous group) made discrete: can $G$ act faithfully on a countable set?

*Comment of 2021*: an affirmative answer was obtained for the nilpotent case (N. Monod, *J. Group Theory*, **25**, no. 5 (2022), 851--865). *Comment of 2025*: this problem is reduced to the case of simple Lie groups; in particular, the solvable case is settled (A. Conversano, N. Monod, *J. Algebra*, **640** (2024), 106--116).

[Archived solved part(s) of 15.8:] a) (S. M. Ulam). Consider the usual compact group $SO(3)$ of all rotations of a 3-dimensional Euclidean space, and let $G$ denote this group viewed as a discrete group. Can $G$ act non-trivially on a countable set?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03659 | KOU-15.80

### Kourovka Notebook Problem 15.80

S / solved | Kourovka 群论问题集

A sequence $\{ F_n\}$ of pairwise disjoint finite subsets of a topological group is called *expansive* if for every open subset $U$ there is a number $m$ such that $ F_n\cap U\ne \varnothing$ for all $n>m$. Suppose that a group $G$ can be partitioned into countably many dense subsets. Is it true that in $ G$ there exists an expansive sequence?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03660 | KOU-15.81

### Kourovka Notebook Problem 15.81

S / solved | Kourovka 群论问题集

Let $G$ be a finite non-supersoluble group. Is it true that $G$ has a non-cyclic Sylow subgroup $P$ such that some maximal subgroup of $P$ has no proper complement in $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03661 | KOU-15.82

### Kourovka Notebook Problem 15.82

O / open | Kourovka 群论问题集

Suppose that a periodic group $G$ contains a strongly isolated $2$-subgroup $U$. Is it true that either $G$ is locally finite, or $U$ is a normal subgroup of $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03662 | KOU-15.83

### Kourovka Notebook Problem 15.83

O / open | Kourovka 群论问题集

(Yu. I. Merzlyakov). Does there exist a rational number $\alpha $ such that $|\alpha |<2$ and the matrices {\small $\left( \begin{array}{cc}1&\alpha \\0&1\end{array}\right)$} and {\small $\left( \begin{array}{cc}1&0\\\alpha &1\end{array}\right)$} generate a free group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03663 | KOU-15.84

### Kourovka Notebook Problem 15.84

O / open | Kourovka 群论问题集

Yu. I. Merzlyakov (*Sov. Math. Dokl.*, **19** (1978), 64--68) proved that if the complex numbers $\alpha ,\beta ,\gamma $ are each at least 3 in absolute value, then the matrices {\small $\left( \begin{array}{cc}1&\alpha \\0&1\end{array}\right)$}, \ {\small $\left( \begin{array}{cc}1&0 \\\beta&1\end{array}\right)$}, \ and \ {\small $\left( \begin{array}{cc}1-\gamma &-\gamma \\\gamma &1+\gamma \end{array}\right)$} generate a free group of rank three. Are there rational numbers $\alpha ,\beta , \gamma $, each less than 3 in absolute value, with the same property?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03664 | KOU-15.85

### Kourovka Notebook Problem 15.85

O / open | Kourovka 群论问题集

A torsion-free group all of whose subgroups are subnormal is nilpotent (H. Smith, *Arch. Math.*, **76**, no. 1 (2001), 1--6). Is a torsion-free group with the normalizer condition

a) hyperabelian?

b) hypercentral?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03665 | KOU-15.86

### Kourovka Notebook Problem 15.86

S / solved | Kourovka 群论问题集

A group $G$ is called *discriminating* if for any finite set of nontrivial elements of the direct square $G\times G$ there is a homomorphism $G\times G\rightarrow G$ which does not annihilate any of them (G. Baumslag, A. G. Myasnikov, V. N. Remeslennikov). A group $G$ is called *squarelike* if $G$ is universally equivalent (in the sense of first order logic) to a discriminating group (B. Fine, A. M. Gaglione, A. G. Myasnikov, D. Spellman). Must every squarelike group be elementarily equivalent to a discriminating group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03666 | KOU-15.87

### Kourovka Notebook Problem 15.87

O / open | Kourovka 群论问题集

Suppose that a $2$-group $G$ admits a regular periodic locally cyclic group of automorphisms that transitively permutes the set of involutions of $G$. Is $G$ locally finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03667 | KOU-15.88

### Kourovka Notebook Problem 15.88

O / open | Kourovka 群论问题集

Let ${\mathfrak{A}}$ and ${\mathfrak{N}}_c$ denote the varieties of abelian groups and nilpotent groups of class $\leq c$, respectively. Let $F_r= F_r({\mathfrak{A}}{\mathfrak{N}}_c)$ be a free group of rank $r$ in $ {\mathfrak{A}}{\mathfrak{N}}_c$. An element of $ F_r$ is called *primitive* if it can be included in a basis of $ F_r$. Does there exist an algorithm recognizing primitive elements in $ F_r$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03668 | KOU-15.89

### Kourovka Notebook Problem 15.89

S / solved | Kourovka 群论问题集

Let $\Gamma$ be an infinite undirected connected vertex-symmetric graph of finite valency without loops or multiple edges. Is it true that every complex number is an eigenvalue of the adjacency matrix of $\Gamma$ under its natural action as a linear operator on the complex vector space of all complex-valued functions on the vertex set of $\Gamma$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03669 | KOU-15.9

### Kourovka Notebook Problem 15.9

O / open | Kourovka 群论问题集

An automorphism $\varphi$ of the free group $F_n$ on the free generators $x_1, x_2,\ldots , x_n$ is called *conjugating* if $x_i^{\varphi}= t_i^ {-1} x_ {\pi (i)} t_i$, $i=1, 2,\ldots ,n$, for some permutation $\pi \in {\mathbb{S}}_n$ and some elements $t_i\in F_n$. The set of conjugating automorphisms fixing the product $x_1x_2\cdots x_n$ forms the braid group $B_n$. The group $B_n$ is linear for any $n\geq 2$, while the group of all automorphisms $\mathrm{Aut} F_n$ is not linear for $n\geq 3$. Is the group of all conjugating automorphisms linear for $n\geq 3$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03670 | KOU-15.90

## Kourovka Notebook Problem 15.90

O / open | Kourovka 群论问题集

Let $\Gamma$ be an infinite directed graph, and $\overline{ \Gamma}$ the underlying undirected graph. Suppose that the graph $\Gamma$ admits a vertex-transitive group of automorphisms, and the graph $ \overline{ \Gamma}$ is connected and of finite valency. Does there exist a positive integer $k$ (possibly depending on $\Gamma$) such that for any positive integer $n$ there is a directed path of length at most $k\cdot n$ in the graph $\Gamma$ whose initial and terminal vertices are at distance at least $n$ in the graph $\overline{ \Gamma}$?

*Comment of 2005:* It is proved that there exists a positive integer $k'$ (depending only on the valency of $\overline{ \Gamma}$) such that for any positive integer $n$ there is a directed path of length at most $k'\cdot n^2$ in the graph $\Gamma$ whose initial and terminal vertices are at distance at least $n$ in the graph $\overline{ \Gamma}$ (V. I. Trofimov, *Europ. J. Combinatorics*, **27**, no. 5 (2006), 690--700).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03671 | KOU-15.91

## Kourovka Notebook Problem 15.91

O / open | Kourovka 群论问题集

Is it true that any irreducible faithful representation of a linear group $G$ of finite rank over a finitely generated field of characteristic zero is induced from an irreducible representation of a finitely generated dense subgroup of the group $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03672 | KOU-15.92

## Kourovka Notebook Problem 15.92

O / open | Kourovka 群论问题集

A group $\left< x,y\mid x^l=y^m=(xy)^n\right>$ is called the *triangle group* with parameters $(l, m, n)$. It is proved in (A. M. Brunner, R. G. Burns, J. Wiegold, *Math. Scientist*, **4** (1979), 93--98) that the triangle group $ (2, 3, 30)$ has uncountably many non-isomorphic homomorphic images that are residually finite alternating groups. Is the same true for the triangle group $(2, 3, n)$ for all $ n > 6$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03673 | KOU-15.93

## Kourovka Notebook Problem 15.93

O / open | Kourovka 群论问题集

Let $G$ be a pro-$p$ group, $p>2$, and $\varphi$ an automorphism of $G$ of order 2. Suppose that the centralizer $C_G(\varphi )$ is abelian. Is it true that $G$ satisfies a pro-$p$ identity?

An affirmative answer would generalize a result of A. N. Zubkov (*Siberian Math. J.*, **28**, no. 5 (1987), 742--747) saying that non-abelian free pro-$p$ groups cannot be represented by $2\times 2$ matrices.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03674 | KOU-15.94

## Kourovka Notebook Problem 15.94

O / open | Kourovka 群论问题集

Define the *weight* of a group $G$ to be the minimum number of generators of $G$ as a normal subgroup of itself. Let $G=G_1*\cdots *G_n$ be a free product of $n$ nontrivial groups.

a) Is it true that the weight of $G$ is at least $n/2$?

b) Is it true if the $G_i$ are cyclic?

c) Is it true for $n=3$ (without assuming the $G_i$ cyclic)?

Problem 5.53 in Archive (now answered in the affirmative) is the special case with $n=3$ and all the $G_i$ cyclic.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03675 | KOU-15.95

## Kourovka Notebook Problem 15.95

O / open | Kourovka 群论问题集

(A. Mann, Ch. Praeger). Suppose that all fixed-point-free elements of a transitive permutation group $G$ have prime order $p$. If $G$ is a finite $p $-group, must $G$ have exponent $p$?

This is true for $p=2, 3$; it is also known that the exponent of $G$ is bounded in terms of $p$ (A. Hassani, M. Khayaty, E. I. Khukhro, C. E. Praeger, *J. Algebra*, **214**, no. 1 (1999), 317--337).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03676 | KOU-15.96

## Kourovka Notebook Problem 15.96

O / open | Kourovka 群论问题集

An automorphism $\varphi $ of a group $G$ is called a *splitting automorphism of order $n$* if $\varphi ^n=1$ and $xx^{\varphi}x^{\varphi ^{2}}\cdots x^{\varphi ^{n-1}}=1$ for any $x\in G$.

a) Is it true that the derived length of a $d$-generated nilpotent $p $-group admitting a splitting automorphism of order $p^n$ is bounded by a function of $d$, $p$, and $n$? This is true for $n=1$, see Archive, 7.53.

b) The same question for $p^n=4$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03677 | KOU-15.97

## Kourovka Notebook Problem 15.97

O / open | Kourovka 群论问题集

Let $p$ be a prime. A group $G$ satisfies the *$p $-minimal condition* if there are no infinite descending chains $G_1>G_2>\ldots $ of subgroups of $G$ such that each difference $G_i\setminus G_{i+1}$ contains a $p $-element (S. N. Chernikov). Suppose that a locally finite group $G$ satisfies the $p $-minimal condition and has a subnormal series each of whose factors is finite or a $p'$-group. Is it true that all $p $-elements of $G$ generate a Chernikov subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03678 | KOU-15.98

### Kourovka Notebook Problem 15.98

O / open | Kourovka 群论问题集

Let $\mathfrak{F}$ be a saturated formation, and $G$ a finite soluble minimal non-$\mathfrak{F}$-group such that $G^{\mathfrak{F}}$ is a Sylow $p $-subgroup of $G$. Is it true that $G^{\mathfrak{F}}$ is isomorphic to a factor group of the Sylow $p $-subgroup of $PSU(3, p^{2n})$? This is true for the formation of finite nilpotent groups (V. D. Mazurov, S. A. Syskin, *Math. Notes*, **14**, no. 2 (1973), 683--686; \ A. Kh. Zhurtov, S. A. Syskin, *Siberian Math. J.*, **26**, no. 2 (1987), 235--239).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03679 | KOU-15.99

### Kourovka Notebook Problem 15.99

O / open | Kourovka 群论问题集

Let $f(n)$ be the number of isomorphism classes of finite groups of order $n$. Is it true that the equation $f(n)=k$ has a solution for any positive integer $k$? The answer is affirmative for all $k \leq 1000$ (G. M. Wei, *Southeast Asian Bull. Math.*, **22**, no. 1 (1998), 93--102).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03680 | KOU-16.1

### Kourovka Notebook Problem 16.1

P / partially_solved | Kourovka 群论问题集

Let $G$ be a finite non-abelian group, and $Z(G)$ its centre. One can associate a graph $\Gamma_G$ with $G$ as follows: take $G\backslash Z(G)$ as vertices of $\Gamma_G$ and join two vertices $x$ and $y$ if $xy\not=yx$. Let $H$ be a finite non-abelian group such that $\Gamma_G \cong \Gamma_H$.

b) If $H$ is nilpotent, is it true that $G$ is nilpotent? \ \

*Comment of 2009:* This is true if $|H|=|G|$ (A. Abdollahi, S. Akbari, H. R. Maimani, *J. Algebra*, **298** (2006), 468--492).

*Comments of 2023:* This is true if $|Z(H)|\geq |Z(G)|$ (H. Shahverdi, *J. Algebra*, **642** (2024), 60--64), using the earlier result under the additional assumption that the centralizer of every non-central element in $H$ is abelian (V. Grazian, C. Monetta, *J. Algebra*, **633** (2023), 389--402).

c) If $H$ is solvable, is it true that $G$ is solvable?

[Archived solved part(s) of 16.1:] a) Let $G$ be a finite non-abelian group, and $Z(G)$ its centre. One can associate a graph $\Gamma_G$ with $G$ as follows: take $G\backslash Z(G)$ as vertices of $\Gamma_G$ and join two vertices $x$ and $y$ if $xy\not=yx$. Let $H$ be a finite non-abelian group such that $\Gamma_G \cong \Gamma_H$. If $H$ is simple, is it true that $G\cong H$? \ *Comment of 2009:* This is true for groups with disconnected prime graphs (L. Wang, W. Shi, *Commun. Algebra*, **36** (2008), 523--528).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03681 | KOU-16.10

### Kourovka Notebook Problem 16.10

O / open | Kourovka 群论问题集

Is there an algorithm for finding the primitive length of a given element of $F_n$? The definition of a primitive element is given in 14.84; the primitive length is defined similarly to the palindromic length in 16.9.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03682 | KOU-16.100

### Kourovka Notebook Problem 16.100

O / open | Kourovka 群论问题集

Is there an (infinite) 2-generator simple group $G$ such that $\mathrm{Aut} F_2$ is transitive on the set of normal subgroups $N$ of the free group $F_2$ such that $F_2/N\cong G$? Cf. 6.45.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03683 | KOU-16.101

### Kourovka Notebook Problem 16.101

S / solved | Kourovka 群论问题集

Do there exist uncountably many infinite 2-groups that are quotients of the group $\left<x, y \mid x^2 = y^4 = (xy)^8 = 1\right>$? There certainly exists one, namely the subgroup of finite index in Grigorchuk's first group generated by $b$ and $ad$; see (R. I. Grigorchuk, *Functional Anal. Appl.*, **14** (1980), 41--43).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03684 | KOU-16.102

### Kourovka Notebook Problem 16.102

O / open | Kourovka 群论问题集

We say that a group $G$ is *rational* if any two elements $x,y\in G$ satisfying $\left< x \right>=\left< y \right>$ are conjugate. Is it true that for any $d$ there exist only finitely many finite rational $d$-generated $2$-groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03685 | KOU-16.103

### Kourovka Notebook Problem 16.103

S / solved | Kourovka 群论问题集

Is there a rank analogue of the Leedham-Green--McKay--Shepherd theorem on $p $-groups of maximal class? More precisely, suppose that $P$ is a 2-generator finite $p $-group whose lower central quotients $\gamma _i(P)/\gamma _{i+1}(P)$ are cyclic for all $i\geq 2$. Is it true that $P$ contains a normal subgroup $N$ of nilpotency class $\leq 2$ such that the rank of $P/N$ is bounded in terms of $p$ only?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03686 | KOU-16.104

### Kourovka Notebook Problem 16.104

S / solved | Kourovka 群论问题集

If \( G \) is a finite group, then every element \( a \) of the rational group algebra \( {\mathbb{Q}}[G] \) has a unique Jordan decomposition \( a =a _{s}+a _{n} \), where \( a _{n}\in {\mathbb{Q}}[G] \) is nilpotent, \( a _{s}\in {\mathbb{Q}}[G] \) is semisimple over \( \mathbb{Q} \), and \( a _{s}a _{n}=a _{n}a _{s} \). The integral group ring \( {\mathbb{Z}}[G] \) is said to have the *additive Jordan decomposition* property (AJD) if \( a _{s}, a _{n} \in {\mathbb{Z}}[G] \) for every \( a \in {\mathbb{Z}}[G] \). If \( a \in {\mathbb{Q}}[G] \) is invertible, then \( a _{s} \) is also invertible and so \( a =a _{s}a _{u} \) with \( a _{u}=1+a _{s}^{-1}a _{n} \) unipotent and \( a _{s}a _{u}=a _{u}a _{s} \). Such a decomposition is again unique. We say that \( {\mathbb{Z}}[G] \) has *multiplicative Jordan decomposition* property (MJD) if \( a _{s}, a _{u} \in {\mathbb{Z}}[G] \) for every invertible \( a \in {\mathbb{Z}}[G] \). See the survey (A. W. Hales, I. B. S. Passi, in: *Algebra, Some Recent Advances*, Birkh\"{a}user, Basel, 1999, 75--87).

Is it true that there are only finitely many isomorphism classes of finite $2$-groups $G$ such that ${\mathbb{Z}}[G]$ has MJD but not AJD?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03687 | KOU-16.105

### Kourovka Notebook Problem 16.105

O / open | Kourovka 群论问题集

Is it true that a locally graded group which is a product of two almost polycyclic subgroups (equivalently, of two almost soluble subgroups with the maximal condition) is almost polycyclic?

By (N. S. Chernikov, *Ukrain. Math. J.*, **32** (1980) 476--479) a locally graded group which is a product of two Chernikov subgroups (equivalently, of two almost soluble subgroups with the minimal condition) is Chernikov.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03688 | KOU-16.106

### Kourovka Notebook Problem 16.106

S / solved | Kourovka 群论问题集

Let $\pi _e(G)$ denote the set of orders of elements of a group $G$, and $h(\Gamma )$ the number of non-isomorphic finite groups $G$ with $\pi _e(G)=\Gamma$. Do there exist two finite groups $G_1, G_2$ such that $\pi _e(G_1)=\pi _e(G_2)$, \ $h(\pi _e(G_1)) < \infty$, and neither of the two groups $G_1, G_2$ is isomorphic to a subgroup or a quotient of a normal subgroup of the other?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03689 | KOU-16.107

### Kourovka Notebook Problem 16.107

S / solved | Kourovka 群论问题集

Is it true that almost every alternating group ${A}_n$ is uniquely determined in the class of finite groups by its set of element orders, i. e. that $h(\pi _e({ A}_n))=1$ for all large enough $n$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03690 | KOU-16.108

### Kourovka Notebook Problem 16.108

O / open | Kourovka 群论问题集

Do braid groups $B_n,\; n>4$, have non-elementary hyperbolic factor groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03691 | KOU-16.109

### Kourovka Notebook Problem 16.109

S / solved | Kourovka 群论问题集

Is there a polynomial time algorithm for solving the word problem in the group $\mathrm{Aut} F_n$ (with respect to some particular finite presentation), where $F_n$ is the free group of rank $n \geq 2$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03692 | KOU-16.11

### Kourovka Notebook Problem 16.11

S / solved | Kourovka 群论问题集

Let $G$ be a finite $p$-group. Does there always exist a finite $p$-group $H$ such that $\Phi(H)\cong [G,G]$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03693 | KOU-16.110

### Kourovka Notebook Problem 16.110

O / open | Kourovka 群论问题集

(I. Kapovich, P. Schupp). Is there an algorithm which, when given two elements $u, v$ of a free group $F_n$, decides whether or not the cyclic length of $\phi(u)$ equals the cyclic length of $\phi(v)$ for every automorphism $\phi$ of $F_n$?

*Comment of 2009:* The answer was shown to be positive for $n=2$ in (D. Lee, *J. Group Theory*, **10** (2007), 561--569).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03694 | KOU-16.111

### Kourovka Notebook Problem 16.111

O / open | Kourovka 群论问题集

Must an infinite simple periodic group with a dihedral Sylow 2-subgroup be isomorphic to $L_2(P)$ for a locally finite field $P$ of odd characteristic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03695 | KOU-16.12

### Kourovka Notebook Problem 16.12

S / solved | Kourovka 群论问题集

Given a finite $p $-group $G$, we define a *$\Phi$-extension* of $G$ as any finite $p $-group $H$ containing a normal subgroup $N$ of order $p$ such that $H/N\cong G$ and $N\leq\Phi(H)$. Is it true that for every finite $p $-group $G$ there exists an infinite sequence $G=G_1, G_2,\ldots $ such that $G_{i+1}$ is a $\Phi$-extension of $G_i$ for all $i=1, 2,\ldots$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03696 | KOU-16.13

### Kourovka Notebook Problem 16.13

S / solved | Kourovka 群论问题集

Does there exist a finite $p$-group $G$ all of whose maximal subgroups $H$ are special, that is, satisfy $Z(H)=[H,H]=\Phi(H)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03697 | KOU-16.14

### Kourovka Notebook Problem 16.14

O / open | Kourovka 群论问题集

Let $G$ be a finite $2$-group such that $\Omega_1(G)\leq Z(G)$. Is it true that the rank of $G/G^2$ is at most double the rank of $Z(G)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03698 | KOU-16.15

### Kourovka Notebook Problem 16.15

P / partially_solved | Kourovka 群论问题集

An element $g$ of a group $G$ is an *Engel element* if for every $h\in G$ there exists $k$ such that $[h,g,\ldots ,g]=1$, where $g$ occurs $k$ times; if there is such $k$ independent of $h$, then $g$ is said to be *boundedly Engel*.

b) Does the set of boundedly Engel elements form a subgroup in a torsion-free group?

c) The same question for right-ordered groups.

d) The same question for linearly ordered groups.

[Archived solved part(s) of 16.15:] An element $g$ of a group $G$ is an *Engel element* if for every $h\in G$ there exists $k$ such that $[h,g,\ldots ,g]=1$, where $g$ occurs $k$ times; if there is such $k$ independent of $h$, then $g$ is said to be *boundedly Engel*.

a) (B. I. Plotkin). Does the set of boundedly Engel elements of a group form a subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03699 | KOU-16.16

### Kourovka Notebook Problem 16.16

O / open | Kourovka 群论问题集

{a)} Does the set of (not necessarily boundedly) Engel elements of a group without elements of order 2 form a subgroup?

b) The same question for torsion-free groups.

c) The same question for right-ordered groups.

d) The same question for linearly ordered groups.

There are examples of 2-groups where a product of two (unboundedly) Engel elements is not an Engel element.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03700 | KOU-16.17

### Kourovka Notebook Problem 16.17

S / solved | Kourovka 群论问题集

Is it true that a non-abelian simple group cannot contain Engel elements other than the identity element?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03701 | KOU-16.18

### Kourovka Notebook Problem 16.18

O / open | Kourovka 群论问题集

Does there exist a linearly orderable soluble group of derived length exactly $n$ that has a single proper normal relatively convex subgroup

a) for $n=3$?

b) for $n > 4$?

Such groups do exist for $ n=2$ and for $n=4$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03702 | KOU-16.19

### Kourovka Notebook Problem 16.19

O / open | Kourovka 群论问题集

Is the variety of lattice-ordered groups generated by nilpotent groups finitely based? (Here the variety is considered in the signature of group and lattice operations.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03703 | KOU-16.2

### Kourovka Notebook Problem 16.2

S / solved | Kourovka 群论问题集

A group $G$ is *subgroup-separable* if for any subgroup $H\leq G$ and element $x\in G\setminus H$ there is a homomorphism to a finite group $f: G\rightarrow F$ such that $f(x)\not\in f(H)$. Is it true that a finitely generated solvable group is locally subgroup-separable if and only if it does not contain a solvable Baumslag--Solitar group? Solvable Baumslag--Solitar groups are $BS(1,n)= \left< a, b \mid bab^{-1}=a^n\right>$ for $n>1$.

Background: It is known that a finitely generated solvable group is subgroup-separable if and only if it is polycyclic (R. C. Alperin, in: *Groups--Korea'98 (Pusan)*, de Gruyter, Berlin, 2000, 1--5).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03704 | KOU-16.20

### Kourovka Notebook Problem 16.20

O / open | Kourovka 群论问题集

Let $M$ be a quasivariety of groups. The *dominion* $\mathrm{dom}^M_A(H)$ of a subgroup $H$ of a group $A$ (in $M$) is the set of all elements $a\in A$ such that for any two homomorphisms $f,g:A\rightarrow B\in M$, if $f,g$ coincide on $H$, then $f(a)=g(a)$. Suppose that the set $\{ \mathrm{dom}^N_A(H)\mid N \mbox{ is a quasivariety}, N\subseteq M\} $ forms a lattice with respect to set-theoretic inclusion. Can this lattice be modular and non-distributive?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03705 | KOU-16.21

### Kourovka Notebook Problem 16.21

O / open | Kourovka 群论问题集

Given a non-central matrix $\alpha \in SL_n(F)$ over a field $F$ for $n>2$, is it true that every non-central matrix in $SL_n(F)$ is a product of $n$ matrices, each similar to $\alpha $?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03706 | KOU-16.22

### Kourovka Notebook Problem 16.22

O / open | Kourovka 群论问题集

(Well-known problem.) Let $E_n(A)$ be the subgroup of $GL_n(A)$ generated by elementary matrices. Is $SL_2(A) = E_2(A)$ when $A= {\mathbb{Z}[x, 1/x]$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03707 | KOU-16.23

### Kourovka Notebook Problem 16.23

O / open | Kourovka 群论问题集

Is there, for some $n>2$ and a ring $A$ with $1$, a matrix in $E_n(A)$ that is nonscalar modulo any proper ideal and is not a commutator?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03708 | KOU-16.24

### Kourovka Notebook Problem 16.24

S / solved | Kourovka 群论问题集

The *spectrum* of a finite group is the set of orders of its elements. Does there exist a finite group $G$ whose spectrum coincides with the spectrum of a finite simple exceptional group $L$ of Lie type, but $G$ is not isomorphic to $L$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03709 | KOU-16.25

### Kourovka Notebook Problem 16.25

S / solved | Kourovka 群论问题集

Do there exist three pairwise non-isomorphic finite non-abelian simple groups with the same spectrum?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03710 | KOU-16.26

### Kourovka Notebook Problem 16.26

O / open | Kourovka 群论问题集

We say that the prime graphs of finite groups $G$ and $H$ coincide if the sets of primes dividing their orders are the same, $\pi (G)=\pi (H)$, and for any distinct $p,q\in \pi (G)$ there is an element of order $pq$ in $G$ if and only if there is such an element in $H$. Does there exist a positive integer $k$ such that there are no $k$ pairwise non-isomorphic finite non-abelian simple groups with the same graphs of primes? *Conjecture:* $k=5$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03711 | KOU-16.27

### Kourovka Notebook Problem 16.27

S / solved | Kourovka 群论问题集

Suppose that a finite group $G$ has the same spectrum as an alternating group. Is it true that $G$ has at most one non-abelian composition factor?

For any finite simple group other than alternating the answer to the corresponding question is affirmative (mod CFSG).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03712 | KOU-16.28

### Kourovka Notebook Problem 16.28

P / partially_solved | Kourovka 群论问题集

Let $G$ be a connected linear reductive algebraic group over a field of positive characteristic, $X$ a closed subset of $G$, and let $X^k=\{x_1\cdots x_k\mid x_i\in X\}$.

a) Is it true that there always exists a positive integer $c=c(X)>1$ such that $X^c$ is closed?

b) If $X$ is a conjugacy class of $G$ such that $X^2$ contains an open subset of $G$, then is $X^2=G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03713 | KOU-16.29

### Kourovka Notebook Problem 16.29

O / open | Kourovka 群论问题集

Which finite simple groups of Lie type $G$ have the following property: for every semisimple abelian subgroup $A$ and proper subgroup $H$ of $G$ there exists $x\in G$ such that $A^x\cap H=1$?

This property holds when $A$ is a cyclic subgroup (J. Siemons, A. Zalesskii, *J. Algebra*, **256** (2002), 611--625), as well as when $A$ is contained in some maximal torus and $G=PSL_n(q)$ (J. Siemons, A. Zalesskii, *J. Algebra*, **226** (2000), 451--478). Note that if $G=L_2(5)$, $A=2\times 2$, and $H=5:2$ (in Atlas notation), then $A^x\cap H>1$ for every $x\in G$ (this example was communicated to the author by V. I. Zenkov).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03714 | KOU-16.3

### Kourovka Notebook Problem 16.3

O / open | Kourovka 群论问题集

Is it true that if $G$ is a finite group with all conjugacy classes of distinct sizes, then $G\cong {\mathbb{S}}_3$?

This is true if $G$ is solvable (J. Zhang, *J. Algebra*, **170** (1994), 608--624); it is also known that $F(G)$ is nontrivial (Z. Arad, M. Muzychuk, A. Oliver, *J. Algebra*, **280** (2004), 537--576).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03715 | KOU-16.30

### Kourovka Notebook Problem 16.30

O / open | Kourovka 群论问题集

Suppose that $A$ and $B$ are subgroups of a group $G$ and $G = AB$. Will $G$ have composition (principal) series if $A$ and $B$ have composition (respectively, principal) series?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03716 | KOU-16.31

### Kourovka Notebook Problem 16.31

S / solved | Kourovka 群论问题集

Suppose that a group $G$ has a composition series and let ${\mathfrak{F}}(G)$ be the formation generated by $G$. Is the set of all subformations of ${\mathfrak{F}}(G)$ finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03717 | KOU-16.32

### Kourovka Notebook Problem 16.32

O / open | Kourovka 群论问题集

Suppose that a group $G$ has a composition series and let $\mathrm{Fit}(G)$ be the Fitting class generated by $G$. Is the set of all Fitting subclasses of $\mathrm{Fit}(G)$ finite? Cf. 14.31.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03718 | KOU-16.33

### Kourovka Notebook Problem 16.33

O / open | Kourovka 群论问题集

Suppose that a finite $p $-group $G$ has an abelian subgroup $A$ of order $p^n$. Does $G$ contain an abelian subgroup $B$ of order $p^n$ that is normal in $\langle B^G\rangle$

a) if $p=3$?

b) if $p=2$?

c) If $p=3$ and $A$ is elementary abelian, does $G$ contain an elementary abelian subgroup $B$ of order $3^n$ that is normal in $\langle B^G\rangle$?

The corresponding results have been proved for greater primes $p$, based on extensions of Thompson's Replacement Theorem (and the dihedral group of order $32$ shows that the third question has negative answer for $p=2$). See (G. Glauberman, *J. Algebra*, **196** (1997), 301--338; \ J. Alperin, G. Glauberman, *J. Algebra*, **203** (1998), 533--566; \ G. Glauberman, *J. Algebra*, **272** (2004), 128--153).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03719 | KOU-16.34

### Kourovka Notebook Problem 16.34

S / solved | Kourovka 群论问题集

Suppose that $G$ is a finitely generated group acting faithfully on a regular rooted tree by finite-state automorphisms. Is the conjugacy problem decidable for $G$?

See the definitions in (R. I. Grigorchuk, V. V. Nekrashevich, V. I. Sushchanskiĭ,

*Proc. Steklov Inst. Math.*, **2000**, no. 4 (231), 128--203).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03720 | KOU-16.35

### Kourovka Notebook Problem 16.35

S / solved | Kourovka 群论问题集

Is every finitely presented soluble group nilpotent-by-nilpotent-by-finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03721 | KOU-16.36

### Kourovka Notebook Problem 16.36

S / solved | Kourovka 群论问题集

(Well-known problem). We call a finite group *rational* if all of its ordinary characters are rational-valued. Is every Sylow 2-subgroup of a rational group also a rational group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03722 | KOU-16.37

### Kourovka Notebook Problem 16.37

S / solved | Kourovka 群论问题集

Let $G$ be a solvable rational finite group with an extra-special Sylow 2-subgroup. Is it true that either $G$ is a 2-nilpotent group, or there is a normal subgroup $E$ of $G$ such that $G/E$ is an extension of a normal 3-subgroup by an elementary abelian 2-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03723 | KOU-16.38

### Kourovka Notebook Problem 16.38

S / solved | Kourovka 群论问题集

Let $G$ be a soluble group, and let $A$ and $B$ be periodic subgroups of $G$. Is it true that any subgroup of $G$ contained in the set $AB=\{ ab\;|\; a\in A,\; b\in B\}$ is periodic? This is known to be true if $AB$ is a subgroup of $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03724 | KOU-16.39

### Kourovka Notebook Problem 16.39

O / open | Kourovka 群论问题集

(J. E. Humphreys, D. N. Verma). Let $G$ be a semisimple algebraic group over an algebraically closed field $k$ of characteristic $p>0$. Let ${\mathfrak{g}}$ be the Lie algebra of $G$ and let $u=u({\mathfrak{g}})$ be the restricted enveloping algebra of ${\mathfrak{g}}$. By a theorem of Curtis every irreducible restricted $u$-module (i. e. every irreducible restricted ${\mathfrak{g}}$-module) is the restriction to ${\mathfrak{g}}$ of a (rational) $G$-module. Is it also true that every projective indecomposable $u$-module is the restriction of a rational $G$-module? This is true if $p\geq 2h-2$ (where $h$ is the Coxeter number of $G$) by results of Jantzen.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03725 | KOU-16.4

### Kourovka Notebook Problem 16.4

O / open | Kourovka 群论问题集

Let $G$ be a finite group with $C,D$ two nontrivial conjugacy classes such that $CD$ is also a conjugacy class. Can $G$ be a non-abelian simple group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03726 | KOU-16.40

### Kourovka Notebook Problem 16.40

O / open | Kourovka 群论问题集

Let $\Delta$ be a subgroup of the automorphism group of a free pro-$p$ group of finite rank $F$ such that $\Delta$ is isomorphic (as a profinite group) to the group $Z_p$ of $ p $-adic integers. Is the subgroup of fixed points of $\Delta$ in $F$ finitely generated (as a profinite group)?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03727 | KOU-16.41

### Kourovka Notebook Problem 16.41

O / open | Kourovka 群论问题集

Let $F$ be a free pro-$p$ group of finite rank $n>1$. Does $\mathrm{Aut} F$ possess an open subgroup of finite cohomological dimension?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03728 | KOU-16.42

### Kourovka Notebook Problem 16.42

S / solved | Kourovka 群论问题集

Is a topological Abelian group $(G,\tau)$ compact if every group topology $\tau '\subseteq\tau$ on $G$ is complete? (The answer is yes if every continuous homomorphic image of $(G,\tau)$ is complete.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03729 | KOU-16.43

### Kourovka Notebook Problem 16.43

S / solved | Kourovka 群论问题集

Is there a partition of the group $\bigoplus_{\omega_1}({\mathbb{Z}}/3{\mathbb{Z}})$ into three subsets whose complements do not contain cosets modulo infinite subgroups? (There is a partition into two such subsets.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03730 | KOU-16.44

### Kourovka Notebook Problem 16.44

O / open | Kourovka 群论问题集

Is there in ZFC a countable non-discrete topological group not containing discrete subsets with a single accumulation point? (Such a group is known to exist under Martin's Axiom.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03731 | KOU-16.45

### Kourovka Notebook Problem 16.45

S / solved | Kourovka 群论问题集

Let $G$ be a permutation group on a set $\Omega$. A sequence of points of $\Omega$ is a *base* for $G$ if its pointwise stabilizer in $G$ is the identity; it is *minimal* if no point may be removed. Let $b(G)$ be the maximum, over all permutation representations of the finite group $G$, of the maximum size of a minimal base for $G$. Let $\mu'(G)$ be the maximum size of an *independent set* in $G$, a set of elements with the property that no element belongs to the subgroup generated by the others. Is it true that $b(G)=\mu'(G)$? (It is known that $b(G)\le\mu'(G)$, and that equality holds for the symmetric groups.)

*Remark.* An equivalent question is the following. Suppose that the Boolean lattice $B(n)$ of subsets of an $n$-element set is embeddable as a meet-semilattice of the subgroup lattice of $G$, and suppose that $n$ is maximal with this property. Is it true that then there is such an embedding of $B(n)$ with the property that the least element of $B(n)$ is a normal subgroup of $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03732 | KOU-16.46

### Kourovka Notebook Problem 16.46

O / open | Kourovka 群论问题集

Among the finitely-presented groups that act arc-transitively on the (infinite) 3-valent tree with finite vertex-stabilizer are the two groups $G_3 = \left< h, a, P, Q \mid h^3\right.$, $ a^2$, $ P^2$, $ Q^2$, $ [P,Q]$, $ [h,P]$, $ (hQ)^2$, $\left. a^{-1}PaQ \right>$ \ and \ $G_4 = \left< h, a, p, q, r \mid h^3\right.$, $ a^2$, $ p^2$, $ q^2$, $ r^2$, $ [p,q]$, $ [p,r]$, $ p(qr)^2$, $ h^{-1}phq$, $ h^{-1}qhpq$, $ (hr)^2$, $ [a,p]$, $\left. a^{-1}qar \right>$, each of which contains the modular group $G_1 = \left< h,a \mid h^3, a^2 \right> \cong PSL_2({\mathbb{Z}})$ as a subgroup of finite index. The free product of $G_3$ and $G_4$ with subgroup $G_1 = \left< h,a \right>$ amalgamated has a normal subgroup $K$ of index 8 generated by $A = h$, \ $B = aha$, \ $C = p$, \ $D = PpP$, \ $E = QpQ$, and $F = PQpPQ$, with dihedral complement $\left< a, P, Q \right>$. The group $K$ has presentation $\left< A, B, C, D, E, F \mid A^3\right.$, $ B^3$, $ C^2$, $ D^2$, $ E^2$, $ F^2$, $ (AC)^3$, $ (AD)^3$, $ (AE)^3$, $ (AF)^3$, $ (BC)^3$, $ (BD)^3$, $ (BE)^3$, $ (BF)^3$, $ (ABA^{-1}C)^2$, $ (ABA^{-1}D)^2$, $ (A^{-1}BAE)^2$, $ (A^{-1}BAF)^2$, $ (BAB^{-1}C)^2$, $ (B^{-1}ABD)^2$, $ (BAB^{-1}E)^2$, $\left. (B^{-1}ABF)^2 \right>$. Does this group have a non-trivial finite quotient?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03733 | KOU-16.47

### Kourovka Notebook Problem 16.47

O / open | Kourovka 群论问题集

(P. Conrad). Is it true that every torsion-free abelian group admits an Archimedean lattice ordering?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03734 | KOU-16.48

### Kourovka Notebook Problem 16.48

S / solved | Kourovka 群论问题集

A group $H$ is said to have *generalized torsion* if there exists an element $h\ne 1$ such that $h^{x_1}h^{x_2}\cdots h^{x_n}= 1$ for some $n$ and some $x_i\in H$. Is every group without generalized torsion right-orderable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03735 | KOU-16.49

### Kourovka Notebook Problem 16.49

S / solved | Kourovka 群论问题集

Is it true that a free product of groups without generalized torsion is a group without generalized torsion?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03736 | KOU-16.5

### Kourovka Notebook Problem 16.5

O / open | Kourovka 群论问题集

A group is said to be *perfect* if it coincides with its derived subgroup. Does there exist a perfect locally finite $p$-group

a) all of whose proper subgroups are hypercentral?

b) all of whose proper subgroups are solvable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03737 | KOU-16.50

### Kourovka Notebook Problem 16.50

S / solved | Kourovka 群论问题集

Do there exist simple finitely generated right-orderable groups?

There exist finitely generated right-orderable groups coinciding with the derived subgroup (G. M. Bergman).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03738 | KOU-16.51

### Kourovka Notebook Problem 16.51

S / solved | Kourovka 群论问题集

Do there exist groups that can be right-ordered in infinitely countably many ways?

There exist groups that can be fully ordered in infinitely countably many ways (R. Buttsworth).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03739 | KOU-16.52

### Kourovka Notebook Problem 16.52

S / solved | Kourovka 群论问题集

Is every finitely presented elementary amenable group solvable-by-finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03740 | KOU-16.53

### Kourovka Notebook Problem 16.53

S / solved | Kourovka 群论问题集

Let $d(G)$ denote the smallest cardinality of a generating set of the group $G$. Suppose that $G=\left<A,B\right>$, where $A$ and $B$ are two $d$-generated finite groups of coprime orders. Is it true that $d(G)\leq d+1$?

See Archive 12.71 and (A. Lucchini, *J. Algebra*, **245** (2001), 552--561).

*Comment of 2021*: the answer is yes if $A$ and $B$ have prime power order (E. Detomi, A. Lucchini, *J. London Math. Soc. (2)*, **87**, no. 3 (2013), 689--706).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03741 | KOU-16.54

### Kourovka Notebook Problem 16.54

S / solved | Kourovka 群论问题集

We say that a group $G$ acts *freely* on a group $V$ if $vg\ne v$ for any nontrivial elements $g\in G,\; v\in V$. Is it true that a group $G$ that can act freely on a non-trivial abelian group is embeddable in the multiplicative group of some skew-field?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03742 | KOU-16.55

### Kourovka Notebook Problem 16.55

S / solved | Kourovka 群论问题集

(Well-known problem). Let $V$ be a faithful absolutely irreducible module for a finite group $G$. Is it true that $\dim H^1(G,V)\leq 2$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03743 | KOU-16.56

### Kourovka Notebook Problem 16.56

O / open | Kourovka 群论问题集

The *spectrum* $\omega(G)$ of a group $G$ is the set of orders of elements of $G$. Suppose that $\omega(G)=\{1,2,3,4,5,6\}$. Is $G$ locally finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03744 | KOU-16.57

### Kourovka Notebook Problem 16.57

S / solved | Kourovka 群论问题集

Suppose that $\omega(G)=\omega(L_2(7))=\{1,2,3,4,7\}$. Is $G\cong L_2(7)$? This is true for finite $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03745 | KOU-16.58

### Kourovka Notebook Problem 16.58

S / solved | Kourovka 群论问题集

Is $SU_2({\mathbb{C}})$ the only group that has just one irreducible complex representation of dimension $n$ for each $n = 1, 2,\ldots $?

(If $R[n]$ is the $n$-dimensional irreducible complex representation of $SU_2({\mathbb{C}})$, then $R[2]$ is the natural two-dimensional representation, and $R[2]\otimes R[n] = R[n-1] + R[n+1]$ for $n > 1$.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03746 | KOU-16.59

### Kourovka Notebook Problem 16.59

S / solved | Kourovka 群论问题集

Given a finite group $K$, does there exist a finite group $G$ such that $K\cong \mathrm{Out }G=\mathrm{Aut } G/\mathrm{Inn }G$? (It is known that an infinite group $G$ exists with this property.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03747 | KOU-16.6

### Kourovka Notebook Problem 16.6

O / open | Kourovka 群论问题集

Can a perfect locally finite $p $-group be generated by a subset of bounded exponent

a) if all of its proper subgroups are hypercentral?

b) if all of its proper subgroups are solvable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03748 | KOU-16.60

### Kourovka Notebook Problem 16.60

O / open | Kourovka 群论问题集

If $G$ is a finite group, let $T(G)$ be the sum of the degrees of the irreducible complex representations of $G$, $T(G)=\sum\limits_{i=1}^{k(G)}d_i$, where $G$ has $k(G)$ conjugacy classes. If $\alpha \in \mathrm{Aut } G$, let $S_{\alpha }=\{g\in G\mid \alpha (g)=g^{-1}\}$. Is it true that $T(G)\geq |S_{\alpha }|$ for all $\alpha \in \mathrm{Aut } G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03749 | KOU-16.61

### Kourovka Notebook Problem 16.61

S / solved | Kourovka 群论问题集

A subgroup $H$ of a group $G$ is *fully invariant* if $\vartheta (H)\leq H$ for every endomorphism $\vartheta$ of $G$. Let $G$ be a finite group such that $G$ has a fully invariant subgroup of order $d$ for every $d$ dividing $|G|$. Must $G$ be cyclic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03750 | KOU-16.62

### Kourovka Notebook Problem 16.62

S / solved | Kourovka 群论问题集

Let $G$ be a group such that every $\alpha \in \mathrm{Aut } G$ fixes every conjugacy class of $G$ (setwise). Must $\mathrm{Aut } G =\mathrm{Inn }G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03751 | KOU-16.63

### Kourovka Notebook Problem 16.63

O / open | Kourovka 群论问题集

Is there a non-trivial finite $p$-group $G$ of odd order such that $|\mathrm{Aut } G| =|G|$?

See also Archive 12.77.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03752 | KOU-16.64

### Kourovka Notebook Problem 16.64

O / open | Kourovka 群论问题集

A non-abelian variety in which all finite groups are abelian is called *pseudo-abelian*. A group variety is called a *$t$-variety* if for all groups in this variety the relation of being a normal subgroup is transitive. By (O. Macedo\'nska, A. Storozhev, *Commun. Algebra*, **25**, no. 5 (1997), 1589--1593) each non-abelian $t$-variety is pseudo-abelian, and the pseudo-abelian varieties constructed in (A. Yu. Olshanskii, *Math. USSR--Sb.*, **54** (1986), 57--80) are $t$-varieties. Is every pseudo-abelian variety a $t$-variety?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03753 | KOU-16.65

### Kourovka Notebook Problem 16.65

S / solved | Kourovka 群论问题集

Does there exist a finitely presented residually torsion-free nilpotent group with a free presentation $G=F/R$ such that the group $F/[F,R]$ is not residually nilpotent?

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03754 | KOU-16.66

### Kourovka Notebook Problem 16.66

O / open | Kourovka 群论问题集

For a group $G$, let $D_n(G)$ denote the $n$-th dimension subgroup of $G$, and $\zeta_n(G)$ the $n$-th term of its upper central series. For a given integer $n\geq 1$, let $f(n)=\max\{ m\mid \exists\;\text{a nilpotent group}\;G\;\text{of class}\;n\;\text{with}\;D_m(G)\neq 1\}$ and $g(n)=\max\{ m\ |\ \exists\ \text{a nilpotent group}\ G\ \text{of\ class}\ n\ \text{such that}\ D_n(G)\not\subseteq \zeta_m(G)\}.$

a) What is $f(3)$?

b) (B. I. Plotkin). Is it true that $f(n)$ is finite for all $n$?

c) Is the growth of $f(n)$ and $g(n)$ polynomial, exponential, or intermediary?

It is known that both $f(n)-n$ and $g(n)$ tend to infinity as $n\rightarrow \infty$ (N. Gupta, Yu. V. Kuz'min, *J. Pure Appl. Algebra*, **104** (1995), 191--197).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03755 | KOU-16.67

### Kourovka Notebook Problem 16.67

S / solved | Kourovka 群论问题集

*Conjecture:* Given any integer $k$, there exists an integer $n_0=n_0(k)$ such that if $n\geq n_0$ then the symmetric group of degree $n$ has at least $k$ different ordinary irreducible characters of equal degrees.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03756 | KOU-16.68

### Kourovka Notebook Problem 16.68

S / solved | Kourovka 群论问题集

Let $W(x, y)$ be a non-trivial reduced group word, and $G$ one of the groups $PSL(2,{\mathbb{R}})$, \ $PSL(2, {\mathbb{C}})$, or $SO(3, {\mathbb{R}})$. Are all the maps $W:G \times G\rightarrow G$ surjective?

For $SL(2, {\mathbb{R}})$, \ $SL(2, {\mathbb{C}})$, \ $GL(2, {\mathbb{C}})$, and for the group $S^3$ of quaternions of norm 1 there exist non-surjective words; see (J. Mycielski, *Amer. Math. Monthly*, **84** (1977), 723--726; \ **85** (1978), 263--264).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03757 | KOU-16.69

### Kourovka Notebook Problem 16.69

S / solved | Kourovka 群论问题集

Let $W$ be as above.

a) Must the range of the function $\mathrm{Tr}(W(x, y))$ for $x, y\in GL(2, {\mathbb{R}})$ include the interval $[-2, +\infty)$?

b) For $x, y$ being non-zero quaternions, must the range of the function $\mathrm{Re}(W(x, y))$ include the interval $[-5/27, 1]$?

(For some comments see *ibid*.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03758 | KOU-16.7

### Kourovka Notebook Problem 16.7

O / open | Kourovka 群论问题集

Is it true that the membership problem is undecidable for any semidirect product $F_n \leftthreetimes F_n$ of non-abelian free groups $F_n$?

An affirmative answer would imply an answer to 6.24. Recall that the membership problem is decidable for $F_n$ (M. Hall, 1949) and undecidable for $F_n \times F_n$ (K. A. Mikhailova, 1958).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03759 | KOU-16.70

### Kourovka Notebook Problem 16.70

O / open | Kourovka 群论问题集

Suppose that a finitely generated group $G$ acts freely on an $A$-tree, where $A$ is an ordered abelian group. Is it true that $G$ acts freely on a ${\mathbb{Z}^n$-tree for some $n$?

*Comment of 2013:* It was proved that every finitely presented group acting freely on an $A$-tree acts freely on some $\mathbb{R}^n$-tree for a suitable $n$, where $\mathbb{R}$ has the lexicographical order (O. Kharlampovich, A. Myasnikov, D. Serbin, *Int. J. Algebra Comput.*, **23** (2013), 325--345).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03760 | KOU-16.71

### Kourovka Notebook Problem 16.71

S / solved | Kourovka 群论问题集

Is the elementary theory of a torsion-free hyperbolic group decidable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03761 | KOU-16.72

### Kourovka Notebook Problem 16.72

O / open | Kourovka 群论问题集

Does there exist an exponential-time algorithm for obtaining a JSJ-decomposition of a finitely generated fully residually free group?

Some algorithm was found in (O. Kharlampovich, A. Myasnikov, in: *Contemp. Math. AMS $($Algorithms, Languages, Logic*), **378** (2005), 87--212).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03762 | KOU-16.73

### Kourovka Notebook Problem 16.73

P / partially_solved | Kourovka 群论问题集

b) Let $G$ be a group generated by a finite set $S$, and let $l(g)$ denote the word length function of $g\in G$ with respect to $S$. The group $G$ is said to be *contracting* if there exist a faithful action of $G$ on the set $X^*$ of finite words over a finite alphabet $X$ and constants $0<\lambda <1$ and $C>0$ such that for every $g\in G$ and $x\in X$ there exist $h\in G$ and $y\in X$ such that $l(h)<\lambda l(g)+C$ and $g(xw)=yh(w)$ for all $w\in X^*$.

Do there exist non-amenable contracting groups?

[Archived solved part(s) of 16.73:] a) Let $G$ be a group generated by a finite set $S$, and let $l(g)$ denote the word length function of $g\in G$ with respect to $S$. The group $G$ is said to be *contracting* if there exist a faithful action of $G$ on the set $X^*$ of finite words over a finite alphabet $X$ and constants $0<\lambda <1$ and $C>0$ such that for every $g\in G$ and $x\in X$ there exist $h\in G$ and $y\in X$ such that $l(h)<\lambda l(g)+C$ and $g(xw)=yh(w)$ for all $w\in X^*$.

Can a contracting group have a non-abelian free subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03763 | KOU-16.74

### Kourovka Notebook Problem 16.74

P / partially_solved | Kourovka 群论问题集

b) Is it true that all groups generated by automata of polynomial growth in the sense of S. Sidki (*Geom. Dedicata*, **108** (2004), 193--204) are amenable?

[Archived solved part(s) of 16.74:] a) Let $G=\left<\alpha,\beta\right>$ be the group generated by the following two permutations of ${\mathbb{Z}}$: \ $\alpha (n)=n+1$; \ $\beta (0)=0$, \ $\beta (2^km)=2^k(m+2)$, where $m$ is odd and $n$ is a positive integer. Is $G$ amenable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03764 | KOU-16.75

### Kourovka Notebook Problem 16.75

S / solved | Kourovka 群论问题集

Can a non-abelian one-relator group be the group of all automorphisms of some group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03765 | KOU-16.76

### Kourovka Notebook Problem 16.76

O / open | Kourovka 群论问题集

We call a group $G$ *strictly real* if each of its non-trivial elements is conjugate to its inverse by some involution in $G$. In which groups of Lie type over a field of characteristic 2 the maximal unipotent subgroups are strictly real?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03766 | KOU-16.77

### Kourovka Notebook Problem 16.77

O / open | Kourovka 群论问题集

It is known that in every Noetherian group the nilpotent radical coincides with the collection of all Engel elements (R. Baer, *Math. Ann.*, **133** (1957), 256--270; B. I. Plotkin, *Izv. Vyssˇ. Učebn. Zaved. Mat.*, **1958**, no. 1(2), 130--135 (Russian)). It would be nice to find a similar characterization of the solvable radical of a finite group. More precisely, let $u=u(x,y)$ be a sequence of words satisfying 15.75. We say that an element $g\in G$ is $u$-Engel if there exists $n=n(g)$ such that $u_n(x,g)=1$ for every element $x\in G$. Does there exist a sequence $u=u(x,y)$ such that the solvable radical of a finite group coincides with the set of all $u$-Engel elements?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03767 | KOU-16.78

### Kourovka Notebook Problem 16.78

O / open | Kourovka 群论问题集

Do there exist linear non-abelian simple groups without involutions?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03768 | KOU-16.79

### Kourovka Notebook Problem 16.79

S / solved | Kourovka 群论问题集

Is it true that in any finitely generated $AT$-group over a sequence of cyclic groups of uniformly bounded orders all Sylow subgroups are locally finite? For the definition of an *$AT $-group* see (A. V. Rozhkov, *Math. Notes*, **40** (1986), 827--836)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03769 | KOU-16.8

### Kourovka Notebook Problem 16.8

S / solved | Kourovka 群论问题集

The *width* $w(G')$ of the derived subgroup $G'$ of a finite non-abelian group $G$ is the smallest positive integer $m$ such that every element of $G'$ is a product of $\leq m$ commutators. Is it true that the maximum value of the ratio $w(G')/|G|$ is $1/6$ (attained at the symmetric group ${\mathbb{S}_3$)?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03770 | KOU-16.80

### Kourovka Notebook Problem 16.80

O / open | Kourovka 群论问题集

Suppose that a group $G$ is obtained from the free product of torsion-free groups $A_1, \ldots , A_n$ by imposing $m$ additional relations, where $m<n$. Is it true that the free product of some $n-m$ of the $A_i$ embeds into $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03771 | KOU-16.82

### Kourovka Notebook Problem 16.82

S / solved | Kourovka 群论问题集

Let $\mathcal{X}$ be a non-empty class of finite groups closed under taking homomorphic images, subgroups, and direct products. With every group $G\in\mathcal{X}$ we associate some set $\tau(G)$ of subgroups of $G$. We say that $\tau$ is a *subgroup functor on* $\mathcal{X}$ if:

1) $G\in\tau(G)$ for all $G\in\mathcal{X}$, and

2) for each epimorphism $\varphi :A\mapsto B$, where $A,B\in\mathcal{X}$, and for any $H\in\tau(A)$ and

$T\in\tau(B)$ we have $H^{\varphi}\in\tau(B)$ and $T^{\varphi^{-1}}\in\tau(A)$.

A subgroup functor $\tau$ is *closed* if for each group $G\in{\mathcal{X}}$ and for every subgroup $H\in {\mathcal{X}} \cap \tau (G)$ we have $\tau (H)\subseteq \tau (G)$. The set $F(\mathcal {X})$ consisting of all closed subgroup functors on $\mathcal{X}$ is a lattice (in which $\tau _1 \leq \tau _2$ if and only if $\tau_1 (G) \subseteq \tau_2 (G)$ for every group $G\in\mathcal{X}$). It is known that $F(\mathcal {X})$ is a chain if and only if $\mathcal{X}$ a class of $p$-groups for some prime $p$ (Theorem 1.5.17 in S. F. Kamornikov and M. V. Sel'kin, *Subgroups functors and classes of finite groups*, Belaruskaya Navuka, Minsk, 2001 (Russian)).

Is there a non-nilpotent class $\mathcal{X}$ such that the width of the lattice $F(\mathcal {X})$ is at most $|\pi(\mathcal{X})|$ where $\pi (\mathcal{X})$ is the set of all prime divisors of the orders of the groups in $\mathcal{X}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03772 | KOU-16.83

### Kourovka Notebook Problem 16.83

P / partially_solved | Kourovka 群论问题集

b) Let $E_n$ be a free locally nilpotent $n$-Engel group on countably many generators, and let $\pi(E_n)$ be the set of prime divisors of the orders of elements of the periodic part of $E_n$. It is known that $2, 3, 5\in \pi(E_4)$.

Is it true that $\pi(E_n)=\pi(E_{n+1}) $ for all sufficiently large $ n$?

[Archived solved part(s) of 16.83:] a) Let $E_n$ be a free locally nilpotent $n$-Engel group on countably many generators, and let $\pi(E_n)$ be the set of prime divisors of the orders of elements of the periodic part of $E_n$. It is known that $2, 3, 5\in \pi(E_4)$.

Does there exist $n$ for which $ 7\in \pi(E_n)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03773 | KOU-16.84

### Kourovka Notebook Problem 16.84

O / open | Kourovka 群论问题集

Can the braid group ${\mathfrak{B}}_n$, $n\geq 4$, act faithfully on a regular rooted tree by finite-state automorphisms? Such action is known for ${\mathfrak{B}}_3$.

See the definitions in (R. I. Grigorchuk, V. V. Nekrashevich, V. I. Sushchanskiĭ,

*Proc. Steklov Inst. Math.*, **2000**, no. 4 (231), 128--203).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03774 | KOU-16.85

### Kourovka Notebook Problem 16.85

S / solved | Kourovka 群论问题集

Suppose that groups $G, H$ act faithfully on a regular rooted tree by finite-state automorphisms. Can their free product $G*H$ act faithfully on a regular rooted tree by finite state automorphisms?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03775 | KOU-16.86

### Kourovka Notebook Problem 16.86

S / solved | Kourovka 群论问题集

Does the group of all finite-state automorphisms of a regular rooted tree possess an irreducible system of generators?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03776 | KOU-16.87

### Kourovka Notebook Problem 16.87

S / solved | Kourovka 群论问题集

Let ${\mathfrak{M}}$ be a variety of groups and let $G_r$ be a free $r$-generator group in $\mathfrak{M}$. A subset $S\subseteq G_r$ is called a *test set* if every endomorphism of $G_r$ identical on $S$ is an automorphism. The minimum of the cardinalities of test sets is called the test rank of $G_r$. Suppose that the test rank of $G_r$ is $r$ for every $r \geq 1$.

a) Is it true that ${\mathfrak{M}}$ is an abelian variety?

b) Suppose that ${\mathfrak{M}}$ is not a periodic variety. Is it true that ${\mathfrak{M}}$ is the variety of all abelian groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03777 | KOU-16.88

### Kourovka Notebook Problem 16.88

O / open | Kourovka 群论问题集

(G. M. Bergman). The *width* of a group $G$ with respect to a generating set $X$ means the supremum over all $g\in G$ of the least length of a group word in elements of $X$ expressing $g$. A group $G$ has *finite width* if the width of $G$ with respect to every generating set is finite. Does there exist a countably infinite group of finite width?

All known infinite groups of finite width (infinite permutation groups, infinite-dimensional general linear groups, and some other groups) are uncountable (G. M. Bergman, *Bull. London Math. Soc.*, **38** (2006), 429--440, and references therein).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03778 | KOU-16.89

### Kourovka Notebook Problem 16.89

O / open | Kourovka 群论问题集

(G. M. Bergman). Is it true that the automorphism group of an infinitely generated free group $F$ has finite width? The answer is affirmative if $F$ is countably generated.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03779 | KOU-16.9

### Kourovka Notebook Problem 16.9

O / open | Kourovka 群论问题集

An element $g$ of a free group $F_n$ on the free generators $x_1,\ldots,x_n$ is called a *palindrome* with respect to these generators if the reduced word representing $g$ is the same when read from left to right or from right to left. The *palindromic length* of an element $w \in F_n$ is the smallest number of palindromes in $F_n$ whose product is $w$. Is there an algorithm for finding the palindromic length of a given element of $F_n$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03780 | KOU-16.90

### Kourovka Notebook Problem 16.90

O / open | Kourovka 群论问题集

Is it true that the automorphism group $\text{Aut} F$ of an infinitely generated free group $F$ is

a) the normal closure of a single element?

b) the normal closure of some involution in $\text{Aut} F$?

For a free group $F_n$ of finite rank $n$ \ M. Bridson and K. Vogtmann have recently shown that $\text{Aut} F_n$ is the normal closure of some involution, which permutes some basis of $F_n$. It is also known that the automorphism groups of infinitely generated free nilpotent groups have such involutions.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03781 | KOU-16.91

### Kourovka Notebook Problem 16.91

O / open | Kourovka 群论问题集

Let $F$ be an infinitely generated free group. Is there an $IA$-automorphism of $F$ whose normal closure in $\text{Aut} F$ is the group of all $IA$-automorphisms of $F?$

问题集入口：https://kourovkanotebookorg.wordpress.com/

03782 | KOU-16.92

### Kourovka Notebook Problem 16.92

O / open | Kourovka 群论问题集

Let $F$ be an infinitely generated free group. Is $\text{Aut} F$ equal to its derived subgroup? This is true if $F$ is countably generated (R. Bryant, V. A. Roman'kov, *J. Algebra*, **209** (1998), 713--723).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03783 | KOU-16.93

### Kourovka Notebook Problem 16.93

O / open | Kourovka 群论问题集

Let $F_n$ be a free group of finite rank $n \geq 2$. Is the group $\text{Inn} F_n$ of inner automorphisms of $F_n$ a first-order definable subgroup of $\text{Aut} F_n$? It is known that the set of inner automorphisms induced by powers of primitive elements is definable in $\text{Aut} F_n$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03784 | KOU-16.94

### Kourovka Notebook Problem 16.94

O / open | Kourovka 群论问题集

If $G=[G,G]$, then the *commutator width* of the group $G$ is its width relative to the set of commutators. Let $V$ be an infinite-dimensional vector space over a division ring. It is known that the commutator width of $GL(V)$ is finite. Is it true that the commutator width of $GL(V)$ is one?

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03785 | KOU-16.95

### Kourovka Notebook Problem 16.95

O / open | Kourovka 群论问题集

*Conjecture:* If $F$ is a field and $A$ is in $GL(n,F)$, then there is a permutation matrix $P$ such that $AP$ is cyclic, that is, the minimal polynomial of $AP$ is also its characteristic polynomial.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03786 | KOU-16.96

### Kourovka Notebook Problem 16.96

O / open | Kourovka 群论问题集

Let $G$ be a locally finite $n$-Engel $p $-group where $p$ a prime greater than $n$. Is $G$ then a Fitting group? (Examples of N. Gupta and F. Levin show that the condition $p>n$ is necessary in general.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03787 | KOU-16.97

### Kourovka Notebook Problem 16.97

O / open | Kourovka 群论问题集

Let $G$ be a torsion-free group with all subgroups subnormal of defect at most $n$. Must $G$ then be nilpotent of class at most $n$? (This is known to be true for $n < 5$).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03788 | KOU-16.98

### Kourovka Notebook Problem 16.98

O / open | Kourovka 群论问题集

Suppose that $G$ is a solvable finite group and $A$ is a group of automorphisms of $G$ of relatively prime order. Is there a bound for the Fitting height $h(G)$ of $G$ in terms of $A$ and $h(C_G(A))$, or even in terms of the length $l(A)$ of the longest chain of nested subgroups of $A$ and $h(C_G(A))$?

When $A$ is solvable, it is proved in (A. Turull, *J. Algebra*, **86** (1984), 555--566) that $h(G)\leq h(C_G(A)) + 2 l(A)$ and this bound is best possible for $h(C_G(A))> 0$. For $A$ non-solvable some results are in (H. Kurzweil, *Manuscripta Math.*, **41** (1983), 233--305).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03789 | KOU-16.99

### Kourovka Notebook Problem 16.99

O / open | Kourovka 群论问题集

Suppose that $G$ is a finite solvable group, $A \leq \mathrm{Aut} G$, \ $C_G(A)=1$, the orders of $G$ and $A$ are coprime, and let $l(A)$ be the length of the longest chain of nested subgroups of $A$. Is the Fitting height of $G$ bounded above by $l(A)$?

For $A$ solvable the question coincides with 5.30. It is proved that for any finite group $A$, first, there exist $G$ with $h(G)=l(A)$ and, second, there is a finite set of primes $\pi$ (depending on $A$) such that if $|G|$ is coprime to each prime in $\pi$, then $h(G)\leq l(A)$. See (A. Turull, *Math. Z.*, **187** (1984), 491--503).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03790 | KOU-17.1

### Kourovka Notebook Problem 17.1

O / open | Kourovka 群论问题集

(I. M. Isaacs). Does there exist a finite group partitioned by subgroups of equal order not all of which are abelian? (Cf. 15.26.) It is known that such a group must be of prime exponent (I. M. Isaacs, *Pacific J. Math.*, **49** (1973), 109--116).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03791 | KOU-17.10

### Kourovka Notebook Problem 17.10

O / open | Kourovka 群论问题集

Is there a group containing a right $4$-Engel element which does not belong to the Hirsch--Plotkin radical?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03792 | KOU-17.100

### Kourovka Notebook Problem 17.100

O / open | Kourovka 群论问题集

*Conjecture*: A finite group is not simple if it has an irreducible complex character of odd degree vanishing on a class of odd length.

If true, this implies the solvability of groups of odd order, so a proof independent of CFSG is of special interest.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03793 | KOU-17.101

### Kourovka Notebook Problem 17.101

O / open | Kourovka 群论问题集

According to P. Hall, a group $G$ is said to be *homogeneous* if every isomorphism of its finitely generated subgroups is induced by an automorphism of $G$. It is known (Higman--Neumann--Neumann) that every group is embeddable into a homogeneous one. Is the same true for the category of representations of groups? A representation $(V,G)$ is finitely generated if $G$ is a finitely generated group, and $V$ a finitely generated module over the group algebra of $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03794 | KOU-17.102

### Kourovka Notebook Problem 17.102

S / solved | Kourovka 群论问题集

We say that two subsets $A, B$ of an infinite group $G$ are *separated* if there exists an infinite subset $X$ of $G$ such that $1\in X$, $X=X^{-1}$, and $XAX\cap B=\varnothing$. Is it true that any two disjoint subsets $A, B$ of an infinite group $G$ satisfying $|A|<|G|$, $|B|<|G|$ are separated? This is so if $A,B$ are finite, or $G$ is Abelian.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03795 | KOU-17.103

### Kourovka Notebook Problem 17.103

O / open | Kourovka 群论问题集

Does there exist a continuum of sets $\pi$ of primes for which every finite group possessing a Hall $\pi$-subgroup is a $D_\pi$-group?

*Comment of 2025*: It has been proved that, for any integer $x$, if $\pi$ is the set of all prime numbers $p>x$, then any finite group containing a Hall $\pi$-subgroup is a $D_\pi$-group (K. A. Ilenko, N. V. Maslova, *Siberian Math. J.*, **62**, no. 1 (2021), 44--51).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03796 | KOU-17.104

### Kourovka Notebook Problem 17.104

O / open | Kourovka 群论问题集

Let $\Gamma$ be a finite non-oriented graph on the set of vertices $\{x_1,\dots,x_n\}$ and let $S_\Gamma=\langle x_1,\dots,x_n\mid x_ix_j=x_jx_i \Leftrightarrow (x_i,x_j) \in \Gamma;\; {\mathfrak{A}}^2\rangle$ be a presentation of a partially commutative metabelian group $S_\Gamma$ in the variety of all metabelian groups. Is the universal theory of the group $S_\Gamma$ decidable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03797 | KOU-17.105

## Kourovka Notebook Problem 17.105

O / open | Kourovka 群论问题集

An equation over a pro-$p$-group $G$ is an expression $v(x)=1$, where $v(x)$ is an element of the free pro-$p$-product of $G$ and a free pro-$p$-group with basis $\{ x_1, \ldots , x_n \}$; solutions are sought in the affine space $G^n$. Is it true that a free pro-$p$-group is equationally Noetherian, that is, for any $n$ every system of equations in $x_1, \ldots , x_n$ over this group is equivalent to some finite subsystem of it?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03798 | KOU-17.107

## Kourovka Notebook Problem 17.107

O / open | Kourovka 群论问题集

Does $G=\mathrm{SL}_2(\mathbb{C})$ contain a 2-generated free subgroup that is conjugate in $G$ to a proper subgroup of itself?

A 6-generated free subgroup with this property has been found (D. Calegari, N. M. Dunfield, *Proc. Amer. Math. Soc.*, **134**, no. 11 (2006), 3131--3136).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03799 | KOU-17.108

## Kourovka Notebook Problem 17.108

S / solved | Kourovka 群论问题集

Is the group $\langle a,b,t \mid a^t=ab,\; b^t=ba\rangle$ linear?

If not, this would be an easy example of a non-linear hyperbolic group.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03800 | KOU-17.109

## Kourovka Notebook Problem 17.109

O / open | Kourovka 群论问题集

A non-trivial group word $w$ is *uniformly elliptic* in a class $\mathcal{C}$ if there is a function $f:\mathbb{N}\rightarrow\mathbb{N}$ such that the width of $w$ in every $d$-generator $\mathcal{C}$-group $G$ is bounded by $f(d)$ (i. e. every element of the verbal subgroup $w(G)$ is equal to a product of $f(d)$ values of $w$ or their inverses). If $\mathcal{C}$ is a class of finite groups, this is equivalent to saying that in every finitely generated pro-$\mathcal{C}$ group $G$ the verbal subgroup $w(G)$ is closed. A. Jaikin-Zapirain (*Revista Mat. Iberoamericana*, **24** (2008), 617--630) proved that $w$ is uniformly elliptic in finite $p$-groups if and only if $w\notin F^{\prime\prime}(F^{\prime})^{p}$, where $F$ is the free group on the variables of $w$. Is it true that $w$ is uniformly elliptic in $\mathcal{C}$ if and only if $w\notin F^{\prime\prime}(F^{\prime})^{p}$ for every prime $p$ in the case where $\mathcal{C}$ is the class of all

a) finite soluble groups?

b) finite groups?

See also Ch. 4 of (D. Segal, *Words: notes on verbal width in groups*, LMS Lect. Note Series, **361**, Cambridge Univ. Press, 2009.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03801 | KOU-17.11

## Kourovka Notebook Problem 17.11

O / open | Kourovka 群论问题集

Is there a group containing a left $3$-Engel element which does not belong to the Hirsch--Plotkin radical?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03802 | KOU-17.110

## Kourovka Notebook Problem 17.110

O / open | Kourovka 群论问题集

a) Is it true that for each word $w$, there is a function $h:\mathbb{N}\times \mathbb{N}\rightarrow\mathbb{N}$ such that the width of $w$ in every finite $p$-group of Pr\"{u}fer rank $r$ is bounded by $h(p,r)$?

b) If so, can $h(p,r)$ be made independent of $p$?

An affirmative answer to a) would imply A. Jaikin-Zapirain's result (*ibid.*) that every word has finite width in each $p$-adic analytic pro-$p$ group, since a pro-$p$ group is $p$-adic analytic if and only if the Pr\"{u}fer ranks of all its finite quotients are uniformly bounded.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03803 | KOU-17.111

### Kourovka Notebook Problem 17.111

S / solved | Kourovka 群论问题集

Let $G$ be a finite group, and $p$ a prime divisor of $|G|$. Suppose that every maximal subgroup of a Sylow $p$-subgroup of $G$ has a $p$-soluble supplement in $G$. Must $G$ be $p$-soluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03804 | KOU-17.112

### Kourovka Notebook Problem 17.112

O / open | Kourovka 群论问题集

A subgroup $A$ of a group $G$ is said to be *$G$-permutable in $G$* if for every subgroup $B$ of $G$ there exists an element $x\in G $ such that $AB^x=B^xA$. A subgroup $A$ is said to be *hereditarily $G$-permutable in $G$* if $A$ is $E$-permutable in every subgroup $E$ of $G$ containing $A$. Which finite non-abelian simple groups $G$ possess

a) a non-trivial $G$-permutable subgroup?

b) a non-trivial hereditarily $G$-permutable subgroup?

*Comments of 2025*: Among sporadic groups, only $J_1$ has a proper $G$-permutable subgroup (A. A. Galt, V. N. Tyutyanov, *Siberian Math. J.*, **63**, no. 4 (2022), 691--698). Sporadic, alternating, and exceptional groups of Lie type have no proper hereditarily $G$-permutable subgroups (A. A. Galt, V. N. Tyutyanov, *Siberian Math. J.*, **63**, no. 4 (2022), 691--698; \ A. F. Vasilyev, V. N. Tyutyanov, *Izv. Gomel. Gos. Univ. F. Skoriny*, **2012**, no. 5(74) (2012), 148--150 (Russian)); \ A. A. Galt, V. N. Tyutyanov, *Siberian Math. J.*, **64**, no. 5 (2023), 1110--1116).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03805 | KOU-17.113

### Kourovka Notebook Problem 17.113

O / open | Kourovka 群论问题集

Do there exist for $p>3$ \ 2-generator finite $p$-groups with deficiency zero (see 8.12) of arbitrarily high nilpotency class?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03806 | KOU-17.114

### Kourovka Notebook Problem 17.114

O / open | Kourovka 群论问题集

Does every generalized free product (with amalgamation) of two non-trivial groups have maximal subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03807 | KOU-17.115

### Kourovka Notebook Problem 17.115

O / open | Kourovka 群论问题集

Can a locally free non-abelian group have non-trivial Frattini subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03808 | KOU-17.116

### Kourovka Notebook Problem 17.116

S / solved | Kourovka 群论问题集

Let $n(G)$ be the maximum of positive integers $n$ such that the $n$-th direct power of a finite simple group $G$ is 2-generated. Is it true that $n(G)\geq \sqrt{|G|}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03809 | KOU-17.117

### Kourovka Notebook Problem 17.117

O / open | Kourovka 群论问题集

(Well-known problem). If groups $A$ and $B$ have decidable elementary theories $Th(A)$ and $Th(B)$, must $Th(A*B)$ be decidable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03810 | KOU-17.118

### Kourovka Notebook Problem 17.118

O / open | Kourovka 群论问题集

Suppose that a finite $p$-group $G$ has a subgroup of exponent $p$ and of index $p$. Must $G$ also have a characteristic subgroup of exponent $p$ and of index bounded in terms of $p$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03811 | KOU-17.119

### Kourovka Notebook Problem 17.119

O / open | Kourovka 群论问题集

Suppose that a finite soluble group $G$ admits a soluble group of automorphisms $A$ of coprime order such that $C_G(A)$ has rank $r$. Let $|A|$ be the product of $l$ not necessarily distinct primes. Is there a linear function $f$ such that $G/F_{f(l)}(G)$ has $(|A|,r)$-bounded rank, where $F_{f(l)}(G)$ is the $f(l)$-th Fitting subgroup?

An exponential function $f$ with this property was found in (E. Khukhro, V. Mazurov, *Groups St. Andrews 2005*, vol. II, Cambridge Univ. Press, 2007, 564--585). It is also known that $|G/F_{2l+1}(G)|$ is bounded in terms of $|C_G(A)|$ and $|A|$ (Hartley--Isaacs).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03812 | KOU-17.12

### Kourovka Notebook Problem 17.12

S / solved | Kourovka 群论问题集

Are there functions $e,c:\mathbb{N}\rightarrow \mathbb{N}$ such that if in a nilpotent group $G$ a normal subgroup $H$ consists of right $n$-Engel elements of $G$, then $H^{e(n)}\leq \zeta_{c(n)}(G)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03813 | KOU-17.120

### Kourovka Notebook Problem 17.120

O / open | Kourovka 群论问题集

Is a residually finite group all of whose subgroups of infinite index are finite necessarily cyclic-by-finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03814 | KOU-17.121

### Kourovka Notebook Problem 17.121

O / open | Kourovka 群论问题集

Let $ G $ be a group whose set of elements is the real numbers and which is ``nicely definable'' (see below). Does $ G $ being $ { \aleph_\omega} $-free imply it being free if ``nicely definable'' means

a) being $ F_ \sigma $?

b) being Borel?

c) being analytic?

d) being projective $ {\mathbb{L} [ {\mathbb{R} ] $?

See https://arxiv.org/pdf/math/0212250.pdf for justification of the restrictions.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03815 | KOU-17.122

### Kourovka Notebook Problem 17.122

O / open | Kourovka 群论问题集

The same questions as 17.121 for an abelian group $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03816 | KOU-17.123

### Kourovka Notebook Problem 17.123

O / open | Kourovka 群论问题集

Do there exist finite groups $G_1, G_2$ such that $\pi _e(G_1)=\pi _e(G_2)$, \ $h(\pi _e(G_1))<\infty$, and each non-abelian composition factors of each of the groups $G_1, G_2$ is not isomorphic to a section of the other?

For definitions see Archive, 13.63.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03817 | KOU-17.124

### Kourovka Notebook Problem 17.124

O / open | Kourovka 群论问题集

Is the set of finitely presented metabelian groups recursively enumerable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03818 | KOU-17.125

### Kourovka Notebook Problem 17.125

O / open | Kourovka 群论问题集

Does every finite group $G$ contain a pair of conjugate elements $a,b$ such that $\pi(G)=\pi(\langle a,b\rangle)$? This is true for soluble groups.

*Comment of 2013:* it was proved in (A. Lucchini, M. Morigi, P. Shumyatsky, *Forum Math.*, **24** (2012), 875--887) that every finite group $G$ contains a 2-generator subgroup $H$ such that $\pi(G)=\pi(H)$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03819 | KOU-17.126

### Kourovka Notebook Problem 17.126

O / open | Kourovka 群论问题集

Suppose that $G$ is a residually finite group satisfying the identity $[x,y]^n=1$. Must $[G,G]$ be locally finite?

An equivalent question: let $G$ be a finite soluble group satisfying the identity $[x,y]^n=1$; is the Fitting height of $G$ bounded in terms of $n$? Cf. Archive, 13.34

问题集入口：https://kourovkanotebookorg.wordpress.com/

03820 | KOU-17.127

### Kourovka Notebook Problem 17.127

O / open | Kourovka 群论问题集

Suppose that a finite soluble group $G$ of derived length $d$ admits an elementary abelian $p$-group of automorphisms $A$ of order $p^n$ such that $C_G(A)=1$. Must $G$ have a normal series of $n$-bounded length with nilpotent factors of $(p,n,d)$-bounded nilpotency class?

This is true for $p=2$. An affirmative answer would follow from an affirmative answer to 11.125.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03821 | KOU-17.128

### Kourovka Notebook Problem 17.128

O / open | Kourovka 群论问题集

Let $T$ be a finite $p$-group admitting an elementary abelian group of automorphisms $A$ of order $p^{2}$ such that in the semidirect product $P=TA$ every element of $P\setminus T$ has order $p$. Does it follow that $T$ is of exponent $p$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03822 | KOU-17.13

### Kourovka Notebook Problem 17.13

O / open | Kourovka 群论问题集

Let $G$ be a totally imprimitive $p$-group of finitary permutations on an infinite set. Suppose that the support of any cycle in the cyclic decomposition of every element of $G$ is a block for $G$. Does $G$ necessarily contain a minimal non-$FC$-subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03823 | KOU-17.14

## Kourovka Notebook Problem 17.14

S / solved | Kourovka 群论问题集

Can the braid group $B_n$ for $n \geq 4$ be embedded into the automorphism group $Aut (F_{n-2})$ of a free group $F_{n-2}$ of rank $n-2$?

Artin's theorem implies that $B_n$ can be embedded into $Aut (F_n)$, and an embedding of $B_n$, $n \geq 3$, into $Aut (F_{n-1})$ was constructed in (B. Perron, J. P. Vannier, *Math. Ann.*, **306** (1996), 231--245).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03824 | KOU-17.15

## Kourovka Notebook Problem 17.15

O / open | Kourovka 群论问题集

Construct an algorithm which, for a given polynomial $f \in {\mathbb{Z}}[x_1, x_2, \ldots, x_n]$, finds (explicitly, in terms of generators) the maximal subgroup $G_f$ of the group $Aut ({\mathbb{Z}}[x_1, x_2, \ldots, x_n])$ that leaves $f$ fixed. This question is related to description of the solution set of the Diophantine equation $f = 0$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03825 | KOU-17.16

## Kourovka Notebook Problem 17.16

O / open | Kourovka 群论问题集

Let $A$ be an Artin group of finite type, that is, the corresponding Coxeter group is finite. It is known that the group $A$ is linear (S. Bigelow, *J. Amer. Math. Soc.*, **14** (2001), 471--486; \ D. Krammer, *Ann. Math.*, **155** (2002), 131--156; \ F. Digne, *J. Algebra*, **268** (2003), 39--57; \ A. M. Cohen, D. B. Wales, *Israel J. Math.*, ** 131** (2002), 101--123). Is it true that the automorphism group $Aut (A)$ is also linear?

An affirmative answer for $A$ being a braid group $B_n$, $n \geq 2$, was obtained in (V. G. Bardakov, O. V. Bryukhanov, *Vestnik Novosibirsk. Univ. Ser. Mat. Mekh. Inform.*, **7**, no. 3 (2007), 45--58 (Russian)).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03826 | KOU-17.17

## Kourovka Notebook Problem 17.17

S / solved | Kourovka 群论问题集

If a finitely generated group $G$ has $n<\infty$ maximal subgroups, must $G$ be finite? In particular, what if $n=3$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03827 | KOU-17.18

## Kourovka Notebook Problem 17.18

O / open | Kourovka 群论问题集

Let $\mathbf{A}$ be the class of compact groups $A$ with the property that whenever two compact groups $B$ and $C$ contain $A$, they can be embedded in a common compact group $D$ by embeddings agreeing on $A$. I showed (*Manuscr. Math.*, **58** (1987) 253--281) that all members of $\mathbf{A}$ are (not necessarily connected) finite-dimensional compact Lie groups satisfying a strong ``local simplicity'' property, and that all finite groups do belong to $\mathbf{A}$.

a) Is it true that $\mathbb{R}/\mathbb{Z}\in\mathbf{A}$?

b) Do any *nonabelian connected* compact Lie groups belong to $\mathbf{A}$?

c) If $A$ belongs to $\mathbf{A}$, must the connected component of the identity in $A$ belong to $\mathbf{A}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03828 | KOU-17.19

## Kourovka Notebook Problem 17.19

S / solved | Kourovka 群论问题集

If $F$ is a free group of finite rank, $R$ a retract of $F$, and $H$ a subgroup of $F$ of finite rank, must $H\cap R$ be a retract of $H$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03829 | KOU-17.2

## Kourovka Notebook Problem 17.2

S / solved | Kourovka 群论问题集

(P. Schmid). Does there exist a finite non-abelian $p$-group $G$ such that $H^1({G}/{\Phi(G)}, Z(\Phi(G) ) )=0$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03830 | KOU-17.20

## Kourovka Notebook Problem 17.20

S / solved | Kourovka 群论问题集

If $M$ is a real manifold with nonempty boundary, and $G$ the group of self-homeomorphisms of $M$ which fix the boundary pointwise, is $G$ right-orderable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03831 | KOU-17.21

## Kourovka Notebook Problem 17.21

S / solved | Kourovka 群论问题集

a) If $A, B, C$ are torsion-free abelian groups with $A \cong A\oplus B\oplus C$, must $A\cong A\oplus B$?

b) What if, furthermore, $B\cong C$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03832 | KOU-17.22

### Kourovka Notebook Problem 17.22

O / open | Kourovka 群论问题集

Suppose $A$ is a group which belongs to a variety $\mathbf V$ of groups, and which is embeddable in the full symmetric group $S$ on an infinite set. Must the coproduct in $\mathbf V$ of two copies of $A$ also be embeddable in $S$? (N. G. de Bruijn proved that this is true if $\mathbf V$ is the variety of all groups.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03833 | KOU-17.23

### Kourovka Notebook Problem 17.23

O / open | Kourovka 群论问题集

Suppose the full symmetric group $S$ on a countably infinite set is generated by the union of two subgroups $G$ and $H$. Must $S$ be finitely generated over one of these subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03834 | KOU-17.24

### Kourovka Notebook Problem 17.24

S / solved | Kourovka 群论问题集

(A. Blass, J. Irwin, G. Schlitt). Does $\mathbb{Z}^\omega$ have a subgroup whose dual is free of uncountable rank?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03835 | KOU-17.25

### Kourovka Notebook Problem 17.25

O / open | Kourovka 群论问题集

(S. P. Farbman). Let $X$ be the set of complex numbers $\alpha$ such that the group generated by the two $2\times 2$ matrices $I+\alpha e_{12}$ and $I+ e_{21}$ is not free on those generators.

a) Does $X$ contain all rational numbers in the interval $(-4,4)$?

b) Does $X$ contain any rational number in the interval $[27/7,4)$?

Cf. 4.41 in Archive, 15.83, and (S. P. Farbman, *Publ. Mat.*, **39** (1995) 379--391).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03836 | KOU-17.26

### Kourovka Notebook Problem 17.26

O / open | Kourovka 群论问题集

Are the classes of right-orderable and right-ordered groups closed under taking solutions of equations $w(a_1,\dots,a_k,x_1,\dots,x_n)=1$? (Here the closures are under group embeddings and order-preserving embeddings, respectively.) This is true for equations with a single constant, when $k=1$ (*J. Group Theory*, **11** (2008), 623--633).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03837 | KOU-17.27

### Kourovka Notebook Problem 17.27

O / open | Kourovka 群论问题集

Can the free product of two ordered groups with order-isomorphic amalgamated subgroups be lattice orderable but not orderable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03838 | KOU-17.28

### Kourovka Notebook Problem 17.28

S / solved | Kourovka 群论问题集

Is there a soluble right-orderable group with insoluble word problem?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03839 | KOU-17.29

### Kourovka Notebook Problem 17.29

S / solved | Kourovka 群论问题集

Construct a finitely presented orderable group with insoluble word problem.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03840 | KOU-17.3

### Kourovka Notebook Problem 17.3

S / solved | Kourovka 群论问题集

Let $G$ be a group in which every $4$-element subset contains two elements generating a nilpotent subgroup. Is it true that every $2$-generated subgroup of $G$ is nilpotent?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03841 | KOU-17.30

### Kourovka Notebook Problem 17.30

O / open | Kourovka 群论问题集

Is there a constant $l$ such that every finitely presented soluble group has a subgroup of finite index of nilpotent length at most $l$? Cf. 16.35 in Archive.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03842 | KOU-17.31

### Kourovka Notebook Problem 17.31

O / open | Kourovka 群论问题集

Can a soluble right-orderable group have finite quotient by the derived subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03843 | KOU-17.32

### Kourovka Notebook Problem 17.32

S / solved | Kourovka 群论问题集

Is the following analogue of the Cayley--Hamilton theorem true for the free group $F_n$ of rank $n$: If $w\in F_n$ and $\varphi\in \mathrm{Aut }F_n$ are such that $\langle w,\varphi (w),\dots,\varphi^{n}(w)\rangle= F_n$, then $\langle w,\varphi (w),\dots,\varphi^{n-1}(w)\rangle=F_n$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03844 | KOU-17.33

### Kourovka Notebook Problem 17.33

O / open | Kourovka 群论问题集

Can the quasivariety generated by the group $\langle a,b\mid a^{-1}ba=b^{-1}\rangle$ be defined by a set of quasi-identities in a finite set of variables? The answer is known for all other groups with one defining relation.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03845 | KOU-17.34

### Kourovka Notebook Problem 17.34

O / open | Kourovka 群论问题集

Let $N_c$ be the quasivariety of nilpotent torsion-free groups of class at most $c$. Is it true that the dominion in $N_c$ (see the definition in 16.20) of a divisible subgroup in every group in $N_c$ is equal to this subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03846 | KOU-17.35

### Kourovka Notebook Problem 17.35

S / solved | Kourovka 群论问题集

Suppose we have a finite two-colourable triangulation of the sphere, with triangles each coloured either black or white so that no pair of triangles with the same colour share an edge. On each vertex we write a generator, and we assume the generators commute. We use these generators to generate an abelian group $G_W$ with relations stating that the generators around each white triangle add to zero. Doing the same thing with the black triangles, we generate a group $G_B$.

*Conjecture:* $G_W$ is isomorphic to $G_B$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03847 | KOU-17.36

### Kourovka Notebook Problem 17.36

S / solved | Kourovka 群论问题集

Two groups are called *isospectral* if they have the same set of element orders. Find all finite non-abelian simple groups $G$ for which there is a finite group $H$ isospectral to $G$ and containing a proper normal subgroup isomorphic to $G$. For every simple group $G$ determine all groups $H$ satisfying this condition.

It is easy to show that a group $H$ must satisfy the condition $G<H\leqslant\operatorname{Aut}G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03848 | KOU-17.37

### Kourovka Notebook Problem 17.37

S / solved | Kourovka 群论问题集

Is there an integer $n$ such that for all $m > n$ the alternating group $A_{m}$ has no non-trivial $A_{m}$-permutable subgroups? (See the definition in 17.112.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03849 | KOU-17.38

### Kourovka Notebook Problem 17.38

O / open | Kourovka 群论问题集

A formation $\mathfrak{F}$ is called {\it radical in a class $\mathfrak{H}$} if $\mathfrak{F} \subseteq \mathfrak{H}$ and in every $\mathfrak{H}$-group the product of any two normal $\mathfrak{F}$-subgroups belongs to $\mathfrak{F}$. Let $\mathfrak{M}$ be the class of all saturated hereditary formations of finite groups such that the formation of all finite supersoluble groups is radical in every element of $\mathfrak{M}$. Is it true that $\mathfrak{M}$ has the largest (by inclusion) element?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03850 | KOU-17.39

### Kourovka Notebook Problem 17.39

O / open | Kourovka 群论问题集

Is there a positive integer $n$ such that the hypercenter of any finite soluble group coincides with the intersection of $n$ system normalizers of that group? What is the least number with this property?

*Comment of 2017*: This is proved for groups with trivial Frattini subgroup (A. Ballester-Bolinches, J. Cossey, S. F. Kamornikov, H. Meng, *J. Algebra*, **499** (2018), 438--449).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03851 | KOU-17.4

### Kourovka Notebook Problem 17.4

O / open | Kourovka 群论问题集

Let $x$ be a right $4$-Engel element of a group $G$.

a) Is it true that the normal closure $\langle x \rangle ^G$ of $x$ in $G$ is nilpotent if $G$ is locally nilpotent?

b) If the answer to a) is affirmative, is there a bound on the nilpotency class of $\langle x \rangle ^G$?

c) Is it true that $\langle x \rangle ^G$ is always nilpotent?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03852 | KOU-17.40

### Kourovka Notebook Problem 17.40

S / solved | Kourovka 群论问题集

Let $N$ be a nilpotent subgroup of a finite group $G$. Do there always exist elements $x,y\in G$ such that $N\cap N^x\cap N^y\leq F(G)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03853 | KOU-17.41

## Kourovka Notebook Problem 17.41

O / open | Kourovka 群论问题集

Let $S$ be a solvable subgroup of a finite group $G$ that has no nontrivial solvable normal subgroups.

(a) (L. Babai, A. J. Goodman, L. Pyber). Do there always exist seven conjugates of $S$ whose intersection is trivial?

(b) Do there always exist five conjugates of $S$ whose intersection is trivial?

*Editors' comments*: Reduction of part (b) to almost simple group $G$ is obtained in (E. P. Vdovin, *J. Algebra Appl.*, **11**, no. 1 (2012), 1250015 (14 pages)). An affirmative answer to (b) has been obtained for almost simple groups with socle isomorphic to an alternating group (A. A. Baikalov, *Algebra Logic*, **56** (2017), 87--97), to a sporadic simple group (T. C. Burness, *Israel J. Math.*, **254** (2023), 313--340), to $PSL(n,q)$ (A. A. Baykalov, *J. Group Theory*, **28** (2025), 1003--1077), \ to $PSU(n,q)$ or $PSp(n,q)'$ (A. A. Baykalov, *Int. J. Algebra Comput.*, **35**, no. 06 (2025), 823--908). Part (b) also has an affirmative answer if $S$ is a maximal subgroup of $G$ (T. C. Burness, *Algebra Number Theory*, **15**, no. 7 (2021), 1755--1807).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03854 | KOU-17.42

## Kourovka Notebook Problem 17.42

S / solved | Kourovka 群论问题集

Let $\overline{G}$ be a simple algebraic group of adjoint type over the algebraic closure $\overline{F}_p$ of a finite field $F_p$ of prime order $p$, and $\sigma$ a Frobenius map (that is, a surjective homomorphism such that $\overline{G}_\sigma=C_{\overline{G}}(\sigma)$ is finite). Then $G=O^{p'}(\overline{G}_\sigma)$ is a finite group of Lie type. For a maximal $\sigma$-stable torus $\overline{T}$ of $\overline{G}$, let $N=N_{\overline{G}}(\overline{T})\cap G$. Assume also that $G$ is simple and $G\not\cong \mathrm{SL}_3(2)$. Does there always exist $x\in G$ such that $N\cap N^x$ is a $p$-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03855 | KOU-17.43

## Kourovka Notebook Problem 17.43

O / open | Kourovka 群论问题集

Let $\pi$ be a set of primes. Find all finite simple $D_\pi$-groups (see Archive, 3.62) in which

(a) every subgroup is a $D_\pi$-group (H. Wielandt);

(b) every subgroup possessing a Hall $\pi$-subgroup is a $D_\pi$-group.

All finite simple $D_\pi$-groups are known (D. O. Revin, *Algebra Logic*, **47**, no. 3 (2008), 210--227). *Comment of 2013:* Alternating and sporadic groups for both parts are found in (N. Ch. Manzaeva, *Siberian Electron. Math. Rep.*, **9** (2012), 294--305 (Russian)). *Comment of 2025*: Simple groups of Lie type of rank 1 satisfying condition (a) are known (D. O. Revin, V. D. Shepelev, *Siberian Math. J.*, **65**, no. 5 (2024), 1187--1194, V. D. Shepelev, *Siberian Math. J.*, **66**, no. 4 (2025), 1049--1062).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03856 | KOU-17.44

## Kourovka Notebook Problem 17.44

S / solved | Kourovka 群论问题集

Let $\pi$ be a set of primes. A finite group is called a *$C_\pi$-group* if it possesses exactly one class of conjugate Hall $\pi$-subgroups. A finite group is called a {\it $D_{\pi}$-group} if any two of its maximal $\pi$-subgroups are conjugate.

a) In a $C_\pi$-group, is an overgroup of a Hall $\pi$-subgroup always a $C_\pi$-group?

An affirmative answer in the case $2\not\in\pi$ follows mod CFSG from (F. Gross, *Bull. London Math. Soc.*, **19**, no. 4 (1987), 311--319).

b) In a $D_\pi$-group, is an overgroup of a Hall $\pi$-subgroup always a $D_\pi$-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03857 | KOU-17.45

### Kourovka Notebook Problem 17.45

P / partially_solved | Kourovka 群论问题集

c) A subgroup $H$ of a group $G$ is called *pronormal* if $H$ and $H^g$ are conjugate in $\langle H,H^g\rangle$ for every $g\in G$. We say that $H$ is *strongly pronormal* if $L^g$ is conjugate to a subgroup of $H$ in $\langle H,L^g\rangle$ for every $L\leq H$ and $g\in G$.

In a finite group, is a Hall subgroup with a Sylow tower always strongly pronormal?

Notice that there exist finite (non-simple) groups with a non-pronormal Hall subgroup. Hall subgroups with a Sylow tower are known to be pronormal.

[Archived solved part(s) of 17.45:] A subgroup $H$ of a group $G$ is called *pronormal* if $H$ and $H^g$ are conjugate in $\langle H,H^g\rangle$ for every $g\in G$. We say that $H$ is *strongly pronormal* if $L^g$ is conjugate to a subgroup of $H$ in $\langle H,L^g\rangle$ for every $L\leq H$ and $g\in G$.

a) In a finite simple group, are Hall subgroups always pronormal?

An affirmative answer is known modulo CFSG for Hall subgroups of odd order (F. Gross, *Bull. London Math. Soc.*, **19**, no. 4 (1987), 311--319).

b) In a finite simple group, are Hall subgroups always strongly pronormal?

Notice that there exist finite (non-simple) groups with a non-pronormal Hall subgroup. Hall subgroups with a Sylow tower are known to be pronormal.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03858 | KOU-17.46

### Kourovka Notebook Problem 17.46

S / solved | Kourovka 群论问题集

Let $G$ be a finite $p$-group in which every 2-generator subgroup has cyclic derived subgroup. Is the derived length of $G$ bounded?

If $p\ne 2$, then $G^{(2)}=1$ (J. Alperin), but for $p=2$ there are examples with $G^{(2)}$ cyclic and elementary abelian of arbitrary order.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03859 | KOU-17.47

### Kourovka Notebook Problem 17.47

S / solved | Kourovka 群论问题集

Let $G$ be a nilpotent group in which every coset $x[G,G]$ for $x\not\in [G,G]$ coincides with the conjugacy class $x^G$. Is there a bound for the nilpotency class of $G$?

If $G$ is finite, then its class is at most 3 (R. Dark, C. Scoppola, *J. Algebra*, **181**, no. 3 (1996), 787--802).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03860 | KOU-17.48

### Kourovka Notebook Problem 17.48

O / open | Kourovka 群论问题集

Does the free product of two groups with stable first-order theory also have stable first-order theory?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03861 | KOU-17.49

### Kourovka Notebook Problem 17.49

O / open | Kourovka 群论问题集

Is it consistent with ZFC that every non-discrete topological group contains a nonempty nowhere dense subset without isolated points? Under Martin's Axiom, there are countable non-discrete topological groups in which each nonempty nowhere dense subset has an isolated point.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03862 | KOU-17.5

### Kourovka Notebook Problem 17.5

O / open | Kourovka 群论问题集

Is the nilpotency class of a nilpotent group generated by $d$ left $3$-Engel elements bounded in terms of $d$? A group generated by two left $3$-Engel elements is nilpotent of class at most $4$ (*J. Pure Appl. Algebra*, **188** (2004), 1--6.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03863 | KOU-17.50

### Kourovka Notebook Problem 17.50

S / solved | Kourovka 群论问题集

Is it true that for every finite group $G$, there is a finite group $F$ and a surjective homomorphism $f: F \to G$ such that for each nontrivial subgroup $H$ of $F$, the restriction $f|_H$ is not injective?

It is known that for every finite group $G$ there is a finite group $F$ and a surjective homomorphism $f: F \to G$ such that for each subgroup $H$ of $F$ the restriction $f|_H$ is not bijective.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03864 | KOU-17.51

### Kourovka Notebook Problem 17.51

O / open | Kourovka 群论问题集

Is it true that every non-discrete topological group containing no countable open subgroup of exponent 2 can be partitioned into two (infinitely many) dense subsets? If the group is Abelian or countable, the answer is yes.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03865 | KOU-17.52

### Kourovka Notebook Problem 17.52

O / open | Kourovka 群论问题集

(N. Eggert). Let $R$ be a commutative associative finite-dimensional nilpotent algebra over a finite field $F$ of characteristic $p$. Let $R^{(p)}$ be the subalgebra of all elements of the form $r^p$ ($r\in R$). Is it true that $\dim R\geq p \dim R^{(p)}$?

An affirmative answer gives a solution to 11.44 for the case where the factors $A$ and $B$ are abelian.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03866 | KOU-17.53

### Kourovka Notebook Problem 17.53

O / open | Kourovka 群论问题集

A finite group $G$ is called *simply reducible* (SR-group) if every element of $G$ is conjugate to its inverse, and the tensor product of any two irreducible representations of $G$ decomposes into a sum of irreducible representations of $G$ with coefficients $0$ or $1$.

a) Is it true that the nilpotent length of a soluble SR-group is at most $5$?

b) Is it true that the derived length of a soluble SR-group is bounded by some constant $c$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03867 | KOU-17.54

### Kourovka Notebook Problem 17.54

O / open | Kourovka 群论问题集

Does there exist a non-hereditary local formation ${\mathfrak{F}}$ of finite groups such that in every finite group the set of all ${\mathfrak{F}}$-subnormal subgroups is a sublattice of the subgroup lattice?

A negative answer would solve 9.75, since all the hereditary local formations with the same property are already known.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03868 | KOU-17.55

### Kourovka Notebook Problem 17.55

S / solved | Kourovka 群论问题集

Does there exist an absolute constant $k$ such that for any prefrattini subgroup $H$ in any finite soluble group $G$ there exist $k$ conjugates of $H$ whose intersection is $\Phi(G)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03869 | KOU-17.56

### Kourovka Notebook Problem 17.56

S / solved | Kourovka 群论问题集

Suppose that a subgroup $H$ of a finite group $G$ is such that $HM=MH$ for every minimal non-nilpotent subgroup $M$ of $G$. Must $H/H_G$ be nilpotent, where $H_G$ is the largest normal subgroup of $G$ contained in $H$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03870 | KOU-17.57

### Kourovka Notebook Problem 17.57

O / open | Kourovka 群论问题集

Let $r(m)=\{r+km\mid k\in \mathbb{Z}\}$ for integers $0\leq r<m$. For \(r_1(m_1)\cap r_2(m_2) = \varnothing\) define the *class transposition* \(\tau_{r_1(m_1),r_2(m_2)}\) as the involution which interchanges \(r_1+km_1\) and \(r_2+km_2\) for each integer \(k\) and fixes everything else. The group \(\mathrm{CT}(\mathbb{Z})\) generated by all class transpositions is simple (*Math. Z.*, **264**, no. 4 (2010), 927--938). Is \(\mathrm{Out}(\mathrm{CT}(\mathbb{Z})) = \langle \sigma \mapsto \sigma^{n \mapsto -n-1} \rangle \cong \mathrm{C}_2\)?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03871 | KOU-17.58

### Kourovka Notebook Problem 17.58

O / open | Kourovka 群论问题集

Does \(\mathrm{CT}(\mathbb{Z})\) have subgroups of intermediate (word-) growth?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03872 | KOU-17.59

### Kourovka Notebook Problem 17.59

O / open | Kourovka 群论问题集

\ A permutation of \(\mathbb{Z}\) is called *residue-class-wise affine* if there is a positive integer \(m\) such that its restrictions to the residue classes (mod \(m\)) are all affine. Is \(\mathrm{CT}(\mathbb{Z})\) the group of all residue-class-wise affine permutations of \(\mathbb{Z}\) which fix the nonnegative integers setwise?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03873 | KOU-17.6

### Kourovka Notebook Problem 17.6

O / open | Kourovka 群论问题集

a) Is there a function $f:\mathbb{N}\times\mathbb{N}\rightarrow \mathbb{N}$ such that every nilpotent group generated by $d$ left $n$-Engel elements is nilpotent of class at most $f(n,d)$?

b) Is there a function $g:\mathbb{N}\times\mathbb{N}\rightarrow \mathbb{N}$ such that every nilpotent group generated by $d$ right $n$-Engel elements is nilpotent of class at most $g(n,d)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03874 | KOU-17.60

### Kourovka Notebook Problem 17.60

O / open | Kourovka 群论问题集

Given a set \(\mathcal{P}\) of odd primes, let \(\mathrm{CT}_{\mathcal{P}}(\mathbb{Z})\) denote the subgroup of \(\mathrm{CT}(\mathbb{Z})\) which is generated by all class transpositions which interchange residue classes whose moduli have only prime factors in \(\mathcal{P} \cup \{2\}\). The groups \(\mathrm{CT}_{\mathcal{P}}(\mathbb{Z})\) are simple (*Math. Z.*, **264**, no. 4 (2010), 927--938). Are they pairwise nonisomorphic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03875 | KOU-17.61

### Kourovka Notebook Problem 17.61

O / open | Kourovka 群论问题集

The group \(\mathrm{CT}_{\mathcal{P}}(\mathbb{Z})\) is finitely generated if and only if \(\mathcal{P}\) is finite. If \(\mathcal{P} = \varnothing\), then it is isomorphic to the finitely presented (first) Higman--Thompson group (J. P. McDermott, see Remark 1.4 in S. Kohl, *J. Group Theory*, **20**, no. 5 (2017), 1025--1030). Is it always finitely presented if \(\mathcal{P}\) is finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03876 | KOU-17.62

### Kourovka Notebook Problem 17.62

S / solved | Kourovka 群论问题集

Given a free group $F_n$ and a proper characteristic subgroup $C$, is it ever possible to generate the quotient $F_n/C$ by fewer than $n$ elements?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03877 | KOU-17.63

### Kourovka Notebook Problem 17.63

O / open | Kourovka 群论问题集

Prove that if $p$ is an odd prime and $s$ is a positive integer, then there are only finitely many $p$-adic space groups of finite coclass with point group of coclass $s$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03878 | KOU-17.64

### Kourovka Notebook Problem 17.64

O / open | Kourovka 群论问题集

Say that a group $G$ is an *$n$-approximation* to the Nottingham group $J=N(p)$ (as defined in Archive, 12.24) if $G$ is an infinite pro-$p$ group, and $G/\gamma_n(G)$ is isomorphic to $J/\gamma_n(J)$. Does there exist a function $f(p)$ such that, if $G$ is an $f(p)$-approximation to the Nottingham group, then $\gamma_i(G)/\gamma_{i+1}(G)$ is isomorphic to $\gamma_i(J)/\gamma_{i+1}(J)$ for all $i$?

Cf. 14.56. Note that an affirmative solution to this problem trivially implies the now known fact that $J$ is finitely presented as a pro-$p$ group (if $p>2$), see 14.55.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03879 | KOU-17.65

### Kourovka Notebook Problem 17.65

O / open | Kourovka 群论问题集

If Problem 17.64 has an affirmative answer, does it follow that, for some function $g(p)$, the number of isomorphism classes of $g(p)$-approximations to $J$ is

a) countable?

b) one?

Note that if, for some $g(p)$, there are only finitely many isomorphism classes, then for some $h(p)$ there is only one.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03880 | KOU-17.66

### Kourovka Notebook Problem 17.66

O / open | Kourovka 群论问题集

(R. Guralnick). Does there exist a positive integer $d$ such that $\dim H^1(G,V)\leq d$ for any faithful absolutely irreducible module $V$ for any finite group $G$? Cf. 16.55 in Archive.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03881 | KOU-17.67

### Kourovka Notebook Problem 17.67

S / solved | Kourovka 群论问题集

(H. Zassenhaus). *Conjecture*: Every invertible element of finite order of the integral group ring ${\mathbb{Z}}G$ of a finite group $G$ is conjugate in the rational group ring ${\mathbb{Q}}G$ to an element of $\pm G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03882 | KOU-17.68

### Kourovka Notebook Problem 17.68

O / open | Kourovka 群论问题集

Let $C$ be a cyclic subgroup of a group $G$ and $G=CFC$ where $F$ is a finite cyclic subgroup. Is it true that $|G:C|$ is finite? Cf. 12.62.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03883 | KOU-17.69

### Kourovka Notebook Problem 17.69

O / open | Kourovka 群论问题集

Let $G$ be a group of prime exponent acting freely on a non-trivial abelian group. Is $G$ cyclic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03884 | KOU-17.7

### Kourovka Notebook Problem 17.7

O / open | Kourovka 群论问题集

a) Is there a group in which the set of right Engel elements does not form a subgroup?

b) Is there a group in which the set of bounded right Engel elements does not form a subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03885 | KOU-17.70

### Kourovka Notebook Problem 17.70

S / solved | Kourovka 群论问题集

Let $\alpha$ be an automorphism of prime order $q$ of an infinite free Burnside group $G=B(q,p)$ of prime exponent $p$ such that $\alpha$ cyclically permutes the free generators of $G$. Is it true that $\alpha$ fixes some non-trivial element of $G$?

*Editors' comment*: this is proved for $q=2$ in (V. S. Atabekyan, H. T. Aslanyan, *Proc. Yerevan State Univ. Physic. Math. Sci.*, **53**, no. 3 (2019), 147--149).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03886 | KOU-17.71

### Kourovka Notebook Problem 17.71

P / partially_solved | Kourovka 群论问题集

b) Let $\alpha$ be a fixed-point-free automorphism of prime order $p$ of a periodic group $G$. Suppose that $G$ does not contain a non-trivial $p$-element. Is $\alpha$ a splitting automorphism?

[Archived solved part(s) of 17.71:] a) Let $\alpha$ be a fixed-point-free automorphism of prime order $p$ of a periodic group $G$. Is it true that $G$ does not contain a non-trivial $p$-element?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03887 | KOU-17.72

### Kourovka Notebook Problem 17.72

P / partially_solved | Kourovka 群论问题集

b) Let $AB$ be a Frobenius group with kernel $A$ and complement $B$. Suppose that $AB$ acts on a finite group $G$ so that $GA$ is also a Frobenius group with kernel $G$ and complement $A$. Is the exponent of $G$ bounded in terms of $|B|$ and the exponent of $C_G(B)$?

[Archived solved part(s) of 17.72:] a) Let $AB$ be a Frobenius group with kernel $A$ and complement $B$. Suppose that $AB$ acts on a finite group $G$ so that $GA$ is also a Frobenius group with kernel $G$ and complement $A$. Is the nilpotency class of $G$ bounded in terms of $|B|$ and the class of $C_G(B)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03888 | KOU-17.73

### Kourovka Notebook Problem 17.73

S / solved | Kourovka 群论问题集

Let $G$ be a finite simple group of Lie type defined over a field of characteristic $p$, and $V$ an absolutely irreducible $G$-module over a field of the same characteristic. Is it true that in the following cases the split extension of $V$ by $G$ must contain an element whose order is distinct from the order of any element of $G$?

a) $G=U_4(q)$;

b) $G=S_{2n}(q), n\geq 3$; \

c) $G=O_{2n+1}(q), n\geq 3$; \

d) $G=O_{2n}^+(q), n\geq 4$; \

e) $G=O_{2n}^-(q), n\geq 4$; \

f) $G= ^3D_{4}(q)$, $q\ne 2$; \

g) $G=E_6(q)$; \

h) $G= ^2E_6(q)$; \

i) $G=E_7(q)$; \

j) $G=G_2(2^m)$. \

问题集入口：https://kourovkanotebookorg.wordpress.com/

03889 | KOU-17.74

### Kourovka Notebook Problem 17.74

S / solved | Kourovka 群论问题集

Let $G$ be a finite simple group of Lie type defined over a field of characteristic $p$ whose Lie rank is at least three, and $V$ an absolutely irreducible $G$-module over a field of characteristic that does not divide $p$. It is true that the split extension of $V$ by $G$ must contain an element whose order is distinct from the order of any element of $G$? The case of $G=U_n(p^m)$ is of special interest.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03890 | KOU-17.75

### Kourovka Notebook Problem 17.75

S / solved | Kourovka 群论问题集

Can the Monster $M$ act on a nontrivial finite $3$-group $V$ so that all elements of $M$ of orders $41$, $47$, $59$, $71$ have no nontrivial fixed points in $V$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03891 | KOU-17.76

### Kourovka Notebook Problem 17.76

S / solved | Kourovka 群论问题集

Does there exist a finite group $G$, with $|G| > 2$, such that there is exactly one element in $G$ which is not a commutator?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03892 | KOU-17.77

### Kourovka Notebook Problem 17.77

S / solved | Kourovka 群论问题集

Let $k$ be a positive integer such that there is an insoluble finite group with exactly $k$ conjugacy classes. Is it true that a finite group of maximal order with exactly $k$ conjugacy classes is insoluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03893 | KOU-17.78

### Kourovka Notebook Problem 17.78

O / open | Kourovka 群论问题集

Does there exist a finitely generated group without free subsemigroups generating a proper variety {containing ${\mathfrak{A}}_p{\mathfrak{A}}$?}

问题集入口：https://kourovkanotebookorg.wordpress.com/

03894 | KOU-17.79

### Kourovka Notebook Problem 17.79

S / solved | Kourovka 群论问题集

Does there exist a finitely generated torsion group of unbounded exponent generating a proper variety?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03895 | KOU-17.8

### Kourovka Notebook Problem 17.8

O / open | Kourovka 群论问题集

a) Is there a group containing a right Engel element which is not a left Engel element?

b) Is there a group containing a bounded right Engel element which is not a left Engel element?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03896 | KOU-17.80

### Kourovka Notebook Problem 17.80

O / open | Kourovka 群论问题集

Let $[u, _nv]:=[u,\overbrace{v,\dots ,v}^{n} ]$. Is the group $M_n=\langle x, y\mid x=[x, _ny],\; y=[y, _nx] \rangle$ infinite for every $n>2$? (See also 11.18.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03897 | KOU-17.81

### Kourovka Notebook Problem 17.81

O / open | Kourovka 群论问题集

Given normal subgroups $R_1,\dots, R_n$ of a group $G$, let $\displaystyle [[R_1,\dots, R_n]]:=\prod \Big[\bigcap_{i\in I}R_i,\bigcap_{j\in J}R_j\Big] $, where the product is over all $I\cup J=\{1,\dots,n\}$, $ I\cap J=\varnothing$. Let $G$ be a free group, and let $R_i=\langle r_i\rangle^G$ be the normal closures of elements $r_i\in G$. It is known (B. Hartley, Yu. Kuzmin, *J. Pure Appl. Algebra*, **74** (1991), 247--256) that the quotient $(R_1\cap R_2)/{[R_1, R_2]}$ is a free abelian group. On the other hand, the quotient $(R_1\cap \dots \cap R_n)/[[R_1,\dots, R_n]] $ has, in general, non-trivial torsion for $n\geq 4$. Is this quotient always torsion-free for $n=3$? This is known to be true if $r_1,r_2,r_3$ are not proper powers in $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03898 | KOU-17.82

### Kourovka Notebook Problem 17.82

S / solved | Kourovka 群论问题集

Is it true that in every finitely presented group the intersection of the derived series has trivial abelianization?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03899 | KOU-17.83

### Kourovka Notebook Problem 17.83

O / open | Kourovka 群论问题集

Does there exist a group such that every central extension of it is residually nilpotent, but there exists a central extension of a central extension of it which is not residually nilpotent?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03900 | KOU-17.84

### Kourovka Notebook Problem 17.84

O / open | Kourovka 群论问题集

An associative algebra $A$ is said to be *Calabi--Yau of dimension $d$* (for short, CY$d$) if there is a natural isomorphism of $A$-bimodules $\mathrm{Ext}_{A\text{-bimod}}^d(A, A\otimes A)\cong A$ and $\mathrm{Ext}_{A\text{-bimod}}^n(A, A\otimes A)=0$ for $n\ne d$. By Kontsevich's theorem, the complex group algebra ${\mathbb{C}G$ of the fundamental group $G$ of a 3-dimensional aspherical manifold is CY$3$. Is every group with CY$3$ complex group algebra residually finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03901 | KOU-17.85

### Kourovka Notebook Problem 17.85

O / open | Kourovka 群论问题集

Let $\mathcal{V}$ be a variety of groups such that any free group in $\mathcal{V}$ has torsion-free integral homology groups in all dimensions. Is it true that $\mathcal{V}$ is abelian?

It is known that free metabelian groups may have 2-torsion in the second homology groups, and free nilpotent groups of class 2 may have 3-torsion in the fourth homology groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03902 | KOU-17.86

### Kourovka Notebook Problem 17.86

P / partially_solved | Kourovka 群论问题集

(Simplest questions related to the Whitehead asphericity conjecture). Let $\langle x_1,x_2,x_3\mid r_1,r_2,r_3\rangle$ be a presentation of the trivial group.

a) Prove that the group $\langle x_1,x_2,x_3\mid r_1,r_2\rangle$ is torsion-free.

[Archived solved part(s) of 17.86:] (Simplest questions related to the Whitehead asphericity conjecture). Let $\langle x_1,x_2,x_3\mid r_1,r_2,r_3\rangle$ be a presentation of the trivial group.

b) Let $F=F(x_1,x_2,x_3)$ and $R_i=\langle r_i\rangle^F$. Is it true that the group $F/[R_1,R_2]$ is residually soluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03903 | KOU-17.87

### Kourovka Notebook Problem 17.87

O / open | Kourovka 群论问题集

Construct a group of intermediate growth with finitely generated Schur multiplier.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03904 | KOU-17.88

### Kourovka Notebook Problem 17.88

O / open | Kourovka 群论问题集

Compute $K_0({\mathbb{F}_2G)$ and $K_1({\mathbb{F}_2G)$, where $G$ is the first Grigorchuk group, ${\mathbb{F}_2G$ its group algebra over the field of two elements, and $K_0, K_1$ the zeroth and first $K$-functors.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03905 | KOU-17.89

### Kourovka Notebook Problem 17.89

O / open | Kourovka 群论问题集

By Bousfield's theorem, the free pronilpotent completion of a non-cyclic free group has uncountable Schur multiplier. Is it true that the free prosolvable completion (that is, the inverse limit of quotients by the derived subgroups) of a non-cyclic free group has uncountable Schur multiplier?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03906 | KOU-17.9

### Kourovka Notebook Problem 17.9

O / open | Kourovka 群论问题集

Is there a group containing a left Engel element whose inverse is not a left Engel element?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03907 | KOU-17.90

### Kourovka Notebook Problem 17.90

O / open | Kourovka 群论问题集

(G. Baumslag). A group is *parafree* if it is residually nilpotent and has the same lower central quotients as a free group. Is it true that $H_2(G)=0$ for any finitely generated parafree group $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03908 | KOU-17.91

### Kourovka Notebook Problem 17.91

O / open | Kourovka 群论问题集

Let $d(X)$ denote the derived length of a group $X$.

a) Does there exist an absolute constant $k$ such that $d(G)-d(M)\leq k$ for every finite soluble group $G$ and any maximal subgroup $M$?

b) Find the minimum $k$ with this property.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03909 | KOU-17.92

### Kourovka Notebook Problem 17.92

S / solved | Kourovka 群论问题集

What are the non-abelian composition factors of a finite non-soluble group all of whose maximal subgroups are Hall subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03910 | KOU-17.93

### Kourovka Notebook Problem 17.93

O / open | Kourovka 群论问题集

(Well-known problem). Let $G$ be a compact topological group which has elements of arbitrarily high orders. Must $G$ contain an element of infinite order?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03911 | KOU-17.94

### Kourovka Notebook Problem 17.94

O / open | Kourovka 群论问题集

(Well-known problem). Can the free product $Z*G$ of the infinite cyclic group $Z$ and a nontrivial group $G$ be the normal closure of a single element?

*Editors' comments*: the negation is also known as Kervaire's conjecture. It was proved for torsion-free groups (A. Klyachko, *Commun. Algebra*, **21**, no. 7 (1993), 2555--2575) and for residually finite groups (M. Gerstenhaber, O. S. Rothaus, *Proc. Nat. Acad. Sci. U.S.A.*, **48** (1962) 1531--1533).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03912 | KOU-17.95

### Kourovka Notebook Problem 17.95

O / open | Kourovka 群论问题集

Let $G$ be a permutation group on the finite set $\Omega$. A partition $\rho$ of $\Omega$ is said to be $G$-*regular* if there exists a subset $S$ of $\Omega$ such that $S^g$ is a transversal of $\rho$ for all $g \in G$. The group $G$ is said to be *synchronizing* if $|\Omega| > 2$ and there are no non-trivial proper $G$-regular partitions on $\Omega$.

a) Are the following affine-type primitive groups synchronizing:

$2^{p}.\mathrm{PSL}(2,2p+1)$ where both $p$ and $2p + 1$ are prime, $p \equiv 3\;(\mathrm{mod}\;4)$ and $p > 23$?

$2^{101}.He$?

b) For which finite simple groups $S$ are the groups $S \times S$ acting on $S$ by $(g,h): x \mapsto g^{-1}xh$ non-synchronizing?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03913 | KOU-17.96

### Kourovka Notebook Problem 17.96

O / open | Kourovka 群论问题集

Does there exist a variety of groups which contains only countably many subvarieties but in which there is an infinite properly descending chain of subvarieties?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03914 | KOU-17.97

### Kourovka Notebook Problem 17.97

O / open | Kourovka 群论问题集

Is every variety of groups of exponent $4$ finitely based?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03915 | KOU-17.98

### Kourovka Notebook Problem 17.98

O / open | Kourovka 群论问题集

A variety of groups is said to be *small* if it contains only countably many non-isomorphic finitely generated groups.

a) Is it true that if $G$ is a finitely generated group and the variety $\mathrm{Var}(G)$ it generates is small then $G$ satisfies the maximal condition on normal subgroups?

b) Is it true that a variety is small if and only if all its finitely generated groups have the Hopf property?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03916 | KOU-17.99

### Kourovka Notebook Problem 17.99

S / solved | Kourovka 群论问题集

Consider the group $B = \langle a,b \mid (bab^{-1}) a (bab^{-1})^{-1} = a^2\rangle $ introduced by Baumslag in 1969. The same relation is satisfied by the functions $ f(x) = 2x$ and $g(x) = 2^x$ under the operation of composition in the group of germs of monotonically increasing to $\infty $ continuous functions on $(0, \infty )$, where two functions are identified if they coincide for all sufficiently large arguments. Is the representation $a\rightarrow f$, $b\rightarrow g$ of the group $B$ faithful?

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03917 | KOU-18.1

### Kourovka Notebook Problem 18.1

S / solved | Kourovka 群论问题集

Given a group $G$ of finite order $n$, does there necessarily exist a bijection $f$ from $G$ onto a cyclic group of order $n$ such that for each element $x\in G$, the order of $x$ divides the order of $f(x)$? An affirmative answer in the case where $G$ is solvable was given by F. Ladisch.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03918 | KOU-18.10

### Kourovka Notebook Problem 18.10

O / open | Kourovka 群论问题集

A formation $\mathfrak{F}$ of finite groups satisfies the *Wielandt property for residuals* if whenever $U$ and $V$ are subnormal subgroups of $\langle U, V\rangle $ in a finite group $G$, then the $\mathfrak{F}$-residual $\langle U, V\rangle ^{\mathfrak{F}}$ of $\langle U, V\rangle $ coincides with $\langle U^{\mathfrak{F}}, V^{\mathfrak{F}}\rangle $. Does every Fitting formation $\mathfrak{F}$ satisfy the Wielandt property for residuals?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03919 | KOU-18.100

### Kourovka Notebook Problem 18.100

O / open | Kourovka 群论问题集

For a given class of groups $\mathfrak{X}$, let $(\mathfrak{X},\infty)^{\ast }$ denote the class of groups in which every infinite subset contains two distinct elements $x$ and $y$ that satisfy $\langle x,x^{y}\rangle\in\mathfrak{X}$. Let $G$ be a finitely generated soluble-by-finite group in the class $(\mathfrak{X},\infty)^{\ast }$, let $m$ be a positive integer, and let $\mathfrak{F}$, $\mathfrak{E}_{m}$, $\mathfrak{E}$, $\mathfrak{N}$, and $\mathfrak{P}$ denote the classes of finite groups, groups of exponent dividing $m$, groups of finite exponent, nilpotent groups, and polycyclic groups, respectively.

a) If $\mathfrak{X}=\mathfrak{E}\mathfrak{N}$, then is $G$ in $\mathfrak{E}\mathfrak{N}$?

b) If $\mathfrak{X}=\mathfrak{E}_{m}\mathfrak{N}$, then is $G$ in $\mathfrak{E}_{m}(\mathfrak{F}\mathfrak{N})$?

c) If $\mathfrak{X}=\mathfrak{N}(\mathfrak{P}\mathfrak{F})$, then is $G$ in $\mathfrak{N}(\mathfrak{P}\mathfrak{F})$?

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03920 | KOU-18.102

### Kourovka Notebook Problem 18.102

O / open | Kourovka 群论问题集

(E. C. Dade). Let $C$ be a Carter subgroup of a finite solvable group $G$, and let $\ell(C)$ be the number of primes dividing $|C|$ counting multiplicities. It was proved in (E. C. Dade, *Illinois J. Math.*, **13** (1969), 449--514) that there is an exponential function $f$ such that the nilpotent length of $G$ is at most $f(\ell (C))$. Is there a linear (or at least a polynomial) function $f$ with this property?

*Editors' comment*: A quadratic function with that property was proved to exist in the special case where $G=H\rtimes C$ and $C$ is a cyclic subgroup such that $C_H(C)=1$, (E. Jabara, *J. Algebra*, **487** (2017), 161--172).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03921 | KOU-18.103

### Kourovka Notebook Problem 18.103

O / open | Kourovka 群论问题集

A group is said to be *minimax* if it has a finite subnormal series each of whose factors satisfies either the minimum or the maximum condition on subgroups. Is it true that in the class of nilpotent minimax groups only finitely generated groups may have faithful irreducible primitive representations over a finitely generated field of characteristic zero?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03922 | KOU-18.104

### Kourovka Notebook Problem 18.104

O / open | Kourovka 群论问题集

Construct an example of a pro-$p$ group $G$ and a proper abstract normal subgroup $K$ such that $G/K$ is perfect.

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03923 | KOU-18.105

### Kourovka Notebook Problem 18.105

O / open | Kourovka 群论问题集

Let $R$ be the formal power series algebra in commuting indeterminates $x_1,\dots x_n, \dots$ over the field with $p$ elements, with $p$ a prime. (Thus its elements are (in general infinite) linear combinations of monomials $x_1^{a_1}\dots x_n^{a_n}$ with $n, a_1,\dots,a_n$ non-negative integers.) Let $I$ be the maximal ideal of $R$ and $J$ the (abstract) ideal generated by all products of two elements of $I$. Is there an ideal $U$ of $R$ such that $U<I$ and $U+J=I$?

If so, then $\mathrm{SL}_3(R)$ and the kernel of the map to $\mathrm{SL}_3(R/U)$ provide an answer to the previous question.

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03924 | KOU-18.106

### Kourovka Notebook Problem 18.106

O / open | Kourovka 群论问题集

Let $R$ be a group that acts coprimely on the finite group $G$. Let $p$ be a prime, let $P$ be the unique maximal $RC_{G}(R)$-invariant $p$-subgroup of $G$ and assume that $C_{G}(O_{p}(G)) \leq O_{p}(G)$. If $p>3$ then $P$ contains a nontrivial characteristic subgroup that is normal in $G$ (P. Flavell, *J. Algebra*, **257** (2002), 249--264). Does the same result hold for the primes 2 and 3?

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03925 | KOU-18.107

### Kourovka Notebook Problem 18.107

O / open | Kourovka 群论问题集

Is there a finitely generated infinite residually finite $p$-group such that every subgroup of infinite index is cyclic?

The answer is known to be negative for $p=2$. Note that in (M. Ershov, A. Jaikin-Zapirain, *J. Reine Angew. Math.*, **677** (2013), 71--134) it is shown that for every prime $p$ there is a finitely generated infinite residually finite $p$-group such that every finitely generated subgroup of infinite index is finite.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03926 | KOU-18.108

### Kourovka Notebook Problem 18.108

O / open | Kourovka 群论问题集

A group $G$ is said to have *property $(\tau )$* if its trivial representation is an isolated point in the subspace of irreducible representations with finite images (in the topological unitary dual space of $G$). Is it true that there exists a finitely generated group satisfying property $(\tau )$ that homomorphically maps onto every finite simple group?

The motivation is the theorem in (E. Breuillard, B. Green, M. Kassabov, A. Lubotzky, N. Nikolov, T. Tao) saying that there are $\epsilon>0$ and $k$ such that every finite simple group $G$ contains a generating set $S$ with at most $k$ elements such that the Cayley graph of $G$ with respect to $S$ is an $\epsilon$-expander. In (M. Ershov, A. Jaikin-Zapirain, M. Kassabov, *Mem. Amer. Math. Soc.*, **1186**, 2017) it is proved that there exists a group with Kazhdan's property (T) that maps onto every simple group of Lie type of rank ${}\geq 2$. (A group $G$ is said to have *Kazhdan's property* (T) if its trivial representation is an isolated point in the unitary dual space of $G$.)

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03927 | KOU-18.109

### Kourovka Notebook Problem 18.109

O / open | Kourovka 群论问题集

Is it true that a group satisfying Kazhdan's property (T) cannot homomorphically map onto infinitely many simple groups of Lie type of rank 1?

It is known that a group which maps onto $PSL_2(q)$ for infinitely many $q$ does not have Kazhdan's property (T).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03928 | KOU-18.11

### Kourovka Notebook Problem 18.11

S / solved | Kourovka 群论问题集

Let $G$ and $H$ be subgroups of the automorphism group $Aut (F_n)$ of a free group $F_n$ of rank $n \geq 2$. Is it true that the free product $G * H$ embeds in the automorphism group $Aut (F_m)$ for some $m$?

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03929 | KOU-18.110

### Kourovka Notebook Problem 18.110

O / open | Kourovka 群论问题集

The *non-$p$-soluble length* of a finite group $G$ is the number of non-$p$-soluble factors in a shortest normal series each of whose factors either is $p$-soluble or is a direct product of non-abelian simple groups of order divisible by $p$. For a given prime $p$ and a given proper group variety ${\mathfrak{V}}$, is there a bound for the non-$p$-soluble length of finite groups whose Sylow $p$-subgroups belong to ${\mathfrak{V}}$?

*Comment of 2021*: the existence of such a bound is proved for $p=2$ (F. Fumagalli, F. Leinen, O. Puglisi, *Proc. London Math. Soc. (3)*, **125**, no. 5 (2022), 1066--1082).

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03930 | KOU-18.111

### Kourovka Notebook Problem 18.111

S / solved | Kourovka 群论问题集

Let $G$ be a discrete countable group, given as a central extension $0\to {\mathbb{Z}}\to G\to Q\to 0$. Assume that either $G$ is quasi-isometric to $\mathbb{Z}\times Q$, or that $\mathbb{Z}\to G$ is a quasi-isometric embedding. Does that imply that $G$ comes from a bounded cocycle on $Q$?

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03931 | KOU-18.112

### Kourovka Notebook Problem 18.112

S / solved | Kourovka 群论问题集

Is it true that the orders of all elements of a finite group $G$ are powers of primes if, for every divisor $d$ ($d>1$) of $|G|$ and for every subgroup $H$ of $G$ of order coprime to $d$, the order $|H|$ divides the number of elements of $G$ of order $d$?

The converse is true (W. J. Shi, *Math. Forum (Vladikavkaz)*, **6** (2012), 152--154).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03932 | KOU-18.113

### Kourovka Notebook Problem 18.113

O / open | Kourovka 群论问题集

Let $\mathfrak{M}$ be a finite set of finite simple nonabelian groups. Is it true that a periodic group saturated with groups from $\mathfrak{M}$ (see 14.101) is isomorphic to one of groups in $\mathfrak{M}$? The case where $\mathfrak{M}$ is one-element is of special interest.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03933 | KOU-18.114

### Kourovka Notebook Problem 18.114

O / open | Kourovka 群论问题集

Does there exist an irreducible $5'$-subgroup $G$ of $GL(V)$ for some finite ${\mathbb{F} _5$-space $V$ such that the number of conjugacy classes of the semidirect product $VG$ is equal to $|V|$ but $G$ is not cyclic?

Such a subgroup exists if 5 is replaced by $p=2, 3$ but does not exist for primes $p>5$ (*J. Group Theory*, **14** (2011), 175--199).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03934 | KOU-18.115

### Kourovka Notebook Problem 18.115

S / solved | Kourovka 群论问题集

Let $G$ be a finite simple group, and let $X, Y$ be isomorphic simple maximal subgroups of $G$. Are $X$ and $Y$ conjugate in $\mathrm{Aut} G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03935 | KOU-18.116

### Kourovka Notebook Problem 18.116

O / open | Kourovka 群论问题集

Given a finite $2$-group $G$ of order $2^n$, does there exist a finite set $S$ of rational primes such that $|S|\le n$ and $G$ is a quotient group of the absolute Galois group of the maximal $2$-extension of $\mathbb{Q}$ unramified outside $S\cup \{\infty \}$?

This is true for $p$-groups, $p$ odd, as noted by Serre.

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03936 | KOU-18.117

### Kourovka Notebook Problem 18.117

O / open | Kourovka 群论问题集

Is every element of a nonabelian finite simple group a commutator of two elements of coprime orders?

The answer is known to be ``yes'' for the alternating groups (P. Shumyatsky, *Forum Math.*, **27**, no. 1 (2015), 575--583) and for $PSL(2,q)$ (M. A. Pellegrini, P. Shumyatsky, *Arch. Math.*, **99** (2012), 501--507). Without the coprimeness condition, this is Ore's conjecture proved in (M. W. Liebeck, E. A. O'Brien, A. Shalev, Pham Huu Tiep, *J. Eur. Math. Soc.*, **12**, no. 4 (2010), 939--1008).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03937 | KOU-18.118

### Kourovka Notebook Problem 18.118

O / open | Kourovka 群论问题集

Let $G$ be a profinite group such that $G/Z(G)$ is periodic. Is $[G,G]$ necessarily periodic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03938 | KOU-18.119

### Kourovka Notebook Problem 18.119

O / open | Kourovka 群论问题集

A multilinear commutator word is any commutator of weight $n$ in $n$ distinct variables. Let $w$ be a multilinear commutator word and let $G$ be a finite group. Is it true that every Sylow $p$-subgroup of the verbal subgroup $w(G)$ is generated by $w$-values?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03939 | KOU-18.12

### Kourovka Notebook Problem 18.12

O / open | Kourovka 群论问题集

Let $G$ be a finitely generated group of intermediate growth. Is it true that there is a positive integer $m$ such that every element of the derived subgroup $G'$ is a product of at most $m$ commutators?

This assertion is valid for groups of polynomial growth, since they are almost nilpotent by Gromov's theorem. On the other hand, there are groups of exponential growth (for example, free groups) for which this assertion is not true.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03940 | KOU-18.120

### Kourovka Notebook Problem 18.120

O / open | Kourovka 群论问题集

Let $P=AB$ be a finite $p$-group factorized by an abelian subgroup $A$ and a class-two subgroup $B$. Suppose, if necessary, $A \cap B=1$. Then is it true that $ \langle A, [B,B] \rangle =AB_{0}$, where $B_{0}$ is an abelian subgroup of $ B$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03941 | KOU-18.13

### Kourovka Notebook Problem 18.13

O / open | Kourovka 群论问题集

(D. B. A. Epstein). Is it true that the group $ H = (\mathbb{Z}_3 \times \mathbb{Z}) * (\mathbb{Z}_2 \times \mathbb{Z}) = \langle x, y, z, t \mid x^3 = z^2 = [x, y] = [z, t] = 1 \rangle$ cannot be defined by three relators in the generators $x, y, z, t$?

It is known that the relation module of the group $H$ has rank 3 \ (K. W. Gruenberg, P. A. Linnell, *J. Group Theory*, **11**, no. 5 (2008), 587--608). An affirmative answer would give a solution of the relation gap problem.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03942 | KOU-18.14

### Kourovka Notebook Problem 18.14

S / solved | Kourovka 群论问题集

For an automorphism $\varphi \in Aut(G)$ of a group $G$, let $[e]_{\varphi} =\{g^{-1}g^{\varphi}\mid g\in G\}$.

*Conjecture*: If $[e]_{\varphi}$ is a subgroup for every $\varphi \in Aut(G)$, then the group $G$ is nilpotent. If in addition $G$ is finitely generated, then $G$ is abelian.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03943 | KOU-18.15

### Kourovka Notebook Problem 18.15

S / solved | Kourovka 群论问题集

For an automorphism $\varphi \in Aut(G)$ of a group $G$, let $[e]_{\varphi} =\{g^{-1}g^{\varphi}\mid g\in G\}$. Is it true that if a group $G$ has trivial center, then there is an inner automorphism $\varphi$ such that $[e]_{\varphi}$ is not a subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03944 | KOU-18.16

### Kourovka Notebook Problem 18.16

S / solved | Kourovka 群论问题集

Is it true that any definable endomorphism of any ordered abelian group is of the form $x\mapsto rx$, for some rational number $r$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03945 | KOU-18.17

### Kourovka Notebook Problem 18.17

S / solved | Kourovka 群论问题集

Is there a torsion-free group which is finitely presented in the quasi-variety of torsion-free groups but not finitely presentable in the variety of all groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03946 | KOU-18.18

### Kourovka Notebook Problem 18.18

S / solved | Kourovka 群论问题集

Mal'cev proved that the set of sentences that hold in all finite groups is not computably enumerable, although its complement is. Is it true that both the set of sentences that hold in almost all finite groups and its complement are not computably enumerable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03947 | KOU-18.19

### Kourovka Notebook Problem 18.19

O / open | Kourovka 群论问题集

Is any torsion-free, relatively free group of infinite rank not $\aleph_1$-homogeneous? This is true for group varieties in which all free groups are residually finite (O. Belegradek, *Arch. Math. Logic*, **51** (2012), 781--787).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03948 | KOU-18.2

### Kourovka Notebook Problem 18.2

O / open | Kourovka 群论问题集

Let the group $G=AB$ be the product of two Chernikov subgroups $A$ and $B$ each of which has an abelian subgroup of index at most 2. Is $G$ soluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03949 | KOU-18.20

### Kourovka Notebook Problem 18.20

O / open | Kourovka 群论问题集

Characters $\varphi$ and $\psi$ of a finite group $G$ are said to be *semiproportional* if they are not proportional and there is a normal subset $M$ of $G$ such that $\varphi|_M$ is proportional to $\psi|_M$ and $\varphi|_{G\setminus M}$ is proportional to $\psi|_{G\setminus M}$.

*Conjecture*: If $\varphi$ and $\psi$ are semiproportional irreducible characters of a finite group, then $\varphi(1) = \psi(1)$.

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03950 | KOU-18.21

### Kourovka Notebook Problem 18.21

O / open | Kourovka 群论问题集

(B. H. Neumann and H. Neumann). Fix an integer $d\geq 2$. If $\mathfrak{V}$ is a variety of groups such that all $d$-generated groups in $\mathfrak{V}$ are finite, must $\mathfrak{V}$ be locally finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03951 | KOU-18.22

### Kourovka Notebook Problem 18.22

O / open | Kourovka 群论问题集

(H. Neumann). Is the Kostrikin variety of all locally finite groups of given prime exponent $p$ determined by finitely many identities?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03952 | KOU-18.23

## Kourovka Notebook Problem 18.23

S / solved | Kourovka 群论问题集

The normal covering number of the symmetric group $S_n$ of degree $n$ is the minimum number $\gamma(S_n)$ of proper subgroups $H_1,\ldots,H_{\gamma(S_n)}$ of $S_n$ such that every element of $S_n$ is conjugate to an element of $H_i$, for some $i=1,\dots, \gamma(S_n)$. Write $n=p_1^{\alpha_1}\cdots p_r^{\alpha_r}$ for primes $p_1<\cdots<p_r$ and positive integers $\alpha_1,\ldots,\alpha_r$.

*Conjecture*: \[ \gamma(S_n)=\left\{ \begin{array}{lcl} \frac{n}{2}\left(1-\frac{1}{p_1}\right)&&\textrm{if }r=1 \textrm{and} \alpha_1=1\\ \frac{n}{2}\left(1-\frac{1}{p_1}\right)+1&&\textrm{if }r=1 \textrm{and} \alpha_1\geq 2\\ \frac{n}{2}\left(1-\frac{1}{p_1}\right)\left(1-\frac{1}{p_2}\right)+1&&\textrm{if }r=2 \textrm{and} \alpha_1+\alpha_2=2\\ \frac{n}{2}\left(1-\frac{1}{p_1}\right)\left(1-\frac{1}{p_2}\right)+2&&\textrm{if }r\geq 2 \textrm{and} \alpha_1+\cdots+\alpha_r\geq 3\\ \end{array} \right. \] This is the strongest form of the conjecture. We would be also interested in a proof that this holds for $n$ sufficiently large. The result for $r\leq 2$, which includes the first three cases above, is proved for $n$ odd (D. Bubboloni, C. E. Praeger, *J. Combin. Theory (A)*, **118** (2011), 2000--2024). When $r\geq3$ we know that $cn\leq \gamma(S_n)\leq \frac{2}{3}n$ for some positive constant $c$ (D. Bubboloni, C. E. Praeger, P. Spiga, *J. Algebra*, **390** (2013) 199--215). We showed that the conjectured value for $\gamma(S_n)$ is an upper bound, by constructing a normal covering for $S_n$ with this number of conjugacy classes of maximal subgroups, and gave further evidence for the truth of the conjecture in other cases (D. Bubboloni, C. E. Praeger, P. Spiga, *Int. J. Group Theory*, **3**, no. 2 (2014), 57--75).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03953 | KOU-18.24

## Kourovka Notebook Problem 18.24

O / open | Kourovka 群论问题集

For a group $G$, a function $\phi : G\to {\mathbb{R}$ is a *quasimorphism* if there is a least non-negative number $D(\phi)$ (called the *defect*) such that $|\phi(gh) -\phi(g) - \phi(h)| \leq D(\phi)$ for all $g, h\in G$. A quasimorphism is *homogeneous* if in addition $\phi(g^n) = n\phi(g)$ for all $g\in G$. Let $\phi$ be a homogeneous quasimorphism of a free group $F$. For any quasimorphism $\psi$ of $F$ (not required to be homogeneous) with $|\phi - \psi| < \mathrm{const}$, we must have $D(\psi) \ge D(\phi)/2$. Is it true that there is some $\psi$ with $D(\psi) = D(\phi)/2$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03954 | KOU-18.25

## Kourovka Notebook Problem 18.25

O / open | Kourovka 群论问题集

Let $\mathrm{form}(G)$ be the formation generated by a finite group $G$. Suppose that $G$ has a unique composition series $ 1\vartriangleleft G_1\vartriangleleft G_2\vartriangleleft G$ and the consecutive factors of this series are $Z_p$, $X$, $Z_q$, where $p$ and $q$ are primes and $X$ is a non-abelian simple group. Is it true that $\mathrm{form}(G)$ has infinitely many subformations if and only if $p=q$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03955 | KOU-18.26

## Kourovka Notebook Problem 18.26

O / open | Kourovka 群论问题集

Suppose that a finite group $G$ has a normal series $ 1< G_1<G_2<G $ such that the groups $G_1$ and $G_2/G_1$ are elementary abelian $p$-groups, $G/G_2\cong A_5\times A_5$ (where $A_5$ is the alternating group of degree 5), $G_1$ and $G_2/G_1$ are minimal normal subgroups of $G$ and $G/G_1$, respectively. Is it true that $\mathrm{form}(G)$ has finitely many subformations?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03956 | KOU-18.27

## Kourovka Notebook Problem 18.27

O / open | Kourovka 群论问题集

Does there exist an algorithm that determines whether there are finitely many subformations in $\mathrm{form}(G)$ for a given finite group $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03957 | KOU-18.28

## Kourovka Notebook Problem 18.28

O / open | Kourovka 群论问题集

Is it true that any subformation of every one-generator formation $\mathrm{form}(G)$ is also one-generator?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03958 | KOU-18.29

## Kourovka Notebook Problem 18.29

O / open | Kourovka 群论问题集

Let $\mathfrak {F}$ be a Fitting class of finite soluble groups which contains every soluble group $G=AB$, where $A$ and $B$ are abnormal $\mathfrak {F}$-subgroups of $G$. Is $\mathfrak {F}$ a formation?

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03959 | KOU-18.3

### Kourovka Notebook Problem 18.3

O / open | Kourovka 群论问题集

Let the group $G=AB$ be the product of two central-by-finite subgroups $A$ and $B$. By a theorem of N. Chernikov, $G$ is soluble-by-finite. Is $G$ metabelian-by-\text{finite?} \text{}

问题集入口：https://kourovkanotebookorg.wordpress.com/

03960 | KOU-18.30

### Kourovka Notebook Problem 18.30

S / solved | Kourovka 群论问题集

A subgroup $H$ of a group $G$ is called \emph{$\mathbb{P}$-subnormal} in $G$ if either $H=G$, or there is a chain of subgroups $H_0\subset H_1\subset \cdots\subset H_{n-1}\subset H_{n} =G$ such that $|H_i : H_{i-1}|$ is a prime for all $i=1,\ldots, n$. Must a finite group be soluble if every Shmidt subgroup of it is $\mathbb{P}$-subnormal?

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03961 | KOU-18.31

### Kourovka Notebook Problem 18.31

S / solved | Kourovka 群论问题集

Let $\pi$ be a set of primes. Is it true that in any $D_\pi$-group $G$ (see Archive, 3.62) there are three Hall $\pi$-subgroups whose intersection coincides with $O_\pi(G)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03962 | KOU-18.32

### Kourovka Notebook Problem 18.32

S / solved | Kourovka 群论问题集

Is every Hall subgroup of a finite group pronormal in its normal closure?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03963 | KOU-18.33

### Kourovka Notebook Problem 18.33

S / solved | Kourovka 群论问题集

A group in which the derived subgroup of every 2-generated subgroup is cyclic is called an *Alperin group*. Is there a bound for the derived length of finite Alperin groups?

G. Higman proved that finite Alperin groups are soluble, and finite Alperin $p$-groups have bounded derived length, see Archive 17.46.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03964 | KOU-18.34

### Kourovka Notebook Problem 18.34

O / open | Kourovka 群论问题集

Let $G$ be a topological group and $S={G_0\leq \dots \leq G_n=G}$ a subnormal series of closed subgroups. The *infinite-length* of $S$ is the cardinality of the set of infinite factors $G_i/G_{i-1}$. The *virtual length* of $G$ is the supremum of the infinite lengths taken over all such series of $G$. It is known that if $G$ is a pronilpotent group with finite virtual length then every closed subnormal subgroup is topologically finitely generated (N. Gavioli, V. Monti, C. M. Scoppola, *J. Austral. Math. Soc.*, **95**, no. 3 (2013), 343--355). Let $G$ be a pro-$p$ group (or more generally a pronilpotent group). If every closed subnormal subgroup of $G$ is tolopologically finitely generated, is it true that $G$ has finite virtual length?

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03965 | KOU-18.35

### Kourovka Notebook Problem 18.35

O / open | Kourovka 群论问题集

A family ${\mathscr F}$ of group homomorphisms $A\to B$ is *separating* if for every nontrivial $a\in A$ there is $f\in {\mathscr F}$ such that $f(a)\ne 1$, and *discriminating* if for any finitely many nontrivial elements $a_1,\dots ,a_n\in A$ there is $f\in {\mathscr F}$ such that $f(a_i)\ne 1$ for all $i=1,\dots , n$. Let $G$ be a relatively free group of rank 2 in the variety of metabelian groups. Let $T$ be a metabelian group in which the centralizer of every nontrivial element is abelian. If $T$ admits a separating family of surjective homomorphisms $T\to G$ must it also admit a discriminating family of surjective homomorphisms $T\to G$?

A positive answer would give a metabelian analogue of a classical theorem of B. Baumslag.

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03966 | KOU-18.36

### Kourovka Notebook Problem 18.36

S / solved | Kourovka 群论问题集

There are groups of cardinality at most $2^{\aleph_{0}}$, even nilpotent of class $2$, that cannot be embedded in $S:=Sym(\mathbb{N})$ (V. A. Churkin, *Algebra and model theory* (Novosibirsk State Tech. Univ.), **5** (2005), 39--43 (Russian)). One condition which might characterize subgroups of $S$ is the following. Consider the metric $d$ on $S$ where for all $x\neq y$ we define $d(x,y):=2^{-k}$ when $k$ is the least element of the set $\mathbb{N}$ such that $k^{x}\neq k^{y}$. Then $(S,d)$ is a separable topological group, and so each subgroup of $S$ (not necessarily a closed subgroup of $(S,d)$) is a separable topological group under the induced metric. Is it true that every group $G$ for which there is a metric $d^{\prime}$ such that $(G,d^{\prime})$ is a separable topological group is (abstractly) embeddable in $S$?

Note that any such group $G$ can be embedded as a *section* in $S$.

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03967 | KOU-18.37

### Kourovka Notebook Problem 18.37

O / open | Kourovka 群论问题集

(Well-known problem). Is every locally graded group of finite rank almost locally soluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03968 | KOU-18.38

### Kourovka Notebook Problem 18.38

S / solved | Kourovka 群论问题集

Let $E_\pi$ denote the class of finite groups that contain a Hall $\pi$-subgroup. Does the inclusion $E_{\pi_1} \cap E_{\pi_2} \subseteq E_{\pi_1 \cap \pi_2} $ hold for arbitrary sets of primes $\pi_1$ and $\pi_2$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03969 | KOU-18.39

### Kourovka Notebook Problem 18.39

O / open | Kourovka 群论问题集

*Conjecture*: Let $G$ be a hyperbolic group. Then every 2-dimensional rational homology class is virtually represented by a sum of closed surface subgroups, that is, for any $\alpha \in H_2(G;\mathbb{Q})$ there are finitely many closed oriented surfaces $S_i$ and injective homomorphisms $\rho_i:\pi_1(S_i) \to G$ such that $\sum_i [S_i] = n \alpha$, where $[S_i]$ denotes the image of the fundamental class of $S_i$ in $H_2(G)$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03970 | KOU-18.4

### Kourovka Notebook Problem 18.4

O / open | Kourovka 群论问题集

(A. Rhemtulla, S. Sidki). a) Is a group of the form $G=ABA$ with cyclic subgroups $A$ and $B$ always soluble? (This is known to be true if $G$ is finite.)

b) The same question when in addition $A$ and $B$ are conjugate in $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03971 | KOU-18.40

### Kourovka Notebook Problem 18.40

O / open | Kourovka 群论问题集

The commutator length $cl(g)$ of $g \in [G,G]$ is the least number of commutators in $G$ whose product is $g$, and the stable commutator length is $scl(g):= \lim_{n \to \infty} cl(g^n)/n$.

*Conjecture*: Let $G$ be a hyperbolic group. Then the stable commutator length takes on rational values on $[G,G]$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03972 | KOU-18.41

### Kourovka Notebook Problem 18.41

O / open | Kourovka 群论问题集

Let $F$ be a free group of rank 2.

a) It is known that $scl(g)=0$ only for $g=1$, and $scl(g)\ge 1/2$ for all $1\ne g\in [F,F]$. Is $1/2$ an isolated value?

b) Are there any intervals $J$ in ${\mathbb{R}}$ such that the set of values of $scl$ on $[F,F]$ is dense in $J$?

c) Is there some $T\in {\mathbb{R}}$ such that every rational number ${}\ge T$ is a value of $scl(g)$ for $g\in [F,F]$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03973 | KOU-18.42

### Kourovka Notebook Problem 18.42

O / open | Kourovka 群论问题集

For an orientation-preserving homeomorphism $f:S^1\to S^1$ of a unit circle $S^1$, let $\tilde f$ be its lifting to a homeomorphism of ${\mathbb{R}}$; then the *rotation number* of $f$ is defined to be the limit $\lim _{n\to \infty} (\tilde f^n(x)-x)/n$ (which is independent of the point $x\in S^1$). Let $F$ be a free group of rank 2 with generators $a,b$. For $ w\in F$ and $r,s\in {\mathbb{R}}$, let $R(w,r,s)$ denote the maximum value of the (real-valued) rotation number of $w$, under all representations from $F$ to the universal central extension of $Homeo^+(S^1)$ for which the rotation number of $a$ is $r$, and the rotation number of $b$ is $s$.

a) If $r,s$ are rational, must $R(w,r,s)$ be rational?

b) Let $R(w,r-,s-)$ denote the supremum of the rotation number of $w$ (as above) under all representations for which $a$ and $b$ are conjugate to rotations through $r$ and $s$, respectively (in the universal central extension of $Homeo^+(S^1)$). Is $R(w,r-,s-)$ always rational if $r$ and $s$ are rational?

c) *Weak Slippery Conjecture*: Let $w$ be a word containing only positive powers of $a$ and $b$. A pair of values $(r,s)$ is *slippery* if there is a strict inequality $R(w,r',s') < R(w,r-,s-)$ for all $r'<r$, $s'<s$. Is it true that then $R(w,r-,s-) = h_a(w)r + h_b(w)s$, where $h_a(w)$ counts the number of $a$'s in $w$, and $h_b(w)$ counts the number of $b$'s in $w$?

d) *Slippery Conjecture*: More precisely, is it always (without assuming $(r,s)$ being slippery) true that if $w$ has the form $w=a^{\alpha_1}b^{\beta_1}\cdots a^{\alpha_m}b^{\beta_m}$ (all positive) and $R(w,r,s) = p/q$ where $p/q$ is reduced, then $|p/q - h_a(w)r - h_b(w)s| \le m/q$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03974 | KOU-18.43

### Kourovka Notebook Problem 18.43

O / open | Kourovka 群论问题集

a) Do there exist elements $u,v$ of a 2-generator free group $F(a,b)$ such that $u,v$ are not conjugate in $F(a,b)$ but for any matrices $A,B\in GL(3, \mathbb{C})$ we have $trace(u(A,B))= trace(v(A,B))$?

b) The same question if we replace $GL(3,\mathbb{C})$ by $SL(3,\mathbb{C})$.

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03975 | KOU-18.44

### Kourovka Notebook Problem 18.44

S / solved | Kourovka 群论问题集

Let $G$ be a finite non-abelian group and $V$ a finite faithful irreducible $G$-module. Suppose that $M=|G/G'|$ is the largest orbit size of $G$ on $V$, and among orbits of $G$ on $V$ there are exactly two orbits of size $M$. Does this imply that $G$ is dihedral of order 8, and $|V|=9$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03976 | KOU-18.45

### Kourovka Notebook Problem 18.45

O / open | Kourovka 群论问题集

Let $G$ be a finite $p$-group of maximal class such that all its irreducible characters are induced from linear characters of normal subgroups. Let $S$ be the set of the derived subgroups of the members of the central series of $G$. Is there a logarithmic bound for the derived length of $G$ in terms of $|S|$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03977 | KOU-18.46

### Kourovka Notebook Problem 18.46

O / open | Kourovka 群论问题集

Every finite group $G$ can be embedded in a group $H$ in such a way that every element of $G$ is a square of an element of $H$. The overgroup $H$ can be chosen such that $|H| \le 2|G|^2$. Is this estimate sharp?

It is known that the best possible estimate cannot be better than $|H| \le |G|^2$ (D. V. Baranov, Ant. A. Klyachko, *Siber. Math. J.*, **53**, no. 2 (2012), 201--206).

问题集入口：https://kourovkanotebookorg.wordpress.com/

03978 | KOU-18.47

### Kourovka Notebook Problem 18.47

O / open | Kourovka 群论问题集

Is it algorithmically decidable whether a group generated by three given class transpositions (for the definition, see 17.57)

a) has only finite orbits on \(\mathbb{Z}\)?

b) acts transitively on the set of nonnegative integers in its support?

A difficult case is the group \(\langle \tau_{1(2),4(6)}, \tau_{1(3),2(6)}, \tau_{2(3),4(6)} \rangle\), which acts transitively on \(\mathbb{N} \setminus 0(6)\) if and only if Collatz' \(3n+1\) conjecture is true.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03979 | KOU-18.48

### Kourovka Notebook Problem 18.48

S / solved | Kourovka 群论问题集

Is it true that there are only finitely many integers which occur as orders of products of two class transpositions? (For the definition, see 17.57.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

03980 | KOU-18.49

### Kourovka Notebook Problem 18.49

S / solved | Kourovka 群论问题集

Let \(n \in \mathbb{N}\). Is it true that for any \(a, b, c \in \mathbb{N}\) satisfying \(1 < a, b, c \leq n-2\) the symmetric group \(\mathrm{S}_n\) has elements of order \(a\) and \(b\) whose product has order \(c\)?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03981 | KOU-18.5

### Kourovka Notebook Problem 18.5

O / open | Kourovka 群论问题集

Let the (soluble) group $G=AB$ with finite torsion-free rank $r_0(G)$ be the product of two subgroups $A$ and $B$. Is the equation $r_0(G) = r_0(A) + r_0(B) - r_0(A \cap B)$ valid in this case? It is known that the inequality $\leq$ always holds.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03982 | KOU-18.50

### Kourovka Notebook Problem 18.50

S / solved | Kourovka 群论问题集

Let \(n \in \mathbb{N}\). Is it true that for every \(k \in \{1, \dots, n!\}\) there is some group \(G\) and pairwise distinct elements \(g_1, \dots, g_n \in G\) such that the set \(\{g_{\sigma(1)} \cdots g_{\sigma(n)} \ | \ \sigma \in \mathrm{S}_n\}\) of all products of the \(g_i\) obtained by permuting the factors has cardinality \(k\)?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03983 | KOU-18.51

### Kourovka Notebook Problem 18.51

O / open | Kourovka 群论问题集

Given a prime \(p\) and \(n \in \mathbb{N}\), let \(f_p(n)\) be the smallest number such that there is a group of order \(p^{f_p(n)}\) into which every group of order \(p^n\) embeds. Is it true that \(f_p(n)\) grows faster than polynomially but slower than exponentially when \(n\) tends to infinity?

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03984 | KOU-18.52

### Kourovka Notebook Problem 18.52

S / solved | Kourovka 群论问题集

Is every finite simple group generated by two elements of prime-power orders $m$, $n$? (Here numbers $m,n$ may depend on the group.) The work of many authors shows that it remains to verify this property for a small number of finite simple groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03985 | KOU-18.53

### Kourovka Notebook Problem 18.53

O / open | Kourovka 群论问题集

Is there a non-linear simple locally finite group in which every centralizer of a non-trivial element is almost soluble, that is, has a soluble subgroup of finite index?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03986 | KOU-18.54

### Kourovka Notebook Problem 18.54

O / open | Kourovka 群论问题集

Can a group be equal to the union of conjugates of a proper finite nonabelian simple subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03987 | KOU-18.55

### Kourovka Notebook Problem 18.55

O / open | Kourovka 群论问题集

a) Can a locally finite $p$-group $G$ of finite exponent be the union of conjugates of an abelian proper subgroup?

b) Can this happen when $G$ is of exponent $p$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03988 | KOU-18.56

### Kourovka Notebook Problem 18.56

O / open | Kourovka 群论问题集

Let $G$ be a finite 2-group, of order greater than 2, such that $|H/H_G| \le 2$ for all $H\le G$, where $H_G$ denotes the largest normal subgroup of $G$ contained in $H$. Must $G$ have an abelian subgroup of index 4?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03989 | KOU-18.57

### Kourovka Notebook Problem 18.57

S / solved | Kourovka 群论问题集

Let $G$ be a finite 2-group generated by involutions in which $[x,u,u]=1$ for every $x\in G$ and every involution $u\in G$. Is the derived length of $G$ bounded?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03990 | KOU-18.58

### Kourovka Notebook Problem 18.58

O / open | Kourovka 群论问题集

Let $G$ be a group generated by finite number $n$ of involutions in which $(uv)^4=1$ for all involutions $u,v\in G$. Is it true that $G$ is finite? is a 2-group? This is true for $n\leq 3$.

*Editors' comment of 2021*: the previously published claim of an affirmative solution of this problem proved to be erroneous.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03991 | KOU-18.59

### Kourovka Notebook Problem 18.59

S / solved | Kourovka 群论问题集

Does there exist a periodic group $G$ such that $G$ contains an involution, all involutions in $G$ are conjugate, and the centralizer of every involution $i$ is isomorphic to $\langle i\rangle \times L_2(P)$, where $P$ is some infinite locally finite field of characteristic $2$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03992 | KOU-18.6

### Kourovka Notebook Problem 18.6

O / open | Kourovka 群论问题集

Is every finitely generated one-relator group residually amenable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03993 | KOU-18.60

### Kourovka Notebook Problem 18.60

S / solved | Kourovka 群论问题集

Let $V$ be an infinite countable elementary abelian additive 2-group. Does $\mathrm{Aut} V$ contain a subgroup $G$ such that

(a) $G$ is transitive on the set of non-zero elements of $V$, and

(b) if $H$ is the stabilizer in $G$ of a non-zero element $v\in V$, then $V=\langle v\rangle \oplus V_v$, where $V_v$ is $H$-invariant, $H$ is isomorphic to the multiplicative group $P^*$ of a locally finite field $P$ of characteristic $2$, and the action $H$ on $V_v$ is similar to the action of $P^*$ on $P$ by multiplication?

*Conjecture:* such a group $G$ does not exist. If so, then the group $G$ in the previous problem does not exist too.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03994 | KOU-18.61

### Kourovka Notebook Problem 18.61

O / open | Kourovka 群论问题集

Is a 2-group nilpotent if all its finite subgroups are nilpotent of class at most 3? This is true if 3 is replaced by 2.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03995 | KOU-18.62

### Kourovka Notebook Problem 18.62

O / open | Kourovka 群论问题集

The *spectrum* of a finite group is the set of orders of its elements. Let $\omega$ be a finite set of positive integers. A group $G$ is said to be *$\omega$-critical* if the spectrum of $G$ coincides with $\omega$, but the spectrum of every proper section of $G$ is not equal to $\omega$.

a) Does there exist a number $n$ such that for every finite simple group $G$ the number of $\omega(G)$-critical groups is less than $n$?

b) For every finite simple group $G$, find all $\omega(G)$-critical groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03996 | KOU-18.63

### Kourovka Notebook Problem 18.63

O / open | Kourovka 群论问题集

Let $G$ be a periodic group generated by two fixed-point-free automorphisms of order 5 of an abelian group. Is $G$ finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

03997 | KOU-18.64

### Kourovka Notebook Problem 18.64

O / open | Kourovka 群论问题集

(K. Harada). *Conjecture*: Let $G$ be a finite group, $p$ a prime, and $B$ a $p$-block of $G$. If $J$ is a non-empty subset of $Irr(B)$ such that $\sum\limits_{\chi \in J}\chi (1)\chi (g)=0$ for every $p$-singular element $g\in G$, then $J=Irr(B)$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03998 | KOU-18.65

### Kourovka Notebook Problem 18.65

O / open | Kourovka 群论问题集

(R. Guralnick, G. Malle). *Conjecture*: Let $p$ be a prime different from 5, and $C$ a class of conjugate $p$-elements in a finite group $G$. If $[c,d]$ is a $p$-element for any $c,d\in C$, then $C\subseteq O_p(G)$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

03999 | KOU-18.66

### Kourovka Notebook Problem 18.66

O / open | Kourovka 群论问题集

Suppose that a finite group $G$ admits a Frobenius group of automorphisms $FH$ with kernel $F$ and complement $H$ such that $GF$ is also a Frobenius group with kernel $G$ and complement $F$. Is the derived length of $G$ bounded in terms of $|H|$ and the derived length of $C_G(H)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04000 | KOU-18.67

### Kourovka Notebook Problem 18.67

O / open | Kourovka 群论问题集

Suppose that a finite group $G$ admits a Frobenius group of automorphisms $FH$ with kernel $F$ and complement $H$ such that $C_G(F)=1$. Is the exponent of $G$ bounded in terms of $|F|$ and the exponent of $C_G(H)$?

This was proved when $F$ is cyclic (E. I. Khukhro, N. Y. Makarenko, P. Shumyatsky, *Forum Math.*, **26** (2014), 73--112) and for $FH\cong {\mathbb{A}_4$ when $(|G|,3)=1$ (P. Shumyatsky, *J. Algebra*, **331** (2011), 482--489).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04001 | KOU-18.68

### Kourovka Notebook Problem 18.68

O / open | Kourovka 群论问题集

What are the nonabelian composition factors of a finite nonsoluble group all of whose maximal subgroups have complements?

Note that finite simple groups with all maximal subgroups having complements are, up to isomorphism, $L_2(7)$, $L_2(11)$, and $L_5(2)$ (V. M. Levchuk, A. G. Likharev, *Siberian Math. J.*, **47**, no. 4 (2006), 659--668). The same groups exhaust finite simple groups with Hall maximal subgroups and composition factors of groups with Hall maximal subgroups (N. V. Maslova, *Siberian Math. J.*, **53**, no. 5 (2012), 853--861). It is also proved that in a finite group with Hall maximal subgroups all maximal subgroups have complements (N. V. Maslova, D. O. Revin, *Siberian Adv. Math.*, **23**, no. 3 (2013), 196--209).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04002 | KOU-18.69

### Kourovka Notebook Problem 18.69

O / open | Kourovka 群论问题集

Does there exist a relatively free group $G $ containing a free subsemigroup and having $[G,G]$ finitely generated?

Note that $G$ cannot be locally graded (*Publ. Math. Debrecen*, **81**, no. 3-4 (2012), 415--420.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

04003 | KOU-18.7

### Kourovka Notebook Problem 18.7

O / open | Kourovka 群论问题集

Let $n\geq 665$ be an odd integer. Is it true that the group of outer automorphisms $\mathrm{Out}(B(m,n))$ of the free Burnside group $B(m,n)$ for $m>2$ is complete, that is, has trivial center and all its automorphisms are inner?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04004 | KOU-18.70

### Kourovka Notebook Problem 18.70

O / open | Kourovka 群论问题集

Is every finitely generated Coxeter group conjugacy separable?

Some special cases were considered in (P.-E. Caprace, A. Minasyan, *Illinois J. Math.*, **57**, no. 2 (2013), 499--523).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04005 | KOU-18.71

### Kourovka Notebook Problem 18.71

S / solved | Kourovka 群论问题集

(N. Aronszajn). Suppose that $W(x, y) = 1$ in an open subset of $G\times G$, where $G$ is a connected topological group. Must $W(x, y) = 1$ for all $(x, y)\in G\times G$?

For locally compact groups the answer is yes.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04006 | KOU-18.72

### Kourovka Notebook Problem 18.72

O / open | Kourovka 群论问题集

Is it true that an existentially closed subgroup of a nonabelian free group of finite rank is a nonabelian free factor of this group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04007 | KOU-18.73

### Kourovka Notebook Problem 18.73

P / partially_solved | Kourovka 群论问题集

a) Does every finitely generated solvable group of derived length $l\ge 2$ embed into a $2$-generated solvable group of length $l+1$?

*Comment of 2021*: It is proved that any countable solvable group of derived length $l$ with torsion-free abelianization embeds in a 2-generated solvable group of derived length $l+1$ (V. A. Roman'kov, *Proc. Amer. Math. Soc.*, **149** (2021), 4133--4143).

[Archived solved part(s) of 18.73:] b) Does every finitely generated solvable group of derived length $l\ge 2$ embed into some $k$-generated $(l+1)$-solvable group, where $k=k(l)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04008 | KOU-18.74

### Kourovka Notebook Problem 18.74

O / open | Kourovka 群论问题集

Let $G$ be a finitely generated elementary amenable group which is not virtually nilpotent. Is there a finitely generated metabelian non-virtually-nilpotent section in $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04009 | KOU-18.75

### Kourovka Notebook Problem 18.75

S / solved | Kourovka 群论问题集

Does every finite solvable group $G$ have the following property: there is a number $d=d(G)$ such that $G$ is a homomorphic image of every group with $d$ generators and one relation?

This property holds for finite nilpotent groups and does not hold for every non-solvable finite group; see (S. A. Zaĭtsev,

*Moscow Univ. Math. Bull.*, **52**, no. 4 (1997), 42--44).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04010 | KOU-18.76

### Kourovka Notebook Problem 18.76

S / solved | Kourovka 群论问题集

Let $A$ be a division ring, $G$ a subgroup of the multiplicative group of $A$, and $E$ an extension of the additive group of $A$ by $G$ such that $G$ acts by multiplication in $A$. Is it true that $E$ splits? This is true if $G$ is finite.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04011 | KOU-18.77

### Kourovka Notebook Problem 18.77

O / open | Kourovka 群论问题集

Let $G$ be a finite $p$-group and let $p^e$ be the largest degree of an irreducible complex representation of $G$. If $p>e$, is it necessarily true that $\bigcap \ker \Theta= 1$, where the intersection runs over all irreducible complex representations $\Theta$ of $G$ of degree $p^e$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04012 | KOU-18.78

### Kourovka Notebook Problem 18.78

O / open | Kourovka 群论问题集

Let $K^tG$ be a twisted group algebra of the finite group $G$ over the field $K$. If $K^tG$ is a simple $K$-algebra, is $G$ necessarily solvable? This is known to be true if $K^tG$ is central simple.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04013 | KOU-18.79

### Kourovka Notebook Problem 18.79

O / open | Kourovka 群论问题集

Let $K[G]$ be the group algebra of the finitely generated group $G$ over the field $K$. Is the Jacobson radical ${\mathcal{J}}K[G]$ equal to the join of all nilpotent ideals of the ring? This is known to be true if $G$ is solvable or linear.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04014 | KOU-18.8

### Kourovka Notebook Problem 18.8

O / open | Kourovka 群论问题集

Let $\mathcal{X}$ be a class of finite simple groups such that $\pi(\mathcal{X}) = char(\mathcal{X})$. A formation of finite groups $\mathfrak{F}$ is said to be $\mathcal{X}$-saturated if a finite group $G$ belongs to $\mathfrak{F}$ whenever the factor group $G/\Phi(O_{\mathcal{X}}(G))$ is in $\mathfrak{F}$, where $O_{\mathcal{X}}(G)$ is the largest normal subgroup of $G$ whose composition factors are in $\mathcal{X}$. Is every $\mathcal{X}$-saturated formation $\mathcal{X}$-local in the sense of F\"orster? (See the definition in (P. F\"orster, *Publ. Sec. Mat. Univ. Aut\`onoma Barcelona*, **29**, no. 2--3 (1985), 39--76).)

问题集入口：https://kourovkanotebookorg.wordpress.com/

04015 | KOU-18.80

### Kourovka Notebook Problem 18.80

O / open | Kourovka 群论问题集

(I. Kaplansky). For $G\neq 1$, show that the augmentation ideal of the group algebra $K[G]$ is equal to the Jacobson radical of the ring if and only if $\mathrm{char} K=p>0$ and $G$ is a locally finite $p$-group.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04016 | KOU-18.81

### Kourovka Notebook Problem 18.81

O / open | Kourovka 群论问题集

Let $G$ be a finitely generated $p$-group that is residually finite. Are all maximal subgroups of $G$ necessarily normal?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04017 | KOU-18.82

### Kourovka Notebook Problem 18.82

S / solved | Kourovka 群论问题集

Is there a function $f:{\mathbb{N} \to {\mathbb{N}}$ such that for any prime $p$, if $p^{f(n)}$ divides the order of a finite group $G$, then $p^n$ divides the order of $\mathrm{Aut} G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04018 | KOU-18.83

### Kourovka Notebook Problem 18.83

O / open | Kourovka 群论问题集

A generating system $X$ of a group $G$ is *fast* if there is an integer $n$ such that every element of $G$ can be expressed as a product of at most $n$ elements of $X$ or their inverses. If not, we say that it is *slow*. For instance, in $(\mathbb{Z} ,+)$, the squares are fast, but the powers of 2 are slow.

Do there exist countable infinite groups without an infinite slow generating set? Uncountable ones do exist.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04019 | KOU-18.84

### Kourovka Notebook Problem 18.84

O / open | Kourovka 群论问题集

Let $\pi$ be a set of primes. We say that a finite group is a $BS_\pi$-group if every conjugacy class in this group any two elements of which generate a $\pi$-subgroup itself generates a $\pi$-subgroup. Is every normal subgroup of a $BS_\pi$-group a $BS_\pi$-group?

In the case $2\not\in\pi$, an affirmative answer follows from (D. O. Revin, *Siberian Math. J.*, **52**, no. 2 (2011), 340--347).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04020 | KOU-18.85

### Kourovka Notebook Problem 18.85

O / open | Kourovka 群论问题集

A subset of a group is said to be *rational* if it can be obtained from finite subsets by finitely many rational operations, that is, taking union, product, and the submonoid generated by a set.

*Conjecture*: every finitely generated solvable group in which all rational subsets form a Boolean algebra is virtually abelian.

This is known to be true if the group is metabelian, or polycyclic, or of finite rank (G. A. Bazhenova).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04021 | KOU-18.86

### Kourovka Notebook Problem 18.86

S / solved | Kourovka 群论问题集

Is the group $G = \langle a, b \mid [[a,b],b]=1\rangle$, which is isomorphic to the group of all unitriangular automorphisms of the free group of rank 3, linear?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04022 | KOU-18.87

### Kourovka Notebook Problem 18.87

O / open | Kourovka 群论问题集

A system of equations with coefficients in a group $G$ is said to be *independent* if the matrix composed of the sums of exponents of the unknowns has rank equal to the number of equations.

a) The *Kervaire--Laudenbach Conjecture (KLC)*: every independent system of equations with coefficients in an arbitrary group $G$ has a solution in some overgroup $\bar{G}$. This is true for every locally residually finite group $G$ (M. Gerstenhaber and O. S. Rothaus).

b) KLC --- nilpotent version: every independent system of equations with coefficients in an arbitrary nilpotent group $G$ has a solution in some nilpotent overgroup $\bar{G}$.

c) KLC --- solvable version: every independent system of equations with coefficients in an arbitrary solvable group $G$ has a solution in some solvable overgroup $\bar{G}$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04023 | KOU-18.88

### Kourovka Notebook Problem 18.88

O / open | Kourovka 群论问题集

Can a finitely generated infinite group of finite exponent be the quotient of a residually finite group by a locally finite normal subgroup?

If not, then there exists a hyperbolic group that is not residually finite.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04024 | KOU-18.89

### Kourovka Notebook Problem 18.89

S / solved | Kourovka 群论问题集

Consider the set of balanced presentations $\langle x_1,\dots ,x_n\mid r_1,\dots, r_n\rangle $ with fixed $n$ generators $x_1,\dots, x_n$. By definition, the group ${AC}_n$ of Andrews--Curtis moves on this set of balanced presentations is generated by the Nielsen transformations together with conjugations of relators. Is ${AC}_n$ finitely presented?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04025 | KOU-18.9

### Kourovka Notebook Problem 18.9

S / solved | Kourovka 群论问题集

Does there exist a subgroup-closed saturated formation $\mathfrak{F}$ of finite groups properly contained in $\mathfrak{E}_{\pi}$, where $\pi = char(\mathcal{F})$, satisfying the following property: if $G \in \mathfrak{F}$, then there exists a prime $p$ (depending on the group $G$) such that the wreath product $C_p\wr G$ belongs to $\mathfrak{F}$, where $C_p$ is the cyclic group of order $p$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04026 | KOU-18.90

### Kourovka Notebook Problem 18.90

O / open | Kourovka 群论问题集

Prove that the group $Y(m,n) =\langle a_{1},a_{2},\dots,a_{m} \mid a_{k}^{n} =1 \text{ }(1\leq k\leq m)\text{, } ( a_{k}^{i}a_{l}^{i}) ^{2} =e\text{ }(1\leq k<l\leq m\text{, } 1\leq i\leq \frac{n}{2})\rangle $ is finite for every pair $( m,n) $.

Some known cases are (where $C_n$ denotes a cyclic group of order $n$): $Y(m,2)=C_{2}^{m}$; $Y(m;3)={A}_{m+2}$ (presentation by Carmichael); $Y(m;4)$ has order $2^{\frac{m(m+3)}{2}}$, nilpotency class $3$ and exponent $4$; \ $Y(2,n)$ is the natural extension of the augmentation ideal $\omega ( C_{n})$ of $GF(2)[C_{n}]$ by $C_{n}$ (presentation by Coxeter); $Y(3,n)\cong SL(2,\omega ( C_{n}) )$.

These groups have connections with classical groups in characteristic $2$. When $n$ is odd, the group $Y(m,n)$ has the following presentation, where $\tau _{ij}$ are transpositions: $y(m,n) =\langle a,\text{ }{\mathbb{S}}_{m}\text{ }\mid a^{n}=e, [ \tau _{12}^{a^{i}},\tau _{12}] =e\; ( 1\leq i\leq \frac{n}{2}) , \tau _{12}^{1+a+\dots+a^{n-1}} =e,\text{ } \tau _{i,i+1}a\tau _{i,i+1} =a^{-1}\text{ }( 2\leq i\leq m-1) \rangle$, and it is known that, for example, $y(3,5) \cong SL(2,16)\cong \Omega ^{-}(4,4)$, $y(4,5)\cong Sp(2,16)\cong \Omega (5,4)$, $y(5,5) \cong SU(4,16)\cong \Omega ^{-}(6,4)$, $y(6,5)\cong 4^{6}\Omega ^{-}(6,4)$, $ y(3,7) \cong SL(2,8)^{2}\cong \Omega ^{+}(4,8)$, $y(4,7)\cong Sp(4,8)\cong \Omega (5,8)$. Extensive computations by Felsch, Neub\"user and O'Brien confirm this trend. Different types reveal Bott periodicity and a connection with Clifford algebras.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04027 | KOU-18.91

### Kourovka Notebook Problem 18.91

S / solved | Kourovka 群论问题集

A subgroup $H$ of a group $G$ is said to be *propermutable* in $G$ if there is a subgroup $B\leq G$ such that $G=N_{G}(H)B$ and $H$ permutes with every subgroup of $B$.

a) Is there a finite group $G$ with subgroups $A\leq B\leq G$ such that $A$ is propermutable in $G$ but $A$ is not propermutable in $B$?

b) Let $P$ be a non-abelian Sylow 2-subgroup of a finite group $G$ with $|P|=2^{n}$. Suppose that there is an integer $k$ such that $1 < k < n$ and every subgroup of $P$ of order $2^{k}$ is propermutable in $G$, and also, in the case of $k=1$, every cyclic subgroup of $P$ of order 4 is propermutable in $G$. Is it true that then $G$ is 2-nilpotent?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04028 | KOU-18.92

### Kourovka Notebook Problem 18.92

S / solved | Kourovka 群论问题集

A non-empty set $\theta$ of formations is called a *complete lattice* of formations if the intersection of any set of formations in $\theta$ belongs to $\theta$ and $\theta$ has the largest element (with respect to inclusion). If $L$ is a complete lattice, then an element $a\in L$ is said to be compact if $ a\leq \bigvee X$ for any $X\subseteq L$ implies that $a\leq\bigvee X_1$ for some finite $X_1\subset X$. A complete lattice is called *algebraic* if every element is the join of a (possibly infinite) set of compact elements.

a) Is there a non-algebraic complete lattice of formations of finite groups?

b) Is there a non-modular complete lattice of formations of finite groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04029 | KOU-18.93

### Kourovka Notebook Problem 18.93

O / open | Kourovka 群论问题集

Let $\mathcal{M}$ be a one-generated saturated formation, that is, the intersection of all saturated formations containing some fixed finite group. Let $\mathcal{F}$ be a subformation of $\mathcal{M}$ such that $\mathcal{F} \ne {\mathcal{F}}{\mathcal{F}}$.

a) Is it true that then $\mathcal{F}$ can be written in the form ${\mathcal{F}}={\mathcal{F}}_{1} \cdots {\mathcal{F}}_{t}$, where ${\mathcal{F}}_{i}$ is a non-decomposable formation for every $i=1, \ldots , t$?

b) Suppose that ${\mathcal{F}}={\mathcal{F}}_{1} \cdots {\mathcal{F}}_{t}$, where ${\mathcal{F}}_{i}$ is a non-decomposable formation for every $i=1, \ldots , t$. Is it true that then all factors ${\mathcal{F}}_{i}$ are uniquely determined?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04030 | KOU-18.94

### Kourovka Notebook Problem 18.94

S / solved | Kourovka 群论问题集

Let $G$ be a group without involutions, $a$ an element of it that is not a square of any element of $G$, and $n$ an odd positive integer. Is it true that the quotient $G/\langle (a^{n})^{G}\rangle $ does not contain involutions?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04031 | KOU-18.95

### Kourovka Notebook Problem 18.95

S / solved | Kourovka 群论问题集

Suppose that a group $G=AB$ is a product of an abelian subgroup $A$ and a locally quaternion group $B$ (that is, $B$ is a union of an increasing chain of finite generalized quaternion groups). Is $G$ soluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04032 | KOU-18.96

### Kourovka Notebook Problem 18.96

O / open | Kourovka 群论问题集

Suppose that a periodic group $G$ contains involutions and the centralizer of each involution is locally finite. Is it true that $G$ has a nontrivial locally finite normal subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04033 | KOU-18.97

### Kourovka Notebook Problem 18.97

O / open | Kourovka 群论问题集

Let $G$ be a periodic Zassenhaus group, that is, a two-transitive permutation group with trivial stabilizer of every three points. Suppose that the stabilizer of a point is a Frobenius group with locally finite kernel $U$ containing an involution. Is it true that $U$ is a $2$-group? This was proved for finite groups by Feit.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04034 | KOU-18.98

### Kourovka Notebook Problem 18.98

O / open | Kourovka 群论问题集

The work of many authors shows that most finite simple groups are generated by two elements of orders 2 and 3; for example, see the survey http://math.nsc.ru/conference/groups2013/slides/MaximVsemirnov_slides.pdf. Which finite simple groups cannot be generated by two elements of orders 2 and 3?

In particular, is it true that, among classical simple groups of Lie type, such exceptions, apart from $PSU(3,5^2)$, arise only when the characteristic is $2$ or $3$?

The question of which finite simple groups are (2,3)-generated remains open only for the orthogonal groups of even dimension $2m>8$ (M. A. Pellegrini, M. C. Tamburini Bellani, *J. Austral. Math. Soc.*, **117** (2024), 130--148).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04035 | KOU-18.99

### Kourovka Notebook Problem 18.99

O / open | Kourovka 群论问题集

Let $\mathrm{cd}(G)$ be the set of all complex irreducible character degrees of a finite group $G$. *Huppert's Conjecture:* if $H$ is a nonabelian simple group and $G$ is any finite group such that $\mathrm{cd}(G)=\mathrm{cd(H)}$, then $G\cong H\times A$, where $A$ is an abelian group.

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04036 | KOU-19.1

### Kourovka Notebook Problem 19.1

O / open | Kourovka 群论问题集

An element $g$ of a group $G$ is a *non-near generator* of $G$ if for every subset $S\subseteq G$ such that $|G:\langle g,S\rangle|<\infty$ it follows that $|G:\langle S\rangle|<\infty$. The set of all non-near generators of $G$ forms a characteristic subgroup of $G$ called the *lower near Frattini subgroup* of $G$, denoted by $\lambda(G)$. A subgroup $M\leq G$ is *nearly maximal* in $G$ if it is maximal with respect to being of infinite index in $G$. The intersection of all nearly maximal subgroups forms a characteristic subgroup called the *upper near Frattini subgroup* of $G$, denoted by $\mu(G)$. In general, $\lambda(G)\leq\mu(G)$. If $\lambda(G)=\mu(G)$, then this subgroup is called the *near Frattini subgroup* of $G$, denoted by $\psi(G)$.

a) Is it true that $\psi(G)=1$ if $G$ is the knot group of any product of knots?

b) Is it true that $\psi(G)=1$ if $G$ is a cable knot group?

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04037 | KOU-19.10

### Kourovka Notebook Problem 19.10

P / partially_solved | Kourovka 群论问题集

a) Is it possible to embed a finitely generated non-abelian free pro-$p$ group as an open subgroup of a simple totally disconnected locally compact group?

b) If the answer is yes, can we require that the simple envelope is also compactly generated?

*Comment of 2025*: Partial progress on part b) is made in (Y. Barnea, M. Ershov, A. Le Boudec, C. D. Reid, M. Vannacci, T. Weigel, *Preprint*, 2025, https://arxiv.org/pdf/2507.04120).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04038 | KOU-19.100

### Kourovka Notebook Problem 19.100

S / solved | Kourovka 群论问题集

Suppose that a finite group $G$ admits a factorization $G=AB=BC=CA$, where $A, B, C$ are abnormal supersoluble subgroups. Is $G$ supersoluble?

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04039 | KOU-19.101

### Kourovka Notebook Problem 19.101

S / solved | Kourovka 群论问题集

The maximum length of a chain of nested centralizers of a group is called its $c$-dimension. Let $G$ be a locally finite group of finite $c$-dimension $k$, and let $S$ be the preimage in $G$ of the socle of $G/R$, where $R$ is the locally solvable radical of $G$. Is it true that the factor group $G/S$ contains an abelian subgroup of index bounded by a function of $k$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04040 | KOU-19.102

### Kourovka Notebook Problem 19.102

S / solved | Kourovka 群论问题集

A subgroup $H$ of a free group $F$ is called *inert* if $r(H\cap K)\leq r(K)$ for every $K\leq F$; and *compressed* if $r(H)\leq r(K)$ for every $H\leq K\leq F$.

Is it true that compressed subgroups are inert?

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04041 | KOU-19.103

### Kourovka Notebook Problem 19.103

S / solved | Kourovka 群论问题集

Is it true that an intersection of compressed subgroups is compressed?

It is known that arbitrary intersections of inert subgroups are inert.

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04042 | KOU-19.104

### Kourovka Notebook Problem 19.104

S / solved | Kourovka 群论问题集

Is there an algorithm which decides whether a given subgroup of $F$ is inert?

An algorithm to decide whether a given $H$ is compressed is known.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04043 | KOU-19.105

### Kourovka Notebook Problem 19.105

S / solved | Kourovka 群论问题集

Is it true that the fixed subgroups of endomorphisms of $F$ are inert?

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04044 | KOU-19.106

### Kourovka Notebook Problem 19.106

O / open | Kourovka 群论问题集

Let $G$ be a uniformly locally finite group (which means that there is a function $f$ on the natural numbers such that any subgroup generated by $n$ elements has size at most $f(n)$). Suppose that for any two definable subgroups $H$ and $K$, the intersection $H\cap K$ has finite index either in $H$ or in $K$. Is $G$ necessarily nilpotent-by-finite? Or even finite-by-abelian-by-finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04045 | KOU-19.107

### Kourovka Notebook Problem 19.107

O / open | Kourovka 群论问题集

Let $\mathscr{AE}$ be the smallest class of locally compact groups such that \begin{itemize} \item[(1)] $\mathscr{AE}$ contains the compact groups and the discrete amenable groups; \item[(2)] $\mathscr{AE}$ is closed under taking closed subgroups, Hausdorff quotients, directed unions of open subgroups, and group extensions. \end{itemize} Is every amenable locally compact group an element of $\mathscr{AE}$?

(A negative answer would follow from a positive answer to 19.21.)

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04046 | KOU-19.108

### Kourovka Notebook Problem 19.108

O / open | Kourovka 群论问题集

Let $P$ be a finite $p$-group, where $p$ is an odd prime. Let $\chi$ be a complex irreducible character of $P$. If $\chi(x)\ne 0$ for some $x\in P$, is it true that the order of $x$ must divide $|P|/\chi(1)^2$?

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04047 | KOU-19.109

### Kourovka Notebook Problem 19.109

S / solved | Kourovka 群论问题集

A subgroup $H$ of a finite group $G$ is called pronormal if for any $g\in G$ the subgroups $H$ and $H^g$ are conjugate by an element of $\langle H,H^g\rangle$. A maximal subgroup of a maximal subgroup is called second maximal. Is it true that in a non-abelian finite simple group $G$ all maximal subgroups are Hall subgroups if and only if every second maximal subgroup of $G$ is pronormal in $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04048 | KOU-19.11

### Kourovka Notebook Problem 19.11

O / open | Kourovka 群论问题集

Does there exist a constant $c$ such that the number of conjugacy classes in a finite group $G$ is always at least $c \log_2 |G|$?

*Editors' comment of 2021*: It is proved that every group $G$ contains at least $\varepsilon \log |G|/(\log \log |G|)^8$ conjugacy classes for some fixed $\varepsilon > 0$ (L. Pyber, *J. London Math. Soc. (2)*, **46**, no. 2 (1992), 239--249). It is also proved that for every $\varepsilon > 0$ there exists $\delta > 0$ such that every finite group $G$ of order at least 3 has at least $\delta \log _2|G|/(\log_2 \log _2|G|)^{3+\varepsilon}$ conjugacy classes (B. Baumeister, A. Mar\'oti, H. P. Tong-Viet, *Forum Math.*, **29**, no. 2 (2017), 259--275).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04049 | KOU-19.110

### Kourovka Notebook Problem 19.110

O / open | Kourovka 群论问题集

(G. M. Bergman and A. Magidin). Do there exist varieties of groups in which the relatively free group of rank 2 is finite, and the relatively free group of rank 3 is infinite?

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04050 | KOU-19.111

### Kourovka Notebook Problem 19.111

O / open | Kourovka 群论问题集

Do there exist infinite groups all of whose proper subgroups are cyclic of order $p$ that do not satisfy any nontrivial group identity except $x^p = 1$ and its consequences?

Note that there exist infinite groups all of whose proper subgroups are infinite cyclic that do not satisfy any nontrivial group identity (due to A. Olshanskii; see P. Zusmanovich, *J. Algebra*, **388** (2013), 268--286, Remark after Theorem 6.1).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04051 | KOU-19.12

### Kourovka Notebook Problem 19.12

O / open | Kourovka 群论问题集

A finite group $G$ is called conjugacy-expansive if for every normal subset $N$ and conjugacy class $C$ of $G$ the normal set $NC$ contains at least as many conjugacy classes of $G$ as $N$ does. Is it true that every finite simple group is conjugacy-expansive?

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04052 | KOU-19.13

### Kourovka Notebook Problem 19.13

O / open | Kourovka 群论问题集

The well-known Baer--Suzuki theorem states that if every two conjugates of an element $a$ of a finite group $G$ generate a finite $p$-subgroup, then $a$ is contained in a normal $p$-subgroup. Does such a theorem hold in the class of periodic groups for $p>2$?

A counterexample is known for $p=2$; see Archive 11.11a).

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04053 | KOU-19.14

### Kourovka Notebook Problem 19.14

O / open | Kourovka 群论问题集

Let $G$ be a finite group such that the quasivariety generated by all its Sylow subgroups contains only finitely many subquasivarieties. Is it true that the quasivariety generated by $G$ also contains only finitely many subquasivarieties?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04054 | KOU-19.15

### Kourovka Notebook Problem 19.15

O / open | Kourovka 群论问题集

For a class of groups $M$, let $L(M)$ denote the class of all groups $G$ in which the normal closure $\langle a^G\rangle $ of any element $a\in G$ is contained in $M$. Let $qA$ be the quasivariety generated by a finite nilpotent group $A$. Can the quasivariety $L(qA)$ contain a non-nilpotent group?

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04055 | KOU-19.16

### Kourovka Notebook Problem 19.16

O / open | Kourovka 群论问题集

(C. Drutu and M. Sapir). Is every free-by-cyclic group of the form $F_n \rtimes \mathbb{Z}$ linear?

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04056 | KOU-19.17

### Kourovka Notebook Problem 19.17

O / open | Kourovka 群论问题集

(Well-known problem). Suppose that $G$ is a finitely presented group such that the set of first Betti numbers over all the finite index subgroups of $G$ is unbounded above. (Here, the first Betti number is the maximum $n$ for which there is a surjective homomorphism onto $\mathbb{Z}^n$.)

a) Must $G$ have a finite index subgroup with a surjective homomorphism to a non-abelian free group?

b) Can $G$ even be soluble?

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04057 | KOU-19.18

### Kourovka Notebook Problem 19.18

O / open | Kourovka 群论问题集

Given an infinite set $\Omega$, define an algebra $A$ (the *reduced incidence algebra of finite subsets*) as follows. Let $V_n$ be the set of functions from the set of $n $-element subsets of $\Omega$ to the rationals $\mathbb{Q}$. Now let $A=\bigoplus V_n$, with multiplication as follows: for $f\in V_n$, $g\in V_m$, and $|X|=m+n$, let $(fg)(X)=\sum f(Y)g(X\setminus Y)$, where the sum is over the $n $-element subsets $Y$ of $X$. If $G$ is a permutation group on $\Omega$, let $A^G$ be the algebra of $G $-invariants in $A$.

Suppose that $G$ has no finite orbits on $\Omega$. It is known that then $A^G$ is an integral domain (M. Pouzet, *Theor. Inf. App.* **42**, no. 1 (2008), 83--103). Is it true that the quotient of $A^G$ by the ideal generated by constant functions is also an integral domain?

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04058 | KOU-19.19

### Kourovka Notebook Problem 19.19

O / open | Kourovka 群论问题集

A finite transitive permutation group $G$ is said to have the *road closure property* if, given any orbit $O$ of $G$ on $2$-sets, and any proper block of imprimitivity for $G$ acting on $O$, the graph with edge set $O\setminus B$ is connected. Such a group must be primitive, and basic (not contained in a wreath product with the product action); it cannot have an imprimitive subgroup of index $2$. In addition, such a group cannot be one of the permutation groups arising from triality (whose socle is $D_4(q)$ and intersects the point stabiliser in the parabolic subgroup corresponding to the three leaves in the Coxeter--Dynkin diagram for $D_4$).

Classify the basic primitive groups $G$ which do not have the road closure property. In particular, is it true that such a group either has a subgroup of index at most 3 or is almost simple?

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04059 | KOU-19.2

### Kourovka Notebook Problem 19.2

O / open | Kourovka 群论问题集

For a subgroup $L$ of a group $G$, let $L_G$ denote the largest normal subgroup of $G$ contained in $L$. If $M\vartriangleleft G$, then we say that $G$ *nearly splits* over $M$ if there is a subgroup $N\leq G$ such that $|G:N|=\infty$, $|G:MN|<\infty$, and $(M\cap N)_G=1$.

Let $G$ be any group, and $H$ a normal subgroup of prime order. Is it true that $\psi(G)\cap H=1$ if and only if $G$ nearly splits over $H$?

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04060 | KOU-19.20

### Kourovka Notebook Problem 19.20

S / solved | Kourovka 群论问题集

For a finite group $G$, let $\mathrm{End}(G)$ denote the semigroup of endomorphisms of $G$, and $\mathrm{PIso}(G)$ the semigroup of partial isomorphisms of $G$ (isomorphisms between subgroups of $G$). If $G$ is abelian, then $|\mathrm{End}(G)|=|\mathrm{PIso}(G)|$. Is the converse true?

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04061 | KOU-19.21

### Kourovka Notebook Problem 19.21

O / open | Kourovka 群论问题集

Can a non-discrete, compactly generated, topologically simple, locally compact group be amenable and non-compact?

(A positive answer implies a negative answer to 19.107.)

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04062 | KOU-19.22

### Kourovka Notebook Problem 19.22

S / solved | Kourovka 群论问题集

Let $\mathscr{S}$ denote the class of nondiscrete compactly generated, topologically simple totally disconnected locally compact groups. Two topological groups are called *locally isomorphic* if they contain isomorphic open subgroups. Is the number of local isomorphism classes of groups in $\mathscr{S}$ uncountable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04063 | KOU-19.23

### Kourovka Notebook Problem 19.23

S / solved | Kourovka 群论问题集

For a group $G$, let $\operatorname{Tor}_{1}(G)$ be the normal closure of all torsion elements of $G$, and then by induction let $\operatorname{Tor}_{i+1}(G)$ be the inverse image of $\operatorname{Tor}_{1}(G/\operatorname{Tor}_{i}(G))$. The torsion length of $G$ is defined to be either the least positive integer $l$ such that $G/\operatorname{Tor}_{l}(G)$ is torsion-free, or $\omega$ if no such integer exists (since $G/\bigcup \operatorname{Tor}_{i}(G)$ is always torsion-free).

Does there exists a finitely generated, or even finitely presented, soluble group with torsion length greater than $2$?

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04064 | KOU-19.24

### Kourovka Notebook Problem 19.24

S / solved | Kourovka 群论问题集

For a group $G$, let $\operatorname{Tor}(G)$ be the normal closure of all torsion elements of $G$. Does there exist a finitely presented group $G$ such that $G/\operatorname{Tor}(G)$ is not finitely presented? Such a group must necessarily be non-hyperbolic.

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04065 | KOU-19.25

### Kourovka Notebook Problem 19.25

S / solved | Kourovka 群论问题集

Let $G$ and $H$ be finite groups of the same order with $\sum_{g\in G}\varphi (|g|) = \sum_{h\in H} \varphi (|h|)$, where $\varphi$ is the Euler totient function. Suppose that $G$ is simple. Is $H$ necessarily simple?

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04066 | KOU-19.26

### Kourovka Notebook Problem 19.26

O / open | Kourovka 群论问题集

(Y. O. Hamidoune). Suppose that $A$ and $B$ are finite subsets of a group $G$ such that $\vert A \vert \ge 2$ and $\vert B \vert \ge 2$, and let $A\cdot_2B$ denote the set of elements of $G$ which can be expressed in the form $ab$ for at least two different $(a,b) \in A \times B$. Is it true that $\vert A \vert + \vert B \vert - (1/2) \vert AB \vert - (1/2) \vert A\cdot_2B\vert \le \max\{2, \vert gH \vert : H \le G,\; g \in G,\; gH \subseteq A\cdot_2B\}$?

This is proved if $G$ is abelian.

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04067 | KOU-19.27

### Kourovka Notebook Problem 19.27

O / open | Kourovka 群论问题集

(Well-known question). A finitely generated group $G$ that acts on a tree in such a way that all vertex and edge stabilizers are infinite cyclic groups is called a generalized Baumslag--Solitar group. The Bass--Serre theory gives finite presentations of such groups. Is the isomorphism problem soluble for generalized Baumslag--Solitar groups?

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04068 | KOU-19.28

### Kourovka Notebook Problem 19.28

O / open | Kourovka 群论问题集

Let $G$ be a group, and $\varphi$ an automorphism of $G$. Elements $x,y\in G$ are said to be $\varphi$-conjugate if $x=z^{-1}y\varphi (z)$ for some $z\in G$. The $\varphi$-conjugacy is an equivalence relation; the number of $\varphi$-conjugacy classes is denoted by $R(\varphi )$.

*Conjecture*: If a finitely generated residually finite group $G$ has an automorphism $\varphi$ such that $R(\varphi )$ is finite, then $G$ has a soluble subgroup of finite index.

The conjecture is proved if $\varphi$ has prime order (E. Jabara, *J. Algebra*, **320**, no. 10 (2008), 3671--3679). If $G$ is infinitely generated, then $G$ does not have to be almost solvable (K. Dekimpe, D. Gonçalves, *Bull. London Math. Soc.*, **46**, no. 4 (2014), 737--746), but if it is of finite upper rank, then it has to be almost solvable (E. Troitsky, *J. Group Theory*, **28**, no. 1 (2025), 151--164).

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04069 | KOU-19.29

### Kourovka Notebook Problem 19.29

O / open | Kourovka 群论问题集

Let $\Phi \in \mathrm{Out } G = \mathrm{Aut} G/\mathrm{Inn} G$. Two automorphisms $\varphi,\psi\in\Phi$ are said to be isogradient if $\varphi=\hat g^{-1}\psi\hat g$ for some inner automorphism $\hat g$. The number of isogradiency classes in $\Phi$ is denoted by $S(\Phi )$.

*Conjecture*: If a finitely generated residually finite group $G$ has an outer automorphism $\Phi$ such that $S(\Phi )$ is finite, then $G$ has a soluble subgroup of finite index.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04070 | KOU-19.3

### Kourovka Notebook Problem 19.3

O / open | Kourovka 群论问题集

Let $G=A\ast_{H}B$ be the generalized free product of groups $A$ and $B$ with amalgamated subgroup $H$. Is $\psi(G)=1$ in the following cases? \begin{itemize} \item[{a)}] $H$ is finite cyclic and $H_G=1$.

\item[{b)}] $H$ is finite cyclic and either $\lambda(A)\cap H_G=1$ or $\lambda(B)\cap H_G=1$.

\item[{c)}] $H_G=1$ and $H$ satisfies the minimum condition on subgroups.

\item[d)] $\lambda(G)\cap H=1$.

\item[{e)}] $A$ and $B$ are free groups, $H$ is finitely generated and at least one of $|A:H|$ or $|B:H|$ is infinite.

\item[{f)}] $H$ is infinite cyclic and is a retract of $A$ and $B$.

\item[{g)}] $G$ nearly splits over $H$, and $H$ is a normal subgroup of $G$ of prime order. \end{itemize}

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04071 | KOU-19.30

### Kourovka Notebook Problem 19.30

O / open | Kourovka 群论问题集

An element $g$ of a finite group $G$ is said to be vanishing if $\chi(g)=0$ for some irreducible complex character $\chi\in \mathrm{Irr}(G)$. Must a finite group and a finite simple group be isomorphic if they have equal orders and the same set of orders of vanishing elements?

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04072 | KOU-19.31

### Kourovka Notebook Problem 19.31

S / solved | Kourovka 群论问题集

Let $\omega(G,S)$ be the exponential growth rate of a group $G$ with a finite generating set $S$ (see 14.7). Let $MCG(\Sigma_g)$ be the mapping class group of the orientable surface $\Sigma_g$ of genus $g$. Is there a constant $C>1$ such that $\omega(MCG(\Sigma_g), S) \ge C$ for every $g >0$ and every finite generating set $S$ of $MCG(\Sigma_g)$?

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04073 | KOU-19.32

### Kourovka Notebook Problem 19.32

O / open | Kourovka 群论问题集

Let $p\geq 673$ be a prime and $r \geq 2$ be an integer. Let $B$ be the free group in the Burnside variety of exponent $p$ on $a_1, \dots , a_r$. Must every non-identical one-variable equation $w(a_1, \dots , a_r, x) = 1$ over $B$ have at most finitely many solutions in $B$?

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04074 | KOU-19.33

### Kourovka Notebook Problem 19.33

O / open | Kourovka 群论问题集

*Conjecture*: Let $G$ be a finite group, $p$ a prime number, and $P$ a Sylow $p$-subgroup of $G$. Suppose that an irreducible ordinary character $\chi$ of $G$ has degree divisible by $p$. If the restriction $\chi_P$ of $\chi$ to $P$ has a linear constituent, then $\chi_P$ has at least $p$ different linear constituents.

This conjecture has been verified for symmetric, alternating, $p$-solvable, and sporadic simple groups.

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04075 | KOU-19.34

### Kourovka Notebook Problem 19.34

O / open | Kourovka 群论问题集

Let $G$ be a finitely generated group such that for every element $g\in G$ the set of commutators $\{[x,g]\mid x\in G\}$ is a subgroup of $G$. Is it true that $G$ is residually nilpotent?

The answer is affirmative if $G$ is finite.

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04076 | KOU-19.35

### Kourovka Notebook Problem 19.35

S / solved | Kourovka 群论问题集

Let $G$ be a finite group of order $n$. Is it true that for every factorization $n=a_1 \cdots a_k$ there exist subsets $A_1,\dots ,A_k$ such that $|A_1|=a_1,\dots ,|A_k|=a_k$ and $G=A_1 \cdots A_k$?

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04077 | KOU-19.36

### Kourovka Notebook Problem 19.36

S / solved | Kourovka 群论问题集

Let $G$ be a periodic group and let $\mathscr{I}=\{ x \in G \mid x^{2}=1 \not =x \}$ be the set of its involutions. Let $D$ be a non-empty set of odd integers greater than $1$; then $G$ is called a group with $D$-involutions if $G=\langle \mathscr{I} \rangle$ and for $x,y \in \mathscr{I}$ the order of $xy$ is in the set $\{1,2\} \cup D$ and all these values actually occur. It is clear that if $G$ is a group with $D$-involutions, then $\mathscr{I}$ is a single conjugacy class.

*Conjecture:* If $G$ is a group with $\{3,5\}$-involutions, then $G \simeq A_{5}$ or $G \simeq PSU(3,4)$.

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04078 | KOU-19.37

### Kourovka Notebook Problem 19.37

S / solved | Kourovka 群论问题集

{a)} Does there exist an absolute constant $k$ such that for any nilpotent injector $H$ of an arbitrary finite group $G$ there are $k$ conjugates of $H$ the intersection of which is equal to the Fitting subgroup $F(G)$ of $G$?

b) Can one choose $k=3$ as such a constant? This is true for finite soluble groups (D. S. Passman, *Trans. Amer. Math. Soc.*, **123**, no. 1 (1966), 99--111; \ A. Mann, *Proc. Amer. Math. Soc.*, **53**, no. 1 (1975), 262--264).

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04079 | KOU-19.38

### Kourovka Notebook Problem 19.38

S / solved | Kourovka 群论问题集

Suppose that $H$ is a subgroup of a finite soluble group $G$ that covers all Frattini chief factors of $G$ and avoids all complemented chief factors of $G$. Is it true that there are elements $x,y\in G$ such that $H\cap H^x\cap H^y=H_G$, where $H_G$ is the largest normal subgroup of $G$ contained in $H$?

This is true if $H$ is a prefrattini subgroup of $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04080 | KOU-19.39

### Kourovka Notebook Problem 19.39

O / open | Kourovka 群论问题集

A group is said to be $\mathfrak{X}$-critical if it does not belong to $\mathfrak{X}$ but all its proper subgroups belong to $\mathfrak{X}$. Let $\mathfrak{F}$ be a soluble hereditary formation of finite groups for which all $\mathfrak{F}$-critical groups are either groups of prime order or $\mathfrak{U}$-critical groups, where $\mathfrak{U}$ is the formation of all supersoluble finite groups. Must $\mathfrak{F}$ be a saturated formation?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04081 | KOU-19.4

### Kourovka Notebook Problem 19.4

O / open | Kourovka 群论问题集

a) If $G$ is the free product of infinitely many finitely generated free groups with cyclic amalgamation, then is $\psi(G)=1$?

b) If $G$ is the free product of infinitely many finitely generated free abelian groups with cyclic amalgamation, then is $\psi(G)=1$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04082 | KOU-19.40

### Kourovka Notebook Problem 19.40

S / solved | Kourovka 群论问题集

Does Thompson's group $F$ (see 12.20) have the Howson property, that is, is the intersection of any two finitely generated subgroups of $F$ finitely generated?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04083 | KOU-19.41

### Kourovka Notebook Problem 19.41

S / solved | Kourovka 群论问题集

Let $\phi$ be an automorphism of a free group $F_n$ of finite rank, and let $F_n\rtimes_\phi \mathbb{Z}$ be the split extension of $F_n$ by $\mathbb{Z}$ with $\mathbb{Z}$ acting as $\langle \phi\rangle$. Is the group $F_n\rtimes_\phi \mathbb{Z}$ conjugacy separable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04084 | KOU-19.42

### Kourovka Notebook Problem 19.42

O / open | Kourovka 群论问题集

Suppose that $H$ is a word-hyperbolic subgroup of a word-hyperbolic group $G$ such that the inclusion of $H$ to $G$ extends to a continuous $H$-equivariant map $j:\partial H\to \partial G$ between their hyperbolic boundaries. If such an extension exists, it is unique and $j$ is called the *Cannon--Thurston map*.

a) Is it true that for every point $p\in \partial G$ its full preimage $j^{-1}(p)$ is finite?

b) Moreover, is it true that there is a number $N=N(G,H)<\infty$ such that for every $p\in \partial G$ the full preimage $j^{-1}(p)$ consists of at most $N$ points?

The answer to both questions is ``yes'' in all the cases where the Cannon--Thurston map is known to exist and where it has been possible to analyze the multiplicity of this map. This includes the original set-up considered by Cannon and Thurston where $H$ is the surface group, and $G$ is the fundamental group of a closed hyperbolic 3-manifold fibering over the circle with that surface as a fiber. In this case, $j:\mathbb{S}^1\to\mathbb{S}^2$ is a uniformly finite-to-one continuous surjective `Peano curve'.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04085 | KOU-19.43

### Kourovka Notebook Problem 19.43

S / solved | Kourovka 群论问题集

Suppose that $\varphi $ is an automorphism of a finite soluble group $G$. Is the Fitting height of $G$ bounded in terms of $|\varphi |$ and $|C_G(\varphi )|$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04086 | KOU-19.44

### Kourovka Notebook Problem 19.44

O / open | Kourovka 群论问题集

By definition a profinite group has finite rank at most $r$ if every subgroup of it can be (topologically) generated by $r$ elements. Suppose that for every element $g$ of a profinite group $G$ there is a closed subgroup $E_g$ of finite rank such that for every $x\in G$ all sufficiently long Engel commutators $[x,g,\dots, g]$ belong to $E_g$, that is, for every $x\in G$ there is a positive integer $n(x,g)$ such that $[x,g,\dots, g]\in E_g$ whenever $g$ is repeated $\geq n(x,g)$ times. Is it true that $G$ has a normal subgroup $N$ of finite rank with locally nilpotent quotient $G/N$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04087 | KOU-19.45

### Kourovka Notebook Problem 19.45

O / open | Kourovka 群论问题集

Let \(G\) be a group generated by 3 class transpositions (see the definition in 17.57), and let \(m\) be the least common multiple of the moduli of the residue classes interchanged by the generators of \(G\). Assume that \(G\) does not setwisely stabilize any union of residue classes modulo \(m\) except for \(\varnothing\) and \(\mathbb{Z}\), and assume that the integers \(0, 1, \dots, 42\) all lie in the same orbit under the action of \(G\) on \(\mathbb{Z}\). Is the action of \(G\) on \(\mathbb{N}\cup \{0\}\) necessarily transitive?

The bound 42 cannot be replaced by a smaller number, since the finite group \(\langle \tau_{0(2),1(2)}, \tau_{0(3),2(3)}, \tau_{0(7),6(7)} \rangle\) acts transitively on the set \(\{0, \dots, 41\}\), as well as on the set of residue classes modulo 42.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04088 | KOU-19.46

### Kourovka Notebook Problem 19.46

O / open | Kourovka 群论问题集

Does the group \(\mathrm{CT}(\mathbb{Z})\) have finitely generated infinite periodic subgroups? (See the definition of \(\mathrm{CT}(\mathbb{Z})\) in 17.57).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04089 | KOU-19.47

### Kourovka Notebook Problem 19.47

O / open | Kourovka 群论问题集

Let $K$ be a finite extension of degree $n$ of a field $k$ of odd characteristic. The multiplicative group $K^{*}$ embeds into the group of all $k$-linear automorphisms $\mathrm{Aut}_{k}(K)$ by the rule $t(x) = tx$ for all $x\in K$. For a fixed basis of $K$ over $k$, the group $\mathrm{Aut}_k(K)$ is isomorphic to $GL(n,k)$. The image of $K^{*}$ is called a *nonsplit maximal torus* corresponding to the extension $K/k$ and is denoted by $T(K/k)$. A subgroup of $GL(n,k)$ is said to be *rich in transvections* if it contains all elementary transvections.

Let $H$ be a subgroup of $GL(n, k)$ which contains $T(K/k)$ and a one-dimensional transformation. Is $H$ rich in transvections?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04090 | KOU-19.48

### Kourovka Notebook Problem 19.48

O / open | Kourovka 群论问题集

A system of additive subgroups $\sigma_{ij}$, $1\leq i,j\leq n$, of a field $K$ is called a *net* (or a *carpet*) of order $n$ if $\sigma_{ir}\sigma_{rj}\subset\sigma_{ij}$ for all $i, r, j$. A net that does not contain the diagonal is called an *elementary net*. A net $\sigma=(\sigma_{ij})$ is said to be *irreducible* if all $\sigma_{ij}$ are nontrivial. An elementary net $\sigma$ is said to be *closed* if the elementary net subgroup $E(\sigma)$ does not contain additional elementary transvections.

Let $R$ be a principal ideals domain with $1\in R$, let $k$ be the field of fractions of $R$, and $K$ an algebraic extension of the field $k$. Let $\sigma=(\sigma_{ij})$ be an irreducible elementary net $\sigma=(\sigma_{ij})$ over $K$ such that all $\sigma_{ij}$ are $R$-modules. Is the net $\sigma$ closed?

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04091 | KOU-19.49

### Kourovka Notebook Problem 19.49

S / solved | Kourovka 群论问题集

A *skew brace* is a set $B$ equipped with two operations $+$ and $\cdot$ such that $(B,+)$ is an additively written (but not necessarily abelian) group, $(B,\cdot )$ is a multiplicatively written group, and $a\cdot (b+c)=ab-a+ac$ for any $a,b,c\in B$.

Let $A$ be a skew brace with left-orderable multiplicative group. Is the additive group of $A$ left-orderable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04092 | KOU-19.5

### Kourovka Notebook Problem 19.5

O / open | Kourovka 群论问题集

If $G$ is the free product of infinitely many finitely generated abelian groups with amalgamated subgroup $H$, then is $\psi(G)$ equal to the torsion subgroup of $H$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04093 | KOU-19.50

### Kourovka Notebook Problem 19.50

S / solved | Kourovka 群论问题集

A finite graph is said to be integral if all eigenvalues of its adjacency matrix are integers.

a) Let $G$ be a finite group generated by a normal subset $R$ consisting of involutions. Is it true that the Cayley graph $Cay(G,R)$ is integral?

b) Let $A_n$ be the alternating group of degree $n$, let $S=\{(123), (124),\dots,(12n)\}$ and $R=S\cup S^{-1}$. Is it true that the Cayley graph $Cay(A_n,R)$ is integral?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04094 | KOU-19.51

### Kourovka Notebook Problem 19.51

O / open | Kourovka 群论问题集

A finite group is *monomial* if each irreducible character of it is induced from a linear character of some subgroup. Monomial groups are soluble (Taketa). The group is *normally* (*subnormally*) *monomial* if each irreducible character is induced from a linear character of some normal (respectively, subnormal) subgroup. Metabelian groups are normally monomial, and abelian-by-nilpotent groups are subnormally monomial. In the other direction, it is known that there exist normally monomial groups of arbitrarily large derived length.

a) For a given prime $p$, do there exist normally monomial finite $p$-groups of arbitrarily large derived length?

b) Do there exist subnormally monomial groups of arbitrarily large nilpotence length?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04095 | KOU-19.52

### Kourovka Notebook Problem 19.52

O / open | Kourovka 群论问题集

The Gruenberg--Kegel graph (or the prime graph) $GK(G)$ of a finite group $G$ has vertex set consisting of all prime divisors of the order of $G$, and different vertices $p$ and $q$ are adjacent in $GK(G)$ if and only if the number $pq$ is an element order in $G$. Is there a finite non-solvable group $G$ such that $GK(G)$ does not contain 3-cocliques and is not isomorphic to the Gruenberg--Kegel graph of any finite solvable group?

It is known that there are no examples of such groups $G$ among almost simple groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04096 | KOU-19.53

### Kourovka Notebook Problem 19.53

O / open | Kourovka 群论问题集

Let $G$ be a group generated by elements $x,y,z$ such that $$x^3=y^2=z^2=(xy)^3=(yz)^3=1\quad \text{and}\ \quad g^{12}=1 \text{ for all }g\in G.$$ Is it true that $|G|\leq 12$?

This is true if $G$ is finite (D. V. Lytkina, V. D. Mazurov, *Siberian Math. J.*, **56**, no. 3 (2015), 471--475).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04097 | KOU-19.54

### Kourovka Notebook Problem 19.54

O / open | Kourovka 群论问题集

What are the chief factors of a finite group in which every 2-maximal subgroup is not $m$-maximal for any $m\ge 3$?

A subgroup $H$ of a group $G$ is said to be $m$-maximal if there is a chain of subgroups $ H=H_0<H_1< \ldots < H_{m-1}< H_m=G$ in which $H_i$ is maximal in $H_{i+1}$ for every $i$. Note that for every $m\ge 3$ there is a finite group in which some 2-maximal subgroup is $m$-maximal.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04098 | KOU-19.55

### Kourovka Notebook Problem 19.55

S / solved | Kourovka 群论问题集

Suppose that in a finite group $G$ every maximal subgroup $M$ is supersoluble whenever $\pi (M)=\pi (G)$, where $\pi (G)$ is the set of all prime divisors of the order of $G$.

a) What are the non-abelian composition factors of $G$?

b) Determine the exact upper bounds for the nilpotency length, the rank, and the $p$-length of $G$ if $G$ is soluble.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04099 | KOU-19.56

### Kourovka Notebook Problem 19.56

O / open | Kourovka 群论问题集

A $\star$-commutator is a commutator $[x,y]$ of two elements $x,y$ of coprime prime-power orders. Is a finite group $G$ soluble if $|ab|\ge |a||b|$ for any $\star$\nobreakdash- commutators $a$ and $b$ of coprime orders?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04100 | KOU-19.57

### Kourovka Notebook Problem 19.57

S / solved | Kourovka 群论问题集

What are the non-abelian composition factors of a finite group $G$ in which every maximal subgroup is simple or $p$\nobreakdash- nilpotent for some fixed odd prime $p\in \pi (G)$?

*Comment of 2026*: These factors are determined in the cases where $G$ is non-simple, or alternating, or sporadic simple (A. Beltr\'an, F. Viudez, *Preprint*, 2026, https://arxiv.org/abs/2607.06434).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04101 | KOU-19.58

### Kourovka Notebook Problem 19.58

S / solved | Kourovka 群论问题集

What are the non-abelian composition factors of a finite group in which every maximal subgroup is simple or $p$-decomposable for some fixed odd prime $p\in \pi (G)$?

*Comment of 2026*: These factors are determined in the cases where $G$ is non-simple, or alternating, or sporadic simple (A. Beltr\'an, F. Viudez, *Preprint*, 2026, https://arxiv.org/abs/2607.06434).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04102 | KOU-19.59

### Kourovka Notebook Problem 19.59

O / open | Kourovka 群论问题集

Does the group of isometries of $\mathbb{Q}^3$ have a free non-abelian subgroup such that every non-trivial element acts without nontrivial fixed points in $\mathbb{Q}^3$?

The answer is yes if the field of rational numbers $\mathbb{Q}$ is replaced by the field $\mathbb{R}$ of real numbers (see G. Tomkowicz, S. Wagon, *The Banach--Tarski Paradox*, Cambridge Univ. Press, 2016.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

04103 | KOU-19.6

### Kourovka Notebook Problem 19.6

O / open | Kourovka 群论问题集

Let $G=A\ast_{H}B$ be the generalized free product of groups $A$ and $B$ with amalgamated subgroup $H$.

a) *Conjecture :* If both $A$ and $B$ are nilpotent, then $\mu(G)\leq H$.

b) *Conjecture :* If $G$ is residually finite, and $H$ satisfies a nontrivial identical relation, then $\lambda(G)\leq H$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04104 | KOU-19.60

### Kourovka Notebook Problem 19.60

O / open | Kourovka 群论问题集

Let $R$ be a principal ideal domain, let $d$ be a positive integer, and let $X_1, \ldots, X_m$ be members of $\mathrm{SL}(d,R)$ (or of $\mathrm{GL}(d,R)$). Is it decidable whether or not these $m$ matrices freely generate a free group? If it is, design an effective algorithm for computing the answer.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04105 | KOU-19.61

### Kourovka Notebook Problem 19.61

S / solved | Kourovka 群论问题集

Let $\mathfrak{A}=\{\mathfrak{A}_r\ |\ r\in \Phi\}$ be an elementary carpet of type $\Phi$ over a commutative ring $K$ (see 7.28), and let $\Phi(\mathfrak{A})=\langle x_r(\mathfrak{A}_r)\ |\ r\in\Phi\rangle$ be its carpet subgroup. Define the *closure* of the carpet $\mathfrak{A}$ to be the set of additive subgroups $\overline{\mathfrak{A}}=\{\overline{\mathfrak{A}}_r\ |\ r\in \Phi\}$, $ where $ \overline{\mathfrak{A}}_r=\{t\in K\ |\ x_r(t)\in \Phi(\mathfrak{A})\}$. $ Is the closure $\overline{\mathfrak{A}}$ of a carpet $\mathfrak{A}$ always a carpet?

An affirmative answer is known if $\Phi=A_l, D_l, E_l$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04106 | KOU-19.62

### Kourovka Notebook Problem 19.62

S / solved | Kourovka 群论问题集

Let $\mathfrak{A}=\{\mathfrak{A}_r\ |\ r\in \Phi\}$ be an elementary carpet of type $\Phi$ of rank $l\geq 2$ (see 7.28). For $p\in \Phi$, define a set of additive subgroups $\mathfrak{B}_p=\sum C_{ij,rs}\mathfrak{A}_r^i\mathfrak{A}_s^j$, where the sum is taken over all natural numbers $i$, $j$ and roots $r,s\in \Phi$ such that $ir+js=p$. It is known that the set $\mathfrak{B}=\{\mathfrak{B}_p\ |\ p\in \Phi\}$ is a carpet called the *derived carpet* of $\mathfrak{A}$. It is also known that for $\Phi=A_l$ the set $\mathfrak{B}$ is a closed (admissible) carpet, which means that its carpet subgroup does not contain new root elements. Is every derived carpet of type $\Phi$ over a commutative ring closed (admissible)?

*Comment of 2025*: An affirmative answer was obtained for $\Phi$ of type $B_l$, $C_l$, or $F_4$ when $\mbox{GCD}(p,2)=1$, and for $\Phi$ of type $G_2$ when $\mbox{GCD}(p,6)=1$ (Ya. N. Nuzhin, *J. Siberian Fed. Univ. Ser. Math. Phys.*, **16**, no. 6 (2023), 732--737).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04107 | KOU-19.63

### Kourovka Notebook Problem 19.63

S / solved | Kourovka 群论问题集

Let $\mathfrak{A}=\{\mathfrak{A}_r\ |\ r\in \Phi\}$ be an elementary carpet of type $\Phi$ over a commutative ring $K$ (see 7.28) and let $\mathfrak{A}_r^2=\{t^2\mid t\in\mathfrak{A}_r\}$. Are the inclusions $\mathfrak{A}_r^2\mathfrak{A}_{-r}\subseteq\mathfrak{A}_r$, $r\in\Phi$, sufficient for the carpet $\mathfrak{A}$ to be closed (admissible)?

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04108 | KOU-19.64

### Kourovka Notebook Problem 19.64

O / open | Kourovka 群论问题集

Let $G$ be a group, and $(g_1, \ldots , g_n)$ a tuple of its elements. The *type* of this tuple in $G$, denoted $Tp^G(g_1, \ldots , g_n)$, is the set of all first order formulas in free variables $x_1, \ldots ,x_n$ in the standard group theory language which are true on $(g_1, \ldots , g_n)$ in $G$. Two groups $G$ and $H$ are called *isotypic* if for every tuple of elements $\bar h=(h_1,\ldots,h_n)$ in $H$ there is a tuple $\bar g =(g_1,\ldots,g_n)$ in $G$, such that $Tp(\bar h)=Tp(\bar g)$ and vice versa, for every tuple $\bar g$ in $G$ there is a tuple $\bar h$ in $H$ such that $Tp(\bar h)=Tp(\bar g)$.

Is it true that every two isotypic finitely generated groups are isomorphic?

The answer is positive if one of the finitely generated groups is free (R. Sklinos), abelian (G. Zhitomirski), virtually polycyclic, metabelian, free solvable (A. Myasnikov, N. Romanovskii), co-Hopfian (in particular homogeneous), finitely presented Hopfian or geometrically Noetherian Hopfian (R. Sklinos). In particular, torsion-free hyperbolic groups, braid groups, linear groups, mapping class groups of compact surfaces, limit groups, quasi-cyclic groups, $Out(F_n)$, $n > 2$, irreducible arithmetic lattices (which are not virtually free groups) in semi-simple Lie groups, are of such kind (R. Sklinos).

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04109 | KOU-19.65

### Kourovka Notebook Problem 19.65

O / open | Kourovka 群论问题集

It is well known that varieties of groups form a free semigroup $N$ (A. L. Shmel'kin, *DAN SSSR*, **149** (1963), 543--545 (Russian)); \ B. H. Neuman, H. Neumann, P. M. Neumann, *Math. Z.*, **80** (1962), 44--62). Varieties of linear representations over an infinite field of zero characteristic also form a free semigroup $M$, and $N$ acts freely on $M$ (B. I. Plotkin, *Siberian Math. J.*, **13**, no. 5 (1972), 713--729) A similar theorem holds for Lie algebras (V. A. Parfenov, Algebra Logika, **6**, no. 4 (1967), 61--73 (Russian); \ L. A. Simonyan, *Siberian Math. J.*, **29**, no. 2 (1988), 276--283).

Are there other varieties of algebras $\Theta$ where the same situation takes place?

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04110 | KOU-19.66

### Kourovka Notebook Problem 19.66

O / open | Kourovka 群论问题集

We say that a variety $\Theta$ is of *Tarski type* if any two non-abelian $\Theta$-free groups of finite rank are elementarily equivalent.

a) Find examples of Tarski type varieties distinct from the variety of all groups.

b) Is it true that the {Burnside variety $B_n$} of all groups of exponent $n$, where $n$ is big enough, is of Tarski type?

c) Is it true that the *$n$-Engel variety $E_n$* of all groups satisfying the identity $[[[x,\underbrace{y],y],\ldots,y}_{n}]\equiv 1$, where $n$ is big enough, is of Tarski type?

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04111 | KOU-19.67

### Kourovka Notebook Problem 19.67

S / solved | Kourovka 群论问题集

Let $G\le \mathrm{Sym}(\Omega)$, where $\Omega$ is finite. The $2$-closure $G^{(2)}$ of the group $G$ is defined to be the largest subgroup of $\mathrm{Sym}(\Omega)$ containing $G$ which has the same orbits as $G$ in the induced action on $\Omega\times\Omega$. Is it true that if $G$ is solvable, then every composition factor of $G^{(2)}$ is either a cyclic or an alternating group?

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04112 | KOU-19.68

### Kourovka Notebook Problem 19.68

O / open | Kourovka 群论问题集

For a finite group $G$ and a permutation group $K$, let $b_G(K)$ denote the number of conjugacy classes of regular subgroups of $K$ isomorphic to $G$. Does there exist a function $f$ such that $b_G(K)\le n^{f(r)}$ for every abelian group $G$ of order $n$ and rank $r$, and every group $K$ such that $K^{(2)}=K$? (See Archive 19.67 for the definition of $K^{(2)}$.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

04113 | KOU-19.69

### Kourovka Notebook Problem 19.69

O / open | Kourovka 群论问题集

(P. Wesolek). The acronym tdlc stands for totally disconnected and locally compact, and tdlcsc for tdlc and second countable. Let $\rm{Res}(G)$ denote the intersection of all open normal subgroups of a topological group $G$. The class of elementary tdlcsc groups is defined as the smallest class $\mathscr{E}$ of tdlcsc groups such that \begin{itemize} \item[(1)] $\mathscr{E}$ contains all second countable profinite groups and countable discrete groups; \item[(2)] $\mathscr{E}$ is closed under taking closed subgroups, Hausdorff quotients, directed unions of open subgroups, and group extensions. \end{itemize} This class admits a well-behaved, ordinal-valued decomposition rank $\xi$ defined recursively as follows: $\xi(\{1\}) = 1$, and if $G \in \mathscr{E}$ is a union of an increasing sequence $(O_i)$ of compactly generated open subgroups, then $\xi(G) = \sup_i \{\xi(\rm{Res}(O_i))\} + 1$.

What is the supremum of the decomposition ranks of elementary tdlcsc groups? In particular, is it countable?

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04114 | KOU-19.7

### Kourovka Notebook Problem 19.7

O / open | Kourovka 群论问题集

The group of virtual pure braids $VP_n$, $n \geq 2$, is generated by elements $\lambda_{ij}$, $1 \leq i \not= j \leq n$, and is defined by the relations $ \lambda_{ij}\lambda_{kl}=\lambda_{kl}\lambda_{ij}$, \ $\lambda_{ki}\lambda_{kj}\lambda_{ij}=\lambda_{ij}\lambda_{kj}\lambda_{ki} $, where different letters denote different indices.

a) Construct a normal form of words in the group $VP_n$ for $n \geq 4$.

b) Is the group $VP_n$ linear for $n \geq 4$? It is known that the group $VP_3$ is linear.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04115 | KOU-19.70

### Kourovka Notebook Problem 19.70

O / open | Kourovka 群论问题集

(P. Wesolek and P.-E. Caprace). Let $\mathscr{S}$ denote the class of nondiscrete compactly generated, topologically simple tdlc groups. Let $G$ be a non-elementary tdlcsc group. Is there a compactly generated closed subgroup $H \leq G$ such that $H$ has a continuous quotient in $\mathscr{S}$?

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04116 | KOU-19.71

### Kourovka Notebook Problem 19.71

O / open | Kourovka 群论问题集

(G. Willis). Can a group $G$ in $\mathscr{S}$ be such that every element of $G$ normalizes a compact open subgroup of $G$?

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04117 | KOU-19.72

### Kourovka Notebook Problem 19.72

O / open | Kourovka 群论问题集

Let $G$ be a group in $\mathscr{S}$. Can every element of $G$ have trivial contraction group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04118 | KOU-19.73

### Kourovka Notebook Problem 19.73

O / open | Kourovka 群论问题集

(P.-E. Caprace and N. Monod). Is there a compactly generated, locally compact group that is topologically simple, but not abstractly simple?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04119 | KOU-19.74

### Kourovka Notebook Problem 19.74

S / solved | Kourovka 群论问题集

A subgroup $H$ of a group $G$ is called *pronormal* if $H$ and $H^g$ are conjugate in $\langle H,H^g\rangle$ for every $g\in G$. A subgroup $H$ of a group $G$ is called *abnormal* if $g\in \langle H,H^g\rangle$ for every $g\in G$.

Does there exist an infinite group that does not contain nontrivial proper pronormal subgroups?

The question is equivalent to the following: does there exist an infinite simple group that does not contain proper abnormal subgroups?

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04120 | KOU-19.75

### Kourovka Notebook Problem 19.75

S / solved | Kourovka 群论问题集

Let $P$ be a finite 2-group of exponent $2^e$ such that the rank of every abelian subgroup is at most $r$. Is it true that $|P|\le 2^{r(e+1)}$? This bound would be sharp (for a direct product of quaternion groups).

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04121 | KOU-19.76

### Kourovka Notebook Problem 19.76

O / open | Kourovka 群论问题集

A semigroup presentation is called *tree-like* if all relations have the form $a=bc$ where $a, b, c$ are letters and no two relations share the left-hand side or the right-hand side. Is it decidable whether the semigroup given by a finite tree-like presentation contains an idempotent?

This is equivalent to the question whether the closure of a finitely generated subgroup of R. Thompson's group $F$ contains an isomorphic copy of $F$.

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04122 | KOU-19.77

### Kourovka Notebook Problem 19.77

O / open | Kourovka 群论问题集

*The Stable Small Cancellation Conjecture*: If $ F$ is a free group of rank $r\geq 2$, then, for any fixed $n > 0$, there exists a generic subset $S$ of $F^n$ such that, for any automorphism $\phi$ of $F$, the set $\phi(S)$ satisfies the small cancellation condition $C'(1/6)$.

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04123 | KOU-19.78

### Kourovka Notebook Problem 19.78

O / open | Kourovka 群论问题集

Does there exist an infinite simple subgroup of $SL(2,{\mathbb{Q}})$?

This is a special case of a question which Serge Cantat asked me about ``non-algebraic'' simple subgroups of $GL(n,{\mathbb{C}})$. It is also a special case of 15.57.

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04124 | KOU-19.79

### Kourovka Notebook Problem 19.79

O / open | Kourovka 群论问题集

(T. Springer). Let $G$ be a group. Suppose that for every integer $n > 0$ the group $G$ has a unique (up to isomorphism) irreducible complex linear representation of dimension $n$. (Note that $G = SL(2,{\mathbb{Q}})$ has these properties, by a theorem of Borel--Tits (*Ann. Math.*, **97** (1973), 499--571).) Is it true that $G$ has a normal subgroup $N$ such that $G/N$ is isomorphic to $SL(2,{\mathbb{Q}})$?

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04125 | KOU-19.8

### Kourovka Notebook Problem 19.8

S / solved | Kourovka 群论问题集

A word in an alphabet $A = \{ a_1, a_2, \ldots, a_n \}$ is called a palindrome if it reads the same from left to right and from right to left. Let $k$ a non-negative integer. A word in the alphabet $A$ is called an *almost $k$-palindrome* if it can be transformed into a palindrome by changing $\leq k$ letters in it. (So an almost $0$-palindrome is a palindrome.) Let elements of a free group $F_2 = \langle x, y \rangle$ be represented as words in the alphabet $\{ x^{\pm 1}, y^{\pm 1} \}$. Do there exist positive integers $m$ and $c$ such that every element in $F_2$ is a product of $\leq c$ almost $m$-palindromes?

It is known that for $m=0$ there is no such a number $c$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04126 | KOU-19.80

### Kourovka Notebook Problem 19.80

S / solved | Kourovka 群论问题集

For a periodic group $G$, let $\pi _e(G)$ denote the set of orders of elements of $G$. A periodic group $G$ is said to be an *$OC_n $-group* if $\pi _e(G)=\{ 1, 2, \ldots , n\}$. Is it true that every $OC_7 $-group is isomorphic to the alternating group $A_7$?

For finite groups, the answer is affirmative.

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04127 | KOU-19.81

### Kourovka Notebook Problem 19.81

S / solved | Kourovka 群论问题集

(Well-known problem). Is the conjugacy problem in the braid group $B_n$ in the class NP (that is, decidable in nondeterministic polynomial time with respect to the maximum of the lengths $|u|$, $|v|$, where $u, v$ are the input braid words)? A stronger question: given two conjugate elements of $B_n$ represented by braid words of lengths $\leq m$, is there a conjugator whose length is bounded by a polynomial function of $m$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04128 | KOU-19.82

### Kourovka Notebook Problem 19.82

O / open | Kourovka 群论问题集

Let $h^*(G)$ denote the generalized Fitting height of a finite group $G$ defined as the minimum number $k$ such that $F^*_k(G)=G$, where $F^*_1(G)=F^*(G)$ is the generalized Fitting subgroup of $G$, and by induction $F^*_{i+1}(G)$ is the inverse image of $F^*(G/F^*_i(G))$. If $G$ is soluble, then $h^*(G)=h(G)$ is the Fitting height of $G$.

Does every finite group $G$ contain a soluble subgroup $K$ such that $h^*(G)=h(K)$?

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04129 | KOU-19.83

### Kourovka Notebook Problem 19.83

O / open | Kourovka 群论问题集

An element $g$ of a group $G$ is almost Engel if there is a finite set ${\mathscr E}(g)$ such that for every $x\in G$ all sufficiently long commutators $[x, {}_n g]$ belong to ${\mathscr E}(g)$, that is, for every $x\in G$ there is a positive integer $n(x,g)$ such that $[x, {}_n g]\in {\mathscr E}(g)$ whenever $n\geq n(x,g)$. By a linear group we understand a subgroup of $GL(m,F)$ for some field $F$ and a positive integer $m$.

Is the set of almost Engel elements in a linear group always a subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04130 | KOU-19.84

### Kourovka Notebook Problem 19.84

S / solved | Kourovka 群论问题集

Let $\mathbb{P}$ be the set of all primes, and let $\sigma =\{\sigma_{i} \mid i\in I\}$ be some partition of $\mathbb{P}$ into disjoint subsets. A finite group $G$ is said to be *$\sigma$-primary* if $G$ is a $\sigma_{i}$-group for some $i$; *$\sigma$-nilpotent* if $G$ is a direct product of $\sigma$-primary groups; *$\sigma$-soluble* if every chief factor of $G$ is $\sigma$-primary. A subgroup $A$ of a finite group $G$ is said to be \emph{${\sigma}$-subnormal in $G$} if there is a chain $A=A_{0} \leq A_{1} \leq \cdots \leq A_{n}=G$ such that for every $i$ either $A_{i-1} \trianglelefteq A_{i}$ or $A_{i}/(A_{i-1})_{A_{i}}$ is ${\sigma}$-primary, where $(A_{i-1})_{A_{i}}$ is the largest normal subgroup of $A_{i}$ contained in $A_{i-1}$.

Suppose that a subgroup $A$ of a finite group $G$ is ${\sigma}$-subnormal in $\langle A, A^{x} \rangle$ for all $x\in G$. Is it true that then $A$ is ${\sigma}$-subnormal in $G$?

An affirmative answer is known if $\sigma =\{\{2\}, \{3 \}, \ldots \}$ (Wielandt).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04131 | KOU-19.85

### Kourovka Notebook Problem 19.85

S / solved | Kourovka 群论问题集

Suppose that every Schmidt subgroup of a finite group $G$ is ${\sigma}$-subnormal in $ G$ (see Archive 19.84). Is it true that then there is a normal $\sigma$-nilpotent subgroup $N\leq G$ such that $G/N$ is cyclic?

An affirmative answer is known if $\sigma =\{\{2\}, \{3 \}, \ldots \}$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04132 | KOU-19.86

### Kourovka Notebook Problem 19.86

O / open | Kourovka 群论问题集

Suppose that a finite group $G$ has a $\sigma_{i}$-Hall subgroup for every $i \in I$. Suppose that a subgroup $A\leq G$ is such that $A\cap H$ is a $\sigma_{i}$-Hall subgroup of $A$ for every $i\in I$ and every $\sigma_{i}$-Hall subgroup $H$ of $G$ (see 19.84 in Archive). Is it true that then $A$ is ${\sigma}$-subnormal in $G$?

An affirmative answer is known if $\sigma =\{\{2\}, \{3 \}, \ldots \}$ and if $G$ is $\sigma$-soluble.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04133 | KOU-19.87

### Kourovka Notebook Problem 19.87

S / solved | Kourovka 群论问题集

Suppose that for every Sylow subgroup $P$ of a finite group $G$ and every maximal subgroup $V$ of $P$ there is a ${\sigma}$-soluble subgroup $T$ such that $VT=G$. Is it true that then $G$ is ${\sigma}$-soluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04134 | KOU-19.88

### Kourovka Notebook Problem 19.88

S / solved | Kourovka 群论问题集

Suppose that for every Sylow subgroup $P$ of a finite group $G$ and every maximal subgroup $V$ of $P$ there is a ${\sigma}$-nilpotent subgroup $T$ such that $VT=G$. Is it true that then $G$ is ${\sigma}$-nilpotent?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04135 | KOU-19.89

### Kourovka Notebook Problem 19.89

S / solved | Kourovka 群论问题集

A permutation group is *subdegree-finite* if every orbit of every point stabiliser is finite. Let $\Omega$ be countably infinite, and let $G$ be a closed and subdegree-finite subgroup of $\mathrm{Sym}(\Omega)$ regarded as a topological group under the topology of pointwise convergence.

*Conjecture*: If the minimal degree of $G$ is infinite, then there is some subset of $\Omega$ whose setwise stabiliser in $G$ is trivial.

Note that this conjecture implies Tom Tucker's well-known Infinite Motion Conjecture for graphs.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04136 | KOU-19.9

## Kourovka Notebook Problem 19.9

O / open | Kourovka 群论问题集

Let $G$ be a finitely generated linear group.

a) Is it true that the non-abelian tensor square $G \otimes G$ is a linear group?

b) In particular, is this true for the braid group $B_n$ for $n > 3$?

It is known that $B_3 \otimes B_3$ is linear, and there is a countable group $G$ such that $G \otimes G$ is not linear. See the definition of the non-abelian tensor square in (R. Brown, J.-L. Loday, *Topology*, **26**, no. 3 (1987), 311--335).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04137 | KOU-19.90

## Kourovka Notebook Problem 19.90

P / partially_solved | Kourovka 群论问题集

A *skew brace* is a set $B$ equipped with two operations $+$ and $\cdot$ such that $(B,+)$ is an additively written (but not necessarily abelian) group, $(B,\cdot )$ is a multiplicatively written group, and $a\cdot (b+c)=ab-a+ac$ for any $a,b,c\in B$.

b) Is there a skew brace with non-soluble additive group but nilpotent multiplicative group?

c) Is there a finite skew brace with soluble additive group but non-soluble multiplicative group?

[Archived solved part(s) of 19.90:] A *skew brace* is a set $B$ equipped with two operations $+$ and $\cdot$ such that $(B,+)$ is an additively written (but not necessarily abelian) group, $(B,\cdot )$ is a multiplicatively written group, and $a\cdot (b+c)=ab-a+ac$ for any $a,b,c\in B$.

a) Is there a skew brace with soluble additive group but non-soluble multiplicative group?

d) Is there a finite skew brace with non-soluble additive group but nilpotent multiplicative group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04138 | KOU-19.91

## Kourovka Notebook Problem 19.91

O / open | Kourovka 群论问题集

Let $G$ be a finite group with an abelian Sylow $p$-subgroup $A$. Suppose that $B$ is a strongly closed elementary abelian subgroup of $A$. Without invoking the Classification Theorem for Finite Simple Groups (CFSG), prove that $G$ has a normal subgroup $N$ such that $B =\Omega _1(A\cap N)$.

For $p = 2$, this is a corollary of a theorem of Goldschmidt. For $p$ odd, this has been proved by Flores and Foote (*Adv. Math.*, **222** (2009), 453--484), but their proof relies on CFSG. A CFSG-free proof in the special case when $p = 3$ and $A$ has 3-rank 3 would already be quite interesting. This case arises in Aschbacher's treatment of the $e(G) = 3$ problem, and a proof would provide an alternative to part of his argument. (For this application, one could assume that all proper simple sections of $G$ are known.)

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04139 | KOU-19.92

## Kourovka Notebook Problem 19.92

S / solved | Kourovka 群论问题集

Let $A$ be the algebra of $3\times 3$ skew-Hermitian matrices over the real octonions, where multiplication is given by bracket product. Determine $\mathrm{Aut}(A)$.

The question might be of interest for octonion algebras over a different base field or ring. The question was inspired by an observation of John Faulkner concerning a possible connection between $A$ or a related algebra and the Dwyer--Wilkerson 2-compact group $BDI(4)$.

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04140 | KOU-19.93

## Kourovka Notebook Problem 19.93

P / partially_solved | Kourovka 群论问题集

*Conjecture*: There exists a function $f:\mathbb{N}\times\mathbb{N}\to \mathbb{N}$ such that, if $G$ is a finite $p$-group with $d$ generators and $G$ has no epimorphic images isomorphic to the wreath product $C_p\wr C_p$, then each factor of the $p$-lower central series of $G$ has order bounded above by $f(p,d)$

a) for $p\geq 7$;

b) for $p =2,3,5$.

It is interesting to compare this conjecture with the celebrated characterization of Aner Shalev of finitely generated $p$-adic analytic pro-$p$-groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04141 | KOU-19.94

### Kourovka Notebook Problem 19.94

S / solved | Kourovka 群论问题集

By definition two elements $g_1$ and $g_2$ of a free metabelian group $F$ with basis $\{x_1,\ldots, x_n\}$ have the same distribution if the equations $g_1(x_1,\ldots, x_n) = g$ and $g_2(x_1,\ldots, x_n) = g$ have the same number of solutions in any finite metabelian group $G$ for every $g \in G$. Is it true that elements $g_1$ and $g_2$ have the same distribution if and only if they are conjugate by some automorphism of $F?$

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04142 | KOU-19.95

### Kourovka Notebook Problem 19.95

O / open | Kourovka 群论问题集

Let $G_\Gamma$ be a partially commutative soluble group of derived length $n \geq 3$ with defining graph $\Gamma$ (the definition is similar to the case of $n=2$, see 17.104). Is it true that the centralizer of any vertex of the graph is generated by its adjacent vertices?

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04143 | KOU-19.96

### Kourovka Notebook Problem 19.96

O / open | Kourovka 群论问题集

Let $M$ be a free metabelian group of finite rank $r \geq 2$, and let $P$ be the set of its primitive elements. (An element of a relatively free group is said to be primitive if it can be included in a basis of the group.)

a) Is $P$ a first-order definable set?

b) Is the set of bases of the group $M$ a first-order definable set?

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04144 | KOU-19.97

### Kourovka Notebook Problem 19.97

O / open | Kourovka 群论问题集

Let $G$ be an $m$-generated group the elementary theory of which $Th(G)$ coincides with the elementary theory $Th(F)$ of a free soluble group $F$ of derived length $n \geq 3$ of finite rank $r \geq 2$. Must the groups $G$ and $F$ be isomorphic?

The answer is affirmative if $m \leq r$ or $n=2$

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04145 | KOU-19.98

### Kourovka Notebook Problem 19.98

S / solved | Kourovka 群论问题集

A connected graph $\Sigma$ is a *symmetrical extension* of a graph $\Gamma$ by a graph $\Delta$ if there exist a vertex-transitive group $G$ of automorphisms of $\Sigma$ and an imprimitivity system $\sigma$ of $G$ on the set of vertices of $\Sigma$ such that the quotient graph $\Sigma/\sigma$ is isomorphic to $\Gamma$ and blocks of $\sigma$ generate in $\Sigma$ subgraphs isomorphic to $\Delta$.

{(a)} Let $\Gamma$ be a locally finite Cayley graph of a finitely presented group, and $\Delta$ a finite graph. Are there only finitely many symmetrical extensions of $\Gamma$ by $\Delta$?

{(b)} Let $\Gamma$ be a locally finite graph which has the property of $k$-contractibility for some positive integer $k$ (see the definition in (V. I. Trofimov, *Proc. Steklov Inst. Math.*, **279**, suppl. 1 (2012), 107--112); note that any $\Gamma$ from (a) is such a graph) and let $\Delta$ be a finite graph. Are there only finitely many symmetrical extensions of $\Gamma$ by $\Delta$?

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04146 | KOU-19.99

### Kourovka Notebook Problem 19.99

O / open | Kourovka 群论问题集

Is it true that for any positive integer $t$ there is a positive integer $n(t)$ such that any finite group with at least $n(t)$ conjugate classes of soluble maximal subgroups of Fitting height at most $t$ is itself a soluble group of Fitting height at most $t$?

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04147 | KOU-2.1

### Kourovka Notebook Problem 2.1

S / solved | Kourovka 群论问题集

Classify the finite groups having a Sylow $p $-subgroup as a maximal subgroup.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04148 | KOU-2.10

### Kourovka Notebook Problem 2.10

S / solved | Kourovka 群论问题集

Prove an analogue of the Remak--Shmidt theorem for decompositions of a group into nilpotent products.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04149 | KOU-2.13

### Kourovka Notebook Problem 2.13

S / solved | Kourovka 群论问题集

(Well-known problem). Let $G$ be a periodic group in which every $\pi$-element commutes with every $\pi '$-element. Does $G$ decompose into the direct product of a maximal $\pi$-subgroup and a maximal $\pi '$-subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04150 | KOU-2.15

### Kourovka Notebook Problem 2.15

S / solved | Kourovka 群论问题集

Does there exist a torsion-free group such that the factor group by some term of its upper central series is nontrivial periodic, with a bound on the orders of the elements?

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04151 | KOU-2.16

### Kourovka Notebook Problem 2.16

S / solved | Kourovka 群论问题集

A group $G$ is called *conjugacy separable* if any two of its elements are conjugate in $G$ if and only if their images are conjugate in every finite homomorphic image of $G$. Is $G$ conjugacy separable in the following cases:

a) $G$ is a polycyclic group,

b) $G$ is a free soluble group,

c) $G$ is a group of (all) integral matrices,

d) $G$ is a finitely generated group of matrices,

e) $G$ is a finitely generated metabelian group?

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04152 | KOU-2.17

### Kourovka Notebook Problem 2.17

S / solved | Kourovka 群论问题集

Is it true that the wreath product $A\wr B$ of two groups that are conjugacy separable is itself conjugacy separable if and only if either $A$ is abelian or $B$ is finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04153 | KOU-2.18

### Kourovka Notebook Problem 2.18

S / solved | Kourovka 群论问题集

Compute the ranks of the factors of the lower central series of a free soluble group.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04154 | KOU-2.19

### Kourovka Notebook Problem 2.19

S / solved | Kourovka 群论问题集

Are finitely generated subgroups of a free soluble group finitely separable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04155 | KOU-2.2

### Kourovka Notebook Problem 2.2

S / solved | Kourovka 群论问题集

A *quasigroup* is a groupoid $Q(\cdot )$ in which the equations $ ax = b$ and $ ya = b$ have a unique solution for any $ a, b\in Q$. Two quasigroups $Q(\cdot )$ and $Q(\circ )$ are *isotopic* if there are bijections $\alpha$, $\beta$, $\gamma$ of the set $Q$ onto itself such that $x \circ y=\gamma (\alpha x\cdot \beta y)$ for all $ x, y \in Q$. It is well-known that all quasigroups that are isotopic to groups form a variety ${\mathfrak{G}}$. Let ${\mathfrak{V}}$ be a variety of quasigroups. Characterize the class of groups isotopic to quasigroups in ${\mathfrak{G}}\cap {\mathfrak{V}}$. For which identities characterizing ${\mathfrak{V}}$ is every group isotopic to a quasigroup in ${\mathfrak{G}}\cap {\mathfrak{V}}$? Under what conditions on ${\mathfrak{V}}$ does any group isotopic to a quasigroup in ${\mathfrak{G}}\cap {\mathfrak{V}}$ consist of a single element?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04156 | KOU-2.20

### Kourovka Notebook Problem 2.20

S / solved | Kourovka 群论问题集

Is it true that the wreath products $A\wr B$ and $A_1\wr B_1$ are elementarily equivalent if and only if $A$, $B$ are elementarily equivalent to $A_1$, $B_1$, respectively?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04157 | KOU-2.21

### Kourovka Notebook Problem 2.21

S / solved | Kourovka 群论问题集

Do the classes of Baer and Fitting groups coincide?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04158 | KOU-2.22

### Kourovka Notebook Problem 2.22

P / partially_solved | Kourovka 群论问题集

a) An abstract group-theoretic property $\Sigma$ is said to be *radical ($in our sense*) if, in any group $G$, the subgroup $\Sigma(G)$ generated by all normal $\Sigma $-subgroups is a $\Sigma $-subgroup itself (called the *$\Sigma $-radical* of $G$). A radical property $\Sigma$ is said to be *strongly radical* if, for any group $G$, the factor-group $G/\Sigma(G)$ contains no non-trivial normal $\Sigma $-subgroups. Is the property $ \overline{\! RN}$ radical? strongly radical?

[Archived solved part(s) of 2.22:] b) An abstract group-theoretical property $\Sigma$ is called *radical ($in our sense$)$* if, in any group $G$, the join $\Sigma(G)$ of all normal $\Sigma$-subgroups is a $\Sigma$-subgroup. Is the property $N^0$ (see Archive, 1.38) radical?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04159 | KOU-2.23

### Kourovka Notebook Problem 2.23

S / solved | Kourovka 群论问题集

a) A subgroup $H$ of a group $G$ is called *quasisubinvariant* if there is a normal system of $G$ passing through $H$. Let ${\mathfrak{K}}$ be a class of groups that is closed with respect to taking homomorphic images. A group $G$ is called an {\it $R^0({\mathfrak{K}})$-group} if each of its non-trivial homomorphic images has a non-trivial quasisubinvariant ${\mathfrak{K}}$-subgroup. Do $R^0({\mathfrak{K}})$ and $ \overline{\! RN}$ coincide when ${\mathfrak{K}}$ is the class of all abelian groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04160 | KOU-2.24

### Kourovka Notebook Problem 2.24

O / open | Kourovka 群论问题集

Are Engel torsion-free groups orderable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04161 | KOU-2.25

### Kourovka Notebook Problem 2.25

P / partially_solved | Kourovka 群论问题集

a) (L. Fuchs). Describe the groups which are linearly orderable in only finitely many ways.

[Archived solved part(s) of 2.25:] b) Do there exist groups that are linearly orderable in countably many ways?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04162 | KOU-2.26

### Kourovka Notebook Problem 2.26

O / open | Kourovka 群论问题集

(L. Fuchs). Characterize as abstract groups the multiplicative groups of orderable skew fields.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04163 | KOU-2.28

### Kourovka Notebook Problem 2.28

O / open | Kourovka 群论问题集

Can every orderable group be embedded in a pro-orderable group? (See 1.35 for the definition of pro-orderable groups.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

04164 | KOU-2.29

### Kourovka Notebook Problem 2.29

S / solved | Kourovka 群论问题集

Does the class of finite groups in which every proper abelian subgroup is contained in a proper normal subgroup coincide with the class of finite groups in which every proper abelian subgroup is contained in a proper normal subgroup of prime index?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04165 | KOU-2.3

### Kourovka Notebook Problem 2.3

S / solved | Kourovka 群论问题集

A finite group is called *quasi-nilpotent* ($\Gamma$-*quasi-nilpotent*) if any two of its (maximal) subgroups $A$ and $B$ satisfy one of the conditions 1) $ A \leq B$, \ \ 2) $ B \leq A$, \ \ 3) $N_A(A\cap B) \ne A\cap B \ne N_B(A\cap B)$. Do the classes of quasi-nilpotent and $\Gamma$-quasi-nilpotent groups coincide?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04166 | KOU-2.30

### Kourovka Notebook Problem 2.30

S / solved | Kourovka 群论问题集

Does there exist a finite group in which a Sylow $p $-subgroup is covered by other Sylow $p $-subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04167 | KOU-2.31

### Kourovka Notebook Problem 2.31

S / solved | Kourovka 群论问题集

Can every group admitting an ordering with only finitely many convex subgroups be represented by matrices over a field?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04168 | KOU-2.33

### Kourovka Notebook Problem 2.33

S / solved | Kourovka 群论问题集

Is a direct summand of a direct sum of finitely generated modules over a Noetherian ring again a direct sum of finitely generated modules?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04169 | KOU-2.35

### Kourovka Notebook Problem 2.35

S / solved | Kourovka 群论问题集

An inverse spectrum $\xi$ of abelian groups is said to be *acyclic* if ${\lim\limits_{\longleftarrow}} ^{(p)} \xi =0$ for $p > 0$. Here ${\lim\limits_{\longleftarrow}}^{(p)}$ denotes the right derived functor of the projective limit functor. Let $\xi$ be an acyclic spectrum of finitely generated groups. Is the spectrum $\bigoplus\xi _{\alpha}$ also acyclic, where each spectrum $\xi _{\alpha}$ coincides with $\xi $?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04170 | KOU-2.36

### Kourovka Notebook Problem 2.36

S / solved | Kourovka 群论问题集

(de Groot). Is the group of all continuous integer-valued functions on a compact space free abelian?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04171 | KOU-2.37

### Kourovka Notebook Problem 2.37

S / solved | Kourovka 群论问题集

Describe the finite simple groups whose Sylow $p $-subgroups are cyclic for all odd $p$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04172 | KOU-2.38

### Kourovka Notebook Problem 2.38

S / solved | Kourovka 群论问题集

(Old problem). The class of rings embeddable in associative division rings is universally axiomatizable. Is it finitely axiomatizable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04173 | KOU-2.39

### Kourovka Notebook Problem 2.39

S / solved | Kourovka 群论问题集

Does there exist a non-finitely-axiomatizable variety of

a) (H. Neumann) groups?

b) of associative rings (the Specht problem)?

c) Lie rings?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04174 | KOU-2.4

### Kourovka Notebook Problem 2.4

S / solved | Kourovka 群论问题集

(S. Chase). Suppose that an abelian group $A$ can be written as the union of pure subgroups $A_{\alpha}$, $\alpha\in\Omega$, where $\Omega$ is the first non-denumerable ordinal, $A_{\alpha}$ is a free abelian group of denumerable rank, and, if $ \beta < \alpha$, then $A_{\beta}$ is a direct summand of $ A_{\alpha}$. Does it follow that $A$ is a free abelian group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04175 | KOU-2.40

### Kourovka Notebook Problem 2.40

P / partially_solved | Kourovka 群论问题集

c) The *$I $-theory* (*$Q $-theory*) of a class ${\mathfrak{K}}$ of universal algebras is the totality of all identities (quasi-identities) that are valid on all algebras in ${\mathfrak{K}}$. Does there exist a finitely axiomatizable variety

(3) of Lie rings

(i) whose $I $-theory is non-decidable?

*Remark:* it is not difficult to find a recursively axiomatizable variety of semigroups with identity whose $ I $-theory is non-recursive (see also A. I. Mal'cev, *Mat. Sbornik*, **69**, no. 1 (1966), 3--12 (Russian)).

[Archived solved part(s) of 2.40:] The *$I$-theory* (*$Q$-theory*) of a class ${\mathfrak{K}}$ of universal algebras is the totality of all identities (quasi-identities) that are true on all the algebras in ${\mathfrak{K}}$. Does there exist a finitely axiomatizable variety of

a) groups,

b) semigroups,

c) (1) rings

(2) of associative rings

(i) whose $I $-theory is non-decidable?

(ii) whose $Q $-theory is non-decidable?

(3) of Lie rings

(ii) whose $Q $-theory is non-decidable?

whose $I$-theory ($Q $-theory) is non-decidable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04176 | KOU-2.41

### Kourovka Notebook Problem 2.41

S / solved | Kourovka 群论问题集

Is the variety generated by \

a) a finite associative ring;

b) a finite Lie ring;

c) a finite quasigroup

finitely axiomatizable?

d) What is the cardinality $n$ of the smallest semigroup generating a non-finitely-axiomatizable variety?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04177 | KOU-2.42

### Kourovka Notebook Problem 2.42

O / open | Kourovka 群论问题集

What is the structure of the groupoid of quasivarieties

a) of all semigroups?

b) of all rings?

c) of all associative rings?

Compare with A. I. Mal'cev, *Siberian Math. J.*, **8**, no. 2 (1967), 254--267).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04178 | KOU-2.43

### Kourovka Notebook Problem 2.43

S / solved | Kourovka 群论问题集

A group $G$ is called an *$FN$-group* if the groups $\gamma_ iG/\gamma _{i+1}G$ are free abelian and $\bigcap_{i=1}^{\infty} \gamma _iG = 1$, where $\gamma _{i+1}G=[\gamma _{i}G, G]$. A variety of groups ${\mathfrak{M}}$ is called a *$\Sigma$-variety* (where $\Sigma$ is an abstract property) if the ${\mathfrak{M}}$-free groups have the property $\Sigma$.

a) Which properties $\Sigma$ are preserved under multiplication and intersection of varieties? Is the $FN$ property preserved under these operations?

b) Are all varieties obtained by multiplication and intersection from the nilpotent varieties ${\mathfrak{N}}_1$, ${\mathfrak{N}}_2, \ldots $ (where ${\mathfrak{N}}_1$ is the variety of abelian groups) $FN$-varieties?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04179 | KOU-2.44

### Kourovka Notebook Problem 2.44

S / solved | Kourovka 群论问题集

Let ${\mathfrak{A}}$ and ${\mathfrak{B}}$ be subvarieties of a variety of groups $ {\mathfrak{M}}$; then $({\mathfrak{A}}{\mathfrak{B}})\cap {\mathfrak{M}}$ is called the {\it ${\mathfrak{M}}$-product of ${\mathfrak{A}}$ by ${\mathfrak{B}}$}, where $ {\mathfrak{A}}{\mathfrak{B}}$ is the usual product. Does there exist a non-abelian variety ${\mathfrak{M}}$ with an infinite lattice of subvarieties and commutative ${\mathfrak{M}}$-multiplication?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04180 | KOU-2.45

### Kourovka Notebook Problem 2.45

P / partially_solved | Kourovka 群论问题集

b) (P. Hall). Prove or refute the following conjecture: If the marginal subgroup $v^*G$ has finite index $m$ in $G$, then the order of $vG$ is finite and divides a power of $m$.

*Editors' comment:* There are examples when $|vG|$ does not divide a power of $m$ (Yu. G. Kleiman, *Trans. Moscow Math. Soc.*, **1983**, no. 2, 63--110), but the question of finiteness remains open.

[Archived solved part(s) of 2.45:] (P. Hall). Prove or refute the following conjectures:

a) If a word $v$ takes only finitely many values in a group $G$, then the verbal subgroup $vG$ is finite.

c) If $G$ satisfies the maximum condition and $vG$ is finite, then $v^*G$ has finite index in $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04181 | KOU-2.46

### Kourovka Notebook Problem 2.46

S / solved | Kourovka 群论问题集

Find conditions under which a finitely-generated matrix group is almost residually a finite $p $-group for some prime $p$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04182 | KOU-2.47

### Kourovka Notebook Problem 2.47

S / solved | Kourovka 群论问题集

In which abelian groups is the lattice of all fully invariant subgroups a chain?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04183 | KOU-2.48

### Kourovka Notebook Problem 2.48

S / solved | Kourovka 群论问题集

(N. Aronszajn). Let $G$ be a connected topological group locally satisfying some identical relation $f|_U= 1$, where $U$ is a neighborhood of the identity element of $G$. Is it then true that $f|_G=1$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04184 | KOU-2.49

### Kourovka Notebook Problem 2.49

S / solved | Kourovka 群论问题集

(A. Selberg). Let $G$ be a connected semisimple linear Lie group whose corresponding symmetric space has rank greater than 1, and let $\Gamma$ be an irreducible discrete subgroup of $G$ such that $ G/\Gamma$ has finite volume. Does it follow that $\Gamma$ is an arithmetic subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04185 | KOU-2.5

### Kourovka Notebook Problem 2.5

O / open | Kourovka 群论问题集

According to Plotkin, a group is called an $NR$-group if the set of its nil-elements coincides with the locally nilpotent radical, or, which is equivalent, if every inner automorphism of it is locally stable. Can an $NR$-group have a nil-automorphism that is not locally stable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04186 | KOU-2.50

### Kourovka Notebook Problem 2.50

S / solved | Kourovka 群论问题集

(A. Selberg). Let $\Gamma$ be an irreducible discrete subgroup of a connected Lie group $G$ such that the factor space $G/\Gamma$ is non-compact but has finite volume in the Haar measure. Prove that $\Gamma$ contains a non-trivial unipotent element.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04187 | KOU-2.51

### Kourovka Notebook Problem 2.51

S / solved | Kourovka 群论问题集

(A. Borel, R. Steinberg). Let $G$ be a semisimple algebraic group and $R_G$ the set of classes of conjugate unipotent elements of $ G$. Is $R_G$ finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04188 | KOU-2.52

### Kourovka Notebook Problem 2.52

S / solved | Kourovka 群论问题集

(F. Bruhat, N. Iwahori, M. Matsumoto). Let $G$ be a semisimple algebraic group over a locally compact, totally disconnected field. Do the maximal compact subgroups of $G$ fall into finitely many conjugacy classes? If so, estimate this number.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04189 | KOU-2.54

### Kourovka Notebook Problem 2.54

S / solved | Kourovka 群论问题集

Can $SL(n,k)$ have maximal subgroups that are not closed in the Zariski topology?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04190 | KOU-2.55

### Kourovka Notebook Problem 2.55

S / solved | Kourovka 群论问题集

b) Does $SL_n(\mathbb{Z} )$, $n\geq 2$, have maximal subgroups of infinite index?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04191 | KOU-2.56

### Kourovka Notebook Problem 2.56

O / open | Kourovka 群论问题集

Classify up to isomorphism the abelian connected algebraic unipotent linear groups over a field of positive characteristic. This is not difficult in the case of a field of characteristic zero. On the other hand, C. Chevalley has solved the classification problem for such groups up to isogeny.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04192 | KOU-2.58

### Kourovka Notebook Problem 2.58

S / solved | Kourovka 群论问题集

Let $G$ be a vector space and $\Gamma$ a group of automorphisms of $G$. $\Gamma$ is called *locally finitely stable* if, for any finitely generated subgroup $\Delta$ of $\Gamma$, \ $G$ has a finite series stable relative to $\Delta$. If the characteristic of the field is zero and $\Gamma$ is locally finitely stable, then $\Gamma$ is locally nilpotent and torsion-free. Is it true that every locally nilpotent torsion-free group can be realized in this way?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04193 | KOU-2.59

### Kourovka Notebook Problem 2.59

S / solved | Kourovka 群论问题集

Let $\Gamma$ be any nilpotent group of class $ n-1$. Does $ \Gamma$ always admit a faithful representation as a group of automorphisms of an abelian group with a series of length $n$ stable relative to $\Gamma$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04194 | KOU-2.6

### Kourovka Notebook Problem 2.6

O / open | Kourovka 群论问题集

In an $NR$-group, the set of generalized central elements coincides with the nil-kernel. Is the converse true, that is, must a group be an $NR$-group if the set of generalized central elements coincides with the nil-kernel?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04195 | KOU-2.61

### Kourovka Notebook Problem 2.61

S / solved | Kourovka 群论问题集

Let $\Gamma$ be a Noetherian group of automorphisms of a vector space $G$ such that every element of $\Gamma$ is unipotent. Is $\Gamma$ necessarily a stable group of automorphisms?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04196 | KOU-2.62

### Kourovka Notebook Problem 2.62

S / solved | Kourovka 群论问题集

If $G$ is a finite-dimensional vector space over a field and $\Gamma$ is a group of automorphisms of $G$ in which every element is stable, then the whole of $\Gamma$ is stable (E. Kolchin). Is Kolchin's theorem true for spaces over skew fields?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04197 | KOU-2.63

### Kourovka Notebook Problem 2.63

S / solved | Kourovka 群论问题集

Let $G$ be a group of automorphisms of a vector space over a field of characteristic zero, and suppose that all elements of $G$ are unipotent, with uniformly bounded unipotency indices. Must such a group be locally finitely stable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04198 | KOU-2.64

### Kourovka Notebook Problem 2.64

S / solved | Kourovka 群论问题集

Does the set of nil-elements of a finite-dimensional linear group coincide with its locally nilpotent radical?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04199 | KOU-2.65

### Kourovka Notebook Problem 2.65

S / solved | Kourovka 群论问题集

Does the adjoint group of a radical ring (in the sense of Jacobson) have a central series?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04200 | KOU-2.66

### Kourovka Notebook Problem 2.66

S / solved | Kourovka 群论问题集

Is an $R$-group determined by its subgroup lattice? Is every lattice isomorphism of $R$-groups induced by a group isomorphism?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04201 | KOU-2.67

### Kourovka Notebook Problem 2.67

O / open | Kourovka 群论问题集

Find conditions which ensure that the nilpotent product of pure nilpotent groups (from certain classes) is determined by the lattice of its subgroups. This is known to be true if the product is torsion-free.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04202 | KOU-2.68

### Kourovka Notebook Problem 2.68

O / open | Kourovka 群论问题集

What can one say about lattice isomorphisms of a pure soluble group? Is such a group strictly determined by its lattice? It is well-known that the answer is affirmative for free soluble groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04203 | KOU-2.69

### Kourovka Notebook Problem 2.69

S / solved | Kourovka 群论问题集

Let a group $G$ be the product of two subgroups $A$ and $B$, each of which is nilpotent and satisfies the minimum condition. Prove or refute the following: \ a) $G$ is soluble; \ b) the divisible parts of $A$ and $B$ commute elementwise.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04204 | KOU-2.7

### Kourovka Notebook Problem 2.7

S / solved | Kourovka 群论问题集

Find the cardinality of the set of all polyverbal operations acting on the class of all groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04205 | KOU-2.70

### Kourovka Notebook Problem 2.70

S / solved | Kourovka 群论问题集

a) Let a group $G$ be the product of two subgroups $A$ and $B$, each of which is locally cyclic and torsion-free. Prove that either $A$ or $B$ has a non-trivial subgroup that is normal in $G$.

b) Characterize the groups that can be factorized in this way.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04206 | KOU-2.71

### Kourovka Notebook Problem 2.71

S / solved | Kourovka 群论问题集

Does there exist a finitely generated right-orderable group which coincides with its derived subgroup and, therefore, does not have the property $RN$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04207 | KOU-2.72

### Kourovka Notebook Problem 2.72

S / solved | Kourovka 群论问题集

(G. Baumslag). Suppose that $F$ is a finitely generated free group, $N$ its normal subgroup and $V$ a fully invariant subgroup of $N$. Is $F/V$ necessarily Hopfian if $F/N$ is Hopfian?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04208 | KOU-2.73

### Kourovka Notebook Problem 2.73

S / solved | Kourovka 群论问题集

(Well-known problem). Does there exist an infinite group all of whose proper subgroups have prime order?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04209 | KOU-2.74

### Kourovka Notebook Problem 2.74

O / open | Kourovka 群论问题集

(Well-known problem). Describe the finite groups all of whose involutions have soluble centralizers.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04210 | KOU-2.75

### Kourovka Notebook Problem 2.75

S / solved | Kourovka 群论问题集

Let $G$ be a periodic group containing an infinite family of finite subgroups whose intersection contains non-trivial elements. Does $G$ contain a non-trivial element with infinite centralizer?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04211 | KOU-2.76

### Kourovka Notebook Problem 2.76

S / solved | Kourovka 群论问题集

Let $\Gamma$ be the holomorph of an abelian group $A$. Find conditions for $A$ to be maximal among the locally nilpotent subgroups of $\Gamma$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04212 | KOU-2.77

### Kourovka Notebook Problem 2.77

S / solved | Kourovka 群论问题集

Let $A$ and $B$ be abelian groups. Find conditions under which every extension of $A$ by $B$ is nilpotent.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04213 | KOU-2.78

### Kourovka Notebook Problem 2.78

O / open | Kourovka 群论问题集

Any set of all subgroups of the same given order of a finite group $G$ that contains at least one non-normal subgroup is called an $IE_{\bar n}$-system of $G$. A positive integer $k$ is called a *soluble* (*non-soluble; \ simple; \ composite; \ absolutely simple*) *group-theoretic number* if every finite group having exactly $ k$ \ $IE_{\bar n}$-systems is soluble (respectively, if there is at least one non-soluble finite group having $k$ \ $IE_{\bar n}$-systems; \ if there is at least one simple finite group having $k$ \ $IE_{\bar n}$-systems; \ if there are no simple finite groups having $k$ \ $IE_{\bar n}$-systems; \ if there is at least one simple finite group having $k$ \ $IE_{\bar n}$-systems and there are no non-soluble non-simple finite groups having $k$ \ $IE_{\bar n}$-systems).

Are the sets of all soluble and of all absolutely simple group-theoretic numbers finite or infinite? Do there exist composite, but not soluble group-theoretic numbers?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04214 | KOU-2.79

### Kourovka Notebook Problem 2.79

S / solved | Kourovka 群论问题集

Do there exist divisible (simple) groups with maximal subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04215 | KOU-2.8

### Kourovka Notebook Problem 2.8

S / solved | Kourovka 群论问题集

(A. I. Mal'cev). Do there exist regular associative operations having the heredity property with respect to transition from the factors to their subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04216 | KOU-2.80

### Kourovka Notebook Problem 2.80

O / open | Kourovka 群论问题集

Does every non-trivial group satisfying the normalizer condition contain a non-trivial abelian normal subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04217 | KOU-2.81

### Kourovka Notebook Problem 2.81

S / solved | Kourovka 群论问题集

a) Does there exist an axiomatizable class of lattices ${\mathfrak{K}}$ such that the lattice of all subsemigroups of a semigroup $S$ is isomorphic to some lattice in ${\mathfrak{K}}$ if and only if $S$ is a free group?

b) The same question for free abelian groups.

Analogous questions have affirmative answers for torsion-free groups, for non-periodic groups, for abelian torsion-free groups, for abelian non-periodic groups, for orderable groups (the corresponding classes of lattices are even finitely axiomatizable). Thus, in posed questions one may assume from the outset that the semigroup $S$ is a torsion-free group (respectively, a torsion-free abelian group).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04218 | KOU-2.82

### Kourovka Notebook Problem 2.82

O / open | Kourovka 群论问题集

Can the class of groups with the $n $th Engel condition $[x,\underbrace{y,\ldots ,y}_{n}]=1$ be defined by identical relations of the form $u=v$, where $u$ and $v$ are words without negative powers of variables? This can be done for $n=1, 2, 3$ (A. I. Shirshov, *Algebra i Logika*, **2**, no. 5 (1963), 5--18 (Russian)).

*Editors' comment:* This has also been done for $n=4$ (G. Traustason, *J. Group Theory*, **2**, no. 1 (1999), 39--46). As remarked by O. Macedo\'nska, this is also true for the class of locally graded $n$-Engel groups because by (Y. Kim, A. Rhemtulla, in *Groups--Korea'94*, de Gruyter, Berlin, 1995, 189--197) such a group is locally nilpotent and then by (R. G. Burns, Yu. Medvedev, *J. Austral. Math. Soc.*, **64** (1998), 92--100) such a group is an extension of a nilpotent group of $n$-bounded class by a group of $n$-bounded exponent; then a classical result of Mal'cev implies that such groups satisfy a positive law.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04219 | KOU-2.83

### Kourovka Notebook Problem 2.83

S / solved | Kourovka 群论问题集

Suppose that a periodic group $G$ is the product of two locally finite subgroups. Is then $G$ locally finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04220 | KOU-2.84

### Kourovka Notebook Problem 2.84

O / open | Kourovka 群论问题集

Suppose that a locally finite group $G$ is a product of two locally nilpotent subgroups. Is $G$ necessarily locally soluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04221 | KOU-2.88

### Kourovka Notebook Problem 2.88

S / solved | Kourovka 群论问题集

Is every Hall $\pi$-subgroup of an arbitrary group a maximal $\pi$-subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04222 | KOU-2.9

### Kourovka Notebook Problem 2.9

O / open | Kourovka 群论问题集

Do there exist regular associative operations on the class of groups satisfying the weakened Mal'cev condition (that is, monomorphisms of the factors of an arbitrary product can be glued together, generally speaking, into a homomorphism of the whole product), but not satisfying the analogous condition for epimorphisms of the factors?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04223 | KOU-20.1

### Kourovka Notebook Problem 20.1

S / solved | Kourovka 群论问题集

For $k\geq1$, a group $G$ is said to be *totally $k$-closed* if in each of its faithful permutation representations, say on a set $\Omega$, $G$ is the largest subgroup of $\mathrm{Sym}(\Omega)$ which leaves invariant each of the $G$-orbits in the induced action on the set of ordered $k$-tuples $\Omega^k$.

Are there any finite insoluble totally $2$-closed groups with nontrivial Fitting subgroup?

The finite totally $2$-closed groups which are either soluble or have trivial Fitting subgroup are known.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04224 | KOU-20.10

## Kourovka Notebook Problem 20.10

P / partially_solved | Kourovka 群论问题集

(a) If ${\mathscr U}$ and ${\mathscr U'}$ are nonprincipal ultrafilters on $\mathbb{N}$, can every group which can be written as a homomorphic image of an ultraproduct of groups with respect to ${\mathscr U}$ also be written as a homomorphic image of an ultraproduct of groups with respect to ${\mathscr U'}$?

(b) If the answer to (a) is negative, is it at least true that for any two nonprincipal ultrafilters ${\mathscr U}$ and ${\mathscr U'}$ on $\mathbb{N}$, there exists a nonprincipal ultrafilter ${\mathscr U''}$ on $\mathbb{N}$ such that every group which can be written as a homomorphic image of an ultraproduct of groups with respect to ${\mathscr U}$ or with respect to ${\mathscr U'}$ can be written as a homomorphic image of an ultraproduct with respect to ${\mathscr U''}$?

A positive answer to (b) would imply that the class of groups which can be written as homomorphic images of nonprincipal countable ultraproducts of groups is closed under finite direct products. See (G. M. Bergman, *Pacific J. Math.*, **274** (2015), 451--495).

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04225 | KOU-20.100

## Kourovka Notebook Problem 20.100

O / open | Kourovka 群论问题集

*Conjecture*: Let $n$ be a positive integer, and let $G$ be a group containing no elements of order among $2,\ldots,n+1$. Then, for any $A\subseteq G$ with $|A|=n$, we may write $A=\{a_1,\ldots,a_n\}$ with $a_1,a_2^2,\ldots,a_n^n$ pairwise distinct.

The conjecture is known to hold for $n\leq 3$.

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04226 | KOU-20.101

## Kourovka Notebook Problem 20.101

S / solved | Kourovka 群论问题集

Let $G$ be a finitely generated branch group. Are all finite-index maximal subgroups of $G$ necessarily normal?

Cf. 18.81.

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04227 | KOU-20.102

## Kourovka Notebook Problem 20.102

O / open | Kourovka 群论问题集

(Y. Barnea, A. Shalev). Let $G$ be a finitely generated pro-$p$ group. Must $G$ be $p$-adic analytic if it has finite Hausdorff spectrum with respect to

a) the $p$-power series?

b) the iterated $p$-power series?

c) the lower $p$-series?

d) the Frattini series?

e) the dimension subgroup series?

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04228 | KOU-20.103

## Kourovka Notebook Problem 20.103

O / open | Kourovka 群论问题集

Let $G_\Gamma$ be a partially commutative soluble group of derived length $n \geq 3$ with defining graph $\Gamma$ (the definition is similar to the case of $n=2$, see 17.104). Is $G_\Gamma$ a torsion-free group?

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04229 | KOU-20.104

## Kourovka Notebook Problem 20.104

O / open | Kourovka 群论问题集

(Well-known questions). Suppose that a group $G$ is finitely generated and decomposable into a direct product $G = G_1 \times G_2$.

a) Is it true that the elementary theory of $G$ is decidable if and only if the elementary theories of the groups $G_1$ and $G_2$ are decidable?

b) Is it true that the universal theory of $G$ is decidable if and only if the universal theories of the groups $G_1$ and $G_2$ are decidable?

An affirmative answer to question a) for almost soluble groups follows from (G. A. Noskov, *Izv. Akad. Nauk SSSR, Ser. Mat.*, **47**, no. 3 (1983), 498--517).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04230 | KOU-20.105

## Kourovka Notebook Problem 20.105

O / open | Kourovka 群论问题集

Is there a perfect locally nilpotent $p$-group, for some prime $p$, whose proper subgroups are hypercentral?

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04231 | KOU-20.106

### Kourovka Notebook Problem 20.106

O / open | Kourovka 群论问题集

Let $G$ be a residually finite $2$-group and let $x\in G$ be a left 3-Engel element of order 2. Is $\langle x^{G}\rangle$ locally nilpotent?

It is known that in any group a 3-Engel element of odd order belongs to the locally nilpotent radical (E. Jabara, G. Traustason, *Proc. Amer. Math. Soc.*, **147**, no. 5 (2019), 1921--1927).

*Comment of 2025*: An element $a\in G$ is called a strong left $3$-Engel element if $\langle a,a^{g}\rangle$ is nilpotent of class at most $2$ and $\langle a,a^{g},a^{h}\rangle$ is nilpotent of class at most $3$ for all $g,h\in G$. (This is equivalent to $a$ being left $3$-Engel when $a$ is of odd order.) It is proved that if $a$ is a strong left $3$-Engel element in an arbitrary group $G$, then $\langle a\rangle^{G}$ is locally nilpotent (A. Hadjievangelou, G. Traustason, *Proc. Amer. Math. Soc.*, **152**, no. 4 (2024), 1467--1477).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04232 | KOU-20.107

### Kourovka Notebook Problem 20.107

O / open | Kourovka 群论问题集

Let $G$ be a group of exponent 8 and $x\in G$ a left 3-Engel element of order 2. Is $\langle x^{G}\rangle$ locally finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04233 | KOU-20.108

### Kourovka Notebook Problem 20.108

P / partially_solved | Kourovka 群论问题集

The *holomorph* $\mathrm{Hol}(G)$ of a group $G$ can be defined as the normalizer of the subgroup of left translations in the group of all permutations of the set $G$. The *multiple holomorph* $\mathrm{NHol}(G)$ of $G$ is the normalizer of the holomorph. Set $T(G) = \mathrm{NHol}(G)/\mathrm{Hol}(G)$.

a) Is there a centerless group $G$ for which $T(G)$ is not an elementary abelian 2-group?

b) Is there a finite centerless group $G$ for which $T(G)$ is not an elementary abelian 2-group?

c) (A. Caranti). Is there a finite $p$-group $G$ for which the order of $T(G)$ has a prime divisor not dividing $(p-1)p$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04234 | KOU-20.109

### Kourovka Notebook Problem 20.109

S / solved | Kourovka 群论问题集

A *skew brace* is a set $B$ equipped with two operations $+$ and $\cdot$ such that $(B,+)$ is an additively written (but not necessarily abelian) group, $(B,\cdot )$ is a multiplicatively written group, and $a\cdot (b+c)=ab-a+ac$ for any $a,b,c\in B$.

Is there a finite skew brace with perfect additive group and non-perfect almost simple multiplicative group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04235 | KOU-20.11

### Kourovka Notebook Problem 20.11

O / open | Kourovka 群论问题集

Let $F \leq H$ be free groups such that there exists a free group $G$ of finite rank with $F\leq G\leq H$, and let $r$ be the least of the ranks of such groups $G$. Which, if any, of the following statements must hold?

(i) There is a largest $G$ of rank $r$ between $F$ and $H$.

(i$'$) For any two $G_1$, $G_2$ of rank $r$ between $F$ and $H$, the subgroup $\langle G_1, G_2\rangle$ has rank $r$.

(ii) There is a smallest $G$ of rank $r$ between $F$ and $H$.

(ii$'$) For any two $G_1$, $G_2$ of rank $r$ between $F$ and $H$, the subgroup $G_1\cap G_2$ has rank $r$.

If (i) holds for all such $F$ and $H$, then so does (i$'$). If (ii) holds for all $F$, $H$, then so does (ii$'$). The converse of the former implication holds because subgroups of a free group of any fixed finite rank satisfy ACC, but I don't see a way to get the converse of the other implication.

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04236 | KOU-20.110

### Kourovka Notebook Problem 20.110

O / open | Kourovka 群论问题集

Are there residually finite hereditarily just infinite groups that are

a) amenable but not solvable?

b) amenable but not elementary amenable?

c) of intermediate word growth?

d) of intermediate subgroup growth?

All examples that we know are either linear (hence Tits Alternative applies), or have a quotient with property (T) (hence cannot be amenable).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04237 | KOU-20.111

### Kourovka Notebook Problem 20.111

O / open | Kourovka 群论问题集

Are there residually finite hereditarily just infinite groups admitting a self-similar action on a rooted tree that are not linear?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04238 | KOU-20.112

### Kourovka Notebook Problem 20.112

O / open | Kourovka 群论问题集

Let $\mathfrak{F}$ be a hereditary saturated formation and let $\mathrm{w}^{*}\mathfrak{F}$ denote the class of all finite groups $G$ for which $\pi(G)\subseteq\pi(\mathfrak{F})$ and the normalizers of all Sylow subgroups of $G$ are $\mathfrak{F}$-subnormal in $G$. It is known that $\mathrm{w}^{*}\mathfrak{F}$ is a formation. Must $\mathrm{w}^{*}\mathfrak{F}$ be a saturated formation?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04239 | KOU-20.113

### Kourovka Notebook Problem 20.113

O / open | Kourovka 群论问题集

Let $H(q,c) = \langle a,b,c,d \mid [b,a]=[d,c], \text{ of exponent } q, \text{ nilpotent of class } c\rangle$.

a) Is it true that the Schur multiplier $M(H(8,12))$ has exponent 32?

b) Is it true that the Schur multiplier $M(H(7,13))$ has exponent 49?

The difficulty is that these groups are too big to compute using current versions of the $p$-Quotient Algorithm, which use 32 bit arithmetic. So to tackle these groups it would help to have a version of the $p$-Quotient Algorithm using 64 bit arithmetic.

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04240 | KOU-20.114

### Kourovka Notebook Problem 20.114

O / open | Kourovka 群论问题集

Is there a positive integer $c$ such that for any $q$ the exponent of the Schur multiplier of a finite group of exponent $q$ divides $q^c$?

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04241 | KOU-20.115

### Kourovka Notebook Problem 20.115

O / open | Kourovka 群论问题集

Let $\chi$ be a complex irreducible character of a finite group $G$. If $\chi(x)\ne 0$ for some $x\in G$, must the order $o(x)$ of $x$ divide $|G|/\chi(1)$?

This is known to be true if $G$ is solvable, and it is known that $(o(x)\chi(1))^4$ divides $|G|^5$ for arbitrary $G$.

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04242 | KOU-20.116

### Kourovka Notebook Problem 20.116

O / open | Kourovka 群论问题集

(Well-known problem). Does a profinite torsion group have finite exponent?

This problem reduces to the case of pro-$p$ groups (cf. W. Herfort, *Arch. Math.*, **33** (1980), 404--410). Also cf. Archive 3.41.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04243 | KOU-20.117

### Kourovka Notebook Problem 20.117

O / open | Kourovka 群论问题集

Let $G$ be a pro-$p$ group that is a 3-dimensional Poincar\'e duality group. Is $G$ coherent?

A group is said to be *coherent* if each of its finitely generated subgroups is finitely presented, and in the question the coherency is used in the pro-$p$ sense. The question is a famous problem for abstract groups, but has a positive answer for 3-manifold groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04244 | KOU-20.118

### Kourovka Notebook Problem 20.118

O / open | Kourovka 群论问题集

Let $G$ be a pro-$p$ group that is a 3-dimensional Poincar\'e duality group. Can it contain a direct product $F_2\times F_2$ of free pro-$p$ groups of rank 2?

If it can, then the answer to 20.117 is negative, but in the abstract case it is known that it can not (P. H. Kropholler, M. A. Roller, *J. London Math. Soc. (2)*, **39** (1989), 271--284).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04245 | KOU-20.119

### Kourovka Notebook Problem 20.119

O / open | Kourovka 群论问题集

(G. Wilkes). A pro-$p$ group is said to be *accessible* if there is a number $n = n(G)$ such that any finite proper reduced graph of pro-$p$ groups with finite edge groups having fundamental group isomorphic to $G$ has at most $n$ edges (*J. Algebra*, **525** (2019), 1--18).

Is every finitely presented pro-$p$ group accessible?

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04246 | KOU-20.12

### Kourovka Notebook Problem 20.12

O / open | Kourovka 群论问题集

Let us say that a group $G$ has the *unique $n$-fold product* property (u.-$n$-p.) if for every $n$-tuple of finite nonempty subsets $A_1, \dots , A_n\subseteq G$ there exists $g\in G$ which can be written in one and only one way as $g = a_1\cdots a_n$ with $a_i\in A_i$. It is easy to see that u.-$n$-p. implies u.-$m$-p. for $n \geq m$ (since some of the $A_i$ can be $\{1\}$).

Are the conditions u.-$n$-p. ($n\geq 2$) all equivalent to u.-2-p., the usual unique product condition?

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04247 | KOU-20.120

### Kourovka Notebook Problem 20.120

O / open | Kourovka 群论问题集

Let $G$ be a locally nilpotent group with an automorphism (of infinite order). Can $G$ be simple as a group with automorphism?

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04248 | KOU-20.121

### Kourovka Notebook Problem 20.121

S / solved | Kourovka 群论问题集

Let $G$ be a finite almost simple group such that $Soc(G)$ is a non-soluble group of Lie type over a field of characteristic $p$. Let $R$ be a Sylow $q$-subgroup of $G$ (for some $q$) and let $\operatorname{Min}_G(R)$ be the subgroup of $R$ generated by all minimal by inclusion intersections of the form $R\cap R^g$, where $g\in G$. Is it true that for $p>3$ the subgroup $\operatorname{Min}_G(R)$ is non-trivial if and only if the following hold: $G=\operatorname{Aut}(L_2(p))$, where $p$ is a Mersenne prime, $\operatorname{Min}_G(R)=R$, and $q=2$.

It is known that this is not always the case for $p=2,3$.

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04249 | KOU-20.122

### Kourovka Notebook Problem 20.122

S / solved | Kourovka 群论问题集

For nilpotent subgroups $A,B, C$ of a finite group $G$, let $\operatorname{Min}_G(A,B,C)$ be the subgroup of $A$ generated by all minimal by inclusion intersections of the form $A\cap B^x\cap C^y$, where $ x, y\in G$, and let $\operatorname{min}_G(A,B,C)$ be the subgroup of $\operatorname{Min}_G(A,B,C)$ generated by all intersections of this kind of minimal order.

a) Is it true that $\operatorname{min}_G(A,B,C)\leq F(G)$?

b) Is it true that $\operatorname{Min}_G(A,B,C)\leq F(G)$?

c) The same questions for soluble groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04250 | KOU-20.123

### Kourovka Notebook Problem 20.123

P / partially_solved | Kourovka 群论问题集

A finite group is called a {\it $D_{\pi}$-group} if any two of its maximal $\pi $-subgroups are conjugate.

a) Is it true that for any finite $D_\pi$-group $G$ and a $\pi$-Hall subgroup $H$ of $G$, there are elements $x,y,z\in G$ such that $O_{\pi }(G)=H\cap H^x\cap H^y\cap H^z$?

b) Suppose that $G$ is a finite $D_{\pi}$-group in which all simple non-abelian composition factors are sporadic or alternating groups, and let $H$ be a Hall $\pi $-subgroup of $G$. Is it true that $H\cap H^x\cap H^y= O_\pi(G)$ for some $x,y\in G$?

c) Suppose that $G$ is a finite $D_{\pi}$-group with trivial soluble radical in which all simple non-abelian composition factors are sporadic groups, and let $H$ be a Hall $\pi $-subgroup of $G$. Is it true that $H\cap H^g= O_\pi(G)$ for some $g\in G$?

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04251 | KOU-20.124

### Kourovka Notebook Problem 20.124

O / open | Kourovka 群论问题集

A *Rota--Baxter operator* on a group $G$ is a mapping $B : G \to G$ such that $B(g) B(h) = B\left(gB(g)hB(g)^{-1} \right)$ for all $g, h \in G$. Let $F$ be a non-abelian free group. Is there a Rota--Baxter operator on $F$ such that its image is equal to the derived subgroup $[F, F]$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04252 | KOU-20.125

### Kourovka Notebook Problem 20.125

S / solved | Kourovka 群论问题集

Does there exist a non-abelian group $G$ and a Rota--Baxter operator $B : G \to G$ such that $B$ is surjective but not injective?

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04253 | KOU-20.126

### Kourovka Notebook Problem 20.126

O / open | Kourovka 群论问题集

A *brace* $(G; +, \circ)$ is non-empty set $G$ with two binary operations $+$, $\circ$ such that $(G, +)$ is an additively written abelian group, $(G, \circ)$ is a multiplicatively written group, and $a \circ (b + c) + a = (a \circ b) + (a \circ c)$ for all $a, b, c \in G$. Does there exist a brace with finitely generated group $(G, +)$ such that

a) the group $(G, \circ)$ is non-solvable?

b) the group $(G, \circ)$ contains a non-abelian free group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04254 | KOU-20.13

## Kourovka Notebook Problem 20.13

S / solved | Kourovka 群论问题集

{(a)} If an abelian group can be written as a homomorphic image of a nonprincipal countable ultraproduct of *not necessarily* abelian groups $G_i$, must it be a homomorphic image of a nonprincipal countable ultraproduct of abelian groups? See (G. M. Bergman, *Pacific J. Math.*, **274** (2015), 451--495).

(b) If an abelian group can be written as a homomorphic image of a *direct product* of an infinite family of not necessarily abelian *finite* groups, can it be written as a homomorphic image of a direct product of finite abelian groups?

To see that neither question is trivial, choose for each $n > 0$ a finite group $G_n$ which is perfect but has elements which cannot be written as products of fewer than $n$ commutators. Then both the direct product of the $G_n$ and any nonprincipal ultraproduct of those groups will have elements which are not products of commutators; hence its abelianization $A$ will be nontrivial. There is no evident family of abelian groups from which to obtain $A$ as an image of a nonprincipal ultraproduct, nor a family of finite abelian groups from which to obtain $A$ as an image of a direct product.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04255 | KOU-20.14

## Kourovka Notebook Problem 20.14

O / open | Kourovka 群论问题集

(a) Do there exist a variety $\mathfrak{V}$ of groups and a group $G\in \mathfrak{V}$ such that the coproduct in $\mathfrak{V}$ of two copies of $G$ is embeddable in $G$, but the coproduct of three such copies is not? See (G. M. Bergman, *Indag. Math.*, **18** (2007), 349--403).

Given an embedding $G *_\mathfrak{V} G \to G$, one might expect the induced map \begin{center} $G*_\mathfrak{V} (G *_\mathfrak{V} G) \to G *_\mathfrak{V} G \to G $ \end{center} to be an embedding. But this is not automatic, because in a general group variety $\mathfrak{V}$, a map $G *_\mathfrak{V} A \to G *_\mathfrak{V} B$ induced by an embedding $A\to B$ is not necessarily again an embedding.

(b) If there exist $\mathfrak{V}$ and $G$ as in (a), does there in fact exist an example with $\mathfrak{V}$ the variety of groups generated by $G$? For this and related questions, see (G. M. Bergman, *Algebra Number Theory*, **3** (2009), 847--879).

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04256 | KOU-20.15

## Kourovka Notebook Problem 20.15

O / open | Kourovka 群论问题集

Let $\kappa$ be an infinite cardinal. If a residually finite group $G$ is embeddable in the full permutation group of a set of cardinality $\kappa$, must it be embeddable in the direct product of $\kappa$ finite groups? See (G. M. Bergman, *Indag. Math.*, **18** (2007), 349--403).

The converse is true: any such direct product is residually finite and embeddable in the indicated permutation group. Also, the statement asked for becomes true if one replaces ``direct product of $\kappa$ finite groups'' by ``direct product of $2^\kappa$ finite groups'', since $G$ has cardinality at most $2^\kappa$, hence homomorphisms to that many finite groups can be chosen which, together, separate each element of $G$ from $e$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04257 | KOU-20.16

## Kourovka Notebook Problem 20.16

O / open | Kourovka 群论问题集

By Hartley's theorem (based on CFSG), if a simple locally finite group does not contain a particular finite group as a section, then it is isomorphic to a group of Lie type over a locally finite field.

a) Is the same true for an infinite definably simple locally finite group? A group is said to be *definably simple* if it does not contain proper definable normal subgroups.

b) The same question for an infinite definably simple locally finite group with bounded derived lengths of its solvable subgroups.

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04258 | KOU-20.17

## Kourovka Notebook Problem 20.17

P / partially_solved | Kourovka 群论问题集

The *normal covering number* $\gamma(G)$ of a finite non-cyclic group $G$ is the minimum number of proper subgroups of $G$ such that $G$ is the union of their conjugates.

(ae) What is the exact value of \ $\displaystyle\liminf_{n\ \mathrm{even}}\gamma(S_n)/n$?

(ao) What is the exact value of \ $\displaystyle\liminf_{n\ \mathrm{odd}}\gamma(S_n)/n$?

*Comment of 2025*: It belongs to $[1/12, 1/6]$ for $n$ odd (S. Eberhard, C. Mellon, *Bull. London Math. Soc.* (2025), https://doi.org/10.1112/blms.70154).

(b) What are the exact values of \ $\displaystyle\liminf_{n\ \mathrm{even}}\gamma(A_n)/n$ \ and \ $\displaystyle\liminf_{n\ \mathrm{odd}}\gamma(A_n)/n$?

*Comment of 2025*: These belong to $[1/18, 1/6]$ for $n$ even, and to $[1/6, 4/15]$ for $n$ odd (S. Eberhard, C. Mellon, *Bull. London Math. Soc.* (2025), https://doi.org/10.1112/blms.70154).

(c) What is the exact value of $\displaystyle\limsup_{n\ \mathrm{even}}\gamma(S_n)/n$?

It is known that $\displaystyle\limsup_{n\ \mathrm{odd}}\gamma(S_n)/n=1/2$.

(d) What are the exact values of \ $\displaystyle\limsup_{n\ \mathrm{odd}}\gamma(A_n)/n$ \ and \ $\displaystyle\limsup_{n\ \mathrm{even}}\gamma(A_n)/n$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04259 | KOU-20.18

## Kourovka Notebook Problem 20.18

O / open | Kourovka 群论问题集

Let $R_{p^k}$ be the variety of class $2$ nilpotent groups of exponent $p^k$, where $p$ is a prime number and $k\geq 2$. It is true that for every $p$ and $k$ there are infinitely many subquasivarieties of $R_{p^k}$ each of which is generated by a finite group with derived subgroup of exponent $p^k$ and does not have an independent basis of quasi-identities?

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04260 | KOU-20.19

## Kourovka Notebook Problem 20.19

O / open | Kourovka 群论问题集

A subgroup $H$ of a group $G$ is called *commensurated* if for all $g\in G$, the index $|H : H \cap gHg^{-1}|$ is finite. Can a non-abelian free group (or a non-elementary hyperbolic group) contain two infinite commensurated subgroups $A$, $B$ with a trivial intersection? The answer is negative if $A$ or $B$ is normal.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04261 | KOU-20.2

## Kourovka Notebook Problem 20.2

S / solved | Kourovka 群论问题集

Are there any nonabelian simple groups of Lie type which are totally $3$-closed?

There are exactly six nonabelian simple totally $2$-closed groups --- all are sporadic groups, the largest being the Monster.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04262 | KOU-20.20

## Kourovka Notebook Problem 20.20

O / open | Kourovka 群论问题集

Let $E(l,m,n)$ be a group on generators $a,b$ with the presentation $\langle a,b\mid a^l= 1,\;(ab)^m=b^n\rangle$, where $l,m,n$ satisfy $1/l + 1/m + 1/n < 1$. Are there any values for $l,m,n$ such that $E(l,m,n)$ is shortlex automatic?

It is known that $E(6,2,6)$ is not shortlex automatic.

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04263 | KOU-20.21

## Kourovka Notebook Problem 20.21

O / open | Kourovka 群论问题集

(G. Verret). Does there exist a finite group $G$ with two normal subgroups $K$ and $L$, each with index 12 in $G$, such that $K$ is isomorphic to $L$, but $G/K$ is isomorphic to $C_{12}$, while $G/L$ is isomorphic to $A_4$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04264 | KOU-20.22

## Kourovka Notebook Problem 20.22

S / solved | Kourovka 群论问题集

Let $G$ be a non-free and non-cyclic one-relator group in which every subgroup of infinite index is free. Is $G$ a surface group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04265 | KOU-20.23

## Kourovka Notebook Problem 20.23

S / solved | Kourovka 群论问题集

Is there a constant $\delta$ with $0 < \delta < 1$ and an integer $N$ such that whenever $A$, $B$, and $C$ are conjugacy classes in the alternating group $Alt(n)$ each of size at least $|Alt(n)|^{\delta}$ with $n \geq N$, then $ABC = Alt(n)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04266 | KOU-20.24

### Kourovka Notebook Problem 20.24

O / open | Kourovka 群论问题集

Does the group $SL_2(\mathbb{Z} [\sqrt 2])$ admit a faithful transitive amenable action?

This is the same as asking if it admits any co-amenable subgroup except the finite index subgroups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04267 | KOU-20.25

### Kourovka Notebook Problem 20.25

O / open | Kourovka 群论问题集

Let $n=k^2$. Let $\sigma\in S_n$ be the product of $k$ disjoint cycles, of lengths $1,3,5,\dots , 2k-1$. Find the limiting proportion of odd entries in the $\sigma$-column of the character table of $S_n$.

For some background, see pages 1005--1006 in (D. Gluck, *Proc. Amer. Math. Soc.*, **147** (2019), 1005--1011).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04268 | KOU-20.26

### Kourovka Notebook Problem 20.26

O / open | Kourovka 群论问题集

Let $\phi$ be the Euler totient function. Does there exist a constant $a > 0$ such that $| Aut(G)| \geq \phi (|G|)^a$

a) for every finite group $G$?

b) for every finite nilpotent group $G$?

If such a constant $a$ exists, then $a\le \frac{40}{41}$ (J. Gonz\'alez-S\'anchez, A. Jaikin-Zapirain, *Forum Math. Sigma*, **3**, Article ID e7, 11 p., electronic only (2015)). Also cf. Archive 12.77.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04269 | KOU-20.27

### Kourovka Notebook Problem 20.27

S / solved | Kourovka 群论问题集

Let $G$ be a finite group, $p$ a prime number, and let $|x^G|_p$ denote the maximum power of $p$ that divides the class size of an element $x\in G$. Suppose that there exists a $p$-element $g\in G$ such that $|g^G|_p=\max_{x\in G}|x^G|_p$. Is it true that $G$ has a normal $p$-complement?

A partial answer is in (https://arxiv.org/abs/1812.03641).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04270 | KOU-20.28

### Kourovka Notebook Problem 20.28

O / open | Kourovka 群论问题集

Let $L$ be a non-abelian finite simple group, and let $H(L)=M(L).L$ be the universal perfect central extension, where $M(L)$ is the Schur multiplier of $L$. Suppose that $G$ is a finite group such that the set of class sizes of $G$ is the same as the set of class sizes of $H(L)$. Is it true that $G\simeq H(L)\times A$, where $A$ is an abelian group?

This is proved for $L=Alt_5$ in (*J. Algebra Appl.*, **21**, no. 11 (2022), Article ID 2250226, 8 p.).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04271 | KOU-20.29

### Kourovka Notebook Problem 20.29

O / open | Kourovka 群论问题集

Let $S$ be a non-abelian finite simple group. Is it true that for any $n\in \mathbb{N}$, if the set of class sizes of a centreless finite group $G$ is the same as the set of class sizes of the direct power $S^n$, then $G\simeq S^n$?

This is proved for $S=Alt_5^2$ in (*J. Algebra Appl.*, **21**, no. 11 (2022), Article ID 2250226, 8 p.).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04272 | KOU-20.30

### Kourovka Notebook Problem 20.30

S / solved | Kourovka 群论问题集

(P. M. Neumann and M. R. Vaughan-Lee). Let $G$ be a perfect and centreless finite group, and let $n$ be the maximum size of a conjugate class in $G$. Is it true that $|G|\leq n^2$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04273 | KOU-20.31

### Kourovka Notebook Problem 20.31

O / open | Kourovka 群论问题集

Let $\omega(G)$ denote the set of element orders of a finite group $G$, and $h(G)$ the number of pairwise nonisomorphic finite groups $H$ with $\omega(H)=\omega(G)$. Find $h(L)$, where

a)\; $L$ is the symmetric group of degree $10$;

b)\; $L$ is the automorphism group of the simple sporadic Janko group $J_2$;

c)\; $PSL(2,q)<L\leq \operatorname{Aut}(PSL(2,q))$.

See the current status in (M. A. Grechkoseeva, V. D. Mazurov, W. J. Shi, A. V. Vasil'ev, N. Yang, *Commun. Math. Stat.*, **11**, no. 2 (2023), 169--194; Subsection 4.2).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04274 | KOU-20.32

### Kourovka Notebook Problem 20.32

O / open | Kourovka 群论问题集

(E. Rapaport Strasser). The Lov\'asz conjecture states that a vertex-transitive connected graph is Hamiltonian. In the special case of Cayley graphs, one can ask the following. Consider a set $A$ which generates a group $G$ and is symmetric ($x\in A$ implies $x^{-1} \in A$). Is there a list $a_1,a_2,\dots, a_n$ of elements of $A$ such that $a_1, a_1a_2, a_1a_2a_3,\dots, a_1a_2\cdots a_n$ is a complete list of the elements of $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04275 | KOU-20.33

### Kourovka Notebook Problem 20.33

O / open | Kourovka 群论问题集

(A. Bauer). Does Higman's Embedding Theorem relativize in the following way? Is it the case that for every subset $X \subseteq \mathbb{N}$, there is a finitely generated group $G_X$ that has an $X$-computable presentation (that is, there is a finite generating set relative to which the set of relations is computably enumerable with an $X$ oracle), and such that any finitely generated group has an $X$-computable presentation if and only if it can be embedded as a finitely generated subgroup of a quotient of a free product of finitely many copies of $G_X$ by the normal closure of a finite subset?

Higman's Embedding Theorem says that for computable $X$, one may take $G_X=\mathbb{Z}$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04276 | KOU-20.34

### Kourovka Notebook Problem 20.34

O / open | Kourovka 群论问题集

Let $f$ be an inner screen of a saturated formation $\mathfrak{F}$ and suppose that a finite group $A$ acts faithfully and $f$-hypercentrally on a finite group $G$. Is it true that $G\rtimes A\in \mathfrak{F}$ for all $G\in \mathfrak{F}$ if and only if $\mathfrak{F}$ is a Fitting class?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04277 | KOU-20.35

### Kourovka Notebook Problem 20.35

O / open | Kourovka 群论问题集

(Well-known problem). Let $S$ be a connected orientable hyperbolic surface of finite type and complexity at least 2. Does its mapping class group ${MCG}(S)$ have a non-elementary hyperbolic quotient?

An affirmative answer is known when $S$ is a closed genus-2 surface (https://arxiv.org/pdf/2005.00567.pdf), and in the same paper it is shown that the answer will be affirmative provided certain hyperbolic groups are residually finite. See also a closely related question 16.108 about braid groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04278 | KOU-20.36

### Kourovka Notebook Problem 20.36

O / open | Kourovka 群论问题集

The free Burnside group of exponent four on three generators, $B(3,4)$, has order $2^{69}$ as shown by Bayes, Kautsky, and Wamsley (1974). Their proof is based on a theorem of Sanov (1940) which shows that $B(n,4)$ is finite. Sanov's proof for $B(3,4)$ uses more than $2^{32}$ fourth powers, because the subgroup of $B(3,4)$ generated by two of its generators and the square of the third has order $2^{32}$. It is also known that $B(3,4)$ needs at least 105 relations to define it, as shown by Havas and Newman (1980).

a) Can $B(3,4)$ be defined with fewer than a million fourth powers?

b) Can $B(3,4)$ be defined with fewer than a thousand fourth powers?

c) What is the smallest number of fourth powers which define $B(3,4)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04279 | KOU-20.37

### Kourovka Notebook Problem 20.37

O / open | Kourovka 群论问题集

Is it true that for every finite group $G$ and every factorization $|G|=ab$ there exist subsets $A,B\subseteq G$ with $|A|=a$ and $|B|=b$ such that $G=AB$? Cf. 19.35 in Archive.

*Comment of 2025*: any minimal counterexample $G$ must be a simple group without prime-index subgroups (R. R. Bildanov, V. A. Goryachenko, A. V. Vasil'ev, *Siberian Electron. Math. Rep.*, **17** (2020), 683--689; \ M. H. Hooshmand, *Commun. Algebra*, **49**, no. 7 (2021), 2927--2933).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04280 | KOU-20.38

### Kourovka Notebook Problem 20.38

O / open | Kourovka 群论问题集

Suppose that $H$ is an almost simple group of Lie type and $G$ is a finite group such that $G$ and $H$ have the same sets of degrees of irreducible complex characters. Must there exist an abelian normal subgroup $A$ of $G$ such that $G/A$ is isomorphic to $H$?

*Comment of 2025*: The conjecture was confirmed for projective general linear and unitary groups of dimension 3 (F. Shirjian, A. Iranmanesh, *Illinois J. Math.*, **64**, no. 1 (2020), 49--69).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04281 | KOU-20.39

### Kourovka Notebook Problem 20.39

O / open | Kourovka 群论问题集

Let $F$ be a non-abelian finitely generated free group, $1\ne w\in F$, and $n\ge 1$. Is the group $\langle F,t \mid t^n=w \rangle $ linear of degree 2 over a field of characteristic 0 if $w$ is not a proper power in $F$?

This question is motivated by the following well-known question: is it true that the free $\mathbb{Q}$-group $F^\mathbb{Q}$ is linear over a field? (See 13.39(b).)

问题集入口：https://kourovkanotebookorg.wordpress.com/

04282 | KOU-20.4

### Kourovka Notebook Problem 20.4

O / open | Kourovka 群论问题集

A finite group $G$ is said to be *cut* (or *inverse semi-rational*) if $\langle x \rangle = \langle y \rangle$ implies that $x$ is conjugate to $y$ or to $y^{-1}$ for all $x, y \in G$.

a) Let $\mathbb{Q}(G)$ denote the field extension of the rationals obtained by adjoining all entries of the ordinary character table of $G$. Is there $c > 0$ such that $|\mathbb{Q}(G):\mathbb{Q}| \leqslant c$ for all cut groups? This is true if one assumes in addition that $G$ is solvable (J. F. Tent, *J. Algebra*, **363** (2012), 73--82).

b) Is a Sylow $3$-subgroup of a cut group also a cut group?

c) Let $\operatorname{O}_p(G)$ denote the largest normal $p$-subgroup of $G$. Let $G$ be a solvable cut group. Is it true that for $p \in \{5, 7\}$ the exponent of $\operatorname{O}_p(G)$ divides $p$?

For additional information and some positive results see (*Adv. Group Theory Appl.*, **8**B, 2020, 157--160 or \texttt{https://arxiv.org/abs/2001.02637}).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04283 | KOU-20.40

### Kourovka Notebook Problem 20.40

O / open | Kourovka 群论问题集

Let $G$ be a $\kappa$-existentially closed group of cardinality $\lambda > \kappa$, where $\kappa$ is a regular cardinal. Is it true that $|Aut(G)|=2^{\lambda}$?

The answer is known to be affirmative if $\lambda=\kappa$ (B. Kaya, M. Kuzucuo\u{g}lu, P. Longobardi, M. Maj, *J. Algebra*, **666** (2025), 840--849).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04284 | KOU-20.41

### Kourovka Notebook Problem 20.41

S / solved | Kourovka 群论问题集

(L. Ciobanu, B. Fine, G. Rosenberger). Suppose that $G$ is a non-cyclic residually finite group in which every subgroup of finite index (including the group itself) is defined by a single defining relation, while all infinite index subgroups are free. Is it true that $G$ is either free or isomorphic to the fundamental group of a compact surface?

See Archive 7.36 for a negative solution of a similar question without the assumption on infinite index subgroups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04285 | KOU-20.42

### Kourovka Notebook Problem 20.42

O / open | Kourovka 群论问题集

(Y. Cornulier, A. Mann, A. Thom). Does there exist a finitely generated residually finite group that is not (elementary) amenable and satisfies a nontrivial group law?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04286 | KOU-20.43

### Kourovka Notebook Problem 20.43

O / open | Kourovka 群论问题集

Is every family of finite groups satisfying a common group law uniformly amenable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04287 | KOU-20.44

### Kourovka Notebook Problem 20.44

O / open | Kourovka 群论问题集

The definition of $\mathrm{CT}(\mathbb{Z})$ is given in 17.57. Is it true that a finitely generated subgroup of $\mathrm{CT}(\mathbb{Z})$ either has only finitely many orbits on $\mathbb{Z}$ or there is a set of representatives for its orbits on $\mathbb{Z}$ which has positive density?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04288 | KOU-20.45

### Kourovka Notebook Problem 20.45

O / open | Kourovka 群论问题集

Let $n$ be a positive integer, and let $G \leq \mathrm{GL}(n,\mathbb{Z})$ be finitely generated. Given a bound $b \in \mathbb{N}$, let $e_b$ be the number of elements of $G$ all of whose matrix entries have absolute value $\leq b$. Does the limit $\displaystyle\lim_{b \rightarrow \infty} \ln{e_b}/\ln{b}$ always exist?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04289 | KOU-20.46

### Kourovka Notebook Problem 20.46

O / open | Kourovka 群论问题集

A group action on a compact space is said to be *topologically free* if the set of points with trivial stabilizer is dense. Let $G$ be a locally compact group, and $\partial_{sp} G$ its Furstenberg boundary (the largest minimal and strongly proximal compact $G$-space). Let $\Gamma_1$ and $\Gamma_2$ be two lattices in $G$, both acting faithfully on $\partial_{sp} G$.

a) Is it possible that the $\Gamma_1$-action on $\partial_{sp} G$ is topologically free, but the $\Gamma_2$-action on $\partial_{sp} G$ is not topologically free?

b) If yes, can this also happen if $\partial_{sp} G = G/H$ is a homogeneous $G$-space?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04290 | KOU-20.47

### Kourovka Notebook Problem 20.47

O / open | Kourovka 群论问题集

Let $G = F_n$ be a finitely generated free group. Let $X$ be a compact $G$-space on which the $G$-action is faithful, minimal, and strongly proximal. Does it follow that the action is topologically free?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04291 | KOU-20.48

### Kourovka Notebook Problem 20.48

O / open | Kourovka 群论问题集

Suppose that $P$ is a finite $p$-group with a non-trivial partition (which is equivalent to having proper Hughes subgroup $H_p(P):=\langle g\in P\mid g^p\ne 1\rangle\ne P$). If $P$ admits a fixed-point-free automorphism of prime order, must $P$ be of exponent $p$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04292 | KOU-20.49

### Kourovka Notebook Problem 20.49

O / open | Kourovka 群论问题集

Is it true that any finite group contains a 2-generated subgroup with the same exponent?

An affirmative answer is known for soluble groups. It is also known that any finite group contains a 3-generated subgroup with the same exponent (E. Detomi, A. Lucchini, *J. London Math. Soc. (2)*, **87**, no. 3 (2013), 689--706).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04293 | KOU-20.5

### Kourovka Notebook Problem 20.5

O / open | Kourovka 群论问题集

(Well-known problem). Let $G$ be a finite group, and $\mathrm{V}(\mathbb{Z}G)$ the group of normalized units of the integral group ring of $G$. Do the spectra of $G$ and $\mathrm{V}(\mathbb{Z} G)$ coincide? That is, is it true that, for any integer $n$, there is an element of order $n$ in $\mathrm{V}(\mathbb{Z} G)$ if and only if there is an element of order $n$ in $G$?

The answer is ``yes'' if $G$ is solvable (M. Hertweck, *Comm. Algebra* **36** (2008), 3585--3588).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04294 | KOU-20.50

### Kourovka Notebook Problem 20.50

O / open | Kourovka 群论问题集

For a positive integer $m$, let $G_m$ be the largest group generated by $m$ involutions such that $(xy)^4=1$ for any two involutions $x,y\in G_m$. What is the order of $G_4$?

It is known that the order of $G_3$ is equal to $2^{11}$. Also cf. 18.58.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04295 | KOU-20.51

### Kourovka Notebook Problem 20.51

O / open | Kourovka 群论问题集

Is it true that for every odd integer $t > 3$ there exists a finite non-abelian group $G$ of odd order with exactly $t$ conjugacy classes?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04296 | KOU-20.52

### Kourovka Notebook Problem 20.52

O / open | Kourovka 群论问题集

A famous result of Burnside states that if $k(G)$ is the number of conjugacy classes of a finite group $G$ of odd order, then $|G|- k(G)$ is divisible by 16. Is it true that for every integer $m > 0$ there exists a finite non-abelian group $G(m)$ of odd order such that $|G(m)| - k(G(m)) = 16m$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04297 | KOU-20.53

### Kourovka Notebook Problem 20.53

O / open | Kourovka 群论问题集

A nonisotropic unitary graph $\Gamma$ is distance-regular with intersection array $\{q(q-1), (q+1)(q-2), q+1; 1, 1, q(q-2)\}$ for some prime power $q$. The group $G=Aut (\Gamma)$ acts transitively on the vertex set and on the edge set of $\Gamma$. It is known that $\Gamma$ is distance-transitive if $q=3$. Does there exist a distance-regular graph with such an intersection array if $q$ is not a prime power?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04298 | KOU-20.54

### Kourovka Notebook Problem 20.54

O / open | Kourovka 群论问题集

A distance-regular graph of diameter 3 with the second eigenvalue $\theta_1=a_3$ is called a *Shilla graph*. For a Shilla graph $\Gamma$ the number $a=a_3$ divides $k$ and we set $b=b(\Gamma)=k/a$. Koolen and Park proved that there are 12 feasible intersection arrays of Shilla graphs with $b=3$. At present it is proved that a Shilla graph with $b=3$ has intersection array $\{12,10,3;1,3,8\}$ (Doro graph), $\{12,10,5;1,1,8\}$ (nonisotropic unitary graph for $q=4$), or $\{15,12,6;1,2,10\}$. The automorphisms of the last graph were found by A. Makhnev and N. Zyulyarkina (*Doklady Maths.*, **84**, no. 1 (2011), 510--514). Does the graph with intersection array $\{15,12,6;1,2,10\}$ exist?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04299 | KOU-20.55

### Kourovka Notebook Problem 20.55

O / open | Kourovka 群论问题集

Are there soluble finite groups $G$ and $H$, of derived lengths 2 and 4 and having identical character tables?

Pairs of nonisomorphic soluble finite groups with identical character tables and with derived lengths $n$ and $n+1$ for any $n\ge 2$ were constructed in (S. Mattarei, *J. Algebra*, **175** (1995) 157--178).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04300 | KOU-20.56

### Kourovka Notebook Problem 20.56

O / open | Kourovka 群论问题集

Is a periodic group $G$ locally finite if it is generated by involutions and the centralizer of every involution in $G$ is locally finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04301 | KOU-20.57

### Kourovka Notebook Problem 20.57

O / open | Kourovka 群论问题集

Is a periodic group $G$ locally finite if every finite subgroup of $G$ is contained in a subgroup of $G$ isomorphic to a finite alternating group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04302 | KOU-20.58

### Kourovka Notebook Problem 20.58

P / partially_solved | Kourovka 群论问题集

Let $\omega(G)$ denote the set of element orders of a finite group $G$. A finite group $G$ is said to be *recognizable* (*by spectrum*) if every finite group $H$ with $\omega(H)=\omega(G)$ is isomorphic to $G$.

a) Is it true that for every $n$ there is a recognizable group that is the $n$-th direct power of a nonabelian simple group?

b) Is it true that there is a nonabelian simple group $L$ such that for every $n$ there is a recognizable group whose socle is the $k$-th direct power of $L$ for some $k\geqslant n$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04303 | KOU-20.59

### Kourovka Notebook Problem 20.59

O / open | Kourovka 群论问题集

A subgroup $H$ is called a *virtual retract* of a group $G$ if $H$ is a retract of a finite-index subgroup of $G$. Is it true that every finitely generated subgroup of a finitely generated virtually free group is a virtual retract?

For motivation and partial results see (A. Minasyan, *Int. Math. Res. Notes*, **2021**, no. 17 (2021), 13434--13477).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04304 | KOU-20.6

### Kourovka Notebook Problem 20.6

O / open | Kourovka 群论问题集

(W. Kimmerle). The prime graph (or Gruenberg--Kegel graph) $\Gamma(X)$ of a group $X$ has vertices labeled by primes appearing as orders of elements in $X$; two distinct primes $p$ and $q$ are adjacent in $\Gamma(X)$ if and only if $X$ contains an element of order $pq$. Denote by $\mathrm{V}(\mathbb{Z}G)$ the group of normalized units of the integral group ring of a group $G$. Is it true that for each finite group $G$ the prime graphs of $G$ and $\mathrm{V}(\mathbb{Z}G)$ coincide?

This question has been reduced to almost simple groups (W. Kimmerle, A. Konovalov, *Internat. J. Algebra Comput.*, **27** (2017), 619--631).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04305 | KOU-20.60

### Kourovka Notebook Problem 20.60

O / open | Kourovka 群论问题集

We say that $G$ is a *virtually compact special group* if $G$ has a finite-index subgroup which is isomorphic to the fundamental group of a compact special complex (in the sense of F. Haglund, D. T. Wise, *Geom. Funct. Anal.*, **17**, no. 5 (2008), 1551--1620). Let $G$ be a virtual retract of a finitely generated right-angled Artin group. Must $G$ be a virtually compact special group?

An affirmative answer would provide an algebraic characterization of the class of virtually compact special groups as the class groups admitting finite index subgroups that are virtual retracts of right-angled Artin groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04306 | KOU-20.61

### Kourovka Notebook Problem 20.61

O / open | Kourovka 群论问题集

Suppose that $G$ is a virtually compact special group. Is it true that the centralizer of any element in $G$ is itself virtually compact special?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04307 | KOU-20.62

### Kourovka Notebook Problem 20.62

O / open | Kourovka 群论问题集

(J. Dixmier) A group is said to be *unitarisable* if its every uniformly bounded representation on a Hilbert space is unitarisable. Is every unitarisable group amenable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04308 | KOU-20.63

### Kourovka Notebook Problem 20.63

O / open | Kourovka 群论问题集

a) Prove that every unitarisable group has trivial cost.

This is known to be true for residually finite groups (I. Epstein, N. Monod, *Int. Math. Res. Notes*, **2009**, no. 22 (2009), 4336--4353).

b) At least prove that every unitarisable group has vanishing first $L^2$ Betti number.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04309 | KOU-20.64

### Kourovka Notebook Problem 20.64

O / open | Kourovka 群论问题集

Let $G$ be a non-amenable group.

a) Prove that $G^n$ is non-unitarisable for some $n$.

b) At least prove that $G^{\infty}$, the direct sum (restricted product), is non-unitarisable.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04310 | KOU-20.65

### Kourovka Notebook Problem 20.65

O / open | Kourovka 群论问题集

Given a complex irreducible character $\chi\in \rm{Irr}(G)$ of a finite group $G$, let $\mathbb{Q}(\chi)$ denote the field extension of $\mathbb{Q}$ obtained by adjoining to $\mathbb{Q}$ all the values of $\chi$. We say that a finite group $G$ is $k$-*rational* if $|\mathbb{Q}(\chi):\mathbb{Q}|$ divides $k$ for every $\chi\in \rm{Irr}(G)$. Does there exist a real-valued function $f$ such that if $p$ is the order of a cyclic composition factor of a $k$-rational group $G$, then $p\leq f(k)$?

If $k=1$, then we know that $p\leq 11$ by a theorem of J. G. Thompson (*J. Algebra*, **319** (2008), 558--594).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04311 | KOU-20.66

### Kourovka Notebook Problem 20.66

O / open | Kourovka 群论问题集

A Schmidt $(p, q)$-group is a finite non-nilpotent group all of whose proper subgroups are nilpotent and whose Sylow $p$-subgroup is normal. The $N$-critical graph $\Gamma_{Nc}(G)$ of a finite group $G$ is a directed graph on the vertex set of all prime divisors of $|G|$ in which $(p, q)$ is an edge of $\Gamma_{Nc}(G)$ if and only if $G$ has a Schmidt $(p, q)$-subgroup. Suppose that a finite group $ G$ is such that $G=AB=AC=BC$, where $ A, B, C$ are subgroups of $G$. Is $\Gamma_{Nc}(G)=\Gamma_{Nc}(A)\cup\Gamma_{Nc}(B)\cup\Gamma_{Nc}(C)$? This is true if $A, B, C$ are soluble.

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04312 | KOU-20.67

### Kourovka Notebook Problem 20.67

O / open | Kourovka 群论问题集

Suppose that $G$ is a finite group, $p$ is a prime, and $B$ is a Brauer $p$-block of $G$ with defect group $D$. Let $\mathrm{cd}(B)$ be the set of degrees of the irreducible complex characters in $B$.

a) Is it true that the derived length of $D$ is bounded by $|\mathrm{cd}(B)|$?

b) (A. Jaikin-Zapirain). Is is even true that $|\mathrm{cd}(D)|\le |\mathrm{cd}(B)|$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04313 | KOU-20.68

### Kourovka Notebook Problem 20.68

O / open | Kourovka 群论问题集

The *submonoid membership problem* for a group $G$ generated by an alphabet $A$ asks, for given words $x_1, x_2, \dots , x_n$ and a word $w$ over $A$, whether $w$ belongs to the submonoid of $G$ generated by the $x_i$. Does there exist a hyperbolic one-relator group with undecidable submonoid membership problem?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04314 | KOU-20.69

### Kourovka Notebook Problem 20.69

O / open | Kourovka 群论问题集

Is the submonoid membership problem decidable for every one-relator group with torsion?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04315 | KOU-20.7

### Kourovka Notebook Problem 20.7

O / open | Kourovka 群论问题集

(W. Boone and G. Higman) Does every finitely generated group with solvable word problem embed into a finitely presented simple group? It is known that every such group embeds into a simple subgroup of a finitely presented group (W. Boone, G. Higman, *J. Austral. Math. Soc.*, **18**, no. 1 (1974), 41--53).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04316 | KOU-20.70

### Kourovka Notebook Problem 20.70

O / open | Kourovka 群论问题集

As a strengthening of the Burnside restriction, for every pair $(k,n)$ of positive integers, let a group $G$ satisfy condition $C_{k,n}$ if every $k$-generated subgroup of $G$ is finite of order at most $n$.

a) Does there exist $k\ge 2$ such that for any $n$ all groups with condition $C_{k,n}$ are locally finite?

b) In particular, is it true that for any $n$ the condition $C_{2,n}$ implies local finiteness?

c) Find possibly more pairs $(k,n)$ for which groups with condition $C_{k,n}$ are locally finite. (For example, all groups with condition $C_{2,20}$ are metabelian, and therefore locally finite.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

04317 | KOU-20.71

### Kourovka Notebook Problem 20.71

P / partially_solved | Kourovka 群论问题集

(J. Lauri). A *card* of a finite simple undirected graph $G$ of order $n=|V(G)|$ is an induced subgraph of order $n-1$. For a connected graph $G$ with $k$ isomorphism types of cards, can $Aut(G)$ have more than $k$ orbits on the vertex set $V(G)$

a) for $k=2$?

b) for $k=3$?

c) for $k=4$?

If a graph $G$ has $k$ isomorphism types of cards, then the group $Aut(G)$ of automorphisms of $G$ has obviously at least $k$ orbits on $V(G)$. It is known that if all cards are mutually isomorphic, then $Aut(G)$ is transitive on $V(G)$. Examples are known of graphs $G$ with 5 isomorphism types of cards for which $Aut(G)$ has $6$ orbits on $V(G)$, and of graphs with 6 isomorphism types for which $Aut(G)$ has 7 orbits on $V(G)$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04318 | KOU-20.72

### Kourovka Notebook Problem 20.72

S / solved | Kourovka 群论问题集

Is there a variety of groups $\Theta$ such that the group of automorphisms of the category of free finitely generated groups $\Theta^0$ contains outer automorphisms?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04319 | KOU-20.73

### Kourovka Notebook Problem 20.73

O / open | Kourovka 群论问题集

Can every countable group $G$ be factorized $G=AB$ into infinite subsets $A, B$ such that every element $g\in G$ has a unique representation $g = ab$ for $a\in A$, $b \in B$?

This is true if $G$ is topologizable.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04320 | KOU-20.74

### Kourovka Notebook Problem 20.74

S / solved | Kourovka 群论问题集

a) *Conjecture*: there are only finitely many nonabelian finite simple groups $G$ that have a normal subset $S$ closed under inversion such that $|S | > |G|/\log_2|G|$ and $S^2\ne G$. *Comment of 2024*: This is true for the class of alternating groups (M. Larsen, P. H. Tiep, *Preprint*, 2023, arXiv:2305.04806) and for any class of groups of Lie type of bounded Lie rank (S. V. Skresanov, *Bull. London Math. Soc.*, **57**, no. 11 (2025), 3429--3440).

b) The same conjecture for the groups $PSL(2,p)$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04321 | KOU-20.75

### Kourovka Notebook Problem 20.75

S / solved | Kourovka 群论问题集

Let $G$ be a finite $p$-group and assume that all abelian normal subgroups of $G$ can be generated by $k$ elements. Is it true that every abelian subgroup of $G$ can be generated by $2k$ elements?

The $k$-th direct power of $D_{16}$ shows that this bound would be best possible.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04322 | KOU-20.76

### Kourovka Notebook Problem 20.76

S / solved | Kourovka 群论问题集

Let $G$ be a finite $p$-group and assume that all abelian normal subgroups of $G$ have order at most $p^k$. Is it true that every abelian subgroup of $G$ has order at most $p^{2k}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04323 | KOU-20.77

### Kourovka Notebook Problem 20.77

O / open | Kourovka 群论问题集

(O. I. Tavgen'). A group is said to be *boundedly generated* if it is a product of finitely many cyclic groups. Is it true that every boundedly generated residually finite group is linear over a field of characteristic zero?

This is true for residually finite-soluble groups (L. Pyber, D. Segal, *J. Reine Angew. Math.*, **612** (2007), 173--211).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04324 | KOU-20.78

### Kourovka Notebook Problem 20.78

O / open | Kourovka 群论问题集

For an irreducible complex character $\chi$ of a finite group $G$, the *codegree* of $\chi$ is defined by $\mathrm{cod}(\chi)=|G:\ker\chi|/\chi(1)$. Let $\mathrm{Cod}(G)$ be the set of irreducible character codegrees of $G$. *Conjecture*: If $G$ has an element of order $m$, then $m$ divides some member of $\mathrm{Cod}(G)$.

The conjecture is proved when $m$ is a prime power (G. Qian, *Arch. Math.*, **97** (2011), 99--103); when $m$ is square-free (I. M. Isaacs, *Arch. Math.*, **97** (2011), 499--501); or when $G$ is solvable (G. Qian, *Bull. London Math. Soc.*, **53** (2021) 820--824); or when $G$ is a symmetric or alternating group (E. Giannelli (*J. Algebra Appl.*, **23**, no. 9 (2024), article ID 2450144); or when $G$ is an almost simple group (S. Y. Madanha, *Commun. Algebra*, **51**, no. 7 (2023), 3143--3151); or when $\mathbf{F}(G)=1$ (Z. Akhlaghi, E. Pacifici, L. Sanus, *J. Algebra*, **644** (2024), 428--441).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04325 | KOU-20.79

### Kourovka Notebook Problem 20.79

O / open | Kourovka 群论问题集

*Conjecture*: Suppose that $G$ is a non-abelian simple group and $H$ is a finite group such that $\mathrm{Cod}(G)=\mathrm{Cod}(H)$. Then $G\cong H$.

*Comment of 2025*: The conjecture is proved for $G$ of type $\mathrm{PSL}(2,q)$ (A. Bahri, Z. Akhlaghi, B. Khosravi, *Bull. Austral. Math. Soc.*, **104**, no. 2 (2021), 278--286); \ $\mathrm{PSL}(3, q)$ or $\mathrm{PSU}(3, q)$ (Y. Liu, Y. Yang, *Results Math.*, **78**, no. 1 (2023), article No. 7); \ a sporadic simple group (M. Dolorfino, L. Martin, Z. Slonim, Y. Sun, Y. Yang, *Bull. Austral. Math. Soc.*, **109**, no. 1 (2024), 57--66); \ an alternating group (M. Dolorfino, L. Martin, Z. Slonim, Y. Sun, Y. Yang, *Bull. Austral. Math. Soc.*, **110**, no. 1 (2024), 115--120); \ $^2F_4(q^2)$ (Y. Yang, *J. Group Theory*, **27**, no. 1 (2024), 141--155); \ $\mathrm{PSL}(n,q)$ or a simple exceptional group of Lie type (H. P. Tong Viet, *Math. Nachr.*, **298**, no. 4 (2025), 1356--1369).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04326 | KOU-20.8

### Kourovka Notebook Problem 20.8

O / open | Kourovka 群论问题集

Suppose $K < H < F$ are free groups of finite rank such that $\operatorname{rank}(H) < \operatorname{rank}(K)$, but all proper subgroups of $H$ which contain $K$ have ranks ${}\geq \operatorname{rank}(K)$. Then is the inclusion of $H$ in $F$ the only homomorphism $H\rightarrow F$ fixing all elements of $K$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04327 | KOU-20.80

### Kourovka Notebook Problem 20.80

O / open | Kourovka 群论问题集

Let $G$ be a finite group, and $\chi$ an irreducible complex character of $G$. We call $\chi$ a $\mathcal{P}$-character if $\chi$ is a constituent of $(1_H)^G$ for some maximal subgroup $H$ of $G$; and $\chi$ is said to be monomial if $\chi$ is induced by a linear character of a subgroup of $G$.

*Conjecture*: $G$ is solvable if and only if all its $\mathcal{P}$-characters are monomial.

The necessity part of the conjecture is true (G. Qian, Y. Yang, *Commun. Algebra*, **46**, no. 1 (2018), 167--175).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04328 | KOU-20.81

## Kourovka Notebook Problem 20.81

O / open | Kourovka 群论问题集

Does there exist a class $\mathfrak{X}$ of finite groups satisfying the following conditions:

(1) $\mathfrak{X}$ is closed with respect to taking subgroups and homomorphic images;

(2) the product of two normal $\mathfrak{X}$-subgroups of an arbitrary group is always an $\mathfrak{X}$-group;

(3) for every positive integer $n$, there exist a finite group and its conjugacy class $D$ such that any $n$ elements of $D$ generate an $\mathfrak{X}$-subgroup, whereas $\langle D\rangle\notin\mathfrak{X}$.

It is known that there are no classes $\mathfrak{X}$ satisfying (1)--(3) that are closed with respect to extensions (D. O. Revin, *Algebra i Analiz*, **37**, no. 1 (2025), 141--176 (Russian)).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04329 | KOU-20.82

## Kourovka Notebook Problem 20.82

O / open | Kourovka 群论问题集

Two groups are said to be *isospectral* if they have the same set of element orders. Suppose that $G$ is a finite group such that every finite group isospectral to $G$ is isomorphic to $G$. Is it true that the quotient of $G$ by its socle is solvable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04330 | KOU-20.83

## Kourovka Notebook Problem 20.83

O / open | Kourovka 群论问题集

Let $\mathfrak{X}$ be a class of finite groups that is closed with respect to taking subgroups, homomorphic images, and extensions. A subgroup $H$ of a finite group $G$ is said to be \emph{$\mathfrak{X}$-submaximal} if there exists an embedding of $G$ into a group $G^*$ such that $G$ is subnormal in $G^*$ and $H$ coincides with the intersection of $G$ and an $\mathfrak{X}$-maximal subgroup of $G^*$. Suppose that all $\mathfrak{X}$-submaximal subgroups of a characteristic subgroup $N$ of a finite group $G$ are conjugate in $N$. Does it follow that $HN/N$ is an $\mathfrak{X}$-submaximal subgroup of $G/N$ for every $\mathfrak{X}$-submaximal subgroup $H$ of $G$?

For normal subgroups $N$, this is not true even if $N$ is an $\mathfrak{X}$-group or if $N$ does not contain nontrivial $\mathfrak{X}$-subgroups. A positive answer is known in the case where $N$ coincides with the $\mathfrak{F}$-radical of $G$ for a Fitting class $\mathfrak{F}$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04331 | KOU-20.84

## Kourovka Notebook Problem 20.84

O / open | Kourovka 群论问题集

Does there exist a nilpotent group of class 3 with a non-trivial 5-th dimension subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04332 | KOU-20.85

## Kourovka Notebook Problem 20.85

O / open | Kourovka 群论问题集

Let $F$ be a free group. An element $\omega\in F$ is said to be *primitive* if there is a minimal generating system of $F$ that contains $\omega$, *almost primitive* if it is primitive in each finitely generated proper subgroup of $F$ containing $\omega$, *tame almost primitive* if, whenever $\omega^\alpha$ is contained in a subgroup $H$ of $F$ with $\alpha\ge 1$ minimal, either $\omega^\alpha$ is primitive in $H$ or the index of $H$ in $F$ is just $\alpha$. In (B. Fine, A. Moldenhauer, G. Rosenberger, L. Wienke, *Topics in Infinite Group Theory: Nielsen Methods, Covering Spaces, and Hyperbolic Groups*, De Gruyter, Berlin, 2021) it is shown that $u=[a_1,b_1][a_2,b_2]\cdots [a_g,b_g]$ is tame almost primitive in the free group on $a_1, b_1,\dots ,a_g,b_g$ with $g\ge 1$, and $v=c_1 ^2\cdots c_p ^2$ is tame almost primitive in the free group on $c_1,\dots,c_p$ with $p\ge 2$.

Are there tame almost primitive elements in free groups other than $u$, $v$, and their product $uv$ in the free group on $a_1, b_1,\dots ,a_g,b_g,c_1,\dots,c_p$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04333 | KOU-20.86

## Kourovka Notebook Problem 20.86

O / open | Kourovka 群论问题集

Suppose $G$ is a finite group and $p$ a prime such that the number $s_p(G)=1+kp$ of Sylow $p$-subgroups $P$ of $G$ is greater than 1. By a theorem of Frobenius the set $G_p$ of all $p$-elements in $G$ has cardinality $|G_p|=|P|\cdot f_p(G)$ for some positive integer $f_p(G)$, and one easily gets that $f_p(G)=1+\ell(p-1)\le k-{k-1\over p-1}$. Usually, $\ell <k$ (but $\ell =k$ if $k<p$). Is always $\ell^p\ge k^{p-1}$?

Recently P. Gheri showed that $f_p(G)^p\ge s_p(G)^{p-1}$ if $G$ is $p$-solvable (*Ann. Mat. Pura Appl. (4)*, **200** (2021), 1231--1243).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04334 | KOU-20.87

## Kourovka Notebook Problem 20.87

O / open | Kourovka 群论问题集

What are the non-abelian composition factors of finite groups in which the order of every element is divisible by at most two primes?

For the case of one prime, see (https://arxiv.org/pdf/2003.09445.pdf).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04335 | KOU-20.88

### Kourovka Notebook Problem 20.88

S / solved | Kourovka 群论问题集

Let $u$ be an element of a free group $F_r$. Is it true that there is $v \in F_r$ (that depends on $u$) that cannot be a subword of any cyclically reduced word $\varphi(u)$, where $\varphi$ is an automorphism of $F_r$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04336 | KOU-20.89

### Kourovka Notebook Problem 20.89

O / open | Kourovka 群论问题集

An element $g$ of a group $G$ is said to be *almost Engel* if there is a finite subset $E(g)$ of $G$ such that for every $x\in G$ all sufficiently long commutators $[...[[x, g], g], \dots , g]$ belong to $E(g)$, that is, there is a positive integer $n(x, g)$ such that $[...[[x, g], g], \dots ,g]\in E(g)$ if $g$ is repeated ${}\geq n(x, g)$ times. An element $g$ is Engel if we can take $E(g)=\{1\}$. The set of Engel elements of a linear group is a subgroup by a well-known result of Gruenberg (*J. Algebra*, **3** (1966), 291--303). Is the set of almost Engel elements of a linear group a subgroup?

A linear group in which all elements are almost Engel is finite-by-hypercentral (P. Shumyatsky, *Monatsh. Math.*, **186** (2018), 711--719).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04337 | KOU-20.9

### Kourovka Notebook Problem 20.9

O / open | Kourovka 群论问题集

In the group algebra of a free group over a field, does every element whose support in the group has cardinality more than 1 generate a proper 2-sided ideal?

This question is one of several related questions in (G. M. Bergman, *Commun. Algebra*, **49**, no. 9 (2021), 3760--3776).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04338 | KOU-20.90

### Kourovka Notebook Problem 20.90

S / solved | Kourovka 群论问题集

A profinite group in which all centralizers of non-trivial elements are pronilpotent is called a CN-group. Find an example of a finitely generated infinite profinite CN-group which is not prosoluble.

The structure of profinite CN-groups is described in (P. Shumyatsky, *Israel J. Math.*, **235**, no. 1 (2020), 325--347).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04339 | KOU-20.91

### Kourovka Notebook Problem 20.91

O / open | Kourovka 群论问题集

A group $G$ is said to be *stable* if for any first-order formula $\phi(\bar{x}, \bar{y})$ (in the first-order language of groups $\{ \cdot , {}^{-1}, 1\}$) there exists a natural number $n$ such that whenever there exist sequences of tuples of $G$, $(a_i)_{i<m}$, $(b_i)_{i<m}$ with $G\models \phi(a_i, b_j)$ if and only if $i<j$, then $m\leq n$. Sela proved that all torsion-free hyperbolic groups are stable.

a) Is there a non-stable hyperbolic group?

b) Is there a non-stable virtually free group?

c) Is $SL_2(\mathbb{Z})$ non-stable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04340 | KOU-20.92

## Kourovka Notebook Problem 20.92

P / partially_solved | Kourovka 群论问题集

This question is about existence of an analogue of the Lazard correspondence for pre-Lie algebras and braces. A *pre-Lie algebra* $A$ is a vector space with a bilinear operation $(x, y) \rightarrow xy$ satisfying $(xy)z -x(yz) = (yx)z - y(xz)$ for every $x,y,z\in A$. A pre-Lie algebra $A$ is said to be *left nilpotent* if, for some $n\in \mathbb{N}$, $A\cdot (A\cdot( A\cdots A)))=0$ where $A$ appears $n$ times in the product. Recall that a set $A$ with binary operations $+$ and $\circ $ is a *left brace* if $(A, +)$ is an abelian group, $(A, \circ )$ is a group, and $a\circ (b+c)+a=a\circ b+a\circ c$ for every $a,b,c\in A$. Let $p$ be a prime, and $\mathbb{F}_{p}$ the field of $p$ elements. A left brace $A$ is called an $\mathbb{F}_p$-brace if its additive group is an $\mathbb{F}_p$-vector space such that $a*({\alpha }b)={\alpha }(a* b)$ for all $a,b\in A$, $ {\alpha }\in \mathbb{F}_p$, where $ a*b=a \circ b -a -b$. The idea of a connection between braces and pre-Lie algebras comes from a paper by W. Rump (2014).

a) Let $A$ be an $\mathbb{F}_{p}$-brace of cardinality $p^{k}$ for some $k$. Is it true that when $p$ is sufficiently large relative to $k$, the set $A$ with the same additive operation $+$ and with the operation $\cdot $ defined as $a\cdot b=-\sum_{i=0}^{p-2}{\frac 1{2^{i}}}((2^{i}a)* b)$ is a pre-Lie algebra?

*Comment of 2025*: If $2\equiv \xi ^{p^{n-1}}$ when $\xi $ is a primitive root modulo $p$, and if $A$ is strongly nilpotent of nilpotency index less than $p$, then the result follows by (A. Smoktunowicz, *Adv. Math.*, **409**, part B (2022), 108683). It is not known if the result follows when $A$ is not strongly nilpotent. b) Let $k$ be a natural number, and let $p$ be a prime number such that $p>2^{k}$. Is there a bijective correspondence between $\mathbb{F}_{p}$-braces of cardinality $p^{k}$ and left nilpotent pre-Lie algebras over ${\mathbb{F}}_{p}$ of cardinality $p^{k}$?

An affirmative answer to any of the above questions would have consequences for the theory of set-theoretic solutions of the Yang--Baxter equation and for the theory of Hopf--Galois extensions.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04341 | KOU-20.93

## Kourovka Notebook Problem 20.93

O / open | Kourovka 群论问题集

A group $G$ is called a *Shunkov group* if for any finite subgroup $H\leq G$ any two conjugate elements of prime order in $N_G(H)/H$ generate a finite subgroup. Are the following well-known results of the theory of finite groups true in the class of (periodic) Shunkov groups?

a) The Baer--Suzuki theorem (see 11.11 in Archive).

b) The Burnside--Brauer--Suzuki theorem on the existence of a normal section of order 2 in a group with a non-trivial Sylow 2-subgroup containing only one involution (see 4.75).

c) Glauberman's $Z^*$-theorem (see 10.62 and Archive, 11.13).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04342 | KOU-20.94

## Kourovka Notebook Problem 20.94

O / open | Kourovka 群论问题集

Does there exist an infinite periodic simple group saturated (see the definition in 14.101) with finite Frobenius groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04343 | KOU-20.95

## Kourovka Notebook Problem 20.95

O / open | Kourovka 群论问题集

Is a periodic group a Frobenius group (see 6.53) if it is saturated with finite Frobenius groups, contains an involution, and does not contain non-cyclic subgroups of order 4?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04344 | KOU-20.96

## Kourovka Notebook Problem 20.96

S / solved | Kourovka 群论问题集

Is a periodic group a Frobenius group if it has a proper non-trivial normal abelian subgroup that contains the centralizer of each of its non-identity elements?

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04345 | KOU-20.97

## Kourovka Notebook Problem 20.97

O / open | Kourovka 群论问题集

Is a 2-group locally finite if the centralizer of every involution is locally finite?

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04346 | KOU-20.98

### Kourovka Notebook Problem 20.98

S / solved | Kourovka 群论问题集

Let $G$ be a group of permutations of the set of positive integers $\mathbb{N}$ isomorphic to the additive group of rational numbers. Must there be an element $g\in G$ such that the set $\{a-ag\mid a\in \mathbb{N}\}$ is infinite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04347 | KOU-20.99

### Kourovka Notebook Problem 20.99

O / open | Kourovka 群论问题集

*Conjecture*: Let $a_1G_1,\ldots,a_kG_k$, $k>1$, be finitely many pairwise disjoint left cosets in a group $G$ with $[G:G_i]<\infty$ for all $i=1,\ldots,k$. Then

$\gcd([G:G_i],[G:G_j])\ge k\qquad \text{for some }1\leq i<j\leq k$.

The conjecture is known to hold for $k=2,3,4$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04348 | KOU-21.1

### Kourovka Notebook Problem 21.1

P / partially_solved | Kourovka 群论问题集

Let $n$ be a positive integer. For a finite group $K$ and an automorphism $\varphi$ of $K$ of order dividing $n$, let

$$ X_{n,\varphi}(K):=\{x\in K\mid x x^{\varphi}\cdots x^{\varphi^{n-1}}=1\}. $$

Let $c_n$ be the supremum of the ratios $|X_{n,\varphi}(H)|/|H|$ over all finite groups $H$ and their automorphisms $\varphi\in\operatorname{Aut}(H)$ such that $\varphi^n=\operatorname{id}$ and $X_{n,\varphi}(H)\ne H$.

(a) Let $n>1$ be a positive integer such that $c_d<1$ for all prime power divisors $d$ of $n$. Is it true that $c_n<1$?

(b) For a finite group $G$ and a positive integer $n$, the generalized Hughes--Thompson subgroup is defined as $H_n(G)=\langle x\in G\mid x^n\ne 1\rangle$. Suppose that $n$ is a positive integer for which there is a positive integer $k_n$ depending only on $n$ such that $|G:H_n(G)|\leqslant k_n$ for all finite groups $G$ with $H_n(G)\ne 1$. Is it true that then $c_n<1$?

来源字段：https://arxiv.org/abs/1401.0300

04349 | KOU-21.10

### Kourovka Notebook Problem 21.10

S / solved | Kourovka 群论问题集

We call a group presentation finite if it represents a finite group. We say that a presentation is just finite if it is finite and is no longer finite on removal of any relation from it. Is it true that every finite group has a just finite presentation?

来源字段：https://arxiv.org/abs/1401.0300

04350 | KOU-21.100

### Kourovka Notebook Problem 21.100

O / open | Kourovka 群论问题集

Suppose that $A$ and $G$ are finite groups such that $A$ acts coprimely on $G$ by automorphisms. Let $C=C_G(A)$ be the fixed-point subgroup, and let $C'$ denote its derived subgroup. Is it true that the number of $A$-invariant irreducible characters $\chi$ of $G$ whose restriction $\chi_C$ is never zero is exactly $|C/C'|$?

来源字段：https://arxiv.org/abs/1401.0300

04351 | KOU-21.101

### Kourovka Notebook Problem 21.101

P / partially_solved | Kourovka 群论问题集

Which finite almost simple groups are the automorphism groups of regular polytopes of rank 3? In other words, which finite almost simple groups are generated by three involutions two of which commute?

来源字段：https://arxiv.org/abs/1401.0300

04352 | KOU-21.102

### Kourovka Notebook Problem 21.102

P / partially_solved | Kourovka 群论问题集

Let $V$ be a variety generated by a finite group, and let $f(n)$ be the order of the free group in $V$ on $n$ generators. Is it true that the sequence $\sqrt[n]{\log f(n)}$ has a limit as $n\to\infty$, and this limit is an integer?

来源字段：https://arxiv.org/abs/1401.0300

04353 | KOU-21.103

### Kourovka Notebook Problem 21.103

P / partially_solved | Kourovka 群论问题集

A Hausdorff topological group G is called minimal if it does not admit a strictly coarser Hausdorff group topology. A topological group is called Raikov complete if its two-sided uniform structure is complete. Is it true that an arbitrary Cartesian product of Raikov complete minimal topological groups remains minimal?

来源字段：https://arxiv.org/abs/1401.0300

04354 | KOU-21.104

### Kourovka Notebook Problem 21.104

P / partially_solved | Kourovka 群论问题集

For a group word $w(x_1,\ldots,x_n)$ on $n$ letters, define $e_0(x_1,\ldots,x_n)=x_1$ and $e_{k+1}(x_1,\ldots,x_n)=w(e_k(x_1,\ldots,x_n),\ldots,x_n)$ for all $k\in\mathbb N$. A group $G$ is said to satisfy the Engel type iterated identity $w$ if for all $x_1,\ldots,x_n\in G$ there exists $m\in\mathbb N$ such that $e_m(x_1,\ldots,x_n)=1$.

Conjecture: For every non-trivial word $w$, if a finitely generated branch group $G$ (see 15.12) satisfies the iterated identity $w$, then $G$ is a torsion group.

来源字段：https://arxiv.org/abs/1401.0300

04355 | KOU-21.105

### Kourovka Notebook Problem 21.105

P / partially_solved | Kourovka 群论问题集

A group word $w$ is said to be concise in a class $\mathcal C$ of groups if for every group $G$ in $\mathcal C$ such that the set $G_w$ of word values of $w$ in $G$ is finite, the verbal subgroup $w(G)=\langle G_w\rangle$ is also finite. Is every word concise in the class of residually finite groups?

来源字段：https://arxiv.org/abs/1401.0300

04356 | KOU-21.106

### Kourovka Notebook Problem 21.106

S / solved | Kourovka 群论问题集

A first order formula $\phi(x)$ in the group language with one free variable is said to be concise in a class $\mathcal C$ of groups if for every group $G$ in $\mathcal C$ such that the set $G_\phi$ of elements in $G$ satisfying $\phi$ is finite, the subgroup $\phi(G)$ generated by $G_\phi$ is also finite. Is every formula with one free variable concise in the class of residually finite groups?

来源字段：https://arxiv.org/abs/1401.0300

04357 | KOU-21.107

### Kourovka Notebook Problem 21.107

O / open | Kourovka 群论问题集

A sequence $\{F_n\}$ of pairwise disjoint finite subsets of a topological group is called expansive if for every open subset $U$ there is a number $m$ such that $F_n\cap U\ne\emptyset$ for all $n>m$. Suppose that a countable group $G$ can be partitioned into countably many dense subsets. Is it true that in $G$ there exists an expansive sequence?

来源字段：https://arxiv.org/abs/1401.0300

04358 | KOU-21.108

### Kourovka Notebook Problem 21.108

P / partially_solved | Kourovka 群论问题集

For a finite group $G$ let $\operatorname{Cod}(G)$ denote the set of irreducible character codegrees of $G$ (see 20.78). Define $\sigma(G)=\max\{|\pi(m)|:m\in\operatorname{Cod}(G)\}$, where $\pi(m)$ denotes the set of prime divisors of an integer $m$. Can the constant $k$ be taken as 4?

来源字段：https://arxiv.org/abs/1401.0300

04359 | KOU-21.109

### Kourovka Notebook Problem 21.109

P / partially_solved | Kourovka 群论问题集

Conjecture: The derived length of a finite solvable group $G$ does not exceed $|\operatorname{Cod}(G)|-1$.

来源字段：https://arxiv.org/abs/1401.0300

04360 | KOU-21.11

### Kourovka Notebook Problem 21.11

P / partially_solved | Kourovka 群论问题集

Can some or all groups of the following sorts be written as homomorphic images of nonprincipal ultraproducts of countable families of groups?

来源字段：https://arxiv.org/abs/1401.0300

04361 | KOU-21.110

### Kourovka Notebook Problem 21.110

O / open | Kourovka 群论问题集

Let $S$ be a nonabelian finite simple group, and $x$ a nonidentity automorphism of $S$. Let $\alpha(x)$ be the smallest number of conjugates of $x$ in $G=\langle x,\operatorname{Inn}S\rangle$ that generate $G$.

(a) Conjecture: If $S$ is an exceptional group of Lie type, then $\alpha(x)\leqslant 5$ for every nonidentity automorphism $x$ of $S$.

(b) For each exceptional group $S$ of Lie type, find the largest value of $\alpha(x)$.

来源字段：https://arxiv.org/abs/1401.0300

04362 | KOU-21.111

### Kourovka Notebook Problem 21.111

O / open | Kourovka 群论问题集

Let $S$ be a finite simple nonabelian group that is not isomorphic to any group ${}^2B_2(q)$. A nonidentity automorphism $x$ of $S$ is called a $\tau$-automorphism if every two conjugates of $x$ in $\langle x,\operatorname{Inn}(S)\rangle$ generate a subgroup of order not divisible by 3. If $S$ admits a $\tau$-automorphism, we call $S$ a $\tau$-group.

(a) List all $\tau$-groups up to isomorphism.

(b) Do $\tau$-automorphisms of odd order exist?

来源字段：https://arxiv.org/abs/1401.0300

04363 | KOU-21.112

### Kourovka Notebook Problem 21.112

P / partially_solved | Kourovka 群论问题集

A nonempty class $\mathcal X$ of finite groups is said to be complete if $\mathcal X$ is closed under taking subgroups, homomorphic images, and extensions. The symmetric boundary of a complete class $\mathcal X$ other than the class of all finite groups is defined as the largest integer $n$ such that $S_n\in\mathcal X$. For every positive integer $n\ne 3$, let $f_+(n)$ and $f_-(n)$ be respectively the maximum and the minimum of $\operatorname{BS}(\mathcal X)$, where $\mathcal X$ runs over all complete classes of symmetric boundary $n$.

(a) Find $f_+(n)$ for $n=4,5,6$.

(b) Is it true that $f_-(n)=n$ for all $n\ne 3$?

来源字段：https://arxiv.org/abs/1401.0300

04364 | KOU-21.113

### Kourovka Notebook Problem 21.113

P / partially_solved | Kourovka 群论问题集

Let $G$ be a finite group and $p$ be a prime. Let $\Psi_{p,G}$ be the class function of $G$ which vanishes on all $p$-singular elements of $G$ and whose value at each $p$-regular element $x$ of $G$ is the number of $p$-elements of $C_G(x)$.

(a) Is it true that $\Psi_{p,G}$ is a character of $G$?

(b) If yes, can $\Psi_{p,G}$ be afforded by a projective $RG$-module, where $R$ is a complete discrete valuation ring of characteristic zero such that the field of fractions of $R$ is a splitting field for $G$ and its subgroups, and the residue field $R/J(R)$ is a splitting field of characteristic $p$ for $G$ and its subgroups?

来源字段：https://arxiv.org/abs/1401.0300

04365 | KOU-21.114

### Kourovka Notebook Problem 21.114

P / partially_solved | Kourovka 群论问题集

A finite group G is called weakly ab-maximal if |H : [H, H]| $\leqslant$ |G : [G, G]| for all H $\leqslant$ G. Do weakly ab-maximal groups have bounded derived length?

来源字段：https://arxiv.org/abs/1401.0300

04366 | KOU-21.115

### Kourovka Notebook Problem 21.115

S / solved | Kourovka 群论问题集

Let $C_1,\ldots,C_n$ be (left or right) cosets of a finite group $G$ such that $U:=C_1\cup\cdots\cup C_n$ is not $G$. Is it always true that $|G\setminus U|\geqslant |G|/2^n$?

来源字段：https://arxiv.org/abs/1401.0300

04367 | KOU-21.116

### Kourovka Notebook Problem 21.116

O / open | Kourovka 群论问题集

A group is boundedly acyclic if its bounded cohomology with trivial real coefficients vanishes in all positive degrees. Is every branch group boundedly acyclic?

来源字段：https://arxiv.org/abs/1401.0300

04368 | KOU-21.117

### Kourovka Notebook Problem 21.117

S / solved | Kourovka 群论问题集

(a) Does there exist a finitely generated simple group that is of exponential growth but not of uniformly exponential growth?

(b) Does there exist a finitely generated hereditarily just-infinite group that is of exponential growth but not of uniformly exponential growth?

来源字段：https://arxiv.org/abs/1401.0300

04369 | KOU-21.118

### Kourovka Notebook Problem 21.118

O / open | Kourovka 群论问题集

Is there any group which is not isomorphic to the quotient of a residually finite group by an amenable normal subgroup?

来源字段：https://arxiv.org/abs/1401.0300

04370 | KOU-21.119

### Kourovka Notebook Problem 21.119

O / open | Kourovka 群论问题集

Does there exist a group $G$ that contains a family $(G_n)_{n\in\mathbb N}$ of finite-index subgroups such that for every $n$ there is a homomorphism $f_n:G_n\to\mathbb Z$ whose kernel is of type $F_n$, but not of type $F_{n+1}$?

来源字段：https://arxiv.org/abs/1401.0300

04371 | KOU-21.12

### Kourovka Notebook Problem 21.12

S / solved | Kourovka 群论问题集

Suppose that $U$ is a nonprincipal ultrafilter on $\omega$, and $B$ is a group such that every element $b\in B$ belongs to a subgroup of $B$ that is a homomorphic image of $\mathbb Z^\omega/U$. Must $B$ then be a homomorphic image of an ultraproduct group $(\prod_{i\in\omega}G_i)/U$ for some groups $G_i$?

来源字段：https://arxiv.org/abs/1401.0300

04372 | KOU-21.120

### Kourovka Notebook Problem 21.120

P / partially_solved | Kourovka 群论问题集

A pro-p group is (relatively) strictly finitely presented if it is the pro-p completion of a group that is finitely presented (respectively, finitely presented in some finitely-based variety of groups). A pro-p group is finitely axiomatizable if it is determined up to isomorphism by a single sentence in the first-order language of group theory.

(a) Does there exist a (relatively) strictly finitely presented pro-p group that is not finitely axiomatizable in the class of all pro-p groups?

(b) In particular, is every finitely generated free pro-p group finitely axiomatizable?

来源字段：https://arxiv.org/abs/1401.0300

04373 | KOU-21.121

### Kourovka Notebook Problem 21.121

P / partially_solved | Kourovka 群论问题集

Let $p$ be a prime number. A group $\Gamma$ is called $p$-Jordan if there exist constants $J$ and $e$ such that any finite subgroup $G\subset\Gamma$ contains a normal abelian subgroup of order coprime to $p$ and of index at most $J\cdot |G(p)|^e$. (For example by the results of Brauer--Feit and Larsen--Pink, for any field $K$ of characteristic $p$ the group $\operatorname{GL}_n(K)$ is $p$-Jordan with $e=3$.) Let the $p$-Jordan exponent $e(\Gamma)$ of the group $\Gamma$ be the infimum of all constants $e$ for which the above bound holds for some constant $J=J(e)$.

a) Is it true that this infimum is always attained?

b) Is it true that $e(\Gamma)\leqslant 3$ for any $p$-Jordan group $\Gamma$?

来源字段：https://arxiv.org/abs/1401.0300

04374 | KOU-21.122

### Kourovka Notebook Problem 21.122

P / partially_solved | Kourovka 群论问题集

Let w be a group word, and G a profinite group. Is it true that the cardinality of the set of w-values in G is either finite or at least continuum?

来源字段：https://arxiv.org/abs/1401.0300

04375 | KOU-21.123

### Kourovka Notebook Problem 21.123

O / open | Kourovka 群论问题集

Is it true that the extension of the A. Agrachev--R. Gamkrelidze construction of groups from pre-Lie rings suggested in Definition 66 produces groups from pre-Lie rings? If this extended construction does give a group, then it also gives a brace, and so an affirmative answer to this question would have consequences for the theory of set-theoretic solutions of the Yang--Baxter equation and for the theory of braces.

来源字段：https://arxiv.org/abs/1401.0300

04376 | KOU-21.124

### Kourovka Notebook Problem 21.124

O / open | Kourovka 群论问题集

A group G is said to be virtually special if G has a finite-index subgroup isomorphic to the fundamental group of a special complex. A group G is called a CAT(0) group if it acts properly discontinuously and cocompactly by isometries on a CAT(0) metric space.

a) Is every CAT(0) free-by-cyclic group virtually special?

b) A weaker question: does every CAT(0) free-by-cyclic group virtually embed into a right-angled Artin group?

来源字段：https://arxiv.org/abs/1401.0300

04377 | KOU-21.125

### Kourovka Notebook Problem 21.125

O / open | Kourovka 群论问题集

Let $F_m$ be a free group of rank $m$ and let $\varphi\in\operatorname{Aut}(F_m)$ be a polynomially growing automorphism of maximal degree $m-1$, which means that for some (equivalently, any) free basis $\{x_1,\ldots,x_m\}$ of $F_m$, the sequence $\max_i |\varphi^n(x_i)|$ grows at the rate of $n^{m-1}$, where $|g|$ denotes the minimal length of $g$ in the $x_i$ and their inverses.

a) Is the free-by-cyclic group $F_m\rtimes_\varphi\mathbb Z$ virtually special?

b) In particular, are the Hydra groups $G_m=F_m\rtimes\mathbb Z=\langle a_1,\ldots,a_m,t\mid t^{-1}a_1t=a_1,\ t^{-1}a_it=a_ia_{i-1}\text{ for all }i>1\rangle$ virtually special?

来源字段：https://arxiv.org/abs/1401.0300

04378 | KOU-21.126

### Kourovka Notebook Problem 21.126

P / partially_solved | Kourovka 群论问题集

Do there exist finitely presented subgroups of right-angled Artin groups whose Dehn functions are super-exponential, or sub-exponential but not polynomial?

来源字段：https://arxiv.org/abs/1401.0300

04379 | KOU-21.127

### Kourovka Notebook Problem 21.127

P / partially_solved | Kourovka 群论问题集

Let G be a right-angled Artin group. Is the stable commutator length scl(g) a rational number for every g $\in$ [G, G]?

来源字段：https://arxiv.org/abs/1401.0300

04380 | KOU-21.128

### Kourovka Notebook Problem 21.128

O / open | Kourovka 群论问题集

Two groups $G_1$ and $G_2$ are said to be commensurable if there exist finite-index subgroups $H_1\leqslant G_1$ and $H_2\leqslant G_2$ (not necessarily of the same index) such that $H_1\cong H_2$. Let $A[F_4]$ and $A[H_4]$ denote the Artin groups of spherical types $F_4$ and $H_4$, respectively. Are these two groups commensurable?

来源字段：https://arxiv.org/abs/1401.0300

04381 | KOU-21.129

### Kourovka Notebook Problem 21.129

O / open | Kourovka 群论问题集

If two Artin groups of spherical type are quasi-isometric, must they be commensurable? (This is not true for right-angled Artin groups.)

来源字段：https://arxiv.org/abs/1401.0300

04382 | KOU-21.13

### Kourovka Notebook Problem 21.13

P / partially_solved | Kourovka 群论问题集

Does $\mathbb Z^\omega$ have a subgroup whose dual is free abelian of still larger rank (the largest possible being $2^{2^{\aleph_0}}$)?

来源字段：https://arxiv.org/abs/1401.0300

04383 | KOU-21.130

### Kourovka Notebook Problem 21.130

O / open | Kourovka 群论问题集

Conjecture: Let $G$ be a finite additive abelian group with $|G|$ odd. Then any subset $A$ of $G$ with $|A|=n>2$ can be written as $\{a_1,\ldots,a_n\}$ in such a way that all the sums $a_1+a_2,a_2+a_3,\ldots,a_{n-1}+a_n,a_n+a_1$ are distinct.

来源字段：https://arxiv.org/abs/1401.0300

04384 | KOU-21.131

### Kourovka Notebook Problem 21.131

O / open | Kourovka 群论问题集

Construct a homomorphism of a subgroup of a Golod group onto an infinite AT-group.

来源字段：https://arxiv.org/abs/1401.0300

04385 | KOU-21.132

### Kourovka Notebook Problem 21.132

O / open | Kourovka 群论问题集

Based on the development of E. S. Golod's construction, for each prime number p, construct a finitely generated residually finite p-group with a non-trivial finite centre.

来源字段：https://arxiv.org/abs/1401.0300

04386 | KOU-21.133

### Kourovka Notebook Problem 21.133

O / open | Kourovka 群论问题集

Does a group need to have a subnormal abelian series if every countable subgroup of it has such a series?

来源字段：https://arxiv.org/abs/1401.0300

04387 | KOU-21.134

### Kourovka Notebook Problem 21.134

S / solved | Kourovka 群论问题集

For a finite group $G$, let the type of $G$ be the function on positive integers whose value at $n$ is the number of solutions of the equation $x^n=1$ in $G$.

a) Is it true that a group having the same type as a group with trivial solvable radical must also have trivial solvable radical?

b) Is it true that a group having the same type as an almost simple group must be isomorphic to it?

来源字段：https://arxiv.org/abs/1401.0300

04388 | KOU-21.135

### Kourovka Notebook Problem 21.135

O / open | Kourovka 群论问题集

For a finite group $G$, let $\chi_1(G)$ denote the totality of the degrees of all irreducible complex characters of $G$ with allowance for their multiplicities. Suppose that $H$ is a finite group with $\chi_1(H)=\chi_1(G)$. If $G$ has trivial solvable radical, must $H$ also have trivial solvable radical?

来源字段：https://arxiv.org/abs/1401.0300

04389 | KOU-21.136

### Kourovka Notebook Problem 21.136

P / partially_solved | Kourovka 群论问题集

Let G be a profinite group with fewer than $2^{\aleph_0}$ conjugacy classes of elements of infinite order. Must G be a torsion group?

来源字段：https://arxiv.org/abs/1401.0300

04390 | KOU-21.137

### Kourovka Notebook Problem 21.137

S / solved | Kourovka 群论问题集

If the $p$-th powers in a finite $p$-group form a subgroup, must that subgroup be powerful? That is, for $p\ne 2$, if the $p$-th powers in a $p$-group of exponent $p^2$ form a subgroup, must that subgroup be abelian? For a 2-group of exponent 8, if the squares form a subgroup, must that subgroup be abelian?

来源字段：https://arxiv.org/abs/1401.0300

04391 | KOU-21.138

### Kourovka Notebook Problem 21.138

O / open | Kourovka 群论问题集

Let G be an infinite finitely presented group such that every subgroup of infinite index is free. Must G be isomorphic to either a free group or a surface group?

来源字段：https://arxiv.org/abs/1401.0300

04392 | KOU-21.139

### Kourovka Notebook Problem 21.139

O / open | Kourovka 群论问题集

Let G be a hyperbolic group which is virtually compact special in the sense of Haglund--Wise. Suppose that the set of second Betti numbers of the finite-index subgroups of G is bounded. Must G be virtually either a free group or a surface group?

来源字段：https://arxiv.org/abs/1401.0300

04393 | KOU-21.14

### Kourovka Notebook Problem 21.14

S / solved | Kourovka 群论问题集

Suppose $\alpha$ is an endomorphism of a group G such that for every group H and every homomorphism $f:G\to H$, there exists an endomorphism $\beta_f$ of $H$ such that $\beta_f f=f\alpha$. Must $\alpha$ then be either an inner automorphism of G or the trivial endomorphism?

来源字段：https://arxiv.org/abs/1401.0300

04394 | KOU-21.140

### Kourovka Notebook Problem 21.140

O / open | Kourovka 群论问题集

Let G be a torsion-free group of type $F_\infty$ of infinite cohomological dimension. Must G contain a copy of Thompson's group F?

来源字段：https://arxiv.org/abs/1401.0300

04395 | KOU-21.141

### Kourovka Notebook Problem 21.141

P / partially_solved | Kourovka 群论问题集

Let $G=G_1\amalg_H G_2$ be a free pro-$p$ product of coherent pro-$p$ groups with polycyclic amalgamation. Is $G$ coherent?

来源字段：https://arxiv.org/abs/1401.0300

04396 | KOU-21.142

### Kourovka Notebook Problem 21.142

S / solved | Kourovka 群论问题集

A group $G$ is said to be invariably generated by $a$ and $b$ if $G$ is generated by the conjugates $a^g,b^h$ for every $g,h$. Let $p\ne q$ be fixed primes. Does every finite group embed into a finite group invariably generated by an element of order $p$ and an element of order $q$?

来源字段：https://arxiv.org/abs/1401.0300

04397 | KOU-21.143

### Kourovka Notebook Problem 21.143

O / open | Kourovka 群论问题集

(Well-known problem). Is Thompson's group F automatic?

来源字段：https://arxiv.org/abs/1401.0300

04398 | KOU-21.144

### Kourovka Notebook Problem 21.144

O / open | Kourovka 群论问题集

Conjecture: Every subgroup of Thompson's group F is either elementary amenable or else contains a subgroup isomorphic to F.

来源字段：https://arxiv.org/abs/1401.0300

04399 | KOU-21.145

### Kourovka Notebook Problem 21.145

O / open | Kourovka 群论问题集

Is Thompson's group F quasi-isometric

(a) to F $\times$ Z?

(b) to F $\times$ F?

来源字段：https://arxiv.org/abs/1401.0300

04400 | KOU-21.146

### Kourovka Notebook Problem 21.146

O / open | Kourovka 群论问题集

(Well-known problem). A classifying space for a group $G$ is a connected CW-complex with fundamental group $G$ and all higher homotopy groups trivial. A group is of type $F_n$ if it has a classifying space with finite $n$-skeleton. For example, type $F_1$ is equivalent to finite generation, and type $F_2$ is equivalent to finite presentability. Type $F_\infty$ means type $F_n$ for all $n$. For $n\geqslant 3$, does every group of type $F_{n-1}$ embed as a subgroup of a group of type $F_n$? Or even in a group of type $F_\infty$?

来源字段：https://arxiv.org/abs/1401.0300

04401 | KOU-21.147

### Kourovka Notebook Problem 21.147

S / solved | Kourovka 群论问题集

A subgroup H of a right-orderable group G is said to be right-relatively convex if it is convex under some right ordering on G. Is the lattice of right-relatively convex subgroups of a right-orderable group always a sublattice of the lattice of its subgroups?

来源字段：https://arxiv.org/abs/1401.0300

04402 | KOU-21.148

### Kourovka Notebook Problem 21.148

S / solved | Kourovka 群论问题集

Is it true that the lattice of right-relatively convex subgroups of a right-orderable group is distributive if and only if it is a chain?

来源字段：https://arxiv.org/abs/1401.0300

04403 | KOU-21.149

### Kourovka Notebook Problem 21.149

O / open | Kourovka 群论问题集

Are there order automorphisms of Dlab groups that are not inner automorphisms?

来源字段：https://arxiv.org/abs/1401.0300

04404 | KOU-21.15

### Kourovka Notebook Problem 21.15

S / solved | Kourovka 群论问题集

Suppose B is a subgroup of the symmetric group $S_\Omega$ on an infinite set $\Omega$. Will the amalgamated free product $S_\Omega *_B S_\Omega$ of two copies of $S_\Omega$ with amalgamation of B be embeddable in $S_\Omega$?

来源字段：https://arxiv.org/abs/1401.0300

04405 | KOU-21.150

### Kourovka Notebook Problem 21.150

S / solved | Kourovka 群论问题集

Let $G$ be an extension of a normal elementary abelian subgroup $A$ by an elementary abelian group $B\cong G/A$ such that $A$ contains an element $a$ with $C_B(a)=1$. Is it true that the rank of the subgroup $Z(\langle a,B\rangle)\cap\langle a,B\rangle'$ is at most the rank of $B$?

来源字段：https://arxiv.org/abs/1401.0300

04406 | KOU-21.16

### Kourovka Notebook Problem 21.16

O / open | Kourovka 群论问题集

Let the width of a group (respectively, a monoid) H with respect to a generating set X mean the supremum over h $\in$ H of the least length of a group word (respectively, a monoid word) in elements of X expressing h. A group (or monoid) is said to have finite width if its width with respect to every generating set is finite. (A common finite bound for these widths is not required.) Do there exist groups G having finite width as groups, but not as monoids?

来源字段：https://arxiv.org/abs/1401.0300

04407 | KOU-21.17

### Kourovka Notebook Problem 21.17

O / open | Kourovka 群论问题集

If $X$ is a class of groups, let $H(X)$ denote the class of homomorphic images of groups in $X$, let $S(X)$ denote the class of groups isomorphic to subgroups of groups in $X$, let $P(X)$ denote the class of groups isomorphic to (unrestricted) direct products of families of groups in $X$, and let $P_f(X)$ denote the class of groups isomorphic to direct products of finite families of groups in $X$. By Birkhoff's theorem, $H(S(P(X)))$ is the variety of groups generated by $X$. If $M$ is a class of metabelian groups, must $H(S(P_f(M)))\subseteq S(H(P(S(M))))$?

来源字段：https://arxiv.org/abs/1401.0300

04408 | KOU-21.18

### Kourovka Notebook Problem 21.18

S / solved | Kourovka 群论问题集

Suppose that G is a finite group, and $A_1,A_2,A_3$ are subsets of G such that the multiplication map $A_1\times A_2\times A_3\to G$ is bijective. Must the subgroup $\langle A_2\rangle$ generated by $A_2$ have order divisible by the cardinality $|A_2|$?

来源字段：https://arxiv.org/abs/1401.0300

04409 | KOU-21.19

### Kourovka Notebook Problem 21.19

O / open | Kourovka 群论问题集

Suppose that S and M are groups of finite Morley rank, S is an infinite group, and M is a non-trivial connected group definably and faithfully acting on S. This action is said to be irreducible if M does not leave invariant any definable non-trivial proper subgroup of S. Prove that if S is a simple group such that every proper definable subgroup of S is nilpotent, and the action of M on S is irreducible, then this action is equivalent to the action of S on itself by conjugation.

来源字段：https://arxiv.org/abs/1401.0300

04410 | KOU-21.2

### Kourovka Notebook Problem 21.2

O / open | Kourovka 群论问题集

Let $S$ be a finite simple group, and let $G$ be a finite group for which there exists a bijection $f:G\to S$ such that $|x|$ divides $|f(x)|$ for all $x\in G$. Must $G$ necessarily be simple?

来源字段：https://arxiv.org/abs/1401.0300

04411 | KOU-21.20

### Kourovka Notebook Problem 21.20

O / open | Kourovka 群论问题集

Prove that a simple group of finite Morley rank without involutions cannot act definably, faithfully, and irreducibly on a connected group other than on itself acting by conjugation.

来源字段：https://arxiv.org/abs/1401.0300

04412 | KOU-21.21

### Kourovka Notebook Problem 21.21

O / open | Kourovka 群论问题集

Prove that a simple (that is, without proper non-trivial connected normal subgroups) algebraic group M over an algebraically closed field cannot act definably, faithfully, and irreducibly on a simple group of finite Morley rank other than M/Z(M).

来源字段：https://arxiv.org/abs/1401.0300

04413 | KOU-21.22

### Kourovka Notebook Problem 21.22

P / partially_solved | Kourovka 群论问题集

Is the (standard, restricted) wreath product $G\wr H$ of two finitely generated Hopfian groups Hopfian?

来源字段：https://arxiv.org/abs/1401.0300

04414 | KOU-21.23

### Kourovka Notebook Problem 21.23

P / partially_solved | Kourovka 群论问题集

A graph is called a cograph if it has no induced subgraph isomorphic to a path with 4 vertices. A graph is said to be chordal if it has no induced cycles with n vertices for every $n\geqslant 4$. For a finite group G, the enhanced power graph E(G) is the graph with vertex set G and edges \{x, y\} for all $x\ne y\in G$ such that $\langle x,y\rangle$ is cyclic.

(a) For a given integer $n\geqslant 4$, determine the set of all finite nonabelian simple groups G such that E(G) has no induced cycles with n vertices.

(b) Determine the set of all finite nonabelian simple groups G such that E(G) is chordal.

来源字段：https://arxiv.org/abs/1401.0300

04415 | KOU-21.24

### Kourovka Notebook Problem 21.24

S / solved | Kourovka 群论问题集

For a finite group G, the power graph P(G) is the graph with vertex set G and edges \{x, y\} for all $x\ne y\in G$ such that either $x\in\langle y\rangle$ or $y\in\langle x\rangle$. Is it true that, for every finite group G, if P(G) is a cograph, then P(G) is chordal?

来源字段：https://arxiv.org/abs/1401.0300

04416 | KOU-21.25

### Kourovka Notebook Problem 21.25

P / partially_solved | Kourovka 群论问题集

Let $G$ be a finite simple group and let $p_1,p_2$ be any (not necessarily distinct) prime divisors of $|G|$. Then can we always find Sylow $p_i$-subgroups $H_i$ such that $G=\langle H_1,H_2\rangle$?

来源字段：https://arxiv.org/abs/1401.0300

04417 | KOU-21.26

### Kourovka Notebook Problem 21.26

P / partially_solved | Kourovka 群论问题集

Let $G$ be a non-trivial finite group and let $p_1,\ldots,p_k$ be the distinct prime divisors of $|G|$. For each $i$, let $H_i$ be a Sylow $p_i$-subgroup of $G$. Is it true that there exists an element $x\in G$ such that for all $i$ the subgroup $H_i\cap H_i^x$ is inclusion-minimal in $\{H_i\cap H_i^g:g\in G\}$?

来源字段：https://arxiv.org/abs/1401.0300

04418 | KOU-21.27

### Kourovka Notebook Problem 21.27

P / partially_solved | Kourovka 群论问题集

A permutation on a set $\Omega$ is called a derangement if it has no fixed points in $\Omega$. Let G be a finite simple transitive permutation group. Is it true that every element in G is the product of two derangements?

来源字段：https://arxiv.org/abs/1401.0300

04419 | KOU-21.28

### Kourovka Notebook Problem 21.28

P / partially_solved | Kourovka 群论问题集

Let $G$ be a finite simple transitive permutation group, and let $\delta(G)$ be the proportion of derangements in $G$. Is it true that $\delta(G)\geqslant 89/325$?

来源字段：https://arxiv.org/abs/1401.0300

04420 | KOU-21.29

### Kourovka Notebook Problem 21.29

S / solved | Kourovka 群论问题集

Let $G\leqslant\operatorname{Sym}(\Omega)$ be a finite primitive permutation group with a regular suborbit (that is, $G$ has a trivial 2-point stabiliser). Then is it true that for all $\alpha,\beta\in\Omega$, there exists $\gamma\in\Omega$ such that the 2-point stabilisers $G_{\alpha,\gamma}$ and $G_{\beta,\gamma}$ are both trivial?

来源字段：https://arxiv.org/abs/1401.0300

04421 | KOU-21.3

### Kourovka Notebook Problem 21.3

P / partially_solved | Kourovka 群论问题集

Let $G=A_n$ or $S_n$ and let $H,K$ be soluble subgroups of $G$. For all sufficiently large $n$, can we always find an element $x\in G$ such that $H\cap K^x=1$? Does this hold for all $n\geqslant 21$?

来源字段：https://arxiv.org/abs/1401.0300

04422 | KOU-21.30

### Kourovka Notebook Problem 21.30

P / partially_solved | Kourovka 群论问题集

(Well-known question). A discrete group G is said to have the Haagerup property (also known as Gromov's a-T-menability property) if there exists a metrically proper isometric action of G on a (possibly infinite-dimensional) Hilbert space. Are all 1-relator groups Haagerup groups?

来源字段：https://arxiv.org/abs/1401.0300

04423 | KOU-21.31

### Kourovka Notebook Problem 21.31

P / partially_solved | Kourovka 群论问题集

Conjecture: If N is a finite soluble group, then any regular subgroup in the holomorph Hol(N) of N is also soluble.

来源字段：https://arxiv.org/abs/1401.0300

04424 | KOU-21.32

### Kourovka Notebook Problem 21.32

S / solved | Kourovka 群论问题集

Is the following problem decidable, and if so, what is its complexity? Given a finite group G, is there a finite group H such that the derived subgroup of H is isomorphic to G?

来源字段：https://arxiv.org/abs/1401.0300

04425 | KOU-21.33

### Kourovka Notebook Problem 21.33

O / open | Kourovka 群论问题集

Does an analogue of Dunwoody's theorem hold for totally disconnected locally compact groups, that is, must a tdlc group of rational discrete cohomological dimension at most 1 be topologically isomorphic to the fundamental group of a graph of profinite groups?

来源字段：https://arxiv.org/abs/1401.0300

04426 | KOU-21.34

### Kourovka Notebook Problem 21.34

O / open | Kourovka 群论问题集

(Well-known problem). A group $G$ is a unique product group if, for any nonempty finite subsets $A,B$ of $G$, there exists an element of $G$ which can be written uniquely as $ab$ with $a\in A$ and $b\in B$. A group $G$ is locally invariant orderable if $G$ admits a partial order $<$ such that for all $g,h\in G$ with $h\ne 1$, we have either $gh>g$ or $gh^{-1}>g$. Does there exist a unique product group which is not locally invariant orderable?

来源字段：https://arxiv.org/abs/1401.0300

04427 | KOU-21.35

### Kourovka Notebook Problem 21.35

P / partially_solved | Kourovka 群论问题集

Let $G$ be a finite group, $w$ a multilinear commutator group-word, and $p$ a prime. Suppose that $p$ divides the order $|xy|$ whenever $x$ is a $w$-value of $p'$-order in $G$ and $y$ is a $w$-value in $G$ of order divisible by $p$. Is it true that then the verbal subgroup $w(G)$ must be $p$-nilpotent?

来源字段：https://arxiv.org/abs/1401.0300

04428 | KOU-21.36

### Kourovka Notebook Problem 21.36

P / partially_solved | Kourovka 群论问题集

Let a hierarchy of tdlc groups $\mathbf H\mathcal K$ be defined analogously to Kropholler's hierarchy in 15.45, with $\mathcal K$ being the class of profinite groups and with the cell stabilisers of the admissible action required to be open. Is it true that $\mathbf H\mathcal K$ is closed under profinite extensions, that is, $(\mathbf H_\alpha\mathcal K)\mathcal K\subseteq \mathbf H_\alpha\mathcal K$ for every $\alpha$?

来源字段：https://arxiv.org/abs/1401.0300

04429 | KOU-21.37

### Kourovka Notebook Problem 21.37

P / partially_solved | Kourovka 群论问题集

By definition, a constructible totally disconnected, locally compact (tdlc) group is the result of a sequence of profinite extensions and ascending HNN-extensions starting from the trivial group. Are soluble tdlc groups of type $FP_\infty$ constructible?

来源字段：https://arxiv.org/abs/1401.0300

04430 | KOU-21.38

### Kourovka Notebook Problem 21.38

S / solved | Kourovka 群论问题集

The spread of a group $G$ is the greatest nonnegative integer $k$ such that for all nontrivial elements $x_1,\ldots,x_k\in G$ there exists $y\in G$ such that $\langle x_1,y\rangle=\cdots=\langle x_k,y\rangle=G$, or is $\infty$ in case there is no such maximum. Does there exist a group with spread equal to 1?

来源字段：https://arxiv.org/abs/1401.0300

04431 | KOU-21.39

### Kourovka Notebook Problem 21.39

P / partially_solved | Kourovka 群论问题集

Are there any locally finite, characteristically simple groups with finitely many orbits under automorphisms that are not residually finite?

来源字段：https://arxiv.org/abs/1401.0300

04432 | KOU-21.4

### Kourovka Notebook Problem 21.4

O / open | Kourovka 群论问题集

Let $G$ be a finite group with trivial solvable radical and let $H_1,\ldots,H_5$ be solvable subgroups of $G$. Then do there always exist elements $x_i\in G$ such that $\bigcap_i H_i^{x_i}=1$?

来源字段：https://arxiv.org/abs/1401.0300

04433 | KOU-21.40

### Kourovka Notebook Problem 21.40

O / open | Kourovka 群论问题集

Let G be a subgroup of GL(n, Q) with finitely many orbits under automorphisms. Is G a virtually soluble group?

来源字段：https://arxiv.org/abs/1401.0300

04434 | KOU-21.41

### Kourovka Notebook Problem 21.41

P / partially_solved | Kourovka 群论问题集

A group is said to be self-similar if it admits a faithful state-closed representation by automorphisms of a regular one-rooted m-tree for some m. Can a torsion-free finitely presented metabelian group which is self-similar contain a subgroup isomorphic to the restricted wreath product H = $\mathbb Z\wr\mathbb Z$?

来源字段：https://arxiv.org/abs/1401.0300

04435 | KOU-21.42

### Kourovka Notebook Problem 21.42

P / partially_solved | Kourovka 群论问题集

Let $\mathcal T_{d,c}$ denote the class of $d$-generated, torsion-free nilpotent groups having class $c$. Are there $\mathcal T_{3,3}$-groups that are not self-similar?

来源字段：https://arxiv.org/abs/1401.0300

04436 | KOU-21.43

### Kourovka Notebook Problem 21.43

S / solved | Kourovka 群论问题集

Conjecture: Suppose that for a fixed positive integer $k$ at least half of the elements of a finite group $G$ have order $k$. Then $G$ is solvable.

来源字段：https://arxiv.org/abs/1401.0300

04437 | KOU-21.44

### Kourovka Notebook Problem 21.44

S / solved | Kourovka 群论问题集

Let $W_n=A_5\wr\cdots\wr A_5$ be the $n$-times iterated permutational wreath product of $A_5$ in its natural action (so $W_n$ acts on $5^n$ points), and let $W=\varprojlim W_n$ be the inverse limit (infinite iterated wreath product of $A_5$). Does $W$ contain a finitely generated dense subgroup of subexponential growth?

来源字段：https://arxiv.org/abs/1401.0300

04438 | KOU-21.45

### Kourovka Notebook Problem 21.45

P / partially_solved | Kourovka 群论问题集

(Well-known problem). Does there exist a finitely presented (infinite) simple group requiring more than two generators?

来源字段：https://arxiv.org/abs/1401.0300

04439 | KOU-21.46

### Kourovka Notebook Problem 21.46

P / partially_solved | Kourovka 群论问题集

(Well-known problem). Does there exist a finitely presented (infinite) simple group of finite cohomological dimension greater than 2?

来源字段：https://arxiv.org/abs/1401.0300

04440 | KOU-21.47

### Kourovka Notebook Problem 21.47

P / partially_solved | Kourovka 群论问题集

(Well-known problem). Does there exist a finitely presented group $G$ such that $G\cong G\times H$ for some non-trivial group $H$?

来源字段：https://arxiv.org/abs/1401.0300

04441 | KOU-21.48

### Kourovka Notebook Problem 21.48

P / partially_solved | Kourovka 群论问题集

A quasimorphism on a group $G$ is a function $f:G\to\mathbb R$ such that the quantity $\sup_{g,h}|f(g)+f(h)-f(gh)|$ is finite. A quasimorphism is homogeneous if it restricts to a homomorphism on every cyclic subgroup of $G$. Let $G$ be a group admitting an unbounded homogeneous quasimorphism $G\to\mathbb R$ that is not a homomorphism. Must $G$ contain a non-abelian free subgroup?

来源字段：https://arxiv.org/abs/1401.0300

04442 | KOU-21.49

### Kourovka Notebook Problem 21.49

P / partially_solved | Kourovka 群论问题集

An isometric action of a group G on a metric space S is called acylindrical if for every $\varepsilon>0$ there exist R, N > 0 such that for every two points x, y with $d(x,y)\geqslant R$, there are at most N elements $g\in G$ satisfying $d(x,gx)\leqslant\varepsilon$ and $d(y,gy)\leqslant\varepsilon$. A group is said to be acylindrically hyperbolic if it is not virtually cyclic and admits an acylindrical action on a hyperbolic space with unbounded orbits. Is the automorphism group of a finitely generated acylindrically hyperbolic group also acylindrically hyperbolic?

来源字段：https://arxiv.org/abs/1401.0300

04443 | KOU-21.5

### Kourovka Notebook Problem 21.5

P / partially_solved | Kourovka 群论问题集

Let $p$ be a prime. Let $G$ be a transitive subgroup of the group of finitary permutations $\operatorname{FSym}(\Omega)$ of a set $\Omega$, let $N$ be a normal subgroup of $G$, and let $S$ be a transitive Sylow $p$-subgroup of $G$.

(a) Is it true that $S\cap N$ is a Sylow $p$-subgroup of $N$?

(b) Is it true that $SN/N$ is a Sylow $p$-subgroup of $G/N$?

(c) Are any two transitive Sylow $p$-subgroups of $G$ locally conjugate in $G$?

Two subgroups $X,Y$ of a group $G$ are said to be locally conjugate if there is a locally inner automorphism $\phi$ of $G$ such that $X^\phi=Y$. An automorphism $\phi$ of $G$ is said to be locally inner if for every finite subset $A\subseteq G$ there is an element $g=g(A)\in G$ such that $a^\phi=g^{-1}ag$ for all $a\in A$.

来源字段：https://arxiv.org/abs/1401.0300

04444 | KOU-21.50

### Kourovka Notebook Problem 21.50

P / partially_solved | Kourovka 群论问题集

Does every finite 3-group $T$ have a nontrivial characteristic subgroup $C$ such that if $T$ is a Sylow 3-subgroup of a finite group $G$, then $T\cap G'=T\cap H'$, where $H=N_G(C)$?

The group $S_4$ shows that no such characteristic subgroups can be found in some Sylow 2-subgroups.

来源字段：https://arxiv.org/abs/1401.0300

04445 | KOU-21.51

### Kourovka Notebook Problem 21.51

P / partially_solved | Kourovka 群论问题集

Let $p$ be a prime, and $P$ a finite $p$-group.

(a) Suppose that $P$ has an abelian subgroup of order $p^n$. For which $n$ does $P$ necessarily have a normal abelian subgroup of order $p^n$?

(b) Suppose that $P$ has an elementary abelian subgroup of order $p^n$. For which $n$ does $P$ necessarily have a normal elementary abelian subgroup of order $p^n$?

来源字段：https://arxiv.org/abs/1401.0300

04446 | KOU-21.52

### Kourovka Notebook Problem 21.52

O / open | Kourovka 群论问题集

Let $L$ be a finite non-abelian simple group, and let $D$ be a conjugacy class of involutions in $L$. Consider the complete graph $\Gamma$ with vertex set $D$. Define an equivalence relation $\sim$ (graph coloring) on the set of edges as follows: $(a,b)\sim(c,d)$ if and only if $|ab|=|cd|$. An automorphism of the coloured graph $\Gamma$ is a permutation $\tau\in S_D$ such that $(a,b)\sim(a^\tau,b^\tau)$ for every edge $(a,b)$. Is it true that the automorphism group of $\Gamma$ is a subgroup of $\operatorname{Aut}(L)$?

来源字段：https://arxiv.org/abs/1401.0300

04447 | KOU-21.53

### Kourovka Notebook Problem 21.53

S / solved | Kourovka 群论问题集

In the notation of 21.52, let $\operatorname{Aut}_t(\Gamma)$ be the set of permutations $\tau\in S_D$ such that $(a,b)\sim(a^\tau,b^\tau)$ whenever $|ab|=t$ for $a,b\in D$. Clearly, $\operatorname{Aut}(\Gamma)=\bigcap_t\operatorname{Aut}_t(\Gamma)$. Is it true that for every finite simple group $G$ we have $\operatorname{Aut}(\Gamma)=\operatorname{Aut}_2(\Gamma)\cap\operatorname{Aut}_p(\Gamma)$, where $\{2,p\}$ are the two minimal prime divisors of $|G|$?

来源字段：https://arxiv.org/abs/1401.0300

04448 | KOU-21.54

### Kourovka Notebook Problem 21.54

O / open | Kourovka 群论问题集

Let $G$ be a finite soluble group with triality, which means that $G$ admits a group of automorphisms $S$ isomorphic to the symmetric group of degree 3 given by the presentation $S=\langle\sigma,\rho\mid \sigma^2=\rho^3=1;\ \sigma\rho\sigma=\rho^2\rangle$ such that $m\cdot m^\rho\cdot m^{\rho^2}=1$ for all $m$ in the set of commutators $M(G):=\{[g,\sigma]\mid g\in G\}$. Suppose in addition that $G=[G,S]$, the group $G$ is generated by $d$ elements of $M(G)$ and their images under $S$, and $x^n=1$ for all $x\in M(G)$. Is it true that the Fitting height of $G$ is bounded in terms of $d$ and $n$?

来源字段：https://arxiv.org/abs/1401.0300

04449 | KOU-21.55

### Kourovka Notebook Problem 21.55

O / open | Kourovka 群论问题集

Let $q$ be a power of a prime $p$, and let $m_n(q)$ be the maximum $p$-length of $p$-solvable subgroups of $\operatorname{GL}(n,q)$. Is it true that $\lim_{n\to\infty}m_n(q)/\log_2 n=1$?

来源字段：https://arxiv.org/abs/1401.0300

04450 | KOU-21.56

### Kourovka Notebook Problem 21.56

P / partially_solved | Kourovka 群论问题集

Let $\ell(X)$ denote the composition length of a finite group $X$. Let $A$ be a finite nilpotent group acting by automorphisms on a finite soluble group $G$. Let $c(G,A)$ be the number of trivial $A$-modules in a given $A$-composition series of $G$. (Note that $c(G,A)=\ell(C_G(A))$ if $(|A|,|G|)=1$.)

Conjecture: there are absolute constants $C_1$ and $C_2$ such that the Fitting height of $G$ is at most $C_1\ell(A)+C_2c(G,A)$.

来源字段：https://arxiv.org/abs/1401.0300

04451 | KOU-21.57

### Kourovka Notebook Problem 21.57

P / partially_solved | Kourovka 群论问题集

Let X be a non-empty class of finite groups of odd order closed under taking subgroups, homomorphic images, and extensions. Let H be an X-maximal subgroup of a finite group G, and N a normal subgroup of G. Must H $\cap$ N be an X-maximal subgroup of N?

来源字段：https://arxiv.org/abs/1401.0300

04452 | KOU-21.58

### Kourovka Notebook Problem 21.58

S / solved | Kourovka 群论问题集

We say that a product $XY=\{xy\mid x\in X,\ y\in Y\}$ of two subsets $X,Y$ of a group $G$ is direct if for every $z\in XY$ there are unique $x\in X$, $y\in Y$ such that $z=xy$. Is there an infinite group $G$ such that every subset $A\subseteq G$ satisfies the following property: all the maximal subsets $B$ for which the product $AB$ is direct have the same cardinality?

来源字段：https://arxiv.org/abs/1401.0300

04453 | KOU-21.59

### Kourovka Notebook Problem 21.59

P / partially_solved | Kourovka 群论问题集

For a finite group $G$, let $\chi_1(G)$ denote the totality of the degrees of all irreducible complex characters of $G$ with allowance for their multiplicities. Suppose that $H$ is a finite group with $\chi_1(H)=\chi_1(G)$.

a) If $G$ is an almost simple group, must $H$ be isomorphic to $G$?

b) If $G$ is a quasisimple group, must $H$ be isomorphic to $G$?

来源字段：https://arxiv.org/abs/1401.0300

04454 | KOU-21.6

### Kourovka Notebook Problem 21.6

O / open | Kourovka 群论问题集

Let p be a prime. A totally imprimitive p-group H of finitary permutations is said to have the cyclic-block property if in the cycle decomposition of every element the support of every cycle is a block for H. Let G be a transitive subgroup of the group of finitary permutations $\operatorname{FSym}(\Omega)$ of a set $\Omega$. Does every transitive Sylow p-subgroup of G contain a transitive subgroup which has the cyclic-block property?

来源字段：https://arxiv.org/abs/1401.0300

04455 | KOU-21.60

### Kourovka Notebook Problem 21.60

S / solved | Kourovka 群论问题集

Let $G$ be a finite group, $\mathbb Z_{(p)}$ the localization at $p$, and $\mathbb F_p$ the field of $p$ elements. Let $\mathcal X$ be the class of $\mathbb F_pG$-modules obtained by reduction of simple $\mathbb QG$-modules. Is it true that $\mathbb Z_{(p)}G$ is semiperfect if and only if each projective indecomposable $\mathbb F_pG$-module can be written as an $\mathbb N$-linear combination of modules in $\mathcal X$ inside the Grothendieck group of $\mathbb F_pG$?

来源字段：https://arxiv.org/abs/1401.0300

04456 | KOU-21.61

### Kourovka Notebook Problem 21.61

O / open | Kourovka 群论问题集

For a fixed (finitely generated free)-by-cyclic group $G=F_n\rtimes\mathbb Z$, is there an algorithm that, given a finite subset $S$ of $G$, finds a finite presentation for the subgroup $H=\langle S\rangle$?

来源字段：https://arxiv.org/abs/1401.0300

04457 | KOU-21.62

### Kourovka Notebook Problem 21.62

O / open | Kourovka 群论问题集

Is the uniform subgroup membership problem decidable for (finitely generated free)-by-cyclic groups? That is, for a fixed group $G=F_n\rtimes\mathbb Z$, is there an algorithm that, given elements $w,h_1,\ldots,h_k\in G$, decides whether or not $w$ belongs to the subgroup $H=\langle h_1,\ldots,h_k\rangle$?

来源字段：https://arxiv.org/abs/1401.0300

04458 | KOU-21.63

### Kourovka Notebook Problem 21.63

P / partially_solved | Kourovka 群论问题集

Let $F$ be a field of characteristic $p>0$, and let $\Gamma$ be the principal congruence subgroup of $\operatorname{Aut}(F[x_1,\ldots,x_n])$ consisting of all automorphisms that send each variable $x_i$ to $x_i$ modulo terms of higher degree. Then $\Gamma$ is a residually-$p$ group. Does $\Gamma$ satisfy a pro-$p$ identity?

来源字段：https://arxiv.org/abs/1401.0300

04459 | KOU-21.64

### Kourovka Notebook Problem 21.64

O / open | Kourovka 群论问题集

Is it true that if a normal subgroup $A$ of a Sylow $p$-subgroup of a $p$-soluble finite group $G$ has exponent $p^e$, then the normal closure of $A$ in $G$ has $(p,e)$-bounded (or even $e$-bounded) $p$-length?

来源字段：https://arxiv.org/abs/1401.0300

04460 | KOU-21.65

### Kourovka Notebook Problem 21.65

P / partially_solved | Kourovka 群论问题集

Suppose that $\phi$ is an automorphism of a finite soluble group $G$. Must $G$ contain a subgroup of index bounded in terms of $|\phi|$ and $|C_G(\phi)|$ whose Fitting height is bounded

(a) in terms of $|\phi|$?

(b) or even in terms of the composition length of $\langle\phi\rangle$?

来源字段：https://arxiv.org/abs/1401.0300

04461 | KOU-21.66

### Kourovka Notebook Problem 21.66

P / partially_solved | Kourovka 群论问题集

Suppose that A is a nilpotent group of automorphisms of a finite soluble group G. Is the Fitting height of G bounded in terms of |A| and |CG(A)|?

Isaacs), when A is cyclic (see 19.43), or when CG(A) = 1 (E. C. Dade).

来源字段：https://arxiv.org/abs/1401.0300

04462 | KOU-21.67

### Kourovka Notebook Problem 21.67

P / partially_solved | Kourovka 群论问题集

Suppose that $\phi$ is an automorphism of a finite soluble group $G$, and let $r$ be the (Pr\"ufer) rank of the fixed-point subgroup $C_G(\phi)$. Is the Fitting height of $G$ bounded in terms of $|\phi|$ and $r$?

来源字段：https://arxiv.org/abs/1401.0300

04463 | KOU-21.68

### Kourovka Notebook Problem 21.68

S / solved | Kourovka 群论问题集

A finite group $G$ is said to be semi-abelian if it has a sequence of subgroups $1=G_0\leqslant G_1\leqslant\cdots\leqslant G_n=G$ such that for every $i$ the subgroup $G_{i+1}$ is isomorphic to a quotient of a semidirect product $A_i\rtimes G_i$ for some abelian group $A_i$.

Conjecture: Semi-abelian finite groups are monomial.

来源字段：https://arxiv.org/abs/1401.0300

04464 | KOU-21.69

### Kourovka Notebook Problem 21.69

O / open | Kourovka 群论问题集

Is there an algorithm deciding if a given one-relator group is hyperbolic?

来源字段：https://arxiv.org/abs/1401.0300

04465 | KOU-21.7

### Kourovka Notebook Problem 21.7

O / open | Kourovka 群论问题集

(Well-known problem). A finite group G is called an IYB-group if it is isomorphic to the permutation group of a finite involutive non-degenerate set-theoretic solution of the Yang--Baxter equation, or equivalently, G is isomorphic to the multiplicative group of a finite left brace. Assume that the Sylow subgroups of a finite soluble group G are IYB-groups. Is G an IYB-group?

来源字段：https://arxiv.org/abs/1401.0300

04466 | KOU-21.70

### Kourovka Notebook Problem 21.70

O / open | Kourovka 群论问题集

A group $G$ is called an orientable Poincar\'e duality group of dimension $n$ over a ring $R$ if it is of type $FP$ over $R$ and $H^i(G;RG)=0$ for $i\ne n$, while $H^n(G;RG)=R$ as an $RG$-module, where the action on $R$ is trivial. (Note that $G$ is not required to be finitely presented.) If $G$ is an orientable Poincar\'e duality group of dimension $n$ over all fields, is it an orientable Poincar\'e duality group over the integers?

来源字段：https://arxiv.org/abs/1401.0300

04467 | KOU-21.71

### Kourovka Notebook Problem 21.71

O / open | Kourovka 群论问题集

For a ring $R$, we say that a group $G$ is of type $FL(R)$ if the trivial $RG$-module $R$ admits a finite resolution by finitely generated free modules. If $R$ is a field, we define the Euler characteristic of $G$ over $R$ to be the alternating sum of $R$-ranks of homology groups $H_i(G;R)$. Does there exist a group of type $FL(F_i)$ for two fields $F_1$ and $F_2$ such that the Euler characteristics of the group over the fields $F_1$ and $F_2$ differ?

来源字段：https://arxiv.org/abs/1401.0300

04468 | KOU-21.72

### Kourovka Notebook Problem 21.72

O / open | Kourovka 群论问题集

We say that a group is a Tarski monster if it is finitely generated, not cyclic, and all of its proper non-trivial subgroups are isomorphic to each other.

a) Do there exist amenable torsion-free Tarski monsters?

b) Do there exist amenable Tarski monsters of prime exponent?

来源字段：https://arxiv.org/abs/1401.0300

04469 | KOU-21.73

### Kourovka Notebook Problem 21.73

O / open | Kourovka 群论问题集

Is the conjugacy problem in $\operatorname{CT}(\mathbb Z)$ algorithmically decidable?

来源字段：https://arxiv.org/abs/1401.0300

04470 | KOU-21.74

### Kourovka Notebook Problem 21.74

O / open | Kourovka 群论问题集

Is it algorithmically decidable whether a given element $g\in\operatorname{CT}(\mathbb Z)$

(a) permutes a nontrivial partition of $\mathbb Z$ into residue classes?

(b) has only finite cycles?

(c) has no finite cycles?

来源字段：https://arxiv.org/abs/1401.0300

04471 | KOU-21.75

### Kourovka Notebook Problem 21.75

S / solved | Kourovka 群论问题集

Given two distinct sets $P_1$ and $P_2$ of odd primes none of which is a subset of the other, is it true that $\langle \operatorname{CT}_{P_1}(\mathbb Z),\operatorname{CT}_{P_2}(\mathbb Z)\rangle \neq \operatorname{CT}_{P_1\cup P_2}(\mathbb Z)$?

来源字段：https://arxiv.org/abs/1401.0300

04472 | KOU-21.76

### Kourovka Notebook Problem 21.76

S / solved | Kourovka 群论问题集

Let $\sigma=(\sigma_{ij})$, $1\leqslant i\ne j\leqslant n$, be an irreducible elementary net (carpet) of order $n\geqslant 3$ over a field $K$ (see 19.48). The net $\sigma$ is said to be closed if the elementary net subgroup $E(\sigma)$ does not contain new elementary transvections. The net $\sigma$ is said to be completable if its diagonal can be supplemented with subgroups to a complete net. Do there exist irreducible closed elementary nets of order $n\geqslant 3$ over a field of odd characteristic that are not completable?

来源字段：https://arxiv.org/abs/1401.0300

04473 | KOU-21.77

### Kourovka Notebook Problem 21.77

O / open | Kourovka 群论问题集

Let $d$ be an integer that is not divisible by $n$-th powers of primes, let $x^n-d$ be an irreducible polynomial over $\mathbb Q$, let $\theta=\sqrt[n]{d}$, and let $K=\mathbb Q(\theta)$ be the radical extension of degree $n$ of the field $\mathbb Q$. The multiplicative group $K^*$ of the field $K$ is canonically embedded into the group $\operatorname{Aut}_{\mathbb Q}(K)$ of all invertible $\mathbb Q$-linear mappings of the $\mathbb Q$-space $K$; let $T$ be the image of $K^*$ under this embedding. In the natural basis $1,\theta,\theta^2,\ldots,\theta^{n-1}$ of the $\mathbb Q$-space $K$ the group $\operatorname{Aut}_{\mathbb Q}(K)$ corresponds to $G=\operatorname{GL}(n,\mathbb Q)$, and the subgroup $T$ to a subgroup $T(d)$ (unsplit maximal torus). Every subgroup $H$ of $G$ containing $T(d)$ and some one-dimensional transformation is rich in elementary transvections and thus defines a net $\sigma=\sigma(H)$. Let $E(\sigma)$ denote the subgroup generated by all transvections in the net group $G(\sigma)$. Is it true that $H\leqslant N_G(E(\sigma))$?

来源字段：https://arxiv.org/abs/1401.0300

04474 | KOU-21.78

### Kourovka Notebook Problem 21.78

P / partially_solved | Kourovka 群论问题集

Let $p$ be a prime and let $G$ be a pro-$p$ group. Suppose that all of the (continuous Galois) cohomology groups $H^n(G,\mathbb F_p)$ of $G$ with coefficients in the field of $p$ elements are finite. Does it necessarily follow that the cohomology ring $H^*(G,\mathbb F_p)$ is finitely generated?

来源字段：https://arxiv.org/abs/1401.0300

04475 | KOU-21.79

### Kourovka Notebook Problem 21.79

O / open | Kourovka 群论问题集

Let $G$ be a finitely generated group with a fixed finite generating set $S$ and the corresponding word metric $L_S(*)$. An element $g$ is said to be distorted in $G$ if $L_S(g^n)/n\to 0$ as $n\to\infty$; this notion is independent of the choice of the generating set $S$. For any, not necessarily finitely generated, group $H$, an element $g\in H$ is said to be distorted if there is a finitely generated subgroup $G$ of $H$ containing $g$ in which $g$ is distorted. Do there exist finitely generated left-orderable groups in which every nontrivial element is distorted?

来源字段：https://arxiv.org/abs/1401.0300

04476 | KOU-21.8

### Kourovka Notebook Problem 21.8

S / solved | Kourovka 群论问题集

As in 17.57, let $r(m)=\{r+km\mid k\in\mathbb Z\}$ for integers $0\leqslant r<m$; for $r_1(m_1)\cap r_2(m_2)=\emptyset$ let the class transposition $\tau_{r_1(m_1),r_2(m_2)}$ be the involution which interchanges $r_1+tm_1$ and $r_2+tm_2$ for each integer t and fixes everything else, and let $\operatorname{CT}(\mathbb Z)$ be the group generated by all class transpositions. Let $\operatorname{CT}_k$ be the subgroup of $\operatorname{CT}(\mathbb Z)$ generated by the class transpositions $\tau_{r_1(k),r_2(k)}$ for $0\leqslant r_1\ne r_2<k$. Since $\tau_{r_1(k),r_2(k)}$ permutes the residue classes modulo k, the group $\operatorname{CT}_k$ is isomorphic to the symmetric group $S_k$. Let $\operatorname{CT}(k)=\langle\operatorname{CT}_2,\operatorname{CT}_3,\ldots,\operatorname{CT}_k\rangle$. Is it true that for $k>3$ the group $\operatorname{CT}(k)$ is isomorphic to the symmetric group $S_N$, where N is the least common multiple of the numbers $2,3,\ldots,k$?

来源字段 : https://arxiv.org/abs/1401.0300

04477 | KOU-21.80

### Kourovka Notebook Problem 21.80

O / open | Kourovka 群论问题集

Do there exist finitely generated left-orderable groups with only one nontrivial conjugacy class?

来源字段 : https://arxiv.org/abs/1401.0300

04478 | KOU-21.81

### Kourovka Notebook Problem 21.81

P / partially_solved | Kourovka 群论问题集

Let $\Gamma$ be a finite simple group and let $N_n(\Gamma)$ denote the set of normal subgroups of the free group $F_n$ of rank $n$ whose quotient is isomorphic to $\Gamma$.

Conjecture: $\operatorname{Aut}(F_n)$ acts transitively on $N_n(\Gamma)$ for $n\geqslant 3$.

来源字段 : https://arxiv.org/abs/1401.0300

04479 | KOU-21.82

### Kourovka Notebook Problem 21.82

P / partially_solved | Kourovka 群论问题集

Conjecture: For $n\geqslant 3$, there are no finite simple characteristic quotients of the free group $F_n$.

来源字段 : https://arxiv.org/abs/1401.0300

04480 | KOU-21.83

### Kourovka Notebook Problem 21.83

O / open | Kourovka 群论问题集

Conjecture: Metabelian groups are permutation-stable.

来源字段 : https://arxiv.org/abs/1401.0300

04481 | KOU-21.84

### Kourovka Notebook Problem 21.84

O / open | Kourovka 群论问题集

For $\sigma\in S_n$ and $\tau\in S_m$, where $n\leqslant m$, let $d_n^{\mathrm{flex}}(\sigma,\tau)=(1/n)\cdot(|\{x\in\{1,\ldots,n\}\mid \sigma(x)\ne\tau(x)\}|+(m-n))$. An almost-homomorphism $\{f_n\}$ is said to be flexibly close to a homomorphism if there is a sequence of group homomorphisms $\rho_n:G\to S_{m_n}$ with $n\leqslant m_n$ such that $d_n^{\mathrm{flex}}(\rho_n(g),f_n(g))\to 0$ as $n\to\infty$ for all $g\in G$. The group $G$ is said to be flexibly permutation stable if every almost-homomorphism of $G$ is flexibly close to a homomorphism. Is $\operatorname{SL}_n(\mathbb Z)$ flexibly permutation-stable?

来源字段 : https://arxiv.org/abs/1401.0300

04482 | KOU-21.85

### Kourovka Notebook Problem 21.85

O / open | Kourovka 群论问题集

Is a flexibly permutation-stable group always permutation-stable?

来源字段 : https://arxiv.org/abs/1401.0300

04483 | KOU-21.86

### Kourovka Notebook Problem 21.86

S / solved | Kourovka 群论问题集

A group $G$ is said to be sofic if for every finite set $F\subseteq G$ containing $1$ and every $\varepsilon>0$ there exist $n\in\mathbb N$ and a map $\phi:F\to S_n$ such that $\phi(1)=1$, $d(\phi(gh),\phi(g)\phi(h))<\varepsilon$ for all $g,h$ such that $gh\in F$, and $\phi(g)$ does not have fixed points for every $g\in F\setminus\{1\}$. Is every group sofic?

来源字段 : https://arxiv.org/abs/1401.0300

04484 | KOU-21.87

### Kourovka Notebook Problem 21.87

S / solved | Kourovka 群论问题集

Assume that a finite group G has a family of d-generator subgroups whose indices have no common divisor. Is it true that G can be generated by d+1 elements?

来源字段 : https://arxiv.org/abs/1401.0300

04485 | KOU-21.88

### Kourovka Notebook Problem 21.88

S / solved | Kourovka 群论问题集

Is there a finite non-abelian group $G$ of odd order, with $k(G)$ conjugacy classes, such that $k(G)/|G|=1/17$?

来源字段 : https://arxiv.org/abs/1401.0300

04486 | KOU-21.89

### Kourovka Notebook Problem 21.89

O / open | Kourovka 群论问题集

For $n>39$, is it true that the number of conjugacy classes in the symmetric group $S_n$ of degree $n$ is never a divisor of the order of $S_n$? In other words, is it true that, for $n>39$, the number $p(n)$ of integer partitions of $n$ is never a divisor of $n!$?

来源字段：https://arxiv.org/abs/1401.0300

04487 | KOU-21.9

### Kourovka Notebook Problem 21.9

O / open | Kourovka 群论问题集

Let $F$ be a non-abelian free pro-$p$ group of finite rank. Can one find a finite collection $U_1,\ldots,U_n$ of open subgroups of $F$, including $F$ itself, such that the only subgroup of $F$ which is contained in $U_i$ and is characteristic in $U_i$ for every $i$ is the trivial subgroup?

来源字段：https://arxiv.org/abs/1401.0300

04488 | KOU-21.90

### Kourovka Notebook Problem 21.90

O / open | Kourovka 群论问题集

Let $\Gamma$ be a graph of diameter $d$. For $i\in\{1,2,\ldots,d\}$, let $\Gamma_i$ be the graph on the same vertex set as $\Gamma$ with vertices $u,w$ adjacent in $\Gamma_i$ if and only if $d_\Gamma(u,w)=i$. Does there exist a $Q$-polynomial distance-regular graph $\Gamma$ of diameter 3 such that $\Gamma_2$ and $\Gamma_3$ are strongly regular?

来源字段：https://arxiv.org/abs/1401.0300

04489 | KOU-21.91

### Kourovka Notebook Problem 21.91

P / partially_solved | Kourovka 群论问题集

Conjecture: The sum of squares of the degrees of the irreducible $p$-Brauer characters of a finite group $G$ is at least the $p'$-part of $|G|$.

来源字段：https://arxiv.org/abs/1401.0300

04490 | KOU-21.92

### Kourovka Notebook Problem 21.92

O / open | Kourovka 群论问题集

Conjecture: The number of irreducible $p$-Brauer characters of a finite group $G$ is bounded above by the maximum of the number of conjugacy classes $k(H)$ in $p'$-subgroups $H$ of $G$.

来源字段：https://arxiv.org/abs/1401.0300

04491 | KOU-21.93

### Kourovka Notebook Problem 21.93

P / partially_solved | Kourovka 群论问题集

Let $G$ be a group and let $k\geqslant 2$. Let $H_1,\ldots,H_k$ be subgroups of $G$, and $g_1,\ldots,g_k$ elements of $G$ such that the cosets $g_1H_1,\ldots,g_kH_k$ form a partition of $G$. Is it true that $|G:H_i|=|G:H_j|$ for some $i\ne j$?

来源字段：https://arxiv.org/abs/1401.0300

04492 | KOU-21.94

### Kourovka Notebook Problem 21.94

O / open | Kourovka 群论问题集

The Gruenberg--Kegel graph (or the prime graph) GK(G) of a finite group G is a labelled graph with vertex set consisting of all prime divisors of the order of G in which different vertices p and q are adjacent if and only if G contains an element of order pq. Let GK(G) denote the abstract graph obtained from GK(G) by removing all labels. A finite group G is said to be recognizable by the isomorphism type of its Gruenberg--Kegel graph if there are no finite groups H $\ncong$ G with GK(H) isomorphic to GK(G). Are there infinitely many (pairwise non-isomorphic) finite groups which are recognizable by the isomorphism type of the Gruenberg--Kegel graph?

来源字段：https://arxiv.org/abs/1401.0300

04493 | KOU-21.95

### Kourovka Notebook Problem 21.95

O / open | Kourovka 群论问题集

Is there an almost simple but not simple group which is recognizable by the isomorphism type of its Gruenberg--Kegel graph?

来源字段：https://arxiv.org/abs/1401.0300

04494 | KOU-21.96

### Kourovka Notebook Problem 21.96

O / open | Kourovka 群论问题集

Is it true that a periodic group containing an involution is locally finite if the centralizer of every element of even order is locally finite?

来源字段：https://arxiv.org/abs/1401.0300

04495 | KOU-21.97

### Kourovka Notebook Problem 21.97

S / solved | Kourovka 群论问题集

Is it true that for every positive rational number $r$ there exists a finite group $G$ such that $|\operatorname{Aut}(G)|/|G|=r$?

来源字段：https://arxiv.org/abs/1401.0300

04496 | KOU-21.98

### Kourovka Notebook Problem 21.98

P / partially_solved | Kourovka 群论问题集

Let w be a multilinear commutator word, and assume that G is a group where the set of w-values is covered by finitely many cyclic subgroups. Is it true that the verbal subgroup w(G) is finite-by-cyclic?

来源字段：https://arxiv.org/abs/1401.0300

04497 | KOU-21.99

### Kourovka Notebook Problem 21.99

S / solved | Kourovka 群论问题集

Conjecture: If $G$ is a transitive permutation group on a finite set $\Omega$, then for any distinct $\alpha,\beta\in\Omega$ there is an element $g\in G$ with $\alpha^g=\beta$ whose number of fixed points is different from 1.

来源字段：https://arxiv.org/abs/1401.0300

04498 | KOU-3.10

### Kourovka Notebook Problem 3.10

S / solved | Kourovka 群论问题集

Need the number of finite non-abelian simple groups contained in a proper variety of groups be finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04499 | KOU-3.11

### Kourovka Notebook Problem 3.11

S / solved | Kourovka 群论问题集

An element $g$ of a group $G$ is said to be *generalized periodic* if there exist $ x_1,\ldots ,x_n\in G$ such that $ x_1^{-1}gx_1\cdots x_n^{-1}gx_n= 1$. Does there exist a finitely generated torsion-free group all of whose elements are generalized periodic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04500 | KOU-3.12

### Kourovka Notebook Problem 3.12

S / solved | Kourovka 群论问题集

(Well-known problem). Is a locally finite group with a full Sylow basis locally soluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04501 | KOU-3.13

### Kourovka Notebook Problem 3.13

S / solved | Kourovka 群论问题集

Is the elementary theory of subgroup lattices of finite abelian groups decidable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04502 | KOU-3.14

### Kourovka Notebook Problem 3.14

S / solved | Kourovka 群论问题集

Let $G$ be a finite group of $ n\times n$ matrices over a skew field $T$ of characteristic zero. Prove that $G$ has a soluble normal subgroup $H$ whose index in $G$ is not greater than some number $f(n)$ not depending on $G$ or $T$. This problem is related to the representation theory of finite groups (the Schur index).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04503 | KOU-3.15

### Kourovka Notebook Problem 3.15

S / solved | Kourovka 群论问题集

A group $G$ is said to be *$U$-embeddable* in a class $ {\mathfrak{K}}$ of groups if, for any finite submodel $M\subset G$, there is a group $ A\in {\mathfrak{K}}$ such that $M$ is isomorphic to some submodel of $A$. Are the following groups $U$-embeddable in the class of finite groups:

a) every group with one defining relation?

b) every group defined by one relation in the variety of soluble groups of a given derived length?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04504 | KOU-3.16

### Kourovka Notebook Problem 3.16

O / open | Kourovka 群论问题集

The word problem for a group admitting a single defining relation in the variety of soluble groups of derived length $n$, $n \geq 3$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04505 | KOU-3.17

### Kourovka Notebook Problem 3.17

S / solved | Kourovka 群论问题集

a) Is a non-abelian group with a unique linear ordering necessarily simple?

b) Is a non-commutative orderable group simple if it has no non-trivial normal relatively convex subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04506 | KOU-3.18

### Kourovka Notebook Problem 3.18

S / solved | Kourovka 群论问题集

(B. H. Neumann). A group $U$ is called *universal* for a class ${\mathfrak{K}}$ of groups if $U$ contains an isomorphic image of every member of ${\mathfrak{K}}$. Does there exist a countable group that is universal for the class of countable orderable groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04507 | KOU-3.19

### Kourovka Notebook Problem 3.19

S / solved | Kourovka 群论问题集

(A. I. Mal'cev). For an arbitrary linearly ordered group $G$, does there exist a linearly ordered abelian group with the same order-type as $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04508 | KOU-3.2

### Kourovka Notebook Problem 3.2

S / solved | Kourovka 群论问题集

Classify the faithful irreducible (infinite-dimensional) representations of the nilpotent group defined by generators $a,b,c$ and relations $[a,b]=c$, $ac=ca$, $bc=cb$. (A condition for a representation to be monomial is given in (A. E. Zalesskiĭ, *Math. Notes*, **9** (1971), 117--123).)

问题集入口：https://kourovkanotebookorg.wordpress.com/

04509 | KOU-3.20

### Kourovka Notebook Problem 3.20

O / open | Kourovka 群论问题集

Does the class of orderable groups coincide with the smallest axiomatizable class containing pro-orderable groups? (See 1.35.)

问题集入口：https://kourovkanotebookorg.wordpress.com/

04510 | KOU-3.21

### Kourovka Notebook Problem 3.21

S / solved | Kourovka 群论问题集

Let ${\mathfrak{K}}$ be the class of one-based models of signature $\sigma$, \ $\vartheta$ a property that makes sense for models in ${\mathfrak{K}}$, and ${\mathfrak{K}}_0$ the class of two-based models whose first base is a set $M$ taken from ${\mathfrak{K}}$ and whose second base consists of all submodels of $M$ with property $\vartheta$ and whose signature consists of the symbols in $\sigma$ together with $\in$ and $\subseteq$ in the usual set-theoretic sense. The elementary theory of ${\mathfrak{K}}_0$ is called the *element-$\vartheta$-submodel theory* of ${\mathfrak{K}}$. Are either of the following theories decidable:

a) the element-pure-subgroup theory of abelian groups?

b) the element-pure-subgroup theory of abelian torsion-free groups?

c) the element-$\vartheta$-subgroup theory of abelian groups, when the set of $\vartheta$-subgroups is linearly ordered by inclusion?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04511 | KOU-3.22

### Kourovka Notebook Problem 3.22

S / solved | Kourovka 群论问题集

Let $\xi = \{ G_{\alpha}, \;\pi_{\beta}^{\alpha}\mid \alpha ,\beta \in I\}$ be a projective system (over a directed set $I$) of finitely generated free abelian groups. If all the projections $ \pi_{\beta}^{\alpha}$ are epimorphisms and all the $G_{\alpha}$ are non-zero, does it follow that $ \lim\limits _{\longleftarrow}\xi \ne 0$? Equivalently, suppose every finite set of elements of an abelian group $A$ is contained in a pure finitely-generated free subgroup of $A$. Then does it follow that $A$ has a direct summand isomorphic to the infinite cyclic group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04512 | KOU-3.26

### Kourovka Notebook Problem 3.26

S / solved | Kourovka 群论问题集

(F. Gross). Is it true that finite groups of exponent $p^{\alpha}q^{\beta}$ have nilpotent length $\leq\alpha +\beta$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04513 | KOU-3.27

### Kourovka Notebook Problem 3.27

S / solved | Kourovka 群论问题集

(J. G. Thompson). Is every finite simple group with a nilpotent maximal subgroup isomorphic to some $PSL_2(q)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04514 | KOU-3.28

### Kourovka Notebook Problem 3.28

S / solved | Kourovka 群论问题集

If $G$ is a finite 2-group with cyclic centre and every abelian normal subgroup 2-generated, then is every abelian subgroup of $G$ \ $3$-generated?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04515 | KOU-3.29

### Kourovka Notebook Problem 3.29

S / solved | Kourovka 群论问题集

Under what conditions can a wreath product of matrix groups over a field be represented by matrices over a field?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04516 | KOU-3.3

### Kourovka Notebook Problem 3.3

O / open | Kourovka 群论问题集

(Well-known problem). Describe the automorphism group of the free associative algebra with $n$ generators, $n \geq 2$.

It is known that all automorphisms are tame for $n=2$ (A. J. Czerniakiewicz, *Trans. Amer. Math. Soc.*, **160** (1971), 393--401; **171** (1972), 309--315; \ L. G. Makar-Limanov, *Funct. Anal. Appl.*, **4** (1971), 262--264). For $n=3$, the Anick automorphism is wild (U. U. Umirbaev, *J. Reine Angew. Math.*, **605** (2007), 165--178). There is an algorithm for recognizing automorphisms among endomorphisms of free associative algebras of finite rank over constructive fields (A. V. Jagzhev, *Siberian. Math. J.*, **21** (1980), 142--146).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04517 | KOU-3.30

### Kourovka Notebook Problem 3.30

S / solved | Kourovka 群论问题集

A torsion-free abelian group is called *factor-decomposable* if, in all its factor groups, the periodic part is a direct summand. Characterize these groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04518 | KOU-3.31

### Kourovka Notebook Problem 3.31

S / solved | Kourovka 群论问题集

Find necessary and sufficient conditions under which every pure subgroup of a completely decomposable torsion-free abelian group is itself completely decomposable.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04519 | KOU-3.33

### Kourovka Notebook Problem 3.33

S / solved | Kourovka 群论问题集

Are two groups necessarily isomorphic if each of them can be defined by a single relation and is a homomorphic image of the other one?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04520 | KOU-3.34

### Kourovka Notebook Problem 3.34

O / open | Kourovka 群论问题集

(Well-known problem). The conjugacy problem for groups with a single defining relation.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04521 | KOU-3.35

### Kourovka Notebook Problem 3.35

S / solved | Kourovka 群论问题集

(K. Ross). Suppose that a group $G$ admits two topologies $\sigma$ and $\tau$ yielding locally compact topological groups $G_{\sigma}$ and $ G_{\tau}$. If the sets of closed subgroups in $G_{\sigma}$ and $G_{\tau}$ are the same, does it follow that $G_{\sigma}$ and $G_{\tau}$ are topologically isomorphic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04522 | KOU-3.37

### Kourovka Notebook Problem 3.37

S / solved | Kourovka 群论问题集

Suppose that every finitely generated subgroup of a locally compact group $G$ is pronilpotent. Then is it true that every maximal closed subgroup of $G$ contains the derived subgroup $G'$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04523 | KOU-3.38

### Kourovka Notebook Problem 3.38

O / open | Kourovka 群论问题集

Describe the topological groups which have no proper closed subgroups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04524 | KOU-3.39

### Kourovka Notebook Problem 3.39

S / solved | Kourovka 群论问题集

Describe the finite groups with a self-centralizing subgroup of prime order.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04525 | KOU-3.4

### Kourovka Notebook Problem 3.4

S / solved | Kourovka 群论问题集

(Well-known problems). a) Is there an algorithm that decides, for any set of group words $f_1,\ldots ,f_m$ (in a fixed set of variables $ x_1,x_2,\ldots $) and a separate word $f$, whether $f=1$ is a consequence of $ f_1=1,\ldots , f_m=1$?

b) Given words $f_1,\ldots ,f_m$, is there an algorithm that decides, for any word $f$, whether $ f = 1$ is a consequence of $ f_1=1,\ldots , f_m=1$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04526 | KOU-3.40

### Kourovka Notebook Problem 3.40

S / solved | Kourovka 群论问题集

(I. R. Shafarevich). Let $SL_2(\mathbb{Z} ) \hat{}$ and $SL_2(\mathbb{Z} )^{-}$ denote the completions of $SL_2(\mathbb{Z} )$ determined by all subgroups of finite index and all congruence subgroups, respectively, and let $\psi : SL_2(\mathbb{Z} ) \hat{}\rightarrow SL_2(\mathbb{Z} )^{-}$ be the natural homomorphism. Is $\mathrm{Ker} \psi$ a free profinite group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04527 | KOU-3.41

### Kourovka Notebook Problem 3.41

S / solved | Kourovka 群论问题集

Is every compact periodic group locally finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04528 | KOU-3.42

### Kourovka Notebook Problem 3.42

S / solved | Kourovka 群论问题集

(Kneser--Tits conjecture). Let $G$ be a simply connected $k$-defined simple algebraic group, and $E_k(G)$ the subgroup generated by unipotent $k$-elements. If $E_k(G)\ne 1$, then $G_k=E_k(G)$. The proof is known for $k$-decomposable groups (C. Chevalley) and for local fields (V. P. Platonov).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04529 | KOU-3.43

### Kourovka Notebook Problem 3.43

O / open | Kourovka 群论问题集

Let $\mu$ be an infinite cardinal number. A group $G$ is said to be $\mu $-*overnilpotent* if every cyclic subgroup of $G$ is a member of some ascending normal series of length less than $\mu$ reaching $G$. It is not difficult to show that the class of $\mu $-overnilpotent groups is a radical class. Is it true that if $\mu _1 < \mu _2$ for two infinite cardinal numbers $\mu _1$ and $\mu _2$, then there exists a group $G$ which is $\mu _2 $-overnilpotent and $\mu _1 $-semisimple?

*Remark of 2001:* In (S. Vovsi, *Sov. Math. Dokl.*, **13** (1972), 408--410) it was proved that for any two infinite cardinals $\mu _1 < \mu _2$ there exists a group that is $\mu _2$-overnilpotent but not $\mu _ 1$-overnilpotent.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04530 | KOU-3.44

### Kourovka Notebook Problem 3.44

O / open | Kourovka 群论问题集

Suppose that a group is generated by its ascendant soluble subgroups. Is it necessarily locally soluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04531 | KOU-3.45

### Kourovka Notebook Problem 3.45

O / open | Kourovka 群论问题集

Let ${\mathfrak{X}}$ be a hereditary radical. Is it true that, in a locally nilpotent torsion-free group $G$, the subgroup ${\mathfrak{X}}(G)$ is isolated?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04532 | KOU-3.46

### Kourovka Notebook Problem 3.46

S / solved | Kourovka 群论问题集

Does there exist a group having more than one, but finitely many maximal locally soluble normal subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04533 | KOU-3.47

### Kourovka Notebook Problem 3.47

O / open | Kourovka 群论问题集

(Well-known problem of A. I. Maltsev). Is it true that every locally nilpotent group is a homomorphic image of some torsion-free locally nilpotent group?

*Editors' comment:* An affirmative answer is known for periodic groups (E. M. Levich, A. I. Tokarenko, *Siberian Math. J.*, **11** (1970), 1033--1034) and for countable groups (N. S. Romanovskiĭ, Preprint, 1969).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04534 | KOU-3.48

### Kourovka Notebook Problem 3.48

O / open | Kourovka 群论问题集

It can be shown that hereditary radicals form a semigroup with respect to taking the products of classes. It is an interesting problem to find all indecomposable elements of this semigroup. In particular, we point out the problem of finding all indecomposable radicals contained in the class of locally finite $p $-groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04535 | KOU-3.49

### Kourovka Notebook Problem 3.49

O / open | Kourovka 群论问题集

Does the semigroup generated by all indecomposable radicals satisfy any identity?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04536 | KOU-3.5

### Kourovka Notebook Problem 3.5

O / open | Kourovka 群论问题集

Can a subgroup of a relatively free group be radicable? In particular, can a verbal subgroup of a relatively free group be radicable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04537 | KOU-3.50

### Kourovka Notebook Problem 3.50

S / solved | Kourovka 群论问题集

Let $G$ be a group of order $p^{\alpha} \cdot m$, where $p$ is a prime, $p$ and $m$ are coprime, and let $k$ be an algebraically closed field of characteristic $p$. Is it true that if the indecomposable projective module corresponding to the 1-representation of $G$ has $k $-dimension $p^{\alpha}$, then $G$ has a Hall $p' $-subgroup? The converse is trivially true.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04538 | KOU-3.51

### Kourovka Notebook Problem 3.51

S / solved | Kourovka 群论问题集

Is it true that every finite group with a group of automorphisms $\Phi$ which acts regularly on the set of conjugacy classes of $G$ (that is, leaves only the identity class fixed) is soluble? The answer is known to be affirmative in the case where $\Phi$ is a cyclic group generated by a regular automorphism.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04539 | KOU-3.52

### Kourovka Notebook Problem 3.52

S / solved | Kourovka 群论问题集

Can the quasivariety generated by the free group of rank 2 be defined by a system of quasi-identities in finitely many variables?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04540 | KOU-3.53

### Kourovka Notebook Problem 3.53

S / solved | Kourovka 群论问题集

Let $L({\mathfrak{N}}_4)$ denote the lattice of subvarieties of the variety ${\mathfrak{N}}_4$ of nilpotent groups of class at most $4$. Is $L({\mathfrak{N}}_4)$ distributive?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04541 | KOU-3.55

### Kourovka Notebook Problem 3.55

O / open | Kourovka 群论问题集

Is every binary soluble group all of whose abelian subgroups have finite rank, locally soluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04542 | KOU-3.56

### Kourovka Notebook Problem 3.56

S / solved | Kourovka 群论问题集

Is a 2-group with the minimum condition for abelian subgroups locally finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04543 | KOU-3.57

### Kourovka Notebook Problem 3.57

O / open | Kourovka 群论问题集

Determine the laws of distribution of non-soluble and simple group-theoretic numbers in the sequence of natural numbers. See 2.78.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04544 | KOU-3.58

### Kourovka Notebook Problem 3.58

S / solved | Kourovka 群论问题集

Let $G$ be a compact $0$-dimensional topological group all of whose Sylow $p $-subgroups are direct products of cyclic groups of order $p$. Then is every normal subgroup of $G$ complementable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04545 | KOU-3.59

### Kourovka Notebook Problem 3.59

S / solved | Kourovka 群论问题集

Let $H$ be an insoluble minimal normal subgroup of a finite group $G$, and suppose that $H$ has cyclic Sylow $p $-subgroups for every prime $p$ dividing $ |G : H|$. Prove that $H$ has at least one complement in $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04546 | KOU-3.6

### Kourovka Notebook Problem 3.6

S / solved | Kourovka 群论问题集

Describe the insoluble finite groups in which every soluble subgroup is either $2$-closed, $2'$-closed, or isomorphic to $\mathbb{S} _4$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04547 | KOU-3.60

### Kourovka Notebook Problem 3.60

O / open | Kourovka 群论问题集

The notion of the *$p $-length* of an arbitrary finite group was introduced in (L. A. Shemetkov, *Math. USSR Sbornik*, **1** (1968), 83--92). Investigate the relations between the $p $-length of a finite group and the invariants $c_p$, $d_p$, $e_p$ of its Sylow $p $-subgroup.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04548 | KOU-3.61

### Kourovka Notebook Problem 3.61

S / solved | Kourovka 群论问题集

Let $\sigma$ be an automorphism of prime order $p$ of a finite group $G$, which has a Hall $\pi $-subgroup with cyclic Sylow subgroups. Suppose that $p \in \pi$. Does the centralizer $C_G(\sigma )$ have at least one Hall $\pi $-subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04549 | KOU-3.62

### Kourovka Notebook Problem 3.62

S / solved | Kourovka 群论问题集

(Well-known problem). A finite group is said to be a {\it $D_{\pi}$-group} if any two of its maximal $\pi$-subgroups are conjugate. Is an extension of a $D_{\pi}$-group by a $D_{\pi }$-group always a $D_{\pi}$-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04550 | KOU-3.64

### Kourovka Notebook Problem 3.64

S / solved | Kourovka 群论问题集

Describe the finite simple groups with a Sylow 2-subgroup of the following type: $\langle a,t\mid a^{2^n}=t^2=1,\;\; tat=a^{2^{n-1}-1}\rangle$, \ $n>2$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04551 | KOU-3.7

### Kourovka Notebook Problem 3.7

S / solved | Kourovka 群论问题集

An automorphism $\sigma$ of a group $G$ is called *algebraic* if, for any $g\in G$, the minimal $\sigma$-invariant subgroup of $G$ containing $g$ is finitely generated. An automorphism $\sigma$ of $G$ is called an *$e$-automorphism* if, for any $\sigma$-invariant subgroups $A$ and $B$, where $A$ is a proper subgroup of $B$, $A\setminus B$ contains an element $ x$ such that $[x, \sigma ] \in B$ . Is every $e$-automorphism algebraic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04552 | KOU-3.8

### Kourovka Notebook Problem 3.8

S / solved | Kourovka 群论问题集

Let $G$ be the free product of free groups $A$ and $B$, and $V$ the verbal subgroup of $G$ corresponding to the equation $ x^4 = 1$. Is it true that, if $ a\in A\setminus V$ and $ b\in B\setminus V$, then $ (ab)^2\notin V$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04553 | KOU-3.9

### Kourovka Notebook Problem 3.9

S / solved | Kourovka 群论问题集

Does there exist an infinite periodic group with a finite maximal subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04554 | KOU-4.1

### Kourovka Notebook Problem 4.1

S / solved | Kourovka 群论问题集

Find an infinite finitely generated group with an identical relation of the form $x^{2^n}=1$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04555 | KOU-4.10

### Kourovka Notebook Problem 4.10

S / solved | Kourovka 群论问题集

A group $G$ is called *locally indicable* if every non-trivial finitely generated subgroup of $G$ has an infinite cyclic factor group. Is every torsion-free one-relator group locally indicable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04556 | KOU-4.11

### Kourovka Notebook Problem 4.11

O / open | Kourovka 群论问题集

Let $F$ be the free group of rank 2 in some variety of groups. If $F$ is not finitely presented, is the multiplicator of $F$ necessarily infinitely generated?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04557 | KOU-4.12

### Kourovka Notebook Problem 4.12

S / solved | Kourovka 群论问题集

Let $G$ be a finite group and $A$ a group of automorphisms of $ G$ stabilizing a series of subgroups beginning with $G$ and ending with its Frattini subgroup. Then is $A$ nilpotent?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04558 | KOU-4.13

### Kourovka Notebook Problem 4.13

O / open | Kourovka 群论问题集

Prove that every finite non-abelian $p$-group admits an automorphism of order $p$ which is not an inner one.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04559 | KOU-4.14

### Kourovka Notebook Problem 4.14

S / solved | Kourovka 群论问题集

Let $p$ be a prime number. What are necessary and sufficient conditions for a finite group $G$ in order that the group algebra of $G$ over a field of characteristic $p$ be indecomposable as a two-sided ideal? There exist some nontrivial examples, for instance, the group algebra of the Mathieu group $M_{24}$ is indecomposable when $p$ is $2$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04560 | KOU-4.16

### Kourovka Notebook Problem 4.16

S / solved | Kourovka 群论问题集

Suppose that ${\mathfrak{K}}$ is a class of groups meeting the following requirements: 1) subgroups and epimorphic images of ${\mathfrak{K}}$-groups are ${\mathfrak{K}}$-groups; \ 2) if the group $ G = UV$ is the product of $ {\mathfrak{K}}$-subgroups $U$ and $V$ (neither of which need be normal), then $ G\in {\mathfrak{K}}$. If $\pi$ is a set of primes, then the class of all finite $\pi$-groups meets these requirements. Are these the only classes ${\mathfrak{K}}$ with these properties?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04561 | KOU-4.17

### Kourovka Notebook Problem 4.17

O / open | Kourovka 群论问题集

If $ {\mathfrak{V}}$ is a variety of groups, denote by $\widetilde{{\mathfrak{V}}}$ the class of all finite ${\mathfrak{V}} $-groups. How to characterize the classes of finite groups of the form $\widetilde{{\mathfrak{V}}}$ for ${\mathfrak{V}}$ a variety?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04562 | KOU-4.18

### Kourovka Notebook Problem 4.18

O / open | Kourovka 群论问题集

Characterize the classes ${\mathfrak{K}}$ of groups, meeting the following requirements: subgroups, epimorphic images and groups of automorphisms of ${\mathfrak{K}} $-groups are $ {\mathfrak{K}} $-groups, but not every countable group is a ${\mathfrak{K}} $-group. Note that the class of all finite groups and the class of all almost cyclic groups meet these requirements.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04563 | KOU-4.19

### Kourovka Notebook Problem 4.19

O / open | Kourovka 群论问题集

Denote by ${\mathfrak{C}}$ the class of all groups $G$ with the following property: if $U$ and $V$ are maximal locally soluble subgroups of $G$, then $U$ and $V$ are conjugate in $G$ (or at least isomorphic). It is almost obvious that a finite group $G$ belongs to ${\mathfrak{C}}$ if and only if $G$ is soluble. What can be said about the locally finite groups in ${\mathfrak{C}}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04564 | KOU-4.2

### Kourovka Notebook Problem 4.2

O / open | Kourovka 群论问题集

a) Find an infinite finitely generated group of exponent $< 100$.

b) Do there exist such groups of exponent 5?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04565 | KOU-4.20

### Kourovka Notebook Problem 4.20

S / solved | Kourovka 群论问题集

a) Let $F$ be a free group, and $N$ a normal subgroup of it. Is it true that the Cartesian square of $N$ is $m$-reducible to $N$ (that is, there is an algorithm that from a pair of words $w_1,w_2\in F$ constructs a word $w\in F$ such that $w_1\in N$ and $w_2\in N$ if and only if $w\in N$)?

b) (Well-known problem). Do there exist finitely presented groups in which the word problem has an arbitrary pre-assigned recursively enumerable $m$-degree of unsolvability?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04566 | KOU-4.21

### Kourovka Notebook Problem 4.21

S / solved | Kourovka 群论问题集

Let $G$ be a finite group, $p$ an odd prime number, and $P$ a Sylow $p $-subgroup of $G$. Let the order of every non-identity normal subgroup of $G$ be divisible by $p$. Suppose $P$ has an element $x$ that is conjugate to no other from $P$. Does $x$ belong to the centre of $G$? For $ p = 2$, the answer is positive (G. Glauberman, *J. Algebra*, **4** (1966), 403--420).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04567 | KOU-4.22

### Kourovka Notebook Problem 4.22

S / solved | Kourovka 群论问题集

(J. G. Thompson). Let $G$ be a finite group, $A$ a group of automorphisms of $G$ such that $|A|$ and $|G|$ are coprime. Does there exist an $A $-invariant soluble subgroup $H$ of $G$ such that $C_A(H)=1$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04568 | KOU-4.23

### Kourovka Notebook Problem 4.23

S / solved | Kourovka 群论问题集

Let $G$ be a finite simple group, $\tau$ some element of prime order, and $\alpha$ an automorphism of $G$ whose order is coprime to $ |G|$. Suppose $\alpha$ centralizes $C_G(\tau )$. Is $\alpha = 1$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04569 | KOU-4.24

### Kourovka Notebook Problem 4.24

P / partially_solved | Kourovka 群论问题集

Suppose $T$ is a non-abelian Sylow 2-subgroup of a finite simple group $G$.

a) Suppose $T$ has nilpotency class $n$. The best possible bound for the exponent of the center of $T$ is $2^{n-1}$. This easily implies the bound $2^{n(n-1)}$ for the exponent of $T$, however, this is almost certainly too crude. What is the best possible bound?

d) Find a ``small number'' of subgroups $T_1,\ldots ,T_n$ of $T$ which depend only on the isomorphism class of $T$ such that $\{ N_G(T_1), \ldots , N_G(T_n)\}$ together control fusion in $T$ with respect to $G$ (in the sense of Alperin).

[Archived solved part(s) of 4.24:] Suppose that $T$ is a non-abelian Sylow 2-subgroup of a finite simple group $G$.

b) Is it possible that $T$ is the direct product of two proper subgroups?

c) Is $T'=\Phi (T)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04570 | KOU-4.27

### Kourovka Notebook Problem 4.27

S / solved | Kourovka 群论问题集

Describe all finite simple groups $G$ which can be represented in the form $G=ABA$, where $A$ and $B$ are abelian subgroups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04571 | KOU-4.28

### Kourovka Notebook Problem 4.28

S / solved | Kourovka 群论问题集

For a given field $k$ of characteristic $p > 0$, characterize the locally finite groups with semisimple group algebras over $k$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04572 | KOU-4.29

### Kourovka Notebook Problem 4.29

S / solved | Kourovka 群论问题集

Classify the irreducible matrix groups over a finite field that are generated by reflections, that is, by matrices with Jordan form $\mathrm{diag}(-1, 1, \ldots , 1)$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04573 | KOU-4.3

### Kourovka Notebook Problem 4.3

S / solved | Kourovka 群论问题集

Construct a finitely presented group with insoluble word problem and satisfying a non-trivial law.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04574 | KOU-4.30

### Kourovka Notebook Problem 4.30

O / open | Kourovka 群论问题集

Describe the groups (finite groups, abelian groups) that are the full automorphism groups of topological groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04575 | KOU-4.31

### Kourovka Notebook Problem 4.31

O / open | Kourovka 群论问题集

Describe the lattice of quasivarieties of nilpotent groups of nilpotency class 2.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04576 | KOU-4.32

### Kourovka Notebook Problem 4.32

S / solved | Kourovka 群论问题集

The conjugacy problem for metabelian groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04577 | KOU-4.33

### Kourovka Notebook Problem 4.33

O / open | Kourovka 群论问题集

Let ${\mathfrak{K}}_n$ be the class of all groups with a single defining relation in the variety of soluble groups of derived length $n$.

a) Under what conditions does a ${\mathfrak{K}}_n $-group have non-trivial center? Can a $ {\mathfrak{K}}_n $-group, $n \geq 2$, that cannot be generated by two elements have non-trivial centre? *Editors' comment (1998):* These questions were answered for $n=2$ (E. I. Timoshenko, *Siberian Math. J.*, **14**, no. 6 (1973), 954--957; *Math. Notes*, **64**, no. 6 (1998), 798--803).

b) Describe the abelian subgroups of ${\mathfrak{K}}_n $-groups.

c) Investigate the periodic subgroups of ${\mathfrak{K}}_n $-groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04578 | KOU-4.34

### Kourovka Notebook Problem 4.34

O / open | Kourovka 群论问题集

Let $v$ be a group word, and let ${\mathfrak{K}}_v$ be the class of groups $G $ such that there exists a positive integer $n=n(G)$ such that each element of the verbal subgroup $vG$ can be represented as a product of $n$ values of the word $v$ on the group $G$.

a) For which $v$ do all finitely generated soluble groups belong to the class ${\mathfrak{K}}_v$?

b) Does the word $v(x,y)=x^{-1}y^{-1}xy$ satisfy this condition?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04579 | KOU-4.35

### Kourovka Notebook Problem 4.35

S / solved | Kourovka 群论问题集

(Well-known problem). Is there an infinite locally finite simple group satisfying the minimum condition for $p $-subgroups for every prime $p$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04580 | KOU-4.36

### Kourovka Notebook Problem 4.36

S / solved | Kourovka 群论问题集

Is there an infinite locally finite simple group $G$ with an involution $i$ such that the centralizer $C_G(i)$ is a Chernikov group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04581 | KOU-4.37

### Kourovka Notebook Problem 4.37

S / solved | Kourovka 群论问题集

Is there an infinite locally finite simple group $G$ that cannot be represented by matrices over a field and is such that for some prime $p$ the $p $-subgroups of $G$ are either of bounded derived length or of finite exponent?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04582 | KOU-4.38

### Kourovka Notebook Problem 4.38

S / solved | Kourovka 群论问题集

Classify the composition factors of automorphism groups of finite (nilpotent of class 2) groups of prime exponent.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04583 | KOU-4.39

### Kourovka Notebook Problem 4.39

S / solved | Kourovka 群论问题集

A countable group $U$ is said to be {\it ${\sc SQ}$-universal} if every countable group is isomorphic to a subgroup of a quotient group of $ U$. Let $G$ be a group that has a presentation with $ r \geq 2$ generators and at most $r-2$ defining relations. Is $G$ \ ${\sc SQ}$-universal?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04584 | KOU-4.4

### Kourovka Notebook Problem 4.4

S / solved | Kourovka 群论问题集

Construct a finitely presented group with undecidable word problem all of whose non-trivial defining relations have the form $A^2=1$. This problem is interesting for topologists.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04585 | KOU-4.40

### Kourovka Notebook Problem 4.40

O / open | Kourovka 群论问题集

Let $C$ be a fixed non-trivial group (for instance, $C={\mathbb{Z}/2{\mathbb{Z}$). As it is shown in (Yu. I. Merzlyakov, *Algebra and Logic*, **9**, no. 5 (1970), 326--337), for any two groups $A$ and $B$, all split extensions of $B$ by $A$ can be imbedded in a certain unified way into the direct product $A \times \mathrm{Aut}(B\wr C)$. How are they situated in it?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04586 | KOU-4.41

### Kourovka Notebook Problem 4.41

S / solved | Kourovka 群论问题集

A point $z$ in the complex plane is called *free* if the matrices {\small $\left( \begin{array}{cc}1&2\\0&1\end{array}\right)$} and {\small $\left( \begin{array}{cc}1&0\\z&1\end{array}\right)$} generate a free group. Are all points outside the rhombus with vertices $\pm 2$, $\pm i$ free?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04587 | KOU-4.42

### Kourovka Notebook Problem 4.42

O / open | Kourovka 群论问题集

For what natural numbers $n$ does the following equality hold: $$GL_n({\mathbb{R})=D_n({\mathbb{R})\cdot O_n({\mathbb{R})\cdot UT_n({\mathbb{R})\cdot GL_n({\mathbb{Z})? $$ For notation, see, for instance, (M. I. Kargapolov, Ju. I. Merzljakov, *Fundamentals of the Theory of Groups*, Springer, New York, 1979). For given $n$ this equality implies the affirmative solution of Minkowski's problem on the product of $n$ linear forms (A. M. Macbeath, *Proc. Glasgow Math. Assoc.*, **5**, no. 2 (1961), 86--89), which remains open for $n \geq 6$. It is known that the equality holds for $n \leq 3$ (Kh. N. Narzullayev, *Math. Notes*, **18** (1975), 713--719); on the other hand, it does not hold for all sufficiently large $n$ (N. S. Akhmedov, *Zapiski Nauchn. Seminarov LOMI*, **67** (1977), 86--107 (Russian)).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04588 | KOU-4.44

### Kourovka Notebook Problem 4.44

O / open | Kourovka 群论问题集

(Well-known problem). Describe the groups whose automorphism groups are abelian.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04589 | KOU-4.45

### Kourovka Notebook Problem 4.45

S / solved | Kourovka 群论问题集

Let $G$ be a free product amalgamating proper subgroups $H$ and $K$ of $A$ and $B$, respectively.

a) Suppose that $H,K$ are finite and $|A:H| > 2$, \ $|B:K| \geq 2$. Is $G$ ${\sc SQ} $-universal?

b) Suppose that $A,B,H,K$ are free groups of finite ranks. Can $G$ be simple?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04590 | KOU-4.46

### Kourovka Notebook Problem 4.46

P / partially_solved | Kourovka 群论问题集

a) We call a variety of groups a *limit variety* if it cannot be defined by finitely many laws, while each of its proper subvarieties has a finite basis of identities. It follows from Zorn's lemma that every variety that has no finite basis of identities contains a limit subvariety. Give explicitly (by means of identities or by a generating group) at least one limit variety.

[Archived solved part(s) of 4.46:] b) We call a variety of groups a *limit variety* if it cannot be defined by finitely many laws, while each of its proper subvarieties has a finite basis of identities. It follows from Zorn's lemma that every variety that has no finite basis of identities contains a limit subvariety. Is the set of limit varieties countable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04591 | KOU-4.47

### Kourovka Notebook Problem 4.47

S / solved | Kourovka 群论问题集

Does there exist a countable family of groups such that every variety is generated by a subfamily of it?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04592 | KOU-4.48

### Kourovka Notebook Problem 4.48

O / open | Kourovka 群论问题集

A locally finite group is said to be an *$A $-group* if all of its Sylow subgroups are abelian. Does every variety of $A $-groups possess a finite basis of identities?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04593 | KOU-4.49

### Kourovka Notebook Problem 4.49

S / solved | Kourovka 群论问题集

Let $G$ and $H$ be finitely generated torsion-free nilpotent groups such that $\mathrm{Aut} G\cong \mathrm{Aut} H$. Does it follow that $G\cong H$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04594 | KOU-4.5

### Kourovka Notebook Problem 4.5

P / partially_solved | Kourovka 群论问题集

b) Is it true that an arbitrary finitely presented group has either polynomial or exponential growth?

[Archived solved part(s) of 4.5:] a) (J. Milnor). Is it true that an arbitrary finitely generated group has either polynomial or exponential growth?

c) Is it true that every finitely generated group with undecidable word problem has exponential growth?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04595 | KOU-4.50

### Kourovka Notebook Problem 4.50

O / open | Kourovka 群论问题集

What are the soluble varieties of groups all of whose finitely generated subgroups are residually finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04596 | KOU-4.51

### Kourovka Notebook Problem 4.51

S / solved | Kourovka 群论问题集

(Well-known problem). Are knot groups residually finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04597 | KOU-4.52

### Kourovka Notebook Problem 4.52

S / solved | Kourovka 群论问题集

Let $G$ be a finitely generated torsion-free group that is an extension of an abelian group by a nilpotent group. Then is $G$ almost a residually finite $p $-group for almost all primes $p$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04598 | KOU-4.53

### Kourovka Notebook Problem 4.53

S / solved | Kourovka 群论问题集

P. F. Pickel has proved that there are only finitely many non-isomorphic finitely-generated nilpotent groups having the same family of finite homomorphic images. Can Pickel's theorem be extended to polycyclic groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04599 | KOU-4.54

### Kourovka Notebook Problem 4.54

S / solved | Kourovka 群论问题集

Are any two minimal relation-modules of a finite group isomorphic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04600 | KOU-4.55

### Kourovka Notebook Problem 4.55

S / solved | Kourovka 群论问题集

Let $G$ be a finite group, and ${\mathbb{Z}_{(p)}$ the localization at $p$. Can every projective $\mathbb{Z}_{(p)}G$-module be uniquely, up to permutations and isomorphisms, written as a direct sum of indecomposable projective $\mathbb{Z}_{(p)}G$-modules?

It is known that the category of finitely generated projective $Z_{(p)}G$-modules is not a Krull--Schmidt category (S. M. Woods, *Canadian J. Math.*, **26**, no. 1 (1974), 121--129). See also (D. Johnston, D. Rumynin, *J. Algebra, to appear*, https://arxiv.org/pdf/2507.21316).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04601 | KOU-4.56

### Kourovka Notebook Problem 4.56

O / open | Kourovka 群论问题集

Let $R$ be a commutative Noetherian ring with 1, and $\Lambda$ an $ R $-algebra, which is finitely generated as $R $-module. Put $T=\{ U \in \mathrm{Mod} \Lambda \mid \exists \mbox{ an exact}\; \Lambda \mbox{-sequence} \; 0 \rightarrow P \rightarrow \Lambda ^{(n)} \rightarrow U \rightarrow 0\mbox{ for some}\;n,\; \mbox{with}\;P_m \cong \Lambda _m^{(n)} \;\mbox{for every maximal ideal}\;m \mbox{ of}\;R\} $. Denote by ${\mathbb{G}(T)$ the Grothendieck group of $T$ relative to short exact sequences.

a) Describe ${\mathbb{G}(T)$, in particular, what does it mean: $[U]=[V]$ in ${\mathbb{G}(T)$?

b) *Conjecture:* if $\dim (\max (R))=d < \infty $, and there are two epimorphisms $\varphi \! : \Lambda ^{(n)} \rightarrow U$, \ $\psi \! : \Lambda ^{(n)} \rightarrow V$, \ $n > d$, \ and \ $[U]=[V]$ \ in \ ${\mathbb{G}(T)$, then $\mathrm{Ker} \varphi =\mathrm{Ker} \psi $.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04602 | KOU-4.57

### Kourovka Notebook Problem 4.57

S / solved | Kourovka 群论问题集

Let a group $G$ be the product of two of its abelian minimax subgroups $A$ and $B$. Prove or refute the following statements:

a) $A_0B_0\ne 1$, where $A_0=\bigcap_{x\in G}A^x$ and similarly for $ B_0$;

b) the derived subgroup of $G$ is a minimax subgroup.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04603 | KOU-4.58

### Kourovka Notebook Problem 4.58

S / solved | Kourovka 群论问题集

Let a finite group $G$ be the product of two subgroups $A$ and $B$, where $A$ is abelian and $B$ is nilpotent. Find the dependence of the derived length of $G$ on the nilpotency class of $B$ and the order of its derived subgroup.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04604 | KOU-4.59

### Kourovka Notebook Problem 4.59

S / solved | Kourovka 群论问题集

(P. Hall). Find the smallest positive integer $n$ such that every countable group can be embedded in a simple group with $n$ generators.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04605 | KOU-4.6

### Kourovka Notebook Problem 4.6

O / open | Kourovka 群论问题集

(P. Hall). Are the projective groups in the variety of metabelian groups free?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04606 | KOU-4.60

### Kourovka Notebook Problem 4.60

S / solved | Kourovka 群论问题集

(P. Hall). What is the cardinality of the set of simple groups generated by two elements, one of order 2 and the other of order 3?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04607 | KOU-4.61

### Kourovka Notebook Problem 4.61

S / solved | Kourovka 群论问题集

Does there exist a linear function $f$ with the following property: if every abelian subgroup of a finite 2-group $G$ is generated by $n$ elements, then $G$ is generated by $f(n)$ elements?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04608 | KOU-4.62

### Kourovka Notebook Problem 4.62

S / solved | Kourovka 群论问题集

Does there exist a finitely based variety of groups whose universal theory is undecidable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04609 | KOU-4.63

### Kourovka Notebook Problem 4.63

S / solved | Kourovka 群论问题集

Does there exist a non-abelian variety of groups (in particular, one that contains the variety of all abelian groups) whose elementary theory is decidable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04610 | KOU-4.64

### Kourovka Notebook Problem 4.64

S / solved | Kourovka 群论问题集

Does there exist a variety of groups that does not admit an independent system of defining identities?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04611 | KOU-4.65

### Kourovka Notebook Problem 4.65

O / open | Kourovka 群论问题集

*Conjecture:* $\displaystyle{\frac{p^q-1}{p-1}}$ never divides $\displaystyle{\frac{q^p-1}{q-1}}$ if $p$, $q$ are distinct primes. The validity of this conjecture would simplify the proof of solvability of groups of odd order (W. Feit, J. G. Thompson, *Pacific J. Math.*, **13**, no. 3 (1963), 775--1029), rendering unnecessary the detailed use of generators and relations.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04612 | KOU-4.66

### Kourovka Notebook Problem 4.66

O / open | Kourovka 群论问题集

Let $P$ be a presentation of a finite group $G$ on $m_p$ generators and $r_p$ relations. The *deficiency* $\mathrm{def} (G)$ is the maximum of $m_p - r_p$ over all presentations $P$. Let $G$ be a finite group such that $G=G'\ne 1$ and the multiplicator $M(G)=1$. Prove that $\mathrm{def} (G^n) \rightarrow - \infty $ as $n \rightarrow \infty $, where $G^n$ is the $n$th direct power of $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04613 | KOU-4.67

### Kourovka Notebook Problem 4.67

S / solved | Kourovka 群论问题集

Let $G$ be a finite $p $-group. Show that the rank of the multiplicator $M(G)$ of $G$ is bounded in terms of the rank of $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04614 | KOU-4.68

### Kourovka Notebook Problem 4.68

S / solved | Kourovka 群论问题集

Construct a finitely generated (infinite) characteristically simple group that is not a direct power of a simple group.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04615 | KOU-4.69

### Kourovka Notebook Problem 4.69

S / solved | Kourovka 群论问题集

Let $G$ be a finite $p $-group, and suppose that $|G'| > p^{n(n-1)/2}$ for some non-negative integer $n$. Prove that $G$ is generated by the elements of breadths $\geq n$. The *breadth* of an element $x$ of $G$ is $b(x)$ where $|G:C_G(x)|=p^{b(x)}$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04616 | KOU-4.7

### Kourovka Notebook Problem 4.7

O / open | Kourovka 群论问题集

(Well-known problem). For which ring epimorphisms $R \rightarrow Q$ is the corresponding group homomorphism $SL_n(R) \rightarrow SL_n(Q)$ an epimorphism (for a fixed $n \geq 2$)? In particular, for what rings $R$ does the equality $SL_n(R)=E_n(R)$ hold?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04617 | KOU-4.70

### Kourovka Notebook Problem 4.70

S / solved | Kourovka 群论问题集

Let $k$ be a field of characteristic different from $2$, and $G_k$ the group of transformations $A = (a, \alpha) : x \rightarrow ax +\alpha$ \ ($a, \alpha \in k$, \ $a\ne 0$). Extend $G_k$ to the projective plane by adjoining the symbols $ (0, \alpha )$ and a line at infinity. Then the lines are just the centralizers $C_G(A)$ of elements $A\in G_k$ and their cosets. Do there exist other groups $G$ complementable to the projective plane such that the lines are just the cosets of the centralizers of elements of $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04618 | KOU-4.71

### Kourovka Notebook Problem 4.71

S / solved | Kourovka 群论问题集

Let $A$ be a group of automorphisms of a finite group $G$ which has a series of $A$-invariant subgroups $G = G_0 > \cdots > G_k = 1$ such that every $|G_i:G_{i+1}|$ is prime. Prove that $A$ is supersoluble.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04619 | KOU-4.72

### Kourovka Notebook Problem 4.72

O / open | Kourovka 群论问题集

Is it true that every variety of groups whose free groups are residually nilpotent torsion-free, is either soluble or coincides with the variety of all groups? For an affirmative answer, it is sufficient to show that every variety of Lie algebras over the field of rational numbers which does not contain any finite-dimensional simple algebras is soluble.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04620 | KOU-4.73

### Kourovka Notebook Problem 4.73

S / solved | Kourovka 群论问题集

(Well-known problem). Does there exist a non-abelian variety of groups

a) all of whose finite groups are abelian?

b) all of whose periodic groups are abelian?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04621 | KOU-4.74

### Kourovka Notebook Problem 4.74

P / partially_solved | Kourovka 群论问题集

b) Is every binary-finite 2-group of order greater than 2 non-simple?

[Archived solved part(s) of 4.74:] a) Is every 2-group of order greater than 2 non-simple?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04622 | KOU-4.75

### Kourovka Notebook Problem 4.75

S / solved | Kourovka 群论问题集

Let $G$ be a periodic group containing an involution $i$ and suppose that the Sylow 2-subgroups of $G$ are either locally cyclic or generalized quaternion. Does the element $ iO_{2'}(G)$ of the factor-group $G/O_{2'}(G)$ always lie in its centre?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04623 | KOU-4.76

### Kourovka Notebook Problem 4.76

S / solved | Kourovka 群论问题集

Let $G$ be a locally finite group containing an element $a$ of prime order such that the centralizer $C_G(a)$ is finite. Is $G$ almost soluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04624 | KOU-4.77

### Kourovka Notebook Problem 4.77

S / solved | Kourovka 群论问题集

In 1972, A. Rudvalis discovered a new simple group $R$ of order $2^{14}\cdot 3^3 \cdot 5^3 \cdot 7\cdot 13\cdot 29$. He has shown that $R$ possesses an involution $i$ such that $C_R(i)=V \times F$, where $V$ is a 4-group (an elementary abelian group of order $4$) and $ F\cong Sz(8)$.

a) Show that $R$ is the only finite simple group $G$ that possesses an involution $i$ such that $C_G(i)=V\times F$, where $V$ is a 4-group and $F\cong Sz(8)$.

b) Let $G$ be a non-abelian finite simple group that possesses an involution $i$ such that $C_G(i)=V\times F$, where $V$ is an elementary abelian 2-group of order $2^n$, $ n \geq 1$, and $F\cong Sz(2^m)$, $ m \geq 3$. Show that $ n = 2$ and $ m = 3$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04625 | KOU-4.8

### Kourovka Notebook Problem 4.8

S / solved | Kourovka 群论问题集

Suppose $G$ is a finitely generated free-by-cyclic group. Is $G$ finitely presented?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04626 | KOU-4.9

### Kourovka Notebook Problem 4.9

O / open | Kourovka 群论问题集

Let $G$ be a finitely generated torsion-free nilpotent group. Are there only a finite number of non-isomorphic groups in the sequence $\alpha G$, $\alpha ^2G, \ldots $ ? Here $\alpha G$ denotes the automorphism group of $G$ and $\alpha ^{n+1}G=\alpha (\alpha ^nG)$ for $n=1, 2,\ldots $

问题集入口：https://kourovkanotebookorg.wordpress.com/

04627 | KOU-5.1

### Kourovka Notebook Problem 5.1

P / partially_solved | Kourovka 群论问题集

b) Is every locally finite minimal non-$FC $-group distinct from its derived subgroup? \ The question has an affirmative answer for minimal non-$BFC $-groups.

[Archived solved part(s) of 5.1:] a) Is every locally finite minimal non-$FC $-group non-simple?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04628 | KOU-5.10

### Kourovka Notebook Problem 5.10

S / solved | Kourovka 群论问题集

Let $E=H*K$ and denote the augmentation ideals of the groups $E$, $H$, $K$ by $ {\mathfrak{e}}$, ${\mathfrak{h}}$, ${\mathfrak{k}}$, respectively. If $I$ is a right ideal in ${\mathbb{Z}E$, let $d_E(I)$ denote the minimum number of generators of $I$ as a right ideal. Assuming $H$ and $K$ finitely generated, is it true that $d_E({\mathfrak{e}})=d_E({\mathfrak{h}}E) + d_E({\mathfrak{k}}E)?$ Here ${\mathfrak{h}}E$ is, of course, the right ideal generated by $ {\mathfrak{h}}$; similarly for ${\mathfrak{k}}E$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04629 | KOU-5.11

### Kourovka Notebook Problem 5.11

S / solved | Kourovka 群论问题集

Let $G$ be a finite group and suppose that there exists a non-empty proper subset $\pi$ of the set of all primes dividing $|G|$ such that the centralizer of every non-trivial $\pi$-element is a $\pi$-subgroup. Does it follow that $G$ contains a subgroup $U$ such that $U^g\cap U=1$ or $U$ for every $g\in G$, and the centralizer of every non-trivial element of $U$ is contained in $U$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04630 | KOU-5.12

### Kourovka Notebook Problem 5.12

S / solved | Kourovka 群论问题集

Let $G$ be a finite group with trivial soluble radical in which there are Sylow 2-subgroups having non-trivial intersection. Suppose that, for any two Sylow 2-subgroups $P$ and $Q$ of $G$ with $P \cap Q \ne 1$, the index $|P:P \cap Q|$ does not exceed $2^n$. Is it then true that $|P| \leq 2^{2n}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04631 | KOU-5.13

### Kourovka Notebook Problem 5.13

S / solved | Kourovka 群论问题集

Suppose that $K$ and $L$ are distinct conjugacy classes of involutions in a finite group $G$ and $\left< x,y\right>$ is a $2$-group for all $x\in K$ and $y\in L$. Does it follow that $G\ne [K,L]$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04632 | KOU-5.14

### Kourovka Notebook Problem 5.14

O / open | Kourovka 群论问题集

An intersection of some Sylow 2-subgroups is called a *Sylow intersection* and an intersection of a pair of Sylow 2-subgroups is called a *paired Sylow intersection*.

a) Describe the finite groups all of whose 2-local subgroups have odd indices.

b) Describe the finite groups all of whose normalizers of Sylow intersections have odd indices.

c) Describe the finite groups all of whose normalizers of paired Sylow intersections have odd indices.

d) Describe the finite groups in which for any two Sylow 2-subgroups $P$ and $Q$, the intersection $P \cap Q$ is normal in some Sylow 2-subgroup of $\left< P, Q\right> $.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04633 | KOU-5.15

### Kourovka Notebook Problem 5.15

O / open | Kourovka 群论问题集

Do there exist finitely presented residually finite groups with recursive, but not primitive recursive, solution of the word problem?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04634 | KOU-5.16

### Kourovka Notebook Problem 5.16

O / open | Kourovka 群论问题集

Is every countable locally linear group embeddable in a finitely presented group? In (G. Baumslag, F. B. Cannonito, C. F. Miller III, *Math. Z.*, **153** (1977), 117--134) it is proved that every countable group which is locally linear of bounded degree can be embedded in a finitely presented group with solvable word problem.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04635 | KOU-5.17

### Kourovka Notebook Problem 5.17

S / solved | Kourovka 群论问题集

If the finite group $G$ has the form $G=AB$ where $A$ and $B$ are nilpotent of classes $\alpha $ and $\beta $, respectively, then $G$ is soluble. Is $G^{(\alpha +\beta )}=1$? One can show that $G^{(\alpha +\beta )}$ is nilpotent (E. Pennington, 1973). One can ask the same question for infinite groups (or Lie algebras), but there is nothing known beyond Ito's theorem: $A'=B'=1$ implies $G^{(2)}=1$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04636 | KOU-5.18

### Kourovka Notebook Problem 5.18

S / solved | Kourovka 群论问题集

Let $G$ be an infinite locally finite simple group. Is the centralizer of every element of $G$ infinite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04637 | KOU-5.19

### Kourovka Notebook Problem 5.19

S / solved | Kourovka 群论问题集

a) Let $G$ be an infinite locally finite simple group satisfying the minimum condition for 2-subgroups. Is $ G = PSL_2(F)$, $F$ some locally finite field of odd characteristic, if the centralizer of every involution of $G$ is almost locally soluble?

b) Can one characterize the simple locally finite groups with the min-2 condition containing a maximal radical non-trivial 2-subgroup of rank $\leq 2$ as linear groups of small rank?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04638 | KOU-5.2

### Kourovka Notebook Problem 5.2

S / solved | Kourovka 群论问题集

Is it true that the growth function $f$ of any infinite finitely generated group satisfies the inequality $f(n)\leq (f(n-1)+f(n+1))/2$ for all sufficiently large $n$ (for a fixed finite system of generators)?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04639 | KOU-5.20

### Kourovka Notebook Problem 5.20

S / solved | Kourovka 群论问题集

Is the elementary theory of lattices of $l $-ideals of lattice-ordered abelian groups decidable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04640 | KOU-5.21

### Kourovka Notebook Problem 5.21

S / solved | Kourovka 群论问题集

Can every torsion-free group with solvable word problem be embedded in a group with solvable conjugacy problem? An example due to A. Macintyre shows that this question has a negative answer when the condition of torsion-freeness is omitted.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04641 | KOU-5.22

### Kourovka Notebook Problem 5.22

S / solved | Kourovka 群论问题集

Does there exist a version of the Higman embedding theorem in which the degree of unsolvability of the conjugacy problem is preserved?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04642 | KOU-5.23

### Kourovka Notebook Problem 5.23

S / solved | Kourovka 群论问题集

Is it true that a free lattice-ordered group of the variety of lattice-ordered groups defined by the law $x^{-1}|y|x \ll |y|^2$ (or, equivalently, by the law $|[x,y]| \ll |x|$), is residually linearly ordered nilpotent?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04643 | KOU-5.24

### Kourovka Notebook Problem 5.24

S / solved | Kourovka 群论问题集

Is it true that a free lattice-ordered group of the variety of the lattice-ordered groups which are residually linearly ordered, is residually soluble linearly ordered?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04644 | KOU-5.25

### Kourovka Notebook Problem 5.25

O / open | Kourovka 群论问题集

Prove that the factor-group of any soluble linearly ordered group by its derived subgroup is non-periodic.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04645 | KOU-5.26

### Kourovka Notebook Problem 5.26

O / open | Kourovka 群论问题集

Let $G$ be a finite $p $-group with the minimal number of generators $d$, and let $r_1$ (respectively, $r_2$) be the minimal number of defining relations on $d$ generators in the sense of representing $G$ as a factor-group of a free discrete group (pro-$p $-group). It is well known that always $r_2 > d^2/4$. For each prime number $p$ denote by $c(p)$ the exact upper bound for the numbers $b(p)$ with the property that $r_2 \geq b(p)d^2$ for all finite $p $-groups.

a) It is obvious that $r_1 \geq r_2$. Find a $p $-group with $r_1 > r_2$.

b) *Conjecture:* $\lim\limits_{p\rightarrow \infty} c(p)=1/4$. It is proved (J. Wisliceny, *Math. Nachr.*, **102** (1981), 57--78) that $\lim\limits_{d\rightarrow \infty} r_2\left/ d^2\right. =1/4$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04646 | KOU-5.27

### Kourovka Notebook Problem 5.27

O / open | Kourovka 群论问题集

Prove that if $G$ is a torsion-free pro-$p $-group with a single defining relation, then $\mathrm{cd} G=2$. This has been proved for a large class of groups in (J. P. Labute, *Inv. Math.*, **4**, no. 2 (1967), 142--158).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04647 | KOU-5.28

### Kourovka Notebook Problem 5.28

S / solved | Kourovka 群论问题集

Let $G$ be a group and $H$ a torsion-free subgroup of $G$ such that the augmentation ideal $I_G$ of the integral group-ring $\mathbb{Z} G$ can be decomposed as $ I_G=I_H \mathbb{Z} G\oplus M$ for some $\mathbb{Z} G$-submodule $ M$. Prove that $G$ is a free product of the form $ G = H*K$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04648 | KOU-5.29

### Kourovka Notebook Problem 5.29

S / solved | Kourovka 群论问题集

Consider a group $G$ given by a presentation with $m$ generators and $n$ defining relations, where $m\geq n$. Do some $m-n$ of the given generators generate a free subgroup of $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04649 | KOU-5.3

### Kourovka Notebook Problem 5.3

S / solved | Kourovka 群论问题集

(Well-known problem). Can every finite lattice $L$ be embedded in the lattice of subgroups of a finite group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04650 | KOU-5.30

### Kourovka Notebook Problem 5.30

O / open | Kourovka 群论问题集

(Well-known problem). Suppose that $G$ is a finite soluble group, $A \leq \mathrm{Aut} G$, $C_G(A)=1$, the orders of $G$ and $A$ are coprime, and let $|A|$ be the product of $n$ not necessarily distinct prime numbers. Is the nilpotent length of $G$ bounded above by $n$? This is proved for large classes of groups (E. Shult, F. Gross, T. Berger, A. Turull), and there is a bound in terms of $n$ if $A$ is soluble (J. G. Thompson's $\leq 5^n$, H. Kurzweil's $\leq 4n$, A. Turull's $\leq 2n$), but the problem remains open.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04651 | KOU-5.32

### Kourovka Notebook Problem 5.32

S / solved | Kourovka 群论问题集

Let $p$ be a prime, $C$ a conjugacy class of $ p $-elements of a finite group $G$ and suppose that for any two elements $x$ and $y$ of $C$ the product $xy^{-1}$ is a $p $-element. Is the subgroup generated by the class $C$ a $p $-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04652 | KOU-5.33

### Kourovka Notebook Problem 5.33

O / open | Kourovka 群论问题集

(Y. Ihara). Consider the quaternion algebra $Q$ with norm $f=x^2 - \tau y^2 - \rho z^2 + \rho \tau u^2$, $\rho , \tau \in {\mathbb{Z}}$. Assume that $f$ is indefinite and of ${\mathbb{Q}}$-rank 0, i. e. $f=0$ for $x,y,z,u \in {\mathbb{Q}}$ implies $ x=y=z=u=0$. Consider $Q$ as the algebra of the matrices $$X=\left(\!\!\begin{array}{cc}x+\sqrt{\tau}y&\rho (z+\sqrt{\tau }u)\cr z-\sqrt{\tau}u&x-\sqrt{\tau}y\end{array}\!\!\right)$$ with $x,y,z,u \in {\mathbb{Q}}$. Let $p$ be a prime, $p\nmid \rho \tau$. Consider the group $G$ of all $X$ with $x,y,z,u \in {\mathbb{Z}^{(p)}}$, $\mathrm{det} X=1$, where ${\mathbb{Z}^{(p)}=\{ m/p^t\mid m,t \in {\mathbb{Z}}\} $.

*Conjecture:* $G$ has the congruence subgroup property, i. e. every non-central normal subgroup $N$ of $G$ contains a full congruence subgroup $N({\mathfrak{A}})=\{ X \in G\mid X \equiv E (\mathrm{mod} {\mathfrak{A}})\} $ for some ${\mathfrak{A}}$. Notice that the congruence subgroup property is independent of the matrix representation of $Q$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04653 | KOU-5.34

### Kourovka Notebook Problem 5.34

S / solved | Kourovka 群论问题集

Let ${\mathfrak{o}}$ be a commutative ring with identity in which 2 is invertible and which is not generated by zero divisors. Do there exist non--standard automorphisms of $GL_n({\mathfrak{o}})$ for $ n \geq 3$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04654 | KOU-5.35

### Kourovka Notebook Problem 5.35

O / open | Kourovka 群论问题集

Let $V$ be a vector space of dimension $n$ over a field. A subgroup $G$ of $GL_n(V)$ is said to be *rich in transvections* if $n \geq 2$ and for every hyperplane $H \subseteq V$ and every line $L \subseteq H$ there is at least one transvection in $G$ with residual line $ L$ and fixed space $H$. Describe the automorphisms of the subgroups of $GL_2(V)$ which are rich in transvections.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04655 | KOU-5.36

### Kourovka Notebook Problem 5.36

O / open | Kourovka 群论问题集

What profinite groups satisfy the maximum condition for closed subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04656 | KOU-5.38

### Kourovka Notebook Problem 5.38

O / open | Kourovka 群论问题集

Is it the case that if $A, B$ are finitely generated soluble Hopfian groups then $A \times B$ is Hopfian?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04657 | KOU-5.39

### Kourovka Notebook Problem 5.39

O / open | Kourovka 群论问题集

Prove that every countable group can act faithfully as a group of automorphisms of a finitely generated soluble group (of derived length at most 4). For background to this problem, in particular its relationship with problem 8.50, see (P. M. Neumann, *in: Groups--Korea, Pusan, 1988 $($Lect. Notes Math.*, **1398**), Springer, Berlin, 1989, 124--139).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04658 | KOU-5.4

### Kourovka Notebook Problem 5.4

S / solved | Kourovka 群论问题集

Let $g$ and $h$ be positive elements of a linearly ordered group $G$. Can one always embed $G$ in a linearly ordered group $ \overline{\! G}$ in such a way that $g$ and $h$ are conjugate in $ \overline{\! G}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04659 | KOU-5.40

### Kourovka Notebook Problem 5.40

S / solved | Kourovka 群论问题集

Let $G$ be a countable group acting on a set $\Omega$. Suppose that $ G$ is $k$-fold transitive for every finite $k$, and $G$ contains no non-trivial permutations of finite support. Is it true that $\Omega$ can be identified with the rational line $\mathbb{Q}$ in such a way that $G$ becomes a group of autohomeomorphisms?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04660 | KOU-5.41

### Kourovka Notebook Problem 5.41

S / solved | Kourovka 群论问题集

Does every non-trivial finite group, which is free in some variety, contain a non-trivial abelian normal subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04661 | KOU-5.42

### Kourovka Notebook Problem 5.42

O / open | Kourovka 群论问题集

Does the free group of rank 2 have an infinite ascending chain of verbal subgroups each being generated as a verbal subgroup by a single element?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04662 | KOU-5.43

### Kourovka Notebook Problem 5.43

S / solved | Kourovka 群论问题集

Does there exist a soluble variety of groups that is not generated by its finite groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04663 | KOU-5.44

### Kourovka Notebook Problem 5.44

O / open | Kourovka 群论问题集

A union ${\mathfrak{A}}=\bigcup\limits_{\alpha} {\mathfrak{V}}_{\alpha}$ of varieties of groups (in the lattice of varieties) is called *irreducible* if $\bigcup\limits_{\beta\ne \alpha} {\mathfrak{V}}_{\beta} \ne {\mathfrak{A}}$ for each index $\alpha $. Is every variety an irreducible union of (finitely or infinitely many) varieties each of which cannot be decomposed into a union of two proper subvarieties?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04664 | KOU-5.46

### Kourovka Notebook Problem 5.46

S / solved | Kourovka 群论问题集

Is every recursively presented soluble group embeddable in a group finitely presented in the variety of all soluble groups of derived length $n$, for suitable $n$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04665 | KOU-5.47

### Kourovka Notebook Problem 5.47

S / solved | Kourovka 群论问题集

Is every countable abelian group embeddable in the center of some finitely presented group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04666 | KOU-5.48

### Kourovka Notebook Problem 5.48

O / open | Kourovka 群论问题集

Suppose that $G$ and $H$ are finitely generated residually finite groups having the same set of finite homomorphic images. Are $G$ and $H$ isomorphic if one of them is a free (free soluble) group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04667 | KOU-5.49

### Kourovka Notebook Problem 5.49

S / solved | Kourovka 群论问题集

Let $A_{m,n}$ be the group of all automorphisms of the free soluble group of derived length $n$ and rank $m$. Is it true that

a) $ A_{m,n}$ is finitely generated for any $m$ and $n$?

b) every automorphism in $A_{m,n}$ is induced by some automorphism in $A_{m,n+1}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04668 | KOU-5.5

### Kourovka Notebook Problem 5.5

O / open | Kourovka 群论问题集

If $G$ is a finitely generated abelian-by-polycyclic-by-finite group, does there exist a finitely generated metabelian group $M$ such that $G$ is isomorphic to a subgroup of the automorphism group of $M$? If so, many of the tricky properties of $G$ like its residual finiteness would become transparent.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04669 | KOU-5.50

### Kourovka Notebook Problem 5.50

S / solved | Kourovka 群论问题集

Is there a finite group whose set of quasi-laws does not have an independent basis?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04670 | KOU-5.51

### Kourovka Notebook Problem 5.51

S / solved | Kourovka 群论问题集

Does a non-abelian free group have an independent basis for its quasi-laws?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04671 | KOU-5.52

### Kourovka Notebook Problem 5.52

S / solved | Kourovka 群论问题集

It is not hard to show that a finite perfect group is the normal closure of a single element. Is the same true for infinite finitely generated groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04672 | KOU-5.53

### Kourovka Notebook Problem 5.53

S / solved | Kourovka 群论问题集

(P. Scott). Let $p,q,r$ be distinct prime numbers. Prove that the free product $G=C_p*C_q*C_r$ of cyclics of orders $p,q,r$ is not the normal closure of a single element. By Lemma 3.1 of (J. Wiegold, *J. Austral. Math. Soc.*, **17**, no. 2 (1974), 133--141), every soluble image, and every finite image, of $G$ is the normal closure of a single element.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04673 | KOU-5.54

### Kourovka Notebook Problem 5.54

O / open | Kourovka 群论问题集

Let $p,q,r$ be distinct primes and $ u(x,y,z)$ a commutator word in three variables. Prove that there exist (infinitely many?) natural numbers $n$ such that the alternating group ${\mathbb{A}_n$ can be generated by three elements $\xi , \eta , \zeta$ satisfying $\xi ^p= \eta ^q= \zeta ^r=\xi \eta \zeta \cdot u(\xi , \eta , \zeta )=1$

问题集入口：https://kourovkanotebookorg.wordpress.com/

04674 | KOU-5.55

### Kourovka Notebook Problem 5.55

O / open | Kourovka 群论问题集

Find a finite $p $-group that cannot be embedded in a finite $p $-group with trivial multiplicator. Notice that every finite group can be embedded in a group with trivial multiplicator.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04675 | KOU-5.56

### Kourovka Notebook Problem 5.56

P / partially_solved | Kourovka 群论问题集

a) Let $p$ be a prime greater than 3. Is it true that every finite group of exponent $p$ can be embedded in the commutator subgroup of a finite group of exponent $p$?

[Archived solved part(s) of 5.56:] b) Does there exist a locally nilpotent group of prime exponent that coincides with its derived subgroup (and hence has no maximal subgroups)?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04676 | KOU-5.59

### Kourovka Notebook Problem 5.59

O / open | Kourovka 群论问题集

Let $G$ be a locally finite group which is the product of a $p $-subgroup and a $q $-subgroup, where $p$ and $q$ are distinct primes. Is $G$ a $\{ p,q\} $-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04677 | KOU-5.6

### Kourovka Notebook Problem 5.6

S / solved | Kourovka 群论问题集

If $R$ is a finitely generated integral domain of characteristic $ p > 0$, does the profinite (ideal) topology of $R$ induce the profinite topology on the group of units of $R$? It does for $ p = 0$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04678 | KOU-5.60

### Kourovka Notebook Problem 5.60

S / solved | Kourovka 群论问题集

Is an arbitrary soluble group that satisfies the minimum condition for normal subgroups countable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04679 | KOU-5.61

### Kourovka Notebook Problem 5.61

S / solved | Kourovka 群论问题集

(Well-known problem). Does an arbitrary uncountable locally finite group have only one end?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04680 | KOU-5.62

### Kourovka Notebook Problem 5.62

S / solved | Kourovka 群论问题集

Is a group locally finite if it contains infinite abelian subgroups and all of them are complementable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04681 | KOU-5.63

### Kourovka Notebook Problem 5.63

S / solved | Kourovka 群论问题集

Prove that a finite group is not simple if it contains two non-identity elements whose centralizers have coprime indices.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04682 | KOU-5.64

### Kourovka Notebook Problem 5.64

S / solved | Kourovka 群论问题集

Suppose that a finite group $G$ is the product of two subgroups $A_1$ and $A_2$. Prove that if $A_i$ contains a nilpotent subgroup of index $\leq 2$ for $ i = 1, 2$, then $G$ is soluble.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04683 | KOU-5.65

### Kourovka Notebook Problem 5.65

S / solved | Kourovka 群论问题集

Is the class of all finite groups that have Hall $\pi $-subgroups closed under taking finite subdirect products?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04684 | KOU-5.67

### Kourovka Notebook Problem 5.67

O / open | Kourovka 群论问题集

Is a periodic residually finite group finite if it satisfies the weak minimum condition for subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04685 | KOU-5.68

### Kourovka Notebook Problem 5.68

S / solved | Kourovka 群论问题集

Let $G$ be a finitely-presented group, and assume that $G$ has polynomial growth in the sense of Milnor. Show that $G$ has soluble word problem.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04686 | KOU-5.69

### Kourovka Notebook Problem 5.69

S / solved | Kourovka 群论问题集

Is every lattice isomorphism between torsion-free groups having no non-trivial cyclic normal subgroups induced by a group isomorphism?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04687 | KOU-5.7

### Kourovka Notebook Problem 5.7

S / solved | Kourovka 群论问题集

An algebraic variety $X$ over a field $k$ is called *rational* if the field of functions $k(X)$ is purely transcendental over $k$, and it is called *stably rational* if $k(X)$ becomes purely transcendental after adjoining finitely many independent variables. Let $T$ be a stably rational torus over a field $k$. Is $ T$ rational? Other formulations of this question and some related results see in (V. E. Voskresenskiĭ, *Russ. Math. Surveys*, **28**, no. 4 (1973), 79--105).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04688 | KOU-5.8

### Kourovka Notebook Problem 5.8

S / solved | Kourovka 群论问题集

Let $G$ be a torsion-free soluble group of finite cohomological dimension $\mathrm{cd} G$. If $hG$ denotes the Hirsch number of $G$, then it is known that $ hG\leq \mathrm{cd} G\leq hG+1$. Find a purely group-theoretic criterion for $hG=\mathrm{cd} G$.

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04689 | KOU-5.9

### Kourovka Notebook Problem 5.9

S / solved | Kourovka 群论问题集

Let $1\rightarrow R_i \stackrel{\pi _i}{\longrightarrow}F\rightarrow G\rightarrow 1$, \ $i=1, 2$, be two exact sequences of groups with $G$ finite and $F$ free of finite rank $d(F)$. If we assume that $d(F)=d(G)+1$ (where $d(G)$ is the minimum number of generators of $G$), are the corresponding abelianized extensions isomorphic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04690 | KOU-6.1

### Kourovka Notebook Problem 6.1

O / open | Kourovka 群论问题集

A subgroup $H$ of an arbitrary group $G$ is said to be *$ C $-closed* if $H=C^2(H)=C(C(H))$, and *weakly $C $-closed* if $C^2(x) \leq H$ for any element $x$ of $H$. The structure of the finite groups all of whose proper subgroups are $ C $-closed was studied by Gasch\"utz. Describe the structure of the locally finite groups all of whose proper subgroups are weakly $C $-closed.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04691 | KOU-6.10

### Kourovka Notebook Problem 6.10

O / open | Kourovka 群论问题集

Let $p$ be a prime and $n$ be an integer with $p > 2n + 1$. Let $x, y$ be $p $-elements in $GL_n({\mathbb{C}})$. If the subgroup $\left< x,y\right>$ is finite, then it is an abelian $p $-group (W. Feit, J. G. Thompson, *Pacif. J. Math.*, **11**, no. 4 (1961), 1257--1262). What can be said about $\left< x,y\right> $ in the case it is an infinite group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04692 | KOU-6.11

### Kourovka Notebook Problem 6.11

O / open | Kourovka 群论问题集

Let ${\mathfrak{L}}$ be the class of locally compact groups with no small subgroups (see D. Montgomery, L. Zippin, *Topological transformation groups*, New York, 1955; V. M. Glushkov, *Uspekhi Matem. Nauk*, **12**, no. 2 (1957), 3--41 (Russian)). Study extensions of groups in this class with the objective of giving a direct proof of the following: For each $G \in {\mathfrak{L}}$ there exists an $H \in {\mathfrak{L}}$ and a (continuous) homomorphism $\vartheta \!:G \rightarrow H$ with a discrete kernel and an image $\mathrm{Im} \vartheta$ satisfying $\mathrm{Im} \vartheta \cap Z(H)=1$ ($Z(H)$ is the center of $H$). This result follows from the Gleason--Montgomery--Zippin solution of Hilbert's 5th Problem (since ${\mathfrak{L}}$ is the class of finite-dimensional Lie groups and the latter are locally linear). On the other hand a direct proof of this result would give a substantially shorter proof of the 5th Problem since the adjoint representation of $H$ is faithful on $\mathrm{Im} \vartheta$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04693 | KOU-6.12

### Kourovka Notebook Problem 6.12

S / solved | Kourovka 群论问题集

a) Is a metabelian group countable if it satisfies the weak minimum condition for normal subgroups?

b) Is such a group minimax if it is torsion-free?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04694 | KOU-6.13

### Kourovka Notebook Problem 6.13

S / solved | Kourovka 群论问题集

Is it true that if a non-abelian Sylow 2-subgroup of a finite group $G$ has a non-trivial abelian direct factor, then $G$ is not simple?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04695 | KOU-6.14

### Kourovka Notebook Problem 6.14

S / solved | Kourovka 群论问题集

Are the following lattices locally finite: the lattice of all locally finite varieties of groups? the lattice of all varieties of groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04696 | KOU-6.15

### Kourovka Notebook Problem 6.15

S / solved | Kourovka 群论问题集

A variety is said to be *pro-locally-finite* if it is not locally finite while all of its proper subvarieties are locally finite. An example --- the variety of abelian groups. How many pro-locally-finite varieties of groups are there?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04697 | KOU-6.16

### Kourovka Notebook Problem 6.16

S / solved | Kourovka 群论问题集

A variety is called *sparse* if it has at most countably many subvarieties. How many sparse varieties of groups are there?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04698 | KOU-6.17

### Kourovka Notebook Problem 6.17

S / solved | Kourovka 群论问题集

Is every variety of groups generated by its finitely generated groups that have soluble word problem?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04699 | KOU-6.18

### Kourovka Notebook Problem 6.18

S / solved | Kourovka 群论问题集

(Well-known problem). Suppose that a class ${\mathfrak{K}}$ of 2-generator groups generates the variety of all groups. Is a non-cyclic free group residually in ${\mathfrak{K}}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04700 | KOU-6.19

### Kourovka Notebook Problem 6.19

S / solved | Kourovka 群论问题集

Let $R$ be a nilpotent associative ring. Are the following two statements for a subgroup $H$ of the adjoint group of $R$ always equivalent: \ 1) $H$ is a normal subgroup; \ 2) $H$ is an ideal of the groupoid $R$ with respect to Lie multiplication?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04701 | KOU-6.2

### Kourovka Notebook Problem 6.2

O / open | Kourovka 群论问题集

The totality of $C $-closed subgroups of an arbitrary group $G$ is a complete lattice with respect to the operations $\; A \wedge B=A \cap B,\; \; \; A\vee B=C(C(A) \cap C(B)).$ Describe the groups whose lattice of $C $-closed subgroups is a sublattice of the lattice of all subgroups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04702 | KOU-6.20

### Kourovka Notebook Problem 6.20

S / solved | Kourovka 群论问题集

Does there exist a supersoluble group of odd order, all of whose automorphisms are inner?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04703 | KOU-6.21

### Kourovka Notebook Problem 6.21

O / open | Kourovka 群论问题集

G. Higman proved that, for any prime number $p$, there exists a natural number $\chi (p)$ such that the nilpotency class of any finite group $G$ having an automorphism of order $p$ without non-trivial fixed points does not exceed $\chi (p)$. At the same time he showed that $\chi (p) \geq (p^2 - 1)/4$ for any such *Higman's function* $\chi$. Find the best possible Higman's function. Is it the function defined by equalities $\chi (p)=(p^2- 1)/4$ for $p > 2$ and $\chi (2)=1$? This is known to be true for $p\leq 7$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04704 | KOU-6.22

### Kourovka Notebook Problem 6.22

S / solved | Kourovka 群论问题集

Construct a braid that belongs to the derived subgroup of the braid group but is not a commutator.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04705 | KOU-6.23

### Kourovka Notebook Problem 6.23

S / solved | Kourovka 群论问题集

A braid $K$ of the braid group ${\mathfrak{B}}_{n+1}$ is said to be *smooth* if removing any of the threads in $K$ transforms $K$ into a braid that is equal to $1$ in ${\mathfrak{B}}_{n}$. It is known that smooth braids form a free subgroup. Describe generators of this subgroup.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04706 | KOU-6.24

### Kourovka Notebook Problem 6.24

O / open | Kourovka 群论问题集

The membership problem for the braid group on four strings. It is known (T. A. Makanina, *Math. Notes*, **29**, no. 1 (1981), 16--17) that the membership problem is undecidable for braid groups with more than four strings.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04707 | KOU-6.25

### Kourovka Notebook Problem 6.25

S / solved | Kourovka 群论问题集

(Well-known problem). Find an algorithm for calculating the rank of coefficient-free equations in a free group. The *rank* of an equation is the maximal rank of the free subgroup generated by a solution of this equation.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04708 | KOU-6.26

### Kourovka Notebook Problem 6.26

O / open | Kourovka 群论问题集

Let $D$ be a normal set of involutions in a finite group $G$ and let $\Gamma (D)$ be the graph with vertex set $D$ and edge set $\{ (a,b)\mid a,b \in D,\; ab=ba \ne 1\} $. Describe the finite groups $G$ with non-connected graph $\Gamma (D)$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04709 | KOU-6.28

### Kourovka Notebook Problem 6.28

O / open | Kourovka 群论问题集

Let $A$ be an elementary abelian 2-group which is a $ TI $-subgroup of a finite group $G$. Investigate the structure of $G$ under the hypothesis that the weak closure of $A$ in a Sylow 2-subgroup of $G$ is abelian.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04710 | KOU-6.29

### Kourovka Notebook Problem 6.29

O / open | Kourovka 群论问题集

Suppose that a finite group $A$ is isomorphic to the group of all topological automorphisms of a locally compact group $G$. Does there always exist a discrete group whose automorphism group is isomorphic to $A$? This is true if $A$ is cyclic; the condition of local compactness of $G$ is essential (R. J. Wille, *Indag. Math.*, **25**, no. 2 (1963), 218--224).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04711 | KOU-6.3

### Kourovka Notebook Problem 6.3

O / open | Kourovka 群论问题集

A group $G$ is called {\it type $(FP)_{\infty}$} if the trivial $G $-module ${\mathbb{Z}}$ has a resolution by finitely generated projective $G $-modules. The class of all groups of type $(FP)_{\infty}$ has a couple of excellent closure properties with respect to extensions and amalgamated products (R. Bieri, *Homological dimension of discrete groups*, Queen Mary College Math. Notes, London, 1976). Is every periodic group of type $(FP)_{\infty}$ finite? This is related to the question whether there is an infinite periodic group with a finite presentation.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04712 | KOU-6.30

### Kourovka Notebook Problem 6.30

S / solved | Kourovka 群论问题集

Let $G$ be a residually finite Hopfian group, and let $\widehat{G}$ be its profinite completion. Is $\widehat{G}$ necessarily Hopfian (in topological sense)?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04713 | KOU-6.31

## Kourovka Notebook Problem 6.31

P / partially_solved | Kourovka 群论问题集

b) Let $G$ be a finitely generated residually finite group, $d(G)$ the minimal number of its generators, and $\delta (G)$ the minimal number of topological generators of the profinite completion of $G$. It is known that there exist groups $G$ for which $d(G) > \delta (G)$ (G. A. Noskov, *Math. Notes*, **33**, no. 4 (1983), 249--254). Is the function $d$ bounded on the set of groups $G$ with the fixed value of $\delta (G) \geq 2$?

[Archived solved part(s) of 6.31:] a) Suppose that $G$ is a finitely-generated residually-finite group, $d(G)$ the minimal number of generators of $G$, and $\delta (G)$ the minimal number of topological generators of the profinite completion of $G$. Is $\delta (G)=d(G)$ always true?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04714 | KOU-6.32

## Kourovka Notebook Problem 6.32

O / open | Kourovka 群论问题集

Let $F_n$ be the free profinite group of finite rank $n > 1$. Is it true that for each normal subgroup $N$ of the free profinite group of countable rank, there exists a normal subgroup of $F_n$ isomorphic to $N$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04715 | KOU-6.34

## Kourovka Notebook Problem 6.34

S / solved | Kourovka 群论问题集

Let ${\mathfrak{o}}$ be an associative ring with identity. A system of its ideals ${\mathfrak{A}}=\{ {\mathfrak{A}}_{ij}\mid i,j \in {\mathbb{Z}}\}$ is called a *carpet of ideals* if $ {\mathfrak{A}}_{ik}{\mathfrak{A}}_{kj} \subseteq {\mathfrak{A}}_{ij}$ for all $i,j,k \in {\mathbb{Z}}$. If ${\mathfrak{o}}$ is commutative, then the set $ \Gamma_n(A)=\{ x \in SL_n({\mathfrak{o}})\mid x_{ij} \equiv \delta _{ij} (\mathrm{mod} {\mathfrak{A}}_{ij})\} $ is a group, the *$($special$)$ congruenz-subgroup modulo the carpet* ${\mathfrak{A}}$ (the *``carpet subgroup"*). Under quite general conditions, it was proved in (Yu. I. Merzlyakov, *Algebra i Logika*, **3**, no. 4 (1964), 49--59 (Russian); see also M. I. Kargapolov, Yu. I. Merzlyakov, *Fundamentals of the Theory of Groups*, 3rd Ed., Moscow, Nauka, 1982, p. 145 (Russian)) that in the groups $GL_n$ and $SL_n$ the mutual commutator subgroup of the congruenz-subgroups modulo a carpet of ideals shifted by $k$ and $l$ steps is again the congruenz-subgroup modulo the same carpet shifted by $k + l$ steps. Prove analogous theorems \ a) for orthogonal groups; b) for unitary groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04716 | KOU-6.35

## Kourovka Notebook Problem 6.35

S / solved | Kourovka 群论问题集

(R. Bieri, R. Strebel). Let ${\mathfrak{o}}$ be an associative ring with identity distinct from zero. A group $G$ is said to be *almost finitely presented over* ${\mathfrak{o}}$ if it has a presentation $G=F/R$ where $F$ is a finitely generated free group and the ${\mathfrak{o}}G $-module $R/[R,R] \otimes _{{\mathbb{Z}} {\mathfrak{o}}$ is finitely generated. It is easy to see that every finitely presented group $G$ is almost finitely presented over ${\mathbb{Z}}$ and therefore also over an arbitrary ring ${\mathfrak{o}}$. Is the converse true?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04717 | KOU-6.36

## Kourovka Notebook Problem 6.36

S / solved | Kourovka 群论问题集

(J. W. Grossman). The *nilpotent-completion diagram* of a group $G$ is as follows: $G/\gamma _1G\leftarrow G/\gamma _2G\leftarrow \cdots $, where $\gamma _iG$ is the $i$th term of the lower central series and the arrows are natural homomorphisms. It is easy to see that every nilpotent-completion diagram $G_1\leftarrow G_2 \leftarrow \cdots $ is a *$\gamma$-diagram*, that is, every sequence $1\rightarrow \gamma _sG_{s+1}\rightarrow G_{s+1}\rightarrow G_s\rightarrow 1$, \ $s=1, 2,\ldots$, is exact. Do $ \gamma$-diagrams exist that are not nilpotent-completion diagrams?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04718 | KOU-6.37

## Kourovka Notebook Problem 6.37

S / solved | Kourovka 群论问题集

(H. Wielandt, O. H. Kegel). Is a finite group $G$ soluble if it has soluble subgroups $A$, $B$, $C$ such that $G=AB=AC=BC$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04719 | KOU-6.38

### Kourovka Notebook Problem 6.38

P / partially_solved | Kourovka 群论问题集

b) Is it true that an arbitrary subgroup of $GL_n(k)$ that intersects every conjugacy class is parabolic? How far is the same statement true for subgroups of other groups of Lie type?

[Archived solved part(s) of 6.38:] a) Let $k$ be a (commutative) field. Find all irreducible subgroups $G$ of $GL_n(k)$ having the property that $G \cap C \ne \varnothing$ for every conjugacy class $C$ of $GL_n(k)$. I conjecture that $ G=GL_n(k)$ except in case $n=\mathrm{char} k=2$, the field $k$ is quadratically closed, and $G$ is conjugate to the group of all matrices of the form $\left(\!\!\!\begin{array}{cc}\alpha &0\cr 0&\beta\end{array}\!\!\!\right)$, $\left(\!\!\!\begin{array}{cc}0&\alpha \cr \beta &0\end{array}\!\!\!\right)$ where $\alpha \ne 0$ and $\beta \ne 0$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04720 | KOU-6.39

### Kourovka Notebook Problem 6.39

O / open | Kourovka 群论问题集

A class of groups ${\mathfrak{K}}$ is said to be *radical* if it is closed under taking homomorphic images and normal subgroups and if every group generated by its normal ${\mathfrak{K}} $-subgroups also belongs to ${\mathfrak{K}}$. The question proposed relates to the topic ``Radical classes and formulae of Narrow (first order) Predicate Calculus (NPC)''. One can show that the only non-trivial radical class definable by universal formulae of NPC is the class of all groups. Recently (in a letter to me), G. M. Bergman constructed a family of locally finite radical classes definable by formulae of NPC. Do there exist similar classes which are not locally finite and different from the class of all groups? In particular, does there exist a radical class of groups which is closed under taking Cartesian products, contains an infinite cyclic group, and is different from the class of all groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04721 | KOU-6.4

### Kourovka Notebook Problem 6.4

S / solved | Kourovka 群论问题集

A group $G$ is called {\it of type $(FP)_{\infty}$} if the trivial $G $-module ${\mathbb{Z}$ has a resolution by finitely generated projective $G $-modules. Is it true that every torsion-free group of type $(FP)_{\infty}$ has finite cohomological dimension?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04722 | KOU-6.42

### Kourovka Notebook Problem 6.42

S / solved | Kourovka 群论问题集

Let $H$ be a strongly 3-embedded subgroup of a finite group $G$. Suppose that $Z(H/O_{3'}(H))$ contains an element of order 3. Does $Z(G/O_{3'}(G))$ necessarily contain an element of order 3?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04723 | KOU-6.43

### Kourovka Notebook Problem 6.43

S / solved | Kourovka 群论问题集

Does the set of quasi-identities holding in the class of all finite groups possess a basis in finitely many variables?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04724 | KOU-6.44

### Kourovka Notebook Problem 6.44

S / solved | Kourovka 群论问题集

Construct a finitely generated infinite simple group requiring more than two generators.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04725 | KOU-6.45

### Kourovka Notebook Problem 6.45

S / solved | Kourovka 群论问题集

Construct a characteristic subgroup $N$ of a finitely generated free group $F$ such that $F/N$ is infinite and simple. If no such exists, it would follow that $d(S^2)=d(S)$ for every infinite finitely generated simple group $S$. There is reason to believe that this is false.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04726 | KOU-6.46

### Kourovka Notebook Problem 6.46

S / solved | Kourovka 群论问题集

If $G$ is $d $-generator group having no non-trivial finite homomorphic images (in particular, if $G$ is an infinite simple $d $-generator group) for some integer $d \geq 2$, must $G \times G$ be a $d $-generator group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04727 | KOU-6.47

### Kourovka Notebook Problem 6.47

O / open | Kourovka 群论问题集

(C. D. H. Cooper). Let $G$ be a group, $v$ a group word in two variables such that the operation $x \odot y=v(x,y)$ defines the structure of a new group $G_v=\left< G,\odot \right>$ on the set $G$. Does $G_v$ always lie in the variety generated by $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04728 | KOU-6.48

### Kourovka Notebook Problem 6.48

O / open | Kourovka 群论问题集

Is every infinite binary finite $p $-group non-simple? Here $p$ is a prime number.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04729 | KOU-6.49

### Kourovka Notebook Problem 6.49

S / solved | Kourovka 群论问题集

Is the minimal condition for abelian normal subgroups inherited by subgroups of finite index? This is true for the minimal condition for (all) abelian subgroups (J. S. Wilson, *Math. Z.*, **114** (1970), 19--21).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04730 | KOU-6.5

### Kourovka Notebook Problem 6.5

O / open | Kourovka 群论问题集

Is it the case that every soluble group $G$ of type $(FP)_{\infty}$ is *constructible* in the sense of (G. Baumslag, R. Bieri, *Math. Z.*, **151**, no. 3 (1976), 249--257)? In other words, can $G$ be built up from the trivial group in terms of finite extensions and $HNN $-extensions?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04731 | KOU-6.51

### Kourovka Notebook Problem 6.51

O / open | Kourovka 群论问题集

Let ${\mathfrak{F}}$ be a local subformation of some formation ${\mathfrak{X}}$ of finite groups and let $\Omega$ be the set of all maximal homogeneous ${\mathfrak{X}} $-screens of ${\mathfrak{F}}$. Find a way of constructing the elements of $\Omega$ with the help of the maximal inner local screen of ${\mathfrak{F}}$. What can be said about the cardinality of $\Omega$? For definitions see (L. A. Shemetkov, *Formations of finite groups*, Moscow, Nauka, 1978 (Russian)).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04732 | KOU-6.52

### Kourovka Notebook Problem 6.52

S / solved | Kourovka 群论问题集

Let $f$ be a local screen of a formation which contains all finite nilpotent groups and let $A$ be a group of automorphisms of a finite group $G$. Suppose that $A$ acts $ f $-stably on the socle of $G/\Phi (G)$. Is it true that $A$ acts $f $-stably on $\Phi (G)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04733 | KOU-6.53

### Kourovka Notebook Problem 6.53

S / solved | Kourovka 群论问题集

A group $G$ of the form $G=F\leftthreetimes H$ is said to be a *Frobenius group with kernel $F$ and complement* $H$ if $H \cap H^g=1$ for any $g \in G\setminus H$ and $F\setminus \{ 1\} =G\setminus \bigcup\limits_{g\in G} H^g$. What can be said about the kernel and the complement of a Frobenius group? In particular, which groups can be kernels? complements?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04734 | KOU-6.54

### Kourovka Notebook Problem 6.54

S / solved | Kourovka 群论问题集

Are there infinite finitely generated Frobenius groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04735 | KOU-6.55

### Kourovka Notebook Problem 6.55

O / open | Kourovka 群论问题集

A group $G$ of the form $G=F\leftthreetimes H$ is said to be a *Frobenius group with kernel $F$ and complement* $H$ if $H \cap H^g=1$ for any $g \in G\setminus H$ and $F\setminus \{ 1\} =G\setminus \bigcup\limits_{g\in G} H^g$. Do there exist Frobenius $p $-groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04736 | KOU-6.56

### Kourovka Notebook Problem 6.56

O / open | Kourovka 群论问题集

Let $G=F\cdot \left< a\right>$ be a Frobenius group with the complement $\left< a\right> $ of prime order.

a) Is $G$ locally finite if it is binary finite?

b) Is the kernel $F$ locally finite if the groups $\left< a, a^g\right> $ are finite for all $g \in G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04737 | KOU-6.57

### Kourovka Notebook Problem 6.57

S / solved | Kourovka 群论问题集

A group $G$ is said to be (*conjugacy, $p $-conjugacy*) *biprimitively finite* if, for any finite subgroup $H$, any two elements of prime order (any two conjugate elements of prime order, of prime order $p$) in $N_G(H)/H$ generate a finite subgroup. Do the elements of finite order in a (conjugacy) biprimitively finite group $G$ form a subgroup (the periodic part of $G$)?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04738 | KOU-6.58

### Kourovka Notebook Problem 6.58

S / solved | Kourovka 群论问题集

Are

a) the Al\"eshin $p $-groups and

b) the 2-generator Golod $p $-groups

conjugacy biprimitively finite groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04739 | KOU-6.59

### Kourovka Notebook Problem 6.59

O / open | Kourovka 群论问题集

A group $G$ is said to be (*conjugacy, $p$-conjugacy*) *biprimitively finite* if, for any finite subgroup $H$, any two elements of prime order (any two conjugate elements of prime order, of prime order $p$) in $N_G(H)/H$ generate a finite subgroup. Prove that an arbitrary periodic (conjugacy) biprimitively finite group (in particular, having no involutions) of finite rank is locally finite.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04740 | KOU-6.6

### Kourovka Notebook Problem 6.6

S / solved | Kourovka 群论问题集

Let $P, Q$ be permutation representations of a finite group $G$ with the same character. Suppose $P(G)$ is a primitive permutation group. Is $Q(G)$ necessarily primitive? The answer is known to be affirmative if $G$ is soluble.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04741 | KOU-6.60

### Kourovka Notebook Problem 6.60

O / open | Kourovka 群论问题集

Do there exist infinite finitely generated simple periodic (conjugacy) biprimitively finite groups which contain both involutions and non-trivial elements of odd order?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04742 | KOU-6.61

### Kourovka Notebook Problem 6.61

O / open | Kourovka 群论问题集

Is every infinite periodic (conjugacy) biprimitively finite group without involutions non-simple?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04743 | KOU-6.62

### Kourovka Notebook Problem 6.62

O / open | Kourovka 群论问题集

Is a (conjugacy, $p $-conjugacy) biprimitively finite group finite if it has a finite maximal subgroup ($p $-subgroup)? There are affirmative answers for conjugacy biprimitively finite $p $-groups and for 2-conjugacy biprimitively finite groups (V. P. Shunkov, *Algebra and Logic*, **9**, no. 4 (1970), 291--297; **11**, no. 4 (1972), 260--272; **12**, no. 5 (1973), 347--353), while the statement does not hold for arbitrary periodic groups, see Archive, 3.9.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04744 | KOU-6.63

### Kourovka Notebook Problem 6.63

S / solved | Kourovka 群论问题集

An infinite group $G$ is called a *monster of the first kind* if it has elements of order $> 2$ and for any such an element $a$ and for any proper subgroup $H$ of $G$, there is an element $g$ in $G\setminus H$, such that $\left< a, a^g\right> =G$. Classify the monsters of the first kind all of whose proper subgroups are finite.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04745 | KOU-6.64

### Kourovka Notebook Problem 6.64

S / solved | Kourovka 群论问题集

A group $G$ is called a *monster of the second kind* if it has elements of order $> 2$ and if for any such element $a$ and any proper subgroup $H$ of $G$ there exists an infinite subset ${\mathfrak{M}}_{a,H}$ consisting of conjugates of $a$ by elements of $G\setminus H$ such that $ \left< a,c\right> =G$ for all $ c\in {\mathfrak{M}}_{a,H}$. Do mixed monsters (that is, with elements of both finite and infinite orders) of the second kind exist? Do there exist torsion-free monsters of the second kind?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04746 | KOU-6.7

### Kourovka Notebook Problem 6.7

S / solved | Kourovka 群论问题集

Suppose that $P$ is a finite 2-group. Does there exist a characteristic subgroup $L(P)$ of $P$ such that $L(P)$ is normal in $H$ for every finite group $H$ that satisfies the following conditions: 1) $P$ is a Sylow 2-subgroup of $H$, 2) $H$ is ${\mathbb{S}}_4 $-free, and 3) $C_H(O_2(H)) \leq O_2(H)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04747 | KOU-6.8

### Kourovka Notebook Problem 6.8

S / solved | Kourovka 群论问题集

Can a residually finite locally normal group be embedded in a Cartesian product of finite groups in such a way that each element of the group has at most finitely many central projections?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04748 | KOU-6.9

### Kourovka Notebook Problem 6.9

S / solved | Kourovka 群论问题集

Is the derived subgroup of any locally normal group decomposable into a product of at most countable subgroups which commute elementwise?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04749 | KOU-7.1

### Kourovka Notebook Problem 7.1

S / solved | Kourovka 群论问题集

The free periodic groups $B(m,p)$ of prime exponent $p > 665$ are known to possess many properties similar to those of absolutely free groups (see S. I. Adian, *The Burnside Problem and Identities in Groups*, Springer, Berlin, 1979). Is it true that all normal subgroups of $B(m,p)$ are not free periodic groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04750 | KOU-7.12

### Kourovka Notebook Problem 7.12

S / solved | Kourovka 群论问题集

Find all groups with a Hall $2'$-subgroup.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04751 | KOU-7.15

### Kourovka Notebook Problem 7.15

S / solved | Kourovka 群论问题集

Prove that if $F^{\displaystyle{*}} (G)$ is quasisimple and $\alpha \in \mathrm{Aut} G$, $|\alpha |=2$, then $C_G(\alpha )$ contains an involution outside $Z(F^{\displaystyle{*}}(G))$, except when $F^{\displaystyle{*}}(G)$ has quaternion Sylow 2-subgroups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04752 | KOU-7.16

### Kourovka Notebook Problem 7.16

S / solved | Kourovka 群论问题集

If $H$ is a proper subgroup of the finite group $G$, is there always an element of prime-power order not conjugate to an element of $H$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04753 | KOU-7.17

### Kourovka Notebook Problem 7.17

S / solved | Kourovka 群论问题集

Is the number of maximal subgroups of the finite group $G$ at most $|G| - 1$?

*Editors' remarks:* This is proved for $G$ soluble (G. E. Wall, *J. Austral. Math. Soc.*, **2** (1961--62), 35--59) and for symmetric groups ${\mathbb{S} _n$ for sufficiently large $n$ (M. Liebeck, A. Shalev, *J. Combin. Theory, Ser. A*, **75** (1996), 341--352); it is also proved (M. W. Liebeck, L. Pyber, A. Shalev, *J. Algebra*, **317** (2007), 184--197) that any finite group $G$ has at most $2C|G|^{3/2}$ maximal subgroups, where $C$ is an absolute constant.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04754 | KOU-7.19

### Kourovka Notebook Problem 7.19

O / open | Kourovka 群论问题集

Construct an explicit example of a finitely presented simple group with word problem not solvable by a primitive recursive function.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04755 | KOU-7.2

### Kourovka Notebook Problem 7.2

S / solved | Kourovka 群论问题集

Prove that the free periodic groups $B(m, n)$ of odd exponent $n\geq 665$ with $ m \geq 2$ generators are non-amenable and that random walks on these groups do not have the recurrence property.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04756 | KOU-7.21

### Kourovka Notebook Problem 7.21

O / open | Kourovka 群论问题集

Is there an algorithm which decides if an arbitrary finitely presented solvable group is metabelian?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04757 | KOU-7.22

### Kourovka Notebook Problem 7.22

S / solved | Kourovka 群论问题集

Suppose that a finite group $G$ is realized as the automorphism group of some torsion-free abelian group. Is it true that for every infinite cardinal $\mathfrak{m}$ there exist $2^{\mathfrak{m}}$ non-isomorphic torsion-free abelian groups of cardinality $\mathfrak{m}$ whose automorphism groups are isomorphic to $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04758 | KOU-7.23

### Kourovka Notebook Problem 7.23

O / open | Kourovka 群论问题集

Does there exist an algorithm which decides, for a given list of group identities, whether the elementary theory of the variety given by this list is soluble, that is, by (A. P. Zamyatin, *Algebra and Logic*, **17**, no. 1 (1978), 13--17), whether this variety is abelian?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04759 | KOU-7.24

### Kourovka Notebook Problem 7.24

S / solved | Kourovka 群论问题集

We say that a group is *sparse* if the variety generated by it has at most countably many subvarieties. Does there exist a finitely generated sparse group that has undecidable word problem?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04760 | KOU-7.25

### Kourovka Notebook Problem 7.25

O / open | Kourovka 群论问题集

We associate with words in the alphabet $x$, $x^{-1}$, $y$, $y^{-1}$, $z$, $z^{-1},\ldots $ operations which are understood as functions in variables $x, y, z,\ldots $. We say that a word $A$ *is expressible in words $B_1,\ldots ,B_n$ on the group $G$* if $A$ can be constructed from the words $B_1,\ldots ,B_n$ and variables by means of finitely many substitutions of words one into another and replacements of a word by another word which is identically equal to it on $G$. A list of words is said to be *functionally complete on $G$* if every word can be expressed on $G$ in words from this list. A word $B$ is called a *Schaeffer word on $G$* if every word can be expressed on $G$ in $B$ (compare with A. V. Kuznetsov, *Matem. Issledovaniya*, Kishin\"ev, **6**, no. 4 (1971), 75--122 (Russian)). Does there exist an algorithm which decides

a) by a word $B$, whether it is a Schaeffer word on every group? Compare with the problem of describing all such words in (A. G. Kurosh, *Theory of Groups*, Moscow, Nauka, 1967, p. 435 (Russian)); here are examples of such words: $xy^{-1}$, $x^{-1}y^2z$, $x^{-1}y^{-1}zx$.

b) by a list of words, whether it is functionally complete on every group?

c) by words $A, B_1,\ldots ,B_n$, whether $A$ is expressible in $ B_1,\ldots ,B_n$ on every group? (This is a problem from A. V. Kuznetsov, *ibid.*, p. 112.)

d) the same for every finite group? (For finite groups, for example, $x^{-1}$ is expressible in $xy$.) For a fixed finite group an algorithm exists (compare with A. V. Kuznetsov, *ibid.*, $\S$ 8).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04761 | KOU-7.26

### Kourovka Notebook Problem 7.26

O / open | Kourovka 群论问题集

The *ordinal height* of a variety of groups is, by definition, the supremum of order types (ordinals) of all well-ordered by inclusion chains of its proper subvarieties. It is clear that its cardinality is either finite or countably infinite, or equal to $\omega _1$. Is every countable ordinal number the ordinal height of some variety?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04762 | KOU-7.27

### Kourovka Notebook Problem 7.27

O / open | Kourovka 群论问题集

Is it true that the group $SL_n(q)$ contains, for sufficiently large $q$, a diagonal matrix which is not contained in any proper irreducible subgroup of $SL_n(q)$ with the exception of block-monomial ones? For $n=2, 3$ the answer is known to be affirmative (V. M. Levchuk, *in: Some problems of the theory of groups and rings*, Inst. of Physics SO AN SSSR, Krasnoyarsk, 1973 (Russian)). Similar problems are interesting for other Chevalley groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04763 | KOU-7.28

### Kourovka Notebook Problem 7.28

O / open | Kourovka 群论问题集

Let $G(K)$ be the Chevalley group over a commutative ring $K$ associated with the root system $\Phi$ as defined in (R. Steinberg, *Lectures on Chevalley groups*, Yale Univ., New Haven, Conn., 1967). This group is generated by the root subgroups $x_r(K)$, $r \in \Phi $. We define an *elementary carpet of type $\Phi$ over $K$* to be any collection of additive subgroups $\{ {\mathfrak{A}}_r\mid r \in \Phi \}$ of $K$ satisfying the condition $$c_{ij,rs}{\mathfrak{A}}^i_r{\mathfrak{A}}^j_s \subseteq {\mathfrak{A}}_{ir+js} \quad \quad \mbox{ for}\; r,s,ir + js \in \Phi , \;\; i > 0,\;\; j > 0,$$ where $c_{ij,rs}$ are constants defined by the Chevalley commutator formula and ${\mathfrak{A}}_r^i=\{ a^i\mid a \in {\mathfrak{A}}_r\} $. What are necessary and sufficient conditions (in terms of the ${\mathfrak{A}}_r$) on the elementary carpet to ensure that the subgroup $\left< x_r({\mathfrak{A}}_r)\mid r \in \Phi \right>$ of $G(K)$ intersects with $x_r(K)$ in $x_r({\mathfrak{A}}_r)$? See also 15.46.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04764 | KOU-7.3

### Kourovka Notebook Problem 7.3

O / open | Kourovka 群论问题集

Prove that the periodic product of odd exponent $n \geq 665$ of non-trivial groups $F_1,\ldots ,F_k$ which do not contain involutions cannot be generated by less than $k$ elements. This would imply, on the basis of (S. I. Adian, *Sov. Math. Doklady*, **19**, (1978), 910--913), the existence of $ k $-generated simple groups which cannot be generated by less than $k$ elements, for any $k > 0$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04765 | KOU-7.30

### Kourovka Notebook Problem 7.30

S / solved | Kourovka 群论问题集

Which finite simple groups can be generated by three involutions, two of which commute?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04766 | KOU-7.31

### Kourovka Notebook Problem 7.31

S / solved | Kourovka 群论问题集

Let $A$ be a group of automorphisms of a finite non-abelian 2-group $G$ acting transitively on the set of involutions of $G$. Is $A$ necessarily soluble?

Such groups $G$ were divided into several classes in (F. Gross, *J. Algebra*, **40**, no. 2 (1976), 348--353). For one of these classes a positive answer was given by E. G. Bryukhanova (*Algebra and Logic*, **20**, no. 1 (1981), 1--12).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04767 | KOU-7.33

### Kourovka Notebook Problem 7.33

O / open | Kourovka 群论问题集

An elementary $TI $-subgroup $V$ of a finite group $G$ is said to be a *subgroup of non-root type* if $1 \ne N_V(V^g) \ne V$ for some $g \in G$. Describe the finite groups $G$ which contain a 2-subgroup $V$ of non-root type such that $[V, V^g]=1$ implies that all involutions in $VV^g$ are conjugate to elements of $V$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04768 | KOU-7.34

### Kourovka Notebook Problem 7.34

O / open | Kourovka 群论问题集

In many of the sporadic finite simple groups the 2-ranks of the centralizers of 3-elements are at most 2. Describe the finite groups satisfying this condition.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04769 | KOU-7.35

### Kourovka Notebook Problem 7.35

O / open | Kourovka 群论问题集

What varieties of groups ${\mathfrak{V}}$ have the following property: the group $G/{\mathfrak{V}}(G)$ is residually finite for any residually finite group $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04770 | KOU-7.36

### Kourovka Notebook Problem 7.36

S / solved | Kourovka 群论问题集

Is it true that every residually finite group in which every subgroup of finite index (including the group itself) is defined by a single defining relation is either free or isomorphic to the fundamental group of a compact surface?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04771 | KOU-7.37

### Kourovka Notebook Problem 7.37

S / solved | Kourovka 群论问题集

We say that a profinite group is *strictly complete* if each of its subgroups of finite index is open. It is known (B. Hartley, *Math. Z.*, **168**, no. 1 (1979), 71--76) that finitely generated profinite groups having a finite series with pronilpotent factors are strictly complete. Is a profinite group strictly complete if it is \ \

a) finitely generated? \

b) finitely generated and prosoluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04772 | KOU-7.38

### Kourovka Notebook Problem 7.38

P / partially_solved | Kourovka 群论问题集

A *variety of profinite groups* is a non-empty class of profinite groups closed under taking subgroups, factor-groups, and Tikhonov products. A subvariety ${\mathfrak{V}}$ of the variety ${\mathfrak{N}}$ of all pronilpotent groups is said to be *$($locally$)$ nilpotent* if all (finitely generated) groups in ${\mathfrak{V}}$ are nilpotent.

a) Is it true that any non-nilpotent subvariety of ${\mathfrak{N}}$ contains a non-nilpotent locally nilpotent subvariety?

b) The same problem for the variety of all pro-$p $-groups (for a given prime $p$).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04773 | KOU-7.39

### Kourovka Notebook Problem 7.39

O / open | Kourovka 群论问题集

Let $G=\left< a, b\mid a^p=(ab)^3=b^2=(a^{\sigma}ba^{2/\sigma }b)^2=1\right> ,$ where $p$ is a prime, $\sigma$ is an integer not divisible by $p$. The group $PSL_2(p)$ is a factor-group of $G$ so that there is a short exact sequence $1 \rightarrow N \rightarrow G \rightarrow PSL_2(p) \rightarrow 1 .$ For each $p > 2$ there is $\sigma$ such that $N=1$, for example, $\sigma =4$. Let $N^{ab}$ denote the factor-group of $N$ by its commutator subgroup. It is known that for some $p$ there is $\sigma$ such that $N^{ab}$ is infinite (for example, for $p=41$ one can take $\sigma ^2 \equiv 2 (\mathrm{mod} 41)$), whereas for some other $p$ (for example, for $p=43$) the group $N^{ab}$ is finite for every $\sigma$.

a) Is the set of primes $p$ for which $N^{ab}$ is finite for every $\sigma$ infinite?

b) Is there an arithmetic condition on $\sigma$ which ensures that $N^{ab}$ is finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04774 | KOU-7.4

### Kourovka Notebook Problem 7.4

S / solved | Kourovka 群论问题集

Is a finitely generated group with quadratic growth almost abelian?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04775 | KOU-7.40

### Kourovka Notebook Problem 7.40

S / solved | Kourovka 群论问题集

Describe the (lattice of) subgroups of a given classical matrix group over a ring which contain the subgroup consisting of all matrices in that group with coefficients in some subring (see Ju. I. Merzljakov, *J. Soviet Math.*, **1** (1973), 571--593).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04776 | KOU-7.41

### Kourovka Notebook Problem 7.41

O / open | Kourovka 群论问题集

(John S. Wilson). Is every linear $ \overline{\! Sl\vphantom{(}} $-group an $ \overline{\! SN\vphantom{(}} $-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04777 | KOU-7.42

### Kourovka Notebook Problem 7.42

S / solved | Kourovka 群论问题集

A group $U$ is called an *$F_q $-group* (where $q \in \pi (U)$) if, for each finite subgroup $K$ of $U$ and for any two elements $a,b$ of order $q$ in $T=N_U(K)/K$, there exists $c \in T$ such that the group $\left< a, b^c\right>$ is finite. A group $U$ is called an {\it $F ^{\displaystyle{*}} $-group} if each subgroup $H$ of $U$ is an $F_q $-group for every $q \in \pi (H)$ (V. P. Shunkov, 1977).

a) Is every primary $F ^{\displaystyle{*}} $-group satisfying the minimum condition for subgroups almost abelian?

b) Does every $F ^{\displaystyle{*}} $-group satisfying the minimum condition for (abelian) subgroups possess the radicable part?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04778 | KOU-7.44

### Kourovka Notebook Problem 7.44

S / solved | Kourovka 群论问题集

Does a normal subgroup $H$ in a finite group $G$ possess a complement in $G$ if each Sylow subgroups of $H$ is a direct factor in some Sylow subgroup of $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04779 | KOU-7.45

### Kourovka Notebook Problem 7.45

O / open | Kourovka 群论问题集

Does the $Q $-theory of the class of all finite groups (in the sense of 2.40) coincide with the $Q $-theory of a single finitely presented group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04780 | KOU-7.48

### Kourovka Notebook Problem 7.48

S / solved | Kourovka 群论问题集

(Well-known problem). Suppose that, in a finite group $G$, each two elements of the same order are conjugate. Is then $|G| \leq 6$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04781 | KOU-7.49

### Kourovka Notebook Problem 7.49

O / open | Kourovka 群论问题集

Let $G$ be a finitely generated group, and $N$ a minimal normal subgroup of $G$ which is an elementary abelian $p $-group. Is it true that either $N$ is finite or the growth function for the elements of $N$ with respect to a finite generating set of $G$ is bounded below by an exponential function?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04782 | KOU-7.5

### Kourovka Notebook Problem 7.5

O / open | Kourovka 群论问题集

We say that a group is *indecomposable* if any two of its proper subgroups generate a proper subgroup. Describe the indecomposable periodic metabelian groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04783 | KOU-7.50

### Kourovka Notebook Problem 7.50

O / open | Kourovka 群论问题集

Study the structure of the primitive permutation groups (finite and infinite), in which the stabilizer of any three pairwise distinct points is trivial. This problem is closely connected with the problem of describing those group which have a Frobenius group as one of maximal subgroups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04784 | KOU-7.51

### Kourovka Notebook Problem 7.51

O / open | Kourovka 群论问题集

What are the primitive permutation groups (finite and infinite) which have a regular sub-orbit, that is, in which a point stabilizer acts faithfully and regularly on at least one of its orbits?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04785 | KOU-7.52

### Kourovka Notebook Problem 7.52

O / open | Kourovka 群论问题集

Describe the locally finite primitive permutation groups in which the centre of any Sylow 2-subgroup contains involutions stabilizing precisely one symbol. The case of finite groups was completely determined by D. Holt in 1978.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04786 | KOU-7.53

### Kourovka Notebook Problem 7.53

S / solved | Kourovka 群论问题集

Let $p$ be a prime. The law $ x\cdot x^{\varphi}\cdots x^{\varphi^{p-1}}=1$ from the definition of a splitting automorphism (see Archive, 1.10) gives rise to a variety of groups with operators $\left< \varphi\right>$ consisting of all groups that admit a splitting automorphism of order $p$. Does the analogue of Kostrikin's theorem hold for this variety, that is, do the locally nilpotent groups in this variety form a subvariety?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04787 | KOU-7.54

### Kourovka Notebook Problem 7.54

O / open | Kourovka 群论问题集

Does there exist a group of infinite special rank which can be represented as a product of two subgroups of finite special rank?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04788 | KOU-7.55

### Kourovka Notebook Problem 7.55

O / open | Kourovka 群论问题集

Is it true that a group which is a product of two almost abelian subgroups is almost soluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04789 | KOU-7.56

### Kourovka Notebook Problem 7.56

O / open | Kourovka 群论问题集

(B. Amberg). Does a group satisfy the minimum (respectively, maximum) condition on subgroups if it is a product of two subgroups satisfying the minimum (respectively, maximum) condition on subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04790 | KOU-7.57

### Kourovka Notebook Problem 7.57

P / partially_solved | Kourovka 群论问题集

a) A set of generators of a finitely generated group $G$ consisting of the least possible number $d(G)$ of elements is called a *basis of $G$*. Let $r_M(G)$ be the least number of relations necessary to define $G$ in the basis $M$ and let $r(G)$ be the minimum of the numbers $r_M(G)$ over all bases $M$ of $G$. It is known that $r_M(G) \leq d(G) + r(G)$ for any basis $M$. Does there exist a finitely presented group $G$ for which the inequality becomes equality for some basis $M$?

[Archived solved part(s) of 7.57:] A set of generators of a finitely presented group $G$ that consists of the least possible number $d(G)$ of generators is called a *basis* for $G$. Let $r_M(G)$ be the least number of relations necessary to define $G$ in the basis $M$, and $r(G)$ the minimum of $r_M(G)$ over all bases $M$ for $G$. Let $G_1$, $G_2$ be any non-trivial groups.

b) Is it true that $r_{M_1\cup M_2}(G_1*G_2)= r_{M_1}G_1+r_{M_2}(G_2)$ for any bases $M_1$, $M_2$ of $G_1$, $G_2$, respectively?

c) Is it true that $r(G_1*G_2)=r(G_1)+r(G_2)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04791 | KOU-7.58

### Kourovka Notebook Problem 7.58

O / open | Kourovka 群论问题集

Suppose that $F$ is an absolutely free group, $R$ a normal subgroup of $F$, and let ${\mathfrak{V}}$ be a variety of groups. It is well-known (H. Neumann, *Varieties of Groups*, Springer, Berlin, 1967) that the group $F/{\mathfrak{V}}(R)$ is isomorphically embeddable in the ${\mathfrak{V}} $-verbal wreath product of a ${\mathfrak{V}} $-free group of the same rank as $F$ with $F/R$. Find a criterion indicating which elements of this wreath product belong to the image of this embedding. A criterion is known in the case where ${\mathfrak{V}}$ is the variety of all abelian groups (V. N. Remeslennikov, V. G. Sokolov, *Algebra and Logic*, **9**, no. 5 (1970), 342--349).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04792 | KOU-7.6

### Kourovka Notebook Problem 7.6

S / solved | Kourovka 群论问题集

Describe the infinite simple locally finite groups with a Chernikov Sylow 2-subgroup. In particular, are such groups the Chevalley groups over locally finite fields of odd characteristic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04793 | KOU-7.7

### Kourovka Notebook Problem 7.7

S / solved | Kourovka 群论问题集

(Well-known problem). Is the group $G=\left< a,b\mid a^9=1,\;\; ab= b^2a^2\right>$ finite? This group contains $F(2, 9)$, the only Fibonacci group for which it is not yet known whether it is finite or infinite.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04794 | KOU-7.8

### Kourovka Notebook Problem 7.8

S / solved | Kourovka 群论问题集

Suppose that $H$ is a normal subgroup of a group $G$, where $H$ and $G$ are subdirect products of the same $n$ groups $G_1,\ldots ,G_n$. Does the nilpotency class of $G/H$ increase with $n$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04795 | KOU-8.1

### Kourovka Notebook Problem 8.1

O / open | Kourovka 群论问题集

Characterize all groups (or at least all soluble groups) $G$ and fields $F$ such that every irreducible $ FG $-module has finite dimension over the centre of its endomorphism ring. For some partial answers see (B. A. F. Wehrfritz, *Glasgow Math. J.*, **24**, no. 1 (1983), 169--176).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04796 | KOU-8.10

### Kourovka Notebook Problem 8.10

S / solved | Kourovka 群论问题集

Is the group $G=\left< a,b\mid a^n=1,\; ab=b^3a^3\right>$ finite or infinite for $n=7$, $n=9$, and $n=15$? All other cases known, see Archive, 7.7.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04797 | KOU-8.11

### Kourovka Notebook Problem 8.11

S / solved | Kourovka 群论问题集

Consider the group $$M=\left< x,y,z,t\mid [x,y]=[y,z]=[z,x]=(x,t)=(y,t)=(z,t)=1\right> $$

where $[x, y]=x^{-1}y^{-1}xy$ and $(x,t)=x^{-1}t^{-1}x^{-1}txt$. The subgroup $H=\left< x,t,y\right>$ is isomorphic to the braid group ${\mathfrak{B}}_4$, and is normally complemented by $N=\left< (zx^{-1})^M\right> $. Is $N$ a free group (?) of countably infinite rank (?) on which $H$ acts faithfully by conjugation?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04798 | KOU-8.12

### Kourovka Notebook Problem 8.12

P / partially_solved | Kourovka 群论问题集

Let $D0$ denote the class of finite groups of deficiency zero, i. e. having a presentation $\left< X\mid R\right>$ with $|X|=|R|$.

a) Does $D0$ contain any 3-generator $p $-group for $p \geq 5$?

c) Do soluble $D0 $-groups have bounded derived length?

d) Which non-abelian simple groups can occur as composition factors of $D0 $-groups?

[Archived solved part(s) of 8.12:] b) Let $D0$ denote the class of finite groups of deficiency zero, i. e. having a presentation $\left< X\mid R\right>$ with $|X|=|R|$. Are the central factors of nilpotent $D0 $-groups 3-generated?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04799 | KOU-8.13

### Kourovka Notebook Problem 8.13

S / solved | Kourovka 群论问题集

Let $G$ be a simple algebraic group over an algebraically closed field of characteristic $p$ and ${\mathfrak{G}}$ be the Lie algebra of $G$. Is the number of orbits of nilpotent elements of ${\mathfrak{G}}$ under the adjoint action of $G$ finite? This is known to be true if $p$ is not too small (i. e. if $p$ is not a ``bad prime'' for $G$).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04800 | KOU-8.14

### Kourovka Notebook Problem 8.14

P / partially_solved | Kourovka 群论问题集

b) Assume a group $G$ is existentially closed in one of the classes $L{\mathfrak{N}}^{+}$, \ $L{\mathfrak{S}}_{\pi}$, \ $L{\mathfrak{S}}^{+}$, \ $L{\mathfrak{S}}$ of, respectively, all locally nilpotent torsion-free groups, all locally soluble $\pi $-groups, all locally soluble torsion-free groups, or all locally soluble groups. Is it true that $G$ is characteristically simple?

The existential closedness of $G$ is a local property, thus it seems difficult to obtain global properties of $G$ from it. See also Archive, 8.14 a.

[Archived solved part(s) of 8.14:] a) Assume a group $G$ is existentially closed in the class $L{\mathfrak{N}}_p$ of all locally finite $p $-groups. Is it true that $G$ is characteristically simple? This is true for $G$ countable in $L{\mathfrak{N}}_p$ (Berthold Maier, Freiburg); in fact, up to isomorphism, there is only one such countable locally finite $ p $-group.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04801 | KOU-8.15

### Kourovka Notebook Problem 8.15

O / open | Kourovka 群论问题集

(Well-known problem). Is every $\widetilde{N} $-group a $ \overline{\! Z} $-group? For definitions, see (M. I. Kargapolov, Yu. I. Merzlyakov, *Fundamentals of the Theory of Groups*, 3rd Edition, Moscow, Nauka, 1982, p. 205 (Russian)).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04802 | KOU-8.16

### Kourovka Notebook Problem 8.16

S / solved | Kourovka 群论问题集

Is the class of all $ \overline{\! Z} $-groups closed under taking normal subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04803 | KOU-8.17

### Kourovka Notebook Problem 8.17

S / solved | Kourovka 群论问题集

An {\it $\widetilde{RN}$-group} is one whose every homomorphic image is an $RN$-group. Is the class of $\widetilde{RN}$-groups closed under taking normal subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04804 | KOU-8.18

### Kourovka Notebook Problem 8.18

S / solved | Kourovka 群论问题集

Is every countably infinite abelian group a verbal subgroup of some finitely generated (soluble) relatively free group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04805 | KOU-8.19

### Kourovka Notebook Problem 8.19

O / open | Kourovka 群论问题集

Which of the following properties of (soluble) varieties of groups are equivalent to one another:

1) to satisfy the minimum condition for subvarieties;

2) to have at most countably many subvarieties;

3) to have no infinite independent system of identities?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04806 | KOU-8.2

### Kourovka Notebook Problem 8.2

O / open | Kourovka 群论问题集

Let $G$ be the amalgamated free product of two polycyclic groups, amalgamating a normal subgroup of each. Is $G$ isomorphic to a linear group? $G$ is residually finite (G. Baumslag, *Trans. Amer. Math. Soc.*, **106**, no. 2 (1963), 193--209) and the answer is ``yes'' if the amalgamated part is torsion-free, abelian (B. A. F. Wehrfritz, *Proc. London Math. Soc.*, **27**, no. 3 (1973), 402--424) and nilpotent (M. Shirvani, 1981, *unpublished*).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04807 | KOU-8.20

### Kourovka Notebook Problem 8.20

S / solved | Kourovka 群论问题集

What is the cardinality of the set of all varieties covering an abelian (nilpotent? Cross? hereditarily finitely based?) variety of groups? The question is related to 4.46 and 4.73.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04808 | KOU-8.21

### Kourovka Notebook Problem 8.21

O / open | Kourovka 群论问题集

If the commutator subgroup $G'$ of a relatively free group $G$ is periodic, must $G'$ have finite exponent?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04809 | KOU-8.22

### Kourovka Notebook Problem 8.22

S / solved | Kourovka 群论问题集

If $G$ is a (non-abelian) finite group contained in a join ${\mathfrak{A}} \vee {\mathfrak{B}}$ of two varieties ${\mathfrak{A}}, {\mathfrak{B}}$ of groups, must there exist finite groups $A \in {\mathfrak{A}}$, $B \in {\mathfrak{B}}$ such that $G$ is a section of the direct product $A \times B$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04810 | KOU-8.23

### Kourovka Notebook Problem 8.23

O / open | Kourovka 群论问题集

If the dihedral group $D$ of order 18 is a section of a direct product \mbox{$A \times B$}, must at least one of $A$ and $B$ have a section isomorphic to $D$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04811 | KOU-8.24

### Kourovka Notebook Problem 8.24

O / open | Kourovka 群论问题集

Prove that every linearly ordered group of finite rank is soluble.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04812 | KOU-8.25

### Kourovka Notebook Problem 8.25

O / open | Kourovka 群论问题集

Does there exist an algorithm which recognizes by an identity whether it defines a variety of groups that has at most countably many subvarieties?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04813 | KOU-8.26

### Kourovka Notebook Problem 8.26

S / solved | Kourovka 群论问题集

We call a variety *passable* if there exists an unrefinable chain of its subvarieties which is well-ordered by inclusion. For example, every variety generated by its finite groups is passable --- this is an easy consequence of (H. Neumann, *Varieties of groups*, Berlin et al., Springer, 1967, Chapter 5). Do there exist non-passable varieties of groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04814 | KOU-8.27

### Kourovka Notebook Problem 8.27

O / open | Kourovka 群论问题集

Does the lattice of all varieties of groups possess non-trivial automorphisms?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04815 | KOU-8.28

### Kourovka Notebook Problem 8.28

S / solved | Kourovka 群论问题集

Is the variety of groups finitely based if it is generated by a finitely based quasivariety of groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04816 | KOU-8.29

### Kourovka Notebook Problem 8.29

O / open | Kourovka 群论问题集

Do there exist locally nilpotent groups with trivial centre satisfying the weak maximal condition for normal subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04817 | KOU-8.3

### Kourovka Notebook Problem 8.3

O / open | Kourovka 群论问题集

Let $m$ and $n$ be positive integers and $p$ a prime. Let $P_n({\mathbb{Z}_{p^m})$ be the group of all $n \times n$ matrices $(a_{ij})$ over the integers modulo $p^m$ such that $a_{ii} \equiv 1$ for all $i$ and $a_{ij} \equiv 0 (\mathrm{mod} p)$ for all $i > j$. The group $P_n({\mathbb{Z}_{p^m})$ is a finite $p$-group. For which $m$ and $n$ is it regular?

*Editors' comment (2001):* The group $P_{n}({\mathbb{Z}_{p^m})$ is known to be regular if $mn<p$ (Yu. I. Merzlyakov, *Algebra i Logika*, **3**, no. 4 (1964), 49--59 (Russian)). The case $m=1$ is completely done by A. V. Yagzhev in (*Math. Notes*, **56**, no. 5--6 (1995), 1283--1290), although this paper is erroneous for $m>1$. The case $m=2$ is completed in (S. G. Kolesnikov, *Issledovaniya in analysis and algebra*, no. 3, Tomsk Univ., Tomsk, 2001, 117--125 (Russian)). *Editors' comment (2009):* The group $P_{n}({\mathbb{Z}_{p^m})$ was shown to be regular for any $m$ if $n^2<p$ (S. G. Kolesnikov, *Siberian Math. J.*, **47**, no. 6 (2006), 1054--1059).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04818 | KOU-8.30

### Kourovka Notebook Problem 8.30

O / open | Kourovka 群论问题集

Let ${\mathfrak{X}}$, ${\mathfrak{Y}}$ be Fitting classes of soluble groups which satisfy the Lockett condition, i. e. ${\mathfrak{X}} \cap {\mathfrak{S}}_{\displaystyle{*}}= {\mathfrak{X}}_{\displaystyle{*}}$, \ ${\mathfrak{Y}} \cap {\mathfrak{S}}_ {\displaystyle{*}} = {\mathfrak{Y}}_{\displaystyle{*}}$ where ${\mathfrak{S}}$ denotes the Fitting class of all soluble groups and the lower star the bottom group of the Lockett section determined by the given Fitting class. Does ${\mathfrak{X}} \cap {\mathfrak{Y}}$ satisfy the Lockett condition?

*Editors' comment (2001):* The answer is affirmative if ${\mathfrak{X}}$ and ${\mathfrak{Y}}$ are local (A. Grytczuk, N. T. Vorob'ev, *Tsukuba J. Math.*, **18**, no. 1 (1994), 63--67).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04819 | KOU-8.31

### Kourovka Notebook Problem 8.31

S / solved | Kourovka 群论问题集

Describe the finite groups in which every proper subgroup has a complement in some larger subgroup. Among these groups are, for example, $PSL_2(7)$ and all Sylow subgroups of symmetric groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04820 | KOU-8.32

### Kourovka Notebook Problem 8.32

S / solved | Kourovka 群论问题集

Suppose $G$ is a finitely generated group such that, for any set $\pi$ of primes and any subgroup $H$ of $G$, if $G/\left< H^G\right>$ is a finite $\pi $-group then $|G:H|$ is a finite $\pi $-number. Is $G$ nilpotent? This is true for finitely generated soluble groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04821 | KOU-8.33

### Kourovka Notebook Problem 8.33

O / open | Kourovka 群论问题集

Let $a, b$ be two elements of a group, $a$ having infinite order. Find a necessary and sufficient condition for $\bigcap_{n=1}^{\infty} \left< a^n, b\right> =\left< b\right> .$

问题集入口：https://kourovkanotebookorg.wordpress.com/

04822 | KOU-8.34

### Kourovka Notebook Problem 8.34

S / solved | Kourovka 群论问题集

Let $G$ be a finite group. Is it true that indecomposable projective ${\mathbb{Z}G $-modules are finitely generated (and hence locally free)?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04823 | KOU-8.35

### Kourovka Notebook Problem 8.35

P / partially_solved | Kourovka 群论问题集

c) Determine the conjugacy classes of maximal subgroups in the sporadic simple group $F_1$. See the current status in *Atlas of Finite Group Representations* (http://brauer.maths.qmul.ac.uk/Atlas/spor/M/).

[Archived solved part(s) of 8.35:] Determine the conjugacy classes of maximal subgroups in the sporadic simple groups \

a) $F'_{24} $; \ \

b) $F_2$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04824 | KOU-8.37

### Kourovka Notebook Problem 8.37

S / solved | Kourovka 群论问题集

(R. Griess). \ a) Is $M_{11}$ a section in $O'N$?

b) The same question for $M_{24}$ in $F_2$; \ $J_1$ in $F_1$ and in $F_2$; \ $J_2$ in $F_{23}$, $F_{24}'$, and $F_2$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04825 | KOU-8.39

### Kourovka Notebook Problem 8.39

S / solved | Kourovka 群论问题集

b) Describe the irreducible subgroups of $SL_6(q)$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04826 | KOU-8.4

### Kourovka Notebook Problem 8.4

O / open | Kourovka 群论问题集

Construct a finite nilpotent loop with no finite basis for its laws.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04827 | KOU-8.40

### Kourovka Notebook Problem 8.40

O / open | Kourovka 群论问题集

Describe the finite groups generated by a conjugacy class $D$ of involutions which satisfies the following property: if $a,b \in D$ and $|ab|=4$, then $[a,b] \in D$. This condition is satisfied, for example, in the case where $D$ is a conjugacy class of involutions of a known finite simple group such that $\left< C_D(a)\right>$ is a 2-group for every $a \in D$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04828 | KOU-8.41

### Kourovka Notebook Problem 8.41

O / open | Kourovka 群论问题集

What finite groups $G$ contain a normal set of involutions $D$ which contains a non-empty proper subset $T$ satisfying the following properties:

1) $C_D(a) \subseteq T$ for any $a \in T$;

2) if $a,b \in T$ and $ab=ba \ne 1$, then $C_D(ab) \subseteq T$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04829 | KOU-8.42

### Kourovka Notebook Problem 8.42

O / open | Kourovka 群论问题集

Describe the finite groups all of whose soluble subgroups of odd indices have 2-length 1. For example, the groups $L_n(2^m)$ are known to satisfy this condition.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04830 | KOU-8.43

### Kourovka Notebook Problem 8.43

O / open | Kourovka 群论问题集

(F. Timmesfeld). Let $T$ be a Sylow 2-subgroup of a finite group $G$. Suppose that $\left< N(B)\mid B \right.$ is a non-trivial characteristic subgroup of $\left. T\right> $ is a proper subgroup of $G$. Describe the group $G$ if $F^{\displaystyle{*}}(M)=O_2(M)$ for any 2-local subgroup $M$ containing $T$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04831 | KOU-8.44

### Kourovka Notebook Problem 8.44

O / open | Kourovka 群论问题集

Prove or disprove that for all, but finitely many, primes $p$ the group $$G_p=\left< a, b\mid a^2=b^p=(ab)^3=(b^rab^{-2r}a)^2=1\right> ,$$ where $r^2 + 1 \equiv 0 (\mathrm{mod} p)$, is infinite. A solution of this problem would have interesting topological applications. It was proved with the aid of computer that $G_p$ is finite for $p \leq 17$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04832 | KOU-8.45

### Kourovka Notebook Problem 8.45

O / open | Kourovka 群论问题集

When it follows that a group is residually a $({\mathfrak{X}} \cap {\mathfrak{Y}} )$-group, if it is both residually a ${\mathfrak{X}} $-group and residually a ${\mathfrak{Y}} $-group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04833 | KOU-8.46

### Kourovka Notebook Problem 8.46

S / solved | Kourovka 群论问题集

Describe the automorphisms of the symplectic group $Sp_{2n}$ over an arbitrary commutative ring. *Conjecture:* they are all standard.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04834 | KOU-8.47

### Kourovka Notebook Problem 8.47

S / solved | Kourovka 群论问题集

Do there exist finitely presented soluble groups in which the maximum condition for normal subgroups fails but all central sections are finitely generated?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04835 | KOU-8.48

### Kourovka Notebook Problem 8.48

S / solved | Kourovka 群论问题集

If a finite group $G$ can be written as the product of two soluble subgroups of odd index, then is $G$ soluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04836 | KOU-8.49

### Kourovka Notebook Problem 8.49

S / solved | Kourovka 群论问题集

Let $G$ be a $p $-group acting transitively as a permutation group on a set $\Omega$, let $F$ be a field of characteristic $p$, and regard $F\Omega$ as an $FG$-module. Then do the descending and ascending Loewy series of $F\Omega$ coincide?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04837 | KOU-8.5

### Kourovka Notebook Problem 8.5

S / solved | Kourovka 群论问题集

Prove that if $X$ is a finite group, $F$ is any field, and $M$ is a non-trivial irreducible $FX $-module then $\frac{1}{|X|}\sum_{x\in X} \dim \mathrm{fix} (x) \leq \frac{1}{2}\dim M$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04838 | KOU-8.50

### Kourovka Notebook Problem 8.50

O / open | Kourovka 群论问题集

At a conference in Oberwolfach in 1979 I exhibited a finitely generated soluble group $G$ (of derived length 3) and a non-zero cyclic ${\mathbb{Z}G $-module $V$ such that $V \cong V \oplus V$. Can this be done with $G$ metabelian? I conjecture that it cannot. For background of this problem and for details of the construction see (P. M. Neumann, *in: Groups--Korea, Pusan, 1988 $($Lect. Notes Math.*, **1398**), Springer, Berlin, 1989, 124--139).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04839 | KOU-8.51

### Kourovka Notebook Problem 8.51

S / solved | Kourovka 群论问题集

(J. McKay). If $G$ is a finite group and $p$ a prime let $ m_p(G)$ denote the number of ordinary irreducible characters of $G$ whose degree is not divisible by $p$. Let $P$ be a Sylow $p $-subgroup of $G$. Is it true that $m_p(G)=m_p(N_G(P))$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04840 | KOU-8.52

### Kourovka Notebook Problem 8.52

O / open | Kourovka 群论问题集

(Well-known problem). Do there exist infinite finitely presented periodic groups? Compare with 6.3.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04841 | KOU-8.53

### Kourovka Notebook Problem 8.53

P / partially_solved | Kourovka 群论问题集

a) Let $n$ be a sufficiently large odd number. Describe the automorphisms of the free Burnside group $B(m,n)$ of exponent $n$ on $m$ generators.

[Archived solved part(s) of 8.53:] b) Let $n$ be a sufficiently large odd number. Is it true that every non-cyclic subgroup of $B(m, n)$ has a subgroup isomorphic to $B(2, n)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04842 | KOU-8.54

### Kourovka Notebook Problem 8.54

O / open | Kourovka 群论问题集

a) (Well-known problem). Classify metabelian varieties of groups (or show that this is, in a certain sense, a ``wild'' problem).

b) Describe the identities of 2-generated metabelian groups, that is, classify varieties generated by such groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04843 | KOU-8.55

### Kourovka Notebook Problem 8.55

O / open | Kourovka 群论问题集

It is easy to see that the set of all quasivarieties of groups, in each of which quasivarieties a non-trivial identity holds, is a semigroup under multiplication of quasivarieties. Is this semigroup free?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04844 | KOU-8.56

### Kourovka Notebook Problem 8.56

S / solved | Kourovka 群论问题集

Let $X$ be a finite set and $f$ a mapping from the set of subsets of $X$ to the positive integers. Under the requirement that in a group generated by $X$, every subgroup $\left< Y\right>$, \ $Y\subseteq X$, be nilpotent of class $\leq f(Y)$, is it true that the free group $G_f$ relative to this condition is torsion-free?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04845 | KOU-8.58

### Kourovka Notebook Problem 8.58

O / open | Kourovka 群论问题集

Does a locally compact locally nilpotent nonabelian group without elements of finite order contain a proper closed isolated normal subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04846 | KOU-8.59

### Kourovka Notebook Problem 8.59

O / open | Kourovka 群论问题集

Suppose that, in a locally compact locally nilpotent group $G$, all proper closed normal subgroups are compact. Does $G$ contain an open compact normal subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04847 | KOU-8.6

### Kourovka Notebook Problem 8.6

S / solved | Kourovka 群论问题集

(M. M. Day). Do the classes of amenable groups and elementary groups coincide? The latter consists of groups that can be obtained from commutative and finite groups by forming subgroups, factor-groups, extensions, and direct limits.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04848 | KOU-8.60

### Kourovka Notebook Problem 8.60

O / open | Kourovka 群论问题集

Describe the locally compact primary locally soluble groups which are covered by compact subgroups and all of whose closed abelian subgroups have finite rank.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04849 | KOU-8.61

### Kourovka Notebook Problem 8.61

S / solved | Kourovka 群论问题集

Suppose that a locally compact group $G$ contains a subgroup that is topologically isomorphic to the additve group of the field of real numbers with natural topology. Is the space of all closed subgroups of $G$ connected in the Chabauty topology?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04850 | KOU-8.62

### Kourovka Notebook Problem 8.62

O / open | Kourovka 群论问题集

Describe the locally compact locally pronilpotent groups for which the space of closed normal subgroups is compact in the $ E $-topology. The corresponding problem has been solved for discrete groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04851 | KOU-8.63

### Kourovka Notebook Problem 8.63

S / solved | Kourovka 群论问题集

Suppose that the space of all closed subgroups of a locally compact group $G$ is $\sigma$-compact in the $E$-topology. Is it true that the set of closed non-compact subgroups of $G$ is at most countable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04852 | KOU-8.64

### Kourovka Notebook Problem 8.64

O / open | Kourovka 群论问题集

Does the class of all finite groups possess an independent basis of quasiidentities?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04853 | KOU-8.66

### Kourovka Notebook Problem 8.66

S / solved | Kourovka 群论问题集

Construct examples of residually finite groups which would separate Shunkov's classes of groups with $(a,b) $-finiteness condition, (weakly) conjugacy biprimitively finite groups and (weakly) biprimitively finite groups (see, in particular, 6.57). Can one derive such examples from Golod's construction?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04854 | KOU-8.67

### Kourovka Notebook Problem 8.67

O / open | Kourovka 群论问题集

Do there exist Golod groups all of whose abelian subgroups have finite ranks? Here a *Golod group* means a finitely generated non-nilpotent subgroup of the adjoint group of a nil-ring. The answer is negative for the whole adjoint group of a nil-ring (Ya. P. Sysak, *Abstracts of the 17th All--Union Algebraic Conf., part 1*, Minsk, 1983, p. 180 (Russian)).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04855 | KOU-8.68

### Kourovka Notebook Problem 8.68

S / solved | Kourovka 群论问题集

Let $G=\left< a,b\mid r=1\right> $ where $r$ is a cyclically reduced word that is not a proper power of any word in $a,b$. If $G$ is residually finite, is $G_t=\left<a,b\mid r^t=1\right> $, $t > 1$, residually finite?

*Editors' comment:* An analogous fact was proved for relative one-relator groups (S. J. Pride, *Proc. Amer. Math. Soc.*, **136**, no. 2 (2008), 377--386).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04856 | KOU-8.69

### Kourovka Notebook Problem 8.69

S / solved | Kourovka 群论问题集

Is every 1-relator group with non-trivial torsion conjugacy separable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04857 | KOU-8.7

### Kourovka Notebook Problem 8.7

S / solved | Kourovka 群论问题集

Does there exist a non-amenable finitely presented group which has no free subgroups of rank 2?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04858 | KOU-8.70

### Kourovka Notebook Problem 8.70

S / solved | Kourovka 群论问题集

Let $A, B$ be polycyclic-by-finite groups. Let $G=A*_H B$ where $H$ is cyclic. Is $G$ conjugacy separable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04859 | KOU-8.71

### Kourovka Notebook Problem 8.71

S / solved | Kourovka 群论问题集

Is every countable conjugacy-separable group embeddable in a 2-generator conjugacy-separable group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04860 | KOU-8.72

### Kourovka Notebook Problem 8.72

O / open | Kourovka 群论问题集

Does there exist a finitely presented group $G$ which is not free or cyclic of prime order, having the property that every proper subgroup of $G$ is free?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04861 | KOU-8.73

### Kourovka Notebook Problem 8.73

S / solved | Kourovka 群论问题集

We say that a finite group $G$ *separates cyclic subgroups* if, for any cyclic subgroups $A$ and $B$ of $G$, there is a $g\in G$ such that $ A\cap B^g=1$. Is it true that $G$ separates cyclic subgroups if and only if $G$ has no non-trivial cyclic normal subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04862 | KOU-8.74

### Kourovka Notebook Problem 8.74

S / solved | Kourovka 群论问题集

A subnormal subgroup $H\mbox{$\lhd\lhd$} G$ of a group $G$ is said to be *good* if and only if $\left< H,J\right> \mbox{$\lhd\lhd$} G$ whenever $J \mbox{$\lhd\lhd$} G$. Is it true that when $H$ and $K$ are good subnormal subgroups of $G$, then $H \cap K$ is good?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04863 | KOU-8.75

### Kourovka Notebook Problem 8.75

S / solved | Kourovka 群论问题集

(A known problem). Suppose $G$ is a finite primitive permutation group on $\Omega$, and $\alpha , \beta $ are distinct points of $\Omega$. Does there exist an element $g \in G $ such that $\alpha g=\beta $ and $g$ fixes no point of $\Omega$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04864 | KOU-8.76

### Kourovka Notebook Problem 8.76

S / solved | Kourovka 群论问题集

Give a realistic upper bound for the torsion-free rank of a finitely generated nilpotent group in terms of the ranks of its abelian subgroups. More precisely, for each integer $n$ let $f(n)$ be the largest integer $h$ such that there is a finitely generated nilpotent group of torsion-free rank $h$ with the property that all abelian subgroups have torsion-free rank at most $n$. It is easy to see that $f(n)$ is bounded above by $n(n + 1)/2$. Describe the behavior of $f(n)$ for large $n$. Is $f(n)$ bounded below by a quadratic in $n$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04865 | KOU-8.77

### Kourovka Notebook Problem 8.77

O / open | Kourovka 群论问题集

Do there exist strongly regular graphs with parameters $\lambda =0$, \ $\mu =2$ of degree $k > 10$? Such graphs are known for $k=5$ and $ k=10$, their automorphism groups are primitive permutation groups of rank 3.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04866 | KOU-8.78

### Kourovka Notebook Problem 8.78

O / open | Kourovka 群论问题集

It is known that there exists a countable locally finite group that contains a copy of every other countable locally finite group. For which other classes of countable groups does a similar ``largest'' group exist? In particular, what about periodic locally soluble groups? periodic locally nilpotent groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04867 | KOU-8.79

### Kourovka Notebook Problem 8.79

O / open | Kourovka 群论问题集

Does there exist a countable infinite locally finite group $G$ that is complete, in the sense that $G$ has trivial centre and no outer automorphisms? An uncountable one exists (K. K. Hickin, *Trans. Amer. Math. Soc.*, **239** (1978), 213--227).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04868 | KOU-8.8

### Kourovka Notebook Problem 8.8

P / partially_solved | Kourovka 群论问题集

b) (D. V. Anosov). Does there exist a non-cyclic finitely presented group $G$ which contains an element $a$ such that each element of $G$ is conjugate to some power of $a$?

[Archived solved part(s) of 8.8:] (D. V. Anosov). a) Does there exist a non-cyclic finitely-generated group $G$ containing an element $a$ such that every element of $G$ is conjugate to a power of $a$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04869 | KOU-8.80

### Kourovka Notebook Problem 8.80

S / solved | Kourovka 群论问题集

Let $G$ be a locally finite group containing a maximal subgroup which is Chernikov. Is $G$ almost soluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04870 | KOU-8.81

### Kourovka Notebook Problem 8.81

S / solved | Kourovka 群论问题集

Let $G$ be a finite $p $-group admitting an automorphism $\alpha $ of prime order $q$ with $|C_G(a)|\leq n$.

a) If $ p = q$, then does $ G$ have a nilpotent subgroup of class at most 2 and index bounded by a function of $n$?

b) If $p\ne q$, then does $G$ have a nilpotent subgroup of class bounded by a function of $q$ and index bounded by a function of $n$ (and, possibly, $q$)?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04871 | KOU-8.82

## Kourovka Notebook Problem 8.82

O / open | Kourovka 群论问题集

Let ${\mathfrak{H}}={\mathbb{C} \times {\mathbb{R} =\{ (z,r)\mid z \in {\mathbb{C},\; r > 0\}$ be the three-dimensional Poincar\'e space which admits the following action of the group $SL_2({\mathbb{C})$: $$(z,r) \left(\!\!\begin{array}{cc}a&b\cr c&d\end{array}\!\!\right)=\left( \frac{(az+b)(\overline{cz+d})+a \overline{\! c}r^2}{|cz+d|^2+|c|^2 r^2} ,\; \frac{r}{|cz+d|^2+|c|^2r^2}\right) .$$ Let ${\mathfrak{o}}$ be the ring of integers of the field $K={\mathbb{Q}(\sqrt{D})$, where $D < 0$, \ $\pi$ a prime such that $\pi \overline{\pi}$ is a prime in ${\mathbb{Z}$ and let $\displaystyle{\Gamma=\left\{ \left.\left(\!\!\begin{array}{cc}a&b\cr c&d\end{array}\!\!\right) \in SL_2({\mathfrak{o}})\;\right|\; b \equiv 0 (\mathrm{mod} \pi )\right\} }$. Adding to the space $\Gamma \setminus {\mathfrak{H}}^3$ two vertices we get a three-dimensional compact space $\overline{\Gamma \setminus {\mathfrak{H}}^3}$.

Calculate $r(\pi )=\mathrm{dim}_{ {\mathbb{Q}} H_1( \overline{\Gamma \setminus {\mathfrak{H}}^3},{\mathbb{Q}})=\mathrm{dim}_{ {\mathbb{Q}} (\Gamma ^{ab} \otimes {\mathbb{Q}}).$

For example, if $D=-3$ then $r(\pi )$ is distinct from zero for the first time for $\pi \mid 73$ (then $r(\pi )=1$), and if $D=-4$ then $r(\pi )=1$ for $\pi \mid 137$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04872 | KOU-8.83

## Kourovka Notebook Problem 8.83

O / open | Kourovka 群论问题集

In the notation of 8.82, for $r(\pi ) > 0$, one can define via Hecke algebras a formal Dirichlet series with Euler multiplication (see G. Shimura, *Introduction to the arithmetic theory of automorphic functions*, Princeton Univ. Press, 1971). Does there exist an algebraic Hasse--Weil variety whose $ \zeta $-function is this Dirichlet series? There are several conjectures.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04873 | KOU-8.84

## Kourovka Notebook Problem 8.84

S / solved | Kourovka 群论问题集

We say that an automorphism $\varphi$ of the group $G$ is a *pseudo-identity* if, for all $x \in G$, there exists a finitely generated subgroup $K_x$ of $G$ such that $x \in K_x$ and $\varphi |_{K_{\scriptstyle{x}}}$ is an automorphism of $K_x$. Let $G$ be generated by subgroups $H, K$ and let $G$ be locally nilpotent. Let $\varphi \! : G \rightarrow G$ be an endomorphism such that $\varphi |_H$ is pseudo-identity of $H$ and $\varphi |_K$ is an automorphism of $K$. Does it follow that $\varphi$ is an automorphism of $G$? It is known that $\varphi$ is a pseudo-identity of $G$ if, additionally, $\varphi |_K$ is pseudo-identity of $K$; it is also known that $\varphi$ is an automorphism if, additionally, $K$ is normal in $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04874 | KOU-8.85

## Kourovka Notebook Problem 8.85

O / open | Kourovka 群论问题集

Construct a finite $p $-group $G$ whose Hughes subgroup $H_p(G)=\left< x \in G\mid \right.$ $\left. |x| \ne p\right> $ is non-trivial and has index $p^3$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04875 | KOU-8.86

## Kourovka Notebook Problem 8.86

O / open | Kourovka 群论问题集

(Well-known problem). Suppose that all proper closed subgroups of a locally compact locally nilpotent group $G$ are compact. Is $G$ abelian if it is non-compact?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04876 | KOU-8.87

## Kourovka Notebook Problem 8.87

S / solved | Kourovka 群论问题集

Find all hereditary local formations ${\mathfrak{F}}$ of finite groups satisfying the following condition: every finite minimal non-${\mathfrak{F}} $-group is biprimary. A finite group is said to be *biprimary* if its order is divisible by precisely two distinct primes.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04877 | KOU-8.9

## Kourovka Notebook Problem 8.9

O / open | Kourovka 群论问题集

(C. Chou). We say that a group $G$ has *property $P$* if for every finite subset $F$ of $G$, there is a finite subset $S \supset F$ and a subset $X \subset G$ such that $x_1S \cap x_2S$ is empty for any $x_1, x_2 \in X$, $x_1 \ne x_2$, and $G=\bigcup\limits_{x\in X}Sx$. Does every group have property $P$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04878 | KOU-9.1

### Kourovka Notebook Problem 9.1

O / open | Kourovka 群论问题集

A group $G$ is said to be *potent* if, to each $x\in G$ and each positive integer $n$ dividing the order of $x$ (we suppose $ \infty$ is divisible by every positive integer), there exists a finite homomorphic image of $G$ in which the image of $x$ has order precisely $ n$. Is the free product of two potent groups again potent?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04879 | KOU-9.10

### Kourovka Notebook Problem 9.10

S / solved | Kourovka 群论问题集

Do there exist finitely generated groups different from ${\mathbb{Z}/2{\mathbb{Z}$ which have precisely two conjugacy classes?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04880 | KOU-9.11

### Kourovka Notebook Problem 9.11

O / open | Kourovka 群论问题集

Is an abelian minimal normal subgroup $A$ of a group $G$ an elementary $p $-group if the factor-group $ G/A$ is a soluble group of finite rank? The answer is affirmative under the additional hypothesis that $ G/A$ is locally polycyclic (D. I. Zaitsev, *Algebra and Logic*, **19**, no. 2 (1980), 94--105).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04881 | KOU-9.12

### Kourovka Notebook Problem 9.12

S / solved | Kourovka 群论问题集

Is there a soluble group with the following properties: torsion-free of finite rank, not finitely generated, and having a faithful irreducible representation over a finite field?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04882 | KOU-9.13

### Kourovka Notebook Problem 9.13

O / open | Kourovka 群论问题集

Is a soluble torsion-free group minimax if it satisfies the weak minimum condition for normal subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04883 | KOU-9.14

### Kourovka Notebook Problem 9.14

O / open | Kourovka 群论问题集

Is a group locally finite if it decomposes into a product of periodic abelian subgroups which commute pairwise? The answer is affirmative in the case of two subgroups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04884 | KOU-9.15

### Kourovka Notebook Problem 9.15

O / open | Kourovka 群论问题集

Describe, without using CFSG, all subgroups $ L$ of a finite Chevalley group $G$ such that $G=PL$ for some parabolic subgroup $P$ of $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04885 | KOU-9.16

### Kourovka Notebook Problem 9.16

S / solved | Kourovka 群论问题集

(Well-known problem). The *prime graph* of a finite group $G$ is the graph with vertex set $\pi (G)$ and an edge joining $p$ and $q$ if and only if $G$ has an element of order $pq$. Describe all finite Chevalley groups over a field of characteristic $2$ whose prime graph is not connected and describe the connected components.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04886 | KOU-9.17

### Kourovka Notebook Problem 9.17

P / partially_solved | Kourovka 群论问题集

b) Let $G$ be a locally normal residually finite group. Does there exist a normal subgroup $H$ of $G$ which is embeddable in a direct product of finite groups and is such that $G/H$ is a divisible abelian group?

[Archived solved part(s) of 9.17:] a) Let $G$ be a locally normal residually finite group. Can $G$ be embedded in a direct product of finite groups when the factor group $G/[G, G]$ is a direct product of cyclic groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04887 | KOU-9.18

### Kourovka Notebook Problem 9.18

S / solved | Kourovka 群论问题集

Let ${\mathfrak{S}}_{\displaystyle{*}}$ be the smallest normal Fitting class. Are there Fitting classes which are maximal in ${\mathfrak{S}}_{\displaystyle{*}}$ (with respect to inclusion)?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04888 | KOU-9.19

### Kourovka Notebook Problem 9.19

P / partially_solved | Kourovka 群论问题集

a) Let $n(X)$ denote the minimum of the indices of proper subgroups of a group $X$. A subgroup $A$ of a finite group $G$ is called *wide* if $A$ is a maximal element by inclusion of the set $\{ X\mid X \mbox{ is a proper subgroup of }G\mbox{ and }n(X)=n(G)\} $. Find all wide subgroups in finite projective special linear, symplectic, orthogonal, and unitary groups.

[Archived solved part(s) of 9.19:] b) Let $n(X)$ denote the minimum of the indices of proper subgroups of a group $X$. A subgroup $A$ of a finite group $G$ is called *wide* if $A$ is a maximal element by inclusion of the set $\{ X\mid X \mbox{ is a proper subgroup of}G\mbox{ and }n(X)=n(G)\} $. Prove, without using CFSG, that $n(F_1)=|F_1:2F_2|$, where $F_1$ and $F_2$ are the Fischer simple groups and $2F_2$ is an extension of a group of order 2 by $F_2$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04889 | KOU-9.2

### Kourovka Notebook Problem 9.2

S / solved | Kourovka 群论问题集

Is $G=\left<a,b\mid a^l=b^m=(ab)^n=1\right> $ conjugacy separable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04890 | KOU-9.21

### Kourovka Notebook Problem 9.21

S / solved | Kourovka 群论问题集

Let $P$ be a maximal parabolic subgroup of the smallest index in a finite group $G$ of Lie type $ E_6$, $ E_7$, $ E_8$, or $^2E_6$ and let $X$ be a subgroup such that $PX=G$. Is it true that $X=G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04891 | KOU-9.23

### Kourovka Notebook Problem 9.23

P / partially_solved | Kourovka 群论问题集

Let $G$ be a finite group, $B$ a block of characters of $ G$, $D(B)$ its defect group and $k(B)$ (respectively, $k_0(B)$) the number of all irreducible complex characters (of height 0) lying in $B$. *Conjectures:*

a) (R. Brauer) \ $k(B) \leq |D(B)|$; this has been proved for $p $-soluble groups (D. Gluck, K. Magaard, U. Riese, P. Schmid, *J. Algebra*, **279** (2004), 694--719);

b) (J. B. Olsson) \ $k_0(B) \leq |D(B) : D(B)'|$, where $D(B)'$ is the derived subgroup of $D(B)$;

c) (R. Brauer) \ $D(B)$ is abelian if and only if $k_0(B)=k(B)$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04892 | KOU-9.24

### Kourovka Notebook Problem 9.24

O / open | Kourovka 群论问题集

(J. G. Thompson). *Conjecture:* every finite simple non-abelian group $G$ can be represented in the form $ G=CC$, where $C$ is some conjugacy class of $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04893 | KOU-9.25

### Kourovka Notebook Problem 9.25

S / solved | Kourovka 群论问题集

Find an algorithm which recognizes, by an equation $w(x_1,\ldots ,x_n)=1$ in a free group $F$ and by a list of finitely generated subgroups $H_1,\ldots ,H_n$ of $ F$, whether there is a solution of this equation satisfying the condition $x_1 \in H_1$, $\ldots$ , $x_n \in H_n$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04894 | KOU-9.26

### Kourovka Notebook Problem 9.26

P / partially_solved | Kourovka 群论问题集

b) Describe the finite groups of 2-local 3-rank 1 which have non-cyclic Sylow 3-subgroups.

[Archived solved part(s) of 9.26:] a) Describe the finite groups of 2-local 3-rank 1 which have 3-rank at least 3.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04895 | KOU-9.27

### Kourovka Notebook Problem 9.27

S / solved | Kourovka 群论问题集

Let $M$ be a subgroup of a finite group $G$, \ $A$ an abelian 2-subgroup of $M$, and suppose that $A^g$ is not contained in $M$ for some $g$ from $G$. Determine the structure of $G$ under the hypothesis that $\left< A,A^x\right> =G$ whenever the subgroup $A^x$, $x \in G$, is not contained in $M$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04896 | KOU-9.28

### Kourovka Notebook Problem 9.28

O / open | Kourovka 群论问题集

Suppose that a finite group $G$ is generated by a conjugacy class $D$ of involutions and let $D_i=\{ d_1\cdots d_i\mid d_1,\ldots ,d_i$ are different pairwise commuting elements of $ D\} $. What is $G$, if $ D_1,\ldots ,D_n$ are all its different conjugacy classes of involutions? For example, the Fischer groups $F_{22}$ and $F_{23}$ satisfy this condition with $n=3$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04897 | KOU-9.29

### Kourovka Notebook Problem 9.29

O / open | Kourovka 群论问题集

(Well-known problem). According to a classical theorem of Magnus, the word problem is soluble in 1-relator groups. Do there exist 2-relator groups with insoluble word problem?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04898 | KOU-9.3

### Kourovka Notebook Problem 9.3

S / solved | Kourovka 群论问题集

Suppose that a countable locally finite group $G$ contains no proper subgroups isomorphic to $G$ itself and suppose that all Sylow subgroups of $G$ are finite. Does $G$ possess a non-trivial finite normal subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04899 | KOU-9.30

### Kourovka Notebook Problem 9.30

S / solved | Kourovka 群论问题集

(Well-known problem). A finite set of reductions $u_i \rightarrow v_i$ of words on a finite alphabet $\Sigma =\Sigma ^{-1}$ is called a *group set of reductions* if $\mbox{*length*} (u_i) > \mbox{*length*} (v_i)$ or $\mbox{*length*} (u_i)=\mbox{*length*} (v_i)$ and $u_i > v_i$ in the lexicographical ordering, and every word in $\Sigma$ can be reduced to the unique reduced form which does not depend on the sequence of reductions. Do there exist group sets of reductions satisfying the condition $\mbox{*length*} (v_i) \leq 1$ for all $i$, which are different from 1) sets of trivial reductions $x^{-\varepsilon }x^{\varepsilon} \rightarrow 1$, $\varepsilon =\pm 1$, 2) multiplication tables $xy \rightarrow z$ of finite groups, and 3) their finite unions?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04900 | KOU-9.31

### Kourovka Notebook Problem 9.31

O / open | Kourovka 群论问题集

Let $k$ be a field of characteristic 0. According to (Yu. I. Merzlyakov, *Proc. Steklov Inst. Math.*, **167** (1986), 263--266), the family $\mathrm{Rep}_k(G)$ of all canonical matrix representations of a $k$-powered group $G$ over $ k$ may be regarded as an affine $ k$-variety. Find an explicit form of equations defining this variety.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04901 | KOU-9.32

### Kourovka Notebook Problem 9.32

S / solved | Kourovka 群论问题集

What locally compact groups satisfy the following condition: the product of any two closed subgroups is also a closed subgroup? Abelian groups with this property were described in (Yu. N. Mukhin, *Math. Notes*, **8** (1970), 755--760).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04902 | KOU-9.33

### Kourovka Notebook Problem 9.33

S / solved | Kourovka 群论问题集

(F. K\"ummich, H. Scheerer). If $H$ is a closed subgroup of a connected locally-compact group $G$ such that $ \overline{\! HX}= \overline{\! H} \overline{\! X}$ for every closed subgroup $X$ of $G$, then is $H$ normal?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04903 | KOU-9.34

### Kourovka Notebook Problem 9.34

S / solved | Kourovka 群论问题集

(S. K. Grosser, W. N. Herfort). Does there exist an infinite compact $p $-group in which the centralizers of all elements are finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04904 | KOU-9.35

### Kourovka Notebook Problem 9.35

O / open | Kourovka 群论问题集

A topological group is said to be *inductively compact* if any finite set of its elements is contained in a compact subgroup. Is this property preserved under lattice isomorphisms in the class of locally compact groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04905 | KOU-9.36

### Kourovka Notebook Problem 9.36

O / open | Kourovka 群论问题集

Characterize the lattices of closed subgroups in locally compact groups. In the discrete case this was done in (B. V. Yakovlev, *Algebra and Logic*, **13**, no. 6 (1974), 400--412).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04906 | KOU-9.37

### Kourovka Notebook Problem 9.37

O / open | Kourovka 群论问题集

Is a compactly generated inductively prosoluble locally compact group prosoluble?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04907 | KOU-9.38

### Kourovka Notebook Problem 9.38

O / open | Kourovka 群论问题集

A group is said to be *compactly covered* if it is the union of its compact subgroups. In a null-dimensional locally compact group, are maximal compactly covered subgroups closed?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04908 | KOU-9.39

## Kourovka Notebook Problem 9.39

O / open | Kourovka 群论问题集

Let $\Omega$ be a countable set and ${\mathfrak{m}}$ a cardinal number such that $\aleph _0 \leq {\mathfrak{m}} \leq 2^{\aleph _0}$ (we assume Axiom of Choice but not Continuum Hypothesis). Does there exist a permutation group $G$ on $\Omega$ that has exactly ${\mathfrak{m}}$ orbits on the power set ${\mathfrak{P}}(\Omega )$?

*Comment of 2001:* It is proved (S. Shelah, S. Thomas, *Bull. London Math. Soc.*, **20**, no. 4 (1988), 313--318) that the answer is positive in set theory with Martin's Axiom. The question is still open in ZFC.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04909 | KOU-9.4

## Kourovka Notebook Problem 9.4

S / solved | Kourovka 群论问题集

It is known that if ${\mathfrak{M}}$ is a variety (quasivariety, pseudovariety) of groups, then the class $I{\mathfrak{M}}$ of all quasigroups that are isotopic to groups in ${\mathfrak{M}}$ is also a variety (quasivariety, pseudovariety), and $I{\mathfrak{M}}$ is finitely based if and only if ${\mathfrak{M}}$ is finitely based. Is it true that if ${\mathfrak{M}}$ is generated by a single finite group then $I{\mathfrak{M}}$ is generated by a single finite quasigroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04910 | KOU-9.40

## Kourovka Notebook Problem 9.40

O / open | Kourovka 群论问题集

Let $\Omega$ be a countably infinite set. Define a *moiety* of $\Omega$ to be a subset $\Sigma$ such that both $\Sigma$ and $\Omega \setminus \Sigma$ are infinite. Which permutation groups on $ \Omega$ are transitive on moieties?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04911 | KOU-9.41

## Kourovka Notebook Problem 9.41

P / partially_solved | Kourovka 群论问题集

Let $\Omega$ be a countably infinite set. For $k \geq 2$ define a *$k$-section* of $\Omega$ to be a partition of $\Omega$ as union of $k$ infinite sets.

b) Does there exist a group that is transitive on $k$-sections but not on $(k + 1)$-sections?

c) Does there exist a transitive permutation group on $\Omega$ that is transitive on $\aleph _0$-sections but which is a proper subgroup of $\mathrm{Sym}(\Omega )$?

[Archived solved part(s) of 9.41:] a) Let $\Omega$ be a countably infinite set. For $k\geq 2$, we define a *$k$-section* of $\Omega$ to be a partition of $\Omega$ into a union of $k$ infinite subsets. Then does there exist a transitive permutation group on $\Omega$ that is transitive on $k$-sections but intransitive on ordered $k$-sections?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04912 | KOU-9.42

## Kourovka Notebook Problem 9.42

S / solved | Kourovka 群论问题集

Let $\Omega$ be a countable set and let $D$ be the set of total order relations on $ \Omega$ for which $ \Omega$ is order-isomorphic with ${\mathbb{Q}}$. Does there exist a transitive proper subgroup $G$ of $\mathrm{Sym} (\Omega )$ which is transitive on $D$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04913 | KOU-9.43

## Kourovka Notebook Problem 9.43

P / partially_solved | Kourovka 群论问题集

b) The group $G$ indicated in (N. D. Podufalov, *Abstracts of the 9th All--Union Symp. on Group Theory*, Moscow, 1984, 113--114 (Russian)) allows us to construct a projective plane of order 3: one can take as lines any class of subgroups of order $2\cdot 5\cdot 11\cdot 17$ conjugate under $ S$ and add four more lines in a natural way. In a similar way, one can construct projective planes of order $p^n$ for any prime $p$ and any natural $ n$. Could this method be adapted for constructing new planes?

[Archived solved part(s) of 9.43:] a) The group $G$ described in the solution of 8.73 (Archive) enables us to construct a projective plane of order 3 in which the lines are the elements of any conjugacy class of subgroups of order $2\cdot 5\cdot 7\cdot 11$ together with four lines added in a natural way. In a similar way, we can construct a projective plane of order $p^n$ for any prime $p$ and any positive integer $n$. Does the resulting projective plane have the Galois property?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04914 | KOU-9.44

### Kourovka Notebook Problem 9.44

O / open | Kourovka 群论问题集

A topological group is said to be *layer compact* if the full inverse images of all of its compacts under mappings $x \rightarrow x^n$, $n=1, 2, \ldots $, are compacts. Describe the locally compact locally soluble layer compact groups.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04915 | KOU-9.45

### Kourovka Notebook Problem 9.45

O / open | Kourovka 群论问题集

Let $a$ be a vector in ${\mathbb{R}^n$ with rational coordinates and set $$S(a)=\{ ka + b\mid k \in {\mathbb{Z}, b \in {\mathbb{Z}^n\} .$$ It is obvious that $S(a)$ is a discrete subgroup in ${\mathbb{R}^n$ of rank $n$. Find necessary and sufficient conditions in terms of the coordinates of $a$ for $S(a)$ to have an orthogonal basis with respect to the standard scalar product in ${\mathbb{R}^n$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04916 | KOU-9.46

### Kourovka Notebook Problem 9.46

S / solved | Kourovka 群论问题集

Let $G$ be a locally compact group of countable weight and $L(G)$ the space of all its closed subgroups equipped with the $E$-topology. Then is $L(G)$ a $k$-space?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04917 | KOU-9.47

### Kourovka Notebook Problem 9.47

O / open | Kourovka 群论问题集

Is it true that every scattered compact can be homeomorphically embedded into the space of all closed non-compact subgroups with $ E$-topology of a suitable locally compact group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04918 | KOU-9.48

### Kourovka Notebook Problem 9.48

S / solved | Kourovka 群论问题集

In a null-dimensional locally compact group, is the set of all compact elements closed?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04919 | KOU-9.49

### Kourovka Notebook Problem 9.49

S / solved | Kourovka 群论问题集

Let $G$ be a compact group of weight $> \omega _2$. Is it true that the space of all closed subgroups of $G$ with respect to $E$-topology is non-dyadic?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04920 | KOU-9.5

### Kourovka Notebook Problem 9.5

O / open | Kourovka 群论问题集

A variety of groups is called *primitive* if each of its subquasivarieties is a variety. Describe all primitive varieties of groups. Is every primitive variety of groups locally finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04921 | KOU-9.50

### Kourovka Notebook Problem 9.50

S / solved | Kourovka 群论问题集

Is every 4-Engel group

a) without elements of order 2 and 5 necessarily soluble?

b) satisfying the identity $x^5=1$ necessarily locally finite?

c) (R. I. Grigorchuk) satisfying the identity $x^8=1$ necessarily locally finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04922 | KOU-9.51

### Kourovka Notebook Problem 9.51

O / open | Kourovka 群论问题集

Does there exist a finitely presented soluble group satisfying the maximum con- dition on normal subgroups which has insoluble word problem?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04923 | KOU-9.52

### Kourovka Notebook Problem 9.52

O / open | Kourovka 群论问题集

Does a finitely presented soluble group of finite rank have soluble conjugacy problem? Note: there is an algorithm to decide conjugacy to a given element of the group.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04924 | KOU-9.53

### Kourovka Notebook Problem 9.53

O / open | Kourovka 群论问题集

Is the isomorphism problem soluble for finitely presented soluble groups of finite rank?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04925 | KOU-9.54

### Kourovka Notebook Problem 9.54

S / solved | Kourovka 群论问题集

If $G=HK$ is a soluble group and $ H$, $ K$ are minimax groups, is it true that $G$ is a minimax group?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04926 | KOU-9.55

### Kourovka Notebook Problem 9.55

O / open | Kourovka 群论问题集

Does there exist a finite $p $-group $G$ and a central augmented automorphism $\varphi$ of ${\mathbb{Z}G$ such that $\varphi $, if extended to $\widehat{{\mathbb{Z}}}_pG$, the $p $-adic group ring, is *not* conjugation by unit in $\widehat{{\mathbb{Z}}}_pG$ followed by a homomorphism induced from a group automorphism?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04927 | KOU-9.56

### Kourovka Notebook Problem 9.56

S / solved | Kourovka 群论问题集

Find all finite groups with the property that the tensor square of any ordinary irreducible character is multiplicity free.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04928 | KOU-9.57

### Kourovka Notebook Problem 9.57

O / open | Kourovka 群论问题集

The set of subformations of a given formation is a lattice with respect to operations of intersection and generation. What formations of finite groups have distributive lattices of subformations?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04929 | KOU-9.58

### Kourovka Notebook Problem 9.58

S / solved | Kourovka 群论问题集

Can a product of non-local formations of finite groups be local?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04930 | KOU-9.59

### Kourovka Notebook Problem 9.59

S / solved | Kourovka 群论问题集

(W. Gasch\"utz). Prove that the formation generated by a finite group has finite lattice of subformations.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04931 | KOU-9.6

### Kourovka Notebook Problem 9.6

O / open | Kourovka 群论问题集

Is it true that an independent basis of quasiidentities of any finite group is finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04932 | KOU-9.60

### Kourovka Notebook Problem 9.60

S / solved | Kourovka 群论问题集

Let ${\mathfrak{F}}$ and ${\mathfrak{H}}$ be local formations of finite groups and suppose that ${\mathfrak{F}}$ is not contained in ${\mathfrak{H}}$. Does ${\mathfrak{F}}$ necessarily have at least one minimal local non-${\mathfrak{H}}$-subformation?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04933 | KOU-9.61

### Kourovka Notebook Problem 9.61

O / open | Kourovka 群论问题集

Two varieties are said to be *$S$-equivalent* if they have the same Mal'cev theory (D. M. Smirnov, *Algebra and Logic*, **22**, no. 6 (1983), 492--501). What is the cardinality of the set of $S$-equivalent varieties of groups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04934 | KOU-9.62

### Kourovka Notebook Problem 9.62

S / solved | Kourovka 群论问题集

In any group $G$, the cosets of all of its normal subgroups together with the empty set form the *block lattice* $C(G)$ with respect to inclusion, which, for infinite $|G|$, is subdirectly irreducible and, for finite $|G| \geq 3$, is even simple (D. M. Smirnov, A. V. Reibol'd, *Algebra and Logic*, **23**, no. 6 (1984), 459--470). How large is the class of such lattices? Is every finite lattice embeddable in the lattice $C(G)$ for some finite group $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04935 | KOU-9.63

### Kourovka Notebook Problem 9.63

S / solved | Kourovka 群论问题集

Is a finite group of the form $G=ABA$ soluble if $A$ is an abelian subgroup and $B$ is a cyclic subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04936 | KOU-9.64

### Kourovka Notebook Problem 9.64

S / solved | Kourovka 群论问题集

Is it true that, in a group of the form $ G=AB$, every subgroup $N$ of $A \cap B$ which is subnormal both in $A$ and in $B$ is subnormal in $G$? The answer is affirmative in the case of finite groups (H. Wielandt).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04937 | KOU-9.65

### Kourovka Notebook Problem 9.65

O / open | Kourovka 群论问题集

Is a locally soluble group periodic if it is a product of two periodic subgroups?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04938 | KOU-9.66

### Kourovka Notebook Problem 9.66

S / solved | Kourovka 群论问题集

a) *B. Jonsson's Conjecture:* elementary equivalence is preserved under taking free products in the class of all groups, that is, if $Th(G_1)=Th(G_2)$ and $Th(H_1)=Th(H_2)$ for groups $G_1$, $G_2$, $H_1$, $H_2$, then $Th(G_1*H_1)=Th(G_2*H_2)$.

b) It may be interesting to consider also the following *weakened conjecture:* if $Th(G_1)=Th(G_2)$ and $Th(H_1)=Th(H_2)$ for countable groups $G_1$, $G_2$, $H_1$, $H_2$, then for any numerations of the groups $G_i$, $ H_i$ there are $m$-reducibility $T_1\mathop{\equiv}\limits_{m}T_2$ and Turing reducibility $T_1\mathop{\equiv}\limits_{T}T_2$, where $T_i$ is the set of numbers of all theorems in $Th(G_i*H_i)$. A proof of this weakened conjecture would be a good illustration of application of the reducibility theory to solving concrete mathematical problems.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04939 | KOU-9.67

### Kourovka Notebook Problem 9.67

S / solved | Kourovka 群论问题集

(A. Tarski). Let $F_n$ be a free group of rank $ n$; is it true that $ Th(F_2)=Th(F_3)$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04940 | KOU-9.68

### Kourovka Notebook Problem 9.68

O / open | Kourovka 群论问题集

Let ${\mathfrak{V}}$ be a variety of groups which is not the variety of all groups, and let $p$ be a prime. Is there a bound on the $ p $-lengths of the finite $p $-soluble groups whose Sylow $ p $-subgroups are in ${\mathfrak{V}}$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04941 | KOU-9.69

### Kourovka Notebook Problem 9.69

S / solved | Kourovka 群论问题集

(P. Cameron). Let $G$ be a finite primitive permutation group and suppose that the stabilizer $G_{\alpha}$ of a point $\alpha$ induces on some of its orbits $\Delta \ne \{ \alpha \}$ a regular permutation group. Is it true that $|G_{\alpha}|=|\Delta |$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04942 | KOU-9.7

### Kourovka Notebook Problem 9.7

O / open | Kourovka 群论问题集

(A. M. St\"epin). Does there exist an infinite finitely generated amenable group of bounded exponent?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04943 | KOU-9.70

### Kourovka Notebook Problem 9.70

O / open | Kourovka 群论问题集

Prove or disprove the conjecture of P. Cameron (*Bull. London Math. Soc.*, **13**, no. 1 (1981), 1--22): if $G$ is a finite primitive permutation group of subrank $m$, then either the rank of $G$ is bounded by a function of $m$ or the order of a point stabilizer is at most $m$. The *subrank* of a transitive permutation group is defined to be the maximum rank of the transitive constituents of a point stabilizer.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04944 | KOU-9.71

### Kourovka Notebook Problem 9.71

O / open | Kourovka 群论问题集

Is it true that every infinite 2-transitive permutation groups with locally soluble point stabilizer has a non-trivial irreducible finite-dimensional representation over some field?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04945 | KOU-9.72

### Kourovka Notebook Problem 9.72

O / open | Kourovka 群论问题集

Let $G_1$ and $G_2$ be Lie groups with the following property: each $G_i$ contains a nilpotent simply connected normal Lie subgroup $B_i$ such that $G_i/B_i \cong SL_2(K)$, where $K={\mathbb{R}$ or ${\mathbb{C}$. Assume that $G_1$ and $G_2$ are contained as closed subgroups in a topological group $G$, that $G_1 \cap G_2 \geq B_1B_2$, and that no non-identity Lie subgroup of $B_1 \cap B_2$ is normal in $G$. Can it then be shown (perhaps by using the method of ``amalgams'' from the theory of finite groups) that the nilpotency class and the dimension of $B_1B_2$ is bounded?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04946 | KOU-9.73

### Kourovka Notebook Problem 9.73

S / solved | Kourovka 群论问题集

Let ${\mathfrak{F}}$ be any local formation of finite soluble groups containing all finite nilpotent groups. Prove that $H^{{\mathfrak{F}}}K=KH^{{\mathfrak{F}}}$ for any two subnormal subgroups $H$ and $K$ of an arbitrary finite group $G$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04947 | KOU-9.74

### Kourovka Notebook Problem 9.74

S / solved | Kourovka 群论问题集

Find all local formations ${\mathfrak{F}}$ of finite groups such that every finite minimal non-${\mathfrak{F}}$-group is either a Shmidt group (that is, a non-nilpotent finite group all of whose proper subgroups are nilpotent) or a group of prime order.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04948 | KOU-9.75

### Kourovka Notebook Problem 9.75

O / open | Kourovka 群论问题集

Find all local formations ${\mathfrak{F}}$ of finite groups such that, in every finite group, the set of ${\mathfrak{F}}$-subnormal subgroups forms a lattice.

*Comment of 2015*: Subgroup-closed formations with this property are described in (Xiaolan Yi, S. F. Kamornikov, *J. Algebra*, **444** (2015), 143--151).

问题集入口：https://kourovkanotebookorg.wordpress.com/

04949 | KOU-9.76

### Kourovka Notebook Problem 9.76

O / open | Kourovka 群论问题集

We define a *Golod group* to be the $r$-generated, $r \geq 2$, subgroup $\left< 1 + x_1+I,\right.$ $\left. 1+x_2+I,\ldots ,1+x_r+I\right>$ of the adjoint group $1 + F/I$ of the factor-algebra $F/I$, where $F$ is a free algebra of polynomials without constant terms in non-commuting variables $x_1, x_2, \ldots , x_r$ over a field of characteristic $p>0$, and $I$ is an ideal of $F$ such that $F/I$ is a non-nilpotent nil-algebra (see E. S. Golod, *Amer. Math. Soc. Transl. (2)*, **48** (1965), 103--106). Prove that in Golod groups the centralizer of every element is infinite.

Note that Golod groups with infinite centre were constructed by A. V. Timofeenko (*Math. Notes*, **39**, no. 5 (1986), 353--355); independently by other methods the same result was obtained in the 90s by L. Hammoudi.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04950 | KOU-9.77

### Kourovka Notebook Problem 9.77

O / open | Kourovka 群论问题集

Does there exist an infinite finitely generated residually finite binary finite group all of whose Sylow subgroups are finite? A. V. Rozhkov (Dr. of Sci. Disser., 1997) showed that such a group does exist if the condition of finiteness of the Sylow subgroups is weakened to local finiteness.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04951 | KOU-9.78

### Kourovka Notebook Problem 9.78

O / open | Kourovka 群论问题集

Does there exist a periodic residually finite $ F^{\displaystyle{*}}$-group (see Archive, 7.42) all of whose Sylow subgroups are finite and which is not binary finite?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04952 | KOU-9.79

### Kourovka Notebook Problem 9.79

S / solved | Kourovka 群论问题集

(A. G. Kurosh). Is every group with the minimum condition countable?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04953 | KOU-9.8

### Kourovka Notebook Problem 9.8

S / solved | Kourovka 群论问题集

Does there exist a finitely generated simple group of intermediate growth?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04954 | KOU-9.80

### Kourovka Notebook Problem 9.80

S / solved | Kourovka 群论问题集

Are the 2-elements of a group with the minimum condition contained in its locally finite radical?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04955 | KOU-9.81

### Kourovka Notebook Problem 9.81

S / solved | Kourovka 群论问题集

Does there exist a simple group with the minimum condition possessing a non-trivial quasi-cyclic subgroup?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04956 | KOU-9.82

### Kourovka Notebook Problem 9.82

S / solved | Kourovka 群论问题集

An infinite group, all of whose proper subgroups are finite, is called *quasi-finite*. Is it true that an element of a quasi-finite group $G$ is central if and only if it is contained in infinitely many subgroups of $G$?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04957 | KOU-9.83

### Kourovka Notebook Problem 9.83

O / open | Kourovka 群论问题集

Suppose that $G$ is a (periodic) $ p $-conjugacy biprimitively finite group (see 6.59) which has a finite Sylow $ p $-subgroup. Is it then true that all Sylow $p $-subgroups of $G$ are conjugate?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04958 | KOU-9.84

### Kourovka Notebook Problem 9.84

O / open | Kourovka 群论问题集

a) Is every binary finite 2-group of finite exponent locally finite?

b) The same question for $p $-groups for $p > 2$.

问题集入口：https://kourovkanotebookorg.wordpress.com/

04959 | KOU-9.9

### Kourovka Notebook Problem 9.9

O / open | Kourovka 群论问题集

Does there exist a finitely generated group which is not nilpotent-by-finite and whose growth function has as a majorant a function of the form $c^{\sqrt{n}}$ where $ c$ is a constant greater than 1?

问题集入口：https://kourovkanotebookorg.wordpress.com/

04960 | AIM-GEOMETRIC_GROUP_THEORY-0032

### Lamination depth is a commensurability invariant, with rank bounds

S / solved | AIM 研讨会问题汇编

Every outer automorphism $\phi\in\operatorname{Out}(F_n)$ admits a poset $P$ of attracting laminations. What is the length of the longest chain in $P$? Is the length an invariant of $G$?

来源字段：http://aimpl.org/freebycyclic/7/

04961 | OWR-4428-010

### Lamplighter Subgroups and Irrational $\ell^2$-Betti Numbers

O / open | Oberwolfach 报告问题汇编

Is $C(G)\not\subseteq\mathbb Q$ equivalent to $\mathbb Z/p\wr\mathbb Z\subseteq G$ for some prime $p$? More generally, is it equivalent to $\mathbb Z/p\wr K\subseteq G$ for some prime $p$ and some infinite finitely generated group $K$?

来源字段：https://doi.org/10.4171/owr/2010/40

04962 | OWR-14602-011

### Langlands Packets Across Extended Pure Inner Forms

P / partially_solved | Oberwolfach 报告问题汇编

For a discrete Langlands parameter and all extended pure inner twists of a quasi-split group $G$, should the union of the associated $L$-packets be parametrized by irreducible representations of the parameter centralizer, compatibly with Kottwitz's bijection between basic isocrystals and the character group of the dual center, with the quasi-split packet containing a unique Whittaker-generic constituent and satisfying the expected endoscopic identities?

来源字段：https://doi.org/10.4171/owr/2016/20

04963 | AIM-GEOMETRIC_GROUP_THEORY-0157

### Large-group transfer and parity obstructions for power quotients of Out(F_n)

P / partially_solved | AIM 研讨会问题汇编

**Reconstructed Problem 3.6.** Does there exist a positive integer \(p\) such that \[ B_p(\operatorname{Out}(F_n)):= \operatorname{Out}(F_n)/ \left\langle\!\left\langle\gamma^p: \gamma\in\operatorname{Out}(F_n)\right\rangle\!\right\rangle \] is infinite?

The PDF itself, not merely the JSON extraction, prints \[ \text{“Does there exist }p\to0\text{ such that } \operatorname{Out}(F_n)/\langle\!\langle\gamma^p\rangle\!\rangle, \text{ all }\gamma\in\operatorname{Out}(F_n),\text{ is infinite?”} \] The superscript \(p\), the double normal-closure brackets, and the scope “all \(\gamma\in\operatorname{Out}(F_n)\)” are visible in the PDF. Thus the intended denominator is \[ \left\langle\!\left\langle \gamma^p\mid\gamma\in\operatorname{Out}(F_n) \right\rangle\!\right\rangle. \] It is nevertheless not meaningful after the existential quantifier if \(p\) is an integer exponent. Taking \(p=0\) would make every relator \(\gamma^0=1\), leave \(\operatorname{Out}(F_n)\) unchanged, and fail to address the preceding sentence, which asks for an infinite torsion quotient. The source therefore contains an unresolved typo.

The mathematically coherent reconstruction used here is:

This is the maximal quotient of \(\operatorname{Out}(F_n)\) of exponent dividing \(p\), a Burnside-type quotient. Two plausible stronger repairs are “\(p\gg0\)” and “as \(p\to\infty\).” The results below also give infinitely many exponents in ranks two and three, so they address those readings partially. The literal printed reading has no nontrivial mathematical content and is not silently adopted.

来源字段：https://aimath.org/WWN/outerauto/outerauto.pdf

04964 | AIM-GEOMETRIC_GROUP_THEORY-0156

### Largeness of Out(F_n): complete rank classification and quantitative transfer consequences

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.5.** Does there exist a finite-index subgroup \(\Gamma < \operatorname{Out}(F_n)\) such that \(\Gamma \to F_2\)?

It then records discussion about applicable techniques, train tracks, an infinite torsion quotient, non-bounded generation, and bounded cohomology.

Immediately before Problem 3.5 it says: “if \(n\ge4\), is \(\operatorname{Out}(F_n)\) large? Meaning, does there exist a finite-index subgroup which acts on a tree, thus maps onto a free group?” Therefore:

- the intended rank range is \(n\ge4\); - \(F_2\) is the nonabelian free group of rank two; - the arrow is intended to be an **epimorphism**, so the reconstructed formula is \(\Gamma\twoheadrightarrow F_2\).

This is the standard definition that \(\operatorname{Out}(F_n)\) is **large**. The phrase about acting on a tree is motivation, not a general equivalence: an epimorphism to \(F_2\) gives an action on its Cayley tree, but an arbitrary nontrivial tree action need not produce a free quotient.

来源字段：https://aimath.org/WWN/outerauto/outerauto.pdf

04965 | AIM-GEOMETRIC_GROUP_THEORY-0059

### Largest actions from universally elliptic Deligne stabilizers

P / partially_solved | AIM 研讨会问题汇编

Is there a ``largest'' acylindrically hyperbolic space on which these Artin groups act that sees all loxodromic elements?

The exact AIM record asks:

It is Problem 4.5 in the “Non-positive curvature” section of the AIM workshop list *Geometry and topology of Artin groups*. The immediately preceding records ask whether all irreducible nonspherical Artin groups are acylindrically hyperbolic and ask for a description of the hyperbolic space. Thus the broad reading concerns irreducible nonspherical Artin groups for which acylindrical hyperbolicity is known or expected.

There are two notions in the question that must be separated. In the terminology now standard after Abbott--Balasubramanya--Osin (ABO), “largest” means largest in the domination poset of **cobounded** acylindrical actions. “Sees all loxodromic elements” means **universal**: every element that is loxodromic in some acylindrical action is loxodromic in the chosen action. A largest action is universal, but universality alone does not prove largestness.

The broad Artin-group question remains beyond what is proved here. The main result below gives a positive answer, with an explicit natural space, for the two-dimensional hyperbolic-type class studied by Martin--Przytycki under their finite-neighbor hypothesis.

来源字段：http://aimpl.org/geomartingp/4/

04966 | OWR-10252930-007

### Largest Product-Free Subsets of Alternating Groups

P / partially_solved | Oberwolfach 报告问题汇编

What is the maximum size of a product-free subset of the alternating group $A_n$?

来源字段：https://doi.org/10.4171/owr/2022/22

04967 | OWR-12697684-008

### Largest Product-Free Subsets of Alternating Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a product-free subset $A\subset A_n$, determine the largest possible value of $|A|$.

来源字段：https://doi.org/10.4171/owr/2023/1

04968 | OWR-1784-005

### Lattices and Covolumes in Davis-Complex Automorphism Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a Davis complex and $G=\operatorname{Aut}(X)$, with lattices defined as discrete subgroups $\Gamma<G$ of finite covolume $\mu(\Gamma\backslash G)$. Does $G$ admit uniform or nonuniform lattices, and how can they be constructed? What is the set of covolumes $\mu(\Gamma\backslash G)$; in particular, is it discrete?

来源字段：https://doi.org/10.4171/owr/2008/03

04969 | OWR-14298160-008

### Layer Growth in the Coxeter Galaxy

P / partially_solved | Oberwolfach 报告问题汇编

Determine how the dimensions of the layers of the Coxeter galaxy grow with rank, and whether the defining Coxeter graph determines the number of vertices in the intersection of its connected component with a given layer.

来源字段：https://doi.org/10.4171/owr/2024/4

04970 | AIM-GEOMETRIC_GROUP_THEORY-0114

### LEF obstruction and a two-gate audit of the Penrose candidate

P / partially_solved | AIM 研讨会问题汇编

Is there a f.p. amenable infinite simple group? A construction of Juschenko and Monod (2012) gives a finitely generated infinite simple amenable group. **Conjecture:** The derived subgroup of the full group of the Penrose tiling is a candidate.

来源字段：http://aimpl.org/amenablediscrete/4/

04971 | OWR-1589-002

### Lifting Mapping-Class Subgroups to Surface Homeomorphisms

O / open | Oberwolfach 报告问题汇编

Let $S$ be a closed oriented surface of genus $g\geq2$, and let $\operatorname{MCG}(S)$ be its mapping class group. Which subgroups $\Gamma\leq\operatorname{MCG}(S)$ admit a lift, namely a homomorphism $\Gamma\to\operatorname{Homeo}(S)$ or $\Gamma\to\operatorname{Diff}(S)$ whose composition with the natural map to $\operatorname{MCG}(S)$ is the inclusion?

来源字段：https://doi.org/10.4171/owr/2007/33

04972 | OWR-1784-003

### Linear Divergence of Higher-Rank Lattices

P / partially_solved | Oberwolfach 报告问题汇编

Does every lattice in a semisimple Lie group of higher rank have linear divergence, where divergence measures the length of paths joining two points while avoiding a sufficiently large ball about a third point?

来源字段：https://doi.org/10.4171/owr/2008/03

04973 | OWR-4411-017

### Linear Representations of Ascending Free-Group Mapping Tori

O / open | Oberwolfach 报告问题汇编

Let $F_n$ be a free group of rank at least two, let $\phi:F_n\to F_n$ be irreducible, proper, and injective, and let $G$ be its mapping-torus fundamental group. Can $G$ be embedded in $SL_2(\mathbb C)$?

来源字段：https://doi.org/10.4171/owr/2010/30

04974 | AIM-GEOMETRIC_GROUP_THEORY-0031

### Linearity and two-dimensional representations of free-by-cyclic groups

P / partially_solved | AIM 研讨会问题汇编

Are free-by-cyclic groups linear? When do they embed in $\mathrm{SL}(2,\mathbb C)$?

来源字段：http://aimpl.org/freebycyclic/7/

04975 | OWR-14297743-003

### Linearity of Mapping Class Groups

O / open | Oberwolfach 报告问题汇编

Is the mapping class group of a surface $\Sigma$ linear, and can the question be formulated intrinsically in terms of $\Sigma$?

来源字段：https://doi.org/10.4171/owr/2023/33

04976 | AIM-GEOMETRIC_GROUP_THEORY-0074

### Local ellipticity and spherical parabolic containers

P / partially_solved | AIM 研讨会问题汇编

Let $A$ be an Artin group with subgroup $H$. If for all $h\in H$ there is a (spherical) parabolic containing $h$, does there exist a (spherical) parabolic containing $H$? Geometric version: If a group $G$ acts on $X$ and $H<G$ with each $h\in H$ fixing a point of $X$, then does $H$ fix a point? What about when $X$ has non-positive curvature?

来源字段：http://aimpl.org/geomartingp/7/

04977 | OWR-17140-003

### Localization of Types in Bruhat-Tits Buildings

O / open | Oberwolfach 报告问题汇编

Let $G$ be a connected reductive $p$-adic group, let $\mathcal B(G)$ be its Bruhat-Tits building, and let $x\in\mathcal B(G)$. For the finite-length representation $\tau(y,g)$ arising in the Mackey decomposition of a type, is there a bounded convex neighborhood $\Gamma$ of $x$, containing $\mathcal B(G)^H$, such that every geodesic $[x,z]$ with $z\notin\Gamma$ meets either $$\operatorname{proj}_0^{-1}(\mathcal B(G^0)\setminus\{x\})$$ or $$\Omega_{t+}\setminus(\Omega_{t+}\cap\Xi_t)\quad\text{for some }0<t<s_0,$$ thereby excluding additional types far from $x$?

来源字段：https://doi.org/10.4171/owr/2019/44

04978 | AIM-GEOMETRIC_GROUP_THEORY-0042

### Lorentzian weighted-star Artin groups with solvable word problem

P / partially_solved | AIM 研讨会问题汇编

We call an Artin group *hyperbolic* if the quadratic form of the associated Coxeter group is rank $(n,1)$.

Find more examples of $\geq 3$-dimensional hyperbolic Artin groups (whose diagrams have no edges labeled $\infty$) where the word problem is solvable.

来源字段：http://aimpl.org/geomartingp/2/

04979 | AIM-GEOMETRIC_GROUP_THEORY-0101

### Low complexity does not have a single amenability threshold

P / partially_solved | AIM 研讨会问题汇编

**Are full groups of labeled graphs of "low" complexity (and "low" growth) amenable? (E.g. "low" = polynomial.)**

**What about different interpretations of complexity?**

来源字段：http://aimpl.org/amenablediscrete/2/

04980 | OWR-11583-009

### Low-Degree Cohomology of Finite Groups of Lie Type

P / partially_solved | Oberwolfach 报告问题汇编

Compute the first and second cohomology groups of finite groups of Lie type with coefficients in simple modules whose highest weight is fundamental dominant.

来源字段：https://doi.org/10.4171/owr/2012/16

04981 | AIM-GEOMETRIC_GROUP_THEORY-0044

### Low-rank closure of spherical Artin groups under abstract isomorphism

P / partially_solved | AIM 研讨会问题汇编

Is the class of spherical Artin groups closed under isomorphism? In other words, if an Artin group $A$ is isomorphic to a spherical Artin group, must $A$ be spherical?

来源字段：http://aimpl.org/geomartingp/3/

04982 | OWR-14749-010

### Lück Approximation for Kernels over Arbitrary Fields

O / open | Oberwolfach 报告问题汇编

Let $F$ be a finitely generated free group, let $K\subseteq\mathbb C$ be a field, and let $$ F\triangleright N_1\triangleright N_2\triangleright\cdots $$ be normal subgroups with $N=\bigcap_i N_i$, $G_i=F/N_i$, and $G=F/N$. For every $A\in\operatorname{Mat}_{n\times m}(K[F])$, is $$ \lim_{i\to\infty}\dim_{G_i}\ker\phi_{G_i}^A=\dim_G\ker\phi_G^A $$ hold?

来源字段：https://doi.org/10.4171/owr/2016/41

04983 | AIM-GEOMETRIC_GROUP_THEORY-0011

### Marked profinite data, fibre rank, and rational BNS transfer

P / partially_solved | AIM 研讨会问题汇编

**Profinite invariants of free-by-cyclic groups.** Which properties of free-by-cyclic groups are invariants of the profinite completion? How does the BNS invarian behave under profinite isomorphism of free-by-cyclic groups?

来源字段：http://aimpl.org/freebycyclic/2/

04984 | AIM-GEOMETRIC_GROUP_THEORY-0046

### Matching transversals reduce the mixed-label infinity-free case

P / partially_solved | AIM 研讨会问题汇编

Solve the isomorphism problem for $\infty$-free 2-dimensional Artin groups.

来源字段：http://aimpl.org/geomartingp/3/

04985 | AIM-GEOMETRIC_GROUP_THEORY-0040

### Maximal-core compiler for Artin-group word algorithms

P / partially_solved | AIM 研讨会问题汇编

Is there an explicit algorithm to solve the word problem for all Artin groups?

来源字段：http://aimpl.org/geomartingp/2/

04986 | OWR-17472-016

### Maximality of Normalizers of Quasisimple Subgroups

O / open | Oberwolfach 报告问题汇编

In the Aschbacher-Scott setting, when is the normalizer $M=N_G(S)$ of a quasisimple subgroup $S<G$ acting irreducibly on $V$ a maximal subgroup of the classical group $G$?

来源字段：https://doi.org/10.4171/owr/2020/9

04987 | OWR-4333-006

### Measure Equivalence of Limit and Free Groups

O / open | Oberwolfach 报告问题汇编

Are all limit groups measure equivalent to a free group?

来源字段：https://doi.org/10.4171/owr/2010/20

04988 | OWR-4411-012

### Measure-Independence of Positive Random-Walk Entropy

O / open | Oberwolfach 报告问题汇编

Let $G$ be finitely generated, and let $\mu_1$ and $\mu_2$ be symmetric finitely supported measures generating $G$. If the entropy $h(\mu_1)$ is positive, must $h(\mu_2)$ also be positive?

来源字段：https://doi.org/10.4171/owr/2010/30

04989 | OWR-14299911-022

### Median Actions and Kazhdan Property for Brin Group 2V

O / open | Oberwolfach 报告问题汇编

Does Brin's higher-dimensional Thompson group $2V$ admit an action with unbounded orbits on a median graph? Does $2V$ have Kazhdan's property $(T)$?

来源字段：https://doi.org/10.4171/owr/2026/8

04990 | OWR-14299911-012

### Median Actions and Residual Finiteness for Hyperbolic Groups with Sphere Boundary

O / open | Oberwolfach 报告问题汇编

Let $G$ be a hyperbolic group with boundary homeomorphic to $S^2$. Can one prove that $G$ admits an action with unbounded orbits on a median space, equivalently that $G$ does not have Kazhdan's property $(T)$? More generally, is every medianized hyperbolic group cubulable, or can one at least prove that it is residually finite?

来源字段：https://doi.org/10.4171/owr/2026/8

04991 | OWR-14299911-021

### Metric Properness of Wreath-Product Actions on Median Graphs

O / open | Oberwolfach 报告问题汇编

Let $G$ act properly on a median graph and let $N\triangleleftefteq G$. If $G/N$ does not act metrically properly on any median graph, can the permutational wreath product $\mathbb{Z}_2\wr_{G/N}G$ nevertheless act metrically properly on a median graph? Characterize when proper median-graph actions of such wreath products can be upgraded to metrically proper actions.

来源字段：https://doi.org/10.4171/owr/2026/8

04992 | OWR-1703863-008

### Minimal Fold of Kazhdan Hyperbolic Triangle Groups

O / open | Oberwolfach 报告问题汇编

What is the smallest $k$ for which there exists a generalized $k$-fold triangle group that is hyperbolic and has Kazhdan property (T)?

来源字段：https://doi.org/10.4171/owr/2020/16

04993 | OWR-793-006

### Minimal Subdegrees of Twisted-Wreath Permutation Groups

O / open | Oberwolfach 报告问题汇编

For every finite primitive permutation group $G=\operatorname{Soc}(G)\rtimes P\leq H\wr S_k$ of O'Nan-Scott twisted-wreath type acting on $\Delta^k$, with primitive component $H\leq\operatorname{Sym}(\Delta)$, is $\operatorname{MinSubDeg}(G)<k\,\operatorname{MinSubDeg}(H)$?

来源字段：https://doi.org/10.4171/owr/2005/12

04994 | OWR-4535-014

### Minimal-Dimensional Cocompact Models for Mapping Class Groups

O / open | Oberwolfach 报告问题汇编

Does there exist a contractible cell complex of dimension $4g-5$ with a proper cocompact action of the mapping class group $\operatorname{MCG}(S_g)$, and are there obstructions to its existence?

来源字段：https://doi.org/10.4171/owr/2010/53

04995 | OWR-12491-002

### Minimal-Genus Surface Subgroups of Random Groups

O / open | Oberwolfach 报告问题汇编

For random groups at density $D<1/2$, do the simplest homologically essential closed surface subgroups have genus $O(n/\log n)$ as the relator length $n$ tends to infinity?

来源字段：https://doi.org/10.4171/owr/2013/28

04996 | OWR-14299097-014

### Mixed and Nonsingular Identities in Finite Lie-Type Groups

P / partially_solved | Oberwolfach 报告问题汇编

For the remaining families of finite groups of Lie type, determine the length of a shortest mixed identity and the length of a shortest nonsingular identity.

来源字段：https://doi.org/10.4171/owr/2025/16

04997 | OWR-14298365-009

### Mixing Times and Character Bounds for Simple Groups

P / partially_solved | Oberwolfach 报告问题汇编

For finite simple groups, determine sharp mixing-time and McKay-graph diameter bounds in terms of character ratios, including the Lubotzky-Shalev and Liebeck-Shalev-Tiep conjectures.

来源字段：https://doi.org/10.4171/owr/2024/17

04998 | OWR-14299292-007

### Modular Isomorphism for Class-Two Exponent-p Groups

O / open | Oberwolfach 报告问题汇编

Let $G$ and $H$ be finite $p$-groups of exponent $p$ and nilpotency class two over a field $k$ of characteristic $p$. If the group algebras $kG$ and $kH$ are isomorphic, must $G$ and $H$ be isomorphic? Resolve the modular isomorphism problem for this group class over arbitrary fields.

来源字段：https://doi.org/10.4171/owr/2025/27

04999 | OWR-8415341-002

### Modular Structure of Exceptional Finite Lie-Type Groups

O / open | Oberwolfach 报告问题汇编

Determine the radical subgroups of exceptional groups of Lie type in bad characteristic; determine $p$-modular decomposition matrices for finite groups of Lie type, including basic sets, unitriangularity, and complete results for exceptional families; and determine the modular Harish-Chandra series for groups of Lie type.

来源字段：https://doi.org/10.4171/owr/2021/38

05000 | AIM-GEOMETRIC_GROUP_THEORY-0099

### Morse solved; factor reductions for Chacon and Rudin--Shapiro

P / partially_solved | AIM 研讨会问题汇编

Does the topological full group of the (Morse, Chacon, Rudin-Shapiro) substitution subshift contain a subgroup of intermediate growth?

来源字段：http://aimpl.org/amenablediscrete/2/

05001 | AIM-GEOMETRIC_GROUP_THEORY-0006

### Morse subgroups of free-by-cyclic groups

P / partially_solved | AIM 研讨会问题汇编

**Which subgroups of free-by-cyclic groups are Morse?**

来源字段：http://aimpl.org/freebycyclic/1/

05002 | OWR-793-002

### Moufangness of Two-Moufang and Half-Moufang Polygons

P / partially_solved | Oberwolfach 报告问题汇编

Is every $2$-Moufang generalized $n$-gon, and every half-Moufang polygon, a Moufang polygon?

来源字段：https://doi.org/10.4171/owr/2005/12

05003 | OWR-12697693-011

### Multiplicative p-Groups of Braces

O / open | Oberwolfach 报告问题汇编

Which finite $p$-groups occur as multiplicative groups of braces; in particular, what is a smallest counterexample and what necessary or sufficient conditions characterize such groups?

来源字段：https://doi.org/10.4171/owr/2023/9

05004 | AIM-GEOMETRIC_GROUP_THEORY-0091

### No atomless invariant measure on the perfect kernel of Sub(F)

S / solved | AIM 研讨会问题汇编

Is there an invariant \emph{continuous} probability measure for the action of \(F\) acting on its perfect kernel (by conjugation)?

The record is Problem 1.6 in the workshop section *Thompson group F and groups of homeomorphisms of the interval and the circle*. The immediately preceding Problem 1.5 defines transfinite Cantor--Bendixson derivatives and asks for the Cantor--Bendixson rank of the space of subgroups of \(F\). This fixes the intended object: “its perfect kernel” means the Cantor--Bendixson perfect kernel \[ P\subseteq \operatorname{Sub}(F), \] not the derived group \(F'\), despite the fact that \(F'\) is itself a perfect group.

In this descriptive/topological setting, a *continuous probability measure* means an atomless (nonatomic) Borel probability measure: \(\mu(\{H\})=0\) for every subgroup \(H\). It cannot mean merely a measure defining a continuous functional on continuous functions, since every Borel probability on a compact metrizable space does so and a Dirac mass at an invariant subgroup would make the question trivial. “A measure on \(P\)” is taken to mean \(\mu(P)=1\), not that its topological support must equal all of \(P\). The negative result below is stronger and therefore also rules out the full-support version.

The current AIM workshop page still identifies the September 2016 workshop and links to its open-problem list.

Under this reconstruction, the answer is **no**.

来源字段：http://aimpl.org/amenablediscrete/1/

05005 | OWR-16160-008

### Non-Bilipschitz Nets and Lattices in Amenable Groups

O / open | Oberwolfach 报告问题汇编

Does every finitely generated amenable group $\Gamma$ contain a coarsely dense net not bi-Lipschitz equivalent to $\Gamma$? Does some solvable or nilpotent Lie group contain two cocompact lattices that are not bi-Lipschitz equivalent?

来源字段：https://doi.org/10.4171/owr/2018/19

05006 | OWR-12697693-006

### Non-Desarguesian Plane Lattices as Bricks

O / open | Oberwolfach 报告问题汇编

Can the subspace lattice of a non-Desarguesian projective plane be the underlying lattice of a brick?

来源字段：https://doi.org/10.4171/owr/2023/9

05007 | OWR-4990377-004

### Non-Helly Coarse-Helly Actions and Coarse-Median Artin Groups

O / open | Oberwolfach 报告问题汇编

Does there exist a group acting geometrically on a coarse Helly graph that is not Helly, and which Artin groups are coarse median?

来源字段：https://doi.org/10.4171/owr/2021/8

05008 | AIM-GEOMETRIC_GROUP_THEORY-0085

### Non-linear residual-finiteness certificates for Coxeter groups

P / partially_solved | AIM 研讨会问题汇编

Prove Coxeter groups are residually finite without appealing to linearity.

来源字段：http://aimpl.org/geomartingp/8/

05009 | OWR-14299097-004

### Non-Locally-Finite Simple Proper Localities

O / open | Oberwolfach 报告问题汇编

Does there exist a simple proper locality that is not locally finite?

来源字段：https://doi.org/10.4171/owr/2025/16

05010 | OWR-16406-008

### Nonarithmetic Lattices in Higher-Dimensional Complex Hyperbolic Space

P / partially_solved | Oberwolfach 报告问题汇编

Do nonarithmetic lattices exist in $\operatorname{Isom}(\mathbb H_{\mathbb C}^n)\cong\operatorname{SU}(n,1)$ for $n\geq3$?

来源字段：https://doi.org/10.4171/owr/2018/31

05011 | OWR-1319-018

### Noncoherence of Higher-Dimensional Hyperbolic Lattices

P / partially_solved | Oberwolfach 报告问题汇编

Are all lattices in $\operatorname{Isom}(\mathbb H^n)$ noncoherent for $n\geq4$?

来源字段：https://doi.org/10.4171/owr/2006/33

05012 | OWR-4411-004

### Noncoherence of Higher-Rank Semisimple Lattices

O / open | Oberwolfach 报告问题汇编

Let $G$ be a connected semisimple real Lie group that is not locally isomorphic to $\operatorname{SL}_2(\mathbb R)$ or $\operatorname{SL}_2(\mathbb C)$, and let $\Gamma<G$ be a lattice. Is every infinite-index subgroup of $\Gamma$ of infinite covolume?

来源字段：https://doi.org/10.4171/owr/2010/30

05013 | OWR-16415-017

### Noncongruence Normal Subgroups of Mapping Class Groups

P / partially_solved | Oberwolfach 报告问题汇编

What are the proper normal subgroups of a mapping class group that are not contained in any congruence subgroup, or alternatively in the union of the congruence subgroups?

来源字段：https://doi.org/10.4171/owr/2018/40

05014 | OWR-12007-016

### Nonconjugate $\operatorname{SL}_3$-Character-Equivalent Free-Group Words

O / open | Oberwolfach 报告问题汇编

Do there exist elements $g,g'\in F_2$ that are $\operatorname{SL}_3(\mathbb C)$-character equivalent, meaning that all corresponding representation traces agree, but are not conjugate?

来源字段：https://doi.org/10.4171/owr/2012/35

05015 | OWR-13750328-018

### Nonexoticity of Block Fusion Systems

O / open | Oberwolfach 报告问题汇编

Prove that the saturated fusion system determined by any $\ell$-block of a finite group is nonexotic.

来源字段：https://doi.org/10.4171/owr/2023/19

05016 | OWR-13750328-019

### Nonnegativity of Fusion-System Multiplicities

O / open | Oberwolfach 报告问题汇编

For every saturated fusion system $\mathcal F$ and every nonnegative integer $d$, prove $m(\mathcal F,d)\geq 0$.

来源字段：https://doi.org/10.4171/owr/2023/19

05017 | OWR-2053-018

### Nontrivial Floyd Boundaries Beyond Relative Hyperbolicity

O / open | Oberwolfach 报告问题汇编

Does there exist a group $G$ with nontrivial Floyd boundary $\partial_fG$ that is not relatively hyperbolic with respect to any family of finitely generated subgroups? For the known infinitely presented candidate group $H$, is $\partial_fH$ trivial, and is $H$ non-relatively-hyperbolic with respect to every such family?

来源字段：https://doi.org/10.4171/owr/2008/42

05018 | OWR-14299580-007

### Normal Complements in Integral Group-Ring Units

O / open | Oberwolfach 报告问题汇编

For a finite nilpotent group $G$, does $G$ have a normal complement in the normalized unit group $U_1(\mathbb Z[G])$? If so, can the complement be chosen torsion-free?

来源字段：https://doi.org/10.4171/owr/2025/49

05019 | AIM-GEOMETRIC_GROUP_THEORY-0122

### Nucleus-core reduction for amenability of contracting groups

P / partially_solved | AIM 研讨会问题汇编

Let \(X\) be a finite alphabet and let \(G\leq \operatorname{Aut}(X^*)\) be a contracting self-similar group. Must \(G\) be amenable?

来源字段：http://aimpl.org/amenablediscrete/4/

05020 | AIM-GEOMETRIC_GROUP_THEORY-0113

### Obstruction package for amenable hereditary just-infinite groups

P / partially_solved | AIM 研讨会问题汇编

**Definition:** A *hereditary just-infinite* group $G$ is a residually finite group such that every finite index subgroup of it is just-infinite.

Is there an amenable hereditary just-infinite group that is not elementary amenable? Is there a finitely presented one?

来源字段：http://aimpl.org/amenablediscrete/4/

05021 | AIM-GEOMETRIC_GROUP_THEORY-0028

### Open-facet conventions, free-by-cyclic obstructions, and a cross-polytope realization family

P / partially_solved | AIM 研讨会问题汇编

If $P$ is a polytope in $\mathbb{R}^n$ with $n \geq 2$ with a marking on its top dimensional faces, determine whether the cone over the marked faces of $P$ corresponds to a BNS invariant of a (free-by-cyclic) group.

来源字段：http://aimpl.org/freebycyclic/6/

05022 | OWR-14299097-016

### Optimal Character Bounds for Almost Quasisimple Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a finite almost quasisimple group and $g\in G\setminus Z(G)$. Find an explicit exponent $0\leq\alpha(g)<1$, as small as possible, such that $|\chi(g)|\leq\chi(1)^{\alpha(g)}$ for every irreducible complex character $\chi$ of $G$.

来源字段：https://doi.org/10.4171/owr/2025/16

05023 | OWR-13750328-007

### Optimal Character-Ratio Bounds for Almost Quasisimple Groups

O / open | Oberwolfach 报告问题汇编

For a finite almost quasisimple group $G$ and $g\in G\setminus Z(G)$, find the smallest explicit $\gamma(g)\in(0,1)$ satisfying $|\chi(g)|/\chi(1)\leq\gamma(g)$ for every nonlinear $\chi\in\operatorname{Irr}(G)$, and the smallest explicit $\alpha(g)\in(0,1)$ satisfying $|\chi(g)|\leq\chi(1)^{\alpha(g)}$ for every $\chi\in\operatorname{Irr}(G)$.

来源字段：https://doi.org/10.4171/owr/2023/19

05024 | OWR-14298365-008

### Optimal Character-Ratio Bounds for Almost Quasisimple Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a finite almost quasisimple group $G$ and $g\notin Z(G)$, determine optimal explicit bounds for $|\chi(g)|/\chi(1)$ and for exponents $\alpha(g)$ satisfying $|\chi(g)|\leq\chi(1)^{\alpha(g)}$ for all irreducible characters $\chi$.

来源字段：https://doi.org/10.4171/owr/2024/17

05025 | OWR-11101920-011

### Optimal Lower Bounds for Character Proportions

O / open | Oberwolfach 报告问题汇编

What are the greatest lower bounds $\inf_G\theta(G)$ and $\inf_G\theta'(G)$ in the Thompson and Gallagher character-proportion bounds?

来源字段：https://doi.org/10.4171/owr/2022/39

05026 | AIM-GEOMETRIC_GROUP_THEORY-0094

### Optimal two-element invariable generation of Thompson's group F

S / solved | AIM 研讨会问题汇编

Solved. \(F\) is invariably generated by a finite set (Golan and Juschenko).

A finite subset \(S=\{x_1,\dots,x_n\}\subseteq G\) invariably generates \(G\) if \[ \left\langle g_1^{-1}x_1g_1,\dots,g_n^{-1}x_ng_n\right\rangle=G \] for every independently chosen \(g_1,\dots,g_n\in G\). Is Thompson's group \(F\) finitely invariably generated?

来源字段：http://aimpl.org/amenablediscrete/1/

05027 | OWR-8415341-008

### Orbit Computation for Constructible Maximal-Class p-Groups

O / open | Oberwolfach 报告问题汇编

For the infinite group $G$ acting on the finite quotients $(\theta-1)^n\Gamma/(\theta-1)^{n+e}\Gamma$ arising from constructible $p$-groups of maximal class, develop computational methods to determine its orbits as $n$ and $e$ vary.

来源字段：https://doi.org/10.4171/owr/2021/38

05028 | AIM-GEOMETRIC_GROUP_THEORY-0143

### Orbit equivalence, simplicial volume, and dimension detection by L2-support

P / partially_solved | AIM 研讨会问题汇编

Do closed, aspherical manifolds with OE fundamental groups have the same simplicial volume?

If, in addition, the simplicial volumes are positive, are the dimensions of the manifolds equal?

来源字段：http://aimpl.org/l2invariantsgroups/3/

05029 | AIM-GEOMETRIC_GROUP_THEORY-0068

### Order-two nonembedding and the equivariant Burau implication

P / partially_solved | AIM 研讨会问题汇编

Does the $Br_4$ triangle complex embed in a (locally finite?) $\widetilde A_2$ building?

Related to the Burau representation; if it doesn't embed, then the Burau rep is not faithful over a finite field.

来源字段：http://aimpl.org/geomartingp/5/

05030 | AIM-GEOMETRIC_GROUP_THEORY-0043

### Orthogonal fractions and minimal counterexample supports

P / partially_solved | AIM 研讨会问题汇编

If \[ ab^{-1}=cd^{-1}\quad\text{in }A_S, \qquad a\widetilde\wedge b=c\widetilde\wedge d=1, \tag{1.2} \] must $a=c$ and $b=d$?

来源字段：http://aimpl.org/geomartingp/2/

05031 | AIM-GEOMETRIC_GROUP_THEORY-0128

### Pair correlations in inverted orbits

P / partially_solved | AIM 研讨会问题汇编

\(G\curvearrowright X\), \(p,x,y\in X\). What can be said on \(P(x\in O_\Lambda(p)\operatorname{ and }y\in O_\Lambda(p))\)? Are these events positively/negatively correlated? (Here \(O_\Lambda(p)\) is the inverted orbit of \(p\).)

来源字段：http://aimpl.org/amenablediscrete/5/

05032 | OWR-1593-004

### Pálfy-Pudlák Finite-Lattice Representation Problem

O / open | Oberwolfach 报告问题汇编

For every nonempty finite lattice $\Lambda$, do there exist a finite group $G$ and a subgroup $H\leq G$ such that $\Lambda$ is isomorphic to the lattice $O_G(H)$ of overgroups of $H$ in $G$?

来源字段：https://doi.org/10.4171/owr/2007/37

05033 | AIM-GEOMETRIC_GROUP_THEORY-0107

### Percolation below one on groups of intermediate growth

S / solved | AIM 研讨会问题汇编

**Recovered question.** For every Cayley graph of every finitely generated group of intermediate growth, is the critical occupation probability $p_c$ for Bernoulli percolation strictly less than one?

来源字段：http://aimpl.org/amenablediscrete/3/

05034 | OWR-1965-002

### Perfect Images of Prosoluble Groups

O / open | Oberwolfach 报告问题汇编

Does there exist a prosoluble group with a nontrivial perfect image?

来源字段：https://doi.org/10.4171/owr/2008/20

05035 | OWR-13349-004

### Periodic Topologically Simple Locally Compact Groups

O / open | Oberwolfach 报告问题汇编

Does there exist a compactly generated totally disconnected locally compact group that is topologically simple, non-discrete, and periodic?

来源字段：https://doi.org/10.4171/owr/2014/47

05036 | OWR-15987-021

### Periodicity of Twisted Outer-Automorphism-Group Cohomology

O / open | Oberwolfach 报告问题汇编

For each fixed $i$ and prime $p$, determine whether $H^i(\operatorname{Out}(F_n),(\mathbb Z/p\mathbb Z)^n)$ is eventually periodic as $n$ increases; if so, ask the same question for coefficients given by a polynomial functor or a VIC-module.

来源字段：https://doi.org/10.4171/owr/2018/2

05037 | OWR-11583-011

### Perverse Deligne-Lusztig Equivalences for Abelian Defect Blocks

P / partially_solved | Oberwolfach 报告问题汇编

Prove that the Deligne-Lusztig-derived equivalence predicted by Broué's abelian-defect-group conjecture for suitable finite groups of Lie type is a perverse equivalence.

来源字段：https://doi.org/10.4171/owr/2012/16

05038 | OWR-1596-003

### Polyhedrality of Novikov Invertibility Loci

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Novikov-Invertibility Polyhedrality Question [OWR-1596-014]

Let $G$ be a finitely generated group, let $A$ be a $p\times p$ matrix over $\mathbb Z[G]$, and let $\mathcal N$ be the set of nonzero real characters $\xi\in\operatorname{Hom}(G,\mathbb R)$ for which $A$ is invertible over the Novikov completion $\mathbb Z[G]_\xi$ or $\mathbb Q[G]_\xi$. Is $\mathcal N$ defined by finitely many integral linear inequalities?

来源字段：https://doi.org/10.4171/owr/2007/40

05039 | OWR-15987-003

### Polynomial Degree of IA-Group Homology Functors

O / open | Oberwolfach 报告问题汇编

For every $k\geq0$, is the functor $H_k(\operatorname{IA}_\bullet)$ from the category $S(\mathbb Z)$ to abelian groups weakly polynomial of degree at most $3k$?

来源字段：https://doi.org/10.4171/owr/2018/2

05040 | OWR-722-001

### Positive Characters of Simply Connected Groups

O / open | Oberwolfach 报告问题汇编

Let $G$ be a connected simply connected compact Lie group, and let $f$ be a virtual character such that $f(x)\in\mathbb R_{\geq0}$ for every $x\in G$ and $\langle f,1\rangle=1$. Must $f=\chi\overline{\chi}$ for an irreducible complex character $\chi$ of $G$?

来源字段：https://doi.org/10.4171/owr/2004/12

05041 | OWR-5154-005

### Possible $L^{2}-Betti$ Numbers of a Fixed Group

P / partially_solved | Oberwolfach 报告问题汇编

For a countable discrete group $G$, determine the set $$\mathcal C(G)=\left\{\dim_{\mathcal N(G)}\ker T:k\geq1,\ T\in M_k(\mathbb QG)\right\}$$ of possible $L^2$-Betti numbers arising from matrices over its rational group ring.

来源字段：https://doi.org/10.4171/owr/2011/26

05042 | OWR-8415341-006

### Post Correspondence and Matrix Commutators in Infinite Groups

O / open | Oberwolfach 报告问题汇编

Determine whether the Post correspondence problem is decidable for free groups, even in rank $2$, and for an infinite linear group $G$ such as $\operatorname{SL}(n,\mathbb{Z})$ or $\operatorname{GL}(n,\mathbb{Z})$, determine whether a given matrix $M\in G$ can be written as a commutator $M=[A,B]$ with $A,B\in G$.

来源字段：https://doi.org/10.4171/owr/2021/38

05043 | OWR-2053-001

### Power Laws for Stable Commutator Length Frequencies

P / partially_solved | Oberwolfach 报告问题汇编

For a free group $F$ and words $w$ of fixed length, do the frequencies of reduced rational values $\operatorname{scl}(w)=p/q$ obey a power law $\operatorname{freq}(p/q)\asymp q^{-\delta}$ for some $\delta$?

来源字段：https://doi.org/10.4171/owr/2008/42

05044 | AIM-GEOMETRIC_GROUP_THEORY-0083

### Power rigidity and affine localization for spherical-parabolic centers

P / partially_solved | AIM 研讨会问题汇编

Compute the centralizer of an arbitrary element.

A more specific question: in a Euclidean Artin group, if two elements are contained in centers of spherical parabolic subgroups, when do the two elements commute?

来源字段：http://aimpl.org/geomartingp/8/

05045 | AIM-GEOMETRIC_GROUP_THEORY-0049

### Powered-generator Artin subgroups and a parabolicity obstruction

P / partially_solved | AIM 研讨会问题汇编

For an Artin group $A$, which Artin groups $A'$ embed in $A$ such that $A'$ is not a parabolic subgroup?

来源字段：http://aimpl.org/geomartingp/3/

05046 | OWR-15216-012

### Practical Criteria for Metrizable Universal Minimal Flows

O / open | Oberwolfach 报告问题汇编

Find a verifiable sufficient condition on a topological group $G$ that implies metrizability of its universal minimal flow $M(G)$ and is easier to check than the existence of a closed, co-precompact, extremely amenable subgroup of $G$. In particular, if $G$ is the automorphism group of a countable structure homogeneous in a finite relational language, must $M(G)$ be metrizable?

来源字段：https://doi.org/10.4171/owr/2017/11

05047 | OWR-14298365-004

### Probabilistic Growth Methods for Linear Algebraic Groups

O / open | Oberwolfach 报告问题汇编

Can probabilistic arguments be used to prove growth theorems for linear algebraic groups over finite fields?

来源字段：https://doi.org/10.4171/owr/2024/17

05048 | OWR-11101920-008

### Probability of Vanishing Character Values

O / open | Oberwolfach 报告问题汇编

For a finite group $G$, what is the probability that a randomly chosen character value $\chi(g)$ equals $0$?

来源字段：https://doi.org/10.4171/owr/2022/39

05049 | OWR-5154-009

### Profinite Closure Obstructions in Free Groups

O / open | Oberwolfach 报告问题汇编

For a finitely generated free group $F$, do there exist $g_1,\ldots,g_k\in F$ such that $\overline{[g_1]^F\cdots[g_k]^F}\not\subseteq N(g_1,\ldots,g_k)$, where the bar denotes closure in the profinite topology, $[g]^F$ is the conjugacy class of $g$, and $N(g_1,\ldots,g_k)$ is their normal closure?

来源字段：https://doi.org/10.4171/owr/2011/26

05050 | OWR-14299086-006

### Profinite Genera and Simple Kähler Groups

P / partially_solved | Oberwolfach 报告问题汇编

Is the Kähler-profinite genus of every residually finite Kähler group finite? Does there exist a simple Kähler group?

来源字段：https://doi.org/10.4171/owr/2025/6

05051 | OWR-14298004-006

### Profinite Invariance among Finite-Rank Coxeter Groups

O / open | Oberwolfach 报告问题汇编

Which homological and cohomological properties or invariants of finite-rank Coxeter groups are determined by their profinite completions?

来源字段：https://doi.org/10.4171/owr/2023/55

05052 | OWR-14299086-005

### Profinite Reconstruction of Complex Projective Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a smooth complex projective variety. Can its homeomorphism type be recovered from its algebraic fundamental group $\pi_1^{\mathrm{alg}}(X)$? Can its topological fundamental group $\pi_1(X)$ be recovered up to isomorphism from $\pi_1^{\mathrm{alg}}(X)$?

来源字段：https://doi.org/10.4171/owr/2025/6

05053 | OWR-14299086-012

### Profinite Rigidity of Coxeter Groups

P / partially_solved | Oberwolfach 报告问题汇编

Write $\mathcal W$ for all Coxeter groups. If $W_\Gamma,W_\Lambda\in\mathcal W$ have isomorphic profinite completions, must $W_\Gamma\cong W_\Lambda$?

来源字段：https://doi.org/10.4171/owr/2025/6

05054 | OWR-14298004-016

### Profinite Rigidity of Rational Euler Characteristics for Coxeter Groups

O / open | Oberwolfach 报告问题汇编

When are Coxeter groups virtually RFRS? Do there exist Coxeter groups $W_1,W_2$ with isomorphic profinite completions but $\chi(W_1)\neq\chi(W_2)$? For the Neretin group $N_q$, do $H^k(N_q,B_i(N_q))$ and $H_c^k(E_{\mathcal O}N_q,\mathbb Q)$ vanish for every $k$? Do non-good Coxeter groups exist?

来源字段：https://doi.org/10.4171/owr/2023/55

05055 | OWR-14299086-007

### Profinite Rigidity with Free Edge Groups

O / open | Oberwolfach 报告问题汇编

Let $\Lambda$ be an amalgamated product $F_1*_H F_2$ or an HNN extension $F_1*_H$ of free groups, where $H$ is an arbitrary finitely generated free group. If the profinite completion of $\Lambda$ is isomorphic to that of a free group or a surface group $\Gamma$, must $\Lambda\cong\Gamma$?

来源字段：https://doi.org/10.4171/owr/2025/6

05056 | OWR-13750328-013

### Projective Characters and Square Roots in Real Blocks

O / open | Oberwolfach 报告问题汇编

Let $B$ be a real nonprincipal $2$-block with defect pair $(D,E)$, and let $(x,b)$ be a $B$-subsection whose defect pair is $(C_D(x),C_E(x))$ and whose block $b$ has unique projective indecomposable character $\Phi$. Prove that every $y\in G$ with $y^2=x$ satisfies $[\Phi_{C_G(y)},1_{C_G(y)}]=|y^{C_G(x)}\cap(E\setminus D)|$.

来源字段：https://doi.org/10.4171/owr/2023/19

05057 | OWR-14298004-002

### Projectivity and Cohomological Dimension without Boundedness or Unimodularity

O / open | Oberwolfach 报告问题汇编

If $P$ is projective in the category of discrete $\mathbb Q[G]$-modules and $M$ is a rational discrete $G$-module, must $P\otimes_{\mathbb Q}M$ be projective? Separately, does the Stallings-Swan theorem for compactly generated totally disconnected locally compact groups of rational cohomological dimension at most one remain valid without compact-open boundedness or unimodularity?

来源字段：https://doi.org/10.4171/owr/2023/55

05058 | OWR-16162-010

### Properness of Localization Maps for Algebraic Groups

P / partially_solved | Oberwolfach 报告问题汇编

For which fields $K$ can one choose a natural set $V$ of discrete valuations such that, for every absolutely almost simple adjoint $K$-group $G$ and every finite $S\subset V$, the localization map $$ \omega_{S,G}:H^1(K,G)\longrightarrow\prod_{v\in V\setminus S}H^1(K_v,G) $$ is proper? In particular, is this true for finitely generated fields of good characteristic?

来源字段：https://doi.org/10.4171/owr/2018/21

05059 | AIM-GEOMETRIC_GROUP_THEORY-0152

### Property (T), Property (tau), and quantitative finite-quotient transfer for Out(F_n)

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.1. Does $\operatorname{Out}(F_n)$ have Property (T) or $(\tau)$ if $n\geq4$?**

$\operatorname{Out}(F_3)$ does not have Property (T) because it has a finite-index subgroup $\Gamma$ with $|H_1(\Gamma;\mathbb Z)|=\infty$. This motivates the question,

来源字段：https://aimath.org/WWN/outerauto/outerauto.pdf

05060 | OWR-15433-004

### Property (TWN+) for Reductive Groups

P / partially_solved | Oberwolfach 报告问题汇编

Does every reductive group $G$ over a number field satisfy property (TWN+)?

来源字段：https://doi.org/10.4171/owr/2017/25

05061 | AIM-GEOMETRIC_GROUP_THEORY-0041

### Property H versus a word-problem algorithm

P / partially_solved | AIM 研讨会问题汇编

(See Question 1.4 of Dehornoy--Godelle (*J. Pure Appl. Algebra* 217, 2013, MR2983847, on greedy normal-form reduction $w \rightsquigarrow_R^{0,1,2} \varepsilon$ for Artin--Tits groups) for the algorithm.)

Does Dehornoy-Godelle's algorithm solve the word problem for all Artin groups?

来源字段：http://aimpl.org/geomartingp/2/

05062 | AIM-GEOMETRIC_GROUP_THEORY-0034

### Pseudo-Anosov analogues for handlebodies and connected sums of S2 x S1

S / solved | AIM 研讨会问题汇编

Does there exist an analogue of pseudo-Anosov maps for homeomorphisms of handlebodies or connect sums of $\mathbb S^2\times\mathbb S^1$?

来源字段：http://aimpl.org/freebycyclic/7/

05063 | OWR-16929-002

### Pseudo-Arithmeticity of Higher-Dimensional Real-Hyperbolic Lattices

O / open | Oberwolfach 报告问题汇编

For every lattice $\Gamma\subset\operatorname{PO}(n,1)$ with $n\geq4$, is $\Gamma$ pseudo-arithmetic?

来源字段：https://doi.org/10.4171/owr/2019/17

05064 | OWR-9790354-004

### Quadratic Schreier Growth for Torsion-Free Solvable Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a finitely generated torsion-free solvable group. Prove that either $G$ is virtually abelian or every faithful $G$-set $X$ satisfies $\operatorname{vol}_{G,X}(n)\succeq n^2$.

来源字段：https://doi.org/10.4171/owr/2022/3

05065 | AIM-GEOMETRIC_GROUP_THEORY-0106

### Quantitative density-one Følner behavior of standard balls in the first Grigorchuk group

P / partially_solved | AIM 研讨会问题汇编

Is the sequence of balls $B_n(e)$ in Grigorchuk's group $G$ a Folner sequence?

来源字段：http://aimpl.org/amenablediscrete/3/

05066 | AIM-GEOMETRIC_GROUP_THEORY-0170

### Quantitative higher-commutator tails and an RD window for return probabilities

P / partially_solved | AIM 研讨会问题汇编

I checked the AIM source and recovered the following five questions.

Let \(\Gamma\) be a discrete group with a proper length function \(L\), let \[ D\delta_g=L(g)\delta_g\qquad(g\in\Gamma), \] and let \(\tau\) be the canonical trace on \(C_r^*(\Gamma)\).

1. For which groups is \[ \mathcal L=\{a\in C_r^*(\Gamma):\tau(a)=0,\ \|[D,a]\|\leq 1\} \] norm precompact? Rieffel's criterion says this is equivalent to the associated Connes metric inducing the weak-* topology on the state space. The source records the affirmative results for \(\mathbb Z^d\) and hyperbolic groups and asks particularly about RD groups. It also records Antonescu--Christensen's result that, for an RD group, the analogous ball \(\mathcal L_k\), defined using the \(k\)-fold commutator with \(D\), is precompact for all sufficiently large \(k\). 2. For which groups does Roe's equality \[ C_r^*(\Gamma)=C_u^*(|\Gamma|)\cap L(\Gamma) \tag{ITAP} \] hold? Here \(C_u^*(|\Gamma|)\) is the uniform Roe algebra and \(L(\Gamma)\) is the group von Neumann algebra. The source observes that a Haagerup-style argument proves this when \(\Gamma\) has RD for a conditionally negative length. 3. Prove Kaplansky's zero-divisor conjecture, \(\mathbb C\Gamma\) is a domain for torsion-free finitely generated \(\Gamma\), under the additional hypothesis RD. 4. If \(\nu\) is a finitely supported symmetric probability measure, is \[ \nu^{*(2n)}(e)\sim c(\nu)n^{-d(\nu)}\rho^{2n}, \tag{RW} \] where \(\rho\) is the spectral radius of convolution by \(\nu\) on \(\ell^2(\Gamma)\)? 5. Is there a relation between RD and entropy of ergodic actions?

来源字段：https://aimath.org/WWN/rapiddecay/rapiddecay.pdf

05067 | OWR-1703863-006

### Quasi-Isometric Rigidity of Acylindrical Hyperbolicity

O / open | Oberwolfach 报告问题汇编

Is acylindrical hyperbolicity invariant under quasi-isometry among finitely generated groups? More broadly, are finitely generated acylindrically hyperbolic groups quasi-isometrically rigid?

来源字段：https://doi.org/10.4171/owr/2020/16

05068 | OWR-14298004-013

### Quasi-Isometry Invariance and RAAG Analogues for TDLC Groups

O / open | Oberwolfach 报告问题汇编

If compactly generated totally disconnected locally compact groups $G$ and $H$ are quasi-isometric and have finite rational cohomological dimension, must $\operatorname{cd}_{\mathbb Q}G=\operatorname{cd}_{\mathbb Q}H$? Is there a useful totally disconnected locally compact analogue of right-angled Artin groups?

来源字段：https://doi.org/10.4171/owr/2023/55

05069 | OWR-16167-020

### Quasi-Retracts of Thompson's Group $F$

O / open | Oberwolfach 报告问题汇编

Does Thompson's group $F$ have any interesting proper nonabelian quasi-retracts; in particular, is $F$ a quasi-retract of $T$, or is $F\times F$ a quasi-retract of $F$?

来源字段：https://doi.org/10.4171/owr/2018/26

05070 | AMR-010-0101

### Questions in Geometric Group Theory - Q 1.1

S / solved | AMR 问题清单汇编

Suppose $G$ admits a finite $K(G,1)$. If $G$ does not contain any Baumslag-Solitar subgroups $BS(m,n)$, is $G$ necessarily hyperbolic? If $G$ embeds in a hyperbolic group, is it hyperbolic?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05071 | AMR-010-0110

### Questions in Geometric Group Theory - Q 1.10

P / partially_solved | AMR 问题清单汇编

(Canary) Let $G$ be word-hyperbolic and $H$ a finitely presented subgroup of $G$. Suppose that for every $g\in G$ there is $n>0$ such that $g^n\in H$. Does it follow that $H$ has finite index in $G$?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05072 | AMR-010-0111

### Questions in Geometric Group Theory - Q 1.11

O / open | AMR 问题清单汇编

(Whyte) Let $\Gamma$ be a 1-ended hyperbolic group which is not virtually a surface group. Can every infinite index subgroup be free?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05073 | AMR-010-0112

### Questions in Geometric Group Theory - Q 1.12

S / solved | AMR 问题清单汇编

(Whyte) Let $\Gamma$ be a 1-ended hyperbolic group. Can a finite index subgroup of $\Gamma$ be isomorphic to a subgroup of $\Gamma$ of infinite index?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05074 | AMR-010-0113

### Questions in Geometric Group Theory - Q 1.13

P / partially_solved | AMR 问题清单汇编

(Swarup) Prove the combination theorem for relatively hyperbolic groups.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05075 | AMR-010-0115

### Questions in Geometric Group Theory - Q 1.15

O / open | AMR 问题清单汇编

Is every word-hyperbolic group residually finite?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05076 | AMR-010-0116

### Questions in Geometric Group Theory - Q 1.16

S / solved | AMR 问题清单汇编

(Dani Wise) Let $G^n$ denote the Cartesian product of $n$ copies of $G$, and let $\operatorname{rank}(G^n)$ be its smallest number of generators. If $G$ is word-hyperbolic, is $\lim_{n\to\infty}\operatorname{rank}(G^n)=\infty$?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05077 | AMR-010-0117

### Questions in Geometric Group Theory - Q 1.17

S / solved | AMR 问题清单汇编

(Dani Wise) Find 'nice' groups, for example CAT(0) or automatic groups, for which $\operatorname{rank}(G^n)$ does not tend to infinity as $n\to\infty$.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05078 | AMR-010-0118

### Questions in Geometric Group Theory - Q 1.18

P / partially_solved | AMR 问题清单汇编

(Epstein) Let $G$ be a word-hyperbolic group and $\partial G$ its boundary. Is there an algorithm to compute $\check H^i(\partial G)\cong H^{i+1}(G,\mathbb{Z}G)$? In particular, is there an algorithm to decide whether $\check H^i(\partial G)\cong\check H^i(S^2)$ for all $i$?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05079 | AMR-010-0119

### Questions in Geometric Group Theory - Q 1.19

P / partially_solved | AMR 问题清单汇编

**Q 1.19 (M. Mitra).** Let $G$ be a word-hyperbolic group and $H$ a word-hyperbolic subgroup. Does the inclusion $H \to G$ extend to a continuous map between the boundaries $\partial H \to \partial G$?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05080 | AMR-010-0102

### Questions in Geometric Group Theory - Q 1.2

S / solved | AMR 问题清单汇编

Suppose G admits a finite K(G, 1), does not contain Z × Z, and whenever x ∈ G is an infinite order element such that xm and xn are conjugate, then |m| = |n|. Is G hyperbolic?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05081 | AMR-010-0120

### Questions in Geometric Group Theory - Q 1.20

P / partially_solved | AMR 问题清单汇编

(Swarup) Suppose $G$ is a hyperbolic group which is a graph of hyperbolic groups such that all edge-to-vertex inclusions are quasi-isometric embeddings. Mitra shows that each vertex-group inclusion $V\hookrightarrow G$ induces a continuous Cannon-Thurston map $\partial V\to\partial G$. Describe its point-preimages; in particular, show that the map is finite-to-one.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05082 | AMR-010-0121

### Questions in Geometric Group Theory - Q 1.21

S / solved | AMR 问题清单汇编

(Thurston) Is every closed hyperbolic 3-manifold finitely covered by one that fibers over the circle?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05083 | AMR-010-0123

### Questions in Geometric Group Theory - Q 1.23

O / open | AMR 问题清单汇编

(Ian Leary) Is there a version of the Kan-Thurston theorem using only CAT(-1) groups, or word hyperbolic groups? (The statement should be: for any finite simplicial complex X, there is a locally CAT(-1) polyhedral complex Y and a map Y →X that is surjective on fundamental groups and induces an isomorphism on homology for any local coefficients on X.)

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05084 | AMR-010-0105

### Questions in Geometric Group Theory - Q 1.5

O / open | AMR 问题清单汇编

(Davis) If $G$ is word-hyperbolic, does the Rips complex $P_d(G)$ have an equivariant negatively curved metric for $d$ sufficiently large?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05085 | AMR-010-0106

### Questions in Geometric Group Theory - Q 1.6

O / open | AMR 问题清单汇编

(Gromov) Does every one-ended word-hyperbolic group contain a closed hyperbolic surface subgroup?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05086 | AMR-010-0107

### Questions in Geometric Group Theory - Q 1.7

O / open | AMR 问题清单汇编

(Gromov) For a given n is there an example of a hyperbolic group of dimension n in which every infinite index subgroup is free? Or in which there are no (quasi-convex) subgroups with codimension ≤k for a given k ≤n-2.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05087 | AMR-010-0108

### Questions in Geometric Group Theory - Q 1.8

P / partially_solved | AMR 问题清单汇编

(Swarup) Suppose $H$ is a finitely presented subgroup of a word-hyperbolic group $G$ and has finite index in its normalizer. Assume that there is $n>0$ such that the intersection of $n$ distinct conjugates of $H$ is always finite. Is $H$ quasiconvex in $G$?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05088 | AMR-010-0109

### Questions in Geometric Group Theory - Q 1.9

P / partially_solved | AMR 问题清单汇编

**Q 1.9 (Mitra).** Let $X_G$ be a finite 2-complex with fundamental group $G$. Let $X_H$ be a cover corresponding to the f.p. subgroup $H$. Let $I(x)$ denote the injectivity radius of $X_H$ at $x$. Does $I(x)\to\infty$ as $x\to\infty$ imply that $H$ is quasi-isometrically embedded in $G$? A positive answer to the above question for $G$ hyperbolic would imply a positive answer to Q 1.8.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05089 | AMR-010-1001

### Questions in Geometric Group Theory - Q 10.1

O / open | AMR 问题清单汇编

Let $X$ be a non-positively curved symmetric space. Find conditions on a group $\Gamma$ so that the space of conjugacy classes of faithful discrete representations of $\Gamma$ into $\operatorname{Isom}(X)$ is compact (noncompact).

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05090 | AMR-010-1002

### Questions in Geometric Group Theory - Q 10.2

O / open | AMR 问题清单汇编

(Belegredek) Is there a 3-complex X (not necessarily aspherical) which is not homotopy equivalent to a 2-complex but $H^3(X; \{G\}) = 0$ for all local coefficients?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05091 | AMR-010-1101

### Questions in Geometric Group Theory - Q 11.1

O / open | AMR 问题清单汇编

Study the quasi-isometry group QI(Rn). How big is it?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05092 | AMR-010-1102

### Questions in Geometric Group Theory - Q 11.2

S / solved | AMR 问题清单汇编

(Kleiner) What are the quasi-isometries of the 3-dimensional group Sol?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05093 | AMR-010-1103

### Questions in Geometric Group Theory - Q 11.3

O / open | AMR 问题清单汇编

(Kleiner) What are the q.i's of the Gromov-Thurston examples of negatively pinched manifolds?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05094 | AMR-010-1104

### Questions in Geometric Group Theory - Q 11.4

O / open | AMR 问题清单汇编

(Feighn) Let $\phi:F_n\to F_n$ be an automorphism and let $M_\phi$ be its mapping torus. Classify these groups up to quasi-isometry.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05095 | AMR-010-1105

### Questions in Geometric Group Theory - Q 11.5

P / partially_solved | AMR 问题清单汇编

(Bridson) Classify the mapping tori of automorphisms $\mathbb{Z}^n\to\mathbb{Z}^n$ up to quasi-isometry.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05096 | AMR-010-1106

### Questions in Geometric Group Theory - Q 11.6

O / open | AMR 问题清单汇编

**(Bridson)** Let $F$ be R. Thompson's group $F$. Is $F \times \mathbb{Z}$ quasi-isometric to $F$? Does there exist *any* finitely generated group $G$ quasi-isometric to $G \times \mathbb{Z}$?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05097 | AMR-010-1201

### Questions in Geometric Group Theory - Q 12.1

S / solved | AMR 问题清单汇编

(Levitt) Can every measured geodesic lamination with 2-sided leaves on a non-orientable compact hyperbolic surface be approximated by a simplicial measured geodesic lamination with 2-sided leaves?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05098 | AMR-010-1211

### Questions in Geometric Group Theory - Q 12.11

P / partially_solved | AMR 问题清单汇编

(Grigorchuk) Aut(Fn) acts on the space ∂3Fn of triples of distinct ends of Fn. Denote by Yn the compact space (Cantor set) the quotient space of ∂3Fn by the group of inner automorphisms. Thus Out(Fn) acts on Yn. Describe the dynamics of this action; in particular the dynamics of any individual outer automorphism.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05099 | AMR-010-1212

### Questions in Geometric Group Theory - Q 12.12

O / open | AMR 问题清单汇编

Do there exist f.g. free subgroups of MCG(S) consisting of identity and pseudo-Anosov mapping classes which are not Schottky?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05100 | AMR-010-1213

### Questions in Geometric Group Theory - Q 12.13

P / partially_solved | AMR 问题清单汇编

Do there exist non-free pseudo-Anosov subgroups?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05101 | AMR-010-1214

### Questions in Geometric Group Theory - Q 12.14

O / open | AMR 问题清单汇编

For $\operatorname{Out}(F_n)$, do there exist finitely generated free subgroups consisting of the identity and irreducible automorphisms which are not Schottky? Do there exist nonfree subgroups all of whose nonidentity elements are irreducible?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05102 | AMR-010-1216

### Questions in Geometric Group Theory - Q 12.16

P / partially_solved | AMR 问题清单汇编

Does every automorphism h : Fn →Fn leave invariant a finite index subgroup K such that hab : K/K′ →K/K′ has an eigenvalue which is a root of unity?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05103 | AMR-010-1217

### Questions in Geometric Group Theory - Q 12.17

S / solved | AMR 问题清单汇编

Let $M$ be a closed 3-manifold which fibers over $S^1$ with fiber a closed surface $F$ of genus $\ge 2$. Does $M$ have a finite-sheeted cover $\widetilde M$ with $b_1(\widetilde M) > 1$?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05104 | AMR-010-1218

### Questions in Geometric Group Theory - Q 12.18

S / solved | AMR 问题清单汇编

Is MCG(Sg) →QI(MCG(Sg)) an isomorphism for g ≥3?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05105 | AMR-010-1219

### Questions in Geometric Group Theory - Q 12.19

P / partially_solved | AMR 问题清单汇编

Suppose that $\varphi : \mathrm{MCG}(S_g) \to \mathrm{MCG}(S_g)$ is a quasi-isometry. Does $\varphi$ map maximal flats to maximal flats ("maximal flats" come from maximal rank abelian subgroups)?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05106 | AMR-010-1202

### Questions in Geometric Group Theory - Q 12.2

P / partially_solved | AMR 问题清单汇编

(Lubotzky) Does Out(Fn) have the congruence subgroup property?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05107 | AMR-010-1220

### Questions in Geometric Group Theory - Q 12.20

O / open | AMR 问题清单汇编

For $\operatorname{Out}(F_n)$, is the natural map $\operatorname{Out}(F_n)\to QI(\operatorname{Out}(F_n))$ an isomorphism? Does every quasi-isometry of $\operatorname{Out}(F_n)$ map maximal flats to maximal flats?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05108 | AMR-010-1221

### Questions in Geometric Group Theory - Q 12.21

S / solved | AMR 问题清单汇编

If Γ is an irreducible uniform lattice in a higher rank connected semisimple Lie group, does every homomorphism Γ →Out(Fn) necessarily have finite image?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05109 | AMR-010-1222

### Questions in Geometric Group Theory - Q 12.22

P / partially_solved | AMR 问题清单汇编

Is MCG(Sg,b,n) linear?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05110 | AMR-010-1223

### Questions in Geometric Group Theory - Q 12.23

S / solved | AMR 问题清单汇编

Let $SB_n$ be the singular braid monoid generated by $\sigma_i^{\pm1}$ and singular generators $\tau_i$. Define $\Phi:SB_n\to\mathbb{Z}B_n$ by $\Phi(\sigma_i)=\sigma_i$ and $\Phi(\tau_i)=\sigma_i-\sigma_i^{-1}$. Conjecture: $\Phi$ is injective.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05111 | AMR-010-1203

### Questions in Geometric Group Theory - Q 12.3

P / partially_solved | AMR 问题清单汇编

(Kapovich) (a) For closed orientable surfaces $\Sigma_g$ and $\Sigma_h$, is there a faithful representation $\pi_1(\Sigma_g)\to\operatorname{MCG}(\Sigma_h)$ whose image consists of the identity and pseudo-Anosov classes, for some $g,h>1$? (b) Is there a four-manifold $M$ which is a surface bundle over a surface such that $\pi_1(M)$ is word-hyperbolic, or such that $M$ is hyperbolic?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05112 | AMR-010-1204

### Questions in Geometric Group Theory - Q 12.4

S / solved | AMR 问题清单汇编

(Bowditch) Is the Weil-Petersson metric on Teichmüller space hyperbolic? Is it quasi-isometric to the curve complex?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05113 | AMR-010-1205

### Questions in Geometric Group Theory - Q 12.5

P / partially_solved | AMR 问题清单汇编

(Brock) What is the rank of the Weil-Petersson metric? What is the rank of the mapping class group?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05114 | AMR-010-1206

### Questions in Geometric Group Theory - Q 12.6

P / partially_solved | AMR 问题清单汇编

Assuming that the fixed subgroup Fix(α) is cyclic, find a bound on the length of a generator of Fix(α) in terms of the complexity of α.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05115 | AMR-010-1207

### Questions in Geometric Group Theory - Q 12.7

P / partially_solved | AMR 问题清单汇编

Which α preserve an order (invariant under right translations) on Fn? If α has periodic elements it cannot preserve an order. Are there other obstructions?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05116 | AMR-010-1208

### Questions in Geometric Group Theory - Q 12.8

P / partially_solved | AMR 问题清单汇编

Does Out(Fn) (n > 2) have a right orderable subgroup of finite index?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05117 | AMR-010-1280

### Questions in Geometric Group Theory - Q 12.80

O / open | AMR 问题清单汇编

**(Yves de Cornulier)** Let $G$ be residually torsion-free nilpotent. Is $G$ Haagerup (= a-(T)-menable)?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05118 | AMR-010-1209

### Questions in Geometric Group Theory - Q 12.9

P / partially_solved | AMR 问题清单汇编

Can the mapping class group (or a finite index subgroup of it) of a closed surface be embedded into Out(Fn)? Into the mapping class group of a punctured surface?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05119 | AMR-010-1301

### Questions in Geometric Group Theory - Q 13.1

O / open | AMR 问题清单汇编

(Kevin Whyte) Let K be a finite complex with π = π(K) amenable. Is there a uniform bound to the betti numbers of finite covers of K?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05120 | AMR-010-1310

### Questions in Geometric Group Theory - Q 13.10

O / open | AMR 问题清单汇编

Let $G$ be a one-relator group whose relator $W$ is a cyclically reduced word in the generators, and let $P$ be the submonoid of $G$ generated by all prefixes of $W$. Is the membership problem for $P$ in $G$ decidable?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05121 | AMR-010-1302

### Questions in Geometric Group Theory - Q 13.2

P / partially_solved | AMR 问题清单汇编

(Kevin Whyte) Is every solvable PD(n) group polycyclic?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05122 | AMR-010-1303

### Questions in Geometric Group Theory - Q 13.3

O / open | AMR 问题清单汇编

(Henry Glover) Does for every finite graph G the following 1-2-∞- conjecture hold: G is planar, a double cover of G is planar, or no finite cover is planar?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05123 | AMR-010-1304

### Questions in Geometric Group Theory - Q 13.4

O / open | AMR 问题清单汇编

(S. Ivanov) Is there a f.p. slender group which is not polycyclic-by-finite?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05124 | AMR-010-1305

### Questions in Geometric Group Theory - Q 13.5

P / partially_solved | AMR 问题清单汇编

(Seymour Bachmut) Is SL2(K) finitely generated for K = Z[X, X-1] or K = F[X, X-1, Y, Y -1] for a field F?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05125 | AMR-010-1306

### Questions in Geometric Group Theory - Q 13.6

S / solved | AMR 问题清单汇编

Are 1-relator groups coherent?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05126 | AMR-010-1307

### Questions in Geometric Group Theory - Q 13.7

O / open | AMR 问题清单汇编

Let $L$ be a flag triangulation of $S^{2k-1}$, let $f_i$ be the number of $i$-simplices of $L$, and define $$\chi=1-\sum_{i=0}^{2k-1}(-1)^i\frac{f_i}{2^{i+1}}.$$ Conjecture: $(-1)^k\chi\geq0$.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05127 | AMR-010-1308

### Questions in Geometric Group Theory - Q 13.8

O / open | AMR 问题清单汇编

**Q 13.8.** Do there exist groups $G$ with balanced presentation (same number of generators and relations), with $H_1(G) = 0$ and with unsolvable word problem?

**Note:** The standard examples of groups with unsolvable word problem have more relations than generators. The condition $H_1(G)=0$ is added to rule out counterexamples obtained by adding silly generators. Any other condition that rules this out is acceptable.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05128 | AMR-010-1309

### Questions in Geometric Group Theory - Q 13.9

O / open | AMR 问题清单汇编

Is there a sequence of (perfect, of course) groups with balanced presentations among which one cannot recognize trivial groups?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05129 | AMR-010-0201

### Questions in Geometric Group Theory - Q 2.1

P / partially_solved | AMR 问题清单汇编

(Swarup) Is there a proof of Johannson's theorem that Out(π1M) is virtually generated by Dehn twists for M a Haken 3-manifold along the lines of Rips-Sela's theorem that Out(G) is virtually generated by Dehn twists for torsion-free 1-ended hyperbolic G? Is this true for CAT(0) groups? In particular, if G is a CAT(0) group and Out(G) is infinite, does G admit a Dehn twist of infinite order?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05130 | AMR-010-0211

### Questions in Geometric Group Theory - Q 2.11

P / partially_solved | AMR 问题清单汇编

(Eric Swenson) Let $X$ be a proper CAT(0) metric space and $G$ a finitely generated group acting properly discontinuously by isometries on $X$. (1) Can $G$ be an infinite torsion group? (2) If the action is cocompact, can $G$ contain an infinite torsion subgroup?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05131 | AMR-010-0212

### Questions in Geometric Group Theory - Q 2.12

P / partially_solved | AMR 问题清单汇编

(Kim Ruane) Let $G$ be a Coxeter group, for example a right-angled Coxeter group, acting properly discontinuously by isometries on a CAT(0) space $X$. How is $X$ different from the Coxeter complex? Specifically, if $H$ is a special subgroup of $G$, is there a closed convex subset of $X$ on which $H$ acts cocompactly?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05132 | AMR-010-0213

### Questions in Geometric Group Theory - Q 2.13

O / open | AMR 问题清单汇编

(Ruth Charney) Classify Coxeter groups up to isomorphism.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05133 | AMR-010-0214

### Questions in Geometric Group Theory - Q 2.14

O / open | AMR 问题清单汇编

(Ruth Charney) Classify Artin groups up to isomorphism.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05134 | AMR-010-0216

### Questions in Geometric Group Theory - Q 2.16

O / open | AMR 问题清单汇编

(Ruth Charney) Are all [finite type] Artin groups CAT(0)?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05135 | AMR-010-0217

### Questions in Geometric Group Theory - Q 2.17

O / open | AMR 问题清单汇编

(Ruth Charney) Are all Artin groups automatic?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05136 | AMR-010-0218

### Questions in Geometric Group Theory - Q 2.18

S / solved | AMR 问题清单汇编

(Ross Geoghegan) A proper CAT(0) space $M$ is almost geodesically complete if there is $R\geq0$ such that for all $a,b\in M$ there is an infinite geodesic ray starting at $a$ and passing within $R$ of $b$. If $\operatorname{Isom}(M)$ acts cocompactly on $M$, is $M$ almost geodesically complete?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05137 | AMR-010-0219

### Questions in Geometric Group Theory - Q 2.19

S / solved | AMR 问题清单汇编

(Dani Wise) A triplane is a CAT(0) space obtained by gluing three Euclidean half-planes along their boundaries, and a CAT(0) space $X$ has isolated flats if it contains no isometrically embedded triplane. A subgroup $H$ is quasiconvex relative to an action on $X$ if, for some $x\in X$ and $K$, every geodesic joining two points of $Hx$ lies in the $K$-neighborhood of $Hx$. Conjecture: if $G$ acts properly discontinuously and cocompactly on a CAT(0) space $X$ with isolated flats and $H\leq G$ is finitely generated, then $H\hookrightarrow G$ is a quasi-isometric embedding if and only if $H$ is quasiconvex relative to the action of $G$ on $X$.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05138 | AMR-010-0202

### Questions in Geometric Group Theory - Q 2.2

O / open | AMR 问题清单汇编

(Gromov) If $G$ admits a finite-dimensional $K(G,1)$, does $G$ act properly discontinuously by isometries on a complete CAT(0) space?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05139 | AMR-010-0203

### Questions in Geometric Group Theory - Q 2.3

O / open | AMR 问题清单汇编

**(Eilenberg-Ganea)** Is there a group $G$ of cohomological dimension $2$ and geometric dimension $3$?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05140 | AMR-010-0205

### Questions in Geometric Group Theory - Q 2.5

S / solved | AMR 问题清单汇编

(Exercise in [BGS85, p.2]) Take a closed surface S of genus ≥2. Let V = S × S and let Σ ⊂V denote the diagonal. Let ˜V be a nontrivially ramified finite cover of V along Σ. Then ˜V has a natural piecewise hyperbolic CAT(0) metric. Show that ˜V admits no C2-smooth Riemannian metric with curvature K ≤0.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05141 | AMR-010-0206

### Questions in Geometric Group Theory - Q 2.6

P / partially_solved | AMR 问题清单汇编

Suppose a group G acts properly discontinuously and cocompactly by isometries on two CAT(0) spaces X and Y . Croke-Kleiner have examples where the boundaries ∂X and ∂Y are not equivariantly homeomorphic. Is there a compact metric space Z and cell-like maps Z →∂X, Z →∂Y ?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05142 | AMR-010-0207

### Questions in Geometric Group Theory - Q 2.7

P / partially_solved | AMR 问题清单汇编

**Q 2.7 (D. Wise).** Let $G$ act properly discontinuously and cocompactly on a CAT(0) space (or let $G$ be automatic). Consider two elements $a, b$ of $G$. Does there exist $n > 0$ such that either the subgroup $\langle a^n, b^n\rangle$ is free or $\langle a^n, b^n\rangle$ is abelian?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05143 | AMR-010-0208

### Questions in Geometric Group Theory - Q 2.8

O / open | AMR 问题清单汇编

Do CAT(0) (or bi)automatic) groups satisfy the Tits alternative?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05144 | AMR-010-0209

### Questions in Geometric Group Theory - Q 2.9

O / open | AMR 问题清单汇编

Does every Artin group have a finite $K(G,1)$?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05145 | AMR-010-0301

### Questions in Geometric Group Theory - Q 3.1

S / solved | AMR 问题清单汇编

(Hanna Neumann Conjecture) If A and B are nontrivial subgroups of a free group, then rk(A ∩B) -1 ≤(rk(A) -1)(rk(B) -1).

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05146 | AMR-010-0310

### Questions in Geometric Group Theory - Q 3.10

S / solved | AMR 问题清单汇编

Conjecture. The only definable subgroups of Fn are cyclic and the entire group.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05147 | AMR-010-0311

### Questions in Geometric Group Theory - Q 3.11

P / partially_solved | AMR 问题清单汇编

For $x\in[F_n,F_n]$, define its genus as the least $g$ such that $x$ is a product of $g$ commutators. Let $f(g)$ be a uniform bound, independent of $x$, on the number of Nielsen-equivalence classes of expressions of a genus-$g$ element as a product of $g$ commutators. Give an explicit upper bound for $f(g)$.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05148 | AMR-010-0302

### Questions in Geometric Group Theory - Q 3.2

P / partially_solved | AMR 问题清单汇编

**(Swarup)** For a torsion-free Fuchsian group $A$, let $\operatorname{area}(A)$ be the area of the convex core of $\mathbb{H}^2/A$. Since $\operatorname{area}(A)=2\pi(\operatorname{rk}(A)-1)$ for free Fuchsian $A$, the Hanna Neumann Conjecture can be phrased as $2\pi\,\operatorname{area}(A\cap B)\le \operatorname{area}(A)\operatorname{area}(B)$ for nontrivial free subgroups $A,B$ of a Fuchsian group. **Prove such inequalities (possibly with a worse constant) for (not necessarily free) torsion-free quasiconvex subgroups $A,B$ of a quasiconvex (convex cocompact) Kleinian group in $\mathbb{H}^n$, with area replaced by the $n$-dimensional volume of the convex core:** $$\operatorname{vol}_n(\operatorname{core}(A\cap B))\ \le\ C\cdot \operatorname{vol}_n(\operatorname{core}(A))\ \cdot\operatorname{vol}_n(\operatorname{core}(B)),$$ with $C$ a universal constant (perhaps depending only on $n$).

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05149 | AMR-010-0303

### Questions in Geometric Group Theory - Q 3.3

P / partially_solved | AMR 问题清单汇编

(J. Cornick) If G is f.g. and the homological dimension hd G = 1, is G free?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05150 | AMR-010-0305

### Questions in Geometric Group Theory - Q 3.5

P / partially_solved | AMR 问题清单汇编

Conjecture. Limit groups are CAT(0).

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05151 | AMR-010-0306

### Questions in Geometric Group Theory - Q 3.6

P / partially_solved | AMR 问题清单汇编

**Q 3.6.** Characterize groups of the form $F_m \ast_{\mathbb{Z}} F_n$ which are limit groups.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05152 | AMR-010-0307

### Questions in Geometric Group Theory - Q 3.7

O / open | AMR 问题清单汇编

Characterize 1-relator groups which are limit groups.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05153 | AMR-010-0309

### Questions in Geometric Group Theory - Q 3.9

S / solved | AMR 问题清单汇编

Conjecture: for $n>1$, the set $$\{(x_1,\ldots,x_n)\in F_n^n:x_1,\ldots,x_n\text{ is a basis of }F_n\}$$ is not definable.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05154 | AMR-010-0401

### Questions in Geometric Group Theory - Q 4.1

S / solved | AMR 问题清单汇编

(de la Harpe) Is PSL2(R) a maximal closed subgroup of Homeo+(S1)?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05155 | AMR-010-0402

### Questions in Geometric Group Theory - Q 4.2

P / partially_solved | AMR 问题清单汇编

Is there a proper closed subgroup of Homeo+(S1) that acts transitively on (unordered) 4-tuples? Or k-tuples (k ≠ 3)? Relation to earthquakes?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05156 | AMR-010-0404

### Questions in Geometric Group Theory - Q 4.4

O / open | AMR 问题清单汇编

(Swarup) Suppose G is a 1-ended finitely presented group that acts on a compact connected metric space X as a convergence group. What can be said about G if X has cut points? Does X have to be locally connected? The model theorem of Bowditch [Bow99] and Swarup [Swa96] says that if G is word hyperbolic, then X is locally connected and doesn’t have cut points. In another interesting case, when G is a geometrically finite Kleinian group and X its limit set, cut points can arise, for example if G splits over a parabolic subgroup.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05157 | AMR-010-0501

### Questions in Geometric Group Theory - Q 5.1

O / open | AMR 问题清单汇编

**Q 5.1. (Jim Anderson)** If $G$ is a group of isometries of $\mathbb{H}^n$, denote by $\mathrm{Ax}(G)$ the set of axes of the elements of $G$. If $G_1$ and $G_2$ are finitely generated and discrete, does $\mathrm{Ax}(G_1) = \mathrm{Ax}(G_2)$ imply that $G_1$ and $G_2$ are commensurable?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05158 | AMR-010-0503

### Questions in Geometric Group Theory - Q 5.3

S / solved | AMR 问题清单汇编

**Q 5.3. (Ed Taylor)** Does there exist a constant $c>0$ such that the limit set of every non-classical Schottky group has Hausdorff dimension $\ge c$?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05159 | AMR-010-0505

### Questions in Geometric Group Theory - Q 5.5

P / partially_solved | AMR 问题清单汇编

(Misha Kapovich) Suppose $G$ is a finitely generated Kleinian group in $\operatorname{Isom}(\mathbb{H}^n)$. Is $$\delta(G)\leq\operatorname{vcd}(G),$$ with equality if and only if $G$ preserves a totally geodesic subspace $\mathbb{H}^k\subset\mathbb{H}^n$ such that $\mathbb{H}^k/G$ is compact?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05160 | AMR-010-0506

### Questions in Geometric Group Theory - Q 5.6

S / solved | AMR 问题清单汇编

(Misha Kapovich) Is there $\epsilon > 0$ so that $\delta(G) < \epsilon$ implies that G is virtually free?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05161 | AMR-010-0507

### Questions in Geometric Group Theory - Q 5.7

O / open | AMR 问题清单汇编

(Misha Kapovich) Is there a finitely-generated discrete subgroup of SO(n, 1) whose action on the limit set is not ergodic? Is not recurrent?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05162 | AMR-010-0601

### Questions in Geometric Group Theory - Q 6.1

S / solved | AMR 问题清单汇编

**(Ian Leary)** Suppose $G$ is virtually of type $FP$ over the field $\mathbb{F}_p$ of $p$ elements, and let $g$ be an element of order $p$. Is the centralizer of $g$ in $G$ also virtually of type $FP$ over $\mathbb{F}_p$?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05163 | AMR-010-0602

### Questions in Geometric Group Theory - Q 6.2

O / open | AMR 问题清单汇编

(Ian Leary) Is there a group of finite vcd that does not act with finite stabilizers on an acyclic complex of dimension equal to its vcd?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05164 | AMR-010-0603

### Questions in Geometric Group Theory - Q 6.3

O / open | AMR 问题清单汇编

(Peter Kropholler) If G is FP over the rationals, is there a bound on the orders of finite subgroups of G?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05165 | AMR-010-0702

### Questions in Geometric Group Theory - Q 7.2

P / partially_solved | AMR 问题清单汇编

(Swarup) Starting with a finitely presented group $G$, take maximal graph-of-groups decompositions alternately over finite and over two-ended edge groups, each time continuing with the resulting vertex groups. Conjecture 1: there is a finitely presented group for which this process never terminates. Conjecture 2 (Strong Accessibility): for hyperbolic groups, and for CAT(0) groups, this process always terminates.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05166 | AMR-010-0703

### Questions in Geometric Group Theory - Q 7.3

O / open | AMR 问题清单汇编

(Sageev) Is there a f.p. 1-ended group G with G ~= G∗Z?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05167 | AMR-010-0705

### Questions in Geometric Group Theory - Q 7.5

O / open | AMR 问题清单汇编

(Papasoglou) Is there a f.p. torsion-free group G that does not split over a virtually abelian subgroup, but has infinitely many splittings over F2?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05168 | AMR-010-0810

### Questions in Geometric Group Theory - Q 8.10

P / partially_solved | AMR 问题清单汇编

Suppose $H=F/N$ is finitely presented and contains $C^n$ for every $n$. Write $G_n=H*_{C^n}H=F*F/N_n$. If $C$ is nontrivial and finite, is $$\lim_{n\to\infty}d_{F*F}(N_n)=\infty?$$

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05169 | AMR-010-0802

### Questions in Geometric Group Theory - Q 8.2

P / partially_solved | AMR 问题清单汇编

(Shalen) If a finitely presented group G acts nontrivially (i.e. without global fixed points) on an R-tree, does it act nontrivially on a simplicial tree?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05170 | AMR-010-0803

### Questions in Geometric Group Theory - Q 8.3

P / partially_solved | AMR 问题清单汇编

(Mohan Ramachandran) For a finitely presented group $G$, consider: (A) some finite-index subgroup of $G$ admits a nontrivial action on a simplicial tree; (B) if $X$ is a finite complex with fundamental group $G$, then some covering space of $X$ has at least two ends. To what extent are (A) and (B) equivalent?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05171 | AMR-010-0805

### Questions in Geometric Group Theory - Q 8.5

S / solved | AMR 问题清单汇编

**Q 8.5 (Noel Brady).** Are there groups of type $F_n$ but not $F_{n+1}$ ($n \geq 3$) which do not contain $\mathbb{Z}\times\mathbb{Z}$? All known examples contain $\mathbb{Z}^{n-1}$.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05172 | AMR-010-0806

### Questions in Geometric Group Theory - Q 8.6

P / partially_solved | AMR 问题清单汇编

(Olympia Talelli) Is there a torsion-free group $G$ of infinite cohomological dimension such that there is $n_0$ with the property that if $H$ is a subgroup of $G$ with finite cohomological dimension $\operatorname{cd} H$, then $\operatorname{cd} H \le n_0$?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05173 | AMR-010-0807

### Questions in Geometric Group Theory - Q 8.7

P / partially_solved | AMR 问题清单汇编

**Q 8.7.** Can $\mathbb{Z}_{p^\infty}$ (the Prüfer $p$-group $\mathbb{Z}[1/p]/\mathbb{Z} = C_{p^\infty}$) be embedded in an $FP_\infty$-group? Or in an $FP_3$-group? Can an $F_n$-group be embedded in an $F_{n+1}$-group ($n \geq 2$)?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05174 | AMR-010-0808

### Questions in Geometric Group Theory - Q 8.8

P / partially_solved | AMR 问题清单汇编

Compute the asymptotic dimension of CAT(0) groups, Out(Fn), mapping class groups, nonuniform lattices, Thompson’s group. Is there a group of finite type whose asymptotic dimension is infinite?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05175 | AMR-010-0809

### Questions in Geometric Group Theory - Q 8.9

O / open | AMR 问题清单汇编

Is there a finitely presented group $G=F/N$, with $F$ a finite-rank free group, such that $$d_G\!\left(N/[N,N]\right)<d_F(N)<\infty?$$ Here $d_F(N)$ is the least number of $G$-orbits of $2$-cells needed to make the associated cover simply connected, while $d_G(N/[N,N])$ is the least number needed to kill its first homology.

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05176 | AMR-010-0902

### Questions in Geometric Group Theory - Q 9.2

O / open | AMR 问题清单汇编

**Question (Bestvina's list, Q 9.2, attributed to Andrews-Curtis).** If $K$ and $L$ are simple homotopy equivalent finite 2-dimensional CW-complexes, can one transform $K$ into $L$ by a sequence of elementary collapses and expansions of 1- and 2-cells, and by sliding 2-cells (reattaching them by maps homotopic to the old attaching maps)?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05177 | AMR-010-0903

### Questions in Geometric Group Theory - Q 9.3

P / partially_solved | AMR 问题清单汇编

(Wise) Is there a finite aspherical 2-complex X with π1(X) coherent and with χ(X) ≥2?

背景链接：https://www.math.utah.edu/~bestvina/eprints/questions-updated.pdf

05178 | OWR-11786-012

### Quillen's $p$-Subgroup Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a finite group and $p$ a prime. If the order complex of the poset of nontrivial $p$-subgroups of $G$ is contractible, is it strongly collapsible, equivalently, does $G$ have a nontrivial normal $p$-subgroup?

来源字段：https://doi.org/10.4171/owr/2012/24

05179 | OWR-14298365-015

### Random Expanders and Expanding Powers of Alternating Groups

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Alternating-Group Expander-Power Question [OWR-14298365-016]

Do random bounded generating sets produce expanders for alternating groups, and for which powers of alternating groups can one construct uniform expander families?

来源字段：https://doi.org/10.4171/owr/2024/17

05180 | OWR-1274-021

### Rank Growth and the Tame Fontaine-Mazur Conjecture

O / open | Oberwolfach 报告问题汇编

If $S$ contains no primes above $p$ and $G_{K,S}$ is infinite, must its rank-growth exponent satisfy $\gamma(G_{K,S})\geq1/2$? Prove or disprove that this lower bound is group-theoretically equivalent to the Tame Fontaine-Mazur conjecture.

来源字段：https://doi.org/10.4171/owr/2006/25

05181 | AIM-GEOMETRIC_GROUP_THEORY-0029

### Rank obstruction and a characteristic-fiber normalizer reduction

P / partially_solved | AIM 研讨会问题汇编

A group \(G\) is said to be of type \(VF\) if there exists a finite index subgroup \(G'\) of \(G\) which admits a finite classifying space.

If \(G\) is a free-by-cyclic group then \(\operatorname{Out}(G)\) is of type \(VF\).

The repository record itself is uncorrupted. More importantly, the workshop literature fixes the convention: a free-by-cyclic group is \[ G=F_n\rtimes_\phi\mathbb Z \] with \(F_n\) of **finite rank** and the quotient infinite cyclic. Andrew--Martino use exactly this convention in the first sentence of their setup. Thus the intended conjecture concerns finite \(n\). If arbitrary-rank free fibers are allowed, the assertion is false (Proposition 5.1 below).

We use the presentation \[ G_\phi=\langle F_n,t\mid txt^{-1}=\phi(x)\ (x\in F_n)\rangle \] and write \(\Phi=[\phi]\in\operatorname{Out}(F_n)\).

来源字段：http://aimpl.org/freebycyclic/7/

05182 | OWR-14298365-011

### Rank-Metric Stability and Finite-Group Exponent Problems

P / partially_solved | Oberwolfach 报告问题汇编

Are almost commuting invertible matrices close to commuting invertible matrices in the normalized rank metric? Also, for each $k\geq2$ and prime $n$, does there exist $N=N(k,n)$ such that every finite group generated by $A=\{a_1,\ldots,a_k\}$ has exponent at most $n$ whenever $(x_1\cdots x_N)^n=1$ for all $x_1,\ldots,x_N\in A\cup\{1\}$?

来源字段：https://doi.org/10.4171/owr/2024/17

05183 | AIM-GEOMETRIC_GROUP_THEORY-0007

### Rank-two verification and peripheral reductions for the relatively hyperbolic LERF conjecture

P / partially_solved | AIM 研讨会问题汇编

**Relatively hyperbolic case.** If \(G\) is a LERF relatively hyperbolic free-by-cyclic group which is not hyperbolic then \(G\) is the fundamental group of a hyperbolic 3-manifold.

Here “hyperbolic” in the hypothesis means word-hyperbolic, while “hyperbolic 3-manifold” in the conclusion means a manifold with a complete hyperbolic metric.

The workshop report supplies important intended context: it discusses free-by-cyclic groups hyperbolic relative to a **nonempty collection of \(\mathbb Z^2\) subgroups**, and identifies such groups as analogues of cusped finite-volume hyperbolic 3-manifold groups. Accordingly, throughout the mathematical results below, “relatively hyperbolic” means non-elementary strong relative hyperbolicity with respect to a finite nonempty family of proper subgroups isomorphic to \(\mathbb Z^2\). This convention is necessary: allowing the degenerate structure relative to the group itself would make the statement ill-posed. The intended conclusion is a finite-volume cusped hyperbolic 3-manifold. The source does not specify orientability, so none is silently imposed.

Write \[ G_\phi=F_n\rtimes_\phi\mathbb Z =\langle F_n,t\mid txt^{-1}=\phi(x),\ x\in F_n\rangle, \] where \(\Phi=[\phi]\in\operatorname{Out}(F_n)\). The exact conclusion is an isomorphism \(G_\phi\cong\pi_1(M)\), not merely a virtual identification.

来源字段：http://aimpl.org/freebycyclic/2/

05184 | AIM-GEOMETRIC_GROUP_THEORY-0171

### Rapid decay status and a quantitative central certificate for braid groups

P / partially_solved | AIM 研讨会问题汇编

**8 Which one of these groups have RD?**

1. \(\operatorname{Out}(\text{free group})\), 2. \(\operatorname{Aut}(\text{free group})\), 3. braid groups \(B_n\) (\(B_3\) has RD), 4. mapping class group, 5. Artin groups (right angled groups have RD since they act freely on cube complexes).

Note that Coxeter groups have RD.

来源字段：https://aimath.org/WWN/rapiddecay/rapiddecay.pdf

05185 | OWR-14298004-003

### Rational Duality among Crystallographic Coxeter Groups

O / open | Oberwolfach 报告问题汇编

Which crystallographic Coxeter groups are rational duality groups of dimension $d\geq1$?

来源字段：https://doi.org/10.4171/owr/2023/55

05186 | OWR-12007-008

### Rationality and Computation of Free-Group Rotation Suprema

O / open | Oberwolfach 报告问题汇编

Let $F$ be the rank-two free group, and let $R(w,r,s)$ be the supremum of the lifted rotation number of $\rho(w)$ over representations with rotation numbers $r$ and $s$ on the generators. Is $R(w,r,s)\in\mathbb Q$ for every $w\in F$ and $r,s\in\mathbb Q$, and can one give a practical method to compute or approximate it?

来源字段：https://doi.org/10.4171/owr/2012/35

05187 | OWR-14743-002

### Real Baum-Connes Isomorphism Conjecture

O / open | Oberwolfach 报告问题汇编

For every countable discrete group $G$, is the real Baum-Connes assembly map $\operatorname{KO}_*^G(\underline EG)\to \operatorname{KO}_*(C_r^*(G;\mathbb R))$ an isomorphism?

来源字段：https://doi.org/10.4171/owr/2016/35

05188 | OWR-14299097-011

### Realizability and Tameness of Fusion Systems

P / partially_solved | Oberwolfach 报告问题汇编

Characterize when a saturated fusion system $\mathcal{F}$ on a Sylow $p$-subgroup $S$ is realized by a finite group: determine conditions guaranteeing a finite group $G$ with a normal subgroup $H\trianglelefteq G$, $S\in\operatorname{Syl}_p(G)$, and $\mathcal{F}=\mathcal{F}_S(G)$. Clarify the relation between realizability and tameness in full generality.

来源字段：https://doi.org/10.4171/owr/2025/16

05189 | OWR-2053-016

### Realization of Rapidly Growing Følner Functions

O / open | Oberwolfach 报告问题汇编

Let $f:\mathbb{N}\to\mathbb{N}$ be nondecreasing with $f(n)\geq\exp(n)$. Is there a finitely generated group $G$ whose Følner function satisfies $\operatorname{Fol}_G(n)\asymp f(n)$?

来源字段：https://doi.org/10.4171/owr/2008/42

05190 | AIM-GEOMETRIC_GROUP_THEORY-0164

## Recovering indecomposable vertex blocks from dual laminations

P / partially_solved | AIM 研讨会问题汇编

Relate a fine/graph-of-actions decomposition theory for \(F_N\)-trees to the tree's dual algebraic lamination, in both directions, and relate both descriptions to finite systems of partial isometries.

It contains the same unexplained string "GL-theory," the same conspicuous capitalization "FINE DECOMPOSITION," and the same horofunction prose. Three source-recovery conclusions are important.

1. **"GL" is not expanded in the source.** In the relevant literature it is genuinely ambiguous. Gaboriau--Levitt is the standard author pair behind the geometric index and a finite-tree/partial-isometry construction [GL95]. Guirardel--Levitt is another standard "GL" author pair in the Outer-space and JSJ literature, while Guirardel introduced the indecomposability notion used in later decomposition results. The exact phrase "fine decomposition" was not found as a named theorem in the primary sources checked. Later papers instead speak of Levitt's graph-of-actions or Levitt decomposition, transverse families, indecomposable trees, compact hearts, and Rips or splitting induction. Because the source itself does not decide between these usages, this report does not silently expand "GL." The strongest contextual evidence for the initials alone is Gaboriau--Levitt: [GL95] both has the conventional citation key GL and directly treats geometric index and systems of partial isometries. The decomposition language, however, is closer to Levitt's graph-of-actions theory and Guirardel's refinements. "FINE DECOMPOSITION" is therefore treated as unresolved workshop shorthand or an editorial placeholder, not as a verified proper noun.

2. **The horofunction paragraph introduces Problem 5.3.** The paragraph changes topic immediately after Problem 5.2 and defines the object whose closure is asked for in the next numbered problem: "Problem 5.3. What is the closure \(X\)? What is \(X\setminus X\), the horofunction boundary ...?" It is transitional prose for 5.3, not an additional clause of Problem 5.2.

3. **The mathematical reconstruction of its OCR is** \[ x\longmapsto [d(\,\cdot\,,x)]\in C^0(X_n)/\mathbb R, \qquad f\sim g\iff f-g\text{ is constant}, \] with uniform convergence on compact sets. Thus "7 →" is a broken \(\mapsto\) glyph, \(C0\) is \(C^0\), and "homotopy to a constant" must be corrected to "differ by an additive constant." This is a mathematically compelled reconstruction, not literal source text. It is not used below because it belongs with Problem 5.3.

Accordingly, the recovered Problem 5.2 is the broad comparison problem:

来源字段：https://aimath.org/WWN/outerauto/outerauto.pdf

05191 | OWR-9790362-006

## Recursive Homological Filling and the Word Problem

O / open | Oberwolfach 报告问题汇编

Let $H$ be a finitely presented group whose homological filling function $\operatorname{FA}_H$ is bounded by a recursive function. Must $H$ have a solvable word problem?

来源字段：https://doi.org/10.4171/owr/2022/11

05192 | OWR-13675-010

## Reduction of Torsion Endotrivial Modules to Trivial p-Prime Core

O / open | Oberwolfach 报告问题汇编

Can determination of the torsion subgroup $TT(G)$ of the endotrivial-module group $T(G)$ be reduced essentially to groups satisfying $O_{p'}(G)=1$?

来源字段：https://doi.org/10.4171/owr/2015/18

05193 | OWR-14602-010

## Refined Local Langlands Correspondence for Quasi-Split Groups

P / partially_solved | Oberwolfach 报告问题汇编

别名：Refined Local Langlands Parametrization of Tempered Packets [OWR-14602-009]

For a quasi-split connected reductive group $G$ over a local field and a tempered Langlands parameter $\varphi$, does the associated packet admit an injective Whittaker-dependent parametrization by the irreducible representations of the component group of the centralizer of $\varphi$, bijective over non-Archimedean local fields, with the trivial representation corresponding to the unique Whittaker-generic constituent and with the expected endoscopic character identities?

来源字段：https://doi.org/10.4171/owr/2016/20

05194 | OWR-12007-009

## Refined Slippery Bounds for Positive-Word Rotation Numbers

O / open | Oberwolfach 报告问题汇编

For a positive word $w=a^{\alpha_1}b^{\beta_1}\cdots b^{\beta_m}$ with $R(w,r,s)=p/q$, does

$$R(w,r,s)-h_a(w)r-h_b(w)s\leq \frac mq$$

hold?

来源字段：https://doi.org/10.4171/owr/2012/35

05195 | AIM-GEOMETRIC_GROUP_THEORY-0136

### Regularized determinant approximation and the zero-spectrum obstruction

P / partially_solved | AIM 研讨会问题汇编

The following is equivalent to the previous problem.

Let $A\in M_l(\mathbb{Z} \Gamma)$, denote $A_k=A/\Gamma_k$. Is it true that $$tr_{L\Gamma}(\log A^*A)=\lim \frac{tr(\log A_k^*A_k)}{|\Gamma:\Gamma_k|}?$$

Its exact problem text is:

The adjacent Problem 11.2 asks whether normalized Ray--Singer torsions of a tower $M_k=\widetilde M/\Gamma_k$ converge to the analytic $L^2$-torsion of $\widetilde M$. Thus the intended standing assumptions are a decreasing normal finite-index chain \[ \Gamma=\Gamma_0\supseteq\Gamma_1\supseteq\cdots, \qquad \bigcap_k\Gamma_k=\{e\}, \] and $q_k=[\Gamma:\Gamma_k]$.

The notation $A/\Gamma_k$ means entrywise application of $\mathbb Z\Gamma\to\mathbb Z[\Gamma/\Gamma_k]$.

There is one essential convention missing from the displayed formula. The ordinary functional-calculus logarithm has $\log 0=-\infty$, whereas torsion uses the **regular Fuglede--Kadison determinant**, i.e. the logarithmic integral over the positive spectrum with the kernel deleted. With a literal logarithm the displayed assertion is false (and often not a finite equality); with the regular/pseudodeterminant convention it is the standard determinant approximation conjecture.

来源字段：http://aimpl.org/l2invariantsgroups/1/

05196 | OWR-1703863-007

### Residual Finiteness and Minimal Presentations of Hyperbolic Groups

O / open | Oberwolfach 报告问题汇编

Is every hyperbolic group residually finite? Separately, what is the smallest total relator length in a presentation of an infinite hyperbolic group with Kazhdan property (T)?

来源字段：https://doi.org/10.4171/owr/2020/16

05197 | OWR-13681-015

### Restriction Criteria for Endotrivial Modules

P / partially_solved | Oberwolfach 报告问题汇编

Let $S$ be a Sylow $p$-subgroup of a finite group $G$ and define $\rho_1(Q)=N_G(Q)'$ and $\rho_i(Q)=\langle N_G(Q)\cap\rho_{i-1}(R):1\ne R\subseteq S\rangle$, with $\rho^\infty(S)=\bigcup_i\rho_i(S)$. The conjecture is that a one-dimensional $kN_G(S)$-module $L$ is the restriction, modulo projectives, of an endo-trivial $kG$-module if and only if $\rho^\infty(S)\subseteq\ker L$; equivalently, the image of $K(G)\to K(N_G(S))$ is the dual of $N_G(S)/\rho^\infty(S)$.

来源字段：https://doi.org/10.4171/owr/2015/24

05198 | OWR-14749-005

### Return Probabilities of Amenable Groups with Exponential Growth

P / partially_solved | Oberwolfach 报告问题汇编

Which finitely generated amenable groups of exponential growth satisfy $p^G_{2n}\asymp\exp(-n^{1/3})$?

来源字段：https://doi.org/10.4171/owr/2016/41

05199 | OWR-4333-003

### Rigidity and Assembly Conjectures for Discrete Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a torsion-free group $G$ and integral domain $R$, are $0$ and $1$ the only idempotents of $RG$? Do the negative K-groups of $\mathbb{Z}G$, the reduced group $K_0(\mathbb{Z}G)$, and $\operatorname{Wh}(G)$ vanish? Are higher signatures homotopy invariant, closed aspherical manifolds topologically rigid, and groups of type FP of type FF? Is every finitely presented Poincaré-duality group of dimension at least five the fundamental group of an aspherical homology ANR-manifold? Is every hyperbolic group with boundary $S^n$ the fundamental group of a closed aspherical manifold?

来源字段：https://doi.org/10.4171/owr/2010/20

05200 | OWR-13686-003

### Rigidity of Continuous Orbit Equivalence

P / partially_solved | Oberwolfach 报告问题汇编

For which topologically free actions of countable groups on compact spaces does continuous orbit equivalence imply conjugacy? In particular, which hypotheses beyond the known duality-group-versus-solvable-group criterion rule out the counterexamples that occur for higher-rank free abelian groups and nonabelian free groups?

来源字段：https://doi.org/10.4171/owr/2015/29

05201 | OWR-12697693-013

### Semi-Affine Structures and Skew Brace Constructions

P / partially_solved | Oberwolfach 报告问题汇编

Classify cancellative semi-affine structures on groups, construct skew braces that are not bi-skew braces from groupal semi-affine structures, and construct two-sided skew braces by the same method.

来源字段：https://doi.org/10.4171/owr/2023/9

05202 | OWR-14213-003

### Serre Conjecture II for Type $E_7$ Groups

O / open | Oberwolfach 报告问题汇编

Serre's conjecture II in type $E_7$: for a perfect field $k$ of cohomological dimension at most 2 and a semisimple simply connected group $G$ of type $E_7$ over $k$, is $H^1(k,G)$ trivial?

来源字段：https://doi.org/10.4171/owr/2016/5

05203 | OWR-13675-006

### Sharp Group-Order Bounds from Small Brauer Degrees

O / open | Oberwolfach 报告问题汇编

Let $G$ be a finite group with $O_p(G)=1$ such that $\beta(1)_p\leq p$ for every irreducible $p$-Brauer character $\beta$. Is $\lvert G\rvert_2\leq2^7$ when $p=2$? Is $\lvert G\rvert_p\leq p^2$ when $p\geq5$, or when $p=3$ and $A_7$ is not involved in $G$? Is $\lvert G\rvert_3\leq3^3$ when $p=3$ and $A_7$ is involved in $G$?

来源字段：https://doi.org/10.4171/owr/2015/18

05204 | OWR-14298365-013

### Shortest Laws in Symmetric Groups

O / open | Oberwolfach 报告问题汇编

What is the asymptotic length of the shortest nontrivial law in the symmetric group $S_n$?

来源字段：https://doi.org/10.4171/owr/2024/17

05205 | OWR-14299097-013

### Shortest Mixed Identities in Finite Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a finite group $G$, determine the minimum length of a nontrivial mixed identity in the free product $G*F_r$, and obtain sharp values or asymptotic bounds across the principal families of finite groups.

来源字段：https://doi.org/10.4171/owr/2025/16

05206 | OWR-14604-007

### Shortest Nontrivial Laws for Symmetric Groups

O / open | Oberwolfach 报告问题汇编

If a word in the free group on two letters is a law for $S_n$ when every pair of permutations evaluates to the identity, how short can a nontrivial such law be?

来源字段：https://doi.org/10.4171/owr/2016/22

05207 | OWR-2491-011

### Simple Groups Acting on Nonspherical Phan Geometries

P / partially_solved | Oberwolfach 报告问题汇编

In the nonspherical case, what are the groups $G_\sigma$ acting on Phan geometries associated with good flips $\sigma$? In particular, are these groups simple?

来源字段：https://doi.org/10.4171/owr/2008/53

05208 | OWR-14299292-013

### Simple Invariant Distinguishing Two Class-Two Exponent-$p$ Groups

O / open | Oberwolfach 报告问题汇编

Let $p$ be an odd prime and let $\omega$ generate $\mathbb{F}_p^\times$. Among groups of exponent $p$ and nilpotency class two, define $G_1=\langle a,b,c,d,e\mid[c,b]=[d,b]=[e,c]=[e,d]=1,\ [d,c]=[b,a],\ [e,b]=[c,a],\ [e,a]=[d,a]^\omega\rangle$ and $G_2=\langle a,b,c,d,e\mid[d,a]=[e,a]=[c,b]=[d,b]=[e,c]=1,\ [d,c]=[b,a],\ [e,b]=[c,a]\rangle$. Find an invariant simpler than automorphism-group order that proves $G_1$ and $G_2$ are nonisomorphic.

来源字段：https://doi.org/10.4171/owr/2025/27

05209 | OWR-11583-014

### Simple Rational Modules in Positive Characteristic

P / partially_solved | Oberwolfach 报告问题汇编

Determine the dimensions and characters of simple rational modules for a split reductive group in positive characteristic, and give a uniform construction or description of them.

来源字段：https://doi.org/10.4171/owr/2012/16

05210 | OWR-4990377-011

### Simultaneous Regularization of Cremona Subgroups

O / open | Oberwolfach 报告问题汇编

Let $G$ be a finitely generated subgroup of the plane Cremona group such that every element of $G$ is regularizable. Must $G$ itself be regularizable, equivalently conjugate to a subgroup of the automorphism group of a surface? Equivalently, must its locally elliptic action on the blow-up complex be globally elliptic?

来源字段：https://doi.org/10.4171/owr/2021/8

05211 | OWR-8415341-010

### Smallest Colored Graph Representations of Permutation Groups

O / open | Oberwolfach 报告问题汇编

For $G\leq\operatorname{Sym}(\Omega)$, determine the smallest vertex-colored and edge-colored graph $\Gamma$ representing $G$, where size is measured by $|V|+|E|$.

来源字段：https://doi.org/10.4171/owr/2021/38

05212 | OWR-16167-004

### Smoothability of Group Actions on One-Manifolds

O / open | Oberwolfach 报告问题汇编

Let $M$ be a connected one-dimensional manifold and let $G<\operatorname{Homeo}^+(M)$ act on $M$. Does the abstract group $G$ admit a faithful action by $C^r$ diffeomorphisms on $M$? Is the given action topologically conjugate to a $C^r$ action, and what obstructs such actions on one-dimensional manifolds? Also, does an infinite Kazhdan group admit a faithful action by homeomorphisms on the circle?

来源字段：https://doi.org/10.4171/owr/2018/26

05213 | OWR-5154-011

### Soficity, Hyperlinearity, and Surjunctivity of Major Group Classes

O / open | Oberwolfach 报告问题汇编

Determine whether Thompson's group $F$, one-relator groups, Burnside groups, Burger-Mozes groups, Gromov monster groups, word-hyperbolic groups, and random finitely presented groups are sofic or hyperlinear. If $G_1$ and $G_2$ are sofic or hyperlinear and a residually finite group $K$ embeds in both, must $G_1*_K G_2$ also be sofic or hyperlinear? If finitely generated groups $G_1$ and $G_2$ are surjunctive, must $G_1\times G_2$ and $G_1*G_2$ be surjunctive? Classify the sofic approximations of a direct product of two nonabelian free groups.

来源字段：https://doi.org/10.4171/owr/2011/26

05214 | OWR-17294-009

### Solvability and Realizability of Finite Skew Braces

P / partially_solved | Oberwolfach 报告问题汇编

Is the multiplicative group of a finite skew left brace with solvable additive group solvable? Is every finite nilpotent group of class two the multiplicative group of a two-sided skew brace of abelian type? Is there a non-IYB finite group all of whose Sylow subgroups are IYB groups?

来源字段：https://doi.org/10.4171/owr/2019/51

05215 | OWR-13112-001

### Solvability of Group Equations and Surjectivity of Word Maps

P / partially_solved | Oberwolfach 报告问题汇编

Can every nonsingular equation $w\in\Gamma*F_n$ with coefficients in a group $\Gamma$ be solved over $\Gamma$? Also, for every nontrivial word $w\in F_2$, does there exist $n_0$ such that the word map $w:\operatorname{PU}(n)\times\operatorname{PU}(n)\to\operatorname{PU}(n)$ is surjective for every $n\geq n_0$?

来源字段：https://doi.org/10.4171/owr/2014/42

05216 | AMR-011-0001

### Some Questions - Question 1

P / partially_solved | AMR 问题清单汇编

Can the odometer acting on the rooted binary tree be embedded in a nonabelian free pro-$2$ group?

背景链接：https://www.renyi.hu/~abert/questions.pdf

05217 | AMR-011-0010

### Some Questions - Question 10

O / open | AMR 问题清单汇编

Let $\Gamma$ be finitely presented with a chain of finite-index normal subgroups having trivial intersection and property $\tau$. Must $\Gamma$ contain a nonabelian free subgroup?

背景链接：https://www.renyi.hu/~abert/questions.pdf

05218 | AMR-011-0002

### Some Questions - Question 2

O / open | AMR 问题清单汇编

Let a finitely presented pro-$p$ group have $r+2$ generators and $r$ relators. Must it have an open subgroup that maps continuously onto a nonabelian free pro-$p$ group?

背景链接：https://www.renyi.hu/~abert/questions.pdf

05219 | AMR-011-0003

### Some Questions - Question 3

O / open | AMR 问题清单汇编

Let $G$ be a closed transitive subgroup of the automorphism group of a rooted tree. Must every level stabilizer of $G$ contain an element acting without fixed points on the boundary?

背景链接：https://www.renyi.hu/~abert/questions.pdf

05220 | AMR-011-0004

### Some Questions - Question 4

P / partially_solved | AMR 问题清单汇编

Let $\Gamma$ be a countable subgroup of $\operatorname{SL}_2(\mathbb{Q}_p)$ containing no parabolic elements, and let $\gamma$ be a random element of $\operatorname{SL}_2(\mathbb{Q}_p)$. Show that $\langle\Gamma,\gamma\rangle$ almost surely contains no parabolic elements.

背景链接：https://www.renyi.hu/~abert/questions.pdf

05221 | AMR-011-0041

### Some Questions - Question 41

P / partially_solved | AMR 问题清单汇编

Does every countable group have fixed price; that is, do all its free probability-measure-preserving actions have the same cost?

背景链接：https://www.renyi.hu/~abert/questions.pdf

05222 | AMR-011-0043

### Some Questions - Question 43

P / partially_solved | AMR 问题清单汇编

Let $\Gamma$ act on $X$ by probability-measure-preserving maps and let $H\leq\Gamma$ have finite index. Is $\operatorname{cost}(H,X)-1=(\operatorname{cost}(\Gamma,X)-1)[\Gamma:H]$?

背景链接：https://www.renyi.hu/~abert/questions.pdf

05223 | AMR-011-0044

### Some Questions - Question 44

O / open | AMR 问题清单汇编

For a free action of $\Gamma$ on $X$, is $\operatorname{cost}(\Gamma,X)=\operatorname{cost}(\Gamma,X\times X)$ for the diagonal action on $X\times X$?

背景链接：https://www.renyi.hu/~abert/questions.pdf

05224 | AMR-011-0045

### Some Questions - Question 45

S / solved | AMR 问题清单汇编

Let an amenable group $\Gamma$ act ergodically and essentially faithfully on $X$. Is the groupoid cost of the action $1$? If $\Gamma$ is finitely presented, is this true for every infinite ergodic action?

背景链接：https://www.renyi.hu/~abert/questions.pdf

05225 | AMR-011-0046

### Some Questions - Question 46

O / open | AMR 问题清单汇编

**Question 46.** Is there a subgraphing whose cost is arbitrarily close to the cost of the relation $\mathrm{cost}(R)$? (i.e. can one always approach $\mathrm{cost}(R)$ by graphings generating sub-relations of $R$?)

背景链接：https://www.renyi.hu/~abert/questions.pdf

05226 | AMR-011-0047

### Some Questions - Question 47

S / solved | AMR 问题清单汇编

Can a nonabelian free group $F$ have a nontrivial pseudocharacter invariant under $\operatorname{Aut}(F)$?

背景链接：https://www.renyi.hu/~abert/questions.pdf

05227 | AMR-011-0048

### Some Questions - Question 48

P / partially_solved | AMR 问题清单汇编

Are two independent random subsets of $\mathbb{Z}$ almost surely quasi-isometric as metric spaces? What is the answer for other Cayley graphs, such as that of $\operatorname{SL}_3(\mathbb{Z})$?

背景链接：https://www.renyi.hu/~abert/questions.pdf

05228 | AMR-011-0049

### Some Questions - Question 49

P / partially_solved | AMR 问题清单汇编

Let $P$ be a finite $p$-group of order $n$ and let $w$ be any word. Is the probability that $w$ is satisfied in $P$ at least $1/n$?

背景链接：https://www.renyi.hu/~abert/questions.pdf

05229 | AMR-011-0005

### Some Questions - Question 5

P / partially_solved | AMR 问题清单汇编

Determine the asymptotic length of a shortest nontrivial group law for the $n$-fold iterated wreath product of $C_2$.

背景链接：https://www.renyi.hu/~abert/questions.pdf

05230 | AMR-011-0050

### Some Questions - Question 50

O / open | AMR 问题清单汇编

Let $M$ be the set of measurable real functions, and define $H_f(x,y)=(x,y+f(x))$ and $V_f(x,y)=(x+f(y),y)$. What group is generated by $\{H_f:f\in M\}$ and $\{V_f:f\in M\}$? Is there an $N$ such that every element is a product of at most $N$ such generators?

背景链接：https://www.renyi.hu/~abert/questions.pdf

05231 | AMR-011-0051

### Some Questions - Question 51

P / partially_solved | AMR 问题清单汇编

Determine the asymptotic length of the shortest nontrivial word that is a law in every group of order $2^n$.

背景链接：https://www.renyi.hu/~abert/questions.pdf

05232 | AMR-011-0009

### Some Questions - Question 9

O / open | AMR 问题清单汇编

Let $\Gamma$ be finitely generated and let $\{H_n:n\geq1\}$ be a property-$\tau$ family of finite-index normal subgroups. Does the chain $\Gamma_n=\bigcap_{k=1}^n H_k$ have property $\tau$?

背景链接：https://www.renyi.hu/~abert/questions.pdf

05233 | AIM-GEOMETRIC_GROUP_THEORY-0142

### Square-mass forestings, invariant random forests, and the Littlewood threshold

P / partially_solved | AIM 研讨会问题汇编

Let $\Gamma$ be a non-amenable group with a probability measure preserving action of $\Gamma$ on $(X,\mu)$.

Is it true that for any $N$ there exist measurable subsets $A_g\subseteq X$ $(g\in\Gamma)$ such that $\prod\limits_{x\in A_g}(x,xg)$ is a forest with $\sum\limits_{g\in\Gamma}\mu^2(A_g)>N$?

来源字段：http://aimpl.org/l2invariantsgroups/2/

05234 | AIM-GEOMETRIC_GROUP_THEORY-0168

### Stabilizer lengths and rapid decay for tree and hyperbolic-graph actions

P / partially_solved | AIM 研讨会问题汇编

This record is the four-question list in Section 5, “RD for groups acting on special metric spaces,” of the AIM workshop notes *The property of rapid decay*.

1. In the \(\widetilde A_2\)-building setting of RRS, for \(g\) supported on elements of shape \((m,n)\), their estimate is \[ \|f*g\|_2\le p(m,n)\|f\|_2\|g\|_2, \qquad p(m,n)=\frac12(m+1)(n+1)(m+n+2)\sqrt{\max\{m,n\}+1}. \] Is the square-root factor necessary? 2. Does Wise's non-Hopfian group have property RD? The presentation intended in the source is \[ W=\langle a,b,s,t\mid [a,b]=1,\; sas^{-1}=(ab)^2,\;tbt^{-1}=(ab)^2\rangle . \] 3. If \(\Gamma\) acts on a simplicial tree, what hypotheses on vertex and edge stabilizers characterize property RD? The workshop notes suggest RD of stabilizers for the lengths induced from \(\Gamma\), with HNN extensions as an intended application. 4. Can the tree be replaced by a bounded-degree \(\delta\)-hyperbolic graph?

The equivalent five-generator presentation used by Barré--Pichot is \[ \langle a,b,c,s,t\mid c=ab=ba,\;c^2=sas^{-1}=tbt^{-1}\rangle . \] Changing both stable letters to their inverses accounts for the opposite conjugation convention sometimes seen in print.

来源字段：https://aimath.org/WWN/rapiddecay/rapiddecay.pdf

05235 | OWR-4411-010

### Stable Commutator Length, Growth, and Subgroup Automorphisms

P / partially_solved | Oberwolfach 报告问题汇编

For stable commutator length on random elements or random walks, what is the constant and error term in the conjectured asymptotic law; can a finitely generated or finitely presented group have intermediate growth; and is the outer automorphism group preserving a finitely generated subgroup of a free group finitely generated or finitely presented?

来源字段：https://doi.org/10.4171/owr/2010/30

05236 | OWR-15433-007

### Stable Prasad Multiplicity Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

别名：Stable Prasad Multiplicity Conjecture [OWR-15433-008]

In the $p$-adic Galois-pair setting, let $\phi$ be a discrete $L$-parameter with stable representation $\Pi_\phi$. Is the multiplicity given by $$m(\Pi_\phi,\chi)=|\ker^1(F;H,G)|^{-1}\sum_\psi\frac{|Z(\phi)|}{|Z(\psi)|},$$ where the sum ranges over compatible lifts $\psi$ through $H^\vee_\chi\to{}^LG$?

来源字段：https://doi.org/10.4171/owr/2017/25

05237 | OWR-17140-010

### Stratified Equivalence for Bernstein Components

P / partially_solved | Oberwolfach 报告问题汇编

For each inertial class $s$, does the ABPS bijection $\mu_s$ between the Bernstein component $\operatorname{Irr}(G)_s$ and its twisted spectral extended quotient arise from a stratified equivalence between $\mathcal O(T_s)\rtimes W_s$ and the Bernstein algebra $\mathcal H(G,s)$?

来源字段：https://doi.org/10.4171/owr/2019/44

05238 | OWR-4428-003

### Strict Cost-$\ell^2$-Betti Inequalities for Free Group Actions

O / open | Oberwolfach 报告问题汇编

Does there exist an infinite group with a free probability-measure-preserving action $\Gamma\curvearrowright(X,\mu)$ such that $\beta_1^{(2)}(\Gamma)<\operatorname{cost}(R_{\Gamma\curvearrowright(X,\mu)})-1$?

来源字段：https://doi.org/10.4171/owr/2010/40

05239 | OWR-14749-009

### Strong Atiyah Conjecture for Free-Group Quotients

O / open | Oberwolfach 报告问题汇编

Let $F$ be finitely generated free, $N\triangleleft F$, $G=F/N$, and $K\subset\mathbb C$. If $G$ has bounded finite-subgroup orders, must every $A\in\operatorname{Mat}_{n\times m}(K[F])$ satisfy $\dim_G\ker\phi_G^A\in(1/\operatorname{lcm}(G))\mathbb Z$?

来源字段：https://doi.org/10.4171/owr/2016/41

05240 | OWR-13675-009

### Structural Explanation of the McKay Conjecture

O / open | Oberwolfach 报告问题汇编

Why should the McKay conjecture hold, namely why should $|\operatorname{Irr}_{\ell'}(G)|=|\operatorname{Irr}_{\ell'}(N_G(P))|$ for a Sylow $\ell$-subgroup $P$?

来源字段：https://doi.org/10.4171/owr/2015/18

05241 | OWR-11101920-001

### Structure and Asymptotics of Finite Group Characters

O / open | Oberwolfach 报告问题汇编

How do tensor products of complex characters decompose; what are their vanishing sets and asymptotic behavior; and what are the dimensions of modular characters?

来源字段：https://doi.org/10.4171/owr/2022/39

05242 | OWR-4535-012

### Structure and Topology of Johnson Cokernels

P / partially_solved | Oberwolfach 报告问题汇编

In the theory of the Johnson filtration of the mapping class group $\mathcal{M}_{g,1}$ of a surface of genus $g$ with one boundary component, let $\tau_k : \mathcal{M}_{g,1}[k] \to \mathfrak{h}_g(k)$ denote the $k$-th Johnson homomorphism into the degree-$k$ part of the derivation Lie algebra $\mathfrak{h}_g$. Determine the algebraic structure and topological meaning of the Johnson cokernels $\mathfrak{h}_g(k) / \tau_k(\mathcal{M}_{g,1}[k])$.

来源字段：https://doi.org/10.4171/owr/2010/53

05243 | OWR-10252930-008

### Structure of Extremal Product-Free Subsets

P / partially_solved | Oberwolfach 报告问题汇编

For a product-free subset $A$ of the alternating group $A_n$, characterize the structure of extremal and near-extremal choices of $A$.

来源字段：https://doi.org/10.4171/owr/2022/22

05244 | OPG-732

### Subgroup formed by elements of order dividing n

P / partially_solved | Open Problem Garden

Conjecture

Suppose $G$ is a finite group, and $n$ is a positive integer dividing $|G|$. Suppose that $G$ has exactly $n$ solutions to $x^{n} = 1$. Does it follow that these solutions form a subgroup of $G$?

背景链接：http://www.openproblemgarden.org/op/subgroup_formed_by_elements_of_order_dividing_n

05245 | AIM-GEOMETRIC_GROUP_THEORY-0030

### Subgroup membership via orbit-generated subgroups of the free kernel

P / partially_solved | AIM 研讨会问题汇编

Solve the subgroup membership problem for free-by-cyclic groups.

来源字段：http://aimpl.org/freebycyclic/7/

05246 | OWR-2053-019

### Subgroup Structure of Interval Exchange Transformations

O / open | Oberwolfach 报告问题汇编

Let $G$ be the group of interval exchange transformations. What can be said about the structure of $G$, and what groups can occur as subgroups of $G$?

来源字段：https://doi.org/10.4171/owr/2008/42

05247 | AIM-GEOMETRIC_GROUP_THEORY-0012

### Subgroup-pattern reduction and a rank-three special case

P / partially_solved | AIM 研讨会问题汇编

Let $G$ and $H$ be free-by-cyclic groups which are quasi-isometric. Then $G$ admits irreducible and atoroidal monodromy if and only if $H$ admits irreducible and atoroidal monodromy.

来源字段：http://aimpl.org/freebycyclic/3/

05248 | OWR-16167-012

### Subgroups of Piecewise-Projective Homeomorphism Groups

O / open | Oberwolfach 报告问题汇编

Is Thompson's group $F$ the only structurally significant subgroup of the group $\operatorname{PL}_o(I)$ of orientation-preserving piecewise-linear homeomorphisms of the interval?

来源字段：https://doi.org/10.4171/owr/2018/26

05249 | OWR-14749-002

### Sublinear Generator Growth of Higher-Rank Lattices

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a higher-rank simple Lie group and $(M_i)$ a sequence of lattices in $G$ with unbounded covolume. Is $\lim_{i\to\infty}(d(M_i)-1)/\operatorname{vol}(G/M_i)=0$?

来源字段：https://doi.org/10.4171/owr/2016/41

05250 | AIM-GEOMETRIC_GROUP_THEORY-0116

### Subquotient normal forms for the SQ-closure of bounded automata groups

P / partially_solved | AIM 研讨会问题汇编

**Definition:** SQ-closure = closure under subgroups and quotients.

1. What are “good” SQ-closed families of amenable groups?
2. What is the SQ-closure of the set of bounded automata groups?

来源字段：http://aimpl.org/amenablediscrete/4/

05251 | OWR-14298365-010

### Super-Expanding Cayley Graphs of Symmetric Groups

O / open | Oberwolfach 报告问题汇编

Can the symmetric groups $\operatorname{Sym}(n)$ be made into a family of super-expanders using a uniformly bounded number of generators?

来源字段：https://doi.org/10.4171/owr/2024/17

05252 | OWR-4411-006

### Surface Subgroups of Word-Hyperbolic Groups

O / open | Oberwolfach 报告问题汇编

Does every word-hyperbolic group $G$ admit an injective homomorphism from the fundamental group of a closed surface of genus at least two?

来源字段：https://doi.org/10.4171/owr/2010/30

05253 | OWR-11101920-004

### Sylow Restrictions and Character Fields of Values

O / open | Oberwolfach 报告问题汇编

Let $P$ be a Sylow $p$-subgroup of a finite group $G$, and let $\chi\in\operatorname{Irr}(G)$ have $p'$-degree and conductor $p^am$ with $p\nmid m$. Is $[\mathbb Q_{p^a}:\mathbb Q(\chi_P)]$ always not divisible by $p$?

来源字段：https://doi.org/10.4171/owr/2022/39

05254 | OWR-16776-002

### Symmetry of Completed Finite-Group Representation Rings

O / open | Oberwolfach 报告问题汇编

Let $G$ be a finite group and let $k$ be a field of characteristic $p$. Write $a_{\mathbb C}(G)=\mathbb C\otimes_{\mathbb Z}a(G)$ for the complexified Green ring of finite-dimensional $kG$-modules, give $\sum_i a_i[M_i]$ the norm $\sum_i|a_i|\dim M_i$, and let $\widehat a(G)$ be its completion with involution induced by module duality. Is $\widehat a(G)$ a symmetric Banach $*$-algebra?

来源字段：https://doi.org/10.4171/owr/2019/14

05255 | OWR-4411-008

### Symplectic Extremal Quasimorphisms on Free Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a free group $G$ and every element $g$ in its commutator subgroup, does there exist an extremal quasimorphism for $g$ arising from symplectic geometry?

来源字段：https://doi.org/10.4171/owr/2010/30

05256 | OWR-12004-002

### Tannakian Lattice Chains for Bruhat-Tits Buildings

O / open | Oberwolfach 报告问题汇编

For a connected reductive group $G$ over a complete discretely valued field and a representation $V$, what lattice chains in $V\otimes K$ should be used to construct a Tannakian description of the Bruhat-Tits building and parahoric group schemes?

来源字段：https://doi.org/10.4171/owr/2012/32

05257 | OWR-14298004-007

### TAP Infinity for Free-by-Cyclic Groups

O / open | Oberwolfach 报告问题汇编

Does every finitely generated free-by-cyclic group belong to $\operatorname{TAP}_\infty(R)$?

来源字段：https://doi.org/10.4171/owr/2023/55

05258 | AIM-GEOMETRIC_GROUP_THEORY-0131

### Tarski number 7: source correction and a sharp seven-piece localization criterion

P / partially_solved | AIM 研讨会问题汇编

Is there a group with Tarski number 7?

If this group is amenable, the Folner function of this group is universal bound.

Is the universal Grigorchuk group amenable?

If this group is amenable, then the Folner function of this group is universal bound on Folner function of $G_\omega$.

来源字段：http://aimpl.org/amenablediscrete/5/

05259 | OWR-14298004-012

### TDLC Analogue of Finite-Subgroup Conjugacy

O / open | Oberwolfach 报告问题汇编

For totally disconnected locally compact groups satisfying appropriate analogues of membership in Kropholler's hierarchy and type $\mathrm{FP}_\infty(\mathbb Z)$, are there only finitely many conjugacy classes of finite elementary abelian $p$-subgroups?

来源字段：https://doi.org/10.4171/owr/2023/55

05260 | OWR-4428-012

### The $\ell^2$-Betti Spectrum of the Lamplighter Group

P / partially_solved | Oberwolfach 报告问题汇编

What is $C(\mathbb Z/2\wr\mathbb Z)$, the set of $\ell^2$-Betti numbers arising from the lamplighter group $\mathbb Z/2\wr\mathbb Z$?

来源字段：https://doi.org/10.4171/owr/2010/40

05261 | AIM-GEOMETRIC_GROUP_THEORY-0039

### The Artin word problem is local on big chunks

P / partially_solved | AIM 研讨会问题汇编

Is the word problem solvable for all Artin groups?

来源字段：http://aimpl.org/geomartingp/2/

05262 | AIM-GEOMETRIC_GROUP_THEORY-0089

### The Brin-Sapir dichotomy and finite embedding obstructions

P / partially_solved | AIM 研讨会问题汇编

We say that a collection of groups $\mathcal C$ is *quasi-ordered* if for any sequence $\{G_i\}_{i\in\mathbb N}$ of groups in $\mathcal C$ there is $i<j$ such that $G_i$ embeds into $G_j$.

Is it true that a subgroup of $F$ is elementary amenable if and only if it does not contain a copy of $F$?

来源字段：http://aimpl.org/amenablediscrete/1/

05263 | AIM-GEOMETRIC_GROUP_THEORY-0067

### The CAT(1) spherical Deligne complex of type B3

P / partially_solved | AIM 研讨会问题汇编

Is the Moussong metric on the spherical Deligne complex for the type \(B_3\) Artin group \(\mathrm{CAT}(1)\)?

来源字段：http://aimpl.org/geomartingp/5/

05264 | OWR-15438-003

### The Dupont-Monod Comparison-Map Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For every connected semisimple Lie group $G$ with finite center, is the comparison map

$$H^n_{\mathrm{cb}}(G;\mathbb R)\longrightarrow H^n_{\mathrm c}(G;\mathbb R)$$

an isomorphism for every degree $n$?

来源字段：https://doi.org/10.4171/owr/2017/30

05265 | AIM-GEOMETRIC_GROUP_THEORY-0154

### The exact Aut-to-Out correction for characteristic congruence kernels

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.3.** What are and what is known about the finite index subgroups of \(\operatorname{Out}(F_n)\)?

One fact about finite-index subgroups concerns principal congruence subgroups. Given \(H\) a characteristic finite-index subgroup of \(F_n\), one has \[ \Gamma=\ker\bigl(\operatorname{Aut}(F_n)\longrightarrow \operatorname{Aut}(F_n/H)\bigr). \]

来源字段：https://aimath.org/WWN/outerauto/outerauto.pdf

05266 | AIM-GEOMETRIC_GROUP_THEORY-0165

### The forward Lipschitz horoboundary and a rose-face orientation test

P / partially_solved | AIM 研讨会问题汇编

Problem 5.3. What is the closure X? What is X\X, the horofunction boundary of outer space in the Lipschitz metric?

Thierry Coulbois asked if we know the horofunction boundary in the simplicial metric, and also if it is reasonable Busemann points could be the whole thing. For the latter, Cormac said yes, as it is true for Teichmüller space. Specifically, a Busemann point \(B_\sigma\) is determined by a 'good' ray

\(\sigma:[0,\infty)\to X_n\) by \(B_\sigma(x)=\lim_{n\to\infty}(d(x,\sigma(n))-n)\). If \(X\) were CAT(0) or Teichmüller space 6 EDITED BY JOHANNA MANGAHAS

with the Lipschitz metric, then X\X would consist entirely of Busemann points. However, warned Cormac, even for normed spaces you can get points that are not Busemann points.

来源字段：https://aimath.org/WWN/outerauto/outerauto.pdf

05267 | AIM-GEOMETRIC_GROUP_THEORY-0148

### The free-factor complex, recursive corank-one links, and a dynamical transfer test

P / partially_solved | AIM 研讨会问题汇编

Construct a \(\delta\)-hyperbolic graph on which \(\operatorname{Out}(F_n)\) acts so that: (1) every iwip \(\gamma\) has positive translation distance \((|\gamma|>0)\); (2) polynomially growing \(\gamma\) are elliptic \((|\gamma|=0)\); (3) the construction is elegant/beautiful/useful/truthful; (4) local structure allows for inductive arguments; and (5) the graph is highly connected, with highly connected links.

来源字段：https://aimath.org/WWN/outerauto/outerauto.pdf

05268 | OWR-4535-003

### The Haagerup Property for Mapping Class Groups

O / open | Oberwolfach 报告问题汇编

To what extent do mapping class groups have the Haagerup property?

来源字段：https://doi.org/10.4171/owr/2010/53

05269 | AIM-GEOMETRIC_GROUP_THEORY-0169

### The Lorentz endpoint behind Lafforgue's finite-set RD condition

P / partially_solved | AIM 研讨会问题汇编

I inspected the official PDF, in particular PDF page 5.

The next line, **7 Applications of RD**, is a new section heading, not part of item 6. Item 6 itself is a remark, not a missing question.

Two reconstructions are necessary for item 4.

* The PDF literally says "finite subspaces"; the only coherent meaning is **finite subsets**. * The letter \(B\) is used both for one of those subsets and for a ball. Below \(B_\ell(e,r)=\{x:\ell(x)\le r\}\) denotes the word-metric ball, so the hypothesis is \(A\subseteq B_\ell(e,r)\).

来源字段：https://aimath.org/WWN/rapiddecay/rapiddecay.pdf

05270 | AIM-GEOMETRIC_GROUP_THEORY-0111

### The missing just-infinite hypothesis and an explicit branch-group counterexample

S / solved | AIM 研讨会问题汇编

**Def:** A subgroup $H$ of a group $G$ is *commensurated* if $\forall g\in G$, $g^{-1}Hg\cap H$ has finite index in $H$.

**Thm (Wesolek):** A finitely generated branch group is just infinite if and only if every commensurated subgroup is either finite or of finite index.

Find a combinatorial proof of the following: every commensurated subgroup of a finitely generated branch group is either finite or of finite index.

来源字段：http://aimpl.org/amenablediscrete/3/

05271 | AIM-GEOMETRIC_GROUP_THEORY-0118

### The non-discrete Day class, half-lamp subgroups, and a Basilica reduction

P / partially_solved | AIM 研讨会问题汇编

Let \(\mathscr{AE}\) be the smallest class of locally compact groups that contains all compact groups, discrete amenable groups, and locally compact abelian groups and that is closed under taking group extensions, closed subgroups, Hausdorff quotients, and directed unions of open subgroups.

来源字段：http://aimpl.org/amenablediscrete/4/

05272 | OWR-14299289-007

### The Numerical Unramified Plancherel Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a split connected reductive group over a nonarchimedean local field and let $X$ be a spherical $G$-variety with hyperspherical cotangent bundle. Is the unramified spherical Hecke module $C_c^\infty(X(F))^{G(\mathcal O)}$, with its $L^2$ inner product and basic vector, isometrically identified with functions on the dual spectral quotient, with Hecke operators acting by multiplication and Plancherel measure determined by the dual representation $V_X$?

来源字段：https://doi.org/10.4171/owr/2025/18

05273 | AIM-GEOMETRIC_GROUP_THEORY-0066

### The open Artin K(pi,1) conjecturette and its exact degree-two obstructions

P / partially_solved | AIM 研讨会问题汇编

**\(K(\pi,1)\) conjecturette:** is \(\pi_2\) of Deligne complex/Salvetti complex/hyperplane complement always trivial? Find a proof using geometric/combinatorial techniques.

来源字段：http://aimpl.org/geomartingp/5/

05274 | AIM-GEOMETRIC_GROUP_THEORY-0061

### The positive-suffix order is a lattice exactly in spherical type

S / solved | AIM 研讨会问题汇编

Nearby items do not define a special group \(A\), and the page has no hidden introduction or attribution. The natural reconstruction is therefore that \(A=A_S\) is a finite-rank Artin group with its standard generating set \(S\), and “positive word” means an element of the standard Artin monoid \(M=A_S^+\).

Two ambiguities materially change the answer.

1. The empty word must be allowed. If “positive” means nonempty, then \(a\nleq a\) for every \(a\), so the relation is not reflexive and cannot define a join-semilattice. Standard monoid usage includes the empty word, representing \(1\); that is the convention used below. 2. The recorded domain is the group \(A\), not the monoid \(A^+\). On \(A^+\) the answer is always yes, by positive-monoid gcds. On the whole group the answer is yes exactly in spherical type.

The issue is under-specification, not damaged text.

来源字段：http://aimpl.org/geomartingp/4/

05275 | AIM-GEOMETRIC_GROUP_THEORY-0108

### The solved volume exponent of the first Grigorchuk group

S / solved | AIM 研讨会问题汇编

Does the limit \[ \lim_{n\rightarrow\infty} \frac{\log \log (B_G(n))}{\log n} \] exist for the Grigorchuk group?

The notation requires recovery. A Wayback Machine snapshot of the actual AIM page dated 2023-12-08 contains the same TeX source `\log \log (B_G(n))`; there are no absolute-value bars in that archived HTML. On the other hand, a logarithm cannot be applied to a set. The immediately preceding AIM item writes $B_n(e)$ for a ball, while the standard scalar object in the literature is the volume growth \[ v_{G,S}(n)=|B_{G,S}(n)|=\#\{g\in G:|g|_S\le n\}. \] The only mathematically coherent recovery is therefore \[ \lim_{n\to\infty}\frac{\log\log |B_{G,S}(n)|}{\log n}. \tag{1.1} \]

The singular phrase “the Grigorchuk group,” together with the neighboring problems' notation $G$, means the **first Grigorchuk group** \[ G=G_{(012)^\infty} \] with the canonical tree-automorphism generators $S=\{a,b,c,d\}$. This is also exactly the convention in Erschler--Zheng. The AIM record does not state $S$; Section 7 proves that the limit and its value are independent of the choice of finite generating set, so this omission does not affect the final answer.

来源字段：http://aimpl.org/amenablediscrete/3/

05276 | AIM-GEOMETRIC_GROUP_THEORY-0132

### The subexponential Folner gap

P / partially_solved | AIM 研讨会问题汇编

**Conjecture:** If the Folner function of a group is sub-exponential, then the group is virtually nilpotent.

* \(G\) is finitely generated and \(S\) is a finite symmetric generating set; * the Følner parameter is reciprocal boundary ratio, not a radius parameter; * “subexponential” means \[ \limsup_{n\to\infty}\frac{\log \operatorname{Fol}_{G,S}(n)}{n}=0, \tag{1.1} \] equivalently, for every \(\varepsilon>0\), \(\operatorname{Fol}_{G,S}(n)\le e^{\varepsilon n}\) for all sufficiently large \(n\).

来源字段：http://aimpl.org/amenablediscrete/5/

05277 | AIM-GEOMETRIC_GROUP_THEORY-0121

### The two-generator alternating-telescope solution

S / solved | AIM 研讨会问题汇编

Is there a 2-generated infinite simple amenable group?

来源字段：http://aimpl.org/amenablediscrete/4/

05278 | AIM-GEOMETRIC_GROUP_THEORY-0125

### The virtually-nilpotent/free-semigroup alternative

P / partially_solved | AIM 研讨会问题汇编

**Conjecture:** A finitely presented group is either virtually nilpotent, or contains a free subsemigroup of order \(2\).

**Grigorchuk’s conjecture.** Every finitely presented group \(G\) is either virtually nilpotent or contains two elements \(a,b\) such that all nonempty positive words in \(a,b\) represent distinct elements of \(G\).

来源字段：http://aimpl.org/amenablediscrete/5/

05279 | OWR-8415353-002

### Thinness of Hyperbolic Hypergeometric Monodromy Groups

O / open | Oberwolfach 报告问题汇编

For hyperbolic hypergeometric monodromy groups $\Gamma_{\alpha,\beta}$, which are arithmetic and which are thin; in particular, are all but finitely many such groups thin?

来源字段：https://doi.org/10.4171/owr/2021/50

05280 | AIM-GEOMETRIC_GROUP_THEORY-0160

### Three obstructions to infinite mapping-class images

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.2.** What is \(\operatorname{Hom}(\operatorname{Out}(F_n),\operatorname{MCG}(S))\)? Can homomorphisms have infinite image if \(n\ge4\)?

**Remark.** Juan warned that you need bounds on \(S\) and \(n\) because otherwise the answer is the infinite abelianization problem.

If there exists a finite index subgroup \(\Gamma<\operatorname{Out}(F_n)\) which surjects onto \(\mathbb Z\), then \(\operatorname{Out}(F_n)\) surjects onto \(A\), a virtually abelian subgroup of \(\operatorname{MCG}(S)\). So a warm-up question would be

**Problem 4.3.** Does every homomorphism \(\phi:\operatorname{Out}(F_n)\to\operatorname{MCG}(S)\) have virtually abelian image?

来源字段：https://aimath.org/WWN/outerauto/outerauto.pdf

05281 | OWR-14298160-004

### Three Structural Problems for Artin Groups

O / open | Oberwolfach 报告问题汇编

Prove that every Artin group is torsion-free, has a solvable word problem, and satisfies the $K(\pi,1)$ conjecture, or determine precisely which Artin groups fail any of these properties.

来源字段：https://doi.org/10.4171/owr/2024/4

05282 | AIM-GEOMETRIC_GROUP_THEORY-0019

### Torsion growth requires genuinely Farber covers

P / partially_solved | AIM 研讨会问题汇编

If \(G\) is a free-by-cyclic group with exponentially growing monodromy then does it have non-vashing homology torsion growth?

来源字段：http://aimpl.org/freebycyclic/4/

05283 | OWR-13750328-023

### Tower-Equivalence for Irreducible Spetsial Groups

O / open | Oberwolfach 报告问题汇编

For every irreducible spetsial group except $G_{32}$, prove that the image of $\operatorname{Id}-f$ equals the space of class functions tower-equivalent to zero.

来源字段：https://doi.org/10.4171/owr/2023/19

05284 | OWR-16929-003

### Trace Fields of Hyperbolic Link Complements

O / open | Oberwolfach 报告问题汇编

Which number fields arise as invariant trace fields of hyperbolic link complements?

来源字段：https://doi.org/10.4171/owr/2019/17

05285 | OWR-1703870-001

### Transfer Operations in Homology of Finite Categories

O / open | Oberwolfach 报告问题汇编

For the homology and cohomology of finite categories, how can restriction, corestriction or transfer, conjugation, and inflation be formulated, and what properties should a categorical transfer map satisfy?

来源字段：https://doi.org/10.4171/owr/2020/23

05286 | OWR-14299097-007

### Transitive Binary Actions of Almost Simple Groups

O / open | Oberwolfach 报告问题汇编

Classify all transitive binary permutation actions of almost simple groups, extending the existing classification of primitive binary actions.

来源字段：https://doi.org/10.4171/owr/2025/16

05287 | AIM-GEOMETRIC_GROUP_THEORY-0070

### Two fixed-surface budget obstructions for Artin group embeddings

P / partially_solved | AIM 研讨会问题汇编

**Which Artin groups embed in the mapping class group of a surface?**

Known: $A_n,B_n,D_n,I_2(m),\widetilde A_n,\widetilde C_n$, RAAGs.

来源字段：http://aimpl.org/geomartingp/6/

05288 | AIM-GEOMETRIC_GROUP_THEORY-0126

### Two generator-rank certificates and an exclusion map

P / partially_solved | AIM 研讨会问题汇编

**Problem 5.2.** Is there a finitely presented simple group that is not 2-generated?

来源字段：http://aimpl.org/amenablediscrete/5/

05289 | OWR-12590-005

### Typical Properties of Marked Grigorchuk Groups

P / partially_solved | Oberwolfach 报告问题汇编

For the compact family $G_p$ of marked Grigorchuk groups parametrized by sequences over $\{0,1,\ldots,p\}$, which group-theoretic properties, especially growth rates, are typical in the invariant-measure sense and in the Baire-category sense?

来源字段：https://doi.org/10.4171/owr/2013/42

05290 | OWR-11129-007

### Undecidable Word Problems in Wreath Product Quotients

P / partially_solved | Oberwolfach 报告问题汇编

Let $F$ be a finite-rank free group, let $S$ be finite, and let $\phi:F\to F\wr_S\operatorname{Sym}(S)$ define iterated actions of $F$ on $S^n$. Let $N$ be the kernel of the action on the disjoint union $\bigsqcup_{n\geq0}S^n$. Can the quotient $F/N$ have an undecidable word problem?

来源字段：https://doi.org/10.4171/owr/2011/37

05291 | OWR-17140-004

### Unified Constructions of Supercuspidal Representations

O / open | Oberwolfach 报告问题汇编

Construct a common framework for supercuspidal representations of reductive $p$-adic groups that simultaneously generalizes the Bushnell-Kutzko and Yu constructions.

来源字段：https://doi.org/10.4171/owr/2019/44

05292 | OWR-14298365-003

### Uniform Diameter Bounds for Finite Simple Groups

O / open | Oberwolfach 报告问题汇编

How can diameter bounds for Cayley graphs of finite simple groups be unified across permutation and linear algebraic groups, and which ideas transfer between $\operatorname{Sym}_n$, $\operatorname{Alt}_n$, and $\operatorname{SL}_n$?

来源字段：https://doi.org/10.4171/owr/2024/17

05293 | OWR-17297-003

### Uniform Doubling for Compact Connected Lie Groups

O / open | Oberwolfach 报告问题汇编

Is every compact connected Lie group $K$ uniformly doubling, meaning that the volume-doubling constants of all left-invariant Riemannian metrics on $K$ have a common finite upper bound?

来源字段：https://doi.org/10.4171/owr/2019/54

05294 | OWR-1738-011

### Uniform Freeness Bounds for Bicyclic Units

O / open | Oberwolfach 报告问题汇编

For a finite non-Hamiltonian group $G$, let $m(G)$ be the least positive integer $m$ such that, for every pair of bicyclic units $b_1,b_2$ whose generated subgroup is not nilpotent, the subgroup generated by $b_1$ and $b_2^m$ is free. Is $\{m(G):G\text{ is finite and non-Hamiltonian}\}$ bounded? Equivalently, for a nonzero algebraic integer $z$, let $m(z)$ be the least $k$ such that $kz$ is a free point, meaning that the group generated by $\begin{bmatrix}1&1\\0&1\end{bmatrix}$ and $\begin{bmatrix}1&z\\0&1\end{bmatrix}$ is free. Is $\{m(z)\}$ bounded?

来源字段：https://doi.org/10.4171/owr/2007/55

05295 | OWR-13675-015

### Uniform Polynomial Bases for Stabilizer Modules

O / open | Oberwolfach 报告问题汇编

Let $\lambda\vdash n$, let $U\leq\operatorname{GL}_n(q)$ be the lower unitriangular subgroup, and decompose the restricted permutation module $\operatorname{Res}^{\operatorname{GL}_n(q)}_U M(\lambda)$ into summands indexed by row-standard $\lambda$-tableaux. For a basis $B$ of the Specht module $S(\lambda)$ adapted to this decomposition, write $B_s$ for the basis vectors whose last nonzero component is indexed by the standard tableau $s\in\operatorname{St}(\lambda)$. Can $B$ be chosen independently of the coefficient field $K$ so that $\lvert B_s\rvert=p_s(q)$ for polynomials $p_s\in\mathbb Z[x]$ with $p_s(1)=1$? Can one moreover choose $p_s\in\mathbb Z_{\geq0}[x-1]$?

来源字段：https://doi.org/10.4171/owr/2015/18

05296 | OWR-16164-021

### Uniform Presentations of Pure Reflection Braid Groups

O / open | Oberwolfach 报告问题汇编

Let $W$ be a finite reflection group with pure braid group $P_c(W)$, let $T$ be the set of squares of the reflection generators, and let $\Delta$ be the full twist. Is $$P_c(W)=\langle T:\operatorname{Red}_T(\Delta)\rangle$$ a presentation for every finite reflection group?

来源字段：https://doi.org/10.4171/owr/2018/23

05297 | OWR-1973-006

### Uniform Representation Zeta Functions of Congruence Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $L$ be a Lie lattice over $\mathbb Z$ such that $\mathbb Q\otimes_{\mathbb Z}L$ is a semisimple $\mathbb Q$-Lie algebra of dimension $d$. For almost all primes $p$, let $G_{p,k}$ be the associated principal congruence groups. Does there exist $W_L(X,Y)\in\mathbb Q(X,Y)$ such that $\zeta_{G_{p,k}}^{\mathrm{irr}}(s)=p^{dk}W_L(p,p^{-s})$ for every $k\geq0$ and almost all $p$?

来源字段：https://doi.org/10.4171/owr/2008/28

05298 | AIM-GEOMETRIC_GROUP_THEORY-0102

### Uniform support separation and a stabilizer obstruction in the hyperfinite full group

P / partially_solved | AIM 研讨会问题汇编

Take a hyperfinite (ergodic) equivalence relation on $[0,1]$. Let $G$ be the full group of all measure-preserving transformations that respect the equivalence relation (i.e. whose graph is contained in the equivalence relation). Does $G$ contain any *discrete* non-amenable subgroups? Here \[ d(a,b)=\mu\{x:ax\ne bx\}. \] **Juschenko:** If the answer is no, then IET is amenable.

来源字段：http://aimpl.org/amenablediscrete/2/

05299 | OWR-1319-024

### Uniqueness of Bi-Invariant Hamiltonian Norms

O / open | Oberwolfach 报告问题汇编

If a $G$-invariant norm on the space of normalized Hamiltonians induces a genuine bi-invariant distance on the Hamiltonian group $G$, is it equivalent to the oscillation norm $\operatorname{osc}(F)=\max F-\min F$?

来源字段：https://doi.org/10.4171/owr/2006/33

05300 | OWR-8415355-011

### Unitarity from Nilpotent-Orbit Boundary Codimension

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a real reductive Lie group and call an irreducible Harish-Chandra module unipotent when it is annihilated by a unipotent ideal associated with a nilpotent orbit $\mathcal O$. If $$\operatorname{codim}(\partial\mathcal O,\mathcal O)\geq4,$$ must every such unipotent module be unitary?

来源字段：https://doi.org/10.4171/owr/2021/52

05301 | OWR-14299518-005

### Valid Orderings of Subsets of Finite Groups

O / open | Oberwolfach 报告问题汇编

For which finite groups $G$ does every subset $S\subseteq G\setminus\{\mathrm{id}\}$ admit an ordering whose successive partial products are all distinct? Does this hold for every finite group?

来源字段：https://doi.org/10.4171/owr/2025/42

05302 | OWR-4333-008

### Valuation Compatibility for Moufang Quadrangles of Type $E_{8}$

O / open | Oberwolfach 报告问题汇编

For a Moufang quadrangle of type $E_8$, do the valuation maps on its root groups satisfy the compatibility condition required for the existence of the corresponding Bruhat-Tits building?

来源字段：https://doi.org/10.4171/owr/2010/20

05303 | OWR-1738-010

### Virtual Cohomological Dimension of Unit Groups

O / open | Oberwolfach 报告问题汇编

Let $\mathbb{Q}G=A_1\times\cdots\times A_n$ for a finite group $G$, let $R_i$ be an order in $A_i$, and let $\operatorname{vcd}(R_i^1)$ be the virtual cohomological dimension of its reduced-norm-one unit group. Classifications are known when $\operatorname{vcd}(R_i^1)\leq2$ for every $i$. Can they be extended to $\operatorname{vcd}(R_i^1)\leq3$, yielding a corresponding classification of $U(\mathbb{Z}G)$?

来源字段：https://doi.org/10.4171/owr/2007/55

05304 | OWR-3477-010

### Virtual Cyclicity from Finite Free Classifying Spaces

O / open | Oberwolfach 报告问题汇编

If a group $G$ admits a finite model for its free classifying space $EG$, must $G$ be virtually cyclic?

来源字段：https://doi.org/10.4171/owr/2009/27

05305 | OWR-9790362-003

### Virtual Fibrations with Precisely Controlled Finiteness Properties

O / open | Oberwolfach 报告问题汇编

Is there a finite-index subgroup $\Lambda_0<\Lambda$ and a homomorphism $\phi:\Lambda_0\to\mathbb{Z}$ whose kernel is of type $F_{n-1}$ but not $F_n$?

来源字段：https://doi.org/10.4171/owr/2022/11

05306 | AIM-GEOMETRIC_GROUP_THEORY-0153

### Virtual first homology of Out(F_n): rank classification and a normal-core certificate

S / solved | AIM 研讨会问题汇编

**Problem 3.2.** Does there exist a finite index subgroup \(\Gamma<\operatorname{Out}(F_n)\) with infinite abelianization?

More generally,

来源字段：https://aimath.org/WWN/outerauto/outerauto.pdf

05307 | OWR-5158-016

### Virtual Nonorderability of Cocompact Arithmetic Groups

S / solved | Oberwolfach 报告问题汇编

Does there exist a cocompact arithmetic subgroup $\Gamma$ of some semisimple Lie group $G$ such that no finite-index subgroup of $\Gamma$ admits a left-invariant total order?

来源字段：https://doi.org/10.4171/owr/2011/30

05308 | AIM-GEOMETRIC_GROUP_THEORY-0075

### Virtual residual finiteness is residual finiteness

P / partially_solved | AIM 研讨会问题汇编

Are all Artin groups virtually residually finite?

来源字段：http://aimpl.org/geomartingp/8/

05309 | OWR-12869-006

### Virtual Second Cohomology of Infinite Kähler Groups

O / open | Oberwolfach 报告问题汇编

Does every infinite Kähler group have a finite-index subgroup whose second cohomology with real coefficients is nonzero?

来源字段：https://doi.org/10.4171/owr/2014/09

05310 | OWR-1703863-004

### Virtual Simplicity of Exotic Affine-Building Lattices

P / partially_solved | Oberwolfach 报告问题汇编

Is every $\widetilde A_2$-lattice whose associated building is not Bruhat-Tits virtually simple?

来源字段：https://doi.org/10.4171/owr/2020/16

05311 | OWR-11101920-015

### Weight Conjectures for Spetsial Reflection Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a spetsial $\mathbb Z_\ell$-reflection group $W$, prove analogues of the Alperin weight conjecture and Robinson's ordinary weight conjecture that relate the unipotent principal-block invariants of $W$ to the fusion-system invariants of $\ell$-compact groups with Weyl group $W$.

来源字段：https://doi.org/10.4171/owr/2022/39

05312 | OWR-16167-015

### Well-Quasi-Ordering of Thompson Subgroups

O / open | Oberwolfach 报告问题汇编

Are the biembeddability classes of subgroups of Thompson's group $F$ well-quasi-ordered, and are there subgroups of $\operatorname{PL}_o(I)$ whose classes violate well-quasi-ordering?

来源字段：https://doi.org/10.4171/owr/2018/26

05313 | OPG-37302

### Which lattices occur as intervals in subgroup lattices of finite groups?

P / partially_solved | Open Problem Garden

Conjecture

There exists a finite lattice that is not an interval in the subgroup lattice of a finite group.

背景链接：http://www.openproblemgarden.org/op/which_lattices_occur_as_intervals_in_finite_groups

05314 | OWR-9790362-008

### Zero Divisors in a Seven-Generator Group Ring

O / open | Oberwolfach 报告问题汇编

Let $$\Delta=\langle a,b,c,d,e,f,g\mid acb,add,afg,bbf,cdg,cef,eeg\rangle,$$ a torsion-free group without the unique-product property. Does $\Delta$ satisfy Kaplansky's zero-divisor conjecture over every field?

来源字段：https://doi.org/10.4171/owr/2022/11

05315 | OWR-16167-006

### Zimmer Rigidity for Homeomorphism Actions

O / open | Oberwolfach 报告问题汇编

Does Zimmer's conjecture remain true for actions by homeomorphisms, rather than diffeomorphisms: namely, do higher-rank lattices have only virtually trivial actions on closed manifolds below the prescribed dimension bound?

来源字段：https://doi.org/10.4171/owr/2018/26

# 资料汇编 04 | 代数几何 / Algebraic Geometry

1,159 条记录。O=开放；P=部分解决；S=已解决。状态与英文/LaTeX 表述均沿用来源。

05316 | OWR-1734-008

### $\lambda$-Independence and Hodge Groups of Motives

O / open | Oberwolfach 报告问题汇编

Let $M$ be a pure polarizable motive with Betti Hodge group $\operatorname{Hg}(M)$ and $\lambda$-adic representations $\rho_\lambda$. Does there exist an algebraic subgroup $H\subseteq\operatorname{GL}(H_B(M))$, defined over the coefficient field $E$ and independent of $\lambda$, such that every $\rho_\lambda$ has Zariski-dense image in $H$ and $H^\circ=\operatorname{Hg}(M)$?

来源字段：https://doi.org/10.4171/owr/2007/51

05317 | OWR-1318-003

### $K$-Theory of $D$-Crystals

O / open | Oberwolfach 报告问题汇编

For every scheme $X$, is the K-theory of the $\infty$-category of D-crystals on $X$ naturally equivalent to $K'(X)$?

来源字段：https://doi.org/10.4171/owr/2006/32

05318 | OWR-16171-001

### $p$-Adic Riemann-Hilbert and de Rham Compatibility

O / open | Oberwolfach 报告问题汇编

别名：Canonical Comparison of Geometric Local-System Realizations [OWR-16171-002]

Let $X$ be a smooth algebraic variety over a number field $E$, let $L$ be a geometric $p$-adic étale local system on $X$, and let $\sigma:E\hookrightarrow\mathbb C$. Are the tensor functors that associate to $L$ a regular integrable connection on $X_\sigma$ through the complex Riemann-Hilbert correspondence and through algebraic de Rham theory canonically isomorphic? Is the filtered de Rham realization a complex variation of Hodge structures?

来源字段：https://doi.org/10.4171/owr/2018/30

05319 | OWR-16628-008

### (-1)-Curve Conjecture for Plane Blowups

O / open | Oberwolfach 报告问题汇编

Let $X_r$ be the blowup of $\mathbb P^2$ at $r\geq0$ general points. If $C\subset X_r$ is an irreducible reduced curve with $C^2<0$, must $C$ be a smooth rational curve with $C^2=-1$?

来源字段：https://doi.org/10.4171/owr/2018/45

05320 | AIM-ARITHMETIC_GEOMETRY-0060

### A bad-surface and Rao-module reduction for smooth limits of complete intersections

P / partially_solved | AIM 研讨会问题汇编

**Problem 13.** Let (C_t) be a family of curves in (mathbf P^3) such that a general curve in this family is a smooth complete intersection, and the special curve (C_0) is smooth. Does it imply that (C_0) is also a complete intersection, assuming that the characteristic is (0)?

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05321 | AIM-ALGEBRAIC_GEOMETRY-0237

### A binary-quartic degeneration certificate and a cominuscule obstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 10.6.** Reprove Mukhin-Tarasov-Varchenko theorem using degenerations. Extend it to cominiscule \(G/P\). Notes by Tony Feng 6

来源字段：https://aimath.org/pastworkshops/degenalggeomproblems.pdf

05322 | AIM-ALGEBRAIC_GEOMETRY-0018

### A block formula and Morse-potential computation for compact monotone Fukaya categories

P / partially_solved | AIM 研讨会问题汇编

“Same questions as above for \(Fuk(X)\) where \(X\) is compact symplectic.”

**Recovered question.** For a well-defined split-closed derived Fukaya category of a compact symplectic manifold \(X\), determine its Rouquier dimension and its full Orlov spectrum, including the occurrence and size of gaps and consecutive sequences.

- 7.1: “What can be said about the Orlov spectrum of a toric variety (or any \(X\))? (e.g. about gaps, sequences, etc.)” - 7.2: “What about Rouquier dimension or Orlov spectrum of \(D^b(\operatorname{Sing}(X)) = D^b(\operatorname{Coh}(X))/D^b(\operatorname{Perf}(X))\)?”

Thus the conservative recovery of 7.3 is:

It takes from 7.1 and 7.2 only the explicitly shared invariants, Rouquier dimension and Orlov spectrum.

来源字段：http://aimpl.org/syzygyms/7/

05323 | AIM-ALGEBRAIC_GEOMETRY-0328

### A boundary-frontier sieve for virtually abelian moduli bases

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.1 (Viehweg).** Let \(M_H\) denote a moduli stack of canonically polarized manifolds. Can \(\pi_1(M_H)\) be almost abelian (almost abelian means having an abelian subgroup of finite index)? More precisely, if \(U\) is a nonsingular variety (not a point) mapping generically finitely to the moduli stack, can \(\pi_1(U)\) be almost abelian. The conjecture is no. The result is known for \(M_g\), and can be verified whenever some sort of local Torelli theorem holds. A \(U\) as above cannot be the complement in \(\mathbb P^n\) of a normal crossings divisor with fewer than \(n\) components. At present this is only known if the number of components of \(H\) is strictly smaller than \(n\) (see [VZ02]). The following two questions are motivated by work of Viehweg and Zuo [VZ02] and [VZ05].

来源字段：https://aimath.org/WWN/birational/birational.pdf

05324 | AIM-ALGEBRAIC_GEOMETRY-0299

### A boundary-saturation obstruction to deforming F-injectivity

P / partially_solved | AIM 研讨会问题汇编

**Recovered deformation question.** Let \((R,\mathfrak m)\) be a commutative Noetherian local ring of characteristic \(p>0\), let \(f\in\mathfrak m\) be a nonzerodivisor, and put \(S=R/fR\). If Frobenius is injective on every \(H^j_{\mathfrak m}(S)\), must Frobenius be injective on every \(H^i_{\mathfrak m}(R)\)?

来源字段：http://aimpl.org/testandmultiplierideals/2/

05325 | AIM-ARITHMETIC_GEOMETRY-0005

### A canonical obstruction to correction-free arithmetic transfer

P / partially_solved | AIM 研讨会问题汇编

In any arithmetic transfer conjectures, could we find perfect test transfer functions for (derived) orbital integrals to match arithmetic intersection numbers with no correction factors?

来源字段：http://aimpl.org/intersectshimura/1/

05326 | AIM-ALGEBRAIC_GEOMETRY-0041

### A canonical two-axis height profile for slice differentials

P / partially_solved | AIM 研讨会问题汇编

**Heights of slice differentials.** The differentials in the slice spectral sequence for $\mathrm{BP}^{((G))}\langle n\rangle$ look like they are stratified by height. Is it possible to make sense of the height of a differential in the slice spectral sequence of $\mathrm{BP}^{((G))}\langle n\rangle$ so that it has a well defined height?

来源字段：http://aimpl.org/equivstable/4/

05327 | AIM-ALGEBRAIC_GEOMETRY-0338

### A clean excess-two intersection of line- and conic-branch loci

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.4.** Find natural loci in moduli spaces of surfaces which would be interesting to intersect with each other.

来源字段：https://aimath.org/WWN/birational/birational.pdf

05328 | AIM-ALGEBRAIC_GEOMETRY-0030

### A coefficient-ring shadow and coherence obstruction for the Pin(2)-Q8 Tate model

P / partially_solved | AIM 研讨会问题汇编

Equip \(\mathrm{Pin}(2)\) (resp. \(Q_8\)) with the canonical \(\mathbb Z/2\)-action (resp. the action given by conjugation by \(i\)). Then the \(E_\infty\)-map \[ L^s(\mathbb Z)_{(2)}\longrightarrow L^n(\mathbb Z) \] can be identified by taking strict \(\mathbb Z/2\)-fixed points of the inclusion-induced map \[ \mathbb Z_{(2)}^{t\mathrm{Pin}(2)}\longrightarrow \mathbb Z^{tQ_8}. \]

This is Conjecture 2.9, “\(L\)-theory of integers,” from the AIM workshop *Equivariant techniques in stable homotopy theory* (October 2022).

Write \[ P=\mathrm{Pin}(2)=S^1\cup jS^1\subset \mathbb H^\times, \qquad Q=Q_8=\{\pm1,\pm i,\pm j,\pm k\}, \] and let \(C=\langle\tau\rangle\cong C_2\) act by \(\alpha=c_i\), conjugation by \(i\). This fixes \(S^1\), sends \(jz\) to \(-jz\), has order two because \(i^2=-1\) is central, and restricts to the stated action on \(Q\). For an extension \(1\to K\to K\rtimes_\alpha C\to C\to1\), the parametrized Tate construction produces a genuine \(C\)-spectrum whose underlying spectrum is the ordinary \(K\)-Tate construction. “Strict fixed points” is interpreted as derived categorical fixed points of this genuine spectrum, not geometric fixed points and not Borel homotopy fixed points. This is the interpretation on which the results below are conditional if the source intended a different model.

来源字段：http://aimpl.org/equivstable/2/

05329 | AIM-ALGEBRAIC_GEOMETRY-0043

### A coherence-and-square-defect gate for power operations in slice spectral sequences

P / partially_solved | AIM 研讨会问题汇编

**Connecting slice differentials via power operations.** We would like a systematic way of understanding slice differentials in the slice spectral sequence for $\mathrm{BP}\langle n\rangle^{((G))}$. Are there power operations connecting differentials in the slice spectral sequence for $\mathrm{BP}\langle n\rangle^{((G))}$?

来源字段：http://aimpl.org/equivstable/4/

05330 | AIM-ALGEBRAIC_GEOMETRY-0315

### A common-SNC carry formula for the negative-perturbation subadditivity question

P / partially_solved | AIM 研讨会问题汇编

Let \((R,\mathfrak m)\) be a regular local ring of characteristic \(p\). Suppose that ideals \(I,\mathfrak a,\mathfrak b\) of \(R\) satisfy \(\mathfrak a\subset I\), \(\mathcal J(I^{-\epsilon}\cdot\mathfrak a^s)\subset\mathfrak m\), and \(\mathcal J(\mathfrak b^t)\subset\mathfrak m\). Then is \[ \mathcal J(I^{-\epsilon}\cdot\mathfrak a^s\cdot\mathfrak b^t) \subset \mathcal J(I^{-\epsilon}\cdot\mathfrak a^s) \mathcal J(\mathfrak b^t)? \]

来源字段：https://aimath.org/WWN/singularvariety/problems_final.pdf

05331 | AIM-ALGEBRAIC_GEOMETRY-0093

### A complete rank-two central reconstruction and a higher-rank extension obstruction

P / partially_solved | AIM 研讨会问题汇编

Can one describe \(\mathrm{HHH}^{a>0}\) using \(\mathrm{HHH}^{a=0}\) after tensoring with a suitable central complex?

来源字段：http://aimpl.org/catheckehilbert/5/

05332 | AIM-ALGEBRAIC_GEOMETRY-0085

### A corrected target and an explicit B2 full-twist benchmark

P / partially_solved | AIM 研讨会问题汇编

Outside of type A is there still a relation between Hochschild cohomology of Rouquier complexes and affine Springer fibres?

来源字段：http://aimpl.org/catheckehilbert/3/

05333 | AIM-ALGEBRAIC_GEOMETRY-0348

### A Cox-restriction bridge for the perfect-cone compactification

P / partially_solved | AIM 研讨会问题汇编

**(Shepherd-Barron) Questions on \(A_g\) and the first Voronoi compactification \(A_g^F\).**

1. Run the MMP on \(A_g^F\) when \(g\leq 10\); in particular, flip its extremal ray. If \(g\geq 12\), \(A_g^F\) is the canonical model of \(A_g\), and when \(g=11\) it is a minimal model. 2. There is a family of natural inclusions \(A_g^F\rightarrow A_{g+1}^F\) parameterized by the \(j\)-line. Does \(H^*(A_g^F,\mathbb C)\) stabilize? To what? 3. Let \(M\) be the bundle of weight 1 modular forms and \(D\) the boundary divisor. Compute the intersection numbers \(M^aD^b\) and the plurigenera of \(A_g^F\). 4. Is the total coordinate ring of \(A_g^F\) finitely generated? 5. Is the divisor \(12M-D\) semi-ample for \(g\geq 12\)? 6. Find the effective cone of \(A_g^F\). This is the well-known question of finding the possible slopes of Siegel cusp forms.

来源字段：https://aimath.org/WWN/birational/birational.pdf

05334 | AIM-ALGEBRAIC_GEOMETRY-0313

### A cyclotomic resolution-Betti obstruction for the global canonical index

P / partially_solved | AIM 研讨会问题汇编

**12. (Shokurov)** Consider projective varieties \(X\) of dimension \(n\) with log-canonical singularities which satisfy \(K_X\equiv 0\). Then is the index of \(K_X\) bounded? This is open for dimension 4.

来源字段：https://aimath.org/WWN/singularvariety/problems_final.pdf

05335 | AIM-ALGEBRAIC_GEOMETRY-0050

### A degree-two saturation test for the unirational fourfold problem

P / partially_solved | AIM 研讨会问题汇编

Does there exist a unirational variety \(X\) of dimension at least \(4\) such that \[ \frac{H^4(X,\mathbb Z)\cap H^{2,2}(X)} {(H^4(X,\mathbb Z)\cap H^{2,2}(X))_{\mathrm{tors}}} \] is non-algebraic? Here, the subscript tors denotes torsion classes. Does there exist such a variety of dimension exactly \(4\)?

来源字段：http://aimpl.org/rationalityag/1/

05336 | AIM-ALGEBRAIC_GEOMETRY-0104

### A Delta-form rank obstruction for pointed elliptic moduli in characteristic p

P / partially_solved | AIM 研讨会问题汇编

When is \(\mathcal{M}_{g,n}\), \(\mathcal{A}_{g}[d]\) or \(\mathcal{K3}_{d}\) unirational over char. \(p\)? When is \(\mathcal{M}_{1,n}\) unirational for \(n\geq 11\)?

来源字段：http://aimpl.org/ratsubvarpos/4/

05337 | AIM-ARITHMETIC_GEOMETRY-0080

### A detachment obstruction and three-generator threshold for generic embedded points

P / partially_solved | AIM 研讨会问题汇编

Problem 33. What is the best pair (d, g ) (in the vague sense that d is as small as possible and g is close to 0 as possible) such that there is a component of Hilb d,g whose general member has an embedded point.

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05338 | AIM-ALGEBRAIC_GEOMETRY-0008

### A direct HHL--Brown--Erman comparison with a corrected Hirzebruch-surface differential

P / partially_solved | AIM 研讨会问题汇编

**Compare \(\Delta\)-resolutions.** Can we directly compare the Hanlon-Hicks-Lazarev and Brown-Erman construction of resolutions?

**Potential mechanism (starting in degree 0):** vertices/points in the stratification of \(T^n\) arising in HHL are in one-to-one correspondence with additional generators for normalization of the semigroup ring, i.e. elements coming from saturation of the semigroup.

来源字段：http://aimpl.org/syzygyms/4/

05339 | AIM-ALGEBRAIC_GEOMETRY-0076

### A finite certified direct model on the two-chart locus

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.4.** To what extend can computers compute HHH\(^{\mathrm{geom}}\) directly from the definition?

来源字段：http://aimpl.org/catheckehilbert/1/

05340 | AIM-ALGEBRAIC_GEOMETRY-0309

### A finite inverse criterion for surface mld divisors

P / partially_solved | AIM 研讨会问题汇编

(1) What kind of exceptional divisor over \(X\) (with mild singularities) gives a minimal log discrepancy which is nonnegative for some \((X,D)\)? (2) What about over a surface \(X\)? Let \(E\) be an exceptional divisor of the minimal resolution \(X'\to X\) of \(X\). Does \(E\) compute some \(\operatorname{mld}\geq 0\)?

来源字段：https://aimath.org/WWN/singularvariety/problems_final.pdf

05341 | AIM-ALGEBRAIC_GEOMETRY-0246

### A finite-group Cartan-Leray spectral sequence for Morel's A1-homology

P / partially_solved | AIM 研讨会问题汇编

**Question (Kirsten Wickelgren).** Is there a Serre spectral sequence for Morel's \(H_*^{\mathbb A^1}(-,\mathbb Z)\)?

来源字段：https://aimath.org/pastworkshops/projectiveA1problems.pdf

05342 | AIM-ARITHMETIC_GEOMETRY-0007

### A finite-slice criterion for numerical modularity of CM-cycle series

P / partially_solved | AIM 研讨会问题汇编

Similar to Kudla program, we could also define a generating series of (derived) complex multiplication cycles on Shimura varieties and conjecture its modularity. Is it possible to compute arithmetic intersection of the generating series with special divisors / Hodge bundle and prove its numerical modularity when the Shimura variety has low dimension e.g. on Siegel threefold or Hilbert modular surface?

来源字段：http://aimpl.org/intersectshimura/2/

05343 | AIM-ALGEBRAIC_GEOMETRY-0171

### A finiteness fracture for infinite-group equivariant spectra

P / partially_solved | AIM 研讨会问题汇编

**What can we say about equivariant stable homotopy theory for infinite discrete groups?**

来源字段：http://aimpl.org/equideralggeom/4/

05344 | AIM-ALGEBRAIC_GEOMETRY-0279

### A fixed-base descending sequence from Monsky quartics

P / partially_solved | AIM 研讨会问题汇编

Fix \(d\). Does there exist a sequence of singular algebras \(\{R_n\}\) over a fixed finite field of characteristic \(p>0\), and of dimension \(d\), such that the \(e_{HK}(R_n)\) descend to some number \(\alpha_d\) from above?

来源字段：http://aimpl.org/testandmultiplierideals/1/

05345 | AIM-ALGEBRAIC_GEOMETRY-0331

### A fixed-base product obstruction to literal uniform isomorphism counts

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.4 (Viehweg).** Does there exist a universal bound for the number of families of minimal manifolds with Hilbert polynomial \(H\) over a curve of genus \(q\) with \(s\) singular fibers depending only on \(H,q,\) and \(s\), and not any more features of the geometry of the fibers?

来源字段：https://aimath.org/WWN/birational/birational.pdf

05346 | AIM-ARITHMETIC_GEOMETRY-0035

### A fixed-degree moduli obstruction for plane-curve Jacobian quotients

P / partially_solved | AIM 研讨会问题汇编

Is every abelian variety the quotient of \(\operatorname{Jac}(C)\), where \(C\) is a smooth plane curve?

Yes, if every surface \(S\) of general type is birational to \(S'\subseteq \mathbb P^3\), where \(S'\) has only isolated singularities.

**Recovered complex statement.** Let \(A/\mathbb C\) be an abelian variety. Does there exist a nonsingular, geometrically integral plane curve \[ C=V(F)\subset \mathbb P^2_{\mathbb C} \] and a surjective homomorphism of abelian varieties \[ q:J(C)\longrightarrow A? \]

来源字段：http://aimpl.org/cohomabelian/1/

05347 | AIM-ARITHMETIC_GEOMETRY-0068

### A fixed-locus criterion and an obstruction in the standard nonreduced toric chart

P / partially_solved | AIM 研讨会问题汇编

**Problem 21.** Is there a multigraded Hilbert scheme with a connected component isomorphic to a fat point?

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05348 | AIM-ALGEBRAIC_GEOMETRY-0174

### A fixed-point obstruction and two boundary cases for equivariant Landweber exactness

P / partially_solved | AIM 研讨会问题汇编

When \(G=C_{2^n}\) what is the \(G\)-equivariant Landweber exact functor theory for \(MU^{(G)}\)?

来源字段：http://aimpl.org/equideralggeom/5/

05349 | AIM-ARITHMETIC_GEOMETRY-0056

### A fixed-polynomial Cartier-divisor bridge for the locally Cohen-Macaulay curve locus

P / partially_solved | AIM 研讨会问题汇编

Problem 9. Is the Hilbert scheme of local Cohen-Macaulay curves in $\mathbb P^3$ connected?

Let $k$ be an algebraically closed field, fix integers $d>0$ and $g$, and let \[ H_{d,g}\subset \operatorname{Hilb}^{P}(\mathbb P^3_k),\qquad P(n)=dn+1-g, \] be the open subscheme parametrizing locally Cohen--Macaulay curves (pure one-dimensional subschemes with no embedded or isolated zero-dimensional components). Whenever $H_{d,g}$ is nonempty, is it connected?

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05350 | AIM-ARITHMETIC_GEOMETRY-0066

### A forbidden strip and a determinantal chart for ICM surfaces in P4

P / partially_solved | AIM 研讨会问题汇编

The AIM workshop list *Components of Hilbert schemes* asks in Problem 19:

The notation is \(\mathbb P^4\). A ``surface'' here is taken to be a closed, pure two-dimensional subscheme \(X\subset\mathbb P^4_k\) that is locally Cohen--Macaulay (ICM); it need not be reduced, irreducible, smooth, or nondegenerate. This interpretation is consistent with the surrounding workshop questions about Hilbert schemes and liaison. The ground field is algebraically closed. Results below involving only Hilbert polynomials and determinantal ideals work in arbitrary characteristic.

For a sufficiently general hyperplane \(H\), no associated component of \(X\) is contained in \(H\), so its equation is a nonzerodivisor on \(X\). The section \(C=X\cap H\subset H\simeq\mathbb P^3\) is therefore an ICM curve. Define \[ d=\deg X=\deg C,\qquad \pi=p_a(C),\qquad \chi=\chi(\mathcal O_X). \] Taking the first difference of the Hilbert polynomial gives the convention \[ P_X(n)=\chi(\mathcal O_X(n)) =\frac d2n^2+\left(\frac d2+1-\pi\right)n+\chi. \tag{1} \] Thus the first concrete geography problem is to determine which lattice triples \((d,\pi,\chi)\) occur. In the convention used by literature writing \(H(d,p,\pi)\), the surface arithmetic genus is \(p=p_a(X)=\chi-1\). For smooth integral surfaces one may further ask for \(K_X^2,c_2,q,p_g\), but smooth adjunction and the double-point formula cannot be applied without qualification to arbitrary reducible or nonreduced ICM schemes.

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05351 | AIM-ALGEBRAIC_GEOMETRY-0336

### A formal local model at a type-2b stable numerical quintic

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.2.** Understand the geometry of the moduli space of stable surfaces better. Here it is probably necessary to find some nice components and study their geometry. In general, after fixing even the differentiable structure of the underlying four-manifolds, the moduli space is disconnected, its connected components are reducible, and its irreducible components may be everywhere nonreduced. In addition, the boundary may not be a divisor, and the next best substitute is not a normal crossings divisor. Also, Vakil shows that the moduli space of canonically polarized surfaces is arbitrarily singular (in a precise sense).

来源字段：https://aimath.org/WWN/birational/birational.pdf

05352 | AIM-ALGEBRAIC_GEOMETRY-0139

### A four-dimensional repair for symmetric surface classes on Mbar_0,7

P / partially_solved | AIM 研讨会问题汇编

What is the effective cone of surfaces \[ \operatorname{Eff}_2\!\left(\overline{\mathcal M}_{0,7}^{\mathfrak S_7}\right), \qquad \overline{\mathcal M}_{0,7}^{\mathfrak S_7} :=\overline{\mathcal M}_{0,7}/\mathfrak S_7? \]

The AIM record asks:

Throughout, \(X=\overline M_{0,7}\) over \(\mathbb C\), \(G=S_7\), \(Y=X/G\), and \(q:X\to Y\) is the finite coarse quotient. Since \(\dim X=4\), surfaces are both dimension-two and codimension-two cycles. We use real numerical classes and write \(\overline{\operatorname{Eff}}_2\) for the closure of the cone generated by effective surfaces. This is the convention called \(\operatorname{Eff}_2\) in Fulger--Lehmann.

There is an important convention warning. The invariant rational Chow/homology group upstairs, the rational Chow group of the quotient stack \([X/G]\), and the rational cycle group of the coarse quotient are related by finite-group transfer, but products of *symmetric divisors* detect only a three-dimensional quotient of the four-dimensional invariant surface space. The latter distinction is essential here.

来源字段：http://aimpl.org/poscycles/1/

05353 | AIM-ALGEBRAIC_GEOMETRY-0211

### A framed quotient and a regular compactification of Carlsson-Weinshall duality

P / partially_solved | AIM 研讨会问题汇编

What is a moduli space description of Carlsson-Weinshall duality? What moduli spaces does this suggest? A framed multiview variety?

来源字段：http://aimpl.org/algvision/5/

05354 | AIM-ALGEBRAIC_GEOMETRY-0055

### A full-exceptional-fiber criterion for uniform rationality under point blow-down

P / partially_solved | AIM 研讨会问题汇编

If \(X\) is a smooth rational variety of dimension \(n\) and \(p\in X\) is a point, is there a Zariski open \(U\subset X\) with \(p\in U\) such that \(U\) is isomorphic to a Zariski open subset of \(\mathbb P^n\)?

来源字段：http://aimpl.org/rationalityag/1/

05355 | AIM-ALGEBRAIC_GEOMETRY-0106

### A genus-three fine-level obstruction and a coarse quotient-curve criterion

P / partially_solved | AIM 研讨会问题汇编

What are the rationality properties of \(\mathcal{A}_{g,s.s.}\)? You can ask the same question for Newton polygon strata.

来源字段：http://aimpl.org/ratsubvarpos/4/

05356 | AIM-ALGEBRAIC_GEOMETRY-0047

### A graph-splitting obstruction to specialization of unirationality

P / partially_solved | AIM 研讨会问题汇编

**Does unirationality specialize in smooth projective families?**

Let \(R\) be a discrete valuation ring (DVR), with fraction field \(K\) and residue field \(k\), and let \(\mathcal X\to\operatorname{Spec}R\) be smooth and projective with geometrically integral fibers. If the geometric generic fiber \(\mathcal X_{\overline K}\) is unirational, must the geometric special fiber \(\mathcal X_{\overline k}\) be unirational?

来源字段：http://aimpl.org/rationalityag/1/

05357 | AIM-ALGEBRAIC_GEOMETRY-0027

### A half-line obstruction and a filtered Poitou-Tate lifting criterion

P / partially_solved | AIM 研讨会问题汇编

**Motivic filtered Poitou-Tate duality.** Is there a duality at the level of motivic filtered spectra \(\mathrm{fil}_{\mathrm{mot}}\mathrm{TC}(\mathcal O_K)\) that recovers Poitou--Tate duality at the level of associated graded?

来源字段：http://aimpl.org/equivstable/2/

05358 | AIM-ARITHMETIC_GEOMETRY-0039

### A hereditary square-closure screen and a quadratic-closure test family

P / partially_solved | AIM 研讨会问题汇编

Find interesting classes $\mathcal F$ of fields such that, for $F\in\mathcal F$, \[ u(F)<\infty\quad\Longrightarrow\quad u(F(t))<\infty. \]

来源字段：http://aimpl.org/deformationbrauer/1/

05359 | AIM-ARITHMETIC_GEOMETRY-0009

### A Hermitian and dual-pair filter for exceptional Siegel--Weil programs

P / partially_solved | AIM 研讨会问题汇编

Could we formulate and prove arithmetic (resp. higher) Siegel--Weil formulas on Shimura varieties (resp. moduli of Shtukas) for exceptional groups?

来源字段：http://aimpl.org/intersectshimura/2/

05360 | AIM-ALGEBRAIC_GEOMETRY-0137

### A high-degree curve-cone counterexample in the weak-Lefschetz range

S / solved | AIM 研讨会问题汇编

In the above cases, assuming \(i:X\hookrightarrow \mathbb P^n\times\mathbb P^m\) or \(i:X\hookrightarrow Gr(k,n)\) is ample then the Lefschetz hyperplane theorems imply that \(i_*:N_k(X)\to N_k(Gr(k,n))\) and \(i_*:N_k(X)\to N_k(\mathbb P^n\times\mathbb P^m)\) are isomorphisms for \(k<\dim(X)/2\). It is natural to ask if there is any Lefschetz type phenomena for effective cones of very general hypersurfaces of sufficiently high degree.

If \(X\subset Y\) is a very general ample divisor of sufficiently high degree, does the pushforward map \(i_*:N_k(X)\to N_k(Y)\) induce an isomorphism of effective cones for \(k\) sufficiently small?

来源字段：http://aimpl.org/poscycles/1/

05361 | AIM-ARITHMETIC_GEOMETRY-0045

### A Laurent-rank counterexample to the proposed period-index formula

S / solved | AIM 研讨会问题汇编

Find reasonable classes \(\mathcal F\) of fields such that \(u(F)\) is a power of 2 for all \(F\in\mathcal F\). For these fields, assume \(k\) is the period-index bound for \(\ell=2\). Does this imply \(u(F)=2^{k+1}\)?

来源字段：http://aimpl.org/deformationbrauer/1/

05362 | AIM-ALGEBRAIC_GEOMETRY-0345

### A level-cover ramification budget for rational and affine coarse curves in M_g

P / partially_solved | AIM 研讨会问题汇编

Does the coarse moduli space \(M_g\) contain a \(\mathbf P^1\)? An \(\mathbf A^1\)? It is known that the moduli stack in this case is algebraically hyperbolic, but the same is not likely to be true for the moduli space.

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来源字段：https://aimath.org/WWN/birational/birational.pdf

05363 | AIM-ALGEBRAIC_GEOMETRY-0339

### A line-arrangement deformation test for surface-source stable maps

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.5 (Alexeev).** Is there a Gromov-Witten theory or quantum cohomology theory arising from stable pairs, perhaps from stable maps \(f:(\mathbb P^2,D=D_1+\cdots+D_n)\to X\), where \(D\) is a union of lines?

来源字段：https://aimath.org/WWN/birational/birational.pdf

05364 | AIM-ALGEBRAIC_GEOMETRY-0296

### A literal obstruction and an exact p-rank criterion for non-nilpotent Frobenius

P / partially_solved | AIM 研讨会问题汇编

Given a smooth, projective variety \(X\) over a number field, does there exist a dense set of primes for which the action of Frobenius on the coherent cohomology \(H^i(X_p,\mathcal O_{X_p})\) is not nilpotent?

来源字段：http://aimpl.org/testandmultiplierideals/2/

05365 | AIM-ARITHMETIC_GEOMETRY-0075

### A local nonsmoothability threshold and very-compressed sieve for Hilb^d(A^3)

P / partially_solved | AIM 研讨会问题汇编

What is the smallest \(d\) such that \(\operatorname{Hilb}^{d}(\mathbb A^{3})\) is reducible?

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05366 | AIM-ALGEBRAIC_GEOMETRY-0038

### A low-stem phase diagram for the classical image of J

P / partially_solved | AIM 研讨会问题汇编

**The image of \(J\) in equivariant truncated Brown--Peterson spectra.** What is the image of the \(J\) homomorphism in the fixed points of \(\mathrm{BP}^{((G))}\langle m\rangle\)?

来源字段：http://aimpl.org/equivstable/4/

05367 | AIM-ARITHMETIC_GEOMETRY-0008

### A mixed-excess obstruction and forced Hodge normalization for U(a,b)

P / partially_solved | AIM 研讨会问题汇编

Could we formulate and prove arithmetic Siegel--Weil formulas of special cycles on unitary Shimura varieties for \(U(a,b)\) (\(a,b>1\)) where there are no special divisors?

来源字段：http://aimpl.org/intersectshimura/2/

05368 | AIM-ALGEBRAIC_GEOMETRY-0162

### A modern étale criterion and an all-characteristic constant-functor test

P / partially_solved | AIM 研讨会问题汇编

**What is an étale map of Green or Tambara functors?**

来源字段：http://aimpl.org/equideralggeom/3/

05369 | AIM-ALGEBRAIC_GEOMETRY-0193

### A normalized essential-matrix orbitope and its density-matrix faces

P / partially_solved | AIM 研讨会问题汇编

What can we say about the convex hull of all the varieties we have considered? Can we give a good description of them?

来源字段：http://aimpl.org/algvision/2/

05370 | AIM-ARITHMETIC_GEOMETRY-0004

### A parahoric Koszul descent criterion at a semistable node

P / partially_solved | AIM 研讨会问题汇编

Derived special cycles have been defined on integral models of certain Shimura varieties (resp. moduli of Shtukas) in recent works of Madapusi (resp. Feng-Yun-Zhang) in the good reduction case. Is it possible to extend their method and define derived special cycles in the bad reduction cases e.g. for some specific parahoric levels?

来源字段：http://aimpl.org/intersectshimura/1/

05371 | AIM-ARITHMETIC_GEOMETRY-0071

### A Pfaffian-colon reducedness certificate for Hilb^8(A^4)

P / partially_solved | AIM 研讨会问题汇编

Is \(\operatorname{Hilb}^{8}(\mathbb A^{4})\) reduced? More generally, develop techniques to prove the reducedness of Hilbert schemes.

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05372 | AIM-ARITHMETIC_GEOMETRY-0020

### A polarized rank-drop bound and the transcendental exponent bottleneck

P / partially_solved | AIM 研讨会问题汇编

Fix a number field and a degree. Look at all K3 surfaces over the number field of degree $2d$. Look at the Brauer group of the K3 surface modulo the Brauer group of the number field. Is there a uniform bound on the size of the quotient? What about the algebraic part? The transcendental part of the Brauer group? Note: the first two groups are known to be finite.

来源字段：http://aimpl.org/brauermoduli/1/

05373 | AIM-ALGEBRAIC_GEOMETRY-0026

### A prime-support reduction for iterated Atiyah--Segal completion

P / partially_solved | AIM 研讨会问题汇编

**Atiyah--Segal completion thoerem for iterated \(K\)-theory.** Can one prove an Atiyah--Segal completion theorem for the algebraic \(K\)-theory of topological \(K\)-theory?

来源字段：http://aimpl.org/equivstable/2/

05374 | AIM-ALGEBRAIC_GEOMETRY-0073

### A purity-collapse criterion and identity-braid audit for the HHH comparison

P / partially_solved | AIM 研讨会问题汇编

Prove that Oblomkov-Rozansky triply graded cohomology \(\mathrm{HHH}^{\mathrm{geom}}\) is isomorphic to Khovanov-Rozansky triply graded cohomology \(\mathrm{HHH}\).

来源字段：http://aimpl.org/catheckehilbert/1/

05375 | AIM-ALGEBRAIC_GEOMETRY-0084

### A radical-free comparison of the affine-Springer and Milnor Alexander constructions

P / partially_solved | AIM 研讨会问题汇编

It is known that the Alexander polynomial can be computed from both the Springer fibre and the Milnor fibre. Understand the relationship between these two constructions.

来源字段：http://aimpl.org/catheckehilbert/3/

05376 | AIM-ALGEBRAIC_GEOMETRY-0091

### A rank-one Morita diagnostic for DAHA and punctured-torus branes

P / partially_solved | AIM 研讨会问题汇编

Is there a wrapped/stopped, possibly singular and \(B\)-field-twisted A-brane category \(\mathcal A_{N,\tau,q}\) of \(\mathfrak X_{N,\tau}\), containing a canonical coisotropic object \(\mathfrak B_{\mathrm{cc}}\), such that \[ \operatorname{RHom}(\mathfrak B_{\mathrm{cc}},-): \mathcal A_{N,\tau,q} \longrightarrow \operatorname{Mod}\bigl(\mathbb{SH}_{q,t}(GL_N)\bigr) \] is an equivalence? Equivalently, does \(\mathfrak B_{\mathrm{cc}}\) generate and have endomorphism algebra \(\mathbb{SH}_{q,t}(GL_N)\)?

来源字段：http://aimpl.org/catheckehilbert/5/

05377 | AIM-ARITHMETIC_GEOMETRY-0062

### A rank-three flat family where the restricted Gröbner fan jumps

S / solved | AIM 研讨会问题汇编

**Problem 15.** Is the Gröbner fan a discrete invariant that distinguishes the components of \(\operatorname{Hilb}^{d}(\mathbb A^{n})\)?

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05378 | AIM-ALGEBRAIC_GEOMETRY-0160

### A rational orientation-kernel presentation of the equivariant sphere ring

P / partially_solved | AIM 研讨会问题汇编

In (Morel 2004, MR2061856) Morel showed that Milnor-Witt K-theory of a field is isomorphic to the graded endomorphism ring of the unit object in the stable homotopy category of motivic spectra over that field.

Can we prove a similar result in the setting of equivariant spectra? This would aid in understanding the tensor triangular geometry of \(Sp^G\).

来源字段：http://aimpl.org/equideralggeom/2/

05379 | AIM-ALGEBRAIC_GEOMETRY-0082

### A recursion bridge, rank-two localization identity, and Demazure obstruction

P / partially_solved | AIM 研讨会问题汇编

In the presence of parity, there are two combinatorial formulas for HHH: one for localization of Hilbert schemes from the GNR conjecture and one from the Elias-Hogencamp-Mellit papers. How are these formulas related? How are they related to Demazure crystals?

来源字段：http://aimpl.org/catheckehilbert/2/

05380 | AIM-ALGEBRAIC_GEOMETRY-0072

### A reduced Merkurjev module and an exponent-p detector

P / partially_solved | AIM 研讨会问题汇编

Given a rationally connected variety which is not universally \(CH_0\)-trivial, determine a nonzero unramified class in some cycle module (in the sense of Rost).

来源字段：http://aimpl.org/rationalityag/1/

05381 | AIM-ALGEBRAIC_GEOMETRY-0077

### A representation-framed stable locus and a type B2 obstruction

P / partially_solved | AIM 研讨会问题汇编

The space \(\mathscr{X}^{\mathrm{big}}\) and category \(\operatorname{MF}(\mathscr{X}^{\mathrm{big}},w)\) makes sense not just in type A. What is the analog of \((\mathscr{X}^{\mathrm{big}})^{\mathrm{stable}}\) in other types? What is the categorical analog of the map \(\operatorname{Br}_{\mathrm{ext}}\to\operatorname{Br}_{\mathrm{fin}}\).

Its exact text is:

The preceding AIM record specifies \[ \mathscr X^{\mathrm{big}} =\mathfrak g\times G\times\mathfrak n\times G\times\mathfrak n. \] This agrees with the “non-reduced” two-fold space \(\mathcal X_2=\mathfrak g\times(G\times\mathfrak n)^2\) in Oblomkov--Rozansky. We work over \(\mathbb C\), take \(G\) to be a connected reductive group, \(B\subset G\) a Borel subgroup, and \(\mathfrak n=\operatorname{Lie}R_u(B)\). Fixing a nondegenerate \(G\)-invariant bilinear form \(\kappa\) on \(\mathfrak g\), the potential is \[ w(X,g_1,Y_1,g_2,Y_2) =\kappa\!\left(X,\operatorname{Ad}_{g_1}Y_1-\operatorname{Ad}_{g_2}Y_2\right). \] No corruption of the source statement was found. “Stable” is interpreted through the type A construction in arXiv:1702.03569, Section 2.6.

来源字段：http://aimpl.org/catheckehilbert/1/

05382 | AIM-ALGEBRAIC_GEOMETRY-0075

### A residual parity-defect criterion for the fourth grading on HHH

P / partially_solved | AIM 研讨会问题汇编

On HHH\(^{\mathrm{geom}}\) of a braid \(\beta\) there is a mysterious 4th \(\mathbb Z\)-grading coming from the \(\mathbb Z/2\)-grading on matrix factorizations. Construct the analog for HHH of \(\beta\).

来源字段：http://aimpl.org/catheckehilbert/1/

05383 | AIM-ALGEBRAIC_GEOMETRY-0039

### A restriction-truncation baseline for fixed-point Hurewicz images

P / partially_solved | AIM 研讨会问题汇编

**The Hurewicz image of equivariant truncated Brown--Peterson spectra.** What is the Hurewicz image of the truncated Brown--Peterson spectrum \(\mathrm{BP}^{((G))}\langle m\rangle\)?

The record comes from the 2022 AIM workshop *Equivariant techniques in stable homotopy theory*, section “Norms and the slice filtration,” Problem 4.3. The linked AIM problem-list page no longer resolves to the problem text. The wording by itself does not specify a map, so it is essential to reconstruct the intended meaning before doing mathematics.

The most strongly supported reconstruction is the **classical unit Hurewicz map into categorical fixed points**

\[ h_{G,m}\colon \pi_k\mathbb S_{(2)}\longrightarrow \pi_k\bigl(\mathrm{BP}^{((G))}\langle m\rangle^G\bigr) =\pi_k^G\mathrm{BP}^{((G))}\langle m\rangle . \tag{1.1} \]

Here \(G=C_{2^n}\) is a cyclic 2-group, as in the surrounding problems and in the Hill--Hopkins--Ravenel construction, and fixed points are genuine/categorical fixed points, not homotopy fixed points. Three pieces of evidence make (1.1) substantially more than a guess:

1. the immediately preceding AIM problem asks for the image of \(J\) “in the fixed points” of the same spectrum; 2. Hill--Shi--Wang--Xu explicitly ask, after computing the slice spectral sequence, “What are the Hurewicz images of the \(C_4\)-fixed points of \(\mathrm{BP}^{((C_4))}\langle2\rangle\)?”; 3. Li--Shi--Wang--Xu use “Hurewicz image” for the unit maps \(\pi_*\mathbb S\to\pi_*BP_{\mathbb R}^{C_2}\) and related fixed-point spectra.

Because the target is 2-local, the unlocalized unit map factors uniquely through \(\mathbb S_{(2)}\), so using the 2-local sphere loses nothing.

Other plausible readings are genuinely different problems:

- the genuine \(RO(G)\)-graded unit map \(\pi_V^G\mathbb S\to\pi_V^G\mathrm{BP}^{((G))}\langle m\rangle\); - the unit into homotopy fixed points rather than categorical fixed points; - the ordinary homology Hurewicz map out of \(\pi_*\mathrm{BP}^{((G))}\langle m\rangle\).

The results below concern (1.1). In particular, the calculation for \(m=0\) below is only an integer-graded fixed-point calculation; it is not an \(RO(G)\)-graded calculation of \(H\underline{\mathbb Z}\).

来源字段：http://aimpl.org/equivstable/4/

05384 | AIM-ALGEBRAIC_GEOMETRY-0266

### A Riemann--Roch obstruction to small very-general Seshadri constants

P / partially_solved | AIM 研讨会问题汇编

For every $n$, is there $c(n)>0$, independent of the smooth projective $n$-fold $X$, the ample Cartier divisor $L$, the uncountable algebraically closed field $k$ of characteristic $p>0$, and $p$, such that $\epsilon(L,x)\geq c(n)$ at a very general point?

来源字段：http://aimpl.org/minimalmodcharp/5/

05385 | AIM-ALGEBRAIC_GEOMETRY-0190

### A root-free focal-positivity certificate for a seven-point pencil

P / partially_solved | AIM 研讨会问题汇编

The focal lengths of a pair of cameras can be written as a function of their fundamental matrix. During reconstruction fundamental matrices are constructed for which these focal lengths are complex. Is there an efficient way to detect if the this will occur before computing the fundamental matrix?

来源字段：http://aimpl.org/algvision/1/

05386 | AIM-ALGEBRAIC_GEOMETRY-0121

### A separability firewall for rationally connected surfaces

P / partially_solved | AIM 研讨会问题汇编

Is being rationally connected = unirational in char $p$?

来源字段：http://aimpl.org/ratsubvarpos/9/

05387 | AIM-ALGEBRAIC_GEOMETRY-0115

### A separable odd-degree parametrization of the characteristic-2 Fermat cubic

P / partially_solved | AIM 研讨会问题汇编

For an algebraically closed field $k$ of characteristic $p>0$, does there exist a smooth cubic threefold $X\subset \mathbf P^4_k$ admitting a dominant rational map $\mathbf P^3_k\dashrightarrow X$ of odd total degree?

来源字段：http://aimpl.org/ratsubvarpos/7/

05388 | OWR-16169-013

### A Serre-Style Proof of the Jordan Property

O / open | Oberwolfach 报告问题汇编

Can the Jordan property for rationally connected varieties over a field of characteristic zero be proved by extending Serre's Minkowski-style method for finite subgroups of rank-two Cremona groups, rather than by using the boundedness argument of Prokhorov and Shramov?

来源字段：https://doi.org/10.4171/owr/2018/28

05389 | AIM-ALGEBRAIC_GEOMETRY-0120

### A sharp failure of supersingularity beyond the q-bic locus

P / partially_solved | AIM 研讨会问题汇编

Find other examples of supersingular hypersurfaces.

Is there a supersingular hypersurface of every degree, dimension and char. $p$?

来源字段：http://aimpl.org/ratsubvarpos/9/

05390 | AIM-ALGEBRAIC_GEOMETRY-0053

### A sharp quantitative obstruction to linear-center constructions on smooth quartics

P / partially_solved | AIM 研讨会问题汇编

Is there a smooth rational quartic hypersurface of some dimension?

来源字段：http://aimpl.org/rationalityag/1/

05391 | AIM-ALGEBRAIC_GEOMETRY-0079

### A sharp two-strand full-twist threshold for HHH-parity

P / partially_solved | AIM 研讨会问题汇编

Call a braid $\beta$ parity if $\operatorname{HHH}(\beta)$ is parity. Find necessary and/or sufficient conditions for a braid to be parity.

Are there operations on braids (for example, adding a positive number of full twists) which transform it into a parity braid or which preserve parity braids?

来源字段：http://aimpl.org/catheckehilbert/2/

05392 | AIM-ARITHMETIC_GEOMETRY-0064

### A sharp uniqueness threshold and conductor-preserving flag criterion

P / partially_solved | AIM 研讨会问题汇编

The primary AIM PDF confirms that the broken expression is \(H_C\), not a product or a new invariant. It means an irreducible component of the Hilbert scheme of curves in \(\mathbb P^3\), with the Hilbert polynomial of \(C\), containing the point \([C]\).

The phrase “double curve, and triple points, pinch point, ...” is the classical description of a surface with **ordinary singularities**. The intended characteristic-zero local models are: two transverse smooth surface branches along the general point of a reduced curve \(D\), ordinary triple points analytically modeled by \(xyz=0\), and finitely many pinch points modeled by \(x^2-y^2z=0\). This interpretation agrees with the later formulation used by Hartshorne--Polini and Zhang.

Two qualifications are logically necessary.

1. If \([C]\) lies on several Hilbert components, “the component” is not defined. The conjecture must be read componentwise, or with the added hypothesis that \([C]\) is a smooth point of the Hilbert scheme (and hence lies on a unique component). 2. The degree of the double curve is not invariant in an arbitrary deformation of hypersurfaces: a ruled cubic with a double line can deform to a smooth cubic. Thus the conclusion concerns a conductor-preserving stratum, not the whole projective space of degree-\(d\) equations.

Throughout the new result below, \(k\) is algebraically closed of characteristic zero, a curve is pure of dimension one, and \(C\) is smooth. The elementary degree estimates themselves are valid over any algebraically closed field.

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05393 | AIM-ALGEBRAIC_GEOMETRY-0232

### A skeleton obstruction and a genus-five enriched tropical certificate

P / partially_solved | AIM 研讨会问题汇编

**Problem 8.3.** Can one give a new proof of Green's theorem by tropical methods?

来源字段：https://aimath.org/pastworkshops/degenalggeomproblems.pdf

05394 | AIM-ARITHMETIC_GEOMETRY-0070

### A stabilizer-aware speciality obstruction to rigid projective curves

P / partially_solved | AIM 研讨会问题汇编

**Problem 23.** (An old question of Joe Harris) Does there exist a nondegenerate rigid curve in \(\mathbf P^n\) other than the rational normal curve? Here rigid means the only deformations of the curve are those induced by automorphisms of \(\mathbf P^n\).

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05395 | AIM-ALGEBRAIC_GEOMETRY-0007

### A Sullivan-Milnor-Moore obstruction screen for Weinstein cocore algebras

P / partially_solved | AIM 研讨会问题汇编

Can the “looking glass” parallel/analogy between rational homotopy theory (= symplectic geometry of cotangent bundles) and local ring theory be generalized/further exploited to other symplectic manifolds (e.g. Weinstein manifolds)?

A) one can attach handles to Weinstein manifolds; there exists a natural “categorical compactification” of Fukaya categories via a stop or Lefschetz fibration.

B) Can rational homotopy theory, e.g. Sullivan minimal models, be used to construct resolutions?

This is Problem 3.1, “Symplectic geometry,” from the AIM workshop list *Syzygies and mirror symmetry*.

The record gives the motivating formulas \[ C_*(\Omega_qM) =\operatorname{Ext}_{C^\bullet(M)}(k,k)=A, \qquad C_*(\Omega_qM)\simeq \operatorname{End}_{\operatorname{Fuk}(T^*M)}(T_q^*M,T_q^*M), \] with \(M\) simply connected.

The original AIM URL was attempted on 22 July 2026 but did not return usable content through the available web reader. No missing mathematical symbol had to be reconstructed. Two notational corrections are essential:

1. at chain level, \(\operatorname{Ext}\) must be replaced by the derived endomorphism dga/\(A_\infty\)-algebra \(\mathbb R\operatorname{Hom}\); \(\operatorname{Ext}\) ordinarily denotes its cohomology; 2. the Floer-theoretic statement is an \(A_\infty\) quasi-isomorphism, with grading, orientation/background-class, and coefficient conventions, not an equality of chain complexes.

With cohomological grading \(C_{-*}^i=C_{-i}\), the precise cotangent pattern used below is \[ CW^*(T_q^*M,T_q^*M;k)\simeq_{A_\infty} C_{-*}(\Omega_qM;k) \simeq_{A_\infty} \mathbb R\operatorname{Hom}_{C^*(M;k)}(k,k). \tag{1} \] Abouzaid proves the first quasi-isomorphism for a closed smooth base (with the appropriate background class, or a corresponding local-system twist). Simple connectedness is not needed for that Floer/loop comparison. It is imposed here for the clean Eilenberg-Moore/Koszul-duality statement and for Sullivan rational homotopy theory.

来源字段：http://aimpl.org/syzygyms/3/

05396 | AIM-ARITHMETIC_GEOMETRY-0067

### A support-deficit criterion and the fold-singularity test family

P / partially_solved | AIM 研讨会问题汇编

**Problem 20.** Let \(C\) be a curve, can \(\operatorname{Hilb}^d(C)\) have a component of dimension \(<d\)?

**Remark.** There exists a non-smoothable component of dimension \(=d>1\).

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05397 | AIM-ALGEBRAIC_GEOMETRY-0057

### A surface criterion for total CH0-triviality and its separation from L-rationality

P / partially_solved | AIM 研讨会问题汇编

Colliot-Thélène and Pirutka defined the notion of a \(CH_0\)-universally trivial resolution of singularities. As a preliminary question, is admitting a \(CH_0\)-universally trivial resolution an analytic local property for complex varieties? Assuming the answer is yes, we pose the following questions.

In low dimensions, say up to 4, can we give an analytic local classification of \(CH_0\)-universally trivial singularities?

More vaguely, is there a local classification of \(CH_0\)-universally trivial singularities in terms of MMP data?

We can ask the same questions for \(L\)-rational singularities (defined by Nicaise and Shinder), and for \(B\)-rational singularities (defined by Kontsevich and Tschinkel).

来源字段：http://aimpl.org/rationalityag/1/

05398 | AIM-ARITHMETIC_GEOMETRY-0055

### A tangent-weight criterion for rational Hilbert components

P / partially_solved | AIM 研讨会问题汇编

**Problem 8.** If a component of the Hilbert scheme contains a smooth Borel fixed point, does the component have to be rational?

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05399 | AIM-ALGEBRAIC_GEOMETRY-0234

### A topological node-parity reduction for the next-to-minimal Welschinger sign

P / partially_solved | AIM 研讨会问题汇编

**Problem 9.2.** What about when \(d_1=2\) or \(3\)? (Conjecturally, [the sign] is \((-1)^{(d-1)(d-2)/2+1}\) for \(d\gg0\).)

来源字段：https://aimath.org/pastworkshops/degenalggeomproblems.pdf

05400 | AIM-ARITHMETIC_GEOMETRY-0026

### A toric-pencil obstruction to irrational curve subfields

P / partially_solved | AIM 研讨会问题汇编

Consider $\mathbb{C}(u,v)=\mathbb{C}(\mathbb{P}^2)$. Look at the cyclic algebra $A=(u,v)_n$, defined to be $\mathbb{C}(u,v)\langle x,y \rangle/(x^n-u, y^n-v,xy-\zeta yx)$ with $\zeta$ a primitive $n$th root of unity. Does there exist a transcendence degree 1 subfield of $A$ (also a $\mathbb{C}$-subalgebra) which is not rational? Geometrically, can we find a branched cover $S$ of $\mathbb{P}^2$ corresponding to a maximal subfield of the cyclic algebra, which has a rational map to an irrational curve?

来源字段：http://aimpl.org/brauermoduli/2/

05401 | AIM-ARITHMETIC_GEOMETRY-0031

### A torsion and multiplication sieve for rational multisections

P / partially_solved | AIM 研讨会问题汇编

Let \(X/\mathbb C\) be a surface of general type. Then Lang's conjecture implies that there exists a constant \(N\), depending only on \(X\), such that if \(C\subseteq X\) is rational, then \(K_X\cdot C\leq N\).

What happens if \(X\) is an elliptic surface? Is there a bound on \(K_X\cdot C\) as \(C\) varies through rational curves?

来源字段：http://aimpl.org/cohomabelian/1/

05402 | AIM-ALGEBRAIC_GEOMETRY-0070

### A torsion-sheaf and finite-coefficient reduction for top unramified cohomology

P / partially_solved | AIM 研讨会问题汇编

Let \(X\) be a rationally connected smooth projective variety of dimension \(n\). Is \[ H^n_{nr}(k(X)/k,\mathbb Q/\mathbb Z)=0? \]

来源字段：http://aimpl.org/rationalityag/1/

05403 | AIM-ALGEBRAIC_GEOMETRY-0062

### A transcendental-rank obstruction to surface hosts

P / partially_solved | AIM 研讨会问题汇编

For a very general cubic fourfold \(X\), is there a surface \(S\) such that the Kuznetsov component \(Ku(X)\subset D(X)\) has an embedding \(Ku(X)\hookrightarrow D(S)\)?

来源字段：http://aimpl.org/rationalityag/1/

05404 | AIM-ALGEBRAIC_GEOMETRY-0080

### A two-channel parity reduction for algebraic cabling

P / partially_solved | AIM 研讨会问题汇编

Is the triply graded HOMFLY-PT/Khovanov-Rozansky homology of every plane-curve singularity link supported in one homological parity, and can its Poincaré series be computed recursively from the singularity’s Puiseux/cabling data?

来源字段：http://aimpl.org/catheckehilbert/2/

05405 | AIM-ALGEBRAIC_GEOMETRY-0033

### A two-primary Euler-divisibility defect for Mahowald redshift

P / partially_solved | AIM 研讨会问题汇编

**Mahowald invariants increase chromatic height.** Given a \(v_n\)-periodic family in the stable homotopy groups of spheres, does its Mahowald invariant give rise to an (eventually) \(v_{n+1}\)-periodic family?

来源字段：http://aimpl.org/equivstable/3/

05406 | AIM-ALGEBRAIC_GEOMETRY-0258

### A two-source specialization-defect formula and the post-2025 status of plurigenera

P / partially_solved | AIM 研讨会问题汇编

**Invariance of plurigenera.** What is the status of invariance of plurigenera?

It is known that \(h^0(mK_X)\) need not be constant in flat families if \(m=1\). For \(m\geq 2\) it remains a possibility.

**Recovered present-day answer.** The optimistic \(m\geq2\) statement is false in general. It is false logarithmically already for smooth families of surface pairs in every residue characteristic, by Brivio's published 2023 examples. A June 2025 revision of a separate Brivio preprint states that it is also false without boundary for smooth projective families of threefolds with semiample canonical divisor and residue characteristic \(p>2\). Important positive regimes remain, particularly in characteristic zero, for many surface families, and for stable families satisfying base-change and vanishing conditions.

来源字段：http://aimpl.org/minimalmodcharp/4/

05407 | AIM-ALGEBRAIC_GEOMETRY-0210

### A typed Carlsson-Weinshall extension for line correspondences

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** Is there a Carlsson-Weinshall duality for point and line correspondence?

来源字段：http://aimpl.org/algvision/5/

05408 | AIM-ALGEBRAIC_GEOMETRY-0028

### A uniform-connectivity criterion for the spectral motivic filtration on p-typical TR

P / partially_solved | AIM 研讨会问题汇编

**Motivic filtration on \(\mathrm{TR}\).** \([\mathrm{HRW}]\) gives a motivic filtration on \(\mathrm{THH},\mathrm{TC},\mathrm{TC}^{-},\mathrm{TP}\) of ring spectra, generalizing previously defined motivic filtrations such as \([\mathrm{BMS}]\). Remark 1.13 of that work says that its techniques can additionally define a motivic filtration on \(\mathrm{TR}\) of discrete rings. Can one define a well-behaved motivic filtration on the \(\mathrm{TR}\) of ring spectra? Can one compute it in good cases?

来源字段：http://aimpl.org/equivstable/2/

05409 | AIM-ALGEBRAIC_GEOMETRY-0267

### A universal counterexample, semiampleness persistence, and degree escape

P / partially_solved | AIM 研讨会问题汇编

Suppose that \(X\) is a variety over \(k\), with \(\operatorname{char} k=0\), and \(L\) is a nef divisor on \(X\). Must \(L_p\) be nef on \(X_p\) for infinitely many \(p\)? What about in the case \(L=K_X\)? What if “nef” is replaced by “semiample”?

来源字段：http://aimpl.org/minimalmodcharp/5/

05410 | AIM-ALGEBRAIC_GEOMETRY-0340

### A wall-free KSBA sector of quartic K3 degenerations

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.6.** Can one work out a geometric compactification of polarized K3 surfaces or Calabi-Yau manifolds in general using the framework of Alexeev/Kollár/Shepherd-Barron? For example, let \((X,H)\) be a pair where \(X\) is a K3 surface and \(H\) is a very ample divisor with \(H^2=4\). Can the stable pairs occurring as degenerations of these objects be classified? Compare the resulting singularities with the work of Shah on GIT-semistable quartic surfaces. Also compare with Martin Olsson’s work on moduli of log K3 surfaces.

来源字段：https://aimath.org/WWN/birational/birational.pdf

05411 | AIM-ALGEBRAIC_GEOMETRY-0151

### A Z/2 computation in the motivic Picard problem

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.7.** What can we say about the motivic Picard group?

来源字段：http://aimpl.org/equideralggeom/1/

05412 | OWR-11139-018

### Abelian-Quotient Models from High-Rank Group Actions

O / open | Oberwolfach 报告问题汇编

Let $X$ be a projective minimal terminal variety of dimension $n$, and let $G\leq\operatorname{Aut}(X)$ act so that its induced action on $\operatorname{NS}_{\mathbb C}(X)$ is not virtually solvable. Suppose that, after passage to finite index and quotient by the radical $R(G)$, the group embeds as a Zariski-dense subgroup of an almost simple real linear algebraic group $H$ of real rank $n-1$. Is $X$ birational to a quotient of an abelian variety of dimension $n$?

来源字段：https://doi.org/10.4171/owr/2011/47

05413 | OWR-14222-014

### Absolute Linear Harbourne Constants

P / partially_solved | Oberwolfach 报告问题汇编

Determine or estimate the absolute linear Harbourne constant $H(d)$, defined as the minimum of $H(K,L)$ over all fields $K$ and all configurations $L$ of $d$ lines.

来源字段：https://doi.org/10.4171/owr/2016/14

05414 | AIM-ALGEBRAIC_GEOMETRY-0206

### Active algebraic-boundary factors for dual visual and convex hulls

P / partially_solved | AIM 研讨会问题汇编

In silhouette reconstruction a duality appears between the visual hull and the convex hull. Is there an algebraic version of this?

来源字段：http://aimpl.org/algvision/4/

05415 | AIM-ALGEBRAIC_GEOMETRY-0149

### Adding transfers in motivic homotopy theory and a quadratic framing-loss obstruction

P / partially_solved | AIM 研讨会问题汇编

**Long Term.** Do we have a motivic description akin to adding transfers?

Can a motivic or equivariant-motivic stable homotopy category be constructed from a more naive motivic category by freely adjoining coherent additive transfers, in analogy with spectral Mackey descriptions of genuine equivariant spectra?

来源字段：http://aimpl.org/equideralggeom/1/

05416 | OWR-2049-010

### Additional Symmetries of Invariant Complex Domains

O / open | Oberwolfach 报告问题汇编

Let $K$ be a connected compact Lie group, let $L$ be a closed subgroup, set $X=K/L$ and $X_{\mathbb{C}}=K_{\mathbb{C}}/L_{\mathbb{C}}$, and let $D\subset X_{\mathbb{C}}$ be a $K$-invariant domain. Besides the evident $K$-symmetries, can $\operatorname{Aut}(D)$ have additional positive-dimensional symmetries?

来源字段：https://doi.org/10.4171/owr/2008/38

05417 | AIM-ALGEBRAIC_GEOMETRY-0155

### Additive realization and a trivial-action gluing obstruction for a C3 analogue of Real bordism

P / partially_solved | AIM 研讨会问题汇编

In order to solve the \(3\)-primary Kervaire invariant one problem using analogous methods to those in [Hill--Hopkins--Ravenel], one would need a \(C_3\)-equivariant version of \(MU_{\mathbb R}\).

Can we describe such a \(C_3\)-commutative ring spectrum \(MU_{\mathbb R}\) so that its underlying Bousfield type is \(MU\) and \[ \left\langle MU^{\Phi C_3}\right\rangle =\left\langle H\mathbb F_3\right\rangle? \]

来源字段：http://aimpl.org/equideralggeom/2/

05418 | OWR-14299908-003

### Adic Effective-Period Complex Conjecture

O / open | Oberwolfach 报告问题汇编

Let $K=\mathbb Q$ with a $p$-adic valuation, or let $K=k(\pi)$ with the $\pi$-adic valuation, where $k$ has characteristic zero, and write $\widehat K$ for the completion of $K$. For the infinite-dimensional Tate ball $B_\infty$ over $\widehat K$, let $\widetilde{\Omega}_{\infty-\bullet}^{K\text{-}\mathrm{alg}}(B_\infty)$ be the boundary-vanishing complex whose coefficients are power series algebraic over $K(z_1,z_2,\ldots)$. Is this complex acyclic outside degree zero, and is formal integration $H_0\!\left(\widetilde{\Omega}_{\infty-\bullet}^{K\text{-}\mathrm{alg}}(B_\infty)\right)\to\widehat K$ injective?

来源字段：https://doi.org/10.4171/owr/2026/3

05419 | AIM-ALGEBRAIC_GEOMETRY-0037

### Adjacent localized slice fibers as homotopy-orbit layers

S / solved | AIM 研讨会问题汇编

The localized slice spectral sequence gives a way to break up the slice spectral sequence into pieces that see phenomena only on certain geometric fixed points. For the group $C_{2^n}$, there is a localized slice spectral sequence for each subgroup.

For the group $C_{2^n}$ one understand more easily the fibres between the localized slice spectral sequence for two adjacent subgroups $H$ and $H'$?

**Reconstructed question.** For $G=C_{2^n}$, can one understand more easily the fibers between the localized slice spectral sequences associated to two adjacent subgroups $H<H'$?

来源字段：http://aimpl.org/equivstable/4/

05420 | AIM-ALGEBRAIC_GEOMETRY-0081

### Adjacent-power compression and gap states for nested full twists

P / partially_solved | AIM 研讨会问题汇编

There is a commutative family of braids built as products of \(\mathrm{FT}_k\), the full twist on the first \(k\) strands, for \(k\leq n\). Are the positive products in this family parity? If so, find a (recursive) formula for \(\mathrm{HHH}\).

来源字段：http://aimpl.org/catheckehilbert/2/

05421 | AIM-ALGEBRAIC_GEOMETRY-0311

### Adjunction beyond l.c.i. embeddings: status, a quantifier obstruction, and an exact slice identity

P / partially_solved | AIM 研讨会问题汇编

**Problem 8 (Schwede).** Let \(Y\) be a smooth projective variety and let \(X\subset Y\) be an irreducible subvariety of codimension \(r\). Assume that \(X\) is normal and \(\mathbb Q\)-Gorenstein, and let \(\mathcal I=\mathcal I_{X/Y}\). If \(X\) is locally a complete intersection, then \[ (Y,\mathcal I^r)\text{ is log canonical} \quad\Longleftrightarrow\quad X\text{ is log canonical}. \] This is not true with the ordinary-power pair when \(X\) is not l.c.i. Is \[ (Y,\mathcal I^{(r)})\text{ log canonical} \quad\Longleftrightarrow\quad X\text{ log canonical}? \] Here \(\mathcal I^{(r)}\) is the \(r\)-th symbolic power.

**Problem 9 (Mustaţă).** Is there an adjunction formula for multiplier ideals under restriction to subvarieties that are not defined by a regular sequence?

**Problem 10 (de Fernex).** Let \(X\) be smooth and \(B\subset X\) a proper closed subscheme. Suppose that a prime divisor \(E\) over \(X\), with center \(P\), satisfies \[ a_E(X,cB)\leq 0 \] for some \(c>0\). Fix \(1\leq e<\dim X\). Does there exist a smooth subvariety \(Y\subset X\) of codimension \(e\), with \(P\subset Y\) and \(Y\not\subset B\), and a divisor \(F\) over \(Y\), centered at \(P\), such that \[ a_F\bigl(Y,cB|_Y-eP\bigr)\leq 0? \]

来源字段：https://aimath.org/WWN/singularvariety/problems_final.pdf

05422 | OWR-3384-004

### Admissible Gelfand-Zetlin Faces Representing Smooth Schubert Cycles

O / open | Oberwolfach 报告问题汇编

别名：Duplicate: Smooth Schubert Cycles from Gelfand-Zetlin Faces [OWR-3384-005]

For $G=GL_n(\mathbb C)$, $X=G/B$, and the Gelfand-Zetlin polytope $Q_\lambda$, can every Schubert cycle defined by a Schubert cell with smooth closure be represented by an admissible face $\Gamma$ of $Q_\lambda$, so that the face combinatorics captures the cycle’s geometry?

来源字段：https://doi.org/10.4171/owr/2009/01

05423 | AIM-ARITHMETIC_GEOMETRY-0029

### Affine automorphism-orbit finiteness and a two-class torus example

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** Fix a variety $X$ and $\alpha\in\operatorname{Br}(X)$. Is \[ \mathcal E_X(\alpha)= \{\beta\in\operatorname{Br}(X):D^b(X,\beta)\simeq D^b(X,\alpha)\} \] finite?

来源字段：http://aimpl.org/brauermoduli/2/

05424 | OWR-1452-005

### Affine Fibrations as Affine-Space Bundles

O / open | Oberwolfach 报告问题汇编

Let $f:X\to Y$ be an affine, faithfully flat $\mathbb{A}^n$-fibration whose closed fibers are geometrically integral. Must $f$ be an $\mathbb{A}^n$-bundle?

来源字段：https://doi.org/10.4171/owr/2007/01

05425 | OWR-824-006

### Affine Pseudocoverings and Their Endomorphisms

O / open | Oberwolfach 报告问题汇编

Let $\varphi:\mathbb{A}^2\to\mathbb{A}^2$ be an étale endomorphism that factors through a smooth affine surface $X$ as $\varphi=\varphi_2\circ\varphi_1$, where $\varphi_1:\mathbb{A}^2\to X$ and $\varphi_2:X\to\mathbb{A}^2$ are étale. What properties must $X$, $\varphi_1$, and $\varphi_2$ have?

来源字段：https://doi.org/10.4171/owr/2005/19

05426 | AIM-ALGEBRAIC_GEOMETRY-0252

### Affirmative for p at least 5 and an all-characteristic adjunction-surface reduction

P / partially_solved | AIM 研讨会问题汇编

**Basepoint free theorem.** Suppose that \(k=\bar k\) and \(\operatorname{char} k=p>0\). Let \(X/k\) be a terminal threefold, and let \(A\) be an ample \(\mathbb Q\)-divisor on \(X\). If \(K_X+A\) is nef, must it be semiample?

来源字段：http://aimpl.org/minimalmodcharp/1/

05427 | AMR-148-0001

### AGNES Open Problems in Algebraic Geometry - Q1

O / open | AMR 问题清单汇编

Does $\pi_*[\overline{M}_g(X,\beta)]^{\operatorname{vir}}\in H^*(\overline{M}_g)$ lie in the tautological ring in cohomology?

**Q2**. When does $\pi_*[\overline{M}_g(X,\beta)]^{\operatorname{vir}}\in A^*(\overline{M}_g)$ lie in the tautological ring in Chow?

I would guess the answer to $**Q1**$ is yes. If $X$ is a curve, an affirmative answer to $**Q1**$ follows from the results of [fabp]. We know $\pi_*[\overline{M}_g(X,\beta)]^{\operatorname{vir}}$ does not always lie in the tautological ring in Chow --- counterexamples can be found when $X$ is a curve. A wild speculation, motivated by the Bloch-Beilinson conjecture, is that the answer to **Q2** is yes when $X$ is defined over over $\overline{\mathbb{Q}}$.

背景链接：https://arxiv.org/abs/1004.3259

05428 | AMR-148-0002

### AGNES Open Problems in Algebraic Geometry - Q3

O / open | AMR 问题清单汇编

Prove the Virasoro constraints in case $X$ is an Enriques surface.

A study of the Gromov-Witten theory of the Enriques surface, closely related to modular forms, has been started in [mp2]. The Enriques surfaces is perhaps the most basic variety where new techniques are required to establish the Virasoro constraints.

背景链接：https://arxiv.org/abs/1004.3259

05429 | AMR-148-0003

### AGNES Open Problems in Algebraic Geometry - Q4

O / open | AMR 问题清单汇编

Prove the GW/DT correspondence for $3$-folds.

The toric cases of **Q4** are known [moop]. Algebraic cobordism results [levp] suggest the possibility of reducing to the toric case using degeneration methods.

背景链接：https://arxiv.org/abs/1004.3259

05430 | AMR-148-0004

### AGNES Open Problems in Algebraic Geometry - Q5

O / open | AMR 问题清单汇编

Donaldson--Thomas invariants are defined only in dimension $3$ because a virtual fundamental class for the moduli space of sheaves is required. Deformations are given by $\operatorname{Ext}^1(E,E)$, obstructions by $\operatorname{Ext}^2(E,E)$, and to define the virtual fundamental class we need (roughly) the vanishing $$\operatorname{Ext}^i(E,E)=0 \ \ \text{for}\ \ i>2 \ .$$ On $3$-folds, the vanishing can often be obtained using Serre duality and stability. However, there are parallel examples of enumerative computations in higher dimensions in Gromov-Witten theory [klp,pz]. Moreover, many aspects of Joyce's counting theory are valid in higher dimensions [J].

Define Donaldson--Thomas invariants in dimensions $>3$.

背景链接：https://arxiv.org/abs/1004.3259

05431 | AMR-148-0005

### AGNES Open Problems in Algebraic Geometry - Q6

O / open | AMR 问题清单汇编

What is the analogue of the tautological ring for $M^{K3}_{2n}$?

A natural guess for **Q6** is the subring generated by the classes of the Noether--Lefschetz loci. The Noether-Lefschetz loci parameterize $K3$ surfaces with higher rank Picard lattices.

背景链接：https://arxiv.org/abs/1004.3259

05432 | AMR-148-0006

### AGNES Open Problems in Algebraic Geometry - Q7

O / open | AMR 问题清单汇编

Do the Noether-Lefschetz divisors span $\text{Pic}(M^{K3}_{2n}) \otimes_{\mathbb{Z}}{\mathbb{Q}}$?

Let $X$ be a compact Calabi-Yau 3-fold expressed as $K3$-fibration over $\mathbb{P}^1$, $$\pi:X\rightarrow \mathbb{P}^1\ .$$ Given an ample line bundle $L$ on $X$, the family $\pi$ determines a morphism of the base $\mathbb{P}^1$ to the moduli of polarized $K3$ surfaces. Via [mpnl], the Gromov-Witten theory of $X$ in $\pi$-fiber classes is calculated in terms of the Noether--Lefschetz numbers of $\pi$ and the Katz-Klemm-Vafa [kkv] conjecture concerning $\lambda_g$ integrals in the reduced Gromov-Witten theory of a fixed $K3$ surface. The KKV conjecture is proven for all classes in genus 0 in [kmps] and all genera in primitive classes in [mpt].

背景链接：https://arxiv.org/abs/1004.3259

05433 | AMR-148-0007

### AGNES Open Problems in Algebraic Geometry - Q8

O / open | AMR 问题清单汇编

Prove the Katz-Klemm-Vafa conjecture for in all genera and in all classes on $K3$ surfaces.

A solution to **Q8** would provide a large class of exact formulas for higher genus Gromov-Witten invariants of compact Calabi-Yau 3-folds. Unlike the local toric cases, mathematical results for higher genus Gromov-Witten invariants have been difficult to obtain, see [zing] for the genus 1 theory of the quintic 3-fold.

背景链接：https://arxiv.org/abs/1004.3259

05434 | OWR-13860-006

### Agreement of Affine Deligne-Lusztig Stratifications

P / partially_solved | Oberwolfach 报告问题汇编

For affine Deligne-Lusztig varieties, does the new invariant $f_g$ induce the same stratification as the generalized Ekedahl-Oort invariant in every case studied by Görtz and He?

来源字段：https://doi.org/10.4171/owr/2015/39

05435 | OWR-1318-002

### Albanese Kernels over $p$-Adic Fields

O / open | Oberwolfach 报告问题汇编

Let $V$ be a smooth projective variety over a $p$-adic local field $k$, and let $\varphi_V : \mathrm{CH}_0(V)_{\deg 0} \to A_V(k)$ be the Albanese map on zero-cycles of degree zero. Is the Albanese kernel $\ker(\varphi_V)$ the direct sum of a finite group and a divisible group?

来源字段：https://doi.org/10.4171/owr/2006/32

05436 | OWR-1465-004

### Algebraic Complexity Bounds for Real Varieties

O / open | Oberwolfach 报告问题汇编

If $Y$ is a real algebraic variety with $\dim Y\geq2$, must $\beta(Y)\leq\dim Y-1$? In particular, is $\beta(S^n)=n-1$ for every $n\geq2$, where $S^n$ is the unit $n$-sphere?

来源字段：https://doi.org/10.4171/owr/2007/14

05437 | OWR-12865-008

### Algebraic Cycles Spanning Predicted Tate Classes

P / partially_solved | Oberwolfach 报告问题汇编

For the basic Newton stratum of a Hodge-type Shimura variety or a moduli space of shtukas, families of cycles $\{Z_\gamma\}$ are parameterized by Mirković-Vilonen cycles. Do the classes of the cycles $Z_\gamma$ span all Tate classes predicted by the Langlands conjecture?

来源字段：https://doi.org/10.4171/owr/2014/05

05438 | OWR-2049-006

### Algebraic Dimension and Reduction of Hyperkähler Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a hyperkähler manifold of complex dimension $2n$. Is its algebraic dimension necessarily in $\{0,n,2n\}$? If $a(X)=n$, does the algebraic reduction admit a holomorphic model $f:X\to B$, where $B$ is a normal projective variety of dimension $n$, and hence define a Lagrangian fibration? Does parabolic $\operatorname{NS}(X)$ imply $a(X)=n$?

来源字段：https://doi.org/10.4171/owr/2008/38

05439 | OWR-704-006

### Algebraic Dimension from Positive Normal Bundles

P / partially_solved | Oberwolfach 报告问题汇编

Let $Z$ be a compact connected complex manifold of dimension $n+p$ containing a $(p-1)$-dimensional submanifold $Y$ with positive normal bundle and a covering compact family of $n$-cycles. Must the algebraic dimension of $Z$ be at least $p$?

来源字段：https://doi.org/10.4171/owr/2004/21

05440 | OWR-14299908-002

### Algebraic Effective-Period Complex Conjecture

O / open | Oberwolfach 报告问题汇编

Let $D=\{z\in\mathbb C:\lvert z\rvert\leq1\}$, and let $\widetilde{\Omega}_{\infty-\bullet}^{\mathrm{alg}}(D_\infty)$ be the boundary-vanishing complex of codegree forms on $D_\infty$ whose coefficients are holomorphic near the closed polydiscs and algebraic over $\mathbb Q(z_1,z_2,\ldots)$. Is this complex acyclic outside degree zero, and is the formal integration map $H_0\!\left(\widetilde{\Omega}_{\infty-\bullet}^{\mathrm{alg}}(D_\infty)\right)\to\mathbb C$ injective?

来源字段：https://doi.org/10.4171/owr/2026/3

05441 | OWR-1730-008

### Algebraic Fundamental Groups Below the Noether Line

O / open | Oberwolfach 报告问题汇编

If $S$ is a surface with $K_S^2<3\chi(S)$ and no irregular étale cover, is it true that, up to finitely many exceptions, $\pi_1^{\mathrm{alg}}(S)$ is a subgroup of $(\mathbb Z/2\mathbb Z)^2$?

来源字段：https://doi.org/10.4171/owr/2007/47

05442 | OWR-1971-009

### Algebraic Realizability of Tropical Varieties

P / partially_solved | Oberwolfach 报告问题汇编

What are the appropriate tropical counterparts of algebraic varieties, including a precise correspondence between tropical cycles/varieties and algebraic varieties or limits of amoebas?

来源字段：https://doi.org/10.4171/owr/2008/26

05443 | OWR-13750341-003

### Algebraicity of Endoscopic Tate Classes

P / partially_solved | Oberwolfach 报告问题汇编

Do all endoscopic Tate classes in the middle-degree cohomology of products $S_K(\operatorname{GSp}_4)\times X_0(N)$ arise from algebraic cycles, including the classes attached to non-generic members of endoscopic $L$-packets?

来源字段：https://doi.org/10.4171/owr/2023/28

05444 | OWR-14298588-013

### Algebraicity of Tate Classes on GSp4 Shimura Varieties

O / open | Oberwolfach 报告问题汇编

Determine when the Tate classes in the relevant degree-three etale cohomology of the Shimura threefold $S_K(\mathrm{GSp}_4)$ arise from algebraic cycles, as predicted by the Tate conjecture.

来源字段：https://doi.org/10.4171/owr/2024/33

05445 | OWR-14298162-004

### Algebro-Geometric Meaning of Noncommutative Volume Bounds

P / partially_solved | Oberwolfach 报告问题汇编

Find an algebro-geometric interpretation of the noncommutative volume term that replaces the Euler characteristic in the analogue of the non-symmetric Mahler inequality for singularities with noncommutative crepant resolutions.

来源字段：https://doi.org/10.4171/owr/2024/14

05446 | OWR-10252925-002

### Algorithms for Multigraded Castelnuovo-Mumford Regularity

O / open | Oberwolfach 报告问题汇编

Develop an effective algorithm computing all minimal elements of the multigraded Castelnuovo-Mumford regularity of a coherent sheaf or module on a product of projective spaces.

来源字段：https://doi.org/10.4171/owr/2022/17

05447 | OWR-14299284-006

### All-Genus Gromov-Witten Invariants of Hilbert Schemes

P / partially_solved | Oberwolfach 报告问题汇编

Give a method to compute, for every genus $g$, the equivariant Gromov-Witten invariants of $\operatorname{Hilb}^d(\mathbb C^2)$ in curve classes $\beta_n$.

来源字段：https://doi.org/10.4171/owr/2025/28

05448 | OWR-4086-010

### Allowed Differential Forms and Monodromy Eigenvalues

O / open | Oberwolfach 报告问题汇编

Can one choose allowed differential forms $\omega$ so that the poles of the associated topological zeta functions correspond precisely to the monodromy eigenvalues of $f$, including all eigenvalues and no extraneous poles?

来源字段：https://doi.org/10.4171/owr/2009/43

05449 | OWR-4427-003

### Almost Aleph-One-Categoricity of Compact Kähler Spaces

O / open | Oberwolfach 报告问题汇编

Is the first-order theory of the many-sorted structure formed by compact complex spaces of Kähler type almost $\aleph_1$-categorical, in the sense that it is stable and nonmultidimensional and every $U$-rank-one type is nonorthogonal to a Morley-rank-one type?

来源字段：https://doi.org/10.4171/owr/2010/39

05450 | OWR-4527-003

### Alpha Bounds for Uniform Fat-Point Schemes

P / partially_solved | Oberwolfach 报告问题汇编

If $X$ is a finite set of points in $\mathbb{P}^{n}$ and $Y$ is the fat-point scheme on $X$ with every point of multiplicity $rn-n+1$, is $\alpha(Y)\geq r\alpha(X)+(r-1)(n-1)$?

来源字段：https://doi.org/10.4171/owr/2010/45

05451 | OWR-14299281-009

### Alternating Signs in Equivariant Quantum K-Theory

O / open | Oberwolfach 报告问题汇编

Do the structure constants of the Schubert-basis product in $T$-equivariant quantum $K$-theory $QK_T(G/P)$ satisfy the conjectured alternating-sign property? Equivalently, does the corresponding Pontryagin product in the localized equivariant $K$-theory of the affine Grassmannian have this alternating-sign behavior?

来源字段：https://doi.org/10.4171/owr/2025/20

05452 | AIM-ALGEBRAIC_GEOMETRY-0289

### Ambient adjoint Cartier test ideals and an intrinsic curve model

P / partially_solved | AIM 研讨会问题汇编

Define a ``test ideal" that relates to the Mather discrepancy multiplier ideal, in the non-\(\mathbb Q\)-Gorenstein case.

来源字段：http://aimpl.org/testandmultiplierideals/2/

05453 | OWR-1971-011

### Ample and Extremal Classes on K3-Type Hyperkähler Manifolds

S / solved | Oberwolfach 报告问题汇编

Let $(F,g)$ be a polarized holomorphic symplectic manifold deformation-equivalent to $S^{[n]}$ for a K3 surface $S$. Is a divisor class $h$ ample exactly when $h\cdot R>0$ for every curve class $R$ with $g\cdot R>0$ and $(R,R)\geq-(n+3)/2$? Do extremal rays satisfying $-(r+3)/2\leq(R,R)$ for some $1\leq r\leq n$ correspond to contractions whose exceptional locus is birationally fibered by $\mathbb{P}^j$ with codimension $j\leq r$, with $R$ represented by a line in a fiber?

来源字段：https://doi.org/10.4171/owr/2008/26

05454 | AIM-ALGEBRAIC_GEOMETRY-0326

### An effective non-nef fixed-parent point-blowup locus

P / partially_solved | AIM 研讨会问题汇编

The paragraph beginning “Kollár's class of examples” is adjacent explanatory text after Problem 1.7 and before Problem 1.8; it is relevant motivation, but not part of the grammatical question.

In the proved result below it means **linearly effective**: \(K_X\sim D\) for an effective integral divisor \(D\), equivalently \(H^0(X,\omega_X)\ne 0\). This is stronger than numerical pseudoeffectivity. No result for the merely pseudoeffective case is claimed.

来源字段：https://aimath.org/WWN/birational/birational.pdf

05455 | AIM-ALGEBRAIC_GEOMETRY-0158

### An equivariant Thomason theorem and a naive-promotion obstruction

S / solved | AIM 研讨会问题汇编

In (Thomason 1995, MR1337494), Thomason showed that connective spectra can be modeled by symmetric monoidal categories.

Is a similar statement true equivariantly? That is, is there an equivalence between \(G\)-symmetric monoidal categories and the homotopy category of connective \(G\)-spectra?

来源字段：http://aimpl.org/equideralggeom/2/

05456 | AIM-ARITHMETIC_GEOMETRY-0074

### An exact ACM certificate for reverse-lexicographic gins of locally Cohen-Macaulay space curves

P / partially_solved | AIM 研讨会问题汇编

**Problem 27.** Are there necessary or sufficient conditions on Borel fixed monomial ideals such that they are the generic initial ideals of local Cohen-Macaulay curves?

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05457 | AIM-ALGEBRAIC_GEOMETRY-0100

### An exact obstruction count for sections of index-one hypersurface families

P / partially_solved | AIM 研讨会问题汇编

\(\mathcal{X}\longrightarrow \mathcal{B}\) be a family of smooth hypersurfaces of degree \(d=n\) in \(\mathbb{P}^{n}\). Is there more than one rational section? Does weak approximation hold?

来源字段：http://aimpl.org/ratsubvarpos/3/

05458 | AIM-ALGEBRAIC_GEOMETRY-0049

### An exact product formula for torsion orders

P / partially_solved | AIM 研讨会问题汇编

Does there exist a rationally connected smooth complex projective variety, not stably rational, that has an integral decomposition of the diagonal?

来源字段：http://aimpl.org/rationalityag/1/

05459 | AIM-ARITHMETIC_GEOMETRY-0041

### An explicit cyclic cubic counterexample from an A4-extension

P / partially_solved | AIM 研讨会问题汇编

Let \[ \nu(F)=\max\{n\mid I^n\ne0\}. \] If \(\nu(F)<\infty\), is it true that \(\nu(F')<\nu(F)+1\) for all finite \(F'/F\)?

来源字段：http://aimpl.org/deformationbrauer/1/

05460 | AIM-ALGEBRAIC_GEOMETRY-0051

### An explicit Frobenius bound for triple products of elliptic curves

P / partially_solved | AIM 研讨会问题汇编

Is there a smooth projective \(3\)-fold \(X\) over a finite field for which

\[ H^3_{nr}(k(X)/k,\mathbb Q/\mathbb Z(2)) \]

is not zero?

来源字段：http://aimpl.org/rationalityag/1/

05461 | AIM-ALGEBRAIC_GEOMETRY-0016

### An explicit interval in the Orlov spectrum of products of projective lines

P / partially_solved | AIM 研讨会问题汇编

What can be said about the Orlov spectrum of a toric variety (or any \(X\))? (e.g. about gaps, sequences, etc.)

Known: - \(OrSpec(\mathbb P^1)=\{1,2\}\). - On a toric surface, there exists full arbitrary large consecutive sequence. - More generally, after replacing \(D^b(X)\) by a category \(\mathcal C\), there is \(OrSpec(\mathcal C)\); spectra for representations of quivers of types A/D have been computed.

来源字段：http://aimpl.org/syzygyms/7/

05462 | AIM-ALGEBRAIC_GEOMETRY-0292

### An explicit one-root certificate for generalized test ideals

P / partially_solved | AIM 研讨会问题汇编

**AIM-ALGEBRAIC_GEOMETRY-0292.** “Realize effective computations of test ideals.”

来源字段：http://aimpl.org/testandmultiplierideals/2/

05463 | AIM-ALGEBRAIC_GEOMETRY-0048

### An explicit smooth locus for the coprime-degree cubic family

P / partially_solved | AIM 研讨会问题汇编

**Does there exist an irrational unirational variety with unirational parametrizations of coprime degrees?**

**Recovered answer.** Yes. In August 2025, Song Yang, Xun Yu, and Zigang Zhu produced a two-dimensional family of smooth cubic threefolds with parametrizations of degrees $2$ and $3$. Every smooth complex cubic threefold is irrational by Clemens--Griffiths. Thus their examples answer the AIM question affirmatively.

来源字段：http://aimpl.org/rationalityag/1/

05464 | AIM-ALGEBRAIC_GEOMETRY-0280

### An extremal bracket, a Segre correction, and a cubic-scroll family

P / partially_solved | AIM 研讨会问题汇编

Fix \(d\). What is the maximum value of the \(F\)-pure threshold of \(\mathfrak m\) of a non-\(\mathbb Q\)-Gorenstein local ring \((R,\mathfrak m)\) of dimension \(d\)?

来源字段：http://aimpl.org/testandmultiplierideals/1/

05465 | AIM-ALGEBRAIC_GEOMETRY-0126

### An integral finite-motive criterion and new product-quotient examples

P / partially_solved | AIM 研讨会问题汇编

Can we check Bloch's conjecture for known examples of super-singular surfaces?

来源字段：http://aimpl.org/ratsubvarpos/9/

05466 | AIM-ALGEBRAIC_GEOMETRY-0307

### An order-profile mld bound from a quasi-etale lci cover

P / partially_solved | AIM 研讨会问题汇编

**Problem 4 (Mustaţă).** Let \(X\) be a \(\mathbb Q\)-Gorenstein variety. A conjecture of Shokurov (Conjecture 2.2 in the introductory notes) says \[ \operatorname{mld}(P;X)\leq \dim X \] for all \(P\in X\). As a weaker form of this question, is there any universal upper bound for \(\operatorname{mld}(P;X)\) after fixing \(\dim X\)?

来源字段：https://aimath.org/WWN/singularvariety/problems_final.pdf

05467 | AIM-ALGEBRAIC_GEOMETRY-0059

### An oriented preservation calculus for categorical representability in codimension two

P / partially_solved | AIM 研讨会问题汇编

Which birational transformations preserve the following property?

\[ D(X)=\langle \mathscr A_1,\ldots,\mathscr A_m\rangle \] has a semiorthogonal decomposition with \(\mathscr A_i\hookrightarrow D(Y_i)\) such that \({\rm dim}\,Y_i\leq {\rm dim}\,X-2\). For future reference, we call this property (*).

Its problem text is:

The standard precise reading is the following. For a smooth projective variety \(X\), \(D(X)\) means \(D^b_{\mathrm{coh}}(X)\). Each \(\mathscr A_i\) is an admissible component, and \(\mathscr A_i\hookrightarrow D(Y_i)\) means that \(\mathscr A_i\) is equivalent to an admissible subcategory of \(D^b_{\mathrm{coh}}(Y_i)\), where \(Y_i\) is smooth and projective. This is categorical representability in dimension at most \(\dim X-2\), or in codimension at least two. Without admissibility the stated property is too weak for the usual formalism and does not agree with the cited definition.

Throughout the mathematical result below, the ground field is \(\mathbb C\), varieties are smooth and projective, and \(n=\dim X\geq2\). The categorical preservation statements themselves work over any field for which the invoked semiorthogonal decompositions hold; \(\mathbb C\) is used for the Hochschild--Hodge interpretation and avoids field qualifications.

来源字段：http://aimpl.org/rationalityag/1/

05468 | AIM-ALGEBRAIC_GEOMETRY-0347

### Andreotti-Mayer slope squeeze and boundary-blind complete subvarieties

P / partially_solved | AIM 研讨会问题汇编

**Problem 6.1 (Grushevsky).** (1) Construct divisors on \(\mathcal A_g\) of small slope. (2) Find explicit complete subvarieties of \(\mathcal A_g\). There is a bound on the codimension of such a variety in [KS03].

来源字段：https://aimath.org/WWN/birational/birational.pdf

05469 | OWR-4086-002

### Applications of Wild Geometry to Algebraic Geometry

O / open | Oberwolfach 报告问题汇编

What is the usefulness for algebraic geometry of wild geometry for differential equations with irregular singularities?

来源字段：https://doi.org/10.4171/owr/2009/43

05470 | AIM-ALGEBRAIC_GEOMETRY-0300

### Arbitrary-ideal alteration formula, finite-principalization obstruction, and exact DVR audit

P / partially_solved | AIM 研讨会问题汇编

Given a log pair \((X,\Delta)\), we know that there exists a finite map \(\phi:Y\to X\) such that \[ \operatorname{im}\!\left( \phi_*\mathcal O_Y\!\left(K_Y-\phi^*(K_X+\Delta)\right) \overset{\mathrm{trace}}{\longrightarrow}\mathcal O_X \right)=\tau(X,\Delta). \] Is this statement true if further decorated by \(\mathfrak a^t\)? What about for alterations?

来源字段：http://aimpl.org/testandmultiplierideals/2/

05471 | AIM-ALGEBRAIC_GEOMETRY-0310

### Arc-space termination and a Betti budget for standard smooth flips

P / partially_solved | AIM 研讨会问题汇编

**7. (Alexeev, Mustaţă)** Give one (non-trivial) application of motivic integration or arc spaces to termination of flips. What about under the assumption that the varieties in a sequence of flips \[ X_1\longrightarrow X_2\longrightarrow\cdots \] are all smooth?

来源字段：https://aimath.org/WWN/singularvariety/problems_final.pdf

05472 | AMR-071-0022

### Are infinite sequences of flips possible in dimensions greater than 3

O / open | AMR 问题清单汇编

Are infinite sequences of flips possible in dimensions greater than 3?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

05473 | AIM-ALGEBRAIC_GEOMETRY-0096

### Arithmetic descent counterexample and bounded graph-complexity transfer

P / partially_solved | AIM 研讨会问题汇编

How does rationality behave in a family over \(\operatorname{Spec}\mathbb Z\) or \(\operatorname{Spec}\mathbb Z[\frac1n]\). If \(X\) is rational / unirational for almost all \(p\), does it follow \(X\) is rational / unirational in char \(0\)?

来源字段：http://aimpl.org/ratsubvarpos/3/

05474 | OWR-14611-005

### Arithmetic Finiteness of Gushel-Mukai Threefolds

O / open | Oberwolfach 报告问题汇编

Let $K$ be a number field, let $S$ be a finite set of finite places of $K$, and set $B=\operatorname{Spec}(\mathcal O_K[S^{-1}])$. Are there only finitely many $B$-isomorphism classes of smooth Fano threefolds of Picard rank one? In particular, is this true for Gushel-Mukai Fano threefolds?

来源字段：https://doi.org/10.4171/owr/2016/29

05475 | AIM-ALGEBRAIC_GEOMETRY-0063

### Arithmetic rigidity and marked Torelli for categorical partners of cubic fourfolds

P / partially_solved | AIM 研讨会问题汇编

Let $X$ and $Y$ be cubic fourfolds. If $Ku(X) \simeq Ku(Y)$, is $X$ birational to $Y$?

来源字段：http://aimpl.org/rationalityag/1/

05476 | AIM-ALGEBRAIC_GEOMETRY-0005

### Arithmetic toric exceptional collections and the anisotropic conic

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** If \(X\) is a smooth projective arithmetic toric variety over a field \(k\), does \(D^b(\operatorname{coh}X)\) have a full exceptional collection when “exceptional” is allowed to mean division-exceptional? What can be said about collections of sheaves or vector bundles?

来源字段：http://aimpl.org/syzygyms/2/

05477 | AIM-ALGEBRAIC_GEOMETRY-0159

### Arithmetic-fracture coefficient decompletions of TP(F_p)

P / partially_solved | AIM 研讨会问题汇编

**Long Term.** Can we construct a decompleted version of the Tate spectrum?

Construct a spectrum mapping to \(TP(\mathbf F_p)\), preferably an \(E_\infty\)-ring, whose homotopy ring is \(\mathbf Z_{(p)}[v^{\pm1}]\) or \(\mathbf Z_{(p)}^h[v^{\pm1}]\), and whose derived \(p\)-completion recovers \(TP(\mathbf F_p)\).

来源字段：http://aimpl.org/equideralggeom/2/

05478 | OWR-13109-007

### Ascending Chain Condition for Log Discrepancies

O / open | Oberwolfach 报告问题汇编

Fix a dimension $n$, a descending-chain-condition set $I\subseteq[0,1]$, and a descending-chain-condition set $J\subseteq\mathbb R_{\geq 0}$. Does there exist a finite subset $J_0\subseteq J$ such that, whenever $(X,\Delta)$ is an $n$-dimensional log canonical pair whose coefficients lie in $I$ and whose log discrepancy lies in $J$, that log discrepancy belongs to $J_0$?

来源字段：https://doi.org/10.4171/owr/2014/39

05479 | OWR-3472-008

### Ascending Chains of Log Canonical Thresholds

S / solved | Oberwolfach 报告问题汇编

For every $n$, does the set of log canonical thresholds of ideals on smooth complex $n$-dimensional varieties satisfy the ascending chain condition; equivalently, does it contain no infinite strictly increasing sequence?

来源字段：https://doi.org/10.4171/owr/2009/22

05480 | OWR-15177-012

### Asymptotic Point Counts on Curve Moduli Spaces

O / open | Oberwolfach 报告问题汇编

For fixed $q$, does the ratio of the number of $\mathbb F_q$-valued points of the coarse moduli space $\mathcal M_g$ to $q^{3g-3}$ have a finite positive limit as $g\to\infty$?

来源字段：https://doi.org/10.4171/owr/2016/56

05481 | OWR-11139-009

### Asymptotics of Intermediate Jet-Bundle Cohomology

O / open | Oberwolfach 报告问题汇编

How does

$$h^0\bigl(X_k,\mathcal O_{X_k}(m)\otimes\pi_k^*\mathcal O(-A)\bigr)$$

behave asymptotically when intermediate cohomology groups are present, and can this prove the generalized Green-Griffiths-Lang conjecture?

来源字段：https://doi.org/10.4171/owr/2011/47

05482 | OWR-14299284-002

### Atomic Categories and Stable Rationality

P / partially_solved | Oberwolfach 报告问题汇编

For a smooth projective surface $X$ over a field $k$, is $X$ rational if and only if every atom of $D^b(X)$ is equivalent to $D^b(L)$ for some finite field extension $L/k$?

来源字段：https://doi.org/10.4171/owr/2025/28

05483 | OWR-14299284-012

### Atomic Decomposition of Rank-Two Bundle Moduli

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Atomic-Decomposition Extraction [OWR-14299284-011]

Is the proposed semiorthogonal decomposition of $SU_C(2,\Lambda)$ atomic, meaning that it is compatible with the decomposition of $H^*(SU_C(2,\Lambda),\mathbb C)$ into generalized eigenspaces of quantum multiplication by $c_1$?

来源字段：https://doi.org/10.4171/owr/2025/28

05484 | OWR-15219-010

### Atomic Measures on Irreducible Truncated Moment Varieties

O / open | Oberwolfach 报告问题汇编

Let $V=Z(p)$ be an irreducible real algebraic set. Must every recursively generated truncated moment matrix $M^{(n)}$ associated with $V$ admit a representing measure with $\operatorname{rank}M^{(n)}$ atoms? How can the variety change when $M^{(n)}$ is extended to $M^{(n+1)}$?

来源字段：https://doi.org/10.4171/owr/2017/14

05485 | OWR-1452-016

### Automorphisms of Rational Surfaces with Antipluricanonical Curves

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a smooth projective complex rational surface. If $\operatorname{Aut}^*(X)=\operatorname{Im}(\operatorname{Aut}(X)\to\operatorname{Aut}(\operatorname{Pic}(X)))$ is infinite, does there exist a birational morphism $X\to Y$ such that $Y$ has an antipluricanonical curve and an infinite subgroup of $\operatorname{Aut}^*(Y)$ lifts to $X$?

来源字段：https://doi.org/10.4171/owr/2007/01

05486 | OWR-13750327-002

### Ax-Schanuel and Zilber-Pink for Abelian-Differential Strata

O / open | Oberwolfach 报告问题汇编

Prove the Ax-Schanuel conjecture for a stratum $S_\alpha$ of abelian differentials. Does $S_\alpha$ contain only finitely many arithmetic points of rank at least $3$?

来源字段：https://doi.org/10.4171/owr/2023/18

05487 | OWR-1737-002

### Backward-Stable Numerical Implicitization of Rational Parametrizations

P / partially_solved | Oberwolfach 报告问题汇编

Develop and analyze numerical implicitization for rational parametrizations whose coefficients are floating-point data with limited precision: determine the best possible implicit equation under those limits, using a structurally compatible exact solution with minimal backward error, especially when the degree of the implicit equation is not known in advance.

来源字段：https://doi.org/10.4171/owr/2007/54

05488 | OWR-12697688-018

### Banach-Colmez Proofs of the Fargues-Fontaine C-One Property

P / partially_solved | Oberwolfach 报告问题汇编

Can Tsen's dimension-count proof of the $C_1$ property for function fields of algebraic curves be adapted to the Fargues-Fontaine curve using Banach-Colmez spaces?

来源字段：https://doi.org/10.4171/owr/2023/5

05489 | AMR-071-0002

### Bass conjecture

O / open | AMR 问题清单汇编

Bass conjecture: for every finitely generated $\mathbb Z$-algebra $A$ and every $n\geq0$, is the G-theory group $K'_n(A)$ finitely generated? Equivalently, when $A$ is regular, is $K_n(A)$ finitely generated?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

05490 | AMR-071-0003

### Bass-Quillen conjecture

P / partially_solved | AMR 问题清单汇编

Bass-Quillen conjecture: if $A$ is a regular Noetherian ring, is every finitely generated projective module over $A[t_1,\ldots,t_n]$ extended from a projective $A$-module?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

05491 | OWR-13101-006

### Beauville-Voisin Conjecture for Hyperkähler Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a projective hyperkähler manifold. Is every polynomial $z$ in first Chern classes of line bundles and in the Chern classes of $T_X$ that is numerically trivial also rationally equivalent to zero?

来源字段：https://doi.org/10.4171/owr/2014/31

05492 | OWR-762-004

### Beilinson-Soulé Vanishing for Smooth Schemes

O / open | Oberwolfach 报告问题汇编

For every smooth scheme $X$ over a field $k$, must motivic cohomology $H^p(X,\mathbb Z(q))$ vanish for $p<0$?

来源字段：https://doi.org/10.4171/owr/2004/48

05493 | OWR-15586-005

### Big Divisor Geometry Relative to Subvarieties

O / open | Oberwolfach 报告问题汇编

Let $X$ be a smooth complex projective variety, let $V\subset X$ be a subvariety, and let $D$ be a big $\mathbb Q$- or $\mathbb R$-divisor. Determine the positivity and birational geometry of the numerical class of $D$ relative to $V$.

来源字段：https://doi.org/10.4171/owr/2017/43

05494 | OWR-4340-001

### Bigness of Subvarieties with Ample Normal Bundle

P / partially_solved | Oberwolfach 报告问题汇编

If $Y$ is a smooth subvariety of $X$ whose normal bundle $N_{Y/X}$ is ample, is the cycle class $[Y]$ big?

来源字段：https://doi.org/10.4171/owr/2010/27

05495 | OWR-1971-002

### Bigness of Very Moving Algebraic Cycles

O / open | Oberwolfach 报告问题汇编

If $W$ is a codimension-$k$ subvariety of a smooth projective variety $Y$ that is very moving-its deformations can be required to pass through a generic point with a generically prescribed tangent space-is the algebraic-cycle class $[W]$ big, meaning that it lies in the interior of the cone generated by effective codimension-$k$ cycle classes?

来源字段：https://doi.org/10.4171/owr/2008/26

05496 | OWR-13946-004

### Birational Geometry of Ball-Quotient Moduli Spaces

P / partially_solved | Oberwolfach 报告问题汇编

For the complex ball quotient moduli spaces of $K3^{[2]}$-type fourfolds with a nonsymplectic automorphism of prime order, determine their Kodaira dimensions and identify the rational cases.

来源字段：https://doi.org/10.4171/owr/2015/51

05497 | OWR-16768-014

### Birational Models with Irreducible Arc Spaces

O / open | Oberwolfach 报告问题汇编

Given a variety $X$, does there exist a proper birational morphism $Y\to X$ such that the arc space $Y_\infty$ is irreducible?

来源字段：https://doi.org/10.4171/owr/2019/6

05498 | OWR-13500-011

### Birational Modifications Proving Rationality of a Ueno Variety

O / open | Oberwolfach 报告问题汇编

The Ueno-type variety $X_{4,6}$ is birational to a diagonal cubic surface over $k(t_1,t_2)$. Can that cubic surface be modified birationally, analogously to the conic-bundle treatment of $X_{3,4}$, to prove that $X_{4,6}$ is rational?

来源字段：https://doi.org/10.4171/owr/2015/15

05499 | OWR-14221-006

### Birational Sequences for Plabic Newton-Okounkov Bodies

O / open | Oberwolfach 报告问题汇编

Can every Newton-Okounkov body arising from a plabic graph be obtained from a birational sequence of a Grassmannian together with a suitable total order?

来源字段：https://doi.org/10.4171/owr/2016/13

05500 | OWR-1318-001

### Bloch-Beilinson Torsion Conjecture for Albanese Kernels

O / open | Oberwolfach 报告问题汇编

For a smooth projective variety $V$ over a number field $k$, is the kernel of its Albanese map $\varphi_V$ torsion?

来源字段：https://doi.org/10.4171/owr/2006/32

05501 | OWR-17139-004

### Blue Sheaf Cohomology for Toric and Tropical Geometry

P / partially_solved | Oberwolfach 报告问题汇编

For distinguished generators in toric sheaf cohomology, determine when they form a basis, what structure they induce on the Cox ring, and how they can be used for explicit computations. Extend blue sheaf cohomology to toroidal, logarithmic, and tropical schemes.

来源字段：https://doi.org/10.4171/owr/2019/43

05502 | OWR-13500-003

### Bogomolov-Gieseker Inequality for Tilt-Semistable Threefold Complexes

S / solved | Oberwolfach 报告问题汇编

BG inequality conjecture: let $X$ be a smooth projective threefold over $\mathbb C$, let $B,\omega\in\operatorname{NS}(X)_\mathbb Q$ with $\omega$ ample, and let $E$ be tilt-semistable with $\nu_{\omega,B}(E)=0$. Then $\operatorname{ch}^{B}_{3}(E)\leq\frac{1}{18}\operatorname{ch}^{B}_{1}(E)\cdot\omega^{2}$. The conjecture is known for $X=\mathbb P^{3}$, smooth quadrics, and étale quotients of abelian threefolds.

来源字段：https://doi.org/10.4171/owr/2015/15

05503 | OWR-14299288-010

### Bondal-Thomsen Polytopes within Lattice-Polytope Theory

P / partially_solved | Oberwolfach 报告问题汇编

Place the finite collection of Bondal-Thomsen polytopes within the general theory of lattice polytopes: characterize their intrinsic combinatorial properties, denominator bounds, and relations to zonotopes and toric arrangements.

来源字段：https://doi.org/10.4171/owr/2025/19

05504 | AIM-ALGEBRAIC_GEOMETRY-0165

### Borel blindness and a geometric-fixed etale site

P / partially_solved | AIM 研讨会问题汇编

Long Term

How best should we be making sense of equivariant derived algebraic geometry? (e.g. What is Spec? What site to use? What sheaf? etc.)

来源字段：http://aimpl.org/equideralggeom/3/

05505 | AIM-ALGEBRAIC_GEOMETRY-0127

### Boundary contact of rational curves in characteristic p

P / partially_solved | AIM 研讨会问题汇编

Let \((X,D)\) snc. How can rational curves in \(X\) meet \(D\)? Are there any restrictions?

来源字段：http://aimpl.org/ratsubvarpos/9/

05506 | OWR-1703871-010

### Boundary-Corrected Ampleness of Extended Hodge Bundles

P / partially_solved | Oberwolfach 报告问题汇编

Let $B$ be smooth and projective, let $Z=\bigcup_iZ_i$ be a simple normal-crossing divisor, and let $\Lambda^e$ be the canonical extension of the Hodge line bundle for a period map whose logarithmic differential is injective. Do there exist integers $a_i\geq0$ and $m_0$ such that $$ m\Lambda^e-\sum_i a_i[Z_i] $$ is ample for every $m\geq m_0$?

来源字段：https://doi.org/10.4171/owr/2020/24

05507 | OWR-1730-006

### Boundary-Intersection Vanishing on Abelian-Variety Moduli

O / open | Oberwolfach 报告问题汇编

Let $\mathcal A_g$ be a toroidal compactification of the moduli space of principally polarized abelian varieties, let $G=g(g+1)/2$, and let $a_N$ be the intersection number of the Hodge class and boundary divisor. Must $a_N$ vanish unless $G-N=k(k+1)/2$ for some $k\leq g$? Does the same vanishing pattern hold for other reasonable toroidal compactifications?

来源字段：https://doi.org/10.4171/owr/2007/47

05508 | OWR-733-007

### Bounded Complements for $\varepsilon-Log-Canonical$ Varieties

P / partially_solved | Oberwolfach 报告问题汇编

For every dimension bound $d$ and $\varepsilon\geq 0$, does there exist a finite set of complementary indices $N_d(\varepsilon)$ such that every $\varepsilon$-log-canonical variety $X$ of dimension at most $d$, with $-K_X$ nef and big over $Z$, has a general bounded complement $(X,D/n)$ that is $\varepsilon$-log-canonical?

来源字段：https://doi.org/10.4171/owr/2004/31

05509 | AIM-ALGEBRAIC_GEOMETRY-0013

### Bounded moduli charts and ghost directions for virtual resolutions

P / partially_solved | AIM 研讨会问题汇编

Observation: The current definition of virtual resolution is so broad and it is not meaningful to parameterize, then.

Is there a well-behaved moduli theory of virtual resolutions equipped with extra structure? e.g. condition under which there exists a finite set? or space/ stacks/...? (c.f. [Boij-Sölderberg] theory constructs a meaningful cone of Betti tables of all free complexes. Look at subcone of virtual resolutions of a given \(M\))

来源字段：http://aimpl.org/syzygyms/4/

05510 | OWR-17296-011

### Bounded Negativity Conjecture for Complex Projective Surfaces

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Bounded Negativity Conjecture [OWR-17296-018]

Does every smooth projective surface $X$ over the complex numbers have bounded negativity; that is, does there exist $b(X)$ such that $C^2\geq-b(X)$ for every reduced irreducible curve $C\subset X$?

来源字段：https://doi.org/10.4171/owr/2019/53

05511 | OWR-14222-011

### Bounded Negativity Conjecture for Projective Surfaces

O / open | Oberwolfach 报告问题汇编

Bounded Negativity Conjecture: for every smooth complex projective surface X, there is $b(X)$ such that every irreducible curve C on X satisfies $C^2\geq-b(X)$.

来源字段：https://doi.org/10.4171/owr/2016/14

05512 | OWR-4990374-007

### Bounded Negativity for Rational Surfaces

O / open | Oberwolfach 报告问题汇编

For every smooth projective rational surface $X$ over a field, is there a constant $B_X$ such that every reduced irreducible curve $C\subset X$ satisfies $C^2\geq B_X$?

来源字段：https://doi.org/10.4171/owr/2021/5

05513 | OWR-17296-019

### Bounded Negativity for Root of Unity Blowups

O / open | Oberwolfach 报告问题汇编

Let $\varepsilon$ be a primitive $m$th root of unity, let $Z(m)$ be the points $(1:\varepsilon^\alpha:\varepsilon^\beta)$ in $\mathbb{P}^2$ for $1\leq\alpha,\beta\leq m$, and let $X(m)$ be the blowup of $\mathbb{P}^2$ at $Z(m)$. Is negativity bounded on $X(m)$, and if so what is its bound $b(X(m))$?

来源字段：https://doi.org/10.4171/owr/2019/53

05514 | OWR-12870-001

### Bounded Negativity on Smooth Projective Surfaces

O / open | Oberwolfach 报告问题汇编

For every smooth projective complex surface $X$, does there exist a constant $b_X$ such that $C^2\geq-b_X$ for every reduced effective curve $C\subset X$?

来源字段：https://doi.org/10.4171/owr/2014/10

05515 | AIM-ALGEBRAIC_GEOMETRY-0323

### Boundedness after fixing the birational model

S / solved | AIM 研讨会问题汇编

**Problem 1.4 (Alexeev).** Are the following classes of varieties bounded? Here bounded means that all members of the class can be put into a family over a base of finite type.

1. Semi-log canonical varieties of general type with fixed \(K^n\). 2. \(\epsilon\)-log terminal Fano varieties.

The words “1.3. Positive characteristic issues” begin the following subsection and are not part of Problem 1.4.

Two parameters have to be regarded as fixed: the dimension \(n\), and in (2) the real number \(\epsilon>0\). Without fixed dimension even the smooth projective spaces would make (2) unbounded. The source does not explicitly state a ground field in this sentence. Its placement immediately before the positive-characteristic subsection, and the later theorems that resolved it, support the usual characteristic-zero reading.

来源字段：https://aimath.org/WWN/birational/birational.pdf

05516 | AIM-ALGEBRAIC_GEOMETRY-0058

### Branches, exceptional cycles, and tangent-cone comparisons among three rationality conditions

P / partially_solved | AIM 研讨会问题汇编

**Compare \(CH_0\)-universally trivial, \(B\)-rational, and \(L\)-rational singularities.**

来源字段：http://aimpl.org/rationalityag/1/

05517 | OWR-1703871-002

### Brauer Groups of Smooth Models of Symmetric Powers

O / open | Oberwolfach 报告问题汇编

Let $X_m=\operatorname{Sym}^m(X)$ and let $X'_m$ be a smooth model of $X_m$. Can $\operatorname{Br}(X'_m)$ be estimated in terms of $\operatorname{Br}(X)$?

来源字段：https://doi.org/10.4171/owr/2020/24

05518 | OWR-14298584-008

### Brauer Groups of Tame Stacky Curves

P / partially_solved | Oberwolfach 报告问题汇编

Compute the Brauer group of an arbitrary tame stacky curve over an algebraically closed field.

来源字段：https://doi.org/10.4171/owr/2024/40

05519 | AIM-ARITHMETIC_GEOMETRY-0019

### Brauer lifts and Beauville--Mukai torsors

P / partially_solved | AIM 研讨会问题汇编

Let $X$ be a K3 surface of degree $2d$. Consider the Hilbert scheme $\mathscr{H}$ of length $d+1$ zero-dimensional subschemes of $X$. $\mathrm{Br}(X)=\mathrm{Br}(\mathscr{H})$. There is a map $\mathscr{H}\to\mathbb P^{d+1}$ with generic fiber an abelian variety. Does every $\alpha\in\mathrm{Br}(X)$ be correspond to a torsor for the relative Jacobian of this vibration in such a way that the order of the torsor associated to $\alpha$ times the order of $\alpha$ is the order of the torsor $\mathscr H$? In this question we are using the same correspondence between principal homogeneous spaces and Brauer classes as in Bianca's talk.

If we consider the moduli space of $\alpha$-twisted sheaves on $X$ of dimension 0 and length $d+1$, does this realize the torsors of the previous part of this question?

来源字段：http://aimpl.org/brauermoduli/1/

05520 | OWR-733-002

### Brauer Obstructions to Rational Points over Surface Function Fields

P / partially_solved | Oberwolfach 报告问题汇编

For a variety over the function field of a surface, what hypotheses on its geometric generic fiber ensure that the Brauer obstruction is the only obstruction to the existence of a rational point?

来源字段：https://doi.org/10.4171/owr/2004/31

05521 | OWR-739-005

### Brauer Obstructions to Rational Sections

O / open | Oberwolfach 报告问题汇编

Let $f:Y\to S$ be a dominant projective morphism with $S$ a surface and with rationally $1$-connected generic fiber $Y_{k(S)}$. Is there a class $\alpha\in\operatorname{Br}(k(S))$ such that, for every dominant morphism of surfaces $S'\to S$, one has $Y(k(S'))\ne\varnothing$ if and only if $\alpha|_{k(S')}=0$?

来源字段：https://doi.org/10.4171/owr/2004/37

05522 | OWR-16158-011

### Brauer-Manin Density in Rationally Connected Fibrations

O / open | Oberwolfach 报告问题汇编

Let $p:Z\to B$ be a dominant morphism of smooth proper varieties over a number field $k$, with $B$ and the generic fiber rationally connected. Suppose $B(k)$ is dense in $B(\mathbb A_k)^{\operatorname{Br}(B)}$ and, for rational points $b$ in a dense open subset of $B$, each $Z_b(k)$ is dense in $Z_b(\mathbb A_k)^{\operatorname{Br}(Z_b)}$. Must $Z(k)$ be dense in $Z(\mathbb A_k)^{\operatorname{Br}(Z)}$?

来源字段：https://doi.org/10.4171/owr/2018/17

05523 | OWR-15174-008

### Brauer-Manin Sufficiency for Smooth Fano Hypersurfaces

O / open | Oberwolfach 报告问题汇编

For smooth Fano hypersurfaces, is the Brauer-Manin obstruction the only obstruction to the Hasse principle, as predicted by the Colliot-Thélène conjecture? Equivalently in the source's density formulation, is $\delta_{\mathrm{global}}=\delta_{\mathrm{local}}$?

来源字段：https://doi.org/10.4171/owr/2016/53

05524 | OWR-13946-011

### Brill-Noether Intersections of K3 Moduli Images

P / partially_solved | Oberwolfach 报告问题汇编

What can be said about $\operatorname{Im}(\psi_{X,n,\delta})\cap M^r_{g,d}$ and $\operatorname{Im}(\psi_{p,n,\delta})\cap M^r_{g,d}$, and what consequences does the first intersection have for the Mori theory of $X^{[d]}$?

来源字段：https://doi.org/10.4171/owr/2015/51

05525 | OWR-1971-008

### Bøgvad's Normal-Koszul Conjecture

O / open | Oberwolfach 报告问题汇编

If $X$ is a smooth projective toric variety with ample line bundle $L$, is the section ring $R(X,L)$ normal and Koszul? More broadly, under what hypotheses on a polarized toric variety is its section ring normal or Koszul?

来源字段：https://doi.org/10.4171/owr/2008/26

05526 | OWR-13750327-001

### Campana-Peternell Conjecture for Fano Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Is every Fano manifold with nef tangent bundle a rational homogeneous variety?

来源字段：https://doi.org/10.4171/owr/2023/18

05527 | AIM-ALGEBRAIC_GEOMETRY-0198

### Canonical curves versus plane differential signatures

P / partially_solved | AIM 研讨会问题汇编

“From a differential geometry perspective we can assign invariants to a smooth curve or surface. This allows us to assign a signature curve to each curve.”

**Problem 3.3.** What is the signature curve of a general canonical curve?

Interpret “signature curve” as the Kogan--Olver rational differential signature of a **plane** algebraic curve, most naturally for the \(\operatorname{PGL}(3)\)-action. Interpret “canonical curve” in its standard algebraic-geometric sense, namely the canonical image of a smooth nonhyperelliptic curve.

来源字段：http://aimpl.org/algvision/3/

05528 | AIM-ALGEBRAIC_GEOMETRY-0312

### Canonical indices, crepant centres, and an exact quotient-family calibration

P / partially_solved | AIM 研讨会问题汇编

**1. (Shokurov)** Fix \(n\) and consider projective varieties \(X\) with an exact canonical singularity of dimension \(n\). Is the index of \(K_X\) at the singularity bounded? What about the case \(\dim X=3\)? More generally, for a fixed minimal log discrepancy of \((X,0)\), is the index of \(K_X\) at any point with such an mld, bounded?

Fix \(n\) and consider projective varieties \(X\) with **strictly canonical singularities** of dimension \(n\). Is the index of \(K_X\) bounded?

来源字段：https://aimath.org/WWN/singularvariety/problems_final.pdf

05529 | OWR-11133-006

### Canonical Maps from Varieties to Okounkov Bodies

P / partially_solved | Oberwolfach 报告问题汇编

Does every projective variety $X$ with a very ample line bundle $L$ admit a reasonable map to its Okounkov body?

来源字段：https://doi.org/10.4171/owr/2011/41

05530 | AIM-ALGEBRAIC_GEOMETRY-0209

### Carlsson-Weinshall duality is Gale-trivial on the fiducial torus

P / partially_solved | AIM 研讨会问题汇编

Is there a relationship between Carlsson-Weinshall and Gale duality?

来源字段：http://aimpl.org/algvision/5/

05531 | OWR-788-003

### Carrell Minimality Conjecture

S / solved | Oberwolfach 报告问题汇编

If a smooth projective variety $X$ of general type admits a nowhere-vanishing holomorphic one-form, then $X$ is minimal.

来源字段：https://doi.org/10.4171/owr/2005/07

05532 | OWR-14299283-013

### Catalan Formulas for Ekedahl-Oort Intersection Cohomology

P / partially_solved | Oberwolfach 报告问题汇编

Generalize the existing method to obtain explicit Catalan-combinatorial formulas for the local intersection cohomology of every Ekedahl-Oort space $\operatorname{EO}(r,h)$.

来源字段：https://doi.org/10.4171/owr/2025/25

05533 | AIM-ALGEBRAIC_GEOMETRY-0003

### Categorical fingerprints for symplectic mirrors of blow-ups

P / partially_solved | AIM 研讨会问题汇编

Probe what is known about symplectic mirrors to algebro-geometric operations (not necessarily toric), such as blow-ups.

来源字段：http://aimpl.org/syzygyms/1/

05534 | OWR-4338-001

### Categorical Gap of Cubic Fourfolds

P / partially_solved | Oberwolfach 报告问题汇编

For a smooth projective variety $X$ with bounded derived category of coherent sheaves $D^b(X)$, Orlov's spectrum (or gap) measures the lengths of gaps in generation times (Rouquier dimensions) of $D^b(X)$. Is the gap of a smooth projective $4$-dimensional cubic hypersurface $X \subset \mathbb{P}^5$ strictly greater than $1$?

来源字段：https://doi.org/10.4171/owr/2010/25

05535 | OWR-5148-004

### Categorical Gaps and Wall-Crossings from Toric Degenerations

O / open | Oberwolfach 报告问题汇编

For a toric degeneration parameter space $Z$ with discriminant $D$, does the rational homotopy type of $Z\setminus D$ determine the gaps in the category spectrum, and do the big loops in $\pi_1(Z\setminus D)$ determine wall-crossings of the category?

来源字段：https://doi.org/10.4171/owr/2011/20

05536 | OWR-14299516-007

### Categorical Künneth Formula for Étale Motives

P / partially_solved | Oberwolfach 报告问题汇编

If $Y$ is smooth and proper over $k$, does exterior product induce an equivalence $\mathrm{DM}(X)^{\prime}\otimes_{\mathrm{DM}(k)}\mathrm{DM}(Y)^{\prime}\simeq\mathrm{DM}(X\times_kY)^{\prime}$?

来源字段：https://doi.org/10.4171/owr/2025/36

05537 | OWR-14299288-006

### Categorical Noncommutative Neighborhoods of Toric Subvarieties

O / open | Oberwolfach 报告问题汇编

For a broad class of subvarieties $Y\subset X_\Sigma$, construct a deformation of categories whose generic fiber is $\operatorname{Coh}_{\mathrm{dg}}(X_\Sigma)$, an object in the special fiber deforming to $\mathcal{O}_Y$, a possibly singular nonpositively curved space $L_V$, and a noncommutative neighborhood described by a functor to $\operatorname{Perf}(\mathbb{C}[\pi_1(L_V)])$, suitably modified for singularities.

来源字段：https://doi.org/10.4171/owr/2025/19

05538 | OWR-1734-010

### Centers of Irreducible Hodge Groups

O / open | Oberwolfach 报告问题汇编

For an irreducible Hodge structure $V$, is the center of its Hodge group $\operatorname{Hg}(V)$ equal to $\mathbb G_m$?

来源字段：https://doi.org/10.4171/owr/2007/51

05539 | AIM-ARITHMETIC_GEOMETRY-0010

### Central normalization and an explicit quaternionic reduced locus

P / partially_solved | AIM 研讨会问题汇编

Could we describe the reduced locus of Rapoport--Zink spaces (integral local Shimura varieties) for inner forms of general linear groups? It will be useful for studying arithmetic transfer conjectures in the context of linear arithmetic fundamental lemmas.

来源字段：http://aimpl.org/intersectshimura/2/

05540 | AIM-ALGEBRAIC_GEOMETRY-0191

### Certified epipolar consistency for disk-valued image points

P / partially_solved | AIM 研讨会问题汇编

How can we account for uncertainty, given the geometric structure of these problems? What if we replace points with disks?

来源字段：http://aimpl.org/algvision/1/

05541 | AIM-ALGEBRAIC_GEOMETRY-0274

### Certified finite computation of F-pure thresholds

P / partially_solved | AIM 研讨会问题汇编

**AIM-ALGEBRAIC_GEOMETRY-0274.** “Realize effective computations of numerical \(F\)-invariants.”

来源字段：http://aimpl.org/testandmultiplierideals/1/

05542 | OWR-14298795-018

### Characteristic Classes of Hodge-Tate Local Systems

O / open | Oberwolfach 报告问题汇编

Let $X$ be a smooth algebraic variety over a finite extension $K/\mathbb Q_p$, and let $L$ be a Hodge-Tate local system on $X$. Is

$$\alpha_X(\ell_i(L))=(i-1)!\sum_{m\in\mathbb Z}m\,\operatorname{ch}_{i-1}(\operatorname{gr}_mD_{\mathrm{HT}}(L))?$$

This asks for the expected Hodge-Tate characteristic-class identity.

来源字段：https://doi.org/10.4171/owr/2024/45

05543 | OWR-15955-013

### Characteristic Varieties of Divisors Beyond Line Arrangements

P / partially_solved | Oberwolfach 报告问题汇编

Does the characterization of high-dimensional essential components of the characteristic variety for a union of lines in $\mathbb P^2$ extend to more general divisors on smooth projective surfaces?

来源字段：https://doi.org/10.4171/owr/2017/50

05544 | OWR-13682-007

### Characteristic Varieties of Rigid-Analytic D-Modules

O / open | Oberwolfach 报告问题汇编

Develop a satisfactory theory of characteristic varieties for co-admissible D-modules on smooth rigid-analytic spaces, analogous to the characteristic-variety theory for algebraic D-modules.

来源字段：https://doi.org/10.4171/owr/2015/25

05545 | AIM-ALGEBRAIC_GEOMETRY-0098

### Characteristic-zero specialization and an odd-index theorem for cubic fourfolds

P / partially_solved | AIM 研讨会问题汇编

“If the general member of a family of smooth cubic is rational, is the special fiber also rational?”

The statement leaves four mathematically important choices unspecified:

1. “cubic” could mean a cubic curve, surface, threefold, fourfold, or an arbitrary cubic hypersurface; 2. “general” could mean the generic fiber, the geometric generic fiber, every point of a dense open set, a very general point, or merely a dense set of points; 3. rationality could mean rationality over the stated residue field or geometric rationality after algebraic closure; 4. the base could be a curve over a field or a mixed-characteristic trait.

The immediately preceding AIM Problem 3.07 explicitly concerns reductions of a cubic fourfold, and the workshop is about positive characteristic. It is therefore plausible that cubic fourfolds in positive or mixed characteristic were intended, but the source does not verify this. This report treats the two most useful precise readings:

- a smooth proper family over a characteristic-zero curve whose (geometric) generic fiber is rational; - a DVR family of smooth cubic fourfolds, possibly in positive or mixed characteristic, for which rationality is witnessed by projection from a plane and an odd-degree multisection.

Dimension checks confirm why the fourfold reading is natural. A smooth plane cubic is geometrically a genus-one curve and is not rational. A smooth cubic surface over an algebraically closed field is rational, so the geometric question is automatic in dimension two. Over \(\mathbb C\), every smooth cubic threefold is irrational. Cubic fourfold rationality is the first classical case in this sequence for which the general rationality problem remains genuinely subtle.

来源字段：http://aimpl.org/ratsubvarpos/3/

05546 | AIM-ALGEBRAIC_GEOMETRY-0322

### Characteristic-zero status and a fixed-volume bridge for stable n-folds

S / solved | AIM 研讨会问题汇编

**Problem 1.3 (Higher dimensions).** In [Kar00], Karu proves that the semistable minimal model program in dimension \(n+1\) can be used (together with the weak semistable reduction of Abramovich and Karu) to construct the moduli space of smoothable stable \(n\)-folds. See the previous problem for problems with this approach. This is what needs to be done to avoid the MMP:

1. Prove boundedness of slc \(n\)-folds with fixed \(K^n\) (see Alexeev's problems below). The semistable MMP is not strong enough to do this for non-smoothable \(n\)-folds anyway. 2. Prove that small \(\mathbb Q\)-Gorenstein deformations of slc singularities are slc. It is true that small \(\mathbb Q\)-Gorenstein deformations of slt singularities are slt. Again, the MMP in one dimension higher would verify this. 3. Verify the valuative criterion for properness for non-smoothable \(n\)-folds. The MMP gives the valuative criterion in the smoothable case.

来源字段：https://aimath.org/WWN/birational/birational.pdf

05547 | OWR-1703867-001

### Characterization of Integral Models of Shimura Varieties

O / open | Oberwolfach 报告问题汇编

What is a good general characterization of integral models of Shimura varieties?

来源字段：https://doi.org/10.4171/owr/2020/20

05548 | OWR-17296-007

### Characterizing Projective Space by Tangent Bundle Seshadri Constants

O / open | Oberwolfach 报告问题汇编

Is projective space $\mathbb{P}^n$ the only $n$-dimensional projective manifold $X$ for which the Seshadri constant $\varepsilon(T_X;x_0)$ of the tangent bundle is positive at some point $x_0\in X$?

来源字段：https://doi.org/10.4171/owr/2019/53

05549 | OWR-13750339-008

### Chern and Degeneracy Classes of Coinvariant Bundles

O / open | Oberwolfach 报告问题汇编

For coinvariant bundles not arising from a cohomological field theory, compute their Chern classes; characterize when higher Chern classes are tautological or effective; realize extremal effective rays as first Chern classes; and determine degeneracy classes of natural maps between such bundles.

来源字段：https://doi.org/10.4171/owr/2023/27

05550 | OWR-14222-016

### Chern Slope Accumulation for Complex Line Arrangements

O / open | Oberwolfach 报告问题汇编

What are the accumulation points of the logarithmic Chern-slope ratio $\bar c_1^2/\bar c_2$ satisfying $5/2<\bar c_1^2/\bar c_2\leq 8/3$ for strictly complex line arrangements, and are they related to the fields of definition?

来源字段：https://doi.org/10.4171/owr/2016/14

05551 | OWR-744-003

### Chern-Class Generators for Wonderful Compactifications

P / partially_solved | Oberwolfach 报告问题汇编

Find explicit Chern-class generators for the rational cohomology ring of the wonderful compactification of an adjoint symmetric space.

来源字段：https://doi.org/10.4171/owr/2004/42

05552 | OWR-13101-008

### Chow Decompositions of Hilbert Schemes of K3 Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

For $X=S^{[m]}$ with $S$ a projective K3 surface, does the Chow decomposition $CH^i_{(j)}(X)$ satisfy $CH^i_{(j)}(X)=0$ for $j<0$, and does the cycle-class map $CH^i_{(0)}(X)\to H^{2i}(X)$ remain injective?

来源字段：https://doi.org/10.4171/owr/2014/31

05553 | OWR-13750339-006

### Chow Vanishing for Lagrangian-Fibered Hyperkähler Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a projective hyperkähler variety of dimension $n$ with a Lagrangian fibration induced by $L$, and let $c_I(X)$ be a weighted polynomial of degree $2i$ in the even Chern classes. Prove $c_I(X)L^{n+1-k}=0$ in $\operatorname{CH}(X)_{\mathbb Q}$ whenever $i\geq k$.

来源字段：https://doi.org/10.4171/owr/2023/27

05554 | AIM-ALGEBRAIC_GEOMETRY-0116

### Church surfaces and an all-dimensional slope-Kunneth propagation theorem

P / partially_solved | AIM 研讨会问题汇编

Can we find a simply-connected variety with supersingular cohomology which is not rationally-connected?

来源字段：http://aimpl.org/ratsubvarpos/9/

05555 | OWR-11695864-001

### Classification and Grid Structure of Geproci Sets

P / partially_solved | Oberwolfach 报告问题汇编

别名：Nonexistence of Geproci Sets in Linear General Position [OWR-11695864-006]

Classify nontrivial geproci point sets in projective three-space. Must every geproci set of at least five points contain three collinear points, and can a general finite subset of a smooth plane curve ever be geproci?

来源字段：https://doi.org/10.4171/owr/2022/54

05556 | OWR-11139-001

### Classification of Class-VII Complex Surfaces

O / open | Oberwolfach 报告问题汇编

Classify compact complex surfaces of class $\mathrm{VII}_0^+$, in particular determining the structure of this still-mysterious class.

来源字段：https://doi.org/10.4171/owr/2011/47

05557 | OWR-719-005

### Classification of General-Type Surfaces with Small Invariants

O / open | Oberwolfach 报告问题汇编

Classify algebraic surfaces of general type with small invariants, including surfaces with geometric genus zero or one.

来源字段：https://doi.org/10.4171/owr/2004/09

05558 | OWR-14298593-004

### Classification of Noncommutative Hypersurfaces

O / open | Oberwolfach 报告问题汇编

Classify noncommutative hypersurfaces $\operatorname{qgr}(A/(f))\subseteq\operatorname{qgr}A$, where $A$ is Artin-Schelter regular and $f\in Z(A)$ is homogeneous, and describe their singularities and derived categories beyond the known quadratic and finite-over-the-center cases.

来源字段：https://doi.org/10.4171/owr/2024/44

05559 | OWR-4990374-001

### Classification of Projective Line Arrangements with Only Triple Points

O / open | Oberwolfach 报告问题汇编

Classify all line arrangements in the projective plane whose singular points are exclusively triple points.

来源字段：https://doi.org/10.4171/owr/2021/5

05560 | OWR-12018-010

### Classification of Simple Normal Surface Singularities

O / open | Oberwolfach 报告问题汇编

What are the simple normal surface singularities? The conjecture is that they are exactly the rational singularities whose resolution graphs arise from graphs of rational double points and rational triple points by making some vertex weights more negative; in particular, rigid normal surface singularities do not exist.

来源字段：https://doi.org/10.4171/owr/2012/46

05561 | OWR-14222-006

### Classification of Unexpected Curves Through Point Sets

P / partially_solved | Oberwolfach 报告问题汇编

Which curves containing a finite point set $Z$ and vanishing to order $m_Z$ at a general point are unexpected curves?

来源字段：https://doi.org/10.4171/owr/2016/14

05562 | OWR-2492-005

### Classification of Wild Quotient Surface Singularities

P / partially_solved | Oberwolfach 报告问题汇编

Let $Y=X/G$ be the quotient of a regular two-dimensional scheme by a finite group action in characteristic $p>0$, and consider its isolated wild singularities, where $p$ divides $|G|$. Classify these singularities up to analytic isomorphism, equivalently by the resolution graphs and combinatorial data of their minimal normal-crossing resolutions.

来源字段：https://doi.org/10.4171/owr/2008/54

05563 | OWR-12334-007

### Classification Problems for Del Pezzo Fibrations

O / open | Oberwolfach 报告问题汇编

For del Pezzo fibrations, are there fibrations with square-free discriminant of height two and with height six and maximal monodromy; are those of a given height irreducible; how can one characterize their Prym varieties; and, in the height-twelve example $X\to\mathbb P^1$, are there sections other than the canonical section?

来源字段：https://doi.org/10.4171/owr/2013/06

05564 | OWR-13946-016

### Closedness of the K3 Locus and Fake K3 Surfaces

O / open | Oberwolfach 报告问题汇编

For $g\geq12$, is the locus $(K3)_g\subset M_g$ of curves lying on K3 surfaces closed? Can Lazarsfeld's proof of the Gieseker-Petri theorem be generalized to curves on fake K3 surfaces?

来源字段：https://doi.org/10.4171/owr/2015/51

05565 | AIM-ARITHMETIC_GEOMETRY-0032

### CM factors force infinitely many nonordinary primes

P / partially_solved | AIM 研讨会问题汇编

Let $K$ be a number field, and let $A/K$ be an abelian variety. Do there exist infinitely many places $v$ of $K$ such that the reduction $A/\mathbb F_v$ is not ordinary? More strongly, do there exist infinitely many places $v$ of $K$ such that the reduction $A/\mathbb F_v$ is supersingular?

来源字段：http://aimpl.org/cohomabelian/1/

05566 | OWR-4990375-014

### Codimension-Two Effective Cones of Principally Polarized Abelian Varieties

O / open | Oberwolfach 报告问题汇编

For $g\geq3$, is the cone $\operatorname{Eff}_2(\mathcal A_g)$ generated by the five surface classes $S_A$, $S_F$, $S_D$, $S_1$, and $S_2$?

来源字段：https://doi.org/10.4171/owr/2021/6

05567 | OWR-15987-009

### Coefficient Formulas for Polynomial Roots

P / partially_solved | Oberwolfach 报告问题汇编

Find and understand formulas for the roots of $P(z)=z^n+a_1z^{n-1}+\cdots+a_n$ in terms of its coefficients.

来源字段：https://doi.org/10.4171/owr/2018/2

05568 | OWR-16408-016

### Cohomological Dimension of Automorphism Groups of Singular K3 Surfaces

O / open | Oberwolfach 报告问题汇编

Does the automorphism group of every singular K3 surface of discriminant 7 or 8 have cohomological dimension 2?

来源字段：https://doi.org/10.4171/owr/2018/33

05569 | OWR-14298581-003

### Cohomological Dimension of Meromorphic Function Fields

P / partially_solved | Oberwolfach 报告问题汇编

Let $S$ be a connected normal Stein space of complex dimension $n$. Is the cohomological dimension of its field of meromorphic functions equal to $n$, that is, does $\operatorname{cd}(\mathcal M(S))=n$ hold? The question remains open in dimensions $n\geq 3$.

来源字段：https://doi.org/10.4171/owr/2024/32

05570 | OWR-13750339-007

### Cohomological Dimension of Stein Meromorphic Function Fields

P / partially_solved | Oberwolfach 报告问题汇编

For a Stein compact set $S$ of complex dimension $n$, prove that the meromorphic function field $\mathcal M(S)$ has cohomological dimension $n$.

来源字段：https://doi.org/10.4171/owr/2023/27

05571 | OWR-16158-004

### Cohomological Obstructions to Rational Points and Weak Approximation

O / open | Oberwolfach 报告问题汇编

For which classes of varieties are the known cohomological obstructions the only obstructions to a $k$-point or to weak approximation? How do these methods relate to patching over function fields, and what is the defect of weak approximation for arbitrary noncommutative algebraic groups?

来源字段：https://doi.org/10.4171/owr/2018/17

05572 | OWR-12334-013

### Cohomological Stability for Moduli of Curves with Group Actions

O / open | Oberwolfach 报告问题汇编

Fix a finite group $G$ and a topological action type $\rho$ on curves, and let $g'$ denote the genus of the quotient curve. Do the cohomology groups of the corresponding moduli spaces $\mathcal M_{g,G,\rho}$ stabilize as $g'\to\infty$?

来源字段：https://doi.org/10.4171/owr/2013/06

05573 | OWR-11580-002

### Cohomology and Classification of Group Schemes over DVRs

O / open | Oberwolfach 报告问题汇编

Describe the coherent cohomology ring of group schemes over a discrete valuation ring, and classify anti-affine group schemes over a discrete valuation ring.

来源字段：https://doi.org/10.4171/owr/2012/13

05574 | OWR-14222-004

### Cohomology of General Sheaves on Polarized Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

For a general sheaf E in the moduli space $M_H(v)$ on a smooth polarized surface, what is the cohomology of E? In particular, when does E have nonzero cohomology in at most one degree, and if $\chi(v)=0$, does E have no nonzero cohomology?

来源字段：https://doi.org/10.4171/owr/2016/14

05575 | OWR-16770-001

### Cohomology of the Drinfeld Tower

O / open | Oberwolfach 报告问题汇编

For the Drinfeld tower $\mathcal M_\infty$ over a $p$-adic field, describe its $p$-adic étale cohomology and coherent cohomology, beyond the known compactly supported $\ell$-adic étale cohomology.

来源字段：https://doi.org/10.4171/owr/2019/8

05576 | OWR-4527-008

### Collisions of Zero-Dimensional Schemes on Surfaces

O / open | Oberwolfach 报告问题汇编

For collisions of zero-dimensional schemes on a smooth surface $S$, construct useful collisions sufficient to estimate dimensions of linear series, and classify collisions involving a small number of points.

来源字段：https://doi.org/10.4171/owr/2010/45

05577 | OWR-17139-001

### Combinatorial Deformations of Toric Fano Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Let $P$ be a reflexive polytope and let $X_P$ be the toric Fano variety associated with its face fan. Characterize the combinatorial data on $P$ that produce deformations or smoothings of $X_P$, and determine the general fibers of the resulting families.

来源字段：https://doi.org/10.4171/owr/2019/43

05578 | OWR-4990374-016

### Combinatorial Invariance of Hyperplane Arrangement Freeness

O / open | Oberwolfach 报告问题汇编

Over a fixed field $K$, does freeness of a central essential hyperplane arrangement $\mathcal A$ depend only on its intersection lattice $L(\mathcal A)$?

来源字段：https://doi.org/10.4171/owr/2021/5

05579 | OWR-14222-002

### Combinatorial Invariance of Strong Lefschetz Properties

O / open | Oberwolfach 报告问题汇编

Suppose $I=(\ell_1^b,\ldots,\ell_{2b+1}^b)$ is minimally generated and has the Strong Lefschetz Property at range 2 in degree $b-2$. If $J=(h_1^b,\ldots,h_{2b+1}^b)$ is minimally generated and the arrangements defined by the $\ell_i$ and $h_i$ have the same combinatorics, must $J$ also have the Strong Lefschetz Property at range 2 in degree $b-2$?

来源字段：https://doi.org/10.4171/owr/2016/14

05580 | OWR-14299288-015

### Combinatorial Proof of Hurwitz and Discriminant Weight-Polytope Equality

O / open | Oberwolfach 报告问题汇编

Give a purely combinatorial proof that the weight polytope $W(\Delta_{X\times\mathbb{P}^{n-1}})$ of the discriminant coincides with the weight polytope $W(\operatorname{Hu}_A)$ of the Hurwitz form.

来源字段：https://doi.org/10.4171/owr/2025/19

05581 | OWR-14299288-009

### Combinatorics of Bondal-Thomsen Polytopes

P / partially_solved | Oberwolfach 报告问题汇编

Determine the combinatorial properties and significance of the Bondal-Thomsen polytopes, including their face structure, lattice points, adjacency, and relation to toric-arrangement strata and derived categories.

来源字段：https://doi.org/10.4171/owr/2025/19

05582 | OWR-1188-006

### Common Branched Covers from Equal Hasse-Weil Zeta Functions

P / partially_solved | Oberwolfach 报告问题汇编

If smooth projective varieties $X,X'$ over $\mathbb{Q}$ have equal Hasse-Weil zeta functions $\zeta_X^{\mathrm{HW}}(s)=\zeta_{X'}^{\mathrm{HW}}(s)$, must they have a common finite branched cover?

来源字段：https://doi.org/10.4171/owr/2006/06

05583 | AIM-ALGEBRAIC_GEOMETRY-0168

### Compact-Lie norms and a fixed-point obstruction

P / partially_solved | AIM 研讨会问题汇编

In genuine \(G\)-spectra we have multiplicative norm and transfer maps when \(G\) has finitely indexed subgroups. Furthermore, (via Ando-Morava) there are objects that look like what may arise from a multiplicative norm for compact Lie groups.

Are there analogues of multiplicative norms or multiplicative transfers for compact Lie groups?

来源字段：http://aimpl.org/equideralggeom/4/

05584 | OWR-13946-018

### Compactifiable Lagrangian Fibrations from Abelian Fibrations

O / open | Oberwolfach 报告问题汇编

Given an abelian fibration, when does its total space admit a symplectic structure making it a Lagrangian fibration compactifiable to a possibly singular irreducible symplectic variety? Separately, let a family of curves over a polydisk in $\mathbb C^k$ be a multiversal unfolding of the singularities of its central fiber, all of type $A_{\mu_i}$ with $1\leq\mu_i\leq5$ or $D_{\mu_i}$ with $4\leq\mu_i\leq5$, and suppose $\sum_i\mu_i\leq5$. Is its relative compactified Jacobian toroidal?

来源字段：https://doi.org/10.4171/owr/2015/51

05585 | OWR-4535-016

### Compactifications and Stein Structures of Thick Teichmüller Parts

O / open | Oberwolfach 报告问题汇编

Which classes of the Deligne-Mumford compactification come from the Penner compactification; is the Sullivan space homotopic to a suitably compactified moduli space; is the thick part $T^\varepsilon(S_g)$ a Stein manifold with a plurisubharmonic exhaustion; and what is its Dolbeault cohomology?

来源字段：https://doi.org/10.4171/owr/2010/53

05586 | OWR-1060-007

### Compactifications of Nonabelian Algebraic Groups

P / partially_solved | Oberwolfach 报告问题汇编

Construct compactifications of nonabelian algebraic groups over $\mathbb{C}$.

来源字段：https://doi.org/10.4171/owr/2005/39

05587 | OWR-16417-003

### Compactified Universal Jacobians over Stable Curves

P / partially_solved | Oberwolfach 报告问题汇编

How can a degree-$d$ universal Jacobian over the moduli space of smooth marked curves be extended over the moduli space of stable marked curves by adding well-behaved compactified Jacobian fibers, and which degenerate line bundles or multidegrees should be included?

来源字段：https://doi.org/10.4171/owr/2018/42

05588 | OWR-3481-002

### Comparison Maps in Motivic Cohomology of Central Simple Algebras

O / open | Oberwolfach 报告问题汇编

For a central simple algebra $A$, compare the motivic-cohomology maps arising from the étale Bloch-Lichtenbaum spectral sequence, generalized Severi-Brauer varieties, and $\operatorname{SL}_1(A)$. When do these three constructions yield the same map?

来源字段：https://doi.org/10.4171/owr/2009/31

05589 | AIM-ALGEBRAIC_GEOMETRY-0147

### Comparison-gated cellular motivic Eilenberg-Moore spectral sequences

P / partially_solved | AIM 研讨会问题汇编

**Can we construct an Eilenberg-Moore spectral sequence in motivic homotopy theory?**

来源字段：http://aimpl.org/equideralggeom/1/

05590 | OWR-1542-014

### Compatibility of Nef Reduction and Iitaka Fibrations

O / open | Oberwolfach 报告问题汇编

For a smooth projective algebraic $K(\pi,1)$ space $X$ in characteristic zero, do the nef reduction of $\omega_X$, the Iitaka fibration, assuming the standard Kodaira-dimension-zero conjecture, and the pluricanonical maps, assuming abundance, agree after the finite étale cover that makes the almost-abelian fibration an abelian scheme, with common base $Y'$ of general type?

来源字段：https://doi.org/10.4171/owr/2007/26

05591 | AIM-ALGEBRAIC_GEOMETRY-0185

### Compatibility via the epipolar fiber product and a gcd component law

P / partially_solved | AIM 研讨会问题汇编

Given an algebraic curve in two views, when are they compatible? This is interesting when the cameras are fixed beforehand.

来源字段：http://aimpl.org/algvision/1/

05592 | OWR-14298593-003

### Compatible Feigin-Odesskii Poisson Brackets

O / open | Oberwolfach 报告问题汇编

For coprime integers $r,d$ with $r>0$, determine the maximal dimension of a vector space of compatible Poisson brackets on $\mathbb P^{d-1}$ that contains a fixed Feigin-Odesskii bracket of type $(d,r)$.

来源字段：https://doi.org/10.4171/owr/2024/44

05593 | AIM-ALGEBRAIC_GEOMETRY-0231

### Complementary tropical minors for relative Koszul complexes

P / partially_solved | AIM 研讨会问题汇编

**Problem 8.2.** Use degeneration techniques (tropical independence or limit linear series or both) to understand higher syzygies.

This is known by Green's conjectures for generic curves under the canonical embed-ding, by non-tropical techniques.

来源字段：https://aimath.org/pastworkshops/degenalggeomproblems.pdf

05594 | OWR-16417-002

### Complete Curves and Positivity in Differential Strata

P / partially_solved | Oberwolfach 报告问题汇编

For strata of holomorphic differentials, can a stratum contain a complete curve; which powers of the tautological class $\eta$ vanish on the stratum; and how can positivity of other divisor classes on compactified strata be determined?

来源字段：https://doi.org/10.4171/owr/2018/42

05595 | AIM-ARITHMETIC_GEOMETRY-0058

### Complete reduced component decomposition for Rao functions (p,1,r)

P / partially_solved | AIM 研讨会问题汇编

Problem 11. Describe the irreducible component of Eρ, the moduli of finite length graded modules of function ρ.

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05596 | AIM-ALGEBRAIC_GEOMETRY-0203

### Complete regular-jet invariants for an irregular Euclidean tile groupoid

P / partially_solved | AIM 研讨会问题汇编

Find local invariants for a natural (Lie) groupoid action. (e.g. bathroom tiles in an irregular bathroom). How is this related to quasi-invariants?

来源字段：http://aimpl.org/algvision/3/

05597 | AIM-ALGEBRAIC_GEOMETRY-0117

### Complete subvarieties at the Diaz bound and a genus-four boundary obstruction

P / partially_solved | AIM 研讨会问题汇编

Find a complete subvariety of \(\mathcal M_g\) of \(\operatorname{codim} 2g-1\).

来源字段：http://aimpl.org/ratsubvarpos/9/

05598 | OWR-15177-013

### Complete Surfaces inside the Genus-Four Moduli Space

O / open | Oberwolfach 报告问题汇编

Does the moduli space $\mathcal M_4$ of smooth genus-$4$ curves contain a complete complex subvariety of complex dimension $2$?

来源字段：https://doi.org/10.4171/owr/2016/56

05599 | OWR-11695864-005

### Complete-Intersection Realizations of Matroids

P / partially_solved | Oberwolfach 报告问题汇编

Which matroids admit a realization by a point configuration that is a complete intersection?

来源字段：https://doi.org/10.4171/owr/2022/54

05600 | OWR-7155449-015

### Completed $\Theta$-Twisted Volumes of Bordered-Curve Moduli

O / open | Oberwolfach 报告问题汇编

Prove the identity $\widehat V_{g,n}(L_1,\ldots,L_n)=V^\Theta_{g,n}(L_1,\ldots,L_n)$ between the completed super Weil-Petersson volume polynomial and the $\Theta$-twisted volume polynomial of bordered-curve moduli.

来源字段：https://doi.org/10.4171/owr/2021/28

05601 | OWR-7155449-014

### Completed Masur-Veech Volumes of Odd Strata

P / partially_solved | Oberwolfach 报告问题汇编

Prove the conjectural identities expressing completed Masur-Veech volumes of odd quadratic-differential strata as their actual volumes plus weighted contributions from adjacent strata.

来源字段：https://doi.org/10.4171/owr/2021/28

05602 | OWR-15170-021

### Completeness Tests for Integral Subvarieties

P / partially_solved | Oberwolfach 报告问题汇编

Given a smooth projective variety $X$ over $\mathbb C$ of dimension greater than two and $\omega\in H^0(X,L\otimes S^r\Omega^1_{X/\mathbb C})$, how can one check that a listed collection of $\omega$-integral curves or hypersurfaces contains all such curves or hypersurfaces of $X$?

来源字段：https://doi.org/10.4171/owr/2016/49

05603 | AIM-ALGEBRAIC_GEOMETRY-0196

### Complex completeness and real descent for differential curve signatures

P / partially_solved | AIM 研讨会问题汇编

How complete are these invariants? If two curves have the same signature what can we say about them?

来源字段：http://aimpl.org/algvision/3/

05604 | OWR-4990374-010

### Complex Line Arrangements without Double Points

O / open | Oberwolfach 报告问题汇编

Classify complex line arrangements with no double points, that is, with $t_2=0$, where $t_2$ is the number of points of multiplicity two.

来源字段：https://doi.org/10.4171/owr/2021/5

05605 | AIM-ALGEBRAIC_GEOMETRY-0170

### Component-group blindness and twisted ambidexterity for compact Lie groups

P / partially_solved | AIM 研讨会问题汇编

**Long Term**

How should we think about equivariant stable homotopy theory in the context of infinite compact Lie groups?

来源字段：http://aimpl.org/equideralggeom/4/

05606 | OWR-15588-005

### Components of Rational-Curve Mapping Spaces

O / open | Oberwolfach 报告问题汇编

For a smooth projective variety $X$, how many irreducible components does $\operatorname{Mor}(\mathbb P^1,X)$ have?

来源字段：https://doi.org/10.4171/owr/2017/45

05607 | OWR-10252925-007

### Computation and Comparison of Toric Degenerations

P / partially_solved | Oberwolfach 报告问题汇编

Develop general methods to compute toric degenerations of a projective variety $X$ and characterize the relationships among distinct toric degenerations of $X$.

来源字段：https://doi.org/10.4171/owr/2022/17

05608 | OWR-14222-005

### Conditions Imposed by Fat Points on Curve Systems

O / open | Oberwolfach 报告问题汇编

Determine the number of independent conditions imposed by a general collection of fat points on a linear system of curves of fixed degree containing a prescribed finite point set $Z$, possibly with prescribed multiplicities at $Z$.

来源字段：https://doi.org/10.4171/owr/2016/14

05609 | OWR-4990374-003

### Congruence Normality in Simplicial Arrangements

O / open | Oberwolfach 报告问题汇编

For simplicial arrangements, classify the congruence-normal ones; determine the relation between congruence normality and freeness; find useful arrangement-theoretic properties of the shard digraph; find the smallest number of hyperplanes in a tight arrangement that is not congruence normal; and determine the behavior of congruence normality under restrictions and other arrangement operations.

来源字段：https://doi.org/10.4171/owr/2021/5

05610 | OWR-11139-012

### Conical Kähler-Einstein Metrics and Ricci Lower Bounds

S / solved | Oberwolfach 报告问题汇编

Let $X$ be a Fano manifold, let $D$ be a smooth anticanonical divisor, and let $R(X)$ be the supremum of $t$ for which some Kähler metric $\omega\in c_1(X)$ satisfies $\operatorname{Ric}(\omega)>t\omega$. Does

$$\operatorname{Ric}(\omega_\beta)=\beta\omega_\beta+(1-\beta)\{D\}$$

admit a cone-singularity solution for $\beta\in(0,R(X))$, and no solution for $\beta\in(R(X),1)$ when $R(X)<1$?

来源字段：https://doi.org/10.4171/owr/2011/47

05611 | AIM-ALGEBRAIC_GEOMETRY-0152

### Connected comparison fibers with isotropic antichains

P / partially_solved | AIM 研讨会问题汇编

What more can we say about the tensor triangular geometry of the motivic stable homotopy category? (See (\text{arxiv:1608.02876)}\) for a list of questions.)

来源字段：http://aimpl.org/equideralggeom/1/

05612 | OWR-15955-006

### Connected Monodromy Groups of Kodaira Fibrations

P / partially_solved | Oberwolfach 报告问题汇编

Which groups occur as connected monodromy groups of Kodaira fibrations?

来源字段：https://doi.org/10.4171/owr/2017/50

05613 | AIM-ARITHMETIC_GEOMETRY-0072

### Connected toric and square-zero classes in three variables

P / partially_solved | AIM 研讨会问题汇编

**Problem 25.** For \(R=k[x_1,x_2,x_3]\), is every multigraded Hilbert scheme connected?

**Remark.** For \(R=k[x_1,\ldots,x_{26}]\), there are non-connected examples, cf. Francisco Santos, *Non-connected toric Hilbert schemes*. Find examples with fewer variables.

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05614 | AIM-ALGEBRAIC_GEOMETRY-0184

### Constraint budgets for partial calibration and the five-center obstruction

P / partially_solved | AIM 研讨会问题汇编

In between the calibrated and uncalibrated cases there are partially calibrated cameras, say with both focal lengths known or with the center position known. What can we say about these situations?

来源字段：http://aimpl.org/algvision/1/

05615 | OWR-14754-011

### Contact Loci and Nash Sets of Prime Divisors

O / open | Oberwolfach 报告问题汇编

Let $p_{i_0}$ be a free infinitely near point used to obtain the minimal model of a prime divisor $E$. Is $\bigcup_{E'\equiv_{i_0}E}\mathcal N_{E'}=\operatorname{Cont}_q(E_{p_{i_0}})$ for some $q$?

来源字段：https://doi.org/10.4171/owr/2016/46

05616 | AIM-ALGEBRAIC_GEOMETRY-0166

### Continuous Galois spectra and an infinite-rank finite-field obstruction

P / partially_solved | AIM 研讨会问题汇编

For the profinite absolute Galois group \(\Gamma=G_K=\operatorname{Gal}(K_s/K)\), construct a genuine continuous stable category \(\operatorname{Sp}_\Gamma\) and a motivic comparison which, for every open subgroup \(U\leq \Gamma\), sends \(\Sigma^\infty \Gamma/U_+\) to \(\Sigma^\infty\operatorname{Spec}(K_s^U)_+\). Determine in what precise senses this comparison retains restrictions, transfers, norms, fixed-point information, or full faithfulness.

来源字段：http://aimpl.org/equideralggeom/4/

05617 | OWR-1452-023

### Contractibility of Quotient Curves from Additive-Group Actions

O / open | Oberwolfach 报告问题汇编

Let $X$ be a smooth affine contractible complex threefold with a nontrivial $\mathbb{G}_a$-action, let $\pi:X\to S=X//\mathbb{G}_a$ be the quotient, and let $\Gamma\subset S$ be a minimal closed curve such that $\pi^{-1}(S\setminus\Gamma)\simeq(S\setminus\Gamma)\times\mathbb{C}$ over $S\setminus\Gamma$. If $\Gamma$ is connected, is $\Gamma$ contractible?

来源字段：https://doi.org/10.4171/owr/2007/01

05618 | OWR-15177-020

### Convex Exhaustions of Curve Moduli Spaces

O / open | Oberwolfach 报告问题汇编

Does the moduli space $\mathcal M_g$ admit a proper, bounded-below, orbifold-smooth, $(g-2)$-convex function $f:\mathcal M_g\to\mathbb R$?

来源字段：https://doi.org/10.4171/owr/2016/56

05619 | OWR-15586-002

### Convex Geometry and Limits of Asymptotic Higher Cohomology

O / open | Oberwolfach 报告问题汇编

Is there a convex-geometric analogue of the Newton-Okounkov interpretation of volume for the asymptotic higher-cohomology functions $$\widehat h^i(D)=(\dim X)!\limsup_{m\to\infty}\frac{\dim H^i(X,\mathcal O_X(mD))}{m^{\dim X}}?$$ In particular, can $\widehat h^i(D)$ be realized as a Euclidean volume, and can the limsup in its definition always be replaced by a limit?

来源字段：https://doi.org/10.4171/owr/2017/43

05620 | OWR-12697710-006

### Convexity of Odd-Power Sum-of-Squares Cones

O / open | Oberwolfach 报告问题汇编

Is $\Sigma_{n,m}(2k+1)$ a closed convex cone?

来源字段：https://doi.org/10.4171/owr/2023/14

05621 | AIM-ALGEBRAIC_GEOMETRY-0241

### Coordinate-skeleton curves, transverse coinvariant algebras, and the cohomological-dimension gap

P / partially_solved | AIM 研讨会问题汇编

Let \(k\) be a field, \(R=k[x_1,\ldots,x_n]\), and let \(I\subset R\) be a radical ideal whose minimal primes all have height \(n-1\). Must there exist \(f_1,\ldots,f_{n-1}\in R\) such that \[ \sqrt{(f_1,\ldots,f_{n-1})}=I? \]

来源字段：https://aimath.org/pastworkshops/projectiveA1problems.pdf

05622 | AIM-ALGEBRAIC_GEOMETRY-0128

### Coprime alteration degrees reduce to one prime-to-p alteration

P / partially_solved | AIM 研讨会问题汇编

Can we get alterations of coprime degrees?

来源字段：http://aimpl.org/ratsubvarpos/9/

05623 | OWR-13946-012

### Coranks of Modified Wahl Maps for Nodal Curves

P / partially_solved | Oberwolfach 报告问题汇编

For a general nodal curve $C$ in a component of $\mathcal V_{p,n,\delta}$, what are the coranks of the Kemeny-Wahl map $kw_C$ and the Halic-Wahl map $hw_C$?

来源字段：https://doi.org/10.4171/owr/2015/51

05624 | AIM-ALGEBRAIC_GEOMETRY-0285

### Corrected normalization and an exact non-Q-Gorenstein determinantal limit

P / partially_solved | AIM 研讨会问题汇编

For \((R,\mathfrak m)\) a local, normal domain of characteristic \(p>0\) and dimension \(d\), let \(\mu_e\) denote the minimal number of generators of \(\operatorname{Hom}_R(R^{1/p^e},R)\) as an \(R\)-module. Does \[ \lim_{e\to\infty}\frac{\mu_e}{p^{ed}} \] exist? If so, what is it?

来源字段：http://aimpl.org/testandmultiplierideals/1/

05625 | AIM-ALGEBRAIC_GEOMETRY-0222

### Counterexamples to naive balance and a fixed-multidegree interpolation family

P / partially_solved | AIM 研讨会问题汇编

Let \(X=Z_1\cup Z_2\) be obtained by gluing two rational curves at \(g+1\) general pairs of points. Does \(X\) have a \(g^r_{d_1,d_2}\) if and only if \(\rho(g,r,d_1+d_2)\geq 0\), subject to a condition that \((d_1,d_2)\) be “relatively balanced”?

来源字段：https://aimath.org/pastworkshops/degenalggeomproblems.pdf

05626 | AIM-ALGEBRAIC_GEOMETRY-0346

### Covering genus and a contact-adjusted level-signature obstruction for moduli of curves

P / partially_solved | AIM 研讨会问题汇编

**Problem 5.4. (Farkas)** For \(g\) large, what is the minimal genus of a curve passing through a general point of \(M_g\)? This can be asked for the moduli space or the moduli stack. For low \(g\), the answer for the moduli stack is zero.

One could ask this question for moduli spaces of canonically polarized manifolds, although I think the point of making \(g\) large is to ensure that \(M_g\) is not uniruled. There is nothing even approximating such a condition to ensure that \(M_H\) is not unirational.

来源字段：https://aimath.org/WWN/birational/birational.pdf

05627 | AIM-ALGEBRAIC_GEOMETRY-0012

### Cox-chart Morse theory for cellular virtual resolutions

P / partially_solved | AIM 研讨会问题汇编

**Topology/Morse theory of resolutions.** Is there an analogue of “Morse theory for cellular resolutions” for virtual resolutions?

来源字段：http://aimpl.org/syzygyms/4/

05628 | OWR-14299574-005

### Crepant Resolutions and Smoothings of Toroidal Fano Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Do log schemes constructed from compatible facetwise zero-mutable Laurent polynomials admit both a log-crepant resolution and a smoothing?

来源字段：https://doi.org/10.4171/owr/2025/45

05629 | AIM-ALGEBRAIC_GEOMETRY-0092

### Crossed-product HHH enhancements and the component-collapse obstruction

P / partially_solved | AIM 研讨会问题汇编

Is there a braid invariant extension of HHH which replaces bigraded vector spaces with bigraded \(S_n\)-modules suitably compatible with the action of \(R=\mathbb{Q}[x_1,\dots,x_n]\) and the action of \(\mathbb{Q}[y_1,\dots,y_n]\) coming from \(y\)-ification?

来源字段：http://aimpl.org/catheckehilbert/5/

05630 | AIM-ALGEBRAIC_GEOMETRY-0229

### Cubic maximal rank after the classical theorem

S / solved | AIM 研讨会问题汇编

Problem 8

(Suggested by David Jensen) The Maximal Rank Conjecture predicts the Hilbert fuction of a general curve (for fixed g, r, d such that ρ(g, r, d) ≥ 0 and r ≥ 3). The Hilbert function measures the rank of the map Sym m H0(X, L) → H0(X, L⊗m). The Maximal Rank Conjecture predicts that this has maximal rank for general X and L.Concretely, this means that it is either injective or surjective. We proposed a combinatorial approach via the notion of tropical independence. How-ever, we can only solve the combinatorial problem for h(2) (the case of quadrics). The next step would be h(3), i.e. cubics.

**Recovered Problem 8.** Determine \(h_V(3)\) for a general Brill-Noether curve/linear series with \(r\geq3\), and, in the direction proposed at the workshop, prove the predicted maximal-rank statement by a tropical-independence certificate.

来源字段：https://aimath.org/pastworkshops/degenalggeomproblems.pdf

05631 | AIM-ALGEBRAIC_GEOMETRY-0208

### Cuspidal silhouettes via Segre adjoints and finite first-jet tests

P / partially_solved | AIM 研讨会问题汇编

Which cuspidal plane curves occur as the apparent contours, equivalently the branch curves, of generic projections of algebraic surfaces?

来源字段：http://aimpl.org/algvision/4/

05632 | OWR-14298588-014

### Cyclic Lifting Property for Moduli Stacks

O / open | Oberwolfach 报告问题汇编

Characterize the algebraic stacks that satisfy the cyclic lifting property. Extend the associated $p$-adic volume formula to moduli spaces with non-quotient singularities, including moduli of Higgs bundles, coherent sheaves, and quiver representations.

来源字段：https://doi.org/10.4171/owr/2024/33

05633 | AIM-ALGEBRAIC_GEOMETRY-0083

### Cyclotomic monodromy reconstruction and a bare-Milnor-fibre obstruction

P / partially_solved | AIM 研讨会问题汇编

To an algebraic link one can associate a Springer fibre and a Milnor fibre. Can one compute HHH from the Milnor fibre?

来源字段：http://aimpl.org/catheckehilbert/3/

05634 | OWR-12592-004

### D-Brane Equivalence Across Calabi-Yau Phases

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Linear $\sigma$-Model Phase Equivalence [OWR-12592-005]

For the compact Calabi-Yau threefolds and their Pfaffian, determinantal, or hybrid Landau-Ginzburg phases constructed from the non-abelian linear sigma models, are the corresponding D-brane categories equivalent?

来源字段：https://doi.org/10.4171/owr/2013/44

05635 | OWR-12697710-009

### Dagger Closure of Trace-Positive Noncommutative Polynomials

O / open | Oberwolfach 报告问题汇编

For the cone $K$ of trace-positive noncommutative polynomials, is $K=K^\dagger$, where $K^\dagger=\{f:\exists g\text{ such that }f+\varepsilon g\in K\text{ for every }\varepsilon>0\}$?

来源字段：https://doi.org/10.4171/owr/2023/14

05636 | OWR-16408-004

### Debarre’s Minimal Cohomology-Class Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $(A,\Theta)$ be an indecomposable principally polarized abelian variety of dimension $g$, and let $X\subset A$ be a reduced pure $d$-dimensional subvariety with $1\leq d\leq g-2$ and minimal cohomology class

$$[X]=\frac{\theta^{g-d}}{(g-d)!}.$$

Must either $(A,\Theta)$ be the Jacobian of a curve $C$ with $X$ a translate of $W_d(C)$, or $g=5$, $d=2$, and $X$ be a translate of the Fano surface in the intermediate Jacobian of a smooth cubic threefold?

来源字段：https://doi.org/10.4171/owr/2018/33

05637 | OWR-4527-002

### Decidability of Semieffective Divisors on Blown-Up Projective Space

O / open | Oberwolfach 报告问题汇编

Given $D=dH-\sum_i m_iE_i$ on the blowup $X\to\mathbb{P}^{N}$ of finitely many points, is there a procedure to decide whether $D$ is semi-effective, i.e. whether $h^{0}(X,kD)>0$ for some $k>0$? Equivalently, can one decide whether a degree-$td$ form vanishes at each point $p_i$ to order at least $tm_i$ for some $t$?

来源字段：https://doi.org/10.4171/owr/2010/45

05638 | OWR-15428-010

### Defining Equations for Group-Based Phylogenetic Network Models

P / partially_solved | Oberwolfach 报告问题汇编

For each of the CFN, Jukes-Cantor, K2P, and K3P models and every phylogenetic network $N$, determine defining polynomials for the variety $V_N$.

来源字段：https://doi.org/10.4171/owr/2017/20

05639 | OWR-14298581-011

### Deformation Equivalence of Categorical Hyperkähler Constructions

O / open | Oberwolfach 报告问题汇编

Let $g\geq 0$ and let $w\in\mathbb Z$ be coprime to $v$. Prove that the reduced category $T_S(v)_w^{\mathrm{red}}$ is deformation equivalent to $D^b(\operatorname{Hilb}(S,g'))$, where $g'=(\langle v,v\rangle+2)/2$.

来源字段：https://doi.org/10.4171/owr/2024/32

05640 | OWR-14299288-014

### Deformation Spaces of Smooth Complete Toric Varieties

O / open | Oberwolfach 报告问题汇编

Is the deformation space of a smooth complete toric variety determined by its tangent cone? Is the number of its irreducible components related to the number of distinct simplicial complexes representing first-order deformations?

来源字段：https://doi.org/10.4171/owr/2025/19

05641 | OWR-1730-013

### Deformations of Complex Nilmanifolds and Torus Bundles

P / partially_solved | Oberwolfach 报告问题汇编

Let $M=\Gamma\backslash G$ be a nilmanifold, meaning a compact quotient of a nilpotent real Lie group, with a left-invariant complex structure. Determine its small deformations; determine when it admits a geometric description as an iterated principal holomorphic torus bundle; and determine all large deformations of such bundles.

来源字段：https://doi.org/10.4171/owr/2007/47

05642 | OWR-17139-002

### Deformations of Non-Gorenstein Toric Fano Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Develop a combinatorial and deformation-theoretic description of $\mathbb Q$-Gorenstein deformations of toric Fano varieties with non-Gorenstein singularities.

来源字段：https://doi.org/10.4171/owr/2019/43

05643 | OWR-14298376-007

### Degeneration and Moduli of Logarithmic Linear Series

P / partially_solved | Oberwolfach 报告问题汇编

When a smooth curve $C$ degenerates to a nodal curve $C_0$, how do the line bundle $L$ and subspace $V$ of a linear series $(L,V)$ degenerate? Describe the moduli space of linear series on $C_0$. How can one detect whether a linear series on $C_0$ is the limit of linear series on smooth curves?

来源字段：https://doi.org/10.4171/owr/2024/23

05644 | OWR-14298795-003

### Degeneration and Purity of Local-Field Adic Systems

O / open | Oberwolfach 报告问题汇编

Let $X$ be a smooth variety over a $p$-adic local field $K$, and let $\ell\neq p$. For pure $\mathbb Q_\ell$-adic local systems on $X$, do the systems degenerate to limiting mixed structures at ramification points, and is their cohomology pure?

来源字段：https://doi.org/10.4171/owr/2024/45

05645 | OWR-13946-015

### Degenerations and Compactifications of Hyperkähler Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

For hyperkähler manifolds of dimension greater than two, how can their degenerations be classified, what examples are known, what is known about their versal deformation spaces, and when do their moduli spaces admit partial compactifications?

来源字段：https://doi.org/10.4171/owr/2015/51

05646 | OWR-1971-010

### Degree Bounds for Canonical Covers of Minimal-Degree Surfaces

O / open | Oberwolfach 报告问题汇编

Let $\varphi:X\to W$ be the canonical map of a surface of general type with canonical singularities and ample base-point-free $K_X$, where $W$ is a singular surface of minimal degree. Must $\deg\varphi\leq 4$?

来源字段：https://doi.org/10.4171/owr/2008/26

05647 | OWR-4990374-004

### Degree Bounds for Generic Initial Ideals of Arrangements

O / open | Oberwolfach 报告问题汇编

Let $\mathcal A$ be a central arrangement in $K^\ell$, let $S=K[x_1,\ldots,x_\ell]$, and set $$ d_0=\min\{d:x_1^d\in\operatorname{rgin}(J(\mathcal A))\}. $$ If a minimal generator $t$ of $\operatorname{rgin}(J(\mathcal A))$ involves $x_3$, must $\deg(t)\geq d_0$?

来源字段：https://doi.org/10.4171/owr/2021/5

05648 | AIM-ALGEBRAIC_GEOMETRY-0197

### Degree, symmetry quotient, and coordinate dependence of differential signature curves

S / solved | AIM 研讨会问题汇编

What can we say about the degree of a signature curve? How is it related to the automorphism group of the original curve?

来源字段：http://aimpl.org/algvision/3/

05649 | OWR-11783-009

### Degrees and Finiteness of Stringy $E$-Polynomials

O / open | Oberwolfach 报告问题汇编

For a Gorenstein polytope $P$ of index $r$, is the degree of its stringy $E$-function $E_{\mathrm{st}}(P,t)$ equal to $\dim(P)+1-2r$, and are there, up to multiples, only finitely many such stringy $E$-polynomials with fixed value of $\dim(P)+1-2r$?

来源字段：https://doi.org/10.4171/owr/2012/21

05650 | AMR-071-0005

### Deligne's conjecture on Hochschild cohomology about the operadic structure on Hochschild cochain complex

S / solved | AMR 问题清单汇编

Deligne's conjecture on Hochschild cohomology about the operadic structure on Hochschild cochain complex.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

05651 | OWR-14746-012

### Density and Closure of Mixed Hodge Loci

P / partially_solved | Oberwolfach 报告问题汇编

For a variation $V$ of mixed Hodge structures over a smooth quasi-projective variety $S$, what geometric constraints exist on the Zariski closure of its Hodge locus $\operatorname{HL}(S,V)$? Which pairs $(S,V)$ have $\operatorname{HL}(S,V)$ Zariski dense in $S$?

来源字段：https://doi.org/10.4171/owr/2016/38

05652 | OWR-13946-014

### Density of Nodal Curves in Equigeneric Families

P / partially_solved | Oberwolfach 报告问题汇编

Let $V_g^\xi$ parametrize reduced curves of genus $g$ in class $\xi$ on a smooth projective surface. Is the locus of curves having only ordinary nodes dense in $V_g^\xi$? Can this be proved using the criterion that the equisingular ideal $I$ is properly contained in $A$ exactly for singularities worse than nodes?

来源字段：https://doi.org/10.4171/owr/2015/51

05653 | OWR-16165-003

### Derived Decomposition of Fixed-Determinant Bundle Moduli

P / partially_solved | Oberwolfach 报告问题汇编

Let $C$ be a smooth projective curve of genus $g$, let $L$ be a line bundle of degree one, and let $M_C(2,L)$ be the moduli space of rank-two bundles with determinant $L$. Does its bounded derived category admit the semiorthogonal decomposition $$ D^b(M_C(2,L))=\left\langle D^b(k),D^b(k),D^b(C),D^b(C),\ldots,D^b(\operatorname{Sym}^{g-2}C),D^b(\operatorname{Sym}^{g-2}C),D^b(\operatorname{Sym}^{g-1}C)\right\rangle $$ ?

来源字段：https://doi.org/10.4171/owr/2018/24

05654 | AIM-ARITHMETIC_GEOMETRY-0025

### Derived equivalence preserves all splitting fields but not the genus-one curve

S / solved | AIM 研讨会问题汇编

Suppose we have two genus one curves $C$ and $C'$ over a field $F$ which are derived-equivalent. Is $C(F) \neq \emptyset$ equivalent to $C'(F) \neq \emptyset$? Are $C$ and $C'$ isomorphic over $F$?

来源字段：http://aimpl.org/brauermoduli/2/

05655 | AIM-ARITHMETIC_GEOMETRY-0015

### Derived invariance of the stable reduction profile and Kulikov type of K3 surfaces

P / partially_solved | AIM 研讨会问题汇编

Say we have a K3 surface over a number field. Are the places of bad reduction of the surface determined by the derived category? What about over $\mathbb{C}((t))$?

来源字段：http://aimpl.org/brauermoduli/1/

05656 | AIM-ARITHMETIC_GEOMETRY-0006

### Derived-excess reduction for commutativity of unitary RZ Hecke operators

P / partially_solved | AIM 研讨会问题汇编

In the recent work of Li--Rapoport--Zhang on arithmetic fundamental lemma conjecture for spherical Hecke algebras, there is a conjecture on commutativity of certain Hecke operators on Grothendieck groups of coherent sheaves on unitary Rapoport--Zink spaces. How to prove this conjecture?

**Recovered LRZ Conjecture 5.5.1.** For distinct even types \(t,t'\) and every closed \(A\subset X\),

来源字段：http://aimpl.org/intersectshimura/1/

05657 | OWR-16162-006

### Descent of Hereditary Orders with Involution

O / open | Oberwolfach 报告问题汇编

Let $R$ be a discrete valuation ring with fraction field $F$ in which $2$ is invertible, and let $(A,\tau)$ and $(A',\tau')$ be hereditary $R$-orders with involution that become isomorphic over $F$ and over a finite étale $R$-algebra. If the induced involutions on the semisimple quotients by the Jacobson radicals are anisotropic, must the orders with involution already be isomorphic over $R$? More generally, does this conclusion hold when every closed fiber of the automorphism group scheme is almost anisotropic, and can every central simple $F$-algebra with involution be extended from such an order?

来源字段：https://doi.org/10.4171/owr/2018/21

05658 | OWR-14613-008

### Descent of Intermediate Jacobians

O / open | Oberwolfach 报告问题汇编

Let $X$ be a smooth projective variety over a subfield $K\subseteq\mathbb C$, and assume $H^{2n+1}(X(\mathbb C),\mathbb Z)$ has Hodge level $1$. Does the intermediate Jacobian abelian variety $J^{2n+1}(X_\mathbb C)$ admit a model over $K$?

来源字段：https://doi.org/10.4171/owr/2016/31

05659 | AIM-ALGEBRAIC_GEOMETRY-0297

### Determinantal FFRT off the rank-(t-3) stratum

P / partially_solved | AIM 研讨会问题汇编

Do determinantal rings have finite \(F\)-representation type?

The \(2\times2\) minors of a generic matrix has been done by Smith and Van den Bergh; try \(3\times3\) minors and higher.

来源字段：http://aimpl.org/testandmultiplierideals/2/

05660 | OWR-11695864-004

### Determinantal Quartics as Weddle Surfaces

O / open | Oberwolfach 报告问题汇编

Which quartic surfaces arise as the Weddle surface of a three-dimensional linear system of quadrics in $\mathbb P^3$? In particular, is a general determinantal quartic such a Weddle surface?

来源字段：https://doi.org/10.4171/owr/2022/54

05661 | OWR-13750327-005

### Determination of Hitchin Complexes from Open Restrictions

P / partially_solved | Oberwolfach 报告问题汇编

Is the complex $R\chi(n,0)_*\operatorname{IC}_{M_{\mathrm{Dol}}(n,0)}$ determined by its restriction to every nonempty open subset of $A_n$, in particular the smooth locus of the Hitchin fibration?

来源字段：https://doi.org/10.4171/owr/2023/18

05662 | OWR-12867-009

### Determination of Moduli Maps by Symplectic Monodromy

O / open | Oberwolfach 报告问题汇编

Let $X$ be a finite-type Riemann surface and let $\varphi:X\to\mathcal M_g$ be holomorphic, with induced symplectic monodromy $\rho'_\varphi:\pi_1(X)\to\operatorname{Mod}_g\to\operatorname{Sp}(2g,\mathbb Z)$. If two such maps $\varphi_1$ and $\varphi_2$ satisfy $\rho'_{\varphi_1}=\rho'_{\varphi_2}$, must $\varphi_1=\varphi_2$?

来源字段：https://doi.org/10.4171/owr/2014/07

05663 | OWR-698-004

### Diagonal Decomposition of Smooth Projective Varieties

O / open | Oberwolfach 报告问题汇编

For every smooth projective variety $X$ over a field, its diagonal class should decompose, modulo an adequate equivalence relation, into orthogonal projectors carrying respectively the even and odd Weil cohomology of $X$.

来源字段：https://doi.org/10.4171/owr/2004/15

05664 | AIM-ALGEBRAIC_GEOMETRY-0019

### Diagonal resolutions versus homotopy-colimit depth

P / partially_solved | AIM 研讨会问题汇编

There are two known ways to bound Rouquier dimension of a category: (1) bound length of a resolution of a diagonal; (2) the minimal depth of a presentation of \(\mathcal C\) as hocolim (cats with Rouquier dimension \(Q\)). Meaning if \(F:I^{\leq}\to Cat\) [is a] diagram with \(Rdim F(i)=0\), then \(Rdim(hocolim F i)\leq depth(I)\). Can (1) be used to bound (2) or vice-versa?

来源字段：http://aimpl.org/syzygyms/7/

05665 | OWR-15170-020

### Differential Criteria for Low-Genus Curves on Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a smooth projective surface over $\mathbb C$, let $\pi:X\to\mathbb P^2$ be a finite morphism with branch divisor $B$, and fix $g\geq0$. If $L$ is an invertible sheaf of minimal degree among those admitting a nonzero section vanishing on the image of every curve on $X$ of geometric genus at most $g$, does there exist $$ \omega\in H^0\bigl(\mathbb P^2,L\otimes S^r\Omega^1_{\mathbb P^2/\mathbb C}\bigr) $$ such that every component of $B$ and every image of such a low-genus curve is $\omega$-integral?

来源字段：https://doi.org/10.4171/owr/2016/49

05666 | AIM-ALGEBRAIC_GEOMETRY-0201

### Differential signatures as a generic coarse moduli quotient

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.6.** Can we use differential (integral, other…) invariants to construct a moduli space?

**Yes, on a dense specialization-good fixed-degree locus, as a generic coarse geometric quotient and as a flat family of target signature curves. No, in general, as a fine moduli scheme or as the quotient stack: signature curves forget stabilizers and twisted families.**

来源字段：http://aimpl.org/algvision/3/

05667 | OWR-13750327-014

### Dimension Bounds for Theta Divisor Singularities

P / partially_solved | Oberwolfach 报告问题汇编

For every indecomposable principally polarized abelian variety $(A,\Theta)$ and every $m\geq2$, is $\dim\operatorname{Sing}_m(\Theta)\leq\dim A-2m+1$?

来源字段：https://doi.org/10.4171/owr/2023/18

05668 | OWR-14299088-010

### Dimension-Free Separation Exponents for Affine and Positive Semidefinite Sets

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Dimension-Free Separation-Exponent Problem [OWR-14299088-015]

Find an algebraic-geometric invariant that yields a dimension-free exponent in the separation inequality between an affine subspace and the cone $S_+^n$ of positive semidefinite matrices.

来源字段：https://doi.org/10.4171/owr/2025/9

05669 | OWR-11695864-003

### Dimensions of General Weddle Loci

P / partially_solved | Oberwolfach 报告问题汇编

What is the dimension of the $d$-Weddle locus of a general set of $r$ points in $\mathbb P^n$?

来源字段：https://doi.org/10.4171/owr/2022/54

05670 | OWR-7155446-003

### Directed Green-Griffiths-Lang Conjecture Beyond Rank One

O / open | Oberwolfach 报告问题汇编

Does the generalized Green-Griffiths-Lang conjecture hold for directed manifolds of general type beyond rank one, perhaps after imposing a suitable semistability condition on the directed subbundle $V$?

来源字段：https://doi.org/10.4171/owr/2021/25

05671 | OWR-15428-008

### Distinguishability of Cycle and Level-One Phylogenetic Networks

P / partially_solved | Oberwolfach 报告问题汇编

For fixed $n$, classify the pairs of $k$-cycle phylogenetic networks that are distinguishable under the CFN, K2P, and K3P group-based models, and extend the classification from $k$-cycle networks to level-one networks.

来源字段：https://doi.org/10.4171/owr/2017/20

05672 | OWR-15174-007

### Distribution of Minimal Rational-Point Heights on Fano Intersections

O / open | Oberwolfach 报告问题汇编

For smooth Fano complete intersections $V/\mathbb Q$ parametrized by $\mathcal V_{d,n}$, let $H(V)$ be the parameter height and $M(V)$ the least height of a rational point. For $0<t<1$, is

$$\limsup_{A\to\infty}\frac{\#\{V\in\mathcal V_{d,n}:H(V)\leq A,\ M(V)\leq tH(V)\}}{\#\{V\in\mathcal V_{d,n}:H(V)\leq A\}}=O(t)$$

as $t\downarrow0$?

来源字段：https://doi.org/10.4171/owr/2016/53

05673 | AIM-ALGEBRAIC_GEOMETRY-0250

### Divisible Chern data are realizable on finite-dimensional CW complexes

P / partially_solved | AIM 研讨会问题汇编

**Question 13. Question (Madhav Nori).** Given positive integers \(d,r\) and \(X\) a CW complex with \(\dim X=d\) and classes \(c_i\in H^{2i}(X,\mathbb Z)\) for \(i=1,2,\ldots,r\), such that the image of \(c_i\) is \(0\) in \(H^{2i}(X,\mathbb Z/m\mathbb Z)\) for all \(m\), does there exist a complex vector bundle \(E\) with \(\operatorname{rank}(E)=r\) and \(c_i(E)=c_i\)?

来源字段：https://aimath.org/pastworkshops/projectiveA1problems.pdf

05674 | AIM-ALGEBRAIC_GEOMETRY-0264

### Divisorial finite generation, Q-Cartierizations, and a two-ray reduction

P / partially_solved | AIM 研讨会问题汇编

Suppose that \(X\) is an \(F\)-regular variety (maybe not \(\mathbb Q\)-Gorenstein). If \(D\) is a Weil divisor, must \(\bigoplus_{m\geq 0}\mathcal O_X(mD)\) be finitely generated?

来源字段：http://aimpl.org/minimalmodcharp/4/

05675 | OWR-13946-009

### Divisorial Images of Universal K3 Moduli Maps

P / partially_solved | Oberwolfach 报告问题汇编

Can the image of the universal K3 moduli map $\psi_{p,n,\delta}$ be a divisor when $g\leq11$? Can some images $\operatorname{Im}(\psi_{p,n,\delta})$ be described by higher-rank Brill-Noether theory?

来源字段：https://doi.org/10.4171/owr/2015/51

05676 | OWR-2046-005

### Dominance of Pure-Cycle Hurwitz-Space Components

O / open | Oberwolfach 报告问题汇编

Fix $r,g\geq0$, $d\geq1$, and $(e_1,\ldots,e_r)$ satisfying $2d-2-g=\sum_i(e_i-1)$. Let $\mathcal H$ be the Hurwitz space of genus-$g$, degree-$d$ pure-cycle covers branched to orders $e_i$ over $r$ marked points and simply branched over $3g$ additional points. Does every component of $\mathcal H$ map dominantly to $\mathcal M_{g,r}$ after forgetting the covering map and the $3g$ simple ramification points?

来源字段：https://doi.org/10.4171/owr/2008/35

05677 | OWR-16770-007

### Dual Complexes of Maximally Degenerate Hyperkähler Varieties

O / open | Oberwolfach 报告问题汇编

Let $\mathcal X$ be a maximally unipotent good minimal dlt degeneration of compact hyperkähler manifolds over $\mathbb C((t))$. Is the dual complex of its special fiber homeomorphic to $\mathbb P^n(\mathbb C)$?

来源字段：https://doi.org/10.4171/owr/2019/8

05678 | OWR-13946-005

### Dual Graphs of Negative Curves on Enriques Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Can the abstract good dual graphs in the Enriques lattice be classified, and can Enriques surfaces realizing the corresponding $(-2)$-configurations be found?

来源字段：https://doi.org/10.4171/owr/2015/51

05679 | OWR-12334-005

### Dual-Graph Generation of Tautological Relations

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathcal S_{g,n}$ be the strata algebra and let $\mathcal R\subset\mathcal S_{g,n}$ be the ideal generated by pullbacks and pushforwards of the dual-graph relations $R(g,n,r)$. Is $\mathcal R$ exactly the kernel of the natural surjection $\mathcal S_{g,n}\to R^*(\overline{\mathcal M}_{g,n})$? Do the analogous completeness statements hold after restriction to compact-type and rational-tail loci?

来源字段：https://doi.org/10.4171/owr/2013/06

05680 | AIM-ALGEBRAIC_GEOMETRY-0192

### Duality, epipoles, and a Segre cubic from four correspondences

P / partially_solved | AIM 研讨会问题汇编

For any of the standard varieties considered it might be possible to study the projective dual. What can we learn from studying these objects?

Source: <http://aimpl.org/algvision/2/>. The live page was checked on 2026-07-23; it has exactly this wording and no status, introduction, or remarks.

“The standard varieties” is not enumerated on that page. Nearby records in the preceding “Reconstruction” section explicitly mention the essential variety, the trifocal tensor variety, the fundamental matrix, and the Hurwitz form of the essential variety. I therefore interpret the prompt as a program for such camera and tensor varieties, not as a request about every projective variety.

To obtain a rigorous result rather than a broad catalogue, this report focuses on the uncalibrated two-view **fundamental-matrix variety** \[ X=\{[F]\in\mathbb P(\operatorname{Mat}_{3\times3}(\mathbb C)): \det F=0\}\subset\mathbb P^8. \] Its projective dual can be determined completely. More importantly, its dual points are exactly the rank-one coefficient matrices of point-correspondence constraints. This leads to an explicit theorem: four point correspondences that form projective frames cut \(X\) in a Segre cubic threefold with ten ordinary double points, split canonically into six rank-one inherited nodes and four rank-two dual-contact nodes.

This is a developed special case and not a solution of the entire open-ended program for essential, trifocal, quadrifocal, and multiview varieties.

来源字段：http://aimpl.org/algvision/2/

05681 | OWR-12697688-017

### Dualizing Complexes and p-Torsion on the Fargues-Fontaine Curve

O / open | Oberwolfach 报告问题汇编

Describe the dualizing complex of the Fargues-Fontaine curve and determine the corresponding cohomological behavior for $p$-torsion coefficients.

来源字段：https://doi.org/10.4171/owr/2023/5

05682 | OWR-13109-008

### Economical Toroidal Resolutions of Log Canonical Pairs

O / open | Oberwolfach 报告问题汇编

For every $n>0$, does there exist $s=s(n)$ such that each $n$-dimensional log canonical pair $(X,\Delta)$ admits a projective birational morphism $\pi:Y\to X$ with the following properties? Writing $\Gamma=\widetilde\Delta+E$, the pair $(Y,\Gamma)$ is quotient-dlt near the generic point of every non-klt center, $\pi$ has at most $s$ exceptional divisors, and every exceptional divisor has log discrepancy zero.

来源字段：https://doi.org/10.4171/owr/2014/39

05683 | AIM-ALGEBRAIC_GEOMETRY-0204

### Edge and tritangent surfaces are the only generic smooth visual-hull event types

P / partially_solved | AIM 研讨会问题汇编

**Which visual event surfaces appear on visual hulls of algebraic surfaces?**

来源字段：http://aimpl.org/algvision/4/

05684 | OWR-4127-005

### Effective and Movable Slopes of Curve Moduli

O / open | Oberwolfach 报告问题汇编

Estimate $s(\mathcal M_g)=\inf_{D\in\operatorname{Eff}(\mathcal M_g)}s(D)$ and $s'(\mathcal M_g)=\inf_{D\in\operatorname{Mov}(\mathcal M_g)}s(D)$.

来源字段：https://doi.org/10.4171/owr/2009/44

05685 | AIM-ALGEBRAIC_GEOMETRY-0071

### Effective certificates and the witness gap for rationality and unirationality

P / partially_solved | AIM 研讨会问题汇编

Let \(X\) be a variety over \(\overline{\mathbb Q}\). Is there an algorithm to determine whether \(X\) is rational?

Same question for unirationality.

来源字段：http://aimpl.org/rationalityag/1/

05686 | OWR-12172-004

### Effective Cones of Wonderful Arrangement Compactifications

O / open | Oberwolfach 报告问题汇编

For a wonderful compactification $Y$ of a hyperplane-arrangement complement, can the tropical iterative procedure determine its effective cone and provide a method to study the Castravet-Tevelev conjecture for $M_{0,n}$?

来源字段：https://doi.org/10.4171/owr/2012/49

05687 | OWR-4990375-013

### Effective Surface Cone of Principally Polarized Abelian Threefolds

O / open | Oberwolfach 报告问题汇编

Is the cone $\operatorname{Eff}_2(\mathcal A_3)$ of effective surfaces on the toroidal compactification of the moduli space of principally polarized abelian threefolds equal to $$ \mathbb R_{\geq0}S_A+\mathbb R_{\geq0}S_F+\mathbb R_{\geq0}S_D+\mathbb R_{\geq0}\sigma_{K3+1}+\mathbb R_{\geq0}\sigma_{C4} $$ ?

来源字段：https://doi.org/10.4171/owr/2021/6

05688 | AIM-ALGEBRAIC_GEOMETRY-0242

### Efficient generation and a completable-coordinate strong-lifting criterion

P / partially_solved | AIM 研讨会问题汇编

**Related question (S. M. Bhatwadekar).** More specifically, let \[ \mathbb C[X_1,\ldots,X_5]\xrightarrow{\alpha} \mathbb C[Y_1,Y_2] \] be a surjective \(k\)-algebra homomorphism, and let \(I=\ker(\alpha)\). We know \(\mu(I/I^2)=3\). Is \(\mu(I)=3\)?

来源字段：https://aimath.org/pastworkshops/projectiveA1problems.pdf

05689 | OWR-1971-004

### Eisenbud's Fiber-Regularity Conjecture

O / open | Oberwolfach 报告问题汇编

Let $X\subset\mathbb{P}^r$ be a smooth projective variety of dimension $n$, and let $\pi:X\to\mathbb{P}^{n+c}$ be a general projection. Is $\operatorname{reg}(\pi^{-1}(y))\leq n/c+1$ for every $y\in\mathbb{P}^{n+c}$? If $Q_y$ is the obstruction sheaf of the fiber, is $c^{-1}\operatorname{length}(Q_y)$ at least the regularity of the fiber?

来源字段：https://doi.org/10.4171/owr/2008/26

05690 | OWR-719-004

### Elementary Examples of Algebraic Surface Phenomena

O / open | Oberwolfach 报告问题汇编

Construct elementary examples of algebraic surfaces, beyond hypersurfaces in projective $3$-space, exhibiting a wide range of surface phenomena.

来源字段：https://doi.org/10.4171/owr/2004/09

05691 | AIM-ALGEBRAIC_GEOMETRY-0161

### Elementary-abelian Balmer intervals factor as a subspace lattice times a chain

P / partially_solved | AIM 研讨会问题汇编

What more can we say about the tensor triangular geometry of the motivic stable homotopy category? (See (\mathrm{arxiv:1608.02876)}\) for a list of questions.)

Can we get a better understanding of the tensor triangular geometry of \(Sp^G\)? (See Problem 1.6)

来源字段：http://aimpl.org/equideralggeom/2/

05692 | OWR-12869-003

### Eliminating Whitney Umbrellas in Projective Group Realization

O / open | Oberwolfach 报告问题汇编

Can Whitney umbrellas be eliminated: does every finitely presented group occur as the fundamental group of an irreducible two-dimensional projective variety with only normal-crossing singularities?

来源字段：https://doi.org/10.4171/owr/2014/09

05693 | OWR-762-007

### Elliptic Zagier Classes and Beilinson Regulators

O / open | Oberwolfach 报告问题汇编

Do suitable nonzero classes in the motivic cohomology of an elliptic curve have Beilinson regulators given by elliptic polylogarithms, as predicted by the elliptic Zagier conjecture?

来源字段：https://doi.org/10.4171/owr/2004/48

05694 | OWR-13750339-003

### Elliptic-Cover Cycles and the Gorenstein Kernel

O / open | Oberwolfach 报告问题汇编

Determine whether the cycles $\operatorname{Tor}^*[\operatorname{NL}_d]$ generated by elliptic-cover Noether-Lefschetz loci account for the entire Gorenstein kernel of $R^*(\mathcal M_g^{\mathrm{ct}})$.

来源字段：https://doi.org/10.4171/owr/2023/27

05695 | OWR-15178-011

### Embedded Resolution of Singularities in Positive Characteristic

O / open | Oberwolfach 报告问题汇编

Does embedded resolution of singularities exist in dimension three or higher, in the sense of resolving $X\subset Z$ by a sequence of blowups in regular centers while preserving the embedding?

来源字段：https://doi.org/10.4171/owr/2016/57

05696 | OWR-14299516-002

### Enriched Gromov-Witten Invariants of Del Pezzo Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Let $k$ be a perfect field of characteristic different from $2$ and $3$, and let $X$ be a rational del Pezzo surface of degree at least $4$ obtained by blowing up $\mathbb{P}^2_k$ along the prescribed finite étale subschemes. Prove the conjectural identity $W^{\mathrm{WD}}(A_0,A_1,\ldots,A_r)=Q_{X,D}(A_0)$ between the enriched curve-counting invariants.

来源字段：https://doi.org/10.4171/owr/2025/36

05697 | OWR-2049-007

### Entire Curves in General Type Orbifolds

O / open | Oberwolfach 报告问题汇编

Let $(X/\Delta)$ be a smooth projective orbifold of general type. Does there exist a proper subvariety $Y\subsetneq X$ such that every orbifold morphism $f:\mathbb{C}\to(X/\Delta)$ has image contained in $Y$?

来源字段：https://doi.org/10.4171/owr/2008/38

05698 | OWR-12592-002

### Entropy of Serre Functors and Algebraicity

O / open | Oberwolfach 报告问题汇编

For general categories of graded matrix factorizations, what is the geometric interpretation of the entropy $h_t(S)$ of the Serre functor $S$? More generally, is $\exp(h_0(F))$ always algebraic, and under what conditions?

来源字段：https://doi.org/10.4171/owr/2013/44

05699 | OWR-14298589-014

### Enumerative Meaning of k-Leaky Double Hurwitz Numbers

O / open | Oberwolfach 报告问题汇编

Find an enumerative interpretation of the $k$-leaky double Hurwitz numbers $H_g^k(A)$ for $k>0$. More broadly, formulate natural enumerative problems for $k$-differentials of arbitrary genus $g$ and zero-pole profile $A$, and determine whether their counts equal or otherwise relate to $H_g^k(A)$.

来源字段：https://doi.org/10.4171/owr/2024/41

05700 | OWR-4340-007

### EPW Sextics from Fano Models of Enriques Surfaces

O / open | Oberwolfach 报告问题汇编

Let $S$ be a general Fano model of an Enriques surface. The linear system $|2D|$ embeds $S$ in $\mathbb P^{19}$ as a surface of degree $40$. Does a suitable projection to $\mathbb P^5$ produce a degree-$40$ surface contained in the singular locus of an EPW sextic?

来源字段：https://doi.org/10.4171/owr/2010/27

05701 | OWR-13750327-012

### Equality Cases for Singularities of Theta Divisors

P / partially_solved | Oberwolfach 报告问题汇编

Let $(A,\Theta)$ be an indecomposable principally polarized abelian variety. If $\dim\operatorname{Sing}_m(\Theta)=\dim A-2m+1$ for some $m\geq2$, must $A$ be either a hyperelliptic Jacobian or the intermediate Jacobian of a cubic threefold?

来源字段：https://doi.org/10.4171/owr/2023/18

05702 | OWR-1971-006

### Equality of Alternate Moduli Spaces of Pointed Elliptic Curves

O / open | Oberwolfach 报告问题汇编

For the alternate birational models $M_{1,n}(m)$ of the moduli space of smooth $n$-pointed elliptic curves and the model obtained as the normalization of the reduced stack, is $M_{1,n}(m)=\widetilde M_{1,n}(m)$ for every $m$?

来源字段：https://doi.org/10.4171/owr/2008/26

05703 | OWR-13952-008

### Equality of Refined Tropical and Algebraic Curve Counts

P / partially_solved | Oberwolfach 报告问题汇编

For each tropical curve $\Gamma$, is its Block-Göttsche refined tropical curve-counting multiplicity equal to the refined curve-counting polynomial $N_{g-\delta}(y)$, with the equality specializing correctly at $y=1$?

来源字段：https://doi.org/10.4171/owr/2015/57

05704 | OWR-12697711-015

### Equality of Tropical and Matroidal Amoeba-Dimension Formulas

P / partially_solved | Oberwolfach 报告问题汇编

For a loopless matroid $M$ on $[n]$ with matroid-fan support $F\subseteq\mathbb R^n$, prove that $\min_U\{2\dim_{\mathbb R}(U+F)-\dim_{\mathbb R}U\}$ equals $\min_{\bigsqcup_{i=1}^kP_i=[n]}\sum_{i=1}^k(2\operatorname{rk}_M(P_i)-1)$, where $U$ ranges over rationally spanned subspaces and every $P_i$ is nonempty.

来源字段：https://doi.org/10.4171/owr/2023/15

05705 | OWR-9790358-015

### Equations of Three-Dimensional Wachspress Varieties

O / open | Oberwolfach 报告问题汇编

For a polytope $P\subset\mathbb R^3$, its Wachspress coordinates define a rational map to projective space with one coordinate for each vertex of $P$. What are the homogeneous equations of the Zariski closure of its image, and what are the higher-dimensional analogues?

来源字段：https://doi.org/10.4171/owr/2022/7

05706 | OWR-16406-005

### Equivalence of Affine-Space Embeddings Under Automorphisms

O / open | Oberwolfach 报告问题汇编

Given embeddings $f,g:\mathbb C^d\hookrightarrow\mathbb C^n$, does there exist an automorphism $\Psi:\mathbb C^n\to\mathbb C^n$ such that $\Psi\circ f=g$?

来源字段：https://doi.org/10.4171/owr/2018/31

05707 | OWR-1452-022

### Equivalence of Explicit Affine Hypersurfaces

O / open | Oberwolfach 报告问题汇编

Let $H_0=Y$, $H_1=X$, and $H_{n+1}=(r^{a_n}+H_n^3)/H_{n-1}$ for $n\geq1$, where $r=X^3+XYZ-Y^3$ and $a_n=\deg H_n$. Let $U_n=V(H_n-1)$ and $V_n=V(XY-Z^{a_n}-1)\subset\mathbb{A}^3$. For $n\geq3$, does a holomorphic automorphism of $\mathbb{C}^3$ map $U_n$ onto $V_n$? What about a real diffeomorphism or homeomorphism?

来源字段：https://doi.org/10.4171/owr/2007/01

05708 | OWR-1452-026

### Equivariant Ax Problem for Affine Varieties

O / open | Oberwolfach 报告问题汇编

Let $X$ be a normal affine $\mathbb{C}$-variety with an effective algebraic action of $G$, let $\varphi:X\to X$ be a $G$-equivariant endomorphism, and let $\psi:X//G\to X//G$ be induced. If $\psi$ is an automorphism, must $\varphi$ be an automorphism without assuming $\varphi$ is unramified?

来源字段：https://doi.org/10.4171/owr/2007/01

05709 | OWR-1326-001

### Equivariant Completions of Log-Homogeneous Spaces

P / partially_solved | Oberwolfach 报告问题汇编

For a log homogeneous variety $(X,D)$, let $\sigma:X\to S(X)$ be the Stein factorization of its Tits map. Classify, in arbitrary dimension, the homogeneous spaces that admit a log homogeneous equivariant completion. Must $S(X)$ have only quotient singularities?

来源字段：https://doi.org/10.4171/owr/2006/40

05710 | OWR-1703866-005

### Equivariant Derived Categories of Symplectic Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

If a finite group acts suitably on the derived category of a symplectic surface $S$, is the equivariant category equivalent, possibly with a Brauer twist, to the derived category of a symplectic surface $S'$?

来源字段：https://doi.org/10.4171/owr/2020/19

05711 | OWR-14299291-002

### Essential Images in p-Adic Simpson Theory

O / open | Oberwolfach 报告问题汇编

Does every semistable Higgs bundle of degree zero on $X$ lie in $\mathcal{H}_{X,\mathrm{Exp}}$, equivalently in the essential image of the p-adic Simpson correspondence?

来源字段：https://doi.org/10.4171/owr/2025/29

05712 | AIM-ALGEBRAIC_GEOMETRY-0179

### Euler inversion, punctured p-torsion, and equivariant chromatic primes

P / partially_solved | AIM 研讨会问题汇编

We may examine how viewing the Tate spectrum of a given \(G\)-spectrum is related to chromatic localizations. For example, given the spectrum \(KO\) with trivial \(C_2\)-action, we have \[ (KO_{(2)})^{tC_2}\simeq\bigvee_{k\in\mathbb Z}\Sigma^{4k}H\mathbb Q_{(2)}. \] Why should taking the Tate spectrum bring an object down in chromatic height (i.e. why do we see a blue shift)? More generally, we'd like to look at “chromatic primes” and “equivariant primes” in the same place and understand how they interact.

来源字段：http://aimpl.org/equideralggeom/5/

05713 | AIM-ALGEBRAIC_GEOMETRY-0233

### Euler parity and complete cubic cancellation profiles for plane Welschinger counts

P / partially_solved | AIM 研讨会问题汇编

**Recovered statement.** Fix a generic conjugation-invariant configuration \(\boldsymbol p\subset \mathbb P^2\) consisting of \(d_1\) real points and \(d_2\) pairs of complex-conjugate points, where \[ d_1+2d_2=3d-1. \] Let \(\mathcal R_d(\boldsymbol p)\) be the finite set of irreducible real rational degree-\(d\) plane curves through \(\boldsymbol p\). For generic constraints these curves are nodal. If \(s(C)\) is the number of solitary (elliptic) real nodes of \(C\), define \[ W_2(d,d_2)=\sum_{C\in\mathcal R_d(\boldsymbol p)}(-1)^{s(C)}. \] Welschinger's theorem makes this signed count independent of the generic configuration. The question in Remark 9.1 is to understand its sign as a function of \(d_1,d_2\); the source suggests a sign-reversing injection as a possible explanation.

来源字段：https://aimath.org/pastworkshops/degenalggeomproblems.pdf

05714 | AIM-ALGEBRAIC_GEOMETRY-0239

### Euler--Chow--Witt comparison, real realization, and a sphere test

P / partially_solved | AIM 研讨会问题汇编

**Question 2 (by popular demand).** What is the connection between Euler class groups and Chow--Witt groups?

来源字段：https://aimath.org/pastworkshops/projectiveA1problems.pdf

05715 | OWR-1272-009

### Euler-Characteristic Growth of Stable Quiver Moduli

P / partially_solved | Oberwolfach 报告问题汇编

For a quiver $Q$ and generic stability $\Theta$, does there exist $C_Q\in\mathbb R$ such that, for large $\Theta$-coprime dimension vectors $d$, $\log\chi(M_d^{\mathrm{st}}(Q))\sim C_Q\bigl(\dim M_d^{\mathrm{st}}(Q)-1\bigr)$?

来源字段：https://doi.org/10.4171/owr/2006/23

05716 | OWR-1703876-015

### Exact Absolute Linear Harbourne Constants

O / open | Oberwolfach 报告问题汇编

For the absolute linear Harbourne constant $H(d)=\min_K\min_{|L|=d}H(K,L)$ of $d$ lines in the projective plane, compute $H(d)$ for every positive integer $d$.

来源字段：https://doi.org/10.4171/owr/2020/30

05717 | AIM-ALGEBRAIC_GEOMETRY-0143

### Exact boundary mobility counts and explicit mixed-face estimates

P / partially_solved | AIM 研讨会问题汇编

Develop better estimates for the growth rate of \(mc(\alpha)\) for \(\alpha\) a numerical class on \(\mathrm{Gr}(2,4)\) or \(\mathbb P^2\times\mathbb P^2\).

来源字段：http://aimpl.org/poscycles/2/

05718 | AIM-ARITHMETIC_GEOMETRY-0077

### Exact deformation tangent spaces for radical-square-zero local algebras

P / partially_solved | AIM 研讨会问题汇编

**Recovered local question.** Over an algebraically closed field \(k\), is there a nontrivial finite local \(k\)-algebra \(A\) with \(T^1_{A/k}=0\)?

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05719 | AIM-ALGEBRAIC_GEOMETRY-0136

### Exact edge cones and coordinate-ray obstructions for biprojective hypersurfaces

P / partially_solved | AIM 研讨会问题汇编

If \(X\subset \mathbb P^n\times\mathbb P^m\) is a very general hypersurface, what is the effective cone of \(k\)-cycles \(\operatorname{Eff}_k(X)\)?

来源字段：http://aimpl.org/poscycles/1/

05720 | AIM-ALGEBRAIC_GEOMETRY-0298

### Exact FFRT classification of the (2,3,7) hypersurface and an all-iterate Fedder obstruction

S / solved | AIM 研讨会问题汇编

For \(p \geq 11\), does \[ \mathbb F_p[x,y,z]/(x^2+y^3+z^7) \] have finite \(F\)-representation type?

来源字段：http://aimpl.org/testandmultiplierideals/2/

05721 | AIM-ALGEBRAIC_GEOMETRY-0286

### Exact Frobenius block formula for a radical extension

P / partially_solved | AIM 研讨会问题汇编

For \((R,\mathfrak m)\) a local, normal domain of characteristic \(p>0\), let \(x\) be a minimal generator of \(\mathfrak m\), and let \(v\) denote a \(t^{\mathrm{th}}\) root of \(x\). Relate \(e_{HK}(R)\) and \(e_{HK}(R[v])\).

来源字段：http://aimpl.org/testandmultiplierideals/1/

05722 | AIM-ARITHMETIC_GEOMETRY-0079

### Exact incidence and complete-intersection interpolation

P / partially_solved | AIM 研讨会问题汇编

**Problem 32.** Fix (d,g). Let (H\subset \operatorname{Hilb}_{d,g}(\mathbf P^3)) be an irreducible component. What is the largest number of points in general position you can make these curves pass through?

**Remark.** This is a hard problem. If the curves are arithmetically Cohen--Macaulay, it should be treatable.

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05723 | AIM-ALGEBRAIC_GEOMETRY-0224

### Exact line-arrangement operation formulas and a rich-line construction barrier

P / partially_solved | AIM 研讨会问题汇编

For a reduced singular plane curve $C \subset \mathbb{P}^2_{\mathbb{C}}$, let $\pi: \widetilde{\mathbb{P}^2} \to \mathbb{P}^2$ be an embedded resolution of singularities of $C$, let $\widetilde{C}$ be the strict transform of $C$, and define $h(C) = (\widetilde{C})^2 / r$, where $r$ is the number of points blown up.

**Problem 5.1 (Harbourne).** Is $h(C)$ bounded below uniformly (i.e., does there exist a real constant $B$ independent of the singular reduced curve $C \subset \mathbb{P}^2_{\mathbb{C}}$ such that $h(C) \geq B$)? If this holds, then the Bounded Negativity Conjecture holds for rational surfaces in characteristic $0$. (In characteristic $p > 0$, $h(C)$ is not bounded below; for unions of lines in characteristic $0$, it is known that $h(C) > -4$, while the best known example achieves approximately $-3.36$.)

来源字段：https://aimath.org/pastworkshops/degenalggeomproblems.pdf

05724 | AIM-ALGEBRAIC_GEOMETRY-0271

### Exact lognormal projections and regular misspecified asymptotics

P / partially_solved | AIM 研讨会问题汇编

As a concrete example, suppose the data is generated from a log normal distribution, and suppose we try to fit two model classes -

a) a normal distribution

b) a gamma distribution

Understand the behavior of asymptotics in this case.

来源字段：http://aimpl.org/singularlearning/1/

05725 | AIM-ALGEBRAIC_GEOMETRY-0046

### Exact low-degree minima and a sharp linear-space gate

P / partially_solved | AIM 研讨会问题汇编

Given \(d\), what is the smallest \(n\) such that there is some smooth unirational hypersurface of degree \(d\) in \(\mathbb P^n_{\mathbb C}\)?

来源字段：http://aimpl.org/rationalityag/1/

05726 | AIM-ALGEBRAIC_GEOMETRY-0284

### Exact nonzero second coefficient for a non-Q-Gorenstein toric family

P / partially_solved | AIM 研讨会问题汇编

If \(a_e\) denote of \(F\)-splitting numbers of a strongly \(F\)-regular local ring \(R\) (i.e., \(R^{1/p^e}\cong R^{a_e}\oplus M\) as \(R\)-modules, and \(M\) contains no free \(R\)-module summands), then is \[ a_e=s(R)p^{ed}+Cp^{e(d-1)}+O\!\left(p^{e(d-2)}\right) \] for some constant \(C\)?

来源字段：http://aimpl.org/testandmultiplierideals/1/

05727 | AIM-ALGEBRAIC_GEOMETRY-0017

### Exact pointwise levels for a uniserial hypersurface singularity category

P / partially_solved | AIM 研讨会问题汇编

**Rdim of \(D^b Sing\)**

What about Rouquier dimension or orlov spectrum of \(D^b(Sing(X))=D^b(Coh(X))/D^b(Perf(X))\)?

来源字段：http://aimpl.org/syzygyms/7/

05728 | OWR-4990374-017

### Exact Sequences for Logarithmic Derivation Sheaves

P / partially_solved | Oberwolfach 报告问题汇编

Are there short exact sequences relating the sheaves of logarithmic derivations $D$, $D'$, $D''$, and $(D_H,m_H)$ associated with an arrangement and its deletion, restriction, and Ziegler restriction?

来源字段：https://doi.org/10.4171/owr/2021/5

05729 | AIM-ALGEBRAIC_GEOMETRY-0020

### Exact strict-subdimension cotangent families and a mirror obstruction

P / partially_solved | AIM 研讨会问题汇编

What can be said about the Rouquier dimension of non-toric varieties and/for their symplectic mirrors?

- Toric varieties of dim \(n\) \(\longleftrightarrow\) (mirror) \((T^*T^n,\text{some stop})\). - What is known symplectically for \((T^*M,\text{stops})\) can be found in Bai-Côté, Hanlon-Hicks-Lazarev, Favero-Huang.

E.g. is there a symplectic \((X^{2n},f)\) such that

\[ \operatorname{Rdim}\operatorname{Fuk}(X,f)\lneq n? \]

(For instance by finding \((X,f)\) mirror to a singular \(Y\) with \(\operatorname{Rdim}D^b(Y)>\dim Y\).)

来源字段：http://aimpl.org/syzygyms/7/

05730 | AIM-ALGEBRAIC_GEOMETRY-0288

### Exact subadditivity defect on non-Q-Gorenstein curve cones

P / partially_solved | AIM 研讨会问题汇编

Investigate the subadditivity of (Hacon-de Fernex) multiplier ideals in the non-\(\mathbb Q\)-Gorenstein case.

来源字段：http://aimpl.org/testandmultiplierideals/2/

05731 | AIM-ALGEBRAIC_GEOMETRY-0216

### Exact vertical tangent defects for canonical genus-four curves

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.2.** Find examples where we can understand behavior distinct from the expected one. There is basically no hope of a general answer, so I mean this as an example-hunting suggestion.

A specific application would be to extend the range in which we can say when \(\pi\) is dominant. If you have examples where the tangent space dimension is wrong, then you can degenerate to a curve where the tangent space dimension is wrong again.

来源字段：https://aimath.org/pastworkshops/degenalggeomproblems.pdf

05732 | AIM-ALGEBRAIC_GEOMETRY-0205

### Exact visual-event equations, camera-line specialization, and a diagonal-cubic formula

S / solved | AIM 研讨会问题汇编

**Can we find equations of visual event surfaces?**

来源字段：http://aimpl.org/algvision/4/

05733 | OWR-13500-007

### Exceptional Collections and Twisted Cubics on Fano Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a Küchle fourfold of type $(c5)$, realized as a half-anticanonical section of the associated fivefold $Y$. Does $Y$ have a full exceptional collection of length $12$ with a rectangular Lefschetz structure? Is the Hilbert scheme of rational twisted cubic curves on $X$ birational to a hyperkähler fourfold?

来源字段：https://doi.org/10.4171/owr/2015/15

05734 | OWR-14298593-001

### Exceptional Collections on Noncommutative Del Pezzo Surfaces

O / open | Oberwolfach 报告问题汇编

Prove the following conjectures for noncommutative del Pezzo surfaces: (1) an exceptional object is determined by its Grothendieck-group class and full exceptional collections are mutation-equivalent; (2) every pure helix deforms to the noncommutative surface; and (3) two marked surfaces are isomorphic exactly when their markings lie in the same affine-Weyl-group orbit.

来源字段：https://doi.org/10.4171/owr/2024/44

05735 | OWR-11792-002

### Exceptional Dynamics of Polarized Surface Endomorphisms

O / open | Oberwolfach 报告问题汇编

For regular polarized self-maps of smooth projective surfaces over a number field, can the exceptional cases with no periodic point having multiplicatively independent derivative eigenvalues be classified, and can the resulting invariant algebraic curves be naturally parameterized by elements of $\overline{\mathbb Q}$ rather than $\mathbb Q$?

来源字段：https://doi.org/10.4171/owr/2012/30

05736 | OWR-3472-004

### Exceptional Sequences on Algebraic Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Does every algebraic variety $X$ admit a full exceptional sequence, or a strongly exceptional sequence, of coherent sheaves? In particular, when $X$ is a smooth complete rational surface, does it admit such a sequence consisting entirely of invertible sheaves?

来源字段：https://doi.org/10.4171/owr/2009/22

05737 | OWR-1971-003

### Existence Criteria for Good Moduli Spaces

S / solved | Oberwolfach 报告问题汇编

For good moduli spaces of Artin stacks, determine: (1) whether there is a Hilbert-Mumford-type criterion; (2) whether topological criteria on an Artin stack, such as a weak valuative criterion, imply existence of a good moduli space; and (3) whether good moduli spaces can be constructed locally.

来源字段：https://doi.org/10.4171/owr/2008/26

05738 | AIM-ALGEBRAIC_GEOMETRY-0276

### Existence for Noetherian filtrations and monomial graded families

P / partially_solved | AIM 研讨会问题汇编

For a graded sequence of ideals \(I_m\) of a ring \(R\), does \[ \lim_{m\to\infty}m\operatorname{fpt}(I_m) \] exist?

来源字段：http://aimpl.org/testandmultiplierideals/1/

05739 | AIM-ALGEBRAIC_GEOMETRY-0277

### Existence is settled and F-thresholds are invariant under nilpotent quotients

P / partially_solved | AIM 研讨会问题汇编

Investigate the existence and rationality of \(F\)-thresholds.

来源字段：http://aimpl.org/testandmultiplierideals/1/

05740 | OWR-14602-002

### Existence of Irreducible Hecke Eigensheaves

O / open | Oberwolfach 报告问题汇编

For every irreducible local system $E$ for the Langlands dual group on a smooth projective curve, does there exist a nonzero Hecke eigensheaf $F$ on $\operatorname{Bun}_G$ with eigenvalue $E$ whose restriction to every irreducible component of $\operatorname{Bun}_G$ is an irreducible perverse sheaf?

来源字段：https://doi.org/10.4171/owr/2016/20

05741 | AIM-ALGEBRAIC_GEOMETRY-0109

### Expected dimension and Frobenius defects for rational curves on hypersurfaces

P / partially_solved | AIM 研讨会问题汇编

Let \(k\) be an algebraically closed field of characteristic \(p>0\), let \(X=X_d\subset\mathbb P^n_k\) be a smooth degree-\(d\) hypersurface, and fix \(e=\deg f^*\mathcal O_X(1)\). Determine the dimensions of \(\operatorname{Hom}_e(\mathbb P^1,X)\), \(\overline{\mathcal M}_{0,0}(X,e)\), and, where relevant, the Hilbert locus of embedded smooth rational curves. Determine which characteristic-zero arguments survive.

来源字段：http://aimpl.org/ratsubvarpos/5/

05742 | OWR-4527-005

### Expected Dimensions of Equisingular Plane-Curve Systems

P / partially_solved | Oberwolfach 报告问题汇编

For equisingular types $S_1,\ldots,S_r$ with three largest associated multiplicities $m_1,m_2,m_3$, and a general proximity-compatible blowup $\pi:X\to\mathbb{P}^{2}$, is it true that, for every $d\geq m_1+m_2+m_3$, one has $\dim\left|d\pi^{*}L-\sum_{i,j}(\operatorname{mult}_{p_{ij}} S_i)E_{ij}\right|=\max\left\{-1,\binom{d+2}{2}-C(S_1,\ldots,S_r)-1\right\}$?

来源字段：https://doi.org/10.4171/owr/2010/45

05743 | OWR-14611-001

### Expected Dimensions of Rational-Curve Spaces on Hypersurfaces

P / partially_solved | Oberwolfach 报告问题汇编

If $X\subset\mathbb P^n$ is a very general hypersurface of degree $d$, does the space $R_e(X)$ of degree-$e$ integral rational curves have dimension $e(n-d+1)+n-4$, except when $(n,d)=(3,4)$?

来源字段：https://doi.org/10.4171/owr/2016/29

05744 | AIM-ALGEBRAIC_GEOMETRY-0221

### Expected-dimension generality of binary curves and two explicit boundary diagnostics

S / solved | AIM 研讨会问题汇编

(Suggested by Brian Osserman) Consider forming a nodal curve \(X\) by gluing together two rational curves at \(g+1\) generic points. Is \(X\) Brill--Noether general?

来源字段：https://aimath.org/pastworkshops/degenalggeomproblems.pdf

05745 | AIM-ARITHMETIC_GEOMETRY-0022

### Explicit addition of fineness obstructions on a Picard-rank-one cyclic Brauer ray

P / partially_solved | AIM 研讨会问题汇编

How does the group structure of the Brauer group relate to Brauer classes as obstructions to fineness of moduli spaces? Given a K3 surface $X$ and two K3 surfaces $S_1$ and $S_2$ such that $X$ is a moduli space of sheaves on $S_1$ and $S_1$, can we explicitly construct a third K3 surface or variety $Y$ such that $X$ is a moduli space of sheaves on $Y$ and the obstruction class to fineness of this moduli space is the sum of the obstruction classes coming from $S_1$ and $S_2$?

Given identifications $X\simeq M_{S_i,H_i}(v_i)$ for $i=1,2$, with fineness obstructions $\alpha_i\in\operatorname{Br}(X)$, construct $Y$ and $v_Y$ such that $X\simeq M_Y(v_Y)$ and the obstruction is $\alpha_1+\alpha_2$.

来源字段：http://aimpl.org/brauermoduli/1/

05746 | AIM-ALGEBRAIC_GEOMETRY-0333

### Explicit diagonal 1-PS bounds and the current hypersurface GIT classification status

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.2.** Determine the GIT (semi)stable quintic surfaces, quartic threefolds, quintic threefolds. For hypersurfaces of a given dimension and degree, is there a bound on the exponents appearing in the diagonal 1-PS that need to be checked? All smaller cases have been checked by various authors.

来源字段：https://aimath.org/WWN/birational/birational.pdf

05747 | OWR-788-007

### Explicit Flips on Voronoi Compactifications

O / open | Oberwolfach 报告问题汇编

For $6\leq g\leq10$, the ray $\mathbb{R}_+[C_1]$ defines a flipping contraction on the first Voronoi compactification $\mathcal{A}_g^F$. Can the predicted flip, and more generally the minimal model program on $\mathcal{A}_g^F$, be carried out explicitly?

来源字段：https://doi.org/10.4171/owr/2005/07

05748 | OWR-14222-015

### Explicit Formula for Absolute Linear Harbourne Constants

P / partially_solved | Oberwolfach 报告问题汇编

For $d\geq 2$, let $q=q(d)$ be the least prime power satisfying $d\leq q^2+q+1$, and put $i=q^2+q+1-d$. Define $m_1(i)=q+1-i$, $m_2(i)=2q+1-i$, $\varepsilon_1(i)=1$ for $0\leq i\leq q-1$ and $0$ otherwise, and $\varepsilon_2(i)=1$ for $i>q+1$ and $0$ otherwise. Also define $t_{q-1}(i)=qi-q^2-q$ for $i>q+1$ and $0$ otherwise, $t_q(i)=qi$ for $i\leq q+1$ and $2q^2-(i-2)q-1$ otherwise, and $t_{q+1}(i)=q^2+q-iq$ for $i\leq q+1$ and $0$ otherwise. With $$h(d)=\frac{q^2+q+1-i-\varepsilon_1(i)m_1(i)-\varepsilon_2(i)m_2(i)-t_{q-1}(i)(q-1)-t_q(i)q-t_{q+1}(i)(q+1)}{\varepsilon_1(i)+\varepsilon_2(i)+t_{q-1}(i)+t_q(i)+t_{q+1}(i)},$$ is the absolute linear Harbourne constant $H(d)$ equal to $h(d)$ whenever $i\leq 2q-2$, and is $H(d)=-(q^3-q^2+2q-2)/(q^2+q-1)$ when $i=2q-1$?

来源字段：https://doi.org/10.4171/owr/2016/14

05749 | OWR-16169-003

### Explicit Generators for Higher-Dimensional Cremona Groups

O / open | Oberwolfach 报告问题汇编

For the Cremona group $Bir(\mathbb{P}^n)$ in dimensions $n\ge 4$, can one find a simple explicit generating set, i.e. an analogue of the Noether-Castelnuovo theorem?

来源字段：https://doi.org/10.4171/owr/2018/28

05750 | AIM-ALGEBRAIC_GEOMETRY-0112

### Explicit inseparably unirational cyclic covers with regular forms

S / solved | AIM 研讨会问题汇编

Can we find new exmaples of unirational varieties that have non-vanishing differential forms?

来源字段：http://aimpl.org/ratsubvarpos/6/

05751 | AIM-ALGEBRAIC_GEOMETRY-0061

### Explicit K3 models and fine-moduli equivalences for discriminants 26 and 38

P / partially_solved | AIM 研讨会问题汇编

There have been new rationality constructions of special cubic fourfolds $X$ due to Russo and Staglianò, for discriminant 26 or 38. Using these explicit constructions, write down an explicit K3 surface $S$ and an equivalence $\operatorname{Ku}(X)\simeq D(S)$.

来源字段：http://aimpl.org/rationalityag/1/

05752 | AIM-ALGEBRAIC_GEOMETRY-0163

### Explicit rational Picard generators and the prime-cyclic index-two obstruction

P / partially_solved | AIM 研讨会问题汇编

What are the explicit generators of $\operatorname{Pic}(\operatorname{Sp}^G)$?

来源字段：http://aimpl.org/equideralggeom/3/

05753 | OWR-14298584-004

### Explicit Semistable Reduction and Galois Actions for Curves

P / partially_solved | Oberwolfach 报告问题汇编

Given a smooth projective curve $Y$ over a discretely valued field $K$, compute a finite extension $L/K$, a semistable model of $Y_L$, and the action of $\operatorname{Gal}(L/K)$ on its special fiber. Develop an explicit method in arbitrary genus, beyond the classes for which such computations are already known.

来源字段：https://doi.org/10.4171/owr/2024/40

05754 | AIM-ALGEBRAIC_GEOMETRY-0272

### Exponential truth fitted by Gaussian linear regression

P / partially_solved | AIM 研讨会问题汇编

As another example, let us assume the true model is a linear regression. $Y=\exp(X)+\epsilon$ where $\epsilon\sim N(\mu,\sigma^2)$. Let us assume that we try to fit a linear regression model to the data, i.e. $y=bx+\epsilon$.

What is the behavior of the stochastic complexity in this case?

Data have conditional mean $\exp(X)$, plus Gaussian noise, but are fitted with a Gaussian linear regression. Determine the KL projection and the asymptotic Bayesian stochastic complexity (negative log marginal likelihood).

来源字段：http://aimpl.org/singularlearning/1/

05755 | AIM-ALGEBRAIC_GEOMETRY-0207

### Extended-Kruppa constraints, realized conic ambiguity, and invariant eliminants for algebraic silhouettes

P / partially_solved | AIM 研讨会问题汇编

Can we find the fundamental matrix from silhouettes? If the silhouettes are bounded by algebraic curves are there algebraic constraints on the fundamental matrix?

来源字段：http://aimpl.org/algvision/4/

05756 | AIM-ALGEBRAIC_GEOMETRY-0243

### Extending A1-homotopy theory over a DVR

P / partially_solved | AIM 研讨会问题汇编

**Question 6 (Anastasia Stavrova).** Can \(\mathbb A^1\)-homotopy be extended over general smooth schemes, e.g. over a DVR? Else explain the issues.

**Remark (Aravind Asok).** Many basic things fail, a specific example is the stable \(\mathbb A^1\)-connectivity theorem.

来源字段：https://aimath.org/pastworkshops/projectiveA1problems.pdf

05757 | OWR-4990374-008

### Extremal H-Constants of Line Arrangements

O / open | Oberwolfach 报告问题汇编

For line arrangements, determine the most negative possible H-constant among arrangements defined over $\mathbb Q$, among complex arrangements, and among complex supersolvable arrangements. In particular, is $H(C)>-3$ for every complex supersolvable line arrangement?

来源字段：https://doi.org/10.4171/owr/2021/5

05758 | OWR-11139-006

### Extremality of Hodge-Tate Laurent Polynomials

O / open | Oberwolfach 报告问题汇编

Is every Laurent polynomial that is Hodge-Tate at infinity extremal?

来源字段：https://doi.org/10.4171/owr/2011/47

05759 | AIM-ALGEBRAIC_GEOMETRY-0135

### Extreme effective cones and stable Schubert bounds for Plucker hypersurfaces

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** If \(X\subset Gr(m,n)\) is a very general hypersurface, what is \(\operatorname{Eff}_k(X)\)?

Its final sentence is

There is no object \(K\) in the record. The preceding clause introduces \(X\), and the question asks for cycles on the hypersurface. Thus the only coherent recovery is

The replacement \(K\rightsquigarrow X\) is a correction of an apparent typo, not a change to the mathematical question. The source also does not specify whether \(Gr(m,n)\) parametrizes projective \(m\)-planes or \(m\)-dimensional vector subspaces. In this report \[ G=\operatorname{Gr}(r,V)=\operatorname{Gr}(r,n) \] parametrizes \(r\)-dimensional vector subspaces of an \(n\)-dimensional complex vector space, with \(1\leq r\leq n-1\). Thus \[ g:=\dim G=r(n-r). \] Write \(H=c_1(\mathcal O_G(1))\) for the Plücker hyperplane class. A degree-\(d\) hypersurface means a divisor \[ X\in |dH|,\qquad d\geq 1. \] We assume that \(X\) is smooth and put \(m=\dim X=g-1\). The conclusions below hold for every smooth member, and hence for a very general member when the smooth locus of the linear system is nonempty.

The word “effective cone” is interpreted as the **closed pseudoeffective cone** \[ \overline{\operatorname{Eff}}_k(X)\subset N_k(X)_{\mathbb R}, \] the closure of the cone generated by effective \(k\)-dimensional algebraic cycles. This is the convention used in the modern higher-cycle literature. For all cones computed exactly below, taking the closure makes no difference.

来源字段：http://aimpl.org/poscycles/1/

05760 | OWR-12490-002

### Étale Covers of Smooth Loci of Singular Varieties

P / partially_solved | Oberwolfach 报告问题汇编

For a singular complex algebraic variety $X$, what obstructs extension of finite étale covers of its smooth locus $X_{\mathrm{reg}}$ to $X$, and how do the étale fundamental groups of $X$ and $X_{\mathrm{reg}}$ differ?

来源字段：https://doi.org/10.4171/owr/2013/27

05761 | OWR-1452-024

### Étale-Local Equivalence of Fiberwise Isomorphic Families

O / open | Oberwolfach 报告问题汇编

Let $p:S\to X$ and $q:T\to X$ be morphisms of varieties over an algebraically closed field $k$, and suppose every schematic fiber $S_x$ is isomorphic to $T_x$. Does there always exist a dominant étale morphism $U\to X$ such that $S\times_XU\simeq T\times_XU$ over $U$? If not, can one give counterexamples over fields such as $\mathbb{F}_p$ or $\mathbb{Q}$?

来源字段：https://doi.org/10.4171/owr/2007/01

05762 | AIM-ALGEBRAIC_GEOMETRY-0330

### Factor persistence and the limits of single-summand Arakelov equality

P / partially_solved | AIM 研讨会问题汇编

Let \(U\) be a smooth variety, étale over a moduli stack \(M_H\) of polarized manifolds. Let \(Y\) be a log compactification of \(U\), and put \(\Gamma=Y\setminus U\).

Assume that \(M_H\) is a moduli stack for abelian varieties or for a class of manifolds for which local Torelli holds. There is a variation of Hodge structures. Take an irreducible subvariation with Higgs bundle \(E^{1,0}\oplus E^{0,1}\). Then \[ \frac{\deg E^{1,0}}{\operatorname{rk}E^{1,0}} - \frac{\deg E^{0,1}}{\operatorname{rk}E^{0,1}} \leq \frac{\deg\Omega_Y^1(\log\Gamma)}{\dim Y}, \] where \(\omega_Y(\Gamma)\) is nef and big and degree is taken with respect to this sheaf. Determine when equality holds. A conjecture in this direction is that \(U\) is a Shimura variety. The source says that this is true when \(Y\) is a curve and for \(A_g\) for small \(g\).

来源字段：https://aimath.org/WWN/birational/birational.pdf

05763 | AIM-ARITHMETIC_GEOMETRY-0030

### Factorwise transfer of density-one ordinary reduction

P / partially_solved | AIM 研讨会问题汇编

Let \(A\) be an abelian variety over a number field \(K\). Does there exist a finite extension \(K'/K\) such that for a set of primes \(v\) of \(K'\) of density \(1\), \(A/\mathbb F_v\) is ordinary?

来源字段：http://aimpl.org/cohomabelian/1/

05764 | OWR-14605-003

### Failure of Faber’s Conjecture in Genus Five

O / open | Oberwolfach 报告问题汇编

For $g\geq5$, determine whether the Faber conjecture fails; in particular, proving $\operatorname{RH}^4(M_5,V_{2222})\ne0$, together with the expected $\operatorname{RH}^2(M_5,V_{2222})=0$, would produce a counterexample.

来源字段：https://doi.org/10.4171/owr/2016/23

05765 | OWR-14298376-009

### Fano Varieties with Maximal Log-Smooth Degenerations

O / open | Oberwolfach 报告问题汇编

Characterize the smooth complex projective Fano varieties $Y$ for which there exist a smooth anticanonical divisor $D\in|-K_Y|$ and a maximal log-smooth degeneration $f:(\mathcal Y,\mathcal D)\to(S,0)$ with generic fiber $(Y,D)$.

来源字段：https://doi.org/10.4171/owr/2024/23

05766 | AIM-ALGEBRAIC_GEOMETRY-0257

### Fano-type threefolds in characteristic p>5 are Mori dream spaces

P / partially_solved | AIM 研讨会问题汇编

**Fano and MDS.** Is every Fano variety a Mori Dream Space?

来源字段：http://aimpl.org/minimalmodcharp/4/

05767 | OWR-12697688-020

### Fargues Eigensheaves for General Langlands Parameters

P / partially_solved | Oberwolfach 报告问题汇编

Extend Fargues's eigensheaf construction from supercuspidal Langlands parameters to general semisimple $L$-parameters, beginning with parameters induced from a torus.

来源字段：https://doi.org/10.4171/owr/2023/5

05768 | OWR-14602-015

### Fargues-Fontaine Hecke Eigensheaves for General Linear Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a discrete Weil parameter $\varphi:W_{\mathbb{Q}_p}\to GL_n$, does there exist, up to shifts and twists, a sheaf $F_{\varphi}$ on $\operatorname{Bun}_{GL_n}$ in the Fargues-Fontaine setting whose Hecke transform is $F_{\varphi}\otimes\varphi$ and whose pullbacks to the basic points correspond to the local Langlands representations attached to $\varphi$?

来源字段：https://doi.org/10.4171/owr/2016/20

05769 | OWR-12697688-016

### Fargues' Cohomological Conjectures for the Curve

P / partially_solved | Oberwolfach 报告问题汇编

For an étale abelian torsion sheaf $\mathcal F$ on $X_{E,F}^{\mathrm{alg}}$, prove that $H^i(X_{E,F}^{\mathrm{alg}},\mathcal F)=0$ for $i\geq 3$; that adification gives $H^i(X_{E,F}^{\mathrm{alg}},\mathcal F)\cong H^i(X_{E,F}^{\mathrm{ad}},\mathcal F^{\mathrm{ad}})$ for $i\geq 0$; and that the function field $E(X_{E,F}^{\mathrm{alg}})$ is $C_1$.

来源字段：https://doi.org/10.4171/owr/2023/5

05770 | AIM-ALGEBRAIC_GEOMETRY-0273

### Fermat counterexamples and a one-step theorem for monomial thickenings

S / solved | AIM 研讨会问题汇编

Given an ideal \(I\) of \(\mathbb Z[x]\), the HSL numbers are the "indices of nilpotency" of the action of Frobenius on \[ H^{\mathrm{top}}_{(\underline{x})} \left(\mathbb Z_p[\underline{x}]/I_p\right), \] for primes \(p\). Is the limsup of the HSL numbers \(1\)?

来源字段：http://aimpl.org/testandmultiplierideals/1/

05771 | AIM-ALGEBRAIC_GEOMETRY-0325

### Fibers of integral moduli spaces: tame compatibility and a wild obstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.6. (Kollár)** Consider the moduli space of canonically polarized surfaces over \(\operatorname{Spec}\mathbb Z\). Is the fiber over the prime \(p\) the moduli space of surfaces over \(\mathbb F_p\)? A similar result for \(M_g\) may be due to Oort. What about other moduli problems? \(A_g\)? K3 surfaces?

**1.4. Other moduli problems.** An article of Schumacher and Tsuji ...

来源字段：https://aimath.org/WWN/birational/birational.pdf

05772 | OWR-1276-010

### Fields of Definition and Galois Orbits of Beauville Surfaces

O / open | Oberwolfach 报告问题汇编

For a Beauville surface $S$, determine a field of definition of $S$ and the $\operatorname{Gal}(\overline{\mathbb Q}/\mathbb Q)$-orbit of its moduli point $[S]$.

来源字段：https://doi.org/10.4171/owr/2006/27

05773 | AIM-ALGEBRAIC_GEOMETRY-0010

### Filtered flow data and stable Betti spectra

P / partially_solved | AIM 研讨会问题汇编

**Topology/Morse theory of cellular resolutions.** "The Morse-theoretic description of $H_*(M)$ can be refined (at e.g. level of 'framed flow categories'). can this refined decomposition tell us something on the algebraic side?"

来源字段：http://aimpl.org/syzygyms/4/

05774 | AIM-ALGEBRAIC_GEOMETRY-0202

### Finite algebraic certificates for recognition by invariant signatures

P / partially_solved | AIM 研讨会问题汇编

**How can we use these signatures for object recognition?**

The archived section introduction says: "From a differential geometry perspective we can assign invariants to a smooth curve or surface. This allows us to assign a signature curve to each curve." The neighboring questions ask about completeness, degree, automorphism groups, realizability, and moduli of signature curves. The workshop report identifies the relevant day as "Shapes, silhouettes, and contours," with talks by Matthew Trager, Peter Olver, and Irina Kogan. Thus "these signatures" means invariant signature curves of geometric boundaries, especially differential signatures, with algebraic signatures as a central workshop specialization.

For an oriented regular planar curve under the proper Euclidean group \(SE(2)\), the classical smooth signature is \[ s\longmapsto \bigl(\kappa(s),\kappa_s(s)\bigr), \] where \(s\) is Euclidean arc length. Analogous signatures use equi-affine or projective differential invariants. For algebraic curves it is advantageous to use a *rational classifying pair*. For example, Kogan--Ruddy--Vinzant use \((\kappa^2,\kappa_s)\) for the complexified \(SE(2)\)-action. They also give rational pairs for \(SA(2)\), \(A(2)\), and \(PGL(3)\).

"Recognition" is interpreted mathematically as deciding whether two boundary curves lie in the same orbit of a specified transformation group \(G\). This is not the same as assigning a semantic label such as "car" or "chair." A semantic classifier may use invariant signatures as features, but its statistical generalization is a separate question.

The original live `aimpl.org` URL now redirects to a missing page.

来源字段：http://aimpl.org/algvision/3/

05775 | AIM-ALGEBRAIC_GEOMETRY-0141

### Finite complete-intersection mobility bounds for the line class in projective three-space

P / partially_solved | AIM 研讨会问题汇编

Let \(X\) be an \(n\)-dimensional variety. Given an effective integral \(k\)-cycle \(\alpha\in N_k(X)_{\mathbb Z}\), the mobility count \(mc(\alpha)\) is the maximum number of general points that can be imposed on an effective cycle of class \(\alpha\). It then displays formulas for mobility and an Iitaka-type exponent and asks, after introducing the line class \([\ell]\in N_1(\mathbb P^3)_{\mathbb Z}\), “What is \(\operatorname{mob}([\alpha])\)?”

Over \(\mathbb C\), let \(\ell\) be the numerical class of a line in \(\mathbb P^3\). Determine \[ \operatorname{mob}(\ell) =\limsup_{m\to\infty}\frac{6\,mc(m\ell)}{m^{3/2}}. \]

1. The displayed mobility formula in the record has no limiting operation. The standard definition in [Leh16, Definition 1.1] is \[ \operatorname{mob}(\alpha) := \limsup_{m\to\infty} \frac{mc(m\alpha)} {m^{\,n/(n-k)}/n!} = \limsup_{m\to\infty} \frac{n!\,mc(m\alpha)}{m^{\,n/(n-k)}}. \tag{1.1} \] The denominator \(m^{n/(n-k)}/n!\) is the standard normalization. 2. The last \([\alpha]\) is reconstructed as \([\ell]\). This is confirmed by the fuller AIM problem-list transcript, which asks for the mobility when \(\alpha\) is the line class on \(\mathbb P^3\). 3. The AIM record calls the unrescaled growth exponent \[ K_{\mathrm{src}}(\alpha) := \sup\left\{r\geq0: \limsup_{m\to\infty}\frac{mc(m\alpha)}{m^r}>0 \right\}. \tag{1.2} \] Lehmann's later convention [Leh19] multiplies this exponent by the codimension: \[ \kappa(\alpha)=(n-k)K_{\mathrm{src}}(\alpha). \tag{1.3} \] Both conventions will be reported explicitly.

Thus the recovered problem is:

来源字段：http://aimpl.org/poscycles/2/

05776 | AIM-ALGEBRAIC_GEOMETRY-0131

### Finite étale and tame obstruction collapse for SRC varieties

P / partially_solved | AIM 研讨会问题汇编

For separably rationally connected varieties, is the Brauer-Manin obstruction the only obstruction for the Hasse principle?

来源字段：http://aimpl.org/ratsubvarpos/9/

05777 | OWR-15588-008

### Finite Factorization Bases for Breaking Maps

O / open | Oberwolfach 报告问题汇编

Let $X$ be a smooth uniruled projective variety. Is there a finite set of breaking maps $f_i:Y_i\to X$ such that every breaking map to $X$ factors rationally through one of the $f_i$?

来源字段：https://doi.org/10.4171/owr/2017/45

05778 | OWR-14221-008

### Finite Generation of Birational-Sequence Monoids

O / open | Oberwolfach 报告问题汇编

For every birational sequence and proper total order in the PBW/Newton-Okounkov construction, is the associated monoid $\Gamma$ finitely generated?

来源字段：https://doi.org/10.4171/owr/2016/13

05779 | OWR-14298795-013

### Finite Index of Arithmetic Dehn-Twist Subgroups

O / open | Oberwolfach 报告问题汇编

For a $p$-adic field $K$, does the profinite subgroup generated by all arithmetic Dehn twists have finite index in $\operatorname{Out}(G_K(p))$, where $G_K(p)$ is the maximal pro-$p$ quotient of the absolute Galois group?

来源字段：https://doi.org/10.4171/owr/2024/45

05780 | AIM-ALGEBRAIC_GEOMETRY-0238

### Finite-colength covectors and a critical-rank punctured-spectrum counterexample

P / partially_solved | AIM 研讨会问题汇编

1. Let \((R,\mathfrak m)\) be a local ring of dimension \(d\), and let \(U=\operatorname{Spec}(R)\setminus\{\mathfrak m\}\). What is an obstruction theory for splitting vector bundles on \(U\), especially bundles of rank \(d-1\)? 2. For a scheme \(X\) and vector bundle \(E\), what obstructs a surjection \(E\twoheadrightarrow\mathcal O_X\)? 3. Can affine obstruction theory for projective modules be extended to more general classes of modules? 4. Is there an Euler class group for reflexive modules over a regular local ring? 5. What role does \(A^1\)-homotopy theory play in the non-affine case?

The associated remarks ask whether a Jouanolou affine replacement is useful, warn that isomorphism classes of vector bundles on non-affine schemes are not in general \(A^1\)-invariant, and ask whether pullback along the incidence Jouanolou device \[ p_1:J(V):=\mathbf P(V)\times\mathbf P(V^\vee)\setminus H \longrightarrow \mathbf P(V) \] is surjective on isomorphism classes of vector bundles. They also ask about Nisnevich/étale topology and realization to equivariant topological bundles.

来源字段：https://aimath.org/pastworkshops/projectiveA1problems.pdf

05781 | AIM-ARITHMETIC_GEOMETRY-0042

### Finite-field counterexamples and a low-u torsion-index obstruction

S / solved | AIM 研讨会问题汇编

For a quadratic form \(q\) over \(F\), define \[ \operatorname{sd}_F(q)=\min\{[L:F]:q_L\text{ is a direct sum of hyperbolic planes}\}. \] The torsion index \(\tau_F\) is the maximum of these splitting degrees over all even-dimensional \(q\). Is \[ \tau_F<2^{u(F)/2-1}? \]

来源字段：http://aimpl.org/deformationbrauer/1/

05782 | AIM-ALGEBRAIC_GEOMETRY-0167

### Finite-index Wirthmuller duality and the compact profinite Adams boundary

P / partially_solved | AIM 研讨会问题汇编

For a profinite group \(G\) (having subgroups of arbitrary index), are there Wirthm\"uller/Adams isomomorphisms in \(Sp^G\)?

来源字段：http://aimpl.org/equideralggeom/4/

05783 | AIM-ALGEBRAIC_GEOMETRY-0306

### Finite-jet detection of log canonical thresholds with quantitative bounds

S / solved | AIM 研讨会问题汇编

**3. (Mustaţă)** We have the ACC conjecture for log canonical thresholds (Conjecture 2.5 of [6], see also [3]). When we consider only smooth varieties, the conjecture is equivalent to the following question:

Fix \(n\). \(\forall c\in\mathbb R\), does there exist \(k=k(c)\) such that \[ \{f\in\mathbb C[[x_1,\ldots,x_n]]\mid \operatorname{lct}(f)\geq c\} \] depends on \(k\)-th truncations of \(f\)?

A weaker question is: Does there exist \(k=k(f)\) such that \(\operatorname{lct}(f+g)\geq\operatorname{lct}(f)\) for all \(g\) with \(\operatorname{ord}(g)\geq k\)?

来源字段：https://aimath.org/WWN/singularvariety/problems_final.pdf

05784 | AIM-ARITHMETIC_GEOMETRY-0002

### Finite-level geometric refinement of Li-Zhang local modularity

P / partially_solved | AIM 研讨会问题汇编

In recent proof of Kudla-Rapoport conjectures by Li--Zhang, numerical local modularity results for local Kudla--Rapoport divisors have been discovered and proved. Is it possible to formulate local modularity geometrically e.g. using moduli of p-adic Shtukas? Could the method of Feng-Yun-Zhang for global modularity be adapted to prove geometric local modularity?

来源字段：http://aimpl.org/intersectshimura/1/

05785 | OWR-14298795-020

### Finiteness for Subvarieties of Abelian Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Extend arithmetic-finiteness and big-monodromy results from high-codimension subvarieties with ample normal bundle to smaller-codimension subvarieties of abelian varieties and, more generally, to varieties admitting a finite morphism to an abelian variety.

来源字段：https://doi.org/10.4171/owr/2024/45

05786 | OWR-14746-013

### Finiteness of Atypical Hodge Subvarieties

O / open | Oberwolfach 报告问题汇编

For an irreducible smooth quasi-projective variety $S$ with a variation $V$ of mixed rational Hodge structures, is the atypical locus $S_{\mathrm{atyp}}(V)$ a finite union of maximal atypical subvarieties, equivalently a proper algebraic subset and hence not Zariski dense; and, in the generalized André-Oort case, does Zariski density of CM points force $V$ to arise by pullback from a connected mixed Shimura variety?

来源字段：https://doi.org/10.4171/owr/2016/38

05787 | OWR-1104-002

### Finiteness of Log Rational Maps to Log-General-Type Varieties

O / open | Oberwolfach 报告问题汇编

Let $k$ be an algebraically closed field with a fine log structure, and let $(X,M_X)$ and $(Y,M_Y)$ be proper semistable log varieties smooth and integral over $(\operatorname{Spec}k,M_k)$, with $(Y,M_Y)$ of log general type. Are there only finitely many dominant separable log rational maps $(X,M_X)\dashrightarrow(Y,M_Y)$?

来源字段：https://doi.org/10.4171/owr/2005/43

05788 | OWR-12870-008

### Finiteness of Minimal Models of Terminal Threefolds

O / open | Oberwolfach 报告问题汇编

For every terminal projective threefold $Y$, are there only finitely many minimal models reachable by the $K_Y$-minimal model program? In particular, are there only finitely many when the program is run with scaling by an ample divisor $H$ on $Y$?

来源字段：https://doi.org/10.4171/owr/2014/10

05789 | OWR-15430-014

### Finiteness of Optimal Points in Shimura Subvarieties

O / open | Oberwolfach 报告问题汇编

Let $S$ be a Shimura variety and $V\subseteq S$ a subvariety. Call a subvariety $W\subseteq V$ optimal in $V$ if every subvariety $Y$ with $W\subsetneq Y\subseteq V$ has larger defect $\delta(Y)=\dim\langle Y\rangle-\dim Y$, where $\langle Y\rangle$ is the smallest special subvariety containing $Y$. Does $V$ contain only finitely many points that are optimal in $V$?

来源字段：https://doi.org/10.4171/owr/2017/22

05790 | OWR-16768-013

### Finiteness of Singular-Centered Arc-Space Components

O / open | Oberwolfach 报告问题汇编

If $X$ is an irreducible variety over a perfect field of characteristic $p>0$, does the subspace of the arc space $X_\infty$ consisting of arcs centered in $\operatorname{Sing}(X)$ have finitely many irreducible components?

来源字段：https://doi.org/10.4171/owr/2019/6

05791 | AIM-ALGEBRAIC_GEOMETRY-0123

### Fixed points in dimension two and wild fixed-scheme multiplicities

P / partially_solved | AIM 研讨会问题汇编

Does an automorphism of a separably rationally connected variety necessarily have a fixed point?

来源字段：http://aimpl.org/ratsubvarpos/9/

05792 | AIM-ALGEBRAIC_GEOMETRY-0164

### Fixed-point marks refute the real-signature and plane-branch comparisons

P / partially_solved | AIM 研讨会问题汇编

The record asks:

1. If a finite group \(G\) acts linearly on \(V=\mathbb C^n\) and \(f:V\to V\) is a \(G\)-equivariant polynomial map with an isolated zero at \(0\), is the \(G\)-module \[ Q(f)=\mathbb C[[x_1,\ldots,x_n]]/(f_1,\ldots,f_n) \] equal in \(R_{\mathbb C}(G)\) to the permutation representation obtained by linearizing the Burnside-ring-valued stable local degree? 2. Over \(\mathbb R\), is the \(G\)-signature of the Eisenbud--Levine--Khimshiashvili (ELK) form on \(Q(f)\) equal to the real linearization of that stable local degree? 3. If \(f=\nabla F:\mathbb C^2\to\mathbb C^2\) for a \(G\)-invariant equation \(F=0\) of an isolated plane-curve singularity, is the \(G\)-set of analytic branches equal, as a Burnside class, to the stable local degree of \(\nabla F\)?

来源字段：http://aimpl.org/equideralggeom/3/

05793 | AIM-ALGEBRAIC_GEOMETRY-0040

### Fixed-point Postnikov windows and a two-term model for 1-slices

P / partially_solved | AIM 研讨会问题汇编

(\mathrm{arXiv):1711.03472}\) gives a description of \(n\)-slices, the objects that appear in the associated graded of the slice filtration. Is there a way to get a good computational grasp of \(n\)-slices? For example understanding their Postnikov towers, homotopy groups etc.

来源字段：http://aimpl.org/equivstable/4/

05794 | OWR-7155450-006

### Flag-Variety Line-Bundle Cohomology in Positive Characteristic

P / partially_solved | Oberwolfach 报告问题汇编

Over a field of positive characteristic, determine the cohomology groups of line bundles on the complete flag variety $\operatorname{Flag}(V)$ as $\operatorname{GL}(V)$-representations, or at least characterize which such cohomology groups vanish.

来源字段：https://doi.org/10.4171/owr/2021/29

05795 | OWR-13682-010

### Forgetful Functors for Mixed Perverse Sheaves

O / open | Oberwolfach 报告问题汇编

Does there exist a forgetful functor $\operatorname{For}:DS^{\mathrm{mix}}(X,F)\to DS^b(X,F)$ that makes $\operatorname{Perv}^{\mathrm{mix}}_S(X,F)$ a graded version of $\operatorname{Perv}_S(X,F)$?

来源字段：https://doi.org/10.4171/owr/2015/25

05796 | OWR-2492-002

### Formal Jacobians of Universal Artin-Schreier Deformations

O / open | Oberwolfach 报告问题汇编

For a wildly ramified Artin-Schreier cover $\pi:C\to\mathbb{P}^1$ with a single branch point at infinity, describe the formal completion of the Jacobian of its universal deformation in formal-group-theoretic terms.

来源字段：https://doi.org/10.4171/owr/2008/54

05797 | AIM-ALGEBRAIC_GEOMETRY-0213

### Formal velocities and obstructions for time-varying epipolar matrices

P / partially_solved | AIM 研讨会问题汇编

Is there a differentiable structure for essential varieties or fundamental varieties over time. Is there a formal structure? Can we use deformation theory and cohomological techniques?

来源字段：http://aimpl.org/algvision/6/

05798 | AIM-ALGEBRAIC_GEOMETRY-0182

### Four ambient trifocal point equations have generic restricted rank three

S / solved | AIM 研讨会问题汇编

How many linear constraints does a point correspondence in three views determine on the trifocal tensor variety?

Numerical results suggest that there are 4 constraints but they only intersect the trifocal tensor variety in codimension three.

来源字段：http://aimpl.org/algvision/1/

05799 | AIM-ALGEBRAIC_GEOMETRY-0314

### Four F-pure-threshold questions and a sharp perturbation cylinder

P / partially_solved | AIM 研讨会问题汇编

**13. (Mustaţă)** Let \(k\) be a field of characteristic \(p\).

1. For \(f\in k[[x_1,\ldots,x_n]]\), is the \(F\)-pure threshold \(\operatorname{fpt}(f)\) a rational number? 2. Let \(c=\operatorname{fpt}(f)\). Is \(\tau(f^c)\) a radical ideal? 3. If the zero set of \(\tau(f^c)\) is zero-dimensional, does there exist \(N>1\) such that for every \(g\in k[[x_1,\ldots,x_n]]\) of order at least \(N\), \[ \operatorname{fpt}(f)=\operatorname{fpt}(f+g)? \]

**14.** Does the set \[ \{\operatorname{fpt}(f)\mid f\in k[[x_1,\ldots,x_n]]\} \] have the ascending chain condition?

来源字段：https://aimath.org/WWN/singularvariety/problems_final.pdf

05800 | OWR-1452-002

### Fourier-Mukai Kernels from Polyhedral Divisors

O / open | Oberwolfach 报告问题汇编

Let $K$ be an algebraically closed field of characteristic zero, let $T$ be a torus with character lattice $M$ and cocharacter lattice $N$, and let $\mathcal D=\sum_i\Delta_i\otimes D_i$ be a polyhedral divisor on a normal semiprojective $K$-variety $Y$, with tail cone $\sigma$. Put $\Delta=\sum_i\Delta_i$ and $Z=\operatorname{TV}(N(\Delta),M)$. Each $\Delta_i$ determines a semiample divisor $E_i$ on $Z$, so regard $\mathcal D$ also as $\sum_iE_i\otimes D_i$. For $u\in\sigma^\vee\cap M$, view $u$ as the corresponding germ of a curve $u:(\mathbb{A}^1_K,0)\to Z$ and let $u_*\mathcal O_{\mathbb{A}^1_K,0}$ denote the direct image of its structure sheaf. Set $\mathcal U=\bigoplus_{u\in\sigma^\vee\cap M}u_*\mathcal O_{\mathbb{A}^1_K,0}$. Does $\mathcal D$ induce an object $\mathcal K_{\mathcal D}\in D^b(Y\times Z)$ such that the graded sheaf $\mathcal O_Y(\mathcal D)=\bigoplus_{u\in\sigma^\vee\cap M}\mathcal O_Y(\mathcal D(u))$ is the Fourier-Mukai transform of $\mathcal U\in D^b(Z)$ with kernel $\mathcal K_{\mathcal D}$?

来源字段：https://doi.org/10.4171/owr/2007/01

05801 | AIM-ALGEBRAIC_GEOMETRY-0316

### Frobenius contraction and a log-canonical non-F-pure stress test for the symbolic cube

P / partially_solved | AIM 研讨会问题汇编

Let \((R,\mathfrak m)\) be a regular local ring of characteristic \(p\) and \(\mathfrak p\subset R\) a prime ideal of height \(2\). If \(\operatorname{Spec}R/\mathfrak p\) has only log canonical singularities, then \(\mathfrak p^{(3)}\subset \mathfrak p^2\)?

来源字段：https://aimath.org/WWN/singularvariety/problems_final.pdf

05802 | AIM-ALGEBRAIC_GEOMETRY-0002

### Frobenius on the affine node: a bimodule ladder and mirror obstruction

P / partially_solved | AIM 研讨会问题汇编

Given a singular variety in char. p, $X$, $\mathcal F:=Frobenius$, then $\mathcal F^{>>0}(Perf(X))=Coh(X)$ (as dg enhanced categories of complexes) [B-I-L-M-P].

Can we see the mirror to this fact explicitly in case e.g. of mirrors to $X=\{xy=0\}\subset \mathbb A_{\mathbb F_p}$? More generally, what is the symplectic mirror to Frobenius for each $p$?

来源字段：http://aimpl.org/syzygyms/1/

05803 | AIM-ALGEBRAIC_GEOMETRY-0265

### Frobenius-polarized cones with double poles for (n-1)-forms

S / solved | AIM 研讨会问题汇编

Suppose that \(X\) is log canonical, and that there exists a log resolution \(f:\widetilde X\to X\) which is an isomorphism over the smooth locus. Let \(\omega\) be an \((n-1)\)-form on \(X\). Does \(f^*\omega|_{X_{\mathrm{smooth}}}\) extend to an \((n-1)\)-form on \(\widetilde X\) with log poles along the exceptional locus?

来源字段：http://aimpl.org/minimalmodcharp/5/

05804 | AIM-ALGEBRAIC_GEOMETRY-0260

### Frobenius-stable adjoint sections in families

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** Let \(f:\mathcal X\to T\) be a flat projective family in characteristic \(p>0\), let \(\mathcal A\) be a fixed relatively ample Cartier divisor, and let \(m\) be fixed so that \(M_t=mK_{X_t}+A_t\) is Cartier. Is \[ t\longmapsto \dim_{k(t)}S^0(X_t,M_t) \] invariant, semicontinuous, or otherwise better behaved than ordinary plurigenera?

来源字段：http://aimpl.org/minimalmodcharp/4/

05805 | AIM-ALGEBRAIC_GEOMETRY-0290

### Frobenius-stable lifting and a sharp curve threshold

P / partially_solved | AIM 研讨会问题汇编

**Recovered working question.** Let \(X\) be projective in characteristic \(p>0\), let \(Z\subseteq X\) be a divisor, fiber, or \(F\)-pure center, and let \(L\) be positive relative to an adjoint divisor. Which trace-stable sections on \(Z\) lift to \(X\)? Under what additional hypotheses is the stable subspace all of \(H^0(Z,L|_Z)\)? What is the exact ordinary-cohomology obstruction when one twists by a test ideal?

来源字段：http://aimpl.org/testandmultiplierideals/2/

05806 | OWR-1703867-004

### Frobenius-Stable Loci and Dense Arithmetic Points

O / open | Oberwolfach 报告问题汇编

Let $X/\overline{\mathbb F}_p$ be smooth, connected, and of finite type; let $S_{\bar\rho}$ parametrize semisimple $\overline{\mathbb Q}_\ell$-local systems with residual representation $\bar\rho$; and let $A_{\bar\rho}$ be its arithmetic points. If a Zariski-closed subset $Z\subset S_{\bar\rho}$ is stable under a power of Frobenius, is $A_{\bar\rho}\cap Z$ Zariski dense in $Z$?

来源字段：https://doi.org/10.4171/owr/2020/20

05807 | AIM-ALGEBRAIC_GEOMETRY-0261

### Frobenius-stable pluricanonical systems on products of curves

P / partially_solved | AIM 研讨会问题汇编

**Embedding by \(S^0(mK_X)\).** Suppose that \(X\) is of general type. Is there an effective constant \(m=m(n)\) such that \(S^0(mK_X)\) (the canonical linear system of Schwede) defines a birational map? What about the usual linear system \(|mK_X|\)?

来源字段：http://aimpl.org/minimalmodcharp/4/

05808 | AMR-071-0006

### Fröberg conjecture on the Hilbert functions of a set of forms

O / open | AMR 问题清单汇编

Fröberg conjecture on the Hilbert functions of a set of forms.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

05809 | OWR-3471-003

### Fujiki-Class Limits of Kähler Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Let $\pi:X\to\Delta$ be a complex-analytic family of compact complex manifolds such that $X_t$ is Kähler for every $t\in\Delta^*=\Delta\setminus\{0\}$. Must the central fiber $X_0$ belong to Fujiki class $\mathcal{C}$?

来源字段：https://doi.org/10.4171/owr/2009/21

05810 | OWR-17296-005

### Fujita's Conjecture for Effective Global Generation

O / open | Oberwolfach 报告问题汇编

For an ample line bundle $L$ on a projective variety $X$, what tensor power $L^{\otimes\ell}$ is guaranteed to be globally generated or very ample? In particular, does Fujita's conjecture hold for a smooth $n$-dimensional projective $X$, with $\omega_X\otimes L^{\otimes\ell}$ globally generated for $\ell\geq n+1$ and very ample for $\ell\geq n+2$?

来源字段：https://doi.org/10.4171/owr/2019/53

05811 | OWR-12870-009

### Fujita's Global Generation and Very-Ampleness Conjecture

O / open | Oberwolfach 报告问题汇编

Let $L$ be an ample line bundle on a complex manifold $X$. Is $K_X\otimes L^m$ globally generated for $m\geq\dim X+1$ and very ample for $m\geq\dim X+2$?

来源字段：https://doi.org/10.4171/owr/2014/10

05812 | AIM-ALGEBRAIC_GEOMETRY-0009

### Fukaya interpretations and the limits of cellular minimization

P / partially_solved | AIM 研讨会问题汇编

**Topology/Morse theory of cellular resolutions.** Can we interpret resolutions via Fukaya categories? (e.g. minimizing resolutions through “cellular” operations)

来源字段：http://aimpl.org/syzygyms/4/

05813 | OWR-13750327-004

### Full Support for Singular Hitchin Systems

P / partially_solved | Oberwolfach 报告问题汇编

Does $R\chi(n,0)_*\operatorname{IC}_{M_{\mathrm{Dol}}(n,0)}$ have full support on the entire Hitchin base $A_n$, meaning that none of its direct summands is properly supported on a subvariety of $A_n$?

来源字段：https://doi.org/10.4171/owr/2023/18

05814 | OWR-11792-005

### Functorial Chow Rings for Singular Varieties

O / open | Oberwolfach 报告问题汇编

For a projective variety $X$ over an algebraically closed field, can one define a Chow ring $CH^*(X)$ with the expected contravariant functoriality and cycle-class and Chern-class properties, including a rational Chern-character isomorphism?

来源字段：https://doi.org/10.4171/owr/2012/30

05815 | OWR-13750327-010

### Fundamental Groups of Five-Dimensional Log-Canonical Singularities

O / open | Oberwolfach 报告问题汇编

Does every finitely presented group occur as the local fundamental group of a five-dimensional log-canonical singularity?

来源字段：https://doi.org/10.4171/owr/2023/18

05816 | OWR-12869-002

### Fundamental Groups of Singular Irreducible Projective Varieties

O / open | Oberwolfach 报告问题汇编

Which groups occur as fundamental groups of complex projective varieties when the varieties are required to be irreducible and their singularities are suitably controlled?

来源字段：https://doi.org/10.4171/owr/2014/09

05817 | AIM-ALGEBRAIC_GEOMETRY-0011

### Fundamental-group information: a torus group-algebra answer and an invisibility counterexample

P / partially_solved | AIM 研讨会问题汇编

\(T^n=K(\pi,1)\)

Given a cellular resolution, is there a meaningful way to algebraically interpret topology of the cell complex beyond homology e.g. \(\pi_1\)?

来源字段：http://aimpl.org/syzygyms/4/

05818 | OWR-4990374-002

### Further Rank-Three Simplicial Arrangements

O / open | Oberwolfach 报告问题汇编

Do rank-three simplicial arrangements include further examples or infinite series beyond those currently known?

来源字段：https://doi.org/10.4171/owr/2021/5

05819 | OWR-1730-005

### Galois Actions on Moduli of General-Type Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

For the absolute Galois group $\operatorname{Gal}(\overline{\mathbb Q}/\mathbb Q)$ acting on moduli spaces of minimal complex surfaces of general type, which topological invariants of a variety over $\overline{\mathbb Q}$ are preserved under Galois conjugation? Is the action on the union $\mathcal M$ of these moduli spaces faithful, and is it faithful on the rigid surfaces?

来源字段：https://doi.org/10.4171/owr/2007/47

05820 | OWR-1703867-008

### Galois Descent of Special Hodge Subvarieties

O / open | Oberwolfach 报告问题汇编

别名：Galois Arithmetic of Special Subvarieties in Hodge Variations [OWR-1703871-003]

If a polarized variation of integral Hodge structure is defined over a number field $K$, is every associated special subvariety defined over a finite extension of $K$, with all of its finitely many $\operatorname{Gal}(\overline{\mathbb Q}/K)$-conjugates again special?

来源字段：https://doi.org/10.4171/owr/2020/20

05821 | OWR-11127-008

### Galois Images in Genus-Two Real-Multiplication Families

P / partially_solved | Oberwolfach 报告问题汇编

For the family of genus-$2$ curves

$$C_t:\quad y^2=x^5-5x^3+5x+t,$$

which has real multiplication by $\mathbb Z[(1+\sqrt5)/2]$, in what sense is this one-dimensional family special inside the two-dimensional real-multiplication moduli space? Which level structure does it fix, can an exceptional integer $N$ make its mod-$N$ Galois image smaller than expected, and what is the image on the Tate module?

来源字段：https://doi.org/10.4171/owr/2011/35

05822 | AIM-ARITHMETIC_GEOMETRY-0053

### Galois-moving point-Hilbert components via constant-field transfer

P / partially_solved | AIM 研讨会问题汇编

**Problem 6.** (1) Give an explicit example (or show it does not happen) of a geometrically irreducible component of \(\operatorname{Hilb}^{d}(\mathbb A^{n})\), which is not fixed under the action of \(\operatorname{Gal}(\overline{\mathbb Q}/\mathbb Q)\). (2) Same question for the Hilbert schemes of curves in \(\mathbb P^{3}\).

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05823 | AIM-ALGEBRAIC_GEOMETRY-0189

### Gauge-correct cheirality, convex separation, and finite certificates

S / solved | AIM 研讨会问题汇编

**For a reconstruction what does it mean for all the points to be “in front” of the cameras? Can this easily be detected.**

来源字段：http://aimpl.org/algvision/1/

05824 | AIM-ARITHMETIC_GEOMETRY-0050

### Gauss-inseparability strata and a Frobenius plane-curve family

P / partially_solved | AIM 研讨会问题汇编

Fix \(d,g,r,e>0\) in characteristic \(p\). Let \[ \operatorname{Hilb}^{\mathrm{sm}}_{d,g}(\mathbb P^r) \subset \operatorname{Hilb}_{d,g}(\mathbb P^r) \] be the open subscheme parametrizing smooth curves. For \([C]\in\operatorname{Hilb}^{\mathrm{sm}}_{d,g}(\mathbb P^r)\), let \[ \gamma_C:C\longrightarrow \operatorname{Gr}(1,r) \] be the Gauss map sending a point to its embedded tangent line, and define \[ Z_e=\left\{[C]\in\operatorname{Hilb}^{\mathrm{sm}}_{d,g}(\mathbb P^r): \deg_{\mathrm{ins}}(\gamma_C)=p^e\right\}. \] Then \[ \bigcup_{e\ge 0}Z_e=\operatorname{Hilb}^{\mathrm{sm}}_{d,g}(\mathbb P^r). \] The questions are: what can be said about \(Z_e\); can this stratification construct exotic components existing only in characteristic \(p\); and how does \[ \operatorname{Gal}(\overline{\mathbb F}_p/\mathbb F_p) \] act? The source also suggests stratification by the topological type of the ramification divisor.

The phrase “smooth curves” is interpreted below as smooth, geometrically integral, nondegenerate curves. This is the standard setting in which the Gauss map has one function-field extension and hence one inseparable degree. For a disconnected smooth curve the definition must instead be made componentwise.

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05825 | AMR-071-0008

### General elephant problem

O / open | AMR 问题清单汇编

General elephant problem: do general elephants have at most Du Val singularities?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

05826 | AIM-ALGEBRAIC_GEOMETRY-0259

### General type in families: a counterexample and a compatible-index nef criterion

S / solved | AIM 研讨会问题汇编

**General type in families.** Suppose that \(\mathcal X\to\Delta\) is a family, and a special fiber is of general type. Does it follow that a general fiber is of general type?

来源字段：http://aimpl.org/minimalmodcharp/4/

05827 | OWR-2049-003

### General Type of Orthogonal Modular Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Let $L$ be an even integral lattice of signature $(2,n)$, and let $\mathcal{F}(L)$ be its modular variety. Is $\mathcal{F}(L)$ of general type for all sufficiently large $n$, in particular for every $n\geq36$?

来源字段：https://doi.org/10.4171/owr/2008/38

05828 | OWR-1971-001

### Generalized Hodge Conjecture

O / open | Oberwolfach 报告问题汇编

Let $X$ be a smooth complex projective variety, and let $L\subset H^k(X,\mathbb Q)$ be a rational sub-Hodge structure of coniveau $c$. Does there exist a closed algebraic subset $Z\subset X$ of codimension $c$ such that $L$ maps to zero under $H^k(X,\mathbb Q)\to H^k(X\setminus Z,\mathbb Q)$?

来源字段：https://doi.org/10.4171/owr/2008/26

05829 | OWR-1452-025

### Generalized Tameness of an Explicit Polynomial Automorphism

O / open | Oberwolfach 报告问题汇编

Let $\Delta=x^2y+z^2$ and define $\gamma\in\operatorname{Aut}\mathbb{C}[x,y,z]$ by $\gamma(x)=x$, $\gamma(z)=z+x\Delta$, and $\gamma(y)=y-2xyz-\Delta^2+2(3z^2\Delta+3xz\Delta^2+x^2\Delta^3)$. Is $\gamma$ generalized tame, that is, in the subgroup generated by tame and exponential automorphisms?

来源字段：https://doi.org/10.4171/owr/2007/01

05830 | OWR-15178-017

### Generator Bounds for Gorenstein Monomial Curve Ideals

O / open | Oberwolfach 报告问题汇编

Are the numbers of generators of the defining ideals of affine Gorenstein monomial curves in $\mathbb{A}^n$ bounded independently of n and the curve?

来源字段：https://doi.org/10.4171/owr/2016/57

05831 | OWR-14754-013

### Generators and Support Loci of Bernstein-Sato Ideals

O / open | Oberwolfach 报告问题汇编

For $F=(f_1,\ldots,f_r):(\mathbb C^n,0)\to(\mathbb C^r,0)$, is the Bernstein-Sato ideal $B_F$ generated by products of linear polynomials $a_1s_1+\cdots+a_rs_r+b$, where $a_i\in\mathbb Q_{\geq0}$ and $b\in\mathbb Q_{>0}$? Is $\operatorname{Exp}(Z(B_F))$ equal to the cohomology-support locus $V_F$?

来源字段：https://doi.org/10.4171/owr/2016/46

05832 | OWR-13946-008

### Generic Finiteness of K3 Severi Moduli Maps

P / partially_solved | Oberwolfach 报告问题汇编

For a general polarized K3 surface $(X,L)$ and an irreducible component $V\subset V_{n,\delta}(X)$, is the moduli map $\psi_{X,n,\delta}|_V$ generically finite, or birational when $g>1$, onto its image in $M_g$?

来源字段：https://doi.org/10.4171/owr/2015/51

05833 | OWR-14299574-002

### Generic Linear Projections and Dimensionality Type

O / open | Oberwolfach 报告问题汇编

For singularities of dimensionality type at least three, is a generic linear projection sufficiently generic that its discriminant computes the dimensionality type?

来源字段：https://doi.org/10.4171/owr/2025/45

05834 | AIM-ARITHMETIC_GEOMETRY-0061

### Generic points via elementary support factorization

P / partially_solved | AIM 研讨会问题汇编

Give a geometric and algebraic description of the generic points of the irreducible components of \(\operatorname{Hilb}^d(\mathbb A^n)\).

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05835 | OWR-1276-004

### Generic Smoothness of Homomorphism Schemes

P / partially_solved | Oberwolfach 报告问题汇编

For a smooth curve $X$ of general moduli and a general member $V'$ of the Kuranishi family of $V$, is every component of $\operatorname{Hom}(X,V')$ generically smooth? Is every component of $\operatorname{Hom}(X,G/P)$ generically smooth?

来源字段：https://doi.org/10.4171/owr/2006/27

05836 | OWR-16162-018

### Generic Splitting Varieties for Pointed Functors

O / open | Oberwolfach 报告问题汇编

Let $k$ be a field, let $S$ be a functor from field extensions of $k$ to pointed sets, and let $s\in S(k)$. Can one construct a smooth geometrically integral $k$-variety $X$ such that, for every field extension $l/k$, the restriction $s_l$ is trivial if and only if $X(l)$ is nonempty, with $X(l)$ Zariski-dense in $X$ whenever it is nonempty?

来源字段：https://doi.org/10.4171/owr/2018/21

05837 | AIM-ALGEBRAIC_GEOMETRY-0249

### Generic-point abelianity and the naive-versus-genuine SU(2) correction

P / partially_solved | AIM 研讨会问题汇编

**Question (Anastasia Stavrova).** When \(G\) is a linear reduced/algebraic group, in a large number of cases, it is known on a case-by-case basis that \(\Pi^{\mathbb A^1}_0(G)\) is abelian. Is this always true? Can this be obtained in a uniform way when it is true?

**Remark (Madhav Nori).** This is not always true e.g. \(G=SU(2,\mathbb R)\). In fact, the same reasoning should show that \(\Pi^{\mathbb A^1}_0(G)\) is not abelian for any anisotropic group \(G\) over \(\mathbb R\).

来源字段：https://aimath.org/pastworkshops/projectiveA1problems.pdf

05838 | AIM-ALGEBRAIC_GEOMETRY-0086

### Generic-support obstruction and rank-one central Rouquier powers

P / partially_solved | AIM 研讨会问题汇编

What is the Drinfeld center of \(K^b(\mathbb{S}\mathrm{Bim}_{\mathrm{fin}})\)? \(K^b(\mathbb{S}\mathrm{Bim}_{\mathrm{aff}})\)? Are there additional complexes in \(K^b(\mathbb{S}\mathrm{Bim})\) which commute with \(\mathbb{S}\mathrm{Bim}\)?

来源字段：http://aimpl.org/catheckehilbert/4/

05839 | AIM-ARITHMETIC_GEOMETRY-0052

### Generically nonreduced components of Hilbert schemes of points

S / solved | AIM 研讨会问题汇编

Is there a nonreduced component of \(\operatorname{Hilb}^{d}(\mathbb A^n)\)? If so, find it.

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05840 | AIM-ALGEBRAIC_GEOMETRY-0172

### Genuine versus normed equivariant elliptic cohomology and a cyclic Frobenius constraint

P / partially_solved | AIM 研讨会问题汇编

For a finite group \(G\), Lurie constructs equivariant elliptic cohomology in commutative naive \(G\)-spectra and in genuine \(G\)-spectra compatibly (Lurie 2009, MR2597740).

Can this construction be extended to commutative genuine \(G\)-spectra or \(G\)-commutative genuine \(G\)-spectra?

来源字段：http://aimpl.org/equideralggeom/5/

05841 | AIM-ALGEBRAIC_GEOMETRY-0107

### Genus-two density and a fine-level Hecke propagation criterion

P / partially_solved | AIM 研讨会问题汇编

**Problem.** Are complete rational curves in \(\mathcal A_g\) dense in characteristic \(p\)?

来源字段：http://aimpl.org/ratsubvarpos/4/

05842 | AIM-ARITHMETIC_GEOMETRY-0033

### Geometric 1-motives versus integral Jacobian splittings

P / partially_solved | AIM 研讨会问题汇编

(a) Let \(K\) be an algebraically closed field, \(X/K\) an abelian variety. The Voevodsky motive \(F\) is given as a sheaf of groups \(F(S)=\operatorname{Mor}_K(S,X)\). Is \(F\) geometric (i.e. in the triangulated subcategory generated by smooth projective varieties)?

(b) (This is a stronger variant of (a)): Is every principally polarized abelian variety a direct summand of a Jacobian integrally (as an abelian variety)?

来源字段：http://aimpl.org/cohomabelian/1/

05843 | OWR-16158-016

### Geometric Chern Classes in Arakelov Cobordism

O / open | Oberwolfach 报告问题汇编

Give a geometric interpretation of the Chern classes in Arakelov algebraic cobordism.

来源字段：https://doi.org/10.4171/owr/2018/17

05844 | OWR-13946-003

### Geometric Description of Involutions on K3 Hilbert Squares

P / partially_solved | Oberwolfach 报告问题汇编

Let $S$ be a generic projective K3 surface with $\operatorname{Pic}(S)=\mathbb ZH$ and $H^2=20$. Find a geometric description of the nontrivial nonsymplectic involution on the Hilbert square $S^{[2]}$.

来源字段：https://doi.org/10.4171/owr/2015/51

05845 | OWR-3471-005

### Geometric Information Encoded by Seshadri Constants

O / open | Oberwolfach 报告问题汇编

Determine what geometric information Seshadri constants encode, and find upper and lower bounds or explicit values for them on surfaces and higher-dimensional projective varieties.

来源字段：https://doi.org/10.4171/owr/2009/21

05846 | OWR-13952-009

### Geometric Langlands Sheaves for Discrete Parameters

O / open | Oberwolfach 报告问题汇编

For a discrete Langlands parameter $\varphi:W_E\to{}^LG$, does there exist a perverse Weil sheaf $F_\varphi$ on $\operatorname{Bun}_G\otimes\mathbb F_q$ with an $S_\varphi$-action satisfying the prescribed central-character, L-packet, Hecke-eigenvalue, and local-global compatibility properties?

来源字段：https://doi.org/10.4171/owr/2015/57

05847 | AIM-ALGEBRAIC_GEOMETRY-0042

### Geometric-fixed-point sparsity in the orbit--Koszul Bockstein filtration

P / partially_solved | AIM 研讨会问题汇编

**The $(G\cdot\overline v_n)$-Bockstein spectral sequence for $\mathrm{BP}^{((G))}\langle n\rangle$.** $\mathrm{BP}^{((G))}\langle n-1\rangle$ is the quotient of $\mathrm{BP}^{((G))}\langle n\rangle$ by $(G\cdot\overline v_n)$. Can one understand the $(G\cdot\overline v_n)$-Bockstein spectral sequence for this quotient map?

The preceding problem in the same source section explicitly works with $C_{2^r}$. Accordingly, and consistently with the primary literature, I recover $G=C_{2^r}$, with $r\geq 1$. I use $m$ for the truncation index so that it is not confused with the exponent $r$ in the group order. Put

$$ q=[G:C_2]=2^{r-1},\qquad R_m=\mathrm{BP}^{((G))}\langle m\rangle. $$

$$ \mathrm{BP}^{((G))}=N_{C_2}^G\mathrm{BP}_{\mathbb R} $$

and one can choose classes

$$ \bar t_i^G\in \pi_{(2^i-1)\rho_2}^{C_2}\mathrm{BP}^{((G))} $$

whose restrictions $t_i^G$ give polynomial generators. Modern papers often write $\bar t_i^G$, while the AIM record writes $\bar v_i$. The precise construction below only requires a chosen generator of the corresponding permutation summand; it does not require identifying every possible choice of $\bar v_i$ with $\bar t_i^G$.

If $\gamma$ generates $G$, then

$$ G\cdot\bar t_i^G= \{\bar t_i^G,\gamma\bar t_i^G,\ldots,\gamma^{q-1}\bar t_i^G\}. $$

This notation denotes a **list of $q$ orbit generators**, not their product. The signed relation is $\gamma^q\bar t_i^G=-\bar t_i^G$, so it does not add another polynomial generator. By contrast, $N_{C_2}^G(\bar t_i^G)$ denotes the multiplicative norm, whose underlying class is the product of the orbit elements up to the standard norm conventions.

The consistent truncation convention is

$$ R_m=\mathrm{BP}^{((G))}/ \bigl(G\cdot\bar t_{m+1}^G,G\cdot\bar t_{m+2}^G,\ldots\bigr), \qquad R_{m-1}\simeq R_m/(G\cdot\bar t_m^G). \tag{1.1} $$

Carrick--Hill, arXiv:2507.07051v1, gives (1.1) in its introductory definition and again states the quotient relation after Definition 4.3. The displayed formula inside Definition 4.3 itself starts the colimit at $\bar t_m^G$; this is an apparent off-by-one typo, since it contradicts both the earlier definition and the immediately following quotient relation. I use the internally consistent convention (1.1).

来源字段：http://aimpl.org/equivstable/4/

05848 | OWR-11133-001

### Geometry Encoded by Newton-Okounkov Bodies

O / open | Oberwolfach 报告问题汇编

What geometric data of a pair $(X,D)$ are encoded, and how, by the combinatorics of its Newton-Okounkov bodies?

来源字段：https://doi.org/10.4171/owr/2011/41

05849 | OWR-11783-011

### Geometry of Cones of $q$-Ample Line Bundles

O / open | Oberwolfach 报告问题汇编

What is a useful geometric description of the generally nonconvex cone of $q$-ample line bundles on a variety?

来源字段：https://doi.org/10.4171/owr/2012/21

05850 | OWR-2049-009

### Global Generation of Relative Pluricanonical Pushforwards

P / partially_solved | Oberwolfach 报告问题汇编

Let $f:X\to Y$ be an algebraic fiber space between smooth projective varieties, with $f$ surjective and fibers connected. Is $f_*\mathcal{O}_X(mK_{X/Y})$ globally generated outside the discriminant locus of $f$ for every sufficiently large $m$?

来源字段：https://doi.org/10.4171/owr/2008/38

05851 | OWR-12334-010

### Global Quotient Presentations of Algebraic Stacks

O / open | Oberwolfach 报告问题汇编

Which algebraic stacks are global quotient stacks, and in particular is every separated Deligne-Mumford stack a global quotient stack?

来源字段：https://doi.org/10.4171/owr/2013/06

05852 | OWR-15987-011

### Global Resolvent Degree of General Polynomial Equations

O / open | Oberwolfach 报告问题汇编

Determine the global resolvent degree of solving general polynomial equations, including whether the parameter reductions conjectured by Hilbert are optimal.

来源字段：https://doi.org/10.4171/owr/2018/2

05853 | OWR-11101913-006

### Global Weinstein Splitting for Poisson Manifolds

O / open | Oberwolfach 报告问题汇编

Under what conditions does a Poisson manifold $X$ decompose globally, perhaps after passing to a suitable covering space, as a product of Poisson manifolds with a symplectic leaf as one factor?

来源字段：https://doi.org/10.4171/owr/2022/32

05854 | OWR-824-005

### Gorenstein Liaison of ACM Subschemes

P / partially_solved | Oberwolfach 报告问题汇编

Is every ACM subscheme of $\mathbb{P}^n$ in the Gorenstein liaison class of a complete intersection, that is, glicci?

来源字段：https://doi.org/10.4171/owr/2005/19

05855 | OWR-15178-016

### Gorenstein Monomial Curves from Principal Matrices

P / partially_solved | Oberwolfach 报告问题汇编

Can one characterize Gorenstein monomial curves in $\mathbb{A}^5$ defined by relatively prime positive integers $a_1,\ldots,a_5$ in terms of their principal matrix?

来源字段：https://doi.org/10.4171/owr/2016/57

05856 | OWR-14601-005

### Gorenstein Secants, Normal Ball Polytopes, and Gradient Images

O / open | Oberwolfach 报告问题汇编

Is the affine cone over the second secant variety of $(P^1)^5$ Gorenstein? Is the convex hull of all lattice points in any ball a normal polytope? For a homogeneous polynomial $f$ in $n+1$ variables and a generic subspace $L$ of $P^n$, is $L^\perp$ intersected with the gradient image of $L$ empty?

来源字段：https://doi.org/10.4171/owr/2016/19

05857 | OWR-1730-002

### Graded Dimensions of Tautological Rings of General Curves

O / open | Oberwolfach 报告问题汇编

For a general curve $C$ of genus $g$, does the tautological ring $R$ satisfy $\dim_{\mathbb Q}R=p(g+1)$ and $\dim R^i_{(j)}=p(i,g+1-i,j)$, where $p(i,g+1-i,j)$ counts partitions of $i$ with $i-j$ parts, all at most $g+1-i$?

来源字段：https://doi.org/10.4171/owr/2007/47

05858 | AIM-ARITHMETIC_GEOMETRY-0065

### Grading and resolution-dependence obstructions to higher-dimensional Nakajima operators

P / partially_solved | AIM 研讨会问题汇编

**Problem 18.** Is it possible to define analogues of the Nakajima operators for the cohomology of a desingularization of the smoothable component of \(\operatorname{Hilb}^{d}(\mathbb A^{n})\)?

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05859 | AIM-ALGEBRAIC_GEOMETRY-0214

### Graph spaces and multi-Rees models for continuous multiview geometry

P / partially_solved | AIM 研讨会问题汇编

What is the proper algebraic formulation for a continuous multiview variety? \(\mathbb{P}^1 \times \mathbb{P}^3 \to (\mathbb{P}^2)^n\)? \(Hom(\mathbb{A}^1,\text{essential matrices})\)? Kontsevich spaces? Or could we use points and optical flow data?

来源字段：http://aimpl.org/algvision/6/

05860 | OWR-13101-001

### Green's Syzygy Conjecture for Arbitrary Curves

O / open | Oberwolfach 报告问题汇编

Does Green's conjecture on the higher syzygies of canonical ideals of curves hold for all curves, beyond the generic cases?

来源字段：https://doi.org/10.4171/owr/2014/31

05861 | AIM-ALGEBRAIC_GEOMETRY-0154

### Green-functor Witt vectors for relative THH

S / solved | AIM 研讨会问题汇编

Hesselholt and Madsen proved that, for a commutative ring \(R\), \[ \pi _0\bigl(\operatorname{THH}(HR)^{C_{p^n}}\bigr) \cong W_{n+1}(R). \] Angeltveit--Blumberg--Gerhardt--Hill--Lawson--Mandell constructed the cyclotomic structure using the circle norm \(N_e^{S^1}\). If \(\underline R\) is a commutative \(C_p\)-Green functor, what is a Witt-vector model for \[ \pi _0\!\left(\left(N_{C_p}^{S^1}H\underline R\right)^{C_{p^n}}\right)? \]

来源字段：http://aimpl.org/equideralggeom/2/

05862 | OWR-7155446-002

### Green-Griffiths-Lang Conjecture for General-Type Varieties

P / partially_solved | Oberwolfach 报告问题汇编

If $X$ is a complex projective variety of general type, does there exist a proper algebraic subvariety $Y$ of $X$ containing every nonconstant holomorphic map $f:\mathbb{C}\to X$, with the smallest such $Y$ expected over number fields to coincide with the Mordell locus?

来源字段：https://doi.org/10.4171/owr/2021/25

05863 | OWR-11139-008

### Green-Griffiths-Lang for Directed Varieties

O / open | Oberwolfach 报告问题汇编

If $(X,V)$ is a projective directed manifold whose canonical sheaf $K_V$ is big, must there exist a proper algebraic subvariety $Y\subset X$ containing the image of every nonconstant entire curve $\mathbb C\to X$ tangent to $V$?

来源字段：https://doi.org/10.4171/owr/2011/47

05864 | OWR-4527-004

### Green-Lazarsfeld Gonality Conjecture

S / solved | Oberwolfach 报告问题汇编

For a sufficiently positive line bundle $L$ on a smooth curve $C$ of gonality $\delta$, do the nonzero linear syzygies $K_{p,1}(C,L)$ occur exactly for $1\leq p\leq r-1-\delta$, where $r=h^{0}(C,L)-1$ (the Green-Lazarsfeld gonality conjecture)?

来源字段：https://doi.org/10.4171/owr/2010/45

05865 | OWR-16408-002

### Green’s Canonical Syzygy Conjecture for Arbitrary Curves

O / open | Oberwolfach 报告问题汇编

For every nonhyperelliptic canonically embedded smooth curve $C$ of genus $g$, is

$$K_{i,1}(C,\omega_C)=0$$

equivalent to $i\geq g-\operatorname{Cliff}(C)-1$?

来源字段：https://doi.org/10.4171/owr/2018/33

05866 | OWR-12010-003

### Grothendieck Section Conjecture for Hyperbolic Curves

O / open | Oberwolfach 报告问题汇编

For a number field $k$ and a smooth, geometrically connected curve $X$ over $k$ with nonabelian geometric fundamental group, is the map from $X(k)$ to conjugacy classes of sections of the fundamental-group exact sequence a bijection onto the noncuspidal sections?

来源字段：https://doi.org/10.4171/owr/2012/38

05867 | OWR-14211-011

### Grothendieck-Serre Conjecture in Mixed Characteristic

P / partially_solved | Oberwolfach 报告问题汇编

Grothendieck-Serre conjecture: if $R$ is a regular local integral domain with fraction field $F$, then for every smooth reductive group scheme $G$ over $R$, the map $H^1_{\mathrm{et}}(R,G)\to H^1_{\mathrm{et}}(F,G)$ is injective.

来源字段：https://doi.org/10.4171/owr/2016/3

05868 | OWR-16158-006

### Grothendieck’s Anabelian Conjectures over Finitely Generated Fields

P / partially_solved | Oberwolfach 报告问题汇编

Let $k/\mathbb Q$ be a finitely generated field extension. Are smooth hyperbolic curves over $k$ anabelian? Does every point of a smooth $k$-variety have a basis of anabelian Zariski neighborhoods? Can $X(k)$ be reconstructed from $\pi_1^{\acute et}(X)$ together with $G_k$?

来源字段：https://doi.org/10.4171/owr/2018/17

05869 | OWR-14298795-004

### Grothendieck’s Section Conjecture

O / open | Oberwolfach 报告问题汇编

Let $X$ be a proper hyperbolic curve over a finitely generated extension $k/\mathbb Q$. Is the canonical map from $X(k)$ to the conjugacy classes of sections of

$$1\to\pi_1^{\mathrm{et}}(X_{\bar k})\to\pi_1^{\mathrm{et}}(X)\to\operatorname{Gal}(\bar k/k)\to1$$

a bijection?

来源字段：https://doi.org/10.4171/owr/2024/45

05870 | OWR-15955-005

### Group Extensions Realized by Kodaira Fibrations

P / partially_solved | Oberwolfach 报告问题汇编

Characterize the extensions $$1\longrightarrow\pi_1(F_b)\longrightarrow G\longrightarrow\pi_1(B)\longrightarrow1,$$ where $B$ and $F_b$ are curves, for which $G$ is the fundamental group of a Kodaira fibration.

来源字段：https://doi.org/10.4171/owr/2017/50

05871 | OWR-12697689-004

### Gröbner-Cell Parametrization of Punctual Hilbert Schemes

O / open | Oberwolfach 报告问题汇编

Let $E\subset k[[x,y]]$ be a zero-dimensional monomial ideal, let $T'(E)$ be the parameter space of admissible perturbation matrices of its canonical Hilbert-Burch matrix $H$, and let $V(E)$ be the Gröbner cell of ideals with initial ideal $E$. Is the map $$\Phi_E:T'(E)\longrightarrow V(E),\qquad N\longmapsto I_t(H+N),$$ a bijection?

来源字段：https://doi.org/10.4171/owr/2023/6

05872 | OWR-17296-016

### Harbourne Constants, Symbolic Powers, and Resurgence of Line Arrangements

O / open | Oberwolfach 报告问题汇编

For complex line arrangements $C$, must $H(C)=\widetilde C^2$ divided by the number of singular points be at least $-225/67$, and what is the infimum of $H(C)$ over rational line arrangements? Also, which finite sets $Z\subset\mathbb{P}^2$ satisfy $I_Z^{(3)}\not\subset I_Z^2$, and can the resurgence of the singular-point ideal be computed for line arrangements, at least for supersolvable ones?

来源字段：https://doi.org/10.4171/owr/2019/53

05873 | OWR-1703867-002

### Hard Lefschetz for Arbitrary $\ell$-Adic Sheaves

O / open | Oberwolfach 报告问题汇编

Does hard Lefschetz hold for every $\ell$-adic local system $\mathcal F$ on a polarized smooth projective $d$-fold $X$; namely, is $\eta^i\cup-:H^{d-i}(X,\mathcal F)\to H^{d+i}(X,\mathcal F)$ an isomorphism?

来源字段：https://doi.org/10.4171/owr/2020/20

05874 | OWR-17289-013

### Hasse-Graph Descriptions of Valuation Semigroups

O / open | Oberwolfach 报告问题汇编

Is the weighted-Hasse-graph description of the valuation semigroup $\Gamma_C$ valid whenever, for every cover relation $p\gtrdot q$ of weight $b$, there is an $\eta\in K(\widehat Y_p)$ satisfying $$ \eta^b=\frac{f_p}{f_q} $$ in $K(\widehat Y_p)$?

来源字段：https://doi.org/10.4171/owr/2019/46

05875 | OWR-3393-001

### Heart-Independent Hochschild Characteristic Homomorphisms

P / partially_solved | Oberwolfach 报告问题汇编

If $\mathcal A$ is the heart of a $t$-structure in a $K$-linear triangulated category $\mathcal T$, does there exist a characteristic homomorphism from $HH^*(\mathcal A)$ to the graded centre of $\mathcal T$, and is it heart-independent? In the geometric setting, does the analogue hold for an analytic space or scheme $X$, replacing $D(\mathcal A)$ by $D(X)$ and using $HH^*(X)$?

来源字段：https://doi.org/10.4171/owr/2009/10

05876 | OWR-12697710-013

### Heat Flows into Sum-of-Squares Cones

P / partially_solved | Oberwolfach 报告问题汇编

For which $d\in\mathbb{N}$ does heat flow send every positive binary form of degree $2d$ into the sum-of-squares cone after a uniform time, and which linear flows have the analogous property for general $(n,d)$?

来源字段：https://doi.org/10.4171/owr/2023/14

05877 | OWR-12697688-006

### Hecke-Orbit Density in Hodge-Type Newton Strata

P / partially_solved | Oberwolfach 报告问题汇编

For a Hodge-type Shimura variety with good reduction at $p$, prove that every isogeny class is Zariski dense in its Newton stratum, without restrictions relating $p$ to the Shimura datum.

来源字段：https://doi.org/10.4171/owr/2023/5

05878 | OWR-15582-004

### Heegner Divisors and the Pseudo-Effective Cone

O / open | Oberwolfach 报告问题汇编

Is the cone generated by primitive Heegner divisors on $\mathcal F_{2d}$ a strict subcone of the pseudo-effective cone $\operatorname{Eff}(\mathcal F_{2d})$, or do the two cones coincide?

来源字段：https://doi.org/10.4171/owr/2017/39

05879 | OWR-11580-012

### Height Bounds for Covering Relations in Wonderful Varieties

O / open | Oberwolfach 报告问题汇编

Prove that the positive height $\operatorname{ht}^+(\gamma)$ of every covering relation $\gamma\in\mathbb N[\Sigma]$ is at most $2$.

来源字段：https://doi.org/10.4171/owr/2012/13

05880 | AIM-ARITHMETIC_GEOMETRY-0044

### Henselian reduction and exact value-group obstructions for the torsion index

P / partially_solved | AIM 研讨会问题汇编

When \(F=\mathbb R\), \(u(F)=\infty\) but \(\tau_F<\infty\). Is there an example where this holds for \(F\) not formally real?

来源字段：http://aimpl.org/deformationbrauer/1/

05881 | OWR-14298588-003

### Higgs Bundles Arising from p-Adic Galois Representations

O / open | Oberwolfach 报告问题汇编

Under the stated $p$-adic Simpson equivalence, characterize the Higgs bundles that correspond to $p$-adic Galois representations through the designated embedding, beyond the known cases of line bundles and abeloid varieties.

来源字段：https://doi.org/10.4171/owr/2024/33

05882 | OWR-9790356-001

### Higgs Bundles Arising from p-Adic Representations

P / partially_solved | Oberwolfach 报告问题汇编

Characterize the Higgs bundles that correspond to genuine representations under the $p$-adic Simpson correspondence.

来源字段：https://doi.org/10.4171/owr/2022/5

05883 | OWR-14298162-005

### Higher Fano Manifolds as Higher Rational-Connectedness Conditions

O / open | Oberwolfach 报告问题汇编

For every $k$, prove that $k$-Fano manifolds are covered by rational $k$-folds and that a proper family over a $k$-dimensional base whose fibers are $k$-Fano admits a rational section, modulo the Brauer obstruction.

来源字段：https://doi.org/10.4171/owr/2024/14

05884 | AIM-ARITHMETIC_GEOMETRY-0040

### Higher u-invariants and a flat-profile criterion

P / partially_solved | AIM 研讨会问题汇编

Define \[ u_k(F)=\max\{\dim(q)\mid q\in I^k\}. \] Then \(u_0=u\). Can we compute \(u_k(F)\)? How about \(u_3(F)\)?

来源字段：http://aimpl.org/deformationbrauer/1/

05885 | OWR-12870-002

### Higher-Dimensional Bounded Negativity for Algebraic Cycles

O / open | Oberwolfach 报告问题汇编

Which natural higher-dimensional generalizations of the Bounded Negativity Conjecture hold for cycles on smooth projective varieties?

来源字段：https://doi.org/10.4171/owr/2014/10

05886 | OWR-1452-014

### Higher-Dimensional Cancellation for Affine Domains

P / partially_solved | Oberwolfach 报告问题汇编

For an algebraically closed field $k$ and a finitely generated $k$-domain $A$ of dimension $d>2$, does $A^{[m]}\simeq k^{[m+d]}$ for some $m>0$ imply $A\simeq k^{[d]}$?

来源字段：https://doi.org/10.4171/owr/2007/01

05887 | AIM-ALGEBRAIC_GEOMETRY-0212

### Higher-dimensional Carlsson-Weinshall duality and its rational-normal center fibers

P / partially_solved | AIM 研讨会问题汇编

What can we say about Carlsson-Weinshall duality for other dimensions like \(\mathbb P^4\to\mathbb P^3\)?

来源字段：http://aimpl.org/algvision/5/

05888 | OWR-14222-009

### Higher-Dimensional Theory of Unexpected Curves

O / open | Oberwolfach 报告问题汇编

Develop a higher-dimensional analogue of the logarithmic-vector-field exact sequence used to characterize unexpected plane curves, and determine whether it yields corresponding characterizations of unexpected hypersurfaces.

来源字段：https://doi.org/10.4171/owr/2016/14

05889 | OWR-14299284-005

### Higher-Genus Gromov-Witten Invariants of Elliptic Targets

P / partially_solved | Oberwolfach 报告问题汇编

How can the higher-genus descendant Gromov-Witten invariants $\langle\Lambda\tau_k(0)\rangle^{\pi,d}_{g,1}$ of the universal elliptic curve be computed when degeneration of the target to a rational curve is unavailable?

来源字段：https://doi.org/10.4171/owr/2025/28

05890 | AIM-ARITHMETIC_GEOMETRY-0001

### Higher-weight special cycles: known cohomological modularity and a rational non-neat descent criterion

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.1.** Is it possible to define a generating series of special cycles over Shimura varieties related to automorphic representations of higher weight? Do we have similar geometric/arithmetic modularity and Siegel-Weil formulas as in the Kudla program?

来源字段：http://aimpl.org/intersectshimura/1/

05891 | HIL-015

### Hilbert's 15th Problem: Schubert's Enumerative Calculus

P / partially_solved | Hilbert 问题

别名：Hilbert's Fifteenth Problem [ALG-018]

Rigorously justify Schubert's enumerative geometry.

背景链接：https://bookstore.ams.org/CTM/11

05892 | OWR-4127-003

### Hilbert-Square Zero Loci of Homogeneous Bundles

P / partially_solved | Oberwolfach 报告问题汇编

Let $V_{10}$ be a $10$-dimensional complex vector space, let $\sigma \in \bigwedge^3 V_{10}^*$ be a general $3$-form defining a Debarre--Voisin fourfold $Y_\sigma \subset G(6, V_{10})$ (the zero locus of the global section $\sigma$ of $\bigwedge^3 \mathcal{U}^*$), and let $S^{[2]}$ be the Hilbert square of a degree-$22$ K3 surface $S$. Find explicit sections $\sigma$ for which $Y_\sigma \cong S^{[2]}$, and find other globally generated homogeneous vector bundles on rational homogeneous varieties whose general zero locus is an irreducible holomorphic symplectic manifold.

来源字段：https://doi.org/10.4171/owr/2009/44

05893 | AMR-114-0001

### Hirzebruch's 1954 Problem List - Problem 1

O / open | AMR 问题清单汇编

At the beginning of June,1953, Hirzebruch proved what he then called the ``index theorem'': $$ \tau (M) = \langle L(p_1,\ldots,p_k),[M]\rangle \ , $$ where $M$ is any closed smooth oriented manifold, $\tau (M)$ is its index or signature, and $L(p_1,\ldots,p_k)$ is the $L$-polynomial in the Pontrygain classes of $M$. Later on the term ``index'' became monopolized by the Fredholm index appearing in the Atiyah--Singer index theorem, and so $\tau (M)$ is now usually called the signature, and the ``index theorem'' has become the Hirzebruch signature theorem.

Give an elementary proof of the index theorem and explain the geometrical meaning of the functional equation valid for $\tau$.

背景链接：https://arxiv.org/abs/1305.4623

05894 | AMR-114-0002

### Hirzebruch's 1954 Problem List - Problem 2

O / open | AMR 问题清单汇编

A Pfaffian equation in $M^{2k+1}$ can be locally reduced to the normal form $$ dx_{2k+1}+x_1dx_{k+1}+\ldots+x_rdx_{k+r}=0\ , \ \ \ \ 0\leq r\leq k\ , $$ where $r$ is the only local invariant, $2r+1$ being what is usually called the class of the local Pfaffian equation. The number $r$ being fixed, give a criterion to determine whether $M^{2k+1}$ admits a Pfaffian equation in the large whose class is $2r+1$ everywhere.

背景链接：https://arxiv.org/abs/1305.4623

05895 | AMR-114-0003

### Hirzebruch's 1954 Problem List - Problem 3

O / open | AMR 问题清单汇编

Let $n+d(n)$ be the minimum dimension of an Euclidean space in which the $n$-dimensional real projective space $\mathbb{P}^n$ can be differentiably imbedded. H. Hopf proved that (even for topological imbedding) $d(n)>1$; Chern proved that $d(n)>2$ if $n\neq 2^k-1$ ($k\geq 2$) and $n\neq 2^k-2$ ($k\geq 2$). Can these bounds be improved? In particular, can $\mathbb{P}^6$ be imbedded in an Euclidean space of $8$ dimensions?

背景链接：https://arxiv.org/abs/1305.4623

05896 | AMR-114-0004

### Hirzebruch's 1954 Problem List - Problem 4

O / open | AMR 问题清单汇编

A closed surface $M$ in the $4$-dimensional Euclidean space ${\mathbb{E}}^4$ has an induced Riemannian metric and hence a Gaussian curvature $K$. Does there exist such a surface for which $K<0$ everywhere? In particular, does there exist a surface of genus $2$ in ${\mathbb{E}}^4$ for which $K<0$ everywhere?

背景链接：https://arxiv.org/abs/1305.4623

05897 | AMR-114-0005

### Hirzebruch's 1954 Problem List - Problem 5

O / open | AMR 问题清单汇编

Let $M^n$ be an almost-complex manifold. Choose an Hermitian structure and consider the numbers $h^{p,q}$ defined as above. Is $h^{p,q}$ independent of the choice of the Hermitian structure? If not, give some other definition of the $h^{p,q}$ of $M^n$ which depends only on the almost-complex structure and which generalizes the $h^{p,q}$ of a complex manifold.

背景链接：https://arxiv.org/abs/1305.4623

05898 | AMR-114-0006

### Hirzebruch's 1954 Problem List - Problem 6

O / open | AMR 问题清单汇编

Consider the complex projective space $\mathbb{P}_n$ as a differentiable manifold with the usual differentiable structure. Determine all complex structures of $\mathbb{P}_n$. \mathrm{(See Problem 14.)} In particular (28*), determine all complex structures of $\mathbb{P}_n$ which can carry a K\"ahler metric.

背景链接：https://arxiv.org/abs/1305.4623

05899 | AMR-114-0007

### Hirzebruch's 1954 Problem List - Problem 7

O / open | AMR 问题清单汇编

This brings us to a specifically four-dimensional problem:

Determine all complex structures of $S^2\times S^2$.

背景链接：https://arxiv.org/abs/1305.4623

05900 | AMR-114-0008

### Hirzebruch's 1954 Problem List - Problem 8

O / open | AMR 问题清单汇编

Are the $h^{p,q}$ and the Chern characteristic numbers of an algebraic variety $V_n$ topological invariants of $V_n$? If not, determine all those linear combinations of the $h^{p,q}$ and the Chern characteristic numbers which are topological invariants.

背景链接：https://arxiv.org/abs/1305.4623

05901 | OWR-11139-015

### Hochschild Homology of Landau-Ginzburg Mirrors

P / partially_solved | Oberwolfach 报告问题汇编

For a mirror pair $S$ and $(S^\vee,F_{S^\vee})$, is the Hochschild homology $HH^k(S)$ isomorphic to the cohomology $H^k(S^\vee,\mathbb C)$?

来源字段：https://doi.org/10.4171/owr/2011/47

05902 | AIM-ALGEBRAIC_GEOMETRY-0066

### Hochschild invariants and a projective Calabi-Yau nonembedding obstruction

P / partially_solved | AIM 研讨会问题汇编

Investigate \(Ku(X)\) for \(X\) a smooth quartic double 5 fold, that is, a double cover of \(\mathbb P^5\) branched along a quartic. This variety is Calabi-Yau and it is never of the form \(D(Y)\) for \(Y\) a variety. Further, the Serre functor of \(X\) is equal to shift by 3; in other words, the category \(Ku(X)\) is 3-Calabi Yau.

来源字段：http://aimpl.org/rationalityag/1/

05903 | OWR-12490-005

### Hodge Algebras of Conjugate Varieties

P / partially_solved | Oberwolfach 报告问题汇编

For which subfields $K\subset\mathbb C$ is the graded algebra of $K$-rational Hodge classes $H^{*,*}(X,K)$ isomorphic to $H^{*,*}(X^\sigma,K)$ for every pair of conjugate varieties $X$ and $X^\sigma$?

来源字段：https://doi.org/10.4171/owr/2013/27

05904 | OWR-13679-003

### Hodge and Order Filtrations on GKZ Modules

P / partially_solved | Oberwolfach 报告问题汇编

For the GKZ modules $M_A^0$ and $M_A^{-c}$, the Hodge filtration equals the order filtration up to the shifts $F^H_\bullet M_A^0=F^{\mathrm{ord}}_{\bullet+d}M_A^0$ and $F^H_\bullet M_A^{-c}=F^{\mathrm{ord}}_{\bullet-n-c_0}M_A^{-c}$, where $c_0$ is the first component of $c\in NA\subset\mathbb Z^{d+1}$.

来源字段：https://doi.org/10.4171/owr/2015/22

05905 | MPP-006

### Hodge Conjecture

O / open | 千禧年问题

别名：The Hodge Conjecture [AG-005]; The Hodge Conjecture [OPG-1803]

On a projective non-singular algebraic variety over $\mathbb{C}$, any Hodge class is a rational linear combination of classes of algebraic cycles.

背景链接：https://www.claymath.org/millennium-problems/

05906 | AIM-ALGEBRAIC_GEOMETRY-0067

### Hodge correction and signed weak-factorization constraints for quartic double fivefolds

P / partially_solved | AIM 研讨会问题汇编

**Problem.** Show that smooth quartic double 5-folds are irrational.

来源字段：http://aimpl.org/rationalityag/1/

05907 | OWR-13750339-002

### Hodge Integrals over Genus-One Admissible Covers

P / partially_solved | Oberwolfach 报告问题汇编

For degree $d>2$, find a closed formula for the Hodge-integral generating series over admissible covers of a genus-one curve; identify the rational function of $q=-e^{iu}$ that it defines.

来源字段：https://doi.org/10.4171/owr/2023/27

05908 | OWR-1114-007

### Hodge Isometries from K3 Derived Autoequivalences

P / partially_solved | Oberwolfach 报告问题汇编

For an algebraic K3 surface $X$, determine the image of the natural homomorphism from the group of exact autoequivalences of $D^b(X)$ to the group of integral Hodge isometries of the Mukai lattice; in particular, is the orientation-reversing isometry acting as minus the identity on $H^2(X)$ in the image?

来源字段：https://doi.org/10.4171/owr/2005/53

05909 | OWR-16169-012

### Hodge Numbers from Landau-Ginzburg Fiber Components

P / partially_solved | Oberwolfach 报告问题汇编

If $X$ is a smooth Fano variety of dimension $n$ with toric Landau-Ginzburg model $f_X$, and $k_{f_X}$ is the number of components of reducible compactification fibers minus the number of reducible fibers, is $h^{1,n-1}(X)=k_{f_X}$?

来源字段：https://doi.org/10.4171/owr/2018/28

05910 | OWR-11785-001

### Hodge Structures on Periodic Cyclic Homology

P / partially_solved | Oberwolfach 报告问题汇编

For a smooth complex algebraic variety $X$, does $\operatorname{Perf}(X)$ remember any part of the mixed Hodge structure on its de Rham cohomology? More generally, does the periodic cyclic homology of every differential graded category carry a functorial Hodge-like structure?

来源字段：https://doi.org/10.4171/owr/2012/23

05911 | OWR-11101913-001

### Holomorphic Anomaly Equations for Elliptic Threefolds

P / partially_solved | Oberwolfach 报告问题汇编

For an elliptically fibered threefold $\pi:X\to B$ satisfying $c_3(T_X\otimes\omega_X)=0$, do its normalized relative Pandharipande-Thomas descendant series satisfy the two Oberdieck-Schimpf holomorphic anomaly equations?

来源字段：https://doi.org/10.4171/owr/2022/32

05912 | OWR-14298795-017

### Holomorphic Convexity of Universal Covers

O / open | Oberwolfach 报告问题汇编

Is the universal cover of every smooth projective variety holomorphically convex?

来源字段：https://doi.org/10.4171/owr/2024/45

05913 | OWR-744-008

### Holomorphic Structures on Vector Bundles over Complex Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Which differentiable vector bundles on a complex surface $X$ admit holomorphic structures; equivalently, which triples $(r,c_1,c_2)$ occur as Chern data of holomorphic rank-$r$ bundles?

来源字段：https://doi.org/10.4171/owr/2004/42

05914 | AMR-118-0010

### Holomorphic Symplectic Geometry - Conjecture 10

O / open | AMR 问题清单汇编

Let $D_1,\ldots ,D_k$ in $\operatorname{Pic}(X)$, and let $z\in CH(X)$ be a class which is a polynomial in $D_1,\ldots ,D_k$ and the Chern classes $c_i(X)$. If $z=0$ in $H^*(X,\mathbb{Z})$, then $z=0$.

背景链接：https://arxiv.org/abs/1002.4321

05915 | AMR-118-0011

### Holomorphic Symplectic Geometry - Conjecture 11

O / open | AMR 问题清单汇编

\[X=\sd_{s\ \mathrm{ even} }X_s \quad\hbox{where }\ X_s :=\{x\in X\ |\ \operatorname{rk} \tau (x)=s\}\ . \]

The following conjecture is due to Bondal ([Bd], see also [P]):

If $X$ is Fano and $s$ even, $X_{\leq s}:=\sd_{k\leq s}X_k$ contains a component of dimension $>s$.

背景链接：https://arxiv.org/abs/1002.4321

05916 | AMR-118-0012

### Holomorphic Symplectic Geometry - Conjecture 12

O / open | AMR 问题清单汇编

This is much larger than one would expect from a naive dimension count. It implies for instance that a Poisson field which vanishes at some point must vanish along a curve. The condition ``$X$ Fano" is probably far too strong. In fact an optimistic modification would be :

If $X_s$ is non-empty, it contains a component of dimension $>s$.

背景链接：https://arxiv.org/abs/1002.4321

05917 | AMR-118-0013

### Holomorphic Symplectic Geometry - Conjecture 13

O / open | AMR 问题清单汇编

There are only two classes of compact holomorphic contact manifolds known so far: a) The projective cotangent bundle $\mathbb{P} T^*_M$, where $M$ is any compact complex manifold; b) Let $\mathfrak{g}$ be a simple complex Lie algebra. The action of the adjoint group on $\mathbb{P}(\mathfrak{g})$ has a unique closed orbit $X_\mathfrak{g}$: every other orbit contains $X_\mathfrak{g}$ in its closure. $X_\mathfrak{g}$ is a contact Fano manifold.

The following conjecture is folklore:

Any projective contact manifold is of type \mathrm{a)} or \mathrm{b)}.

背景链接：https://arxiv.org/abs/1002.4321

05918 | AMR-118-0004

### Holomorphic Symplectic Geometry - Conjecture 4

O / open | AMR 问题清单汇编

Having so few examples leads naturally to the following question:

There are finitely many hyperk\"ahler manifolds (up to deformation) in each dimension.

背景链接：https://arxiv.org/abs/1002.4321

05919 | AMR-118-0006

### Holomorphic Symplectic Geometry - Conjecture 6

O / open | AMR 问题清单汇编

\mathrm{a)} Let $L$ be a nontrivial nef line bundle on $X$ with $q(L)=0$. There exists a Lagrangian fibration $f:X\rightarrow \mathbb{P}^r$ such that $L=f^*\mathcal{O}_{\mathbb{P}^r}(k)$ for some $k\geq 1$. \mathrm{b)} There exists a hyperk\"ahler manifold $X'$ bimeromorphic to $X$ and a Lagrangian fibration $X'\rightarrow \mathbb{P}^r$ if and only if $X$ admits a line bundle $L\neq \mathcal{O}_S$ with $q(L)=0$.

背景链接：https://arxiv.org/abs/1002.4321

05920 | AMR-118-0001

### Holomorphic Symplectic Geometry - Question 1

O / open | AMR 问题清单汇编

In the two last examples it would seem simpler to start with a moduli space $\mathcal{M}$ for which the natural compactification $\overline{\mathcal{M}}$ is smooth; in that case $\overline{\mathcal{M}}$ is a hyperk\"ahler manifold [M], but it turns out that it is a deformation of $S^{[r]}$ or $K_r(T)$ (G\"ottsche-Huybrechts, O'Grady, Yoshioka ...). On the other hand, when $\overline{\mathcal{M}}$ is singular, it admits a hyperk\"ahler desingularization only in the two cases considered by O'Grady [KLS]. Thus it seems that a new idea is required to answer our first problem:

Find new examples of hyperk\"ahler manifolds.

背景链接：https://arxiv.org/abs/1002.4321

05921 | AMR-118-0002

### Holomorphic Symplectic Geometry - Question 2

O / open | AMR 问题清单汇编

Let $X$, $X'$ be two hyperk\"ahler manifolds of the same dimension. If there is a Hodge isometry $\lambda :H^2(X,\mathbb{Z})\vbox{\hbox to .8cm{\hfill{$\scriptstyle\sim$}\hfill} \nointerlineskip\hbox to .8cm{{\hfill$\longrightarrow $\hfill}} } H^2(X',\mathbb{Z})$, what can we say of $X$ and $X'$? Can we conclude that $X$ and $X'$ are isomorphic by imposing that $\lambda $ preserves some extra structure?

背景链接：https://arxiv.org/abs/1002.4321

05922 | AMR-118-0003

### Holomorphic Symplectic Geometry - Question 3

O / open | AMR 问题清单汇编

Thus the subalgebra $\mathcal{H}$ is completely determined by the form $q$ and the dimension of $X$. In contrast, not much is known about the $\mathcal{H}$-module $\mathcal{H}^\perp$. Note that it is nonzero for the examples $a)$ and $b)$ of 1.2, with the exception of $S^{[2]}$ for a K3 surface $S$. We do not know much about the quadratic form $q$ either. For the two infinite series of (1.2) we have lattice isomorphisms [B1] \[H^2(S^{[r]},\mathbb{Z})=H^2(S,\mathbb{Z}) \buildrel {\scriptscriptstyle\perp}\over {\oplus} \langle 2-2r\rangle\qquad H^2(K_r(T),\mathbb{Z})=H^2(T,\mathbb{Z}) \buildrel {\scriptscriptstyle\perp}\over {\oplus} \langle -2-2r\rangle\ ; \] The lattices of O'Grady's two examples are computed in [R]; they are also even.

Is the quadratic form $q$ always even? More generally, what are the possibilities for $q$? What are the possibilities for the Fujiki index $f_X$ (see $1.3$)?

背景链接：https://arxiv.org/abs/1002.4321

05923 | AMR-118-0005

### Holomorphic Symplectic Geometry - Question 5

O / open | AMR 问题清单汇编

Note that $b_2$ is $23$ for $S^{[2]}$ and $7$ for $K_2(T)$ (1.2). [G] contains some more information on the other Betti numbers.

Can we exclude some more cases, in particular $b_2=3$? If $b_2=23$, can we conclude that $X$ is deformation equivalent to $S^{[2]}$?

背景链接：https://arxiv.org/abs/1002.4321

05924 | AMR-118-0007

### Holomorphic Symplectic Geometry - Question 7

O / open | AMR 问题清单汇编

Does every hyperk\"ahler manifold admit a deformation with a Lagrangian fibration?

背景链接：https://arxiv.org/abs/1002.4321

05925 | AMR-118-0008

### Holomorphic Symplectic Geometry - Question 8

O / open | AMR 问题清单汇编

Let $X$ be a hyperk\"ahler manifold, and $T\subset X$ a Lagrangian submanifold which is a complex torus. Is it the fibre of a Lagrangian fibration $X\rightarrow \mathbb{P}^r$?

背景链接：https://arxiv.org/abs/1002.4321

05926 | AMR-118-0009

### Holomorphic Symplectic Geometry - Question 9

O / open | AMR 问题清单汇编

Describe the general projective deformation of $S^{[2]}$, for $S$ a polarized K3 surface of genus $1$, $2$, $3$, ... (and for some choice of polarization on $S^{[2]}$); or at least find more examples of locally complete projective families. Same question with $S^{[r]}$ for $r\geq 3$.

背景链接：https://arxiv.org/abs/1002.4321

05927 | OWR-12018-003

### HOMFLY Polynomials from Nested Hilbert Schemes

S / solved | Oberwolfach 报告问题汇编

别名：Duplicate: HOMFLY Polynomials from Nested Hilbert Schemes [OWR-12018-005]

Let $C=\{E(x,y)=0\}$ be a planar curve singularity at $(0,0)$, with Milnor number $$\mu=\dim_{\mathbb C}\frac{\mathbb C[[x,y]]}{(\partial E/\partial x,\partial E/\partial y)}.$$ Does the HOMFLY-PT polynomial of its algebraic link satisfy $$\overline P(L_{C,(0,0)})=(a/q)^{\mu-1}\sum_{\ell,m}q^{2\ell}(-a^2)^m\chi\!\left(C_{(0,0)}^{[\ell,\ell+m]}\right)$$?

来源字段：https://doi.org/10.4171/owr/2012/46

05928 | OWR-12018-006

### Homological Hilbert-Scheme Conjecture for Algebraic Links

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate: Homological Hilbert-Scheme Conjecture [OWR-12018-007]

Let $C$ be a planar curve singularity at $(0,0)$ with Milnor number $\mu$. Does the triply graded HOMFLY homology of its algebraic link satisfy $$\left(\frac aq\right)^{\mu-1}\sum_{\ell,m}q^{2\ell}a^{2m}t^{m^2}P_t^{\mathrm{vir}}\!\left(C_{(0,0)}^{[\ell,\ell+m]}\right)=P(L_{C,(0,0)})$$?

来源字段：https://doi.org/10.4171/owr/2012/46

05929 | OWR-4990374-011

### Homotopy Generators from Elliptic Artin Noncancellativity

O / open | Oberwolfach 报告问题汇编

Let $A(R,G)^+$ be the elliptic Artin monoid and let $S_{\mathrm{ell}}\setminus D_{\mathrm{ell}}$ be its elliptic discriminant complement. For pairs of noncancellative quadruples $(a,b,c,d)$ and $(a',b,c,d')$, do all difference classes $$ a\backslash R(abd,acd)/d-a'\backslash R(a'bd',a'cd')/d' $$ generate $\pi_2(S_{\mathrm{ell}}\setminus D_{\mathrm{ell}},*)$?

来源字段：https://doi.org/10.4171/owr/2021/5

05930 | OWR-7155450-003

### Hyperkähler Compactification of Projective Fibrations

P / partially_solved | Oberwolfach 报告问题汇编

Given a smooth projective fibration $M_U\to U$, where $U\subseteq\mathbb{P}^n$, when does there exist a smooth hyper-Kähler compactification $M$ of $M_U$ with a regular morphism $\pi:M\to\mathbb{P}^n$ extending $\pi_U$?

来源字段：https://doi.org/10.4171/owr/2021/29

05931 | AIM-ARITHMETIC_GEOMETRY-0059

### Hyperplane lifting defects and a Gorenstein Rao-module obstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 12.** What do properties of the Rao modules imply about $C$? For example, if $M_C$ is Gorenstein or annihilated by a linear form, does $C$ have any nice properties?

**Remark.** We might have to require $C$ to be minimal.

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05932 | AIM-ALGEBRAIC_GEOMETRY-0054

### Hyperplane-pencil transfer of stable irrationality

P / partially_solved | AIM 研讨会问题汇编

Can we use the degeneration method to obstruct stable rationality for \(4\)-folds fibered by \(3\)-folds over \(\mathbb P^1\)?

来源字段：http://aimpl.org/rationalityag/1/

05933 | OWR-4086-009

### Igusa Monodromy Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For a polynomial $f$ in $n$ variables over $\mathbb{Q}$, for all but finitely many primes $p$, does every pole $s_0$ of the $p$-adic Igusa zeta function $Z_p(f;s)$ satisfy $\exp(2\pi i s_0)$ equal to a monodromy eigenvalue of $f:\mathbb{C}^n\to\mathbb{C}$ at a point of $\{f=0\}$?

来源字段：https://doi.org/10.4171/owr/2009/43

05934 | OWR-11101913-002

### Iitaka Subadditivity Conjecture for Kodaira Dimension

P / partially_solved | Oberwolfach 报告问题汇编

别名：Known Cases of Iitaka Subadditivity [OWR-11101913-003]

For an algebraic fiber space $f:X\to Y$ with smooth projective complex varieties and general fiber $F$, is $\kappa(X)\geq\kappa(F)+\kappa(Y)$?

来源字段：https://doi.org/10.4171/owr/2022/32

05935 | OWR-12697710-014

### Images of Principal Minor Maps for Square Matrices

P / partially_solved | Oberwolfach 报告问题汇编

What equations describe the image of $\mathbb{C}^{5\times5}$ under the principal-minor map, and can the image for $\mathbb{C}^{n\times n}$ be cut out using the action of $\operatorname{SL}_2(\mathbb{C})^n\rtimes S_n$ and compatible combinatorial structures?

来源字段：https://doi.org/10.4171/owr/2023/14

05936 | AIM-ALGEBRAIC_GEOMETRY-0022

### Immaculate line bundles as unit-orthogonal toric branes

P / partially_solved | AIM 研讨会问题汇编

[Wang, Borisov--Wang] found a condition on the underlying fan for a toric stack to admit infinitely many acyclic line bundles, where “acyclic” means a line bundle \(\mathcal L\) with \(H^i(\mathcal L)=0\) for every \(i\) (for example \(\mathcal O(-1)\) on projective space). In dimension two, the sample condition is the presence of collinear rays. (1) Understand the symplectic mirror of these criteria. (2) Understand consequences of generalizations, for example involving a toric subvariety whose fan is contained in a linear subspace.

来源字段：http://aimpl.org/syzygyms/9/

05937 | OWR-16408-006

### Implications Among Conjectures on Principally Polarized Abelian Varieties

O / open | Oberwolfach 报告问题汇编

Let $(A,\Theta)$ be an indecomposable principally polarized abelian variety, and let $X\subset A$ be a geometrically nondegenerate subvariety for which $\mathcal I_X(\Theta)$ is a generic-vanishing sheaf. Quantify a strengthening of this hypothesis sufficient to prove the bound

$$\operatorname{mult}_x\Theta\leq\frac{\dim A+1}{2}$$

for every $x\in\Theta$. Conversely, determine how that multiplicity bound can be used toward classifying $X$ as a minimal-class subvariety, namely a translate of $W_d(C)$ in a Jacobian or of the Fano surface in a five-dimensional intermediate Jacobian.

来源字段：https://doi.org/10.4171/owr/2018/33

05938 | AMR-071-0010

### In spherical or hyperbolic geometry, must polyhedra with the same volume and Dehn invariant be scissors-congruent

P / partially_solved | AMR 问题清单汇编

In spherical or hyperbolic geometry, must polyhedra with the same volume and Dehn invariant be scissors-congruent?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

05939 | OWR-14299088-016

### Incident-Variety Description of Euler Discriminants

P / partially_solved | Oberwolfach 报告问题汇编

Find an intrinsic incident-variety description of the Euler discriminant analogous to the established geometric picture for classical discriminants.

来源字段：https://doi.org/10.4171/owr/2025/9

05940 | OWR-13109-004

### Indeterminacy Locus of the Perfect Cone Prym Map

O / open | Oberwolfach 报告问题汇编

For the extension $P_g^{\mathrm{Perf}}$ of the Prym map to the perfect-cone compactification, is its indeterminacy locus exactly $FS_2\cup FS_3$?

来源字段：https://doi.org/10.4171/owr/2014/39

05941 | OWR-14298795-011

### Index Divisibility in p-Power Pullbacks of Curves

O / open | Oberwolfach 报告问题汇编

Let $X$ be a smooth projective curve of genus at least $2$ embedded in an abelian variety $A$ over a finite extension $k/\mathbb Q_p$, and assume that $X$ contains no torsion point of order prime to $p$. For all sufficiently large $n$, is the index of the inverse image of $X$ under $[p^n]:A\to A$ divisible by $p$?

来源字段：https://doi.org/10.4171/owr/2024/45

05942 | AIM-ALGEBRAIC_GEOMETRY-0108

### Index-gap obstruction for line covers on Fano hypersurfaces in characteristic p

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** Let \(k\) be an algebraically closed field of characteristic \(p>0\), let \(X_d\subset \mathbb P^n_k\) be a general degree-\(d\) hypersurface with \(n\geq d+2\), and let \(e\geq1\). Is the genus-zero Kontsevich space \[ \overline{\mathcal M}_{0,0}(X,e) \] irreducible of dimension \[ E_e:=e(n+1-d)+n-4? \] Is this true simultaneously for every \(e\geq1\)?

来源字段：http://aimpl.org/ratsubvarpos/5/

05943 | AIM-ALGEBRAIC_GEOMETRY-0052

### Index-two obstruction for maximal-plane projections of odd-dimensional cubics

P / partially_solved | AIM 研讨会问题汇编

Is there a smooth rational cubic hypersurface of odd dimension?

来源字段：http://aimpl.org/rationalityag/1/

05944 | AIM-ALGEBRAIC_GEOMETRY-0069

### Infinite degree-three unramified cohomology and a rationally connected coefficient sieve

S / solved | AIM 研讨会问题汇编

Let \(X\) be a smooth projective complex variety. For a finite abelian group \(M\), is unramified cohomology \[ H^i_{\mathrm{nr}}(k(X)/k,M) \] always finite?

来源字段：http://aimpl.org/rationalityag/1/

05945 | OWR-1703871-006

### Infinite Transverse Actions on Codimension-One Foliations

O / open | Oberwolfach 报告问题汇编

If $X$ is a projective manifold, $\mathcal F$ is a codimension-one foliation on $X$, and $f\in\operatorname{End}(X,\mathcal F)$ has infinite transverse action, is $\mathcal F$ virtually transversely additive?

来源字段：https://doi.org/10.4171/owr/2020/24

05946 | OWR-4086-007

### Infinitesimal-Fiber Criteria for Isomorphisms of Scheme Germs

P / partially_solved | Oberwolfach 报告问题汇编

If a morphism of scheme germs induces isomorphisms on every scheme-theoretic infinitesimal fiber $X_m(x)\to Y_m(y)$, must the morphism itself be an isomorphism for noetherian schemes?

来源字段：https://doi.org/10.4171/owr/2009/43

05947 | OWR-16768-016

### Injective Projective Maps of Curves and Segre-Veronese Varieties

O / open | Oberwolfach 报告问题汇编

Can every smooth complex projective curve be injected into $\mathbb P^2$? More generally, can every smooth $d$-dimensional projective variety be injected into $\mathbb P^{2d}$? Can every Segre-Veronese variety be injectively projected into a projective space of twice its dimension?

来源字段：https://doi.org/10.4171/owr/2019/6

05948 | AIM-ALGEBRAIC_GEOMETRY-0095

### Inseparability-robust irrationality bounds over small fields

P / partially_solved | AIM 研讨会问题汇编

Can we give lower bounds on the degree of irrationality / covering gonality.... of hypersurfaces in char \(p\)? How do we make sense of these notions over countable / finite fields?

来源字段：http://aimpl.org/ratsubvarpos/2/

05949 | OWR-9790356-003

### Integral Geometrization with Nilpotent Singular Support

P / partially_solved | Oberwolfach 报告问题汇编

For quasi-split $G$, let $W_\psi$ be the Whittaker sheaf associated with a nondegenerate character $\psi$. Prove that the spectral-action functor $F\mapsto F*W_\psi$ extends to an equivalence from $\operatorname{Coh}_{\mathrm{Nilp}}(\operatorname{LocSys}_{\widehat G}/\mathbb Z_\ell)$ to $D_{\mathrm{lis}}(\operatorname{Bun}_G,\mathbb Z_\ell)$, compatible with the spectral action.

来源字段：https://doi.org/10.4171/owr/2022/5

05950 | OWR-8415342-017

### Integral Shimura Models with Universal Shtukas

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathcal G$ be a parahoric group scheme. Construct a pro-system of normal flat integral Shimura-variety models $\mathcal S_K$ over the reflex integers, with finite étale transition maps, such that the valuative extension property holds, the generic-fiber $\mathcal G$-shtuka extends to $\mathcal S_K$, and every formal completion is identified with the corresponding integral local Shimura variety compatibly with its tautological shtuka.

来源字段：https://doi.org/10.4171/owr/2021/39

05951 | OWR-4990381-008

### Integrality of Rigid Irreducible Local Systems

O / open | Oberwolfach 报告问题汇编

Let $X$ be a connected smooth quasi-projective variety over $\mathbb C$ with a normal-crossing compactification. Is every rigid irreducible representation $\rho:\pi_1^{\mathrm{top}}(X,x)\to\operatorname{GL}_n(\mathbb C)$ with quasi-unipotent local monodromies conjugate to a representation in $\operatorname{GL}_n(\mathcal O_K)$ for the ring of integers $\mathcal O_K$ of some number field $K$?

来源字段：https://doi.org/10.4171/owr/2021/12

05952 | OWR-2049-004

### Intersection Numbers on First Voronoi Compactifications

O / open | Oberwolfach 报告问题汇编

别名：Vanishing Pattern for Voronoi Compactification Intersections [OWR-2049-005]

On the first Voronoi compactification $\mathcal{A}_g^F$, let $L$ be the Hodge divisor and $D$ the boundary divisor. Is $\langle L^nD^{g(g+1)/2-n}\rangle_{\mathcal{A}_g^F}=0$ unless $n=k(k+1)/2$ for some integer $k$ with $0\leq k\leq g$?

来源字段：https://doi.org/10.4171/owr/2008/38

05953 | OWR-12697710-012

### Intersection of Sum-of-Squares and SONC Cones

O / open | Oberwolfach 报告问题汇编

Characterize the intersection $\Sigma_{n,2d}\cap C_{n,2d}$ of the sum-of-squares and SONC cones.

来源字段：https://doi.org/10.4171/owr/2023/14

05954 | AIM-ARITHMETIC_GEOMETRY-0054

### Irrational Hilbert-scheme components and a threshold-10 MRC strengthening

S / solved | AIM 研讨会问题汇编

**Problem 7.** Does there exist a nonrational component of \(\operatorname{Hilb}^{d}(\mathbb A^{n})\)?

12RECORDED ON WHITEBOARD BY IZZET COSKUN, AND EDITED BY LI LI

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05955 | AIM-ALGEBRAIC_GEOMETRY-0278

### Irrational non-Q-Gorenstein F-jumps and boundary-certificate escape

P / partially_solved | AIM 研讨会问题汇编

For an \(F\)-finite normal domain \(R\), not assumed \(\mathbb Q\)-Gorenstein, and a nonzero ideal \(\mathfrak a\), are the jumping numbers of the intrinsic filtration \(\tau_b(R;\mathfrak a^t)\) discrete, and are they rational?

来源字段：http://aimpl.org/testandmultiplierideals/1/

05956 | OWR-16628-010

### Irrational Seshadri Constants from the Nagata Conjecture

O / open | Oberwolfach 报告问题汇编

For $r\geq9$, does the Nagata conjecture for the blowup $X_r$ of $\mathbb P^2$ at $r$ general points imply the existence of an ample line bundle on $X_r$ whose Seshadri constant at a very general point is irrational?

来源字段：https://doi.org/10.4171/owr/2018/45

05957 | OWR-16628-001

### Irrationality of Seshadri Constants

O / open | Oberwolfach 报告问题汇编

Can a Seshadri constant be irrational?

来源字段：https://doi.org/10.4171/owr/2018/45

05958 | OWR-13946-007

### Irreducibility of Nodal-Curve Spaces on K3 Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Let $(X,L)$ be a general primitively polarized K3 surface of genus $p$, and let $0<\delta<n^2(p-1)+1$. Is the Severi variety $V_{n,\delta}(X)$ of irreducible $\delta$-nodal curves in $|nL|$ irreducible?

来源字段：https://doi.org/10.4171/owr/2015/51

05959 | OWR-13946-006

### Irreducibility of Severi Varieties on Polarized K3 Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathcal V_{p,n,\delta}$ be the universal Severi variety of triples $((X,L),C)$, where $(X,L)$ is a primitively polarized K3 surface of genus $p$ and $C\in|nL|$ is irreducible with exactly $\delta$ nodes. Is $\mathcal V_{p,n,\delta}$ irreducible?

来源字段：https://doi.org/10.4171/owr/2015/51

05960 | OWR-14299511-008

### Irreducible Forest Decomposition of Consistency-Equation Varieties

O / open | Oberwolfach 报告问题汇编

Prove that the forest decomposition of the algebraic variety defined by the fusion consistency equations is its irreducible decomposition.

来源字段：https://doi.org/10.4171/owr/2025/34

05961 | AIM-ALGEBRAIC_GEOMETRY-0199

### Iterated differential signatures as a quotient tower

P / partially_solved | AIM 研讨会问题汇编

**AIM Problem 3.4 (Invariants).** “What can we say about the signature of the signature of a curve?”

The workshop schedule places talks by Peter Olver and Irina Kogan on the same day, and the participant compilation records Olver's interest in differential-invariant signatures; this supports the reconstructed setting without identifying an author for the problem.

The following reconstruction is therefore used.

Let \(G\subset \operatorname{PGL}_3(\mathbb C)\) be a positive-dimensional closed algebraic group acting rationally on the affine plane. Let \[ I=(K_1,K_2) \] be a rational classifying pair of differential invariants for the prolonged \(G\)-action. For a suitable irreducible plane curve \(X\), restriction gives the rational signature map \[ \sigma_X=(K_1|_X,K_2|_X):X\dashrightarrow \mathbb A^2, \] and the Zariski closure \[ \mathcal S_I(X)=\overline{\sigma_X(X)} \] is its signature curve. The question asks what happens when the signature construction is applied again to \(\mathcal S_I(X)\).

来源字段：http://aimpl.org/algvision/3/

05962 | OWR-12015-001

### Jet Curvature and the First Chern Class

O / open | Oberwolfach 报告问题汇编

If a complex manifold $X$ has nondegenerate negative $k$-jet curvature, must its real first Chern class satisfy $c_1(X)_{\mathbb R}\ne0$?

来源字段：https://doi.org/10.4171/owr/2012/43

05963 | OWR-4527-007

### Jet Spanning by Nef Toric Vector Bundles

S / solved | Oberwolfach 报告问题汇编

For a nef toric vector bundle $E$ on a smooth projective toric variety, does $\tau(E)\geq k$ imply that $E$ is $k$-jet spanned for every $k\geq1$? Equivalently, does $\varepsilon(E)\geq k$ imply $k$-jet spanning?

来源字段：https://doi.org/10.4171/owr/2010/45

05964 | OWR-7155446-004

### Jet-Differential Base Loci for Green-Griffiths-Lang

O / open | Oberwolfach 报告问题汇编

Can one prove the Green-Griffiths-Lang conjecture by showing that the projection to $X$ of the base locus of Green-Griffiths jet differentials is a proper subvariety, using further differential equations from sections on that base locus?

来源字段：https://doi.org/10.4171/owr/2021/25

05965 | AIM-ALGEBRAIC_GEOMETRY-0188

### Joint-monodromy degree classification and branch-overlap smoothness budget

P / partially_solved | AIM 研讨会问题汇编

Fix two cameras and two image curves. Is there is a curve in \(\mathbb{P}^3\) that projects onto these two curves? What if we put restriction on degrees or smoothness?

来源字段：http://aimpl.org/algvision/1/

05966 | OWR-14746-011

### Jump Loci for Motivic Complexity in Families

O / open | Oberwolfach 报告问题汇编

Given a smooth morphism $f$ from $X$ to $S$ of quasi-projective varieties, can one describe the locus of points $s\in S$ where the motive of the fiber $X_s$ is simpler than that of a very general fiber, in the sense that $X_s$ and its powers contain more algebraic cycles?

来源字段：https://doi.org/10.4171/owr/2016/38

05967 | OWR-14298589-015

### k-Differential Interpretation of Leaky Double Hurwitz Numbers

P / partially_solved | Oberwolfach 报告问题汇编

For arbitrary genus $g$ and zero-pole profile $A$, formulate a natural enumerative problem for $k$-differentials with prescribed periods or residues. Determine its relationship to the $k$-leaky double Hurwitz number $H_g^k(A)$, extending the similarities observed in genus zero.

来源字段：https://doi.org/10.4171/owr/2024/41

05968 | OWR-13109-005

### K-Moduli and Minimal Fano Fillings

O / open | Oberwolfach 报告问题汇编

Consider families of Fano varieties with log-terminal singularities and a fibre-tangent holomorphic vector field $V$, with every pair $(X_t,V_t)$ modified K-semistable. Does their moduli functor admit a proper coarse algebraic moduli space $M$ parametrizing modified K-polystable Fano varieties, with Kähler connected components? Do the limits of the classes $L_{T,k}$ define a Kähler class, or an ample real line bundle $L_T$, on $M$? For a family over a punctured curve, does a modified K-polystable Fano filling minimize $\deg L_T$ among all Fano fillings?

来源字段：https://doi.org/10.4171/owr/2014/39

05969 | AIM-ALGEBRAIC_GEOMETRY-0064

### K_0 under K-equivalence: codimension-one invariance and the threefold case

P / partially_solved | AIM 研讨会问题汇编

If two smooth projective varieties \(X\) and \(Y\) are K-equivalent, is the Grothendieck group \(K_0(X)\) isomorphic to \(K_0(Y)\)?

来源字段：http://aimpl.org/rationalityag/1/

05970 | OWR-1114-001

### Kawamata Derived-Equivalence Conjecture

O / open | Oberwolfach 报告问题汇编

Birational smooth projective varieties with trivial canonical class should be $D$-equivalent; that is, their bounded derived categories of coherent sheaves should be equivalent.

来源字段：https://doi.org/10.4171/owr/2005/53

05971 | AIM-ALGEBRAIC_GEOMETRY-0319

### Kähler invariance of plurigenera: current boundary and a pseudoeffective-chamber reduction

P / partially_solved | AIM 研讨会问题汇编

(Siu) Let \(\pi:X\to\Delta\) be a holomorphic family of compact Kähler manifolds over the open unit \(1\)-disk \(\Delta\) with fiber \(X_t\). Then is the dimension of \(H^0(X_t,mK_{X_t})\) independent of \(t\in\Delta\)? This is the Kähler case of Siu's famous invariance of plurigenera theorem for projective smooth varieties. One of the difficulties in using similar ideas from the algebraic case is that we cannot use an auxiliary ample line bundle.

来源字段：https://aimath.org/WWN/singularvariety/problems_final.pdf

05972 | OWR-3471-004

### Kähler Limits and Transcendental Morse Inequalities

P / partially_solved | Oberwolfach 报告问题汇编

Let $\pi:X\to\Delta$ be a proper holomorphic submersion whose fibers are compact complex manifolds. If $X_t$ is Kähler for every $t\in\Delta\setminus\{0\}$, must the central fiber $X_0$ belong to Fujiki class $\mathcal C$? More generally, does Demailly's transcendental Morse conjecture hold: if a compact complex $n$-fold $Y$ carries a $d$-closed positive $(1,1)$-current $T$ satisfying $\int_Y(T_{\mathrm{ac}})^n>0$, must the de Rham class of $T$ contain a Kähler current without assuming its integrality?

来源字段：https://doi.org/10.4171/owr/2009/21

05973 | OWR-11101913-007

### Kähler Poisson Manifolds with Infinite-Fundamental-Group Leaves

O / open | Oberwolfach 报告问题汇编

Describe compact Kähler Poisson manifolds having a compact leaf with infinite fundamental group.

来源字段：https://doi.org/10.4171/owr/2022/32

05974 | OWR-14298581-006

### Kähler Shafarevich-Tate Twists

P / partially_solved | Oberwolfach 报告问题汇编

Let $\pi:X\to B$ be a Lagrangian fibration on an irreducible holomorphic symplectic manifold, and let $X^\phi\to B$ be the Shafarevich-Tate twist associated with a class $\phi$. Characterize the classes $\phi$ for which $X^\phi$ is Kähler.

来源字段：https://doi.org/10.4171/owr/2024/32

05975 | OWR-15955-007

### Kähler, Projective, and Quasiprojective Group Realizability

O / open | Oberwolfach 报告问题汇编

For a natural class of groups, characterize which members are Kähler groups, projective groups, or quasiprojective groups.

来源字段：https://doi.org/10.4171/owr/2017/50

05976 | OWR-14298795-014

### Kim’s Nonabelian Chabauty Conjecture

O / open | Oberwolfach 报告问题汇编

For a hyperbolic curve $X/\mathbb Q$ and a prime $p$ of good reduction, do the refined Chabauty-Kim sets eventually equal the rational points; that is, is $X(\mathbb Q_p)_n=X(\mathbb Q)$ for all sufficiently large $n$?

来源字段：https://doi.org/10.4171/owr/2024/45

05977 | AIM-ALGEBRAIC_GEOMETRY-0269

### KL projection under Bayesian misspecification

P / partially_solved | AIM 研讨会问题汇编

Let us assume that the true probability distribution \(q\) of a data generating process is outside the family of models \(\mathcal M\) that are being considered. Under what conditions does the posterior distribution converge to a distribution with smallest KL divergence to the true distribution \(q\)?

来源字段：http://aimpl.org/singularlearning/1/

05978 | AIM-ARITHMETIC_GEOMETRY-0057

### Known irreducibility regions and the explicit q=1 component decomposition

P / partially_solved | AIM 研讨会问题汇编

**Problem 10.** Let \(\rho=(p,q,r,0,0,\ldots)\) and define \(E_{p,q,r}:=E_\rho\) to be the moduli space of finite-length graded modules of function \(\rho\). For which \(p,q,r\) is \(E_{p,q,r}\) irreducible?

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05979 | OWR-15582-007

### Kobayashi Pseudometric Vanishing on Calabi-Yau Manifolds

O / open | Oberwolfach 报告问题汇编

Does the Kobayashi pseudometric vanish identically on every compact Calabi-Yau manifold? In particular, must every such manifold be Kobayashi non-hyperbolic and contain an entire curve?

来源字段：https://doi.org/10.4171/owr/2017/39

05980 | OWR-13946-010

### Kodaira Dimension of K3 Severi Images

O / open | Oberwolfach 报告问题汇编

For a maximal-rank component of a K3 Severi moduli map, what is the Kodaira dimension of its image in $M_g$?

来源字段：https://doi.org/10.4171/owr/2015/51

05981 | OWR-4990375-010

### Kodaira Dimension of Odd Minimal Strata

P / partially_solved | Oberwolfach 报告问题汇编

别名：Kodaira Dimension of Odd Minimal Strata [OWR-4990375-011]

For $g\geq12$, do the odd components of the minimal strata $\mathbb P\Omega\mathcal M_g(2g-2)^{\mathrm{odd}}$ have maximal Kodaira dimension?

来源字段：https://doi.org/10.4171/owr/2021/6

05982 | OWR-7155450-004

### Kodaira Dimensions of Abelian-Differential Moduli Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Determine the behavior of the Kodaira dimension of moduli spaces of Abelian differentials as the genus $g$ becomes large, especially for strata with few marked points, including the subcanonical case $n=1$ with $\mu=(2g-2)$.

来源字段：https://doi.org/10.4171/owr/2021/29

05983 | AIM-ALGEBRAIC_GEOMETRY-0344

### Kodaira dimensions of the stable-curve moduli spaces in the transition range

P / partially_solved | AIM 研讨会问题汇编

**Problem 5.2.** Determine the Kodaira dimension of \(\overline{\mathcal M}_g\) when \(g=15\) or \(17\leq g\leq22\). For \(g>22\), \(\overline{\mathcal M}_g\) is of general type, and in the other known cases, the Kodaira dimension is negative.

来源字段：https://aimath.org/WWN/birational/birational.pdf

05984 | OWR-11101913-008

### Kontsevich-Soibelman Conjecture for Calabi-Yau Degenerations

P / partially_solved | Oberwolfach 报告问题汇编

For a Calabi-Yau degeneration, do the base of the SYZ fibration and the essential skeleton coincide as metric integral-affine manifolds away from their singular loci?

来源字段：https://doi.org/10.4171/owr/2022/32

05985 | AIM-ALGEBRAIC_GEOMETRY-0248

### Koszul complexes as derived intersections and splitting obstructions

P / partially_solved | AIM 研讨会问题汇编

**Question 11 (Paul Balmer).** Using operations like \(\otimes\), \(\wedge\) and taking cones, is there a construction so that one obtains the Koszul complex \(K_\bullet(s)\) from the complex \[ \cdots\longrightarrow 0\longrightarrow P\xrightarrow{s}A \longrightarrow 0\longrightarrow\cdots? \] Does this generalize to tensor triangulated categories?

**Remark (Paul Balmer).** Let \(K\) be a tensor triangulated category and let \(X\in\operatorname{Ob}(K)\). Suppose there is a map \(X\xrightarrow{s}\mathbf 1\). If the above construction works, then one can obtain a class \(c(X)\in GW(K)\). This class should then define an obstruction class for \(X\) to split a free summand \(\mathbf 1\).

**Remark (Madhav Nori).** There is such a construction.

来源字段：https://aimath.org/pastworkshops/projectiveA1problems.pdf

05986 | OWR-14299516-008

### Künneth Equivalence for Enlarged Motivic Subcategories

O / open | Oberwolfach 报告问题汇编

If $Y$ is smooth and proper over $k$, does exterior product induce an equivalence $\mathrm{DM}(X)^{\prime\prime}\otimes_{\mathrm{DM}(k)}\mathrm{DM}(Y)^{\prime\prime}\simeq\mathrm{DM}(X\times_kY)^{\prime\prime}$ for the enlarged motivic subcategories?

来源字段：https://doi.org/10.4171/owr/2025/36

05987 | OWR-10252932-007

### Künneth Equivalence for Weil Motivic Tate Categories

O / open | Oberwolfach 报告问题汇编

For algebraic varieties $X_1,X_2$ over $\mathbb F_p$, does exterior product induce an equivalence $$\operatorname{DTM}(X_1^{\mathrm{Weil}})\otimes_{\operatorname{DTM}(\mathbb F_p)}\operatorname{DTM}(X_2^{\mathrm{Weil}})\simeq\operatorname{DTM}(X_1^{\mathrm{Weil}}\times_{\mathbb F_p}X_2^{\mathrm{Weil}})$$?

来源字段：https://doi.org/10.4171/owr/2022/24

05988 | OWR-15588-002

### L-Equivalence of Derived-Equivalent Simply Connected Varieties

S / solved | Oberwolfach 报告问题汇编

If $X$ and $Y$ are smooth projective simply connected D-equivalent varieties over a field of characteristic zero, must they be L-equivalent, i.e. must $([X]-[Y])\mathbb L^r=0$ in $K_0(\operatorname{Var}/k)$ for some $r\geq0$?

来源字段：https://doi.org/10.4171/owr/2017/45

05989 | OWR-14298795-019

### Lang-Vojta and Shafarevich Finiteness Conjectures

O / open | Oberwolfach 报告问题汇编

Does every hyperbolic variety over a number field have only finitely many integral points? More generally, for which classes of canonically polarized varieties are there only finitely many isomorphism classes with good reduction outside a fixed finite set of places?

来源字段：https://doi.org/10.4171/owr/2024/45

05990 | OWR-12495-004

### Lefschetz-Class Properties Under CM Reduction

P / partially_solved | Oberwolfach 报告问题汇编

Let $A$ be a CM-type abelian variety over $\overline{\mathbb Q}$, and let $A_0$ be a reduction of $A$. Does the property that all Hodge classes on all powers of $A$ are Lefschetz classes imply that all Tate classes on all powers of $A_0$ are Lefschetz classes? Is there a class of CM-type abelian varieties for which the converse implication holds?

来源字段：https://doi.org/10.4171/owr/2013/32

05991 | OWR-3481-004

### Left Adjoints for n-Motivic Sheaf Inclusions

O / open | Oberwolfach 报告问题汇编

Does the inclusion $HI^{\leq n}(k)\hookrightarrow HI(k)$ admit a left adjoint $(-)^{\leq n}:HI(k)\to HI^{\leq n}(k)$?

来源字段：https://doi.org/10.4171/owr/2009/31

05992 | OWR-3481-003

### Left Exactness of n-Motivic Sheaf Inclusions

O / open | Oberwolfach 报告问题汇编

In Voevodsky's theory of homotopy invariant Nisnevich sheaves with transfers over a perfect field $k$, let $HI(k)$ denote the abelian category of homotopy invariant sheaves with transfers and let $HI^{\leq n}(k)$ be the full subcategory of $n$-motivic (slice-filtration $\leq n$) sheaves. Is the inclusion functor $HI^{\leq n}(k) \hookrightarrow HI(k)$ left exact?

来源字段：https://doi.org/10.4171/owr/2009/31

05993 | OWR-11792-006

### Leray-Hodge Filtrations and Chern Classes on Singular Varieties

O / open | Oberwolfach 报告问题汇编

For a singular complex projective variety $X$, is the Leray filtration on $H^n(X,\mathbb Z)$ a filtration by sub-mixed-Hodge structures, and does the image of the $p$th Chern class map

$$K_0(X)\otimes\mathbb Q\to H^{2p}(X,\mathbb Q)$$

equal

$$\operatorname{Hom}_{\mathrm{MHS}}(\mathbb Z(-p),L^pH^{2p}(X,\mathbb Z))\otimes\mathbb Q?$$

Equivalently, are these precisely the rational Hodge classes in the $p$th Leray step?

来源字段：https://doi.org/10.4171/owr/2012/30

05994 | OWR-16417-006

### Lifting Tropical Divisors with Prescribed Rank

O / open | Oberwolfach 报告问题汇编

Given a tropical curve $\Gamma$ and a divisor $D=\sum_{i=1}^n a_i p_i$ of degree $d$ and rank $r$ on $\Gamma$, does there exist a smooth projective curve $X$ over a non-Archimedean field $K$ with dual tropical curve $\Gamma$ and a divisor $\widetilde D$ on $X$ of multiplicity profile $(a_1,\ldots,a_n)$ and rank $r$ whose specialization is $D$?

来源字段：https://doi.org/10.4171/owr/2018/42

05995 | OWR-13678-004

### Linear Precision of Toric Bezier Patches and Volumes

P / partially_solved | Oberwolfach 报告问题汇编

Which toric Bezier patches have linear precision, which toric Bezier volumes have rational linear precision, and how can birational maps of projective 3-space be classified?

来源字段：https://doi.org/10.4171/owr/2015/21

05996 | AIM-ALGEBRAIC_GEOMETRY-0140

### Linear-center effective cones and the vertical exceptional face

P / partially_solved | AIM 研讨会问题汇编

Let \(Z\subset \mathbb P^n\) be a smooth subvariety with \(1\leq \dim Z\leq n-2\). What is the effective cone of \(k\)-cycles \(\operatorname{Eff}_k(X)\)?

Its mathematical prompt is:

The immediately preceding sentences ask for the relationship between the effective cones of a variety and “its blowup,” and then discuss a positive-dimensional blowup center in \(\mathbb P^n\). Thus throughout this report the reconstructed meaning is

\[ X=\operatorname{Bl}_Z(\mathbb P^n),\qquad \pi:X\longrightarrow \mathbb P^n. \]

This reconstruction is necessary and is not silently attributed to the source. The source remark proposes as first examples a small union of lines and low-degree, low-codimension complete intersections. A union of pairwise disjoint lines is a smooth disconnected center; a union of intersecting lines is singular and lies outside the displayed smooth-center hypothesis.

Unless otherwise stated, varieties are over an algebraically closed field. The comparison with the complex literature is made over \(\mathbb C\). The linear-center calculation below is characteristic-independent.

来源字段：http://aimpl.org/poscycles/1/

05997 | AIM-ARITHMETIC_GEOMETRY-0069

### Linear-span lifting and exact plane-curve component dimensions

P / partially_solved | AIM 研讨会问题汇编

Fix integers \(d,g,n\). Find a useful, preferably sharp, lower bound for the dimensions of irreducible components of \(\operatorname{Hilb}_{d,g}(\mathbf P^n)\), the Hilbert scheme of curves with Hilbert polynomial \(dm+1-g\). The usual expected-dimension bounds are \(4d\) in \(\mathbf P^3\) and \(5d+1-g\) in \(\mathbf P^4\); the latter is uninformative for fixed \(d\) and large \(g\).

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

05998 | AIM-ALGEBRAIC_GEOMETRY-0122

### Literal projective-plane example and a logarithmic BMY obstruction

P / partially_solved | AIM 研讨会问题汇编

Can a surface with \(c_{1}^{2}=3c_{2}\) contain a rational curve in char. \(p\)?

Let \(X\) be a smooth projective minimal surface of general type over an algebraically closed field of characteristic \(p>0\), with \(K_X\) nef (or, in the ball-quotient analogue, ample) and \(K_X^2=3c_2(X)\). Can \(X\) contain an integral curve whose normalization is \(\mathbf P^1\)?

来源字段：http://aimpl.org/ratsubvarpos/9/

05999 | AIM-ALGEBRAIC_GEOMETRY-0302

### Local F-Bertini theorems, inseparability obstructions, and simultaneous good flags

P / partially_solved | AIM 研讨会问题汇编

**Problem.** Investigate possible Bertini theorems for \(F\)-singularities.

来源字段：http://aimpl.org/testandmultiplierideals/2/

06000 | OWR-13750327-008

### Local Fundamental Groups of Algebraic Singularities

P / partially_solved | Oberwolfach 报告问题汇编

For each natural class $\mathcal C_n$ of $n$-dimensional complex algebraic singularities, characterize the finitely presented groups that occur as local fundamental groups $\pi_1(X;x)$ of members of $\mathcal C_n$.

来源字段：https://doi.org/10.4171/owr/2023/18

06001 | OWR-13750327-009

### Local Fundamental Groups of Rational Singularities

P / partially_solved | Oberwolfach 报告问题汇编

Which finitely presented groups occur as local fundamental groups of rational complex algebraic singularities?

来源字段：https://doi.org/10.4171/owr/2023/18

06002 | OWR-16768-015

### Local-Cohomology Obstructions to Regular Maps and Projections

O / open | Oberwolfach 报告问题汇编

Can local-cohomology obstructions rule out injective or $k$-regular maps of complex varieties, or low-dimensional projections defined by separating invariants?

来源字段：https://doi.org/10.4171/owr/2019/6

06003 | OWR-12015-004

### Log Abundance for Semi-Log-Canonical Pairs

P / partially_solved | Oberwolfach 报告问题汇编

Let $(X,\Delta)$ be a projective semi-log-canonical pair with $\Delta$ a $\mathbb Q$-divisor. If $K_X+\Delta$ is nef, must $K_X+\Delta$ be semi-ample?

来源字段：https://doi.org/10.4171/owr/2012/43

06004 | AIM-ALGEBRAIC_GEOMETRY-0308

### Log canonical implies Du Bois, with a sharp star-graph converse obstruction

S / solved | AIM 研讨会问题汇编

**Recovered question.** Let \(X\) be a complex normal variety, assume \(K_X\) is \(\mathbb Q\)-Cartier, and assume that \(X\) has log-canonical singularities. Is the natural map \[ \mathcal O_X\longrightarrow \underline{\Omega}^{0}_{X} \] a quasi-isomorphism?

来源字段：https://aimath.org/WWN/singularvariety/problems_final.pdf

06005 | OWR-1276-002

### Log Canonical Thresholds and Modular $F$-Thresholds

O / open | Oberwolfach 报告问题汇编

For a polynomial $f$, are there infinitely many primes $p$ for which the log canonical threshold $\operatorname{lc}_0(f)$ equals the $F$-threshold $c^{\mathfrak m}(f_p)$?

来源字段：https://doi.org/10.4171/owr/2006/27

06006 | OWR-16942-005

### Log K-Stability versus Geometric Invariant Theory

P / partially_solved | Oberwolfach 报告问题汇编

For a K-polystable Fano variety $X$ and sufficiently large $\beta$, is log K-polystability of the pair $(X,(1-\beta)D)$ equivalent to geometric invariant theory stability for the natural $\operatorname{Aut}(X)$-action on $H^0(X,K_X^{-l})$?

来源字段：https://doi.org/10.4171/owr/2019/30

06007 | OWR-13750339-013

### Logarithmic Hilbert Schemes for Divisor-Stratified Varieties

P / partially_solved | Oberwolfach 报告问题汇编

For a smooth projective variety $Y$ with a simple-normal-crossings divisor $D$, construct and characterize a logarithmic Hilbert scheme encoding how subschemes meet $D$ and its strata.

来源字段：https://doi.org/10.4171/owr/2023/27

06008 | AIM-ALGEBRAIC_GEOMETRY-0329

### Logarithmic positivity and a two-point configuration-space test

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.2 (Viehweg).** Let \(U\) be a smooth variety, étale over a moduli stack of polarized manifolds \(M_H\). Let \(Y\) be a log compactification of \(U\) and \(\Gamma=Y\setminus U\).

1. Is \(\Omega_Y^1(\log \Gamma)\) weakly positive with respect to \(U\)? 2. Is \(\omega_Y(\Gamma)\) ample with respect to \(U\)?

Both of these statements are known for period domains, and it is known that the second implies the first. These problems may be hard, but similar questions would be interesting and perhaps more tractable for configuration spaces.

来源字段：https://aimath.org/WWN/birational/birational.pdf

06009 | OWR-12875-007

### Looijenga's Dolbeault Dimension Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For the moduli space $\mathcal M_g$ of Riemann surfaces of genus $g\geq2$, is its Dolbeault cohomological dimension equal to $g-2$?

来源字段：https://doi.org/10.4171/owr/2014/15

06010 | OWR-1276-006

### Low-Degree Curves Through Points in Symmetric Position

P / partially_solved | Oberwolfach 报告问题汇编

If $\Gamma\subset\mathbb P^{n-1}$ is a set of $d\geq2n+2m-1$ points in symmetric position, $h_\Gamma(2)\geq2n+m-2$, and $1\leq m\leq n-3$, must $\Gamma$ lie on a curve of degree at most $n+m-2$?

来源字段：https://doi.org/10.4171/owr/2006/27

06011 | AIM-ALGEBRAIC_GEOMETRY-0236

### Low-degree smoothness and a Wronskian kernel criterion for osculating Schubert curves

P / partially_solved | AIM 研讨会问题汇编

Let $E = H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(d))$, $n = d+1$, and $k = r+1$, so that linear series $g^r_d$ on $\mathbb{P}^1$ correspond to points of the Grassmannian $G = G(k, E)$ of dimension $N = k(n-k)$. For $x \in \mathbb{P}^1$, let $F_\bullet(x)$ be the complete osculating flag defined by order of vanishing at $x$. Given pairwise distinct real points $x_1, \ldots, x_s \in \mathbb{P}^1(\mathbb{R})$ and partitions $\lambda_i \subseteq k \times (n-k)$ satisfying $\sum_{i=1}^s |\lambda_i| = N - 1$, let $$ S = S(\lambda_\bullet; x_\bullet) := \bigcap_{i=1}^s \Omega_{\lambda_i}(F_\bullet(x_i)) \subseteq G $$ be the scheme-theoretic intersection of the corresponding Schubert varieties, which by the Eisenbud--Harris proper-intersection theorem is a projective Cohen--Macaulay curve (when nonempty) whose real locus is smooth.

**Problem 10.4 (Levinson).** Is the entire complex curve $S_{\mathbb{C}}$ smooth? Equivalently, can $S$ have a conjugate pair of nonreal singular points even though every real point of $S$ is smooth?

来源字段：https://aimath.org/pastworkshops/degenalggeomproblems.pdf

06012 | OWR-3471-006

### Lower Bounds for Seshadri Constants on Minimal Surfaces

O / open | Oberwolfach 报告问题汇编

If $S$ is a minimal projective surface, $L$ is ample on $S$, and $x\in S$, is $\varepsilon(S,L;x)\geq1/\sqrt[4]{2+|K_S^2|}$?

来源字段：https://doi.org/10.4171/owr/2009/21

06013 | OWR-16171-006

### Lower Bounds for Symplectic Resolvent Degrees

O / open | Oberwolfach 报告问题汇编

For $n\geq6$, choose an embedding $A_n\hookrightarrow\operatorname{Sp}_{2g}(\mathbb F_2)$ for which $A_n\cap U$ is a maximal elementary abelian $2$-subgroup of $A_n$, where $U$ is the unipotent radical of the Siegel parabolic, and put $e_{A_n}=\operatorname{rank}_{\mathbb F_2}(A_n\cap U)$. Let $\widetilde A_{g,2}\to A_g$ be the Galois closure of the universal $2$-torsion cover, and let $A_g^{A_n}=\widetilde A_{g,2}/A_n$. Is $$ \operatorname{RD}(\widetilde A_{g,2}\to A_g^{A_n})\geq e_{A_n} $$ ?

来源字段：https://doi.org/10.4171/owr/2018/30

06014 | AIM-ALGEBRAIC_GEOMETRY-0283

### Lower semicontinuity and finite-Frobenius open certificates

P / partially_solved | AIM 研讨会问题汇编

Investigate the lower semicontinuity of the map \[ \mathfrak p\in\operatorname{Spec}R\longmapsto s(R_{\mathfrak p}). \]

来源字段：http://aimpl.org/testandmultiplierideals/1/

06015 | OWR-13750339-012

### Manin Components of Section Spaces of Fano Fibrations

O / open | Oberwolfach 报告问题汇编

For a Fano fibration $\pi:X\to B$, prove uniqueness of the nonpathological component of $\operatorname{Sec}(X/B,\alpha)$ for every sufficiently positive section class $\alpha$, and prove homological or motivic stability of these Manin components as degree grows.

来源字段：https://doi.org/10.4171/owr/2023/27

06016 | OWR-15588-009

### Manin-Type Asymptotics for Rational-Curve Components

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a smooth projective Fano variety and let $N^\circ(X,q,d)$ sum $q^{\dim W}$ over dominant components $W$ of $\operatorname{Mor}(\mathbb P^1,X)$ of anticanonical degree $d$ whose family maps do not rationally factor through breaking maps. Must $N^\circ(X,q,d)\sim Cq^d d^{\rho(X)-1}$ as $d\to\infty$ for some $C>0$?

来源字段：https://doi.org/10.4171/owr/2017/45

06017 | OWR-11783-007

### Maps from Varieties to Okounkov Bodies

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist, in general, a reasonable map from a variety $X$ to its Okounkov body $\Delta(X,L,\nu)$, and under what hypotheses can such a map be constructed?

来源字段：https://doi.org/10.4171/owr/2012/21

06018 | OWR-12592-003

### Mass Growth Versus Categorical Entropy

P / partially_solved | Oberwolfach 报告问题汇编

For an exact endofunctor $F$ of a triangulated category equipped with a Bridgeland stability condition $\sigma$, under what natural hypotheses does the mass-growth entropy $h_\sigma(F)$ equal the categorical entropy $h_0(F)$?

来源字段：https://doi.org/10.4171/owr/2013/44

06019 | OWR-1114-005

### Mathematical Construction of Gopakumar-Vafa Invariants

P / partially_solved | Oberwolfach 报告问题汇编

Give a direct mathematical construction of Gopakumar-Vafa invariants for Calabi-Yau threefolds, including the requisite moduli spaces and Lefschetz actions, and prove their relation to Gromov-Witten invariants and integrality.

来源字段：https://doi.org/10.4171/owr/2005/53

06020 | AIM-ALGEBRAIC_GEOMETRY-0218

### Matroid classification and a sharp four-factor Cohen-Macaulay obstruction

P / partially_solved | AIM 研讨会问题汇编

Let $X$ be a variety in a generalized flag manifold $G/P$, with Schubert varieties $X_w$ ($w \in W/W_P$). Say that $X$ is *multiplicity-free* if in cohomology $[X] = \sum_{s \in S} [X_s]$ for a subset $S \subset W/W_P$. By a theorem of Brion, if $X$ is reduced, irreducible, and multiplicity-free, then $X$ admits a flat degeneration to the reduced union $\bigcup_{s \in S} X_s$, and this union is Cohen--Macaulay.

**Problem 2.2 (Knutson).** Classify subsets $S \subset W/W_P$ that deform to irreducibles (i.e., such that $\bigcup_{s \in S} X_s$ is the special fiber of a flat degeneration of an irreducible subvariety $X \subset G/P$). For example, is it sufficient that $\bigcup_{s \in S} X_s$ be Cohen--Macaulay? The first case to try is $G/P \cong (\mathbb{P}^1)^n$.

来源字段：https://aimath.org/pastworkshops/degenalggeomproblems.pdf

06021 | OWR-11695864-002

### Matroid Invariance and Reflection Constructions for Geproci Sets

O / open | Oberwolfach 报告问题汇编

Can irreducible complex reflection groups be used to classify nontrivial geproci sets? Which matroid properties are invariant under the resulting reflection constructions?

来源字段：https://doi.org/10.4171/owr/2022/54

06022 | OWR-14299288-002

### Matroids as Smooth Projective Toric Varieties

O / open | Oberwolfach 报告问题汇编

Develop a precise theory explaining the extent to which matroids behave like smooth projective toric varieties, including analogues for sheaf cohomology, vector bundles, intersection theory, and Hodge-theoretic structures.

来源字段：https://doi.org/10.4171/owr/2025/19

06023 | AMR-071-0012

### Maulik-Nekrasov-Okounkov-Pandharipande conjecture on an equivalence between Gromov-Witten theory and Donaldson-Thomas theory

P / partially_solved | AMR 问题清单汇编

Maulik-Nekrasov-Okounkov-Pandharipande conjecture on an equivalence between Gromov-Witten theory and Donaldson-Thomas theory

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

06024 | OWR-14298593-002

### Maximal Modifying Algebras for Canonical Threefold Singularities

O / open | Oberwolfach 报告问题汇编

Prove that every local three-dimensional Gorenstein canonical singularity admits a maximal modifying algebra.

来源字段：https://doi.org/10.4171/owr/2024/44

06025 | OWR-16169-016

### Maximum Simple Singularities of Plane Curves

O / open | Oberwolfach 报告问题汇编

For a plane curve of degree $d$, what is the largest $k$ for which an $A_k$ singularity can occur? More generally, for $(a,b)\in\mathbb N^2$, what is the largest $k$ for which a curve in $\mathbb A^2(\mathbb C)$ of bidegree $(a,b)$ can have an $A_k$ singularity?

来源字段：https://doi.org/10.4171/owr/2018/28

06026 | OWR-9790358-016

### Mean-Value Coordinate Equations and Wachspress-Surface Moduli

P / partially_solved | Oberwolfach 报告问题汇编

For an $n$-gon, determine the homogeneous equations of the Zariski closure of the image of the mean-value barycentric-coordinate map to $\mathbb P^{n-1}$. For Wachspress surfaces arising from $n$-gons, describe flat limits under polygon degeneration and construct a natural compactified moduli space.

来源字段：https://doi.org/10.4171/owr/2022/7

06027 | OWR-1060-006

### Meromorphic Function Fields on Locally Symmetric Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Determine the transcendence degree of the field of meromorphic functions $\mathcal{M}(\Gamma\backslash X)$ for a noncompact Hermitian locally symmetric space $\Gamma\backslash X$.

来源字段：https://doi.org/10.4171/owr/2005/39

06028 | OWR-788-006

### Metrics and Coordinate Rings on Voronoi Compactifications

P / partially_solved | Oberwolfach 报告问题汇编

Let $U_g\to\mathcal{A}_g$ be the universal abelian $g$-fold, let $\overline{U}_g$ be its compactification as the boundary divisor $D\subset\mathcal{A}_{g+1}^F$, and let $L=D|_{\overline{U}_g}$. Does $L$ admit a metric whose restriction to the generic fiber of $U_g\to\mathcal{A}_g$ has translation-invariant curvature? Is $\operatorname{TC}(\mathcal{A}_g^F)$, or its subring $\bigoplus_{a,b\geq0}H^0(\mathcal{A}_g^F,\mathcal{O}(aM-bD))$, finitely generated?

来源字段：https://doi.org/10.4171/owr/2005/07

06029 | OWR-16166-004

### Milnor K-Theory Reconstruction of Function Fields and Varieties

O / open | Oberwolfach 报告问题汇编

In reconstruction from Milnor $K$-theory, can field-reconstruction counterexamples arise from function fields of curves over algebraically closed or finite fields? Can reconstruction be extended to function fields of transcendence degree at least $2$ over more general base fields? What minimal data determine a normal, proper, geometrically integral variety over a perfect field up to isomorphism rather than merely birational equivalence?

来源字段：https://doi.org/10.4171/owr/2018/25

06030 | AIM-ARITHMETIC_GEOMETRY-0034

### Minimal finite fields and a complete dimension-two analysis

P / partially_solved | AIM 研讨会问题汇编

Let $p$ be a prime number. For a fixed dimension $g$ and a fixed $p$-rank $f$, what is the smallest field of definition of a simple abelian variety over $\overline{\mathbb F_p}$ with dimension $g$ and $p$-rank $f$?

来源字段：http://aimpl.org/cohomabelian/1/

06031 | OWR-11783-006

### Minimal Models Near Stable Surface Limits

P / partially_solved | Oberwolfach 报告问题汇编

For a nonisotrivial $\mathbb Q$-Gorenstein deformation of projective surfaces whose special fiber $X$ is an irreducible normal surface with $T$-singularities, what are the smooth minimal models of the singular and nonsingular surfaces in the moduli space near the stable limit of $X$?

来源字段：https://doi.org/10.4171/owr/2012/21

06032 | OWR-14601-007

### Minimal Rational Curves on Almost Homogeneous Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Describe the minimal families of rational curves on every smooth projective almost homogeneous variety $X$, meaning a variety equipped with an action of a connected linear algebraic group having an open orbit.

来源字段：https://doi.org/10.4171/owr/2016/19

06033 | OWR-13682-009

### Minimal Rational Curves on Complete Symmetric Varieties

O / open | Oberwolfach 报告问题汇编

Develop a theory of minimal rational curves on complete symmetric varieties, namely toroidal compactifications of symmetric spaces $G/H$ with $G$ reductive, extending the theory for wonderful group compactifications.

来源字段：https://doi.org/10.4171/owr/2015/25

06034 | OWR-11454-001

### Minimal Singularity Classes for Functorial Resolution

P / partially_solved | Oberwolfach 报告问题汇编

What is the smallest class $\mathcal S$ of singularities containing all normal-crossings singularities such that every variety $X$ admits a proper birational or bimeromorphic morphism from a variety $X'$ having only singularities in $\mathcal S$, with the morphism an isomorphism over the normal-crossings locus of $X$?

来源字段：https://doi.org/10.4171/owr/2011/56

06035 | OWR-13500-013

### Mirror Classification of Orbifold Del Pezzo Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Are mutation-equivalence classes of Fano polygons in bijection with $\mathbb Q$-Gorenstein-deformation-equivalence classes of locally $\mathbb Q$-Gorenstein-rigid class-TG del Pezzo surfaces with cyclic quotient singularities, with a polygon $P$ corresponding to a generic $\mathbb Q$-Gorenstein deformation of its toric surface $X_P$?

来源字段：https://doi.org/10.4171/owr/2015/15

06036 | OWR-13500-009

### Mirror Conifold Transitions and Analytic Smoothings

P / partially_solved | Oberwolfach 报告问题汇编

For a maximally degenerating projective conifold $Y$ with mirror $\check Y$ and conifold-point sets $S$ and $\check S$, are the following conditions equivalent: $Y$ has an analytic smoothing; some $\alpha\in H^1(Y\setminus S,\Omega^2)$ maps nontrivially to every summand of $H^2_S(Y,\Omega^2)\cong\mathbb C^S$; an analogous class in $H^1(\check Y\setminus\check S,\Omega^1)$ maps nontrivially to every summand of $H^2_{\check S}(\check Y,\Omega^1)\cong\mathbb C^{\check S}$; and $\check Y$ has a projective resolution?

来源字段：https://doi.org/10.4171/owr/2015/15

06037 | OWR-16169-004

### Mirror Families from Log Calabi-Yau Pairs

O / open | Oberwolfach 报告问题汇编

For a simple normal-crossings Calabi-Yau pair $(Y,D_Y)$ with maximal boundary and $D_Y$ supporting an ample divisor, does the Gross-Hacking-Keel curve-counting algebra construction produce a flat family of affine log Calabi-Yau varieties whose torus quotient depends only on $U=Y\setminus D_Y$ and is its mirror family?

来源字段：https://doi.org/10.4171/owr/2018/28

06038 | OWR-11139-014

### Mirror Symmetry for Relative Hodge Numbers

P / partially_solved | Oberwolfach 报告问题汇编

Mirror Hodge-number conjecture: For a mirror pair $(\widetilde X_\sigma,w)$ and $(\widetilde X_{\sigma^\vee},w^\vee)$,

$$h^{p,q}(\widetilde X_\sigma,w)=h^{n-p,q}(\widetilde X_{\sigma^\vee},w^\vee)$$

when both spaces are smooth, with orbifold Hodge numbers used in the orbifold case.

来源字段：https://doi.org/10.4171/owr/2011/47

06039 | AIM-ALGEBRAIC_GEOMETRY-0270

### Misspecified Bayesian stochastic complexity and competing pseudo-truths

P / partially_solved | AIM 研讨会问题汇编

Understand asymptotics of the stochastic complexity in the case where the true distribution is not in any model under consideration.

来源字段：http://aimpl.org/singularlearning/1/

06040 | AIM-ALGEBRAIC_GEOMETRY-0291

### Mixed test ideals need not have polyhedral constancy regions

S / solved | AIM 研讨会问题汇编

Given ideals \(I_1,\ldots,I_n\) of a regular, \(F\)-finite ring \(R\), consider the map sending an \(r\)-tuple of nonnegative real numbers \((\lambda_1,\ldots,\lambda_r)\) to \(\tau(I_1^{\lambda_1}\cdots I_n^{\lambda_n})\), the mixed test ideal. If we bound all the \(\lambda_i\) by some fixed \(M\), does there exist a rational polyhedral decomposition such that the function described is constant on the interior of each region?

来源字段：http://aimpl.org/testandmultiplierideals/2/

06041 | AIM-ALGEBRAIC_GEOMETRY-0294

### Mixed-characteristic tight closure as a plural theory, with an exact DVR comparison

P / partially_solved | AIM 研讨会问题汇编

**Extend tight closure to mixed characteristic.**

来源字段：http://aimpl.org/testandmultiplierideals/2/

06042 | OWR-13101-011

### Mixed-Spin Fields and All-Genus Quintic Invariants

P / partially_solved | Oberwolfach 报告问题汇编

Do moduli spaces of mixed-spin-$P$ fields provide an effective algorithm for computing all-genus FJRW invariants of the Fermat quintic with multiple insertions and all-genus Gromov-Witten invariants $N_g(d)$ of the quintic threefold?

来源字段：https://doi.org/10.4171/owr/2014/31

06043 | OWR-1971-005

### Modular Compactifications from Curve Singularities

P / partially_solved | Oberwolfach 报告问题汇编

Given a deformation-open collection of isolated curve singularities, does there exist a corresponding stability condition whose moduli space gives a proper compactification of the moduli space of smooth curves?

来源字段：https://doi.org/10.4171/owr/2008/26

06044 | OWR-11792-004

### Modular Interpretations of Log Canonical Curve-Moduli Models

P / partially_solved | Oberwolfach 报告问题汇编

Does the log canonical model $\overline{\mathcal M}_g(\alpha)$ inherit a moduli interpretation as a moduli space of singular curves?

来源字段：https://doi.org/10.4171/owr/2012/30

06045 | OWR-16935-005

### Moduli Spaces of Higgs Harder-Narasimhan Strata

P / partially_solved | Oberwolfach 报告问题汇编

Can coarse moduli spaces be constructed for Higgs Harder-Narasimhan strata and ordinary Harder-Narasimhan strata? Do the refined strata carry comparably rich geometry, including compatible group actions and Hitchin fibrations?

来源字段：https://doi.org/10.4171/owr/2019/23

06046 | AIM-ALGEBRAIC_GEOMETRY-0244

### Mohan Kumar dimensions and an odd-rank prime-divisor propagation theorem

P / partially_solved | AIM 研讨会问题汇编

**Question 7. Question (Aravind Asok).** Fix an integer \(d\). For certain \(d\), Mohan Kumar constructs projective modules of rank \(d-2\) which are stably free but not free over a smooth affine variety \(X\) of dimension \(d\) over an algebraically closed field. Is this known for all \(d\)? Is there a pattern?

来源字段：https://aimath.org/pastworkshops/projectiveA1problems.pdf

06047 | OWR-17475-006

### Moment Inequalities from Positive Semidefinite Localization Matrices

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Moment-Matrix Question [OWR-17475-007]

What inequalities among moments follow from positive semidefiniteness of moment matrices? More generally, for measures supported on a basic semialgebraic set, what additional inequalities follow from positive semidefiniteness of the associated localization matrices?

来源字段：https://doi.org/10.4171/owr/2020/12

06048 | OWR-12975-006

### Moment Vanishing and Polynomial Abel Equations

O / open | Oberwolfach 报告问题汇编

Let $\Omega\subset\mathbb R^n$ be a bounded domain and let $P$ and $q$ be polynomials. Determine necessary and sufficient conditions for all moments $$m_k=\int_\Omega P(x)^kq(x)\,dx,\qquad k=0,1,\ldots,$$ to vanish. In particular, when $P'=p$, determine the corresponding closed-solution conditions for the polynomial Abel equation $y'=p(x)y^2+q(x)y^3$.

来源字段：https://doi.org/10.4171/owr/2014/17

06049 | OWR-14754-010

### Mond's Conjecture for Finite Map Germs

O / open | Oberwolfach 报告问题汇编

For an $\mathcal A$-finite map germ $f:(\mathbb C^n,S)\to(\mathbb C^{n+1},0)$ in nice dimensions, is the $\mathcal A_e$-codimension of $f$ at most its image Milnor number $\mu_I(f)$, with equality when $f$ is weighted homogeneous?

来源字段：https://doi.org/10.4171/owr/2016/46

06050 | AIM-ARITHMETIC_GEOMETRY-0063

### Monomial signatures distinguish components through affine length eight

P / partially_solved | AIM 研讨会问题汇编

The AIM record (Problem 16 in *Components of Hilbert schemes*) asks:

For an irreducible component \(C\), define its full monomial signature by \[ \operatorname{Mon}(C)=\{I:[I]\in C\text{ and }I\text{ is monomial}\}. \] The question is whether \(C\mapsto\operatorname{Mon}(C)\) is injective. It is important that the question concerns equality of the entire sets, not whether two components share one monomial ideal.

The source remarks that Richard Liebling might have given a counterexample for a Hilbert scheme of curves in his 1996 thesis. The thesis and later citations were identified, but accessible sources did not verify this precise assertion. A statement about generic initial ideals would not by itself establish equality of the full monomial signatures.

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

06051 | AIM-ALGEBRAIC_GEOMETRY-0105

### Moret--Bailly rational curves with Hodge degree p-1

S / solved | AIM 研讨会问题汇编

**AIM Problem 4.2.** Find rational curves in \(\mathcal A_g\) in characteristic \(p\) that do not lift to characteristic \(0\).

来源字段：http://aimpl.org/ratsubvarpos/4/

06052 | OWR-4427-006

### Morrison’s Cone Conjecture for Calabi-Yau Pairs

P / partially_solved | Oberwolfach 报告问题汇编

For a klt Calabi-Yau pair $(X,\Delta)$, do the automorphism and pseudo-automorphism groups admit rational polyhedral fundamental domains on the effective nef cone and effective movable cone, respectively?

来源字段：https://doi.org/10.4171/owr/2010/39

06053 | AIM-ALGEBRAIC_GEOMETRY-0004

### Morse filtrations versus algebraic decompositions

P / partially_solved | AIM 研讨会问题汇编

Explore the mirror symmetry correspondence between Morse-theoretic decompositions and algebraic/algebro-geometric decompositions.

来源字段：http://aimpl.org/syzygyms/1/

06054 | OWR-14613-006

### Motives and Higher Chow Groups with Modulus

O / open | Oberwolfach 报告问题汇编

For motives with modulus, can the isomorphism $\operatorname{Hom}(M(Y,E),M(X,0)[-j])\cong CH^d(X\times(Y\setminus|E|),j)$, where $X$ is smooth, proper, equidimensional, and of dimension $d$, and $(Y,E)$ is a modulus pair, be generalized by replacing the zero divisor on $X$ with a nonzero divisor, possibly using higher Chow groups with modulus? For a curve $C$ with modulus $D$, can $i(M(C,D))\in D(\mathrm{MNST})$ be described explicitly as a complex of MNST objects?

来源字段：https://doi.org/10.4171/owr/2016/31

06055 | OWR-3481-005

### Motivic Albanese and Walker Abel-Jacobi Targets

O / open | Oberwolfach 报告问题汇编

For a complex algebraic variety $X$, is $\operatorname{Alb}(CH_g^p(X))(\mathbb C)$ canonically isomorphic to the target of Walker's morphic Abel-Jacobi map?

来源字段：https://doi.org/10.4171/owr/2009/31

06056 | OWR-16162-016

### Motivic Equivalence and Similarity of Quadratic Forms

P / partially_solved | Oberwolfach 报告问题汇编

Let $q$ and $q'$ be quadratic forms of the same dimension whose projective quadrics have isomorphic Chow motives. Is equality of all Witt indices over every field extension equivalent to similarity of $q$ and $q'$? Can the motivic criterion be obtained using Grassmannians of isotropic subspaces instead of projective quadrics, and can it be extended to all semisimple groups through motives of projective homogeneous varieties?

来源字段：https://doi.org/10.4171/owr/2018/21

06057 | OWR-14299574-007

### Motivic Igusa Series of Plane Curves

O / open | Oberwolfach 报告问题汇编

For a smooth curve $f$, compute the motivic Igusa series $Q_f(s,T)$; ultimately compute it for every reduced plane curve. For a reduced plane curve, also find an exponential relation between the cohomology of its restricted contact loci $X_{f,n}^0$ and its punctual Hilbert schemes $\operatorname{Hilb}_n(f)_0$.

来源字段：https://doi.org/10.4171/owr/2025/45

06058 | AIM-ALGEBRAIC_GEOMETRY-0240

### Motivic local systems and the Atiyah-class blindness of Artin monodromy

P / partially_solved | AIM 研讨会问题汇编

**Question 3. Question (Christian Haesemeyer). What are “motivic local systems”? Can one define motivic Atiyah classes?**

来源字段：https://aimath.org/pastworkshops/projectiveA1problems.pdf

06059 | OWR-16941-007

### Motivic Origins of Étale Height Pairings

O / open | Oberwolfach 报告问题汇编

For homologically trivial cycles on a smooth projective variety over a function field, does the generalized Beilinson étale height pairing arise from a motivic pairing with values in the rational Picard group of the base?

来源字段：https://doi.org/10.4171/owr/2019/29

06060 | OWR-14754-012

### Motivic Poles and Bernstein-Sato Roots

O / open | Oberwolfach 报告问题汇编

For a polynomial $f$ in $n$ variables, do the poles of its motivic zeta function give zeros of its Bernstein-Sato polynomial?

来源字段：https://doi.org/10.4171/owr/2016/46

06061 | OWR-4990381-007

### Motivic Realization of Rigid Local Systems

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a connected smooth quasi-projective variety over $\mathbb C$ with a smooth projective normal-crossing compactification, and let $\rho:\pi_1^{\mathrm{top}}(X,x)\to\operatorname{GL}_n(\mathbb C)$ be irreducible, rigid, and quasi-unipotent along the boundary. After restriction to a dense open subset $U\subseteq X$, must $\rho$ occur as a summand of $R^mf_*\mathbb C$ for a smooth projective family $f:Y\to U$?

来源字段：https://doi.org/10.4171/owr/2021/12

06062 | OWR-1318-006

### Motivic Sheaves over Dual Numbers

O / open | Oberwolfach 报告问题汇编

What are motivic sheaves over $k[t]/(t^2)$, and what is the tangent space to the space of motives?

来源字段：https://doi.org/10.4171/owr/2006/32

06063 | OWR-12334-004

### Motivic Stabilization of Symmetric Powers

O / open | Oberwolfach 报告问题汇编

Motivic stabilization conjecture: If $X$ is geometrically irreducible, then

$$\frac{[\operatorname{Sym}^nX]}{\mathbb L^{n\dim X}}$$

tends to a limit as $n\to\infty$.

来源字段：https://doi.org/10.4171/owr/2013/06

06064 | AIM-ALGEBRAIC_GEOMETRY-0150

### Motivic suspension, framed BPQ, and a quadratic zero-stem obstruction

P / partially_solved | AIM 研讨会问题汇编

In classical homotopy theory, there are, among many others, three important notions/results:

(1) The Freudenthal suspension theorem

(2) The Barratt-Priddy-Quillen theorem

(3) Infinite loop space theory

Can we construct/prove the above notions in the context of motivic homotopy theory?

来源字段：http://aimpl.org/equideralggeom/1/

06065 | AIM-ALGEBRAIC_GEOMETRY-0317

### Movable versus divisorial canonicity in degrees four and five

P / partially_solved | AIM 研讨会问题汇编

The record comes from Problem 17 (Cheltsov) in the AIM workshop list *Numerical invariants of singularities and higher-dimensional algebraic varieties*.

Let \(X\) be a smooth Fano variety with \(\operatorname{Pic}X=\mathbb Z\). Is birational superrigidity equivalent to

\[ \tag{1} \text{for every movable }\mathcal M\subset |-mK_X|,\qquad \left(X,\frac1m\mathcal M\right)\text{ is canonical} \]

for every \(m>0\)? Here “movable” is the phrase “with no fixed components” in the source. A second condition is

\[ \tag{2} \text{for every }D\in |-mK_X|,\qquad \left(X,\frac1mD\right)\text{ is canonical (or plt)} \]

for every \(m>0\). The note then says that if (1) and (2) hold, the only Mori fibre structures on \(X\times\mathbb P^1\) are the two projections, and adds: “More precisely, every birational map from \(X\times\mathbb P^1\) to another Mori fiber space is an isomorphism.” Finally it cites Pukhlikov for (1) and (2) on a sufficiently general hypersurface

\[ X_N\subset\mathbb P^N,\qquad \deg X_N=N,\qquad N\geq 6, \]

and asks what happens for \(N=4\) and \(N=5\). Thus \(X_4\) is a quartic threefold and \(X_5\) is a quintic fourfold.

They are treated separately below. All varieties are over \(\mathbb C\).

来源字段：https://aimath.org/WWN/singularvariety/problems_final.pdf

06066 | OWR-1737-001

### Moving Ideals of Rationally Parametrized Curves and Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

For rationally parametrized curves and surfaces, what is the structure of the remaining moving-curve ideal, in connection with Cox's conjecture and adjoint linear systems? What can be said about the moving-surface ideal for tensor-product surfaces of bidegree $(2,1)$, and, when three bihomogeneous polynomials of bidegree $(2,1)$ have no common zero on $\mathbb P^1\times\mathbb P^1$, what is the free resolution of the ideal they generate?

来源字段：https://doi.org/10.4171/owr/2007/54

06067 | OWR-15177-010

### Multiple Kodaira Fiberings and Relative Irregularity

O / open | Oberwolfach 报告问题汇编

Do compact complex surfaces exist with three or more Kodaira fiberings? Can one obtain better or asymptotically sharp upper bounds, as the fiber genus $g(F)$ grows, for the relative irregularity $q_f$ of a nonisotrivial fibered Kähler surface, especially a Kodaira fibration?

来源字段：https://doi.org/10.4171/owr/2016/56

06068 | OWR-5155-007

### Multiplicative Generation of Rational K-Theory

O / open | Oberwolfach 报告问题汇编

For $X$ smooth and proper over a field $F$ of characteristic $p$, is rational $K$-theory multiplicatively generated by $K_0(X)_{\mathbb Q}$ and Milnor $K$-theory of $F$; equivalently, what is the kernel of the surjection $CH^n(X)_{\mathbb Q}\otimes K^M_{i-n}(F)_{\mathbb Q}\twoheadrightarrow H_M^{i+n}(X,\mathbb Q(n))$?

来源字段：https://doi.org/10.4171/owr/2011/27

06069 | OWR-14299284-004

### Multiplicativity of Tautological Chow Projections

P / partially_solved | Oberwolfach 报告问题汇编

Is the tautological projection from the Chow ring to its tautological subspace a ring homomorphism?

来源字段：https://doi.org/10.4171/owr/2025/28

06070 | OWR-16408-005

### Multiplicity Bounds for Principal Theta Divisors

P / partially_solved | Oberwolfach 报告问题汇编

For every point $x\in\Theta$ of the theta divisor of an indecomposable principally polarized abelian variety of dimension $g$, is

$$\operatorname{mult}_x\Theta\leq\frac{g+1}{2}$$

always hold?

来源字段：https://doi.org/10.4171/owr/2018/33

06071 | OWR-15178-015

### Multiplicity Reduction by Regular Equimultiple Blowups

O / open | Oberwolfach 报告问题汇编

Can reduction of the multiplicity of a variety in positive characteristic be proved by blowing up along regular equimultiple centers, in particular for $X=\operatorname{Spec}(B)\to W=\operatorname{Spec}(S)$ with $S\subset B$ purely inseparable and S regular?

来源字段：https://doi.org/10.4171/owr/2016/57

06072 | OWR-4427-004

### Multiplier-Ideal Divisors from Kähler-Ricci Flow Limits

O / open | Oberwolfach 报告问题汇编

Let $X$ be an $n$-dimensional Fano manifold, let $\phi_\infty$ be a subsequential limit of its normalized Kähler-Ricci flow, and let $V(\gamma\phi_\infty)$ be the associated multiplier-ideal subscheme. Is $V(\gamma\phi_\infty)$ a divisor? For every $\varepsilon>0$ and all sufficiently large $m$, choose a $G$-invariant linear system $|\Sigma|\subset|-mK_X|$ containing a divisor $D$ for which the pair $\left(X,\frac{n+\varepsilon}{(n+1)m}D\right)$ is not Kawamata log terminal. How do $V(\gamma\phi_\infty)$ and $D$ compare, and which one violates Ross-Thomas stability of $X$?

来源字段：https://doi.org/10.4171/owr/2010/39

06073 | OWR-4527-001

### Multipoint Seshadri Constants and Nagata-Type Conjectures

O / open | Oberwolfach 报告问题汇编

For sufficiently many sufficiently general points $Z$ on a polarized variety $(X,L)$, is the Seshadri constant $\varepsilon(X,L;Z)$ maximal possible? In $\mathbb{P}^{2}$ this includes Nagata's conjecture for $r\geq10$ general points and the SHGH prediction for plane linear systems.

来源字段：https://doi.org/10.4171/owr/2010/45

06074 | OWR-14754-014

### Multivariable Motivic Poles and Monodromy

P / partially_solved | Oberwolfach 报告问题汇编

For a holomorphic germ $F=(f_1,\ldots,f_r):(\mathbb C^n,0)\to(\mathbb C^r,0)$, do all polar hyperplanes of its multivariable motivic zeta function lie in the zero locus $Z(B_F)$ of the Bernstein-Sato ideal? Does the corresponding multivariable monodromy conjecture hold?

来源字段：https://doi.org/10.4171/owr/2016/46

06075 | OWR-11101913-009

### Mumford-Tate Conjectures for Gushel-Mukai Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Do the Mumford-Tate, motivated Mumford-Tate, Hodge, and Tate conjectures hold for Gushel-Mukai varieties and their generalized partners or duals?

来源字段：https://doi.org/10.4171/owr/2022/32

06076 | OWR-17473-004

### Mutable Laurent Polynomials and Toric Smoothing Components

O / open | Oberwolfach 报告问题汇编

Let $F$ be a $3$-dimensional lattice polytope and let $U_F$ be the Gorenstein toric affine $3$-fold obtained from the cone over $F$ at height $1$. Are the smoothing components of the miniversal deformation of $U_F$ in one-to-one correspondence with the $0$-mutable Laurent polynomials having Newton polytope $F$?

来源字段：https://doi.org/10.4171/owr/2020/10

06077 | OWR-10252925-008

### Mutations of Higher Grassmannian Degenerations

O / open | Oberwolfach 报告问题汇编

Characterize the combinatorial mutations relating toric degenerations arising from distinct maximal cones in tropical Grassmannians $\operatorname{Gr}(k,n)$ for $k>2$.

来源字段：https://doi.org/10.4171/owr/2022/17

06078 | OWR-17296-002

### Nagata Conjecture for Multipoint Seshadri Constants

O / open | Oberwolfach 报告问题汇编

Nagata's conjecture asks whether, for $r\geq 10$ and sufficiently general points $x_1,\ldots,x_r$ in $\mathbb{P}^2$, the multipoint Seshadri constant of $\mathcal{O}_{\mathbb{P}^2}(1)$ equals $1/\sqrt r$.

来源字段：https://doi.org/10.4171/owr/2019/53

06079 | AMR-071-0015

### Nakai conjecture

O / open | AMR 问题清单汇编

Nakai conjecture: if a complex algebraic variety has a ring of differential operators generated by its contained derivations, then it must be smooth.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

06080 | OWR-1703871-008

### Natural Completions of Period Maps

P / partially_solved | Oberwolfach 报告问题汇编

Given a smooth projective variety $B$ with reduced normal-crossing divisor $Z$, a period map $\Phi:B\setminus Z\to\Gamma\backslash D$, and image $\mathcal P=\Phi(B\setminus Z)$, what natural completions of $\Phi$ to a map from $B$ exist, and what properties do they have?

来源字段：https://doi.org/10.4171/owr/2020/24

06081 | OWR-11139-005

### Necessary Conditions for Laurent-Polynomial Fano Mirrors

P / partially_solved | Oberwolfach 报告问题汇编

What necessary conditions must a Laurent polynomial $f$ satisfy in order to be mirror to a Fano manifold?

来源字段：https://doi.org/10.4171/owr/2011/47

06082 | OWR-8415350-007

### Negativity of Log Divisors on Q-Acyclic Surfaces

O / open | Oberwolfach 报告问题汇编

别名：Unspecified Rank-One Del Pezzo Surface Conjecture [OWR-8415350-008]

If $S$ is a complex $\mathbb Q$-acyclic surface and $(X,D)$ is a log-smooth completion, is $$ \kappa\!\left(K_X+\frac12D\right)=-\infty $$ ?

来源字段：https://doi.org/10.4171/owr/2021/47

06083 | AIM-ALGEBRAIC_GEOMETRY-0275

### Newton nondegeneracy needs base-p arithmetic, with a disjoint-variable cap lemma

S / solved | AIM 研讨会问题汇编

Do Howald's nondegeneracy conditions with respect to the Newton polyhedron on the term ideal provide conditions under which \[ \operatorname{fpt}(\text{polynomial}) = \operatorname{fpt}(\text{associated term ideal})? \]

来源字段：http://aimpl.org/testandmultiplierideals/1/

06084 | OWR-11133-003

### Newton-Okounkov Bodies of Bott-Samelson Varieties

O / open | Oberwolfach 报告问题汇编

For a Bott-Samelson variety $X=R(i_1,\ldots,i_l)$ of $\operatorname{GL}_n$, does the Newton-Okounkov body for the indicated flag equal

$$P=A_{i_1}T_{v_1}A_{i_2}\cdots T_{v_{l-1}}A_{i_l}(p)$$

for suitable $p\in\mathbb R^d$ and $v_j\in\mathbb R^{d_{i_j}}$?

来源字段：https://doi.org/10.4171/owr/2011/41

06085 | OWR-15586-004

### Newton-Okounkov Geometry for Positive Curve Classes

O / open | Oberwolfach 报告问题汇编

Can the positivity theory for curve classes in $N_1(X)$ be expressed through Newton-Okounkov-type convex bodies?

来源字段：https://doi.org/10.4171/owr/2017/43

06086 | AIM-ALGEBRAIC_GEOMETRY-0078

### Nilpotent critical support and Hilbert-scheme Jordan strata

P / partially_solved | AIM 研讨会问题汇编

What is the interaction between cell theory and the Hilbert scheme? Matrix factorizations?

In the affine Steinberg setting this corresponds to the stratification by nilpotent orbits. Does this have an analog for \(\mathscr X^{\mathrm{big}}\)?

Is there a nilpotent-orbit support filtration of the Oblomkov--Rozansky matrix-factorization convolution category, does it descend to a partition stratification on the Hilbert-scheme model, and does that filtration agree with the Kazhdan--Lusztig cell filtration?

来源字段：http://aimpl.org/catheckehilbert/1/

06087 | AIM-ARITHMETIC_GEOMETRY-0027

### No essential transcendental Brauer-Manin obstruction on abelian torsors

S / solved | AIM 研讨会问题汇编

Does any Abelian variety have a transcendental Brauer class which obstructs the existence of rational points?

A theorem of Manin says that if an Abelian variety has a transcendental Brauer class, and $\mathrm{Sha}(A)$ is finite, $E$ a torsor for $A$, and $E(\mathbb{A})^{\mathrm{Br}_{\mathrm{alg}}} \neq \emptyset$, then $E(K) \neq \emptyset$. Can we show that the existence of a transcendental obstruction implies the existence of algebraic obstructions which rule out the same adelic points as coming from global points?

The stored source URL is http://aimpl.org/brauermoduli/2/; the surviving official workshop page is https://aimath.org/pastworkshops/brauermoduli.html.

The text needs two repairs before it becomes a coherent mathematical question.

1. An abelian variety $A/k$, as opposed to a torsor under it, has its identity $0\in A(k)$. Therefore no Brauer class can obstruct the *existence* of a rational point on $A$. The diagonal adelic identity is orthogonal to every Brauer class. 2. The clause “if an Abelian variety has a transcendental Brauer class” plays no role in the Manin statement. The intended non-pointed object is a torsor $X$ (called $E$ in the record) under an abelian variety $A$ over a number field $k$.

The intended problem has three logically different readings.

- **Essential transcendental obstruction:** can \[ X(\mathbb A_k)^{\operatorname{Br}_1X}\ne\varnothing \quad\text{but}\quad X(\mathbb A_k)^{\operatorname{Br}X}=\varnothing? \] - **Obstruction-level replacement:** if one or more possibly transcendental classes make a Brauer--Manin set empty, is there an algebraic class (or algebraic subgroup) that also makes it empty? - **Pointwise/classwise simulation:** can the Brauer--Manin evaluation function or the zero set of each individual transcendental class always be reproduced by algebraic classes?

The first two readings are solved affirmatively in the desired direction: essential transcendental obstructions do not occur, and an emptiness obstruction is captured by a single locally constant algebraic class. The third, much stronger reading is not what the known theorem says.

All results below concern number fields. For a genuine Brauer--Manin obstruction to the Hasse principle we assume $X(\mathbb A_k)\ne\varnothing$.

来源字段：http://aimpl.org/brauermoduli/2/

06088 | OWR-15586-006

### Non-Bigness of Diagonals in Projective Varieties

O / open | Oberwolfach 报告问题汇编

Let $X$ be a smooth projective variety. Develop criteria proving that the diagonal $\Delta_X\subset X\times X$ is not big.

来源字段：https://doi.org/10.4171/owr/2017/43

06089 | OWR-16158-014

### Nonabelian Torsors and Essentially Finite Parabolic Bundles

O / open | Oberwolfach 报告问题汇编

Does the equivalence between tamely ramified finite abelian $G$-torsors with prescribed ramification data and essentially finite parabolic bundles with abelian monodromy extend to finite nonabelian group schemes $G$ over $k$?

来源字段：https://doi.org/10.4171/owr/2018/17

06090 | AIM-ARITHMETIC_GEOMETRY-0018

### Noncyclic discriminant kernels obstruct prescribed fineness classes

P / partially_solved | AIM 研讨会问题汇编

Given a K3 surface $X$ with a Brauer class, is that Brauer class the obstruction to fineness as a moduli space of sheaves?

来源字段：http://aimpl.org/brauermoduli/1/

06091 | OWR-10252925-004

### Nondivisorial Valuations in Explicit Fano Degenerations

P / partially_solved | Oberwolfach 报告问题汇编

Construct an explicit nontrivial canonical degeneration of a Fano variety whose entropy-minimizing valuation is nondivisorial.

来源字段：https://doi.org/10.4171/owr/2022/17

06092 | OWR-1784-002

### Nonemptiness of Affine Deligne-Lusztig Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathbb F_q$ be finite, let $k=\overline{\mathbb F_q}$ and $L=k((\varepsilon))$, and let $G$ be split, semisimple, and simply connected over $\mathbb F_q$. Let $\mathbf a$ be the base alcove, $I$ its standard Iwahori subgroup, $W_a$ the affine Weyl group with finite Weyl group $W$ and simple reflections $S$, and $\sigma$ the Frobenius. For $b\in G(L)$ and $x\in W_a$, set $X_x(b)=\{g\in G(L)/I:g^{-1}b\sigma(g)\in IxI\}$. Let $\eta_1:W_a\to W$ be projection, and define $\eta_2(x)$ as the unique $w\in W$ for which $w^{-1}x\mathbf a$ lies in the dominant Weyl chamber. Call $x$ shrunken if $U_\alpha\cap xI\neq U_\alpha\cap I$ for every finite root $\alpha$. For shrunken $x$, is $X_x(1)$ nonempty exactly when $\eta_2(x)^{-1}\eta_1(x)\eta_2(x)\notin\bigcup_{T\subsetneq S}W_T$, and in that case is $\dim X_x(1)=\tfrac12\bigl(\ell(x)+\ell(\eta_2(x)^{-1}\eta_1(x)\eta_2(x))\bigr)$? For every $\sigma$-conjugacy class $[b]$, does there exist $n_0$ such that $X_x(b)$ is nonempty exactly when $X_x(1)$ is nonempty for all $x$ with $\ell(x)\geq n_0$?

来源字段：https://doi.org/10.4171/owr/2008/03

06093 | AIM-ALGEBRAIC_GEOMETRY-0334

### Nonnested span ranks and corrected jet-span estimates for Hilbert weights

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.3 (Morrison).** For a variety \(X\) and ample line bundle \(L\), can we work harder on the combinatorics of the monomial subspaces of \(H^0(X,L^m)\) to get easier estimates for the weights \(r_{\lambda,X}\) of the Hilbert point of \(X\) with respect to the filtration given by some one-parameter subgroup \(\lambda\)? For example:

1. Find more flexible filtrations of \(H^0(X,L^m)\) with stages of the type occurring in geometric estimates (e.g. cusps are not semistable) and compute the leading term of the resulting estimates for \(r_{\lambda,X}\). 2. Find methods for producing estimates for \(r_{\lambda,X}\) which use estimates for non-nested collections of subspaces of \(H^0(X,L)\).

来源字段：https://aimath.org/WWN/birational/birational.pdf

06094 | AIM-ALGEBRAIC_GEOMETRY-0256

### Nonnormal plt centers and their isolated Hartogs defect

S / solved | AIM 研讨会问题汇编

**Normality of plt pairs.** Suppose that \((X,D=S+B)\) is a plt pair. Must \(S\) be normal?

If this isn't true, there may be an obstruction to existence of flips.

来源字段：http://aimpl.org/minimalmodcharp/2/

06095 | OWR-17296-010

### Nonreduced Special Linear Systems on Plane Blowups

O / open | Oberwolfach 报告问题汇编

Let $X$ be the blowup of $\mathbb{P}^2$ at $r$ very general points and let $D=dH-\sum_i m_iE_i$. If the linear system $|D|$ is special, meaning its dimension exceeds $\max\{\binom{d+2}{2}-\sum_i\binom{m_i+1}{2}-1,-1\}$, must every divisor in $|D|$ be nonreduced?

来源字段：https://doi.org/10.4171/owr/2019/53

06096 | OWR-7155446-005

### Nonsmooth Homogeneous Complex Monge-Ampère Solutions

O / open | Oberwolfach 报告问题汇编

How can one construct a solution $u$ of the homogeneous complex Monge-Ampère equation on $\mathbb{P}^1$ times the unit disc that fails to be $C^2$, including at an interior point?

来源字段：https://doi.org/10.4171/owr/2021/25

06097 | AIM-ALGEBRAIC_GEOMETRY-0332

### Nonstabilization of pluricanonical GIT and a finite-wall diagnostic

S / solved | AIM 研讨会问题汇编

**Problem 3.1 (from Kollár's lecture).** Fix a class of (pluri)canonically embedded manifolds in \(\mathbf P^n\). There is a notion of GIT semistability for Hilbert points in the closure of the Hilbert points of the class of manifolds considered. Now embed the family in another \(\mathbf P^m\) using a higher power of \(K_X\) and consider the GIT semistability conditions in that projective space. Is there an integer \(N\) such that the class of GIT semistable surfaces stabilizes for embeddings by \(M K_X\) for \(M>N\)? This problem is fundamental, and the answer for curves is that \(N=5\).

来源字段：https://aimath.org/WWN/birational/birational.pdf

06098 | OWR-1452-004

### Nontame Automorphisms with Point-Centered Sarkisov Factorizations

O / open | Oberwolfach 报告问题汇编

Does there exist an essentially nontame automorphism $\theta$ of affine $3$-space $\mathbb{C}^3$ such that the Sarkisov factorization of the induced Cremona transformation $\Phi_\theta$ of $\mathbb{P}^3$ begins with a possibly weighted blowup of a point on the hyperplane at infinity?

来源字段：https://doi.org/10.4171/owr/2007/01

06099 | OWR-1326-002

### Nonvanishing or Uniruledness for Smooth Projective Varieties

O / open | Oberwolfach 报告问题汇编

Does every smooth projective variety $X$ either have a nonzero pluricanonical section in $H^0(X,\mathcal O_X(mK_X))$ for some $m>0$, or is it covered by rational curves?

来源字段：https://doi.org/10.4171/owr/2006/40

06100 | AIM-ALGEBRAIC_GEOMETRY-0153

### Norm-compatible Tate gluing and a non-normable E-infinity triple

P / partially_solved | AIM 研讨会问题汇编

When \(G=C_p\), any genuine \(G\)-spectrum can be described via gluing data of the form \((X,Y,X\rightarrow Y^{tG})\) where \(X\) is a spectrum, \(Y\) is a Borel spectrum, and \(Y^{tG}\) denotes the Tate spectrum. An analogous pasting construction can be used to describe genuine \(G\)-spectra for any finite group \(G\).

What is the description of \(Comm_G(Sp^G)\) using the above model?

来源字段：http://aimpl.org/equideralggeom/2/

06101 | AIM-ALGEBRAIC_GEOMETRY-0341

### Normal directions and forced nodes for two explicit stable limits

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.7.** Can one similarly give explicit examples of moduli spaces of surfaces of general type? For example, it would be interesting to understand the stable degenerations of (1) octic double planes, (2) quintic hypersurfaces, (3) bidouble covers, and other abelian covers, following Catanese, Manetti, Pardini, etc.

来源字段：https://aimath.org/WWN/birational/birational.pdf

06102 | AIM-ALGEBRAIC_GEOMETRY-0006

### Normal-bundle criteria for exceptional sheaves on toric strata

P / partially_solved | AIM 研讨会问题汇编

Does every smooth toric variety admits an exceptional collection of sheaves ... More specifically, is it possible to find an exceptional collection from (push-forwards of) line bundles on toric strata?

**Recovered question.** Over an algebraically closed field of characteristic zero, does every smooth projective toric variety \(X\) admit a full exceptional collection whose objects are coherent sheaves? Can the sheaves be chosen as \(i_*\mathcal L\), where \(i:V(\sigma)\hookrightarrow X\) includes a torus-orbit closure (allowing \(V(0)=X\)) and \(\mathcal L\) is a line bundle on \(V(\sigma)\)?

来源字段：http://aimpl.org/syzygyms/2/

06103 | OWR-11139-010

### Normality of Varieties of Minimal Rational Tangents

O / open | Oberwolfach 报告问题汇编

Is the variety of minimal rational tangents $\mathcal C_x$ normal at a general point $x\in X$?

来源字段：https://doi.org/10.4171/owr/2011/47

06104 | AIM-ALGEBRAIC_GEOMETRY-0146

### Normed motivic spectra and incomplete Galois-window norms

P / partially_solved | AIM 研讨会问题汇编

In (Blumberg--Hill 2015, MR3406512), Blumberg and Hill define \(N_{\infty}\)-operads, equivariant generalizations of \(E_{\infty}\)-operads.

What is the motivic analogue of an \(N_{\infty}\)-operad or in other words, what are the analogues of \(G\)-commutative ring spectra in motivic homotopy theory?

来源字段：http://aimpl.org/equideralggeom/1/

06105 | OWR-704-002

### Numerical and Nef Dimensions of Line Bundles

P / partially_solved | Oberwolfach 报告问题汇编

For a nef line bundle $L$ on a smooth projective variety $X$, what geometric mechanisms explain strict inequality between its numerical and nef dimensions?

来源字段：https://doi.org/10.4171/owr/2004/21

06106 | OWR-4340-009

### Numerical Characterization of $q$-Ampleness

O / open | Oberwolfach 报告问题汇编

Give a numerical characterization of $q$-ampleness that generalizes Kleiman's theorem for $q=0$.

来源字段：https://doi.org/10.4171/owr/2010/27

06107 | OWR-13852-001

### Numerical Criterion for Deformed Hermitian-Yang-Mills Metrics

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Deformed Hermitian-Yang-Mills Stability Criterion [OWR-13852-002]

For a holomorphic line bundle $L$ over a compact Kähler manifold $X$, does the deformed Hermitian-Yang-Mills equation admit a solution if and only if every irreducible holomorphic subvariety $V\subset X$ satisfies $\operatorname{Arg}\left(\int_V e^{-i\omega}\operatorname{ch}(L)\right)>-\operatorname{Arg}\left(\int_X e^{-i\omega}\operatorname{ch}(L)\right)$?

来源字段：https://doi.org/10.4171/owr/2015/31

06108 | AIM-ALGEBRAIC_GEOMETRY-0001

### Numerical mirror data and a torsion-sector obstruction

P / partially_solved | AIM 研讨会问题汇编

How to find (algebraically, say) a mirror construction for fake projective planes (certain rigid surface of general type with same Hodge numbers as projective space, otherwise very different).

- There exist 100 of them, but only 22 have known explicit equations. - Not complete intersections.

来源字段：http://aimpl.org/syzygyms/1/

06109 | OWR-1971-007

### Oda and Sturmfels Conjectures for Smooth Toric Varieties

O / open | Oberwolfach 报告问题汇编

Let $X$ be a smooth projective toric variety, $L$ ample, and $L'$ nef. Is $H^0(X,L)\otimes H^0(X,L')\to H^0(X,L\otimes L')$ always surjective? If the section ring $R(X,L)$ is normal, is its ideal of relations generated in degree two?

来源字段：https://doi.org/10.4171/owr/2008/26

06110 | OWR-14299288-011

### Oda’s Strong Factorization Conjecture for Toric Varieties

O / open | Oberwolfach 报告问题汇编

Can any two smooth birational toric varieties be connected by a sequence of smooth blowups followed by a sequence of smooth blowdowns?

来源字段：https://doi.org/10.4171/owr/2025/19

06111 | AIM-ARITHMETIC_GEOMETRY-0014

### Odd and transcendental 3-torsion Brauer--Manin obstructions on K3 surfaces

S / solved | AIM 研讨会问题汇编

Does there exist a K3 surface $X$ and a Brauer class $\alpha$ of odd order $n$ defined over a number field $k$ such that $\alpha$ gives an obstruction to the existence of rational points?

What about an example with $n>2$?

What about a transcendental example?

来源字段：http://aimpl.org/brauermoduli/1/

06112 | AIM-ALGEBRAIC_GEOMETRY-0110

### Odd-characteristic stable irrationality and bounded-complexity reduction for ruledness

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** Fix \(n\geq1\) and \(d\geq 2\lceil(n+3)/3\rceil\). Is there a number \(P(n,d)\) such that, for every prime \(p>P(n,d)\), the very general smooth degree-\(d\), \(n\)-dimensional hypersurface \(X_d\subset\mathbb P^{n+1}\) in characteristic \(p\) is (i) irrational, (ii) not stably rational, and (iii) not ruled?

来源字段：http://aimpl.org/ratsubvarpos/6/

06113 | AIM-ALGEBRAIC_GEOMETRY-0245

### Odd-parity and Euler-realizability reduction for top-Chern splitting

P / partially_solved | AIM 研讨会问题汇编

**Question 8. Question (Jean Fasel).** Let \(X\) be a smooth affine \(k\)-variety of odd dimension \(d\). Let \(P\) be a vector bundle of rank \(d\). Does \(c_d(P)\in CH^d(X)\) detect existence of a free summand of rank 1 for \(P\)?

来源字段：https://aimath.org/pastworkshops/projectiveA1problems.pdf

06114 | AIM-ALGEBRAIC_GEOMETRY-0035

### Odd-primary Wood filtrations and a chromatic lower bound

P / partially_solved | AIM 研讨会问题汇编

The Wood cofibre sequence is the cofibre sequence \(\Sigma KO\xrightarrow{\eta}KO\to KU\) (and similarly for the connective versions). There is also a version relating \(BP_{\mathbb R}\langle n\rangle^{C_2}\) to \(BP_{\mathbb R}\langle n\rangle\).

What are odd primary analogs of the Wood cofibre sequence? For example, one might expect a \(p\)-stage filtration on \(E_{p-1}\) with associated graded \(E_{p-1}^{hC_p}\). Is there an analog for \(E_{(p-1)p^{n-1}m}\)?

来源字段：http://aimpl.org/equivstable/3/

06115 | AIM-ALGEBRAIC_GEOMETRY-0029

### Odd-prime solution and completion-transfer reduction

P / partially_solved | AIM 研讨会问题汇编

Fix a prime \(p\). Let \[ j_p=\tau_{\geq 0}L_{K(1)}\mathbb S. \] Is there an equivalence \[ \operatorname{TP}(\mathbb Z_p)\simeq (j_p^{\mathrm{triv}})^{tS^1}, \] where the circle acts trivially on \(j_p\)?

The record adds that this would follow, for example, from \(\operatorname{TR}(\mathbb Z_p)\simeq j_p\). There is no visible corruption in the record. The citation `MR1317575` is Hesselholt--Madsen, *The \(S^1\)-Tate spectrum for \(J\)*.

There is one convention that must be made explicit. In this fixed-prime context, the natural modern reading is continuous (or \(p\)-complete) periodic topological cyclic homology \[ \operatorname{TP}(R;\mathbb Z_p) :=\bigl(\operatorname{THH}(R)^{tS^1}\bigr)^\wedge_p. \] The right side is built from the \(p\)-complete image-of-\(J\) spectrum. The theorem below proves the conjecture under this reading. If the displayed source statement is read as an assertion about *uncompleted* \(\operatorname{TP}(H\mathbb Z_p)\), only its \(p\)-completion is addressed by the available theorem.

来源字段：http://aimpl.org/equivstable/2/

06116 | AIM-ALGEBRAIC_GEOMETRY-0044

### Odd-stem exponent two at C2 and a C4 transfer-kernel reduction

P / partially_solved | AIM 研讨会问题汇编

**Relations in \(\mathrm{BP}^{((G))}\).** Is \(8\sigma=0\) in the fixed points of \(\mathrm{BP}^{((G))}\)?

来源字段：http://aimpl.org/equivstable/4/

06117 | AIM-ALGEBRAIC_GEOMETRY-0087

### One-strand generation dichotomy and a six-strand decategorified obstruction

P / partially_solved | AIM 研讨会问题汇编

Is \(Z(K^b(\mathbb{S}\mathrm{Bim}_{\mathrm{ext}}))\) generated by Gaitsgory central sheaves? Is \(Z(K^b(\mathbb{S}\mathrm{Bim}_{\mathrm{fin}}))\) generated by images of Gaitsgory central sheaves? Is \(Z(K^b(\mathbb{S}\mathrm{Bim}_{\mathrm{fin}}))\) split generated by powers of \(\mathrm{FT}_n\)? Is \(Z(K^b(\mathbb{S}\mathrm{Bim}_{\mathrm{fin}}))\) symmetric monoidal?

来源字段：http://aimpl.org/catheckehilbert/4/

06118 | OWR-12697688-005

### Oort’s Hecke-Orbit Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $x=(A_x,\lambda)$ be an $\mathbb F_p$-point of $\mathcal A_g$. Prove that the Hecke orbit of $x$ under symplectic isogenies is Zariski dense in the Newton stratum containing $x$.

来源字段：https://doi.org/10.4171/owr/2023/5

06119 | AMR-146-0008

### Open Problems in Algebraic Statistics - Bonus Problem 8

O / open | AMR 问题清单汇编

} {\em Does every strict CI model have a $\mathbb{Q}$-rational point?}

背景链接：https://arxiv.org/abs/0707.4558

06120 | AMR-146-0002

### Open Problems in Algebraic Statistics - General Problem 2

O / open | AMR 问题清单汇编

This particular open problem appears in [ASCB, Conjecture 3.24], and it here serves as a placeholder for the following broader direction of inquiry.

*Study the geometry and commutative algebra of graphical models with hidden random variables. Construct these varieties by gluing familiar secant varieties, and by applying representation theory.*

背景链接：https://arxiv.org/abs/0707.4558

06121 | AMR-146-0004

### Open Problems in Algebraic Statistics - General Problem 4

O / open | AMR 问题清单汇编

*Study the geometry of maximum likelihood estimation for algebraic statistical models. *

背景链接：https://arxiv.org/abs/0707.4558

06122 | AMR-146-0006

### Open Problems in Algebraic Statistics - General Problem 6

O / open | AMR 问题清单汇编

*Study the geometry of conditional independence models for multivariate Gaussian random variables.*

背景链接：https://arxiv.org/abs/0707.4558

06123 | AMR-146-0007

### Open Problems in Algebraic Statistics - General Problem 7

O / open | AMR 问题清单汇编

} {\em Characterize the image of the entropy map $ H $ and how it intersects the various faces of ${\tt SubMod}_n$. Study the fibers of this map.}

背景链接：https://arxiv.org/abs/0707.4558

06124 | AMR-146-0001

### Open Problems in Algebraic Statistics - Specific Problem 1

O / open | AMR 问题清单汇编

Our first question concerns three-dimensional contingency tables $(p_{ijk})$ whose indices $i,j,k$ range over a set of four elements, such as the set $ \{{\tt A},{\tt C},{\tt G},{\tt T}\}$ of DNA bases.

} {\em Consider the variety of $4 {\times} 4 {\times} 4$-tables of tensor rank at most $4$. There are certain known polynomials of degree at most nine which vanish on this variety. Do they suffice to cut out the variety?}

背景链接：https://arxiv.org/abs/0707.4558

06125 | AMR-146-0003

### Open Problems in Algebraic Statistics - Specific Problem 3

O / open | AMR 问题清单汇编

} *Find a geometric characterization of those projective varieties whose maximum likelihood degree (ML degree) is equal to one.* This question and others raised in [CHKS, HKS] are just the tip of an iceberg:

背景链接：https://arxiv.org/abs/0707.4558

06126 | AMR-146-0005

### Open Problems in Algebraic Statistics - Specific Problem 5

O / open | AMR 问题清单汇编

{\em Which sets of almost-principal minors can be zero for a positive definite symmetric $5 {\times} 5$-matrix?}

背景链接：https://arxiv.org/abs/0707.4558

06127 | OWR-15586-001

### Openness of Cones Generated by Ample Subvarieties

O / open | Oberwolfach 报告问题汇编

Let $X$ be a projective variety. Is the convex cone in $N_d(X)$ generated by numerical classes of ample $d$-dimensional subvarieties open?

来源字段：https://doi.org/10.4171/owr/2017/43

06128 | OWR-16768-017

### Optimal Dimensions for Complex k-Regular Maps

O / open | Oberwolfach 报告问题汇编

Does a continuous $k$-regular map $\mathbb C^m\to\mathbb C^N$ exist if and only if $N\geq m(k-1)+1$?

来源字段：https://doi.org/10.4171/owr/2019/6

06129 | OWR-12490-003

### Optimal Pluricanonical Bounds for Stable Log Surfaces

O / open | Oberwolfach 报告问题汇编

For a stable log surface $(X,\Delta)$ with Gorenstein locus $U$, what is the optimal integer $r$ such that $rK_U$ induces a birational map or an embedding? Does there exist a stable log surface for which $5l(K_X+\Delta)$ is not very ample?

来源字段：https://doi.org/10.4171/owr/2013/27

06130 | OWR-2091-007

### Orbifold Lifts of Deligne Mostow Forgetful Maps

O / open | Oberwolfach 报告问题汇编

For two Deligne-Mostow weight systems $\mu$ and $\nu$ satisfying the integrality condition, consider the forgetful map between their open moduli spaces of weighted configurations. When does this map extend to the metric completions of these moduli spaces? When an extension exists, can it be lifted to a map of orbifolds, equivalently to suitable torsion-free manifold covers?

来源字段：https://doi.org/10.4171/owr/2008/45

06131 | OWR-14299281-012

### Orbit Decompositions in Wonderful Group Schemes

O / open | Oberwolfach 报告问题汇编

Let $x$ lie in a chamber $C$ of the Bruhat-Tits building, let $G(k)_{C,0}$ be the associated Iwahori subgroup, and let $G_{x,f}$ be the corresponding wonderful group scheme. Classify the $G(k)_{C,0}\times G(k)_{C,0}$-orbits in $G_{x,f}(\mathfrak{o})$, and describe its diagonal-stable pieces, in analogy with the Bruhat decomposition and stable-piece decomposition of the wonderful compactification.

来源字段：https://doi.org/10.4171/owr/2025/20

06132 | OWR-13860-007

### Orbit Structure of Generic A-Number-One Loci

O / open | Oberwolfach 报告问题汇编

For the affine Deligne-Lusztig invariant $f_g$, does the locus of $p$-divisible groups with generic $a$-number $1$ form a single $J_b(F)$-orbit of strata even when $b$ is not basic?

来源字段：https://doi.org/10.4171/owr/2015/39

06133 | AIM-ARITHMETIC_GEOMETRY-0016

### Order-three fineness obstructions on degree-two and degree-four K3 surfaces

P / partially_solved | AIM 研讨会问题汇编

Can we identify 3-torsion elements of the Brauer group of a degree 2 K3 surface as obstructions to the fineness of some moduli space?

Can we do this for K3 surfaces of other degrees? What about degree 4?

来源字段：http://aimpl.org/brauermoduli/1/

06134 | OWR-14222-013

### Ordinary Degree-d Curves Through Finite Point Sets

P / partially_solved | Oberwolfach 报告问题汇编

For a finite set $P\subset\mathbb{P}^2(\mathbb{R})$ and a positive integer $d$, must either $P$ lie on a degree-$d$ curve or some degree-$d$ curve contain exactly $(d^2+3d)/2$ points of $P$ and be uniquely determined by them?

来源字段：https://doi.org/10.4171/owr/2016/14

06135 | AIM-ALGEBRAIC_GEOMETRY-0176

### Ordinary obstruction theory and a C2 norm-lift calculation

P / partially_solved | AIM 研讨会问题汇编

Is there a Goerss-Hopkins obstruction theory for \(Comm(Sp_G)\)?

来源字段：http://aimpl.org/equideralggeom/5/

06136 | AIM-ALGEBRAIC_GEOMETRY-0157

### Ordinary thick closure of representation spheres for finite groups

P / partially_solved | AIM 研讨会问题汇编

Can we understand the thick subcategory of \(Sp^G\) generated by representation spheres?

来源字段：http://aimpl.org/equideralggeom/2/

06137 | OWR-16628-011

### Origins of Unexpected Projective Varieties

O / open | Oberwolfach 报告问题汇编

What general geometric or algebraic mechanism produces unexpected varieties and unexpected hypersurfaces in projective space?

来源字段：https://doi.org/10.4171/owr/2018/45

06138 | OWR-14746-005

### p-Adic Period Maps and Hodge-Theoretic Variations

P / partially_solved | Oberwolfach 报告问题汇编

What is the p-adic analogue of the classical theory of period maps and variations of Hodge structure?

来源字段：https://doi.org/10.4171/owr/2016/38

06139 | AIM-ARITHMETIC_GEOMETRY-0048

### Packed square-fat-point lower bounds for tangent spaces on the smoothable component

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.** Describe the singularities of the smoothable component of \(\operatorname{Hilb}^{d}(\mathbb A^{n})\). To be more specific,

- How large can the dimension of the Zariski tangent space to this component get? - Does the maximum occur in the intersection of components? Does it occur in the smoothable component?

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

06140 | OWR-1326-005

### Packing Numbers of Automorphism Graphs on Curves

O / open | Oberwolfach 报告问题汇编

Let $B$ be a curve of genus $b\geq2$. A finite set $S\subset\operatorname{Aut}(B)$ is good when the union of the graphs of its elements is a smooth divisor in $B\times B$; define its packing number by $\alpha=|S|/(b-1)$. What is the maximal possible value of $\alpha$?

来源字段：https://doi.org/10.4171/owr/2006/40

06141 | OWR-8415350-004

### Pair-of-Pants Products on Contact Loci of Jets

O / open | Oberwolfach 报告问题汇编

Under the isomorphisms $HF^*(\phi_f^d)\cong H_c^{*+2nd+n-1}(\chi_d(f))$, which operation on $\bigoplus_dH_c^*(\chi_d(f))$ corresponds to the pair-of-pants product on $\bigoplus_dHF^*(\phi_f^d)$?

来源字段：https://doi.org/10.4171/owr/2021/47

06142 | AMR-071-0016

### Parshin's conjecture

P / partially_solved | AMR 问题清单汇编

Parshin's conjecture: the higher algebraic K-groups of any smooth projective variety defined over a finite field must vanish up to torsion.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

06143 | OWR-5155-005

### Parshin's Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

If $X$ is smooth and proper over a finite field, is $K_i(X)$ torsion for every $i>0$, equivalently, does the corresponding rational motivic cohomology vanish outside the predicted degree (Parshin's conjecture)?

来源字段：https://doi.org/10.4171/owr/2011/27

06144 | OWR-1730-001

### Partition Formulas for Tautological Rings of General Curves

O / open | Oberwolfach 报告问题汇编

For a general curve $C$ of genus $g$, does its tautological ring $R$ satisfy $\dim_{\mathbb Q}R=p(g+1)$ and $\dim R^i_{(j)}=p(i,g+1-i,j)$, where the latter number counts partitions of $i$ into $i-j$ parts, each at most $g+1-i$, assuming Polishchuk's conjectural identification of the auxiliary ring $S$ with $R$?

来源字段：https://doi.org/10.4171/owr/2007/47

06145 | OWR-13750334-003

### Patchworking and Twisted Real Quintics

O / open | Oberwolfach 报告问题汇编

Does Viro patchworking of the Newton polytope of a quintic in $\mathbb P^4$, with the rays on its two-dimensional faces assigned the same sign as the interior point, produce a real quintic diffeomorphic to the connected $(M-2)$ real quintic obtained by twisting the corresponding real Lagrangian torus fibration?

来源字段：https://doi.org/10.4171/owr/2023/23

06146 | OWR-16171-015

### Period Conjecture for Nori Motives

P / partially_solved | Oberwolfach 报告问题汇编

For every motive $M$ in Nori's category, is the evaluation map from the formal period algebra $\widetilde P_{\langle M\rangle}$ to the period algebra $P_{\langle M\rangle}$ injective? Equivalently, does the associated formal period scheme have the conjectured connectedness and transcendence-degree dimension?

来源字段：https://doi.org/10.4171/owr/2018/30

06147 | OWR-13500-014

### Period Conjectures for Fano Polygons and Del Pezzo Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Let $P$ be a Fano polygon and $X$ a generic $\mathbb Q$-Gorenstein deformation of its toric surface. Is there an affine-linear mirror map $\phi:L_P^T\to H_X^{\mathrm{ts}}$ satisfying $\widehat G_X\circ\phi=\pi_P$? For Fano polygons with the same singularity content, does equality of classical periods under such an identification imply mutation equivalence? For class-TG del Pezzo surfaces with the same $\mathbb Q$-Gorenstein-rigid cyclic quotient singularities, does equality of quantum periods imply $\mathbb Q$-Gorenstein deformation equivalence?

来源字段：https://doi.org/10.4171/owr/2015/15

06148 | OWR-11133-005

### Periodicity and Definability of Newton-Okounkov Bodies

O / open | Oberwolfach 报告问题汇编

Is the volume of every line bundle on a smooth projective variety a period? For every integral divisor, is there a flag for which its Newton-Okounkov body is defined by polynomial inequalities with rational coefficients?

来源字段：https://doi.org/10.4171/owr/2011/41

06149 | OWR-733-006

### Perverse Point Sheaves and the Category $D(X,B)$

P / partially_solved | Oberwolfach 报告问题汇编

For a Kawamata log terminal (KLT) projective pair $(X, B)$, give a rigorous definition of the derived category $D(X, B)$ associated to the pair and an intrinsic definition of perverse point sheaves in the context of flops and derived categories of stacks.

来源字段：https://doi.org/10.4171/owr/2004/31

06150 | OWR-14298368-002

### Perverse-Weight Correspondence for Singular Dolbeault Moduli Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Does the $PI=WI$ conjecture hold for the intersection cohomology of singular Dolbeault moduli spaces $M_{\mathrm{Dol}}(n,d)$? In particular, does curious hard Lefschetz hold for arbitrary degree $d$?

来源字段：https://doi.org/10.4171/owr/2024/16

06151 | AIM-ARITHMETIC_GEOMETRY-0024

### Phantoms on projective space and the failure of a surface rationality criterion

P / partially_solved | AIM 研讨会问题汇编

Consider $D^b(\mathbb{P}^n)$. Take an admissible thick subcategory, $\mathcal{A}$, i.e. a triangulated subcategory closed under taking summands and such that the inclusion has a left (equivalently right) adjoint. If $K_0(\mathcal{A})=0$, is $\mathcal{A}=0$? Such an $\mathcal{A}$ is called a phantom. When $n=1$, it is known that there is no such phantom subcategory. This is related to Caldararu's conjecture on the existence of exceptional collections on projective homogeneous varieties. The answer is no if we replace $\mathbb{P}^n$ by an arbitrary variety (there is a surface counterexample).

If we replace $\mathbb{P}^n$ by another regular surface, is the condition of not having phantom subcategories equivalent to rationality? What if we take $K_0(X) \otimes \mathbb{Q}$?

来源字段：http://aimpl.org/brauermoduli/2/

06152 | OWR-14221-005

### Plabic Newton-Okounkov Bodies and FFLV Polytopes

P / partially_solved | Oberwolfach 报告问题汇编

For the distinguished non-mirror plabic graph $G_0$ associated with $\operatorname{Gr}_{k,n}(\mathbb C)$, is $\operatorname{NO}_{G_0}^r$ unimodularly equivalent to $\operatorname{FFLV}(r\varpi_k)$? This is known for $\operatorname{Gr}_{3,6}$.

来源字段：https://doi.org/10.4171/owr/2016/13

06153 | OWR-14299289-014

### Plancherel Algebras of Dual Hyperspherical Varieties

O / open | Oberwolfach 报告问题汇编

Under derived geometric Satake, is the internal endomorphism object of the basic sheaf on a hyperspherical variety-the Plancherel algebra-sent to the function algebra $\mathcal O(M^\vee)$ of the dual Hamiltonian variety?

来源字段：https://doi.org/10.4171/owr/2025/18

06154 | OWR-14222-003

### Point Configurations Producing Unexpected Curves

P / partially_solved | Oberwolfach 报告问题汇编

Characterize the finite point sets $Z$ for which a general fat point imposes dependent conditions on the curves containing $Z$, thereby producing unexpected curves.

来源字段：https://doi.org/10.4171/owr/2016/14

06155 | OWR-16169-009

### Polar Cylinders and Affine Complements of Del Pezzo Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

For a del Pezzo surface $S$ in its ambient weighted projective space $P$, is the absence of a $(-K_S)$-polar cylinder equivalent to the affine complement $P\setminus S$ being non-cylindrical? The implication from no polar cylinder to non-cylindricity is known.

来源字段：https://doi.org/10.4171/owr/2018/28

06156 | OWR-16169-010

### Polar Cylinders and Complement Automorphisms

O / open | Oberwolfach 报告问题汇编

For a del Pezzo surface $S$, does $S$ contain no $(-K_S)$-polar cylinder if and only if $\operatorname{Aut}(P\setminus S)=\operatorname{Aut}(P,S)$?

来源字段：https://doi.org/10.4171/owr/2018/28

06157 | OWR-15586-007

### Polyhedral Global Newton-Okounkov Bodies for Mori Dream Spaces

O / open | Oberwolfach 报告问题汇编

Does every smooth Mori dream space $X$ admit an admissible flag for which its global Newton-Okounkov body is rational polyhedral?

来源字段：https://doi.org/10.4171/owr/2017/43

06158 | OWR-15586-003

### Polyhedrality of Movable Cycle Cones on Toric Varieties

O / open | Oberwolfach 报告问题汇编

If $X$ is a smooth toric variety and $\operatorname{Mov}_d(X)$ is the closed cone generated by numerical classes of movable $d$-dimensional subvarieties, is $\operatorname{Mov}_d(X)$ rational polyhedral?

来源字段：https://doi.org/10.4171/owr/2017/43

06159 | OWR-15588-006

### Polynomial Bounds for Dominant Rational-Curve Components

P / partially_solved | Oberwolfach 报告问题汇编

Fix an ample divisor $L$ on a smooth projective Fano variety $X$. Is the number of dominant components of $\operatorname{Mor}(\mathbb P^1,X)$ of $L$-degree at most $d$ bounded above by a polynomial in $d$?

来源字段：https://doi.org/10.4171/owr/2017/45

06160 | OWR-788-002

### Polynomial Bounds for Rational Maps Between Curves

O / open | Oberwolfach 报告问题汇编

For curves $X$ and $Y$, is the number $m(X,Y)$ of rational maps from $X$ to $Y$ polynomially bounded in the genus of $X$?

来源字段：https://doi.org/10.4171/owr/2005/07

06161 | AIM-ARITHMETIC_GEOMETRY-0012

### Positive semidecision and a rank-one geometric decision algorithm

P / partially_solved | AIM 研讨会问题汇编

Is there an algorithm to compute derived equivalence for K3 surfaces over a finitely generated field?

来源字段：http://aimpl.org/brauermoduli/1/

06162 | AIM-ALGEBRAIC_GEOMETRY-0254

### Positive-characteristic connectedness and an amplified Frobenius certificate

P / partially_solved | AIM 研讨会问题汇编

**Connectedness.** Is there a positive-characteristic analog of the connectedness lemma?

**Recovered strong connectedness statement.** Let \(f\colon (X,B)\to S\) be a contraction, meaning that \(f\) is projective and \(f_*\mathcal O_X=\mathcal O_S\). If \[ -(K_X+B) \] is \(f\)-nef and \(f\)-big, then the reduced non-klt locus \(\operatorname{Nklt}(X,B)\) is connected in a neighborhood of every fiber. Over a non-algebraically-closed residue field, the precise fiberwise conclusion is that each nonempty fiber is geometrically connected.

来源字段：http://aimpl.org/minimalmodcharp/1/

06163 | AIM-ALGEBRAIC_GEOMETRY-0301

### Positive-characteristic discrepancy candidates and an exact SNC calibration

P / partially_solved | AIM 研讨会问题汇编

**Identify a possible positive characteristic analog of minimal log discrepancy.**

来源字段：http://aimpl.org/testandmultiplierideals/2/

06164 | AIM-ALGEBRAIC_GEOMETRY-0102

### Positive-density rational reductions and a finite-monodromy density criterion

P / partially_solved | AIM 研讨会问题汇编

4th day

Do there exist a cubic \(4\)-fold defined over number fields which is conjecured to be irrational but many reductions are conjectured to be ratioal?

Does there exist a cubic fourfold \(Y\) over a number field \(K\) which is conjectured to be irrational in characteristic zero but for which many reductions \(Y_v\) are rational?

来源字段：http://aimpl.org/ratsubvarpos/3/

06165 | AIM-ALGEBRAIC_GEOMETRY-0097

### Positive-density rational reductions and unbounded K3 certificate complexity

P / partially_solved | AIM 研讨会问题汇编

Can we find a cubic fourfold which is conjectually irratioal, but (most of ) its reductions are rational?

来源字段：http://aimpl.org/ratsubvarpos/3/

06166 | OWR-14298581-008

### Positive-Entropy Automorphisms of Strict Calabi-Yau Threefolds

P / partially_solved | Oberwolfach 报告问题汇编

Construct further smooth strict Calabi-Yau threefolds admitting primitive biregular automorphisms of positive entropy.

来源字段：https://doi.org/10.4171/owr/2024/32

06167 | AMR-140-0012

### Positively Curved Projective Varieties - Conjecture 12

O / open | AMR 问题清单汇编

The MRC fibration is defined only by $X$, and thus it does not reflect information on the boundary $\Delta$. On the other hand, the slope rationally connected quotient (sRC quotient for short), introduced in [Cam16], gives a generalization of MRC fibrations reflecting the boundary $\Delta$ (see [Cam04, Cam16]). The following conjecture concerns a structure theorem for sRC quotients. This conjecture was solved when $(X, \Delta)$ is a log smooth surface in [CCM19, Theorem 1.6], but it is still open in the general case.

[{[CCM19, Conjecture 1.5]}]

Let $(X, \Delta)$ be a klt pair such that $X$ is smooth and $-(K_X +\Delta)$ is nef. Then there exists an orbifold morphism $\rho : (X, \Delta)\rightarrow (R, \Delta_R)$ with the following properties$:$

$\bullet$ $(R, \Delta_R)$ is a klt pair such that $R$ is smooth and $c_1(K_R+ \Delta_R)=0$.

$\bullet$ General orbifold fibers $(X_r, \Delta_r )$ are slope rationally connected.

$\bullet$ The fibration is locally trivial with respect to pairs, namely, for any small open set $U \subset R$, we have the isomorphism over $U \subset Y$$:$ $$ (\rho^{-1} (U), \Delta)\cong (U, \Delta_R |_U) \times (X_r, \Delta_{X_r}), $$ where $X_r$ is a general fiber of $\rho$.

背景链接：https://arxiv.org/abs/2211.09153

06168 | AMR-140-0002

### Positively Curved Projective Varieties - Conjecture 2

O / open | AMR 问题清单汇编

The following conjecture posed by Yau can be regarded as an analogy of Mok's result for holomorphic sectional curvature.

[Yau's conjecture, {[Yau82, Problem 47]}] If a compact K\"ahler manifold $X$ has positive holomorphic sectional curvature, then $X$ is projective and rationally connected.

背景链接：https://arxiv.org/abs/2211.09153

06169 | AMR-140-0001

### Positively Curved Projective Varieties - Problem 1

O / open | AMR 问题清单汇编

MRC fibrations are almost holomorphic by definition, but they are not necessarily represented by holomorphic maps. In fact, there is a projective variety $Z$ with Picard number one, such that $Z$ is uniruled but not rationally connected, which was constructed by Fujita (see [EIM20, Example 6.5] for details). The variety $Z$ admits no non-trivial fibrations (in particular, no holomorphic MRC fibrations) since the Picard number of $Z$ is one. Note that $Z$ has bad singularities: $Z$ is neither log canonical nor $\mathbb{Q}$-factorial. The author does not know an example of smooth projective varieties (or varieties with mild singularities) admitting no holomorphic MRC fibration. Now we suggest the following problem:

When do we choose an MRC fibration to be holomorphic from the ambiguities in the choices of $Y$?

背景链接：https://arxiv.org/abs/2211.09153

06170 | AMR-140-0010

### Positively Curved Projective Varieties - Problem 10

O / open | AMR 问题清单汇编

$\bullet$ Does the blow-up of Hirzebruch surfaces at general four points have the pseudo-effective tangent bundle? \\ $\bullet$ What can we say for the blow-up of Hirzebruch surfaces at special points?

背景链接：https://arxiv.org/abs/2211.09153

06171 | AMR-140-0011

### Positively Curved Projective Varieties - Problem 11

O / open | AMR 问题清单汇编

Under the above assumption for the relative anti-canonical divisor, can we obtain a structure theorem? Specifically, can we choose a holomorphic morphism $X \to Y'$ that is birationally equivalent to the original $\phi: X \dashrightarrow Y$?

背景链接：https://arxiv.org/abs/2211.09153

06172 | AMR-140-0003

### Positively Curved Projective Varieties - Problem 3

O / open | AMR 问题清单汇编

[{[Mat18b, Problem 5.1]}]

Let $(X, g)$ be a compact K\"ahler manifold with semi-positive holomorphic sectional curvature. Assume that $n_{{\rm{tf}}}{(X, g)} = \dim X$.

$\bullet$ Does it hold that $h^{0}(X, \Omega_X^q)=0$ for any $q>0$?

$\bullet$ Is $X$ automatically projective?

背景链接：https://arxiv.org/abs/2211.09153

06173 | AMR-140-0004

### Positively Curved Projective Varieties - Problem 4

O / open | AMR 问题清单汇编

[cf. {[Mat18b, Problem 5.2]}]

Let $(X, g)$ be a compact K\"ahler manifold with semi-positive holomorphic sectional curvature.

$\bullet$ Is the fundamental group of $X$ is an extension of a finite group by $\mathbb{Z}^{\oplus 2m}$?

For an appropriate finite \'etale cover $X^* \to X$, we consider its Albanese map $\alpha: X^* \to \operatorname{Alb}(X^*)$.

$\bullet$ Does the fiber $F$ admit a K\"ahler metric $g_F$ such that $n_{{\rm{tf}}}(F, g_F)=\dim F$?

$\bullet$ Is the fiber $F$ projective and rationally connected?

背景链接：https://arxiv.org/abs/2211.09153

06174 | AMR-140-0005

### Positively Curved Projective Varieties - Problem 5

O / open | AMR 问题清单汇编

Let $(X, g)$ be a compact K\"ahler manifold with semi-positive $($or semi-negative$)$ holomorphic sectional curvature. Let $V \subset T_X$ be the set of ``appropriate" truly flat tangent vectors.

$\bullet$ Does $V$ determine an integrable subbundle of $T_X$?

$\bullet$ Can we find a subbundle $W \subset T_X$ such that $T_X \cong W \oplus V $?

$\bullet$ In the case of semi-positive holomorphic sectional curvatures, can we construct an MRC fibration $f: X \to Y$ such that $W=T_{X/Y}$?

$\bullet$ In the case of semi-negative holomorphic sectional curvatures, can we construct the Iitaka fibration $f: X \to Z$ associated with the canonical bundle $K_X$ such that $V=T_{X/Z}$?

背景链接：https://arxiv.org/abs/2211.09153

06175 | AMR-140-0006

### Positively Curved Projective Varieties - Problem 6

O / open | AMR 问题清单汇编

[cf.{[Mat18b, Problem 5.3]}]

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$\bullet$ Does the blow-up of Hirzebruch surfaces admit a positive holomorphic sectional curvature?

$\bullet$ Can we classify all the varieties of fixed dimension $($e.g., dimension two or three$)$ with semi-positive $($or positive$)$ holomorphic sectional curvature?

$\bullet$ Can we find an analog of semi-positive $($or positive$)$ holomorphic sectional curvatures in algebraic geometry?

背景链接：https://arxiv.org/abs/2211.09153

06176 | AMR-140-0007

### Positively Curved Projective Varieties - Problem 7

O / open | AMR 问题清单汇编

Let $X$ be a smooth projective variety with pseudo-effective tangent bundle. Then is the MRC fibration $\phi : X \to Y$ locally trivial?

背景链接：https://arxiv.org/abs/2211.09153

06177 | AMR-140-0008

### Positively Curved Projective Varieties - Problem 8

O / open | AMR 问题清单汇编

The second difference concerns the positivity of fibers $F$ of $\phi:X \to Y$. It was proved in [DPS94] that rationally connected manifolds with nef tangent bundle are always Fano manifolds. However, the same conclusion does not hold for pseudo-effective tangent bundles, since the tangent bundle of the Hirzebruch surfaces is pseudo-effective (see [HIM19, Proposition 4.5]). Then we suggest the following problem:

[{[HIM19, Problem 3.13]}]

Let $X$ be a rationally connected manifold with pseudo-effective tangent bundle. Then is the anti-canonical divisor $-K_X$ big?

背景链接：https://arxiv.org/abs/2211.09153

06178 | AMR-140-0009

### Positively Curved Projective Varieties - Problem 9

O / open | AMR 问题清单汇编

Let $X$ be a compact K\"ahler manifold with pseudo-effective tangent bundle. After we replace $X$ with an appropriate finite \'etale covering of $X$, we consider the Albanese map $X \to \operatorname{Alb}(X)$.

$\bullet$ Can we prove that the anti-canonical divisor $-K_F$ is big or $F$ is projective?

$\bullet$ Is the Albanese map $X \to \operatorname{Alb}(X)$ an MRC fibration of $X$?

背景链接：https://arxiv.org/abs/2211.09153

06179 | OWR-740-001

### Positivity of Landau-Ginzburg TERP Structures

O / open | Oberwolfach 报告问题汇编

Let $f:Y\to\mathbb C$ be an algebraic function on an affine algebraic manifold, with only isolated singularities and tame at infinity. Is its associated TERP structure positive-definite, pure, and polarized?

来源字段：https://doi.org/10.4171/owr/2004/38

06180 | OWR-1703871-011

### Positivity of Log Canonical Bundles under Local Torelli

P / partially_solved | Oberwolfach 报告问题汇编

Does generic local Torelli for the period map imply that $K_B+[Z]$ is nef and big, and does local Torelli imply that $K_B+[Z]$ is free?

来源字段：https://doi.org/10.4171/owr/2020/24

06181 | OWR-12697710-003

### Positivity of Noncommutative State Polynomials on Convex Sets

O / open | Oberwolfach 报告问题汇编

Are matrix evaluations sufficient to test positivity of noncommutative state polynomials on a convex set $K_C$, and is there a Krivine-Stengle-type Positivstellensatz using evaluations on unbounded self-adjoint operators?

来源字段：https://doi.org/10.4171/owr/2023/14

06182 | AIM-ALGEBRAIC_GEOMETRY-0101

### Positivity, separability, and weak approximation for low-degree hypersurfaces

P / partially_solved | AIM 研讨会问题汇编

If \(d^{2}\leq n\), does weak approximation hold in char. \(p\) for hypersurfaces? (it does in char. \(0\))

来源字段：http://aimpl.org/ratsubvarpos/3/

06183 | AIM-ARITHMETIC_GEOMETRY-0021

### Potential descent, odd-class lower bounds, and the Galois-orbit obstruction

P / partially_solved | AIM 研讨会问题汇编

If we have a K3 surface $X$ defined over some number field $k$, what can we say about the fields of definition of its Fourier-Mukai partners (partners over $\mathbb{C}$)? What about twisted K3 surfaces? What about Abelian varieties? What about other varieties?

If we look at real quadratic extensions of $\mathbb{Q}$, can the order of the odd part of the class group go to $\infty$? Does this answer the previous question? How does this relate to the previous problem?

来源字段：http://aimpl.org/brauermoduli/1/

06184 | OWR-16158-007

### Precise Reconstruction Criteria for Anabelian Varieties

P / partially_solved | Oberwolfach 报告问题汇编

For an anabelian variety $X$ over a field $k$, which isomorphisms or morphisms of $X$ can be reconstructed from its étale fundamental group $\pi_1^{\acute et}(X)$?

来源字段：https://doi.org/10.4171/owr/2018/17

06185 | AIM-ARITHMETIC_GEOMETRY-0023

### Presentation gauge, a binary complement, and the two Clifford K3 surfaces

P / partially_solved | AIM 研讨会问题汇编

Pick three quadrics, \(Q_0,Q_1,Q_2\). Put \[ X=V(x_0Q_0+x_1Q_1+x_2Q_2)\subset \mathbf P^5. \] Then \(X\) contains the plane \(P=V(x_0,x_1,x_2)\). The net generated by the \(Q_i\) is a family of quadric fourfolds over \(\mathbf P^2\); the Stein factorization of its relative variety of maximal isotropic spaces is a \(\mathbf P^3\)-family over a double-plane K3 surface and gives a Brauer class. Projection of \(X\) from \(P\) gives a family of residual quadric surfaces over the \(\mathbf P^2\) parametrizing \(3\)-planes containing \(P\); its maximal isotropic spaces similarly give a projective family over another double-plane K3 surface. Is there an intrinsic description of the relation between these twisted K3 surfaces?

来源字段：http://aimpl.org/brauermoduli/1/

06186 | AIM-ALGEBRAIC_GEOMETRY-0169

### Prime support of possible odd equivariant MU-classes

P / partially_solved | AIM 研讨会问题汇编

Quillen's theorem for \(MU\) states that the map from the Lazard ring \(L\) to \(\pi_*MU\) is an isomorphism and so, in particular, the homotopy of \(MU\) is concentrated in even degrees. Furthermore, when \(G\) is a finite abelian group, it is known that the \(G\)-equivariant homotopy of \(MU\) is also even.

For any finite or compact Lie group \(G\), is it also the case that \(\pi_*^G(MU)\) is even?

来源字段：http://aimpl.org/equideralggeom/4/

06187 | AIM-ALGEBRAIC_GEOMETRY-0175

### Prime-to-order orbit diagrams and a C2 theta--Tambara coefficient model

P / partially_solved | AIM 研讨会问题汇编

What is an algebraic model for \(L_{K(1)}Comm(Sp_G)\) (for even small groups \(G\))?

来源字段：http://aimpl.org/equideralggeom/5/

06188 | OWR-2492-001

### Principality of Artin-Schreier Deformation Ideals

O / open | Oberwolfach 报告问题汇编

Let $\pi:C\to\mathbb{P}^1$ be a wildly ramified Artin-Schreier cover with a single branch point at infinity, and let $R_\pi$ be its universal deformation ring. When $p=3$, one has $R_\pi=R_\infty=W[[t_1,t_2]]/I$ with $I\neq0$. Is $I$ principal?

来源字段：https://doi.org/10.4171/owr/2008/54

06189 | OWR-16941-008

### Prismatic Cohomology from Topological Cyclic Homology

O / open | Oberwolfach 报告问题汇编

Let $R$ be an $S_n$-algebra. Do the groups $$ \operatorname{Ext}_{TP_0(R/S_n^{\otimes2})}^j\bigl(TP_0(R/S_n),TP_i(R/S_n)\bigr) $$ identify with the prismatic cohomology of $R$, and is the spectral sequence $$ \operatorname{Ext}_{TP_0(R/S_n^{\otimes2})}^j\bigl(TP_0(R/S_n),TP_i(R/S_n)\bigr)\Longrightarrow TP_{i-j}(R) $$ the Bhatt-Morrow-Scholze spectral sequence?

来源字段：https://doi.org/10.4171/owr/2019/29

06190 | OWR-16941-009

### Prismatic Cohomology of Local Complete Intersections

O / open | Oberwolfach 报告问题汇编

Let $Y\subset\mathbb P^n_{\mathbb Z_p}$ be closed with $Y_{\mathbb F_p}$ a local complete intersection, and let $T_n$ denote the ambient infinitesimal object in the relative periodic-cyclic-homology setup. Is $$ \operatorname{Ext}_{\mathrm{P\text{-}Mod}}^j\bigl(TP_0(\mathcal O_Y/\mathcal O_{T_n}),TP_0(\mathcal O_Y/\mathcal O_{T_n})\{i\}\bigr) $$ the prismatic cohomology of $Y$, and is the resulting spectral sequence to $TP_{2i-j}(Y)$ the Bhatt-Morrow-Scholze spectral sequence?

来源字段：https://doi.org/10.4171/owr/2019/29

06191 | OWR-14299291-005

### Prismatic Extension of Generic-Point Vanishing

O / open | Oberwolfach 报告问题汇编

Does generic-point vanishing for de Rham and étale cohomology extend to prismatic cohomology?

来源字段：https://doi.org/10.4171/owr/2025/29

06192 | OWR-4340-003

### Projection-Locus Smoothness and Cohomology Vanishing

P / partially_solved | Oberwolfach 报告问题汇编

Let $X\subset\mathbb P^{n+c}$ be a smooth $n$-dimensional variety and put $N=n+c$. Are the following assertions true? (1) $H^p(\mathcal J_X(q))=0$ whenever $p+(q+1)(c-1)\leq n$ and $p<N-1$. (2) If $k(c-1)<n$, then the normalization $\widetilde X_k$ of the multiplicity-at-least-$k$ locus of a general projection is smooth and connected. (3) If $k(c-1)\leq n$ and $\widetilde X_k$ is connected, then $H^1(\mathcal J_X(k-1))=0$.

来源字段：https://doi.org/10.4171/owr/2010/27

06193 | OWR-1595-002

### Projective Normality of Smooth Toric Varieties

O / open | Oberwolfach 报告问题汇编

Let $X_\Sigma$ be a smooth projective toric variety and let $L$ be ample. Is the embedding defined by the complete linear series of $L$ projectively normal? Equivalently, is $\operatorname{Sym}^kH^0(X_\Sigma,L)\to H^0(X_\Sigma,L^k)$ surjective for every $k\geq1$?

来源字段：https://doi.org/10.4171/owr/2007/39

06194 | OWR-1326-004

### Projective Subspaces with Trivial Normal Bundle in Fano Manifolds

O / open | Oberwolfach 报告问题汇编

If $X$ is a Fano manifold with second Betti number $b_2(X)=1$ and $X$ is not projective space, does $X$ contain no immersed projective space $\mathbb P^k$ with trivial normal bundle for every $1<k<\dim X$?

来源字段：https://doi.org/10.4171/owr/2006/40

06195 | AIM-ALGEBRAIC_GEOMETRY-0060

### Projective-space stabilization gives rationally connected stably irrational examples with property (*)

S / solved | AIM 研讨会问题汇编

Is there a rationally connected irrational variety \(X\) with property (*)?

Property (*) means that there is a semiorthogonal decomposition \[ D(X)=\langle \mathscr A_1,\ldots,\mathscr A_m\rangle \] with fully faithful embeddings \(\mathscr A_i\hookrightarrow D(Y_i)\) for varieties satisfying \(\dim Y_i\leq \dim X-2\).

Together they ask:

As is standard in the cited categorical-representability literature, this report interprets \(D(X)\) as \(D^b_{\mathrm{coh}}(X)\) and works with smooth projective varieties over \(\mathbb C\). In that literature the image in \(D(Y_i)\) is required to be admissible. The construction below satisfies this stronger formulation: every component is equivalent to the whole derived category of a smooth projective \(Y_i\).

来源字段：http://aimpl.org/rationalityag/1/

06196 | OWR-1703871-009

### Projectivity of the Stein-Factorized Period-Map Completion

P / partially_solved | Oberwolfach 报告问题汇编

For the Stein factorization $B\xrightarrow{\widehat{\Phi}}\widehat{\mathcal{P}}\to\mathcal{P}$ of the period map, where $\widehat{\mathcal{P}}$ is open in a compact complex analytic variety $\widehat{\mathcal{P}}^{T}$, is $\widehat{\mathcal{P}}^{T}$ a projective completion of $\widehat{\mathcal{P}}$?

来源字段：https://doi.org/10.4171/owr/2020/24

06197 | OWR-14299288-016

### Projectivized Toric Vector Bundles as Toric Varieties

O / open | Oberwolfach 报告问题汇编

Characterize the ways in which the projectivization of a toric vector bundle behaves like a smooth projective toric variety. Determine which toric-geometric structures, invariants, and vanishing properties extend to such projectivizations.

来源字段：https://doi.org/10.4171/owr/2025/19

06198 | OWR-15428-007

### Proper Intersections of Phylogenetic Network Varieties

P / partially_solved | Oberwolfach 报告问题汇编

For two distinct phylogenetic networks $N_1$ and $N_2$, let $V_N$ denote the Zariski closure of the model variety associated with $N$. Characterize when $V_{N_1}\cap V_{N_2}$ is a proper subvariety of both $V_{N_1}$ and $V_{N_2}$.

来源字段：https://doi.org/10.4171/owr/2017/20

06199 | AIM-ARITHMETIC_GEOMETRY-0003

### Proper support and clean excess for improper AFL intersections

P / partially_solved | AIM 研讨会问题汇编

In the formulation of the arithmetic fundamental lemma, arithmetic intersection numbers are only formulated for regular semi-simple orbits because of proper intersections of cycles. Is it possible to define arithmetic intersection numbers in case of improper intersections of cycles (and formulate a similar analytic side)?

来源字段：http://aimpl.org/intersectshimura/1/

06200 | AIM-ALGEBRAIC_GEOMETRY-0225

### Proper-point defects and the degree-21 bottleneck

P / partially_solved | AIM 研讨会问题汇编

**Problem 5.3.** For irreducible curves, the best known examples have \(h\to-2\). Can we do better?

来源字段：https://aimath.org/pastworkshops/degenalggeomproblems.pdf

06201 | OWR-11783-002

### Property $N_{3d-3}$ for Veronese Embeddings

P / partially_solved | Oberwolfach 报告问题汇编

Does the $d$-uple Veronese embedding of projective space satisfy property $N_{3d-3}$?

来源字段：https://doi.org/10.4171/owr/2012/21

06202 | OWR-2492-004

### Prorepresentable Obstructions for Local Action Deformations

O / open | Oberwolfach 报告问题汇编

别名：Smooth Decomposition of Local Group Action Deformations [OWR-2492-003]

For a local $G$-action $\rho$ on $k[[t]]$, with subgroup $N$, let $\Theta=\operatorname{Der}_k(k[[t]])$ and $\Theta^\sharp=\operatorname{Der}_k(k[[t]]^N)$. Does there exist a pro-representable functor $F$ with tangent space $H^1(G/N,\Theta^\sharp/\Theta^N)$ and maps $D_N^{G/N}\to F$ and $D_{G/N}\to F$ such that $\psi:D_G\to D_N^{G/N}\times D_{G/N}$ factors through $D_N^{G/N}\times_FD_{G/N}$ and the induced map $D_G\to D_N^{G/N}\times_FD_{G/N}$ is smooth?

来源字段：https://doi.org/10.4171/owr/2008/54

06203 | AMR-071-0023

### Prove resolution of singularities for algebraic varieties over fields of positive characteristic in arbitrary dimension

P / partially_solved | AMR 问题清单汇编

Prove resolution of singularities for algebraic varieties over fields of positive characteristic in arbitrary dimension.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

06204 | OWR-13489-006

### Pseudoconvex Domains and Their Cores

P / partially_solved | Oberwolfach 报告问题汇编

For a domain $\Omega$ in a complex manifold, let $c(\Omega)$ be the set where the Levi form of every bounded-above plurisubharmonic function on $\Omega$ degenerates. How are geometric properties of $\Omega$ related to those of $c(\Omega)$; in particular, how is pseudoconvexity of $\Omega$ related to pluripolarity of its core?

来源字段：https://doi.org/10.4171/owr/2015/4

06205 | OWR-11792-001

### Pure Resolutions of Paracanonical Level Curves

O / open | Oberwolfach 报告问题汇编

For a general level-$\ell$ curve $[C,\eta]$ of even genus $g\geq6$, does the paracanonically embedded curve $C\hookrightarrow\mathbb P^{g-2}$ have a pure homogeneous coordinate-ring resolution; equivalently, is $K_{g/2-2,1}(C,K_C\otimes\eta)=0$?

来源字段：https://doi.org/10.4171/owr/2012/30

06206 | AIM-ALGEBRAIC_GEOMETRY-0320

### Q-Cartierness versus pluricanonical base change

P / partially_solved | AIM 研讨会问题汇编

Let \(f:X\to B\) be a family of surfaces. For a base change \(q:B'\to B\), with \(q_X:X'=X\times_BB'\to X\), say that \(\omega_{X/B}^{[m]}\) commutes with base change if the natural pluricanonical comparison is an isomorphism. Equivalently (using base change for \(\omega\) on the relative Gorenstein locus), \[ (q_X^*\omega_{X/B})^{[m]}\ \cong\ q_X^*(\omega_{X/B}^{[m]}). \] Here \(\mathcal F^{[m]}=(\mathcal F^{\otimes m})^{**}\). Kollár's condition asks for this for every \(m>0\) and every base change. Is it equivalent to \(\omega_{X/B}\) being \(\mathbb Q\)-Cartier?

The displayed question itself says only “a family of surfaces,” however. This omitted hypothesis matters. It is also essential to distinguish:

1. the assumption that the original base \(B\) is reduced; and 2. the quantifier over test morphisms \(B'\to B\), where \(B'\) may be nonreduced.

The answer below treats these readings separately. Throughout the characteristic-zero assertions, schemes are locally Noetherian and the families are flat and of finite type.

来源字段：https://aimath.org/WWN/birational/birational.pdf

06207 | OWR-14299516-005

### q-de Rham Cohomology over the Habiro Ring

P / partially_solved | Oberwolfach 报告问题汇编

Construct a canonical version of $q$-de Rham cohomology with coefficients in the Habiro ring $\mathcal H=\varprojlim_{m\in\mathbb N}\mathbb{Z}[q]^{\wedge}_{(q^m-1)}$, rather than only in $\mathbb{Z}[[q-1]]$.

来源字段：https://doi.org/10.4171/owr/2025/36

06208 | OWR-14299291-006

### q-Hodge Filtrations and Habiro Descent

P / partially_solved | Oberwolfach 报告问题汇编

Does the Hodge filtration on de Rham cohomology admit a $q$-deformation? Can $q$-de Rham cohomology be descended from $\mathbb{Z}[[q-1]]$ to the Habiro ring $\mathcal H=\varprojlim_{m\in\mathbb N}\mathbb{Z}[q]^{\wedge}_{(q^m-1)}$?

来源字段：https://doi.org/10.4171/owr/2025/29

06209 | OWR-14299516-003

### Quadratic Curve Counts on Low-Degree Del Pezzo Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

For a rational del Pezzo surface $X$ of degree at least $4$ over a perfect field of characteristic different from $2$ and $3$, prove $W^{\mathrm{WD}}(A_0,A_1,\ldots,A_r)=Q_{X,D}(A_0)$ for the incidence data specified by the blown-up finite étale subschemes.

来源字段：https://doi.org/10.4171/owr/2025/36

06210 | OWR-1187-001

### Quadratic Toric Ideals and Unimodular Triangulations

P / partially_solved | Oberwolfach 报告问题汇编

If $P$ is a lattice polytope whose projective toric variety $X_P$ is smooth, is its defining ideal $I_P$ generated by quadrics? Also, for each dimension $d$, is there an integer $n(d)$ such that every $d$-dimensional lattice polytope has a regular unimodular triangulation after multiplication by $n(d)$?

来源字段：https://doi.org/10.4171/owr/2006/05

06211 | OWR-13750334-004

### Quadratically Enriched Tropical Curve Invariants

S / solved | Oberwolfach 报告问题汇编

For all integers $a,b\geq0$, do the quadratically enriched tropical invariants satisfy $$N^{\mathbb A^1}_{a,a+b}=\sum_{k\geq0}\binom{b+2k}{k}N^{\mathbb A^1}_{\triangle,a-k,b+2k}?$$ Establish or disprove this identity.

来源字段：https://doi.org/10.4171/owr/2023/23

06212 | OWR-14299284-010

### Quantum-Monodromy Computation of Mutation Braids

O / open | Oberwolfach 报告问题汇编

Let a smooth Fano variety $X$ admit extremal contractions to smooth Fano varieties $Y$ and $Z$, and let the pullback-compatible semiorthogonal decompositions of $D^b(X)$ be related by a mutation braid. Is this braid equal to the monodromy braid of the eigenvalues of quantum multiplication by $c_1(X)$ as the complexified quantum parameter moves toward the walls of the ample cone?

来源字段：https://doi.org/10.4171/owr/2025/28

06213 | OWR-7155449-005

### Quasi-Jacobi Structure of K3 Hilbert-Scheme Invariants

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate K3 Hilbert-Scheme Conjecture Suite [OWR-7155449-006]

For the enumerative series $F_{g,m}(\alpha;\gamma_1,\ldots,\gamma_N)$ associated with Hilbert schemes of points on a K3 surface, prove the conjectured quasi-Jacobi-form structure with its predicted index and weight, the multiple-cover formula via formal Hecke operators, and the holomorphic-anomaly equation.

来源字段：https://doi.org/10.4171/owr/2021/28

06214 | OWR-13750339-001

### Quasi-Modularity of Elliptic-Curve Stable-Map Classes

O / open | Oberwolfach 报告问题汇编

For the universal family $\pi:\mathcal E\to\overline{\mathcal M}_{1,1}$, prove the quasi-modularity of $\sum_{d=0}^\infty q^d[\overline{\mathcal M}_{g,n}(\pi,d)]^{\mathrm{vir}}$ as a cohomology-valued generating series.

来源字段：https://doi.org/10.4171/owr/2023/27

06215 | OWR-4427-005

### Quasimonomial Valuations Computing Asymptotic Log Canonical Thresholds

P / partially_solved | Oberwolfach 报告问题汇编

For a graded sequence of ideals $\mathfrak a_\bullet$ on $X$ and a fixed ideal $\mathfrak q$, does there always exist a quasimonomial valuation $\nu$ that computes the log canonical threshold $\operatorname{lct}_{\mathfrak q}(\mathfrak a_\bullet)$?

来源字段：https://doi.org/10.4171/owr/2010/39

06216 | AIM-ALGEBRAIC_GEOMETRY-0327

### Quotient stacks, inertia obstructions, and polarized point blowups

P / partially_solved | AIM 研讨会问题汇编

Kollár's class of examples comes from the observation that the blowups of points on a smooth variety \(X\) are parameterized by \(X\) modulo the automorphisms of \(X\).

**Problem 1.8.** More generally, is every stack which is a quotient of a quasi-projective scheme by a group action a moduli stack for some class of polarized manifolds?

来源字段：https://aimath.org/WWN/birational/birational.pdf

06217 | AIM-ALGEBRAIC_GEOMETRY-0251

### Rank correction, top-Chern criterion, and an affine-surface counterexample

S / solved | AIM 研讨会问题汇编

Let \(X\) be a \(d\)-dimensional smooth affine variety. Let \(\alpha\in F^{d-1}K_0(X)\). Then does there exist a vector bundle \(E\) of rank \(d-1\) such that \([E]=\alpha\)?

来源字段：https://aimath.org/pastworkshops/projectiveA1problems.pdf

06218 | AIM-ALGEBRAIC_GEOMETRY-0074

### Rank-one monoidal equivalence and the graph-generation bottleneck

P / partially_solved | AIM 研讨会问题汇编

Prove that MF$((\mathscr{X}^{\mathrm{big}})^{\mathrm{stable}},w)$ is equivalent as a monoidal category to K$^b(\mathbb{S}\mathrm{Bim}_n)$, where $\mathscr{X}^{\mathrm{big}}=\mathfrak g\times G\times\mathfrak n \times G\times\mathfrak n$.

来源字段：http://aimpl.org/catheckehilbert/1/

06219 | AIM-ALGEBRAIC_GEOMETRY-0088

### Rank-two equality and the all-rank coherence obstruction

P / partially_solved | AIM 研讨会问题汇编

Is the flattening of the Gaitsgory complex for the standard representation equal to the tautological sheaf from the GNR conjecture?

来源字段：http://aimpl.org/catheckehilbert/4/

06220 | OWR-739-004

### Rational 1-Connectedness Under Higher Fano Conditions

O / open | Oberwolfach 报告问题汇编

Suppose that $X$ is Fano and that $c_2(\Omega_X^1)-\frac12K_X^2$ is negative, for example, its intersection product with every surface on $X$ is negative. Does it follow that $X$ is rationally $1$-connected?

来源字段：https://doi.org/10.4171/owr/2004/37

06221 | OWR-2049-002

### Rational Algebraic Realizations of Compact Manifolds

O / open | Oberwolfach 报告问题汇编

Can every compact differentiable manifold be realized as a connected component of the real locus $W(\mathbb{R})$ of a smooth real rational algebraic variety $W$? In higher dimensions, what topological restrictions do Kodaira dimension $-\infty$ or rational connectedness impose on such components?

来源字段：https://doi.org/10.4171/owr/2008/38

06222 | OWR-1703866-001

### Rational Cubic Fourfolds and K3 Categories

O / open | Oberwolfach 报告问题汇编

For a cubic fourfold $Y$, is $Y$ rational if and only if its Kuznetsov component $\operatorname{Ku}(Y)$ is equivalent to the derived category of a K3 surface?

来源字段：https://doi.org/10.4171/owr/2020/19

06223 | OWR-12015-002

### Rational Curves on Calabi-Yau Threefolds with Nef Divisors

P / partially_solved | Oberwolfach 报告问题汇编

If a Calabi-Yau threefold $X$ has a nonzero nef, non-ample line bundle, must $X$ contain a rational curve?

来源字段：https://doi.org/10.4171/owr/2012/43

06224 | OWR-13109-003

### Rational Curves on Compact Kähler Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a compact Kähler manifold. Is $X$ covered by rational curves if and only if $K_X$ is not pseudoeffective? If $K_X$ is pseudoeffective but not nef, must $X$ contain a rational curve?

来源字段：https://doi.org/10.4171/owr/2014/39

06225 | AIM-ALGEBRAIC_GEOMETRY-0024

### Rational cyclotomic invariants and a connective-boundary obstruction

P / partially_solved | AIM 研讨会问题汇编

**The \(K\)-theory of Johnson--Wilson theory.** The algebraic \(K\)-theory of Johnson--Wilson theory \(E(n)\) is closely related to the topological cyclic homology of the connective cover of the two periodic Johnson--Wilson theory \(E'(n)\). Compute something about \(\mathrm{TC}\) (and related invariants such as \(\mathrm{TP},\mathrm{THH}\)) of \(\tau_{\geq0}E'(n)\), where \(E'(n)\) is the \(2\)-periodic Johnson--Wilson theory.

来源字段：http://aimpl.org/equivstable/2/

06226 | AIM-ALGEBRAIC_GEOMETRY-0304

### Rational ideal compression and the non-Q-Gorenstein sign defect

S / solved | AIM 研讨会问题汇编

**1. (Alexeev)** Give a definition of log canonical, log terminal etc for pairs \((X,Y)\) where \(X\) is a normal variety which is not necessarily \(\mathbb Q\)-Gorenstein and \(Y\) is a formal sum of subschemes of \(X\).

来源字段：https://aimath.org/WWN/singularvariety/problems_final.pdf

06227 | AIM-ALGEBRAIC_GEOMETRY-0103

### Rational reduction and a prime-to-characteristic sieve for CTO examples

P / partially_solved | AIM 研讨会问题汇编

**Plausible reconstruction.** Can an irrational variety over a **number field** have a rational specialization in characteristic \(p>0\)? What happens for the examples of Colliot-Thélène and Ojanguren?

It is Problem 3.78 in the section “Rationality in a family” of the AIM workshop list *Rational subvarieties in positive characteristic*. The immediately preceding Problem 3.72 asks about cubic fourfolds “defined over number fields” and their reductions. On that contextual evidence, the most plausible reconstruction is:

This reconstruction is not source-verified, so the exact wording above is preserved. There is a second important ambiguity. “Irrational over a number field” can mean either:

1. not rational over the number field, perhaps becoming rational over its algebraic closure (arithmetic irrationality); or 2. geometrically irrational, and in the parenthetical CTO setting, not even stably rational after base change to an algebraic closure.

The first reading has an elementary affirmative answer. The second is the mathematically substantive reading suggested by the parenthesis.

Throughout the main result below, specialization means a smooth proper model \(\mathcal X\to\operatorname{Spec}R\) over a mixed-characteristic valuation ring, or a smooth proper spread over an open subscheme of the ring of integers of a number field. Rationality of a geometric fiber means rationality after base change to an algebraic closure of its residue field.

来源字段：http://aimpl.org/ratsubvarpos/3/

06228 | OWR-17300-014

### Rational Singularities of Integral Modular Curve Models

P / partially_solved | Oberwolfach 报告问题汇编

For which congruence subgroups $\Gamma$ does the modular curve model $X_\Gamma$ have rational singularities, and more generally for which primes $p$ and groups $\Gamma$ does its model $(X_\Gamma)_{\mathbb{Z}_{(p)}}$ have rational singularities?

来源字段：https://doi.org/10.4171/owr/2019/57

06229 | AIM-ALGEBRAIC_GEOMETRY-0173

### Rational weak relative Gorenstein duality at level 3 without perfectness

P / partially_solved | AIM 研讨会问题汇编

Can we find explicit ways to write \(S\in R\text{-Alg}\) as an \(R\)-module? What are objects that “untwist” when smashed together? (e.g. \(KU\simeq KO\wedge C(\eta)\), \(H\mathbb F_2\simeq ko\wedge\mathcal A(1)\)).

A slightly easier question: Is \(tmf^{(n)}_0\to tmf^{(n)}_1\) relatively Gorenstein?

来源字段：http://aimpl.org/equideralggeom/5/

06230 | AIM-ALGEBRAIC_GEOMETRY-0305

### Rationality and a sharp finite-jet radius for formal log-canonical thresholds

S / solved | AIM 研讨会问题汇编

Let \(f\in\mathbb C[[x_1,\ldots,x_n]]\). If \(f\) is convergent near \(0\), its log-canonical threshold \(\operatorname{lct}_0(f)\) is defined in the usual way and is rational because it is computed on a log resolution. For an arbitrary formal power series, the results of Demailly--Kollár allow one to define \(\operatorname{lct}(f)\) as the limit of the log-canonical thresholds of its truncations. Is this limit rational?

来源字段：https://aimath.org/WWN/singularvariety/problems_final.pdf

06231 | OWR-1276-001

### Rationality of Moduli Spaces of Algebraic Curves

O / open | Oberwolfach 报告问题汇编

Determine which moduli spaces of algebraic curves are rational.

来源字段：https://doi.org/10.4171/owr/2006/27

06232 | OWR-11139-004

### Rationality of Plane-Curve Moduli Spaces

O / open | Oberwolfach 报告问题汇编

Are the moduli spaces

$$\mathbb P(\operatorname{Sym}^d(\mathbb C^3)^*)/SL_3(\mathbb C)$$

of plane curves rational for the $15$ values of $d$ for which rationality remains unknown?

来源字段：https://doi.org/10.4171/owr/2011/47

06233 | OWR-17296-014

### Rationality of Seshadri Constants from Plane-Curve Singularities

O / open | Oberwolfach 报告问题汇编

Let $C$ be a reduced plane curve, let $Z$ be its singular-point set, and let $\widetilde C$ be its proper transform on the blowup of $\mathbb{P}^2$ at $Z$. If $\widetilde C^2<-1$, must the Seshadri constant $\varepsilon(Z)$ be rational, and can it be computed?

来源字段：https://doi.org/10.4171/owr/2019/53

06234 | OWR-17296-009

### Rationality of Seshadri Constants on Plane Blowups

O / open | Oberwolfach 报告问题汇编

For a smooth projective surface $X$ obtained by blowing up $\mathbb{P}^2$ at very general points and an ample line bundle $L$ on $X$, is the global Seshadri constant $\varepsilon(X,L)$ always rational, or can it be irrational?

来源字段：https://doi.org/10.4171/owr/2019/53

06235 | OWR-13500-010

### Rationality of Ueno-Type Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Let $E_4$ and $E_6$ be the anharmonic and equianharmonic elliptic curves with their standard automorphisms $g_4$ and $g_6$, and let $X_{n,m}$ be the minimal resolution of $E_m^n/\langle g_m\rangle$ for the diagonal action. Is $X_{4,6}$ rational? Is $X_{5,6}$ unirational?

来源字段：https://doi.org/10.4171/owr/2015/15

06236 | OWR-13109-009

### Rationally Connected Cycles on Hyperkähler Manifolds

O / open | Oberwolfach 报告问题汇编

Let $X$ be a projective hyperkähler manifold of dimension $2n$. Does $X$ contain a rationally connected subscheme of pure dimension $n$? If $Y_1,Y_2\subset X$ are rationally connected subschemes of dimension $n$, do arbitrary points $y_i\in Y_i$ determine the same class $c_X\in CH_0(X)$?

来源字段：https://doi.org/10.4171/owr/2014/39

06237 | OWR-13101-009

### Rationally Connected Fibration of a Brill-Noether Moduli Space

O / open | Oberwolfach 报告问题汇编

Does the maximal rationally connected fibration of the moduli space $M_{15,16}$ of curves of genus 15 and degree 16 have a three-dimensional base?

来源字段：https://doi.org/10.4171/owr/2014/31

06238 | OWR-14299088-013

### Real Components of Two Multi-Affine Polynomial Zero Sets

O / open | Oberwolfach 报告问题汇编

Is the number of connected components of the common real zero set of two multi-affine polynomials bounded solely in terms of their degrees, independently of the number $n$ of variables?

来源字段：https://doi.org/10.4171/owr/2025/9

06239 | AIM-ALGEBRAIC_GEOMETRY-0181

### Real essential matrices in the calibrated five-point problem

P / partially_solved | AIM 研讨会问题汇编

The essential variety is a 5-dimensional variety of degree 10. A point correspondence gives a linear constraint so a five point correspondence should give 10 complex solutions.

Of the ten possibilities, how many of the solutions can be real?

来源字段：http://aimpl.org/algvision/1/

06240 | OWR-11780-003

### Real Orbit Counts on Wonderful Varieties

O / open | Oberwolfach 报告问题汇编

Let $X$ be a wonderful $G$-variety with canonical $\sigma$-equivariant real structure $\mu$, let $X(\mathbb{R})=\{x\in X:\mu(x)=x\}$, and let $G_0^\sigma$ be the identity component of the fixed-point group. What is the exact number of $G_0^\sigma$-orbits in $X(\mathbb{R})$?

来源字段：https://doi.org/10.4171/owr/2012/18

06241 | AIM-ALGEBRAIC_GEOMETRY-0156

### Real spectra and the hidden geometric stratum of a naive C2-spectrum

S / solved | AIM 研讨会问题汇编

**Problem 2.4.** What should the correct notion of a Real spectrum be? Is it ``more" than a genuine \(C_2\)-spectrum? (see (arXiv:0908.3724) Sec B12)

来源字段：http://aimpl.org/equideralggeom/2/

06242 | OWR-13750332-001

### Real Subspaces in Zeros of Elementary Symmetric Polynomials

O / open | Oberwolfach 报告问题汇编

If $r$ is even, does the zero set $e_r^{-1}(0)$ of the elementary symmetric polynomial contain no real vector subspace of dimension $r$?

来源字段：https://doi.org/10.4171/owr/2023/21

06243 | OWR-744-001

### Reality of Beauville Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Is the Beauville surface $S$ biholomorphic to its complex conjugate $\overline S$, and is $S$ real?

来源字段：https://doi.org/10.4171/owr/2004/42

06244 | OWR-17135-004

### Realizability of Balanced Complexes as Tropical Varieties

O / open | Oberwolfach 报告问题汇编

Characterize the balanced polyhedral complexes and fans that arise as tropical varieties of classical ideals.

来源字段：https://doi.org/10.4171/owr/2019/39

06245 | OWR-14299089-003

### Realizability of Blade-Induced Matroid Subdivisions

O / open | Oberwolfach 报告问题汇编

Characterize when a subdivision induced by an arrangement of blades is realizable, equivalently when it is induced by a realizable tropical Plücker vector.

来源字段：https://doi.org/10.4171/owr/2025/12

06246 | OWR-13489-002

### Realizing Complex Manifolds Transversely to Algebraic Foliations

O / open | Oberwolfach 报告问题汇编

For any compact complex manifold $(X,J)$, does there exist a smooth complex projective variety Z, an algebraic foliation D on Z, and an isotopy class $\alpha$ of transverse embeddings $X\to Z$ such that $J=J_f$ for some $f\in\alpha$?

来源字段：https://doi.org/10.4171/owr/2015/4

06247 | OWR-14298376-006

### Reconstruction of Real Gromov-Witten Invariants for Semisimple Targets

P / partially_solved | Oberwolfach 报告问题汇编

Does the real Gromov-Witten theory of every real-orientable target with semisimple quantum cohomology satisfy the proposed reconstruction formula for a suitable series $U_{t,q}(\psi)$?

来源字段：https://doi.org/10.4171/owr/2024/23

06248 | OWR-11695865-008

### Recursive Degree Formulas in Likelihood Geometry

O / open | Oberwolfach 报告问题汇编

Find recursive formulas for the SSL-degree, ML-degree, and dual-ML-degree sequences of the unrestricted $n\times n$ correlation model.

来源字段：https://doi.org/10.4171/owr/2022/55

06249 | OWR-11580-008

### Reducedness of the Commuting Scheme

O / open | Oberwolfach 报告问题汇编

Determine whether the commuting scheme $\{(x,y)\in\mathfrak g\times\mathfrak g:[x,y]=0\}$ is reduced.

来源字段：https://doi.org/10.4171/owr/2012/13

06250 | OWR-1596-009

### Reducible-Fiber Bounds for Higher-Dimensional Linear Systems

O / open | Oberwolfach 报告问题汇编

For pencils of plane curves, at most four completely reducible fibers can occur. What analogous upper bounds hold for linear systems in higher dimensions, and what analogous bounds hold for curves of degree greater than one in the space of hypersurfaces or homogeneous polynomials?

来源字段：https://doi.org/10.4171/owr/2007/40

06251 | AIM-ALGEBRAIC_GEOMETRY-0287

### Reduction of non-Q-Gorenstein multiplier ideals and finite toric divisor witnesses

P / partially_solved | AIM 研讨会问题汇编

Let \(X\) be a normal variety in characteristic zero, let \(\mathfrak a\subseteq\mathcal O_X\) be a nonzero ideal, and let \(t\in\mathbb Q_{\geq0}\). After choosing a model over a finitely generated \(\mathbb Z\)-algebra \(A\), is there a dense open \(U\subseteq\operatorname{Spec}A\) such that \[ \mathcal J_{\mathrm{dFH}}(X,\mathfrak a^t)_\mu =\tau_b(X_\mu,\mathfrak a_\mu^t) \] for every closed point \(\mu\in U\)? Does this hold when \(K_X\) is numerically \(\mathbb Q\)-Cartier?

来源字段：http://aimpl.org/testandmultiplierideals/2/

06252 | OWR-14298376-005

### Refined Donaldson-Thomas Meaning of Block-Göttsche Polynomials

O / open | Oberwolfach 报告问题汇编

Do the generalized Block-Göttsche polynomials $\operatorname{BPS}_{\beta_S}^S(q)$ admit an interpretation in refined Donaldson-Thomas theory for the Calabi-Yau threefold $K_S$, the total space of the canonical bundle of $S$?

来源字段：https://doi.org/10.4171/owr/2024/23

06253 | OWR-1465-005

### Regular Representatives of Homotopy Classes Between Spheres

O / open | Oberwolfach 报告问题汇编

For every pair of nonnegative integers $(n,p)$, is every continuous map $S^n\to S^p$ homotopic to a regular map?

来源字段：https://doi.org/10.4171/owr/2007/14

06254 | OWR-1730-007

### Reid’s Fundamental-Group Conjecture for General-Type Surfaces

O / open | Oberwolfach 报告问题汇编

If $S$ is a minimal complex surface of general type with $K_S^2<4\chi(S)$, must its algebraic fundamental group be, up to finite group extensions, the fundamental group of a curve? When $K_S^2<3\chi(S)$ and $S$ has no irregular étale covers, what explicit upper bounds hold for $|\pi_1^{\mathrm{alg}}(S)|$?

来源字段：https://doi.org/10.4171/owr/2007/47

06255 | OWR-9790358-018

### Relations among Adjoint Curves from Cyclic Point Orders

O / open | Oberwolfach 报告问题汇编

Given $n$ points in $\mathbb P^2$, each cyclic ordering up to reversal defines an $n$-gon and hence an adjoint curve. How are the adjoint curves arising from the $(n-1)!/2$ orderings related?

来源字段：https://doi.org/10.4171/owr/2022/7

06256 | OWR-10252929-007

### Relative Bogomolov Conjecture for Abelian Schemes

P / partially_solved | Oberwolfach 报告问题汇编

Let $\pi:A\to S$ be an abelian scheme and let $X\subset A$ be irreducible with $\pi(X)=S$. Must either the points of arbitrarily small canonical height fail to be Zariski dense in $X$, or $X$ have codimension at most $\dim S$ in a horizontal torsion coset?

来源字段：https://doi.org/10.4171/owr/2022/21

06257 | OWR-11125-003

### Relative K-Stability and Extremal Kähler Metrics

P / partially_solved | Oberwolfach 报告问题汇编

If a polarized projective manifold $(M,L)$ is $K$-stable relative to a maximal algebraic torus $T$ in $\operatorname{Aut}(M)$, does the polarization class $c_1(L)$ contain an extremal Kähler metric?

来源字段：https://doi.org/10.4171/owr/2011/33

06258 | OWR-14299288-008

### Resolutions of Zero-Mutable Log Structures

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Zero-Mutable Log-Resolution Conjecture [OWR-14299288-007]

Prove that every zero-mutable log structure admits a log-crepant log resolution.

来源字段：https://doi.org/10.4171/owr/2025/19

06259 | OWR-16171-005

### Resolvent Degree of Abelian-Variety Torsion Covers

O / open | Oberwolfach 报告问题汇编

If $(g,p)\neq(2,2)$, then the resolvent degree of the $p$-torsion cover of the moduli space $A_g$ of principally polarized abelian varieties is $\operatorname{RD}(A_{g,p}\to A_g)=\dim A_g=g(g+1)/2$.

来源字段：https://doi.org/10.4171/owr/2018/30

06260 | OWR-15987-010

### Resolvent Degree of Polynomial Root Covers

O / open | Oberwolfach 报告问题汇编

Let $\widetilde{\mathcal P}_n\to\mathcal P_n$ be the finite cover from monic degree-$n$ polynomials with a chosen root to all monic degree-$n$ polynomials. Determine its resolvent degree $\operatorname{RD}(\widetilde{\mathcal P}_n/\mathcal P_n)$; in particular, decide Hilbert's sextic, thirteenth, and octic conjectures.

来源字段：https://doi.org/10.4171/owr/2018/2

06261 | OWR-16171-004

### Resolvent Degrees of Lines and Bitangents

O / open | Oberwolfach 报告问题汇编

Conjecturally, the resolvent degree of the 27-sheeted cover parametrizing lines on smooth cubic surfaces is $\operatorname{RD}(H_{3,3}^{(1)}\to H_{3,3})=3$, and the resolvent degree of the 28-sheeted cover parametrizing bitangents to smooth plane quartics is $\operatorname{RD}(H_{4,2}^{(1)}\to H_{4,2})=\dim H_{4,2}=6$.

来源字段：https://doi.org/10.4171/owr/2018/30

06262 | OWR-16171-003

### Resolvent Degrees of Universal Polynomials

O / open | Oberwolfach 报告问题汇编

Hilbert's conjecture asserts that for $n\geq 6$ the resolvent degree of the universal degree-$n$ polynomial cover $\widetilde{P}_n\to P_n$ satisfies $\operatorname{RD}(\widetilde{P}_n\to P_n)\geq 2$; for $n=6,7,8,9$ the values are conjectured to be $2,3,4,4$, respectively.

来源字段：https://doi.org/10.4171/owr/2018/30

06263 | OWR-14299288-013

### Restrictions of Toric Vector Bundles to Embedded Varieties

O / open | Oberwolfach 报告问题汇编

Characterize the vector bundles $E$ on a variety $Y$ for which there exist a toric embedding $i:Y\to X_\Sigma$ and a toric vector bundle $F$ on $X_\Sigma$ with $E\cong i^*F$.

来源字段：https://doi.org/10.4171/owr/2025/19

06264 | OWR-9790356-002

### Riemann-Hilbert Correspondence for Étale and v-Sheaves

P / partially_solved | Oberwolfach 报告问题汇编

Extend the $p$-torsion Riemann-Hilbert correspondence from overconvergent étale $\mathbb F_p$-sheaves to all étale $\mathbb F_p$-sheaves, and ultimately to $v$-sheaves.

来源字段：https://doi.org/10.4171/owr/2022/5

06265 | OWR-1326-003

### Rigid Fano Targets of Holomorphic Maps

P / partially_solved | Oberwolfach 报告问题汇编

Which Fano manifolds $X$ are rigid targets of surjective holomorphic maps? Here $X$ is a rigid target if every projective subvariety of $T_X$ dominating $X$ is a section; equivalently, every deformation of a surjective holomorphic map $Y\to X$ arises from an automorphism of $X$.

来源字段：https://doi.org/10.4171/owr/2006/40

06266 | OWR-16169-001

### Rigidity of Hirzebruch-Kummer Coverings

P / partially_solved | Oberwolfach 报告问题汇编

Let $L$ be a rigid line configuration and let $S(n,L)$ be its Hirzebruch-Kummer covering. Is $S(n,L)$ rigid for all sufficiently large exponents $n$?

来源字段：https://doi.org/10.4171/owr/2018/28

06267 | OWR-17300-005

### S-Unit Differences of Singular Moduli

O / open | Oberwolfach 报告问题汇编

For a fixed set $S$ of primes, are there finitely many differences of singular moduli that are $S$-units?

来源字段：https://doi.org/10.4171/owr/2019/57

06268 | OWR-14213-008

### Schubert Classes in the Gamma Filtration

O / open | Oberwolfach 报告问题汇编

For $n\geq 8$, the special Schubert class $e_n\in K(X_{F(S)})$ does not belong to term $n+1-\lfloor(n+1)/2\rfloor$ of the gamma filtration.

来源字段：https://doi.org/10.4171/owr/2016/5

06269 | AIM-ALGEBRAIC_GEOMETRY-0142

### Schubert-cycle mobility exponents and general incidence lower bounds

P / partially_solved | AIM 研讨会问题汇编

What is $K(\alpha)$ for $\alpha\in N_k(\operatorname{Gr}(k,n))$ a Schubert cycle?

The preceding record on the same AIM page defines $$ \operatorname{mc}(\alpha) = \max\{\text{number of general points imposed on an effective cycle of class }\alpha\} $$ and displays $$ K_{\mathrm{AIM}}(\alpha) = \max\left\{a\in\mathbb R: \limsup_{m\to\infty}\frac{\operatorname{mc}(m\alpha)}{m^a}>0 \right\}. \tag{1} $$ For logical safety, throughout this report “max” in (1) is interpreted as “sup”; the endpoint need not be attained for an arbitrary sequence. This does not alter any exact value proved below.

There are two source ambiguities.

1. The symbol $k$ is used both for the dimension of the planes parametrized by the Grassmannian and for the dimension of the numerical cycles. These are independent quantities. I write $$ G=\operatorname{Gr}(d,V)=\operatorname{Gr}(d,N),\qquad \dim V=N,\qquad q=N-d,\qquad D=\dim G=dq, $$ and use $p$ for the dimension of a cycle. 2. The nearby AIM Problem 2.4 says that $K$ is known to be integral for curves and for $\operatorname{Gr}(2,N)$. That statement is incompatible with the unscaled exponent (1): for example a big curve on a $D$-fold has exponent $D/(D-1)$. It agrees instead with Brian Lehmann's later normalized cycle Iitaka dimension $$ \kappa_{\mathrm L}(\alpha) =(D-p)\sup\left\{a\geq 0: \limsup_{m\to\infty}\frac{\operatorname{mc}(m\alpha)}{m^a}>0 \right\}. \tag{2} $$

Thus, if $c=D-p>0$ is the codimension, $$ \boxed{\kappa_{\mathrm L}(\alpha) =cK_{\mathrm{AIM}}(\alpha)}. \tag{3} $$ Both conventions are reported below. Formula (3), rather than an unacknowledged change of definition, resolves the discrepancy.

The full-dimensional class is exceptional: a positive multiple of $[G]$ contains arbitrarily many points, so its mobility count is infinite. All finite-exponent statements below concern $0\leq p<D$.

The page currently has the same wording and the same displayed definition (1).

来源字段：http://aimpl.org/poscycles/2/

06270 | AIM-ALGEBRAIC_GEOMETRY-0148

### Scoped motivic Serre sequences and a finite-etale monodromy obstruction

P / partially_solved | AIM 研讨会问题汇编

Can we construct a Serre spectral sequence in motivic homotopy theory?

来源字段：http://aimpl.org/equideralggeom/1/

06271 | OWR-733-001

### Scroll-Locus and Veronese-Fano Surface Correspondences

O / open | Oberwolfach 报告问题汇编

Is there a geometric correspondence between rational normal scrolls and $W_d$ loci, and between Veronese surfaces and Fano surfaces of lines on cubic threefolds; what Castelnuovo-type properties are shared?

来源字段：https://doi.org/10.4171/owr/2004/31

06272 | OWR-15428-002

### Secant Varieties of Gaussian Moment Varieties

P / partially_solved | Oberwolfach 报告问题汇编

For all $k,n,d\geq1$, determine the dimensions, degrees, and defining equations of the secant varieties $\operatorname{Sec}_k(G_{n,d})$ of Gaussian moment varieties.

来源字段：https://doi.org/10.4171/owr/2017/20

06273 | OWR-14298581-010

### Second-Chern Orthogonality and Calabi-Yau Automorphisms

O / open | Oberwolfach 报告问题汇编

Let $X$ be a strict Calabi-Yau threefold with infinite automorphism group. Must $c_2(X)^\perp\cap\operatorname{Amp}(X)_{\mathbb Q}$ contain a nonzero class? Also, does any such threefold other than the known examples $X_3$ and $X_7$ admit a primitive automorphism of positive entropy?

来源字段：https://doi.org/10.4171/owr/2024/32

06274 | AIM-ALGEBRAIC_GEOMETRY-0045

### Seed-preserving explicit bounds for unirational hypersurfaces

P / partially_solved | AIM 研讨会问题汇编

Problem 1

Find more integers $d$ and $n$ such that a general hypersurface of degree $d$ in $\mathbb P^n_{\mathbb C}$ is unirational. Similarly, find more integers $d$ and $n$ such that every smooth hypersurface of degree $d$ in $\mathbb P^n_{\mathbb C}$ is unirational.

来源字段：http://aimpl.org/rationalityag/1/

06275 | OWR-788-008

### Semiampleness and Ampleness on Abelian Moduli Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathcal{A}_g^F$ be the first Voronoi (perfect-cone) toroidal compactification of the moduli space $\mathcal{A}_g$ of principally polarized abelian varieties of dimension $g$, let $E$ be the Hodge bundle on $\mathcal{A}_g^F$ with first Chern class $M = c_1(E)$, and let $D$ be the boundary divisor. Is the divisor $12M - D$ semi-ample on $\mathcal{A}_g^F$? Furthermore, for which ample line bundles $L$ on $\mathcal{A}_g^F$ is the vector bundle $E \otimes L$ ample?

来源字段：https://doi.org/10.4171/owr/2005/07

06276 | OWR-4086-003

### Semicontinuity of the $M$-Number in Plane-Curve Deformations

P / partially_solved | Oberwolfach 报告问题汇编

Is the $M$-number upper-semicontinuous under deformations of plane-curve singularities, particularly under $\delta$-constant deformations?

来源字段：https://doi.org/10.4171/owr/2009/43

06277 | AIM-ALGEBRAIC_GEOMETRY-0263

### Semigroup-gap counterexamples on weighted projective threefolds

S / solved | AIM 研讨会问题汇编

Suppose that \(A\) is Cartier and ample, and \(K_X+A\) is nef. Must \(H^0(X,K_X+A)\) be nonzero?

来源字段：http://aimpl.org/minimalmodcharp/4/

06278 | OWR-14299284-009

### Semiorthogonal Decompositions Along Extremal Contractions

O / open | Oberwolfach 报告问题汇编

Suppose a smooth Fano variety $X$ has extremal contractions to smooth Fano varieties $Y$ and $Z$. Do there exist semiorthogonal decompositions of $D^b(X)$ compatible with pullback from $Y$ and $Z$, with the two decompositions related by mutation?

来源字段：https://doi.org/10.4171/owr/2025/28

06279 | OWR-1703866-003

### Semiorthogonal Decompositions in Algebraic Families

O / open | Oberwolfach 报告问题汇编

How do semiorthogonal decompositions of derived categories behave in algebraic families?

来源字段：https://doi.org/10.4171/owr/2020/19

06280 | OWR-17296-003

### Seshadri Constants of Line Arrangement Singularities

O / open | Oberwolfach 报告问题汇编

Let $x_1,\ldots,x_r$ be the singular points of a line configuration in $\mathbb{P}^2$, and let $k$ be the maximum number of these points on one line. Must the multipoint Seshadri constant $\varepsilon(\mathcal{O}_{\mathbb{P}^2}(1);x_1,\ldots,x_r)$ equal $1/k$?

来源字段：https://doi.org/10.4171/owr/2019/53

06281 | AIM-ALGEBRAIC_GEOMETRY-0318

### Sharp factoriality threshold and defect-one equality cases

P / partially_solved | AIM 研讨会问题汇编

It comes from Problem 8 (Cheltsov) in the AIM workshop list *Numerical invariants of singularities and higher-dimensional algebraic varieties*.

Let \[ X_n\subset \mathbf P^4 \] be a hypersurface of degree \(n\) with finitely many isolated ordinary double points. The record asks:

1. When is \(X_n\) factorial, equivalently \(\operatorname{Cl}(X_n)=\mathbf Z\)? 2. Is \(X_n\) factorial when \[ |\operatorname{Sing}X_n|<(n-1)^2? \] 3. For a general projection \[ \varphi:\mathbf P^4\dashrightarrow\mathbf P^2, \] can at most \(k(n-1)\) points of \(\varphi(\operatorname{Sing}X_n)\) lie on a plane curve of degree \(k\)?

The source also gives the sharp family \[ X_n=\{x f_{n-1}+y g_{n-1}=0\}, \] where \(f_{n-1},g_{n-1}\) are general forms, and states that it has class group \(\mathbf Z\oplus\mathbf Z\). It records the then-known bound \(|\operatorname{Sing}X_n|\leq \frac23(n-1)^2\).

Two corrections are necessary before doing mathematics.

* The source says that factoriality is equivalent to independence of the nodes on forms of degree \(n\). The correct degree is \[ 2n-5. \] This is Theorem 1.1 and the paragraph following it in Cheltsov--Park [CP06], based on the defect formula of Dimca. * The isolated line “\(S\subset X_n\) smooth of codimension one \(\Rightarrow S\) Cartier” cannot be unconditional: the displayed sharp example contains the smooth plane \(\{x=y=0\}\), which is not Cartier. Cheltsov--Park state the intended scoped version: if \(|\operatorname{Sing}X_n|<(n-1)^2\), every smooth surface in \(X_n\) is cut by an ambient hypersurface.

来源字段：https://aimath.org/WWN/singularvariety/problems_final.pdf

06282 | OWR-14299574-006

### Sharp Homological Bounds for Realizable Monomial Ideals

O / open | Oberwolfach 报告问题汇编

Is the homological-dimension lower bound for the number of realizations sharp for every realizable monomial ideal in $k[x_1,x_2]$?

来源字段：https://doi.org/10.4171/owr/2025/45

06283 | OWR-3471-007

### SHGH Conjecture for Linear Systems on Blown-Up Planes

O / open | Oberwolfach 报告问题汇编

Let $p_1,\ldots,p_n$ be general points of $\mathbb{P}^2$, and let $X$ be their blow-up with line class $H$ and exceptional divisors $E_i$. Is $|dH-\sum_{i=1}^nm_iE_i|$ of dimension $\max\{-1,d(d+3)/2-\sum_{i=1}^nm_i(m_i+1)/2\}$ exactly when no $(-1)$-curve $C\subset X$ satisfies $C\cdot(dH-\sum_{i=1}^nm_iE_i)\leq-2$?

来源字段：https://doi.org/10.4171/owr/2009/21

06284 | OWR-788-005

### Shokurov Finite-Generation Conjecture

O / open | Oberwolfach 报告问题汇编

Let $(X,B)$ be a pair and let $A=A(X,B)$ be its discrepancy b-divisor. A system $D_\bullet$ of b-divisors on $X$ is asymptotically canonically saturated if, for all $i,j$, there is a model $Y(i,j)\to X$ such that $$\operatorname{Mob}\left\lceil jD_i+A\right\rceil_Y\leq j(D_j)_Y$$ on every model $Y\to Y(i,j)$. Let $(X,B)$ be klt and let $f:X\to Z$ be a pl flipping contraction for $K_X+S+B$, with $Z$ affine. If $D_\bullet$ is an asymptotically canonically saturated, positive, convex, bounded characteristic system of b-divisors on $X$, is the pbd-algebra $R(X,D_\bullet)$ finitely generated?

来源字段：https://doi.org/10.4171/owr/2005/07

06285 | OWR-1276-008

### Simple Connectivity of Godeaux Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Let $S$ be a Godeaux surface with $p_g=0$ and $K_S^2=1$. If its bicanonical pencil $|2K_S|$ has four distinct base points and every member of $|2K_S|$ is $2$-connected, must $S$ be simply connected?

来源字段：https://doi.org/10.4171/owr/2006/27

06286 | OWR-15428-003

### Single Polynomial Description of Binary Tensor Gram Loci

P / partially_solved | Oberwolfach 报告问题汇编

For real binary tensors of fixed format, can the Gram locus be described semialgebraically by the nonnegativity of a single polynomial?

来源字段：https://doi.org/10.4171/owr/2017/20

06287 | AIM-ALGEBRAIC_GEOMETRY-0090

### SL_N fixed-ray quotients and minimal hook complexes

P / partially_solved | AIM 研讨会问题汇编

A theorem of Morton-Samuelson connected the Elliptic Hall algebra at \(q=t\) to the Skein algebra of the torus. Categorify this result.

Are there simplifications of the above problem for the \(\mathfrak{sl}_n\)-quotient invariant?

“The above problem” is Problem 5.1:

Morton-Samuelson [MS, Theorem 5.6] work over \[ \mathscr R=\mathbb Z[v^{\pm1},s^{\pm1},(s^r-s^{-r})^{-1}:r\geq 1] \] with HOMFLYPT relation \[ L_+-L_-=(s-s^{-1})L_0 \] and framing curls \(v^{-1}\) and \(v\). In their elliptic-Hall convention, which inverts the second Burban-Schiffmann parameter, their isomorphism is at \[ q_{\mathrm E}=t_{\mathrm E}=s^{-2}. \tag{1.1} \] It sends the torus power-sum curve \(P_{\mathbf x}\) to their normalized elliptic-Hall generator \(w_{\mathbf x}\).

Write \(N\) for the rank parameter, reserving \(m\) for the power-sum degree. In the convention above, the quantum \(\mathfrak{sl}_N\) specialization is \[ v=s^{-N}. \tag{1.2} \] Indeed, the contractible unknot then has value \[ \frac{v^{-1}-v}{s-s^{-1}}=\frac{s^N-s^{-N}}{s-s^{-1}}=[N]_s. \tag{1.3} \] Equivalently, in the common Khovanov-Rozansky notation \(a=v^{-1}\), one sets \(a=s^N\). In the Heisenberg-category notation of Mousaid-Savage [MouSa], this is \[ z=s-s^{-1},\qquad t_{\mathrm H}=v^{-1}=s^N. \tag{1.4} \] The elliptic-Hall parameters in (1.1) do **not** change. Thus the \(\mathfrak{sl}_N\) operation is not another \(q_{\mathrm E},t_{\mathrm E}\) specialization: it is first the coefficient specialization (1.2), and then a Reshetikhin-Turaev tensor quotient (or representation) of the specialized skein category.

Specialization alone is insufficient. At generic \(s\), the actual \(\mathfrak{sl}_N\) ribbon category imposes \[ \bigwedge^{N+1}V=0,\qquad \bigwedge^N V\cong\mathbf 1, \tag{1.5} \] where \(V\) is the vector representation. The second relation distinguishes \(\mathfrak{sl}_N\) from the polynomial \(\mathfrak{gl}_N\) quotient, in which the determinant remains a nontrivial one-dimensional object.

The invariant in (1.3) is the **unreduced**, framed invariant. A reduced \(\mathfrak{sl}_N\) homology requires a marked component and normalizes the unknot differently; it is not naturally an algebra-valued invariant of unmarked torus links. All algebra statements below are therefore unreduced.

来源字段：http://aimpl.org/catheckehilbert/5/

06288 | SMA-005

### Smale's 5th Problem: Height Bounds for Diophantine Curves

O / open | Smale 问题

Find effective uniform bounds for the heights of rational points on algebraic curves.

背景链接：https://mathworld.wolfram.com/SmalesProblems.html

06289 | OWR-14299574-008

### Small Resolutions Detected by Symplectic Cohomology

P / partially_solved | Oberwolfach 报告问题汇编

Let $F$ be the Milnor fiber of an isolated compound Du Val singularity. Does the singularity admit a small resolution with $\ell$ irreducible exceptional components if and only if $\operatorname{rank}SH^k(F)=\ell$ in every negative degree $k$?

来源字段：https://doi.org/10.4171/owr/2025/45

06290 | AIM-ALGEBRAIC_GEOMETRY-0125

### Smooth hypersurfaces beyond the Chevalley--Warning range

P / partially_solved | AIM 研讨会问题汇编

For every \(p,d\) with \(d\geq n+1\), can we find hypersurface over \(\mathbb F_p\) of degree \(d\) in \(\mathbb P^n\) such that either \(\#X\not\equiv 1\pmod p\) or \(X\) is not rationally connected?

来源字段：http://aimpl.org/ratsubvarpos/9/

06291 | OWR-7155446-010

### Smooth Interior Regularity for Complex Monge-Ampère Equations

O / open | Oberwolfach 报告问题汇编

If $\Omega$ is a bounded hyperconvex domain in $\mathbb{C}^n$ and $u$ solves $(dd^c u)^n=f\,d\lambda$ with $u=0$ on the boundary, does $f\in C^\infty(\Omega)$ and $f>0$ imply $u\in C^\infty(\Omega)$?

来源字段：https://doi.org/10.4171/owr/2021/25

06292 | OWR-16939-003

### Smooth Modular Compactifications of Abelian-Differential Strata

P / partially_solved | Oberwolfach 报告问题汇编

Construct a proper smooth compactification of the stratum $\mathbb P\Omega\mathcal M_g(\mu)$ of abelian differentials that has a natural modular interpretation.

来源字段：https://doi.org/10.4171/owr/2019/27

06293 | AIM-ALGEBRAIC_GEOMETRY-0321

### Smoothable closure and stable reduction for stable surfaces

S / solved | AIM 研讨会问题汇编

(1) The moduli space \(\overline{M^{\mathrm{sm}}_{K^2,\chi}}\) of smoothable stable surfaces has no natural scheme structure at the boundary if one simply throws away components parametrizing surfaces with worse than rational double points. Can the notion of smoothability be organized so as to give a good scheme structure?

(2) If \(X\to\Delta'\) is a family of stable surfaces over a punctured disk satisfying Kollár's condition, can it, possibly after base change, be completed to a family of stable surfaces over the disk that also satisfies Kollár's condition?

来源字段：https://aimath.org/WWN/birational/birational.pdf

06294 | OWR-13860-003

### Smoothness of Ekedahl-Kottwitz-Oort-Rapoport Strata

P / partially_solved | Oberwolfach 报告问题汇编

For a parahoric-level Shimura variety, is the finite morphism from every Kottwitz-Rapoport stratum at Iwahori level to its corresponding EKOR stratum finite étale, and consequently are all EKOR strata smooth?

来源字段：https://doi.org/10.4171/owr/2015/39

06295 | OWR-14299289-015

### Spectral Exchange of Period and L-Sheaves

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Global Geometric Duality Conjecture [OWR-14299289-016]

Under geometric Langlands duality, are the spectral projection of the period sheaf and the $L$-sheaf exchanged?

来源字段：https://doi.org/10.4171/owr/2025/18

06296 | AIM-ALGEBRAIC_GEOMETRY-0178

### Spectral formal-group moduli and a square-zero comparison obstruction

P / partially_solved | AIM 研讨会问题汇编

**Long Term.** Can we describe the moduli stack of formal groups in the derived setting?

来源字段：http://aimpl.org/equideralggeom/5/

06297 | AIM-ALGEBRAIC_GEOMETRY-0180

### Spectral local systems and augmentation-Bott orthogonality for S[BU]

P / partially_solved | AIM 研讨会问题汇编

What can we say about modules over \(S[BU_+]\)?

来源字段：http://aimpl.org/equideralggeom/5/

06298 | OWR-12697711-010

### Spectrality of Infinite Symmetric Orbit-Space Quotients

O / open | Oberwolfach 报告问题汇编

Is the Kolmogorov quotient $\operatorname{KQ}(A_{\mathbb K,\infty}^n/G)$ a spectral space?

来源字段：https://doi.org/10.4171/owr/2023/15

06299 | AIM-ALGEBRAIC_GEOMETRY-0021

### Spherical twists must be formed after Cox localization

P / partially_solved | AIM 研讨会问题汇编

Let $X$ be a smooth projective toric variety with Cox ring $S$ and irrelevant ideal $B$. Can a spherical twist be implemented on bounded graded free $S$-complexes so that it acts on virtual resolutions, and what obstruction is caused by $B$-torsion?

来源字段：http://aimpl.org/syzygyms/8/

06300 | AIM-ALGEBRAIC_GEOMETRY-0335

### Split-branch degenerations as a stable-surface boundary application

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.1.** Find applications of moduli of stable surfaces to surface theory. For example degenerations of curves are used to prove Brill-Noether type statements.

来源字段：https://aimath.org/WWN/birational/birational.pdf

06301 | AIM-ALGEBRAIC_GEOMETRY-0118

### Square-field superspecial point growth and a Betti-complexity barrier

P / partially_solved | AIM 研讨会问题汇编

For each of \(\mathcal M_g\) and \(\mathcal A_g\), with \(q\) fixed and \(g\to\infty\), which natural sublocus accounts for the largest number of \(\mathbb F_q\)-points?

来源字段：http://aimpl.org/ratsubvarpos/9/

06302 | OWR-14298581-007

### Stability Conditions from Surface Degenerations

O / open | Oberwolfach 报告问题汇编

Construct new Bridgeland stability conditions on surfaces by degenerating the geometric stability conditions associated with ample classes, particularly when contracting several rational curves or passing to mildly singular surfaces.

来源字段：https://doi.org/10.4171/owr/2024/32

06303 | OWR-14298160-003

### Stability Manifolds for Threefold Flops

O / open | Oberwolfach 报告问题汇编

For the category $\mathcal D$ associated with a threefold flopping contraction, determine whether its distinguished stability-manifold component regularly covering the associated hyperplane complement is the universal cover, whether the mutation action is faithful, whether $\operatorname{Stab}(\mathcal D)$ is connected, and whether the distinguished component is contractible; equivalently, determine whether the hyperplane complement is a $K(\pi,1)$ space.

来源字段：https://doi.org/10.4171/owr/2024/4

06304 | OWR-13750339-011

### Stability of Brill-Noether Triples

P / partially_solved | Oberwolfach 报告问题汇编

Prove that all but finitely many Brill-Noether triples $(d,g,r)$ with $g\geq 2$ are stable; more strongly, classify the exceptional triples and prove stability for every other triple in characteristic zero.

来源字段：https://doi.org/10.4171/owr/2023/27

06305 | OWR-1452-015

### Stable Coordinate Conjecture for Complex Polynomials

P / partially_solved | Oberwolfach 报告问题汇编

If a polynomial $P\in\mathbb{C}[x_1,\ldots,x_n]$ becomes equivalent to $x_1$ after adjoining finitely many variables, must $P$ already be a coordinate?

来源字段：https://doi.org/10.4171/owr/2007/01

06306 | OWR-1275-008

### Stable Homology Generators for Compactified Moduli Spaces

P / partially_solved | Oberwolfach 报告问题汇编

What are generators for the stable rational homology of the Deligne-Mumford compactification of the moduli space of curves?

来源字段：https://doi.org/10.4171/owr/2006/26

06307 | AIM-ALGEBRAIC_GEOMETRY-0113

### Stable irrationality of cyclic covers of Grassmannians

P / partially_solved | AIM 研讨会问题汇编

Look for new applications of Koll\'ar / Totaro's techique.

**Recovered research task.** Find a class of varieties, different from the adjacent lower-degree and unirationality projects, to which the Kollár--Totaro positive-characteristic degeneration method gives a rigorous new irrationality or stable-irrationality theorem.

来源字段：http://aimpl.org/ratsubvarpos/6/

06308 | OWR-11101913-005

### Stable Limits of Smooth Plane Curves

P / partially_solved | Oberwolfach 报告问题汇编

Is every smooth limit of a family of plane curves of prime degree a Cartier divisor in a log-terminal $\mathbb Q$-Gorenstein degeneration of $\mathbb P^2$?

来源字段：https://doi.org/10.4171/owr/2022/32

06309 | OWR-11139-003

### Stable Rationality of Generically Free Simple-Group Quotients

O / open | Oberwolfach 报告问题汇编

For simply connected simple groups, are generically free linear quotients stably rational in the $\operatorname{Spin}$ series and for $E_8$?

来源字段：https://doi.org/10.4171/owr/2011/47

06310 | OWR-14298584-005

### Stable Reduction of Plane Quartics from Cayley Octads

O / open | Oberwolfach 报告问题汇编

Prove that the combinatorial data of the configuration of the eight points in a Cayley octad determines the stable reduction type of the associated smooth plane quartic, analogously to the way a cluster picture determines stable reduction in the hyperelliptic case.

来源字段：https://doi.org/10.4171/owr/2024/40

06311 | OWR-1703866-006

### Stable-Pairs and Gromov-Witten Series for K3 Times Elliptic Curves

S / solved | Oberwolfach 报告问题汇编

For a K3 surface $S$ and an elliptic curve $E$, does the Igusa cusp-form identity $F_{PT}=F_{GW}+F$ hold, where $F_{PT}$ and $F_{GW}$ are the stable-pairs and Gromov-Witten generating series and $F$ is an explicit correction term?

来源字段：https://doi.org/10.4171/owr/2020/19

06312 | AIM-ALGEBRAIC_GEOMETRY-0324

### Stable-surface properness and an optimal universal tame Kummer index

P / partially_solved | AIM 研讨会问题汇编

Problem 1. 5. (Hassett) Prove the valuative criterion for properness for the moduli stack of stable surfaces in positive characteristic.

**Problem 1.5 (Hassett).** Prove the valuative criterion for properness for the moduli stack of stable surfaces in positive characteristic.

The exact mathematical sentence is:

The source did not fix a characteristic, a volume, or a smoothability component. Thus its literal scope is all positive characteristics and the full stable-surface moduli problem.

In a modern precise formulation one fixes numerical data, most commonly the volume $$ v=K_S^2 $$ (and, for pairs, a coefficient set and corresponding Hilbert data), and uses the Kollár condition in the definition of a family. For a discrete valuation ring $R$ with fraction field $K$, the existence part of the valuative criterion asks that a stable surface over $K$ extend to a stable family after replacing $R$ by the integral closure in a finite extension of $K$. The uniqueness part is separatedness. Properness is therefore stronger than algebraicity, finite type, separatedness, or projectivity of subspaces already known to be proper.

来源字段：https://aimath.org/WWN/birational/birational.pdf

06313 | AIM-ALGEBRAIC_GEOMETRY-0056

### Stein covers, MRC quotients, and the finite-cover diagonal obstruction

P / partially_solved | AIM 研讨会问题汇编

Let $X$ be a rationally connected $3$-fold which is a del Pezzo fibration over $\mathbb P^1$. Is there a component $M$ of the space of sections of the map $X \rightarrow \mathbb P^1$ such that the Stein factorization of $\alpha: M \rightarrow J(X)$ is the MRC fibration of $M$?

Can we find an example of such a rationally connected $3$-fold $X$ with does not admit a decomposition of the diagonal and with the above property for the components of spaces of rational curves on X?

The phrase “with does not” is visibly a typographical error; below it is read as “which does not.” Nothing else in the statement is changed. The record's “Sho” is identified as Sho Tanimoto through Lehmann--Tanimoto's theorem on Stein-factor stabilization [LT24].

There are two minor interpretive points.

1. A component of the section space is usually quasi-projective and may be singular. “Its MRC fibration” therefore means the MRC quotient of a smooth projective model of its Hilbert-scheme closure. 2. “The Stein factorization is the MRC fibration” means that the **connected-fiber morphism** in the Stein factorization is the MRC quotient. It cannot mean the finite morphism to $J(X)$. This distinction is exactly what allows a nontrivial finite cover to evade Voisin's obstruction.

The second question does not specify whether “a decomposition” is meant in the Chow or integral cohomological sense. These are distinguished below. The concrete example of [LT24, Example 10.5] has neither an integral Chow decomposition nor an integral homological (equivalently, in this complex setting, integral cohomological) decomposition.

来源字段：http://aimpl.org/rationalityag/1/

06314 | OWR-11580-009

### Stringy Euler Criteria for Smooth Spherical Varieties

O / open | Oberwolfach 报告问题汇编

For every locally factorial spherical variety $X$, prove or disprove that $e_{\mathrm{st}}(X)=e(X)$ is a smoothness criterion.

来源字段：https://doi.org/10.4171/owr/2012/13

06315 | OWR-14754-017

### Stringy Euler Numbers under Divisorial Contractions

P / partially_solved | Oberwolfach 报告问题汇编

If $f:X\to X'$ is a divisorial Mori contraction, is $e_{\mathrm{alg}}^{\mathrm{str}}(X)>e_{\mathrm{alg}}^{\mathrm{str}}(X')$?

来源字段：https://doi.org/10.4171/owr/2016/46

06316 | OWR-14754-016

### Stringy Euler Numbers under Mori Flips

O / open | Oberwolfach 报告问题汇编

If $g:X\dashrightarrow X^+$ is a Mori flip, is $e_{\mathrm{alg}}^{\mathrm{str}}(X)>e_{\mathrm{alg}}^{\mathrm{str}}(X^+)$?

来源字段：https://doi.org/10.4171/owr/2016/46

06317 | OWR-16158-005

### Stronger Obstructions to Zero-Cycles Beyond Étale Homotopy

O / open | Oberwolfach 报告问题汇编

Can one construct an obstruction to zero-cycles that is strictly stronger than the obstruction arising from stable étale homotopy?

来源字段：https://doi.org/10.4171/owr/2018/17

06318 | AIM-ALGEBRAIC_GEOMETRY-0099

### Supersingular dimension ladder and unirational bounds for smooth quartic K3 surfaces

P / partially_solved | AIM 研讨会问题汇编

Let \(\mathbb P\) be the parametrizing space of higher surfaces of degree \(d\) in \(\mathbb P^n\). What is the largest dimensional family of unirational / supersinglar hypersurfaces?

来源字段：http://aimpl.org/ratsubvarpos/3/

06319 | OWR-4990374-015

### Supersolvability of Low-Exponent Free Hyperplane Arrangements

O / open | Oberwolfach 报告问题汇编

Is every central free hyperplane arrangement whose exponents consist of ones and twos, with at most one exponent equal to three, supersolvable?

来源字段：https://doi.org/10.4171/owr/2021/5

06320 | OWR-17296-017

### Supersolvable Line Arrangements with One Modular Point

O / open | Oberwolfach 报告问题汇编

Can supersolvable line arrangements be classified, in particular can complex supersolvable arrangements with exactly one modular point be classified?

来源字段：https://doi.org/10.4171/owr/2019/53

06321 | OWR-4990374-009

### Supersolvable Line Arrangements with One Modular Point

O / open | Oberwolfach 报告问题汇编

Classify all complex supersolvable line arrangements having exactly one modular singular point.

来源字段：https://doi.org/10.4171/owr/2021/5

06322 | OWR-14299291-009

### Surjectivity of the p-Adic Hitchin Morphism

O / open | Oberwolfach 报告问题汇编

Is the p-adic Hitchin morphism $\widetilde{H}_n:\operatorname{Rep}_{n,X}\to A_{n,X}$ surjective?

来源字段：https://doi.org/10.4171/owr/2025/29

06323 | OWR-12176-001

### Surjectivity on Component Groups of Semistable Curves

O / open | Oberwolfach 报告问题汇编

Let $f:X'\to X$ be a finite morphism of smooth, proper, geometrically connected curves over $K$, with induced map $f_*:\Phi_{X'}\to\Phi_X$. Is $f_*$ surjective when the minimal regular model of $X'$ consists of two rational components meeting transversally and $X$ is a Mumford curve of genus at least $2$?

来源字段：https://doi.org/10.4171/owr/2012/53

06324 | OWR-16628-005

### Symbolic-Power Noncontainments for Reflection Arrangements

O / open | Oberwolfach 报告问题汇编

For which irreducible complex reflection groups does the triple-point ideal $I_X$ of the reflection arrangement satisfy $I_X^{(3)}\not\subseteq I_X^2$?

来源字段：https://doi.org/10.4171/owr/2018/45

06325 | OWR-13679-002

### Symmetric Landau-Ginzburg Mirrors of Homogeneous Spaces

P / partially_solved | Oberwolfach 报告问题汇编

For a homogeneous space $X$ and its Landau-Ginzburg mirror, can the relationship be made more symmetric? Can the mirror be expressed in coordinates naturally related to the cohomology of $X$? Does the map from cohomology to twisted de Rham cohomology extend to an isomorphism of bundles with connection?

来源字段：https://doi.org/10.4171/owr/2015/22

06326 | OWR-12015-007

### Symplectic Moduli Spaces of Sheaves

O / open | Oberwolfach 报告问题汇编

Let $X$ be an irreducible holomorphic symplectic manifold and $M$ a moduli space of sheaves on $X$. Must $M$ also be an irreducible holomorphic symplectic manifold?

来源字段：https://doi.org/10.4171/owr/2012/43

06327 | OWR-4086-005

### Symplectic Singularities from Symplectic Reduction

P / partially_solved | Oberwolfach 报告问题汇编

When does symplectic reduction produce a symplectic singularity?

来源字段：https://doi.org/10.4171/owr/2009/43

06328 | AIM-ALGEBRAIC_GEOMETRY-0177

### Tame normalization and two-axis integrality for level TMF

P / partially_solved | AIM 研讨会问题汇编

What would “rings of integers” in \(tmf\) with level structure be? Also, is there a TAF analogue of such objects with connections to TMF?

来源字段：http://aimpl.org/equideralggeom/5/

06329 | AIM-ARITHMETIC_GEOMETRY-0049

### Tangent spaces and prescribed-jet smoothings on the monogenic chart

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.** Can you describe the Zariski tangent space to the smoothable component of \(\operatorname{Hilb}^d(\mathbb A^n)\)?

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

06330 | AIM-ALGEBRAIC_GEOMETRY-0023

### Tate-completion obstruction and finite torsion windows for the motivic filtration on TC of the sphere

P / partially_solved | AIM 研讨会问题汇编

The motivic filtration on the algebraic \(K\)-theory of the sphere spectrum

Due to work of Rognes, Blumberg--Mandell, and others, we understand the algebraic K-theory of the sphere spectrum fairly well. Recent work of Hahn--Raksit--Wilson gives a motivic filtration on TC of \(\mathbb E_\infty\)-rings which is closely related to the motivic filtration on algebraic K-theory in motivic homotopy theory.

Can one understand the motivic filtration (or motivic spectral sequence) on the \(\mathrm{TC}\) of the sphere spectrum? What about for the motivic filtration on related invariants such as \(\mathrm{THH}\), \(\mathrm{TP}\)?

来源字段：http://aimpl.org/equivstable/2/

06331 | OWR-14222-007

### Terao Conjecture for Line Arrangements

O / open | Oberwolfach 报告问题汇编

Is Terao's conjecture true for line arrangements; equivalently, is freeness determined by the incidence lattice?

来源字段：https://doi.org/10.4171/owr/2016/14

06332 | AIM-ALGEBRAIC_GEOMETRY-0255

### Terminal singularities need not be Cohen-Macaulay: the sharp threefold threshold and a cone-Veronese audit

S / solved | AIM 研讨会问题汇编

**Terminal and Cohen-Macaulay.** Are (log) terminal singularities Cohen-Macaulay? Rational?

来源字段：http://aimpl.org/minimalmodcharp/2/

06333 | AIM-ALGEBRAIC_GEOMETRY-0253

### Termination of klt threefold flips in positive characteristic

P / partially_solved | AIM 研讨会问题汇编

**Termination in dimension 3.** Do klt flips terminate in dimension 3?

The supplied literature note says:

The record itself does not specify the ground field, projectivity, a base, $\mathbb Q$-factoriality, the coefficient field, or whether “terminate” means every sequence or only a specially directed MMP. These omissions matter. The preceding problem 1.1 explicitly works over an algebraically closed field of characteristic $p>0$, and the AIM workshop description confirms that positive characteristic is the intended setting. I therefore use the following precise absolute version as the main modern reading:

$$\tag{P} \begin{gathered} k\text{ is algebraically closed of characteristic }p>5,\\ (X,B)\text{ is a projective }\mathbb Q\text{-factorial klt threefold with a }\mathbb Q\text{-boundary},\\ \text{and }X=X_0\dashrightarrow X_1\dashrightarrow\cdots \text{ is an arbitrary sequence of }(K_X+B)\text{-flips.} \end{gathered}$$

The question is whether every such sequence is finite. Relative and $\mathbb R$-boundary versions are recorded separately below. Hypothesis $p>5$ is not silently read into the historical question; it is the range where the modern threefold MMP used in the analysis is available in its cleanest form.

来源字段：http://aimpl.org/minimalmodcharp/1/

06334 | AIM-ALGEBRAIC_GEOMETRY-0303

### Test ideals and component-weighted symbolic interpolation on projective space

P / partially_solved | AIM 研讨会问题汇编

Investigate the applications of the topics of the conference (e.g., test ideals) to projective geometry.

来源字段：http://aimpl.org/testandmultiplierideals/2/

06335 | AG-002

### The Abundance Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Abundance Conjecture [ALGGEOM-001]; Abundance Conjecture for Projective Minimal Models [OWR-13109-001]

For a minimal model $X$ of non-negative Kodaira dimension, the canonical divisor $K_X$ is semi-ample.

背景链接：https://en.wikipedia.org/wiki/Abundance_conjecture

06336 | OWR-14299284-001

### The Bondal-Orlov D-Equivalence Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ and $X'$ be smooth projective complex Calabi-Yau varieties. If $X$ and $X'$ are birational, must their bounded derived categories $D^b(X)$ and $D^b(X')$ be equivalent?

来源字段：https://doi.org/10.4171/owr/2025/28

06337 | AIM-ALGEBRAIC_GEOMETRY-0034

### The C2-equivariant even filtration and its comparison problem

S / solved | AIM 研讨会问题汇编

**Equivariant and synthetic.** Can one profitably mix synthetic spectra with equivariant homotopy theory?

Along these lines, what is the $\mathbf Z/2$-equivariant analogue of the even filtration (where $\tau_{\geq 2*}$ is replaced by $\tau_{\geq \rho *}$, and evenness is replaced by slice-even)?

来源字段：http://aimpl.org/equivstable/3/

06338 | AIM-ALGEBRAIC_GEOMETRY-0195

### The calibrated camera functor and an infinitesimal essential-boundary obstruction

S / solved | AIM 研讨会问题汇编

Define, over arbitrary base schemes and compatibly with base change, the moduli functor of calibrated multiview camera configurations, and determine whether it is representable and how it maps to a Hilbert scheme or to the usual calibrated tensor varieties.

来源字段：http://aimpl.org/algvision/2/

06339 | AIM-ALGEBRAIC_GEOMETRY-0114

### The Cartier obstruction and a cone-K3 Brauer escape

P / partially_solved | AIM 研讨会问题汇编

Find an example of varieties which are degenerations of Fano hypersurfaces with $H^{i}(\mathcal O_X)\neq 0$ for some $i$. Try to construct varieties with non-trivial Brawer classes.

来源字段：http://aimpl.org/ratsubvarpos/6/

06340 | AIM-ALGEBRAIC_GEOMETRY-0343

### The curve cone of the stable genus-zero moduli space and the first symmetric F-curve redundancy

P / partially_solved | AIM 研讨会问题汇编

**Problem 5.1. Determine the cone of curves of \(\boldsymbol{\overline M_{0,n}}\).**

Inspection of the PDF content stream resolves the typography: immediately before the glyph \(M\), the stream draws a short filled horizontal rectangle at the height of an overline. Thus the mathematical source has an overline that was stored as a vector rule rather than as a font glyph, and both text extractions lost it. The recovered statement is therefore

This reading is also forced by context. The section is headed “Moduli of curves-Birational geometry,” and cites Farkas-Gibney and Gibney-Keel-Morrison, both of which study the proper stable-curve compactification. The usual Kleiman-Mori cone is a cone for a proper projective variety; the open configuration space \(M_{0,n}\) is not the intended object.

For \(n\geq 3\), every stable labeled genus-zero curve has trivial automorphism group: its dual tree is fixed by its labeled leaves, and every component has at least three fixed special points. Consequently \(\overline{\mathcal M}_{0,n}\) is represented by the smooth projective scheme \(\overline M_{0,n}\). There is no stack/coarse numerical discrepancy for the space in the recovered problem. The symmetric quotient \(\overline M_{0,n}/S_n\), used below, does have finite quotient stabilizers; it is treated through pullback and pushforward on real numerical spaces.

来源字段：https://aimath.org/WWN/birational/birational.pdf

06341 | AIM-ALGEBRAIC_GEOMETRY-0186

### The degree-30 essential Hurwitz divisor and a harmonic-cubic pullback

P / partially_solved | AIM 研讨会问题汇编

**What is the Hurwitz form of the essential variety?**

来源字段：http://aimpl.org/algvision/1/

06342 | AIM-ARITHMETIC_GEOMETRY-0076

### The degree-two branchcurve locus is an explicit connected quotient

P / partially_solved | AIM 研讨会问题汇编

Fix a Hilbert polynomial (P), consider the moduli space \(\mathrm{BrV}^{P}\mathbf P^n\) of branchvarieties that are equidimensional and connected in codimension (1). Is \(\mathrm{BrV}^{P}(\mathbf P^n)\) connected when non-empty?

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

06343 | AIM-ALGEBRAIC_GEOMETRY-0293

### The direct summand theorem and trace-free monic towers

S / solved | AIM 研讨会问题汇编

**Recovered direct summand conjecture (now theorem).** Let \(R\) be a regular Noetherian commutative ring and let \(i:R\hookrightarrow S\) be a module-finite extension of commutative rings. Then there is an \(R\)-linear map \(\rho:S\to R\) such that \(\rho\circ i=\operatorname{id}_R\).

来源字段：http://aimpl.org/testandmultiplierideals/2/

06344 | AIM-ALGEBRAIC_GEOMETRY-0124

### The E8 quadric and the geometric boundary

P / partially_solved | AIM 研讨会问题汇编

Is there an irrational smooth proper variety over \(\operatorname{Spec}\mathbb Z\)? (Can we find such a hypersurface?)

来源字段：http://aimpl.org/ratsubvarpos/9/

06345 | AIM-ALGEBRAIC_GEOMETRY-0111

### The exact one-cover differential-form barrier

P / partially_solved | AIM 研讨会问题汇编

Can we improve those techniques to find differential form for lower degree hypersurfaces?

The record has no remarks or bibliography. The original problem-list URL now returns a missing-page response, so the intended antecedents have to be recovered from the section title, the neighboring problem 6.1, the 2016 workshop report, and the papers current at the time of the workshop.

The neighboring problem 6.1 asks about a “very general hypersurface of degree \(d\geq 2\lceil(n+3)/3\rceil\)” in characteristic \(p\). This is Kollár's 1995 bound for an \(n\)-dimensional hypersurface \[ X_d^n\subset \mathbf P^{\,n+1}. \] The section title is “New applications of Kollar's and Totaro's techniques.” Thus “those techniques” means:

1. Kollár's mixed-characteristic degeneration of a degree-\(2a\) hypersurface to an inseparable cyclic cover in characteristic \(2\), and the resulting invertible subsheaf of differential forms on a resolution; 2. Totaro's use of the same form, together with a universally \(CH_0\)-trivial resolution and specialization of the decomposition of the diagonal, to obstruct stable rationality.

The intended range is \(n\geq3\), principally the Fano range \(d\leq n+1\). In Kollár's argument the line bundle must be **big**, giving \(d\geq2\lceil(n+3)/3\rceil\). Totaro needs only a nonzero section, giving \[ d\geq2\left\lceil\frac{n+2}{3}\right\rceil . \] Accordingly, “lower degree” most naturally means below these linear bounds.

A crucial clarification is that the differential form is not asserted to live on a smooth very general Fano hypersurface. It lives on a smooth resolution \(\widetilde Y\) of a singular inseparable-cover special fiber \(Y\). Passing from such a form to a statement about a different smooth fiber requires a separate specialization theorem and control of the resolution.

来源字段：http://aimpl.org/ratsubvarpos/6/

06346 | AIM-ALGEBRAIC_GEOMETRY-0032

### The filtered geometric-fixed-point model and an odd-primary answer for a-inverted real Artin--Tate spectra

P / partially_solved | AIM 研讨会问题汇编

**\(a\) inverted Artin--Tate \(\mathbb R\)-motives.** The category of Artin--Tate \(\mathbb R\)-motives has a topological model as a \(1\)-parameter deformation of \(C_2\)-equivariant homotopy theory [arXiv:2010.10325]. Let \(\mathrm{Sp}_{\mathbb R}^{AT}\) denote this category. What category do you get when you invert the class \(a\) in \(\mathrm{Sp}_{\mathbb R}^{AT}\)?

来源字段：http://aimpl.org/equivstable/3/

06347 | OWR-14298795-005

### The Generalized Étale-Homotopy Section Conjecture

O / open | Oberwolfach 报告问题汇编

Determine the classes of varieties $X/k$ for which the canonical map $X(k)\to\Gamma^{\mathrm{et}}(X/k)$ is a bijection, where $\Gamma^{\mathrm{et}}(X/k)$ is the set of homotopy classes of sections of the étale-homotopy fibration $\Pi^{\mathrm{et}}(X)\to\Pi^{\mathrm{et}}(k)$.

来源字段：https://doi.org/10.4171/owr/2024/45

06348 | AIM-ALGEBRAIC_GEOMETRY-0342

### The generic hyperelliptic boundary of Hacking's quartic stack

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.8.** Study the geometry of Hacking's moduli space of plane curves. Are there applications to questions about families of smooth plane curves? Similarly for configuration spaces or moduli spaces of marked del Pezzo surfaces.

来源字段：https://aimath.org/WWN/birational/birational.pdf

06349 | AIM-ARITHMETIC_GEOMETRY-0078

### The generic very-compressed point for Hilbert function (1,4,10,a)

P / partially_solved | AIM 研讨会问题汇编

Problem 31. Consider 1, 4, 10, a where 6 ≤ a ≤ 10. The general Artinian algebra with this Hilbert function is nonsmoothable. What is the generic point of the component it lies on?

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

06350 | OWR-4427-002

### The Global Monodromy Conjecture for Calabi-Yau Varieties

P / partially_solved | Oberwolfach 报告问题汇编

For every Calabi-Yau variety $X/K$, does its motivic zeta function $Z_X(T)$ admit a rational expression with finitely many denominator factors $1-\mathbb L^aT^b$ such that, for each occurring pair $(a,b)$, the cyclotomic polynomial whose order is the order of $a/b$ in $\mathbb Q/\mathbb Z$ divides the characteristic polynomial of tame monodromy on some $\ell$-adic cohomology group of $X$?

来源字段：https://doi.org/10.4171/owr/2010/39

06351 | OWR-11139-007

### The Green-Griffiths-Lang Entire-Curve Conjecture

O / open | Oberwolfach 报告问题汇编

If $X$ is a projective variety of general type, must there exist a proper algebraic subvariety $Y\subset X$ containing the image of every nonconstant entire curve $\mathbb C\to X$?

来源字段：https://doi.org/10.4171/owr/2011/47

06352 | OWR-14298795-008

### The Grothendieck-Katz p-Curvature Conjecture

O / open | Oberwolfach 报告问题汇编

Let $A$ be a matrix with entries in $\mathbb Q(z)$. Are all solutions of $(\partial/\partial z-A)\mathbf f(z)=0$ algebraic if and only if the reduction of $(\partial/\partial z-A)^p$ vanishes for almost all primes $p$?

来源字段：https://doi.org/10.4171/owr/2024/45

06353 | AIM-ALGEBRAIC_GEOMETRY-0138

### The Hilbert--Chow zero-degree face of the surface cone of a Hilbert square

P / partially_solved | AIM 研讨会问题汇编

Effective cones of Hilbert schemes and Moduli Spaces

It is natural to try to compute effective cones in the case of Hilbert schemes or moduli spaces. One natural Hilbert scheme to consider first is the Hilbert scheme of length \(n\) subscheme of a smooth surface \(S\), denoted by \(S^{[n]}\). A great deal is known about these Hilbert schemes. In recent years there has been a lot of exciting work on the effective cones of divisors and curves of the Hilbert schemes \(\operatorname{Eff}^1(S^{[n]})\) and \(\operatorname{Eff}_1(S^{[n]})\). See for example Bayer and Macri's work on Hilbert schemes of K3 surfaces or Huizenga's work on Hilbert schemes of \(\mathbb P^2\). It is then natural to ask how to compute the effective cone of surfaces or the codimension 2 effective cones.

What is the effective cone of surfaces \(\operatorname{Eff}_2(S^{[n]})\)? What is the codimension 2 effective cone \(\operatorname{Eff}^2(S^{[n]})\)? What about higher dimension/codimension?

来源字段：http://aimpl.org/poscycles/1/

06354 | AIM-ALGEBRAIC_GEOMETRY-0036

### The known C-motivic lift and a tau-torsion nonuniqueness obstruction

P / partially_solved | AIM 研讨会问题汇编

**Motivic modular forms (mmf).** Is there a connective version of motivic modular forms, which should be named \(\mathrm{mmf}\), and is a lift of \(\mathrm{tmf}\)?

来源字段：http://aimpl.org/equivstable/3/

06355 | AIM-ALGEBRAIC_GEOMETRY-0031

### The L-theoretic obstruction to hermitian devissage

P / partially_solved | AIM 研讨会问题汇编

**Devissage for equivariant/hermitian \(K\)-theory**

We understand devissage for the \(K\)-theory of stable categories fairly well now due to work of Quillen, Barwick, and Burklund--Levy (see (arXiv:2112.14723)). See section 2.2 of (arXiv:2009.07225) for what is known for hermitian \(K\)-theory.

What is the optimal devissage result for equivariant/hermitian \(K\)-theory?

来源字段：http://aimpl.org/equivstable/2/

06356 | OWR-14299289-008

### The Local Geometric Langlands Conjecture

O / open | Oberwolfach 报告问题汇编

For a Hamiltonian $G$-space $M$, can one construct a dual Hamiltonian $G^\vee$-space $M^\vee$ so that local geometric Langlands duality matches the boundary conditions associated with $M$ and $M^\vee$? In the unramified cotangent case $M=T^*X$, can this be realized by an equivalence involving $G(\mathcal D)$-equivariant sheaves on $X(\mathring{\mathcal D})$?

来源字段：https://doi.org/10.4171/owr/2025/18

06357 | OWR-11792-003

### The Minimal Model Program for Stable-Curve Moduli

P / partially_solved | Oberwolfach 报告问题汇编

Can the minimal model program be run on $\overline{\mathcal M}_g$, the moduli space of Deligne-Mumford stable curves?

来源字段：https://doi.org/10.4171/owr/2012/30

06358 | OWR-14299291-003

### The Nekovář-Scholl Plectic Cohomology Conjecture

O / open | Oberwolfach 报告问题汇编

Does $R\Gamma(\operatorname{Sh}_\infty,L_\xi)\in D^b(\Gamma_E,\mathbb{Q}_\ell)$ lift canonically, via $[\mu]$, to an object of $D^b(\Gamma_{F/\mathbb{Q}},\mathbb{Q}_\ell)$ compatibly with the Hecke actions?

来源字段：https://doi.org/10.4171/owr/2025/29

06359 | AIM-ALGEBRAIC_GEOMETRY-0220

### The one-point count for a plane cubic and the degree-nine log-BPS answer

P / partially_solved | AIM 研讨会问题汇编

Let $X \subset \mathbb{P}^2$ be a smooth plane cubic (elliptic curve).

**Problem 3.1.** What is the number of degree-$d$ rational curves in $\mathbb{P}^2$ meeting $X$ at a single point (with multiplicity $3d$)? This is known for $d \leq 8$ by Takahashi, and even $d = 9$ would be new and interesting. One possible approach is to degenerate $X$ to a union of $3$ lines; is a tropical approach also possible?

来源字段：https://aimath.org/pastworkshops/degenalggeomproblems.pdf

06360 | OWR-5159-006

### The p-Adic Variational Hodge Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $W$ be the Witt ring of a perfect field of characteristic $p>0$, let $X$ be smooth and projective over $W$ with special fiber $X_1$, and let $\xi_1\in CH^r(X_1)_{\mathbb Q}$. If its crystalline cycle class lies in the Hodge filtration $F^rH^{2r}_{\mathrm{dR}}(X/W)_{\mathbb Q}$, must there exist $\xi\in CH^r(X)_{\mathbb Q}$ whose restriction to $X_1$ has crystalline cycle class equal to that of $\xi_1$?

来源字段：https://doi.org/10.4171/owr/2011/31

06361 | AIM-ARITHMETIC_GEOMETRY-0037

### The polarization-orbit ring detects the full cohomological spectral radius

S / solved | AIM 研讨会问题汇编

Let $A$ be an abelian variety over a finite field $K$, and $u:A\to A$ a fixed automorphism. Then $u$ induces $u^*:H^*(\overline A,\mathbf Q_\ell)\to H^*(\overline A,\mathbf Q_\ell)$. Let $\alpha_i$ be its eigenvalues, and let $|\alpha_i|$ denote the archimedean size of $\alpha_i$. Take the subring in $H^*(\overline A,\mathbf Q_\ell)$ generated by a polarization and stabilized by $u$. Is the maximum archimedean size of the eigenvalues of the subring equal to the maximum archimedean size of the eigenvalues of $H^*(\overline A,\mathbf Q_\ell)$?

来源字段：http://aimpl.org/cohomabelian/1/

06362 | AIM-ALGEBRAIC_GEOMETRY-0094

### The Procesi bundle as a traced and Koszul-dual Soergel unit

P / partially_solved | AIM 研讨会问题汇编

**Problem 5.6.** How does one describe the Procesi bundle in the other settings, like $K^b(\mathbb S\mathrm{Bim}_{\mathrm{fin}})$?

来源字段：http://aimpl.org/catheckehilbert/5/

06363 | AIM-ALGEBRAIC_GEOMETRY-0015

### The projective-dimension sign and a Cox-depth criterion

P / partially_solved | AIM 研讨会问题汇编

Given $M$ an $S$-module, $S=\operatorname{Cox}(X)$,

1. can we always find another $S$-module $M'$ satisfying - $\widetilde M=\widetilde {M'}$, - $\operatorname{pdim}(M')\geq \dim(X)$?

This is known to be true if $X=\mathbb P^{n_1}\times\cdots\times\mathbb P^{n_r}$. 2. What is the relation of this to examples of Chardin-D'Cruz where adding embedded components drops regularity? 3. Potential mechanism by truncating $M$ to $M_{\geq d}$? (This works for products of projective spaces.) 4. Alternatively, could there be an invariant of $\widetilde M$ which obstructs existence of such an $M'$?

来源字段：http://aimpl.org/syzygyms/6/

06364 | AIM-ALGEBRAIC_GEOMETRY-0065

### The quadric residual of a bidegree (d,2) hypersurface

P / partially_solved | AIM 研讨会问题汇编

Investigate $Ku(X)$ for a Fano hypersurface $X$ in a product of projective spaces. Specifically, investigate this for $X \subset \mathbb P^2 \times \mathbb P^3$ of bidegree $(d,2)$ for $d \geq 2$.

来源字段：http://aimpl.org/rationalityag/1/

06365 | OWR-17475-002

### The Real-Zero Polynomial Amalgamation Conjecture

S / solved | Oberwolfach 报告问题汇编

Let $p\in\mathbb R[x,y]$ and $q\in\mathbb R[x,z]$ be real-zero polynomials of degree at most $d\in\mathbb N_0$ with $p(x,0)=q(x,0)$. Does there exist a real-zero polynomial $r\in\mathbb R[x,y,z]$ of degree at most $d$ such that $r(x,y,0)=p(x,y)$ and $r(x,0,z)=q(x,z)$?

来源字段：https://doi.org/10.4171/owr/2020/12

06366 | AIM-ARITHMETIC_GEOMETRY-0073

### The resolved four-ruling-lines degeneration and an explicit liaison audit

S / solved | AIM 研讨会问题汇编

**Problem 26.** (i) Let $C$ be of bidegree $(3,7)$ on a nonsingular quadric surface in $\mathbf P^3$. Can $C$ be connected to an extremal curve in $\operatorname{Hilb}_{10,12}(\mathbf P^3)$?

(ii) Given four skew lines $C_1$ on a nonsingular quadric $Q_1$, does there exist a family $Q_t\rightsquigarrow 2H$, and a family $C_t\subset Q_t$, such that $C_0\subset Q_0=2H$ is locally Cohen--Macaulay?

**Remark.** (ii) implies (i).

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

06367 | OWR-8415342-018

### The Restricted Geometric Langlands Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $k$ be algebraically closed, let $X/k$ be a smooth projective geometrically connected curve, and let $G$ be reductive, with $\operatorname{char}(k)$ greater than the Coxeter number of $G$. In the corresponding derived $\ell$-adic setting, prove the canonical equivalence $\operatorname{Shv}_{\operatorname{Nilp}}(\operatorname{Bun}_G)\simeq\operatorname{IndCoh}_{\operatorname{Nilp}}(\operatorname{LocSys}^{\operatorname{restr}}_{\widehat G})$.

来源字段：https://doi.org/10.4171/owr/2021/39

06368 | AIM-ALGEBRAIC_GEOMETRY-0295

### The solved F-pure multiplicity bound and a squarefree tangent-cone deficit theorem

S / solved | AIM 研讨会问题汇编

Suppose that $R$ is a local, equidimensional, $F$-pure ring of dimension $d$ with embedding dimension $n$. Is the multiplicity of $R$ at most $\binom{n}{d}$?

来源字段：http://aimpl.org/testandmultiplierideals/2/

06369 | AIM-ARITHMETIC_GEOMETRY-0028

### The spin^c obstruction rules out the proposed threefold

S / solved | AIM 研讨会问题汇编

Does there exist a smooth projective complex threefold $X$ and $\gamma\in H^2(X,\mathbb Z/2)$ for which $[\beta_2(\gamma^2)]\ne0$ in $\mathcal Q_{\beta_2(\gamma)}(X)$?

The stored AIM record asks for a smooth projective complex threefold $X$ and $\gamma\in H^2(X,\mathbb Z/2)$ such that $\beta(\gamma^2)\in H^5(X,\mathbb Z)$ remains nonzero after quotienting by

$$ H^5(X,\mathbb Z)/(H^2(X,\mathbb Z)\,\mathbb Cup\,\beta(\gamma)). $$

The source has two corruptions.

1. The integral Bockstein is the connecting map for $$ 0\longrightarrow\mathbb Z\xrightarrow{\times2}\mathbb Z \longrightarrow\mathbb Z/2\longrightarrow0. $$ The multiplication-by-$2$ label is missing from the AIM text. 2. Comparison with Antieau--Williams, *The topological period-index problem over 6-complexes*, identifies the string “$\mathbb{C}up$” as a corrupted cup product.

Put $\alpha=\beta_2(\gamma)\in H^3(X,\mathbb Z)$. The intended quotient is

$$ \mathcal Q_\alpha(X):= \frac{H^5(X,\mathbb Z)} {H^2(X,\mathbb Z)\smile\alpha}, \qquad H^2(X,\mathbb Z)\smile\alpha =\{u\smile\alpha:u\in H^2(X,\mathbb Z)\}. $$

The recovered question is:

The coefficient-sensitive Antieau--Williams operation uses the Pontryagin square $P_2(\gamma)\in H^4(X,\mathbb Z/4)$:

$$ \widetilde Q(\gamma)= \bigl[\beta_4(P_2(\gamma))\bigr]\in\mathcal Q_\alpha(X). $$

Naturality of the coefficient sequences for $4$ and $2$, together with $P_2(\gamma)\bmod2=\gamma^2$, gives

$$ \beta_2(\gamma^2)=2\beta_4(P_2(\gamma)). \tag{1} $$

Thus the displayed AIM class is $2\widetilde Q(\gamma)$, not the entire secondary obstruction. If it is nonzero, then $\alpha$ is automatically a nonzero class of exact period $2$: if $\beta_2(\gamma)=0$, then $\gamma$ lifts integrally, so its square lifts integrally and $\beta_2(\gamma^2)=0$.

来源字段：http://aimpl.org/brauermoduli/2/

06370 | AG-001

### The Standard Conjectures on Algebraic Cycles

P / partially_solved | 独立 / 未分配集合

A collection of conjectures about algebraic cycles on smooth projective varieties, including Lefschetz standard conjecture and Künneth standard conjecture.

背景链接：https://en.wikipedia.org/wiki/Standard_conjectures_on_algebraic_cycles

06371 | AIM-ALGEBRAIC_GEOMETRY-0133

### The surface cone for split projective-plane bundles over K3 surfaces

P / partially_solved | AIM 研讨会问题汇编

What is the effective cone of surfaces \(\operatorname{Eff}^2(X)\) where \(X\) is a \(\mathbb P^2\)-bundle over a K3 surface?

The preceding record says that \(X\) is a projective bundle \(\mathbb P(\mathcal E)\), and the following record separately asks about split vector bundles. I therefore use the following recovered interpretation.

- \(S\) is a complex projective K3 surface. - \(E\) is a rank-three vector bundle on \(S\). - \(X=\mathbb P_S(E)\) is the Grothendieck projectivization of one-dimensional **quotients**, with projection \(\pi:X\to S\) and tautological quotient \(\pi^*E\twoheadrightarrow \mathcal O_X(1)\). - Since \(\dim X=4\), surfaces have both dimension and codimension two, so \(N_2(X)=N^2(X)\) and the notations \(\operatorname{Eff}_2(X)\) and \(\operatorname{Eff}^2(X)\) refer to the same numerical cone.

A Brauer--Severi \(\mathbb P^2\)-fibration not arising as \(\mathbb P(E)\) is not covered; the surrounding AIM text supports the vector-bundle interpretation. I distinguish the cone generated by effective cycles from its closure, denoting the latter by \(\overline{\operatorname{Eff}}_2(X)\). The workshop summary uses “pseudo-effective” for the closed cone, and this is the convention most likely intended by the problem.

The general problem for arbitrary \(E\) is not solved here. The main proof-level result is a complete answer for split rank-three bundles, followed by a three-ray formula when the K3 surface has Picard rank one.

来源字段：http://aimpl.org/poscycles/1/

06372 | AIM-ALGEBRAIC_GEOMETRY-0132

### The surface cone of a split rank-three bundle over the projective plane

P / partially_solved | AIM 研讨会问题汇编

In (Fulger 2011, MR2836078) Fulger showed how to compute the effective cone of \(k\)-cycles \(\operatorname{Eff}^k(\mathbb P(\mathcal E)\) of a projective bundle over a smooth curve. In this case the answer can be given in terms of data coming from the Harder-Narasimhan filtration of \(\mathcal E\). However, even in the case where \(X\) is a \(\mathbb P^2\)-bundles over a surface \(S\), it is not known how to compute \(\operatorname{Eff}_2(X)\). It is natural to first consider the case where \(S\) is the projective plane or a K3 surface.

**What is the effective cone of surfaces \(\operatorname{Eff}_2(X)\) where \(X\) is a \(\mathbb P^2\)-bundle over \(\mathbb P^2\)?**

来源字段：http://aimpl.org/poscycles/1/

06373 | AG-004

### The Tate Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Generalized Sato-Tate conjecture [AMR-093-0084]

For varieties over finite fields, are the $\ell$-adic representations arising from étale cohomology related to algebraic cycles in the expected way?

背景链接：https://en.wikipedia.org/wiki/Tate_conjecture

06374 | AIM-ARITHMETIC_GEOMETRY-0017

### The universal-sheaf gerbe and a finite descent-spectrum test

P / partially_solved | AIM 研讨会问题汇编

If $X$ is a non-fine moduli space of sheaves on a K3 surface $Y$, and the dimension of $X$ is greater than 2, can the corresponding Brauer class obstruct the Hasse principle?

来源字段：http://aimpl.org/brauermoduli/1/

06375 | OWR-11139-011

### The Yau-Tian-Donaldson Correspondence for Polarized Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Does a polarized manifold $(X,L)$ admit a constant-scalar-curvature Kähler metric in $c_1(L)$ if and only if $(X,L)$ is K-stable?

来源字段：https://doi.org/10.4171/owr/2011/47

06376 | OWR-11792-007

### The Zariski Multiplicity Problem

P / partially_solved | Oberwolfach 报告问题汇编

Do hypersurfaces with topologically equivalent isolated singularities necessarily have the same multiplicities at their singular points?

来源字段：https://doi.org/10.4171/owr/2012/30

06377 | OWR-4990374-005

### Third Weak Lefschetz Property for Arrangement Jacobian Quotients

O / open | Oberwolfach 报告问题汇编

For every central arrangement $\mathcal A$ in $K^3$, does the quotient $S/J(\mathcal A)$ have the third weak Lefschetz property?

来源字段：https://doi.org/10.4171/owr/2021/5

06378 | AIM-ALGEBRAIC_GEOMETRY-0194

### Three multiview functors and a collision obstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.3.** What is the functor of points of multiview geometry?

来源字段：http://aimpl.org/algvision/2/

06379 | OWR-8415350-009

### Topological Determination of Polar Curve Positions

O / open | Oberwolfach 报告问题汇编

Does the topological type of a complex surface germ $(X,0)$ restrict the possible relative positions of its polar-curve families? For which classes of surface germs are those relative positions determined by the topological type?

来源字段：https://doi.org/10.4171/owr/2021/47

06380 | OWR-1270-001

### Topological Mirror Symmetry for Calabi-Yau Threefolds

P / partially_solved | Oberwolfach 报告问题汇编

Every family $M$ of Calabi-Yau threefolds has a mirror family $W$ with $h^{1,1}(W)=h^{2,1}(M)$, $h^{2,1}(W)=h^{1,1}(M)$, and corresponding Type-II prepotentials identified.

来源字段：https://doi.org/10.4171/owr/2006/21

06381 | OWR-4086-001

### Toric Embedded Resolution of Equicharacteristic Local Domains

P / partially_solved | Oberwolfach 报告问题汇编

Given a Noetherian excellent equicharacteristic local domain $R$ with algebraically closed residue field $k$, does a toric birational map $Z\to\mathbb A_k^N$ exist such that the strict transform of $\operatorname{Spec}R$ is nonsingular and transverse to every nondense toric orbit over the closed point?

来源字段：https://doi.org/10.4171/owr/2009/43

06382 | OWR-11783-003

### Toric Hypersurface Implicitization from Syzygies

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a toric variety, $D$ a divisor, and $U\subseteq H^0(X,\mathcal O_X(D))$ a basepoint-free subspace with $\dim U=\dim X+2$ whose image is a hypersurface in $\mathbb P(U)$. How can its implicit equation be computed efficiently from an approximation complex, and how do the multigraded syzygies of $I_U$ control the geometry of the hypersurface?

来源字段：https://doi.org/10.4171/owr/2012/21

06383 | OWR-14601-006

### Toric Local Uniformization Without Embedded Resolution

O / open | Oberwolfach 报告问题汇编

When embedded resolution is not known to exist for a variety $X$, how can one find local re-embeddings and toric birational maps that uniformize a chosen valuation, making the strict transform of $X$ nonsingular and transverse to the toric boundary at the valuation-selected point?

来源字段：https://doi.org/10.4171/owr/2016/19

06384 | OWR-15178-013

### Toric Reembeddings for Embedded Local Uniformization

O / open | Oberwolfach 报告问题汇编

Given an affine variety $X\subset\mathbb{A}^n(k)$ without a prescribed embedded resolution, how can one construct suitable re-embeddings $X\subset\mathbb{A}^N(k)$ and toric maps that yield embedded local uniformizations?

来源字段：https://doi.org/10.4171/owr/2016/57

06385 | OWR-14299291-004

### Torsion of Strongly p-Divisible Cohomology Classes

O / open | Oberwolfach 报告问题汇编

Let $U$ be a smooth rigid analytic (or dagger/affoid) variety over a $p$-adic field, and let $H^i(\widehat{U})$ denote the completed de Rham / pro-étale cohomology of the completion $\widehat{U}$. In the $p$-adic Green--Griffiths--Hawking--Scholze ($p$-GGHS) setting, are all strongly $p$-divisible elements of $H^i(\widehat{U})$ torsion?

来源字段：https://doi.org/10.4171/owr/2025/29

06386 | OWR-14603-016

### Torsion Specializations of Sections of Abelian Schemes

S / solved | Oberwolfach 报告问题汇编

Let $A\to C$ be a family of abelian varieties over a complex algebraic curve, of relative dimension at least two, and let $s:C\to A$ be a section. If $s(C)$ is not contained in a proper algebraic subgroup scheme, are there only finitely many $p\in C$ for which $s(p)$ is torsion in the fiber $A_p$?

来源字段：https://doi.org/10.4171/owr/2016/21

06387 | OWR-14299578-007

### Torsion-Anomalous Conjecture for Semiabelian Varieties

O / open | Oberwolfach 报告问题汇编

Let $V$ be an irreducible subvariety of a semiabelian variety $G$. Prove that $V$ contains only finitely many maximal $V$-torsion-anomalous subvarieties.

来源字段：https://doi.org/10.4171/owr/2025/52

06388 | AIM-ALGEBRAIC_GEOMETRY-0200

### Torsion-logarithmic obstructions and realizable Euclidean signature lines

P / partially_solved | AIM 研讨会问题汇编

Which irreducible curves \(S\subset\mathbb A^2_{u,v}\) are the Zariski closures of the standard \(SE(2,\mathbb C)\)-signature of an irreducible nonline complex plane algebraic curve?

来源字段：http://aimpl.org/algvision/3/

06389 | AIM-ALGEBRAIC_GEOMETRY-0134

### Torus formula for effective cycles on split projective bundles

S / solved | AIM 研讨会问题汇编

If \(\mathcal E\cong\mathcal L_1\oplus\cdots\oplus\mathcal L_r\) is a split vector bundle on a smooth projective surface \(S\), what are the effective cones of \(k\)-cycles \(\operatorname{Eff}_k(\mathbb P(\mathcal E))\)?

来源字段：http://aimpl.org/poscycles/1/

06390 | OWR-12490-004

### Torus Quotients among Simple Compact Kähler Threefolds

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be either a smooth compact Kähler threefold or a normal $\mathbb Q$-factorial compact Kähler threefold with at most terminal singularities. If no positive-dimensional subvariety passes through a very general point, must $X$ be bimeromorphic to $T/G$ for a torus $T$ and a finite group $G$ acting on it?

来源字段：https://doi.org/10.4171/owr/2013/27

06391 | OWR-11568-005

### Torus Quotients with Second Betti Number One

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist an algebraic torus quotient $Y=A/G$ by a freely acting finite group $G$ such that $b_2(Y)=1$?

来源字段：https://doi.org/10.4171/owr/2012/01

06392 | AIM-ALGEBRAIC_GEOMETRY-0089

### Trace categorification and a fixed-slope hook-complex lift

P / partially_solved | AIM 研讨会问题汇编

A theorem of Morton-Samuelson connected the Elliptic Hall algebra at \(q=t\) to the Skein algebra of the torus. Categorify this result.

来源字段：http://aimpl.org/catheckehilbert/5/

06393 | OWR-704-009

### Transcendental Holomorphic Morse Inequalities

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a compact complex manifold and $\alpha$ a closed $(1,1)$-form. If the integral of $\alpha^{\dim X}$ over the locus where $\alpha$ has at most one negative eigenvalue is positive, then $[\alpha]$ should contain a Kähler current, and $\operatorname{vol}([\alpha])$ should be at least that integral.

来源字段：https://doi.org/10.4171/owr/2004/21

06394 | AIM-ARITHMETIC_GEOMETRY-0011

### Triality and a middle-dimensional G2 cycle for O(4,4)

P / partially_solved | AIM 研讨会问题汇编

Study non-algebraic locally symmetric spaces for \(O(4,4)\) and natural special cycles on them. What kind of structures do we expect?

来源字段：http://aimpl.org/intersectshimura/2/

06395 | OWR-4427-011

### Triple-Point Theta Divisors at Odd Two-Torsion Points

O / open | Oberwolfach 报告问题汇编

For every genus $g$, is the locus $G(g)\subset\mathcal A_g$ of principally polarized abelian varieties whose theta divisor has multiplicity at least three at an odd two-torsion point pure of codimension $g$?

来源字段：https://doi.org/10.4171/owr/2010/39

06396 | OWR-12865-009

### Triviality of Stratified Bundles on Étale-Simply-Connected Varieties

O / open | Oberwolfach 报告问题汇编

Let $U$ be a smooth quasi-projective variety over an algebraically closed field $k$ of characteristic $p>0$. Does $\pi_1^{\mathrm{\acute et}}(U)=0$ imply that every stratified bundle on $U$ is trivial?

来源字段：https://doi.org/10.4171/owr/2014/05

06397 | OWR-13750334-005

### Tropical Corrections to P equals W on Mumford Curves

O / open | Oberwolfach 报告问题汇编

Does a $P=W$ phenomenon with a tropical correction occur in other non-Archimedean settings, such as Mumford curves?

来源字段：https://doi.org/10.4171/owr/2023/23

06398 | OWR-13750334-006

### Tropical Curve Counts for Smooth Cubic Log Divisors

O / open | Oberwolfach 报告问题汇编

For a smooth cubic plane curve $E\subset\mathbb P^2$, construct a tropical space whose curve counts agree with logarithmic Gromov-Witten invariants of the pair $\mathbb P^2(\log E)$.

来源字段：https://doi.org/10.4171/owr/2023/23

06399 | OWR-16770-008

### Tropical Dolbeault Cohomology of Non-Archimedean Varieties

O / open | Oberwolfach 报告问题汇编

For a variety $X$ over an algebraically closed complete non-Archimedean field, determine the tropical Dolbeault cohomology groups $H^{p,q}(X^{\mathrm{an}})$ and characterize their finiteness, duality, and relation to tropicalizations.

来源字段：https://doi.org/10.4171/owr/2019/8

06400 | AIM-ALGEBRAIC_GEOMETRY-0226

### Tropical Hasse-jet minors and a layered confluent-Vandermonde family

P / partially_solved | AIM 研讨会问题汇编

The source PDF is <https://aimath.org/pastworkshops/degenalggeomproblems.pdf>.

The mathematical request is the following. Given a degree \(d\) and multiplicities \(m_1,\ldots,m_s\) at general points of \(\mathbb P^2\), try to prove the predicted dimension of the interpolation system by adding point conditions until its expected vector-space dimension is one, and then find a single isolated and liftable tropical solution.

There is a convention issue in the source. Put \[ N=\binom{d+2}{2},\qquad C=\sum_{i=1}^s\binom{m_i+1}{2},\qquad q=N-C. \] The vector-space virtual dimension is \(q\), whereas the projective virtual dimension is \(q-1\). Thus “expected dimension is 1” must mean vector-space dimension one, equivalently projective dimension zero. When \(q>0\), this is reached by adding \(q-1\) independent simple point conditions.

The source's informal identification with the SHGH conjecture also needs a qualification: the uncorrected assertion “positive expected dimension implies the expected answer” is false in the presence of fixed \((-1)\)-curve components. Standard formulations of SHGH incorporate those components (or first put the system in an appropriate standard form). Section 8 gives the explicit counterexample \(\mathcal L_4(3,3)\) to the literal uncorrected slogan.

来源字段：https://aimath.org/pastworkshops/degenalggeomproblems.pdf

06401 | OWR-13101-003

### Tropical-Disc Mirror Algebras for Log Calabi-Yau Varieties

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Summary: Canonical Bases for Log Calabi-Yau Varieties [OWR-13101-002]

Let $U$ be an affine log Calabi-Yau variety with maximal boundary, and let $$A=\bigoplus_{p\in U^t(\mathbb Z)}\mathbb C\vartheta_p.$$ Does $A$ have a natural finitely generated commutative associative algebra structure whose integer structure constants in $$\vartheta_p\vartheta_q=\sum_r\alpha(p,q,r)\vartheta_r$$ count tropical discs? Is $U^\vee=\operatorname{Spec}(A)$ an affine log Calabi-Yau mirror with maximal boundary, is mirror duality involutive, and do the induced canonical bases restrict to bases of regular functions on every partial minimal model?

来源字段：https://doi.org/10.4171/owr/2014/31

06402 | AIM-ALGEBRAIC_GEOMETRY-0281

### Two-minor complementarity and rank-stratum reduction for determinantal F-invariants

P / partially_solved | AIM 研讨会问题汇编

Given a determinantal ring \(R\), compute \(e_{HK}(R)\) and \(s(R)\).

来源字段：http://aimpl.org/testandmultiplierideals/1/

06403 | AIM-ALGEBRAIC_GEOMETRY-0187

### Two-view reconstruction from conics and synchronized rational spline curves

S / solved | AIM 研讨会问题汇编

Can we use conic correspondence between two images to do reconstruction? Can we use point and curve correspondences to do reconstruction? Can we use cubic spline correspondences to do reconstruction?

来源字段：http://aimpl.org/algvision/1/

06404 | OWR-13101-004

### Ulrich Bundles on Smooth Projective Varieties

O / open | Oberwolfach 报告问题汇编

Does every smooth projective variety $X$ of dimension $n$ admit an Ulrich vector bundle? Equivalently, does its cone of cohomology tables coincide with that of $\mathbb P^n$?

来源字段：https://doi.org/10.4171/owr/2014/31

06405 | OWR-2654827-003

### Unbounded Non-Grid Geproci Point Configurations

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a family of non-grid point configurations in $\mathbb{P}^3$ with an unbounded number of points having the geproci property, and are there analogous examples in higher-dimensional projective spaces?

来源字段：https://doi.org/10.4171/owr/2020/31

06406 | OWR-17296-012

### Unbounded Semi Effective Orders on Projective Surfaces

O / open | Oberwolfach 报告问题汇编

Does there exist a smooth projective surface $X$ carrying primitive classes $F_n$ whose semi-effective orders $k_n$, defined as the least positive $k$ for which $kF_n$ is numerically equivalent to an effective divisor, tend to infinity? Is the analogous phenomenon possible when the effective divisors are irreducible curves?

来源字段：https://doi.org/10.4171/owr/2019/53

06407 | OWR-5148-006

### Undetermined Coefficients in Universal Residual Polynomials

P / partially_solved | Oberwolfach 报告问题汇编

For the Kleiman-Piene universal residual polynomials $R_n(u,c_1,c_2)$, determine the coefficients left undetermined by the proposed linear-system algorithm for $n\geq20$, and explain why the system ceases to have full rank.

来源字段：https://doi.org/10.4171/owr/2011/20

06408 | OWR-4990374-014

### Unexpected Curves from Plus-One Generated Arrangements

O / open | Oberwolfach 报告问题汇编

Can one find combinatorial conditions on plus-one generated line arrangements, analogous to those for nearly free arrangements, that imply that the dual point set admits unexpected curves?

来源字段：https://doi.org/10.4171/owr/2021/5

06409 | OWR-7155446-008

### Uniform Bounded Solutions of the d-Bar Equation

O / open | Oberwolfach 报告问题汇编

Given a bounded $(0,1)$-form $f$ on a domain $\Omega\subset\mathbb{C}^n$ with $\bar\partial f=0$, does there exist a bounded function $u$ such that $\bar\partial u=f$ and $\|u\|_\infty\leq C(\Omega)\|f\|_\infty$, where $C(\Omega)$ depends only on the domain?

来源字段：https://doi.org/10.4171/owr/2021/25

06410 | AIM-ARITHMETIC_GEOMETRY-0038

### Uniform finite-extension envelopes and valuation invariance

P / partially_solved | AIM 研讨会问题汇编

If \(u(F)<\infty\), does there exist \(B\in\mathbb N\) such that \(u(F')<\infty\) for all \(F'/F\) finite? In particular, is \(B=2u(F)\) sufficient?

来源字段：http://aimpl.org/deformationbrauer/1/

06411 | AIM-ALGEBRAIC_GEOMETRY-0129

### Uniform Frobenius-summand ranks for arithmetic curves

P / partially_solved | AIM 研讨会问题汇编

Characterize a variety \(X\) over \(\operatorname{Spec}\mathbb Z\) such that the indecomposable summands of \(F_*\mathcal O_X\) have universally bounded rank.

来源字段：http://aimpl.org/ratsubvarpos/9/

06412 | AIM-ALGEBRAIC_GEOMETRY-0262

### Uniform Fujita vanishing via flatness strata

S / solved | AIM 研讨会问题汇编

**Recovered question.** Let \(S\) be noetherian, let \(f:X\to S\) be projective (and, as in the source, possibly flat), let \(\mathcal F\) be coherent, and let \(\mathcal L\) be \(f\)-ample. Is there one \(m_0\), independent of \(s\) and of the global \(f\)-nef line bundle \(\mathcal M\), for which (1) holds for every \(m\geq m_0\)?

来源字段：http://aimpl.org/minimalmodcharp/4/

06413 | OWR-14222-010

### Uniform Integral Multiples of Zariski Decompositions

O / open | Oberwolfach 报告问题汇编

Does every projective surface $X$ admit an integer $m$, depending only on $X$, such that the Zariski decomposition of $mD$ is integral for every effective divisor $D$ on $X$?

来源字段：https://doi.org/10.4171/owr/2016/14

06414 | OWR-14298581-005

### Uniform Primary Brauer Bounds for K3 Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Let $K$ be a number field and $p$ a prime. Is there a uniform upper bound, depending only on $K$ and $p$, for $|\operatorname{Br}'(X)[p^\infty]|$ as $X$ ranges over all K3 surfaces over $K$?

来源字段：https://doi.org/10.4171/owr/2024/32

06415 | OWR-12975-012

### Uniform Sum-of-Squares Multipliers on Real Projective Curves

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be an irreducible nondegenerate real projective curve with dense real points, and fix an even degree $2s$. Find an explicit integer $k$, depending only on $s$ and invariants of $X$, such that every nonnegative degree-$2s$ form $p$ on $X$ has a sum-of-squares multiplier $q$ of degree $2k$ for which $pq$ is a sum of squares of degree $2s+2k$.

来源字段：https://doi.org/10.4171/owr/2014/17

06416 | OWR-1595-006

### Unimodular Coverings of Adjoint Toric Polytopes

O / open | Oberwolfach 报告问题汇编

Let $X$ be a smooth projective toric variety of dimension $d$, let $A$ be ample and $B$ nef, and set $D=(d+2)A+B+K_X$. Assuming that the lattice polytope associated with $D$ is normal, must it also have a unimodular covering?

来源字段：https://doi.org/10.4171/owr/2007/39

06417 | OWR-14299288-012

### Unimodular Placing Triangulations after Lattice Dilation

O / open | Oberwolfach 报告问题汇编

For every lattice polytope $P$, prove that there exist a positive integer $n$ and a total ordering of the lattice points of $nP$ for which the induced placing triangulation of $nP$ is unimodular.

来源字段：https://doi.org/10.4171/owr/2025/19

06418 | OWR-14746-006

### Unique Harmonic Metrizations of Line Bundles

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a projective variety over an algebraically closed, complete, nontrivially valued field, and let $\omega$ be a normalized positive form on $X$. Does every line bundle $M$ on $X$ admit a difference-of-semipositive metrization $\overline M$, unique up to a scalar multiple, such that $c_1(\overline M)$ is $\omega$-bounded and

$$c_1(\overline M)\,\omega^{n-1}=\lambda(M)\omega^n,$$

where $n=\dim X$ and $\lambda(M)=c_1(M)\cdot[\omega]^n$?

来源字段：https://doi.org/10.4171/owr/2016/38

06419 | OWR-12592-001

### Uniqueness and Existence of Link-Invariant Categorifications

O / open | Oberwolfach 报告问题汇编

Do categorifications of a given link invariant become unique, and does the Kauffman polynomial admit a categorification?

来源字段：https://doi.org/10.4171/owr/2013/44

06420 | OWR-3471-008

### Uniqueness in Linear Programs for Binomial Multiplier Ideals

S / solved | Oberwolfach 报告问题汇编

Let $k$ have characteristic zero and let $\mathfrak a\subseteq k[x_1,\ldots,x_n]$ be a binomial ideal containing no monomial, with minimal binomial generators $f_i=x^{a_i}-\gamma_i x^{b_i}$ for $1\leq i\leq r$. Put $$A=\begin{pmatrix}a_1&\cdots&a_r&b_1&\cdots&b_r\\ l_r&l_r\end{pmatrix}$$ and maximize $\sum_{i=1}^r(\mu_i+\nu_i)$ subject to $A(\mu_1,\ldots,\mu_r,\nu_1,\ldots,\nu_r)^{\mathsf T}\leq\mathbf1$ and $\mu_i,\nu_i\in\mathbb Q_{\geq0}$. Must there exist an optimal solution $(\mu,\nu)$ whose image $A(\mu,\nu)^{\mathsf T}$ differs from the image of every other optimal solution?

来源字段：https://doi.org/10.4171/owr/2009/21

06421 | AIM-ARITHMETIC_GEOMETRY-0013

### Uniqueness, counting, and a descent sieve for the degree-eight K3 partner

P / partially_solved | AIM 研讨会问题汇编

If you fix a degree 2 K3 surface over $\mathbb{Q}$ with a 2-torsion Brauer class $\alpha$ (over $\overline{\mathbb{Q}}$), van Geeman tells us that we can find a variety $Y$ such that $X$ is a moduli space of sheaves on $Y$ and such that $\alpha$ is the obstruction for the moduli space to be fine.

Is $Y$ unique? How many such $Y$ are there? If $X$ is defined over $\mathbb{Q}$ and $Y$ is defined over some nontrivial extension of $\mathbb{Q}$, is it possible that $\alpha$ is defined over $\mathbb{Q}$, i.e. in the image of $\operatorname{Br}(X)\to\operatorname{Br}(\overline X)$?

来源字段：http://aimpl.org/brauermoduli/1/

06422 | OWR-1730-011

### Uniruledness and Generic Nefness of Tangent Bundles

O / open | Oberwolfach 报告问题汇编

For a projective manifold $X$, if a maximal covering family of curves $C_t$ has $\Omega_X^1|_{C_t}$ non-nef for general $t$, must $X$ be uniruled? Also, if $-K_X$ is nef, is $T_X$ generically nef with respect to some or all polarizations, where generic nefness is tested on Mehta-Ramanathan-general complete-intersection curves?

来源字段：https://doi.org/10.4171/owr/2007/47

06423 | AIM-ALGEBRAIC_GEOMETRY-0025

### Unit propagation and a trace-Bockstein reduction for Greek-letter redshift

P / partially_solved | AIM 研讨会问题汇编

**Detecting Greek letter families in algebraic \(K\)-theory.** Prove Conjecture 1.1 of arXiv:1810.10088. This is a version of redshift for the Greek letter family in the homotopy groups of spheres: it asks that the \(K\)-theory of an \(\mathbb E_\infty\)-ring detecting the \(\alpha^{(n)}\)-family detects the \(\alpha^{(n+1)}\)-family.

**Conjecture 1.1 (Angelini-Knoll).** If \(R\) is a commutative ring spectrum that detects the \(n\)-th Greek-letter family, then \(K(R)\) detects the \((n+1)\)-st Greek-letter family.

来源字段：http://aimpl.org/equivstable/2/

06424 | OWR-15582-006

### Universal-Cover Decomposition for Nef Anticanonical Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

If $X$ is a compact Kähler manifold with nef anticanonical class, does its universal cover decompose as

$$\widetilde X\simeq\mathbb C^q\times\prod_jY_j\times\prod_kS_k\times Z,$$

where each $Y_j$ is an irreducible Calabi-Yau manifold, each $S_k$ is an irreducible hyperkähler manifold, and $Z$ is rationally connected?

来源字段：https://doi.org/10.4171/owr/2017/39

06425 | OWR-14299284-007

### Universal-Cover Decomposition of Klt Pairs with Nef Anticanonical Class

O / open | Oberwolfach 报告问题汇编

Describe the universal-cover decomposition of a klt pair with nef anticanonical class.

来源字段：https://doi.org/10.4171/owr/2025/28

06426 | OWR-1452-027

### Unramified Quotients of Equivariant Endomorphisms

O / open | Oberwolfach 报告问题汇编

Let $X$ be a normal affine $\mathbb{C}$-variety with an effective action of an algebraic group $G$, let $\varphi:X\to X$ be a $G$-equivariant unramified endomorphism, and let $\psi:X//G\to X//G$ be induced. Must $\psi$ be unramified?

来源字段：https://doi.org/10.4171/owr/2007/01

06427 | OWR-12495-006

### Unstable Motivic Homotopy Exact-Sequence Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For every $n\geq 4$, is there an exact sequence of strictly $\mathbb A^1$-invariant sheaves

$$K^M_{n+2}/24\longrightarrow \pi_n^{\mathbb A^1}(\mathbb A^n\setminus\{0\})\longrightarrow GW_{n+1}^n,$$

which becomes short exact after $n$-fold contraction?

来源字段：https://doi.org/10.4171/owr/2013/32

06428 | AIM-ALGEBRAIC_GEOMETRY-0282

### Upper semicontinuity, uniform Frobenius bounds, and a reduced non-equidimensional counterexample

P / partially_solved | AIM 研讨会问题汇编

Investigate the upper semicontinuity of the map \(\mathfrak p\in\operatorname{Spec}R\mapsto e_{HK}(R_{\mathfrak p})\).

来源字段：http://aimpl.org/testandmultiplierideals/1/

06429 | OWR-14298162-002

### Valuation Bounds When the Center Meets the Negative Part

O / open | Oberwolfach 报告问题汇编

Let $X$ be a smooth Fano threefold, let $P\in S\subset X$, and for $u\in[0,\tau]$ write the Zariski decomposition $-K_X-uS=P(u)+N(u)$. If $P\in\operatorname{Supp}N(u)$ for some $u$, find $K>0$ such that $\frac{3}{(-K_X)^3}\int_0^\tau P(u)^2\cdot S\cdot\operatorname{ord}_F(N(u)|_S)\,du\leq K A_S(F)$ for every prime divisor $F$ over $X$.

来源字段：https://doi.org/10.4171/owr/2024/14

06430 | OWR-1452-012

### Vanishing Conjecture for Powers of Homogeneous Polynomials

P / partially_solved | Oberwolfach 报告问题汇编

Let $\Delta=\sum_{i=1}^n\partial^2/\partial X_i^2$. For every homogeneous polynomial $P\in\mathbb{C}[X_1,\ldots,X_n]$ of degree $4$, does $\Delta^m(P^m)=0$ for all $m\geq1$ imply $\Delta^m(P^{m+1})=0$ for all sufficiently large $m$?

来源字段：https://doi.org/10.4171/owr/2007/01

06431 | OWR-13750339-004

### Vanishing Conjectures for Birational Cohomology of Projective Space

O / open | Oberwolfach 报告问题汇编

For every finite field $F$ of characteristic $p$ and every prime $\ell\neq p$, prove $H^{2i}(F_0\mathbb P_{\overline F}^n,\mathbb Q_\ell(i+1))^{G_F}=0$ and $H^{2i-1}(F_0\mathbb P_{\overline F}^n,\mathbb Q_\ell(i))^{G_F}=0$ for all $i,n\geq 2$.

来源字段：https://doi.org/10.4171/owr/2023/27

06432 | OWR-11790-006

### Vanishing Exceptional Curves in Class-VII Surface Degenerations

P / partially_solved | Oberwolfach 报告问题汇编

In the study of complex surfaces (such as minimal class $\mathrm{VII}$ surfaces with small $b_2$), consider a holomorphic deformation family $X \to D$ over the unit disk whose general fibers $X_t$ ($t \neq 0$) contain exceptional curves $E_{i,t}$, whereas the central fiber $X_0$ does not. What is the geometric limiting behavior of the exceptional curves $E_{i,t}$ as $t \to 0$ (for example, when their areas with respect to a family of Hermitian/Gauduchon metrics diverge to infinity), and why do they disappear in the central fiber?

来源字段：https://doi.org/10.4171/owr/2012/28

06433 | OWR-3384-003

### Vanishing Theorems for Complete Intersections in Homogeneous Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Can the vanishing theory for complete intersections in toric varieties be generalized to complete intersections in algebraic homogeneous spaces and their equivariant compactifications?

来源字段：https://doi.org/10.4171/owr/2009/01

06434 | OWR-14299288-003

### Vanishing Theorems for Divisors on Matroid Wonderful Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Let $W_L$ be the wonderful variety of a realizable matroid and let $D$ be a nef divisor on the ambient permutohedral variety. Prove that $H^i(W_L,\mathcal{O}_{W_L}(D|_{W_L}))=0$ for every $i>0$; if $D|_{W_L}$ is also big, prove that $H^i(W_L,\mathcal{O}_{W_L}(K_{W_L}+D|_{W_L}))=0$ for every $i>0$. Also determine whether $H^i(W_L,\mathcal{O}_{W_L}(-K_{W_L}-\partial W_L))=0$ for every $i<\dim W_L$.

来源字段：https://doi.org/10.4171/owr/2025/19

06435 | AIM-ARITHMETIC_GEOMETRY-0051

### Vertical Hilbert components and a Borel-fixed boundary reduction

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.** (1) Is there a component of \(\operatorname{Hilb}^d(\mathbb A^n)\) that exists only in characteristic \(p\) for some \(p\)? (2) Same question for the Hilbert schemes of curves in \(\mathbb P^3\).

来源字段：https://aimath.org/WWN/hilbertschemes/hilbertschemes.pdf

06436 | AIM-ALGEBRAIC_GEOMETRY-0215

### Vertical tangents and the cubic jumping wall for plane curves

P / partially_solved | AIM 研讨会问题汇编

The displayed map is \[ \pi:\mathcal M_{g,n}(\mathbf P^r,\beta)\longrightarrow \operatorname{Conf}(\mathbf P^r,n), \] and the question asks for the possible dimensions of its fibers and vertical tangent spaces, especially when the map is known to be dominant.

There is a genuine notational inconsistency in the PDF: the sentence before the display says that the curve lies in \(\mathbf P^n\), whereas the display, the degree \(d\), and the rest of the problem use the ambient dimension \(r\). I therefore recover the intended ambient space as \(\mathbf P^r\) and write \(\beta=d[\text{line}]\). Since the marked points are ordered, I interpret \(\operatorname{Conf}(\mathbf P^r,n)\) as the open subset of \((\mathbf P^r)^n\) consisting of ordered pairwise-distinct points. Passing to an unordered configuration space is a finite étale quotient away from coincident points and does not change the local tangent calculation.

The source does not specify a compactification or a particular component. The dominance papers cited below use the Brill--Noether component of the stable-map space. On this locus the stable-map moduli problem is locally the incidence problem for embedded curves and marked points. No assertion below extends without modification to nodal maps, ramified maps, multiple covers, or contracted components.

来源字段：https://aimath.org/pastworkshops/degenalggeomproblems.pdf

06437 | AIM-ARITHMETIC_GEOMETRY-0036

### Very general complex ppav's have no Chow-theoretic half theta square

S / solved | AIM 研讨会问题汇编

Let $X$ be a principally polarized abelian variety of dimension $\geq 4$. Is $\theta^2\in CH^2(X)/\operatorname{tors}$ divisible by $2$?

来源字段：http://aimpl.org/cohomabelian/1/

06438 | AIM-ALGEBRAIC_GEOMETRY-0145

### Very moving cycle classes: Chow-family and positive-form reductions

P / partially_solved | AIM 研讨会问题汇编

Let \(X\) be a projective variety. Suppose that there is a class \(\alpha\in \operatorname{Eff}_k(X)_{\mathbb Z}\) such that for a general point \(x\in X\) and a general \(k\)-plane \(V\subset T_xX\) there is an irreducible subvariety \(Y\subset X\) with \([V]=\alpha\) and \(T_xY=V\).

Is \(\alpha\) in the interior of the effective cone?

**Recovered problem.** Let \(X\) be a smooth irreducible projective complex variety of dimension \(n\), and let \(\alpha\in N_k(X)_{\mathbb Z}\) be an effective integral numerical class. Assume that for a general pair \[ (x,V)\in G:=\operatorname{Gr}_X(k,T_X) \] there is an irreducible \(k\)-dimensional subvariety \(Y\subset X\) such that \[ [Y]=\alpha,\qquad x\in Y_{\mathrm{reg}},\qquad T_xY=V. \tag{VM} \] Must \(\alpha\) lie in \(\operatorname{int}\overline{\operatorname{Eff}}_k(X)\)?

来源字段：http://aimpl.org/poscycles/2/

06439 | AMR-071-0020

### Virasoro conjecture

P / partially_solved | AMR 问题清单汇编

Virasoro conjecture: a certain generating function encoding the Gromov-Witten invariants of a smooth projective variety is fixed by an action of half of the Virasoro algebra.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

06440 | AIM-ALGEBRAIC_GEOMETRY-0337

### Virtual classes for KSBA stable-surface moduli

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.3. (Hassett)** Does the moduli space of stable surfaces have a virtual fundamental class?

来源字段：https://aimath.org/WWN/birational/birational.pdf

06441 | OWR-16408-015

### Virtual Cohomological Dimension of Surface Automorphism Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a Coble surface, an Enriques surface, or an elliptic K3 surface $S$, is

$$\operatorname{vcd}(\operatorname{Aut}S)=\max_f\operatorname{MW-rk}(f),$$

where the maximum ranges over genus-one fibrations $f:S\to\mathbb P^1$ and $\operatorname{MW-rk}(f)$ is the Mordell-Weil rank of the Jacobian fibration?

来源字段：https://doi.org/10.4171/owr/2018/33

06442 | OWR-13750327-011

### Virtual Nilpotence of Rational Log-Canonical Local Groups

O / open | Oberwolfach 报告问题汇编

For every rational log-canonical singularity $(X;x)$, is the local fundamental group $\pi_1(X;x)$ virtually nilpotent?

来源字段：https://doi.org/10.4171/owr/2023/18

06443 | OWR-4527-006

### Volumes of Line Bundles as Periods

O / open | Oberwolfach 报告问题汇编

Is the volume of a line bundle on every smooth projective variety always a period?

来源字段：https://doi.org/10.4171/owr/2010/45

06444 | OWR-13946-013

### Wahl's Conjecture for Nodal Curves

P / partially_solved | Oberwolfach 报告问题汇编

Can Wahl's conjecture for smooth curves and the Arbarello-Bruno-Sernesi theorem be extended to nodal curves using the Kemeny-Wahl map $kw_C$, the Halic-Wahl map $hw_C$, or both?

来源字段：https://doi.org/10.4171/owr/2015/51

06445 | OWR-17296-015

### Waldschmidt Constants of Plane Curve Singular Sets

O / open | Oberwolfach 报告问题汇编

For a reduced plane curve $C$ with singular set $Z$, must the Waldschmidt constant of the ideal $I_Z$ be rational, and can it be computed, for example for Klein's arrangement of $21$ lines?

来源字段：https://doi.org/10.4171/owr/2019/53

06446 | OWR-17473-001

### Wall Crossing Between Adjacent Tropical Toric Degenerations

O / open | Oberwolfach 报告问题汇编

Given two adjacent maximal-dimensional prime cones in a tropicalization sharing a facet, how are the two associated toric degenerations related, including their wall-crossing maps and semigroups?

来源字段：https://doi.org/10.4171/owr/2020/10

06447 | AIM-ALGEBRAIC_GEOMETRY-0130

### Weak finite-set interpolation on unirational surfaces

P / partially_solved | AIM 研讨会问题汇编

If \(X\) is separably rationally connected, there exists \(N(\dim X,\deg X,m)\) such that, for any \(p_1,\ldots,p_m\in X(\mathbb F_p)\), there exists a rational curve \(C\) defined over \(\mathbb F_{p^N}\) containing all \(p_i\). (due to Kollar)

Can you do this without lifting the cardinality of field tend to \(\infty\)?

Let \(X\) be a cubic surface over \(\mathbb F_q\). Can you find a rational curve defined over \(\mathbb F_q\) through each point?

来源字段：http://aimpl.org/ratsubvarpos/9/

06448 | AIM-ALGEBRAIC_GEOMETRY-0227

### Weak non-bigness versus strict Nagata

P / partially_solved | AIM 研讨会问题汇编

**7 Problem 7** *(Suggested by Omid Amini)* Here is a “weak version” of Nagata’s conjecture. **Problem 7.1.** When Nagata predicts \(h^0=0\), show that \[ dL-m_1E_1-\cdots-m_sE_s \] is not big by using “convex methods.” **Remark 7.2.** This might actually imply the strong form of Nagata’s conjecture.

来源字段：https://aimath.org/pastworkshops/degenalggeomproblems.pdf

06449 | OWR-4127-002

### Weak Zariski Decompositions of Pseudo-Effective Divisors

S / solved | Oberwolfach 报告问题汇编

Does every pseudo-effective relative $\mathbb R$-Cartier divisor $D$ on a normal variety $X/Z$ admit a weak Zariski decomposition over $Z$?

来源字段：https://doi.org/10.4171/owr/2009/44

06450 | OWR-11127-019

### Weighted-Projective Moduli of Marked Elliptic Curves

O / open | Oberwolfach 报告问题汇编

How close are we to classifying all pairs $(G,M)$ for which the moduli space of marked elliptic curves is a weighted projective space, or a complete intersection in weighted projective space, and what further applications do these moduli spaces have?

来源字段：https://doi.org/10.4171/owr/2011/35

06451 | OWR-14298376-004

### Welschinger Specialization of Generalized Block-Göttsche Polynomials

P / partially_solved | Oberwolfach 报告问题汇编

For a surface $S$ and curve class $\beta_S$, does the generalized Block-Göttsche polynomial specialize at $q=-1$ to the real Welschinger invariant, so that $\operatorname{BPS}_{\beta_S}^S(-1)=W_{\beta_S}^S$?

来源字段：https://doi.org/10.4171/owr/2024/23

06452 | AIM-ALGEBRAIC_GEOMETRY-0014

### What diagonal and toric-correspondence transforms remember

P / partially_solved | AIM 研讨会问题汇编

What does the Fourier-Mukai transform by either resolution of $\Delta$ applied to $M$ tell us about $M$? More generally what does applying Fourier-Mukai transforms of resolutions of toric subvarieties tell us about the modules apply them to? [e.g. in the spirit of the way we obtain Eagon-Northcott-type complexes for modules]

The archive of the preceding section also resolves an important ambiguity: its Problem 4.1 explicitly calls the two constructions the **Hanlon--Hicks--Lazarev (HHL)** and **Brown--Erman (BE)** resolutions. Thus “either resolution” means those two diagonal resolutions, rather than the later Anderson construction.

The record does not define $X$ or $M$. In the workshop context, the intended setup is a smooth projective toric variety $X$, its Cox ring $S$ with irrelevant ideal $B$, and a finitely generated $\operatorname{Pic}(X)$-graded $S$-module $M$. Write $\widetilde M$ for its associated coherent sheaf. The intrinsic Fourier--Mukai statement concerns $\widetilde M$; a free Cox complex obtained by applying $\Gamma_*$ is additional, presentation-sensitive data. This distinction is essential.

来源字段：http://aimpl.org/syzygyms/5/

06453 | OWR-1703867-003

### Whittaker-Normalized Functors for Reductive Groups

P / partially_solved | Oberwolfach 报告问题汇编

For every reductive triple $(G,B,T)$ with Whittaker datum $\psi$, does there exist a fully faithful $Z_G$-linear functor $$ R_G^\psi:D^+(\operatorname{Rep}_{[T,1]}G)\longrightarrow D^+\operatorname{QCoh}([\mathcal X_{\check G}/\check G]) $$ that is compatible with parabolic induction for every $B\subset P\subset G$, agrees with the canonical equivalence in the torus case, and sends the compactly induced Whittaker object to the structure sheaf?

来源字段：https://doi.org/10.4171/owr/2020/20

06454 | AIM-ALGEBRAIC_GEOMETRY-0183

### Why six trifocal point triplets have algebraic degree three

S / solved | AIM 研讨会问题汇编

It is known that the trifocal tensor variety is dimension 18 with degree 297 inside of \(\mathbb P^{28}\). It is also known that a six point correspondence in three views is a minimal problem.

How do the 24 linear constraints lead a minimal problem? How many complex solutions are there?

来源字段：http://aimpl.org/algvision/1/

06455 | AMR-071-0013

### Wikipedia geometry item 13: Nagata's conjecture on curves, specifically the minimal degree required for a plane algebraic c…

O / open | AMR 问题清单汇编

Nagata's conjecture on curves, specifically the minimal degree required for a plane algebraic curve to pass through a collection of very general points with prescribed multiplicities.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

06456 | AMR-071-0014

### Wikipedia geometry item 14: Nagata-Biran conjecture that if $X$ is a smooth algebraic surface and $L$ is an ample line bund…

O / open | AMR 问题清单汇编

Nagata-Biran conjecture that if $X$ is a smooth algebraic surface and $L$ is an ample line bundle on $X$ of degree $d$, then for sufficiently large $r$, the Seshadri constant satisfies $\varepsilon(p_1,\ldots,p_r;X,L) = d/\sqrt{r}$.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

06457 | AMR-071-0017

### Wikipedia geometry item 17: Section conjecture on splittings of group homomorphisms from fundamental groups of complete smo…

O / open | AMR 问题清单汇编

Section conjecture on splittings of group homomorphisms from fundamental groups of complete smooth curves over finitely-generated fields $k$ to the Galois group of $k$.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

06458 | AMR-071-0007

### Wikipedia geometry item 7: Fujita conjecture regarding the line bundle $K_{M} \otimes L^{\otimes m}$ constructed from a po…

O / open | AMR 问题清单汇编

Fujita conjecture regarding the line bundle $K_{M} \otimes L^{\otimes m}$ constructed from a positive holomorphic line bundle $L$ on a compact complex manifold $M$ and the canonical line bundle $K_{M}$ of $M$

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

06459 | OWR-12490-008

### Wild McKay Identity for Crepant Resolutions

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a quotient by a faithful linear action of a finite group $\Gamma$, and let $Y\to X$ be a crepant resolution with exceptional fiber $E$ over the origin. Is the wild McKay identity $\#E(\kappa)=M(K,\Gamma,-w)$ valid?

来源字段：https://doi.org/10.4171/owr/2013/27

06460 | AIM-ALGEBRAIC_GEOMETRY-0119

### Wild-degree supersingular Klein-loop hypersurfaces

P / partially_solved | AIM 研讨会问题汇编

**Problem 9.2.** Is there a supersingular hypersurface of every degree, dimension and char. \(p\)?

Fix \(n,d,q\). How many supersingular hypersurfaces \(X\subset \mathbf P^n\) exist of degree \(d\) over \(\mathbf F_q\)? In particular, does such a hypersurface exist for every degree, dimension, and characteristic?

来源字段：http://aimpl.org/ratsubvarpos/9/

06461 | OWR-14222-012

### Wiseman-Wilson Theorem for Complex Conics

O / open | Oberwolfach 报告问题汇编

Let $P\subset\mathbb{P}^2(\mathbb{C})$ be a finite point set not contained in a conic. Must there be a conic determined by exactly five points of $P$ and containing no other point of $P$, as in the real Wiseman-Wilson theorem?

来源字段：https://doi.org/10.4171/owr/2016/14

06462 | AIM-ALGEBRAIC_GEOMETRY-0068

### Witt-index and odd-degree criteria for intersections of two quadrics

P / partially_solved | AIM 研讨会问题汇编

Hassett and Tschinkel showed that a complete intersection of two quadrics in \(\mathbb P^5\) over an arbitrary field \(k\) is rational if and only if it contains a line over \(k\).

Give criteria for other geometrically rational Fano \(3\)-folds to be rational over a field \(k\).

Give criteria for the complete intersection of two quadrics in \(\mathbb P^n\) to be rational over \(k\).

来源字段：http://aimpl.org/rationalityag/1/

06463 | AIM-ALGEBRAIC_GEOMETRY-0268

### Witt-slope congruences, global F-regularity, and Fano cone obstructions

P / partially_solved | AIM 研讨会问题汇编

Let \(X/\mathbb F_q\) be a normal, projective, geometrically connected variety. Which local or global singularity/positivity hypotheses force \(\#X(\mathbb F_{q^r})\equiv1\pmod{q^r}\), a rational point, or a quantitative error bound, and what fails without those hypotheses?

来源字段：http://aimpl.org/minimalmodcharp/5/

06464 | OWR-2043-008

### Yau Stability Conjecture for Constant Scalar Curvature Metrics

P / partially_solved | Oberwolfach 报告问题汇编

Let $(X,L)$ be a polarized compact Kähler manifold with Kähler class $[\omega_0]$. Is $[\omega_0]$ represented by a Kähler metric of constant scalar curvature if and only if the polarization $L\to X$ is stable in the sense of geometric invariant theory?

来源字段：https://doi.org/10.4171/owr/2008/32

06465 | OWR-13109-010

### Yau-Tian-Donaldson Conjecture for Polarized Varieties

O / open | Oberwolfach 报告问题汇编

Let $(X,L)$ be a smooth polarized complex projective variety with finite automorphism group. Does $c_1(L)$ contain a constant-scalar-curvature Kähler metric if and only if $(X,L)$ is K-stable?

来源字段：https://doi.org/10.4171/owr/2014/39

06466 | OWR-16941-010

### Zariski Descent for the Milnor-Witt Rost-Schmid Complex

O / open | Oberwolfach 报告问题汇编

Does the Bloch-Levine-Rost-Schmid complex $C^*(-,K_q^{MW},q)$ satisfy Zariski descent?

来源字段：https://doi.org/10.4171/owr/2019/29

06467 | AMR-071-0021

### Zariski multiplicity conjecture on the topological equisingularity and equimultiplicity of varieties at singular points

O / open | AMR 问题清单汇编

Zariski multiplicity conjecture on the topological equisingularity and equimultiplicity of varieties at singular points

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

06468 | OWR-8415350-011

### Zariski's Multiplicity Conjecture

O / open | Oberwolfach 报告问题汇编

Let $f,g:(\mathbb C^n,0)\to(\mathbb C,0)$ be reduced complex analytic functions. If there is a homeomorphism $\phi:(\mathbb C^n,0)\to(\mathbb C^n,0)$ with $\phi(V(f))=V(g)$, must $m(V(f),0)=m(V(g),0)$?

来源字段：https://doi.org/10.4171/owr/2021/47

06469 | OWR-14613-005

### Zariski-Nisnevich Comparison for Relative Milnor K-Theory

O / open | Oberwolfach 报告问题汇编

Let $X$ be a smooth $d$-dimensional variety over a perfect field of characteristic $p$, and let $Y\hookrightarrow X$ be a normal-crossing divisor with infinitesimal thickenings $Y_s$. Is the canonical map $\lim_s H^i_{\mathrm{Zar}}(X,K_n^M(X,Y_s))\to\lim_s H^i_{\mathrm{Nis}}(X,K_n^M(X,Y_s))$ an isomorphism, in particular over a finite base field when $i=n=d$, as conjectured by Kato and Saito?

来源字段：https://doi.org/10.4171/owr/2016/31

06470 | OWR-12975-007

### Zero Thresholds for Positive Semidefinite Ternary Forms

S / solved | Oberwolfach 报告问题汇编

Let $\alpha(k)$ be the least integer such that every positive semidefinite ternary form $p$ of degree $2k$ with more than $\alpha(k)$ projective zeros can be written $p=h^2q$ with $h$ indefinite. Is $\alpha(k)\geq k^2+1$ for every $k\geq3$?

来源字段：https://doi.org/10.4171/owr/2014/17

06471 | OWR-13750327-006

### Zero-Free One-Forms on Compact Kähler Varieties

O / open | Oberwolfach 报告问题汇编

Does the threefold classification for smooth projective varieties whose underlying differentiable manifolds fiber over the circle extend to compact Kähler threefolds? More generally, does every smooth complex projective variety of nonnegative Kodaira dimension with a nowhere-vanishing holomorphic one-form have a birational model admitting a finite étale cover with a positive-dimensional abelian variety as a factor?

来源字段：https://doi.org/10.4171/owr/2023/18

06472 | OWR-14299574-004

### Zero-Mutable Laurent Polynomials as Cluster Variables

O / open | Oberwolfach 报告问题汇编

Is every zero-mutable Laurent polynomial in two variables a cluster variable for a suitable quiver?

来源字段：https://doi.org/10.4171/owr/2025/45

06473 | OWR-10252929-008

### Zilber-Pink for Powers of the Modular Curve

P / partially_solved | Oberwolfach 报告问题汇编

别名：Zilber-Pink Unlikely Intersection Conjecture [OWR-10252929-001]

For an irreducible Hodge-generic subvariety $V\subset Y(1)^n$, is the union of the atypical components of intersections of $V$ with proper special subvarieties non-Zariski-dense in $V$?

来源字段：https://doi.org/10.4171/owr/2022/21

06474 | OWR-5158-008

### Zucker-Type Cohomology for Satake-Compactified Curve Moduli

O / open | Oberwolfach 报告问题汇编

For the moduli space $M_g$, with the Siegel metric pulled back by the Torelli map, and its Satake compactification $M_g^*$, does the analogue of Zucker's conjecture hold, namely that its $L^2$-cohomology is the middle-perversity intersection cohomology of $M_g^*$?

来源字段：https://doi.org/10.4171/owr/2011/30

# 资料汇编 05 | 几何 / Geometry

1,854 条记录。O=开放；P=部分解决；S=已解决。状态与英文/LaTeX 表述均沿用来源。

06475 | AMR-108-0001

### 1.1 (Agol) - Strictly convex projective manifolds and cubulation

O / open | AMR 问题清单汇编

If $M^n$ is a closed manifold with a strictly convex projective structure, is it cubulated?

背景链接：https://arxiv.org/pdf/1512.04620#page=1

06476 | AMR-108-0002

### 1.2 (Choi) - Convex projective deformations from a CR structure

O / open | AMR 问题清单汇编

Suppose a hyperbolic $3$-manifold $M$ admits a CR structure, not necessarily a spherical one. Can the deformation theory of convex real projective structures on $M$ be understood in terms of the CR structure? The source suggests hyperbolic Coxeter $3$-orbifolds as a potentially easier case.

背景链接：https://arxiv.org/pdf/1512.04620#page=1

06477 | AMR-108-0003

### 1.3 (Cooper) - Convexity of projective structures on hyperbolic 3-manifolds

O / open | AMR 问题清单汇编

If $M$ is a closed hyperbolic $3$-manifold, is every projective structure on $M$ convex?

背景链接：https://arxiv.org/pdf/1512.04620#page=1

06478 | AMR-108-0004

### 1.4 (Danciger) - Convex projective structures on glued figure-eight complements

P / partially_solved | AMR 问题清单汇编

Let $N$ be the closed $3$-manifold obtained by gluing two copies of the figure-eight knot complement along their torus boundaries by a homeomorphism. Does $N$ admit a convex projective structure?

背景链接：https://arxiv.org/pdf/1512.04620#page=1

06479 | AMR-108-0005

### 1.5 (Danciger) - Convex projective structures and hyperbolic JSJ pieces

P / partially_solved | AMR 问题清单汇编

Let $N$ be a closed $3$-manifold whose JSJ decomposition contains only hyperbolic pieces. Does $N$ admit a convex projective structure?

背景链接：https://arxiv.org/pdf/1512.04620#page=1

06480 | AMR-108-0006

### 2.1 (Leitner) - Limits between Thurston geometries

P / partially_solved | AMR 问题清单汇编

Geometric transitions are continuous paths of geometries that abruptly change type in the limit. Understand all transitions between the eight Thurston geometries. More generally, how can one tell when one geometry is a limit of another, and what properties must a limiting geometry satisfy?

背景链接：https://arxiv.org/pdf/1512.04620#page=2

06481 | AMR-054-0011

### 3SUM Hard Problems

P / partially_solved | AMR 问题清单汇编

Can the class of 3SUM hard problems be solved in subquadratic time? These problems can be reduced from the problem of determining whether, given three sets of integers, $A$, $B$, and $C$ with total size $n$, there are elements $a \in A$, $b \in B$, and $c \in C$ such that $a+b=c$.

背景链接：https://topp.openproblem.net/p11

06482 | AMR-108-0032

### 5.7 (Agol) - Renormalized volume as a metric

S / solved | AMR 问题清单汇编

The renormalized volume of quasi-Fuchsian groups gives a function $\rho:\mathcal{T}(S)\times\mathcal{T}(S)\to\mathbb{R}$. Is $\rho$ a metric on the Teichmüller space $\mathcal{T}(S)$?

背景链接：https://arxiv.org/pdf/1512.04620#page=4

06483 | AMR-108-0034

### 5.9 (Cooper) - Thurston's Lego sets in dimensions at least four

O / open | AMR 问题清单汇编

Given $R>0$ and an integer $n\geq4$, is there an $\varepsilon>0$ and a finite set of hyperbolic $n$-simplices such that every closed cone $n$-manifold obtained by gluing these simplices with all codimension-two cone angles in $(2\pi-\varepsilon,2\pi+\varepsilon)$ admits a hyperbolic metric, and every closed hyperbolic $n$-manifold with injectivity radius everywhere greater than $R$ is obtained this way?

背景链接：https://arxiv.org/pdf/1512.04620#page=4

06484 | AMR-108-0047

### 6.3 (Maher) - Structure behind Rivin's experimental regularity

O / open | AMR 问题清单汇编

Rivin's experimental results appear extremely regular, possibly indicating additional structure. Investigate this phenomenon.

背景链接：https://arxiv.org/pdf/1512.04620#page=5

06485 | AIM-GEOMETRY-0337

### A 20-dimensional holonomy shadow of the tropical K3 structure space

P / partially_solved | AIM 研讨会问题汇编

**What can one say about the geometry or topology of the set of tropical Calabi--Yau structures on \(S^2\)?**

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

06486 | AIM-GEOMETRY-0356

### A Bach-tensor certificate for information transverse to conformal classes

P / partially_solved | AIM 研讨会问题汇编

Problem 1a: Given any functional of the metric that is well understood conformally, is there information that can arise going across conformal classes?

Problem 1c: Study variational problems arising from conformally invariant problems.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

06487 | AIM-GEOMETRY-0067

### A BCFT shortcut on equilibrium slopes and an obstruction at infinity

P / partially_solved | AIM 研讨会问题汇编

Can one find a simplified proof of Burns-Gelfert in Burns--Gelfert (*Discrete Contin. Dyn. Syst.* 34, 2014, MR3124716, on the Lyapunov spectrum and thermodynamics of geodesic flows of rank-$1$ nonpositively curved surfaces) using Burns--Climenhaga--Fisher--Thompson (*Geom. Funct. Anal.* 28, 2018, MR3856792, on unique equilibrium states for geodesic flows in nonpositive curvature under the pressure-gap condition $P_{\mathrm{sing}}(\varphi) < P(\varphi)$), possibly with stronger conclusions? Similarly for the results of Paulin-Pollicott-Schapira in Paulin--Pollicott--Schapira (*Astérisque* 373, 2015, MR3444431, on equilibrium states in negative curvature).

来源字段：http://aimpl.org/equibdynsysgeom/4/

06488 | AIM-GEOMETRY-0217

### A Bezout obstruction to the printed divisibility model

O / open | AIM 研讨会问题汇编

It also shows that the page header and all the subsequent text beginning “4. Curvature conditions and hyperbolicity of the geodesic flow” belong to the next section, not to Conjecture 3.2.2.

The immediately preceding paragraph is indispensable context. It defines \[ \alpha_{g,G}=\lim_{\deg Y\to\infty}\frac{\deg Y}{\operatorname{cr}(Y)} \] for nonzero free-loop homology classes over a field \(G\), and then supposes, as a “little example,” that for a generator \(X\) of an integral rank-one homology group there are \(a<b<c<d\) such that \[ \operatorname{cr}(mX)= \begin{cases} d,&(m,3)=(m,7)=1,\\ c,&3\mid m,\ (m,7)=1,\\ b,&7\mid m,\ (m,3)=1,\\ a,&21\mid m. \end{cases} \tag{1} \] The displayed “conjecture” then gives the proposed coefficient pattern \[ \alpha_{g,\mathbb Q}=a,\qquad \alpha_{g,\mathbb F_3}=b,\qquad \alpha_{g,\mathbb F_7}=c,\qquad \alpha_{g,\mathbb F_p}=a\quad(p\ne3,7). \tag{2} \]

As printed, (2) is not a quantified proposition: it specifies neither a metric nor an existence or universal quantifier, and its constants come only from the hypothetical (1). Moreover, (1) is algebraically impossible. Thus this record cannot conservatively be classified as an open conjecture with a determinate truth value.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06489 | AIM-GEOMETRY-0037

### A boundary-homology certificate for quasi-cluster automorphisms

P / partially_solved | AIM 研讨会问题汇编

Given some automorphism of the braid variety, does symplectic geometry help inform if it's a quasi cluster automorphism?

来源字段：http://aimpl.org/clusterbraid/8/

06490 | AIM-GEOMETRY-0289

### A boundary-transfer obstruction for 3-Sasakian Kirwan surjectivity

P / partially_solved | AIM 研讨会问题汇编

G

Conjecture 2.1 (C. Boyer) Let S be a complete compact 3-Sasakian manifold with a con-nected compact Lie group G ⊂ Aut (S) acting on S. Consider the 3-Sasakian reduction Sred

(reduction at 0). Then the 3-Sasakian Kirwan map

(S) → H∗(Sred )

is surjective up to the middle dimension. Here we take either Q or R coefficients.

Comment 2.2 (C. Boyer) We know the conjecture is true for S1-reductions of spheres. Note also that in odd dimension, "up to the middle dimension" is all we can hope for. Boyer, Galicki, and Piccinni [BGP02] have constructed some examples of 3-Sasakian quotients for which they don't know how to compute cohomology, and Kirwan surjectivity would be a nice way to get at some cohomology.

Comment 2.3 (C. Boyer and T. Hausel) We might need some additional hypotheses for the nonabelian case, but T. Hausel has some ideas about how to prove something like this for abelian reductions. Namely, the surjectivity in the 3-Sasakian case should be related to surjectivity for hyperK¨ ahler quotients. 3We consider a more specific example. Suppose there is an action of Sp (n + 1) acting on S4n+3. Restrict to an action of a compact subgroup G of this Sp (n + 1). Consider the 3-Sasakian reduction S4n+3 ///G, and additionally assume that this is orbifold. We conjecture that, in this case, the 3-Sasakian Kirwan map is surjective to the middle dimension of the quotient. Here is a tentative idea of the proof. Look at the correponding hyperK¨ ahler quotient. Consider the same group G acting now on Hn+1, which containes S4n+3 as its sphere. Let X

denote now the hyperK¨ ahler quotient. The space X might be singular since G might not have a center. But now suppose there exists some ξ in the center of G so that the hyperK¨ ahler quotient is orbifold. Assume also that the quotient is hypercompact (see Definition 6.9), so the core is middle-dimensional. Then, in this case, the surjectivity of the hyperK¨ ahler Kirwan map should imply the surjectivity (up to middle dimension) of the 3-Sasakian Kirwan map. T. Hausel thinks this should be an elementary argument. C. Boyer and T. Hausel will be pursuing this line of thought after the AIM Workshop ends.

Let \(S\) be a compact complete 3-Sasakian manifold, let a connected compact Lie group \(G\subset\operatorname{Aut}(S)\) act, and let \(S_{\mathrm{red}}\) be the 3-Sasakian reduction at zero. The conjecture is that \[ \kappa_{3S}\colon H_G^*(S;K)\longrightarrow H^*(S_{\mathrm{red}};K),\qquad K=\mathbb Q\ \text{or}\ \mathbb R, \] is surjective up to the middle dimension. Comment 2.3 specializes to \(S=S^{4n+3}\subset\mathbb H^{n+1}\), \(G\subset Sp(n+1)\), and proposes passing through a nonzero central-level hyperkähler quotient having a middle-dimensional core.

If \(L=S^{4n+3}///G\) has dimension \(4m-1\), this report uses “through the lower middle” to mean degrees \[ q\leq \left\lfloor\frac{4m-1}{2}\right\rfloor=2m-1. \] The phrase in the source does not separately specify whether its endpoint is inclusive; the result below isolates exactly what happens in that endpoint degree.

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

06491 | AIM-GEOMETRY-0042

### A canonical Poisson compactification and a boundary test for all-braid correspondences

P / partially_solved | AIM 研讨会问题汇编

Braid varieties are compactified by Brick manifolds. The strata here relate to positive braids.

Is there a compactification of the cotangent bundle of Bott-Samelson varieties or the cotangent bundle of Brick manifolds related to all braids?

来源字段：http://aimpl.org/clusterbraid/10/

06492 | AIM-GEOMETRY-0178

### A centered product-torus obstruction to spectral improvement by mean curvature flow

P / partially_solved | AIM 研讨会问题汇编

Is there a flow related to Skeklov eigenvalue maximization? For closed surfaces in $S^3$, does Mean Curvature Flow improve $\lambda_1 \cdot Area$? What about balanced flow that keeps the center of mass?

来源字段：http://aimpl.org/flowriemannian/5/

06493 | AIM-GEOMETRY-0096

### A certified autocorrelation-SDP formulation of the hyperbolic packing LP bound

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.4.** Numerical computation of LP bound in $\mathbb H^n$.

来源字段：http://aimpl.org/discreteaf/1/

06494 | AIM-GEOMETRY-0435

### A cohomological target and periodic certificates for contact dissipation

P / partially_solved | AIM 研讨会问题汇编

Let \((M,\xi)\) be a closed contact manifold and let \(\varphi\in\operatorname{Cont}(M,\xi)\). Suppose that the induced action on contact homology \(CH(M,\xi)\) is hyperbolic, in a sense still to be defined. Then \(\varphi\) is strongly dissipative: there is no smooth \(\varphi\)-invariant volume form.

The bracketed comments and, in particular, the warning that “hyperbolic” still has to be formulated are in the source itself.

The recovered conjecture is:

The source then places the conjecture in the “functor” from smooth topology to contact topology: a manifold gives its unit cotangent bundle, a submanifold gives its unit conormal, and a diffeomorphism gives a lifted contactomorphism. The question asks how much smooth information survives these constructions.

Throughout the mathematical results below, \(M^{2n+1}\) is closed and \(\xi\) is cooriented, \(\alpha\) is a positive contact form with \(\ker\alpha=\xi\), and \(\varphi\) preserves the coorientation. This is the standard setting for the contact-homology algebra and for Polterovich's paper. For a coorientation-reversing map, an invariant volume for \(\varphi\) would also be invariant under \(\varphi^2\), so obstructions may instead be applied to the square.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

06495 | AIM-GEOMETRY-0305

### A collision-stable global-residue reduction for the Higgs-moduli Bethe formula

P / partially_solved | AIM 研讨会问题汇编

For the fixed-determinant, trace-free Higgs moduli space \(\mathcal M^d_{\mathrm{Dol}}(SL(n,\mathbb C))\), its circle-equivariant volume \[ \oint_{\mathcal M}e^\alpha :=\sum_{F\subset \mathcal M^{\mathbb C^*}} \int_F\frac{e^{\alpha|_F}}{e_{\mathbb C^*}(N_F)}, \] and, more generally, all equivariant intersection numbers, admit an iterated-residue expression \(BA(y_1,\ldots,y_n)\). Its poles divide into poles indexed by ordered partitions \(n=\lambda_1+\cdots+\lambda_k\) and poles satisfying Bethe-Ansatz equations. The residue at a partition pole equals the localization contribution of the corresponding fixed component, while minus the sum of the Bethe residues equals the equivariant volume.

Here the curve is a smooth projective complex curve of genus \(g\), the Higgs field is scaled by \(\mathbb C^*\), and one normally assumes \(\gcd(n,d)=1\) to stay in the smooth stable setting. The source calls the first quantity an integral even though the space is noncompact; it is the formal equivariant integral defined by its fixed-point sum.

The database field accidentally continues past the heading “6.5 Arithmetic approach” and includes Theorem 6.17 and later definitions. Comparison with the source PDF shows that this material is not part of Conjecture 6.16 and is not treated here. The source also writes “Nekrasov--Shatasvili--Moore”; the primary paper and later references use **Moore--Nekrasov--Shatashvili**.

The AIM statement does not display the higher-rank function \(BA\), specify an order for the iterated residues, state a contour/summation prescription for the generally infinite Bethe set, or give a compactification and its boundary terms. Those omissions matter mathematically.

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

06496 | AIM-GEOMETRY-0108

### A compact finite-shell reduction for lattice soft packings

P / partially_solved | AIM 研讨会问题汇编

**Lattice Packings**

It is know that the problem of finding the densest lattice packing of spheres in a fixed dimension is a solvable by a finite algorithm (Voronoi, Minkowski).

Can the lattice soft ball packing problem be solved by a finite algorithm? Amongst lattice packings of unit balls, which maximizes the density with respect to soft balls?

来源字段：http://aimpl.org/softpack/1/

06497 | AIM-GEOMETRY-0274

### A compact flat counterexample from odd holonomy

P / partially_solved | AIM 研讨会问题汇编

Suppose a closed manifold \((M,g)\) admits a nonzero \((1,1)\)-tensor field that is self-adjoint with respect to \(g\), parallel and nilpotent. Does this manifold or its double cover admit a nonzero light-like parallel vector field?

It is transition text leading to Problem 12.0.4 on affinely equivalent Finsler metrics.

The metric in Question 12.0.3 must be indefinite. For a positive-definite metric, a self-adjoint endomorphism is diagonalizable over the reals, so a nilpotent self-adjoint endomorphism is zero. Below “light-like” means nonzero and null.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06498 | AIM-GEOMETRY-0263

### A compact normalized-null-cone obstruction

P / partially_solved | AIM 研讨会问题汇编

Can there exist a complete semi-Riemannian metric \(g\) of indefinite signature and a nontrivial one-form \(\eta\) on a closed manifold such that, for every lightlike geodesic \(\gamma(t)\), there are constants \(C_1\ne0,C_2\) for which \[ \eta_{\gamma(t)}(\dot\gamma(t))=C_1t+C_2? \]

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06499 | AIM-GEOMETRY-0027

### A complete tropical and theta calculation for the braid variety X(sigma_1^3)

P / partially_solved | AIM 研讨会问题汇编

For each cluster torus, take a co-character lattice and glue them by piecewise linear maps coming from exchange relations. These tropical points label the Gross-Hacking-Keel-Kontsevich $\Theta$-basis.

**What are the tropical points of the braid variety? What are the $\Theta$-functions on the braid variety?**

来源字段：http://aimpl.org/clusterbraid/4/

06500 | AIM-GEOMETRY-0312

### A concrete obstruction to resolution-independent products on intersection cohomology

P / partially_solved | AIM 研讨会问题汇编

**Question 9.3 (T. Holm).** Is there a natural ring structure on the intersection cohomology of singular Kähler quotients?

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

06501 | AIM-GEOMETRY-0376

### A conformal-factor cocycle proof of fact 2

S / solved | AIM 研讨会问题汇编

**Problem 18.** Deduce fact 2 from fact 1 or vice versa. Alternatively, construct a Lie algebraic proof of fact 2.

The beginning of that field is

The two facts, supplied by the immediately preceding source record and verified against the official source, are as follows. Let \(n=2m\), let \(\delta\) be a flat Euclidean metric, and let \(\widehat\delta=\Omega^2\delta\).

- **Fact 1.** The critical power \(P_\delta=\Delta_\delta^m\), regarded as an operator from weight-zero functions to conformal densities of weight \(-n\), is conformally invariant on flat space. In scalar representatives, \[ P_{\Omega^2\delta}f=\Omega^{-n}P_\delta f \] whenever both sides are expressed in flat scales. - **Fact 2.** If \(\Omega^2\delta\) is itself flat, then \[ \Delta_\delta^m\log\Omega=0. \]

The AIM source works locally, with \(\Omega>0\), and calls \(\Delta\) the ordinary Euclidean Laplacian. These are the assumptions used below. The statement only has the displayed differential-operator meaning in even dimension; odd-dimensional fractional operators are outside its scope.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

06502 | OPG-357

### A conjecture on iterated circumcentres

O / open | Open Problem Garden

Conjecture Let $p_1,p_2,p_3,\ldots$ be a sequence of points in ${\mathbb R}^d$ with the property that for every $i \ge d+2$, the points $p_{i-1}, p_{i-2}, \ldots p_{i-d-1}$ are distinct, lie on a unique sphere, and further, $p_i$ is the center of this sphere. If this sequence is periodic, must its period be $2d+4$?

背景链接：http://www.openproblemgarden.org/op/a_conjecture_on_iterated_circumcentres

06503 | AIM-GEOMETRY-0270

### A convention-safe reduction for cubic integrals on the two-torus

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 10.2.1 ([Bo-Ko-Fo1995]).** If the geodesic flow of a Riemannian metric on the torus \(T^2\) admits an integral that is polynomial of degree 3 in the velocities, then the metric admits a Killing vector field.

If a smooth Riemannian metric on \(T^2\) has a nonzero homogeneous cubic first integral, then it has a nonzero homogeneous linear first integral.

The mathematical record begins with the following statement.

The paragraphs in the record through the proposed odd/even degree analogues belong to Section 10.2 of the source. They explain that a Killing field gives a linear integral whose cube is cubic, that homogeneous velocity components of a geodesic-flow integral are separately integrals, and that “degree” in the accompanying table means the *smallest degree of a nontrivial integral*. The material beginning “10.3.

There are two suppressed conventions in the displayed conjecture. The intended integral is nontrivial (additional to the energy), and “degree 3” means exact fibre degree, not degree at most three. Equivalently, after taking the top homogeneous part, the intended assertion is:

For a linear function \(L=X^i p_i\) on \(T^*T^2\), the identity \(\{H,L\}=0\) is equivalent to \(\mathcal L_Xg=0\), so this is exactly the Killing-vector conclusion. Smoothness is enough for the elementary results below. Analyticity is stated separately where a cited global theorem needs it.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06504 | AIM-GEOMETRY-0423

### A convex support-function obstruction for holomorphic curves in R x T^3

P / partially_solved | AIM 研讨会问题汇编

At the end of his talk, Hutchings mentioned that one could try to generalize the methods therein to compute the Embedded Contact Homology of \(S^1\times S^2\), torus bundles over \(S^1\), or unit cotangent bundles of surfaces of genus \(g>1\). It might also be interesting to try to further understand the holomorphic curves in \(\mathbb R\times T^3\) in terms of amoebas etc.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

06505 | AIM-GEOMETRY-0438

### A cotangent-graph calibration for Lagrangian Floer spectra

P / partially_solved | AIM 研讨会问题汇编

Question 1.42. Peter Ozsvath said that it would be nice to see a spectrum theory (a la C. Manolescu) for Lagrangian intersections.

Peter **Ozsváth** said that it would be nice to see a spectrum theory (**à la** C. Manolescu) for Lagrangian intersections.

Construct stable-homotopy refinements of Lagrangian intersection Floer theory, recovering Floer (co)homology and invariant under the appropriate choices and Hamiltonian isotopies.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

06506 | AIM-GEOMETRY-0213

### A critical horocycle energy level without closed magnetic geodesics

S / solved | AIM 研讨会问题汇编

**Question 2.4.1 (Paternain).** Is there at least one closed magnetic geodesic in every energy level?

The exact question is:

The next paragraph states that Contreras, Macarini, and Paternain proved an affirmative answer when the magnetic form is exact; explains the lack of a global Lagrangian when the form is non-exact; suggests proving contact type; and cites Ginzburg's survey.

No unprinted restriction such as exactness, positive genus exclusion, or contact type is inferred. As printed, the question ranges over closed surfaces and closed magnetic two-forms.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06507 | AIM-GEOMETRY-0198

### A decomposition-group criterion and a local-global ramification gap

P / partially_solved | AIM 研讨会问题汇编

**11. (Owen) Specialization:** When is the Galois group of a particular rigid framework a subgroup of the Galois group of the graph? When is \(\mathcal G(G,p)\subseteq \mathcal G(G)\) for any \(p\) with \((G,p)\) isostatic? (Isostatic is sufficient but not necessary). Here \(p\) generic means \(p=x_1,\ldots,x_n\) are algebraically independent over \(\mathbb Q\).

来源字段：https://aimath.org/pastworkshops/linkagesproblems.pdf

06508 | AIM-GEOMETRY-0033

### A derived exact filling from a 3D plabic graph

P / partially_solved | AIM 研讨会问题汇编

Is there a symplectic geometric application of 3d-plabic graphs? Do 3d-plabic graphs give some filling of a Legendrian link?

来源字段：http://aimpl.org/clusterbraid/7/

06509 | AIM-GEOMETRY-0043

### A derived-collapse obstruction for the Soergel-braid-variety program

P / partially_solved | AIM 研讨会问题汇编

What can we say about Soergel bimodules and braid varieties?

来源字段：http://aimpl.org/clusterbraid/10/

06510 | AIM-GEOMETRY-0163

### A dimension-filtered quotient reduction for a malformed Ricci-flow classification prompt

O / open | AIM 研讨会问题汇编

Classify singularities modulo singularities that in bounded scalar curvature setting.

来源字段：http://aimpl.org/flowriemannian/1/

06511 | AIM-GEOMETRY-0087

### A dimension-linearized E6 reduction for A_+'(12)

P / partially_solved | AIM 研讨会问题汇编

For the real-dimension Hankel continuation of the $+1$ sign-uncertainty constant, prove that the derivative at $d=12$ exists and equals \[ \frac{\sqrt8}{63}=\frac{(\sqrt2)^9}{504}. \]

来源字段：http://aimpl.org/discreteaf/1/

06512 | AIM-GEOMETRY-0286

### A dual heat-tail certificate and exact zero-free Epstein zeta examples

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.** Does there exist in each dimension a lattice $L$ for which $E(L,s)$ has no zeroes in $(0,\infty)$? *(Note: One can show that if a lattice $L'$ is a local minimizer of $E(L,s)$ for all positive $s$, then at least $E(L',s)$ has no odd order zero in $s$ on $(0,\infty)$.)*

来源字段：https://aimath.org/WWN/spherepacking/spherepacking.pdf

06513 | AMR-037-0014

### A dynamic-programming interval problem

P / partially_solved | AMR 问题清单汇编

Given a sorted list of $n$ real numbers, find for every $1\le k\le n$ the shortest interval containing exactly $k$ entries. Find a subquadratic algorithm or prove a superlinear lower bound.

背景链接：https://jeffe.cs.illinois.edu/open/

06514 | AIM-GEOMETRY-0317

### A finite-area counterexample on the cotangent cylinder

P / partially_solved | AIM 研讨会问题汇编

Let \(M\) be a compact manifold, and consider \(T^*M\). Let \(\sigma\) be the canonical symplectic form on \(T^*M\). For any other exact symplectic form \(\omega=d\alpha\) on \(T^*M\), with \(\alpha|_M\) exact, is it possible to find a diffeomorphism on \(T^*M\) preserving \(M\) sending \(\omega\) to \(\pm\sigma\)?

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

06515 | AIM-GEOMETRY-0047

### A finite-total-curvature gate for unbounded negative cusps

P / partially_solved | AIM 研讨会问题汇编

Let \(M\) be a finite volume, complete, negatively curved manifold. Is the Liouville measure an equilibrium state? The case when the curvature is uniformly bounded below is known, so the case of curvature is unbounded is the remaining case.

来源字段：http://aimpl.org/equibdynsysgeom/1/

06516 | AIM-GEOMETRY-0251

### A first-integral obstruction and finite Liouville-barrier dichotomy

P / partially_solved | AIM 研讨会问题汇编

**Question 8.4.2.** Assuming \(n=\dim M\geq 3\), do the metrics for which the geodesic flow is transitive form a dense subset of \(\mathcal G\)? Recall that a geodesic flow is transitive if there exists a geodesic \(\gamma\), with \(|\dot\gamma|=1\), such that the set of its tangent vectors is dense in \(SM\). If \(n=\dim M=2\), the answer is negative: if we take a metric with integrable geodesic flow such that certain Liouville tori are irrational, any small perturbation of the metric has nontransitive geodesic flow by KAM theory. This question is also closely related to the famous Arnold diffusion conjecture.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06517 | AIM-GEOMETRY-0303

### A fixed-component formula for the hyper-compact signature

P / partially_solved | AIM 研讨会问题汇编

The recovered statement of Conjecture 6.10 is as follows. Put

\[ K=\mathbb Q(u),\qquad \widehat H^*_{S^1}(M)=H^*_{S^1}(M;\mathbb Q)\otimes_{\mathbb Q[u]}K, \]

and order $K$ by declaring a nonzero rational function positive when its leading coefficient (equivalently its sign for $u\gg0$) is positive. For a circle-compact $M$, Hausel--Proudfoot define integration by fixed-set localization and the pairing

\[ B_M(\alpha,\beta)=\int_M\alpha\beta. \]

The intended conjecture is that if $M^{4n}$ is hyper-compact and \(\sigma(M)\) is the signature of this localized pairing, then \((-1)^n\sigma(M)\ge0\).

Two corrections are necessary. First, the AIM Definition 6.9 immediately preceding the record says “finitely many critical points.” Read literally, this is inconsistent in positive dimension with the weight-one symplectic hypothesis and the existence of a minimum (Proposition 5.1 below). Hausel's primary 2003 paper instead defines hyper-compactness using a proper Hamiltonian with a minimum and explicitly allows fixed components $F_1,\ldots,F_f$; the later semiprojective formulation also has projective fixed components. Second, on odd cohomology cup product is alternating, not symmetric. The abelianization Theorems 6.12--6.13 are boundary context, not additional problems addressed here.

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

06518 | AIM-GEOMETRY-0155

### A fixed-space orbifold-link obstruction to Ricci smoothing

P / partially_solved | AIM 研讨会问题汇编

Let $K$ be a simplicial complex with metric $g$ such that each simplex has a metric with positive (or flat) curvature and $K$ has positive (or nonnegative) curvature in the Alexander sense. Does a Ricci flow $(K,g(t))$ exist such that for each $t>0$, $g(t)$ is a smooth orbifold metric and $$\lim_{t\to0}g(t)=g$$ in the Gromov--Hausdorff topology. Can one get rigidity in the nonnegative case?

来源字段：http://aimpl.org/flowriemannian/1/

06519 | AIM-GEOMETRY-0295

### A flag certificate for abstract Kirwan surjectivity

P / partially_solved | AIM 研讨会问题汇编

**Recovered Conjecture 5.2 (G. Landweber).** Let a torus \(T\) act on a smooth manifold \(M\), and let \(\Phi:M\to\mathfrak t^*\) be an abstract moment map: for every closed subgroup \(H\subset T\), the projection \(\Phi_H:M\to\mathfrak h^*\) is constant on each connected component of \(M^H\). Assume \(\Phi\) is nondegenerate, meaning that for every \(\xi\in\mathfrak t\), \[ \operatorname{Crit}(\Phi^\xi)=\{x\in M:\xi_M(x)=0\} \] and \(\Phi^\xi=\langle\Phi,\xi\rangle\) is Morse--Bott. For a regular value, translated to \(0\), the conjecture asks whether restriction \[ \kappa:H_T^*(M)\longrightarrow H_T^*(\Phi^{-1}(0)) \] is surjective.

**Recovered Comments 5.3--5.5.** The workshop notes say that the compact \(S^1\) case is proved in GGK, while the GGK proof for higher-dimensional tori has an error. They ask whether local symplectic forms can replace global symplectic geometry. GGK Theorem G.22 gives such a form near an orbit when the abstract moment map is nondegenerate and the symplectic slice admits an invariant complex structure.

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

06520 | AIM-GEOMETRY-0046

### A flow-defect criterion for Brownian harmonic measure

P / partially_solved | AIM 研讨会问题汇编

Is the harmonic measure from Brownian motion in the finite-volume noncompact setting an equilibrium state?

来源字段：http://aimpl.org/equibdynsysgeom/1/

06521 | AIM-GEOMETRY-0398

### A form-Q uniqueness problem and a flat-torus dichotomy

P / partially_solved | AIM 研讨会问题汇编

Problem 35: There are analogues for the operators M gk of most of the Problems in Sections II. and III. for the Q-curvature.

Chapter D: Reference list

A list of references related to the topic of the workshop is available at http://www.und.nodak.edu/instruct/lapeters/bib

**Problem 35. There are analogues for the operators \(M_k^g\) of most of the problems in Sections II and III for the \(Q\)-curvature.**

On a connected closed oriented Riemannian four-manifold, for a fixed closed one-form \(c\), when does the value \(Q_1^g c\) determine \(g\) up to constant rescaling inside \([g]\)?

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

06522 | AIM-GEOMETRY-0350

### A four-dimensional curvature-quadratic slice of the global conformal invariant decomposition

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 2.** Any \(S\) as above may be written \[ \mathrm{const}\cdot Q+L+V, \] where \(V\) is an exact divergence.

The words “as above” are essential. The surrounding text first asks in Conjecture 1 for a **natural \(n\)-form** \(S\) whose integral is conformally invariant, and then introduces Conjecture 2 as the stronger assertion that the third term can be chosen to be an exact divergence. The PDF also makes the following conventions explicit.

* \(M\) has even dimension \(n\). The primer starts with \(M\) oriented; equivalently, one can use scalar densities and omit orientability. * All local invariants in this discussion are treated as density-valued. In scalar notation they have conformal weight \(-n\), so that \(P(g)dV_g\) has weight zero under constant rescaling. The filtration immediately after Conjecture 2 uses monomials in curvature and covariant derivatives satisfying \(k_\nabla+2k_R=n\). * For \(\widehat g=e^{2\omega}g\), the critical \(Q\)-curvature density satisfies \(\widehat Q=Q+P_n\omega\), with \(P_n\) formally self-adjoint and annihilating constants. Hence \(\int_M Q\) is conformally invariant on a closed manifold. * A local conformal invariant \(L\) is a natural density fixed pointwise by conformal rescaling. An exact divergence is \[ V=(\nabla_iT^i)dV_g=d(\iota_TdV_g) \] for a natural vector field \(T\).

The conjecture itself does not repeat “compact without boundary.” This is recovered from the surrounding integral discussion and is necessary for an exact divergence to integrate to zero without a boundary correction. In this report, **closed** means compact without boundary. The exact literal phrase “natural \(n\)-form” is not given a formal category in the workshop notes; this ambiguity matters for the present-day status in Section 2.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

06523 | AIM-GEOMETRY-0116

### A four-disk obstruction between bond energy and container density

P / partially_solved | AIM 研讨会问题汇编

**Crystal Growth From Clusters**

Can connections be drawn between cluster structures that appear during complex crystal growth, and finite clusters that are optimal under other conditions? Are these finite clusters energy minimizers? Do they instead (or in addition) represent especially dense configurations under some packing constraints? Drawing these connections would help us build physical intuition for cluster growth mechanisms.

来源字段：http://aimpl.org/softpack/2/

06524 | AIM-GEOMETRY-0044

### A frozen-gauge seed criterion for parabolic braid models

P / partially_solved | AIM 研讨会问题汇编

Can we study parabolic braid varieties, ie. in \(G/P\)?

Do these have cluster structures? Do we need to change the definition of a seed?

来源字段：http://aimpl.org/clusterbraid/10/

06525 | AIM-GEOMETRY-0083

### A genus-two feasibility certificate for extremal even unimodular lattices

P / partially_solved | AIM 研讨会问题汇编

Use Siegel modular forms to improve bounds for the maximal dimension of an extremal even lattice.

**Recovered problem.** Use degree at least two Siegel modular forms to lower the upper bound on the rank of an extremal positive-definite even unimodular lattice.

来源字段：http://aimpl.org/discreteaf/1/

06526 | AMR-063-0003

### A Geometrization Picture for Smooth Four-Manifolds

O / open | AMR 问题清单汇编

Find a structure or conjectural decomposition for smooth four-manifolds that could play the guiding role that Thurston's Geometrization Conjecture plays for three-manifolds.

背景链接：https://wwwf.imperial.ac.uk/~skdona/NONLINEARITYPROBLEMS.PDF

06527 | AIM-GEOMETRY-0009

## A global quantitative square-stability bound for rectangles

P / partially_solved | AIM 研讨会问题汇编

Determine the best lower bound on \[ J(\Omega):=\frac{\lambda _1(\Omega)}{h(\Omega)^2} \] among planar convex open sets, conjecturing \[ J(\Omega)\geq J(Q):=\frac{2\pi^2}{(2+\sqrt\pi)^2} =1.3870172735\ldots, \] with equality, up to similarities, only for a square. It also asks whether a minimizer is polygonal and suggests first studying polygons through the geometry of their Cheeger sets.

Two notational repairs are necessary. For a measurable set \(E\), \(|\partial E|\) must mean the ambient De Giorgi perimeter \(P(E;\mathbb R^n)\), not the measure of the topological boundary and not the relative perimeter in \(\Omega\). Thus \[ h(\Omega)=\inf_{\substack{E\subset\Omega,\ |E|>0\\E\text{ of finite perimeter}}} \frac{P(E;\mathbb R^n)}{|E|}. \] The functional formulation in the source also needs either nonnegative test functions or an absolute value in the denominator: \[ h(\Omega)=\inf_{0\neq u\in C_c^\infty(\Omega)} \frac{\int_\Omega |\nabla u|}{\int_\Omega |u|}. \] Without this repair, sign-changing functions can make the displayed denominator zero or negative. The spectral quantity is \[ \lambda _1(\Omega)=\inf_{0\neq u\in H_0^1(\Omega)} \frac{\int_\Omega |\nabla u|^2}{\int_\Omega u^2}. \]

The source calls the square prediction "Ilias Ftouhi's conjecture." The primary literature checked attributes the conjecture to Enea Parini; Ftouhi subsequently improved the universal planar-convex lower bound.

来源字段：http://aimpl.org/symmetrybreaking/3/

06528 | AIM-GEOMETRY-0362

## A Helmholtz test and a scale-descent obstruction for conformal primitives

P / partially_solved | AIM 研讨会问题汇编

**Problem 5b.** What characterizes such curvature invariants? A related problem is posed by T. Branson: On \(M^n\), \(Q\)-curvature is a local invariant (of density weight \(-n\)) which does not have a conformal primitive. The local invariants that have conformal primitives form a vector subspace, say \(\mathcal L'\), of the space of local invariants \(\mathcal L\). Thus the quotient space \(\mathcal L/\mathcal L'\) is the space which measures “how many things” do not have a conformal primitive. There are also local conformal invariants, \(\mathcal L''\), say.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

06529 | AIM-GEOMETRY-0117

## A holonomy profile for frustration in nested cluster growth

P / partially_solved | AIM 研讨会问题汇编

**Frustration** What, precisely, is geometric frustration? More specifically, how does it manifest itself in the context of nested cluster growth models of complex crystals? While the concept makes intuitive sense, it has evaded attempts at formalization.

来源字段：http://aimpl.org/softpack/2/

06530 | AIM-GEOMETRY-0209

## A hyperbolic geodesic and the robust infinite branch

S / solved | AIM 研讨会问题汇编

**Conjecture 2.2.2 (Long).** The existence of one hyperbolic prime closed geodesic on a Finsler $S^2$ implies the existence of infinitely many distinct prime closed geodesics.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06531 | AIM-GEOMETRY-0290

## A lift-dependent Martin formula and a sharp middle-degree obstruction

P / partially_solved | AIM 研讨会问题汇编

**Question 2.4 (T. Hausel)** Is there a Martin theorem for 3-Sasakian quotients? What about for Sasakian quotients? Getting a Martin-type formula requires Kirwan surjectivity, but perhaps surjectivity "up to middle dimension" would be enough to get a Martin-type theorem for the 3-Sasakian case.

**Comment 2.5 (J. Munn)** Martin's theorem probably should be extendable analytically in the polysymplectic case. Since 3-Sasakian manifolds can arise as boundaries of hyperK¨ ahler manifolds, we can look at compact cohomology based on the 3-Sasakian manifold (regarded as a boundary). Here we can ignore cone points and do integration theory.

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

06532 | AIM-GEOMETRY-0215

## A local-critical-homology reduction for exponential closed-geodesic growth

P / partially_solved | AIM 研讨会问题汇编

**Question 3.1.2 (Gromov).** Does the number of closed geodesics with length \(\leq T\) grow exponentially as \(T\to\infty\) if the Betti numbers of the loop space grow exponentially?

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06533 | AIM-GEOMETRY-0427

### A mixed gluing ledger for higher-genus relative symplectic field theory

P / partially_solved | AIM 研讨会问题汇编

Can one construct a rigorous relative SFT-type invariant which includes holomorphic curves of arbitrary genus, rather than only genus-zero (“rational”) curves, and can such a theory illuminate Gopakumar--Vafa/BPS structures?

The source record says:

It is not part of Question 1.31.** The statement above is the complete text of Question 1.31 in the official source.

The note itself admits that substantial context is missing. It therefore does not specify:

- the relative geometric object (a Legendrian, a Lagrangian cobordism, a contact manifold with boundary, or something else); - the generators, coefficients, grading, or completion; - the precise relation sought with Gopakumar--Vafa invariants.

The preceding Question 1.30 concerns knots and their Legendrian conormal tori, and other passages in the workshop notes discuss relative contact homology as a module over absolute contact homology. This makes a Legendrian/Lagrangian-relative reading plausible, but it does not prove that this was the unique intended meaning. No stronger reconstruction is asserted.

The minimal justified mathematical question is:

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

06534 | AIM-GEOMETRY-0245

### A moduli correction and fixed-volume factor obstruction

S / solved | AIM 研讨会问题汇编

**Corrected irreducible, unmarked conjecture.** Let \((M,g)=\Gamma\backslash X\) be a closed irreducible locally symmetric manifold of noncompact type with \(\operatorname{rank}_{\mathbb R}X\ge2\). If \(h\) is a smooth Riemannian metric on \(M\) without conjugate points, then \(h=\phi^*(c\,g)\) for some diffeomorphism \(\phi\) and \(c>0\).

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06535 | AIM-GEOMETRY-0222

### A monotone special case of marked-length rigidity

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 5.1.1 ([Bur-Kat1985]).** Two metrics with negative curvature on a compact manifold must be isometric if they have the same marked length spectrum.

Croke and Otal showed that this conjecture is true for metrics on surfaces [Otal1990, Croke1990]. Indeed it is enough to assume that the metrics have non-positive curvature. The problem is open in higher dimensions. Hamenst¨ adt showed that the geodesic flows of the two metrics must be C0-conjugate, thereby reducing the problem to

Conjecture 5.2.1 in the next subsection [Hamenstädt1991].

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06536 | AIM-GEOMETRY-0407

### A Morita shadow of contact homology and a dualizability obstruction

P / partially_solved | AIM 研讨会问题汇编

**Question 1.11.** Generalizing the theme of computing things, one could try to extend symplectic field theory to an “extended field theory”. Recall that a TQFT assigns to an \(n\)-manifold (possibly with some extra structure) a number, and to an \((n-1)\)-manifold a vector space, satisfying various axioms which allow one to compute the invariant of an \(n\)-manifold by cutting it up along \((n-1)\)-manifolds. But then one is left with the problem of understanding the \((n-1)\)-dimensional invariant. In an extended TQFT, one can compute the latter by cutting along \((n-2)\)-dimensional manifolds, to each of which is assigned a category. (One can continue by assigning 2-categories to \((n-3)\)-manifolds and so forth, so that the manifolds get simpler while the theory gets more complicated...) Now the question is, how can one do this for SFT?

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

06537 | AIM-CONVEX_GEOMETRY-0009

### A necessary moment order and a scale obstruction

P / partially_solved | AIM 研讨会问题汇编

Let the $k$-th moment tensor of a convex body be the $k$-th moment tensor of the uniform distribution on it. For any positive integer $t$, is there $k=k(t)$, independent of $d$, such that the map taking a $d$-dimensional polytope with $d^t$ facets to its first $k$ moment tensors is injective?

It then records:

1. [GLPR] an $n$-vertex $d$-polytope can be recovered from $O(dn)$ axial moments along $d$ random directions; 2. [FJK] a parallelepiped can be estimated efficiently from the first four moment tensors; 3. [AGR] a simplex can be estimated efficiently from the first three moment tensors.

Finally it asks for an efficient reconstruction algorithm, for $1/10$ Hausdorff accuracy, both from moment tensors and from polynomially many uniform samples.

The repairs and ambiguities are:

- “$dt$ facets” in the extraction is $d^t$ facets in the PDF. - The PDF prints ordinary $O(dn)$, not an exponent. The underlying Gravin--Lasserre--Pasechnik--Robins theorem is stated as $O(Nd)$ axial moments for $N$ vertices in $d+1$ generic directions; the workshop text says $d$ random directions. - In the algorithmic paragraph the PDF says “time polynomial in $n$” and “sample size polynomial in $n$,” although the ambient dimension in the question is $d$, and $n$ appeared only locally as the number of vertices in item (a). This is preserved as printed. The likely intended complexity variable is $d$ (equivalently polynomial in the facet count $d^t$ for fixed $t$), but that reconstruction is not certain. - Taken literally, the $t=1$ class consists of full-dimensional $d$-polytopes with exactly $d$ facets and is empty: every $d$-polytope has at least $d+1$ facets. The substantive regime is therefore $t\ge2$, unless “$d^t$” was intended only as asymptotic shorthand.

For a full-dimensional compact polytope $P\subset\mathbb R^d$, write $$ M_j(P)=\frac1{\operatorname{vol}(P)}\int_P x^{\otimes j}\,dx \in \operatorname{Sym}^j(\mathbb R^d). $$

来源字段：https://aimath.org/pastworkshops/sectionsconvexproblems.pdf

06538 | AIM-GEOMETRY-0060

### A non-Anosov magnetic horseshoe and a pressure-capture criterion

P / partially_solved | AIM 研讨会问题汇编

Are there examples of non-uniformly hyperbolic magnetic flows for which we can apply the tools of thermodynamical formalism?

来源字段：http://aimpl.org/equibdynsysgeom/3/

06539 | AIM-GEOMETRY-0433

### A non-exact l.c.s. family on a Sol mapping torus

S / solved | AIM 研讨会问题汇编

Question 1.37. Find other constructions of c.s.s. that represent a non zero class in Lichnerowicz cohomology.

The exact question in the official PDF is:

Two textual points matter.

The next unnumbered paragraph observes that compatible almost complex structures can be considered, and Question 1.38 then asks about their holomorphic curves. Typographically that sentence follows Question 1.37, but logically it is transitional exposition motivating Question 1.38, not an additional clause in the construction request of Question 1.37.

Thus the recovered mathematical request is: construct l.c.s. forms whose class in the associated Lichnerowicz (Morse--Novikov) cohomology is nonzero.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

06540 | AIM-GEOMETRY-0298

### A non-Morse obstruction certificate and the contact sphere

P / partially_solved | AIM 研讨会问题汇编

Question 5.8 (E. Lerman) Can we prove Kirwan surjectivity without Morse theory? The motivation for this question comes from the fact that two fundamental results in the theory of symplectic moment maps - connectedness and convexity, which were originally proved using Morse theory have an alternative proof [CDM88]. This alternative approach works well in the contact setting (equivalently in the setting of symplectic cones) where Morse theory fails. The reasons for the failure in the contact setting are due to the fact are that the contact moment maps are not Morse and that there is no relationship between critical points and isotropy groups. In the equivalent setting of symplectic cones the moment maps are not proper.

Comment 5.9 (E. Lerman) As mentioned in

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

06541 | AIM-GEOMETRY-0082

### A normalized lattice transport problem and a product-stability bound

P / partially_solved | AIM 研讨会问题汇编

Choose $\Lambda \subseteq \mathbb R^d$ to minimize the Wasserstein (optimal transport) distance form $\sum_{x \in \Lambda} \delta_x$ to the Lebesque measure.

The exact AIM record asks:

The source remark says that for the Wasserstein 2-distance in $\mathbb R^2$ the unique minimizer is the triangular lattice, citing Bourne--Peletier--Theil [BPT14].

There are two more substantive omissions. Ordinary $W_p$ is a distance between finite measures of the same mass (usually probability measures), whereas both $dx$ and $\sum_{\lambda\in\Lambda}\delta_\lambda$ have infinite mass on $\mathbb R^d$. Moreover, if $V=\operatorname{covol}(\Lambda)$, then the counting measure has mean mass density $1/V$. Thus unit atoms match $dx$ only when $V=1$, and the scale must be fixed or the lattice can simply be shrunk.

The natural recovered problem is therefore the following. For $p\geq1$ and a full-rank lattice $\Lambda\subset\mathbb R^d$ of fixed covolume $V$, let

\[ \mathbb{T}_\Lambda=\mathbb R^d/\Lambda, \qquad m_\Lambda=V^{-1}dx, \]

where the torus has the quotient (periodic/geodesic) Euclidean distance. Minimize

\[ \mathcal E_p(\Lambda) :=W_{p,\mathbb{T}_\Lambda}^{p}(m_\Lambda,\delta_0). \tag{1} \]

Equivalently, (1) is the transport cost per unit volume for a $\Lambda$-equivariant transport from $dx$ to the mass-matched atomic measure

\[ V\sum_{\lambda\in\Lambda}\delta_\lambda. \]

For the unit atoms appearing literally in the AIM record one takes $V=1$. Minimizing $W_p$ or its $p$-th power $\mathcal E_p$ gives the same lattices, but their numerical values are different.

来源字段：http://aimpl.org/discreteaf/1/

06542 | AIM-GEOMETRY-0189

### A one-coordinate unlockability certificate for open chains in three-space

P / partially_solved | AIM 研讨会问题汇编

**2. (Streinu)** Same as the previous question, but with an open arm in \(\mathbb R^3\), and \(\ell_1=\ell_2=\cdots=\ell_n\). **Question:** Is the space of non-intersecting configurations connected?

The original AIM problem-list PDF reads:

The preceding item defines the self-intersection locus for closed polygons. More decisively, the AIM workshop report restates Item 2 as the three-dimensional Carpenter's Rule problem: can an arbitrary open chain of unit-length bars be unfolded to a straight line by non-crossing motions? Thus the intended object is an **equilateral open polygonal chain with universal joints**, not a fixed-angle chain and not merely a topological arc.

The source does not specify whether rigid motions have been quotiented out or whether tangential contact is admitted. The standard strict model is adopted below; the convention issue is discussed in Section 3.

来源字段：https://aimath.org/pastworkshops/linkagesproblems.pdf

06543 | AIM-GEOMETRY-0098

### A one-dimensional counterexample and arithmetic reductions for all-shell 12-design lattices

S / solved | AIM 研讨会问题汇编

**Problem 1.44.** Prove there is no lattice whose shells are spherical $12$-designs.

来源字段：http://aimpl.org/discreteaf/1/

06544 | AIM-GEOMETRY-0202

### A one-point differential criterion for global symmetry persistence

P / partially_solved | AIM 研讨会问题汇编

**15. (Schulze)** Under what conditions is a linkage guaranteed to preserve the original symmetry throughout the configuration space?

来源字段：https://aimath.org/pastworkshops/linkagesproblems.pdf

06545 | AIM-GEOMETRY-0105

## A parameterized twelve-contact case and full dodecahedral shell formula

P / partially_solved | AIM 研讨会问题汇编

Its exact `problem` field is:

Taken by itself, this record does not define a soft packing, state its parameter, or specify which of two density ratios is intended. The section introduction on the original AIM webpage supplies the missing first part: a soft ball is a concentric pair consisting of a unit ball and a ball of radius $1+\lambda$, where $\lambda>0$; the unit balls have disjoint interiors, while density means the fraction of space covered by the radius-$1+\lambda$ balls.

The precise framework was introduced before the 2016 AIM workshop by Bezdek and Lángi. Put $B=B^3=\{x\in\mathbb R^3:|x|\leq1\}$, let $C=\{c_i\}\subset\mathbb R^3$ be locally finite with $|c_i-c_j|\geq2$, and set

$$ P_\lambda=\bigcup_i\bigl(c_i+(1+\lambda)B\bigr). $$

The cores $c_i+B$ form an ordinary unit-ball packing; the enlarged balls form a soft packing with *soft parameter* (originally, *penetrating constant*) $\lambda$. Its upper global soft density is

$$ \overline\delta(P_\lambda) =\limsup_{R\to\infty} \frac{\operatorname{vol}(P_\lambda\cap RB)} {\operatorname{vol}(RB)}, \qquad \overline\delta_3(\lambda)=\sup_P\overline\delta(P_\lambda). $$

For a center translated to the origin, its Voronoi cell is

$$ V=\{x:|x|\leq |x-c_j|\text{ for every other center }c_j\}. $$

Writing $s=1+\lambda$, the local soft density intended by the AIM wording is

$$ \overline\rho_s(V)= \frac{\operatorname{vol}(V\cap sB)}{\operatorname{vol}(V)}. \tag{1} $$

This is distinct from the companion *truncated density*

$$ \rho_s^{\rm tr}(V)= \frac{\operatorname{vol}(B)}{\operatorname{vol}(V\cap sB)}. \tag{2} $$

Let $D$ be the regular dodecahedron circumscribed about $B$, meaning that $B$ is its insphere and its inradius is one. Its circumradius and edge-midpoint radius are

$$ R_D=\sqrt{15-6\sqrt5}=\sqrt3\tan\frac\pi5 =1.258408572\ldots, \qquad q_D=\sqrt{\frac{5-\sqrt5}{2}} =1.175570505\ldots. \tag{3} $$

$$ N_D(s)=\operatorname{vol}(D\cap sB),\qquad \overline\tau_D(\lambda)=\frac{N_D(1+\lambda)}{\operatorname{vol}(D)}. \tag{4} $$

The strongest literal recovery of the AIM statement is the cellwise conjecture

$$ \boxed{\quad \overline\rho_{1+\lambda}(V)\leq\overline\tau_D(\lambda) \quad\text{for every bounded Voronoi cell }V, \quad 0<\lambda<R_D-1. \quad} \tag{SDC-local} $$

Bezdek and Lángi's 2015 paper instead explicitly asks the global inequality

$$ \boxed{\quad \overline\delta_3(\lambda)\leq\overline\tau_D(\lambda). \quad} \tag{SDC-global} $$

The local statement implies the global one for periodic packings (and under standard uniform boundary control more generally), but they should not be silently identified. A regular dodecahedron does not tile space; it is a local comparison cell, not a claim that every cell in an optimizing infinite packing is dodecahedral.

The 2015 source prints $0<\lambda<R_D$, although the inflated radius is $1+\lambda$. Once $\lambda\geq R_D-1$, one has $D\subset(1+\lambda)B$ and $\overline\tau_D(\lambda)=1$, so the soft-density inequality is merely the tautology that a covered fraction is at most one. The 2025 formulation uses the meaningful range $0<\lambda\leq R_D-1=0.258408572\ldots$.

来源字段：http://aimpl.org/softpack/1/

06546 | AIM-GEOMETRY-0311

## A perversity-closure diagnostic for intersection-cohomology rings

P / partially_solved | AIM 研讨会问题汇编

**Question 9.1 (N. Proudfoot).** The intersection cohomology of singular hypertoric varieties has a “seemingly natural” ring structure (as a quotient of the equivariant cohomology $H_T^*(T^*\mathbb C^n)$ of the original space). Is there, in general, a natural ring structure on the intersection cohomology group of a singular hyperkähler quotient?

**Comment 9.2 (N. Proudfoot).** One setting in which one gets a ring structure on intersection cohomology is when one has a small resolution, but in the hypertoric examples, Nick is not aware of any such small resolution.

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

06547 | AIM-GEOMETRY-0331

## A phase-enriched tropical certificate for Kwon's chamberwise real counts

P / partially_solved | AIM 研讨会问题汇编

**A.11 Real Gromov--Witten invariants and tropical geometry.** Background: G. Tian and S. Kwon recently defined a real Gromov--Witten invariant on each chamber in the real Chow cycles' parameter space when the target space is CP 2. That is a real enumerative invariant, counting the number of intersection points of pull back of real Chow cycles in the real part of the Kontsevich's moduli space of stable maps from genus 0 curves. To use Mikhalkin's work on counting plane rational nodal curves, we showed that the classical nodal Severi variety is embedded as a Zariski open dense subset in the Kontsevich's moduli space. Question: It will be interesting to develop techniques to calculate real Gromov--Witten invariants by using tropical geometry. (contributed by Seongchun Kwon)

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

06548 | AIM-GEOMETRY-0231

### A pinching and Riccati-occupation obstruction to typical Lyapunov spread beyond two

P / partially_solved | AIM 研讨会问题汇编

**Question 5.4.2.** Is there a metric of negative curvature for which the ratio of largest positive Lyapunov exponent to smallest positive Lyapunov exponent is greater than 2 for typical geodesics (not just exceptional closed geodesics)?

The Anosov splitting is \(C^\infty\) if and only if the metric is locally symmetric. For surfaces, if one stable or unstable foliation is \(C^{1+o(x\log|x|)}\), then the splitting is \(C^\infty\). In higher dimensions the conjectural implication \(C^2\Rightarrow C^\infty\) was open.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06549 | AIM-GEOMETRY-0266

### A point-critical entropy-carrier obstruction to smooth integrability

P / partially_solved | AIM 研讨会问题汇编

**Question 10.1.1 (Bangert).** Does there exist a Riemannian metric on a closed surface of genus \(\geq 2\) whose geodesic flow is completely integrable?

The answer may depend on what we understand by “completely integrable”: whether the integral is functionally independent of the Hamiltonian on an open everywhere dense subset, or we additionally assume that the subset has full measure.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06550 | AIM-GEOMETRY-0285

### A pole-safe dual-paired Ewald reduction for the remaining D4 conjecture

P / partially_solved | AIM 研讨会问题汇编

**Recovered Conjecture 1.** For every real $s>0$, the covolume-one forms of $D_4,E_8$, and the Leech lattice globally minimize the analytically continued Epstein zeta function among covolume-one lattices in their respective dimensions.

\[ E_L(s)=\sum_{0\ne x\in L}|x|^{-2s}, \]

initially for $\operatorname{Re}s>n/2$ and by meromorphic continuation otherwise. “These lattices” means

\[ D_4\quad(n=4),\qquad E_8\quad(n=8),\qquad \Lambda_{24}\quad(n=24), \]

each scaled to covolume one. Thus the reconstructed conjecture is:

There is an essential qualification. $E_L(s)$ has a common simple pole at $s=n/2$, so it has no finite value there. At the pole, the only mathematically meaningful version of the conjecture is minimization of the finite part

\[ \operatorname{FP}E_L(n/2)=\lim_{s\to n/2} \left(E_L(s)-\frac{\pi^{n/2}/\Gamma(n/2)}{s-n/2}\right). \]

This convention agrees, up to a positive dimension-dependent factor and an additive constant, with the height of the corresponding flat torus.

This report owns Conjecture 1 only. It does not attempt to answer neighboring Question 0 or Problem 2.

来源字段：https://aimath.org/WWN/spherepacking/spherepacking.pdf

06551 | AIM-CONVEX_GEOMETRY-0035

### A polylogarithmic nonsymmetric MM* bound and exact simplex calculation

P / partially_solved | AIM 研讨会问题汇编

Let \(K\) be a symmetric convex body. There exists a linear operator \(T\) such that \[ \int_{S^{n-1}}h_{TK}(\theta)\,d\sigma(\theta)\, \int_{S^{n-1}}h_{(TK)^\circ}(\theta)\,d\sigma(\theta) \le C\log n. \tag{AIM} \] What is the upper bound for the above quantity for nonsymmetric convex bodies \(K\)? (In this case we need to take \(T\) affine.)

来源字段：https://aimath.org/WWN/fourierconvex/fourierconvex.pdf

06552 | AIM-GEOMETRY-0074

### A pressure-normalized potential model and geometric range obstructions

P / partially_solved | AIM 研讨会问题汇编

Build a moduli space for $C^{1,\theta}$-perturbations of geodesic flows using potential functions (following McMullen's approach of expanding maps of the circle). In this case, can one identify the space of geodesic flows by a condition on their corresponding potential function?

来源字段：http://aimpl.org/equibdynsysgeom/5/

06553 | AIM-GEOMETRY-0152

### A product obstruction to coarse higher-dimensional modularity

P / partially_solved | AIM 研讨会问题汇编

Modularity conjecture

Amongst the real 4d complete hyperkahler manifolds (gravitational instantons) all the known examples of type ALG and ALH are modular, for example the ALG examples are all moduli spaces of meromorphic Higgs bundles (although this has not been proved yet---the examples match precisely).

Prove that this holds in higher dimensions, i.e. define classes of complete hyperkahler manifolds (for example by their asymptotics), higherdiemsnional analogues of ALG and ALH, and show that all examples are modular (spaces of solutions of Hitchin's equations, or the Bogomolnyi equations or other reduction of the ASDYM equations)

来源字段：http://aimpl.org/spectralhiggs/1/

06554 | AIM-GEOMETRY-0176

### A quantitative construction calculus for ANCO manifolds

P / partially_solved | AIM 研讨会问题汇编

A closed smooth manifold \(M\) is ANCO if for every \(\epsilon>0\) there is a Riemannian metric \(g_\epsilon\) such that \[ \lambda_{\min}(\mathcal R_{g_\epsilon}) \operatorname{diam}(M,g_\epsilon)^2>-\epsilon, \] where the minimum is over all points and all eigenvalues of \(\mathcal R_g:\Lambda^2TM\to\Lambda^2TM\). Which closed manifolds are ANCO, and which construction operations preserve the property?

来源字段：http://aimpl.org/flowriemannian/4/

06555 | AIM-GEOMETRY-0267

### A quantitative no-focal-point obstruction to zero Liouville entropy

P / partially_solved | AIM 研讨会问题汇编

**Question 10.1.2 (Paternain).** Is there a Riemannian metric on a closed surface of genus \(\ge 2\) whose geodesic flow has zero Liouville entropy?

It immediately adds that Bangert and Paternain had outlined a nonconstructive proof of the analogous assertion for Finsler metrics, and that the Riemannian problem is harder.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06556 | AIM-GEOMETRY-0181

### A radial Ricci accumulation criterion for bounded scalar curvature

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** If $(M^4,g,f)$ is a smooth, connected, complete four-real-dimensional gradient shrinking Ricci soliton \[ \operatorname{Ric}+\nabla^2f=\frac12g, \] must its scalar curvature satisfy $\sup_M R<\infty$?

来源字段：http://aimpl.org/flowriemannian/6/

06557 | AIM-GEOMETRY-0151

### A reduction certificate for a Harnad-dual Lax family

P / partially_solved | AIM 研讨会问题汇编

The Lax project

For each classical integrable system there is often a "Lax pair", which is basically a meromorphic Higgs bundle on the trivial bundle on the Riemann sphere. Thus we can consider the corresponding wild Hitchin moduli space which is a complete hyperkahler manifold. Thus we have a map from integrable systems to complete hyperkahler manifolds.

The project is to go though the integrable systems literature and compile a list Lax pairs and the corresponding irregular curve and the resulting hyperkahler manifold. Further in the cases where integrable systems have alternative Lax pairs, show that the corresponding hyperkahler manifolds are isomorphic.

来源字段：http://aimpl.org/spectralhiggs/1/

06558 | AIM-GEOMETRY-0032

### A removable-edge certificate for positroid-link cobordisms

P / partially_solved | AIM 研讨会问题汇编

Which local edits of a reduced plabic graph can be certified combinatorially to give a directed orientable exact Lagrangian cobordism between the endpoint positroid Legendrians, and which familiar graph moves instead describe Legendrian isotopy or surgery between fillings of the same Legendrian?

来源字段：http://aimpl.org/clusterbraid/7/

06559 | AIM-GEOMETRY-0236

### A second-jet obstruction to a nonperiodic flat geodesic

P / partially_solved | AIM 研讨会问题汇编

**Question 6.2.1 (Burns).** Is there a \(C^\infty\) (or at least \(C^k\), \(k\geq 3\)) metric for which there exists a non closed geodesic along which the Gaussian curvature is everywhere zero?

A negative answer to this question would give a positive answer to Question 6.1.1.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06560 | AIM-GEOMETRY-0030

### A seed dictionary and a rank-one compatibility test

P / partially_solved | AIM 研讨会问题汇编

Plabic graphs and weaves give seeds in cluster structures on braid varieties.

Investigate: (1) cluster variables, (2) quivers, (3) positive parametrizations, (4) twist maps and Donaldson--Thomas transformations, and (5) the relationship between 3d-plabic graphs and weaves.

来源字段：http://aimpl.org/clusterbraid/6/

06561 | AIM-GEOMETRY-0278

### A semi-flat period and weighted-geodesic reduction for fibred special Lagrangians

P / partially_solved | AIM 研讨会问题汇编

**1. Elliptically fibred CY manifolds and their SLAG submanifolds.** What can be said about SLAGs in a CY 3-fold that happens to be elliptically fibred or K3 fibred? Are there nice characterizations? Are such SLAGs fibred? More specifically, if a real six-manifold $M^6$ is a Calabi--Yau threefold fibred over $\mathbb{CP}^1$ by K3 surfaces, can a real three-dimensional special Lagrangian $L^3$ be assembled over a curve $C\subset\mathbb{CP}^1$ from real two-tori in the K3 fibres? In an adiabatically collapsing family $M_t\to B$, what is the limiting base curve, and can the limit control the special Lagrangians?

The primary three-page PDF is an AIM Workshop on Calibrations problem list compiled by Spiro Karigiannis in 2006. Lines 9--30 of page 1 form one numbered item:

The primary source starts a distinct numbered item “2. Yau--Zaslow number for K3 manifolds” at line 31, followed by separately numbered items on Bäcklund transformations, $G_2$ Floer analogues, Bernstein problems, geometric measure theory, special Lagrangian cones, and regularity. Those later items are preserved as source provenance but are outside the mathematical ownership of this attempt.

There is also an apparent source typo in the fifth bullet. It first defines $B$ as the base and $C\subset B$ as the curve, then asks “What is the curve $B$ in the limit?” The coherent reading is “What is the curve $C$ in the limit?” Both the literal wording and this reconstruction are recorded.

来源字段：https://aimath.org/WWN/calibrations/calibrations.pdf

06562 | AIM-GEOMETRY-0308

### A sharp circle criterion for intersecting Kirwan kernels

S / solved | AIM 研讨会问题汇编

Question 7.1 (N. Proudfoot) In this setting, is it true that \[ \ker(f)=\bigcap_\alpha\ker(\kappa_\alpha), \] where the intersection is over central regular values \(\alpha\)?

Comment 7.2 (N. Proudfoot) When \(G=T\) is abelian, the conjecture is true. The proof requires hyperkähler geometry, despite the fact that the statement does not involve anything hyperkähler. Is it possible to prove it directly?

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

06563 | AIM-CONVEX_GEOMETRY-0014

### A sharp product-prism family for Kuperberg's fixed-combinatorial-type question

P / partially_solved | AIM 研讨会问题汇编

**18. (G. Kuperberg)** In \(\mathbb R^3\), consider a class \(\mathcal A\) of centrally symmetric polytopes of a given combinatorial type, excluding the type of the cube and the octahedron. Then the conjecture is that the volume product does not have a local minimum in \(\mathcal A\). Furthermore, one may ask if each such \(\mathcal A\) has a unique critical point up to affine transformations, and it is a local maximum.

来源字段：https://aimath.org/WWN/mahlerduality/mahlerduality.pdf

06564 | AIM-CONVEX_GEOMETRY-0019

### A sharp wedge-orthant family for the spherical simplex volume product

P / partially_solved | AIM 研讨会问题汇编

For \(Q\subset S^{n-1}\), define \[ Q^\circ=\{x\in S^{n-1}:\langle x,y\rangle>0\ \text{for all }y\in Q\} \] and \( P(Q)=\operatorname{vol}(Q)\operatorname{vol}(Q^\circ)\) . For a spherical simplex \(\Delta_n\subset S^{n-1}\) determined by \(n\) distinct hyperplanes through the origin, is \[ P(\Delta_n)\leq P(\mathbb R_+^n\cap S^{n-1})? \] The source adds: "In particular, if \(n=3\), it is true."

来源字段：https://aimath.org/WWN/mahlerduality/mahlerduality.pdf

06565 | AIM-GEOMETRY-0388

### A spectral-gap criterion for the critical Paneitz kernel

P / partially_solved | AIM 研讨会问题汇编

**Problem 28:** When does the equation \(P f=0\) have only constant solutions? On a compact manifold in two dimensions this is always true: harmonic functions are constant. In four dimensions, though there are conditions under which \(P f=0\) has only constant solutions, there are also counterexamples, even on conformally flat manifolds.

III. Problems extracted from the document “Origins, applications, and generalizations of the Q-curvature” ...

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

06566 | AIM-GEOMETRY-0172

### A surface classification and a noncollapse criterion for almost nonnegative curvature operator

P / partially_solved | AIM 研讨会问题汇编

What are closed manifolds with almost nonnegative curvature operator? \[ \mbox{Rm} \cdot \mbox{diam}^2 \geq -\epsilon. \]

A closed smooth manifold $M$ admits **almost nonnegative curvature operator** (ANCO) if, for every $\epsilon>0$, there is a Riemannian metric $g_\epsilon$ on $M$ such that every eigenvalue of \[ \operatorname{Rm}_{g_\epsilon}:\Lambda^2TM\longrightarrow\Lambda^2TM \] satisfies \[ \lambda(\operatorname{Rm}_{g_\epsilon}) \operatorname{diam}(M,g_\epsilon)^2>-\epsilon. \tag{1.1} \]

来源字段：http://aimpl.org/flowriemannian/4/

06567 | AIM-GEOMETRY-0026

### A theta compactification and tropical obstructions for the rank-one braid

P / partially_solved | AIM 研讨会问题汇编

How does this relate to compactifications coming from tropical geometry?

The pronoun “this” is not self-contained. The immediately preceding records ask:

- 4.1 for a natural compactification of a braid variety whose smaller strata have cluster structures; - 4.2 whether brick manifolds can be obtained cluster-theoretically; - 4.3 how the construction relates to subword complexes.

The AIM workshop report confirms this reconstruction. For a positive braid word \(\beta\), it considers \[ X(\overline\beta;\delta(\beta))\subset\operatorname{brick}(\beta), \] and reports a proposed cluster compactification with potential \[ W=z_1+\cdots+z_\ell. \] Thus problem 4.4 asks how the word-relative brick/cluster compactification and its subword strata compare with compactifications built from tropical data.

There are at least four mathematically different meanings of “coming from tropical geometry,” so the comparison is meaningless unless they are separated:

1. Tevelev’s closure of a very affine variety in a toric variety; 2. a GHKK theta compactification obtained by taking Proj of a positive-set theta algebra; 3. a toric degeneration associated with a seed valuation or Newton--Okounkov body; 4. a toric variety constructed from an ordinary Newton, moment, or brick polytope.

The main result below makes all four distinctions explicit for the nontrivial braid \(\beta=\sigma_1^3\).

来源字段：http://aimpl.org/clusterbraid/4/

06568 | AIM-GEOMETRY-0158

### A time-axis and flat-factor audit of higher-dimensional Ricci-flow models

P / partially_solved | AIM 研讨会问题汇编

Understand singularities in Ricci Flow in dimension $n\geq 4$. Examples are $S^3\times \mathbb{R}$, $S^2\times \mathbb{R}^2$, $4d$ Bryant soliton, $3d \mbox{Bryant soliton} \times \mathbb{R}$ and FIK solitons.

来源字段：http://aimpl.org/flowriemannian/1/

06569 | AIM-GEOMETRY-0294

### A topological certificate for Kirwan-type restriction

P / partially_solved | AIM 研讨会问题汇编

Which topological and Morse-theoretic properties of an equivariant map $F:M\to V$ suffice to make the restriction $H_G^*(M)\to H_G^*(F^{-1}(0))$ surjective, and what extra issue arises when $V=\mathfrak g^*\otimes\mathbb R^3$ for a hyperkähler moment map?

This record is Question 5.1, attributed to E. Lerman, in the AIM workshop list *Moment maps and surjectivity in various geometries*.

I reconstruct `hyperK¨ ahler` as *hyperkähler*, `M.Note` as `M. Note`, `6the` as a page-number artifact followed by `the`, and $R2,S1,\mathfrak g\!\ast,M H,M G$ as $\mathbb R^2,S^1,\mathfrak g^*,M^H,M^G$. The last clause should be read componentwise: for $\xi\in\mathfrak g$, the critical set of $\langle\mu,\xi\rangle$ is the zero set of the infinitesimal vector field $\xi_M$, not generally the single set $M^G$. This agrees with Ginzburg--Guillemin--Karshon (GGK).

The precise mathematical question addressed here is:

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

06570 | AIM-GEOMETRY-0273

### A topology-aware phase-box reduction for generic density of closed geodesics

P / partially_solved | AIM 研讨会问题汇编

**Question 11.2.2.** Are the tangent vectors to closed geodesics dense in the tangent bundle for a generic Riemannian metric on a closed manifold?

It is known that this is the case for metrics with nonuniformly hyperbolic geodesic flows.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06571 | AIM-GEOMETRY-0071

### A trapping-tail test for exponential mixing of equilibrium states

P / partially_solved | AIM 研讨会问题汇编

Can one prove exponential decay of correlations for equilibrium states in settings where it it is known for the measure of maximal entropy? (eg, Weil-Petersson geodesic flows, noncompact spaces, non-positive curvature, etc)

Can one prove exponential decay of correlations for equilibrium states in settings where it is known for the measure of maximal entropy? (e.g. Weil--Petersson geodesic flows, noncompact spaces, nonpositive curvature, etc.)

来源字段：http://aimpl.org/equibdynsysgeom/5/

06572 | AIM-GEOMETRY-0361

### A variational test and a family of conformal primitives

P / partially_solved | AIM 研讨会问题汇编

n

Problem 5a: How to decide which curvature invariants have a conformal primitive? For example on manifold M, we have Δ( J n/ 2-1) has 2

Jn/ 2 as a conformal primitive, i.e.

(ω) = Δ( Jn/ 2-1)for all smooth function ω on M, see "Origins, applications and generalizations of the Q-curvature" by T. Branson and R. Gover. Available through http://www.aimath.org.

The displayed formula in the problem list still suppresses a necessary integral pairing on its right-hand side: the derivative of a functional in direction $\omega$ is a number, whereas $\Delta(J^{n/2-1})$ is a function.

The cited Branson--Gover article removes the ambiguity. It states

\[ J^\bullet=-2\omega J+\Delta\omega \]

\[ \left(\int_M J^{n/2}\,dv_g\right)^\bullet(\omega) =\frac n2\int_M\omega\,\Delta(J^{n/2-1})\,dv_g. \]

Thus the verified reconstruction of the example is

\[ \boxed{ \left.\frac{d}{dt}\right|_{0} \int_M\frac2n J_{g_t}^{\,n/2}\,dv_{g_t} = \int_M\omega\,\Delta_g(J_g^{\,n/2-1})\,dv_g, \qquad g_t=e^{2t\omega}g.} \]

The cited article works primarily in even dimension. To make the integration by parts literal, this report assumes that $M^n$ is a smooth closed Riemannian manifold and that $n\geq4$ is even. The conventions are

\[ J=\frac{R}{2(n-1)},\qquad \Delta=-\nabla^i\nabla_i=d^*d,\qquad (dv_g)^\bullet=n\omega\,dv_g. \]

With this nonnegative Laplacian convention, the displayed formula for $J^\bullet$ is correct. All primitive identities below are identities of integral pairings against $\omega$, not pointwise identities between a number and a scalar field.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

06573 | AIM-GEOMETRY-0029

### A weave-to-MV quotient for tropical Lusztig propagation

P / partially_solved | AIM 研讨会问题汇编

How much of the CGGLSS tropical Lusztig propagation is precisely the reduced-word PL atlas of MV polytopes, and what additional structure is introduced by the nonreduced \(0\)-Hecke vertices of a Demazure weave?

来源字段：http://aimpl.org/clusterbraid/4/

06574 | AIM-GEOMETRY-0357

### A Yamabe stability model for conformal variational problems

P / partially_solved | AIM 研讨会问题汇编

**Problem 1c:** Study variational problems arising from conformally invariant problems.

Michael Eastwood.

Study critical points, second variations, compactness, and transverse-to-conformal-class information for variational functionals naturally arising in conformal geometry.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

06575 | AIM-GEOMETRY-0140

### A zero-section brane in a tame parabolic quiver model and its Rees quantization

P / partially_solved | AIM 研讨会问题汇编

Construct and quantize Lagrangian in the moduli space of tame/wild parabolic Higgs bundles/quiver varieties.

来源字段：http://aimpl.org/spectralhiggs/1/

06576 | AMR-065-0010

### A. Haynes: Gaps Problems - Problem

P / partially_solved | AMR 问题清单汇编

Let $1,\alpha,\beta$ be $\mathbb{Q}$-linearly independent, let $Y(\alpha,\beta)$ be their canonical cut-and-project set, and let $\xi_{(\alpha,\beta)}(\Omega)$ be the set of distinct frequencies of patches of shape $\Omega$. Does some $(\alpha,\beta)$ satisfy $\sup_{\Omega\in\mathcal S}\#\xi_{(\alpha,\beta)}(\Omega)=\infty$, where $\mathcal S$ is the family of aligned squares?

背景链接：https://arxiv.org/abs/1604.06280

06577 | AMR-065-0011

### A. Haynes: Gaps Problems - Problem

P / partially_solved | AMR 问题清单汇编

Let $1,\alpha,\beta$ be $\mathbb{Q}$-linearly independent, let $Y(\alpha,\beta)$ be their canonical cut-and-project set, and let $\xi_{(\alpha,\beta)}(\Omega)$ be the set of distinct frequencies of patches of shape $\Omega$. Does some $(\alpha,\beta)$ satisfy $\sup_{\Omega\in\mathcal R}\#\xi_{(\alpha,\beta)}(\Omega)=\infty$, where $\mathcal R$ is the family of aligned rectangles?

背景链接：https://arxiv.org/abs/1604.06280

06578 | AMR-065-0012

### A. Haynes: Gaps Problems - Problem

P / partially_solved | AMR 问题清单汇编

For $1,\alpha,\beta$ linearly independent over $\mathbb{Q}$, does $\liminf_{n\to\infty}n\|n\alpha\|\|n\beta\|=0$ imply that the number of distinct patch frequencies $\#\xi_{(\alpha,\beta)}(\Omega)$ is unbounded over aligned rectangles $\Omega$?

背景链接：https://arxiv.org/abs/1604.06280

06579 | AMR-065-0013

### A. Julien: Relationship between Complexity and Cohomology - Problem

O / open | AMR 问题清单汇编

Let $p(n)$ count radius-$n$ patches in an aperiodic repetitive tiling of dimension $d$, and let $\Omega$ be its tiling space. If $p(n)=O(n^d)$, must the rational cohomology $H^*(\Omega,\mathbb{Q})$ have finite rank?

背景链接：https://arxiv.org/abs/1604.06280

06580 | AMR-065-0014

### A. Navas: A Conjecture on Delone Sets BL to Lattices (after P. Alestalo, D.A. Trotsenko and J. V\"ais\"al\"a). - Problem

S / solved | AMR 问题清单汇编

Let $\mathcal{D}\subset\mathbb{R}^2$ be a Delone set BL to $\mathbb{Z}^2$. Does there exist a bi--Lipschitz map $L:\mathbb{R}^2\mapsto\mathbb{R}^2$ such that $L(\mathcal{D})=\mathbb{Z}^2$?

背景链接：https://arxiv.org/abs/1604.06280

06581 | AIM-GEOMETRY-0345

### Abstract tropical spaces and the actual fibers of phase tropicalization

P / partially_solved | AIM 研讨会问题汇编

This record is Section C.2, “Basic definitions,” in the 14 January 2004 AIM workshop report *Amoebas and Tropical Geometry*. It records workshop proposals and questions, not a settled definition. The questions are:

1. What should an abstract tropical variety or abstract idempotent variety be, including local models and gluing maps? 2. For a non-Archimedean variety, how should valuation, phase, tropical, complex tropical, and real tropical images be distinguished? 3. When is a complex/phase tropical limit homeomorphic or homotopy equivalent to a nearby classical fiber? 4. Is there a tropical Hodge theory compatible with classical limiting Hodge theory? 5. Can patchworking be refined to retain information about an individual hypersurface, and how should maps and bundles be defined?

The source also prints: “The fiber of the projection from such a complex tropical variety to the corresponding tropical variety is a torus.” That statement is false in general. The modern fiber is a coamoeba (or its closure, depending on convention) of an initial degeneration. Section 5 gives the exact binomial case, and Section 7 gives a tropical-line counterexample.

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

06582 | OWR-14299521-006

### Abundance of Negatively Curved Einstein Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

In a meaningful asymptotic or topological sense, how abundant are closed manifolds admitting Einstein metrics with negative sectional curvature?

来源字段：https://doi.org/10.4171/owr/2025/30

06583 | AMR-099-0064

### Accumulation points of packings of $\mathbb{Z}^3$

O / open | AMR 问题清单汇编

Prove that every sphere packing in $\mathbb{R}^3$ whose tangency graph is $\mathbb{Z}^3$ has at most one accumulation point in the one-point compactification $\mathbb{R}^3\cup\{\infty\}$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=8

06584 | OWR-13494-012

### ACM Line Configurations and Symbolic Initial Degrees

O / open | Oberwolfach 报告问题汇编

Which configurations of lines in $\mathbb P^3$ are arithmetically Cohen-Macaulay?

来源字段：https://doi.org/10.4171/owr/2015/9

06585 | AMR-037-0002

### Acute triangulation of the cube

S / solved | AMR 问题清单汇编

Does the three-dimensional cube admit a triangulation into tetrahedra all of whose dihedral angles are acute?

背景链接：https://jeffe.cs.illinois.edu/open/

06586 | AIM-GEOMETRY-0103

### Adjacency maximization on a lattice and its zonotopal limit shape

P / partially_solved | AIM 研讨会问题汇编

Pick \(n\) points in \(\Lambda\) to maximize the number of adjacent pairs. What is the limit shape as \(n\rightarrow\infty\)?”

“We should specify that \(\Lambda\) is a lattice?

来源字段：http://aimpl.org/discreteaf/1/

06587 | AIM-GEOMETRY-0396

### Affine conformal densities from form Q-operators

S / solved | AIM 研讨会问题汇编

**Problem 34:** Construct other natural tensor-densities which transform according to (43). (Note that any solution yields a conformally invariant natural operator \(L\).) Solutions to

Solutions to Problem 34 have a role to play in the problem of characterizing the \(Q\)-curvature and the GJMS operators.

Equation (43) is not contained in this record. The immediately preceding canonical record and the official AIM source verify that it is \[ N^{\widehat g}=N^g+L^g\omega, \qquad \widehat g=e^{2\omega}g. \tag{43} \] Here \(N^g\) is a natural Riemannian tensor-density and \(L^g\) is a universal linear differential operator. The official source verifies that the truncated final words continue as a separate sentence:

The following paragraph, “Generalizations of \(Q\),” is explanatory material, not a further clause in Problem 34. It describes Branson--Gover operators on closed differential forms.

The word “other” means objects besides critical \(Q\)-curvature, whose density form already satisfies (43) with the critical GJMS operator.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

06588 | AIM-GEOMETRY-0346

### Affine dimension of algebraic amoebas via constant characters

P / partially_solved | AIM 研讨会问题汇编

A. How can examples of amoebas and tropical varieties be computed? B. Can one compute complement homology, and is there a Hermitian form whose signature equals the number of complement components? C. Which classical varieties are expansive, meaning that 0 lies in the amoeba? D. Develop the theory of smooth curves. E. What is the complexity of deciding resultant vanishing? F. When does Log(X) lie in a lower-dimensional subspace? G. What is the complexity of distance to the discriminantal variety?

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

06589 | AIM-GEOMETRY-0335

## Affine phase retrieval and a same-matroid obstruction

P / partially_solved | AIM 研讨会问题汇编

- what the fibers are of $\operatorname{Log}:M\to\mathcal A$, where $M=V\cap(\mathbb C^*)^n$ and $\mathcal A=\operatorname{Log}(M)$; - when this map is a homeomorphism; - what can be said about the topology of the amoeba of a linear space; and - how this relates to the description of the tropicalization by the matroid of the associated hyperplane arrangement.

The source then states that for $d=1$, hyperplanes are points in a complex line: three points not lying on one real line make Log injective, while if all points lie on one real line, the fibers are the $\mathbb Z_2$-orbits of reflection in that line. It also gives an injective $3d$-hyperplane product construction in $\mathbb C^d$.

The notation recovered from the AIM source is $$ \mathbb C^n,\qquad (\mathbb C^*)^n,\qquad \mathbb Z_2,\qquad V=\mathbb C^d, $$ with four separate questions A--D.

There is a substantive convention issue. If “linear subspace” literally means a vector subspace $V\subset\mathbb C^n$ through the origin, the restricted coordinate hyperplanes form a central arrangement. For $d=1$ they all meet at the origin, rather than forming arbitrary points. Moreover, $$ \operatorname{Log}(e^{i\theta}v)=\operatorname{Log}(v), $$ so Log can never be injective on a nonempty homogeneous $V\cap(\mathbb C^*)^n$. The one-dimensional discussion therefore forces the intended convention to be either:

1. an **affine** $d$-plane, parameterized by affine forms $$ \Phi:\mathbb C^d\longrightarrow\mathbb C^n,\qquad z\longmapsto(\ell_1(z),\ldots,\ell_n(z)), $$ with $\ell_j(z)=c_j^*\widetilde z$ and $\widetilde z=(z,1)$; or 2. a projective hyperplane complement together with a choice of affine gauge.

This report adopts the first convention. Then $$ M=\mathbb C^d\setminus\bigcup_{j=1}^nH_j,\qquad H_j=\{\ell_j=0\}, \quad \operatorname{Log}_{\mathcal H}(z) =\bigl(\log|\ell_1(z)|,\ldots,\log|\ell_n(z)|\bigr). $$ The associated matroid used below is the matroid of the homogenized forms $\ell_j^h(z,t)$ on $\mathbb C^{d+1}$. This makes the affine/projective convention explicit rather than silently changing the source.

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

06590 | AIM-GEOMETRY-0277

## Affine rigidity and a mobility-two reduction on the Lorentz 3-sphere

P / partially_solved | AIM 研讨会问题汇编

Question 12.0.6. Can a 3-dimensional sphere admit two nonproportional projectively equivalent metrics of indefinite signature?

The text beginning “Acknowledgements” immediately after the question is end matter, not part of the mathematical statement, and is excluded.

In dimension three, an indefinite metric has Lorentz signature, either \((2,1)\) or \((1,2)\) according to convention. The natural literal reading is that both metrics are indefinite. Projective equivalence itself does not require equal signatures; indeed metrics in one projective class can have different signatures. In odd dimension, multiplying one metric by \(-1\) exchanges \((2,1)\) and \((1,2)\) without changing its Levi-Civita connection, so the ordering convention causes no substantive distinction here.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06591 | AIM-GEOMETRY-0353

## Affine tractor factories and Q-like tensor-densities

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.** Construct other natural tensor-densities which transform according to (34). (Any solution yields a conformally invariant natural operator \(L\).)

“It is clear that solutions to Problem 2 have a role to play in the problem of characterising the \(Q\)-curvature and the GJMS operators.”

Immediately before the problem, the notes define a natural tensor-density \(N_g\) to have a linear conformal change law when

\[ N_{\widehat g}=N_g+L_g\omega, \qquad \widehat g=e^{2\omega}g. \tag{34} \]

The problem is therefore:

The printed text says “(43),” but the immediately preceding displayed formula is numbered (34), and equation (43) occurs only later in a compiled list. Thus “(43)” is an internal cross-reference typo. The duplicate phrase “of the the form” is also in the PDF. Finally,

\[ \iota(D)|C|^2I_g=4\Delta |C|^2 \]

The missing end of the record is:

The preceding notation uses the standard tractor bundle \(\mathcal T\), density bundles \(\mathcal E[w]\), the scale-dependent tractor \(I_g\in\mathcal T[-1]\), and the tractor \(D\)-operator. The key identity is

\[ I_{e^{2\omega}g}=I_g+D\omega. \tag{33} \]

If \(a_g\) is the ordinary scalar representative of \(A_g\in\mathcal E[-n]\), then

\[ e^{n\omega}a_{\widehat g}=a_g+L_g\omega. \]

This factor is essential in coordinate checks.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

06592 | AMR-059-0013

### Affine Vertices and Inflection Points

P / partially_solved | AMR 问题清单汇编

For a closed curve on a Riemann surface with projectively flat connection, estimate the number of affine vertices and, assuming isolated inflection points, relate the number of affine vertices to the number of inflection points.

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

06593 | AIM-CONVEX_GEOMETRY-0002

### Affinely equivalent central sections and an ellipsoidal anchor

P / partially_solved | AIM 研讨会问题汇编

Let \(G(n,k)\) denote the set of \(k\)-dimensional subspaces of \(\mathbb R^n\), where \(2\leq k\leq n-1\) is fixed in the following problems. If \(K\) is a set let \(\operatorname{int}K\) denote its interior. Also let \(o\) denote the origin in \(\mathbb R^n\).

(See [G, Problem 7.4 and Note 7.2].) Suppose \(o\in\operatorname{int}K\) and all sections \(K\cap S\), \(S\in G(n,k)\), are affinely equivalent. Does it follow that \(K\) is an ellipsoid? Here \(K\) is a star body but the problem is also open when \(K\) is convex.

来源字段：https://aimath.org/pastworkshops/sectionsconvexproblems.pdf

06594 | AIM-CONVEX_GEOMETRY-0003

### Affinely equivalent projections and an ellipsoidal 2-cover criterion

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.** (See [G, Problem 3.3 and Note 3.2].) Same problem as in (1) but for projections, i.e. if all projections \(K|S\), \(S\in G(n,k)\), are affinely equivalent, is \(K\) an ellipsoid? Here \(K\) is convex.

来源字段：https://aimath.org/pastworkshops/sectionsconvexproblems.pdf

06595 | OWR-1454-002

### Agreement of Mean Curvatures on Overlapping Varifolds

S / solved | Oberwolfach 报告问题汇编

Let $\mu_1$ and $\mu_2$ be integral $n$-varifolds with generalized mean curvatures $H_{\mu_1}\in L^p(\mu_1)$ and $H_{\mu_2}\in L^p(\mu_2)$. Is $H_{\mu_1}(x)=H_{\mu_2}(x)$ for $\mu_1$-almost every $x\in\{\theta^{*n}(\mu_1)>0\}\cap\{\theta^{*n}(\mu_2)>0\}$, where $\theta^{*n}$ denotes the $n$-density?

来源字段：https://doi.org/10.4171/owr/2007/03

06596 | AIM-GEOMETRY-0136

### Airy topological recursion, exact WKB, and the wild Hodge gap

P / partially_solved | AIM 研讨会问题汇编

Describe the relation between the WKB, Eynard-Orantin, Dumitrescu-Mulase topological recursion and the (wild) non-abelian Hodge correspondence, at least in one example.

来源字段：http://aimpl.org/spectralhiggs/1/

06597 | AIM-GEOMETRY-0159

### ALE singularities: Type-II existence and a bounded-scalar scale-energy certificate

S / solved | AIM 研讨会问题汇编

In dimension $4$, can there be an ALE space as singularity for Ricci Flow?

来源字段：http://aimpl.org/flowriemannian/1/

06598 | AIM-GEOMETRY-0144

### Algebraic and analytic automorphisms in rank one

P / partially_solved | AIM 研讨会问题汇编

Classify algebraic and holomorphic automorphisms of the de Rham and Betti moduli spaces, with appropriate fixed rank, topological data, and determinant conditions.

来源字段：http://aimpl.org/spectralhiggs/1/

06599 | DARPA-010

### Algorithmic Origami and Biology

P / partially_solved | DARPA 挑战

Strengthen mathematical theory for isometric and rigid embedding relevant to protein folding.

背景链接：https://web.math.utk.edu/~vasili/refs/darpa07.MathChallenges.html

06600 | OWR-12697711-005

### Algorithmic Recognition of Zonoids

O / open | Oberwolfach 报告问题汇编

Given a polynomial $p$ for which $K=\{x\in\mathbb R^d:p(x)\geq 0\}$ is a centrally symmetric convex body, decide algorithmically whether $K$ is a zonoid.

来源字段：https://doi.org/10.4171/owr/2023/15

06601 | AIM-GEOMETRY-0319

### All integer volume-growth exponents from flat hyperkähler quotients

P / partially_solved | AIM 研讨会问题汇编

Let \(M\) be a connected noncompact hyperkähler manifold. Fix a point \(p\in M\). Consider the open ball \(B(p,r)\) of radius \(r\) around \(p\) in \(M\). Describe the asymptotic behavior of \(\operatorname{Vol}(B(p,r))\) as \(r\to\infty\). (Fact: this is independent of the choice of \(p\in M\).) It would be interesting to search for examples of hyperkähler manifolds with different volume growth.

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

06602 | AIM-GEOMETRY-0304

### All-root abelianization for cotangent Grassmannians

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.14 (Hausel 2000).** Suppose that both \(T^*A////G\) and \(T^*A////T\) are hyper-compact. Then \[ H^*(T^*A////G)\cong \frac{H^*(T^*A////T)^W}{\operatorname{Ann}(\widetilde e)}, \qquad \widetilde e=\prod_{\alpha\in\Delta}\alpha \in (\operatorname{Sym}\mathfrak t^*)^W \cong H_T^*(\mathrm{pt})^W. \]

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

06603 | OWR-3385-008

### Alternating Cyclic-Quotient Identities over Commutative Fields

O / open | Oberwolfach 报告问题汇编

Let $K$ be a commutative field and let $A_1,\ldots,A_n,A'_1,\ldots,A'_n\in K^2$. Define $$\alpha(A_1,\ldots,A_nmid A'_1,\ldots,A'_n)=\frac{[A_1A_2]\cdots[A_nA_1]}{[A'_1A'_2]\cdots[A'_nA'_1]}.$$ Is $$\sum_{\substack{\pi=(1,\pi_2,\ldots,\pi_n)\in S_n}}\operatorname{sgn}(\pi)\,\alpha(\pi(A_1,\ldots,A_n)mid\pi(A'_1,\ldots,A'_n))=0$$ for every $n$?

来源字段：https://doi.org/10.4171/owr/2009/02

06604 | AIM-GEOMETRY-0019

### Alternative weave tori and the cup obstruction

P / partially_solved | AIM 研讨会问题汇编

What can we say if $X$ is the cluster variety of a $(-1)$-closure, but $T$ comes from some weave other than the obvious/natural one?

The exact AIM record asks:

This is Problem 2.4 in the workshop section “Stratifications of braid varieties.” The preceding item asks about the geometry of $X\setminus T$ for a general cluster variety and cluster torus. Thus the intended question appears to be: for the cluster/augmentation variety attached to the Legendrian $(-1)$-closure of a positive braid, what is the status and complement of the torus furnished by a non-natural weave?

The words “weave” and “comes from” need a modern distinction that was not explicit in the AIM sentence:

1. For a positive braid word $\beta$ with Demazure product $w_0$, a **Demazure weave** $W:\beta\to\Delta$ uses braid moves, commuting moves, and folds $\sigma_i^2\to\sigma_i$, but no cups or caps. It determines a full-dimensional open torus $T_W\subset X(\beta)$. 2. A **simplifying weave** may also use cups $\sigma_i^2\to 1$ (but no caps). Its algebraic correspondence gives a locally closed piece $S_W\cong\mathbb A^c\times(\mathbb G_m)^t$, where $c$ and $t$ are the numbers of cups and trivalent vertices. This piece is generally not a torus and need not be open. 3. A Legendrian weave or filling can also give a local-system torus in an augmentation variety. The assertion that every such torus is a cluster seed torus is stronger than the theorem for Demazure weave tori and is not established by the papers checked.

In type $A$, with the usual marking convention, the relevant modern identification is \[ X(\beta)\cong \operatorname{Aug}(\Lambda(\beta\Delta)), \] where $\Lambda(\beta\Delta)$ is the Legendrian $(-1)$-closure. The precise placement of $\Delta$ varies among sources; all results below are stated on the braid-variety side, where the conventions are unambiguous.

来源字段：http://aimpl.org/clusterbraid/2/

06605 | AIM-GEOMETRY-0352

### Ambient tractor Laplacians and affine conformal laws

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.** Give general formulae or inductive formulae for the operators \(\square_{2\ell}\).

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

06606 | AIM-GEOMETRY-0126

### An ambiguous quantization diagram and a projective-flatness obstruction

O / open | AIM 研讨会问题汇编

Let $X$ be a surface and fix topological data. Then: \[ X\rightarrow Bun_{G}\left(X\right)\rightarrow H^{0}\left( Bun_{G},\alpha\right) \] \[ \underline{X} \rightarrow Bun_{G}\left(X\right)\rightarrow U \qquad (\star) \] Is $(\star)$ naturally projectively flat?

来源字段：http://aimpl.org/spectralhiggs/1/

06607 | AIM-GEOMETRY-0241

### An angular Jacobi barrier for divergence of geodesic rays

P / partially_solved | AIM 研讨会问题汇编

Let \(\alpha\) and \(\beta\) be two geodesics in a complete simply connected Riemannian manifold without conjugate points.

If \(X\) is complete and simply connected without conjugate points, \(p\in X\), and \(v\ne w\) are unit vectors in \(T_pX\), must the unit rays \(\gamma_v(t)=\exp_p(tv)\) and \(\gamma_w(t)=\exp_p(tw)\) satisfy \(d(\gamma_v(t),\gamma_w(t))\to\infty\)?

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06608 | AIM-CONVEX_GEOMETRY-0032

### An attained symmetry-reduced cosine program for Schneider's zonoid-sandwich parameter

P / partially_solved | AIM 研讨会问题汇编

For an origin-symmetric convex body \(K\subset\mathbb R^n\), let \(\lambda(K)\) be the smallest number \(\lambda\geq1\) such that there is a zonoid \(Z\) with \[ K\subseteq Z\subseteq \lambda K. \] For the cross-polytope \(C_n=\operatorname{conv}\{\pm e_1,\ldots,\pm e_n\}\), \[ \lambda(C_n)=\frac{n}{2^{n-1}} \binom{n-1}{\lfloor (n-1)/2\rfloor} \sim \sqrt{\frac{2n}{\pi}}. \] Is \(\lambda(K)\) maximal for the cross-polytope?

来源字段：https://aimath.org/WWN/fourierconvex/fourierconvex.pdf

06609 | AIM-GEOMETRY-0100

### An exact E8 shell moment diagnostic and audit of the Lehmer equivalence

P / partially_solved | AIM 研讨会问题汇编

Is it true that no shell of \(E_8\) is a \(8\)-design? (This is equivalent to Lehman's conjecture)

来源字段：http://aimpl.org/discreteaf/1/

06610 | AIM-CONVEX_GEOMETRY-0031

### An exact one-harmonic threshold for polar-zonoid candidates

P / partially_solved | AIM 研讨会问题汇编

**Problem 1 (R. Schneider).** Let \(Z_n\subset\mathbb R^n\) be a zonoid whose polar \(Z_n^\circ\) is also a zonoid. Let \(d\) denote the Banach--Mazur distance. Does \[ d(Z_n,B_2^n)\longrightarrow 1 \] as \(n\to\infty\)?

来源字段：https://aimath.org/WWN/fourierconvex/fourierconvex.pdf

06611 | AIM-CONVEX_GEOMETRY-0018

### An exact quantitative shadow-system model

P / partially_solved | AIM 研讨会问题汇编

2. (S. Reisner) Quantify the shadow movement theorem (see [MR]).

**22. (S. Reisner) Quantify the shadow movement theorem (see [MR]).**

Here [MR] is Meyer and Reisner, *Shadow systems and volumes of polar convex bodies*. Their Theorem 1 is the precise qualitative theorem meant by the AIM question: if `(K_t)_{t in I}` is a nondegenerate shadow system, then

\[ F(t):=|K_t^*|^{-1} \]

is convex. The star denotes polarity about the Santaló point, not polarity about a fixed origin in general. The AIM phrase "shadow movement theorem" is informal; it is not the title of a theorem in [MR].

This report interprets "quantify" as asking for a positive lower bound on a Jensen deficit, after normalizing the shadow speed and removing the affine equality directions. It gives an exact nontrivial model calculation and explains why those qualifications are unavoidable. It does **not** claim a general quantitative Meyer--Reisner theorem.

来源字段：https://aimath.org/WWN/mahlerduality/mahlerduality.pdf

06612 | AIM-GEOMETRY-0201

### An expansive tangent obstruction to symmetry breaking

P / partially_solved | AIM 研讨会问题汇编

4. (Schulze) Suppose a symmetric framework (linkage) has a 1DOF expansive mecha-nism. Does the mechanism preserve the symmetry?

1

**14. (Schulze)** Suppose a symmetric framework (linkage) has a 1DOF expansive mechanism. Does the mechanism preserve the symmetry?

来源字段：https://aimath.org/pastworkshops/linkagesproblems.pdf

06613 | AIM-CONVEX_GEOMETRY-0025

### An explicit iid-uniform counterexample to scalar entropy-power supermodularity

S / solved | AIM 研讨会问题汇编

The AIM list records Problem 6, attributed to Mokshay Madiman. In the source's notation, \(H\) is differential entropy and the variables are independent, real-valued random variables with densities. The question is whether

\[ e^{2H(X_1+X_2+X_3)}+e^{2H(X_2)} \mathrel{\geq?} e^{2H(X_1+X_2)}+e^{2H(X_2+X_3)}. \tag{1} \]

The typography and exponent in the AIM PDF are exactly \(H\) and \(e^{2H}\); there is no missing dimension factor because the variables in (1) are scalar. Below, \(h\) denotes the same differential entropy and

\[ N(Y):=e^{2h(Y)}. \tag{2} \]

The source remarks that equality holds for Gaussian variables and proposes a related inequality for subset sums \(X_S=\sum_{i\in S}X_i\). It calls the weights in that remark a fractional *covering*. Section 7 below explains a necessary correction: the published entropy-power theorem uses a fractional partition (and consequently also gives the weaker assertion for a fractional packing), whereas unrestricted covering weights cannot satisfy the stated lower bound.

来源字段：https://aimath.org/WWN/convexgeometry/convexgeometry.pdf

06614 | AMR-046-0023

### An extended Poncelet problem I

S / solved | AMR 问题清单汇编

Do there exist two irreducible algebraic curves of degrees $n$ and $m$, with $n+m>4$, each having an oval, for which the Poncelet map is well defined and conjugate to a rotation of the circle?

背景链接：https://arxiv.org/abs/2012.02524

06615 | AMR-046-0024

### An extended Poncelet problem II

O / open | AMR 问题清单汇编

Let $\gamma=\{x^2+y^2-1=0\}$ and $\Gamma_\varepsilon=\{p_2(x,y)+\varepsilon p_m(x,y)=0\}$, where $\Gamma_0$ is an ellipse surrounding $\gamma$, the curve $p_2+\varepsilon p_m=0$ is irreducible, $\deg p_m=m\geq3$, and $\varepsilon$ is small. Is the Poncelet map for the two ovals conjugate to a rotation if and only if $\varepsilon=0$?

背景链接：https://arxiv.org/abs/2012.02524

06616 | AIM-GEOMETRY-0301

### An integral Kirwan presentation for cotangent Grassmannians

P / partially_solved | AIM 研讨会问题汇编

**AIM Conjecture 6.1 (recovered).** Let a compact group \(G\) act unitarily on a finite-dimensional complex vector space \(A\), and let \(\xi\) be a regular central value for the hyperkähler moment map on \(T^*A\). Then the hyperkähler Kirwan map \[ \kappa:H_G^*(T^*A)\cong H^*(BG)\longrightarrow H^*(T^*A///_\xi G) \] is surjective.

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

06617 | AIM-GEOMETRY-0021

### An inverted complement and a braid-link codimension criterion

O / open | AIM 研讨会问题汇编

**Reconstructed question.** When \(P\ne\varnothing\), when should the peripheral locus \(P=X\setminus M\) itself have codimension two?

来源字段：http://aimpl.org/clusterbraid/3/

06618 | AIM-GEOMETRY-0412

### An orbit-category and Rees-deformation model for Legendrians in open books

P / partially_solved | AIM 研讨会问题汇编

Auroux asked whether there is a category for Legendrians in open books, analogous to the Fukaya category of Lagrangians in a symplectic manifold. Morphisms between two Legendrians cannot in general be isolated from their self-chords, so coefficients in the contact homology of the individual Legendrians may be needed. The open-book angular direction suggests a filtration by the rotation of Reeb chords. If two Legendrians lie on one page, their contact-homological morphisms should involve all iterated monodromy images rather than only their ordinary Lagrangian Floer homology. Restoring the binding might be a deformation of an \(A_\infty\)-category. Relative contact homology should also be a module over absolute contact homology. Eliashberg suggested studying the gluing by rounding the convex, flat, and concave boundary pieces in the symplectization of the binding complement and then partially gluing along the resulting convex boundary.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

06619 | OWR-1195-005

### Analytic Functions on Flat-Connection Representation Varieties

O / open | Oberwolfach 报告问题汇编

Does a gauge-equivariant analytic function on flat connections of a flat bundle $E\to M$ induce an analytic function on the representation variety $\operatorname{Hom}(\pi_1(M),\mathbb{C})$?

来源字段：https://doi.org/10.4171/owr/2006/13

06620 | OWR-16942-003

### Ancient Lagrangian Mean-Curvature Flows with Planar Blow-Downs

P / partially_solved | Oberwolfach 报告问题汇编

Can one classify the exact, almost calibrated, ancient solutions to Lagrangian mean curvature flow in $\mathbb{C}^2$ whose blow-down is a pair of planes intersecting in a line?

来源字段：https://doi.org/10.4171/owr/2019/30

06621 | AIM-GEOMETRY-0195

### Angle-data reconstruction and the 6g genus deficit

P / partially_solved | AIM 研讨会问题汇编

(Hempel) Consider a simplicial polyhedra with fixed combinatorics in \(\mathbb R^3\). Question: characterize those collections of dihedral angles and face angles that can be realized by a polyhedron with these combinatorics. Conjecture: dimension = \(E-1\), where \(E\) is the number of edges. Needs more clarification.

来源字段：https://aimath.org/pastworkshops/linkagesproblems.pdf

06622 | OWR-1394-010

### Approximation of Smooth Convex Bodies by Inscribed Polytopes

O / open | Oberwolfach 报告问题汇编

How well can a smooth convex body in $\mathbb{R}^d$ be approximated by an inscribed polytope of restricted complexity, especially a prescribed number of vertices, measured by affine-invariant quantities such as Banach-Mazur distance or volume difference?

来源字段：https://doi.org/10.4171/owr/2006/56

06623 | AIM-GEOMETRY-0167

### Arbitrary asymptotic separation of graphical mean curvature flow and heat flow

S / solved | AIM 研讨会问题汇编

For \(d=1\), if we have a bounded \(C^{2,\alpha}\) graph, let \(c(t)\) be the curve shortening flow and let \(u(t)\) be the solutions to be heat equation. Nara and Taniguchi proved that \[ |c(t)-u(t)|\to0\quad\text{as }t\to\infty. \] Is there a higher dimension analog? The rotationally symmetric case is known.

来源字段：http://aimpl.org/flowriemannian/2/

06624 | AIM-GEOMETRY-0179

### Arbitrary-genus realization and a topology-curvature tradeoff

P / partially_solved | AIM 研讨会问题汇编

Existence of high genus free boundary minimal surface in $B^3$.

The exact AIM record (Geometry, “Geometric flows and Riemannian geometry,” section “Eigenvalue Estimates,” Problem 5.2) is:

I use the standard strong reading:

* $B^3=\{x\in\mathbb R^3:|x|\leq 1\}$ is the closed unit ball; * $\Sigma\subset B^3$ is a smooth, compact, connected, orientable, properly embedded surface, so $\Sigma\cap\partial B^3=\partial\Sigma$; * its mean curvature vanishes, and it meets $S^2=\partial B^3$ orthogonally (the free-boundary condition); * “genus” is orientable genus, and “high genus” can mean either genera tending to infinity, every sufficiently large genus, or the strongest version, every prescribed genus; * unless stated otherwise, the number $b$ of connected components of $\partial\Sigma$ is not prescribed.

The source does not say “embedded,” “orientable,” or “connected boundary.” Modern results solve the question with all of those additional properties: one may prescribe every genus, and even every positive number of boundary components. Nothing below silently changes the source; these are explicit strengthened readings. Nonorientable crosscap genus is a different realization problem and is not asserted here.

来源字段：http://aimpl.org/flowriemannian/5/

06625 | AMR-067-0008

### Area minimizing projective spaces in the projective space with the Berger metric

P / partially_solved | AMR 问题清单汇编

For ${2n+1}>3$ and $0<k<{2n+1}$, are projective subspaces obtained by projection of the $k$-dimensional equatorial spheres minimal submanifolds of the Berger projective space $( \mathbb{R} P^{2n+1}, g_\mu)$? Are they the only $k$-dimensional volume minimizing cycle in their homology classes?

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

06626 | OWR-4334-002

### Area-Minimizing Intrinsic Graphs in the Heisenberg Group

O / open | Oberwolfach 报告问题汇编

Classify the complete area-minimizing intrinsic graphs in the sub-Riemannian Heisenberg group $\mathbb H^1$. In particular, does the known classification of complete stable area-stationary intrinsic graphs extend to all complete area-minimizing intrinsic graphs?

来源字段：https://doi.org/10.4171/owr/2010/21

06627 | OWR-14298170-015

### Area-Nonincreasing Smoothing of Lipschitz Maps

O / open | Oberwolfach 报告问题汇编

Let $f:M\to S^n$ be Lipschitz and satisfy $\Lambda^2f^*g_{S^n}\leq\Lambda^2g$ almost everywhere. Can $f$ be smoothed to a map $F$ satisfying $\Lambda^2F^*g_{S^n}\leq\Lambda^2g$ everywhere?

来源字段：https://doi.org/10.4171/owr/2024/9

06628 | OWR-3389-006

### Area-Refined Spectral Gap Bounds for Convex Domains

P / partially_solved | Oberwolfach 报告问题汇编

For a planar convex domain $K$ with diameter $d$, spectral gap $\gamma(K)=\lambda_2(K)-\lambda_1(K)$, area $A$, and $B_1$ the unit-area disk, is $$\gamma(K)\geq\frac{12\pi^2\gamma(B_1)}{3\pi^3d^2+[4\gamma(B_1)-3\pi^3]\sqrt{d^4-16A^2/\pi^2}}?$$ Is equality attained only by the disk, or asymptotically by domains converging to an infinite strip?

来源字段：https://doi.org/10.4171/owr/2009/06

06629 | OWR-14299512-002

### Area-Spectral Rigidity of Convex Planar Domains

P / partially_solved | Oberwolfach 报告问题汇编

If two convex planar domains $\Omega$ and $\Omega'$ have equal area and the same billiard length spectrum, must they differ by a unitary affine transformation?

来源字段：https://doi.org/10.4171/owr/2025/31

06630 | AIM-GEOMETRY-0348

### Arithmetic tau interpretation and flat tropical Monge-Ampere rigidity

P / partially_solved | AIM 研讨会问题汇编

A. From number theory: Consider one polynomial in one variable with integer coeffi-cients, written as a straight line program of complexity τ. How many roots are there which are congruent to 1 modulo p, as a function of τ?There exist estimates using amoeba theory, by work of Maurice Rojas. B. Prove Calabi conjecture for tropical Calabi-Yaus: in each metric class there is a unique metric which solves the tropical Monge-Amp` ere measure. In the real case this corresponds to det(Hessian) = const.

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

06631 | OWR-13951-006

### Associativity in Operations on Convex Bodies

O / open | Oberwolfach 报告问题汇编

What is the role of associativity in classifying operations between arbitrary compact convex sets?

来源字段：https://doi.org/10.4171/owr/2015/56

06632 | AIM-GEOMETRY-0418

### Asymptotic spectral invariants and field-factor slopes

P / partially_solved | AIM 研讨会问题汇编

Polterovich mentioned spectral invariants in Floer homology. Namely, for (f\in \operatorname{Ham}(M)) and \(\alpha\in QH^*(M)\), define \(c(f,\alpha)\) to be the smallest \(c\) such that \(\alpha\) appears in \(HF^*(\mathcal A\leq c)\), where \(\mathcal A\) is the action functional on a suitable cover of the loop space. Question: study the asymptotic properties of \(c(f^n,\alpha)\).

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

06633 | OWR-4136-012

### Asymptotic Volume Parameters for Euclidean Balls

O / open | Oberwolfach 报告问题汇编

For Euclidean balls, is the sequence $\Theta(B_2^n)$ bounded, and is it true that for every $A>0$ one has $\operatorname{vol}(\widetilde{E}_A)=1-o(1)$ as $n\to\infty$?

来源字段：https://doi.org/10.4171/owr/2009/53

06634 | OWR-1319-022

### Asymptotically Equivalent Cocompact Metrics

S / solved | Oberwolfach 报告问题汇编

Let a group $\Gamma$ act cocompactly by isometries on the complete length spaces $(M,d_1)$ and $(M,d_2)$, with $d_1(x,y)/d_2(x,y)\to1$ as $d_1(x,y)\to\infty$. Must $|d_1-d_2|$ be uniformly bounded?

来源字段：https://doi.org/10.4171/owr/2006/33

06635 | OWR-12867-004

### Asymptotics Along Rays in Hitchin Components

P / partially_solved | Oberwolfach 报告问题汇编

Let $S$ be a closed oriented surface of genus at least $2$, let $(E,\varphi)$ be a stable $\operatorname{SL}(n,\mathbb C)$-Higgs bundle, and let $(E,t\varphi)$ determine flat connections $\nabla_t$, representations $\rho_t$, and parallel-transport matrices $T_{P,Q}(t)$. Determine the asymptotic behavior of $\rho_t$ and $T_{P,Q}(t)$ as $t\to\infty$, particularly along rays in the Hitchin component.

来源字段：https://doi.org/10.4171/owr/2014/07

06636 | OWR-1275-003

### Asymptotics of Stretch Lines in Teichmüller Space

P / partially_solved | Oberwolfach 报告问题汇编

Does a stretch line in Teichmüller space converge to a point of Thurston's boundary, and what is its asymptotic behavior?

来源字段：https://doi.org/10.4171/owr/2006/26

06637 | AIM-GEOMETRY-0008

### Attainment and a quantitative segment bound for planar p-capacity

P / partially_solved | AIM 研讨会问题汇编

The source asks the following. For \(1<p<2\) and a bounded convex set \(K\subset\mathbb R^2\), define the homogeneous variational capacity \[ \operatorname{Cap}_p(K)=\inf\left\{ \int_{\mathbb R^2}|\nabla u|^p\,dx: u\in C_c^1(\mathbb R^2),\quad u\geq1 \text{ on a neighbourhood of }K\right\}. \] Among convex sets of prescribed perimeter, is this capacity minimized by a segment, with a segment's perimeter counted twice?

There is one necessary correction to the source text. At \(p=2\) the displayed infimum is mathematically well-defined, but it is degenerate: it equals zero for every compact planar set. Indeed, if \(K\subset B_R\), logarithmic radial cutoffs supported in \(B_A\) have energy tending to \(2\pi/\log(A/R)\to0\). Logarithmic capacity is the nondegenerate replacement.

To include flat limits without ambiguity, throughout this report the perimeter of every nonempty compact convex set is \[ P(K):=\int_0^\pi w_K(\theta)\,d\theta, \qquad w_K(\theta)=h_K(e_\theta)+h_K(-e_\theta). \] This is the usual boundary length when \(K\) has interior, while \(P([a,b])=2|a-b|\). It is translation invariant, homogeneous, Minkowski additive, and continuous for Hausdorff convergence.

来源字段：http://aimpl.org/symmetrybreaking/2/

06638 | AIM-GEOMETRY-0007

### Attainment and normalization transfer for Steklov polygons

P / partially_solved | AIM 研讨会问题汇编

Given an integer \(n\ge 3\), what is the maximal first positive Steklov eigenvalue \(\sigma_1\) among \(n\)-sided polygonal sets of fixed area, or of fixed perimeter? Even \(n=3,4\) are said to be open, and corners must be treated carefully in a shape derivative.

The exact AIM record, attributed there to Iosif Polterovich, asks:

It leaves several mathematically consequential choices unstated:

- Is a polygon required to be simple, convex, or merely connected? - Does “\(N\)-gon” mean exactly \(N\) genuine corners, at most \(N\) sides, or \(N\) labelled vertices with collinear/repeated labels allowed? - Are holes, self-intersections, disconnected unions, or zero-area degeneracies allowed, and if so how are their sides and boundary lengths counted? - Is the perimeter the total length of every boundary component?

For status statements about the customary conjecture, this report reads an \(N\)-gon as a bounded simply connected Lipschitz domain whose boundary is a simple polygonal Jordan curve. The proved existence theorem uses the more precise closed class \(\mathcal C_N\) of interiors of compact **convex** polygons with nonempty interior and at most \(N\) maximal sides, modulo rigid motions. Convexity is a substantive restriction: Steklov eigenvalues are not domain-monotone, so one cannot pass to a convex hull without proof.

Using “at most \(N\)” is essential for compactness. If collinear subdivision of an edge is permitted, it is also exactly the geometric closure of the class of labelled \(N\)-gons. If exactly \(N\) noncollinear, distinct vertices are required, the admissible class is not closed and the existence conclusion below need not follow.

来源字段：http://aimpl.org/symmetrybreaking/2/

06639 | OWR-12581-002

### Auslander Conjecture for Complete Affine Manifolds

O / open | Oberwolfach 报告问题汇编

Is every closed complete affine manifold finitely covered by a complete affine solvmanifold $G/\Gamma$, where $G$ is a closed solvable subgroup of the affine group?

来源字段：https://doi.org/10.4171/owr/2013/33

06640 | OWR-782-004

### Auxiliary Functions for Optimal Leech and $E_8$ Packings

S / solved | Oberwolfach 报告问题汇编

In dimensions $24$ and $8$, construct a nonzero admissible function $f$ with $f(0)=\widehat f(0)>0$, $f(x)\leq0$ for $|x|\geq2$, and $\widehat f(t)\geq0$ for all $t$, thereby proving the sharp linear-programming bounds for the Leech lattice and $E_8$ sphere packings.

来源字段：https://doi.org/10.4171/owr/2005/01

06641 | OPG-605

### Average diameter of a bounded cell of a simple arrangement

P / partially_solved | Open Problem Garden

Conjecture The average diameter of a bounded cell of a simple arrangement defined by $n$ hyperplanes in dimension $d$ is not greater than $d$.

背景链接：http://www.openproblemgarden.org/op/average_diameter_of_a_bounded_cell_of_a_simple_arrangement

06642 | AIM-GEOMETRY-0080

### Average optimality versus a smallest individual Voronoi cell

P / partially_solved | AIM 研讨会问题汇编

Are the Voronoi cells of the optimal $8$ or $24$ dimensional sphere packings as small as possible by volume?

来源字段：http://aimpl.org/discreteaf/1/

06643 | GREEN-051

### Axis-Parallel Rectangles in Dense Sets

O / open | Ben Green 问题

Suppose $A$ is an open subset of $[0, 1]^2$ with measure $\alpha$. Are there four points in $A$ determining an axis-parallel rectangle with area $\geq c\alpha^2$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

06644 | AMR-065-0020

### B. Weiss - Problem

P / partially_solved | AMR 问题清单汇编

Let $E\subset\mathbb{R}^k$ be a totally irrational subspace of dimension $d\ge 1$, and let $Y$ be a cut--and--project set obtained from $E$ using a bounded window $\mathcal{W}$ with non--empty interior and with the property that the $(k-d)$--dimensional Lebesgue measure of $\partial\mathcal{W}$ is zero. Is such a set $Y$ always BL to a lattice in $E$?

背景链接：https://arxiv.org/abs/1604.06280

06645 | AMR-005-0007

### Baker's Dozen - A converse Desargues theorem

P / partially_solved | AMR 问题清单汇编

Let $f(x,y)$ be a polynomial for which $0$ is a nonsingular value, let $\gamma$ be an oval component of $f(x,y)=0$, and assume that $\gamma_\varepsilon=\{f(x,y)=\varepsilon\}$ for $\varepsilon>0$ foliates an outer neighborhood of $\gamma$. Suppose that for every tangent line $\ell$ to $\gamma$, its intersections with the curves $\gamma_\varepsilon$ define a local projective involution on $\ell$. Prove that $\gamma$ is an ellipse and the curves $\gamma_\varepsilon$ form a pencil of conics.

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/14-01/

06646 | AMR-005-0014

### Baker's Dozen - A totally skew disc

O / open | AMR 问题清单汇编

Does there exist a totally skew embedded $3$-disc in $\mathbb{R}^7$?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/14-01/

06647 | AMR-005-0008

### Baker's Dozen - Chains of null geodesics

P / partially_solved | AMR 问题清单汇编

For the ellipsoid $x^2/a+y^2/b+z^2/c=1$ in Minkowski space with metric $dx^2+dy^2-dz^2$, find conditions on $a,b,c>0$ ensuring the existence of an $(n,r)$-chain of alternating left and right null geodesics.

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/14-01/

06648 | AMR-005-0001

### Baker's Dozen - Commuting billiard maps

S / solved | AMR 问题清单汇编

Consider two nested convex plane domains, and let $T_1,T_2$ be their billiard ball maps on the oriented lines intersecting both domains. If $T_1\circ T_2=T_2\circ T_1$, are the two domains bounded by confocal ellipses? Resolve the corresponding multidimensional question for inner and outer billiards.

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/14-01/

06649 | AMR-005-0005

### Baker's Dozen - Completely periodic hyperbolic outer billiards

O / open | AMR 问题清单汇编

Another problem is to describe polygonal outer billiard tables in the hyperbolic plane for which all orbits are periodic. For example, right-angled regular $n$-gons (with $n \geq 5$) have this property (Dogru and Tabachnikov [2003]). In the affine plane, every outer billiard orbit about a lattice polygon is periodic.

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/14-01/

06650 | AMR-005-0013

### Baker's Dozen - Configuration theorems

S / solved | AMR 问题清单汇编

Find conceptual proofs and possible generalizations of the new projective-geometry configuration theorems described in the source. In particular, prove that when the outer dodecagon in Figure 5 is inscribed in a conic, the inner dodecagon is also inscribed in a conic.

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/14-01/

06651 | AMR-005-0011

### Baker's Dozen - Convex tangent-segment iteration

P / partially_solved | AMR 问题清单汇编

Given an oriented oval $\gamma$, let $\gamma_1$ be the locus of endpoints of its oriented unit tangent segments, and iterate this construction. If every iterate is convex, must $\gamma$ be a circle?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/14-01/

06652 | AMR-005-0003

### Baker's Dozen - Lorentzian Birkhoff theorem

P / partially_solved | AMR 问题清单汇编

For billiards inside an oval in the Lorentz plane with metric $ds^2=dx^2-dy^2$, is there an analogue of Birkhoff's theorem, with separate existence statements for space-like and time-like periodic trajectories? What happens for multidimensional pseudo-Euclidean billiards?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/14-01/

06653 | AMR-005-0009

### Baker's Dozen - Origami hyperbolic paraboloids

S / solved | AMR 问题清单汇编

Assume the origami hyperbolic-paraboloid pattern has invisible straight folds along a chosen diagonal of each elementary trapezoid. What is the shape of the piecewise-linear surface obtained by folding? What is obtained from analogous constructions with other patterns of fold lines?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/14-01/

06654 | AMR-005-0002

### Baker's Dozen - Periodic billiard trajectories

P / partially_solved | AMR 问题清单汇编

Are there smooth convex curves, other than ellipses, simultaneously admitting one-parameter families of $p$- and $q$-periodic billiard trajectories for $p\ne q$? Ask the analogous question for outer billiards.

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/14-01/

06655 | AMR-005-0004

### Baker's Dozen - Periodic hyperbolic outer billiards

O / open | AMR 问题清单汇编

Does every polygonal outer billiard in the hyperbolic plane have periodic orbits, possibly lying on the circle at infinity?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/14-01/

06656 | AMR-005-0006

### Baker's Dozen - Periodic multidimensional outer-billiard trajectories

P / partially_solved | AMR 问题清单汇编

For a smooth strictly convex hypersurface $M\subset\mathbb{R}^{2n}$, find lower bounds for the number of $p$-periodic outer-billiard orbits for values of $p$ other than $3$.

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/14-01/

06657 | AMR-005-0015

### Baker's Dozen - Relations among triangle areas

S / solved | AMR 问题清单汇编

For a fixed combinatorial dissection of a square into $n$ triangles, the ordered triangle areas satisfy a polynomial relation depending only on the combinatorics. What can be said about this relation, and how can its least degree be found from the combinatorics of the dissection?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/14-01/

06658 | AMR-005-0012

### Baker's Dozen - Self-dual curves and surfaces

P / partially_solved | AMR 问题清单汇编

Extend the known results on projectively self-dual polygons and curves to projectively self-dual polyhedra and to projectively self-dual polygons in multidimensional projective spaces.

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/14-01/

06659 | AMR-005-0010

### Baker's Dozen - Unbounded unicycle tracks

P / partially_solved | AMR 问题清单汇编

Let $\gamma$ be a smooth arc agreeing to all orders with the $x$-axis at endpoints $(0,0)$ and $(1,0)$, and iterate $T(\gamma)=\gamma+\gamma'$. Unless $\gamma$ is a straight segment, is the resulting infinite curve unbounded in amplitude, not a graph, and not embedded?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/14-01/

06660 | GREEN-048

### Balanced Ham Sandwich Line

O / open | Ben Green 问题

Let $X \subset \mathbb{R}^2$ be a set of $n$ points. Does there exist a line $\ell$ through at least two points of $X$ such that the numbers of points on either side of $\ell$ differ by at most 100?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

06661 | OWR-1394-011

### Banach-Mazur Bounds and Mean Width of Inscribed Polytopes

O / open | Oberwolfach 报告问题汇编

Let $P_n\subset B^d$ be an inscribed polytope with $n=d+k$ vertices, where $1\leq k\leq d$. Conjecturally, $\delta_{\mathrm{BM}}(P_n,B^d)\geq\widetilde{c}d/\sqrt{k}$ for an absolute constant $\widetilde{c}>0$. Also, is the mean width of an inscribed $(d+1)$-vertex polytope $P_{d+1}$ maximal for the regular simplex?

来源字段：https://doi.org/10.4171/owr/2006/56

06662 | OWR-14299583-005

### Banach-Mazur Diameter of Nonsymmetric Convex Bodies

P / partially_solved | Oberwolfach 报告问题汇编

Determine the diameter of the Banach-Mazur compactum of $n$-dimensional convex bodies, equivalently the maximum possible Banach-Mazur distance between two nonsymmetric convex bodies in dimension $n$.

来源字段：https://doi.org/10.4171/owr/2025/54

06663 | OWR-13110-004

### Bang Plank Conjecture for Nonsymmetric Convex Bodies

O / open | Oberwolfach 报告问题汇编

If a convex body $K\subset\mathbb{R}^n$ is covered by planks $P_1,\ldots,P_m$, must the sum of their relative widths, measured in the norm whose unit ball is $K-K$, be at least $1$?

来源字段：https://doi.org/10.4171/owr/2014/40

06664 | AIM-GEOMETRY-0212

### Bangert's asymmetric Finsler cylinder and the choice of volume

P / partially_solved | AIM 研讨会问题汇编

**Question 2.3.2 (Bangert).** Does there exist an irreversible Finsler metric of finite volume on \(\mathbb R\times S^1\) with no closed geodesics?

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06665 | OWR-1703876-007

### Barycentric Hyperplanes Through Convex Body Centroids

O / open | Oberwolfach 报告问题汇编

If $K\subset\mathbb{R}^n$ is a convex body, does the barycenter of $K$ always lie on at least $n+1$ distinct barycentric hyperplanes, meaning hyperplanes through it for which it is the barycenter of the corresponding section $K\cap h$?

来源字段：https://doi.org/10.4171/owr/2020/30

06666 | OWR-4798-007

### Bárány-Larman Colored Tverberg Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For integers $d\geq1$ and $r\geq2$, determine the least $n$ such that every affine map $f$ from the simplex $\Delta^{n-1}$ to $\mathbb{R}^{d}$, with its vertices partitioned into $d+1$ color classes each of size at least $r$, admits $r$ pairwise disjoint rainbow faces whose images have a common point; is the Bárány-Larman conjecture $n(d,r)=(d+1)r$ true? In the alternative colored Tverberg problem, determine the least $t$ such that the analogous assertion holds for equal color classes of size $t$, equivalently, whether $t(d,r)=r$.

来源字段：https://doi.org/10.4171/owr/2011/08

06667 | GEO-009

### Bellman's Lost in a Forest Problem

P / partially_solved | 独立 / 未分配集合

别名：Bellman's Lost-in-a-Forest Problem [GEOM-025]

What is the shortest path that guarantees escape from a forest of known shape and size, starting from an unknown location?

背景链接：https://en.wikipedia.org/wiki/Lost_in_a_forest_problem

06668 | AIM-GEOMETRY-0054

### Bernoulli mixing and the spectral-gap bottleneck for a branched hyperbolic gluing

P / partially_solved | AIM 研讨会问题汇编

Glue together two hyperbolic manifolds along some geodesic segment, and consider the corresponding \(\operatorname{CAT}(-1)\) geodesic flow. Does the Bowen-Margulis measure satisfy exponential decay of correlations?

来源字段：http://aimpl.org/equibdynsysgeom/2/

06669 | AIM-GEOMETRY-0269

### Berwald, conjugate-point, and curvature-sign obstructions for higher-genus Landsberg surfaces

P / partially_solved | AIM 研讨会问题汇编

**Question 10.1.4.** Does there exist a nonriemannian Finsler metric satisfying the Landsberg condition on a surface of genus at least \(2\)?

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06670 | AIM-GEOMETRY-0329

### Betti bounds and a defect-vector formula for real tropical patchworks

P / partially_solved | AIM 研讨会问题汇编

Real tropical hypersurfaces are directly related to T-hypersurfaces (piecewise-linear hypersurfaces arising in the combinatorial patchworking). Many restrictions on the topology of T-hypersurfaces are known. It would be interesting to look at these restrictions from the point of view of tropical geometry and to study the topology of real tropical varieties. For example, the following question arises. Question: What can be said about Betti numbers of a real tropical variety? (contributed by Ilia Itenberg)

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

06671 | AMR-067-0009

### Bi-invariant metrics and multiplicity of conjugate points

O / open | AMR 问题清单汇编

Assume that a left-invariant Riemannian metric is given on a compact connected Lie group $G$ such that the index of any geodesic segment is even. Must the metric be bi-invariant?

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

06672 | OWR-4990377-001

### Bicombing Convexity and Consistency in Metric Spaces

O / open | Oberwolfach 报告问题汇编

Does every metric space with a bicombing admit a convex bicombing, and does every metric space with a convex bicombing admit a consistent bicombing?

来源字段：https://doi.org/10.4171/owr/2021/8

06673 | OPG-588

### Big Line or Big Clique in Planar Point Sets

P / partially_solved | Open Problem Garden

Let $S$ be a set of points in the plane. Two points $v$ and $w$ in $S$ are visible with respect to $S$ if the line segment between $v$ and $w$ contains no other point in $S$.

Conjecture For all integers $k,\ell\geq2$ there is an integer $n$ such that every set of at least $n$ points in the plane contains at least $\ell$ collinear points or $k$ pairwise visible points.

背景链接：http://www.openproblemgarden.org/op/big_line_or_big_clique_in_planar_point_sets

06674 | OWR-14298589-001

### Bijections for Self-Overlapping Polygons in Other Topologies

O / open | Oberwolfach 报告问题汇编

Extend the known bijective approach to self-overlapping polygons from its present topological setting to other surface topologies, and characterize the resulting enumerative formulas.

来源字段：https://doi.org/10.4171/owr/2024/41

06675 | AMR-129-0013

### Billiards and Quantitative Symplectic Geometry - Alfonso Sorrentino - Problem 1

O / open | AMR 问题清单汇编

**Problem:** *Provide examples of real-world systems (from physics, mechanics, biology, economics, etc\ldots) that can be modeled as locally conformal symplectic flows or maps.*

In the following, we provide some material that could be useful to address this question. We begin by recalling the formal setting, starting from the well-understood *global* conformally symplectic case, namely when the ambient manifold is a symplectic manifold. We then introduce the *local* conformally symplectic case and provide some remarks and computations that may be useful for identifying and constructing the sought-after examples.

背景链接：https://arxiv.org/abs/2602.12896

06676 | AMR-129-0023

### Billiards and Quantitative Symplectic Geometry - Bernhard Albach, Umberto Hryniewicz, and Alexandru Oancea - Problem 1

O / open | AMR 问题清单汇编

**Conjecture 1.** For a Reeb flow on a closed connected $3$-manifold, either there are exactly two periodic orbits, or $$ \liminf_{T\to+\infty} \frac{\log P(T)}{\log T} \geq 2 $$ holds.

The conjecture above, which is also cited in [DanCG], has been confirmed in [Albach] for geodesic flows of reversible Finsler metrics on $S^2$. This significantly improves the best known growth rate $$ \liminf_{T\to+\infty} P(T)\frac{\log T}{T} > 0 $$ which was due to Hingston [Hingston1993]. In fact, in this case there is always quadratic growth since the number of closed geodesics is always infinite. On closed connected orientable reversible Finsler surfaces of genus $\geq 1$, Conjecture 1 is long known to be true for the geodesic flow. Hence, there is always quadratic growth of closed geodesics in any reversible Finsler closed surface -- orientable or not. Geodesic flows on $S^2$ can be lifted to Reeb flows on the standard contact $3$-sphere. It should be noted that the results from [Albach] cover a significantly larger class of such Reeb flows.

One might wonder if the same holds for a general low-dimensional symplectic dynamical system, which means either a symplectic diffeomorphism on a compact symplectic surface (with or without boundary), or the Hamiltonian flow on a $3$-dimensional compact regular energy level without boundary. In [Albach] it is also proved, via a simple argument based on the transverse foliations by Le Calvez and Tal [LeCalvez-Tal], that the same dichotomy as in Conjecture 1 holds for any symplectic diffeomorphism on $S^2$.

The confirmation of Conjecture 1 is, however, **not** the problem proposed here. Instead, we would like to propose the problem of investigating whether the same phenomenon holds in any dimension.

背景链接：https://arxiv.org/abs/2602.12896

06677 | AMR-129-0024

### Billiards and Quantitative Symplectic Geometry - Bernhard Albach, Umberto Hryniewicz, and Alexandru Oancea - Problem 2

O / open | AMR 问题清单汇编

**Problem 1.** Is it true that for a Hamiltonian diffeomorphism on a closed symplectic manifold, either the number of periodic orbits is finite, or periodic orbits grow at least quadratically in the sense that $\liminf_{T\to+\infty} \frac{\log P(T)}{\log T} \geq 2$?

Due to the lack of counterexamples, we suspect the answer might be positive. For instance, as test cases, we propose two particular instances of the above.

背景链接：https://arxiv.org/abs/2602.12896

06678 | AMR-129-0025

### Billiards and Quantitative Symplectic Geometry - Bernhard Albach, Umberto Hryniewicz, and Alexandru Oancea - Problem 3

O / open | AMR 问题清单汇编

**Problem 2.** Is it true that $\liminf_{T\to+\infty} \frac{\log P(T)}{\log T} \geq 2$ holds for every Hamiltonian diffeomorphism on a standard symplectic torus? Or in an arbitrary aspherical closed symplectic manifold?

背景链接：https://arxiv.org/abs/2602.12896

06679 | AMR-129-0026

### Billiards and Quantitative Symplectic Geometry - Bernhard Albach, Umberto Hryniewicz, and Alexandru Oancea - Problem 4

O / open | AMR 问题清单汇编

**Problem 3.** Is it true that for a Hamiltonian diffeomorphism on $\mathbf{C}P^n$ with infinitely many periodic orbits the inequality $\liminf_{T\to+\infty} \frac{\log P(T)}{\log T} \geq 2$ holds?

In low-dimensions, integrable Hamiltonian systems indicate that quadratic growth should be sharp. To justify this, if one would count invariant tori foliated by periodic orbits or isolated closed geodesics on a closed Riemannian $2$-sphere of revolution, instead of only closed geodesics, then it is simple to present examples where the growth is exactly quadratic. One would conjecture that it is possible to perturb such a system to get existence of systems with exact quadratic growth of periodic orbits. The corresponding statement is also expected to be true for integrable annulus maps, as pointed out by Gerhard Knieper. The construction of such a perturbation is not trivial and can potentially be a very difficult task.

The question that we are asking in higher dimensions is to be understood in the context of the question asked by Hofer and Zehnder in [HZ, p. 263], whether every Hamiltonian map on a compact symplectic manifold $(M,\omega)$ possessing more fixed points than necessarily required by the Arnold conjecture possesses always infinitely many periodic orbits. This question came to be known as the Hofer-Zehnder conjecture.

Shelukhin proved this conjecture in [Shelukhin] for closed monotone symplectic manifolds with semi-simple even-dimensional quantum homology ring, a class that includes complex projective spaces. His proof provides the estimate $\liminf_{T\to+\infty} P(T) \frac{\log T}{T}>0$, i.e., the growth rate of the number of periodic points is at least the growth rate of the prime numbers, just like in [Hingston1993].

Ginzburg and G\"urel interpreted more broadly the question of Hofer and Zehnder, in the sense that the existence of a fixed point not strictly required by Floer theory forces the existence of infinitely many periodic points, for example the existence of a hyperbolic fixed point [Ginzburg-Gurel2014], or the existence of a non-contractible fixed point [Gurel2013,Ginzburg-Gurel2016]. Notably, the estimate $\liminf_{T\to+\infty} P(T) \frac{\log T}{T}>0$ is already proved by G\"urel in [Gurel2013]. One could venture to ask whether quadratic growth holds in such cases as well.

背景链接：https://arxiv.org/abs/2602.12896

06680 | AMR-129-0007

### Billiards and Quantitative Symplectic Geometry - Boris Khesin - Problem 1

O / open | AMR 问题清单汇编

**1. Find physically interesting examples of infinite-dimensional billiards**

**Motivation:** It is a well-known folklore (made into a theorem under certain conditions) that billiard and geodesic flows are closely related by taking appropriate limits, see e.g. [Giannetto, Lange]. There are various physical problems leading to the construction of geodesics on infinite-dimensional spaces. One of the sources is the Euler-Arnold equation describing geodesics on infinite-dimensional groups. In particular, solutions of the Euler equation of incompressible hydrodynamics can be regarded as geodesics on the group of volume-preserving diffeomorphisms with respect to the right-invariant $L^2$-metric on the group. A similar framework exists for the KdV equation, magnetohydrodynamics, Heisenberg chain, etc. [Arnold66, ArnKh] A natural question is whether there exist interesting infinite-dimensional billiards, possibly related to equations of mathematical physics.

One possible approach would be to consider not one or several particles hitting a wall with a billiard reflection, but a continuum of particles doing that. The difficulty is that then the reflected flow will be passing through the same domain as the flow before reflection, and hence there will be two different flows passing simultaneously through each point in the vicinity of the boundary. However, this is exactly the setting handled by multi-phase flows and by the corresponding groupoid structure for generalized flows, see [Brenier, IzKh]. This way infinite-dimensional billiards in fluids might correspond to a choice of appropriate boundary conditions in the diffeomorphism groupoid for two-phase fluids. It would be very interesting to develop the corresponding theory for multi-phase fluids with billiard reflections from the boundary.

背景链接：https://arxiv.org/abs/2602.12896

06681 | AMR-129-0008

## Billiards and Quantitative Symplectic Geometry - Boris Khesin - Problem 2

O / open | AMR 问题清单汇编

**3. Is there a fluid of negative density?**

**Motivation:** A mild reformulation of the question above is as follows: *Study continuous versions for the interactions of particles of positive and negative mass. Explore the relation with pseudo-Euclidean metrics.*

To motivate this question, we start with the motion of three point-like beads of masses $m_1, m_2$, and $m_3$ on a frictionless ring, making elastic collisions with each other. By using the momentum and energy conservation laws and quotienting out the rotational symmetry, the motion of the beads is equivalent to a billiard trajectory in an acute triangle with angles given by $$ \tan \alpha_1=\sqrt{\frac{m_1(m_1+m_2+m_3)}{m_2 m_3}}, $$ as well as respective cyclic permutations of indices. (Note that if one of the masses, say $m_3$, goes to infinity, this corresponds to a billiard in a right triangle, $\alpha_3=\pi/2$, which is equivalent to two beads constrained to a segment between two walls.) This equivalence allows one to answer various questions on the existence of periodic orbits, etc.

A natural question: ``Does there exist a mechanical system corresponding to a billiard in an obtuse triangle?'' has a curious answer. This would be a system where one allows beads of *negative mass*, as long as they satisfy $(m_1 + m_2 + m_3)m_1 m_2 m_3 > 0$, see [GlashowMittag]. Thus, if the masses have the same sign, then the associated triangle is acute; otherwise, it is obtuse. In this way, the obtuse triangular billiard tables suggest considering the one-dimensional dynamics of particles with positive and negative mass. For instance, in a nice paper [Artigue], Artigue presents a study of all possible cases of positive and negative masses. One introduces a small particle of negative mass, ``graviton'', which via elastic collisions, pulls together other positive masses (and might be the source of gravity in general!), cf. [Wiki].

However, there is a different point of view: instead of changing the angles, one can consider the billiard in a fixed triangle on the plane with an appropriately rescaled Euclidean metric. In the case of negative mass(es), the metric becomes pseudo-Euclidean. Billiards in pseudo-Euclidean (or, more generally, in pseudo-Riemannian) metrics were described in [KhTab]. The corresponding theory and applications to gravitons are described in [Artigue]. The problem is to construct a continuous version of this theory of particles of negative mass. Are there some interesting phenomena related to the corresponding Euler-like equations for the ``fluid of negative density''?

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06682 | AMR-129-0009

## Billiards and Quantitative Symplectic Geometry - Jean-Fran\c{c - Problem 1

O / open | AMR 问题清单汇编

**Problem statement** Let $M$ be a closed manifold that is not a $K(\pi,1)$, and endow the cotangent bundle $T^*M$ with its standard symplectic form. We conjecture that, for any closed and connected hypersurface $\Sigma\subset T^*M$ and any Hamiltonian $H:T^*M\to\mathbf{R}$ that admits $\Sigma$ as a regular energy level $\Sigma=H^{-1}(c)$, there exists $\varepsilon>0$ such that almost all energy levels of $H$ in the interval $(c-\varepsilon,c+\varepsilon)$ carry a contractible periodic orbit of $H$.

**Context and motivation**.

The study of closed characteristics on hypersurfaces of symplectic manifolds is a fundamental topic in symplectic geometry. When such a closed hypersurface is of contact type, the famous Weinstein conjecture [Weinstein79], first proved by Viterbo in $\mathbf{R}^{2n}$ [Viterbo-WeinsteinR2n], claims the existence of at least one closed characteristic. Without assuming the contact type hypothesis, given some symplectic manifold $(W,\omega)$ and some proper autonomous smooth Hamiltonian $H:W\to \mathbf{R}$, one can ask which regular levels $H^{-1}(c)$ contain at least one closed characteristic. The question of whether *all* the regular levels of any Hamiltonian satisfy this property is known as the Hamiltonian Seifert conjecture. It was disproved already for $W=\mathbf{R}^{2n}$ by Ginzburg [Ginzburg-Ham-Seifert1] [Ginzburg-Ham-Seifert2] and Herman [Herman] for $2n\geq6$; G\"urel and Ginzburg later provided a ${\mathcal{C}^2$ counterexample for $2n=4$ [Ginzburg-Gurel1], [Ginzburg-Gurel2]. Once the conjecture had been disproved, it is natural to ask whether at least some of the regular levels $H^{-1}(c)$ contain closed characteristics. In the particular case $W=\mathbf{R}^{2n}$ endowed with the standard symplectic form it actually turns out that many of them do. More precisely, a celebrated result of Hofer-Zehnder [HZ] and Struwe [Struwe] asserts that almost all (regular) levels of a smooth proper Hamiltonian $H:\mathbf{R}^{2n}\to \mathbf{R}$ carry at least one closed characteristic. This result is known as *the almost existence property*.

In a cotangent bundle any closed and connected hypersurface separates $T^*M$ into a bounded component and an unbounded component. Hofer and Viterbo [HV88] proved that the above property holds for all hypersurfaces that enclose the $0$-section, without contractibility. Albers-Frauenfelder-Oancea [AFO17] proved the above property under the assumption that the Hurewicz map $\pi_2(M)\to H_2(M)$ does not vanish. Their proof uses the multiplicative structure on symplectic homology twisted by suitable rank 1 local coefficients. Frauenfelder-Pajitnov [Frauenfelder-Pajitnov] proved this property for rationally inessential manifolds $M$, i.e., manifolds whose fundamental class with rational coefficients vanishes under their classifying map to $B\pi_1$. Their proof involves $S^1$-equivariant homology and localization.

**Remarks, partial results**

In our recent paper [BDHO-cotangent] we have proved this property for any $M$ that is not a $K(\pi,1)$, but for a somewhat restricted class of hypersurfaces $\Sigma$. Our proof uses a new version of symplectic homology with differential graded local coefficients.

Our hope is that the methods of [AFO17, Frauenfelder-Pajitnov, BDHO-cotangent], which are complementary to each other, can be combined in order to prove the above conjecture.

In all cases in which it is known to hold for all closed hypersurfaces, the contractible almost-existence property is a consequence of the finiteness of the so-called *$\pi_1$-sensitive Hofer-Zehnder capacity* for the disc cotangent bundle $D^*Q$. One could also conjecture that this capacity is finite for all manifolds $Q$ that are not $K(\pi,1)$.

The assumption of not being $K(\pi,1)$ seems reasonable as a borderline between existence and non-existence of contractible periodic Hamiltonian orbits: the standard class of manifolds that *are* $K(\pi,1)$ is that of manifolds that carry a Riemannian metric of non-positive sectional curvature, and such metrics do not admit contractible closed geodesics. In particular, none of the positive radius sphere bundles for such a Riemannian metric carries any contractible closed characteristic. Thus, in the above conjecture, the contractibility requirement is intimately tied to the non-$K(\pi,1)$ assumption.\\

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06683 | AMR-129-0005

## Billiards and Quantitative Symplectic Geometry - Johanna Bimmermann - Problem 1

O / open | AMR 问题清单汇编

**Question:.** \begin{center} Let $(N,g)$ be an orientable closed hyperbolic surface.\\ Does $c_{\mathrm{HZ}}\bigl(D^{*}N,d\lambda\bigr)<\infty$ hold? \end{center}

We included the word ``orientable'' for the following reason. If the Euler characteristic $\chi(N)$ is divisible by $4$ and $N$ is non-orientable, then there exists a Lagrangian embedding $N\hookrightarrow (\mathbb{C}^{2},\omega_{0})$ [Giv86]. By Weinstein's tubular neighborhood theorem this implies that, for some $\varepsilon>0$, $(D^{*}_{\varepsilon}N,d\lambda)\hookrightarrow (\mathbb{C}^{2},\omega_{0})$ embeds symplectically, which implies finiteness.\\

We now present two arguments that might hint finiteness and two arguments that might hint infiniteness. These are not mathematically worked out and are based on discussions with Gabriele Benedetti, Dylan Cant, Stefan Nemirovski and Alexandru Oancea.

**Finiteness.**

- *Capacity relative to the zero section is finite.*

There is a modification of the definition of $c_{\mathrm{HZ}}$ for which the answer to (Q) is yes. Require the Hamiltonians $H\in C_c^\infty(D^{*}N,\mathbb{R}_{\ge 0})$ to attain their maximum along the zero section $N\subset D^{*}N$, i.e.,\ $H|_{N}=const.=\max H$. It is a direct consequence of [Wbr06] that $$ c_{\mathrm{HZ}}(D^{*}N,N,d\lambda)=\operatorname{sys}<\infty, $$ where $\operatorname{sys}$ denotes the length of the shortest closed (non-contractible) geodesic.

- *Displaceability of the zero section after removing a point.*

Displacement energy gives an upper bound for the Hofer--Zehnder capacity, and if the zero section would be displaceable then so is $D^{*}_{\varepsilon}N$ for $\varepsilon$ small. In particular, displaceability of the zero section yields finiteness of $c_{\mathrm{HZ}}(D^{*}N,d\lambda)$. Actually, the full zero-section $N\subset T^{*}N$ of a closed manifold $N$ is never displaceable, but it becomes displaceable if one removes an arbitrarily small disc $D_{\varepsilon}\subset N$.

To see this, take a Morse function $f:N\to\mathbb{R}$ with all critical points in $D_{\varepsilon/2}$. Then there exists some $\delta>0$ such that $\vert \nabla f\vert \geq \delta$ on $N\setminus D_{\varepsilon}$. Hence, the Hamiltonian flow of $H(q,p)=\frac{1}{2}\vert p\vert^2+f(q)$ displaces $N\setminus D_{\varepsilon}\subset T^{*}N$. We obtain: $$ c_{HZ}(D^*(N\setminus D_\varepsilon), d\lambda)<\infty,\quad \forall\varepsilon>0. $$

**Infiniteness.**

- \emph{Lack of pseudoholomorphic curves in $T^{*}N$.}

Most known bounds for the Hofer--Zehnder capacity ultimately come from pseudoholomorphic curves, as first explained by Hofer--Viterbo [HV92]. Indeed one needs a form of uniruledness, morally, a pseudoholomorphic curve through every point. For all closed surfaces $N$ for which finiteness of $c_{\mathrm{HZ}}(D^{*}N,d\lambda)$ is known, uniruledness is ensured by the existence of a Lagrangian embedding $N\hookrightarrow \Sigma$ into a ruled complex surface $\Sigma$; for example $$ S^{2}\subset \mathbb{CP}^{1}\times\mathbb{CP}^{1},\qquad \mathbb{RP}^{2}\subset \mathbb{CP}^{2},\qquad T^{2}\subset \mathbb{CP}^{2},\ \ldots $$ For orientable hyperbolic surfaces, however, no such embedding exists. Indeed, suppose that an orientable surface $N$ of genus $g\ge 2$ was embedded as a Lagrangian into a ruled complex surface $\Sigma$. Then the self-intersection satisfies $$ [N]^{2} \;=\; -\chi(N) \;=\; 2g-2 \;>\; 0, $$ since for a Lagrangian surface $L$ in a complex surface one has $[L]^{2}=-\chi(L)$. On the other hand, if $[\omega]$ denotes the Kähler class of $\Sigma$, the Lagrangian condition $\omega|_{N}=0$ implies $$ [\omega]\cdot \mathrm{PD}[N] \;=\; 0. $$ Thus $[\omega]$ and $\mathrm{PD}[N]$ are orthogonal classes of positive square in $H^{2}(\Sigma;\mathbb{R})$, which is impossible as the intersection form has exactly one positive eigenvalue [Bea96, Prop.\ III.18]. Consequently, an orientable hyperbolic surface cannot embed as a Lagrangian into a ruled complex surface.

- *No upper bounds from symplectic homology.* The other existing strategies for proving finiteness of the Hofer--Zehnder capacity (or uniruledness) of cotangent bundles rely on symplectic homology. In particular, the existence of a *dilation class* $\Delta x = 1$ [BC25], a nontrivial *pair-of-pants product* $x \star y = 1$ [Iri14], or the vanishing of symplectic homology with (differential graded) local coefficients [AFO17,BDHO-cotangent] implies such finiteness. Except for Irie's pair-of-pants product, all these criteria provide upper bounds for the $\pi_1$-sensitive Hofer--Zehnder capacity, which only requires all contractible orbits to be slow. It is easily seen that for a hyperbolic surface, $c_{HZ}^0(D^*N, d\lambda)=\infty$. For example, the geodesic flow, generated by the Hamiltonian $E(q,p)=\tfrac{1}{2}\lvert p\rvert_q^2$, has no contractible periodic orbits. Hence, none of these criteria can be used to prove finiteness for hyperbolic surfaces.\\

For cotangent bundles there is a ring isomorphism between symplectic homology and the homology of the free loop space [Abo15]. For a closed hyperbolic surface $N$, the latter can be computed explicitly since $N$ is a $K(\pi,1)$ [LO15, Ch.\ 1.4]: $$ H_*(\mathcal{L}N,\mathbb{Z}) \cong \bigoplus_{[g]\in \mathrm{Conj}} H_*(K(C_\pi(g),1),\mathbb{Z}), $$ where $C_\pi(g)$ denotes the centralizer of $g\in\pi=\pi_1(N)$. Here $C_\pi(e)=\pi$ and $C_\pi(g)\cong\mathbb{Z}$ for $g\neq e$, giving $$ H_*(\mathcal{L}N,\mathbb{Z}) \cong H_*(N,\mathbb{Z}) \oplus \bigoplus_{[g]\neq[e]} H_*(S^1,\mathbb{Z}). $$ This simple structure immediately rules out the existence of a nontrivial loop product $x\star y = 1$, as $\star: H_i(\mathcal{L}N,\mathbb{Z})\times H_j(\mathcal{L}N,\mathbb{Z})\to H_{i+j-2}(\mathcal{L}N,d\lambda)$, and the unit is the only degree $2$ element.

We conclude with a possible strategy to prove infiniteness by linking quantitative symplectic geometry with billiards.

**Lower bounds via hyperbolic billiards.** If $\Omega\subset N$ is open and simply connected, it lifts to an open subset of the hyperbolic plane $\mathbb{H}^{2}$ with compact closure, let $\Omega\text{-}\mathrm{sys}$ denote the length of the shortest billiard trajectory in $\Omega$. Then $$ \Omega\text{-}\mathrm{sys}\le c_{\mathrm{HZ}}(D^{*}N,d\lambda). $$ Note that the notion of “billiard trajectory’’ here might be slightly relaxed: the lower bound only requires closed billiard trajectories to have length $\ge \Omega\text{-}\mathrm{sys}$ when they arise as limits of approximate solutions in the sense of approximation schemes developed in [AM11,Voc21].

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06684 | AMR-129-0006

## Billiards and Quantitative Symplectic Geometry - Johanna Bimmermann - Problem 2

O / open | AMR 问题清单汇编

**Question:.** \begin{center} Is there a sequence of embedded billiard tables $\Omega_i\subset N$, $i\in\mathbb{N}$, such that $$ \Omega_i\text{-}\mathrm{sys}\longrightarrow \infty \quad \text{as } i\to\infty\ ? $$ \end{center} A positive answer would, to the best of the author’s knowledge, give the first example of a disk cotangent bundle with infinite Hofer--Zehnder capacity.

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06685 | AMR-129-0001

## Billiards and Quantitative Symplectic Geometry - Misha Bialy - Problem 1

O / open | AMR 问题清单汇编

**Magnetic billiards for strong magnetic fields..**

Magnetic billiards were introduced and studied in [berry-robnik]. In this model, the billiard ball moves under the influence of a constant magnetic field along Larmor circles (for a magnetic field of magnitude $B$, the Larmor circles have radius $1/B$). The only known example of an integrable magnetic billiard is a circular billiard. An algebraic approach to this problem was discussed in [magnetic1], [magnetic2]. Numerical experiments were performed in [albers]. Moreover, the approach of total integrability works well for convex magnetic billiard in a *weak* constant magnetic field [bialy-magnetic-hopf].

**Question 1.** {\it Is it true that the only totally integrable magnetic billiard in a *strong* constant magnetic field is necessarily circular?}

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06686 | AMR-129-0002

### Billiards and Quantitative Symplectic Geometry - Misha Bialy - Problem 2

O / open | AMR 问题清单汇编

**Minkowski billiard..**

The model of Minkowski and, more generally, Finsler billiards, was introduced and studied by Gutkin and Tabachnikov in [GUTKIN2002277]. It plays an important role in a recent study by Artstein-Avidan, Karasev, and Ostrover [Artstein_Avidan_2014] on Mahler's conjecture. Consider a smooth convex curve $\gamma\subset \mathbb{R}^2$, and a Minkowski (not necessarily symmetric) norm $N$. The Minkowski billiard in $\gamma$ with respect to the norm $N$ is defined as a twist map that corresponds to the generating function \[L(s_0,s_1)=N(\gamma(s_1)-\gamma(s_0)) ,\] and the corresponding variational principle $ \mathcal{L}\{s_n\}=\sum_{n} L(s_n,s_{n+1}),\ $ where $\gamma$ is parametrized by $N$-arc length parameter (meaning, $N(\dot{\gamma}(s))=1$). Thus, the Birkhoff billiard is a special case of the Minkowski billiard when $N$ is chosen to be a Euclidean norm.

**Question 2.** (Communicated by Yaron Ostrover) *Suppose that the Minkowski billiard, corresponding to the norm $N$, in a unit ball of $N$ is totally integrable. Does this imply that the norm $N$ is necessarily Euclidean?*

This question was answered affirmatively in [IMRN] for billiard tables with a finite symmetry group.

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06687 | AMR-129-0003

### Billiards and Quantitative Symplectic Geometry - Misha Bialy - Problem 3

O / open | AMR 问题清单汇编

**Newton equations with $1\frac 12$ degrees of freedom, with periodic potential..**

Newton's equation $\ddot u=-V'_u(u,t)$ is the Euler-Lagrange equation associated with the Lagrangian $L(u,\dot u, t)=\frac 12\dot u^2-V(u,t)$. Assume that $V$ is periodic both in $u$ and in $t$. The corresponding Hamiltonian flow acts on $T^*\mathbb{S}^1\times\mathbb{S}^1$. One can prove that this flow is totally integrable on the whole phase space if and only if $V$ does not depend on $u$. Moreover, if $V=V(nu+mt)$, with $ m\neq0, n\in\mathbb{Z}$, then there exists an integral of motion that is quadratic in momentum $p$, and hence the system is totally integrable in a neighborhood of infinity, $|p|>c$, for sufficiently large $c$.

**Question 3.** *Are there other potential functions $V$ such that the system is totally integrable at infinity?*

背景链接：https://arxiv.org/abs/2602.12896

06688 | AMR-129-0004

### Billiards and Quantitative Symplectic Geometry - Misha Bialy - Problem 4

O / open | AMR 问题清单汇编

**An elliptic PDE.**

Consider an elliptic PDE $\Delta u=-V'_u(u, x_1, \ldots ,x_n)$, where $\Delta=\sum_{i=1}^{n}\partial^2_{x_i}$ and $V$ is a 1-periodic function of $u, x_1,\ldots ,x_n$. This equation is the Euler-Lagrange equation for the functional $\int \left( \frac 12|\nabla u|^2-V(u, x_1, \ldots, x_n)\right) dx_1 \ldots dx_n$. Aubry-Mather theory, as well as KAM theory for this equation, were developed by J. Moser [moser] and V. Bangert [bangert]. If $V$ does not depend on $u$, we have the equation $\Delta u=0$. In this case, for any rotation vector $\alpha\in \mathbb{R}^n$, there exists a family of minimal solutions of the form $u=\alpha\cdot x+c$. Thus, for every $\alpha$, the graphs of these solutions form a foliation of $\mathbb{T}^{n+1}(u,x_1, \ldots, x_n)$ by minimals (hyperplanes of slope $\alpha$).

**Question 4.** {\it Let $V$ be a 1-periodic potential. Suppose that for every $\alpha \in \mathbb{R}^n$ there exists a foliation of $\mathbb{T}^{n+1}$ by graphs of minimal solutions of slope $\alpha$. Is it true that, in this case, $V$ does not depend on $u$?}

For compactly supported potentials $V$, a variational version of this question was approached in [bialy-mackay].

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06689 | AMR-129-0010

## Billiards and Quantitative Symplectic Geometry - Olga Paris-Romaskevich - Problem 1

O / open | AMR 问题清单汇编

**Tiling billiards.** A tiling billiard is a dynamical system associated with a plane polygonal tiling. Here is the rule: a particle moves in a straight line until it reaches an edge of a polygon; then it crosses into a neighboring polygon with its velocity direction changed according to Snell's law of refraction with refraction coefficient $-1$. Several families of tiling billiards have been studied since $2015$. Here is the full (as far as I know) list of them. For this list, a tiling billiard's name is followed by a description of the underlying tiling, the generic behavior of trajectories, and some bibliography.

1. **Trihexagonal tiling billiard** (a periodic tiling by regular hexagons and triangles) has orbits that are dense in an unbounded periodic open set, as proven in the work of Diana Davis and Pat Hooper [DH18].

2. **Triangle tiling billiards** (a periodic tiling by one triangular tile via central symmetries) have orbits that are typically either periodic or escape linearly to infinity (we call this **integrable behavior**). There exists a zero-measure set of exceptional triangle tilings (parametrized by the Rauzy gasket (The Rauzy gasket has now been extensively studied since it appears as the parameter set of exceptional trajectories in various dynamical and topological settings, see, in particular, [AR91, AHS, DDe]. For a definition of the Rauzy gaskets, an attractor of a multidimensional continued fraction algorithm, and an introduction to their dynamical properties, see [AS].)) that permit trajectories escaping non-linearly (These trajectories necessarily pass by the circumcenters of all tiles that they cross.). This has been shown in a series of works [BDFI18, TAF, H_Paro], see also [highlights_O]. For a quick introduction to the triangle tiling billiards and their properties, I recommend a short animation movie by Ofir David [Ofir], based on the research results in [TAF].

3. **Cyclic quadrilateral tiling billiards** (a periodic tiling by one cyclic quadrilateral via central symmetries) have integrable behavior, see [H_Paro, TAF]. The set of exceptional tilings (admitting non-linearly escaping trajectories) has been recently classified in [Dynnikovetal] and is parametrized by a fractal set named Novikov's gasket.

4. **Double triangle tiling billiards** (a periodic tiling by a pair of triangles $\Delta$ and $\Delta'$ with one common edge $a$, and such that their angles $\alpha$ and $\alpha'$ opposite to this edge are equal) are generalizations of triangle tiling billiards. Partial results on the dynamics have been obtained by Magali Jay in her thesis [Jay_thesis]. Probably, their behavior is integrable, as in the two previous examples.

5. **Refractive wind-tree model** (the tiling is defined by rectangles of size $a \times b$ with centers belonging to a lattice $\Lambda$, and the complement of their union (an \guillemotleft{} unbounded polygon \guillemotright{})) has its generic trajectories trapped in bands, see [Jay_thesis].

6. **Generalized tiling billiards** are a generalization of a tiling billiard corresponding to one fixed polygon that does not necessarily tile the plane. A trajectory starting in a copy of $P$ can still be followed by reflecting $P$ by central symmetry with respect to the midpoint of a crossed side, thus producing a local refraction of a trajectory with coefficient $-1$. **Cyclic pentagonal generalized tiling billiards** show a behavior different from triangle and cyclic quadrilateral tiling billiards. In particular, the orbits may escape in a non-linear way in open sets: they have a principal direction but deviate from it, see [Jay_pentagons] and [Jay_thesis]. The generic behavior is not yet known.

7. **Brick tiling billiards** (a periodic tiling defined by two rows of $1 \times 1$ squares that are shifted relative to one another by some parameter $\theta \in (0,1)$) have all of their orbits periodic when $\theta \in \mathbb{Q}$, as shown in [thesis_park]. The behavior for $\theta \notin \mathbb{Q}$ has not yet been studied.

A tiling of the plane by polygons is **locally foldable** if it is $2$-colorable and the **defect of a vertex** (the alternating sum of angles in that vertex) is equal to $0$. This property implies (see [TAF]) strong restrictions on the dynamics of the corresponding billiard: all trajectories are either periodic or escaping, and any trajectory passes through each tile at most once. Moreover, the study of the dynamics of tiling billiards in locally foldable tilings can be related to Novikov's problem in topology. See [Dynnikov_main] for an introduction to Novikov's problem and [PaRoH, Dynnikovetal] for its link to tiling billiards. This is the case for examples $2., 3.$, and $4.$ above.

It is worth mentioning that all of the work on tiling billiards mentioned above was accomplished by reducing their dynamics to that of linear flows on certain families of translation or half-translation surfaces, although the construction method is ad hoc for each family.

背景链接：https://arxiv.org/abs/2602.12896

06690 | AMR-129-0011

## Billiards and Quantitative Symplectic Geometry - Olga Paris-Romaskevich - Problem 2

O / open | AMR 问题清单汇编

**Two questions.** In the following, I ask two questions, related in spirit -- one general and one specific.

背景链接：https://arxiv.org/abs/2602.12896

06691 | AMR-129-0012

### Billiards and Quantitative Symplectic Geometry - Olga Paris-Romaskevich - Problem 3

O / open | AMR 问题清单汇编

\begin{question} In the class of tiling billiards in periodic tilings, is there a finite list of generic archetypal behaviors? For instance, the tiling billiards for which generic behavior is understood fall into one of two categories: either their typical trajectories are dense in open sets of the plane, as in cases 1. and 5., or the behavior is integrable (typical trajectories are either periodic or linearly escaping), as in cases 2. and 3. (and conjecturally in case 4. as well). One could ask whether such a dichotomy holds in general. \end{question}

Locally foldable periodic tiling billiards, up to some technical regularity conditions, are integrable. This is not at all obvious but follows from fine Morse theory developed by Ivan Dynnikov, see in particular Theorem $1$ in [Dynnikov_main], and [PaRoH]. The first ingredient in the proof comes from [BDFI18] and was further developed in [TAF, PaRoH]. It states that for any locally foldable tiling there exists a **global folding map** $\mathcal{F}$ defined on a tiled plane such that, on the restriction of two neighboring tiles, it is a folding of one of them onto the other along the common edge (which is an axial symmetry with respect to this edge).

It seems that the tilings on which one could start testing Question $1$ should be the simplest not-locally foldable tilings - for example, irrational brick tiling billiards or the parallelogram tiling billiards that we discuss in Question $2$.

背景链接：https://arxiv.org/abs/2602.12896

06692 | AMR-129-0014

### Billiards and Quantitative Symplectic Geometry - Serge Tabachnikov - Problem 1

O / open | AMR 问题清单汇编

**Projective billiards..** Projective billiards were defined in [Ta97]. A projective billiard table is a convex domain with smooth boundary equipped with a transverse field of directions; the phase space is the same as that of the conventional billiard, and the reflection law is such that the tangential component of the incoming velocity vector is preserved, whereas the transverse component is reversed.

Problem: *when does a projective billiard possess an invariant symplectic form?* And, related, *when does it have a variational description?*

One such situation is described in [Ta97-1]: the billiard table is bounded by a sphere, and the transverse field satisfies a certain *exactness* condition. The invariant symplectic form on oriented lines that intersect the sphere is induced by the hyperbolic metric in its interior (the projective, or Beltrami-Cayley-Klein model). Another example, in dimension 2, is when the billiard curve is an arbitrary oval, but the lines of the transverse field are all concurrent; the invariant area form comes from the standard area form in the plane via polar duality (see [Ta97]).

In recent papers [Glu,GM], a somewhat related question was studied: when is a Minkowski billiard projective? (In which case it has a variational description and an invariant symplectic form). The answer is that such a billiard is the conventional one in some Euclidean metric.

背景链接：https://arxiv.org/abs/2602.12896

06693 | AMR-129-0015

### Billiards and Quantitative Symplectic Geometry - Serge Tabachnikov - Problem 2

O / open | AMR 问题清单汇编

**Inner and outer periodic orbits on immersed submanifolds..** Let $M$ be a closed immersed submanifold of Euclidean space. A generalized $n$-periodic billiard trajectory is an $n$-gon inscribed in $M$ (that is, the vertices lie on $M$) that has an extremal perimeter.

Problem: *estimate from below the number of $n$-periodic trajectories in terms of the topology of $M$ and the immersion*.

The case of $n=2$ was studied in [Pu], where the answer is given in terms of the topology of $M$ (but not of the immersion). For example, for a 2-dimensional torus, the number of such *diameters* is not less than 10, and this estimate is sharp. For higher periods, see [Du1,Du2].

A similar question can be asked about periodic trajectories of the outer symplectic billiard about $M$; see [ACT] for their recent study.

背景链接：https://arxiv.org/abs/2602.12896

06694 | AMR-129-0016

### Billiards and Quantitative Symplectic Geometry - Serge Tabachnikov - Problem 3

O / open | AMR 问题清单汇编

**Complexity of polygonal symplectic billiards..** Symplectic billiards were introduced in [AT]; the polygonal ones were studied in [ABSST,ALW].

Label the sides of a polygon by some symbols and assign to a billiard trajectory its symbolic code, the corresponding bi-infinite word in these symbols. The set of all trajectories yields a collection of such words. The complexity function $p(n)$ is the number of different subwords of length $n$ in this collection.

For the conventional polygonal billiards, it is known that the complexity function is sub-exponential (the literature on this subject is huge, starting with [Ka]), and it has polynomial, actually cubic, growth for rational billiards (whose angles are rational multiples of $\pi$). For polygonal outer billiards, the growth of the complexity function is polynomial, see [GT].

Problem: *study the complexity of polygonal symplectic billiards*.

Note that there are examples of such billiards with all trajectories being periodic; in such cases, the complexity function has a zero growth rate.

背景链接：https://arxiv.org/abs/2602.12896

06695 | AMR-129-0017

### Billiards and Quantitative Symplectic Geometry - Serge Tabachnikov - Problem 4

O / open | AMR 问题清单汇编

**Chaotic symplectic billiards?.** Problem: *do chaotic symplectic billiards in convex $C^1$-smooth domains exist?*

A natural candidate is the stadium; there is numerical evidence that this might be the case.

背景链接：https://arxiv.org/abs/2602.12896

06696 | AMR-129-0018

### Billiards and Quantitative Symplectic Geometry - Serge Tabachnikov - Problem 5

O / open | AMR 问题清单汇编

**Planar characteristic curves..** Let $M$ be (a germ of) a smooth hypersurface in the standard symplectic space.

Problem: *how restrictive is the condition that the characteristic curves on $M$ are planar?*

If $M$ is closed and convex, the situation is rigid: $M$ is an affine symplectic image of a round ball; see [KS].

背景链接：https://arxiv.org/abs/2602.12896

06697 | AMR-129-0019

### Billiards and Quantitative Symplectic Geometry - Serge Tabachnikov - Problem 6

O / open | AMR 问题清单汇编

**Density of reflected wave fronts..** Consider a domain with smooth boundary in Euclidean space and let $P$ be a generic point in its interior. Let $P$ be a source of light, and consider the time $t$-front, that is, the locus of ``photons'' emanating from $P$ in all directions, undergoing optical reflection off the boundary, and traveling for time $t$.

Problem: *does this time $t$-front become dense in the limit $t \to \infty$?*

This problem, and its analog for smooth Riemannian manifolds, is discussed in [KK]. Even the case of a round circle is open. The length growth of wave fronts is studied in [dV,dV2].

背景链接：https://arxiv.org/abs/2602.12896

06698 | AMR-129-0020

### Billiards and Quantitative Symplectic Geometry - Serge Tabachnikov - Problem 7

O / open | AMR 问题清单汇编

**Ivrii's conjecture for outer symplectic billiards..** The Ivrii conjecture for conventional billiards asserts that the set of periodic orbits has zero measure (with respect to the measure associated with the invariant symplectic structure of the phase space). A similar conjecture applies to the outer symplectic billiards.

For conventional billiards, this conjecture is proved for periods $n=2,3$ in all dimensions, and for $n=4$ in dimension 2. For planar outer billiards, there are proofs for $2n+1$-periodic orbits with rotation number $n$ and for $2n$-periodic orbits with rotation number $n-1$. See [Sh] and the references therein about these results.

Problem: *prove Ivrii's conjecture for 3-periodic multidimensional outer symplectic billiards with respect to a strictly convex closed hypersurface.*

背景链接：https://arxiv.org/abs/2602.12896

06699 | AMR-129-0021

### Billiards and Quantitative Symplectic Geometry - Serge Tabachnikov - Problem 8

O / open | AMR 问题清单汇编

**Approximating a convex curve by polygons with respect to the symmetric difference..** A popular topic in convex geometry is the approximation of smooth convex closed curves by polygons, see [Gr]. The quality of approximation by $n$-gons may be measured by the difference of perimeters or areas, and the polygons may be inscribed or circumscribed. This leads to four functions of $n$, and these four functions correspond to the minimal actions of four billiards: the conventional, Birkhoff, billiards, the outer billiards, the symplectic billiards, and the outer length billiards (see [AT24,BBF] about these lesser studied billiards).

Consider the approximation of an oval by $n$-gons where the quality of approximation is measured by the area of the symmetric difference.

Problem: *is the resulting function of $n$ related to some billiard-like dynamical system?*

背景链接：https://arxiv.org/abs/2602.12896

06700 | AMR-129-0022

### Billiards and Quantitative Symplectic Geometry - Serge Tabachnikov - Problem 9

O / open | AMR 问题清单汇编

**Optical transformations of the space of oriented lines..** The billiard (optical) reflection is a (local) symplectic transformation of the space of oriented lines (rays of light); this space carries a symplectic structure induced by the canonical symplectic structure of the cotangent bundle of the ambient $n$-dimensional space via symplectic reduction.

The space of oriented lines has dimension $2n-2$, and its symplectic transformation depends on a smooth function of $2n-2$ variables (for example, an infinitesimal symplectic transformation, i.e., a Hamiltonian vector field, is defined by its Hamiltonian function).

On the other hand, a smooth mirror is a hypersurface, locally the graph of a function of $n-1$ variables. Such functions constitute a minuscule part of the space of smooth functions of $2n-2$ variables. Thus, one expects the *optical* set of (local) symplectic transformations of the space of rays that are realized by a finite collection of mirrors to be a very small part of the set of all (local) symplectic transformations of the space of rays.

Problem: *how to characterize this optical symplectic transformation within the space of all symplectic transformations of the space of oriented lines?*

See [PTT,Pe,Ta20,Gl1] for some results in this direction.

背景链接：https://arxiv.org/abs/2602.12896

06701 | AMR-067-0004

### Biorthogonal curvature

P / partially_solved | AMR 问题清单汇编

Does $S^2\times T^2$ admit a Riemannian metric with positive biorthogonal curvature?

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

06702 | OWR-16168-005

### Bipolar Comparison and Transport Continuity

O / open | Oberwolfach 报告问题汇编

Does $(4,1)$-bipolar comparison imply the transport continuity property, and are these two properties equivalent?

来源字段：https://doi.org/10.4171/owr/2018/27

06703 | OWR-762-002

### Bloch-Group Formulas for Odd-Dimensional Hyperbolic Volumes

P / partially_solved | Oberwolfach 报告问题汇编

If $M^{2n-1}$ is a finite-volume hyperbolic manifold, must $\operatorname{vol}(M)=\sum_i P_n(z_i)$ for elements $z_i$ in the Bloch group satisfying $\delta_n(\sum_i z_i)=0$?

来源字段：https://doi.org/10.4171/owr/2004/48

06704 | AIM-GEOMETRY-0073

### Block-pressure ties and matrix equilibrium coexistence

P / partially_solved | AIM 研讨会问题汇编

Describe the set of matrix cocycles which have multiple eqilibrium states. For simplicity, consider a subshift of finite type, and a locally constant, fiber bunched cocycle.

来源字段：http://aimpl.org/equibdynsysgeom/5/

06705 | OWR-11569-001

### Bolza Surface Maximization of the First Eigenvalue

O / open | Oberwolfach 报告问题汇编

Is the first eigenvalue on the Teichmüller space of genus-$2$ surfaces maximized by the Bolza surface?

来源字段：https://doi.org/10.4171/owr/2012/02

06706 | GEOM-026

### Borromean Rings Question

P / partially_solved | 独立 / 未分配集合

Can three unknotted space curves (not all circles) be arranged as Borromean rings?

背景链接：https://en.wikipedia.org/wiki/Borromean_rings

06707 | OWR-14298163-006

### Borsuk Partition Problem in Dimensions Four Through Sixty-Three

P / partially_solved | Oberwolfach 报告问题汇编

For each $4\leq d\leq 63$, determine whether every bounded set $S\subseteq\mathbb R^d$ can be partitioned into at most $d+1$ subsets of diameter strictly smaller than $\operatorname{diam}(S)$.

来源字段：https://doi.org/10.4171/owr/2024/3

06708 | OWR-2654829-006

### Bott-Chern and Aeppli Cohomology for Almost-Complex Manifolds

O / open | Oberwolfach 报告问题汇编

For an almost-complex manifold, which Bott-Chern and Aeppli cohomology groups complete the vertical row of the Dolbeault-de Rham diagram while recovering the classical integrable case, respecting the natural maps and spectral sequences, carrying a real structure, and possessing the expected algebra and module structures?

来源字段：https://doi.org/10.4171/owr/2020/33

06709 | AMR-061-0001

### Boundaries of Groups and Kleinian Groups - Problem 1

P / partially_solved | AMR 问题清单汇编

What spaces can arise as boundaries of hyperbolic groups? As a sub-problem: For which k do k-dimensional stable Menger spaces appear as boundaries?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06710 | AMR-061-0010

### Boundaries of Groups and Kleinian Groups - Problem 10

O / open | AMR 问题清单汇编

Does there exist a Coxeter group Gn with n-dimensional boundary ∂Gn, so that the rational homological dimension of ∂Gn equals 1?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06711 | AMR-061-0100

### Boundaries of Groups and Kleinian Groups - Problem 100

O / open | AMR 问题清单汇编

Is there a similar statement to this inflexibility result this with no group specified-that is, for subsets Λ ⊂ S2 of the boundary sphere of H3?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06712 | AMR-061-0101

### Boundaries of Groups and Kleinian Groups - Problem 101

O / open | AMR 问题清单汇编

Given p ∈ H3, estimate the biLipschitz constant of QΛ near p in terms of the distance d from p to the exterior of the convex hull of Λ. More concretely: if Λ is a quasicircle, is the decay exponential in d?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06713 | AMR-061-0102

### Boundaries of Groups and Kleinian Groups - Problem 102

P / partially_solved | AMR 问题清单汇编

别名：Eliminating the center in the CAT(0) braid-group problem [AIM-GEOMETRIC_GROUP_THEORY-0052]

Are braid groups CAT(0)?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06714 | AMR-061-0103

### Boundaries of Groups and Kleinian Groups - Problem 103

O / open | AMR 问题清单汇编

Extend Rips' theory to higher-dimensional buildings, e.g. products ofR-trees. Rank rigidity. Let X be a CAT (0) metric space. The space X is said to be or rank ≥ n if every geodesic segment in X is contained in a subset E which is isometric to a flat n-dimensional parallelepiped. If Y is a locally CAT (0) metric space, then Y is said to have rank ≥ n if its universal cover is of rank ≥ n. The rank rigidity theorem proven by Ballmann [1] and by Burns and Spatzier [15, 16] states that: If M is a compact nonpositively curved Riemannian manifold of rank ≥ 2, then either M admits a finite cover the universal cover of M splits (nontrivially) as a Riemannian direct product or M is a locally symmetric space.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06715 | AMR-061-0104

### Boundaries of Groups and Kleinian Groups - Problem 104

P / partially_solved | AMR 问题清单汇编

Suppose that Y is a compact finitedimensional locally CAT (0) metric space of rank n ≥ 2. Then either the universal cover of Y splits (nontrivially) as a Riemannian direct product or it is isometric to a Euclidean building. This problem is most natural in the context of piecewise-Euclidean metric cell complexes. The conjecture was proven in the case of 2-dimensional and 3-dimensional complexes by Ballmann and Brin [2, 3]. Cogrowth. Let H ⊂ G be a subgroup of a finitely-generated group G. The cogrowth of H in G is the growth of the Shreier graph Γ G/H, where Γ G is a Cayley graph of G.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06716 | AMR-061-0105

### Boundaries of Groups and Kleinian Groups - Problem 105

P / partially_solved | AMR 问题清单汇编

Compute cogrowth for notable subgroups $H\subset G$, where cogrowth is the growth of the Schreier graph $\Gamma_{G/H}$. In particular: prove that the cogrowth of $\operatorname{SL}(n,\mathbb Z)$ in $\operatorname{SL}(n+1,\mathbb Z)$ is exponential; compute cogrowth of special subgroups in Coxeter groups; and, if $\Gamma_{G/H}$ is Gromov-hyperbolic, decide whether its cogrowth is necessarily constant, linear, or exponential.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06717 | AMR-061-0107

### Boundaries of Groups and Kleinian Groups - Problem 107

O / open | AMR 问题清单汇编

Under the above assumptions, is it true that Y has coarsely trivial πm for m ≥ 2?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06718 | AMR-061-0108

### Boundaries of Groups and Kleinian Groups - Problem 108

O / open | AMR 问题清单汇编

Does the Coarse Whitehead Conjecture hold if G is hyperbolic?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06719 | AMR-061-0011

### Boundaries of Groups and Kleinian Groups - Problem 11

O / open | AMR 问题清单汇编

Under which conditions on the Coxeter diagram of G, the boundary of a Coxeter group is n-connected and locally n-connected?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06720 | AMR-061-0012

### Boundaries of Groups and Kleinian Groups - Problem 12

O / open | AMR 问题清单汇编

Can exotic homology manifolds as in [14] appear as ideal boundaries of Coxeter groups?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06721 | AMR-061-0013

### Boundaries of Groups and Kleinian Groups - Problem 13

P / partially_solved | AMR 问题清单汇编

Find further universality phenomena: classes of groups or spaces of different nature whose ideal boundaries are nevertheless all homeomorphic, beyond the source's examples involving Menger compacta, right-angled hyperbolic buildings, and Davis--Vinberg complexes.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06722 | AMR-061-0014

### Boundaries of Groups and Kleinian Groups - Problem 14

P / partially_solved | AMR 问题清单汇编

Let $N$ be a closed $n$-manifold and $\Delta$ a flag, no-square triangulation, and let $C(N,\Delta)$ be the associated Davis--Vinberg complex. Is $\partial_\infty C(N,\Delta)$ a topological invariant of $N$, independent of $\Delta$?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06723 | AMR-061-0015

### Boundaries of Groups and Kleinian Groups - Problem 15

O / open | AMR 问题清单汇编

Suppose that ( N1, Δ1) and (N2, Δ2) are closed 3-manifolds equipped with flag-triangulations, so that ∂∞C(N1, Δ1) = ∂∞C(N2, Δ2). Does it follow that every prime connected sum summand of Ni appears as a connected sum summand of Ni+1, i = 1, 2? What can be said in higher dimensions?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06724 | AMR-061-0016

### Boundaries of Groups and Kleinian Groups - Problem 16

P / partially_solved | AMR 问题清单汇编

Let Z be a compactum which is a Z-boundary of a group G. Then Z is never a Boltyansky compactum. In the special case when Z is an Markov compactum, so that all building blocks Kσ → σ are isomorphic, it was proven in [29] that Z cannot be a Boltyansky compactum.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06725 | AMR-061-0017

### Boundaries of Groups and Kleinian Groups - Problem 17

O / open | AMR 问题清单汇编

Examples of Kleiner and Croke [22], [23] of non-unique boundaries are badly non-locally-connected. Is that essential in having the “flexibility” to have many boundaries? That is, does local connectedness imply uniqueness of the boundary (in the 1-ended case) for CAT(0) groups?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06726 | AMR-061-0018

### Boundaries of Groups and Kleinian Groups - Problem 18

O / open | AMR 问题清单汇编

Suppose that G is a CAT (0) group which does not split over a small subgroup. Does it follow that ∂∞G is unique?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06727 | AMR-061-0019

### Boundaries of Groups and Kleinian Groups - Problem 19

O / open | AMR 问题清单汇编

Is the boundary well-defined for groups acting geometrically on CAT (0)-cube complexes? More precisely, suppose that X1, X2 are cube complexes which admit geometric actions of a group G. Does it follow that ∂∞X1 = ∂∞X2?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06728 | AMR-061-0002

### Boundaries of Groups and Kleinian Groups - Problem 2

O / open | AMR 问题清单汇编

Can one remove the “right-angled” assumption in Osajda result?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06729 | AMR-061-0020

### Boundaries of Groups and Kleinian Groups - Problem 20

O / open | AMR 问题清单汇编

What topological invariants distinguish boundaries? In particular, what topological properties of boundaries are quasi-isometry invariants? Does something coarser than the topology stay invariant?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06730 | AMR-061-0021

### Boundaries of Groups and Kleinian Groups - Problem 21

P / partially_solved | AMR 问题清单汇编

If G acts geometrically on two CAT(0) spaces, are the resulting boundaries cell-like equivalent? (That is, does there exist a space Z with cell-like maps to each of the two spaces?)

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06731 | AMR-061-0022

### Boundaries of Groups and Kleinian Groups - Problem 22

P / partially_solved | AMR 问题清单汇编

Is there a convex core for the diagonal action of G on X1 × X2? (A special case is surface groups G with X1 and X2 corresponding to different hyperbolic structures.) If there is a convex core, can Z (the space with cell-like maps to X1 and X2) be taken to be the boundary of the core?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06732 | AMR-061-0023

### Boundaries of Groups and Kleinian Groups - Problem 23

O / open | AMR 问题清单汇编

Define a topology on the set of quasi geodesics in a (proper geodesic, or coarsely homogeneous, or cocompact) CAT (0) space which (1) has a description as an increasing union of compact metrizable spaces (2) has an inclusion of its visual boundary ∂∞X into it (3) is quasi-isometry invariant (4) has reasonable measure classes which are quasipreserved According to a theorem by Brian Bowditch and Gadde Swarup [13, 56], if G is a 1-ended hyperbolic group then ∂∞G has no cut points. For G a CAT(0) group, a theorem of Eric Swenson says that if c ∈ ∂∞G is a cut point, then there is an infinite-torsion subgroup of G fixing c.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06733 | AMR-061-0024

### Boundaries of Groups and Kleinian Groups - Problem 24

P / partially_solved | AMR 问题清单汇编

Any CAT (0) group has no infinite-torsion subgroups. A Euclidean retract is a compact space that embeds into some Rn as a retract. A compact metrizable space Z is a Z-set in ˜X if it is “homotopically negligible” (for every open U ⊂ ˜X, the inclusion U \mathbb{Z} in U is a homotopy equivalence). A Z-structure on a group G is a pair ( ˜X, Z ) such that • ˜X is a Euclidean retract, • Z is a Z-set in ˜X, • X:= ˜X \ Z admits a covering space action of G with X/G compact, • the set of translates of any compact set K ⊂ X is a null sequence in ˜X (that is, for each ϵ > 0 there are only finitely many translates with diam > ϵ). Finally, Z is a boundary of G (or Z-structure boundary) if there exists a Z-structure ( ˜X, Z ) on G. The above notion boundary of G was generalized by T. Farrell and J. Lafont as follows: An EZ-boundary of a group G is a boundary Z = ∂EZ G so that the action of G on X extends to topological action of G on Z.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06734 | AMR-061-0025

### Boundaries of Groups and Kleinian Groups - Problem 25

O / open | AMR 问题清单汇编

Let G be a hyperbolic group and ∂EZ G be its EZ boundary. Is it true that ∂EZ G is equivariantly homeomorphic to the Gromov boundary of G?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06735 | AMR-061-0026

### Boundaries of Groups and Kleinian Groups - Problem 26

O / open | AMR 问题清单汇编

Can there be two different boundaries in the sense of Z-structures for a group G that are not cell-like equivalent?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06736 | AMR-061-0027

### Boundaries of Groups and Kleinian Groups - Problem 27

O / open | AMR 问题清单汇编

Is the property of splitting over a 2-ended subgroup an invariant of Bestvina boundaries? Some necessary conditions are known for compact, metrizable spaces X to be the boundary of some proper cocompact CAT (0) space: (1) X should have 1,2, or infinitely many components (2) X is finite dimensional (Theorem of Swenson) (3) X has nontrivial top ˇCech cohomology (Geoghegan-Ontaneda) In the case when X admits a cocompact free(?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06737 | AMR-061-0028

### Boundaries of Groups and Kleinian Groups - Problem 28

P / partially_solved | AMR 问题清单汇编

Extend the known necessary conditions for a compact metrizable space $X$ to be the boundary of a proper cocompact CAT(0) space, or give a complete classification. The listed conditions are that $X$ have one, two, or infinitely many components, be finite dimensional, and have nontrivial top Čech cohomology; if a discrete isometry group acts freely and cocompactly, every nonempty open subset must have the same dimension as $X$.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06738 | AMR-061-0029

### Boundaries of Groups and Kleinian Groups - Problem 29

P / partially_solved | AMR 问题清单汇编

Does every CAT(0) group have finite asymptotic dimension?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06739 | AMR-061-0003

### Boundaries of Groups and Kleinian Groups - Problem 3

P / partially_solved | AMR 问题清单汇编

What 2-dimensional spaces arise as boundaries of hyperbolic groups?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06740 | AMR-061-0030

### Boundaries of Groups and Kleinian Groups - Problem 30

P / partially_solved | AMR 问题清单汇编

Do Papasoglu's three results for finitely presented one-ended groups extend to all finitely generated groups: quasi-isometry invariance of the JSJ decomposition; characterization of coarse separation by a quasiline via splitting over a virtually cyclic group or virtual surface-group structure; and nonseparation by quasi-rays?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06741 | AMR-061-0031

### Boundaries of Groups and Kleinian Groups - Problem 31

O / open | AMR 问题清单汇编

Are splittings overZ2 (orZn) invariant under quasiisometry? The analogous problem also makes sense for the JSJ decompositions.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06742 | AMR-061-0032

### Boundaries of Groups and Kleinian Groups - Problem 32

P / partially_solved | AMR 问题清单汇编

Suppose G is finitely generated and there is a sequence of quasicircles that separate its Cayley graph. Is G virtually a surface group?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06743 | AMR-061-0033

### Boundaries of Groups and Kleinian Groups - Problem 33

O / open | AMR 问题清单汇编

If G is finitely generated with asymptotic dimension ≥ n, and X is a subset of the Cayley graph with asymptotic dimension ≤ n - 2 that coarsely separates the Cayley graph, then G splits over some subgroup H ≤ G with asymptotic dimension ≤ n - 1. A homogeneous continuum is a locally connected compact metric space whose group of homeomorphisms acts transitively. Papasoglu showed that every simply connected homogeneous continuum has the property that no simple arc separates it.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06744 | AMR-061-0034

### Boundaries of Groups and Kleinian Groups - Problem 34

P / partially_solved | AMR 问题清单汇编

Do all homogeneous continua (with dimension greater than 2) have this property?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06745 | AMR-061-0036

### Boundaries of Groups and Kleinian Groups - Problem 36

O / open | AMR 问题清单汇编

Let f: Bn → Bn be a quasiconformal homeomorphism. Can f be approximated by globally quasiconformal diffeomorphisms fj: Bn → Bn? Can this be done so that fj's are K-quasiconformal for all j?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06746 | AMR-061-0037

### Boundaries of Groups and Kleinian Groups - Problem 37

O / open | AMR 问题清单汇编

Find good classes of spaces such that the infinitesimal metric condition (for quasiconformality) implies the local condition. (This is generally true in Loewner spaces.)

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06747 | AMR-061-0038

### Boundaries of Groups and Kleinian Groups - Problem 38

O / open | AMR 问题清单汇编

Outside of the boundaries of Fuchsian buildings, what boundaries have the Loewner property?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06748 | AMR-061-0039

### Boundaries of Groups and Kleinian Groups - Problem 39

O / open | AMR 问题清单汇编

Let X be a non-smoothable closed simply connected 4-manifold. Does it admit an Ahlfors 4-regular linearly locally contractible metric? This is wide open; unknown even for examples, like E8.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06749 | AMR-061-0004

### Boundaries of Groups and Kleinian Groups - Problem 4

O / open | AMR 问题清单汇编

Are there torsion-free hyperbolic groups G with cdQ(G)/cdZ(G) < 2/3?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06750 | AMR-061-0040

### Boundaries of Groups and Kleinian Groups - Problem 40

P / partially_solved | AMR 问题清单汇编

Develop a theory for analysis on the ideal boundaries of relatively hyperbolic groups, as it is done for hyperbolic groups.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06751 | AMR-061-0041

### Boundaries of Groups and Kleinian Groups - Problem 41

P / partially_solved | AMR 问题清单汇编

In what generality does quasiconformal imply quasisymmetric?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06752 | AMR-061-0042

### Boundaries of Groups and Kleinian Groups - Problem 42

O / open | AMR 问题清单汇编

Take your favorite metric fractal. Is it quasisymmetrically cohopfian? What about the boundaries of hyperbolic groups?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06753 | AMR-061-0043

### Boundaries of Groups and Kleinian Groups - Problem 43

O / open | AMR 问题清单汇编

If ∂∞G is Loewner, then it is quasisymmetrically cohopfian. (Boundaries of Fuchsian buildings provide a good test case for this conjecture.)

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06754 | AMR-061-0044

### Boundaries of Groups and Kleinian Groups - Problem 44

O / open | AMR 问题清单汇编

If G is a hyperbolic group and ∂∞G is connected with no local cut points, is there a natural measure class which is quasisymmetrically invariant?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06755 | AMR-061-0045

### Boundaries of Groups and Kleinian Groups - Problem 45

O / open | AMR 问题清单汇编

Can you do analysis on CAT(0) boundaries? With no natural metric, is there any structure beyond topology?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06756 | AMR-061-0046

### Boundaries of Groups and Kleinian Groups - Problem 46

O / open | AMR 问题清单汇编

G = Isom( X) acts on ∂∞(X). Is this action “nice” with respect to the metrics in the previous remark?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06757 | AMR-061-0047

### Boundaries of Groups and Kleinian Groups - Problem 47

O / open | AMR 问题清单汇编

If D is the boundary of a hyperbolic group and D is connected, has no local cut points, and is not Loewner, is there a quasisymmetrically invariant nontrivial closed equivalence relation ∼ on D so that D/ ∼ is Hausdorff and is a boundary of G relative to a collection of parabolic subgroups?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06758 | AMR-061-0048

### Boundaries of Groups and Kleinian Groups - Problem 48

O / open | AMR 问题清单汇编

Study relationships between different notions of conformal structure on ∂∞(G) for hyperbolic G. Here is an (incomplete) list of such notions: (1) 1-quasiconformal in the metric sense, i.e. H(f ) = 1. (2) preserving modulus of curves joining two compacta. (3) η-quasisymmetric with η as close to linear as we like. ( η are functions of the point x ∈ ∂∞G where we test f for conformality) (4) if Poincar´ e inequality holds for ∂∞G, then, using Cheeger cotangent bundle T ∗∂∞G, can give a notion of measurable bounded conformal structure µ such that Conf(∂∞G, µ) is a convergence group.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06759 | AMR-061-0049

### Boundaries of Groups and Kleinian Groups - Problem 49

O / open | AMR 问题清单汇编

For homeomorphisms of Hilbert spaces, do the Euclidean implications 'quasiconformal implies quasisymmetric implies mapping balls to quasiballs' continue to hold?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06760 | AMR-061-0005

### Boundaries of Groups and Kleinian Groups - Problem 5

O / open | AMR 问题清单汇编

What can be said about boundaries arising from strict hyperbolization constructions of Charney and Davis, [18]?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06761 | AMR-061-0050

### Boundaries of Groups and Kleinian Groups - Problem 50

O / open | AMR 问题清单汇编

Can one do this with smaller m? Say m = n + 1? (Same problem valid for complex hyperbolic space.) Subproblem (Misha Kapovich): Consider X = ∂∞HHn sitting inside of Y = ∂∞HHn+1. Is X locally quasi-symmetrically rigid in Y? More precisely, is it true that each quasisymmetric embedding f: X → Y which is sufficiently close to the identity is induced by an isometry of HHn+1?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06762 | AMR-061-0051

### Boundaries of Groups and Kleinian Groups - Problem 51

P / partially_solved | AMR 问题清单汇编

Are all quasi-isometric embeddings between higher-rank symmetric spaces either isometries or algebraic in this way?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06763 | AMR-061-0052

### Boundaries of Groups and Kleinian Groups - Problem 52

O / open | AMR 问题清单汇编

Let G be a hyperbolic group. Is it true that G admit a uniformly quasiconformal discrete action on Sn (for some n)?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06764 | AMR-061-0006

### Boundaries of Groups and Kleinian Groups - Problem 6

O / open | AMR 问题清单汇编

Is there an example of a group G which is hyperbolic relative to some parabolic subgroups that are nilpotent of class ≥ 3 whose Bowditch boundary is homeomorphic to some n-sphere?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06765 | AMR-061-0061

### Boundaries of Groups and Kleinian Groups - Problem 61

O / open | AMR 问题清单汇编

For a hyperbolic group G, ACD (∂∞G) = inf G↷X {Hdim(∂∞X, visual)}, where the infimum is taken over all geometric actions of G on metric spaces X. A bolder conjecture would be that, when the infimum is attained, it is attained by a visual metric.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06766 | AMR-061-0062

### Boundaries of Groups and Kleinian Groups - Problem 62

O / open | AMR 问题清单汇编

What is ACD of the standard Sierpinski carpet? In particular, does the above conjecture hold?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06767 | AMR-061-0063

### Boundaries of Groups and Kleinian Groups - Problem 63

O / open | AMR 问题清单汇编

Under what assumptions on hyperbolic groups G with Q-Loewner boundary ∂∞G does it admit a 1-Poincar´ e inequality for the boundary?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06768 | AMR-061-0064

### Boundaries of Groups and Kleinian Groups - Problem 64

S / solved | AMR 问题清单汇编

[Cannon-Thurston] Is this action conjugate to a conformal action?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06769 | AMR-061-0065

### Boundaries of Groups and Kleinian Groups - Problem 65

S / solved | AMR 问题清单汇编

The limit set of the Kleinian group ι(G) is locally connected. In the presence of two geodesic laminations, the limit set of ι(G) is the entire 2-sphere, so local connectedness is meaningless. Then the correct reformulation is as follows:

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06770 | AMR-061-0066

### Boundaries of Groups and Kleinian Groups - Problem 66

S / solved | AMR 问题清单汇编

Is there an equivariant continuous map (called Cannon-Thurston map) from the unit circle S1 (the ideal boundary of G as an abstract group) to S2? Then Problem 64 is equivalent to 66.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06771 | AMR-061-0067

### Boundaries of Groups and Kleinian Groups - Problem 67

S / solved | AMR 问题清单汇编

Let H ⊂ G be a hyperbolic subgroup of a hyperbolic group (we do not assume that H is quasiconvex). Is it true that there exists an equivariant continuous map ∂∞H → ∂∞G. See [46] for partial results in this direction.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06772 | AMR-061-0068

### Boundaries of Groups and Kleinian Groups - Problem 68

P / partially_solved | AMR 问题清单汇编

What is the Poisson boundary of the free group with an arbitrary measure?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06773 | AMR-061-0069

### Boundaries of Groups and Kleinian Groups - Problem 69

S / solved | AMR 问题清单汇编

There almost surely exists a horofunction h such that lim n→∞ - 1 n h(xn) = A, where A = lim n→∞ 1 n d(x0, xn). A theorem of Karlsson states that ∀ϵ > 0 there exists a horofunction hϵ such that A - ϵ ≤ - 1 n hϵ(xn) ≤ A + ϵ for all n ≥ N (ϵ).

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06774 | AMR-061-0007

### Boundaries of Groups and Kleinian Groups - Problem 7

O / open | AMR 问题清单汇编

Suppose that Z is a compact metrizable topological space, G↷ Z is a convergence action which is topologically transitive, i.e. each G- orbit is dense in Z. Is there a Gromov-hyperbolic space X with the ideal boundary Z so that the action G ↷ Z extends to a uniformly quasi-isometric quasi-action G↷ X?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06775 | AMR-061-0070

### Boundaries of Groups and Kleinian Groups - Problem 70

O / open | AMR 问题清单汇编

For any proper metric space it is possible to associate a kind of incidence geometry at infinity via horofunctions, halfspaces and their limits called stars. For the CAT(0) case, this structure is intimately connected with the Tits geometry, and for Teichm¨ uller space it should relate well with the curve complex. In which situations do homomorphisms induce “incidence preserving” maps between these geometries at infinity? Same problem for quasi-isometries.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06776 | AMR-061-0071

### Boundaries of Groups and Kleinian Groups - Problem 71

O / open | AMR 问题清单汇编

Consider the compactification of a finitely generated group constructed in the usual Stone- ˇCech way using the first l2 (or some other function space) cohomology. Is the associated incidence geometry at infinity always trivial (i.e., hyperbolic)?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06777 | AMR-061-0072

### Boundaries of Groups and Kleinian Groups - Problem 72

P / partially_solved | AMR 问题清单汇编

Characterize the hitting measure ν on PMF obtained from the random walk by mapping classes on Teichm¨ uller space. Is it absolutely continuous with respect to visual measure (that is, Lebesgue measure on the visual sphere of directions)?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06778 | AMR-061-0073

### Boundaries of Groups and Kleinian Groups - Problem 73

S / solved | AMR 问题清单汇编

What is the Poisson boundary of Outer space?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06779 | AMR-061-0074

### Boundaries of Groups and Kleinian Groups - Problem 74

P / partially_solved | AMR 问题清单汇编

Is there are meaningful structure theory for lacunary hyperbolic groups? Can one define a useful boundary for such groups? Is it true that either Out(G) is finite or G splits over a virtually cyclic subgroup? 1Such groups are never finitely-presented

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06780 | AMR-061-0075

### Boundaries of Groups and Kleinian Groups - Problem 75

P / partially_solved | AMR 问题清单汇编

Every relatively hyperbolic group has cut points in all of its asymptotic cones. To what extent does the converse hold? Characterize the finitely generated groups all of whose asymptotic cones have cut points.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06781 | AMR-061-0076

### Boundaries of Groups and Kleinian Groups - Problem 76

O / open | AMR 问题清单汇编

The study of asymptotic cones has been non-analytic (they have been studied up to homeomorphism). What analytic tools could be developed?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06782 | AMR-061-0077

### Boundaries of Groups and Kleinian Groups - Problem 77

P / partially_solved | AMR 问题清单汇编

For the fundamental group G of a closed hyperbolic n-manifold consider a short exact sequence 1 →Zp → Γ → G → 1. Is the group Γ residually finite? In other words, is there a finite-index subgroup G′ in G so that the restriction map H 2(G,Zp) → H 2(G′,Zp) is zero?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06783 | AMR-061-0078

### Boundaries of Groups and Kleinian Groups - Problem 78

P / partially_solved | AMR 问题清单汇编

Let $G$ be the fundamental group of a closed hyperbolic $n$-manifold. Is there a finite-index subgroup G′ ⊂ G so that the restriction map H 3(G,Z2) → H 3(G′,Z2) is zero?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06784 | AMR-061-0079

### Boundaries of Groups and Kleinian Groups - Problem 79

P / partially_solved | AMR 问题清单汇编

Let G be a Gromov-hyperbolic Coxeter group. Does G admit a discrete embedding in Isom(Hn) for large n?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06785 | AMR-061-0008

### Boundaries of Groups and Kleinian Groups - Problem 8

O / open | AMR 问题清单汇编

Find topological restrictions on the ideal boundaries of CAT (-1) cubical complexes.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06786 | AMR-061-0080

### Boundaries of Groups and Kleinian Groups - Problem 80

O / open | AMR 问题清单汇编

Let G ⊂ P U(2, 1) be a convex-cocompact subgroup of isometries of complex-hyperbolic 2-space. Can the limit set of G be homeomorphic to the Sierpinski carpet?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06787 | AMR-061-0081

### Boundaries of Groups and Kleinian Groups - Problem 81

P / partially_solved | AMR 问题清单汇编

Let G ⊂ Isom(Hn) be a discrete torsion-free finitelygenerated subgroup without abelian subgroups of rank ≥ 2. Is it true that (a) cdZ(G) ≤ Hdim(Λc(G)) + 1? Here Λ c is the conical limit set. The answer is known [36] to be positive if one considers homological rather than cohomological dimension. (b) In the case of equality, is it true that the limit set of G is the round sphere and G? This is known to be true in the case when G is geometrically finite [36]. (c) If Hdim(Λc(G)) < 2, is it true that G is geometrically finite? (d) If Hdim(Λc(G)) < 1, does it follow that G is a classical Schottky-type group?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06788 | AMR-061-0082

### Boundaries of Groups and Kleinian Groups - Problem 82

S / solved | AMR 问题清单汇编

Let G ⊂ Isom(H4) be a Schottky group (or, more generally, a free convex-cocompact group). Can Hausdorff dimension of the limit set of G be arbitrarily close to 3?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06789 | AMR-061-0083

### Boundaries of Groups and Kleinian Groups - Problem 83

P / partially_solved | AMR 问题清单汇编

Let G be a finitely-generated discrete group of isometries of a Gromov-hyperbolic space X so that the limit set of G is connected. Is it true that the limit set of G is locally connected?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06790 | AMR-061-0084

### Boundaries of Groups and Kleinian Groups - Problem 84

O / open | AMR 问题清单汇编

Let $G$ be a group and let $\rho_1,\rho_2:G\to\operatorname{Isom}(\mathbb{H}^n)$ be discrete faithful representations; write $\ell_{\rho}(g)=\inf_x d(\rho(g)x,x)$. Suppose that ρ1, ρ2 are discrete and faithful representations so that there exists C > 0 for which we have C -1 ≤ ℓρ1(g) ℓρ2(g) ) ≤ C, ∀g ∈ G. Does it follow that there exists a quasiconformal map f: Λ(ρ1(G)) → Λ(ρ2(G)) which is equivariant with respect to the isomorphism ρ2 ◦ ρ-1 1? Can one choose f which is K-quasiconformal for K = K(C)?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06791 | AMR-061-0085

### Boundaries of Groups and Kleinian Groups - Problem 85

P / partially_solved | AMR 问题清单汇编

Find a “constructive” proof of the above theorem. More precisely, consider a finite presentation ⟨g1,.., gk|R1,.., Rm⟩ of G. Given [ ρ] ∈ D n(G) define Bn([ρ]):= inf x∈Hn max i=1,...,k d(x, ρ(gi)(x)). Find an explicit constant C, which depends on n, k, m and the lengths of the words Ri, so that the function Bn: Dn(G) →R is bounded from above by C.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06792 | AMR-061-0086

### Boundaries of Groups and Kleinian Groups - Problem 86

O / open | AMR 问题清单汇编

Find new restrictions on Kleinian groups. Recall that a group G is called coherent if every finitely-generated subgroup of G is finitely-presented.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06793 | AMR-061-0087

### Boundaries of Groups and Kleinian Groups - Problem 87

P / partially_solved | AMR 问题清单汇编

Prove that every arithmetic lattice in Isom( Hn) ( n ≥ 4) is non-coherent. See [38] for some partial results in this direction. It is well-known that every lattice in Isom( HHn) ( n ≥ 2) has Property T.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06794 | AMR-061-0088

### Boundaries of Groups and Kleinian Groups - Problem 88

O / open | AMR 问题清单汇编

Suppose that G ⊂ Isom(HHn) is a discrete subgroup satisfying Property T. Does it follow that G preserves a totally-geodesic subspace H in HHn and acts on H as a lattice?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06795 | AMR-061-0089

### Boundaries of Groups and Kleinian Groups - Problem 89

O / open | AMR 问题清单汇编

Suppose that Δ is a developable triangle of groups, where all the cellgroups have Property T and so that all the links in the universal cover of T have λ1 > 1/2. Does it follow that π1(Δ) has Property T?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06796 | AMR-061-0009

### Boundaries of Groups and Kleinian Groups - Problem 9

O / open | AMR 问题清单汇编

Is it true that isomorphic Coxeter groups have homeomorphic boundaries?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06797 | AMR-061-0090

### Boundaries of Groups and Kleinian Groups - Problem 90

P / partially_solved | AMR 问题清单汇编

Generalize Bestvina-Feighn combination theorem from graphs of groups to complexes of groups.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06798 | AMR-061-0091

### Boundaries of Groups and Kleinian Groups - Problem 91

O / open | AMR 问题清单汇编

Generalize Vinberg’s finiteness theorem for reflection groups to complex-hyperbolic reflection groups, i.e., prove that there exists a number N such that for n ≥ N, there are no lattices in P U(n, 1) which are generated by reflections.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06799 | AMR-061-0092

### Boundaries of Groups and Kleinian Groups - Problem 92

P / partially_solved | AMR 问题清单汇编

There is a theory of quasi-convex groups acting on Gromov hyperbolic spaces, generalizing the theory of convex-compact groups of isometries of the real hyperbolic space. Develop a theory of geometric finiteness in CAT(0) spaces.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06800 | AMR-061-0093

### Boundaries of Groups and Kleinian Groups - Problem 93

P / partially_solved | AMR 问题清单汇编

Extend this relation of Anosov structure and dynamics on the limit set to representations of other hyperbolic groups.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06801 | AMR-061-0094

### Boundaries of Groups and Kleinian Groups - Problem 94

O / open | AMR 问题清单汇编

Generalize holomorphic chain patterns in ∂∞CHn in order to prove rigidity results for embeddings of lattices in P U(n, 1) into other higher rank Lie groups.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06802 | AMR-061-0095

### Boundaries of Groups and Kleinian Groups - Problem 95

P / partially_solved | AMR 问题清单汇编

Obtain new rigidity results for embeddings of realhyperbolic lattices into higher-rank semisimple Lie groups in terms of the boundary maps.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06803 | AMR-061-0096

### Boundaries of Groups and Kleinian Groups - Problem 96

O / open | AMR 问题清单汇编

If X is a compact polyhedron and G is a discrete group of simple homotopy equivalences X → X, is there a compact space X ′, homotopy equivalent to X, such that G can be realized as a group of homeomorphisms of X ′.

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06804 | AMR-061-0098

### Boundaries of Groups and Kleinian Groups - Problem 98

O / open | AMR 问题清单汇编

For a word-hyperbolic G not splitting over any virtually cyclic group, can an infinite-index subgroup and a finite-index subgroup be isomorphic?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06805 | AMR-061-0099

### Boundaries of Groups and Kleinian Groups - Problem 99

S / solved | AMR 问题清单汇编

Consider Teichm¨ uller spaceT (S) with Teichm¨ uller metric. Does it have quadratic isoperimetric inequality?

背景链接：https://www.math.ucdavis.edu/~kapovich/EPR/problems.pdf

06806 | OWR-761-006

### Boundary and Hyperbolic Dimensions of Symmetric Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Is $\dim\partial_\infty\Gamma=\operatorname{mdim}\partial_\infty\Gamma$ for a visual metric on the boundary of a Gromov-hyperbolic group? What is $\operatorname{hdim}(\mathbb{H}^2\times\mathbb{H}^2)$? More generally, is the hyperbolic dimension of every noncompact symmetric space without Euclidean factors at least its dimension?

来源字段：https://doi.org/10.4171/owr/2004/47

06807 | AMR-058-0012

### Boundary Singularities of Soap Films

P / partially_solved | AMR 问题清单汇编

Is the known list of boundary singularities of soap films in $\mathbb{R}^3$ complete? Classify what happens for singular or immersed boundaries and when a soap film may touch only part of its boundary wire.

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

06808 | AIM-GEOMETRY-0123

### Boundary stretching and a semiclassical quiver-metric limit

P / partially_solved | AIM 研讨会问题汇编

What is the precise relation between the complete hyperkahler metrics on the moduli spaces $\mathcal{M^{\ast }}$ and those on the full (wild) Hitchin moduli space $\mathcal{M}$? Can one view the simpler metric on $\mathcal{M^{\ast }}$ as an approximation?

来源字段：http://aimpl.org/spectralhiggs/1/

06809 | OWR-13108-003

### Bounded Negativity and Infinite Families of Negative Curves

O / open | Oberwolfach 报告问题汇编

Bounded-negativity conjecture: for every smooth projective surface $X$ over $\mathbb C$, the self-intersections of reduced irreducible curves on $X$ are bounded below; additionally, for every $g>1$, does some $X$ have infinitely many curves with $C^2=-1$ and arithmetic genus $g$?

来源字段：https://doi.org/10.4171/owr/2014/38

06810 | OWR-16412-008

### Bounded Plurisubharmonic Functions and Manifold Cores

O / open | Oberwolfach 报告问题汇编

For a complex manifold $M$ and $w_0\in M$, is $A_{cb}(w_0)=A_{cb}^e(w_0)$, is $A_{cb}(w_0)\subseteq c(M)$, and does there exist a nonnegative function $u\in\operatorname{PSH}_{cb}(M)$ with zero set $\{u=0\}=c(M)$?

来源字段：https://doi.org/10.4171/owr/2018/37

06811 | AMR-038-0002

### Bounded-degree triangulations

O / open | AMR 问题清单汇编

Can every convex polytope be triangulated so that every vertex degree, or every edge degree, is bounded by a constant or by a polylogarithmic function of the input size?

背景链接：https://ics.uci.edu/~eppstein/junkyard/open.html

06812 | AMR-067-0005

### Branch points of area-minimizing surfaces

O / open | AMR 问题清单汇编

1. Does ${\rm Sing}_b (T)$ have zero $(m-1)$-dimensional Hausdorff measure? 2. If yes, does ${\rm Sing}_b (T)$ have (Hausdorff) dimension at most $m-2$? 3. A related question is: can we find an example as in where the sequence of accumulating singularities $\{P_k\}\subset {\rm Sing}_b (T)$ are of branching type? If yes, then one could imagine a Cantor type construction giving a negative answer to 2. 4. White in conjectures that if a $2$-dimensional area-minimizing currents spans a real analytic closed curve, then it has finitely many (interior and boundary) singularities and hence finite topological type. 5. The example in is topologically a disk. Is it possible to give an example of a smooth closed curve $\Gamma \subset \mathbb R^{2+n}$ which bounds an area-minimizing $2$-dimensional current with infinite topology?

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

06813 | AMR-142-0002

### Branko Grunbaum in Many Dimensions - Conjecture 2

O / open | AMR 问题清单汇编

The Four Color Theorem for planar graphs is equivalent to the statement that the dual graph of every triangulated $2$-sphere is $3$-edge colorable. This is essentially Tait's 1880 observation that the Four Color Theorem is equivalent to showing that every cubic bridgeless planar graph is $3$-edge colorable [Tait1880].

In 1968, Gr\"unbaum conjectured a beautiful generalization.

If $ G $ is a simple loopless triangulation of an orientable surface $S$, then the dual graph of $ G $ is 3-edge-colorable.

背景链接：https://arxiv.org/abs/1901.08622

06814 | AMR-142-0003

### Branko Grunbaum in Many Dimensions - Conjecture 3

O / open | AMR 问题清单汇编

Still, this leaves the question open for genus $g$ with $1 \le g \le 4$. In particular, the following seems to be open.

If $ G $ is a simple loopless triangulation of a torus, then the dual graph of $ G $ is 3-edge-colorable.

背景链接：https://arxiv.org/abs/1901.08622

06815 | AMR-142-0006

### Branko Grunbaum in Many Dimensions - Conjecture 6 (Ramos's conjecture)

O / open | AMR 问题清单汇编

According to Florian Frick, there is reason to think that the answer is yes. Let $\Delta(j,k)$ denote the minimum dimension $d$ such that every set of $j$ objects in $\mathbb{R}^d$ can be simultaneously partitioned into $2^k$ equal pieces by $k$ hyperplanes.

For every $j, k \ge 1$, $$\Delta(j,k) = \left\lceil \frac{2^k - 1}{k} j \right\rceil.$$

背景链接：https://arxiv.org/abs/1901.08622

06816 | AMR-142-0007

### Branko Grunbaum in Many Dimensions - Conjecture 7

O / open | AMR 问题清单汇编

[Gr\"unbaum--Kalai--Sarkaria] For every $d \ge 1$ there is a constant $C_d$ such that $$f_d (S)\le C_d f_{d-1}(S)$$ for every $d$-complex $S$ which is embeddable in $\mathbb{R}^{2d}$.

背景链接：https://arxiv.org/abs/1901.08622

06817 | AMR-142-0008

### Branko Grunbaum in Many Dimensions - Conjecture 8

O / open | AMR 问题清单汇编

Various formulations of this conjecture have apparently been discussed for some time---see, for example, Dey [Dey93]. Work of Kalai and Sarkaria [Kalai91,Sarkaria92] suggests a precise formulation, which would in particular give the best possible constant $C_d$ in every dimension. Gr\"unbaum wrote that this question is ``still'' open in his 1970 article, but so far I have not been able to find any earlier reference to it in the literature.

Another possible generalization of the inequality for planar graphs $ e \le 3v$ would be to show that

For every $d \ge 1$ there is a constant $C_d$ such that $f_d \le C_d f_0^d$

背景链接：https://arxiv.org/abs/1901.08622

06818 | AMR-142-0001

### Branko Grunbaum in Many Dimensions - Question 1

O / open | AMR 问题清单汇编

There is still more to do in this area, however. A Venn diagram is said to be *simple* if no more than two curves intersect at any point.

Do simple, symmetric Venn diagrams with $n$ curves exist for every prime number $n$?

背景链接：https://arxiv.org/abs/1901.08622

06819 | AMR-142-0004

### Branko Grunbaum in Many Dimensions - Question 4

O / open | AMR 问题清单汇编

A $2$-dimensional simplicial complex is said to be *polyhedral* if it admits an embedding in $\mathbb{R}^3$ with every vertex corresponding to a point, every edge corresponding to a straight line segment, and every triangle corresponding to a flat triangle contained in an affine plane. (It must also actually be a topological embedding, i.e.\ a continuous, injective, map on the geometric realization of the complex.)

Is every triangulation of the $2$-dimensional torus polyhedral?

背景链接：https://arxiv.org/abs/1901.08622

06820 | AMR-142-0005

### Branko Grunbaum in Many Dimensions - Question 5

O / open | AMR 问题清单汇编

Can every object in $\mathbb{R}^4$ be partitioned into $16$ equal parts with $4$ hyperplanes? In particular, can every set of $16n$ points in general position in $\mathbb{R}^4$ be separated into $16$ subsets of $n$ points, using 4 hyperplanes?

背景链接：https://arxiv.org/abs/1901.08622

06821 | AIM-GEOMETRY-0385

### Branson's cocycle as an action, polarization, and kernel detector

P / partially_solved | AIM 研讨会问题汇编

**Conundrum.** Are there any deeper properties of Branson's cocycle \(\mathcal H[\widehat g,g]\)?

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

06822 | OWR-1319-005

### Bubbling-Invariant Counts of J-Holomorphic Disks

P / partially_solved | Oberwolfach 报告问题汇编

Construct an invariant count of $J$-holomorphic disks through cycles despite bubbling, using cluster moduli spaces or another suitable theory.

来源字段：https://doi.org/10.4171/owr/2006/33

06823 | AMR-037-0020

### Building convex polytopes

P / partially_solved | AMR 问题清单汇编

Develop exact polynomial-time algorithms for the constructive forms of Aleksandrov's, Cauchy's, Minkowski's, Steinitz's, and Koebe's polytope-realization theorems: reconstruct a convex polytope from a net, facets with adjacency, area-weighted normals, or a 3-connected planar edge graph.

背景链接：https://jeffe.cs.illinois.edu/open/

06824 | AMR-067-0001

### Can you hear an orbifold singularity?

P / partially_solved | AMR 问题清单汇编

Can one hear the presence of an orbifold singularity, i.e. whether or not there exists a pair of isospectral orbifolds, one of which has singular points whereas the other does not and is therefore a manifold.

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

06825 | OWR-717-005

### Cannon's Conjecture

O / open | Oberwolfach 报告问题汇编

Cannon's conjecture asserts that every Gromov-hyperbolic group whose boundary is homeomorphic to the $2$-sphere is standard, equivalently that it acts geometrically on hyperbolic $3$-space.

来源字段：https://doi.org/10.4171/owr/2004/07

06826 | OWR-1319-010

### Cannon's Conjecture

O / open | Oberwolfach 报告问题汇编

If the ideal boundary of a hyperbolic group $\Gamma$ is homeomorphic to $S^2$, must $\Gamma$ act properly discontinuously and cocompactly by isometries on hyperbolic three-space?

来源字段：https://doi.org/10.4171/owr/2006/33

06827 | OWR-4990377-002

### Canonical Bicombings and Group Actions on Injective Spaces

O / open | Oberwolfach 报告问题汇编

For an injective metric space $X$, consider its canonical embedding $X=E(X)\to\Delta(X)$ and retraction $\Delta(X)\to E(X)$. Is the induced bicombing convex or consistent? Can a compact metric space admit distinct consistent bicombings? Must a group acting geometrically on an injective space be Helly, CAT(0), or biautomatic?

来源字段：https://doi.org/10.4171/owr/2021/8

06828 | AIM-GEOMETRY-0220

### Canonical endpoint stability and an explicit no-focal-points bound

P / partially_solved | AIM 研讨会问题汇编

Is the fundamental group of a closed manifold without conjugate points semihyperbolic?

The record reads:

It adds that semihyperbolicity was introduced by Alonso and Bridson to describe non-positive curvature on a large scale. The wording is unambiguous; the AIM spelling is “semihyperbolic” (some later papers write “semi-hyperbolic”). Here “closed” means compact without boundary, the metric is Riemannian, and “without conjugate points” applies to that metric.

Let \(M\) be such a manifold, let \(X=\widetilde M\), and let \(\Gamma=\pi_1(M)\) act on \(X\) by deck transformations. A standard consequence of compactness and absence of conjugate points is that \(\exp_x:T_xX\to X\) is a diffeomorphism for every \(x\in X\). Thus any two points of \(X\) are joined by a unique minimizing geodesic. This fact is recorded explicitly by Ivanov--Kapovitch.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06829 | OWR-14297743-004

### Canonical Harmonic Metrics on Noncompact Riemann Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Given holomorphic differentials $\vec q=(q_2,\ldots,q_n)$ on a noncompact Riemann surface $X$, does there exist a compatible harmonic metric on the associated Higgs bundle, and is there a canonical best such metric that exists uniquely?

来源字段：https://doi.org/10.4171/owr/2023/33

06830 | AIM-CONVEX_GEOMETRY-0004

### Canonical homothety fields and rigidity of centered sections

P / partially_solved | AIM 研讨会问题汇编

Let \(K,L\subset\mathbb R^n\) be star bodies with respect to the origin. Assume that, for every \(S\in G(n,k)\), the central sections \(K\cap S\) and \(L\cap S\) are homothetic. Must \(K\) and \(L\) be homothetic?

来源字段：https://aimath.org/pastworkshops/sectionsconvexproblems.pdf

06831 | OWR-13347-012

### Canonical Left-Invariant Metrics on Lie Groups

O / open | Oberwolfach 报告问题汇编

Does there exist a best left-invariant Riemannian metric on a given Lie group, and if so, how is its geometry related to the Lie-group structure?

来源字段：https://doi.org/10.4171/owr/2014/45

06832 | OWR-2654830-004

### Capture Strategies in the Circular Lion-and-Man Game

P / partially_solved | Oberwolfach 报告问题汇编

In the discrete equal-speed lion-and-man pursuit-evasion game on a closed circular arena, what tactics should the lion use to guarantee capture, or equivalently, under what geometric conditions does the lion win?

来源字段：https://doi.org/10.4171/owr/2020/34

06833 | OWR-11125-002

### Cardinality and Rigidity of Metric Tangent Spaces

O / open | Oberwolfach 报告问题汇编

For a geodesic metric space $X$ and basepoint $x_0$, how large can the set $\operatorname{Tan}(X,x_0)$ of pointed-isometry classes of tangents be, and what structural consequences follow when the tangent is unique?

来源字段：https://doi.org/10.4171/owr/2011/33

06834 | AIM-GEOMETRY-0064

### CAT(0) orbit decompositions and a zero-core pressure principle

P / partially_solved | AIM 研讨会问题汇编

Define a decomposition and pressure gap for \(\mbox{CAT}(0)\) geodesic flows, and extend Burns--Climenhaga--Fisher--Thompson (*Geom. Funct. Anal.* 28, 2018, MR3856792, on unique equilibrium states for geodesic flows in nonpositive curvature under the pressure-gap condition $P_{\mathrm{sing}}(\varphi) < P(\varphi)$) for these flows.

来源字段：http://aimpl.org/equibdynsysgeom/4/

06835 | AIM-GEOMETRY-0371

### Cauchy saturation and spin-structure obstruction for Lorentzian spinor kernels

P / partially_solved | AIM 研讨会问题汇编

**Problem 13.** Find all Lorentzian conformal structures \((M,[g])\) with \(t(g)>0\) or \(h(g)>0\).

Fix a connected, oriented and time-oriented Lorentzian conformal spin manifold \((M^n,[g],\mathfrak s)\), \(n\geq3\), including a chosen spin structure \(\mathfrak s\). For smooth complex spinors, classify those for which \(\ker D_g\neq0\) or \(\ker P_g\neq0\).

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

06836 | AIM-GEOMETRY-0153

### Central twists and the G2 locus of Painleve VI surfaces

P / partially_solved | AIM 研讨会问题汇编

It asks for a theory of the different wild-Hitchin realizations of one abstract hyperkähler manifold and proposes, as a first case, the complex two-dimensional \(D_4\)/Painlevé VI space: rank-two logarithmic connections on \(\mathbb P^1\) with four simple poles. The record asks in particular to “find all the possible representations of this space.” Its remarks propose rank-changing realizations via Fourier-Laplace transform or Katz middle convolution and cite the \(G_2\) realization in arXiv:1305.6594.

The request is a classification program, not a yes/no conjecture. In particular, “realization” and its equivalence relation are not defined in the source. This report therefore does **not** claim an exhaustive classification. It gives (i) a hierarchy of structures that a realization may preserve, (ii) a proved eightfold nonuniqueness obstruction for the ordered Painlevé VI trace data, and (iii) an exact trace-parameter criterion for reaching the symmetric Fricke subfamily, and hence the known \(G_2\) Betti realization, by central twists.

来源字段：http://aimpl.org/spectralhiggs/1/

06837 | OWR-15587-014

### Centrally Symmetric Two-Level Reflexive Polytopes

O / open | Oberwolfach 报告问题汇编

What are the centrally symmetric 2-level reflexive polytopes? Can the characterization of the exhibited examples be generalized to larger classes of 2-level polytopes, or to all 2-level polytopes?

来源字段：https://doi.org/10.4171/owr/2017/44

06838 | OWR-15219-006

### Certificates of Global Optimality for Thomson Configurations

O / open | Oberwolfach 报告问题汇编

For Thomson's problem of minimizing $$\sum_{1\leq i<j\leq N}\frac{1}{\lVert x_i-x_j\rVert}$$ over $N$ distinct points on $S^2$, and for its Riesz-energy variants, develop rigorous certificates that a candidate configuration attains the global minimum.

来源字段：https://doi.org/10.4171/owr/2017/14

06839 | AIM-GEOMETRY-0084

### Certifying complex-Gaussian LP auxiliary functions

P / partially_solved | AIM 研讨会问题汇编

For fixed $n$, characterize finite conjugation-symmetric sums of radial Gaussians $e^{-\pi a|x|^2}$ with $\operatorname{Re}a>0$, formulate a rigorous finite certificate for the Cohn--Elkies sign conditions, and test whether certified sums improve existing numerical LP bounds.

来源字段：http://aimpl.org/discreteaf/1/

06840 | OWR-12697711-016

### Characterizations of Symmetrically Hyperbolic Hook-Shaped Polynomials

O / open | Oberwolfach 报告问题汇编

For a diagonal map $T:\mathbb R[t]_{n,0}\to\mathbb R[t]_{d,0}$, prove that $T((x-1)^{n-1}(x+n-1))$ is real-rooted with $d-1$ roots of one sign exactly when $T$ preserves zero-sum hyperbolicity. Also prove that a hook-shaped symmetric polynomial $p$ is symmetrically hyperbolic exactly when $\Delta_{\mathbf 1,\mathbf 1}p$ is a sum of squares.

来源字段：https://doi.org/10.4171/owr/2023/15

06841 | AIM-GEOMETRY-0383

### Characterizing Riemannian Q-curvature

P / partially_solved | AIM 研讨会问题汇编

Find a geometrically meaningful, noncircular package of properties that uniquely selects Branson's critical \(Q\)-curvature from other natural weight-\(-n\) Riemannian scalar densities. Determine which familiar properties fail to give uniqueness and what additional normalization removes the ambiguity.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

06842 | GEO-016

### Chern's Affine Conjecture

P / partially_solved | 独立 / 未分配集合

文本重复：GEOM-016

Does the Euler characteristic of a compact affine manifold vanish?

背景链接：https://doi.org/10.1090/S0002-9904-1975-13891-0

06843 | GEOM-016

### Chern's Conjecture (Affine Geometry)

O / open | 独立 / 未分配集合

文本重复：GEO-016

Does the Euler characteristic of a compact affine manifold vanish?

背景链接：https://en.wikipedia.org/wiki/Chern_conjecture_(affine_geometry

06844 | GEO-017

### Chern's Conjecture for Hypersurfaces in Spheres

P / partially_solved | 独立 / 未分配集合

What minimal hypersurfaces in spheres have constant mean curvature?

背景链接：https://arxiv.org/abs/2504.02621

06845 | OPG-59984

### Chromatic number of associahedron

P / partially_solved | Open Problem Garden

Conjecture Associahedra have unbounded chromatic number.

背景链接：http://www.openproblemgarden.org/op/chromatic_number_of_associahedron

06846 | AMR-038-0003

### Chromatic number of the plane

O / open | AMR 问题清单汇编

Determine the least number of colors needed to color the Euclidean plane so that points at unit distance receive different colors.

背景链接：https://ics.uci.edu/~eppstein/junkyard/open.html

06847 | OWR-11136-008

### Circle Free Convex Bodies Under Minkowski Addition

O / open | Oberwolfach 报告问题汇编

If $K$ is a circle-free convex body in $\mathbb{R}^3$ and $B$ is a ball, is the Minkowski sum $K+B$ also circle-free?

来源字段：https://doi.org/10.4171/owr/2011/44

06848 | OWR-2090-023

### Circumscribed $2n$-Facet Polytopes around the Unit Ball

O / open | Oberwolfach 报告问题汇编

Let $B^n$ be the Euclidean unit ball and let $P_{2n}\supseteq B^n$ be an $n$-dimensional polytope with $2n$ facets. Is $\max_{x\in P_{2n}}d(0,x)\geq\sqrt n$, with equality exactly when $P_{2n}$ is a cube circumscribed about $B^n$?

来源字段：https://doi.org/10.4171/owr/2008/44

06849 | AMR-063-0004

### Classification at Symplectic Kodaira Invariant Zero

P / partially_solved | AMR 问题清单汇编

Extend Liu's classification of compact symplectic four-manifolds with positive numerical invariant $\kappa$ to the borderline case $\kappa=0$; determine whether the only examples are the K3 surface and torus bundles.

背景链接：https://wwwf.imperial.ac.uk/~skdona/NONLINEARITYPROBLEMS.PDF

06850 | OWR-1275-006

### Classification of Algebraic Teichmüller Curves

O / open | Oberwolfach 报告问题汇编

What types of algebraic curves arise as Teichmüller curves?

来源字段：https://doi.org/10.4171/owr/2006/26

06851 | OWR-14298170-003

### Classification of Bad Sets in Scalar-Curvature Limits

P / partially_solved | Oberwolfach 报告问题汇编

In the Dong-Song positive-mass stability theorem, determine whether the bad sets consist only of splines, bubbles, and drawstrings.

来源字段：https://doi.org/10.4171/owr/2024/9

06852 | OWR-7155451-016

### Classification of Complete Positive-Scalar-Curvature Three-Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Can complete 3-manifolds admitting metrics of positive scalar curvature be classified up to diffeomorphism, extending the known classification of the closed orientable case?

来源字段：https://doi.org/10.4171/owr/2021/30

06853 | OWR-14298364-007

### Classification of Conformally Einstein Kähler Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Which Kähler manifolds are conformally Einstein? More precisely, classify Kähler manifolds $(M,g_0)$ for which there exist a smooth function $\sigma:M\to\mathbb{R}$ and an Einstein metric $g$ satisfying $g=e^\sigma g_0$.

来源字段：https://doi.org/10.4171/owr/2024/28

06854 | OWR-14219-005

### Classification of Elliptic Affine Spheres and Hemispheres

O / open | Oberwolfach 报告问题汇编

Why are complete elliptic affine spheres much rarer than hyperbolic affine spheres, and can replacing completeness by the affine-hemisphere condition provide a meaningful classification?

来源字段：https://doi.org/10.4171/owr/2016/11

06855 | AMR-040-0001

### Classification of Finite Metric Spaces and Combinatorics of Convex Polytopes

O / open | AMR 问题清单汇编

Let $(X,\rho)$ be a finite metric space. Its fundamental polytope $R_{X,\rho}$ is the convex hull of the vectors $e_{x,y}=(\delta_x-\delta_y)/\rho(x,y)$ for distinct $x,y\in X$; its combinatorial structure is the isomorphism class of the face poset of $R_{X,\rho}$. (1) Express this combinatorial structure, including the $f$-vector, directly in terms of linear inequalities in the metric $\rho$. (2) For $|X|=n$, estimate the number of combinatorial structures and its asymptotic growth, especially the number of open (generic) types. (3) Give sufficient conditions for two finite metric spaces to have the same combinatorial structure. (4) Describe the combinatorial types of finite metric spaces that embed isometrically into a Euclidean or Hilbert space; do all combinatorial types occur? (5) Is the stratification of the cone of distance matrices into combinatorial types universal, or are there restrictions on the topological types of its open components?

背景链接：https://armj.math.stonybrook.edu/html-articles/Files-2015-2024/14-05/

06856 | OWR-11125-006

### Classification of Finite-Energy Gravitational Instantons

S / solved | Oberwolfach 报告问题汇编

Classify complete four-dimensional gravitational instantons with $\operatorname{Rm}\in L^2$ that are neither ALE nor ALF. Can such metrics have slow curvature decay, and can they occur as singularity dilations?

来源字段：https://doi.org/10.4171/owr/2011/33

06857 | OWR-14299521-005

### Classification of Four-Dimensional Ricci-Flow $\kappa$-Solutions

O / open | Oberwolfach 报告问题汇编

Classify all $\kappa$-solutions of four-dimensional Ricci flow, up to scaling and finite quotients, as follows: the shrinkers $S^4$, $\mathbb{CP}^2$, $S^2\times S^2$, $\mathbb{R}\times S^3$, and $\mathbb{R}^2\times S^2$; the four-dimensional Bryant soliton, the three-dimensional Bryant soliton times a line, and Lai’s one-parameter family of $\mathbb{Z}_2\times O(3)$-symmetric steady solitons; and the known four-dimensional ovals, products of three-dimensional ovals with a line, and the one-parameter family of $\mathbb{Z}_2^2\times O(3)$-symmetric ovals.

来源字段：https://doi.org/10.4171/owr/2025/30

06858 | OWR-13347-003

### Classification of Noncompact Homogeneous Einstein Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Classify the noncompact, nonflat homogeneous spaces $G/K$ that admit a $G$-invariant Einstein metric.

来源字段：https://doi.org/10.4171/owr/2014/45

06859 | OWR-11789-006

### Classification of Ricci-Flat ALE Four-Manifolds

O / open | Oberwolfach 报告问题汇编

Do Ricci-flat ALE four-manifolds exist beyond Kronheimer’s gravitational instantons and their known finite quotients?

来源字段：https://doi.org/10.4171/owr/2012/27

06860 | AMR-036-0011

### Classification of spherical quadrilaterals

P / partially_solved | AMR 问题清单汇编

A spherical quadrilateral is a disk with four marked boundary vertices, curvature-one metric, geodesic sides, and interior angles $\pi\alpha_j>0$. Classify such quadrilaterals up to isometry: determine which angle quadruples with $\sum_j\alpha_j>2$ occur, which conformal moduli occur for prescribed angles, and when existence or uniqueness fails once angles larger than $\pi$ are allowed.

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

06861 | AMR-067-0015

### Classification problems and Poisson structures

O / open | AMR 问题清单汇编

Explain the existence and the role of the symplectic nature of the groupoid/algebroid and its relevance for the geometry of the moduli spaces of geometric structures of finite type.

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

06862 | AMR-064-0006

### Closed Curves with a Nondegenerate Frenet Frame

P / partially_solved | AMR 问题清单汇编

Let $\mu(n)$ be the least $m$ such that a convex plane curve traversed $m$ times has a regular small perturbation in $\mathbb{R}^n$. Determine $\mu(n)$ for $n>3$, and decide whether the Frenet-frame length of every regular curve in $\mathbb{R}^n$ exceeds the length of $\mathrm{SO}(2)\subset\mathrm{O}(n)$ multiplied by $\mu(n)$.

背景链接：https://arxiv.org/abs/1304.2590

06863 | AIM-GEOMETRY-0211

### Closed geodesics in a finite-volume torus ansatz

P / partially_solved | AIM 研讨会问题汇编

Question 2.3.1 (Bangert). Does every complete Riemannian manifold with finite volume have at least one closed geodesic?

The question was answered affirmatively for dimension 2. Moreover, in dimen-sion 2 a complete Riemannian manifold of finite volume even has infinitely many geometrically different geodesics [Bangert1980]. The argument that was used in the proof is based on the Birkhoff minimax procedure we recalled in the beginning of §2, and does not work in dimensions ≥ 3. One can even hope to construct coun-terexamples in the class of Liouville-integrable geodesic flows. In this case, most orbits of the geodesic flow are rational or irrational windings on the Liouville tori; they are closed, if all of the corresponding frequencies are rational. Since there are essentially ( n - 1) frequencies in dimension n, one can hope that if n > 2 it would be possible to ensure that there is always at least one irrational frequency. Initial attempts to find a counterexample on T 2 × R with the metric of the form

a(r)dφ 2 + b(r)dψ 2 + dr 2 were, however, unsuccessful. A more difficult problem would be to prove that on any complete Riemannian manifold of finite volume there exist infinitely many closed geodesics.Questions of this nature are also interesting in the realm of Finsler geometry. One would expect the results for reversible Finsler metrics to be very similar to those for Riemannian metrics. For irreversible Finsler metrics, Katok's example shows that there are compact Finsler manifolds with only finitely many closed geodesics, but it is still possible that all non compact Finsler manifolds with finite volume might have infinitely many closed geodesics. On the other hand, the following question is completely open:

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06864 | AIM-GEOMETRY-0403

### Closed overtwisted contact manifolds and negative stabilization

S / solved | AIM 研讨会问题汇编

Paul Biran asked, following up some results presented by Emmanuel Giroux, if one can give a definition of "overtwisted" in higher dimensions in terms of open books. [Giroux proposed some definition, see also what he said on the sixth day.]

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

06865 | AIM-GEOMETRY-0425

### Closed three-manifold HF=HM and the residual naturality problem

P / partially_solved | AIM 研讨会问题汇编

Question 1.29. A big open problem is to prove the conjectured equivalence between Ozsvath-Szabo theory and Seiberg-Witten theory. Yi-Jen Lee discussed a possible approach to this.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

06866 | AIM-GEOMETRY-0246

### Closed-one-form twists obstruct the asserted cyclic-cover product

P / partially_solved | AIM 研讨会问题汇编

Let \(M^2\) be a closed surface of genus at least two, let \(S^1\) be the circle, and let \(g\) be a Riemannian metric without conjugate points on \(M\times S^1\).

**Conjecture 7.3.1 (Burago--Kleiner).** The metric on the \(\mathbb Z\)-cover corresponding to the \(S^1\) factor is a direct product \[ (M\times\mathbb R,g)=(M,g_1)\times(\mathbb R,dt^2). \]

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06867 | AIM-CONVEX_GEOMETRY-0038

### Closedness, dimensional correction, and a dual certificate for polar-zonoid intersection bodies

P / partially_solved | AIM 研讨会问题汇编

`0. (R. Schneider) Show that for most origin-symmetric convex K, the intersection body of K is not the polar of a zonoid. ("Most" means in the Baire category sense). 1`

**Problem 10 (R. Schneider).** Show that for most origin-symmetric convex bodies \(K\), the intersection body of \(K\) is not the polar of a zonoid. Here “most” is meant in the Baire-category sense.

来源字段：https://aimath.org/WWN/fourierconvex/fourierconvex.pdf

06868 | AMR-099-0060

### Closest finite vertex-transitive graph to the round sphere

O / open | AMR 问题清单汇编

Among all finite connected vertex-transitive graphs rescaled by their diameters, which one minimizes Gromov-Hausdorff distance to the round sphere $S^2$? Is the minimizer the one-skeleton of the truncated icosahedron?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=6

06869 | AIM-GEOMETRY-0166

### Closure repair and instantaneous smoothing of the topologist's sine curve

O / open | AIM 研讨会问题汇编

Let $C\subset\mathbb{R}^2$ be the topologist's sine curve. More precisely, \[ C=\left\{\left(x,\sin\frac1x\right):x\in\left(0,\frac1{2\pi}\right)\right\}\cup\Gamma, \] where $\Gamma$ is a smooth curve connecting the origin and $(1/(2\pi),0)$ that does not intersect $C\setminus\Gamma$. What happens under level-set flow? Does it become instantly smooth? More generally, what happens to a compact set in $\mathbb R^2$ under level-set flow?

There is a genuine formulation defect. Put \[ a=\frac1{2\pi},\qquad G=\left\{\left(x,\sin\frac1x\right):0<x<a\right\},\qquad V=\{0\}\times[-1,1]. \] Then \[ \overline G=G\cup V\cup\{(a,0)\}, \] and, because $(a,0)\in\Gamma$, \[ \overline{C}=G\cup\Gamma\cup V. \tag{1.1} \] Thus the displayed set is bounded but is not closed, and hence is not compact, unless the unmentioned exceptional possibility $V\subset\Gamma$ occurs. Under the source-verified intended reading, $\Gamma$ is an embedded arc disjoint from $G\cup V$ except at its endpoints, so the literal displayed set definitely omits $V$ and is noncompact.

This matters because Lam--Lauer's geometric definition begins: “Let $K\subset\mathbb R^{n+1}$ be compact,” both for weak-set flow and level-set flow. The literal $C$ is outside that definition.

The intended repair is nevertheless verified, not guessed: Lam--Lauer Definition 1.3 defines their topologist's sine curve $T$ as the union of the oscillating graph, the full compactifying segment $V$, and a smooth connecting arc. Therefore this report separates:

1. **Literal reading:** $C_{\mathrm{lit}}=G\cup\Gamma$, not admissible compact data. 2. **Closure reading:** $T=\overline{C_{\mathrm{lit}}}=G\cup\Gamma\cup V$. 3. **Verified intended reading:** exactly the Lam--Lauer compact set $T$, up to the inessential choice of graph cutoff and connecting arc.

The exact stored formulation is classified as invalid, but its verified intended problem is solved.

来源字段：http://aimpl.org/flowriemannian/2/

06870 | AMR-058-0006

### Clusters with Unequal Surface Tensions

P / partially_solved | AMR 问题清单汇编

Determine the shapes of minimizing clusters of immiscible fluids when different interfaces have different surface tensions, even for planar double bubbles.

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

06871 | AMR-058-0027

### CMC Graphs Spanning Convex Planar Curves

P / partially_solved | AMR 问题清单汇编

If a convex planar curve $\Gamma$ has length less than $2\pi$, does it bound a graph of constant mean curvature one? Determine the volume threshold $V(\Gamma)$ below which any CMC spanning surface is a graph, and decide whether the disk threshold for a unit circle remains valid for higher topology.

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

06872 | OWR-10252937-007

### CMC Min-Max Width Under Nonnegative Scalar Curvature

O / open | Oberwolfach 报告问题汇编

Let $(M^3,g)$ be complete, asymptotically flat, without boundary, with nonnegative scalar curvature, and not isometric to $\mathbb R^3$. For $c>0$, is the one-parameter min-max value $\omega_c$ of

$$\mathcal A_c(\Omega)=\operatorname{Area}(\partial\Omega)-c\operatorname{Vol}(\Omega)$$

always less than $16\pi/(3c^2)$?

来源字段：https://doi.org/10.4171/owr/2022/28

06873 | AMR-058-0028

### CMC Surfaces with Circular Boundary

P / partially_solved | AMR 问题清单汇编

Is every embedded constant-mean-curvature surface, or every immersed constant-mean-curvature disk, with boundary a round circle necessarily a spherical cap?

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

06874 | AMR-067-0012

### Coarse embeddings

O / open | AMR 问题清单汇编

Find more numerical invariants of metric spaces that are nondecreasing under coarse embeddings.

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

06875 | AMR-067-0013

### Coarse embeddings - Question 2

O / open | AMR 问题清单汇编

Find applications of the harmonic map approximation of coarse embeddings.

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

06876 | OWR-16160-005

### Coarse Hilbert Containment and Expander Embeddings

O / open | Oberwolfach 报告问题汇编

Does there exist a superreflexive Banach space that does not coarsely contain Hilbert space? If $Y$ is weak Hilbert or asymptotically Hilbertian, must every expander sequence fail to equi-coarsely embed into $Y$?

来源字段：https://doi.org/10.4171/owr/2018/19

06877 | OWR-11101915-002

### Coarse Median Spaces with $\delta$-Median Metrics

O / open | Oberwolfach 报告问题汇编

Does every coarse median space admit a $\delta$-median metric?

来源字段：https://doi.org/10.4171/owr/2022/34

06878 | OWR-12007-020

### Cocompact Hyperbolic Actions on CAT Spaces

O / open | Oberwolfach 报告问题汇编

Can one classify the $\operatorname{CAT}(0)$ or $\operatorname{CAT}(-1)$ spaces on which $\operatorname{Isom}(\mathbb H^n)$ acts cocompactly?

来源字段：https://doi.org/10.4171/owr/2012/35

06879 | AIM-GEOMETRY-0326

### Codimension-one connectivity and robust facet chains

S / solved | AIM 研讨会问题汇编

Let $k$ be an algebraically closed non-Archimedean field, let \[ R=k[x_1^{\pm1},\ldots,x_d^{\pm1}], \] and let $\mathfrak p\subset R$ be prime. The non-Archimedean amoeba of $\mathfrak p$ is connected in $\mathbb R^d$. If it is regarded as an $r$-dimensional polyhedral complex, where $r=\dim(R/\mathfrak p)$, is it connected in dimension $r-1$? Equivalently, is the graph whose vertices are the $r$-dimensional faces, with two vertices joined when the corresponding faces share an $(r-1)$-dimensional face, connected?

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

06880 | OWR-12339-005

### Coefficients in Signed Simplex Decompositions

O / open | Oberwolfach 报告问题汇编

For the signed simplex decomposition of the uniform measure on $P$, is every nonzero coefficient $\alpha_T$ always $\pm1$? If not, characterize the possible coefficients.

来源字段：https://doi.org/10.4171/owr/2013/11

06881 | OWR-14299521-007

### Coexistence of Einstein Metrics with Different Constants

O / open | Oberwolfach 报告问题汇编

Can odd-dimensional closed manifolds carry Einstein metrics with different Einstein constants? In particular, can Ricci-flat and positive-Einstein metrics coexist, and can special-holonomy methods decide this?

来源字段：https://doi.org/10.4171/owr/2025/30

06882 | AIM-GEOMETRY-0280

### Cohn--Elkies magic functions solve sphere packing in dimensions 8 and 24

S / solved | AIM 研讨会问题汇编

Question 1. Can the technique of Cohn-Elkies prove that the E8 and Leech lattices have the highest sphere packing densities in dimensions 8 and 24, respectively?

The wording is unambiguous. The preceding source text explicitly discusses packings “lattice or not,” and the sentence following the question says that a positive answer gives the complete sphere-packing solution in dimensions \(8\) and \(24\). Thus this asks for optimality among all packings, not merely among lattice packings.

来源字段：https://aimath.org/WWN/spherepacking/spherepacking.pdf

06883 | AIM-GEOMETRY-0282

### Cohn--Miller determinant approximants and a controlled-limit rigidity theorem

P / partially_solved | AIM 研讨会问题汇编

**Question 3. Further conjectures of Cohn-Miller:** A: Can their determinants be rescaled according to their conjecture 3.2? B: Do their rescaled determinants converge uniformly on compact sets to a limiting function? C: Does this limiting function have any unexpected sign changes (i.e. sign changes other than the forced ones). If 'yes', 'yes', and 'no', respectively, then the answer to

If ‘yes’, ‘yes’, and ‘no’, respectively, then the answer to Question 2 is ‘yes’.

来源字段：https://aimath.org/WWN/spherepacking/spherepacking.pdf

06884 | GREEN-084

### Cohn-Elkies Scheme for Circle Packings

O / open | Ben Green 问题

Can the Cohn-Elkies scheme be used to prove the optimal bound for circle-packings?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

06885 | OWR-2654829-009

### Cohomology Rings of Almost-Complex Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Which rational cohomology rings can be realized by closed manifolds that admit almost-complex structures?

来源字段：https://doi.org/10.4171/owr/2020/33

06886 | AIM-GEOMETRY-0316

### Cohomology targets and sectorwise Kirwan surjectivity

P / partially_solved | AIM 研讨会问题汇编

Question 10.3, the issue is not just one of sur-jectivity but also of the choice of the cohomology theory for the reduced space. I.e. should one take Chen-Ruan orbifold cohomology? Or that of Haefliger? One approach to addressing the question would be to look for a good (= optimal) condition on critical sets of the norm-square of the moment map.

11 Localization formulas for non-compact groups

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

06887 | AMR-065-0007

### Coincidence Rank Conjecture

P / partially_solved | AMR 问题清单汇编

The coincidence rank of a one--dimensional Pisot inflation tiling must divide the algebraic norm of $\lambda$.

背景链接：https://arxiv.org/abs/1604.06280

06888 | OWR-11786-015

### Collapsibility of Linear Triangulations of Convex Balls

P / partially_solved | Oberwolfach 报告问题汇编

Is every linear triangulation of a convex ball in Euclidean space collapsible?

来源字段：https://doi.org/10.4171/owr/2012/24

06889 | GREEN-044

### Collinear 4-tuples Force Collinear 5-tuples

P / partially_solved | Ben Green 问题

Suppose $A \subset \mathbb{R}^2$ is a set of size $n$ with $cn^2$ collinear 4-tuples. Does it contain 5 points on a line?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

06890 | GREEN-042

### Collinear Triples and Cubic Curves

P / partially_solved | Ben Green 问题

Fix $k$. Let $A \subset \mathbb{R}^2$ be a set of $n$ points with no more than $k$ on any line. Suppose at least $\delta n^2$ pairs $(x, y) \in A \times A$ have the line $xy$ containing a third point of $A$. Is there a cubic curve containing at least $cn$ points of $A$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

06891 | OWR-11136-026

### Colored Transversals for Translates of Convex Sets

O / open | Oberwolfach 报告问题汇编

If $F_1,F_2,F_3$ are pairwise intersecting families of translates of a convex compact planar set, must some $F_i$ have a $3$-transversal?

来源字段：https://doi.org/10.4171/owr/2011/44

06892 | AIM-GEOMETRY-0424

### Combinatorial Khovanov-to-Floer differentials and a determinant rank budget

S / solved | AIM 研讨会问题汇编

Peter Ozsvath discussed the problem of computing Ozsvath and Szabo's invariants combinatorially or axiomatically. Their theory lies somewhere between combina-torial invariants, which are easy to compute but not very useful, and invariants defined in terms of PDE's, which are very useful but hard to compute. "Right next door" is Khovanov's categorification of the Jones polynomial, which looks like a Floer theory but is defined purely combinatorially. In fact there is the following bridge between them: given a link K in S3,there is a spectral sequence whose E2 term is the Khovanov homology of K (with Z/2 co-efficients) and which converges toˆ HF (Σ 2(K)), where Σ2(K) denotes the double cover of S3 branched along K. Question: can one compute the differentials in the spectral sequence combinatorially? Also,ˆ HF (Σ 2(K)) has applications to slice genus bounds: can Khovanov homology say anything about the 4-ball genus?

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

06893 | OWR-12985-001

### Combinatorial Positivity of Schubert CSM Coefficients

O / open | Oberwolfach 报告问题汇编

Is there a combinatorial proof that the coefficients $\gamma_{\alpha,\beta}$ in the Chern-Schwartz-MacPherson expansion of a Schubert cell are nonnegative?

来源字段：https://doi.org/10.4171/owr/2014/27

06894 | AMR-058-0021

### Combinatorial Types of Equal-Pressure Foam Cells

O / open | AMR 问题清单汇编

Are there only finitely many combinatorial types of cells in equal-pressure foams in $\mathbb{R}^3$? In particular, can tetrahedra or dodecahedra occur?

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

06895 | OWR-2090-028

### Common Tangent Loci of Three Convex Bodies

O / open | Oberwolfach 报告问题汇编

For pairwise disjoint convex sets $A,B,C\subset\mathbb{R}^3$, let $S(A,B,C)$ be the points through which a line tangent to all three sets passes. Must $S(A,B,C)$ be contained in a measure-zero set?

来源字段：https://doi.org/10.4171/owr/2008/44

06896 | OWR-13498-006

### Commuting Billiard Maps and Confocal Ellipses

P / partially_solved | Oberwolfach 报告问题汇编

For two nested smooth convex plane domains with billiard maps $T_1,T_2$ acting on oriented lines meeting both domains, does $T_1\circ T_2=T_2\circ T_1$ imply that the two domains are bounded by confocal ellipses?

来源字段：https://doi.org/10.4171/owr/2015/13

06897 | AIM-GEOMETRY-0343

### Compact tropicalization, Monge-Ampère, curve realizability, and a sharp circle model

P / partially_solved | AIM 研讨会问题汇编

This record is item A.23, contributed by Ilia Zharkov, in the AIM workshop notes *Amoebas and tropical geometry*. Comparison with the source PDF shows that “Chapter B: Snapshot of the pre-open problem session” begins a new chapter and is not part of A.23. The present job therefore owns only A.23, not the trailing chapter material.

1. **Compact tropical varieties.** For a compact complex algebraic variety, can one naturally modify the valuation map to obtain a compact tropical variety, retaining as much affine structure as possible? The text anticipates a role for a polarization (a tropical Kähler class) but expects the affine structure itself to be algebro-geometric. 2. **Tropical Monge-Ampère / Calabi problem.** In an affine chart \(U\subset B_0\), a potential \(K\) is represented by the graph \[ dK\subset \mathbb R^n\times(\mathbb R^n)^*. \] The smooth graph should be replaced tropically by a piecewise-linear one and the Monge-Ampère equation interpreted distributionally. Is there a tropical Monge-Ampère operator whose solution produces a unique Monge-Ampère measure in a polarization class \[ H^1(B,\operatorname{Aff}(B,\mathbb R))? \] 3. **Curve counting and realizability.** Can tropical Gromov-Witten calculations be performed on a Calabi-Yau? Is there a simple rule deciding which tropical curves are limits of classical curves, given that in higher dimensions some tropical curves are not realizable?

The PDF itself contains “deleting pseudo-pods”; this unusual phrase is not silently corrected. The malformed strings “Monge-Amp` ere,” “K¨ ahler,” “C∞,” “Rn,” and “( Rn)∗” have been recovered as “Monge-Ampère,” “Kähler,” \(C^\infty\), \(\mathbb R^n\), and \((\mathbb R^n)^*\).

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

06898 | AIM-GEOMETRY-0051

### Compact-core transfer of Lyapunov multifractal spectra

P / partially_solved | AIM 研讨会问题汇编

Extend the results of Burns--Gelfert (*Discrete Contin. Dyn. Syst.* 34, 2014, MR3124716, on the Lyapunov spectrum and thermodynamics of geodesic flows of rank-$1$ nonpositively curved surfaces) to noncompact spaces.

来源字段：http://aimpl.org/equibdynsysgeom/1/

06899 | OWR-11125-007

### Compactifiable Gravitational Instantons

O / open | Oberwolfach 报告问题汇编

Classify complex surfaces $X\setminus D$ carrying gravitational-instanton metrics, up to biholomorphism and hyperkähler rotation.

来源字段：https://doi.org/10.4171/owr/2011/33

06900 | OWR-16933-003

### Compactifying Lagrangian Lifts of Tropical Hypersurfaces

O / open | Oberwolfach 报告问题汇编

When a tropical hypersurface in a toric moment polytope meets the boundary, determine whether its Lagrangian lift compactifies to a smooth Lagrangian submanifold without boundary.

来源字段：https://doi.org/10.4171/owr/2019/21

06901 | AMR-063-0009

### Compactness for Calibrated Submanifolds

P / partially_solved | AMR 问题清单汇编

Develop compactness and singularity theories for special Lagrangian, associative, and co-associative calibrated submanifolds that are strong enough to define enumerative invariants.

背景链接：https://wwwf.imperial.ac.uk/~skdona/NONLINEARITYPROBLEMS.PDF

06902 | OWR-13347-019

### Compactness of Positively Curved Shrinking Ricci Solitons

S / solved | Oberwolfach 报告问题汇编

Must every gradient shrinking Ricci soliton with bounded positive curvature operator be compact?

来源字段：https://doi.org/10.4171/owr/2014/45

06903 | AMR-038-0014

### Comparing sums of square roots

O / open | AMR 问题清单汇编

Can sums of square roots of integers be compared in polynomial time on a Turing machine? Equivalently, obtain effective polynomial bit bounds for a nonzero difference of two such sums, with consequences for placing Euclidean optimization problems in NP.

背景链接：https://ics.uci.edu/~eppstein/junkyard/open.html

06904 | AIM-GEOMETRY-0068

### Compatibility of Gibbs leaf cocycles for commuting Anosov maps

P / partially_solved | AIM 研讨会问题汇编

Let \(f_1,f_2\) be commuting hyperbolic diffeomorphisms of a compact manifold \(M\). One may construct coarse Lyapunov foliations \(\mathcal W\). Given a potential \(\varphi\), can one construct leaf-wise conditional measures \(\mu^{\mathcal W}_\varphi\) which satisfy Margulis-like cocycle properties?

来源字段：http://aimpl.org/equibdynsysgeom/5/

06905 | OWR-15208-002

### Completeness of Periodic Data and Marked Length Spectra

O / open | Oberwolfach 报告问题汇编

In dimensions greater than two, are periodic-orbit data and the marked length spectrum complete invariants of Anosov geodesic flows?

来源字段：https://doi.org/10.4171/owr/2017/3

06906 | OWR-14298808-010

### Complex and Quaternionic Affine Quermassintegral Inequalities

O / open | Oberwolfach 报告问题汇编

Let $\mathbb F\in\{\mathbb C,\mathbb H\}$, put $p=\dim_{\mathbb R}\mathbb F$, let $1\leq m<n$, and let $K\subset\mathbb F^n$ be a convex body with nonempty interior. Write $\kappa_j$ for the volume of the Euclidean unit ball in $\mathbb R^j$, equip the Grassmannian $\operatorname{Gr}_m(n,\mathbb F)$ with Haar probability measure $dE$, and let $P_E$ be orthogonal projection onto $E$. Prove

$$|K|^{-m}\geq\frac{(\kappa_{mp})^n}{(\kappa_{np})^m}\int_{\operatorname{Gr}_m(n,\mathbb F)}|P_EK|^{-n}\,dE,$$

with equality exactly for real, complex, or quaternionic ellipsoids, and prove the associated projection inequality

$$\frac{\kappa_{np}}{\kappa_{mp}}\int_{\operatorname{Gr}_m(n,\mathbb F)}|P_EK|\,dE\geq |K|^{m/n}$$.

来源字段：https://doi.org/10.4171/owr/2024/58

06907 | AMR-037-0012

### Complex collinearities

P / partially_solved | AMR 问题清单汇编

Given $n$ points in $\mathbb C^2$, determine in quadratic time whether three lie on a complex line, or prove a quadratic lower bound; the known algorithm takes $O(n^2\log n)$ time.

背景链接：https://jeffe.cs.illinois.edu/open/

06908 | OWR-12581-004

### Complex Extension Conjecture for Polyhedral Kähler Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Let $M^{2n}$ be a polyhedral Kähler manifold. Does its flat complex structure away from the metric singularities extend across all of $M$, making $M$ a normal complex analytic variety whose complex singularities have codimension $3$?

来源字段：https://doi.org/10.4171/owr/2013/33

06909 | OWR-761-005

### Complex Hyperbolic Plane Embeddings into Four Trees

P / partially_solved | Oberwolfach 报告问题汇编

Does the complex hyperbolic plane $\mathbb H^2_{\mathbb C}$ admit a quasi-isometric embedding into a product of four metric trees?

来源字段：https://doi.org/10.4171/owr/2004/47

06910 | AMR-063-0006

### Complex Jörgens-Calabi-Pogorelov Theorem

P / partially_solved | AMR 问题清单汇编

Prove an appropriate complex analogue of the Jörgens--Calabi--Pogorelov theorem: classify global solutions on $\mathbb{C}^n$ of the complex Monge--Ampère equation corresponding to determinant one.

背景链接：https://wwwf.imperial.ac.uk/~skdona/NONLINEARITYPROBLEMS.PDF

06911 | AIM-GEOMETRY-0175

### Complex sectional curvature blow-up at high-codimension fixed sets

P / partially_solved | AIM 研讨会问题汇编

Let $M_\epsilon =M\times_{S^1} (\epsilon S^1)$. As $\epsilon \to 0$, is Riem(Curvature Operator) bounded below?

Is there a constant $C<\infty$, independent of sufficiently small $\epsilon$, such that every complex sectional curvature of $(M,g_\epsilon)$ is at least $-C$? If not, which orbit types obstruct such a bound?

This sentence is incomplete in isolation.

A circle acts isometrically on a Riemannian manifold $(M,g)$, and $g_\epsilon$ is the Cheeger-deformed metric induced by the Riemannian submersion

\[ \pi_\epsilon:(M,g)\times(\epsilon S^1)\longrightarrow M\times_{S^1}(\epsilon S^1)\cong M. \tag{1.1} \]

Here the diagonal circle action is free because it translates the second factor, and the generator of the second circle has length $\epsilon$. The report asks whether the curvature operators of $g_\epsilon$ have a lower bound independent of $\epsilon$. It records positive regimes when the action is almost free or its fixed-point set has codimension two, and reports a negative flat linear model when the fixed-set codimension is greater than two.

I use the Riemannian complex-sectional-curvature convention of Ni--Wolfson: the tangent space and curvature tensor are complexified, and the curvature is tested on every complex decomposable two-vector. This convention is defined precisely in Section 3. A competing convention tests only totally isotropic complex two-planes. The counterexample below uses a totally isotropic plane, so it is negative under both conventions. “Holomorphic sectional curvature” on a manifold equipped with an almost-complex or complex structure is a different notion and cannot be the intended general question because the AIM setup assumes no such structure.

来源字段：http://aimpl.org/flowriemannian/4/

06912 | OWR-11568-004

### Complex Structures on Polyhedral Kähler Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

If a nonnegatively curved, nonflat polyhedral manifold $M$ has irreducible holonomy and $b_2(M)>0$, does $M$ have a natural holomorphic structure, is it biholomorphic to $\mathbb{CP}^n$, and is its metric a singular Kähler metric? More generally, does every polyhedral Kähler manifold $M^{2n}$ have a natural complex structure, specifically one making it PL-diffeomorphic to a complex analytic space with complex singularities of complex codimension at least $3$ and biholomorphic outside the metric singularities?

来源字段：https://doi.org/10.4171/owr/2012/01

06913 | OWR-4078-005

### Complex-Hyperbolic Quasi-Fuchsian Representation Components

P / partially_solved | Oberwolfach 报告问题汇编

Do the minimal Lagrangian surface solutions in complex hyperbolic two-space parametrize a region of $\operatorname{Hom}(\pi_1\Sigma,\operatorname{SU}(2,1))/\operatorname{SU}(2,1)$ analogous to the quasi-Fuchsian component of $\operatorname{Hom}(\pi_1\Sigma,\operatorname{PSL}(2,\mathbb C))/\operatorname{PSL}(2,\mathbb C)$?

来源字段：https://doi.org/10.4171/owr/2009/35

06914 | AIM-GEOMETRY-0017

### Component obstructions to cluster augmentation varieties

P / partially_solved | AIM 研讨会问题汇编

For more general Legendrians, is the augmentation variety cluster?

**Recovered question.** Let $\Lambda\subset(\mathbb R^3,\xi_{\mathrm{st}})$ be a Legendrian link, let $T$ contain one marked point on each component, fix a Maslov grading and a field $k$, and let $$\operatorname{Aug}^{\mathrm{gr}}_k(\Lambda,T)$$ be the classical graded augmentation scheme of its marked Chekanov--Eliashberg DGA. For which $\Lambda$ is its coordinate ring a Fomin--Zelevinsky cluster algebra, preferably equal to its upper cluster algebra, with filling or weave tori as cluster charts?

来源字段：http://aimpl.org/clusterbraid/2/

06915 | AIM-GEOMETRY-0283

### Conditional identification of the Cohn--Miller limits and exact quadratic moments

P / partially_solved | AIM 研讨会问题汇编

Question 4. What are the purported limits of the Cohn-Miller sequences? Can the observed rationalities in their Taylor coefficients be explained? Are there others outside from the quadratic terms? Contributed by Peter Sarnak and Andreas Str¨ ombergsson [followed by an Epstein-zeta paragraph].

Question 4 ends after “quadratic terms?” The contributor line and the Epstein-zeta paragraph begin a new item and are not part of Question 4. Accordingly, this report treats only the three questions about the Cohn--Miller sequences.

In Cohn and Miller [CM], the relevant objects are radial polynomial-times-Gaussian auxiliary functions in dimensions $8$ and $24$. The finite functions have prescribed roots inspired by the nonzero vector lengths of $E_8$ and the Leech lattice. A naive exact-root sequence fails to converge numerically, so [CM, (3.5)] perturbs the outer roots. For every fixed lattice shell, however, the perturbed root eventually equals the true shell radius. Cohn and Miller conjecture that the resulting functions $f_k$ and their Fourier transforms converge locally uniformly in a complex neighborhood of the real axis [CM, Conjecture 4.1].

Throughout, the Fourier transform is

$$ \widehat f(y)=\int_{\mathbb R^d}f(x)e^{-2\pi i\langle x,y\rangle}\,dx. $$

来源字段：https://aimath.org/WWN/spherepacking/spherepacking.pdf

06916 | AIM-GEOMETRY-0205

### Conference-matrix reduction and a strict coherence-gap certificate

P / partially_solved | AIM 研讨会问题汇编

Let $v(n,r,\mathbb K)$ denote the maximum possible number of equi-isoclinic $n$-planes that can be embedded in $\mathbb K^r$, where $\mathbb K=\mathbb R,\mathbb C$, or $\mathbb H$. (a) For $r\ge4$, find $v(2,2r,\mathbb R)$. (b) Find $v(2,7,\mathbb R)$. (c) For $r\ge3$, find $v(3,3r,\mathbb R)$. (d) Find all regular $v$-tuples in $G(4,4v)$ whose symmetry groups are isomorphic to $S_v$. (e) Find $v(3,3r,\mathbb K)$ for $\mathbb K=\mathbb C$ or $\mathbb H$. (f) Find all complex equiangular tight frames with parameters $(2M,M)$.

来源字段：https://aimath.org/pastworkshops/frametheorygeomproblems.pdf

06917 | AMR-068-0001

### Configuration Spaces of Tensegrities - Problem 1

P / partially_solved | AMR 问题清单汇编

Describe the combinatorics of B2(K6); B3(K4) and B3(K5).

背景链接：https://livrepository.liverpool.ac.uk/3022838/1/stratum.pdf

06918 | AMR-068-0002

### Configuration Spaces of Tensegrities - Problem 2

O / open | AMR 问题清单汇编

Describe all the possible different types of strata for 10 points.

背景链接：https://livrepository.liverpool.ac.uk/3022838/1/stratum.pdf

06919 | AMR-068-0003

### Configuration Spaces of Tensegrities - Problem 3

O / open | AMR 问题清单汇编

Compute the number of different types of strata for n points with arbitrary n. 4 OLEG KARPENKOV v1 v2 v3v4 v5 v6 K3;3 q1 q2 q3 p1 p2 p3 p4 p5 p6 q3 q2 q1 p1 p2 p3 p4 p5 p6 Figure 2.

背景链接：https://livrepository.liverpool.ac.uk/3022838/1/stratum.pdf

06920 | AMR-068-0004

### Configuration Spaces of Tensegrities - Problem 4

O / open | AMR 问题清单汇编

Which subgraphs of Kn define the same stratifications?

背景链接：https://livrepository.liverpool.ac.uk/3022838/1/stratum.pdf

06921 | AMR-068-0005

### Configuration Spaces of Tensegrities - Problem 5

O / open | AMR 问题清单汇编

Find all strata of codimension more than 1 that are not defined as an intersection of the closure of several codimension 1 strata.

背景链接：https://livrepository.liverpool.ac.uk/3022838/1/stratum.pdf

06922 | AMR-068-0006

### Configuration Spaces of Tensegrities - Problem 6

O / open | AMR 问题清单汇编

Which Cayley algebra systems define the same strata?

背景链接：https://livrepository.liverpool.ac.uk/3022838/1/stratum.pdf

06923 | AMR-068-0007

### Configuration Spaces of Tensegrities - Problem 7

P / partially_solved | AMR 问题清单汇编

Given a graph G. Does there exist a Cayley algebra system (or several systems) describing the union of the codimension 1 tensegrity strata in the plane (i.e., the union of the codimension 1 strata of B2(Kn))?

背景链接：https://livrepository.liverpool.ac.uk/3022838/1/stratum.pdf

06924 | AMR-068-0008

### Configuration Spaces of Tensegrities - Problem 8

O / open | AMR 问题清单汇编

Write (if exist) Cayley algebra systems defining the strata for the following graph: Currently this example is a strong candidate for a counterexample to Problem 7.

背景链接：https://livrepository.liverpool.ac.uk/3022838/1/stratum.pdf

06925 | AMR-068-0009

### Configuration Spaces of Tensegrities - Problem 9

O / open | AMR 问题清单汇编

Develop theory of geometric conditions for strata in multidimensional case.

背景链接：https://livrepository.liverpool.ac.uk/3022838/1/stratum.pdf

06926 | OWR-2090-007

### Conflict-Free Chromatic Numbers of Rectangle Hypergraphs

O / open | Oberwolfach 报告问题汇编

For the geometric hypergraph formed by $n$ axis-parallel rectangles in the plane, what is the best possible asymptotic upper bound for its conflict-free chromatic number?

来源字段：https://doi.org/10.4171/owr/2008/44

06927 | OWR-1115-008

### Conformal Dimension and $\ell^p$-Cohomology Critical Exponents

O / open | Oberwolfach 报告问题汇编

For a compact uniformly perfect metric space $(X,d)$, is its conformal dimension $\operatorname{Cdim}(X,d)$ always equal to the critical exponent $p(\mathcal{J}(X,d))$ of first $\ell^p$-cohomology over its conformal gauge?

来源字段：https://doi.org/10.4171/owr/2005/54

06928 | AMR-059-0007

### Conformal Flatness and Geometric Divergence

O / open | AMR 问题清单汇编

Investigate the relationship between the geometry of a conformally flat Riemannian manifold and Matsuzoe's geometric divergence for a conformally-projectively flat statistical manifold.

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

06929 | OWR-13857-003

### Conformal Positivity of Weighted $\sigma_2$-Curvature

O / open | Oberwolfach 报告问题汇编

Let $(M^n,g)$ be compact. If $\sup_{\Gamma_2^\infty}W_2<\infty$, does the conformal class $[g]$ contain a metric with positive weighted $\sigma_2$-curvature? In particular, does every compact four-dimensional gradient Ricci soliton satisfy $\chi(M^4)>\tfrac32|\tau(M^4)|$?

来源字段：https://doi.org/10.4171/owr/2015/36

06930 | OWR-14297736-006

### Conformal Spinorial Eigenvalue Infimum on the Torus

O / open | Oberwolfach 报告问题汇编

For the trivial spin structure on the torus $T$, prove $\inf_{g\in[g_{a,b}]}\overline\mu_1(g)=2\pi/\sqrt b$ when $b>\pi$ and equals $2\sqrt\pi$ when $b\leq\pi$.

来源字段：https://doi.org/10.4171/owr/2023/36

06931 | OWR-1460-005

### Conformal Type of Surfaces Spread over the Sphere

O / open | Oberwolfach 报告问题汇编

For a surface spread over the Riemann sphere, given by $(X,f)$ with associated conformal Riemann surface $X_f$, how do the topological and metric properties of $f$ determine whether $X_f$ is parabolic or hyperbolic and determine the value-distribution properties of $f$?

来源字段：https://doi.org/10.4171/owr/2007/09

06932 | AIM-GEOMETRY-0349

### Conformal-primitivity obstruction and curvature-only rigidity on surfaces

P / partially_solved | AIM 研讨会问题汇编

Let \(M^n\) be a closed manifold, with orientation if density-valued invariants are written as \(n\)-forms. Closedness is inherited from the surrounding compact-manifold discussion; the displayed one-line conjecture does not repeat it. All local invariants in this passage are being treated as densities. If \(\mathbf S_g\) is a natural \(n\)-density and \[ \int_M \mathbf S_g \] is invariant under \(g\mapsto e^{2\omega}g\), then the conjecture asserts that \[ \mathbf S= a\mathbf Q+\mathbf L+\mathbf G, \] where \(a\) is constant, \(\mathbf Q\) is a chosen Branson \(Q\)-curvature density, \(\mathbf L\) is a local conformal invariant density, and \(\mathbf G\) has a **local conformal primitive**. The last condition means that there is a natural local density \(\mathbf F_g\) such that, for every smooth \(\varphi\), \[ \left.\frac{d}{dt}\right|_{t=0} \int_M \mathbf F_{e^{2t\varphi}g} =\int_M \varphi\,\mathbf G_g. \]

The nearby text first defines a conformal primitive more generally as a two-metric functional on a fixed conformal class. If \(\hat g=e^{2\omega}g\), then \[ c(\hat g,g):=\omega. \] A two-metric primitive \(\mathcal A(\hat g,g)\) is required to be alternating, cocyclic, \[ \mathcal A(g_2,g_0)=\mathcal A(g_2,g_1)+\mathcal A(g_1,g_0), \] and to have the required first variation when its first metric is varied and its second is held fixed. A **local** primitive is the special coboundary \[ \mathcal A_F(\hat g,g)=\int_M(\mathbf F_{\hat g}-\mathbf F_g) \] arising from one local invariant \(\mathbf F\).

The extracted formula following the conjecture is \[ \frac12\int_M c(\hat g,g) \bigl(\mathbf L_{\hat g}+\mathbf L_g\bigr). The phrase “the The \(Q\)” is a genuine duplicated-word typo in the PDF and is read as “The \(Q\).” The text also says that the chosen \(Q\) is not unique: changing it by a local conformal invariant merely changes \(\mathbf L\).

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

06933 | AMR-054-0073

### Congruent Partitions of Polygons

O / open | AMR 问题清单汇编

Partition a given polygon $P$ into $n$ mutually congruent pieces so that the area of $P$ not covered by the union of the pieces is as small as possible. A partition which leaves out the least area is an optimal congruent partition for that $n$. If a congruent partition is a perfect cover, leaving no area uncovered, then it is called a perfect congruent partition. Two polygons are congruent if one can be made to coincide with the other by translation, rotation, or reflection (flipping over).

背景链接：https://topp.openproblem.net/p73

06934 | OWR-16412-002

### Connected Interiors of Conservative Hénon Filled Sets

O / open | Oberwolfach 报告问题汇编

For a volume-preserving Hénon map $f$ with $|\delta|=1$, can the interior of the forward filled set $K^+$ be connected and satisfy $\overline{\operatorname{int}K^+}=K^+$?

来源字段：https://doi.org/10.4171/owr/2018/37

06935 | AIM-GEOMETRY-0258

### Connected Morse fibers, Reeb graphs, and pants width

P / partially_solved | AIM 研讨会问题汇编

Is there a universal constant \(C\) such that \[ L_b(M,g)\le C\sqrt{\operatorname{area}(M,g)}? \]

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

06936 | AMR-058-0005

### Connectedness of Regions in Minimizing Clusters

P / partially_solved | AMR 问题清单汇编

Are all regions in every area-minimizing bubble cluster in $\mathbb{R}^n$ connected?

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

06937 | OWR-4075-003

### Connections Between Simplex and Torus Translations

O / open | Oberwolfach 报告问题汇编

What relates the piecewise-translation and torus-translation roles of the simplex map, and is the resulting invariant fundamental set connected with a possibly non-convex polytope shape?

来源字段：https://doi.org/10.4171/owr/2009/32

06938 | OWR-1319-014

### Connectivity and Geometry of Hofer-Metric Annuli

O / open | Oberwolfach 报告问题汇编

For the Hofer-metric annulus $A_t=\{f\in\operatorname{Ham}:t\leq\rho(\mathrm{id},f)\leq2t\}$, is $A_t$ path connected for large $t$? What is the large-$t$ geometry of its intrinsic length metric $\rho_t$; in particular, is $\operatorname{diam}(A_t,\rho_t)$ finite?

来源字段：https://doi.org/10.4171/owr/2006/33

06939 | AMR-071-0062

### Connelly’s blooming conjecture

O / open | AMR 问题清单汇编

Connelly’s blooming conjecture: Does every net of a convex polyhedron have a blooming?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

06940 | OWR-8415335-002

### Consistent Conical Bicombings in Metric Spaces

O / open | Oberwolfach 报告问题汇编

Does every metric space with a conical bicombing also admit a consistent conical bicombing?

来源字段：https://doi.org/10.4171/owr/2021/32

06941 | AIM-GEOMETRY-0127

### Constant abelian systems from a free torus reduction

P / partially_solved | AIM 研讨会问题汇编

Let $X$ be a complex symplectic algebraic manifold and $G$ a complex reductive group with $G\curvearrowright X$ in a Hamiltonian fashion, $\mu:X\rightarrow \mathfrak{g}^{\ast}$ a moment map and $Y=X//_{\lambda}G=\mu^{-1}(\lambda)/G$.

Find examples where:

i) $Y$ is an algebraic completely integrable Hamiltonian system.

ii) $Y\ncong \mathcal{M^{\ast}}(\Sigma)$ (some moduli of tame/wild Higgs on decorated surface)

来源字段：http://aimpl.org/spectralhiggs/1/

06942 | AMR-067-0021

### Constant mean curvature in homogeneous $3$-manifolds

P / partially_solved | AMR 问题清单汇编

; Do CMC spheres about a point $x$ in such a space form a foliation of $X-\{x\}$?; Could this be a way of proving embeddedness of CMC spheres in general?

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

06943 | AMR-067-0022

### Constant mean curvature in homogeneous $3$-manifolds - Question 2

P / partially_solved | AMR 问题清单汇编

; Calabi-Yau problem. For an embedded minimal surface in $\mathbb{R}^3$, does complete imply proper?; Hoffman-Meeks conjecture. For a complete embedded minimal surface of genus $g$ and $k$ ends and finite total curvature, $k\leq g+2$.

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

06944 | AIM-GEOMETRY-0310

### Contact-form scaling obstruction and the Sasakian Higgs repair

P / partially_solved | AIM 研讨会问题汇编

For a three-manifold, Corlette's theorem relates \(SL(2,\mathbb C)\)-representations of the fundamental group to harmonic data. If the three-manifold has a contact structure, does the corresponding moduli space acquire additional structure analogous to the Higgs-bundle moduli of a surface? In particular, is there a contact interpretation and critical-point description of the \(L^2\)-norm of the Higgs field, and can this help produce representations into \(SU(2)\) or \(PSL(2,\mathbb R)\)?

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

06945 | OPG-610

### Continous analogue of Hirsch conjecture

P / partially_solved | Open Problem Garden

Conjecture The order of the largest total curvature of the primal central path over all polytopes defined by $n$ inequalities in dimension $d$ is $n$.

背景链接：http://www.openproblemgarden.org/op/continous_analogue_of_hirsch_conjecture

06946 | OWR-4339-010

### Continuation of Kähler-Ricci Flow through Singularities

P / partially_solved | Oberwolfach 报告问题汇编

For the Kähler-Ricci flow, what is the behavior as $t$ approaches the singular time $T$? In particular, can the flow pass through the singularity and continue on a new manifold?

来源字段：https://doi.org/10.4171/owr/2010/26

06947 | OWR-1703876-018

### Continuous Selection of Piecewise Linear Planar Extensions

O / open | Oberwolfach 报告问题汇编

For a fixed reference triangle $ABC$, does there exist a continuous function assigning to every PL one-to-one boundary map $\partial ABC\to\mathbb{R}^2$ a PL one-to-one extension $ABC\to\mathbb{R}^2$?

来源字段：https://doi.org/10.4171/owr/2020/30

06948 | OWR-13110-019

### Contractibility of Convex-Position Polygon Spaces

O / open | Oberwolfach 报告问题汇编

Call a convex polygon with at most $k$ vertices a $k$-gon. Is the space of ordered arrangements of $n$ such $k$-gons in convex position, modulo affine transformations, contractible?

来源字段：https://doi.org/10.4171/owr/2014/40

06949 | OWR-7155451-017

### Contractible Positive-Scalar-Curvature Three-Manifolds

O / open | Oberwolfach 报告问题汇编

Is every complete contractible $3$-manifold with positive scalar curvature homeomorphic to $\mathbb{R}^3$?

来源字段：https://doi.org/10.4171/owr/2021/30

06950 | AIM-GEOMETRY-0360

### Convention-safe six-dimensional Q-curvature decomposition

S / solved | AIM 研讨会问题汇编

Problem 4: Can we explicitly write Q in dimension 6 uniquely as constant times Pff (R)plus local conformally invariant plus divergence?

Answer to problem 4: Robin Graham reports the answer to be YES.

Alice Chang. General problems in conformal geometry:

**Problem 4.** Can we explicitly write \(Q\) in dimension 6 uniquely as a constant times \(\operatorname{Pff}(R)\), plus a local conformal invariant, plus a divergence?

**Answer to Problem 4.** Robin Graham reports the answer to be yes.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

06951 | OWR-13347-010

### Convergence and Curvature of Homogeneous Ricci Flows

P / partially_solved | Oberwolfach 报告问题汇编

For a general solvable Lie group, does the bracket flow converge after suitable normalization to a Ricci soliton, and is the limit unique? If a homogeneous space admits homogeneous metrics with positive scalar curvature, must the scalar curvature of every homogeneous Ricci-flow solution on that space eventually become positive?

来源字段：https://doi.org/10.4171/owr/2014/45

06952 | OWR-12182-003

### Convergence of Arbitrary Steiner-Symmetrization Sequences

O / open | Oberwolfach 报告问题汇编

For every compact set $K\subset\mathbb R^n$ and every sequence of directions $(u_m)\subset S^{n-1}$, do the iterated Steiner symmetrals $S_{u_m}\cdots S_{u_1}K$ always converge in shape, without any assumption on the directions?

来源字段：https://doi.org/10.4171/owr/2012/59

06953 | OWR-12867-005

### Convergence of Discrete to Classical Conformality

P / partially_solved | Oberwolfach 报告问题汇编

Does discrete conformality converge to classical conformality as the triangulations of a surface become arbitrarily fine?

来源字段：https://doi.org/10.4171/owr/2014/07

06954 | OPG-720

### Convex 'Fair' Partitions Of Convex Polygons

P / partially_solved | Open Problem Garden

Basic Question: Given any positive integer n, can any convex polygon be partitioned into n convex pieces so that all pieces have the same area and same perimeter?

Definitions: Define a Fair Partition of a polygon as a partition of it into a finite number of pieces so that every piece has both the same area and the same perimeter. Further, if all the resulting pieces are convex, call it a Convex Fair Partition.

Questions: 1. (Rephrasing the above 'basic' question) Given any positive integer n, can any convex polygon be convex fair partitioned into n pieces?

2. If the answer to the above is "Not always", how does one decide the possibility of such a partition for a given convex polygon and a given n? And if fair convex partition is allowed by a specific convex polygon for a give n, how does one find the optimal convex fair partition that minimizes the total length of the cut segments?

3. Finally, what could one say about higher dimensional analogs of this question?

Conjecture: The authors tend to believe that the answer to the above 'basic' question is "yes". In other words they guess: Every convex polygon allows a convex fair partition into n pieces for any n

背景链接：http://www.openproblemgarden.org/op/textbf_convex_fair_partitions_of_convex_polygons

06955 | OWR-4136-006

### Convex Body Uniqueness from Tangential Hyperplane Sections

P / partially_solved | Oberwolfach 报告问题汇编

If convex bodies $K,L\subset\mathbb{R}^n$ contain a sphere of radius $t$ in their interiors and have equal $(n-1)$-dimensional volumes on every hyperplane tangent to that sphere, must $K=L$?

来源字段：https://doi.org/10.4171/owr/2009/53

06956 | OWR-14298007-003

### Convex Equipartitions in Tautological Vector Bundles

P / partially_solved | Oberwolfach 报告问题汇编

Let $d,j,n,k\geq 1$, let $\mathcal C$ be a collection of $j$ proper convex bodies in $\mathbb R^d$ containing the origin in their interiors, and let $\mathcal P$ be a prescribed class of convex partitions of $\mathbb R^k$. Determine whether there are a $k$-dimensional linear subspace $L\subseteq\mathbb R^d$ and a partition $(A_1,\ldots,A_n)$ of $L$ from $\mathcal P$ such that $\operatorname{vol}_k(C\cap L\cap A_1)=\cdots=\operatorname{vol}_k(C\cap L\cap A_n)$ for every $C\in\mathcal C$.

来源字段：https://doi.org/10.4171/owr/2023/58

06957 | OPG-60010

### Convex Equipartitions with Extreme Perimeter

P / partially_solved | Open Problem Garden

To divide a given 2D convex region C into a specified number n of convex pieces all of equal area (perimeters could be different) such that the total perimeter of pieces is (1) maximized (2) minimized.

Remark: It appears maximizing the total perimeter is the easier problem.

背景链接：http://www.openproblemgarden.org/op/convex_equipartitions_with_extreme_perimeter

06958 | OWR-17135-017

### Convex Geometry of General Polytope Face Vectors

O / open | Oberwolfach 报告问题汇编

For $d\geq4$, is $$\operatorname{Conv}(f(\mathcal P^d))=\operatorname{Cone}_{f(\sigma^d)}(f(\mathcal P^d))$$? Is this cone finitely generated, are its rays dense, and which polytopes determine its boundary?

来源字段：https://doi.org/10.4171/owr/2019/39

06959 | OWR-14298163-029

### Convex Hulls of Selected Barycenters

O / open | Oberwolfach 报告问题汇编

Given $n$ points in $\mathbb R^d$, determine the maximum possible number of vertices of the convex hull of an arbitrary subset of their $k$-point barycenters.

来源字段：https://doi.org/10.4171/owr/2024/3

06960 | OWR-12484-005

### Convex Spheres with an Isolated Index-Three-Halves Umbilic

O / open | Oberwolfach 报告问题汇编

Does there exist a strictly convex closed smooth sphere in $\mathbb R^3$ with an isolated umbilic point of index $3/2$?

来源字段：https://doi.org/10.4171/owr/2013/21

06961 | OPG-37456

### Convex uniform 5-polytopes

O / open | Open Problem Garden

Problem Enumerate all convex uniform 5-polytopes.

背景链接：http://www.openproblemgarden.org/op/convex_uniform_5_polytopes

06962 | OWR-1394-012

### Convex-Body Reconstruction from Seven Parallel X-Rays

O / open | Oberwolfach 报告问题汇编

Are convex bodies in $\mathbb{R}^3$ uniquely determined by their parallel X-rays in every set of seven directions in general position?

来源字段：https://doi.org/10.4171/owr/2006/56

06963 | OWR-1194-006

### Convex-Hull-Preserving Bloomings of Polytope Boundaries

O / open | Oberwolfach 报告问题汇编

Can one cut a convex polytope boundary along edges and begin an infinitesimal unfolding motion during which every face remains on the convex hull?

来源字段：https://doi.org/10.4171/owr/2006/12

06964 | AIM-GEOMETRY-0062

### Convexity obstruction and a fat-Cantor geometric embedding

P / partially_solved | AIM 研讨会问题汇编

Can one identify a geometric example of a system for which the set of phase transitions has positive Lebesgue measure?

It does not name a parameter or define “phase transition.” The immediately preceding problem 3.3 asks about the inverse-temperature family $\{q\varphi\}_{q\in\mathbb R}$ and the pressure of the geometric potential for a nonpositively curved surface. The best-supported reconstruction is therefore: for a geometrically defined system and a distinguished potential $\varphi$, can the exceptional set for $q\mapsto P(q\varphi)$ have positive one-dimensional Lebesgue measure?

Four inequivalent exceptional sets must be separated:

1. **first-order:** $P(q\varphi)$ is not differentiable; 2. **analytic:** $P(q\varphi)$ is not real analytic in any neighborhood of $q$; 3. **coexistence:** $q\varphi$ has multiple equilibrium states; 4. **recurrence/existence:** an equilibrium state appears, disappears, or changes recurrence class.

来源字段：http://aimpl.org/equibdynsysgeom/3/

06965 | AMR-058-0008

### Convexity of a Crystal on a Table

P / partially_solved | AMR 问题清单汇编

Is a crystal resting on a table under gravity necessarily convex?

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

06966 | OWR-2042-007

### Convexity of Injectivity Loci Under MTW

P / partially_solved | Oberwolfach 报告问题汇编

On a Riemannian manifold, does the Ma-Trudinger-Wang condition automatically imply that all tangent injectivity loci are convex?

来源字段：https://doi.org/10.4171/owr/2008/31

06967 | OWR-7155451-013

### Core Metrics and Positive Ricci Curvature on Simply Connected Manifolds

O / open | Oberwolfach 报告问题汇编

Does every closed simply connected manifold admitting a positive-Ricci-curvature metric admit a core metric, and does every closed simply connected 6-manifold admit a positive-Ricci-curvature metric?

来源字段：https://doi.org/10.4171/owr/2021/30

06968 | OWR-14299911-029

### Corner Peeling in Cubical Subcomplexes of Integer Three-Space

O / open | Oberwolfach 报告问题汇编

Let $G$ be a finite connected subgraph of $\mathbb{Z}^3$, and let $X(G)$ be the cubical complex containing exactly those squares and three-cubes whose vertices and edges all belong to $G$. Assume that $X(G)$ is simply connected and that every connected component of the intersection of $X(G)$ with a coordinate hyperplane is simply connected. Must $G$ admit a corner peeling, that is, an ordering $v_1,\ldots,v_n$ in which $v_i$ belongs to a unique maximal cell of the subcomplex induced by $v_1,\ldots,v_i$?

来源字段：https://doi.org/10.4171/owr/2026/8

06969 | AIM-GEOMETRY-0191

### Corner structure and chamberwise product topology for variable-length planar polygons

P / partially_solved | AIM 研讨会问题汇编

**4. (Holmes-Cerfon)** Consider the configuration space of planar \(n\)-gons such that \[ L_i-\varepsilon\leq \ell_i\leq L_i+\varepsilon. \] \(M_{L,\varepsilon}\) is a manifold with boundary. Question: Understand its topology and volume.

来源字段：https://aimath.org/pastworkshops/linkagesproblems.pdf

06970 | OWR-13687-009

### Counting Hyperbolic Manifolds with Fixed Cusp Number

P / partially_solved | Oberwolfach 报告问题汇编

What is known about $\rho_n^c(V)$, the number of complete hyperbolic n-manifolds with exactly $c>0$ cusps and volume less than V?

来源字段：https://doi.org/10.4171/owr/2015/30

06971 | OWR-4136-003

### Counting Integer Points in Polytopes

P / partially_solved | Oberwolfach 报告问题汇编

How can one compute or estimate the number of integer points in a polytope $P$, namely $|P\cap\mathbb{Z}^n|$?

来源字段：https://doi.org/10.4171/owr/2009/53

06972 | OWR-1795-009

### Counting Lattice Oval Configurations by Weight

P / partially_solved | Oberwolfach 报告问题汇编

Let $\Lambda$ be a planar lattice, let $S$ be a closed bounded convex region containing the origin, and set $J(r,P)=\Lambda\cap S(r,P)$ and $N(r,P)=|J(r,P)|$. Up to translation by a lattice vector, how many configurations satisfy $N(r,P)=n$? How many satisfy $N(r,P)\leq N$?

来源字段：https://doi.org/10.4171/owr/2008/14

06973 | OWR-16637-002

### Covariance Determinant Bounds for Polar Convex Bodies

S / solved | Oberwolfach 报告问题汇编

For every convex body $K\subseteq\mathbb R^n$ with barycenter at the origin and polar body $K^\circ$, does a universal constant $C>0$ exist such that

$$\det\!\bigl(\operatorname{Cov}(K)\operatorname{Cov}(K^\circ)\bigr)\leq(C/n)^{2n}.$$

Does every such $K$ satisfy the displayed bound?

来源字段：https://doi.org/10.4171/owr/2018/54

06974 | OWR-1394-005

### Covariogram Determination of Convex Bodies

P / partially_solved | Oberwolfach 报告问题汇编

For a convex body $K\subset\mathbb{R}^n$, let $g_K(x)=\operatorname{vol}(K\cap(K+x))$. Does the covariogram $g_K$ determine $K$ among all convex bodies, up to translations and point reflections?

来源字段：https://doi.org/10.4171/owr/2006/56

06975 | OWR-2090-017

### Cover Decomposability of Planar Convex Sets

S / solved | Oberwolfach 报告问题汇编

A planar set $P$ is cover-decomposable if there is an integer $k$ such that every $k$-fold covering of the plane by translates of $P$ can be decomposed into two coverings. Is every planar convex set cover-decomposable?

来源字段：https://doi.org/10.4171/owr/2008/44

06976 | OPG-37327

### Covering a square with unit squares

P / partially_solved | Open Problem Garden

Conjecture For any integer $n \geq 1$, it is impossible to cover a square of side greater than $n$ with $n^2+1$ unit squares.

背景链接：http://www.openproblemgarden.org/op/covering_a_square_with_unit_squares

06977 | OWR-15427-012

### Covering Balls by Translated Slabs

O / open | Oberwolfach 报告问题汇编

For $d>2$, does there exist a constant $C(d)$ such that, whenever $n_i\in S^{d-1}$ are unit directions and $\varepsilon_i>0$ satisfy $\sum_i\varepsilon_i>C(d)$, suitable translates of the slabs $$\{x\in\mathbb R^d:a_i\leq\langle x,n_i\rangle\leq a_i+\varepsilon_i\}$$ cover the unit ball?

来源字段：https://doi.org/10.4171/owr/2017/19

06978 | AMR-038-0004

### Covering points by congruent rectangles

S / solved | AMR 问题清单汇编

Given a finite planar point set and a prescribed rectangle, approximate efficiently the minimum number of congruent copies of the rectangle needed to cover the points; determine the best achievable approximation ratio.

背景链接：https://ics.uci.edu/~eppstein/junkyard/open.html

06979 | AMR-037-0017

### Crashing motorcycles efficiently

S / solved | AMR 问题清单汇编

Given motorcycles moving simultaneously along fixed rays and crashing upon reaching another track, determine the motorcycle graph in near-linear time. Can one decide which motorcycles survive, or the fate of a single motorcycle, faster? Prove sharper lower bounds.

背景链接：https://jeffe.cs.illinois.edu/open/

06980 | AIM-GEOMETRY-0156

### Critical escape and collapse-at-infinity obstructions

P / partially_solved | AIM 研讨会问题汇编

Does there exist a complete Ricci Flow that starts with uniformly bounded curvature that instantly has unbounded curvature?

来源字段：http://aimpl.org/flowriemannian/1/

06981 | OWR-1115-009

### Critical Exponents of Sierpiński Carpets and Gaskets

P / partially_solved | Oberwolfach 报告问题汇编

Determine the conformal-dimension and $\ell^p$-cohomology critical-exponent behavior for the Sierpiński carpet and the Sierpiński gasket, in particular the relevant critical exponents $p(\mathcal{J}(X,d))$.

来源字段：https://doi.org/10.4171/owr/2005/54

06982 | OWR-12485-006

### Critical Integral Menger Curvature

P / partially_solved | Oberwolfach 报告问题汇编

For the integral Menger curvature $M_p$ of a fixed-length space curve, is the round circle the unique minimizer? Does $M_3(\gamma)<\infty$ imply that $\gamma'$ exists everywhere? What regularity do local minimizers and stationary points have?

来源字段：https://doi.org/10.4171/owr/2013/22

06983 | OWR-14297736-034

### Critical Points of Second Neumann Eigenfunctions on Acute Triangles

O / open | Oberwolfach 报告问题汇编

Prove that every second Neumann eigenfunction of an acute planar triangle has at most one nonvertex critical point.

来源字段：https://doi.org/10.4171/owr/2023/36

06984 | OWR-14299290-002

### Critical Points of Symmetric Geometric Energies

P / partially_solved | Oberwolfach 报告问题汇编

How do the symmetry-based minimization methods extend from minimizers to critical points and from the original polygon class to polygons with more sides? What is the appropriate notion of a maximally symmetric competitor, and is its Wulff shape the minimizer for sufficiently small masses?

来源字段：https://doi.org/10.4171/owr/2025/23

06985 | AIM-GEOMETRY-0406

### Critical Weinstein surgery, cyclic chord words, and finite action windows

P / partially_solved | AIM 研讨会问题汇编

This is Question 1.10 in the AIM workshop list *Holomorphic curves in contact geometry*. It asks how contact homology changes under the handle decomposition of a Stein filling. The critical case is attachment of a middle-index Weinstein handle along a Legendrian sphere \(L\); the source observes that new closed Reeb orbits should be concatenations of Reeb chords of \(L\), and asks for a surgery formula in terms of the relative contact homology of \(L\).

The symbols in the official source are \(W\) for the Stein filling, \(\varphi:W\to\mathbb R\) for a pseudoconvex function, and \(L\) for the Legendrian attaching sphere. It is the running page number at the PDF page break. AIM's official HTML transcription omits it and continues directly with the critical-handle paragraph. No mathematical clause is missing there.

In modern language, take a \(2n\)-dimensional Weinstein domain \(X_0\), a Legendrian attaching link \(\Lambda\subset\partial X_0\) of \((n-1)\)-spheres, and let \(X=X_0\cup H^n_\Lambda\). The one-sphere case in the AIM question is included by taking one component.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

06986 | AIM-GEOMETRY-0180

### Critical-catenoid status and an explicit rotational endpoint rigidity proof

P / partially_solved | AIM 研讨会问题汇编

Is the catenoid the only embedded free boundary minimal annulus in $B^3$ ?

来源字段：http://aimpl.org/flowriemannian/5/

06987 | OWR-8415335-007

### Critical-Exponent Geometry of Negatively Pinched Hadamard Quotients

P / partially_solved | Oberwolfach 报告问题汇编

Let $\Gamma$ be a finitely generated discrete isometry group of a Hadamard manifold $X$ with $-b^2\leq K_X\leq-1$ and critical exponent $\delta_\Gamma\leq2$. Must $X/\Gamma$ have finitely many cusps and bounded orders of torsion elements in $\Gamma$? Also, do there exist four-dimensional examples $(X_k,\Gamma_k)$ with $\delta_{\Gamma_k}\to2$ from above and $b_k\to1$?

来源字段：https://doi.org/10.4171/owr/2021/32

06988 | AIM-GEOMETRY-0367

### Critical-weight conformal invariants built from Schouten jets

P / partially_solved | AIM 研讨会问题汇编

Problem 10: If n ≥ 4 is even, is there a nonzero scalar conformal invariant of weight -n

which is expressible as a linear combination of complete contractions of the tensors ∇lP,

l ≥ 0? If the answer to this question is no, then the Q-curvature defined via the ambient metric construction is uniquely determined by its transformation law in terms of the GJMS operator

n = 4, as is the norm squared of the ambient obstruction tensor in higher even dimensions.

Its mathematical typography is \(\nabla^lP\), meaning the \(l\)-th covariant derivative of the Schouten tensor, and \(P_n\), meaning the critical order-\(n\) GJMS operator. No missing qualifier was found.

The source alternates between scalar and density language elsewhere in the workshop document. Here “scalar conformal invariant of weight \(-n\)” is recovered as a scalar function \(I_g\) satisfying

\[ I_{e^{2\omega}g}=e^{-n\omega}I_g. \]

Equivalently, \(I_g\,dv_g\) is a conformally invariant density. For scalar \(Q\)-curvature, the corresponding transformation law is

\[ e^{n\omega}Q_{e^{2\omega}g}=Q_g+\mathcal P_{n,g}\omega, \]

where \(\mathcal P_n\) is used in this report for the critical GJMS operator, to distinguish it from the Schouten tensor \(P_{ij}\). In density notation the factor \(e^{n\omega}\) is absorbed into the volume form.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

06989 | OWR-4199-003

### Crofton Measures in Hilbert Geometry

O / open | Oberwolfach 报告问题汇编

Prove or disprove: in the Hilbert geometry of a convex body $C\subset\mathbb{R}^n$, a positive Crofton measure for Holmes-Thompson curve length $\operatorname{vol}_1$ exists only when the geometry is hyperbolic, equivalently when $C$ is an ellipsoid.

来源字段：https://doi.org/10.4171/owr/2010/04

06990 | OWR-12985-002

### CSM Positivity Beyond Grassmannian Schubert Cells

S / solved | Oberwolfach 报告问题汇编

Are the analogous Chern-Schwartz-MacPherson coefficients nonnegative for Schubert cells in flag varieties other than Grassmannians?

来源字段：https://doi.org/10.4171/owr/2014/27

06991 | GREEN-041

### Cubic Curves in F_p^2

P / partially_solved | Ben Green 问题

Suppose $A \subset \mathbb{F}_p^2$ is a set meeting every line in at most 2 points. Is it true that all except $o(p)$ points of $A$ lie on a cubic curve?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

06992 | OWR-17135-027

### Cubical Facet Counts and Nonintersecting Spatial Triangles

O / open | Oberwolfach 报告问题汇编

Let $C(d,n)$ be the maximum number of facets of a cubical $d$-polytope with $n$ vertices. Is $C(d,n)=o(n^2)$ for every fixed $d$? Separately, let $t(n)$ be the maximum number of triangles with vertices in an $n$-point subset of $\mathbb R^3$ such that two triangles meet only in a common vertex or are disjoint. Determine the growth of $t(n)$.

来源字段：https://doi.org/10.4171/owr/2019/39

06993 | OWR-4535-006

### Curvature Negativity of the Takhtajan-Zograf Metric

O / open | Oberwolfach 报告问题汇编

Is the curvature of the Takhtajan-Zograf metric negative?

来源字段：https://doi.org/10.4171/owr/2010/53

06994 | OWR-15208-010

### Curvature Rigidity of Complex Projective 3-Space Bundles

O / open | Oberwolfach 报告问题汇编

别名：Curvature Detection of Exotic Complex Projective Space Bundles [OWR-15208-028]

Let $M$ be a smooth fiber bundle over $S^1$ with fiber $\mathbb{CP}^3$. If $M$ admits a metric of nonnegative sectional curvature, must $M$ be diffeomorphic to $S^1\times\mathbb{CP}^3$ or $S^1\times_{\tau}\mathbb{CP}^3$? Does the same conclusion follow if its fibers admit a smoothly varying family of positively curved metrics?

来源字段：https://doi.org/10.4171/owr/2017/3

06995 | OWR-13347-013

### Curvature-Derivative Invariants of Homogeneous Spaces

P / partially_solved | Oberwolfach 报告问题汇编

For a homogeneous Riemannian manifold, let $k_o$ denote its constant Jacobi osculating rank, let $k_t$ be the least $k\geq0$ for which the completely traceless Cartan component $Y_0^{(k+3,2)}R$ vanishes, and let $k_S$ be its Singer invariant. Determine $k_t$ for $\operatorname{Sp}(2)/\operatorname{SU}(2)$, $(\operatorname{SO}(3)\times\operatorname{SU}(3))/U^\bullet(2)$, $\operatorname{SU}(3)/T^2$, and Kaplan's six-dimensional geodesic-orbit nilmanifold, and compute these invariants for further homogeneous spaces. Is $k_t$ finite for every naturally reductive, geodesic-orbit, or homogeneous Riemannian space, and does finite $k_t$ characterize complete homogeneous spaces? Is $k_o$ finite for every homogeneous Einstein manifold? When does equality hold in $k_t\leq k_o$, $k_S\leq k_o$, or, in the Einstein case, $k_S\leq k_t$?

来源字段：https://doi.org/10.4171/owr/2014/45

06996 | AMR-059-0009

### Curvature-Type Tensors of Statistical Manifolds

S / solved | AMR 问题清单汇编

Let $K(X,Y,Z,W)=h(R(X,Y)Z,W)$ be the curvature $(0,4)$-tensor of a statistical manifold. Investigate the structure of all $(0,4)$-tensors satisfying $K(X,Y,Z,W)=-K(Y,X,Z,W)$, the first Bianchi identity, and $K(X,Y,Z,W)+K(Y,W,Z,X)+K(W,X,Z,Y)=0$.

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

06997 | AIM-GEOMETRY-0113

### Curved soft Rogers bounds after the planar solution

P / partially_solved | AIM 研讨会问题汇编

**Spherical and Hyperbolic Soft Packings** The methods of Rogers for hard spheres can be extended to curved space. Can the soft Rogers bound also be extended?

来源字段：http://aimpl.org/softpack/1/

06998 | OWR-14297736-028

### Curved-Waveguide Simplicity and Surface-Curvature Optimization

O / open | Oberwolfach 报告问题汇编

For soft or hard-wall curved waveguides, determine whether every discrete eigenvalue above the ground state is simple. Separately, among connected compact smooth surfaces $\Sigma\subset\mathbb R^3$ of fixed area and genus $g\geq 1$, minimize $H_\Sigma=\|k_1-k_2\|_\infty$ and characterize minimizers.

来源字段：https://doi.org/10.4171/owr/2023/36

06999 | OWR-13494-013

### Curves with Minimal Symbolic Initial-Degree Gap

O / open | Oberwolfach 报告问题汇编

Which reduced, possibly irreducible, curves $C$ in $\mathbb P^3$ have ideal $I(C)$ satisfying $\alpha(I(C)^{(2)})-\alpha(I(C))=1$?

来源字段：https://doi.org/10.4171/owr/2015/9

07000 | OWR-4333-016

### Cusp Covering Conjecture for Hyperbolic Manifolds

O / open | Oberwolfach 报告问题汇编

Let $M$ be a finite-volume hyperbolic $n$-manifold. For each cusp $C$, does there exist a sublattice $\Lambda_C\subset\pi_1(C)$ such that, for every choice of sublattices $\Lambda'_C\subset\Lambda_C$, there is a finite cover $M'\to M$ whose cusps over $C$ are exactly the covers determined by $\Lambda'_C$?

来源字段：https://doi.org/10.4171/owr/2010/20

07001 | AMR-064-0002

### Cutting Corners in Sub-Riemannian Spaces

S / solved | AMR 问题清单汇编

Let $\gamma_i:[0,1]\to M$, $i=0,1$, be smooth admissible paths of a sub-Riemannian structure with $\gamma_0(0)=\gamma_1(0)=q_0$ and $\dot\gamma_0(0)\wedge\dot\gamma_1(0)\ne0$. Does there exist an admissible path connecting $\gamma_0(1)$ to $\gamma_1(1)$ that is strictly shorter than the concatenation of $\gamma_0$ and $\gamma_1$?

背景链接：https://arxiv.org/abs/1304.2590

07002 | AIM-GEOMETRY-0122

### Cycles as rings, winding classes, and defect charges

P / partially_solved | AIM 研讨会问题汇编

**Cycles in skeletal complexes.** What are the applications and physical significance of the cycles in skeletal polygonal complexes?

来源字段：http://aimpl.org/softpack/3/

07003 | AMR-065-0002

### D. Damanik: Quantum Mechanics and Quasicrystals - Conjecture

P / partially_solved | AMR 问题清单汇编

Let $\bm{x}\in\{0,1\}^{\mathbb{Z}}$ be a pattern-Sturmian sequence and define $[H\psi](m)=\psi(m+1)+\psi(m-1)+\lambda x_m\psi(m)$ on $\ell^2(\mathbb{Z})$. Prove that $\sigma(H)$ is a zero-Lebesgue-measure Cantor set and that all spectral measures are singular continuous.

背景链接：https://arxiv.org/abs/1604.06280

07004 | AMR-065-0004

### D. Damanik: Quantum Mechanics and Quasicrystals - Conjecture

O / open | AMR 问题清单汇编

There exist values of $\lambda_1$ and $\lambda_2$ such that the spectrum $\sigma(H)$ of $H$ is a Cantorval; that is, the spectrum is the closure of its interior and no connected component is isolated.

背景链接：https://arxiv.org/abs/1604.06280

07005 | AMR-065-0001

### D. Damanik: Quantum Mechanics and Quasicrystals - Problem

P / partially_solved | AMR 问题清单汇编

Determine all single sided (resp. double sided) sequences that are pattern Sturmian.

背景链接：https://arxiv.org/abs/1604.06280

07006 | AMR-065-0003

### D. Damanik: Quantum Mechanics and Quasicrystals - Problem

P / partially_solved | AMR 问题清单汇编

For the graph $(V,E)$ of a Penrose tiling, define $H$ on $\ell^2(V)$ by $[H\psi](v)=\sum_{w:(v,w)\in E}(\psi(w)-\psi(v))$. Determine the spectrum $\sigma(H)$.

背景链接：https://arxiv.org/abs/1604.06280

07007 | GEOM-027

### Danzer's Problem

O / open | 独立 / 未分配集合

Do Danzer sets of bounded density or bounded separation exist?

背景链接：https://en.wikipedia.org/wiki/Danzer_set

07008 | OWR-14297736-004

### de Gennes Bound for Magnetic Neumann Eigenvalues

O / open | Oberwolfach 报告问题汇编

For every bounded smooth planar domain $\Omega$ and constant magnetic field $\beta$, prove $\lambda_1^N(\Omega,\beta)\leq\Theta_0\beta$, where $\Theta_0$ is the de Gennes constant.

来源字段：https://doi.org/10.4171/owr/2023/36

07009 | OWR-16629-009

### Decomposition of Equiangular Tight Frames

P / partially_solved | Oberwolfach 报告问题汇编

别名：Decomposition of Equiangular Tight Frames [OWR-16629-002]

If $\Phi$ is an equiangular tight frame, when can it be decomposed nontrivially into a disjoint union of equiangular tight frames for their respective spans?

来源字段：https://doi.org/10.4171/owr/2018/46

07010 | AIM-GEOMETRY-0022

### Deep points, matroid connectivity, and stabilizer components in the top positroid stratum

P / partially_solved | AIM 研讨会问题汇编

Let \(X\) be really full rank. This implies there is a torus of dimension the number of frozen variables acting on \(X\) preserving cluster structure. \(S\) acts freely on each cluster torus so any point with nontrivial \(S\)-stabilizers is in \(P\).

In \(G(k,n)\), nontrivial stabilizers correspond to disconnected matroids. So if \([v_1,\dots,v_n]\) is a \(k\times n\) matrix where the matroid of \(v_1,\dots,v_n\) is disconnected, this point of \(G(k,n)\) is in \(P\).

Is the converse also true (Is \(P\) for the big positroid cell exactly the points with disconnected matroid)?

来源字段：http://aimpl.org/clusterbraid/3/

07011 | AIM-GEOMETRY-0102

### Defect charge and a metric-Delaunay reduction for optimal spherical codes

P / partially_solved | AIM 研讨会问题汇编

Prove/Disprove that for optimal $N$-point codes on $S^2$ non-hexagonal Voronoi cells are dense in $S^2$ as $N \rightarrow \infty$.

来源字段：http://aimpl.org/discreteaf/1/

07012 | OWR-12975-013

### Definite Determinantal Multiples of Hyperbolic Polynomials

O / open | Oberwolfach 报告问题汇编

If $h$ is hyperbolic with respect to $e$, does there exist a hyperbolic polynomial $q$ with $C_h(e)\subseteq C_q(e)$ such that $qh$ has a definite determinantal representation?

来源字段：https://doi.org/10.4171/owr/2014/17

07013 | OWR-4078-003

### Deformations of Canonical Projective Structures on Hyperbolic Three-Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Can the canonical projective structure on a hyperbolic three-manifold be nontrivially deformed?

来源字段：https://doi.org/10.4171/owr/2009/35

07014 | AMR-037-0003

### Degenerate facets of polytopes

P / partially_solved | AMR 问题清单汇编

A facet of a $d$-polytope is degenerate if it has more than $d$ vertices. Determine the maximum number of degenerate facets of an $n$-vertex $d$-polytope; in particular, can a four-polytope have at least $2n$ degenerate facets?

背景链接：https://jeffe.cs.illinois.edu/open/

07015 | OWR-2042-006

### Degenerate Versus Full Ma-Trudinger-Wang Conditions

O / open | Oberwolfach 报告问题汇编

For the Riemannian squared-distance cost, does $\operatorname{MTW}^{\perp}\geq0$ imply the stronger condition $\operatorname{MTW}\geq0$?

来源字段：https://doi.org/10.4171/owr/2008/31

07016 | OWR-4535-005

### Degeneration Proofs of Index Theorems for Punctured Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Can the index theorem for punctured surfaces be derived from the index theorem for compact surfaces through degeneration?

来源字段：https://doi.org/10.4171/owr/2010/53

07017 | AIM-GEOMETRY-0119

### Delaunay and cutoff skeletal complexes on Delone sets

P / partially_solved | AIM 研讨会问题汇编

**Building skeletal complexes**

The notion of a skeletal complex can be naturally extended to Delone sets.

Study skeletal complexes based on Delone sets.

来源字段：http://aimpl.org/softpack/3/

07018 | OWR-1194-004

### Delaunay Unfoldings of Convex Polytope Boundaries

O / open | Oberwolfach 报告问题汇编

Can a convex polytope boundary be unfolded without overlap by cutting along its Delaunay triangulation; in particular, when all faces are acute triangles, by cutting along polytope edges?

来源字段：https://doi.org/10.4171/owr/2006/12

07019 | AIM-GEOMETRY-0088

### Deletion frontiers for the 24-cell, 600-cell, E8, and Leech spherical codes

P / partially_solved | AIM 研讨会问题汇编

If the vertices of the 24-cell, 600-cell, the normalized $E_8$ roots, or the normalized Leech minimal vectors are viewed as spherical codes, does deleting the indicated number of vertices leave an optimal code of the smaller cardinality?

来源字段：http://aimpl.org/discreteaf/1/

07020 | AIM-GEOMETRY-0272

### Dense periodic directions force dense stationary-net crossing vertices

P / partially_solved | AIM 研讨会问题汇编

We call a stationary net nontrivial if it has at least one vertex of valency \(\geq 3\), in other words if it is not a disjoint union of simple closed geodesics.

**Question 11.2.1 (Gromov).** Are nontrivial stationary nets dense on any closed Riemannian surface? In other words, for each nonempty open subset \(U\), is there a stationary net intersecting \(U\)?

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07021 | AIM-GEOMETRY-0254

### Dense projected geodesics for cyclic torus metrics

P / partially_solved | AIM 研讨会问题汇编

**Question 8.5.2.** Does every metric on a compact surface of positive genus have a geodesic that is dense in the surface?

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07022 | OWR-14297736-021

### Dense Pure Point Spectrum on Quasi-Conical Domains

O / open | Oberwolfach 报告问题汇编

Choose the hole sizes in a connected quasi-conical domain so that its Dirichlet Laplacian has neither absolutely continuous nor singular continuous spectrum, and hence has dense pure point spectrum.

来源字段：https://doi.org/10.4171/owr/2023/36

07023 | OPG-1761

### Dense rational distance sets in the plane

P / partially_solved | Open Problem Garden

Problem Does there exist a dense set $S \subseteq {\mathbb R}^2$ so that all pairwise distances between points in $S$ are rational?

背景链接：http://www.openproblemgarden.org/op/dense_rational_distance_sets_in_the_plane

07024 | OWR-12012-008

### Density and Kissing Numbers for Binary Sphere Packings

O / open | Oberwolfach 报告问题汇编

For packings of Euclidean balls of radii $r_1$ and $r_2$ occurring in a prescribed proportion, what are the maximal packing density and mean kissing numbers $k_{11},k_{12},k_{21},k_{22}$? When $r_1$ and $r_2$ are close, does the density equal that for congruent balls, so that the two sizes do not mix?

来源字段：https://doi.org/10.4171/owr/2012/40

07025 | AIM-GEOMETRY-0296

### Density fails by compactness for locally free circle actions

S / solved | AIM 研讨会问题汇编

**Question 5.6 (G. Landweber)** In standard Morse theory, every function is arbitrarily close to a Morse function. In the space of abstract moment maps, is every abstract moment map arbitrarily close to a non-degenerate abstract moment map?

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

07026 | OWR-2090-012

### Density Thresholds for Finite-Field Configurations

P / partially_solved | Oberwolfach 报告问题汇编

Let $E\subseteq\mathbb{F}_q^d$, where $d\geq2$, and let $V=\{0,v_1,\ldots,v_{k-1}\}$ be a nondegenerate $k$-point configuration whose nonzero vectors span a $(k-1)$-dimensional subspace. How large must $E$ be to guarantee that it contains a congruent copy $\tau+O(V)$ for some $\tau\in\mathbb{F}_q^d$ and $O\in\operatorname{SO}_d(\mathbb{F}_q)$?

来源字段：https://doi.org/10.4171/owr/2008/44

07027 | OWR-14297732-011

### Desingularization of Kähler-Einstein Metrics

O / open | Oberwolfach 报告问题汇编

Can every singular Kähler-Einstein metric be desingularized by Einstein four-manifolds, possibly using Einstein metrics that are not Kähler?

来源字段：https://doi.org/10.4171/owr/2023/29

07028 | OWR-1595-007

### Detecting Nonnormal Smooth Lattice Polytopes

O / open | Oberwolfach 报告问题汇编

For lattice polytopes $P$, find examples for which the number of lattice points in $P$ is substantially smaller than the number in $2P$, preferably with $P$ smooth; test smooth polytopes arising from smooth Fano toric varieties and small projective embeddings; and determine whether useful invariants, including Ehrhart functions, detect nonnormality.

来源字段：https://doi.org/10.4171/owr/2007/39

07029 | AIM-GEOMETRY-0249

### Deterministic lift escape and the centered torus cocycle

P / partially_solved | AIM 研讨会问题汇编

**Question 8.3.1 (Benjamin Schmidt).** Which of these sets has full measure?

Let \((M^2,g)\) be a closed surface whose geodesic flow \(\phi^t:SM\to SM\) is ergodic for Liouville measure. For \(v\in SM\), let \(\gamma_v:\mathbb R\to M\) be its unit-speed geodesic and let \(\widetilde\gamma_v:\mathbb R\to\widetilde M\) be any lift. The source defines \[ \begin{aligned} T_b&=\{v\in SM:\widetilde\gamma_v(\mathbb R) \text{ is contained in a bounded subset of }\widetilde M\},\\ T_p&=\{v\in SM:\widetilde\gamma_v \text{ is unbounded and approaches infinity properly}\},\\ T_i&=SM\setminus(T_b\cup T_p). \end{aligned} \] The PDF prints \(TM\) rather than \(SM\) in the last line, but the paragraph begins with \(v\in SM\) and all three sets partition unit tangent vectors, so \(SM\) is the consistent reading. The source states that the sets are measurable and invariant, hence ergodicity makes exactly one full measure.

The exact problem is:

The acknowledgements identify the participating Schmidt as Benjamin Schmidt. The following source commentary predicts \(T_i\) on the torus and \(T_p\) on surfaces of genus at least two, discusses minimizing geodesics, and says the question makes sense in all dimensions. The page header “KEITH BURNS AND VLADIMIR S. MATVEEV” is not part of the problem. All text beginning with “8.4. Generic metrics” is spill from the next section and is excluded.

The source does not formalize “approaches infinity properly.” Here it has its standard two-sided meaning: \(\widetilde\gamma_v:\mathbb R\to\widetilde M\) is a proper map, equivalently \[ d(\widetilde\gamma_v(0),\widetilde\gamma_v(t)) \longrightarrow\infty\quad\text{as }|t|\to\infty. \tag{1.1} \] All results below also have the evident one-sided version if only \(t\to+\infty\) was intended.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07030 | AIM-GEOMETRY-0320

### Diagonal Hodge type under regular Kähler reduction

P / partially_solved | AIM 研讨会问题汇编

**Question 13.1 (J. Weitsman).** Suppose $M$ is a compact Kähler manifold and a Hamiltonian $G$-space. Suppose all the ordinary cohomology is of type $(p,p)$. Can we say anything about the cohomology of the quotient?

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

07031 | OWR-15427-013

### Dihedral Angle Comparison for Combinatorially Equivalent Polytopes

O / open | Oberwolfach 报告问题汇编

Let $P_1,P_2\subset\mathbb R^3$ be combinatorially equivalent convex polytopes. Must there be corresponding edges $e_1\subset P_1$ and $e_2\subset P_2$ such that the dihedral angle at $e_1$ is no greater than the angle at $e_2$, unless all corresponding dihedral angles are equal?

来源字段：https://doi.org/10.4171/owr/2017/19

07032 | OWR-13498-009

### Dihedral Rigidity and Offsets of Nonconvex Polytopes

P / partially_solved | Oberwolfach 报告问题汇编

For a polytope with fixed combinatorics, are all face angles determined by its dihedral angles in the nonconvex case? If $P$ is a nonconvex three-dimensional polytope with spherical boundary, does an offset polytope $P_\varepsilon$ exist for every sufficiently small $\varepsilon>0$, with each face translated outward by $\varepsilon$ and no additional faces?

来源字段：https://doi.org/10.4171/owr/2015/13

07033 | OWR-14298163-028

### Dihedral-Angle Gaps and Box-Graph Independence

O / open | Oberwolfach 报告问题汇编

Resolve the following two questions. (1) Determine the largest $\varepsilon_d$ for which infinitely many $d$-polytopes have every dihedral angle at most $\pi-\varepsilon_d$. (2) Does some $c>0$ guarantee an independent set of size at least $cn$ in every intersection graph of $n$ axis-parallel boxes in $\mathbb R^3$ with clique number $2$?

来源字段：https://doi.org/10.4171/owr/2024/3

07034 | AIM-GEOMETRY-0366

### Dimension four bounds total Q above and determines its exact range

S / solved | AIM 研讨会问题汇编

Problem 9: Given a compact manifold of even dimension \(>2\), show that there exists a sequence of metrics such that \(\int Q\to+\infty\).

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

07035 | OWR-11136-024

### Dimension of Convex Partition Spaces in Three Dimensions

O / open | Oberwolfach 报告问题汇编

What is the dimension of the space of all partitions of $\mathbb{R}^3$ into $n>3$ convex pieces?

来源字段：https://doi.org/10.4171/owr/2011/44

07036 | AIM-GEOMETRY-0368

### Dimension-four Schouten-jet uniqueness and a two-obstruction reduction

P / partially_solved | AIM 研讨会问题汇编

If the answer is yes, then this gives a characterization of the GJMS operator \(P_n\). Combined with a negative answer to Problem 11, this would provide a unique specification of \(Q\).

This reconstruction is source-verified, but the self-reference is almost certainly a typographical error: the preceding Problem 10 asks whether a weight-\(-n\) scalar conformal invariant can be made purely from \(\nabla^IP\), and that is exactly the complementary condition needed to specify \(Q\). Replacing the printed second “Problem 11” by “Problem 10” is therefore a logical inference, not verified wording.

Here \(P_{ij}\) is the Schouten tensor, not the GJMS operator; to prevent collision this report writes the Schouten tensor as \(\mathsf P\) and the critical GJMS operator as \(P_n\).

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

07037 | OWR-14297736-007

### Dimension-Free Neumann Eigenvalue-Ratio Bounds

O / open | Oberwolfach 报告问题汇编

For every bounded convex domain $\Omega$ and every $k$, prove a dimension-free bound $\mu_{k+1}(\Omega)\lesssim\mu_k(\Omega)$; find an analytic proof; and extend the universal eigenvalue-ratio bound to convex domains in manifolds of nonnegative Ricci curvature.

来源字段：https://doi.org/10.4171/owr/2023/36

07038 | OWR-11695863-001

### Dirac Rayleigh-Faber-Krahn Inequality

O / open | Oberwolfach 报告问题汇编

For every smooth bounded domain $\Omega\subset\mathbb R^2$, is $E_1(\Omega)\geq\sqrt{\pi/|\Omega|}\,E_1(\mathbb D)$, with equality only for disks?

来源字段：https://doi.org/10.4171/owr/2022/53

07039 | OPG-59888

### Dirac's Conjecture

P / partially_solved | Open Problem Garden

Conjecture For every set $P$ of $n$ points in the plane, not all collinear, there is a point in $P$ contained in at least $\frac{n}{2}-c$ lines determined by $P$, for some constant $c$.

背景链接：http://www.openproblemgarden.org/op/diracs_conjecture

07040 | OWR-13498-002

### Discrete Riemann Mapping by Conformal Triangulations

O / open | Oberwolfach 报告问题汇编

Discrete Riemann mapping conjecture: let $\Omega$ be a Jordan domain with distinct boundary points $p,q,r$. For each $\varepsilon>0$, triangulate a maximal simply connected $\Omega_\varepsilon\subset\Omega$ by equilateral triangles of edge length $\varepsilon$, choose boundary vertices $p_\varepsilon,q_\varepsilon,r_\varepsilon$ within $2\varepsilon$ of $p,q,r$, and form its metric double. After discrete conformal uniformization to area $3/2$ with curvature $4\pi/3$ at those three vertices and zero elsewhere, does the resulting polyhedral sphere converge as $\varepsilon\to0$ to the metric double of the unit equilateral triangle?

来源字段：https://doi.org/10.4171/owr/2015/13

07041 | OWR-14299290-001

### Discrete Symmetries in Geometric Set Minimization

O / open | Oberwolfach 报告问题汇编

How does imposing a discrete symmetry group change a geometric set-minimization problem? Does the analogous conclusion hold when the competitors are planar polygons?

来源字段：https://doi.org/10.4171/owr/2025/23

07042 | OWR-13498-010

### Discrete-Conformal Metric Subdivision Schemes

O / open | Oberwolfach 报告问题汇编

Let $T$ be a triangulated surface and let discrete conformally equivalent metrics $\lambda,\mu$ satisfy $\mu_{ij}=e^{(u_i+u_j)/2}\lambda_{ij}$. Does there exist a metric subdivision scheme producing subdivided metrics $\lambda',\mu'$ that remain discretely conformally equivalent?

来源字段：https://doi.org/10.4171/owr/2015/13

07043 | OWR-14212-003

### Discretization of Elastic Curves in Shape Spaces

P / partially_solved | Oberwolfach 报告问题汇编

How can elastic curves be discretized in infinite-dimensional shape spaces in a way compatible with the continuous elastic-bending variational problem?

来源字段：https://doi.org/10.4171/owr/2016/4

07044 | OWR-13110-005

### Displacement-Energy Bound for Plank Coverings

O / open | Oberwolfach 报告问题汇编

Let $K$ be a convex body covered by finitely many planks whose relative widths sum to $w$, and let $B^\circ$ be the polar of the unit ball for the norm used to measure those widths. Must the displacement energy satisfy $e(K\times B^\circ)\leq 2w$?

来源字段：https://doi.org/10.4171/owr/2014/40

07045 | OWR-8415336-017

### Displacement-Energy Bounds for Poisson Brackets of Open Covers

P / partially_solved | Oberwolfach 报告问题汇编

Let $(M,\omega)$ be a closed symplectic manifold and let $\mathcal U=\{U_i\}_{i\in I}$ be a finite open cover by displaceable sets. Is there a constant $C(M,\omega)>0$ such that $$ pb(\mathcal U)e(\mathcal U)\geq C(M,\omega), $$ where $e(\mathcal U)=\max_i e(U_i)$?

来源字段：https://doi.org/10.4171/owr/2021/33

07046 | AMR-071-0064

### Dissection into orthoschemes

P / partially_solved | AMR 问题清单汇编

Dissection into orthoschemes - is it possible for simplices of every dimension?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

07047 | AMR-063-0001

### Distinguishing Fintushel-Stern Manifolds

P / partially_solved | AMR 问题清单汇编

If knots $K$ and $K'$ have the same Alexander polynomial, are the corresponding Fintushel--Stern four-manifolds $X_K$ and $X_{K'}$ diffeomorphic?

背景链接：https://wwwf.imperial.ac.uk/~skdona/NONLINEARITYPROBLEMS.PDF

07048 | AMR-071-0085

### Does every convex polyhedron have Rupert's property

P / partially_solved | AMR 问题清单汇编

Does every convex polyhedron have Rupert's property?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

07049 | AMR-071-0079

### Does every higher-dimensional tiling by translations of convex polytope tiles have an affine transformation taking it to a Voronoi diagram

P / partially_solved | AMR 问题清单汇编

Does every higher-dimensional tiling by translations of convex polytope tiles have an affine transformation taking it to a Voronoi diagram?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

07050 | OWR-14298163-008

### Dolnikov’s Colorful Transversal Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $K\subseteq\mathbb R^2$ be compact and convex, and let $\mathcal F_1,\mathcal F_2,\mathcal F_3$ be finite families of translates of $K$. If $A\cap B\neq\varnothing$ whenever $A\in\mathcal F_i$, $B\in\mathcal F_j$, and $i\neq j$, prove that some $\mathcal F_j$ can be pierced by three points.

来源字段：https://doi.org/10.4171/owr/2024/3

07051 | OWR-14610-011

### Donaldson Symplectic Calabi-Yau Equation

O / open | Oberwolfach 报告问题汇编

Given a compact symplectic 4-manifold $(M,J,\omega)$ with $b_2^+(M)=1$ and $F\in C^\infty(M)$ satisfying $\int_M e^F\omega^2=\int_M\omega^2$, does there exist a symplectic form $\widetilde{\omega}\in[\omega]$ such that $\widetilde{\omega}^2=e^F\omega^2$?

来源字段：https://doi.org/10.4171/owr/2016/28

07052 | OWR-4339-006

### Donaldson's Almost-Complex Calabi-Yau Conjecture

O / open | Oberwolfach 报告问题汇编

Let $(M,\omega)$ be a compact symplectic four-manifold, let $J$ be an almost-complex structure tamed by $\omega$, and let $\sigma$ be a smooth volume form with $\int_M\sigma=\int_M\omega^2$. If a symplectic form $\widetilde\omega$ is cohomologous to $\omega$, compatible with $J$, and satisfies $\widetilde\omega^2=\sigma$, do $C^\infty$ a priori bounds for $\widetilde\omega$ follow from bounds depending only on $(M,\omega,J,\sigma)$?

来源字段：https://doi.org/10.4171/owr/2010/26

07053 | OWR-14297745-002

### Double Cap Conjecture

O / open | Oberwolfach 报告问题汇编

Is the largest measurable subset of $S^{n-1}$ containing no orthogonal pair the union of two open antipodal spherical caps of angular radius $\pi/4$?

来源字段：https://doi.org/10.4171/owr/2023/35

07054 | AMR-058-0007

### Double Crystals with Anisotropic Surface Energy

P / partially_solved | AMR 问题清单汇编

Determine double crystals in $\mathbb{R}^3$ minimizing orientation-dependent surface energy for a cubic Wulff shape and for general Wulff shapes. Must the minimizer use only orientations appearing on the Wulff shape? What changes if the three interfaces have different cost functions?

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

07055 | AIM-CONVEX_GEOMETRY-0013

### Dual random inner products: status split and Hanner closure

P / partially_solved | AIM 研讨会问题汇编

Let \(K\) be a symmetric convex body in \(\mathbb R^n\), and let random vectors \(X\) and \(Y\) be distributed in \(K\) and \(K^*\), respectively. Is the variance of \(\langle X,Y\rangle\) largest for ellipsoids? A weaker conjecture asks whether \(\int_{K\times K^*}|\langle x,y\rangle|^2\,dx\,dy\) is largest for ellipsoids. Paouris asks the corresponding questions with exponent \(p>0\). Given \(0<c<1\), one may also ask about either the probability that \(\langle X,Y\rangle=c\), or the volume of the region where this is so.

来源字段：https://aimath.org/WWN/mahlerduality/mahlerduality.pdf

07056 | AMR-059-0003

### Dually Flat Structures on Riemannian Manifolds

P / partially_solved | AMR 问题清单汇编

Given a Riemannian manifold $(M,g)$, can one always introduce a symmetric $(0,3)$-tensor $T$ so that $(M,g,\nabla,\nabla^*)$ is dually flat? If the construction is not unique, characterize the class of resulting spaces.

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

07057 | OPG-37459

### Durer's Conjecture

O / open | Open Problem Garden

别名：Edge-Unfolding Convex Polyhedra [OPG-60037]

Conjecture Every convex polytope has a non-overlapping edge unfolding.

背景链接：http://www.openproblemgarden.org/op/d_urers_conjecture

07058 | OPG-37084

### Edge-Colouring Geometric Complete Graphs

P / partially_solved | Open Problem Garden

Question What is the minimum number of colours such that every complete geometric graph on $n$ vertices has an edge colouring such that:

\item[Variant A] crossing edges get distinct colours,
\item[Variant B] disjoint edges get distinct colours,
\item[Variant C] non-disjoint edges get distinct colours,
\item[Variant D] non-crossing edges get distinct colours.

背景链接：http://www.openproblemgarden.org/op/edge_colouring_geometric_complete_graphs

07059 | AMR-054-0064

### Edge-Unfolding Polycubes

P / partially_solved | AMR 问题清单汇编

Is there any genus-zero orthogonal polyhedron $P$ built by gluing together cubes face-to-face that cannot be edge-unfolded, where all cube edges on the surface of $P$ are considered edges available for cutting? These orthogonal polyhedra are sometimes known as polycubes, 3D versions of 2D polyominoes.

背景链接：https://topp.openproblem.net/p64

07060 | OWR-14298163-024

### Edges of Centrally Symmetric Four-Polytopes

O / open | Oberwolfach 报告问题汇编

Determine the maximum number of edges of a centrally symmetric four-dimensional polytope with $n$ vertices.

来源字段：https://doi.org/10.4171/owr/2024/3

07061 | AIM-GEOMETRY-0199

### Effective moduli symmetry and the obstruction from gauge fixing

P / partially_solved | AIM 研讨会问题汇编

**12. (Whiteley) Conjecture:** Given a symmetric bar-joint framework \((G,p)\), the configuration space of \((G,p)\) (with appropriate parts of the frame of reference fixed), has the symmetry of the most symmetric individual realization in the configuration space.

来源字段：https://aimath.org/pastworkshops/linkagesproblems.pdf

07062 | AIM-GEOMETRY-0053

### Effective simple-geodesic growth and sharp metric stability

S / solved | AIM 研讨会问题汇编

Let $\varphi_t$ be geodesic flow on the unit tangent bundle of a compact surface of constant negative curvature. Let $S_n$ denote the number of simple closed geodesics with length $\le n$. Find precise growth rates for $S_n$. More generally, what can be said about variable negative curvature?

来源字段：http://aimpl.org/equibdynsysgeom/2/

07063 | OWR-1106-009

### Effective Thurston Fixed-Point Theorems for Skinning Maps

O / open | Oberwolfach 报告问题汇编

Find an effective proof of Thurston's fixed-point theorem for the skinning map.

来源字段：https://doi.org/10.4171/owr/2005/45

07064 | GEO-028

### Ehrhart's Volume Conjecture

P / partially_solved | 独立 / 未分配集合

Does a convex body in $\mathbb{R}^n$ with one interior lattice point at its center of mass have volume at most $(n+1)^n/n!$?

背景链接：https://cdn.openai.com/pdf/ten-proofs-oai.pdf

07065 | OWR-16415-010

### Eigenfunction Averages Along Simple Closed Geodesics

O / open | Oberwolfach 报告问题汇编

Let $S$ be a compact hyperbolic surface with Laplace eigenfunctions $\phi_n$ and eigenvalues $\lambda_n$. Do constants $\alpha,\beta>0$ exist such that

$$\frac{\sum_{\ell(C)\leq T}\int_C\phi_n\,ds}{\#\{C:\ell(C)\leq T\}}=O(\lambda_n^{-\alpha}T^{-\beta})$$

when the sum ranges over simple closed geodesics $C$?

来源字段：https://doi.org/10.4171/owr/2018/40

07066 | AIM-GEOMETRY-0364

### Einstein factorization and the trace-free-Ricci obstruction to Paneitz positivity

P / partially_solved | AIM 研讨会问题汇编

Problem 7: On M 4, Gursky ("The principal eigenvalue of a conformally invariant dif-ferential operator, with an application to semilinear elliptic PDE." Comm. Math. Phys., 207(1):131-143, 1999.) proved that if Sc > 0, and if \(\int Q>0\), then the Paneitz operator \(P_4\) is positive with its kernel consisting of constants. The original proof given by Gursky depends on estimates of solution of some non-linear PDE. Can one also see this fact from the construction method of the general GJMS operators? Claude LeBrun.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

07067 | OWR-12581-003

### Einstein Metric Gluing by Reverse Bubbling

P / partially_solved | Oberwolfach 报告问题汇编

Can Einstein metrics be glued across connected-sum pieces by starting from a limit space and attaching a bubble, thereby producing a critical point of a Riemannian functional that generalizes the Einstein-Hilbert functional?

来源字段：https://doi.org/10.4171/owr/2013/33

07068 | OWR-13347-004

### Einstein Solvable Factors and Orthogonal Levi Orbits

O / open | Oberwolfach 报告问题汇编

For a noncompact nonflat homogeneous Einstein space $G/K$ with Levi decomposition $G=G_1\ltimes G_2$ and $G_2=AN$, is the geometry induced on $G_2$ as a subspace of $G/K$ Einstein? If $G_1=H_1\cdots H_k$ is a product of simple subgroups, are the $H_i$-orbits through $[e]$ pairwise orthogonal?

来源字段：https://doi.org/10.4171/owr/2014/45

07069 | AIM-GEOMETRY-0120

### Element-order reduction and short-span bounds for Delone cluster groups

P / partially_solved | AIM 研讨会问题汇编

For $d \ge 4$, given a Delone set $(R,r)$ for which all $2R$-clusters are pairwise congruent, find upper bound on the order of the symmetry group of the clusters which does not depend on $R/r$.

**Recovered Group Order Problem.** Let $X\subset\mathbb R^d$, $d\geq 4$, be a Delone set with exact packing and covering parameters $(r,R)$. Suppose that $X$ is $2R$-regular: for every $x,x'\in X$ there is an ambient Euclidean isometry taking $x$ to $x'$ and \[ C_x(2R):=X\cap B_x(2R) \quad\text{to}\quad C_{x'}(2R). \] For \[ S_x(2R):=\{g\in\operatorname{Iso}(\mathbb R^d):g(x)=x, \ g(C_x(2R))=C_x(2R)\}, \] find a number $h_d$, depending only on $d$, such that $|S_x(2R)|\leq h_d$.

来源字段：http://aimpl.org/softpack/3/

07070 | OWR-16637-012

### Ellipsoid Rigidity from Constant Central-Section Products

P / partially_solved | Oberwolfach 报告问题汇编

If an origin-symmetric convex body $K\subseteq\mathbb R^n$, with $n\geq3$, satisfies

$$h_K(\theta)\operatorname{vol}_{n-1}(K\cap\theta^\perp) =c$$

for every $\theta\in S^{n-1}$ and a constant $c$ independent of $\theta$, must $K$ be an ellipsoid?

来源字段：https://doi.org/10.4171/owr/2018/54

07071 | OWR-11568-007

### Elliptic Genera of Positively Curved Spin Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Let $(M,g)$ be a closed Riemannian spin manifold with $\operatorname{sec}(M)>0$. Is its elliptic genus constant as a modular function, namely $\phi(M)=\operatorname{sign}(M)$?

来源字段：https://doi.org/10.4171/owr/2012/01

07072 | OWR-16933-007

### Embedding Spaces of Divisors on Real Cubics

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a plane projective $M$-curve of degree $3$, and let $A$ be a transverse real divisor of degree $9$. Is $\pi_1(\operatorname{Emb}(A,X))$ trivial, and how does the answer depend on the distribution of $A$ among the connected components of $X(\mathbb R)$?

来源字段：https://doi.org/10.4171/owr/2019/21

07073 | AMR-038-0006

### Embedding the hyperbolic plane

P / partially_solved | AMR 问题清单汇编

Does the hyperbolic plane admit a smooth isometric immersion into $\mathbb R^4$? More generally, determine the least Euclidean dimension for such an immersion under natural regularity assumptions.

背景链接：https://ics.uci.edu/~eppstein/junkyard/open.html

07074 | AIM-GEOMETRY-0318

### End-area moduli on the cotangent cylinder

P / partially_solved | AIM 研讨会问题汇编

Question 11.2 (M. Libine) If there is no such diffeomorphism, how can we parametrize symplectic forms on T ∗M up to this equivalence?

Comment 11.3 (M. Libine) Answering this question would provide further extensions to the Berligne-Vergne localization formula extended to non-compact (reductive) group actions.

12 Volume growth of hyperK¨ ahler manifolds

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

07075 | AIM-GEOMETRY-0161

### Endpoint Ricci-energy balance and Weyl-dominated blow-ups under bounded scalar curvature

P / partially_solved | AIM 研讨会问题汇编

Let $(M^n,g_t)_{t\in[0,T)}$ be a Ricci flow with $T<\infty$. Assume $R<1$ on $M\times[0,T)$. Characterize $g_t$ as $t\to T$.

来源字段：http://aimpl.org/flowriemannian/1/

07076 | AIM-GEOMETRY-0408

### Energy and spectral-gap gates for Lagrangian corners

P / partially_solved | AIM 研讨会问题汇编

**Question 1.12.** In setting up such an extended field theory picture, it is important to formulate holomorphic curve theory for manifolds with boundary, or open manifolds with some asymptotic conditions. The basic idea is that when you replace boundary conditions with asymptotic conditions, you get more information about and control over how you approach the boundary. For example, if you are looking at holomorphic curves with boundary in a Lagrangian, it is sometimes better to consider curves with asymptotics in the unit cotangent bundle of the Lagrangian. More generally, for the extended field theory picture, one may need to consider holomorphic curves with boundary along some Lagrangian cylinders, and then you have holomorphic curves with corners, two different types of asymptotic conditions...

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

07077 | AIM-GEOMETRY-0247

### Energy-normalized Hopf defect and a curvature-sign rigidity theorem

P / partially_solved | AIM 研讨会问题汇编

Question 7.4.1 (Paternain). Consider a magnetic flow on a closed surface of genus

≥ 2 and an energy level such that the topological entropy vanishes. Suppose that the magnetic geodesics on this level do not have conjugate points. Is the system locally symmetric in the sense that the metric has constant curvature and the magnetic form is a constant multiple of the volume form?

Many of the earlier questions in this section can also be asked about magnetic geodesics lying on a certain (possibly sufficiently high) energy level. Let us men-tion [Bialy2000], where a natural analog of Theorem 7.0.2 was proved under the assumption that the metric is conformally flat, and it was conjectured that this assumption is not essential. 8. Unrestricted curvature

8.1. Ergodic geodesic flows. Does every closed manifold (with dimension ≥ 2)admit a metric (Riemannian or Finsler) with ergodic geodesic flow?

This is known for surfaces [Donnay1988II], for 3-manifolds [Katok1994], for prod-uct manifolds in which the factors have dimenson ≤ 3 [Bur-Ger1994], and for spheres [Bur-Ged]. Donnay and Pugh have shown that any embedded surface can be perturbed, in the C0 topology to an embedded surface whose geodesic flow is ergodic [Don-Pugh2004]. These constructions all make essential use of the focusing caps introduced by Donnay in [Donnay1988I]. The general problem is still wide open despite some reports of its solution (p. 87 of [Berger2000] and Section 10.9 of [Berger2003]). 8.2. Positive curvature.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07078 | AIM-GEOMETRY-0048

### Entrance-normalized extreme values and a universal hyperbolic-cusp residence factor

P / partially_solved | AIM 研讨会问题汇编

What can one say about extreme value theory for (non-uniformly) hyperbolic flows on noncompact spaces?

来源字段：http://aimpl.org/equibdynsysgeom/1/

07079 | AIM-GEOMETRY-0070

### Entropy--SRB coincidence does not force smooth linearity

P / partially_solved | AIM 研讨会问题汇编

Let \(f:\mathbb T^2\to\mathbb T^2\) be a nonuniformly hyperbolic diffeomorphism. Let \(\mu\) be the unstable SRB measure, \(\nu\) the measure of maximal entropy, and \(m\) the measure of maximal dimension (make assumptions on \(f\) so that these exist and are unique). If \(\nu=\mu\) or \(m\), is \(f\) smoothly conjugate to a linear automorphism?

来源字段：http://aimpl.org/equibdynsysgeom/5/

07080 | AIM-GEOMETRY-0284

### Epstein zeta minimizers and a two-sided theta transfer principle

P / partially_solved | AIM 研讨会问题汇编

Question 0: For a fixed s, which L yields the minimum value of E(L, s )? For s → ∞ this becomes the familiar densest lattice packing problem. For n = 2 Rankin and Cassels showed that the hexagonal lattice is the unique minimizer for all s > 0. What we can show is that D4 (when n = 4), L = E8 (when n = 8) and the Leech lattice (when n = 24) are local optima for E(L, s ), for every s > 0.

来源字段：https://aimath.org/WWN/spherepacking/spherepacking.pdf

07081 | OWR-3389-016

### Equality Cases for Universal Eigenvalue Mean Inequalities

O / open | Oberwolfach 报告问题汇编

Let $E_1\leq E_2\leq\cdots$ be Dirichlet eigenvalues and, for $p>0$, define $$M_p(J)=\left(\frac{n+2p}{nJ}\sum_{j=1}^J E_j^p\right)^{1/p}.$$ Does there exist a domain $\Omega$ and an integer $J$ for which $$M_1(J)^2-M_2(J)^2\geq\frac{(E_{J+1}-E_J)^2}{4}$$ is attained with equality?

来源字段：https://doi.org/10.4171/owr/2009/06

07082 | OWR-14299905-034

### Equilateral Sets in Finite-$p$ Metric Tori

O / open | Oberwolfach 报告问题汇编

For each finite $p\geq1$, is there a constant $C_p$ such that the largest equilateral subset of the metric torus $T_p^n$ has size at most $C_p n$? If $2n+1$ is prime, is this maximum exactly $2n+1$?

来源字段：https://doi.org/10.4171/owr/2026/1

07083 | OWR-1535-001

### Equilibrium Supports on Solids of Revolution

P / partially_solved | Oberwolfach 报告问题汇编

Let $\Gamma(A)$ be the set of revolution generated by a compact set $A$, and let $\mu_{s,\Gamma(A)}$ be its $s$-equilibrium measure for $0<s<1$. Is it true that, for every $0<s<1$, there exists a compact set $A$ for which the support of $\mu_{s,\Gamma(A)}$ is a proper subset of the outer boundary of $\Gamma(A)$?

来源字段：https://doi.org/10.4171/owr/2007/19

07084 | AMR-054-0076

### Equiprojective Polyhedra

P / partially_solved | AMR 问题清单汇编

Identify or construct all $k$-equiprojective polyhedra. A polyhedron $P$ is $k$-equiprojective if its orthogonal projection to a plane is a $k$-gon in every direction not parallel to a face of $P$. Thus a cube is 6-equiprojective.

背景链接：https://topp.openproblem.net/p76

07085 | OWR-7155451-010

### Equivalence of Weak Scalar-Curvature Notions for Continuous Metrics

O / open | Oberwolfach 报告问题汇编

Is the Ricci-flow-based $\beta$-weak notion of lower scalar-curvature bounds for $C^0$ metrics equivalent to Gromov's polyhedral formulation, locally or globally? For a metric near the origin in $\mathbb{R}^n$ of the form $g_{ij}(x)=\delta_{ij}+r^2G_{ij}(x/r)+O(|x|^{2+\eta})$, where $x^ix^jG_{ij}(x)=0$ on $S^{n-1}$, characterize explicitly when it has nonnegative scalar curvature at the origin in this weak sense.

来源字段：https://doi.org/10.4171/owr/2021/30

07086 | OWR-15439-002

### Equivariant Minimal Surfaces for Semisimple Representations

P / partially_solved | Oberwolfach 报告问题汇编

Let $\Sigma$ be a closed surface of genus at least $2$, let $\rho:\pi_1(\Sigma)\to G$ be a representation into a semisimple Lie group, and let $X=G/K$. Does there exist a smooth $\rho$-equivariant map $\widetilde\Sigma\to X$ whose image is a minimal surface?

来源字段：https://doi.org/10.4171/owr/2017/31

07087 | OWR-17135-023

### Equivariant Pancake Partitions of Spheres

O / open | Oberwolfach 报告问题汇编

Let $n>k\geq1$, let $p$ be an odd prime, and let $\mathbb Z/p$ act freely and orientation-preservingly on $S^n$. Suppose $c:\operatorname{CO}(S^n)\to S^n$ is a center map, $V$ is a real $k$-dimensional $\mathbb Z/p$-representation with $$e(V)\notin\operatorname{Index}_{\mathbb Z/p}(SO(n+1);\mathbb F_p),$$ and $f:S^n\to V$ is equivariant. For every $\varepsilon>0$ and all sufficiently large $i$, is there an invariant convex equal-volume partition of $S^n$ into $p^{i+1}$ pieces whose centers map to $0$ and whose pieces are $(k,\varepsilon)$-pancakes?

来源字段：https://doi.org/10.4171/owr/2019/39

07088 | OWR-13106-008

### Equivariant Yamabe Minimizers for Isometric Group Actions

P / partially_solved | Oberwolfach 报告问题汇编

Given $G\subseteq\operatorname{Isom}(M,g)$, does the $G$-invariant conformal class $[g]_G$ contain a $G$-invariant metric that minimizes the Yamabe functional among all metrics in $[g]_G$?

来源字段：https://doi.org/10.4171/owr/2014/36

07089 | OPG-2400

### Erdös-Szekeres conjecture

P / partially_solved | Open Problem Garden

Conjecture Every set of $2^{n-2} + 1$ points in the plane in general position contains a subset of $n$ points which form a convex $n$-gon.

背景链接：http://www.openproblemgarden.org/op/erdos_szekeres_conjecture

07090 | EP-100

### Erdős Problem #100

O / open | Erdős 问题

Let $A$ be a set of $n$ points in $\mathbb{R}^2$ such that all pairwise distances are at least $1$ and if two distinct distances differ then they differ by at least $1$. Is the diameter of $A$ $\gg n$?

来源字段：https://www.erdosproblems.com/100

07091 | EP-103

### Erdős Problem #103

O / open | Erdős 问题

Let $h(n)$ count the number of incongruent sets of $n$ points in $\mathbb{R}^2$ which minimise the diameter subject to the constraint that $d(x,y)\geq 1$ for all points $x\neq y$. Is it true that $h(n)\to \infty$?

来源字段：https://www.erdosproblems.com/103

07092 | EP-105

### Erdős Problem #105: Geometry

S / solved | Erdős 问题

Let $A,B\subset \mathbb{R}^2$ be disjoint sets of size $n$ and $n-3$ respectively, with not all of $A$ contained on a single line. Is there a line which contains at least two points from $A$ and no points from $B$?

来源字段：https://www.erdosproblems.com/105

07093 | EP-1069

### Erdős Problem #1069: Geometry

S / solved | Erdős 问题

Given any $n$ points in $\mathbb{R}^2$, the number of $k$-rich lines (lines which contain $\geq k$ of the points) is, provided $k\leq n^{1/2}$,\[\ll \frac{n^2}{k^3}.\]

来源字段：https://www.erdosproblems.com/1069

07094 | EP-106

### Erdős Problem #106: Geometry

S / solved | Erdős 问题

Draw $n$ squares inside the unit square with no common interior point. Let $f(n)$ be the maximum possible sum of the side-lengths of the squares. Is $f(k^2+1)=k$?

来源字段：https://www.erdosproblems.com/106

07095 | EP-107

### Erdős Problem #107: Geometry, Convex

O / open | Erdős 问题

别名：The Happy Ending Problem [GEO-035]; Happy Ending Problem [GEOM-020]

Let $f(n)$ be minimal such that any $f(n)$ points in $\mathbb{R}^2$, no three on a line, contain $n$ points which form the vertices of a convex $n$-gon. Prove that $f(n)=2^{n-2}+1$.

来源字段：https://www.erdosproblems.com/107

07096 | EP-1082

### Erdős Problem #1082: Geometry, Distances

O / open | Erdős 问题

Let $A\subset \mathbb{R}^2$ be a set of $n$ points with no three on a line. Does $A$ determine at least $\lfloor n/2\rfloor$ distinct distances? In fact, must there exist a single point from which there are at least $\lfloor n/2\rfloor$ distinct distances?

来源字段：https://www.erdosproblems.com/1082

07097 | EP-1090

### Erdős Problem #1090: Geometry, Ramsey Theory

S / solved | Erdős 问题

Let $k\geq 3$. Does there exist a finite set $A\subset \mathbb{R}^2$ such that, in any $2$-colouring of $A$, there exists a line which contains at least $k$ points from $A$, and all the points of $A$ on the line have the same colour?

来源字段：https://www.erdosproblems.com/1090

07098 | EP-1121

### Erdős Problem #1121: Geometry

S / solved | Erdős 问题

If $C_1,\ldots,C_n$ are circles in $\mathbb{R}^2$ with radii $r_1,\ldots,r_n$ such that no line disjoint from all the circles divides them into two non-empty sets then the circles can be covered by a circle of radius $r=\sum r_i$.

来源字段：https://www.erdosproblems.com/1121

07099 | EP-1124

### Erdős Problem #1124: Geometry

S / solved | Erdős 问题

Can a square and a circle of the same area be decomposed into a finite number of congruent parts?

来源字段：https://www.erdosproblems.com/1124

07100 | EP-1127

### Erdős Problem #1127: Geometry, Distances

S / solved | Erdős 问题

Can $\mathbb{R}^n$ be decomposed into countably many sets, such that within each set all the pairwise distances are distinct?

来源字段：https://www.erdosproblems.com/1127

07101 | EP-1129

### Erdős Problem #1129

S / solved | Erdős 问题

For $x_1,\ldots,x_n\in [-1,1]$ let\[l_k(x)=\frac{\prod_{i\neq k}(x-x_i)}{\prod_{i\neq k}(x_k-x_i)},\]which are such that $l_k(x_k)=1$ and $l_k(x_i)=0$ for $i\neq k$.

Describe which choice of $x_i$ minimise\[\Lambda(x_1,\ldots,x_n)=\max_{x\in [-1,1]} \sum_k \lvert l_k(x)\rvert.\]

来源字段：https://www.erdosproblems.com/1129

07102 | EP-1207

### Erdős Problem #1207: Geometry, Distances

O / open | Erdős 问题

Let $P_d(n)$ be such that in any set of $n$ points in $\mathbb{R}^d$ there exist at least $P_d(n)$ many points which do not contain an isosceles triangle. Estimate $P_d(n)$ - in particular, is it true that\[P_2(n)<n^{1-c}\]for some constant $c>0$?

来源字段：https://www.erdosproblems.com/1207

07103 | EP-1208

### Erdős Problem #1208: Geometry, Distances

O / open | Erdős 问题

For $d\geq 2$ let $F_d(n)$ be minimal such that every set of $n$ points in $\mathbb{R}^d$ contains a set of $F_d(n)$ points with distinct distances. Estimate $F_d(n)$ for fixed $d$ as $n\to \infty$.

来源字段：https://www.erdosproblems.com/1208

07104 | EP-135

### Erdős Problem #135: Must $A$ determine $\gg n^2$ many distances?

S / solved | Erdős 问题

Let $A\subset \mathbb{R}^2$ be a set of $n$ points such that any subset of size $4$ determines at least $5$ distinct distances. Must $A$ determine $\gg n^2$ many distances?

来源字段：https://www.erdosproblems.com/135

07105 | EP-189

### Erdős Problem #189: Geometry, Ramsey Theory

S / solved | Erdős 问题

If $\mathbb{R}^2$ is finitely coloured then must there exist some colour class which contains the vertices of a rectangle of every area?

来源字段：https://www.erdosproblems.com/189

07106 | EP-209

### Erdős Problem #209: Geometry

S / solved | Erdős 问题

Let $A$ be a finite collection of $d\geq 4$ non-parallel lines in $\mathbb{R}^2$ such that there are no points where at least four lines from $A$ meet. Must there exist a 'Gallai triangle' (or 'ordinary triangle'): three lines from $A$ which intersect in three points, and each of these intersection points only intersects two lines from $A$?

来源字段：https://www.erdosproblems.com/209

07107 | EP-210

### Erdős Problem #210: Geometry

S / solved | Erdős 问题

Let $f(n)$ be minimal such that the following holds. For any $n$ points in $\mathbb{R}^2$, not all on a line, there must be at least $f(n)$ many lines which contain exactly 2 points (called 'ordinary lines'). Does $f(n)\to \infty$? How fast?

来源字段：https://www.erdosproblems.com/210

07108 | EP-211

### Erdős Problem #211: Geometry

S / solved | Erdős 问题

Let $1\leq k<n$. Given $n$ points in $\mathbb{R}^2$, at most $n-k$ on any line, there are $\gg kn$ many lines which contain at least two points.

来源字段：https://www.erdosproblems.com/211

07109 | EP-214

### Erdős Problem #214: Must the complement of $S$ contain four points which form a unit square?

S / solved | Erdős 问题

Let $S\subset \mathbb{R}^2$ be such that no two points in $S$ are distance $1$ apart. Must the complement of $S$ contain four points which form a unit square?

来源字段：https://www.erdosproblems.com/214

07110 | EP-215

### Erdős Problem #215: Geometry

S / solved | Erdős 问题

Does there exist $S\subseteq \mathbb{R}^2$ such that every set congruent to $S$ (that is, $S$ after some translation and rotation) contains exactly one point from $\mathbb{Z}^2$?

来源字段：https://www.erdosproblems.com/215

07111 | EP-216

### Erdős Problem #216: e. with no point in the interior). Does $g(k)$ exist? If so, estimate $g(k)$

S / solved | Erdős 问题

Let $g(k)$ be the smallest integer (if any such exists) such that any $g(k)$ points in $\mathbb{R}^2$ contains an empty convex $k$-gon (i.e. with no point in the interior). Does $g(k)$ exist? If so, estimate $g(k)$.

来源字段：https://www.erdosproblems.com/216

07112 | EP-223

### Erdős Problem #223: Geometry, Distances

S / solved | Erdős 问题

Let $d\geq 2$ and $n\geq 2$. Let $f_d(n)$ be maximal such that there exists some set of $n$ points $A\subseteq \mathbb{R}^d$, with diameter $1$, in which the distance 1 occurs between $f_d(n)$ many pairs of points in $A$. Estimate $f_d(n)$.

来源字段：https://www.erdosproblems.com/223

07113 | EP-224

### Erdős Problem #224: Geometry

S / solved | Erdős 问题

If $A\subseteq \mathbb{R}^d$ is any set of $2^d+1$ points then some three points in $A$ determine an obtuse angle.

来源字段：https://www.erdosproblems.com/224

07114 | EP-232

### Erdős Problem #232: Geometry, Distances

S / solved | Erdős 问题

For $A\subset \mathbb{R}^2$ we define the upper density as\[\overline{\delta}(A)=\limsup_{R\to \infty}\frac{\lambda(A \cap B_R)}{\lambda(B_R)},\]where $\lambda$ is the Lebesgue measure and $B_R$ is the ball of radius $R$.

Estimate\[m_1=\sup \overline{\delta}(A),\]where $A$ ranges over all measurable subsets of $\mathbb{R}^2$ without two points distance $1$ apart. In particular, is $m_1\leq 1/4$?

来源字段：https://www.erdosproblems.com/232

07115 | EP-353

### Erdős Problem #353: Geometry

S / solved | Erdős 问题

Let $A\subseteq \mathbb{R}^2$ be a measurable set with infinite measure. Must $A$ contain the vertices of an isosceles trapezoid of area $1$? What about an isosceles triangle, or a right-angled triangle, or a cyclic quadrilateral, or a convex polygon with congruent sides?

来源字段：https://www.erdosproblems.com/353

07116 | EP-502

### Erdős Problem #502: Geometry, Distances

S / solved | Erdős 问题

What is the size of the largest $A\subseteq \mathbb{R}^n$ such that there are only two distinct distances between elements of $A$? That is,\[\# \{ \lvert x-y\rvert: x\neq y\in A\} = 2.\]

来源字段：https://www.erdosproblems.com/502

07117 | EP-504

### Erdős Problem #504: Geometry

S / solved | Erdős 问题

Let $\alpha_n$ be the supremum of all $0\leq \alpha\leq \pi$ such that in every set $A\subset \mathbb{R}^2$ of size $n$ there exist three distinct points $x,y,z\in A$ such that the angle determined by $xyz$ is at least $\alpha$. Determine $\alpha_n$.

来源字段：https://www.erdosproblems.com/504

07118 | EP-505

### Erdős Problem #505: Geometry

S / solved | Erdős 问题

别名：Borsuk's Conjecture [GEO-029]; Bing-Borsuk conjecture [AMR-112-0001]

Is every set of diameter $1$ in $\mathbb{R}^n$ the union of at most $n+1$ sets of diameter $<1$?

来源字段：https://www.erdosproblems.com/505

07119 | EP-506

### Erdős Problem #506: Geometry

O / open | Erdős 问题

What is the minimum number of circles determined by any $n$ points in $\mathbb{R}^2$, not all on a circle?

来源字段：https://www.erdosproblems.com/506

07120 | EP-507

### Erdős Problem #507

P / partially_solved | Erdős 问题

别名：The Heilbronn Triangle Problem [GEO-024]; Heilbronn Triangle Problem [GEOM-021]

Let $\alpha(n)$ be such that every set of $n$ points in the unit disk contains three points which determine a triangle of area at most $\alpha(n)$. Estimate $\alpha(n)$.

来源字段：https://www.erdosproblems.com/507

07121 | EP-526

### Erdős Problem #526: Probability, Geometry

S / solved | Erdős 问题

Let $a_n\geq 0$ with $a_n\to 0$ and $\sum a_n=\infty$. Find a necessary and sufficient condition on the $a_n$ such that, if we choose (independently and uniformly) random arcs on the unit circle of length $a_n$, then all the circle is covered with probability $1$.

来源字段：https://www.erdosproblems.com/526

07122 | EP-605

### Erdős Problem #605: Geometry, Distances

S / solved | Erdős 问题

Is there some function $f(n)\to \infty$ as $n\to\infty$ such that there exist $n$ distinct points on the surface of a two-dimensional sphere with at least $f(n)n$ many pairs of points whose distances are the same?

来源字段：https://www.erdosproblems.com/605

07123 | EP-606

### Erdős Problem #606: Geometry

S / solved | Erdős 问题

Given any $n$ distinct points in $\mathbb{R}^2$ let $f(n)$ count the number of distinct lines determined by these points. What are the possible values of $f(n)$?

来源字段：https://www.erdosproblems.com/606

07124 | EP-607

### Erdős Problem #607: Geometry

S / solved | Erdős 问题

For a set of $n$ points $P\subset \mathbb{R}^2$ let $\ell_1,\ldots,\ell_m$ be the lines determined by $P$, and let $A=\{\lvert \ell_1\cap P\rvert,\ldots,\lvert \ell_m\cap P\rvert\}$.

Let $F(n)$ count the number of possible sets $A$ that can be constructed this way. Is it true that\[F(n) \leq \exp(O(\sqrt{n}))?\]

来源字段：https://www.erdosproblems.com/607

07125 | EP-651

### Erdős Problem #651: Geometry, Convex

S / solved | Erdős 问题

Let $f_k(n)$ denote the smallest integer such that any $f_k(n)$ points in general position in $\mathbb{R}^k$ contain $n$ which determine a convex polyhedron. Is it true that\[f_k(n) > (1+c_k)^n\]for some constant $c_k>0$?

来源字段：https://www.erdosproblems.com/651

07126 | EP-652

### Erdős Problem #652

S / solved | Erdős 问题

Let $x_1,\ldots,x_n\in \mathbb{R}^2$ and let $R(x_i)=\#\{\lvert x_j-x_i\rvert: j\neq i\}$, where the points are ordered such that\[R(x_1)\leq \cdots \leq R(x_n).\]Let $\alpha_k$ be minimal such that, for all large enough $n$, there exists a set of $n$ points with $R(x_k)<\alpha_kn^{1/2}$. Is it true that $\alpha_k\to \infty$ as $k\to \infty$?

来源字段：https://www.erdosproblems.com/652

07127 | EP-653

### Erdős Problem #653

O / open | Erdős 问题

Let $x_1,\ldots,x_n\in \mathbb{R}^2$ and let $R(x_i)=\#\{\lvert x_j-x_i\rvert: j\neq i\}$, where the points are ordered such that\[R(x_1)\leq \cdots \leq R(x_n).\]Let $g(n)$ be the maximum number of distinct values the $R(x_i)$ can take. Is it true that $g(n) \geq (1-o(1))n$?

来源字段：https://www.erdosproblems.com/653

07128 | EP-655

### Erdős Problem #655

P / partially_solved | Erdős 问题

Let $x_1,\ldots,x_n\in \mathbb{R}^2$ be such that no circle whose centre is one of the $x_i$ contains three other points. Are there at least\[(1+c)\frac{n}{2}\]distinct distances determined between the $x_i$, for some constant $c>0$ and all $n$ sufficiently large?

来源字段：https://www.erdosproblems.com/655

07129 | EP-659

### Erdős Problem #659: Geometry, Distances

S / solved | Erdős 问题

Is there a set of $n$ points in $\mathbb{R}^2$ such that every subset of $4$ points determines at least $3$ distances, yet the total number of distinct distances is\[\ll \frac{n}{\sqrt{\log n}}?\]

来源字段：https://www.erdosproblems.com/659

07130 | EP-661

### Erdős Problem #661

O / open | Erdős 问题

Are there, for all large $n$, some points $x_1,\ldots,x_n,y_1,\ldots,y_n\in \mathbb{R}^2$ such that the number of distinct distances $d(x_i,y_j)$ is\[o\left(\frac{n}{\sqrt{\log n}}\right)?\]

来源字段：https://www.erdosproblems.com/661

07131 | EP-662

### Erdős Problem #662

P / partially_solved | Erdős 问题

Consider the triangular lattice with minimal distance between two points $1$. Denote by $f(t)$ the number of distances from any points $\leq t$. For example $f(1)=6$, $f(\sqrt{3})=12$, and $f(3)=18$.

Let $x_1,\ldots,x_n\in \mathbb{R}^2$ be such that $d(x_i,x_j)\geq 1$ for all $i\neq j$. Is it true that, provided $n$ is sufficiently large depending on $t$, the number of distances $d(x_i,x_j)\leq t$ is less than or equal to $f(t)$ with equality perhaps only for the triangular lattice?

In particular, is it true that the number of distances $\leq \sqrt{3}-\epsilon$ is less than $1$?

来源字段：https://www.erdosproblems.com/662

07132 | EP-669

### Erdős Problem #669

P / partially_solved | Erdős 问题

Let $F_k(n)$ be minimal such that for any $n$ points in $\mathbb{R}^2$ there exist at most $F_k(n)$ many distinct lines passing through at least $k$ of the points, and $f_k(n)$ similarly but with lines passing through exactly $k$ points.

Estimate $f_k(n)$ and $F_k(n)$ - in particular, determine $\lim F_k(n)/n^2$ and $\lim f_k(n)/n^2$.

来源字段：https://www.erdosproblems.com/669

07133 | EP-733

### Erdős Problem #733: Combinatorics, Geometry

S / solved | Erdős 问题

Call a sequence $1<X_1\leq\cdots X_m\leq n$ line-compatible if there is a set of $n$ points in $\mathbb{R}^2$ such that there are $m$ lines $\ell_1,\ldots,\ell_m$ containing at least two points, and the number of points on $\ell_i$ is exactly $X_i$.

Prove that there are at most\[\exp(O(n^{1/2}))\]many line-compatible sequences.

来源字段：https://www.erdosproblems.com/733

07134 | EP-735

### Erdős Problem #735: Geometry

S / solved | Erdős 问题

Given any $n$ points in $\mathbb{R}^2$ when can one give positive weights to the points such that the sum of the weights of the points along every line containing at least two points is the same?

来源字段：https://www.erdosproblems.com/735

07135 | EP-754

### Erdős Problem #754:

S / solved | Erdős 问题

Let $f(n)$ be maximal such that there exists a set $A$ of $n$ points in $\mathbb{R}^4$ in which every $x\in A$ has at least $f(n)$ points in $A$ equidistant from $x$.

Is it true that $f(n)\leq \frac{n}{2}+O(1)$?

来源字段：https://www.erdosproblems.com/754

07136 | EP-755

### Erdős Problem #755: Geometry

S / solved | Erdős 问题

The number of equilateral triangles of size $1$ formed by any set of $n$ points in $\mathbb{R}^6$ is at most $(\frac{1}{27}+o(1))n^3$.

来源字段：https://www.erdosproblems.com/755

07137 | EP-756

### Erdős Problem #756: Geometry, Distances

S / solved | Erdős 问题

Let $A\subset \mathbb{R}^2$ be a set of $n$ points. Can there be $\gg n$ many distinct distances each of which occurs for more than $n$ many pairs from $A$?

来源字段：https://www.erdosproblems.com/756

07138 | EP-798

### Erdős Problem #798: Geometry

S / solved | Erdős 问题

Let $t(n)$ be the minimum number of points in $\{1,\ldots,n\}^2$ such that the $\binom{t}{2}$ lines determined by these points cover all points in $\{1,\ldots,n\}^2$.

Estimate $t(n)$. In particular, is it true that $t(n)=o(n)$?

来源字段：https://www.erdosproblems.com/798

07139 | EP-831

### Erdős Problem #831

O / open | Erdős 问题

Let $h(n)$ be maximal such that in any $n$ points in $\mathbb{R}^2$ (with no three on a line and no four on a circle) there are at least $h(n)$ many circles of different radii passing through three points. Estimate $h(n)$.

来源字段：https://www.erdosproblems.com/831

07140 | EP-898

### Erdős Problem #898: Geometry

S / solved | Erdős 问题

If $A,B,C\in \mathbb{R}^2$ form a triangle and $P$ is a point in the interior then, if $N$ is where the perpendicular from $P$ to $AB$ meets the triangle, and similarly for $M$ and $L$,\[\overline{PA}+\overline{PB}+\overline{PC}\geq 2(\overline{PM}+\overline{PN}+\overline{PL}).\]

来源字段：https://www.erdosproblems.com/898

07141 | EP-93

### Erdős Problem #93: Geometry, Convex

S / solved | Erdős 问题

If $n$ distinct points in $\mathbb{R}^2$ form a convex polygon then they determine at least $\lfloor \frac{n}{2}\rfloor$ distinct distances.

来源字段：https://www.erdosproblems.com/93

07142 | EP-94

### Erdős Problem #94: Geometry, Convex

S / solved | Erdős 问题

Suppose $n$ points in $\mathbb{R}^2$ determine a convex polygon and the set of distances between them is $\{u_1,\ldots,u_t\}$. Suppose $u_i$ appears as the distance between $f(u_i)$ many pairs of points. Then\[\sum_i f(u_i)^2 \ll n^3.\]

来源字段：https://www.erdosproblems.com/94

07143 | EP-957

### Erdős Problem #957: Geometry, Distances

S / solved | Erdős 问题

Let $A\subset \mathbb{R}^2$ be a set of size $n$ and let $\{d_1<\ldots<d_k\}$ be the set of distinct distances determined by $A$. Let $f(d)$ be the number of times the distance $d$ is determined. Is it true that\[f(d_1)f(d_k) \leq (\tfrac{9}{8}+o(1))n^2?\]

来源字段：https://www.erdosproblems.com/957

07144 | EP-958

### Erdős Problem #958: Distances, Geometry

S / solved | Erdős 问题

Let $A\subset \mathbb{R}^2$ be a finite set of size $n$, and let $\{d_1,\ldots,d_k\}$ be the set of distances determined by $A$. Let $f(d)$ be the multiplicity of $d$, that is, the number of ordered pairs from $A$ of distance $d$ apart.

Is it true that $k=n-1$ and $\{f(d_i)\}=\{n-1,\ldots,1\}$ if and only if $A$ is a set of equidistant points on a line or a circle?

来源字段：https://www.erdosproblems.com/958

07145 | EP-95

### Erdős Problem #95: Geometry, Convex

S / solved | Erdős 问题

Let $x_1,\ldots,x_n\in\mathbb{R}^2$ determine the set of distances $\{u_1,\ldots,u_t\}$. Suppose $u_i$ appears as the distance between $f(u_i)$ many pairs of points. Then for all $\epsilon>0$\[\sum_i f(u_i)^2 \ll_\epsilon n^{3+\epsilon}.\]

来源字段：https://www.erdosproblems.com/95

07146 | EP-97

### Erdős Problem #97: Geometry, Distances

O / open | Erdős 问题

Does every convex polygon have a vertex with no other $4$ vertices equidistant from it?

来源字段：https://www.erdosproblems.com/97

07147 | EP-98

### Erdős Problem #98

O / open | Erdős 问题

Let $h(n)$ be such that any $n$ points in $\mathbb{R}^2$, with no three on a line and no four on a circle, determine at least $h(n)$ distinct distances. Does $h(n)/n\to \infty$?

来源字段：https://www.erdosproblems.com/98

07148 | EP-982

### Erdős Problem #982: Geometry, Convex

O / open | Erdős 问题

If $n$ distinct points in $\mathbb{R}^2$ form a convex polygon then some vertex has at least $\lfloor \frac{n}{2}\rfloor$ different distances to other vertices.

来源字段：https://www.erdosproblems.com/982

07149 | GREEN-043

### Erdős-Szekeres with Visibility

P / partially_solved | Ben Green 问题

Fix integers $k, \ell$. Given $n \geq n_0(k, \ell)$ points in $\mathbb{R}^2$, is there either a line containing $k$ of them, or $\ell$ of them that are mutually visible?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

07150 | GEO-033

### Erdős-Ulam Problem

P / partially_solved | 独立 / 未分配集合

Is there a dense set of points in the plane with all pairwise distances rational?

背景链接：https://doi.org/10.1007/s10711-025-01019-0

07151 | AMR-017-0002

### Eremenko-Gabrielov secant conjecture

P / partially_solved | AMR 问题清单汇编

Let $m,p\ge 2$, put $d=m+p-1$, and let $F(x)=(1,x,\ldots,x^d)$ be the rational normal curve. For $j=1,\ldots,mp$, let $X_j$ be the $p$-plane spanned by $F(x_{j,1}),\ldots,F(x_{j,p})$, where all $x_{j,k}$ are real and the $mp$ sets of parameters lie in pairwise disjoint intervals. Are all complex $m$-planes meeting every $X_j$ real?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-23/

07152 | AIM-GEOMETRY-0256

### Escape obstructions for a monotone round-to-Calabi-Croke path

P / partially_solved | AIM 研讨会问题汇编

**Question 8.7.1 (Babenko).** Does there exist, in the space of metrics with conical singularities, a (continuous or smooth) family of metrics \(g_t\), \(0\leq t\leq1\), such that \(g_0=g_{\mathrm{rd}}\), \(g_1=g_{\mathrm{CC}}\), and \(\sigma(g_t)\) is a decreasing function of \(t\)?

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07153 | AIM-GEOMETRY-0365

### Euler integrands, Schouten symmetric functions, and total Q-curvature

P / partially_solved | AIM 研讨会问题汇编

In even dimension \(n\), express the Chern--Gauss--Bonnet scalar as a multiple of \(\sigma_{n/2}(P)\) plus contractions containing the Weyl tensor, and use this to clarify the relation between critical \(Q\)-curvature and topology.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

07154 | AIM-GEOMETRY-0031

### Euler rigidity and handle-minimal cobordisms for positroid closure relations

P / partially_solved | AIM 研讨会问题汇编

For each positroid stratum of a Grassmannian, there is a Legendrian link whose augmentation variety is that stratum. Positroid strata are also ordered with respect to closure. Is this partial order recoverable on the Legendrian links in terms of topology, for instance by (decomposable) Lagrangian cobordisms? Could the difference in dimension be related to the minimal genus of the cobordisms?

来源字段：http://aimpl.org/clusterbraid/7/

07155 | AIM-GEOMETRY-0142

### Euler-characteristic boundary sieve for Hilbert schemes of Painleve surfaces

P / partially_solved | AIM 研讨会问题汇编

Hilbert schemes of points on \(K2\)-surfaces.

There are \(12\) known deformation classes of complete hyperkahler manifolds of real dimension four that arise as moduli spaces of meromorphic Higgs bundles. Lets call them \(K2\) surfaces. For each of these \(K2\) surfaces there is a sequence of Higgs bundle moduli space of dimension \(4n\) for each \(n\).

Is each of these \(4n\) dimensional hyperkahler manifolds diffeomorphic to (or a deformation of) the Hilbert scheme of \(n\)-points on the underlying \(K2\) surface?

**Recovered question.** For generic admissible parameters and each of the twelve two-dimensional tame or wild meromorphic Hitchin moduli spaces \(S\) listed by Boalch, is the corresponding higher-dimensional moduli space \(\mathcal H_n\) diffeomorphic to \(S^{[n]}=\operatorname{Hilb}^n(S)\), or related to it by a topology-preserving deformation?

来源字段：http://aimpl.org/spectralhiggs/1/

07156 | AIM-GEOMETRY-0309

### Euler-ideal kernel for the locally-free locus of a circle representation

P / partially_solved | AIM 研讨会问题汇编

The final line is the heading of the next section and is not part of Question 7.3. The hyphen in “representations” is a line-break artifact.

The preceding page supplies the intended context. A compact Lie group \(G\) acts linearly and Hamiltonianly on \(V=\mathbb C^n\). For central regular values \(\alpha\), the list first considers the usual Kirwan maps \[ \kappa_\alpha:H_G^*(V)\cong H^*(BG)\longrightarrow H^*(V/\!/_\alpha G). \] It then prints \[ X:=\bigl(V\setminus\{\text{points where \(G\) fails to act locally freely}\}\bigr)/G_{\mathbb C}. \] This printed definition is inconsistent with the very next display, \(H^*(X/G)=H_G^*(X)\): if \(X\) has already been divided by \(G_{\mathbb C}\), there is no such residual \(G\)-action in general. Moreover the map induced by inclusion is defined on the prequotient open set. The only coherent recovery is therefore to set \[ \widetilde X:=\{v\in V:G_v\text{ is finite}\}, \] the **locally-free locus**, and to recover the question as: \[ \boxed{\text{Is }i^*:H_G^*(V)\cong H^*(BG)\longrightarrow H_G^*(\widetilde X)\text{ surjective, and what is }\ker i^*?} \tag{Q} \] The slash notation in the source may then be interpreted as the quotient orbifold/stack \([\widetilde X/G]\). For a locally free compact action, \(H_G^*(\widetilde X;\mathbb Q)\) is the ordinary cohomology of the orbifold classifying space and agrees with the rational cohomology of the coarse orbit space under the standard finite-isotropy comparison. It is not the Chen--Ruan cohomology ring, which has additional twisted sectors.

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

07157 | AIM-GEOMETRY-0414

### Even-dimensional intersection theorem and conditional odd case for Lagrangian spheres in quadrics

P / partially_solved | AIM 研讨会问题汇编

**Question 1.18 (Polterovich).** For $n \geq 2$, let $$ Q^n = \{[z_0 : \cdots : z_{n+1}] \in \mathbb{CP}^{n+1} : z_0^2 + \cdots + z_{n+1}^2 = 0\} $$ be the smooth complex $n$-dimensional quadric hypersurface. Does $Q^n$ contain two disjoint Lagrangian spheres? (Conjecturally, $Q^n$ does not contain two disjoint Lagrangian spheres; this is immediate when $n$ is even.)

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

07158 | AIM-GEOMETRY-0416

### Even-dimensional knotted tori and two isotopy reductions

P / partially_solved | AIM 研讨会问题汇编

**Question 1.20.** (a) If \(L\subset T^*\Sigma_g\) is a Lagrangian that is homologous to the \(0\)-section, and \(g>1\), must \(L\) be Lagrangian isotopic to the \(0\)-section? (This is now known for \(g=0\) and \(g=1\) by the works of Richard Hind and Alex Ivrii respectively.) (b) Is there a Lagrangian \(T^n\subset\mathbb R^{2n}\) which is not Lagrangian isotopic to the Clifford torus when \(n>2\)? The answer is no for \(n=2\) by Ivrii. However current holomorphic curve techniques do not seem applicable to higher \(n\). We do not even know: (c) Are there any local Lagrangian knots in \(\mathbb R^{2n}\) for \(n>2\), i.e. Lagrangian submanifolds that are asymptotic to a Lagrangian \(n\)-plane but not Lagrangian isotopic to one? (Eliashberg and Polterovich showed that the answer is no when \(n=2\).)

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

07159 | OWR-14298007-012

### Even-Dimensional Stress-Space Reconstruction

P / partially_solved | Oberwolfach 报告问题汇编

Let $d=2k\geq 4$, and let $P$ be a simplicial $d$-polytope in its natural embedding with no missing faces of dimension at least $d-k+1$. Prove that the space of affine $k$-stresses determines $P$ up to affine equivalence.

来源字段：https://doi.org/10.4171/owr/2023/58

07160 | AIM-GEOMETRY-0063

### Exact cylinder pressure and explicit Hölder gap certificates

P / partially_solved | AIM 研讨会问题汇编

Let $S$ be a surface which has a flat cylinder of parallel periodic orbits. In Burns--Climenhaga--Fisher--Thompson (*Geom. Funct. Anal.* 28, 2018, MR3856792, on unique equilibrium states for geodesic flows in nonpositive curvature under the pressure-gap condition $P_{\mathrm{sing}}(\varphi) < P(\varphi)$), a criterion for the existence and uniqueness of equilibrium states includes a pressure gap, $P_{\operatorname{sing}}(\varphi)<P(\varphi)$. Can this be made explicit for the surface $S$?

来源字段：http://aimpl.org/equibdynsysgeom/4/

07161 | OWR-1740-002

### Exact Dimensions of Secant Cones

O / open | Oberwolfach 报告问题汇编

Let $A\subset\mathbb{N}^m$ have cardinality $n$ and lie on an affine hyperplane, let $f=(x^\alpha)_{\alpha\in A}$ be the associated monomial map, and let $C$ be the closure of its image. For a $k$-tuple $\ell=(\ell_1,\ldots,\ell_k)$ of affine-linear forms, set $C_i(\ell)=\{\alpha\in A:\ell_i(\alpha)<\ell_j(\alpha)\text{ for every }j\neq i\}$. The tropical method gives $\dim(kC)\geq\sum_{i=1}^k(1+\dim\operatorname{Aff}_{\mathbb{R}}C_i(\ell))$. Does maximizing the right-hand side over all such $\ell$ always give $\dim(kC)$?

来源字段：https://doi.org/10.4171/owr/2007/57

07162 | AIM-GEOMETRY-0003

### Exact directional energies and sharp Hessian comparison for boxes

P / partially_solved | AIM 研讨会问题汇编

For a bounded convex \(\Omega\subset\mathbb R^n\), let \[ -\Delta u=\lambda_1(\Omega)u,\qquad u|_{\partial\Omega}=0, \qquad u>0,\qquad \|u\|_{L^2(\Omega)}=1, \] and, for \(|\xi|=1\), define \[ P(\xi)=\int_{\partial\Omega}|\xi\cdot\nabla u|^2\,dS, \qquad Q(\xi)=\int_\Omega|\xi\cdot\nabla u|^2\,dx. \tag{1} \] The first question asks for a relation between these boundary and interior directional energies. The second is David Jerison's dimension-uniform Hessian conjecture: if \(p^*\) is the maximum point of \(u\), is there \(C_n\) such that, for every \(p\) with \(u(p)>u(p^*)/2\), \[ \frac1{C_n}\bigl(-\nabla^2\log u(p)\bigr) \preceq -\nabla^2\log u(p^*) \preceq C_n\bigl(-\nabla^2\log u(p)\bigr)? \tag{2} \] Here \(\preceq\) is Loewner order on symmetric matrices.

The source has three textual issues.

1. “Dirichet” is only a spelling error for “Dirichlet.” 2. After defining \(Q(\xi)\), it states \(Q(\Omega)\le c/\operatorname{inradius}(\Omega)^2\), but \(Q(\Omega)\) is never defined. The safe repair is either the pointwise assertion \[ Q(\xi)\le \frac{c_n}{r(\Omega)^2}\quad(|\xi|=1), \tag{3} \] or the same assertion after defining \(Q(\Omega):=\sup_{|\xi|=1}Q(\xi)\). Both follow immediately from \[ Q(\xi)\le\int_\Omega|\nabla u|^2=\lambda_1(\Omega) \le\lambda_1(B_{r(\Omega)}) =\frac{j_{n/2-1,1}^2}{r(\Omega)^2}. \tag{4} \] The middle inequality is Dirichlet domain monotonicity because an inball lies in \(\Omega\). I do not claim to know which of the two repairs the author intended. 3. The sentence that \(Q\) is “the projection body function for volume” is terminologically unclear. Extended off the unit sphere, \(Q\) is a quadratic form in \(\xi\), whereas the classical projection-body support function is degree-one homogeneous. No argument below relies on this sentence.

来源字段：http://aimpl.org/symmetrybreaking/1/

07163 | OWR-7155448-003

### Exact Flat Choir Numbers and Dimension-Sixteen Tori

O / open | Oberwolfach 报告问题汇编

What is the exact flat choir number $b_n$ in each dimension $n$, and in particular what is the largest collection of mutually isospectral, nonisometric flat tori in dimension $16$?

来源字段：https://doi.org/10.4171/owr/2021/27

07164 | AIM-CONVEX_GEOMETRY-0024

### Exact half-volume graph theorem for symmetric planar rectangles

P / partially_solved | AIM 研讨会问题汇编

For a symmetric convex body $K$, let $S$ be a least area surface that divides the volume of $K$ into two halves (or any other fixed proportion). Is the surface a graph? (The surface in question may not be a hyperplane in general.)

来源字段：https://aimath.org/WWN/convexgeometry/convexgeometry.pdf

07165 | AIM-GEOMETRY-0323

### Exact membership of the origin in an ideal amoeba

S / solved | AIM 研讨会问题汇编

**A.3 Membership problems.** For every ideal $\mathfrak a$ in \[ R_d=\mathbb Z[x_1^{\pm1},\ldots,x_d^{\pm1}] \] there is a related dynamical system generated by $d$ commuting automorphisms of a compact abelian group via Pontryagin duality. For dynamics it is very important to determine when such systems have a finiteness condition called expansiveness. A theorem of Klaus Schmidt states in effect that when $\mathfrak a$ contains no nonzero integers, then the system is expansive if and only if the complex amoeba of $\mathfrak a$ does not contain the origin.

**Question.** Is there an algorithm to determine whether the complex amoeba of an ideal $\mathfrak a$ in $R_d$ contains the origin? (Contributed by Manfred Einsiedler and Doug Lind.)

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

07166 | AIM-GEOMETRY-0106

### Exact pair-overlap regime and an FCC-BCC crossing bracket for soft ball packings

P / partially_solved | AIM 研讨会问题汇编

**Phase transitions.** The best packing of hard spheres in \(\mathbb R^3\) is known to be achieved by the FCC lattice, while the thinnest covering is conjectured to be given by the BCC lattice. This implies that as \(\lambda\) is varied, for \(0<\lambda<\sqrt{5/3}-1\) we expect there to be an FCC--BCC transition.

Describe the behavior of optimal arrangements of soft balls as \(\lambda\) varies.

来源字段：http://aimpl.org/softpack/1/

07167 | AIM-GEOMETRY-0347

### Exact phase recognition for a complex hyperplane

P / partially_solved | AIM 研讨会问题汇编

**C.4 Recognition problems**

A. How to recognize whether a a pure d-complex is a tropical variety?

B. Characterize the image of linear subspaces w.r.t. under the phase map (“phlat”). i.e., characterize π(L) where L ⊂ C∗ is a linear subspace, and π is the phase map

π : C∗ → T n (T n: n-dimensional torus).

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

07168 | AIM-CONVEX_GEOMETRY-0015

### Exact radial gluing and symmetric star-body rigidity

P / partially_solved | AIM 研讨会问题汇编

**Rogers's theorem (1965).** Suppose that \(2\leq k\leq n-1\) and that \(K,L\) are compact convex sets in \(\mathbb R^n\) containing the origin in their relative interiors. If, for every \(S\in G(n,k)\), the corresponding sections \(K\cap S\) and \(L\cap S\) are homothetic (respectively, translates), then \(K,L\) are homothetic (respectively, translates).

**Problem.** Does the theorem remain true for star bodies, or perhaps more generally?

来源字段：https://aimath.org/WWN/mahlerduality/mahlerduality.pdf

07169 | AIM-GEOMETRY-0373

### Exact spin-tractor holonomy formula for the twistor-spinor dimension

S / solved | AIM 研讨会问题汇编

**Problem 15.** Relate \(t(g)\) to the holonomy of conformal Cartan connections.

For a connected, oriented conformal spin manifold \((M^n,c,\mathfrak s)\), \(n\geq3\), express the complex dimension of the global twistor-spinor space in terms of the holonomy of the normal conformal Cartan connection and its spin lift. Distinguish full from restricted holonomy and local/universal-cover from global spinors.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

07170 | OWR-11136-027

### Exact Transversals for Families with the (p,q)-Property

O / open | Oberwolfach 报告问题汇编

For a family of hyperplanes in $\mathbb{R}^d$ with the $(p,q)$-property, $d+1\leq q\leq p$, must there be a $(p-q+1)$-transversal; and does the analogous statement hold for families of $r$-element sets with $r+1\leq q\leq p$?

来源字段：https://doi.org/10.4171/owr/2011/44

07171 | AIM-GEOMETRY-0426

### Exactification and the marked/unmarked cotangent boundary

P / partially_solved | AIM 研讨会问题汇编

**Question 1.30 (Eliashberg).** Do symplectic and contact invariants of the cotangent bundle of a smooth manifold \(M\) remember the topology of \(M\)? For example, if \(M_1\) and \(M_2\) are homeomorphic but not diffeomorphic (e.g. exotic spheres), can one detect this by showing that \(T^*M_1\) and \(T^*M_2\) are not symplectomorphic, or that \(ST^*M_1\) and \(ST^*M_2\) are not contactomorphic? Related question: is the natural map from knots in \(S^3\) to Legendrian tori in \(ST^*S^3\cong S^3\times S^2\) injective? Ng's work is the first result in this direction.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

07172 | AIM-GEOMETRY-0333

### Exactness and topology in the higher-genus adiabatic limit

P / partially_solved | AIM 研讨会问题汇编

**A.13 Moduli space of holomorphic polygons.** Background: In the paper with Fukaya "Zero loop open strings in the cotangent bundle and Morse homotopy", Asian J. Math. 1 (1997), 96--180, we proved that "The moduli space of holomorphic polygons with boundary lying on k-tuples of La-grangian graphs of k-Morse functions is diffeomorphic to that of graph flows of the Morse functions in the adiabatic limit or (in the large complex structure limit). The projections, near the limit, of the holomorphic polygons on the base of the cotangent bundle resembles amoeba-type shapes and it shrinks to the graphs of Morse flows in the limit." In the paper, we dealt with the case of discs, i.e., open Riemann surfaces of genus zero. **Problem:** Study the similar degeneration problem for the higher genus case.

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

07173 | OWR-15575-010

### Exceptional Rational Directions of Zero-Entropy Finsler Tori

O / open | Oberwolfach 报告问题汇编

For a zero-topological-entropy Finsler metric on $T^2$, let $E_\xi$ be the set of nonminimal geodesics having rational asymptotic direction $\xi$. Does some rational $\xi$ satisfy $E_\xi=\varnothing$? Is $\{\xi\in S^1:\xi\text{ is rational and }E_\xi\neq\varnothing\}$ finite?

来源字段：https://doi.org/10.4171/owr/2017/32

07174 | AIM-GEOMETRY-0281

### Existence and forced structure of the 8- and 24-dimensional magic functions

S / solved | AIM 研讨会问题汇编

Question 2. Settle the Cohn-Elkies conjecture: that there exist 'optimal functions' for use in their theorem in these dimensions. (If yes, the answer to

Question 2. Settle the Cohn-Elkies conjecture: that there exist ‘optimal functions’ for use in their theorem in these dimensions. (If yes, the answer to Question 1 is ‘yes’.)

On page 2 it gives the complete text:

Here “these dimensions” refers to dimensions 8 and 24 in the immediately preceding workshop text. This reconstruction is source-verified, not an inferred completion. AIM-GEOMETRY-0281 owns Question 2 only; the adjacent Questions 1 and 3 are separate records.

来源字段：https://aimath.org/WWN/spherepacking/spherepacking.pdf

07175 | OWR-3389-010

### Existence and Smoothness of Dirichlet Eigenvalue Minimizers

O / open | Oberwolfach 报告问题汇编

Let $M$ be a smooth complete two-dimensional manifold. Does there exist a smooth domain minimizing the first Dirichlet eigenvalue among domains of the same area? Is every local minimizer, for the symmetric-difference metric on fixed-area plane domains, necessarily smooth?

来源字段：https://doi.org/10.4171/owr/2009/06

07176 | OWR-1060-010

### Existence and Uniqueness of Extremal Kähler Metrics

P / partially_solved | Oberwolfach 报告问题汇编

Given a Kähler class $\Omega$ on $M$, does it contain an extremal Kähler metric, and if so, is that representative unique?

来源字段：https://doi.org/10.4171/owr/2005/39

07177 | AMR-058-0015

### Existence of Equal-Volume Least-Area Partitions

P / partially_solved | AMR 问题清单汇编

Do least-area partitions of $\mathbb{R}^n$ into regions of unit volume exist? Give the correct definition and determine their regularity.

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

07178 | OWR-4412-008

### Existence of Perfect Billiard Retroreflectors

O / open | Oberwolfach 报告问题汇编

Do billiard retroreflectors exist, meaning bodies that reverse the direction of every incident particle or ray?

来源字段：https://doi.org/10.4171/owr/2010/31

07179 | AIM-GEOMETRY-0052

### Exit measures, Gibbs densities, and rank obstructions

P / partially_solved | AIM 研讨会问题汇编

Let $\tilde{M}$ be contractible negatively curved Riemannian manifold, and $\Gamma \subset \operatorname{Isom}(\tilde{M})$ be a discrete group of isometries which acts cocompactly.

1. Consider a nearest-neighbor random walk on $\Gamma$, which produces an exit measure on the boundary $\partial\Gamma$. Is this measure a Gibbs state? 2. If $\tilde{M}$ is nonpositively curved, what can be said about the measures induced on the boundary by Brownian motion? Is it a Gibbs state? What else can be said about it? (In the negative curvature case, this is known by //////)

来源字段：http://aimpl.org/equibdynsysgeom/2/

07180 | OWR-14299092-002

### Expander Degree Under Boundary Connected Sums

O / open | Oberwolfach 报告问题汇编

How are $\deg_{\mathrm{exp}}(X_1)$, $\deg_{\mathrm{exp}}(X_2)$, and $\deg_{\mathrm{exp}}(X_1\mathbin{\#_\partial}X_2)$ related? In particular, what is $\deg_{\mathrm{exp}}(D^3\times S^1)$?

来源字段：https://doi.org/10.4171/owr/2025/11

07181 | OWR-16160-006

### Expanders Avoiding All CAT-Zero Spaces

O / open | Oberwolfach 报告问题汇编

Does there exist an expander that does not coarsely embed into any CAT(0) space?

来源字段：https://doi.org/10.4171/owr/2018/19

07182 | OWR-15219-005

### Expected Degrees of Random Real Grassmannians

P / partially_solved | Oberwolfach 报告问题汇编

Is there a convex body naturally associated with the lines on a random cubic surface, analogous to the convex-geometric objects used to study expected degrees of random real Grassmannians?

来源字段：https://doi.org/10.4171/owr/2017/14

07183 | AMR-099-0004

### Explicit bound for symmetric point configurations on the sphere

O / open | AMR 问题清单汇编

Call a finite subset $X\subset S^2$ symmetric if a finite group acts transitively on $X$ by isometries. Determine an explicit universal upper bound for $|X|$ under the condition that $X$ is contained in neither a great circle nor the union of two parallel circles. In particular, is the $60$-point vertex set of the truncated icosahedron the largest such configuration?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/equilibrium.pdf

07184 | AIM-CONVEX_GEOMETRY-0010

### Explicit conditioned stability for relative central sections

P / partially_solved | AIM 研讨会问题汇编

**Problem 2 (stability of reconstruction from relative central sectional areas).** For each positive integer $t$, is there a positive integer $t'=t'(t)$ such that whenever $P,Q$ are isotropic centrally symmetric $d$-dimensional polytopes with at most $d^t$ facets and $$ \sup_{\theta\in S^{d-1}} \left| \frac{\operatorname{Vol}_{d-1}(P\cap\theta^\perp)}{\operatorname{Vol}_d(P)}- \frac{\operatorname{Vol}_{d-1}(Q\cap\theta^\perp)}{\operatorname{Vol}_d(Q)} \right|\le d^{-t'}, \tag{1.1} $$ the Hausdorff distance between $P$ and $Q$ is at most $1/10$? If so, can one reconstruct $P$, to Hausdorff error $1/10$, in polynomial time from an oracle returning these relative section areas to additive error $d^{-t'}$?

来源字段：https://aimath.org/pastworkshops/sectionsconvexproblems.pdf

07185 | AIM-GEOMETRY-0001

### Explicit Pólya bounds for thin non-tangential isosceles trapezoids

P / partially_solved | AIM 研讨会问题汇编

It asks about the scale-invariant Pólya functional $$ F(\Omega)=\frac{\lambda _1(\Omega)T(\Omega)}{|\Omega|}, \qquad T(\Omega)=\int_\Omega u_\Omega, $$ where $\lambda _1(\Omega)$ is the first Dirichlet eigenvalue and the torsion function is the weak solution of $$ -\Delta u_\Omega=1\quad\hbox{in }\Omega, \qquad u_\Omega=0\quad\hbox{on }\partial\Omega. $$ For bounded convex planar domains the conjectured sharp bounds are $$ \boxed{\frac{\pi^2}{24}\leq F(\Omega)\leq\frac{\pi^2}{12}}. \tag{1} $$ The lower endpoint is approached by triangles collapsing to an interval, and the upper endpoint by elongating rectangles.

There are two transcription issues in the supplied record. First, its displayed boundary condition is `\nu=0`; direct inspection of the AIM page shows that the intended condition is $u=0$. Second, the AIM heading says “$L^1$ norm of first eigenfunction,” while the body concerns the integral of the torsion function, not an eigenfunction.

来源字段：http://aimpl.org/symmetrybreaking/1/

07186 | AIM-GEOMETRY-0110

### Explicit saturated aperiodic locally jammed ball packings

P / partially_solved | AIM 研讨会问题汇编

**Aperiodic Jammed Packings** Find aperiodic jammed packings in $\mathbb{R}^2$. In $\mathbb{R}^3,$ find a saturated jammed maximal packing that is aperiodic in all directions.

Construct congruent-unit-ball packings with trivial translation stabilizer that are locally jammed; in dimension $3$, also require saturation, equivalently inclusion-maximality.

来源字段：http://aimpl.org/softpack/1/

07187 | OWR-13951-010

### Exponential Maps for Convex Body Geometric Means

O / open | Oberwolfach 报告问题汇编

Does there exist an exponential map $E:K_0^n\to K_0^n$ with $E(\{0\})=D$ and $E((A+B)/2)=G(E(A),E(B))$ for all convex bodies A and B?

来源字段：https://doi.org/10.4171/owr/2015/56

07188 | OWR-14742-004

### Exponential Multiplicity of Minimal Disks Spanning Helical Curves

O / open | Oberwolfach 报告问题汇编

Let $\Gamma_n$ be the simple closed curve in the boundary of a solid cylinder $B\times I$ consisting of two nonparallel diameters joined by two helical arcs winding approximately $n$ times. What embedded minimal disks does $\Gamma_n$ bound; in particular, does the number of such disks grow at least like $2^n$?

来源字段：https://doi.org/10.4171/owr/2016/34

07189 | OWR-1597-004

### Exponential-Map Asymptotics Near Algebraic Surface Singularities

O / open | Oberwolfach 报告问题汇编

For general real algebraic surfaces with isolated singularities, do the normal-form metrics yield analogues of the higher-order asymptotic and regularity results for the exponential map proved for the modeled quasihomogeneous singularities, after suitable resolution or blow-up?

来源字段：https://doi.org/10.4171/owr/2007/41

07190 | OWR-12697711-017

### Extendability of Operators Associated with Hook-Shaped Polynomials

O / open | Oberwolfach 报告问题汇编

If $p$ is hook-shaped and symmetrically hyperbolic, prove that its associated operator is extendable if and only if $p$ is weakly sum-of-squares hyperbolic.

来源字段：https://doi.org/10.4171/owr/2023/15

07191 | OWR-17135-036

### Extending Combinatorial Spheres on the Same Vertex Set

P / partially_solved | Oberwolfach 报告问题汇编

If $S$ is a combinatorial $d$-sphere on $n\geq d+3$ vertices, does there exist a combinatorial $(d+1)$-sphere $S'$ on the same vertex set such that $S$ is a subcomplex of $S'$?

来源字段：https://doi.org/10.4171/owr/2019/39

07192 | OPG-37341

### Extension complexity of (convex) polygons

P / partially_solved | Open Problem Garden

The extension complexity of a polytope $P$ is the minimum number $q$ for which there exists a polytope $Q$ with $q$ facets and an affine mapping $\pi$ with $\pi(Q) = P$.

Question Does there exists, for infinitely many integers $n$, a convex polygon on $n$ vertices whose extension complexity is $\Omega(n)$?

背景链接：http://www.openproblemgarden.org/op/extension_complexity_of_convex_polygons

07193 | AMR-055-0010

### Extension into a Complete Hermitian Ball

S / solved | AMR 问题清单汇编

Let $\Delta$ be a ball in $\mathbb{C}^n$, $n\geq2$, and let $F$ be a complete Hermitian manifold. Does every holomorphic map from $\Delta\setminus\{0\}$ to $F$ extend holomorphically over the origin?

背景链接：https://people.math.harvard.edu/~hirolee/pdfs/2014-fall-230a-icm1970-chern-differential-geometry.pdf

07194 | OWR-13857-006

### Extensions of CMC Gluing Constructions

O / open | Oberwolfach 报告问题汇编

For constant-mean-curvature hypersurfaces built by gluing spheres and Delaunay pieces along a balanced graph, can this gluing scheme be extended to more general ambient manifolds or to hypersurfaces of infinite topology? Are the resulting hypersurfaces nondegenerate, so that nearby graph deformations produce curves in a smooth moduli space?

来源字段：https://doi.org/10.4171/owr/2015/36

07195 | AIM-GEOMETRY-0006

## Exterior Robin optimization at the Steklov threshold

P / partially_solved | AIM 研讨会问题汇编

The AIM record asks, for a fixed volume and dimension \(n\geq 3\), whether the bottom of the Robin spectrum in the exterior of a smooth bounded obstacle has a global maximizer, and whether the fact that a ball is locally but not globally maximizing forces another critical obstacle by a mountain-pass argument.

Several corrections are needed before the question is mathematically unambiguous.

* The opening condition \(\Omega\subset\mathbb R^3\) conflicts with the exterior \(\mathbb R^n\setminus\overline\Omega\); the consistent reading is \(\Omega\subset\mathbb R^n\). * Put \(D=\Omega^{\rm ext}\) and let \(\nu_\Omega\) point outward from the obstacle and into \(D\). The quadratic form in the record, \[ q_{\alpha,\Omega}[u]=\int_D|\nabla u|^2\,dx+\alpha\int_{\partial\Omega}|u|^2\,dS, \qquad u\in H^1(D), \] produces \(\partial_{\nu_\Omega}u=\alpha u\). If \(\nu\) instead denotes the outward normal of the exterior domain, the sign is reversed. * The displayed ``minimum'' is an infimum. It is attained when the spectral bottom is a negative discrete eigenvalue, but in general the bottom is \(0\), the lower edge of the essential spectrum, and need not be an eigenvalue. * There is no scale-free dimensional constant \(\alpha_*(n)\). In \(n\geq3\), the threshold is the domain-dependent number \[ \alpha_*(D)=-\sigma_1(D), \] where \(\sigma_1(D)\) is the first exterior Steklov eigenvalue.

Thus the recovered problem is to maximize \[ \lambda_1^\alpha(D):=\inf_{0\ne u\in H^1(D)} \frac{q_{\alpha,\Omega}[u]}{\|u\|_{L^2(D)}^2} \tag{1} \] over a specified class of smooth bounded obstacles of prescribed volume, while distinguishing optimization of the *spectral bottom* from optimization restricted to obstacles for which that bottom is a negative eigenvalue.

For \(\alpha\geq0\), the quadratic form is nonnegative and the essential spectral edge is \(0\), so \(\lambda_1^\alpha(D)=0\) for every obstacle. Thus the global question is trivial for nonnegative parameters; the nontrivial and discrete-eigenvalue regimes have \(\alpha<0\).

来源字段：http://aimpl.org/symmetrybreaking/2/

07196 | OWR-13494-019

## Extremal Fat-Point Initial Degrees and Star Configurations

P / partially_solved | Oberwolfach 报告问题汇编

Let $Z$ be a finite set of points in $\mathbb P^n$. If $d=\alpha(nZ)=\alpha(Z)+n-1$, must $Z$ either lie in a hyperplane or be the star configuration from $d$ general hyperplanes, with every degree-$d$ hypersurface vanishing to order $n$ on $Z$ splitting into those hyperplanes? In $(\mathbb P^1)^n$, if $d=\alpha(nZ)=\alpha(Z)$, must $Z$ be a $d$ by $\cdots$ by $d$ grid?

来源字段：https://doi.org/10.4171/owr/2015/9

07197 | OWR-11136-018

## Extremal Geometric Graphs with Forbidden Configurations

O / open | Oberwolfach 报告问题汇编

Given a family $H$ of forbidden geometric configurations, what is the maximum number of edges in an $n$-vertex geometric graph containing no configuration from $H$?

来源字段：https://doi.org/10.4171/owr/2011/44

07198 | OWR-7155449-010

## Extremal Invariants of Bounded-Volume Hyperbolic Manifolds

O / open | Oberwolfach 报告问题汇编

For connected closed hyperbolic $n$-manifolds of volume at most $v$, determine or maximize the systole, kissing number, first positive Laplace eigenvalue, and multiplicity of that eigenvalue; in dimension two, volume is equivalent to fixing the genus.

来源字段：https://doi.org/10.4171/owr/2021/28

07199 | OWR-822-009

## Extremal Projections and Simplices in Hypercubes

P / partially_solved | Oberwolfach 报告问题汇编

For the unit $n$-cube $I^n$, determine the maximum and minimum $k$-dimensional volumes of its central sections and orthogonal projections, and determine the maximum volume $\gamma(n,k)$ of a $k$-simplex inscribed in it.

来源字段：https://doi.org/10.4171/owr/2005/17

07200 | AIM-CONVEX_GEOMETRY-0045

### Extremal slabs, spherical negative moments, and cross-polytope formulas

P / partially_solved | AIM 研讨会问题汇编

This record is the continuation of Problem 23 in the AIM workshop list *Fourier analytic methods in convex geometry*.

For \(t>0\) and \(a\in S^{n-1}\), the PDF defines

\[ \operatorname{Sl}(a,t)=\{x\in\mathbb R^n:|x|_\infty\leq 1, |\langle x,a\rangle|\leq t\}, \qquad S(a,t)=\{x:|x|_\infty\leq1,\ \langle x,a\rangle=t\}, \]

and writes \(V(a,t)=\operatorname{vol}_n\operatorname{Sl}(a,t)\) and \(A(a,t)=\operatorname{vol}_{n-1}S(a,t)\). It also sets

\[ f_m=\frac1{\sqrt m}(\underbrace{1,\ldots,1}_{m},0,\ldots,0). \]

1. Is the minimum of \(V(a,t)\), over \(a\in S^{n-1}\) for fixed \(t\), always attained at some \(f_m\)? 2. For \(t\leq2\sqrt2-2\), is the minimum attained at \(f_1\)? The source records the real result for \(t\leq3/4\), the complex result for \(t\leq4/5\), and asks whether the complex range extends to \(t_0\simeq0.867\). 3. If \(X_j\) are independent and uniform on \(S^{m-1}\subset\mathbb R^m\), determine the best constants in \[ c_p^{-1}|a|_2\leq \left\|\sum_{j=1}^n a_jX_j\right\|_{L_p}, \qquad -m+1<p<0. \] 4. Find an integral formula for symmetric slab volumes and noncentral hyperplane sections of \(B_1^n\), and find the extremal slab directions.

* The PDF writes the cube as \([-1,1]^n\) but immediately quotes \(1\leq A(a,0)\leq\sqrt2\). With ordinary Euclidean section measure the correct bounds for that cube are \(2^{n-1}\leq A(a,0)\leq2^{n-1}\sqrt2\). The displayed bounds are for the volume-one cube \([-\tfrac12,\tfrac12]^n\), or for section volume in \([-1,1]^n\) divided by \(2^{n-1}\). Extremal directions are unaffected by this common factor. * The OCR has \(t(t^2-1)\) in the equation for \(t_0\). This cannot have a root near \(0.867\). The geometrically correct equation, also consistent numerically with the value printed in the PDF, is \[ \pi t^2=t(t^2+1)\sqrt{2-t^2} +(2t^2-1)\arcsin\!\bigl(t\sqrt{2-t^2}\bigr). \tag{1} \] A bisection check gives \(t_0=0.8671146218586\ldots\). For \(0\leq t\leq1\), (1) is equivalently obtained by replacing the last arcsine by \(2\arcsin(t/\sqrt2)\).

来源字段：https://aimath.org/WWN/fourierconvex/fourierconvex.pdf

07201 | AMR-037-0013

### Extreme points

O / open | AMR 问题清单汇编

For fixed $d>3$, determine whether every point of an $n$-point set in $\mathbb R^d$ is a convex-hull vertex faster than the best known near-$n^{2\lfloor d/2\rfloor/(\lfloor d/2\rfloor+1)}$ algorithm, or prove a matching lower bound.

背景链接：https://jeffe.cs.illinois.edu/open/

07202 | OWR-14298010-003

### Extremizers of Bézout Mixed-Volume Invariants

P / partially_solved | Oberwolfach 报告问题汇编

Characterize the convex bodies $K\subset\mathbb R^n$ satisfying $b(K)=n$. Is the $n$-simplex the only minimizer of the invariant $b_2$ among convex bodies?

来源字段：https://doi.org/10.4171/owr/2023/51

07203 | OWR-1455-003

### Face Bounds for Minkowski Sums of Polytopes

S / solved | Oberwolfach 报告问题汇编

Let $P=P_1+\cdots+P_n$ be the Minkowski sum of convex $d$-polytopes, and let $v_i=f_0(P_i)$ be the number of vertices of $P_i$. Determine tight upper bounds for every face number $f_k(P)$, where $0\leq k\leq d-1$, in terms of $v_1,\ldots,v_n$.

来源字段：https://doi.org/10.4171/owr/2007/04

07204 | OWR-14298163-018

### Face-Size Distribution in Random Pseudoline Arrangements

O / open | Oberwolfach 报告问题汇编

For a uniformly random simple pseudoline arrangement on $n$ pseudolines and a uniformly random face, prove that the limiting probabilities $p_k$ of having $k$ sides exist and determine them, beginning with the triangle probability $p_3$.

来源字段：https://doi.org/10.4171/owr/2024/3

07205 | AMR-037-0004

### Faces of intricate polytopes

O / open | AMR 问题清单汇编

Determine the maximum total number of faces of a $d$-dimensional convex polytope with $n$ vertices and $n$ facets. In dimension four, do such fat-lattice polytopes have superlinear complexity? Are joins of polygons asymptotically optimal in dimensions at least six?

背景链接：https://jeffe.cs.illinois.edu/open/

07206 | AIM-GEOMETRY-0091

### Facet-vector bases from obtuse superbases

P / partially_solved | AIM 研讨会问题汇编

Does every lattice have a $\mathbb{Z}$-basis of Voronoi vectors? (I.e., vectors defining facets of Voronoi cells)

来源字段：http://aimpl.org/discreteaf/1/

07207 | OWR-1455-015

### Facial Structure of Intersections of Unit Balls

O / open | Oberwolfach 报告问题汇编

Let $P$ be the intersection of finitely many unit balls in $\mathbb{R}^d$. What can be said about the facial structure of $P$?

来源字段：https://doi.org/10.4171/owr/2007/04

07208 | OWR-2090-014

### Falconer Distance Conjecture

O / open | Oberwolfach 报告问题汇编

If $E\subseteq\mathbb{R}^d$ has Hausdorff dimension $\dim_H(E)>d/2$, must its distance set $\Delta(E)=\{|x-y|:x,y\in E\}$ have positive one-dimensional Lebesgue measure?

来源字段：https://doi.org/10.4171/owr/2008/44

07209 | OWR-15952-006

### Falconer Distance Conjecture in the Plane

P / partially_solved | Oberwolfach 报告问题汇编

If $K\subset\mathbb R^2$ is a Borel set with Hausdorff dimension greater than $1$, must its distance set $$D(K)=\{|x-y|:x,y\in K\}$$ have positive Lebesgue measure?

来源字段：https://doi.org/10.4171/owr/2017/47

07210 | GEO-011

### Falconer's Conjecture

O / open | 独立 / 未分配集合

If a compact set in $\mathbb{R}^d$ has Hausdorff dimension greater than $d/2$, must it determine a set of distances with positive Lebesgue measure?

背景链接：https://arxiv.org/abs/1808.09346

07211 | OWR-14298588-008

### Fargues-Scholze Categorical Local Langlands Equivalence

O / open | Oberwolfach 报告问题汇编

别名：Categorical Local Langlands Program Summary [OWR-14298588-006]; Equivalent Categorical Local Langlands Formulation [OWR-14298588-007]

Let $E/\mathbb{Q}_p$ be finite and let $G/E$ be quasisplit reductive. Construct the conjectural equivalence between $D_{\mathrm{lis}}(\operatorname{Bun}_G,\overline{\mathbb{Q}}_\ell)$ and the ind-coherent sheaves on the stack of $L$-parameters, normalized by a Whittaker datum, and prove that it intertwines Hecke operators with tensoring by the corresponding vector bundles.

来源字段：https://doi.org/10.4171/owr/2024/33

07212 | OPG-316

### Fat 4-polytopes

O / open | Open Problem Garden

The fatness of a 4-polytope $P$ is defined to be $(f_1 + f_2)/(f_0 + f_3)$ where $f_i$ is the number of faces of $P$ of dimension $i$.

Question Does there exist a fixed constant $c$ so that every convex 4-polytope has fatness at most $c$?

背景链接：http://www.openproblemgarden.org/op/fat_4_polytopes

07213 | OWR-1540-001

### Fatou Theory for Entire Minimal Graphs in $\mathbb{H}^2\times\mathbb{R}$

P / partially_solved | Oberwolfach 报告问题汇编

For an entire minimal graph in $\mathbb{H}^2\times\mathbb{R}$, classify its asymptotic values and establish appropriate Fatou-type boundary theorems. In particular, if the graphing function is bounded, do radial limits exist almost everywhere?

来源字段：https://doi.org/10.4171/owr/2007/24

07214 | AIM-CONVEX_GEOMETRY-0017

### Few-vertex Mahler bounds and pyramid cores

P / partially_solved | AIM 研讨会问题汇编

**Problem 21 (S. Reisner).** Improve the following theorem by finding a number \(N(n)>n+3\).

**Theorem (M. Meyer--S. Reisner).** Let \(K\) be a convex polytope in \({\mathbb R}^n\), with at most \(n+3\) vertices (or, alternatively, at most \(n+3\) facets) and nonempty interior. Then \[ {\cal P}(K)\geq \frac{(n+1)^{n+1}}{(n!)^2}, \] with equality if and only if \(K\) is an \(n\)-dimensional simplex.

来源字段：https://aimath.org/WWN/mahlerduality/mahlerduality.pdf

07215 | OWR-13680-004

### Fiber Genus of Rational Maps from Newton Hypersurfaces

O / open | Oberwolfach 报告问题汇编

Let $\Delta\subset\mathbb Z^n$ be a Newton polyhedron and $H$ a generic hypersurface with Newton polyhedron $\Delta$. For a dominant rational map $f:H\dashrightarrow\mathbb P^{n-2}$, determine the genus of the generic fiber $f^{-1}(x)$ for $x\in\mathbb P^{n-2}$.

来源字段：https://doi.org/10.4171/owr/2015/23

07216 | AIM-GEOMETRY-0192

### Fiber products and exact Betti calculations for partial 2-tree linkages

P / partially_solved | AIM 研讨会问题汇编

`5. (Sitharam) Consider two polygons that share a "chain" (see below), or graphs of tree-width 2. Question: Apply Morse Theory \` a la Farber to derive the Betti numbers: a) Chambers b) Homologies`

**5. (Sitharam)** Consider two polygons that share a “chain” (see below), or graphs of tree-width 2. **Question:** Apply Morse Theory à la Farber to derive the Betti numbers: a) Chambers b) Homologies.

来源字段：https://aimath.org/pastworkshops/linkagesproblems.pdf

07217 | AIM-GEOMETRY-0253

### Fiber-gap and bounded-period criteria for density of closed geodesics

P / partially_solved | AIM 研讨会问题汇编

Let \((M,g)\) be a connected closed smooth Riemannian manifold. Is the union of the images of all nonconstant closed \(g\)-geodesics dense in \(M\), for every \(g\)?

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07218 | OWR-7155451-009

### Fillability of Bartnik Boundary Data Under Scalar Curvature Bounds

P / partially_solved | Oberwolfach 报告问题汇编

Given a closed $(n-1)$-dimensional Riemannian manifold $(Y,h)$, a smooth function $\varphi$ on $Y$, and $\sigma$, find necessary and sufficient conditions for a complete $n$-dimensional manifold $(X,g)$ with boundary $Y$ and scalar curvature $R_g\geq\sigma$ to realize the Bartnik data $(g^\top,H_g)=(h,\varphi)$.

来源字段：https://doi.org/10.4171/owr/2021/30

07219 | AMR-071-0056

### Finding matching upper and lower bounds for k-sets and halving lines

P / partially_solved | AMR 问题清单汇编

Finding matching upper and lower bounds for k-sets and halving lines

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

07220 | OWR-14297732-004

### Finite Generation of Fundamental Groups Under Nonnegative Ricci Curvature

O / open | Oberwolfach 报告问题汇编

Let $(M^n,g)$ be a smooth complete Riemannian manifold with $\operatorname{Ric}_g\geq 0$. Must $\pi_1(M)$ be finitely generated when $n\in\{4,5,6\}$? In arbitrary dimension, must $\pi_1(M)$ be finitely generated if one additionally assumes, respectively, that $g$ is Ricci-flat, that $(M,g)$ is Kähler, or that the universal cover $(\widetilde M,\widetilde g)$ has Euclidean volume growth?

来源字段：https://doi.org/10.4171/owr/2023/29

07221 | AIM-GEOMETRY-0224

### Finite marked-length calibration for branched hyperbolic triangulations

P / partially_solved | AIM 研讨会问题汇编

Question 5.1.2 (Schmidt). Does the marked length spectrum determine such a metric (in the class of all locally CAT \((-1)\) metrics on this space, or in the class of all 2-dimensional (metrical) simplicial complexes with the above properties homeomorphic to the given complex)?

The antecedent “such a metric” occurs in the immediately preceding source paragraph (record AIM-GEOMETRY-0223). It means the intrinsic length metric on a two-dimensional metrical simplicial complex \(K\) satisfying:

1. every two-simplex is a hyperbolic geodesic triangle; 2. the metric link condition giving local CAT\((-1)\) holds at every vertex; and 3. every edge belongs to at least two two-simplices.

Thus the complex has no one-sided edge boundary, but it can branch along an edge contained in three or more triangles. The source does not explicitly repeat “finite” or “compact” in this paragraph. The intended marked-length setting, and all later rigidity results discussed below, concern finite/compact quotients; this extra hypothesis is stated whenever used here.

There are two comparison categories in Schmidt's question:

- all locally CAT\((-1)\) metrics on the fixed underlying topological space; or - all piecewise hyperbolic two-dimensional metrical simplicial complexes with the displayed properties and homeomorphic to the given complex.

The second version needs a **marking**. For metrics on the same \(K\), equality means equality as functions on conjugacy classes of \(\pi_1(K)\). For spaces \(K_0,K_1\), one must specify a homeomorphism \(f:K_0\to K_1\) (or an identified fundamental-group isomorphism) and require \[ \ell_{g_1}(f_*[\gamma])=\ell_{g_0}([\gamma]) \quad\text{for every conjugacy class }[\gamma]. \] An isometry proving marked rigidity should induce that marking, normally up to homotopy or isotopy. “Homeomorphic” without this datum is not enough to define equality of marked spectra.

The sentences beginning “The questions above are closely related to the boundary rigidity problem” are a transition to Question 5.1.3, not part of Schmidt's marked-length-spectrum question. The standard intended definition is that another **metric** \(g'\) with the same boundary-distance function must make \((M,g')\) isometric to \((M,g)\), through a diffeomorphism fixing the boundary pointwise. This correction is recorded only to disambiguate the transition; boundary rigidity is not the problem studied in this attempt.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07222 | AMR-058-0029

### Finite Total Scalar Curvature and Planarity

O / open | AMR 问题清单汇编

If an area-minimizing $k$-dimensional submanifold of $\mathbb{R}^n$ has finite total scalar curvature and $k>n/2$, must it be planar?

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

07223 | AIM-GEOMETRY-0160

### Finite traceless-Ricci budget and Ricci-flat bubbles under bounded scalar curvature

P / partially_solved | AIM 研讨会问题汇编

Let $(M^n,g(t))$, $0\leq t<T<\infty$, be the maximal smooth unnormalised Ricci flow on a closed manifold, \[ \partial_tg=-2\operatorname{Ric}(g(t)). \] Must \[ \limsup_{t\nearrow T}\max_M R(\cdot,t)=+\infty? \]

来源字段：http://aimpl.org/flowriemannian/1/

07224 | AIM-GEOMETRY-0104

### Finite, periodic, and spherical-code reductions for unequal-sphere average contacts

P / partially_solved | AIM 研讨会问题汇编

(G. Kuperberg - Schramm) How large can the average kissing number be for packing with spheres of varying radii? It is known that in $\mathbb{R}^3$ it is larger than $12$ and at most $24$. Is it true that it is strictly larger than the kissing number for dimension $4, 8,$ and $24$?

来源字段：http://aimpl.org/discreteaf/1/

07225 | OWR-12875-004

### Finite-Area Infinite-Genus Surfaces with Lattice Veech Groups

O / open | Oberwolfach 报告问题汇编

Does there exist an infinite-genus, finite-area translation surface $S$ whose Veech group is a lattice?

来源字段：https://doi.org/10.4171/owr/2014/15

07226 | AIM-CONVEX_GEOMETRY-0016

### Finite-dimensional local rigidity from central-section perimeters

P / partially_solved | AIM 研讨会问题汇编

**20. (R.J. Gardner) [G, Problem 7.6]** If \(K\) and \(L\) are origin-symmetric star bodies in \(\mathbb R^3\) whose sections by every plane through the origin have equal perimeters, is \(K=L\)? In particular, This was proved by R. Howard, F. Nazarov, D. Ryabogin and A. Zvavitch for bodies of revolution and by V. Yaskin for polytopes.

来源字段：https://aimath.org/WWN/mahlerduality/mahlerduality.pdf

07227 | OWR-17288-004

### Finite-Dimensional Uniform Embeddings of Compact Metric Spaces

O / open | Oberwolfach 报告问题汇编

A loose embedding of a metric space $(X,d)$ in $\mathbb R^N$ is a continuous map that preserves equality and inequality of pairwise distances. If every finite subspace of a compact metric space $(X,d)$ is loosely embeddable in the same $\mathbb R^N$, is $X$ itself loosely embeddable there? Does every compact Riemannian manifold with its geodesic distance have this property?

来源字段：https://doi.org/10.4171/owr/2019/45

07228 | OWR-16629-010

### Finite-Field Constructions of Complex Equiangular Tight Frames

P / partially_solved | Oberwolfach 报告问题汇编

Can difficult existence problems for complex equiangular tight frames, including the Fickus and Zauner conjectures, be solved over finite fields, and can finite-field solutions be lifted back to the complex setting?

来源字段：https://doi.org/10.4171/owr/2018/46

07229 | AIM-GEOMETRY-0252

### Finite-resolution density of periodic geodesic directions

P / partially_solved | AIM 研讨会问题汇编

**Question 8.4.3.** Is there a dense subset of metrics in $\mathcal G$ for which the tangent vectors to the closed geodesics are dense in $SM$?

It is relatively easy to construct, see e.g. [Weinstein1970], a metric on any manifold such that a certain open subset of $SM$ contains no vectors tangent to a closed geodesic.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07230 | AIM-GEOMETRY-0293

### Finite-sector additive nonabelian orbifold Kirwan surjectivity

P / partially_solved | AIM 研讨会问题汇编

**Question 4.3 (M. Pflaum)** Is there an analogous theorem for compact non-abelian Lie group \(G\)?

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

07231 | OWR-16415-009

### Finite-Sided Weil-Petersson Dirichlet Domains

O / open | Oberwolfach 报告问题汇编

Does the mapping class group action on the Weil-Petersson completion of Teichmüller space admit a finite-sided Dirichlet fundamental domain?

来源字段：https://doi.org/10.4171/owr/2018/40

07232 | AIM-GEOMETRY-0235

### Finite-time angular screening for the all-flat set

P / partially_solved | AIM 研讨会问题汇编

**Question 6.1.1.** Is the geodesic flow of a metric of nonpositive curvature on a closed surface of genus at least \(2\) ergodic with respect to the Liouville measure?

Matveev, *Open problems and questions about geodesics*, §6.1, Question 6.1.1. Its mathematical question is:

The source explains that the flow is ergodic on the invariant set of geodesics meeting the negative-curvature locus. Its complement consists of vectors \(v\) such that

\[ K(\gamma_v(t))=0\qquad\text{for every }t\in\mathbb R. \]

It asks whether this complementary set has Liouville measure zero. It also records the analogous higher-dimensional rank-one question, warns that several 1980s claims of a solution used an incorrect proof that the higher-rank set has measure zero, and distinguishes Liouville measure from Knieper's unique measure of maximal entropy.

Zero curvature geodesics and flat strips. Comparison with the original PDF shows that §6.2 starts there. It is not part of Question 6.1.1.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07233 | OWR-4128-001

### Finite-Topology Embedded Minimal Surfaces

O / open | Oberwolfach 报告问题汇编

For every nonnegative integer $g$, does there exist a unique non-planar complete embedded minimal surface in $\mathbb{R}^3$ with genus $g$ and one end; more generally, does a finite-topology surface of genus $g$ with $n\geq2$ ends embed minimally and completely in $\mathbb{R}^3$ exactly when $n\leq g+2$?

来源字段：https://doi.org/10.4171/owr/2009/45

07234 | OWR-16417-008

### Finite-Volume Symplectic Quotients of Hitchin Components

O / open | Oberwolfach 报告问题汇编

Is there a group of symplectomorphisms acting properly discontinuously on the Hitchin component with a finite-volume quotient?

来源字段：https://doi.org/10.4171/owr/2018/42

07235 | OWR-4535-013

### Finsler Realizations of the Bers Compactification

O / open | Oberwolfach 报告问题汇编

Does there exist a Finsler metric on Teichmüller space $T(S)$ whose horofunction compactification is the Bers compactification?

来源字段：https://doi.org/10.4171/owr/2010/53

07236 | AIM-GEOMETRY-0086

### First variation of the real-dimensional Cohn--Elkies radius problem

P / partially_solved | AIM 研讨会问题汇编

**Recovered problem.** For real \(d>0\), define \(A_-(d)\) as the infimum of the last-sign radius over normalized radial Schwartz profiles, using the dimension-\(d\) Fourier--Bessel transform. Determine \(A_-(2)\), study regularity of \(d\mapsto A_-(d)\), and prove or disprove the numerically conjectured derivatives at \(d=8\) and \(d=24\).

Its problem text is:

There is one obvious transcription error: “minimum or \(r\)” means “minimum of \(r\).” There is also a substantive compactness issue: an optimizer is not known to exist in the strict Schwartz class for every dimension. Accordingly the safe recovered statement replaces *minimum* by *infimum*:

Only the phrase “minimum or \(r\)” is silently repaired to “minimum of \(r\)” when quoting a recovered grammatical reading. The further change from minimum to infimum is explicitly declared and is not attributed to the source.

来源字段：http://aimpl.org/discreteaf/1/

07237 | AIM-GEOMETRY-0095

### Five-point Riesz phase-transition status and an explicit high-exponent competitor

P / partially_solved | AIM 研讨会问题汇编

Minimize energy for \(5\) points on \(S^2\) under \(f(r)=r^{-s}\), for the values of \(s\) where this problem is still open.

来源字段：http://aimpl.org/discreteaf/1/

07238 | OWR-14298808-003

### Fixed-Delta Convolution-Body Extremizers

O / open | Oberwolfach 报告问题汇编

Fix $\delta\in(0,1)$. Among convex bodies $K$ of a prescribed volume, characterize the maximizers of the volume $|C_\delta K|$ of the associated convolution body.

来源字段：https://doi.org/10.4171/owr/2024/58

07239 | AIM-GEOMETRY-0093

### Fixed-density small-Gaussian asymptotics and the dual Hermite boundary layer

P / partially_solved | AIM 研讨会问题汇编

Minimize energy for $f(r) = e^{-\alpha r^2}$ in $\mathbb{R}^n$ as $\alpha \rightarrow 0$

Fix density $\rho>0$. Minimize the lower Gaussian energy per particle among discrete configurations of density $\rho$, and determine both the minimum value and the structure of minimizers as $\alpha\downarrow0$.

来源字段：http://aimpl.org/discreteaf/1/

07240 | AIM-GEOMETRY-0411

### Fixed-end connectivity and Morse walls for Weinstein potentials

P / partially_solved | AIM 研讨会问题汇编

(a) Make a theory of “Weinstein spaces”, as opposed to Weinstein manifolds, by analogy with Stein manifolds and Stein spaces. (b) Is the space of psh functions on a Weinstein manifold connected? (No critical points at infinity.) What if we assume the same contact structure at infinity? There exists an example of a manifold diffeomorphic to \(\mathbb C^n\) for which there is more than 1 critical point for any psh function, and this bears on uniqueness.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

07241 | AIM-GEOMETRY-0076

### Fixed-length closing for subadditive orbit weights

P / partially_solved | AIM 研讨会问题汇编

Develop the theory for weighting orbits in a subadditive (rather than additive) way. <code>{\rm Ask Kiho for references}</code>

来源字段：http://aimpl.org/equibdynsysgeom/5/

07242 | AIM-GEOMETRY-0174

### Fixed-point curvature-operator blow-up under circle Cheeger deformation

S / solved | AIM 研讨会问题汇编

Let \(M_\epsilon=M\times_{S^1}(\epsilon S^1)\). As \(\epsilon\to0\), is Riem(Curvature Operator) bounded below?

来源字段：http://aimpl.org/flowriemannian/4/

07243 | AIM-GEOMETRY-0004

### Fixed-volume degeneration of Maxwell cavity eigenvalues

P / partially_solved | AIM 研讨会问题汇编

The AIM problem asks about the first three eigenvalues of the perfectly conducting electric cavity problem \[ \operatorname{curl}\operatorname{curl}E=\lambda E, \qquad \operatorname{div}E=0\quad\hbox{in }\Omega, \qquad E\times\nu=0\quad\hbox{on }\partial\Omega, \] and asks which shapes optimize \(\lambda_k(\Omega)\), for \(k=1,2,3\), under a volume constraint, either without or with convexity. It asks in particular whether the ball should be expected to be optimal.

There are three necessary repairs to the wording.

1. A nonempty domain is open and therefore is not compact. Here “smooth compact domain” is read as a bounded connected open set with compact closure and smooth boundary. 2. The last displayed constraint on the AIM page says “in \(\partial\Omega\)”; it must read “on \(\partial\Omega\).” 3. The displayed \(C^1\) Rayleigh quotient is only formal. Its natural completion is a subspace of \(H_0(\operatorname{curl};\Omega)\cap H(\operatorname{div};\Omega)\). Also, zero-frequency harmonic fields have to be removed or excluded by a topological hypothesis before the positive eigenvalues can be numbered unambiguously.

The strongest result below concerns the standard convex Lipschitz relaxation of the question. Cuboids are not smooth, so the distinction from the literal smooth class is recorded rather than hidden.

来源字段：http://aimpl.org/symmetrybreaking/2/

07244 | OWR-15208-019

### Flat Bands on Nonpositively Curved Surfaces

O / open | Oberwolfach 报告问题汇编

Let $(S,g)$ be a nonpositively curved surface of genus at least 2, and let $K$ be the set of rank-2 unit tangent vectors whose geodesics encounter only zero curvature. Must $K$ have measure zero, and must it consist of finitely many flat bands of closed geodesics?

来源字段：https://doi.org/10.4171/owr/2017/3

07245 | OWR-830-004

### Flat-Side Evolution Under Gauss Curvature Flow

O / open | Oberwolfach 报告问题汇编

Determine the final shape of the shrinking flat side in the Gauss-curvature flow of a weakly convex surface with flat sides.

来源字段：https://doi.org/10.4171/owr/2005/25

07246 | OWR-1319-021

### Flatness from Unique Parallel Geodesics on the Plane

O / open | Oberwolfach 报告问题汇编

If a complete metric $g$ on $\mathbb R^2$ has the unique-parallel-geodesic property, must $g$ be flat?

来源字段：https://doi.org/10.4171/owr/2006/33

07247 | OWR-14298808-009

### Floating-Area and Centro-Affine Entropy Inequalities

P / partially_solved | Oberwolfach 报告问题汇编

Prove the conjectured isoperimetric inequalities $\Omega_1(K)\leq\Omega_1(C_K)$ for floating area, $E_{\mathrm{PW}}(K)\geq E_{\mathrm{PW}}(B_K)$ for centro-affine entropy, and $E^s(K)\geq E^s(C_K)$ for the spherical analogue.

来源字段：https://doi.org/10.4171/owr/2024/58

07248 | OWR-14297732-007

### Flow Limits of Local Mass Invariants

P / partially_solved | Oberwolfach 报告问题汇编

How does $\lim_{r\to\infty}M_C^0(g,\Phi,\varphi_r,r)$ relate to Huisken's isoperimetric mass and Jauregui's isocapacitary mass?

来源字段：https://doi.org/10.4171/owr/2023/29

07249 | AIM-GEOMETRY-0287

### Flow-closed hyperkähler norm squares and an equivariantly perfect exhaustion criterion

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 1.1.** Let $X$ be a hyperkähler manifold with a hyper-Hamiltonian action of a compact Lie group $G$. Let $f$ denote the norm-square of the hyperkähler moment map. Suppose that for every $x\in X$, the forward trajectory of $x$ under the negative gradient flow of $f$ is contained in a compact subset of $X$. Then there is a surjection \[ H_K^*(X)\longrightarrow H_K^*(\mu_{HK}^{-1}(0)) \] in $K$-equivariant cohomology.

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

07250 | OWR-13857-005

### Flows for Weighted $\sigma_2$-Entropy

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Weighted-Curvature Extraction [OWR-13857-004]

For a compact Riemannian manifold $(M^n,g)$, is at least one of the following true: $\sup_{\Gamma_2^\infty}W_2$ is monotone along Ricci flow, or the negative gradient flow of $W_2$ is well posed within $\Gamma_2^\infty$?

来源字段：https://doi.org/10.4171/owr/2015/36

07251 | AIM-GEOMETRY-0187

### Flying wings refute uniqueness of the collapsed product soliton

S / solved | AIM 研讨会问题汇编

Is $\mathbb{R} \times \Sigma$, for $\Sigma$ the cigar soliton, the only collapsed $3$-dimension steady soliton?

来源字段：http://aimpl.org/flowriemannian/6/

07252 | AMR-071-0057

### For each arrangement of points in which the rectilinear crossing number is minimized, is the number of halving lines maximized

O / open | AMR 问题清单汇编

For each arrangement of points in which the rectilinear crossing number is minimized, is the number of halving lines maximized?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

07253 | AMR-037-0009

### Forced convex subsets

P / partially_solved | AMR 问题清单汇编

Determine the exact Erdős-Szekeres number $f(n)$, the least number of planar points in general position forcing a convex $n$-gon. Also determine sharp bounds for empty convex polygons; the source's former empty-hexagon subproblem has since been resolved.

背景链接：https://jeffe.cs.illinois.edu/open/

07254 | AIM-GEOMETRY-0012

### Forest-only mutation components and plabic-link HOMFLY

P / partially_solved | AIM 研讨会问题汇编

Is there an association between the mutation equivalence of plabic graphs or the mutation equivalence of quivers to the invariants of isotopy classes of their associated links? Can these be used to produce invariants of quivers up to mutation equivalence?

**Recovered reading.** Use the oriented smooth plabic link \(L_G^{\mathrm{plab}}\subset S^3\) and the mutable interior-face quiver \(Q_G\) of a simple plabic graph \(G\). Determine when an oriented smooth link invariant such as \[ P(L_G^{\mathrm{plab}};a,z) \] depends only on the Fomin--Zelevinsky mutation class of \(Q_G\).

来源字段：http://aimpl.org/clusterbraid/1/

07255 | AIM-GEOMETRY-0354

### Form Q-operators: recovered status and a two-gate cohomology descent criterion

P / partially_solved | AIM 研讨会问题汇编

The official HTML transcription and the source PDF resolve several important ambiguities.

The opening sentence, “Problem 2 have a role to play in the problem of characterising the \(Q\)-curvature and the GJMS operators,” is the tail of the preceding item and is not part of the present problem. The heading “A generalisation: maps like \(Q\)” begins an expository construction. The displayed item labelled Exercise 7 is genuinely an exercise. The statement about \(S^p\times S^q\) is explicitly labelled a fact, not a question. Only the final sentence is the open-ended prompt:

The official recompiled HTML labels this “Problem 35: Generalizations of \(Q\).” It is a research program rather than a single proposition with a yes-or-no answer.

Let \(M\) be compact, oriented, without boundary, not necessarily connected, and of even dimension \(n\). Write \[ \mathcal E^k=\Omega^k(M),\qquad \mathcal C^k=\ker(d:\Omega^k\longrightarrow\Omega^{k+1}). \] Weighted forms are denoted \(\mathcal E^k[w]\). Orientation and the conformal Hodge operator identify the appropriate dual weighted \(k\)-form bundle with \((n-k)\)-forms. The kernel and range of an operator are denoted \(\mathcal N(\cdot)\) and \(\mathcal R(\cdot)\).

The record first views ordinary \(Q\)-curvature as multiplication \[ Q^g:\mathcal C^0\longrightarrow\mathcal E^n, \] where \(\mathcal C^0\) consists of functions constant on each connected component. Its five listed properties are:

1. \(\widehat Q=Q+P_n\omega\) for \(\widehat g=e^{2\omega}g\), with \(P_n=dMd\) formally self-adjoint. 2. \(Q:\mathcal C^0\to H^n(M)\) is conformally invariant. 3. If \(c\in\mathcal C^0\) and \(u\in\ker P_n\), then \(\int_MuQc\) is conformally invariant. 4. Multiplication by \(Q\), viewed as \(\mathcal E^0\to\mathcal E[-n]\), is formally self-adjoint. 5. \(Q1\) is \(Q\)-curvature.

The middle-degree model in the clean source is \[ M^g=d\delta+2J-4\mathsf P^\sharp: \mathcal E^{n/2-1}\longrightarrow\mathcal E^{n/2-1}[-2], \tag{35} \] where \(\delta\) is the metric codifferential, \(\mathsf P\) is the Schouten tensor acting as an endomorphism of forms, and \(J=\operatorname{tr}\mathsf P\). The target is conformally identified with \(\mathcal E^{n/2+1}\). Exercise 7 asks for the transformation law on closed forms \[ M^{\widehat g}=M^g+\beta\,\delta d\,\omega , \] where \(\omega\) acts by multiplication and \(\beta\ne0\) depends on conventions. The induced detour operator \[ \delta M^g d:\mathcal E^{n/2-2}\longrightarrow \mathcal E^{n/2-2}[-4] \] is conformally invariant; in dimension four it is the Paneitz operator, up to normalization and sign conventions.

The general construction asserted in the record consists of operators \[ M_k^g:\mathcal E^k\longrightarrow\mathcal E^{n-k}, \qquad 0\le k\le n/2-1, \] with:

- on \(c\in\mathcal C^k\), \[ M_k^{\widehat g}c=M_k^gc+L_k^g(\omega c), \qquad L_k^g\ \text{formally self-adjoint}; \] - \(L_k^g\) is a nonzero constant multiple of \(dM_{k+1}^gd\), in the complementary-degree realization; - \(\mathcal H_g^k:=\ker(dM_k^g:\mathcal C^k\to\mathcal E^{n-k+1})\) is conformally invariant and \[ M_k:\mathcal H^k\longrightarrow H^{n-k}(M) \] is conformally invariant; - \(\int_M\langle u,M_k^gc\rangle\) is conformally invariant for \(c\) closed and \(u\in\ker L_k\); - \(M_k^g\) is formally self-adjoint; - the clean source says \(M_0^g1\) is \(Q\)-curvature. The OCR string “\(M^g>0\,1\)” is corrupt.

In the weighted-form presentation used in the literature, codifferentials occur on the left, for example \(L_k=\delta Q_{k+1}d\). After the conformal Hodge identification with complementary-degree true forms, the left \(\delta\) becomes, up to conventional sign, an exterior \(d\). These are two presentations of the same structure and must not be mixed in one formula.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

07256 | AIM-GEOMETRY-0128

### Formal Hamiltonian exponential and algebraic rigidity for the Hitchin section

P / partially_solved | AIM 研讨会问题汇编

Extend Hitchin section \(B\rightarrow\mathcal{M}_{H}\) to a map \(T^{\ast}B\rightarrow\mathcal{M}_{H}\).

Does there exist a formal version?

来源字段：http://aimpl.org/spectralhiggs/1/

07257 | AIM-GEOMETRY-0196

### Formation diameter after deleting one bar

P / partially_solved | AIM 研讨会问题汇编

**9. (St. John)** Consider a multi-robot formation that is a generically minimally rigid framework \(G\), with diameter \[ D=\max_{(p_i,p_j)}\|p_i-p_j\|. \] We remove an edge and obtain \(\bar G\) that is flexible. Let \[ d=\min \operatorname{diameter}(\text{configuration space of }\bar G). \] Question: Understand the relationship between \(D\) and \(d\), and find algorithms to detect what edge to delete for maximal change in diameter. More precisely, find find bar \(e\) and “positioning” \(q\) such that the diameter of \((G\setminus e,q)\) is minimum under the constraints that bar lengths in \(G\setminus e\) are maintained.

来源字段：https://aimath.org/pastworkshops/linkagesproblems.pdf

07258 | OWR-3389-012

### Fractal Optimizers in Spectral Isoperimetric Problems

O / open | Oberwolfach 报告问题汇编

Can an isoperimetric problem for eigenvalues have a fractal minimizer or other fractal solution?

来源字段：https://doi.org/10.4171/owr/2009/06

07259 | OWR-15208-020

### Frame Bundle Rigidity of Negatively Curved Manifolds

O / open | Oberwolfach 报告问题汇编

Let $M$ and $N$ be closed negatively curved manifolds with isomorphic fundamental groups. Must their $k$-frame bundles be isomorphic? This is known for $k=1,2$.

来源字段：https://doi.org/10.4171/owr/2017/3

07260 | AIM-GEOMETRY-0261

### Free Fubini--Study quotients refute the literal global-CROSS conclusion

S / solved | AIM 研讨会问题汇编

**Question 8.9.1.** Let \(M\) be a closed manifold not covered by a sphere. Let \(g\) be a metric on \(M\) for which all geodesics are closed. Is \((M,g)\) a CROSS (=compact rank one symmetric) manifold (with the standard metric)?

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07261 | AMR-054-0035

### Freeze-Tag: Optimal Strategies for Awakening a Swarm of Robots

S / solved | AMR 问题清单汇编

An optimization problem that naturally arises in the study of ``swarm robotics'' is to wake up a set of ``asleep'' robots, starting with only one ``awake'' robot. One robot can only awaken another when they are in the same location. As soon as a robot is awake, it may assist in waking up other robots. The goal is to compute an optimal awakening schedule such that all robots are awake by time $t^*$, for the smallest possible value of $t^*$ (the optimal makespan). The $n$ robots are initially at $n$ points of a metric space. The problem is equivalent to finding a spanning tree with maximum out-degree two that minimizes the radius from a fixed source. Is it NP-hard to determine an optimal awakening schedule for robots in the Euclidean (or $L_1$) plane? In more general metric spaces, can one obtain an approximation algorithm with better than $O(\log n)$ performance ratio?

背景链接：https://topp.openproblem.net/p35

07262 | AIM-GEOMETRY-0148

### From a measured foliation to rank-two Hitchin data

P / partially_solved | AIM 研讨会问题汇编

Given a measured foliation can we construct the Hitchin system?

来源字段：http://aimpl.org/spectralhiggs/1/

07263 | AIM-GEOMETRY-0330

### From tropical Grassmannians to Chow quotients

P / partially_solved | AIM 研讨会问题汇编

**A.10 The tropical Grassmannian**

The tropical Grassmannian, studied by Speyer and Sturmfels, turns out, at least in the cases they study, (2,n) and (3,6) to have strong combinatorial connections with the Kapranov's Chow quotient G(r,n)//(C*)^n.

Question (Eugene Tevelev and Sean Keel): Try to understand the precise relationship. (contributed by Sean Keel)

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

07264 | AMR-059-0006

### Fundamental Groups of Compact Affine Flat Manifolds

P / partially_solved | AMR 问题清单汇编

Determine the fundamental group of a compact affine flat manifold.

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

07265 | AIM-GEOMETRY-0170

### Gauge-corrected uniqueness for conical Yang-Mills shrinkers

P / partially_solved | AIM 研讨会问题汇编

Must two Yang-Mills self-shrinkers that are asymptotic to the conical connection be actually identical?

If two smooth solutions of (1) approach a fixed cone connection in specified asymptotic gauges and at a specified rate, must they be gauge equivalent? If a global gauge normalization is also imposed, must the normalized representatives coincide?

来源字段：http://aimpl.org/flowriemannian/3/

07266 | OWR-14299521-008

### Gauge-Theoretic and Calibrated-Submanifold Counting Invariants

O / open | Oberwolfach 报告问题汇编

Define counting invariants for gauge-theoretic objects and calibrated submanifolds in $G_2$ geometry, and prove an equality between the two invariants.

来源字段：https://doi.org/10.4171/owr/2025/30

07267 | AMR-055-0004

### Gauss-Bonnet Defect of a Complete Manifold

P / partially_solved | AMR 问题清单汇编

Let $M$ be a complete Riemannian manifold of even dimension. Suppose that its Euler characteristic $\chi(M)$ and the convergent Gauss--Bonnet curvature integral $C(M)$ both exist. Find a geometrical interpretation of the difference $\delta(M)=\chi(M)-C(M)$.

背景链接：https://people.math.harvard.edu/~hirolee/pdfs/2014-fall-230a-icm1970-chern-differential-geometry.pdf

07268 | OPG-59914

### General position subsets

P / partially_solved | Open Problem Garden

Question What is the least integer $f(n)$ such that every set of at least $f(n)$ points in the plane contains $n$ collinear points or a subset of $n$ points in general position (no three collinear)?

背景链接：http://www.openproblemgarden.org/op/general_position_subsets

07269 | AIM-GEOMETRY-0395

### General tractor Laplacians and their Einstein factorization

P / partially_solved | AIM 研讨会问题汇编

**Problem 33. Give general formulae or inductive formulae for the operators \(\Box_{2\ell}\).**

Its exact `problem` field begins

and then records the affine transformation law labelled (43).

The sentence beginning “In another direction” starts the discussion leading to Problem 34 and is not part of Problem 33.

The preceding source context is essential. On a conformal \(n\)-manifold the operators \(\Box_{2\ell}\) are conformally invariant, even-order, tractor-coupled analogues of the GJMS operators. They arise by applying powers of the ambient Laplacian to homogeneous ambient tensor fields corresponding to tractors. In even dimension, the particular operator \(\Box_{n-2}\) on the weighted standard tractor bundle has leading part \(\Delta^{n/2-1}\) and obeys \[ \Box_{n-2}I^g=(0,0,Q_n^g)^T, \qquad \Box_{n-2}D\omega=(0,0,P_n^g\omega)^T. \] Thus a formula for the full tractor operator gives a route to critical \(Q\)-curvature and the critical GJMS operator.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

07270 | AMR-054-0043

### General Unfoldings of Nonconvex Polyhedra

O / open | AMR 问题清单汇编

Can every closed polyhedron be cut along its surface and unfolded into one piece in the plane without overlap? Such an unfolding is called a general unfolding to distinguish from edge-unfoldings (see Problem 9) and vertex-unfoldings (see Problem 42).

背景链接：https://topp.openproblem.net/p43

07271 | OPG-59923

### Generalised Empty Hexagon Conjecture

P / partially_solved | Open Problem Garden

Conjecture For each $\ell\geq3$ there is an integer $f(\ell)$ such that every set of at least $f(\ell)$ points in the plane contains $\ell$ collinear points or an empty hexagon.

背景链接：http://www.openproblemgarden.org/op/generalised_empty_hexagon_conjecture

07272 | OWR-1383-003

### Generalized and Standard Almost-Nonnegative Curvature

O / open | Oberwolfach 报告问题汇编

Does every closed smooth manifold that is almost nonnegatively curved in the generalized sense also admit almost nonnegative curvature in the standard, strict $k=0$ sense? In the generalized sense, the defining collapsing sequence is allowed on $M\times\mathbb{R}^k$ for some $k\geq0$.

来源字段：https://doi.org/10.4171/owr/2006/45

07273 | OWR-1394-006

### Generalized Covariograms of Planar Convex Bodies

O / open | Oberwolfach 报告问题汇编

For a planar convex body $K$ and a strictly monotone, translation-invariant valuation $\varphi$, for example perimeter, does the generalized covariogram $x\mapsto\varphi(K\cap(K+x))$ determine $K$ up to translations and reflections?

来源字段：https://doi.org/10.4171/owr/2006/56

07274 | AIM-GEOMETRY-0401

### Generating families, augmentations, and the missing quantifiers

P / partially_solved | AIM 研讨会问题汇编

(a) Is Linearized Contact Homology always equal to Generating Function Homology [definition?] for a (compact?) Legendrian \(L\subset J^1(\mathbb R^n)\)? As evidence for this conjecture, it was proved in the case where \(L\) is the double of a compact \(n\)-manifold with boundary \(M\subset\mathbb R^n\).

(b) For a Legendrian \(L\subset\mathbb R^n\), does \(L\) admit a generating function [satisfying what restrictions?] if / only if the Linearized Contact Homology of \(L\) is (uniquely) defined?

The official AIM PDF uses the convention \[ J^1(M)=\mathbb R_u\times T^*M, \qquad \alpha=du-p\,dq, \] which is the usual \(T^*M\times\mathbb R_z\) with contact form \(dz-p\,dq\), up to the order and name of the coordinates. It prints Question 1.4 as follows (typographical line-break hyphens suppressed, but the tentative brackets preserved):

Literally, however, that phrase does not specify a contact structure or the appropriate Legendrian dimension. The surrounding definition and part (a) strongly suggest the intended ambient space was \(J^1(\mathbb R^n)\). All mathematical conclusions below explicitly state their ambient space.

The appended “Theme 1.5” begins on the next page and expressly says that it is not asking a question. It is neighboring workshop context, not part of Question 1.4.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

07275 | AMR-037-0019

### Generating random simple polygons

P / partially_solved | AMR 问题清单汇编

Given a planar point set $P$, sample uniformly from the simple polygons with vertex set $P$ in polynomial time, or determine the complexity of counting them. Determine the maximum possible number of such polygons and analogous bounds for triangulations, paths, and simple spanning trees.

背景链接：https://jeffe.cs.illinois.edu/open/

07276 | AIM-GEOMETRY-0194

### Generic boundary anchoring, arbitrary pin cuts, and an equilateral six-triangle singularity

P / partially_solved | AIM 研讨会问题汇编

**7. (Thorpe)** Consider a network of corner-sharing triangles in the plane, with holes of size 5, 6, 7, 8, 9 (and an average hole size of 6). If this is an infinite network, it is isostatic.

Now take a large finite piece of the framework. Experimentally, if we pin every other triangle boundary vertex (vertex of degree two) and run the pebble game, we get an isostatic network. Question: prove that this approach works in general and find other distributions of pins that also work.

a) generic

b) equilateral triangles

Question (generalization): What about the class of graphs that have no proper rigid subgraphs, but that have more than two bodies at a pin? The underlying body-pin graph is no longer 3-regular.

来源字段：https://aimath.org/pastworkshops/linkagesproblems.pdf

07277 | OWR-16629-003

### Generic Recovery from Group Orbits on Subspaces

O / open | Oberwolfach 报告问题汇编

Given a representation $\rho:G\to\operatorname{GL}(\mathbb F^d)$, characterize the subspaces $S\subseteq\mathbb F^d$ for which the orbit $\{\rho(g)z:g\in G\}$ determines almost every $z\in S$ up to the unavoidable scalar ambiguities.

来源字段：https://doi.org/10.4171/owr/2018/46

07278 | OWR-12869-004

### Generic Simplicity of Hyperbolic Dirichlet Tilings

O / open | Oberwolfach 报告问题汇编

Let $\Gamma<\operatorname{PO}(3,1)$ be discrete and torsion-free. For a generic base point $x\in\mathbb H^3$, is its Dirichlet tiling $D_x$ simple, with exactly three facets meeting along every edge and exactly four facets meeting at every vertex?

来源字段：https://doi.org/10.4171/owr/2014/09

07279 | OWR-2654829-012

### Generic Vanishing of Anti-Invariant Cohomology in Dimension Four

P / partially_solved | Oberwolfach 报告问题汇编

For a generic almost-complex structure $J$ on a compact $4$-manifold $M$, is the dimension $h_J^-$ of the space of closed $J$-anti-invariant $2$-forms zero? If $h_J^-\geq3$, must $J$ be integrable?

来源字段：https://doi.org/10.4171/owr/2020/33

07280 | AIM-GEOMETRY-0184

### Genuine non-Kähler shrinking solitons

S / solved | AIM 研讨会问题汇编

Are there any (genuine) examples of shrinkers which are not Kahler?

来源字段：http://aimpl.org/flowriemannian/6/

07281 | OWR-704-005

### Genus Bounds from Positive Closed Currents

P / partially_solved | Oberwolfach 报告问题汇编

If a positive closed current $T$ with potential $\varphi$ satisfies $T[\varphi] \geq \lambda[C]$ for a curve $C$ and $\lambda>0$, must the geometric genus of $C$ be less than $2$?

来源字段：https://doi.org/10.4171/owr/2004/21

07282 | OWR-14297735-007

### Genus Monotonicity and Klein-Bottle Willmore Minimizers

O / open | Oberwolfach 报告问题汇编

Is the minimum of the Willmore functional among orientable surfaces of fixed genus monotone in the genus? Also prove that $\widetilde\tau_{3,1}$ is the unique Willmore minimizer among all Klein bottles.

来源字段：https://doi.org/10.4171/owr/2023/32

07283 | OWR-13494-017

### Genus Thresholds for Curves of Fixed Gonality

O / open | Oberwolfach 报告问题汇编

Does there exist an affine linear function $f$ such that for every $g\geq f(\gamma)$ there is an algebraic curve over the rational numbers with genus $g$ and gonality $\gamma$?

来源字段：https://doi.org/10.4171/owr/2015/9

07284 | OWR-4078-008

### Genus-One Helicoid Classification and End Bounds

P / partially_solved | Oberwolfach 报告问题汇编

Is every complete embedded minimal surface of genus one with one end congruent to the genus-one helicoid? Does every complete embedded minimal surface of finite total curvature and genus $g$ have at most $g+2$ ends?

来源字段：https://doi.org/10.4171/owr/2009/35

07285 | OWR-2053-002

### Geodesic Expansions of Off-Diagonal Spectral Functions

P / partially_solved | Oberwolfach 报告问题汇编

Let $M$ be a compact Riemannian surface, let $N_{x,y}(\lambda)$ be its spectral function, and let $\Gamma_{x,y}$ be the geodesic segments from $x$ to $y$. Does there exist $p\geq1$ such that, for every $x\in M$ and almost every $y\in M$, the function $N_{x,y}(\lambda)/\sqrt{1+\lambda}$ has in the Besicovitch space $B^p$ the asymptotic expansion obtained by summing over $\gamma\in\Gamma_{x,y}$, with frequencies $l(\gamma)$, phases determined by the Morse indices $m(\gamma)$, and amplitudes determined by $l(\gamma)$ and the orthogonal Jacobi field $J$ satisfying $J(0)=0$ and $J'(0)=1$?

来源字段：https://doi.org/10.4171/owr/2008/42

07286 | OWR-1188-005

### Geodesic Zeta Functions and Commensurability

P / partially_solved | Oberwolfach 报告问题汇编

If $M$ and $M'$ are compact negatively curved Riemannian manifolds with equal geodesic zeta functions $\zeta_M(s)=\zeta_{M'}(s)$, must $M$ and $M'$ be commensurable?

来源字段：https://doi.org/10.4171/owr/2006/06

07287 | OWR-1726-007

### Geometric Applications of Secondary Rho Invariants

O / open | Oberwolfach 报告问题汇编

Can secondary $\rho$-invariants for Dirac operators in noncommutative or foliated settings yield new geometric applications?

来源字段：https://doi.org/10.4171/owr/2007/43

07288 | OWR-14297736-002

### Geometric Bounds for Magnetic-Laplacian Ground States

O / open | Oberwolfach 报告问题汇编

Determine sharp relationships between the first magnetic Neumann eigenvalue and planar-domain geometry for constant magnetic fields and Aharonov-Bohm potentials.

来源字段：https://doi.org/10.4171/owr/2023/36

07289 | OWR-13346-008

### Geometric Characterization and Removal of Stickiness

P / partially_solved | Oberwolfach 报告问题汇编

Find descriptors for the degree of full stickiness. Can stickiness always be removed by changing the geometry, and which classes cannot be so treated? Can only subsets of $Q_0$ be sticky? Characterize when stickiness is possible, for example in terms of the tangent cone.

来源字段：https://doi.org/10.4171/owr/2014/44

07290 | OWR-15587-003

### Geometric Criteria for Ehrhart Positivity

O / open | Oberwolfach 报告问题汇编

Which families of integral polytopes are Ehrhart positive, and what geometric methods can prove Ehrhart positivity?

来源字段：https://doi.org/10.4171/owr/2017/44

07291 | OWR-12581-001

### Geometric Decomposition from Long-Time Ricci Flow

O / open | Oberwolfach 报告问题汇编

For a Ricci flow with surgery on a closed $3$-manifold that exists for all sufficiently large times, determine whether and how its behavior as $t\to\infty$ realizes the full geometric decomposition of the manifold.

来源字段：https://doi.org/10.4171/owr/2013/33

07292 | OWR-830-003

### Geometric Gradient Flows for Elastic Rods

P / partially_solved | Oberwolfach 报告问题汇编

Starting from a rod configuration $\Gamma\subset\mathbb{R}^3$, can one find critical points of a Kirchhoff elastic energy, the so-called elasticae, by means of geometric gradient flows?

来源字段：https://doi.org/10.4171/owr/2005/25

07293 | OWR-13347-015

### Geometric Invariant Theory for Homogeneous Spaces

O / open | Oberwolfach 报告问题汇编

What other tools from Geometric Invariant Theory can be applied to obtain new information about the geometry of homogeneous spaces?

来源字段：https://doi.org/10.4171/owr/2014/45

07294 | OWR-14298170-018

### Geometric Noncollapse and Hybrid Convergence

O / open | Oberwolfach 报告问题汇编

Develop noncollapsing hypotheses based on constant-mean-curvature surfaces or isoperimetric regions. Also construct a notion of convergence for varying manifolds that implies volume-preserving intrinsic-flat convergence while retaining $W^{1,p}$ information on good regions, possibly through a metric on Riemannian manifolds.

来源字段：https://doi.org/10.4171/owr/2024/9

07295 | OWR-4332-001

### Geometric Quotients of K-Sheets

O / open | Oberwolfach 报告问题汇编

Do $K$-sheets have a geometric quotient, and, if so, can the quotient be described using the $p$-Slodowy slice?

来源字段：https://doi.org/10.4171/owr/2010/19

07296 | OWR-11101915-003

### Geometry and Actions of $\delta$-Median Spaces

O / open | Oberwolfach 报告问题汇编

Determine whether actions on uniformly locally finite $\delta$-median spaces induce proper actions on $L^p$-spaces, whether such spaces admit strongly bolic metrics, and whether $\delta$-median and hierarchically hyperbolic spaces coincide.

来源字段：https://doi.org/10.4171/owr/2022/34

07297 | OWR-14299914-001

### Geometry and Sampling on Probability Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Compute geometric quantities, including Riemannian and sectional curvatures, for general thermodynamical probability manifolds arising from master equations, develop the associated network analysis of curvature tensors, and use these calculations to design sampling algorithms for distributions on simplices.

来源字段：https://doi.org/10.4171/owr/2026/13

07298 | AMR-018-0007

### Geometry of Continued Fractions - Combinatorial structure of sails

O / open | AMR 问题清单汇编

Describe all finite two-dimensional sails (and the corresponding continued fractions).

背景链接：https://arxiv.org/abs/1712.01450

07299 | AMR-018-0008

### Geometry of Continued Fractions - Combinatorial structure of sails

O / open | AMR 问题清单汇编

{\bf (Multidimensional IKEA problem.)} Describe the collections of the sails of the cones for all polytopes of a given combinatorial type.

背景链接：https://arxiv.org/abs/1712.01450

07300 | AMR-018-0009

### Geometry of Continued Fractions - Combinatorial structure of sails

O / open | AMR 问题清单汇编

{\bf (V. Arnold.)} Does there exist an algorithm to decide whether a given type of fundamental domain is realizable by a periodic continued fraction?

背景链接：https://arxiv.org/abs/1712.01450

07301 | AMR-018-0010

### Geometry of Continued Fractions - Combinatorial structure of sails

O / open | AMR 问题清单汇编

{\bf (V. Arnold.)} Torus decompositions of integer noncongruent Klein sails are distinct.

背景链接：https://arxiv.org/abs/1712.01450

07302 | AMR-018-0011

### Geometry of Continued Fractions - Combinatorial structure of sails

O / open | AMR 问题清单汇编

{\bf (V. Arnold.)} Describe all torus decompositions that are realized by periodic two-dimensional continued fractions.

背景链接：https://arxiv.org/abs/1712.01450

07303 | AMR-018-0013

### Geometry of Continued Fractions - Combinatorial structure of sails

O / open | AMR 问题清单汇编

{\bf (V. Arnold.)} Classify continued fractions that correspond to the same cubic extension of the field of rational numbers.

背景链接：https://arxiv.org/abs/1712.01450

07304 | AMR-018-0014

### Geometry of Continued Fractions - Combinatorial structure of sails

O / open | AMR 问题清单汇编

Prove the existence of a cone for a single non-periodic combinatorial structure ($n\ge 3$).

背景链接：https://arxiv.org/abs/1712.01450

07305 | AMR-018-0003

### Geometry of Continued Fractions - Faces of sails

O / open | AMR 问题清单汇编

Classify all combinatorial possible types of faces.

背景链接：https://arxiv.org/abs/1712.01450

07306 | AMR-018-0004

### Geometry of Continued Fractions - Faces of sails

P / partially_solved | AMR 问题清单汇编

Classify all empty simplices of dimension $n$ up to lattice congruence.

背景链接：https://arxiv.org/abs/1712.01450

07307 | AMR-018-0005

### Geometry of Continued Fractions - Faces of sails

O / open | AMR 问题清单汇编

Which $n$-gons are realizable as faces of an $m$-dimensional continued fraction? Here are two essentially geometrically different subcases: ; {\bf Faces at integer distance 1 to the origin:} this is a question of description of integer convex polyhedra that are inscribed to simplices of full dimension in $\r^n$. ; {\bf Faces at integer distance greater than 1 to the origin:} here we have a typical view-obstacle problem, one avoids integer points that are view-obstacles for the corresponding faces.

背景链接：https://arxiv.org/abs/1712.01450

07308 | AMR-018-0019

### Geometry of Continued Fractions - Further open questions

P / partially_solved | AMR 问题清单汇编

Find a natural generalization of the Farey tessellation to higher-dimensional hyperbolic geometry.

背景链接：https://arxiv.org/abs/1712.01450

07309 | AMR-018-0020

### Geometry of Continued Fractions - Further open questions

P / partially_solved | AMR 问题清单汇编

{\bf (Jacobi's last theorem.)} Let $K$ be a totally real cubic number field. Consider arbitrary elements $y$ and $z$ of $K$ such that $0<y,z<1$ (here we assume that $1$, $y$, and $z$ are independent over $\mathbb \q$). Is it true that the Jacobi-Perron algorithm generates an eventually periodic continued fraction with starting data $v=(1,y,z)$.

背景链接：https://arxiv.org/abs/1712.01450

07310 | AMR-018-0021

### Geometry of Continued Fractions - Further open questions

O / open | AMR 问题清单汇编

Study geometric properties of Markov spectrum.

背景链接：https://arxiv.org/abs/1712.01450

07311 | AMR-018-0022

### Geometry of Continued Fractions - Further open questions

P / partially_solved | AMR 问题清单汇编

Generalize continued fractions to describe 3-bridge knots.

背景链接：https://arxiv.org/abs/1712.01450

07312 | AMR-018-0001

### Geometry of Continued Fractions - Integer trigonometry and IKEA problem

O / open | AMR 问题清单汇编

Find an integer cosine rule for integer triangles in integer trigonometry.

背景链接：https://arxiv.org/abs/1712.01450

07313 | AMR-018-0002

### Geometry of Continued Fractions - Integer trigonometry and IKEA problem

S / solved | AMR 问题清单汇编

{\bf(IKEA problem.)} Classify all $n$-tuples of LLS-sequences for the angles that form integer $n$-gons.

背景链接：https://arxiv.org/abs/1712.01450

07314 | AMR-018-0015

### Geometry of Continued Fractions - Sail statistics

O / open | AMR 问题清单汇编

Find frequencies on $n$-dimensional continued fractions with the highest relative frequencies.

背景链接：https://arxiv.org/abs/1712.01450

07315 | AMR-018-0016

### Geometry of Continued Fractions - Sail statistics

O / open | AMR 问题清单汇编

For every positive integer constant $C$ there exist only finitely many pairwise integer non-congruent faces with frequencies exceeding $C$.

背景链接：https://arxiv.org/abs/1712.01450

07316 | AMR-018-0017

### Geometry of Continued Fractions - Sail statistics

O / open | AMR 问题清单汇编

Is that true that sum of all relative frequencies for all possible faces is finite for higher dimensions $(n\ge 3)$?

背景链接：https://arxiv.org/abs/1712.01450

07317 | AMR-018-0018

### Geometry of Continued Fractions - Sail statistics

O / open | AMR 问题清单汇编

In case of positive answer to the above question find the generalization of the Gauss map and compare the corresponding frequencies of faces with the related frequencies coming from M\"obius geometry.

背景链接：https://arxiv.org/abs/1712.01450

07318 | AMR-069-0001

### Geometry of Curves and Surfaces - Problem 1.1

O / open | AMR 问题清单汇编

Does there exist a closed C2 surface in Euclidean space R3 which is flexible?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07319 | AMR-069-0002

### Geometry of Curves and Surfaces - Problem 1.2

O / open | AMR 问题清单汇编

Are all smooth tight surfaces in R3 rigid?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07320 | AMR-069-0003

### Geometry of Curves and Surfaces - Problem 1.3

O / open | AMR 问题清单汇编

Are negatively curved annuli bounded by a pair of fixed convex planar curves rigid?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07321 | AMR-069-0004

### Geometry of Curves and Surfaces - Problem 1.4

P / partially_solved | AMR 问题清单汇编

Let Γ be a smooth closed curve immersed in R3. Suppose that Γ has a continuous binormal vector field B which is one-to-one. Does it follow then that the ribbon (Γ,B ) is twisted?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07322 | AMR-069-0005

### Geometry of Curves and Surfaces - Problem 1.5

O / open | AMR 问题清单汇编

Given a metric of positive curvature on the disk what is the condition on a space curve to form the boundary of an isometric embedding of the disk?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07323 | AMR-069-0006

### Geometry of Curves and Surfaces - Problem 1.6

S / solved | AMR 问题清单汇编

Does every curve bounding a surface of positive curvature in 3-space have (at least) four points where the torsion vanishes?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07324 | AMR-069-0007

### Geometry of Curves and Surfaces - Problem 1.7

O / open | AMR 问题清单汇编

Are there some nonconvex surfaces which remain rigid after finitely many points of them have been deleted. For instance, are punctured analytic tight surfaces, such as a torus of revolution, rigid?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07325 | AMR-069-0008

### Geometry of Curves and Surfaces - Problem 1.8

O / open | AMR 问题清单汇编

(The global isometric embedding problem, Yau [189] 1993; Gromov [82]). Can everyC∞ 2-dimensional Riemannian manifold be isometrically embedded in R4?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07326 | AMR-069-0009

### Geometry of Curves and Surfaces - Problem 1.9

P / partially_solved | AMR 问题清单汇编

Given a C∞ metric in a neighborhood of a point in a 2-dimensional Riemannian manifold, does there exist an isometric embedding of some neighborhood of that point into R3?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07327 | AMR-069-0010

### Geometry of Curves and Surfaces - Problem 2.1

O / open | AMR 问题清单汇编

For which setsA⊂ Sn is there an immersionf: M→ Rn+1 such that Gf(M)⊂A?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07328 | AMR-069-0011

### Geometry of Curves and Surfaces - Problem 2.2

S / solved | AMR 问题清单汇编

Does connectedness of the shadows imply that f(M) is convex?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07329 | AMR-069-0012

### Geometry of Curves and Surfaces - Problem 2.3

O / open | AMR 问题清单汇编

LetM,M′⊂ R3 be smooth orientable closed surfaces. Suppose there exists a diffeomorphism f: M→ M′ which preserved the Gauss curvature and the Gauss map. Does it follow then that M and M′ are congruent?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07330 | AMR-069-0013

### Geometry of Curves and Surfaces - Problem 2.4

P / partially_solved | AMR 问题清单汇编

LetP, P′⊂ R3 be polyhedral surfaces. Suppose that the faces of P and P′ are parallel and have the same area. Does it follow then that P and P′ are congruent?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07331 | AMR-069-0014

### Geometry of Curves and Surfaces - Problem 3.1

O / open | AMR 问题清单汇编

Is every convex polytope unfoldable?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07332 | AMR-069-0015

### Geometry of Curves and Surfaces - Problem 3.2

O / open | AMR 问题清单汇编

Does there exist a reasonably simple algorithm for detecting the edges of a convex polyhedron intrinsically?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07333 | AMR-069-0016

### Geometry of Curves and Surfaces - Problem 3.3

S / solved | AMR 问题清单汇编

Does there exist a convex polyhedron with a pseudo edge graph which is not unfoldable.

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07334 | AMR-069-0017

### Geometry of Curves and Surfaces - Problem 4.1

P / partially_solved | AMR 问题清单汇编

Of all convex surfaces with a fixed intrinsic diameter, is the one with the greatest area a doubled disk?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07335 | AMR-069-0018

### Geometry of Curves and Surfaces - Problem 4.2

O / open | AMR 问题清单汇编

Let S ⊂ R3 be a closed surface of constant width and fixed area. How small can the volume of S be?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07336 | AMR-069-0019

### Geometry of Curves and Surfaces - Problem 4.3

P / partially_solved | AMR 问题清单汇编

LetS⊂ R3 be a closed surface of diameter d. Suppose that there exists a constant h < dso that whenever a pair of planes separated by a distance of h intersect S, the area of S contained between these planes is constant. Does it then follow that S is a sphere?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07337 | AMR-069-0020

### Geometry of Curves and Surfaces - Problem 5.1

P / partially_solved | AMR 问题清单汇编

What is the shortest curve in R3 with a given width or inradius?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07338 | AMR-069-0021

### Geometry of Curves and Surfaces - Problem 5.2

O / open | AMR 问题清单汇编

Let $\Gamma$ be a closed curve of fixed length $L$ in $\mathbb{R}^3$. Determine the maximum possible volume of the convex hull of $\Gamma$.

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07339 | AMR-069-0022

### Geometry of Curves and Surfaces - Problem 5.3

O / open | AMR 问题清单汇编

Let $\Gamma$ be a closed curve of fixed length $L$ in $\mathbb{R}^3$, and let $A$ be the area of its convex hull. Prove that $A$ is maximized when $\Gamma$ is a circle, treating its convex hull as a doubly covered disk.

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07340 | AMR-069-0023

### Geometry of Curves and Surfaces - Problem 6.1

O / open | AMR 问题清单汇编

Is every compact connected minimal surface bounded by a pair of convex planar curves topologically an annulus?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07341 | AMR-069-0024

### Geometry of Curves and Surfaces - Problem 6.2

P / partially_solved | AMR 问题清单汇编

Does there exist an embedded compact surface of constant mean curvature which is bounded by a circle, but is not a piece of a sphere.

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07342 | AMR-069-0025

### Geometry of Curves and Surfaces - Problem 6.3

P / partially_solved | AMR 问题清单汇编

Show that any compact embedded CMC surface which is bounded by a convex planar curve, and lies on one side of the boundary plane, is topologically a disk.

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07343 | AMR-069-0026

### Geometry of Curves and Surfaces - Problem 7.1

S / solved | AMR 问题清单汇编

Are there any complete surfaces of negative curvature in Euclidean 3-space whose principal curvatures are bounded away from zero?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07344 | AMR-069-0027

### Geometry of Curves and Surfaces - Problem 7.2

O / open | AMR 问题清单汇编

Are there any complete negatively curved surfaces embedded in the unit ball?

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07345 | AMR-069-0028

### Geometry of Curves and Surfaces - Problem 7.3

O / open | AMR 问题清单汇编

Does there exist any complete negatively curved surfaces with negative Euler characteristic contained in between a pair of parallel planes in R3.

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07346 | AMR-069-0029

### Geometry of Curves and Surfaces - Problem 8.2

P / partially_solved | AMR 问题清单汇编

Show that the index of any singularity of a principal line fields on a surface is at most one.

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07347 | AMR-069-0030

### Geometry of Curves and Surfaces - Problem 8.3

O / open | AMR 问题清单汇编

Let M be a complete noncompact convex surface in R3, with principal curvatures k1, k2, then show that inf M|k1-k2| = 0.

背景链接：https://people.math.gatech.edu/~ghomi/Papers/op.pdf

07348 | AMR-059-0018

### Geometry on Sample-Parameter Product Spaces

O / open | AMR 问题清单汇编

For a statistical family $f(x;\theta)$, find and study a useful geometry on the product of the sample space and parameter space, accounting for the choice of a statistically meaningful $\theta$-coordinate system.

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

07349 | OWR-14298170-002

### Geroch Stability with a Good-Set Bad-Set Decomposition

P / partially_solved | Oberwolfach 报告问题汇编

Formulate and prove a version of Geroch stability analogous to the good-set bad-set positive-mass stability theorem of Dong and Song.

来源字段：https://doi.org/10.4171/owr/2024/9

07350 | OWR-14298170-001

### Geroch Stability Without Bubbles and Drawstrings

P / partially_solved | Oberwolfach 报告问题汇编

Formulate and prove a Geroch stability theorem in which assumptions exclude bubbles and drawstrings and the conclusion is volume-preserving Sormani-Wenger intrinsic-flat convergence to a flat torus.

来源字段：https://doi.org/10.4171/owr/2024/9

07351 | AMR-059-0014

### Global Affine Curve Flow

P / partially_solved | AMR 问题清单汇编

Prove existence of a time-global solution to the affine-plane curve evolution equation $\partial x/\partial u=(1+kp)x''$, where $u$ is time, $x''$ is the affine normal, $k$ is affine curvature, and $p$ is the affine support function.

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

07352 | AIM-GEOMETRY-0324

### Global balancing as a recognition obstruction

P / partially_solved | AIM 研讨会问题汇编

Given such a homogeneous, value-group-rational, totally concave polyhedral complex, what further conditions are necessary for it to be the amoeba of a prime Laurent ideal?

The AIM PDF and HTML recover this as follows. Let \(k\) be an algebraically closed non-archimedean field and let \[ \mathfrak p\subset k[x_1^{\pm1},\ldots,x_d^{\pm1}] \] be prime. Its non-archimedean amoeba agrees with the Bieri--Groves set of \[ A=k[x_1^{\pm1},\ldots,x_d^{\pm1}]/\mathfrak p. \] The source says that this set is a “homogeneous polyhedral complex,” has dimension \(\dim A\), is rational over the value group, and is totally concave. It asks:

The historical terminology matters. Bieri and Groves define a polyhedron to be homogeneous when it admits a finite decomposition into convex polyhedra all having the dimension of the whole set. Thus “homogeneous” means pure-dimensional here, not invariant under scaling and not “homogeneous ideal.” Their total concavity condition says that at every \(u\) the convex hull of the local cone \(LC_u(S)\) is an affine subspace. Gubler later uses exactly this local-cone formulation.

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

07353 | AIM-GEOMETRY-0351

### Global conformal invariants and the four-dimensional Q-Pfaffian bridge

S / solved | AIM 研讨会问题汇编

Conjecture 3. Any \(S\) as above may be written \[ \mathrm{const}\cdot\mathrm{Pff}+L+V, \] where \(V\) is an exact divergence.

If \(S\) is a natural \(n\)-form and \(\int S\) is conformally invariant, then ...

**Recovered Conjecture 3.** Let \(M^n\) be a closed oriented Riemannian manifold of even dimension. If \(S(g)\) is a natural metric \(n\)-form whose integral is unchanged by every conformal rescaling \(\widehat g=e^{2\omega}g\), then \[ S(g)=c\,\operatorname{Pff}(g)+L(g)+d\beta(g), \] where \(L\) is a local conformal invariant \(n\)-form and \(\beta\) is a natural \((n-1)\)-form.

The missing phrase “\(S\) as above” is recovered from Conjecture 1 on the preceding page of the AIM source:

The surrounding chapter assumes an oriented Riemannian manifold of even dimension \(n\), regards scalar densities as \(n\)-forms, and suppresses some density weights. For the global assertion and for an exact divergence to integrate to zero without boundary terms, the standard hypothesis is that \(M^n\) is closed. Thus the intended conjecture is:

For comparison with Alexakis, write \(S(g)=P(g)dV_g\). The precise Deser-Schwimmer class consists of scalar Riemannian invariants \(P(g)\) that are finite linear combinations of complete contractions of covariant derivatives of the Riemann tensor and have constant-rescaling weight \(-n\).

Only the displayed Conjecture 3 is the assigned conjecture. The subsequent \(\mathcal P\)- and \(\mathcal L\)-filtrations, Exercise 6, and the tractor construction of \(Q\) are explanatory material and further questions, not additional clauses of Conjecture 3.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

07354 | OWR-4535-007

### Global Formulas and Laplacians for the Takhtajan-Zograf Form

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Takhtajan-Zograf Representation Question [OWR-4535-008]

Does the Takhtajan-Zograf Kähler form have a global representation formula, and does its Laplace operator have a self-adjoint extension?

来源字段：https://doi.org/10.4171/owr/2010/53

07355 | OWR-16412-003

### Global Linearization of Rank-One Rotation Domains

O / open | Oberwolfach 报告问题汇编

If $\Omega$ is a rank-one rotation domain, is $(f|_\Omega,\Omega)$ biholomorphically conjugate to $(L,D)$, where $D\subset\mathbb C^2$ and $$L=\operatorname{diag}(\alpha^p,\alpha^q),\qquad\gcd(p,q)=1,$$ with $|\alpha|=1$ and $\alpha$ not a root of unity?

来源字段：https://doi.org/10.4171/owr/2018/37

07356 | AMR-063-0008

### Global Moduli of G2 Metrics

P / partially_solved | AMR 问题清单汇编

For a compact seven-manifold $M$ admitting holonomy-$G_2$ metrics, describe their moduli space modulo diffeomorphisms isotopic to the identity. If $\pi:\mathcal M\to H^3(M;\mathbb{R})$ maps a metric to the cohomology class of its defining three-form and is locally a diffeomorphism, is $\pi$ globally a diffeomorphism onto its image?

背景链接：https://wwwf.imperial.ac.uk/~skdona/NONLINEARITYPROBLEMS.PDF

07357 | OWR-14298364-002

### Global Ricci Bounds along Kähler-Ricci Flow

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a compact Kähler manifold admitting a holomorphic fibration onto a lower-dimensional base (so that the normalized Kähler--Ricci flow $\partial_t \omega(t) = -\operatorname{Ric}(\omega(t)) - \omega(t)$ is immortal and collapses the fibers as $t \to \infty$). Along the Kähler--Ricci flow on $X$, does the Ricci curvature norm $|\operatorname{Ric}(\omega(t))|_{\omega(t)}$ remain uniformly bounded for all time $t \geq 0$ and globally on $X$?

来源字段：https://doi.org/10.4171/owr/2024/28

07358 | OWR-12484-004

### Global Willmore Minimizers Near the Clifford Torus

P / partially_solved | Oberwolfach 报告问题汇编

For conformal classes sufficiently near that of the Clifford torus, are the constrained Willmore Hopf tori that minimize the constrained energy also the global minimizers of the Willmore energy in their respective conformal classes?

来源字段：https://doi.org/10.4171/owr/2013/21

07359 | AIM-GEOMETRY-0432

### Globality, Lee-cover volume, and mapping-torus resonance for lcs forms

P / partially_solved | AIM 研讨会问题汇编

**Question 1.36.** Given a c.s.s. is it global?

There is the following result: a c.s.s. \(\Omega\) is global if and only if \(\omega\) is exact. Are there other conditions?

Banyaga has constructed c.s.s. for which \(\Omega\) is non zero in the Lichnerowicz cohomology, but such examples seem few and far between.

The acronym `c.s.s.` is the notes' abbreviation for a “(local) conformal symplectic structure”; the standard modern term and abbreviation are **locally conformally symplectic (lcs)**. The immediately preceding source text fixes the convention \[ d\Omega=-\omega\wedge\Omega, \qquad d_\omega=d+\omega\wedge, \qquad d_\omega\Omega=0, \] where \(\omega\) is a closed Lee form. To avoid confusing the Lee form with a symplectic form, this report renames it \(\theta\).

With this sign convention, if \(\theta=dh\), then \[ d(e^h\Omega)=e^h(dh\wedge\Omega+d\Omega)=0. \] Thus the global symplectic representative is \(e^h\Omega\). Conversely, if \(\Omega=f\sigma\), where \(f>0\) and \(d\sigma=0\), then \(\theta=-d\log f\). This verifies the sign in the source's “if and only if” rather than silently importing the opposite convention common in some papers.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

07360 | OWR-15579-001

### Goldberg Conjecture in Four Dimensions

O / open | Oberwolfach 报告问题汇编

If a closed almost Kähler four-manifold $(M,g,J,\omega)$ is Einstein, must its almost complex structure $J$ be integrable?

来源字段：https://doi.org/10.4171/owr/2017/36

07361 | OWR-8415362-007

### Gradient Flows and Ball Topology of Amplituhedra

O / open | Oberwolfach 报告问题汇编

Can one classify gradient flows preserving $A_{n,k,m}(Z)$, and does such a flow exist for every $Z$, implying that $A_{n,k,m}(Z)$ is homeomorphic to a closed ball?

来源字段：https://doi.org/10.4171/owr/2021/57

07362 | OWR-11789-007

### Gradient Lower Bounds for Shrinking Ricci Solitons

O / open | Oberwolfach 报告问题汇编

For a complete shrinking Ricci soliton $(M,g,f)$ with basepoint $p$ minimizing $f$, does there exist $C>0$ such that $|\nabla f|(x)\geq C^{-1}$ whenever $d(x,p)\geq C$?

来源字段：https://doi.org/10.4171/owr/2012/27

07363 | OWR-17135-006

### Grassmann Polytopes as Positive Geometries

O / open | Oberwolfach 报告问题汇编

Are all Grassmann polytopes positive geometries? Determine their topology and combinatorics in a form analogous to the theory of ordinary convex polytopes.

来源字段：https://doi.org/10.4171/owr/2019/39

07364 | AIM-GEOMETRY-0075

### Grazing periodic orbits and the negative-temperature pressure wall

P / partially_solved | AIM 研讨会问题汇编

Consider a finite-horizon billiards, and let \(\varphi\) be the geometric potential. Is there a jump in pressure \(P(q\varphi)\) and \(q \to 0^-\)? Can one find perioidic orbits with arbitrarily large (but finite) Lyapunov exponent?

来源字段：http://aimpl.org/equibdynsysgeom/5/

07365 | OWR-13106-011

### Gromov Macroscopic Dimension Conjecture

O / open | Oberwolfach 报告问题汇编

If a closed $n$-manifold $M$ admits a metric of positive scalar curvature, must the macroscopic dimension of its universal cover satisfy $\dim_{\mathrm{mc}}(\widetilde M)\leq n-2$?

来源字段：https://doi.org/10.4171/owr/2014/36

07366 | OWR-11101916-009

### Gromov's Scalar Curvature Band-Width Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $M$ be a closed connected manifold of dimension $n-1\neq4$ that admits no metric of positive scalar curvature. If $g$ is a metric on $V=M\times[-1,1]$ with scalar curvature at least $n(n-1)$, must $\operatorname{width}(V,g)\leq2\pi/n$, where $\operatorname{width}(V,g)=\operatorname{dist}_g(\partial_-V,\partial_+V)$?

来源字段：https://doi.org/10.4171/owr/2022/35

07367 | AMR-067-0026

### Gromov-Hausdorff convergence of K\"ahler Ricci flow

P / partially_solved | AMR 问题清单汇编

Does the normalized Ricci flow converge in Gromov-Hausdorff sense to a generalized K\"ahler-Einstein space?

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

07368 | AMR-058-0023

### Gromov-Knothe Isoperimetry for Multiple Regions

P / partially_solved | AMR 问题清单汇编

Extend Gromov's Knothe-based proof of the isoperimetric inequality to multiple regions, and determine whether its vector field can be chosen canonically.

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

07369 | OWR-14211-004

### Ground States of the Gaussian Core Model

O / open | Oberwolfach 报告问题汇编

For the Gaussian core model in Euclidean space with pair potential $r\mapsto e^{-\alpha r^2}$ and fixed particle density 1, what are the ground-state point configurations minimizing the average energy, particularly for values of $\alpha\geq 4\pi/e$?

来源字段：https://doi.org/10.4171/owr/2016/3

07370 | OWR-14298808-011

### Growing-Parameter Limits for Convex-Body Functionals

O / open | Oberwolfach 报告问题汇编

Let $(d_j)$ and $(k_j)$ satisfy $0\leq k_j\leq d_j-1$, $2k_j>d_j+1$, and $d_j\to\infty$. Determine the asymptotic behavior of the standardized random variables $Y^*_{d_j,k_j}$. Find conditions, such as $k_j/d_j\to 1/2$, that imply convergence in distribution, and determine whether any limit is Gaussian.

来源字段：https://doi.org/10.4171/owr/2024/58

07371 | AMR-037-0006

### Halving lines and k-sets

P / partially_solved | AMR 问题清单汇编

For an $n$-point planar set, determine the maximum number of halving lines. More generally, determine the maximum number of $k$-sets, subsets obtained by intersecting the point set with a half-plane, and extend the sharp bounds to higher-dimensional halving hyperplanes.

背景链接：https://jeffe.cs.illinois.edu/open/

07372 | OWR-11783-004

### Hamiltonian Displaceability of Toric Fibers

P / partially_solved | Oberwolfach 报告问题汇编

For a $2n$-dimensional toric symplectic manifold $(M,\omega)$ with moment map $\mu$ and Lagrangian torus fibers $T_u=\mu^{-1}(u)$, which fibers can be displaced by a Hamiltonian isotopy; equivalently, for which $u$ does there exist $\phi\in\operatorname{Ham}(M,\omega)$ with $\phi(T_u)\cap T_u=\varnothing$?

来源字段：https://doi.org/10.4171/owr/2012/21

07373 | OWR-15575-011

### Hamiltonian Pseudo-Rotations of the Complex Projective Plane

O / open | Oberwolfach 报告问题汇编

Can the Anosov-Katok conjugation method be used to construct genuine Hamiltonian pseudo-rotations of $\mathbb{CP}^2$ with exactly three periodic points that are not true rotations?

来源字段：https://doi.org/10.4171/owr/2017/32

07374 | OWR-15208-029

### Harmonic Fibrations of Exotic Complex Projective Space Bundles

O / open | Oberwolfach 报告问题汇编

For a smooth manifold homeomorphic to $S^1\times\mathbb{CP}^3$ but diffeomorphic to an exotic connected sum, can the harmonic map to $S^1$ arising from a Riemannian metric be a fibration?

来源字段：https://doi.org/10.4171/owr/2017/3

07375 | OWR-14297736-005

### Heat-Content Asymptotics for Polyhedra and Other Planar Domains

O / open | Oberwolfach 报告问题汇编

Determine small-time heat-content asymptotics for broader classes of planar domains and for polyhedra in $\mathbb R^3$, including the exact angle contributions, and determine how much the first three asymptotic terms distinguish planar domains.

来源字段：https://doi.org/10.4171/owr/2023/36

07376 | OWR-14298588-010

### Hecke Eigensheaves for Arthur Parameters

P / partially_solved | Oberwolfach 报告问题汇编

别名：Arthur-Eigensheaf Spectral-Side Continuation [OWR-14298588-012]

For every Arthur parameter $\psi$, construct an associated perverse eigensheaf $F_\psi$ on $\operatorname{Bun}_G$ with the predicted Hecke eigenproperty and, when $G$ is semisimple, the predicted decomposition over basic strata into the Arthur packets of the pure inner forms of $G$.

来源字段：https://doi.org/10.4171/owr/2024/33

07377 | OWR-14298588-009

### Hecke Eigensheaves for Discrete L-Parameters

P / partially_solved | Oberwolfach 报告问题汇编

For every discrete $L$-parameter $\phi$ of $G$, construct a perverse Hecke eigensheaf $F_\phi\in D_{\mathrm{lis}}(\operatorname{Bun}_G,\overline{\mathbb{Q}}_\ell)$ whose restriction to each basic stratum realizes the local Langlands packet of the corresponding inner form with the predicted multiplicities.

来源字段：https://doi.org/10.4171/owr/2024/33

07378 | AIM-GEOMETRY-0018

### Height-one valuations and a nonreduced A3 seed-torus boundary

P / partially_solved | AIM 研讨会问题汇编

If $X$ is a general cluster variety, and $T\subset X$ is a cluster torus. What can we say about $X - T$ (Eg. what are the irreducible components, intersections, etc.)?

来源字段：http://aimpl.org/clusterbraid/2/

07379 | OWR-2090-019

### Helly Numbers for Isolated Line Transversals

O / open | Oberwolfach 报告问题汇编

For an isolated line transversal to a finite collection of pairwise disjoint convex sets in $\mathbb R^d$, does an analogue of the Helly theorem for disjoint balls hold? In particular, is every such transversal already an isolated transversal to a subcollection of bounded size, for example at most $2d-1$?

来源字段：https://doi.org/10.4171/owr/2008/44

07380 | OWR-1703876-005

### Helly-Radon-Tverberg Theory for Nonconvex Systems

P / partially_solved | Oberwolfach 报告问题汇编

For nonconvex set systems or their associated closure operators, what general Helly-Radon-Tverberg, fractional-Helly, weak-$\varepsilon$-net, and $(p,q)$-type theorems and quantitative bounds can be obtained?

来源字段：https://doi.org/10.4171/owr/2020/30

07381 | OWR-11101921-002

### Hermitian Volume Invariants for Plurisigned Metrics

O / open | Oberwolfach 报告问题汇编

For a plurisigned Hermitian metric $\omega$, is $v_+(\omega)$ always finite and is $v_-(\omega)$ always positive?

来源字段：https://doi.org/10.4171/owr/2022/40

07382 | OWR-13750327-007

### Hermitian-Symplectic Structures on Non-Kähler Manifolds

O / open | Oberwolfach 报告问题汇编

Does any non-Kähler compact complex manifold admit a Hermitian-symplectic structure?

来源字段：https://doi.org/10.4171/owr/2023/18

07383 | OWR-12723-008

### Hexagonal Extremizers for a Geometric Uncertainty Principle

O / open | Oberwolfach 报告问题汇编

For a finite measurable partition $[0,1]^2=\bigcup_{i=1}^N\Omega_i$, define

$$\mathcal A(\Omega_i)=\inf_{|B|=|\Omega_i|}\frac{|\Omega_i\mathbin\triangle B|}{|\Omega_i|},\qquad \mathcal D(\Omega_i)=\frac{|\Omega_i|-\min_j|\Omega_j|}{|\Omega_i|},$$

where the infimum is over disks $B$. Is the minimum of

$$\sum_{i=1}^N|\Omega_i|\mathcal A(\Omega_i)+\sum_{i=1}^N|\Omega_i|\mathcal D(\Omega_i)$$

asymptotically attained by a hexagonal partition, with suitable modifications along the boundary?

来源字段：https://doi.org/10.4171/owr/2013/49

07384 | AMR-054-0027

### Hexahedral Meshing

P / partially_solved | AMR 问题清单汇编

Can the interior of every simply connected polyhedron whose surface is meshed by an even number of quadrilaterals be partitioned into a hexahedral mesh compatible with the surface meshing?

背景链接：https://topp.openproblem.net/p27

07385 | OWR-11136-031

### High Incidence Partitions and Minimum Ellipsoid Coresets

O / open | Oberwolfach 报告问题汇编

For every $c$, does there exist a locally finite convex-cell partition of $\mathbb{R}^3$ in which every cell meets at least $c$ vertices and every vertex meets at least $c$ cells? Separately, for a finite set $S\subset\mathbb{R}^d$, let $\operatorname{MEE}(S)$ be its minimum enclosing ellipsoid, let $\operatorname{DBM}$ be the Banach-Mazur distance, and define $f(d,\varepsilon)$ as the supremum over $S$ of the minimum cardinality of $X\subseteq S$ satisfying $\operatorname{DBM}(\operatorname{MEE}(X),\operatorname{MEE}(S))\leq\varepsilon$. Is $f(d,\varepsilon)=O(d/\varepsilon^2)$?

来源字段：https://doi.org/10.4171/owr/2011/44

07386 | AIM-GEOMETRY-0094

### High-dimensional Gaussian energy and a tensor-product obstruction

P / partially_solved | AIM 研讨会问题汇编

Behavior of energy as $n \rightarrow \infty$ with $\alpha$ fixed. (Gaussian core model). Known for small $\alpha$ by the paper of Cohn and de Courcy-Ireland.

This one-line record does not specify the energy normalization, the particle density, or whether competitors are restricted to lattices. The cited primary paper makes the intended problem precise [CDI18]. Its setup is as follows. For \[ f_\alpha(r)=e^{-\alpha r^2},\qquad \alpha>0, \] and a discrete set $\mathcal C\subset\mathbb R^n$ of density $\rho>0$, meaning \[ \#(\mathcal C\cap B_R)=\bigl(\rho+o(1)\bigr)\operatorname{vol}(B_R), \] the lower energy is \[ \underline E_\alpha(\mathcal C) =\liminf_{R\to\infty} \frac{1}{\#(\mathcal C\cap B_R)} \sum_{\substack{x,y\in\mathcal C\cap B_R\\x\ne y}} e^{-\alpha\lVert x-y\rVert^2}. \tag{1} \] The sum counts ordered pairs. Thus this is energy per particle without the physical convention of dividing pair energy by two. The quantity under study is \[ M_n(\alpha,\rho) =\inf\bigl\{\underline E_\alpha(\mathcal C): \mathcal C\subset\mathbb R^n\text{ discrete of density }\rho\bigr\}. \tag{2} \] Competitors are arbitrary configurations, not merely lattices or periodic sets.

For a lattice $\Lambda$ of density $\rho=1/\operatorname{covol}(\Lambda)$, the limit exists and \[ E_\alpha(\Lambda) =\sum_{x\in\Lambda\setminus\{0\}}e^{-\alpha\lVert x\rVert^2} =\Theta_\Lambda(\alpha i/\pi)-1. \tag{3} \] All results below use exactly this reconstruction. A different convention with unordered pairs would multiply every energy by $1/2$.

来源字段：http://aimpl.org/discreteaf/1/

07387 | OWR-12486-002

### High-Dimensional Limits of Metric-Measure Spaces

O / open | Oberwolfach 报告问题汇编

How should sequences of Riemannian or metric-measure spaces be studied when their dimensions tend to infinity, particularly with respect to concentration convergence, curvature-dimension conditions, and Laplacian eigenvalues?

来源字段：https://doi.org/10.4171/owr/2013/23

07388 | OWR-14297736-032

### High-Index Spectral Optimizers Among Polygons

O / open | Oberwolfach 报告问题汇编

For polygons with a fixed number $n$ of sides, prove that optimizers of normalized Dirichlet and Neumann eigenvalues converge to the regular $n$-gon as the eigenvalue index tends to infinity.

来源字段：https://doi.org/10.4171/owr/2023/36

07389 | OWR-14610-004

### Higher-Codimension Weyl Laws and Equidistributed Min-Max Surfaces

O / open | Oberwolfach 报告问题汇编

Determine the constants $a(n)$; decide whether the Weyl law for $p$-widths extends to higher codimension, namely whether $\lim_{p\to\infty}\omega_p^k(M)p^{-(n+1-k)/(n+1)}=a(n,k)\operatorname{vol}(M)^{k/(n+1)}$; understand the analogous limits for Steenrod-power widths; and determine whether $p$-parameter min-max families yield minimal surfaces equidistributed as $p\to\infty$.

来源字段：https://doi.org/10.4171/owr/2016/28

07390 | AIM-GEOMETRY-0255

### Higher-dimensional asymptotic geodesics and the homology obstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 8.6.1 (Schmidt).** Generalize this statement to higher dimensions.

Let \(M\) be an oriented closed Riemannian surface of genus at least one, let \(\Gamma\) be a nontrivial free homotopy class, and let \(p\in M\). There are a closed geodesic \(\gamma:\mathbb R\to M\) in \(\Gamma\) and a geodesic half-orbit \(\beta:[0,\infty)\to M\), with \(\beta(0)=p\), such that, after choosing the origins of the two unit-speed parametrizations, \[ d(\beta(t),\gamma(t))\longrightarrow 0. \]

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07391 | OWR-14298170-009

### Higher-Dimensional Hamilton-Lott Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $(M^n,g)$ be complete, Ricci-pinched, and $2$-pinched, so that $\operatorname{Ric}(g)\geq cR_g g\geq 0$ and the sum of the two smallest curvature-operator eigenvalues is at least $cR_g$. Without assuming noncollapse, prove that $(M,g)$ is either flat or compact.

来源字段：https://doi.org/10.4171/owr/2024/9

07392 | AIM-GEOMETRY-0389

### Higher-even-dimensional Polyakov formulae

P / partially_solved | AIM 研讨会问题汇编

Problem 29: 3 Is (40) true in higher even dimensions? The following conjecture would be enough to answer the previous problem.

This record cannot be read in isolation. With \(\widehat g=e^{2\omega}g\), the recovered display is \[ -\log\frac{\det \widehat A}{\det A} =\alpha\left\{\frac12\int_M\omega P_g\omega\,dv_g +\int_M\omega Q_g\,dv_g\right\} +\left(\int_M F_{\widehat g}\,dv_{\widehat g} -\int_M F_g\,dv_g\right)+\mathcal H . \tag{40} \] Here \(A\) is a natural differential operator with positive-definite leading symbol, described as a positive power of a conformally invariant operator; the source gives a GJMS operator and the square of the Dirac operator as examples. It says that (40) is known in dimensions \(2,4,6\), that \(F\) is a local scalar invariant, and that \(\mathcal H\) depends on the nullspace of \(A\). The determinant is zeta regularized.

Two repairs are necessary for mathematical well-posedness.

1. If \(\ker A\ne0\), the displayed determinant must be read as the reduced determinant \(\det' A\) over the nonzero spectrum (or as an equivalent determinant-line norm). Otherwise the determinant vanishes and its logarithm is undefined. This is precisely the regime for which \(\mathcal H\) is present. 2. “Positive power” has at least two meanings. A literal spectral power \(B^r\) is different from a natural conformal power such as a GJMS operator with leading term \(\Delta^r\). Conformal covariance between weighted bundles does not in general survive naive composition after a metric trivialization. The results below keep these meanings separate.

The problem is therefore recovered as follows: **on a closed manifold of even dimension \(n\ge8\), under what hypotheses does the finite conformal variation of a zeta determinant have the form (40)?**

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

07393 | OWR-14212-001

### Higher-Order Principal Components on Riemannian Manifolds

O / open | Oberwolfach 报告问题汇编

Can higher-order derivative constructions of barycentric or affine subspaces, and the associated principal-component analysis, be generalized from spheres to arbitrary Riemannian manifolds using sub-Riemannian geometry?

来源字段：https://doi.org/10.4171/owr/2016/4

07394 | HIL-016

### Hilbert's 16th Problem: Topology of Algebraic Curves and Limit Cycles

P / partially_solved | Hilbert 问题

别名：Hilbert's Sixteenth Problem [ALG-019]

Determine the maximum number and relative positions of limit cycles for polynomial vector fields of degree $n$, and investigate the topology of real algebraic curves and surfaces.

背景链接：http://www.scholarpedia.org/article/Limit_cycles_of_planar_polynomial_vector_fields

07395 | HIL-018

### Hilbert's 18th Problem: Polyhedra and Space-Filling

P / partially_solved | Hilbert 问题

Are there only finitely many essentially different space-filling convex polyhedra? Is there a polyhedron which tiles space but not in a lattice arrangement?

背景链接：https://encyclopediaofmath.org/wiki/Hilbert_problems

07396 | OWR-16415-021

### Hitchin Curves and Weil-Petersson Boundary Dynamics

O / open | Oberwolfach 报告问题汇编

Find an ordinary differential equation describing at least some finite-dimensional families of $C^1$ curves arising from Hitchin representations. For universal Weil-Petersson Teichmüller space, determine the limit points in the Thurston boundary of Weil-Petersson geodesics, or construct analogues of earthquake paths.

来源字段：https://doi.org/10.4171/owr/2018/40

07397 | OWR-14298010-004

### Hodge-Riemann Relations for Convex Valuations

P / partially_solved | Oberwolfach 报告问题汇编

Do the hard Lefschetz theorem and the Hodge-Riemann bilinear relations hold in the algebra $\operatorname{Val}^\infty$ with respect to every element of the cone $\mathcal K=\{V(C,\mathbin{\bullet}[n-1]):C\in\mathcal K_+^\infty\}$ of valuations associated with smooth strictly convex bodies?

来源字段：https://doi.org/10.4171/owr/2023/51

07398 | OWR-16412-006

### Holomorphic Homogeneous Regularity of Infinite-Dimensional Convex Domains

O / open | Oberwolfach 报告问题汇编

Is every bounded convex domain in an infinite-dimensional Hilbert space holomorphic homogeneous regular?

来源字段：https://doi.org/10.4171/owr/2018/37

07399 | OWR-12727-003

### Homogeneity from Legendrian Varieties of Minimal Rational Tangents

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a Fano manifold with $b_2(X)=1$. If its variety of minimal rational tangents at a general point is a connected nonlinear Legendrian submanifold of the contact projective space, must $X$ be homogeneous, or at least quasi-homogeneous? Likewise, if the variety is a nonlinear smooth hypersurface, must $X$ be homogeneous, or at least quasi-homogeneous?

来源字段：https://doi.org/10.4171/owr/2013/53

07400 | OWR-12727-001

### Homogeneity of Fano Contact Manifolds

O / open | Oberwolfach 报告问题汇编

Is every Fano contact manifold homogeneous?

来源字段：https://doi.org/10.4171/owr/2013/53

07401 | OWR-13347-007

### Homogeneous Einstein Metrics on Flag and Stiefel Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Find all homogeneous Einstein metrics on generalized flag manifolds $G/K$ and on Stiefel manifolds $V_k\mathbb R^n=\operatorname{SO}(n)/\operatorname{SO}(n-k)$; and determine whether the number of homogeneous Einstein metrics is finite for homogeneous spaces with nonequivalent irreducible isotropy components and for compact semisimple Lie groups.

来源字段：https://doi.org/10.4171/owr/2014/45

07402 | AMR-067-0020

### Homogeneous Riemannian manifolds with nontrivial nullity

P / partially_solved | AMR 问题清单汇编

1. If the normal holonomy group of an irreducible and full homogeneous submanifold $M^n$ of the sphere with $n \geq 2$ does not act transitively, then $M$ is the orbit of an $s$-representation. 2. The index of an irreducible symmetric space which is different from $G_2/SO(4)$ (or its symmetric dual) coincides with its reflective index,. 3. Open questions about homogeneous Riemannian manifolds with nontrivial nullity.; Are there examples which are not topologically trivial?; Are there examples $M = G/H$, with $G$ non-solvable?; Are there K\"ahler examples?; Are there examples in any dimension $d \geq 5$?

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

07403 | OWR-4334-003

### Homological Area Minimizers in Compact Flat Three-Manifolds

O / open | Oberwolfach 报告问题汇编

In the Tri, Tetra, and Hexa Cosms, is a horizontal planar surface the mod-two area minimizer for one range of the height parameter and, for the complementary range, respectively the catenoid-like surface rPD, tP, or H? In the two remaining nonzero homology classes of the Tetra Cosm, are the minimizers respectively the CLP and tD minimal surfaces?

来源字段：https://doi.org/10.4171/owr/2010/21

07404 | AMR-065-0006

### Homological Pisot Conjecture

P / partially_solved | AMR 问题清单汇编

A one--dimensional, unimodular Pisot inflation tiling has pure point spectrum if its first rational \v{C}ech cohomology group has rank equal to the algebraic degree of $\lambda$.

背景链接：https://arxiv.org/abs/1604.06280

07405 | OWR-3385-006

### Homological Structure of Planar Self-Stress Identities

O / open | Oberwolfach 报告问题汇编

For the self-stress identities on planar $K_4$ and wheel graphs, including $\sum_{i<j}\omega_{ij}f_{ij}=1$, determine whether they are instances of a general homological or cohomological construction.

来源字段：https://doi.org/10.4171/owr/2009/02

07406 | OWR-7155451-019

### Homotopy Invariance of Positive-Scalar-Curvature Metric Spaces

O / open | Oberwolfach 报告问题汇编

Let $M$ and $N$ be closed $d$-manifolds with $d\geq 5$ and the same normal $2$-type. If both admit metrics of positive scalar curvature, prove that the spaces $R^+(M)$ and $R^+(N)$ are homotopy equivalent.

来源字段：https://doi.org/10.4171/owr/2021/30

07407 | OWR-11568-006

### Hopf Conjecture

O / open | Oberwolfach 报告问题汇编

Does every closed even-dimensional Riemannian manifold with positive sectional curvature have positive Euler characteristic?

来源字段：https://doi.org/10.4171/owr/2012/01

07408 | OWR-12007-019

### Horosphere Geometry Under Quasi-Isometries of Universal Covers

O / open | Oberwolfach 报告问题汇编

In dimension $3$, are the horospheres for the lifted hyperbolic metric $\widetilde g_0$ and another negatively curved metric $\widetilde g$ quasi-isometric? More generally, what does a quasi-isometry between the universal covers say about their horospheres, and under what conditions do the lower and upper polynomial volume-growth exponents $\alpha$ and $\beta$ coincide?

来源字段：https://doi.org/10.4171/owr/2012/35

07409 | AIM-GEOMETRY-0430

### Hurwitz stabilizers and open-book centralizers

O / open | AIM 研讨会问题汇编

Question 1.34. Auroux discussed how, roughly, a certain subgroup of the braid group gives automorphisms of a symplectic Lefschetz pencil, asked whether we can do something like this with open books or contact pencils, and in connection with this explained how Giroux's question (a) above has a positive answer in a certain precise case.

[The following is from the notes of John Etnyre and David Farris, since I wasn't there.]

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

07410 | OWR-13347-011

### Hyperbolic Conformal Boundaries of Poincaré-Einstein Manifolds

S / solved | Oberwolfach 报告问题汇编

Does there exist a complete conformally compactifiable Poincaré-Einstein manifold whose boundary at infinity is a compact connected manifold of constant sectional curvature $-1$?

来源字段：https://doi.org/10.4171/owr/2014/45

07411 | OWR-12007-011

### Hyperbolic Orbifold Realizations of Compact PL Manifolds

O / open | Oberwolfach 报告问题汇编

For every compact PL $n$-manifold $M$ without boundary, does there exist a cocompact lattice $\Gamma\subset O(n,1)$ such that $\mathbb H^n/\Gamma$ is PL-isomorphic to $M$?

来源字段：https://doi.org/10.4171/owr/2012/35

07412 | OWR-1784-004

### Hyperbolic Rank of Affine Buildings

O / open | Oberwolfach 报告问题汇编

For an affine building $X$, determine its hyperbolic rank. In particular, for an affine building of type $\widetilde A_2$, does there exist a quasi-isometric embedding $\mathbb H^2\to X$?

来源字段：https://doi.org/10.4171/owr/2008/03

07413 | OWR-4333-007

### Hyperbolic-Group Boundaries Homeomorphic to Trees of $\mathbb{CP}^{2}$

O / open | Oberwolfach 报告问题汇编

Is the tree of manifolds $X(\mathbb{CP}^{2})$ homeomorphic to the Gromov boundary $\partial\Gamma$ of some hyperbolic group $\Gamma$?

来源字段：https://doi.org/10.4171/owr/2010/20

07414 | AIM-CONVEX_GEOMETRY-0006

### Hyperplane counterexamples and a difference-body symmetry criterion

S / solved | AIM 研讨会问题汇编

**Problem 5.** Suppose \(K\) and \(L\) are convex bodies and \(K\mid S\) is congruent to \(L\mid S\) for every \(S\in G(n,k)\). Does it follow that \(K\) is a translate of \(\pm L\)?

The source cites Gardner's Problem 3.2 and Note 3.1, and records an affirmative answer for \(k=2\) when the projections are related by rotations about the origin, due to Ryabogin.

来源字段：https://aimath.org/pastworkshops/sectionsconvexproblems.pdf

07415 | AIM-GEOMETRY-0332

### Idempotent geometry and the ideal--congruence divide

P / partially_solved | AIM 研讨会问题汇编

The record is item A.12, “Idempotent geometry,” contributed by G. L. Litvinov in cooperation with G. B. Shpiz to the AIM workshop list *Amoebas and tropical geometry*. The source asks a six-part foundational program:

1. Can algebraic geometry be constructed over a class of algebraically closed idempotent semifields, beyond only tropical semifields? 2. Can one define abstract, rather than only affine or projective, varieties over tropical and idempotent semifields? 3. Can regular functions and regular maps be defined so as to obtain a natural category of affine varieties and a correspondence with idempotent semirings of functions? 4. Can equations and ideals be defined so that points and subvarieties correspond to suitable analogues of ideals? 5. What should prime ideals, irreducible varieties, and decomposition into irreducible components mean? 6. Can dequantization produce a natural correspondence, perhaps a functor or “almost functor,” from traditional algebraic varieties to tropical varieties?

This is a recovered statement, not a single yes/no theorem. The isolated “8” is the PDF page number, not mathematics.

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

07416 | OWR-1275-007

### Imaginary-Axis Geodesics in Period-Matrix Loci

O / open | Oberwolfach 报告问题汇编

In the universal Teichmüller space $T(1)$, let $H^+(U)$ denote the period-matrix locus in the Siegel upper half-space associated to the unitary-type subspace $U$. Is the imaginary axis in $H^+(U)$ the image under the period mapping of a Teichmüller geodesic?

来源字段：https://doi.org/10.4171/owr/2006/26

07417 | OWR-15427-018

### Incidence Lower Bounds for Dense Points and Lines

O / open | Oberwolfach 报告问题汇编

Let $P\subset\mathbb R^2$ consist of $n$ points with minimum pairwise distance $1$ and diameter $O(\sqrt n)$, and let $L$ be a family of $n$ pairwise essentially different lines, where essential difference is measured by angular separation $1/n$ or disjoint $1/\sqrt n$-neighborhoods inside $\operatorname{conv}(P)$. Must $$\bigl|\{(p,\ell)\in P\times L:\operatorname{dist}(p,\ell)\leq1/\sqrt n\}\bigr|=\Omega(n^{4/3})$$?

来源字段：https://doi.org/10.4171/owr/2017/19

07418 | OWR-1394-004

### Incidence Problems for Families of Convex Bodies

O / open | Oberwolfach 报告问题汇编

Extend the classical sphere-through-points and hyperplane-through-points incidence problems to convex bodies: determine existence and uniqueness of spheres, balls, or hyperplanes simultaneously tangent to, or slicing prescribed volume fractions from, well-separated families of convex bodies.

来源字段：https://doi.org/10.4171/owr/2006/56

07419 | OWR-13857-002

### Index Bounds for Minimal-Hypersurface Volume

O / open | Oberwolfach 报告问题汇编

Let $M^n$ be a closed minimal hypersurface in a Riemannian manifold $N$ with strictly positive sectional curvature. If $\operatorname{index}(M)=I$, is $\mathcal H^n(M)\leq C(N,I)$? Consequently, must bounded-index sequences have uniformly bounded volume and volume-divergent sequences have index tending to infinity?

来源字段：https://doi.org/10.4171/owr/2015/36

07420 | OWR-1726-002

### Index Bundles for General Foliations

P / partially_solved | Oberwolfach 报告问题汇编

In the theory of foliations, develop a general index theory that incorporates index bundles whenever they exist, extending the usual family-index framework to the foliation case.

来源字段：https://doi.org/10.4171/owr/2007/43

07421 | OPG-36901

### Inequality of the means

P / partially_solved | Open Problem Garden

Question Is is possible to pack $n^n$ rectangular $n$-dimensional boxes each of which has side lengths $a_1,a_2,\ldots,a_n$ inside an $n$-dimensional cube with side length $a_1 + a_2 + \ldots a_n$?

背景链接：http://www.openproblemgarden.org/op/inequality_of_the_means

07422 | AIM-GEOMETRY-0041

### Inertia classification of real two-strand braid varieties

P / partially_solved | AIM 研讨会问题汇编

How can we understand the connected components, topology, etc. of the real points in braid varieties?

来源字段：http://aimpl.org/clusterbraid/9/

07423 | OWR-13106-005

### Infinite Values of the $kappa$-Invariant

O / open | Oberwolfach 报告问题汇编

For a closed manifold $M$, define $\kappa(M)$ to be the least integer $k$ such that, for every $\varepsilon>0$, there is a Riemannian metric $g_\varepsilon$ whose Dirac eigenvalues satisfy $\mu_k(g_\varepsilon)=1$ and $|\mu_i(g_\varepsilon)|<\varepsilon$ for $0\leq i<k$; set $\kappa(M)=\infty$ if no such $k$ exists. Can $\kappa(M)=\infty$ occur?

来源字段：https://doi.org/10.4171/owr/2014/36

07424 | OWR-4128-002

### Infinite-Topology Types of Embedded Minimal Surfaces

O / open | Oberwolfach 报告问题汇编

Which non-compact orientable surfaces of infinite topology occur as topological types of properly embedded minimal surfaces in $\mathbb{R}^3$?

来源字段：https://doi.org/10.4171/owr/2009/45

07425 | AMR-059-0012

### Inflection Points of Projectively Flat Curves

P / partially_solved | AMR 问题清单汇编

Estimate the minimum number of inflection points of a closed curve on a Riemann surface with a projectively flat connection.

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

07426 | OWR-1383-004

### Inherited Curvature Bounds for Convex and Saddle Boundaries

O / open | Oberwolfach 报告问题汇编

In Alexandrov geometry, does the boundary of a convex subset of a space with curvature bounded below inherit the same lower curvature bound? Do saddle surfaces in a space with curvature bounded above inherit the ambient upper curvature bound?

来源字段：https://doi.org/10.4171/owr/2006/45

07427 | GEO-005

### Inscribed Square Problem (Toeplitz Conjecture)

O / open | 独立 / 未分配集合

别名：The Inscribed Square Problem [GEO-034]; Inscribed Square Problem [OPG-757]

Does every simple closed curve in the plane contain all four vertices of some square?

背景链接：https://en.wikipedia.org/wiki/Inscribed_square_problem

07428 | OWR-13106-010

### Instability of Nontrivial Compact Ricci Solitons

O / open | Oberwolfach 报告问题汇编

For every nontrivial compact Ricci soliton, does the second variation of Perelman's $\nu$-entropy have a positive eigenvalue?

来源字段：https://doi.org/10.4171/owr/2014/36

07429 | AMR-038-0008

### Integer-distance point sets

S / solved | AMR 问题清单汇编

Do there exist seven planar points in general position-no three collinear and no four concyclic-such that every pairwise distance is an integer?

背景链接：https://ics.uci.edu/~eppstein/junkyard/open.html

07430 | AIM-GEOMETRY-0099

### Integer-weight latitude reduction and rigidity at the one-half barrier

P / partially_solved | AIM 研讨会问题汇编

Prove there exist spherical $t$-designs on $S^2$ with $(\frac{1}{2}+o(1))t^2$ points.

来源字段：http://aimpl.org/discreteaf/1/

07431 | OWR-2053-003

### Integrable Convex Hamiltonians on Exotic Tori

O / open | Oberwolfach 报告问题汇编

If $\Sigma$ is an exotic $n$-torus, does there exist a completely integrable convex Hamiltonian on its cotangent bundle $T^*\Sigma$?

来源字段：https://doi.org/10.4171/owr/2008/42

07432 | AMR-059-0011

### Integrable Radial Distributions and One-Conformal Flatness

P / partially_solved | AMR 问题清单汇编

On a statistical manifold, let $D$ be the rank-$(n-1)$ distribution orthogonal to the velocity of the $\nabla$-geodesic from a fixed point. If every such distribution is integrable, must the dual statistical manifold be one-conformally flat?

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

07433 | AIM-GEOMETRY-0315

### Integral and mod-p Kirwan maps for weighted projective orbifolds

P / partially_solved | AIM 研讨会问题汇编

For a Hamiltonian presentation of a symplectic toric orbifold, when is the Kirwan restriction map surjective with integral coefficients? What is the answer with \(\mathbb F_p\) coefficients, and how does the answer change when the target is ordinary coarse-space cohomology, Haefliger/Borel orbifold cohomology, or Chen--Ruan cohomology?

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

07434 | AIM-GEOMETRY-0216

## Integral divisibility profiles and a gcd obstruction for minimax levels

P / partially_solved | AIM 研讨会问题汇编

**Question 3.2.1 (Hingston).** Does there exist a metric on \(S^n\) and a class \(X\in H_*(\Lambda S^n;\mathbb Z)\) such that \[ 0<\operatorname{cr}_{\mathbb Z}(mX) <\operatorname{cr}_{\mathbb Z}(X) \] for some \(m\in\mathbb N\)? In the simplest case, can one find a metric on \(S^2\) and \(m\in\mathbb N\) with this strict inequality when \(X\) generates \(H_1(\Lambda S^2;\mathbb Z)\)?

It comes from Section 3.2, “Stability of minimax levels,” of Keith Burns and Vladimir S. Matveev, *Open problems and questions about geodesics*.

<https://aimath.org/pastworkshops/geodesicsproblems.pdf>.

The preceding paragraph in the PDF fixes a compact Riemannian (or Finsler) manifold \(M\), writes \(\Lambda M\) for its free loop space, and sets \[ \Lambda^a M=\{\gamma\in\Lambda M:\operatorname{Length}(\gamma)\leq a\}. \] For \(0\neq X\in H_*(\Lambda M;G)\), its critical level is \[ \operatorname{cr}_G(X) =\inf\{a:X\in\operatorname{im}(H_*(\Lambda^aM;G) \longrightarrow H_*(\Lambda M;G))\}. \] Equivalently it is the infimum, over singular cycles representing \(X\), of the largest loop length on the cycle.

The record then explains the coefficient question. For a coefficient field \(G\), it writes the global mean frequency as \[ \alpha_{g,G} =\lim_{\deg X\to\infty}\frac{\deg X}{\operatorname{cr}_G(X)} \tag{3.1} \] over nonzero \(X\in H_*(\Lambda S^n;G)\), and cites the resonance theorem of Hingston--Rademacher for existence of the relevant asymptotic quantity on spheres. It observes that the nonzero integral homology groups of \(\Lambda S^n\) are \(\mathbb Z\) or \(\mathbb Z_2=\mathbb Z/2\mathbb Z\) (and are torsion-free for odd spheres), then specializes to \(H_k(\Lambda S^n;\mathbb Z)=\mathbb Z X\). For \(j,m\in\mathbb N\), \[ \operatorname{cr}_{\mathbb Z}(jmX) \leq\operatorname{cr}_{\mathbb Z}(mX) \] because every filtered image is an additive subgroup.

Finally, to illustrate a possible coefficient effect, the source supposes that there are real numbers \(a<b<c<d\) for which \[ \operatorname{cr}_{\mathbb Z}(mX)= \begin{cases} d,&\gcd(m,3)=\gcd(m,7)=1,\\ c,&3\mid m\text{ and }\gcd(m,7)=1,\\ b,&7\mid m\text{ and }\gcd(m,3)=1,\\ a,&21\mid m. \end{cases} \tag{1.1} \]

The source itself alternates between \(M\) and \(S^n\); in the question \(M=S^n\).

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07435 | AIM-GEOMETRY-0313

## Integral Kirwan surjectivity after stabilizer localization

P / partially_solved | AIM 研讨会问题汇编

**Problem 10.1 (R. Goldin, S. Tolman, J. Weitsman).** Describe all the conditions under which Kirwan surjectivity holds for symplectic reductions over the integers. What are the most general conditions under which it holds?

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

07436 | AIM-GEOMETRY-0297

## Intrinsic contact moment maps and a realizability sieve

P / partially_solved | AIM 研讨会问题汇编

We can model the local structure on \(M\) using abstract moment maps. Is there an analogous statement for contact moment maps? Is there a theory of abstract contact moment maps?

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

07437 | OWR-1056-002

## Intrinsic Definitions of Asymptotically Hyperbolic Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Can conformally compact manifolds and asymptotically hyperbolic manifolds be defined intrinsically, and does a hyperbolic rigidity theorem remain valid in the resulting intrinsic category?

来源字段：https://doi.org/10.4171/owr/2005/35

07438 | OWR-14298170-017

## Intrinsic-Flat and Sobolev Limit Spaces

O / open | Oberwolfach 报告问题汇编

Under the hypotheses of the IAS minimum-area compactness conjecture, if $g_j\to g_\infty$ in $W^{1,p}$, determine whether the intrinsic-flat limit is the metric completion of the nonsingular part of $(M,g_\infty)$; if not, characterize their relation and the consequences of distributional nonnegative scalar curvature.

来源字段：https://doi.org/10.4171/owr/2024/9

07439 | OWR-14219-009

## Intrinsic-Volume Deficits of Random Spherical Polytopes

P / partially_solved | Oberwolfach 报告问题汇编

Let $K$ be a spherically convex body in an open hemisphere and $P_n$ the spherical convex hull of $n$ independent uniform random points of $K$. Is $\mathbb E[U_j(K)-U_j(P_n)]$ of order $n^{-(d+1-j)}$ as $n\to\infty$ for every $j=1,\ldots,d$?

来源字段：https://doi.org/10.4171/owr/2016/11

07440 | OWR-4199-001

### Invariant Homogeneous Valuations on Convex Bodies

O / open | Oberwolfach 报告问题汇编

Characterize all upper-semicontinuous, $\mathrm{SL}(n)$-invariant, homogeneous valuations $\Phi:K_0^n\to\mathbb{R}_+\cup\{\infty\}$, where $K_0^n$ denotes convex bodies containing the origin in their interiors.

来源字段：https://doi.org/10.4171/owr/2010/04

07441 | OWR-13106-014

### Invariant Positive-Scalar-Curvature Metrics

P / partially_solved | Oberwolfach 报告问题汇编

Let a compact connected Lie group $G$ act effectively on a closed manifold $M$. Must $M$ admit a $G$-invariant Riemannian metric of positive scalar curvature?

来源字段：https://doi.org/10.4171/owr/2014/36

07442 | OWR-4075-002

### Invariant Sets of Simplex Piecewise Translations

O / open | Oberwolfach 报告问题汇编

Can the invariant fundamental set for piecewise translations associated with simplices be characterized in the unresolved obtuse and unrestricted-translation cases, and is it connected and a possibly non-convex polytope?

来源字段：https://doi.org/10.4171/owr/2009/32

07443 | OWR-13951-005

### Invariant Valuations for Noncompact Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a noncompact transformation group $G$, such as $O(p,q)$, acting on convex bodies, what generalized valuation framework and kinematic formulas replace the compact-group theory?

来源字段：https://doi.org/10.4171/owr/2015/56

07444 | OWR-12697711-006

### Irreducibility of Exposed-Point Varieties of Generic Discotopes

O / open | Oberwolfach 报告问题汇编

Let $D=D_1+\cdots+D_N$ be a generic discotope with $\dim D_i\geq 2$, and let $E$ be the Zariski closure of its exposed points. Prove that $E$ is irreducible.

来源字段：https://doi.org/10.4171/owr/2023/15

07445 | OWR-13494-010

### Irreducible Curves with Strongly Negative Harbourne Constants

P / partially_solved | Oberwolfach 报告问题汇编

For a reduced plane curve $C \subset \mathbb{P}^2_{\mathbb{C}}$, let $\widetilde{\mathbb{P}^2} \to \mathbb{P}^2$ be the blow-up of the singular points of $C$, let $\widetilde{C}$ be its strict transform, and let $H(C) = (\widetilde{C})^2 / s$ be the Harbourne constant (where $s$ is the number of singular points blown up). Do there exist irreducible complex plane curves $C \subset \mathbb{P}^2_{\mathbb{C}}$ with $H(C) \leq -2$, and what analogous bounded-negativity questions and constructions arise on other algebraic surfaces?

来源字段：https://doi.org/10.4171/owr/2015/9

07446 | AMR-071-0083

### Is the upper bound of a closed knot's minimum ropelength linear to its crossing number

P / partially_solved | AMR 问题清单汇编

Is the upper bound of a closed knot's minimum ropelength linear to its crossing number?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

07447 | AMR-071-0084

### Is there a general expression for how much the ends of a long rope of radius 1 get closer when a tight open knot is tied into it

O / open | AMR 问题清单汇编

Is there a general expression for how much the ends of a long rope of radius 1 get closer when a tight open knot is tied into it?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

07448 | AMR-071-0081

### Is there a general expression for the minimum ropelength of an arbitrary closed knot

O / open | AMR 问题清单汇编

Is there a general expression for the minimum ropelength of an arbitrary closed knot?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

07449 | AMR-071-0087

### Is there a non-convex polyhedron without self-intersections with more than seven faces, all of which share an edge with each other

O / open | AMR 问题清单汇编

Is there a non-convex polyhedron without self-intersections with more than seven faces, all of which share an edge with each other?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

07450 | OWR-14299092-006

### Isoperimetric Inequalities Between Total Q-Curvatures

P / partially_solved | Oberwolfach 报告问题汇编

For $0<\ell<k<n/2$, establish the sharp isoperimetric comparison between the total $k$-th and $\ell$-th $Q$-curvatures of metrics $g=u^{4/(n-2k)}g_{S^n}$ with positive $k$-th $Q$-curvature, and prove that equality holds exactly when $u(\xi)=a(1+b\cdot\xi)^{-(n-2k)/2}$ with $a\in\mathbb{R}$ and $b\in B^{n+1}$.

来源字段：https://doi.org/10.4171/owr/2025/11

07451 | AMR-067-0018

### Isoperimetric Problem in $\mathbb{C} P^2$

P / partially_solved | AMR 问题清单汇编

Prove that geodesic spheres provide the least-perimeter way to enclose prescribed volume in $CP^2$.

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

07452 | OWR-13857-001

### Isoperimetric Regions of Fixed Volume in General Spaces

P / partially_solved | Oberwolfach 报告问题汇编

For a space X and fixed volume v, what is the minimum area needed to enclose v, does an optimal shape exist, and if so can it be described?

来源字段：https://doi.org/10.4171/owr/2015/36

07453 | OWR-3389-008

### Isoperimetric-Defect Corrections to Faber-Krahn Bounds

S / solved | Oberwolfach 报告问题汇编

For every planar convex domain $\Omega$ of area $A$ and perimeter $L$, does there exist a universal constant $K>0$ such that $$\lambda_1(\Omega)\geq\frac{\pi j_{0,1}^2}{A}+K\frac{L^2-4\pi A}{A^2}?$$ Or must the optimal coefficient of the isoperimetric-defect term be zero?

来源字段：https://doi.org/10.4171/owr/2009/06

07454 | AMR-059-0008

### Isothermal Coordinates for Affine Minimal Surfaces

P / partially_solved | AMR 问题清单汇编

Do affine minimal surfaces admit isothermal coordinates with respect to their affine fundamental form, analogously to minimal surfaces in Euclidean three-space?

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

07455 | AMR-065-0019

### J. Marklof

P / partially_solved | AMR 问题清单汇编

Determine all $SL_d(\mathbb{R})$--invariant Borel probability measures on $\mathbf{Cl}(\mathbb{R}^d)$ and similarly for the $ASL_d(\mathbb{R})$ action.

背景链接：https://arxiv.org/abs/1604.06280

07456 | OWR-761-011

### Jacobi Fields and Neck Sizes of Coplanar Unduloids

P / partially_solved | Oberwolfach 报告问题汇编

For coplanar $k$-unduloids, are the upper bounds on the dimension of the space of $L^2$ Jacobi fields sharp? In particular, does a nonzero $L^2$ Jacobi field exist on a triunduloid when $n_1+n_2+n_3=2\pi$ or when $n_i+n_j=n_k$? Can these techniques be extended to other constant-mean-curvature surfaces to bound neck sizes or the dimension of their Jacobi-field spaces? Is every tempered Jacobi field on a constant-mean-curvature surface integrable?

来源字段：https://doi.org/10.4171/owr/2004/47

07457 | GEO-008

### Kakeya Needle Problem

P / partially_solved | 独立 / 未分配集合

What is the minimum area of a region in the plane in which a unit line segment can be continuously rotated through 360 degrees?

背景链接：https://en.wikipedia.org/wiki/Kakeya_set

07458 | GEO-025

### Kalai's $3^d$ Conjecture

O / open | 独立 / 未分配集合

别名：Kalai's 3^d Conjecture [GEOM-022]

Does every centrally symmetric $d$-dimensional polytope have at least $3^d$ faces?

背景链接：https://arxiv.org/abs/1711.09310

07459 | OWR-8415364-006

### Kannan-Lovász-Simonovits Isoperimetric Conjecture

O / open | Oberwolfach 报告问题汇编

别名：Kannan-Lovasz-Simonovits Spectral Gap Conjecture [OWR-8415364-001]; KLS Isoperimetric Coefficient Bound for Log-Concave Measures [OWR-8415364-005]; KLS Isoperimetric Coefficient Bound for Log-Concave Measures [OWR-8415364-007]

Is there a universal constant $c>0$ such that every isotropic log-concave probability density $p$ satisfies $\psi(p)\geq c/\sqrt{\rho(p)}$, and hence has an isoperimetric coefficient bounded below by a universal positive constant?

来源字段：https://doi.org/10.4171/owr/2021/59

07460 | AIM-GEOMETRY-0228

### Katok entropy rigidity and an exact time-change obstruction

P / partially_solved | AIM 研讨会问题汇编

“Flaminio showed that Conjecture 5.3.1 holds for deformations of constant curvature metrics [Flaminio1995].”

For a smooth closed negatively curved Riemannian manifold \((M^n,g)\), the entropy of its geodesic flow with respect to normalized Liouville measure equals its topological entropy if and only if \(\nabla R_g=0\).

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07461 | OWR-2654829-002

### Kähler Criteria from Nonintegrable Bundle Spectral Theory

P / partially_solved | Oberwolfach 报告问题汇编

For compact complex manifolds of complex dimension greater than two, develop an analogue of the Lamari-Buchdahl criteria for the existence of Kähler metrics using the spectral theory of nonintegrable complex vector bundles.

来源字段：https://doi.org/10.4171/owr/2020/33

07462 | AIM-GEOMETRY-0370

### Kernel moments, renormalized volume, and a fixed-boundary obstruction for Q3

P / partially_solved | AIM 研讨会问题汇编

**Problem 12.** What sort of information is encoded by \(Q_3\) and/or \(\int Q_3\)?

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

07463 | AIM-GEOMETRY-0292

### Kernel, cokernel, and a top-degree obstruction for contact Kirwan maps

P / partially_solved | AIM 研讨会问题汇编

**Recovered Question 3.1 (E. Lerman).** Let \[ S^1\curvearrowright S^3\subset\mathbb C^2,\qquad \lambda\cdot(z_1,z_2)=(\lambda z_1,\lambda^{-1}z_2). \] The action is free, so \(H^*_{S^1}(S^3)\) is the ordinary cohomology of \(S^3/S^1\cong\mathbb P^1\). For the standard invariant contact form, the contact moment map is \[ \mu(z_1,z_2)=|z_1|^2-|z_2|^2, \] and the contact quotient is \(\mu^{-1}(0)/S^1\cong S^1\). Thus the restriction map cannot be surjective. What are its kernel and cokernel?

**Recovered Comment 3.2 (C. Boyer).** Determine conditions under which a contact Kirwan surjectivity theorem might hold, possibly for a class of 3-Sasakian spaces and perhaps only through a dimension range.

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

07464 | OWR-12333-001

### Kinematic Formulas for DC-Boundary Sets

O / open | Oberwolfach 报告问题汇编

Do kinematic formulas hold for sets whose boundaries are described by differences of convex functions and for which normal cycles can be constructed?

来源字段：https://doi.org/10.4171/owr/2013/05

07465 | OWR-10252928-005

### Kissing Number of Regular Tetrahedra

O / open | Oberwolfach 报告问题汇编

What is the maximum number of pairwise interior-disjoint regular tetrahedra in three-dimensional Euclidean space that can share a common vertex?

来源字段：https://doi.org/10.4171/owr/2022/20

07466 | AMR-037-0018

### Klee's measure problem

P / partially_solved | AMR 问题清单汇编

Determine the optimal complexity of computing the volume of the union of axis-aligned boxes in fixed dimension at least three. In particular, is there a near-linear three-dimensional algorithm or an $\Omega(n^{3/2})$ lower bound, and can fat or equal-sized boxes be handled faster?

背景链接：https://jeffe.cs.illinois.edu/open/

07467 | AIM-GEOMETRY-0218

### Klingenberg's conjecture after the 2025 hyperbolicity advance

P / partially_solved | AIM 研讨会问题汇编

Heintze took two cusped hyperbolic 3-manifolds, cut off the cusps, glued the two pieces together along their boundary tori, and then smoothed out the metric to obtain a manifold with nonpositive curvature.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07468 | OPG-2089

### Kneser-Poulsen conjecture

P / partially_solved | Open Problem Garden

Conjecture If a finite set of unit balls in $\mathbb{R}^n$ is rearranged so that the distance between each pair of centers does not decrease, then the volume of the union of the balls does not decrease.

背景链接：http://www.openproblemgarden.org/op/kneser_poulsen_conjecture

07469 | AIM-GEOMETRY-0221

### Known genuine examples and a contact-symplectic surface obstruction

S / solved | AIM 研讨会问题汇编

If \(\phi^t:SM\to SM\) is the geodesic flow, it asks whether there is a \(D\phi^t\)-invariant splitting \[ T(SM)=E^s\oplus E^c\oplus E^u \] and constants \(C\geq 1\) and \(\lambda>\mu>0\) such that, for \(t\geq0\), \[ \|D\phi^t\xi\|\leq Ce^{-\lambda t}\|\xi\|\quad(\xi\in E^s), \qquad \|D\phi^{-t}\eta\|\leq Ce^{-\lambda t}\|\eta\|\quad(\eta\in E^u), \tag{4.1} \] and, for every \(t\in\mathbb R\) and \(\zeta\in E^c\), \[ C^{-1}e^{-\mu |t|}\|\zeta\| \leq \|D\phi^t\zeta\| \leq Ce^{\mu |t|}\|\zeta\|. \tag{4.2} \] The question is: **is there an example of a geodesic flow satisfying this condition?** The source then observes that an Anosov geodesic flow is a degenerate positive answer and says that genuine examples were constructed by Carneiro and Pujals.

1. The words beginning “5. Negative curvature and hyperbolicity…” and the marked-length-spectrum paragraph belong to the next section of the PDF, not to Question 4.1.4. They are excluded here. 2. The workshop list describes the Carneiro--Pujals starting metric as a “higher rank locally symmetric space.” Their primary paper instead starts with a **compact locally symmetric manifold of nonconstant negative sectional curvature** (hence a rank-one noncompact-type model). Indeed, the same paper proves that a compact locally symmetric space of nonpositive curvature whose geodesic flow is partially hyperbolic must already have an Anosov geodesic flow, and that any nonpositively curved partially hyperbolic example has rank one. Thus “higher rank” in the workshop sentence should not be used to describe their construction.

Throughout, “genuine” means partially hyperbolic but not Anosov, and the stable and unstable bundles are required to be nonzero. This convention is essential; see Section 8.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07470 | AMR-065-0015

### L. Sadun - Problem

O / open | AMR 问题清单汇编

Classify tilings having a geometric property such as bounded-displacement equivalence (BD), bi-Lipschitz equivalence (BL), or linear repetitivity (LR), for which every tiling that is MLD, topologically conjugate, or homeomorphic to it has the same property.

背景链接：https://arxiv.org/abs/1604.06280

07471 | AMR-065-0016

### L. Sadun - Problem

O / open | AMR 问题清单汇编

Develop and study new geometric properties, analogous but not identical to BD, BL, etc., that are invariant under MLD, topological conjugacy, or homeomorphism.

背景链接：https://arxiv.org/abs/1604.06280

07472 | AMR-065-0017

### L. Sadun - Problem

P / partially_solved | AMR 问题清单汇编

Find matching rules in dimension two or three satisfying both: (A) every tile-type discrepancy in a finite patch is bounded by a constant times the boundary measure; and (B) the same bound holds for every patch satisfying the matching rules.

背景链接：https://arxiv.org/abs/1604.06280

07473 | AMR-065-0018

### L. Sadun - Problem

P / partially_solved | AMR 问题清单汇编

Find matching rules in dimension two satisfying condition (A): for every tile type $\mathfrak t$ and finite region $\mathcal R$, the discrepancy $|N_{\mathfrak t}(\mathcal R)-d(\mathfrak t)\operatorname{vol}(\mathcal R)|$ is bounded by $c_{\mathfrak t}|\partial\mathcal R|$.

背景链接：https://arxiv.org/abs/1604.06280

07474 | OWR-14297736-025

### Landau-Level Multiplicities and Sign-Changing Spherical Interfaces

O / open | Oberwolfach 报告问题汇编

Determine the multiplicities of Landau levels under delta-interactions supported on general finite $C^{1,1}$ curves. Separately, characterize interfaces on $S^2$ for which a sign-changing elliptic problem has no eigenvalues $-1/2+i\eta$ with nonzero real $\eta$ when the material contrast is below $-1$.

来源字段：https://doi.org/10.4171/owr/2023/36

07475 | OWR-13346-004

### Landmark Kernel Approximation on Algebraic Manifolds

O / open | Oberwolfach 报告问题汇编

Let $\mathcal M\subset\mathbb R^n$ be an algebraic manifold of dimension $D$ and let $k$ be a degree-$d$ polynomial kernel. For fixed $x,y\in\mathcal M$ and i.i.d. standard-normal landmark points $z_1,\ldots,z_m$ with $m\geq\lambda Dd\log n$, prove an asymptotic bound of compressed-sensing type for $k(x,y)-K(x,Z)K(Z,Z)^{-1}K(Z,y)$, such as $\Pr\bigl(\lvert k(x,y)-K(x,Z)K(Z,Z)^{-1}K(Z,y)\rvert\geq\varepsilon\bigr)\leq C\varepsilon^{-c\lambda}$.

来源字段：https://doi.org/10.4171/owr/2014/44

07476 | AMR-059-0004

### Large Deviations and Dual Connections

O / open | AMR 问题清单汇编

Investigate the relationship between the large-deviation principle, whose rate functions are relative entropies, and dual-connection structures in information geometry.

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

07477 | OWR-4798-010

### Large Inscribed Lattice Squares in Polyomino Boundaries

O / open | Oberwolfach 报告问题汇编

Let $P$ be a polyomino whose boundary is a closed Jordan curve $J$ and whose unit-square vertices lie in the lattice. If the greatest axis-parallel open square in the bounded component of $\mathbb{R}^{2}\setminus J$ has side length $s$, must $J$ contain four lattice points forming a square of side at least $s/\sqrt{2}$ when $s$ is even, and at least $\sqrt{m^{2}+(m+1)^{2}}$ when $s=2m+1$?

来源字段：https://doi.org/10.4171/owr/2011/08

07478 | OWR-14299586-002

### Large-Genus Hyperbolic Surfaces Above the One-Quarter Spectral Threshold

O / open | Oberwolfach 报告问题汇编

Do there exist closed hyperbolic surfaces of arbitrarily large genus whose first positive Laplace eigenvalue is greater than $1/4$?

来源字段：https://doi.org/10.4171/owr/2025/57

07479 | AIM-CONVEX_GEOMETRY-0041

### Lattice facet-count constraints and nonsymmetric projection reconstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 13 (S. Robins), Minkowski's theorem in \(\mathbb Z^d\).** Given finitely many vectors \(\{\vec n_i\}\subset\mathbb Z^d\) and integers \(\alpha_i\in\mathbb Z\), is there a polytope such that the \(\vec n_i\) are normals to its facets and the number of integer points on the \(i\)-th facet is \(\alpha_i\)? Are extra constraints needed?

**Problem 14.** Which projection and/or section data are needed to determine nonsymmetric convex bodies? Several known cases are listed: volumes and centers of gravity of sections through a fixed interior point; mean widths and Steiner points; brightness and illumination. The questions are: (a) (Schneider) Do the areas of projections together with their centers of gravity determine a body? (b) What is the dual version of the third known case (brightness plus illumination)?

来源字段：https://aimath.org/WWN/fourierconvex/fourierconvex.pdf

07480 | OWR-14299583-004

### Lattice Packing Conjecture for Origin-Symmetric Convex Bodies

O / open | Oberwolfach 报告问题汇编

Prove that there is a universal constant $c>0$ such that every origin-symmetric convex body $K\subset\mathbb{R}^n$ with $\operatorname{Vol}_n(K)=cn^2$ admits a covolume-one lattice $L\subset\mathbb{R}^n$ satisfying $L\cap K=\{0\}$.

来源字段：https://doi.org/10.4171/owr/2025/54

07481 | AIM-GEOMETRY-0079

### Lattice Riesz polarization and the quantitative covering limit

P / partially_solved | AIM 研讨会问题汇编

Which lattices $\Lambda$ in $\mathbb R^d$ of determinant $1$ maximize \[ \min_{y \in \mathbb R^d \setminus \Lambda} \sum_{x \in \Lambda} \frac{1}{|x-y|^s}? \] For $s \to \infty$ this is the covering problem. On $S^{n-1}$ this is called polarization.

来源字段：http://aimpl.org/discreteaf/1/

07482 | AIM-GEOMETRY-0193

### Leading asymptotics for polygonal-linkage chamber orbits

P / partially_solved | AIM 研讨会问题汇编

**6. (Farber) Question:** Find asymptotic behaviour of \(C_n\) (the number of orbits of cham-bers in the case of polygonal linkages).

来源字段：https://aimath.org/pastworkshops/linkagesproblems.pdf

07483 | OWR-761-009

### Least Dimension for Curvature-Finiteness Counterexamples

O / open | Oberwolfach 报告问题汇编

What is the least dimension $n$ for which there are infinitely many pairwise homotopy-equivalent but mutually nonhomeomorphic closed simply connected $n$-manifolds admitting Riemannian metrics with $0\leq\operatorname{sec}\leq1$, positive Ricci curvature, and diameter bounded by a constant depending only on $n$?

来源字段：https://doi.org/10.4171/owr/2004/47

07484 | AMR-058-0025

### Least Soap Films on Tetrahedral and Open-Book Frames

P / partially_solved | AMR 问题清单汇编

Is the cone over the regular tetrahedron the smallest soap film having the entire tetrahedral frame as boundary? For a frame of two rectangles sharing an edge with small exterior dihedral angle, is the obvious soap film minimizing?

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

07485 | AMR-071-0072

### Lebesgue's universal covering problem on the minimum-area convex shape in the plane that can cover any shape of diameter one

P / partially_solved | AMR 问题清单汇编

Lebesgue's universal covering problem on the minimum-area convex shape in the plane that can cover any shape of diameter one

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

07486 | AIM-GEOMETRY-0434

### Lee charge, weighted area, and holomorphic curves in exact l.c.s. manifolds

P / partially_solved | AIM 研讨会问题汇编

What can be said about \(J\)-holomorphic curves for an almost complex structure compatible with a locally conformal symplectic form? Can those curves be used to study l.c.s. geometry?

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

07487 | OWR-14299288-005

### Legendrian Augmentations and Matroid Realizations

O / open | Oberwolfach 报告问题汇编

For every matroid $M$, construct a Legendrian $\Lambda_M$ whose augmentations correspond naturally to realizations of $M$, or characterize precisely which matroids admit such a Legendrian model.

来源字段：https://doi.org/10.4171/owr/2025/19

07488 | AIM-GEOMETRY-0243

### Length rigidity in a free homotopy class

S / solved | AIM 研讨会问题汇编

**Recovered Question 7.2.1.** Let \(M\) be a connected complete Riemannian manifold without conjugate points. If two nonconstant closed geodesics are freely homotopic, must their lengths be equal?

Matveev's problem list, contains the following question and then a transition paragraph:

Inspection of page 16 of the displayed journal pagination (page 15 of the PDF text indexing) confirms this wording. The singular phrase “no conjugate point” means the standard condition “without conjugate points.” The subsequent paragraph beginning “One can still hope for a version of the theorem...” is contextual prose. Its latter part introduces the Euclidean parallel-postulate problem that is separately labeled Question 7.2.2 in the source. It imposes no extra hypothesis on Question 7.2.1.

The natural meaning of “homotopic” here is **freely homotopic**: the closed curves need not have a common basepoint. This agrees with the source's use of the free loop space and with the later published formulation. A based-homotopy reading is a special case. The standard setting of Section 7 is a smooth complete Riemannian manifold without conjugate points; in particular, a closed manifold is included. The section introduction explicitly recalls that the universal cover is homeomorphic to Euclidean space and that any two of its points are joined by a unique geodesic.

Thus the recovered problem is:

The answer is yes. If “closed geodesic” is regarded as an unoriented geometric curve, the same answer holds when one loop is freely homotopic to the orientation reversal of the other.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07489 | OWR-4330-003

### Length Spectra of Periodic Convex Billiard Trajectories

O / open | Oberwolfach 报告问题汇编

For a bounded strictly convex smooth planar domain $\Omega$, determine the length spectrum of periodic billiard trajectories, including the asymptotic behavior of $T_{m,n}=\sup L(\Gamma(m,n))$ and $t_{m,n}=\inf L(\Gamma(m,n))$ as $n\to\infty$.

来源字段：https://doi.org/10.4171/owr/2010/17

07490 | OWR-4535-015

### Length-Convex Hulls in Teichmüller Space

O / open | Oberwolfach 报告问题汇编

For length-convex sets in Teichmüller space defined from sublevel sets of curve-length functions, what is the length-convex hull of two points $\{p,q\}$?

来源字段：https://doi.org/10.4171/owr/2010/53

07491 | OWR-13106-007

### Lichnerowicz Conjecture for Conformal Isometry Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $(M,g)$ be a compact Riemannian $n$-manifold, $n\geq 3$, that is not conformal to the round sphere. Does its conformal class contain a constant-scalar-curvature metric $\widetilde g$ satisfying $\operatorname{Isom}(M,\widetilde g)=\operatorname{Conf}(M,g)$?

来源字段：https://doi.org/10.4171/owr/2014/36

07492 | OWR-17135-008

### Lifting Hypersimplicial Subdivisions to Zonotopal Tilings

P / partially_solved | Oberwolfach 报告问题汇编

For which triples $(n,d,k)$ does every hypersimplicial subdivision of the cyclic polytope $C_{n,d}$ extend to a zonotopal tiling of the cyclic zonotope $Z(C_{n,d})$?

来源字段：https://doi.org/10.4171/owr/2019/39

07493 | OWR-17135-018

### Limit Rays and Fatness of Polytope Face Vector Cones

O / open | Oberwolfach 报告问题汇编

For $d>4$, is every ray of $\operatorname{Cone}_{f(\sigma^d)}(f(\mathcal P^d))$ a limit ray? In dimension $4$, is the fatness ratio $$(f_1+f_4)/(f_0+f_3)$$ bounded above by an absolute constant?

来源字段：https://doi.org/10.4171/owr/2019/39

07494 | OWR-13110-021

### Line Discrepancy Centers and Orthic-Triangle Coverings

O / open | Oberwolfach 报告问题汇编

Two independent questions are posed. First, for every $\alpha>0$, does there exist $\beta<\alpha$ such that every two-colored $n$-point set $P\subset\mathbb{R}^2$ with line discrepancy at most $\alpha n$ has a point $q$ through which every line has discrepancy at most $\beta n$? Second, if an acute triangle $T$ is covered by acute triangles $S_1,\ldots,S_N$, must $\xi(T)\leq\sum_{i=1}^N\xi(S_i)$, where $\xi$ denotes the perimeter of the orthic triangle?

来源字段：https://doi.org/10.4171/owr/2014/40

07495 | AIM-GEOMETRY-0185

### Line splitting: literal counterexamples and a quotient criterion

P / partially_solved | AIM 研讨会问题汇编

Do complete shrinking solitons split off a line?

来源字段：http://aimpl.org/flowriemannian/6/

07496 | OWR-15427-016

### Linear Crossing Families in Dense Point Sets

P / partially_solved | Oberwolfach 报告问题汇编

Let $P\subset\mathbb R^2$ consist of $n$ points with minimum pairwise distance $1$ and diameter $O(\sqrt n)$. Must $P$ contain a crossing family of size $\Omega(n)$?

来源字段：https://doi.org/10.4171/owr/2017/19

07497 | OWR-13498-004

### Linear Sum-of-Squares Representations on Grassmann Orbitopes

O / open | Oberwolfach 报告问题汇编

For every Grassmann orbitope $G_{k,n}$, is every calibration representable as a sum of squares of linear polynomials relative to the vanishing ideal $I_{k,n}$? Equivalently, if $\delta-\ell$ is the supporting nonnegative function on $G_{k,n}$, does there exist a representation $\delta-\ell(p)=\sum_i h_i(p)^2$ with all $h_i$ linear modulo $I_{k,n}$?

来源字段：https://doi.org/10.4171/owr/2015/13

07498 | OWR-822-002

### Linear-Programming Sharpness and Universal Lattice Optimality

P / partially_solved | Oberwolfach 报告问题汇编

Does the linear-programming auxiliary-function method prove universal optimality for a finite spherical configuration exactly when the linear-programming bound for spherical codes is sharp for that configuration; and does its Euclidean analogue prove universal optimality for the hexagonal lattice, the $E_8$ root lattice, and the Leech lattice?

来源字段：https://doi.org/10.4171/owr/2005/17

07499 | OWR-3385-009

### Linked Skeletons of Convex Four-Polytopes

O / open | Oberwolfach 报告问题汇编

Does a convex four-polytope $P$ exist whose one-skeleton is linked in $\partial P\cong S^3$ but admits another embedding that is not linked? Here an embedding is linked if it contains two disjoint linked cycles.

来源字段：https://doi.org/10.4171/owr/2009/02

07500 | AIM-GEOMETRY-0045

### Liouville equilibrium and a cusp-expansion pressure test

P / partially_solved | AIM 研讨会问题汇编

What can be said about equilibrium states for the geometric potential associated to geodesic flows on finite volume, complete surfaces? More generally, manifolds?

来源字段：http://aimpl.org/equibdynsysgeom/1/

07501 | OWR-4807-006

### Lipschitz Equivalence of Dust-Like Self-Similar Sets

O / open | Oberwolfach 报告问题汇编

Let $\rho=(\rho_1,\ldots,\rho_m)\in(0,1)^m$ satisfy $\sum_{j=1}^m\rho_j^d<1$, and let $D(\rho)$ denote any dust-like self-similar attractor in $\mathbb{R}^d$ with contraction ratios $\rho$. Given contraction vectors $\rho$ and $\tau$, characterize nontrivial necessary and sufficient conditions for $D(\rho)$ and $D(\tau)$ to be Lipschitz equivalent.

来源字段：https://doi.org/10.4171/owr/2011/17

07502 | OWR-12007-021

### Lipschitz Realization of Surface Homotopy Equivalences

P / partially_solved | Oberwolfach 报告问题汇编

For compact negatively curved surfaces $M_1$ and $M_2$, does Thurston's length-spectrum criterion imply that every homotopy equivalence $f:M_1\to M_2$ is homotopic to an $L$-Lipschitz homeomorphism?

来源字段：https://doi.org/10.4171/owr/2012/35

07503 | AIM-GEOMETRY-0363

### Literal quotient is two-dimensional in the four-dimensional parity-even model

P / partially_solved | AIM 研讨会问题汇编

Problem 6: Is \(\mathcal L/(\mathcal L'+\mathcal L'')\) one-dimensional and generated by the class of \(Q\)?

The exact AIM record is:

The notation is recovered from the immediately preceding Problems 5a--5b and from Branson--Gover's workshop notes. On a closed even-dimensional Riemannian manifold, write a critical-weight local scalar invariant as the density \[ \mathbf S_g=S_g\,dv_g. \] Problem 5b literally calls the space of such local invariants \(\mathcal L\). The subspace \(\mathcal L'\) consists of those \(\mathbf S\) having a **local conformal primitive**: there is a critical-weight local density \(\mathbf F\) such that, for every background metric and every \(\omega\), \[ \left.\frac d{dt}\right|_{0}\int_M\mathbf F_{e^{2t\omega}g} =\int_M\omega\,\mathbf S_g. \tag{1} \] The space \(\mathcal L''\) consists of pointwise local conformal invariant densities. The source's displayed equation in Problem 5a suppresses the integral against \(\omega\); (1) is the unambiguous formula in the cited Branson--Gover notes.

There is a consequential ambiguity. The same workshop document later asks, in Problem 30, for the corresponding decomposition only for natural \(n\)-forms \(\mathbf S\) whose integrals are conformally invariant. Branson's precise invariant-theoretic treatment also uses the smaller formally-self-adjoint space \(\mathcal X_{\rm FSA}\). Thus this report treats separately:

1. the literal Problem 5b reading, where \(\mathcal L\) is the full critical-weight local-invariant space; and 2. the conformal-index/formally-self-adjoint reading that appears to be the intended conjecture.

All statements below concern densities, not bare scalars. The calculation is fixed-dimension, parity-even, polynomial invariant theory in dimension four. It does not silently assert a stable-dimensional or non-polynomial classification.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

07504 | AIM-GEOMETRY-0015

### Local acyclicity and the residual deep locus of plabic cluster varieties

P / partially_solved | AIM 研讨会问题汇编

When is the plabic cluster algebra locally acyclic? If it is not locally acyclic, is it still a polynomial count and can we say anything about a mixed Hodge structure?

来源字段：http://aimpl.org/clusterbraid/1/

07505 | AIM-GEOMETRY-0147

### Local Airy recursion and handle recurrences for type A1 Verlinde theories

P / partially_solved | AIM 研讨会问题汇编

Is there a topological recursion formula for the \(SL(2,\mathbb C)\)-Verlinde's formula?

来源字段：http://aimpl.org/spectralhiggs/1/

07506 | AMR-099-0062

### Local metric homogeneity forcing periodic triangulations

O / open | AMR 问题清单汇编

Let the Euclidean plane or hyperbolic plane have a triangulation whose triangles have diameter at most $r$. Suppose that for every pair of radius-$r$ metric balls centered at vertices, an ambient isometry maps one ball to the other while respecting the triangulation. Must the triangulation be periodic?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=7

07507 | OWR-12007-006

### Local Middle-Dimensional Symplectic Nonsqueezing

O / open | Oberwolfach 报告问题汇编

Let $P_k$ be orthogonal projection onto the standard $\mathbb R^{2k}\subset\mathbb R^{2n}$. Two local middle-dimensional nonsqueezing questions remain open. First, for every symplectomorphism $\phi:\mathbb R^{2n}\to\mathbb R^{2n}$ and every $x\in\mathbb R^{2n}$, is

$$\operatorname{vol}_{2k}\bigl(P_k\phi(B_r(x))\bigr)\geq\omega_{2k}r^{2k}$$

for all sufficiently small $r>0$? Second, if $\{\phi_t\}_{t\in[0,1]}$ is a smooth family of symplectomorphisms with $\phi_0$ linear, is

$$\operatorname{vol}_{2k}\bigl(P_k\phi_t(B_1)\bigr)\geq\omega_{2k}$$

for all sufficiently small $t\geq0$?

来源字段：https://doi.org/10.4171/owr/2012/35

07508 | AIM-GEOMETRY-0390

### Local primitives versus divergence terms

P / partially_solved | AIM 研讨会问题汇编

The following conjecture would be enough to answer the previous problem.

Problem 30: If S is a natural n-form and ∫ S is conformally invariant, then

S = const · Q + L + G,

where L is a local conformal invariant and G has a local conformal primitive. That is, there is a local invariant F for which the conformal variation of ∫ F is ∫ ωG.

Problem 29 asks whether a Polyakov-type determinant formula known in dimensions \(2,4,6\) holds in higher even dimensions. A footnote in the official PDF says explicitly that Problems 29--32 are conjectures. Thus Problem 30 is a conjectural assertion, not a question despite its label “Problem.”

The source uses boldface \(n\)-forms; equivalently one can use scalar densities. On an oriented manifold write

\[ \mathbf S_g=S_g\,dv_g. \]

The intended meanings, reconstructed from Branson's source article, are:

- “natural” means a universal local Riemannian invariant of critical weight \(-n\), constructed from a finite jet of the metric; the source's invariant-theoretic discussion uses polynomial contractions of curvature and its covariant derivatives; - \(\mathbf L\) is pointwise conformally invariant: \(\mathbf L_{e^{2\omega}g}=\mathbf L_g\); - \(\mathbf G\) has a local conformal primitive if there is another natural local critical density \(\mathbf F\) such that, on a closed manifold, \[ \left.\frac d{dt}\right|_{0}\int_M\mathbf F_{e^{2t\omega}g} =\int_M\omega\,\mathbf G_g \tag{1.1} \] for every smooth \(\omega\).

The adjective “local” in the last item is strong: \(\mathbf F\) must itself be a natural finite-jet expression. A reference-metric-dependent path integral on one conformal class does not by itself meet this requirement.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

07509 | AIM-GEOMETRY-0417

### Local toric bookkeeping and a focus-focus monodromy obstruction

O / open | AIM 研讨会问题汇编

**Question 1.21.** Margaret Symington asked a series of questions centered around the question of whether a locally toric structure is helpful for counting holomorphic curves. This is inspired by the success of using amoebas in Mikhalkin's work; locally toric pictures have a lot of the same structure Mikhalkin used. [Unfortunately I do not have good notes on this.]

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

07510 | AIM-GEOMETRY-0118

### Local-to-global gap criteria for exotic-order design

P / partially_solved | AIM 研讨会问题汇编

Engineering structures by carefully designing the interactions of the particles making up the system, is seldom straightforward and often requires trial and error, as unintended crystal structures might still be favored. It is not clear which clusters frustrate the formation of simple crystals and favor exotic order. This mechanism for stabilization and growth of exotic order has recently been explored with some success in a lattice gas model, and we will work to explore the possibility of this scenario in off-lattice models of real materials.

来源字段：http://aimpl.org/softpack/2/

07511 | AIM-GEOMETRY-0055

### Localized equilibrium states with strict intermediate entropy

P / partially_solved | AIM 研讨会问题汇编

Consider the geodesic flow on a translation surface. The Lebesgue measure is a 0-entropy measure, and the measure of maximal entropy sits on saddle connections. Are there other interesting invariant measures with intermediate entropy?

来源字段：http://aimpl.org/equibdynsysgeom/2/

07512 | OWR-7155451-015

### Localized Scalar Curvature Conditions Forcing Nonnegative ADM Mass

P / partially_solved | Oberwolfach 报告问题汇编

For an asymptotically flat Riemannian manifold $X$ with compact boundary and nonnegative scalar curvature, can metric conditions involving a positive scalar-curvature lower bound in a region and the distance from that region to the boundary force the ADM mass of $X$ to be nonnegative?

来源字段：https://doi.org/10.4171/owr/2021/30

07513 | OWR-14298808-006

### Log-Concave Busemann-Petty Problems

O / open | Oberwolfach 报告问题汇编

Resolve the following measure-comparison questions. (1) For $n=5$ or $6$, let $\mu$ be a rotation-invariant log-concave measure with strictly decreasing density, and let $K,L\subset\mathbb R^n$ be origin-symmetric convex bodies satisfying $\mu(tK\cap\theta^\perp)\leq\mu(tL\cap\theta^\perp)$ for every $t>0$ and $\theta\in S^{n-1}$. Must $\mu(K)\leq\mu(L)$? (2) Does the same conclusion hold in arbitrary dimension when $d\mu(x)=e^{-\phi(\|x\|_M)}dx$ and $M$ is an intersection body?

来源字段：https://doi.org/10.4171/owr/2024/58

07514 | OWR-15587-007

### Log-Concavity of Cohen-Macaulay H-Vectors

P / partially_solved | Oberwolfach 报告问题汇编

If $R$ is a standard graded Cohen-Macaulay domain with normal $\operatorname{Proj}(R)$, must its $h$-vector be log-concave?

来源字段：https://doi.org/10.4171/owr/2017/44

07515 | OWR-13951-011

### Logarithmic Linearization of Convex-Body Means

O / open | Oberwolfach 报告问题汇编

Does there exist a logarithmic map $L$ on a natural class of convex bodies satisfying $L(G(A,B))=(L(A)+L(B))/2$ for the geometric mean $G$? Is $G(M_{\mathrm{ag}}(A,B),M_{\mathrm{gh}}(A,B))=G(A,B)$, where $M_{\mathrm{ag}}$ and $M_{\mathrm{gh}}$ are the arithmetic-geometric and geometric-harmonic iterations?

来源字段：https://doi.org/10.4171/owr/2015/56

07516 | OWR-14610-001

### Long-Time Asymptotics of Homogeneous Ricci Flows

P / partially_solved | Oberwolfach 报告问题汇编

For an immortal homogeneous Ricci flow $(M,g(t))$ defined for all $t\geq 0$, what is the asymptotic behavior of $g(t)$ as $t\to\infty$?

来源字段：https://doi.org/10.4171/owr/2016/28

07517 | OWR-4799-001

### Loop-Space Localization and Witten-Type Interpolation

O / open | Oberwolfach 报告问题汇编

Can the lattice-sum evaluation of the heat kernel on a maximal torus be related to Duistermaat-Heckman and Berline-Vergne localization, and can an analogous Witten-type interpolation between Hodge theory and Morse theory be constructed on loop space for a Lagrangian such as the energy?

来源字段：https://doi.org/10.4171/owr/2011/09

07518 | AIM-GEOMETRY-0129

### Low-dimensional Krull dimensions and obstructions to cotangent integrability

P / partially_solved | AIM 研讨会问题汇编

i) What is the Krull dimension of $\mathcal{O}(T^{\ast}\overline{\mathcal{M}}_{g,n})$?

ii) Is the affinization map an algebraic completely integrable Hamiltonian system?

来源字段：http://aimpl.org/spectralhiggs/1/

07519 | OWR-7155449-011

### Low-Volume and Asymptotic Hyperbolic-Invariant Maximizers

O / open | Oberwolfach 报告问题汇编

For connected oriented closed hyperbolic $n$-manifolds of volume at most $v$, determine the low-volume maximizers and large-$v$ asymptotics of the maximal systole, kissing number, first positive Laplace eigenvalue, and its multiplicity. In dimension two, does the Bolza surface maximize the first positive eigenvalue and its multiplicity among genus-two surfaces?

来源字段：https://doi.org/10.4171/owr/2021/28

07520 | OWR-15427-015

### Lower Bounds for Convex Decompositions of Crossing Curves

P / partially_solved | Oberwolfach 报告问题汇编

Let $M(d)$ be the least integer such that every $(d+1)$-crossing curve in $\mathbb R^d$ can be decomposed into $M(d)$ convex curves. Establish nontrivial lower bounds for $M(d)$.

来源字段：https://doi.org/10.4171/owr/2017/19

07521 | OWR-1595-011

### M-Sequence Conditions for Smooth Lattice Polytopes

O / open | Oberwolfach 报告问题汇编

The $h^*$-vector of a normal lattice polytope is necessarily an $M$-sequence. Is the $h^*$-vector of every smooth lattice polytope an $M$-sequence?

来源字段：https://doi.org/10.4171/owr/2007/39

07522 | OWR-3389-019

### Magnetic Lieb-Thirring Constants and Geometric Critical Exponents

O / open | Oberwolfach 报告问题汇编

For magnetic Schrödinger operators, can the Li-Yau/Lieb-Thirring bound hold for arbitrary magnetic fields when $1\leq\gamma<3/2$? Can a magnetic field change the optimal constant, in particular for $n=3$ and $\gamma=1$? More broadly, do Lieb-Thirring inequalities hold on manifolds, and do their critical exponents depend on the geometry?

来源字段：https://doi.org/10.4171/owr/2009/06

07523 | OWR-3389-024

### Magnetic, Robin, and Hot-Spots Spectral Gap Questions

P / partially_solved | Oberwolfach 报告问题汇编

Open spectral-gap questions include: can a valid magnetic gap inequality be obtained by subtracting a term depending on the magnetic potential $A$; is the ground state of $-\Delta$ log-concave; does the Dirichlet eigenfunction ratio $u_2/u_1$ attain its extrema on the boundary of a convex domain (the analogue of Hot Spots); and, for Robin boundary conditions $\partial u/\partial\nu=-\alpha u$, is $\lambda_2-\lambda_1$ minimized when $V=0$ and the domain degenerates to a segment of the same diameter, and can Lavine's methods be adapted in one dimension?

来源字段：https://doi.org/10.4171/owr/2009/06

07524 | OWR-14297736-030

### Magnetic-Laplacian Eigenvalue Infima on Closed Surfaces

O / open | Oberwolfach 报告问题汇编

For a closed Riemannian surface $(M,g)$ and $i>1$, prove $\inf_A\overline\lambda_i(M,g,A)\geq C(M,g)>0$ over all real magnetic potentials $A$, and develop a natural construction of critical metric-potential pairs.

来源字段：https://doi.org/10.4171/owr/2023/36

07525 | OWR-4136-014

### Mahler Volume Product Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For every origin-symmetric convex body $K\subset\mathbb{R}^n$, is the Mahler volume product $P(K)=\operatorname{vol}_n(K)\operatorname{vol}_n(K^*)$ at least $P(B_\infty^n)$, where $B_\infty^n=\{x:|x_i|\leq1\}$?

来源字段：https://doi.org/10.4171/owr/2009/53

07526 | GEO-006

### Mahler's Conjecture

P / partially_solved | 独立 / 未分配集合

Among all centrally symmetric convex bodies in $\mathbb{R}^n$, does the cube (or cross-polytope) minimize the product of the body's volume and the volume of its polar dual?

背景链接：https://en.wikipedia.org/wiki/Mahler_volume

07527 | OWR-1394-007

### Mahler's Symmetric Convex-Body Conjecture

O / open | Oberwolfach 报告问题汇编

For every centrally symmetric convex body $K\subset\mathbb{R}^n$, one should have $|K||K^*|\geq|B_1^n||B_\infty^n|=4^n/n!$, where $K^*$ is the polar body.

来源字段：https://doi.org/10.4171/owr/2006/56

07528 | OWR-16637-003

### Mahler’s Volume Conjecture

O / open | Oberwolfach 报告问题汇编

For every convex body $K\subseteq\mathbb R^n$, is its Mahler volume bounded by

$$\bar s(K)=\lvert K\rvert\inf_{x\in K}\lvert(K-x)^\circ\rvert\geq\frac{(n+1)^{n+1}}{(n!)^2},$$

with equality for a simplex?

来源字段：https://doi.org/10.4171/owr/2018/54

07529 | AMR-067-0006

### Manifolds modelled on flag manifolds

O / open | AMR 问题清单汇编

Which manifolds can be modeled on an orbit of a real form in a space of flags?

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

07530 | AMR-067-0007

### Manifolds modelled on flag manifolds - Question 2

O / open | AMR 问题清单汇编

What is the homotopy classification of totally real immersions of real $3$-manifolds in the complex full flag manifold $F_{12}$?

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

07531 | OWR-16415-019

### Mapping-Class Orbit Counting in the Thurston Metric

P / partially_solved | Oberwolfach 报告问题汇编

For $x,y$ in Teichmüller space, determine as $R\to\infty$ the number of points in the mapping-class-group orbit of $y$ lying in the Thurston-metric ball of radius $R$ about $x$. How is this asymptotic related to orbit counting by extremal length rather than hyperbolic length?

来源字段：https://doi.org/10.4171/owr/2018/40

07532 | OWR-11695856-001

### Mapping-Class-Invariant Parametrization of Hitchin Components

P / partially_solved | Oberwolfach 报告问题汇编

Construct a mapping-class-group-invariant parametrization of higher-rank Hitchin and maximal components that remains valid after nonuniqueness of equivariant conformal maps invalidates Labourie's parametrization.

来源字段：https://doi.org/10.4171/owr/2022/47

07533 | OWR-16942-001

### Marked Length Spectrum Rigidity

P / partially_solved | Oberwolfach 报告问题汇编

Let a group $G$ act by isometries on metric spaces $X$ and $Y$, and define the marked length spectrum by $\ell_X(g)=\inf_{x\in X}d(x,gx)$. Under which hypotheses does $\ell_X=\ell_Y$ imply that $X$ and $Y$ are $G$-equivariantly isometric?

来源字段：https://doi.org/10.4171/owr/2019/30

07534 | OWR-16411-006

### Mass Curvature Relations in Asymptotically Hyperbolic Geometry

O / open | Oberwolfach 报告问题汇编

Extend conformal and Weyl mass covariants to broader classes of asymptotically hyperbolic manifolds, and establish a sharp relation between each mass and an associated curvature operator.

来源字段：https://doi.org/10.4171/owr/2018/36

07535 | OWR-3389-001

### Mass-Constrained Neumann and Steklov Eigenvalue Maximization

O / open | Oberwolfach 报告问题汇编

For a membrane of prescribed mass, how large can the $k$th nonzero Neumann eigenvalue $\mu_k$ and the $k$th Steklov eigenvalue $\sigma_k$ be?

来源字段：https://doi.org/10.4171/owr/2009/06

07536 | OWR-11788-006

### Mass-Decreasing Flows and the Positive Mass Theorem

O / open | Oberwolfach 报告问题汇编

Can the mass-decreasing flow on asymptotically flat three-manifolds be used to give an independent proof of the positive mass theorem?

来源字段：https://doi.org/10.4171/owr/2012/26

07537 | OWR-13347-016

### Maximal Isometry Groups of Solvable Ricci Solitons

P / partially_solved | Oberwolfach 报告问题汇编

To what extent do the maximal-isometry-group results for Einstein metrics extend to Ricci soliton metrics on general solvable Lie groups?

来源字段：https://doi.org/10.4171/owr/2014/45

07538 | OWR-13680-005

### Maximal Real Betti Numbers of Tridegree Surfaces

O / open | Oberwolfach 报告问题汇编

For a nonsingular real algebraic surface $X\subset(\mathbb{CP}^1)^3$ of a prescribed multidegree, what is the maximal possible first Betti number $b_1(\mathbb RX)$ of its real part?

来源字段：https://doi.org/10.4171/owr/2015/23

07539 | AIM-GEOMETRY-0002

### Maximal torsion gradient, flat fail points, and a no-plateau obstruction

P / partially_solved | AIM 研讨会问题汇编

For a bounded convex planar domain \(\Omega\), let \(u_\Omega\) solve \[ -\Delta u_\Omega=1\quad\text{in }\Omega, \qquad u_\Omega=0\quad\text{on }\partial\Omega. \] Determine \[ C_*:=\sup\left\{\|\nabla u_\Omega\|_{L^\infty(\Omega)}: \Omega\subset\mathbb R^2\text{ convex},\ |\Omega|=1\right\}. \] Is a point where the maximal gradient is reached necessarily on a flat boundary part, in a quantitatively controlled sense?

\[ \nu_\Omega=0\qquad\hbox{on }\partial\Omega, \]

This correction is verified directly against the paper on which the AIM entry is based, Hoskins--Steinerberger (2021), and against the subsequent papers of Huang (2025) and Burdzy--Ftouhi--Mariano (2025). Thus the intended problem is the following.

The normalization is natural. If \(t>0\), then \[ u_{t\Omega}(x)=t^2u_\Omega(x/t),\qquad \|\nabla u_{t\Omega}\|_\infty=t\|\nabla u_\Omega\|_\infty, \qquad |t\Omega|^{1/2}=t|\Omega|^{1/2}. \] Hence \[ J(\Omega):=\frac{\|\nabla u_\Omega\|_\infty}{|\Omega|^{1/2}} \] is scale invariant and \(C_*=\sup J(\Omega)\).

来源字段：http://aimpl.org/symmetrybreaking/1/

07540 | OWR-2654831-006

### Maximum Central Points in Cube-Move $\alpha$-Immersions

O / open | Oberwolfach 报告问题汇编

For $\alpha$-immersions with $\alpha>1$, is three the maximal possible number of potential central points produced by the cube-move construction?

来源字段：https://doi.org/10.4171/owr/2020/35

07541 | OWR-1703876-009

### Maximum Complexity of Hyperplane Arrangement Levels

P / partially_solved | Oberwolfach 报告问题汇编

Determine the correct asymptotic order of the maximum size $f_k^{(d)}(n)$ of the $k$-level in an arrangement of $n$ hyperplanes in $\mathbb R^d$.

来源字段：https://doi.org/10.4171/owr/2020/30

07542 | OWR-15427-011

### Maximum Diameters of Lattice Polytopes in Cubes

P / partially_solved | Oberwolfach 报告问题汇编

For positive integers $d$ and $k$, let $\delta(d,k)$ be the largest diameter of a lattice polytope contained in $[0,k]^d$. Is this maximum, up to translation, attained by a Minkowski sum of lattice vectors? In particular, when $k<2d$, is $$\delta(d,k)=\left\lfloor\frac{(k+1)d}{2}\right\rfloor$$?

来源字段：https://doi.org/10.4171/owr/2017/19

07543 | OWR-1183-002

### Maximum Edge Counts of Geometric Polyhedral Surfaces

O / open | Oberwolfach 报告问题汇编

For geometric polyhedral surfaces with $n$ vertices and $f_1$ edges, is the upper bound $f_1=O(n^2)$ asymptotically tight?

来源字段：https://doi.org/10.4171/owr/2006/01

07544 | OWR-16938-004

### Maximum Equiangular Line Systems in Euclidean Space

O / open | Oberwolfach 报告问题汇编

Determine the maximum number of pairwise equiangular lines in $\mathbb R^n$, both without fixing the common angle and for each prescribed common angle.

来源字段：https://doi.org/10.4171/owr/2019/26

07545 | AMR-058-0020

### Maximum Shear Modulus of Planar Froths

P / partially_solved | AMR 问题清单汇编

Prove or disprove that no two-dimensional froth of average bubble area one has shear modulus in any direction exceeding that of the regular hexagonal foam.

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

07546 | OWR-15587-001

### Maximum Volume of Canonical Fano Polytopes

O / open | Oberwolfach 报告问题汇编

For $d\geq4$, let $P$ be a $d$-dimensional canonical Fano lattice polytope, and define the Sylvester sequence by $s_1=2$ and $s_{i+1}=s_1\cdots s_i+1$. Is $$\operatorname{vol}(P)\leq\frac{2(s_d-1)^2}{d!}, $$ with equality only for the canonical Fano simplex $R_{(d)}$?

来源字段：https://doi.org/10.4171/owr/2017/44

07547 | OWR-8415364-014

### Maximum-Volume Sections of $ell_p$ Balls

O / open | Oberwolfach 报告问题汇编

For $2<p<\infty$, which linear sections of an $\ell_p$ ball have maximum volume?

来源字段：https://doi.org/10.4171/owr/2021/59

07548 | OWR-16415-011

### McShane Identities for Laplace Spectra

O / open | Oberwolfach 报告问题汇编

Construct a McShane-type identity involving the eigenvalues of the Laplacian of a compact hyperbolic surface.

来源字段：https://doi.org/10.4171/owr/2018/40

07549 | AIM-GEOMETRY-0169

### Mean-convex tubes and curvature-driven centerline drift

P / partially_solved | AIM 研讨会问题汇编

Recall that the mean curvature flow of a “thin” torus of revolution in $\mathbb R^3$ goes singular everywhere simultaneously as it converges to a round circle. Under what conditions on a smooth embedded closed curve $\gamma\subset\mathbb R^3$ does there exist a toroidal mean convex mean curvature flow $M_t$ that Hausdorff converges to $\gamma$ at the blow-up time?

来源字段：http://aimpl.org/flowriemannian/2/

07550 | OWR-14299092-005

### Mean-Curvature Blowup at Intermediate-Dimensional Singularities

O / open | Oberwolfach 报告问题汇编

At every finite-time singularity of mean-curvature flow in dimensions $3<n<7$, must the mean curvature become unbounded?

来源字段：https://doi.org/10.4171/owr/2025/11

07551 | OWR-2045-001

### Mean-Curvature Flow from Isolated Conical Singularities

P / partially_solved | Oberwolfach 报告问题汇编

Let $M_0\subset\mathbb R^{n+1}$ be a hypersurface with isolated point singularities modeled on regular hypercones, and let $(M_t)_{t\geq0}$ be a mean-curvature flow with initial data $M_0$. In dimensions $n\geq7$, does $M_t$ evolve by self-similar expansion asymptotically near each singular point and become smooth for a short positive time?

来源字段：https://doi.org/10.4171/owr/2008/34

07552 | OWR-15575-009

### Measure of Rational-Direction Geodesics on Finsler Tori

O / open | Oberwolfach 报告问题汇编

For a zero-topological-entropy Finsler metric on $T^2$, is $$\mu_L\left(\bigcup_{\substack{\xi\in S^1\\ \xi\text{ rational}}}E_\xi\right)<\mu_L(ST^2), $$ where $\mu_L$ is Liouville measure and $E_\xi$ is the set of geodesics with rational asymptotic direction $\xi$?

来源字段：https://doi.org/10.4171/owr/2017/32

07553 | AMR-037-0008

### Medial surfaces and Voronoi diagrams of lines

P / partially_solved | AMR 问题清单汇编

Determine the worst-case complexity of the medial surface and of an offset surface of an $n$-feature polyhedron, and of the Voronoi diagram of $n$ lines in three-space. The known upper bounds are superquadratic while quadratic lower-bound constructions are known.

背景链接：https://jeffe.cs.illinois.edu/open/

07554 | OWR-3389-005

### Meissner Bodies as Minimum-Volume Constant-Width Bodies

O / open | Oberwolfach 报告问题汇编

Are the three-dimensional convex bodies of constant width that minimize volume exactly Meissner’s bodies?

来源字段：https://doi.org/10.4171/owr/2009/06

07555 | AMR-058-0024

### Melzak's Shortest-Edge Polyhedron Conjecture

P / partially_solved | AMR 问题清单汇编

Prove that the unit-volume polyhedron with shortest total edge length is an equilateral triangular prism, and establish existence of a minimizer.

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

07556 | AIM-GEOMETRY-0214

### Metric comparison for filtered loop and path homology

P / partially_solved | AIM 研讨会问题汇编

Question 3.1.1 (Paternain). How do the sums of the Betti numbers for ΛT M and

Ω( T (p, q ) grow as T → ∞? Does the growth depend on the metric?

It was shown by Gromov in [Gromov2001] that if M is simply connected there is a constant C such that sum of the Betti numbers of Λ CN is at least ∑Ni=0 bi(Λ).

The exact `problem` field supplied for this attempt is:

The original PDF and adjacent text were checked. Immediately before the question, the source defines $\Lambda^T M$ to be the free loops of length at most $T$ and $\Omega^T(p,q)$ to be the paths from $p$ to $q$ of length at most $T$. Thus `ΛT M` and `Ω( T (p, q )` are damaged renderings of \[ \Lambda^T M, \qquad \Omega^T(p,q). \] The PDF itself has a stray parenthesis in the second notation on the question line; the preceding definition verifies the intended expression.

Its mathematically verified reconstruction is \[ \sum_{j\ge0} b_j(\Lambda^{CN}M) \;\ge\; \sum_{i=0}^{N}b_i(\Lambda M). \tag{1.1} \] Indeed, Gromov's stronger statement says that, for a closed simply connected $M$, every class in $H_i(\Lambda M)$ has a representative supported on loops of length at most $Ci$. Equivalently, $H_i(\Lambda^{CN}M)\to H_i(\Lambda M)$ is surjective for $i\le N$, which implies (1.1). This stronger formulation is reproduced with proof in Heistercamp's Appendix C [Hei]. The compactness/closedness assumption, suppressed on the question line, is required by the cited theorem and by the uniform metric comparison below.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07557 | AIM-GEOMETRY-0090

### Midpoint projection and modular-period reductions for the E8 and Leech magic functions

P / partially_solved | AIM 研讨会问题汇编

Let $f$ be the magic function in $8$-dimension and $M_f(s) = \int_0^\infty f(x) x^{s-1} dx$. Conjecture of Cohn and Miller: $M_f(4) = \frac{1}{15} = M_{\hat{f}}(4)$. Can we prove this? The same question for the magic function in $24$-dimension, where $M_f(12) = M_{\hat{f}}(12) = 0.17786094729650\ldots$?

来源字段：http://aimpl.org/discreteaf/1/

07558 | OWR-1383-001

### Minimal-Filling Conjecture for Simple Riemannian Manifolds

O / open | Oberwolfach 报告问题汇编

Every simple compact Riemannian manifold $M$ should be a minimal filling: for every orientable $M'$ with $\partial M'=\partial M$, if $d_{M'}(x,y)\geq d_M(x,y)$ for all $x,y\in\partial M$, then $\operatorname{vol}(M')\geq\operatorname{vol}(M)$. Moreover, equality should imply that $M'$ is boundary-preservingly isometric to $M$. Here simple means that the boundary is strictly convex, any two points are joined by a unique geodesic, and there are no conjugate points.

来源字段：https://doi.org/10.4171/owr/2006/45

07559 | AMR-067-0024

### Minimax minimal surfaces

O / open | AMR 问题清单汇编

Prove the lower bound \[ d\le \mbox{Index}(\Phi_{\mathcal A})+\mbox{Null}(\Phi_{\mathcal A}), \] where $\mbox{Null}(\Phi_{\mathcal A})$ is the {\it nullity} of $\Phi_{\mathcal A}$ that is the dimension of the space of its Jacobi fields.

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

07560 | AMR-067-0025

### Minimax minimal surfaces - Question 2

P / partially_solved | AMR 问题清单汇编

Prove that there exists infinitely many distinct minimal branched 2-dimensional immersions in $N^n$.

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

07561 | OWR-2654833-001

### Minimizers of Nonlocal Knot Self-Avoidance Energies

P / partially_solved | Oberwolfach 报告问题汇编

For nonlocal self-avoidance energies on embeddings, do minimizers or stationary points exist within prescribed isotopy classes?

来源字段：https://doi.org/10.4171/owr/2020/37

07562 | AMR-058-0009

### Minimizing Polyhedral Soap-Film Cones

O / open | AMR 问题清单汇编

Classify minimizing polyhedral soap-film cones in dimensions five through seven and in all higher dimensions.

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

07563 | OWR-4412-005

### Minimum Density of Area-Minimizing Hypersurface Singularities

P / partially_solved | Oberwolfach 报告问题汇编

What is the infimum of the density $\Theta(M,x)$ over all area-minimizing hypersurfaces $M$ in Riemannian manifolds and all interior singular points $x$ of $M$?

来源字段：https://doi.org/10.4171/owr/2010/31

07564 | OWR-4412-006

### Minimum Density of Singular Area-Minimizing Hypercones

P / partially_solved | Oberwolfach 报告问题汇编

What is the infimum of the density $\Theta(C)$ over all area-minimizing hypercones $C$ with an isolated singularity at the origin?

来源字段：https://doi.org/10.4171/owr/2010/31

07565 | OWR-12012-001

### Minimum Distance in Spherical Fibonacci Lattices

O / open | Oberwolfach 报告问题汇编

For the spherical Fibonacci point set $Z_{F_n}$, is $|z_0-z_1|=2/\sqrt{F_n}$ the least pairwise distance among its $F_n$ points?

来源字段：https://doi.org/10.4171/owr/2012/40

07566 | OWR-15952-005

### Minimum Distinct Distances Among Planar Point Sets

P / partially_solved | Oberwolfach 报告问题汇编

Determine the minimum number $f(n)$ of distinct distances determined by $n$ distinct points in the Euclidean plane.

来源字段：https://doi.org/10.4171/owr/2017/47

07567 | AMR-054-0006

### Minimum Euclidean Matching in 2D

P / partially_solved | AMR 问题清单汇编

What is the complexity of computing a minimum-cost Euclidean matching for $2n$ points in the plane? The cost of a matching is the total length of the edges in the matching.

背景链接：https://topp.openproblem.net/p6

07568 | OWR-3385-004

### Minimum Maximum Degree of Centrally Symmetric Sphere Triangulations

S / solved | Oberwolfach 报告问题汇编

What is the smallest possible maximum vertex degree $f(d)$ of a centrally symmetric triangulation of the $d$-sphere, equivalently of a simplicial $(d+1)$-polytope? The known bounds are $f(1)=2$, $f(2)=4$, and $d+1<f(d)\leq2d$.

来源字段：https://doi.org/10.4171/owr/2009/02

07569 | OWR-11451-010

### Minimum Size of Cone Lifts

O / open | Oberwolfach 报告问题汇编

Given a nested family of cones $(K_k)$, determine the least $k$ for which a cone $C$ can be represented as $C=\pi(K_k\cap L)$ for an affine subspace $L$ and a linear map $\pi$.

来源字段：https://doi.org/10.4171/owr/2011/53

07570 | OWR-13110-018

### Minimum Volume of Bodies with Unit Width

O / open | Oberwolfach 报告问题汇编

What is the minimum volume among all convex sets in $\mathbb{R}^d$ whose width is at least 1 in every direction?

来源字段：https://doi.org/10.4171/owr/2014/40

07571 | AMR-037-0011

### Minimum-area triangles

P / partially_solved | AMR 问题清单汇编

Given $n$ planar points, find a subquadratic algorithm for the minimum-area triangle or prove a quadratic lower bound in a suitable computation model. More generally, close the gap for minimum-volume simplices in fixed dimension.

背景链接：https://jeffe.cs.illinois.edu/open/

07572 | AMR-054-0022

### Minimum-Link Path in 2D

P / partially_solved | AMR 问题清单汇编

Can a minimum-link path among polygonal obstacles be found in subquadratic time?

背景链接：https://topp.openproblem.net/p22

07573 | OWR-14298808-016

### Minkowski Problem for Asymptotic Convex Sets

P / partially_solved | Oberwolfach 报告问题汇编

Characterize the Borel measures on the spherical domain $\Omega$ that arise as surface-area measures of $C$-asymptotic convex sets in dimensions greater than two. In particular, prove or disprove the conjectured necessary condition $\int_0^{\pi/2}S_K(\omega_\alpha)^{1/(n-1)}\,d\alpha<\infty$.

来源字段：https://doi.org/10.4171/owr/2024/58

07574 | OWR-17473-002

### Mirror Symmetry for Maximally Degenerate Positive Calabi-Yau Pairs

O / open | Oberwolfach 报告问题汇编

Formulate and prove mirror symmetry for every maximally degenerate positive Calabi-Yau pair $(M,D)$.

来源字段：https://doi.org/10.4171/owr/2020/10

07575 | AMR-038-0009

### Mirrored-room illumination

S / solved | AMR 问题清单汇编

Given a polygonal room with perfectly reflecting sides and a point light source, characterize when every point of the room is illuminated. In particular, can a polygonal mirrored room contain a dark region?

背景链接：https://ics.uci.edu/~eppstein/junkyard/open.html

07576 | AIM-GEOMETRY-0014

### Mixed Hodge structures and the Hochschild-zero link-homology slice for plabic graphs

P / partially_solved | AIM 研讨会问题汇编

In high levels of generality, varieties pick up a mixed Hodge structure (Lam-Speyer). When the cluster variety is adequately nice, the mixed Hodge structure has only two dimensions, and the point count polynomial corresponds to a certain polynomial, but not the HOMFLY-PT.

What is the relationship between the mixed Hodge structure on the cohomology of a cluster variety coming from a plabic graph and knot invariants (eg. the Khovanov-Rozansky homology)?

来源字段：http://aimpl.org/clusterbraid/1/

07577 | AIM-GEOMETRY-0322

### Mixed-cell multiplicity and the real parity obstruction

P / partially_solved | AIM 研讨会问题汇编

For a connected compact component \(K\) of an intersection of amoebas, does the associated mixed Monge--Ampère mass equal the volume of the mixed cell corresponding to \(K\)? Are the roots in \(\operatorname{Log}^{-1}(K)\) in one-to-one correspondence with the roots of the binomial system attached to that cell, in a way preserving real roots?

This is Question A.2 from the AIM workshop list *Amoebas and tropical geometry* (January 2004), contributed by F. Bihan.

In particular, `Monge-Amp\` ere`, `Rullg˚ ard`, and `Log -1(K)` mean Monge--Ampère, Rullgård, and \(\operatorname{Log}^{-1}(K)\), respectively. With those repairs, the question is:

Here \[ \operatorname{Log}(z_1,\ldots,z_n) =(\log|z_1|,\ldots,\log|z_n|). \]

There is an important correction to the background premise. Passare and Rullgård do not identify the local mixed Monge--Ampère mass with the root count of the one fixed, unrotated system. Their Theorem 5 identifies it with an **average** of root counts for independently phase-rotated systems. This distinction is central to both questions.

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

07578 | OWR-4334-004

### Mod-Two Area Minimizers in Compact Flat Three-Manifolds

O / open | Oberwolfach 报告问题汇编

Classify the mod-two area-minimizing surfaces in the Tri, Tetra, and Hexa Cosms, the compact orientable flat three-manifolds not covered by the known planar-minimizer results for the three-torus, Di Cosm, and Didi Cosm.

来源字段：https://doi.org/10.4171/owr/2010/21

07579 | AMR-055-0007

### Moduli of Minimal Sphere Immersions

S / solved | AMR 问题清单汇编

Consider minimal immersions $S^n\to S^N(1)$ with total area at most a fixed constant $A$, identifying immersions which differ by a motion of the ambient space. Is the resulting set a finite-dimensional space with some natural topology?

背景链接：https://people.math.harvard.edu/~hirolee/pdfs/2014-fall-230a-icm1970-chern-differential-geometry.pdf

07580 | OWR-15208-018

### Moduli of Positively Curved Metrics on Projective Spaces

O / open | Oberwolfach 报告问题汇编

What can be said about the moduli space of positively curved metrics when the sphere $S^d$ is replaced by a complex projective space?

来源字段：https://doi.org/10.4171/owr/2017/3

07581 | OWR-15208-015

### Moduli of Singular Metrics Under Homeomorphisms

O / open | Oberwolfach 报告问题汇编

Can diffeomorphisms be replaced by homeomorphisms and Riemannian metrics by singular metrics in the study of moduli spaces?

来源字段：https://doi.org/10.4171/owr/2017/3

07582 | OWR-4799-002

### Moment Polyhedra of Noncompact Coadjoint Orbits

P / partially_solved | Oberwolfach 报告问题汇编

Given a coadjoint orbit $O$ of a connected Lie group $G$ and a compact subgroup $K$, can one determine explicit equations for the moment polyhedron $\Delta_{K}(O)$ of the orbit projection?

来源字段：https://doi.org/10.4171/owr/2011/09

07583 | OWR-14221-009

### Moment-Map Images as Duistermaat-Heckman Supports

O / open | Oberwolfach 报告问题汇编

Can one choose a connection $\nabla$ such that the image of the moment map $\Phi$ is the support of the Duistermaat-Heckman measure?

来源字段：https://doi.org/10.4171/owr/2016/13

07584 | OPG-2435

### Monochromatic empty triangles

P / partially_solved | Open Problem Garden

If $X \subseteq {\mathbb R}^2$ is a finite set of points which is 2-colored, an empty triangle is a set $T \subseteq X$ with $|T|=3$ so that the convex hull of $T$ is disjoint from $X \setminus T$. We say that $T$ is monochromatic if all points in $T$ are the same color.

Conjecture There exists a fixed constant $c$ with the following property. If $X \subseteq {\mathbb R}^2$ is a set of $n$ points in general position which is 2-colored, then it has $\ge cn^2$ monochromatic empty triangles.

背景链接：http://www.openproblemgarden.org/op/monochromatic_empty_triangles

07585 | AMR-058-0004

### Monotonicity and Concavity of Bubble Area

P / partially_solved | AMR 问题清单汇编

Let $A_X(v_1,\ldots,v_k)$ be the area of a minimizing $k$-bubble of volumes $v_i$ in a Riemannian manifold $X$. For $X=\mathbb{R}^n$, is $A_X$ strictly increasing in each volume, and is it strictly concave?

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

07586 | OWR-15215-006

### Monotonicity Constants for Expected Random-Simplex Volumes

P / partially_solved | Oberwolfach 报告问题汇编

Let $K\subseteq L\subset\mathbb{R}^d$ be convex bodies, and let $P_{d+1}^K$ and $P_{d+1}^L$ be simplices generated by $d+1$ independent uniform points in $K$ and $L$, respectively. Does there exist $c(d)$ with $\limsup_{d\to\infty}c(d)^{1/d}<\infty$ such that $\mathbb{E}V_d(P_{d+1}^K)\leq c(d)\mathbb{E}V_d(P_{d+1}^L)$? Determine lower bounds for the optimal $c(d)$.

来源字段：https://doi.org/10.4171/owr/2017/10

07587 | AMR-067-0016

### Morse index of embedded minimal surfaces

P / partially_solved | AMR 问题清单汇编

; Do there exist embedded minimal surfaces with finite genus and Morse index $4$?; More focussed: in the $1$-parameter deformation of Costa's surface, does the index stay constant?

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

07588 | AMR-064-0003

### Morse-Sard Questions for Endpoint Maps

O / open | AMR 问题清单汇编

For the endpoint map from the $H^1$ Hilbert manifold of admissible paths starting at $q_0$ to $M$, can the singular curves starting at $q_0$ fill all of $M$? Can the optimal singular curves starting at $q_0$ fill a positive-measure subset of $M$?

背景链接：https://arxiv.org/abs/1304.2590

07589 | AMR-071-0075

### Moser's worm problem

P / partially_solved | AMR 问题清单汇编

Moser's worm problem - what is the smallest area of a shape that can cover every unit-length curve in the plane?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

07590 | AMR-054-0059

### Most Circular Partition of a Square

O / open | AMR 问题清单汇编

What is the optimal partition of a square into convex pieces such that the circularity of the pieces is optimized? The circularity of a polygon is the ratio of the radius of its smallest circumscribing circle to the radius of its largest inscribed circle. Thus circular pieces have circularity near $1$, and noncircular pieces have circularity greater than $1$. An optimal partition minimizes the maximum ratio over all pieces in the partition.

背景链接：https://topp.openproblem.net/p59

07591 | OWR-16933-006

### Motion and Trace of Real Divisors

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a nonsingular projective real algebraic curve and let $A$ be a simple real divisor. How can $A$ move within its linear-equivalence class while remaining simple and real? What is $\pi_1(\operatorname{Emb}(A,X))$, and what can be the image of its trace homomorphism to $H_1(X)$?

来源字段：https://doi.org/10.4171/owr/2019/21

07592 | OWR-15575-006

### Movable Invariant Neighborhoods of Compact Flow Sets

O / open | Oberwolfach 报告问题汇编

Let $\varphi$ be a flow on $\mathbb R^3$ with compact invariant set $A$. Does every neighborhood of $A$ contain a movable compact invariant set that contains $A$?

来源字段：https://doi.org/10.4171/owr/2017/32

07593 | OWR-4412-007

### Multidirectional Invisible and Zero Resistance Bodies

P / partially_solved | Oberwolfach 报告问题汇编

Do there exist bodies invisible or of zero resistance in three or more directions, and can one prove that no such body is invisible or of zero resistance in a set of directions of positive measure?

来源字段：https://doi.org/10.4171/owr/2010/31

07594 | OWR-11136-013

### Multiple Cover Decomposition Thresholds for Planar Sets

O / open | Oberwolfach 报告问题汇编

A planar set $P \subset \mathbb{R}^2$ is called *cover-decomposable* if there exists an integer $m = m_2(P)$ such that every $m$-fold covering of the plane (or any subset) by translates of $P$ can be decomposed into $2$ coverings. For every cover-decomposable planar set $P$ and every integer $k \geq 2$, does the $k$-decomposition number $m_k(P)$ (such that every $m_k(P)$-fold covering by translates of $P$ decomposes into $k$ coverings) exist?

来源字段：https://doi.org/10.4171/owr/2011/44

07595 | OWR-14299521-009

### Multiple Minimal Surfaces in Riemannian Three-Spheres

P / partially_solved | Oberwolfach 报告问题汇编

Beyond the established positive-Ricci-curvature cases, does every closed Riemannian three-sphere contain at least four minimal two-spheres and five minimal tori? More generally, must it contain multiple minimal surfaces of every genus?

来源字段：https://doi.org/10.4171/owr/2025/30

07596 | OWR-14298808-015

### Multiplicity of Centroidal Hyperplane Sections

P / partially_solved | Oberwolfach 报告问题汇编

For a convex body $K\subset\mathbb R^n$, must at least $n+1$ distinct hyperplane sections through the centroid $c(K)$ also have centroid $c(K)$? More generally, determine the minimum number $\mu(n)$ of such sections over all convex bodies, especially for $n=3$ and $4$.

来源字段：https://doi.org/10.4171/owr/2024/58

07597 | OWR-4799-003

### Multiplicity-Free Quantization and Orbit Counts

O / open | Oberwolfach 报告问题汇编

For a coadjoint orbit $O_{G}$ with quantization $Q(O_{G})$ and a subgroup $H$, is the restriction $Q(O_{G})|_{H}$ multiplicity-free exactly when $n(O_{G},O_{H})=\#(\mu^{-1}(O_{H})/H)\leq1$ for every $H$-coadjoint orbit $O_{H}$; and for a family $O_{G}^{\lambda}$, does multiplicity-freeness of the restrictions imply this orbit-count bound for every parameter $\lambda$?

来源字段：https://doi.org/10.4171/owr/2011/09

07598 | OWR-15988-005

### Multiscale Principal Flows on Riemannian Manifolds

O / open | Oberwolfach 报告问题汇编

For principal flows on Riemannian manifolds, can a scale parameter describe a flow that dies out, and is extra regularization needed to prevent self-intersection or winding? Can persistence diagrams of the Fréchet potential detect local means at different scales, can one scale encode both starting points and local covariance, and are there canonical higher-dimensional principal submanifolds?

来源字段：https://doi.org/10.4171/owr/2018/3

07599 | AIM-GEOMETRY-0028

### MV versus theta bases on unipotent groups and cells

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.6.** Let \(U\) be a unipotent cell. Then \(U\) can be thought of as a braid variety. Also \(\mathbb C[U]\) has a Mirkovi\'c--Vilonen basis. Does the MV basis equal the \(\Theta\)-basis?

**Comment.** See the appendix of Baumann--Kamnitzer--Knutson.

来源字段：http://aimpl.org/clusterbraid/4/

07600 | AIM-GEOMETRY-0248

### Necessary gates for positive-curvature ergodicity

P / partially_solved | AIM 研讨会问题汇编

**Question 8.2.1.** Is there a Riemannian metric on a closed manifold with everywhere positive sectional curvatures and ergodic geodesic flow?

Metrics close to the standard metric on \(S^2\) would be especially interesting.

**8.3. The measure of transitive and recurrent sets.**

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07601 | OWR-13347-018

### Negative Ricci Curvature on Solvable Lie Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathfrak{g}$ be a solvable Lie algebra with nilradical $\mathfrak{n}$. Does $\mathfrak{g}$ admit an inner product with $\operatorname{Ric}<0$ if and only if some $X\in\mathfrak{g}\setminus\mathfrak{n}$ has $\left.\operatorname{ad}X\right|_{\mathfrak{n}}$ satisfying linear inequalities determined solely by $\mathfrak{n}$?

来源字段：https://doi.org/10.4171/owr/2014/45

07602 | OWR-7155451-014

### Negative-Mass Ends of Nonnegative-Scalar-Curvature Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Let $(M,g)$ be complete with nonnegative scalar curvature and let $E$ be an asymptotically flat end of $M$. Can $E$ have negative ADM mass?

来源字段：https://doi.org/10.4171/owr/2021/30

07603 | OWR-15219-008

### Negatively Curved Barriers Beyond Hyperbolicity Cones

O / open | Oberwolfach 报告问题汇编

Does there exist an algebraic convex cone, defined by homogeneous multivariate polynomial inequalities, that admits a normal barrier of negative curvature but is not a hyperbolicity cone?

来源字段：https://doi.org/10.4171/owr/2017/14

07604 | OWR-1455-010

### Neighborliness of Centrally Symmetric Polytopes

O / open | Oberwolfach 报告问题汇编

How neighborly can a centrally symmetric polytope be as a function of its dimension and number of vertices?

来源字段：https://doi.org/10.4171/owr/2007/04

07605 | AMR-099-0063

### Nerve graphs of Euclidean sphere packings

O / open | AMR 问题清单汇编

Characterize the graphs that occur as tangency, or nerve, graphs of sphere packings with disjoint interiors in $\mathbb{R}^d$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=7

07606 | AIM-GEOMETRY-0321

### Newton subdivisions as combinatorial models for tropical hypersurfaces

P / partially_solved | AIM 研讨会问题汇编

We know that the Bergman complex of a variety (as defined in Chapter 9 of Sturmfels' book on “Solving systems of polynomial equations”), i.e. the tropical variety, is a polyhedral complex. For linear subspaces, the paper of Ardila and Klivans shows that this Bergman complex has a very nice combinatorial structure: a subdivision of it is the order complex of the lattice of flats of the associated matroid, a very well understood combinatorial object. (As a corollary we get the topology, etc.) Question: Can we find a similar combinatorial description for other classes of Bergman complexes? (contributed by Federico Ardila)

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

07607 | AMR-154-0001

### Nijenhuis Geometry and Integrable Systems - Problem 1

O / open | AMR 问题清单汇编

For the KdV system, this was reported to be done already in [veselov], though we did not find detailed formulas. The booklet [moser81] relates stationary solutions of the KdV to the solutions of the Neumann systems. Namely, it is shown that the potential energy of the Neumann system evaluated along the trajectory and suitably scaled is a solution of the KdV system. Moreover, the solutions of the KdV equations obtained by this method are related to potentials in the Hills-Schr\"odinger equation whose spectrum has finitely many bands. See also [DKN, p. 219, eqn (2.162), pp. 247, 248].

What is special in the potential generated by $V(t)= C(t)/m(t)$?

背景链接：https://arxiv.org/abs/2410.04276

07608 | AMR-154-0010

### Nijenhuis Geometry and Integrable Systems - Problem 10

O / open | AMR 问题清单汇编

Are geometric Poisson structures of any order greater than 1 projectively invariant? Can one interpret the leading coefficient $g^{ij}$ as a projectively invariant object?

背景链接：https://arxiv.org/abs/2410.04276

07609 | AMR-154-0011

### Nijenhuis Geometry and Integrable Systems - Problem 11

O / open | AMR 问题清单汇编

Certain Frobenius manifolds seem to give examples of geometric Poisson bracket of order 3 (at least, the WDVV equation does it). Is Frobenius structure projectively invariant? Can one replace, in the definition of Frobenius structure, the flat metric by a metric of constant curvature?

背景链接：https://arxiv.org/abs/2410.04276

07610 | AMR-154-0012

### Nijenhuis Geometry and Integrable Systems - Problem 12

O / open | AMR 问题清单汇编

\begin{Problem} Certain Frobenius manifolds seem to give examples of geometric Poisson bracket of order 3 (at least, the WDVV equation does it). Is Frobenius structure projectively invariant? Can one replace, in the definition of Frobenius structure, the flat metric by a metric of constant curvature? \end{Problem}

As we mentioned above, geometric Poisson brackets of order 3 are essentially the same as Killing tensors whose covariant derivative satisfies a certain relation, and certain Forbenius manifolds were used to construct such Killing tensors. Does there exists a construction of a Killing tensor by a Frobenius manifold?

背景链接：https://arxiv.org/abs/2410.04276

07611 | AMR-154-0013

### Nijenhuis Geometry and Integrable Systems - Problem 13

O / open | AMR 问题清单汇编

\begin{Problem} As we mentioned above, geometric Poisson brackets of order 3 are essentially the same as Killing tensors whose covariant derivative satisfies a certain relation, and certain Forbenius manifolds were used to construct such Killing tensors. Does there exists a construction of a Killing tensor by a Frobenius manifold? \end{Problem}

Can two different Poisson structures of order 3, one of which is not Darboux-Poisson, be compatible?

背景链接：https://arxiv.org/abs/2410.04276

07612 | AMR-154-0014

### Nijenhuis Geometry and Integrable Systems - Problem 14

O / open | AMR 问题清单汇编

Describe all geometric Poisson structures of the first order compatible with a geometric Poisson structure of the third order which is not Darboux-Poisson. How big can the compatible pencil of such first order structures be?

背景链接：https://arxiv.org/abs/2410.04276

07613 | AMR-154-0002

### Nijenhuis Geometry and Integrable Systems - Problem 2

O / open | AMR 问题清单汇编

\begin{Problem} Can one write the system of equations in such a way that its coefficients depend on $L$ only (at least in the $\mathrm{gl}$-regular case)? \end{Problem} Note that in the $\mathrm{gl}$-regular case any two pairs $(L_1,e_1)$ and $(L_2, e_2)$ with identical Jordan blocks of $L_1$ and $L_2$ are isomorphic by a coordinate transformation. Moreover, $\alpha$ is just the dual form for $e$. This makes us hope that the system can be written in terms of $L$ only.

Are the compatibility conditions for this system precisely the conditions $\mathcal{N}_L=0$? They must be geometric conditions on $L$ which vanish if $L$ is a Nijenhuis operator.

背景链接：https://arxiv.org/abs/2410.04276

07614 | AMR-154-0003

### Nijenhuis Geometry and Integrable Systems - Problem 3

O / open | AMR 问题清单汇编

Why is the WDVV equation interesting? What are its applications? Is it possible to reformulate the part which is interesting for application by replacing the solution of these equations with a pair $(L,e)$, at least in the $\mathrm{gl}$-regular case? What pairs $(L,e)$ appear in known examples of Frobenius manifolds?

背景链接：https://arxiv.org/abs/2410.04276

07615 | AMR-154-0004

### Nijenhuis Geometry and Integrable Systems - Problem 4

O / open | AMR 问题清单汇编

Given a Nijenhuis operator $L$, describe all of its (local) conservation laws (of course, the same question can be asked for any operator, not necessarily Nijenhuis, see below discussion on superintegrable potentials).

背景链接：https://arxiv.org/abs/2410.04276

07616 | AMR-154-0005

### Nijenhuis Geometry and Integrable Systems - Problem 5

O / open | AMR 问题清单汇编

Which Nijenhuis operators admit (The answer is known near almost every points and follows from the local description of geodesically compatible pairs in [splitting2, splitting1]. The question is most interesting near singular points of the Nijenhuis operator.) at least one geodesically compatible metric $g$? Given $L$, describe all the metrics geodesically compatible with it. Under some additional assumptions ($L$ has $n$ different eigenvalues at generic points and $g$ is Riemannian), the problem was solved in [Matveev2006]. The case of metrics of non-Riemannian signatures, even in dimension 2, is still open. The Riemannian case is solved in [BMF], and the description of 2-dimensional geodesically compatible pairs such that $g$ has signature $(+,-)$ near regular points is in [BMP,japan].

背景链接：https://arxiv.org/abs/2410.04276

07617 | AMR-154-0006

### Nijenhuis Geometry and Integrable Systems - Problem 6

O / open | AMR 问题清单汇编

We have a complete classification of $\mathrm{gl}$-regular Nijenhuis operators in dimension 2 as a list of explicit normal forms. Describe all conservation laws for each of these operators. Or at least, find one regular conservation law for each of them.

背景链接：https://arxiv.org/abs/2410.04276

07618 | AMR-154-0007

### Nijenhuis Geometry and Integrable Systems - Problem 7

O / open | AMR 问题清单汇编

Next, if we have a regular conservation law $f$ for $L$, then we should be able to construct a companion coordinate system and then get a metric $g$ geodesically compatible with $L$, by using the above mentioned explicit formula from [nijappl5].

For each $\mathrm{gl}$-regular Nijenhuis operator $L$ in dimension 2, construct (explicitly) a geodesically compatible metric.

背景链接：https://arxiv.org/abs/2410.04276

07619 | AMR-154-0008

### Nijenhuis Geometry and Integrable Systems - Problem 8

O / open | AMR 问题清单汇编

\begin{Problem} For each $\mathrm{gl}$-regular Nijenhuis operator $L$ in dimension 2, construct (explicitly) a geodesically compatible metric. \end{Problem}

Generalise the above mentioned formula from [nijappl5] to an arbitrary coordinate system $x^1,\dots, x^n$ assuming that in this coordinate system we have a conservation law given as an explicit function $f(x^1,\dots, x^n)$.

背景链接：https://arxiv.org/abs/2410.04276

07620 | AMR-154-0009

### Nijenhuis Geometry and Integrable Systems - Problem 9

O / open | AMR 问题清单汇编

\begin{Problem} Generalise the above mentioned formula from [nijappl5] to an arbitrary coordinate system $x^1,\dots, x^n$ assuming that in this coordinate system we have a conservation law given as an explicit function $f(x^1,\dots, x^n)$. \end{Problem}

Construct new explicit examples of metrics geodesically compatible with $\mathrm{gl}$-regular operators (at singular points!) in arbitrary dimension.

背景链接：https://arxiv.org/abs/2410.04276

07621 | GREEN-047

### No 5 Points on 2-plane in [N]^d

O / open | Ben Green 问题

What is the largest subset of $[N]^d$ with no 5 points on a 2-plane?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

07622 | AIM-GEOMETRY-0219

### No conjugate points: flat-growth rigidity and the three-dimensional residue

P / partially_solved | AIM 研讨会问题汇编

**Question 4.1.2.** If \((M,g)\) has no conjugate points, does \(M\) carry a metric with nonpositive sectional curvature?

Matveev's problem list *Open problems and questions about geodesics*. Its text is:

The record continues by saying that uniformisation gives an affirmative answer in dimension two, that the problem is open even in dimension three, and that Croke--Schroeder, Lebedeva, and Ivanov--Kapovitch establish similarities between fundamental groups in the no-conjugate-points and nonpositively curved settings.

The surrounding section discusses **compact** manifolds, and every cited group-theoretic theorem relevant to the question is stated for compact manifolds. Accordingly, throughout this report:

* \(M\) is a connected, closed, smooth manifold of positive dimension; * \(g\) is a complete Riemannian metric without conjugate points; and * the requested nonpositively curved metric is allowed to differ from \(g\).

This closedness convention is substantive, not cosmetic. Lebedeva's rigidity theorem and the comparison between group growth and volume growth both use compactness.

There is a source-version discrepancy. A later short version of the list says that uniformisation and geometrisation make the answer affirmative in dimensions two and three. No proof or citation accompanies that stronger sentence. By contrast, Ivanov--Kapovitch's primary paper explicitly leaves the no-conjugate-points question open in dimension three and isolates a graph-manifold test case. Searches through 8 August 2026 did not locate a primary source resolving that remaining case. I therefore retain the conservative status “open,” including in dimension three.

The PDF writes “non positive”; below this is silently standardised to “nonpositive.” No mathematical symbols in the question required reconstruction.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07623 | GREEN-045

### No Three in Line in [N]^2

P / partially_solved | Ben Green 问题

What is the largest subset of the grid $[N]^2$ with no three points on a line? In particular, for $N$ sufficiently large, is it impossible to have a set of size $2N$ with this property?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

07624 | AIM-GEOMETRY-0130

### Node residues, spectral gluing, and rank-one affinization

P / partially_solved | AIM 研讨会问题汇编

What happens on singular curves? e.g.

i)$\mathcal{O}(T^{\ast}Bun_{G}(X))$

ii) relate to meromorphic Higgs on $\tilde{X}$

iii) spectral data

iv) extending moduli space $/\mathcal{M}_{g,n}$ to $/\mathcal{\overline{M}}_{g,n}$

来源字段：http://aimpl.org/spectralhiggs/1/

07625 | OWR-13494-011

### Non-ACM Line Configurations with Minimal Symbolic Initial-Degree Gap

O / open | Oberwolfach 报告问题汇编

Does there exist a non-ACM configuration of lines $L$ in $\mathbb P^3$ whose ideal $I$ satisfies $\alpha(I^{(2)})-\alpha(I)=1$?

来源字段：https://doi.org/10.4171/owr/2015/9

07626 | AIM-GEOMETRY-0143

### Non-GL compact Hitchin deformations and a grading obstruction

P / partially_solved | AIM 研讨会问题汇编

For $G$ other than $GL(n)$, do we have a smoothing of $\overline{G-Hitchin}$?

来源字段：http://aimpl.org/spectralhiggs/1/

07627 | OWR-12987-004

### Non-Kähler Analogues of Yau's Monge-Ampère Theorem

P / partially_solved | Oberwolfach 报告问题汇编

Are there non-Kähler analogues of Yau's solution of the complex Monge-Ampère equation and of the resulting resolution of the Calabi conjecture?

来源字段：https://doi.org/10.4171/owr/2014/29

07628 | OWR-13347-002

### Non-Kähler Shrinking Ricci Solitons

O / open | Oberwolfach 报告问题汇编

Do compact non-Kähler shrinking Ricci solitons exist? Do complete noncompact non-Kähler shrinkers exist that are not of Gaussian type?

来源字段：https://doi.org/10.4171/owr/2014/45

07629 | OWR-13347-008

### Non-Naturally Reductive Einstein Metrics on Compact Lie Groups

P / partially_solved | Oberwolfach 报告问题汇编

Find further non-naturally reductive Einstein metrics on compact simple Lie groups $G$ by using homogeneous spaces $G/H$.

来源字段：https://doi.org/10.4171/owr/2014/45

07630 | OWR-15587-002

### Non-Unimodal Ehrhart Polynomials of Reflexive Simplices

P / partially_solved | Oberwolfach 报告问题汇编

Among the reflexive simplices $\Delta_{(1,q)}=\operatorname{conv}(e_1,\ldots,e_n,-q)$ having the integer-decomposition property, which have a nonunimodal Ehrhart $h^*$-polynomial?

来源字段：https://doi.org/10.4171/owr/2017/44

07631 | AIM-GEOMETRY-0302

### Noncompact equivariant integration and the boundary at infinity

P / partially_solved | AIM 研讨会问题汇编

Let \((M,\omega)\) be a noncompact real-algebraic symplectic manifold. Let a compact torus \(T\), with Lie algebra \(\mathfrak t\), act Hamiltonianly on \(M\), with proper semialgebraic moment map \(\mu:M\to\mathfrak t^*\), and suppose the fixed set \(M^T\) is compact. For a smooth equivariant differential form \(\alpha:\mathfrak t\to\Omega^*(M)\) having the intended semialgebraic dependence (not necessarily polynomial) on the variables, can one geometrically define \(\int_M\alpha\), valued in smooth functions on \(\mathfrak t\), so that fixed-point localization holds?

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

07632 | AIM-GEOMETRY-0188

### Noncontractible equilateral trefoil components after rotation reduction

P / partially_solved | AIM 研讨会问题汇编

**1. (Farber) Configuration spaces of closed polygons in \(\mathbb R^3\).** Let \(\ell=(\ell_1,\ldots,\ell_n)\) be the length vector defining a closed polygon. The configuration space is \[ M_\ell\setminus\Sigma=\bigsqcup U_i, \] where \(U_i\) are the connected components, and \(\Sigma\) is the set of configurations with self-intersections. Questions: Are the \(U_i\) contractible? What can we say about their topology?

来源字段：https://aimath.org/pastworkshops/linkagesproblems.pdf

07633 | OWR-15579-002

### Nonexistence of Edge-Cone Yamabe Metrics at Small Cone Angles

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Edge-Cone Yamabe Nonexistence Conjecture [OWR-15579-006]

For $1<\beta<2$, does the conformal class $[h_\beta]$ of the standard edge-cone metric of cone angle $2\pi\beta$ on $(S^n,S^{n-2})$ admit no edge-cone Yamabe metric?

来源字段：https://doi.org/10.4171/owr/2017/36

07634 | AIM-GEOMETRY-0206

### Nonexistence of the complex (8,3) ETF and a triangle-balance certificate

S / solved | AIM 研讨会问题汇编

**20.** Do complex ETF's exist for the parameters \(M=3\), \(N=8\)?

**21.** Construct a table/catalogue whose entries correspond to pairs \((N,M)\) for small \(N,M\), indicating whether complex equiangular frames are known to exist, not to exist, or remain open, and, when settled, the method. The source gives \((7,3)\) as existing via difference sets and \((8,3)\) as open.

来源字段：https://aimath.org/pastworkshops/frametheorygeomproblems.pdf

07635 | OWR-14299519-002

### Nonfattening and Intersection Principles for Level-Set Flows

O / open | Oberwolfach 报告问题汇编

For a level-set mean-curvature flow $(M_t)_{t\geq0}$, to what extent is nonfattening equivalent to nondiscrepancy and to the intersection principle with every smooth closed mean-curvature flow?

来源字段：https://doi.org/10.4171/owr/2025/33

07636 | OWR-4136-005

### Nonmonotonicity of Expected Random Simplex Volume

P / partially_solved | Oberwolfach 报告问题汇编

In $\mathbb{R}^3$, let $L$ be the convex hull of the half unit ball $\{x:\|x\|_2\leq1,\ x_1\geq0\}$ and the point $(-\varepsilon,0,0)$, and let $K=L\cap\{x:x_1\geq-\varepsilon+\delta\}$, where $\varepsilon,\delta>0$ are small. Is $K\subset L$ a counterexample to monotonicity of expected random-simplex volume; that is, is the expected volume of the convex hull of four independent uniform points of $K$ greater than the corresponding expectation for $L$?

来源字段：https://doi.org/10.4171/owr/2009/53

07637 | OWR-1194-003

### Nonoverlapping Edge Unfoldings of Convex Polytopes

O / open | Oberwolfach 报告问题汇编

Can the boundary of every convex 3-polytope be unfolded into the plane without overlap by cutting only along edges?

来源字段：https://doi.org/10.4171/owr/2006/12

07638 | OWR-13951-007

### Nonpointwise, Nonassociative Operations on Convex Objects

O / open | Oberwolfach 报告问题汇编

Find classification results for operations on convex functions or convex sets that are neither necessarily pointwise nor associative.

来源字段：https://doi.org/10.4171/owr/2015/56

07639 | AMR-058-0010

### Nonpolyhedral Soap-Film Cones

O / open | AMR 问题清单汇编

Do nonpolyhedral minimizing soap-film cones exist in dimensions four through seven? Construct higher-dimensional examples using triple junctions which are not cones over smooth manifolds.

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

07640 | OWR-14299288-004

### Nonpositively Curved Cusped Manifolds from Tropical Curves

O / open | Oberwolfach 报告问题汇编

Given a tropical curve represented by $\Sigma^{(1)}\subset\Sigma_{\mathbb{P}^n}$, construct a nonpositively curved cusped manifold $L_{\Sigma^{(1)}}$ whose boundary is a union of $n+1$ real tori of dimension $n-1$.

来源字段：https://doi.org/10.4171/owr/2025/19

07641 | AMR-059-0017

### Nonproduct Compact Hessian Manifolds

P / partially_solved | AMR 问题清单汇编

Construct compact Hessian manifolds other than direct products of hyperbolic and flat compact Hessian manifolds; in particular, construct one with no fiber-bundle structure using deformation theory of affine structures.

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

07642 | AIM-GEOMETRY-0380

### Nonuniqueness of Q-curvature extensions to Weyl structures

P / partially_solved | AIM 研讨会问题汇编

How is \(Q\)-curvature related to Weyl structures?

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

07643 | AIM-GEOMETRY-0171

### Normal stability and scale selection for inverse Gauss curvature flow near the round sphere

P / partially_solved | AIM 研讨会问题汇编

Let $S^2$ be the 2 dimension unit sphere and $g$ be a metric on $S^2$ with positive Gauss curvature. Take $g=e^{2u} ds^2$. Consider the flow $$\partial_t u=-1+\frac{\overline{K_u}}{K_u}$$ where $\overline{K_u}$ is the average of $K_u$. Known: long time existence; some weak convergence (a priori estimate degenerate at $t=\infty$).

Question: Does the flow converge to round? (If the genus is greater than 1, the analogous flow is understood). What about higher dimensions? More information for $n=2$ can be found on the paper on arxiv.

来源字段：http://aimpl.org/flowriemannian/3/

07644 | OWR-1595-005

### Normality and Smoothness of Truncated Hadamard Simplices

O / open | Oberwolfach 报告问题汇编

Let $C=[-1,1]^d$, let $\Delta$ be a Hadamard simplex (a regular simplex whose vertices are vertices of $C$), and let $m,n$ be odd natural numbers with $n<m<nd$. In the affine lattice $\Lambda=(n+2\mathbb{Z})^d$, is $m\Delta\cap nC$ a lattice polytope? For which values of $m$ and $n$ is it normal, and for which is it smooth?

来源字段：https://doi.org/10.4171/owr/2007/39

07645 | OWR-15587-006

### Normality and Very Ampleness of Lattice Polytopes

O / open | Oberwolfach 报告问题汇编

If $B\subset\mathbb R^n$ is a ball, is $\operatorname{conv}(B\cap\mathbb Z^n)$ normal? Does there exist a very ample nonnormal simple lattice polytope, or even a very ample nonnormal lattice simplex?

来源字段：https://doi.org/10.4171/owr/2017/44

07646 | OWR-15587-008

### Normality of Smooth Lattice Polytopes

O / open | Oberwolfach 报告问题汇编

Is every smooth lattice polytope normal?

来源字段：https://doi.org/10.4171/owr/2017/44

07647 | AIM-GEOMETRY-0154

### Normalized extinction-gap budget and the correct C0 topology

P / partially_solved | AIM 研讨会问题汇编

Consider $(S^3,g)$ with scalar curvature $R\geq 6$. The Ricci Flow exists on the maximal interval $(0,T)$. The scalar curvature evolves by $$\partial_t R = \Delta R + 2 |Rc|^2.$$ One can estimate that $$\partial_t R = \Delta R + 2 |Rc|^2 = \Delta R + \frac 2 3 R^2 + |\mathring{Rc}|^2 \geq \Delta R + \frac 2 3 R^2 .$$ It follows from the maximum principle that $$R\geq \frac{6}{1-4t} \mbox{ and } T \leq \frac 1 4.$$

If $T$ is close to $\frac 1 4$, is $g(0)$ close to the round metric in the Gromov-Hausdorff sense and in the $C^0$ sense? One can also ask the same question for other flows such as mean curvature flow. What would be a suitable topology?

来源字段：http://aimpl.org/flowriemannian/1/

07648 | OWR-1726-004

### Novikov-Shubin Positivity and Foliated Index Characters

O / open | Oberwolfach 报告问题汇编

For Riemannian foliations, are the Novikov-Shubin invariants always positive? More generally, whenever the index bundle of a leafwise elliptic operator exists, is its Chern-Connes character always equal to the expected characteristic classes?

来源字段：https://doi.org/10.4171/owr/2007/43

07649 | AIM-GEOMETRY-0265

### Null completeness, the surface case, and a flat-holonomy obstruction

P / partially_solved | AIM 研讨会问题汇编

**Question 9.3.2 ([Sanchez2013]).** Assume that a compact Lorentzian manifold is globally conformal to a manifold of constant curvature. Must it be geodesically complete?

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07650 | AMR-058-0011

### Number of Regions Meeting in a Minimizing Partition

P / partially_solved | AMR 问题清单汇编

Let $k(n)$ be the maximum number of regions meeting at one point in an area-minimizing partition in $n$ dimensions. Determine the growth of $k(n)$ and whether it is strictly increasing; investigate the analogue for other surface energies.

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

07651 | AMR-038-0010

### Odd rep-tiling by a 14-omino

O / open | AMR 问题清单汇编

Can the $3\times6$ rectangle with a $2\times2$ corner removed tile a rectangle using an odd number of congruent copies?

背景链接：https://ics.uci.edu/~eppstein/junkyard/open.html

07652 | AMR-067-0017

### On the Hodge spectra of lens spaces

P / partially_solved | AMR 问题清单汇编

- Construct congruence lattices which are norm$_1$ and norm$_1*$- isospectral in all dimensions (see ). - Are there families of $p$-isospectral lens spaces for all $p$, with more than two elements? - Give a procedure to go from $p$-isospectral lens spaces (for all $p$) of dimension $m$ to dimension $m + 1$. - Establish connections with toric geometry (see ).

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

07653 | AIM-GEOMETRY-0011

### One-dimensional radial power kernels and an Ehrhard Gaussian obstruction

P / partially_solved | AIM 研讨会问题汇编

If \(F:\mathbb R\to\mathbb R\) is even and convex, then for any \(f,g\in L^1(\mathbb R^n)\), \[ \iint F(x\cdot y)f(x)g(y)\,dx\,dy \geq \iint F(x\cdot y)f^*(x)g^*(y)\,dx\,dy, \] and asks whether convexity of \(F\) can be weakened and whether there is a Gaussian version.

There is a minor transcription error in the displayed domain (one occurrence of `R^n` lacks the LaTeX backslash), but there is also a major mathematical omission. Xi--Zhao's actual Theorem 1.2 assumes that \(f\) and \(g\) are **nonnegative, quasi-concave, and integrable**, not arbitrary members of \(L^1\). Their theorem says that, for such \(f,g:\mathbb R^n\to[0,\infty)\) and every finite even convex \(F:\mathbb R\to\mathbb R\), \[ \mathcal E_F(f,g):=\int_{\mathbb R^n}\int_{\mathbb R^n} F(x\cdot y)f(x)g(y)\,dx\,dy \geq \mathcal E_F(f^*,g^*). \tag{1} \] For strictly convex \(F\), their equality statement describes almost all superlevel-set closures of \(f\) and \(g\) as dilates of a common pair of polar-reciprocal origin-symmetric ellipsoids.

The qualification matters. Symmetric decreasing rearrangement is canonically defined here for nonnegative functions; for signed functions, writing \(f^*\) without specifying whether one rearranges \(|f|\), positive and negative parts, or a signed distribution is ambiguous. If one adopts the common convention \(f^*=(|f|)^*\), even the constant kernel \(F=1\) fails for \(f=-\mathbf1_{[0,1]}\), \(g=\mathbf1_{[0,1]}\): the two sides are \(-1\) and \(+1\). Quasi-concavity is also structural in Xi--Zhao: it makes the positive superlevel sets convex, allowing their convex-body symmetrization argument. The AIM workshop report states the indicator-function version for convex sets, consistent with this scope.

There is a second formulation issue: \(L^1\) alone does not make the displayed energies finite when \(F\) grows. A safe extended-real reading of (1) is obtained by setting \[ \Phi(t)=F(t)-F(0)\geq0, \qquad \mathcal E_F(f,g)=F(0)\|f\|_1\|g\|_1+ \iint \Phi(x\cdot y)f(x)g(y)\,dx\,dy. \tag{2} \] The last integral is defined by Tonelli in \([0,\infty]\), so there is no \(\infty-\infty\) ambiguity. Since rearrangement preserves both masses, the constant term cancels. For a non-vacuous finite-energy assertion one should additionally assume either side, or preferably both sides, finite. For example, \(F(t)=t^2\) and an integrable quasi-concave function with an infinite second moment already produce an infinite energy in dimension one.

来源字段：http://aimpl.org/symmetrybreaking/3/

07654 | AMR-071-0053

### Opaque forest problem on finding opaque sets for various planar shapes

O / open | AMR 问题清单汇编

Opaque forest problem on finding opaque sets for various planar shapes

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

07655 | AMR-149-0001

### Open Problems and Questions about Geodesics - Conjecture 1 (Long, Bangert, Problem 15 from [Alvarez2006])

O / open | AMR 问题清单汇编

V. Bangert and Y. Long in [Ban-Lon2010] and [Long2006] proved that there are always at least 2 distinct prime closed geodesics for every irreversible Finsler metric on $S^2$.

*Every irreversible Finsler metric on $S^2$ has either exactly $2$ or infinitely many distinct prime closed geodesics.*

背景链接：https://arxiv.org/abs/1308.5417

07656 | AMR-149-0011

### Open Problems and Questions about Geodesics - Conjecture 11 (Hingston)

O / open | AMR 问题清单汇编

\begin{align*} \alpha_{g,G} & =a\text{ if }G=\mathbb{Q}\\ \alpha_{g,G} & =b\text{ if }G = \mathbb{Z}_{3}\\ \alpha_{g,G} & =c\text{ if }G = \mathbb{Z}_{7}\\ \alpha_{g,G} & =a\text{ if }G= \mathbb{Z}_{p},p\neq3,7. \end{align*}

背景链接：https://arxiv.org/abs/1308.5417

07657 | AMR-149-0012

### Open Problems and Questions about Geodesics - Conjecture 12 (Klingenberg)

O / open | AMR 问题清单汇编

The class of compact manifolds that support metrics with variable negative curvature is much larger than the class that support hyperbolic metrics (of constant negative curvature). The earliest examples of manifolds that support variable but not constant negative curvature were given by Mostow-Siu [Mos-Siu1980] and Gromov-Thurston [Gro-Thu1987]. Recent work of Ontaneda has vastly increased the supply of examples [Ontaneda2011, Ontaneda2014].

*If a closed manifold admits a metric with Anosov geodesic flow, then it admits a metric with negative sectional curvature.*

背景链接：https://arxiv.org/abs/1308.5417

07658 | AMR-149-0015

### Open Problems and Questions about Geodesics - Conjecture 15 ([Bur-Kat1985])

O / open | AMR 问题清单汇编

Suppose $(M, g)$ is a closed Riemannian manifold. The marked length spectrum of $M$ is the function that assigns to each free homotopy class of loops the infimum of the length of loops in the class (i.e. the length of the shortest closed geodesic lying in this class).

*Two metrics with negative curvature on a compact manifold must be isometric if they have the same marked length spectrum.*

背景链接：https://arxiv.org/abs/1308.5417

07659 | AMR-149-0018

### Open Problems and Questions about Geodesics - Conjecture 18

O / open | AMR 问题清单汇编

Two Riemannian manifolds $(M_1,g_1)$ and $(M_2,g_2)$ have $C^k$-conjugate geodesic flows if there is an invertible map $h: S^M_1 \to S^M_2$ such that $h$ and $h^{-1}$ are $C^k$ and $h \circ \phi^1_t = \phi^2_t \circ h$ for all $t$. Here $\phi^i_t$ is the geodesic flow for the metric $g_i$. A long standing conjecture is

Compact Riemannian manifolds with negative curvature must be isometric if they have $C^0$-conjugate geodesic flows.

背景链接：https://arxiv.org/abs/1308.5417

07660 | AMR-149-0019

### Open Problems and Questions about Geodesics - Conjecture 19 ([Katok1982])

O / open | AMR 问题清单汇编

Katok [Katok1982] showed that if $g$ is an arbitrary metric and $g_0$ a metric of constant negative curvature on a surface of genus $\geq 2$ such that $\text{area}(g) = \text{area}(g_0)$, then $$ h_{Liou}(g) \leq h_{Liou}(g_0) =h_{top}(g_0) \leq h_{top}(g) $$ and both inequalities are strict unless $g$ also has constant negative curvature. He explicitly formulated the following influential and still open conjecture:

$h_{top} = h_{Liou}$ for a metric of negative curvature on a compact manifold if and only if it is locally symmetric.

背景链接：https://arxiv.org/abs/1308.5417

07661 | AMR-149-0002

### Open Problems and Questions about Geodesics - Conjecture 2 (Long)

O / open | AMR 问题清单汇编

There exist results supporting this conjecture. In particular, H. Hofer, K. Wysocki and E. Zehnder in [Ho-Wy-Ze2003] studied Reeb orbits on contact $S^3$. Their result can be projected down to $S^2$, and implies that the total number of distinct prime closed geodesics for a bumpy Finsler metric on $S^2$ is either 2 or infinite, provided the stable and unstable manifolds of every hyperbolic closed geodesics intersect transversally. See also [Har-Pat2008,Rademacher2016].

The closed geodesics in Katok's example are elliptic.

*The existence of one hyperbolic prime closed geodesic on a Finsler $S^2$ implies the existence of infinitely many distinct prime closed geodesics.*

背景链接：https://arxiv.org/abs/1308.5417

07662 | AMR-149-0022

### Open Problems and Questions about Geodesics - Conjecture 22

O / open | AMR 问题清单汇编

Let $(M, g)$ be a closed Riemannian manifold with negative sectional curvatures. It is known [Margulis2004] that the number $N(T)$ of closed geodesics of length $\le T$ grows asymptotically as ${e^{hT}}/{hT}$.

{\it $N(T)= (1+ O(e^{-\varepsilon T})) \displaystyle \int_2^{e^{hT}} \frac{du}{\log u} $. }

背景链接：https://arxiv.org/abs/1308.5417

07663 | AMR-149-0003

### Open Problems and Questions about Geodesics - Conjecture 3 (Long)

O / open | AMR 问题清单汇编

\begin{con}[Long] *The existence of one hyperbolic prime closed geodesic on a Finsler $S^2$ implies the existence of infinitely many distinct prime closed geodesics.* \end{con}

*Every Finsler $S^2$ has at least one elliptic prime closed geodesic.*

背景链接：https://arxiv.org/abs/1308.5417

07664 | AMR-149-0030

### Open Problems and Questions about Geodesics - Conjecture 30 (Spatzier)

O / open | AMR 问题清单汇编

*Let $(M,g)$ be a closed symmetric space of non-compact type and of higher rank. Then the only metrics on $M$ with no conjugate points are homothetic rescalings of $g$.*

背景链接：https://arxiv.org/abs/1308.5417

07665 | AMR-149-0031

### Open Problems and Questions about Geodesics - Conjecture 31 (Burago-Kleiner)

O / open | AMR 问题清单汇编

Consider the product of a closed surface $M^2$ of genus $\ge 2$ with the circle $S$. Let $g$ be a Riemannian metric on $M\times S$ with no conjugate points.

{\it The metric on the $\mathbb{Z}$-cover corresponding to $S^1$ factor is a direct product: $(M\times \mathbb{R}, g)=(M, g_1)\times (\mathbb{R}, dt^2)$. }

背景链接：https://arxiv.org/abs/1308.5417

07666 | AMR-149-0050

### Open Problems and Questions about Geodesics - Conjecture 51

O / open | AMR 问题清单汇编

In the Lorenzian setting, one can ask to construct all manifolds such that all light-like geodesics are closed, see e.g.\ [Mou-Suh2013, Suhr2013b].

An easier version of the question above is the following conjecture sometimes attributed to Berger:

*Let $(M,g)$ be a closed simply connected manifold such that all geodesics are closed. Then all geodesics have the same length.*

背景链接：https://arxiv.org/abs/1308.5417

07667 | AMR-149-0059

### Open Problems and Questions about Geodesics - Conjecture 60 ([Bo-Ko-Fo1995])

O / open | AMR 问题清单汇编

*If the geodesic flow of a Riemannian metric on the torus $T^2$ admits an integral that is polynomial of degree 3 in the velocities, then the metric admits a Killing vector field. *

背景链接：https://arxiv.org/abs/1308.5417

07668 | AMR-149-0060

### Open Problems and Questions about Geodesics - Conjecture 61

O / open | AMR 问题清单汇编

We ask whether there exists a non-constant smooth function on the tangent bundle which is constant along every trajectory. It is well known, see e.g.\ [Tabachnikov2005], that a billiard in an ellipse is integrable.

*If the billiard in $\Omega$ is integrable, then the boundary of $\Omega$ is an ellipse. *

背景链接：https://arxiv.org/abs/1308.5417

07669 | AMR-149-0039

### Open Problems and Questions about Geodesics - Problem 40

O / open | AMR 问题清单汇编

Flag curvature is the Finslerian analogue of sectional curvature. It was shown in [Akbar-Zadeh1988] that: any closed Finsler surface with constant flag curvature zero is locally Minkowskian; any sufficiently smooth closed Finsler surface with constant negative curvature is Riemannian (there are non-Riemannian examples which are not sufficiently smooth); and any connected closed Finsler surface with constant positive flag curvature is homeomorphic to $S^2$ or $\mathbb{R} P^2$. The result for negative flag curvature also follows from [Foulon1997] and is true in all dimensions.

*Describe Finsler metrics on $S^2$ of flag curvature $1$.*

背景链接：https://arxiv.org/abs/1308.5417

07670 | AMR-149-0040

### Open Problems and Questions about Geodesics - Problem 41

O / open | AMR 问题清单汇编

Moreover, the Finsler metrics satisfying (i) necessary admit a Killing vector field and are quite well understood. Applying the Zermelo transformation to such metrics (see e.g. [Fou-Mat2018]), one obtains non-Zoll metrics satisfying (ii). Moreover, any such metric (of flag curvature $1$ on $S^2$ and satisfying (ii)) admitting a Killing vector field can be obtained by this procedure. Unfortunately, no other examples of non-Zoll metrics satisfying (ii) are known.

*Construct examples of Finsler metrics of constant curvature on $S^2$ that are not Zoll and such that they admit no Killing vector field. *

背景链接：https://arxiv.org/abs/1308.5417

07671 | AMR-149-0043

### Open Problems and Questions about Geodesics - Problem 44 (Benjamin Schmidt)

O / open | AMR 问题清单汇编

We consider a Riemannian metric on an oriented closed surface $M$ of genus $\ge 1$. Let $\Gamma$ be a non-trivial free homotopy class and $p$ a point in $M$. A theorem of Gaidukov [Gaidukov1966] says that there exist a closed geodesic $\gamma:\mathbb{R}\to M$ in $\Gamma$ and a ray $\beta:[0,\infty)\to M$ with $\beta(0)=p$ such that $dist(\beta(t), \gamma(t)) \to 0$ as $t \to \infty$. As explained in [Bia-Pol1986], Gajdukov's results follow easily from the classical results of [Morse1924] and [Hedlund1932].

*Generalize this statement to higher dimensions.*

背景链接：https://arxiv.org/abs/1308.5417

07672 | AMR-149-0064

### Open Problems and Questions about Geodesics - Problem 65

O / open | AMR 问题清单汇编

Also in the Finsler case parameterized geodesics determine the connection. Note though that in the Finsler geometry the connection is non-linear and actually there are more than one popular and widely used connections. We will call two Finsler metrics *affinely equivalent* if any geodesic of the first metric (considered as a curved parametrized such that the length of the velocity vector is a constant) is a geodesic of the second metric; this is equivalent to the Berwald connections of the first and the second metric coinciding in the natural sense.

*Describe all affinely equivalent Finsler metrics.*

背景链接：https://arxiv.org/abs/1308.5417

07673 | AMR-149-0065

### Open Problems and Questions about Geodesics - Problem 66

O / open | AMR 问题清单汇编

*Construct a system of scalar invariants of a projective structure that vanish if and only if there exists (locally, in a neighborhood of almost every point) a metric compatible with a given projective structure.*

背景链接：https://arxiv.org/abs/1308.5417

07674 | AMR-149-0010

### Open Problems and Questions about Geodesics - Question 10 (Hingston)

O / open | AMR 问题清单汇编

Let $M$ be a compact manifold with a Riemannian (or Finsler) metric $g$. \ Let $G$ be a finitely generated abelian group. \ Given a non-trivial homology class $X \in H_{\ast}(\Lambda M;G)$, the critical (minimax) level of $X$ is \begin{align*} cr X & =\inf\{a:X\in\text{Image }H_{\ast}(\Lambda_a M;G)\}\\ & =\underset{x\epsilon X}{\inf}\text{ }\underset{\gamma\in\text{Image}}{\sup}Length({\gamma}) \end{align*} Here $\Lambda_a M$ is the subset of the free loop space $\Lambda M$ consisting of loops whose length is at most $a$. The second definition is the minimax definition: the singular chain $x$ ranges over all representatives of the homology class $X$, and $\gamma$ over all the points in the image of $x$, which are loops in $M$.

*Do there exist a metric on *$S^{n}$* and a homology class *$X \in H_{\ast}(\Lambda M;

\mathbb{Z}

)$* with * \[ 0 < cr(mX) < c r(X) \] *for some *$m \in \mathbb{N}

$?

背景链接：https://arxiv.org/abs/1308.5417

07675 | AMR-149-0013

### Open Problems and Questions about Geodesics - Question 13

O / open | AMR 问题清单汇编

Several examples of manifolds that admit metrics of non-positive curvature but cannot support a metric of negative curvature (or with Anosov geodesic flow) can be found in the introduction to [Ba-Br-Eb1985]. These examples have a copy of ${\mathbb{Z}} \times {\mathbb{Z}}$ in their fundamental group. The simplest of them is due to Heintze who took two cusped hyperbolic 3-manifolds, cut off the cusps, glued the two pieces together along their boundary tori and then smoothed out the metric to obtain a manifold with non-positive curvature.

*If $(M,g)$ has no conjugate points, does $M$ carry a metric with non-positive sectional curvature?*

背景链接：https://arxiv.org/abs/1308.5417

07676 | AMR-149-0014

### Open Problems and Questions about Geodesics - Question 14

O / open | AMR 问题清单汇编

Again it is not difficult to use the uniformization theorem to show that the answer is affirmative in dimension 2. But the problem is still open even in dimension 3. See [Cro-Sch1986], [Lebedeva2002] and [Iva-Kap2014] for results of the nature that the fundamental groups of manifolds with no conjugate points share properties with those of non-positive curvature. Section 8 of [Iva-Kap2014] contains a number of open problems of which we mention:

*Is the fundamental group of a closed manifold without conjugate points semihyperbolic?*

背景链接：https://arxiv.org/abs/1308.5417

07677 | AMR-149-0016

### Open Problems and Questions about Geodesics - Question 16 (Benjamin Schmidt)

O / open | AMR 问题清单汇编

One can ask a similar question for non-manifolds. Consider a 2-dimensional (metrical) simplicial complex such that every simplex is hyperbolic with geodesic edges and such that $CAT(-1)$ condition holds on every vertex. Assume in addition that every edge is contained in at least two simplices.

*Does the marked length spectrum determine such a metric (in the class of all locally $CAT(-1)$ metrics on this space, or in the class of all 2-dimensional (metrical) simplicial complexes with the above properties homeomorphic to the given complex)?*

背景链接：https://arxiv.org/abs/1308.5417

07678 | AMR-149-0017

### Open Problems and Questions about Geodesics - Question 17 (Boundary rigidity problem)

O / open | AMR 问题清单汇编

The questions above are closely related to the boundary rigidity problem, which we now recall. Given a compact manifold $M$ with smooth boundary $N$, a Riemannian metric $g$ on $M$ induces a non-negative real valued function $d$ on $N \times N$ where $d(p, q)$ is the distance in $(M, g)$ between $p$ and $q$. We call $(M,g)$ boundary rigid if a Riemannian manifold $g'$ on $M$ that induces the same function on $N \times N$ must be isometric to $g'$.

*What conditions on $M$, $N$ and $g$ imply boundary rigidity?*

背景链接：https://arxiv.org/abs/1308.5417

07679 | AMR-149-0020

### Open Problems and Questions about Geodesics - Question 20

O / open | AMR 问题清单汇编

An extensive list of results about the regularity of the Anosov splitting for the geodesic flow in negative curvature (and for other Anosov systems) can be found in the introduction to [Hasselblatt1994]. In dimension two, or if the curvature is $1/4$-pinched, the splitting is always at least $C^1$. In higher dimensions it is always H\"older continuous but typically not $C^1$. It is natural to ask:

*Does there exist a closed Riemannian manifold of negative curvature such that $E^s $ and $ E^u$ are $C^1$ smooth, but the dimension is $> 2$ and the metric is not $1/4$-pinched*?

背景链接：https://arxiv.org/abs/1308.5417

07680 | AMR-149-0021

### Open Problems and Questions about Geodesics - Question 21

O / open | AMR 问题清单汇编

Is there a metric of negative curvature for which the ratio of largest positive Lyapunov exponent to smallest positive Lyapunov exponent is greater than $2$ for typical geodesics (not just exceptional closed geodesics)?

背景链接：https://arxiv.org/abs/1308.5417

07681 | AMR-149-0023

### Open Problems and Questions about Geodesics - Question 23 (Miller [Miller2001], Reid)

O / open | AMR 问题清单汇编

*Are there infinitely many simple closed geodesics in every hyperbolic complete three-manifold of finite volume?*

背景链接：https://arxiv.org/abs/1308.5417

07682 | AMR-149-0024

### Open Problems and Questions about Geodesics - Question 24

O / open | AMR 问题清单汇编

*Is the geodesic flow of a metric of non-positive curvature on a closed surface of genus $\ge 2$ ergodic with respect to the Liouville measure? *

背景链接：https://arxiv.org/abs/1308.5417

07683 | AMR-149-0025

### Open Problems and Questions about Geodesics - Question 25

O / open | AMR 问题清单汇编

\ Consider a closed surface of genus $\ge 2$.

[Burns] *Is there a $C^\infty$ (or at least $C^k, \ k\ge 3)$ metric for which there exists a non-closed geodesic along which the Gaussian curvature is everywhere zero?*

背景链接：https://arxiv.org/abs/1308.5417

07684 | AMR-149-0026

### Open Problems and Questions about Geodesics - Question 26

O / open | AMR 问题清单汇编

[Knieper] *Consider a closed surface of genus $\geq 2$ with a metric of non-positive curvature. Let $Z(L)$ be the number of free homotopy classes containing closed geodesics with length $\leq L$ along which the curvature is everywhere zero. Is there a quadratic upper bound for $Z(L)$, i.e.\ is there a constant $C \geq 0$ such that $$ Z(L) \leq C \cdot L^2? $$ *

背景链接：https://arxiv.org/abs/1308.5417

07685 | AMR-149-0027

### Open Problems and Questions about Geodesics - Question 27

O / open | AMR 问题清单汇编

Let $\alpha$ and $\beta$ be two geodesics in a complete simply connected Riemannian manifold without conjugate points.

*Suppose $\alpha(0)=\beta(0)$. Does $dist(\alpha(t), \beta(t)) \to \infty$ as $t\to \infty$?*

背景链接：https://arxiv.org/abs/1308.5417

07686 | AMR-149-0028

### Open Problems and Questions about Geodesics - Question 28

O / open | AMR 问题清单汇编

The answer is positive in dimension 2 [Green1954]. The question is open in higher dimensions; the proof in [Green1956] is incorrect. The answer to the corresponding question about Finsler metrics is negative. The Teichm\"uller metric on Teichm\"uller space has no conjugate points, but Masur showed that there exist two rays through the same point which do not diverge [Masur1975].

A closely related question is:

*Suppose $dist(\alpha(t),\beta(t)) \to 0$ as $t \to -\infty$. Does $dist(\alpha(t), \beta(t)) \to \infty$ as $t\to \infty$?*

背景链接：https://arxiv.org/abs/1308.5417

07687 | AMR-149-0029

### Open Problems and Questions about Geodesics - Question 29 ([Bur-Kni1991])

O / open | AMR 问题清单汇编

The simplest question of this type asks if a Riemannian plane satisfying Euclid's 5th postulate must be flat. Consider the plane $\mathbb{R}^2$ with a complete Riemannian metric. Assume that for every geodesic and for every point not on the geodesic there exists precisely one non-trivial geodesic that passes through the point and does not intersect the geodesic. This assumption implies that the metric has no conjugate points.

*Must a metric satisfying this version of the parallel postulate be flat?*

背景链接：https://arxiv.org/abs/1308.5417

07688 | AMR-149-0032

### Open Problems and Questions about Geodesics - Question 32 (Paternain)

O / open | AMR 问题清单汇编

*Consider a magnetic flow on a closed surface of genus $\ge 2$ and an energy level such that the topological entropy vanishes. Suppose that the magnetic geodesics on this level do not have conjugate points. Is the system locally symmetric in the sense that the metric has constant curvature and the magnetic form is a constant multiple of the volume form?*

背景链接：https://arxiv.org/abs/1308.5417

07689 | AMR-149-0033

### Open Problems and Questions about Geodesics - Question 33

O / open | AMR 问题清单汇编

The general problem is still wide open despite some reports of its solution (p. 87 of [Berger2000] and Section 10.9 of [Berger2003]).

*Is there a Riemannian metric on a closed manifold with everywhere positive sectional curvatures and ergodic geodesic flow?*

背景链接：https://arxiv.org/abs/1308.5417

07690 | AMR-149-0034

### Open Problems and Questions about Geodesics - Question 34 (Benjamin Schmidt)

O / open | AMR 问题清单汇编

\begin{align*} T_b &:=\{ v\in SM\mid \textrm{ any lift of $\gamma_v$ stays in a bounded subset of $\widetilde M$}\}, \\ T_p &:=\{ v\in SM\mid \textrm{ any lift of $\gamma_v$ is unbounded, and approaches infinity properly}\}, \\ T_i &:=TM\setminus (T_b \cup T_p). \end{align*}

All three sets are measurable and invariant under the geodesic flow. By ergodicity one of them has full measure.

*Which of these sets has full measure?*

背景链接：https://arxiv.org/abs/1308.5417

07691 | AMR-149-0035

### Open Problems and Questions about Geodesics - Question 35

O / open | AMR 问题清单汇编

We use $C^k$-topology either locally or on closed manifolds, where we may assume that the atlas is finite and the gluing functions have all derivatives bounded. Then the convergence in the topology means the following: a sequence $g^{(1)},g^{(2)},\dots$ of metrics converges to a metric $g$ in the topology $C^k$ if in local charts the entries $g^{(m)}_{ij}$ (viewed as local functions in this chart) and their partial derivatives $\frac{\partial g^{(m)}_{ij}}{\partial x_\ell}, \frac{\partial^2 g^{(m)}_{ij}}{\partial x_\ell \partial x_s},\ldots $ up to order $k$ converge to those of $g_{ij}$.

Assume $M$ is closed. Do the metrics with positive entropy form a $C^k$-dense subset of $\mathcal{G}(M)$?

背景链接：https://arxiv.org/abs/1308.5417

07692 | AMR-149-0036

### Open Problems and Questions about Geodesics - Question 36

O / open | AMR 问题清单汇编

Assume $M$ is closed. If $\dim(M)\ge 3$, do the metrics for which the geodesic flow is transitive (i.e.\ has an orbit whose closure is $SM$) form a dense subset of $\mathcal{G}(M)$?

背景链接：https://arxiv.org/abs/1308.5417

07693 | AMR-149-0037

### Open Problems and Questions about Geodesics - Question 37

O / open | AMR 问题清单汇编

This question is closely related to the famous Arnold diffusion conjecture. If $\dim(M)=2$ and $k$ is large enough, the answer is negative: if we take a metric with integrable geodesic flow such that certain Liouville tori are irrational, and any small perturbation of the metric has non-transitive geodesic flow by the KAM theory.

{\it Assume $M$ is closed. Is there a dense subset of metrics in $\mathcal{G}$ for which the tangent vectors to the closed geodesics are dense in $SM$? }

背景链接：https://arxiv.org/abs/1308.5417

07694 | AMR-149-0038

### Open Problems and Questions about Geodesics - Question 38

O / open | AMR 问题清单汇编

In the previous section we discussed the unit tangent bundle $SM$. In this section we ask similar questions about $M$ itself, and we will not assume that the metrics are generic.

*Is the union of the closed geodesics always dense in the manifold?*

背景链接：https://arxiv.org/abs/1308.5417

07695 | AMR-149-0004

### Open Problems and Questions about Geodesics - Question 4

O / open | AMR 问题清单汇编

[Bangert] *Does every complete Riemannian manifold with finite volume have at least one closed geodesic?*

背景链接：https://arxiv.org/abs/1308.5417

07696 | AMR-149-0041

### Open Problems and Questions about Geodesics - Question 42

O / open | AMR 问题清单汇编

*Does Frankel's Theorem remain true for Finsler manifolds? More precisely, must two closed totally geodesic submanifolds of a closed Finsler manifold with positive flag curvature intersect if the sum of their dimensions is at least the dimension of the ambient manifold? *

背景链接：https://arxiv.org/abs/1308.5417

07697 | AMR-149-0042

### Open Problems and Questions about Geodesics - Question 43

O / open | AMR 问题清单汇编

*Let $(M^3, F)$ be a closed 3-dimensional reversible Finsler manifold of positive flag curvature and $N$ be a closed totally geodesic 2-dimensional submanifold. Is it true that each closed geodesic intersects this submanifold or lies on it? *

背景链接：https://arxiv.org/abs/1308.5417

07698 | AMR-149-0044

### Open Problems and Questions about Geodesics - Question 45 (Babenko)

O / open | AMR 问题清单汇编

*Does there exist, in the space of metrics with conical singularities, a (continuous or smooth) family of metrics $g_t$ such that $g_0$ is the round metric of the sphere, $g_1$ is the metric from the example above, and $\sigma(g_t)$ is a decreasing function of $t$.*

背景链接：https://arxiv.org/abs/1308.5417

07699 | AMR-149-0045

### Open Problems and Questions about Geodesics - Question 46 (Babenko)

O / open | AMR 问题清单汇编

{ \it Does every Riemannian metric on $S^2$ with area $4\pi$ that is close enough to the round metric have a closed geodesic with length $\leq 2\pi$? In other words, is it true that $\sigma \ge 1/\pi$ for metrics close to the round metric? }

背景链接：https://arxiv.org/abs/1308.5417

07700 | AMR-149-0046

### Open Problems and Questions about Geodesics - Question 47 (Guth)

O / open | AMR 问题清单汇编

For the diastole defined by (a), Balacheff and Sabourau showed in [Bal-Sab2010] that there is a constant $C$ such that $$ \textrm{dias}(M,g) \leq C\sqrt{\textrm{genus}(M)}\sqrt{\textrm{area}(M, g)}; $$ the dependence of this inequality on the genus is again optimal.

*Is there a constant $C$ such that the invariant defined by (b) above is bounded from above by* $C\sqrt{\textrm{area}(M,g)}$*?*

背景链接：https://arxiv.org/abs/1308.5417

07701 | AMR-149-0047

### Open Problems and Questions about Geodesics - Question 48 (Question 4.11 of [Gromov2001])

O / open | AMR 问题清单汇编

{ \it Does a Riemannian metric on a real projective space with the same volume as the canonical metric have a closed geodesic with length $\leq \pi$?}

背景链接：https://arxiv.org/abs/1308.5417

07702 | AMR-149-0048

### Open Problems and Questions about Geodesics - Question 49 (\'Alvarez Paiva)

O / open | AMR 问题清单汇编

{ \it Can there be Riemannian metrics on $S^3$ or $S^2 \times S^1$ with unit volume all of whose closed geodesics are long? More specifically, does every metric on these spaces have a closed geodesic with length $\leq 10^{24}$ if the volume is $1$?}

背景链接：https://arxiv.org/abs/1308.5417

07703 | AMR-149-0005

### Open Problems and Questions about Geodesics - Question 5 (Bangert)

O / open | AMR 问题清单汇编

Questions of this nature are also interesting in the realm of Finsler geometry. One would expect the results for reversible Finsler metrics to be very similar to those for Riemannian metrics. For irreversible Finsler metrics, Katok's example shows that there are compact Finsler manifolds with only finitely many closed geodesics, but it is still possible that all non-compact Finsler manifolds with finite volume might have infinitely many closed geodesics. On the other hand, the following question is completely open:

{\it Does there exist an irreversible Finsler metric of finite volume on $\mathbb{R}\times S^1$ with no closed geodesics?}

背景链接：https://arxiv.org/abs/1308.5417

07704 | AMR-149-0049

### Open Problems and Questions about Geodesics - Question 50

O / open | AMR 问题清单汇编

*Let $M$ be a closed manifold not covered by a sphere. Let $g$ be a metric on $M$ for which all geodesics are closed. Is $(M,g)$ a CROSS (=compact rank one symmetric) manifold (with the standard metric)?*

背景链接：https://arxiv.org/abs/1308.5417

07705 | AMR-149-0051

### Open Problems and Questions about Geodesics - Question 52

O / open | AMR 问题清单汇编

To the best of our knowledge, the existence of a closed geodesic on a Lorentzian manifold is a quite complicated problem and nothing is known in dimensions $\ge 3$. In dimension 2, a closed orientable manifold with a metric of signature $(+,-)$ is homeomorphic to the torus. By [Suhr2013a], every Lorentzian 2-dimensional torus has at least two simple closed geodesics one of which is definite, i.e. timelike or spacelike. Moreover, explicit examples show the optimality of this claim.

We therefore ask the following

*Does every closed Lorentzian manifold have at least one closed geodesic?*

背景链接：https://arxiv.org/abs/1308.5417

07706 | AMR-149-0052

### Open Problems and Questions about Geodesics - Question 53

O / open | AMR 问题清单汇编

*Can there exist a complete semi-Riemannian metric $g$ and a non-trivial 1-form $\eta$ on a closed manifold such that for every light-like geodesic $\gamma(t)$ the function $\eta(\dot\gamma(t))$ grows linearly in both directions: i.e. for every light-like geodesic there exist $C_1 \neq 0, C_2 $ such that $\eta(\dot\gamma(t))= C_1\cdot t + C_2$?*

背景链接：https://arxiv.org/abs/1308.5417

07707 | AMR-149-0053

### Open Problems and Questions about Geodesics - Question 54

O / open | AMR 问题清单汇编

It is well-known that a closed Riemannian manifold is geodesically complete, in the sense that every geodesic $\gamma:(a,b)\to {M}$ can be extended to a geodesic $\tilde \gamma:\mathbb{R} \to M$ such that $\tilde \gamma|_{(a,b)}= \gamma$. It is also well known that for any indefinite signature there exist metrics on closed manifolds that are not geodesically complete.

*What geometric assumptions imply that a metric (possibly, of a fixed signature) on a closed manifold is geodesically complete?*

背景链接：https://arxiv.org/abs/1308.5417

07708 | AMR-149-0054

### Open Problems and Questions about Geodesics - Question 55 ([Sanchez2015])

O / open | AMR 问题清单汇编

We refer to the paper [Sanchez2015] for a list of interesting results on this topic and for a list of open questions from which we repeat here only:

*Assume that a compact Lorentzian manifold is globally conformal to a manifold of constant curvature. Must it be geodesically complete?*

背景链接：https://arxiv.org/abs/1308.5417

07709 | AMR-149-0055

### Open Problems and Questions about Geodesics - Question 56 (Bangert)

O / open | AMR 问题清单汇编

*Does there exist a Riemannian metric on a closed surface of genus $\ge 2$ whose geodesic flow is completely integrable? *

背景链接：https://arxiv.org/abs/1308.5417

07710 | AMR-149-0056

### Open Problems and Questions about Geodesics - Question 57 (Paternain)

O / open | AMR 问题清单汇编

Ma\ n\'e showed that a Hamiltonian flow on surfaces is generically Anosov or has zero Liouville exponents [Mane1996]. This suggests that an easier version of the above question would be the following

*Is there a Riemannian metric on a closed surface of genus $\ge 2$ whose geodesic flow has zero Liouville entropy?*

背景链接：https://arxiv.org/abs/1308.5417

07711 | AMR-149-0057

### Open Problems and Questions about Geodesics - Question 58

O / open | AMR 问题清单汇编

*Let $F_1$, $F_2$ be Finsler metrics on a closed surface of genus $\ge 2$. Assume every (unparameterized) $F_1$-geodesic is an $F_2$-geodesic. Must $F_1$ be obtained from $F_2$ by multiplication by a constant and adding a closed form? *

背景链接：https://arxiv.org/abs/1308.5417

07712 | AMR-149-0058

### Open Problems and Questions about Geodesics - Question 59

O / open | AMR 问题清单汇编

The previous question is related to the other questions in this section because of the following observation from [Mat-Top1998]: one can use the second metric to construct an integral of the geodesic flow of the first one. If both metrics are Riemannian, the integral is quadratic in velocities and the affirmative answer follows from [Kolokoltsov1983]; see [Mat-Top2000].

*Does there exist a non-Riemannian Finsler metric satisfying the Landsberg condition on a surface of genus $\ge 2$?*

背景链接：https://arxiv.org/abs/1308.5417

07713 | AMR-149-0006

### Open Problems and Questions about Geodesics - Question 6 (Paternain)

O / open | AMR 问题清单汇编

It is known that the trajectory of a charged particle in the presence of magnetic forces (=``magnetic geodesic'') is described by a Hamiltonian system with the ``kinetic" Hamiltonian of the form $\sum_{i,j} p_ip_jg^{ij}$ on $T^*M$ with the symplectic form $dp\wedge dx +\pi^* \omega$, where $\omega $ is a closed (but not necessarily exact) form on $M$ and $\pi:T^*M\to M$ is the canonical projection.

We assume that our surface $M^2 $ is closed.

*Is there at least one closed magnetic geodesic in every positive energy level?*

背景链接：https://arxiv.org/abs/1308.5417

07714 | AMR-149-0061

### Open Problems and Questions about Geodesics - Question 62 ([Has-Mor1996])

O / open | AMR 问题清单汇编

A $\Theta$-graph is a graph that looks like the Greek letter ``Theta'': two vertices connected by three edges, see the picture below.

*Does every metric on $S^2$ admit a stationary $\Theta$-graph?*

背景链接：https://arxiv.org/abs/1308.5417

07715 | AMR-149-0062

### Open Problems and Questions about Geodesics - Question 63 (Gromov)

O / open | AMR 问题清单汇编

We call a stationary net non-trivial if it has at least one vertex of valency $\ge 3$, in other words if it is not a disjoint union of simple closed geodesics.

*Are non-trivial stationary nets dense on any closed Riemannian surface? In other words, for each non-empty open subset $U$, is there a stationary net intersecting $U$? *

背景链接：https://arxiv.org/abs/1308.5417

07716 | AMR-149-0063

### Open Problems and Questions about Geodesics - Question 64

O / open | AMR 问题清单汇编

*Suppose a closed manifold $(M,g)$ admits a non-zero $(1,1)$-tensor field that is self-adjoint with respect to $g$, parallel and nilpotent. Does this manifold or its double cover admit a non-zero light-like parallel vector field?*

背景链接：https://arxiv.org/abs/1308.5417

07717 | AMR-149-0066

### Open Problems and Questions about Geodesics - Question 67

O / open | AMR 问题清单汇编

It is known that, in dimension 2, locally, each reversible path structure is metrizable [Alv-Ber2010, Theorem B].

*In dimension $2$, is each irreversible path structure metrizable? *

背景链接：https://arxiv.org/abs/1308.5417

07718 | AMR-149-0067

### Open Problems and Questions about Geodesics - Question 68

O / open | AMR 问题清单汇编

There exist local (the first examples are due already to Lagrange, Beltrami and Dini) and global examples of non-proportional projectively equivalent metrics. Locally, in the Riemannian case, a complete description of projectively equivalent metrics is due to Levi-Civita [Levi-Civita1896] and in arbitrary signature is due to [Bol-Mat2015]. Globally, in the Riemannian case, the situation is also pretty clear, see e.g.\ [Mat-Top1998,Matveev2003a,Matveev2003b], but if the metrics have arbitrary signature, virtually nothing is known. A general problem is to understand the topology of closed manifolds admitting non-proportional projectively equivalent metrics of indefinite signature and as the simplest version of this problem we suggest

*Can a 3-dimensional sphere admit two non-proportional projectively equivalent metrics of Lorentzian signature?*

背景链接：https://arxiv.org/abs/1308.5417

07719 | AMR-149-0068

### Open Problems and Questions about Geodesics - Question 69

O / open | AMR 问题清单汇编

In the Finslerian case, the freedom of reconstructing metric from its geodesic can be much higher. For example, adding any sufficiently small closed 1-form to the Finsler metric does not change geodesics of this metric. This transformation of the metric was called a * trivial projective change* in [Matveev2013].

*In dimension $\ge 3$, for a generic Finsler metric $F$, is any metric projectively equivalent to $F$ locally proportional to a trivial projective change of the metric? *

背景链接：https://arxiv.org/abs/1308.5417

07720 | AMR-149-0007

### Open Problems and Questions about Geodesics - Question 7

O / open | AMR 问题清单汇编

Recent interest in Kropina metrics is due to their relations to the so-called chains in CR geometry, see [CMMM2019].

*Does every Kropina metric on a closed manifold have a closed geodesic?*

背景链接：https://arxiv.org/abs/1308.5417

07721 | AMR-149-0008

### Open Problems and Questions about Geodesics - Question 8 (Paternain)

O / open | AMR 问题清单汇编

Let $p, q$ be points in a Riemannian manifold. The space $\Omega_T(p,q)$ of paths from $p$ to $q$ with length $\leq T$ has the homotopy type of a finite complex (see e.g.\ [Milnor1963]), and hence the sum of its Betti numbers is finite for each $T$. The same is true for the space $\Lambda_TM$ of loops with length at most $T$.

*How do the sums of the Betti numbers for $\Lambda_TM$ and $\Omega(_T(p,q)$ grow as $T\to \infty$? Does the growth depend on the metric?*

背景链接：https://arxiv.org/abs/1308.5417

07722 | AMR-149-0009

### Open Problems and Questions about Geodesics - Question 9 (Gromov)

O / open | AMR 问题清单汇编

It was shown by Gromov in [Gromov2001] that if $M$ is simply connected there is a constant $C$ such that the sum of the Betti numbers of $\Lambda_{CN}$ is at least $ \sum_{i=0}^N b_{i}(\Lambda)$.

*Does the number of closed geodesics with length $\leq T$ grow exponentially as $T \to \infty$ if the Betti numbers of the loop space grow exponentially?*

背景链接：https://arxiv.org/abs/1308.5417

07723 | OWR-11786-001

### Open Problems in Triangulation Theory

P / partially_solved | Oberwolfach 报告问题汇编

Determine the following: (1) all facets of the secondary polytope of the $d$-cube; (2) whether a graph of triangulations can be disconnected in dimensions $3$ or $4$; (3) the complexity of counting triangulations and a method for uniformly random generation; (4) whether every smooth polytope has a regular unimodular triangulation; (5) whether every matroid polytope has a regular unimodular triangulation; (6) whether every $3$-polyhedron has a triangulation with Hamiltonian dual graph; and (7) optimal diameter bounds for graphs of triangulations.

来源字段：https://doi.org/10.4171/owr/2012/24

07724 | AIM-GEOMETRY-0404

### Open-book chord dictionaries and a binding-intersection filtration

P / partially_solved | AIM 研讨会问题汇编

There was a fair bit of discussion on how to possibly compute contact homology in terms of (contact) open books. For example, Denis Auroux pointed out that if you have a Lagrangian \(L\) in a page, then this is a Legendrian in the open book, and the Reeb chords are the intersections of \(L\) with \(\phi^k(L)\), where \(\phi\) is the monodromy. This might be a starting point for computing the relative contact homology of \(L\), in the complement of the binding, in terms of things related to symplectic Floer homology of Lagrangians. There is then the problem of understanding what happens when one puts back the binding, but this might correspond to an “\(A_\infty\) deformation”.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

07725 | OWR-14299905-020

### Optimal Covering Density of Euclidean Balls

O / open | Oberwolfach 报告问题汇编

Determine the optimal asymptotic covering density of Euclidean space by congruent balls.

来源字段：https://doi.org/10.4171/owr/2026/1

07726 | OWR-16412-009

### Optimal Dimensions for Polynomially Convex Embeddings

O / open | Oberwolfach 报告问题汇编

Determine the least integer $n=n(m)$ such that every compact smooth real $m$-manifold admits a polynomially convex smooth embedding into $\mathbb C^n$.

来源字段：https://doi.org/10.4171/owr/2018/37

07727 | OWR-8415364-011

### Optimal Great-Subsphere Concentration for the Negative-$p$ Aleksandrov Problem

O / open | Oberwolfach 报告问题汇编

For $p\leq-1$, what is the optimal upper bound on $\mu(\xi)/\mu(S^{n-1})$, over great subspheres $\xi\subset S^{n-1}$, that guarantees a convex body whose $L_p$ integral curvature measure is $\mu$?

来源字段：https://doi.org/10.4171/owr/2021/59

07728 | OWR-11136-016

### Optimal Homothety Factor for Three Intersecting Disks

O / open | Oberwolfach 报告问题汇编

If $D\subset\mathbb{R}^2$ is a disk, is the optimal homothety factor $\lambda_h(D,3)=\sqrt{3}$ for families with property $T(3)$?

来源字段：https://doi.org/10.4171/owr/2011/44

07729 | DARPA-011

### Optimal Nanostructures

P / partially_solved | DARPA 挑战

Develop mathematics for creating optimal symmetric structures through nanoscale self-assembly.

背景链接：https://web.math.utk.edu/~vasili/refs/darpa07.MathChallenges.html

07730 | OWR-822-001

### Optimal Point Distributions on High-Dimensional Spheres

P / partially_solved | Oberwolfach 报告问题汇编

What is the best way to distribute $N$ points on the unit sphere $S^{n-1}\subset\mathbb R^n$?

来源字段：https://doi.org/10.4171/owr/2005/17

07731 | OWR-1394-013

### Optimal Stability Exponents in Convex-Body Tomography

O / open | Oberwolfach 报告问题汇编

Determine the best possible Hölder exponents in stability inequalities for reconstructing convex bodies from tomographic data derived from sections, projections, and their averages. In particular, determine the optimal dependence on the dimension $n$, conjectured to be linear in $1/n$ rather than quadratic.

来源字段：https://doi.org/10.4171/owr/2006/56

07732 | OWR-15575-003

### Optimal Systolic Ratio of the Two-Sphere

O / open | Oberwolfach 报告问题汇编

For the Riemannian systolic ratio $\rho(S^2,\cdot)$, is the supremum equal to $2\sqrt3$?

来源字段：https://doi.org/10.4171/owr/2017/32

07733 | AIM-CONVEX_GEOMETRY-0034

### Optimal vector balancing extends from zonotopes to all zonoids

P / partially_solved | AIM 研讨会问题汇编

**Problem 5 (G. Schechtman).** A theorem of Spencer states that if \(\{X_i\}_{i=1}^n\) are in \(B_\infty^n\), then \[ \min_{\varepsilon_i\in\{-1,1\}} \left\|\sum_{i=1}^n\varepsilon_iX_i\right\|_\infty \leq C\sqrt n. \] Does the same hold for any \(n\)-dimensional zonoid? That is, does there exist a universal constant \(C\) such that for every \(n\)-dimensional centered zonoid \(Z\), if \(X_1,\ldots,X_n\in Z\), then there are signs \(\varepsilon_1,\ldots,\varepsilon_n\) with \[ \sum_{i=1}^n\varepsilon_iX_i\in C\sqrt n\,Z? \] If one can remove the logarithmic factor in Problem 4, then the answer is positive. As stated in the 2007 list, the best known substitute for \(C\sqrt n\) was \(C\sqrt n\log\log n\).

来源字段：https://aimath.org/WWN/fourierconvex/fourierconvex.pdf

07734 | AMR-058-0017

### Optimality and Existence of the Weaire-Phelan Foam

P / partially_solved | AMR 问题清单汇编

Is the Weaire--Phelan A15 foam the optimal partition of three-space into equal volumes, and can the existence of a foam in the A15 pattern be proved?

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

07735 | OWR-13498-011

### Optimality of Greedy Circle Packings

O / open | Oberwolfach 报告问题汇编

Place infinitely many nonoverlapping disks tangent to the $x$-axis in the curvilinear triangular region bounded by the $x$-axis and two unit disks centered at $(\pm1,1)$, maximizing their total area. Does the greedy procedure that successively inserts a disk tangent to two previously placed disks attain the maximum area?

来源字段：https://doi.org/10.4171/owr/2015/13

07736 | OWR-15208-014

### Orbifold Singularities in Riemannian Metric Moduli Spaces

O / open | Oberwolfach 报告问题汇编

When the diffeomorphism action on the space of Riemannian metrics is not free and the moduli space has orbifold singularities, what mathematical framework or construction should be used to analyze the resulting moduli space?

来源字段：https://doi.org/10.4171/owr/2017/3

07737 | GEOM-023

### Orchard-Planting Problem

P / partially_solved | 独立 / 未分配集合

What is the maximum number of 3-point lines attainable by a configuration of $n$ points in the plane?

背景链接：https://en.wikipedia.org/wiki/Orchard-planting_problem

07738 | AIM-GEOMETRY-0225

### Ordered boundary rigidity from quadratically spanning minimizing geodesics

P / partially_solved | AIM 研讨会问题汇编

**Question 5.1.3 ([Michel1981]).** What conditions on (M,N) and (g) imply boundary rigidity?

In the official PDF, the preceding paragraph defines the data. If (M) is a compact manifold with smooth boundary (N=\partial M), a Riemannian metric (g) determines \[ d_g^\partial(p,q)=d_g(p,q),\qquad (p,q)\in N\times N. \] The question itself is:

It then notes three variants: placing an additional assumption on a competing metric (g'), asking for restrictions on the allowed metric class, and replacing Riemannian metrics by Finsler metrics.

Two source issues matter.

1. The official PDF, not merely the JSON extraction, says that another metric with the same data “must be isometric to (g').” This is tautological. The standard and logically coherent reconstruction is “must be isometric to (g),” with the boundary gauge made explicit: there must be a diffeomorphism \(\psi:M\to M\) satisfying \(\psi|_N=\mathrm{Id}\) and, in one consistent pullback convention, \(g'=\psi^*g\). This correction is a labeled reconstruction, not a silent alteration of the source. 2. Everything in the canonical record beginning with “5.2. The conjugacy problem” belongs to the next section and next problem. The expressions OCRed as `SM >1`, `φ1 >t`, and `Ck` concern that separate problem. They are not analyzed here.

Thus the precise reading used below is: find geometrically meaningful hypotheses under which equality of boundary distance functions determines (g) up to a boundary-fixing diffeomorphism.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07739 | OWR-3385-010

### Origami, Triangle-Area, and Circle-Packing Rigidity Problems

P / partially_solved | Oberwolfach 报告问题汇编

Can the standard origami model of a hyperbolic paraboloid be realized with straight creases and developable faces isometric to plane faces, or must the paper stretch? For a partition of the unit square into triangles, what relation among their signed areas exists besides their sum being one? Is every doubly periodic hexagonal-lattice circle packing with prescribed neighboring inversive distances rigid?

来源字段：https://doi.org/10.4171/owr/2009/02

07740 | OWR-12012-010

### Orthogonality-Avoiding Sets and Coulomb Local Minima

P / partially_solved | Oberwolfach 报告问题汇编

For a measurable set $A\subset S^2$ with $\langle x,y\rangle\ne0$ for all $x,y\in A$, what is the largest possible measure of $A$, and can the upper bound $1/3$ be improved? Separately, if $L_n$ is the number of local minima of the Coulomb potential for $n$ points on $S^2$, is $L_n>1$ for every sufficiently large $n$?

来源字段：https://doi.org/10.4171/owr/2012/40

07741 | AIM-GEOMETRY-0059

### Oseledets integrability can fail in smooth no-cusp dispersing billiards

S / solved | AIM 研讨会问题汇编

Consider a dynamical billiard without cusps. Do (non-periodic) invariant probability measures for the billiard map satisfy the integrability condition for Oseledet's theorem?

来源字段：http://aimpl.org/equibdynsysgeom/3/

07742 | AMR-054-0015

### Output-sensitive Convex Hull in $\mathbb{R}^d$

P / partially_solved | AMR 问题清单汇编

What is the best output-sensitive convex hull algorithm for $n$ points in $\mathbb{R}^d$?

背景链接：https://topp.openproblem.net/p15

07743 | AMR-038-0015

### Packing reciprocal rectangles in a square

P / partially_solved | AMR 问题清单汇编

For every positive integer $k$, let $R_k$ be a $1/k$ by $1/(k+1)$ rectangle. Can the entire collection $(R_k)_{k\ge1}$ be packed without overlap into the unit square?

背景链接：https://ics.uci.edu/~eppstein/junkyard/open.html

07744 | AMR-054-0055

### Pallet Loading

O / open | AMR 问题清单汇编

What is the complexity of the pallet loading problem? Given two pairs of numbers, $(A,B)$ and $(a,b)$, and a number $n$, decide whether $n$ small rectangles of size $a \times b$, in either axis-parallel orientation, can be packed into a large rectangle of size $A \times B$. This problem is not even known to be in NP, because of the compact input description, and the possibly complicated structure of a packing, if there is one.

背景链接：https://topp.openproblem.net/p55

07745 | OWR-12007-007

### Parallel-Axiom Rigidity of Complete Riemannian Planes

O / open | Oberwolfach 报告问题汇编

If a complete Riemannian plane $P$ satisfies the parallel axiom that every geodesic and every point off it determine a unique disjoint geodesic through that point, must $P$ be isometric to the Euclidean plane?

来源字段：https://doi.org/10.4171/owr/2012/35

07746 | OWR-13347-014

### Parity and Ricci-Flow Evolution of Curvature Invariants

O / open | Oberwolfach 报告问题汇编

With $k_o$, $k_t$, and $k_S$ as the curvature-derivative, traceless-curvature, and Singer invariants of a homogeneous Riemannian manifold, is finite $k_o$ always even? If $k_t=1$, what can be deduced about the rough-Laplacian eigenvalue equation $\nabla^*\nabla R=\lambda R$? How do the defining traceless-curvature equation and the invariants $k_t$, $k_o$, and $k_S$ evolve under Ricci flow?

来源字段：https://doi.org/10.4171/owr/2014/45

07747 | OPG-37086

### Partition of Complete Geometric Graph into Plane Trees

P / partially_solved | Open Problem Garden

Conjecture Every complete geometric graph with an even number of vertices has a partition of its edge set into plane (i.e. non-crossing) spanning trees.

背景链接：http://www.openproblemgarden.org/op/partition_of_complete_geometric_graph_into_plane_trees

07748 | OPG-56328

### Partitioning the Projective Plane

P / partially_solved | Open Problem Garden

Throughout this post, by projective plane we mean the set of all lines through the origin in $\mathbb{R}^3$.

Definition Say that a subset $S$ of the projective plane is octahedral if all lines in $S$ pass through the closure of two opposite faces of a regular octahedron centered at the origin.

Definition Say that a subset $S$ of the projective plane is weakly octahedral if every set $S'\subseteq S$ such that $|S'|=3$ is octahedral.

Conjecture Suppose that the projective plane can be partitioned into four sets, say $S_1,S_2,S_3$ and $S_4$ such that each set $S_i$ is weakly octahedral. Then each $S_i$ is octahedral.

背景链接：http://www.openproblemgarden.org/op/partitioning_the_projective_plane

07749 | OWR-1972-002

### Perelman Functionals for Coupled Geometric Flows

P / partially_solved | Oberwolfach 报告问题汇编

For geometric flows of the form $\partial_t g_{ij}=-2S_{ij}$, including Ricci flow coupled to harmonic-map or other fields, to what extent do Perelman's other monotone functionals-especially the reduced volume and its associated non-collapsing consequences-remain monotone?

来源字段：https://doi.org/10.4171/owr/2008/27

07750 | AMR-038-0012

### Perfect rational triangles

P / partially_solved | AMR 问题清单汇编

Does there exist a nondegenerate triangle whose side lengths, three medians, three altitudes, and area are all rational?

背景链接：https://ics.uci.edu/~eppstein/junkyard/open.html

07751 | OWR-3389-009

### Perimeter-Normalized Neumann Eigenvalue Maximizers

P / partially_solved | Oberwolfach 报告问题汇编

Among convex planar domains, maximize $\mu_1L^2$, where $\mu_1$ is the first positive Neumann eigenvalue and $L$ is the perimeter. Are the maximizers the equilateral triangle and the square, and is their equal value explained by an underlying principle?

来源字段：https://doi.org/10.4171/owr/2009/06

07752 | OWR-14299905-036

### Periodic Constructions for Colored-Lattice Packing Constants

O / open | Oberwolfach 报告问题汇编

For every dimension $d$ and word $w$, is the extremal word-occurrence constant $C_d(w)$ attained asymptotically by a periodic grid? What are $C_d(ABB)$ for $d>2$ and $C_2(ABBB)$?

来源字段：https://doi.org/10.4171/owr/2026/1

07753 | AIM-GEOMETRY-0020

### Peripheral locus of the top open Grassmannian positroid stratum

P / partially_solved | AIM 研讨会问题汇编

When is \(P\) empty (in \(G(2,2k+1)\))?

What about the big open positroid cell in \(G(k,n)\)?

来源字段：http://aimpl.org/clusterbraid/3/

07754 | OWR-14298170-016

### Persistence of Nonnegative Scalar Curvature under Intrinsic-Flat Limits

O / open | Oberwolfach 报告问题汇编

If three-manifolds $(M_j,g_j)$ with nonnegative scalar curvature converge in the Sormani-Wenger intrinsic-flat sense to a smooth Riemannian manifold $(M_0,g_0)$, must $g_0$ have nonnegative scalar curvature? Does the conclusion hold when the sequence also satisfies a uniform positive lower bound on minimal-surface area?

来源字段：https://doi.org/10.4171/owr/2024/9

07755 | OWR-13951-015

### Petty's Projection-Body Conjecture

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Petty-Conjecture Extraction [OWR-13951-014]

For every convex body $K\subset\mathbb{R}^d$ with $d\geq3$, is the affine invariant $P(K)=|\Pi K|/|K|^{d-1}$ minimized exactly by ellipsoids?

来源字段：https://doi.org/10.4171/owr/2015/56

07756 | AIM-GEOMETRY-0393

### Pfaffian decomposition of global conformal invariants

S / solved | AIM 研讨会问题汇编

Problem 32: Is it possible to write any S, as in Problem 30, in the form \[ \operatorname{const}\cdot \mathrm{Pff}+L+V? \]

If \(\mathbf S\) is a natural critical \(n\)-density and \(\int_M\mathbf S\) is conformally invariant, can one write, pointwise and universally, \[ \mathbf S=c\,\mathbf{Pf}_g+\mathbf L_g+\mathbf V_g, \] where \(\mathbf L\) is a local conformal invariant and \(\mathbf V\) is the exact divergence of a natural vector field?

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

07757 | OWR-11136-005

### Piercing Large Convex Sets in the Unit Square

O / open | Oberwolfach 报告问题汇编

If $f(\varepsilon)$ is the minimum number of points in the unit square meeting every compact convex set of area at least $\varepsilon$, is $f(\varepsilon)=O(1/\varepsilon)$ as $\varepsilon\to0$?

来源字段：https://doi.org/10.4171/owr/2011/44

07758 | OWR-15427-002

### Piercing Line Families with the (p,3)-Property

S / solved | Oberwolfach 报告问题汇编

Let $p\geq3$, and let $\mathcal F$ be a finite family of lines with the $(p,3)$-property: among every $p$ lines of $\mathcal F$, some three are concurrent. Must there be a set $U$ of at most $p-2$ points such that every line in $\mathcal F$ contains a point of $U$?

来源字段：https://doi.org/10.4171/owr/2017/19

07759 | AMR-054-0049

### Planar Euclidean Maximum TSP

O / open | AMR 问题清单汇编

What is the complexity of finding a tour of maximum Euclidean length for a planar point set?

背景链接：https://topp.openproblem.net/p49

07760 | OWR-16637-006

### Planar Log-Brunn-Minkowski After Translation

O / open | Oberwolfach 报告问题汇编

For planar convex bodies $K,L\in\mathcal K_2$, can the origin be positioned and $L$ translated so that $K$ and the translate satisfy the logarithmic Brunn-Minkowski inequality?

来源字段：https://doi.org/10.4171/owr/2018/54

07761 | OWR-13110-010

### Planar Regions Avoiding Prescribed Interior Lattice Point Counts

O / open | Oberwolfach 报告问题汇编

Does there exist, for each natural number n, a planar region of area n such that no congruent copy of the region contains exactly n interior lattice points?

来源字段：https://doi.org/10.4171/owr/2014/40

07762 | AIM-GEOMETRY-0244

### Playfair rigidity without focal points

P / partially_solved | AIM 研讨会问题汇编

Question 7.2.2 ([Bur-Kni1991]). Must a metric satisfying this version of the par-allel postulate be flat?

The question looks like a question in the synthetic geometry, but it is not, since we do not require a priori that the other axioms of the Euclidean geometry are fulfilled (for example the congruence axioms). 7.2.1. Higher rank rigidity.

Consider the plane \(\mathbb R^2\) with a complete Riemannian metric. Assume that for every geodesic and for every point not on the geodesic there exists precisely one nontrivial geodesic that passes through the point and does not intersect the geodesic. This assumption implies that the metric has no conjugate points.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07763 | OWR-17471-023

### Pleating-Ray Tilings of the Representation-Space Complement

O / open | Oberwolfach 报告问题汇编

Let $\mathcal R$ be the Riley slice for the marked groups generated by

$$A=\begin{pmatrix}1&1\\0&1\end{pmatrix},\qquad B_\omega=\begin{pmatrix}1&0\\\omega&1\end{pmatrix}.$$

Is $\mathbb C^*\setminus\overline{\mathcal R}$ naturally tiled by quadrilaterals $\mathcal R_c(r)$ indexed by $r\in\mathbb Q/2\mathbb Z$, each containing the extended pleating ray of slope $r$ as a diagonal and arising from restrictions of holonomy representations of Dehn fillings of $S^3\setminus(K(r^*)\cup\alpha_r)$?

来源字段：https://doi.org/10.4171/owr/2020/8

07764 | OWR-14297736-018

### Pleijel Constants and Steklov-Robin Duality

O / open | Oberwolfach 报告问题汇编

Determine whether the Robin and Dirichlet Pleijel constants always coincide on a fixed domain; bound nodal counts for Dirichlet-to-Neumann eigenfunctions; and find further consequences of Steklov-Robin duality.

来源字段：https://doi.org/10.4171/owr/2023/36

07765 | OWR-2042-003

### Point Incidence on One-Lipschitz Graphs

O / open | Oberwolfach 报告问题汇编

Given a finite set of points in the plane or in three-dimensional space, what is the maximum number that can lie on a graph of a $1$-Lipschitz function?

来源字段：https://doi.org/10.4171/owr/2008/31

07766 | OPG-37286

### Point sets with no empty pentagon

P / partially_solved | Open Problem Garden

Problem Classify the point sets with no empty pentagon.

背景链接：http://www.openproblemgarden.org/op/point_sets_with_no_empty_pentagon

07767 | AMR-037-0005

### Point-hyperplane incidences

P / partially_solved | AMR 问题清单汇编

Given $n$ points and $m$ hyperplanes in $\mathbb R^d$ whose incidence graph contains no $K_{s,t}$, determine the maximum number of incidences. Of special interest is the case $n=m$ and $s=t=d$, where the known bounds are far apart.

背景链接：https://jeffe.cs.illinois.edu/open/

07768 | OWR-1782-007

### Point-Set Reconstruction from Directional X-Rays

O / open | Oberwolfach 报告问题汇编

Let $A$ be a finite point set in the plane, and call $A$ reconstructible from directions $u_1,\ldots,u_k$ if its discrete X-rays in those directions uniquely determine $A$. Define $f(u_1,\ldots,u_k)$ as the largest $n$ such that every $n$-point set is reconstructible, and $F(k)=\max f(u_1,\ldots,u_k)$ over $k$ distinct directions. Determine or estimate $F(k)$.

来源字段：https://doi.org/10.4171/owr/2008/01

07769 | OWR-14298163-016

### Point-to-Nonincident-Line Separation

O / open | Oberwolfach 报告问题汇编

Let $p_1,\ldots,p_n\in[0,1]^2$ and let each line $\ell_i$ pass through $p_i$. Determine the largest possible value of $\delta=\min_{i\neq j}d(p_i,\ell_j)$. In particular, prove or disprove that $\delta\leq n^{-1+\varepsilon}$ for every $\varepsilon>0$ and all sufficiently large $n$.

来源字段：https://doi.org/10.4171/owr/2024/3

07770 | OWR-3385-011

### Polygon Development and Higher-Dimensional Discrete Integrability

O / open | Oberwolfach 报告问题汇编

Given a simple piecewise-geodesic curve on $S^2$, when does its planar development with unchanged lengths and angles avoid self-intersection, and what happens for star-shaped polygons? Do discrete integrable systems exist in dimensions at least four, with consistent restrictions to every $n$-face of an $(n+1)$-cube?

来源字段：https://doi.org/10.4171/owr/2009/02

07771 | AMR-054-0024

### Polygonal Curve Simplification

O / open | AMR 问题清单汇编

Can an $n$-vertex polygonal curve be simplified in time nearly linear in $n$?

背景链接：https://topp.openproblem.net/p24

07772 | OWR-12339-004

### Polyhedra with Vertex-Factored Fantappiè Denominators

O / open | Oberwolfach 报告问题汇编

Give a geometric characterization of compact polyhedra $P$ whose Fantappiè-transform denominator is $\Omega(u)=\prod_{v\in V(P)}(1-\langle v,u\rangle)$.

来源字段：https://doi.org/10.4171/owr/2013/11

07773 | OWR-12581-005

### Polyhedral Frankel Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Is every orientable nonnegatively curved polyhedral manifold with irreducible holonomy and $b_2(M)>0$ biholomorphic to $\mathbb{CP}^n$ in a natural holomorphic structure, with a singular Kähler metric?

来源字段：https://doi.org/10.4171/owr/2013/33

07774 | AMR-054-0025

### Polyhedral Surface Approximation

O / open | AMR 问题清单汇编

How efficiently can one compute a polyhedral surface that is an $\epsilon$-approximation of a given triangulated surface in $\mathbb{R}^3$?

背景链接：https://topp.openproblem.net/p25

07775 | OWR-8415362-003

### Polyhedrality of Totally Nonnegative Critical Varieties

O / open | Oberwolfach 报告问题汇编

In the study of scattering amplitudes and positive geometries on the totally nonnegative Grassmannian $\mathrm{Gr}_{\geq 0}(k, n)$, let $\operatorname{Crit}^{\geq 0}_f$ denote a totally nonnegative critical variety (the closure of the critical locus of the master function $f$ on a positroid cell, as studied by Galashin). Is every totally nonnegative critical variety $\operatorname{Crit}^{\geq 0}_f$ homeomorphic to a polytope?

来源字段：https://doi.org/10.4171/owr/2021/57

07776 | AMR-054-0072

### Polyhedron with Regular Pentagon Faces

O / open | AMR 问题清单汇编

Let $M$ be a closed polyhedral surface homeomorphic to $S^2$ which is entirely composed of equal regular pentagons. If $M$ is immersed in 3-space, is it necessarily the boundary of a union of solid dodecahedra that are glued together at common facets?

背景链接：https://topp.openproblem.net/p72

07777 | AIM-CONVEX_GEOMETRY-0044

### Polynomial ridge completeness and switching obstructions for the four AIM directions

P / partially_solved | AIM 研讨会问题汇编

A measurable function $f:\mathbb R^2\to\mathbb R$ is a ridge function if $f(x)=g(\langle x,v\rangle)$ for a measurable $g:\mathbb R\to\mathbb R$ and a direction $v$. A superlevel set of a sum of ridge functions is uniquely determined among measurable sets by the corresponding X-rays. The four directions \[ v_1=(1,0),\qquad v_2=(0,1),\qquad v_3=(2,1),\qquad v_4=(-1,2) \] determine every planar convex body by X-rays, whereas three directions never suffice universally. Is every planar convex body $K$ representable as \[ K=\left\{x\in\mathbb R^2:\sum_{i=1}^4 f_i(x)\geq 0\right\}, \qquad f_i(x)=g_i(\langle x,v_i\rangle)? \]

来源字段：https://aimath.org/WWN/fourierconvex/fourierconvex.pdf

07778 | OWR-14298163-002

### Polynomial-Homology Families and the p-q Theorem

O / open | Oberwolfach 报告问题汇编

Fix constants $c$ and $t$. Let $\mathcal F$ be a family of subsets of $\mathbb R^d$ such that $\sum_{i=1}^{\lfloor d/2\rfloor}\widetilde\beta_i(\bigcap\mathcal G)\leq c|\mathcal G|^t$ for every finite $\mathcal G\subseteq\mathcal F$. Prove that $\mathcal F$ satisfies a $(p,q)$-theorem for every $p\geq q\geq d+1$.

来源字段：https://doi.org/10.4171/owr/2024/3

07779 | OWR-14219-006

### Polytope Approximation in Surface-Area Deviation

P / partially_solved | Oberwolfach 报告问题汇编

How well can a convex body be approximated by a polytope, particularly in symmetric surface-area deviation as a function of the number of vertices?

来源字段：https://doi.org/10.4171/owr/2016/11

07780 | AIM-GEOMETRY-0040

### Positive cluster charts cannot cover a nodal deep point

P / partially_solved | AIM 研讨会问题汇编

Can we use different cluster structures or parametrizations to cover braid varieties or other interesting spaces by their different positive parts?

来源字段：http://aimpl.org/clusterbraid/9/

07781 | OWR-16168-007

### Positive Curvature on Higher Rank Symmetric Spaces

O / open | Oberwolfach 报告问题汇编

Does any compact, simply connected Riemannian symmetric space of rank greater than $1$ admit a metric of positive sectional curvature? In particular, does $S^2\times S^2$ admit such a metric?

来源字段：https://doi.org/10.4171/owr/2018/27

07782 | OWR-7155451-006

### Positive Energy Conditions and Spaces of Initial Data

O / open | Oberwolfach 报告问题汇编

Do known results for nonnegative or positive scalar curvature generalize to initial data sets $(g,k)$ satisfying the strict dominant energy condition? In particular, do nontrivial homotopy classes detected for the space of positive-scalar-curvature metrics yield nontrivial classes in the corresponding space of initial data sets?

来源字段：https://doi.org/10.4171/owr/2021/30

07783 | AIM-GEOMETRY-0250

### Positive entropy and the topology of smooth metrics

P / partially_solved | AIM 研讨会问题汇编

Question 8.4.1. Do the metrics with positive entropy form a dense subset of G?This question can be asked for any k. For k = 2 it was recently positively answered in all dimensions in [Contreras2010]; see also [Con-Pat2002]. For k = ∞

and dimension 2, it was answered positively in [Kni-Wei2002].

The missing definitions occur in §8.4 immediately before this record. Let $M$ be a smooth closed manifold and let

$$ \mathcal G=\operatorname{Met}^{\infty}(M) $$

be the set of all smooth Riemannian metrics on $M$. For a finite integer $k$, $\mathcal G$ is given the topology induced by the $C^k$-norms of the coefficient matrices $g_{ij}$ in a finite atlas. The source explains that this topology is independent of the chosen finite atlas. The phrase “positive entropy” means

$$ h_{\mathrm{top}}(\phi_g)>0, $$

where $\phi_g^t:S_gM\to S_gM$ is the geodesic flow and entropy is the topological entropy of its time-one map. This interpretation is explicit in all three cited primary papers.

For $k=\infty$, the correct standard interpretation on a compact manifold is the Whitney $C^\infty$, equivalently Fréchet, topology on smooth sections of $S^2T^*M$. It is generated by the seminorms $\|\cdot\|_{C^r}$, $r=0,1,\ldots$. A basic neighborhood controls only finitely many derivative orders; there is no single finite-valued “$C^\infty$-norm.”

The question concerns Riemannian metrics. Contreras discusses why analogous perturbation problems are more flexible for Finsler metrics, but his stated Theorem A and the space $\mathcal G$ above are Riemannian. No Finsler conclusion is silently imported here.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07784 | AIM-GEOMETRY-0039

### Positive interiors and nonnegative compactifications of braid varieties

P / partially_solved | AIM 研讨会问题汇编

Define an open positive locus on a braid variety which recovers the positive part of an open positroid variety, and define a closed nonnegative locus in a specified compactification which recovers the closure of that positive positroid part. State precisely which choices (cluster seed, weave orientation, braid presentation, and compactification) the construction does or does not depend on.

来源字段：http://aimpl.org/clusterbraid/9/

07785 | AIM-GEOMETRY-0264

### Positive Killing tensors as coercive completeness certificates

P / partially_solved | AIM 研讨会问题汇编

**Question 9.3.1.** What geometric assumptions imply that a metric (possibly, of a fixed signature) on a closed manifold is geodesically complete?

We of course are interested in geometric assumptions that are easy to check or which are fulfilled for many interesting metrics.

The source PDF has the section heading “9.3. Completeness of closed manifolds of arbitrary signature (communicated by H. After removing a page header and line-break hyphenation, the exact question belonging to this record is:

The following source paragraph recalls the results of Marsden, Carrière, and Klingler and then ends with a colon.

This question is deliberately broad. It asks for useful sufficient geometric criteria rather than proposing a single yes/no conjecture. It remains an active research program rather than a fully classified problem.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07786 | OWR-13106-003

### Positive Mass on Simply Connected Nonspin Manifolds

S / solved | Oberwolfach 报告问题汇编

For every $n\geq 8$, does there exist a closed simply connected nonspin $n$-manifold $M$ such that every admissible metric on $M$ has nonnegative mass?

来源字段：https://doi.org/10.4171/owr/2014/36

07787 | OWR-15208-011

### Positive Ricci Metric Spaces on Nonspin Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

What can be said about the spaces of Riemannian metrics, especially positive-Ricci metrics, on a simply connected non-spin manifold or on a manifold whose universal cover is non-spin?

来源字段：https://doi.org/10.4171/owr/2017/3

07788 | OWR-14298364-004

### Positive Scalar Curvature on Aspherical Manifolds

O / open | Oberwolfach 报告问题汇编

Can a compact aspherical smooth manifold admit a Riemannian metric of positive scalar curvature?

来源字段：https://doi.org/10.4171/owr/2024/28

07789 | OWR-3385-007

### Positive Self-Stress Functionals on Planar $K_{4}$ Embeddings

O / open | Oberwolfach 报告问题汇编

For a general-position planar embedding of $K_4$, let $\omega$ be its self-stress and let $f$ assign a number to each segment. Characterize the functions $f$ for which $\sum_{i<j}\omega_{ij}f_{ij}>0$ for every such embedding. Are all of them obtained, up to positive scaling and addition of a function in the rigidity-map range, from $f_{ij}=[ap_ip_j][bp_jp_i]$?

来源字段：https://doi.org/10.4171/owr/2009/02

07790 | AIM-GEOMETRY-0341

### Positive tropicalization, cluster fans, and a finite wall-detection criterion

P / partially_solved | AIM 研讨会问题汇编

**Question.** What is the connection between the tropicalization of the totally positive part of a variety and the cluster algebra structure of the variety?

**Background.** Speyer and Williams had described the tropicalization of the totally positive part of the Grassmannian \(G(k,n)\). For \(k=2\), the resulting fan is related to the type-\(A\) associahedron; for \(G(3,6)\) and \(G(3,7)\), the fans are related to the type-\(D_4\) and type-\(E_6\) associahedra. Scott had shown that the cluster algebra structures of \(G(2,n)\), \(G(3,6)\), and \(G(3,7)\) have the corresponding types.

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

07791 | AIM-GEOMETRY-0034

### Positive-braid seed realization and a mutable-rank obstruction

P / partially_solved | AIM 研讨会问题汇编

For ADE types, all seeds in the cluster algebra can be realized by Lagrangian fillings. Can this be generalized?

来源字段：http://aimpl.org/clusterbraid/7/

07792 | OWR-11568-002

### Positive-Ricci Cohomogeneity Manifolds with Exceptional Orbits

O / open | Oberwolfach 报告问题汇编

Can one find manifolds with more than two non-principal orbits that admit an invariant metric of positive Ricci curvature in cohomogeneities other than 3 and 5?

来源字段：https://doi.org/10.4171/owr/2012/01

07793 | OWR-16933-002

### Positivity Criteria for Tropical Tori

O / open | Oberwolfach 报告问题汇编

For a tropical torus $X=\mathbb{R}^n/\Lambda$, determine whether the positivity criterion for representing an integral (1,1)-class by an effective tropical cycle can be inverted, whether tropical analogues of the Nakai-Moishezon and Kleiman criteria exist, and which hypersurface combinatorial types allow large or small ranges of Picard ranks.

来源字段：https://doi.org/10.4171/owr/2019/21

07794 | OWR-14297736-026

### Pólya-Type Spectral Inequalities for Manifold Families

O / open | Oberwolfach 报告问题汇编

Develop Pólya-type lower eigenvalue inequalities, with appropriate correction terms, that hold uniformly over natural families of Riemannian manifolds.

来源字段：https://doi.org/10.4171/owr/2023/36

07795 | OWR-1054-005

### Prescribed Q-Curvature in Conformal Classes

P / partially_solved | Oberwolfach 报告问题汇编

Given $K$ on a Riemannian manifold $(M,g)$, does the conformal class $[g]$ contain a metric $\widetilde g$ whose $Q$-curvature equals $K$?

来源字段：https://doi.org/10.4171/owr/2005/33

07796 | AIM-GEOMETRY-0386

### Prescribing Q as a density and as a scalar

P / partially_solved | AIM 研讨会问题汇编

Problem 26: On a given manifold \(M\), can one find a metric with specified \(Q\)? One can also ask this question within a given conformal class or within the realm of conformally flat metrics though, of course, if \(M\) is compact, then \(\int_M Q\) must be as specified by the conformal class and the topology of \(M\).

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

07797 | OWR-7155451-011

### Preservation of Positive-Scalar-Curvature Obstructions Under Surgery

O / open | Oberwolfach 报告问题汇编

If a smooth manifold $M$ admits no metric of positive scalar curvature, under what kinds of surgery does the resulting manifold also admit no positive-scalar-curvature metric?

来源字段：https://doi.org/10.4171/owr/2021/30

07798 | AIM-GEOMETRY-0232

### Prime closed geodesics with exponential error

P / partially_solved | AIM 研讨会问题汇编

The original AIM PDF was checked directly. Lines 564--580 give the missing context and the intact typeset display. Let $(M,g)$ be a closed Riemannian manifold with negative sectional curvature, let $h=h_{\mathrm{top}}(\phi)$ be the topological entropy of its geodesic flow, and let $N(T)$ count closed geodesics of length at most $T$. The recovered conjecture is

\[ N(T)=\left(1+O(e^{-\varepsilon T})\right) \int_2^{e^{hT}}\frac{du}{\log u} =\left(1+O(e^{-\varepsilon T})\right)\operatorname{Li}(e^{hT}) \tag{1} \]

for some $\varepsilon>0$.

The source does not explicitly say “primitive” or “oriented.” Its formula and the literature it invokes determine the natural convention:

* a **primitive** orbit traverses its image once and has least period $\ell(\gamma)$; * a periodic orbit of the geodesic flow is **oriented** by increasing flow time; * the Ruelle product and the prime-orbit theorem count primitive oriented periodic orbits.

Giulietti--Liverani--Pollicott explicitly use “oriented” in their prime-geodesic theorem. With that convention the leading coefficient in (1) is $1$. Time reversal pairs the two orientations of each geometric geodesic, so the unoriented primitive count has leading term $\frac12\operatorname{Li}(e^{hT})$. Counting all iterates instead of only primitive orbits does not change the leading term, but it introduces an error of exact exponential scale $e^{hT/2}$; this is proved below. Thus the coherent recovery of $N$ in (1) is the **primitive oriented** count.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07799 | AIM-GEOMETRY-0115

### Prime clusters as finite structural certificates

P / partially_solved | AIM 研讨会问题汇编

Which clusters in complex crystals, periodic or aperiodic, are useful for understanding their growth? Which are useful for describing structure?

来源字段：http://aimpl.org/softpack/2/

07800 | AIM-GEOMETRY-0325

### Prime integral counterexamples to adelic halfspace intersection

S / solved | AIM 研讨会问题汇编

Let \(\mathfrak p\) be a prime ideal of \(R_d\) with \(\mathfrak p\cap\mathbb Z=\{0\}\), and let \(r=\dim_{\mathrm{Krull}}(R_d/\mathfrak p)\). For every subspace of \(\mathbb R^d\) of dimension \(d-r+1\), does every halfspace in that subspace intersect the adelic amoeba of \(\mathfrak p\)?

The AIM PDF and the exact HTML page confirm that the displayed ring is \[ R_d=\mathbb Z[x_1^{\pm1},\ldots,x_d^{\pm1}]. The recovered statement is:

The published follow-up uses **rational open halfspaces** \[ H=\partial H+\mathbb R_{>0}v, \] where \(\partial H\) is a rational linear subspace and \(v\notin\partial H\). This is the convention used below.

There is a dimension convention that must not be lost. Since \(\mathfrak p\cap\mathbb Z=0\), the generic fiber \[ X=V(\mathfrak p\otimes_{\mathbb Z}\mathbb Q) \subset\mathbb G_{m,\mathbb Q}^{d} \] has \[ \dim X=r-1. \] Thus the AIM number \(d-r+1\) is \(d-\dim X\), the complementary dimension used in the global-field formulation. Payne denotes \(\dim X\), not the dimension of the integral model, by \(r\).

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

07801 | AMR-038-0011

### Prince Rupert ratio for tetrahedra

S / solved | AMR 问题清单汇编

What is the largest possible ratio between the sum of edge lengths of a tetrahedron that can pass through or fit inside another tetrahedron and the sum of edge lengths of the containing tetrahedron?

背景链接：https://ics.uci.edu/~eppstein/junkyard/open.html

07802 | AMR-059-0002

### Probability Densities and Equiaffine Transformations

P / partially_solved | AMR 问题清单汇编

The smooth positive probability densities on $S^1$ are diffeomorphic to the manifold of smooth equiaffine transformations of $S^1$. Find the corresponding description for $S^n$ and for $\mathbb{R}^n$.

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

07803 | AMR-058-0019

### Product Partitions of Slabs and Long Cylinders

P / partially_solved | AMR 问题清单汇编

If an optimal planar cluster is crossed with a short interval, is the resulting partition optimal in the slab? Is a horizontal mid-height slice optimal for dividing a very long cylinder or prism into two equal-volume halves?

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

07804 | AIM-CONVEX_GEOMETRY-0043

### Projection-body eigenbodies: exact planar classification and product formulas

P / partially_solved | AIM 研讨会问题汇编

**Problem 21 (W. Weil).** Let \(\Pi K\) be the projection body of \(K\). It is known that if \(K\) is a polytope and \(\Pi\Pi K\) is homothetic to \(K\), then \(K\) is a direct sum of planar polygons (in even dimensions) or a direct sum of planar polygons and a segment (in odd dimensions). If \(K\) is an arbitrary convex body and \(\Pi\Pi K\) is homothetic to \(K\), what can we conclude about \(K\)?

来源字段：https://aimath.org/WWN/fourierconvex/fourierconvex.pdf

07805 | OWR-12869-001

### Projective Fundamental Groups Among Kähler Groups

O / open | Oberwolfach 报告问题汇编

Does there exist a Kähler group that is not the fundamental group of any smooth complex projective variety?

来源字段：https://doi.org/10.4171/owr/2014/09

07806 | OWR-12869-007

### Projective Realization of Torsion-Free Kähler Groups

P / partially_solved | Oberwolfach 报告问题汇编

Is every torsion-free Kähler group the fundamental group of a smooth projective variety?

来源字段：https://doi.org/10.4171/owr/2014/09

07807 | OWR-11136-019

### Projectively Unique Polytopes and Their Realization Spaces

O / open | Oberwolfach 报告问题汇编

For a fixed dimension $d$, are there only finitely many combinatorial types of projectively unique $d$-dimensional convex polytopes? More generally, how does the dimension of the realization space of a $d$-polytope grow with its size?

来源字段：https://doi.org/10.4171/owr/2011/44

07808 | OWR-12006-001

### Projectively Unique Polytopes in Intermediate Dimensions

P / partially_solved | Oberwolfach 报告问题汇编

For which dimensions $4\leq d\leq80$ are there infinitely many combinatorial types of projectively unique $d$-polytopes?

来源字段：https://doi.org/10.4171/owr/2012/34

07809 | AIM-GEOMETRY-0036

### Proper cluster modular actions and a braid-centralizer kernel obstruction

P / partially_solved | AIM 研讨会问题汇编

Does the cluster modular group act properly discontinuously on the positive real part of the braid variety? Does this group contain the centralizer of our braid?

来源字段：http://aimpl.org/clusterbraid/8/

07810 | OWR-4334-001

### Proper Minimal Embeddings of Open Surfaces

O / open | Oberwolfach 报告问题汇编

Let $M$ be an open surface. Does $M$ admit a complete proper minimal embedding in some smooth bounded domain $D\subset\mathbb R^3$ if and only if $M$ has finitely many nonorientable ends and every end of $M$ has infinite genus?

来源字段：https://doi.org/10.4171/owr/2010/21

07811 | OWR-15427-008

### Proper Three-Colorings for Homothets of Convex Sets

S / solved | Oberwolfach 报告问题汇编

For every plane convex body $D$, does there exist an integer $m$ such that every finite point set admits a three-coloring in which each homothetic copy of $D$ containing at least $m$ points contains two points of different colors?

来源字段：https://doi.org/10.4171/owr/2017/19

07812 | OWR-15427-009

### Proper Three-Colorings for Pseudo-Disk Arrangements

S / solved | Oberwolfach 报告问题汇编

Does every finite point set admit a three-coloring such that each member of every pseudo-disk family containing at least four points contains two points of different colors?

来源字段：https://doi.org/10.4171/owr/2017/19

07813 | AIM-GEOMETRY-0419

### Pullback Hamiltonians, maximality, and toric Floer resonances

O / open | AIM 研讨会问题汇编

**Question 1.23.** Polterovich also suggested that it if \(\Phi:M\to B\) is a [what kind of?] fibration, then the set \(\{\Phi^*H\mid H:M\to\mathbb R\}\) is a “maximal torus” in \(\operatorname{Ham}(M)\) and should provide a good source of examples for calculations in Floer homology. [There was then some further discussion of locally toric fibrations by various people of which I do not have good notes.]

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

07814 | OWR-11136-004

### Purdy Plane Line Incidence Conjecture

O / open | Oberwolfach 报告问题汇编

For sufficiently large $n$ points in $\mathbb{R}^3$, not all on two skew lines and with no plane containing $n-1$ points, is the number $p$ of determined planes at least the number $l$ of determined lines?

来源字段：https://doi.org/10.4171/owr/2011/44

07815 | GREEN-083

### Pyjama Set Covering

P / partially_solved | Ben Green 问题

How many rotated (about the origin) copies of the "pyjama set" $\{(x, y) \in \mathbb{R}^2 : \operatorname{dist}(x, \mathbb{Z}) \leq \varepsilon\}$ are needed to cover $\mathbb{R}^2$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

07816 | AIM-GEOMETRY-0381

### Q-curvature for Weyl structures and the Faraday ambiguity

S / solved | AIM 研讨会问题汇编

On an oriented conformal manifold of general even dimension, construct a natural invariant of a Weyl structure, preferably an \(n\)-form or weight-\(-n\) density, which reduces to Branson's Riemannian \(Q\)-curvature form when the Weyl connection is Levi-Civita. Determine what such an extension says about the choice of Riemannian \(Q\).

With its line breaks displayed, it reads:

The preceding text constructs, in dimension four, the invariant four-form \[ \boldsymbol Q^{\rm ES}_{(g,\alpha)} = \left[ Q_g-S_g^a\alpha_a +4\nabla^a\bigl(\alpha^b\nabla_{[a}\alpha_{b]}\bigr) \right]dv_g, \] which reduces to the Riemannian \(Q\)-curvature form \(Q_g\,dv_g\) when \(\alpha=0\). Problem 22 asks for an analogue on Weyl structures in all even dimensions.

The source corrections are:

- “Slov'ack” is Jan **Slovák**. - The attached “27” is a page/footer extraction marker, not part of the mathematical sentence; it is absent from the official HTML. - Boldface was lost. The four-dimensional contextual formula is \[ \boxed{ \boldsymbol Q =2\boldsymbol P^2 -2\boldsymbol P^{ab}\boldsymbol P_{ba} -D^aD_a\boldsymbol P.} \] Here \(\boldsymbol P_{ab}\) is the Rho/Schouten tensor of the Weyl connection, \(\boldsymbol P=\boldsymbol P_a{}^a\), and \(D\) is the Weyl covariant derivative. The reversed contraction \(\boldsymbol P^{ab}\boldsymbol P_{ba}\) is intentional.

The original primer places Problem 22 before this boldface formula. It then says that the already constructed invariant is “not manifestly so” and rewrites it in the displayed form. Thus the formula is a four-dimensional answer/context immediately following the question, not an additional clause of the question. It also motivates Problem 23, which asks whether the detailed calculation was necessary.

The recovered problem is:

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

07817 | AIM-GEOMETRY-0239

### Quadratic counting of closed flat geodesic classes

P / partially_solved | AIM 研讨会问题汇编

Consider a closed surface of genus at least two with a metric of nonpositive curvature. Let \(N(T)\) be the number of free homotopy classes containing closed geodesics along which the Gaussian curvature is everywhere zero.

The PDF then asks: \[ \lim_{T\to\infty}\frac{\log N(T)}{T}=0 \quad\Longrightarrow\quad N(T)\leq C T^2\ ? \] It attributes the question to Gerhard Knieper, says the answer is positive for the torus, and suggests a Finsler analogue.

The only coherent reconstruction is therefore \[ N(T)=\#\{\text{eligible free homotopy classes whose geodesic length is at most }T\}. \] This reconstruction is also consistent with standard closed-geodesic counting notation and with the claimed torus case. In a nonpositively curved metric, all closed geodesics in one free homotopy class have the same length: their lifts are axes of one deck transformation, and the period is its translation length. Thus the threshold remains unambiguous when a class contains a whole flat cylinder.

The question ends with “One can also study this problem in the Finsler category.” The running header and everything from “7. In particular, Theorems 7.0.2 and 7.0.3 are not part of Question 6.5.1.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07818 | OWR-15427-017

### Quadratically Many Essentially Different Lines in Dense Point Sets

P / partially_solved | Oberwolfach 报告问题汇编

Let $P\subset\mathbb R^2$ consist of $n$ points with minimum pairwise distance $1$ and diameter $O(\sqrt n)$. Must $P$ determine $\Omega(n^2)$ pairwise essentially different lines, where two lines are essentially different if their directions differ by at least $1/n$ or their $1/\sqrt n$-neighborhoods do not intersect inside $\operatorname{conv}(P)$?

来源字段：https://doi.org/10.4171/owr/2017/19

07819 | OWR-17135-016

### Qualitative Differences Among Polytope and Sphere Face Vectors

O / open | Oberwolfach 报告问题汇编

For $d\geq4$, identify qualitative differences between the $f$-vectors of simplicial and general $d$-polytopes, and between the flag $f$-vectors of $d$-polytopes and regular CW $(d-1)$-spheres.

来源字段：https://doi.org/10.4171/owr/2019/39

07820 | AIM-GEOMETRY-0168

### Quantitative accounting for a many-turn spiral under curve shortening

P / partially_solved | AIM 研讨会问题汇编

It makes sense to consider the weak limit of manifolds with lower sectional curvature bounds by passing to a sequence of covers if the sequence collapses. Is the universal cover of an almost flat manifold uniformly locally $\mathrm{CAT}(0)$ (after rescaling the metric)? Is there a uniform bound on the injectivity radius of the universal cover of manifolds with $|K| \le 1$?

来源字段：http://aimpl.org/flowriemannian/2/

07821 | AIM-GEOMETRY-0065

### Quantitative bad-tail control from the BCFT singular pressure gap

P / partially_solved | AIM 研讨会问题汇编

Can one simplify the assumptions and/or arguments of Climenhaga--Thompson (*J. Lond. Math. Soc.* 87, 2013, MR3046278, on equilibrium states beyond specification and the Bowen property) to the setting of Burns--Climenhaga--Fisher--Thompson (*Geom. Funct. Anal.* 28, 2018, MR3856792, on unique equilibrium states for geodesic flows in nonpositive curvature under the pressure-gap condition $P_{\mathrm{sing}}(\varphi) < P(\varphi)$)?

来源字段：http://aimpl.org/equibdynsysgeom/4/

07822 | AIM-GEOMETRY-0242

### Quantitative opposite-end divergence under convex distance

P / partially_solved | AIM 研讨会问题汇编

If \(\alpha\ne\beta\) are complete unit-speed geodesics in a complete simply connected Riemannian manifold without conjugate points and \[ d(\alpha(t),\beta(t))\longrightarrow0\qquad(t\to\infty), \] must their equal-time distance tend to infinity as \(t\to+\infty\)?

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07823 | AIM-GEOMETRY-0328

### Quantitative tropical-theta approximation on an affine circle

P / partially_solved | AIM 研讨会问题汇编

In particular, the affine transition group is \[ \mathbb R^n\rtimes \operatorname{GL}_n(\mathbb Z), \] the singular set is $\Delta=B\setminus B_0$, the dual lattice is $\check\Lambda\subset \mathcal T^*_{B_0}$, and the equation is \[ \det\left(\frac{\partial^2s_i}{\partial y_j\partial y_k}\right)=\text{constant}.

1. If $B$ has an integral-affine structure on a dense open $B_0$, with $B\setminus B_0$ a locally finite union of locally closed submanifolds, should $B$ be called a tropical Calabi--Yau variety, and are those singularities natural tropically? 2. With $\operatorname{Aff}(B,\mathbb R)$ the sheaf of continuous functions that are integral affine on $B_0$, a tropical line bundle is defined as a class in $H^1(B,\operatorname{Aff}(B,\mathbb R))$. If it is represented by a Cech cocycle $(\alpha_{ij})$, a section is a family of tropical functions satisfying ordinary differences $s_i-s_j=\alpha_{ij}$. What are tropical Riemann--Roch, ampleness, very ampleness, and projective embeddings? 3. For the printed tropical cubic \[ -6x^3-4x^2y-3xy^2-6y^3-4y^2z-3yz^2-0xyz-3x^2z-1xz^2-3z^3, \] the source asks the reader to check that its compact polygonal core has affine perimeter $13$, and then asks for an embedding $\mathbb R/13\mathbb Z\to\mathbb{TP}^2$ by three sections of a degree-five line bundle. It asks whether contraction observed for a degree-three construction means that degree three is not very ample. Proposition 2 below finds a discrepancy: the exact printed coefficients give perimeter $15$ under the standard projective lattice convention. 4. For the torus bundles $X(B_0)=\mathcal T_{B_0}/\Lambda$ and $\check X(B_0)=\mathcal T^*_{B_0}/\check\Lambda$, can smooth local potentials be found that solve the real Monge--Ampere equation? More specifically, can tropical sections $s_n\in\Gamma(B,\mathcal L^{\otimes n})$ be chosen so that $n^{-1}s_n$ converges to a classical solution, and can this be used numerically to visualize a Ricci-flat metric?

The HTML displays an index mismatch near the last question, writing $s_i\in\Gamma(B,\mathcal L^n)$ and then $s_n$. The sequence notation $s_n\in\Gamma(B,\mathcal L^{\otimes n})$ used above is an explicit reconstruction from the stated scaling $\hbar=1/n$.

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

07824 | AIM-GEOMETRY-0133

### Quantitative two-pole confluence and the hyperkähler energy bottleneck

P / partially_solved | AIM 研讨会问题汇编

Study confluence of simple poles, e.g. hyperkahler metric on limit as limit of hyperkahler metrics.

来源字段：http://aimpl.org/spectralhiggs/1/

07825 | OWR-4799-008

### Quantization near Singular Faces of Delzant Polytopes

P / partially_solved | Oberwolfach 报告问题汇编

What is the behavior of geometric quantization at the edges and vertices, including singular elliptic points, of a Delzant polytope?

来源字段：https://doi.org/10.4171/owr/2011/09

07826 | OWR-4799-004

### Quantization of Nilpotent Coadjoint Orbits by Degeneration

O / open | Oberwolfach 报告问题汇编

For a nilpotent coadjoint orbit $O_{\mathrm{nilp}}$ approached by semisimple orbits $O_{\nu}$, can one construct the representation $Q(O_{\mathrm{nilp}})$ from the geometric quantizations $Q(O_{\nu})$?

来源字段：https://doi.org/10.4171/owr/2011/09

07827 | OWR-4799-005

### Quantization of Quasi-Hamiltonian Moduli Spaces

P / partially_solved | Oberwolfach 报告问题汇编

For a pre-quantized quasi-Hamiltonian $G$-space $M_{G}(\Sigma;C_{1},\ldots,C_{r})$, compute its quantization $Q(M_{G}(\Sigma;C_{1},\ldots,C_{r}))$, especially for non-simply connected $G$ where pre-quantization has obstructions.

来源字段：https://doi.org/10.4171/owr/2011/09

07828 | OWR-12007-018

### Quasi-Isometry of Horospheres in Negative Curvature

O / open | Oberwolfach 报告问题汇编

If $(X^{n+1},g)$ is closed with negative sectional curvature, are all horospheres $(H,\widetilde g|_H)$ in its universal cover quasi-isometric to one another, and if so uniformly quasi-isometric?

来源字段：https://doi.org/10.4171/owr/2012/35

07829 | OWR-1106-004

### Quasiconformal Homogeneity Gaps for Hyperbolic Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a constant $K_2>1$ such that every $K$-quasiconformally homogeneous hyperbolic surface other than $\mathbb{H}^2$ satisfies $K\geq K_2$?

来源字段：https://doi.org/10.4171/owr/2005/45

07830 | AIM-CONVEX_GEOMETRY-0033

### Radial harmonic universality and linear-size zonotope approximation

P / partially_solved | AIM 研讨会问题汇编

**Problem 3 (R. Gardner).** Let \(\mathcal K^n\) be the convex bodies in \(\mathbb R^n\). A Minkowski class is closed in the Hausdorff metric and under Minkowski linear combinations and translations. A body \(K\) is a generalized \(\mathcal M\)-body if \(K+M_1=M_2\) for some \(M_1,M_2\in\mathcal M\). If \(\mathcal M_{B,G}\) is the smallest \(G\)-invariant Minkowski class containing \(B\), Schneider and F. Schuster's theorem says that, for nonsymmetric \(B\), every neighborhood of \(B\) contains an affine image \(B'\) of \(B\) for which the generalized \(\mathcal M_{B',SO(n)}\)-bodies are dense in \(\mathcal K^n\). The problem asks for a dual theorem, replacing generalized zonoids/Minkowski sums by generalized intersection bodies/radial sums.

**Problem 4.** If \(K\subset\mathbb R^n\) is a zonoid and \(Z\) is a zonotope that is a sum of \(M\) segments, Talagrand obtained \(M=C(\varepsilon)n\log n\) under multiplicative distance \(d(Z,K)\leq 1+\varepsilon\), while the Euclidean ball admits \(M=C(\varepsilon)n\). Is the extra \(\log n\) necessary?

来源字段：https://aimath.org/WWN/fourierconvex/fourierconvex.pdf

07831 | AIM-GEOMETRY-0238

### Radial zero curvature as a finite Jacobi-operator condition

P / partially_solved | AIM 研讨会问题汇编

Its boundaries matter.

1. The opening paragraph is a **remark continuing Question 6.2.1**, whose actual question occurs in the preceding canonical records. It asks what can be inferred, without assuming nonpositive curvature and possibly in higher dimensions, from either \[ K_{\gamma(t)}(\sigma)=0\quad\text{for every two-plane }\sigma\subset T_{\gamma(t)}M, \] or the weaker radial condition \[ K_{\gamma(t)}(\dot\gamma(t),v)=0\quad(v\perp\dot\gamma(t)). \tag{R} \] 2. Section 6.3 asks Eberlein's flat-closing question: if a compact rank-one manifold of nonpositive curvature contains an isometrically immersed Euclidean plane, must it contain a totally geodesically immersed flat torus? 3. Section 6.4 asks whether the Besson--Courtois--Gallot degree/volume-entropy rigidity theorem extends from negatively curved locally symmetric targets to higher-rank symmetric spaces of noncompact type. 4. The final two sentences merely begin Section 6.5 by defining \(N(T)\), the number of free homotopy classes containing zero-curvature closed geodesics. The actual Knieper question and its formula occur in the next canonical record, AIM-GEOMETRY-0239, and are not claimed here.

The input's garbled expression `h4 > vol Vol` is reconstructed from the original PDF and Merlin's paper as \[ h_{\mathrm{vol}}(g)^4\operatorname{Vol}(M,g), \] the four-dimensional instance of the scale-invariant normalized volume entropy \(h_{\mathrm{vol}}(g)^n\operatorname{Vol}(M,g)\).

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07832 | OWR-4078-004

### Radius and Fixed Points of Convex Building Subsets

S / solved | Oberwolfach 报告问题汇编

Let $C$ be a closed convex subset of a spherical building $B$ that is not a subbuilding. Is $\operatorname{rad}(C)\leq\pi/2$, and does the isometry group of $C$ have a fixed point?

来源字段：https://doi.org/10.4171/owr/2009/35

07833 | OWR-13490-006

### Ramos Conjecture for Equipartitions by Hyperplanes

O / open | Oberwolfach 报告问题汇编

For $j,k\geq 1$, let $\Delta(j,k)$ be the least dimension in which any j masses can be simultaneously cut into $2^k$ equal orthants by k affine hyperplanes. The Ramos conjecture asserts that the known lower bound for $\Delta(j,k)$ is tight for all j and k.

来源字段：https://doi.org/10.4171/owr/2015/5

07834 | AIM-CONVEX_GEOMETRY-0021

### Random central planes and approximate convexity

P / partially_solved | AIM 研讨会问题汇编

**Conjecture.** Let \(\mathcal K\) be the family consisting of the empty set and all compact convex subsets of \({\mathbb R}^n\) with nonempty interior. For compact \(S\subset{\mathbb R}^n\), choose \[ C\in\operatorname*{arg\,min}_{D\in\mathcal K} |\Delta(S,D)|. \] Call \(S\) \(\varepsilon\)-convex when \[ |\Delta(S,C)|\leq\varepsilon |S|. \] Assume that the center of gravity of \(S\) is the origin. Draw \(x,y\) independently and uniformly from \(S\), put \[ H(x,y)=\operatorname{span}\{x,y\},\qquad P(x,y)=S\cap H(x,y), \] and let \(\mu\) be the resulting distribution of central two-dimensional sections. The conjecture is \[ \Pr_\mu\{P(x,y)\text{ is convex}\}\geq1-\varepsilon \quad\Longrightarrow\quad S\text{ is }O(n\varepsilon)\text{-convex}. \tag{AIM} \]

The big-\(O\) means a universal constant. To make the sampling statement well-defined one must add \(n\geq2\), \(0<|S|<\infty\), and, as in the property-testing context, \(0<\varepsilon\leq1\). Compactness gives measurability and finite volume. Positive volume makes the uniform distribution and center of gravity meaningful. It also implies that \(x,y\) are nonzero and linearly independent almost surely, so \(H(x,y)\) is two-dimensional almost surely.

来源字段：https://aimath.org/WWN/convexgeometry/convexgeometry.pdf

07835 | AIM-GEOMETRY-0109

### Random jammed sphere packings: a rattler-corrected contact bound and a density-contact counterexample

P / partially_solved | AIM 研讨会问题汇编

**Random packings**

It is know from simulations and experiments that a "random" packing of spheres that cannot be "locally improved" achieves a density of 64%, well short of the maximal density of 74...%.

What is a reasonable definition of a random jammed sphere packing? Can the experimental density of 64% in $\mathbb{R}^3$ be justified? How is density related to contact number(is it)?

How, if at all, is density related to contact number?

来源字段：http://aimpl.org/softpack/1/

07836 | AIM-GEOMETRY-0024

### Rank-one brick pairs and an obstruction to one-seed toric recovery

P / partially_solved | AIM 研讨会问题汇编

**Can we obtain brick manifolds by cluster-theoretic means?**

来源字段：http://aimpl.org/clusterbraid/4/

07837 | AIM-GEOMETRY-0038

### Rank-one comparison of cluster, braid-coordinate, and TNN positivity

P / partially_solved | AIM 研讨会问题汇编

Describe the positive real/totally non negative part of the braid variety. Can this be done using motivation from the perspective of algebraic geometry? symplectic topology? linear algebra?

来源字段：http://aimpl.org/clusterbraid/9/

07838 | AIM-GEOMETRY-0137

### Rank-one deformation families for Hitchin branes

P / partially_solved | AIM 研讨会问题汇编

Determine the deformation families for branes in the Hitchin system (complex Lagrangian or hyper-holomorphic?).

来源字段：http://aimpl.org/spectralhiggs/1/

07839 | AIM-GEOMETRY-0141

### Rank-one handlebody Lagrangians versus Hitchin fibres

P / partially_solved | AIM 研讨会问题汇编

Understand the intersection of Lagrangians coming from 3-manifolds bounding the surface with fibres of the Hitchin fibration.

来源字段：http://aimpl.org/spectralhiggs/1/

07840 | AIM-GEOMETRY-0139

### Rank-one hyperholomorphic lines, Fourier-Mukai graph mirrors, and a proper-support obstruction

P / partially_solved | AIM 研讨会问题汇编

Ways to construct hyper-holomorphic sheaves on branes, their mirrors and spectral data.

来源字段：http://aimpl.org/spectralhiggs/1/

07841 | AIM-GEOMETRY-0138

### Rank-one meromorphic Hitchin duality and a marked-data obstruction

P / partially_solved | AIM 研讨会问题汇编

Langlands duality for meromorphic Higgs bundles

The existence of the hyperkahler metrics on the wild Hitchin spaces means the moduli spaces of meromorphic connection on curves (and the corresponding wild character varieties) have natural special Lagrangian torus fibrations, by hyperkahler rotation of the Hitchin fibration (shown to be proper in this generality by Nitsure).

Study the SYZ mirror symmetry proposal in this context. For example suppose we take the dual holomorphic Lagrangian fibration of the smooth fibres of the meromorphic Higgs bundle moduli space. Is the result another Hitchin fibration? What is the involution on the set of data (group, irregular curve, residue data) that corresponds to this T-duality?

For example for holomorphic Higgs bundles it is known one should just replace the group by the Langland dual group, so we are asking for an involution generalizing this.

The approach of Donagi-Pantev works for the Poisson moduli spaces of meromorphic Higgs bundles, so the question is about matching symplectic leaves.

Gukov-Witten studied the tame case and described interesting phenomenon related to rigid coadjoint orbits, but it does not seem that a precise involution was established

来源字段：http://aimpl.org/spectralhiggs/1/

07842 | AIM-GEOMETRY-0306

### Rank-one normalization and status audit for the HRV mixed-Hodge formula

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.18 (Hausel--Rodriguez-Villegas).** For a compact Riemann surface of genus $g$, coprime integers $(n,d)=1$, and $\zeta_n = \exp(2\pi i/n)$, let $$ \mathcal{M}_B^d(\mathrm{GL}_n) = \left\{(A_1, B_1, \ldots, A_g, B_g) \in \mathrm{GL}_n(\mathbb{C})^{2g} : \prod_{i=1}^g [A_i, B_i] = \zeta_n^d I_n\right\} \!\!/\!\!\, \mathrm{PGL}_n $$ be the twisted $\mathrm{GL}_n(\mathbb{C})$ character variety, with ordinary-cohomology mixed Hodge polynomial $H_n^{\mathrm{geom}}(x,y,t) = \sum_{p,q,k} h^{p,q;k}(\mathcal{M}_B^d(\mathrm{GL}_n)) x^p y^q t^k$, and let $\mathbb{H}_n(q,t)$ be the combinatorial rational function defined via the partition-hook plethystic product. Then $$ H_n^{\mathrm{geom}}(\sqrt{q}, \sqrt{q}, t) = \mathbb{H}_n(q, t). $$

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

07843 | AMR-092-0004

### Rational distances from the vertices of a square

O / open | AMR 问题清单汇编

Given a unit square, does there exist a point in its plane, inside or outside the square, whose distances from all four vertices are rational? Equivalently after scaling, can a square $ABCD$ and a coplanar point $P$ be found such that $AB,PA,PB,PC,PD$ are all integers?

背景链接：https://unsolvedproblems.org/index_files/RationalDistance.htm

07844 | OWR-5146-002

### Rational Ellipticity, Entropy, Euler Characteristic, and Formality

O / open | Oberwolfach 报告问题汇编

Are almost nonnegatively curved manifolds rationally elliptic; do they admit Riemannian metrics with vanishing topological entropy; do nonnegatively curved manifolds have nonnegative Euler characteristic; and are positively curved manifolds formal?

来源字段：https://doi.org/10.4171/owr/2011/18

07845 | OWR-14299513-002

### Rational Points Near Submanifolds

O / open | Oberwolfach 报告问题汇编

Given a submanifold $M$, how many rational points of height at most $H$ can lie in a prescribed small neighborhood of $M$?

来源字段：https://doi.org/10.4171/owr/2025/32

07846 | AMR-038-0007

### Rationality of Hermite constants

P / partially_solved | AMR 问题清单汇编

Are the Hermite constants associated with densest lattice sphere packings always rational? Determine their arithmetic nature in dimensions where the exact value is unknown.

背景链接：https://ics.uci.edu/~eppstein/junkyard/open.html

07847 | OWR-4416-013

### Real Places of Mapping Torus Trace Fields

P / partially_solved | Oberwolfach 报告问题汇编

For every surface type $(-\chi,p)$ with $-\chi>1$, does there exist a hyperbolic mapping torus of that type whose trace field has a real place?

来源字段：https://doi.org/10.4171/owr/2010/35

07848 | OWR-15587-012

### Real-Rootedness and Log-Concavity of Ehrhart Vectors

O / open | Oberwolfach 报告问题汇编

Can the implication diagram for Ehrhart $h^*$-vectors$ and properties such as smoothness, very ampleness, and spanning be extended to include real-rootedness or log-concavity?

来源字段：https://doi.org/10.4171/owr/2017/44

07849 | OWR-2491-005

### Realizing Convex Building Subcomplexes by Destabilizing Loci

O / open | Oberwolfach 报告问题汇编

Given a convex contractible subcomplex $Y$ of a building $X$, can one find an action of $G$ on a variety $V$ and a point $v\in V$ such that $C_0(v)\subseteq Y$ and the stabilizer of $Y$ in $\operatorname{Aut}(X)$ stabilizes $C_0(v)$?

来源字段：https://doi.org/10.4171/owr/2008/53

07850 | AMR-059-0001

### Realizing Statistical Manifolds in Dually Flat Manifolds

P / partially_solved | AMR 问题清单汇编

For a statistical manifold $(M,g,\nabla,\nabla^*)$, find conditions under which it can be realized as an $n$-dimensional submanifold of an $m$-dimensional, $m>n$, dually flat manifold. If this is not always possible, determine what additional quantity ensures realization in finite dimension.

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

07851 | OWR-1795-008

### Recognition of Lattice Point Configurations as Convex Ovals

P / partially_solved | Oberwolfach 报告问题汇编

Let $\Lambda$ be a planar lattice, let $S$ be a closed bounded convex region containing the origin, let $S(r,P)$ be its scale-$r$ translate, and set $J(r,P)=\Lambda\cap S(r,P)$. How efficiently can one decide whether a finite lattice configuration is an $S$-oval of this form?

来源字段：https://doi.org/10.4171/owr/2008/14

07852 | OWR-13951-013

### Reconstruction from Directly Congruent Convex Projections

P / partially_solved | Oberwolfach 报告问题汇编

For $2\leq k\leq n-1$, if convex bodies $K,L\subseteq\mathbb{R}^n$ have directly congruent projections onto every $H\in G(n,k)$, with the congruences induced by elements of $SO(k,H)$, must K be a translate of $\pm L$?

来源字段：https://doi.org/10.4171/owr/2015/56

07853 | OWR-13951-012

### Reconstruction of Convex Bodies from Congruent Projections

P / partially_solved | Oberwolfach 报告问题汇编

For $2\leq k\leq n-1$, if convex bodies $K,L\subseteq\mathbb{R}^n$ have congruent projections onto every $H\in G(n,k)$, with the congruences induced by elements of $O(k,H)$, must K be a translate of $\pm L$?

来源字段：https://doi.org/10.4171/owr/2015/56

07854 | AIM-GEOMETRY-0146

### Recovered vortex quotient and its one-vortex quantum K-ring

P / partially_solved | AIM 研讨会问题汇编

Compute \(R=K_{\mathbb{C}^{\ast}}^{quant}(V^{\otimes k}\oplus Adj//_{U(N)})\) and describe what this ring is.

来源字段：http://aimpl.org/spectralhiggs/1/

07855 | AIM-GEOMETRY-0226

### Recovering the marking from a flow conjugacy

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 5.2.1.** Compact Riemannian manifolds with negative curvature must be isometric if they have C0-conjugate geodesic flows. As with

As with Conjecture 5.1.1 it may be possible to relax the hypothesis of negative curvature to no conjugate points, but some restriction on the metrics is required. Weinstein pointed out that all of the Zoll metrics on \(S^2\) have conjugate geodesic flows. Using the explicit rotationally-symmetric examples of Zoll metrics (Tannery metrics in the terminology of [Besse1978, Chapter 4]) Croke and Kleiner constructed examples of different metrics on an arbitrary closed manifold such that their geodesic flows are conjugate and their marked length spectra coincide, see [Cro-Klei1994].

The conjecture is true in dimension two; indeed it is enough to assume that one of the manifolds has nonpositive curvature [Cr-Fa-Fe1992]. It follows from the theorem of Besson--Courtois--Gallot discussed in the next subsection that the conjecture holds if one of the metrics is locally symmetric. Croke and Kleiner showed that \(C^1\) conjugacy implies that the volumes of the manifolds are the same with no restrictions on the metrics and implies isometry of the metrics if one of them has a global parallel vector (as is the case in a Riemannian product) [Cro-Klei1994].

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07856 | AMR-054-0078

### Rectangling a Rectangle

P / partially_solved | AMR 问题清单汇编

Do there exist rectangles that may be partitioned into a finite number $n$ of rectangular pieces of equal area but with all perimeters different?

背景链接：https://topp.openproblem.net/p78

07857 | OWR-16415-016

### Rectifiable Nonsmooth Limit Curves of Anosov Representations

O / open | Oberwolfach 报告问题汇编

For the fundamental group $\Gamma$ of a closed orientable surface of genus $g\geq2$, does there exist a $1$-Anosov representation $\Gamma\to\operatorname{PGL}(d,\mathbb R)$ whose circular limit set in $\mathbb P(\mathbb R^d)$ is rectifiable but not $C^{1,\alpha}$?

来源字段：https://doi.org/10.4171/owr/2018/40

07858 | AMR-067-0003

### Reducibility of the holonomy of flat manifolds

O / open | AMR 问题清单汇编

Give an alternative, geometric proof that the holonomy representation of a closed flat manifold is reducible.

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

07859 | OWR-3389-023

### Refinements and Degenerations in Schrödinger Spectral Gaps

P / partially_solved | Oberwolfach 报告问题汇编

For one-dimensional Schrödinger operators, characterize the largest class of potentials for which the fundamental-gap estimate holds. Are the normalized potential means $\langle Vu_j^2\rangle$ increasing with the eigenvalue index $j$, and can the gap estimate be sharpened by a correction term involving $V$? Can Lavine-type, Brascamp-Lieb, or Korevaar methods yield analogous or stronger estimates in higher dimensions?

来源字段：https://doi.org/10.4171/owr/2009/06

07860 | AMR-054-0066

### Reflexivity of Point Sets

P / partially_solved | AMR 问题清单汇编

Let $\rho(S)$ be the fewest number of reflex vertices in a polygonization of a 2D point set $S$, i.e., the fewest reflexivities of any simple polygon whose vertex set is $S$. Let $\rho(n)$ be the maximum of $\rho(S)$ over all sets $S$ with $n$ points. What is $\rho(n)$?

背景链接：https://topp.openproblem.net/p66

07861 | AIM-CONVEX_GEOMETRY-0005

### Refutation by congruent hyperplane sections and rigidity for off-center balls

S / solved | AIM 研讨会问题汇编

Let \(G(n,k)\) be the Grassmannian of \(k\)-dimensional linear subspaces of \(\mathbb R^n\), where \(2\leq k\leq n-1\) is fixed. Suppose \(K\) and \(L\) are star bodies and, for every \(S\in G(n,k)\), the sections \(K\cap S\) and \(L\cap S\) are congruent. Must \(K=\pm L\)?

The source adds that the answer is affirmative for \(k=2\) when “congruent” is replaced by “a rotation of,” citing [R].

来源字段：https://aimath.org/pastworkshops/sectionsconvexproblems.pdf

07862 | OWR-17135-009

### Regular Ball Decompositions of Totally Nonnegative Fibers

O / open | Oberwolfach 报告问题汇编

Are the natural cell decompositions of every fiber of the maps to totally nonnegative spaces regular CW complexes homeomorphic to closed balls?

来源字段：https://doi.org/10.4171/owr/2019/39

07863 | OWR-3385-015

### Regular Octahedra Inscribed in Smooth Two-Spheres

O / open | Oberwolfach 报告问题汇编

Does every $C^\infty$-embedded two-sphere in $\mathbb R^3$ contain six points spanning a regular octahedron?

来源字段：https://doi.org/10.4171/owr/2009/02

07864 | AIM-GEOMETRY-0276

### Regular-stratum curvature-jet tests for projective metrizability

P / partially_solved | AIM 研讨会问题汇编

Construct a system of scalar invariants of a projective structure that vanish if and only if there exists (locally, in a neighborhood of almost every point) a metric compatible with a given projective structure.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

07865 | OWR-2042-004

### Regularity of Minimal Surfaces in the Heisenberg Group

O / open | Oberwolfach 报告问题汇编

What is the regularity of critical points of the area functional (equivalently, minimal surfaces in the relevant sense) in the Heisenberg group? In particular, are such critical points smooth under natural hypotheses?

来源字段：https://doi.org/10.4171/owr/2008/31

07866 | OWR-16412-004

### Reinhardt Models of Rank-Two Rotation Domains

O / open | Oberwolfach 报告问题汇编

Which Reinhardt domains occur as linear models of rank-two rotation domains?

来源字段：https://doi.org/10.4171/owr/2018/37

07867 | AMR-071-0029

### Reinhardt's conjecture

P / partially_solved | AMR 问题清单汇编

Reinhardt's conjecture: the smoothed octagon has the lowest maximum packing density of all centrally-symmetric convex plane sets

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

07868 | OWR-14298364-005

### Relative Homology Obstructions to Complete Positive Scalar Curvature

P / partially_solved | Oberwolfach 报告问题汇编

Let $Y$ be a manifold with boundary and classifying map $\nu$. Is nonvanishing of $\nu_*[Y,\partial Y]\in H_n(B\Gamma,B\Gamma^{\prime};\mathbb{Q})$ an obstruction to a complete positive-scalar-curvature metric on $\operatorname{int}(Y)$ when $\Gamma^{\prime}\to\Gamma$ is injective or the metric has finite volume?

来源字段：https://doi.org/10.4171/owr/2024/28

07869 | OWR-731-005

### Relative Isoperimetric Partitions of Convex Regions

O / open | Oberwolfach 报告问题汇编

For a convex region $\Omega$, is the minimizer of the relative isoperimetric partition problem a single disk?

来源字段：https://doi.org/10.4171/owr/2004/29

07870 | OWR-1106-010

### Relative Rigidity of Geometrically Finite Hyperbolic Three-Manifolds

O / open | Oberwolfach 报告问题汇编

Prove that geometrically finite hyperbolic 3-manifolds are locally rigid relative to the bending lamination on their convex-hull boundaries.

来源字段：https://doi.org/10.4171/owr/2005/45

07871 | OWR-15579-003

### Removal of Conical Singularities in Higher-Dimensional Vacuum Metrics

P / partially_solved | Oberwolfach 报告问题汇编

For a prescribed rod structure in the higher-dimensional stationary vacuum construction, can the conical singularities along the bounded axis rods be removed by adjusting the parameters?

来源字段：https://doi.org/10.4171/owr/2017/36

07872 | OWR-14297736-011

### Removing Complementary-Cap Restrictions for Spherical Neumann Eigenvalues

P / partially_solved | Oberwolfach 报告问题汇编

For domains in $S^2$ of area at most half the sphere, prove that a spherical cap maximizes the second Neumann eigenvalue without assuming that the competing domain lies outside a complementary cap.

来源字段：https://doi.org/10.4171/owr/2023/36

07873 | AIM-GEOMETRY-0078

### Renormalized and cone-relative universal optimality

P / partially_solved | AIM 研讨会问题汇编

Can we extend the definition of universal optimality to potentials with slow decay (or functions that are not completely monotonic)?

来源字段：http://aimpl.org/discreteaf/1/

07874 | AIM-GEOMETRY-0391

### Replacing the Pfaffian by Q in global conformal invariants

S / solved | AIM 研讨会问题汇编

**Problem 31.** Is it possible to write any \(\mathbf S\), as in Problem 30, in the form \[ \operatorname{const}\cdot\mathbf Q+\mathbf L+\mathbf V, \] where \(\mathbf V\) is an exact divergence?

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

07875 | OWR-1540-003

### Rescaled Reflective Submanifolds of Symmetric Spaces

O / open | Oberwolfach 报告问题汇编

Let $N=\exp(\mathfrak p)$ be a simply connected immersed reflective submanifold of a semisimple Riemannian symmetric space $\overline N$. For the metric rescaling $N_R$, is there a local or global isometric immersion of $N_R$ into $\overline N$ that preserves normal curvature and is a $V_{\mathfrak p}$-submanifold? Ask this for $R>1$ when $\overline N$ is of compact type and for $R<1$ when it is of noncompact type.

来源字段：https://doi.org/10.4171/owr/2007/24

07876 | AMR-058-0018

### Restricted Optimality of Kelvin's Foam

P / partially_solved | AMR 问题清单汇编

Prove or disprove successively that the Kelvin cell is the least-area fundamental domain for the BCC torus, that it minimizes among all unit-volume flat tori, that Kelvin's foam minimizes among partitions with congruent cells, and that it minimizes among equal-pressure foams.

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

07877 | AIM-GEOMETRY-0072

### Restricted spectral radii and zeta separation for a pressure gap

P / partially_solved | AIM 研讨会问题汇编

Can one find a dynamical interpretation of the pressure gap? For instance, can one relate the pressure gap to an essential spectral radius for a dynamical operator? Or some dynamically defined \(\zeta\)-function?

来源字段：http://aimpl.org/equibdynsysgeom/5/

07878 | OWR-13494-007

### Resurgence Bounds for Linear Arrangements in Multiprojective Spaces

O / open | Oberwolfach 报告问题汇编

For ideals of line arrangements, which further arrangements admit useful resurgence bounds? Can analogous containment and resurgence results be extended to higher-dimensional linear arrangements in products of projective spaces, and what upper bounds hold for concrete subschemes of multiprojective spaces?

来源字段：https://doi.org/10.4171/owr/2015/9

07879 | OWR-2654832-001

### Retracted versus Intrinsic Thin Plate Splines on Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

How does a retracted thin plate spline on a Riemannian manifold differ from the intrinsic manifold thin plate spline, particularly in the interpolation case $\lambda=\infty$ and as a function of the manifold's closeness to Euclidean space?

来源字段：https://doi.org/10.4171/owr/2020/36

07880 | OWR-1726-006

### Rho Invariants for Foliated Dirac Operators

P / partially_solved | Oberwolfach 报告问题汇编

For Dirac operators on foliations and on group coverings, can secondary invariants of $\rho$-type be defined, are they mathematically useful, and can they yield purely geometric applications?

来源字段：https://doi.org/10.4171/owr/2007/43

07881 | AMR-067-0014

### Ricci pinching on solvable Lie groups

P / partially_solved | AMR 问题清单汇编

For solvable Lie groups $G$, show that solvsolitons are the only local maxima of the Ricci pinching functional $g\mapsto F(g)=\frac{Scal(g)^2}{|Ric(g)|^2}$ on left-invariant Riemannian metrics on $G$.

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

07882 | OWR-14298364-009

### Ricci-Flat Metrics on Spheres and Invariant Einstein Metrics

O / open | Oberwolfach 报告问题汇编

Does $S^4$ admit a Ricci-flat metric? For each $(d_1,d_2)\in\{(2,7),(3,6),(4,5)\}$, are the three constructed metrics the only non-round $\operatorname{SO}(d_1+1)\times\operatorname{SO}(d_2+1)$-invariant Einstein metrics on $S^{10}$?

来源字段：https://doi.org/10.4171/owr/2024/28

07883 | AIM-GEOMETRY-0177

### Ricci-flow smoothing at a pointwise curvature-volume radius

P / partially_solved | AIM 研讨会问题汇编

Can we smooth metrics with $\mbox{Riem} \geq -\mbox{Id}$ on volume scale, i.e., want nearby metrics with $|\nabla^k Rm| \leq C_k C_p^{-k-2}$?

来源字段：http://aimpl.org/flowriemannian/4/

07884 | OWR-13347-017

### Ricci-Tensor Fibrations for Generalized Flag Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Let $K\subset L\subset G$ define a fibration $G/K\to G/L$ with fiber $L/K$, where $G/K$ and $G/L$ are generalized flag manifolds. A Kähler-Einstein metric and the homogeneous-fibration Ricci-tensor formula determine the Ricci tensor when $b_2(G/K)=2$, $b_2(G/L)=1$, and the isotropy decomposition has at most six summands. Does the homogeneous-fibration Ricci-tensor method extend to arbitrary generalized flag manifolds?

来源字段：https://doi.org/10.4171/owr/2014/45

07885 | OWR-4078-006

### Riemann-Type Minimal Surfaces with One Handle

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a one-parameter family of Riemann-type properly embedded minimal surfaces obtained by adding one handle between two consecutive planar ends?

来源字段：https://doi.org/10.4171/owr/2009/35

07886 | AMR-067-0002

### Riemannian manifolds with curvature bounds

O / open | AMR 问题清单汇编

For every $\ell,k>0$, there exist $C,L,K>0$ with the following effect. Let $(M,g)$ be a complete Riemannian manifold with injectivity radius $inj(M,g)\geq \ell$ and Ricci curvature $Ric(g)\geq k$. Then there exists a metric $h$ on $M$ with $inj(M,h)\geq L$ and $|Ric(h)|\leq K$ such that $$ \frac{1}{C}\,g\leq h\leq C\,g. $$

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

07887 | OWR-13498-003

### Rigid Miura Tessellations Approximating Curved Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Given an arbitrary surface with intrinsic curvature, does there exist a Miura-like tessellation of the plane whose folded realization approximates that surface, and can the tessellation be made rigidly foldable with one degree of freedom?

来源字段：https://doi.org/10.4171/owr/2015/13

07888 | AIM-CONVEX_GEOMETRY-0023

### Rigidity and a sharp halfspace benchmark for Gaussian cylinders

P / partially_solved | AIM 研讨会问题汇编

Let \(K\subseteq\mathbb R^n\) be a convex set, and let \(X\) be a Gaussian random variable conditioned to lie in \(K\). Does \(\operatorname{Var}(X_\theta)=1\) imply that \(K\) is a cylinder in direction \(\theta\)? If \(\operatorname{Var}(X_\theta)=1-\varepsilon\), then does it mean that \(K\) has a small symmetric difference with a cylinder in direction \(\theta\)?

来源字段：https://aimath.org/WWN/convexgeometry/convexgeometry.pdf

07889 | OWR-14297735-005

### Rigidity and Symmetry of Steady Ricci Solitons

O / open | Oberwolfach 报告问题汇编

Prove that two three-dimensional flying-wing steady gradient Ricci solitons with the same asymptotic cone angle are isometric up to rescaling; determine whether $O(2)$-symmetry holds for three-dimensional ancient Ricci flows and higher-dimensional positive-curvature steady solitons, and whether all noncollapsed four-dimensional steady solitons have $O(3)$-symmetry.

来源字段：https://doi.org/10.4171/owr/2023/32

07890 | OWR-16160-012

### Rigidity from Coarse Embeddings of Warped-Cone Levels

P / partially_solved | Oberwolfach 报告问题汇编

Does a rigidity theorem hold for quasi-isometric embeddings, or even coarse embeddings, between corresponding warped-cone level sequences?

来源字段：https://doi.org/10.4171/owr/2018/19

07891 | OWR-7155451-008

### Rigidity in Scalar-Curvature Band Width Inequalities

P / partially_solved | Oberwolfach 报告问题汇编

Let $Y$ be a closed $(n-1)$-manifold, with $n-1\neq4$, that admits no positive-scalar-curvature metric; set $X=Y\times[-1,1]$ and $c_n=\sqrt{(n-1)/n}$. Suppose $\operatorname{Sc}(X,g)\geq\sigma>0$, $-\pi c_n/\sqrt{\sigma}<\ell_-<\ell_+<\pi c_n/\sqrt{\sigma}$, $H(\partial_-X)\geq\sqrt{\sigma}c_n\tan(\sqrt{\sigma}\ell_-/(2c_n))$, $H(\partial_+X)\geq-\sqrt{\sigma}c_n\tan(\sqrt{\sigma}\ell_+/(2c_n))$, and $\operatorname{width}(X,g)\geq\ell_+-\ell_-$. Must $(X,g)$ be isometric to $Y\times[\ell_-,\ell_+]$ with metric $\varphi(t)^2g_Y+dt^2$, where $\varphi(t)=\cos(\sqrt{\sigma}t/(2c_n))^{n/2}$ and $g_Y$ is Ricci-flat?

来源字段：https://doi.org/10.4171/owr/2021/30

07892 | AIM-GEOMETRY-0186

### Rigidity in stronger curvature cones and exact defect identities

P / partially_solved | AIM 研讨会问题汇编

If a compact gradient shrinking Ricci soliton $((M^n,g,f)$ satisfies $(\operatorname{Sec}_g>0)$, must $(g)$ be Einstein?

来源字段：http://aimpl.org/flowriemannian/6/

07893 | OWR-4128-003

### Rigidity of Complete Embedded Minimal Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Does every intrinsic isometry of a connected complete embedded minimal surface $M\subset\mathbb{R}^3$ extend to an ambient isometry, with minimal rigidity unless $M$ is a helicoid?

来源字段：https://doi.org/10.4171/owr/2009/45

07894 | OWR-5158-011

### Rigidity of Large-Radius Convex Subsets in Buildings

P / partially_solved | Oberwolfach 报告问题汇编

Are convex subsets of spherical buildings with circumradius greater than $\pi/2$ necessarily rigid, in the sense that they are subbuildings?

来源字段：https://doi.org/10.4171/owr/2011/30

07895 | AMR-055-0008

### Rigidity of Smooth Isometric Families of Compact Surfaces

O / open | AMR 问题清单汇编

Let $M$ be a compact surface, $I=(-1,1)$, and $f:M\times I\to\mathbb{R}^3$ a differentiable map such that each $f_t$ is an immersion and its induced metric is independent of $t$. Must there be a family of rigid motions $g(t)$ with $f_t=g(t)f_0$?

背景链接：https://people.math.harvard.edu/~hirolee/pdfs/2014-fall-230a-icm1970-chern-differential-geometry.pdf

07896 | AIM-GEOMETRY-0336

### Robust compact certificates and a countable hierarchy for higher-codimension amoebas

P / partially_solved | AIM 研讨会问题汇编

Item A.16, “Nullstellensatz for amoebas,” was contributed by Kevin Purbhoo to the AIM list *Amoebas and tropical geometry*.

\[ f(z_1,\ldots,z_r)=\sum_{i=1}^{\ell}m_i(z) \]

be a Laurent polynomial, where the \(m_i\) are the nonzero monomial terms of \(f\), not “of \(x\).” Equal exponents are first collected. For \(a\in\mathbb R^r\), define

\[ f\{a\}= \bigl[\,|m_1(z)|,\ldots,|m_\ell(z)|\,\bigr], \qquad \operatorname{Log}(z)=a. \]

This is independent of the choice of \(z\) in the logarithmic torus fiber. A list satisfies the polygon condition when no entry is strictly greater than the sum of the other entries. Equivalently, it is not lopsided.

**Theorem 1 (Purbhoo’s amoeba Nullstellensatz).** If \(I\) is an ideal and \(\mathcal A(I)\) is the amoeba of its zero set in \((\mathbb C^\times)^r\), then

\[ a\in\mathcal A(I) \quad\Longleftrightarrow\quad f\{a\}\text{ satisfies the polygon condition for every }f\in I. \]

\[ P(f)=\{a\in\mathbb R^r:f\{a\}\text{ satisfies the polygon condition}\}. \]

For a hypersurface polynomial \(f\) and a positive integer \(m\), define its cyclic resultant

\[ \widetilde f_m(z)= \prod_{\substack{u_1^m=1\\ \vdots\\u_r^m=1}} f(u_1z_1,\ldots,u_rz_r). \]

**Theorem 2 (Purbhoo).** The outer approximations \(P(\widetilde f_m)\) converge uniformly, in the Euclidean metric, to \(\mathcal A(f)\).

The questions ask for: (A) an explicit analogous family in higher codimension; (B) a context more general than the Archimedean and non-Archimedean cases; (C) typical, rather than worst-case, constants in the \(O(\log m/m)\) rate; and (D) applications to other open problems.

The condition rendered as \(u_i^{m_i}=1\) is \(u_i^m=1\). The isolated “10” after the cyclic product is the PDF page number. The AIM HTML page retains some of the same image-conversion typos, so Purbhoo’s arXiv source was also checked before using the product.

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

07897 | OWR-15988-007

### Robustness and Tree Topology in Tropical Principal Components

P / partially_solved | Oberwolfach 报告问题汇编

For tropical PCA and tropical line segments in tree space, quantify sensitivity to outliers and perturbations, describe the changes of tree topology along a tropical line segment, and determine whether tropical geometry provides useful connections to deep learning.

来源字段：https://doi.org/10.4171/owr/2018/3

07898 | AIM-GEOMETRY-0203

### Role-compatible symmetry reduction for persistent formations

P / partially_solved | AIM 研讨会问题汇编

**16. (St. John/Schulze)** Persistence theory: Group of connected agents, every agent has out-degree 2 (except for 2 agents, the leader and the co-leader). **Question:** Can symmetry of the configuration be exploited to reduce the computation? What about a body-CAD version?

来源字段：https://aimath.org/pastworkshops/linkagesproblems.pdf

07899 | OWR-1394-001

### Root Bounds for Steiner Polynomials of Convex Bodies

S / solved | Oberwolfach 报告问题汇编

Let $K$ be a convex body in $\mathbb{R}^n$, with inradius $r$ and circumradius $R$. If $a_1\leq\cdots\leq a_n$ are the real parts of the roots of the Steiner polynomial of $K$, then one conjectures that $a_1\leq-R\leq-r\leq a_n\leq0$.

来源字段：https://doi.org/10.4171/owr/2006/56

07900 | AIM-CONVEX_GEOMETRY-0037

### Rotation-equivariant Minkowski-additive operators

P / partially_solved | AIM 研讨会问题汇编

Classify all continuous \(T:\mathcal K^n\to\mathcal K^n\) with the following properties: \[ h_{T(K+L)}=h_{TK}+h_{TL},\qquad T(\theta K)=\theta TK\quad\text{for every rotation }\theta. \]

来源字段：https://aimath.org/WWN/fourierconvex/fourierconvex.pdf

07901 | AIM-GEOMETRY-0016

### Ruling and weave stratifications for Legendrian (-1)-closures

P / partially_solved | AIM 研讨会问题汇编

Given a positive braid, the front projection of a Legendrian link which is its rainbow closure has an augmentation variety with a ruling stratification into \((\mathbb C^k)\times(\mathbb C^*)^l\) pieces. This stratification is the same as the one coming from the cluster variety.

Let \(\beta\in\operatorname{Br}_n^+\) have Demazure product \(w_0\), and let \(\Lambda(\beta\Delta)\) be the Legendrian \((-1)\)-closure appearing in the braid-variety/augmentation-variety correspondence, where \(\Delta\) is the positive half twist. Under the isomorphism between \(X(\beta)\) and \(\operatorname{Aug}(\Lambda(\beta\Delta))\), does the normal-ruling decomposition of the augmentation variety coincide piece by piece with a weave/cluster decomposition of the braid variety?

来源字段：http://aimpl.org/clusterbraid/2/

07902 | AIM-CONVEX_GEOMETRY-0020

### Sample-only volume estimation and a sharp box threshold

P / partially_solved | AIM 研讨会问题汇编

**Problem 1 (Santosh Vempala).** Let `K subset R^n` be a convex body. How many uniformly randomly chosen points from `K` are needed to estimate the volume of `K` within a factor of 2?

来源字段：https://aimath.org/WWN/convexgeometry/convexgeometry.pdf

07903 | AMR-066-0010

### Scalar Curvature Question [?10]: Extend the concept ofSc > 0 to singular Fano Varieties

P / partially_solved | AMR 问题清单汇编

Problem Extend the concept ofSc > 0 to singular Fano Varieties. For example, work out a definition ofSc(X) along the lines suggested in Question 1 of the previous section, such that the non-strict positivitySc(X) ≥0 of generalised scalar curvature would be stable under deformations of smooth Fano varieties, e.g. of hypersurfacesXreg ⊂CPm+1 of low degrees, to singular ones. Noticethatthevolumesofsmallballsatsingularpointsinalgebraicvarieties, especially where they are not normal (locally reducible) are significantly greater thanthevolumesofballsattheregularpointswhichindicateofnon-applicability of the definition of scalar curvature via volumes of balls to singular spaces. (Compare with [11]-Basilio+ Sewing Riemannian Manifolds 2017].) Further example of manifolds withSc > 0 are obtained with codimension 2 surgeryon n-dimensional manifolds, i.e. surgery based on submanifolds (e.g embedded spheres) of dimensions≤n-3, since codimension 2 surgery can be(rather naturally)performed in the Riemannian category of manifolds withSc > σ. (See [122]-Walsh Metrics of positive scalar 2008] and references therein.) For instance,connected sumsof n-manifolds withSc > 0 carry metrics with Sc > 0 for alln ≥3 and all orientable manifolds withSc > 0 of dimension ≥4 can made simply connectedby attaching2-handles, while keepingSc > 0. To see geometrically how this works, look at a smallε-neighbourhood Xε = Uε(P) of a compact smooth submanifoldP in a Riemannin manifoldW.6 Ifcodim(P) ≥3 then the boundary∂Uε(P) is fibered byε-spheres of dimensionsk ≥2 the scalar curvatures of which are approximatelyk(k-1) ε2 which blows up to+∞for ε→0 and Sc(Xε) ≍ε-2→+∞ as well. 5A complex manifoldX is Fanoif theanticanonical line bundleLac(X), i.e. the top exterior power of the tangent bundleT(X), isample: some powerLN ac is generated by holomorphic sections. 6Topologically, a surgery over a manifoldX results in a manifoldW with two boundary components where the the first one isX and the second one is the result of the surgery. The geometric construction we describe may be performed in thisW with a cylindrical Riemannian metric nearX ⊂W. 9 Now, more generally, letP ⊂W be a piecewise smooth polyhedral subset,...

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07904 | AMR-066-0011

### Scalar Curvature Question [?11]: What could be a, possibly non-geometric, extension of the concept ofSc ≥0, where one would be able perform sym

O / open | AMR 问题清单汇编

Question. What could be a, possibly non-geometric, extension of the concept ofSc ≥0, where one would be able perform symmetrization and reduce the case of general neighbourhoodsV to that ofO(n-1)-symmetric ones?

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07905 | AMR-066-0012

### Scalar Curvature Question [?12]: Q-Non-Essentiality of Manifolds with Sc > 0

P / partially_solved | AMR 问题清单汇编

Conjecture: Q-Non-Essentiality of Manifolds with Sc > 0. No rational homology class14 in the classifying spaceBΓ of a discrete groupΓ can be realised by a continuous map from a closed oriented (spin or non-spin) Riemannian manifoldX with Sc(X) > 0 to BΓ. In particular, 13Notice that the set{0, 1, 2, 4}is multiplicatively closed mod 8. 14This is false forinteger homologywhere the simplest examples are lens spaces. 14

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07906 | AMR-066-0013

### Scalar Curvature Question [?13]: [∗] no closed aspherical15 manifold admits a metric withSc > 0

P / partially_solved | AMR 问题清单汇编

[∗] no closed aspherical15 manifold admits a metric withSc > 0.

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07907 | AMR-066-0014

### Scalar Curvature Question [?14]: How common are Ricci flat metrics on compact simply connected manifolds X which admit metrics with positive sca

O / open | AMR 问题清单汇编

Question. How common are Ricci flat metrics on compact simply connected manifolds X which admit metrics with positive scalar curvatures?

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07908 | AMR-066-0015

### Scalar Curvature Question [?15]: Connected sums of sufficiently many copies of compact manifolds that are not homotopy spheres carry no Ricci-f

O / open | AMR 问题清单汇编

Conjecture. Connected sums of sufficiently many copies of compact manifolds that are not homotopy spheres carry no Ricci-flat metrics.

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07909 | AMR-066-0016

### Scalar Curvature Question [?16]: Singularities are Unstable

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Conjecture. Singularities are Unstable. Brian White told me about 30 years ago that he believed that Volume minimising hypersurfaces in generic Riemannian manifoldsX are non-singular: singularities disappear under arbitrarily small smooth perturbations of metrics inX. This was confirmed in 1993 by Nathan Smale, [118] forn= 8, which extends the Schoen-Yau theorem ton= 8.

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07910 | AMR-066-0017

### Scalar Curvature Question [?17]: 6

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Conjecture 6. ISC: Singularities are Irrelevant. Schoen and Yau announced 35 years ago [110], [114] that their descent metod extends to singular minimal subvarieties In a series of papers over the last decade, Lohkamp suggested an approach to the solution of this conjecture(see [86]-Lohkamp The Higher Dimensional Positive Mass Theorem II 2016where one can find references to his earlier papers). Recently, Schoen and Yau published an alternative proof of a version of the irrelevance conjecture [115]. 27 7 Flatly Twisted Spinors over Tori and their Precursors in Algebraic Topology. The kernels ofD+ andD-on the flat even dimensional torus consist of parallel spinors, where both spaces haveequal dimensions(= 2n-1); hence the ordinary index of the Dirac operator vanishes on the torus. However, these parallel spinors make the followingK-theoretic indexof D non zero not only on tori, but also on what we callover-torical manifolds X, which admit maps to then-torus of non-zero degrees, or equivalently, admitsn homology classes withnon-zero intersection index. The index of the Dirac operatorD = D+⊕D-on a Riemannian manifold X of even dimensionn we speak about takes the values in theK-theory of the torus Tm which comes about as the space of theunitary charactersofπ1(X), i.e. of homomorphismsτ :π1(X)→T, which is them-torus form=rankQH1(X). These homomorphisms define flat unitary bundlesIτ, τ ∈Tm, overX which are used to "twist" the spinor bundleS = S+⊕S-overX by taking the tensor products S±⊗Iτ and accordingly twist the Dirac operator on the sections of these bundles, D± τ =D± ⊗Iτ :C∞(S±⊗Iτ)→C∞(S∓⊗Iτ) 27I have not studied the papers by Lohkamp and Schoen-Yau in depth. 21 The kernels ofD+ τ andD- τ, even though their dimensions, in general, depend onτ, may be regarded as vector bundles overTm the difference of which defines the K-theoretic index of Atiyah-Singer. ⋆If X admits homology classes h1,h 2,...,h n ∈H1(X) with non zero intersection index, h1 ⌢h2 ⌢...⌢hn ≠0 or, equivalently, if X admits a continuous map to then-torus with non-zero degree, then thisK-theoretic index is non-zero. This was shown in 1972 by Lustig [?

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07911 | AMR-066-0018

### Scalar Curvature Question [?18]: Let $X_{\mathrm{fl}}=\mathbb{R}^n/\Gamma$ be a complete flat manifold whose group $\Gamma$ acts by parallel tr

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Let $X_{\mathrm{fl}}=\mathbb{R}^n/\Gamma$ be a complete flat manifold whose group $\Gamma$ acts by parallel translations. If a complete Riemannian manifold $X$ satisfies $\operatorname{Sc}(X)\geq 0$ and is isometric to $X_{\mathrm{fl}}$ at infinity, must $X$ be flat?

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07912 | AMR-066-0019

### Scalar Curvature Question [?19]: Probably, one can fully determine assumptions onπ1(X) depending on Xfl needed for this conclusion for all flat

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Probably, one can fully determine assumptions onπ1(X) depending on Xfl needed for this conclusion for all flat manifoldsXfl.

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07913 | AMR-066-0001

### Scalar Curvature Question [?1]: ○What arepossible topologiesof manifolds whichadmit Riemannin metrics with scalar curvaturesSc > 0

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○What arepossible topologiesof manifolds whichadmit Riemannin metrics with scalar curvaturesSc > 0?

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07914 | AMR-066-0020

### Scalar Curvature Question [?20]: Also one can possibly relax theisometry at infinitycondition by some "asymptotic flatness" and negativity of a s

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Also one can possibly relax theisometry at infinitycondition by some "asymptotic flatness" and negativity of a suitable "energy at infinity".

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07915 | AMR-066-0022

### Scalar Curvature Question [?21]: Evaluate σ○(X0) and σ□(X0) for "simple" Riemannian manifolds X0 = (X,g 0)

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Problem. Evaluate σ○(X0) and σ□(X0) for "simple" Riemannian manifolds X0 = (X,g 0). ###□Dirac operators, because they are invariant under isometries, often deliver optimal geometric inequalitiesfor manifolds withSc ≥σ. For instance, σ○(X0) and σ□(X0) are equal toSc(X0) for many (conjecturally for all) compact symmetric spaces(X0), see sections 17, but this seems hard, if possible at all, to prove with minimal hypersurfaces. Even in the most transparent case, where we know by Llarull's theorem, that if a smooth Riemannian metricg on X = Sn is greater than the spherical metricg (the differenceg-g is positive semidefinite) then there is a pointx∈X, where Sc(g)(x) ≤n(n-1) =Sc(Sn), there is no proof of this by means of minimal hypersurfaces forn≥3. (Maybe, such a proof is possible, it seems realistic forn = 3, but this is unlikely for complex and quaternionic projective spaces instead ofSn, where the Dirac operator works with no problem). On the other hand, there are incomplete manifoldsX0 where a sharp evaluation is possible by means of minimal hypersurfaces, but not by the Dirac operator methods, see section 21. Thomas Schick Example [1998].[105]. LetX =XSch be obtained from the n-torus Tn) by attaching the 2-handle to the circle representing the triplymultiple of one of the generators ofπ1(Tn). This is a spin Schoe-Yau-Schick manifold X for alln ≥4 with the fundamental groupsZn-1×Z/3Z, where all (known?

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07916 | AMR-066-0021

### Scalar Curvature Question [?21]: Remove the spin an the uniform positivity conditions, relax completeness and determine the sharp value ofconst

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Problem. Remove the spin an the uniform positivity conditions, relax completeness and determine the sharp value ofconstn (depending on theK-theory class ofL) in the inequality[estimate].

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07917 | AMR-066-0023

### Scalar Curvature Question [?22]: It seems not impossible, at least for compactX, that, in fact, K-area(X×R) =K-area(X×S1)

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It seems not impossible, at least for compactX, that, in fact, K-area(X×R) =K-area(X×S1).

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07918 | AMR-066-0024

### Scalar Curvature Question [?23]: Is the residual finiteness of the fundamental group essential

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Question. Is the residual finiteness of the fundamental group essential?

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07919 | AMR-066-0025

### Scalar Curvature Question [?24]: (i) It is unclear if the last step in the above argument is truly needed: conceivably, maps Φ :Sn-1→U(N) with

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(i) It is unclear if the last step in the above argument is truly needed: conceivably, maps Φ :Sn-1→U(N) with Lip(Φ) < 1 2 are contractible to constant onescontinuously inΦ for alln.

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07920 | AMR-066-0026

### Scalar Curvature Question [?25]: (iii) TheSn- andS2-product inequalities seems to hold fornon-trivial sphere fibrations, but I have not checked

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(iii) TheSn- andS2-product inequalities seems to hold fornon-trivial sphere fibrations, but I have not checked this carefully.

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07921 | AMR-066-0027

### Scalar Curvature Question [?27]: On the other hand, if theδ-neighbourhoods Uδ(S) ⊂X of allT(X)- non-spin surfacesS in a Riemannian manifoldX ar

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On the other hand, if theδ-neighbourhoods Uδ(S) ⊂X of allT(X)- non-spin surfacesS in a Riemannian manifoldX are "large" then the spin area of X must be also large.

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07922 | AMR-066-0029

### Scalar Curvature Question [?28]: Besides, similar results are expected for allsingular Alexandrov spaces with lower curvature boundsbut the Alm

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Besides, similar results are expected for allsingular Alexandrov spaces with lower curvature boundsbut the Almgren’s regularity theory has not been developed even for Alexandrov spaces withconical singularities, where the only apparent, yet instructive, case is that ofisolated singularities. ●≤κ0. If the injectivity radius ofX at a pointx∈Ym min ⊂X is ≥R and if the sectional curvatures ofX in theR-ball Bx(R) ⊂X are bounded from aboveby κ0 then them-volumevol(Ym min⋂Bx(R)) is bounded from below by the volume of theR-ballBm(R,κ 0) in the standardm-space with constant curvatureκ0 by the monotonicity formulafor minimal subvarieties. About the Balls.Neither●κ≥1 nor●≤κ0 directly apply to the ballsBn(R,κ 0) with constant curvatures; yet, by comparing their waists to those of spheres by means of suitableO(n)-equivariant mapsBn(R,κ 0)→Sn, one arrives at the expected values ( see section 3 in [?

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07923 | AMR-066-0028

### Scalar Curvature Question [?28]: For instance, letX be homomorphic to CP 2 and letvol(Uδ(S)) ≥δ2 for allT(X)-non-spin surfacesS ⊂X and 0<δ ≤1

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For instance, letX be homomorphic to CP 2 and letvol(Uδ(S)) ≥δ2 for allT(X)-non-spin surfacesS ⊂X and 0<δ ≤1. Is then spin-area(X) ≥1/1 000 000?

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07924 | AMR-066-0002

### Scalar Curvature Question [?2]: ○What are topologies ofspaces of metricsg with Sc(g)>0

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○What are topologies ofspaces of metricsg with Sc(g)>0?

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07925 | AMR-066-0030

### Scalar Curvature Question [?30]: On the other hand, there probably exist compact simply connected n-dimensional manifolds for alln ≥4 with arbi

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On the other hand, there probably exist compact simply connected n-dimensional manifolds for alln ≥4 with arbitrarily prescribed (finite) values of theK-area and the slice-area.

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07926 | AMR-066-0031

### Scalar Curvature Question [?31]: the sharp values ofwidthn-m for these solids remains problematic for m≥2, (unless I missed some paper)

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the sharp values ofwidthn-m for these solids remains problematic for m≥2, (unless I missed some paper).

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07927 | AMR-066-0032

### Scalar Curvature Question [?32]: Waist-Width Inequality

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Conjecture: Waist-Width Inequality. All complete Riemannian n-manifolds X satisfy widthn-1(X) ≤constn·waistn-k+1(X). Contractibility Radius. This "radius", denoted contr(X,r), r > 0, of a metric spaceX is the infimum of the numbersR, such that everyr-ballBx(r) ⊂ X is contractible within the concentric ballBx(R) ⊃Bx(r) of radiusR. It is (almost) obvious that [41]: A. Complete Riemannian manifolds with cocompact isometry groups, e.g universal coverings of compact aspherical manifolds, havecontr(X,r) = ∞for all r > 0. B. If a completen-dimensional manifoldX satisfies contr(X,ri) ≤ri+1 for i= 1, 2,...,n and r1 ≤r2 ≤...≤rn+1 <∞, then fil.rad[X]≥r1/(n+ 1)!. C. It follows that the universal coverings ˜X of compact aspherical manifoldsX have fil.rad( ˜X) =∞. Consequently, these ˜X satisfy waistm( ˜X) =∞,m = 1, 2,...,n =dim(X), as well. 16 Standard Geometric and Topological Conjectures on Complete Manifolds with Positive Scalar Curvatures.

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07928 | AMR-066-0033

### Scalar Curvature Question [?34]: Bounds on Width and on the Macroscopic Dimension

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Conjecture. Bounds on Width and on the Macroscopic Dimension. Complete n-dimensional Riemannian manifoldsX with the scalar curvaturesSc(X) ≥σ > 0 satisfy macr.dim(X) ≤n-2. Moreover, m2, width n-2(X) ≤constnσ-1 2. where, in fact proving even the weaker inequality m1 widthn-1(X) ≤constnσ-1 2. would make one happy.

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07929 | AMR-066-0034

### Scalar Curvature Question [?35]: Bound on the Filling Radius forSc ≥σ > 0

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Conjecture. Bound on the Filling Radius forSc ≥σ > 0., fil.rad[X]≤constn·( inf x∈X Sc(X)(x))-2. This, in view ofA,B,C from the previous section, yields the following. [?

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07930 | AMR-066-0035

### Scalar Curvature Question [?38]: Asphericity⇒K-Area =∞

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Conjecture. Asphericity⇒K-Area =∞. The universal coverings ˜X of compact aspherical manifoldsX satisfy K-area( ˜X) =∞. Notice that this inequality, even in a stabilised form, implies the strong Novikov conjecturefor π1(X), which is stronger than the non-asphericity for Sc ≥0 [101]. This makes it too good to be true; yet, no candidate for a counterexample is anywhere in sight. (e) Recall that a more comprehensive form of the asphericity conjecture [?

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07931 | AMR-066-0036

### Scalar Curvature Question [?39]: Let $B=B\Gamma$ be the classifying space of a discrete countable group, and let $f:X\to B$ be a continuous map

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Let $B=B\Gamma$ be the classifying space of a discrete countable group, and let $f:X\to B$ be a continuous map from a Riemannian manifold. Does there exist a compact subset $B_0\subset B$ containing $f(X)$ such that the Fredholm coareas of certain nonzero multiples of pullback bundles become arbitrarily small? More precisely, given $\varepsilon>0$, do there exist an integer $N\ne0$ and a Fredholm bundle $(L,\nabla)$ over $X$ with $\|\operatorname{curv}(\nabla)\|\leq\varepsilon$, such that $L$ is K-theoretically equivalent to the $N$th Whitney power of a pullback bundle $f^*(L_0)$?

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07932 | AMR-066-0003

### Scalar Curvature Question [?3]: ○What are geometries ofindividual manifoldswith Sc > σ

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○What are geometries ofindividual manifoldswith Sc > σ?

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07933 | AMR-066-0037

### Scalar Curvature Question [?40]: Area Extremality and Rigidity of Symmetric and Einstein Spaces

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Conjecture Area Extremality and Rigidity of Symmetric and Einstein Spaces. All Riemannin manifolds with positive and parallel Ricci tensor, in particular all Symmetric and all Einstein SpacesX are area extremal and those of them which contain no local flatfactors are area rigid. For Einstein spaces, this agrees with local extremality lemma in [39], while an essential class of examples where the available proofs do not work are compact Lie groupsX with biinvarinat metricsg, where the tangent bundles are trivial and can not force non-vanishing of indices of natural Dirac operators over theseX.

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07934 | AMR-066-0039

### Scalar Curvature Question [?41]: Are there compact manifoldsX which support metricsg withSc(g) > 0 but admit no area extremal or length extrema

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Are there compact manifoldsX which support metricsg withSc(g) > 0 but admit no area extremal or length extremal metricsg?

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07935 | AMR-066-0038

### Scalar Curvature Question [?41]: For instance, ifX =SO(n) withn≥5, then no known method can rule out metricsg ≥g on X with Sc(g) > Sc(g)

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For instance, ifX =SO(n) withn≥5, then no known method can rule out metricsg ≥g on X with Sc(g) > Sc(g).

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07936 | AMR-066-0040

### Scalar Curvature Question [?42]: Can one "effectively" evaluate the minimal constantλ=λ(X,g)„ such that a given Riemannin manifoldX = (X,g), e

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Can one "effectively" evaluate the minimal constantλ=λ(X,g)„ such that a given Riemannin manifoldX = (X,g), e.g wheresect.curv(g) > 0, would support an area extremal (or a length extremal) metricg which would beλbiLipschitz equivalent tog, where such aλ were expressible in terms of the pinching constant in the case wheresect.curv(g) > 0?

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07937 | AMR-066-0041

### Scalar Curvature Question [?43]: Would it be more prudent to replace the conditionSc(g) > 0 byRicci> 0

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Would it be more prudent to replace the conditionSc(g) > 0 byRicci> 0?

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07938 | AMR-066-0042

### Scalar Curvature Question [?45]: Spin Problem

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Spin Problem. All of the above only applies to spin maps f :X→X, for which the required twisted Dirac operator defined, and, as on similar occasions we met earlier, the necessity of the spin condition, say for equidimensional maps of degrees≠0, remains unsettled. Extremality via Lipschitz Maps.If a manifoldX admits noC2smooth length decreasing map X →X of degree≠0 which also strictly decreases the scalar curvature, then it admits no such Lipschitz map either my a simple approximation argument.

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07939 | AMR-066-0043

### Scalar Curvature Question [?46]: But it is unclear if this remain true with "area" in place of "length"

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But it is unclear if this remain true with "area" in place of "length".

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07940 | AMR-066-0044

### Scalar Curvature Question [?47]: Stabilisation of Extremality

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Conjecture: Stabilisation of Extremality.Let X0 be a compact area extremal Riemannin manifold. Then A. X0× Rm is area gap extremal for allm. 53 B. X0× R is area extremal. This can be confirmed for all known examples ofX0 the area extremality of which was established by a Dirac operator arguments sketched in the previous section. Let us do it in two cases. Start with A for X0 = Sn, let X be a complete orientable spin Riemannian manifold, letf :X→Sn× Rm be a smooth proper map. All we shall need ofRm for our purpose is thatSc(Rm) = 0 and K-Area(Rm) =∞. Assumeforsimplicity’ssakethat misandlet Lε0 bea Q-homologically essential unitary vector bundle withε0-flat connection which is flat at infinity. Assume for the same reason thatn is even and let S+ be the "positive" spinor bundle onSn. Let L! ε0 be the f-pullback of the tensor product S+ ⊗Lε0 and observe - this needs looking at the Llarull’s computation forRL!ε0 - that if the scalar curvature ofX at infinity is≥(n(n-1)+ε > 0 and if ε0 > 0 is much smaller thanε, then the twisted Dirac operatorD⊗Lε! 0 onX is strictly positive at infinity and ifdeg(f) ≠0 it has non-zero index relative the Dirac operator twisted with the pullback ofS+. Thus,X carries non-trivial twisted harmonic spinors. On the other hand, if the scalar curvature ofX at infinity is ≥(n(n-1)+ε> 0 on all ofX and εo <ε, then again, by looking at looking at LLarull’sRL!ε0 and sees thatX can’t carry such spinors and the proof follows. "Subcomplete" Extremality. LetXo be an open, not necessarily complete Riemannin manifold.

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07941 | AMR-066-0045

### Scalar Curvature Question [?48]: When does such anX0 is area extremal in the category of complete manifolds

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Question. When does such anX0 is area extremal in the category of complete manifolds?

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07942 | AMR-066-0046

### Scalar Curvature Question [?49]: The sphereSn minus Σo is area extremal in the "subcomplete" sense for all closed subsetsΣo ⊂Sn of topological

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Conjecture. The sphereSn minus Σo is area extremal in the "subcomplete" sense for all closed subsetsΣo ⊂Sn of topological dimensions k ≤1. 20 Lengths, Widths and Areas of Non-Complete Manifolds withSc > 0. To develop an adequate picture ofcomplete manifolds with scalar curvatures ≥σ one needs to understand geometric constraints imposed by the inequalitySc ≥σ on (bounded) domains in these manifolds. Thus, we look at possible sizes ofincomplete manifoldsX, e.g. (compact) manifolds with boundaries, where we do not, a priori, assume they are contained in complete manifolds with lower bounds on Sc. (We shall say something about theshapes rather than mere sizes of theseX in section 22) Asimpleexampleshowingwhatcanandwhatcannotbeexpected in this regard is the universal covering of the 2-sphere minus 2 opposite points times Rn-2, denoted ˜Σn π = the universal cover of S2\{··}× Rn-2 55 which satisfies: ●Sc( ˜Σn π) =Sc(S2) = 2, ●waist1[˜Σn π]=π and width1( ˜Σn π) ≤π. ●waistm[˜Σn π]=widthn-m( ˜Σn π) =∞for m= 2,...,n and K-area[˜Σn π]=∞as well. ●The r-interior ˜(Σn π)-r ⊂Σn π, that is the set of pointsx which are r-far from infinity, i.e. such that the closedr′-balls Bx(r′) ⊂Σn π for r′< r are compact, hassmall all waistm and K-area, actually, zero in the present example. r-Interior and Completeness. Metric completeness of anX in this terms is equivalent to non-emptiness of ther-interior ofX for r =+∞.

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07943 | AMR-066-0004

### Scalar Curvature Question [?4]: ○What are effect of lower boundsSc ≥σ on the topology and geometry of maps between manifolds

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○What are effect of lower boundsSc ≥σ on the topology and geometry of maps between manifolds?

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07944 | AMR-066-0047

### Scalar Curvature Question [?50]: All of the above is satisfied, modulo constants, for alln-manifolds, possibly incomplete and/or with boundaries

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Conjecture. All of the above is satisfied, modulo constants, for alln-manifolds, possibly incomplete and/or with boundaries, withSc(X) ≥σ > 0. Namely mbnd 1 widthn-1(X) ≤constnσ-1 2, mbnd 2 widthn-2(X-r) ≤constnσ-1 2 for r ≥constnσ-1 2, -bnd 2 waist2(X-r) ≤constnσ-1 for r ≥constnσ-1 2, [boundary estimate] K-area(X-r) ≤constnσ-1 for r ≥constnσ-1 2, where, of course, all theseconstn, especially the optimal ones, may be different. Among the first three inequalities, which generalise the corresponding conjecturesin section 16, a definite result is available only for mbnd 2 for 3-manifolds, which, similarly tom2, is proven with a use of minimal surfaces [54]. On the other hand, the inequality [boundary estimate], which, in the case whereX is complete spin, easily follows from the index theorem for the twisted Dirac operator, remains problematic fornon-complete, let them be spin,n-manifolds starting fromn = 3. (To make sense of this forn = 3 one should replaceK-area(X3 -r), which is zero by definition for oddn, byK-area(X3 -r× R).) It is tempting to try to reduce these conjectures, especially [boundary estimate], to the case of complete manifolds by extending (the metric on)X (onX× RN?

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07945 | AMR-066-0048

### Scalar Curvature Question [?51]: Extension Problem

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Extension Problem.LetX be a Riemanniann-manifold withSc(X) ≥σ > 0 and letσ-≤σ,r andr+ ≥r be positive numbers. Whendoesthereexistan n-dimensionalmanifold X+ withSc(X+) ≥ σ-, such that ther-interior X-r ⊂X isometrically embeds into the r+-interior (X+)-r+ ⊂X+?

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07946 | AMR-066-0049

### Scalar Curvature Question [?52]: Completion by Extension

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Conjecture. Completion by Extension.If σ > σ-and r ≥constn(σ -σ-)-1 2 for some (large) constant constn, then the extension problem is solvable withr+ =∞: there exists a completeX+ withSc(X+) ≥σ-which isometrically containsX-r Let us introduce geometrically more transparent geometric invariants which may serve as lower bound to theK-area.

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07947 | AMR-066-0050

### Scalar Curvature Question [?53]: Sharp Spherical Length Comparison Inequality

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Conjecture. Sharp Spherical Length Comparison Inequality. Spheres with finitely many punctures are length extremal. In fact - this is, probably equivalent- All Riemanninn-manifolds X, possibly non-complete and with boundaries, which haveSc(X) ≥Sc(Sn) =n(n-1) satisfy co-s.leng(X) ≤2π In plain words, ifSc(X) ≥n(n-1), then there is nono strictly distance decreasingproper maps fromX to Sn with non-zero degrees.54

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07948 | AMR-066-0051

### Scalar Curvature Question [?54]: ExtremalityofConcaveSphericalBalls

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Conjecture: ExtremalityofConcaveSphericalBalls. The balls B(R) ⊂Sn of radiiR≥π 2 are length extremal: no Riemannian metricg on such a ball which is greater than the spherical one (of constant curvature 1) can haveSc(g) > n(n-1) = Sc(Sn). [?

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07949 | AMR-066-0052

### Scalar Curvature Question [?55]: 18

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Conjecture 18. Interior Hemi-Spherical Area Inequality. The r-interiors of all compact Riemanninn-manifolds X with boundaries and withSc(X) ≥Sc(Sn) =n(n-1) satisfy for all r ≥π/2, co-s+ar(X-r) ≤2π: no strictly area decreasing proper mapX-r→Sn + of non-zero degree for r > π/2 exists. It is even unclear what happens in this regard to domains in closed spin manifoldsX. For instance:

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07950 | AMR-066-0053

### Scalar Curvature Question [?56]: What are possible values of the co-s+areas of the complements ofr-balls in compact spin manifoldsX withSc(X) ≥

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What are possible values of the co-s+areas of the complements ofr-balls in compact spin manifoldsX withSc(X) ≥n(n-1)?

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07951 | AMR-066-0054

### Scalar Curvature Question [?58]: there is no apparent non-trivial bound on the width ofX = Σn-1× [-1, 1]even we assume that thesectional curvat

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there is no apparent non-trivial bound on the width ofX = Σn-1× [-1, 1]even we assume that thesectional curvatureof X is = 1.

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07952 | AMR-066-0055

### Scalar Curvature Question [?59]: what is the (asymptotically forn→∞and/or fork→∞) sharp inequality for immersions of theseΣn-1 to spheres

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what is the (asymptotically forn→∞and/or fork→∞) sharp inequality for immersions of theseΣn-1 to spheres

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07953 | AMR-066-0005

### Scalar Curvature Question [?5]: An optimist would expect similar inequalities distg(∂-,∂+) <δ =δ(Y ) <∞ (ideally withδ = 2π dim(Y )+1) for met

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An optimist would expect similar inequalities distg(∂-,∂+) <δ =δ(Y ) <∞ (ideally withδ = 2π dim(Y )+1) for metricsg on Y ×[-1,+1]with Sc(g) ≥n(n-1) for all closed manifoldsY of dimensions≠4, which themselvesadmit no metrics withSc > 0.

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07954 | AMR-066-0056

### Scalar Curvature Question [?60]: Also it is unclear if there are (non-trivial) inequalities of this kind for other exotic spheres

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Also it is unclear if there are (non-trivial) inequalities of this kind for other exotic spheres.

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07955 | AMR-066-0057

### Scalar Curvature Question [?61]: Is then every immersion fromXj to the unit ball in RN satisfies supcurv(Xj↪BN(1) ⊂RN)) ≥ √ k for allN ≥n1+

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Is then every immersion fromXj to the unit ball in RN satisfies supcurv(Xj↪BN(1) ⊂RN)) ≥ √ k for allN ≥n1+....+nj+ 1?

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07956 | AMR-066-0058

### Scalar Curvature Question [?62]: But it is also not impossible that all manifolds admit immersions into the unit ball in the Hilbert spaceR∞wit

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But it is also not impossible that all manifolds admit immersions into the unit ball in the Hilbert spaceR∞with principal curvatures bounded by a universal constant, say by 1 000 000.

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07957 | AMR-066-0059

### Scalar Curvature Question [?63]: Identify combinatorial typesPextr of convex polyhedra where all representativeP ∈P are extremal and also descr

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Problem. Identify combinatorial typesPextr of convex polyhedra where all representativeP ∈P are extremal and also describe extremal P of non-extremal typesPnonextr. Next, callP mean convexly extremalif there is noP′⊂Rn diffeomorphic toP and such that ●thefaces Q′ i ⊂P′correspondingtoall Qi ⊂P havemean.curv(Q′ i) ≥ 0, ●the dihedral angles ofP′, that are the angles between the tangent spacesTp′(Q′ i) andTp′(Q′ j) at the pointsp′on the(n-2)-faces Q′ ij =Q′ i∩Q′ j, satisfy ∠ij(P′) ≤∠ij(P), ●this angle inequality is strict at some point, i.e. there exits p′ 0 ∈Q′ ij in someQ′ ij, such that ∠(Tp′ 0(Q′ i),Tp′ 0(Q′ j)) <∠ij(P).

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07958 | AMR-066-0060

### Scalar Curvature Question [?64]: Are all extremal convex polyhedraP are mean convexly extremal

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Question. Are all extremal convex polyhedraP are mean convexly extremal?

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07959 | AMR-066-0061

### Scalar Curvature Question [?65]: Is the regular Euclidean $3$-simplex mean-convexly extremal

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Is the regular Euclidean $3$-simplex mean-convexly extremal? Equivalently, can a simplex mapped facewise to it without decreasing distances have nonnegative face mean curvatures and no larger dihedral angles, with at least one angle strictly smaller?

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07960 | AMR-066-0062

### Scalar Curvature Question [?66]: Probably, these equalities imply thatP is isometric to a Euclidean rectangular solidbut the approximation/smoo

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Probably, these equalities imply thatP is isometric to a Euclidean rectangular solidbut the approximation/smoothing is no good for proving this kind of rigidity.

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07961 | AMR-066-0063

### Scalar Curvature Question [?67]: This suggests a possibility of definingSc(X) ≥0 for some singular spaces, X, e

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This suggests a possibility of definingSc(X) ≥0 for some singular spaces, X, e.g. for manifolds with continuous (bounded measurable?

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07962 | AMR-066-0064

### Scalar Curvature Question [?68]: Let $\widetilde X$ be the universal cover of a Riemannian $n$-manifold $X$ homeomorphic to the $n$-torus

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Let $\widetilde X$ be the universal cover of a Riemannian $n$-manifold $X$ homeomorphic to the $n$-torus. Conjecture that $\widetilde X$ has non-positive scalar curvature at infinity: it can be exhausted by overcubic domains $P_i\subset \widetilde X$ with corners whose codimension-one faces have positive mean curvature and whose dihedral angles are at most $\pi/2$. Here overcubic means admitting a degree-one map to the $n$-cube that sends each $k$-face to a $k$-face.

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07963 | AMR-066-0065

## Scalar Curvature Question [?69]: Shrinking of Singularities

O / open | AMR 问题清单汇编

Conjecture. Shrinking of Singularities. Let X be a compact orientable Riemanninn-manifold, f0 :X →Tn be a continuous map of non-zero degree, hi, i=0,1,2,..., be 1-dimensional homology classes in Tn which generate H1(Tn) and let εi →0 be positive numbers. Then the above (regularised for prudence) spaces Xi satisfy diam(X′ i)→0 for i→∞. 64 Moreover, theminimalhypersurfaces Yi ⊂X′ i becomenon-singular for sufficiently largei and the manifoldsX′ i as well asXi admit λibi-Lipschitz homeomorphisms to flatn-toriYi withλi→1 fori→∞. This conjecture implies that overtorical manifoldsX, (i.e. admitting maps toTn with non-zero degrees) withSc(X) ≥0 are, in fact, flat, since the scalar curvature is semicontinuous under "Lipschitz limits". (The proof of this semicontinuity forC0-convergence given in [49] automatically extends to the Lipschitz convergence.) (The arguments used in in [86] and those in [115], for the proof of this "non-positivity"Sc(X) >/0 probably yield the proof of the above Conjecture as well.) 24 Scalar Curvature and Mean Curvature. One may think of positive scalar curvature asRiemannian internalisation of the concept ofmean convexity, where a Riemannian manifoldY with boundary, e.g. a smooth domainY in a larger Riemannian manifold, is calledmean convexif the boundary∂Y has positive mean curvature. This "internalisation": is motivated by the following. Doubling Lemma[52]. The naturalC0-Riemannian metricg0 on the double56 X =Y+∂Y of a manifold(Y,∂Y )withSc(Y ) ≥0 and withmean.curv(∂Y ) > 0 along the boundary can beC0-approximated by C2-metrics g with withSc(g) > 0, where, moreover, this approximation isC2 away from the "∂-edge"Z∂ ⊂X where the two copies of Y meet inX. This is achieved by (1) Smoothingg0 with Sc > 0 close toZ∂ by rescaling along the geodesics normal toZ∂ (a five line argument) + (2) redistribution of positivity ofSc on all ofX by aC2-small conformal deformation (another 5 lines). Also there is a similarity between the following "thin" spaces with mean.curv > 0 and withSc > 0. Thinmean> 0. Givenaclosedsubset Y inaRiemannian n-manifold X, definevol∂(Y ) as the infimum of the(n-1)-volumes of the boundaries of arbitrarily small neighbourhoodsU ⊃Y of Y...

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07964 | AMR-066-0006

## Scalar Curvature Question [?6]: that the surface-tangent-bundle condition in Llarull's scalar-curvature rigidity theorem is redundant

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Conjecture that the surface-tangent-bundle condition in Llarull's scalar-curvature rigidity theorem is redundant. Specifically, let $X$ be a closed orientable Riemannian $n$-manifold with $\operatorname{Sc}(X)\geq n(n-1)$. If a $C^1$ map $f:X\to S^n$ has nonzero degree and does not increase the area of any surface, must $f$ be an isometry without assuming that $T(X)$ restricts trivially to every closed surface?

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07965 | AMR-066-0066

### Scalar Curvature Question [?70]: Let a domainY ⊂Rn havemean

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Conjecture Let a domainY ⊂Rn havemean.curv(∂Y ) ≥ n-k+ε for someε> 0 and k = 2,...,n -1. ThenY-1 admits a continuous map onto a(k-1)-dimensional polyhedral space space, say Δ:Y-1→Pk-1, such that the pullbacks of all points are uniformly bounded, diam(Δ-1(p)) ≤const=const(n,ε). Thus, the macroscopic dimension (see below) ofY-1 is ≤k-1. (The extremal case whereε= 0 is seen inY =Bn-k(1)× Rk ⊂Rn, where Rk admits no continuous map to any Pk-1 with uniformly bounded pullbacks of allp∈P by Lebesgue Lemma.) In particular, If Y ⊂Rn is a connected domain withmean.curv(∂Y ) > n-2+ε then the subsetY-1 ⊂Y is bounded. (Recall that the macroscopic dimension of a metric spaceM is the minimal dimension of polyhedral spacesP, for whichM admits a continuous map Δ :M →P, such thatdiamM(Δ-1(p)) ≤d for some constantd=d(M).) Now let us formulate the scalar curvature conjecture of the above conjecture. Start with a few definitions. Length Metrics in Spaces of Maps.The space Φ of mapsφ from X to S, whereS is a metric space comes with thesup-metric dist(φ1,φ 2) = sup x∈X dist(φ1(x),φ 2(x)). Let us endow subsets inΦ with the correspondinglength metrics: such a metric on aΨ⊂Φ is the supremum of the metrics which locally agree with the supmetric onΨ, which the same (except for irrelevant pathological cases) as the metric defined via the length of curves inΨ. What is interesting is that this length metric in a Ψ may be significantly greater than the sup-metric, which happens whenΨ is distorted inside Φ. Examples. (A) Continuous Maps Let Ψ = C(X→S) ⊂Φ be the space ofcontinuous maps X→S with the above length metric and let ˜S →S be a locally isometric covering map.65 Then the corresponding mapC(X→˜S)→C(X→S) is an isometry. It follows that the space of maps from then-ballBn to a compact spaceS with an infinitefundamental group hasinfinite diameter. 65Here and below we assume that our spaces are locally contractible, e.g. manifolds or cell complexes and thatS is alength metric space, e.g. a Riemannian manifold. 70 But ifS has finite fundamental group, then the spaceC(Bn→S) has finite diameter. For instance, diam(C(Bn→Sm(1)))=diam(Sm(1)) =π for m> n. Somewhat less obviously, diam(C(Bn→Sn(1)))≤3π,...

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07966 | AMR-066-0067

### Scalar Curvature Question [?71]: [a] Are the diameters diamc(Lip1(Bn(R)→S))) bounded for a large fixedc and R →∞if Hi(S, R) = 0 for i = 1, 2,

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[a] Are the diameters diamc(Lip1(Bn(R)→S))) bounded for a large fixedc and R →∞if Hi(S, R) = 0 for i = 1, 2,...,n.

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07967 | AMR-066-0068

### Scalar Curvature Question [?72]: [b] What is the asymptotics of the diameters diamc(Lip1(Bn H(R)→S))) for the hyperbolic ballsBn H(R) and R→∞

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[b] What is the asymptotics of the diameters diamc(Lip1(Bn H(R)→S))) for the hyperbolic ballsBn H(R) and R→∞?

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07968 | AMR-066-0069

### Scalar Curvature Question [?73]: [c] LetS be a Riemannian manifold homeomorphic to the connected sum of twenty copies ofS2× S2

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[c] LetS be a Riemannian manifold homeomorphic to the connected sum of twenty copies ofS2× S2. Are there 1-Lipschitz maps fR :B4(R)→S, R→∞, such thath(fR) ≥const·R4 for a cocycle h (e.g. a closed 4-form) which represents the fundamental cohomology class[S]∈H4(S; R), and someconst=const(S) > 0?

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07969 | AMR-066-0070

### Scalar Curvature Question [?74]: Parametric Hypersphericity

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Conjecture. Parametric Hypersphericity. Let X be a complete oriented Riemanniann-manifold and letΨ(X) ⊂Lipλ(X →Sn(1))) be the space of 1-Lipschitz locally constant at infinity maps66 of degree one fromX to the unit sphere. If Sc(X) ≥m(m-1) +ε, m ≥2, ε > 0, then the macroscopic dimension of Ψ(X) is ≤n-m-1.

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07970 | AMR-066-0071

### Scalar Curvature Question [?75]: If m = n-1 then, conjecturally, this is the only manifold with this property: the inequalities macr

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If m = n-1 then, conjecturally, this is the only manifold with this property: the inequalities macr.dim(Ψ(X)) ≥1 and Sc(X) ≥(n-1)(n-2) should imply thatX =Sn-1× R.

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07971 | AMR-066-0072

### Scalar Curvature Question [?76]: Stability of Periodic Slabs

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Conjecture. Stability of Periodic Slabs. The only Zn-3-invariantmeanconvexdomainsin Rn withdisconnectedboundaries are slabs between parallel hyperplanes. Besides "thin and narrow" mean convex domainsY which surround codimension two subsets in Riemannian manifoldsX, there are also "thick" ones. For instance, let X be a complete Riemanniann-manifold, such that ○∞ X is connected at infinity. ∞ voln-1(∂∞X) =∞, that is every proper continuous function f :X→R+ satisfies lim sup t→∞ voln-1(f-1(t))→∞. ¯●X is locally "(n-2)-thick": There existε> 0,α> 1 andc> 0, such that all(n-2)-cycles B ⊂X with diametersdiam(B) ≤ε and with voln-2(B) ≤εn-2 bound (n-1)-chains C, i.e. B = ∂C, such that voln-1(C) ≤c·voln-2(B)α. (An X which isuniformly bi-Lipschitz homeomorphic to Rn, or to any Riemannian homogeneous space, satisfies these three conditions.) Then, granted○∞, ∞, ¯●, every compact subsetY0 ⊂X is contained in asmooth compact mean convexdomain Y1 ⊂X. Sketch of the

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07972 | AMR-066-0073

### Scalar Curvature Question [?77]: Describe "Remnants of Collapse" of Hypersurfaces with Scalar Curvatures Blowing-up to+∞

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Problem. Describe "Remnants of Collapse" of Hypersurfaces with Scalar Curvatures Blowing-up to+∞. Namely, decide when a closed subsetY in aC2-smooth Riemannian manifold(W,g) appears as alimit of smooth domainsVi ⊂W, i ∈I, with Sc(∂Vi) →∞, where "limit" means that Y = ⋂iVi, where, ifY is non-compact, one may additionally insist thatVj ⊂Vi for j > i, or moreover, thatVi eventually become smaller than any given neighbourhoodU ⊃Y. (For this one needs uncountableI.) The following conjectures may give you a feeling of what this description might tell you. [?

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07973 | AMR-066-0074

### Scalar Curvature Question [?78]: Subsets with Low Hausdorff Dimensions are Remains of Scalar Curvature Blow-ups

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Conjecture. Subsets with Low Hausdorff Dimensions are Remains of Scalar Curvature Blow-ups.All closed subset Y ⊂W with dimHau(Y ) <n-1=dim(W)-2, are intersections of decreasing families of smooth domainsVi ⊂W with Sc(∂Vi)→+∞ This is obvious forn = 2. Also subsets Y ⊂W which are contained in smooth hypersurfacesZn ⊂W and which havezero Hausdorffmesn-1, are representable as such intersections by a simple argument. It is convenient at this point defineScg(∂↓Y ) as the supremum of the numbersσ such that every neighbourhoodU ⊃Y contains a smaller smoothV ⊃Y such that the scalar curvature of∂V for the metric induced from the metricg in the ambient manifoldW.

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07974 | AMR-066-0077

### Scalar Curvature Question [?79]: C0-closeness of the spaces ofC0-metrics withVolumicallyPositiveScalarCurvatures

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Conjecture. C0-closeness of the spaces ofC0-metrics withVolumicallyPositiveScalarCurvatures. IfaRiemannian C0-metricg onan n-dimensionalmanifold X canbe C0-approximated byC0-metrics g′with Scvoln(g′) ≥κ then Scvoln(g) ≥κ. This conjecture is motivated by the corresponding property of smooth metrics: C0-Closure Theorem.If a RiemannianC2-metric g on X can beC0-approximated byC2-metricsg′withSc(g′) ≥κ thenSc(g) ≥κ [49, 10]. Test. Check [?

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07975 | AMR-066-0075

### Scalar Curvature Question [?79]: InvarianceandNon-invarianceof Scn(Y ) = +∞

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Conjecture. InvarianceandNon-invarianceof Scn(Y ) = +∞. The inequalitySc[n] gn(Y ) =+∞is independent of the Riemannian metric g in W ⊃Y Moreover it is a bi-Lipschitz invariant. But it is not a topological invariant. There is no serious evidence here, but there are a few examples. For instance, givenk = 1, 2,...,n -2 and μ > 0, one can arrange nested neighbourhoodsVεi, i= 1, 2,..., of flatk-subtori inW = Tn+1 such that Sc(∂Vεi) →∞and such that the Lebesgue measure of their (solenoidal) intersectionY will be equalmes(W)-μ. (With a little effort, one can make a similar construction with all Vε homeomorphic to thek-ball.) On the other hand, there, probably, existcompact zero dimensional (Cantor) setsY ⊂Rn+1 with Sc[n] n(Y ) ≠+∞. (Compare 5.3 in [48].) 78

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07976 | AMR-066-0007

### Scalar Curvature Question [?7]: But deeper structures (if they exist at all) that lie at the roots of Dirac operators and of minimal hypersurf

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But deeper structures (if they exist at all) that lie at the roots of Dirac operators and of minimal hypersurfaces are yet to be revealed.

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07977 | AMR-066-0078

### Scalar Curvature Question [?80]: C2-Smoothing of Continuous Metrics with Volumically Positive Scalar Curvatures

P / partially_solved | AMR 问题清单汇编

Conjecture. C2-Smoothing of Continuous Metrics with Volumically Positive Scalar Curvatures.All continuous Riemannian metricsg on a smoothn-dimensional manifoldX which satisfy Scvoln(g) =def Scvoln(X,distg,volg) > κ 81 for a givenκ ∈(-∞,+∞) can be uniformly (i.e.C0) approximated byC2-metrics g′with Sc(g′) > κ. An obvious topological corollary of this reads:

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07978 | AMR-066-0076

### Scalar Curvature Question [?80]: Stabilisation under Cartesian Products

O / open | AMR 问题清单汇编

Conjecture. Stabilisation under Cartesian Products. [Sc[n] gn(Y ) =+∞]⇔[Sc[n+k] g⊕gkn(Y × Xk) =+∞], where Xk = (Xk,gk) is a compact Riemannian manifold of dimension k and where W × Xk ⊃Y × Xk is endowed with the metric g⊕gk. Notice that the implication [Sc[n] gn(Y ) =+∞]⇒[Sc[n+k] g⊕gkn(Y × Xk) =+∞] is obvious for compact manifoldsXk without boundaryas well as for comple tenon-compactmanifoldswiththescalarcurvaturesbounded from below where the leading example isXk = Rk. Possibly, this remains true for compact manifoldswith boundary. The reverse implication [Sc[n+k] g⊕gkn(Y × Xk) =+∞]⇒[Sc[n] gn(Y ) =+∞], probably, fails to be true forn = 2 and, possibly, forn = 3, 4 but it is plausible fornn≥5. Relaxing C2-Smoothness of ∂Vε to C1. The requirement for the existence metrics withSc > σ →∞on the boundaries of domainsVε ⊂W whichapproximate Y ⊂W intheabovedefinitionof Sc[n] gn(Y ) =+∞iscompoundedwiththenecessityoftheseboundaries to be isometrically embeddable to(W,g).67 This complication can be removed by applying to the NashKuipertheoremonisometric C1-embeddingsandallowing Vε tohave C1-smoothboundaries, yet withC2-smooth, rather than continuous, induced metrics, where it would be even more natural to admitcontinuous metrics on ∂Vε. However the extension of the inequality Sc ≥σ to continuous metrics is a non-trivial matter. 26 Manifolds with Small Balls. The definition of the scalar curvature in terms of the volumes of small balls in section 1 suggests the following. Say that ametric measure spaceX = (X,dist,vol ) where vol = volX is the measure onX, called herevolume, islocally volume-wise smaller than another such spaceX′= (X′,dist′,vol′) and write X ≺volX′ 67There is variety of obstructions to embeddablity of surfaces to 3-spaces (see a brief overview of basic examples in section 3.2.3 in [42]) but amazingly little is known fornn≥3. 79 if allε-balls inX are smaller than theε-balls inX′, vol(Bx(ε)) ≺vol′(Bx′(ε)) for all x ∈X,x′∈X′and some continuous positive functionε = ε(x,x′). Cartesian Additivity. Observe that X ≺volX′and Y ≺volY ′ imply that X× Y ≺volX′× Y ′, where the product spaces are endowed withvolX×Y =def volX⊕volY and with the Pythagorean product metrics,...

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07979 | AMR-066-0079

### Scalar Curvature Question [?81]: Topological Equivalence of Different Scalar Curvatures

O / open | AMR 问题清单汇编

Conjecture. Topological Equivalence of Different Scalar Curvatures. If a smoothn-manifold admits acontinuous metricg1 withScvoln(g1) > 0 then it also admits aC2-smooth metric g2 with Sc(g1) > 0. This says in other words that if a closed n-manifold X carries no smooth metric with Sc > 0, then every continuous Riemannian metric g on X admits balls B(X,g;R) of arbitrarily small radii R > 0 such that B(X,g;R) ≥ vol(Bn Eucl(R)). Inviewofthis, onehasthefollowing, probablyunrealistic, strengthening of theQ-non-essentiality conjecture [?

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07980 | AMR-066-0080

### Scalar Curvature Question [?82]: C0-ContinuousGuth-GerochLowerVolume Bound for Balls in the Coverings of Essential Manifolds

P / partially_solved | AMR 问题清单汇编

Conjecture. C0-ContinuousGuth-GerochLowerVolume Bound for Balls in the Coverings of Essential Manifolds. [57]. The universal coverings ˜X of Q-essential n-manifolds X with continuous Riemannian metrics containR-balls ˜B(R) of all radii R> 0 such that vol( ˜B(R)) ≥vol(Bn Eucl(R)). The main justification for5 is the following rough version of it. Corollary to Guth' Mesoscopic Filling Radius Theorem [55, 57]. The universal coverings ˜X of Q-essential Riemanniannmanifolds X containR-balls ˜B(R) of all radiiR> 0 such that vol( ˜B(R)) ≥εnvol(Bn Eucl(R)) for some universal constantεn > 0. (d) The sharp bound here, i.e. withεn = 1 is available forlarge balls by the following result. Burago-Ivanov Asymptotic Ball Volume Theorem[21].68 If the universal cover ˜X of an Q-essential manifold X admits a sequence of balls ˜B(Ri) ⊂ ˜X,Ri →∞, (where Ri depend on the metric inX), such that vol( ˜B(Ri)) ≤vol(Bn Eucl(Ri)), then X is flat. 68This paper is aboutX homeomorphic Tn; the general case reduces to that by the classification of groups of polynomial growth. 82 (e) The Burago-Ivanov argument automatically extends tononRiemannian manifolds and pseudomanifolds (X,dist) with arbitrarymetricsandsuitablydefined"volumes"onthem, namely Hilbert volumes Hilbert-volume (see [46]). (These "volumes" are not, a priori, additive.) Guth' theorem also remains valid for all(X,dist, Hilbert-volume) which motivates the following.

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07981 | AMR-066-0081

### Scalar Curvature Question [?83]: Non-Riemannian Guth-Geroch

O / open | AMR 问题清单汇编

Conjecture. Non-Riemannian Guth-Geroch. Let X be an n-dimensional Q-essential pseudomanifold (e.g. manifold) with an arbitrary metric. Then the universal covering˜X of X contains balls of all radiiR the Hilbert volumes of which are≥than these of the EuclideanR-balls, Hilbert-volume( ˜B(R)) ≥vol(Bn Eucl(R)).

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07982 | AMR-066-0082

### Scalar Curvature Question [?84]: Non-Riemannianε-Llarull

O / open | AMR 问题清单汇编

Conjecture Non-Riemannianε-Llarull. Let a compact n-dimen sionalpseudomanifoldhastheHilbertvolumesofallitsballs of radii≤ε0 smaller than the volumes of such balls inSn. Then all λ-Lipschitz maps from X to the sphere Sn are contractible, say, starting fromε0 = π 4 and λ≤1 2. The conjectures [79] - [84] are on the side of wishful thinking - something quite opposite may be true, e.g. the following.

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07983 | AMR-066-0083

### Scalar Curvature Question [?85]: C0-Density of C0-metrics with Volumically Positive Scalar Curvatures

P / partially_solved | AMR 问题清单汇编

Conjecture. C0-Density of C0-metrics with Volumically Positive Scalar Curvatures. Continuous Riemannian metrics withScvoln > 0 on anX are dense in the spaces of all Riemannian metrics onX for alln-dimensional manifoldsX forn≥3.69 (This may be compared withLohkampC0-Approximation Theorem [82]: C2-metrics with Sc ≤-1 are C0-dense in the spaces of all Riemannian metrics on n-manifolds X for n≥3.) On the other hand, it seems probable that most known properties of smooth manifolds withSc > 0 generalise to spaces with "benign singularities", e.g. toAlexandrov spacesX with sectional curvatures bounded from below by-1. (Ageodesicmetricspace 70 isAlexander Kapovitch Petrunin Alexandrov geometry 2017with sect.curv ≥-1 if, for every quadruple of points xi ∈X, i = 1, 2, 3, 4, there exists a quadruple of pointsx′ i in the hyperbolic planeH2 with sectional curvature -1, such that 69Conformal representation of metrics on surfaces makes this approximation unlikely for n=2. In fact, it may be safer to assumen≥4. 70"Geodesic" means that every two pointsx1 andx2 inX can be joined by a path of length d=dist(x1,x 2). 83 distH 2(x′ i,x′ j) =distX(xi,xj) for i,j = 1, 2, 3, while distH 2(x′ i,x′ 4) ≤ distX(xi,x 4), see [4] and references therein.) The simplest among expected results is the following.

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07984 | AMR-066-0084

### Scalar Curvature Question [?86]: Geroch for Alexandrov Spaces

O / open | AMR 问题清单汇编

Conjecture. Geroch for Alexandrov Spaces.If anndimensionalAlexandrovspace X withsect.curv ≥-1andScvoln(X) ≥ 0 admits a continuous mapΦ with non-zero degree(i.e. the homology homomorphism Φ∗does not vanish onHn(X)) to then-torus, then the universal covering ofX is isometric toRn. Moral in Conclusion. The definition ofScvol is not supposed to answer the question"What is scalar curvature", but rather to inspire a quest for a true definition. 27 Fredholm Coarea and StableK-Area. Let X be a Riemannin manifold or a more general (metric) space where one may speak of length of curves and areas of surfaces and define the FredholmK-area on its homology similarly to how it was done for K-area+ in sections 12,13 but now allowinginfinite dimensional bundles L overX. Namely, given a complex Hilbertian vector bundle L over X, introduce the concept of a unitary connection∇in it via parallel transport over certain curves inX and define the norm of the curvature of∇as in section 9: ||curv(∇)||(x) is the infimum of positive functionsC(x) such that the maximal rotation angles α ∈[-π,π ] of the parallel transports along the boundaries of "nice" (smooth in the Riemannin case) surfacesS in X satisfy |α|≤∫S C(s)ds 84 Represent elementsκ of theK-cohomologygroup ofX by pairs of complex Hilbertian vector bundleL= (L1,L 2) overX withFredholm homomorphisms71 and it is also useful for certain non-compact ones.Φ:L1→L2 and define the coarea (norm) ofκ as the infimum of numbersc, such thatκ admits a representation byΦ:L1→L2, whereL1 and L2 are endowed with connections with ||curv||≤c. Do the same for theK-cohomology with compact supports(say, on locally compact X), where the homomorphismsΦ are required to be unitaryisomorphisms at infinity, i.e. outside compact subsets inX and such thatΦ must be connection preserving at infinity Define the coarea "norm" onκ in the so definedKcomp(X) as the infimum of the numbersc, such thatκ admits a representation byΦ :L1→L2, where L1 and L2 are endowed with connections with||curv||≤c and such that Φ is connection preserving at infinity. DefineFredholmK-areaon theK-homology ofX, on the ordinary homology and on the homology with infinite supports by linear duality: for...

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07985 | AMR-066-0085

### Scalar Curvature Question [?87]: but there are no apparent examples (if any) where these inequalities are strict

P / partially_solved | AMR 问题清单汇编

but there are no apparent examples (if any) where these inequalities are strict. Everything we know aboutK-area+ easily extends to the the FredholmKarea where the main gain iscovariant functorialityof the FredholmK-area, such as follows. Fredholm Push-forwards of under possibly infinite covering maps.Let f : X→Y be a covering between oriented Riemannin manifolds. There is an obviouspush forward mapfrom Hilbertian bundlesL overX to Hilbertian bundles overY, sayL↦M =f∗(L), where the fiber of the bundle M over y ∈Y equals the Hilbertian sum of the fibers ofL over the pullback f-1(y) ⊂X, My = ⊗.big.disp x∈f-1(y) Lx. For instance, ifX =Y ×Σ→Y is the trivial covering with all fibers equal to a given countable setΣ and L is the trivial line bundle, thenM is the trivial bundle with the fiberl2(Σ). Now if Φ:L1→L2 is a Fredholm homomorphism which is an isomorphism at infinity, then the corresponding homomorphism between the pushed forward bundles, say Ψ:M1→M2 is also Fredholm as well as isomorphic at infinity. And since, clearly, the pushed bundles have the same curvatures, ||curv(M1,2)||= ||curv(L1,2),|| 71This is usually defined for compact spacesX where the operators are required to be bounded. We shall use below unbounded operators and our spaces may be sometimes noncompact. 85 the FredholmK-areas of the fundamental homology classes ofX andY, assuming these are manifolds or pseudomanifolds, satisfy:
FredholmK-area[X1]≥FredholmK-area[X2]. More generally letf :X→Y be a fibration where the fibersZy ⊂X have positive dimension and let them carry Riemannian structures continuous in y ∈Y. Let Θ(Zy) besomebundleover Zy associatedwiththetangentbundle T(Zy) and let the Θ-push-forward L↦ Θ M = f∗Θ(L) be defined by taking the space Hy of the square integrable sections ofL|Zy ⊗Θ(Zy), (whereL|Zy denotes the restriction ofL to Zy) for the fiberMy. Assume the fibersZy are closed even dimensional spin manifolds, let the restrictions L|Zy be endowed with unitary connections continuous iny ∈Y and let Dy :S+⊗My→S-⊗My be the Dirac operators onZy twisted withL|Zy. If L is a finite dimensional bundle then the operatorsDy are Fredholm72 and the resulting Fredholm bundle overY serve asf∗(L). (The spinor...

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07986 | AMR-066-0008

### Scalar Curvature Question [?8]: Find a useful local geometric definition of a scalar-curvature lower bound $\operatorname{Sc}\geq\sigma$ that

P / partially_solved | AMR 问题清单汇编

Find a useful local geometric definition of a scalar-curvature lower bound $\operatorname{Sc}\geq\sigma$ that supports global theorems and extends to singular spaces, for example through local spectral invariants or localized minimal-hypersurface methods rather than only small-ball volumes.

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07987 | AMR-066-0086

### Scalar Curvature Question [?90]: Hyperbolic Volume Inequality

P / partially_solved | AMR 问题清单汇编

Conjecture. Hyperbolic Volume Inequality.Then every continuous mapf0:X→X0 is homotopic to a mapf, such that voln(f(X)) ≤vol(X) where, moreover, this inequality can be made strict, unlessX has a constant negative curvature and the mapf0 is homotopic to a locally isometric map. Notice that ifX as well asX0 has constant sectional curvature -1, this conjecture, which generalisesthe Mostow's rigidity theorem, is proved by evaluating the simplicial volumeof X. Recall thatsimpl.vol(X) is a non-negative numerical invariant of compact orientable topological manifoldsX such that continuous maps between equidimensional manifolds,X→Y of degreed, satisfy simpl.vol(X) ≥d·simpl.vol(Y ), withtheequality simpl.vol(X) =d·simpl.vol(Y ) ford-sheetedcoverings X→Y. The simplicial volume of anX is known to be non-zero ifX admits a metric with negative sectional curvatures.

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07988 | AMR-066-0087

### Scalar Curvature Question [?91]: Prove that there is a dimension-dependent constant $c_n$ such that every compact Riemannian $n$-manifold $X$ w

P / partially_solved | AMR 问题清单汇编

Prove that there is a dimension-dependent constant $c_n$ such that every compact Riemannian $n$-manifold $X$ with $\operatorname{Sc}(X)\geq-\sigma^2$ satisfies $\lVert X\rVert\leq c_n\sigma^n\operatorname{vol}(X)$, where $\lVert X\rVert$ is the simplicial volume.

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07989 | AMR-066-0009

### Scalar Curvature Question [?9]: Identify the most general classes of geometric objects having properties analogous to those of $C^2$ Riemannia

O / open | AMR 问题清单汇编

Identify the most general classes of geometric objects having properties analogous to those of $C^2$ Riemannian manifolds with $\operatorname{Sc}\geq\sigma$.

背景链接：https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/101-problemsOct1-2017.pdf

07990 | OWR-7155451-004

### Scalar-Curvature Rigidity on Wrapped Sphere Complements

P / partially_solved | Oberwolfach 报告问题汇编

Let $\Sigma$ be a subset of the standard unit sphere $S^n$ with the wrapping property, and let $X=S^n\setminus\Sigma$ with its standard metric $g_0$. If a possibly incomplete metric $g$ on $X$ satisfies $g\geq g_0$ and $\operatorname{Sc}(g)\geq n(n-1)=\operatorname{Sc}(g_0)$, must $g=g_0$?

来源字段：https://doi.org/10.4171/owr/2021/30

07991 | OWR-14299521-002

### Scalar-Curvature Smoothing at Critical-Codimension Singular Sets

O / open | Oberwolfach 报告问题汇编

For $n\geq 3$, does there exist an $L^\infty$ Riemannian metric on $\mathbb{R}^n$, uniformly bilipschitz to the Euclidean metric and smooth away from a singular set of finite $(n-2)$-dimensional Minkowski content, for which nonnegative-scalar-curvature smoothing fails? Does such a counterexample exist when that Minkowski content is zero, or does smoothing still hold?

来源字段：https://doi.org/10.4171/owr/2025/30

07992 | AIM-GEOMETRY-0164

### Scalar-free backwards pseudolocality with an explicit noncollapse dictionary

P / partially_solved | AIM 研讨会问题汇编

R. Bamler and Q. Zhang proved a backwards pseudolocality theorem: Assume \(R<1\). For \(r\in(0,1)\), if \(|Rm|<r^{-2}\) on \(B(x,t,r)\), then \[ |Rm|<(\epsilon r)^{-2} \] on \[ P(x,t,\epsilon r,-(\epsilon r)^2) =B(x,t,\epsilon r)\times[t-(\epsilon r)^2,t). \] Can we remove the \(R<1\) assumption in the backwards pseudolocality theorem?

来源字段：http://aimpl.org/flowriemannian/1/

07993 | AIM-GEOMETRY-0162

### Scalar-Type-I dilation limits and an entropy-defect certificate

P / partially_solved | AIM 研讨会问题汇编

Let $(M^n,g_t)_{t\in[0,T)}$ be a Ricci flow with $T<\infty$. Assume $R< \frac{C}{T-t}$. Characterize $(T-t)^{-1} g_t$ as $t\to T$. This is related to the Hamilton-Tian conjecture. Note that the Kahler Ricci flow on Fano manifolds satisfies $R< \frac{C}{T-t}$.

来源字段：http://aimpl.org/flowriemannian/1/

07994 | AIM-GEOMETRY-0165

### Scalar-upper-free heat-center estimates and the worldline obstruction

P / partially_solved | AIM 研讨会问题汇编

Can we remove $R<1$ assumption in R. Bamler and Q. Zhang's heat kernel bound and distance bound? For more information, see their papers on arxiv: HEAT KERNEL AND CURVATURE BOUNDS IN RICCI FLOWS WITH BOUNDED SCALAR CURVATURE, and HEAT KERNEL AND CURVATURE BOUNDS IN RICCI FLOWS WITH BOUNDED SCALAR CURVATURE-PART II.

来源字段：http://aimpl.org/flowriemannian/1/

07995 | OWR-14212-002

### Scale-Invariant Distances between Curves in Shape Space

P / partially_solved | Oberwolfach 报告问题汇编

How should shapes such as cardiac configurations at different times or different hand postures be compared, and what scale-invariant distance measure should be used along the resulting curves in shape space?

来源字段：https://doi.org/10.4171/owr/2016/4

07996 | OWR-14298808-007

### Schneider Higher-Order Difference-Body Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $n\geq 3$ and $m\geq 2$. Prove that the minimizers of the normalized volume $\delta_m(K)=|D^m(K)|/|K|^m$ among convex bodies $K\subset\mathbb R^n$ are precisely the ellipsoids.

来源字段：https://doi.org/10.4171/owr/2024/58

07997 | OWR-14297736-029

### Second Neumann Eigenvalues of Convex Subdomains of a Square

O / open | Oberwolfach 报告问题汇编

Let $D$ be the unit square. Prove that every convex subdomain $\Omega\subset D$ satisfies $\mu_2(\Omega)\geq\mu_2(D)$.

来源字段：https://doi.org/10.4171/owr/2023/36

07998 | AIM-CONVEX_GEOMETRY-0011

### Sections, projections, polarity, and an ellipsoid obstruction

P / partially_solved | AIM 研讨会问题汇编

The PDF declares \(IK\) to be the intersection body, \(\Pi K\) the projection body, \(\mathcal P(K)\) the volume product, \(K|u^\perp\) the orthogonal projection, \(\operatorname{St}_u\) Steiner symmetrization with respect to \(u^\perp\), and \(B_p^n\) the unit ball of \(\ell_p^n\). With symbols restored, its twelve questions are:

1. If \(IK\subset IL\) and \(\Pi K\subset\Pi L\), must \(K\subset L\)? 2. Under the same inclusions, must \(\operatorname{vol}(K)\leq\operatorname{vol}(L)\)? The PDF says this is true for \(n\leq4\). 3. For every convex (symmetric) body \(K\), is there \(u\in S^{n-1}\) such that \[ \operatorname{vol}\bigl(\operatorname{St}_u(K)^*\bigr) \leq C\operatorname{vol}(K^*)? \] 4. Is the averaged version \[ \int_{S^{n-1}}\operatorname{vol} \bigl(\operatorname{St}_u(K)^*\bigr)\,du \leq C\operatorname{vol}(K^*)? \] 5. If \(K\) is the symmetric convex hull of a fixed number of independent Gaussian vectors, is \(\mathbb E\mathcal P(K)\geq\mathcal P(B_\infty^n)\)? 6. Given an even log-concave probability measure \(\mu\), estimate \(\mu(K)\mu(K^*)\) from above. The PDF prints, with a single factor on the right, that \(\mu(K)\mu(K^*)\leq\mu(B_2^n)\) is known if \(\mu\) is unconditional. 7. In Question 6, characterize equality and find a lower bound in the unconditional case. 8. Consider \(\mu^*\) in the sense of Artstein--Milman and ask the analogous questions for \(\mu(K)\mu^*(K)\). 9. For symmetric convex bodies \(K,L\subset\mathbb R^n\), is \[ N(K,L)\leq\bigl[N(L^*,cK^*)\bigr]^\delta \] for absolute constants \(c,\delta\)? 10. For independent uniform \(X\in K\) and \(Y\in K^*\), compare the first “100” moments of \(\langle X,Y\rangle\) with Gaussian moments up to a factor \(1+\varepsilon\). 11. Prove Mahler's conjecture for finite-dimensional normed spaces embedding in \(L_p\), \(p<1\), in particular for convex intersection bodies (associated in the PDF with \(p=-1\)). 12. If \(1<p<2\) and the \(n\)-dimensional space \(X\) embeds in \(L_p\), is \[ \mathcal P(B_X)\geq\mathcal P(B_p^n)? \]

Direct inspection shows no equal-volume hypothesis in Question 1 and no printed square on \(\mu(B_2^n)\) in Question 6. Since the latter display lacks a scale convention, it may be workshop shorthand; it is reproduced faithfully rather than silently amended.

Several questions remain underspecified even after faithful recovery. Questions 3--4 do not say whether \(C\) is dimension-free, how the polar is centered for nonsymmetric \(K\), or whether \(du\) is normalized. Questions 6--8 do not state a scale or normalization for \(\mu\) or \(\mu^*\). Question 10 does not specify covariance normalization, an asymptotic parameter, or what “same” means. These omissions are not silently repaired below.

来源字段：https://aimath.org/WWN/mahlerduality/mahlerduality.pdf

07999 | OWR-13347-005

### Self-Adjointness of the Isotropy Action on Nilradicals

O / open | Oberwolfach 报告问题汇编

For a group $U$ acting on $\operatorname{Lie}(N)$, is the restricted adjoint action $\operatorname{Ad}(U)|_{\operatorname{Lie}(N)}$ closed under transpose; equivalently, is the action self-adjoint?

来源字段：https://doi.org/10.4171/owr/2014/45

08000 | AIM-GEOMETRY-0400

### Semi-local versus tame-at-infinity generating families

P / partially_solved | AIM 研讨会问题汇编

**Question 1.2.** Which \(L\) admit generating functions? One wants to impose some restrictions on the generating function. Typically one wants the generating function to be quadratic at infinity, although it might be interesting to explore other conditions at infinity. The notes add that stabilizations (zig-zags) are “somehow prohibited” in dimension \(3\), and report a theorem of Giroux when no condition at infinity is imposed.

Its question is deliberately tentative:

The following paragraph says that a suitable generating function supports Morse-theoretic invariants. The record then starts a separate **Theme 1.3**, concerning Lisa Traynor's generating-function polynomials and linearized contact homology. I therefore use the comparison only to illuminate existence obstructions and do not treat it as a second problem to solve here.

The preceding source record fixes the convention

\[ J^1(M)=T^*M\times\mathbb R_z,\qquad \alpha=dz-p\,dq, \]

and describes the front generated by fiber-critical values of a function on a bundle over \(M\).

Two qualifications are essential to recover a precise modern question.

1. “Generating function” is now usually called a **generating family**. 2. The verified Giroux theorem concerns a **germ, or semi-local generating family, near its fiber-critical locus**. It is not by itself a theorem producing a global function with prescribed behavior at infinity.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

08001 | OWR-12697711-008

### Semialgebraicity of Finite Rank-One Convex Hulls

P / partially_solved | Oberwolfach 报告问题汇编

Prove that the rank-one convex hull of every finite set of matrices is semialgebraic.

来源字段：https://doi.org/10.4171/owr/2023/15

08002 | OWR-11136-030

### Semicircle Packings and Colorings of Simplicial 4-Polytopes

O / open | Oberwolfach 报告问题汇编

For a plane curve $\gamma$ and $r>0$, let $f_c(r,\gamma)$ be the maximum density of a congruent packing by copies of the Minkowski sum of $\gamma$ and the closed disk of radius $r$. If $\gamma_1$ is a semicircle, is $f_c(r,\gamma_1)=\Theta(\sqrt{r})$? Separately, does every simplicial 4-polytope admit a coloring with a bounded number of colors and no monochromatic 2-face?

来源字段：https://doi.org/10.4171/owr/2011/44

08003 | AIM-GEOMETRY-0049

### Separate criteria for metric and measure entropy attainment

P / partially_solved | AIM 研讨会问题汇编

MR1348316 is M. Handel and B. Kitchens, *Metrics and entropy for non-compact spaces*, Israel Journal of Mathematics 91 (1995), 253--271, with an appendix by D. J. Rudolph.

whether the two extrema \[ h_{\mathrm{Borel}}(f) =\sup_{\mu\in\mathcal M_f}h_\mu(f), \qquad h_{\mathrm{top}}(f) =\inf_{d\in\mathcal D_X}h_d(f) \] can be attained simultaneously for a geodesic flow on a noncompact metric space. Here \[ h_d(f)=\sup_{K\subset X\ \mathrm{compact}}h_d(f,K), \] \(\mathcal M_f\) is the set of invariant Borel probability measures, and \(\mathcal D_X\) is the set of distances inducing the topology of \(X\). The record says that the Borel and topological entropies coincide for locally compact spaces by the unresolved key \(\mathrm{MR1348316}\).

That key can be resolved precisely:

There are, however, three points requiring correction or interpretation.

1. Handel--Kitchens prove the general inequality \[ \sup_{\mu\in\mathcal M_f}h_\mu(f) \leq \inf_{d\in\mathcal D_X}h_d(f) \tag{1.1} \] and give an example in which it is strict. Thus the record's sentence attributing equality in full locally compact generality to MR1348316 is inaccurate as written. 2. Later work of Patrão and of Caldas--Patrão introduced a locally compact topological entropy for which \[ \sup_\mu h_\mu(f)=h(f)=\min_{d\in\mathcal D_X}h_d(f) \tag{1.2} \] on locally compact separable metrizable spaces. Caldas--Patrão allow an arbitrary continuous map. 3. For a geodesic flow \(g_t\), the displayed discrete-time notation is read as \(f=g_1\). The distance \(d\) is a compatible distance on the phase space \(T^1M\), not necessarily the Sasaki distance or a distance induced by varying the Riemannian metric on the base manifold.

Under the entropy convention in (1.2), metric attainment is now known very generally. Measure attainment is the substantive dynamical issue.

来源字段：http://aimpl.org/equibdynsysgeom/1/

08004 | OWR-11136-017

### Sequence Bounds for the Function r of m

O / open | Oberwolfach 报告问题汇编

For $m \geq 1$, let $r(m)$ denote the minimum number of parts into which a convex body in $\mathbb{R}^d$ (or planar convex set) must be partitioned in the $m$-step covering/Tverberg-type geometric partition problem of Discrete Geometry (OWR 2011/44). Does the OEIS sequence A059522 ($\lfloor n \alpha \rfloor$ with $\alpha = 2 - \sqrt{2}$) give a valid lower bound for $r(m)$ for all $m$?

来源字段：https://doi.org/10.4171/owr/2011/44

08005 | OWR-13106-006

### Sharp $\hat A$-Genus Bound for the $\kappa$-Invariant

O / open | Oberwolfach 报告问题汇编

For a compact spin manifold $M$ of dimension $4m$, the known estimate is $|\hat A(M)|\leq 2^{2m-1}\kappa(M)$. Can the factor $2^{2m-1}$ be replaced by $2^m$?

来源字段：https://doi.org/10.4171/owr/2014/36

08006 | AIM-GEOMETRY-0259

### Sharp conformal projective systole and a volume-preserving Crofton obstruction

P / partially_solved | AIM 研讨会问题汇编

**Question 8.8.1 (Question 4.11 of [Gromov2001]).** Does a Riemannian metric on a real projective space with the same volume as the canonical metric have a closed geodesic with length $\leq \pi$?

In dimension two, an affirmative answer is in [Gromov2001, Proposition 4.10].

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

08007 | OWR-1588-003

### Sharp Dimension Threshold for Ricci-Flow Curvature Conditions

O / open | Oberwolfach 报告问题汇编

Write a curvature operator as $R=R_I+R_{\operatorname{Ric}_0}+R_W$. For $c>0$, consider the cone defined by $\operatorname{scal}(R)\geq0$ and $\lVert R_W\rVert^2\leq c\lVert R_I\rVert^2$. Let $n_0$ be the least threshold such that, in every dimension $n\geq n_0$, the Böhm-Wilking classification determines exactly which values of $c$ make this cone invariant under the curvature ODE $R'=R^2+R^\#$. It is known that $n_0\geq12$; is $n_0=12$?

来源字段：https://doi.org/10.4171/owr/2007/32

08008 | OWR-761-004

### Sharp Dimensions for Totally Skew Embeddings

O / open | Oberwolfach 报告问题汇编

For a given manifold, is the current dimension lower bound for a Euclidean space admitting a totally skew embedding sharp? In particular, does there exist a totally skew embedding of the three-dimensional disk in $\mathbb{R}^7$?

来源字段：https://doi.org/10.4171/owr/2004/47

08009 | AIM-GEOMETRY-0145

### Sharp genus-rank classification and local obstruction for SL_n Higgs moduli

S / solved | AIM 研讨会问题汇编

Does \(\mathcal{M}_{Higgs}^{ss}(SL(n,\mathbb{C}))\) have a symplectic resolution?

来源字段：http://aimpl.org/spectralhiggs/1/

08010 | OWR-3389-013

### Sharp Geometric Constants for Simply Connected Planar Domains

O / open | Oberwolfach 报告问题汇编

For arbitrary simply connected planar domains, identify the optimal universal constants $C_1,C_2,C_3,C_4$ and the geometry of extremal domains in the four inequalities relating, respectively, the Hardy integral to the distance-to-boundary function, the first Dirichlet eigenvalue to the inradius, the expected Brownian exit time to the inradius, and the hyperbolic metric density to the inradius.

来源字段：https://doi.org/10.4171/owr/2009/06

08011 | AMR-058-0030

### Sharp Interior Curvature Bound for Minimizing Hypersurfaces

P / partially_solved | AMR 问题清单汇编

Find the best constant $C$ such that the principal curvatures of an area-minimizing hypersurface in low dimensions are bounded by $C/r$ at a point whose distance from the boundary is $r$.

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

08012 | AMR-058-0022

### Sharp Isoperimetric Constant for Bubble Clusters

P / partially_solved | AMR 问题清单汇编

Find the best constant $C$ in the inequality $A\leq C(\int_B H+L)^2$ for a bubble cluster of area $A$, mean curvature $H$, and boundary length $L$. Determine the analogues for other norms and higher dimensions.

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

08013 | OWR-10252928-006

### Sharp Linear-Programming Bounds for Sphere Packing

O / open | Oberwolfach 报告问题汇编

Does the Cohn-Elkies linear-programming method yield an asymptotic sphere-packing upper bound $$ 2^{-(\lambda+o(1))n} $$ for some $0.604<\lambda<0.605$? Is the strengthened three-point lattice bound sharp in dimension $4$?

来源字段：https://doi.org/10.4171/owr/2022/20

08014 | AIM-GEOMETRY-0257

### Sharp local systolic inequality near the round two-sphere

S / solved | AIM 研讨会问题汇编

**Question 8.7.2 (Babenko).** Does every Riemannian metric on $S^2$ with area $4\pi$ that is close enough to the round metric have a closed geodesic with length $\le 2\pi$?

$\operatorname{dias}(M,g)\le C\sqrt{\operatorname{genus}(M))}\sqrt{\operatorname{area}(M,g)}$,

The source first defines $\operatorname{sys}(S^2,g)$ as the least length of a nontrivial (equivalently here, nonconstant) closed geodesic. The exact question is:

The paragraph beginning “For surfaces of higher genus Gromov proved ...” follows the question and supplies context for Section 8.7; it is not an additional hypothesis or part of the requested conclusion.

with an extra closing parenthesis after $\operatorname{genus}(M)$. The same extra parenthesis is visible in the source PDF; the intended formula is evidently $C\sqrt{\operatorname{genus}(M)}\sqrt{\operatorname{area}(M,g)}$. Nothing below depends on that contextual formula.

Let $g_0$ denote the unit round metric, so

$$ \operatorname{Area}(S^2,g_0)=4\pi, \qquad \ell_{\min}(g_0)=2\pi. $$

The phrase “close enough” has no topology attached to it and therefore is not literally a complete mathematical proposition. The strongest conservative recovery is:

- in the $C^2$ topology on smooth Riemannian metrics, and hence in every $C^r$ topology with $r\ge2$ and in the smooth Fréchet topology, the answer is yes; - an independent contact/Finsler theorem gives a $C^3$ neighborhood of every Zoll Finsler metric; - the sources checked do not justify replacing “close” by $C^1$, $C^0$, or merely bilipschitz closeness.

Thus “solved” below always means the natural $C^2/C^\infty$ reading, which is also the reading explicitly identified as the Babenko--Balacheff conjecture in the solving paper.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

08015 | OWR-2045-003

### Sharp Ricci-Flow-Invariant Curvature Cones

P / partially_solved | Oberwolfach 报告问题汇编

For $\mathcal C_d=\{R:\operatorname{scal}(R)>d\lVert R_W\rVert\}$ with $d(n)=\sqrt{2(n-2)(n-1)}$, for which values of $d$ is $\mathcal C_d$ invariant under Ricci flow? Does the high-dimensional invariance result hold in every dimension $n\geq12$?

来源字段：https://doi.org/10.4171/owr/2008/34

08016 | AIM-GEOMETRY-0260

### Sharp volume bounds for circle-warped metrics on S2 x S1

P / partially_solved | AIM 研讨会问题汇编

**Question 8.8.2 (Alvarez Paiva).** Can a Riemannian metric on $S^3$ or on $S^2\times S^1$ have all closed geodesics long? Specifically, for volume one, must one of these spaces have a closed geodesic of length at most $10^{24}$?

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

08017 | OWR-7155451-007

### Sharp Width Bounds for Positive-Scalar-Curvature Bands

P / partially_solved | Oberwolfach 报告问题汇编

If $Y$ is a closed smooth $(n-1)$-manifold, with $n-1\neq4$, that admits no positive-scalar-curvature metric, and $g$ on $Y\times[-1,1]$ has scalar curvature at least $\sigma>0$, prove that $\operatorname{dist}_g(Y\times\{-1\},Y\times\{1\})\leq 2\pi\sqrt{(n-1)/(\sigma n)}$.

来源字段：https://doi.org/10.4171/owr/2021/30

08018 | OWR-16629-008

### Sharper Paulsen Bounds from Frame Potential Flows

P / partially_solved | Oberwolfach 报告问题汇编

Can the gradient flow of the frame potential prove a Paulsen bound of the form $$\|\Phi-\Psi\|_2^2\leq C\varepsilon d^{2-\alpha}$$ for some $0<\alpha<1$?

来源字段：https://doi.org/10.4171/owr/2018/46

08019 | OWR-14298170-007

### Sharper Positive-Scalar-Curvature Face Gluing

P / partially_solved | Oberwolfach 报告问题汇编

For positive-scalar-curvature manifolds with corners, determine whether isometric faces can be attached without requiring one of the two face mean-curvature functions to have constant sign, whether it suffices that the dihedral angles sum to $\pi$ pointwise, and what sharp hypotheses permit attachment of noncylindrical distinguished faces while preserving the required scalar and boundary curvature conditions.

来源字段：https://doi.org/10.4171/owr/2024/9

08020 | OWR-13110-020

### Sharpness of the One-Round Voronoi Game Bound

O / open | Oberwolfach 报告问题汇编

In the one-round Voronoi game in which the first player places t facilities and the second places one, is the bound $VR_{t:1}\leq\frac{t-1}{t+1}$ sharp?

来源字段：https://doi.org/10.4171/owr/2014/40

08021 | OWR-17135-028

### Shellability and Homology of Totally Mixed Face Posets

O / open | Oberwolfach 报告问题汇编

Let $P_1,P_2\subset\mathbb R^n$. A face $F=F_1+F_2$ of $P_1+P_2$ is totally mixed if $F_i\subset P_i$, $\dim F_i>0$, and $\dim F=\dim F_1+\dim F_2$. Is the poset of totally mixed faces shellable, and does its geometric realization have the rational homology of a wedge of $(n-2)$-spheres?

来源字段：https://doi.org/10.4171/owr/2019/39

08022 | OWR-11136-028

### Shift Chain Colorings and Disk Covering Paths

P / partially_solved | Oberwolfach 报告问题汇编

A shift-chain is a $k$-uniform hypergraph whose edges are comparable coordinatewise after their vertices are written in increasing order. Are all shift-chains 2-colorable for sufficiently large $k$? Separately, is there a constant $c$ such that every closed-unit-disc covering of the plane admits, between any two points at distance $d$, a path of length at most $\sqrt{2}\,d+c$ contained in the region covered at least twice?

来源字段：https://doi.org/10.4171/owr/2011/44

08023 | OWR-8415362-002

### Shift Maps Between Totally Nonnegative Grassmannians

O / open | Oberwolfach 报告问题汇编

Is the shift map a bijection $\operatorname{Gr}_{\geq 0}(k,n)\dashrightarrow\operatorname{Gr}_{\geq 0}(k-1,n)$ between the specified subsets of adjacent totally nonnegative Grassmannians, sending each positroid cell to its shifted cell?

来源字段：https://doi.org/10.4171/owr/2021/57

08024 | AMR-049-0007

### Short geodesics on the regular dodecahedron

O / open | AMR 问题清单汇编

On a regular dodecahedron, unfold a geodesic beginning at a vertex $v$ through successive faces. Call it short if it ends at a vertex and meets no vertex in between, and call it type $A_0$ when the endpoint-angle parameters satisfy $\beta-\alpha=2\pi/5$. Prove that such a type-$A_0$ short geodesic beginning at $v$ never ends at a vertex at graph distance $2$ from $v$ in the dodecahedron's edge graph.

背景链接：https://amj.math.stonybrook.edu/pdf-Springer-final/020-0170.pdf

08025 | OWR-12012-009

### Short Hamiltonian Cycles and Matchings in Convex Bodies

O / open | Oberwolfach 报告问题汇编

If $B$ is the unit disk and $P$ is a finite point set in $B$, must $P$ admit a Hamiltonian cycle whose sum of squared edge lengths is at most $8$? If $|P|$ is even, must it admit a perfect matching with sum of squared edge lengths at most $4$? What analogues hold for other convex bodies $C$?

来源字段：https://doi.org/10.4171/owr/2012/40

08026 | OWR-16942-007

### Short-Time PIC1 Ricci Flow

P / partially_solved | Oberwolfach 报告问题汇编

Let $(M,g_0)$ be a smooth complete $n$-dimensional manifold with $n\geq3$ satisfying PIC1. Does there exist $T>0$ and a smooth PIC1 Ricci flow $g(t)$ on $M$ for $0\leq t<T$ such that $g(0)=g_0$?

来源字段：https://doi.org/10.4171/owr/2019/30

08027 | OWR-13110-001

### Shortest Billiard Trajectories in Constant Width Bodies

O / open | Oberwolfach 报告问题汇编

A shortest closed billiard trajectory in a Euclidean convex body of constant width 1 in the plane has period 2; does the same period-2 conclusion hold for constant-width bodies in every higher dimension?

来源字段：https://doi.org/10.4171/owr/2014/40

08028 | OWR-13110-002

### Shortest Billiards Beyond Euclidean Convex Bodies

O / open | Oberwolfach 报告问题汇编

Can the Bezdek-Bezdek approach to shortest billiard trajectories be extended from Euclidean convex bodies to hyperbolic or spherical geometry, or to Riemannian and Finsler manifolds?

来源字段：https://doi.org/10.4171/owr/2014/40

08029 | AMR-037-0015

### Shortest paths in line arrangements

O / open | AMR 问题清单汇编

Given lines in the plane and two vertices $s,t$ of their arrangement, find a subquadratic algorithm for the shortest $s$-$t$ path along arrangement edges, or prove a quadratic lower bound.

背景链接：https://jeffe.cs.illinois.edu/open/

08030 | OWR-15988-004

## Shortest Piecewise-Smooth Scaling-Rotation Curves

O / open | Oberwolfach 报告问题汇编

For symmetric positive-definite matrices $X$ and $Y$, does the scaling-rotation advantage persist when smooth curves are replaced by piecewise-smooth curves? Give algebraic criteria for the minimal piecewise-smooth scaling-rotation curves to be shortest among all such curves.

来源字段：https://doi.org/10.4171/owr/2018/3

08031 | AIM-GEOMETRY-0005

## Signed curl spectra, component purification, and connected minimizers

P / partially_solved | AIM 研讨会问题汇编

The canonical AIM record 2.2, “Curl eigenvalue,” considers a bounded Lipschitz set \(\Omega\subset\mathbb R^3\) of volume \(1\) and formally writes \[ \operatorname{curl}u=\mu u,\qquad u\cdot\nu=0\text{ on }\partial\Omega, \qquad \int_\Omega u\cdot w=0 \quad\text{for every }w\in L^2(\Omega)\text{ with }\operatorname{curl}w=0. \tag{1} \] It asks to minimize the first positive eigenvalue \(\mu _1(\Omega)\) under the volume constraint. It also gives the squared variational formula \[ \min\{\mu _1(\Omega)^2,\mu _{-1}(\Omega)^2\} =\inf_{u\ne0} \frac{\int_\Omega|\operatorname{curl}u|^2} {\int_\Omega|u|^2}, \tag{2} \] with the stated constraints.

The following repairs and qualifications are necessary.

1. The operator domain, not merely \(C^1(\Omega;\mathbb R^3)\), is essential in (2). The precise zero-flux Giga--Yoshida realization is given in Section 3 below. 2. The spectrum is ordered as \[ \cdots\le\mu_{-3}\le\mu_{-2}\le\mu_{-1}<0 <\mu_1\le\mu_2\le\mu_3\le\cdots, \tag{3} \] with \(\mu_k\to+\infty\) as \(k\to+\infty\) and \(\mu_k\to-\infty\) as \(k\to-\infty\). The source’s final “\(\cdots\to\infty\)” describes only the positive tail; read as a statement about both tails it is wrong. 3. Formula (2) characterizes the least absolute eigenvalue \[ \beta(\Omega):=\min\{\mu_1(\Omega),-\mu_{-1}(\Omega)\}, \tag{4} \] not \(\mu_1(\Omega)\) alone. The two can differ on an individual domain. 4. A “domain” is often connected by definition, while the source begins with “set.” Component arguments below concern the relaxed class of finite disjoint unions of Lipschitz domains. They are not silently applied to an operator on an infinitely disconnected open set. 5. The source’s unconditional phrase that a smooth optimum is not axisymmetric is stronger than the cited theorem. Enciso--Peralta-Salas exclude \(C^{2,\alpha}\) axisymmetric local optima under a mild additional condition; in particular, they exclude all such domains with convex meridional section and hence the ball. Gerner later broadened the rotational obstruction, but the papers checked do not exclude every imaginable axisymmetric geometry without qualification.

The live AIM page itself was unavailable in this run, so no unrecorded source wording is assumed.

来源字段：http://aimpl.org/symmetrybreaking/2/

08032 | AIM-GEOMETRY-0429

## Signed variation of cotangent Dehn-twist powers and the five-dimensional negative stabilization

P / partially_solved | AIM 研讨会问题汇编

This is Question 1.33 (Giroux), under “Sixth day,” in the AIM workshop notes *Holomorphic Curves in Contact Geometry*.

1. **Generation and the boundary twist.** A contact structure on a closed manifold is represented by a Stein/Weinstein open book, hence by a boundary-relative symplectomorphism of its page, modulo stabilization. Are such symplectomorphisms generated by Dehn twists about parametrized Lagrangian spheres? For an integral closed symplectic manifold \((W,\omega)\), a high-degree Donaldson hypersurface \(H_k\) has Stein complement \(F_k=W\setminus H_k\). The annulus-fiber twist near \(\partial F_k\), equivalently the canonical fibered boundary twist, is boundary-relative. Is it a product of ordinary Lagrangian-sphere twists? If \(F_k\) is subcritical and contains no Lagrangian sphere, is this monodromy isotopic, even topologically, to the identity? 2. **Odd powers and the Ustilovsky examples.** The standard contact structure on \(S^{2n+1}\) is supported by the positive Dehn twist \(\phi\) on the cotangent page. Do \(\phi^3,\phi^5,\ldots\) produce Ustilovsky's contact structures on \(S^{4m+1}\)? Can \(\phi\) and \(\phi^3\) be distinguished by Floer homology, perhaps after stabilization, and how does that compare with contact homology? More generally, what happens when each Dehn twist in an open-book factorization is replaced by an odd power? 3. **Left-handed stabilization.** Attach a Weinstein handle to a Stein page along the boundary of a Lagrangian disk, obtaining a Lagrangian sphere, and compose the old monodromy with a left-handed Dehn twist about that sphere. The underlying closed manifold is unchanged. Is the resulting contact structure different and is it unfillable? The notes specifically propose \(S^5\) and one left-handed twist “in \(UT^*(S^2)\).”

There is one necessary dimensional correction to that last phrase and to the informal use of \(T^*S^n\): the compact page is \[ P_n=D^*S^n, \] the disk cotangent bundle, while \(UT^*S^n=\partial D^*S^n\) is its boundary and becomes the binding. The monodromy is compactly supported, hence the identity on a collar of the boundary.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

08033 | OWR-1455-014

## Simple Hyperplane Sections of Real Algebraic Varieties

O / open | Oberwolfach 报告问题汇编

For a finite set $Y\subset\mathbb{R}^d$, call a hyperplane $H$ simple if all but one point of $H\cap Y$ lie on a $(d-2)$-flat in $H$, and not too simple if $H\cap Y$ affinely spans $H$. Find suitable analogues of these notions for a $k$-dimensional real algebraic variety $X\subseteq\mathbb{R}^d$ spanning $\mathbb{R}^d$, and determine conditions on the degree of $X$, $k$, and $d$ that guarantee such a hyperplane section.

来源字段：https://doi.org/10.4171/owr/2007/04

08034 | AMR-054-0016

### Simple Polygonalizations

O / open | AMR 问题清单汇编

Can the number of simple polygonalizations of a set of $n$ points in the plane be computed in polynomial time? A simple polygonalization is a simple polygon whose vertices are the points.

背景链接：https://topp.openproblem.net/p16

08035 | OWR-1703876-012

### Simple Spatial Polygon Realizations from Angle Sequences

O / open | Oberwolfach 报告问题汇编

Let $A=(\alpha_0,\ldots,\alpha_{n-1})$, $n\geq 3$, be realizable by a polygon in $\mathbb{R}^3$. Can $A$ be realized without self-intersections if and only if $n$ is even or $\sum_{i=0}^{n-1}(\pi-\alpha_i)\neq\pi$?

来源字段：https://doi.org/10.4171/owr/2020/30

08036 | AIM-GEOMETRY-0111

### Simplex cavities and vacant percolation thresholds

P / partially_solved | AIM 研讨会问题汇编

**Percolation**

Maximal density packings have connected interstices in dimensions greater than 2. Is there a critical $\lambda$ such that optimal soft packings have path connected interstices? What about the existence of an infinite component?

The path connected part of this question can be addressed by considering the circumcenters of the faces of the regular simplex of the appropriate dimension.

来源字段：http://aimpl.org/softpack/1/

08037 | OWR-4136-011

### Simplex Equality Cases in Convex Geometric Inequalities

P / partially_solved | Oberwolfach 报告问题汇编

Let $n>2$, and let $K\subset\mathbb{R}^n$ be a convex body whose extreme points have Euclidean norm at least $r$. Write $g_K$ for its centroid and $C_2(K)=|K|^{-1}\int_K|x\|_2^2\,dx$. If equality holds in the sharp bound $C_2(K)\geq\bigl(r^2+(n+1)\|g_K\|_2^2\bigr)/(n+2)$, must $K$ be a simplex whose vertices lie on the Euclidean sphere of radius $r$?

来源字段：https://doi.org/10.4171/owr/2009/53

08038 | AIM-GEOMETRY-0334

### Simplex solidness and a sharp coefficient-robustness radius

P / partially_solved | AIM 研讨会问题汇编

**Question.** Does every maximally sparse Laurent polynomial have a solid amoeba?

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

08039 | OPG-1820

### Simplexity of the n-cube

P / partially_solved | Open Problem Garden

Question What is the minimum cardinality of a decomposition of the $n$-cube into $n$-simplices?

背景链接：http://www.openproblemgarden.org/op/simplexity_of_the_cube

08040 | OWR-13750335-006

### Singer’s Unobstructedness Conjecture for Anti-Self-Dual Four-Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Let $(M^4,g)$ be anti-self-dual. Prove that positivity of its Yamabe invariant implies $\ker D_g^*=\{0\}$, equivalently that $(M^4,g)$ is unobstructed.

来源字段：https://doi.org/10.4171/owr/2023/24

08041 | AIM-GEOMETRY-0061

### Singular maximal-entropy measure and its exact dimension reduction

P / partially_solved | AIM 研讨会问题汇编

Study the measure of maximal entropy for the Handel-Thurston Anosov flow. Does it coincide with Lebesgue measure? If not, what is its dimension?

来源字段：http://aimpl.org/equibdynsysgeom/3/

08042 | AMR-064-0001

### Singularities of Time-Optimal Trajectories

O / open | AMR 问题清单汇编

Let $f,g$ be smooth vector fields on an $n$-dimensional manifold $M$, and consider $\dot q=f(q)+ug(q)$, $|u|\leq1$, with fixed endpoint. For a generic pair in dimension $3$, is every individual time-optimal trajectory piecewise smooth, and is $\operatorname{sw}(q)<\infty$ for every $q\in M$? For real-analytic $f,g$, can any two points joined by a time-optimal trajectory be joined by one with at most countably many switching points?

背景链接：https://arxiv.org/abs/1304.2590

08043 | OWR-1394-014

### Slicing Problem for Symmetric Convex Bodies

S / solved | Oberwolfach 报告问题汇编

Does there exist a universal constant $C$ such that every origin-symmetric convex body $K\subset\mathbb{R}^n$ satisfies $\operatorname{vol}(K)^{(n-1)/n}\leq C\max_{\xi\in S^{n-1}}\operatorname{vol}_{n-1}(K\cap\xi^\perp)$?

来源字段：https://doi.org/10.4171/owr/2006/56

08044 | SMA-010

### Smale's 10th Problem: The Pugh Closing Lemma

P / partially_solved | Smale 问题

Is the $C^r$ closing lemma true for dynamical systems?

背景链接：https://arxiv.org/abs/2004.06855

08045 | SMA-012

### Smale's 12th Problem: Centralizers of Diffeomorphisms

P / partially_solved | Smale 问题

Determine the structure of centralizers of generic diffeomorphisms.

背景链接：https://arxiv.org/abs/0804.1416

08046 | SMA-006

### Smale's 6th Problem: Finiteness of Central Configurations

P / partially_solved | Smale 问题

For the Newtonian $n$-body problem with positive masses, are there only finitely many central configurations (relative equilibria) for each $n$?

背景链接：https://doi.org/10.1007/s10569-020-09993-1

08047 | SMA-007

### Smale's 7th Problem: Distribution of Points on the 2-Sphere

P / partially_solved | Smale 问题

What is the optimal arrangement of $n$ points on the 2-sphere to minimize energy for various potential functions?

背景链接：https://mathworld.wolfram.com/SmalesProblems.html

08048 | AIM-GEOMETRY-0077

### Smale's seventh problem and finite-precision stability

P / partially_solved | AIM 研讨会问题汇编

Smale's problem: Rapidly generate $N$ points on $S^2$ whose logarithmic energy differs from the optimal energy by $O(log N)$.

来源字段：http://aimpl.org/discreteaf/1/

08049 | AIM-CONVEX_GEOMETRY-0039

### Small Fourier data do not force simplicial facets without local normalization

P / partially_solved | AIM 研讨会问题汇编

**11. (S. Robins)** **(a)** Let \(K\) be a polytope and consider the expansion of \(h_K\) in spherical harmonics. If all of the coefficients have small absolute value, are all the facets of \(K\) simplices or close to simplices? **(b)** If, for each integer frequency, the Fourier transform of the indicator function \(1_K\) is small, does the same conclusion hold? (This conjecture is made by analogy with pseudorandom number sequences which exhibit this behavior in dimension 1, and the intuition here is that the given hypothesis on the coefficients should exhibit a “random” polytope in some sense.)

来源字段：https://aimath.org/WWN/fourierconvex/fourierconvex.pdf

08050 | OWR-14299905-025

### Small Maximal Partial Ovoids in Polar Spaces

O / open | Oberwolfach 报告问题汇编

For odd $q$, do the hyperbolic quadrics $Q^+(2n+1,q)$, parabolic quadrics $Q(2n,q)$, and elliptic quadrics $Q^-(2n+1,q)$ have no maximal partial ovoids of size $O(q)$?

来源字段：https://doi.org/10.4171/owr/2026/1

08051 | GREEN-050

### Small Triangles in the Unit Disc

P / partially_solved | Ben Green 问题

Given $n$ points in the unit disc, must there be a triangle of area at most $n^{-2+o(1)}$ determined by them?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

08052 | AMR-058-0013

### Smallest Density of a Nonflat Minimal Cone

P / partially_solved | AMR 问题清单汇编

Determine the smallest density greater than one of a minimal cone in $\mathbb{R}^n$, with variants imposing area-minimizing or isolated-singularity hypotheses.

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

08053 | OWR-16412-011

### Smooth Levi-Flat Hypersurfaces in Complex Projective Planes

O / open | Oberwolfach 报告问题汇编

Does $\mathbb{CP}^2$ contain a smooth Levi-flat real hypersurface?

来源字段：https://doi.org/10.4171/owr/2018/37

08054 | OWR-2053-004

### Smooth Obstructions to Geometrically Semisimple Hamiltonians

O / open | Oberwolfach 报告问题汇编

What are the smooth obstructions to the existence of a geometrically semisimple convex Hamiltonian?

来源字段：https://doi.org/10.4171/owr/2008/42

08055 | AIM-GEOMETRY-0268

### Smooth reversible counterexamples and an exact symmetrization diagnostic

S / solved | AIM 研讨会问题汇编

**Question 10.1.3.** Let $F_1,F_2$ be Finsler metrics on a closed surface of genus $\ge 2$. Assume every (unparameterized) $F_1$-geodesic is an $F_2$-geodesic. Must $F_1$ be obtained from $F_2$ by multiplication by a constant and adding a closed form?

The next paragraph asks the analogous question on arbitrary closed manifolds admitting a hyperbolic metric, records Matveev's affirmative Riemannian result, and points to the first integral associated to trajectory-equivalent metrics.

There is an orientation convention hidden in “unparameterized.” In modern Finsler projective geometry, and in Lang's theorem resolving this question, it means the same **oriented** curves up to orientation-preserving reparameterization. This matters because a nonreversible Finsler metric need not have the reversed curve as a geodesic. Under this standard convention, the apparently one-way hypothesis already gives equality: through every $(x,[v]_+)$ (a point and positive tangent ray), each strongly convex Finsler spray has one local oriented unparameterized geodesic. If the $F_1$-curve is an $F_2$-curve, uniqueness makes it the $F_2$-curve with that ray; varying the ray identifies both families. If instead one reads “geodesic” as an unoriented point set, the statement is genuinely ambiguous for asymmetric metrics. The counterexamples below are reversible, so they refute either reading.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

08056 | OWR-16412-005

### Smooth Stein Neighborhood Domains Beyond One H Convexity

O / open | Oberwolfach 报告问题汇编

Does there exist a bounded pseudoconvex domain $\Omega\subset\mathbb C^2$ with $C^2$ or $C^\infty$ boundary and a Stein neighborhood basis that is not $1$-$H$-convex?

来源字段：https://doi.org/10.4171/owr/2018/37

08057 | GREEN-046

### Smooth Surfaces Intersecting 2-planes

P / partially_solved | Ben Green 问题

Let $\Gamma$ be a smooth codimension 2 surface in $\mathbb{R}^n$. Must $\Gamma$ intersect some 2-dimensional plane in 5 points, if $n$ is sufficiently large?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

08058 | OWR-13852-004

### Smoothing and Deforming Metric Cones by Ricci Expanders

P / partially_solved | Oberwolfach 报告问题汇编

Which metric cones can be smoothed by gradient Ricci expanders, can conical gradient Ricci expanders be deformed implicitly, and are such deformations globally unique?

来源字段：https://doi.org/10.4171/owr/2015/31

08059 | AMR-058-0026

### Soap Film on a Regular Octahedral Frame

P / partially_solved | AMR 问题清单汇编

For a regular octahedral wire frame, determine the least-area soap film separating the eight regions, both with ordinary and fractional-density soap films.

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

08060 | AMR-058-0014

### Soap-Film Singularities in Orbifolds

O / open | AMR 问题清单汇编

Classify the singularities allowed in soap films in three-dimensional orbifolds, and more generally in cone manifolds.

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

08061 | OWR-10252937-003

### Sobolev-Trace Minimizers in All Dimensions

P / partially_solved | Oberwolfach 报告问题汇编

Let $p\in(1,n)$, let $\Omega\subset\mathbb R^n$ be a bounded domain with $C^2$ boundary, and let $\Phi_\Omega(T)$ denote the optimal Sobolev-trace variational problem with prescribed boundary norm $T>0$. Does $\Phi_\Omega(T)$ admit a minimizer for every $T>0$ without the known restriction $n>2p$?

来源字段：https://doi.org/10.4171/owr/2022/28

08062 | AIM-GEOMETRY-0131

### Solid-torus loci in the genus-one Hitchin system

P / partially_solved | AIM 研讨会问题汇编

Let $\Sigma$ a surface and $H$ a handlebody. Then: $$ \mathcal{L}\subset \mathcal{M}_{B} \text{ (holomorphic)} \Rightarrow \mathcal{L}\subset \mathcal{M}_{Dol} \text{ (real, Lagrangian)} $$ for $X^{3},\partial(X)=\Sigma$.

Questions:

i) $H(\mathcal{L}_X)$ is a subvariety of $B$ of half dimension. What is the intersection of $\mathcal{L}$ and fibres of the Hitchin map, i.e. the connected components of $H^{-1}(b)\cap \mathcal{L}_X$? [do genus 1 at least]

ii) $RFM(\mathcal{L})=?$

来源字段：http://aimpl.org/spectralhiggs/1/

08063 | AIM-GEOMETRY-0421

### Solved Audin conjecture and a partial structural result on Maslov-two areas

S / solved | AIM 研讨会问题汇编

**Audin's conjecture.** If \(L\) is a Lagrangian torus in \(\mathbb C^n\), then its minimal Maslov number is \(N_L=2\).

Here “torus” is understood as a smooth closed embedded Lagrangian diffeomorphic to \(T^n\), in the standard symplectic vector space

\[ (\mathbb C^n,\omega_0),\qquad \omega_0=\sum_{j=1}^n dx_j\wedge dy_j. \]

For the Maslov homomorphism

\[ \mu_L:\pi_2(\mathbb C^n,L)\longrightarrow \mathbb Z, \]

\(N_L\) is the positive generator of its image (with \(N_L=\infty\) sometimes used when the image is zero).

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

08064 | AIM-GEOMETRY-0240

### Solved cylinder rigidity and invariance of sublinear opening

S / solved | AIM 研讨会问题汇编

- “Riemann-ian” is the line-broken word “Riemannian”; - the basepoint is \(p_0\), not \(p\,0\); - the denominator is \(\operatorname{dist}(p,p_0)=d(p,p_0)\); - “with the based at \(p\)” means “based at \(p\)”; - the sentence beginning “7.1. Divergence of geodesics” belongs to the next section and is not part of Conjecture 7.0.4.

For comparison with the primary solution, \(l(p)\) should be read as the length of a shortest **noncontractible** loop based at \(p\). Such a minimizing based loop is geodesic away from its basepoint. Equivalently, after lifting \(p\) to the universal cover, it is represented by a minimizing segment to a suitable nontrivial deck translate.

The displayed limit is a full limit: \(p\) ranges over all points of the cylinder with \(d(p,p_0)\to\infty\). Since a complete connected Riemannian manifold is proper, this is escape from every compact set. Since a cylinder has two ends, the formula simultaneously quantifies over both ends and over every escaping sequence, not just over one chosen ray or one sequence in each end.

The basepoint \(p_0\) is arbitrary. Proposition 5.1 below proves directly that the condition does not depend on it.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

08065 | AIM-GEOMETRY-0207

### Solved Finsler two-or-infinity conjecture and an isometry-orbit certificate

S / solved | AIM 研讨会问题汇编

**Conjecture 2.2.1 (Long, Bangert, Problem 15 from [Álvarez2006]).** Every irreversible Finsler metric on \(S^2\) has either exactly \(2\) or infinitely many distinct prime closed geodesics.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

08066 | AIM-GEOMETRY-0066

### Solved no-focal extension and robust cap obstructions

S / solved | AIM 研讨会问题汇编

Extend Burns--Climenhaga--Fisher--Thompson (*Geom. Funct. Anal.* 28, 2018, MR3856792, on unique equilibrium states for geodesic flows in nonpositive curvature under the pressure-gap condition $P_{\mathrm{sing}}(\varphi) < P(\varphi)$) to the case of no focal points. In particular, can one apply this to the case of the geometric potential? In particular, the Donnay-type sphere examples of surfaces with ``spherical'' caps.

来源字段：http://aimpl.org/equibdynsysgeom/4/

08067 | AIM-GEOMETRY-0089

### Solved uniqueness of the four-dimensional kissing configuration and a cross-polytope completion theorem

S / solved | AIM 研讨会问题汇编

Uniqueness of \(4\)-dimensional kissing configuration.

**Recovered problem.** Prove or disprove that every 24-point set \(C\subset S^3\) with \(\langle x,y\rangle\leq1/2\) for distinct \(x,y\in C\) is congruent, under an element of \(O(4)\) and relabeling, to the normalized \(D_4\) root system, equivalently to the vertices of the regular 24-cell.

来源字段：http://aimpl.org/discreteaf/1/

08068 | AIM-GEOMETRY-0101

### Source correction, the E8 origin ratio, and a shell-distribution certificate

P / partially_solved | AIM 研讨会问题汇编

Suppose we have a radial function \(f\) on \(\mathbb{R}^n\) (\(0 1\), and \(\hat f\) has double roots at \(\sqrt{2k}\) for \(k \geq 1\). Then, \[ \frac{f(0)}{\hat f(0)} =-\frac{(n^4-56n^3+1184n^2-11200n+40320} {16(n-10)(n-14)(n-18)}. \]

Suppose we have a radial function \(f\) on \(\mathbb{R}^n\) (\(0<n<10\)) such that \(f\) has a single root at \(\sqrt2\), double roots at \(\sqrt{2k}\) for \(k>1\), and \(\hat f\) has double roots at \(\sqrt{2k}\) for \(k\ge1\). Then, \[ \frac{f(0)}{\hat f(0)} =-\frac{(n^4-56n^3+1184n^2-11200n+40320} {16(n-10)(n-14)(n-18)}. \]

Rendered mathematically, the corrupted field says:

This loses most of the hypotheses between the opening parenthesis and the phrase “and \(\hat f\),” and the numerator contains an unmatched parenthesis.

The saved original AIM page recovers the following text and attributes it to Cohn:

Thus the missing root conditions and the range \(0<n<10\) are verified. The AIM page itself still has the unmatched parenthesis and does not specify the Fourier normalization, the function class, whether \(n\) must be integral, or the nonvanishing of \(\hat f(0)\). The database labels the item “1.5”; its position between 1.48 and 1.52 makes this numerically the same label as 1.50, not a change in mathematical content.

The exact polynomial and the intended analytical context occur in Henry Cohn and Stephen D. Miller, *Some properties of optimal functions for sphere packing in dimensions 8 and 24*, arXiv:1603.04759, Section 6. Their conjecture is \[ R(n):=\frac{f(0)}{\widehat f(0)} =-\frac{n^4-56n^3+1184n^2-11200n+40320} {16(n-10)(n-14)(n-18)} \qquad(0<n<10). \tag{1} \] This verifies that the extra opening parenthesis in AIM is a typo and that the numerator closes after \(40320\).

Crucially, Cohn--Miller do not start with an arbitrary radial function. For each finite \(K\), they construct a unique normalized polynomial--Gaussian \[ f_K(x)=q_K(|x|^2)e^{-\pi|x|^2},\qquad f_K(0)=1, \tag{2} \] of degree \(4K-1\), with prescribed simple and double zeros for \(f_K\) and \(\widehat f_K\). The largest third of the finite set of root locations is moved outward by a specified \(K\)-dependent perturbation. They conjecture that \(f_K\) and \(\widehat f_K\) converge locally uniformly near the real axis, that the limiting functions have no unexpected real roots, and then conjecture (1) for this particular limit. They explicitly use \[ \widehat f(\xi)=\int_{\mathbb R^n}f(x)e^{-2\pi i x\cdot\xi}\,dx. \tag{3} \] They also explain that the parameter \(n\) need not be an integer in their Laguerre/Bessel-transform formulation. By contrast, the literal phrase “on \(\mathbb R^n\)” only makes sense for integral \(n\).

Accordingly, there are two distinct readings:

1. **Literal AIM reading:** for an integer \(1\le n\le9\), every sufficiently regular radial function with the listed zeros has ratio (1). 2. **Cohn--Miller reading:** for a real parameter \(0<n<10\), the specifically normalized limiting sequence (2), if it converges in the conjectured manner, has ratio (1).

These readings are not known to be equivalent. The second is the source-supported intended conjecture. The first requires additional uniqueness and well-posedness hypotheses that the AIM page omits.

来源字段：http://aimpl.org/discreteaf/1/

08069 | AMR-128-0001

### Spaces with Lower Ricci Curvature - Conjecture 1

O / open | AMR 问题清单汇编

Let $\big(M^n_i,g_i,p_i\big)\to (X,d,p)$ satisfy $\mathrm{Ric}_i\geq -\lambda$ and $\mathrm{Vol}(B_1(p_i))>v>0$, then $X$ is homeomorphic to a manifold away from a set of codimension three.

背景链接：https://arxiv.org/abs/2010.10031

08070 | AMR-128-0002

### Spaces with Lower Ricci Curvature - Conjecture 2

O / open | AMR 问题清单汇编

Let $\big(M^n_i,g_i,p_i\big)\to (X,d,p)$ satisfy $\mathrm{Ric}\geq -(n-1)$ and $\mathrm{Vol}(B_1(p_i))>v>0$, then the tangent cones of $X$ are unique away from a set of codimension three.

背景链接：https://arxiv.org/abs/2010.10031

08071 | AMR-128-0003

### Spaces with Lower Ricci Curvature - Conjecture 3

O / open | AMR 问题清单汇编

Let $\big(M^n_i,g_i,p_i\big)\to (X,d,p)$ satisfy $\mathrm{Ric}\geq -(n-1)$ and $\mathrm{Vol}(B_1(p_i))>v>0$, then the tangent cones of $X$ are all homoemorphic at a fixed point away from a set of codimension five.

背景链接：https://arxiv.org/abs/2010.10031

08072 | AMR-128-0004

### Spaces with Lower Ricci Curvature - Conjecture 4

O / open | AMR 问题清单汇编

Let $\big(M^n_i,g_i,p_i\big)\to (X,d,p)$ satisfy $\mathrm{Ric} \geq -\lambda$ and $\mathrm{Vol}(B_1(p_i))>v>0$, then $\mathcal{S}^k$ is contained in a union of $k$-dimensional submanifolds. More precisely, there exist a countable collection of bi-H\"older maps $\varphi_i\colon B_1\big(0^k\big)\to M$ such that if $\hat \mathcal{S}^k(X)\equiv \bigcup_i \varphi_i\big(B_1\big(0^k\big)\big)$ then

\itemsep=0pt

$1)$ $\mathcal{S}^k(X)\subseteq \hat \mathcal{S}^k(X)$,

$2)$ for each $x\in \hat \mathcal{S}^k(X)$ we have that every tangent cone is at least $k$-symmetric.

背景链接：https://arxiv.org/abs/2010.10031

08073 | AMR-128-0005

### Spaces with Lower Ricci Curvature - Conjecture 5

O / open | AMR 问题清单汇编

Let $\big(M^n_i,g_i,p_i\big)\to (X,d,p)$ satisfy $\mathrm{Ric}\geq -\lambda$ and $\mathrm{Vol}(B_1(p_i))>v>0$, then for each $k\leq n-2$ there exists $\hat\mathcal{S}^k$ with $\dim\hat\mathcal{S}^k\leq k$ such that for each $x\in X\setminus \hat\mathcal{S}^k$ we have that *every* tangent cone is at least $k+1$-symmetric.

背景链接：https://arxiv.org/abs/2010.10031

08074 | AMR-128-0006

### Spaces with Lower Ricci Curvature - Conjecture 6 (noncollapsing Yau's conjecture)

O / open | AMR 问题清单汇编

Let $\big(M^n,g,x\big)$ satisfy \[ \mathrm{Ric}\geq -\lambda \qquad \text{and} \qquad \mathrm{Vol}(B_1(p))>v>0,\] then $\exists$ $C(n,\lambda,v)>0$ such that \begin{gather*} \fint_{B_1(x)} R < C . \end{gather*}

背景链接：https://arxiv.org/abs/2010.10031

08075 | AMR-128-0007

### Spaces with Lower Ricci Curvature - Conjecture 7 (energy identity)

O / open | AMR 问题清单汇编

Let $(M^n_j,g_j,x_j)\to (X,d,x)$ satisfy $\mathrm{Ric}_j\geq -\lambda$ and the noncollapsing condition $\mathrm{Vol}(B_1(x_j))\geq v>0$, then as measures the scalar curvature converges \begin{gather*} R_j \mathrm{d}v_j\to R_X \mathcal{H}^n+2\theta_x \mathcal{H}^{n-1}_\Sigma+2\pi(1-r_x)\mathcal{H}^{n-2}_S , \end{gather*} where $R_X\colon X\to {\mathbb{R}}$ is a locally $L^1$ function which is bounded from below, $\Sigma$ is an $n-1$ rectifiable set with $\theta_x\leq 1$, and $2\pi(1-r_x)\mathcal{H}^{n-2}_S$ is an $n-2$-rectifiable measure supported on the top stratum of the singular set $\operatorname{Sing}(X)$.

背景链接：https://arxiv.org/abs/2010.10031

08076 | AMR-128-0008

### Spaces with Lower Ricci Curvature - Conjecture 8

O / open | AMR 问题清单汇编

Let $\gamma\colon (-2,2)\to X$ be a minimizing geodesic, then for $s,t\in (-1,1)$ we have that

\itemsep=0pt

$1)$ $($Lipschitz rate$)$ $\frac{\nu(B_r(\gamma(s)))}{\nu(B_r(\gamma(t)))}\leq C(n,\lambda)|t-s|$,

$2)$ $($constant density$)$ if $X$ is noncollapsed, then $\lim\limits_{r\to 0}\frac{\nu(B_r(\gamma(s)))}{\nu(B_r(\gamma(t)))} = 1$.

背景链接：https://arxiv.org/abs/2010.10031

08077 | AMR-128-0009

### Spaces with Lower Ricci Curvature - Conjecture 9 (local Yau conjecture)

O / open | AMR 问题清单汇编

Finally, let us end by discussing the regularity of manifolds with lower Ricci bounds, potentially including those with small volume. The main conjecture out there is one due to Yau, and we state a local version of it below:

Let $\big(M^n,g\big)$ satisfy $\mathrm{Ric}\geq -\lambda$, then $\int_{B_1(p)} R \leq C(n,\lambda)$.

背景链接：https://arxiv.org/abs/2010.10031

08078 | OWR-17135-034

### Spanning Rank of Lattice Polytopes

O / open | Oberwolfach 报告问题汇编

For a $d$-dimensional lattice polytope $P$ with $\operatorname{aff}(P\cap\mathbb Z^d)=\mathbb Z^d$, let $\operatorname{SR}(P)$ be the least cardinality of a subset $B\subseteq P\cap\mathbb Z^d$ satisfying $\operatorname{aff}(B)=\mathbb Z^d$, and let $\operatorname{SR}(d)=\max_P\operatorname{SR}(P)$. Determine $\operatorname{SR}(4)$ and the asymptotic growth of $\operatorname{SR}(d)$.

来源字段：https://doi.org/10.4171/owr/2019/39

08079 | OWR-2090-027

### Spanning-Tree Counts versus Triangulation Counts

O / open | Oberwolfach 报告问题汇编

For a finite planar point set $P$, let $\operatorname{tr}(P)$ and $\operatorname{st}(P)$ count its triangulations and crossing-free spanning trees. Does there exist $c>1$ such that $\operatorname{st}(P)\geq c^{|P|}\operatorname{tr}(P)$ for every sufficiently large $P$?

来源字段：https://doi.org/10.4171/owr/2008/44

08080 | GREEN-049

### Sparse Hitting Set for Rectangles

O / open | Ben Green 问题

Let $A$ be a set of $n$ points in the plane. Can one select $A' \subset A$ of size $n/2$ such that any axis-parallel rectangle containing 1000 points of $A$ contains at least one point of $A'$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

08081 | AIM-CONVEX_GEOMETRY-0001

### Sparse-normal formulas for perimeters and related cube-section functionals

P / partially_solved | AIM 研讨会问题汇编

On page 2 it gives the title **“Max/min Perimeter of central cross-sections,”** proposed by Hermann König, and asks:

1. Find the central cross section \[ u^\perp=\{x\in\mathbb R^n:x\cdot u=0\} \] of \[ B_\infty^n=\{x\in\mathbb R^n:|x_i|\leq 1\} \] with the largest perimeter. 2. Which central cross section has smallest perimeter? 3. What happens for mean width or intrinsic volume instead of perimeter? 4. What happens in other codimensions? 5. What happens for other measures, such as Gaussian measures? 6. What happens for the perimeter of projections?

Only these six questions are treated here.

For \(n\geq3\), write \(Q_n=[-1,1]^n\), \(H_u=u^\perp\), and use \[ P_n(u):=\mathcal H^{n-2}\!\left(\partial_{H_u}(Q_n\cap H_u)\right) =\mathcal H^{n-2}(\partial Q_n\cap H_u). \] This is the convention used in the later solution of the maximum problem. For \(n=2\), the corresponding \(\mathcal H^0\)-measure of the two boundary points is \(2\), independent of \(u\). The case \(n=1\) is excluded because the relative boundary convention degenerates.

来源字段：https://aimath.org/pastworkshops/sectionsconvexproblems.pdf

08082 | OWR-8415335-008

### Special Lagrangian Fibrations in Calabi-Yau Degenerations

P / partially_solved | Oberwolfach 报告问题汇编

In a large-complex-structure degeneration of Calabi-Yau $n$-folds, does the Ricci-flat metric admit a special Lagrangian $T^n$-fibration on a generic region, as predicted by the Strominger-Yau-Zaslow conjecture?

来源字段：https://doi.org/10.4171/owr/2021/32

08083 | OWR-12975-003

### Spectrahedrality of Hyperbolicity Cones

O / open | Oberwolfach 报告问题汇编

Is every hyperbolicity cone of a real homogeneous hyperbolic polynomial spectrahedral?

来源字段：https://doi.org/10.4171/owr/2014/17

08084 | OWR-14610-003

### Spectral Convergence Under Lower Ricci Bounds

O / open | Oberwolfach 报告问题汇编

Can the spectral, tensor, and related convergence results for noncollapsed Ricci-bounded sequences be extended from $|\operatorname{Ric}_{X_i}|\leq K$ to the weaker lower bound $\operatorname{Ric}_{X_i}\geq-K$?

来源字段：https://doi.org/10.4171/owr/2016/28

08085 | OWR-1597-002

### Spectral Remainder Bounds for Liouville Tori

O / open | Oberwolfach 报告问题汇编

Let $M$ be a torus with a Liouville metric. Do the Weyl-law remainder $R(\lambda)$, the pointwise remainder $R_x(\lambda)$, and the off-diagonal spectral function $N_{x,y}(\lambda)$ all satisfy $O(\lambda^{1/2+\varepsilon})$ for every $\varepsilon>0$ and every pair $x\neq y$ in $M$?

来源字段：https://doi.org/10.4171/owr/2007/41

08086 | OWR-1597-001

### Spectral Remainder Bounds on Surfaces of Revolution

O / open | Oberwolfach 报告问题汇编

Let $M$ be a generic Riemannian $2$-sphere with a metric of revolution. For the Weyl-law remainder $R(\lambda)$, the pointwise remainder $R_x(\lambda)$ away from a pole, and the off-diagonal spectral function $N_{x,y}(\lambda)$ when at least one of $x,y$ is not a pole, do the bounds $R(\lambda)=O(\lambda^{1/2+\varepsilon})$, $R_x(\lambda)=O(\lambda^{1/2+\varepsilon})$, and $N_{x,y}(\lambda)=O(\lambda^{1/2+\varepsilon})$ hold for every $\varepsilon>0$?

来源字段：https://doi.org/10.4171/owr/2007/41

08087 | OWR-1597-003

### Spectral Remainders on Negatively Curved Surfaces

O / open | Oberwolfach 报告问题汇编

Let $M$ be a generic negatively curved Riemannian surface. Does the Weyl-law remainder satisfy $R(\lambda)=O(\lambda^\varepsilon)$, while the pointwise remainder and off-diagonal spectral function satisfy $R_x(\lambda)=O(\lambda^{1/2+\varepsilon})$ and $N_{x,y}(\lambda)=O(\lambda^{1/2+\varepsilon})$, for every $\varepsilon>0$ and all $x\neq y$ in $M$?

来源字段：https://doi.org/10.4171/owr/2007/41

08088 | OWR-3389-011

### Spectral Rigidity of Unit-Area Riemannian Two-Spheres

O / open | Oberwolfach 报告问题汇编

Let $(M,g)$ be a unit-area Riemannian topological $2$-sphere, and let $\lambda_1(g)$ be its first nonzero Laplace eigenvalue. Is $(M,g)$ isometric to $(S^2,s g_0)$ for some positive conformal factor $s$ satisfying $\lVert s\rVert_{H^{-1}(S^2,g)}\leq C\lambda_1(g)$ for a universal constant $C$?

来源字段：https://doi.org/10.4171/owr/2009/06

08089 | GEO-032

### Sphere Packing in High Dimensions

P / partially_solved | 独立 / 未分配集合

What is the densest packing of unit spheres in dimensions other than 1, 2, 3, 8, and 24?

背景链接：https://doi.org/10.1007/s00454-026-00841-x

08090 | GEO-002

### Sphere Packing in Higher Dimensions

P / partially_solved | 独立 / 未分配集合

别名：Sphere Packing Problem in Higher Dimensions [GEO-001]; Sphere Packing Problem Higher Dimensions [GEOM-001]

What is the densest packing of congruent spheres in $n$ dimensions for $n \geq 4$?

背景链接：https://arxiv.org/abs/1603.04246

08091 | OWR-15579-005

### Sphere Rigidity from Scalar Curvature and Injectivity Radius

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist $\varepsilon=\varepsilon(n)>0$ such that every closed $n$-dimensional Riemannian manifold $M$ with $\operatorname{scal}_M\geq n(n-1)$ and injectivity radius at least $\pi-\varepsilon(n)$ is diffeomorphic to the standard $n$-sphere?

来源字段：https://doi.org/10.4171/owr/2017/36

08092 | OWR-14297736-015

### Spherical Robin-Eigenvalue Maximization Beyond Known Regimes

O / open | Oberwolfach 报告问题汇编

Prove spherical-cap maximality of the second Robin eigenvalue for hyperbolic domains with positive Robin parameter and in the regime where the cap's second eigenfunction is radial.

来源字段：https://doi.org/10.4171/owr/2023/36

08093 | AMR-067-0023

### Spherical submetries

S / solved | AMR 问题清单汇编

Is every Laplacian algebra of polynomials maximal?

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

08094 | OWR-4796-002

### Spin Transformations of Discrete Conformal Structures

P / partially_solved | Oberwolfach 报告问题汇编

Can the notion of discrete conformal equivalence, in which edge lengths differ by positive scale factors at vertices, be extended to all spin transformations?

来源字段：https://doi.org/10.4171/owr/2011/07

08095 | AIM-GEOMETRY-0374

### Spin-lifted null dynamics and a parity correction on Lorentzian tori

P / partially_solved | AIM 研讨会问题汇编

**Problem 16:** Relate \(h(g)\) to the dynamic of null geodesics.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

08096 | AIM-GEOMETRY-0372

### Spinorial nullities, Yamabe sign, and Lorentzian finiteness

P / partially_solved | AIM 研讨会问题汇编

Problem 14: How t(g) and h(g) relate to other conformal invariants?

On a spin manifold \((M,g)\) with complex spinor bundle \(S\), the spin connection decomposes into the Dirac and twistor components. The note sets

\[ h(g)=\dim_{\mathbb C}\ker D_g,\qquad t(g)=\dim_{\mathbb C}\ker P_g, \]

where \(D_g\) is the Dirac operator and

\[ (P_g\psi)(X)=\nabla_X^S\psi+\frac1nX\mathbin{\cdot}D_g\psi \tag{1.1} \]

is the twistor (Penrose) operator, using the Clifford convention \(X\mathbin{\cdot}X=-g(X,X)\). The source explicitly says that both dimensions are conformal invariants, that the Riemannian case was comparatively well studied, and that the surrounding questions concern the Lorentzian case. Problem 13 asks for a classification with \(t>0\) or \(h>0\), Problem 15 asks specifically about conformal Cartan holonomy, and Problem 16 asks about null-geodesic dynamics.

The question omits four hypotheses that materially change its meaning: signature, compactness, spin structure, and the function space/domain used for \(D_g\). This report fixes those hypotheses in each theorem. All dimensions are complex dimensions and the spin structure is held fixed under conformal rescaling.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

08097 | OWR-14298170-004

### Spinorial Scalar-Curvature Stability

P / partially_solved | Oberwolfach 报告问题汇编

Use spinors to establish Geroch stability and positive-mass-theorem stability in dimensions $n\geq 3$, and generalize Llarull stability to the scalar-curvature rigidity settings of Goette and Semmelmann.

来源字段：https://doi.org/10.4171/owr/2024/9

08098 | AIM-GEOMETRY-0314

### Split integral Kirwan surjectivity for smooth compact toric reductions

P / partially_solved | AIM 研讨会问题汇编

Let a compact connected torus \(T\) act linearly and Hamiltonianly on \(\mathbb C^n\). At a regular value \(\alpha\), suppose the level \(Z=\mu^{-1}(\alpha)\) is compact and \(T\) acts freely, so that \(X=Z/T\) is a smooth compact toric manifold. Is the integral Kirwan map \[ \kappa_{\mathbb Z}:H_T^*(\mathbb C^n;\mathbb Z) \cong H^*(BT;\mathbb Z) \longrightarrow H^*(X;\mathbb Z) \] surjective? How far does this extend beyond the smooth compact free case?

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

08099 | AIM-GEOMETRY-0050

### SPR stability for path-dependent compact-window potentials

P / partially_solved | AIM 研讨会问题汇编

Give a complete picture of existence and uniqueness for H\"older potentials on the space of geodesics for the geodesic flow on noncompact $\operatorname{CAT}(-1)$ spaces. In particular, consider potentials which depend not only the basepoint of a geodesic, but the geodesic itself.

来源字段：http://aimpl.org/equibdynsysgeom/1/

08100 | AIM-GEOMETRY-0275

### Spray and holonomy classifications for affinely equivalent Finsler metrics

P / partially_solved | AIM 研讨会问题汇编

**Problem 12.0.4.** Describe all affinely equivalent Finsler metrics.

The primary Burns--Matveev PDF first defines affine equivalence in the preceding lines: two Finsler metrics are affinely equivalent when a geodesic of the first, with its constant-speed affine parametrization, is a geodesic of the second. It then states the entire target:

Affine changes of a parameter along one geodesic are harmless because a spray is two-homogeneous. In particular, multiplying a Finsler metric by a positive constant does not change its spray.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

08101 | AMR-071-0031

### Square packing in a square

P / partially_solved | AMR 问题清单汇编

Square packing in a square: what is the asymptotic growth rate of wasted space?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

08102 | AIM-GEOMETRY-0057

### Square-locus transfer and the canonical-cover obstruction

P / partially_solved | AIM 研讨会问题汇编

Generalize the results of Bufetov and Gurevich in Bufetov--Gurevich (*Sb. Math.* 202, 2011, MR2857792, on existence and uniqueness of the measure of maximal entropy for the Teichmüller flow on strata of Abelian differentials) to the space of quadratic differentials.

A. I. Bufetov and B. M. Gurevich, “Existence and uniqueness of the measure of maximal entropy for the Teichmüller flow on the moduli space of Abelian differentials,” *Sbornik: Mathematics* 202 (2011), no. 7, 935-970, MR2857792, DOI 10.1070/SM2011v202n07ABEH004172.

For every connected component \(\mathcal C\) of a unit-area stratum of quadratic differentials, prove that the normalized Masur-Veech probability is the unique invariant Borel probability of maximal entropy for the Teichmüller flow on \(\mathcal C\), and identify its entropy.

来源字段：http://aimpl.org/equibdynsysgeom/3/

08103 | OWR-3385-013

### Square-Peg Problem for Jordan Curves

O / open | Oberwolfach 报告问题汇编

Does every continuously embedded circle in the plane contain four points spanning a square?

来源字段：https://doi.org/10.4171/owr/2009/02

08104 | OWR-4798-011

### Square-Peg Problem for Polyomino Boundaries

O / open | Oberwolfach 报告问题汇编

For every polyomino boundary $J$ that is a closed Jordan curve, do there exist four points of $J$ that are the vertices of a square?

来源字段：https://doi.org/10.4171/owr/2011/08

08105 | OWR-16412-007

### Squeezing Constants of Bounded Homogeneous Domains

P / partially_solved | Oberwolfach 报告问题汇编

Can the constant value of the squeezing function be computed for every bounded homogeneous domain?

来源字段：https://doi.org/10.4171/owr/2018/37

08106 | OWR-10252937-010

### Stability of Almost-Einstein Metrics

P / partially_solved | Oberwolfach 报告问题汇编

If a manifold $M$ admits a metric $\bar g$ that is almost Einstein in a suitable sense, does there exist an Einstein metric on $M$ close to $\bar g$?

来源字段：https://doi.org/10.4171/owr/2022/28

08107 | AMR-059-0016

### Stability of Hessian Metrics

P / partially_solved | AMR 问题清单汇编

Let $M$ be a compact Hessian manifold and deform its affine structure. Does every sufficiently small deformation admit a Hessian metric?

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

08108 | OWR-1703876-014

### Stability of Longest Edge Simplex Refinement

O / open | Oberwolfach 报告问题汇编

Start with a sufficiently generic $d$-dimensional simplex, repeatedly bisect its longest edge and subdivide the simplex into two equal-volume simplices. Does the resulting subdivision stabilize after finitely many iterations for every starting simplex in every dimension, or can the simplices become arbitrarily badly shaped with inradius-to-circumradius ratio tending to $0$?

来源字段：https://doi.org/10.4171/owr/2020/30

08109 | AMR-058-0003

### Stability of Spherical Plateau Clusters

O / open | AMR 问题清单汇编

Is every bubble cluster in $\mathbb{R}^3$ made of spherical pieces meeting according to Plateau's rules necessarily stable, in the sense of having nonnegative second variation?

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

08110 | AMR-058-0001

### Stable Bubble Cluster with a Toroidal Region

P / partially_solved | AMR 问题清单汇编

Is there a stable cluster of bubbles in $\mathbb{R}^3$ in which some bubble is topologically a torus?

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

08111 | AIM-GEOMETRY-0338

### Stable contour graphs and their toric ends

P / partially_solved | AIM 研讨会问题汇编

For \(f\in\mathbb C[x_1,x_2]\), let \(C_f\subset\mathbb R^2\) denote the contour of the amoeba of \(f\). The singular points \(V\) on \(C_f\) divide it into arcs \(E\), giving a planar graph \((V,E)\). What combinatorial properties does this graph have, and which graphs can be realized by a polynomial? How does this generalize to higher dimension?

The AIM source (problem A.18 from the workshop *Amoebas and tropical geometry*) asks:

The contour lies in \(\mathbb R^2\), whereas critical points lie on the complex curve. The standard reconstruction is as follows. Regard \(f\) as a Laurent polynomial on the algebraic torus (ordinary polynomials are included), put \[ C=\{f=0\}\subset(\mathbb C^*)^2, \qquad \operatorname{Log}(z,w)=(\log|z|,\log|w|), \] and define \[ \Sigma=\operatorname{Crit}(\operatorname{Log}|_C), \qquad \mathcal K_f=\operatorname{Log}(\Sigma). \] Thus the contour \(\mathcal K_f\) is the set of **critical values**. If \(C\) is smooth, its logarithmic Gauss map is \[ \gamma:C\longrightarrow\mathbb {CP}^1, \qquad \gamma(z,w)=[z f_z:w f_w], \] and \(\Sigma=\gamma^{-1}(\mathbb {RP}^1)\). Critical points of \(\gamma\) are instead its ramification points and form a different set.

The PDF's broken words “Com-puting” and “nu-merically” are line-wrap artifacts. Its 2003 Passare--Tsikh preprint appeared in 2005. Monomial factors do not change the curve in the torus, so Laurent notation is the natural one.

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

08112 | OWR-14742-002

### Stable Embedded Minimal Hypersurfaces Below Dimension Eight

P / partially_solved | Oberwolfach 报告问题汇编

Are the only stable embedded minimal hypersurfaces in $\mathbb R^n$ hyperplanes when $n\leq7$?

来源字段：https://doi.org/10.4171/owr/2016/34

08113 | OWR-13103-003

### Stable Minimal Hypersurfaces and Entire Minimal Graphs

P / partially_solved | Oberwolfach 报告问题汇编

Are there complete stable minimal hypersurfaces in asymptotically flat manifolds, and are affine functions the only entire minimal graphs over $\mathbb{R}^{n-1}$ in $\mathbb{R}^n$?

来源字段：https://doi.org/10.4171/owr/2014/33

08114 | AMR-058-0002

### Standard k-Bubble Conjectures

P / partially_solved | AMR 问题清单汇编

For $k\leq n+1$ and prescribed volumes in $\mathbb{R}^n$, prove that the standard $k$-bubble is uniquely area-minimizing. For $n=3$, are standard clusters the only strictly stable $k$-bubbles for $k\leq4$? Is there a strictly stable six-bubble cluster with a nonspherical interface but no such cluster with fewer bubbles?

背景链接：http://torus.math.uiuc.edu/jms/Papers/foams/soap-prob.pdf

08115 | AIM-GEOMETRY-0271

### Stationary theta graphs on every circle-symmetric Riemannian two-sphere

P / partially_solved | AIM 研讨会问题汇编

A graph \(G\) in a Riemannian surface \((S,g)\) is called a stationary net if every edge is a geodesic and if at every vertex the sum of diverging unit vectors is zero. Vertices must have valence at least \(3\).

**Question 11.1.1 ([Has--Mor1996]).** Does every metric on \(S^2\) admit a stationary \(\Theta\)-graph?

That material begins the next problem and is not part of Question 11.1.1.

The intended convention is recovered from Hass--Morgan's source paper. Their geodesic net is a graph **embedded** in the surface. Thus the target is an embedded copy of the abstract theta multigraph: two distinct vertices \(p\ne q\), three positive-length embedded geodesic arcs joining them, no self-intersection of an edge, and pairwise disjoint edge interiors. The three edges are distinct geometric arcs. Loops, coincident vertices, zero-length edges, repeated/overlapping edges, or an immersed three-cage with extra crossings do not answer the original question.

All edges have unit weight. At each vertex one takes the three unit tangents directed outward from that vertex and requires their vector sum to vanish. For three unit vectors in a Euclidean plane this is equivalent to pairwise angles exactly \(2\pi/3\). Later authors sometimes use “geodesic net” for an immersed weighted multigraph, allowing loops and intersections; results under that convention must not silently be substituted for the Hass--Morgan question.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

08116 | AIM-GEOMETRY-0342

### Statistical algebraic geometry and an exact random-binomial law

P / partially_solved | AIM 研讨会问题汇编

This record is Section A.22, “Statistical algebraic geometry,” contributed by Steve Zelditch to the AIM workshop list *Amoebas and Tropical Geometry*. It is a collection of four related research questions rather than a single assertion.

1. For random real plane algebraic curves of degree \(N\), determine the most probable and expected number of connected components. 2. For random spherical harmonics of degree \(N\), determine the most probable and expected number of nodal domains. 3. For \(m\) random real \(f\)-fewnomials of degree \(N\), with spectra chosen among \(f\)-element subsets of \(\mathbb Z_+^m\cap N\Sigma\), determine the expected number of real common zeros. Variants keep Newton polytopes fixed and dilate them, or use Kac coefficients instead of projectively normalized coefficients. 4. For a random holomorphic section \(s\in H^0(M,L)\), count and describe the critical points solving \(\nabla s=0\), noting their dependence on the Hermitian metric and connection.

The fewnomial upper bound displayed in the original slide is \[ \nu(m,f)\le 2^{m^2f(f-1)/2}(m+1)^f. \] The source also displays a preliminary zero-density formula formed from an average over spectra \(S\in\mathcal S(N,f)\) and the restricted kernel \[ \Pi_{N\mid S}(x,y)=\sum_{\beta\in S}\binom N\beta x^\beta y^\beta. Accordingly, no claim below uses that normalization.

There is a further mathematical ambiguity in “the \(O(m+1)\)-invariant measure.” Invariance alone does not determine a unique Gaussian ensemble on real homogeneous polynomials: the representation decomposes into harmonic summands, whose variances can be weighted differently. Thus the first question has no unique numerical answer until the covariance is specified. The Kostlan and real Fubini-Study ensembles already give different growth orders.

This report gives a status update on all four strands and proves a complete fixed-support special case of the third strand.

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

08117 | AMR-059-0005

### Statistical Manifolds from Probability Families

P / partially_solved | AMR 问题清单汇编

Given a statistical manifold, determine conditions for the existence of a family of probability distributions whose induced statistical manifold coincides with it. Determine the analogous conditions for dually flat manifolds.

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

08118 | AIM-GEOMETRY-0405

### Stein fillability, augmentation, and the curvature defect of cylindrical contact homology

P / partially_solved | AIM 研讨会问题汇编

Conjecture: for any Stein fillable contact manifold, the cylindrical contact homology is defined. (As some have used it in the past, CCH is defined if there exists a contact form with no contractible Reeb orbits of certain Maslov indices. If such a form exists, then any two such forms give the same CCH. For this conjecture, one might not be able to eliminate bad contractible orbits, but one could hope that one can make them somehow “algebraically cancel”.) If so, then counting pairs of pants etc. gives operations on the CCH, as in Floer theory of symplectomorphisms.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

08119 | AIM-GEOMETRY-0413

### Stein nonfillability of a torus prequantization

S / solved | AIM 研讨会问题汇编

**Question 1.17.** Let \(P^5\) be the contact \(S^1\)-bundle over \(T^4\) with \(c_1\) equal to the class of the symplectic form on \(T^4\). Conjecture: \(P\) has no Stein filling.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

08120 | AMR-059-0015

### Stein Tangent Bundles of Complete Hessian Manifolds

P / partially_solved | AMR 问题清单汇编

If $(M,g)$ is a complete Hessian manifold, is its tangent bundle with the natural complex structure a Stein manifold? In particular, if a convex domain $\Omega\subset\mathbb{R}^n$ contains no complete line and a discrete affine group $\Gamma$ acts freely and properly discontinuously, is $T(\Omega/\Gamma)$ Stein?

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

08121 | OWR-4199-002

### Steiner and Kinematic Formulas for Affine Surface Area

P / partially_solved | Oberwolfach 报告问题汇编

Determine whether affine surface area admits a Steiner formula for $K+\varepsilon B^n$, including its maximal variant; determine explicitly the valuation $\Phi(K)=\int\Omega(K|E)\,dE$; and decide whether affine kinematic formulae exist.

来源字段：https://doi.org/10.4171/owr/2010/04

08122 | OWR-15988-008

### Stochastic Completeness and Bridge Continuity on Landmark Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Is the LDDMM landmark manifold stochastically complete? When logarithm maps are used to construct bridges, how should continuity across the cut locus be defined and proved?

来源字段：https://doi.org/10.4171/owr/2018/3

08123 | AMR-037-0016

### Straight skeleton of a simple polygon

S / solved | AMR 问题清单汇编

Is there a near-linear-time algorithm to construct the straight skeleton of a simple polygon? Determine the optimal complexity, including for polygons with reflex angles bounded away from zero.

背景链接：https://jeffe.cs.illinois.edu/open/

08124 | AIM-GEOMETRY-0197

### Strict Delaunay realization spaces and a stellar simplex cell

P / partially_solved | AIM 研讨会问题汇编

**10. (Theran)** Consider a Delaunay triangulation (no vertex is inside the circumcircle of any triangle). Fix a combinatorial type of a triangulation. Question: What is the configuration space?

\(d=2\) it is a ball

\(d=3\) is it universal?

来源字段：https://aimath.org/pastworkshops/linkagesproblems.pdf

08125 | OWR-1740-001

### Strominger-Yau-Zaslow Mirror-Symmetry Conjecture

O / open | Oberwolfach 报告问题汇编

Given a manifold with a singular special Lagrangian torus fibration over a base $B$, should its mirror be the dual torus fibration over the same base, equipped with a holomorphic complex-valued superpotential $W$ that appropriately counts holomorphic discs with boundary on the fibres of the original fibration?

来源字段：https://doi.org/10.4171/owr/2007/57

08126 | OWR-2053-005

### Strong Intersection Conjecture for Hamiltonian Isotopies

O / open | Oberwolfach 报告问题汇编

Are Hamiltonian isotopies naturally strongly intersecting in the sense required by the symplectic homoclinic Floer invariant?

来源字段：https://doi.org/10.4171/owr/2008/42

08127 | OWR-13951-003

### Strong Reverse Minkowski Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For a symmetric convex body $K$ and an $n$-dimensional lattice $L$ in $\mathbb{R}^n$, is $|K\cap L|\leq M(O(\log n)K,L)$? For the Euclidean unit ball $B_2^n$, is $|B_2^n\cap L|\leq M(O(\sqrt{\log n})B_2^n,L)$?

来源字段：https://doi.org/10.4171/owr/2015/56

08128 | AIM-GEOMETRY-0409

### Strong Weinstein uniqueness and a stabilization-chain audit

P / partially_solved | AIM 研讨会问题汇编

**Question 1.13.** How unique are the open books corresponding to contact manifolds? Giroux explained that the open books produced by Donaldson's construction, which depend on a positive integer parameter \(k\), are unique up to stabilization when \(k\) is sufficiently large.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

08129 | AIM-GEOMETRY-0069

### Strong-coding obstructions and a transverse Hölder bootstrap

P / partially_solved | AIM 研讨会问题汇编

For metric Anosov flows (Smale flows), under what conditions does orbit equivalence imply Hölder orbit equivalence? Can one find a Smale flow which is not Hölder covered by the suspension of a symbolic dynamical system?

来源字段：http://aimpl.org/equibdynsysgeom/5/

08130 | OWR-13951-009

### Structural Properties of Geometric Means of Convex Bodies

O / open | Oberwolfach 报告问题汇编

Let A and B be convex bodies containing the origin, let $A^\circ$ be the polar body, D the Euclidean unit ball, and G and $G_p$ their geometric means. Determine whether $G(A,B)=D$ implies $B=A^\circ$; whether $A+B=A^\circ+B^\circ$ implies $B=A^\circ$; whether a convex body P representable as $G(A,B)$ for every A must be an ellipsoid; whether $G_p(A,B)=G_1(A,B)$; whether the upper elliptic envelope $E_u(A,B)$ equals $G(A,B)$; whether $|G(A,B)|^2\geq|A||B|$ under identifiable conditions; and whether an exponential map or the stated homogeneity identity exists.

来源字段：https://doi.org/10.4171/owr/2015/56

08131 | OWR-13110-006

### Subadditivity of Symplectic Capacities Under Convex Covers

O / open | Oberwolfach 报告问题汇编

If a convex body $X\subset\mathbb{R}^{2n}$ is covered by finitely many convex bodies $X_i$, is there a symplectic capacity $c$ for which $\sum_i c(X_i)\geq c(X)$?

来源字段：https://doi.org/10.4171/owr/2014/40

08132 | OWR-4339-003

### Submetries of Compact Ptolemy Spaces

O / open | Oberwolfach 报告问题汇编

Let $X$ be a compact Ptolemy space satisfying these $R$-circle axioms: any two points lie on an $R$-circle; a fourth point satisfying Ptolemy equality with three points on an $R$-circle lies on that circle; closest-point projection to an $R$-circle is self-dual; and every Möbius map between $R$-circles extends to a Möbius automorphism of $X$. For every $\omega\in X$, does there exist a submetry $\pi_\omega:X\setminus\{\omega\}\to B_\omega$, where $B_\omega$ is isometric to $\mathbb R^k$, such that Möbius automorphisms induce compatible homotheties of the bases? The completed fibers $F\cup\{\omega\}$ should be $K$-circles homeomorphic to $S^p$, with $k+p=\dim_\partial X$, and satisfy: two distinct points determine a unique $K$-circle; a $K$-circle and an $R$-circle meet in at most two points; if $F\cup\{\omega\}$ is a $K$-circle and $x\notin F\cup\{\omega\}$, a unique $R$-circle through $x$ and $\omega$ meets $F$; and, for two distinct fibers $F,F'$ and two $R$-circles through $\omega$ meeting $F,F'$, any third fiber that meets one of those $R$-circles also meets the other.

来源字段：https://doi.org/10.4171/owr/2010/26

08133 | AIM-GEOMETRY-0025

### Subword complexes as boundary complexes of brick compactifications

P / partially_solved | AIM 研讨会问题汇编

For the brick (cluster) compactification of a braid variety, which boundary strata are present, what are their closure relations, and when does this combinatorics agree with cluster-complex or brick-polytope combinatorics?

The exact AIM record asks:

The demonstrative “this” is ambiguous in isolation. The two immediately preceding records in the same section ask:

1. whether a braid variety has a natural compactification whose smaller strata also carry cluster structures; and 2. whether brick manifolds can be obtained by cluster-theoretic means.

The 2023 AIM workshop summary confirms this reading: it considers the inclusion of the braid variety \(X(\mathbf i)\) into its brick variety \(\operatorname{brick}(\mathbf i)\), reports that the latter is a cluster compactification, and asks about further polytopal and positive-geometric structure. Thus the recovered question is:

This report answers the first two parts exactly, gives the precise finite-type case in which the subword and cluster complexes agree, and exhibits a small obstruction to any word-independent identification in general.

来源字段：http://aimpl.org/clusterbraid/4/

08134 | AIM-CONVEX_GEOMETRY-0027

### Super-Gaussian directions and the compact-set ambiguity

P / partially_solved | AIM 研讨会问题汇编

**Problem 8 (Grigoris Paouris).** Is it true that every compact set is anti-\(\psi _2\) in some direction?

来源字段：https://aimath.org/WWN/convexgeometry/convexgeometry.pdf

08135 | OWR-16160-002

### Superexpansion and Coarse Classification of Expanders

O / open | Oberwolfach 报告问题汇编

What notion of coarse equivalence or coarse distinction gives a useful classification of expander sequences?

来源字段：https://doi.org/10.4171/owr/2018/19

08136 | AMR-054-0026

### Surface Reconstruction

S / solved | AMR 问题清单汇编

Given a sufficiently dense sample of points on a surface (technically, an $\epsilon$-sample), reconstruct a surface homeomorphic to the original.

背景链接：https://topp.openproblem.net/p26

08137 | AMR-063-0002

### Surgery Generators for a Four-Manifold Homotopy Type

O / open | AMR 问题清单汇编

Is there a useful list of surgery procedures which generates all smooth four-manifolds of a given homotopy type?

背景链接：https://wwwf.imperial.ac.uk/~skdona/NONLINEARITYPROBLEMS.PDF

08138 | OWR-7155451-012

### Surgery Stability of Positive Ricci Curvature

O / open | Oberwolfach 报告问题汇编

Does a surgery theorem for positive Ricci curvature hold in the same generality as the Schoen-Yau and Gromov-Lawson surgery theorem for positive scalar curvature?

来源字段：https://doi.org/10.4171/owr/2021/30

08139 | OWR-13494-009

### Symbolic Containment for Roulleau-Urzúa Configurations

O / open | Oberwolfach 报告问题汇编

For the Roulleau-Urzúa configurations $C_n$ of cubics, with $I$ the ideal of their singular points, does $I^{(3)}\subseteq I^2$ hold for all $n$; and are there interesting analogous configurations of conics?

来源字段：https://doi.org/10.4171/owr/2015/9

08140 | AMR-064-0005

### Symmetries of Vector Distributions

O / open | AMR 问题清单汇编

A distribution is singular transitive if any two points can be connected by a concatenation of singular curves. Does singular transitivity imply that its symmetry group is a finite-dimensional Lie group?

背景链接：https://arxiv.org/abs/1304.2590

08141 | AIM-GEOMETRY-0200

### Symmetry in Carpenter's-rule motions and pointed pseudotriangulations

S / solved | AIM 研讨会问题汇编

**13. (Schulze)** Understand the following question: does the pseudo triangulation algorithm for the Carpenter's Rule give some unfolding that preserves symmetries? (Note that Connelly, Demaine, Rote have non-algorithmic positive solution).

来源字段：https://aimath.org/pastworkshops/linkagesproblems.pdf

08142 | AIM-GEOMETRY-0092

### Symmetry obstruction and invariant-subprogram collapse for the AIM lattice three-point bound

P / partially_solved | AIM 研讨会问题汇编

\[ \inf\Big\{f(0,0): f\in\mathcal S(\mathbb R^{2n}),\ \widehat f(0,0)=1,\ \widehat f\geq0,\ f\leq0\text{ on }C_2\Big\}^{1/2}, \]

\[ C_2=\Big\{(x,y):\|x\|,\|y\|,\|x-y\| \in\{0\}\cup[1,\infty)\Big\}\setminus\{(0,0)\}. \]

It says that this bounds optimal lattice sphere-packing center density, reports numerical equality with the Cohn--Elkies bound and apparent invariance under $O(n)\times O(n)$, and asks for an explanation and maps between the two programs.

1. **Why “three-point”?** The three points are $0,x,y$, whose three pairwise differences have norms $\|x\|,\|y\|,\|x-y\|$. Thus a scalar function on $\mathbb R^n\times\mathbb R^n$ can encode a three-point constraint. 2. **The square root is structural.** Poisson summation is over $\Lambda\times\Lambda$, so it produces the square of the reciprocal covolume. Taking a square root is correct. 3. **Density convention.** With minimum lattice-vector length normalized to 1, the displayed objective directly bounds the number density $\operatorname{covol}(\Lambda)^{-1}$. The conventional scale-invariant center density is $2^{-n}\operatorname{covol}(\Lambda)^{-1}$, and packing density is $\operatorname{vol}(B_{1/2}^n)\operatorname{covol}(\Lambda)^{-1}$. Thus factors must be restored according to convention. 4. **The stated group is not a symmetry.** The condition on $\|x-y\|$ is invariant under simultaneous rotations $(x,y)\mapsto(Ux,Uy)$, but not under independent rotations $(x,y)\mapsto(Ux,Vy)$. Therefore independent averaging is not legitimate for this program.

The capitalization of “Lattice” has no mathematical effect. A later primary source uses a related but different constraint that also requires $\|x+y\|\in\{0\}\cup[1,\infty)$; this is not silently inserted into the recovered AIM statement below.

We use the Fourier convention

\[ \widehat f(\xi,\eta)=\int_{\mathbb R^{2n}}f(x,y)\, e^{-2\pi i(x\cdot\xi+y\cdot\eta)}\,dx\,dy. \]

来源字段：http://aimpl.org/discreteaf/1/

08143 | AIM-GEOMETRY-0035

### Symplectic DWZ mutation for braid curve QPs and a quadratic-rank obstruction

P / partially_solved | AIM 研讨会问题汇编

Can mutations of quivers with potentials be realized as a symplectic operation? Could we use this to write down potentials for quivers from braid varieties?

来源字段：http://aimpl.org/clusterbraid/7/

08144 | OWR-12485-005

### Symplectic Measure of Two-Component Link Types

O / open | Oberwolfach 报告问题汇编

For a two-component link type $[L]$, let $A([L])$ be the infimum of the symplectic measure $A(K_1,K_2)$ over its representatives. Determine $A([L])$. Is it positive for every hyperbolic link?

来源字段：https://doi.org/10.4171/owr/2013/22

08145 | OWR-14297732-006

### Symplectic Structures and Einstein Metrics

O / open | Oberwolfach 报告问题汇编

For which additional four-manifolds can a symplectic structure reveal information about the existence or structure of Einstein metrics?

来源字段：https://doi.org/10.4171/owr/2023/29

08146 | OWR-5148-001

### SYZ Fourier Transforms between Mirror Calabi-Yau Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

For mirror Calabi-Yau manifolds, is there an explicit SYZ or fiberwise Fourier transformation that interchanges their symplectic and complex geometries?

来源字段：https://doi.org/10.4171/owr/2011/20

08147 | GEOM-013

### Tammes Problem

P / partially_solved | 独立 / 未分配集合

For n > 14 points (except n=24), what is the maximum minimum distance between points on a unit sphere?

背景链接：https://en.wikipedia.org/wiki/Tammes_problem

08148 | OWR-16942-002

### Tangent Flows and Type-Two Blow-Ups

P / partially_solved | Oberwolfach 报告问题汇编

When can tangent flows be related to blow-downs of Type II blow-ups in zero-Maslov Lagrangian mean curvature flow?

来源字段：https://doi.org/10.4171/owr/2019/30

08149 | AMR-037-0007

### Tangent pairs of pseudocircles

P / partially_solved | AMR 问题清单汇编

For $n$ pseudocircles in general position, determine the maximum number of tangent pairs and the maximum number of digon cells. Determine whether the worst-case bounds for tangencies and digons are equal for genuine circles.

背景链接：https://jeffe.cs.illinois.edu/open/

08150 | AMR-059-0010

### Tangent-Bundle Symplectic and Almost Complex Structures

O / open | AMR 问题清单汇编

Characterize the symplectic manifolds with compatible almost complex structure which are locally obtained from a tangent bundle $T(M)$ by combining the canonical symplectic form induced by a Riemannian metric with the almost complex structure determined by a compatible torsion-free affine connection.

背景链接：https://www.jstage.jst.go.jp/article/iis/4/2/4_2_125/_pdf/-char/en

08151 | AIM-GEOMETRY-0422

### Targeted punctured curves versus full SFT, with a Liouville-period bridge to Clifford boundary rigidity

P / partially_solved | AIM 研讨会问题汇编

With those corrections, the question asks how much of Eliashberg's proposed SFT proofs of (i) the Audin conjecture and (ii) weak boundary rigidity of the Clifford torus in \(S^3\) truly needs the then-unfinished foundations of symplectic field theory. It suggests replacing holomorphic curves with boundary on \(L\) by punctured curves asymptotic in \(T^*L\), reducing a recurring part of the analysis to curves in \(T^*(T^n)\). It also asks, without formulating a precise conjecture, whether CR structures can be useful.

The preceding questions fix the intended statements:

1. **Audin conjecture.** Every Lagrangian torus \(L\subset\mathbb C^n\) has minimal Maslov number \(N_L=2\). 2. **Weak boundary rigidity in dimension four.** Let \[ \Lambda=S^1(1/\sqrt2)\times S^1(1/\sqrt2) \subset \partial\overline B^4(1)=S^3. \] If a Lagrangian torus \(K\subset\overline B^4(1)\) is Hamiltonian isotopic to \(\Lambda\) through the ball, must \(K\subset S^3\)?

The official PDF and the nearby Questions 1.19 and 1.25 were checked directly. The phrase about CR structures has no sufficiently precise mathematical target, so no theorem is claimed about it here.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

08152 | OWR-16417-007

### Teichmüller Projections of Hitchin Components

O / open | Oberwolfach 报告问题汇编

Does there exist a mapping-class-group-equivariant projection from the Hitchin component for $\operatorname{PSL}(n,\mathbb{R})$ to Teichmüller space $T(\Sigma_g)$ whose fiber over a Riemann surface $X$ is $\bigoplus_{i=3}^n H^0(X,K_X^i)$?

来源字段：https://doi.org/10.4171/owr/2018/42

08153 | OWR-13346-006

### Tensor-Bundle Parametrization of Diffusion Densities

O / open | Oberwolfach 报告问题汇编

For diffusion PCA on a Riemannian manifold, prove or disprove the conjecture that the family $\Gamma$ of point-sourced diffusion densities can be parametrized by the bundle of symmetric positive-definite covariant tensors of order $2$.

来源字段：https://doi.org/10.4171/owr/2014/44

08154 | AIM-GEOMETRY-0010

### Tetrahedra are sharp among three-dimensional pyramids

P / partially_solved | AIM 研讨会问题汇编

Suppose \(\Omega\subset\mathbb R^3\) is convex, has volume \(1\), and has center of mass at the origin. If \[ A_{ij}=\int_\Omega x_i x_j\,dx, \] the ball minimizes \(\det A\). Which shape maximizes \(\det A\)? The scale-invariant functional in dimension \(n\) is \(\det A/|\Omega|^{n+2}\). In dimension two the maximizer is a triangle; is a simplex the maximizer in dimension three?

Here and below a convex body is compact, convex, and has nonempty interior. The archived source URL is <http://aimpl.org/symmetrybreaking/3/>.

There are two normalization points worth making explicit. For a convex body \(K\subset\mathbb R^n\), put \[ V_K=|K|,\qquad b_K=\frac1{V_K}\int_Kx\,dx, \] \[ M_K=\int_K(x-b_K)(x-b_K)^T\,dx, \qquad \Sigma_K=\frac{M_K}{V_K}. \] Then the source matrix \(A\) is \(M_K=\Sigma_K\) only when \(V_K=1\) and \(b_K=0\). The invariant objective is \[ \mathcal I(K):=\frac{\det M_K}{V_K^{n+2}} =\frac{\det\Sigma_K}{V_K^2}=:L_K^{2n}. \] For every invertible affine map \(x\mapsto Tx+a\), \[ M_{TK+a}=|\det T|\,T M_KT^T, \qquad \det M_{TK+a}=|\det T|^{n+2}\det M_K, \] so \(\mathcal I\), equivalently the isotropic constant \(L_K\), is invariant under *all* invertible affine maps. Thus equality shapes can only be unique up to affine equivalence.

来源字段：http://aimpl.org/symmetrybreaking/3/

08155 | OWR-8415364-010

### The $L_p$ Aleksandrov Problem for Integral Curvature Measures

P / partially_solved | Oberwolfach 报告问题汇编

For each real $p$, characterize the finite Borel measures on $S^{n-1}$ that occur as $L_p$ integral curvature measures of convex bodies.

来源字段：https://doi.org/10.4171/owr/2021/59

08156 | OWR-16637-008

### The $L_p$ Brunn-Minkowski Conjecture

O / open | Oberwolfach 报告问题汇编

For $0\leq p<1$, origin-symmetric convex bodies $K,C\subseteq\mathbb R^n$, and every $\lambda\in[0,1]$, is

$$V\bigl((1-\lambda)K+_p\lambda C\bigr)\geq V(K)^{1-\lambda}V(C)^\lambda,$$

where $+_p$ denotes the $L_p$-linear combination?

来源字段：https://doi.org/10.4171/owr/2018/54

08157 | AIM-GEOMETRY-0262

### The 2024 Lorentzian refutation and a proved symmetry-forced Morse criterion

S / solved | AIM 研讨会问题汇编

**Question 9.1.1.** Must every compact boundaryless Lorentzian manifold possess at least one nonconstant affinely closed geodesic?

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

08158 | AIM-GEOMETRY-0121

### The 6R lower bound and a three-contact regularity criterion

P / partially_solved | AIM 研讨会问题汇编

In 3 dimensions, it is known that if all $10R$ clusters are equivalent, the Delone set is regular.

Can this result be improved for $\rho < 10R$, e.g. $\rho = 4R$? Is it possible to show similar results for full dimensional $r$-sets with conditions weaker than the $R$ condition?

来源字段：http://aimpl.org/softpack/3/

08159 | AMR-071-0059

### The Atiyah conjecture on configurations on the invertibility of a certain $n$-by-$n$ matrix depending on $n$ points in $\mathbb{R}^{3}$

O / open | AMR 问题清单汇编

The Atiyah conjecture on configurations on the invertibility of a certain $n$-by-$n$ matrix depending on $n$ points in $\mathbb{R}^{3}$

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

08160 | AIM-GEOMETRY-0182

### The BCCD shrinker and a topological separation from FIK

P / partially_solved | AIM 研讨会问题汇编

**Reconstructed reading.** Are there complete, smooth, noncompact, non-rigid gradient shrinking Ricci solitons of real dimension four other than the Feldman--Ilmanen--Knopf (FIK) shrinker on \[ \widehat{\mathbb C^2}=\operatorname{Bl}_0(\mathbb C^2)? \]

来源字段：http://aimpl.org/flowriemannian/6/

08161 | AIM-GEOMETRY-0384

### The Branson cocycle and its Weyl-scale obstruction

P / partially_solved | AIM 研讨会问题汇编

Do Weyl structures help in this regard? Tom Branson has suggested that, for two metrics \(g\) and \(\widehat g=\Omega^2g\) in the same conformal class on a compact manifold \(M\), one should consider \[ \mathcal H[\widehat g,g]=\int_M (\log\Omega)(\widehat Q+Q). \] The cocycle identity \[ \mathcal H[\widehat{\widehat g},\widehat g] +\mathcal H[\widehat g,g] =\mathcal H[\widehat{\widehat g},g] \] is easily seen to be equivalent to self-adjointness of the GJMS operator \(P\).

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

08162 | GEO-014

### The Carathéodory Conjecture

O / open | 独立 / 未分配集合

别名：Carathéodory Conjecture [GEOM-014]

Does every convex, closed, twice-differentiable surface in $\mathbb{R}^3$ have at least two umbilical points?

背景链接：https://arxiv.org/abs/0808.0851

08163 | GEO-015

### The Cartan-Hadamard Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Cartan-Hadamard Conjecture [GEOM-015]

Does the isoperimetric inequality hold for Cartan-Hadamard manifolds?

背景链接：https://cordis.europa.eu/project/id/101001677/reporting

08164 | AIM-GEOMETRY-0234

### The closed case remains open and a finite-deck descent obstruction

P / partially_solved | AIM 研讨会问题汇编

**Question 5.6.1 (Miller, Reid).** Are there infinitely many simple closed geodesics in every hyperbolic three-manifold of finite volume?

One can ask the same question for manifolds of variable negative curvature in any dimension. The answer is positive for surfaces, since in this case a free homotopy class contains a simple closed geodesic if and only if it contains a simple closed curve. The answer is also positive for generic metrics of negative curvature. Reid has examples of (arithmetic) hyperbolic manifolds of finite volume in which every closed geodesic is simple.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

08165 | GEO-018

### The Closed Curve Problem

P / partially_solved | 独立 / 未分配集合

What are necessary and sufficient conditions for an integral curve defined by two periodic functions to be closed?

背景链接：https://doi.org/10.1007/s00022-025-00792-3

08166 | AMR-071-0025

### The covering problem of Rado

O / open | AMR 问题清单汇编

The covering problem of Rado: if the union of finitely many axis-parallel squares has unit area, how small can the largest area covered by a disjoint subset of squares be?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

08167 | AIM-GEOMETRY-0387

### The critical Q-density fibre and the kernel of P

P / partially_solved | AIM 研讨会问题汇编

Problem 27: When does Q determine the metric up to constant rescaling within a given conformal class? Since we know how Q changes under conformal rescaling:̂

Q = Q + P log Ω,

where P is a linear differential operator from functions to n-forms whose symbol is Δ n/ 2 this question is equivalent to

Problem 28: When does the equation \(Pf=0\) have only constant solutions?

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

08168 | AIM-CONVEX_GEOMETRY-0026

### The cube-projection core of the zonoid psi_2 problem

P / partially_solved | AIM 研讨会问题汇编

**Problem 7 (Grigoris Paouris).** Which bodies are \(\psi_2\)-bodies? Are zonoids \(\psi_2\)-bodies? Are projections of \(\psi_2\)-bodies \(\psi_2\)-bodies?

**Remark.** It is known that the \(B_p^n\) for \(p\geq2\) is \(\psi_2\).

来源字段：https://aimath.org/WWN/convexgeometry/convexgeometry.pdf

08169 | AMR-071-0026

### The Erdős-Oler conjecture

O / open | AMR 问题清单汇编

The Erdős-Oler conjecture: when $n$ is a triangular number, packing $n-1$ circles in an equilateral triangle requires a triangle of the same size as packing $n$ circles.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

08170 | AIM-GEOMETRY-0378

### The exact Pfaffian coefficient in conformally flat Q-curvature

S / solved | AIM 研讨会问题汇编

**Problem 20.** Find a direct link between \(Q\) and the Pfaffian in the conformally flat case. Prove directly that \(\int_M Q\) is a topological invariant in this case.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

08171 | GEO-019

### The Filling Area Conjecture

P / partially_solved | 独立 / 未分配集合

Does a hemisphere have minimum area among shortcut-free surfaces with a given boundary length?

背景链接：https://www.ihes.fr/~gromov/wp/site/wp-content/uploads/2018/08/fillingRiemannianManifolds.pdf

08172 | AIM-GEOMETRY-0013

### The Galashin-Lam comparison and a q=1 obstruction

P / partially_solved | AIM 研讨会问题汇编

To a quiver, we can associate a point count polynomial defined recursively (Galashin-Lam). This is sometimes equal to the HOMFLY-PT polynomial of the related link. For any simple plabic graphs, are these polynomials always equivalent?

来源字段：http://aimpl.org/clusterbraid/1/

08173 | OWR-14299092-007

### The General Total-Q-Curvature Inequality

P / partially_solved | Oberwolfach 报告问题汇编

Does a sharp isoperimetric inequality relate total $k$th and $\ell$th $Q$-curvatures whenever $0<\ell<k\leq n/2$?

来源字段：https://doi.org/10.4171/owr/2025/11

08174 | AIM-GEOMETRY-0058

### The geometric-potential transition and a pressure secant diagnostic

P / partially_solved | AIM 研讨会问题汇编

What can be said about interesting phase transitions for potentials $\{q\varphi\}_{q\in\mathbb R}$? In particular, let $M$ be a compact surface of nonpositive curvature, and $\varphi$ be the geometric potential, so that $q = 1$ is a phase transition.

来源字段：http://aimpl.org/equibdynsysgeom/3/

08175 | DARPA-015

### The Geometry of Genome Space

P / partially_solved | DARPA 挑战

Establish appropriate distance metrics on genome space incorporating biological utility.

背景链接：https://web.math.utk.edu/~vasili/refs/darpa07.MathChallenges.html

08176 | GEO-023

### The Hadwiger Covering Conjecture

O / open | 独立 / 未分配集合

别名：Hadwiger Conjecture (Covering) [GEOM-019]; Fractional Hadwiger [OPG-59908]

Can every $n$-dimensional convex body be covered by at most $2^n$ smaller homothetic copies?

背景链接：https://doi.org/10.1137/22M1490314

08177 | AIM-CONVEX_GEOMETRY-0028

### The half-mass hitting-time endpoint for lazy reversible chains

P / partially_solved | AIM 研讨会问题汇编

The PDF asks about lazy random walks on graphs or reversible Markov chains. Its intended worst-case total-variation mixing time is

\[ \tau_{\mathrm{TV}}(1/2) :=\min\left\{t\ge0: \max_{x\in\Omega}\lVert P^t(x,\cdot)-\mu\rVert_{\mathrm{TV}} \le\frac12\right\}. \tag{1} \]

The displayed question is whether a universal constant \(c>0\) exists such that

\[ \tau_{\mathrm{TV}}(1/2) \le c\max_{x,A:\,\mu(A)\ge1/2}\mathbb E_x T_A, \tag{2} \]

where \(T_A\) is the first hitting time of \(A\) from \(x\). We use the standard convention

\[ T_A=\inf\{t\ge0:X_t\in A\}; \tag{3} \]

thus \(T_A=0\) if the starting state already belongs to \(A\). The PDF writes this random time as \(T_{A,x}\) and then places an expectation in front of it.

The source sentence leaves the quantifier on \(x\) implicit; the standard mixing-time convention and the displayed maximum on the right make the worst-case interpretation (1) unambiguous. The finite-state, irreducible, discrete-time formulation is the one used in the paper that resolved the problem. Its precise laziness hypothesis is

\[ P(x,x)\ge\frac12 \quad\text{for every }x. \tag{4} \]

The PDF literally places \(\mu(A)\) before a maximum in the expression \(c\,\mu(A)\max_{x,A}\mathbb E T_{A,x}\), which leaves \(A\) free in a factor outside its own maximum. It is not a subscripted constant \(c_{\mu(A)}\). Aldous's theorem, and all later precise formulations, show that the intended quantity is

\[ t_{\mathrm{prod}} :=\max_{x,A:\,\mu(A)>0}\mu(A)\mathbb E_xT_A. \tag{5} \]

来源字段：https://aimath.org/WWN/convexgeometry/convexgeometry.pdf

08178 | AIM-GEOMETRY-0135

### The handlebody vacuum from meridian spectral projectors

P / partially_solved | AIM 研讨会问题汇编

Find the vector \(v_H\in Z(\Sigma)\) associated to a handlebody \(H\) bounding \(\Sigma\).

来源字段：http://aimpl.org/spectralhiggs/1/

08179 | OWR-3389-002

### The Hexagonal Conjecture for Spectral Minimal Partitions

O / open | Oberwolfach 报告问题汇编

Let $\mathcal L_k(S^2)$ be the optimal maximal first Dirichlet eigenvalue over $k$-partitions of the standard unit sphere, and let $\mathcal L_{k,1}(S^2)$ be the corresponding optimal arithmetic-mean energy. If $\operatorname{Hexa}_1$ is the unit-area regular hexagon, is the scale-corrected Hexagonal Conjecture true, namely $$\lim_{k\to\infty}\frac{\mathcal L_k(S^2)}{k}=\lim_{k\to\infty}\frac{\mathcal L_{k,1}(S^2)}{k}=\frac{\lambda(\operatorname{Hexa}_1)}{4\pi}$$?

来源字段：https://doi.org/10.4171/owr/2009/06

08180 | AIM-GEOMETRY-0420

### The Hofer diameter of equators and a bounded graphical chamber

P / partially_solved | AIM 研讨会问题汇编

This is Question 1.24 in Michael Hutchings's AIM outline *Holomorphic curves in contact geometry*, written with help from Yasha Eliashberg and John Etnyre.

\[ \operatorname{length}\{L_t\} :=\int_0^1\left(\max_{L_t}F_t-\min_{L_t}F_t\right)dt, \] and \[ d(L_0,L_1):=\inf_{\gamma}\operatorname{length}(\gamma), \] where \(F(x,t)\) generates the Hamiltonian isotopy \(\{L_t\}\) and the infimum is over Hamiltonian paths from \(L_0\) to \(L_1\). The source then asks for the diameter of the space \(\mathscr L\) of smooth simple closed curves in \(S^2\) dividing it into two equal-area regions. It also asks about the analogous orbit of \(\mathbb RP^2\subset\mathbb CP^2\) and records the graph map \[ \operatorname{Ham}(M,\omega)\longrightarrow \mathscr L(M\times M,\omega\oplus-\omega). \]

No mean-zero normalization of \(F_t\) is needed for the displayed length: adding any time-dependent spatial constant changes neither maximum minus minimum. More generally, two generators of the same unparametrized Lagrangian motion have restrictions to the connected curve \(L_t\) that differ by a constant, so the integrand is intrinsic to the exact Lagrangian path.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

08181 | AIM-GEOMETRY-0377

### The holographic conformally flat Q-curvature formula and a finite coefficient recurrence

S / solved | AIM 研讨会问题汇编

**Problem 19.** Find a formula for \(Q\) in the conformally flat case. Show that the procedure outlined by Eastwood and Slovák produces a formula for \(Q\).

In the conformally flat case, it follows from a theorem of Branson, Gilkey, and Pohjanpelto that \(Q\) must be a multiple of the Pfaffian plus a divergence.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

08182 | AIM-GEOMETRY-0358

### The holographic Q-curvature-Pfaffian relation and its four-dimensional defect

S / solved | AIM 研讨会问题汇编

**Problem 2:** Find an explicit relation between Q and Pff (R) in the conformally flat case.

“Problem 2: Find an explicit relation between \(Q\) and \(\operatorname{Pff}(R)\) in the conformally flat case.”

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

08183 | GEO-020

### The Hopf Conjectures

O / open | 独立 / 未分配集合

别名：Hopf Conjectures [GEOM-017]

What is the relationship between curvature and Euler characteristic for even-dimensional Riemannian manifolds?

背景链接：https://arxiv.org/abs/2507.16936

08184 | OWR-12337-001

### The Hyperplane Slicing Conjecture

S / solved | Oberwolfach 报告问题汇编

Hyperplane slicing conjecture: There exists a universal constant $c>0$ such that every convex body $K\subset\mathbb R^n$ of volume $1$ has a hyperplane section of $(n-1)$-dimensional volume at least $c$.

来源字段：https://doi.org/10.4171/owr/2013/09

08185 | GEO-007

### The Illumination Conjecture

O / open | 独立 / 未分配集合

别名：Illumination Problem [GEOM-008]

Can every convex body in $n$-dimensional space be illuminated by at most $2^n$ point light sources?

背景链接：https://en.wikipedia.org/wiki/Illumination_problem

08186 | GEO-003

### The Kakeya Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Kakeya Conjecture [GEOM-007]

A Kakeya set (containing a unit line segment in every direction) in $\mathbb{R}^n$ must have Hausdorff dimension $n$.

背景链接：https://www.polytechnique.edu/en/news/closer-look-kakeyas-conjecture

08187 | GEO-030

### The Kissing Number Problem

P / partially_solved | 独立 / 未分配集合

别名：Kissing Number Problem [GEOM-010]

What is the maximum number of non-overlapping unit spheres that can touch a central unit sphere in $n$ dimensions?

背景链接：https://doi.org/10.1007/s00454-026-00841-x

08188 | GEO-010

### The Knaster Problem

P / partially_solved | 独立 / 未分配集合

Can a solid cube be completely covered by finitely many smaller homothetic cubes with ratio less than 1, such that the interiors are disjoint?

背景链接：https://en.wikipedia.org/wiki/Cube

08189 | AMR-071-0050

### The Kobon triangle problem on triangles in line arrangements

O / open | AMR 问题清单汇编

The Kobon triangle problem on triangles in line arrangements

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

08190 | AMR-071-0051

### The Kusner conjecture

O / open | AMR 问题清单汇编

The Kusner conjecture: at most $2d$ points can be equidistant in $L^1$ spaces

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

08191 | AMR-071-0052

### The McMullen problem on projectively transforming sets of points into convex position

O / open | AMR 问题清单汇编

The McMullen problem on projectively transforming sets of points into convex position

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

08192 | AIM-GEOMETRY-0150

### The mirror of Higgs-field scaling on the regular Hitchin locus

P / partially_solved | AIM 研讨会问题汇编

What is the mirror of the action $\mathbb{C}^{\ast}\curvearrowright \mathcal{M}_{G_{\mathbb{C}}}$ in the Langlands dual side $\mathcal{M}_{^{L}G_{\mathbb{C}}}$?

来源字段：http://aimpl.org/spectralhiggs/1/

08193 | GEO-004

### The Moving Sofa Problem

P / partially_solved | 独立 / 未分配集合

What is the largest area of a shape that can be maneuvered through an L-shaped corridor of unit width?

背景链接：https://arxiv.org/abs/2407.02587

08194 | OWR-1056-004

### The Non-Locally-Conformally-Flat $\sigma_k$ Yamabe Problem

O / open | Oberwolfach 报告问题汇编

Solve the $\sigma_k$-Yamabe problem for $2<k\leq n/2$ on compact manifolds that are not locally conformally flat.

来源字段：https://doi.org/10.4171/owr/2005/35

08195 | GEO-021

### The Osserman Conjecture

P / partially_solved | 独立 / 未分配集合

Is every Osserman manifold either flat or locally isometric to a rank-one symmetric space?

背景链接：https://arxiv.org/abs/math/0204258

08196 | AIM-GEOMETRY-0382

### The Paneitz factor as a conformal gauge companion

P / partially_solved | AIM 研讨会问题汇编

**Problem 23.** Did we really need to go through that detailed calculation? What are the implications, if any, for the operator \(S:\) 1-forms \(\to\) 4-forms?

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

08197 | AIM-GEOMETRY-0379

### The Pfaffian decomposition of critical Q-curvature

S / solved | AIM 研讨会问题汇编

See Problem 4 for the 6 dimensional case. Also, T. Branson has appointed that if it is true that any local invariant \(L\) of density weight \(-n\) has the form \[ \operatorname{constant}_L\operatorname{Pff} +\operatorname{divergence}_L +(\operatorname{local\ conformal\ invariant})_L, \] where \(L\) signals the dependence on \(L\), then, in this decomposition for \(Q\), we have \(\operatorname{constant}_Q\ne0\). In fact we know \(\operatorname{constant}_Q\) exactly, since we know (the constant values of) \(Q\) and \(\operatorname{Pff}\) on the sphere.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

08198 | AIM-CONVEX_GEOMETRY-0022

### The planar resolution and a one-dimensional obstruction

P / partially_solved | AIM 研讨会问题汇编

Let \(K\subset\mathbb R^d\) (\(d\) fixed dimension) be a convex body. Let \(x_1,\ldots,x_n\) be uniformly random points from \(K\), and let \[ X:=\operatorname{vol}(\operatorname{conv}(x_1,\ldots,x_n)). \] Does \(X\) satisfy the central limit theorem?

来源字段：https://aimath.org/WWN/convexgeometry/convexgeometry.pdf

08199 | AIM-GEOMETRY-0125

### The punctured Hitchin-TUY comparison and a boundary residue test

S / solved | AIM 研讨会问题汇编

Extend Laszlo's proof of the relation between Hitchin's connection and TUY connection, to the case with punctures. Laszlo looks at the case with no marked points and group \(SU_n\), genus at least 3.

来源字段：http://aimpl.org/spectralhiggs/1/

08200 | AIM-GEOMETRY-0230

### The quarter-pinching endpoint and a spread obstruction

P / partially_solved | AIM 研讨会问题汇编

**Question 5.4.1.** Does there exist a closed Riemannian manifold of negative curvature such that \(E^s\) and \(E^u\) are \(C^1\) smooth, but the dimension is \(>2\) and the metric is not \(1/4\)-pinched?

The record is Question 5.4.1 in Keith Burns and Vladimir Matveev, *Open problems and questions about geodesics*.

The remainder of the record explains the motivation from Hasselblatt--Wilkinson's perturbative construction and observes that complex hyperbolic space has Lyapunov exponents \(1\) and \(2\), arising from parallel planes of curvature \(-1\) and \(-4\), respectively.

No mathematical content was silently changed. The following question about typical Lyapunov ratios and the subsequent Pollicott subsection are outside this record.

There is one important semantic ambiguity. If “\(1/4\)-pinched” means the **strict** inequality, then compact complex-hyperbolic quotients already answer the literal question affirmatively. The only nontrivial reading, and the reading used below, is that “not \(1/4\)-pinched” means worse than the **weak** quarter-pinching bound.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

08201 | AIM-GEOMETRY-0210

### The residual all-hyperbolic branch of Long's elliptic geodesic conjecture

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 2.2.3 (Long).** Every Finsler \(S^2\) has at least one elliptic prime closed geodesic.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

08202 | AIM-GEOMETRY-0157

### The scale-invariant stability functional and factor-splitting instability

P / partially_solved | AIM 研讨会问题汇编

Given a hypersurface $M\subset \mathbb{R}^{n+1}$, define $$F(M,x_0,t_0)=(4\pi t_0)^{-\frac n 2} \int_M e^{-\frac{|x-x_0|^2}{4t_0}} dH^{n},$$ $$\lambda(M) = \sup_{(x_0,t_0)} F(M,x_0,t_0).$$ Question: Is there a useful notion of $\mu$-stability for Ricci Flow self-shrinkers analogous to Colding-Minicozzi for the $\lambda$ entropy in mean curvature flow? What are the stable solitons?

来源字段：http://aimpl.org/flowriemannian/1/

08203 | AIM-CONVEX_GEOMETRY-0030

### The sharp Gaussian perimeter bound for ellipsoids

S / solved | AIM 研讨会问题汇编

**Problem 11 (Ryan O'Donnell).** Is it true that every ellipsoid has Gaussian surface area bounded by a universal constant?

**Remark.** Keith Ball has shown that for a general convex body the answer is \(O(n^{1/4})\); this bound is attained by intersection of random half-spaces.

来源字段：https://aimath.org/WWN/convexgeometry/convexgeometry.pdf

08204 | AIM-CONVEX_GEOMETRY-0007

### The sharp symmetric 2n-vertex case of the Euclidean-ball vertex-index conjecture

P / partially_solved | AIM 研讨会问题汇编

Prove that the vertex index of the \(n\)-dimensional Euclidean ball \(B_2^n\) is \(2n^{3/2}\), where the vertex index of a centrally-symmetric convex body \(K=-K\subset\mathbb R^n\) is defined as \[ \operatorname{vein}(K)=\inf\left\{\sum_i|p_i|_K:K\subset\operatorname{conv}\{p_i\}\right\}. \]

来源字段：https://aimath.org/pastworkshops/sectionsconvexproblems.pdf

08205 | GEO-012

### The Shephard's Problem

P / partially_solved | 独立 / 未分配集合

Can the unit ball in $\mathbb{R}^n$ be illuminated by fewer than $2^n$ directions?

背景链接：https://ems.press/content/serial-article-files/52513?nt=1

08206 | AIM-GEOMETRY-0132

### The solid-torus quantum handlebody ideal

P / partially_solved | AIM 研讨会问题汇编

Let $\Sigma, H$ as above.

i) $\mathcal{O}(\mathcal{M}_{B})\xrightarrow{def. quant.}\mathcal{A}_{q}$

ii) $\mathcal{L}\subset\mathcal{M}_{B}\Rightarrow \mathcal{I_{L}}\subset\mathcal{O}$ $(\mathcal{M}_{B})\xrightarrow{def. quant.}\mathcal{I_{L}^{q}}<\mathcal{A}_{q}$ (left ideal)

iii) $\mathcal{I_{L}^{q}}$ acts on $H^{0}(\mathbb{L}^{k},Bun_{G}(\Sigma))$ and this algebra annihilates $v_{H}\in H^{0}(\mathbb{L}^{k},Bun_{G}(\Sigma))$.

Therefore the following reconstruction is explicit rather than silently substituted.

* $\Sigma=\partial H$ is a closed oriented surface and $H$ is a handlebody. * $\mathcal M_B(\Sigma,G)=\operatorname{Hom}(\pi_1\Sigma,G)//G$ is the Betti character variety. * $\mathcal L_H\subset\mathcal M_B(\Sigma,G)$ is the locus of boundary local systems that extend over $H$. * One seeks a deformation quantization $\mathcal A_q$ of $\mathcal O(\mathcal M_B)$ and a quantum **left** ideal $\mathcal I_H^q$ deforming the classical vanishing ideal $\mathcal I_H$. * Under the quantum algebra action on the level-$k$ state space, normally written $H^0(Bun_G(\Sigma),\mathbb L^k)$, the ideal should annihilate the handlebody state $v_H$.

This report proves that program in the smallest nontrivial case: $G=\mathrm{SL}_2(\mathbb C)$ and $H=S^1\times D^2$. It also states carefully what transfers to a root-of-unity TQFT state and what remains unproved about the holomorphic-section realization.

来源字段：http://aimpl.org/spectralhiggs/1/

08207 | AIM-GEOMETRY-0208

### The solved Finsler two-or-infinity conjecture and an orbit-covering audit

S / solved | AIM 研讨会问题汇编

**Conjecture 2.2.1 (Long, Bangert, Problem 15 from [Alvarez2006]).** Every irreversible Finsler metric on \(S^2\) has either exactly two or infinitely many distinct prime closed geodesics.

Matveev, *Open problems and questions about geodesics*, an extended report from the August 2010 International Workshop on Geodesics [BM21]. The paper later appeared in *Ergodic Theory and Dynamical Systems* **41** (2021), 641--684.

After repairing line breaks and typography, the source statement is:

The following source paragraph says that Hofer--Wysocki--Zehnder's work on contact \(S^3\) implies this dichotomy for a bumpy Finsler metric on \(S^2\), provided stable and unstable manifolds of every hyperbolic closed geodesic intersect transversely; it also cites Harris--Paternain and notes that the two closed geodesics in Katok's example are elliptic [HWZ03, HP08].

- The corpus string `irre-versible` is only a line-break hyphen; the word is **irreversible**. - `S2` is \(S^2\); the 2 is a superscript in the PDF. - The citation key is `[Alvarez2006]`. Its bibliography entry is Juan Carlos **Álvarez Paiva**, *Some problems on Finsler geometry*, Handbook of Differential Geometry, vol. II (2006), pp. 1--33 [AP06]. The displaced acute accents near the name and just after the closing parenthesis in the PDF are typesetting artifacts, not mathematical notation. - The corpus ends with `4 KEITH BURNS AND VLADIMIR S. MATVEEV`. This is the running footer/header created at the page break, not part of the problem. - The paper calls the item Conjecture 2.2.1 and attributes it to Long and Bangert while identifying it with Problem 15 in Álvarez Paiva's list. The canonical number `15` should not be confused with the paper's section number.

The count is orientation-sensitive in the standard Finsler convention used by Bangert--Long: a prime closed geodesic is considered up to time translation and is not an iterate. For an irreversible metric, the reversed parametrization need not be a geodesic. If it is, it is counted separately [BL10]. “Prime” does not mean embedded in the base; a prime geodesic may self-intersect.

来源字段：https://aimath.org/pastworkshops/geodesicsproblems.pdf

08208 | GEO-013

### The Spherical Bernstein Problem

P / partially_solved | 独立 / 未分配集合

What is the classification of complete minimal hypersurfaces in spheres of all dimensions?

背景链接：https://people.math.harvard.edu/~hirolee/pdfs/2014-fall-230a-icm1970-chern-differential-geometry.pdf

08209 | AIM-GEOMETRY-0190

### The symplectic-volume proportion of unknotted spatial polygons

P / partially_solved | AIM 研讨会问题汇编

Problem 1 in that document supplies the missing notation. For a length vector $\ell=(\ell_1,\ldots,\ell_n)\in\mathbb R_{>0}^n$, let

$$ M_\ell= \left\{(u_1,\ldots,u_n)\in(S^2)^n: \sum_{i=1}^n\ell_i u_i=0\right\}/SO(3). $$

This is the moduli space of labeled closed spatial polygons with those edge lengths, modulo orientation-preserving Euclidean motions. Let $\Sigma$ be the locus of non-embedded polygons. The source writes

$$ M_\ell\setminus\Sigma=\bigsqcup_i U_i, $$

where the $U_i$ are connected components. Klyachko's symplectic reduction gives a symplectic form $\omega$ on a generic $M_\ell$, and hence Liouville volume. The intended question is to understand

$$ P_0(\ell):= \frac{\operatorname{Vol}_\omega \left(\bigsqcup_{i\in I_0}U_i\right)} {\operatorname{Vol}_\omega(M_\ell)}, \qquad I_0=\{i:U_i\text{ consists of unknots}\}. \tag{1} $$

Here “unknot” must mean an embedded polygon ambient-isotopic to a round circle. The equality printed in the PDF, $\ell_1+\cdots+\ell_{n-1}=\ell_n$, is genuinely an equality, not a lost inequality sign. It is nevertheless a degenerate boundary case, and its assertion that the proportion is exactly one is incompatible with the preceding embedded-polygon definition; see Section 8.

来源字段：https://aimath.org/pastworkshops/linkagesproblems.pdf

08210 | OWR-2654829-004

### The Tamed-to-Compatible Symplectic Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

If an almost-complex structure $J$ on a compact $4$-manifold is tamed by a symplectic form, does there exist a symplectic form compatible with $J$?

来源字段：https://doi.org/10.4171/owr/2020/33

08211 | AMR-071-0088

### The Thomson problem

P / partially_solved | AMR 问题清单汇编

The Thomson problem - what is the minimum energy configuration of $n$ mutually-repelling particles on a unit sphere?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

08212 | AMR-071-0067

### The values of the Hermite constants for dimensions other than 1-8 and 24

P / partially_solved | AMR 问题清单汇编

The values of the Hermite constants for dimensions other than 1-8 and 24

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

08213 | OWR-16417-004

### Thin Kontsevich-Zorich Monodromy Groups

O / open | Oberwolfach 报告问题汇编

How frequently are Kontsevich-Zorich monodromy groups thin, especially for square-tiled surfaces of genus 3; in particular, can one find thin monodromy groups among genus-3 square-tiled surfaces without one-cylinder decompositions?

来源字段：https://doi.org/10.4171/owr/2018/42

08214 | OWR-14219-010

### Thin-Shell Conjecture for Schatten Balls

O / open | Oberwolfach 报告问题汇编

For the unit balls of the Schatten classes $S_p^n$, does the thin-shell conjecture hold for the remaining values of $p$ not covered by the known large-$p$ results?

来源字段：https://doi.org/10.4171/owr/2016/11

08215 | OWR-14297736-033

### Third Neumann Eigenvalue Maximization Under Curvature Bounds

P / partially_solved | Oberwolfach 报告问题汇编

Among simply connected surfaces with curvature bounded above by $K$, prove that the third Neumann eigenvalue is maximized by a disjoint union of two constant-curvature-$K$ disks when $K\neq 0$.

来源字段：https://doi.org/10.4171/owr/2023/36

08216 | OWR-14297736-003

### Three Magnetic-Laplacian Optimization Problems

O / open | Oberwolfach 报告问题汇编

Determine the maximal domain for the second magnetic Neumann eigenvalue with Aharonov-Bohm flux; prove that a disk maximizes the constant-field ground state among simply connected domains of fixed area; and determine the ground-state asymptotics as an Aharonov-Bohm pole approaches the boundary.

来源字段：https://doi.org/10.4171/owr/2023/36

08217 | AIM-GEOMETRY-0359

### Three meanings of a global ambient metric

P / partially_solved | AIM 研讨会问题汇编

**Problem 3: Is there a global ambient metric construction?**

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

08218 | OWR-14298163-005

### Three Problems on Triangulations

O / open | Oberwolfach 报告问题汇编

Resolve the following triangulation problems: (1) do two triangulations with the same support always have a common iterated stellar subdivision; (2) are any two point configurations in $\mathbb R^3$ or $\mathbb R^4$ connected by bistellar flips; and (3) for every $d$, is there a constant $c_d$ such that $c_dP$ has a unimodular triangulation for every lattice polytope $P\subseteq\mathbb R^d$?

来源字段：https://doi.org/10.4171/owr/2024/3

08219 | OWR-15428-006

### Three-Component Tropical Approximation of Ultrametric Data

P / partially_solved | Oberwolfach 报告问题汇编

Given ultrametrics $d_i\in U_m$, choose $D^{(1)},D^{(2)},D^{(3)}\in U_m$ to minimize $$\sum_i d_{\mathrm{tr}}(d_i,d_i'),$$ where $d_i'$ is the tropical convex combination of the three $D^{(j)}$ with coefficients $\lambda_{ij}=\min(d_i-D^{(j)})$, and $$d_{\mathrm{tr}}(v,w)=\max_{k<\ell}|v_k-w_k-v_\ell+w_\ell|.$$ Characterize the minimizers and the minimum value.

来源字段：https://doi.org/10.4171/owr/2017/20

08220 | AMR-099-0030

### Three-dimensional sphere packing from a planar height function

O / open | AMR 问题清单汇编

Let $G$ be a planar graph circle-packed in $\mathbb{R}^2$ and let $f:V(G)\to\mathbb{Z}$ change by at most one across every edge. Add from each vertex $v$ an edge to a Euclidean-nearest vertex $u$ with $f(u)<f(v)$. Is the resulting graph the tangency graph of a sphere packing in $\mathbb{R}^3$?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=53

08221 | AIM-CONVEX_GEOMETRY-0008

### Three-dimensional vertex index and the choice of center

P / partially_solved | AIM 研讨会问题汇编

**Section 9, Problem 2.** Find the best possible universal upper bound for the vertex index of a three-dimensional centrally symmetric convex body.

For an origin-symmetric convex body \(K\subset\mathbb R^n\), the preceding Problem 1 defines \[ \operatorname{vein}(K) =\inf\left\{ \sum_{i=1}^{N}\|p_i\|_K: N<\infty,\quad K\subseteq\operatorname{conv}\{p_1,\ldots,p_N\} \right\}, \tag{1.1} \] where \[ \|x\|_K=\inf\{t>0:x\in tK\} \] is the norm whose unit ball is \(K\).

**Section 9, Problem 3.** Provide an upper bound, for a general convex body \(K\subset\mathbb R^n\), for the quantity displayed in the PDF as \[ \inf\left\{ \sum_{i=1}^{N}\|p_i\|_{K-a}: a\in K,\quad K-a\subseteq\operatorname{conv}\{p_1,\ldots,p_N\} \right\}. \tag{1.2} \] The PDF really says \(a\in K\), not \(a\in\operatorname{int}K\).

The heading “10 Reconstruction of Polytopes with Few Facets (or Few Vertices)” and the following sentence about estimating polytopes from random points begin the next section. They are context spill, not part of either vertex-index problem.

There is a genuine definitional issue in (1.2). If \(a\in\partial K\), then \(0\notin\operatorname{int}(K-a)\), so the usual Minkowski functional of \(K-a\) is not a finite gauge on all of \(\mathbb R^n\), much less a norm. We therefore keep two readings separate:

1. the **standard repaired reading**, with \(a\in\operatorname{int}K\); 2. the **literal boundary-allowed reading**, only after defining the functional as extended-valued, possibly \(+\infty\).

All positive structural results below use the standard repaired reading.

来源字段：https://aimath.org/pastworkshops/sectionsconvexproblems.pdf

08222 | OWR-1052-004

### Tire-Track Geometry and Cylindrical Logarithms

P / partially_solved | Oberwolfach 报告问题汇编

Find further examples in tire-track geometry and determine the two-dimensional analogue of the uniform cylindrical logarithm.

来源字段：https://doi.org/10.4171/owr/2005/31

08223 | OWR-793-009

### Tits’ Center Conjecture for Spherical Buildings

S / solved | Oberwolfach 报告问题汇编

Does every convex contractible subcomplex $Y$ of a spherical building $X$ have a point fixed by every automorphism of $X$ that stabilizes $Y$?

来源字段：https://doi.org/10.4171/owr/2005/12

08224 | OWR-2654829-008

### Topological Obstructions to Integrability of Geometric Structures

O / open | Oberwolfach 报告问题汇编

For a closed manifold equipped with an almost-complex structure $J$ or a nondegenerate $2$-form $\omega$, what additional topological restrictions follow from integrability? In particular, what topological restrictions does integrability of $J$ impose?

来源字段：https://doi.org/10.4171/owr/2020/33

08225 | OWR-17135-022

### Topology Algorithms and Curvature of Tropical Bisectors

O / open | Oberwolfach 报告问题汇编

Determine the topology of tropical bisectors of $k$ points in every dimension $d$, improve the expected $O(nd\log n)$ construction algorithm when possible, and define a useful tropical curvature for arbitrary polygonal space curves.

来源字段：https://doi.org/10.4171/owr/2019/39

08226 | OWR-1056-003

### Topology and Constant $\sigma_k$-Curvature Metrics

P / partially_solved | Oberwolfach 报告问题汇编

For $2\leq k<n/2$, characterize compact manifolds admitting a $k$-positive metric, identify topological obstructions to such metrics, and determine when a conformal class containing one also contains a metric of constant $\sigma_k$-curvature.

来源字段：https://doi.org/10.4171/owr/2005/35

08227 | OWR-14298163-020

### Topology Near the Systolic-Diameter Bound

O / open | Oberwolfach 报告问题汇编

For each $\varepsilon>0$, are there only finitely many topological types of Riemannian manifolds satisfying $\operatorname{systole}(M)/\operatorname{diam}(M)\geq 2-\varepsilon$?

来源字段：https://doi.org/10.4171/owr/2024/3

08228 | OWR-1275-010

### Topology of Baby Teichmüller Spaces

O / open | Oberwolfach 报告问题汇编

Let $B_n$ be the quotient, by $\operatorname{SL}_2(\mathbb R)$, of functions $f:\mathbb Z/n\mathbb Z\to\mathbb{RP}^1$ such that $f(i)\neq f(i+1)$ and whose image contains at least three points. What is the cohomology of $B_n$, and how many connected components does it have?

来源字段：https://doi.org/10.4171/owr/2006/26

08229 | AMR-063-0007

### Topology of Compact Manifolds with Holonomy G2

P / partially_solved | AMR 问题清单汇编

Which compact seven-manifolds admit a Riemannian metric with holonomy $G_2$?

背景链接：https://wwwf.imperial.ac.uk/~skdona/NONLINEARITYPROBLEMS.PDF

08230 | AMR-067-0010

### Toral manifolds and positive scalar curvature

P / partially_solved | AMR 问题清单汇编

Let $M$ be a connected closed manifold with finite fundamental group of odd order. Assume that the universal cover of $M$ admits a metric of positive scalar curvature. Does $M$ admit a metric of positive scalar curvature?

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

08231 | AMR-067-0011

### Toral manifolds and positive scalar curvature - Question 2

O / open | AMR 问题清单汇编

Let $M$ be a connected closed manifold admitting a metric of positive scalar curvature. Does this imply that $M$ is $p$-atoral for all odd $p$?

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

08232 | AMR-067-0027

### Totally geodesic submanifolds and positive curvature

P / partially_solved | AMR 问题清单汇编

Does Frankel's theorem hold for symmetric Finsler metrics?

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

08233 | AMR-054-0060

### Transforming Polygons via Vertex-Centroid Moves

P / partially_solved | AMR 问题清单汇编

Given an arbitrary polygon, transform it by a finite sequence of ``vertex-centroid'' moves to a regular polygon. A vertex-centroid move is a translation of a vertex $v$ along the line $vm$, where $m$ is the centroid of the vertices of the polygon, i.e., $1/n$-th of the sum of the vertex coordinates. Vertices may move only one at a time, but in any order and any number of times.

背景链接：https://topp.openproblem.net/p60

08234 | AIM-GEOMETRY-0402

### Transverse invariants via negative-stabilization coequalizers

P / partially_solved | AIM 研讨会问题汇编

Ko Honda spoke on results with Etnyre that certain knot types are not transversally simple. The proof is indirect and there is no invariant here. Eliashberg asked if an invariant can be constructed [to distinguish transversal knots].

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

08235 | OWR-1726-003

### Transverse Smoothness of Leafwise Harmonic Projections

O / open | Oberwolfach 报告问题汇编

For a Riemannian foliation, does the projection onto the leafwise harmonic forms always have a Schwartz kernel that is smooth in the transverse directions?

来源字段：https://doi.org/10.4171/owr/2007/43

08236 | AMR-054-0031

### Trapping Light Rays with Segment Mirrors

O / open | AMR 问题清单汇编

Is it possible to trap all the light from one point source by a finite collection of two-sided disjoint segment mirrors? A light ray is trapped if it includes no point strictly exterior to the convex hull of the mirrors. The source point is disjoint from the mirrors. Although several versions of the problem are possible, it seems to make the most sense to treat the mirrors as open segments (i.e., not including their endpoints), but demand that they are disjoint as closed segments.

背景链接：https://topp.openproblem.net/p31

08237 | AMR-054-0054

### Traveling Salesman Problem in Solid Grid Graphs

O / open | AMR 问题清单汇编

What is the complexity of finding a shortest tour in a solid planar grid graph? A planar grid graph is a graph whose vertices are any set of points on the planar integer lattice and whose edges connect every pair of vertices at unit distance. Distances between nodes correspond to induced shortest-path distances in the graph, which corresponds to ``Manhattan'' distances. A grid graph is solid if it does not have any holes, i.e., its complement in the planar integer lattice is connected.

背景链接：https://topp.openproblem.net/p54

08238 | OWR-17470-008

### Travelling-Salesman Girth and Metric Ordering Ratios

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Ordering-Ratio Dichotomy [OWR-17470-009]

For a metric space $M$ of infinite Assouad-Nagata dimension, is its travelling-salesman girth infinite? Consequently, for a uniformly discrete metric space $M$, is the following dichotomy true: either every order $T$ satisfies $\operatorname{OR}_{M,T}(k)=k-1$ for all $k\geq 1$, or some order $T$ satisfies $\operatorname{OR}_{M,T}(k)\leq C\log k$ for all $k\geq 1$ and some constant $C$?

来源字段：https://doi.org/10.4171/owr/2020/7

08239 | AMR-038-0005

### Triangulating a hypercube

P / partially_solved | AMR 问题清单汇编

Determine the minimum number of $d$-simplices needed to triangulate the $d$-dimensional cube, and its asymptotic growth with $d$.

背景链接：https://ics.uci.edu/~eppstein/junkyard/open.html

08240 | AMR-038-0016

### Triangulations with many distinct areas

O / open | AMR 问题清单汇编

Find the largest function $t(n)$ such that every convex $n$-gon has a triangulation containing at least $t(n)$ distinct triangle areas; also determine the lattice-vertex special case.

背景链接：https://ics.uci.edu/~eppstein/junkyard/open.html

08241 | AMR-067-0019

### Triple Bubble in $\mathbb{R}^3$

S / solved | AMR 问题清单汇编

Prove that the pictured standard triple soap bubble is the least-perimeter way to enclose and separate three given volumes in $\mathbb{R}^3$.

背景链接：https://www.imo.universite-paris-saclay.fr/~pansu/problems_MTDG.tex

08242 | AMR-071-0058

### Tripod packing

P / partially_solved | AMR 问题清单汇编

Tripod packing: how many tripods can have their apexes packed into a given cube?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

08243 | AIM-GEOMETRY-0340

### Tropical computation and spin-structure covariance of real curve counts in projective three-space

S / solved | AIM 研讨会问题汇编

Can this invariant be computed using tropical algebraic geometry?

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

08244 | AIM-GEOMETRY-0339

### Tropical determinantal bases and a 16-coordinate certificate quotient

P / partially_solved | AIM 研讨会问题汇编

Let \(I\subset \mathbb C[x_1,\ldots,x_n]\) be an ideal. Characterize or compute subsets \(J\subset I\) for which \[ \mathcal T(I)=\bigcap_{f\in J}\mathcal T(f). \] It records that the \(3\times3\) minors of a generic \(n\times n\) matrix suffice to define the tropical variety of their ideal, and asks (Sturmfels) whether the \(4\times4\) minors of a generic \(5\times5\) matrix likewise suffice. It then asks for a characterization of a smaller sufficient set that is easier or better to compute.

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

08245 | OWR-5148-002

### Tropical Limits of Spectral Curves in Mirror Symmetry

O / open | Oberwolfach 报告问题汇编

What is the mathematical meaning of the tropical limit of spectral curves arising in local mirror symmetry or large-$N$ Chern-Simons-matter theories?

来源字段：https://doi.org/10.4171/owr/2011/20

08246 | AIM-GEOMETRY-0327

### Tropical Riemann--Roch and a genus-one residual test

P / partially_solved | AIM 研讨会问题汇编

**A.7 What does the Riemann-Roch theorem say in the tropical world?**

Background: One way to approach this would be to build up the machinery of line bundles (or maybe coherent sheaves?). A different, more immediately geometric approach might be called the "Brill-Noether" approach. This requires only two ingredients: A. "plane curve with ordinary nodes" and B. If A, B and C are curves with a common point of intersection, what is "A intersect B minus A intersect C", the "residual intersection of C in A intersect B". If O_x is the local ring of x on A, and f_B, f_C are the images in O_x of the equations of B and C, then in the classical case the parts of the intersections A intersect B and A intersect C supported at x are represented by the ideals (f_B) and (f_C) in O_x, and the residual is represented by the ideal

(f_B:f_C) := {g in O_x | g f_C is in (f_B)}.

The early work on Riemann-Roch treated only the case where A is smooth at x. Then A intersect B at x is represented just by a multiplicity, and residuation is just subtraction. When A is arbitrary, things still work because O_x is a Gorenstein ring for any smooth curve, no matter how singular. Question: How do these notions play out for tropical plane curves? (contributed by David Eisenbud)

来源字段：https://aimath.org/WWN/amoebas/amoebas.pdf

08247 | OWR-1703876-008

### Tverberg Depth Bounds from Tukey Depth

O / open | Oberwolfach 报告问题汇编

For a finite point set $S\subset\mathbb{R}^d$ and $q\in\mathbb{R}^d$, is the Tverberg depth conjecturally bounded below by $\operatorname{TvD}(S,q)\geq\frac{1}{d-1}\operatorname{TD}(S,q)$ for $d\geq 2$, with this bound being tight?

来源字段：https://doi.org/10.4171/owr/2020/30

08248 | OWR-14298163-022

### Tverberg Regions and Random Great-Circle Cells

O / open | Oberwolfach 报告问题汇编

Resolve the following two questions. (1) For which $k$ is the region $D_k=\{q\in\mathbb R^d:\operatorname{TvD}(P,q)\geq k\}$ contractible for every $n$-point set $P\subseteq\mathbb R^d$? (2) For $n$ great circles in general position on $S^2$, choose a cell $c$ uniformly, remove the circles touching $c$, and let $x$ be the number of edges of the resulting cell containing $c$; how large can $\mathbb E[x]$ be?

来源字段：https://doi.org/10.4171/owr/2024/3

08249 | AIM-GEOMETRY-0183

### Twisted cylindrical quotients obstruct global line splitting

S / solved | AIM 研讨会问题汇编

For any dimension, is it true that a shrinker \((M,g)\) either is asymptotically conical, or it must split off a line?

来源字段：http://aimpl.org/flowriemannian/6/

08250 | AIM-GEOMETRY-0173

### Twisted-product pieces and the missing global-split branch

P / partially_solved | AIM 研讨会问题汇编

Let \(M^3\) be a closed manifold. Outside flat points, it splits locally as \(\Sigma\times\mathbb R\). Does there exist a collection of totally geodesic flat surfaces \(S_1,\ldots,S_k,\ldots\) such that \[ M\setminus(S_1\cup\cdots\cup S_k\cup\cdots) \text{ is }(\Sigma\times\mathbb R)/\text{group}? \] where \(\Sigma\) is a surface (with possibly finitely or infinitely many components) with boundary a union of closed geodesics.

Moreover, if \(M\) has nonnegative curvature, then is it diffeomorphic to a lens space?

来源字段：http://aimpl.org/flowriemannian/4/

08251 | OWR-14299905-031

### Two Low-Multiplicity Distances in Planar Point Sets

O / open | Oberwolfach 报告问题汇编

Given $n>4$ distinct points in the plane, must there be two distinct distances that each occur among at most $n$ pairs of points?

来源字段：https://doi.org/10.4171/owr/2026/1

08252 | AIM-GEOMETRY-0355

### Two-channel metric variation and a transverse Einstein certificate

P / partially_solved | AIM 研讨会问题汇编

**Problem 1a:** Given any functional of the metric that is well understood conformally, is there information that can arise going across conformal classes? If the functional is the integral of a local invariant we can obtain information by computing its anti-conformal variation. If the functional is a nonlocal spectral invariant, like the functional determinant, then it is even a challenge to compute the anti-conformal deformation.

来源字段：https://aimath.org/WWN/confstruct/confstruct.pdf

08253 | AIM-GEOMETRY-0056

### Two-observable pressure calibration and covariance rank

P / partially_solved | AIM 研讨会问题汇编

Let $\varphi_1,\varphi_2 : X \to \R$ be two potentials. What interesting analysis can be said about the 2-parameter family of equilibrium states $\mu_{q_1\varphi_1 + q_2\varphi_2}$? Can we find useful applications for these results?

来源字段：http://aimpl.org/equibdynsysgeom/3/

08254 | OWR-5158-003

### Typical Geometry and Systoles of Large-Genus Surfaces

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Teichmüller-Metric Question [OWR-5158-004]

What does a surface of large genus, or a typical such surface, look like? Which hyperbolic surface of a given topological type has the largest possible systole? What can be said about metrics on Teichmüller space from the geometry of hyperbolic surfaces?

来源字段：https://doi.org/10.4171/owr/2011/30

08255 | AMR-065-0008

### U. Grimm: Diffraction of a Pinwheel Tiling - Problem

O / open | AMR 问题清单汇编

Determine the position of sharp rings in the diffraction measure of a Pinwheel Tiling and their intensity.

背景链接：https://arxiv.org/abs/1604.06280

08256 | AMR-065-0009

### U. Grimm: Diffraction of a Pinwheel Tiling - Problem

O / open | AMR 问题清单汇编

Does the diffraction measure of the Pinwheel Tiling contain an absolutely continuous component?

背景链接：https://arxiv.org/abs/1604.06280

08257 | GEO-031

### Ulam's Packing Conjecture

P / partially_solved | 独立 / 未分配集合

Is the sphere the worst-packing convex solid?

背景链接：https://doi.org/10.1016/j.aim.2014.07.015

08258 | AMR-037-0001

### Unfolding convex polytopes

P / partially_solved | AMR 问题清单汇编

Does every three-dimensional convex polytope have a non-self-intersecting edge unfolding? Does a minimum spanning tree of the dual edge graph, with a natural dihedral-angle weighting, always specify such an unfolding? Does every convex polytope have a creased unfolding that cannot be refolded into a different convex polytope?

背景链接：https://jeffe.cs.illinois.edu/open/

08259 | AMR-064-0004

### Unfolding the Sub-Riemannian Distance

O / open | AMR 问题清单汇编

Find a $C^1$-classification of the germs of sub-Riemannian spheres at points of optimal singular curves for generic metrics. In particular, obtain such a classification for generic Martinet metrics in $\mathbb{R}^3$ and for the Engel distribution in $\mathbb{R}^4$.

背景链接：https://arxiv.org/abs/1304.2590

08260 | OWR-1194-005

### Unfoldings Along Arbitrary Geodesic Triangulations

O / open | Oberwolfach 报告问题汇编

For an arbitrary geodesic triangulation $T$ of a convex polytope boundary using its vertices, can the surface be unfolded without overlap by cutting along edges of $T$?

来源字段：https://doi.org/10.4171/owr/2006/12

08261 | OWR-14298589-013

### Uniform Bounds across Strata of Flat Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Extend the known lower bound to every hyperelliptic stratum. Establish corresponding upper and lower bounds for all strata. Determine whether the almost-everywhere qualification in the lower bound can be removed, or exhibit counterexamples.

来源字段：https://doi.org/10.4171/owr/2024/41

08262 | OWR-8415364-009

### Uniform Bounds for Volume-Decomposition Functionals

O / open | Oberwolfach 报告问题汇编

Let $P\subset\mathbb R^n$ be a polytope whose centroid is the origin. For $1\leq k\leq n-1$, does a constant $c(n,k)$ exist such that $$ \frac{X_k(P)}{V_n(P)}\leq c(n,k) $$ for every such $P$?

来源字段：https://doi.org/10.4171/owr/2021/59

08263 | OWR-1319-020

### Uniform Bounds on Closed Geodesics in Torus Homotopy Classes

O / open | Oberwolfach 报告问题汇编

Does there exist a Riemannian metric on $T^2$ and $n_0\in\mathbb N$ such that every homotopy class contains fewer than $n_0$ closed geodesics?

来源字段：https://doi.org/10.4171/owr/2006/33

08264 | OWR-13110-013

### Uniform Cover Decomposition Bounds for Convex Polygons

S / solved | Oberwolfach 报告问题汇编

For every integer $n>3$, does there exist an integer $m(n)$ such that every convex n-gon Q satisfies $m(Q)\leq m(n)$, where $m(Q)$ is the least multiplicity for which every m-fold covering of the plane by translates of Q decomposes into two coverings?

来源字段：https://doi.org/10.4171/owr/2014/40

08265 | AIM-CONVEX_GEOMETRY-0036

### Uniform point configurations and exact Fourier certificates for convexity

P / partially_solved | AIM 研讨会问题汇编

**7. (S. Robins)** Given \(N\), can we find explicitly \(N\) points on \(S^{n-1}\) that are “uniformly” arranged? Are they the same as the vertices of the polytope with \(N\) vertices and maximum volume that is contained in \(S^{n-1}\)?

**8.** Let \(f\) be a function on \(S^{n-1}\). Can we decide from the Fourier transform of \(f\) if \(f\) is the support function of a convex body? Or if \(f\) is the radial function of a convex body?

来源字段：https://aimath.org/WWN/fourierconvex/fourierconvex.pdf

08266 | OWR-15587-016

### Unimodality and Log-Concavity under Integer Decomposition

P / partially_solved | Oberwolfach 报告问题汇编

For a lattice polytope $P$, does the integer-decomposition property imply that $h^*(P;z)$ is unimodal or log-concave? Do either of these conclusions follow if $P$ is both Gorenstein and has the integer-decomposition property?

来源字段：https://doi.org/10.4171/owr/2017/44

08267 | OWR-15587-015

### Unimodality of Gorenstein Hilbert Series

P / partially_solved | Oberwolfach 报告问题汇编

If $R$ is a standard graded Gorenstein domain, is the $h$-polynomial of its Hilbert series unimodal? Does the same conclusion hold for every standard graded Cohen-Macaulay domain, or with log-concavity replacing unimodality?

来源字段：https://doi.org/10.4171/owr/2017/44

08268 | OWR-1595-010

### Unimodular Coverings from Face and Facet Conditions

O / open | Oberwolfach 报告问题汇编

Does pairwise face unimodularity of a lattice polytope, which implies normality, also imply the existence of a unimodular covering? Does facet unimodularity, which implies a regular unimodular triangulation, imply the existence of such a covering?

来源字段：https://doi.org/10.4171/owr/2007/39

08269 | OWR-11786-024

### Unimodular Triangulations of Smooth Lattice Polytopes

O / open | Oberwolfach 报告问题汇编

Does every smooth lattice polytope have a unimodular triangulation? For the specified smooth reflexive $7$-polytope with facets $x_2-2x_6-x_7\leq 1$, $2x_6-x_1-x_2\leq 1$, $2x_1-x_2-x_3-x_4-x_5+x_6\leq 1$, $x_2-x_6\leq 1$, and $x_i\leq 1$ for $i=1,\ldots,7$, does it have a regular unimodular triangulation, and is its toric ideal generated by quadrics?

来源字段：https://doi.org/10.4171/owr/2012/24

08270 | AIM-GEOMETRY-0124

### Uniqueness and descent in the Hitchin-Faltings projective-connection comparison

S / solved | AIM 研讨会问题汇编

Prove: [Hitchin's projectively flat connection over $\mathcal{M_{g}}$] $\cong$ [Faltings' projectively flat connection over $\mathcal{M_{g}}$]

来源字段：http://aimpl.org/spectralhiggs/1/

08271 | OWR-12182-004

### Uniqueness for Symmetric Cone-Volume Measures

O / open | Oberwolfach 报告问题汇编

For origin-symmetric convex bodies $K$ and $C$ with equal volume and cone-volume measures $V_K=V_C$, must there be direct-sum decompositions $K=K_1\oplus\cdots\oplus K_m$ and $C=C_1\oplus\cdots\oplus C_m$ with $K_i=\lambda_i C_i$ for some $\lambda_i>0$?

来源字段：https://doi.org/10.4171/owr/2012/59

08272 | OWR-4078-007

### Uniqueness of Genus-$g$ Helicoids

P / partially_solved | Oberwolfach 报告问题汇编

For every $g\geq1$, is there, up to congruence and scaling, exactly one properly embedded minimal helicoid of genus $g$?

来源字段：https://doi.org/10.4171/owr/2009/35

08273 | OWR-16942-006

### Uniqueness of Scalar-Flat Kähler ALE Metrics

O / open | Oberwolfach 报告问题汇编

For an ALE Kähler manifold admitting scalar-flat Kähler metrics, is the scalar-flat Kähler metric in a given Kähler class unique?

来源字段：https://doi.org/10.4171/owr/2019/30

08274 | AMR-063-0005

### Uniqueness of Symplectic Structures on Four-Manifolds

P / partially_solved | AMR 问题清单汇编

Is a symplectic structure $\omega$ on a four-manifold unique up to diffeomorphism when the elementary topological invariants $[\omega]$ and $c_1(M)$ are fixed?

背景链接：https://wwwf.imperial.ac.uk/~skdona/NONLINEARITYPROBLEMS.PDF

08275 | OWR-13493-001

### Uniqueness of the Bryant-Salamon $G_{2}$ Metric on the Spin Bundle

O / open | Oberwolfach 报告问题汇编

Is the asymptotically conical holonomy-$G_2$ metric on the spin bundle $S(S^3)$ unique?

来源字段：https://doi.org/10.4171/owr/2015/8

08276 | AIM-GEOMETRY-0097

### Universal equatorial critical points and the local-minimum correction

S / solved | AIM 研讨会问题汇编

Is it true that for $1 \leq k \leq d$, the only critical points for $d+k$ particles in $S^{d-1} \subset \mathbb{R}^d$ under some potential functions (logarthmic + others) are orthogonal unions of $k$ regular simplices?

来源字段：http://aimpl.org/discreteaf/1/

08277 | OWR-4075-004

### Universal Integral Relations for Closed Polygons

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathcal C_n$ be the space of closed $n$-gons, with monodromy integrals $I_j,J_j$, and define $w(j)=3j-k$ when $n=2k$ and $w(j)=3j-k+1$ when $n=2k+1$. Besides $\sum_j I_j=\sum_j J_j=3$, $\sum_j w(j)I_j=\sum_j w(j)J_j=0$, and $\sum_j w(j)^2(I_j-J_j)=0$, are there any universal relations among the $I_j,J_j$ on $\mathcal C_n$? For an inscribed polygon, is $I_j=J_j$ for every $j$? Is the induced Poisson map on twisted $n$-gons in $\mathbb{RP}^1$ completely integrable?

来源字段：https://doi.org/10.4171/owr/2009/32

08278 | OWR-14297736-027

### Universal Lower Bounds for Biharmonic Steklov Eigenvalues

O / open | Oberwolfach 报告问题汇编

Given $\alpha,M>0$, prove that constants $a,b>0$ depending only on $\alpha,M,n$ satisfy $\rho_k(\Omega)\geq ak^{2/(2n-1)}$ for every convex $\Omega\subset\mathbb R^n$ of volume $M$, and $\rho_k(\Omega)\geq bk^{1/n}$ for every such general domain.

来源字段：https://doi.org/10.4171/owr/2023/36

08279 | AIM-GEOMETRY-0081

### Universal optimality and magic-function obstructions for A2

P / partially_solved | AIM 研讨会问题汇编

Prove $A_2$ is universally optimal. Or find magic function for the sphere packing poblem in $\mathbb R^2$.

来源字段：http://aimpl.org/discreteaf/1/

08280 | OWR-2090-018

### Universal Optimality of Exceptional Spherical Codes

O / open | Oberwolfach 报告问题汇编

Are the experimentally identified balanced spherical codes with $40$ points in $\mathbb R^{10}$ and $64$ points in $\mathbb R^{14}$ universally optimal, meaning that they minimize every completely monotonic potential? Are there further universal optima?

来源字段：https://doi.org/10.4171/owr/2008/44

08281 | OWR-15219-007

### Universal Sharpness of Second-Level Five-Point Energy Relaxations

O / open | Oberwolfach 报告问题汇编

Is the second-level relaxation $E_2$ sharp for the energy minimization problem of five points on $S^2$ for every pair potential in the prescribed class?

来源字段：https://doi.org/10.4171/owr/2017/14

08282 | OWR-11786-025

### Universal Triangulations with Prescribed Boundary

O / open | Oberwolfach 报告问题汇编

For a simplicial $(d-1)$-sphere $T$ linearly embedded in $\mathbb{R}^d$, characterize universal triangulations, construct algorithmically a universal $d$-ball $U$ with prescribed boundary $T$, and bound the size of $U$. The case $d=2$ is known, but higher dimensions remain open.

来源字段：https://doi.org/10.4171/owr/2012/24

08283 | AIM-GEOMETRY-0107

### Unknown minimum dimension and a finite periodic witness reduction

P / partially_solved | AIM 研讨会问题汇编

**Recovered problem.** What is the **lowest** dimension $d$ for which there is a $d$-dimensional convex body $K$ whose translative packing density is strictly greater than its lattice packing density, i.e.

\[ \delta_T(K)>\delta_L(K)? \]

来源字段：http://aimpl.org/softpack/1/

08284 | OWR-14297736-013

### Unrestricted Spherical-Cap Maximization of the Second Neumann Eigenvalue

O / open | Oberwolfach 报告问题汇编

Prove that among all simply connected domains of any prescribed area in $S^2$, a spherical cap maximizes the second Neumann eigenvalue.

来源字段：https://doi.org/10.4171/owr/2023/36

08285 | AIM-GEOMETRY-0112

### Value stability without structural stability

P / partially_solved | AIM 研讨会问题汇编

Do results for soft packings stabilize to results for hard packings as $\lambda \rightarrow 0$?

来源字段：http://aimpl.org/softpack/1/

08286 | OWR-13106-004

### Vanishing of the $hat A$-Genus Under Nonnegative Curvature

O / open | Oberwolfach 报告问题汇编

Does the $\hat A$-genus vanish for every spin manifold that admits a metric of nonnegative sectional curvature?

来源字段：https://doi.org/10.4171/owr/2014/36

08287 | OWR-12875-003

### Veech Groups with Prescribed End Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a closed subspace of the Cantor set and let $G\leq\operatorname{GL}^+(2,\mathbb R)$ be a countable subgroup without contracting elements. Can $G$ be realized as the Veech group of a tame infinite-genus translation surface $S$ without planar ends and with $\operatorname{Ends}(S)\cong X$?

来源字段：https://doi.org/10.4171/owr/2014/15

08288 | AMR-054-0023

### Vertex $\pi$-Floodlights

P / partially_solved | AMR 问题清单汇编

How many $\pi$-floodlights are always sufficient to illuminate any polygon of $n$ vertices, with at most one floodlight placed at each vertex? An $\alpha$-floodlight is a light of aperture $\alpha$. (We consider here ``inward-facing'' floodlights, whose defining halfspace lies inside the polygon, locally in the neighborhood of the vertex. Other models of the problem allow general orientations of floodlights or restricted orientations (e.g., ``edge-aligned'').)

背景链接：https://topp.openproblem.net/p23

08289 | AMR-054-0042

### Vertex-Unfolding Polyhedra

P / partially_solved | AMR 问题清单汇编

Consider a polyhedron with simply connected facets (no holes on a facet) and without boundary (every edge is incident to exactly two facets). Can the polyhedron be cut along potentially all of its edges, but leaving certain faces connected at vertices, and unfolded into one piece in the plane without overlap? Such an unfolding is called a vertex-unfolding, to distinguish from widely studied edge-unfoldings (see Problem 9) and general unfoldings. An important subproblem here is whether all convex polyhedra have vertex-unfoldings; a negative answer would also resolve Problem Problem.9.

背景链接：https://topp.openproblem.net/p42

08290 | AIM-CONVEX_GEOMETRY-0029

### Vertex-weight perturbations do not preserve lazy mixing time

S / solved | AIM 研讨会问题汇编

\[ P_W(u,v)=\frac{W_v}{\sum_{z\in N(u)}W_z}\qquad(v\in N(u)). \]

The question is whether changing all vertex weights by a bounded multiplicative factor can change the mixing time by more than a bounded factor. The source remark says that the analogous assertion for the spectral gap is true (the PDF has the typographical error “spetral”).

The source does not specify aperiodicity. Ordinary discrete-time total-variation mixing is infinite on every connected bipartite graph for this non-lazy walk, regardless of the positive weights. We therefore use the standard \(1/2\)-lazy kernel

\[ L_W=\frac{I+P_W}{2} \]

and the worst-start total-variation mixing time at threshold \(1/4\). Changing the fixed threshold in \((0,1/2)\), or using continuous time, changes the quantities below only by universal factors. Under this standard interpretation the answer is **no**.

来源字段：https://aimath.org/WWN/convexgeometry/convexgeometry.pdf

08291 | OWR-4411-005

### Virtual Fibering of Arithmetic Hyperbolic Three-Manifolds

S / solved | Oberwolfach 报告问题汇编

Every arithmetic hyperbolic 3-manifold of quaternionic origin has a finite cover that fibers over the circle.

来源字段：https://doi.org/10.4171/owr/2010/30

08292 | AMR-037-0010

### Visibility complex of disjoint unit spheres

S / solved | AMR 问题清单汇编

Determine the combinatorial complexity of the visibility complex of $n$ pairwise disjoint unit spheres in three-dimensional space.

背景链接：https://jeffe.cs.illinois.edu/open/

08293 | OWR-15575-001

### Viterbo's Symplectic Capacity-Volume Conjecture

S / solved | Oberwolfach 报告问题汇编

For every convex domain $X\subset\mathbb R^{2n}$, is $$ c_{\mathrm{EHZ}}(X)^n\leq n!\operatorname{vol}(X) $$?

来源字段：https://doi.org/10.4171/owr/2017/32

08294 | OWR-15587-013

### Volume Bounds for Clean Lattice Tetrahedra

P / partially_solved | Oberwolfach 报告问题汇编

For a clean tetrahedron in $\mathbb{R}^3$ with $k>0$ interior lattice points, is Duong's volume conjecture true when the interior points are collinear? If they are not collinear, is the volume then strictly below the maximum?

来源字段：https://doi.org/10.4171/owr/2017/44

08295 | OWR-1595-012

### Volume Bounds for Holes in Very Ample Semigroups

O / open | Oberwolfach 报告问题汇编

Suppose a set of lattice points, not necessarily all lattice points of its convex hull, gives a very ample embedding, and let $P$ be the resulting lattice polytope. Can the height of the highest hole in the associated semigroup be bounded in terms of the normalized volume of $P$?

来源字段：https://doi.org/10.4171/owr/2007/39

08296 | OWR-1394-015

### Volume Bounds for Symmetric Lattice Polytopes

O / open | Oberwolfach 报告问题汇编

If $P$ is a $0$-symmetric lattice polytope in $\mathbb{R}^n$, then one conjectures that $\operatorname{vol}(P)\geq(2^{n-1}/n!)(G(\operatorname{int}P)+1)$, where $G(\operatorname{int}P)$ is the number of interior lattice points of $P$.

来源字段：https://doi.org/10.4171/owr/2006/56

08297 | OWR-15581-011

### Volume Envelopes of Cusped Hyperbolic Three-Manifolds

O / open | Oberwolfach 报告问题汇编

Can one prove an explicit upper bound for the asymptotic envelope function $\Omega_{\mathrm{cm}}(t)$ of normalized volumes of cusped hyperbolic $3$-manifolds?

来源字段：https://doi.org/10.4171/owr/2017/38

08298 | AMR-054-0062

### Volume Maximizing Convex Shape

O / open | AMR 问题清单汇编

Let $C$ be a convex piece of paper; its boundary may be a smooth curve, or a polygon. A perimeter halving folding is a folding of $C$ obtained by identifying two points $x$ and $y$ on the boundary of $C$ that halve the perimeter, and then folding $C$ by ``gluing'' $xy$ to $yx$. This always results in a unique convex shape in 3D, a polyhedron if $C$ is a convex polygon. What unit-area shape $C$ achieves the maximum volume possible via a perimeter-halving folding?

背景链接：https://topp.openproblem.net/p62

08299 | OWR-15587-017

### Volume of Minkowski Sums with Fixed Mixed Volume

P / partially_solved | Oberwolfach 报告问题汇编

For $d$-dimensional lattice polytopes $P_1,\ldots,P_d\subset\mathbb R^d$ with normalized mixed volume $V(P_1,\ldots,P_d)=m$, is the sharp upper bound for the normalized volume of $P_1+\cdots+P_d$ equal to $(m+d-1)^d$?

来源字段：https://doi.org/10.4171/owr/2017/44

08300 | OWR-14298170-005

### Volume-Above Distance-Below Without Fixed Topology

P / partially_solved | Oberwolfach 报告问题汇编

Suppose compact oriented Riemannian manifolds have a uniform diameter bound, volumes converging to that of a compact limit manifold, and metric tensors bounded below by factors converging to one relative to the limit metric. Prove volume-preserving intrinsic-flat convergence without assuming that the manifolds are diffeomorphic to the limit.

来源字段：https://doi.org/10.4171/owr/2024/9

08301 | AMR-054-0003

### Voronoi Diagram of Lines in 3D

O / open | AMR 问题清单汇编

What is the combinatorial complexity of the Voronoi diagram of a set of lines (or line segments) in three dimensions?

背景链接：https://topp.openproblem.net/p3

08302 | OWR-16160-009

### Warped-Cone Quasi-Isometries and Commensuration of Actions

O / open | Oberwolfach 报告问题汇编

Let $\Gamma\curvearrowright M$ and $\Lambda\curvearrowright N$ be finitely presented, free, minimal, isometric actions on closed manifolds, with neither group commensurable to a group having a nontrivial free abelian factor and with $\Gamma$ having Property FSL. If there are uniform quasi-isometries $M_{t_n}\to N_{s_n}$ for $t_n,s_n\to\infty$, must the two actions admit a commensuration?

来源字段：https://doi.org/10.4171/owr/2018/19

08303 | OWR-16160-011

### Warped-Cone Rigidity Without Property FSL

O / open | Oberwolfach 报告问题汇编

In coarse geometry, let $\mathcal{O}_\Gamma Y$ denote the warped cone associated to a measure-preserving action of a finitely generated group $\Gamma$ on a compact metric space $Y$. Can the technical condition $\mathrm{FSL}$ (finite subset of lengths / translation-like control) be removed from the warped-cone quasi-isometric rigidity theorem (Vigolo)?

来源字段：https://doi.org/10.4171/owr/2018/19

08304 | AIM-GEOMETRY-0415

### Weak boundary rigidity and strict escape for extremal ellipsoid tori

P / partially_solved | AIM 研讨会问题汇编

**Question 1.19.** (a) Let \(M\) be the unit ball in \(\mathbb R^4\) and let \[ L=\{p_1^2+q_1^2=p_2^2+q_2^2=1/2\}\subset\mathbb R^4 \] be the Clifford torus. Is \(L\) weakly boundary rigid? That is, if \(K\) is Hamiltonian isotopic to \(L\) in \(M\), then must \(K\subset\partial M\)? (There are several proofs that one cannot have \(K\subset M\setminus\partial M\).) [See Eliashberg's discussion at the end of the day.]

(b) Similarly investigate Lagrangian tori in other ellipsoids in \(\mathbb R^{2n}\), and explore nonremovable intersections in this setting.

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

08305 | OWR-4798-016

### Weak Knaster Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist an integer $n=n(\ell)$ such that for every set of $\ell$ points $X=\{x_{1},\ldots,x_{\ell}\}$ on $S^{n-1}$ and every continuous $f:S^{n-1}\to\mathbb{R}$, some rotation $\rho\in O(n)$ satisfies $f(\rho x_{1})=\cdots=f(\rho x_{\ell})$?

来源字段：https://doi.org/10.4171/owr/2011/08

08306 | OWR-15208-009

### Weak-Topology Dependence for Geometric Flows

O / open | Oberwolfach 报告问题汇编

Can continuous dependence on initial data for geometric flows be proved in a weak topology such as the Gromov-Hausdorff or $C^{\alpha}$ topology?

来源字段：https://doi.org/10.4171/owr/2017/3

08307 | OWR-12339-006

### Weakening General Position in Signed Simplex Representations

O / open | Oberwolfach 报告问题汇编

Can the requirement that every $(d+2)$-subset of the vertex set $V$ affinely span $\mathbb R^d$ be removed or weakened while retaining a signed representation of a compact polyhedron’s uniform measure by uniform measures on $(d+1)$-vertex simplices?

来源字段：https://doi.org/10.4171/owr/2013/11

08308 | AIM-GEOMETRY-0149

### What a parabolic de Rham Torelli theorem can recover

P / partially_solved | AIM 研讨会问题汇编

Torelli theorems on \(\mathcal M_{dR}\) with parabolic structure.

Under explicit hypotheses, does an **algebraic** isomorphism between two spaces (1.1), perhaps preserving the algebraic symplectic form or a natural determinant class, force an isomorphism of pointed curves \((X,D)\cong(Y,D')\)? Which of the rank, flags, residues, stability chamber, and determinant data can also be recovered?

It is a research program rather than a proposition with a truth value: it does not specify the curve, group, residues, weights, determinant, stability condition, or the structure that an isomorphism must preserve.

Here is a precise version of the program. This is an **added formulation**, not a silent replacement of the source. Let

- \(X\) be a smooth projective complex curve of genus \(g\), - \(D=p_1+\cdots+p_n\) a reduced marked divisor, - \(r\) the rank and \(\xi\) a determinant line bundle, - \(E_{p_i}=F_{i,0}\supsetneq\cdots\supsetneq F_{i,\ell_i}=0\) prescribed flag types, - \(\boldsymbol\lambda=(\lambda_{i,j})\) prescribed residue eigenvalues on the graded pieces, satisfying the trace/degree constraint, - \(\boldsymbol\alpha\) auxiliary generic stability weights, and - \(\nabla_\xi\) a fixed logarithmic determinant connection.

\[ \mathcal M_{dR}(X,D;r,\boldsymbol\alpha, \boldsymbol\lambda,\xi,\nabla_\xi) \tag{1.1} \]

for a coarse moduli space of stable logarithmic connections \((E,F_\bullet,\nabla)\) with those data. A meaningful marked Torelli question is:

The word “algebraic” is essential. A bare biholomorphism is too weak by the Riemann--Hilbert correspondence. Rank \(r=1\) must also be excluded from any unqualified marked Torelli theorem; Section 5 gives a sharp proof.

来源字段：http://aimpl.org/spectralhiggs/1/

08309 | AMR-071-0082

### What constant $1.1 < a \leq 10.76$ governs the lower bound of a closed knot $K$'s minimum ropelength $L(K) \geq a\operatorname{Cr}(K)^{3/4}$

P / partially_solved | AMR 问题清单汇编

What constant $1.1 < a \leq 10.76$ governs the lower bound of a closed knot $K$'s minimum ropelength $L(K) \geq a\operatorname{Cr}(K)^{3/4}$?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

08310 | AMR-071-0068

### What is the lowest number of faces possible for a holyhedron

O / open | AMR 问题清单汇编

What is the lowest number of faces possible for a holyhedron?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

08311 | AIM-GEOMETRY-0134

### What the Hitchin-Witten connection acts on

P / partially_solved | AIM 研讨会问题汇编

Describe the Hitchin-Witten connection on the space of holomorphic sections on the moduli space of Higgs bundles.

来源字段：http://aimpl.org/spectralhiggs/1/

08312 | AIM-GEOMETRY-0114

### Which potentials are mixtures of soft-union densities?

P / partially_solved | AIM 研讨会问题汇编

**Potentials**

There are other potentials that could reasonably describe a "soft packing." Can we fit them into this framework?

来源字段：http://aimpl.org/softpack/1/

08313 | AIM-GEOMETRY-0291

### Why fixed subtori miss the regular 3-Sasakian zero set

P / partially_solved | AIM 研讨会问题汇编

S1

Question 2.6 (C. Boyer) In the case of 3-Sasakian reduction by abelian groups, would it be possible to extract information about the cohomology of the reductions by using fixed points of smaller tori? The question is motivated by the fact that some cases of 3-Sasakian reductions of spheres by S1 is well-understood. Suppose there is a circle action on S4n+3, where the action is locally free on the whole sphere, and free on the zero level set. Then one can see explicitly that H∗

(S4n+3; Q) surjects onto H∗(S4n+3 ///S 1; Q), up to the middle dimension. The idea would be to look at the fixed point set M H for subtori H ⊂ T, where we start with the abelian group T acting on the 3-Sasakian manifold M. Note that the fixed point sets M H are contact, and for the Sasakian case, known examples of M H are spheres.

Comment 2.7 (C. Boyer) Main references for 3-Sasakian toral reductions are [BG99, BGM94, BGMR98].

3 Kirwan surjectivity for contact quotients

The official PDF, page 4, confirms the intended formula \[ H_{S^1}^*(S^{4n+3};\mathbb Q)\longrightarrow H^*(S^{4n+3}///S^1;\mathbb Q) \] and the notation \(M^H\). The final line “3 Kirwan surjectivity for contact quotients” is the next section heading, not part of Question 2.6.

The recovered question is therefore: can fixed sets \(M^H\) of connected subtori \(H\subset T\) be used to compute ordinary rational cohomology of a 3-Sasakian torus reduction, extending the understood circle reductions of spheres?

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

08314 | AMR-071-0027

### Wikipedia geometry item 27: The disk covering problem about finding the smallest real number $r(n)$ such that $n$ disks of…

O / open | AMR 问题清单汇编

The disk covering problem about finding the smallest real number $r(n)$ such that $n$ disks of radius $r(n)$ can be arranged in such a way as to cover the unit disk.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

08315 | AMR-071-0071

### Wikipedia geometry item 71: The Kelvin problem on minimum-surface-area partitions of space into equal-volume cells, and the…

P / partially_solved | AMR 问题清单汇编

The Kelvin problem on minimum-surface-area partitions of space into equal-volume cells, and the optimality of the Weaire-Phelan structure as a solution to the Kelvin problem

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

08316 | AMR-071-0078

### Wikipedia geometry item 78: Can every spherical non-convex polyhedron that tiles space by translation have its faces groupe…

O / open | AMR 问题清单汇编

Can every spherical non-convex polyhedron that tiles space by translation have its faces grouped into patches with the same combinatorial structure as a parallelohedron?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Geometry

08317 | OWR-11454-002

### Witten Spinor Proofs on Nonspin Asymptotically Flat Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Can Witten's proof of the positive mass theorem be adapted to asymptotically flat nonspin manifolds using singular spin structures and Witten spinors?

来源字段：https://doi.org/10.4171/owr/2011/56

08318 | AIM-GEOMETRY-0023

### Word-relative brick compactifications and cluster strata

P / partially_solved | AIM 研讨会问题汇编

Is there a natural way to compactify a braid variety so that the smaller strata also have cluster structures?

来源字段：http://aimpl.org/clusterbraid/4/

08319 | OWR-17473-003

### Wrapped Fukaya Categories as Crepant Resolutions

O / open | Oberwolfach 报告问题汇编

For a maximally degenerate positive Calabi-Yau pair $(M,D)$ admitting a weak section, do the functors $\pi^*$ and $\pi_*$ make $\mathcal W(X)$ a categorical crepant resolution of $SH^0(X)$? In particular, is $\mathcal W(X)$ relatively Calabi-Yau over $SH^0(X)$?

来源字段：https://doi.org/10.4171/owr/2020/10

08320 | AMR-055-0009

### Wu's Higher-Dimensional Picard Conjecture

S / solved | AMR 问题清单汇编

Let $B$ be a set of $n+2$ hyperplanes in general position in complex projective space $\mathbb{P}^n(\mathbb{C})$. Prove that there is no nondegenerate holomorphic map $\mathbb{C}^n\to\mathbb{P}^n(\mathbb{C})\setminus B$.

背景链接：https://people.math.harvard.edu/~hirolee/pdfs/2014-fall-230a-icm1970-chern-differential-geometry.pdf

08321 | OWR-10252937-004

### Yamabe Metrics with Prescribed Boundary Volume Ratio

O / open | Oberwolfach 报告问题汇编

Can the analysis be refined to produce a one-parameter family $\{g_T\}_{T>0}$ of Yamabe metrics satisfying $\operatorname{vol}(\partial M,g_T)/\operatorname{vol}(M,g_T)=T$?

来源字段：https://doi.org/10.4171/owr/2022/28

08322 | OWR-13106-009

### Yamabe Solutions on Products of Spheres

O / open | Oberwolfach 报告问题汇编

For $n,m\geq 2$ and $T>0$, consider the conformal class $[g_0^n+Tg_0^m]$ on $S^n\times S^m$. As $T$ varies, is the full space of solutions of the Yamabe equation connected? Does every solution lie on a branch bifurcating from a constant solution? Are there solutions that depend nontrivially on both sphere variables?

来源字段：https://doi.org/10.4171/owr/2014/36

08323 | GEOM-018

### Yau's Conjecture on First Eigenvalue

P / partially_solved | 独立 / 未分配集合

Is the first eigenvalue of the Laplace-Beltrami operator on an embedded minimal hypersurface of $S^{n+1}$ equal to $n$?

背景链接：https://arxiv.org/abs/1201.0666

08324 | GEO-022

### Yau's Conjecture on First Eigenvalues

P / partially_solved | 独立 / 未分配集合

Is the first eigenvalue of the Laplace-Beltrami operator on a minimal hypersurface in $S^{n+1}$ equal to $n$?

背景链接：https://arxiv.org/abs/2508.06123

08325 | AIM-GEOMETRY-0307

### Zero-Higgs restriction and the sharp rank-two pure ring

P / partially_solved | AIM 研讨会问题汇编

It also appends the beginning of Section 7, “Kernel computations for Kirwan maps,” which starts immediately after Example 6.22 in the source PDF and is not part of Conjecture 6.20.

The ambient assumptions are stated at the start of Section 6.5 of the PDF: \(C\) is the fixed complex curve (denoted \(\Sigma\) there), and the rank \(n\) and degree \(d\) are relatively prime. For the cohomological results below, assume that \(C\) is smooth projective of genus \(g\geq2\).

\[ \mathcal M^d_{\mathrm{Dol}}(GL_n) \quad\text{and}\quad \mathcal N^d(GL_n) \]

for the moduli spaces of stable rank-\(n\), degree-\(d\) Higgs bundles and stable vector bundles, respectively. With the standard normalized universal classes \(a_i\in H^{2i}\), define

\[ P_{\mathrm{Dol}}(n,d):=\mathbb Q[a_2,\ldots,a_n] \subset H^*(\mathcal M^d_{\mathrm{Dol}}(GL_n);\mathbb Q), \qquad P_{\mathrm{bun}}(n,d):=\mathbb Q[a_2,\ldots,a_n] \subset H^*(\mathcal N^d(GL_n);\mathbb Q). \]

The recovered conjecture says that these two graded rings are isomorphic, that this pure ring is independent of the coprime degree \(d\), and that its Poincaré polynomial \(PP_n(t)\) is determined by a partition generating series. A direct transcription of the typeset display, retaining its notation, is

\[ PV_n(t)=PP_n(t)\frac{t^{2(1-g)n(n-1)}}{t^2-1}, \qquad PZ_n(t,T)=\exp\!\left(\sum_{r\geq1}\frac{PV_n(t^r)T^r}{r}\right), \]

\[ PH^g_\lambda(t)=t^{4(1-g)n(\lambda')} \prod_{\substack{x\in d(\lambda)\\a(x)=0}} \frac{1}{1-t^{2h(x)}}, \qquad n(\lambda')=\sum_{z\in d(\lambda)}\ell(z), \]

\[ \prod_{n=1}^{\infty}PZ_n(t,T^n) =\sum_{\lambda\in\mathcal P}PH^g_\lambda(t)T^{|\lambda|}. \]

The PDF does not define every partition statistic in this immediate display, and \(n\) is used both for rank and as a product index. I therefore preserve the formula as a source transcription and do not use it in the proof below. In particular, I do not silently repair the hook-length convention or claim that a later formula in the literature is identical to this display without a term-by-term comparison.

来源字段：https://aimath.org/WWN/momentmaps/momentmaps.pdf

08326 | AIM-GEOMETRY-0399

### Zero-section intersections in a 1-jet space

P / partially_solved | AIM 研讨会问题汇编

**Question 1.1.** Tobias Ekholm asked [something approximating] the following: If \(L\subset J^1(M)\) is \(\{0\}\) cross the 0-section of \(T^*M\) and \(L'\subset J^1(M)\) is Legendrian isotopic to \(L\), must the projections of \(L\) and \(L'\) to \(T^*M\) intersect?

来源字段：https://aimath.org/WWN/contactgeom2/contactgeom2.pdf

08327 | AMR-054-0077

### Zipper Unfoldings of Convex Polyhedra

O / open | AMR 问题清单汇编

Does every convex polyhedron $P$ have a zipper unfolding? A zipper unfolding cuts open $P$ via a single path, necessarily a Hamiltonian path (to span all vertices), and unfolds the surface to a non-overlapping polygon in the plane. The segments of the path need not lie along edges of $P$.

背景链接：https://topp.openproblem.net/p77

08328 | AIM-CONVEX_GEOMETRY-0040

### Zonoid ratio in proportional-dimensional sections

P / partially_solved | AIM 研讨会问题汇编

Let \(K\) be a symmetric convex body. Define \[ \operatorname{zr}(K)= \min_{\substack{Z\in\mathcal Z\\ Z\subset K}} \frac{|K|^{1/n}}{|Z|^{1/n}}, \qquad \operatorname{vr}(K)= \min_{\substack{E\in\mathrm{ellipsoid}\\ E\subset K}} \frac{|K|^{1/n}}{|E|^{1/n}}. \] Clearly \(\operatorname{vr}(K)\geq \operatorname{zr}(K)\). Gordon and Junge proved that there is \(C_0>0\) such that every centrally symmetric \(K\subset\mathbb R^n\) has a section \(L\) of dimension \([n/2]\) with \[ \operatorname{zr}(L)\geq C_0\operatorname{vr}(K). \] Given \(0<\lambda<1\), what is the greatest \(f(\lambda)>0\) such that every \(n\geq3\) and every \(n\)-dimensional convex body \(K\) have a section \(L\) of dimension \([\lambda n]\) for which \[ \operatorname{zr}(L)\geq f(\lambda)\operatorname{vr}(K)? \]

来源字段：https://aimath.org/WWN/fourierconvex/fourierconvex.pdf

# 资料汇编 06 | 拓扑 / Topology

1,974 条记录。O=开放；P=部分解决；S=已解决。状态与英文/LaTeX 表述均沿用来源。

08329 | OWR-14299915-008

### $E_\infty$ Structures from Frobenius Lifts

P / partially_solved | Oberwolfach 报告问题汇编

For a flat spectrum $A$, characterize which commutative ring structures on $\pi_0(A)$ equipped with Frobenius lifts $\varphi_p:\pi_0(A)\to\pi_0(A)^\wedge_p$ arise from $E_\infty$ ring structures, and determine whether such extensions are unique.

来源字段：https://doi.org/10.4171/owr/2026/9

08330 | OWR-8415349-005

### $I$-Chain Models of Spaces in Coalgebras and Lie Algebras

O / open | Oberwolfach 报告问题汇编

Can $I$-chains be used to construct models of spaces in differential graded cocommutative coalgebras and Lie algebras?

来源字段：https://doi.org/10.4171/owr/2021/46

08331 | OWR-4341-003

### $L^2$ Eta Forms Without Determinant-Class Fibers

O / open | Oberwolfach 报告问题汇编

For a family of normal coverings and the rescaled Bismut superconnection $A_t$ of the family of signature operators, is $\operatorname{tr}_\Gamma\!\left(\tau\,\frac{dA_t}{dt}e^{-A_t^2}\right)$ integrable for $t\geq1$, so that the $L^2$ eta form is well-defined and continuous, without assuming that the fibers are of determinant class?

来源字段：https://doi.org/10.4171/owr/2010/28

08332 | OWR-1191-004

### $L^2$-Invariants of Finite-Volume Locally Symmetric Spaces

O / open | Oberwolfach 报告问题汇编

For finite-volume locally symmetric spaces, do the $L^2$-Betti numbers and $L^2$-torsion come entirely from the complete interior? More generally, extend the existing calculations first to rank-one and then to higher-rank locally symmetric spaces.

来源字段：https://doi.org/10.4171/owr/2006/09

08333 | OWR-5147-003

### $L^{2}$ Cohomology and Intersection Cohomology of Projective Varieties

P / partially_solved | Oberwolfach 报告问题汇编

For every projective variety $Y\subset\mathbb{CP}^N$, is the $L^2$ de Rham cohomology of its regular part $Y\setminus\operatorname{Sing}Y$, with the restricted Fubini-Study metric, naturally isomorphic to the middle-perversity intersection cohomology $IH^k(Y)$?

来源字段：https://doi.org/10.4171/owr/2011/19

08334 | AMR-108-0010

### 3.2 (Agol) - A minimal-Thurston-norm surface from a tree action

O / open | AMR 问题清单汇编

Let $M$ be a $3$-manifold whose fundamental group acts on a simplicial tree without global fixed points. In the covering space associated to an edge stabilizer, there is a unique $2$-dimensional homology class separating the two ends corresponding to the two ends of the tree minus that edge. Is there a surface in this class, of minimal Thurston norm, that embeds in $M$?

背景链接：https://arxiv.org/pdf/1512.04620#page=2

08335 | AMR-108-0011

### 3.3 (Agol) - Injective surfaces with only double curves

O / open | AMR 问题清单汇编

Does every closed hyperbolic $3$-manifold contain a closed $\pi_1$-injective surface with only double curves of intersection?

背景链接：https://arxiv.org/pdf/1512.04620#page=2

08336 | AMR-108-0012

### 3.4 (Agol) - Injective surfaces with the 1-line property

O / open | AMR 问题清单汇编

Does every closed hyperbolic $3$-manifold contain a closed $\pi_1$-injective surface with the 1-line property: in the universal cover, every pair of preimages of the surface intersects in a single line, with intersection of stabilizers isomorphic to $\mathbb{Z}$?

背景链接：https://arxiv.org/pdf/1512.04620#page=2

08337 | AMR-108-0013

### 3.5 (Agol) - Quasi-Fuchsian surfaces cubulating away from cusps

S / solved | AMR 问题清单汇编

Do cusped finite-volume hyperbolic $3$-manifolds have closed quasi-Fuchsian surfaces that cubulate except for the cusps? Equivalently in the stated geometric formulation, for every pair of points of $\partial_\infty\mathbb{H}^3$, should some lift of such a surface's limit set separate the pair?

背景链接：https://arxiv.org/pdf/1512.04620#page=2

08338 | AMR-108-0014

### 3.6 (Agol) - Virtual semi-fibering

O / open | AMR 问题清单汇编

Are finite-volume hyperbolic $3$-manifolds virtually semi-fibered?

背景链接：https://arxiv.org/pdf/1512.04620#page=2

08339 | AMR-108-0016

### 3.8 (Agol) - Twisted homology products after cutting along a surface

P / partially_solved | AMR 问题清单汇编

Let $M$ be a $3$-manifold, let $\phi:\pi_1M\to\mathbb{Z}$ be dual to $(\Sigma,\partial\Sigma)\subset(M,\partial M)$, and let $N$ be obtained by cutting $M$ along $\Sigma$. For which $\alpha\in\operatorname{Hom}(\pi_1M,SL_2(\mathbb{C}))$ is $N$ an $\alpha$-twisted homology product?

背景链接：https://arxiv.org/pdf/1512.04620#page=3

08340 | AMR-108-0026

### 5.1 (Agol) - Hyperbolic 3-manifolds with infinitely generated fundamental group

S / solved | AMR 问题清单汇编

Characterize hyperbolic $3$-manifolds with infinitely generated fundamental group. In particular, is there a $3$-manifold that is locally hyperbolic, has no infinitely divisible subgroup of its fundamental group such as $\mathbb{Q}$, but is not hyperbolic? Here locally hyperbolic means that every cover with finitely generated fundamental group admits a complete hyperbolic metric and is therefore tame.

背景链接：https://arxiv.org/pdf/1512.04620#page=4

08341 | AMR-108-0036

### 5.11 (Futer) - A combinatorial model with explicit bilipschitz constants

P / partially_solved | AMR 问题清单汇编

Build a combinatorial model for hyperbolic $3$-manifolds with explicit bilipschitz constants.

背景链接：https://arxiv.org/pdf/1512.04620#page=4

08342 | AMR-108-0040

### 5.15 (Gabai, Trnkova) - Ideal triangulations with arbitrarily many positive tetrahedra

O / open | AMR 问题清单汇编

Let $M$ be a hyperbolic $3$-manifold and let $n$ be any positive integer. Does $M$ admit an ideal triangulation with $m\geq n$ positively oriented tetrahedra?

背景链接：https://arxiv.org/pdf/1512.04620#page=5

08343 | AMR-108-0027

### 5.2 (Agol) - Large injectivity radius in hyperbolic homology manifolds

O / open | AMR 问题清单汇编

Do there exist fibered hyperbolic $3$-manifolds that are homology $S^2\times S^1$ and have arbitrarily large injectivity radius? Are there hyperbolic homology spheres of arbitrarily large injectivity radius?

背景链接：https://arxiv.org/pdf/1512.04620#page=4

08344 | AMR-108-0028

### 5.3 (Agol) - Thurston norm polytopes

P / partially_solved | AMR 问题清单汇编

Characterize the Thurston norm polytopes of finite-volume hyperbolic $3$-manifolds.

背景链接：https://arxiv.org/pdf/1512.04620#page=4

08345 | AMR-108-0029

### 5.4 (Agol) - Virtual CAT(0) cubical manifold models

O / open | AMR 问题清单汇编

Does every hyperbolic $3$-manifold have a finite-sheeted cover homeomorphic to a CAT(0) cube complex?

背景链接：https://arxiv.org/pdf/1512.04620#page=4

08346 | AMR-108-0030

### 5.5 (Agol) - Asymptotic frequency of small drilled manifolds

O / open | AMR 问题清单汇编

Fix $\mu$ below the three-dimensional Margulis constant. For hyperbolic $3$-manifolds of volume less than $V$, drill all closed geodesics of length less than $\mu$, discard duplicate resulting manifolds, and let $s(V)$ be the fraction of this finite collection that are small, meaning they contain no closed incompressible non-boundary-parallel surface. What is the limiting behavior of $s(V)$ as $V\to\infty$, and how does it depend on $\mu$?

背景链接：https://arxiv.org/pdf/1512.04620#page=4

08347 | AMR-108-0031

### 5.6 (Agol) - Cusp-preserving virtual domination

S / solved | AMR 问题清单汇编

If $M_1$ and $M_2$ are cusped hyperbolic $3$-manifolds, does there exist a cover $M'_1\to M_1$ and a nonzero-degree map $M'_1\to M_2$ taking cusps to cusps?

背景链接：https://arxiv.org/pdf/1512.04620#page=4

08348 | AMR-108-0054

### 7.6 (McMullen) - Totally geodesic surfaces and arithmeticity

S / solved | AMR 问题清单汇编

Let $M$ be a finite-volume hyperbolic $3$-manifold. If $M$ contains infinitely many immersed totally geodesic surfaces, must $M$ be arithmetic?

背景链接：https://arxiv.org/pdf/1512.04620#page=6

08349 | AMR-108-0055

### 8.1 (Agol) - Strongly irreducible Heegaard splittings of Haken manifolds

O / open | AMR 问题清单汇编

Do Haken hyperbolic $3$-manifolds have strongly irreducible Heegaard splittings?

背景链接：https://arxiv.org/pdf/1512.04620#page=6

08350 | AMR-108-0056

### 8.2 (Dunfield) - Profinite detection of knot complements

P / partially_solved | AMR 问题清单汇编

For a hyperbolic $3$-manifold with torus boundary, does its profinite completion determine whether it is a knot complement?

背景链接：https://arxiv.org/pdf/1512.04620#page=6

08351 | AMR-108-0059

### 8.5 (Schleimer) - Classification of strongly irreducible Heegaard splittings

P / partially_solved | AMR 问题清单汇编

Is there a classification of the strongly irreducible Heegaard splittings of a given $3$-manifold?

背景链接：https://arxiv.org/pdf/1512.04620#page=6

08352 | AMR-108-0061

### 8.7 (Tillmann) - Higher-dimensional multisections and stabilization

P / partially_solved | AMR 问题清单汇编

Do higher-dimensional smooth manifolds always admit multisections? What is the correct generalization of uniqueness up to stabilization for multisections of smooth $n$-manifolds when $n\geq5$?

背景链接：https://arxiv.org/pdf/1512.04620#page=6

08353 | AMR-108-0062

### 8.8 (Taylor) - Hyperbolic knots not arising from complicated bands

O / open | AMR 问题清单汇编

A band joining the components of a two-component link $L\subset S^3$ is called complicated if either its core cannot be isotoped to meet a splitting sphere in fewer than three points when $L$ is split, or its core cannot be isotoped disjoint from any minimal-genus Seifert surface for $L$. Give an example of a hyperbolic knot that cannot be created by attaching a complicated band to a two-component link.

背景链接：https://arxiv.org/pdf/1512.04620#page=6

08354 | AIM-TOPOLOGY-0187

### A base-visitation criterion for dense directed Thurston geodesics

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.9.** Is there a dense Thurston geodesic in moduli space?

Masur used closed Teichmüller geodesics to approximate a Teichmüller geodesic which is dense in moduli space.

Does there exist a **directed** geodesic line $\gamma:\mathbb R\to\mathcal T(S)$, meaning $d_{\rm Th}(\gamma(s),\gamma(t))=t-s$ whenever $s<t$, such that $\pi(\gamma(\mathbb R))$ is topologically dense in $\mathcal M(S)$?

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08355 | AIM-TOPOLOGY-0062

### A block-peeling recursion for an algebraic family of torus-knot cables

P / partially_solved | AIM 研讨会问题汇编

Compute the Khovanov-Rozansky homology $HHH$ of cables of torus knots and develop a recursion for this class of knots.

来源字段：http://aimpl.org/agclinkhom/2/

08356 | AIM-TOPOLOGY-0026

### A boundary-relative obstruction to a Liouville existence h-principle

P / partially_solved | AIM 研讨会问题汇编

**Problem (attributed on the source page to Oleg Lazarev).** Do Liouville domains of dimension at least six satisfy an existence h-principle?

来源字段：http://aimpl.org/highdimcontacttop/10/

08357 | AIM-TOPOLOGY-0164

### A braided Thompson counterexample to the quasimorphism--amenability dichotomy

S / solved | AIM 研讨会问题汇编

What about bounded cohomology? In $\mathbb{R}^2-C$, is it true that every subgroup of $\operatorname{Map}(S)$ has either infinite-dimensional space of quasimorphisms or is amenable?

来源字段：http://aimpl.org/genusinfinity/4/

08358 | AIM-TOPOLOGY-0050

### A Cartan--Perron constraint in rank two

P / partially_solved | AIM 研讨会问题汇编

**Classification of non-semi simple modular categories**

Classify non semi-simple modular categories for low rank (rank $\geq 2$) where the rank denotes the number of simple objects.

来源字段：http://aimpl.org/quantumlowdimtop/3/

08359 | AIM-TOPOLOGY-0182

### A certified short-curve criterion and punctured-torus itinerary

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.4 (K. Rafi).** Identify curves that are short along the preferred path (G_{X,Y}).

F. Kassel and K. Rafi prove that, for any (X,Y\in\mathcal T(S)), there is a Thurston geodesic (G_{X,Y}), parametrized linearly in Thurston's shear (cataclysm) coordinates associated with a canonical lamination (\lambda(X,Y)), such that lengths of all simple closed curves along (G_{X,Y}) are convex functions (up to reparametrization).

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08360 | AIM-TOPOLOGY-0136

### A character-corrected bipolar-depth metric on topologically slice concordance

P / partially_solved | AIM 研讨会问题汇编

**Metrics on $\mathcal T$.** Construct an interesting (i.e. non-discrete, perhaps nice with respect to the bipolar filtration) metric on the set of topologically slice knots.

来源字段：http://aimpl.org/concordsliceknot/5/

08361 | AIM-TOPOLOGY-0037

### A coefficient-diagonal volume conjecture for the Gukov-Manolescu invariant

P / partially_solved | AIM 研讨会问题汇编

**State and prove the volume conjecture for the Gukov-Manolescu invariant.**

来源字段：http://aimpl.org/quantumlowdimtop/1/

08362 | AIM-TOPOLOGY-0065

### A coefficient-sensitive integral computation for positive two-strand torus links

P / partially_solved | AIM 研讨会问题汇编

Determine \(Kh^{i,j}(T^+_{m,n};R)\), including torsion when \(R=\mathbb Z\), for all positive integers \(m,n\), and relate mirrors by duality.

来源字段：http://aimpl.org/agclinkhom/3/

08363 | AIM-TOPOLOGY-0124

### A compression-ladder reduction and category audit for stable four-genus

P / partially_solved | AIM 研讨会问题汇编

**Stable 4-genus and 2-torsion.** Find a knot \(K\) which has infinite order in \(\mathcal C\) and yet which has \(g_4^{st}(K)=0\).

The preceding canonical record defines \[ g_4^{st}(K)=\lim_{n\to\infty}\frac{g_4(nK)}{n} \] “in either category.” This creates a category issue that must be resolved explicitly. Standard usage in this workshop makes unadorned \(\mathcal C\) and \(g_4\) smooth. Thus the intended reading is:

* \(K\) has infinite order in the smooth concordance group \(\mathcal C_{\mathrm{sm}}\); and * its smooth stable four-genus \(g_{4,\mathrm{sm}}^{st}(K)\) is zero.

There is also a coherent topological version, with both terms interpreted locally flat topologically, and an incoherent “mixed” version using smooth order but topological four-genus. The mixed version has easy explicit solutions (Section 5); they do not solve the intended problem.

来源字段：http://aimpl.org/concordsliceknot/3/

08364 | AIM-TOPOLOGY-0055

### A concrete grading bridge between quantum gl(1|1) homology and knot Floer homology

P / partially_solved | AIM 研讨会问题汇编

Make \(gl(1|1)\)-homology concrete and relate it to knot/tangle Floer homology.

来源字段：http://aimpl.org/agclinkhom/1/

08365 | AIM-TOPOLOGY-0053

### A cone-lift criterion for categorifying the lens-space skein quotient

P / partially_solved | AIM 研讨会问题汇编

**Categorification of Kauffman Bracket Skein Module**

Categorify Kauffman Bracket Skein Module of Lens space L(p,1)

来源字段：http://aimpl.org/quantumlowdimtop/4/

08366 | OPG-37540

### A conjecture about direct product of funcoids

P / partially_solved | Open Problem Garden

Conjecture Let $f_1$ and $f_2$ are monovalued, entirely defined funcoids with $\operatorname{Src}f_1=\operatorname{Src}f_2=A$. Then there exists a pointfree funcoid $f_1 \times^{\left( D \right)} f_2$ such that (for every filter $x$ on $A$ ) $$\left\langle f_1 \times^{\left( D \right)} f_2 \right\rangle x = \bigcup \left\{ \langle f_1\rangle X \times^{\mathsf{FCD}} \langle f_2\rangle X \hspace{1em} | \hspace{1em} X \in \mathrm{atoms}^{\mathfrak{A}} x \right\}.$$ (The join operation is taken on the lattice of filters with reversed order.)

A positive solution of this problem may open a way to prove that some funcoids-related categories are cartesian closed.

背景链接：http://www.openproblemgarden.org/op/a_conjecture_about_direct_product_of_funcoids

08367 | AIM-TOPOLOGY-0088

### A convention-robust rank-two superspace coinvariant basis

P / partially_solved | AIM 研讨会问题汇编

Find a basis of the coinvariant ring of \(\mathbb C[x_1,\ldots,x_n,y_1,\ldots,y_n,\theta_1,\ldots,\theta_n,\phi_1,\ldots,\phi_n]\) and a combinatorial model for the Frobenius characteristic in the \(\mathbb C[x_1,\ldots,x_n,y_1,\ldots,y_n,\theta_1,\ldots,\theta_n]\) case; does the basis proposed by Haglund--Sergel work?

来源字段：http://aimpl.org/agclinkhom/9/

08368 | AIM-TOPOLOGY-0052

### A cyclic Euler--Fourier obstruction for the volume conjecture

P / partially_solved | AIM 研讨会问题汇编

**Categorification of CJP**

Categorify Colored Jones Polynomial to study the volume conjecture.

来源字段：http://aimpl.org/quantumlowdimtop/4/

08369 | AIM-TOPOLOGY-0015

### A decorated cocore criterion for trivial bypasses

P / partially_solved | AIM 研讨会问题汇编

**Can we characterize trivial bypasses in terms of the belt sphere of \(n\)-handle and the attaching sphere of the \(n+1\) handle?**

Let \((M^{2n+1},\xi)\) have Weinstein-convex boundary \(\Sigma^{2n}=R_+\cup\Gamma\cup R_-\). Attach a contact \(n\)-handle \(H_n\), followed by a contact \((n+1)\)-handle \(H_{n+1}\), in Honda--Huang bypass position. Which extra contact-geometric decoration on the smooth pair \((B^n,A^n)\) detects that the bypass data is of type (TB1) or (TB2), and hence that the resulting layer is vertically invariant relative to the boundary? Does the unadorned one-point intersection suffice?

来源字段：http://aimpl.org/highdimcontacttop/7/

08370 | AIM-TOPOLOGY-0142

### A determinant-sensitive transfer theorem for directional measures

P / partially_solved | AIM 研讨会问题汇编

Given a pseudo-Anosov acting on a translation surface, what can be said about the measured laminations in the stable and in the unstable direction? In general, "How much of Thurston's notes goes through?"

\(f\) is an affine automorphism of an infinite-type translation surface whose derivative has a real expanding and a real contracting eigendirection.

来源字段：http://aimpl.org/genusinfinity/1/

08371 | AIM-TOPOLOGY-0086

### A diagonal-normal Ext calculation and diagrammatic gate

P / partially_solved | AIM 研讨会问题汇编

Let $B_w\in$ SBim$_n$ be the indecomposable Soergel bimodule associated to $w\in S_n$. Compute Ext$_{R-R \text{ bimod}}(B_v,B_w)$ and develop its diagrammatics.

来源字段：http://aimpl.org/agclinkhom/9/

08372 | OPG-60024

### A diagram about funcoids and reloids

P / partially_solved | Open Problem Garden

Define for posets with order $\sqsubseteq$:

- $\Phi_{\ast} f = \lambda b \in \mathfrak{B}: \bigcup \{ x \in \mathfrak{A} \mid f x \sqsubseteq b \}$;

- $\Phi^{\ast} f = \lambda b \in \mathfrak{A}: \bigcap \{ x \in \mathfrak{B} \mid f x \sqsupseteq b \}$.

Note that the above is a generalization of monotone Galois connections (with $\max$ and $\min$ replaced with suprema and infima).

Then we have the following diagram:

What is at the node "other" in the diagram is unknown.

Conjecture "Other" is $\lambda f\in\mathsf{FCD}: \top$.

Question What repeated applying of $\Phi_{\ast}$ and $\Phi^{\ast}$ to "other" leads to? Particularly, does repeated applying $\Phi_{\ast}$ and/or $\Phi^{\ast}$ to the node "other" lead to finite or infinite sets?

背景链接：http://www.openproblemgarden.org/op/a_diagram_about_funcoids_and_reloids

08373 | AIM-TOPOLOGY-0132

### A diagrammatic smoothly non-approximable topological slice disk

S / solved | AIM 研讨会问题汇编

**Understanding topological slice discs.** Explicitly describe some non-smooth topologically slice disc.

来源字段：http://aimpl.org/concordsliceknot/5/

08374 | AIM-TOPOLOGY-0191

### A directional reduction for the Thurston quasi-isometry group

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.13.** What is the quasi-isometric group of Teichmüller space equipped with the Thurston metric?

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08375 | AIM-TOPOLOGY-0067

### A double-line test package for geometric Khovanov homology

P / partially_solved | AIM 研讨会问题汇编

For Khovanov homology develop analogues of

a) The Oblomkov--Rasmussen--Shende conjecture.

b) Braid varieties.

c) \(\operatorname{Hilb}^n(\mathbb C^2)\) should be \(\operatorname{Hilb}^n(x^2=0)\).

来源字段：http://aimpl.org/agclinkhom/3/

08376 | AIM-TOPOLOGY-0242

### A factor-support classification for one-dimensional totally geodesic fibers

P / partially_solved | AIM 研讨会问题汇编

**Problem 29.** Classify Riemannian submersions from a Lie group with a bi-invariant metric. (As was observed at the workshop, they are not necessarily biquotient submersions.)

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08377 | AIM-TOPOLOGY-0123

### A finite certificate for crossing the one-half stable 4-genus barrier

P / partially_solved | AIM 研讨会问题汇编

**Small stable 4-genera.** The stable 4-genus of a knot \(K\), in either category, is defined as \[ g_4^{\mathrm{st}}(K)=\lim_{n\to\infty}\frac{g_4(nK)}{n}, \] see Livingston. Find a knot \(K\) with \(0<g_4^{\mathrm{st}}(K)<1/2\).

来源字段：http://aimpl.org/concordsliceknot/3/

08378 | AIM-TOPOLOGY-0131

### A finite compatibility target for slice-ribbon

P / partially_solved | AIM 研讨会问题汇编

Find new ways to build knots which are by construction slice but not by construction ribbon.

来源字段：http://aimpl.org/concordsliceknot/5/

08379 | AIM-TOPOLOGY-0032

### A finite-channel BB-BLWY comparison and conservative ADO/modified-TV audit

P / partially_solved | AIM 研讨会问题汇编

Understanding connections between Baseilhac-Benedetti invariants and other quantum invariants

What are the connections between Baseilhac-Benedetti invariants and 1. ADO invariant? 2. BWY invariant? 3. modified TV invariant?

来源字段：http://aimpl.org/quantumlowdimtop/1/

08380 | AIM-TOPOLOGY-0169

### A finite-cover test and a periodic-cover obstruction for Veech dichotomy

P / partially_solved | AIM 研讨会问题汇编

In which cases do we have a Veech dichotomy on infinite translation surfaces? Are there any cases at all?

来源字段：http://aimpl.org/genusinfinity/5/

08381 | AIM-TOPOLOGY-0157

### A finite-target phase diagram for pure and full big mapping class groups

P / partially_solved | AIM 研讨会问题汇编

Are there any finite non-abelian quotients of $\operatorname{PMap}(S)$ or $\operatorname{Map}(S)$?

来源字段：http://aimpl.org/genusinfinity/4/

08382 | AIM-TOPOLOGY-0203

### A first-kind regular cover with variable critical exponent

P / partially_solved | AIM 研讨会问题汇编

M. Kapovich suggested that the above question maybe related to Problem 4.1. He guessed that the construction depends on the ergodic properties of Γ0.

**Problem 4.3.** Construct an infinitely generated Fuchsian group \(\Gamma _0\) of the first kind for which the critical exponent function is nonconstant on \(\mathcal T_{qc}(\Gamma _0)\).

Thus the recovered question is:

“Some elements” means some marked hyperbolic structures (equivalently, Fuchsian representations) in the quasiconformal Teichmüller space; it does not mean individual group elements. For a Fuchsian group \(\Gamma\),

\[ \delta(\Gamma)=\inf\left\{s>0: \sum_{\gamma\in\Gamma}e^{-s d(o,\gamma o)}<\infty\right\}. \]

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08383 | AIM-TOPOLOGY-0118

### A fixed-boundary group-spectrum reduction for homology slice versus contractibly slice knots

P / partially_solved | AIM 研讨会问题汇编

**Homology slice vs. contractibly slice.** Fix an integer homology sphere \(Y\) which bounds a contractible 4-manifold \(W\). Is there a knot \(K\) in \(Y\) such that \(K\) bounds a disc in some integer homology ball \(W'\), but \(K\) does not bound a disc in any contractible 4-manifold?

来源字段：http://aimpl.org/concordsliceknot/2/

08384 | AIM-TOPOLOGY-0103

### A flag-complex model for closure manifolds and cobordism

P / partially_solved | AIM 研讨会问题汇编

What are manifolds in Cech closure spaces? What is the notion of cobordism in Cech closure spaces?

来源字段：http://aimpl.org/combhomotop/4/

08385 | AIM-TOPOLOGY-0235

### A flat counterexample and an exact symmetry-degree formula

S / solved | AIM 研讨会问题汇编

Let M n be a manifold with sec > 0 or sec ≥ 0 or almost non-negative curvature. Does M n have a positive symmetry degree, i.e. is there an S1 ⊂ Diff( M n)?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08386 | AIM-TOPOLOGY-0102

### A four-point failure of inductive cubical excision

P / partially_solved | AIM 研讨会问题汇编

**Cubical homology**

Is the Excision Axiom true for cubical homology of Cech closure spaces with respect to interior covers?

来源字段：http://aimpl.org/combhomotop/4/

08387 | OPG-60019

### A funcoid related to directed topological spaces

S / solved | Open Problem Garden

Conjecture Let $R$ be the complete funcoid corresponding to the usual topology on extended real line $[-\infty,+\infty] = \mathbb{R}\cup\{-\infty,+\infty\}$. Let $\geq$ be the order on this set. Then $R\sqcap^{\mathsf{FCD}}\mathord{\geq}$ is a complete funcoid.

Proposition It is easy to prove that $\langle R\sqcap^{\mathsf{FCD}}\mathord{\geq}\rangle \{x\}$ is the infinitely small right neighborhood filter of point $x\in[-\infty,+\infty]$.

If proved true, the conjecture then can be generalized to a wider class of posets.

背景链接：http://www.openproblemgarden.org/op/a_funcoid_related_to_directed_topological_spaces

08388 | AIM-TOPOLOGY-0027

### A Gromoll sieve for contact loops on the standard contact 5-sphere

P / partially_solved | AIM 研讨会问题汇编

What do CHT techniques tell us about the map \[ \operatorname{Cont}(M,\xi)\longrightarrow \operatorname{Diff}(M)? \] Is \[ \pi _1\operatorname{Cont}(S^5,\xi)\longrightarrow \pi _1\operatorname{Diff}(S^5) \] a surjection?

来源字段：http://aimpl.org/highdimcontacttop/10/

08389 | AIM-TOPOLOGY-0232

### A homological endpoint obstruction for torus collapse models

S / solved | AIM 研讨会问题汇编

**Problem 19.** Can an \(n\)-dimensional torus collapse to an interval? (The answer to this question is “No”, essentially settled at the workshop.)

If \(M\cong T^n\), \(g_i\) are Riemannian metrics with \(\sec_{g_i}\geq-1\), and \(L>0\), then \((M,g_i)\) cannot converge in Gromov--Hausdorff distance to \([0,L]\).

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08390 | AIM-TOPOLOGY-0061

### A Jacobian criterion for the Soergel counit on Hochschild (co)homology

P / partially_solved | AIM 研讨会问题汇编

If \(w^2=1\), we have a map \(B_w\rightarrow R\). Is the induced map \(HH(B_w)\rightarrow HH(R)\) injective? Given that the Hochschild homology is defined as \(HH(\beta):=\operatorname{Ext}_{R\text{-bimod}}(R,B)\).

来源字段：http://aimpl.org/agclinkhom/2/

08391 | AIM-TOPOLOGY-0006

### A Lagrangian-locus reduction for stratified Liouville skeleta

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** Let \((X^{2n},\lambda)\) be a Liouville manifold of the half-dimensional Morse/homotopy type contemplated in problem 3.2. Suppose, for the given Liouville form \(\lambda\), that its skeleton (core) is stratified by smooth manifolds of dimension at most \(n\). Is \(\lambda\) Liouville homotopic to a Weinstein structure? Is there a one-parameter version?

来源字段：http://aimpl.org/highdimcontacttop/3/

08392 | AIM-TOPOLOGY-0016

### A local-capacity obstruction for trivial bypass detection

P / partially_solved | AIM 研讨会问题汇编

**Problem 7.4.** Can triviality be detected via capacities on balls around the intersection point?

The record occurs in the section “Explicit contact handlebodies” of the April 2024 AIM workshop *Higher-dimensional contact topology*. The immediately preceding Problem 7.3 asks whether a trivial bypass can be characterized using the belt sphere of the contact (n)-handle and the attaching sphere of the contact ((n+1))-handle. This gives the most likely antecedent of “the intersection point”: the unique geometric intersection which makes those two handles smoothly cancel.

There is a second, closely related intersection built into the bypass attaching data. In the notation now used by Breen--Christian--Honda--Huang (BCHH), a convex hypersurface has a decomposition

\[ \Sigma^{2n}=R_+\cup \Gamma\cup R_-, \]

and bypass data is a tuple

\[ \mathcal D=(\Lambda_-,\Lambda_+;D_-,D_+), \]

where (D_\pm\subset R_\pm) are properly embedded Lagrangian disks with Legendrian boundaries (\Lambda_\pm\subset\Gamma), and (\Lambda_-) and (\Lambda_+) have one (\xi_\Gamma)-transverse intersection. Thus “intersection point” can also naturally refer to (p=\Lambda_-\cap\Lambda_+). The two readings lead to the same conclusion for ordinary ball capacities, and both are treated below.

BCHH’s current definition says that the data is trivial of type (TB1) if ((\Lambda_+,D_+)) is a standard Legendrian unknot with its standard Lagrangian disk and (\Lambda_+) is below (\Lambda_-); type (TB2) is the sign-reversed condition on ((\Lambda_-,D_-)). Their Proposition A.26 realizes every bypass as a smoothly canceling contact (n)- and ((n+1))-handle pair and proves that trivial data gives a vertically invariant contact cobordism. In particular, “triviality” is not merely smooth handle cancellation: every bypass pair in that proposition cancels smoothly.

The ambiguity above is retained rather than silently resolved.

来源字段：http://aimpl.org/highdimcontacttop/7/

08393 | AIM-TOPOLOGY-0007

### A matched-region criterion for tight convex hypersurface germs

P / partially_solved | AIM 研讨会问题汇编

Can we characterize convex hypersurfaces with tight neighbourhoods?

来源字段：http://aimpl.org/highdimcontacttop/4/

08394 | AIM-TOPOLOGY-0274

### A measured Toledo bound and obstructions to maximality

P / partially_solved | AIM 研讨会问题汇编

**Question 4.1 (Marc Burger).** Let \(M\) be a compact manifold foliated by surfaces, with transverse measure \(\mu\). Denote the foliation by \(\mathcal F\). Let \(X\) be a Hermitian symmetric space, \(G=\operatorname{Isom}(X)^0\), and \(\rho:\pi_1(M)\to G\). Given a \(\rho\)-equivariant map \(f:\widetilde M\to X\), we have \[ f^*\omega\in H^2(\mathcal F), \] and let \(C\) be a Ruelle--Sullivan cycle. Find bounds on \[ \langle f^*\omega,C\rangle\in\mathbb R \] and study the maximal representations.

来源字段：https://aimath.org/WWN/surfacegroups/surfacegroups.pdf

08395 | AIM-TOPOLOGY-0138

### A metric obstruction and an invariant lamination for big multitwists

P / partially_solved | AIM 研讨会问题汇编

Does every big mapping class either preserve a hyperbolic metric or preserve a lamination?

来源字段：http://aimpl.org/genusinfinity/1/

08396 | AIM-TOPOLOGY-0087

### A Morita gate for gl(m|n) evaluations of the derived horizontal trace

P / partially_solved | AIM 研讨会问题汇编

Find functors from derived horizontal trace computing $\mathcal{gl}(m|n)$-homology.

来源字段：http://aimpl.org/agclinkhom/9/

08397 | AIM-TOPOLOGY-0031

### A multi-cusp formulation and analytic reduction

P / partially_solved | AIM 研讨会问题汇编

**Teichmüller TQFT volume conjecture for Fundamental Shadow Link complements**

Formulate and prove the volume conjecture coming from the Teichmüller TQFT for Fundamental Shadow Link complements

来源字段：http://aimpl.org/quantumlowdimtop/1/

08398 | AIM-TOPOLOGY-0269

### A negative answer and persistence of exceptional components under finite covers

S / solved | AIM 研讨会问题汇编

**Question 1.1 (Bill Goldman).** Can every maximal representation in \(\operatorname{Sp}(4,\mathbb R)\) be deformed to a proper Zariski closed subgroup?

**Comment 1.2 (Bill Goldman).** Every maximal representation of the unitary group can be deformed in this way.

**Comment 1.3 (Anna Wienhard).** The problem should be easier for \(\operatorname{Sp}(2n,\mathbb R)\) with \(n>2\) (it is still not known though). For \(n=2\) progress was made at the conference on many components, however there exist \(2(g-2)\) “exotic” components for which nothing is currently known.

来源字段：https://aimath.org/WWN/surfacegroups/surfacegroups.pdf

08399 | AIM-TOPOLOGY-0001

### A normal-twisted h-principle for contact submanifolds

P / partially_solved | AIM 研讨会问题汇编

**Taxonomize h-principles for contact submanifolds:** Write down a full h-principle for twisted contact submanifolds for some (or any) definition of twisted.

来源字段：http://aimpl.org/highdimcontacttop/1/

08400 | AIM-TOPOLOGY-0083

### A one-strand specialization gate for braid-like models of super and Khovanov homology

P / partially_solved | AIM 研讨会问题汇编

Can we determine $gl(m|n)$ / Khovanov homology using braid-like varieties?

来源字段：http://aimpl.org/agclinkhom/6/

08401 | AIM-TOPOLOGY-0265

### A quadratic-relation sieve for formality in dimension seven

P / partially_solved | AIM 研讨会问题汇编

**Problem 52.** Does positive sectional curvature imply that the manifold is formal (in the sense of Sullivan's minimal model)?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08402 | AIM-TOPOLOGY-0231

### A quantitative nonnegatively curved collapse to a ray

P / partially_solved | AIM 研讨会问题汇编

**Problem 18. Study collapse to a ray.**

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08403 | AIM-TOPOLOGY-0074

### A rank-two discriminant obstruction for categorical Schur expansions

P / partially_solved | AIM 研讨会问题汇编

**Problem.** What are Schur expansions of projector closures at the category level and what are their connections to Macdonald theory?

来源字段：http://aimpl.org/agclinkhom/5/

08404 | AIM-TOPOLOGY-0038

### A relative gauge-quotient criterion for extending Teichmüller TQFT

P / partially_solved | AIM 研讨会问题汇编

Generalize Teichmüller TQFT to wider families e.g. cone manifolds or fundamental shadow link complements

Generalize Teichmüller TQFT to wider families e.g. cone manifolds or fundamental shadow link complements.

来源字段：http://aimpl.org/quantumlowdimtop/1/

08405 | AIM-TOPOLOGY-0035

### A shadow-descent criterion for Teichmüller TQFT

P / partially_solved | AIM 研讨会问题汇编

Find a shadow formula for Teichmüller TQFT

来源字段：http://aimpl.org/quantumlowdimtop/1/

08406 | AIM-TOPOLOGY-0248

### A sharp cap-profile classification for the standard sphere join action

P / partially_solved | AIM 研讨会问题汇编

**Problem 35.** Classify cohomogeneity-one manifolds with \(\sec\geq 0\) and at least one totally geodesic principal orbit.

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08407 | AIM-TOPOLOGY-0239

### A sharp ordered-eigenvalue criterion for nonnegative curvature on SU(2)

P / partially_solved | AIM 研讨会问题汇编

Problem 26. Given a homogeneous space \(G/H\), with \(G\) compact, classify all homogeneous metrics with \(\sec\geq 0\).

Problem 26. Given a homogeneous space G/H, with G compact, classify all homogeneous metrics with sec ≥ 0. 3

The final 3 is not mathematical text. In the AIM PDF it is the page number at the page boundary: the line ends with the footnote-style page marker immediately before the next page begins. The recovered statement is therefore:

Source: AIM workshop, Manifolds with nonnegative sectional curvature, Problem 26, [official PDF](https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf).

This report preserves the source record and explicitly removes only that verified layout artifact.

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08408 | AIM-TOPOLOGY-0246

### A sharp prescribed-radius criterion for diagonal join metrics

P / partially_solved | AIM 研讨会问题汇编

**Problem 33.** Study the existence and non-existence of metrics with non-negative curvature on manifolds of cohomogeneity-one.

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08409 | AIM-TOPOLOGY-0194

### A sign-splitting reduction for length-spectrum isometries

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.3 (A. Papadopoulos).** Is \[ \operatorname{Isom}(\mathcal T(S),d_{ls})=\operatorname{Mod}(S)? \]

Inspection of page 3 of the source PDF recovers the mathematical statement as

The dots are a separator. “4 WEIXU SU” is the page number and running header on the following page, not part of the problem. The preceding text says that Sorvali's length-spectrum metric is a symmetrization of the Thurston metric.

There is a genuine convention issue in the recovered statement. In much of the literature, \(\operatorname{Mod}(S)\) means orientation-preserving mapping classes, whereas an isometry theorem is normally stated using the extended group \(\operatorname{Mod}^{\pm}(S)\). The 2010 Oberwolfach formulation by Papadopoulos asks explicitly whether the isometry group *strictly contains the extended mapping class group*. Thus the intended question is best read as

\[ \operatorname{Isom}(\mathcal T(S),d_{ls}) \stackrel{?}{=} \operatorname{Mod}^{\pm}(S)/K(S), \tag{1} \]

where \(K(S)\) is the pointwise kernel of the action on Teichmüller space. Without the effective quotient, equality is not literally possible in the familiar low-complexity cases.

This report treats connected orientable finite-type surfaces \(S=S_{g,n}\) of negative Euler characteristic, without geodesic boundary, and with \(3g-3+n>0\). The zero-dimensional pair of pants is a separate degenerate case: its Teichmüller space is a point and its isometry group is trivial.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08410 | AIM-TOPOLOGY-0064

### A signed Hochschild formula and scope audit for HHH annulus maps

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.5.** Describe annulus maps in Khovanov-Rozansky homology \(HHH\).

来源字段：http://aimpl.org/agclinkhom/2/

08411 | AIM-TOPOLOGY-0045

### A specialization firewall for comparing four SL_n skein constructions

P / partially_solved | AIM 研讨会问题汇编

**$SL_n$ skein modules definition equivalence.** Compare the definitions of $SL_n$ skein modules/algebra for either a generic $q$ or $q$ being a root of unity.

来源字段：http://aimpl.org/quantumlowdimtop/2/

08412 | AIM-TOPOLOGY-0133

### A split-target partial order and the exact composition defect

P / partially_solved | AIM 研讨会问题汇编

**Topological ribbon concordance.** Gordon asked whether two knots which are mutually ribbon concordant must be isotopic. This problem offers a topological version.

Define knots $K$ and $J$ to be topologically homotopy ribbon concordant, and write $K\leq_{thrc}J$, if $K$ and $J$ are topologically concordant via an annulus $A$ such that the inclusion-induced map $\pi_1(X_K)\to\pi_1(X_A)$ is a surjection and the inclusion-induced map $\pi_1(X_J)\to\pi_1(X_A)$ is an injection. Is $\leq_{thrc}$ a partial order on the collection of knots in $S^3$?

来源字段：http://aimpl.org/concordsliceknot/5/

08413 | AIM-TOPOLOGY-0145

### A Thurston bordification and a periodic-end obstruction

P / partially_solved | AIM 研讨会问题汇编

Is there a natural bordification/boundary of Teichmüller space for big surfaces?

来源字段：http://aimpl.org/genusinfinity/3/

08414 | AIM-TOPOLOGY-0023

### A topology ledger for splitting-surface decompositions

P / partially_solved | AIM 研讨会问题汇编

Can we restrict the topology of symplectic fillings using convex surfaces in their boundaries? (c.f. filling by holomorphic curves)

来源字段：http://aimpl.org/highdimcontacttop/9/

08415 | AIM-TOPOLOGY-0119

### A two-branch reduction for torsion in the homology-concordance cokernel

P / partially_solved | AIM 研讨会问题汇编

**Torsion in the cokernel.** Does \(\operatorname{coker}(\phi_{\mathrm{smooth}})\) have torsion?

来源字段：http://aimpl.org/concordsliceknot/2/

08416 | AIM-TOPOLOGY-0117

### A two-defect completion obstruction to topological homology concordance

P / partially_solved | AIM 研讨会问题汇编

Concordance of knots in homology spheres.

Is every knot in every integer homology sphere topologically concordant in some integer homology cobordism to a knot in $S^3$? (i.e. is $\phi_{top}$ surjective?)

In the smooth category, the answer is `No' due to work of Levine (Levine 2016, MR3589337) The answer is 'Yes' modulo every term of the $n$-solvable filtration, due to work of Davis (arXiv:1803.01086).

来源字段：http://aimpl.org/concordsliceknot/2/

08417 | AIM-TOPOLOGY-0281

### A two-gate length criterion for proper energy

P / partially_solved | AIM 研讨会问题汇编

**Question 5.5 (Bill Goldman).** Let \(S\) be a closed orientable surface with \(\chi(S)<0\), and \(G\) a semisimple Lie group. Define \(E_\rho:\mathcal T_S\to\mathbb R\) to be the energy function associated to a representation \(\rho:\pi_1(S)\to G\). What are the most general conditions on \(\rho\) for which \(E_\rho\) is proper?

来源字段：https://aimath.org/WWN/surfacegroups/surfacegroups.pdf

08418 | AIM-TOPOLOGY-0121

### A two-gate obstruction to negative-amphichiral representatives

P / partially_solved | AIM 研讨会问题汇编

**2-torsion in $\mathcal C$.** Can one detect whether $K$ is concordant to a negative amphichiral knot?

来源字段：http://aimpl.org/concordsliceknot/3/

08419 | AIM-TOPOLOGY-0018

### A two-gate reduction for connected-sum tightness

P / partially_solved | AIM 研讨会问题汇编

**Does connected sum preserve tightness?**

来源字段：http://aimpl.org/highdimcontacttop/7/

08420 | AIM-TOPOLOGY-0101

### A two-point obstruction to uniqueness of closure-space homology

P / partially_solved | AIM 研讨会问题汇编

**Eilenberg-Steenrod axioms**

Do the Eilenberg-Steenrod axioms or a variant thereof determine the homology theory uniquely for closure spaces (or graphs)?

来源字段：http://aimpl.org/combhomotop/4/

08421 | AIM-TOPOLOGY-0057

### A two-row link-homology sector of multivariate diagonal coinvariants

P / partially_solved | AIM 研讨会问题汇编

Do \(d\)-agonal coinvariants relate to link homology?

来源字段：http://aimpl.org/agclinkhom/1/

08422 | AIM-TOPOLOGY-0162

### Action-produced quasimorphisms and a uniform-perfectness obstruction

P / partially_solved | AIM 研讨会问题汇编

For surfaces for which this is not known, can we produce quasimorphisms using actions of $\operatorname{Map}(S)$?

Its problem text is:

The supplied literature note is:

The immediately preceding records concern homomorphisms and lattice maps, while item 4.7 asks about subgroups having infinite-dimensional spaces of quasimorphisms. The literature note also points to the ray and loop graph constructions. The safest reconstruction is therefore:

Here and below “nontrivial” means nonzero after quotienting by homomorphisms and bounded functions. This qualification is essential: a homomorphism is itself a quasimorphism, but it represents zero in the reduced space.

Unless continuity is explicitly mentioned, groups and quasimorphisms are treated algebraically. Big mapping class groups carry a natural Polish topology, so continuous quasimorphisms form a genuinely different problem.

来源字段：http://aimpl.org/genusinfinity/4/

08423 | OWR-1381-003

### Additivity of Knot Crossing Number

P / partially_solved | Oberwolfach 报告问题汇编

For all oriented knots $K_1$ and $K_2$, the crossing number should be additive under connected sum: $c(K_1\mathbin{\#}K_2)=c(K_1)+c(K_2)$.

来源字段：https://doi.org/10.4171/owr/2006/43

08424 | OWR-11786-002

### Additivity of Triangulation Complexity under Connected Sum

O / open | Oberwolfach 报告问题汇编

Let $M$ be a compact $(d-1)$-dimensional manifold with boundary, where $d\geq4$, and define $$\Gamma(M)=\min_{|\Delta|=M}\bigl(h_2(\Delta)-h_{d-1}(\Delta)-d h_d(\Delta)\bigr)=\min_{|\Delta|=M}\bigl(h_2(\Delta)-f_0^{\mathrm{int}}(\Delta)\bigr).$$ For a closed manifold, set $\Gamma(M)=\Gamma(M\setminus B^{d-1})$. Is $\Gamma(M_1\mathbin{\#}M_2)=\Gamma(M_1)+\Gamma(M_2)$?

来源字段：https://doi.org/10.4171/owr/2012/24

08425 | AIM-TOPOLOGY-0167

### Affine realizations from balanced infinite Penner systems

P / partially_solved | AIM 研讨会问题汇编

**Problem.** Which mapping classes are realized by affine automorphisms on some translation surface?

We can ask the same question, but for Penner's construction: Given a mapping class that is obtained by Penner's construction, does it fix some flat metric?

来源字段：http://aimpl.org/genusinfinity/5/

08426 | OWR-11126-003

### Algebraic and Geometric Overtwistedness

S / solved | Oberwolfach 报告问题汇编

Is algebraic overtwistedness equivalent to overtwistedness for contact manifolds, in particular does algebraic overtwistedness imply overtwistedness?

来源字段：https://doi.org/10.4171/owr/2011/34

08427 | OWR-1190-004

### Algebraic Models for Rational Compact Lie-Group Spectra

P / partially_solved | Oberwolfach 报告问题汇编

别名：Repeated Compact Lie-Group Algebraic-Model Conjecture [OWR-1190-005]

For every compact Lie group $G$, does there exist an abelian category $\mathcal{A}(G)$ of injective dimension $\operatorname{rank}(G)$ and a Quillen equivalence $G\text{-spectra}/\mathbb{Q}\simeq\mathrm{dg}\mathcal{A}(G)$, where objects of $\mathcal{A}(G)$ are sheaves of modules over the space of closed subgroups of $G$, assembled from Borel-theory fibers by localization and inflation, and the model of a cohomology theory is determined by its values on spheres together with limited extra structure?

来源字段：https://doi.org/10.4171/owr/2006/08

08428 | AIM-TOPOLOGY-0155

### Algebraic rigidity and a finite-subgroup genus detector

S / solved | AIM 研讨会问题汇编

If $S_1,S_2$ are infinite-type surfaces and $\operatorname{Map}(S_1)\cong\operatorname{Map}(S_2)$ as abstract groups, must $S_1\cong S_2$?

来源字段：http://aimpl.org/genusinfinity/4/

08429 | OWR-8415349-008

### Algebraic Structures and Applications of Two-Dimensional Field Theories

P / partially_solved | Oberwolfach 报告问题汇编

For two-dimensional topological conformal or homological conformal field theories, what are the appropriate algebraic structures, do nontrivial examples exist, and what applications do they have?

来源字段：https://doi.org/10.4171/owr/2021/46

08430 | OWR-15951-006

### Algebraic Topology of Mathematical Quasicrystals

O / open | Oberwolfach 报告问题汇编

What algebro-topological tools should be employed to study mathematical quasicrystals and their tiling spaces?

来源字段：https://doi.org/10.4171/owr/2017/46

08431 | OWR-13108-001

### Algebraicity of Perturbative Fiber-Knot Invariants

O / open | Oberwolfach 报告问题汇编

For any Seifert space $M$ with $\chi(M)>0$, does there exist a finite-degree extension $L(M)$ of $\mathbb Q(e^u)$ containing all genus-$g$ $A$-series perturbative invariants of fiber knots in $M$?

来源字段：https://doi.org/10.4171/owr/2014/38

08432 | AIM-TOPOLOGY-0216

### Almost half-maximal diameter and threshold realizability

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.** Let $M^n$ be a manifold for which $\sec M\geq 1$ and $\operatorname{diam}(M)\geq \pi/2-\varepsilon(n)$. Is $M$ homeomorphic to a manifold $M'$ where $\sec M'\geq 1$ and $\operatorname{diam}(M')\geq \pi/2$? This problem may be easier to address if we also assume that $\operatorname{Vol}(M)\geq V$ and $\varepsilon=\varepsilon(n,V)$.

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08433 | AIM-TOPOLOGY-0220

### Almost spherical rank under non-collapsed convergence

P / partially_solved | AIM 研讨会问题汇编

**Problem 7.** Suppose $X$ is the non-collapsed Gromov--Hausdorff limit of \((M_i^n,g_i)\), where \(|\sec M_i|\le1\), and that along every geodesic on $M_i$ one hits a conjugate point before $t=\pi+1/i$. Is $X$ rigid in any sense?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08434 | AIM-TOPOLOGY-0266

### Almost-cyclic fundamental groups: normality, linear models, and dimension three

P / partially_solved | AIM 研讨会问题汇编

**Problem 53.** For compact, odd dimensional, positively curved manifolds is there a cyclic subgroup of the fundamental group whose index is bounded only in terms of the dimension?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08435 | AIM-TOPOLOGY-0046

### Ambient-torus mutation and a local bifiltration for BKL pants moves

P / partially_solved | AIM 研讨会问题汇编

**Quantum trace for closed surfaces.** For Bloomquist--Karuo--Lê quantum trace for closed surface, what happens when we change the pants decomposition? (Hatcher--Thurston move)

来源字段：http://aimpl.org/quantumlowdimtop/2/

08436 | AIM-TOPOLOGY-0229

### An additive calculus for almost submetries

P / partially_solved | AIM 研讨会问题汇编

**Problem 16.** Study Alexandrov (almost) submetries.

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08437 | AIM-TOPOLOGY-0171

### An arithmetic and angular reduction for the 3-4-5 triangular billiard

P / partially_solved | AIM 研讨会问题汇编

**Problem.** Is the billiard flow on the triangle with side lengths $3$, $4$, and $5$ ergodic?

来源字段：http://aimpl.org/genusinfinity/5/

08438 | AIM-TOPOLOGY-0021

### An augmentation-rigid-boundary sieve for distinct Weinstein fillings

P / partially_solved | AIM 研讨会问题汇编

Given a Weinstein domain \(W\) with contact boundary \((Y,\xi)\), construct either (i) another Weinstein filling of \((Y,\xi)\) inequivalent to \(W\), or (ii) a nontrivial boundary contactomorphism which preserves the augmentation of the rational full contact-homology DGA induced by a filling; ideally do both so that two inequivalent fillings induce the same augmentation after identifying their boundaries.

来源字段：http://aimpl.org/highdimcontacttop/9/

08439 | AIM-TOPOLOGY-0011

### An equivariant prequantization reduction for a CHT proof of Donaldson divisors

P / partially_solved | AIM 研讨会问题汇编

**Recovered question (hyperplane-section reading).** Can convex hypersurface theory give a proof that if \((X^{2n},\omega)\) is closed and \([\omega/2\pi]\) is integral, then, for all sufficiently large integers \(k\), there is a symplectic hypersurface \(W\subset X\) Poincaré dual to \(k[\omega/2\pi]\), with the connectivity and Weinstein-complement properties of a Donaldson--Giroux hyperplane section?

来源字段：http://aimpl.org/highdimcontacttop/6/

08440 | AIM-TOPOLOGY-0014

### An essential-tight bypass ladder on contact S2 x S3

P / partially_solved | AIM 研讨会问题汇编

**Find other interesting explicit bypass decompositions of contact manifolds.**

The record has no remarks or literature field. The prompt is deliberately open-ended: “other” refers to the preceding Problem 7.1, which asks for a bypass description of the layers between two Darboux balls in

\[ (S^5,\xi_k)=\operatorname{OB}(T^*S^2,\tau^{2k+1}). \]

The official April 2024 AIM workshop report confirms this reading. Its “Explicit contact handlebodies” group sought those \(S^5\) layers and, as a three-dimensional stepping stone, computed that two overtwisted bypasses in \(\operatorname{OB}(T^*S^1,\tau^{-1})\) realize a negative plus a positive stabilization. The report says the higher-dimensional target was not completed.

This report preserves the broad prompt and develops a different explicit family: a finite ladder of nontrivial, non-overtwisted bypasses between convex \(S^2\times S^2\) levels in a tight contact \(S^2\times S^3\). It does not claim to settle every possible meaning of “find other.”

来源字段：http://aimpl.org/highdimcontacttop/7/

08441 | AIM-TOPOLOGY-0051

### An explicit cross-dimensional extension gap for non-semisimple TQFTs

P / partially_solved | AIM 研讨会问题汇编

**Non semi-simple categories to TQFTS** Construct non semi-simple categories that give rise to certain TQFTS.

来源字段：http://aimpl.org/quantumlowdimtop/3/

08442 | AIM-TOPOLOGY-0247

### An explicit infinite nonsymmetric cohomogeneity-one family

P / partially_solved | AIM 研讨会问题汇编

Find cohomogeneity-one manifolds with "interesting" topology, in particular not homeomorphic to a symmetric space. Study curvature properties of these manifolds.

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08443 | AIM-TOPOLOGY-0091

### An explicit rank-two test for the trace--Hilbert--quasi-invariant bridge

P / partially_solved | AIM 研讨会问题汇编

a) Relate \(tr(T(m,n))\) to rings of quasi-invariants and to A. Wilson's conjecture on \(\nabla_{p_1}^n\).

b) Describe \(H^*(\operatorname{Hilb}(x^{nd}=y^n))\) as a module over rational Double Affine Hecke algebra (DAHA) and relate it to the above.

来源字段：http://aimpl.org/agclinkhom/9/

08444 | AIM-TOPOLOGY-0044

### An explicit SL_N quantum-torus summand for the -I torus bundle

P / partially_solved | AIM 研讨会问题汇编

**\(SL_n\) skein modules.** Explore \(SL_n\) skein modules for \(n\geq 3\). For example, the dimension \(<\infty\) is proved by Gunningham-Jordan-Safranov.

来源字段：http://aimpl.org/quantumlowdimtop/2/

08445 | AIM-TOPOLOGY-0043

### An integral free core and cyclotomic quotient for connected sums of lens spaces

P / partially_solved | AIM 研讨会问题汇编

**KBSM of connected sum of Lens space**

Compute the Kauffman Bracket Skein Module of $\#$ Lens space (Haken manifold of finite type) over $\mathbb{Z}[A^{\pm}]$.

**Recovered problem.** For a finite connected sum \[ M=\mathop{\#}_{i=1}^{r}L(p_i,q_i),\qquad r\geq 2, \] of standard closed lens spaces, with $p_i>0$ and $\gcd(p_i,q_i)=1$, compute the Kauffman bracket skein module over \[ R=\mathbb Z[A,A^{-1}]=\mathbb Z[A^{\pm1}]. \]

来源字段：http://aimpl.org/quantumlowdimtop/2/

08446 | AIM-TOPOLOGY-0270

### An invariant Kähler structure on the SL(3,R) Hitchin component

S / solved | AIM 研讨会问题汇编

**Question 2.1 (John Loftin).** Does there exist a mapping class group-invariant Kähler structure on the Hitchin component for \(\mathrm{SL}(3,\mathbb R)\)?

**Comment 2.2 (Richard Wentworth).** Theorem 1.0.2 of [Labourie 2007] shows that the Hitchin component for \(\mathrm{SL}(3,\mathbb R)\) is given by the bundle of cubic differentials on Teichmüller space. There is a mapping class group-invariant complex structure on this space.

**Comment 2.3 (John Loftin).** There is evidence for a Kähler structure, since transverse to the fibres there is a Kähler metric (Weil--Petersson), and on the fibres there is a Kähler metric.

来源字段：https://aimath.org/WWN/surfacegroups/surfacegroups.pdf

08447 | AIM-TOPOLOGY-0175

### An invariant ledger for the Thurston infinitesimal norm

S / solved | AIM 研讨会问题汇编

**Problem 1.2 (K. Rafi).** What properties of the hyperbolic surface are determined by the Thurston infinitesimal norm?

On its first page, under “Infinitesimal properties of the Thurston metric,” the source reads:

The PDF declares that \(S=S_{g,n}\) is a genus-\(g\) hyperbolic surface with \(n\) punctures and that \(\mathcal T(S)\) is its Teichmüller space. The question immediately follows Problem 1.1, which asks whether the norm at \(X\) determines \(X\), but Problem 1.2 asks for a broader inventory of recoverable properties.

This report interprets the datum as the **abstract real asymmetric normed vector space**

\[ N_X=(T_X\mathcal T(S),\|\cdot\|_{\mathrm{Th}},X}) \]

up to surjective real-linear isometry. This convention is essential. A norm written in already marked coordinates, or a dual sphere supplied with curve labels, contains extra data not present in an abstract norm.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08448 | AIM-TOPOLOGY-0029

### An odd-subsequence counterexample to the literal ratio-one Volume Conjecture

S / solved | AIM 研讨会问题汇编

**The Colored Jones Polynomial Volume Conjecture for odd root of unity.** Let \(L\in S^3\) be a hyperbolic link. For \(n\) odd, let \(J_n(L)\in\mathbb Z[t^\pm]\) be the \(n\)-th normalized Colored Jones Polynomial. Then, as \(n\to\infty\), \[ J_n\!\left(L,t=e^{2\pi i/n}\right) \sim \exp\!\left[\frac{n}{2\pi} \bigl(\operatorname{vol}(S^3-L)+i\,CS(S^3-L)\bigr)\right]. \]

来源字段：http://aimpl.org/quantumlowdimtop/1/

08449 | OWR-14743-005

### Analytic Continuation of Topological Zeta Functions

O / open | Oberwolfach 报告问题汇编

For a manifold $X$ and the topological zeta function values $\zeta_X(m_n)$ defined from limiting homological densities, does $\zeta_X(s)$ admit an analytic continuation to noninteger values of $s$?

来源字段：https://doi.org/10.4171/owr/2016/35

08450 | OWR-15208-022

### Anosov Geodesic Flows Without Negative Curvature Metrics

O / open | Oberwolfach 报告问题汇编

Does there exist a closed manifold that admits no metric of negative curvature but does admit a metric whose geodesic flow is Anosov?

来源字段：https://doi.org/10.4171/owr/2017/3

08451 | OPG-59970

### Another conjecture about reloids and funcoids

P / partially_solved | Open Problem Garden

Definition $\square f = \bigcap^{\mathsf{RLD}} \mathrm{up}^{\Gamma (\operatorname{Src} f; \operatorname{Dst} f)} f$ for reloid $f$.

Conjecture $(\mathsf{RLD})_{\Gamma} f = \square (\mathsf{RLD})_{\mathrm{in}} f$ for every funcoid $f$.

Note: it is known that $(\mathsf{RLD})_{\Gamma} f \ne \square (\mathsf{RLD})_{\mathrm{out}} f$ (see below mentioned online article).

背景链接：http://www.openproblemgarden.org/op/another_conjecture_about_reloids_and_funcoids

08452 | AMR-110-0025

### Applications 1 - Introduce stable homotopy theory into the world of C^*-algebras, like Voevodsky has done in algebraic…

P / partially_solved | AMR 问题清单汇编

Introduce stable homotopy theory into the world of C^*-algebras, like Voevodsky has done in algebraic geometry. More specifically, find a model structure on some category of C^*-algebras that is useful. Then stabilize it somehow to find a stable homotopy category of C^*-algebras where Kasparov's KK theory is just Hom in the category. This will probably be the same as finding a category where K-theory of C^*-algebras is representable. I have thought about this a little--almost the first thing you realize is that C^*-algebras are not closed under inverse limits, so you need pro-C^*-algebras. These have been considered by Chris Phillips in N. C. Phillips, Inverse limits of $C\sp *$-algebras, J. Operator Theory {\bf 19} (1988), no.~1, 159--195; MR 90c:46090. Another thing you realize is that it is easier to map into C^*-algebras than it is to map out of them, and so it seems likely that any model category will be fibrantly generated instead of cofibrantly generated. But not much else is clear to me. One of the people who knows the most about the combination of C^*-algebras and homotopy theory is Claude Schochet. The viewpoint I am advocating here is shared by Jim McClure, and probably Marius Dadarlat, and they may know more about this. In particular, C^*-algebraists often make an assumption about their cohomology theories which basically forces them to be K-theory--I think it is Bott periodicity. To do what I am saying, you would have to drop this assumption.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08453 | AMR-110-0026

### Applications 2 - Neil Strickland points out that several different moduli spaces are used in differential geometry and…

O / open | AMR 问题清单汇编

Neil Strickland points out that several different moduli spaces are used in differential geometry and physics. For example, there is the moduli space people are always talking about in every talk on gauge theory I have ever been to. I can't remember what this moduli space is now, unfortunately--something to do with connections and the gauge group? Anyway, the geometers tend to think only about the rational homology of these spaces. What about their Morava K-theory? (Neil also mentioned solitons and Seiberg-Witten theory).

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08454 | AMR-110-0027

### Applications 3 - Investigate Voevodsky's stable homotopy category of schemes from a homotopy theorist's point of view.

P / partially_solved | AMR 问题清单汇编

Investigate Voevodsky's stable homotopy category of schemes from a homotopy theorist's point of view. This is obviously a huge, unstructured problem, but I think there is room for a serious algebraic topologist to make some inroads. In particular, Bousfield localization is really a great thing, and they have not exploited this at all so far, I believe. The drawback is having to learn so much algebraic geometry before you have a chance.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08455 | AMR-110-0028

### Applications 4 - Stefan Stolz showed that a simply connected Spin manifold of dimension at least 5 admits a metric of p…

P / partially_solved | AMR 问题清单汇编

Stefan Stolz showed that a simply connected Spin manifold of dimension at least 5 admits a metric of positive scalar curvature if and only if its image under the orientation MSpin --> KO is 0. But the situation is not completely understand when the manifolds are not simply connected. It seems to get involved with the Novikov conjecture. Oh yes, I remember--it is actually false for some fundamental groups, but the trouble appears to be all with products with the Bott manifold. That is, the general conjecture is: suppose M is a Spin manfold of dimension at least 5. Then a product of M with some finite number of Bott manifolds admits a metric of positive scalar curvature.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08456 | AMR-110-0029

### Applications 5 - Try to carry out Stolz's plan for metrics of positive Ricci curvature.

P / partially_solved | AMR 问题清单汇编

Try to carry out Stolz's plan for metrics of positive Ricci curvature. Here we expect the obstruction to lie in elliptic cohomology rather than K-theory, and the manifolds should be MO8 manifolds. But hardly anything is known here, I think.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08457 | AMR-110-0030

### Applications 6 - Improve on Benson-Carlson-Rickard.

S / solved | AMR 问题清单汇编

Improve on Benson-Carlson-Rickard. Recall their theorem: if G is a finite p-group and k is an algebraically closed field, then thick subcategories in the stable k[G]-module category (= category obtained by killing projectives = injectives = frees) are in 1-1 correspondence with subsets of Proj H^*(G,k) closed under specialization. There are several open questions related to this. The most obvious one is to remove the algebraically closed requirement. This may involve understanding some Galois theory of stable homotopy categories, but may not be too difficult. I sincerely hope that anybody who wants to work on the stable module category will make the effort to understand my work with Palmieri and Strickland on this; our setup makes things considerably simpler, in my opinion.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08458 | AMR-110-0031

### Applications 7 - Classify the localizing subcategories of the stable k[G]-module category.

S / solved | AMR 问题清单汇编

Classify the localizing subcategories of the stable k[G]-module category. These should be in 1-1 correspondence with arbitrary subsets of Proj H^*(G,k). I have no idea how to do this one.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08459 | AMR-110-0032

### Applications 8 - Extend the results of Benson-Carlson-Rickard to connected, cocommutative Hopf algebras over a field, l…

P / partially_solved | AMR 问题清单汇编

Extend the results of Benson-Carlson-Rickard to connected, cocommutative Hopf algebras over a field, like A(n). Hovey-Palmieri have achieved some partial success here; we reduce the calculation of thick subcategories to the case when the Hopf algebra is quasi-elementary. For k[G], quasi-elementary subHopf algebras look like k[E] for elementary abelian p-groups E, and the group cohomology is a polynomial algebra, so Benson-Carlson-Rickard use algebraic geometry to finish the proof (this is where they require the field to be algebraically closed). For A(n), the quasi-elementary subHopf algebras are also exterior, but the Ext groups are bigraded. We don't know how to do bigraded algebraic geometry, so we are stuck there. But I think something should be doable here. Anyway, here is the abstract of the Hovey-Palmieri paper , and here is the dvi file.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08460 | OWR-5154-004

### Approximation Conjectures for $L^2$-Torsion and Homology

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a finite connected CW complex with a $G$-cover $\widetilde X$, and let $(G_i)$ be normal finite-index subgroups of $G$ with trivial intersection. Do the normalized torsion invariants of $G_i\backslash\widetilde X$ converge to the corresponding $L^2$-torsion of $\widetilde X$? Establish the analogous approximation for analytic torsion of closed Riemannian manifolds and for integral homology torsion when all $L^2$-Betti numbers vanish. For an aspherical closed $d$-manifold, determine the limiting normalized Betti numbers and torsion growth, including the middle-dimensional cases governed by $d$ and $\chi(M)$.

来源字段：https://doi.org/10.4171/owr/2011/26

08461 | OWR-8415350-005

### Arc-Space Models of Contact Homology

O / open | Oberwolfach 报告问题汇编

Can the contact-homology algebra be computed from the space of short arcs? Is there an arc-space version of its spectral sequence? For a two-dimensional singularity, can the embedded contact homology of its link be described in terms of arcs?

来源字段：https://doi.org/10.4171/owr/2021/47

08462 | AIM-TOPOLOGY-0147

### Area-rooted Weil-Petersson limits lose their genus

P / partially_solved | AIM 研讨会问题汇编

Consider a geometric invariant for which we know the asymptotics in genus. This describes the generic shape of a surface of high genus. Can we use this information to obtain information about the generic shape of an infinite-genus surface?

来源字段：http://aimpl.org/genusinfinity/3/

08463 | DARPA-018

### Arithmetic Langlands, Topology, and Geometry

P / partially_solved | DARPA 挑战

Explore homotopy theory's role in Langlands programs.

背景链接：https://web.math.utk.edu/~vasili/refs/darpa07.MathChallenges.html

08464 | OWR-4416-001

### Arithmetic Volume Conjecture for Kashaev Invariants

P / partially_solved | Oberwolfach 报告问题汇编

Does the Volume Conjecture hold for Kashaev invariants, including its arithmetic refinement relating root-of-unity asymptotics to the knot trace field and modular transformations?

来源字段：https://doi.org/10.4171/owr/2010/35

08465 | OWR-17127-009

### Arnold Fixed Point Bound by Cup Length

O / open | Oberwolfach 报告问题汇编

Does every Hamiltonian diffeomorphism of a closed connected symplectic manifold $M$ have at least $\operatorname{cl}(M)$ fixed points, where $\operatorname{cl}(M)$ denotes its cup length?

来源字段：https://doi.org/10.4171/owr/2019/31

08466 | AIM-TOPOLOGY-0017

### Asymmetric filling obstructions to cancellation on the contact 5-sphere

P / partially_solved | AIM 研讨会问题汇编

Understand contact connected sum in high dimensions: can non-standard structures on $S^5$ connect sum to give $\xi_{std}$?

来源字段：http://aimpl.org/highdimcontacttop/7/

08467 | OWR-17471-015

### Asymptotic Prevalence of Nonlayered Veering Triangulations

O / open | Oberwolfach 报告问题汇编

As the number of tetrahedra tends to infinity among veering triangulations, does the proportion of non-layered veering triangulations tend to one?

来源字段：https://doi.org/10.4171/owr/2020/8

08468 | OWR-820-003

### Atiyah Conjecture on $L^2$-Betti Numbers

P / partially_solved | Oberwolfach 报告问题汇编

If $\Gamma$ is torsion-free, then every $L^2$-Betti number of a finite CW complex with fundamental group $\Gamma$ is an integer.

来源字段：https://doi.org/10.4171/owr/2005/15

08469 | OPG-37386

### Atomicity of the poset of completary multifuncoids

S / solved | Open Problem Garden

Conjecture The poset of completary multifuncoids of the form $(\mathscr{P}\mho)^n$ is for every sets $\mho$ and $n$:

- atomic; - atomistic.

See below for definition of all concepts and symbols used to in this conjecture.

Refer to this Web site for the theory which I now attempt to generalize.

背景链接：http://www.openproblemgarden.org/op/atomicity_of_the_poset_of_multifuncoids

08470 | OPG-37388

### Atomicity of the poset of multifuncoids

S / solved | Open Problem Garden

Conjecture The poset of multifuncoids of the form $(\mathscr{P}\mho)^n$ is for every sets $\mho$ and $n$:

- atomic; - atomistic.

See below for definition of all concepts and symbols used to in this conjecture.

Refer to this Web site for the theory which I now attempt to generalize.

背景链接：http://www.openproblemgarden.org/op/atomicity_of_the_poset_of_multifuncoids_0

08471 | AIM-TOPOLOGY-0166

### Automatic continuity is surface-dependent

S / solved | AIM 研讨会问题汇编

For any homomorphism from $\operatorname{Map}(S)$ to a separable topological group $G$, is this automatically continuous?

If $M$ is a compact manifold then $\operatorname{Homeo}(M)$ has this property.

来源字段：http://aimpl.org/genusinfinity/4/

08472 | OWR-13490-018

### Averaged Morse Numbers and Lefschetz Theory

O / open | Oberwolfach 报告问题汇编

How is Bagchi’s lower-bound conjecture for the $g$-vector related to the $g$-conjecture? How are the averaged discrete Morse invariants $\mu_i$ related to discrete Morse theory, and is $\mu_1$ a lower bound for the minimum number of generators of the fundamental group?

来源字段：https://doi.org/10.4171/owr/2015/5

08473 | AMR-110-0033

### Axiomatic stable homotopy 1 - In our memoir, we give a conjecture for the thick subcategories in a Noetherian stable homotopy catego…

P / partially_solved | AMR 问题清单汇编

In our memoir, we give a conjecture for the thick subcategories in a Noetherian stable homotopy category C--they should be in 1-1 correpondence with subsets of Spec pi_* S closed under specialization. Prove this conjecture. (This seems out of reach to me, but you never know).

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08474 | AMR-110-0042

### Axiomatic stable homotopy 10 - What is the endomorphism ring of the identity functor on the stable homotopy category?

O / open | AMR 问题清单汇编

What is the endomorphism ring of the identity functor on the stable homotopy category? The ring Z splits off this ring, including by multiples of the identity, and projecting off by observing what the natural transformation does to the identity map of S^0. If we assume the generating hypothesis, any element of the kernel is a natural self-phantom map of X for all X. This must force it to be 0, but why? And do we really need the generating hypothesis?

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08475 | AMR-110-0034

### Axiomatic stable homotopy 2 - Characterize the stable homotopy category up to equivalence.

P / partially_solved | AMR 问题清单汇编

Characterize the stable homotopy category up to equivalence. This has been done for categories that are homotopy categories of model categories by Schwede, I think, and a lot is known in general. Margolis characterized the quotient category when phantoms are killed, and Christensen-Strickland characterized the phantom subcategory. So the category is determined up to a square 0 extension, but maybe there are tons of different such square 0 extensions.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08476 | AMR-110-0035

### Axiomatic stable homotopy 3 - Show that there is only a set of localizing subcategories.

S / solved | AMR 问题清单汇编

Show that there is only a set of localizing subcategories. It is known that there is only a set of Bousfield classes (Ohkawa; Strickland simplified his proof a bit, and then Dwyer and Palmieri have an even simpler proof). Of course, we expect there to be a 1-1 correspondence between Bousfield classes, localizing subcategories, and colocalizing subcategories, but perhaps that is out of reach.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08477 | AMR-110-0036

### Axiomatic stable homotopy 4 - In one of Bob Thomason's last papers, he determined the thick subcategories of finite objects in the d…

P / partially_solved | AMR 问题清单汇编

In one of Bob Thomason's last papers, he determined the thick subcategories of finite objects in the derived category of a scheme. For the derived category of a ring R, they are in 1-1 correspondence with unions of closed sets X_i in Spec R such that Spec R - X_i is quasi-compact. How about localizing subcategories? What about applying his ideas to more general non-Noetherian stable homotopy categories?

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08478 | AMR-110-0037

### Axiomatic stable homotopy 5 - John Palmieri has determined the E_2 term of the Adams spectral sequence up to nilpotence--at least he…

P / partially_solved | AMR 问题清单汇编

John Palmieri has determined the E_2 term of the Adams spectral sequence up to nilpotence--at least he has found a computable ring which is f-isomorphic to the E_2 term. I think this is only at p=2. Will his methods work in other specific non-Noetherian stable homotopy categories? Haynes Miller suggested trying to use John's methods to calculate group cohomology of GL(infinity) up to f-isomorphism.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08479 | AMR-110-0038

### Axiomatic stable homotopy 6 - My general feeling about stable homotopy categories is that they are like commutative rings.

P / partially_solved | AMR 问题清单汇编

My general feeling about stable homotopy categories is that they are like commutative rings. Follow this up; define Spec C for example, for a stable homotopy category C. Several people have had an idea like this; Jack Morava said something to me about it that I forgot.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08480 | AMR-110-0039

### Axiomatic stable homotopy 7 - The equivariant stable homotopy category is not treated very well in our memoir.

P / partially_solved | AMR 问题清单汇编

The equivariant stable homotopy category is not treated very well in our memoir. That is, we assume that the generators have to be dualizable. This is not true unless you use the complete universe. Peter May tried to talk us out of this at the time, and I think he was right. So try to understand what happens when the generators are not dualizable. Is there some other condition that does hold in the equivariant stable homotopy category over an incomplete universe that replaces this? There is a notion of weakly dualizable, for example, which is just that D^2 X = X. Maybe the generators are weakly dualizable?

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08481 | AMR-110-0040

### Axiomatic stable homotopy 8 - From an axiomatic point of view, I don't understand Grojnowski's equivariant elliptic cohomology.

P / partially_solved | AMR 问题清单汇编

From an axiomatic point of view, I don't understand Grojnowski's equivariant elliptic cohomology. This theory takes values in an abelian category that is not modules over a ring--I think it is sheaves over a scheme or something. Is there some way to understand homology theories that land in, say, Grothendieck categories, from an axiomatic point of view?

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08482 | AMR-110-0041

### Axiomatic stable homotopy 9 - Suppose G is a self-equivalence of the stable homotopy category.

P / partially_solved | AMR 问题清单汇编

Suppose G is a self-equivalence of the stable homotopy category. Must G be some iterate of the suspension functor? If G commutes with the suspension, I can prove that GS^0 = S^n for some n, but this is almost all that I know. One could ask the same question for the K(n) local category or the E(n) local category.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08483 | OWR-13490-017

### Bagchi’s Lower Bound for Triangulated Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

For a $d$-dimensional triangulated homology manifold and every relevant index $i$, is $g_i\geq\binom{d+1}{i}\left((-1)^i+\sum_{j=1}^{i}(-1)^{i-j}\mu_{j-1}\right)$, where the $\mu_j$ are the averaged discrete Morse numbers of the triangulation?

来源字段：https://doi.org/10.4171/owr/2015/5

08484 | OWR-4798-013

### Ball-Product Structure of Cross-Polytope Subcomplexes

O / open | Oberwolfach 报告问题汇编

Let $B(i,d)$ be the subcomplex of the boundary of the $d$-dimensional cross-polytope generated by facets whose encoding words have at most $i$ switches. Is $B(i,d)$ a combinatorial triangulation of $S^{i}\times B^{d-i-1}$?

来源字段：https://doi.org/10.4171/owr/2011/08

08485 | OWR-14297741-006

### Balmer Spectra for Constant Mackey Functors

O / open | Oberwolfach 报告问题汇编

For $G=C_p$ and $R$ a constant Mackey functor associated with a commutative ring, describe $\operatorname{Spc}(\operatorname{Perf}_{C_p}(HR))$.

来源字段：https://doi.org/10.4171/owr/2023/40

08486 | OWR-14297741-005

### Balmer Spectra of Equivariant Eilenberg-MacLane Modules

O / open | Oberwolfach 报告问题汇编

For a finite group $G$ and a Mackey functor $M$, describe the Balmer spectrum $\operatorname{Spc}(\operatorname{Perf}_{G(HM)})$ of compact modules over the equivariant Eilenberg-MacLane spectrum $HM$.

来源字段：https://doi.org/10.4171/owr/2023/40

08487 | TOP-007

### Baum-Connes Conjecture

P / partially_solved | 独立 / 未分配集合

Is the assembly map in K-theory an isomorphism for all locally compact groups?

背景链接：https://arxiv.org/abs/1905.10081

08488 | OWR-16407-009

### Baum-Connes Assembly for Minimal Exact Crossed Products

P / partially_solved | Oberwolfach 报告问题汇编

For the minimal exact crossed-product functor $\rtimes_E$, understand the reformulated Baum-Connes assembly map $K_*^{\mathrm{top}}(G;A)\to K_*(A\rtimes_E G)$, including concrete realization and computation of the crossed product.

来源字段：https://doi.org/10.4171/owr/2018/32

08489 | OWR-1060-009

### Baum-Connes Conjecture for Torsion-Free Groups

O / open | Oberwolfach 报告问题汇编

For a torsion-free discrete group $G$, the Baum-Connes assembly map from $K$-homology $K_*(BG)$ to $K_*(C_r^*(G))$ should be an isomorphism.

来源字段：https://doi.org/10.4171/owr/2005/39

08490 | OWR-4084-007

### Baum-Connes Theory for Non-Hausdorff Groupoids

P / partially_solved | Oberwolfach 报告问题汇编

Why does Tu's Baum-Connes result for amenable groupoids require the groupoid $G$ to be Hausdorff, and how can crossed products and equivariant bivariant $KK$-theory be extended to non-Hausdorff, locally Hausdorff actions?

来源字段：https://doi.org/10.4171/owr/2009/41

08491 | TOP-008

### Berge Conjecture

P / partially_solved | 独立 / 未分配集合

Are Berge knots the only knots in $S^3$ admitting lens space surgeries?

背景链接：https://doi.org/10.4007/annals.2013.177.2.3

08492 | OWR-17470-015

### Bestvina Complexes Beyond Compact Strict Fundamental Domains

O / open | Oberwolfach 报告问题汇编

For which group actions with noncompact or non-strict fundamental domains can a Bestvina complex be constructed?

来源字段：https://doi.org/10.4171/owr/2020/7

08493 | OWR-15581-010

### Betti Sequences of Closed Smooth Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Which sequences $b_0=1,b_1,\ldots,b_{4k}=1$ with $b_{2k}$ odd can occur as the Betti numbers of a closed smooth $4k$-manifold?

来源字段：https://doi.org/10.4171/owr/2017/38

08494 | OWR-15427-003

### Binomial Bounds for Gamma Vectors of Flag Spheres

P / partially_solved | Oberwolfach 报告问题汇编

Let $\Delta$ be a flag homology sphere with gamma-vector $(\gamma_0,\gamma_1,\ldots)$ and $\gamma_1=\ell$. Must $$0\leq\gamma_i\leq\binom{\ell}{i}\qquad\text{for every }i\geq2$$?

来源字段：https://doi.org/10.4171/owr/2017/19

08495 | OWR-8415338-002

### Bogomolov-Miyaoka-Yau Inequality for Smooth Four-Manifolds

O / open | Oberwolfach 报告问题汇编

Let $X$ be a closed, oriented, smooth four-manifold with $b_1(X)=0$, odd $b^+(X)\geq3$, Seiberg-Witten simple type, and a nonzero Seiberg-Witten invariant. Must $$c_1(X)^2\leq3c_2(X)$$ hold?

来源字段：https://doi.org/10.4171/owr/2021/35

08496 | OWR-15208-008

### Borderline Continuity of Conformal Metric Parametrizations

P / partially_solved | Oberwolfach 报告问题汇编

For a simply connected surface $V$ equal to $\mathbb{R}^2$ or $S^2$, does the parametrization $(u,\phi)\mapsto\phi^*e^{-2u}g_\kappa$ remain a homeomorphism in the borderline $C^{r,0}$ metric, function, and diffeomorphism topologies?

来源字段：https://doi.org/10.4171/owr/2017/3

08497 | OWR-1319-004

### Bordism-Valued Moduli Spaces of Morse Flow Lines

O / open | Oberwolfach 报告问题汇编

For nonconsecutive critical points $P$ and $Q$ in a Morse flow, how should the noncompact moduli space $M(P,Q)$ of flow lines be measured or assigned a framed-bordism value?

来源字段：https://doi.org/10.4171/owr/2006/33

08498 | TOP-005

### Borel Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Borel Conjecture [TOP-012]

Are aspherical closed manifolds determined up to homeomorphism by their fundamental groups?

背景链接：https://doi.org/10.4007/annals.2012.175.2.5

08499 | AIM-TOPOLOGY-0199

### Boundary-sensitive extensions of the Thurston metric

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.8.** Let \(S\) be a hyperbolic surface with ideal boundary. Can we define the Thurston metric on the non-reduced Teichmüller space of \(S\)?

There is no omitted attribution or remark in the PDF.

The nearby context is essential. Immediately before this question, Problems 3.6--3.7 discuss the **reduced**, finite-dimensional Teichmüller space of a compact bordered surface and its arc metric. Problem 3.8 changes both the boundary geometry and the equivalence relation:

- The **ideal boundary** is the conformal boundary of a hyperbolic Riemann surface. In a Fuchsian model it is \(\Omega_\Gamma/\Gamma\). Equivalently, in the 2024 terminology of Alekseev--Meinrenken, the interior carries a conformally compact hyperbolic or hyperbolic \(0\)-metric; its ends are funnels or trumpets and their boundary circles lie at infinite hyperbolic distance. - A reduced Teichmüller point is a marked surface modulo a conformal comparison map in the ordinary homotopy class. In the **non-reduced** (or unreduced) space, that homotopy is required to be relative to the ideal boundary. Thus boundary parametrizations are retained. For nonempty ideal boundary the resulting space is infinite-dimensional.

These meanings agree with the explicit definitions in Fortier Bourque and with the hyperbolic \(0\)-metric model of Alekseev--Meinrenken. They are distinct from the geodesic-boundary arc-metric theory of Liu--Papadopoulos--Su--Théret.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08500 | OWR-15177-015

### Boundedness of Tautological Kappa Classes

P / partially_solved | Oberwolfach 报告问题汇编

Are the tautological kappa classes on mapping class groups bounded cohomology classes?

来源字段：https://doi.org/10.4171/owr/2016/56

08501 | OWR-12984-009

### Bounding Immersed Curves by Stable Singular Disk Maps

O / open | Oberwolfach 报告问题汇编

Given a generically immersed closed curve $\gamma$ in an oriented surface $S_g$, does there exist a compact $3$-manifold $M$ with $\partial M=S_g$ and a proper map $D\to M$ from a disk, having only stable singularities, such that $\partial D=D\cap S_g=\gamma$?

来源字段：https://doi.org/10.4171/owr/2014/26

08502 | AIM-TOPOLOGY-0126

### Branched covers do not determine knot concordance

S / solved | AIM 研讨会问题汇编

**Branched covers and concordance.** Suppose \(K\) and \(J\) are knots in \(S^3\) such that for every prime power \(q\), the \(q\)th cyclic branched covers \(\Sigma_q(K)\) and \(\Sigma_q(J)\) are rationally homology cobordant. Must \(K\) and \(J\) be concordant?

来源字段：http://aimpl.org/concordsliceknot/4/

08503 | AMR-102-0016

### Branched surfaces and triangulations - Question 7.1

O / open | AMR 问题清单汇编

Characterize branched surfaces embedded in $3$-manifolds which can be non-trivially split to a homeomorphic copy of themselves.

背景链接：https://arxiv.org/abs/math/0209081

08504 | AMR-102-0017

### Branched surfaces and triangulations - Question 7.2

O / open | AMR 问题清单汇编

Develop a theory of hierarchies for branched surfaces.

背景链接：https://arxiv.org/abs/math/0209081

08505 | AMR-102-0018

### Branched surfaces and triangulations - Question 7.3

P / partially_solved | AMR 问题清单汇编

Which boundary slopes are realized by essential laminations carried by a fixed taut ideal triangulation? Give an algorithm.

背景链接：https://arxiv.org/abs/math/0209081

08506 | AMR-102-0019

### Branched surfaces and triangulations - Question 7.4

O / open | AMR 问题清单汇编

When does a Haken sum operation make sense for a pair of laminations in normal form with respect to a fixed triangulation?

背景链接：https://arxiv.org/abs/math/0209081

08507 | AMR-102-0020

### Branched surfaces and triangulations - Question 7.5

O / open | AMR 问题清单汇编

Let $M$ be a $3$-manifold, and $\Lambda$ an essential lamination. Let $C$ be a cycle representing the fundamental class of $M$. Is there a cycle $C'$ with the same Gromov norm as $C$, and another essential lamination $\Lambda'$ which is normal with respect to $C'$?

背景链接：https://arxiv.org/abs/math/0209081

08508 | AMR-102-0021

### Branched surfaces and triangulations - Question 7.7

O / open | AMR 问题清单汇编

Suppose $\mathscr{B}$ is a branched surface in $M$ which is dual to a taut local orientation. Is there a finite cover of $M$ in which the pullback of $\mathscr{B}$ fully carries a lamination? What about an amenable cover?

背景链接：https://arxiv.org/abs/math/0209081

08509 | AMR-102-0022

### Branched surfaces and triangulations - Question 7.8

P / partially_solved | AMR 问题清单汇编

Do branched surfaces without sink disks carry automatic laminations?

背景链接：https://arxiv.org/abs/math/0209081

08510 | AMR-102-0023

### Branched surfaces and triangulations - Question 7.9

O / open | AMR 问题清单汇编

Give a useful definition of thin position for an embedded graph $\Gamma \subset M$ with respect to a taut foliation $\mathscr{F}$. If $\Gamma$ is the $1$-skeleton $\tau^1$ for a triangulation, can one find an isotopy such that the leaves of $\mathscr{F}$ are made up of polyhedral disks of bounded index?

背景链接：https://arxiv.org/abs/math/0209081

08511 | OWR-8415349-001

### Bubble-Free Criterion for Polygraphic and Nerve Homology

O / open | Oberwolfach 报告问题汇编

For a two-dimensional polygraph, does its polygraphic homology coincide with the homology of its nerve if and only if the polygraph is bubble-free?

来源字段：https://doi.org/10.4171/owr/2021/46

08512 | OWR-12697687-012

### Cable Torsion and Sliceness of Knots

P / partially_solved | Oberwolfach 报告问题汇编

Is the $(p,q)$-cable of a knot $K$ torsion in the knot concordance group if and only if $K$ is slice and $|q|=1$?

来源字段：https://doi.org/10.4171/owr/2023/4

08513 | OWR-12697687-008

### Cancellation and Fixed-Form Stable Classes of Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Can boundaries of thickened even-dimensional complexes be used systematically to produce manifolds with arbitrarily large fixed-form homotopy stable class $\mathcal S_{\lambda}^{\mathrm{st},h}(M)$? For which fundamental groups is $\mathcal S_{\lambda}^{\mathrm{st},h}(M)$ trivial, finite, or infinite, particularly for closed four-manifolds?

来源字段：https://doi.org/10.4171/owr/2023/4

08514 | AIM-TOPOLOGY-0042

### Canonical-basis skeins in the bigon

P / partially_solved | AIM 研讨会问题汇编

State Skein algebra of bigons

Find skeins representing the basis for the isomorphism “Stated Skein Algebra of bigon \(\cong O_q(SL_n)\) for \(n\geq 3\)” where \(O_q(SL_n)\) has Kashiwara-Lusztig canonical basis.

来源字段：http://aimpl.org/quantumlowdimtop/2/

08515 | OWR-14298370-007

### Carlsson’s Homology Bound for Free Group Actions

P / partially_solved | Oberwolfach 报告问题汇编

If an elementary abelian $p$-group $G\cong(\mathbb{Z}/p\mathbb{Z})^d$ acts freely on a nontrivial finite CW complex $X$, must $\sum_n\dim_kH_n(X;k)\geq2^d$?

来源字段：https://doi.org/10.4171/owr/2024/20

08516 | OWR-14299911-026

### Cartan-Hadamard Theorem without Uniform Local Medianity

O / open | Oberwolfach 报告问题汇编

Let $X$ be a complete simply connected path-metric space in which every point has a neighborhood where each triple has a unique median. Must $X$ be a median space even when the radii of these neighborhoods are not uniformly bounded below?

来源字段：https://doi.org/10.4171/owr/2026/8

08517 | OWR-15578-008

### Cartan-Hadamard Isoperimetric Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Does every Cartan-Hadamard manifold $(M^{n+1},g)$ satisfy $$ |\Omega|\leq n^{-(n+1)/n}\omega_{n+1}^{-1/n}|\partial\Omega|^{(n+1)/n} $$ for every bounded sufficiently regular open set $\Omega\subset M$?

来源字段：https://doi.org/10.4171/owr/2017/35

08518 | OWR-17470-013

### CAT(0) Geometry and Retraction of the Bestvina Complex

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Bestvina-Complex Retraction Question [OWR-17470-014]

For a group $G$ equipped with the Bestvina and Davis complexes arising from a strictly developable simple complex of groups, does the Bestvina complex admit a $G$-invariant CAT(0) metric? Is it a $G$-equivariant deformation retract of the Davis complex?

来源字段：https://doi.org/10.4171/owr/2020/7

08519 | OWR-11790-003

### Categorification and Higher-Dimensional Extensions of Parity Brackets

O / open | Oberwolfach 报告问题汇编

How can one construct a homological complex whose Euler characteristic is the parity bracket, and how can parity theory and a bracket be generalized to higher-dimensional objects such as surfaces in 4-space?

来源字段：https://doi.org/10.4171/owr/2012/28

08520 | OWR-12014-008

### Categorification of the Quantum Volume Conjecture

O / open | Oberwolfach 报告问题汇编

Does the generalized or quantum volume conjecture for the colored Jones polynomial admit a natural categorification, yielding a two-parameter super-A-polynomial governing the color behavior of HOMFLY homology?

来源字段：https://doi.org/10.4171/owr/2012/42

08521 | AIM-TOPOLOGY-0025

### Category separation and formal-data audit for codimension-two Weinstein embeddings

P / partially_solved | AIM 研讨会问题汇编

**Flexibility.** Do properly embedded codimension 2 Weinstein domains satisfy an existence h-principle?

来源字段：http://aimpl.org/highdimcontacttop/10/

08522 | OWR-1596-004

### Category-Weight Constructions for Nonsimply Connected Spaces

O / open | Oberwolfach 报告问题汇编

Can the category-weight construction that assigns a zero-divisor $\phi(u)\in H^*(X\times X)$ to a sufficiently high-category-weight cohomology class $u\in H^*(X)$, together with its cohomology-weight lower bound, be extended from simply connected spaces to nonsimply connected spaces?

来源字段：https://doi.org/10.4171/owr/2007/40

08523 | AIM-TOPOLOGY-0071

### Centering the double-line Hilbert scheme and a derived-loop model at n=1

P / partially_solved | AIM 研讨会问题汇编

We define Hilb\(^n(x^2=0)=\{\text{codiminsional }n \text{ ideals in }\mathbb{C}[x,y]/(x^2=0)\}\subseteq\) Hilb\(^n(\mathbb{C}^2)\)

Relate Hilb\(^n(x^2=0)\) to \(HH\) of the arc algebra.

来源字段：http://aimpl.org/agclinkhom/4/

08524 | OWR-12014-014

### Chain-Independent Homology Growth in Finite Covers

O / open | Oberwolfach 报告问题汇编

Let $G$ be finitely presented. For every field $K$ and every inverse system of finite-index normal subgroups $(G_i)_{i\in I}$ with $\bigcap_{i\in I}G_i=\{1\}$, does the limit $\lim_{i\in I}b_1(G_i;K)/[G:G_i]$ exist and remain independent of the choices of $(G_i)$ and $K$?

来源字段：https://doi.org/10.4171/owr/2012/42

08525 | OWR-12697687-009

### Characterising and Non-Characterising Slopes of Knots

P / partially_solved | Oberwolfach 报告问题汇编

Given a knot $K\subset S^3$, classify all of its characterizing and noncharacterizing Dehn-surgery slopes.

来源字段：https://doi.org/10.4171/owr/2023/4

08526 | OWR-11787-003

### Cheeger-Müller Gluing Without Product Structures

O / open | Oberwolfach 报告问题汇编

In the Cheeger-Müller theorem for singular manifolds, can the assumption that all structures are products near the gluing boundary be removed?

来源字段：https://doi.org/10.4171/owr/2012/25

08527 | AIM-TOPOLOGY-0236

### Chern-lattice obstructions to positively curved principal two-torus bundles

P / partially_solved | AIM 研讨会问题汇编

**Problem 23.** Is there a principal \(T^2\)-bundle whose total space admits \(\sec>0\)?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08528 | OWR-13499-003

### Chromatic Fiber Structure at Bad Primes

O / open | Oberwolfach 报告问题汇编

At primes p that are bad relative to chromatic level n, give a plausible systematic description of the homotopy type of $\Sigma F_n$, where $F_n$ is the common fiber in the chromatic splitting square for the p-local sphere.

来源字段：https://doi.org/10.4171/owr/2015/14

08529 | OWR-17132-006

### Chromatic Fixed-Point Implications for Finite p-Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $H<G$ be subgroups of a finite $p$-group and let $m,n$ be nonnegative integers. For which pairs $(m,n)$ does $$K(m)_*(\Phi^H X)=0\quad\Longrightarrow\quad K(n)_*(\Phi^G X)=0$$ hold for every finite $G$-spectrum $X$?

来源字段：https://doi.org/10.4171/owr/2019/36

08530 | OWR-10252932-002

### Chromatic Lichtenbaum-Quillen and Redshift Conjectures

P / partially_solved | Oberwolfach 报告问题汇编

How does algebraic $K$-theory shift chromatic height; in particular, is the algebraic $K$-theory of an $L^f_{n-1}$-local ring $T(m+1)$-acyclic for every $m\geq n$?

来源字段：https://doi.org/10.4171/owr/2022/24

08531 | OWR-13499-002

### Chromatic Splitting of the Local Sphere

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Chromatic-Splitting Extraction [OWR-13499-001]

For the $p$-local sphere spectrum, let $L_nS$ denote its chromatic localizations and let $F_n$ be the common fiber in the chromatic fracture square. Does $L_{n-1}L_{K(n)}S\simeq L_{n-1}S\vee\Sigma F_n$, with $\Sigma F_n$ admitting the predicted wedge decomposition into lower chromatic layers?

来源字段：https://doi.org/10.4171/owr/2015/14

08532 | AIM-TOPOLOGY-0048

### Circle gluing as balanced endomorphism extraction

P / partially_solved | AIM 研讨会问题汇编

**Gluing Stated skein algebra.** Is there a formula for gluing stated skein algebra along circles?

来源字段：http://aimpl.org/quantumlowdimtop/2/

08533 | AIM-TOPOLOGY-0151

### Circular order and fixed-puncture subgroup obstructions

P / partially_solved | AIM 研讨会问题汇编

Give an example of a pair $(S,G)$ of a surface and a countable group such that $G$ is not a subgroup of $\operatorname{Map}(S)$. What obstructions exist barring countable groups from being subgroups of families of big mapping class groups?

For example, $S^2-C$ is circularly orderable, so its subgroups must also be circularly orderable.

来源字段：http://aimpl.org/genusinfinity/4/

08534 | AMR-102-0034

### Classical 3-manifold theory - Question 9.1

O / open | AMR 问题清单汇编

Is there a universal transverse surgery description of tautly foliated manifolds, in the sense that there is a fixed $M$ such that for every tautly foliated manifold $N,\mathscr{F}$ there is a link $L \subset N$ transverse to $\mathscr{F}$ so that $M$ is obtained from $N$ by surgery on $L$?

背景链接：https://arxiv.org/abs/math/0209081

08535 | AMR-102-0035

### Classical 3-manifold theory - Question 9.2

O / open | AMR 问题清单汇编

Give a collection of fundamental operations on foliations and an explicit family of base foliations such that every tautly foliated manifold $M,\mathscr{F}$ is obtained from one of the base family by repeated application of fundamental operations.

背景链接：https://arxiv.org/abs/math/0209081

08536 | AMR-102-0036

### Classical 3-manifold theory - Question 9.3

P / partially_solved | AMR 问题清单汇编

What is the most general class of knots to which the techniques of Delman-Roberts (in constructing persistent laminations) can be extended?

背景链接：https://arxiv.org/abs/math/0209081

08537 | AMR-102-0037

### Classical 3-manifold theory - Question 9.4

O / open | AMR 问题清单汇编

Suppose $K$ is a non-torus alternating knot. Then essential laminations can be constructed which realize every (nontrivial) boundary slope. Can essential laminations be constructed with an even sided bundle complementary region containing $K$, so that every nontrivial surgery can be filled in with a monkey saddle?

背景链接：https://arxiv.org/abs/math/0209081

08538 | AMR-102-0038

### Classical 3-manifold theory - Question 9.5

O / open | AMR 问题清单汇编

It is known that if a $3$-manifold $M$ contains an essential surface of genus $g$, the distance of any Heegaard splitting of $M$ has distance at most $2g$. Does the ``distance filtration'' put any useful structure on the essential laminations supported by a given $M$? i.e. if $M$ admits Heegaard splittings of distance at least $2g$, what can one say about the essential laminations $\Lambda$ contained in $M$?

背景链接：https://arxiv.org/abs/math/0209081

08539 | OWR-12984-005

### Classification of Cyclic Covers of Two-Bridge Knots

O / open | Oberwolfach 报告问题汇编

For every two-bridge knot $K$ and integer $n>1$, determine whether the cyclic branched cover $\Sigma_n(K)$ is a total L-space, whether $\pi_1(\Sigma_n(K))$ is left-orderable, and whether $\Sigma_n(K)$ is excellent.

来源字段：https://doi.org/10.4171/owr/2014/26

08540 | OWR-14299915-016

### Classification of Minimal Synthetic Buildings

P / partially_solved | Oberwolfach 报告问题汇编

For each finite group $G$ and prime $p$, prove that there are only finitely many equivariant homotopy types of minimal synthetic buildings; prove that the $p$-subgroup complex $\Delta(\mathcal S_p(G))$ has the equivariant homotopy type of one; and prove that every minimal synthetic building is equivariantly homotopy equivalent to a homomorphic image of such a complex. For groups of Lie type in characteristic $p$, prove that every nontrivial minimal synthetic building is equivariantly homotopy equivalent to the Tits building.

来源字段：https://doi.org/10.4171/owr/2026/9

08541 | OWR-14298591-008

### Classification of Static Near-Horizon Geometries

O / open | Oberwolfach 报告问题汇编

For cosmological constant $\Lambda\geq0$ and horizon dimension $n\geq3$, classify the compact static near-horizon geometries, equivalently the compact solutions of the near-horizon quasi-Einstein equation with closed dual one-form $X^\flat$.

来源字段：https://doi.org/10.4171/owr/2024/36

08542 | AIM-TOPOLOGY-0094

### Clique and cubical-nerve realizations of digital homotopy groups

P / partially_solved | AIM 研讨会问题汇编

This problem assumes that higher homotopy groups of digital spaces have been constructed.

Is there a functor \(F\) from the category of digital images to that of topological spaces such that the digital invariants of a digital image \(X\) agree with the corresponding classical invariants of \(FX\), e.g., \(\pi_n(X,*)\cong \pi_n(FX,*)\)?

来源字段：http://aimpl.org/combhomotop/1/

08543 | AIM-TOPOLOGY-0172

### Closed geodesics from compact trapping and cover monodromy

P / partially_solved | AIM 研讨会问题汇编

What conditions can we put on an infinite translation surface to ensure that it contains a closed geodesic?

来源字段：http://aimpl.org/genusinfinity/5/

08544 | OWR-14299512-005

### Closed-Orbit Multiplicity on Star-Shaped Hypersurfaces

O / open | Oberwolfach 报告问题汇编

Let $M$ be the smooth boundary of a star-shaped domain in $\mathbb R^{2n}$. Must its Reeb flow have at least $n$ prime closed orbits?

来源字段：https://doi.org/10.4171/owr/2025/31

08545 | OPG-48767

### Closing Lemma for Diffeomorphism (Dynamical Systems)

P / partially_solved | Open Problem Garden

Conjecture Let $f\in Diff^{r}(M)$ and $p\in\omega_{f}$. Then for any neighborhood $V_{f}\subset Diff^{r}(M)$ there is $g\in V_{f}$ such that $p$ is periodic point of $g$

There is an analogous conjecture for flows ( $C^{r}$ vector fields. In the case of diffeos this was proved by Charles Pugh for $r = 1$. In the case of Flows this has been solved by Sushei Hayahshy for $r = 1$. But in the two cases the problem is wide open for $r > 1$

背景链接：http://www.openproblemgarden.org/op/closing_lemma_for_diffeomorphism_dynamical_systems

08546 | AIM-TOPOLOGY-0160

### Co-Hopfian and Hopfian big mapping class groups

P / partially_solved | AIM 研讨会问题汇编

Is every injective self-homomorphism from $\operatorname{Map}(S)$ an isomorphism? That is to say, for which $S$ is $\operatorname{Map}(S)$ co-Hopfian (or Hopfian)?

来源字段：http://aimpl.org/genusinfinity/4/

08547 | AIM-TOPOLOGY-0168

### Coarse and conformal diagnostics for the Chamanara surface

P / partially_solved | AIM 研讨会问题汇编

Can you relate the flat and hyperbolic geometry of a given surface? That is, show how to uniformize the flat structure. For instance, describe (up to quasi-isometry) the hyperbolic structure on the Loch Ness monster corresponding to the flat structure given by the Chamanara surface.

来源字段：http://aimpl.org/genusinfinity/5/

08548 | OWR-13488-004

### Coarse Baum-Connes Failure without Embedded Expanders

P / partially_solved | Oberwolfach 报告问题汇编

Let a finitely generated group $G$ act ergodically by measure-preserving Lipschitz homeomorphisms on a compact metric probability space $M$, and suppose the action has a spectral gap. Must the warped cone $\mathcal O_G(M)$ fail the coarse Baum-Connes conjecture? Does there exist such an action for which $\mathcal O_G(M)$ contains no coarsely embedded sequence of expanders?

来源字段：https://doi.org/10.4171/owr/2015/3

08549 | OWR-11137-007

### Coarse Baum-Connes Maps for Large-Girth Graph Spaces

O / open | Oberwolfach 报告问题汇编

For a coarse space formed from bounded-degree finite graphs whose girths tend to infinity, what are the injectivity and surjectivity properties of the coarse Baum-Connes assembly map, especially for expanders and geometric property (T)?

来源字段：https://doi.org/10.4171/owr/2011/45

08550 | AMR-102-0051

### Coarse foliations - Question 12.1

O / open | AMR 问题清单汇编

Suppose $\rho:\pi_1(M) \to \mathbb{R}$ is a $1$-cochain with bounded coboundary; i.e. there is a uniform $C$ so that $$|\rho(\alpha) + \rho(\beta) - \rho(\alpha\beta)|<C$$ for all $\alpha,\beta \in \pi_1(M)$. Consider $\pi_1(M)$ as a metric space by thinking of it as the vertices of some Cayley graph. Let $L_\rho = \rho^{-1}(I) \subset \pi_1(M)$.

- Are the coarse connected components of $L_\rho$ coarsely simply connected?

- If the answer is ``yes'', is there a sense in which $L_\rho$ and its translates by $\pi_1(M)$ can be thought of as coarse minimal planes in $\pi_1(M)$?

背景链接：https://arxiv.org/abs/math/0209081

08551 | AMR-102-0052

### Coarse foliations - Question 12.2

O / open | AMR 问题清单汇编

Does every hyperbolic $3$-manifold admit a taut cone field? That is, a cone field $C$ which is recurrent and supports only homotopically essential loops.

背景链接：https://arxiv.org/abs/math/0209081

08552 | AMR-102-0053

### Coarse foliations - Question 12.3

O / open | AMR 问题清单汇编

What deformations of a foliation or lamination should be thought of as ``inessential''? For instance -- monotone equivalence, cut-and-shear along a surface or transverse lamination, isotopy of branch locus in a branched cover, isomorphic universal circles etc.

背景链接：https://arxiv.org/abs/math/0209081

08553 | OWR-1588-006

### Codimension-Two and Index Obstructions to Positive Curvature

S / solved | Oberwolfach 报告问题汇编

Let $M$ be a closed smooth spin manifold. What is the relation between the codimension-$2$ obstruction to positive scalar curvature and the index obstruction $\alpha(M)$ in real $K$-theory? For manifolds to which the codimension-$2$ method applies, must $\alpha(M)\neq0$?

来源字段：https://doi.org/10.4171/owr/2007/32

08554 | OWR-2491-009

### Cofinite Models for Universal Proper-Action Spaces

O / open | Oberwolfach 报告问题汇编

Let $E\Gamma$ be a universal space for proper $\Gamma$-actions, so every finite-subgroup fixed-point set is nonempty and contractible. Characterize the groups $\Gamma$ for which $\Gamma\backslash E\Gamma$ has finite CW type, and construct explicit cofinite models, especially for arithmetic groups and mapping class groups.

来源字段：https://doi.org/10.4171/owr/2008/53

08555 | OWR-15177-009

### Coherence and Finiteness Properties of Surface-Bundle Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $E$ be the total space of a bundle with closed-surface base $B$ and fiber $F$, and let $\pi=\pi_1(E)$. Is $\pi$ coherent? Does every finitely presented subgroup $G\leq\pi$ admit a finite $K(G,1)$? More generally, is every subgroup of type $\mathrm{FP}^{\lfloor n/2\rfloor}$ of a $\mathrm{PD}^n$ group necessarily of type FP?

来源字段：https://doi.org/10.4171/owr/2016/56

08556 | OWR-14297744-007

### Coherence of Equivariant Formal-Group Moduli Stacks

O / open | Oberwolfach 报告问题汇编

For every compact Lie group $G$, prove that the stacks $\mathcal M_{\mathrm{FG}}^G$ are coherent, meaning that coherent sheaves corresponding to comodules with finitely presented underlying modules form an abelian category.

来源字段：https://doi.org/10.4171/owr/2023/34

08557 | OWR-13687-005

### Cohomological Purity for Finite Index Mapping Class Subgroups

O / open | Oberwolfach 报告问题汇编

For sufficiently natural finite-index subgroups $\Delta_g<\Gamma_g$, is $H^k(M_{\Delta_g})\to H^k(\Delta_g)$ onto for $g\gg 0$, equivalently is $H^k(\Delta_g)$ pure?

来源字段：https://doi.org/10.4171/owr/2015/30

08558 | OWR-12697687-004

### Collapse and Involutions in the Khovanov-Floer Spectral Sequence

O / open | Oberwolfach 报告问题汇编

For which knots does the involutive Khovanov-Floer spectral sequence collapse immediately, and what does its collapse page signify? When can it determine the involutive Heegaard Floer homology of the branched double cover? Does $\widetilde{BN}$ arise from a nontrivial homotopy involution of the Khovanov complex, and what chain endomorphism does reflection followed by crossing reversal induce?

来源字段：https://doi.org/10.4171/owr/2023/4

08559 | OWR-11786-016

### Collapse Preservation under Subdivision

P / partially_solved | Oberwolfach 报告问题汇编

If a simplicial complex $C$ collapses to a subcomplex $C'$, and $D$ is a subdivision of $C$ restricting to a subdivision $D'$ of $C'$, must $D$ collapse to $D'$?

来源字段：https://doi.org/10.4171/owr/2012/24

08560 | OWR-15987-001

### Combinatorial Determination of Arrangement Milnor-Fiber Betti Numbers

O / open | Oberwolfach 报告问题汇编

Are the Betti numbers $b_i(F)$ of the Milnor fiber $F$ of a complex hyperplane arrangement determined by the arrangement's combinatorial data $d_{\mathcal A}$?

来源字段：https://doi.org/10.4171/owr/2018/2

08561 | OWR-15987-002

### Combinatorial Determination of Milnor-Fiber Monodromy

O / open | Oberwolfach 报告问题汇编

Let $\mathcal{A}$ be a central hyperplane arrangement in $\mathbb{C}^n$ with Milnor fiber $F$ and algebraic monodromy homomorphism $h : H^1(F; \mathbb{C}) \to H^1(F; \mathbb{C})$. Is the first-cohomology monodromy homomorphism $H^1(F) \to H^1(F)$ combinatorially determined by the intersection lattice / arrangement data $d_{\mathcal{A}}$?

来源字段：https://doi.org/10.4171/owr/2018/2

08562 | OWR-4416-017

### Commensurability Classes of Hyperbolic Knot Complements

O / open | Oberwolfach 报告问题汇编

For a hyperbolic knot $K$, describe the set $C(K)$ of knots whose complements are commensurable with the complement of $K$, including upper bounds for its cardinality and a characterization of when $|C(K)|>1$.

来源字段：https://doi.org/10.4171/owr/2010/35

08563 | OWR-14299911-002

### Common Refinements of PL-Homeomorphic Simplicial Complexes

P / partially_solved | Oberwolfach 报告问题汇编

Given two PL-homeomorphic simplicial complexes, does there exist a common refinement reachable from both by elementary moves? Determine the corresponding factorization relation and its analogues in toric geometry.

来源字段：https://doi.org/10.4171/owr/2026/8

08564 | AIM-TOPOLOGY-0165

### Compact support is not preserved by arbitrary homomorphisms

S / solved | AIM 研讨会问题汇编

Given a homomorphism $f:\operatorname{Map}(S) \to \operatorname{Map}(S')$, does $f$ preserve the notion of being compactly supported? That is, does $f$ send compactly supported elements to compactly supported elements.

来源字段：http://aimpl.org/genusinfinity/4/

08565 | AIM-TOPOLOGY-0197

### Compact-dual cone decomposition and an escaping-curve certificate

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.6 (F. Guéritaud).** Describe the cone of directions in the tangent space \(T_X\mathcal T(S)\) which shorten the lengths of all simple closed curves on \(S\).

Problem 3.6 (F. Gu´ eritaud). Describe the cone of directions in the tangent space T X T (S) which shorten the lengths of all simple closed curves on S.

The arc metric dA on T (S) is a natural generalization of the Thurston metric By doubling, ...

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08566 | OWR-14297741-009

### Compactly Generated t-Structures on Schemes and dg Rings

P / partially_solved | Oberwolfach 报告问题汇编

Classify suitable compactly generated t-structures on $D(\operatorname{Qcoh}(X))$ for a Noetherian scheme $X$ and on $D(R)$ for a cohomologically nonpositive, bounded commutative dg ring $R$.

来源字段：https://doi.org/10.4171/owr/2023/40

08567 | OWR-15208-012

### Comparing Positive and Nonnegative Ricci Metric Spaces

O / open | Oberwolfach 报告问题汇编

How different are the spaces of positive-Ricci metrics and nonnegative-Ricci metrics, and is the inclusion $R^{>0}(M)\to R^{\geq 0}(M)$ a homotopy equivalence or a surjection on homotopy?

来源字段：https://doi.org/10.4171/owr/2017/3

08568 | AIM-TOPOLOGY-0141

### Completeness and two mapping-torus tests

P / partially_solved | AIM 研讨会问题汇编

**Can we characterize big mapping classes whose mapping tori admit a complete hyperbolic metric?**

来源字段：http://aimpl.org/genusinfinity/1/

08569 | AIM-TOPOLOGY-0245

### Completeness is essential for the fixed-circle length question

P / partially_solved | AIM 研讨会问题汇编

Problem 32. If a group acts isometrically on a manifold of sec \(\ge 1\) and the fixed point set is a circle, is it's length \(\le 2\pi\)?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08570 | OWR-14298592-002

### Completeness of Unzipped-Link Invariants for Spatial Theta Graphs

O / open | Oberwolfach 报告问题汇编

For a spatial theta-graph $\Gamma\subset S^3$, let $L_\Gamma$ consist of its three constituent knots and all two-component links obtained by unzipping one edge with every possible linking number. Does $L_\Gamma$ determine $\Gamma$ up to isotopy, and what minimal sublist of $L_\Gamma$ suffices?

来源字段：https://doi.org/10.4171/owr/2024/43

08571 | OWR-744-006

### Complex Foliations of Odd-Dimensional Spheres

O / open | Oberwolfach 报告问题汇编

For which $n$ does the sphere $S^{2n-1}$ admit a foliation by complex manifolds?

来源字段：https://doi.org/10.4171/owr/2004/42

08572 | OWR-2654829-001

### Complex Structures on the Six-Sphere and Rational Homotopy Types

O / open | Oberwolfach 报告问题汇编

Does the six-sphere $S^6$ admit the structure of a complex manifold? More generally, which rational homotopy types can occur for complex manifolds?

来源字段：https://doi.org/10.4171/owr/2020/33

08573 | AIM-TOPOLOGY-0278

### Complex-structure dependence of cyclotomic Hitchin loci

P / partially_solved | AIM 研讨会问题汇编

**Question 5.1 (Oscar García-Prada).** The cyclic group of order \(n\) acts on the Hitchin component of \(\operatorname{Rep}(\pi_1(S),\operatorname{SL}(n,\mathbb R))\) (multiply the Higgs field by roots of unity). Call the subspace of \(\mathbb Z/n\mathbb Z\)-invariant Higgs fields “cyclotomic Higgs fields,” which are in one-to-one correspondence with \(H^0(S,K^n)\). Is this space independent of the complex structure on \(S\)?

来源字段：https://aimath.org/WWN/surfacegroups/surfacegroups.pdf

08574 | OWR-15581-013

### Complexified Volume Conjecture for Hyperbolic Knots

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Complexified Volume Conjecture [OWR-15581-014]

For every hyperbolic knot $K\subset S^3$, is

$$\lim_{N\to\infty}\frac{2\pi\log J_N\!\left(K;e^{2\pi i/N}\right)}{N}=\operatorname{Vol}(S^3\setminus K)+i\operatorname{CS}(S^3\setminus K),$$

where $J_N$ is the $N$-colored Jones invariant with the indicated root-of-unity specialization?

来源字段：https://doi.org/10.4171/owr/2017/38

08575 | OWR-15208-007

### Components of Nonnegative Curvature Metrics on Homotopy Spheres

P / partially_solved | Oberwolfach 报告问题汇编

别名：Components of Nonnegative Curvature Metrics on Homotopy Spheres [OWR-15208-025]

For every nonnegatively curved homotopy sphere $\Sigma$ of dimension $4n-1\geq 7$, does the moduli space of metrics with nonnegative sectional curvature have infinitely many connected components?

来源字段：https://doi.org/10.4171/owr/2017/3

08576 | OWR-15579-004

### Composition of Positive-Scalar-Curvature Concordance Classes

O / open | Oberwolfach 报告问题汇编

In the category of Lie manifolds, can one define an appropriate composition law on concordance classes of metrics relative to a fixed compatible positive-scalar-curvature metric?

来源字段：https://doi.org/10.4171/owr/2017/36

08577 | AIM-TOPOLOGY-0047

### Compression-body quotients and a classical boundary-support obstruction

P / partially_solved | AIM 研讨会问题汇编

Given a left module \(N\) over the Kauffman bracket skein algebra \(\mathscr A(\Sigma)=\mathscr S(\Sigma\times I)\), find necessary and sufficient conditions for the existence of a compact oriented 3-manifold \(M\), with a parametrized boundary component \(f:\Sigma\to\partial M\), such that \(N\cong\mathscr S(M)\) and the module action is induced by a collar of \(f(\Sigma)\).

来源字段：http://aimpl.org/quantumlowdimtop/2/

08578 | OWR-13940-002

### Computational Topological Similarity of Curves

O / open | Oberwolfach 报告问题汇编

How can topological similarity measures for curves, such as homology area and homotopy area, be applied, computed, and used efficiently in higher-dimensional settings?

来源字段：https://doi.org/10.4171/owr/2015/45

08579 | OWR-1052-006

### Concordance Classes of Finite-Energy Foliations

O / open | Oberwolfach 报告问题汇编

Is there only one concordance class of finite-energy holomorphic foliations on the tight contact three-sphere? Does some overtwisted contact structure on $S^1\times S^2$ have multiple concordance classes?

来源字段：https://doi.org/10.4171/owr/2005/31

08580 | OWR-12697687-005

### Concordance Homomorphisms from Knot Floer Modules

O / open | Oberwolfach 报告问题汇编

How can one extract a concordance homomorphism directly from the knot Floer module $J\cdot K$ without first tensoring with a simpler module?

来源字段：https://doi.org/10.4171/owr/2023/4

08581 | OWR-14298592-014

### Concordance of an Explicit Pair of Positive Three-Braid Knots

O / open | Oberwolfach 报告问题汇编

Are the closures of the positive three-braids $\sigma_1^3\sigma_2^3\sigma_1^6\sigma_2^6$ and $\sigma_1^3\sigma_2^5\sigma_1^3\sigma_2^7$ smoothly concordant?

来源字段：https://doi.org/10.4171/owr/2024/43

08582 | OWR-14298592-013

### Concordance Rigidity of Braid-Positive Knots

O / open | Oberwolfach 报告问题汇编

If two braid-positive knots are smoothly concordant, must they be isotopic?

来源字段：https://doi.org/10.4171/owr/2024/43

08583 | OWR-8415345-009

### Concordance Rigidity of Positive Braid Knots

O / open | Oberwolfach 报告问题汇编

Are concordant positive braid knots in $S^3$ necessarily isotopic?

来源字段：https://doi.org/10.4171/owr/2021/42

08584 | OWR-13673-005

### Conformal Field Theories Representing TMF Classes

O / open | Oberwolfach 报告问题汇编

Under the Stolz-Teichner identification of the moduli space of $\operatorname{Fer}(n)$-twisted extended two-dimensional field theories with $\mathrm{TMF}^{-n}$, how much of the field-theory side can conformal field theories detect, and can they produce interesting classes in $\pi_n(\mathrm{TMF})$? Here $\operatorname{Fer}(n)$ denotes $n$ real chiral free fermions.

来源字段：https://doi.org/10.4171/owr/2015/16

08585 | AMR-103-0019

### Conjecture 1.19 - (The volume conjecture, [198, 296]) For any knot K, 2π·lim N →∞ log|JN (K)| N = v3||S3- K||, (2) where||·||denotes th…

P / partially_solved | AMR 问题清单汇编

(The volume conjecture, [198, 296]) For any knot K, 2π·lim N →∞ log|JN (K)| N = v3||S3- K||, (2) where||·||denotes the simplicial volume and v3 denotes the hyperbolic volume of the regular ideal tetrahedron.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08586 | AMR-103-0021

### Conjecture 1.21 - (H.

P / partially_solved | AMR 问题清单汇编

(H. Murakami, J. Murakami, M. Okamoto, T. Takata, Y. Yokota [297]) For a hyperbolic link L, 2π √ -1·lim N →∞ log JN (L) N = CS(S3- L) + √ -1vol(S3- L) for an appropriate choice of a branch of the logarithm, where CS and vol denote the Chern-Simons invariant and the hyperbolic volume respe ctively. Moreover, lim N →∞ JN +1(L) JN (L) = exp ( 1 2π√-1 ( CS(S3- L) + √ -1vol(S3- L) ) ). (7)

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08587 | AMR-103-0189

### Conjecture 10.14 - The map (50) is an isomorphism.

O / open | AMR 问题清单汇编

The map (50) is an isomorphism. 35The Yd -equivalence is also called the ( d - 1)-equivalence (due to Goussarov) in some literatures. This conjecture might be reduced to Conjecture 10.8 and the f ollowing conjec- ture.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08588 | AMR-103-0190

### Conjecture 10.15 - {M ~ Y2d S3}/ ~ Y2d+1 is torsion free for each d.

O / open | AMR 问题清单汇编

{M ~ Y2d S3}/ ~ Y2d+1 is torsion free for each d.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08589 | AMR-103-0192

### Conjecture 10.17 - (M.

P / partially_solved | AMR 问题清单汇编

(M. Polyak, see [153, “Theorem 4”]) Let F be an oriented compact surface. Two homology cylinders C and C ′ over F are Yd -equivalent if and only if v(C) = v(C ′) for any A-valued finite type invariant v ofF Y ⋆ -degree < d for any abelian group A.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08590 | AMR-103-0182

### Conjecture 10.7 - F as d (ZM)/F as d+1(ZM) (resp.F b d (ZM)/F b d+1(ZM)) is torsion free for each d.

O / open | AMR 问题清单汇编

F as d (ZM)/F as d+1(ZM) (resp.F b d (ZM)/F b d+1(ZM)) is torsion free for each d.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08591 | AMR-103-0183

### Conjecture 10.8 - A(∅; Z) is torsion free.

O / open | AMR 问题清单汇编

A(∅; Z) is torsion free. 10.2.2 Do finite type invariants distinguish homology 3-sph eres?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08592 | AMR-103-0184

### Conjecture 10.9 - Finite type invariants distinguish integral homology 3-spheres.

O / open | AMR 问题清单汇编

Finite type invariants distinguish integral homology 3-spheres. (See Conjecture 11.2.) 10.2.3 Dimensions of spaces of finite type invariants A finite type invariant v is called primitive if v(M1#M2) = v(M1) + v(M2) for any integral homology 3-spheres M1 and M2. We denote by A(∅; R)conn the submodule of A(∅; R) spanned by Jacobi diagrams with connected trivalent graphs. As a graded vector space A(∅; Q) is isomorphic to the symmetric tensor algebra of A(∅; Q)conn.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08593 | AMR-103-0198

### Conjecture 11.2 - The LMO invariant distinguishes integral homology 3-spheres.

O / open | AMR 问题清单汇编

The LMO invariant distinguishes integral homology 3-spheres. (See Conjecture 10.9.)

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08594 | AMR-109-0324

### Conjecture 11.3 - 1.

O / open | AMR 问题清单汇编

1. ˜σ∗(˜t2k+1) is non-trivial in H 2(Hg,1) for any k 2. σ∗(˜t2k+1) is trivial in H 2(Hg,1) for any k. The first part of the above conjecture is the “group version” of Conjecture 6.1 and it should be even more difficult to prove. On the other hand, if the classes σ∗(˜t2k+1) were non-trivial, then they would serve as invariants for certain 4-manifolds (2-dimensional family of homology cylinders). This seems unlikely to be the case. Thus the second part is related to the following problem.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08595 | AMR-103-0215

### Conjecture 12.10 - (D.

O / open | AMR 问题清单汇编

(D. Thurston) The shadow number of a 3-manifold is quasi-linear in its Gromov norm. That is, there exist consta nts c1 and c2 such that c1||M||≤ (shadow number of M )≤ c2||M|| for any 3-manifold M, where ||M|| denotes the Gromov norm of M.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08596 | AMR-103-0221

### Conjecture 12.16 - (X.-S.

O / open | AMR 问题清单汇编

(X.-S. Lin [262]) Any automorphism of G is either the identity or the mirror map, that is, any automorphism of G is induced by a diffeomorphism of the ambient space.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08597 | AMR-103-0208

### Conjecture 12.2 - (R.

O / open | AMR 问题清单汇编

(R. Benedetti) For every W, for every (pL)-class α1 as above, f ∗ 1 is an isomorphism. This means, in particular, that finite typ e invari- ants of Legendrian knots should be definitely not sensitive t o geometric (rigid) properties of the contact structures like “tightness”. See also [50] for a more detailed discussion and related ques tions. 12.2 Knots and finite groups Knot groups are known to be residually finite, that is, any non -trivial element can be detected by a homomorphism to some finite group. Now by Dehn’s lemma and the loop theorem a knot is trivial if an d only if its longitude represents the trivial element of the knot gro up. Consequently for each non-trivial knot there is a homomorphism to some finite group which carries the longitude to a non-trivial element.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08598 | AMR-103-0231

### Conjecture 12.26 - (V.

O / open | AMR 问题清单汇编

(V. Turaev) A pair (a finitely generated abelian group H of rank 1, an element Δ( t)∈ Z[H/TorsH] = Z[t±1]) (where t is a generator of H/TorsH ) can be realized as the pair (H1(M ), the Alexander polynomial Δ M of M ) for a closed connected oriented 3-manifold M if and only if Δ( t) = tkΔ( t-1) with even k∈ Z and Δ(1) = ±|TorsH|.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08599 | AMR-103-0210

### Conjecture 12.4 - (3-move conjecture, Y.

S / solved | AMR 问题清单汇编

(3-move conjecture, Y. Nakanishi [305]) Any link can be related to a trivial link by a sequence of 3-moves.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08600 | AMR-103-0211

### Conjecture 12.5 - (Y.

S / solved | AMR 问题清单汇编

(Y. Nakanishi, T. Harikae [220, Conjecture 1.59 (6)]) Any link can be related to a trivial link by a sequence of (2,2)-mo ves.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08601 | AMR-103-0213

### Conjecture 12.7 - (D.

O / open | AMR 问题清单汇编

(D. Bar-Natan, D. Thurston) For each compact Lie group G, level k, and every KTG K: Γ → R3, there exists a collection of measures μ K on the space of gauge equivalence classes of G-connections on Γ satisfying the following conditions. • It is well-behaved under KTG operations. • It is “localized” near connections that extend to S3- K. • A half-twist framing change acts by e √ -1Hℏ/2, where H is the Schr¨ odinger operator on G. • It recovers quantum invariants by IR(K) = ∫ hR(A)dμ K (A), where hR(A) denotes the holonomy of A in R. Here, R is a set of representations of G associated to edges of Γ and appropriate intertwiners associated to vertices of Γ.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08602 | AMR-103-0023

### Conjecture 2.1 - ([220, Problem 1.92 (N)]) Fd(ZK)/Fd+1(ZK) is torsion free for each d.

O / open | AMR 问题清单汇编

([220, Problem 1.92 (N)]) Fd(ZK)/Fd+1(ZK) is torsion free for each d.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08603 | AMR-109-0186

### Conjecture 2.1 - Let Ω∗(Hom(π,G )/G) be the de Rham algebra consisting of all measurable differential forms on Hom(π,G )/G.

O / open | AMR 问题清单汇编

Let Ω∗(Hom(π,G )/G) be the de Rham algebra consisting of all measurable differential forms on Hom(π,G )/G. Then the symplectic structures ωB generate the subalgebra of Ω∗(Hom(π,G )/G) consisting of ModΣ-invariant forms. Since the μ-measure of Hom( π,G )/G is finite, the representation of ModΣ on H:=L2(Hom(π,G )/G,μ )) is unitary. Andersen has informed me that he has proved vanishing of the first cohomology group H 1(ModΣ, H), and has raised the following conjecture generalizing Conjecture 2.1:: 13. Mapping class group dynamics on surface group representations 209

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08604 | AMR-109-0303

### Conjecture 2.1 - The natural homomorphisms ( Λ∗Λ3HQ )Sp →H ∗(Mg,∗; Q), (Λ∗UQ)Sp→H ∗(Mg; Q) induce isomorphisms ( Λ∗Λ3H ∗ Q/ ( [12]tore…

O / open | AMR 问题清单汇编

The natural homomorphisms ( Λ∗Λ3HQ )Sp →H ∗(Mg,∗; Q), (Λ∗UQ)Sp→H ∗(Mg; Q) induce isomorphisms ( Λ∗Λ3H ∗ Q/ ( [12]torelli⊕ [22] ))Sp ∼=R∗(Mg,∗) ( Λ∗U ∗ Q/([22]) )Sp∼=R∗(Mg). Furthermore, the algebras on the left hand sides are Poincar´ e duality algebras of dimensions 2g-2 and 2g- 4 respectively. Here we mention that for a single Riemann surface X, the cohomology H ∗(Jac(X); Q) is a Poincar´ e duality algebra of dimension 2 g while it can be shown that there exists a canonical isomorphism H ∗(Jac(X); Q)/([12])∼=H ∗(X; Q) which is a Poincar´ e duality algebra of dimension 2. Here [12]⊂H 2(Jac(X); Q) 354 S. Morita denotes the kernel Ker(Λ 2H ∗ Q→Q) of the intersection pairing and ([1 2]) denotes the ideal generated by it. Observe that we can write Λ ∗U ∗ Q = H ∗(PH 3(Jac)) which is a Poincar´ e duality algebra of dimension (2g 3 ) - 2g, where PH 3(Jac) denotes the primitive part of the third cohomology of the Jacobian variety. Hence the above conjecture can be rewritten as ( H ∗(PH 3(Jac))/([22]) )Sp∼=R∗(Mg) so that it could be phrased as the family version of the above simple fact for a single Riemann surface.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08605 | AMR-103-0033

### Conjecture 2.11 - (S.

O / open | AMR 问题清单汇编

(S. Willerton [401]) Let v3 be as above. If a knot K has a diagram with n crossings, then |v3(K)|≤ ⌊ n(n2- 1) 24 ⌋.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08606 | AMR-103-0038

### Conjecture 2.16 - (M.

O / open | AMR 问题清单汇编

(M. Polyak) The following two maps are injective, A(I)(d)-→-→A (I)(d) A(I)(d)/FI-→- →A(I)(d)/- → FI, where they are defined by ↦-→ +.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08607 | AMR-103-0039

### Conjecture 2.17 - [154] Every Vassiliev invariant of classical knots can be extended to a finite type invariant of long virtual knots.

S / solved | AMR 问题清单汇编

[154] Every Vassiliev invariant of classical knots can be extended to a finite type invariant of long virtual knots. (Se e also Problem 3.9.) 2.8 Finite type invariants derived from local moves One aspect of the study of knot invariants is the study of the s et of knots. A local move and finite type invariants derived from it might gi ve an approach of this study. A local move is a move between two knots, which are identical except for a b all, where they differ as shown in both sides of a move in Figure 10. L et R be a commutative ring with 1, and K the set of isotopy classes of oriented knots, as before. For a local move m, we define Fd(RK, m) as follows. Let K be an oriented knot with d disjoint balls B1, B2,···, Bd such that K is as shown in one side of m in each Bi. For any subset S⊂{ 1, 2,···, d}, we denote by KS the knot obtained from K by applying m in each Bi for i∈ S. We define Fd(RK, m) to be the submodule of RK spanned by ∑ S (-1)#S KS (13) for any K with d balls, where # S denotes the number of elements of S, and the sum runs over all subsets S of{1, 2,···, d}. Then, we have a descending series of submodules, RK =F0(RK, m) ⊃ F1(RK, m) ⊃ F2(RK, m) ⊃ ···. Note that Fd(RK) =Fd(RK,×) for a crossing change “ ×”. An R-homomor- phism v: RK→ R is called a finite type invariant of m-degree d, or an m finite type invariant of degree d, if v|Fd+1(RK,m) = 0. A crossing change “ ×”: ←→ A double crossing change “ ××”: ←→ A # move: ←→ A pass move: ←→ A Δ move: ←→ A doubled delta move Δ Δ: ←→ An n-gon move: ←→ Figure 10: Some local moves among oriented knots. The strand s of both sides of a Δ move and an n-gon move have any orientations such that corresponding str ands from opposite sides of the moves are oriented in the same way. Each side of an n-gon move has n strands. It is a fundamental problem of finite type invariants to calcu late the correspond- ing graded spaces, which would enable us to identify finite ty pe invariants in some sense.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08608 | AMR-103-0024

### Conjecture 2.2 - A(S1; Z) is torsion free.

O / open | AMR 问题清单汇编

A(S1; Z) is torsion free.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08609 | AMR-109-0187

### Conjecture 2.2 - Suppose C∞(Hom(π,G )/G) D-→C∞(Hom(π,G )/G) is a differential operator which commutes with the ModΣ-action on Hom(π,G…

O / open | AMR 问题清单汇编

Suppose C∞(Hom(π,G )/G) D- →C∞(Hom(π,G )/G) is a differential operator which commutes with the ModΣ-action on Hom(π,G )/G. Then D is a scalar multiple of the identity operator.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08610 | AMR-103-0043

### Conjecture 2.21 - (A.

O / open | AMR 问题清单汇编

(A. Kricker) Take (M1, K1) and (M2, K2) of the above sort. Then, there exists a (Z/pZ)-equivariant isomorphism φ: H1(Σ p (M1,K1); Z) → H1(Σ p (M2,K2); Z) preserving the linking pairing if and only if (M1, K1) is equivalent to (M2, K2) by a finite sequence of mod p loop moves.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08611 | AMR-103-0044

### Conjecture 2.22 - The map (15) is an isomorphism.

O / open | AMR 问题清单汇编

The map (15) is an isomorphism. This conjecture might be reduced to Conjecture 2.2 and the fo llowing conjec- ture.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08612 | AMR-103-0045

### Conjecture 2.23 - {K~ Cd O}/~ Cd+1 is torsion free for each d.

O / open | AMR 问题清单汇编

{K~ Cd O}/~ Cd+1 is torsion free for each d.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08613 | AMR-103-0046

### Conjecture 2.24 - (K.

S / solved | AMR 问题清单汇编

(K. Habiro [165], see also [153, “Theorem 5”]) Two m-strand string links L and L′ are Cd -equivalent if and only if v(L) = v(L′) for any A-valued finite type invariant v of degree < d for any abelian group A.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08614 | AMR-103-0025

### Conjecture 2.3 - (X.-S.

P / partially_solved | AMR 问题清单汇编

(X.-S. Lin [262]) Let R be a commutative ring with 1, say Z/2Z. Every weight system A(S1; R)(d)/FI→ R is induced by some Vassiliev invariant RK→ R.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08615 | AMR-109-0190

### Conjecture 2.3 - If r≥ 3, the action of Out(π) on Hom(π,G ) is ergodic.

S / solved | AMR 问题清单汇编

If r≥ 3, the action of Out(π) on Hom(π,G ) is ergodic. Using calculations in [ 43], this conjecture has been proved [ 47] when all of the simple factors of G are locally isomorphic to SU(2).

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08616 | AMR-103-0027

### Conjecture 2.5 - Vassiliev invariants distinguish oriented knots.

O / open | AMR 问题清单汇编

Vassiliev invariants distinguish oriented knots. (See Con jec-ture 3.2 for an equivalent statement of this conjecture.)

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08617 | AMR-103-0029

### Conjecture 2.7 - (see [220, Problem 1.89 (B)]) For any oriented knot K, no Vassiliev invariants distinguish K from -K.

O / open | AMR 问题清单汇编

(see [220, Problem 1.89 (B)]) For any oriented knot K, no Vassiliev invariants distinguish K from -K. (See Conjecture 3.4 for an equivalent statement of this conjecture.)

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08618 | AMR-109-0193

### Conjecture 3.1 - Suppose that b = 0 (Σ is closed).

P / partially_solved | AMR 问题清单汇编

Suppose that b = 0 (Σ is closed). For each integer 1≤k≤ 2g +b- 2, the ModΣ-action on the component e-1(2- 2g +b +k) of Hom(π,G ) is ergodic. When b = 0, the component e-1(3-2g) ≈ Σ× R6g-8 represents a 6 g- 6-dimensional thickening of Σ, upon which ModΣ acts. However, Morita [ 81] showed that ModΣ cannot act smoothly on Σ itself inducing the homomorphism Diff(Σ) -→ ModΣ. (Recently Markovic [ 80] has announced that if Σ is a closed surface of genus > 5, then ModΣ cannot even act on Σ by homeomorphisms inducing Homeo(Σ) -→ ModΣ.)

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08619 | AMR-109-0036

### Conjecture 3.15 - (Density of pseudo-Anosovs).

P / partially_solved | AMR 问题清单汇编

(Density of pseudo-Anosovs). LetP denote the set of pseudo-Anosov ele- ments of Modg. Then d(P) = 1. J. Maher [ Mah] has recently proven that a random walk on Mod g lands on a pseudo-Anosov element with probability tending to one as the length of the walk tends to infinity. I. Rivin [ Ri] has proven that a random (in a certain specific sense) element of Mod g is pseudo-Anosov by proving a corresponding result for Sp(2 g, Z). While the methods in [ Mah] and [ Ri] may be relevant, Conjecture 3.15 does not seem to follow directly from these results. As another test we pose the following.

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08620 | AMR-109-0037

### Conjecture 3.16 - d(Ig) = 0.

O / open | AMR 问题清单汇编

d(Ig) = 0. Even better would be to determine dlog(Ig). Conjecture 3.16 would imply that d(Ig(m)) = 0 for each m≥ 2. It is not hard to see that Ig(m) has exponential growth for each g≥ 2,m≥ 1, and one wants to understand how the various exponential growth rates compare to each other. In other words, one wants to know how common an occurence it is, as a function of k, for an element of Mod g to act trivially on the first k terms of the lower central series of π1Σg.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08621 | AMR-109-0199

### Conjecture 3.2 - If k = 1, then U is onto.

P / partially_solved | AMR 问题清单汇编

If k = 1, then U is onto. In general a PSL(2, R)-representation with dense image lies in Image(U). 220 W. Goldman

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08622 | AMR-103-0052

### Conjecture 3.2 - The Kontsevich invariant distinguishes oriented knots.

O / open | AMR 问题清单汇编

The Kontsevich invariant distinguishes oriented knots. (S ee Conjecture 2.5 for an equivalent statement of this conjectu re.)

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08623 | AMR-103-0072

### Conjecture 3.22 - [357, 139] The map (29) is injective.

S / solved | AMR 问题清单汇编

[357, 139] The map (29) is injective. In particular, the map (30) is injective.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08624 | AMR-103-0054

### Conjecture 3.4 - Z(K) = Z(-K) for any oriented knot K, where -K de- notes K with the opposite orientation.

P / partially_solved | AMR 问题清单汇编

Z(K) = Z(-K) for any oriented knot K, where -K de- notes K with the opposite orientation. (See Conjecture 2.7 for an eq uivalent statement of this conjecture.) 3.3 Characterization and interpretation of the Kontsevich in- variant The space A(S1) is an algebra with the product given by connected sum of Jacobi diagrams on S1. Since the Kontsevich invariant Z(K) of a knot K is group-like in A(S1), its logarithm log Z(K) belongs to A(S1)conn, where A(S1)conn denotes the vector subspace of A(S1) spanned by Jacobi diagrams on S1 with connected uni-trivalent graphs.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08625 | AMR-103-0058

### Conjecture 3.8 - (D.

O / open | AMR 问题清单汇编

(D. Bar-Natan, A. Haviv) ι ( Z(O) ) = closure ( exp (1 2 ( - ) ) ), where Z(O) denotes the Kontsevich invariant of the trivial knot (see [3 5]) and ι is the map of Conjecture 2.16.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08626 | AMR-109-0040

### Conjecture 4.1 - (Inhomogeneity of all metrics).

P / partially_solved | AMR 问题清单汇编

(Inhomogeneity of all metrics). Let Teichg denote the Teichm¨ uller space of closed, genus g≥ 2 Riemann surfaces. Let h be any Riemannian metric (or any Finsler metric with some weak regularity conditions) on Teichg which is invariant under the action of the mapping 2. Some problems on mapping class groups and moduli space 31 class group Modg, and for which this action has finite covolume. Then Isom(Teichg,h ) is discrete; even better, it contains Modg as a subgroup of index C =C(g). Royden’s Theorem is the special case when h is the Teichm¨ uller metric (Royden gets C = 2 here). A key philosophical implication of Conjecture 4.1 is that the mechanism behind the inhomogeneity of Teichm¨ uller space is due not to fine regularity properties of the unit ball in Q(X) (as Royden’s proof suggests), but to the global topology of moduli space. This in turn is tightly controlled by the structure of Mod g. As one piece of evidence for Conjecture 4.1, I would like to point out that it would follow if one could extend the main theorem of [ FW1] from the closed to the finite volume case. In some sense looking at Mod g-invariant metrics seems too strong, especially since Mod g has torsion. Perhaps, for example, the inhomogeneity of Teich g is simply caused by the constraints of the torsion in Mod g. Sufficiently large index subgroups of Mod g are torsion free. Thus one really wants to strengthen Conjecture 4.1 by replacing Mod g by any finite index subgroup H, and by replacing the constant C =C(g) by a constant C =C(g, [Modg:H]). After this one can explore metrics invariant by much smaller subgroups, such as Ig, and at least hope for discreteness of the corresponding isometry group (as long as the subgroup is sufficiently large). If one can prove the part of Conjecture 4.1 which gives discreteness of the isometry group of any Modg-invariant metric on Teich g, one can approach the stronger statement that Cg = 1 or Cg = 2 as follows. Take the quotient of Teich g by any group Λ of isometries properly containing Mod g. By discreteness of Λ, the quotient Teich g/Λ is a smooth orbifold which is finitely orbifold-covered byMg.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08627 | AMR-103-0093

### Conjecture 4.15 - (J.

O / open | AMR 问题清单汇编

(J. Przytycki, see [286]) (1) There is a polynomial invariant of unoriented links, P1(L)∈ Z[x, t] which satisfies: (i) Initial conditions: P1(Tn) = tn, where Tn is a trivial link of n components. (ii) Skein relation P1(L0) + xP1(L1)- xP1(L2)-P1(L3) = 0 where L0, L1, L2, L3 is a standard, unoriented skein quadruple ( Li+1 is obtained from Li by a right-handed half twist on two arcs involved in Li; compare Figure 15.) (2) There is a polynomial invariant of unoriented framed links, P2(L) ∈ Z[A±1, t] which satisfies: (i) Initial conditions: P2(Tn) = tn, (ii) Framing relation: P2(L(1)) =-A3P2(L) where L(1) is obtained from a framed link L by a positive half twist on its framing. (iii) Skein relation: P2(L0) + A(A2 + A-2)P2(L1) + (A2 + A-2)P2(L2) + AP2(L3) = 0. (3) There is a rational function invariant of unoriented framed links, P3(L)∈ Z[a±1, x, y, (x + y + xy + y2)-1] which satisfies: (i) Initial conditions: P3(Tn) = ( -a3(x+y+xy+x2)+a7(x+y+1)2-a-1 x+y+xy+y2 )n-1, (ii) Framing relation: P3(L(1)) = aP3(L), (iii) Skein relation: P3(L0)+axP3(L1)+a2yP3(L2)-a3(x+y+1)P3(L3) = 0. (4) The invariant predicted in (1) (respectively (2) and (3)) is not uniquely defined (if it exists). Note that a solution to (3) becomes a solution to (1) under the substitution a = 1, x =-y and that a solution to (3) becomes a solution to (2) under the substitution a =-A3, x =-1- A-4, y = A-4 + A-8. As for the uniqueness of (4), note that all such invariants agree on trivial links and therefore they agree on the space spanned by trivial links in the related cubic ske in module. The above conjectures assume that b∞ = 0 in our skein relation. Let consider the possibility that b∞ is invertible in R. Using the “denominator” of our skein relation (the first line of Figure 16) we get the relation whic h allows to compute the effect of adding a trivial component to a link L (we write tn for the trivial link Tn ): (a-3b3 + a-2b2 + a-1b1 + b0 + b∞t)L = 0. (33) When considering the “numerator” of the relation and its mir ror image (Figure 16) we obtain formulas for Hopf link summands, and because un oriented Hopf link is amphicheiral we can eliminate it from our equations t o get the formula (34): b3(L#H) + (ab2 + b1t + a-1b0 + ab∞)L = 0. b0(L#H) + (a-1b1 + b2t + ab3 + a2b∞)L = 0. ((b0b1-b2b3)t + (a-1b2 0- ab2 3) + (ab0b2- a-1b1b3) + b∞(ab0-a2b3))L = 0. (34) It is possible that (33) and (34) are the only relations in the module. Precisely, we ask whether S4,∞(S3; R) is the quotient ring R[t]/(I) where ti represents the trivial link of i components andI is the ideal generated by (33) and (34) for L = t. The substitution which realizes the relations is: b0 = b3 = a = 1, b1 = b2 = x, b∞ = y. This may lead to the polynomial invariant of unoriented lin ks in S3 with values in Z[x, y] and the skein relation L3+xL2+xL1+L0+yL∞ = 0.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08628 | AMR-109-0041

### Conjecture 4.2 - ( Mg is maximal).

O / open | AMR 问题清单汇编

( Mg is maximal). For g ≥ 3 the smooth orbifold Mg does not finitely orbifold-cover any other smooth orbifold. A much stronger statement, which may be true, would be to prove that if N is any finite cover ofMg, then the only orbifolds which N can orbifold cover are just the covers of Mg. Here is a related basic topology question about Mg.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08629 | AMR-109-0309

### Conjecture 4.2 - The classes μi are non-trivial for all i = 1, 2,⋯.

O / open | AMR 问题清单汇编

The classes μi are non-trivial for all i = 1, 2,⋯. More generally we have the following.

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08630 | AMR-103-0081

### Conjecture 4.3 - If every closed incompressible surface in M is parallel to ∂M, then S2,∞(M ) is torsion free.

O / open | AMR 问题清单汇编

If every closed incompressible surface in M is parallel to ∂M, then S2,∞(M ) is torsion free.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08631 | AMR-109-0043

### Conjecture 4.4 - (Nonpositive curvature).

O / open | AMR 问题清单汇编

(Nonpositive curvature). For g≥ 2 the orbifold Mg admits no complete, finite volume Riemannian metric with nonpositive sectional curvatures uniformly bounded away from-∞. 5As Mg is an orbifold, technically one studies Mod g-invariant metrics on the Teichm¨ uller space Teichg. 32 B. Farb One might be more ambitious in stating Conjecture 4.4, by weakening the finite volume condi- tion, by dropping the uniformity of the curvature bound, or by extending the statement from Mg to any finite cover of Mg (or perhaps even to certain infinite covers). It would also be interesting to extend Conjecture 4.4 beyond the Riemannian realm to that of CAT(0) metrics; see, e.g., [ BrF] for a notion of finite volume which extends to this context. In the end, it seems that we will have to make do with various relative notions of nonpositive or negative curvature, as in [ MM1, MM2 ], or with various weaker notions of nonpositive curvature, such as holomorphic, Ricci, or highly singular versions (see, e.g.,[ LSY]), or isoperimetric type versions such a Kobayashi or Kahler hyperbolicity (see [ Mc2]). Part of the difficulty with trying to fit Mg into the “standard models” seems to be the topological structure of the cusp of Mg. Scalar curvature and Q-rank. Let S be a genus g surface with n punctures, and let M(S) denote the corresponding moduli space. We set d(S) = 3 g- 3 + n The constant d(S) is fundamental in Teichm¨ uller theory: it is the complex dimension of the Teichm¨ uller space Teich(S); it is also the number of curves in any pair-of-pants decomposition of S. While previous results have concentrated on sectional and holomorphic curvatures, Shmuel Weinberger and I have recently proven the following; see [ FW2].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08632 | AMR-109-0252

### Conjecture 4.4 - Let M be a closed hyperbolic 4-manifold.

O / open | AMR 问题清单汇编

Let M be a closed hyperbolic 4-manifold. Then all the Seiberg-Witten invariants of M vanish. The relevance of this is given in the following proposition. We give a proof for hyperbolic manifolds that makes use only of Taubes celebrated paper [ 37] when b+ 2 > 1, and hence avoid complexities that arise when b+ 2 = 1. Proposition 4.5. A postive answer to Conjecture 4.4 implies that a closed hyperbolic 4- manifold M cannot be symplectic. Proof: This follows automatically from [ 37] in the case when b+ 2 > 1, since [ 37] shows that a compact oriented symplectic 4-manifold with b+ 2 > 1 has non-vanishing Seiberg-Witten invariants. Thus assume that b+ 2 (M ) = 1, and M is symplectic (if b2(M ) = 0 there is nothing to prove). From the discussion above, since M is hyperbolic, the signature of M is 0, and so b2(M ) = 2. Also M being hyperbolic implies π1(M ) is residually finite and so M has many finite covers. Let p:M1→M be a cover of degree d> 1. Since M is symplectic, M1 will be symplectic using the pullback of the symplectic form on M. We claim that b+ 2 (M1)> 1, and so we can apply [ 37] to get a contradiction. 282 A.W. Reid For if b2(M1) = 2, then (from the volume formula above) since the Euler characteristic satisfies χ(M1) > 0, it follows that b1(M1) < 2. However, χ(M1) = dχ(M ) and this shows b1(M1) = 2-d(2-b1(M )). These remarks yield the desired contradiction. ⊔ ⊓ On the otherhand, any X that has the description of a surface bundle over a surface where both the base and fiber have genus ≥ 2 is symplectic by an old argument of Thurston (see [ 39] or [ 14] Theorem 10.2.17). Motivated by this discussion, one can ask the following generalization of Question 4.1, which is an analogue of the virtual fibering question in dimension 3.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08633 | AMR-109-0044

### Conjecture 4.6 - LetS be any surface with d(S)≥ 1.

O / open | AMR 问题清单汇编

LetS be any surface with d(S)≥ 1. Then M does not admit a finite volume Riemannian metric of (uniformly bounded) positive scalar curvature in the quasi-isometry class of the Teichm¨ uller metric. The analogue of Conjecture 4.6 in the Γ \G/K case was proven by S. Chang in [ Ch]. The same method of proof as in [ Ch] should reduce Conjecture 4.6 to the following discussion, which seems to be of independent interest, and which came out of discussions with H. Masur. What does Mg, endowed with the Teichm¨ uller metric dTeich, look like from far away? This can be formalized by Gromov's notion of tangent cone at infinity: Cone(Mg):= lim n→∞ (Mg, 1 ndTeich) (10) where the limit is taken in the sense of Gromov-Hausdorff convergence of pointed metric spaces; here we have fixed a basepoint in Mg once and for all. This limit is easily shown to make sense and exist in our context. To state our conjectural answer as to what Cone( Mg) looks like, we will 2. Some problems on mapping class groups and moduli space 33 need the complex of curves on Σ g. Recall that the complex of curves Cg forg≥ 2 is the simplicial complex with one vertex for each nontrivial, nonperipheral isotopy class of simple closed curves on Σ g, and with a k-simplex for every ( k + 1)-tuple of such isotopy classes for which there are mutually disjoint representatives. Note that Mod g acts by simplicial automorphisms on Cg.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08634 | AMR-109-0045

### Conjecture 4.7 - ( Q-rank of moduli space).

S / solved | AMR 问题清单汇编

( Q-rank of moduli space). Cone(Mg) is homeomorphic to the (open) cone on the quotient Cg/ Modg 6. One can pose a stronger version of Conjecture 4.7 that predicts the precise bilipschitz type of the natural metric on Cone( Mg); an analogous statement for quotients Γ \G/K of symmetric spaces by lattices was proven by Hattori [ Hat]. H. Masur and I have identified the right candidate for a coarse fundamental domain needed to prove Conjecture 4.7; its description involves certain length inequalities analogous to those on roots defining Weyl chambers. Further, the (conjectured) dimensions of the corresponding tangent cones are Q-rank(Γ) and d(S), respectively. Thus we propose the following additions to the list of analogies between arithmetic lattices and Mod g. arithmetic lattices Modg Q-rank(Γ) d(S) root lattice {simple closed curves } simple roots top. types of simple closed curves Cone(Γ\G/K) Cone(Mg) As alluded to above, Conjecture 4.7 should imply, together with the methods in [ Ch], the second statement of Conjecture 4.5.

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08635 | AMR-109-0046

### Conjecture 4.8 - (Mod g is Kahler).

P / partially_solved | AMR 问题清单汇编

(Mod g is Kahler). Forg≥ 3, the group Modg is a Kahler group, i.e. it is isomorphic to the fundamental group of a compact Kahler manifold. It was shown in [ V e] that Mod 2 is not a Kahler group. This is proven by reducing (via finite extensions) to the pure braid group case; these groups are not Kahler since they are iterated extensions of free groups by free groups. A natural place to begin proving Conjecture 4.8 is the same strategy that Toledo uses in [ T o] for nonuniform lattices in SU( n, 1),n ≥ 3. The main point is the following. One starts with a smooth open variety V and wants to prove that π1V is a Kahler group. The first step is to find a compactificationV ofV which is projective, and for which V-V has codimension at least 3. This assumption guarantees that the intersection of the generic 2-plane P in projective space with V misses V. The (weak) Lefschetz Theorem then implies that the inclusion i:V∩P ↪→V induces an isomorphism on fundamental groups, thus giving that π1V is a Kahler group. 6Note: This statement is a slight cheat; the actual version requires the language of orbi-complexes. 34 B. Farb One wants to apply this idea to the Deligne-Mumford compactification Mg of moduli space Mg. This almost works, except that there is a (complex) codimension one singular stratum of Mg, so that the above does not apply. Other compactifications of Mg are also problematic in this regard. What about the Torelli group Ig? This group, at least for g≥ 6, is not known to violate any of the known constraints on Kahler groups. Most notably, Hain [ Ha3] proved the deep result that for g≥ 6 the group Ig has a quadratically presented Malcev Lie algebra; this is one of the more subtle properties posessed by Kahler groups. Note Akita's theorem (Theorem 5.13 above) that the classifying space of Ig,g≥ 7 does not have the homotopy type of a finite complex shows that these Ig are not fundamental groups of closed aspherical manifolds. There are of course Kahler groups (e.g. finite Kahler groups) with this property. In contrast to Conjecture 4.8, I have recently proven [ F a3] the following.

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08636 | AMR-109-0093

### Conjecture 4.8 - Each component of Hc g is simply connected.

S / solved | AMR 问题清单汇编

Each component of Hc g is simply connected. This is trivially true in genus 2, where there is one component which is all of h2. If true in genus 3, it implies quite directly the known fact that T3 is generated by 35 = 36 - 1 bounding pair elements. The number 35 is the rank of H1(J3,H3). The inverse images of generators of π1(J3,H3), once oriented, generate T3.10 The conjecture has an equivalent statement in more group theoretic terms. Define the hy- perelliptic Torelli group to be the intersection of the hyperelliptic mapping class group and the Torelli group: T Δg:= Δ g∩Tg = ker{Δg→ Spg(Z)}. It is a subgroup of the Johnson subgroup Kg:= ker{τ:Tg→ Λ3H/H}. Examples of elements in T Δg are Dehn twists on separating simple closed curves that are invariant under the hyperelliptic involutionσ. If Hg,α is a component of Hg, then there is an isomorphism π1(Hg,α,∗)∼=T Δg. The conjugacy classes of twists on a σ-invariant separating SCC correspond to loops about com- ponents of the divisor Hc,red g. Van Kampen's Theorem implies that π1(Hc g,α,∗)∼=π1(Hg,α)/these conjugacy classes ∼=T Δg/these conjugacy classes. The conjecture is thus equivalent to the statement that T Δg is generated by the conjugacy classes of Dehn twists on σ-invariant separating SCCs. It is important to understand the topology of the loci of hyperelliptic curves as it is the branch locus of the period mapping and also because it is important in its own right.

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08637 | AMR-109-0315

### Conjecture 4.9 - The stable rational cohomology of OutFn is trivial.

S / solved | AMR 问题清单汇编

The stable rational cohomology of OutFn is trivial. Namely lim n→∞ ˜H ∗(OutFn; Q) = 0. We can aslo ask how the cohomology of Out Fn with twisted coefficients look like.

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08638 | AMR-103-0108

### Conjecture 5.12 - (R.

S / solved | AMR 问题清单汇编

(R. Fenn, C. Rourke, B. Sanderson) H3(Rp)∼ = Z⊕ Z/pZ for p prime.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08639 | AMR-109-0058

### Conjecture 5.12 - Ig is finitely presented for g≥ 4.

O / open | AMR 问题清单汇编

Ig is finitely presented for g≥ 4. One thing we do know is that, in contrast to Mod g, neither Ig norKg has a classifying space which is homotopy equivalent to a finite complex; indeed Akita proved the following stronger result.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08640 | AMR-109-0068

### Conjecture 5.24 - (Picard number one conjecture for level L structures).

P / partially_solved | AMR 问题清单汇编

(Picard number one conjecture for level L structures). Prove that H2(Modg[L]; Q) = Q when g≥ 3. More generally, compute H2(Modg[L]; Z) for all g≥ 3,L≥ 2. Harer [ Har2] proved this conjecture in the case L = 1. This generalization was stated (for Picard groups) as Question 7.12 in [ HL]. The case L = 2 was claimed in [ F o], but there is apparently an error in the proof. At this point even the ( g,L ) = (3, 2) case is open. Here is a possible approach to Conjecture 5.24 for g≥ 4. First note that, since Mod g[L] is a finite index subgroup of the finitely presented group Mod g, it is finitely presented. As we have a lot of explicit information about the finite group Sp(2 g, Z/LZ), it seems possible in principle to answer the following, which is also a test of our understanding of Mod g[L].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08641 | AMR-103-0099

### Conjecture 5.3 - Let hX be as above.

S / solved | AMR 问题清单汇编

Let hX be as above. Then, log hX is not a Vassiliev invariant, unless it is constant. 5.3 (Co)homology of quandles Second cohomology classes of a quandle are used in order to de fine quandle cocycle invariants of knots. They are introduced as follows. Let A be an abelian group, written additively, and let C n(X; A) be the abelian group consisting of maps X n→ A, where X n denotes the direct product of n copies of X. We put C 1 Q(X; A) = C 1(X; A), C 2 Q(X; A) ={f∈ C 2(X; A)| f (x, x) = 0 for any x∈ X}, C 3 Q(X; A) ={g∈ C 3(X; A)| g(x, x, y) = 0 and g(x, y, y) = 0 for any x, y∈ X}. The coboundary operators di: C i Q(X; A)→ C i+1 Q (X; A) are given by d1f (x, y) = f (x)- f (x∗ y), d2g(x, y, z) = g(x, z)- g(x, y)- g(x∗ y, z) + g(x∗ z, y∗ z), for f∈ C 1 Q(X; A) and g∈ C 2 Q(X; A). We define the second quandle cohomol- ogy group by H 2 Q(X; A) = (kernel d2)/(image d1). It is known that H 2 Q(X; A) is isomorphic to Hom ( H Q 2 (X); A ) by the universal coefficient theorem, noting that H Q 1 (X) is free abelian (see [82]). Here, H Q 2 (X) denotes the second homol- ogy group of the dual complex of {C ⋆ Q(X; Z), d⋆}. See [82] for the definition of the nth quandle (co)homology group. Therefore, to obtain H 2 Q(X; A) for any A, it is sufficient to compute H Q 2 (X). Connected quandle X Order H Q 2 (X) H Q 3 (X) R3 3 0 Z/3Z Z[t±1]/(2, t2 + t + 1) 4 Z/2Z Z/2Z⊕ Z/4Z R5 5 0 Z/5Z Z[t±1]/(5, t- 2) 0 0 R7 0 Z/7Z Z[t±1]/(7, t- 2) 7 0 0 Z[t±1]/(7, t- 3) 0 0 Z[t±1]/(2, t3 + t + 1) 8 0 Z/2Z R9 0 Z/9Z Z[t±1]/(9, t- 2) 0 Z/3Z Z[t±1]/(3, t2 + 1) 9 Z/3Z (Z/3Z)3 Z[t±1]/(3, t2- t + 1) Z/3Z Z/3Z⊕ Z/9Z Z[t±1]/(3, t2 + t- 1) 0 0 Z[t±1]/(p, t- a) p 0for any prime p and any a̸= 0, 1∈ Z/pZ Table 5: The cohomologies of the quandles, except for the las t one, in the table are due to [264]. From a table in [264] we omit one of two dual quandles and quandles that are not connected (see remarks on Problem 5.6). The 2nd homology of Z[t±1]/(p, t- a) is due to [284]. See [264, 284] for computations of cohomolog y groups of some more quandles.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08642 | AMR-109-0052

### Conjecture 5.5 - For each k≥ 1, the group (Ig)k is not finitely generated.

S / solved | AMR 问题清单汇编

For each k≥ 1, the group (Ig)k is not finitely generated. Another test of our understanding of the Johnson filtration is the following.

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08643 | AMR-109-0318

### Conjecture 6.1 - The classes e1,t 3,t 5,⋯ are all non-trivial.

O / open | AMR 问题清单汇编

The classes e1,t 3,t 5,⋯ are all non-trivial. Furthermore they are linearly independent and form a basis of H 2(hQ g,1)Sp.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08644 | AMR-103-0125

### Conjecture 7.10 - (The growth rate conjecture) Let d = max{d0,..., dn}.

P / partially_solved | AMR 问题清单汇编

(The growth rate conjecture) Let d = max{d0,..., dn}. Then|Z G r (M )| = O(rd). It is well known that the quantum invariants only grows like r to some power. The power is bounded from above by some simple function (depe nding on G) of the Heegaard genus of the manifold.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08645 | AMR-103-0126

### Conjecture 7.11 - There is a construct of the right measure, say τM (A)1/2 for A∈M i, from the square root of the Reidemeister torsion…

P / partially_solved | AMR 问题清单汇编

There is a construct of the right measure, say τM (A)1/2 for A∈M i, from the square root of the Reidemeister torsion generaliz ing the non-degenerate case explained above and such that eπ√ -1 ˜Ij/4vj = ∫ A∈Mi eπ√-1(-2IA+h0 A+h1 A)/4τM (A)1/2. Conjectures 7.7 and 7.9 together with Conjecture 7.11 were fi rst proved for mapping tori of all finite order diffeomorphisms of all surfac es of genus at least two in [6]. Recently, Conjecture 7.7 was proved for all Seife rt fibered spaces in [168] by supplementing the calculations in [353] and [354] w ith the need analytic estimates.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08646 | AMR-103-0127

### Conjecture 7.12 - (H.

P / partially_solved | AMR 问题清单汇编

(H. Murakami [294]) For any closed 3-manifold M, 2π √ -1·o-lim N →∞ log τ SU (2) N (M ) N = CS(M ) + √ -1vol(M ), where vol(M ) and CS(M ) denote the hyperbolic volume 23 and the Chern- Simons invariant24 of M respectively, and o-lim denotes the “optimistic limit” introduced in [294]. 23When M is not hyperbolic, we define vol( M ) to be v3||M ||, where ||M || is the simplicial volume and v3 is the hyperbolic volume of the regular ideal tetrahedron. 24It is also conjectured (see Problem 7.16) that there exists a n appropriate definition of CS(M ) of any closed 3-manifold M, though CS( M ) is defined only for hyperbolic 3-manifolds M at present.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08647 | AMR-103-0140

### Conjecture 7.25 - (S.

O / open | AMR 问题清单汇编

(S. Baseilhac, R. Benedetti) (Real Volume Conjecture for QHI) For any triple (W, L, ρ) one has: lim N →∞ (2π/N 2) log(|KN (W, L, ρ)|) = Im R ( cI (W, L, ρ) ).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08648 | AMR-103-0144

### Conjecture 7.29 - (K.

S / solved | AMR 问题清单汇编

(K. Habiro, T. Le) For each g as above, there is a (unique) invariant I g(M )∈ R1 of an integral homology 3-sphere M such that for each root of unity ζ of order r divisible by d we have I g(M ) ı ı q=ζ = τ g ζ (M ).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08649 | AMR-103-0145

### Conjecture 7.30 - (K.

O / open | AMR 问题清单汇编

(K. Habiro) Suppose that Conjecture 7.29 would hold. For a new indeterminate t, set R′ 1 = lim←-nR1[t]/((t- q)(t- q2)⋯(t- qn)) Then there exists an invariant I sl(M )∈ R′ 1 of an integral homology 3-sphere M such that I sl(M )|t=qn = I sln(M ) for any n≥ 1, where we set I sl1(M ) = 1.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08650 | AMR-103-0120

### Conjecture 7.5 - For non-vanishing $\tau_r^G(M)$, the absolute value $|\tau_r^G(M)|$ depends only on the fundamental group $\pi_1(M)$.

P / partially_solved | AMR 问题清单汇编

For non-vanishing $\tau_r^G(M)$, the absolute value $|\tau_r^G(M)|$ depends only on the fundamental group $\pi_1(M)$.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08651 | AMR-103-0121

### Conjecture 7.6 - (The perturbative expansion conjecture) The asymptotic expansion of Z G k (M ) of a closed oriented 3-manifold M is g…

P / partially_solved | AMR 问题清单汇编

(The perturbative expansion conjecture) The asymptotic expansion of Z G k (M ) of a closed oriented 3-manifold M is given by Z G k (M ) ∼ k→∞ e-π√ -1(dim G)(1+b1(M ))/4 × ∫ [A]∈M e2π√-1rCS(A)r(h1 A-h0 A)/2e-2π√ -1(lA/4+(h0 A+h1 A)/8)τM (A)1/2 × exp ( ( ∞∑ l=1 clk-l (2l)!(3l)! ∑ e(Γ)= -l ZΓ (M, A) |Aut(Γ)| ) ), putting r = k + h ν, where the right hand side can be given in the mathemat- ical viewpoint in certain cases, as mentioned above, but whi ch needs further interpretation in general.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08652 | AMR-103-0122

### Conjecture 7.7 - (The asymptotic expansion conjecture, J.E.

P / partially_solved | AMR 问题清单汇编

(The asymptotic expansion conjecture, J.E. Andersen [6]) Let{c0 = 0, c1,···, cm} be the set of values of the Chern-Simons functional of flat G connections on a closed oriented 3-manifold M. There exist dj∈ Q, ˜lj∈ Q/Z, vj∈ R+, and ae j∈ C for j = 0, 1,···, m and e = 1, 2, 3,···such that ( r = k + h ν ) Z G k (M ) ∼ r→∞ m∑ j=0 e2π√ -1rcj rdj eπ√-1 ˜lj/4vj ( 1 + ∞∑ e=1 ae jr-e), that is, for all E = 0, 1, 2,..., there exists a constant cE such that ıı ıZ G k (M )- m∑ j=0 e2π√ -1rcj rdj eπ√-1 ˜lj/4vj ( 1 + E∑ e=1 ae jr-e) ıı ı≤ cErd-E-1 for all r = 2, 3, 4,···. Here, d = max{d0,···, dm}.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08653 | AMR-103-0124

### Conjecture 7.9 - (Topological interpretations of the dj ’s) Let Mj be the union of components of the moduli space of flat connections…

P / partially_solved | AMR 问题清单汇编

(Topological interpretations of the dj ’s) Let Mj be the union of components of the moduli space of flat connections M which has Chern-Simons value cj. Then dj = 1 2 max A∈Mj (h1 A- h0 A), where max here means the maximum value that (h1 A-h0 A) assumes on a Zariski open subset of Mj. Note that this conjecture might be rather optimistic, and ma y only hold in the non-degenerate cases. However, we do not know of any cases wh ere it fails (see [136]).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

08654 | AMR-109-0103

### Conjecture I - f f is pseudo-Anosov element of a mapping class group ModS with sufficiently big dilatation coefficient, then the subgrou…

S / solved | AMR 问题清单汇编

f f is pseudo-Anosov element of a mapping class group ModS with sufficiently big dilatation coefficient, then the subgroup of ModS normally generated by f is a free group having as generators the conjugates of f. More cautiously, one may conjecture that the above holds for a sufficiently high power g =fN of a given pseudo-Anosov element f. This conjecture is motivated by a theorem of M. Gromov (see [ Gr], Theorem 5.3.E). According to this theorem, the subgroup of a hyperbolic group normally generated by a hyperbolic element (with sufficiently big translation length) is a free group having as generators the conjugates of this element. Of course, it is well-known that mapping class groups are not hyperbolic. But, as M. Gromov noticed, what is essential for his proof is not the global negative curvature (hyperbolicity), but the negative curvature around the loop representing the considered element in an Eilenberg- MacLane space of the group in question. So, the hope is that the Teichm¨ uller spaces have enough negative curvature along the axes of pseudo-Anosov elements for a similar conclusion to hold. For a more direct and elementary approach to Gromov’s result see the work of Th. Delzant [ D]. 80 N. Ivanov This conjecture may be relevant to the Problem 3 above, since one may expect that all no- trivial elements of such subgroups are pseudo-Anosov.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08655 | AMR-109-0098

### Conjecture I - f Γ is an irreducible arithmetic group of rank ≥ 2, then every homomorphism Γ→ ModS has finite image.

S / solved | AMR 问题清单汇编

f Γ is an irreducible arithmetic group of rank ≥ 2, then every homomorphism Γ→ ModS has finite image. For many arithmetic groups Γ the conjecture can be proved by combining some well-known information about arithmetic groups with equally well-known properties of Mod S. But, for ex- ample, for cocompact lattices in SU (p,q ) this straightforward approach seems to fail. (I owe this specific example to G. Prasad.) In any case, by now this conjecture is completely proved. V. A. Kaimanovich and H. Masur [ KaM] proved that so-called non-elementary subgroups (subgroups containing a pair pseudo-Anosov elements with disjoint sets of fixed points in Thurston boundary) of Mod S are not isomorphic to irreducible arithmetic groups of rank ≥ 2. The proof is based on the theory of random walks on Mod S developed by Masur and Kaimanovich-Masur and on the results of H. Furstenberg about random walks on arithmetic groups (see the references in [KaM]). If combined with the Margulis finiteness theorem and the techniques of N. V. Ivanov, Subgroups of Teichm¨ uller Modular Groups, Translations of Mathematical Monographs, Vol. 115, American Math. Soc., 1992, this result can be used to prove the conjecture. A proof close in the spirit to the one alluded to in the previous paragraph is due to B. Farb and H. Masur [ FM]. In a addition, S-K. Yeung [ Y] completed the picture by proving that any homomorphism from a lattice Γ in either Sp( m, 1) or F -20 4 (the isometry group of the Cayley plane) into a mapping class group has finite image. The lattices in the remaining simple Lie groups of rank 1 often admit non-trivial homomorphisms to Z, and, therefore, homomorphisms to the mapping class groups with infinite image.

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08656 | AMR-109-0095

### Conjecture - Every subgroup of finite index in ModS contains a congruence subgroup.

P / partially_solved | AMR 问题清单汇编

Every subgroup of finite index in ModS contains a congruence subgroup. V. Voevodsky had indicated (in a personal communication) a beautiful application of this conjecture. Namely, the conjecture implies that a smooth algebraic curve over Q is determined up to an isomorphism by its algebraic fundamental group (which is isomorphic to the profinite completion of π1(S)) considered together with the natural action of the absolute Galois group Gal(Q/Q) on it. This corollary was apparently first conjectured by A. Grothendieck. I am not aware of any publication where this conjecture of Grothendieck is deduced from the solution of the congruence subgroup problem for the mapping class groups. The Grothendieck conjecture itself was proved by S. Mochizuki [ Mo1] after the initial work by A. Tamagawa [ T am]. See also [Mo3] and the expository accounts by S. Mochizuki [ Mo2] and G. Faltings [ F]. The proof of the Grothendieck conjecture lends some additional credibility (beyond the analogy with the classical congruence-subgroup problem) to the above congruence-subgroup conjecture.

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08657 | AMR-109-0107

### Conjecture - For every finitely generated subgroups G of ModS, the group Φf (G) is nilpotent.

O / open | AMR 问题清单汇编

For every finitely generated subgroups G of ModS, the group Φf (G) is nilpotent. For a a little bit more detailed discussion, see [ I4], Section 10.10. I repeated here this old problem in order to stress that our understanding of the subgroups of finite index in Mod S is rather limited.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08658 | AMR-109-0106

### Conjecture - LetS and R be closed surfaces.

O / open | AMR 问题清单汇编

LetS and R be closed surfaces. Let Γ be a subgroup of finite index in ModS. If the genus of R is less than the genus of S, then there is no homomorphism Γ→ ModR having as an image a subgroup of finite index in ModR. In fact, one may hope that any such homomorphism Γ → ModR has a finite image. But while the problem 7 about H 1(Γ) is unresolved, a more cautious conjecture seems to be more appropriate and more accessible, because if H 1(Γ) is infinite, Γ admits a homomorphism onto Z, and therefore, a lot of homomorphisms into Mod R, in partucular, with infinite image. In the special case when Γ = Mod S, the conjecture is recently proved by W. Harvey and M. Korkmaz [ HaK]. Their methods relay on the use of elements of finite order in Γ = Mod S, and therefore cannot be extended to general subgroupsΓ of finite index.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

08659 | OWR-15575-008

### Contact Entropy and Fundamental-Group Growth

O / open | Oberwolfach 报告问题汇编

If a contact $3$-manifold $(Y,\xi)$ has positive entropy, meaning that every Reeb flow has positive topological entropy, must $\pi_1(Y)$ have exponential growth?

来源字段：https://doi.org/10.4171/owr/2017/32

08660 | AIM-TOPOLOGY-0010

### Contact-homology invisibility versus rational-SFT detection

P / partially_solved | AIM 研讨会问题汇编

Do there exist tight but not fillable contact structures whose fillability is not obstructed by SFT?

来源字段：http://aimpl.org/highdimcontacttop/5/

08661 | AIM-TOPOLOGY-0181

### Continuity of Thurston geodesic envelopes and diagonal collapse

P / partially_solved | AIM 研讨会问题汇编

Problem 2.3 (K. Rafi). Does `Env(X,Y)` depend continuously on `X,Y`?

The immediately preceding Problem 2.2 in the AIM source defines

\[ \operatorname{Env}(X,Y)=\bigcup\{G:G\text{ is a Thurston geodesic connecting }X\text{ to }Y\}. \]

Thus the question concerns a set-valued map on \(\mathcal T(S)\times\mathcal T(S)\). The source does not specify a topology on sets. Following Dumas--Lenzhen--Rafi--Tao and Pan--Wolf, the precise reading used here is Hausdorff continuity on nonempty compact subsets, for the symmetric metric

\[ d_s(U,V):=\max\{d_{\rm Th}(U,V),d_{\rm Th}(V,U)\}. \]

The AIM document is the workshop version of Weixu Su's problem list. Its ambient surface is of finite type (genus \(g\), \(n\) punctures, negative Euler characteristic), so a result only for closed surfaces does not literally settle every case of the recovered question.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08662 | OWR-15987-004

### Continuous Choices of Points on Smooth Cubic Curves

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Universal Smooth-Cubic Multisection Problem [OWR-15987-005]

Classify the multisections of the universal family of smooth plane cubic curves: for which integers $n$ can one continuously choose $n$ distinct unlabeled points on every smooth cubic, and how are all such choices classified up to homotopy?

来源字段：https://doi.org/10.4171/owr/2018/2

08663 | OWR-15208-004

### Contractibility of Anosov Metric Spaces on Surfaces

O / open | Oberwolfach 报告问题汇编

For a closed surface $M$ of genus at least 2, is the space of Riemannian metrics whose geodesic flow is Anosov contractible?

来源字段：https://doi.org/10.4171/owr/2017/3

08664 | OWR-14299521-014

### Contractible Ricci-Nonnegative Four-Manifolds with Euclidean Growth

O / open | Oberwolfach 报告问题汇编

Let $(M^4,g)$ be a smooth, complete, contractible four-manifold with $\operatorname{Ric}\geq 0$ and Euclidean volume growth. Must $M$ be homeomorphic to $\mathbb{R}^4$, and must every asymptotic cross-section be homeomorphic to $S^3$?

来源字段：https://doi.org/10.4171/owr/2025/30

08665 | OWR-11101915-006

### Controlled Exotic Smooth Four-Manifolds with Boundary

O / open | Oberwolfach 报告问题汇编

Construct exotic smooth four-manifolds with boundary while fixing the boundary three-manifold and multiple invariants from their topological classification.

来源字段：https://doi.org/10.4171/owr/2022/34

08666 | OWR-12593-002

### Controlled K-Theory for Farrell-Jones Assembly Maps

O / open | Oberwolfach 报告问题汇编

Let $\Gamma$ act asymptotically properly on a compact space $X$. Can controlled $K$-theory prove split injectivity of the Farrell-Jones assembly map in algebraic $K$-theory and thereby establish the Farrell-Jones conjecture for such an action?

来源字段：https://doi.org/10.4171/owr/2013/45

08667 | OWR-1703864-006

### Convergence of Connes-Chern and Higher-Rho Pairings

P / partially_solved | Oberwolfach 报告问题汇编

Do pairings of Connes-Chern characters with higher rho invariants always converge? What further geometric hypotheses guarantee convergence?

来源字段：https://doi.org/10.4171/owr/2020/17

08668 | AIM-TOPOLOGY-0186

### Counting pseudo-Anosov classes by Thurston translation length

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.8 (K. Rafi).** Find the asymptotic behavior for the number of conjugacy classes of pseudo-Anosov elements of the mapping class group whose translation length in the Thurston metric is less than $R$.

The source fixes a finite-type hyperbolic surface $S$, of genus $g$ with $n$ punctures, and its Teichmüller space $\mathcal T(S)$. The statement in the original PDF is:

The original PDF confirms the wording above. The immediately preceding text says that this question is inspired by Eskin-Mirzakhani’s count for the Teichmüller metric.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08669 | AIM-TOPOLOGY-0205

### Critical exponents converge under the metric-ball subgroup exhaustion

S / solved | AIM 研讨会问题汇编

Let \(\Gamma_0\) be an infinitely generated Fuchsian group of the first kind. Denote by \(\Gamma_n\) the fundamental group carried by the \(n\)-ball in \(\mathbb H^2/\Gamma_0\).

**Problem 4.4 (F. Guéritaud).** Does \[ \delta(\Gamma_n)\longrightarrow\delta(\Gamma_0) \qquad(n\to\infty)? \]

It says, with minor grammatical errors:

The recovered problem is:

Gu´ ertitaud” is **François Guéritaud**. Su's updated 2016 list repeats the question as Problem 5.6 with the corrected name and with “infinitely-generated.”

To make “carried” precise, choose \(x\in M:=\mathbb H^2/\Gamma_0\), let \(B_n=B_M(x,n)\), and define \[ \Gamma_n :=\operatorname{im}\!\left[ \pi_1(B_n,x)\longrightarrow\pi_1(M,x)\cong\Gamma_0 \right]. \tag{1.1} \] This is the standard meaning of a subgroup carried by a subset. It is also the only intrinsic meaning under which \(\Gamma_n\) is canonically a Fuchsian subgroup and \(\delta(\Gamma_n)\) is defined. No assumption that \(B_n\hookrightarrow M\) is \(\pi_1\)-injective is needed.

Open rather than closed metric balls give the same conclusion. The intended setting is torsion-free because the source speaks of a surface; the proof also works for hyperbolic orbifolds when orbifold fundamental groups are used.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08670 | OWR-5146-008

### Curvature-Driven Product Splittings and Topological Obstructions

O / open | Oberwolfach 报告问题汇编

Which curvature conditions on a closed manifold $M$ imply that some finite cover $\widetilde M$ splits as $T^k\times N$ with $N$ simply connected and satisfies $b_1(M)\leq k\leq\operatorname{cup}(\widetilde M)$, where $\operatorname{cup}$ denotes rational cuplength?

来源字段：https://doi.org/10.4171/owr/2011/18

08671 | AIM-TOPOLOGY-0228

### Curvature-fidelity gaps in smoothing Alexandrov spaces

P / partially_solved | AIM 研讨会问题汇编

**Problem 15.** Is every finite dimensional Alexandrov space a limit of Riemannian manifolds with \(\sec\ge K\)?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08672 | AIM-TOPOLOGY-0139

### Curve-orbit limits: an exact no-escape and uniqueness criterion

P / partially_solved | AIM 研讨会问题汇编

This was proposed as an "easier version" of the previous problem.

Given an (irreducible) big mapping class \(f\) and a simple closed curve \(\alpha\), do \(f^n(\alpha)\) and \(f^{-n}(\alpha)\) converge to a lamination?

The exact AIM record asks:

Source: AIM workshop *Surfaces of infinite type*, section “Classification of elements of big mapping class groups,” Problem 1.3 (AIM-TOPOLOGY-0139, source index 138).

The preceding record, Problem 1.1, supplies the workshop's provisional meaning of reducibility: a big mapping class is reducible when it preserves a possibly infinite discrete collection of pairwise disjoint essential simple closed curves and proper arcs, where discrete means no accumulation inside the surface.

1. whether curves are replaced by geodesic representatives; 2. which topology is meant by “converge”; 3. whether an empty lamination is allowed.

This report uses the standard geodesic reading made explicit by Bestvina, Fanoni, and Tao: \(S\) carries a complete first-kind hyperbolic metric \(m\) (no funnels or half-planes), each curve is replaced by its \(m\)-geodesic representative, and convergence is local Hausdorff/Chabauty--Fell convergence on finite-type compact windows. A “limit lamination” below is required to be nonempty. With the convention that the empty set is a lamination, an orbit escaping every compact set would otherwise give a vacuous affirmative example.

来源字段：http://aimpl.org/genusinfinity/1/

08673 | OWR-11137-008

### Cyclic-Cohomology Classes of the Radul Cocycle

O / open | Oberwolfach 报告问题汇编

Identify the canonical cyclic-cohomology class $[c]\in HC^1(CS(M))$ of the Radul cocycle, including its image in periodic cyclic cohomology.

来源字段：https://doi.org/10.4171/owr/2011/45

08674 | AIM-TOPOLOGY-0218

### DC stability and a BV compactness obstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 5.** Understand DC-structures on manifolds. In particular, does Perelman's Stability Theorem hold in the DC-category? Is PL = DC always?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08675 | OPG-56573

### Decomposition of completions of reloids

S / solved | Open Problem Garden

Conjecture For composable reloids $f$ and $g$ it holds

- $\operatorname{Compl} ( g \circ f) = ( \operatorname{Compl} g) \circ f$ if $f$ is a co-complete reloid; - $\operatorname{CoCompl} ( f \circ g) = f \circ \operatorname{CoCompl} g$ if $f$ is a complete reloid; - $\operatorname{CoCompl} ( ( \operatorname{Compl} g) \circ f) = \operatorname{Compl} ( g \circ ( \operatorname{CoCompl} f)) = ( \operatorname{Compl} g) \circ ( \operatorname{CoCompl} f)$; - $\operatorname{Compl} ( g \circ ( \operatorname{Compl} f)) = \operatorname{Compl} ( g \circ f)$; - $\operatorname{CoCompl} ( ( \operatorname{CoCompl} g) \circ f) = \operatorname{CoCompl} ( g \circ f)$.

背景链接：http://www.openproblemgarden.org/op/decomposition_of_completions_of_reloids

08676 | OWR-12984-003

### Deformation K-Theory of Artin Braid Groups

O / open | Oberwolfach 报告问题汇编

For an Artin braid group $B_n$, does deformation $K$-theory satisfy $K_{\mathrm{def}}(B_n)\cong H_*(\operatorname{Irr}(B_n))$? Is the deformation $K$-theory of $B_n$ isomorphic to its complex topological $K$-theory?

来源字段：https://doi.org/10.4171/owr/2014/26

08677 | OWR-12984-002

### Deformation K-Theory of Pure Braid Groups

O / open | Oberwolfach 报告问题汇编

For a pure braid group $PB_n$, can the identification $K_{\mathrm{def}}(G)\cong H_*(\operatorname{Irr}(G))$ between deformation $K$-theory and the homology of the topological monoid of irreducible representations be extended to $G=PB_n$?

来源字段：https://doi.org/10.4171/owr/2014/26

08678 | AIM-TOPOLOGY-0201

### Dehn-twist tails obstruct one-sided length shortening

P / partially_solved | AIM 研讨会问题汇编

Its mathematical content can be recovered unambiguously:

Let \(\Gamma _0<\mathrm{PSL}_2(\mathbb R)\) be an infinitely generated Fuchsian group of the first kind (its limit set is the full circle at infinity) and let \(X_0=\mathbb H^2/\Gamma _0\). Does there exist a nontrivial marked quasiconformal deformation \(Y\in\mathcal T_{qc}(X_0)\) such that \[ \ell_\gamma(Y)\leq\ell_\gamma(X_0) \quad\text{for every essential simple closed curve }\gamma? \]

The 2014 PDF adds that Alessandrini and Guéritaud had recent results. No citable joint paper corresponding to that sentence was located. The 2016 update omits it, so it is treated as an informal progress note rather than a verified theorem.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08679 | OWR-8415349-006

### Del Pezzo Surfaces and Iterated Cyclic Loop Spaces

O / open | Oberwolfach 报告问题汇编

Is there an explicit relation between the del Pezzo surfaces $B_k$ and the iterated cyclic loop spaces $L_c^kS^4$ for $0\leq k\leq8$ that matches their $E_k$ symmetry patterns and relates geometric data such as curve volumes?

来源字段：https://doi.org/10.4171/owr/2021/46

08680 | OWR-16407-003

### Delocalized Cyclic Pairings with Structure Sets

O / open | Oberwolfach 报告问题汇编

Let $(M,g)$ be a closed manifold with a metric of positive scalar curvature. Determine, in general, the pairing of delocalized continuous cyclic cocycles with the topological structure set; equivalently, compute the associated pairings $\tau_c\circ\rho(M,g)$.

来源字段：https://doi.org/10.4171/owr/2018/32

08681 | AIM-TOPOLOGY-0257

### Dense positive sectional curvature: carryovers, a localized Frankel criterion, and a Synge counterexample

P / partially_solved | AIM 研讨会问题汇编

**Problem 44.** Which theorems from \(\sec>0\) carry over to positive curvature on an open dense set?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08682 | OWR-14299289-012

### Derived Geometric Satake for the Basic Object

O / open | Oberwolfach 报告问题汇编

Under derived geometric Satake, is the basic object $\delta_X$ sent to the sheared structure sheaf $\mathcal O(M^\vee)$, so that the endomorphism algebra of $\delta_X$ identifies with the sheared invariant function algebra $\mathcal O(M^\vee)^{G^\vee}$?

来源字段：https://doi.org/10.4171/owr/2025/18

08683 | OWR-15955-011

### Derived Pure Braid Groups in the Solvable Filtration

O / open | Oberwolfach 报告问题汇编

For every $n\geq1$, is the $(n+2)$-nd derived subgroup $(P_m)^{(n+2)}$ of the pure braid group contained in $\mathcal F_n^m$?

来源字段：https://doi.org/10.4171/owr/2017/50

08684 | OWR-15208-005

### Detecting Components of Riemannian Metric Moduli Spaces

O / open | Oberwolfach 报告问题汇编

How can one detect the connected components of the space of Riemannian metrics or its moduli space?

来源字段：https://doi.org/10.4171/owr/2017/3

08685 | OWR-1967-004

### Detection and Classification of Virtual Knots by Biquandles

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a nontrivial virtual knot not detected by the linear biquandle, for example the closure of $\kappa\tau\kappa^{-1}\tau^{-1}$ with $\kappa$ in the kernel of the Burau map? Is the fundamental biquandle a complete invariant of oriented virtual knots?

来源字段：https://doi.org/10.4171/owr/2008/22

08686 | OWR-13488-007

### Determinant Approximation for $L^{2}$-Torsion$

O / open | Oberwolfach 报告问题汇编

Let $G$ be residually finite and let $G=G_0\supset G_1\supset\cdots$ be a chain of normal finite-index subgroups with trivial intersection. For a matrix $A$ over $\mathbb QG$, is its Fuglede-Kadison determinant over $G$ the limit of the normalized Fuglede-Kadison determinants of the induced matrices over the finite quotients $G/G_i$?

来源字段：https://doi.org/10.4171/owr/2015/3

08687 | OWR-13108-005

### Determinant Bounds in Terms of Knot-Crossing Number

O / open | Oberwolfach 报告问题汇编

For a knot or link $K \subset S^3$, let $\det(K) = |\Delta_K(-1)|$ be its determinant, let $c(K)$ be its crossing number, and let $v_8 \approx 3.66386$ denote the volume of the regular ideal hyperbolic octahedron. For every knot or link $K$, does the upper bound $$ \frac{2\pi \log \det(K)}{c(K)} \leq v_8 $$ hold?

来源字段：https://doi.org/10.4171/owr/2014/38

08688 | OWR-13108-006

### Diagrammatic and Geometric Maximality

O / open | Oberwolfach 报告问题汇编

For a sequence of knots or links $K_n$ with crossing number $c(K_n)\to\infty$, call it geometrically maximal if $\operatorname{vol}(K_n)/c(K_n)\to v_8$ and diagrammatically maximal if $2\pi\log\det(K_n)/c(K_n)\to v_8$, where $v_8$ is the volume of the regular ideal octahedron. Is every diagrammatically maximal sequence geometrically maximal, and conversely?

来源字段：https://doi.org/10.4171/owr/2014/38

08689 | OWR-14299915-024

### Diffeomorphic Doubles of Homotopy-Equivalent Two-Complexes

O / open | Oberwolfach 报告问题汇编

Let $X$ and $Y$ be finite two-dimensional CW complexes with finite fundamental group, and let $M(X)$ and $M(Y)$ be their smooth four-dimensional doubles. If $X\simeq Y$, must $M(X)$ and $M(Y)$ be diffeomorphic?

来源字段：https://doi.org/10.4171/owr/2026/9

08690 | OWR-15208-017

### Diffeomorphism Group Homotopy in Positive Ricci Metric Spaces

O / open | Oberwolfach 报告问题汇编

Can stable rational homotopy classes of the diffeomorphism group, known to occur in the moduli space of pinched metrics on $S^d$, occur for some manifold in its space of positive-Ricci-curvature metrics?

来源字段：https://doi.org/10.4171/owr/2017/3

08691 | AIM-TOPOLOGY-0092

### Digital pi_2, clique realization, and the octahedral sphere

P / partially_solved | AIM 研讨会问题汇编

**Higher homotopy groups**

Define higher homotopy groups in digital topology. As a test case, one should verify that $\pi_2(S^2, *) = \mathbb{Z}$, where $S^2$ is the digital 2-sphere, i.e., a graph with six vertices appropriately embedded in $\mathbb{Z}^3$.

来源字段：http://aimpl.org/combhomotop/1/

08692 | AIM-TOPOLOGY-0095

### Digital strong homotopy is reflexive-graph times-homotopy

P / partially_solved | AIM 研讨会问题汇编

**Digital topology v \(x\)-homotopy theory.** What is the relation between digital topology and \(\times\)-homotopy theory of reflexive graphs?

来源字段：http://aimpl.org/combhomotop/1/

08693 | AIM-TOPOLOGY-0261

### Dimension-dependent curvature-operator answer for PIC blow-up limits

P / partially_solved | AIM 研讨会问题汇编

**Problem 48.** Let \(M^n\) be a compact manifold with positive isotropic curvature. Does any blow-up limit of the Ricci flow have non-negative curvature operator?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08694 | OPG-59900

### Direct proof of a theorem about compact funcoids

P / partially_solved | Open Problem Garden

Conjecture Let $f$ is a $T_1$-separable (the same as $T_2$ for symmetric transitive) compact funcoid and $g$ is a uniform space (reflexive, symmetric, and transitive endoreloid) such that $( \mathsf{\tmop{FCD}}) g = f$. Then $g = \langle f \times f \rangle^{\ast} \Delta$.

The main purpose here is to find a direct proof of this conjecture. It seems that this conjecture can be derived from the well known theorem about existence of exactly one uniformity on a compact set. But that would be what I call an indirect proof, we need a direct proof instead.

The direct proof may be constructed by correcting all errors an omissions in this draft article.

Direct proof could be better because with it we would get a little more general statement like this:

Conjecture Let $f$ be a $T_1$-separable compact reflexive symmetric funcoid and $g$ be a reloid such that

- $( \mathsf{\tmop{FCD}}) g = f$; - $g \circ g^{- 1} \sqsubseteq g$.

Then $g = \langle f \times f \rangle^{\ast} \Delta$.

背景链接：http://www.openproblemgarden.org/op/direct_proof_of_a_theorem_about_compact_funcoids

08695 | OWR-16415-018

### Disconnected Preimages of Subsurfaces Under Finite Covers

O / open | Oberwolfach 报告问题汇编

Given a finite cover $f:S_h\to S_g$ with $h>g\geq2$, is there always a four-holed sphere $H$ with essential boundary components in $S_g$ such that the interior of $f^{-1}(H)$ is disconnected, and what is the answer for a three-holed sphere?

来源字段：https://doi.org/10.4171/owr/2018/40

08696 | OWR-3384-001

### Displaceability of Lagrangian Toric Fibers

P / partially_solved | Oberwolfach 报告问题汇编

Let $(M,\omega,T)$ be a symplectic toric manifold with moment polytope $\Delta$, and let $L_u=\Phi^{-1}(u)$ be the Lagrangian toric fiber over $u\in\operatorname{int}\Delta$. Which fibers $L_u$ are displaceable; that is, for which $u$ is there a Hamiltonian symplectomorphism $\varphi$ with $\varphi(L_u)\cap L_u=\varnothing$?

来源字段：https://doi.org/10.4171/owr/2009/01

08697 | OWR-4416-008

### Distances Between Products of Torus Knots

O / open | Oberwolfach 报告问题汇编

For coprime pairs $(a,b)$ and $(c,d)$, let $T(a,b)$ and $T(c,d)$ denote the corresponding torus knots, let $d_\chi(T(a,b), T(c,d))$ be the smooth Euler-characteristic cobordism distance between them, and let $E(a \times b, c \times d)$ be the corresponding combinatorial rectangle-transformation cost. What are the precise values of $E(a \times b, c \times d)$ and $d_\chi(T(a,b), T(c,d))$?

来源字段：https://doi.org/10.4171/owr/2010/35

08698 | AIM-TOPOLOGY-0012

### Divisors with Weinstein exteriors in Liouville domains

P / partially_solved | AIM 研讨会问题汇编

Is it true that given a Liouville domain, there is a codimension two Liouville submanifold whose complement is Weinstein?

来源字段：http://aimpl.org/highdimcontacttop/6/

08699 | AIM-TOPOLOGY-0072

### Double-line restriction as the geometric d_2 complex

P / partially_solved | AIM 研讨会问题汇编

Let \(\mathcal F(\beta)\) be the sheaf on \(\operatorname{Hilb}^n(\mathbb C^2)\) associated to \(\beta\). How does \(\mathcal F(\beta)|_{\operatorname{Hilb}^n(x^2=0)}\) relate to Khovanov homology of \(\beta\)?

来源字段：http://aimpl.org/agclinkhom/4/

08700 | OWR-14297732-008

### Doubling and Homotopy of Positive Scalar Curvature Metrics

O / open | Oberwolfach 报告问题汇编

Is the Doubling Conjecture always true? What is the homotopy type of the space of positive-scalar-curvature metrics on a compact manifold with boundary when one imposes either product-type boundary behavior or the corresponding mean-convex boundary condition?

来源字段：https://doi.org/10.4171/owr/2023/29

08701 | AIM-TOPOLOGY-0134

### Dyadic rigidity and non-discreteness for the endpoint grope pseudometric

P / partially_solved | AIM 研讨会问题汇编

**Grope metric.** Understand the grope metric \(d_1\) defined by Cochran--Harvey--Powell (Cochran--Harvey--Powell 2017, MR3665407).

A closely related question is whether there exists a non smoothly slice knot which bounds an infinite height symmetric grope where all surfaces have genus 1, and whether such a knot must be topologically slice.

来源字段：http://aimpl.org/concordsliceknot/5/

08702 | OWR-17471-012

### Effective Subgroup Separability in Hyperbolic Knot Groups

P / partially_solved | Oberwolfach 报告问题汇编

Given a subgroup $H\subset\pi_1(S^3\setminus K)$ and an element $\sigma\notin H$, can one bound the index of a subgroup $G\subset\pi_1(S^3\setminus K)$ such that $H\subset G$ but $\sigma\notin G$?

来源字段：https://doi.org/10.4171/owr/2020/8

08703 | OWR-14298591-006

### Einstein Boundary Regularity with Positive Cosmological Constant

O / open | Oberwolfach 报告问题汇编

Do the generic regularity and local-solvability properties of the Anderson boundary map for Ricci-flat and negative Einstein metrics extend to Einstein metrics with positive Einstein constant?

来源字段：https://doi.org/10.4171/owr/2024/36

08704 | AMR-110-0022

### Elliptic cohomology 2 - Almost everyone who has ever thought about elliptic cohomology ends up thinking it has something to do…

P / partially_solved | AMR 问题清单汇编

Almost everyone who has ever thought about elliptic cohomology ends up thinking it has something to do with 2-categories. If you think about vector bundles, you have a bunch of local trivializations, tied together by these transition functions. The transition functions must satisfy a cocycle condition--here we are thinking of a vector bundle as a locally free sheaf, or some such nonsense. I mean on U the vector bundle is U cross R^n, on V it is V cross R^n, so on U intersect V you have two different trivializations, related by a transition function. Then on U intersect V intersect W, you have three ways to trivialize and you need a cocycle condition to hold relating the three different transtion functions. One of the standard ideas for how to build elliptic cohomology is to only require the cocycle condition to hold up to natural isomorphism rather than on the nose. I believe this requires you to replace vector spaces by an appropriate 2-category. My own idea was to use the 2-category of 2-vector spaces, which I learned about in a paper by Kapranov-Voevodsky. Is this idea worth anything?

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08705 | AMR-110-0023

### Elliptic cohomology 3 - Dennis McLaughlin and Jean-Luc Brylinski also thought along these lines.

P / partially_solved | AMR 问题清单汇编

Dennis McLaughlin and Jean-Luc Brylinski also thought along these lines. They wanted to use gerbes, or 2-gerbes maybe, instead. I could never understand what a gerbe was, but I am sure it is supposed to be a bundle of groupoids. I haven't heard anything about this for a while. Can this idea be made to go somewhere?

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08706 | AMR-110-0024

### Elliptic cohomology 4 - Yet another idea is to go back to a decription of cobordism I once heard.

P / partially_solved | AMR 问题清单汇编

Yet another idea is to go back to a decription of cobordism I once heard. I think this description is in print somewhere, but I don't know where or who wrote it. I believe this idea is a geometric description of MU^* X. Think of X as a simplicial set. Over each vertex of X put a manifold. Over each edge of X put a bordism between the manifolds at the vertices. Over each triangle, put a bordism between the bordisms on the edges--I don't really know what these means, but orientation must be involved. Continue in this way, and that is an element of MU^* X. There must be some equivalence relation you put on these, but I don't know what it is. The problem would then be to figure out whether this works and if it has been published, then to determine how you get K-cohomology from this description--presumably the bordisms between bordisms go away, so are the identity, and this is the cocycle condition for vector bundles. Then figure out how to get elliptic cohomology.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08707 | AIM-TOPOLOGY-0224

### Elliptic-fiber brotherhood and a rational-homotopy obstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 11.** Consider finite towers of fiber bundles, where the fibers \(\{F_1,\ldots,F_k\}\) and \(M_k\) are fixed topological manifolds. Loosen this notion to get a “brotherhood” on the manifolds \(M_0\) and a property of such \(M_0\) not known to be possessed by all manifolds of \(\sec\ge K,\ \operatorname{diam}\le1\).

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08708 | OWR-700-001

### Embedding Calculus for Knots in Three-Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

For a three-dimensional smooth manifold $M$ with nonempty boundary and prescribed endpoint tangent data, is the natural map $\operatorname{Emb}(I,M;\partial)\to\operatorname{Tot}(X^\bullet)$ from long knots to the totalization of the compactified-configuration cosimplicial space a homotopy equivalence?

来源字段：https://doi.org/10.4171/owr/2004/17

08709 | OWR-1455-008

### Embedding Obstructions from Missing Simplicial Faces

P / partially_solved | Oberwolfach 报告问题汇编

Let $K$ be the $d$-skeleton of a triangulated $2d$-sphere, and let $T$ be a missing $d$-face of $K$, meaning that $T\notin K$ while $\partial T\subset K$. Then $K\cup\{T\}$ should not embed in the $2d$-sphere.

来源字段：https://doi.org/10.4171/owr/2007/04

08710 | OWR-12485-007

### End Angles of Tight Open Knots

O / open | Oberwolfach 报告问题汇编

What mathematical model describes a tight open knot whose released ends form a characteristic angle, and how does that angle depend on the knot type?

来源字段：https://doi.org/10.4171/owr/2013/22

08711 | AIM-TOPOLOGY-0005

### End-control and parametric reductions for Liouville-to-Weinstein deformation

P / partially_solved | AIM 研讨会问题汇编

If a Liouville domain \((W^{2n},\lambda)\) has the homotopy type of a half-dimensional CW complex, must \(\lambda\) be Liouville homotopic to a Weinstein structure?

来源字段：http://aimpl.org/highdimcontacttop/3/

08712 | AIM-TOPOLOGY-0192

### Endpoint and intrinsic symmetrizations of the Thurston metric

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.1 (A. Papadopoulos).** What is a good way to symmetrize the Thurston metric? The length-spectrum metric by Sorvali is a symmetrization of the Thurston metric.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08713 | AIM-TOPOLOGY-0034

### Equal volume and the Turaev--Viro discrepancy filtration

P / partially_solved | AIM 研讨会问题汇编

**Pairs of 3-manifold having the same volume but different TV invariants**

For two manifolds with same volume and different Turaev-Viro invariants, how are they related?

来源字段：http://aimpl.org/quantumlowdimtop/1/

08714 | OWR-4798-030

### Equality Cases in Lower Bounds for Triangulated 3-Manifolds

S / solved | Oberwolfach 报告问题汇编

For a closed $3$-manifold $M$, with $\Gamma(M)=\min_{|\Delta|=M}(f_{1}(\Delta)-4f_{0}(\Delta)+10)$, which manifolds besides connected sums of $S^{2}\times S^{1}$ and the nonorientable $S^{2}$-bundle over $S^{1}$ attain equality in the bound $\Gamma(M)\geq10\beta_{1}(M)$?

来源字段：https://doi.org/10.4171/owr/2011/08

08715 | OWR-12697687-013

### Equality of Involutive and Arf Knot Invariants

O / open | Oberwolfach 报告问题汇编

For every torsion knot $K$ in the smooth concordance group, is $A(K)=\operatorname{Arf}(K)$?

来源字段：https://doi.org/10.4171/owr/2023/4

08716 | OWR-16415-008

### Equality of Variationally Minimizing Maps

O / open | Oberwolfach 报告问题汇编

Let $X$ be a finite ideal hyperbolic simplicial complex with vertex set $S$, and let $\sigma,\tau\in\mathcal T(X)$. Let $\mathcal Y$ be the facewise $W^{1,2}$ maps $(X\setminus S,\sigma)\to(X\setminus S,\tau)$ that preserve every edge and face, and let $\mathcal D\subset\mathcal Y$ consist of maps restricting to a diffeomorphism on each face. If $u\in\mathcal Y$ and $\widetilde u\in\mathcal D$ minimize the respective energies, must $u=\widetilde u$? Equivalently, must $u$ be a diffeomorphism on every face?

来源字段：https://doi.org/10.4171/owr/2018/40

08717 | OWR-15577-010

### Equivalence of Hardy Spaces on Noncompact Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

For a complete connected noncompact Riemannian manifold $M$, do the Hardy spaces $H_R^1(M)$, $H_H^1(M)$, and $H_P^1(M)$, defined through the Riesz transform, heat maximal operator, and Poisson maximal operator, respectively, coincide and admit atomic decompositions?

来源字段：https://doi.org/10.4171/owr/2017/34

08718 | OWR-14297744-006

### Equivariant Algebraic K-Theory of Stable h-Cobordisms

O / open | Oberwolfach 报告问题汇编

Describe the stable moduli space $H_G^\infty(M)$ of equivariant $h$-cobordisms in terms of equivariant algebraic $K$-theory.

来源字段：https://doi.org/10.4171/owr/2023/34

08719 | AMR-110-0044

### Equivariant homotopy 2 - Currently we know how to do equivariant stable homotopy theory only when the structure group G is comp...

S / solved | AMR 问题清单汇编

Currently we know how to do equivariant stable homotopy theory only when the structure group G is compact Lie. But I bet we can do it when the group G is profinite as well. The main example I am thinking of is the Morava stabilizer group, though one could warm up with the p-adics. Presumably Morava E-theory should be an object of this category, but how?

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08720 | AMR-110-0045

### Equivariant homotopy 3 - Figure out how to do equivariant stable homotopy theory without restriction on the group.

P / partially_solved | AMR 问题清单汇编

Figure out how to do equivariant stable homotopy theory without restriction on the group. Here you are going to have to change the current setup a lot, I think. Mike Mandell has some ideas about this. He points out that a lot of the crucial problems in mathematics concern infinite discrete groups, like the Novikov conjecture, and so it may be of value to figure out this problem.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08721 | AMR-110-0046

### Equivariant homotopy 4 - As a simpler model of the equivariant stable homotopy category, construct a derived category of Mackey...

P / partially_solved | AMR 问题清单汇编

As a simpler model of the equivariant stable homotopy category, construct a derived category of Mackey functors over a Green functor, and analyze its properties. We know a lot about the derived category of a Noetherian ring--thick, localizing, and colocalizing subcategories are completely classified (See Amnon Neeman's paper on the derived category, though in my opinion you should also see Hovey-Palmieri-Strickland for simpler proofs of most of his results). Can we say analogous things about Mackey functors? Gaunce Lewis has thought about this problem a bit.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

08722 | OWR-1328-001

### Equivariant Nullhomotopy of the Regular-Map Derivative

O / open | Oberwolfach 报告问题汇编

For odd $n$, let $\delta:\operatorname{reg}(D^n\times D^2,D^2)\to\Omega^{n+2}(O(n+2)/O(n))$ send a regular smooth map, equal to the projection near the boundary, to its derivative. Construct a rational nullhomotopy of $\delta$ satisfying the derived $SO(2)$-equivariance condition and extending the preferred rational nullhomotopy on $\operatorname{reg}_1(D^n\times D^2,D^2)=\{f:q_2\circ f=q_{n+2}\}$.

来源字段：https://doi.org/10.4171/owr/2006/42

08723 | OWR-14297744-005

### Equivariant Splitting of A-Theory

P / partially_solved | Oberwolfach 报告问题汇编

For a compact smooth $G$-manifold $M$, prove an equivariant splitting $A_G(M)\simeq\Sigma_G^\infty M_+\wedge\operatorname{Wh}_G(M)$ with $\Omega^{\infty+1}\operatorname{Wh}_G(M)^G\simeq H_G^\infty(M)$.

来源字段：https://doi.org/10.4171/owr/2023/34

08724 | OWR-14297741-001

### Equivariant Stable Connectivity in Higher Representation Dimension

O / open | Oberwolfach 报告问题汇编

Let $G$ be a finite group with an irreducible representation of dimension greater than one. If a spectral presheaf on smooth $G$-schemes is Nisnevich-locally connected, must its equivariant $\mathbb A^1$-localization remain connected?

来源字段：https://doi.org/10.4171/owr/2023/40

08725 | OWR-17132-010

### Equivariant Topological Modular Forms Duality for Compact Lie Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a compact Lie group with adjoint representation $L$. Is $\operatorname{TMF}^G$ a dualizable $\operatorname{TMF}$-module with dual $(\Sigma^{-L}\operatorname{TMF})^G$?

来源字段：https://doi.org/10.4171/owr/2019/36

08726 | AIM-TOPOLOGY-0013

### Essential tight bypasses in Ustilovsky five-sphere layers

P / partially_solved | AIM 研讨会问题汇编

For \(k\ge0\), choose and mark two disjoint small Darboux balls \(B_-,B_+\subset(S^5,\xi_k)\). Give an ordered list of Honda--Huang bypass attachment data on the evolving convex \(S^4\) levels whose stacked contact cobordism is contactomorphic, relative to both marked boundary collars, to \[ L_k=(S^5\setminus\operatorname{Int}(B_-\cup B_+),\xi_k). \]

来源字段：http://aimpl.org/highdimcontacttop/7/

08727 | OWR-17132-012

### Eta-Periodic Slice Spectral Sequences over General Fields

O / open | Oberwolfach 报告问题汇编

Let $k$ be a field of characteristic zero over which $\rho$ is nonnilpotent. Determine the differentials, extensions, and abutment of the $\eta$-periodic slice spectral sequence over $k$.

来源字段：https://doi.org/10.4171/owr/2019/36

08728 | OWR-17470-011

### Euler Characteristic and Integral Approximation from Simplicial Volume

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Integral-Approximation Question [OWR-17470-012]

Let $M$ be an oriented closed connected aspherical manifold with simplicial volume $\lVert M\rVert=0$. Must $\chi(M)=0$? More strongly, if $\pi_1(M)$ is residually finite, does

$$\lVert M\rVert_{\mathbb Z}^{\infty}=\lVert M\rVert$$?

来源字段：https://doi.org/10.4171/owr/2020/7

08729 | OWR-11454-007

### Even-Codimension Stratifications of Codimension-Two Embeddings

O / open | Oberwolfach 报告问题汇编

Let $M$ be a simply connected piecewise-linear manifold with a homotopy equivalence $M\simeq N$, and hence an embedding $M\hookrightarrow N\times D^2$. Can this embedding be stratified using only strata of even codimension?

来源字段：https://doi.org/10.4171/owr/2011/56

08730 | AIM-TOPOLOGY-0040

### Every knot has nontrivial universal peripheral ideal

P / partially_solved | AIM 研讨会问题汇编

Find other classes of knots for which the peripheral ideal is non trivial. For example, trying to find a knot which is not a 2-bridge knot

来源字段：http://aimpl.org/quantumlowdimtop/2/

08731 | OPG-57401

### Every metamonovalued funcoid is monovalued

P / partially_solved | Open Problem Garden

Conjecture Every metamonovalued funcoid is monovalued.

The reverse is almost trivial: Every monovalued funcoid is metamonovalued.

背景链接：http://www.openproblemgarden.org/op/every_metamonovalued_funcoid_is_monovalued

08732 | OPG-57403

### Every metamonovalued reloid is monovalued

O / open | Open Problem Garden

Conjecture Every metamonovalued reloid is monovalued.

背景链接：http://www.openproblemgarden.org/op/every_metamonovalued_reloid_is_monovalued

08733 | AIM-TOPOLOGY-0090

### Exact comparison of the q,t-Catalan and torus-link HHH recursions

S / solved | AIM 研讨会问题汇编

Compare recursions for $q,t$-Catalan numbers $C_n(q,t)$ with the recursions for $HHH$.

来源字段：http://aimpl.org/agclinkhom/9/

08734 | AIM-TOPOLOGY-0108

### Exact distortion of canonical maps between bi-invariant free groups

P / partially_solved | AIM 研讨会问题汇编

**Geometric group theory.** For the free group \(F_k\) on \(k\) generators, consider the generating set given by the closure of the standard generating set under conjugation. When considered with the word metric, are \(F_2\) and \(F_3\) quasi-isometric?

来源字段：http://aimpl.org/combhomotop/6/

08735 | AIM-TOPOLOGY-0244

### Exact metric lifting, orbifold sufficiency, and a singular even-jet obstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 31.** Suppose we have an isometric group action on \(M\). If one changes the metric on the orbit space, does it lift to an invariant metric on \(M\)?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08736 | AIM-TOPOLOGY-0254

### Exact rank-three targets beyond torus reduction

P / partially_solved | AIM 研讨会问题汇编

**Problem 41.** Which vector bundles over \(S^2\times S^2\) or \(\mathbb{CP}^2\#\,\pm\mathbb{CP}^2\) where the structure group does not reduce to a torus admit \(\sec\geq0\)?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08737 | AIM-TOPOLOGY-0113

### Exact splitting criteria for bipolar quotient subgroups

P / partially_solved | AIM 研讨会问题汇编

Bipolar quotients

Does $\mathcal{T}_n/ \mathcal{T}_{n+1}$ contain a $\mathbb{Z}^{\infty}$-summand? What about a $(\mathbb{Z}/2\mathbb{Z})^{\infty}$ subgroup?

In fact, it is unknown in general even if $\mathcal{T}_n/ \mathcal{T}_{n+1}$ contains a $\mathbb{Z}/ 2\mathbb{Z}$ subgroup. Work of Cha-Kim (Cha--Kim 2014, MR3228458) shows that $\mathcal{T}_n/ \mathcal{T}_{n+1}$ contains a $\mathbb{Z}^{\infty}$-subgroup for all $n \in \mathbb{N}$.

来源字段：http://aimpl.org/concordsliceknot/1/

08738 | OWR-14298802-005

### Existence of the Smith-Toda Complex $V(4)$

P / partially_solved | Oberwolfach 报告问题汇编

Does the Smith-Toda complex $V(4)$ exist at any prime?

来源字段：https://doi.org/10.4171/owr/2024/47

08739 | AMR-102-0001

### Existence questions - Question 2.1

P / partially_solved | AMR 问题清单汇编

Which hyperbolic $3$-manifolds admit taut foliations? Give an effective procedure to decide if a hyperbolic $3$-manifold admits a taut foliation. For a useful property $\mathsf{p}$ of a $3$-manifold, give an explicit construction of infinitely many $3$-manifold with property $\mathsf{p}$ with/without a taut foliation. Same question for essential laminations.

背景链接：https://arxiv.org/abs/math/0209081

08740 | AMR-102-0002

### Existence questions - Question 2.2

O / open | AMR 问题清单汇编

Is there an effective algorithmic procedure to produce and recognize a hyperbolic knot of depth $n$ for any given $n$? What about $\ge n$?

背景链接：https://arxiv.org/abs/math/0209081

08741 | AMR-102-0003

### Existence questions - Question 2.3

O / open | AMR 问题清单汇编

Given a collection $\mathscr{C}$ of topological or geometric types of surface, what $3$-manifolds admit a taut foliation $\mathscr{F}$ whose leaves are all homeomorphic or coarsely quasi-isometric to an element of the collection $\mathscr{C}$?

背景链接：https://arxiv.org/abs/math/0209081

08742 | AMR-102-0004

### Existence questions - Question 2.4

O / open | AMR 问题清单汇编

Let $X$ be a vector field on a $3$-manifold. When is there a foliation $\mathscr{F}$ of $M$ transverse to $X$?

背景链接：https://arxiv.org/abs/math/0209081

08743 | OWR-11101915-007

### Exotic Four-Manifolds and the Smooth Poincaré Conjecture

O / open | Oberwolfach 报告问题汇编

Decide whether $S^4$ has an exotic smooth structure, whether $S^1\times S^3$ or a definite four-manifold can be exotic, and whether every genus $g$ has a unique smooth surface $\Sigma\subset S^4$ whose complement satisfies $\pi_1(S^4\setminus\Sigma)\cong\mathbb Z$.

来源字段：https://doi.org/10.4171/owr/2022/34

08744 | AIM-TOPOLOGY-0039

### Explicit peripheral relations for 2-bridge knots

P / partially_solved | AIM 研讨会问题汇编

**Peripheral ideal in Kauffman bracket skein algebra.** Find non-trivial explicit element in the peripheral ideal in the Kauffman bracket skein algebra of the boundary torus for two bridge knots.

来源字段：http://aimpl.org/quantumlowdimtop/2/

08745 | AIM-TOPOLOGY-0196

### Explicit reverse Thurston horofunctions from Dehn twists

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.5 (C. Walsh).** What is the horofunction boundary of the reversed Thurston metric?

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08746 | AIM-TOPOLOGY-0256

### Explicit stable homogeneous representatives over S^2 x S^2

P / partially_solved | AIM 研讨会问题汇编

**Problem 43.** Suppose \(E\to M\) is a vector bundle over a compact, simply-connected manifold \(M\) for which \(\sec M\geq 0\). Does \(E\oplus\mathbb R^k\to M\) have \(\sec\geq0\) for \(k\) large?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08747 | OWR-12984-004

### Étale Homotopy Realization of Khovanov Homology

O / open | Oberwolfach 报告问题汇编

Can the Khovanov homology of a link $K$ be represented by an appropriate étale homotopy type that is itself an invariant of $K$?

来源字段：https://doi.org/10.4171/owr/2014/26

08748 | OWR-4430-004

### Faithful Homotopy Actions and Cohomology of Baut(X)

O / open | Oberwolfach 报告问题汇编

For a finite simply connected CW complex $X$ and subgroups of $\operatorname{Baut}(X)$ corresponding to faithful actions up to homotopy, how many such subgroups are there, how large can one be, and what is the cohomology of $\operatorname{Baut}(X)$?

来源字段：https://doi.org/10.4171/owr/2010/42

08749 | OWR-13680-001

### Family Floer and Kontsevich-Soibelman Wall Crossing

O / open | Oberwolfach 报告问题汇编

How can wall-crossing formulas from family Floer homology be related to the Kontsevich-Soibelman wall-crossing formulas, and thereby to refined tropical counts?

来源字段：https://doi.org/10.4171/owr/2015/23

08750 | AIM-TOPOLOGY-0041

### Finite certificates for Hilbert-basis SL3 webs in the KTGS cone

P / partially_solved | AIM 研讨会问题汇编

For an ideal triangulation \(\lambda\) of a punctured finite-type surface, determine the additive indecomposable nonzero integral points of the \(SL_3\) KTGS cone, and identify the corresponding parallel-equivalence classes of reduced (nonelliptic) \(SL_3\)-webs.

来源字段：http://aimpl.org/quantumlowdimtop/2/

08751 | OWR-11101915-008

### Finite Equivariant Cohomology for Compact Lie Groups

O / open | Oberwolfach 报告问题汇编

Construct equivariant cohomology theories having strong finiteness properties uniformly for all compact Lie groups.

来源字段：https://doi.org/10.4171/owr/2022/34

08752 | OWR-8415335-005

### Finite Euclidean Models of Compact Nonpositively Curved Surfaces

O / open | Oberwolfach 报告问题汇编

Is every compact two-dimensional nonpositively curved space homotopy equivalent to a finite two-dimensional nonpositively curved Euclidean complex?

来源字段：https://doi.org/10.4171/owr/2021/32

08753 | OWR-15987-014

### Finite Generation of Graph-Configuration Homology Functors

O / open | Oberwolfach 报告问题汇编

Does the finite-generation result for the functor $G\mapsto H_q(\operatorname{Conf}_k(G))$ extend from $q\leq1$ to every $q\geq0$?

来源字段：https://doi.org/10.4171/owr/2018/2

08754 | AIM-TOPOLOGY-0159

### Finite quotients, genus-end filling, and coarse type

P / partially_solved | AIM 研讨会问题汇编

Let $S$ have infinite genus and no punctures, with finitely many ends accumulated by genus. Must every homomorphism factor through an abelian subgroup? Can you forget ends accumulated by genus?

Are surfaces with infinite genus and no punctures, with finitely many ends accumulated by genus quasi-isometric?

It is, however, mathematically underspecified in four ways.

1. No codomain is given for “every homomorphism.” Literally, the identity homomorphism of a nonabelian mapping class group is an immediate counterexample. The preceding AIM records ask about finite quotients, so the likely intended reading is “every homomorphism to a finite group.” 2. The source does not say whether the domain is $\operatorname{PMap}(S)$ or $\operatorname{Map}(S)$. These readings have different answers. 3. “Factor through an abelian subgroup” is not standard wording. We read it as “factor through an abelian group (equivalently, have abelian image in the finite-quotient question).” 4. A topological surface has no canonical metric, so its quasi-isometry type is undefined until a metric is chosen. In the surrounding section, the likely object is the mapping class group with its Rosendal quasi-isometry type.

We therefore analyze the readings separately. For the most controlled family, write \[ B_n=\text{the orientable boundaryless infinite-genus surface with exactly $n$ ends, all accumulated by genus}, \] and put $G_n=\operatorname{PMap}(B_n)$ and $\widehat G_n=\operatorname{Map}(B_n)$. The phrase “no punctures” does not by itself exclude nonisolated planar ends, so results stated for $B_n$ are explicitly a special case rather than a silent reconstruction of every possible $S$.

来源字段：http://aimpl.org/genusinfinity/4/

08755 | AIM-TOPOLOGY-0279

### Finite-central transfer and parity for lifted Hitchin components

P / partially_solved | AIM 研讨会问题汇编

Do there exist embeddings $\phi:\mathrm{SL}(2,\mathbb R)\hookrightarrow G$ such that the component in $\operatorname{Rep}(\pi_1(S),G)$ containing $\phi\circ\rho_0$ consists entirely of discrete and faithful representations?

来源字段：https://aimath.org/WWN/surfacegroups/surfacegroups.pdf

08756 | OWR-1539-001

### Finite-Dimensional Integration of Poisson Manifolds

O / open | Oberwolfach 报告问题汇编

For an integrable Poisson manifold $(M,\Pi)$, whose canonical symplectic integration is constructed by symplectic reduction in an infinite-dimensional Banach manifold, under what conditions can the infinite-dimensional setting be bypassed?

来源字段：https://doi.org/10.4171/owr/2007/23

08757 | OWR-13686-008

### Finite-Order Projections outside the Assembly-Map Image

O / open | Oberwolfach 报告问题汇编

For a nonidentity finite-order element $g\in G$ of order $d$, set $p_g=d^{-1}\sum_{k=1}^d g^k\in\mathbb QG$. If $g_1,\ldots,g_n$ have distinct finite orders, do $[p_{g_1}],\ldots,[p_{g_n}]$ generate a rank-$n$ subgroup of the finite part of $K_0(C^*(G))$, every nonzero element of which lies outside the assembly-map image?

来源字段：https://doi.org/10.4171/owr/2015/29

08758 | OWR-11786-011

### Finite-Ring Solutions of Thurston Equations

O / open | Oberwolfach 报告问题汇编

Let $M$ be a compact $3$-manifold. For every nontrivial $\gamma\in\pi_1(M)$, is there a finite commutative ring $R$ with identity and a homomorphism $\pi_1(M)\to\operatorname{PSL}(2,R)$ that does not send $\gamma$ to the identity? If $M$ is closed, oriented, and not $S^3$, does some one-vertex triangulation of $M$ have Thurston equations solvable over such a finite ring?

来源字段：https://doi.org/10.4171/owr/2012/24

08759 | AIM-TOPOLOGY-0096

### Finite-stage and filtered homotopy colimits in A-theory

P / partially_solved | AIM 研讨会问题汇编

The category of graphs with weak equivalences given by the maps inducing isomorphisms on all A-homotopy groups admits (small) homotopy colimits.

“... and allow \(n\) to depend on \(n\).”

Its exact problem statement is:

The record adds the following test problem. For a graph \(X\), \[ \Sigma_nX=(X\square I_n)/{\sim}, \] where all vertices \((x,0)\) are identified and all vertices \((x,n)\) are identified. It asks, for \(n\geq 5\), whether \(\Sigma_{n+1}X\) and \(\Sigma_nX\) are weakly homotopy equivalent, even for \(n=5\) and \(X=C_5\). It then asks whether a level-collapse map \[ c_{n,j}\colon \Sigma_{n+1}X\longrightarrow\Sigma_nX \] is \(m\)-connected, meaning that it induces isomorphisms on \(A_k\) for \(k\leq m\).

It still literally ends:

The most natural repair in context is “allow \(n\) to depend on \(m\),” i.e. ask whether, for every desired connectivity \(m\), sufficiently long suspensions have an \(m\)-connected collapse. The alternative repair “allow \(m\) to depend on \(n\)” asks for a connectivity estimate as a function of length. Both are mathematically sensible, and neither is silently substituted for the source.

To make “collapsing a level” precise, fix \(0\leq j\leq n\) and define \[ q_{n,j}(t)= \begin{cases} t,&t\leq j,\\ t-1,&t\geq j+1. \end{cases} \] This is a graph map \(I_{n+1}\to I_n\), with \(q_{n,j}(j)=q_{n,j}(j+1)\), and it induces \[ c_{n,j}[x,t]=[x,q_{n,j}(t)]. \] Different choices of \(j\) are retained; no equivalence between the resulting maps is assumed.

There is a historical discrepancy worth recording. The summary does not contain the proof, while the live problem page still says that the \(n=5,C_5\) case is unknown. This report treats the workshop statement as reported progress, not as a substitute for a citable detailed proof.

来源字段：http://aimpl.org/combhomotop/2/

08760 | AIM-TOPOLOGY-0099

### Finite-stage classifying graphs and the one-stage shift

P / partially_solved | AIM 研讨会问题汇编

**Classifying spaces.** Do there exist classifying graphs for fibrations with a fixed fiber (up to weak equivalence) in A-theory? Here, by fibrations we mean maps satisfying the graph analogue of the sharp map condition from simplicial sets.

来源字段：http://aimpl.org/combhomotop/2/

08761 | AIM-TOPOLOGY-0146

### Finite-stage fidelity for finite-genus pure mapping class groups

P / partially_solved | AIM 研讨会问题汇编

Are there general methods to take limits, for going from finite-type to infinite-type. What properties hold under limits?

When \(S\) has finite genus, there is an inverse limit structure and some properties for the pure mapping class group, like residual finiteness, can be inherited from finite-type surfaces.

来源字段：http://aimpl.org/genusinfinity/3/

08762 | AIM-TOPOLOGY-0241

### Finite-window largeness of the O'Neill tensor

P / partially_solved | AIM 研讨会问题汇编

Problem 28. Given a Riemannian submersion with positively curved total space, is the dimension of the fiber less than the dimension of the base? It is perhaps simpler to decide if there is a bound on the dimension of the fiber in terms of the dimension of the base. Is the image of the A-tensor large at some point?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08763 | OWR-4798-031

### Finiteness from Face-Number Bounds for Manifolds with Boundary

O / open | Oberwolfach 报告问题汇编

For $d$-manifolds with boundary, is there a finite number of homeomorphism types with the natural invariant $f_{1}-d f_{0}+\binom{d+1}{2}-\#\{\text{interior vertices}\}$ bounded above by any fixed $n$?

来源字段：https://doi.org/10.4171/owr/2011/08

08764 | OWR-1106-005

### Finiteness of Degree-One Three-Manifold Targets

S / solved | Oberwolfach 报告问题汇编

别名：Duplicate: Finiteness of Degree-One Three-Manifold Targets [OWR-1106-006]; Duplicate: Finiteness of Degree-One Three-Manifold Targets [OWR-1106-007]

Given a closed orientable three-manifold $M$, are there, up to homeomorphism, only finitely many closed orientable geometrized three-manifolds $N$ for which there is a degree-one map $M\to N$?

来源字段：https://doi.org/10.4171/owr/2005/45

08765 | OWR-14298592-012

### Finiteness of Fibered Strongly Quasipositive Knots in Concordance Classes

O / open | Oberwolfach 报告问题汇编

Does each smooth concordance class contain only finitely many fibered strongly quasipositive knots?

来源字段：https://doi.org/10.4171/owr/2024/43

08766 | OWR-5158-014

### Finiteness of Hyperbolic-Volume-Generated Homology Seminorms

O / open | Oberwolfach 报告问题汇编

Is the functorial seminorm on singular homology generated by the volumes of oriented closed connected hyperbolic manifolds finite?

来源字段：https://doi.org/10.4171/owr/2011/30

08767 | OWR-1194-002

### Five-or-Six Edge-Degree Triangulations of Three-Manifolds

O / open | Oberwolfach 报告问题汇编

Does every 3-manifold admit a triangulation in which every edge has degree 5 or 6, possibly with no triangle containing more than one degree-6 edge?

来源字段：https://doi.org/10.4171/owr/2006/12

08768 | OWR-1063-004

### Fixed Points from Equivariant Cohomology Divisibility

O / open | Oberwolfach 报告问题汇编

Let $G=(\mathbb Z/p)^r$ act on a projective $\mathbb ZG$-chain complex $C$, and suppose $j^*H^n(C/G)=p^kH^n(C)$. Must some subgroup isomorphic to $(\mathbb Z/p)^{r-k}$ have a fixed point on $C$?

来源字段：https://doi.org/10.4171/owr/2005/42

08769 | AIM-TOPOLOGY-0189

### Fixed-clock boundedness for cylindrical Thurston stretch rays

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.11.** Determine when two Thurston geodesic rays stay bounded distance apart.

The sentence immediately preceding the problem says that Lenzhen and Masur had completely answered the analogous question for Teichmüller geodesic rays. Their formulation compares unit-speed rays at the same time, possibly from distinct basepoints.

There is an essential ambiguity that was absent in the Teichmüller-metric analogue. Thurston's metric is asymmetric, so “distance” may mean the forward distance, the reverse distance, or a symmetrization. Moreover, “rays” may be compared at the same clock time, after reparameterization, or only as image sets. These conventions are inequivalent, as proved below.

The original problem does not restrict to stretch rays. The proved theorem in this report concerns the substantial special class of cylindrical stretch rays on a closed oriented surface of genus at least two, with chain-recurrent complete supports, which is the setting of Théret's theorem used below.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08770 | AIM-TOPOLOGY-0233

### Fixed-concavity collar collapse and failure of the raw double

P / partially_solved | AIM 研讨会问题汇编

**Problem 20. Study the collapse of Riemannian manifolds with boundary which have \(\sec\ge K\) on the interior and controlled boundary concavity.**

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08771 | AIM-TOPOLOGY-0107

### Fixed-diagram-safe reductions for finite equivariant posets

P / partially_solved | AIM 研讨会问题汇编

**Equivariant discrete homotopy theory**

Develop equivariant discrete homotopy theory.

**Scope.** Fix a finite group $G$ and work with finite posets carrying a left $G$-action by order automorphisms. Morphisms are equivariant monotone maps. Homotopies are finite fences of pointwise comparable equivariant maps.

来源字段：http://aimpl.org/combhomotop/6/

08772 | OWR-15427-004

### Flag Sphere Gamma Vectors Beyond Simplicial Polytopes

O / open | Oberwolfach 报告问题汇编

别名：Flag Sphere Gamma Vectors and Polytope Separation [OWR-15427-007]

Does there exist a vector that is the gamma-vector of a flag homology sphere but is not the gamma-vector of any flag simplicial polytope?

来源字段：https://doi.org/10.4171/owr/2017/19

08773 | OWR-15427-006

### Flag-Complex Realization of Gamma Vectors

O / open | Oberwolfach 报告问题汇编

Is the gamma-vector of every flag homology sphere the face vector of a flag simplicial complex?

来源字段：https://doi.org/10.4171/owr/2017/19

08774 | OWR-1319-007

### Floer Homology and Growth of Surface Mapping Classes

S / solved | Oberwolfach 报告问题汇编

For a mapping class $g$ of a closed oriented surface, is there a monotone representative $\phi$ for which $$HF^*(\phi)=H^*(M_{\mathrm{id}},\partial M_{\mathrm{id}};\mathbb Z_2)\oplus\mathbb Z_2^{N(\phi|_{M\setminus M_{\mathrm{id}}})}$$ holds? If $\psi$ is a Thurston canonical representative and $\lambda$ is its largest pseudo-Anosov stretching factor, is the identity $$F^\infty(g)=\lambda=h(\psi)=\limsup_{n\to\infty}|N(\psi^n)|^{1/n}$$ valid?

来源字段：https://doi.org/10.4171/owr/2006/33

08775 | AIM-TOPOLOGY-0222

### Flow-Morse functions and suspension examples

P / partially_solved | AIM 研讨会问题汇编

**Problem 9.** Find an appropriate definition of Morse functions on Alexandrov spaces and construct examples.

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08776 | OWR-2654833-003

### Folded Ribbonlength Bounds and Knot Complexity

P / partially_solved | Oberwolfach 报告问题汇编

For folded ribbon knots and links, can one prove lower bounds for folded ribbonlength; can one find embeddings with $\operatorname{Rib}(K_w)\leq O(\operatorname{Cr}(K)^p)$ for some $1\leq p<3/2$ or families with superlinear folded ribbonlength in crossing number; and how does folded ribbonlength depend on ribbon equivalence and framing?

来源字段：https://doi.org/10.4171/owr/2020/37

08777 | AMR-102-0047

### Foliated Teichmüller theory - Question 11.1

O / open | AMR 问题清单汇编

What kind of nontrivial ``mapping class elements'' are possible for taut foliations?

背景链接：https://arxiv.org/abs/math/0209081

08778 | AMR-102-0048

### Foliated Teichmüller theory - Question 11.2

O / open | AMR 问题清单汇编

A foliation is taut iff it admits a volume-preserving transverse flow. Pseudo-Anosov flows are good candidates for ``best'' such transverse flows, when they exist, which is frequently. Is there an analytic construction of pseudo-Anosov flows, by analogy with Bers' proof of Thurston's classification of surface automorphisms?

背景链接：https://arxiv.org/abs/math/0209081

08779 | AMR-102-0049

### Foliated Teichmüller theory - Question 11.3

O / open | AMR 问题清单汇编

Suppose $M$ is atoroidal and $\mathscr{F}$ arises from a slithering over $S^1$. Let $X$ be pseudo-Anosov transverse to $\mathscr{F}$, such that the time $1$ flow $Z$ takes $\mathscr{F}$ to itself. Lift to $\widetilde{M}$ and let $\lambda, Z^n(\lambda)$ be leaves of $\widetilde{\mathscr{F}}$, both uniformized as $\mathbb{H}^2$ by Candel's theorem. Can $Z$ be approximated by mapping class elements between compact surfaces? That is, are there integers $n_i$, a sequence $\Sigma_i$ of hyperbolic surfaces and $\phi_i:\Sigma_i \to \Sigma_i$ Teichmüller representatives for the isotopy class $[\phi_i]$ such that the composition $$\widetilde{\phi_i^{-1}}Z^{n_i}:\mathbb{H}^2 \to \mathbb{H}^2$$ is a $k_i$-quasi-isometry, where $k_i \to 1$?

背景链接：https://arxiv.org/abs/math/0209081

08780 | AMR-102-0050

### Foliated Teichmüller theory - Question 11.4

O / open | AMR 问题清单汇编

If $\mathscr{F}$ is a taut foliation, one can let $\gamma_i$ be a collection of transverse circles to $\mathscr{F}$ intersecting every leaf and study the space of functions $\mathscr{O}(\sum n_i\gamma_i)$ which are leafwise holomorphic, with poles of order at most $n_i$ along $\gamma_i$. (Here the notation $\sum n_i\gamma_i$ is meant to suggest a divisor on a Riemann surface). How do these function spaces change as a function of $\gamma_i$? What is the effect of certain ``topological'' operations on the $\gamma_i$; e.g. crossing changes, cabling etc.

背景链接：https://arxiv.org/abs/math/0209081

08781 | AMR-115-0001

### Foliations and Diffeomorphism Groups - Question 1 (Virtual Euler class one conjecture)

O / open | AMR 问题清单汇编

Let $M$ be a closed hyperbolic 3-manifold whose first Betti number is at least one. Let $a \in H^2(M ; \mathbb{R})$ be an (even) integral second cohomology class of dual Thurston norm one. Is there a finite cover $p \colon \tilde{M} \rightarrow M$ and a taut foliation $\mathcal{F}$ of $\tilde{M}$ such that the Euler class $e(\mathcal{F})$ of $\mathcal{F}$ is equal to $p^*(a)$? Here $p^* \colon H^1(M) \rightarrow H^1(\tilde{M})$, and we have used Poincar\'{e} duality to identify $H^1(M) \cong H^2(M)$ and $H^1(\tilde{M}) \cong H^2(\tilde{M})$?

背景链接：https://arxiv.org/abs/2503.18274

08782 | AMR-115-0010

### Foliations and Diffeomorphism Groups - Question 10

O / open | AMR 问题清单汇编

Yoshi Mitsumatsu has suggested a variation of this question:

Is the automorphism group of a generic Engel manifold trivial?

背景链接：https://arxiv.org/abs/2503.18274

08783 | AMR-115-0011

### Foliations and Diffeomorphism Groups - Question 11

O / open | AMR 问题清单汇编

Nicola Pia has some work constraining the dynamics of the kernel line field $\mathcal{W}$, but overall we know very little. A crucial remark is that there is no stability for Engel structures, meaning that perturbing an Engel structure will produce some other Engel structure that in general won't be diffeomorphic to the original one. A vague question is then:

Are there any dynamical properties of $\mathcal{W}$ that are persistent under homotopies of the Engel structure?

背景链接：https://arxiv.org/abs/2503.18274

08784 | AMR-115-0012

### Foliations and Diffeomorphism Groups - Question 12

O / open | AMR 问题清单汇编

How does one systematically produce examples of a contact 3-manifold $(M,\xi)$, a line field $L$ tangent to $\xi$, and a non-vanishing closed 2-form $\omega$, of integral class, annihilating $L$?

背景链接：https://arxiv.org/abs/2503.18274

08785 | AMR-115-0013

### Foliations and Diffeomorphism Groups - Question 13

O / open | AMR 问题清单汇编

Another topic that is of interest is whether Engel structures interact with other geometric structures. Here are two questions:

Consider a $4$-manifold endowed with a foliation $\mathcal{F}$ by surfaces. Can the foliation be perturbed to an Engel structure?

背景链接：https://arxiv.org/abs/2503.18274

08786 | AMR-115-0014

### Foliations and Diffeomorphism Groups - Question 14

O / open | AMR 问题清单汇编

Consider a $4$-manifold endowed with a codimension-$1$ foliation $\mathcal{F}$. Suppose moreover that there is a plane field $\xi \subset T\mathcal{F}$ that is a leafwise contact structure. Assume that $\xi$ splits as a sum of line fields. Can this data be perturbed to an Engel structure?

背景链接：https://arxiv.org/abs/2503.18274

08787 | AMR-115-0015

### Foliations and Diffeomorphism Groups - Question 15

O / open | AMR 问题清单汇编

I would like to conquer some doubt on Palmeira's theorem for foliation by planes $\mathcal{F}$ of $\mathbb{R}^3$. In his paper there is a proposition saying that an infinite braid can be extended as an isotopy of plane, which has counter-examples. This is used to show that the leaf space determines the foliation.

Find an alternative argument.

背景链接：https://arxiv.org/abs/2503.18274

08788 | AMR-115-0016

### Foliations and Diffeomorphism Groups - Question 16

O / open | AMR 问题清单汇编

Let $\omega$ be a volume-form on $**S**^2$ and $\mathrm{Diff}_0(**S**^2, \mathrm{vol})$ be the group of volume-preserving diffeomorphisms of the 2-sphere that are in the same component as the identity map. The isomorphism type of the group $\mathrm{Diff}_0(**S**^2, \mathrm{vol})$ does not depend on the choice of $\omega$. Banyaga [banyaga1978structure] showed that $\mathrm{Diff}_0(**S**^2, \mathrm{vol})$ is a simple group. Ghys and Gambaudo [gambaudo2004commutators] showed that there are plenty of non-trivial quasimorphisms on the group $\mathrm{Diff}_0(**S**^2, \mathrm{vol})$, in particular this group is not uniformly perfect. The following question is posed by Burago, Ivanov, and Poltervoich [burago2008conjugation].

- Is $\mathrm{Diff}_0(**S**^3, \mathrm{vol})$ uniformly perfect?

- Does $\mathrm{Diff}_0(**S**^3, \mathrm{vol})$ admit any non-trivial quasimorphism?

背景链接：https://arxiv.org/abs/2503.18274

08789 | AMR-115-0017

### Foliations and Diffeomorphism Groups - Question 17

O / open | AMR 问题清单汇编

Let $\text{Diff}_0(**D**^2, *)$ be the group of smooth orientation-preserving diffeomorphisms of $**D**^2$ that fix a marked point $*$ say at the origin. There are two bounded Euler classes on this group; one is induced by the restriction to the boundary and the other is by taking derivative at the origin. The difference between these two classes gives a non-trivial quasimorphism on $\text{Diff}_0(**D**^2, *)$.

Is the space of homogenous quasimorphisms on $\mathrm{Diff}_0(**D**^2, *)$ one dimensional?

背景链接：https://arxiv.org/abs/2503.18274

08790 | AMR-115-0018

### Foliations and Diffeomorphism Groups - Question 18

O / open | AMR 问题清单汇编

Using the Bott vanishing theorem, Morita showed that the MMM classes $\kappa_i$ for surface bundles vanish in $H^*(\mathrm{Diff}(\Sigma_g);\mathbb{R})$ for $i>2$. Do they also vanish in bounded cohomology $H^*_b(\mathrm{Diff}(\Sigma_g); \mathbb{R})$?

背景链接：https://arxiv.org/abs/2503.18274

08791 | AMR-115-0019

### Foliations and Diffeomorphism Groups - Question 19

O / open | AMR 问题清单汇编

By now, there are several constructions of simple orderable groups [hyde2019finitely,hyde2023finitely, hyde2023two,bon2020groups]. However, only the examples in [hyde2023finitely] of Hyde and Lodha are finitely presented. In fact, they are even of type $F_\infty$. Recall a group $G$ is of type $F_n$ if there is a $K(G,1)$ with finite $n$-skeleton, and it is of type $F_\infty$ if it is of type $F_n$ for any $n$.

For any $n\geq 2$, is there a simple orderable group of type $F_n$ but not $F_{n+1}$?

背景链接：https://arxiv.org/abs/2503.18274

08792 | AMR-115-0002

### Foliations and Diffeomorphism Groups - Question 2

O / open | AMR 问题清单汇编

Let $\varphi$ be a transitive pseudo-Anosov flow on a closed, oriented 3-manifold, and let $\gamma_1,\dots, \gamma_k$ be a collection of closed orbits. Let $Z\subseteq \mathbb{R}^k$ be the closure of the set of multislopes such that simultaneous Dehn surgery on the $\gamma_i$'s results in a pseudo-Anosov flow admitting a transverse codimension 1 foliation. What structure does the set $Z$ have?

背景链接：https://arxiv.org/abs/2503.18274

08793 | AMR-115-0020

### Foliations and Diffeomorphism Groups - Question 20

O / open | AMR 问题清单汇编

If one drops the orderable condition, there are two constructions [skipper2019simple, belk2022twisted]. A related question of Navas [navas2018group, Question 8] was the following: does there exist a finitely generated, left-orderable group $G$ without surjection onto $\mathbb{Z}$ and $H^2_b(G, \mathbb{R}) = 0$? The question was answered positively by Fournier-Facioa and Lodha in [fournier2023second]. In fact, the simple orderable groups of Matte Bon and Triestino are even boundedly acyclic [campagnolo2023algebraic, Corollary 1.16]. But the only known finitely presented simple orderable groups are not boundedly acyclic [hyde2023finitely, Corollary 1.7]. Thus we ask the following.

Is there a finitely presented, or even type $F_\infty$, boundedly acyclic simple orderable group?

背景链接：https://arxiv.org/abs/2503.18274

08794 | AMR-115-0021

### Foliations and Diffeomorphism Groups - Question 21 (Well-Known)

O / open | AMR 问题清单汇编

Another interesting question would be to find simple orderable groups of finite cohomological dimension. A candidate would be the Burger-Mozes groups [burger2000lattices].

Are Burger--Mozes' examples of simple groups orderable?

背景链接：https://arxiv.org/abs/2503.18274

08795 | AMR-115-0022

### Foliations and Diffeomorphism Groups - Question 22 (Freedman)

O / open | AMR 问题清单汇编

Is that true that for small $\epsilon>0$, we have $g_{\mathscr{E}(p)}(\epsilon) \geq \frac{c|\mathscr{E}(p)|}{\epsilon^2+O(\epsilon^3)}$, for some constant $c$ (perhaps $c=2\pi$)?

背景链接：https://arxiv.org/abs/2503.18274

08796 | AMR-115-0023

### Foliations and Diffeomorphism Groups - Question 23

O / open | AMR 问题清单汇编

Suppose $\epsilon>0$ is a small number. We have a representation $\rho\colon \pi_1(\Sigma_g) \to \mathrm{Diff}_0(**S**^1)$ such that for a generating set $S$ of $\pi_1(\Sigma_g)$, the set $\rho(S)$ are diffeomorphisms that are at most $\epsilon$ away from $\mathrm{PSL}_2(\mathbb{R})$. Can we bound the GV class for the representation $\rho$ in terms of $g$ and $\epsilon$?

背景链接：https://arxiv.org/abs/2503.18274

08797 | AMR-115-0024

### Foliations and Diffeomorphism Groups - Question 24

O / open | AMR 问题清单汇编

Is there a geometric proof of the above theorem? i.e. can one find a family of representations of a fundamental group of $4$-manifolds into $\mathrm{Homeo}_0(**S**^3)$ that exhibits the unboundedness of the Euler class?

背景链接：https://arxiv.org/abs/2503.18274

08798 | AMR-115-0025

### Foliations and Diffeomorphism Groups - Question 25 (Freedman)

O / open | AMR 问题清单汇编

Suppose $**S**^3\to **S**^7\to **S**^4$ is a ``generalized'' Hopf fibration. Does there exist a homology $4$-sphere $H$ and a degree one map $H\to **S**^4$ such that the pull-back of the generalized Hopf fibration is a flat $**S**^3$-bundle?

背景链接：https://arxiv.org/abs/2503.18274

08799 | AMR-115-0026

### Foliations and Diffeomorphism Groups - Question 26

O / open | AMR 问题清单汇编

Suppose we are given an $**S**^3$-bundle $**S**^3\to E\xrightarrow{p} M$ over a $4$-manifold $M$ with nontrivial Euler class and $p_1$. Mather-Thurston's theory says we can find a bordism of this bundle to a $C^0$-flat bundle $**S**^3\to E'\to M'$. Can one bound the simplicial volume of $M'$ so that it does not depend on the bundle map $p$?

背景链接：https://arxiv.org/abs/2503.18274

08800 | AMR-115-0027

### Foliations and Diffeomorphism Groups - Question 27

O / open | AMR 问题清单汇编

Let $M$ be a closed hyperbolic 3-manifold whose first Betti number is at least one. Is every integral second cohomology class $a \in H^2(M ; \mathbb{R})$ of dual Thurston norm one realised as the Euler class of some representation $\rho \colon \pi_1(M) \rightarrow \mathrm{Homeo}_+(**S**^1)$?

背景链接：https://arxiv.org/abs/2503.18274

08801 | AMR-115-0028

### Foliations and Diffeomorphism Groups - Question 28

O / open | AMR 问题清单汇编

Hirsch and Thurston [hirsch1975foliated] proved that the Euler class of amenable actions on spheres $**S**^{2n-1}$ should vanish. This is interesting because now using bounded cohomology techniques, we know a non-constructive proof that the Euler class for flat $**S**^{2n-1}$ bundles when $n>1$ is unbounded. However, by Hirsch-Thurston, it vanishes when the holonomy is amenable.

How can one explain this phenomenon that the Euler class in $H^{2n}(\mathrm{Homeo}_0(**S**^{2n-1}))$ is unbounded but it vanishes on amenable subgroups? Is it a ``hyperbolic'' class in the sense of Gromov (see [brunnbauer2024atoroidal])? Could it be that there is non-trivial bounded Euler class in $H^{2n}(\mathrm{Homeo}^{\text{vol}}_0(**S**^{2n-1}); \mathbb{R})$ where $\mathrm{Homeo}^{\text{vol}}_0(**S**^{2n-1})$ means the volume preserving homeomorphisms of the sphere $**S**^{2n-1}$?

背景链接：https://arxiv.org/abs/2503.18274

08802 | AMR-115-0029

### Foliations and Diffeomorphism Groups - Question 29

O / open | AMR 问题清单汇编

In a recent work with Fournier and Monod [fournier2024bounded], we showed that in fact $\text{Homeo}_0(\mathbb{R}^n)$ is bounded acyclic for all $n$. Therefore, in particular, the Euler class for actions on $\mathbb{R}^{2n}$ is unbounded. But we do not know if the analog of Hirsch--Thurston still holds in these cases.

Suppose $G$ is an amenable group acting on $\mathbb{R}^{2n}$ such that the action is at least $C^1$. Does the Euler class of such action vanish?

背景链接：https://arxiv.org/abs/2503.18274

08803 | AMR-115-0003

### Foliations and Diffeomorphism Groups - Question 3

O / open | AMR 问题清单汇编

Establish the following properties of $Z$:

- The corners of $Z$ are rational points.

- The accumulation points of the corners are precisely the $**S**^1\times **S**^2$ surgeries.

背景链接：https://arxiv.org/abs/2503.18274

08804 | AMR-115-0030

### Foliations and Diffeomorphism Groups - Question 30 (Calegari)

O / open | AMR 问题清单汇编

Calegari's paper has many interesting points about the Euler class and planar group actions, but here I want to emphasize one of his questions:

Do analytic actions of surface groups on $\mathbb{R}^2$ satisfy the Milnor--Wood inequality?

背景链接：https://arxiv.org/abs/2503.18274

08805 | AMR-115-0031

### Foliations and Diffeomorphism Groups - Question 31

O / open | AMR 问题清单汇编

Do volume preserving actions of surface groups on $\mathbb{R}^2$ satisfy the Milnor--Wood inequality?

背景链接：https://arxiv.org/abs/2503.18274

08806 | AMR-115-0032

### Foliations and Diffeomorphism Groups - Question 32

O / open | AMR 问题清单汇编

Under what condition on the holonomy of a compact leaf $L$ in a $C^0$ transversally oriented foliation of codimension $>1$, the leaf $L$ is stable? For example, suppose $\mathcal{F}$ is a $C^0$ codimension $2$ foliation and $L$ is a compact leaf whose normal microbundle has a trivial Euler class, its fundamental group is amenable and $H^1(L; \mathbb{R})=0$. Is $L$ stable?

背景链接：https://arxiv.org/abs/2503.18274

08807 | AMR-115-0033

### Foliations and Diffeomorphism Groups - Question 33

O / open | AMR 问题清单汇编

Consider a foliation $\mathcal{F}$ defined by a fibration with closed basis $B$ and fiber $F$ and $\pi_1(F)=\mathbb{Z}^k$. Under what hypothesis does any foliation $\mathcal{G}$ close to $\mathcal{F}$ have a compact leaf close to a fiber?

背景链接：https://arxiv.org/abs/2503.18274

08808 | AMR-115-0034

### Foliations and Diffeomorphism Groups - Question 34

O / open | AMR 问题清单汇编

Let $\mathcal{F}$ be the foliation defined by a fibration $\pi\colon M\to S$ where $S$ is a genus $2$ closed oriented surfaces and the fiber is the torus $T^2$. Under what hypothesis on the monodromy does any foliation $\mathcal{G}$ close to $\mathcal{F}$ have a compact leaf close to a fiber?

背景链接：https://arxiv.org/abs/2503.18274

08809 | AMR-115-0035

### Foliations and Diffeomorphism Groups - Question 35

O / open | AMR 问题清单汇编

One says that two commuting vector fields $(U,V)$ on $\tilde S$ are $\rho$-equivariant if for any $p$ and $\gamma\in\pi_1(S)$ one has \[ (U(\varphi(\gamma) (p)) , V(\varphi(\gamma)(p))= \rho(\gamma)(U(p),V(p)), \] where $A(U,V) =(aU+bV, cU+dV))$ if $A$ is the matrix $$\left( \begin{array}{cc} a & b\\ c & d \end{array} \right).$$

Characterize the representations $\rho$ for which any $\rho$-equivariant commuting vector fields $(U,V)$ have a common zero.

背景链接：https://arxiv.org/abs/2503.18274

08810 | AMR-115-0036

### Foliations and Diffeomorphism Groups - Question 36

O / open | AMR 问题清单汇编

Elon Lima proved that commuting vector fields for the trivial representation have a common zero. I build an example of $C^1$ commuting pair of vector fields with no common zero, equivariant by $\rho$ if $\rho(\pi_1(S))$ is generated by a unique hyperbolic matrix.

This question is essentially equivalent to the following problem:

Consider a torus $T^2$ bundle over S. Under what hypothesis on the monodromy $\rho$ can we find a foliation with no compact leaves close to a fiber?

背景链接：https://arxiv.org/abs/2503.18274

08811 | AMR-115-0037

### Foliations and Diffeomorphism Groups - Question 37

O / open | AMR 问题清单汇编

Prove every leaf of a $C^r$-generic codimension $1$ oriented and co-oriented foliation on a $3$-manifold has genus $<2$.

背景链接：https://arxiv.org/abs/2503.18274

08812 | AMR-115-0038

### Foliations and Diffeomorphism Groups - Question 38

O / open | AMR 问题清单汇编

Consider four local $C^r$ diffeomorphisms $f_a,f_b,f_c,f_d$ of $R$ whose definition domain contains $[-1,1]$ and having $0$ as a common fixed point. Let $h=[f_a,f_b][f_c,f_d]$ be the product of the commutators.

Does there exist $\tilde f_a,\tilde f_b, \tilde f_c,\tilde f_d$ coinciding with $f_a,f_b,f_c,f_d$ out of $[-1,1]$, $C^r$-close to $f_a,f_b,f_c,f_d$, so that $$h=[\tilde f_a\tilde f_b][\tilde f_c\tilde f_d]$$ and so that $$Fix f_a\cap Fix f_b\cap Fix f_c \cap Fix f_d \cap (-1,1)=\emptyset.$$

背景链接：https://arxiv.org/abs/2503.18274

08813 | AMR-115-0039

### Foliations and Diffeomorphism Groups - Question 39

O / open | AMR 问题清单汇编

Let $\mathrm{Hom}_k(\mathbb{Z}^2, \mathrm{Homeo}_0(\mathbb{R}^2))$ be the subspace of $\mathbb{Z}^2$-actions on $\mathbb{R}^2$ whose Euler number is $k$. Are subspaces $\mathrm{Hom}_k(\mathbb{Z}^2, \mathrm{Homeo}_0(\mathbb{R}^2))$ connected for each integer $k$?

背景链接：https://arxiv.org/abs/2503.18274

08814 | AMR-115-0004

### Foliations and Diffeomorphism Groups - Question 4

O / open | AMR 问题清单汇编

Let $(\xi_-, \xi_+)$ be a pair of cooriented contact structures with opposite signs on $M$, and assume that there exists a vector field $Z$ transverse to both $\xi_-$ and $\xi_+$, and that both of $\xi_-$ and $\xi_+$ are tight. Does $M$ admit a cooriented *Reebless* $C^0$-foliation?

背景链接：https://arxiv.org/abs/2503.18274

08815 | AMR-115-0040

### Foliations and Diffeomorphism Groups - Question 40

O / open | AMR 问题清单汇编

Since our knowledge of the commuting diffeomorphisms of the circle has deepened by the works of Navas, Eynard-Bontemps and others, I would like to pose the following question again 40 years after my paper.

For commuting diffeomorphisms $f$ and $g$ of $**S**^1$, to what extent we can say that the 2 cycle $(f,g)-(g,f)$ of $B\mathrm{Diff}(**S**^1)^\delta$ is a boundary ?

背景链接：https://arxiv.org/abs/2503.18274

08816 | AMR-115-0041

### Foliations and Diffeomorphism Groups - Question 41

O / open | AMR 问题清单汇编

For which $f\in\operatorname{Diff}^1(M)$, does one have \[\operatorname{CritCent}(f)>\operatorname{CritReg}(f)?\] What are the possible values of $\operatorname{CritCent}(f)$ in this case?

背景链接：https://arxiv.org/abs/2503.18274

08817 | AMR-115-0042

### Foliations and Diffeomorphism Groups - Question 42

O / open | AMR 问题清单汇编

Consider a group action $\Phi: G\to \text{Homeo}^+(\mathbb{R}^2)$ on the plane preserving 2 transverse (maybe with prong singularities) foliations $\mathcal{F}^+$ and $\mathcal{F}^-$. This action always extends [bonatti2024action] to *the circle at infinity* associated to the pair of foliations $(\mathcal{F}^+,\mathcal{F}^-)$, as an action $\varphi: G\to Homeo^+(**S**^1)$ called *the action induced by $\Phi$*.

Can we see on the induced action $\varphi$ if $\Phi$ is smoothable or not?

背景链接：https://arxiv.org/abs/2503.18274

08818 | AMR-115-0043

### Foliations and Diffeomorphism Groups - Question 43

O / open | AMR 问题清单汇编

Is there an action $\varphi$ on $**S**^1$ which is induced by a topological action $\Phi$, but is not induced by any smooth action $\Psi$ on $\mathbb{R}^2$ preserving two transverse foliations?

背景链接：https://arxiv.org/abs/2503.18274

08819 | AMR-115-0044

### Foliations and Diffeomorphism Groups - Question 44

O / open | AMR 问题清单汇编

Which pairs $\Lambda, \Lambda'$ of (possibly non-Hausdorff) simply connected 1-manifolds can be realized as the leaf spaces of a pair of transverse foliations of the plane? What if you replace "pair" by ``triple"?

One can also specialize this to the pairs of stable/unstable leaf spaces for Anosov flows, it is as yet not even known whether the stable leaf space determines the unstable.

背景链接：https://arxiv.org/abs/2503.18274

08820 | AMR-115-0045

### Foliations and Diffeomorphism Groups - Question 45

O / open | AMR 问题清单汇编

Let $G$ be a finitely generated amenable orderable group, and $\varphi\colon G\to \mathrm{Homeo}_+\mathbb{R}$ an action of $G$ without global fixed points. Is it true that $\varphi$ is either semi-conjugate to an affine action or laminar?

背景链接：https://arxiv.org/abs/2503.18274

08821 | AMR-115-0046

### Foliations and Diffeomorphism Groups - Question 46

O / open | AMR 问题清单汇编

Which pair of prelaminations forces any group action that preserves it to induce an Ansov-like action on the corresponding bifoliated plane?

背景链接：https://arxiv.org/abs/2503.18274

08822 | AMR-115-0047

### Foliations and Diffeomorphism Groups - Question 47

O / open | AMR 问题清单汇编

Related to the Anosov-like action conjecture, Baik-Wu-Zhao [baik2024reconstruction] recently defined a flowable group action on the bifoliated plane, and showed that if a group admits a flowable orientation-preserving action on a bifoliated plane, the group is the fundamental group of a 3-manifold with a pseudo-Anosov flow and the bifoliated plane is the orbit space of the flow. The set of axioms is slightly different, but the most important condition in both settings is the hyperbolicity (topologically expanding one leaf and contracting another leaf when two leaves cross).

Is every Anosov-like action flowable? If not, what are counterexamples and what is the precise difference between two actions?

背景链接：https://arxiv.org/abs/2503.18274

08823 | AMR-115-0048

### Foliations and Diffeomorphism Groups - Question 48

O / open | AMR 问题清单汇编

The known examples of groups with a veering pair of invariant laminations are the fundamental groups of 3-manifolds with either a pseudo-Anosov flow without perfect fits or a veering triangulation. The following is a natural question from this.

Classify all 3-manifolds whose fundamental group admits an action on $**S**^1$ with a veering pair of invariant laminations.

背景链接：https://arxiv.org/abs/2503.18274

08824 | AMR-115-0049

### Foliations and Diffeomorphism Groups - Question 49 (Calegari)

O / open | AMR 问题清单汇编

Calegari once conjectured that laminar groups preserving a very full lamination (a lamination all of whose gaps are finite-sided ideal polygons) should be residually finite. The moral of this conjecture is that preserving a lamination with enough structure should restrict the possible algebraic structure of the group.

Show that a group acts on the circle with a very full invariant lamination is residually finite.

背景链接：https://arxiv.org/abs/2503.18274

08825 | AMR-115-0005

### Foliations and Diffeomorphism Groups - Question 5

O / open | AMR 问题清单汇编

Refinements of the Eliashberg--Thurston theorem by Colin and Bowden imply that any Reebless aspherical $C^0$-foliation approximates to such pairs. As a warm-up, one could investigate the following question raised by Colin and Firmo in [colin2011paires]:

Let $(\xi_-, \xi_+)$ be a tight positive contact pair as in the previous question. Is $M$ irreducible?

背景链接：https://arxiv.org/abs/2503.18274

08826 | AMR-115-0050

### Foliations and Diffeomorphism Groups - Question 50

O / open | AMR 问题清单汇编

A theorem of Michel Herman says that in the group of $\mathrm{Diff}_+^\infty(**S**^1)$, every element with diophantine rotation number is $C^\infty$ conjugate to a rotation. Using that the rotation number is a continuous function, it follows easily that every element is the product of a rotation and of a conjugate to a rotation. In particular, every element is the product of two commutators. However, it is not known if one commutator is enough.

Is every element in $\mathrm{Diff}_+^\infty(**S**^1)$ a commutator?

背景链接：https://arxiv.org/abs/2503.18274

08827 | AMR-115-0051

### Foliations and Diffeomorphism Groups - Question 51

O / open | AMR 问题清单汇编

We can ask the same question, say, in $\mathrm{Diff}_0^\infty(**S**^2)$. Here is a variation on $\mathrm{Diff}_0^\infty(\Sigma_g)$, where $\Sigma_g$ is surface of genus $g \geq 1$. After the work of Bowden, Hensel and Webb using their fine curve graph [bowden2022quasi], we know that the stable commutator length is unbounded on these groups.

Can we find $f \in \mathrm{Diff}_0^\infty(\Sigma_g)$ such that $\mathrm{scl}(f) = 0$ but $f$ is not a commutator? What about $\mathrm{Homeo}_0(\Sigma_g)$?

背景链接：https://arxiv.org/abs/2503.18274

08828 | AMR-115-0052

### Foliations and Diffeomorphism Groups - Question 52

O / open | AMR 问题清单汇编

Likewise, on these groups we may define the *flow length* as the word length for the set $\mathcal{S}$ of all time one map of flows, which generates $\mathrm{Homeo}_0(M)$ or $\mathrm{Diff}_0^\infty(M)$ since these groups are simple. This norm (or its asymptotic version) measures in some sense the dynamical complexity of a map, especially on surfaces where flows are considered to have ``simple" dynamics. The methods used in [bowden2022quasi] for finding elements with positive scl also yield positive stable flow length. Also note that an element which is conjugate to its inverse has vanishing scl.

Can we find an element in $\mathrm{Homeo}_0(\Sigma_g)$ with vanishing scl and positive stable flow length? Does there exist such an element which is conjugate to its inverse?

背景链接：https://arxiv.org/abs/2503.18274

08829 | AMR-115-0053

### Foliations and Diffeomorphism Groups - Question 53

O / open | AMR 问题清单汇编

Let $G$ denote the group of area and orientation preserving homeomorphisms of the 2-sphere, and $[G,G]$ its subgroup generated by commutators. We recently proved ([cristofaro2023pfh], see also [cristofaro2024proof, cristifaro-gardiner2022quantitative]) that $G$ is not perfect, i.e. \[ [G,G]\varsubsetneq G\] However, the group $G$ remains poorly understood and the following questions are wide open:

What is $[G,G]$ ? What is the quotient $G/[G,G]$?

背景链接：https://arxiv.org/abs/2503.18274

08830 | AMR-115-0054

### Foliations and Diffeomorphism Groups - Question 54

O / open | AMR 问题清单汇编

We will call an element that satisfies the conclusion of the proposition non-spreading.

Is the converse true?

背景链接：https://arxiv.org/abs/2503.18274

08831 | AMR-115-0055

### Foliations and Diffeomorphism Groups - Question 55

O / open | AMR 问题清单汇编

Is it true that a non-spreading homeomorphism of the $2$-torus has conjugates arbitrarily close to a rotation ? \\ Is it true that a non-spreading homeomorphism of a higher genus surface has conjugates arbitrarily close to the identity?

背景链接：https://arxiv.org/abs/2503.18274

08832 | AMR-115-0056

### Foliations and Diffeomorphism Groups - Question 56

O / open | AMR 问题清单汇编

Let's consider the group $\mathcal{G}^r$ of germs of $C^r$ diffeomorphisms about the origin in $\mathbb{R}$, where $r$ is at least $1$ (the reason for excluding $r=0$ is that, in the continuous setting, every 1-dimensional germ -or homeomorphism- is a distortion element, because every nontrivial element is conjugate to its square, for example). Here, $r$ can be also equal to $\omega$, which means that we also consider the real-analytic case.

What are the distortion elements of $\mathcal{G}^r$?

背景链接：https://arxiv.org/abs/2503.18274

08833 | AMR-115-0057

### Foliations and Diffeomorphism Groups - Question 57

O / open | AMR 问题清单汇编

For the case of real-analytic codimension 1 foliations, the presence of an exceptional minimal set implies that an extremal leaf (passing through an endpoint of an interval of its complement in a transversal section) has infinitely many ends.

Can the above theory for the analytic group actions be generalized to the case of codimension 1 real analytic foliations? Can it lead to establishing of the Dippolito conjecture for foliations?

背景链接：https://arxiv.org/abs/2503.18274

08834 | AMR-115-0058

### Foliations and Diffeomorphism Groups - Question 58

O / open | AMR 问题清单汇编

Classify the exceptional minimal sets for codimension-$q$ foliations of compact manifolds, for $q \geq 2$ and $r \geq 2$.

背景链接：https://arxiv.org/abs/2503.18274

08835 | AMR-115-0059

### Foliations and Diffeomorphism Groups - Question 59

O / open | AMR 问题清单汇编

Give an example of $\chi^2\ne 0$, namely, construct a flat $**S**^1$-bundle over some oriented closed 4-manifold $M$ with $\chi^2\ne 0 \in H^4(M;{\mathbb{Z}})$.

背景链接：https://arxiv.org/abs/2503.18274

08836 | AMR-115-0006

### Foliations and Diffeomorphism Groups - Question 6

O / open | AMR 问题清单汇编

Let $(\xi_-, \xi_+)$ be a pair of (co)oriented contact structures with opposite signs on $M$ which are homotopic as plane fields, and assume that both of $\xi_-$ and $\xi_+$ are tight. Does $M$ admit a cooriented *Reebless* $C^0$-foliation?

背景链接 : https://arxiv.org/abs/2503.18274

08837 | AMR-115-0060

### Foliations and Diffeomorphism Groups - Question 60

O / open | AMR 问题清单汇编

Nothing is known about the power of the Euler class for real analytic flat $**S**^1$-bundles.

$\chi_{\mathbb{Q}}^k\ne 0 \in H^{2k}(B\mathrm{Diff}_+^\omega(**S**^1)^{\delta}; {\mathbb{Q}})$ for $k\geq 2$ ?

背景链接 : https://arxiv.org/abs/2503.18274

08838 | AMR-115-0061

### Foliations and Diffeomorphism Groups - Question 61

O / open | AMR 问题清单汇编

Consider the subgroup $R_1$ of $G_1$ which consists of rotations. For $n\in \mathbb{N}$, we put $R_n=p_n^{-1}(R_1)$. Then as a topological group, $R_1$ is isomorphic to $\mathbb{R}/\mathbb{Z}$; \[ R_1\cong R_1\cong\cdots \cong R_ n \cdots\cong \mathbb{R}/\mathbb{Z} \] We fix these isomorphisms. For $n\in \mathbb{N}$, let $G^{(n)}$ be the free product of $G_i$ for $(i = 1 ..... n)$ with amalgamation $R_1\cong R_1\cong\cdots \cong R_ n$. Then we have a direct system of groups; $G^{(0)}\hookrightarrow G^{(1)} \hookrightarrow G^{(2)}\hookrightarrow \cdots$. We define $G$ to be the direct limit group; $G:=\varinjlim G^{(n)}$

Can we show that $G$ is dense in $\mathrm{Diff}^\omega (**S**^1)_0$ ?

背景链接 : https://arxiv.org/abs/2503.18274

08839 | AMR-115-0062

### Foliations and Diffeomorphism Groups - Question 62

O / open | AMR 问题清单汇编

Is the subgroup $\langle \mathrm{PSL}(2)^{(k)},\mathrm{PSL}(2)^{(l)}\rangle$ in $\mathrm{Diff}^{\omega}(**S**^1)$ isomorphic to the amalgamated product $\mathrm{PSL}(2)^{(k)}*_{\mathrm{SO}(2)}\mathrm{PSL}(2)^{(l)}$?

背景链接 : https://arxiv.org/abs/2503.18274

08840 | AMR-115-0063

### Foliations and Diffeomorphism Groups - Question 63 (``Haefliger-Thurston conjecture")

O / open | AMR 问题清单汇编

The following questions, which seem to resist to the contemporary attempts, all ask for some triviality in the homotopy of $B\Gamma$, that would imply the existence of foliations, as well as some acyclicity in the homology of some diffeomorphism groups. On the other hand, if the answers were negative, it would mean that Reeb's fundamental problem admits obstructions of a completely new nature, different from the known ones: essentially, Bott's obstructions, and ``secondary characteristic classes" such as the Godbillon-Vey invariant.

\

Is $B\bar\Gamma_q$ $2q$-connected i.e. do $\pi_i(B\bar\Gamma_q)$ vanish for all $i\leq 2q$?

背景链接 : https://arxiv.org/abs/2503.18274

08841 | AMR-115-0064

### Foliations and Diffeomorphism Groups - Question 64 (Hurder-Vogt [hurder1993product,hurder2005problem,langevin1992list])

O / open | AMR 问题清单汇编

Consider the $\bar\Gamma_q$-structure $\gamma$ on the manifold $SO(q)$ whose microfoliation is the suspension of the canonical action of $SO(q)$ on $\mathbb{R}^q$. Is $\gamma$ null-homotopic in $B\bar\Gamma_q$ ?

背景链接 : https://arxiv.org/abs/2503.18274

08842 | AMR-115-0065

### Foliations and Diffeomorphism Groups - Question 65 (``Foliations with only one leaf": see Tsuboi [tsuboi1984gamma1])

O / open | AMR 问题清单汇编

\ Let $\Gamma_q$ be the groupoid of the germs of local diffeomorphisms of $\mathbb{R}^q$, and $G_q\subset\Gamma_q$ be the subgroup fixing $0$. Let $B\bar G_q$ be the homotopy-theoretic fibre of the map $BG_q\to BGL_q(\mathbb{R})$ classifying the normal bundle of the universal $G_q$-structure. Is $B\bar G_q$ homotopically trivial in $B\bar\Gamma_q$ ?

背景链接 : https://arxiv.org/abs/2503.18274

08843 | AMR-115-0066

### Foliations and Diffeomorphism Groups - Question 66 (E. Ghys, appendix to [molino1988riemannian])

O / open | AMR 问题清单汇编

This amounts to asking if any (framed) Haefliger structure whose microfoliation has the zero section as a leaf is homotopically trivial. Tsuboi answered positively for $q=1$.

\

i) Is every codimension-$q$ Riemannian foliation homotopically trivial in $B\Gamma_q$ ?

ii) Is every codimension-$q$ Lie foliation homotopically trivial in $B\bar\Gamma_q$ ?

背景链接：https://arxiv.org/abs/2503.18274

08844 | AMR-115-0067

### Foliations and Diffeomorphism Groups - Question 67

O / open | AMR 问题清单汇编

The subgroups of piecewise linear (PL for short) homeomorphisms of the line have been a rich source for interesting finitely generated groups with surprising algebraic and dynamical properties. However, not much is known about the algebraic properties of the PL homeomorphisms of higher dimensional PL manifolds. Although they are more combinatorial in nature, the analytical tools for diffeomorphism groups and the Mather infinite repetition trick for homeomorphisms are not available for PL homeomorphisms. So the following basic question due to Epstein [epstein1970simplicity] is still open.

(Epstein) Let $M$ be a PL manifold. Is $\mathrm{PL}_0(M)$, the group of PL homeomorphisms of $M$ that are isotopic to the identity, a simple group?

背景链接：https://arxiv.org/abs/2503.18274

08845 | AMR-115-0068

### Foliations and Diffeomorphism Groups - Question 68

O / open | AMR 问题清单汇编

It is likely that using Bowden-Hensel-Webb's work, one could show that ${\textnormal{\text{PL}}}_\circ(\Sigma,\text{rel }\partial)$ is not uniformly perfect for cases where $\text{Homeo}_0(\Sigma,\text{rel }\partial)$ is not uniformly perfect. This gives an h-principle approach for perfectness of ${\textnormal{\text{PL}}}_\circ(\Sigma,\text{rel }\partial)$. I still do not know an algebraic proof and more importantly do not know how to generalize my approach to higher dimensions. In particular, I do not know the following.

Is the space $\overline{\mathrm{B}\Gamma}_n^{\textnormal{\text{PL}}}$ is $n+1$-connected?

背景链接：https://arxiv.org/abs/2503.18274

08846 | AMR-115-0069

### Foliations and Diffeomorphism Groups - Question 69

O / open | AMR 问题清单汇编

Does an analog of discrete $\mathrm{GV}$ exist in higher dimensions? In particular, does there exist a $3$-cocycle in the third group cohomology $H^3(\mathrm{PL}_c(\mathbb{R}^2); \mathbb{R})$ that induces a surjection \[ H_3(\mathrm{PL}_c(\mathbb{R}^2); \mathbb{Z})\twoheadrightarrow \mathbb{R}. \]

背景链接：https://arxiv.org/abs/2503.18274

08847 | AMR-115-0007

### Foliations and Diffeomorphism Groups - Question 7

O / open | AMR 问题清单汇编

Let $(\xi_-, \xi_+)$ be a pair of (co)oriented contact structures with opposite signs on $M$, and assume that their contact invariants pair to $1$: \begin{align} \langle c(\xi_-), c(\xi_+) \rangle = 1. \end{align} Does $M$ admit a Reebless foliation?

背景链接：https://arxiv.org/abs/2503.18274

08848 | AMR-115-0070

### Foliations and Diffeomorphism Groups - Question 70

O / open | AMR 问题清单汇编

Also, it is tempting to think about the analog of Milnor's conjecture for PL homeomorphisms. Recall that for a finite dimensional Lie group $G$, Milnor conjectured that \[ H_*(BG^{\delta}; \mathbb{F}_p)\to H_*(BG; \mathbb{F}_p) \] is an isomorphism in all degrees. Here, $\mathbb{F}_p$ is the finite field with $p$ elements, $BG$ is the classifying space of the Lie group $G$ and $BG^{\delta}$ is the classifying space of $G$ made discrete. The work of Greenberg on PL foliations seems to suggest that the classifying space of PL Haefliger structures is inductively built out of classifying spaces of Lie groups made discrete. So it might be reasonable to expect the following.

Is the map \[ H_*(B\mathrm{PL}(M)^{\delta}; \mathbb{F}_p)\to H_*(B\mathrm{PL}(M); \mathbb{F}_p) \] an isomorphism in all degrees?

背景链接：https://arxiv.org/abs/2503.18274

08849 | AMR-115-0071

### Foliations and Diffeomorphism Groups - Question 71

O / open | AMR 问题清单汇编

As $B\Gamma_H$ has a trivial $\pi_2$ there is no nontrivial spherical cycle in $H_2(B\Gamma_H; {\mathbb{Z}})$. It is also known that all maps from the 2-torus $T^2$ to $B\Gamma_H$ yield trivial cycles in $H_2(B\Gamma_H; {\mathbb{Z}})$.

(Tsuboi [MR3726712]) Determine whether $H_2(B\Gamma_H;{\mathbb{Z}})=0$.

背景链接：https://arxiv.org/abs/2503.18274

08850 | AMR-115-0072

### Foliations and Diffeomorphism Groups - Question 72

O / open | AMR 问题清单汇编

If a loop in $B\Gamma_H$ is freely homotopic to one that is transverse to the $\Gamma^\omega_1$ structure, it presents a non trivial element in $\pi_1(B\overline{\Gamma}^\omega_1)=\Gamma_H$ which is of infinite order.

Does there exist a homotopically non-trivial loop in $B\overline{\Gamma}^\omega_1$ which is not free-homotopic to a transverse one?

背景链接：https://arxiv.org/abs/2503.18274

08851 | AMR-115-0073

### Foliations and Diffeomorphism Groups - Question 73

O / open | AMR 问题清单汇编

If this is not true, we can describe the whole $\Gamma_H$ fairly clearly, but it is not plausible. If it is true, we have a ${\mathbb{Z}}_{\geq 0}$-valued pseudonorm on $\Gamma_H$.

Is the group $\Gamma_H$ simple?

背景链接：https://arxiv.org/abs/2503.18274

08852 | AMR-115-0008

### Foliations and Diffeomorphism Groups - Question 8

O / open | AMR 问题清单汇编

Let $M$ be a closed hyperbolic 3-manifold whose first Betti number is at least one. Is every even integral second cohomology class $a \in H^2(M ; \mathbb{R})$ of dual Thurston norm one realised as the Euler class of some tight contact structure on $M$?

背景链接：https://arxiv.org/abs/2503.18274

08853 | AMR-115-0009

### Foliations and Diffeomorphism Groups - Question 9

O / open | AMR 问题清单汇编

The group of Engel automorphisms of the local model is very rich: it is isomorphic to the group of contactomorphisms of standard tight $\mathbb{R}^3$. However, automorphisms do not globalize (we cannot ``cut them off''). In fact, it is known, due to work of R. Montgomery, that there are Engel manifolds with very small automorphism group. He posed the question of whether there exists an Engel manifold with trivial automorphism group. One such example was recently found by Koji Yamazaki, but the ambient manifold is non-compact. The following question remains open:

Is there a closed Engel manifold with trivial automorphism group?

背景链接：https://arxiv.org/abs/2503.18274

08854 | AIM-TOPOLOGY-0080

### Forgetting full flags and the local geometry of parabolic braid varieties

P / partially_solved | AIM 研讨会问题汇编

Relate ordinary braid varieties built from complete flags in $G/B$ to parabolic (or “singular,” in the singular-Soergel sense) braid varieties built from partial flags in $G/P$.

It is item 6.3 in the section “Braid varieties” of the 2023 AIM workshop *Algebra, geometry, and combinatorics of link homology*. The record gives no definition of “singular braid varieties.” The live problem-list endpoint returned an internal error on 2026-08-13.

The official workshop report removes much of the ambiguity. Its corresponding working group is titled **Parabolic braid varieties** and says that the group defined a parabolic analogue of braid varieties, proved its affineness, and verified that every parabolic Richardson variety is a special case. Smoothness, Deodhar decompositions, and cluster structures were left as future directions. Thus the best-supported reading is:

This terminology must be separated from two other readings:

1. a **scheme-singular closure** of a smooth ordinary braid variety; and 2. the **singular braid monoid** from classical knot theory.

No evidence was found that item 6.3 uses the second meaning. Scheme singularities do arise after taking orbit/Richardson closures and projections, but “singular” in the workshop context primarily means parabolic or nonregular.

来源字段：http://aimpl.org/agclinkhom/6/

08855 | OWR-16927-004

### Formal Hyperplanes in Periodic Bordism

O / open | Oberwolfach 报告问题汇编

For periodic bordism $MP$, does there exist a formal group with underlying formal hyperplane $MP[[t]]$ whose induced formal group over $\pi_0(MP)$ is additive, and do the three formal hyperplanes corepresented by $MP[t^{\pm1}]$ completed at $t^{-1}$, $MP[[t]]$, and $MP^{\mathbb{CP}^\infty}$ agree?

来源字段：https://doi.org/10.4171/owr/2019/15

08856 | AIM-TOPOLOGY-0008

### Formal-neutral amplification of tight contact structures

P / partially_solved | AIM 研讨会问题汇编

**Given the existence of one tight structure in this almost contact class, must there be infinitely many?**

**Must every almost contact class admit a tight structure?**

来源字段：http://aimpl.org/highdimcontacttop/4/

08857 | AIM-TOPOLOGY-0214

### Four-dimensional finiteness and a uniform Gromov--Hausdorff gap

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.** Does the Grove--Petersen--Wu Finiteness Theorem hold in dimension 4, i.e. are there only finitely many diffeomorphism types in the class of 4-dimensional Riemannian manifolds satisfying \[ \sec M\ge -\Lambda^2,\qquad \operatorname{diam}(M)\le D,\qquad \operatorname{Vol}(M)\ge V\,? \]

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08858 | AIM-TOPOLOGY-0033

### Fourier transforms and max-envelope asymptotics for homeomorphic link exteriors

P / partially_solved | AIM 研讨会问题汇编

**For links with diffeomorphic complements, how are their colored Jones polynomial (asymptotics) related?** If two links have diffeomorphic complements in \(S^3\), relate the asymptotics of their colored Jones Polynomial.

来源字段：http://aimpl.org/quantumlowdimtop/1/

08859 | AIM-TOPOLOGY-0120

### Framing-neutral immersed traces and an embedded-annulus sufficient condition

P / partially_solved | AIM 研讨会问题汇编

Does the homotopy class of the infection curve determine the action of a pattern on topological concordance?

Does the Mazur pattern act by the identity on the topological concordance group? More generally, if $\eta_1$ and $\eta_2$ are two homotopic curves in the complement of a slice knot $R$, must the induced maps $R_{\eta_1}$ and $R_{\eta_2}$ on the topological concordance group agree?

来源字段：http://aimpl.org/concordsliceknot/3/

08860 | OWR-758-006

### Free Finite-Group Actions on Products of Spheres

O / open | Oberwolfach 报告问题汇编

Which additional conditions on a finite group $G$ yield a free action by homeomorphisms or diffeomorphisms on a product $(S^n)^k$ with $k>1$?

来源字段：https://doi.org/10.4171/owr/2004/44

08861 | OWR-14299915-022

### Free Finite-Group Actions on Products of Spheres

P / partially_solved | Oberwolfach 报告问题汇编

Prove that a finite group $G$ acts freely on a finite CW complex homotopy equivalent to a product of $r$ spheres if and only if $G$ has no elementary abelian subgroup of rank $r+1$. In particular, determine whether $Qd(p)$ acts freely on a finite complex homotopy equivalent to $S^n\times S^m$ for odd $p$.

来源字段：https://doi.org/10.4171/owr/2026/9

08862 | OWR-17471-010

### Free Subgroups Avoiding All Dehn-Surgery Kernels

O / open | Oberwolfach 报告问题汇编

Let $K\subset S^3$ be a hyperbolic knot with no finite surgery, and let $G(K)=\pi_1(S^3\setminus K)$. Does there exist a rank-two free subgroup $H\subset G(K)$ such that

$$H\cap\bigcup_{r\in\mathbb Q}\langle\!\langle r\rangle\!\rangle=\{1\},$$

where $\langle\!\langle r\rangle\!\rangle$ is the normal closure of the slope $r$?

来源字段：https://doi.org/10.4171/owr/2020/8

08863 | OWR-4413-004

### Free Toral Rank versus Free $p$-Rank

P / partially_solved | Oberwolfach 报告问题汇编

For a finite connected complex $X$ whose fundamental group is abelian and acts trivially on higher homotopy, and whose rational homotopy is finite-dimensional, compare $$\operatorname{rk}_T(X)=\max\{r:(S^1)^r\text{ acts freely on }X\}$$ with $$\operatorname{rk}_p(X)=\max\{r:(\mathbb Z/p)^r\text{ acts freely on }X\}$$.

来源字段：https://doi.org/10.4171/owr/2010/32

08864 | AIM-TOPOLOGY-0176

### From one Thurston co-sphere to pointed global geometry

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.3 (F. Guéritaud).** Does the unit sphere in \(T_X^*\mathcal T(S)\), endowed with the Thurston norm, determine the global behaviour of stretch lines or geodesics?

As shown by Thurston, the behavior of lengths of laminations infinitesimally in measured lamination space gives good global coordinates for Teichmüller space.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08865 | AIM-TOPOLOGY-0085

### Full twist on the derived annular Hecke trace

P / partially_solved | AIM 研讨会问题汇编

What is the action of the full twist on the trace of the Hecke category? Does it relate to $\nabla$? How does it act on Schur objects?

来源字段：http://aimpl.org/agclinkhom/9/

08866 | OWR-14297744-010

### Fully Faithful Enveloping Algebras for Spectral Lie Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Prove that the universal enveloping algebra functor from $T(n)$-local spectral Lie algebras to $T(n)$-local spectral Hopf algebras is fully faithful.

来源字段：https://doi.org/10.4171/owr/2023/34

08867 | OWR-14297744-011

### Fully Two-Segal Waldhausen S-Constructions

O / open | Oberwolfach 报告问题汇编

Characterize the Waldhausen categories whose $S_\bullet$-construction is fully $2$-Segal rather than merely left $2$-Segal, and find natural examples that are left but not fully $2$-Segal.

来源字段：https://doi.org/10.4171/owr/2023/34

08868 | OWR-2089-013

### Functorial Maps Between p-Local Finite-Group Classifying Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Given $p$-local finite groups $(S,\mathcal F,\mathcal L)$ and $(S',\mathcal F',\mathcal L')$ and a fusion-preserving homomorphism $\phi:S\to S'$, does $B\phi:BS\to BS'$ extend, up to homotopy, to a map $|\mathcal L|^\wedge_p\to|\mathcal L'|^\wedge_p$?

来源字段：https://doi.org/10.4171/owr/2008/43

08869 | AIM-TOPOLOGY-0060

### Functoriality depends on the Khovanov--Rozansky flavor

P / partially_solved | AIM 研讨会问题汇编

Does triply graded HOMFLY-PT Khovanov--Rozansky homology, possibly with exterior-power colors, admit maps for arbitrary oriented link cobordisms which depend only on the isotopy class of the cobordism and compose coherently?

来源字段：http://aimpl.org/agclinkhom/2/

08870 | OPG-37151

### Fundamental group torsion for subsets of Euclidean 3-space

O / open | Open Problem Garden

Problem Does there exist a subset of $\mathbb R^3$ such that its fundamental group has an element of finite order?

背景链接：http://www.openproblemgarden.org/op/torsion_for_subsets_of_mathbb_r_3

08871 | OWR-16933-005

### Fundamental Groups of Tropical Curves

O / open | Oberwolfach 报告问题汇编

For a connected abstract tropical curve represented by a weighted finite metric graph, define a fundamental-group theory, including graph-of-groups constructions that classify covers, harmonic morphisms, and realizations of admissible covers.

来源字段：https://doi.org/10.4171/owr/2019/21

08872 | AIM-TOPOLOGY-0259

### Fundamental groups under quasi- and almost-positive curvature

P / partially_solved | AIM 研讨会问题汇编

**Problem 46.** Find new examples of fundamental groups in quasi-positive curvature or positive curvature on an open dense set.

The section heading at lines 140--141 is “Quasi-positive Curvature and Positive Curvature on an Open Dense Set.” Problem 45 asks about homogeneous quotients immediately before it, and Problem 47 asks a stabilization/symmetry question immediately after it. Thus “positive curvature on an open dense set” has the standard pointwise meaning used in the literature, not merely positivity for an open dense subset of the Grassmann bundle.

For a Riemannian manifold \((M,g)\), define \[ \mathcal P(g)=\{p\in M:\sec_g(\sigma)>0 \text{ for every }2\text{-plane }\sigma\subset T_pM\}. \] The metric is **quasi-positively curved** if \(\sec_g\geq0\) and \(\mathcal P(g)\neq\varnothing\). It is **almost positively curved** (the modern name for the second condition in the problem) if \(\sec_g\geq0\) and \(\mathcal P(g)\) is open and dense. Openness is automatic from continuity of sectional curvature and compactness of the Grassmannian of two-planes at a point; density is the extra hypothesis. Almost-positive curvature implies quasi-positive curvature.

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08873 | OWR-1194-009

### Gaps Between Piecewise-Linear and Linear Embedding Dimensions

O / open | Oberwolfach 报告问题汇编

Can the minimum dimensions for piecewise-linear embeddability and linear embeddability of a simplicial complex differ by more than one?

来源字段：https://doi.org/10.4171/owr/2006/12

08874 | OWR-17135-007

### Generalized Baues Problem for Hypersimplicial Subdivisions

P / partially_solved | Oberwolfach 报告问题汇编

For the cyclic polytope $C_{n,d}$ and a hypersimplicial parameter $k$, does the poset of hypersimplicial subdivisions have the homotopy type of $S^{n-d-2}$?

来源字段：https://doi.org/10.4171/owr/2019/39

08875 | OPG-59896

### Generalized path-connectedness in proximity spaces

O / open | Open Problem Garden

Let $\delta$ be a proximity.

A set $A$ is connected regarding $\delta$ iff $\forall X,Y \in \mathscr{P} A \setminus \{ \emptyset \}: \left( X \cup Y = A \Rightarrow X \mathrel{\delta} Y \right)$.

Conjecture The following statements are equivalent for every endofuncoid $\mu$ and a set $U$:

- $U$ is connected regarding $\mu$. - For every $a, b \in U$ there exists a totally ordered set $P \subseteq U$ such that $\min P = a$, $\max P = b$, and for every partion $\{ X, Y \}$ of $P$ into two sets $X$, $Y$ such that $\forall x \in X, y \in Y: x < y$, we have $X \mathrel{[ \mu]^{\ast}} Y$.

背景链接：http://www.openproblemgarden.org/op/generalized_path_connectedness_in_proximity_spaces

08876 | OWR-17132-004

### Generalized Smith Questions for Compact Lie Groups

P / partially_solved | Oberwolfach 报告问题汇编

Given reduced homology theories $h_*$ and $h'_*$ and a compact Lie group $G$, when does there exist a finite based $G$-CW complex $X$ such that $h_*(X)=0$ but $h'_*(X^G)\neq0$?

来源字段：https://doi.org/10.4171/owr/2019/36

08877 | OWR-1190-003

### Generating Hypothesis and Nilpotence for Finite Spectra

O / open | Oberwolfach 报告问题汇编

If $f:X\to Y$ is a map of finite spectra with $\pi_*(f)=0$, must $f$ be null-homotopic? Also, when is such a map nilpotent under composition?

来源字段：https://doi.org/10.4171/owr/2006/08

08878 | AIM-TOPOLOGY-0190

### Generic endpoint mismatch between maximal stretch laminations and Teichmuller foliations

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.12 (K. Rafi).** Given two points in Teichmüller space, do we have a comparison between the maximal stretch laminations (which define the concatenation of stretch segments) and the vertical (horizontal) foliations of quadratic differential for the Teichmüller geodesic connecting them? Minsky has studied the approximate behavior of high-energy harmonic maps when the domain surface is varied, and compared the harmonic maps (which are stretched along the Hopf foliation) with Teichmüller maps and stretch maps.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08879 | OWR-16406-010

### Generic Properties of Random Heegaard Three-Manifolds

O / open | Oberwolfach 报告问题汇编

What properties does a generic closed 3-manifold have when its Heegaard gluing map is chosen by a random walk on the mapping class group?

来源字段：https://doi.org/10.4171/owr/2018/31

08880 | AIM-TOPOLOGY-0276

### Generic specialization and wall cancellation for Witt-Toledo classes

P / partially_solved | AIM 研讨会问题汇编

Let \(k\) be a field and \(V\) a symplectic vector space over \(k\). There exists a central extension \[ 1\longrightarrow W(k)\longrightarrow \widetilde{\operatorname{Sp}}(V) \longrightarrow \operatorname{Sp}(V)\longrightarrow 1, \] where \(W(k)\) is the Witt group. Given a representation \(\rho:\pi_1(S)\to\operatorname{Sp}(V)\), the Toledo invariant \(\tau(\rho)\) is an element of \(W(k)\). Let \(A\) be an algebraic set and consider a map \(u:A\to\operatorname{Rep}(\pi_1(S),\operatorname{Sp}(2n,\mathbb R))\). To this map associate the tautological representation \(\rho:\pi_1(S)\to\operatorname{Sp}(2n,k(A))\), and define the invariant \(\tau(\rho)\in W(k(A))\). Study this.

来源字段：https://aimath.org/WWN/surfacegroups/surfacegroups.pdf

08881 | OWR-13687-010

### Generic Uniqueness for Skein-Algebra Representations

S / solved | Oberwolfach 报告问题汇编

Let $A$ be a primitive $2N$-th root of unity with $N$ odd. Is there a Zariski-dense open subset of the $\mathrm{SL}_2(\mathbb{C})$ character variety of a surface on which an irreducible finite-dimensional representation of its Kauffman bracket skein algebra is uniquely determined by its classical shadow and compatible puncture invariants?

来源字段：https://doi.org/10.4171/owr/2015/30

08882 | OWR-12697686-011

### Genus-Free Intersection Bounds for Embedded Graphs

O / open | Oberwolfach 报告问题汇编

Can two graphs $G_1$ and $G_2$ embedded on a common surface be redrawn with at most $O(|E(G_1)||E(G_2)|)$ intersections, with a constant independent of the genus?

来源字段：https://doi.org/10.4171/owr/2023/3

08883 | AIM-TOPOLOGY-0125

### Genus-g trace boundaries: the zero-genus answer and a marked Alexander obstruction

P / partially_solved | AIM 研讨会问题汇编

**Genus $g$ traces and concordance.** For a knot $K$ in $S^3$, let $X^g(K)$ denote the 4-ball union $\Sigma_g\times D^2$, where $\Sigma_g$ is a genus $g$ surface with one boundary component, attached to $B^4$ along a 0-framed neighborhood of $K$. Are there nonconcordant knots $K$ and $J$ such that $\partial X^g(K)$ and $\partial X^g(J)$ are integrally homology cobordant?

来源字段：http://aimpl.org/concordsliceknot/4/

08884 | OWR-8415345-008

### Genus-Reducing Local-Knot Concordance in Orientable Three-Manifolds

O / open | Oberwolfach 报告问题汇编

Does the genus-reducing concordance phenomenon constructed for local knots in non-orientable 3-manifolds also occur in an orientable 3-manifold; equivalently, can the additional topology of an orientable 3-manifold yield more efficient cobordisms between local knots?

来源字段：https://doi.org/10.4171/owr/2021/42

08885 | OWR-2089-003

### Geodesic Representatives and Lengths of Unknotting Tunnels

P / partially_solved | Oberwolfach 报告问题汇编

Let $M$ be a hyperbolic tunnel-number-one 3-manifold and let $\tau$ be an unknotting tunnel. Is $\tau$ always isotopic to a geodesic? Does every unknotting tunnel have bounded length?

来源字段：https://doi.org/10.4171/owr/2008/43

08886 | OWR-2089-004

### Geodesicity of Core Tunnels in Compression Bodies

O / open | Oberwolfach 报告问题汇编

Let $C$ be a compression body with genus-two positive boundary and torus negative boundary obtained from a hyperbolic tunnel-number-one manifold, and let $\tau$ be the core arc of its compressible handle. Is $\tau$ isotopic to a geodesic for every geometrically finite hyperbolic structure on $C$?

来源字段：https://doi.org/10.4171/owr/2008/43

08887 | OWR-12485-009

### Geometric, Energetic, and Probabilistic Knot Problems

P / partially_solved | Oberwolfach 报告问题汇编

Can complexity methods prove the existence of nontrivial local minima for knot energies? Are the spaces of embedded equilateral arms and unknotted equilateral polygons connected? How many knot types have a given stick number? What are the limiting torsion laws for confined equilateral walks and polygons? Does mutation preserve arc index? What is the Frisch-Wasserman-Delbrück exponent $\varepsilon$?

来源字段：https://doi.org/10.4171/owr/2013/22

08888 | OWR-12007-023

### Geometry and Quasimorphism Extension Along Contactomorphism Inclusions

O / open | Oberwolfach 报告问题汇编

What can be said about the geometry of

$$j:\operatorname{Cont}_0(M,\alpha)\hookrightarrow\operatorname{Cont}_0(M,\xi),$$

and can quasimorphisms from $\operatorname{Ham}(X,\omega)$ extend along $j$ in the prequantization case?

来源字段：https://doi.org/10.4171/owr/2012/35

08889 | OWR-16415-001

### Good Curve Graphs with Unbounded Mapping-Class Orbits

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Good-Curve-Graph Conjecture [OWR-16415-002]

Given a surface $S$, does there exist a connected graph whose vertices are essential simple closed curves, on which the mapping class group acts with unbounded orbits? In terms of the source complexity $f(S)$, is the answer negative when $f(S)=1$, while both positive and negative examples occur when $f(S)\in\{2,3\}$?

来源字段：https://doi.org/10.4171/owr/2018/40

08890 | OWR-12867-002

### Good Spines of Teichmüller Space

O / open | Oberwolfach 报告问题汇编

Construct geometrically natural, mapping-class-group-equivariant spines of Teichmüller space $\mathcal T_g$.

来源字段：https://doi.org/10.4171/owr/2014/07

08891 | OWR-2040-002

### Gradient-Path Uniqueness in Persistence Pairings

O / open | Oberwolfach 报告问题汇编

Let $K$ be a filtered simplicial complex, let $P$ be a persistence pairing on $K$, and let $V_P$ be the discrete vector field formed from persistence pairs in which one simplex is a face of the other. If $\{\sigma,\tau\}$ is a persistence pair and $\sigma$ is not a face of $\tau$, is there exactly one gradient path from $\tau$ to $\sigma$ in $V_P$?

来源字段：https://doi.org/10.4171/owr/2008/29

08892 | OWR-1703868-001

### Graph Complex Invariants and Classical Embedding Invariants

O / open | Oberwolfach 报告问题汇编

For the natural finite-to-one map from components of high-dimensional embedding spaces to Maurer-Cartan elements of the associated hairy graph complex, is it related to classical embedding invariants such as the Boechat-Haefliger and Whitney-Skopenkov invariants and linking invariants?

来源字段：https://doi.org/10.4171/owr/2020/21

08893 | OPG-37389

### Graph product of multifuncoids

P / partially_solved | Open Problem Garden

Conjecture Let $F$ is a family of multifuncoids such that each $F_i$ is of the form $\lambda j \in N \left( i \right): \mathfrak{F} \left( U_j \right)$ where $N \left( i \right)$ is an index set for every $i$ and $U_j$ is a set for every $j$. Let every $F_i = E^{\ast} f_i$ for some multifuncoid $f_i$ of the form $\lambda j \in N \left( i \right): \mathfrak{P} \left( U_j \right)$ regarding the filtrator $\left( \prod_{j \in N \left( i \right)} \mathfrak{F} \left( U_j \right); \prod_{j \in N \left( i \right)} \mathfrak{P} \left( U_j \right) \right)$. Let $H$ is a graph-composition of $F$ (regarding some partition $G$ and external set $Z$ ). Then there exist a multifuncoid $h$ of the form $\lambda j \in Z: \mathfrak{P} \left( U_j \right)$ such that $H = E^{\ast} h$ regarding the filtrator $\left( \prod_{j \in Z} \mathfrak{F} \left( U_j \right); \prod_{j \in Z} \mathfrak{P} \left( U_j \right) \right)$.

背景链接：http://www.openproblemgarden.org/op/graph_product_of_multifuncoids

08894 | AIM-TOPOLOGY-0004

### Graphical Weinstein perturbations of Liouville doubles

P / partially_solved | AIM 研讨会问题汇编

**Explicitly describe Weinstein convex perturbations of Liouville convex hypersurfaces. For instance, doubles of Liouville manifolds.**

来源字段：http://aimpl.org/highdimcontacttop/3/

08895 | OWR-1268-004

### Gromov's Simplicial-Volume Conjecture for $L^2$-Betti Numbers

O / open | Oberwolfach 报告问题汇编

If a closed aspherical manifold has vanishing simplicial volume, must all of its $L^2$-Betti numbers vanish?

来源字段：https://doi.org/10.4171/owr/2006/19

08896 | OWR-11790-004

### Group-Valued Parity and Knots in Three-Manifolds

O / open | Oberwolfach 报告问题汇编

For parity theory, how can group information be encoded at crossings and a group bracket defined; does a parity theory exist for knots in arbitrary 3-manifolds; and are there applications to link homotopy theory?

来源字段：https://doi.org/10.4171/owr/2012/28

08897 | OWR-14298007-010

### Growth of Triangulated Homology Three-Spheres

P / partially_solved | Oberwolfach 报告问题汇编

Let $t_N$ be the number of combinatorial isomorphism classes of triangulations of three-dimensional homology spheres with $N$ facets. Determine whether $t_N$ grows exponentially or superexponentially in $N$.

来源字段：https://doi.org/10.4171/owr/2023/58

08898 | OWR-9790367-003

### Habiro-Ring Refinements of Three-Manifold Quantum Invariants

P / partially_solved | Oberwolfach 报告问题汇编

For every rational homology sphere $Y$, construct elements $I_s(Y)$ of the Habiro ring, indexed by $s\in\operatorname{Spin}^c(Y)$, such that $\widetilde N_r(Y,\omega)=\sum_s C_s(Y,\omega)I_s(Y)|_{q=e^{2\pi i/r}}$ for all admissible $r$ and $\omega$.

来源字段：https://doi.org/10.4171/owr/2022/16

08899 | AMR-112-0002

### Halperin conjecture

P / partially_solved | AMR 问题清单汇编

For every fibration $F\to E\to B$ of simply connected spaces whose fiber $F$ is rationally elliptic with nonzero Euler characteristic, does the rational Serre spectral sequence collapse at the $E_2$ page?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Topology

08900 | OWR-14299521-013

### Harmonic Two-Spheres in Compact Metric Spaces

O / open | Oberwolfach 报告问题汇编

Let $X$ be a compact metric space with a nontrivial $k$th homotopy group for some $k\geq 2$. What additional conditions on $X$ guarantee the existence of a nontrivial harmonic two-sphere?

来源字段：https://doi.org/10.4171/owr/2025/30

08901 | AIM-TOPOLOGY-0179

### Hausdorff noncomputability and a certified punctured-torus classifier

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.1.** Is there an algorithm to find the maximally stretch lamination?

The long row of dots after the question is a printed separator. The following sentence, “Generically, a Thurston geodesic connecting two points in Teichmüller space is not unique,” introduces the envelope question numbered 2.2; it is not part of Problem 2.1.

The implicit input is a pair of marked hyperbolic structures (X,Y) on a finite-type surface (S). The source does not specify either a representation of (X,Y) or a representation of the requested lamination. That omission is mathematically decisive: exact symbolic input, approximate numerical input, and a promised generic stratum lead to different algorithmic questions.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08902 | AIM-TOPOLOGY-0097

### Higher A-group vanishing for finite dihedral 3-parabolic arrangements

P / partially_solved | AIM 研讨会问题汇编

Applications to subspace arrangements

Can earlier results of Barcelo on the relation between A-theory and subspace arrangements be extended beyond $A_1$ by showing that the higher $A$-groups vanish?

来源字段：http://aimpl.org/combhomotop/2/

08903 | OWR-14298005-001

### Higher Associativity Structures on Mapping Cones

O / open | Oberwolfach 报告问题汇编

Develop the theory of $A_3^+$-structures encoding associativity for multiplicative structures on mapping cones, and more generally the theory of $A_\infty^+$-structures encoding associativity up to homotopy.

来源字段：https://doi.org/10.4171/owr/2023/56

08904 | AIM-TOPOLOGY-0105

### Higher homotopy groups in categorical-product graph homotopy

S / solved | AIM 研讨会问题汇编

**Higher homotopy groups.** In ×-homotopy theory, what are the higher homotopy groups?

来源字段：http://aimpl.org/combhomotop/5/

08905 | OWR-4529-003

### Higher Topological Complexity of Spatial Polygon Spaces

S / solved | Oberwolfach 报告问题汇编

For a fixed positive length vector $\ell=(\ell_1,\ldots,\ell_n)$, let

$$N_\ell=\left\{(v_1,\ldots,v_n)\in(S^2)^n:\sum_i\ell_iv_i=0\right\}/SO(3).$$

Compute the higher topological complexities $\operatorname{TC}_n(N_\ell)$.

来源字段：https://doi.org/10.4171/owr/2010/47

08906 | OWR-14610-005

### Higher-Dimensional Smale Conjecture for 2-Convex Spheres

P / partially_solved | Oberwolfach 报告问题汇编

Higher-dimensional Smale conjecture: the moduli space of 2-convex embedded n-spheres in $\mathbb{R}^{n+1}$ is contractible for every n.

来源字段：https://doi.org/10.4171/owr/2016/28

08907 | OWR-1321-006

### Higher-Order Arf Invariants from Whitney Towers

P / partially_solved | Oberwolfach 报告问题汇编

Do the higher-order Arf invariants suggested by Whitney towers define well-defined link-concordance invariants? In the first test case, let $L=l_1\cup l_2$ be the Bing double of the figure-eight knot, bounding embedded disks $D_1,D_2\subset B^4$, with an immersed Whitney disk $W_{(1,2)}$ pairing the intersections of $D_1$ and $D_2$ and a Whitney disk $W_{((1,2),(1,2))}$ pairing the self-intersections of $W_{(1,2)}$. Does the single interior intersection of $W_{((1,2),(1,2))}$ with $W_{(1,2)}$ represent a nontrivial concordance invariant of $L$?

来源字段：https://doi.org/10.4171/owr/2006/35

08908 | AIM-TOPOLOGY-0271

### Higher-rank counterexamples to uniqueness of Hitchin minimal surfaces

S / solved | AIM 研讨会问题汇编

**Question 2.4 (Richard Wentworth, François Labourie).** Let \(S\) be a closed surface, \(\rho:\pi_1(S)\to \mathrm{SL}(n,\mathbb R)\) a representation in the Hitchin component. Given a complex structure \(J\) on \(S\) there exists a unique \(\rho\)-equivariant harmonic map \[ u:(\widetilde S,J)\longrightarrow X=\mathrm{SL}(n,\mathbb R)/\mathrm{SO}(n) \] (Corollary 3.5 of [Cor88]). Is there a unique complex structure \(J\) such that \(u\) is also conformal?

**Comment 2.5 (Richard Wentworth).** This is true for \(n=2,3\) (see Theorem 9.3.2 of [Lab06]).

**Comment 2.6 (Anna Wienhard).** One can ask the same question for maximal representations.

来源字段：https://aimath.org/WWN/surfacegroups/surfacegroups.pdf

08909 | AIM-TOPOLOGY-0161

### Higher-rank lattices embed in the compact-support closure

S / solved | AIM 研讨会问题汇编

Lattices don't map into $\operatorname{Mod}(S_g)$ for a compact surface $S_g$ of genus $g$.

Do lattices map into $\operatorname{Map}(S)$ or $\overline{\operatorname{PMap}_c(S)}$? In particular, can higher-rank lattices map into these groups?

来源字段：http://aimpl.org/genusinfinity/4/

08910 | OWR-11786-010

### Highly Ramified Figure-Eight Covers of Hyperbolic 3-Manifolds

O / open | Oberwolfach 报告问题汇编

For a closed hyperbolic $3$-manifold $M$ and every $L>0$, does there exist a finite cover $\widetilde M\to M$ and a branched covering $p:\widetilde M\to S^3$ whose branch locus is the figure-eight knot and whose ramification index along every component over the knot exceeds $L$?

来源字段：https://doi.org/10.4171/owr/2012/24

08911 | TOP-009

### Hilbert-Smith Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Hilbert-Smith conjecture [OPG-37297]

If a locally compact group acts faithfully and continuously on a manifold, must it be a Lie group?

背景链接：https://en.wikipedia.org/wiki/Hilbert%E2%80%93Smith_conjecture

08912 | OWR-14299911-014

### Hilbert-Space Coarse Embeddings of Coarse Median Spaces

O / open | Oberwolfach 报告问题汇编

Does every coarse median space admit a coarse embedding into a Hilbert space?

来源字段：https://doi.org/10.4171/owr/2026/8

08913 | OWR-12014-003

### Hirota Equations and Knot Meaning of Spectral Curves

O / open | Oberwolfach 报告问题汇编

For the spectral-curve partition function $\widetilde Z$, does it formally satisfy Hirota equations in infinitely many deformation parameters? What is the knot-theoretic meaning of the genus-zero property of the quotient of the $A$-polynomial curve by its involution?

来源字段：https://doi.org/10.4171/owr/2012/42

08914 | OWR-12007-017

### Hofer Norms on Fundamental Groups of Symplectic Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

For a symplectic manifold $(M,\omega)$, a ball $B$, and the induced Hofer norm $\lVert\gamma\rVert_B$ on $\pi_1(M)$, what can be said about this conjugation-invariant norm? Is it bounded, and is it equivalent to word length?

来源字段：https://doi.org/10.4171/owr/2012/35

08915 | AIM-TOPOLOGY-0272

### Holomorphic monodromy as a Teichmüller-section problem and finite-cover defect

P / partially_solved | AIM 研讨会问题汇编

**Question 3.1 (Dieter Kotschick).** Fix a closed Riemann surface \(B\) of genus \(g\geq 3\), and fix \(h\geq 2\). There exist at most finitely many non-isotrivial holomorphic genus \(h\) fibrations over \(B\), \[ F_h\longrightarrow X\longrightarrow B. \] This gives a conjugacy class of representations \[ \rho:\pi_1(B)\longrightarrow \operatorname{MCG}(F_h). \] How to characterise the representations which are the holonomy of a holomorphic fibration?

**Comment 3.2 (Dieter Kotschick).** When the fibration is holomorphic there is a Kähler structure on the surface bundle \(X\), and so the cohomology of \(X\) satisfies certain constraints from Hodge theory (e.g. \(h_1(X)\) is even). These constraints give some restrictions on the representations, but they are not enough to give an “if and only if” statement.

The recovered typography repairs \(Fh\) to \(F_h\), adds the missing space after “\(B\).”, joins the line-break hyphenations “charac-terise” and “struc-ture,” and restores “Kähler.” The source displays \(h_1(X)\); it is the first Betti number in this context.

The intended fibration is a proper holomorphic submersion with connected smooth genus-\(h\) fibers. Ehresmann's theorem then makes it a smooth surface bundle. This is not a singular fibration: if critical values \(D\subset B\) are allowed, geometric monodromy is naturally a representation of \(\pi_1(B\setminus D)\), with additional local monodromy data. Nor does “holomorphic fibration” mean holomorphically locally trivial; a non-isotrivial Kodaira fibration is smoothly locally trivial but its fiber complex structure varies.

来源字段：https://aimath.org/WWN/surfacegroups/surfacegroups.pdf

08916 | OWR-17471-006

### Homological Stability for Stabilized Four-Manifold Diffeomorphisms

O / open | Oberwolfach 报告问题汇编

For each fixed $k$, does there exist $n$ such that

$$H_k\!\left(B\operatorname{Diff}_{\partial}\!\left(\mathop{\natural}^{n}(S^2\times S^2\setminus\mathring D^4)\right);\mathbb Q\right)$$

is equal to the stable-limit homology?

来源字段：https://doi.org/10.4171/owr/2020/8

08917 | OWR-10252932-003

### Homological Stability of Reductive Borel-Serre Categories

P / partially_solved | Oberwolfach 报告问题汇编

What homological-stability properties hold for reductive Borel-Serre categories, and do these categories possess higher homotopy commutativity?

来源字段：https://doi.org/10.4171/owr/2022/24

08918 | OWR-4428-007

### Homological Torsion Growth in Arithmetic Hyperbolic Towers

O / open | Oberwolfach 报告问题汇编

Let $M$ be the arithmetic hyperbolic three-manifold associated with a quaternion division algebra over an imaginary quadratic field, and let $(M_N)$ be any residual chain of congruence covers. Does $$\frac{\log\#H_1(M_N,\mathbb Z)_{\mathrm{tors}}}{\operatorname{vol}(M_N)}\longrightarrow\frac{1}{6\pi}$$ as $N\to\infty$?

来源字段：https://doi.org/10.4171/owr/2010/40

08919 | OWR-15216-005

### Homology and Projective Fraïssé Limits

O / open | Oberwolfach 报告问题汇编

Develop a homology theory for projective Fraïssé limits based on projective-Fraïssé simplices and their boundary maps, and determine how it behaves under the projective Fraïssé limit operation. In particular, can such a theory identify canonical quotients arising from projective-Fraïssé constructions of universal Menger compacta?

来源字段：https://doi.org/10.4171/owr/2017/11

08920 | AIM-TOPOLOGY-0106

### Homology from the edge Hom complex

P / partially_solved | AIM 研讨会问题汇编

Homology theory for graphs with loops

Is there a good homology theory for graphs with loops? How does it interact with homotopy groups of graphs with loops (which also need to be defined). Does it satisfy a variant Eilenberg-Steenrod axioms?

One approach discussed during the workshop was to take homology of $Hom(K_2, X)$.

来源字段：http://aimpl.org/combhomotop/5/

08921 | OWR-13488-009

### Homology Growth in Residual Towers

O / open | Oberwolfach 报告问题汇编

For a residual tower of finite covers $M_i\to M$, how does the size of $H_k(M_i)$ grow as $i\to\infty$, when size is measured by mod-$p$ Betti numbers or by $|\operatorname{tors}H_k(M_i;\mathbb Z)|$?

来源字段：https://doi.org/10.4171/owr/2015/3

08922 | OWR-11138-007

### Homology of Infinite Loop Spaces

O / open | Oberwolfach 报告问题汇编

How can one compute $H_*(\Omega^\infty X)$ from knowledge of $H_*(X)$?

来源字段：https://doi.org/10.4171/owr/2011/46

08923 | AIM-TOPOLOGY-0115

### Homology sliceness versus ordinary sliceness

P / partially_solved | AIM 研讨会问题汇编

**Is homology slice the same as slice?** Is there a non-smoothly slice knot \(K\) in \(S^3\) which is slice in a homology \(B^4\)? (i.e., is \(\phi_{smooth}\) non-injective?)

来源字段：http://aimpl.org/concordsliceknot/2/

08924 | OWR-12697684-032

### Homotopical versus Homological Cycle-Filling Complexity

O / open | Oberwolfach 报告问题汇编

Over finite connected graphs, is the ratio between the minimum total cycle length needed to attach $2$-cells that kill the fundamental group and the minimum total cycle length needed to kill first homology over $\mathbb F_2$ unbounded? Does this remain true for bounded-degree graphs?

来源字段：https://doi.org/10.4171/owr/2023/1

08925 | OWR-14298375-006

### Homotopy Classes from Elliptic Artin Monoids

O / open | Oberwolfach 报告问题汇编

Are the $\Pi$-classes associated with twins of non-cancellative elements in elliptic Artin monoids nontrivial, and are their inertia groups trivial?

来源字段：https://doi.org/10.4171/owr/2024/29

08926 | OWR-15208-003

### Homotopy Equivalence of Negative Curvature and Anosov Metric Spaces

O / open | Oberwolfach 报告问题汇编

Are the spaces of negatively curved metrics and Anosov metrics on a manifold homotopy equivalent?

来源字段：https://doi.org/10.4171/owr/2017/3

08927 | OWR-4196-003

### Homotopy Groups of Definable Quotients

O / open | Oberwolfach 报告问题汇编

Let $X$ be definable in an o-minimal structure and let $E$ be a type-definable equivalence relation of bounded index. Give $X/E$ the logic topology. Suppose $X/E$ is locally contractible and every $E$-class is the intersection of a countable decreasing family of definably contractible sets. Is $\pi_n^{\mathrm{def}}(X)\cong\pi_n(X/E)$ for every $n$?

来源字段：https://doi.org/10.4171/owr/2010/01

08928 | OWR-4084-005

### Homotopy Invariance of Algebraic K-Theory for Group Rings

P / partially_solved | Oberwolfach 报告问题汇编

For every group $\Gamma$, is the functor $X\longmapsto K_0^{\mathrm{alg}}(C(X)[\Gamma])$ homotopy invariant on compact Hausdorff spaces?

来源字段：https://doi.org/10.4171/owr/2009/41

08929 | OWR-15208-027

### Homotopy Types of Curvature-Metric Moduli on Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

What are the homotopy types of the moduli spaces of nonnegative-curvature metrics on $S^2$ and of positive-curvature or nonnegative-curvature metrics on $\mathbb{R}^2$?

来源字段：https://doi.org/10.4171/owr/2017/3

08930 | OWR-17471-005

### Homotopy Types of Four-Manifold Homeomorphism and Diffeomorphism Groups

O / open | Oberwolfach 报告问题汇编

For a fixed $4$-manifold $M$, what are the homotopy types of the topological groups $\operatorname{Homeo}_{\partial}(M)$ and $\operatorname{Diff}_{\partial}(M)$ of boundary-fixing homeomorphisms and diffeomorphisms?

来源字段：https://doi.org/10.4171/owr/2020/8

08931 | OWR-2041-001

### Homotopy-Correct Initialization for Manifold Regression

P / partially_solved | Oberwolfach 报告问题汇编

In nonparametric regression between manifolds $M$ and $N$, with target mapping $\eta$ and training-sample size $k$, can one construct an initial mapping $\phi(0)$ that, for all sufficiently large $k$, is guaranteed to lie in the same homotopy class as $\eta$?

来源字段：https://doi.org/10.4171/owr/2008/30

08932 | OWR-1381-004

### Homotopy-Orbit Constructions for Compact Lie-Group Actions

O / open | Oberwolfach 报告问题汇编

For the chain-level construction of homotopy orbits of an $S^1$-action, do analogues of the primitive and cobar quasi-isomorphism results hold for every compact Lie group? What are the mathematical meaning and content of the resulting higher operations $\omega_k$?

来源字段：https://doi.org/10.4171/owr/2006/43

08933 | OWR-4413-003

### Homotopy-Theoretic Subgroups and Cohomology of $\operatorname{Baut}(X)$

O / open | Oberwolfach 报告问题汇编

For a finite simply connected CW complex $X$ and the monoid $\operatorname{Baut}(X)$ of self-homotopy equivalences, how many homotopy-theoretic subgroups are there, how large can such a subgroup be, and what is the cohomology of $\operatorname{Baut}(X)$?

来源字段：https://doi.org/10.4171/owr/2010/32

08934 | OWR-1728-004

### Hopkins Conjecture for Morava-Stabilizer Subgroups

O / open | Oberwolfach 报告问题汇编

For every finite subgroup $G$ of the Morava stabilizer group $\mathbb{S}_n$, there should be a $G$-equivariant isomorphism $E_{n*}\cong S_{\mathbb{Z}_{p^n}}(M_n)[\Delta^{-1}]^{\wedge}_I$, where $S$ denotes the symmetric algebra, $M_n$ is placed in degree $-2$, $\Delta$ is the $G$-invariant class corresponding to the multiplicative norm over $G$ on $M_n$, and $I$ is an ideal in degree zero.

来源字段：https://doi.org/10.4171/owr/2007/45

08935 | AIM-TOPOLOGY-0185

### Horofunction drift obstructs recurrence and finite invariant measure for the marked harmonic-stretch flow

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.7 (K. Rafi).** Define the geodesic flow for the Thurston metric and study its properties, such as ergodic or mixing.

The exact mathematical question in the source PDF is:

Inspection of the PDF and neighboring records shows that this sentence introduces Problem 2.8, the counting problem for pseudo-Anosov conjugacy classes; it is not part of Problem 2.7. The contextual paragraph immediately before Problem 2.7 compares the desired theory with the Teichmüller geodesic flow on moduli space.

This report treats the recovered one-sentence question above. It distinguishes:

1. a selected-ray flow on marked Teichmüller space; 2. the induced flow on the diagonal mapping-class quotient; and 3. the impossible stronger request for a canonical flow containing every Thurston geodesic direction without an additional selection rule.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

08936 | AIM-TOPOLOGY-0084

### Hurwitz-Grassmann braid representation varieties

P / partially_solved | AIM 研讨会问题汇编

Define the analogues of braid varieties for \(sl_N\) homology, and study their connection with representation varieties.

来源字段：http://aimpl.org/agclinkhom/6/

08937 | OWR-14299911-024

### Hyperbolic Cubulability, Residual Finiteness, and Local Medianity

O / open | Oberwolfach 报告问题汇编

If every infinite-order element of a hyperbolic group has unbounded orbits in some cobounded cubical action with quasiconvex hyperplane stabilizers, must the group be cocompactly cubulable or at least a-T-menable? If a hyperbolic group acts properly and cocompactly on a finite-rank median space, must it be residually finite or linear? If a complete simply connected path-metric space is locally median with no uniform radius, must it be globally median?

来源字段：https://doi.org/10.4171/owr/2026/8

08938 | AMR-102-0039

### Hyperbolic geometry - Question 10.1

O / open | AMR 问题清单汇编

Suppose $\mathscr{F}$ is a taut foliation of a hyperbolic $3$-manifold $M$ with two-sided branching. Must there be a leaf $\lambda$ of $\widetilde{\mathscr{F}}$ whose complement contains an open halfspace of $\mathbb{H}^3$ on either side? We call such a leaf asymptotically separated.

背景链接：https://arxiv.org/abs/math/0209081

08939 | AMR-102-0040

### Hyperbolic geometry - Question 10.2

P / partially_solved | AMR 问题清单汇编

Do leaves of $\widetilde{\Lambda}$ for $\Lambda$ an essential lamination have the continuous extension property? More generally, what is the relationship between the action of $\pi_1(M)$ on various ideal boundaries of $\widetilde{M}$ arising from the foliated structure (e.g. universal circles) and the ideal boundaries arising from the geometry of $\widetilde{M}$.

背景链接：https://arxiv.org/abs/math/0209081

08940 | AMR-102-0041

### Hyperbolic geometry - Question 10.3

O / open | AMR 问题清单汇编

Suppose $\mathscr{F}$ is a finite depth foliation of a hyperbolic $3$-manifold. What is the relationship (if any) between the Hausdorff dimension of the limit set of a leaf $\lambda$ of $\widetilde{\mathscr{F}}$ and the depth of $\mathscr{F}$ or $\lambda$?

背景链接：https://arxiv.org/abs/math/0209081

08941 | AMR-102-0042

### Hyperbolic geometry - Question 10.4

P / partially_solved | AMR 问题清单汇编

Suppose $M$ an atoroidal $3$-manifold admits an essential lamination. Does it admit a (necessarily genuine) lamination with quasi-geodesic leaves?

背景链接：https://arxiv.org/abs/math/0209081

08942 | AMR-102-0043

### Hyperbolic geometry - Question 10.5

O / open | AMR 问题清单汇编

What do short geodesics look like with respect to taut foliations? Is there a universal $\epsilon$ such that for every hyperbolic manifold $M$, every taut foliation $\mathscr{F}$ of $M$, and every geodesic $\gamma$ with $|\gamma|<\epsilon$, $\gamma$ is either isotopic into a leaf of $\mathscr{F}$ or isotopic to be transverse to $\mathscr{F}$? What about homotopic?

背景链接：https://arxiv.org/abs/math/0209081

08943 | AMR-102-0044

### Hyperbolic geometry - Question 10.6

O / open | AMR 问题清单汇编

Is there a uniform bound on the Godbillon-Vey invariants of the taut foliations of a hyperbolic manifold in terms of its volume?

背景链接：https://arxiv.org/abs/math/0209081

08944 | AMR-102-0045

### Hyperbolic geometry - Question 10.7

O / open | AMR 问题清单汇编

Suppose $\mathscr{F}$ is a taut foliation of a hyperbolic $3$-manifold $M$. Let $$\pi:\widetilde{M} \to L$$ be the projection to the leaf space of $\widetilde{\mathscr{F}}$.

- For $\gamma$ a random walk in $\widetilde{M}$ (which is isometric to $\mathbb{H}^3$), what is the typical behaviour of $\pi(\gamma)$?

- Does a random walk in $\widetilde{M}$ converge to a definite end of $L$?

- What if we replace ``random walk'' with ``random geodesic'' in the previous question?

- Is the pushforward of asymptotic behaviour well-defined? That is, is it true that for a set of geodesics $\gamma$ of full measure, for all $\gamma'$ a bounded distance from some $\gamma$ the behaviour of the pushforward of $\gamma$ and $\gamma'$ exit the same end of $L$?

- Suppose $\mathscr{F}$ has one-sided branching. Does a random walk always exit $L$ in the unbranching direction?

- Suppose there is a positive probability of a random walk exiting a proper positive or negative end of $L$. Does this imply $\mathscr{F}$ is $\mathbb{R}$-covered? What about if there is a positive probability of a random walk being recurrent?

背景链接：https://arxiv.org/abs/math/0209081

08945 | AMR-102-0046

### Hyperbolic geometry - Question 10.8

O / open | AMR 问题清单汇编

Suppose $\Lambda$ is an essential lamination of a hyperbolic manifold $M$. Is $\Lambda$ isotopic to a lamination whose curvature is bounded below everywhere by $-2$?

背景链接：https://arxiv.org/abs/math/0209081

08946 | OWR-12590-003

### Hyperbolic Limit Sets and Fibered Four-Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a hyperbolic $3$-manifold group $\Gamma<\operatorname{Isom}(\mathbb H^4)$ whose limit set has Hausdorff dimension arbitrarily close to $3$? Does there exist a real hyperbolic $4$-manifold fibering over a surface with surface fiber?

来源字段：https://doi.org/10.4171/owr/2013/42

08947 | OWR-15177-017

### Hyperbolic Three-Manifold Groups in $\operatorname{SL}_3(\mathbb Z)$

O / open | Oberwolfach 报告问题汇编

Does there exist a finite-volume hyperbolic $3$-manifold $M$ whose fundamental group admits an injective homomorphism $\pi_1(M)\to\operatorname{SL}_3(\mathbb Z)$?

来源字段：https://doi.org/10.4171/owr/2016/56

08948 | OWR-13108-012

### Hyperbolic-Volume Decay of Partition Functions

P / partially_solved | Oberwolfach 报告问题汇编

For the quantum-Teichmüller TQFT state-integral $Z_b(X)$ associated with a triangulated hyperbolic $3$-manifold $X$ whose edges may carry conical singularities, is

$$\lvert Z_b(X)\rvert\sim\exp\left(-\frac{\operatorname{vol}(X)}{2\pi b^2}\right)$$

as $b\to0$?

来源字段：https://doi.org/10.4171/owr/2014/38

08949 | AMR-102-0061

### Immersed objects - Question 14.1

P / partially_solved | AMR 问题清单汇编

Is there a geometric notion for a $3$-manifold analogous to LERFness for foliations? What properties could a manifold have so that immersed essential laminations are virtually embedded?

背景链接：https://arxiv.org/abs/math/0209081

08950 | AMR-102-0062

### Immersed objects - Question 14.2

P / partially_solved | AMR 问题清单汇编

Let $\mathscr{F}$ be a taut foliation of $M$. Can leaves of $\mathscr{F}$ be approximated by compact essential surfaces? That is, given a leaf $\lambda$ of $\mathscr{F}$ and a point $p \in \lambda$, is there a sequence of immersed incompressible surfaces $\phi_i:\Sigma_i \to M$ and points $p_i \in \Sigma_i$ such that the images under $\phi_i$ of the balls of radius $r_i$ about $p_i$ where $r_i \to \infty$ converge on compact sets to $p,\lambda$?

背景链接：https://arxiv.org/abs/math/0209081

08951 | AMR-102-0063

### Immersed objects - Question 14.3

O / open | AMR 问题清单汇编

What $3$-manifolds admit total taut foliations?

背景链接：https://arxiv.org/abs/math/0209081

08952 | AMR-102-0064

### Immersed objects - Question 14.4

O / open | AMR 问题清单汇编

Are there any interesting examples of total genuine laminations?

背景链接：https://arxiv.org/abs/math/0209081

08953 | AMR-102-0065

### Immersed objects - Question 14.5

P / partially_solved | AMR 问题清单汇编

What is the weakest useful $2$-dimensional object that might be present in every atoroidal $3$-manifold? For instance, does every hyperbolic $3$-manifold $M$ contain an immersed quasigeodesic surface $\Sigma$ of amenable growth?

背景链接：https://arxiv.org/abs/math/0209081

08954 | OWR-15208-006

### Infinite Components of Positive Ricci Metric Moduli

P / partially_solved | Oberwolfach 报告问题汇编

别名：Infinite Components of Positive Ricci Metric Moduli [OWR-15208-024]

For every closed spin manifold of dimension $4n-1\geq 7$ that admits positive Ricci curvature, does the moduli space of positive-Ricci-curvature metrics have infinitely many connected components?

来源字段：https://doi.org/10.4171/owr/2017/3

08955 | OPG-60017

### Infinite distributivity of meet over join for a principal funcoid

S / solved | Open Problem Garden

Conjecture $f \sqcap \bigsqcup S = \bigsqcup \langle f \sqcap \rangle^{\ast} S$ for principal funcoid $f$ and a set $S$ of funcoids of appropriate sources and destinations.

背景链接：http://www.openproblemgarden.org/op/infinite_distributivity_of_meet_over_join_for_a_principal_funcoid

08956 | OWR-15177-016

### Infinite Generation in Mapping-Class-Group Cohomology

O / open | Oberwolfach 报告问题汇编

If $\Gamma$ is a torsion-free finite-index subgroup of $\operatorname{Mod}_g$ and $\mathcal C_g$ is the curve complex, is $H^{2g-1}(\Gamma;H_{2g-2}(\mathcal C_g))$ infinitely generated?

来源字段：https://doi.org/10.4171/owr/2016/56

08957 | OWR-16406-002

### Infinite Rank and Torsion in Integral Homology Cobordism

P / partially_solved | Oberwolfach 报告问题汇编

Does the integral homology cobordism group $\Theta^3_{\mathbb Z}$ contain a direct sum of infinitely many copies of $\mathbb Z$? Does it contain any nontrivial torsion?

来源字段：https://doi.org/10.4171/owr/2018/31

08958 | OWR-14299521-012

### Infinitesimal Quasiconformality of Harmonic Spheres

O / open | Oberwolfach 报告问题汇编

Let $X$ be a compact quasiconvex metric space satisfying a local quadratic isoperimetric inequality, and assume that every continuous map $S^2\to X$ of sufficiently small diameter is null-homotopic. Is every harmonic map $u:S^2\to X$ infinitesimally quasiconformal?

来源字段：https://doi.org/10.4171/owr/2025/30

08959 | AIM-TOPOLOGY-0098

### Injective cubes require noninjective relations

P / partially_solved | AIM 研讨会问题汇编

**Computation of cubical homology**

When computing cubical homology of a simple graph \(X\), one builds a cubical set \(MX\) whose \(n\)-cubes are maps \[ I_1^{\square n}\longrightarrow X \] and then takes the usual cubical homology of this cubical set.

Can cubical homology of a graph be computed using only injective maps \(I_1^{\square n}\to X\)? Should this restriction be made at the cubical set or chain complex level?

来源字段：http://aimpl.org/combhomotop/2/

08960 | OWR-13686-007

### Injectivity of the Baum-Connes Assembly Map

P / partially_solved | Oberwolfach 报告问题汇编

For every discrete group $\Gamma$ and $i\in\{0,1\}$, is the Baum-Connes assembly map $\mu:K_i^\Gamma(E\Gamma)\to K_i(C_r^*(\Gamma))$ injective, where $E\Gamma$ is the universal space for proper $\Gamma$-actions?

来源字段：https://doi.org/10.4171/owr/2015/29

08961 | AIM-TOPOLOGY-0249

### Integral and coefficient fingerprints of the Q_k candidates

P / partially_solved | AIM 研讨会问题汇编

**Problem 36.** Compute topological invariants of cohomogeneity-one manifolds. Classify topologically the new candidates for positive curvature.

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08962 | OWR-4413-009

### Integral Skyline Bases for Symmetric-Group Cohomology

O / open | Oberwolfach 报告问题汇编

Does the $2$-local cohomology of symmetric groups have an additive basis consisting of integral skyline cycles together with higher Bocksteins of skyline-basis classes?

来源字段：https://doi.org/10.4171/owr/2010/32

08963 | AIM-TOPOLOGY-0237

### Integral symplectic pencils give compact fat torus bundles

S / solved | AIM 研讨会问题汇编

Given a fat $G$-principal bundle, must $G$ be $S^1$, $S^3$ or $\mathrm{SO}(3)$?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

08964 | OWR-4990381-018

### Integral Universality of Goodwillie-Weiss Knot Invariants

P / partially_solved | Oberwolfach 报告问题汇编

Does the $n$th Goodwillie-Weiss approximation give the universal type-$(n-1)$ invariant of classical knots in $\mathbb R^3$ integrally, without inverting any primes?

来源字段：https://doi.org/10.4171/owr/2021/12

08965 | OWR-13687-006

### Invariant Prym Homology and Arithmetic Monodromy

O / open | Oberwolfach 报告问题汇编

Let $S_K\to S_{g,n}$ be the finite Galois cover defined by a mapping-class-invariant finite-index subgroup $K\subset\pi_1(S_{g,n})$, let $H_K=H_1(\overline{S}_K;\mathbb{Q})$, let $N_K$ be the group of lifted mapping classes, and let $\mathcal G_K$ be the Zariski closure of its monodromy representation $\rho$. For $g\geq3$, is $H_K^{\mathcal G_K^\circ}=0$, and is $\rho(N_K)$ arithmetic in $\mathcal G_K$?

来源字段：https://doi.org/10.4171/owr/2015/30

08966 | OPG-37161

### Is there an algorithm to determine if a triangulated 4-manifold is combinatorially equivalent to the 4-sphere?

P / partially_solved | Open Problem Garden

Problem Is there an algorithm which takes as input a triangulated 4-manifold, and determines whether or not this manifold is combinatorially equivalent to the 4-sphere?

背景链接：http://www.openproblemgarden.org/op/is_there_an_algorithm_to_determine_if_a_triangulated_4_manifold_is_combinatorially_equivalent_to_the_4_sphere

08967 | OWR-12697687-006

### Isomorphism Type of the Khovanov-Complex Concordance Group

P / partially_solved | Oberwolfach 报告问题汇编

Determine the isomorphism type of the abelian group $G$ formed by equivalence classes of the graded Khovanov chain complexes $\llbracket K\rrbracket$ under tensor product.

来源字段：https://doi.org/10.4171/owr/2023/4

08968 | OPG-48770

### Jacob Palis Conjecture(Finitude of Attractors)(Dynamical Systems)

P / partially_solved | Open Problem Garden

Conjecture Let $Diff^{r}(M)$ be the space of $C^{r}$ Diffeomorphisms on the connected, compact and boundaryles manifold M and $\chi^{r}(M)$ the space of $C^{r}$ vector fields. There is a dense set $D\subset Diff^{r}(M)$ ( $D\subset \chi^{r}(M)$ ) such that $\forall f\in D$ exhibit a finite number of attractor whose basins cover Lebesgue almost all ambient space $M$

This is a very Deep and Hard problem in Dynamical Systems. It present the dream of the dynamicist mathematicians.

背景链接：http://www.openproblemgarden.org/op/jacob_palis_conjecture_finitude_of_attractors_dynamical_systems

08969 | OWR-15177-003

### Johnson-Filtration Images of Stable Heegaard Splittings

P / partially_solved | Oberwolfach 报告问题汇编

For the Johnson-filtration images $V_k(3)$ of stable Heegaard splittings of integral homology $3$-spheres, describe $V_k(3)$ and determine whether $V_k(3)=V_1(3)$ for every $k\geq1$. If $\varphi\in A_{g,1}(k)$ and $\psi\in B_{g,1}$ agree modulo $\Gamma_{k+1}$ on $\pi$, does some $\mu\in AB_{g,1}$ satisfy $\mu(x)=\varphi(x)=\psi(x)$ modulo $\Gamma_{k+1}$ for every $x$?

来源字段：https://doi.org/10.4171/owr/2016/56

08970 | AIM-TOPOLOGY-0116

### Jointly slice normal generators for a fixed homology ball

P / partially_solved | AIM 研讨会问题汇编

Let \(K\) be a knot in an integer homology sphere \(Y\), and suppose that \(Y\) bounds an integer homology ball \(W\) such that \(K\) is null-homotopic in \(W\). Must \(K\) be homotopic in \(Y\) to a knot \(K'\) which is smoothly slice in \(W\)?

来源字段：http://aimpl.org/concordsliceknot/2/

08971 | OWR-11138-001

### K-Theoretic Descent from MU to the Sphere

O / open | Oberwolfach 报告问题汇编

Let $S \to MU$ be the unit map from the sphere spectrum $S$ to the complex cobordism spectrum $MU$, viewed as a commutative $MU$-Hopf--Galois extension in the sense of Rognes. Can the algebraic $K$-theory spectrum $K(S)$ of the sphere spectrum be recovered from $K(MU)$ by homotopy fixed-point/cosimplicial descent along $S \to MU$?

来源字段：https://doi.org/10.4171/owr/2011/46

08972 | OWR-1726-008

### Kasparov’s Baum-Connes Method in Higher Rank

O / open | Oberwolfach 报告问题汇编

Can Kasparov's method for proving the Baum-Connes Conjecture-constructing a suitable $\gamma$-element and proving $\gamma=1$ by a homotopy argument-be carried out for higher-rank semisimple Lie groups, possibly using a weaker homotopy because property T obstructs $\gamma=1$ in KK-theory?

来源字段：https://doi.org/10.4171/owr/2007/43

08973 | OWR-17471-022

### Kernels of Nonfaithful Type-Preserving Punctured-Torus Representations

O / open | Oberwolfach 报告问题汇编

Let $T$ be the once-punctured torus. If $\rho:\pi_1(T)\to\operatorname{PSL}(2,\mathbb C)$ is a nonfaithful, discrete, type-preserving representation, is its kernel either

$$\langle\!\langle\beta_r^m\rangle\!\rangle$$

for some $r\in\widehat{\mathbb Q}$ and $m\geq2$, or

$$\langle\!\langle\beta_{r_1}^{m_1},\beta_{r_2}^{m_2}\rangle\!\rangle$$

for distinct $r_1,r_2\in\widehat{\mathbb Q}$ and $m_1,m_2\geq2$?

来源字段：https://doi.org/10.4171/owr/2020/8

08974 | KP-1.1

### Kirby Problem 1.1

P / partially_solved | Kirby 低维拓扑问题

Is the crossing number additive under connected sum: $c(K_{1}\#K_{2}) = c(K_{1}) + c(K_{2})$?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

08975 | KP-1.10

### Kirby Problem 1.10

P / partially_solved | Kirby 低维拓扑问题

(Generalized Property R Conjecture). Let $L \subset S^{3}$ be an $n$- component link such that 0-framed Dehn surgery on $L$ results in $\#^{n}(S^{1} \times S^{2})$. Is there a sequence of handleslides converting $L$ into an $n$-component unlink?

背景链接：http://projecteuclid.org/euclid.jdg/1214441487

08976 | KP-1.11

### Kirby Problem 1.11

P / partially_solved | Kirby 低维拓扑问题

(Cabling conjecture). Let $K \subset S^{3}$ be a knot and $r \in \mathbb{Q}$. If $r$-framed Dehn surgery on $K$ is not prime, then $K$ is a nontrivial cable of a knot $J \subset S^{3}$ and $r$ is the slope of the cabling annulus.

背景链接：http://projecteuclid.org/euclid.jdg/1432842362

08977 | KP-1.12

### Kirby Problem 1.12

P / partially_solved | Kirby 低维拓扑问题

This problem presents several variations on the Cosmetic Surgery Conjecture, discussed in turn below. (a) $($ Cosmetic Surgery Conjecture $)$ Two surgeries along inequivalent slopes are never purely cosmetic. (b) $($ Oriented knot complement conjecture $)$ If $K_{1}$ and $K_{2}$ are knots in a closed, oriented 3-manifold $N$ whose complements are homeomorphic via an orientation-preserving homeomorphism, then there exists a homeomorphism of $N$ tak- ing $K_{1}$ to $K_{2}$. (c) Let $K \subset S^{3}$ be a nontrivial knot, then $K$ admits no purely cosmetic surg- eries. (d) If $K \subset S^{3}$ is not the torus knot $T_{2g+1,2}$, then $K$ admits no chirally cos- metic surgeries with inequivalent slopes.

背景链接：https://projecteuclid.org/euclid.ojm/1539158668

08978 | KP-1.13

### Kirby Problem 1.13

P / partially_solved | Kirby 低维拓扑问题

Let $K$ be a null-homotopic knot in a 3-manifold $Y$ , and let $Y_{0}(K)$ be the manifold obtained by 0-surgery on $K$. (a) Conjecture: Let $F$ be a Seifert surface for $K$, and let $\widetilde{F} \subset Y_{0}(K)$ be the closed surface obtained by capping off $\partial F$ with a disk. If $Y_{0}(K)$ is a surface bundle over $S^{1}$ with fiber in the homology class $[\widetilde{F}]$, then $K$ is a fibered knot with fiber in the homology class $[F]$. (b) Conjecture: If $F$ is taut in the exterior of $K$ in $Y$ , then $\widetilde{F}$ is also taut in $Y_{0}(K)$.

背景链接：http://projecteuclid.org/euclid.jdg/1214441487

08979 | KP-1.14

### Kirby Problem 1.14

P / partially_solved | Kirby 低维拓扑问题

For which nonzero $r \in \mathbb{Q}$ is it true that for every nontrivial knot $K \subset S^{3}$ there is a homomorphism

$$ \pi_{1}(S^{3}_{r}(K)) \to \operatorname{SU}(2) $$

with nonabelian image?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

08980 | KP-1.15

### Kirby Problem 1.15

P / partially_solved | Kirby 低维拓扑问题

(a) Are there integral homology spheres with arbitrarily large (integral) Dehn surgery number? Are there irreducible examples? Does the connected sum of $n$ nontrivial homology spheres have Dehn surgery number at least $n$? (b) Given a 3-manifold $Y$ with a smooth orientation-preserving $\mathbb{Z}/n\mathbb{Z}$ sym- metry $\varphi$, Sakuma [Sak01] (generalizing [PS01]) proved that $Y$ admits a surgery description compatible with the symmetry. Define the equivariant (integral) Dehn surgery number of $(Y, \varphi)$ as the minimal number of con- nected components of any such link. Do there exist $(Y, \varphi)$ whose equivari- ant (integral) Dehn surgery number is arbitrarily larger than the (integral) Dehn surgery number? (c) Following the remarks below, we can consider the following specific case: Does there exist a manifold with a 2-fold symmetry whose equivariant (integral) Dehn surgery number is larger than the (integral) Dehn surgery number?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

08981 | KP-1.16

### Kirby Problem 1.16

P / partially_solved | Kirby 低维拓扑问题

(a) Given a knot $K \subset S^{3}$ determine all knots $K' \subset S^{3}$ for which the branched double covers of $S^{3}$ along $K$ and $K'$ are homeomorphic. (b) Find a set of moves on links in $S^{3}$ so that two links are related by these moves if and only if they have the same branched double cover.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

08982 | KP-1.17

### Kirby Problem 1.17

O / open | Kirby 低维拓扑问题

Can an alternating link and a non-alternating link have home- omorphic branched double covers?

背景链接：http://projecteuclid.org/euclid.pjm/1102700086

08983 | KP-1.18

## Kirby Problem 1.18

P / partially_solved | Kirby 低维拓扑问题

(a) $($ Meridional Rank Conjecture $)$ Is the meridional $\operatorname{rank} \mu(L)$ of every link $L$ equal to its bridge number $b(L)$? (b) Given two knots $K_{1}, K_{2} \subset S^{3}$ such that $\pi_{1}(S^{3}\setminus K_{1})$ and $\pi_{1}(S^{3}\setminus K_{2})$ have the same finite quotients, does $\mu(K_{1}) = \mu(K_{2})$? In particular, is it possible to explicitly detect the meridional $\operatorname{rank} \mu(K)$ of a knot $K \subset S^{3}$ from the finite quotients of $\pi_{1}(S^{3}\setminus K)$? (c) Given two links $L_{1}, L_{2} \subset S^{3}$, if there exists a surjective homomorphism $\pi_{1}(S^{3}\setminus L_{1}) \twoheadrightarrow\pi_{1}(S^{3}\setminus L_{2})$ that sends the meridians of $L_{1}$ to the meridians of $L_{2}$, do the bridge numbers satisfy $b(L_{1}) \geq b(L_{2})$? (d) Given two knots $K_{1}, K_{2} \subset S^{3}$, if there exists a surjective homomorphism $\pi_{1}(S^{3}\setminus K_{1}) \twoheadrightarrow\pi_{1}(S^{3}\setminus K_{2})$ (which may not send meridians to meridians): (i) Do the meridional ranks satisfy $\mu(K_{1}) \geq \mu(K_{2})$? (ii) Do the bridge numbers satisfy $b(K_{1}) \geq b(K_{2})$? (e) Let $K \subset S^{3}$ be a hyperbolic knot. Is every subgroup of $\pi_{1}(S^{3}\setminus K)$ generated by at most $b(K) - 1$ meridians free? See [BJW18].

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

08984 | KP-1.19

## Kirby Problem 1.19

O / open | Kirby 低维拓扑问题

Let $Y = Y_{1}\#Y_{2}$ be a connected sum of 3-manifolds with $Y_{i} \neq$ $S^{3}$, for $i = 1, 2$. Let $\Phi: Y \to Y$ be a Dehn twist around the connected sum separating 2-sphere, and assume $Y_{1}$ and $Y_{2}$ are such that $\Phi$ is not isotopic to the identity. Does there exist a knot $K \subseteq Y$ such that $K$ and $\Phi(K)$ are not ambiently isotopic?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

08985 | KP-1.2

## Kirby Problem 1.2

P / partially_solved | Kirby 低维拓扑问题

(a) Show that if $P$ is a nontrivial satellite operator and $K_{P}$ is a nontrivial satellite of a knot $K$, then

$$ c(K_{P}) \geq c(K), $$

where $c$ denotes the crossing number. (b) More generally, show that if $P$ is a satellite operator then

$$ c(K_{P}) \geq c(K) \cdot w(P)^{2}, $$

where $w(P)$ denotes the geometric winding number.

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

08986 | KP-1.20

## Kirby Problem 1.20

P / partially_solved | Kirby 低维拓扑问题

(a) Are there any null-homologous Floer minimal knots with irreducible com- plements other than the Borromean knots $B_{g}, g \geq 0$, in any 3-manifolds? It is reasonable to make the following more concrete conjectures. (b) Conjecture (Ni): If $b_{1}(Y) < 2$, then the only null-homologous Floer mini- mal knot in $Y$ is the unknot. (c) Conjecture (Ni): If the Thurston norm of $Y$ is not identically zero, then $Y$ does not contain any null-homologous Floer minimal knot with irreducible complement. (d) Conjecture (Hedden, Rasmussen): The only Floer minimal knots in lens spaces are simple knots.

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

08987 | KP-1.21

## Kirby Problem 1.21

P / partially_solved | Kirby 低维拓扑问题

(a) For a given positive integer $g$, are there only finitely many L-space knots of genus $g$? A related but more general question is: (b) Question (Hedden): For which knots $L \subset S^{3}$ are there only finitely many knots $K \subset S^{3}$ with $\widehat{\mathrm{HFK}}(S^{3},K) \cong \widehat{\mathrm{HFK}}(S^{3},L)$?

(c) In some special cases, one expects stronger results saying that a class of knots is characterized by its knot Floer homology. We state 3 conjectures, ordered by their strength. (i) For a knot $K \subset S^{3}$, if $\widehat{\mathrm{HFK}}(S^{3},K) \cong \widehat{\mathrm{HFK}}(S^{3},T_{2g+1,2})$, then $K = T_{2g+1,2}$. (ii) For a knot $K \subset S^{3}$, if $\widehat{\mathrm{HFK}}(S^{3},K) \cong \widehat{\mathrm{HFK}}(S^{3},T_{p,q})$, then $K$ is an iterated torus knot. (iii) Conjecture (Ni): For an L-space knot $K \subset S^{3}$, if all roots of $\Delta_{K}(t)$ lie on the unit circle, then $K$ is an iterated torus knot.

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

08988 | KP-1.22

## Kirby Problem 1.22

P / partially_solved | Kirby 低维拓扑问题

If $K$ is a hyperbolic $L$-space knot, show that its branched cover $\Sigma_{2}(K)$ is not an $L$-space.

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

08989 | KP-1.23

## Kirby Problem 1.23

P / partially_solved | Kirby 低维拓扑问题

Let $K$ be a cubic graph embedded in the plane, and let $\mathrm{Tait}(K)$ be the number of Tait colorings of $K$. (a) Is $\dim J^{7}(K) = \mathrm{Tait}(K)$? (b) Is $J^{5}(K)$ nonzero when $K$ is bridgeless?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

08990 | KP-1.24

### Kirby Problem 1.24

P / partially_solved | Kirby 低维拓扑问题

(Jones Unknot Detection). (a) Is there a nontrivial knot with the same Jones polynomial as the unknot?

(b) Does there exist a nontrivial knot whose colored Jones polynomials are all trivial? (c) Is there a nontrivial knot with trivial HOMFLYPT polynomial? (A knot with trivial HOMFLYPT polynomial would have trivial Jones polynomial.)

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

08991 | KP-1.25

### Kirby Problem 1.25

P / partially_solved | Kirby 低维拓扑问题

(a) Conjecture: The noncommutative $A$-ideal of a knot $K$ is exactly the an- nihilator of the colored Jones polynomial $($ the infinite dimensional vector whose entries are the colored Jones polynomials of $K)$. (b) $($ AJ Conjecture $)$ Let $A(K; L, M)$ be the $A$-polynomial of the knot $K$, and $\alpha(K, q; M, L)$ a polynomial of minimal degree in $L$ and co-prime coeffi- cients that annihilates the colored Jones polynomial $J_{n}(K; q)$. Then

$$ \alpha(K, -1; M, L) = A(K; L, M) $$

up to a factor depending on $M.$

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

08992 | KP-1.26

### Kirby Problem 1.26

P / partially_solved | Kirby 低维拓扑问题

(Jones Slope Conjecture). For a knot $K$, the Jones slopes $js(K)$ are the set of cluster points in

$$ \left\{\frac{4}{n^{2}}\deg_{+}\bigl(J_{n}(K;q)\bigr)\right\}_{n\in\mathbb{N}} \cup \left\{\frac{4}{n^{2}}\deg_{-}\bigl(J_{n}(K;q)\bigr)\right\}_{n\in\mathbb{N}}, $$

where deg $^{+}$ denotes the highest degree and deg $^{-}$ the lowest degree.

背景链接：http://nyjm.albany.edu:

08993 | KP-1.27

### Kirby Problem 1.27

P / partially_solved | Kirby 低维拓扑问题

(Kashaev-Murakami-Murakami Volume Conjecture). For a link $L \subset S^{3}$,

$$ \frac{1}{2\pi}\operatorname{Vol}_{\mathrm{hyp}}(S^{3}-L) =\lim_{n\to\infty}\frac{1}{n}\log\left|J_{n}\bigl(L,e^{2\pi i/n}\bigr)\right|. $$

背景链接：http://projecteuclid.org/euclid.em/1057777432

08994 | KP-1.28

### Kirby Problem 1.28

P / partially_solved | Kirby 低维拓扑问题

Let $L$ be a link in the thickened annulus $S^{1} \times I \times I$. (a) Wrapping conjecture: $w(L)$ is equal to the maximal nonzero annular de- gree of the annular Kauffman bracket of $L$. (b) Categorifed wrapping conjecture: $w(L)$ is equal to the maximal nonzero annular grading of the annular Khovanov homology of $L$.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

08995 | KP-1.29

### Kirby Problem 1.29

P / partially_solved | Kirby 低维拓扑问题

Is the first inequality below true? If so, is the second? (a) $(\operatorname{Vol}$-Det Conjecture $)$ For any alternating hyperbolic knot $K \subset S^{3}$, vol $(K) < 2\pi \log \det(K),$ where vol is the hyperbolic volume of $K$ and det is the determinant. (b) For any hyperbolic knot $K \subset S^{3}$, vol $(K) < 2\pi \log \operatorname{rank} \mathrm{Kh}(K),$ where Kh is the reduced Khovanov homology of $K$?

背景链接：http://nyjm.albany.edu:8000/j/2018/24

08996 | KP-1.3

### Kirby Problem 1.3

P / partially_solved | Kirby 低维拓扑问题

How does unknotting number behave under connected sum and mutation? (a) Does the connected sum of $n$ nontrivial knots have unknotting number at least $n$? (b) Is unknotting number invariant under mutation?

背景链接：https://arxiv.org/abs/2506.24088

08997 | KP-1.30

### Kirby Problem 1.30

P / partially_solved | Kirby 低维拓扑问题

Does the Khovanov homology of every nontrivial knot contain 2-torsion?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

08998 | KP-1.31

### Kirby Problem 1.31

P / partially_solved | Kirby 低维拓扑问题

(a) Compute the Khovanov homology for all torus knots $T(m, n)$. (b) Compute the Khovanov-Rozansky $\mathfrak{g}\mathfrak{l}(N)$ homology for all torus knots.

背景链接：https://knotinfo.org

08999 | KP-1.32

### Kirby Problem 1.32

O / open | Kirby 低维拓扑问题

(a) Recover the Jones polynomial of links $L \subset \mathbb{R}^{3}$ by counting solutions to the Kapustin-Witten equations on $\mathbb{R}^{3} \times \mathbb{R}_{+}$ with Nahm pole boundary conditions; (b) Recover the Khovanov homology of links $L \subset \mathbb{R}^{3}$ by counting solutions to the Haydys-Witten equations on $\mathbb{R} \times \mathbb{R}^{3} \times \mathbb{R}_{+}$ with Nahm pole boundary conditions;

(c) Use the Kapustin-Witten or Haydys-Witten equations to construct invari- ants of links in other 3-manifolds.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09000 | KP-1.33

### Kirby Problem 1.33

P / partially_solved | Kirby 低维拓扑问题

Describe topological necessary or sufficient conditions for a link to have KR-parity. For example: (a) Are all links with KR-parity positive? Quasipositive? (b) Do all algebraic links have KR-parity?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09001 | KP-1.34

### Kirby Problem 1.34

P / partially_solved | Kirby 低维拓扑问题

(a) Khovanov and Rozansky [KR08b] used braid presentations to define a triply graded link homology theory whose Euler characteristic is the HOM- FLYPT polynomial. Find a description of this homology in terms of arbi- trary planar diagrams of the link, and determine whether some flavor of this theory is functorial with respect to link cobordisms. (b) Do the same for Cautis' link homologies from [Cau17], which are also defined in terms of braid presentations.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09002 | KP-1.35

### Kirby Problem 1.35

P / partially_solved | Kirby 低维拓扑问题

(a) Is symplectic Khovanov homology isomorphic to Khovanov homology, over $\mathbb{Z}$? (b) Give a construction of odd symplectic Khovanov homology $\mathrm{Kh}^{\mathrm{odd}}_{\mathrm{symp}}(K)$.

背景链接：https://doi.org/10.1090/jams/902

09003 | KP-1.36

### Kirby Problem 1.36

P / partially_solved | Kirby 低维拓扑问题

Categorify the ($\mathfrak{s}\mathfrak{l}(2), \mathfrak{s}\mathfrak{l}(N)$, HOMFLYPT) skein algebras for surfaces.

背景链接：http://www.numdam.org/item?id=ASENS

09004 | KP-1.37

### Kirby Problem 1.37

P / partially_solved | Kirby 低维拓扑问题

(a) For every link $L \subset \mathbb{R}^{3}$, every simple Lie algebra $\mathfrak{g}$, and every coloring of the components of $L$ with irreducible representations of $\mathfrak{g}$, construct a bigraded link homology theory that is functorial under link cobordisms, and whose Euler characteristic is the Reshetikhin-Turaev link invariant associated to $\mathfrak{g}$ and $L$. (b) For each of the theories above, define an odd version $($ with the same mod 2 reduction $)$ or explain the obstruction to its existence.

背景链接：https://doi-org.stanford.idm.oclc.org/10.4171/QT/78

09005 | KP-1.38

### Kirby Problem 1.38

O / open | Kirby 低维拓扑问题

What is the structure of the smooth knot concordance group? (a) Is there a torsion element of the smooth concordance group $\mathcal{C}$ having order other than two? Is there a torsion element in $\mathcal{C}_{TS}$, the smooth concordance group of topologically slice knots, having order other than two? (b) Is there a knot $K$ having concordance order two in $\mathcal{C}$ that is not concordant to a strongly negatively amphichiral knot? (c) Do there exist infinitely divisible elements in $\mathcal{C}$, that is, does there exist $K$ such that for infinitely many $n \in \mathbb{N}$, there is a $J$ such that $[K] = n[J]$?

背景链接：http://projecteuclid.org/euclid.ojm/1200691730

09006 | KP-1.39

### Kirby Problem 1.39

P / partially_solved | Kirby 低维拓扑问题

(a) Do the algebraic knots freely generate a subgroup of the smooth concor- dance group $\mathcal{C}$? (b) Do the algebraic knots freely generate a subgroup of the topological con- cordance group $\mathcal{C}^{\mathrm{TOP}}$?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09007 | KP-1.4

### Kirby Problem 1.4

P / partially_solved | Kirby 低维拓扑问题

Let $P$ be a nontrivial satellite pattern with winding number $w(P) \neq 0$. Then for any nontrivial knot $K$ and its satellite $K_{P}$ , one has

$$ u(K_{P}) \geq w(P) + 1. $$

背景链接：https://arxiv.org/abs/2206.04196

09008 | KP-1.40

### Kirby Problem 1.40

P / partially_solved | Kirby 低维拓扑问题

A satellite operator $P \subset S^{1} \times D^{2}$ induces an operation $P$ on the concordance group $\mathcal{C}$ [Gor75]. (a) Let $P$ be a winding number one satellite operator with $P(\operatorname{U}) \simeq \operatorname{U}$. If $P(K) \simeq \operatorname{U}$, does this mean that $K \simeq \operatorname{U}$? (b) Is the Whitehead doubling operator injective on $\mathcal{C}$? Can we exhibit any (winding-number-zero) operator that is injective on $\mathcal{C}$? (c) Conjecture (Hedden; see [HPC21]). The only homomorphisms on $\mathcal{C}$ in- duced by satellite operators are the zero map, the identity, and the invo- lution induced by orientation reversal. (d) Conjecture (Hedden, Pinzón Caicedo) Let $P$ be a non-constant satellite operator with winding number zero. Does $\\langle \\operatorname{im} P \\rangle$ necessarily have infinite rank? Some special cases are: (i) Does there always exist a fixed knot $K$ such that $\langle\{P(nK)\}_{n\in\mathbb{Z}}\rangle$ has infinite rank? (ii) If $\{nK\}_{n\in\mathbb{Z}}$ is any rank-one subgroup of $\mathcal{C}$, does $\langle\{P(nK)\}_{n\in\mathbb{Z}}\rangle$ have infinite rank?

背景链接：http://projecteuclid.org/euclid.jdg/1344430825

09009 | KP-1.41

### Kirby Problem 1.41

P / partially_solved | Kirby 低维拓扑问题

This problem is concerned with the stable 4-genus $g_{s}(K)$ of a knot $K$, defined below. (a) Is there a knot $K$ such that $g_{s}(K) \in \mathbb{Q}\setminus\mathbb{Z}$? (b) Is there a knot $K$ such that $g_{s}(K) \notin \mathbb{Q}$? (c) Is there a knot $K$ such that $0 < g_{s}(K) < 1/2$? (d) [Liv10, Question 1] Does $K$ represent a torsion element in the concor- dance group if and only if $g_{s}(K) = 0$?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09010 | KP-1.42

### Kirby Problem 1.42

O / open | Kirby 低维拓扑问题

Do there exist algebraically concordant Seifert forms $V_{1}$ and $V_{2}$ for which there do not exist concordant knots $K_{1}$ and $K_{2}$ with Seifert forms $V_{1}$ and $V_{2}$, respectively?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09011 | KP-1.43

### Kirby Problem 1.43

P / partially_solved | Kirby 低维拓扑问题

Does knot Floer homology give a categorification of the Fox-Milnor condition?

背景链接：http://projecteuclid.org/euclid.ojm/1200691730

09012 | KP-1.44

### Kirby Problem 1.44

O / open | Kirby 低维拓扑问题

(a) If $K \in \mathcal{F}_{n}$ for all $n$, is $K$ topologically slice? (b) If $K \in \mathcal{T}_{n}$ for all $n$, is $K$ smoothly slice?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09013 | KP-1.45

### Kirby Problem 1.45

O / open | Kirby 低维拓扑问题

(a) For arbitrary $n \geq 2.5$ and $g > 1$, does there exist a knot in $\mathcal{F}_{n}$ with topological slice genus at least $g$? (b) For arbitrary $n \geq 0$ and $g > 1$, does there exist a knot in $\mathcal{T}_{n}$ with smooth slice genus at least $g$? (c) Could the $L^{(2)}$-signature invariants used to obstruct membership in deeper levels of the solvable/bipolar filtration be used to give bounds on the slice genus?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09014 | KP-1.46

### Kirby Problem 1.46

P / partially_solved | Kirby 低维拓扑问题

(a) Determine the topological slice genera of torus knots. In particular, does the topological slice genus of a torus knot equal half the absolute value of its maximal Levine-Tristram signature? (b) Is a two-bridge knot topologically slice if and only if it is ribbon, if and only if it is topologically homotopy-ribbon? (c) Are the Casson-Gordon ribbon and topological sliceness obstructions [CG86] complete for two-bridge knots? (d) Can the difference between the smooth and topological slice genus of a two-bridge knot be arbitrarily large?

背景链接：http://projecteuclid.org/euclid.em/1259158428

09015 | KP-1.47

### Kirby Problem 1.47

P / partially_solved | Kirby 低维拓扑问题

Given a smooth knot $K \subset S^{3}$, determine its nonorientable 4- genus $\gamma_{4}$, i.e. the minimal first Betti number for all compact nonorientable smooth surfaces $F$ properly embedded in $B^{4}$ with boundary $K$. This question may be devel- oped in several directions: (1) (Allen's Geography Question [All23]) For a given knot $K \subset S^{3}$, find the pairs $(e, b) \in \mathbb{Z}^{2}$ realized by the normal Euler number and first Betti number of a nonorientable bounding surface $F$. (2) By analogy with the Milnor Conjecture for the orientable 4-genus of torus knots, what is the nonorientable 4-genus of a torus knot $T_{p,q}$?

背景链接：http://nyjm.albany.edu/j/2023/29-41v.pdf

09016 | KP-1.48

### Kirby Problem 1.48

O / open | Kirby 低维拓扑问题

(a) Suppose $K$ and $K\#J$ are (smoothly) doubly slice knots. Must $J$ be a (smoothly) doubly slice knot? (b) Does there exist a knot that is smoothly slice, topologically doubly slice, and not smoothly doubly slice but such that $K\#K$ is smoothly doubly slice? (c) Let $M$ and $N$ be closed 3-manifolds such that $M$ and $M\#N$ embed (smoothly) in $S^{4}$. Must $N$ embed (smoothly) in $S^{4}$?

背景链接：http://projecteuclid.org/euclid.jdg/1456754013

09017 | KP-1.49

### Kirby Problem 1.49

P / partially_solved | Kirby 低维拓扑问题

(a) What is the structure of the equivariant concordance groups? (b) Is the strongly negative amphichiral concordance group abelian? (c) For any type of knot involution, is it possible to exhibit $K_{1}$ and $K_{2}$ that are equivariantly concordant but not standardly equivariantly concordant? (d) Does there exist a freely periodic slice knot that is not equivariantly slice? (e) (Boyle-Rouse) Is every periodic or freely periodic $L$-space knot either a torus knot or an iterated torus knot?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09018 | KP-1.5

### Kirby Problem 1.5

P / partially_solved | Kirby 低维拓扑问题

Is there a relationship between genus and unknotting number for specific classes of knots? Here are two instances of classes of knots for which there might be an interesting answer. (a) Conjecture (Murasugi-Przytycki). For every positive fibered knot $K$, the genus of $K$ equals its unknotting number. (b) Question (Stoimenow). For every alternating fibered knot, is the genus greater than or equal to the unknotting number?

背景链接：https://knotinfo.org

09019 | KP-1.50

### Kirby Problem 1.50

P / partially_solved | Kirby 低维拓扑问题

(a) Is every slice knot a ribbon knot? (b) Is every slice link ribbon? (c) Suppose $K$ is a knot with smooth four-genus $g_{4}(K) = g$. Does $K$ bound a smooth, genus $g$, ribbon surface $F$ in $B^{4}$?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09020 | KP-1.51

### Kirby Problem 1.51

P / partially_solved | Kirby 低维拓扑问题

Does every ribbon knot arise as a symmetric union?

背景链接：https://arxiv.org/abs/2606.02968

09021 | KP-1.52

### Kirby Problem 1.52

O / open | Kirby 低维拓扑问题

Given $K$ in $S^{3}$, is there an algorithm to detect if $K$ is slice? Ribbon?

背景链接：http://projecteuclid.org/euclid.pjm/1102702126

09022 | KP-1.53

### Kirby Problem 1.53

P / partially_solved | Kirby 低维拓扑问题

(a) Which knot properties are hereditary under ribbon concordance? Is the property of being alternating hereditary under ribbon concordance? (b) Which knot properties imply minimality under ribbon concordance? Are positive knots minimal under ribbon concordance? (c) Which knot invariants are monotone under ribbon concordance? Is cross- ing number, bridge number, or braid index monotone under ribbon con- cordance?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09023 | KP-1.54

### Kirby Problem 1.54

P / partially_solved | Kirby 低维拓扑问题

This problem is concerned with the restriction of the partial ordering $\geq$ coming from ribbon concordance to the concordance class $[K]$ of a knot $K$. (a) What can be said about the order type of $[K]$? (b) Does it depend on $[K]$? Must it contain a (unique) minimal element? (c) (Gordon) Does there exist a concordance class of knots that contains two minimal elements with respect to ribbon concordance?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09024 | KP-1.55

### Kirby Problem 1.55

P / partially_solved | Kirby 低维拓扑问题

Suppose that $C$ is a ribbon concordance from a fibered knot $K_{1}$ to a fibered knot $K_{0}$. (a) Does the capped-off monodromy of $K_{1}$ (i.e. extended over a disk) extend to a compression body whose lower boundary gives the capped-off monodromy of $K_{0}$? (b) Is $C$ fibered by compression bodies? (c) Suppose a fibered knot in a homology 3-sphere bounds a homotopy-ribbon disk $D$ in a contractible 4-manifold $V$ . Is $V \setminus\nu(D)$ fibered by handlebodies?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09025 | KP-1.56

### Kirby Problem 1.56

O / open | Kirby 低维拓扑问题

(Hom). If $K_{0}$ and $K_{1}$ are ribbon concordant and

$$ \widehat{\mathrm{HFK}}(K_{0}) \cong \widehat{\mathrm{HFK}}(K_{1}), $$

are $K_{0}$ and $K_{1}$ isotopic?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09026 | KP-1.57

### Kirby Problem 1.57

P / partially_solved | Kirby 低维拓扑问题

(a) In either the smooth or topological settings, are 0-shake slice knots slice? (b) Does there exist a knot $K$ whose topological 0-shake slice genus is strictly less than the topological slice genus?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09027 | KP-1.58

### Kirby Problem 1.58

O / open | Kirby 低维拓扑问题

What concordance information about a knot $K$ is contained in its 0-trace $X_{0}(K)$ and in its 0-surgery $S^{3}_{0}(K)$? Specifically, (a) Suppose $K$ and $K'$ have homeomorphic 0-traces. (i) Are $K$ and $K'$ topologically concordant? (ii) What if one just assumes $K$ and $K'$ have homeomorphic 0-surgeries? (iii) What if one just assumes $K$ and $K'$ have 0-surgeries that are topolog- ically homology cobordant, preserving the homology class of a merid- ian.

(b) Suppose $K$ and $K'$ have homeomorphic 0-surgeries, where the image of the meridian of $K$ is freely homotopic to the meridian of $K'$. Are $K$ and $K'$ smoothly concordant?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09028 | KP-1.59

### Kirby Problem 1.59

P / partially_solved | Kirby 低维拓扑问题

Let $K \subset S^{3}$ be a slice knot. (a) Determine the set $\mathcal{R}(K)$ of ribbon disks bounded by $K$ modulo isotopy. (b) Determine the set $\mathcal{D}(K)$ of slice disks bounded by $K$ modulo isotopy and local knotting. (c) Determine the set $\mathcal{H}\mathcal{R}(K)$ of homotopy-ribbon disks bounded by $K$ modulo isotopy. (d) If $K$ is fibered, determine the set $\mathcal{F}\mathcal{H}\mathcal{R}(K)$ of fibered, homotopy-ribbon disks bounded by $K$ in some homotopy 4-ball, modulo isotopy.

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09029 | KP-1.6

### Kirby Problem 1.6

P / partially_solved | Kirby 低维拓扑问题

Suppose that $V_{1}$ and $V_{2}$ are $S$-equivalent Seifert forms. Does there exist a fixed knot $K$ bounding Seifert surfaces $F_{1}$ and $F_{2}$ for which the as- sociated Seifert forms are $V_{1}$ and $V_{2}$?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09030 | KP-1.60

### Kirby Problem 1.60

O / open | Kirby 低维拓扑问题

Is there a knot in $S^{3}$ that is not smoothly slice in $B^{4}$ but is smoothly slice in an integer homology ball? What about a $\mathbb{Z}$/2-homology ball?

背景链接：https://nyjm.albany.edu/j/2023/29-52p.pdf

09031 | KP-1.61

### Kirby Problem 1.61

O / open | Kirby 低维拓扑问题

A knot in $S^{3}$ bounds a topological disk in $B^{4}$ by coning (not necessarily locally flat); this problem asks about topological disks that a knot in a homology sphere $Y$ might bound in a homology or homotopy ball bounded by $Y$ . (a) Given a locally flat knot in an integer homology 3-sphere, is it concordant to a knot in $S^{3}$, via a locally flat annulus in a topological integer homology cobordism? Variation: Can the homology cobordism be taken to be simply connected? (b) Given an arbitrary knot in an integer homology 3-sphere $Y$ , does it bound an embedded disc (not necessarily locally flat) in $\Delta$, the Freedman filling of $Y$ ? Variation: Is there such a disk in a homology ball bounded by $Y$ ? (c) Is there a link in an integer homology 3-sphere, each of whose components is independently concordant to a knot in $S^{3}$, that is not itself concordant to a link in $S^{3}$. (d) Suppose that $\mathbb{Z}_{n}$ acts semi-freely on the homology sphere $Y$ , with fixed point set a knot $K$. Does the action of $\mathbb{Z}_{n}$ extend semi-freely over $\Delta$?

背景链接：http://projecteuclid.org/euclid.jdg/1214437136

09032 | KP-1.62

### Kirby Problem 1.62

P / partially_solved | Kirby 低维拓扑问题

(a) Are all good boundary links topologically slice? Freely topologically slice? (b) A special case of interest: Is the Whitehead double of the Borromean rings (with any choice of clasps) topologically slice?

背景链接：http://projecteuclid.org/euclid.jdg/1099587731

09033 | KP-1.63

### Kirby Problem 1.63

O / open | Kirby 低维拓扑问题

Is there a knot type with Legendrian representatives that do not destabilize but have arbitrarily negative Thurston-Bennequin number?

背景链接：http://projecteuclid.org/euclid.jsg/1239974381

09034 | KP-1.64

### Kirby Problem 1.64

P / partially_solved | Kirby 低维拓扑问题

(a) Let $L \subset (S^{3}, \xi_{std})$ be a transverse link such that the branched double cover $(\Sigma_{2}(L), \xi_{L})$ is Stein fillable. Is $L$ transversely isotopic to the closure of a quasipositive braid? (b) Is the same true if $(\Sigma_{n}(L), \xi_{L})$ is Stein fillable for some $n > 2$? (c) Does Stein fillability of the branched cover imply that the slice-Bennequin inequality must be sharp for the given link?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09035 | KP-1.65

### Kirby Problem 1.65

O / open | Kirby 低维拓扑问题

Decomposable Lagrangian cobordisms between Legendrian knots or links in $\mathbb{R}^{3}$ are compositions of certain simple pieces admitting diagrammatic de- scriptions [EHK16]. Such cobordisms necessarily have no index 2 critical points. Is every exact Lagrangian cobordism without index 2 critical points Lagrangian iso- topic to a decomposable Lagrangian cobordism?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09036 | KP-1.66

### Kirby Problem 1.66

P / partially_solved | Kirby 低维拓扑问题

For Legendrian links $\Lambda_{1}, \Lambda_{2} \subset (\mathbb{R}^{3}, \xi_{std})$, write $\Lambda_{1} \preceq \Lambda_{2}$ if there is an exact Lagrangian cobordism in the symplectization $\mathbb{R} \times \mathbb{R}^{3}$ from $\Lambda_{1}$ to $\Lambda_{2}$. Is $\preceq$ a partial order?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09037 | KP-1.67

### Kirby Problem 1.67

P / partially_solved | Kirby 低维拓扑问题

Given a Legendrian link in the standard contact $\mathbb{R}^{3}$ besides the standard unknot or Hopf link, classify its exact Lagrangian fillings up to compactly supported Hamiltonian isotopy.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09038 | KP-1.68

### Kirby Problem 1.68

P / partially_solved | Kirby 低维拓扑问题

Determine the smooth knot types that have Legendrian repre- sentatives with orientable exact Lagrangian fillings.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09039 | KP-1.69

### Kirby Problem 1.69

P / partially_solved | Kirby 低维拓扑问题

Let $L \subset (S^{3}, \xi_{std})$ be a transverse link with

$$ sl_{\Sigma}(L) = -\chi(\Sigma), $$

for some Seifert surface $\Sigma$. Must $L$ be strongly quasipositive?

背景链接：http://www.numdam.org/item?id=AIF

09040 | KP-1.7

### Kirby Problem 1.7

P / partially_solved | Kirby 低维拓扑问题

Show that the sequence of absolute values of the coefficients of the Alexander polynomial of a link are: (a) concave $($ Fox’s trapezoidal conjecture $)$, or (b) for an alternating link, log concave $($ Stoimenow’s strong Fox conjecture $)$. (c) For a positive link, is the sequence of the coefficients of the Conway poly- nomial strongly log concave?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09041 | KP-1.70

### Kirby Problem 1.70

P / partially_solved | Kirby 低维拓扑问题

(a) Let $L \subset (S^{3}, \xi_{std})$ be a transverse link with

$$ sl_{\Sigma}(L) = -\chi(\Sigma), $$

for some smooth surface $\Sigma \subset B^{4}$ bounded by $L$. Is $L$ quasipositive? (b) Is the class of strongly quasipositive links equal to the class of quasipositive links whose Seifert and smooth slice genera are equal?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09042 | KP-1.71

### Kirby Problem 1.71

O / open | Kirby 低维拓扑问题

Does a Gordian unknot exist?

背景链接：http://www.springerlink.com/index/10.1007/s00222-002-0234-y

09043 | KP-1.72

### Kirby Problem 1.72

O / open | Kirby 低维拓扑问题

(The equilateral stuck unknots conjecture.). Are there equilat- eral embedded polygons that are unknotted yet cannot be unknotted through polygons preserving edge lengths? Such an unknot is said to be stuck.

背景链接：http://torus.math.uiuc.edu/jms/Papers/scicor/scicor.ps.gzpapers3://publication/uuid/A1C$D^{2}$FF8-BAEE-4CCC-A98C-BB5ECC1DC83A

09044 | KP-1.73

### Kirby Problem 1.73

O / open | Kirby 低维拓扑问题

(The 15 pearls conjecture). Is the pearl number of the trefoil equal to 15?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09045 | KP-1.74

### Kirby Problem 1.74

O / open | Kirby 低维拓扑问题

How does ropelength behave under connected sum of knots? Here are two conjectures, the second a weakening of the first. (a) For any knot or link types $K_{1}$ and $K_{2}$, Rop $(K_{1}\#K_{2}) \leq$ Rop $(K_{1}) +$ Rop $(K_{2}) - (4\pi - 4).$ (b) For any knot or link types $K_{1}$ and $K_{2}$,

$$ Rop(K_{1}\#K_{2}) \leq Rop(K_{1}) + Rop(K_{2}) - c, $$

where $c > 0$ is some constant independent of knot type.

背景链接：http://www.springerlink.com/index/10.1007/s00222-002-0234-y

09046 | KP-1.75

### Kirby Problem 1.75

O / open | Kirby 低维拓扑问题

(a) Find some knot energy on the space of smoothly embedded unknotted circles in $\mathbb{S}^{3}$ for which all unknotted critical points are great circles. (b) Define a gradient flow for this knot energy which yields a deformation retract of the space of unknots to the subspace of great circles.

背景链接：https://ldtopology.wordpress.com/2016/10/02/a-gorgeous-but-incomplete-proof-of-the-smale-conjecture/

09047 | KP-1.76

### Kirby Problem 1.76

P / partially_solved | Kirby 低维拓扑问题

(a) Is there an algorithm to detect the unknot that runs in polynomial time (as a function of the number of crossings in an input diagram)? (b) What is the structure of the unknot Reidemeister graph?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09048 | KP-1.77

### Kirby Problem 1.77

P / partially_solved | Kirby 低维拓扑问题

How many Reidemeister moves are required to relate two dia- grams of a knot (as a function of their numbers of crossings)?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09049 | KP-1.78

### Kirby Problem 1.78

P / partially_solved | Kirby 低维拓扑问题

Let $D$ be any diagram of the unknot with $n$ crossings. Let $h(D)$ be the smallest number such that some series of Reidemeister moves that transforms

$D$ to a 0-crossing diagram has the property that all intermediate diagrams have at most $h(D)$ crossings. Define $h(n)$ to be the maximum of $h(D)$ over all $n$-crossing diagrams of the unknot. A diagram represents a hard unknot if $h(D) > n$. What is $h(n)$?

背景链接：https://doi.org/10.1080/10586458.2022.2161676

09050 | KP-1.79

### Kirby Problem 1.79

O / open | Kirby 低维拓扑问题

Are there additional moves that, when added to the three Rei- demeister moves, allow for strict monotonic descent in the crossing number of an unknot diagram?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09051 | KP-1.8

### Kirby Problem 1.8

P / partially_solved | Kirby 低维拓扑问题

Which multi-variable Laurent polynomials arise as the multi-variable Alexander polynomial of a link in the 3-sphere or, more generally, a ho- mology 3-sphere?

背景链接：http://gateway.proquest.com/openurl?url

09052 | KP-1.80

### Kirby Problem 1.80

O / open | Kirby 低维拓扑问题

Is unknotting number computable? Is there even an algorithm to decide whether a knot has unknotting number one?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09053 | KP-1.81

### Kirby Problem 1.81

O / open | Kirby 低维拓扑问题

(a) Are all knots trivial? (b) Conjecture: The Bing sling is knotted.

背景链接：https://wildandnoncompactknots.wordpress.com

09054 | KP-1.82

### Kirby Problem 1.82

P / partially_solved | Kirby 低维拓扑问题

(a) What is a positive knot? (b) Describe a simple set of moves to convert between two positive diagrams of the same knot or link.

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09055 | KP-1.83

### Kirby Problem 1.83

O / open | Kirby 低维拓扑问题

Determine the algebraic structure of the concordance group $\mathcal{O}$ of open strings. (a) Is it abelian? (b) Does it contain torsion?

背景链接：http://aif.cedram.org/item?id=AIF

09056 | KP-1.84

### Kirby Problem 1.84

P / partially_solved | Kirby 低维拓扑问题

For a classical knot, does its slice genus as a virtual knot agree with its slice genus as a classical knot?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09057 | KP-1.85

### Kirby Problem 1.85

P / partially_solved | Kirby 低维拓扑问题

Let $K$ be a hyperbolic knot in $S^{3}$ and $\chi(K)$ the space of con- jugacy classes of $\operatorname{PSL}_{2}(\mathbb{C})$ representations of $\pi_{1}(S^{3} - K)$. There is a distinguished component of $\chi(K)$ that contains the discrete faithful representation coming from the finite volume hyperbolic structure on the complement of $K$. Does this component always contain an arc of representations to $\operatorname{SO}(3)$?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09058 | KP-1.86

### Kirby Problem 1.86

P / partially_solved | Kirby 低维拓扑问题

(a) Every connected cubic $($ i.e. trivalent $)$ graph has freeness index at least 2. (b) Every graph has freeness index at least two. (c) There is a graph with freeness index two.

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09059 | KP-1.87

### Kirby Problem 1.87

P / partially_solved | Kirby 低维拓扑问题

Is every fibered link in $S^{3}$ realized as the link of an isolated singular point of a polynomial map $\mathbb{R}^{4} \to \mathbb{R}^{2}$?

背景链接：https://www.jstor.org/stable/j.ctt1bd6kvv

09060 | KP-1.88

### Kirby Problem 1.88

P / partially_solved | Kirby 低维拓扑问题

Are there infinitely many congruence arithmetic links in the 3-sphere?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09061 | KP-1.89

### Kirby Problem 1.89

P / partially_solved | Kirby 低维拓扑问题

(a) Fix a long link L. What is the homotopy type of the embedding space of links isotopic to L? (b) Fix a link L in a 3-manifold M. What is the homotopy type of the embed- ding space of links in M isotopic to L?

(c) Given a link $L$ obtained by infecting a link $L_{1}$ by another link $L_{2}$, describe the homotopy type of component of $L$ in the space of links in terms of the components of $L_{1}$ and $L_{2}$.

背景链接：https://pi.math.cornell.edu/„hatcher/Papers/knotspaces.pdf

09062 | KP-1.9

### Kirby Problem 1.9

P / partially_solved | Kirby 低维拓扑问题

If Dehn surgery on a knot $K$ gives a lens space, then $K$ is a Berge knot.

背景链接：http://projecteuclid.org/euclid.pjm/1102969920

09063 | KP-2.1

### Kirby Problem 2.1

P / partially_solved | Kirby 低维拓扑问题

(Ivanov conjecture). Let $S$ be an orientable surface of finite type with genus at least three. If $G \leq \operatorname{Mod}(S)$ is a subgroup of finite index, does $G$ have finite abelianization?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09064 | KP-2.10

### Kirby Problem 2.10

P / partially_solved | Kirby 低维拓扑问题

(a) Is there an efficient algorithm to compute distances in the curve complex of a surface? The input to the algorithm should be the surface and the curves. One may ask similar questions for related complexes, such as the arc complex of a surface with boundary. (b) Is there an algorithm (efficient or otherwise) to compute the distance be- tween quasi-convex subsets of the curve complex? This question is partic- ularly interesting in the case where the quasi-convex subsets are the disk sets of handlebodies.

背景链接：http://gateway.proquest.com/openurl?url

09065 | KP-2.11

### Kirby Problem 2.11

P / partially_solved | Kirby 低维拓扑问题

Find precise estimates for both the extremal and average behav- ior of the simple lifting degree of curves, in terms of combinatorial (e.g., intersection number) and/or geometric (e.g., geodesic length, word length) data.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09066 | KP-2.12

### Kirby Problem 2.12

P / partially_solved | Kirby 低维拓扑问题

(a) What is the maximum number of systoles on a closed, hyperbolic surface of genus $g$? (b) What is the maximum cardinality of a set of pairwise non-isotopic, essen- tial, simple closed curves on a closed surface of genus $g$ with the property that no two curves in the set intersect in more than one point?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09067 | KP-2.13

### Kirby Problem 2.13

P / partially_solved | Kirby 低维拓扑问题

Let $S$ be a surface, and let $\Gamma_{1}, \Gamma_{2}$ be isotopy classes of embedded graphs in $S$. Determine when $\Gamma_{1}$ and $\Gamma_{2}$ are related by a sequence of surgeries.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09068 | KP-2.14

### Kirby Problem 2.14

P / partially_solved | Kirby 低维拓扑问题

(a) Does every Jordan curve in the Euclidean plane contain the vertices of a square? (b) Does every Jordan curve in the Euclidean plane contain the vertices of the affine image of a regular hexagon? (c) Let $\gamma \subset \mathbb{C}$ be a smooth Jordan curve, and let $S \subset \mathbb{C}$ consist of $2n \geq 4$ concyclic points. Does there exist a non-constant polynomial $p(z) \in \mathbb{C}[z]$ of degree $< n$ such that $p(S) \subset \gamma$? (d) Suppose that $h$ is a continuous, real-valued function defined on the unit sphere $S^{2} \subset \mathbb{R}^{3}$ and that points $x_{1},..., x_{4} \in S^{2}$ form the vertices of a square. Is there an isometry $\rho \in \operatorname{SO}(3)$ such that $h$ is constant on $\rho(x_{1}),..., \rho(x_{4})$?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09069 | KP-2.15

### Kirby Problem 2.15

P / partially_solved | Kirby 低维拓扑问题

Is the genus $g$ Goeritz group $\mathcal{G}_{g}$ finitely generated when $g \geq 4$? If so, find a set of generators.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09070 | KP-2.16

### Kirby Problem 2.16

P / partially_solved | Kirby 低维拓扑问题

If two hyperbolic surfaces have the same unmarked simple length spectra (i.e., the same multiset of lengths that correspond to simple closed curves), are they isometric?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09071 | KP-2.17

### Kirby Problem 2.17

O / open | Kirby 低维拓扑问题

Suppose that $c$ is a geodesic current on a hyperbolic surface, and suppose that, on the space of hyperbolic metrics on the surface, $c$ has the same length function as a closed geodesic. Must $c$ be a convex combination of closed curves?

背景链接：http://www.jstor.org/stable/1971388

09072 | KP-2.18

### Kirby Problem 2.18

P / partially_solved | Kirby 低维拓扑问题

What is the best lower bound on the volume of a fibered hyper- bolic 3-manifold one can give in terms of the translation length of the monodromy with respect to various metric spaces, such as Teichmüller space with the Weil-Peterson metric or the pants complex?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09073 | KP-2.19

### Kirby Problem 2.19

O / open | Kirby 低维拓扑问题

Which mapping classes give rise to arithmetic hyperbolic 3-manifolds as their mapping tori?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09074 | KP-2.2

### Kirby Problem 2.2

P / partially_solved | Kirby 低维拓扑问题

(Congruence subgroup problem). Does every finite-index sub- group of the mapping class group of $S$ contain a congruence subgroup?

背景链接：http://nyjm.albany.edu:

09075 | KP-2.20

### Kirby Problem 2.20

P / partially_solved | Kirby 低维拓扑问题

Can one detect holomorphicity from a monodromy factoriza- tion of a Lefschetz pencil, fibration, or a surface bundle over a surface? What are the special properties of monodromy factorizations of holomorphic fibrations?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09076 | KP-2.21

### Kirby Problem 2.21

P / partially_solved | Kirby 低维拓扑问题

What is the minimum number, $m_{g,b}$, of right-handed Dehn twists along essential curves into which the boundary multi-twist, $\Delta:= T_{\delta_{1}} \cdots T_{\delta_{b}}$, can be factorized in $\operatorname{Mod}(S_{g}^{b})$? Find factorizations realizing the minimum.

背景链接：https://msp.org/obs/2022/5-1/p09.xhtml

09077 | KP-2.22

### Kirby Problem 2.22

P / partially_solved | Kirby 低维拓扑问题

Does every Lefschetz fibration over the 2-sphere admit a sec- tion?

背景链接：https://arxiv.org/abs/2506.18066

09078 | KP-2.23

### Kirby Problem 2.23

O / open | Kirby 低维拓扑问题

Does there exist a surface bundle over a surface that admits a complete hyperbolic metric, or, more generally, a complete metric of variable negative curvature?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09079 | KP-2.24

### Kirby Problem 2.24

O / open | Kirby 低维拓扑问题

Does there exist a complex surface $X$ that admits three or more non-isomorphic structures as a surface bundle over a surface?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09080 | KP-2.25

### Kirby Problem 2.25

O / open | Kirby 低维拓扑问题

Consider surface bundles over surfaces where both fiber $F$ and base $B$ have genus $\geq 2$ and where $\pi_{1}(B)$ injects in the mapping class group of the

fiber. Does such a bundle have a $k$-multisection for some $k > 0$ (i.e., a continuous choice of $k$ everywhere distinct points)?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09081 | KP-2.26

### Kirby Problem 2.26

P / partially_solved | Kirby 低维拓扑问题

(Kontsevich-Zorich conjecture). Understand the homotopy types of strata of abelian differentials. Which stratum-components are $K(\pi, 1)$ spaces? What are the fundamental groups? For which stratum-components $\mathcal{H}$ is some ver- sion of the monodromy map $\rho: \pi_{1}(\mathcal{H}) \to \operatorname{Mod}(S_{g})$ injective? Do the answers to these questions change in a predictable way as one moves from the principal stra- tum down to the minimal?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09082 | KP-2.27

### Kirby Problem 2.27

O / open | Kirby 低维拓扑问题

For $n \geq 4$, does the braid group $B_{n}$ admit a finite-index sub- group that embeds in a right-angled Artin group?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09083 | KP-2.28

### Kirby Problem 2.28

O / open | Kirby 低维拓扑问题

Let $\Gamma$ be a graph that is not a nontrivial join, and let $A(\Gamma)$ be the associated right-angled Artin group. Does there exist an injective map from $A(\Gamma)$ to the mapping class group of a surface such that every loxodromic element of $A(\Gamma)$ is sent to a pseudo-Anosov mapping class?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09084 | KP-2.29

### Kirby Problem 2.29

P / partially_solved | Kirby 低维拓扑问题

Determine the Artin groups that can be embedded into a map- ping class group.

背景链接：https://doi.org/10.5565/PUBLMAT

09085 | KP-2.3

### Kirby Problem 2.3

P / partially_solved | Kirby 低维拓扑问题

Is the mapping class group of a surface of finite type linear?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09086 | KP-2.30

### Kirby Problem 2.30

P / partially_solved | Kirby 低维拓扑问题

Which right-angled Artin groups contain closed hyperbolic sur- face groups? Is there an algorithmic or graph-theoretic criterion to decide this?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09087 | KP-2.31

### Kirby Problem 2.31

P / partially_solved | Kirby 低维拓扑问题

Let $S$ be a closed surface of genus at least 2. Show that the stable commutator length is rational on the commutator subgroup of $\pi_{1}(S)$.

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09088 | KP-2.32

### Kirby Problem 2.32

P / partially_solved | Kirby 低维拓扑问题

Does every surface bundle over a surface admit a flat connec- tion? What about surface bundles over 3-manifolds?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09089 | KP-2.33

### Kirby Problem 2.33

O / open | Kirby 低维拓扑问题

Let $S$ be a closed compact surface (without boundary). Is there a finitely generated, torsion-free group $G$ such that $G$ cannot act faithfully by homeomorphisms on $S$?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09090 | KP-2.34

### Kirby Problem 2.34

P / partially_solved | Kirby 低维拓扑问题

Let $S$ be a compact surface. For $0 \leq r < s$, does there exist a nontrivial finitely generated subgroup $G_{r} \leq \operatorname{Diff}^{r}_{0}(S)$ such that every action of $G_{r}$ on $S$ by $C^{s}$ diffeomorphisms is trivial? Can these examples be torsion-free?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09091 | KP-2.35

### Kirby Problem 2.35

O / open | Kirby 低维拓扑问题

Is the first-order theory of the mapping class group of a surface decidable?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09092 | KP-2.36

### Kirby Problem 2.36

O / open | Kirby 低维拓扑问题

Are systems of equations over mapping class groups and braid groups decidable?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09093 | KP-2.37

### Kirby Problem 2.37

P / partially_solved | Kirby 低维拓扑问题

Give a Nielsen-Thurston-type classification for the mapping class groups of infinite-type surfaces. In particular, which homeomorphisms are the appropriate generalization of pseudo-Anosov homeomorphisms in this setting?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09094 | KP-2.38

### Kirby Problem 2.38

P / partially_solved | Kirby 低维拓扑问题

Give an appropriate analogue of the curve graph for infinite-type surfaces, and characterize the surfaces for which no such graph exists.

背景链接：https://lamington.wordpress.com/2009/06/22/big-mapping-class-groups-and-dynamics/

09095 | KP-2.39

### Kirby Problem 2.39

P / partially_solved | Kirby 低维拓扑问题

(a) Does the mapping class group of an infinite-genus surface with no planar ends contain every countable group? (b) Does the mapping class group of the one-ended infinite-genus surface have any proper finite-index subgroups? (c) For an infinite-type surface, describe and characterize the subgroup of mapping classes with quasi-conformal representatives. In particular, when is it a normal subgroup of the mapping class group?

背景链接：https://https://www.patelp.com/uploads/2/5/7/9/25792573/inftypeproblems.pdf

09096 | KP-2.4

### Kirby Problem 2.4

P / partially_solved | Kirby 低维拓扑问题

Let $S_{1}$ and $S_{2}$ be orientable surfaces of finite type. Under what conditions do injective maps from (finite-index subgroups of) the mapping class group of $S_{1}$ to the mapping class group of $S_{2}$ necessarily arise from “manipulations” (e.g., inclusions, (branched) coverings, etc.) taking $S_{1}$ to $S_{2}$?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09097 | KP-2.40

## Kirby Problem 2.40

P / partially_solved | Kirby 低维拓扑问题

(a) Let $S$ be an infinite-type surface and $\varphi$ a mapping class for which there is a (marked) conformal structure $\Sigma$ on $S$ with respect to which $\varphi$ is realized as a quasi-conformal homeomorphism. Then for each quasi-conformal structure in the connected component of Teichmüller space containing $\Sigma$, there is a quasiconformal homeomorphism representing $\varphi$ which minimizes the dilatation. If the infimum of the minimal dilatations for each structure in this component is achieved and is greater than 1, is the minimizing quasiconformal homeomorphism unique? What is the structure of such a minimizer? (b) If $\varphi$ is parabolic (see Remark (1) below) for $T$, must a sequence of con- formal structures with dilatation converging to the infimum degenerate by pinching along some sequence of essential simple closed curves? Is this system unique? Can it depend on the component $T$? (c) If $\varphi$ fixes more than one component $T, T'$ of $\operatorname{Teich}(S)$, can the type be dif- ferent for $T$ and $T'$? In particular, can $\varphi$ be parabolic for $T$ and hyperbolic for $T'$?

背景链接：https://https://www.patelp.com/uploads/2/5/7/9/25792573/infinftypeproblems.pdf

09098 | KP-2.41

## Kirby Problem 2.41

O / open | Kirby 低维拓扑问题

Give a finite list of practically computable invariants of the mapping class group or pure mapping class group of an infinite-type surface $S$ that determine the topology of $S$.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09099 | KP-2.42

## Kirby Problem 2.42

O / open | Kirby 低维拓扑问题

Is the geodesic flow in almost every direction on the Chamanara surface ergodic? What about on the translation surface considered by Bruin and Lukina, which is similar to the Chamanara surface?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09100 | KP-2.43

## Kirby Problem 2.43

P / partially_solved | Kirby 低维拓扑问题

Let $X$ be a compact, totally disconnected subset of $\mathbb{R}^{2}$ with $|X| \geq 2$, and let $\Gamma_{X}$ denote the mapping class group of $\mathbb{R}^{2} - X$. (a) For $Y \subset X$, let $\Gamma_{X,Y}$ be the subgroup of $\Gamma_{X}$ permuting $Y$ . Classify in- variants $q$ that are functorial. (b) If $X \subset \mathbb{R}^{2}$ is totally disconnected and $\alpha$ is a homeomorphism of $\mathbb{R}^{2}$ fixing $X$ as a set, let $C_{\alpha}(X)$ denote the space of $\alpha$-invariant closed subsets of

$X$, in the Hausdorff topology. Let $q$ be functorial as in part (a). Classify those $q$ that are continuous or semi-continuous. (c) What is the relationship between the translation length $\tau_{X}$ and topological entropy? More precisely, is there a positive constant $C$ so that if $\alpha$ is a homeomorphism of $\mathbb{R}^{2}$ permuting $X$ and $g$ is the class of $\alpha$ in $\Gamma_{X}$, then the (topological) entropy of $\alpha$ is at least as big as $C \cdot \tau_{X}(g)$? (d) Suppose $X$ is a Cantor set in the plane, and let $P\Gamma_{X}$ denote the ‘pure’ subgroup of $\Gamma_{X}$ fixing $X$ pointwise. Let $N$ be a subgroup of $P\Gamma_{X}$ that is normal in $\Gamma_{X}$, and for each integer $n$, let $N_{n}$ be the subgroup of $P\Gamma_{n}$ that is the image of $N$ restricted to any $n$-element subset of $X$. What possible sequences of subgroups $N_{n}$ can arise in this way? (e) For which $g \in \Gamma_{X}$ is there an embedding $X \to \mathbb{R}^{2}$ for which $g$ is represented by a diffeomorphism $\alpha$ which is $C^{\infty}$? Or $C^{k}$ for fixed $k$?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09101 | KP-2.44

## Kirby Problem 2.44

P / partially_solved | Kirby 低维拓扑问题

Given an infinite-type surface $S$, which homeomorphisms $f: S \to$ $S$ give rise to mapping tori $M_{f}$ that admit a hyperbolic structure? For those which do admit a hyperbolic metric, is $M_{f}$ homeomorphic to the interior of a compact hyperbolic manifold with totally geodesic boundary?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09102 | KP-2.45

## Kirby Problem 2.45

O / open | Kirby 低维拓扑问题

Compute the end-periodic cobordism group $\Delta^{e}_{2}$ of end-periodic automorphisms (diffeomorphisms or homeomorphisms) of surfaces.

背景链接：http://www.numdam.org/item?id=ASENS

09103 | KP-2.46

## Kirby Problem 2.46

P / partially_solved | Kirby 低维拓扑问题

(a) Which coarsely boundedly generated mapping class groups of infinite-type surfaces are hyperbolic? (b) Consider the class of surfaces with $n \geq 2$ ends, all accumulated by genus. Are the mapping class groups of these surfaces quasi-isometric for different values of $n$?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09104 | KP-2.47

## Kirby Problem 2.47

P / partially_solved | Kirby 低维拓扑问题

(a) Given a mapping class $\psi$ of a based surface $S$, there is an induced endo- morphism of the symmetric product $\operatorname{Sym}^{i}(S)$ and hence an endofunctor $\psi_{*}$ of the partially wrapped Fukaya category $\mathcal{F}(S)$ as constructed, say, by Auroux [Aur10]. Compute the categorical entropy of $\psi_{*}$, in the sense of [DHKK14], for $\psi$ pseudo-Anosov. (b) Given a fibered knot $K$ with pseudo-Anosov monodromy, what is

$$ \limsup_{n\to\infty}\left(\dim \widehat{\mathrm{HFK}}(\Sigma^{n}(K),\widetilde{K};i)\right)^{1/n}, $$

where $\widetilde{K}$ is the branch locus in the $n$-fold cyclic branched cover $\Sigma^{n}(K)$, and $i$ denotes the Alexander grading (in $\mathbb{Z}$) with respect to the fiber surface. Here, $\widehat{\mathrm{HFK}}$ denotes the knot Floer homology of $K$ [OS04c, Ras03]. (c) With notation as in the previous question, what is

$$ \limsup_{n\to\infty}\left(\dim \widehat{\mathrm{HF}}(\Sigma^{n}(K))\right)^{1/n} $$

for the Heegaard Floer invariant $\widehat{\mathrm{HF}}$ [OS04e] of the closed 3-manifold $\Sigma^{n}(K)$? (d) Given a closed surface $S$ and a pseudo-Anosov automorphism $\psi$, let $M_{\psi}$ denote the mapping torus of $\psi$, and $\widehat{\mathrm{HF}}(M_{\psi};i)$ the summand of $\widehat{\mathrm{HF}}(M_{\psi})$ spanned by the spin $^{c}$-structures $\mathfrak{s}$ with $\langle c_{1}(\mathfrak{s}), [F]\rangle = i$. What is

$$ \limsup_{n\to\infty}\left(\dim \widehat{\mathrm{HF}}(M_{\psi^{n}},i)\right)^{1/n}? $$

What about

$$ \limsup_{n\to\infty}\left(\dim_{\mathbb{F}_{2}}\mathrm{HF}^{+}(M_{\psi^{n}},i)\right)^{1/n} $$

for $i \neq 0$ (or for twisted $\mathrm{HF}^{+}$ for $i = 0$)?

背景链接：https://arxiv.org/abs/math/0306378

09105 | KP-2.48

## Kirby Problem 2.48

P / partially_solved | Kirby 低维拓扑问题

The mapping class group of a closed, orientable, genus $g$ sur- face $S$ acts by symplectomorphisms on the symmetric product $\operatorname{Sym}^{g}(S)$. In partic- ular, it acts smoothly on $\operatorname{Sym}^{g}(S)$. (a) What is the kernel of the map from the mapping class group of $S$ to the smooth mapping class group of $\operatorname{Sym}^{g}(S)$? What about in the based case? (b) Given disjoint simple closed curves $\alpha_{1},..., \alpha_{g}$ in $S$, there is a correspond- ing torus $T_{\alpha}$ in $\operatorname{Sym}^{g}(S)$: the image of $\alpha_{1} \times \cdots \times \alpha_{g} \subset S^{g}$. These are the tori that appear in Heegaard Floer theory. When do $\alpha_{1},..., \alpha_{g}$ and $\beta_{1},..., \beta_{g}$ give smoothly isotopic tori? When are they Lagrangian isotopic for an appropriate choice of symplectic form?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09106 | KP-2.49

## Kirby Problem 2.49

P / partially_solved | Kirby 低维拓扑问题

(AMU conjecture). Let $S$ be a surface with negative Euler char- acteristic. If $\varphi \in \operatorname{Mod}(S)$ acts by a pseudo-Anosov on some subsurface (including $S$ itself), show that there exists $p_{0}(\varphi)$ such that the Witten-Reshetikhin-Turaev representation $\rho_{p}(\varphi)$ has infinite order for $p \geq p_{0}(\varphi)$.

背景链接：http://projecteuclid.org/euclid.cmp/1104178138

09107 | KP-2.5

## Kirby Problem 2.5

O / open | Kirby 低维拓扑问题

For $g \geq 3$, determine a finite presentation for the Torelli group $\mathcal{I}_{g}$, or show that no finite presentation exists.

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09108 | KP-2.50

## Kirby Problem 2.50

P / partially_solved | Kirby 低维拓扑问题

(Volume conjecture for surface diffeomorphisms). Let $S$ be a closed oriented surface, let $q=e^{2\pi i/n}$ be a root of unity, and let $\mathcal{K}^{q}(S)$ be the Kauff- man bracket skein algebra. Let $\phi: S \to S$ be a pseudo-Anosov diffeomorphism with mapping torus $M_{\phi}$, and let $r: \pi_{1}(S) \to \operatorname{SL}_{2}(\mathbb{C})$ be a smooth point in the charac- ter variety that is $\phi$-invariant. Finally, let $L$ be the intertwiner realizing an iso- morphism between the representations of $\mathcal{K}^{q}(S)$ corresponding to $[r]$ and $[r \circ \varphi_{*}]$, normalized so that $\det(L) = 1$. Show that

$$ \lim_{n\to\infty}\frac{1}{n}\log|\operatorname{Trace} L|=\frac{1}{4\pi}\operatorname{vol}_{\mathrm{hyp}} M_{\phi}. $$

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09109 | KP-2.6

## Kirby Problem 2.6

P / partially_solved | Kirby 低维拓扑问题

Give a classification or enumeration of the finite-index subgroups of $\operatorname{Mod}(S_{g})$ that are generated by Dehn twists, Dehn multitwists, or powers thereof. Which of these subgroups are “geometrically meaningful”, in the sense that they are given as the stabilizers of some kind of geometric or topological structure on the surface? Conversely, when is a geometrically meaningful subgroup (not necessarily of finite index) generated by powers of Dehn multitwists?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09110 | KP-2.7

## Kirby Problem 2.7

P / partially_solved | Kirby 低维拓扑问题

Classify the homomorphisms from the braid group $B_{n}$ on $n$ strands to the braid group $B_{m}$ on $m$ strands, where $n, m \in \mathbb{N}$ are arbitrary.

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09111 | KP-2.8

## Kirby Problem 2.8

P / partially_solved | Kirby 低维拓扑问题

Fix distinct trivial tangles $\tau_{1}, \tau_{2}$ for which $\tau_{1} \cup\tau_{2}$ is the unknot. Describe the intersection of the associated wicket groups $W_{n}(\tau_{1}) \cap W_{n}(\tau_{2}) \leq B_{2n}$. Is this group finitely generated? Finitely presented?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09112 | KP-2.9

## Kirby Problem 2.9

O / open | Kirby 低维拓扑问题

Is there a nice presentation of the $n$-stranded braid group whose generating set is the set of all positive elementary braid half-twists?

背景链接：http://gateway.proquest.com/openurl?url

09113 | KP-3.1

## Kirby Problem 3.1

P / partially_solved | Kirby 低维拓扑问题

Classify the smallest volume hyperbolic 3-manifolds of various types. In particular:

(a) Determine the nonorientable closed hyperbolic 3-manifolds of least volume.

(b) Determine the n-cusped hyperbolic 3-manifolds of least volume for each $n\geq 3$.

(c) Determine the smallest volume hyperbolic 3-manifolds with n orientable cusps for each n.

(d) Determine the n-cusped orientable hyperbolic 3-manifolds of least volume for each n.

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09114 | KP-3.10

## Kirby Problem 3.10

O / open | Kirby 低维拓扑问题

(a) Are there infinitely many commensurability classes of arithmetic rational homology 3-spheres?

(b) Are there infinitely many arithmetic integral homology 3-spheres?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09115 | KP-3.11

## Kirby Problem 3.11

P / partially_solved | Kirby 低维拓扑问题

Does every hyperbolic knot in the 3-sphere have meridian length at most 4?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09116 | KP-3.12

### Kirby Problem 3.12

P / partially_solved | Kirby 低维拓扑问题

(a) Considering all closed, orientable, $\pi_1$-injective surfaces (possibly non-embedded) in all closed hyperbolic 3-manifolds, what is the infimum of the areas of all these surfaces?

(b) One can ask the same question, but considering just closed, orientable, $\pi_1$-injective surfaces of a fixed genus g.

(c) One can also consider only closed, orientable, embedded $\pi_1$-injective surfaces.

(d) Alternatively, one can consider all closed, orientable, essential surfaces in all hyperbolic 3-manifolds M, allowing M not to be closed.

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09117 | KP-3.13

### Kirby Problem 3.13

P / partially_solved | Kirby 低维拓扑问题

Does every closed hyperbolic 3-manifold admit an immersed $\pi_1$-injective surface with only double points? More precisely, if M is a closed, connected, hyperbolic 3-manifold, is there a closed, connected surface $\Sigma$ (other than a 2-sphere) and an immersion f : $\Sigma$ $\to$ M such that $f_*$ : $\pi_1(\Sigma)$ $\to$ $\pi_1(M)$ is injective and $|f^{-1}(\{x\})|\leq 2$ for all $x\in M$?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09118 | KP-3.14

### Kirby Problem 3.14

P / partially_solved | Kirby 低维拓扑问题

Can a hyperbolic knot complement in the 3-sphere contain a closed, embedded totally geodesic surface?

背景链接：http://projecteuclid.org/euclid.pjm/1102710789

09119 | KP-3.15

### Kirby Problem 3.15

P / partially_solved | Kirby 低维拓扑问题

Let $M$ be a closed hyperbolic 3-manifold with positive first Betti number.

(a) Which elements of $H^{2}(M;\mathbb{R})$ are realized as the Euler classes of taut foliations?

(b) Which elements of $H^{2}(M;\mathbb{R})$ are realized as the Euler classes of tight contact structures?

(c) How about universally tight contact structures?

(d) The question can also be asked for pseudo-Anosov flows...

(e)... and for quasi-geodesic flows. (f ) How about Euler classes of representations into $\operatorname{Homeo}^{+}(S^1)$?

背景链接：http://www.numdam.org/item?id=AIF

09120 | KP-3.16

### Kirby Problem 3.16

P / partially_solved | Kirby 低维拓扑问题

Does every finite-volume hyperbolic 3-manifold contain infinitely many simple closed geodesics?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09121 | KP-3.17

### Kirby Problem 3.17

P / partially_solved | Kirby 低维拓扑问题

Let $M_1$ and $M_2$ be finite-volume hyperbolic n--manifolds. If the length spectra of $M_1$ and $M_2$ coincide, must the two manifolds be commensurable?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09122 | KP-3.18

### Kirby Problem 3.18

O / open | Kirby 低维拓扑问题

Is there a closed hyperbolic 3-manifold that is foliated with minimal leaves?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09123 | KP-3.19

### Kirby Problem 3.19

P / partially_solved | Kirby 低维拓扑问题

(a) Does every closed hyperbolic 3-manifold have a nowhere zero vector field whose lift to the universal cover has proper flow lines?

(b) Can one ensure that the leaf space of the foliation by flow lines in the universal cover is Hausdorff ?

(c) If so, the leaf space in the universal cover is homeomorphic to a plane and one can then ask: can the induced action of $\pi_1(M)$ on a plane be compactified to an action on the closed disk?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09124 | KP-3.2

### Kirby Problem 3.2

O / open | Kirby 低维拓扑问题

Show that the volumes of hyperbolic 3-manifolds are not all rationally related.

背景链接：http://www.numdam.org/item?id=ASNSP

09125 | KP-3.20

### Kirby Problem 3.20

O / open | Kirby 低维拓扑问题

Let $M$ be a closed hyperbolic 3-manifold with a faithful homomorphism $\rho:\pi_1(M)\to \operatorname{Homeo}^{+}(\mathbb{R})$. Prove that $M$ supports a dual transversely orientable, essential lamination $\lambda$.

背景链接：http://aif.cedram.org/item?

09126 | KP-3.21

### Kirby Problem 3.21

P / partially_solved | Kirby 低维拓扑问题

In this problem, all 3-manifolds are orientable, while all flows are considered up to orbit equivalence and are assumed to be transitive.

(a) Are there only finitely many Anosov flows on a given closed 3-manifold?

(b) Are there only finitely many pseudo-Anosov flows on a given closed 3-manifold?

(c) Are there only finitely many quasigeodesic pseudo-Anosov flows on a given closed 3-manifold?

(d) Are there only finitely many pseudo-Anosov flows without perfect fits on a given closed 3-manifold?

(e) Are there only finitely many veering triangulations on a given finite volume cusped hyperbolic 3-manifold?

背景链接：http://gateway.proquest.com/openurl?url

09127 | KP-3.22

### Kirby Problem 3.22

P / partially_solved | Kirby 低维拓扑问题

Let $G=\pi_1(M)$ be the fundamental group of a finite-volume hyperbolic 3-manifold $M$. What is the regularity of the smoothest (virtual) action of $G$ on $S^1$? On the interval?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09128 | KP-3.23

### Kirby Problem 3.23

P / partially_solved | Kirby 低维拓扑问题

What is the Margulis constant in dimension 3? Is it realized uniquely by the Weeks manifold W, where $\mu(W)$ = 0.77442...?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09129 | KP-3.24

### Kirby Problem 3.24

O / open | Kirby 低维拓扑问题

(a) (Cannon Conjecture) If G is a finitely presented, Gromov hyperbolic group with space at infinity equal to the 2-sphere, must G be a cocompact Kleinian group?

(b) More generally, if $G$ is a Gromov hyperbolic group whose boundary is planar, is G virtually a convex cocompact Kleinian group?

背景链接：http://projecteuclid.org/euclid.jdg/1090351321

09130 | KP-3.25

### Kirby Problem 3.25

P / partially_solved | Kirby 低维拓扑问题

(Bending Conjecture). (a) Is a quasi-Fuchsian group determined by the hyperbolic metric on the boundary of its convex core?

(b) Is a quasi-Fuchsian group with parabolics, other than a Fuchsian group, determined by the bending measure on the boundary of its convex core?

背景链接：http://www.numdam.org/item?id=AFST

09131 | KP-3.26

### Kirby Problem 3.26

P / partially_solved | Kirby 低维拓扑问题

Let $M$ be a finite-volume hyperbolic 3-manifold, and let $M^1$ be a minimal-index finite cover of $M$ such that $\pi_1(M^1)$ embeds in a right-angled Artin group. Can one obtain good control over the complexity of the smallest right-angled Artin group containing $\pi_1(M^1)$ from the data of $M$?

背景链接：https://elibm.org/article/10000267

09132 | KP-3.27

### Kirby Problem 3.27

P / partially_solved | Kirby 低维拓扑问题

(a) What is the computational complexity of the homeomorphism problem for compact, orientable 3-manifolds?

(b) Is there a polynomial-time algorithm to recognize the 3-sphere?

(c) What is the computational complexity of the recognition problem for other 3-manifolds?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09133 | KP-3.28

### Kirby Problem 3.28

P / partially_solved | Kirby 低维拓扑问题

How many Pachner moves are needed to pass between two triangulations of a compact 3-manifold?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09134 | KP-3.29

### Kirby Problem 3.29

P / partially_solved | Kirby 低维拓扑问题

(a) Given a closed hyperbolic 3-manifold $M$, can one find an explicit bound on the degree of a finite cover $\widetilde M$ having $b_1(\widetilde M)>0$?

(b) For example, is there such a finite cover with degree at most a polynomial function of the number of tetrahedra in some triangulation of $M$?

(c) Similarly, can one find an explicit upper bound on the degree of a finite cover $\widetilde M_1$ that fibers over the circle?

背景链接：https://elibm.org/article/10000267

09135 | KP-3.3

### Kirby Problem 3.3

P / partially_solved | Kirby 低维拓扑问题

Does every cusped hyperbolic 3-manifold have a geometric ideal triangulation?

背景链接：http://projecteuclid.org/euclid.ojm/1200786689

09136 | KP-3.30

### Kirby Problem 3.30

P / partially_solved | Kirby 低维拓扑问题

(a) What is the computational complexity of determining whether a compact 3-manifold admits a hyperbolic structure?

(b) If a compact 3-manifold does admit a hyperbolic structure, what is the computational complexity of finding it?

(c) In particular, do these problems lie in the complexity classes NP and FNP respectively?

背景链接：http://snappy.computop.org

09137 | KP-3.31

### Kirby Problem 3.31

O / open | Kirby 低维拓扑问题

Suppose M is a closed 3-manifold.

(a) Can one decide if the fundamental group of M is left-orderable?

(b) What is the complexity of a certificate of left-orderability, or of non-left-orderability?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09138 | KP-3.32

### Kirby Problem 3.32

P / partially_solved | Kirby 低维拓扑问题

Is there an algorithm to determine whether two closed, embedded surfaces in $\mathbb{R}^3$ are isotopic?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09139 | KP-3.33

### Kirby Problem 3.33

P / partially_solved | Kirby 低维拓扑问题

Let M and N be closed orientable 3-manifolds. Prove that if there is a degree-1 map $f:M\to N$ then $g(M)\geq g(N)$, where $g(M)$ is the Heegaard genus of $M$.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09140 | KP-3.34

### Kirby Problem 3.34

P / partially_solved | Kirby 低维拓扑问题

Do any two genus-g Heegaard splittings of a closed, orientable 3-manifold M become equivalent after at most g stabilizations?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09141 | KP-3.35

### Kirby Problem 3.35

P / partially_solved | Kirby 低维拓扑问题

Given a compact manifold M, let $r(M)$ denote the rank of its fundamental group and $g(M)$ denote its Heegaard genus.

(a) Does every closed orientable hyperbolic 3-manifold M with $r(M)$ = 2 satisfy $g(M)$ = 2?

(b) Is there a non-Haken manifold $M$ such that $r(M)<g(M)$?

(c) Is there a constant $k>0$ such that $g(M)\leq k r(M)$ for every closed orientable 3-manifold $M$?

(d) If $\{M_i\}$ is the set of congruence covers of an arithmetic hyperbolic 3-manifold M, is the infimum of $r(M_i)/\operatorname{vol}(M_i)$ zero or positive?

(e) Is there a knot $K\subset S^3$ such that $r(S^3\setminus\nu(K))<g(S^3\setminus\nu(K))$?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09142 | KP-3.36

### Kirby Problem 3.36

P / partially_solved | Kirby 低维拓扑问题

(Simple loop conjecture) Let f : F $\to$ M be a 2-sided immersion of a surface into a 3-manifold such that $f_*$ : $\pi_1(F)$ $\to$ $\pi_1(M)$ is not injective. Must some simple closed curve lie in the kernel of $f_*$ ?

背景链接：http://projecteuclid.org/euclid.jdg/1214439470

09143 | KP-3.37

### Kirby Problem 3.37

P / partially_solved | Kirby 低维拓扑问题

(a) Is every finitely generated 3-manifold group linear (over some field with characteristic zero)?

(b) If so, can one bound the dimension of a faithful linear representation?

背景链接：https://elibm.org/article/10000267

09144 | KP-3.38

### Kirby Problem 3.38

P / partially_solved | Kirby 低维拓扑问题

Is every PD$_3$-group the fundamental group of a closed, aspherical 3-manifold?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09145 | KP-3.39

### Kirby Problem 3.39

P / partially_solved | Kirby 低维拓扑问题

(a) Is every finitely generated perfect group the normal closure of a single element?

(b) Is there an integral homology sphere whose fundamental group has weight bigger than one?

(c) For any $n$, is there an integral homology sphere whose fundamental group has weight bigger than $n$?

(d) Specifically, are the weights of the fundamental groups of Seifert fibered homology spheres $\Sigma(a_1,\ldots,a_n)$ unbounded?

背景链接：https://arxiv.org/abs/2510.26073

09146 | KP-3.4

### Kirby Problem 3.4

P / partially_solved | Kirby 低维拓扑问题

(Chen--Yang Volume Conjecture). (a) Prove that, for any hyperbolic 3-manifold $M$,

$$ \lim_{\substack{r\to\infty\\ r\mathrm{odd}}}\frac{1}{r}\log\bigl(TV(M;e^{2\pi i/r})\bigr) =\frac{1}{2\pi}\operatorname{Vol}(M). $$

(b) Prove that, for any closed, oriented, hyperbolic 3-manifold $M$,

$$ \lim_{\substack{r\to\infty\\ r\mathrm{odd}}}\frac{1}{r}\log\bigl(WRT(M;e^{2\pi i/r})\bigr) =\frac{1}{4\pi}\bigl(\operatorname{Vol}(M)-i\operatorname{CS}(M)\bigr). $$

背景链接：https://doi.org/10.4171/QT/111

09147 | KP-3.40

### Kirby Problem 3.40

P / partially_solved | Kirby 低维拓扑问题

Does every closed, orientable, hyperbolic 3-manifold admit a tight contact structure?

背景链接：http://projecteuclid.org/euclid.jdg/1214437784

09148 | KP-3.41

### Kirby Problem 3.41

P / partially_solved | Kirby 低维拓扑问题

Is it true that for every knot $K\subset S^3$, there is an integer $n_K$ such that $S^3_r(K)$ admits a tight contact structure for all $r\geq n_K$?

背景链接：https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf

09149 | KP-3.42

### Kirby Problem 3.42

P / partially_solved | Kirby 低维拓扑问题

Does every tight contact 3-manifold have finite Giroux torsion?

背景链接：https://doi.org/10.2140/gt.2006.10.1749

09150 | KP-3.43

### Kirby Problem 3.43

P / partially_solved | Kirby 低维拓扑问题

Understand how various properties of contact structures behave under different kinds of symplectic cobordism. For instance:

(a) Is tightness preserved under exact symplectic cobordism?

(b) Is weak fillability preserved under weak symplectic cobordism?

(c) How do the support genus and Giroux torsion behave under Legendrian surgery?

背景链接：https://doi.org/10.4007/annals.2015.182.2.8

09151 | KP-3.44

### Kirby Problem 3.44

P / partially_solved | Kirby 低维拓扑问题

Is there an algorithm to decide, given an open book, whether the corresponding contact 3-manifold is tight or fillable?

背景链接：https://doi.org/10.1007/s00222-007-0051-4

09152 | KP-3.45

### Kirby Problem 3.45

O / open | Kirby 低维拓扑问题

(a) Are there contact 3-manifolds with support genus greater than one?

(b) Are there contact 3-manifolds with arbitrarily large support genus?

背景链接：https://doi.org/10.1090/S0002-9947-08-04459-0

09153 | KP-3.46

### Kirby Problem 3.46

P / partially_solved | Kirby 低维拓扑问题

Let $\lambda$ be a contact form on a closed 3-manifold that is not a lens space. Must the associated Reeb flow have infinitely many simple periodic orbits?

背景链接：http://projecteuclid.org/euclid.jdg/1452002876

09154 | KP-3.47

### Kirby Problem 3.47

P / partially_solved | Kirby 低维拓扑问题

(a) Does every Reeb flow on $S^3$, associated to a contact form giving the standard contact structure, have an elliptic periodic orbit?

(b) What about if one restricts to Reeb flows on $S^3$ arising from boundaries of strictly convex domains in $\mathbb{R}^4$ ?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09155 | KP-3.48

### Kirby Problem 3.48

P / partially_solved | Kirby 低维拓扑问题

别名：L-Space Conjecture [OWR-12014-005]

(L-space Conjecture). For prime rational homology 3-spheres Y, are the following equivalent?

(a) $\pi_1(Y)$ is left-orderable.

(b) Y is not an L-space.

(c) Y admits a coorientable taut foliation.

背景链接：http://aif.cedram.org/item?

09156 | KP-3.49

### Kirby Problem 3.49

P / partially_solved | Kirby 低维拓扑问题

Are any of the three conditions in the L-space Conjecture equivalent, for all prime rational homology 3-spheres Y, to the condition that Y admits a contact structure that is filled by a symplectic 4-manifold with $b_2^+>0$.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09157 | KP-3.5

### Kirby Problem 3.5

O / open | Kirby 低维拓扑问题

(a) Do there exist closed non-Haken hyperbolic 3-manifolds with arbitrarily large injectivity radius?

(b) Does there exist a cofinal tower of regular covers of closed hyperbolic 3-manifolds where all of the manifolds in the tower are non-Haken?

(c) Is there a tower of hyperbolic rational homology spheres for which all fundamental groups are not left-orderable, the manifolds contain no coorientable taut foliations, or the manifolds are L-spaces? (The three properties listed are connected via the L-space conjecture. See Problem 3.48)

背景链接：https://elibm.org/article/10000267

09158 | KP-3.50

### Kirby Problem 3.50

P / partially_solved | Kirby 低维拓扑问题

(The Floer Poincaré Conjecture). If Y is an integral homology sphere that is an L-space, show that Y is $S^3$ or the connected sum of some copies of the Poincaré sphere (with arbitrary orientations).

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09159 | KP-3.51

### Kirby Problem 3.51

P / partially_solved | Kirby 低维拓扑问题

Suppose Y is a rational homology 3-sphere such that every homomorphism $\pi_1(Y)$ $\to$ $\operatorname{SU}(2)$ has abelian image. Does it follow that

$$ \dim_{\mathbb{C}} I^{\#}(Y;\mathbb{C})=|H_1(Y;\mathbb{Z})|, $$ i.e., that Y is an instanton L-space?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09160 | KP-3.52

### Kirby Problem 3.52

P / partially_solved | Kirby 低维拓扑问题

(a) Does every closed 3-manifold M besides the 3-sphere admit a nontrivial representation $\pi_1(M)$ $\to$ $\operatorname{SU}(2)$?

(b) For which M with nonabelian fundamental group does every homomorphism $\pi_1(M)$ $\to$ $\operatorname{SU}(2)$ have abelian image?

背景链接：http://projecteuclid.org/euclid.jdg/1274707316

09161 | KP-3.53

### Kirby Problem 3.53

P / partially_solved | Kirby 低维拓扑问题

Are all strong L-spaces branched double covers of alternating links in $S^3$?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09162 | KP-3.54

### Kirby Problem 3.54

O / open | Kirby 低维拓扑问题

(a) Is there a closed 3-manifold $M$ whose Heegaard Floer homology $\widehat{HF}(M;\mathbb{Z})$ has torsion?

(b) Is there a rational homology 3-sphere $M$ for which $\widehat{HF}(M;\mathbb{Z})$ has torsion?

(c) Is there a knot $K\subset S^3$ whose knot Floer homology $\widehat{HFK}(S^3,K;\mathbb{Z})$ has torsion?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09163 | KP-3.55

### Kirby Problem 3.55

P / partially_solved | Kirby 低维拓扑问题

(a) For $K$ a nontrivial knot in $S^3$, does $HFK^-(K)$ always admit an $\mathbb{F}_2$-summand, as an $\mathbb{F}_2[U]$-module?

(b) For $Y$ a rational homology sphere, if $HF_{\mathrm{red}}(Y)$ is nontrivial, does $HF_{\mathrm{red}}(Y)$ always admit an $\mathbb{F}_2$-summand?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09164 | KP-3.56

### Kirby Problem 3.56

P / partially_solved | Kirby 低维拓扑问题

(a) For $Y^3$ a rational homology sphere, is the Seiberg--Witten Floer spectrum $SWF(Y)$ always a wedge of spheres?

(b) Is every monopole Floer homology L-space a Seiberg--Witten Floer homotopy L-space?

(c) Find an example of two rational homology spheres with the same monopole Floer homology but with different Seiberg--Witten Floer spectra.

(d) Develop an algorithm for computing Seiberg--Witten Floer spectra.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09165 | KP-3.57

### Kirby Problem 3.57

P / partially_solved | Kirby 低维拓扑问题

Construct an $S^1$ - or $\operatorname{Pin}(2)$-equivariant lattice homotopy type that computes the Seiberg--Witten Floer homotopy type.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09166 | KP-3.58

### Kirby Problem 3.58

P / partially_solved | Kirby 低维拓扑问题

Prove that there is an isomorphism relating Heegaard Floer homology and monopole Floer homology that commutes with the cobordism maps in the two settings. In particular, prove that the mixed invariant in Heegaard Floer homology is equal to the Seiberg--Witten invariant for closed 4-manifolds.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09167 | KP-3.59

### Kirby Problem 3.59

O / open | Kirby 低维拓扑问题

Prove an isomorphism relating instanton Floer homology and Heegaard Floer homology.

背景链接：http://projecteuclid.org/euclid.jdg/1274707316

09168 | KP-3.6

### Kirby Problem 3.6

P / partially_solved | Kirby 低维拓扑问题

Given a cofinal tower of covers M $\leftarrow$ $M_1$ $\leftarrow$ $M_2$ $\leftarrow$ $\cdots$, is it true that the torsion subgroups $\operatorname{Tor}(M_{n})$ of $H_1(M_{n}, \mathbb{Z})$ and the hyperbolic volumes $\operatorname{vol}(M_{n})$ satisfy

$$ \lim_{n\to\infty}\frac{\log|\operatorname{Tor}(M_n)|}{\operatorname{vol}(M_n)}=\frac{1}{6\pi}\,? $$

背景链接：http://www.math.columbia.edu/„volconf09/notes/leconf.pdf

09169 | KP-3.60

### Kirby Problem 3.60

P / partially_solved | Kirby 低维拓扑问题

Find an algorithm to compute instanton Floer homology of closed 3-manifolds and the Donaldson invariants of closed 4-manifolds.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09170 | KP-3.61

### Kirby Problem 3.61

P / partially_solved | Kirby 低维拓扑问题

Is the dimension of Heegaard Floer homology invariant under genus 2 mutation?

背景链接：http://nyjm.albany.edu:8000/j/2010/16

09171 | KP-3.62

### Kirby Problem 3.62

P / partially_solved | Kirby 低维拓扑问题

How do Floer homological invariants behave under maps of nonzero degree? For instance, let $Y$ and $Z$ be closed, oriented 3-manifolds, and suppose that there is a degree-1 map $f:Y\to Z$. Is it true that \begin{equation}\label{eq:hf-rank} \operatorname{rk} HF(Y)\geq \operatorname{rk} HF(Z), \tag{6} \end{equation} where $HF$ is the $\widehat{HF}$ or $HF_{\mathrm{red}}$ version of Heegaard Floer homology?

背景链接：http://projecteuclid.org/euclid.ijm/1498032026

09172 | KP-3.63

### Kirby Problem 3.63

O / open | Kirby 低维拓扑问题

Give a method for computing the $\eta$ invariant for the Dirac operator, $\eta_{\mathrm{Dirac}}(Y, s)$, associated to a spin structure s on a hyperbolic 3-manifold Y.

背景链接：https://doi.org/10.1007/978-3-642-22842-1

09173 | KP-3.64

## Kirby Problem 3.64

P / partially_solved | Kirby 低维拓扑问题

Let A be a flat connection on the trivial $\operatorname{SU}(2)$ bundle on a closed three-manifold M. The Chern--Simons invariant $\operatorname{CS}(M, A)$ is defined to be

$$ \operatorname{CS}(M,A)=\frac{1}{4\pi^2}\int_M \operatorname{Tr}\!\left(A\wedge dA+\frac{2}{3}A\wedge A\wedge A\right). $$ Is $\operatorname{CS}(M, A)$ always rational?

背景链接：http://projecteuclid.org/euclid.cmp/1104252787

09174 | KP-3.65

## Kirby Problem 3.65

P / partially_solved | Kirby 低维拓扑问题

Let $S_{2,\infty}(Y)$ denote the Kauffman bracket skein module of a closed, oriented 3-manifold $Y$; this is a module over $R=\mathbb{Z}[A,A^{-1}]$. For $Y$ prime and connected, does $S_{2,\infty}(Y)$ having rank one, in the sense that

$$ S_{2,\infty}(Y)\otimes_R \mathbb{Q}(A)\cong \mathbb{Q}(A), $$

imply that $Y$ is $S^3$ or $S^1\times S^2$?

背景链接：https://bpb-us-e2.wpmucdn.com/websites.umass.edu/dist/b/22144/files/2026/04/K3-problem-list-watermarked.pdf

09175 | KP-3.66

## Kirby Problem 3.66

P / partially_solved | Kirby 低维拓扑问题

Suppose that $Y$ is a closed, oriented 3-manifold, and let $S_{2,\infty}(Y)$ denote the Kauffman bracket skein module over $R=\mathbb{Z}[A,A^{-1}]$ as in Problem 3.65.

(a) Is the dimension

$$ \dim_{\mathbb{Q}(A)} S_{2,\infty}(Y)\otimes_R \mathbb{Q}(A) $$

equal to the dimension of the degree 0 part of Abouzaid--Manolescu's sheaf-theoretic Floer homology $HP^{\bullet}_{\#}(Y)$ from [AM20]?

(b) When is $S_{2,\infty}(Y)\otimes_R\mathbb{Q}[A,A^{-1}]$ tame, as defined in Problem 3.65?

背景链接：https://bpb-us-e2.wpmucdn.com/websites.umass.edu/dist/b/22144/files/2026/04/K3-problem-list-watermarked.pdf

09176 | KP-3.67

## Kirby Problem 3.67

P / partially_solved | Kirby 低维拓扑问题

Categorify the Witten--Reshetikhin--Turaev invariants of 3-manifolds.

背景链接：http://projecteuclid.org/euclid.cmp/1104178138

09177 | KP-3.68

## Kirby Problem 3.68

P / partially_solved | Kirby 低维拓扑问题

(a) Give a mathematical definition of the $\widehat{Z}$ invariants for all 3-manifolds.

(b) Categorify the $\widehat{Z}$ invariants.

背景链接：https://bpb-us-e2.wpmucdn.com/websites.umass.edu/dist/b/22144/files/2026/04/K3-problem-list-watermarked.pdf

09178 | KP-3.69

## Kirby Problem 3.69

P / partially_solved | Kirby 低维拓扑问题

(a) What is the isomorphism type of $\Theta^3_{\mathbb{Z}}$?

(b) Does there exist a torsion element $[Y]$ in $\Theta^3_{\mathbb{Z}}$?

(c) Does there exist a torsion element $[Y]\in\Theta^3_{\mathbb{Z}}$ having Rokhlin invariant $\mu([Y])=1$?

(d) Do there exist infinitely-divisible elements in $\Theta^3_{\mathbb{Z}}$; that is, does there exist a homology 3-sphere $Y$ such that for infinitely many $n\in\mathbb{N}$ there exists $[Z_n]\in\Theta^3_{\mathbb{Z}}$ such that $[Y]=n[Z_n]$?

背景链接：http://gateway.proquest.com/openurl?url

09179 | KP-3.7

## Kirby Problem 3.7

P / partially_solved | Kirby 低维拓扑问题

Does every finite-volume hyperbolic 3-manifold admit a finitesheeted cover fibering over the circle with orientable pseudo-Anosov monodromy?

背景链接：https://elibm.org/article/10000267

09180 | KP-3.70

## Kirby Problem 3.70

O / open | Kirby 低维拓扑问题

Is $\Theta^3_{\mathbb{Z}}$ generated by the classes of knot surgeries $[S^3_{1/n}(K)]$, where $n$ ranges over all integers and $K$ ranges over all knots in $S^3$?

背景链接：https://bpb-us-e2.wpmucdn.com/websites.umass.edu/dist/b/22144/files/2026/04/K3-problem-list-watermarked.pdf

09181 | KP-3.71

### Kirby Problem 3.71

O / open | Kirby 低维拓扑问题

Is there a nontrivial element in the kernel of the natural map

$$ \Theta^3_{\mathbb{Z}}\longrightarrow \Theta^3_{\mathbb{Z}/2\mathbb{Z}}; $$

that is, does there exist an integer homology 3-sphere that does not bound a smooth $\mathbb{Z}$-homology 4-ball but does bound a smooth $\mathbb{Z}/2\mathbb{Z}$-homology 4-ball?

背景链接：https://bpb-us-e2.wpmucdn.com/websites.umass.edu/dist/b/22144/files/2026/04/K3-problem-list-watermarked.pdf

09182 | KP-3.72

### Kirby Problem 3.72

O / open | Kirby 低维拓扑问题

(a) Does the kernel of the map $\Theta^3_{\mathbb{Z}}\to\Theta^3_{\mathbb{Q}}$ contain a subgroup that is isomorphic to $\mathbb{Z}^{\infty}$?

(b) If so, does it contain such a subgroup as a summand?

背景链接：https://bpb-us-e2.wpmucdn.com/websites.umass.edu/dist/b/22144/files/2026/04/K3-problem-list-watermarked.pdf

09183 | KP-3.73

### Kirby Problem 3.73

P / partially_solved | Kirby 低维拓扑问题

(a) Calculate $\Theta^{\mathrm{TOP}}_{\mathbb{Z}/p}$.

(b) Calculate $\Theta^{\mathrm{TOP}}_{\mathbb{Q}}$.

(c) Is the linking form homomorphism $[\operatorname{lk}]$ injective?

背景链接：http://projecteuclid.org/euclid.jdg/1214437136

09184 | KP-3.74

### Kirby Problem 3.74

P / partially_solved | Kirby 低维拓扑问题

Let $Y$ be a rational homology sphere and $f:Y\to Y$ be a self-diffeomorphism of $Y$. Suppose $W$ is a 4-manifold with boundary $Y$ such that $f$ extends to a self-diffeomorphism of $W$.

(a) What constraints are there on the intersection form of $W$?

(b) Does there exist a pair $(Y,f)$ such that $Y$ bounds a homology ball but $f:Y\to Y$ does not extend over any definite manifold of either sign?

(c) Develop methods for constraining the intersection form of $W$ that apply to indefinite, non-spin $W$.

背景链接：http://projecteuclid.org/euclid.jdg/1214446320

09185 | KP-3.75

### Kirby Problem 3.75

P / partially_solved | Kirby 低维拓扑问题

Let $Y$ be a rational homology 3-sphere equipped with an action of a cyclic group $\mathbb{Z}/p\mathbb{Z}$. Suppose $W$ is a 4-manifold with boundary $Y$ over which the action of $\mathbb{Z}/p\mathbb{Z}$ extends smoothly.

(a) What constraints are there on the intersection form of $W$?

(b) Does there exist an integer homology sphere $Y$ bounding a homology ball such that $Y$ admits a cyclic group action that does not extend over any definite manifold of either sign?

(c) Develop methods for constraining the intersection form of $W$ that apply to indefinite, non-spin $W$.

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09186 | KP-3.76

### Kirby Problem 3.76

P / partially_solved | Kirby 低维拓扑问题

What is the structure of the equivariant homology cobordism groups?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09187 | KP-3.77

### Kirby Problem 3.77

O / open | Kirby 低维拓扑问题

Does there exist a hyperbolic rational homology 3-sphere that is the totally geodesic boundary of a compact, orientable hyperbolic 4-manifold?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09188 | KP-3.78

### Kirby Problem 3.78

P / partially_solved | Kirby 低维拓扑问题

(a) Is there a non-semisimple 3-TQFT whose mapping class group representation is faithful or has an element in its kernel?

(b) Define a 4-manifold invariant via non-semsimple categories that can distinguish 4-manifolds that are not distinguished by classical or gauge-theoretic invariants.

(c) Is there is a full 3-TQFT extension of the representations, constructed by Bonahon and Wong, of the Kauffman bracket skein algebra? In particular, do non-semisimple TQFTs recover the Bonahon--Wong representations?

背景链接：http://www.numdam.org/item?id=PMIHES

09189 | KP-3.8

### Kirby Problem 3.8

P / partially_solved | Kirby 低维拓扑问题

If $M_1$ and $M_2$ are finite-volume hyperbolic 3-manifolds whose fundamental groups have isomorphic profinite completions, must $M_1$ and $M_2$ be isometric?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09190 | KP-3.9

### Kirby Problem 3.9

P / partially_solved | Kirby 低维拓扑问题

Is being Haken a profinite invariant amongst 3-manifolds? That is, if $M_1$ and $M_2$ are 3-manifolds so that $\pi_1(M_{1})$ and $\pi_1(M_{2})$ have isomorphic profinite completions, and $M_1$ is Haken, must $M_2$ also be Haken?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09191 | KP-4.1

### Kirby Problem 4.1

O / open | Kirby 低维拓扑问题

(4-dimensional Poincaré conjecture). Is there a unique smooth structure on the 4-sphere?

背景链接：http://projecteuclid.org/euclid.jdg/1214437136

09192 | KP-4.10

### Kirby Problem 4.10

P / partially_solved | Kirby 低维拓扑问题

(a) Is every homotopy $B^{4}$ with boundary $S^{3}$ obtained by performing a Gluck twist on some knotted 2-sphere in $B^{4}$?

(b) Suppose a homotopy 4-ball X is obtained by performing a Gluck twist on a knotted 2-sphere in $B^{4}$. Is $X \times I$ necessarily diffeomorphic to $B^{5}$?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09193 | KP-4.100

### Kirby Problem 4.100

P / partially_solved | Kirby 低维拓扑问题

Given a closed, connected, symplectic 4-manifold (X, $\omega)$ and $c \in H_{2}(X,\mathbb{Z})$ represented by an embedded, connected, oriented, smooth surface S such that

(i) $x[\omega]$, cy $>$ 0, and

(ii) $xc_{1}(X, \omega)$, cy $=\chi(S) +S^{2}$, is c represented by an embedded, connected, symplectic surface?

背景链接：http://projecteuclid.org/euclid.jdg/1214425535

09194 | KP-4.101

### Kirby Problem 4.101

P / partially_solved | Kirby 低维拓扑问题

Is every smooth symplectic surface in $(\mathbb{CP}^{2}, \omega_{FS})$ symplectically isotopic to a complex curve? Equivalently, is there a unique symplectic isotopy class of smoothly embedded symplectic surfaces of degree d in $(\mathbb{CP}^{2}, \omega_{FS})$?

背景链接：http://projecteuclid.org/euclid.jdg/1214425276

09195 | KP-4.102

### Kirby Problem 4.102

P / partially_solved | Kirby 低维拓扑问题

Is every symplectic rational cuspidal curve in $(\mathbb{CP}^{2}, \omega_{FS})$ equisingularly symplectically isotopic to a complex curve? More generally, which types of singular symplectic surfaces in $(\mathbb{CP}^{2}, \omega_{FS})$ (where type is specified by the genus of each irreducible component and the topological types of the singularities), admit symplectic representatives that are not equisingularly symplectically isotopic to complex curves?

背景链接：http://www.numdam.org/item?id=BSMF

09196 | KP-4.103

### Kirby Problem 4.103

P / partially_solved | Kirby 低维拓扑问题

(a) What polynomials can occur as the Alexander polynomials of complex plane algebraic curves?

(b) More generally, what are the conditions that must be satisfied by a finitely presented group G, so that there exist a plane algebraic curve having Gas the fundamental group of its complement?

背景链接：http://projecteuclid.org/euclid.dmj/1077315533

09197 | KP-4.104

### Kirby Problem 4.104

O / open | Kirby 低维拓扑问题

Does there exist a transverse link $L \subset (S^{3}, \xi_{std})$ bounding a pair of complex curves in $B^{4} \subset \mathbb{C}^{2}$ that are isotopic through embedded smooth surfaces but not through complex curves? Smoothly isotopic but not symplectically? Can there be infinitely many of them?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09198 | KP-4.105

### Kirby Problem 4.105

P / partially_solved | Kirby 低维拓扑问题

Does there exist a planar contact 3-manifold that has infinitely many distinct Stein fillings?

背景链接：http://projecteuclid.org/euclid.jsg/1339096433

09199 | KP-4.106

### Kirby Problem 4.106

O / open | Kirby 低维拓扑问题

Is the exact symplectomorphism type of $T*X^{4}$ sensitive to the smooth structure on a 4-manifold X, or does it depend only on the simple-homotopy or homeomorphism type of X?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09200 | KP-4.107

### Kirby Problem 4.107

P / partially_solved | Kirby 低维拓扑问题

Problems about contact hypersurfaces:

(a) Let (Y, $\xi)$ be a contact manifold and $(\mathbb{R} \times Y, \omega)$ its symplectization. Let f:Y $\to \mathbb{R} \times Y$ be a smooth embedding such thatf induces an isomorphism on homology. When can we isotope f so that its image is a contact type hypersurface?

(b) Special case: Is every embedded 3-sphere $Y \subset (\mathbb{R}^{4}, \omega_{std})$ smoothly isotopic to a hypersurface of contact type?

(c) Is there any contact rational homology sphere other than $(S^{3}, \xi_{std})$ which embeds as a contact type hypersurface in $(\mathbb{R}^{4}, \omega_{std}))$?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09201 | KP-4.108

### Kirby Problem 4.108

O / open | Kirby 低维拓扑问题

Let $W_{+}$ and $W_{-}$ be two 4-dimensional Liouville domains with a contactomorphism $\Phi :\partial W_{-} \cong \partial W_{+}$. This determines an $\mathbb{R}$-invariant$ contact structure $\xi$ on $\mathbb{R} \times X$ where X is the gluing $X =W_{+} \cup _{\Phi}W_{-}$ Moreover, 0 $\times X \subset \mathbb{R} \times X$ is a convex hypersurface in the sense of Giroux.

(a) Is there a 4-dimensional version of Giroux's criterion in the case where $W_{+}$ and $W_{-}$ are Weinstein? That is, a necessary and sufficient topological criterion on $(\Phi, W_{+}, W_{-})$ for $\xi$ to be overtwisted.

(b) Does every 4-manifold X admit a decomposition $W_{+} \cup _{\Gamma} W_{-}$ so that the corresponding contact structure $\xi$ is tight (i.e. not overtwisted)?

(c) Is there a 4-dimensional Liouville domain W and a contactomorphism $\Phi$ : $\partial W \to \partial W$ that extends to a diffeomorphism $\Psi$ of W, but so that the contact structure corresponding to $(\Phi, W$, W) is overtwisted?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09202 | KP-4.109

### Kirby Problem 4.109

P / partially_solved | Kirby 低维拓扑问题

If $\Sigma \subset$ (X, $\omega)$ is a symplectic surface in a closed symplectic 4-manifold with $[\Sigma] = P D(k[\omega]), k \in \mathbb{Z}$, does (X $\setminus \Sigma, \omega)$ support a Weinstein structure?

背景链接：http://projecteuclid.org/euclid.jdg/1214459407

09203 | KP-4.11

### Kirby Problem 4.11

O / open | Kirby 低维拓扑问题

Let M be a smooth 4-manifold and letf: $S^{2} \to M$ be a smooth embedding with trivial normal bundle. Then let $M_{f}$ denote the result of Gluck twisting on M along f.

(a) Does there exist an orientable M, and an embedding f, such that M and $M_{f}$ are homeomorphic but not diffeomorphic?

(b) Does there exist an orientable M, and smooth, homotopic embeddings f, $g: S^{2} \to M$ with trivial normal bundle such that $M_{f}$ and $M_{g}$ are homeomorphic but not diffeomorphic?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09204 | KP-4.110

### Kirby Problem 4.110

O / open | Kirby 低维拓扑问题

Does there exist a 2-handlebody W that admits an exact symplectic structure with convex contact boundary that does not admit a Weinstein structure filling the same contact boundary?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09205 | KP-4.111

### Kirby Problem 4.111

P / partially_solved | Kirby 低维拓扑问题

Is trisection genus additive? In other words, must it be the case that $g(X\#X1)$ =g(X) +g(X1).

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09206 | KP-4.112

### Kirby Problem 4.112

P / partially_solved | Kirby 低维拓扑问题

Is every trisection of the 4-sphere with positive genus a stabilization of the genus zero trisection?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09207 | KP-4.113

### Kirby Problem 4.113

P / partially_solved | Kirby 低维拓扑问题

Which closed, oriented, smooth 4-manifolds admit genus-3 trisections? Which ones admit genus-3 simplified trisections? How about genus-4?

背景链接：https://projecteuclid.org/euclid.ojm/1579079109

09208 | KP-4.114

### Kirby Problem 4.114

O / open | Kirby 低维拓扑问题

For a given Heegaard splitting of a closed orientable3-manifold, classify self-indexing Morse functions that give the given Heegaard splitting, up to $C\infty$ right-left (or right) equivalence. Likewise, for a given (simplified) trisection of a closed orientable 4-manifold, classify generic maps that give the given trisection, up to right-left (or right) equivalence.

背景链接：https://projecteuclid.org/euclid.ojm/1579079109

09209 | KP-4.115

### Kirby Problem 4.115

O / open | Kirby 低维拓扑问题

(a) Find two diffeomorphic but non-isotopic trisections of the same4-manifold.

(b) Find two non-diffeomorphic balanced trisections of the same genus on a closed simply connected 4-manifold.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09210 | KP-4.116

### Kirby Problem 4.116

P / partially_solved | Kirby 低维拓扑问题

Is there an algorithm to compute ‘distance’ in the cut complex of a trisection? Is the L-invariant computable?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09211 | KP-4.117

### Kirby Problem 4.117

P / partially_solved | Kirby 低维拓扑问题

Let X be a closed, orientable, smooth 4-manifold, with $g(X)$ the trisection genus of X. Does $g(X) =\chi(X) -2+3rk(\pi_{1}(X))$?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09212 | KP-4.118

### Kirby Problem 4.118

P / partially_solved | Kirby 低维拓扑问题

Does every simply connected, closed, smooth 4-manifold admit a handle decomposition without any 1-handles? Without 1-handles and 3-handles?

背景链接：http://projecteuclid.org/euclid.mmj/1029002261

09213 | KP-4.119

### Kirby Problem 4.119

P / partially_solved | Kirby 低维拓扑问题

Is every topological 4-manifold homeomorphic to a CW complex?

背景链接：http://projecteuclid.org/euclid.jdg/1214437139

09214 | KP-4.12

### Kirby Problem 4.12

P / partially_solved | Kirby 低维拓扑问题

For X a closed simply connected smooth 4-manifold, let $g_{X}: H_{2}(X) \to \mathbb{N}$ denote the genus function, which assigns to every homology class the minimal genus of a smooth embedded surface representing that class.

(a) Suppose $f: X_{1} \to X_{2}$ is a homeomorphism between simply connected, smooth, closed 4-manifolds. Let $f_{*}: H_{2}(X_{1}) \to H_{2}(X_{2})$ be the induced map on homology. If for all x in $H_{2}(X_{1})$ we have $g_{X,1}(x) = g_{X,2}(f_{*}(x))$, is $X_{1}$ diffeomorphic to $X_{2}$?

(b) Suppose $f: X_{1} \to X_{2}$ is a homeomorphism between simply connected smooth compact 4-manifolds with the same boundary $\partial X_{1} = \partial X_{2}$, such thatf is the identity on the boundary. $Letf_{*}: H_{2}(X_{1},\partial X_{1}) \to H_{2}(X_{2},\partial X_{2})$ be the induced map on relative homology. If for all knots K and for all x in $H_{2}(X_{1},\partial X_{1})$ we have $g_{X,1,K}(x) =g_{X,2,K}(f_{*}(x))$, is $X_{1}$ diffeomorphic to $X_{2}$ rel. boundary?

背景链接：https://arxiv.org/abs/2307.04202

09215 | KP-4.120

### Kirby Problem 4.120

P / partially_solved | Kirby 低维拓扑问题

Which closed, smooth 4-manifolds admit achiral Lefschetz pencils? Does every simply connected 4-manifold have one?

背景链接：https://arxiv.org/abs/math/0510008

09216 | KP-4.121

### Kirby Problem 4.121

P / partially_solved | Kirby 低维拓扑问题

Which closed, smooth4-manifolds admit open book decompositions? In particular, does every closed, simply connected 4-manifold with signature zero admit one?

背景链接：https://arxiv.org/abs/2504.10975

09217 | KP-4.122

### Kirby Problem 4.122

P / partially_solved | Kirby 低维拓扑问题

Is there a universal branching surface $S \subset S^{4}$ such that every closed, orientable 4-manifold W admits a branched covering $W \to S^{4}$ with branchin g locus S?

背景链接：https://doi.org/10.1112/S0024611504015072

09218 | KP-4.123

## Kirby Problem 4.123

P / partially_solved | Kirby 低维拓扑问题

Is every closed leaf of a two dimensional co-orientable smooth taut foliation of an oriented 4-manifold smoothly genus-minimizing in its homology class?

背景链接：https://doi.org/10.2307/121113

09219 | KP-4.124

## Kirby Problem 4.124

O / open | Kirby 低维拓扑问题

Does there exist a hyperbolic integer homology four-sphere? What about an arithmetic one? Homology four-spheres have Euler characteristic 2, so it makes sense to ask more generally if there exist any closed hyperbolic fourmanifold with Euler characteristic 2.

背景链接：http://projecteuclid.org/euclid.em/1046889595

09220 | KP-4.125

## Kirby Problem 4.125

P / partially_solved | Kirby 低维拓扑问题

Is there a noncompact, finite volume, orientable hyperbolic four-manifold without a spin structure?

背景链接：https://arxiv.org/abs/2510.12657

09221 | KP-4.126

## Kirby Problem 4.126

P / partially_solved | Kirby 低维拓扑问题

(a) If M is a closed, orientable hyperbolic 4-manifold then it always has signature 0, because its Pontryagin class vanishes [Che55]. This implies that M bounds a compact, orientable 5-manifold. Is M always a geometric boundary?

(b) For general n, suppose that M is a closed hyperbolic n-manifold. Is M a geometric boundary?

背景链接：http://projecteuclid.org/euclid.ojm/1200782445

09222 | KP-4.127

## Kirby Problem 4.127

P / partially_solved | Kirby 低维拓扑问题

Given an aspherical closed (or compact and bounded by flat 3-manifolds) 4-manifold M and a self-diffeomorphism f of M, find necessary and sufficient conditions on f so that the resulting 5-dimensional mapping torus $M_{f}$ admits a hyperbolic structure.

背景链接：https://doi.org/10.1007/s00222-022-01141-w

09223 | KP-4.128

## Kirby Problem 4.128

P / partially_solved | Kirby 低维拓扑问题

What is the structure of 4-manifolds that admit a Riemannian metric of positive scalar curvature? There are variations of this problem for different classes of manifolds.

(a) Is every closed simply connected PSC 4-manifold diffeomorphic to a connected sum of copies of $\mathbb{CP}^{2}, \mathbb{CP}^{2}$, and $S^{2} \times S^{2}$?

(b) What is the structure of non-simply connected closed PSC 4-manifolds?

(c) Which 4-manifolds with boundary have a PSC metric?

(d) Which non-compact 4-manifolds have a complete PSC metric with uniformly positive scalar curvature?

背景链接：https://londmathsoc.onlinelibrary.wiley.com/doi/abs/10.1112/topo.12117

09224 | KP-4.129

## Kirby Problem 4.129

P / partially_solved | Kirby 低维拓扑问题

Given a closed, 4-dimensional PSC manifold M, is there a (possibly disconnected) 4-dimensional orbifold $M^{1}$ with isolated singularities such that the following hold. (I) The orbifold M1 also admits a metric of positive scalar curvature. (II) All components of $M^{1}$ have finite orbifold-fundamental group. (III) M can be obtained from $M^{1}$ by a series of 0 and 1 surgeries. Here we also allow 0-surgeries between two orbifold points of the same type, which amounts to a removal of two subsets of the for $m D^{4}/\Gamma$ and an addition of a copy of $S^{3}/\Gamma \times$ [0,1].

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09225 | KP-4.13

## Kirby Problem 4.13

O / open | Kirby 低维拓扑问题

(a) Does every large $\mathbb{R}^{4}-homeomorph$ lie $in\mathcal{R}_{K}$ for some Kthat is not smoothly slice?

(b) Does there exist an infinite sequence of knots ${K_{i}}$ such that $\mathcal{R}_{K,i} \ne \mathcal{R}_{K,j}$ whenever $i \ne j$?

背景链接：http://projecteuclid.org/euclid.jdg/1214439566

09226 | KP-4.130

## Kirby Problem 4.130

O / open | Kirby 低维拓扑问题

Does longitudinal knot surgery using a knot K, along a fiber in a K3 surface always yield a reducible 4-manifold? A completely decomposable 4-manifold?

背景链接：http://projecteuclid.org/euclid.jdg/1214440260

09227 | KP-4.131

### Kirby Problem 4.131

P / partially_solved | Kirby 低维拓扑问题

Does every Lipschitz 4-manifold admit a smooth structure? Is this smooth structure unique if so? Some more specific, related questions are as follows.

(a) Is there a topological, spin, closed, indefinite 4-manifold X that admits a Lipschitz structure and violates the 10/8-inequality $b_{2}(X) \geq 5|\sigma(X)|$ +2 by Furuta [Fur01], or the “10/8+4” inequality $b_{2}(X) \geq 5|\sigma(X)|$ +4 by Hopkins-Lin-Shi-Xu [HLSX22] for $X \ne S^{4}, S^{2} \times S^{2}, K3$.

(b) Let $X \subset \mathbb{R}^{4}$ be a Lipschitz embedded 4-manifold with boundary $\partial X \cong S^{3}$. Does X admit a unique smooth structure?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09228 | KP-4.14

### Kirby Problem 4.14

P / partially_solved | Kirby 低维拓扑问题

Is there a universal cork? More precisely, does there exist some cork (C, f) such that given any pair W and $W^{1}$ of closed, simply connected 4-manifolds that are homeomorphic but not diffeomorphic, there exists a smooth embedding $C \hookrightarrow W$ such that (W $\setminus$ Int C) $\cup _{f} C$ is diffeomorphic to $W^{1}$?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09229 | KP-4.15

### Kirby Problem 4.15

P / partially_solved | Kirby 低维拓扑问题

(11/8 Conjecture). Does every smooth, spin, closed 4-manifold X satisfy $b_{2}(X) \geq 11|\sigma(X)|$, where $\sigma(X)$ is the signature of the intersection form?

背景链接：http://projecteuclid.org/euclid.jdg/1214441485

09230 | KP-4.16

### Kirby Problem 4.16

P / partially_solved | Kirby 低维拓扑问题

(a) Do there exist closed, oriented, smooth, irreducible 4-manifolds with $b^{+}_{2} >$ 1 and $c^{2}_{1}:=2\chi+3\sigma<0$?

(b) Is there an irreducible exotic smooth structure on $\mathbb{CP}^{2}\#_{n}\mathbb{CP}^{2}$ for n>9?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09231 | KP-4.17

### Kirby Problem 4.17

O / open | Kirby 低维拓扑问题

Is there an irreducible, closed, simply connected, oriented 4-manifold with $b^{+}_{2}$ and $b^{-}_{2}$ both even?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09232 | KP-4.18

### Kirby Problem 4.18

O / open | Kirby 低维拓扑问题

(a) Does there exist a pair of smooth, closed 4-manifolds that are homotopy equivalent but not simple homotopy equivalent?

(b) Does there exist a pair of smoothlyh-cobordant, smooth, closed 4-manifolds that are not smoothly s-cobordant?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09233 | KP-4.19

### Kirby Problem 4.19

P / partially_solved | Kirby 低维拓扑问题

What are the possible Euler characteristics of closed, aspherical 4-manifolds? More specifically, we ask the following.

(a) Is it always the case that $\chi \geq |\sigma|$?

(b) What is the smallest Euler characteristic of a closed hyperbolic 4-manifold?

背景链接：http://projecteuclid.org/euclid.em/1046889595

09234 | KP-4.2

### Kirby Problem 4.2

P / partially_solved | Kirby 低维拓扑问题

Does every smooth, closed 4-manifold admit an exotic smooth structure? Infinitely many?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09235 | KP-4.20

### Kirby Problem 4.20

O / open | Kirby 低维拓扑问题

Is $*\mathbb{RP}^{4}\#*\mathbb{RP}^{4}$ smoothable? Is *En\#*En smoothable?

背景链接：http://projecteuclid.org/euclid.jdg/1214437136

09236 | KP-4.21

### Kirby Problem 4.21

P / partially_solved | Kirby 低维拓扑问题

Is every topological closed 4-manifold M the union of submanifolds $Y \cup Z$, where Y is smoothable, Z is acyclic, and $Y \cap Z$ is their common boundary, a homology sphere $\Sigma$?

背景链接：http://projecteuclid.org/euclid.jdg/1214437136

09237 | KP-4.22

### Kirby Problem 4.22

O / open | Kirby 低维拓扑问题

Let $\pi$ be a good group, and let X be a smooth 4-manifold with $\pi_{1}(X) = \pi$. Does $L^{s}_{5}(\mathbb{Z}[\pi])$ act on the smooth structure set of X, in such a way that the action reduces under the forgetful map to the Wall realization action on the topological structure set? In particular, can this be done when $\pi= \mathbb{Z}, \mathbb{Z} \times \mathbb{Z}_{n}$, or $\mathbb{Z}_{2}*\mathbb{Z}_{2}$?

背景链接：http://projecteuclid.org/euclid.jdg/1320067650

09238 | KP-4.23

### Kirby Problem 4.23

P / partially_solved | Kirby 低维拓扑问题

(Schoenflies problem). If $\Sigma$ is a smoothly embedded $S^{3}$ in $S^{4}$, then its closed complements are smooth 4-balls.

背景链接：http://projecteuclid.org/euclid.jdg/1214441488

09239 | KP-4.24

### Kirby Problem 4.24

O / open | Kirby 低维拓扑问题

Let K be a framed knot in $S^{3} = \partial B^{4}$. Let U be a meridian of K. Does there exist a smoothly embedded disk D in $B^{4} \cup _{\nu K} h^{2}$ bounded by U that intersects the cocore of $h^{2}$ algebraically zero times, such that the 4-manifold obtained by removing an open tubular neighborhood of the disk is not diffeomorphic to the 4-ball?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09240 | KP-4.25

### Kirby Problem 4.25

P / partially_solved | Kirby 低维拓扑问题

Under what conditions does a closed, orientable 3-manifold M smoothly embed in $S^{4}$? Is this question algorithmically decidable?

背景链接：http://projecteuclid.org/euclid.jdg/1214437136

09241 | KP-4.26

### Kirby Problem 4.26

P / partially_solved | Kirby 低维拓扑问题

If Y is a homology three-sphere, does the punctured manifold $Y_{0}$ =Y $\setminus \operatorname{Int}(B^{3})$ smoothly embed in $S^{4}$?

背景链接：https://repositories.lib.utexas.edu/server/api/core/bitstreams/32f94088-e19b-4b49-8dd8-7b2320ef72e3/content

09242 | KP-4.27

### Kirby Problem 4.27

O / open | Kirby 低维拓扑问题

Find exotic 3-balls in $S^{4}$, considered up to isotopy rel. boundary. That is, find a pair of 3-balls $B_{1}, B_{2}$ smoothly embedded in $S^{4}$ with the same boundary such that $B_{1}, B_{2}$ are topologically but not smoothly isotopic rel. boundary.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09243 | KP-4.28

### Kirby Problem 4.28

P / partially_solved | Kirby 低维拓扑问题

Every closed, orientable 3-manifold embeds smoothly in some connected sum of copies of $S^{2} \times S^{2}$. Given a closed 3-manifold M, let $s(M) \geq$ 0 denote the smallest integer such that M embeds smoothly (respectively locally flatly) in a connected sum of $s(M)$ copies of $S^{2} \times S^{2}$. Let $s(M) \geq$ 0 denote the smallest integer such that M embeds smoothly (respectively locally $flatly^{~})$ in a connected sum of $s_{~}(M)$ copies of $S^{2} \times _{~}S^{2}$. Compute the functions $s(M)$ and $s_{~}(M)$, in either the smooth or locally flat case, for interesting classes of 3-manifolds. Here are three more specific questions.

(a) What are the $functionssands_{~}for$ lens spaces, and for Brieskorn spheres?

(b) Freedman [Fre82] proved that every $\mathbb{Z}$-homology$ sphere embeds topologically in $S^{4}$. Is there an integer homology 3-sphere such that (smoothly) $s(M)$ =1?

(c) Are the functions s or $s_{~}algorithmically$ computable?

背景链接：http://projecteuclid.org/euclid.jdg/1214437136

09244 | KP-4.29

### Kirby Problem 4.29

P / partially_solved | Kirby 低维拓扑问题

Let $\Sigma$ be a locally flat surface in $S^{4}$ with $\pi_{1}(S^{4} \setminus \Sigma)$ cyclic.

(a) Prove that $\Sigma$ is topologically unknotted.

(b) Assuming that $\Sigma$ is smoothly embedded and orientable, prove that $\Sigma$ is smoothly unknotted (or find an example that is not).

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09245 | KP-4.3

### Kirby Problem 4.3

P / partially_solved | Kirby 低维拓扑问题

Are there exotic smooth structures on the following closed, simply-connected 4-manifolds?

(a) $\#_{k}\mathbb{CP}^{2}$ for any $k \geq$ 1.

(b) $\#_{m}(\mathbb{CP}^{2}\#\overline{\mathbb{CP}}{}^{2})$ and $\#_{n}(S^{2} \times S^{2})$ for $m \leq$ 8, $n \leq$ 10.

(c) $\#_{p}K3 \#_{q}(S^{2} \times S^{2})$ for any $p \geq$ q+5. If so, what are the smallest k, m and n? The largest p and q?

背景链接：https://msp.org/obs/2022/5-1/p09.xhtml

09246 | KP-4.30

### Kirby Problem 4.30

O / open | Kirby 低维拓扑问题

Does there exist a pair of closed, oriented surfaces in $S^{4}$ that are topologically but not smoothly isotopic? If such an exotic pair exists, does there exist an infinite family? Can the surfaces be taken to be 2-spheres? Surfaces of genus-g for any given $g \geq$ 0?

背景链接：http://projecteuclid.org/euclid.jdg/1214441179

09247 | KP-4.31

### Kirby Problem 4.31

P / partially_solved | Kirby 低维拓扑问题

Does every knot in $S^{3}$ bound an exotic pair of orientable surfaces in $B^{4}$?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09248 | KP-4.32

### Kirby Problem 4.32

P / partially_solved | Kirby 低维拓扑问题

Does there exist a locally flat embedding $f: \Sigma \to S^{4}$ for some closed surface $\Sigma$ such that f is not topologically ambiently isotopic to a smooth embedding? Particularly in the case that $\Sigma=S^{2}$?

背景链接：http://muse.jhu.edu/journals/american

09249 | KP-4.33

### Kirby Problem 4.33

P / partially_solved | Kirby 低维拓扑问题

Let $S_{1}, S_{2}$ be two topologically isotopic, smoothly embedded closed surfaces in a closed, oriented, smooth 4-manifold X. When do $S_{1}$ and $S_{2}$ in X become smoothly isotopic after one

(a) external stabilization?

(b) standard internal stabilization?

(c) internal stabilizations? Is there an upper bound depending on X on the number of such stabilizations required to make the surfaces smoothly isotopic?

背景链接：https://doi.org/10.4171/jems/1541

09250 | KP-4.34

### Kirby Problem 4.34

P / partially_solved | Kirby 低维拓扑问题

Let $\Sigma$ be a surface embedded in $S^{4}$. Can $\Sigma$ be unknotted by a sequence of torus surgeries in its complement, such that the ambient manifold is always $S^{4}$?

背景链接：https://math.berkeley.edu/sites/default/files/surv-295-ruberman-watermarked-author-pdf.pdf

09251 | KP-4.35

### Kirby Problem 4.35

P / partially_solved | Kirby 低维拓扑问题

(a) Give an algebraic classification of groups that arise as the fundamental group of the complement of a smooth or locally flat 2-sphere in $S^{4}$.

(b) Let $\Lambda = \mathbb{Z}[t, t-^{1}]$. Classify the $\Lambda-modules$ that arise as the Alexander module of a smooth or locally flat 2-sphere in $S^{4}$.

背景链接：http://www.numdam.org/item?id=BSMF

09252 | KP-4.36

### Kirby Problem 4.36

P / partially_solved | Kirby 低维拓扑问题

Are homotopy types of 2-knot complements determined by their homotopy 2-types?

背景链接：http://projecteuclid.org/euclid.pjm/1102735075

09253 | KP-4.37

### Kirby Problem 4.37

P / partially_solved | Kirby 低维拓扑问题

(Kinoshita conjecture). Does every projective plane in $S^{4}$ decompose as a connected sum of a knotted 2-sphere and an unknotted projective plane?

背景链接：http://projecteuclid.org/euclid.ojm/1200785461

09254 | KP-4.38

### Kirby Problem 4.38

P / partially_solved | Kirby 低维拓扑问题

Let $\Delta \subset B^{4}$ be a ribbon disk. Is $B^{4} \setminus \Delta$ aspherical, i.e. is $\pi_{i}(B^{4} \setminus \Delta)$ =0 for all i>1?

背景链接：http://projecteuclid.org/euclid.pjm/1102735075

09255 | KP-4.39

### Kirby Problem 4.39

P / partially_solved | Kirby 低维拓扑问题

Let K be a closed surface smoothly embedded in a 4-manifold $M^{4}$. Describe the subgroup of the mapping class group $Mod(K)$ of diffeomorphisms of K that extend to diffeomorphisms of M. Conversely, is it possible to characterize when a given subgroup of the mapping class group arises as the group of extendable diffeomorphisms for some K inside some $M^{4}$?

背景链接：https://arxiv.org/abs/2502.17640

09256 | KP-4.4

### Kirby Problem 4.4

O / open | Kirby 低维拓扑问题

Is there an exotic smooth structure on some product 4-manifold $S^{1} \times Y^{3}$ or $\Sigma_{g} \times \Sigma_{h}$? Do they all admit exotic smooth structures?

背景链接：https://msp.org/obs/2022/5-1/p09.xhtml

09257 | KP-4.40

### Kirby Problem 4.40

P / partially_solved | Kirby 低维拓扑问题

Is every link of 2-spheres in $S^{4}$ slice?

背景链接：http://www.numdam.org/item?id=BSMF

09258 | KP-4.41

### Kirby Problem 4.41

P / partially_solved | Kirby 低维拓扑问题

Let $\Sigma$ be a compact surface and let X be a connected 4manifold. Let $f_{0}, f_{1}: \Sigma \to X$ be $\pi_{1}$-negligible$ embeddings that agree on $\partial\Sigma$ and that are homotopic rel. boundary. Give computable invariants that decide whether there is $a \pi_{1}$-negligible$ concordance in $X \times I$ between $f_{0}$ and $f_{1}$.

背景链接：https://doi.org/10.1112/topo.12191

09259 | KP-4.42

### Kirby Problem 4.42

P / partially_solved | Kirby 低维拓扑问题

(a) If X is a smooth, closed, simply connected 4-manifold $withb_{2}(X) \geq$ 2, are all knots slice in X?

(b) In particular, are all knots slice in $\mathbb{CP}^{2}\#\overline{\mathbb{CP}}{}^{2}$? In the K3 surface?

(c) Can the set of slice knots detect exotic smooth structures? That is, do there exist homeomorphic 4-manifolds X and $X^{1}$ and a knot $K \subset S^{3}$ that is slice in X but not in $X^{1}$?

背景链接：http://projecteuclid.org/euclid.pja/1195511928

09260 | KP-4.43

### Kirby Problem 4.43

P / partially_solved | Kirby 低维拓扑问题

Let K be a knot on the boundary of $X \setminus B^{\circ 4}$, where X is a negative definite, smooth 4-manifold. Suppose K bounds a smoothly embedded, nullhomotopic surface $\Sigma$ in $X \setminus B^{\circ 4}$. Is the following inequality $2g(\Sigma) \geq$ |s(K)| (8) always satisfied? Here $s(K)$ denotes Rasmussen's s-invariant [Ras10].

背景链接：https://doi.org/10.1215/00127094-2022-0039

09261 | KP-4.44

### Kirby Problem 4.44

P / partially_solved | Kirby 低维拓扑问题

(a) Let X be a closed, simply connected 4-manifold. Let $g \geq$ 0 and $d \geq$ 0 be integers. Fix $x \in H_{2}(X;\mathbb{Z})$. Does there exist a (smooth, locally flat) generic immersion $f: \Sigma_{g} \to X$ such that $f_{*}[\Sigma_{g}] = x$, with d transverse double points?

(b) An important special case is $d =$ 0. Let $g: H_{2}(X;\mathbb{Z}) \to \mathbb{N}_{0}$ be the genus function assigning to $x \in H_{2}(X;\mathbb{Z})$ the minimal genus of any (smooth, locally flat) embedded surface in X whose fundamental class represents x. The problem is to compute the locally flat genus function for some 4-manifold $X \ne S^{4}$, and to compute the smooth genus function for new 4-manifolds.

(c) For a specific example, is the class (3,2) $\in H_{2}(\mathbb{CP}^{2}\#\mathbb{CP}^{2};\mathbb{Z})$ represented by a sphere?

背景链接：http://muse.jhu.edu/journals/american

09262 | KP-4.45

### Kirby Problem 4.45

P / partially_solved | Kirby 低维拓扑问题

Let X be a closed simply connected smooth 4-manifold and let $\Sigma$ be a closed, orientable surface. Fix a smooth embedding of $f: \Sigma \hookrightarrow X$ to serve as a basepoint. For which $k\geq 0$ does the equality $$ \ker i_{*}=\ker s_{*} $$ hold, where $$ \begin{aligned} i_{*}&:\pi_{k}(\operatorname{Emb}^{DIFF}(\Sigma,X^{\circ});f)\to \pi_{k}(\operatorname{Emb}^{TOP}(\Sigma,X^{\circ});f),\\ s_{*}&:\pi_{k}(\operatorname{Emb}^{DIFF}(\Sigma,X^{\circ});f)\to \operatorname*{colim}_{N\to\infty}\pi_{k}(\operatorname{Emb}^{DIFF}(\Sigma,X^{\circ}\#_{N}S^{2}\times S^{2});f). \end{aligned} $$ Here $i_*$ is induced by inclusion and $s_*$ by stabilization. To make sense of this for k=0, define $keri_{*}$ and $kers_{*}$ to be the preimages under $i_{*}$ and $s_{*}$ respectively of [f].

背景链接：http://muse.jhu.edu/journals/american

09263 | KP-4.46

### Kirby Problem 4.46

P / partially_solved | Kirby 低维拓扑问题

Are all groups good?

背景链接：http://projecteuclid.org/euclid.jdg/1214441179

09264 | KP-4.47

### Kirby Problem 4.47

O / open | Kirby 低维拓扑问题

(Round handle problem). Is there a link $L \subset S^{3}$ with vanishing pairwise linking numbers that is not round handle slice?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09265 | KP-4.48

### Kirby Problem 4.48

P / partially_solved | Kirby 低维拓扑问题

Let $\Delta$ be a contractible, compact 4-manifold. Is the space of homeomorphisms of $\Delta$ that fix the boundary pointwise, with the compact-open topology, a contractible space?

背景链接：http://projecteuclid.org/euclid.jdg/1214440552

09266 | KP-4.49

### Kirby Problem 4.49

P / partially_solved | Kirby 低维拓扑问题

Give an effective necessary and sufficient condition for an open 4-manifold to be homeomorphic to the interior of a compact 4-manifold.

背景链接：http://gateway.proquest.com/openurl?url

09267 | KP-4.5

### Kirby Problem 4.5

P / partially_solved | Kirby 低维拓扑问题

Does every connected, open 4-manifold admit uncountably many smooth structures?

背景链接：http://projecteuclid.org/euclid.jdg/1214437139

09268 | KP-4.50

### Kirby Problem 4.50

P / partially_solved | Kirby 低维拓扑问题

Classify closed topological 4-manifolds (orientable and not) with finite fundamental group, up to homeomorphism. The following infinite families of fundamental groups are of particular interest.

(a) Dihedral groups.

(b) Quaternionic groups.

(c) Abelian groups with two or three generators. A significant starting point would be to complete the classification for 4-manifolds with fundamental group of order at most eight.

背景链接：http://projecteuclid.org/euclid.jdg/1214437136

09269 | KP-4.51

### Kirby Problem 4.51

P / partially_solved | Kirby 低维拓扑问题

Let X be a closed, smooth 4-manifold with fundamental group isomorphic to $\mathbb{Z}$. Is the $\mathbb{Z}[\mathbb{Z}]$-valued$ intersection form on $\pi_{2}(X)$ extended from $\mathbb{Z}$?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09270 | KP-4.52

### Kirby Problem 4.52

P / partially_solved | Kirby 低维拓扑问题

Let M and N be closed, orientable topological 4-manifolds with $\pi_{1}(M) \cong \pi_{1}(N)a$ good group. Suppose that M and N are simple homotopy equivalent and stably homeomorphic. Is there a simple homotopy equivalence $N \to M$ that lies in the orbit of Id $:M \to M$ under the Wall realization action of $L^{s}_{5}(\mathbb{Z}[\pi_{1}(M)])$ on the simple structure set of M?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09271 | KP-4.53

### Kirby Problem 4.53

O / open | Kirby 低维拓扑问题

Does there exist an algorithm that takes as input a closed, triangulated 4-manifold, and outputs in finite time whether or not that 4-manifold is homeomorphic to $S^{4}$?

背景链接：https://celebratio.org/Scharlemann

09272 | KP-4.54

### Kirby Problem 4.54

P / partially_solved | Kirby 低维拓扑问题

The quadratic 2 type of a 4-manifold M is the data $(\pi_{1}(M), \pi_{2}(M), \lambda_{M}, k_{M})$ of the fundamental group $\pi_{1}(M)$, the second homotopy group $\pi_{2}(M)$ considered as $a \mathbb{Z}[\pi_{1}(M)]-module$, the equivariant intersection for $m \lambda_{M}$, and the k-invariant in $k_{M} \in H^{3}(B\pi_{1}(M);\pi_{2}(M))$. Which quadratic 2-types are realized by closed, oriented topological 4-manifolds? Which are realized by closed, oriented, smooth 4-manifolds? Can this problem be solved for specific fundamental groups, for example for certain families of non-cyclic finite groups?

背景链接：http://projecteuclid.org/euclid.jdg/1214437136

09273 | KP-4.55

### Kirby Problem 4.55

P / partially_solved | Kirby 低维拓扑问题

Let M and N be closed, orientable, connected 4-manifolds with isomorphic quadratic 2-types. If $\pi_{1}(M) \cong \pi_{1}(N)are$ finite, are M and N homotopy equivalent?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09274 | KP-4.56

### Kirby Problem 4.56

O / open | Kirby 低维拓扑问题

(4D s-cobordism conjecture). Let $(W^{4};M_{0}^{3}, M_{1}^{3})$ be a smooth 4-dimensionals-cobordism between closed 3-manifolds. Is W diffeomorphic to $M_{0} \times$ [0,1]?

背景链接：http://projecteuclid.org/euclid.jdg/1214441179

09275 | KP-4.57

### Kirby Problem 4.57

P / partially_solved | Kirby 低维拓扑问题

Let X and Y be closed, oriented, smooth 4-manifolds with the same Euler characteristic and signature. Is there a torus link L in X with trivial normal bundle such that some choice of torus surgery along L transforms X into a manifold diffeomorphic to Y? What if we only require that the result be homeomorphic to Y?

背景链接：https://arxiv.org/abs/2506.20950

09276 | KP-4.58

### Kirby Problem 4.58

P / partially_solved | Kirby 低维拓扑问题

(a) Which Seifert fibered homology spheres $\Sigma(a_{1}$, . . . , $a_{n})bound$ acyclic manifolds? Are there any examples with four or more singular fibers that bound acyclic manifolds?

(b) Which Seifert fibered homology spheres bound contractible manifolds? Is there an example that bounds an acyclic manifold but not a contractible manifold?

(c) Are there any Seifert fibered homology spheres that arise as cork boundaries?

背景链接：http://projecteuclid.org/euclid.mmj/1029002261

09277 | KP-4.59

### Kirby Problem 4.59

P / partially_solved | Kirby 低维拓扑问题

Are lens spaces topologically homology cobordant if and only if they are homeomorphic?

背景链接：http://www.numdam.org/item?id=

09278 | KP-4.6

### Kirby Problem 4.6

P / partially_solved | Kirby 低维拓扑问题

Does every closed, orientable 3-manifold bound an absolutely exotic pair of smooth, orientable 4-manifolds?

背景链接：https://bpb-us-e2.wpmucdn.com/websites.umass.edu/dist/b/22144/files/2026/04/K3-problem-list-watermarked.pdf

09279 | KP-4.60

### Kirby Problem 4.60

P / partially_solved | Kirby 低维拓扑问题

(a) Let X be an open, spin, smooth4-manifold. Does X have a proper smooth embedding in $\mathbb{R}^{6}$?

(b) By choosing a proper exhaustion function on W,(a) would follow from an affirmative answer to the following. Let (W;M, N) be a compact, smooth spin cobordism with a smooth embedding f of M in $S^{5}$. Is there a smooth embedding F: (W;M, N) $\to (S^{5} \times I;S^{5} \times$ \{0\}, $S^{5} \times$ \{1\}) whose restriction to M coincides with f?

(c) Let (W;M, N) be a compact, smooth spin cobordism with smooth embeddings f of M in $S^{5}$ and g of N in $S^{5}$ such that $\sigma(N, S^{5}) -\sigma(M, S^{5}) = \sigma(W)$. Is there a smooth embedding F: (W;M, N) $\to (S^{5} \times I;S^{5} \times$ \{0\}, $S^{5} \times$ \{1\}) whose restriction to M coincides with f and whose restriction to N coincides with g?

背景链接：https://bpb-us-e2.wpmucdn.com/websites.umass.edu/dist/b/22144/files/2026/04/K3-problem-list-watermarked.pdf

09280 | KP-4.61

### Kirby Problem 4.61

P / partially_solved | Kirby 低维拓扑问题

What do different 4-manifold gauge theories see?

背景链接：http://projecteuclid.org/euclid.jdg/1214440981

09281 | KP-4.62

### Kirby Problem 4.62

P / partially_solved | Kirby 低维拓扑问题

Let X be a smooth, closed, connected, oriented 4-manifold with $b^{+}_{2}(X)$ >1.

(a) Does X have Donaldson simple type?

(b) Does X have Seiberg-Witten simple type?

背景链接：https://bpb-us-e2.wpmucdn.com/websites.umass.edu/dist/b/22144/files/2026/04/K3-problem-list-watermarked.pdf

09282 | KP-4.63

### Kirby Problem 4.63

P / partially_solved | Kirby 低维拓扑问题

How many independent basic classes can a simply connected smooth 4-manifold X have, as measured $bybr(X)$, the rank of the span of the basic classes?

(a) Is there an upper bound $forbr(X)$ in terms of topological invariants of X?

(b) In particular, is $br(X) \leq b^{+}_{2}(X)$ for all simply connected X with $b^{+}_{2}(X)$ odd?

(c) Is there a smooth indefinite, simply connected 4-manifold X with $b_{2}(X) \geq$ 3 for which the image of $ev_{*}: \operatorname{Diff}(X) \to \operatorname{Aut}(Q_{X})$ is finite, or even trivial?

背景链接：https://bpb-us-e2.wpmucdn.com/websites.umass.edu/dist/b/22144/files/2026/04/K3-problem-list-watermarked.pdf

09283 | KP-4.64

### Kirby Problem 4.64

O / open | Kirby 低维拓扑问题

Find an irreducible, closed, smooth 4-manifold with nontrivial Bauer-Furuta invariant but with trivial Seiberg-Witten invariant.

背景链接：https://bpb-us-e2.wpmucdn.com/websites.umass.edu/dist/b/22144/files/2026/04/K3-problem-list-watermarked.pdf

09284 | KP-4.65

### Kirby Problem 4.65

P / partially_solved | Kirby 低维拓扑问题

Suppose X is a smooth4-manifold with the homology of $S^{1} \times S^{3}$ whose infinite cyclic cover $\widetilde{X}$ has $H^{1}(\widetilde{X})=0$. Furuta and Ohta [FO93] define an invariant $\lambda_{FO}(X)$ as 1/4 of the signed count of irreducible flat $SU(2)$ connections on X; this requires an orientation of X and a specified generator of $H^{1}(X)$.

(a) Does the following hold? The invariant $\lambda_{FO}(X)$ is an integer, and $\lambda_{FO}(X)\equiv \rho(Y,\mathfrak{s})$, where $\rho(Y,\mathfrak{s})$ is the Rokhlin invariant of an oriented spin 3 manifold Y that is Poincaré dual to the generator of $H^{1}(X)$.

(b) Mrowka-Ruberman-Saveliev [MRS11] give an approach, defining an invariant $\lambda_{SW}(X)$ by counting solutions to the Seiberg-Witten equations and adding an index-theoretic correction term. Is $\lambda_{FO}(X)=-\lambda_{SW}(X)$?

背景链接：http://projecteuclid.org/euclid.jdg/1320067650

09285 | KP-4.66

### Kirby Problem 4.66

P / partially_solved | Kirby 低维拓扑问题

Can the skein lasagna module detect exotic smooth structures on closed 4-manifolds?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09286 | KP-4.67

### Kirby Problem 4.67

P / partially_solved | Kirby 低维拓扑问题

(a) Compute $\pi_{0}(\operatorname{Diff}^{+}(S^{4}))$. Do we have $\pi_{0}(\operatorname{Diff}^{+}(S^{4})) =$ \{1\}?

(b) In particular, does some implantation of the barbell map provide a nontrivial element in $\pi_{0}(\operatorname{Diff}^{+}(S^{4}))$?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09287 | KP-4.68

### Kirby Problem 4.68

P / partially_solved | Kirby 低维拓扑问题

Does every closed smooth 4-manifold admit an exotic diffeomorphism? How about the following special cases?

(a) Is there a definite smooth closed 4-manifold that admits an exotic diffeomorphism?

(b) Does $S^{2} \times S^{2}$ or K3 admit an exotic diffeomorphism?

背景链接：http://projecteuclid.org/euclid.jdg/1214440552

09288 | KP-4.69

### Kirby Problem 4.69

O / open | Kirby 低维拓扑问题

Does there exist a diffeomorphism of a closed3-manifoldf: $M \to M$ such that f is topologically but not smoothly pseudo-isotopic to the identity?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09289 | KP-4.7

### Kirby Problem 4.7

P / partially_solved | Kirby 低维拓扑问题

(a) If $M_{1},M_{2}are$ two homeomorphic closed, oriented 4-manifolds, is $M_{1}\#S^{2} \times S^{2}$ diffeomorphic to $M_{2}\#S^{2} \times S^{2}$?

(b) Is there a fixedn>0such that $M_{1}\#_{n}S^{2} \times S^{2}$ is diffeomorphic to $M_{2}\#_{n}S^{2} \times S^{2}$ for every pair $M_{1}, M_{2}$ as above?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09290 | KP-4.70

### Kirby Problem 4.70

P / partially_solved | Kirby 低维拓扑问题

Do there exist $k \geq 0$ and a smooth closed 4-manifold X such that the map $\pi_{k}(\operatorname{Diff}(X)) \to \pi_{k}(\operatorname{Homeo}(X))$ induced by the inclusion $\operatorname{Diff}(X) \hookrightarrow \operatorname{Homeo}(X)$ is an isomorphism?

背景链接：http://projecteuclid.org/euclid.jdg/1214441784

09291 | KP-4.71

### Kirby Problem 4.71

P / partially_solved | Kirby 低维拓扑问题

(a) Do there exist $k \geq 0$ and a smooth closed orientable 4-manifold X such that $\pi_{k}(\operatorname{Diff}(X))$ is finitely generated?

(b) Do there exist $k >$ 0 and a smooth closed orientable 4-manifold X such that $H_{k}(BDiff(X);\mathbb{Z})$ is finitely generated?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09292 | KP-4.72

### Kirby Problem 4.72

P / partially_solved | Kirby 低维拓扑问题

Let X be a closed orientable topological 4-manifold with finite $\pi_{1}(X)$.

(a) Is $\pi_{k}(\operatorname{Homeo}(X))$ finitely generated for every $k \geq 0$?

(b) Is $H_{k}(BHomeo(X);\mathbb{Z})$ finitely generated for every $k \geq 0$?

背景链接：http://projecteuclid.org/euclid.jdg/1214440552

09293 | KP-4.73

### Kirby Problem 4.73

O / open | Kirby 低维拓扑问题

Does the Morlet correspondence $BDiff_{\partial}(D^{n}) \cong \Omega^{n}_{0}(Top(n)/O(n))$ (10) hold for n=4?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09294 | KP-4.74

### Kirby Problem 4.74

O / open | Kirby 低维拓扑问题

Does there exist a closed, smooth 4-manifold X and a diffeomorphism f: $X \to X$ such that f is smoothly pseudo-isotopic to the identity, but f is not stably smoothly isotopic to the identity?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09295 | KP-4.75

### Kirby Problem 4.75

P / partially_solved | Kirby 低维拓扑问题

Let X be a connected smooth 4-manifold with nonempty boundary, with finite $\pi_{1}(X)$, and let $k \geq$ 0. Let $Diff_{\partial}(X)$ denote the group of diffeomorphisms of X that are the identity near $\partial X$. Is the image of the natural map $s_{*}: \pi_{k}(Diff_{\partial}(X)) \to$ colim $\pi_{k}(Diff_{\partial}(X\#_{N}S^{2} \times S^{2}))$ (11) $N \to \infty$ finitely generated? What about the analogous problem in the topological category?

背景链接：http://projecteuclid.org/euclid.jdg/1214441784

09296 | KP-4.76

### Kirby Problem 4.76

O / open | Kirby 低维拓扑问题

Let X be a closed, oriented, simply connected, smooth 4manifold and fix $k >$ 0. Is there $N \geq$ 0 such that, for every $n \geq N$, the natural map $\pi_{k}(\operatorname{Diff}^{+}(X\#_{n}S^{2} \times S^{2})) \to \pi_{k}(\operatorname{Homeo}^{+}(X\#_{n}S^{2} \times S^{2}))$ is surjective?

背景链接：http://projecteuclid.org/euclid.jdg/1214437136

09297 | KP-4.77

### Kirby Problem 4.77

O / open | Kirby 低维拓扑问题

For which $k \geq 0$ and closed smooth 4-manifold X does the equality $\ker(i_{*}: \pi_{k}(Diff_{\partial}(X^{\circ})) \to \pi_{k}(Homeo_{\partial}(X^{\circ}))) =\ker(s_{*}: \pi_{k}(Diff_{\partial}(X^{\circ})) \to$ colim $\pi_{k}(Diff_{\partial}(X^{\circ}\#_{n}S^{2} \times S^{2})))$ (13) $n \to \infty$ hold?

背景链接：http://projecteuclid.org/euclid.jdg/1214440552

09298 | KP-4.78

### Kirby Problem 4.78

O / open | Kirby 低维拓扑问题

Let X be a simply connected closed smooth 4-manifold. Does $BDiff(X)$ satisfy homological stability over $\mathbb{Q}$?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09299 | KP-4.79

### Kirby Problem 4.79

O / open | Kirby 低维拓扑问题

Given $k >$ 0, is there a closed, simply connected, smooth 4manifold X and a nonzero homotopy class $\alpha \in \pi_{k}(Diff_{\partial}(X^{\circ}))$ such that $\alpha \in \ker(i_{*}: \pi_{k}(Diff_{\partial}(X^{\circ})) \to \pi_{k}(Homeo_{\partial}(X^{\circ})))$ (14) and $\alpha \notin \ker(s_{*}: \pi_{k}(Diff_{\partial}(X^{\circ})) \to \pi_{k}(Diff_{\partial}(X^{\circ}\#S^{2} \times S^{2})))$? (15)

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09300 | KP-4.8

### Kirby Problem 4.8

P / partially_solved | Kirby 低维拓扑问题

Let X be a closed, simply connected, smooth 4-manifold, and T a smoothly embedded torus in X with $\pi_{1}(X$ -T) =1 and $[T]^{2}$ =0. Let $X_{K}$ be the result of Fintushel-Stern knot surgery on X along a knot K. If $K_{1}$ and $K_{2}$ are two prime knots such that $X_{K,1}$ and $X_{K,2}$ are diffeomorphic, does it follow that $K_{1}$ is either $K_{2}$ or its mirror?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09301 | KP-4.80

### Kirby Problem 4.80

O / open | Kirby 低维拓扑问题

Is there $n >$ 2 and a smooth closed simply connected 4manifold X for which there is an element of ordernin the $subgroupker(\pi_{0}\operatorname{Diff}(X) \to \pi_{0}\operatorname{Homeo}(X))$? More generally, is there a subgroup of order nin $\ker(\pi_{0}\operatorname{Diff}(X) \to \pi_{0}\operatorname{Homeo}(X))$?

背景链接：http://projecteuclid.org/euclid.jdg/1214437136

09302 | KP-4.81

### Kirby Problem 4.81

O / open | Kirby 低维拓扑问题

Is there a smooth, closed, simply connected 4-manifold X for which the group $\ker(\pi_{0}(\operatorname{Diff}(X)) \to \pi_{0}(\operatorname{Homeo}(X)))$ is finitely generated?

背景链接：http://projecteuclid.org/euclid.jdg/1214440552

09303 | KP-4.82

### Kirby Problem 4.82

O / open | Kirby 低维拓扑问题

Let $\phi$ be a self-diffeomorphism of a closed, simply-connected, smooth 4-manifold X. Suppose that for every smooth surface $\Sigma$ in X, the surfaces $\Sigma$ and $\phi(\Sigma)$ are smoothly isotopic.

(a) Is $\phi$ necessarily smoothly isotopic to the identity map?

(b) Is $\phi smoothly$ isotopic to a diffeomorphism that is supported on $a B^{4} \subset X$?

背景链接：http://projecteuclid.org/euclid.jdg/1214440552

09304 | KP-4.83

### Kirby Problem 4.83

P / partially_solved | Kirby 低维拓扑问题

For which 4-manifolds does there exist a smooth structure such that there exists a non-smoothable homeomorphism with respect to that smooth structure?

背景链接：http://projecteuclid.org/euclid.jdg/1214441784

09305 | KP-4.84

### Kirby Problem 4.84

O / open | Kirby 低维拓扑问题

Is there a closed oriented smooth 4-manifold X for which every finite subgroup Gof the mapping class group $\pi_{0}(\operatorname{Diff}^{+}(X))can$ be realized by a finite group action on X?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09306 | KP-4.85

### Kirby Problem 4.85

O / open | Kirby 低维拓扑问题

Is there a closed orientable smooth 4-manifold X for which the identity component $Diff_{0}(X)$ of the diffeomorphism group is not uniformly perfect?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09307 | KP-4.86

### Kirby Problem 4.86

O / open | Kirby 低维拓扑问题

Is it the case that for every closed, smoothable topological 4manifold X, there exists a locally linear finite group action on X, such that for every smooth structure on X, the action is non-smoothable?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09308 | KP-4.87

### Kirby Problem 4.87

O / open | Kirby 低维拓扑问题

Is there an exotic action of $\mathbb{Z}/n$ on $S^{4}$ with 0-dimensional fixed point set? 1-dimensional? 2-dimensional?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09309 | KP-4.88

### Kirby Problem 4.88

O / open | Kirby 低维拓扑问题

Let $\tau: S^{4} \to S^{4}$ be a free (hence orientation-reversing) involution. Is there an embedded $S^{2} \subset S^{4}$ that is invariant under $\tau$? This is equivalent to the existence of an embedded $\mathbb{RP}^{2}$ in $S^{4}/\tau \cong \mathbb{RP}^{4}$, carrying the nontrivial class in $H_{2}(\mathbb{RP}^{4};\mathbb{Z}/2)$. This condition on the homology class will be assumed for the rest of the problem.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09310 | KP-4.89

### Kirby Problem 4.89

P / partially_solved | Kirby 低维拓扑问题

Classify smooth, effective circle actions on simply connected 4-manifolds with boundary.

(a) Classify simply connected4-manifolds with boundary that admit circle actions.

(b) For each such 4-manifold, classify the circle actions up to conjugation by diffeomorphisms.

(c) Given a 4-manifold X with a circle action on its boundary, when does the action extend over X? When does it extend uniquely?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09311 | KP-4.9

### Kirby Problem 4.9

P / partially_solved | Kirby 低维拓扑问题

Is every Gluck twist in $S^{4}$ standard?

背景链接：http://projecteuclid.org/euclid.jdg/1214437136

09312 | KP-4.90

### Kirby Problem 4.90

P / partially_solved | Kirby 低维拓扑问题

Do the Chern numbers $c^{2}_{1}$ and $c_{2}$ of every closed, symplectic 4-manifold X that is not a ruled surface satisfy the following?

(a) $c^{2}_{1} \leq 3c_{2}$.

(b) $c^{2}_{1} =3c_{2}$ >0 if and only if $X =\mathbb{CP}^{2}$ or X is a complex ball quotient.

(c) $c_{2} \geq$ 0.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09313 | KP-4.91

### Kirby Problem 4.91

O / open | Kirby 低维拓扑问题

Present a topological construction of symplectic fake projective planes. Does there exist a symplectic fake projective plane that is not a complex ball quotient?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09314 | KP-4.92

### Kirby Problem 4.92

P / partially_solved | Kirby 低维拓扑问题

Is every symplectic Calabi-Yau surface diffeomorphic to either the K3 surface, the Enriques surface or $a T^{2}-bundle$ over $T^{2}$?

背景链接：http://projecteuclid.org/euclid.jdg/1175266207

09315 | KP-4.93

### Kirby Problem 4.93

O / open | Kirby 低维拓扑问题

Is every symplectic form on the standard K3 surface symplectomorphic to a Kähler form?

背景链接：https://doi.org/10.1142/9789812772688

09316 | KP-4.94

### Kirby Problem 4.94

O / open | Kirby 低维拓扑问题

Are homotopy equivalent Horikawa surfaces in different deformation classes diffeomorphic as 4-manifolds? Are they symplectomorphic?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09317 | KP-4.95

### Kirby Problem 4.95

P / partially_solved | Kirby 低维拓扑问题

(a) Is there a closed hyperbolic oriented 4-manifold that admits a symplectic structure?

(b) Do the Seiberg-Witten invariants vanish on every closed hyperbolic 4manifold?

背景链接：http://projecteuclid.org/euclid.jdg/1214425535

09318 | KP-4.96

### Kirby Problem 4.96

O / open | Kirby 低维拓扑问题

Does there exist a pair of symplectic 4-manifolds $(X_{1}, \omega_{1})and (X_{2}, \omega_{2})$, where there is a diffeomorphism $f: X_{1} \to X_{2}$ such that $f*(c_{1}(X_{2}, \omega_{2})) = c_{1}(X_{1}, \omega_{1})$ and $f*([\omega_{2}]) = [\omega_{1}] \in H^{2}(X_{1},\mathbb{R})$, but $(X_{1}, \omega_{1})$ is not symplectomorphic to $(X_{2}, \omega_{2})$?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09319 | KP-4.97

### Kirby Problem 4.97

P / partially_solved | Kirby 低维拓扑问题

Let $\lambda:=c^{2}_{1}/c_{2}$ be the Chern slope of a closed, almost complex 4-manifold X. What is the supremum of $\lambda as X$ ranges over the following families?

(a) Symplectic $\Sigma_{g}$-bundles$ over $\Sigma_{h}$, with g, $h \geq$ 2.

(b) Holomorphic $\Sigma_{g}$-bundles$ over $\Sigma_{h}$, with g, $h \geq$ 2.

(c) Symplectic Lefschetz fibrations over $S^{2}$.

(d) Holomorphic Lefschetz fibrations over $S^{2}$. (Here, the Lefschetz fibrations are assumed to have critical points.)

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09320 | KP-4.98

### Kirby Problem 4.98

P / partially_solved | Kirby 低维拓扑问题

Does every closed symplectic 4-manifold admit inequivalent Lefschetz pencils with the same fiber genus g, for sufficiently large g? How about infinitely many?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09321 | KP-4.99

### Kirby Problem 4.99

P / partially_solved | Kirby 低维拓扑问题

Let X be a closed symplectic 4-manifold. Let $T \subset X$ be a symplectic submanifold that is diffeomorphic to a 2-dimensional torus such that $[T]^{2} =$ 0. Let $X_{K}$ be a manifold obtained by Fintushel-Stern knot surgery on T using a knot $K \subset S^{3}$. If $X_{K}$ has a symplectic structure, must K be a fibered knot?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09322 | KP-5.1

### Kirby Problem 5.1

O / open | Kirby 低维拓扑问题

Does every cellular set in the plane have the fixed point property?

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09323 | KP-5.10

### Kirby Problem 5.10

O / open | Kirby 低维拓扑问题

The Andrews--Curtis Conjecture [AC65] for the trivial group: a presentation of the trivial group can be changed to the trivial presentation by Andrews--Curtis moves.

背景链接：https://aimath.org/pastworkshops/kirbylistrep.pdf

09324 | KP-5.11

### Kirby Problem 5.11

P / partially_solved | Kirby 低维拓扑问题

(Whitehead's Asphericity Question). Is every subcomplex of an aspherical 2-complex aspherical?

背景链接：https://arxiv.org/abs/1212.1943

09325 | KP-5.12

### Kirby Problem 5.12

P / partially_solved | Kirby 低维拓扑问题

(Zeeman Conjecture). If $K$ is a finite contractible 2-complex, then $K\times I$ collapses to a point [Zee64, Conjecture (1)].

背景链接：https://gradmath.org/wp-content/uploads/2021/07/GJM2021-Kupers.pdf

09326 | KP-5.13

### Kirby Problem 5.13

P / partially_solved | Kirby 低维拓扑问题

Let $M$ be a finite-volume hyperbolic $n$-manifold.

- Does it always have a finite cover with $b_{1}>0$?

- Does it always have a finite cover with fundamental group that surjects onto a non-abelian free group?

- Does it have a finite cover with cubulated fundamental group?

- When $n$ is odd, does it always have a finite cover that fibers over the circle?

背景链接：https://elibm.org/article/10000267

09327 | KP-5.14

### Kirby Problem 5.14

O / open | Kirby 低维拓扑问题

Does there exist a 1-cusped finite-volume hyperbolic $n$-manifold for any $n\geq 5$?

背景链接：https://doi.org/10.2140/gt.2013.17.905

09328 | KP-5.15

### Kirby Problem 5.15

P / partially_solved | Kirby 低维拓扑问题

Suppose $M$ is a manifold with a complete Riemannian metric with nonnegative Ricci curvature. Is the fundamental group of $M$ finitely generated?

背景链接：http://projecteuclid.org/euclid.jdg/1214501132

09329 | KP-5.16

## Kirby Problem 5.16

P / partially_solved | Kirby 低维拓扑问题

Let $R$ be $\Z$ or a field. Let $A$ and $B$ be differential graded algebras so that either: \begin{itemize}

- As a graded algebra, $A$ (respectively $B$) is isomorphic to the free, non-commutative $R$-algebra on finitely many homogeneous generators $x_{i}$, i.e., to the tensor algebra on the free $R$-module generated by $x_{1},\ldots,x_{n}$.

- As a graded algebra, $A$ (respectively $B$) is isomorphic to a free graded-commutative algebra on finitely many homogeneous generators $x_{i}$. (So, if the $x_{i}$ have even gradings, $A$ is a polynomial algebra.) \end{itemize} (Different generators can have different gradings, $A$ and $B$ may have different numbers of generators, and gradings of generators may be negative.)

Are the following questions decidable?

- Is $A$ stable tame isomorphic to $B$?

- Is $A$ quasi-isomorphic to $B$?

- Is $A$ derived Morita equivalent to $B$?

背景链接：https://arxiv.org/abs/2605.08122

09330 | KP-5.17

## Kirby Problem 5.17

P / partially_solved | Kirby 低维拓扑问题

Let $(W,\omega,V_{i},\phi_{i})$ be two Weinstein structures on a fixed symplectic manifold $(W,\omega)$ (or equivalently consider two Weinstein handle decompositions). Is there a Weinstein homotopy from $(W,\omega,V_{1},\phi_{1})$ to $(W,\omega,V_{2},\phi_{2})$?

背景链接：https://doi.org/10.1090/coll/059

09331 | KP-5.18

## Kirby Problem 5.18

P / partially_solved | Kirby 低维拓扑问题

- In higher dimensions, find the `non-analytic' cohomology module $\HH^{*}_{?}$ analogous to the analytic lattice cohomology $\HH^{*}_{\mathrm{an}}$, and a natural functor $\HH^{*}_{\mathrm{an}}\to \HH^{*}_{?}$ connecting them.

- Define a version $\ECH^{*}$ of Embedded Contact Homology for isolated complex singularity links (in any dimension) associated with the canonical contact structure of the link, together with a natural graded $\Z[U]$-module morphism $\HH^{*}_{\mathrm{an}}\to \ECH^{*}$. More precisely,

- Show that this morphism is injective.

- Fix the diffeomorphism type of a link, and also a contact structure on it that can be realized as the canonical contact structure associated with some singularity analytic structure. Then characterize the family $\{\HH^{*}_{\mathrm{an}}\}$, indexed by all the possible analytic germs inducing this contact link, via those graded sub-$\Z[U]$-modules of $\ECH^{*}$ which satisfy certain specific properties.

背景链接：https://arxiv.org/abs/2109.11266

09332 | KP-5.2

## Kirby Problem 5.2

P / partially_solved | Kirby 低维拓扑问题

(Doubly-Small Morphisms of Manifolds).

- Suppose that $h: \R^{n} \to \R^{n}$ is a homeomorphism (or diffeomorphism) which satisfies two smallness hypotheses:

- every orbit of $h$ (of any $x\in \R^{n}$, under all powers of $h$) is uniformly bounded in diameter (by 1 say), and

- some subsequence of powers of $h$ converges to $\mathrm{Id}_{\R^{n}}$ in (say) the compact-open topology.

Then must $h$ be the identity map?

- A special case of this question is: let $h$ be a homeomorphism of $B^{n}$ that is the identity on $\partial B^{n}$. If there is a subsequence of powers of $h$ which converge to the identity, must $h$ itself be the identity?

背景链接：https://doi.org/10.1016/0040-9383(69

09333 | KP-5.3

## Kirby Problem 5.3

P / partially_solved | Kirby 低维拓扑问题

(Hilbert--Smith Conjecture).

- The Hilbert--Smith Conjecture [Smi41] asserts that a locally compact subgroup of the homeomorphism group of a connected manifold must be a Lie group.

- Conjecture: The free-action subset of any Cantor group action on an ENR is a homology-$Z$-subset (of the ENR).

背景链接：https://doi.org/10.1090/S0894-0347-2013-00766-3

09334 | KP-5.4

### Kirby Problem 5.4

P / partially_solved | Kirby 低维拓扑问题

Is the homeomorphism group of a manifold an absolute neighborhood retract (ANR)?

背景链接：https://bpb-us-e2.wpmucdn.com/websites.umass.edu/dist/b/22144/files/2026/04/K3-problem-list-watermarked.pdf

09335 | KP-5.5

### Kirby Problem 5.5

P / partially_solved | Kirby 低维拓扑问题

Is the connected sum of (locally flat) pairs

$$
(M^{n+k}_{1},N^{n}_{1})\#(M^{n+k}_{2},N^{n}_{2})
$$

well-defined in the topological category?

背景链接：https://bpb-us-e2.wpmucdn.com/websites.umass.edu/dist/b/22144/files/2026/04/K3-problem-list-watermarked.pdf

09336 | KP-5.6

### Kirby Problem 5.6

P / partially_solved | Kirby 低维拓扑问题

- Does every closed, PL, orientable $n$-manifold admit an $n$-fold branched covering map over $S^{n}$?

Assuming that the answer to this problem is "yes", the following follow-up questions would be natural to ask.

- Does every $n$-manifold admit an $n$-fold branched covering over $S^{n}$ with branch locus a codimension-2 embedded submanifold?

- If the answer to (b) is "yes", then one could naturally ask if the cover can additionally be taken to be simple, meaning the covering monodromy sends every meridian of the embedded submanifold to a transposition. For $n=2,3,4$, the above papers show the answer is, "yes".

- If the answer to (b) is "yes", then in a separate direction one could ask when the branch locus can be taken to be orientable.

背景链接：https://bpb-us-e2.wpmucdn.com/websites.umass.edu/dist/b/22144/files/2026/04/K3-problem-list-watermarked.pdf

09337 | KP-5.7

### Kirby Problem 5.7

O / open | Kirby 低维拓扑问题

(Montgomery--Yang problem). Does there exist a pseudo-free, smooth, $S^{1}$ action on $S^{5}$ with more than three multiple orbits?

背景链接：https://bpb-us-e2.wpmucdn.com/websites.umass.edu/dist/b/22144/files/2026/04/K3-problem-list-watermarked.pdf

09338 | KP-5.8

### Kirby Problem 5.8

O / open | Kirby 低维拓扑问题

Is there a closed aspherical 5-manifold that is not triangulable?

背景链接：http://projecteuclid.org/euclid.jdg/1214447212

09339 | KP-5.9

### Kirby Problem 5.9

P / partially_solved | Kirby 低维拓扑问题

Let $M_{1}$ and $M_{2}$ be smooth manifolds of dimension $n$. Suppose $M_{1}$ admits an $S$-map into $\R^{p}$. If $M_{2}$ is homeomorphic to $M_{1}$, then does $M_{2}$ admit an $S$-map as well?

背景链接：http://gateway.proquest.com/openurl?url

09340 | OWR-11790-007

### Kirby’s Whitehead-Double Sliceness Conjecture

O / open | Oberwolfach 报告问题汇编

Is the Whitehead double of a knot $K$ slice if and only if $K$ is slice?

来源字段：https://doi.org/10.4171/owr/2012/28

09341 | OWR-8415345-002

### Knot Complements and Prime Decomposition in General Three-Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

For knots in general 3-manifolds, what is the correct Gordon-Luecke analogue and is it true-for example, does an orientation-preserving homeomorphism of complements imply equivalence of knots in closed oriented irreducible 3-manifolds outside the rational-unknot lens-space exceptions-and does a prime decomposition theory for knots exist?

来源字段：https://doi.org/10.4171/owr/2021/42

09342 | OWR-13108-014

### Knot Volume Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For every knot $K$, does the volume conjecture hold in the form

$$\lim_{N\to\infty}\frac{\log\left\lvert J_N\!\left(K;\exp(2\pi i/N)\right)\right\rvert}{N}=\frac{\operatorname{Vol}(S^3\setminus K)}{2\pi},$$

where $J_N$ is the $N$-colored Jones polynomial and $\operatorname{Vol}$ is the simplicial volume of the knot complement?

来源字段：https://doi.org/10.4171/owr/2014/38

09343 | OWR-11138-004

### Koszul Differentials in Bousfield-Kuhn Homology

P / partially_solved | Oberwolfach 报告问题汇编

Let $E$ be Morava $E$-theory of height $n$, let $q$ be odd, and identify $$E_q^{\wedge}\!\left(\Sigma^kD_{2^k}(\operatorname{Id})(S^q)\right)\cong C[k]_{-q}^{\vee}.$$ Under this isomorphism, is the map $(\Phi_n\beta_k)_*$ on $E$-homology the differential in the Koszul resolution of $\widetilde E_q(S^q)$?

来源字段：https://doi.org/10.4171/owr/2011/46

09344 | OWR-13110-016

### Kühnel Conjecture for Embedded Simplicial Skeletons

P / partially_solved | Oberwolfach 报告问题汇编

Kühnel's conjecture: if the k-skeleton $\Delta_n^{(k)}$ embeds in a compact $(k-1)$-connected 2k-manifold with kth Betti number b over $\mathbb{Z}_2$, then $\binom{n-k-1}{k+1}\leq\binom{2k+1}{k+1}b$.

来源字段：https://doi.org/10.4171/owr/2014/40

09345 | OWR-14298580-004

### L-Space Conjecture for Rational Homology Spheres

O / open | Oberwolfach 报告问题汇编

别名：Taut Foliations on Rational Homology Spheres [OWR-14298580-005]

For every irreducible rational homology sphere $Y$, are the following equivalent: $Y$ supports no taut foliation, $Y$ is an L-space, and $\pi_1(Y)$ is not left-orderable?

来源字段：https://doi.org/10.4171/owr/2024/34

09346 | OWR-1195-006

### Lagrangian Embeddings of Closed Three-Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Does every closed $3$-manifold admit a Lagrangian embedding into $\mathbb{C}^3$?

来源字段：https://doi.org/10.4171/owr/2006/13

09347 | OWR-1056-001

### Lagrangian Matching and Seiberg-Witten Invariants

O / open | Oberwolfach 报告问题汇编

Let $(X, \pi)$ be a closed oriented $4$-manifold equipped with a broken Lefschetz pencil (or near-symplectic structure) $\pi$, and let $S$ be the set of $\mathrm{Spin}^c$ structures on $X$. For every $s \in S$, does Perutz's Lagrangian matching invariant $L_{X,\pi}(s)$ coincide up to sign with the Seiberg--Witten invariant, $L_{X,\pi}(s) = \pm \operatorname{SW}_X(s)$?

来源字段：https://doi.org/10.4171/owr/2005/35

09348 | OWR-1321-002

### Lagrangian Matching and Seiberg-Witten Invariants

O / open | Oberwolfach 报告问题汇编

For a homology-oriented closed $4$-manifold $X$ with $b_2^+(X)>1$ and a broken fibration $f$, the Lagrangian matching invariant $L_{X,f}(s)$ is conjectured to equal the Seiberg-Witten invariant $SW_X(s)$ for every admissible $\operatorname{Spin}^c$ structure $s$; hence $L_{X,f}$ should be independent of $f$.

来源字段：https://doi.org/10.4171/owr/2006/35

09349 | OWR-16933-004

### Lagrangian Sphere Generators for Quintic-Type Fourfolds

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a symplectic four-dimensional manifold diffeomorphic to a quintic in $\mathbb P^4$ and equipped with a Lagrangian fibration. Can tropical hypersurfaces in the base be lifted to enough Lagrangian spheres to generate $H_3(X,\mathbb Z)$, which has rank $204$?

来源字段：https://doi.org/10.4171/owr/2019/21

09350 | OWR-1190-009

### Large Cardinals Supporting Brown-Adams Pairs

O / open | Oberwolfach 报告问题汇编

Are there arbitrarily large cardinals $\alpha$ such that $(\mathcal{T},\alpha)$ is a Brown-Adams pair, and which cardinals $\alpha$ are permissible?

来源字段：https://doi.org/10.4171/owr/2006/08

09351 | AMR-102-0024

### Leaf spaces and transverse structures - Question 8.1

P / partially_solved | AMR 问题清单汇编

Suppose $M$ is irreducible. Suppose further that $\pi_1(M)$ admits a nontrivial action on $\mathbb{R}$. When does $M$ admit a taut foliation with a transverse $(\pi_1(M),\mathbb{R})$ structure?

背景链接：https://arxiv.org/abs/math/0209081

09352 | AMR-102-0033

### Leaf spaces and transverse structures - Question 8.10

O / open | AMR 问题清单汇编

What is the best analytic quality for the action of $\pi_1(M)$ on a universal circle $S^1_\mathrm{univ}$?

背景链接：https://arxiv.org/abs/math/0209081

09353 | AMR-102-0025

### Leaf spaces and transverse structures - Question 8.2

O / open | AMR 问题清单汇编

Suppose $\mathscr{F}$ is an $\mathbb{R}$-covered foliation of an atoroidal $3$-manifold $M$. Is the holonomy representation $\rho_H$ of $\pi_1(M)$ on $\mathbb{R}$ conjugate to a group of coarse $1$-quasi-isometries? i.e. is there a positive valued function $C:\pi_1(M) \to \mathbb{R}$ such that $$| d(\rho_H(\alpha)(p), \rho_H(\alpha)(q)) - d(p,q) | \le C(\alpha)$$ for all $\alpha \in \pi_1(M)$ and $p,q \in \mathbb{R}$?

背景链接：https://arxiv.org/abs/math/0209081

09354 | AMR-102-0026

### Leaf spaces and transverse structures - Question 8.3

P / partially_solved | AMR 问题清单汇编

Suppose $M$ is atoroidal and admits a taut foliation. Must it admit an $\mathbb{R}$-covered foliation?

背景链接：https://arxiv.org/abs/math/0209081

09355 | AMR-102-0027

### Leaf spaces and transverse structures - Question 8.4

O / open | AMR 问题清单汇编

For a fixed manifold $M$, describe the structure of the set of all essential laminations with a transverse $\widetilde{SL(2,\mathbb{R})}$ structure.

背景链接：https://arxiv.org/abs/math/0209081

09356 | AMR-102-0028

### Leaf spaces and transverse structures - Question 8.5

O / open | AMR 问题清单汇编

Suppose $M$ admits a minimal taut foliation. What is the best analytic (transverse) quality of a taut foliation it admits? Can we find a minimal foliation such that the holonomy groupoid is of type $\text{III}_\lambda$ for some algebraic $\lambda$? What about if one asks for a foliation monotone equivalent to the first? Homotopic?

背景链接：https://arxiv.org/abs/math/0209081

09357 | AMR-102-0029

### Leaf spaces and transverse structures - Question 8.6

O / open | AMR 问题清单汇编

Is there a universal constant $c$ such that a hyperbolic $3$-manifold $M$ whose fundamental group $\pi_1(M)$ can be ordered out to radius $c$ can be left-ordered? Or weaker, is there an effective method to compute such a $c(M)$ for a given $M$?

背景链接：https://arxiv.org/abs/math/0209081

09358 | AMR-102-0030

### Leaf spaces and transverse structures - Question 8.7

O / open | AMR 问题清单汇编

Let $\mathsf{T}$ be some class of abstract computers; e.g. finite state automata, Turing machines, Turing machines relative to some oracle $O$, etc. A $\mathsf{T}$-order on a group $G$ is a left-invariant order such that there is a machine $T \in \mathsf{T}$ which recognizes the positive cone $G^+ \subset G$. What kinds of $\mathsf{T}$-orders are possible for fundamental groups $G$ of hyperbolic $3$-manifolds?

背景链接：https://arxiv.org/abs/math/0209081

09359 | AMR-102-0031

### Leaf spaces and transverse structures - Question 8.8

O / open | AMR 问题清单汇编

Let $\Lambda^\pm$ be a pair of laminations of $S^1$ which are transverse to each other and have finite area complementary domains. Suppose $\Gamma$ is a group of automorphisms of $S^1$ which preserves $\Lambda^\pm$ and acts minimally on the leaves of either lamination. When is $\Gamma$ commensurable with $\pi_1(M)$ for $M$ a hyperbolic $3$-manifold?

背景链接：https://arxiv.org/abs/math/0209081

09360 | AMR-102-0032

### Leaf spaces and transverse structures - Question 8.9

P / partially_solved | AMR 问题清单汇编

What possibilities are there for universal circles $S^1_\mathrm{univ}$ for a fixed manifold? For a fixed foliation? For what taut foliations is there a unique minimal universal circle?

背景链接：https://arxiv.org/abs/math/0209081

09361 | OWR-14298592-009

### Leaf Spaces of Explicit Taut Foliations

O / open | Oberwolfach 报告问题汇编

For explicit taut foliations constructed from knot diagrams, determine the leaf space of the lifted foliation well enough to decide whether the induced $\pi_1$-action descends to a nontrivial orientation-preserving action on $\mathbb R$.

来源字段：https://doi.org/10.4171/owr/2024/43

09362 | OWR-13673-006

### Lie-Algebraic Parametrization of TMF Classes

O / open | Oberwolfach 报告问题汇编

Are elements of $\pi_n(\mathrm{TMF})$ naturally parametrized by $n$-dimensional semisimple compact-type Lie algebras through the Stolz-Teichner cocycle construction? Does the Stolz-Teichner cocycle construction hit exactly the classes represented by compact Lie groups with left-invariant framings?

来源字段：https://doi.org/10.4171/owr/2015/16

09363 | OWR-832-007

### Lifting Cohomology Operations to Smooth Extensions

P / partially_solved | Oberwolfach 报告问题汇编

Determine whether cohomology operations lift to smooth extensions of generalized cohomology theories.

来源字段：https://doi.org/10.4171/owr/2005/27

09364 | OWR-15177-019

### Lifting Dehn-Twist Relations to Punctured Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Given a relation $t_{a_1}\cdots t_{a_n}=1$ of Dehn twists in $\operatorname{Mod}_g$, can the curves be moved away from the puncture to curves $\widetilde a_i$ in the punctured surface satisfying $t_{\widetilde a_1}\cdots t_{\widetilde a_n}=1$ in $\operatorname{Mod}_{g,1}$?

来源字段：https://doi.org/10.4171/owr/2016/56

09365 | OWR-12172-006

### Lifting Pentagon-Arrangement Symmetries to Homotopy Invariants

O / open | Oberwolfach 报告问题汇编

For the pentagon hyperplane arrangement whose combinatorial symmetry induces an automorphism of the complement homology, can this automorphism extend to an automorphism of the $E^\infty$-coalgebra, and can it extend to an automorphism of the fundamental group?

来源字段：https://doi.org/10.4171/owr/2012/49

09366 | OWR-12485-003

### Linear Ropelength Bounds in Crossing Number

P / partially_solved | Oberwolfach 报告问题汇编

For every knot $K$, is its ropelength $O(\operatorname{Cr}(K))$, where $\operatorname{Cr}(K)$ is the crossing number? If not, do infinitely many knots have faster growth? Do alternating knots have a ropelength lower bound of linear order in crossing number?

来源字段：https://doi.org/10.4171/owr/2013/22

09367 | OWR-12697687-003

### Lipschitz Concordance Homomorphisms for Braid Quasimorphisms

O / open | Oberwolfach 报告问题汇编

Does there exist a one-Lipschitz concordance homomorphism $I$ such that the braid quasimorphism $f$ is a real scalar multiple of $I$?

来源字段：https://doi.org/10.4171/owr/2023/4

09368 | OWR-17470-007

### Lipschitz Growth of Puncture Classes and Formality

O / open | Oberwolfach 报告问题汇编

Let

$$M=\bigl(S^2\times(\mathbb{CP}^2)^{\#4}\bigr)\setminus\{\text{point}\},$$

and let $\alpha\in\pi_5(M)$ be the puncture class. The known growth of $\operatorname{Lip}(k\alpha)$ is $\omega(k^{1/6})$. Can one prove a sharper lower bound, or an upper bound of order strictly smaller than $k^{1/5}$? Separately, suppose a simply connected space $Y$ has the property that, for every finite simplicial complex $X$, every nullhomotopic $L$-Lipschitz map $X\to Y$ admits a $C(X,Y)(L+1)$-Lipschitz nullhomotopy. Must $Y$ be formal?

来源字段：https://doi.org/10.4171/owr/2020/7

09369 | AIM-TOPOLOGY-0230

### Lipschitz homotopy groups and a quantitative size filtration

P / partially_solved | AIM 研讨会问题汇编

**Problem 17.** Is there an alternate approach to homotopy groups that is adapted to Alexandrov spaces?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

09370 | AIM-TOPOLOGY-0280

### Literal infinitude and a finite height count for integral symplectic surface representations

P / partially_solved | AIM 研讨会问题汇编

Count rational representations whose image lies in the set of integer points of the symplectic group.

来源字段：https://aimath.org/WWN/surfacegroups/surfacegroups.pdf

09371 | AIM-TOPOLOGY-0156

### Literal word metrics fail; maximal CB metrics give the canonical replacement

P / partially_solved | AIM 研讨会问题汇编

Does \(\operatorname{Map}(S)\) have a canonical word metric?

来源字段：http://aimpl.org/genusinfinity/4/

09372 | OWR-14298590-001

### Little Three-Disks Actions on Operadic Homotopy Centers

O / open | Oberwolfach 报告问题汇编

Does the little three-disks operad act on the homotopy center of every nonsymmetric operad?

来源字段：https://doi.org/10.4171/owr/2024/39

09373 | OWR-16415-020

### Local Branching of Holonomy Maps for Geometric Structures

O / open | Oberwolfach 报告问题汇编

For a closed manifold $M$, is the holonomy map from the deformation space of $(G,X)$-structures on $M$ to the representation variety $\operatorname{Rep}(\pi_1(M),G)$, which is an open map with discrete fibers, always a branched local homeomorphism? Find conditions under which it is a local homeomorphism.

来源字段：https://doi.org/10.4171/owr/2018/40

09374 | AIM-TOPOLOGY-0149

### Local finiteness before asymptotics

P / partially_solved | AIM 研讨会问题汇编

Come up with reasonable counting problems and analogues of geodesic currents.

A possible counting problem is as follows. Fix a hyperbolic metric \(R\) on the blooming \(3\)--pod surface. Fix a curve \(\alpha\) and a basepoint \(x \in R\), then does there exists \(p(L)\) such that \[ \frac{ | \phi(\alpha) : \ell_R(\phi(\alpha)) < L | }{p(L)} \simeq \text{Vol}(B_L(x))? \]

来源字段：http://aimpl.org/genusinfinity/3/

09375 | AIM-TOPOLOGY-0177

### Local rigidity of the Thurston metric and a germ-to-global corollary

S / solved | AIM 研讨会问题汇编

**Problem 1.4.** Is each local isometry of Teichmüller space with the Thurston metric induced by an element of the extended mapping class group?

Direct inspection of the source PDF gives:

The problem itself is intact.

There are two meanings of “local isometry.” The published theorem uses an isometric embedding of a connected open set, preserving the ordered distance for every pair in that set. Section 7 also treats the strictly local, germwise meaning.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

09376 | AIM-TOPOLOGY-0093

### Loop objects for digital spaces, with fixed-length and concatenation obstructions

P / partially_solved | AIM 研讨会问题汇编

**Loop spaces.** Is there a good notion of a loop space of a digital space?

来源字段：http://aimpl.org/combhomotop/1/

09377 | OWR-12495-001

### Lott Secondary Index and Transfer Compatibility

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a natural transformation $KR^0\to KR_{\mathbb R/\mathbb Z}^{-1}$ under which analytic push-forward corresponds to the Becker-Gottlieb-Dold transfer?

来源字段：https://doi.org/10.4171/owr/2013/32

09378 | OWR-13686-009

### Lower Bounds for Reduced Structure Groups

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Reduced-Structure-Group Lower Bound [OWR-13686-010]

Let $M$ be a compact oriented manifold of dimension $4k-1$ with $k>1$ and fundamental group $G$. Is $\operatorname{rank}\widetilde{\mathcal S}(M)\geq N_{\mathrm{fin}}(G)$, where $\widetilde{\mathcal S}(M)$ is the structure group modulo classes induced by orientation-preserving self-homotopy equivalences and $N_{\mathrm{fin}}(G)$ is the number of distinct finite orders of nonidentity elements of $G$?

来源字段：https://doi.org/10.4171/owr/2015/29

09379 | AMR-110-0001

### Major problems 1 - The biggest problem, in my opinion, is to come up with a specific vision of where homotopy theory shou…

O / open | AMR 问题清单汇编

The biggest problem, in my opinion, is to come up with a specific vision of where homotopy theory should go, analogous to the Weil conjectures in algebraic geometry or the Ravenel conjectures in our field in the late 70s. You can't win the Fields Medal without a Fields Medal-winning problem; Deligne would not be DELIGNE without the Weil conjectures and Mike Hopkins would not be MIKE HOPKINS without the Ravenel conjectures. We can't all be Deligne or Mike, but making the conjectures requires different talents than proving them, and more of us might have a chance. This was actually my motivation for making this list; to provide a forum for conjectures so that we might collectively be able to form a program analogous to the Weil conjectures. This would make a huge difference to our field, I think. Of course, they have to be somewhat accessible conjectures, which the problems below may not be!

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09380 | AMR-110-0010

### Major problems 10 - Once again, I am not sure whether this problem deserves to be called major, but it is annoying that th…

P / partially_solved | AMR 问题清单汇编

Once again, I am not sure whether this problem deserves to be called major, but it is annoying that the the R. Cohen - Goerss result proving that h_0 h_i is a permanent cycle in the Adams spectral sequence for all primes bigger than 3 is wrong. The flaw was found by Minami, and it appears to be fatal to their proof. So this problem is still open.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09381 | AMR-110-0002

### Major problems 2 - The generating hypothesis, which asserts that the stable homotopy functor is faithful on the category…

O / open | AMR 问题清单汇编

The generating hypothesis, which asserts that the stable homotopy functor is faithful on the category of finite spectra. That is, if f is a map of finite spectra such that pi_* f is 0, then f is nullhomotopic. An unbelievable consequence of this is that the stable homotopy functor is full as well. This conjecture has withstood serious attempts for many years, so be careful! The basic reference is P. Freyd, Stable homotopy, in {\it Proc. Conf. Categorical Algebra (La Jolla, Calif., 1965)}, 121--172, Springer, New York, 1966; MR {\bf 35} \#2280. Devinatz and Hopkins have a program to prove the generating hypothesis when the target is a sphere; see E. S. Devinatz, The generating hypothesis revisited, in {\it Stable and unstable homotopy (Toronto, ON, 1996)}, 73--92, Amer. Math. Soc., Providence, RI, ; CNO CMP 1 622 339.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09382 | AMR-110-0003

### Major problems 3 - Find some geometric meaning for elliptic cohomology.

P / partially_solved | AMR 问题清单汇编

Find some geometric meaning for elliptic cohomology. I believe this problem may be solvable--we keep learning new things about it. One thing I will say here; if I am called to referee a paper on elliptic cohomology that does not deal with the Hopkins viewpoint on elliptic spectra, I will almost surely reject it. The time is gone when one could write papers about the Landweber-Ravenel-Stong elliptic cohomology based on the Jacobi quartic--we now understand that that is only one of many different elliptic cohomology theories, and all papers on elliptic cohomology should now accept that and deal with it. The fundamental reference here is M. J. Hopkins, Topological modular forms, the Witten genus, and the theorem of the cube, in {\it Proceedings of the International Congress of Mathematicians, Vol.\ 1, 2 (Z\"urich, 1994)}, 554--565, Birkh\"auser, Basel, 1995; MR 97i:11043. But one should also see Grojnowski's approach to equivariant elliptic cohomology--unfortunately, this does not seem to be published, but there is a preprint. Matthew Ando has also thought about this, see M. Ando, Power operations in elliptic cohomology and representations of loop groups, Trans. Amer. Math. Soc. ; CNO CMP 1 637 129. I am sure I have left something out here as well.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09383 | AMR-110-0004

### Major problems 4 - On the same theme, find some way of doing index theory related to elliptic cohomology.

P / partially_solved | AMR 问题清单汇编

On the same theme, find some way of doing index theory related to elliptic cohomology. This is not really algebraic topology, but would have a major impact on our field. I don't know much about this, but people who might are Ezra Getzler and Richard Melrose on the analysis side. Richard Melrose has a theory of index theory on manifolds with corners that might possibly be relevant, and Ezra has worked on index theory on certain infinite dimensional manifolds, which again might be relevant. From the algebraic topology side, I know Haynes Miller and Mike Hopkins have thought about this some.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09384 | AMR-110-0005

### Major problems 5 - The chromatic splitting conjecture, which is considerably more complicated to state.

P / partially_solved | AMR 问题清单汇编

The chromatic splitting conjecture, which is considerably more complicated to state. Basically nothing is known about this, and so this one may be more accessible. Besides Hopkins and me, Nori Minami and Ethan Devinatz have both thought about this conjecture, so might be good resources. See M. Hovey, Bousfield localization functors and Hopkins' chromatic splitting conjecture, in {\it The \v Cech centennial (Boston, MA, 1993)}, 225--250, Contemp. Math., 181, Amer. Math. Soc., Providence, RI, 1995; MR 96m:55010 .

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09385 | AMR-110-0007

### Major problems 7 - Classify all finite loop spaces.

S / solved | AMR 问题清单汇编

Classify all finite loop spaces. This is the long term project of Bill Dwyer and Clarence Wilkerson. The theory, I believe, is that the Lie groups are essentially the only examples. So one constructs Weyl groups and maximal tori and the like. But certainly at individual primes there can be other examples, like BD_3 at p=2.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09386 | AMR-110-0008

### Major problems 8 - Say something general about the stable or unstable homotopy groups of spheres.

P / partially_solved | AMR 问题清单汇编

Say something general about the stable or unstable homotopy groups of spheres. For example, Ravenel has suggested that the size of the nth homotopy group of S^k grows polynomially in n, maybe even cubically. I presume he meant stable homotopy, but one could also ask the question unstably.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09387 | AMR-110-0009

### Major problems 9 - Kervaire invariant one in dimension 126

P / partially_solved | AMR 问题清单汇编

Does the possible Kervaire-invariant-one element $\theta_6\in\pi_{126}^{S}$ exist; equivalently, is $h_6^2$ a permanent cycle in the mod-2 Adams spectral sequence?

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09388 | OWR-5158-001

### Mapping-Class-Group Presentations of Surface Steinberg Modules

O / open | Oberwolfach 报告问题汇编

Understand the $\operatorname{Mod}(\Sigma_g^1)$-module structure of the Steinberg module $\operatorname{St}(\Sigma_g^1)$, and give a presentation based on its single module generator.

来源字段：https://doi.org/10.4171/owr/2011/30

09389 | AIM-TOPOLOGY-0049

### Marked-boundary comparison and empty-sector defect for non-semisimple skeins

P / partially_solved | AIM 研讨会问题汇编

**Versions of Stated skeins for non semi-simple categories.** Compare different definitions of the Stated Skeins for non semi-simple categories.

来源字段：http://aimpl.org/quantumlowdimtop/3/

09390 | OWR-740-002

### Maslov-Zero Lagrangian Tori in Quartic K3 Surfaces

S / solved | Oberwolfach 报告问题汇编

If $L_0$ is an oriented Lagrangian torus in a quartic K3 surface and its Maslov class $\mu_{L_0}$ vanishes, must its homology class $[L_0]$ be nonzero?

来源字段：https://doi.org/10.4171/owr/2004/38

09391 | AIM-TOPOLOGY-0127

### Matched filling, the homology-slice reduction, and odd-surgery recovery of the V-profile

P / partially_solved | AIM 研讨会问题汇编

**Surgeries and concordance.** Suppose \(K\) and \(J\) are knots in \(S^3\) such that for all \(p/q\in\mathbb Q\), the Dehn surgeries \(S^3_{p/q}(K)\) and \(S^3_{p/q}(J)\) are integer homology cobordant. Must \(K\) and \(J\) be concordant?

来源字段：http://aimpl.org/concordsliceknot/4/

09392 | OWR-11786-020

### Matveev-Jaco-Rubinstein Lens-Space Complexity Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For coprime integers $p>q>0$ with $p>3$, is the complexity of the lens space $L(p,q)$ equal to $E(p,q)-3$, where $E(p,q)$ is the sum of the partial denominators in the continued-fraction expansion of $p/q$?

来源字段：https://doi.org/10.4171/owr/2012/24

09393 | OWR-4214-012

### Maximal Coarse Baum-Connes Assembly for Expander Quotients

O / open | Oberwolfach 报告问题汇编

For a residually finite, finitely generated group $\Gamma$ with property $\tau$ and finite quotients $X(\Gamma)=\bigsqcup_i \Gamma/\Gamma_i$, is the maximal coarse Baum-Connes assembly map $\mu_{X(\Gamma),*}^{\max}$ an isomorphism?

来源字段：https://doi.org/10.4171/owr/2010/13

09394 | AMR-112-0003

### Mazur's finite-components conjecture for rational points

P / partially_solved | AMR 问题清单汇编

For every algebraic variety $X$ defined over $\mathbb{Q}$, does the closure of $X(\mathbb{Q})$ inside the real locus $X(\mathbb{R})$ have only finitely many connected components?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Topology

09395 | OWR-16406-006

### Minimal Dimensions of Manifold Thickenings

O / open | Oberwolfach 报告问题汇编

Let $X$ be a finite aspherical complex with nonzero $k$th $L^2$-Betti number. Must every manifold thickening of $X$ have dimension at least $2k$? Equivalently, what is the minimum possible dimension of such a thickening?

来源字段：https://doi.org/10.4171/owr/2018/31

09396 | OWR-17471-017

### Minimal Exotic Smooth Structures on Closed Four-Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Determine the simplest closed compact orientable smooth 4-manifold admitting multiple smooth structures, and whether any closed compact orientable definite 4-manifold admits multiple smooth structures.

来源字段：https://doi.org/10.4171/owr/2020/8

09397 | OWR-4798-027

### Minimal Facets in Triangulated Tori

O / open | Oberwolfach 报告问题汇编

If $P$ is a simplicial cell decomposition of the $d$-dimensional torus $T^d=(S^1)^d$, must $P$ have at least $(d+1)!$ facets?

来源字段：https://doi.org/10.4171/owr/2011/08

09398 | OWR-12014-009

### Minimal Pseudo-Anosov Dilatations

P / partially_solved | Oberwolfach 报告问题汇编

What is the behavior of the minimum pseudo-Anosov dilatation $\delta(S_{g,n})$ as a function of genus $g$ and punctures $n$? Is there an accumulation point of normalized dilatations smaller than $\ell_0=((3+\sqrt 5)/2)^2$, and what do small-dilatation mapping classes look like?

来源字段：https://doi.org/10.4171/owr/2012/42

09399 | AMR-102-0007

### Minimal surfaces - Question 4.1

P / partially_solved | AMR 问题清单汇编

Suppose $\mathscr{F}$ is a taut foliation of $M$. Characterize the space of metrics on $M$ for which $\mathscr{F}$ can be isotoped to consist of minimal surfaces.

背景链接：https://arxiv.org/abs/math/0209081

09400 | AMR-102-0008

### Minimal surfaces - Question 4.2

O / open | AMR 问题清单汇编

Given a collection of taut foliations $\mathscr{F}_i$ of $M$, what are the obstructions to finding a metric on $M$ for which the $\mathscr{F}_i$ (after an isotopy) are simultaneously minimal?

背景链接：https://arxiv.org/abs/math/0209081

09401 | OWR-13940-007

### Minimal Triangulated Sphere Eversions

O / open | Oberwolfach 报告问题汇编

For simplicial immersions of triangulated $2$-spheres, which spheres can be turned inside out; what is the smallest sphere that can be everted; and what is the smallest triangulated sphere with $2^{2g}$ discrete regular homotopy classes?

来源字段：https://doi.org/10.4171/owr/2015/45

09402 | AIM-TOPOLOGY-0208

### Minimal-area conformal metrics and a thick-part Thurston bound

P / partially_solved | AIM 研讨会问题汇编

For a fixed Riemann surface \(X\), among conformal metrics \(\rho|dz|\) such that every homotopically nontrivial closed curve \(\gamma\) has \(\ell_\rho(\gamma)\geq1\), is the infimum of area attained? Does this problem have a relation to Thurston's metric?

\[ \inf_{\gamma}\ell_\rho(\gamma)\geq 1. \]

The updated version in Su's 2016 handbook chapter repeats the same question as Problem 6.5. Neither short problem-list version says in the displayed formula which curves \(\gamma\) are allowed. That missing quantifier is fixed by the cited primary source, Wolf--Zwiebach: \(\gamma\) runs through all homotopically nontrivial closed curves on the Riemann surface, with homotopy taken relative to the punctures. It is enough to impose the condition on simple closed curves. Wolf--Zwiebach normalize the lower bound to \(2\pi\); rescaling the conformal factor gives the bound \(1\) in Su's question.

Thus the recovered question is:

There is an important qualification. On a closed surface, “area” is ordinary area. On a punctured surface, the intended Wolf--Zwiebach problem uses complete metrics and a renormalized or *reduced* area. Ordinary area is necessarily infinite when peripheral curves are included; Proposition 5.1 below proves the elementary divergence and exactly recovers Wolf--Zwiebach's subtraction coefficient.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

09403 | AIM-TOPOLOGY-0066

### Minimal-rank locus and exact signed families for Jucys--Murphy braid closures

P / partially_solved | AIM 研讨会问题汇编

For integer exponents \(\boldsymbol\tau=(\tau_1,\ldots,\tau_n)\), compute the ordinary unreduced Khovanov homology of the planar closure of the Jucys--Murphy braid product \(\prod_j JM_j^{\tau_j}\).

The sentence is not a complete mathematical specification. It omits the functor applied to the braid, the closure, coefficients, reduction, grading normalization, the allowed exponents, and the convention for \(JM_j\).

We use \(\tau_j\) rather than \(t_j\) for braid exponents, reserving \(s\) for the homological variable. Our convention is \[ JM_1=1,\qquad JM_j=\sigma_{j-1}\cdots\sigma_2\sigma_1^2 \sigma_2\cdots\sigma_{j-1}\in Br_n\quad(j\ge2). \] This is the positive multiplicative Jucys--Murphy braid used, for example, by Gorsky--Hogancamp. The \(JM_j\) are pure and commute. Let \[ \beta_{\boldsymbol\tau}=\prod_{j=1}^n JM_j^{\tau_j}, \qquad L_{\boldsymbol\tau}=\widehat{\beta_{\boldsymbol\tau}}, \] with the coherent braid orientation. We use ordinary even, unreduced, integral Khovanov cohomology \(Kh^{i,j}\) in Khovanov's normalization; field coefficients are written explicitly. The exponent \(\tau_1\) is inert because \(JM_1=1\).

来源字段：http://aimpl.org/agclinkhom/3/

09404 | OWR-758-011

### Minimum Coincidence Components of Homotopic Map Pairs

P / partially_solved | Oberwolfach 报告问题汇编

Let $f_1, f_2 : M \to N$ be continuous maps between closed manifolds, and let $C(f_1, f_2) = \{x \in M : f_1(x) = f_2(x)\}$ denote their coincidence locus. In Nielsen coincidence theory, determine the minimum number of coincidence path-components $$ \operatorname{MCC}(f_1, f_2) = \min\left\{\#\pi_0(C(f'_1, f'_2)) : f'_1 \simeq f_1,\ f'_2 \simeq f_2\right\}. $$

来源字段：https://doi.org/10.4171/owr/2004/44

09405 | OWR-15427-010

### Minimum Faces in Complexes Not Embeddable in Manifolds

O / open | Oberwolfach 报告问题汇编

Given a $k$-dimensional simplicial complex and a $2k$-dimensional manifold $M$, determine the minimum number of $k$-faces in a simplicial complex that does not embed in $M$.

来源字段：https://doi.org/10.4171/owr/2017/19

09406 | OWR-12697686-016

### Minimum Genus Connected Sums of Complex Plane Curves

P / partially_solved | Oberwolfach 报告问题汇编

For positive integers $n,m$, is the connected sum of smooth complex plane curves of degrees $n$ and $m$, embedded in the two summands of $\mathbb{CP}^2\#\mathbb{CP}^2$, a smoothly minimum-genus representative of its homology class?

来源字段：https://doi.org/10.4171/owr/2023/3

09407 | OWR-14298592-006

### Minimum-Volume Hyperbolic Knots in the Three-Torus

O / open | Oberwolfach 报告问题汇编

Is the topological volume of the three-torus equal to twice the volume $v_8$ of a regular ideal octahedron, with the minimum realized by the three-component link whose complement is homeomorphic to the Borromean-rings complement; that is, is $\operatorname{vol}_t(T^3)=2v_8$?

来源字段：https://doi.org/10.4171/owr/2024/43

09408 | AIM-TOPOLOGY-0188

### Mirzakhani-typical recurrence and equidistribution of classical stretch lines

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.10 (K. Rafi).** Is a stretch line typically recurrent/dense/ equidistributed in moduli space?

Rafi. Inspection of the original PDF gives the question

First, the line break `e-quidistributed` means “equidistributed.” Second, the sentence about Lenzhen and Masur is printed after a dotted separator and introduces Problem 2.11, concerning pairs of Thurston geodesic rays; it is not part of Problem 2.10.

The source works with Teichmüller spaces of finite-type hyperbolic surfaces. The result below directly treats the closed case of genus at least two, which is the setting of the presently available ergodic stretch-flow theorem.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

09409 | AMR-110-0063

### Miscellaneous problems 1 - Build MU from the moduli stack of formal groups.

P / partially_solved | AMR 问题清单汇编

Build MU from the moduli stack of formal groups. This has got to be doable somehow, though it is an old problem (I first heard it in Ravenel's green book). Note that we have some more tools now--the moduli stack is, I think, just a space in Voevodsky's category, so maybe it is an infinite loop space there?

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09410 | AMR-110-0064

### Miscellaneous problems 2 - Classify all possible Bousfield classes of E-infinity ring spectra.

P / partially_solved | AMR 问题清单汇编

Classify all possible Bousfield classes of E-infinity ring spectra. I know very little about this problem. Note that the Spanier-Whitehead dual of the suspension spectrum of a space X is an E-infinity ring spectrum, with multiplication dual to the diagonal map. But you are forgetting the disjoint basepoint!! When you add, as you must, a disjoint basepoint to X, you find that the E-infinity ring spectrum has the Bousfield class of the sphere. Also, I once heard somebody--Jim McClure?--say that if you start with, say, MU, and you mod out by p, you find that you must also mod out by all the higher v's too in order to get an E-infinity ring spectrum. Therefore, I conjecture that if E is an E-infinity ring spectrum that kills a nontrivial finite spectrum X, then E has the Bousfield class of E(n) for some n. (I must be p-local here). We know these Bousfield classes do occur, since Morava E-theory is an E-infinity ring spectrum. (Goerss-Hopkins).

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09411 | AMR-110-0065

### Miscellaneous problems 3 - This one is due to Mike Hopkins.

S / solved | AMR 问题清单汇编

This one is due to Mike Hopkins. Generalize the whole Thom spectrum business as follows. Take an A-infinity ring spectrum E. Look at the space of A-infinity self equivalences of E. I think this has a classifying space B, because of composition of self equivalences. Given a map X --> B you should be able to construct a Thom spectrum, and it should be some kind of half-smash product of E and X. Mike had a more explicit description, which I seem to have forgotten.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09412 | AMR-102-0066

### Miscellaneous - Question 15.1

O / open | AMR 问题清单汇编

What possibilities are there for (co-oriented) laminations in a $3$-manifold whose transverse spaces are well-ordered?

Is there a (useful) theory of branched surfaces with ordinal-valued weights?

背景链接：https://arxiv.org/abs/math/0209081

09413 | AMR-102-0067

### Miscellaneous - Question 15.2

P / partially_solved | AMR 问题清单汇编

Is there a good notion of taut foliated cobordism? Are there numerical invariants of the equivalence classes this induces on taut foliations which are finer than the Godbillon-Vey invariant?

背景链接：https://arxiv.org/abs/math/0209081

09414 | OWR-14299291-008

### Mixed-Chromatic Cohomology Beyond the Extreme Case

P / partially_solved | Oberwolfach 报告问题汇编

In chromatic homotopy theory, let $L_{K(n)} S$ be the $K(n)$-local sphere spectrum at height $n$ and prime $p$. Extend the mixed-chromatic / prismatic cohomology comparison isomorphism for the rationalization of the $K(n)$-local sphere from chromatic level $t = n - 1$ to arbitrary $t$, and remove the completed-perfection term from the $t = n - 1$ comparison isomorphism.

来源字段：https://doi.org/10.4171/owr/2025/29

09415 | OWR-14743-007

### Mod-p Formality of Little-Disks Cochains

P / partially_solved | Oberwolfach 报告问题汇编

Is the nonsymmetric cooperad $C^*(D_n,\mathbb F_p)$ of the little $n$-disks operad formal for every prime $p$ when $n>2$?

来源字段：https://doi.org/10.4171/owr/2016/35

09416 | AMR-110-0047

### Model categories 1 - The safest sort of problem to work on with model categories is building one of interest in applications.

S / solved | AMR 问题清单汇编

The safest sort of problem to work on with model categories is building one of interest in applications. The essential idea is: whenever someone uses the word homology, there ought to be a model category around. I like this idea a great deal, and it might lead to expansion of algebraic topology into many different areas. The simplest example that I personally do not understand is complexes of (quasi-coherent?) sheaves over a scheme. There is certainly a model structure here, and it is probably even known. But I think it would be good to find this out, and find out how the model structure is built. I believe this should be a symmetric monoidal model category. I also have the impression that one can not generalize the usual model structure on chain complexes over a ring, because you won't have projectives. But these two impressions sort of contradict each other, since the second one would lead you to generalize the injective model structure on chain complexes over a ring, but this model structure is not symmetric monoidal. So there is something for me at least to learn here.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09417 | AMR-110-0056

### Model categories 10 - Is every monoidal model category Quillen equivalent to a simplicial monoidal model category?

P / partially_solved | AMR 问题清单汇编

Is every monoidal model category Quillen equivalent to a simplicial monoidal model category? This would remove the loose end in my book on model categories, where I am unable to show that the homotopy category of a monoidal model category is a central algebra over the homotopy category of simplicial sets. The centrality is the problem, and I can cope with this problem for simplicial monoidal model categories.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09418 | AMR-110-0057

### Model categories 11 - Charles Rezk has a homotopy theory of homotopy theories.

S / solved | AMR 问题清单汇编

Charles Rezk has a homotopy theory of homotopy theories. This is just a category, though it is large. The objects are generalizations of categories where composition is not associative on the nose--that is, they are some kind of simplicial spaces. Understand the relationship between Rezk's point of view and mine on the 2-category of model categories. They should be equivalent in some sense.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09419 | AMR-110-0058

### Model categories 12 - In the appendix to my book on model categories, I said maybe what we are doing in associating to a mod...

P / partially_solved | AMR 问题清单汇编

In the appendix to my book on model categories, I said maybe what we are doing in associating to a model category its homotopy category is the wrong thing. Maybe we should be associating to a model category C the homotopy categories of all the diagram categories C^I, together with all the adjunctions induced by functors I --> J. This would make homotopy limits and colimits part of the structure. Does this viewpoint have any value?

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09420 | AMR-110-0059

### Model categories 13 - Find a model category you can prove is not cofibrantly generated.

S / solved | AMR 问题清单汇编

Find a model category you can prove is not cofibrantly generated. This is just an annoyance, not a very significant problem, but it has been bugging me for a while. The obvious candidate for this is the simplest nontrivial model category, the one on chain complexes where weak equivalences are chain homotopy equivalences. Mike Cole is, so far as I know, the first to write down a desciption of this model category, though one certainly has the feeling that Quillen must have known about it. But how do you prove something is not cofibrantly generated?

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09421 | AMR-110-0048

### Model categories 2 - A scheme is a generalization of a ring, in the same way that a manfold is a generalization of R^n.

P / partially_solved | AMR 问题清单汇编

A scheme is a generalization of a ring, in the same way that a manfold is a generalization of R^n. So maybe there is some kind of model structure on sheaves over a manifold? Presumably this is where de Rham cohomology comes from, but I don't know. It doesn't seem like homotopy theory has made much of a dent in analysis, but I think this is partly due to our lack of trying. Floer homology, quantum cohomology--do these things come from model structures?

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09422 | AMR-110-0049

### Model categories 3 - Every stable homotopy category I know of comes from a model category.

S / solved | AMR 问题清单汇编

Every stable homotopy category I know of comes from a model category. Well, that used to be true, but it is no longer. Given a flat Hopf algebroid, Strickland and I have constructed a stable homotopy category of comodules over it. This clearly ought to be the homotopy category of a model structure on the category of chain complexes of comodules, but we have been unable to build such a model structure. My work with Strickland is still in progress, so you will have to contact me for details.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09423 | AMR-110-0050

### Model categories 4 - Given a symmetric monoidal model category C, Schwede and Shipley have given conditions under which the…

S / solved | AMR 问题清单汇编

Given a symmetric monoidal model category C, Schwede and Shipley have given conditions under which the category of monoids in C is again a model category (with underlying fibrations and weak equivalences). On the other hand, the category of commutative monoids seems to be much more subtle. It is well-known that the category of commutative differential graded algebras over Z can not be a model category with uinderlying fibrations and weak equivalences (= homology isos). On the other hand, the solution to this is also pretty well-known--you are supposed to be using E-infinity DGAs, not commutative ones. Find a generalization of this statement. Here is how I think this should go, broken down into steps. The first step: find a model structure on the category of operads on a given model category. (Has this already been done? Charles Rezk is the person I would ask). We probably have to assume the model category is cofibrantly generated.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09424 | AMR-110-0051

### Model categories 5 - The second step: show that the category of algebras over a cofibrant operad admits a model structure,…

S / solved | AMR 问题清单汇编

The second step: show that the category of algebras over a cofibrant operad admits a model structure, where the fibrations and weak equivalences are the underlying ones. Show that a weak equivalence of cofibrant operads induces a Quillen equivalence of the categories of algebras. Show that an E-infinity operad is just a cofibrant approximation to the commutative ring operad. (This latter statement is probably known, since to me it seems to be the whole point of E-infinity).

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09425 | AMR-110-0052

### Model categories 6 - Find conditions under which algebras over a noncofibrant operad admit a model structure that generaliz…

S / solved | AMR 问题清单汇编

Find conditions under which algebras over a noncofibrant operad admit a model structure that generalize the monoid axiom of Schwede-Shipley. This would include the case where everything is fibrant, for example. Show that, under some more conditions, a weak equivalence of operads induces a Quillen equivalence of the algebra categories. Thus, sometimes you can use commutative, sometimes you can't, but you can always use E-infinity. And using E-infinity will not hurt you when you can use commutative.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09426 | AMR-110-0053

### Model categories 7 - Let A be a cofibrant operad as above.

S / solved | AMR 问题清单汇编

Let A be a cofibrant operad as above. Use the above results to construct spectral sequences that converge to the homotopy groups of the space of A-algebra structures on a given object X, and to the homotopy groups of the mapping space of A-algebra maps between two given A-algebras. These spectral sequences for the A-infinity operad are the key formal ingredients to the Hopkins-Miller proof that Morava E-theory admits an action by the stabilizer group.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09427 | AMR-110-0054

### Model categories 8 - My general theory is that the category of model categories is not itself a model category, but a 2-mod…

P / partially_solved | AMR 问题清单汇编

My general theory is that the category of model categories is not itself a model category, but a 2-model category. Weak equivalences of model categories are Quillen equivalences, and weak equivalences of Quillen functors are natural weak equivalences. Define a 2-model category and show the 2-category of model categories is one. Note that the homotopy 2-category at least makes sense (in a higher universe): we can just invert the Quillen equivalences and the natural weak equivalences. This localization process for an n-category has been studied by Andre Hirschowitz and Carlos Simpson in descent pour les n-champs, on xxx.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09428 | AMR-110-0055

### Model categories 9 - The 2-category of simplicial model categories is supposed to be (according to me) 2-Quillen equivalent…

P / partially_solved | AMR 问题清单汇编

The 2-category of simplicial model categories is supposed to be (according to me) 2-Quillen equivalent to the 2-category of model categories. Even without having all the definitions one can try to find out if this is true. For example, Dan Dugger has shown that every model category (with some hypotheses--surely cofibrantly generated at least) is Quillen equivalent to a simplicial model category. Understand his result in the context of the preceding two problems. That is, does Dugger's construction in fact give a 2-functor from model categories to simplicial model categories? Does it preserve enough structure to make it clear that it will induce some kind of equivalences on the homotopy 2-categories?

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09429 | AIM-TOPOLOGY-0264

### Model dependence of generic asphericity and a random-triangulation theorem

P / partially_solved | AIM 研讨会问题汇编

**Problem 51.** Is a “generic” manifold a $K(\pi,1)$-space (where “generic” is to be determined)?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

09430 | OWR-832-002

### Modular Forms from Generalized Cohomology Theories

P / partially_solved | Oberwolfach 报告问题汇编

Stabilizing classifying spaces of elliptic bundles associated with genus-one conformal field theories produces generalized cohomology theories $E$ with natural maps $E\to K[[q]][q^{-1}]$. Are the images of the coefficient rings under these maps always modular forms?

来源字段：https://doi.org/10.4171/owr/2005/27

09431 | OWR-11101915-009

### Modulo-Four Reduced Khovanov Rank of Ribbon Knots

O / open | Oberwolfach 报告问题汇编

Let $R$ be $\mathbb F_p$ or $\mathbb Z$. Prove that $\operatorname{rk}\widetilde{\operatorname{Kh}}_R(K)\pmod4$ is a knot-concordance invariant; equivalently, prove $\operatorname{rk}\widetilde{\operatorname{Kh}}_R(J)\equiv1\pmod4$ for every ribbon knot $J$.

来源字段：https://doi.org/10.4171/owr/2022/34

09432 | OWR-15177-005

### Monodromy of Universal Plane-Curve Bundles

S / solved | Oberwolfach 报告问题汇编

For the universal smooth plane-curve bundle of degree $d\geq4$, is the monodromy group $\Gamma_d$ equal to the stabilizer $\operatorname{Mod}_g[\phi_d]$ of the associated $(d-3)$-spin structure, where $g=(d-1)(d-2)/2$?

来源字段：https://doi.org/10.4171/owr/2016/56

09433 | AMR-110-0011

### Morava K- and E-theory 1 - Show that pi_* L_K(n) S^0 is finitely generated over the p-adics in each degree.

P / partially_solved | AMR 问题清单汇编

Show that pi_* L_K(n) S^0 is finitely generated over the p-adics in each degree. This would follow from the chromatic splitting conjecture, I think. (I believe there is an argument for this in Devinatz' last paper on the generating hypothesis). I believe we know almost nothing about these groups, though we probably know they are pro-p-groups. I am not even sure about that right now. An unfortunate point here is that this is false if * is allowed to range over the Picard group instead of just the integers (Hovey-Strickland, based on Shimomura's calculation of L_2 S^0 at p>3).

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09434 | AMR-110-0020

### Morava K- and E-theory 10 - Bousfield has give a description of the E(1)-local category in terms of algebraic data related to K-th…

P / partially_solved | AMR 问题清单汇编

Bousfield has give a description of the E(1)-local category in terms of algebraic data related to K-theory. Franke claims to have generalized all this, but once again, I do not understand what he does. Bousfield describes the isomorphism classes, but not the maps. In the K(1)-local category, it seems to me, things might be a bit simpler. Is there some more natural description of Bousfield's work in the K(1)-local category? It is always dangerous to try to improve on Bousfield, and certainly no one has succeeded in the past 10 plus years.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09435 | AMR-110-0012

### Morava K- and E-theory 2 - Show that the Picard group is finitely generated over the p-adics.

P / partially_solved | AMR 问题清单汇编

Show that the Picard group is finitely generated over the p-adics. I don't think this is known even for the algebraic Picard group, which is obtained purely through group cohomology. So I think there might be some room for some purely algebraic understanding of profinite group cohomology here.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09436 | AMR-110-0013

### Morava K- and E-theory 3 - Elucidate the connection between the Morava stabilizer groups and the K(n)-local category.

S / solved | AMR 问题清单汇编

Elucidate the connection between the Morava stabilizer groups and the K(n)-local category. The first such problem, which is certainly not very hard and is no doubt known to Hopkins and some others, is to determine the structure of E*E and L_K(n)(E_*E), where E is Morava E-theory. You can take Morava E-theory to be the K(n)-localization of E(n), or you can take it to be the one related to the Lubin-Tate moduli space. The answer is supposed to be: for E^*E, you should get the twisted completed group ring E[[S]] of the stabilizer group--recall the stabilizer group acts on E, and that is where the twisting comes in. The completion affects both E and S and is an inverse limit over open subgroups of finite index. The answer for L_K(n)(E_*E), by which I mean pi_*(L_K(n) (E smash E)), should be C(S,E), continuous functions from S to E. Part of this problem is choosing exactly which E you want to work with and defining all these terms precisely. Actually, it would be better to do this for every reasonable choice of Morava E-theory E. Neil Strickland and I could probably have done this in our memoir on the K(n)-local category, but we thought the paper was long enough. But nobody else has done it, so maybe not. Even though this is folklore, I think it would be of value to get it all straight.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09437 | AMR-110-0014

### Morava K- and E-theory 4 - As a rule, I am not happy about the arbitrary nature of some of the constructions in the K(n)-local ca…

P / partially_solved | AMR 问题清单汇编

As a rule, I am not happy about the arbitrary nature of some of the constructions in the K(n)-local category. Consider the spectral sequence, for example, which relates the continuous cohomology of S with coefficients in pi_*(L_K(n)(E smash X)) to pi_* L_K(n) X, for example. The usual construction of this is to take the Adams-Novikov spectral sequence for E smash X smash a type n, and then take an inverse limit. I have never liked this, and it certainly only works for dualizable X. Is there some more natural construction? I have the same objection to the more general Devinatz-Hopkins construction of fixed point spectra designed to yield a spectral sequence starting with continuous cohomology rather than discrete cohomology.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09438 | AMR-110-0015

### Morava K- and E-theory 5 - Find the shadow of the thick subcategory theorem in the K(n)-local category.

P / partially_solved | AMR 问题清单汇编

Find the shadow of the thick subcategory theorem in the K(n)-local category. There is only one thick subcategory of small spectra in the K(n)-local category, corresponding to the type n finites. But the finite spectra of smaller type are still there--they are dualizable, but not small. My own conjecture, with Strickland, is that the ideals of dualizable spectra give you the expected filtration--an ideal is a thick subcategory closed under smashing with any dualizable spectrum.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09439 | AMR-110-0016

### Morava K- and E-theory 6 - We now know that Morava E-theory admits an action of the stabillizer group S.

S / solved | AMR 问题清单汇编

We now know that Morava E-theory admits an action of the stabillizer group S. This is the famous Hopkins-Miller result, which one day I hope will see the light of day. Their proof relies on calculating the A_infinity automorphism group of E and showing that it is a homotopy discrete group whose pi_0 is isomorphic as an abstract group to the stabilizer group. As a method, this leaves a lot to be desired. The biggest problem here is: give a construction of Morava E-theory that makes it obvious that there is an action of the stabilizer group on it. Presumably this will require going from Lubin-Tate moduli spaces to spectra using some sort of infinite loop space technology. Note that the action should be continuous in an appropriate sense.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09440 | AMR-110-0017

### Morava K- and E-theory 7 - Presumably one should be able to form a category of E-S module spectra; spectra with an action of the…

S / solved | AMR 问题清单汇编

Presumably one should be able to form a category of E-S module spectra; spectra with an action of the ring spectrum E and a compatible action of the group S. Is this possible--and is it worthwhile? This one is going to take you into formal issues that one can spend a lot of time and verbiage on.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09441 | AMR-110-0018

### Morava K- and E-theory 8 - Understand the relationship between the K(n)-local category and some sort of (algebraic) derived categ…

P / partially_solved | AMR 问题清单汇编

Understand the relationship between the K(n)-local category and some sort of (algebraic) derived category of E_*-S-modules. Jens Franke has claimed there is an equivalence there, but I don't really understand what he does. (This is likely a deficiency in me) In particular, I don't even know what the right algebraic category is. E_*E is a Hopf algebroid, so maybe we should consider comodules over it. But it is really pi_* L_K(n)(E smash E) that we should be considering, and I don't know what sort of algebraic gadget that is.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09442 | AMR-110-0019

### Morava K- and E-theory 9 - One of the corollaries of the Hopkins-Miller theorem, together with the Devinatz-Hopkins fixed point b…

P / partially_solved | AMR 问题清单汇编

One of the corollaries of the Hopkins-Miller theorem, together with the Devinatz-Hopkins fixed point business, is that the famous class zeta in continuous H^1 of the stabilizer group survives to give a class in homotopy of degree -1. This is trivial at large primes, where the spectral sequence collapses, but at small primes it was not known. The chromatic splitting conjecture asserts that there should be other such classes, in degree -3, -5,...,-2n+1, so presumably in continuous H^3, H^5,...,H^2n-1 of S with trivial coefficients. Find these classes, and see if Hopkins-Miller allows you to determine that they survive to homotopy classes at all primes.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

09443 | OWR-17471-016

### Motion Groups of Hopf-Tree Links

O / open | Oberwolfach 报告问题汇编

For embedding spaces of Hopf-tree links in $\mathbb{R}^3$, is the motion group isomorphic to the symmetric automorphism group of the right-angled Artin group associated to the corresponding disjoint union of tree graphs?

来源字段：https://doi.org/10.4171/owr/2020/8

09444 | AIM-TOPOLOGY-0154

### Natural generators and exact genus-flow/end-motion budgets

P / partially_solved | AIM 研讨会问题汇编

Are there “nice” or “natural” generators for \(\operatorname{Map}(S)\)?

来源字段：http://aimpl.org/genusinfinity/4/

09445 | AMR-112-0005

### Nearby Lagrangian conjecture

P / partially_solved | AMR 问题清单汇编

Let $M$ be a closed manifold. Is every closed exact Lagrangian submanifold of the cotangent bundle $T^*M$ Hamiltonian isotopic to the zero section?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Topology

09446 | AIM-TOPOLOGY-0122

### Necessary gates and a bounded iteration trap for winding-zero sliceness detectors

P / partially_solved | AIM 研讨会问题汇编

**Injectivity of winding number 0 satellite operators.** Is there a pattern $P$ with winding number 0 such that $P(K)$ is slice if and only if $K$ is slice, in either category?

来源字段：http://aimpl.org/concordsliceknot/3/

09447 | AIM-TOPOLOGY-0273

### Negative curvature on a surface bundle over a surface

P / partially_solved | AIM 研讨会问题汇编

**Question 3.3.** Fix \(g,h\), with \(h\geq 2\). Does there exist a locally trivial fibration of surfaces \[ F_h\longrightarrow X\longrightarrow B_g \] such that \(X\) admits a metric of strictly negative curvature?

来源字段：https://aimath.org/WWN/surfacegroups/surfacegroups.pdf

09448 | AIM-TOPOLOGY-0128

### Nilpotent blindness for meridians in zero-surgery cobordisms

P / partially_solved | AIM 研讨会问题汇编

**0-surgeries and concordance.** Suppose $K$ and $J$ are knots in $S^3$ such that there is an integer homology cobordism $W$ between $S^3_0(K)$ and $S^3_0(J)$ in which the positively oriented meridians $\mu_K$ and $\mu_J$ are freely homotopic. Must $K$ and $J$ be concordant?

来源字段：http://aimpl.org/concordsliceknot/4/

09449 | AIM-TOPOLOGY-0158

### Non-forgetful cyclic quotients and controlled finite-quotient families

P / partially_solved | AIM 研讨会问题汇编

When $S$ is a finite genus surface, there are forgetful homomorphisms from $\operatorname{PMap}(S)$ to the mapping class groups of finite-type surfaces. Those finite-type mapping class groups are residually finite, so there are many further quotients. What other finite quotients can we have for $\operatorname{Map}(S)$ and $\operatorname{PMap}(S)$?

来源字段：http://aimpl.org/genusinfinity/4/

09450 | OWR-15177-011

### Non-Liftable Monodromy of Surface Bundles

O / open | Oberwolfach 报告问题汇编

Does there exist a nonflat surface bundle $\Sigma_g\to E\to\Sigma_h$ for which the monodromy homomorphism from $\pi_1(\Sigma_h)$ to $\operatorname{Mod}_g$ does not lift through $\operatorname{Diff}^+(\Sigma_g)$ in the category of groups?

来源字段：https://doi.org/10.4171/owr/2016/56

09451 | AIM-TOPOLOGY-0019

### Non-minimal connected sums of non-destabilizable open books

P / partially_solved | AIM 研讨会问题汇编

Can non-destabilizable OBDs connect sum to a destabilizable (or otherwise non-minimal) OBD?

来源字段：http://aimpl.org/highdimcontacttop/7/

09452 | OWR-15955-009

### Nonabelian Quotients of the String Link Solvable Filtration

P / partially_solved | Oberwolfach 报告问题汇编

For $m\geq2$ and $n\geq0$, is the quotient $\mathcal F_n^m/\mathcal F_{n.5}^m$ in the $n$-solvable filtration of the $m$-component string-link concordance group nonabelian?

来源字段：https://doi.org/10.4171/owr/2017/50

09453 | OWR-1596-002

### Noncommutative Alexander Polynomials and the Thurston Norm

O / open | Oberwolfach 报告问题汇编

Let $M$ be a closed aspherical $3$-manifold and let $\Lambda=\mathbb Z[H_1(M;\mathbb Z)/\mathrm{torsion}]$. Does there exist a noncommutative Alexander polynomial $\widetilde\Delta(M)$ whose abelianization is the classical Alexander polynomial, which satisfies the natural involution symmetry, whose support width computes the Thurston norm, and for which a cohomology class $\xi$ is represented by a nonsingular one-form exactly when $\widetilde\Delta(M)$ is invertible in the corresponding Novikov completion?

来源字段：https://doi.org/10.4171/owr/2007/40

09454 | OWR-14298007-009

### Noncompact Casacuberta-Dicks Conjecture

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Noncompact Casacuberta-Dicks Question [OWR-14299915-017]

Prove the Casacuberta-Dicks fixed-point conjecture without the compactness assumption: every action of a finite group on a contractible two-dimensional complex has a fixed point.

来源字段：https://doi.org/10.4171/owr/2023/58

09455 | OPG-37295

### Nonseparating planar continuum

P / partially_solved | Open Problem Garden

Conjecture Does any path-connected, compact set in the plane which does not separate the plane have the fixed point property?

A set has the fixed point property if every continuous map from it into itself has a fixed point.

背景链接：http://www.openproblemgarden.org/op/nonseparating_planar_continuum

09456 | OWR-13108-009

### Nonsimple Deformation Varieties and Geometrically Isolated Cusps

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a $k$-cusped hyperbolic $3$-manifold. If the deformation variety of $X$ is not simple, then $X$ has a set of cusps strongly geometrically isolated from the rest.

来源字段：https://doi.org/10.4171/owr/2014/38

09457 | OWR-1106-011

### Nonstabilized Weakly Reducible Heegaard Splittings

O / open | Oberwolfach 报告问题汇编

For the complements of the twisted torus knots $T(7,17,10m-4)$ with sufficiently large $|m|$, is the weakly reducible Heegaard splitting obtained by boundary stabilization of the unique genus-two Heegaard splitting stabilized? More generally, does a three-manifold with one boundary component admit a weakly reducible nonstabilized Heegaard splitting whose weak reduction is nonessential?

来源字段：https://doi.org/10.4171/owr/2005/45

09458 | OWR-15955-004

### Nontorus Abelian Duality Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a closed orientable manifold that is an abelian duality space but is not a real torus?

来源字段：https://doi.org/10.4171/owr/2017/50

09459 | OWR-13687-007

### Nontrivial Actions on Invariant Prym Homology

O / open | Oberwolfach 报告问题汇编

In the Prym-homology setting, let $\mathcal D_K$ be the connected normal subgroup generated by the one-parameter subgroups arising from lifted Dehn twists. Are there examples with $H_K^{\mathcal D_K}\neq0$ on which $\mathcal G_K^\circ/\mathcal D_K$ acts nontrivially?

来源字段：https://doi.org/10.4171/owr/2015/30

09460 | OWR-11454-005

### Nontrivial Witt Pairings in $\mathbb Z_2$-Witt Spaces

O / open | Oberwolfach 报告问题汇编

Does there exist a compact $\mathbb Z$-oriented $\mathbb Z_2$-Witt space of dimension $4j+2$ whose middle-dimensional pairing represents the nontrivial element of the Witt group $W(\mathbb Z_2)$?

来源字段：https://doi.org/10.4171/owr/2011/56

09461 | OWR-1063-006

### Nonvanishing Euler Classes in Topological Bordism

O / open | Oberwolfach 报告问题汇编

Let $G$ be a finite abelian $p$-group for an odd prime $p$, and let $V$ be a complex $G$-representation with no trivial direct summand. Is the Euler class of $V$ nonzero in topological bordism theory?

来源字段：https://doi.org/10.4171/owr/2005/42

09462 | AIM-TOPOLOGY-0210

### Normalized least-Lipschitz geometry beyond the hyperbolic locus

P / partially_solved | AIM 研讨会问题汇编

**Problem 5.4 (D. Dumas).** How does the “Lipschitz constant function” \[ (g,h)\longmapsto\inf_{\phi:g\to h}\operatorname{Lip}(\phi) \] behave on a class of metrics larger than the set of hyperbolic metrics, for example, the set of negatively curved Riemannian metrics or its closure?

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

09463 | AIM-TOPOLOGY-0221

### Normalized obstructions to pointwise quarter-pinched collapse

P / partially_solved | AIM 研讨会问题汇编

**Problem 8.** Is there a sequence of simply-connected, pointwise strictly \(\frac14\)-pinched manifolds \(M_i^n\), \(n>2\), that collapse?

There is a substantive ambiguity that the one-line source does not resolve. “Collapse” normally comes with a scale convention. In the bounded-curvature sense one asks for, after normalization, for example \[ |\sec_{g_i}|\leq 1,\qquad \operatorname{diam}(M_i,g_i)\leq D, \] and Gromov--Hausdorff convergence to a compact space of dimension less than \(n\) (equivalently, in this setting, volume tending to zero). Problem 7 immediately before this one explicitly assumes \(|\sec|\leq1\), which is evidence that a curvature normalization was intended in Problem 8 as well. Without an upper-curvature or diameter normalization, Problem 8 has a trivial affirmative answer by homothety; this is proved below.

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

09464 | OWR-17132-001

### Normed Algebra Lifts of the Motivic Tau Cofiber

O / open | Oberwolfach 报告问题汇编

Can the motivic cofiber $S^0/\tau$, regarded as a commutative algebra in the $2$-complete motivic stable homotopy category over $\mathbb C$, be lifted to a normed algebra?

来源字段：https://doi.org/10.4171/owr/2019/36

09465 | TOP-006

### Novikov Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Novikov Conjecture [TOP-013]; Novikov Homotopy Invariance Conjecture [OWR-4341-001]

Are certain combinations of Pontryagin classes homotopy invariant?

背景链接：https://doi.org/10.1007/s002229900032

09466 | OWR-761-003

### Nowhere-Zero Perturbations of Nonclosed One-Forms

O / open | Oberwolfach 报告问题汇编

Let $M$ be a closed manifold with $\chi(M)=0$, and let $\alpha$ be a one-form on $M$ with $d\alpha\neq0$. Does there exist a smooth function $f$ such that $\alpha+df$ has no zeros?

来源字段：https://doi.org/10.4171/owr/2004/47

09467 | OWR-15575-007

### Number of Simple Reeb Orbits on Three-Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

What can one say about the number of simple Reeb orbits of a contact form on a closed three-manifold?

来源字段：https://doi.org/10.4171/owr/2017/32

09468 | AMR-102-0054

### Numerical invariants - Question 13.1

O / open | AMR 问题清单汇编

Suppose $\mathscr{F}$ is a minimal taut $C^2$ foliation of an atoroidal $3$-manifold $M$ with $$\mathfrak{gv}(\mathscr{F})[M] \ne 0$$ Is there a choice of $1$-form $\alpha$ with $T\mathscr{F} = \ker(\alpha)$ for which the Godbillon-Vey form $\omega$ (where $d\alpha = \alpha \wedge \omega$) has the same sign? i.e. either $\omega \wedge d\omega \ge 0$ everywhere or $\le 0$ everywhere. Say that such a foliation has monotone wobble.

背景链接：https://arxiv.org/abs/math/0209081

09469 | AMR-102-0055

### Numerical invariants - Question 13.2

O / open | AMR 问题清单汇编

For $\mathscr{F}$ as in the previous question, suppose there is a choice of $\alpha$ for which $\omega$ is a contact form. Is the contact structure defined by $\omega$ necessarily tight?

背景链接：https://arxiv.org/abs/math/0209081

09470 | AMR-102-0056

### Numerical invariants - Question 13.3

O / open | AMR 问题清单汇编

Calculate the norm of the fundamental class of a hyperbolic $3$-manifold for some taut foliation $\mathscr{F}$ with two-sided branching.

背景链接：https://arxiv.org/abs/math/0209081

09471 | AMR-102-0057

### Numerical invariants - Question 13.4

O / open | AMR 问题清单汇编

Let $\mathscr{F},\mathscr{G}$ be taut foliations on a hyperbolic manifold $M$. Are there examples where there is a finite cover of $M$ such that a sequence of isotopies of the lift of $\mathscr{G}$ converges geometrically to the lift of $\mathscr{F}$, but no such sequence of isotopies exists in $M$?

背景链接：https://arxiv.org/abs/math/0209081

09472 | AMR-102-0058

### Numerical invariants - Question 13.5

O / open | AMR 问题清单汇编

Suppose $\mathscr{F}$ is a foliation (possibly $\mathbb{R}$-covered) of a hyperbolic $3$-manifold. Define a foliated Gromov norm using cubical chains. Is the value of the foliated norm on the fundamental class always strictly greater than the value of the usual (cubical) Gromov norm?

背景链接：https://arxiv.org/abs/math/0209081

09473 | AMR-102-0059

### Numerical invariants - Question 13.6

O / open | AMR 问题清单汇编

What kinds of local order structure are there on a family of deformations of a (taut) foliation? Can one use such structures to define co-ordinates on the ``space of deformations'' of a taut foliation?

背景链接：https://arxiv.org/abs/math/0209081

09474 | AMR-102-0060

### Numerical invariants - Question 13.7

P / partially_solved | AMR 问题清单汇编

Is there some notion of a Godbillon-Vey invariant for a lamination?

背景链接：https://arxiv.org/abs/math/0209081

09475 | AIM-TOPOLOGY-0200

### Obstructions and a Liouville-domination metric on a proper universal locus

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.9 (D. Dumas).** Can we define the Thurston metric on the universal Teichmüller space?

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

09476 | AIM-TOPOLOGY-0129

### Odd-primary defects in rational slice disks

P / partially_solved | AIM 研讨会问题汇编

**Rational sliceness.** Is every knot in $S^3$ which is slice in some rational homology ball actually slice in some $\mathbb Z[1/2]$-homology ball?

来源字段：http://aimpl.org/concordsliceknot/5/

09477 | AIM-TOPOLOGY-0130

### One sliceness problem resolved and an equivariant half-surgery reduction for the remaining knot cases

P / partially_solved | AIM 研讨会问题汇编

Difficult sliceness problems

Are any of the following slice? 1. The $(2,1)$ cable of the figure-eight knot. 2. The positive Whitehead double of any left-handed torus knot. 3. The positive Whitehead double of the figure-eight knot. 4. The $(+,+,-)$ Whitehead double of the Borromean rings.

来源字段：http://aimpl.org/concordsliceknot/5/

09478 | AMR-137-0001

### Open Problems in Topological Algebra - Problem 1 (Choban)

O / open | AMR 问题清单汇编

Is every topological group a quotient group of a zero-dimensional topological group of the same weight? When an almost metrizable topological group is a quotient of a zero-dimensional group of the same weight?

背景链接：https://arxiv.org/abs/1202.1619

09479 | AMR-137-0010

### Open Problems in Topological Algebra - Problem 10 (Banakh)

O / open | AMR 问题清单汇编

Is there a (metrizable) $o$-bounded topological group $G$ such that the player I has a winning strategy in the game OF($G$)?

背景链接：https://arxiv.org/abs/1202.1619

09480 | AMR-137-0011

### Open Problems in Topological Algebra - Problem 11 (Banakh)

O / open | AMR 问题清单汇编

Such a group, if exists, cannot be analytic and abelian (more generally, a SIN-group). We remind that a topological space $X$ is analytic if it is a metrizable continuous image of a separable complete metric space. On the other hand, the group ${\mathbb{Z}}^\omega$ contains a dense $o$-bounded $G_\delta$-subset $A$ such that the first player has a winning strategy in the game OF($A$), see [BZ].

Let $A$ be a compact subset of the real line $\mathbb{R}$ such that $A-A=\{a-a':a,a'\in A\}$ is a neighborhood of zero in $\mathbb{R}$. Has the sum $A+A+A+A$ non-empty interior in $\mathbb{R}$?

背景链接：https://arxiv.org/abs/1202.1619

09481 | AMR-137-0002

### Open Problems in Topological Algebra - Problem 2 (Choban)

O / open | AMR 问题清单汇编

Under which conditions is the free universal algebra of an uncountable signature over a metrizable space $X$ paracompact? In particular, is the free topological linear space $L(X)$ of $X$ over the discrete field of real numbers paracompact?

背景链接：https://arxiv.org/abs/1202.1619

09482 | AMR-137-0003

### Open Problems in Topological Algebra - Problem 3 (Protasov)

O / open | AMR 问题清单汇编

For a topological space $X$ let $S(X)\subset X^X$ be the semigroup of all continuous selfmappings of $X$ endowed with the topology of pointwise convergence (i.e., the topology inherited from the Tychonov product $X^X$). A topological space $X$ is called *homogeneous* if for any points $x_1,x_2\in X$ there is a homemorphism $h$ of $X$ with $h(x_1)=x_2$.

Is the semigroup $S(X)$ left bounded for every zero-dimensional compact homogeneous space?

背景链接：https://arxiv.org/abs/1202.1619

09483 | AMR-137-0004

### Open Problems in Topological Algebra - Problem 4 (Protasov)

O / open | AMR 问题清单汇编

The answer is positive provided $X$ has a base of the topology, consisting of pairwise homeomorphic clopen subsets, see [Pr1].

Is the semigroup $S(X)$ right bounded for any zero-dimensional homogeneous space $X$?

背景链接：https://arxiv.org/abs/1202.1619

09484 | AMR-137-0005

### Open Problems in Topological Algebra - Problem 5 (Protasov)

O / open | AMR 问题清单汇编

Let $X$ be a topological space. A subgroup $H$ of $S(X)$ is called *distal* if for any distinct points $x_1,x_2\in X$ and any point $x\in X$ there is a neighborhood $U$ of $x$ such that $\{h(x_1),h(x_2)\}\not\subset U$ for all $h\in H$. It is proven in [Pr1, Theorem 4] that a left bounded subgroup $H\subset S(X)$ is distal provided $H$ acts transitively on $X$.

Let $X$ be a compact space and $H$ be a distal subgroup of $S(X)$, acting transitively on $X$. Is $H$ left bounded?

背景链接：https://arxiv.org/abs/1202.1619

09485 | AMR-137-0006

### Open Problems in Topological Algebra - Problem 6 (Protasov)

O / open | AMR 问题清单汇编

It is well known that $\sigma$-compact topological groups have countable cellularity [Tk] while compact topological groups support a strictly positive probability measure (i.e., a Borel probability measure $\mu$ such that $\mu(U)>0$ for any nonempty open subset $U$ of the group). Recently T. Banakh and O. Ravsky [BR] (see also [BR2]) proved that any bounded paratopological group $G$ has countable cellularity (moreover, each cardinal of uncountable cofinality is a precaliber of $G$). On the other hand, according to [Pr2] for every infinite cardinal $\tau$ there is a left bounded left topological group of cellularity $\tau$.

Let $G$ be a bounded semitopological (quasitopological) group. Has $G$ countable cellularity?

背景链接：https://arxiv.org/abs/1202.1619

09486 | AMR-137-0007

### Open Problems in Topological Algebra - Problem 7 (Protasov)

O / open | AMR 问题清单汇编

Does every zero-dimensional compact homogeneous space admit the structure of a left topological group? (This problem was answered in negative by Banakh in [T.Banakh, *A homogeneous first-countable zero-dimensional compactum failing to be a left-topological group*, Mat. Stud.29 (2008) 215--217].) In particular, does the countable power $K^\omega$ of a first-countable zero-dimensional compact space $K$ admit the structure of a left topological group?

背景链接：https://arxiv.org/abs/1202.1619

09487 | AMR-137-0008

### Open Problems in Topological Algebra - Problem 8 (Protasov)

O / open | AMR 问题清单汇编

Can every non-discrete topological group $G$ be algebraically generated by a nowhere dense subset? (The answer to Problem 6 is ``No'': under CH the topological group $\mathbb{R}^{\omega_1}$ contains a dense non-separable subgroup $G$ such that each meager subset of $G$ generates a metrizable separable subgroup of $G$; see Proposition 13.8 in [I.Protasov, T.Banakh. Ball stuctures and colorings of graphs and groups // (Mat. Stud. Monograph Series. 11), VNTL Publ. 2003, 148p.])

背景链接：https://arxiv.org/abs/1202.1619

09488 | AMR-137-0009

### Open Problems in Topological Algebra - Problem 9 (Guran)

O / open | AMR 问题清单汇编

A paratopological group $G$ is *left $\omega$-bounded* (resp. *right $\omega$-bounded*) if for any neighborhood $U\subset G$ of the origin there is a countable subset $F\subset G$ with $G=FU$ (resp. $G=UF$); $G$ is *$\omega$-bounded* if it is both left and right $\omega$-bounded.

Is a paratopological group $G$ left $\omega$-bounded if it is right $\omega$-bounded?

背景链接：https://arxiv.org/abs/1202.1619

09489 | OWR-3478-004

### Open-Closed Conformal-Field Structure in String Topology

O / open | Oberwolfach 报告问题汇编

Does string topology carry the structure of an open-closed topological conformal field theory?

来源字段：https://doi.org/10.4171/owr/2009/28

09490 | AIM-TOPOLOGY-0195

### Opposition-symmetric lifts and boundary scalarizations of the Thurston metric

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.4 (K. Rafi).** Is the Thurston metric a degeneration of some better vector-valued symmetric distance on Teichmüller space?

It is introductory text for Problem 3.5, not part of Problem 3.4. It is nevertheless directly relevant notation.

The source list works with the Teichmüller space \(\mathcal T(S)\) of a finite-type surface \(S\) admitting complete finite-area hyperbolic metrics. We assume that \(\mathcal T(S)\) is nontrivial. With \(\mathcal C(S)\) the set of essential simple closed curves and \(\ell_X(\alpha)\) the \(X\)-length of \(\alpha\), the Thurston metric is \[ d(X,Y)=d_{\rm Th}(X,Y) =\log\sup_{\alpha\in\mathcal C(S)} \frac{\ell_Y(\alpha)}{\ell_X(\alpha)}. \]

The workshop report supplies useful context for the word “vector-valued”: the talk immediately preceding the problem session was M. Kapovich's “Vector-valued metrics on higher-rank Riemannian symmetric spaces (Cartan projections, conditions for triangular inequality).” In that setting reversal is governed by the opposition involution of a Weyl chamber, rather than by literal equality of oriented vector distances. This report makes that analogy precise and also records what cannot hold under a stricter interpretation.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

09491 | OWR-12867-001

### Optimal-Dimensional Spines for Teichmüller and Arithmetic Spaces

O / open | Oberwolfach 报告问题汇编

Construct a spine of Teichmüller space $\mathcal T_g$ of dimension $4g-5$. More generally, if a nonuniform arithmetic subgroup $\Gamma<G$ acts on the symmetric space $X$, construct a $\Gamma$-equivariant spine of dimension $\operatorname{vcd}(\Gamma)=\dim X-\operatorname{rank}_{\mathbb Q}(\Gamma)$.

来源字段：https://doi.org/10.4171/owr/2014/07

09492 | AIM-TOPOLOGY-0073

### Orbifold Hilbert schemes and the missing lens-link sector

P / partially_solved | AIM 研讨会问题汇编

Construct link invariants for links in lens spaces using Hilb\($([\mathbb{C}^2/(\mathbb{Z}/l\mathbb{Z}])\$\) and relate to wreath Macdonald polynomials.

来源字段：http://aimpl.org/agclinkhom/4/

09493 | OWR-10252937-012

### Orbifold Surgery Models for Positive Scalar Curvature Four Manifolds

O / open | Oberwolfach 报告问题汇编

Can a closed oriented topologically positive-scalar-curvature $4$-manifold $M$ be obtained from a topologically positive-scalar-curvature $4$-orbifold $M'$ with isolated singularities and finite orbifold fundamental group on each component by performing $0$- and $1$-surgeries?

来源字段：https://doi.org/10.4171/owr/2022/28

09494 | OWR-5158-010

### Ordinary Versus Immersive Simplicial Volume

O / open | Oberwolfach 报告问题汇编

For an aspherical manifold $M$, does its ordinary simplicial volume $\lVert M\rVert$ equal its immersive simplicial volume $\lVert M\rVert_{\mathrm{imm}}$, defined using fundamental cycles whose singular simplices are immersive?

来源字段：https://doi.org/10.4171/owr/2011/30

09495 | AIM-TOPOLOGY-0036

### Ordinary, inverted, and formal two-variable knot invariants

P / partially_solved | AIM 研讨会问题汇编

**Two variable invariants and their connections.** Are the Habiro 2-variable invariants for knots and Gukov--Manolescu invariant (also known as the F-invariant) the same?

来源字段：http://aimpl.org/quantumlowdimtop/1/

09496 | OWR-14213-013

### Orientable Cohomology Theories with Universal Chern Operations

P / partially_solved | Oberwolfach 报告问题汇编

For a generalized cohomology theory $K$, is there a notion of $K$-orientable theories equipped with $K$-analogue Chern-class operations that freely generate all operations into an oriented theory?

来源字段：https://doi.org/10.4171/owr/2016/5

09497 | OPG-37282

### Outer reloid of restricted funcoid

S / solved | Open Problem Garden

Question $( \mathsf{RLD})_{\mathrm{out}} (f \cap^{\mathsf{FCD}} ( \mathcal{A} \times^{\mathsf{FCD}} \mathcal{B})) = (( \mathsf{RLD})_{\mathrm{out}} f) \cap^{\mathsf{RLD}} ( \mathcal{A} \times^{\mathsf{RLD}} \mathcal{B})$ for every filter objects $\mathcal{A}$ and $\mathcal{B}$ and a funcoid $f\in\mathsf{FCD}(\mathrm{Src}\,f; \mathrm{Dst}\,f)$?

背景链接：http://www.openproblemgarden.org/op/outer_reloid_of_restricted_funcoid

09498 | OPG-60020

### Outward reloid of composition vs composition of outward reloids

S / solved | Open Problem Garden

Conjecture For every composable funcoids $f$ and $g$ $$(\mathsf{RLD})_{\mathrm{out}}(g\circ f)\sqsupseteq(\mathsf{RLD})_{\mathrm{out}}g\circ(\mathsf{RLD})_{\mathrm{out}}f.$$

背景链接：http://www.openproblemgarden.org/op/outward_reloid_of_composition_vs_composition_of_outward_reloids

09499 | AIM-TOPOLOGY-0020

### Overtwisted contact germs from first-kind lcs data

P / partially_solved | AIM 研讨会问题汇编

For an lcs manifold \((M^{2n},\Phi,\vartheta)\), where \(d\vartheta=0\), \(\Phi^n\neq0\), and \[ d\Phi=\vartheta\wedge\Phi, \] find an overtwistedness notion that is intrinsic under the intended lcs equivalences. Compare a local embedded-model definition with a definition using an associated contact germ, and state any extra primitive, covering, or first-kind data needed to construct that germ.

来源字段：http://aimpl.org/highdimcontacttop/8/

09500 | OWR-12697686-005

### Ozsváth-Szabó Conjecture for Integer Homology Spheres

O / open | Oberwolfach 报告问题汇编

If $M$ is an integer homology $3$-sphere, is $M$ a non-L-space if and only if it is neither $S^3$ nor the Poincaré homology sphere $\Sigma(2,3,5)$?

来源字段：https://doi.org/10.4171/owr/2023/3

09501 | AIM-TOPOLOGY-0079

### P-Bruhat transfer and a codimension-two obstruction

P / partially_solved | AIM 研讨会问题汇编

What properties of braid varieties in $G/B$ should survive in $G/P$? In other words, what properties of full flag varieties remain in partial flag varieties.

来源字段：http://aimpl.org/agclinkhom/6/

09502 | AIM-TOPOLOGY-0078

### Parabolic braid strata and singular projected closures

P / partially_solved | AIM 研讨会问题汇编

Develop singular braid varieties and relate them to Richardson varieties in $G/P$ and to colored homology.

来源字段：http://aimpl.org/agclinkhom/6/

09503 | AIM-TOPOLOGY-0082

### Parabolic flag kernels as braid-stratified projectors

P / partially_solved | AIM 研讨会问题汇编

Are there “braid variety analogues” of projectors?

Its mathematical ambiguity is real, however: neither “projector” nor the category in which it should be idempotent is specified. The official workshop report supplies the missing context. It discusses annular closures of Elias-Hogancamp categorified projectors, used to color link components, and separately records the workshop construction of parabolic braid varieties [AIM]. Thus the intended target appears to be a geometric object attached to braid varieties that plays some role analogous to a Jones-Wenzl/Young/Soergel categorical projector.

This report gives a rigorous partial answer in one precise category. For a standard parabolic subgroup, a proper correspondence assembled from braid relative-position strata has an idempotent structure-sheaf kernel under coherent convolution. It also gives the expected absorption law for nested colors. This does **not** identify that finite kernel with the infinite Soergel complexes used in colored HOMFLYPT homology.

来源字段：http://aimpl.org/agclinkhom/6/

09504 | OWR-13346-002

### Parallel Transport Along Sticky Principal Components

P / partially_solved | Oberwolfach 报告问题汇编

How can parallel transport be defined along sticky principal components in stratified spaces such as BHV tree space, so that second principal components and related manifold-statistical techniques can be defined?

来源字段：https://doi.org/10.4171/owr/2014/44

09505 | OWR-17292-009

### Parametrization of Turaev-Viro Three-Manifold Invariants

P / partially_solved | Oberwolfach 报告问题汇编

For the fusion categories $\mathcal C_{N,1,0}$ with $N\in\mathbb N_{>0}$, find a uniform parametrization of the associated Turaev-Viro $3$-manifold invariants as $N$ varies.

来源字段：https://doi.org/10.4171/owr/2019/49

09506 | AIM-TOPOLOGY-0063

### Parity cones and explicit module structure for Jucys--Murphy products

P / partially_solved | AIM 研讨会问题汇编

Compute \(HHH(JM_1^{t_1}\dots JM_n^{t_n})\) "for as many \(t\)'s as possible" is it parity? Describe it as an \(R\)-module.

来源字段：http://aimpl.org/agclinkhom/2/

09507 | OWR-15955-003

### Parity-Refined Heegaard-Floer and Lattice Cohomology Correspondence

S / solved | Oberwolfach 报告问题汇编

别名：Heegaard-Floer Homology and Lattice Cohomology Conjecture [OWR-15955-001]

Let $M$ be a negative-definite plumbed rational homology $3$-sphere and let $\sigma\in\operatorname{Spin}^c(M)$. Are there grading-shifted isomorphisms $$HF^+_{\mathrm{red,even/odd}}(-M,\sigma)\cong\bigoplus_{q\in\mathbb{even/odd}}\mathbb H^q_{\mathrm{red}}(M,\sigma)[-d(M,\sigma)]$$?

来源字段：https://doi.org/10.4171/owr/2017/50

09508 | AIM-TOPOLOGY-0077

### Parking-function Frobenius data and a hook-information obstruction for torus-link traces

P / partially_solved | AIM 研讨会问题汇编

Find the Macdonald/Schur expansion of the $a$-graded Frobenius character of the parking function space. How does the $tr(T(m,n)$ relate to the parking function module?

来源字段：http://aimpl.org/agclinkhom/5/

09509 | AIM-TOPOLOGY-0148

### Periodic classes and translations are exactly the hyperbolic isometries

S / solved | AIM 研讨会问题汇编

In the finite-type setting, a mapping class has a representative that's a hyperbolic isometry for some hyperbolic metric exactly when it is periodic.

Characterize the big mapping classes that can be realized by hyperbolic isometries of some hyperbolic metric on the surface.

来源字段：http://aimpl.org/genusinfinity/3/

09510 | OWR-14299915-007

### Periodic Loci for Derived Permutation Modules

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a finite or profinite group and let $H\leq G$ be a finite or profinite $p$-subgroup. Prove that the point $P(H,p)$ in the spectrum of compact derived permutation modules is periodic if and only if $p$ is periodic; equivalently, prove that the periodic locus is the complement of the closed points.

来源字段：https://doi.org/10.4171/owr/2026/9

09511 | OWR-15575-002

### Periodic Orbits Bounding Global Surfaces of Section

P / partially_solved | Oberwolfach 报告问题汇编

Given a smooth flow $\varphi_t$ on a closed smooth $3$-manifold $M$ and periodic orbits $\gamma_1,\ldots,\gamma_N$, when does their union bound a global surface of section?

来源字段：https://doi.org/10.4171/owr/2017/32

09512 | OWR-11126-002

### Periodic Reeb Orbits and Classification of Contact Structures

O / open | Oberwolfach 报告问题汇编

Does every closed contact-type hypersurface in a symplectic manifold admit a periodic orbit, equivalently does every Reeb vector field on a closed contact manifold admit a periodic orbit; and how are contact structures on a fixed manifold classified up to diffeomorphism?

来源字段：https://doi.org/10.4171/owr/2011/34

09513 | OWR-4413-005

### Persistent Gaps between Free $p$-Rank and Toral Rank

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a finite connected complex $X$ whose fundamental group is abelian and acts trivially on higher homotopy, and whose rational homotopy is finite-dimensional, such that $\operatorname{rk}_p(X)>\operatorname{rk}_T(X)$ for infinitely many primes $p$?

来源字段：https://doi.org/10.4171/owr/2010/32

09514 | DARPA-009

### Physical Consequences of Perelman's Proof

P / partially_solved | DARPA 挑战

Apply Perelman's proof of the Poincaré conjecture to materials fabrication across scales.

背景链接：https://web.math.utk.edu/~vasili/refs/darpa07.MathChallenges.html

09515 | OWR-14297741-002

### Picard and Brauer Groups of Stable Module Blocks

P / partially_solved | Oberwolfach 报告问题汇编

Can the Picard and Brauer groups of principal blocks of stable module categories be computed completely by descent spectral sequences based on their associated spectral schemes?

来源字段：https://doi.org/10.4171/owr/2023/40

09516 | AIM-TOPOLOGY-0267

### Pinching-gap reduction and dimension-six finiteness

P / partially_solved | AIM 研讨会问题汇编

**Problem 54.** Is there a finiteness result for $n$-dimensional, positively curved manifolds with $\pi_1=\pi_2=0$?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

09517 | AIM-TOPOLOGY-0217

### PL and smooth stability under non-collapsed Alexandrov convergence

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.** Perelman's Stability Theorem yields that manifolds in a given sequence of non-collapsing manifolds are eventually pairwise homeomorphic. Are they also PL-homeomorphic or diffeomorphic?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

09518 | AIM-TOPOLOGY-0058

### Polygraph rings and Hopf-link homology

P / partially_solved | AIM 研讨会问题汇编

Do Haiman's polygraph rings relate to link homology? How do they relate to cables of Hopf link?

来源字段：http://aimpl.org/agclinkhom/1/

09519 | OWR-8415345-001

### Polynomial Knot Invariants and Projections in General Three-Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

For knots in general 3-manifolds, do Jones/HOMFLYPT-type polynomial invariants exist; what knot projections have useful computational and invariant-theoretic properties; and is there a Reidemeister theorem for such projections?

来源字段：https://doi.org/10.4171/owr/2021/42

09520 | OWR-15208-016

### Positive Ricci Observer Homotopy Beyond Spheres

O / open | Oberwolfach 报告问题汇编

Can the nontrivial stable-range rational homotopy of the observer moduli space of positive-Ricci-curvature metrics on $S^d$ be extended to other manifolds?

来源字段：https://doi.org/10.4171/owr/2017/3

09521 | OWR-17470-004

### Positive Scalar Curvature for Spin Manifolds with Odd Fundamental Group

O / open | Oberwolfach 报告问题汇编

Let $M$ be a closed connected spin manifold of dimension $d\geq5$ with odd-order abelian fundamental group. Assume that $M$ is $p$-atoral for every prime $p$ dividing $|\pi_1(M)|$: for every $\ell\geq1$ and $c_1,\ldots,c_d\in H^1(M;\mathbb Z/p^\ell)$,

$$\langle c_1\smile\cdots\smile c_d,[M]\rangle=0\in\mathbb Z/p^\ell.$$

Is the vanishing of the Rosenberg $\alpha$-invariant sufficient for $M$ to admit a metric of positive scalar curvature?

来源字段：https://doi.org/10.4171/owr/2020/7

09522 | OWR-1381-009

### Positive Scalar Curvature Information from Rosenberg Indices

O / open | Oberwolfach 报告问题汇编

Let $M$ be a closed smooth spin manifold, with Rosenberg indices $\alpha_{\mathrm{red}}(M)\in KO_*(C^*_{\mathrm{red}}\pi_1M)$ and $\alpha_{\mathrm{max}}(M)\in KO_*(C^*_{\mathrm{max}}\pi_1M)$. How much information do these indices contain about obstructions to positive scalar curvature on $M$?

来源字段：https://doi.org/10.4171/owr/2006/43

09523 | OWR-8415338-009

### Positive Scalar Curvature Obstruction for Aspherical Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

For every closed smooth aspherical manifold $M$ of dimension $n\geq 4$, $M$ admits no Riemannian metric of positive scalar curvature.

来源字段：https://doi.org/10.4171/owr/2021/35

09524 | OWR-17470-003

### Positive Scalar Curvature Through Odd-Order Finite Covers

P / partially_solved | Oberwolfach 报告问题汇编

Let $M$ be a closed connected smooth manifold of dimension at least $5$ whose fundamental group is finite of odd order. If the universal cover of $M$ admits a metric of positive scalar curvature, must $M$ admit one as well?

来源字段：https://doi.org/10.4171/owr/2020/7

09525 | OWR-15581-015

### Positivity and Evenness in HOMFLY-PT Knot Homology

P / partially_solved | Oberwolfach 报告问题汇编

For which knots does the HOMFLY-PT polynomial have positive coefficients when expressed in $q$ and $-a$, and for which knots is triply graded Khovanov-Rozansky homology concentrated in even homological degrees?

来源字段：https://doi.org/10.4171/owr/2017/38

09526 | AIM-TOPOLOGY-0054

### Positivity and split-lift obstructions for categorified colored-Jones recursions

P / partially_solved | AIM 研讨会问题汇编

Lift the recursion relation of the colored Jones Polynomial to categorified CJP.

The record does not specify a knot, a normalization, or which recursion is meant.

There are two mathematically plausible readings.

1. **Color-recursion/AJ reading (used below).** For a fixed knot, the colored Jones sequence is annihilated by a noncommutative polynomial in the color-shift and multiplication operators. Lift such an annihilator to maps, cones, or a filtered acyclic totalization of colored-Jones complexes. 2. **Jones--Wenzl-projector reading.** Lift the recursive construction of the Jones--Wenzl projectors used to define the colored invariant.

The second reading has a substantial known answer: Cooper--Krushkal construct chain complexes categorifying Jones--Wenzl projectors and give an inductive construction modeled on the Frenkel--Khovanov recursion. The first reading remains the sharper unresolved interpretation, and it is the subject of this report.

We fix the reduced normalization, with the colored Jones polynomial of the unknot equal to 1, only to avoid irrelevant units. All statements below apply after fixing any normalization and a particular recurrence presentation.

来源字段：http://aimpl.org/quantumlowdimtop/4/

09527 | OWR-12014-006

### Positivity of Hyperbolic and Seifert Representation Volumes

P / partially_solved | Oberwolfach 报告问题汇编

For which closed oriented $3$-manifolds $M$ is the hyperbolic volume $HV(M)$, or the Seifert volume $SV(M)$, defined as the maximal representation volume for $\operatorname{PSL}(2,\mathbb{C})$ or $\operatorname{Isom}\widetilde{\operatorname{SL}}_2(\mathbb{R})$, respectively, positive?

来源字段：https://doi.org/10.4171/owr/2012/42

09528 | OWR-3384-002

### Probe Displaceability and Floer Detection of Toric Fibers

P / partially_solved | Oberwolfach 报告问题汇编

For two-dimensional moment polytopes containing open sets of points that cannot be displaced by probes, are the corresponding toric fibers genuinely nondisplaceable, and can this be detected by a suitable Floer homology? More generally, for a Fano moment polytope, can every point except its special interior integral point be displaced by a probe?

来源字段：https://doi.org/10.4171/owr/2009/01

09529 | AMR-109-0208

### Problem 1 - (Geodesics on general flat surfaces).

O / open | AMR 问题清单汇编

(Geodesics on general flat surfaces). Describe the behavior of geodesics on general flat surfaces. Prove (or disprove) the conjecture that the geodesic flow is ergodic on a typical (in 248 15. Problems on billiards, flat surfaces and translation surfaces 249 any reasonable sense) flat surface. Does any (almost any) flat surface have at least one closed geodesic which does not pass through singular points? If the answer is positive then one can ask for the asymptotics for the number of closed geodesics of bounded length as a function of the bound. Note that typically a geodesic representative in a homotopy class of a simple closed curve is realized by a broken line containing many geodesic segments going from one conical singularity to the other. The counting problem for regular closed geodesics (ones which do not pass through singularities) is quite different from the counting problem for geodesics realized by broken lines. The following questions treat billiards in arbitrary polygons in the plane.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09530 | AMR-109-0179

### Problem 1 - Develop techniques to describe the sets Cχ,θ.

O / open | AMR 问题清单汇编

Develop techniques to describe the sets Cχ,θ.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09531 | AMR-109-0131

### Problem 1 - Understand either classically or as quantum geometric objects the non-Hausdorff quotients of PL 0(F ) or PL(F ) by MC…

O / open | AMR 问题清单汇编

Understand either classically or as quantum geometric objects the non-Hausdorff quotients of PL 0(F ) or PL(F ) by MC (F ) or PMC (F ). T (F ) has been quantized in [5] and [17] as surveyed in [6] and [35], respectively, and PL 0(F 1 1,0) has been quantized in [6]. In interesting contrast, [21,54] has described a program for studying real quadratic number fields as quantum tori limits of elliptic curves, thus quantizing limiting curves rather than Teichm¨ uller space.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09532 | AMR-103-0001

### Problem 1.1 - ([188, Problem 1]) Find a non-trivial knot K with VK (t) = 1.

O / open | AMR 问题清单汇编

([188, Problem 1]) Find a non-trivial knot K with VK (t) = 1.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09533 | AMR-109-0229

### Problem 1.1 - Given a subgroup G <MCG (S), how is the geometry of ΓG related to the dynamics of the action of G on Thurston ’s comp…

O / open | AMR 问题清单汇编

Given a subgroup G <MCG (S), how is the geometry of ΓG related to the dynamics of the action of G on Thurston ’s compactification of Teichm¨ uller space T (S) = T (S)∪ ∂T (S)? Several of the problems that we propose in Sections 2-4 are taken from the author’s joint paper with Benson Farb [ FM02a].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09534 | AMR-109-0185

### Problem 1.1 - Investigate the dependence of Expρ,⟨f,M ⟩ on the marked Riemann surface ⟨f,M⟩.

O / open | AMR 问题清单汇编

Investigate the dependence of Expρ,⟨f,M ⟩ on the marked Riemann surface ⟨f,M⟩.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09535 | AMR-109-0001

### Problem 1.1 - Study the function Ψ: Mg→ [-log(µg),∞) as a (rational) Morse function.

O / open | AMR 问题清单汇编

Study the function Ψ: Mg→ [-log(µg),∞) as a (rational) Morse function. Classify its rational critical points. Deduce properties of the rational cohomology of Mg (see below for specific statements).

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09536 | AMR-109-0157

### Problem 1.1 - The ideas which were just described relate to the beginning of the lower central series of Ig.

O / open | AMR 问题清单汇编

The ideas which were just described relate to the beginning of the lower central series of Ig. There is also the lower central series of Kg. The correspondence between the group structure of Mg and 3-manifold topology, as regards the subgroups of Mg that have been studied, has been remarkable. It suggests strongly that there is much more to be done, with the possibility of new 3-manifold invariants as a reward.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09537 | AMR-103-0010

### Problem 1.10 - (N.

P / partially_solved | AMR 问题清单汇编

(N. Dunfield) Find the relationship between the hyperbolic volume of knot complements and log VK (-1) (resp. log VK(-1)/ log degVK(t)). 3.5 4 4.5 5 5.5 6 6.5 7 7.5 8 0 5 10 15 20 25 30 Pi*log(J(-1)) Volume of complement 13 crossing alternating knots "13_alt.data" Figure 4: The distribution of pairs of the hyperbolic volume of knot complements and π log VK(-1) for alternating knots with 13 crossings [112].

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09538 | AMR-103-0011

### Problem 1.11 - Understand Khovanov’s categorification of the Jones polyno - mial.

S / solved | AMR 问题清单汇编

Understand Khovanov’s categorification of the Jones polyno - mial.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09539 | AMR-103-0012

### Problem 1.12 - Categorify other knot polynomials.

P / partially_solved | AMR 问题清单汇编

Categorify other knot polynomials.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09540 | AMR-103-0013

### Problem 1.13 - (A.

P / partially_solved | AMR 问题清单汇编

(A. Stoimenow) Does the Jones polynomial V admit only finitely many values of given span? What about the Q polynomia l or the skein, Kauffman polynomials (when fixing the span in both variables)?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09541 | AMR-103-0014

### Problem 1.14 - (A.

O / open | AMR 问题清单汇编

(A. Stoimenow) Why are the unit norm complex numbers α for which the value QK (α) has maximal norm statistically concentrated around e11π√ -1/25?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09542 | AMR-103-0015

### Problem 1.15 - (M.

P / partially_solved | AMR 问题清单汇编

(M. Kidwell, A. Stoimenow) Let K be a non-trivial knot, and let WK be a Whitehead double of K. Is then degm PWK (l, m) = 2 deg z FK (a, z) + 2?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09543 | AMR-103-0016

### Problem 1.16 - (E.

S / solved | AMR 问题清单汇编

(E. Ferrand, A. Stoimenow) Is for any alternating link L, σ(L)≥ min degl ( PL(l, m) ) ≥ min dega ( FL(a-1, z) )?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09544 | AMR-103-0017

### Problem 1.17 - (A.

O / open | AMR 问题清单汇编

(A. Stoimenow) If∇k is the coefficient of zk in the Conway polynomial and c(L) is the crossing number of a link L, is then ı ı∇k(L) ı ı≤ c(L)k 2k k!?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09545 | AMR-103-0018

### Problem 1.18 - (A.

O / open | AMR 问题清单汇编

(A. Stoimenow) Does min deg a ( FL(a-1, z) ) ≤ 1- χ(L) hold for any link L? If u(K) is the unknotting number of a knot K, does min dega ( FK (a-1, z) ) ≤ 2u(K) hold for any knot K?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09546 | AMR-103-0002

### Problem 1.2 - ([188, Problem 2]) Characterize those elements of Z[t, t-1] of the form VK(t).

O / open | AMR 问题清单汇编

([188, Problem 2]) Characterize those elements of Z[t, t-1] of the form VK(t).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09547 | AMR-103-0022

### Problem 1.22 - (H.

S / solved | AMR 问题清单汇编

(H. Murakami) For a torus knot K, calculate CS(S3- K) (giving an appropriate definition of it) and calculate lim log JN (K) N (fixing an appropriate choice of a branch of the logarithm).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09548 | AMR-103-0003

### Problem 1.3 - Find a 3-dimensional topological interpretation of the Jon es polynomial of links.

S / solved | AMR 问题清单汇编

Find a 3-dimensional topological interpretation of the Jon es polynomial of links.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09549 | AMR-103-0004

### Problem 1.4 - (J.

O / open | AMR 问题清单汇编

(J. Roberts) Why is the Jones polynomial a polynomial?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09550 | AMR-103-0005

### Problem 1.5 - (J.

P / partially_solved | AMR 问题清单汇编

(J. Roberts) Is there a relationship between values of Jones polynomials at roots of unity and branched cyclic coverings of a knot?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09551 | AMR-103-0006

### Problem 1.6 - (J.

P / partially_solved | AMR 问题清单汇编

(J. Roberts) Is there a relationship between the Jones polynomial of a knot and the counting of points in varieties defined o ver finite fields?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09552 | AMR-103-0007

### Problem 1.7 - (J.

O / open | AMR 问题清单汇编

(J. Roberts) Define the Jones polynomial intrinsically using homology of local systems.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09553 | AMR-103-0008

### Problem 1.8 - (J.

O / open | AMR 问题清单汇编

(J. Roberts) Study the relation between the Jones polynomial and Gromov-Witten theory.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09554 | AMR-103-0009

### Problem 1.9 - (X.-S.

P / partially_solved | AMR 问题清单汇编

(X.-S. Lin) Describe the set of zeros of the Jones polynomial of all (alternating) knots. -1 -0.5 0.5 1 1.5 -1 -0.5 0.5 1 -1 -0.5 0.5 1 -1 -0.5 0.5 1 -2 -1.5 -1 -0.5 0.5 1 1.5 -1.5 -1 -0.5 0.5 1 1.5 -1 -0.5 0.5 1 -1 -0.5 0.5 1 Figure 3: The upper pictures show the distribution of zeros o f the Jones polynomial for n-twist knots, with n from 1 to 50 and from 51 to 100, respectively [262]. The lower pictures show the distribution of zeros of the Jones po lynomial for (2, 2n- 1) torus knots, with n from 1 to 50 and from 51 to 100, respectively [262]. See [262] f or further pictures for (3, 3n + 1) and (3, 3n + 2) torus knots.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09555 | AMR-109-0217

### Problem 10 - (Ergodic measures).

S / solved | AMR 问题清单汇编

(Ergodic measures). Classify the ergodic measures for the action of SL(2, R) on H1(α) andQ1(β). McMullen [ McM3] has solved Problems 9 and 10 in the case of translation surfaces in genus 2. A subset Ω is called minimal for the action of SL(2, R) if it is closed, invariant, and it has no proper closed invariant subsets. The SL(2, R) orbit of a Veech surface is an example of a minimal set.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09556 | AMR-109-0140

### Problem 10 - Though the (virtual) Euler characteristics are already known [28,56], devise a matrix model using screens to calculat…

O / open | AMR 问题清单汇编

Though the (virtual) Euler characteristics are already known [28,56], devise a matrix model using screens to calculate these invariants for ˜M (F ).

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09557 | AMR-103-0176

### Problem 10.1 - Can the Casson invariant of an integral homology 3-sphere M be characterized by the signature of a certain 4-manifold…

P / partially_solved | AMR 问题清单汇编

Can the Casson invariant of an integral homology 3-sphere M be characterized by the signature of a certain 4-manifold bo unded by M?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09558 | AMR-103-0185

### Problem 10.10 - Determine the dimension of the space of primitive finite type invariants of integral homology 3-spheres of each degre…

P / partially_solved | AMR 问题清单汇编

Determine the dimension of the space of primitive finite type invariants of integral homology 3-spheres of each degree d. Equivalently, deter- mine the dimension of the space A(∅; Q)(d) conn for each d. d 0 1 2 3 4 5 6 7 8 9 10 prime diag. 0 1 0 0 1 0 1 1 1 1 2 dimA(∅)(d) conn 0 1 1 1 2 2 3 4 5 6 8 dimA(∅)(d) 1 1 2 3 6 9 16 25 42 65 105 d 11 12 13 14 prime diag. 1 dimA(∅)(d) conn 9 ≥ 11 ≥ 13 ≥ 15 dimA(∅)(d) 161 ≥ 254 ≥ 386 ≥ 595 Table 7: Some dimensions for Problem 10.10

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09559 | AMR-103-0186

### Problem 10.11 - Describe Vogel’s algebra Λ, say, by giving complete sets of generators and relations of Λ.

O / open | AMR 问题清单汇编

Describe Vogel’s algebra Λ, say, by giving complete sets of generators and relations of Λ.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09560 | AMR-103-0187

### Problem 10.12 - Find a constructive combinatorial presentation of each fini te type invariant of integral homology 3-spheres, and, in…

O / open | AMR 问题清单汇编

Find a constructive combinatorial presentation of each fini te type invariant of integral homology 3-spheres, and, in part icular, of the Casson invariant, by localizing configuration space integrals.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09561 | AMR-103-0188

### Problem 10.13 - (J.

O / open | AMR 问题清单汇编

(J. Roberts) What is the space of 3-manifolds?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09562 | AMR-103-0191

### Problem 10.16 - (T.

O / open | AMR 问题清单汇编

(T. Ohtsuki) Define a product M1◦ M2 of integral homol- ogy 3-spheres M1 and M2 which is related, by (50), to the product of Jacobi diagrams given by their connected sum.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09563 | AMR-103-0193

### Problem 10.18 - (F.

P / partially_solved | AMR 问题清单汇编

(F. Deloup) Classify the monoid (for orthogonal sum) of isomorphism classes of quadratic forms qσ.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09564 | AMR-103-0194

### Problem 10.19 - (G.

P / partially_solved | AMR 问题清单汇编

(G. Massuyeau) Describe the quotient set {spin closed 3-manifolds}/∼ Y s d, in particular, for d = 2, 3.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09565 | AMR-103-0177

### Problem 10.2 - (V.

O / open | AMR 问题清单汇编

(V. Turaev) Relate this surgery formula for the Casson-Walker-Lescop invariant with that of Lescop [251]. 32The normalization here is that λ CW(M ) = 2 λ C(M ) for an integral homology 3-sphere M. 33The normalization here is that λ CWL(M ) = ( |H1(M; Z)|/ 2 ) λ CW(M ) for a rational homology 3-sphere M. (C. Lescop) In 1984, Casson defined his invariant of integral homology 3- spheres as an integer that “counts” the SU (2)-representations of their funda- mental group in an appropriate way (see [3, 159]). Cappell, L ee and Miller [76] showed that the Casson way of counting SU (2)-representations of the π1 works for any compact Lie group and provides other invariants of in tegral homology spheres.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09566 | AMR-103-0195

### Problem 10.20 - (F.

P / partially_solved | AMR 问题清单汇编

(F. Deloup, G. Massuyeau) Describe the quotient set {spin c closed 3-manifolds}/∼ Y c d, in particular, for d = 2, 3.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09567 | AMR-109-0322

### Problem 10.3 - Give examples of odd valent graphs Γ whose associated homology classes Φ(Γ)∈ H∗(OutFn; Q) are non-trivial as many as...

O / open | AMR 问题清单汇编

Give examples of odd valent graphs Γ whose associated homology classes Φ(Γ)∈ H∗(OutFn; Q) are non-trivial as many as possible. Also compare these classes with the homology classes constructed by Conant and Vogtmann [10] as explicit cycles in the moduli space of graphs. Furthermore investigate whether these classes survive in H∗(GL(n, Z); Q), or else come fromH∗(IOutn; Q), or not.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09568 | AMR-103-0179

### Problem 10.4 - (M.

O / open | AMR 问题清单汇编

(M. Polyak) Define an invariant λ of a pair (M, σ) of a closed 3-manifold M and a spin structure σ on M such that λCWL(M ) = ∑ σ λ(M, σ) for any closed 3-manifold M, where the sum runs over all spin structures σ on M. Note that the set of spin structures on M is a torsor over H 1(M; Z/2Z) in the sense that differences of spin structures can be detected by cohomology classes in H 1(M; Z/2Z), while the set of spin c structures on M is a torsor over H 1(M; Z) in a similar sense.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09569 | AMR-103-0181

### Problem 10.6 - (M.

O / open | AMR 问题清单汇编

(M. Polyak) By presenting 3-manifolds by surgery along framed links in S3, we can regard an invariant of 3-manifolds as an invari- ant of framed links. Establish a Gauss diagram formula for th e link invariant derived from each finite type invariant of 3-manifolds.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09570 | AMR-109-0218

### Problem 11 - (Minimal sets).

O / open | AMR 问题清单汇编

(Minimal sets). Describe the minimal sets for the SL(2, R)-action on H1(α) and Q1(β). Since Veech surfaces give rise to minimal sets, this problem generalizes the problem of characterizing Veech surfaces. The problem below is particularly important for numerous applications. One application is to counting problems.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09571 | AMR-109-0141

### Problem 11 - [LevelN Torelli Franchetta Problem] What is the second cohomology group of the levelN Torelli group?

P / partially_solved | AMR 问题清单汇编

[LevelN Torelli Franchetta Problem] What is the second cohomology group of the levelN Torelli group? The cell decomposition of ¯M(F ) described before is compatible with the ideal cell decompositions of Torelli spaces, i.e., the fatgraph dual to an ideal cell decomposition of F admits a “homology marking” in the sense of [48] as well as admitting the structure of screens. There are thus “DM type” boundaries of each Torelli space replete with an ideal cell decomposition. It is natural to try to understand the topology of these DM-type bordifications of Torelli spaces to approach the following class of problems:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09572 | AMR-103-0197

### Problem 11.1 - For each rational homology 3-sphere M, calculate Z L M O(M ) for all degrees.

O / open | AMR 问题清单汇编

For each rational homology 3-sphere M, calculate Z L M O(M ) for all degrees.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09573 | AMR-103-0206

### Problem 11.10 - (D.

O / open | AMR 问题清单汇编

(D. Thurston) Do configuration spaces of [237] have torsion in Z-homology? Does such torsion deduce a torsion invariant of h omology 3-spheres?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09574 | AMR-109-0323

### Problem 11.2 - Study the central extension (20) from the point of view of group cohomology as well as geometric topology.

O / open | AMR 问题清单汇编

Study the central extension (20) from the point of view of group cohomology as well as geometric topology. In particular determine the Euler class of this central extension which is an element of the group H 2(Hg,1; Θ 3 Z)∼= Hom(H2(Hg,1), Θ3 Z)⊕ ExtZ(H1(Hg,1), Θ3 Z). This should be an extremely difficult problem. Here we would like to indicate a possible method of attacking it, and in particular a possible way of obtaining additive invariants for the group Θ 3 Z, very briefly. Details will be given in a forthcoming paper. 372 S. Morita Using the traces, we can define a series of certain cohomology classes ˜t2k+1∈H 2(lim←-Aut0(Γ/Γm)) ( k = 1, 2,··· ). These are the “group version” of the elements t2k+1 ∈ H 2(hQ g,1)Sp defined in §6 (Definition 6.1). The homomorphism σ is trivial on Θ 3 Z so that we have the induced homomorphism ˜ σ: Hg,1→ lim←-Aut0(Γ/Γm).

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09575 | AMR-103-0199

### Problem 11.3 - Does there exist an integral/rational homology 3-sphere M such that Z L M O(M ) = Z L M O(S3)?

O / open | AMR 问题清单汇编

Does there exist an integral/rational homology 3-sphere M such that Z L M O(M ) = Z L M O(S3)? 11.3 Characterization of the image of the LMO invariant

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09576 | AMR-103-0200

### Problem 11.4 - Characterize those elements of ˆA(∅)conn which are of the form log Z L M O(M ) for integral/rational homology 3-spheres.

O / open | AMR 问题清单汇编

Characterize those elements of ˆA(∅)conn which are of the form log Z L M O(M ) for integral/rational homology 3-spheres.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09577 | AMR-109-0325

### Problem 11.4 - Determine the abelianization of the group Hg,1.

P / partially_solved | AMR 问题清单汇编

Determine the abelianization of the group Hg,1. Is it trivial? Also determine the second homology group H2(Hg,1; Z). Is the rank of it equal to 1 given by the signature? If everything will be as expected, we would obtain non-trivial homomorphisms ˆt2k+1: Θ 3 Z-→Z as secondary invariants associated to the cohomology classes ˜t2k+1. There should be both similarity and difference between these cases and the situation where we interpreted the Casson invariant as the secondary invariant associated to the first Mumford-Morita-Miller class e1 (see [ 74]). More precisely, they are similar because they are all related to some cohomology classes in H 2(hQ g,1). They are different because e1 is non-trivial in H 2(Hg,1) whereas we expect that the other classes σ∗(˜t2k+1) would be all trivial in the same group. Also recall here that Matumoto [ 67] and Galewsky and Stern [ 23] proved that every topological manifold (of dimension n≥ 7) is simplicially triangulable if and only if the homomorphism (21) splits. In view of this result, it should be important to investigate the mod 2 structure of the extension (20) keeping in mind the works of Birman and Craggs [ 3] as well as Johnson [ 44] for the case of the mapping class group. However we have come too far and surely many things have to be clarified before we would understand the structure of the group Hg,1. Finally we would like to propose a problem.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09578 | AMR-103-0201

### Problem 11.5 - Construct the LMO invariant with coefficients in a finite field.

O / open | AMR 问题清单汇编

Construct the LMO invariant with coefficients in a finite field.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09579 | AMR-109-0326

### Problem 11.5 - Generalize the infinitesimal presentation of the Torelli Lie algebra given by Hain [29] to the case of the group of h…

O / open | AMR 问题清单汇编

Generalize the infinitesimal presentation of the Torelli Lie algebra given by Hain [29] to the case of the group of homology cobordism classes of homology cylinders. 22. Cohomological structure of the mapping class group and beyond 373

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09580 | AMR-103-0202

### Problem 11.6 - Construct the LMO invariant (or the theory of finite type invariants) in arrow diagrams.

O / open | AMR 问题清单汇编

Construct the LMO invariant (or the theory of finite type invariants) in arrow diagrams. 11.5 Refinements of the LMO invariant (T. Le) As mentioned in a remark in Problem 11.1, the LMO invar iant is a weak invariant when b1(M ) > 0; in particular, Z L M O(M ) = 1 when b1(M ) > 3. The following two problems might give refinements of Z L M O(M ) which would be stronger than Z L M O(M ), in particular, when b1(M ) > 0.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09581 | AMR-103-0203

### Problem 11.7 - (T.

O / open | AMR 问题清单汇编

(T. Le, V. Turaev) Define the LMO invariant Z L M O(M, σ) of the pair of a closed 3-manifold M and a spin structure σ of M such that Z L M O(M ) = ∑ σ Z L M O(M, σ), where the sum runs over all spin structures on M. There is also a similar problem for spin c structures.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09582 | AMR-103-0204

### Problem 11.8 - (T.

O / open | AMR 问题清单汇编

(T. Le, V. Turaev) For every element ξ∈ H 1(M, Z) construct an extension of Z L M O(M, ξ) of the LMO invariant such that when ξ = 0 one recovers the usual LMO invariant. The idea is that the usual LMO invariant corresponds only to t he trivial coho- mology class, and for manifolds with high Betti number, it is equal to 0. K. Habiro has an extension of the LMO invariant that might be a so lution to this problem.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09583 | AMR-109-0142

### Problem 12 - Calculate various group-theoretic boundaries of mapping class and Torelli groups, for instance, Tits boundaries.

O / open | AMR 问题清单汇编

Calculate various group-theoretic boundaries of mapping class and Torelli groups, for instance, Tits boundaries.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09584 | AMR-109-0327

### Problem 12.1 - Prove that the above characteristic classes induce surjective homomorphism H3(BDiffδ +Σg; Z)-→R2 for any g.

S / solved | AMR 问题清单汇编

Prove that the above characteristic classes induce surjective homomorphism H3(BDiffδ +Σg; Z)-→R2 for any g. The cohomology classes in (22) are stable with respect to g. On the other hand, in [ 57][58] we found an interesting interaction between the twisted cohomology group of the mapping class group and some well known concepts in symplectic topology such as the flux homomorphism as well as the Calabi homomorphism (see [ 68] for generalities of the symplectic topology). By making use of this, we defined certain cohomology classes of BSymp δΣg and proved non-triviality of them. In view of the fact that all the known cohomology classes of BDiff δ +Σg as well as BSymp δΣg are stable with respect to the genus, we would like to ask the following problem.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09585 | AMR-103-0216

### Problem 12.11 - (D.

O / open | AMR 问题清单汇编

(D. Thurston) Find a condition on shadow diagrams which is satisfied by shadow diagrams from alternating knots; and g ives a lower bound on the hyperbolic volume. The Reshetikhin-Turaev invariant and the Turaev-Viro-Ocn eanu invariant can be described in terms of the KTG algebra, via I -bundles and S1 -bundles over shadow surfaces respectively. The relation between the two invariants is derived from the relation between the two construction of 3-manifol ds shown in Figure 23. Closed shadow surfaces with 0 gleams by making I -bundles and cap off boundary @ @ @R by making S1 -bundles Closed 3-manifolds Certain closed 3-manifolds - M ↦-→M #M #(S2× S1)’s Figure 23: Two ways to obtain 3-manifolds from shadow surfac es

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09586 | AMR-103-0217

### Problem 12.12 - Construct a universal Reshetikhin-Turaev invariant and a universal Turaev-Viro-Ocneanu invariant of closed 3-mani fol…

O / open | AMR 问题清单汇编

Construct a universal Reshetikhin-Turaev invariant and a universal Turaev-Viro-Ocneanu invariant of closed 3-mani folds, in terms of the KTG algebra.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09587 | AMR-103-0218

### Problem 12.13 - (J.

O / open | AMR 问题清单汇编

(J. Roberts) What are quantum groups?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09588 | AMR-103-0220

### Problem 12.15 - (M.

P / partially_solved | AMR 问题清单汇编

(M. Boileau [220, Problem 1.69 (C)]) Are there mutants of distinct unknotting numbers?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09589 | AMR-103-0222

### Problem 12.17 - (X.-S.

O / open | AMR 问题清单汇编

(X.-S. Lin [262]) What is the homotopy type of the space L(K) of long ropes (as shown in the picture below) with the fixed kno t type K?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09590 | AMR-103-0223

### Problem 12.18 - (J.

O / open | AMR 问题清单汇编

(J. Roberts) Extend Kuperberg’s work on webs.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09591 | AMR-103-0224

### Problem 12.19 - (J.

O / open | AMR 问题清单汇编

(J. Roberts) Extend the theory of measured laminations to higher rank groups.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09592 | AMR-109-0328

### Problem 12.2 - Study whether the homology groups of BDiffδ +Σg stabilize with respect to g or not.

S / solved | AMR 问题清单汇编

Study whether the homology groups of BDiffδ +Σg stabilize with respect to g or not. The same problem for the group SympδΣg. 374 S. Morita Acknowledgments The author would like to express his hearty thanks to R. Hain, N. Kawazumi, D. Kotschick, M. Matsumoto, H. Nakamura, T. Sakasai for enlightening discussions as well as useful informations concerning the problems treated in this paper.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09593 | AMR-103-0225

### Problem 12.20 - (J.

O / open | AMR 问题清单汇编

(J. Roberts) What is the generating function for q -spin net evaluations?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09594 | AMR-103-0226

### Problem 12.21 - (Y.

O / open | AMR 问题清单汇编

(Y. Shinohara [364]) If n = 4 k + 1 with k > 0, is there a knot with determinant n and signature 4?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09595 | AMR-103-0227

### Problem 12.22 - (T.

P / partially_solved | AMR 问题清单汇编

(T. Stanford) IsC2 solvable? Does C2 contain a free group?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09596 | AMR-103-0228

### Problem 12.23 - (A.

S / solved | AMR 问题清单汇编

(A. Stoimenow) Do positive links of given signature σ have bounded (below) maximal Euler characteristic χ?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09597 | AMR-103-0229

### Problem 12.24 - (A.

O / open | AMR 问题清单汇编

(A. Stoimenow) If a prime knot K can be transformed into its mirror image by one crossing change, is K achiral or (algebraically?) slice?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09598 | AMR-103-0230

### Problem 12.25 - (A.

O / open | AMR 问题清单汇编

(A. Stoimenow) Let n be an odd natural number, different from 1, 9, and 49, such that n is the sum of two squares. Is there a prime alternating achiral knot of determinant n?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09599 | AMR-103-0209

### Problem 12.3 - (H.R.

O / open | AMR 问题清单汇编

(H.R. Morton) From a knot diagram find an explicit such homomorphism to some permutation group or establish that th e knot is trivial. Refinements. (1) Give an upper bound in terms of the diagram for the order of the permu- tation groups which need to be considered. (2) See what happens if the meridians (which are all conjugat e) are restricted to map to permutations of some specified cycle type, for examp le, single trans- positions.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09600 | AMR-103-0212

### Problem 12.6 - Find a new proof of the existence of a universal Vassiliev in- variant of knots, presenting them by KTG’s and their o…

O / open | AMR 问题清单汇编

Find a new proof of the existence of a universal Vassiliev in- variant of knots, presenting them by KTG’s and their operati ons.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09601 | AMR-103-0214

### Problem 12.8 - Construct an invariant of KTG’s from configuration space in- tegrals in a natural way.

O / open | AMR 问题清单汇编

Construct an invariant of KTG’s from configuration space in- tegrals in a natural way. Turaev [388] introduced a presentation of 3-manifolds as S1 -bundles over “sha- dow surfaces”, as follows (for details see [388, 40, 384]). A fake surface is a singular surface such that a neighborhood of each point is ho meomorphic to an open subset of the cone over a tetrahedron. A S1 -bundle over a fake surface can appropriately be defined and its isomorphism class is det ermined by the Chern number, which is an integer or half-integer associate d to each face; we call the Chern number the gleam. A shadow surface is a fake surface with gleams associated to the faces. Every (closed) 3-manifold c an be presented by a S1 -bundle over a (closed) shadow surface. The pentagon and hex agon relations (see [388, Figure 1.1 of Chapter VIII]) are moves among shado w surfaces which present a homeomorphic 3-manifold, though they are not enou gh to characterize a homeomorphism class of 3-manifolds. Exercise 12.9 Find a complete set of moves among shadow surfaces which present a homeomorphic 3-manifold. We obtain a shadow surface as a time evolution of a sequence of KTG’s given by KTG operations. Thus, we have relations among links, 3-ma nifolds, KTG’s and shadow surfaces as in the commutative diagram in Figure 2 2; for detailed statements see [40, 384]. Motivated by a complexity of 3-manifolds discussed in [279, 272, 273], D. Thurston introduced the shadow number of 3-manifolds. The shadow num- ber is defined to be the minimal number of vertices of a shadow surf ace. All graph manifolds have shadow number 0 and all surgeries on the Borromean rings have shadow number 1. The volume conjecture might be re lated to the following conjecture. Framed links -exterior Framed link exteriors -surgery Closed 3-manifolds presentation 6 by making S1 -bundle 6 by making S1 -bundle 6 Certain sequences of KTG’s - time evolution Collapsible shadow surfaces - cap off ∂ Closed shadow surfaces Figure 22: Links, 3-manifolds, KTG’s, and shadow surfaces

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09602 | AMR-109-0220

### Problem 13 - (Kernel foliation).

O / open | AMR 问题清单汇编

(Kernel foliation). IsN a complex-analytic (complex-algebraic) orbifold? When is dimCN = dim CO +n-m? On the other hand when does N coincide with the entire connected component of the enveloping stratum H(α′ 1,...,α ′ n)? One of the key properties used in [ McM3] for the classification of the closures of orbits of GL(2, R) in each of H(1, 1) and H(2) was the knowledge that on any translation surface of either stratum, one can find a pair of homologous saddle connections. For example, cutting a surface ( X,ω ) in H(1, 1) along two homologous saddle connections joining distinct zeroes decomposes the surface into two tori, allowing one to apply the machinery of Ratner’s Theorem.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09603 | AMR-109-0143

### Problem 13 - What are the kernels of the Magnus representations?

O / open | AMR 问题清单汇编

What are the kernels of the Magnus representations? Finally in [48] by taking contractions of powers of our canonical one cocycle, new combinatorially explicit cycles and cocycles on M(F ) are constructed which on the other hand generate the tautological algebra. It is natural to wonder about the extension of these classes to the screen model for ¯M(F ) and to the combinatorial compactification, and to revisit [16,19,22] in this context.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09604 | AMR-109-0221

### Problem 14 - (Decomposition of surfaces).

O / open | AMR 问题清单汇编

(Decomposition of surfaces). Given a connected component of the stratum H(α) of Abelian differentials (or of quadratic differentials Q(β) find those configurations of homologous saddle connections (or homologous closed geodesics), which are present on every surface in the stratum. For quadratic differentials the notion of homologous saddle connections (homologous closed geodesics) should be understood in terms of homology with local coefficients, see [ MaZo]. The last two problems in this section concern the Teichm¨ uller geodesic flow on the moduli spaces. This is the flow defined by the 1-parameter subgroup ( et 0 0 e-t ) t∈R onH(α1,...,α m). In any smooth dynamical system the Lyapunov exponents (see [ BaPe], [ F o2]) are important. Recently, A. Avila and M. Viana [ AvVi] have shown the simplicity of the spectrum for the cocycle related to the Teichm¨ uller geodesic flow (strengthening the earlier result of Forni on positivity of the smallest Lyapunov exponent).

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09605 | AMR-109-0222

### Problem 15 - (Lyapunov exponents).

O / open | AMR 问题清单汇编

(Lyapunov exponents). Study individual Lyapunov exponents of the Teichm¨ uller geodesic flow: - for all known SL(2; R)-invariant subvarieties; - for strata; - for strata of large genera as the genus tends to infinity. Are they related to characteristic numbers of any natural bundles over appropriate compacti- fications of the strata? The motivation for this problem is a beautiful formula of Kontsevich [ Ko] representing the sum of the first g Lyapunov exponents in differential-geometric terms. It follows from the Calabi Theorem [ Cb] that given a real closed 1-form ω0 with isolated zeroes Σ (satisfying some natural conditions) on a smooth surface S of real dimension two, one 15. Problems on billiards, flat surfaces and translation surfaces 255 can find a complex structure on S and a holomorphic 1-form ω such that the ω0 is the real part of ω. Consider the resulting point ( X,ω ) in the corresponding stratum. For generic ( X,ω ) the cocycle related to the Teichm¨ uller geodesic flow acting on H 1(X, R) defines a pair of transverse Lagrangian subspaces H 1(X, R) = L0⊕L1 by means of the Oseledets Theorem ([ BaPe]). These subspaces correspond to contracting and to expanding directions. Though the pair ( X,ω ) is not uniquely determined by ω0, the subspace L0⊂H 1(X, R) does not depend on ( X,ω ) for a given ω0. Moreover, L0 does not change under small deformations of ω0 that preserve the cohomology class [ ω]∈H 1(S, Σ; R). We get a topological object L0 defined in implicit dynamical terms.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09606 | AMR-109-0144

### Problem 15 - [Ehrenpreis Conjecture] Given two closed Riemann surfaces, there are finite un- branched covers with homeomorphic tot…

S / solved | AMR 问题清单汇编

[Ehrenpreis Conjecture] Given two closed Riemann surfaces, there are finite un- branched covers with homeomorphic total spaces which are arbitrarily close in the Teichm¨ uller metric.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09607 | AMR-109-0223

### Problem 16 - (Dynamical Hodge decomposition).

O / open | AMR 问题清单汇编

(Dynamical Hodge decomposition). Study properties of distributions of the La- grangian subspaces in H 1(S; R) defined by the Teichm¨ uller geodesic flow, in particular, their con- tinuity. Is there any topological or geometric way to define them? The Lagrangian subspaces are an interesting structure relating topology and geometry.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09608 | AMR-109-0224

### Problem 17 - (Converse to dichotomy).

O / open | AMR 问题清单汇编

(Converse to dichotomy). Characterize translation surfaces for which (1) the set of minimal directions coincides with the set of uniquely ergodic directions; (2) the set of completely periodic directions coincides with the set of non-uniquely ergodic directions. 256 P. Hubert, H. Masur, T. Schmidt and A. Zorich Note that Property (2) implies Property (1). In genus g = 2 it is known that for every translation surface which is not a Veech surface there is a direction θ which is minimal and not uniquely ergodic [ ChMa]. On the other hand using work of Hubert-Schmidt [ HuSt2], B. Weiss has given an example of a surface which is not a Veech surface and yet for which Property (2) holds. (This surface is obtained as a ramified covering over a Veech surface with a single ramification point.) As mentioned above, a closed orbit avoiding the conical singularities of the flat surface deter- mines a cylinder of parallel lines, all of the same length. It is of interest to find the asymptotics for the number of cylinders (in all possible directions) of lengths less than a given number. In the case of the standard flat torus the number of cylinders of length at most L is asymptotic to 1 ζ(2)πL2. Each Veech surface also has quadratic asymptotics [ V e3] and the same is true for generic surfaces in each stratum [ EsMa].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09609 | AMR-109-0225

### Problem 18 - (Quadratic asymptotics for any surface).

P / partially_solved | AMR 问题清单汇编

(Quadratic asymptotics for any surface). Is it true that every translation surface or quadratic differential has exact quadratic asymptotics for the number of saddle connections and for the number of regular closed geodesics?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09610 | AMR-109-0226

### Problem 19 - (Error term for counting functions).

P / partially_solved | AMR 问题清单汇编

(Error term for counting functions). What can be said about the error term in the quadratic asymptotics for counting functions N ((X,ω ),L )∼c·L2 on a generic translation surface (X,ω )? In particular, is it true that lim sup L→∞ log|N (S,L )-c·L2| logL < 2? Is the lim sup the same for almost all flat surfaces in a given connected component of a stratum? The classical Circle Problem gives an estimate for the error term in the case of the torus. Veech proved that for Veech surfaces the limsup in the error term is actually a limit, see [ V e3]. However, nothing is known about the value of this limit. One may ask whether there is a uniform bound for this limit for Veech surfaces in a given stratum or even for the square-tiled surfaces in a given stratum.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09611 | AMR-109-0200

### Problem 1: - Determine the metric completion of the Gromov boundary of C(S) and relate this metric completion to the geometry of C…

P / partially_solved | AMR 问题清单汇编

Determine the metric completion of the Gromov boundary of C(S) and relate this metric completion to the geometry of C(S). There is yet another way to construct unparametrized uniform quasi-geodesics in C(S). Namely, Teichm¨ uller spaceforS is the space Tg,m of marked isometry classes of complete hyperbolic met- rics on S of finite volume. The Teichm¨ uller space can naturally be identified with a domain in C3g-3+m (see Chapter 6 of [IT]). By a classical result of Bers (see [Bu]), there is a number χ> 0 such that for every complete hyperbolic metric h on S there is a pants decomposition for S consisting of simple closed h- geodesics of length at most χ. Moreover, the diameter in C(S) of the set of simple closed curves on S of h-length at most χ is bounded from above by a universal constant D >0. Thus we can define a map Ψ: Tg,m→C (S) by associating to a marked hyperbolic metric h a simple closed curve of h-length at most χ. For any two such maps Ψ, Ψ′ we then have sup h∈Tg,md(Ψ(h), Ψ′(h))≤D. The Teichm¨ uller metriconTg,m is a complete Finsler metric which is just the Kobayashi metric on the domain in C3g-3+m representingTg,m (see [IT]). Through any two distinct points in Tg,m passes a unique Teichm¨ uller geodesic. Each such geodesic line in Tg,m is uniquely determined by its endpoints in the Thurston boundary ofTg,m which is just the space PML of projective measured laminations on S. The supports of the two measured laminations on S defining the endpoints of the geodesic together fill up the surface S, i.e. every simple closed curve on S intersects at least one of the two laminations transversely. These laminations then define a holomorphic quadratic 14. Geometric properties of the mapping class group 235 differential (we refer to Section 4 of [Ke] for a discussion of this fact). The following result is implicitly contained in the paper [MM1] by Masur and Minsky; an explicit proof using a result of Rafi [R] can be found in Section 4 of [H4].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09612 | AMR-107-0001

### Problem 1A - Present explicit obstructions to the Lyashko-Looijenga covering in terms of braid groups.

O / open | AMR 问题清单汇编

Present explicit obstructions to the Lyashko-Looijenga covering in terms of braid groups. Which braids cannot be lifted to the space ${\mathbb{C}}^{\mu}{\setminus}\Sigma$?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09613 | AMR-107-0002

### Problem 1B - Let a non-simple singularity $f$ be given and the Dynkin diagram of it be defined by an easily disting…

O / open | AMR 问题清单汇编

Let a non-simple singularity $f$ be given and the Dynkin diagram of it be defined by an easily distinguished system of paths connecting 0 with critical points of $f_{\lambda}$. Which Dynkin graphs can be obtained from it by a sequence of formal Picard-Lefschetz moves defined by a braid, but cannot appear as Dynkin diagrams of Morsifications $f_{\lambda^{\prime}}$ with the same critical values defined by the same system of paths?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09614 | AMR-107-0003

### Problem 1C - Are there more refined restrictions to the collision of critical values?

O / open | AMR 问题清单汇编

Are there more refined restrictions to the collision of critical values? Is it true that for any two vanishing cycles, whose intersection number is equal to $\pm 1$ or 0, we can lift the collision of the corresponding critical values to $B_{\varepsilon}$ via the Lyashko-Looijenga submersion?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09615 | AMR-107-0004

### Problem 1D - Give more general lower bounds of the dimension of $\mu=const$ strata in terms of intersection forms o…

O / open | AMR 问题清单汇编

Give more general lower bounds of the dimension of $\mu=const$ strata in terms of intersection forms of vanishing cycles.

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09616 | AMR-107-0005

### Problem 1E - What are the obstructions to the realization of these chains of formal changes by paths in the paramet…

P / partially_solved | AMR 问题清单汇编

What are the obstructions to the realization of these chains of formal changes by paths in the parameter space ${\mathbb{R}}^{k}$?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09617 | AMR-107-0006

### Problem 1F - Is there any convenient topological characteristic of the function $f_{-\varepsilon}$ which allows to…

O / open | AMR 问题清单汇编

Is there any convenient topological characteristic of the function $f_{-\varepsilon}$ which allows to predict these indices?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09618 | AMR-109-0209

### Problem 2 - (Billiards in general polygons).

O / open | AMR 问题清单汇编

(Billiards in general polygons). Does every billiard table have at least one regular periodic trajectory? If the answer is affirmative, does this trajectory persist under deformations of the billiard table? If a periodic trajectory exists, find the asymptotics for the number of periodic trajectories of bounded length as a function of the bound. Describe the behavior of a generic regular billiard trajectory in a generic polygon; in particular, prove (or disprove) the assertion that the billiard flow is ergodic. 1 We note that the case of triangles is already highly non-trivial. For recent work on billiards in obtuse triangles see [ Sc1] and [ Sc2]. In the case of triangles the notion of generic can be interpreted as follows. The space of triangles up to similarity can be parametrized as the set of triples ( θ1,θ 2,θ 3) with ∑θi =π and eachθi> 0. It is naturally an open simplex. Generic then refers to the natural Lebesgue measure. To motivate the next problem we note that there is a close connection between the study of interval exchange transformations and billiards in rational polygons (defined below). An important technique in the study of interval exchange maps is that of renormalization. Given an interval exchange on the unit interval one can take the induced transformation on a subinterval. The resulting map is again an interval exchange map, and if one renormalizes so that the new interval has length one, then this gives a transformation on the space of unit interval exchange maps. This transformation is called the Rauzy-Veech induction ([ Ra], [ V e2]), and it has proved to be of fundamental importance. There is a corresponding notion for the renormalization of translation surfaces given by the Teichm¨ uller geodesic flow.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09619 | AMR-109-0132

### Problem 2 - Given a tuple ×N i=1(mi,ti) ∈ ZN, give a tractable expression in terms of Dehn-Thurston or other coordinates for the…

O / open | AMR 问题清单汇编

Given a tuple ×N i=1(mi,ti) ∈ ZN, give a tractable expression in terms of Dehn- Thurston or other coordinates for the number of components of the corresponding weighted family of curves and arcs. 7. Probing mapping class groups using arcs 107 There is an algorithm which leads to a multiply weighted curve from an integral measure on a general train track akin to that on the torus gotten by serially “splitting” the track, cf. [6], but we ask in Problem 2 for a more closed-form expression. See also Problem 3. A related problem which also seems challenging is to describe A′(F ) or Arc′(F ) in Dehn-Thurston coordinates on MF(F ). This class of curve and arc component counting problems might be approachable using the quan- tum path ordering techniques of [5,6] or with standard fermionic statistical physics [57,29].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09620 | AMR-109-0180

### Problem 2 - Show that the inclusion of Proposition 5 is a bijection.

O / open | AMR 问题清单汇编

Show that the inclusion of Proposition 5 is a bijection. This will probably require more thought about the analytical and geometric constructions which underpin the theory. Another question is suggested by the Auroux-Katzarkov doubling formula. Suppose we know one element fk0 of an asymptotic sequence. Can we describe fl for other values of l apart from l = 2 rk0? Or perhaps better

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09621 | AMR-109-0158

### Problem 2.1 - Assume, for this problem, that M is a 3-manifold with non-empty boundary.

O / open | AMR 问题清单汇编

Assume, for this problem, that M is a 3-manifold with non-empty boundary. Then, on one side of the double coset HφH the handlebody subgroup needs to be modified to a ‘compression body subgroup’. Make this precise, by describing how to modify the double coset to take account of the handle decomposition of the compression body. What happens in the case of a knot space?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09622 | AMR-109-0002

### Problem 2.1 - Determine the finiteness properties of Ig.

P / partially_solved | AMR 问题清单汇编

Determine the finiteness properties of Ig. For which k is Hk(Ig) finitely gen- erated? For which k is there a K(Ig, 1) with finite k-skeleton (one says Ig is of type Fk)?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09623 | AMR-103-0032

### Problem 2.10 - (N.

O / open | AMR 问题清单汇编

(N. Okuda [325]) Describe the set {(v2(K) n2, v3(K) n3 ) ∈ R× R ı ı ı K has a knot diagram with n crossings }. (9)

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09624 | AMR-109-0115

### Problem 2.10 - Compute ϕ(g,n ).

P / partially_solved | AMR 问题清单汇编

Compute ϕ(g,n ). Is it constant? If not, for g < h, compare ϕ(g,n ) and ϕ(h,n ). 90 M. Korkmaz

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09625 | AMR-109-0167

### Problem 2.10 - Given a normal subgroup Gg ofMg, what basic properties are needed in a complex G(Sg) of curves on Sg so that Mg will…

P / partially_solved | AMR 问题清单汇编

Given a normal subgroup Gg ofMg, what basic properties are needed in a complex G(Sg) of curves on Sg so that Mg will turn out to be naturally isomorphic to Aut(G(S))? We return to the central theme of this article:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09626 | AMR-109-0168

### Problem 2.11 - Hempel’s distance function was chosen so that it would capture the geometry, and indeed it does that very well, yet i…

S / solved | AMR 问题清单汇编

Hempel’s distance function was chosen so that it would capture the geometry, and indeed it does that very well, yet in some ways it feels unnatural. The Hatcher-Thurston complex HT(S) seems much more natural to us, since and pairs of vertices in the latter determine a Heegaard diagram, and one gets every genus g Heegaard diagram this way. One wonders whether it is possible to redefine Heegaard distance, using HT(S), or perhaps even P(S) or one of the other complexes that has proved to be so useful in studying subgroups of M, and whether new things will be learned that way? We have focussed our discussion, up to now, on the 3-manifold that is determined by a choice of an element φ in the group M via the Heegaard splitting construction. A very different construction which also starts with the choice of an element in the mapping class group, say α∈M g, produces the mapping torus of α, i.e. the surface bundle ( S× [0, 1])/α, defined by setting ( p, 0) = ( α(p), 1). Surface bundle structures on 3-manifolds, when they exist, are also not unique. Two surface bundles ( S×I)/α, (S×I)/α′ are equivalent if and only if α,α ′ are in the same conjugacy class inMg. In [ 1] an interesting description is given of a natural way to produce, for each ( S×I)/α, a related Heegaard splitting HυβH ′. Choose a fiber S of (S×I)/α, say S×{ 0} and choose points p,q∈S, ⁄ϕ=q, ⁄ϕ=α(q). Let P andQ be disjoint closures of regular neighborhoods of p×[0, 1/2] and q× [1/2, 1] respectively. Set H = (S× ([0, 1/2]-Q)∪P, H ′ = (S× [1/2, 1]-P )∪Q. 152 J. Birman Note that H and H ′ are homeomorphic handlebodies of genus 2 g + 1 which are embedded in (S×I)/α and identified along their boundaries, so they give a Heegaard decomposition of ( S×I)/α. We call it the bundle-related Heegaard splitting of ( S×I)/α. It is HυβH ′ for some β∈M 2g+1. We have several problems that relate to this construction:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09627 | AMR-109-0116

### Problem 2.11 - Let g≥ 3 and b≥ 1.

S / solved | AMR 问题清单汇编

Let g≥ 3 and b≥ 1. Let a1,a 2,... be an infinite sequence of nonseparating simple closed curves on an oriented surface S of genus g with b boundary components. In the mapping class group Modb g of S, does the limit lim n→∞ c(ta1ta2··· tan) n exist? If it does, is it always positive? More generally, is there a positive number K such that c(ta1ta2··· tan)≥nK for all n? Suppose that there is a constant K such that c(ta1ta2··· tan) ≥ nK for any sequence of nonseparating simple closed curves a1,a 2,... and for all n. If f ∈ Modb g can be written as a product of right Dehn twists and if f = ta1ta2··· tak, then c(f ) = c(ta1ta2··· tak )≥ kK. Hence, k≤ c(f )/K. Therefore, a positive answer to Problem 2.11, with K independent of sequences, implies a positive answer to Problem 2.9. We would like to note that the hypothesis b≥ 1 in Problem 2.11 is necessary. Otherwise, the identity element in Mod g can be written as a product of right Dehn twists, say 1 = ta1ta2··· tan. Then for the sequence a1,a 2,...,a n,a 1,a 2,...,a n,a 1,a 2,...,a n,... the limit in Problem 2.11 is zero. We would also like to note that when b≥ 1, the identity in Mod b g cannot be expressed as a product of right Dehn twists. By embedding the surface with boundary into a closed surface of bigger genus, the proof of this follows from a theorem of Smith [ 41]; on a closed surface S, if 1 = ta1ta2··· tan, then the curves a1,a 2,...,a n fill up S. (I first learned this fact from Rostislav Matveyev in June 2000.)

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09628 | AMR-103-0034

### Problem 2.12 - Determine the dimension of the space of primitive Vassiliev invariants of each degree d.

P / partially_solved | AMR 问题清单汇编

Determine the dimension of the space of primitive Vassiliev invariants of each degree d. Equivalently, determine the dimension of the space A(S1; Q)(d) conn for each d. d 0 1 2 3 4 5 6 7 8 9 10 dimA(S1)(d) conn 0 1 1 1 2 3 5 8 12 18 27 dimA(S1)(d) 1 1 2 3 6 10 19 33 60 104 184 dimA(S1)(d)/Fl 1 0 1 1 3 4 9 14 27 44 80 d 11 12 13 14 dimA(S1)(d) conn 39 55 ≥ 78 ≥ 108 dimA(S1)(d) 316 548 ≥ 932 ≥ 1591 dimA(S1)(d)/Fl 132 232 ≥ 384 ≥ 659 Table 1: Some dimensions given in [67, 224]

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09629 | AMR-109-0015

### Problem 2.12 - Forg≥ 2, determine the irreducible factors of the graded pieces of the Malcev Lie algebra tg ofIg as Sp-modules.

O / open | AMR 问题清单汇编

Forg≥ 2, determine the irreducible factors of the graded pieces of the Malcev Lie algebra tg ofIg as Sp-modules. While Hain gives in [ Ha3] an explicit and reasonably simple presentation for tg when g >6, Problem 2.12 still seems to be an involved problem in classical representation theory. As Chris Leininger pointed out to me, all of the questions above have natural “virtual versions”. For example, one can ask about the classification of normal subgroups of finite index subgroups of Mod g. Another variation is the classification of virtually normal subgroups of Mod g, that is, subgroups whose normalizers in Mod g have finite index in Mod g.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09630 | AMR-109-0117

### Problem 2.12 - Let r be a positive integer and let Γr be the (normal) subgroup of Modg generated by the rth powers of all Dehn twists.

P / partially_solved | AMR 问题清单汇编

Let r be a positive integer and let Γr be the (normal) subgroup of Modg generated by the rth powers of all Dehn twists. Is Γr of infinite index? If we let Nr to be the normal closure in the mapping class group Mod g of the rth power of a single Dehn twist about a nonseparating simple closed curve, then by a result of Humphries [ 17] the index of Nr is finite in Mod g for all g when r = 2 and for ( g,r ) = (2, 3). Thus, the answer to Problem 2.12 is no in these cases. It is also known from [ 17] and [ 10] that Nr is of infinite index if r= 2, 3, 4, 6, 8, 12. Clearly, Nr is contained in Γ r. Let PMod b g,p denote the subgroup of Mod b g,p consisting of the isotopy classes of those diffeo- morphisms which fix each puncture. If g≥ 4, then the homology group H2(PModb g,p; Z) is iso- morphic to Zp+1. (cf. [ 13], [ 39] and [ 33].) (We note that the roles of the subscript p and the superscript b are interchanged in [ 33].) 5. Problems on homomorphisms of mapping class groups 91

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09631 | AMR-109-0169

### Problem 2.12 - This one is a warm-up.

O / open | AMR 问题清单汇编

This one is a warm-up. Given α∈M g, say as a product of Dehn twists, expressβ∈M 2g+1 as a related product of Dehn twists. With that in hand, observe that if α,α ′ are equivalent in the mapping class group Mg then the Heegaard splittings associated to β,β ′ appear to be equivalent. What about the converse? And how can we tell whether an arbitrary Heegaard splitting of a 3-manifold is the bundle-related splitting of a fibered 3-manifold? What restrictions must we place on β in order to be able to reverse the construction, and produce a surface bundle from a Heegaard splitting?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09632 | AMR-109-0118

### Problem 2.13 - (a) It is known from [33] that H2(Mod3; Z) and H2(Mod1 3; Z) are either Z or Z⊕ Z2.

P / partially_solved | AMR 问题清单汇编

(a) It is known from [33] that H2(Mod3; Z) and H2(Mod1 3; Z) are either Z or Z⊕ Z2. What are they? Also compute H2(PModb 3,p; Z) for all p and b. (b) It is also known that H2(Mod2; Z)~= H2(Mod1 2; Z)~= Z2. Compute H2(PModb 2,p; Z) for b≥ 2 and for all p.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09633 | AMR-109-0170

### Problem 2.13 - In [43] it is proved that in the case of the trivial genus g surface bundle, i.e.

O / open | AMR 问题清单汇编

In [43] it is proved that in the case of the trivial genus g surface bundle, i.e. Sg×S1 the bundle-related splitting is unique, up to equivalence. Are there other cases when it is unique?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09634 | AMR-103-0036

### Problem 2.14 - (M.

P / partially_solved | AMR 问题清单汇编

(M. Polyak) Milnor's μ -invariants of string links can be de- fined similarly as above (see [329]). Find a topological pres entation of a μ - invariant of string links (not assuming the vanishing of the lower μ -invariants). (1) Show that lk (L12···n-1, Ln) is well-defined in an appropriate sense. (2) Identify it with μ 12···n-1,n(L). 2.7 Finite type invariants of virtual knots A virtual knot [203] is defined by a knot diagram with virtual crossings modu lo Reidemeister moves. Finite type invariants of virtual knot s were studied in [154], where their weight systems are defined on the space - →A(X; R)/- → FI of arrow diagrams. Here an arrow diagram [330] is a chord diagram with oriented chords, and- →A (X; R) denotes the module over a commutative ring R spanned by arrow diagrams on X subject to the 6T relation, and - → FI denotes the oriented FI relation (see Figure 9 for these relations). It is known [3 30] that - →A (X; R) is isomorphic to the module spanned by acyclic oriented Jaco bi diagrams on X subject to the relations = 0 = and the -→ AS,--→ IHX, and --→ STU relations (see Figure 9).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09635 | AMR-109-0171

### Problem 2.14 - A 3-manifold is fibered if it admits a surface bundle structure.

S / solved | AMR 问题清单汇编

A 3-manifold is fibered if it admits a surface bundle structure. It is virtually fibered if it has a finite-sheeted cover that admits a surface bundle structure. In [49] Thurston asked whether every finite-volume hyperbolic 3-manifold is virtually fibered. This question has turned out to be one of the outstanding open problems of the post-Thurston period in 3-manifold topology. We ask a vague question: does the distance and the very special nature of the Heegaard splitting that's associated to a 3-manifold which has a surface bundle structure give any hint about the possibility of a 3-manifold which is not fibered being virtually fibered? With regard to Problem 2.14, we remark that the first examples of hyperbolic knots which are virtually fibered but not fibered were discovered 20 years after the question was posed, by Leininger [ 28] even though it seems to us that fibered knots should have been one of the easiest cases to understand. As we write this, in February 2005, there seems to be lots to learn about virtually fibered 3-manifolds.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09636 | AMR-109-0016

### Problem 2.14 - Give a proof of Theorem 2.13 which does not depend on the classification of finite simple groups.

O / open | AMR 问题清单汇编

Give a proof of Theorem 2.13 which does not depend on the classification of finite simple groups. 22 B. Farb To complete the picture, one would like to understand the frequency of those g for which the lower bound in (7) occurs.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09637 | AMR-109-0017

### Problem 2.15 - (Frequency of low symmetry).

O / open | AMR 问题清单汇编

(Frequency of low symmetry). Let H denote the set of integers g≥ 2 such that N (g) = 8( g + 1). Find the s0 for which the series ∑ g∈Hg-s converges absolutely for the real part$\Re$(s) of s satisfying$\Re$(s)>s 0, and has a singularity at s =s0. There are various refinements and variations on Problem 2.15. For example, Accola proves in [Ac] that when g is divisible by 3, then N (g)≥ 8(g + 3), with the bound attained infinitely often. One can try to build on this for other g, and can also ask for the frequency of this occurence. One can begin to refine Hurwitz's Theorem by asking for bounds of orders of groups of auto- morphisms which in addition satisfy various algebraic constraints, such as being nilpotent, being solvable, being a p-group, etc. There already exist a number of theorems of this sort. For example, Zomorrodian [ Zo] proved that if Aut( Xg) is nilpotent then it has order at most 16( g- 1), and if this bound is attained then g- 1 must be a power of 2. One can also ask for lower bounds in this context. As these kinds of bounds are typically attained and not attained for infinitely many g, one then wants to solve the following.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09638 | AMR-103-0037

### Problem 2.15 - Let I denote an oriented interval.

O / open | AMR 问题清单汇编

Let I denote an oriented interval. (1) Determine the dimensions of - →A (S1; Q)(d) and- →A (I; Q)(d) for each d. The 6T relation: + + = + + The- → FI relation: = 0 = The weak- → FI relation: =, = The- → AS relation: =- The--→ IHX relation: = - The--→ STU relation: = - Figure 9: The 6T and the oriented FI, AS, IHX, and STU relation s. Here, a thick dashed line implies the sum of the two orientations, and corr esponding thin dashed lines of pictures in the same formula have the same (arbitrar ily given) orientation. (2) Determine the dimensions of - →A (S1; Q)(d)/- → FI and - →A(I; Q)(d)/- → FI for each d. (3) Determine the dimensions of - →A (S1; Q)(d)/(weak- → FI) and- →A(I; Q)(d)/(weak- → FI) for each d.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09639 | AMR-109-0018

### Problem 2.16 - (Automorphism groups with special properties).

O / open | AMR 问题清单汇编

(Automorphism groups with special properties). LetP be a property of finite groups, for example being nilpotent, solvable, or a p-group. Prove a version of Larsen ’s theorem which counts those g for which the upper bound of | Aut(Xg)| is realized for some Xg withAut(Xg) havingP. Similarly for lower bounds. Determine the least g for which each given bound is realized. Many of the surfaces realizing the extremal bounds in all of the above questions are arithmetic, that is they are quotients of H2 by an arithmetic lattice. Such lattices are well-known to have special properties, in particular they have a lot of symmetry. On the other hand arithmetic surfaces are not typical. Thus to understand the “typical” surface with symmetry, the natural problem is the following.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09640 | AMR-109-0019

### Problem 2.17 - (Nonarithmetic extremal surfaces).

O / open | AMR 问题清单汇编

(Nonarithmetic extremal surfaces). Give answers to all of the above problems on automorphisms of Riemann surfaces for the collection of non-arithmetic surfaces. For example find bounds on orders of automorphism groups which are nilpotent, solvable, p-groups, etc. Prove that these bounds are sharp for infinitely many g. Determine the frequency of those g for which such bounds are sharp. Determine the least genus for which the bounds are sharp. The model result for these kind of problems is the following.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09641 | AMR-103-0040

### Problem 2.18 - CalculateFd(ZK, m)/Fd+1(ZK, m), letting m be a local move such as (1) a # move, (2) a pass move, (3) a Δ move, (4) an…

O / open | AMR 问题清单汇编

CalculateFd(ZK, m)/Fd+1(ZK, m), letting m be a local move such as (1) a # move, (2) a pass move, (3) a Δ move, (4) an n-gon move.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09642 | AMR-109-0020

### Problem 2.19 - (Canonical basepoints for Mg).

O / open | AMR 问题清单汇编

(Canonical basepoints for Mg). Find other properties of automorphisms or automorphism groups that determine a unique point of Mg. For example, is there a unique Riemann surface of genus g≥ 2 whose automorphism group is nilpotent, and is the largest possible order among nilpotent automorphism groups of genus g surfaces? A Hurwitz surface is a hyperbolic surface attaining the bound 84( g- 1). As the quotient of such a surface by its automorphism group is the (2, 3, 7) orbifold, which has a unique hyperbolic metric, it follows that for each g≥ 2 there are finitely many Hurwitz surfaces. As these are the surfaces of maximal symmetry, it is natural to ask precisely how many there are.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09643 | AMR-103-0041

### Problem 2.19 - (Y.

P / partially_solved | AMR 问题清单汇编

(Y. Ohyama) Find necessary and sufficient conditions for two μ -component links ( μ > 2) to be Δ link homotopic.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09644 | AMR-109-0009

### Problem 2.2 - Compute CommModg (Γ) for various subgroups Γ< Modg.

O / open | AMR 问题清单汇编

Compute CommModg (Γ) for various subgroups Γ< Modg. Paris-Rolfsen and Paris (see, e.g., [ Pa]) have proven that most subgroups of Mod g stabiliz- ing a simple closed curve, or coming from the mapping class group of a subsurface of S, are self-commensurating in Mod g. Self-commensurating subgroups, that is subgroups Γ < H with CommH (Γ) = Γ, are particularly important since the finite-dimensional unitary dual of Γ in- jects into the unitary dual of H; in other words, any unitary representation of H induced from a finite-dimensional irreducible unitary representation of Γ must itself be irreducible. V olumes of representations. Consider the general problem of classifying, for a fixed finitely generated group Γ, the set Xg(Γ):= Hom(Γ, Modg)/ Modg of conjugacy classes of representations ρ: Γ → Modg. Here the representations ρ1 and ρ2 are conjugate if ρ1 =Ch◦ρ2, where Ch: Mod g→ Modg is conjugation by some h∈ Modg. Suppose Γ = π1X whereX is, say, a smooth, closed n-manifold. Since Mg is a classifying space for Mod g, we know that for each for each [ ρ]∈X g(Γ) there exists a smooth map f:X→M g with f∗ =ρ, and that f is unique up to homotopy. Eachn-dimensional real cocycle ξ onMg then gives a well-defined invariant νξ:Xg(Γ)→ R defined by νξ([ρ]):= ∫ X f ∗ξ It is clear that νξ([ρ]) does not depend on the choices, and indeed depends only on the cohomology class of ξ. As a case of special interest, let X be a 2 k-dimensional manifold and let ωWP denote the Weil-Petersson symplectic form on Mg. Define the complexk-volume of ρ:π1X→ Modg to be Volk([ρ]):= ∫ X f ∗ωk WP

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09645 | AMR-109-0188

### Problem 2.2 - Decompose the representation on H0 into irreducible representations of ModΣ.

O / open | AMR 问题清单汇编

Decompose the representation on H0 into irreducible representations of ModΣ. When G = U(1), and Σ is the 2-torus, Hom( π,G )/G naturally identifies with T 2, by the functions α,β corresponding to a basis of π1(Σ). The functions φm,n:=αmβn, forms a Hilbert basis of H, indexed by ( m,n )∈ Z2. The ModΣ-representation on H arises from the linear GL(2, Z)-action on its basis Z2. The GL(2, Z)-orbits on Z2 are indexed by integers d≥ 0. The orbit of ( d, 0) consists of all ( m,n )∈ Z2 with gcd(m,n ) = d. These are Hilbert bases for irreducible constituents Cd of H. The irreducible constituents Cd admit an alternate description, as follows. The d-fold covering homomorphism G Φd - - →G induces a covering space Hom(π,G )/G-→ Hom(π,G )/G. Let Ld denote the closure of the image of the induced map H-→ H. Then Ld = ˆ⊕ d′|d Cd′ so Cd consists of the orthocomplement in Ld of the sum of all Ld′ for d′|d but d′=d.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09646 | AMR-109-0230

### Problem 2.2 - Does the converse hold in the above theorem without the assumption that G is free?

P / partially_solved | AMR 问题清单汇编

Does the converse hold in the above theorem without the assumption that G is free? The gist of Problem 2.2 is to find an extension of the Bestvina-Feighn combination theorem beyond the setting of groups acting on trees, to the wider setting of groups acting on Gromov hyperbolic cell complexes. In particular, in the context of Problem 2.2, the group Γ G acts on the 16. Problems in the geometry of surface group extensions 263 Rips complex of G, and one wants to use this information to decide about word hyperbolicity of ΓG. Unfortunately there are at present no examples on which to test Problem 2.2, but we will propose some problems along these lines in Section 4.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09647 | AMR-109-0159

### Problem 2.2 - How is the Nielsen-Thurston trichotomy related to the question of whether the distance is 0, 1, 2 or≥ 3?

O / open | AMR 问题清单汇编

How is the Nielsen-Thurston trichotomy related to the question of whether the distance is 0, 1, 2 or≥ 3? The next 3 problems concern the very non-constructive nature of the definition of d(φ):

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09648 | AMR-109-0108

### Problem 2.2 - Is there an endomorphism of the mapping class group of a (closed) orientable surface onto an infinite index infinite…

S / solved | AMR 问题清单汇编

Is there an endomorphism of the mapping class group of a (closed) orientable surface onto an infinite index infinite subgroup? If there is one such endomorphism f, is it true that f restricted to the image of fn is injective for some n? We conjecture that there is no such f. There are also results on injective homomorphisms from subgroups of finite index into the extended mapping class group Mod ∗ g,p, the group of isotopy classes of all (including orientation- reversing) diffeomorphisms of the surface S of genus g with p punctures. Let Γ be a subgroup of finite index in Mod ∗ g,p. Any injective homomorphism from Γ into Mod ∗ g,p is the restriction of an automorphism of Mod ∗ g,p. This result was proved by Irmak [ 18, 19 ] for g = 2, p≥ 2 and for g≥ 3, and by Bell and Margalit [ 1] for g = 0. One more result about the homomorphisms from the mapping class group we mention is the following.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09649 | AMR-109-0003

### Problem 2.2 - Letγ1,⋯,γ 2g be the standard basis of Z2g.

P / partially_solved | AMR 问题清单汇编

Letγ1,⋯,γ 2g be the standard basis of Z2g. Study the function L = ∑ Lγi:Yg→ [0,∞) as a Morse function. Find critical sets and deduce properties of the topology of Yg. For example, estimate from above and below the homotopy dimension 2 of Yg.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09650 | AMR-103-0042

### Problem 2.20 - Let R be a commutative ring with 1, say, Z or Q.

P / partially_solved | AMR 问题清单汇编

Let R be a commutative ring with 1, say, Z or Q. (1) Describe the spaces Fl(R(M K); loop)/Fl+1(R(M K); loop). (2) Describe the spaces Fl(RK; Δ Δ) /Fl+1(RK; Δ Δ). (3) Describe the image of the above map Fl(RK; Δ Δ) →F l(R(M K); loop).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09651 | AMR-103-0047

### Problem 2.25 - (M.

P / partially_solved | AMR 问题清单汇编

(M. Polyak) Establish the Goussarov-Habiro theory for virtual knots.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09652 | AMR-103-0048

### Problem 2.26 - (K.

O / open | AMR 问题清单汇编

(K. Habiro) Describe the abelian group {(M, K)~ H Ld (S3, unknot)}/ ~ H Ld+1 for each d.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09653 | AMR-103-0049

### Problem 2.27 - (D.

O / open | AMR 问题清单汇编

(D. Bar-Natan) Is there a similar statement for finite type invariants of links? Let I be an ideal in the algebra V of finite type invariants of links. Let Z be the set of links that are annihilated by all members of I, and let J be the ideal in V of all invariants that vanish on Z. Clearly, J always contains the radical of I. Are they always equal?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09654 | AMR-103-0050

### Problem 2.28 - (M.-J.

O / open | AMR 问题清单汇编

(M.-J. Jeong, C.-Y. Park) Find a minimal finite subset An of Vn such that span (An) = Vn.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09655 | AMR-109-0010

### Problem 2.3 - (Volume spectrum).

O / open | AMR 问题清单汇编

(Volume spectrum). Determine for each 1≤ k≤ 3g- 3 the image of Volk: Xg(Γ)→ R. Determine the union of all such images as Γ ranges over all finitely presented groups. It would also be interesting to pose the same problem for representations with special geometric constraints, for example those with holomorphic or totally geodesic (with respect to a fixed metric) representatives. In particular, how do such geometric properties constrain the set of possible volumes? Note that Mirzakhani [ Mir] has given recursive formulas for the Weil-Petersson volumes of moduli spaces for surfaces with nonempty totally geodesic boundary. Invariants from linear representations. Each linear representation ψ: Mod g→ GLm(C) provides us with many invariants for elements of Xg(Γ), simply by composition with ψ followed by taking any fixed class function on GL m(C). One can obtain commensurability invariants for subgroups of Mod g this way as well. While no faithful ψ is known for g ≥ 3 (indeed the existence of such a ψ remains a major open problem), there are many such ψ which give a great 2. Some problems on mapping class groups and moduli space 17 deal of information. Some computations using this idea can be found in [ Su2]. I think further computations would be worthwhile.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09656 | AMR-109-0189

### Problem 2.3 - Find a similar geometric interpretation for the irreducible constituents for compact nonabelian groups G.

O / open | AMR 问题清单汇编

Find a similar geometric interpretation for the irreducible constituents for compact nonabelian groups G.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09657 | AMR-109-0160

### Problem 2.3 - Find an algorithm to compute the distance d(φ) of an arbitrary element φ∈M.

O / open | AMR 问题清单汇编

Find an algorithm to compute the distance d(φ) of an arbitrary element φ∈M. We note that an algorithm to compute shortest paths between fixed vertices v,w in the curve complex has been presented by Shackleton in [45]. That problem is a small piece of the problem of computing the distance.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09658 | AMR-109-0004

### Problem 2.3 - Work out the details of this construction of the completion Yg of Yg.

O / open | AMR 问题清单汇编

Work out the details of this construction of the completion Yg of Yg. Show that L:Yg→ [0,∞) extends to L:Yg→ [0,∞) and is a proper map. Ideally, inclusion Yg→Yg should be a homotopy equivalence (and Yg-Yg should be the boundary of the manifold with corners Yg). Study L as a “Morse” function. Critical points of L are most likely not isolated; however, explain the change in the homotopy type of L -1 [0,t ] as t passes through a critical value. For example, take the case g = 2. Spaces added at infinity are Y 1 1×Y 1 1×S1 = H2× H2×S1 and the restriction of L to such a set factors through H2× H2 where it has a unique minimum and no other critical points. Thus this set contains a “mincircle”. Hope. A point of Y 2 is a local minimum of L iff it belongs to one of these mincircles. In addition there are isolated critical points of index 1 that connect up these circles in a tree-like pattern. Thus Y 2 and Y2 are homotopy equivalent to a wedge of circles, one circle for every splitting, recovering the theorem of Mess mentioned above. 2the smallest dimension of a space homotopy equivalent to Yg 1. Four questions about mapping class groups 7

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09659 | AMR-109-0191

### Problem 2.4 - Determine necessary and sufficient conditions on a general representation ρ for its orbit ModΣ· [ρ] to be dense.

O / open | AMR 问题清单汇编

Determine necessary and sufficient conditions on a general representation ρ for its orbit ModΣ· [ρ] to be dense. The case when G = SU(2) and Σ an n-holed sphere for n> 4 remains open.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09660 | AMR-109-0161

### Problem 2.4 - Knowing that d(φ)≤ 1, can we decide whether d(φ) = 0?

O / open | AMR 问题清单汇编

Knowing that d(φ)≤ 1, can we decide whether d(φ) = 0? Geometrically, if a Heegaard splitting is weakly reducible, can you decide if it’s reducible?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09661 | AMR-109-0192

### Problem 2.5 - Construct an example of a pseudo-Anosov mapping class for a closed surface which is not ergodic on the SU(2)-characte…

O / open | AMR 问题清单汇编

Construct an example of a pseudo-Anosov mapping class for a closed surface which is not ergodic on the SU(2)-character variety.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09662 | AMR-109-0162

### Problem 2.5 - Knowing that d(φ)≥ 1, can we decide whether it is ≥ 2?

O / open | AMR 问题清单汇编

Knowing that d(φ)≥ 1, can we decide whether it is ≥ 2? Knowing that it’s ≥ 2, can we decide whether it is ≥ 3?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09663 | AMR-109-0110

### Problem 2.5 - Let g > hand let Γ be a finite index subgroup of Modg.

O / open | AMR 问题清单汇编

Let g > hand let Γ be a finite index subgroup of Modg. Assume that g≥ 3 and φ: Γ → Modh is a homomorphism. (a) Is the image of φ necessarily finite? (b) (A weaker version of (a)) Is the image of φ necessarily of infinite index in Modh. On the other hand, it is well known that H 1(Modg; Z) = 0 for all g. Another way of saying this is that any homomorphism Mod g→ Z is trivial. A problem of Ivanov in [ 25], Problem 2.11 (A), asks whether the same conclusion holds for finite index subgroups:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09664 | AMR-109-0012

### Problem 2.6 - (Holomorphic representatives).

O / open | AMR 问题清单汇编

(Holomorphic representatives). Find an algorithm or a group-theoretic invariant which determines or detects whether or not a given representation ρ: π1Σh → Modg has a holomorphic representative. 2. Some problems on mapping class groups and moduli space 19 Note that a necessary, but not sufficient, condition for a representation ρ:π1Σh→ Modg to be holomorphic is that it be irreducible, i.e. there is no essential isotopy class of simple closed curve α in Σ g such that ρ(π1Σh)(α) = α. I believe it is not difficult to give an algorithm to determine whether or not any given ρ is irreducible or not. We would like to construct and classify (up to conjugacy) such ρ. We would also like to compute their associated invariants, such as ν(ρ):= ∫ Σh f ∗ωWP whereωWP is the Weil-Petersson 2-form on Mg, and where f: Σ h→M g is any map with f∗ =ρ. This would give information on the signatures of surface bundles over surfaces, and also on the Gromov co-norm of [ ωWP]∈H ∗(Mg, R). The classification question is basically impossible as stated, since e.g. surface groups surject onto free groups, so it is natural to first restrict to injective ρ. Using a technique of Crisp-Wiest, J. Crisp and I show in [ CF] that irreducible, injective ρ are quite common.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09665 | AMR-103-0028

### Problem 2.6 - Does there exists a non-trivial oriented knot which can not b e distinguished from the trivial knot by Vassiliev inva…

O / open | AMR 问题清单汇编

Does there exists a non-trivial oriented knot which can not b e distinguished from the trivial knot by Vassiliev invariant s? (See Problem 3.3 for an equivalent problem.)

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09666 | AMR-109-0111

### Problem 2.6 - Let Γ be a finite index subgroup of Modg.

P / partially_solved | AMR 问题清单汇编

Let Γ be a finite index subgroup of Modg. Is it true that H 1(Γ; Z) = 0? There are some partial answers to Problem 2.6. If g≥ 3 and if Γ contains the Torelli group, it was proved by McCarthy [ 37] that H 1(Γ; Z) = 0. However, if g = 2, this is not true anymore; the first examples of finite index subgroups in Mod 2 with nontrivial first cohomology were constructed by McCarthy in [ 37] and by Taherkhani in [ 43]. Moreover, if g = 1 or 2, it was shown in [ 31] that for each positive integer n≥ 2 the mapping class group Mod g contains a subgroup Γ n of finite index which admits a homomorphism onto a free group of rank n. In particular, the rank of H 1(Γn; Z) is at least n. We note that a positive answer to Problem 1 (a) implies a positive answer to Problem 2.6. Note also that since there is a subgroup of finite index in Mod 2 mapping onto a finite index subgroup of Mod 1, the hypothesis g≥ 3 in Problem 1 is necessary. A special case of Problem 2.6 is still interesting. Suppose that g≥ 3 and that Γ is s subgroup of finite index in Mod g containing the Johnson group (or Johnson kernel) Kg, the subgroup of Modg generated of by Dehn twists about separating simple closed curves. Is it true that any homomorphism Γ → Z is trivial? 1After this paper is written up, Behrstock and Margalit announced in [] that there exists an isomorphism between finite index subgroups of Mod ∗ 1,2 which is not the restriction of an inner automorphism. 88 M. Korkmaz

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09667 | AMR-109-0163

### Problem 2.6 - Schleimer has proved in [42] that each fixed 3-manifold M has a bound on the distances of its Heegaard splittings.

O / open | AMR 问题清单汇编

Schleimer has proved in [42] that each fixed 3-manifold M has a bound on the distances of its Heegaard splittings. Study this bound, with the goal of developing an algorithm for computing it. 150 J. Birman Understanding the handlebody subgroup Hg of the mapping class group is a problem that is obviously of central importance in understanding Heegaard splittings. A finite presentation for Hg was given by Wajnryb in [ 51]. To the best of our knowledge, this presentation has not been simplified, except in the special case g = 2. Very little is known about the structure of Hg, apart from its induced action on H1(2g, Z), which is a rather transparent subgroup of the symplectic group Sp(2 g, Z). We pose the problem:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09668 | AMR-109-0164

### Problem 2.7 - Study the handlebody subgroup of Mg.

O / open | AMR 问题清单汇编

Study the handlebody subgroup of Mg. A simplified presentation which would reveal new things about its structure, and/or anything new about its coset representatives in Mg would be of great interest.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09669 | AMR-109-0112

### Problem 2.7 - Suppose that ta1ta2⋯ tan = 1 in Modg, where n≥ 1.

P / partially_solved | AMR 问题清单汇编

Suppose that ta1ta2⋯ tan = 1 in Modg, where n≥ 1. Let G denote the quotient H1(S)/⟨[a1], [a2],..., [an]⟩, where [ai] denotes the homology class of the simple closed curve ai. Is it true that b1(G)≤g? A positive answer to this problem imply that the total space of every genus g Lefschetz fibration over the 2-sphere has the first Betti number b1≤g, whenever the Lefschetz fibration has at least one singular fiber. To the best knowledge of the author, all known examples of Lefschetz fibrations satisfy this conclusion. Assume that g≥ 2. On the closed connected oriented surface S, let us mark a point P. Let Mod1 g denote the mapping class group S with one puncture P. In this case, by forgetting that P is marked, we get an epimorphism ϕ: Mod 1 g→ Modg, whose kernel is isomorphic to the fundamental group of S at the point P.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09670 | AMR-109-0113

### Problem 2.8 - Given a factorization ta1ta2⋯ tan = 1 of the identity into a product of right Dehn twists in Modg, is it always pos…

P / partially_solved | AMR 问题清单汇编

Given a factorization ta1ta2⋯ tan = 1 of the identity into a product of right Dehn twists in Modg, is it always possible to lift this factorization to a factorization of the identity into a product of n right Dehn twists in Mod1 g? That is, is it always possible to choose simple closed curves b1,b 2,...,b n on S disjoint from P such that tb1tb2⋯ tbn = 1 in Mod1 g and ϕ(tbi) = tai, i.e. bi is isotopic to ai through isotopies not fixing P? The existence of a lifting of the factorization ta1ta2⋯ tan = 1 in Mod g to Mod 1 g implies that the corresponding Lefschetz fibration has a section; by lifting the relation to Mod 1 g, we specify 5. Problems on homomorphisms of mapping class groups 89 a point on each fiber. Thus, Problem 2.8 may be rephrased as follows: Does every Lefschetz fibration over S2 admit a section?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09671 | AMR-109-0165

### Problem 2.8 - Recall that we noted, earlier, that every genus g Heegaard splitting of every homology 3-sphere is obtained by allowi…

P / partially_solved | AMR 问题清单汇编

Recall that we noted, earlier, that every genus g Heegaard splitting of every homology 3-sphere is obtained by allowing ϕ to range over Ig. We also noted that Morita proved in [36] that every genus g Heegaard splitting of every homology 3-sphere is obtained by allowing ϕ to range over Kg. For these reasons it might be very useful to find generators for Hg∩Ig and/or Hg∩Kg. Our next problem is in a different direction. It concerns the classification of Heegaard splittings of graph manifolds:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09672 | AMR-103-0031

### Problem 2.9 - (X.-S.

O / open | AMR 问题清单汇编

(X.-S. Lin [262]) Is the knot signature the limit of a sequence of Vassiliev invariants?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09673 | AMR-109-0114

### Problem 2.9 - Suppose that a mapping class f ∈ Modb g, b≥ 1, is a product of right Dehn twists.

S / solved | AMR 问题清单汇编

Suppose that a mapping class f ∈ Modb g, b≥ 1, is a product of right Dehn twists. Does there exist a constant Cf, depending on f, such that whenever f can be written as a product of N right Dehn twists, then N≤Cf?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09674 | AMR-109-0166

### Problem 2.9 - Uncover the structure in the mapping class group that relates to the classifica- tion theorem for the Heegaard splitt…

O / open | AMR 问题清单汇编

Uncover the structure in the mapping class group that relates to the classifica- tion theorem for the Heegaard splittings of graph manifolds in [44].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09675 | AMR-109-0227

### Problem 20 - (Topology of strata).

P / partially_solved | AMR 问题清单汇编

(Topology of strata). Is it true that the connected components of the strata H(α) and of the strata Q(β) areK(π, 1)-spaces (i.e. their universal covers are contractible)? It is known [ KoZo], [ La] that the strata H(α) and Q(β) need not be connected. With the exception of the four strata listed below, there are intrinsic invariants that allow one to tell which component a given translation surface or quadratic differential belongs to. 15. Problems on billiards, flat surfaces and translation surfaces 257

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09676 | AMR-109-0228

### Problem 21 - (Exceptional Strata).

P / partially_solved | AMR 问题清单汇编

(Exceptional Strata). Find a geometric invariant which distinguishes differ- ent connected components of the four exceptional strata Q(-1, 9),Q(-1, 3, 6),Q(-1, 3, 3, 3) and Q(12). At the moment the known invariant (called the extended Rauzy class) distinguishing connected components is given in combinatorial and not geometric terms. ([ La])

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09677 | AMR-109-0201

### Problem 2: - For a fixed number R > 0, is there is compact subset K(R) of moduli space containing the projection of every Teichm¨…

O / open | AMR 问题清单汇编

For a fixed number R > 0, is there is compact subset K(R) of moduli space containing the projection of every Teichm¨ uller geodesic γ which satisfies d(Ψ(γ(s)), Ψ(γ(t)))≥ |s-t|/R-R for all s,t∈ R? Conversely, is there for a given compact set K in Mod( S) a number R =R(K)> 0 such that d(Ψ(γ(s)), Ψ(γ(t)))≥|s-t|/R-R for every Teichm¨ uller geodesic which projects into K? Analyze the images in C(S) of geodesics in Tg,m determined by minimal geodesic laminations which fill up S and are not uniquely ergodic.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09678 | AMR-107-0007

### Problem 2A - What is the minimal number of open sets $U_{i}$ covering ${\mathbb{R}}^{6}$ such that for any $U_{i}$…

O / open | AMR 问题清单汇编

What is the minimal number of open sets $U_{i}$ covering ${\mathbb{R}}^{6}$ such that for any $U_{i}$ there is a continuous map $\varphi_{i}:U_{i}\to{\mathbb{R}}^{2}$ sending any pair $(a,b)\in U_{i}$ into some solution of the system $\{f_{a}=0,g_{b}=0\}$?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09679 | AMR-107-0008

### Problem 2B - The same questions concerning the approximate solutions.

O / open | AMR 问题清单汇编

The same questions concerning the approximate solutions. That is, for any $i$ and any $(a,b)\in U_{i}$, the value $\varphi_{i}(a,b)$ should be not necessarily a root of the system $\{f_{a}=0,g_{b}=0\}$, but just a point in the $\varepsilon$-neighborhood of such a root for some fixed positive $\varepsilon$.

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09680 | AMR-109-0210

### Problem 3 - (Renormalization of billiards in polygons).

O / open | AMR 问题清单汇编

(Renormalization of billiards in polygons). Is there a natural dynamical system acting on the space of billiards in polygons so as to allow a useful renormalization procedure?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09681 | AMR-109-0133

### Problem 3 - Give a useful (piecewise) tropical description of the two elementary transformations.

O / open | AMR 问题清单汇编

Give a useful (piecewise) tropical description of the two elementary transformations. One thus immediately derives a (piecewise) tropical polynomial representation of the mapping class groups. What properties does it have, for instance under iteration? As alternative coordinates, [9] describes a family of curves whose intersection numbers alone coordinatize measured foliations of compact support (but there are relations), and presumably these intersection numbers could be computed using Theorem 3. We also wonder what are further applications or consequences of all these formulas.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09682 | AMR-109-0181

### Problem 3 - Given a topological description of fk0,fk1 describe fk0+k1.

O / open | AMR 问题清单汇编

Given a topological description of fk0,fk1 describe fk0+k1. A good understanding of this would enable one to drop the rather artificial introduction of the subclass D(0) g,ν,p in the discussion above. More generally still, if we have two pencils on a symplectic manifold X with fibre classes F1,F 2 one can ask for a description of a TLP with fibre F1 +F2 (if such exists). This might 11. Lefschetz pencils and mapping class groups 171 give information about the problem of describing the classes represented by symplectic forms on a fixed 4-manifold. Rather than trying to use the TLP description to reduce questions to combinatorics one can attempt to use it as a tool to prove general properties of symplectic 4-manifolds. So far, this has been more fruitful, giving a new approach to Taubes' results independent of the Seiberg-Witten theory [ 6], [ 13], [ 9]. There is also a generalisation of the TLP description to other 4-manifolds [3] and there are many things one could try here; for example to prove that any 4-manifold has "simple type". One thing that should be important to understand is the role of the canonical class K(ω). Using the Seiberg-Witten theory and pseudo-holomorphic curve techniques, a complete classification is known of symplectic 4-manifolds with ω.K (ω)< 0, i.e. withθ< 0 in our notation above. The only examples are the standard ones given by rational and ruled complex surfaces [ 8]. It would be interesting to derive this by the Lefschtez pencil method:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09683 | AMR-107-0009

### Problem 3 - Is the complement of the essential ramification set in ${\mathbb{R}}^{d}$ a $K(\pi,1)$-space?

O / open | AMR 问题清单汇编

Is the complement of the essential ramification set in ${\mathbb{R}}^{d}$ a $K(\pi,1)$-space?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09684 | AMR-109-0119

### Problem 3.1 - (a) Is it possible to generate the mapping class group of a closed nonori- entable surface by two elements?

P / partially_solved | AMR 问题清单汇编

(a) Is it possible to generate the mapping class group of a closed nonori- entable surface by two elements? (b) Is it possible to generate the mapping class group of a closed nonorientable surface by two torsion elements? three involutions?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09685 | AMR-109-0194

### Problem 3.1 - Determine the smallest dimensional manifold homotopy-equivalent to Σ upon which ModΣ acts compatibly with the outer a…

O / open | AMR 问题清单汇编

Determine the smallest dimensional manifold homotopy-equivalent to Σ upon which ModΣ acts compatibly with the outer action of ModΣ on π1(Σ).

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09686 | AMR-103-0051

### Problem 3.1 - For each oriented knot K, calculate the Kontsevich invariant Z(K) for all degrees.

O / open | AMR 问题清单汇编

For each oriented knot K, calculate the Kontsevich invariant Z(K) for all degrees.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09687 | AMR-109-0124

### Problem 3.1 - Give a homotopy theoretic construction of a map ρh: Ω ∞CP ∞ -1→K Sp(Z) with ρ≃ ρh◦α, at least after localization at a…

O / open | AMR 问题清单汇编

Give a homotopy theoretic construction of a map ρh: Ω ∞CP ∞ -1→K Sp(Z) with ρ≃ ρh◦α, at least after localization at a regular prime. (The 2-local case is of particular interest; cf. [ 12]). The rational cohomology of Ω ∞CP ∞ -1 is easy to list. For each connected component, H ∗(Ω∞ k CP ∞ -1; Q) = Q[κ′ 1,κ ′ 2,... ] 6. The mapping class groups and homotopy theory 97 whereκi =α∗(κ′ i) are the standard Miller-Morita-Mumford classes of degree 2 i, see [ 5], [ 15], [ 17] for further details. The rational cohomology of B Sp(Z) was calculated by Borel: H ∗(B Sp(Z); Q)∼=H ∗(Sp/U; Q)∼= Q[c1,c 3,... ]. where c2i+1 is the image of the (2 i + 1)st Chern class under the natural map from Sp /U to BU. Moreover,ρ∗(c2i+1) is a non-zero multiple of κ′ 2i+1 so that κ2i+1 restricts to zero in the rational cohomology of the Torelli group, [ 17].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09688 | AMR-109-0304

### Problem 3.1 - Prove (or disprove) that the even Mumford-Morita-Miller classes e2i∈H 4i(Ig; Q) are non-trivial, in a suitable stable…

P / partially_solved | AMR 问题清单汇编

Prove (or disprove) that the even Mumford-Morita-Miller classes e2i∈H 4i(Ig; Q) are non-trivial, in a suitable stable range, as cohomology classes of the Torelli group. The difficulty of the above problem comes from the now classical fact, proved by Johnson [ 40], that the abelianization of the Torelli group is very big, namely H1(Ig; Q)∼=UQ (g≥ 3). Observe that if Ig were perfect, then the above problem would have been easily solved by simply applying the Quillen plus construction to each group of the group extension (1) and then looking at the homotopy exact sequence of the resulting fibration. The work of Igusa [ 38] (in particular Corollary 8.5.17) shows a close connection between the above problem with another very important problem (see Problem 4.4 in § 4) of non-triviality of Igusa’s higher Franz-Reidemeister torsion classes in H 4i(IOutn; R) (Igusa uses the notation Out hFn for the group IOut n). We also refer to a recent work of Sakasai [ 95] which is related to the above problem. 22. Cohomological structure of the mapping class group and beyond 355 Next we recall the following two well-known problems about the structure of the Torelli group which are related to a foundational work of Hain [ 29].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09689 | AMR-109-0172

### Problem 3.1 - Study, systematically and with the help of computers, the finite quotients of Mg which do not factor through Sp (2g, Z).

P / partially_solved | AMR 问题清单汇编

Study, systematically and with the help of computers, the finite quotients of Mg which do not factor through Sp (2g, Z). We remark that Problem 3.1 would simply have been impossible in the days before high-speed computers, but it is within reach now. A fairly simple set of defining relations for Mg,0,0 can be found in [ 52]. As for checking whether any homomophism so-obtained factors through Sp(2 g, Z), there is are two additional relations to check, namely the Dehn twist on a genus 1 separating curve for g≥ 2, and the Dehn twist on a genus 1 bounding pair (see [ 23]) for g≥ 3. One method of organization is to systematically study homomorphisms of Mg (maybe starting with g = 3) into the symmetric group Σ n, beginning with low values of n and gradually increasing n. One must check all possible images of the generators of Mg in Σ n, asking (for each choice) whether the defining relations in Mg and Sp(2 g, Z) are satisfied. Note that if one uses Dehn twists on non-separating curves as generators, then they must all be conjugate, which places a big restriction. There are additional restrictions that arise from the orders of various generating sets, for example in [ 7] it is proved that Mg is generated by 6 involutions. Of course, as one proceeds with such an investigation, tools will present themselves and the calculation will organize itself, willy-nilly. We do not mean to suggest that non-finite quotients are without interest, so for completeness we pose a related problem:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09690 | AMR-109-0005

### Problem 3.1 - What is the smallest n = n(g) such that MCG (Sg) admits a properly dis- continuous action on Rn?

P / partially_solved | AMR 问题清单汇编

What is the smallest n = n(g) such that MCG (Sg) admits a properly dis- continuous action on Rn? on a contractible n-manifold? (The answers are expected to be the same.) Of course, MCG (Sg) acts properly discontinuously on Teichm¨ uller space Tg~= R6g-6 so it is a natural guess that n(g) = 6 g- 6 (for g≥ 2; clearly n(1) = 2). The question is motivated by the work in [ BKK02] where a technique is developed for finding lower bounds on the dimension n(Γ) of a contractible manifold where a given group Γ acts properly discontinuously. The optimal n(Γ) is called the action dimension of Γ. For example, n(Fg 2 ) = 2 g and if Γ is a lattice in a connected semisimple Lie group G then n(Γ) = dim G/K (where K is a maximal compact subgroup) [ BF02a]. As observed in [ BKK02], if Bk is the braid group on k strands then n(Bk) = 2 k- 3 ( k≥ 2) which equals the dimension of the corresponding Teichm¨ uller space. The method is this. Find a convenient finite simplicial complex L that does not embed in Rm “for homological reasons”. Such a complex is called an m-obstructor complex. Complexes used in all applications are joins of pointed spheres (a pointed sphere Sk + is a sphere Sk union a disjoint point). E.g. the “utilities graph” is the join of two pointed 0-spheres, it does not embed in the plane for homological reasons, and is a 2-obstructor complex. For example, if L =Sk1 + ∗Sk2 + ∗···∗ Skp + then L is an m-obstructor complex 3 for m =k1 +k2 +··· +kp + 2p- 2 and all obstructor complexes used in [ BKK02] and [ BF02b] have this form. Then one wants to find a map F:L× [0,∞)→X which is proper and expanding and X is a proper metric space on which the given group Γ acts isometrically, properly discontinuously and cocompactly. The map F is expanding if for any two disjoint simplices σ,τ ⊂L we have lim t→∞ d(F (σ× [t,∞)),F (τ× [t,∞))) = ∞ For the case Γ = MCG (Sg) it is natural to take X to be the subset T ≥ϵ g of Teichm¨ uller space consisting of marked surfaces with no closed geodesics of length < ϵ for a certain small ϵ > 0. When such a map is found it then follows from [ BKK02] that n(Γ)≥m + 2.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09691 | AMR-103-0060

### Problem 3.10 - (D.

O / open | AMR 问题清单汇编

(D. Thurston) Construct a series of configuration space integrals whose value is in - →A(I) so that it gives all finite type invariants of virtual knots.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09692 | AMR-109-0031

### Problem 3.10 - Find a finitely presented subgroup H < Modg for which there are infinitely many conjugacy classes of finite subgroups…

S / solved | AMR 问题清单汇编

Find a finitely presented subgroup H < Modg for which there are infinitely many conjugacy classes of finite subgroups in H. The motivation for this problem comes from a corresponding example, due to Bridson [ Br], of such an H in GL( n, Z). One might to solve Problem 3.10 by extending Bridson’s construction to Sp(2 g, Z), pulling back such an H, and also noting that the natural map Mod g→ Sp(2g, Z) is injective on torsion. Another determination of the variety of subgroups of Mod g is the following. Recall that the isomorphism problem for a collection S of finitely presented groups asks for an algorithm which takes as input two presentations for two elements of S and as output tells whether or not those groups are isomorphic.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09693 | AMR-103-0061

### Problem 3.11 - (M.

O / open | AMR 问题清单汇编

(M. Polyak) Find another way to kill the hidden strata, so that the above three approaches can naturally present the ma pping degree of the same map. 12S. Poirier [328] showed the equivalence between the invaria nts derived from the first and second approaches, under the assumption of the vanishing of anomaly, by comparing these invariants for quasi-tangles (see Question 3.12). 13D. Thurston suggests that Etingof-Kazhdan R matrices [117] might be helpful to relate the invariants derived from the first and third approaches. 3.6 The Chern-Simons series of configuration space integral s

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09694 | AMR-103-0063

### Problem 3.13 - Find a combinatorial direct presentation of an associator f or all degrees, in particular, an associator with rationa…

P / partially_solved | AMR 问题清单汇编

Find a combinatorial direct presentation of an associator f or all degrees, in particular, an associator with rational coe fficients.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09695 | AMR-103-0064

### Problem 3.14 - (J.

P / partially_solved | AMR 问题清单汇编

(J. Roberts) Construct a rational Drinfel’d associator in the context of rational homotopy theory.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09696 | AMR-103-0065

### Problem 3.15 - (J.

S / solved | AMR 问题清单汇编

(J. Roberts) What is graph cohomology the cohomology of?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09697 | AMR-103-0066

### Problem 3.16 - (R.

O / open | AMR 问题清单汇编

(R. Bott) Give a geometric construction of these homology classes coming from Lie algebras. The third and currently best interpretation of graph cohomo logy is that it is the cohomology of an infinite-dimensional Lie algebra of formal Hamiltonian vector fields. Kontsevich uses this to explain (and vastly generali se) Rozansky-Witten weight systems in terms of Gelfand-Fuchs cohomology. Can th is interpretation be employed on the topological rather than algebraic side? I n other words, is there a construction involving knots and algebras of formal vector fields which yields the Kontsevich integral?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09698 | AMR-109-0038

### Problem 3.17 - (Logarithmic densities of the Johnson filtration).

O / open | AMR 问题清单汇编

(Logarithmic densities of the Johnson filtration). Determine the asymptotics of dlog(Ig(m)) both as g→∞ and as m→∞. Indeed, as far as I know, even the asymptotics of the (logarithmic) density of the kth term of the lower central series of π1Σg in π1Σg as k→∞ has not been determined. Entropy. The exponential growth rate of a group Γ with respect to a finite generating set S is defined as w(Γ,S ):= lim r→∞ (Br(Γ,S ))1/r 4For a wonderful discussion of this kind of issue, see Barry Mazur’s article [ Maz]. 30 B. Farb where Br(Γ,S ) denotes the cardinality of the r-ball in the Cayley graph of Γ with respect to the generating set S; the limit exists since β is submultiplicative. The entropy of Γ is defined to be ent(Γ) = inf {logw(Γ,S ): S is finite and generates Γ } Among other things, the group-theoretic entropy of ent( π1M ) of a closed, Riemannian mani- fold M gives a lower bound for (the product of diameter times) both the volume growth entropy of M and the topological entropy of the geodesic flow on M. See [ Harp1] for a survey. Eskin-Mozes-Oh [ EMO] proved that nonsolvable, finitely-generated linear groups Γ have positive entropy. Since it is classical that the action of Mod g on H1(Σg, Z) gives a surjection Modg→ Sp(2g, Z), it follows immediately that ent(Mod g)> 0. This method of proving positivity of entropy fails for the Torelli group Ig since it is in the kernel of the standard symplectic repre- sentation of Mod g. However, one can consider the action of Ig on the homology of the universal abelian cover of Σ g, considered as a (finitely generated) module over the corresponding covering group, to find a linear representation of Ig which is not virtually solvable. This is basically the Magnus representation. Again by Eskin-Mozes-Oh we conclude that ent( Ig)> 0.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09699 | AMR-103-0067

### Problem 3.17 - Find a topological construction of the 2-loop polynomial P θ K.

O / open | AMR 问题清单汇编

Find a topological construction of the 2-loop polynomial P θ K.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09700 | AMR-103-0068

### Problem 3.18 - (A.

S / solved | AMR 问题清单汇编

(A. Kricker) Let KT be the knot obtained from a tangle T as shown in Figure 12. Find a presentation of the 2-loop polyn omial P θ KT of KT by using the Kontsevich invariant Z(T ) of T. T K T Figure 12: The knot KT is obtained from the 2-parallel of a 2-strand tangle T by adding the tangle depicted in solid lines in the right pictur e. The dotted lines imply strands possibly knotted and linked in some fashion.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09701 | AMR-109-0039

### Problem 3.18 - Give explicit upper and lower bounds for ent(Modg).

O / open | AMR 问题清单汇编

Give explicit upper and lower bounds for ent(Modg). Compute the asymptotics of ent(Modg) and of ent(Ig) as g→∞. Similarly for ent(Ig(k)) as k→∞.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09702 | AMR-103-0069

### Problem 3.19 - Find a topological construction of the polynomial P ′ K given above.

O / open | AMR 问题清单汇编

Find a topological construction of the polynomial P ′ K given above. = = = = a + b Figure 13: The multi-linear relations. Here, f (t), g(t)∈ S, and a, b are scalars. = Figure 14: The push relation The loop expansion in a general loop-degree is described as f ollows. Let R be a field, say Q, and let S be a subring of R(t) which is invariant under the involution t↦→t-1, where t is an indeterminate. A labeled Jacobi diagram on ∅ is a vertex-oriented trivalent graph, whose edges are label ed by pairs of local orientations and elements of S. We define AS(∅; R) to be the vector space over R spanned by labeled Jacobi diagrams on ∅ subject to the AS, IHX, multilinear, and push relations (see Figures 13 and 14). The loop-degree of a labeled Jacobi diagram is half the number of trivalent vertices of the Jacob i diagram. For a polynomial A(t) with A(1) = 1 and A(t) = A(t-1), we have a map AQ[t±1,1/A(t)](∅; Q)-→B, (29) defined by ↦-→c0 + c1 + c2 +···+ cn +···, where f (t)∈ Q[t±1, 1/A(t)] is written f (eh) = ∑ ∞ k=0 ckhk. In particular, the map AQ[t±1](∅; Q)-→B (30) is defined by ↦-→+ + 1 2 +···+ 1 n! +···. The loop expansion of the Kontsevich invariant is described by the rational Z invariant Z rat(K)∈A Q[t±1,1/Δ K (t)](∅; Q) which is taken to log ⊔ Z(K) by the map (29). In particular, when Δ K(t) = 1, Z rat(K)∈ AQ[t±1](∅; Q). (The existence of Z rat(K) has been shown in [231], and the canonicality of Z rat(K) has been shown in [139].)

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09703 | AMR-109-0023

### Problem 3.2 - (Conjugator length bounds).

S / solved | AMR 问题清单汇编

(Conjugator length bounds). Prove that there exist constants C,K, depend- ing only on S, so that if u,v ∈ Modg are conjugate, then there exists g ∈ Modg with||g||≤ K max{||u||,||v||} +C so that u =gvg -1. Masur-Minsky ([ MM2], Theorem 7.2) solved this problem in the case where u and v are pseudo-Anosov; their method of hierarchies seems quite applicable to solving Problem 3.2 in the general case. While interesting in its own right, even the solution to Problem 3.2 would not answer the following basic problem.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09704 | AMR-109-0120

### Problem 3.2 - Compute the (outer) automorphism group of Mod(N ).

S / solved | AMR 问题清单汇编

Compute the (outer) automorphism group of Mod(N ).

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09705 | AMR-109-0173

### Problem 3.2 - Construct any representations of Mg, finite or infinite, which do not factor through Sp(2g, Z).

P / partially_solved | AMR 问题清单汇编

Construct any representations of Mg, finite or infinite, which do not factor through Sp(2g, Z). In a very different direction, every mathematician would do well to have in his or her pile of future projects, in addition to the usual mix, a problem to dream about. In this category I put:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09706 | AMR-109-0305

### Problem 3.2 - Determine whether the Torelli group Ig (g≥ 3) is finitely presentable or not (note that Ig (g≥ 3) is known to be fini…

O / open | AMR 问题清单汇编

Determine whether the Torelli group Ig (g≥ 3) is finitely presentable or not (note that Ig (g≥ 3) is known to be finitely generated by Johnson [42]).

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09707 | AMR-109-0006

### Problem 3.2 - Find a (6g- 8)-obstructor complex L and a proper expanding map F:L× [0,∞)→T ≥ϵ g 3For concreteness we triangulate eac…

O / open | AMR 问题清单汇编

Find a (6g- 8)-obstructor complex L and a proper expanding map F:L× [0,∞)→T ≥ϵ g 3For concreteness we triangulate each sphere as the join of 0-spheres and we triangulate L as the join. 8 M. Bestvina

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09708 | AMR-109-0195

### Problem 3.2 - Find general conditions which ensure that (10) is proper.

O / open | AMR 问题清单汇编

Find general conditions which ensure that (10) is proper. The level set R3∩κ-1(2) consists of characters of abelian representations, and ModΣ is ergodic on each of the four connected components of the smooth part of R3∩κ-1(2). When 2 <t ≤ 18, the ModΣ-action on R3∩κ-1(t) is ergodic. Fort> 18, the level sets R3∩κ-1(t) display both proper dynamics and chaotic dynamics. The region (-∞,-2]3 consists of characters of discrete embeddings ρ where the quotient hyperbolic surface H2/ρ(π) is homeomorphic to a three-holed sphere. Every homotopy equivalence Σ -→ P, where P is a hyperbolic surface homeomorphic to a three-holed sphere, determines such a character. Furthermore these determine closed triangular regions which are freely permuted by ModΣ. On the complement of these wandering domains the action is ergodic. WhenG = PGL(2, R), the group of (possibly orientation-reversing) isometries of H2, a similar analysis was begun by Stantchev [ 94, 52 ]. One obtains similar dynamical systems, where ModΣ acts now on the space of representations into the group G± = SL(2, C)∩ ( GL(2, R)∪ i GL(2, R) ) 13. Mapping class group dynamics on surface group representations 215 which doubly covers the two-component group PGL(2, R). These G±-representations are again parametrized by traces. They comprise four components, one of which is the subset of R3 parametrizing SL(2, R)-representations discussed above. The other three components are R×iR×iR, i R× R×iR, i R×iR× R respectively. Consider iR× R×iR. For -14≤t< 2, the ModΣ-action is ergodic, but when t< -14, wandering domains appear. The wandering domains correspond to homotopy-equivalences Σ-→P, where P is a hyperbolic surface homeomorphic to a two-holed projective plane. The action is ergodic on the complement of the wandering domains.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09709 | AMR-103-0070

### Problem 3.20 - Find a topological construction of the loop-degree l part of the rational Z invariant Z rat(K)∈A Q[t±1,1/Δ K (t)](∅;…

O / open | AMR 问题清单汇编

Find a topological construction of the loop-degree l part of the rational Z invariant Z rat(K)∈A Q[t±1,1/Δ K (t)](∅; Q) of a knot K, for each l.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09710 | AMR-103-0071

### Problem 3.21 - Find a basis of the space AQ[t±1,1/A(t)](∅; Q)(loop l), for each l, where A(t) is a polynomial with A(1) = 1 and A(t)…

O / open | AMR 问题清单汇编

Find a basis of the space AQ[t±1,1/A(t)](∅; Q)(loop l), for each l, where A(t) is a polynomial with A(1) = 1 and A(t) = A(t-1). In particular, find a basis of the space AQ[t±1](∅; Q)(loop l).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09711 | AMR-103-0073

### Problem 3.23 - (T.

P / partially_solved | AMR 问题清单汇编

(T. Kohno) Construct explicitly a universal invariant of finite type for links in Σ × [0, 1] with values in AΣ. In the case of genus 0 the above problem is solved by Kontsevic h integral. In higher genus case a suggestion for a construction of a univer sal invariant was given by Deligne at Oberwolfach meeting 1995. In the case of a punctured surface the problem was solved by Andersen, Mattes and Reshe tikhin. Let G be a simple Lie group and MG(Σ) the moduli space of G flat connections on Σ. The space of smooth functions on MG(Σ) denoted by C(MG(Σ)) has a structure of a Poisson algebra coming from a symplectic stru cture on MG(Σ). The algebraAΣ has also a Poisson algebra structure (see [8]). If each compo nent ofAΣ is colored by a representation of G, then there is a natural Poisson algebra homomorphism τ:AΣ → C(MG(Σ)). Problem 3.23 is related to the following problem.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09712 | AMR-103-0074

### Problem 3.24 - (T.

P / partially_solved | AMR 问题清单汇编

(T. Kohno) Give a deformation quantization of the Poisson algebraAΣ which descends to a deformation quantization of C(MG(Σ)). The above problem will give a new insight on quantization of MG(Σ). It would also be interesting to investigate a relation to the ge ometric quantization ofMG(Σ).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09713 | AMR-103-0075

### Problem 3.25 - (T.

P / partially_solved | AMR 问题清单汇编

(T. Kohno) Clarify the relation between a deformation quan- tization of C(MG(Σ)) at a special parameter and the space of conformal blocks in WZW models. Section 3.10 was written by T. Kohno.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09714 | AMR-103-0076

### Problem 3.26 - (T.

P / partially_solved | AMR 问题清单汇编

(T. Kohno) Determine the image and the kernel of the above map τ. The space of conformal blocks in WZW model is defined as the spa ce of coin- variant tensors in the following way. Let p1,⋯, pn be marked points on Σ and H1,⋯, Hn be representations of the affine Lie algebra ˆg. The space of conformal blocks is defined to be the set of linear forms φ: H1⊗⋯⊗Hn-→ C invariant under the action of meromorphic functions with va lues in g with poles at most at p1,⋯, pn, where the action is defined by the Laurent expansion at these points. There is a twisted version of the above constru ction, where the above meromorphic functions are replaced by meromorphic se ctions of a g local system.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09715 | AMR-103-0077

### Problem 3.27 - (T.

P / partially_solved | AMR 问题清单汇编

(T. Kohno) Compute the holonomy of the space of conformal blocks of the twisted WZW model. In particular, determine th e action of the braid group of Σ on the space of conformal blocks for each G flat connection on Σ. There is also a notion of the algebra of chord diagrams on n strings with horizontal chord on Σ, which we shall denote by An(Σ).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09716 | AMR-103-0078

### Problem 3.28 - (T.

O / open | AMR 问题清单汇编

(T. Kohno) Let Pn(Σ) denote the pure braid group of Σ with n strings. Does there exist an injective multiplicative homo morphism θ: Pn(Σ) →A n(Σ) defined over Q?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09717 | AMR-109-0024

### Problem 3.3 - (Fast conjugacy problem).

O / open | AMR 问题清单汇编

(Fast conjugacy problem). Find a polynomial time algorithm to solve the con- jugacy problem in Modg. Is there a quadratic time algorithm, as for the word problem? As explained in the example on page 11, a solution to Problem 3.3 would be a major step in finding a polynomial time algorithm to solve the homeomorphism problem for 3-manifolds that fiber over the circle. Almost convexity. In [ Ca] Cannon initiated the beautiful theory of almost convex groups. A group Γ with generating set S is almost convex if there exists C >0 so that for each r >0, and for any two points x,y ∈ Γ on the sphere of radius r in Γ with d(x,y ) = 2, there exists a path γ of length at most C connectingx toy and lying completely inside the ball of radius r in Γ. There is an obvious generalization of this concept from groups to spaces. One strong consequence of the almost convexity of Γ is that for such groups one can recursively build the Cayley graph of Γ near any point x in the n-sphere of Γ by only doing a local computation involving elements of Γ lying (universally) close to x; see [ Ca]. In particular one can build each n-ball, n≥ 0, and so solve the word problem in an efficient way. 3The second ingredient, crucial but ignored by some authors, is the computability of the Thurston norm. 2. Some problems on mapping class groups and moduli space 25

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09718 | AMR-109-0126

### Problem 3.3 - .

O / open | AMR 问题清单汇编

. Find a direct description of some particular simple torsion classes in H ∗(BΓ∞; Z). There is an interesting connection between the higher Reidemeister torsion classes from [ 10] and the classes κi that require further study. Indeed both [ 10] and [ 14] use parametrized Morse theory. Could Problem 3.3 be related to “modular higher Reidemeister torsion”? Let Fg denote the free groups on g generators and Aut( Fg) its automorphism group. It obviously maps into Aut( Fg+1) and one can form BAut(F∞) and also its plus construction BAut(F∞)+. (Aut( F∞) contains a perfect index two subgroup). A. Hatcher has shown that Ω∞S∞ is a direct factor of Z×BAut(F∞)+ (up to homotopy), [ 8].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09719 | AMR-109-0196

### Problem 3.3 - Determine the ergodic behavior of the ModΣ-action on the level sets ( iR× R×iR ) ∩κ-1(t) wheret> 2.

P / partially_solved | AMR 问题清单汇编

Determine the ergodic behavior of the ModΣ-action on the level sets ( iR× R×iR ) ∩κ-1(t) wheret> 2. The level sets for t> 6 contains wandering domains corresponding to Fricke spaces of a one-holed Klein bottle.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09720 | AMR-109-0231

### Problem 3.3 - Does there exist an algorithm which produces the integer M in Theorem 3.1, given φ1,...,φ n?

O / open | AMR 问题清单汇编

Does there exist an algorithm which produces the integer M in Theorem 3.1, given φ1,...,φ n?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09721 | AMR-103-0053

### Problem 3.3 - Does there exists a non-trivial oriented knot K such that Z(K) = Z(O) for the trivial knot O?

O / open | AMR 问题清单汇编

Does there exists a non-trivial oriented knot K such that Z(K) = Z(O) for the trivial knot O? (See Problem 2.6 for an equivalent problem.)

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09722 | AMR-109-0174

### Problem 3.3 - Is there a faithful finite dimensional matrix representation of Mg,b,n for any value of the triplet (g,b,n ) other th…

P / partially_solved | AMR 问题清单汇编

Is there a faithful finite dimensional matrix representation of Mg,b,n for any value of the triplet (g,b,n ) other than (1, 0, 0), (1, 1, 0), (1, 0, 1), (0, 1,n ), (0, 0,n ) or (2, 0, 0)? We have mentioned Problem 3.3 because we believe it has relevance for Problems 3.1 and 3.2, for reasons that relate to the existing literature. To the best of our knowledge there isn’t even a known candidate for a faithful representation of Mg,0,0 for g≥ 3, even though many experts feel thatMg,0,0 is linear. This leads us to ask a question:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09723 | AMR-109-0306

### Problem 3.3 - Let ug denote the graded Lie algebra associated to the prounipotent radical of the relative Malcev completion of Ig d…

O / open | AMR 问题清单汇编

Let ug denote the graded Lie algebra associated to the prounipotent radical of the relative Malcev completion of Ig defined by Hain [29] and let ug→hQ g be the natural homomorphism (here hQ g denotes the graded Lie algebra consisting of symplectic derivations, with positive degrees, of the Malcev Lie algebra of π1Σg). Determine whether this homomorphism is injective or not. In [ 80], we defined a series of secondary characteristic classes for the mapping class group. However there was ambiguity coming from possible odd dimensional stable cohomology classes of the mapping class group. Because of the result of Madsen-Weiss cited above, we can now eliminate the ambiguity and give a precise definition as follows. For each i, we constructed in [ 51][52] explicit group cocycles zi∈ Z2i(Mg; Q) which represent the i-th Mumford-Morita-Miller class ei by making use of the homomorphism Mg→U ⋊ Sp(2g, Z) constructed in [ 77] which extends the (first) Johnson homomorphism Ig→U. These cocycles are Mg-invariant by the definition. Furthermore we proved that such cocycles are unique up to coboundaries. On the other hand, as is well known, any odd class e2i-1 comes from the Siegel modular group Sp(2 g, Z) so that there is a cocycle z′ 2i-1∈ Z4i-2(Mg; Q) which comes from Sp(2 g, Z). This cocycle is uniquely defined up to coboundaries and Mg-invariant. Now consider the difference z2i-1-z′ 2i-1. It is a coboundary so that there exists a cochain yi∈ C4i-3(Mg; Q) such that δyi = z2i-1-z′ 2i-1. Since H 4i-3(Mg; Q) = 0 by [ 64] (in a suitable stable range), the cochain yi is well-defined up to coboundaries. Now let Kg be the kernel of the Johnson homomorphism so that we have an extension 1-→Kg-→Ig-→U-→1. (8) Recall that Johnson [ 43] proved that Kg is the subgroup of Mg generated by Dehn twists along separating simple closed curves on Σ g. The cocycle z′ 2i-1 is trivial on the Torelli group Ig while the cocycle z2i-1 (in fact any zi) vanishes on Kg. It follows that the restriction of the cochain yi toKg is a cocycle. Hence we obtain a cohomology class di = [yi|Kg ]∈H 4i-3(Kg; Q). This cohomology class is Mg-invariant where Mg acts on H ∗(Kg; Q) via outer conjugations. This can be shown as follows. For any element ϕ∈M g, let ϕ∗(yi) be the cochain obtained by applying the conjugation by ϕ on yi. Since both cocycles z2i-1,z ′ 2i-1 areMg-invariant, we have δϕ∗(yi) = δyi. Hence ϕ∗(yi)-yi is a cocycle of Mg. By the result of [ 64] again, we see that ϕ∗(yi)-yi is a coboundary. Hence the restrictions of ϕ∗(yi) and yi toKg give the same cohomology class.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09724 | AMR-109-0121

### Problem 3.3 - Letg >h, and let N andN ′ denote the closed nonorientable surfaces of genera g andh respectively.

S / solved | AMR 问题清单汇编

Letg >h, and let N andN ′ denote the closed nonorientable surfaces of genera g andh respectively. Is it true that any homomorphism φ: Mod(N )→ Mod(N ′) has finite image?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09725 | AMR-109-0175

### Problem 3.4 - Find a candidate for a faithful finite-dimensional matrix representation of Mg orMg,1,0.

P / partially_solved | AMR 问题清单汇编

Find a candidate for a faithful finite-dimensional matrix representation of Mg orMg,1,0.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09726 | AMR-109-0197

### Problem 3.4 - Find a point ρ∈ Hom(π, SL(2, C)) such that the closure of its orbit ModΣ· [ρ] meets both the image of the unitary cha…

O / open | AMR 问题清单汇编

Find a point ρ∈ Hom(π, SL(2, C)) such that the closure of its orbit ModΣ· [ρ] meets both the image of the unitary characters Hom(π, SU(2)) and the closure QΣ of the quasi-Fuchsian characters. Homological actions. The action of ModΣ on the homology of Hom( π,G )/G furnishes another source of possibly interesting linear representations of ModΣ. With Neumann [ 51], we proved that for the relative SL(2, C)-character varieties of the one-holed torus and four-holed sphere, the action of ModΣ factors through a finite group. Atiyah-Bott [ 3] use infinite-dimensional Morse theory to analyze the algebraic topology of Hom(π,G )/G, when G is compact. For the nonsingular components their techniques imply that the ModΣ-action on the rational cohomology of Hom( π,G )/G factors through the symplectic representation of ModΣ on H ∗(Σ). In particular Biswas [ 9] proved that the Torelli group acts trivially on nonsingular components. In contrast, Cappell-Lee-Miller [ 17, 18 ] proved the surprising result that that the Torelli group acts nontrivially on the homology of the SU(2)-character variety when Σ is closed.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09727 | AMR-109-0232

### Problem 3.4 - If H⊂MCG (S) is finite rank free subgroup whose nonidentity elements are pseudo-Anosov, is H a Schottky group?

P / partially_solved | AMR 问题清单汇编

If H⊂MCG (S) is finite rank free subgroup whose nonidentity elements are pseudo-Anosov, is H a Schottky group? For specific examples on which to test the above problem, Whittlesey [ Whi00b] answered a question of Penner by producing a normal subgroup of MCG (S),S of genus 2, whose nonidentity elements are entirely pseudo-Anosov. Moreover, this subgroup is an infinite rank free subgroup [Whi00a].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09728 | AMR-109-0122

### Problem 3.4 - Let φ: Mod( N )→ Mod(N ) be a homomorphism such that the image of φ is of finite index.

P / partially_solved | AMR 问题清单汇编

Let φ: Mod( N )→ Mod(N ) be a homomorphism such that the image of φ is of finite index. Is φ necessarily an automorphism? How about if we take the domain of φ to be a subgroup of finite index of Mod(N )?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09729 | AMR-109-0307

### Problem 3.4 - Prove that all the secondary classes d2,d 3,··· are non-trivial.

O / open | AMR 问题清单汇编

Prove that all the secondary classes d2,d 3,··· are non-trivial. Here is a problem concerning the first class d1. Let C be a separating simple closed curve on Σg which divides Σ g into two compact surfaces of genera h and g-h and let τC ∈K g be the Dehn twist along C. Then we know that the value of d1 onτC∈K g ish(g-h) (up to a constant depending on g). This is a very simple formula. However at present there is no known algorithm to calculate the value d1(ϕ) for a given element ϕ∈K g, say by analyzing the action of ϕ onπ1Σg.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09730 | AMR-103-0055

### Problem 3.5 - Characterize those elements of ˆA(S1)conn of the form log Z(K), or those elements of Bconn of the form log⊔ Z(K).

O / open | AMR 问题清单汇编

Characterize those elements of ˆA(S1)conn of the form log Z(K), or those elements of Bconn of the form log⊔ Z(K).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09731 | AMR-109-0308

### Problem 3.5 - Find explicit way of calculating d1(ϕ) for any given element ϕ ∈ Kg.

O / open | AMR 问题清单汇编

Find explicit way of calculating d1(ϕ) for any given element ϕ ∈ Kg. In particular, determine whether the Magnus representation Ig,1→GL(2g; Z[H]) of the Torelli group detectsd1 or not. Suzuki [ 103] proved that the Magnus representation of the Torelli group mentioned above is not faithful so that it may happen that the intersection of the kernel of the Magnus representation withKg is not contained in the kernel of d1. We may also ask whether the representation of the hyperelliptic mapping class group given by Jones [ 45], restricted to the intersection of this group withKg, detects d1 or not (cf. Kasahara [ 46] for a related work for the case g = 2). There are also various interesting works related to the class d1 such as Endo [ 14] and Morifuji [ 71] treating the hyperelliptic mapping class group, Kitano [ 54] as well as Hain and Reed [ 33]. Recently Biss and Farb [ 5] proved that the group Kg is not finitely generated for all g≥ 3 (K2 is known to be an infinitely generated free group by Mess [ 69]). However it is still not yet known whether the abelianization H1(Kg) is finitely generated or not (cf. Problem 2.2 of [ 80]). Finally we would like to mention that Kawazumi [ 50] is developing a theory of harmonic Mag- nus expansions which gives in particular a system of differential forms representing the Mumford-Morita-Miller classes on the universal family of curves over the moduli space Mg.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09732 | AMR-109-0233

### Problem 3.5 - Is every finite rank subgroup of Whittlesey’s group a Schottky subgroup of MCG (S)?

S / solved | AMR 问题清单汇编

Is every finite rank subgroup of Whittlesey’s group a Schottky subgroup of MCG (S)? As a consequence of Theorem 2.1, if H <MCG (S) has a finite index subgroup which is Schottky, that is, if H is a virtual Schottky subgroup, then Γ H has a finite index word hyperbolic subgroup and so Γ H is itself word hyperbolic.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09733 | AMR-109-0176

### Problem 3.5 - Is there a natural quotient complex of any one of the complexes discussed in §1 which might be useful for the constru…

O / open | AMR 问题清单汇编

Is there a natural quotient complex of any one of the complexes discussed in §1 which might be useful for the construction of non-faithful representations of Mg? Let’s suppose that we have some answers to either Problem 3.1 or 3.2 or 3.5. At that moment, our instincts would lead us right back to a line of investigation that was successful many years ago when, in [ 4], we used the symplectic representation and found an invariant which distinguished inequivalent minimal Heegaard splittings. In the intervening years we suggested that our students try to do something similar with other representations, but that project failed. We propose it anew. Recall that a 3-manifold M may have one or more distinct equivalence classes of Heegaard splittings. It is known that any two become equivalent after some number of stabilizations. There are many interesting unanswered questions about the collection of all equivalence classes 10. 3-manifolds, Heegaard distance and mapping class groups 155 of Heegaard splittings of a 3-manifold, of every genus. Recall that the equivalence class of the Heegaard splitting H∪φH ′ is the double coset HφH inM.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09734 | AMR-109-0123

### Problem 3.5 - Study homomorphisms Modg→ Mod(N ) and Mod(N )→ Modg.

O / open | AMR 问题清单汇编

Study homomorphisms Modg→ Mod(N ) and Mod(N )→ Modg. 92 M. Korkmaz It is known by the work of Birman and Chillingworth [ 3] that the mapping class group Mod( N ) of a closed nonoreintable surface N of genus g is isomorphic to a subgroup of the extended mapping class group Mod ∗ g-1 of the orientation double cover of N modulo an element of order two. It follows that there are finite index subgroups in Mod( N ) which are isomorphic to a subgroup of Mod g-1.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09735 | AMR-109-0027

### Problem 3.6 - (Generalized word problem).

P / partially_solved | AMR 问题清单汇编

(Generalized word problem). Determine the subgroups H in Modg for which the generalized word problem is solvable. Give efficient algorithms to solve the generalized word problem for these subgroups. Find the optimal time bounds for such algorithms. Of course this problem is too broad to solve in complete generality; results even in special cases might be interesting. Some solutions to Problem 3.6 are given by Leininger-McReynolds 26 B. Farb in [ LM]. We also note that Bridson-Miller (personal communication), extending an old result of Baumslag-Roseblade, have proven that any product of finite rank free groups has solvable generalized word problem with respect to any finitely presented subgroup. In light of this result, it would be interesting to determine whether or not there is a finitely presented subgroup of Mod g with respect to which the generalized word problem is not solvable. Distortion and quasiconvexity. There is a refinement of Problem 3.6. Let H be a finitely generated subgroup of a finitely generated group Γ. Fix finite generating sets on both H and Γ. This choice gives a word metric on both H and Γ, where dΓ(g,h ) is defined to be the minimal number of generators of Γ needed to represent gh-1. Let N denote the natural numbers. We say that a function f: N-→ N is a distortion function for H in Γ if for every word w in the generators of Γ, if w represents an element w∈H then dH (1,w)≤f (dΓ(1,w)) In this case we also say that “ H has distortion f (n) in Γ.” It is easy to see that the growth type of f, i.e. polynomial, exponential, etc., does not depend on the choice of generators for either H or Γ. It is also easy to see that f (n) is constant if and only if H is finite; otherwise f is at least linear. It is proved in [ F a1] that, for a group Γ with solvable word problem, the distortion of H in Γ is recursive if and only if H has solvable generalized word problem in Γ. For some concrete examples, we note that the center of the 3-dimensional integral Heisenberg group has quadratic distortion; and the cyclic group generated by b in the group ⟨a,b: aba-1 = b2⟩ has exponential distortion since anba-n =b2n.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09736 | AMR-103-0056

### Problem 3.6 - (J.

O / open | AMR 问题清单汇编

(J. Roberts) Give a good topological construction of the Kontsevich integral.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09737 | AMR-109-0198

### Problem 3.6 - Find a substitute for convex cocompactness in higher rank which includes the above examples of proper ModΣ-actions, a…

S / solved | AMR 问题清单汇编

Find a substitute for convex cocompactness in higher rank which includes the above examples of proper ModΣ-actions, and for which Eρ is proper. The work of Bonahon-Thurston on geometric tameness, and its recent extensions, implies that the energy function of a discrete embedding π-→ PSL(2, C) is proper if and only if it is quasi-Fuchsian [ 53].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09738 | AMR-109-0234

### Problem 3.6 - Give examples and constructions of virtual Schottky subgroups of MCG (S).

O / open | AMR 问题清单汇编

Give examples and constructions of virtual Schottky subgroups of MCG (S). One such construction is due to Honglin Min, currently a doctoral candidate at Rutgers University, Newark:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09739 | AMR-109-0177

### Problem 3.6 - Study the double coset HφH inM, using new finite or infinite quotients of M.

O / open | AMR 问题清单汇编

Study the double coset HφH inM, using new finite or infinite quotients of M. In this regard we stress finite, because a principle difficulty when this project was attempted earlier was in recognizing the image of H in infinite quotients of M, however if the quotient is finite and not too big, it suffices to know generators of H⊂M. Since a presentation for H was found by Waynryb in [51], we can compute the associated subgroup. Some of the open questions which might be revealed in a new light are: (1) How many times must one stabilize before two inequivalent Heegaard splittings become equivalent? (2) How can we tell whether a Heegaard splitting is not of minimal genus? (3) How can we tell whether a Heegaard splitting is stabilized? (4) Are any of the representations that we noted earlier useful in answering (1), (2) or (3) above? While we have stressed the search for good working quotients of Mg, we should not forget that in the case of homology spheres, we have already pointed out that any homology sphere may be defined by a Heegaard splitting with the Heegaard glueing map (now redefined with a new ‘base point’ ) ranging over Ig. Even more, as was proved earlier, Morita has shown in [ 36] that it suffices to let the glueing map range over Kg. This leads us to ask:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09740 | AMR-109-0028

### Problem 3.7 - (Distortion).

P / partially_solved | AMR 问题清单汇编

(Distortion). Find the possible distortions of subgroups in Modg. In particular, compute the distortions of Ig. Determine the asymptotics of the distortion of Ig(k) as k→∞. Is there a subgroup H <Modg that has precisely polynomial distortion of degree d> 1? There are some known results on distortion of subgroups in Mod g. Convex cocompact sub- groups (in the sense of [ FMo]) have linear distortion in Mod g; there are many such examples where H is a free group. Abelian subgroups of Mod g have linear distortion (see [ FLMi]), as do subgroups corresponding to mapping class groups of subsurfaces (see [ MM2, Ham ]). I would guess that Ig has exponential distortion in Mod g. A first step to the question of how the distortion of Ig(k) in Mod g behaves as k→∞ would be to determine the distortion of Ig(k + 1) in Ig(k). The “higher Johnson homomorphisms” (see, e.g., [ Mo3]) might be useful here. A stronger notion than linear distortion is that of quasiconvexity. Let S = S-1 be a fixed generating set for a group Γ, and let π: S∗→ Γ be the natural surjective homomorphism from the free monoid on S to Γ sending a word to the group element it represents. Let σ: Γ →S∗ be a (perhaps multi-valued) section of π; that is, σ is just a choice of paths in Γ from the origin to each g∈ Γ. We say that a subgroup H <Γ is quasiconvex (with respect to σ) if there exists K >0 so that for each h∈ H, each path σ(h) lies in the K-neighborhood of H in Γ. Quasiconvexity is a well-known and basic notion in geometric group theory. It is easy to see that if H is quasiconvex with respect to some collection of quasigeodesics then H has linear distortion in Mod g (see [ F a1]). 2. Some problems on mapping class groups and moduli space 27

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09741 | AMR-109-0178

### Problem 3.7 - Are there quotients of Ig orKg in which the intersection of either Ig orKg with the handlebody group Hg is sufficiently…

O / open | AMR 问题清单汇编

Are there quotients of Ig orKg in which the intersection of either Ig orKg with the handlebody group Hg is sufficiently tractible to allow one to study the double cosets: (Ig∩Hg)(φ)(Ig∩Hg), where φ∈I g, or ( Kg∩Hg)(φ)(Kg∩Hg), where φ∈K g? In regard to Problem 3.7 we note that in [ 36] Morita was seeking to understand how topological invariants of 3-manifolds might lead him to a better understanding of the representations of Ig andKg, but he did not ask about the potential invariants of Heegaard splittings that might, at the same time, be lurking there.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09742 | AMR-103-0057

### Problem 3.7 - Construct the Kontsevich invariant (i.e.

O / open | AMR 问题清单汇编

Construct the Kontsevich invariant (i.e. a universal Vassi liev invariant) with coefficients in a finite field.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09743 | AMR-109-0029

### Problem 3.8 - Determine which subgroups of Modg are quasiconvex with respect to some collection of geodesics.

O / open | AMR 问题清单汇编

Determine which subgroups of Modg are quasiconvex with respect to some collection of geodesics. This question is closely related to, but different than, the question of convex cocompactness of subgroups of Mod g, as defined in [ FMo], since the embedding of Mod g in Teichg via any orbit is exponentially distorted, by [ FLMi].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09744 | AMR-103-0059

### Problem 3.9 - (M.

O / open | AMR 问题清单汇编

(M. Polyak) Construct the “Kontsevich invariant” (i.e. a universal finite type invariant) of virtual knots in - →A (I). (See also Conjecture 2.17.)

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09745 | AMR-109-0202

### Problem 3: - Describe the space of geodesic currents for Mg,m.

O / open | AMR 问题清单汇编

Describe the space of geodesic currents for Mg,m. Is the set of weighted sums of Dirac masses at the pairs of fixed points of pseudo-Anosov elements dense? Is there a geodesic current μ which is absolutely continuous, i.e. such that there is a Mg,m-invariant measure class μ0 on ∂C(S) with the property that for every Borel subset A of ∂C(S) we have μ0(A) = 0 if and only if μ(A×∂C(S)- Δ) = 0? Is there a distinguished absolutely continuous current such that the invariant measure class on ∂C(S) is determined by a Hausdorff measure with respect to one of the distance functions δ on ∂C(S)?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09746 | AMR-109-0211

### Problem 4 - (Characterization of Veech surfaces).

O / open | AMR 问题清单汇编

(Characterization of Veech surfaces). Characterize all Veech surfaces (for each stratum of each genus). This problem is trivial in genus one; in genus two K. Calta [ Ca] and C. McMullen [ McM2] have provided solutions. In the papers [ KnSm] and Puchta [ Pu] the acute rational billiard triangles that give rise to Veech surfaces were classified. In [ SmW e] there is a criterion for a surface ( X,ω ) to be a Veech surface that is given in terms of the areas of triangles embedded in (X,ω ).

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09747 | AMR-107-0010

### Problem 4 - Are these geometric obstructions sufficient to solve the above problem?

P / partially_solved | AMR 问题清单汇编

Are these geometric obstructions sufficient to solve the above problem?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09748 | AMR-109-0134

### Problem 4 - Does the recipe in Theorem 4 give virtually all pA maps?

S / solved | AMR 问题清单汇编

Does the recipe in Theorem 4 give virtually all pA maps? That is, given a pA map f, is there some iterate fn, for n≥ 1, so that fn arises from the recipe? (This question from [23,25] is related [10] to the Ehrenpreis Conjecture, our Problem 14.) In relation to Problem 4, let us mention that there are still other descriptions of pA maps up to iteration, for instance by Mosher [42] and in joint work of the author with Papadopoulos [45]; these descriptions are combinatorial rather than in terms of Dehn twists. For a fixed surface F, consider the set of logarithms of dilatations of all pA maps supported on F. This characteristic “spectrum” Σ( F )⊆ R>0 of F is precisely the Teichm¨ uller geoedsic length spectrum of Riemann’s moduli space M(F ). The spectrum Σ( F ) is discrete. (In fact, dilatations occur as spectral radii of integral-linear Perron-Frobenius symplectomorphisms in a range of dimensions bounded above and below in terms of the topological type of F.)

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09749 | AMR-109-0182

### Problem 4 - Reproduce the classification of manifolds with ω.K (ω)< 0 by studying the sets Cχ,θ for θ< 0.

O / open | AMR 问题清单汇编

Reproduce the classification of manifolds with ω.K (ω)< 0 by studying the sets Cχ,θ for θ< 0. There is a network of interesting questions dealing with the borderline case when K(ω).ω = 0 or, stronger still, K(ω) = 0. In the latter case the only known examples are the the standard complex tori, certain other torus bundles over tori and K3 surfaces. So we have:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09750 | AMR-109-0087

### Problem 4.1 - ( Topological Schottky Problem).

O / open | AMR 问题清单汇编

( Topological Schottky Problem). Understand the homotopy type of Jc g and use it to compute H •(Jc g ) and H • c (Jc g ). The first interesting case is when g = 4, where Jc g is a singular divisor in h4. It would also be interesting and natural to compute the intersection homology of Jc g. 3. Finiteness and Torelli spaces 69 A knowledge of the topology of Jg,Jc g orQc g should help with the computation of the σ-invariant homology H•(Tg)⟨σ⟩.7 When 2 is invertible in the coefficient ring R, we can write H•(Tg;R) = H•(Tg;R)+⊕H•(Tg;R)- where σ acts as the identity on H•(Tg;R)+ and as -1 on H•(Tg;R)-.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09751 | AMR-103-0079

### Problem 4.1 - Calculate S2,∞(M ) for each oriented 3-manifold M.

O / open | AMR 问题清单汇编

Calculate S2,∞(M ) for each oriented 3-manifold M. Find a convenient methodology to calculate it.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09752 | AMR-109-0235

### Problem 4.1 - Do there exist two surfaces S,S ′, closed and of genus ≥ 2, such that MCG (S) contains a subgroup isomorphic to π1(S′…

P / partially_solved | AMR 问题清单汇编

Do there exist two surfaces S,S ′, closed and of genus ≥ 2, such that MCG (S) contains a subgroup isomorphic to π1(S′) all of whose nontrivial elements are pseudo-Anosov? The closest attempt so far is an example of Leininger and Reid [ LR04] in which all closed curves on S′ give pseudo-Anosov elements of MCG (S) with the exception of a single simple closed curve and its iterates. This problem invites the following refinement: 266 L. Mosher

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09753 | AMR-109-0007

### Problem 4.1 - Show that there are many quasihomomorphisms f:MCG (Sg)→ R that satisfy (1) and (2) above plus (3) f is bounded on eve…

O / open | AMR 问题清单汇编

Show that there are many quasihomomorphisms f:MCG (Sg)→ R that satisfy (1) and (2) above plus (3) f is bounded on every G(q). I remark that the consequence about product sets of G(q)'s was shown to be true by a different method (Bestvina-Feighn, unpublished)

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09754 | AMR-109-0047

### Problem 4.10 - (Algorithmic Schottky problem).

O / open | AMR 问题清单汇编

(Algorithmic Schottky problem). Give an algorithm, in the sense of complexity theory over R, which takes as input a 2g×2g symplectic matrix representing a principally polarized abelian variety, and as output tells whether or not that torus lies in the period locus. One might also fix some ϵ = ϵ(g), and then ask whether or not a given principally polarized abelian variety lies within ϵ (in the locally symmetric metric on Ag) of the period locus. It should be noted that S. Grushevsky [ Gr] has made the KP-equations solution to the Schot- tky problem effective in an algebraic sense. This seems to be different than what we have just discussed, though. We now address the question of what the Schottky locus looks like from far away. To make this precise, let Cone( Ag) denote the tangent cone at infinity (defined in (10) above) of the locally symmetric Riemannian orbifold Ag. Hattori [ Hat] proved that Cone( Ag) is homeomorphic to the open cone on the quotient of the Tits boundary of the symmetric space Sp(2 g, R)/U(g); indeed it is isometric to a Weyl chamber in the symmetric space, which is just a Euclidean sector of dimension g.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09755 | AMR-103-0088

### Problem 4.10 - Calculate S3,∞(M ) for each oriented 3-manifold M.

O / open | AMR 问题清单汇编

Calculate S3,∞(M ) for each oriented 3-manifold M. Find a convenient methodology to calculate it.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09756 | AMR-109-0316

### Problem 4.10 - Compute the cohomology of AutFn and OutFn with coefficients in various GL(n, Q)-modules.

O / open | AMR 问题清单汇编

Compute the cohomology of AutFn and OutFn with coefficients in various GL(n, Q)-modules. 360 S. Morita For example, we could ask how Looijenga's result [ 62] for the case of the mapping class group can be generalized in these contexts. We refer to the work of Kawazumi [ 49] and also Satoh [ 97] for recent results concerning the above problem. Finally we recall the following well known problem.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09757 | AMR-109-0048

### Problem 4.11 - (Coarse Schottky problem).

P / partially_solved | AMR 问题清单汇编

(Coarse Schottky problem). Describe, as a subset of a g-dimensional Euclidean sector, the subset of Cone(Ag) determined by the Schottky locus in Ag. Points in Cone(Ag) are recording how the relative sizes of basis vectors of the tori are changing; it is precisely the “skewing parameters” that are being thrown away. It doesn’t seem unreasonable to think that much of the complexity in describing the Schottky locus is coming precisely from these skewing parameters, so that this coarsification of the Schottky problem, unlike the classical version, may have a reasonably explicit solution. There is a well-known feeling that the Schottky locus is quite distorted in Ag. Hain and Toledo (perhaps among others) have posed the problem of determining the second fundamental form of the Schottky locus, although they indicate that this would be a rather difficult computation. We can coarsify this question by extending the definition of distortion of subgroups given in Subsection 3.2 above to the context of subspaces of metric spaces. Here the distortion of a subset S in a metric space Y is defined by comparing the restriction of the metric dY to S versus the induced path metric on S.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09758 | AMR-103-0089

### Problem 4.11 - Calculate the higher skein modules based on the Kauffman skein relation W 3,∞ i (M ) and ˆW 3,∞(M ) (see below for the…

O / open | AMR 问题清单汇编

Calculate the higher skein modules based on the Kauffman skein relation W 3,∞ i (M ) and ˆW 3,∞(M ) (see below for their definitions).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09759 | AMR-109-0317

### Problem 4.11 - Determine whether the natural homomorphisms ˜H ∗(AutF2g; Q)-→˜H ∗(Mg,1; Q) ˜H ∗(OutF2g; Q)-→˜H ∗(Mg,∗; Q) induced by…

O / open | AMR 问题清单汇编

Determine whether the natural homomorphisms ˜H ∗(AutF2g; Q)-→˜H ∗(Mg,1; Q) ˜H ∗(OutF2g; Q)-→˜H ∗(Mg,∗; Q) induced by the inclusions Mg,1→AutF2g, Mg,∗→OutF2g are trivial or not. We refer to a result of Wahl [ 107] for a homotopy theoretical property of the homomorphism Mg,1→AutF2g where g tends to ∞.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09760 | AMR-109-0049

### Problem 4.12 - (Distortion of the Schottky locus).

O / open | AMR 问题清单汇编

(Distortion of the Schottky locus). Compute the distortion of the Schottky locus in Ag. 36 B. Farb A naive guess might be that it is exponential.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09761 | AMR-103-0091

### Problem 4.13 - Calculate HS q(M ) for each 3-manifold M.

O / open | AMR 问题清单汇编

Calculate HS q(M ) for each 3-manifold M.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09762 | AMR-103-0092

### Problem 4.14 - (J.

O / open | AMR 问题清单汇编

(J. Przytycki) (i) Find generators of S4,∞(S3, R). (ii) For which parameters of the (4,∞) skein and framing relations, trivial links are linearly independent in S4,∞(S3; R)? (iii) For which parameters of the (4,∞) skein and framing relations, the trivial knot is not representing a torsion element of S4,∞(S3, R)? A generalization of the Montesinos-Nakanishi conjecture [ 345] said that S4,∞(S3, R) is generated by trivial links and that the (4,∞) skein module (suit- ably defined) for n-tangles is generated by ∏ n-1 i=1 (3i +1) certain basic n-tangles. This would give a generating set for the (4,∞) skein module of S3 or D3 with 2 n boundary points (for n-tangles). However, the Montesinos-Nakanishi 3-move conjecture has been disproved by M.Dabkowski and J.H.Przytycki in February 2002 [99] and [342]. Therefore ∏ n-1 i=1 (3i + 1) is only the lower bound for the number of generators. In [345] we extensively analyze the possibilities that triv ial links are linearly independent; if b∞ = 0, then this may happen only if b0b1 = b2b3. These leads to the following conjecture (cases (1)-(2)):

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09763 | AMR-103-0094

### Problem 4.16 - (J.

O / open | AMR 问题清单汇编

(J. Przytycki) For which coefficients of the (4,∞) skein relation is the number of Fox 7-colorings measured by the (4,∞) skein module? Figure 16

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09764 | AMR-103-0095

### Problem 4.17 - Calculate Wk(M ) for each 3-manifold M.

O / open | AMR 问题清单汇编

Calculate Wk(M ) for each 3-manifold M.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09765 | AMR-103-0096

### Problem 4.18 - Define a skein module of 3-manifolds, and calculate it.

O / open | AMR 问题清单汇编

Define a skein module of 3-manifolds, and calculate it.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09766 | AMR-103-0080

### Problem 4.2 - (J.

S / solved | AMR 问题清单汇编

(J. Przytycki) Incompressible tori and 2-spheres in M yield torsion in S2,∞(M ) [339]. It is a question of fundamental importance whether other surfaces can yield torsion as well.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09767 | AMR-109-0128

### Problem 4.2 - .

P / partially_solved | AMR 问题清单汇编

. Find an analogue of Theorem 4.1 for topological manifolds, and relate it to surgery and pseudo-isotopy theory. Waldhausen's functor A(X) is the algebraic K-theory of the "ring" Ω ∞S∞(ΩX+). It is a ho- motopy theoretic construction which, quite explicitly, contains information about both Diff( X× I,∂X ×I∪X× 0) and Top( X×I,∂X ×I∪X× 0) where X is a compact d-manifold in a range of dimensions that increase with d, cf [ 20], [ 21]. Less explicitly A(X) also relates to Diff( X) and Top(X), cf. [ 22]. The topological cyclic homology functor TC( X), introduced in [ 1], is an enriched version of Connes' cyclic homology. A(X) maps to TC( X) via the cyclotomic trace, but TC( X) and the associated functor for rings (or linear categories) is of interest in its own right, cf. [ 9]. The p-adic completion TC( X)∧ p can be expressed in terms of more standard constructions in homotopy theory involving the free loop space. For X =pt, TC(pt)∧ p≃ (Ω∞S∞× Ω∞-1CP ∞ -1)∧ p. (6)

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09768 | AMR-109-0088

### Problem 4.2 - Determine whether or not H•(Tg; Z[1/2])- is always a finitely generated Z[1/2]-module.

S / solved | AMR 问题清单汇编

Determine whether or not H•(Tg; Z[1/2])- is always a finitely generated Z[1/2]- module. Does the infinite topology of Tg comes from Jg? To get one's hands on H•(Tg)-, it is necessary to better understand the topology of T c g or Sc g. Since Jc g is a Stein space, so is Sc g. Hamm's result [ 9] (see also [ 5, p. 152]) implies that Sc g has the homotopy type of a CW-complex of dimension at most 3 g- 3. Since Sc g is not a rational homology manifold when g≥ 3, it is probably most useful to study the topology of the manifold T c g as Poincar´ e duality will then be available. 8

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09769 | AMR-109-0236

### Problem 4.2 - Do there exist two surfaces S,S ´, closed and of genus ≥ 2, and a subgroup G <MCG (S) isomorphic to π1(S´), so that Γ…

O / open | AMR 问题清单汇编

Do there exist two surfaces S,S ´, closed and of genus ≥ 2, and a subgroup G <MCG (S) isomorphic to π1(S´), so that ΓG is word hyperbolic? Or so that G is convex cocompact? The equivalence of the two questions in Problem 4.2 is still open, as indicated in Problem 2.2.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09770 | AMR-109-0089

### Problem 4.3 - Determine good bounds for the homological dimension (or the CW-dimension) ofT c g.

O / open | AMR 问题清单汇编

Determine good bounds for the homological dimension (or the CW-dimension) ofT c g.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09771 | AMR-109-0237

### Problem 4.3 - Does there exist any non-virtually free, finitely generated subgroup G< MCG (S) whose nontorsion elements are all pse…

P / partially_solved | AMR 问题清单汇编

Does there exist any non-virtually free, finitely generated subgroup G< MCG (S) whose nontorsion elements are all pseudo-Anosov? Does G exist so that ΓG is word hyperbolic? So that G is convex cocompact?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09772 | AMR-109-0310

### Problem 4.3 - Produce non-trivial rational (co)homology classes of OutFn.

S / solved | AMR 问题清单汇编

Produce non-trivial rational (co)homology classes of OutFn. Next we consider the group IOut n. In [ 38] Igusa defined higher Franz-Reidemeister torsion classes τ2i∈H 4i(IOutn; R) as a special case of his general theory. These classes reflect Igusa's result mentioned above (The- orem 4.1) that the pull back of the Borel classes β4i+1∈H 4i+1(GL(n, Z); R) in H 4i+1(OutFn; R) vanish. However it seems to be unknown whether his classes are non-trivial or not.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09773 | AMR-109-0311

### Problem 4.4 - (Igusa).

O / open | AMR 问题清单汇编

(Igusa). Prove that the higher Franz-Reidemeister torsion classes τ2i∈H 4i(IOutn; R) are non-trivial in a suitable stable range. 22. Cohomological structure of the mapping class group and beyond 359 In the unstable range, where the Borel classes vanish in H ∗(GL(n, Z); R), there seem to be certain relations between the classes τ2i, (dual of) μi and unstable cohomology classes in H ∗(GL(n, Z); Q). As the first such example, we would like to ask the following specific problem.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09774 | AMR-103-0082

### Problem 4.4 - (J.

S / solved | AMR 问题清单汇编

(J. Przytycki) Compute S2,∞(F0,3× S1).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09775 | AMR-109-0238

### Problem 4.4 - Does there exist a simple cycle of dihedral subgroups of MCG (S) so that the associated reflection group P injects in…

O / open | AMR 问题清单汇编

Does there exist a simple cycle of dihedral subgroups of MCG (S) so that the associated reflection group P injects in MCG (S)? So that the image of P inMCG (S) answers any of the questions in Problem 4.3?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09776 | AMR-103-0083

### Problem 4.5 - Let F be a surface and I an interval.

O / open | AMR 问题清单汇编

Let F be a surface and I an interval. Describe the algebra S2,∞(F× I).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09777 | AMR-109-0312

### Problem 4.5 - Prove (or disprove) that the natural homomorphism H 4(OutF4; Q)~= Q-→H 4(IOut4; Q)GL is an isomorphism where the righ…

O / open | AMR 问题清单汇编

Prove (or disprove) that the natural homomorphism H 4(OutF4; Q)~= Q-→H 4(IOut4; Q)GL is an isomorphism where the right hand side is generated by (certain non-zero multiple of ) τ2. Here is another very specific problem. We know the following groups explicitly by various authors: H 8(M3,∗; Q)~= Q2 (Looijenga [ 61]) H 8(GL(6, Z); Q)~= Q (Elbaz-Vincent, Gangl, Soul´ e [ 13]) H 8(OutF6; Q)~= Q (Ohashi [ 91]) On the other hand, we have the following natural injection i as well as projection p M3,∗ i-→OutF6 p -→GL(6, Z). (9)

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09778 | AMR-109-0090

### Problem 4.5 - Try to understand the “topology at infinity” of T c g.

O / open | AMR 问题清单汇编

Try to understand the “topology at infinity” of T c g. In particular, try to compute Hk ∞(T c g ) for k in some range k≥do. Alternatively, try to compute the homology H•(T c g ) in lower degrees. Note that T c 2 is a manifold with boundary S5. The boundary of Jc 3 isS11. The first interesting case is in genus 3.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09779 | AMR-103-0084

### Problem 4.6 - Calculate the skein homology based on the Kauffman bracket skein relation.

O / open | AMR 问题清单汇编

Calculate the skein homology based on the Kauffman bracket skein relation.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09780 | AMR-109-0091

### Problem 4.6 - Compute H • ∞(T c 3 ).

O / open | AMR 问题清单汇编

Compute H • ∞(T c 3 ). The homology of T c g is related to that of Tg via the Gysin sequence. In order to apply it, one needs to understand the topology of the divisor T c,red g. This is built up out of products of lower genus Torelli spaces of compact type. The combinatorics of the divisor is given by the complex Ksep(S)/Tg.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09781 | AMR-109-0313

### Problem 4.6 - Determine the homomorphisms H 8(M3,∗; Q) i∗ ←-H 8(OutF6; Q) p∗ ←-H 8(GL(6, Z); Q) (10) induced by the above homomorph…

O / open | AMR 问题清单汇编

Determine the homomorphisms H 8(M3,∗; Q) i∗ ←-H 8(OutF6; Q) p∗ ←-H 8(GL(6, Z); Q) (10) induced by the above homomorphisms in (9).

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09782 | AMR-109-0092

### Problem 4.7 - Compute the Spg(Z)-moduleHk c (T c,red g ) in some range k≥ko.

O / open | AMR 问题清单汇编

Compute the Spg(Z)-moduleHk c (T c,red g ) in some range k≥ko. 3. Finiteness and Torelli spaces 71 We already know that H 6g-7 c (T c,red g ) = H0(Bc g) and that there is a surjection H 6g-8 c (T c,red g )→ H1(Bc g), where Bc g denotes the normalization (i.e., disjoint union of the irreducible components) ofT c,red g. In concrete terms: Bc g = ⌊g/2⌋⨿ h=1 ⨿ φ∈Spg(Z)(T c h,1×T c g-h,1) φ ( T c h,1×T c g-h,1 ). Lurking in the background is the folk conjecture Hk(Tg)is finitely generated when k<g - 1. If true, this places strong conditions on the finiteness of the topology of T c g andT c,red g. It is worthwhile to contemplate (for g≥ 3) the Gysin sequence: ⋯ →→ H 6g-10 c (T c,red g ) →→ H3(Tg) →→ H3(T c g ) ↙↙←←←

09787 | AMR-103-0087

### Problem 4.9 - Let F be a surface and I an interval.

O / open | AMR 问题清单汇编

Let F be a surface and I an interval. Describe the algebra S3(F× I).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09788 | AMR-109-0203

### Problem 4: - Does the above definition of a convex cocompact subgroup of Mg,m coincide with the definition of Farb and Mosher in [...

P / partially_solved | AMR 问题清单汇编

Does the above definition of a convex cocompact subgroup of Mg,m coincide with the definition of Farb and Mosher in [FMo]? Is the natural extension of such a group Γ by the fundamental group π1(S) of S word hyperbolic? Is there a convex cocompact subgroup of Mg,m which is isomorphic to the fundamental group of a closed surface of genus at least 2? A particular interesting class of subgroups of Mg,m arise from Veech surfaces. These surfaces are the projections to moduli space of the stabilizer of a complex geodesic in Teichm¨ uller space (which is a maximal embedded complex disc in the Teichm¨ uller space viewed as a bounded domain in C3g-3+m) with the additional property that this stabilizer is a lattice in PSL (2, R). Veech surfaces are surfaces of finite type with isolated singularities embedded in moduli space; they are never closed [V]. Thus their corresponding subgroup of Mg,m contains a free group of finite index with a distinguished family of conjugacy classes corresponding to the cusps of the curve. Veech surfaces have many beautiful algebraic and geometric properties (see e.g. [McM1], [McM2]). Elementary constructions of such surfaces and their coresponding subgroups of Mg,m are for example discussed in [L]. Veech surfaces can also be used to construct explicit subgroups of mapping class groups with prescribed geometric properties. A particularly beautiful result along this line was recently obtained by Leininger and Reid [LR].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09789 | AMR-109-0212

### Problem 5 - (Fuchsian groups).

P / partially_solved | AMR 问题清单汇编

(Fuchsian groups). Which Fuchsian groups are realized as Veech groups? Which subgroups of the mapping class group appear as Veech groups? This is equivalent to asking which subgroups are the stabilizers of a Teichm¨ uller disc.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09790 | AMR-109-0183

### Problem 5 - Analyse the monodromy of Lefschetz fibrations on manifolds with K(ω) = 0.

O / open | AMR 问题清单汇编

Analyse the monodromy of Lefschetz fibrations on manifolds with K(ω) = 0. Related to this is the general question of understanding the place of complex algebraic surfaces among general symplectic 4-manifolds. One can ask:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09791 | AMR-109-0135

### Problem 5 - For a given surface F, calculate Σ( F ).

O / open | AMR 问题清单汇编

For a given surface F, calculate Σ( F ). More modestly, calculate the least element of Σ( F ) or the least gap among elements of Σ( F ). Characterize the number fields arising as dilatations of pA maps on F. 7. Probing mapping class groups using arcs 111 Problems 4 and 5 are clearly related. For instance, the recipe in Theorem 4 allows one to give estimates on least elements in Problem 5, cf. [26,2]. McMullen [50] has also given estimates and examples.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09792 | AMR-103-0097

### Problem 5.1 - Classify the isomorphism classes of connected quandles of o rder n for each positive integer n.

P / partially_solved | AMR 问题清单汇编

Classify the isomorphism classes of connected quandles of o rder n for each positive integer n. See Table 4 for a list of connected quandles of order n for some n. n # Connected quandles of order n Self-dual Not self-dual 1 1 A trivial quandle 2 0 3 1 R3 4 1 Λ 2/(t2 + t + 1) 5 3 R5 Λ 5/(t- 2), its dual 6 2 2 subquandles of Conj( S4) 7 5 R7 Λ 7/(t- 2), Λ 7/(t- 3), their duals 8 ≥ 3 An abelian extension Λ 2/(t3 + t + 1), its dualof Λ 2/(t2 + t + 1) 9 8 R9, Λ 3/(t2- t + 1), Λ 9/(t- 2), Λ 3/(t2 + t- 1), R3× R3, Λ 3/(t2 + 1) their duals 10 ≥ 1 A subquandle of Conj( S5) 11 9 R11 Λ 11/(t- a) ( a = 2, 3,···, 9) 12 ≥ 2 R3× ( Λ 2/(t2 + t + 1) ), An icosahedral quandle 13 11 R13 Λ 13/(t- a) ( a = 2, 3,···, 11) 14 ≥ 0 15 ≥ 4 R3× R5, R3× ( Λ 5/(t- 2) ), its dualA subquandle of Conj( S5)... Prime p p- 2 Rp Λ p/(t- a) ( a = 2, 3,···, p- 2) Table 4: A table of some connected quandles. The second colum n shows the numbers of isomorphism classes of connected quandles of order n. We denote Z[t±1]/(n) by Λ n. Conj(Sm) denotes the conjugation quandle of the mth symmetric group Sm. An icosahedral quandle is a quandle whose elements are the vertices of an icosahedro n such that Sx of each element x is given by a rotation of the icosahedron centered at x.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09793 | AMR-103-0106

### Problem 5.10 - (M.

O / open | AMR 问题清单汇编

(M. Polyak) Define a quantum quandle.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09794 | AMR-103-0107

### Problem 5.11 - (C.

P / partially_solved | AMR 问题清单汇编

(C. Rourke, B. Sanderson) Is there a natural quandle space whose cohomology groups are the quandle cohomology groups?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09795 | AMR-109-0057

### Problem 5.11 - (Torelli finiteness).

O / open | AMR 问题清单汇编

(Torelli finiteness). Determine the maximal number f (g) for which there is a K(Ig, 1) space with finitely many cells in dimensions ≤f (g). Here is what is currently known about Problem 5.11: (1) f (2) = 0 since I 2 is not finitely generated (McCullough-Miller [ McM]). (2) f (3)≤ 3 (Johnson-Millson, unpublished, referred to in [ Me]). (3) For g≥ 3, combining Johnson's finite generation result [ Jo2] and a theorem of Akita (Theorem 5.13 below) gives 1 ≤f (g)≤ 6g- 5. One natural guess which fits with the (albeit small amount of) known data is that f (g) = g-2. As a special case of Problem 5.11, we emphasize the following, which is a folklore conjecture.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09796 | AMR-109-0059

### Problem 5.14 - Extend Akita's result to 2<g < 7.

S / solved | AMR 问题清单汇编

Extend Akita's result to 2<g < 7. Since Akita's proof produces no explicit homology classes, the following seems fundamental.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09797 | AMR-109-0060

### Problem 5.15 - (Explicit cycles).

P / partially_solved | AMR 问题清单汇编

(Explicit cycles). Explicitly construct infinitely many linearly independent cy- cles in H∗(Ig, Q) and H∗(Kg, Q). So, we are still at the stage of trying to find explicit nonzero cycles. In a series of papers (see [Jo1] for a summary), Johnson proved the quite nontrivial result: H 1(Ig, Z)≈∧3H H ⊕B2 (16) where B2 consists of 2-torsion. While the ∧3H/H piece comes from purely algebraic consider- ations, the B2 piece is "deeper" in the sense that it is purely topological, and comes from the Rochlin invariant (see [ BC] and [ Jo1]); indeed the former appears in H1 of the "Torelli group" in the analogous theory for Out( Fn), while the latter does not. 42 B. Farb

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09798 | AMR-109-0061

### Problem 5.16 - Determine the subalgebras of H ∗(Ig,K ), for K = Q and K = F2, generated by H 1(Ig,K ).

O / open | AMR 问题清单汇编

Determine the subalgebras of H ∗(Ig,K ), for K = Q and K = F2, generated by H 1(Ig,K ). Note that H ∗(Ig,K ) is a module over Sp(2 g,K ). When K = Q this problem has been solved in degree 2 by Hain [ Ha3] and degree 3 (up to one unknown piece) by Sakasai [ Sa]. Symplectic representation theory (over R) is used as a tool in these papers to greatly simplify computations. When K = F2, the seemingly basic facts one needs about representations are either false or they are beyond the current methods of modular representation theory. Thus computations become more complicated. Some progress in this case is given in [ BF a2], where direct geometric computations, evaluating cohomology classes on abelian cycles, shows that each of the images of σ∗:H 2(B3, F2)→H 2(Ig, F2) (σ|Kg )∗:H 2(B2, F2)→H 2(Kg, F2) has dimension at least O(g4).

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09799 | AMR-109-0063

### Problem 5.19 - Give an elementary, purely combinatorial-topological and group-theoretic, proof of Hain 's theorem.

O / open | AMR 问题清单汇编

Give an elementary, purely combinatorial-topological and group-theoretic, proof of Hain 's theorem. It seems that a solution to Problem 5.19 will likely require us to advance our understanding oflg in new ways. It may also give a hint towards attacking the following problem, where mixed Hodge theory does not apply. 44 B. Farb

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09800 | AMR-103-0098

### Problem 5.2 - Describe (the number of) representations of a knot quandle t o a fixed connected quandle of finite order, say, by usi…

O / open | AMR 问题清单汇编

Describe (the number of) representations of a knot quandle t o a fixed connected quandle of finite order, say, by using knot in variants known so far, or by reducing the problem to the case of smaller targe t quandles.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09801 | AMR-109-0064

### Problem 5.20 - (Hain for Aut( Fn)).

O / open | AMR 问题清单汇编

(Hain for Aut( Fn)). Give an explicit finite presentation for the Malcev Lie AlgebraL(IAn), where IAn is the group of automorphisms of the free group Fn acting trivially on H1(Fn, Z). There is a great deal of interesting information at the prime 2 which Hain’s theorem does not address, and indeed which remains largely unexplored. While Hain’s theorem tells us that reduction mod 2 gives us a large subalgebra of L2(Ig,1) coming from L0(Ig,1), the Lie algbera L2(Ig,1) over F2 is much bigger. This can already be seen from (16). As noted above, the 2-torsion B2 exists for “deeper” reasons than the other piece of H1(Ig,1, Z), as it comes from the Rochlin invariant as opposed to pure algebra. Indeed, for the analogous “Torelli group” IA n for Aut(Fn), the corresponding “Johnson homomorphism” gives all the first cohomology. Thus the 2-torsion in H 1(Ig, Z) is truly coming from 3-manifold theory.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09802 | AMR-109-0065

### Problem 5.21 - (Malcev mod 2).

O / open | AMR 问题清单汇编

(Malcev mod 2). Give an explicit finite presentation for the F2-Lie algebra L2(Ig,1). We can also build a Lie algebra using the Johnson filtration. Let hg:= ∞⊕ k=1 Ig(k) Ig(k + 1) ⊗ R Then h is a real Lie algebra. In §14 of [ Ha3], Hain proves that the Johnson filtration is not cofinal with the lower central series of Ig. He also relates hg to tg. The following basic question remains open.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09803 | AMR-109-0067

### Problem 5.23 - Compute H1(Modg[L]; Z).

P / partially_solved | AMR 问题清单汇编

Compute H1(Modg[L]; Z). McCarthy and (independently) Hain proved that H1(Modg[L], Z) is finite for g≥ 3; see, e.g. Proposition 5.2 of [ Ha2]7. As discussed in §5 of [ Ha2], the following conjecture would imply that the (orbifold) Picard group for the moduli spaces of level L structures has rank one; this group is finitely-generated by the Hain and McCarthy result just mentioned. 7Actually, Hain proves a much stronger result, computing H 1(Modg[L], V ) for V any finite-dimensional sym- plectic representation. 2. Some problems on mapping class groups and moduli space 45

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09804 | AMR-109-0069

### Problem 5.25 - (Presentation for level L structures).

O / open | AMR 问题清单汇编

(Presentation for level L structures). Give an explicit finite presentation for Modg[L]. Once one has such a presentation, it seems likely that it would fit well into the framework of Pitsch’s proof [ Pi] that rank( H2(Modg,1, Z))≤ 1 for g≥ 4. Note that Pitsch’s proof was extended to punctured and bordered case by Korkmaz-Stipsicsz; see [ Ko]. What Pitsch does is to begin with an explicit, finite presentation of Mod g,1, and then to apply Hopf’s formula for groups Γ presented as the quotient of a free group F by the normal closure R of the relators: H2(Γ, Z) = R∩ [F,F ] [F,R ] (17) In other words, elements of H2(Γ, Z) come precisely from commutators which are relators, except for the trivial ones. Amazingly, one needs only write the form of an arbitrary element of the numerator in (17), and a few tricks reduces the computation of (an upper bound for) space of solutions to computing the rank of an integer matrix. In our case this approach seems feasible, especially with computer computation, at least for small L. Of course one hopes to find a general pattern.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09805 | AMR-109-0051

### Problem 5.3 - (Interpolations).

P / partially_solved | AMR 问题清单汇编

(Interpolations). Letg≥ 3. For each subgroup L< ∧3H/H, determine whether or not π-1(L) is finitely generated. As for subgroups deeper down than Kg =Ig(2) in the Johnson filtration {Ig(k)}, we would like to record the following.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09806 | AMR-103-0100

### Problem 5.4 - Compute H Q 2 (X) for each connected quandle X.

P / partially_solved | AMR 问题清单汇编

Compute H Q 2 (X) for each connected quandle X. More gener- ally, find a convenient methodology to compute quandle (co)h omology groups. See Table 5 for some quandle homology groups given in [264]; s ee also [284] for computations of quandle cohomology groups of many Alexa nder quandles. There are maple programs [185] for computing quandle cohomo logy groups.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09807 | AMR-109-0239

### Problem 5.4 - Suppose that G <MCG (S) is a finite co-area Veech subgroup.

S / solved | AMR 问题清单汇编

Suppose that G <MCG (S) is a finite co-area Veech subgroup. What can one say about ΓC? In particular, does it contain ΓG with finite index?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09808 | AMR-103-0101

### Problem 5.5 - (J.S.

O / open | AMR 问题清单汇编

(J.S. Carter) Compute H Q i (Sm n ) of Sm n which denotes the quandle of the nth symmetric group with the binary operation given by x∗ y = y-mxym. 5.4 Quandle cocycle invariant The quandle cocycle invariant, introduced in [79, 80], is de fined as follows. For α∈ H 2(X; A) we choose a 2-cocycle φ representing α. Any representation of a knot quandle Q(K) to X is presented by a coloring of a knot diagram of K, where a coloring of an oriented knot diagram is a map of the set of over-arcs of it to X satisfying the condition depicted in the pictures of (35) at each crossing of the knot diagram. We define the weight of a crossing of a colo red diagram by W ( x x∗y y ) = φ(x, y)∈ A, W ( y x x∗y) = φ(x, y)-1∈ A, (35) where we write A multiplicatively here. The quandle cocycle invariant of a knot K is defined by Φ α(K) = ∑ C ∏ τ W (τ,C)∈ Z[A], where the sum runs over all coloring C of a diagram of K, and the product runs over all crossing τ of the diagram, and Z[A] denotes the group ring of A. Φ α(K) only depends on K and α.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09809 | AMR-109-0240

### Problem 5.5 - Explore ΓC for other free subgroups G< MCG (S), for example free subgroups generated by high powers of Dehn twists ab…

O / open | AMR 问题清单汇编

Explore ΓC for other free subgroups G< MCG (S), for example free subgroups generated by high powers of Dehn twists about a pair of filling curves. IfG is not free then [ Mos03a] contains a result more general than Theorem 5.3, with QI(Γ G) replaced by a group QI f (ΓG) constructed from quasi-isometries Γ G→ ΓG which coarsely preserve the family of cosets of π1(S) < ΓG. The point of G being free is that in this case one can use methods of coarse algebraic topology to prove that QI f (ΓG) = QI(Γ G); see e.g. [ FM00]. In the case that G =MCG (S) covered in Theorem 5.2, Whyte's homological methods are used to prove that QI f (ΓG) = QI(Γ G). One might be able to use Whyte's homological methods for other subgroups G <MCG (S), particularly when G is a Bieri-Eckmann duality group which is not a Poincar´ e duality group, such as G =MCG (S) [ Har86]. Unfortunately, none of these methods will work when G is a Leininger-Reid group, because in this case, G is a 2-dimensional Poincar´ e duality group, which rules out the methods of [ FM00] 16. Problems in the geometry of surface group extensions 269 as well as Whyte's homological methods. We do not have any idea of how to overcome this obstacle. Nonetheless it should still be interesting to calculate Γ C = QI f (ΓG) for a Leininger-Reid subgroup G.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09810 | AMR-103-0102

### Problem 5.6 - Compute the quandle cocycle invariant Φ α(K) of each knot K for a second cohomology class α of a connected quandle.

P / partially_solved | AMR 问题清单汇编

Compute the quandle cocycle invariant Φ α(K) of each knot K for a second cohomology class α of a connected quandle.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09811 | AMR-109-0053

### Problem 5.6 - Find H1(Ig(k), Z) for all k≥ 2.

P / partially_solved | AMR 问题清单汇编

Find H1(Ig(k), Z) for all k≥ 2. Generating sets for Ig. One difficulty in working with Ig is the complexity of its generating sets: any such set must have at least 1 3 [4g3-g] elements since Ig has abelian quotients of this rank (see [ Jo5], Corollary after Theorem 5). Compare this with Mod g, which can always be generated by 2 g + 1 Dehn twists (Humphries), or even by 2 elements (Wajnryb [ W a])! How does one keep track, for example, of the (at least) 1330 generators for I 10? How does one even give a usable naming scheme for working with these? Even worse, in Johnson's proof of finite generation of Ig (see [ Jo2]), the given generating set has O(2g) elements. The following therefore seems fundamental; at the very least it seems that solving it will require us to understand the combinatorial topology underlying Ig in a deeper way than we now understand it.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09812 | AMR-109-0054

### Problem 5.7 - (Cubic genset problem).

S / solved | AMR 问题清单汇编

(Cubic genset problem). Find a generating set for Ig withO(gd) many elements for some d≥ 3. Optimally one would like d = 3. In fact in §5 of [ Jo2], Johnson explicitly poses a much harder problem: for g≥ 4 can Ig be generated by 1 3 [4g3-g] elements? As noted above, this would be a sharp result. Johnson actually obtains this sharp result in genus three, by finding ([ Jo2], Theorem 3) an explicit set of 35 generators for Ig. His method of converting his O(2g) generators to O(g3) becomes far too unwieldy when g >3. One approach to Problem 5.7 is to follow the original plan of [ Jo2], but using a simpler generating set for Mod g. This was indeed the motivation for Brendle and me when we found in [ BF a1] a generating set for Mod g consisting of 6 involutions, i.e. 6 elements of order 2. This bound was later improved by Kassabov [ Ka] to 4 elements of order 2, at least when g≥ 7. Clearly Modg is never generated by 2 elements of order two, for then it would be a quotient of the infinite dihedral group, and so would be virtually abelian. Since the current known bounds are so close to being sharp, it is natural to ask for the sharpest bounds.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09813 | AMR-103-0103

### Problem 5.7 - Find relations between quandle cocycle invariants and knot in- variants known so far, such as quantum invariants.

P / partially_solved | AMR 问题清单汇编

Find relations between quandle cocycle invariants and knot in- variants known so far, such as quantum invariants.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09814 | AMR-109-0055

### Problem 5.8 - (Sharp bounds for involution generating sets).

P / partially_solved | AMR 问题清单汇编

(Sharp bounds for involution generating sets). For each g≥ 2, prove sharp bounds for the minimal number of involutions required to generate Modg. In particular, for g≥ 7 determine whether or not Modg is generated by 3 involutions.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09815 | AMR-103-0104

### Problem 5.8 - Compute H 2 Q(X; A) for each X -module A.

O / open | AMR 问题清单汇编

Compute H 2 Q(X; A) for each X -module A.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09816 | AMR-109-0056

### Problem 5.9 - (Cohomological Dimension).

P / partially_solved | AMR 问题清单汇编

(Cohomological Dimension). Compute the cohomological dimension of Ig and ofKg. More generally, compute the cohomological dimension of Ig(k) for all k≥ 1. Note that the cohomological dimension cd( Ig) is bounded above by the (virtual) cohomological dimension of Mod g, which is 4 g- 5. The following is a start on some lower bounds.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09817 | AMR-103-0105

### Problem 5.9 - Let the notation be as above.

P / partially_solved | AMR 问题清单汇编

Let the notation be as above. Then, extending the definition o f the quandle cocycle invariant, define a knot invariant assoc iated with α, which is, roughly speaking, an invariant obtained by counting rep resentations of a knot quandle Q(K) to X with information whether each representation can lift to a representation Q(K)→ Y. 5.5 Quantum quandles

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09818 | AMR-109-0204

### Problem 5: - Develop a theory of geometrically finite subgroups of Mg,m which include the groups defined by Veech surfaces.

O / open | AMR 问题清单汇编

Develop a theory of geometrically finite subgroups of Mg,m which include the groups defined by Veech surfaces. 14. Geometric properties of the mapping class group 241

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09819 | AMR-107-0011

### Problem 5A - Is it true that any hypersurface from the space $P(d;N)$ can be connected with a trivial one by a gene…

P / partially_solved | AMR 问题清单汇编

Is it true that any hypersurface from the space $P(d;N)$ can be connected with a trivial one by a generic path in this space in such a way that it experiences only Morse surgeries, which decrease this complexity measure?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09820 | AMR-107-0012

### Problem 5B - A version of the previous problem, in which the complexity measure is not purely topological: namely,…

O / open | AMR 问题清单汇编

A version of the previous problem, in which the complexity measure is not purely topological: namely, it is the lowest number of critical points of Morse functions defined by restrictions of linear functions ${\mathbb{R}}^{N}\to{\mathbb{R}}$ to our varieties. $($Correspondingly, the surgeries of the variety affecting this measure should be not only those of topological nature but also include bifurcations of the dual variety$)$.

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09821 | AMR-107-0013

### Problem 5C - Give an upper bound for the function $T\mapsto F$.

O / open | AMR 问题清单汇编

Give an upper bound for the function $T\mapsto F$.

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09822 | AMR-107-0014

### Problem 5D - Do non-singular real plane projective curves of an odd degree consisting of a single connected compone…

P / partially_solved | AMR 问题清单汇编

Do non-singular real plane projective curves of an odd degree consisting of a single connected component form a connected set (i.e., are they rigid isotopic)?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09823 | AMR-109-0213

### Problem 6 - (Purely cyclic).

O / open | AMR 问题清单汇编

(Purely cyclic). Is there a Veech group that is cyclic and generated by a single hy- perbolic element? Equivalently, is there a pseudo-Anosov map such that its associated Teichm¨ uller disk, is invariant only under powers of the pseudo-Anosov?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09824 | AMR-109-0136

### Problem 6 - Calculate the topological type (PL-homeomorphism, homotopy, homology...

O / open | AMR 问题清单汇编

Calculate the topological type (PL-homeomorphism, homotopy, homology... type) of the Arc-complexes. The first non-trivial case is the calculation of the topological type of the PL-manifolds Arc(M ) for the four type 1 surfaces M. Arc complexes as stratified spaces conjecturally have specific singularities and topology described recursively as follows. A PL sphere is a type zero space. A closed, connected, and simply connected manifold is a type one space provided it occurs among a list of four specific such (non-spherical) manifolds of respective dimensions 5,7,7, and 9, namely, the arc complexes of the four type one surfaces. For n> 1, define a type n space to be a finite polyhedron, defined up to PL-isomorphism, so that the link of each vertex in any compatible triangulation is PL isomorphic to an iterated suspension of the join of at most two spaces of type less than n. By Theorem 5, many links of simplices are indeed of this type, and we conjecture that any arc complex is of some finite type. (As explained in [46], a specific collapsing argument in the “calculus of mapping cylinders” would give a proof a of this conjecture.) 112 R. Penner The non-Hausdorff space PF (F )/PMC (F ) thus contains the stratified space Arc(F ) as an open dense subset, explaining one classical (i.e., non-quantum) aspect to Problem 1. In light of the stratification of Arc-complexes in general, one might hope to apply techniques such as [1,34] to address parts of Problem 6.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09825 | AMR-109-0184

### Problem 6 - Find special features of the monodromy of algebraic surfaces.

O / open | AMR 问题清单汇编

Find special features of the monodromy of algebraic surfaces. There is some good motivation for this coming from at least three directions • The problem includes (in principle) the well-known problem of describing possible fun- damental groups of algebraic surfaces. • One famous constraint is the “hard Lefschetz” property, which has a well-known trans- lation into the action of the monodromy on homology. • From the Seiberg-Witten theory we know that there are strong restrictions on the basic classes of algebraic surfaces, and these can be translated into the TLP point of view along the lines of [ 6], [ 13].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09826 | AMR-103-0109

### Problem 6.1 - ([188, Problem 3]) Is the representation of the braid group inside the Temperley-Lieb algebra faithful?

P / partially_solved | AMR 问题清单汇编

([188, Problem 3]) Is the representation of the braid group inside the Temperley-Lieb algebra faithful?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09827 | AMR-109-0070

### Problem 6.1 - (Actions on buildings).

S / solved | AMR 问题清单汇编

(Actions on buildings). Determine all isometric actions ψ: Mod g→ Isom(Xn), whereXn is an n-dimensional Euclidean building, and n is sufficiently small compared to g. For example, one would like conditions under which ψ has a global fixed point, that is, a point x∈ Xn such that ψ(Modg)·x = x. One method to attack this problem is the so-called “Helly technique” introduced in [ F a4]. Using standard CAT(0) methods, one can show that each Dehn twist Tα in Mod g has a nontrivial fixed set Fα under the ψ-action; Fα is necessarily convex. Considering the nerve of the collection {Fα} gives a map Cg→ Xn from the complex of curves to Xn. Now Cg has the homotopy type of a wedge of spheres (see, e.g., [ Iv1]), while Xn is contractible. Hence the spheres in the nerve must be filled in, which gives that many more elementsψ(Tα) have common fixed points. The problem now is to understand in an explicit way the spheres inside Cg.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09828 | AMR-109-0241

### Problem 6.1 - Are the Leininger-Reid subgroups geometrically finite, with cusp groups the reducible cyclic subgroups?

O / open | AMR 问题清单汇编

Are the Leininger-Reid subgroups geometrically finite, with cusp groups the reducible cyclic subgroups?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09829 | AMR-109-0242

### Problem 6.2 - If G< MCG (S) is geometrically finite with cusp groups H1,...,H n, what can be said about the geometric properties of...

P / partially_solved | AMR 问题清单汇编

If G< MCG (S) is geometrically finite with cusp groups H1,...,H n, what can be said about the geometric properties of the group ΓG? Does it have useful large scale geometric properties? Does it satisfy some form of quasi-isometric rigidity? Note that Γ G is not hyperbolic relative to its subgroups Γ H1,..., ΓHn, because each of these subgroups and their conjugates contains the kernel π1(S), whereas relative hyperbolicity implies that the intersection of any two of these conjugates is finite.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09830 | AMR-103-0111

### Problem 6.3 - (S.J.

O / open | AMR 问题清单汇编

(S.J. Bigelow) Is the action of B6 on V 6 2 faithful?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09831 | AMR-103-0112

### Problem 6.4 - (S.J.

O / open | AMR 问题清单汇编

(S.J. Bigelow) Generalise Lawrence’s construction to obtain the irreducible representations of the BMW algebra.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09832 | AMR-103-0113

### Problem 6.5 - (S.J.

O / open | AMR 问题清单汇编

(S.J. Bigelow) Find a larger family of irreducible representations of Bn which includes those coming from the BMW algebra.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09833 | AMR-109-0074

### Problem 6.5 - (The sections problem).

O / open | AMR 问题清单汇编

(The sections problem). Determine those subgroups H ≤ Modg for which π has a section over H. Do this as well with Homeo+(Σg) replaced by various subgroups, such as Diffr(S) with r = 1, 2,..., ∞,ω; similarly for the group of area-preserving diffeomorphisms, quasiconformal homeomorphisms, etc.. Answers to Problem 6.5 are known in a number of cases. (1) When H is free then sections clearly always exist over H. (2) Sections to π exist over free abelian H, even when restricted to Diff ∞(Σg). This is not hard to prove, given the classification by Birman-Lubotzky-McCarthy [ BLM] of abelian subgroups of Mod g. (3) Sections exist over any finite group H <Modg, even when restricted to Diff ω(Σg). This follows from the Nielsen Realization Conjecture, proved by Kerckhoff [ Ke], which states that any such H acts as a group of automorphisms of some genus g Riemann surface. 2. Some problems on mapping class groups and moduli space 49 (4) In contrast, Morita showed (see, e.g., [ Mo5]) that π does not have a section with image in Diff 2(Σ2) over all of Mod g when g≥ 5. The C2 assumption is used in a crucial way since Morita uses a putative section to build a codimension 2 foliation on the universal curve over Mg, to whose normal bundle he applies the Bott vanishing theorem, contradicting nonvanishing of a certain (nontrivial!) Miller-Morita-Mumford class. It seems like Morita’s proof can be extended to finite index subgroups of Mod g. (5) Markovic [ Mar] has recently proven that H = Mod g does not even have a section into Homeo(Σg), answering a well-known question of Thurston. As is usual when one studies representations of a discrete group Γ, one really desires a theorem about all finite index subgroups of Γ. One reason for this is that the existence of torsion and special relations in a group Γ often highly constrains its possible representations. Markovic’s proof in [Mar] uses both torsion and the braid relations in what seems to be an essential way; these both disappear in most finite index subgroups of Mod g. Thus it seems that a new idea is needed to answer the following.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09834 | AMR-103-0114

### Problem 6.6 - Classify all irreducible representations of Bn.

O / open | AMR 问题清单汇编

Classify all irreducible representations of Bn.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09835 | AMR-103-0115

### Problem 6.7 - (S.J.

O / open | AMR 问题清单汇编

(S.J. Bigelow) Is there a faithful representation of Bn into a group of matrices over ¯Q?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09836 | AMR-109-0205

### Problem 6: - For a closed surface of genus g≥ 3, is the Torelli subgroup of Mg,m undistorted?

S / solved | AMR 问题清单汇编

For a closed surface of genus g≥ 3, is the Torelli subgroup of Mg,m undistorted? More generally, find a distorted finitely generated subgroup of Mg,m. The splitting sequences on the complex of train tracks can be used to investigate the large-scale geometric behavior of the mapping class group. Note that such a splitting sequence ( τi) is determined by an initial train track and for each i by a choice of a splitting move among a uniformly bounded number of possibilities which transforms the train track τi to the train track τi+1. In other words, it is possible to treat splitting sequences and hence uniform quasi-geodesics inMg,m in an algorithmic way. 14. Geometric properties of the mapping class group 243 Algorithmic calculations in a finitely generated group Γ are very intimately related to two basis decision problems which go back to Dehn and can be formulated as follows (see Chapter III.Γ.1 in [BH]). Word problem: A word w in a fixed system of generators for Γ is given. One is required to find a method to decide in a finite number of steps whether or not this word represents the identity in Γ. Conjugacy problem: Two elements g,h∈ Γ are given. A method is sought to decide in a finite number of steps whether or not the elements g,h are conjugate, i.e. whether there is some u∈ Γ such that h =ugu-1. In the last decade of the twentith century, Epstein, Cannon, Holt, Levy, Paterson, Thurston [E] formulated a property for finitely generated groups which ensures that these problems can be solved in controlled time. Namely, a biautomatic structure for a finitely generated group Γ consists of a finite alphabet A, a (not necessarily injective) map π:A→ Γ and a regular language L over the alphabet A with the following properties. The set π(A) generates Γ, and there is an inversion ι: A→ A (i.e. ι2 = Id) with π(ιa) = π(a)-1 for all a∈ A. In particular, π(A) is a symmetric set of generators for Γ. Via concatenation, every word w in the alphabet A is mapped by π to a word in the generators π(A) of Γ and hence it defines an element π(w)∈ Γ. We require that the restriction of the map π to the set of all words from the language L maps L onto Γ. For all x,y∈A and each word w∈L of length k≥ 0, the word xwy defines via the projection π a path s: [0,k + 2] → Γ. By assumption, there is a word w′∈L of length ℓ >0 with π(w′) = π(xwy). Let s′: [0,ℓ ]→ Γ be the corresponding path in Γ; we require that the distance in Γ between s(i) and s′(i) is bounded by a universal constant which neither depends on i nor on the choice of x,y,w,w ′. Extending earlier work of Mosher [M1], in [H1] the complex of train tracks and its algorithmic properties are used to show.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09837 | AMR-107-0015

### Problem 6A - Is it true that any real Morsification of $f$ can be connected with one of complexity $\rho(f)$ by a g…

O / open | AMR 问题清单汇编

Is it true that any real Morsification of $f$ can be connected with one of complexity $\rho(f)$ by a generic path in the base of a versal deformation, in such a way that all Morse surgeries $[A_{2}]$ in this path only decrease the number of real critical points?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09838 | AMR-107-0016

### Problem 6B - What can be said about the number $\rho(f)$?

O / open | AMR 问题清单汇编

What can be said about the number $\rho(f)$?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09839 | AMR-107-0017

### Problem 6C - Is it true that any component of the complement of the discriminant variety of a versal deformation co…

P / partially_solved | AMR 问题清单汇编

Is it true that any component of the complement of the discriminant variety of a versal deformation contains a Morsification, whose all $\mu(f)$ critical points are real?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09840 | AMR-109-0214

### Problem 7 - (Algorithm for Veech groups).

P / partially_solved | AMR 问题清单汇编

(Algorithm for Veech groups). Is there an algorithm for determining the Veech group of a general translation surface or quadratic differential? An interesting class of Veech surfaces are the square-tiled surfaces. These surfaces can be represented as a union of glued squares all of the same size, see [ Zo3], [ HuLe1].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09841 | AMR-109-0137

### Problem 7 - Devise a matrix model (cf.

O / open | AMR 问题清单汇编

Devise a matrix model (cf. [28]) for the calculation of the Euler characteristics of Arc-complexes.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09842 | AMR-107-0018

### Problem 7 - Give a similar universal estimate of the radius of convergence for multidimensional Newton’s method of…

S / solved | AMR 问题清单汇编

Give a similar universal estimate of the radius of convergence for multidimensional Newton’s method of [Shub and Smale1993].

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/15-11/

09843 | AMR-103-0116

### Problem 7.1 - (see [220, Problem 3.108]) Does there exist a closed 3-manifold M, other than S3, such that τ SO(3) r (M ) = τ SO(3)…

O / open | AMR 问题清单汇编

(see [220, Problem 3.108]) Does there exist a closed 3-manifold M, other than S3, such that τ SO(3) r (M ) = τ SO(3) r (S3) for all odd r≥ 3?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09844 | AMR-103-0128

### Problem 7.13 - (H.

P / partially_solved | AMR 问题清单汇编

(H. Murakami) Calculate o- lim log τ SU (2) N (M ) N for Seifert fibered 3-manifolds M.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09845 | AMR-103-0129

### Problem 7.14 - (D.

P / partially_solved | AMR 问题清单汇编

(D. Thurston) Find a series of invariants of a 3-manifold (depending on roots of unity) that grows as its hyperbolic volume (or its simplicial volume).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09846 | AMR-103-0130

### Problem 7.15 - (D.

O / open | AMR 问题清单汇编

(D. Thurston) Find a correct generalization of the volume conjecture to other non-compact Lie groups.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09847 | AMR-103-0131

### Problem 7.16 - (S.

P / partially_solved | AMR 问题清单汇编

(S. Morita [228]) Define the Chern-Simons invariant CS(M ) as a topological invariant of any closed oriented 3-manifol d M, and of any knot (link) complement M in a closed 3-manifold. This problem includes two problems: to define CS( M ) (topologically or com- binatorially) as a topological invariant, and to define it fo r non-hyperbolic 3- manifolds.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09848 | AMR-103-0132

### Problem 7.17 - (T.

O / open | AMR 问题清单汇编

(T. Ohtsuki) Give a “complex structure” to the set of 3-manifolds. More precisely, find an embedding (or, an immersi on) of the set of 3-manifolds to some complex variety such that its restric tion to the set {NK;(p,q)| p2 + q2 > >0} can be extended to a holomorphic map of the above mentioned complex parameter for any (hyperbolic) knot K in any 3-manifold N. We would expect some structures of the set of 3-manifolds suc h as mentioned in Problems 7.17 and Problem 10.16. Such structures would yiel d new viewpoints in the study of (the set of, and invariants of) 3-manifolds.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09849 | AMR-103-0133

### Problem 7.18 - (S.

P / partially_solved | AMR 问题清单汇编

(S. Baseilhac, R. Benedetti) Generalize the construction of the QHI for flat principal G-bundles, for Lie groups G different from B. Section 7.4 was written by S. Baseilhac and R. Benedetti.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09850 | AMR-103-0134

### Problem 7.19 - (S.

O / open | AMR 问题清单汇编

(S. Baseilhac, R. Benedetti) Fix (W, L) and vary ρ. Study KN as a function of the bundle, that is as a function defined on the character variety of W with respect to B: regularity, fibers, and so on.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09851 | AMR-109-0319

### Problem 7.2 - Find explicit graphs Γ∈G odd such that the corresponding homology classes Φ(Γ) are non-trivial.

O / open | AMR 问题清单汇编

Find explicit graphs Γ∈G odd such that the corresponding homology classes Φ(Γ) are non-trivial.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09852 | AMR-103-0135

### Problem 7.20 - (S.

O / open | AMR 问题清单汇编

(S. Baseilhac, R. Benedetti) Specialize Problem 7.19 to bundles coming from the ordinary cohomology as above. For real a dditive ones, analyze the behaviour of the QHI with respect to Thurston’s n orm. Are they constant on the faces of the corresponding unit sphere?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09853 | AMR-103-0136

### Problem 7.21 - (S.

O / open | AMR 问题清单汇编

(S. Baseilhac, R. Benedetti) Understand the ‘phase factor’ (i.e. the ambiguity due to N -th roots of unity) of the state sum HN (T ). Possi- bly derive from it an invariant for (W, L, ρ) endowed with some extra-structure, thus refining KN (W, L, ρ).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09854 | AMR-103-0137

### Problem 7.22 - (S.

P / partially_solved | AMR 问题清单汇编

(S. Baseilhac, R. Benedetti) Determine a suitable (2 + 1) ‘decorated’ cobordism theory supporting a (non purely topo logical) QFT con- taining the already defined QHI. Study in particular the beha viour of the QHI with respect to connected sums.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09855 | AMR-103-0138

### Problem 7.23 - (S.

O / open | AMR 问题清单汇编

(S. Baseilhac, R. Benedetti) Develop a 4-dimensional theory of QHI based on Turaev’s shadow theory.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09856 | AMR-103-0139

### Problem 7.24 - (S.

S / solved | AMR 问题清单汇编

(S. Baseilhac, R. Benedetti) Determine the actual relation- ship between KN (S3,·) and the coloured Jones polynomial JN (·) (evaluated at ω = exp(2iπ/N) and normalized by JN (unknot) = 1 ), as functions of links.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09857 | AMR-103-0141

### Problem 7.26 - For each rational homology 3-sphere M, calculate τ SO(3)(M ) and τ P SU (N )(M ) for all degrees.

O / open | AMR 问题清单汇编

For each rational homology 3-sphere M, calculate τ SO(3)(M ) and τ P SU (N )(M ) for all degrees.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09858 | AMR-103-0142

### Problem 7.27 - (J.

P / partially_solved | AMR 问题清单汇编

(J. Roberts) Explain the appearance of modular forms in the Witten invariants.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09859 | AMR-103-0143

### Problem 7.28 - Characterize those elements of Z[[q-1]] of the form τ SO(3)(M ) of integral homology 3-spheres M.

O / open | AMR 问题清单汇编

Characterize those elements of Z[[q-1]] of the form τ SO(3)(M ) of integral homology 3-spheres M.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09860 | AMR-103-0118

### Problem 7.3 - (S.K.

P / partially_solved | AMR 问题清单汇编

(S.K. Hansen, T. Takata) Do the family of quantum G invari- ants τ G r or the family of quantum P G invariants τ P G r, G running through all simply connected compact simple Lie groups and r running through all allowed levels, separate rational homology 3-spheres? How well do t hese families of invariants separate closed oriented 3-manifolds?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09861 | AMR-109-0079

### Problem 7.3 - Compute L(Modg) explicitly for small g≥ 2.

P / partially_solved | AMR 问题清单汇编

Compute L(Modg) explicitly for small g≥ 2. In principle L(Modg) can be computed for any given g. The point is that one can first bound the degree of L(Modg), then give bounds on the smallest possible value λ(α), where α ranges over all algebraic integers of a fixed range of degrees, and λ(α) denotes the largest root of the minimal polynomial of α. One then finds all train tracks on Σ g, and starts to list out all pseudo-Anosovs. It is possible to give bounds for when the dilatations of these become large. Now one tries to match up the two lists just created, to find the minimal dilatation pseudo-Anosov on Σ g. Of course actually following out this procedure, even for small g, seems to be impracticable.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09862 | AMR-103-0146

### Problem 7.31 - Characterize those elements of Habiro’s expansion (45) of τ SO(3)(M ) of integral homology 3-spheres M.

O / open | AMR 问题清单汇编

Characterize those elements of Habiro’s expansion (45) of τ SO(3)(M ) of integral homology 3-spheres M.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09863 | AMR-103-0119

### Problem 7.4 - Find a 3-dimensional topological interpretation of quantu m in- variants of 3-manifolds.

O / open | AMR 问题清单汇编

Find a 3-dimensional topological interpretation of quantu m in- variants of 3-manifolds.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09864 | AMR-109-0081

### Problem 7.5 - (Shortest Teichm¨ uller loop in a stratum).

O / open | AMR 问题清单汇编

(Shortest Teichm¨ uller loop in a stratum). For each fixed g≥ 2, and for each r-tuple as above, give upper and lower bounds for λg(k1,...,k r):= inf {logλ(f ): f∈ Modg whose stable foliation has data (k1,...,k r)} This problem is asking for bounds for the shortest Teichm¨ uller loop lying in a given substratum in moduli space (i.e. the projection in Mg of the corresponding substratum in the cotangent bundle). L(Modg) = min {λg(k1,...,k r)} where the min is taken of all possible singularity data.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09865 | AMR-103-0123

### Problem 7.8 - (J.E.

O / open | AMR 问题清单汇编

(J.E. Andersen) If such an expansion exists, understand how it is related to the expansion of Ohtsuki and the expansion of Habiro. It will of course be important to establish, that an expansio n of this type ex- ists, however, of far greater importance will be to give inde pendent topological meaning to the many resulting new invariants, e.g. to prove t hat the phases are the Chern-Simons values cj. From the discussion above on the semi-classical approximation we derive the following conjecture:

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09866 | AMR-109-0084

### Problem 7.8 - Give an algorithm which tells whether or not any given pseudo-Anosov is represented by a simple closed Teichm¨ uller…

O / open | AMR 问题清单汇编

Give an algorithm which tells whether or not any given pseudo-Anosov is represented by a simple closed Teichm¨ uller geodesic, and also whether or not this geodesic lies on a Veech curve. Note that the analogue of Question 7.7 is not known for hyperbolic surfaces, although it is true for a generic set of surfaces in Mg. 52 B. Farb

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09867 | AMR-109-0206

### Problem 7: - Is the mapping class group linear?

P / partially_solved | AMR 问题清单汇编

Is the mapping class group linear? A locally compact group Γ is said to satisfy the Haagerup approximation property or is a-T- menable if there exists a continuous, isometric action α of Γ on some affine Hilbert space H which is metrically proper. This means that for all bounded subsets B ofH, the set {g∈ Γ|α(g)B∩B=∅} is relatively compact in Γ. There are other equivalent characterizations of this property (see [CCJJV]) which can be viewed as a strong negation of the (perhaps more widely know) property (T) of Kazhdan. The class of a-T-menable groups contains for example all amenable groups, Coxeter groups and the isometry groups of real and complex hyperbolic spaces. It is also known that for a-T-menable groups the Baum-Connes conjecture holds (see [CCJJV]).

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09868 | AMR-109-0215

### Problem 8 - (Orbits of square-tiled surfaces).

O / open | AMR 问题清单汇编

(Orbits of square-tiled surfaces). Classify the SL(2, R) orbits of square-tiled sur- faces in any stratum. Describe their Teichm¨ uller discs. A particular case of this problem is that of the stratum H(1, 1). This above problem is solved only for the stratum H(2), see [ HuLe1], [ McM4].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09869 | AMR-109-0138

### Problem 8 - [Contributed by the referee] Does Theorem 5 say anything about the structure of the end of Riemann’s moduli space?

O / open | AMR 问题清单汇编

[Contributed by the referee] Does Theorem 5 say anything about the structure of the end of Riemann’s moduli space? For instance, what is the homology near the end? Take the one-point compactification F × of F = Fs g,r, where all of the s≥ 0 punctures of F are identified to a single point in F ×, so F × =F if and only if s≤ 1. Let ∂ denote the boundary mapping of the chain complex {Cp(Arc): p≥ 0} of Arc = Arc(F ). Suppose that α is an arc family in F with corresponding cell σ[α]∈Cp(Arc). A codimension-one face of σ[α] of course corresponds to removing one arc from α, and there is a dichotomy on such faces σ[β] depending upon whether the rank of the first homology of F ×-∪β agrees with or differs by one from that of F ×-∪α. This dichotomy decomposes ∂ into the sum of two operators ∂ =∂1 +∂2, where ∂2 corresponds to the latter case. The operators ∂1,∂ 2 are a pair of anti-commuting differentials, so there is a spectral sequence converging to H∗(Arc) corresponding to the bi-grading E0 u,v ={chains on σ[α]∈Cp(Arc): v =-rank(H1(Fα)) and u =p-v}, where ∂1:E0 u,v→E0 u-1,v and the differential of the E0 term is ∂2:E0 u,v→E0 u,v-1. It is not quite fair to call it a problem, nor a theorem since the argument is complicated and has not been independently checked, but we believe that this spectral sequence collapses in its E1-term to its top horizontal row except in dimension zero. Thus, the homology of Arc is the ∂1-homology of the ∂2-kernels in the top row, and on the other hand, it follows from [31] that the∂1-homology of the top row itself agrees with that of uncompactified Riemann’s moduli space M(F ) As discussed in [30,46], the stratified structure of the arc complexes for bordered surfaces gives a corresponding stratified structure to A(F ) for punctured F. This may be enough to re-visit the calculations of [19] and [16,22] with an eye towards avoiding technical difficulties with the Deligne-Mumford compactification ¯M(F ).

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09870 | AMR-109-0320

### Problem 8.1 - Determine the image as well as the cokernel of the homomorphism (15) explicitly.

O / open | AMR 问题清单汇编

Determine the image as well as the cokernel of the homomorphism (15) explic- itly. Note that Hain [ 29] proved that the image of (15), after tensored with Q, is precisely the Lie subalgebra generated by the degree 1 part. However it is unclear which part of hQ g,1 belongs to this Lie subalgebra. In relation to this problem, Oda predicted, in the late 1980’s, that there should arise “arith- metic obstructions” to the surjectivity of Johnson homomorphism. More precisely, based on the theory of Ihara in number theory which treated mainly the case g = 0,n = 3, he expected that the absolute Galois group Gal( Q/Q) should “appear” in hg,1⊗ Zℓ outside of the geometric part and which should be Sp-invariant for any genus g and for any prime ℓ. In 1994, Nakamura [ 86] 22. Cohomological structure of the mapping class group and beyond 367 proved, among other results, that this is in fact the case (see also Matsumoto [ 65]). This was the second obstruction to the surjectivity of Johnson homomorphism, the first one being the traces in [76]. This raised the following problem.

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09871 | AMR-103-0147

### Problem 8.1 - Find (and classify) all TQFT’s.

O / open | AMR 问题清单汇编

Find (and classify) all TQFT’s.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09872 | AMR-103-0156

### Problem 8.10 - For a given TQFT (V, Z), determine whether the image of ˜Mg in End ( V (Σ g) ) is finite.

P / partially_solved | AMR 问题清单汇编

For a given TQFT (V, Z), determine whether the image of ˜Mg in End ( V (Σ g) ) is finite.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09873 | AMR-103-0157

### Problem 8.11 - (G.

O / open | AMR 问题清单汇编

(G. Masbaum) Is there a relation between the Nielsen-Thurston classification of mapping classes of Σ g and their images on V (Σ g) for TQFT’s (V, Z)?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09874 | AMR-103-0158

### Problem 8.12 - (T.

P / partially_solved | AMR 问题清单汇编

(T. Kerler) [Cyclotomic integer TQFT's] (1) Find explicit/computable bases for the Vp(Σ g) as free modules over Z[ζp]. (2) Show that Vp can be extended to all cobordisms as a half-projective TQFT with x = (ζp- 1) p-3 2 ∈ R = Z[ζp]. (3) Determine the structure of the V [j] p (M ) and in how far they have lifts from Fp to Z, analogous to the Ohtsuki invariants for closed 3-manifold s. (4) Find a universal TQFT that combines all Vp, at least perturbatively, into one. In the case of p = 5 the program for items (1)-(3) has been mostly carried out in [213], for primes p≥ 7 not much is known though. Some explicit bases have been found for genus g = 1 by Gilmer, but the situation for higher genera g≥ 3 is unknown. An immediate application of item (2) is that the quantum order, as introduced in [92], is also an upper bound for the cu t-number of a 3-manifold. A closely related statement for (2) would also y ield a very different proof for the fact that the Ohtsuki invariants are of finite ty pe. In item (3) the "lift" must depend on p since the dimensions of the vector spaces do, and must also involve further quotients that arise since the irreduc ible TQFT's over Z do not match the required dimensions either, but they become reducible when reduced to Fp. Item (4) is rather vague at this point, indicating for some s ort of infinite filtered space with finite graded components. Any TQFTV: Cob→ R-mod implies a sequence of representation V[g]: Γ g→ GLR(V(Σ g)) of the mapping class groups. We say that a TQFT is homological if each of these representations factors through the quotie nt Γ g-: Sp(2g, Z) (given by the action on H1(Σ g)), and we say it is strictly homological if each of the Sp(2 g, Z)-representations is algebraic, i.e. either faithful or ze ro. A par- ticular example of strictly homological TQFT's over R = Z are the Lefschetz componentsV (j) of the Frohman-Nicas TQFT, see [128, 214]. From these we can generate a larger family Q 0 of such TQFT's by taking all direct sums of V (j) 's. For example all the TQFT's constructed in [110] lie in Q 0. An even larger family Q∗ is found by taking also tensor products and their irreducibl e summands.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09875 | AMR-103-0159

### Problem 8.13 - (T.

O / open | AMR 问题清单汇编

(T. Kerler) [Homological TQFT's] (1) Find the irreducible components and ring structure (w.r.t ⊕ and⊗) of Q∗. (2) Determine whether all strictly homological TQFT's lie in Q ∗. (3) Identify the homological TQFT's that arise from the gauge th eory of higher rank groups (such as P SU (n) in [129]) with elements in Q ∗. (4) Identify the irreducible factors of the constant orders V [0] p of the cyclo- tomic integer expansion of the Reshetikhin-Turaev theory w ith elements in Q∗. The first item is in some sense about finding the representatio n ring of Sp(2, Z)× Sp(4, Z)×...× Sp(2g, Z)×... equipped with further generators and relations given by the standard handle attachments. The constraints g iven by the latter may be just good enough to ensure that the answer to item (2) is positive. The application of (3) is a better understanding and possibl y a closed form for the polynomials from [129] that express the P SU (n)-invariants in terms of the coefficients of the Alexander polynomial. Evidence seems to s uggest that the TQFT's from (4) stem from p-3 2 -fold symmetric products of elements in Q 0. A plausible corollary would be that for a closed manifold wit h b1(M )≥ 1 we have Vp(M ) = ( ζp- 1) p-3 2 P p-3 2 (λCW L(M )) + O((ζp- 1) p-1 2 ), (46) where λCW L is the Casson-Walker-Lescop invariant, and Pj is a polynomial of degree j with integer coefficients. (Note our normalization Vp(S3) = 1). As remarked in [212] the identity in (46) is true for p = 5 and general M with b1(M )≥ 1. Moreover, work in progress shows that (46) holds also for g eneral p if M is a torus-bundle over a circle. The homological TQFT's are the starting point for a more gene ral, pertur- bative view point on TQFT's that should parallel and extend t hat of the fi- nite type theory of homology-3-spheres. At least for fixed p one can under- stand, for example, the Reshetikhin-Turaev theory as defor mation of the Q ∗ - theories. The notion that is somewhat parallel to that of fini te type for closed 3-manifolds is what we shall call finite length. More precisely, the representa- tionsV[g]: Γ g→ GLR(V(Σ g)) of the mapping class groups extend linearly to homomorphismsV[g]: Z[Γ g]→ EndR(V(Σ g)). Denote by IIg⊂ Z[Γ g] the aug- mentation ideal of the Torelli group. The length ofV is the maximal L∈ N such thatV[g]((IIg)L+1) = 0. Clearly, the L = 0-theories are just the homological ones. The L = 1-theories can be thought of as elements of some Ext( V,W) with V,W∈ Q ∗. Restricted to representations of the Γ g 's they factor (in char= 2) through the Johnson-Morita-homomorphism Γ g→ ⋀ 3 H1(Σ g) ⋊ Sp(2g, Z), for which such extension are explicitly constructible [211].

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09876 | AMR-103-0160

## Problem 8.14 - (T.

O / open | AMR 问题清单汇编

(T. Kerler) [Length = 1 TQFT's] (1) Describe and construct algebraic L = 1 -extensions of Γ g -representations to TQFT's, preferably as "simple" generalizations of the Fr ohman-Nicas- U (1)-theory. (2) Produce a classification of L = 1 -TQFT's in the sense of an extension theory of Q ∗. (3) Identify the Γ g -representations on relative SU (2)-moduli space from [77] with these TQFT's, and find similar, higher rank theories. (4) Identify the V [0] p as L = 1 -theories, if possible. The conceivable generalizations of the TQFT construction o f Frohman and Nicas described in (1) include using different, possibly non -compact gauge groups instead of U (1) and using more refined versions of intersection homolo- gies for stratified moduli spaces. Given the theory for Q ∗ the solution to item (2) will lead to well defined problems in sp-invariant theory. Constructions of L = 1-theories follow the schemes from (1) and (3). The identifi cation in (4) is carried out for p = 5 in [211]. The notion of finite length can be refined into the notion of q/l -solvable intro- duced in [212], indicating a TQFT over R = M[y]/yl+1 such that the constant order TQFT over the ground ring M is of length q. This, clearly, defines a special case of a TQFT of length ≤ (q·l + q + l). Murakami's result [291] can be restated as saying that the Reshetikhin-Turaev theor y gives rise to a 1/1-solvable TQFT V [≤1] p with ground ring Fp (i.e. a TQFT of length 3 over Fp[y]/y2 ) such that V [≤1] p (M ) = 1 + y 1 6 λCW L(M ) (47) for any closed homology sphere M. Following Ohtsuki's work Murakami's iden- tity (with some extra renormalizations by the order of H1(M )) extends also to rational homology spheres. Let us call a theory with this pro perty a TQFT of Casson type. Recall, that the similar relation (46) for λCW L for manifolds with b1(M )≥ 1 is already contained in the information of a homological ( L = 0) TQFT, and is indeed a special evaluation of the Turaev-Milnor Torsion, s ee [212]. Given the richer structure of a 1/1-solvable TQFT we will expect new in variants Ξ that are refinements of λCW L and the torsion invariants. To be more precise, note that for a pair ( M, ϕ), where ϕ: π1(M )→ →Z defines a cyclic cover, any TQFT V yields an invariantV(M, ϕ) = trace(V(CΣ )) where CΣ = M- Σ: Σ → Σ and Σ ⊂ M is any surface dual to ϕ. In this way the Frohman Nicas theoriesV (j) yields the coefficients of the Alexander Polynomial, and, as shown in [212], thus also λCW L. A more refined invariant, which, roughly speaking, generali zes the Alexander module, is the Turaev-Viro module MT V (M, ϕ). It is described by Gilmer in [143]. MT V (M, ϕ) is given, up to conjugacy, by V(Σ) / ker(V(CΣ )N ) (with N large enough) together with the action of V(CΣ ) on it. The traces of V(CΣ ) or its powers are the most obvious well defined numerical inv ariants ofMT V (M, ϕ). The dimension of the module is yet another such invariant. For a 1/1-solvable theory V the invariantV(M, ϕ) takes values in M[y]/y2 and can hence be written as V(M, ϕ) = λV ϕ(M ) + y·Ξ V ϕ(M ), where λV and Ξ V are now M-valued invariants. If y coincides with the half projective parameter λV does not depend on ϕ, and we expect it to be some function of λCW L. Moreover, if V descends from a 1/2-solvable TQFT with the same property also Ξ V would be independent of ϕ. For the modular TQFT over F5[y]/y2 obtained from the Reshetikhin Turaev theory this invariant has already been defined in [212], and w e may expect it to lift, similarly, to an invariant Ξ Z over Z. For p > 5 we expect, as in the case of λCW L, the next order terms in the expansions (46) of the Reshetikh in Turaev theories to be polynomial expressions in λCW L and Ξ Z.

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09877 | AMR-103-0161

### Problem 8.15 - (T.

O / open | AMR 问题清单汇编

(T. Kerler) [ q/l -solvable and Casson TQFT's] (1) Lift the 1/1-solvable TQFT's of Casson type over Fp to a universal 1/1- solvable TQFT's of Casson type over Z. (2) Describe the resulting invariant Ξ Z for 3-manifolds with b1(M )≥ 1. (3) Develop a perturbation theory for general q/l -solvable TQFT's. (4) Relate those with the various, standard resolutions of Γ g. (5) Relate them also to the traditional finite type theory for clo sed 3-manifolds. (6) Describe the Reshetikhin-Turaev theories in this pattern. Preparations for item (1) can be found in [212] in which formu lae for the Casson invariant over Z are derived that have the same form as general TQFT formulae. Item (2) is immediate from the preceding discussion. The rem aining items are logical continuations. The category of 3-dim cobordisms Cob• between compact, oriented surfaces with one boundary component has a natural structure of a brai ded tensor cat- egory. Another, category Alg can be defined entirely algebraically in terms of generators and relations with respect to a tensor product an d a composition product. On the level of objects it has exactly one generator, say A, so that all other objects are of the form A⊗g with 1 = A⊗0. The morphisms are given by all words that can be generated by taking composition and t ensor products of elementary morphisms m: A⊗ A→ A, Δ: A→ A⊗ A, e: 1 → A, ε: A→ 1,..., that appear in the definition of a braided, ribbon Hopf algebra with integrals and a non-degenerate pairing. For example, i n [215] a surjective functorAlg-: Cob• is constructed, which, in the genus one restriction in fact an isomorphism.

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09878 | AMR-103-0162

### Problem 8.16 - (T.

P / partially_solved | AMR 问题清单汇编

(T. Kerler) [3-dim cobordisms from Hopf algebras] (1) Find further relations on Alg, besides the ones arising from the axiomat- ics of Hopf algebras, that would make Alg→ Cob• an isomorphism. (2) Find relations on Alg such that the maps AutAlg(A⊗g)→ Γ g~ = AutCob •(Σ g,1) are isomorphisms. (3) Relate this to obstructions, such as Steinberg and Whitehea d groups, via stratified function spaces. (4) What are the analogous algebraic structures in higher dimen sions. The first problem is easily stated, but presumably very difficu lt as it implies a faithful translation of 3-dimensional topology into an alg ebraic gadget. In this respect it is vaguely parallel to the geometrization and Poi ncar´ e conjectures. The easier problem stated in item (2) can, in theory, be attac ked head-on, given the known presentations of the mapping class groups. T he third point hints to the fact that the generators in Alg correspond to Morse-theoretically elementary cobordisms, and the relations can be interprete d, similarly, in terms of handle slides and cancellation. This is, thus, reminisce nt of the definitions of, e.g. Steinberg groups of 3-manifolds. The problem in item (4 ) is, again, easily stated but even in 4 dimensions lingers in almost complete to tal darkness. It is not hard to understand that higher category theory has to b e invoked and not just one "object" A suffices as a "generator". Any partial answers may open the possibility of constructing functorial 4-manifol d invariants by "linear representation" of such structures. In [216] ETQFT's V are defined as double functors from the double category of relative, 2-framed 1+1+1-dim cobordisms Cob∗ to the double category of linear, abelian categories over a perfect field. (The "E" sta nds for "extended to surfaces with boundaries"). Applied to a single circle, t hought of as a 0- object in Cob∗, it yields an abelian category C V = V (S1), which we call the associated circle category. The main result of [216] is a construction of a V C, for each given modular tensor category C (meaning a bounded, ribbon, braided tensor category with some additional properties) s uch that C V C =C. The construction is made for all semisimple C, and is extended, in the case of non-semisimpleC, to both to the situation of connected surfaces with boundar y as well as disconnected, closed surfaces using the previous ly mentioned notion of half-projective TQFT's.

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09879 | AMR-103-0163

### Problem 8.17 - (T.

O / open | AMR 问题清单汇编

(T. Kerler) [Extended and half-projective TQFT's] (1) Describe in how far an ETQFT V with circle category C can differ from V C, thus introducing a equivalence notion that would establis h a bijective correspondence between the class of ETQFT's and the class of modular tensor categories. (2) Find an extended notion of half-projectivity that includes also surfaces that are both disconnected and have boundary. (3) Find constructions and axioms of ETQFT's that apply to more r elaxed notions of boundedness or modularity. The functorAlg→ Cob • already imposes that a circle category C V must fulfill about all axioms of a modular tensor category, and contain a H opf algebra object with properties. Given some rigidity assumption it a ctually must be the same chosen in the construction of V C. What may still differ is the choice of algebra structures of the same object in the same category, which is thus the main source of possible ambiguities. Already in [216] it is clear that there are several choices. The correct axiomatics for item (2) sho uld follow from a careful analysis of the double composition laws for surgery tangles from [216] and generalization of [210]. Item (3) is relevant to include more general notions of TQFT's as they would be of interest in the theory of finite ty pe invariants. The Reshetikhin-Turaev theory typically starts with non-s emisimple modular categoryC, typically the representation category of a non-semisimpl e quantum groups Uq(g), and then considers a canonical semisimple sub-quotient C, see [208]. Thus VC yields a semisimple TQFT. It is known that this is different from the non-semisimple TQFT VC, which in the case of a quantum group is obtained via the Hennings algorithm. TQFT's can also be generated from a rigid, monoidal category B without any braiding. One way is to take the Drinfel'd double D(B), which is then a modular category for some choice of ribbon element, and use VD(B). For semisimple B one can also extract the 6j-symbol data and follow the Turaev -Viro construction to obtain a TQFT WB.

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09880 | AMR-103-0164

### Problem 8.18 - (T.

P / partially_solved | AMR 问题清单汇编

(T. Kerler) [Non-semisimple vs. semisimple TQFT's, the dou - ble conjecture] (1) Clarify the difference in the content of VC andVC! Are there homological TQFT'sH such that VC is in some essential way equivalent to VC⊗H? (2) Find a construction of WB that generalizes the Turaev-Viro TQFT's to non-semisimpleB 's, similar to the way [216] generalized the Reshetikhin- Turaev construction. In the case of quantum groups and close d 3-mani- folds this should reproduce a version of the Kuperberg invar iant. (3) What is the relation between WB andVD(B)? Are they in some sense isomorphic TQFT's? For the case of Uq(sl2) there is evidence from the genus=1 case that such an H is indeed given by the Frohman-Nicas- U (1)-theory. Item (2) is rather natural as a problem. As is apparent in [236] one may expect technical challenges requiring "minimal" cell decompositions of cobordisms, as opposed to general triangulations, as well as "combings" instead of framings. The last conjecture appears also as Question 5 in [209] which was motivated by works of and discussions with D. Kazhdan and S. Gelfand in 1 994. Since it is a rather nearby conjecture from a formal point of view it may have been posed already earlier. For categories arising from subfact ors and closed mani- folds results answering this conjecture have been obtained in [204]. As outlined in [209] further, more general results in this direction sho uld yield a deeper un- derstanding of both TQFT constructions involved as well as e ntail a topological picture for the Drinfel'd double construction.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09881 | AMR-109-0321

### Problem 8.2 - Describe the Galois images in hg,1⊗ Zℓ.

O / open | AMR 问题清单汇编

Describe the Galois images in hg,1⊗ Zℓ. The above result was proved by analyzing the number theoretical enhancement of the Johnson homomorphism where the geometric mapping class group is replaced by the arithmetic mapping class group which is expressed as an extension 1-→ ˆMn g-→πalg 1 Mn g/Q-→Gal(Q/Q)-→1 and studied by Grothendieck, Deligne, Ihara, Drinfel'd and many number theorists. Nakamura continued to study the structure of the mapping class group from the point of view of number theory extensively (see e.g. [ 88][89]). On the other hand, Hain and Matsumoto recently proved remarkable results concerning this subject (see [ 31][32]). In view of deep theories in number theory, as well as the above explicit results, it seems to be conjectured that there should exist an embedding FreeLieZ(σ3,σ 5,⋯ ) ⊂ hg,1 of certain free graded Lie algebra over Z generated by certain elements σ3,σ 5,⋯, corresponding to the Soul´ e elements, into hg,1 such that the tensor product of it with Zℓ coincides with the image of Gal( Q/Q) for any prime ℓ. We expect that the above conjectured free graded Lie algebra (the motivic Lie algebra) can be realized inside hg,1 (in fact inside the commutator ideal [ hg,1, hg,1]) explicitly in terms of the traces. In some sense, the elements σ2k+1 should be decomposable in higher genera. Here we omit the precise form of the expected formula which will be given in a forthcoming paper. Finally we mention the analogue of the Johnson homomorphisms for the group Aut Fn very briefly. Prior to the work of Johnson, Andreadakis [ 1] introduced and studied the filtration on {AutFn(k)}k which is induced from the action of Aut Fn on the lower central series of Fn. The first group Aut Fn(1) in this filtration is nothing but the group IAut n. It can be checked that an analogous procedure as in the case of the mapping class group gives rise to certain homomorphisms τk: Aut Fn(k)-→Hom(Hn,Ln(k + 1)) and the totality {τk}k of these homomorphims induces an injective homomorphism of graded Lie algebras Gr+(AutFn) = ∞⊕ k=1 AutFn(k)/AutFn(k + 1)-→Der+(Ln). (16) We refer to [ 49][93][98] for some of the recent works related to the above homomorphism. 368 S. Morita

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09882 | AMR-103-0148

### Problem 8.2 - Find (and classify) all modular categories.

O / open | AMR 问题清单汇编

Find (and classify) all modular categories. For a TQFT ( V, Z), put P(V,Z )(t) =∑ ∞ g=0 ( dimV (Σ g) ) tg, where Σ g denotes a closed surface of genus g. The following problem is a refinement of Problem 8.1.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09883 | AMR-103-0149

### Problem 8.3 - (1) Characterize the power series of the form P(V,Z )(t).

O / open | AMR 问题清单汇编

(1) Characterize the power series of the form P(V,Z )(t). (2) For each power series P (t) (satisfying the characterization of (1)), classify all TQFT's (V, Z) such that P(V,Z )(t) = P (t).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09884 | AMR-103-0150

### Problem 8.4 - Find other spin TQFT's.

O / open | AMR 问题清单汇编

Find other spin TQFT's.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09885 | AMR-103-0151

### Problem 8.5 - Formulate and find spin c TQFT's.

O / open | AMR 问题清单汇编

Formulate and find spin c TQFT's.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09886 | AMR-103-0152

### Problem 8.6 - (V.

O / open | AMR 问题清单汇编

(V. Turaev) (1) Extend HQFT's to spin and spin c settings. (2) Find algebra structures behind spin and spin c HQFT's in dimension 1+1.

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09887 | AMR-103-0153

### Problem 8.7 - (V.

O / open | AMR 问题清单汇编

(V. Turaev) Study (spin and spin c ) HQFT's with the target space K(H, 2) in dimensions 1 + 1, 2 + 1, and 3 + 1 for H = ZN.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09888 | AMR-103-0154

### Problem 8.8 - Find a geometric construction of a TQFT using H 0(MΣ,L⊗k).

O / open | AMR 问题清单汇编

Find a geometric construction of a TQFT using H 0(MΣ,L⊗k). Namely, find a geometric way to associate a vector in H 0(MΣ,L⊗k) to a 3- manifold M with ∂M = Σ.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09889 | AMR-103-0155

### Problem 8.9 - (G.

O / open | AMR 问题清单汇编

(G. Masbaum) Study this action of the finite group E(Σ) on H 0(MΣ,L⊗k), and describe the induced decompositions of this vector spa ce according to the characters of E(Σ). Also relate these decompositions to de- compositions of V (Σ) for the TQFT (V, Z) derived from the quantum group Uq(sl2) at a (k + N )-th root of unity.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09890 | AMR-109-0207

### Problem 8: - Is the mapping class group a-T-menable?

O / open | AMR 问题清单汇编

Is the mapping class group a-T-menable? 14. Geometric properties of the mapping class group 245

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09891 | AMR-109-0216

### Problem 9 - (Orbit closures for moduli spaces).

S / solved | AMR 问题清单汇编

(Orbit closures for moduli spaces). Determine the closures of the orbits for the GL+(2, R)-action on H(α) andQ(β). Are these closures always complex-analytic (complex-algebraic?) orbifolds? Characterize the closures geometrically. Note that by a theorem of Kontsevich any GL +(2, R)-invariant complex-analytic subvariety is represented by an affine subspace in period coordinates. Consider the subset H1(α)⊂H (α) of translation surfaces of area one. It is a real codimension one subvariety in H(α) invariant under the action of SL(2, R). In the period coordinates it is defined by a quadratic equation (the Riemann bilinear relation). It is often called a unit 15. Problems on billiards, flat surfaces and translation surfaces 253 hyperboloid. It is worth noting that it is a manifold locally modelled on a paraboloid. The invariant measure on H(α) gives a natural invariant measure on the unit hyperboloid H1(α). Similarly one can define the unit hyperboloid Q1(β). It was proved by Masur and Veech that the total measure of any H1(α),Q1(β) is finite.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09892 | AMR-109-0139

### Problem 9 - [Bounded Distortion Conjecture] Given a hyperbolic structure on F, associate its combinatorial invariant, namely, an…

O / open | AMR 问题清单汇编

[Bounded Distortion Conjecture] Given a hyperbolic structure on F, associate its combinatorial invariant, namely, an ideal cell decomposition of F together with the projective simplicial coordinate assigned to each edge. Take these projective simplicial coordinates as Strebel coordinates on the dual fatgraph to build a conformal structure on F. The underlying map on Teichm¨ uller space is of bounded distortion in the Teichm¨ uller metric. As posed by Ed Witten to the author in the early 1990’s, a compelling problem at that time was to find an orbifold compactification of M(F ) which comes equipped with a cellular description in terms of suitably generalized fatgraphs. Calculations such as [16,19,22] and more might then be performed using matrix models derived from the combinatorics of this putative compactification. Perhaps the desired compactification was the Deligne-Mumford compactification or perhaps an- other one. The combinatorial compactification of the previous section fails to provide an orbifold but rather another stratified generalization of manifold. Guidance from Dennis Sullivan has recently led to the following solution:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09893 | AMR-103-0165

### Problem 9.1 - (1) Find (and classify) all semi-simple monoidal categories (w ith finitely many isomorphism classes of simple objects).

O / open | AMR 问题清单汇编

(1) Find (and classify) all semi-simple monoidal categories (w ith finitely many isomorphism classes of simple objects). (2) Find (and classify) (finite dimensional) fusion rule algebr as and sets of 6j -symbols. (3) Find (and classify) all subfactors (of finite depth).

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09894 | AMR-103-0174

### Problem 9.10 - (N.

P / partially_solved | AMR 问题清单汇编

(N. Sato) Construct a well-defined state sum type invariant from a strongly amenable subfactor. Note that, unlike the Ponzano-Regge model, we do not have an a symptotic description of the quantum 6 j -symbols in general. (Recall that 6 j -symbols of SU (2) have an asymptotic description.) Let us consider the Turaev-Viro-Ocneanu invariant for a clo sed 3-manifold constructed from a subfactor. Then, this invariant can be co nsidered as a Reshetikhin-Turaev type invariant constructed from a subf actor by passing the initial subfactor through the Longo-Rehren construction. If we start with a subfactor which has a non-degenerate braiding in particula r, then this Turaev- Viro-Ocneanu invariant splits into a Reshetikhin-Turaev i nvariant and its com- plex conjugate. The following question will open a way to est ablish a theory of the minimal non-degenerate extension of a degenerate braid ing.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09895 | AMR-103-0175

### Problem 9.11 - (N.

O / open | AMR 问题清单汇编

(N. Sato) Let us consider the Turaev-Viro-Ocneanu invariant from a subfactor with a degenerate braiding. Then, find a desc ription of this invariant as a Reshetikhin-Turaev invariant.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09896 | AMR-103-0166

### Problem 9.2 - (Y.

O / open | AMR 问题清单汇编

(Y. Kawahigashi) Suppose we have a three-dimensional TQFT. Can we determine whether it arises from a fusion rule algebra and 6j -symbols? If yes, can we describe all fusion rule algebras with 6j -symbols producing the TQFT?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09897 | AMR-103-0167

### Problem 9.3 - (Y.

O / open | AMR 问题清单汇编

(Y. Kawahigashi) Suppose we have two fusion rule algebras with 6j -symbols and that two TQFT’s arising from them are isomorphi c. What relation do we have for the two sets of 6j -symbols?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09898 | AMR-103-0168

### Problem 9.4 - (Y.

O / open | AMR 问题清单汇编

(Y. Kawahigashi) Suppose we have a TQFT arising from a fusion rule algebra with 6j -symbols. Using a fusion rule subalgebra and 6j - symbols restricted on it, we can construct another TQFT. Wha t relation do we have for these TQFT’s?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09899 | AMR-103-0169

### Problem 9.5 - (Y.

S / solved | AMR 问题清单汇编

(Y. Kawahigashi) Suppose we have a semisimple ribbon category C with finitely many isomorphism classes of simple objects. If the S - matrix is invertible, we can construct the Reshetikhin-Tur aev invariant and the state-sum invariant from C and the latter is the square of the absolute value of the former. If the S -matrix is not invertible, do we still have a similar description of the state-sum invariant?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09900 | AMR-103-0170

### Problem 9.6 - (Y.

P / partially_solved | AMR 问题清单汇编

(Y. Kawahigashi) Suppose we have a semisimple ribbon category C1 with finitely many isomorphism classes of simple objects, bu t the S -matrix is not invertible. Then we can construct a new modula r category C2 containing C1 as a full subcategory by the “quantum double” construction [315, 316, 182], but there may be another extension of C1 to a modular cate- gory. Theorem 2.13 in [315] claims that we have a “minimal” ex tension in an “essentially unique” way. Do we indeed have existence and ce rtain uniqueness of such an extension? If so, what is the relation between the t wo TQFT’s arising from C1 and its minimal extension?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09901 | AMR-103-0171

### Problem 9.7 - (Y.

O / open | AMR 问题清单汇编

(Y. Kawahigashi) Suppose we have a semisimple ribbon category C1 with a degenerate S -matrix as in Problem 9.6. By the method in [288], we can also make a modular tensor category C2 from C1. What is the relation between the two TQFT’s arising from C1 and C2?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09902 | AMR-103-0172

### Problem 9.8 - (Y.

P / partially_solved | AMR 问题清单汇编

(Y. Kawahigashi) There are some fusion rule algebras with 6j -symbols that do not seem to arise from quantum groups in [14] and more conjectured candidates of such examples in [160]. What are t he corresponding TQFT’s? Especially if the series conjectured in [160] does e xist, it would give a parametrized family of TQFT’s. Does a differentiation by a p arameter (af- ter a certain reparametrization) give a more interesting in variant, possibly of Vassiliev type? 9.4 Turaev-Viro-Ocneanu invariants The state-sum invariant of 3-manifolds derived from 6 j -symbols is called the Turaev-Viro-Ocneanu invariant when the set of 6 j -symbols arises from a sub- factor. There are infinitely many subfactors other than thos e derived from quantum groups or finite groups. The Turaev-Viro-Ocneanu in variants derived from such subfactors might be new invariants of 3-manifolds. (N. Sato) The Haagerup subfactor of Jones index 5+ √ 13 2 has the smallest index among finite depth subfactors with Jones index bigger t han 4 and it is expected to have some “exotic” properties from the subfacto r theoretical view- point. However, it does not seem so sensitive to classify 3-m anifolds. The Turaev-Viro-Ocneanu invariant constructed from the Haage rup subfactor can- not distinguish lens spaces L(5, 1) and L(5, 2), as well as L(7, 1) and L(7, 2). On the other hand, generalized E6 -subfactors with the group symmetries Z/3Z and Z/5Z can distinguish L(3, 1) and L(3, 2), L(5, 1) and L(5, 2), respectively.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09903 | AMR-103-0173

### Problem 9.9 - (N.

P / partially_solved | AMR 问题清单汇编

(N. Sato) Find a subfactor which can distinguish lens spaces L(7, 1) and L(7, 2). Moreover, find a subfactor to classify 3-manifolds as well as possible. In the lattice field theory, Ponzano and Regge [332] construc ted a state sum model for SU (2) and investigated an asymptotic behavior of the model. Some infinite depth subfactors are manageable in the sense of growth rate (amenability). Such subfactors are called strongly amenable. The strong amen- ability condition might be enough to control the asymptotic behavior of the state sum model constructed from a strongly amenable subfac tor.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09904 | AIM-TOPOLOGY-0069

### Projector endomorphisms and the correct Hilbert chart

P / partially_solved | AIM 研讨会问题汇编

For each standard tableau $(T)$ of size $(n)$, compute the $((q,t,a))$-graded algebra $(H^*\operatorname{REnd}(P_T))$ in the appropriate completed/bounded-above homotopy category of type-A Soergel complexes, and compare it with $$ \Gamma(\mathring{\operatorname{FH}}_T(\mathbb C), \mathcal O_{\operatorname{FH}^{\mathrm{dg}}_n(\mathbb C)}) \otimes \Lambda^\bullet(\mathcal T_n^\vee). $$

来源字段：http://aimpl.org/agclinkhom/4/

09905 | OWR-15955-012

### Pure Braid Squares in the Symmetric Grope Filtration

O / open | Oberwolfach 报告问题汇编

Is the square subgroup $(P_m)^2$ of the pure braid group contained in the second stage $\mathcal G_2^m$ of the symmetric grope filtration?

来源字段：https://doi.org/10.4171/owr/2017/50

09906 | AMR-112-0004

### Quadrisecants of wild knots

P / partially_solved | AMR 问题清单汇编

Does every wild knot have infinitely many quadrisecants, that is, lines meeting the knot in at least four distinct points?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Topology

09907 | AIM-TOPOLOGY-0144

### Quantifier audit and local stretch-factor reduction

P / partially_solved | AIM 研讨会问题汇编

Let $R$ be a hyperbolic metric on $S$ and $\operatorname{Mod}(R)$ the Teichmüller modular group of $R$.

If $f\in\operatorname{Map}(S)$ satisfies that there exists $\lambda$ such that $\forall\alpha,\beta$ and $n\in\mathbb N$, $$ i(f^n(\alpha),\beta)<c\lambda^n, $$ is it true then that $f\in\operatorname{Mod}(R)$ for some $R$?

来源字段：http://aimpl.org/genusinfinity/3/

09908 | OWR-17470-006

### Quantitative Complexity of Homotopy Representatives and Homotopies

P / partially_solved | Oberwolfach 报告问题汇编

For finite simply connected piecewise Riemannian simplicial complexes, or compact Riemannian manifolds, how can one estimate the least geometric complexity of a representative of a homotopy class from its algebraic size? If two maps are homotopic, how does the least complexity of a homotopy depend on the complexities of the maps? Is scalability an $\mathbb R$-homotopy invariant? Is a space $Y$ scalable exactly when its cohomology algebra admits a differential-graded-algebra embedding

$$H^*(Y;\mathbb R)\hookrightarrow\bigwedge^*\mathbb R^N$$

for some $N$?

来源字段：https://doi.org/10.4171/owr/2020/7

09909 | AIM-TOPOLOGY-0223

### Quantitative submetry models for Alexandrov collapse

P / partially_solved | AIM 研讨会问题汇编

**Problem 10. Study the collapse of Alexandrov spaces.**

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

09910 | OWR-1967-006

### Quantum Invariants and Boundary Character Varieties

O / open | Oberwolfach 报告问题汇编

别名：Duplicate: Quantum Invariants and Boundary Character Varieties [OWR-1967-007]

Let $M$ be a three-manifold with boundary $F$ and let $G$ be semisimple. Do the Witten-Reshetikhin-Turaev vectors $I_{G,r}(M)\in V_{G,r}(F)$ over all levels $r$ determine the image of the character variety $X_G(M)$ in $X_G(\partial M)$?

来源字段：https://doi.org/10.4171/owr/2008/22

09911 | OWR-1967-008

### Quantum Knot Invariants and Boundary Character Varieties

O / open | Oberwolfach 报告问题汇编

别名：Duplicate: Quantum Knot Invariants and Boundary Character Varieties [OWR-1967-009]

For every complex reductive group $G$, do the quantum $G$-invariants of a knot $K\subset S^3$ determine the image of $X_G(\pi_1(S^3\setminus K))$ in the $G$-character variety of the boundary torus? Equivalently, do knots with the same $G$-quantum invariants have the same image?

来源字段：https://doi.org/10.4171/owr/2008/22

09912 | OWR-1596-007

### Quasi-Projective Fundamental Groups of Graph Links

P / partially_solved | Oberwolfach 报告问题汇编

Which graph links, other than Seifert links, have global, equivalently quasi-projective, fundamental groups? Here a graph link is a link in a compact oriented $\mathbb Z$-homology 3-sphere whose exterior has only Seifert-fibered geometric pieces.

来源字段：https://doi.org/10.4171/owr/2007/40

09913 | AMR-109-0150

### Question (Smith) - Let αi, i = 1,...,n be a configuration of curves on a surface S of genus g with one boundary component δ such that ev…

O / open | AMR 问题清单汇编

Let αi, i = 1,...,n be a configuration of curves on a surface S of genus g with one boundary component δ such that every pair of curves intersect in 0 or 1 point and Tα1...T αn =Tδ. What is the maximal length n of such a product? Can it be arbitrarily long? In the case of genus 2 the curves are part of a D6 configuration (with repetitions). Since the Coxeter element for D6 gives a square of the boundary twist we probably cannot use the curve α′ 1 so we are left with A5 configuration. Here we have a simple algebraic question.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09914 | AMR-109-0148

### Question (Smith) - What is the maximal length of such a product?

S / solved | AMR 问题清单汇编

What is the maximal length of such a product? Can it be arbitrarily long?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09915 | AMR-109-0258

### Question 1 - If R = Q(q1,q 2), is the above map from Bn toHn(q1,q 2) injective?

O / open | AMR 问题清单汇编

If R = Q(q1,q 2), is the above map from Bn toHn(q1,q 2) injective?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09916 | AMR-109-0266

### Question 1.1 - Does the Teichm¨ uller space for Sg admit an equivariant deformation retraction onto a cocompact spine whose dimensio…

O / open | AMR 问题清单汇编

Does the Teichm¨ uller space for Sg admit an equivariant deformation retraction onto a cocompact spine whose dimension is equal to 4g- 5, the virtual cohomological dimension of Mod±(Sg)? 1if g ≥ 2 then the curvature will be negative 20. Automorphism groups of free, surface and free abelian groups 321 Further questions of a similar nature are discussed in (2.1). The issues involved in using these symmetric space analogs to prove purely group theoretic properties are illustrated in the proof of the Tits alternative, which holds for all three classes of groups. A group Γ is said to satisfy the Tits alternative if each of its subgroups either contains a non-abelian free group or else is virtually solvable. The strategy for proving this is similar in each of the three families that we are considering: inspired by Tits's original proof for linear groups (such as GL( n, Z)), one attempts to use a ping-pong argument on a suitable boundary at infinity of the symmetric space. This strategy ultimately succeeds but the details vary enormously between the three contexts, and in the case of Out( Fn) they are particularly intricate ([ 4, 3 ] versus [ 9]). One finds that this is often the case: analogies between the three classes of groups can be carried through to theorems, and the architecture of the expected proof is often a good guide, but at a more detailed level the techniques required vary in essential ways from one class to the next and can be of completely different orders of difficulty. Let us return to problems more directly phrased in terms of the geometry of the symmetric spaces. The symmetric space for GL( n, Z) has a left-invariant metric of non-positive curvature, the geometry of which is relevant to many areas of mathematics beyond geometric group theory. Teichm¨ uller space has two natural metrics, the Teichm¨ uller metric and the Weyl-Petersen metric, and again the study of each is a rich subject. In contrast, the metric theory of Outer space has not been developed, and in fact there is no obvious candidate for a natural metric. Thus, the following question has been left deliberately vague:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09917 | AMR-109-0243

### Question 1.1 - Forg≥ 2, does Γg contain a purely pseudo-Anosov surface subgroup?

P / partially_solved | AMR 问题清单汇编

Forg≥ 2, does Γg contain a purely pseudo-Anosov surface subgroup? The paper is organized as follows. In §2, we discuss the existence of surface subgroups in Kleinian groups of finite co-volume. In §3, we discuss surface subgroups of Γ g, and contrast and compare with §2. In §4 we discuss some related topics; for example we discuss the connection of Question 1.1 with some conjectures in 4-manifold topology. 274 17. Surface subgroups of mapping class groups 275

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09918 | AMR-109-0267

### Question 1.2 - Develop a metric theory of Outer space.

O / open | AMR 问题清单汇编

Develop a metric theory of Outer space. The elements of infinite order in GL( n, Z) that are diagonalizable over C act as loxodromic isometries of X. When n = 2, these elements are the hyperbolic matrices; each fixes two points at infinity in X = H2, one a source and one a sink. The analogous type of element in Mod ±(Sg) is a pseudo-Anosov, and in Out( Fn) it is an iwip (irreducible with irreducible powers). In both cases, such elements have two fixed points at infinity (i.e. in the natural boundary of the symmetric space analog), and the action of the cyclic subgroup generated by the element exhibits the north- south dynamics familiar from the action of hyperbolic matrices on the closure of the Poincar´ e disc [62], [ 54]. In the case of Mod ±(Sg) this cyclic subgroup leaves invariant a unique geodesic line in Teichm¨ uller space, i.e. pseudo-Anosov's are axial like the semi-simple elements of infinite order in GL(n, Z). Initial work of Handel and Mosher [ 43] shows that in the case of iwips one cannot hope to have a unique axis in the same metric sense, but leaves open the possibility that there may be a reasonable notion of axis in a weaker sense. (We highlighted this problem in an earlier version of the current article.) In a more recent preprint [ 42] they have addressed this last point directly, defining an axis bundle associated to any iwip, cf. [ 63]. Nevertheless, many interesting questions remain (some of which are highlighted by Handel and Mosher). Thus we retain a modified version of our original question:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09919 | AMR-109-0268

### Question 1.3 - Describe the geometry of the axis bundle (and associated objects) for an iwip acting on Outer Space.

O / open | AMR 问题清单汇编

Describe the geometry of the axis bundle (and associated objects) for an iwip acting on Outer Space. 322 M. Bridson and K. Vogtmann

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09920 | AMR-103-0196

### Question 10.21 - (F.

O / open | AMR 问题清单汇编

(F. Deloup) Is there a lift of arg γ(qσ) to a mod 16 invariant? This would give a finite type invariant of degree 1 in the spin c Goussarov-Habiro theory. 37A quadratic function q is a a map such that q(x + y) - q(x) - q(y) is bilinear in x and y. It is called homogeneous if q(nx) = n2q(x) for any n ∈ Z and x ∈ G. In fact, there is a canonical map σ ↦→qσ from spin c structures to quadratic functions and qσ is homogeneous if and only if σ actually comes from a spin structure. Note that not all spin c structures come from spin structures.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09921 | AMR-103-0178

### Question 10.3 - (C.

P / partially_solved | AMR 问题清单汇编

(C. Lescop) Are the Cappell-Lee-Miller Casson-type SU (n)-invariants of finite type? If so, what are their degrees and th eir weight systems?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09922 | AMR-103-0180

### Question 10.5 - (M.

O / open | AMR 问题清单汇编

(M. Polyak) Is there a "Rokhlin invariant" of a pair (M, α) of a closed 3-manifold M and a spin c structure α on M? (See Question 10.21.)

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09923 | AMR-103-0205

### Question 11.9 - (1) Find a surgery formula for the Kuperberg-Thurston in- variant [237] in terms of the Chern-Simons series of Questi…

O / open | AMR 问题清单汇编

(1) Find a surgery formula for the Kuperberg-Thurston in-variant [237] in terms of the Chern-Simons series of Questio n 3.12 (2) Compare the Kuperberg-Thurston invariant to the LMO invari ant.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09924 | AMR-103-0207

### Question 12.1 - (R.

O / open | AMR 问题清单汇编

(R. Benedetti) Are torsions actually sensitive only to the (pL)-homotopy immersion classes of (pL)-knots? If one fix a C - homotopy immersion class of knots, say α, then one can define the set of finite type invariants F(α) of the C -isotopy classes contained in α. If α0 is a class of Legendrian knots, one can take α1 = f1(α0) and α2 = f2(α1); a finite type invariant for αi lifts to a finite type invariant for αi-1. So one has natural maps F(α2) f ∗ 2 →F (α1) f ∗ 1 →F (α0). It is known [130] that, under certain hypotheses on W (for instance when W is a Z-homology sphere), f ∗ 1◦ f ∗ 2 is a bijection. On the oder hand, one can Section 12.1 was written by R. Benedetti. find in [380] examples where f ∗ 1◦ f ∗ 2 is not surjective and Legendrian finite type invariants can eventually distinguish some Legendrian kno ts which are isotopic as framed knots. In fact one can realize that for these exampl es f ∗ 2 is already not surjective and that (pL)-finite type invariants can even tually distinguish some (pL)-knots which are isotopic as framed knots. The foll owing conjecture is not in contradiction with all these known results on the su bject.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09925 | AMR-109-0259

### Question 2 - What are the equivalence classes of braids modulo the moves • ab↔ba, and • b↔σnι(b)?

O / open | AMR 问题清单汇编

What are the equivalence classes of braids modulo the moves • ab↔ba, and • b↔σnι(b)? In other words, what happens if the Markov move b↔σ-1 n ι(b) is omitted? This question was shown in [ OS03] to be equivalent to the important problem in contact geometry of classifying transversal links up to transversal isotopy. The Bennequin number of a braid b∈ Bn is e-n, where e is the sum of the exponents in a word in the generators σi representingb. The Bennequin number is invariant under the moves in Question 2. Thus it can be used to show that there are braids that are related by Markov moves, but not by the moves in Question 2. Birman and Menasco [ BM] and Etnyre and Honda [ EH] have independently found pairs of braids that are related by Markov moves and have the same Bennequin invariant, but are not related by the moves in Question 2. Their proofs are quite complicated, and it would be nice to have a new invariant that could distinguish their pairs of braids.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09926 | AMR-109-0008

### Question 2.1 - (Ends spectrum).

O / open | AMR 问题清单汇编

(Ends spectrum). What are the possibile values of ends(Modg,H ) for finitely- generated subgroups H <Modg? It is well-known that the moduli space Mg has one end. The key point of the proof is that the complex of curves is connected. This proof actually gives more: any cover of Mg has one end; see, e.g., [ FMa]. However, I do not see how this fact directly gives information about Question 2.1. Commensurators. Asking for two subgroups of a group to be conjugate is often too restrictive a question. A more robust notion is that of commensurability. Subgroups Γ 1, Γ2 of a group H are commensurable if there exists h ∈ H such that hΓ1h-1∩ Γ2 has finite index in both hΓ1h-1 and in Γ 2. One then wants to classify subgroups up to commensurability; this is the natural equivalence relation one studies in order to coarsify the relation of “conjugate” to ignore finite index information. The primary commensurability invariant for subgroups Γ < H is the commensurator of Γ in H, denoted Comm H (Γ), defined as: CommH (Γ):= {h∈H:hΓh-1∩ Γ has finite index in both Γ and hΓh-1} The commensurator has most commonly been studied for discrete subgroups of Lie groups. One of the most striking results about commensurators, due to Margulis, states that if Γ is an irreducible lattice in a semisimple 2 Lie group H then [Comm H (Γ): Γ] = ∞ if and only if Γ is arithmetic. In other words, it is precisely the arithmetic lattices that have infinitely many “hidden symmetries”. 2By semisimple we will always mean linear semisimple with no compact factors. 16 B. Farb

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09927 | AMR-109-0269

### Question 2.1 - Do mapping class groups or Out(Fn) satisfy the Baum-Connes conjecture?

P / partially_solved | AMR 问题清单汇编

Do mapping class groups or Out(Fn) satisfy the Baum-Connes conjecture? Does Out(Fn) satisfy the Novikov conjecture? An approach to proving these conjectures is given by work of Rosenthal [ 75], generalizing results of Carlsson and Pedersen [ 23]. A contractible space on which a group Γ acts properly and for which the fixed point sets of finite subgroups are contractible is called an EΓ. Rosenthal's theorem says that the Baum-Connes map for Γ is split injective if there is a cocompact EΓ = E that admits a compactification X, such that (1) the Γ-action extends to X; (2) X is metrizable; (3) XG is contractible for every finite subgroup G of Γ (4) EG is dense in XG for every finite subgroup G of Γ (5) compact subsets of E become small near Y = X \E under the Γ-action: for every compact K⊂E and every neighborhood U⊂X of y∈Y, there exists a neighborhood V ⊂X of y such that γK∩V=∅ implies γK⊂U. The existence of such a space E also implies the Novikov conjecture for Γ. For Out(Fn) the spine of Outer space mentioned in the previous section is a reasonable candidate for the required EΓ, and there is a similarly defined candidate for Aut( Fn). For mapping class groups of punctured surfaces the complex of arc systems which fill up the surface is a good candidate (note that this can be identified with a subcomplex of Outer space, as in [ 47], section 5).

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09928 | AMR-109-0152

### Question 2.1 - Is the Hurwitz problem for mapping class group factorizations decidable?

O / open | AMR 问题清单汇编

Is the Hurwitz problem for mapping class group factorizations decidable? Are there interesting criteria which can be used to conclude that two given factorizations are equivalent, or inequivalent, up to Hurwitz moves and global conjugation? In broader terms, the question is whether mapping class group factorizations can be used to derive non-trivial and useful invariants of Lefschetz fibrations, or even better, of the underlying symplectic 4-manifolds. At this point, it is worth mentioning two spectacular examples of such invariants which arise from geometric considerations (rather than purely from mapping class group theory). One is Sei- del's construction of a Fukaya-type A∞-category associated to a Lefschetz fibration [ 21], which seems to provide a computationally manageable approach to Lagrangian submanifolds and Fukaya categories in open 4-manifolds equipped with exact symplectic structures. The other is the enu- merative invariant introduced by Donaldson and Smith, which counts embedded pseudoholomor- phic curves in a symplectic 4-manifold by viewing them as sections of a "relative Hilbert scheme" associated to a Lefschetz fibration [ 9].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09929 | AMR-109-0244

### Question 2.1 - Let M = H3/Γ be a finite volume hyperbolic 3-manifold, does Γ contain a surface subgroup.

S / solved | AMR 问题清单汇编

Let M = H3/Γ be a finite volume hyperbolic 3-manifold, does Γ contain a surface subgroup. This was answered in [ 8] for non-compact but finite volume manifolds, and so the remaining cases of Question 2.1 are the closed manifolds, and this seems far from resolution at present. We will discuss [ 8] in more detail below. 2.2. Given an essential surface Σ in a finite volume hyperbolic 3-manifold M, one can at- tempt to understand the surface in terms of how the hyperbolic metric on M restricts to the surface Σ. In a non-compact finite volume hyperbolic 3-manifold there are two possibilities for the geometry of a closed essential surface Σ. Σ is either quasi-Fuchsian or it is said to contain accidental parabolics. In the latter case, as suggested by the name, these surface groups contain parabolic elements, whilst in the former case, all non-trivial elements are hyperbolic. Both these surfaces are geometrically finite. The methods of [ 8] only provide surfaces containing accidental parabolics, and a comment on the construction in [ 8] will be informative for our discussion of Γ g (see also [ 7]). Suppose that M = H3/Γ is a non-compact finite volume hyperbolic 3-manifold. We can assume that M is orientable on passing to a double cover if necessary, and so M is the interior of a compact 3-manifold with boundary consisting of a disjoint union of tori. Using residual finiteness of Γ there is a finite cover, ˆM of M, which has at least 3 boundary components. Standard 3-manifold topology shows that the first betti number of ˆM is at least 3, and that one can find an embedded orientable essential surface F with non-empty boundary in ˆM missing at 276 A.W. Reid least one of the boundary components of ˆM. This surface can then be used to build a finite cyclic cover of ˆM containing a closed embedded essential surface Σ of genus at least 2. Indeed, the construction is explicit, the surface Σ is constructed by taking two copies of F "tubed together" along their boundary. Σ pushes down to M to provide an essential surface in M and so the desired surface subgroup. By construction, these "tubed surfaces" contain accidental parabolic elements corresponding to the peripheral elements from ∂F. The question of existence of closed quasi-Fuchsian surfaces in M as above remains open. However recent work of Masters and Zhang [ 27] appears to make some progress on this. 2.3. The surface subgroups built in [ 8] can also be viewed as being built by repeated appli- cations of the Maskit combination theorems, a version of which is given below (see [ 23] and [ 26] Chapter VII for more details). Suppose G0,G 1,G 2< Γ with G1∩G2 =G0< Γ. If Γ acts on a set X, then we say that a pair of subsets Θ 1, Θ2⊂X is a proper interactive pair for G1,G 2 if (1) Θi=∅ for each i = 1, 2, (2) Θ1∩ Θ2 =∅, (3) G0 leaves Θ i invariant for each i = 1, 2, (4) for every φ1 ∈ G1\G0 we have φ1(Θ2)⊂ Θ1 and for every φ2 ∈ G2\G0 we have φ2(Θ1)⊂ Θ2, and (5) for i = 1, 2, there exists θi∈ Θi, such that for every φi∈Gi\G0, θi∈φi(Θi′) for i′=i. With this notation, we can state the following combination theorem.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09930 | AMR-109-0278

### Question 2.10 - What is the least integer δ such that Out(Fn) acts without a global fixed point on a complete CAT (0) space of dimens…

O / open | AMR 问题清单汇编

What is the least integer δ such that Out(Fn) acts without a global fixed point on a complete CAT (0) space of dimension δ? And what is the least dimension for the mapping class group Mod±(Sg)? The action of Out( Fn) on the first homology of Fn defines a map from Out( Fn) to GL( n, Z) and hence an action of Out( Fn) on the symmetric space of dimension 1 2n(n + 1)- 1. This action does not have a global fixed point and hence we obtain an upper bound on δ. On the other hand, since Out( Fn) has property F A, δ≥ 2. In fact, motivated by work of Farb on GL( n, Z), Bridson [14] has shown that using a Helly-type theorem and the structure of finite subgroups in Out( Fn), one can obtain a lower bound on δ that grows as a linear function of n. Note that a lower bound 2topological covering dimension 20. Automorphism groups of free, surface and free abelian groups 325 of 3 n- 3 on δ would imply that Outer Space did not support a complete Out( Fn)-equivariant metric of non-positive curvature. If X is a CAT(0) polyhedral complex with only finitely many isometry types of cells (e.g. a finite dimensional cube complex), then each isometry of X is either elliptic (fixes a point) or hyperbolic (has an axis of translation) [ 15]. If n≥ 4 then a variation on an argument of Gersten [36] shows that in any action of Out( Fn) on X, no Nielsen generator can act as a hyperbolic isometry.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09931 | AMR-109-0279

### Question 2.11 - If n ≥ 4, then can Out(Fn) act without a global fixed point on a finite-dimensional CAT(0) cube complex?

P / partially_solved | AMR 问题清单汇编

If n ≥ 4, then can Out(Fn) act without a global fixed point on a finite- dimensional CAT(0) cube complex?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09932 | AMR-109-0280

### Question 2.12 - Does Out(F3) have a faithful representation into GL(m, C) for some m∈ N?

O / open | AMR 问题清单汇编

Does Out(F3) have a faithful representation into GL(m, C) for some m∈ N? Note that braid groups are linear [ 8] but it is unknown if mapping class groups of closed surfaces are. Brendle and Hamidi-Tehrani [ 13] showed that the approach of Formanek and Procesi cannot be adapted directly to the mapping class groups. More precisely, they prove that the type of “poison subgroup” described above does not arise in mapping class groups. The fact that the above question remains open is an indication that Out( F3) can behave differently from Out( Fn) for n large; the existence of finite index subgroups mapping onto Z was another instance of this, and we shall see another in our discussion of automatic structures and isoperimetric inequalities.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09933 | AMR-103-0035

### Question 2.13 - (T.

O / open | AMR 问题清单汇编

(T. Stanford) Does Mn have an interesting complementary space in Vn? Consider, for example, the space Nn⊂ Vn of invariants v with the property that v(L) = 0 for any string link L such that π1(B3- L) is free. Is Nn nontrivial? Do Nn and Mn together span Vn? Here is some background and motivation. When considering finite-type invariants of string links, th e first ones that come to mind are the Milnor invariants. These were defined by Milno r [283] in 1954 as numbers associated to links. They are not quite invariant s of links, in the usual sense, because of some indeterminacy. They are, howev er, well-defined as invariants of string links, and this point of view was take n by Habegger and Lin [163]. After Vassiliev’s work appeared, Bar-Natan [ 26] and Lin [261] showed (independently) that the Milnor invariants are finit e-type invariants. Habegger and Masbaum [164] showed that on the chord diagram l evel, the Milnor invariants (including products of Milnor invariant s) are exactly the ones that vanish on Jacobi diagrams that contain internal loops, and also that the Milnor invariants are the only rational-valued finite-type invariants of string links which are also concordance invariants. String links may have local knots in the strands, and such kno ts are not detected by Milnor invariants. If a string link L has local knots, then π1(B3- L) is not free. Hence the question as to whether finite-type invariant s can show that the complement of a string link is not free. (M. Polyak) Let us review the constructions of Milnor μ -invariant in [89]. For a n-component link L = L1∪···∪Ln, regard the homotopy class of Ln as in π1 ( S3- (L1∪···∪Ln-1) ), and write it in terms of meridians m1,···, mn-1 of L1,···, Ln-1. Consider its Magnus expansion putting mi = 1 + Xi for non- commutative variables Xi. Then, Milnor’s μ -invariant μ i1···ik,n(L) is defined to be the coefficient of X i1···X ik in the expansion, which is an invariant under the assumption that the lower μ -invariants vanish. For example, μ 1,2 is equal to the linking number lk( L1, L2) of L1 and L2. Further, if μ i,j(L) = 0 for any i, j, then μ 12,3(L) = lk( L12, L3), where L12 denotes the link which is the intersection of Seifert surfaces of L1 and L2. In general, under the vanishing assumption of the lower μ -invariants, μ 12···n-1,n(L) = lk( L12···n-1, Ln) where L12···k (for k = 2, 3,···, n- 1) denotes the link which is the intersection of Seifert surfaces of L12···k-1 and Lk.

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09934 | AMR-109-0153

### Question 2.2 - (Donaldson).

O / open | AMR 问题清单汇编

(Donaldson). Is it possible to enumerate all matching paths in a Lefschetz fibration with given monodromy factorization?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09935 | AMR-109-0270

### Question 2.2 - Does there exist a compactification of the spine of Outer space satisfying Rosen- thal’s conditions?

O / open | AMR 问题清单汇编

Does there exist a compactification of the spine of Outer space satisfying Rosen- thal’s conditions? Same question for the complex of arc systems filling a punctured surface. 20. Automorphism groups of free, surface and free abelian groups 323 In all of the cases mentioned above, the candidate space E has dimension equal to the virtual cohomological dimension of the group. G. Mislin [ 68] has constructed a cocompact EG for the mapping class group of a closed surface, but it has much higher dimension, equal to the dimension of the Teichm¨ uller space. This leads us to a slight variation on Question 1.1.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09936 | AMR-109-0021

### Question 2.20 - (Number of Hurwitz surfaces).

O / open | AMR 问题清单汇编

(Number of Hurwitz surfaces). Give a formula for the number of Hurwitz surfaces of genus g. What is the frequency of those g for which there is a unique Hurwitz surface?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09937 | AMR-109-0154

### Question 2.3 - (Smith).

S / solved | AMR 问题清单汇编

(Smith). Is there an a priori upper bound on the length of any factorization of the boundary twist δ as a product of positive Dehn twists in Mapg,n? Equivalently: is there an upper bound (in terms of the genus only) on the number of singular fibers of a Lefschetz fibration admitting a section of square -1? (In the opposite direction, various lower bounds have been established, see e.g. [ 28]). Unfortunately, it is hard to quantify the amount of rotation induced by a Dehn twist on the boundary of the hyperbolic disc, so it is not clear whether the approach in [ 26] can shed light on this question. More generally, given an element T ∈ Map+ g,n, we can try to study factorizations of T as a product of positive Dehn twists. Geometrically, such factorizations correspond to Lefschetz fibrations over the disc (with bounded fibers), such that the monodromy along the boundary of the disc is the prescribed element T. The boundary of such a Lefschetz fibration is naturally a contact 3-manifold Y equipped with a structure of open book [10], and the total space of the fibration is a Stein filling of Y [1, 10, 15 ]. Hence the classification of factorizations of T in Map g,n is related to (and a subset of) the classification of Stein fillings of the contact 3-manifold Y. Some remarkable results have been obtained recently concerning the classification of symplec- tic fillings of lens spaces or links of singularities, using tools from symplectic geometry, and in particular pseudo-holomorphic curves (see e.g. [ 14, 18 ]); meanwhile, Lefschetz fibrations have 9. Mapping class group factorizations and symplectic 4-manifolds 137 been used to construct examples with infinitely many inequivalent fillings (see e.g. [ 20]). Hence we may ask:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09938 | AMR-109-0271

### Question 2.3 - Can one construct a cocompact EG with dimension equal to the virtual coho- mological dimension of the mapping class g…

S / solved | AMR 问题清单汇编

Can one construct a cocompact EG with dimension equal to the virtual coho- mological dimension of the mapping class group of a closed surface?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09939 | AMR-103-0026

### Question 2.4 - (T.

O / open | AMR 问题清单汇编

(T. Stanford) The Dogolazky-Kneissler 2-torsion element in A(↓↓, Z) (see Figure 7) can be embedded into a chord diagram in A(S1, Z) in many ways. Such an embedding will always produce an element x∈A (S1, Z) with 2x = 0. Is it possible to produce such an x which is nontrivial? If so, this would give a counterexample to Conjecture 2.3. 2.2 Do Vassiliev invariants distinguish knots?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09940 | AMR-109-0155

### Question 2.4 - For which T∈ Map+ g,n is it possible to classify factorizations of T as a product of positive Dehn twists in Mapg,n?

O / open | AMR 问题清单汇编

For which T∈ Map+ g,n is it possible to classify factorizations of T as a product of positive Dehn twists in Mapg,n? In particular, for which T is there a unique factorization, or only finitely many factorizations, up to Hurwitz equivalence and global conjugation?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09941 | AMR-109-0272

### Question 2.4 - Forn> 3, does Aut(Fn) have property (T)?

S / solved | AMR 问题清单汇编

Forn> 3, does Aut(Fn) have property (T)? The corresponding question for mapping class groups is also open. If Aut( Fn) were to have Property (T), then an argument of Lubotzky and Pak [ 64] would provide a conceptual explanation of the apparently-unreasonable effectiveness of certain algorithms in computer science, specifically the Product Replacement Algorithm of Leedham-Green et al. If a group has Property (T) then it has Serre's property F A: every action of the group on an R-tree has a fixed point. When n≥ 3, GL(n, Z) has property F A, as do Aut( Fn) and Out( Fn), and mapping class groups in genus ≥ 3 (see [ 28]). In contrast, McCool [ 67] has shown that Aut( F3) has a subgroup of finite-index with positive first betti number, i.e. a subgroup which maps onto Z. In particular this subgroup acts by translations on the line and therefore does not have property F A or (T). Since property (T) passes to finite-index subgroups, it follows that Aut( F3) does not have property (T).

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09942 | AMR-109-0011

### Question 2.5 - Does there exist some Modg,g ≥ 2 that contains a subgroup Γ isomorphic to a cocompact (resp.

P / partially_solved | AMR 问题清单汇编

Does there exist some Modg,g ≥ 2 that contains a subgroup Γ isomorphic to a cocompact (resp. noncocompact) lattice in SO(m, 1) with m≥ 5 (resp. m≥ 4)? a cocompact lattice in SU(n, 1),n≥ 2? Must there be only finitely many conjugacy classes of any such fixed Γ in Modg? In light of example (3) above, I would like to specifically ask: can Mod g contain infinitely many isomorphism types of cocompact lattices in SO(4, 1)? Note that when Γ is the fundamental group of a (complex) algebraic variety V, then it is known that there can be at most finitely many representations ρ: Γ → Modg which have holomorphic representatives, by which we mean the unique homotopy class of maps f:V →M g with f∗ =ρ contains a holomorphic map. This result follows from repeatedly taking hyperplane sections and finally quoting the result for (complex) curves. The result for these is a theorem of Arakelov- Parshin (cf. Example 1.2 above, and §2.3 below.) For representations which do not a priori have a holomorphic representative, one might try to find a harmonic representative and then to prove a Siu-type rigidity result to obtain a holomorphic representative. One difficulty here is that it is not easy to find harmonic representatives, since (among other problems) every loop in Mg,g≥ 2 can be freely homotoped outside every compact set. For recent progress, however, see [ DW] and the references contained therein.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09943 | AMR-109-0273

### Question 2.5 - For n > 3, does Aut(Fn) have a subgroup of finite index with positive first betti number?

P / partially_solved | AMR 问题清单汇编

For n > 3, does Aut(Fn) have a subgroup of finite index with positive first betti number? Another finite-index subgroup of Aut( F3) mapping onto Z was constructed by Alex Lubotzky, and was explained to us by Andrew Casson. Regard F3 as the fundamental group of a graph R with one vertex. The single-edge loops provide a basis {a,b,c} for F3. Consider the 2-sheeted covering ˆR → R with fundamental group ⟨a,b,c 2,cac -1,cbc -1⟩ and let G ⊂ Aut(F3) be the stabilizer of this subgroup. G acts on H1( ˆR, Q) leaving invariant the eigenspaces of the involution that generates the Galois group of the covering. The eigenspace corresponding to the eigenvalue -1 is two dimensional with basis {a-cac-1, b-cbc-1}. The action of G with respect to this basis gives an epimorphism G→ GL(2, Z). Since GL(2, Z) has a free subgroup of finite-index, we obtain a subgroup of finite index in Aut( F3) that maps onto a non-abelian free group. One can imitate the essential features of this construction with various other finite-index subgroups of Fn, thus producing subgroups of finite index in Aut( Fn) that map onto GL( m, Z). In each case one finds that m≥n- 1.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09944 | AMR-109-0156

### Question 2.5 - Given two factorizations of the boundary twist δ as a product of positive Dehn twists along nonseparating curves in M…

P / partially_solved | AMR 问题清单汇编

Given two factorizations of the boundary twist δ as a product of positive Dehn twists along nonseparating curves in Mapg,n, such that the total spaces of the corresponding Lefschetz fibrations have the same Euler characteristic and signature, is it always possible to obtain one from the other by a sequence of Hurwitz moves and partial conjugations?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09945 | AMR-109-0274

### Question 2.6 - If there is a homomorphism from a subgroup of finite index in Aut(Fn) onto a subgroup of finite index in GL(m, Z), th…

O / open | AMR 问题清单汇编

If there is a homomorphism from a subgroup of finite index in Aut(Fn) onto a subgroup of finite index in GL(m, Z), then must m≥n- 1? 324 M. Bridson and K. Vogtmann Indeed one might ask:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09946 | AMR-109-0275

### Question 2.7 - If m<n - 1 and H⊂Aut(Fn) is a subgroup of finite index, then does every homomorphism H→ GL(m, Z) have finite image?

O / open | AMR 问题清单汇编

If m<n - 1 and H⊂Aut(Fn) is a subgroup of finite index, then does every homomorphism H→ GL(m, Z) have finite image? Similar questions are interesting for the other groups in our families (cf. section 3). For exam- ple, if m<n - 1 and H⊂Aut(Fn) is a subgroup of finite index, then does every homomorphism H→ Aut(Fm) have finite image? A positive answer to the following question would answer Question 2.5; a negative answer would show that Aut( Fn) does not have property (T).

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09947 | AMR-109-0013

### Question 2.8 - (Normal subgroups).

S / solved | AMR 问题清单汇编

(Normal subgroups). Let Γ be a finitely generated normal subgroup of Modg, whereg≥ 3. Must Γ be commensurable with Modg or with Ig? One way of constructing infinitely generated normal subgroups of Mod g is to take the group generated by the nth powers of all Dehn twists. Another way is to take the normal closure Nφ of a single element φ∈ Modg. It seems unclear how to determine the algebraic structure of these Nφ, in particular to determine whether Nφ is finite index in Mod g, or in one of the Ig(k). The following is a basic test question.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09948 | AMR-103-0030

### Question 2.8 - (T.

O / open | AMR 问题清单汇编

(T. Stanford) Can we approximate hG by Vassiliev invariants for other G than dihedral groups?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09949 | AMR-109-0276

### Question 2.8 - Forn≥ 4, do subgroups of finite index in Aut(Fn) have Property F A?

P / partially_solved | AMR 问题清单汇编

Forn≥ 4, do subgroups of finite index in Aut(Fn) have Property F A? A promising approach to this last question breaks down because we do not know the answer to the following question.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09950 | AMR-109-0277

### Question 2.9 - Fix a basis for Fn and let An-1⊂ Aut(Fn) be the copy of Aut(Fn-1) corresponding to the first n- 1 basis elements.

O / open | AMR 问题清单汇编

Fix a basis for Fn and let An-1⊂ Aut(Fn) be the copy of Aut(Fn-1) corre- sponding to the first n- 1 basis elements. Let φ:Aut(Fn)→G be a homomorphism of groups. If φ(An-1) is finite, must the image of φ be finite? Note that the obvious analog of this question for GL( n, Z) has a positive answer and plays a role in the foundations of algebraic K-theory. A different approach to establishing Property (T) was developed by Zuk [ 85]. He established a combinatorial criterion on the links of vertices in a simply connected G-complex which, if satisfied, implies that G has property (T): one must show that the smallest positive eigenvalue of the discrete Laplacian on links is sufficiently large. One might hope to apply this criterion to one of the natural complexes on which Aut( Fn) and Out( Fn) act, such as the spine of Outer space. But David Fisher has pointed out to us that the results of Izeki and Natayani [ 55] (alternatively, Schoen and Wang - unpublished) imply that such a strategy cannot succeed.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09951 | AMR-109-0014

### Question 2.9 - Is it true that, given any pseudo-Anosov φ∈ Modg, there exists n =n(φ) such that the normal closure of φn is free?

S / solved | AMR 问题清单汇编

Is it true that, given any pseudo-Anosov φ∈ Modg, there exists n =n(φ) such that the normal closure of φn is free? Gromov discovered the analogous phenomenon for elements of hyperbolic type inside nonele- mentary word-hyperbolic groups; see [ Gro], Theorem 5.3.E. One should compare Question 2.8 to the Margulis Normal Subgroup Theorem (see, e.g. [ Ma]), which states that if Λ is any irreducible lattice in a real, linear semisimple Lie group with no compact factors and with R-rank at least 2, then any (not necessarily finitely generated) normal subgroup of Λ is finite and central or has finite index in Λ. Indeed, we may apply this result to analyzing normal subgroups Γ of Mod g. For g≥ 2 the group Sp(2 g, Z) satisfies the hypotheses of Margulis's theorem, and so the image π(Γ) under the natural representation π: Mod g→ Sp(2g, Z) is normal, hence is finite or finite index. This proves the following. Proposition 2.10 (Maximality of Torelli). Any normal subgroup Γ of Modg containingIg is commensurable either with Modg or with Ig. Proposition 2.10 is a starting point for trying to understand Question 2.8. Note too that Mess [Me] proved that the group I 2 is an infinitely generated free group, and so it has no finitely generated normal subgroups. Thus we know that if Γ is any finitely generated normal subgroup of Mod 2, then π(Γ) has finite index in Sp(4, Z). This in turn gives strong information about Γ; see [ F a3]. One can go further by considering the Malcev Lie algebra tg ofIg, computed by Hain in [ Ha3]; cf.§5.4 below. The normal subgroup Γ ∩I g ofIg gives an Sp(2 g, Z)-invariant subalgebra h of tg. LetH =H1(Σg, Z). The Johnson homomorphism τ:Ig→∧ 3H/H is equivariant with respect to the action of Sp(2 g, Z), and ∧3H/H is an irreducible Sp(2 g, Z)-module. It follows that the first quotient in the lower central series of Γ ∩Ig is either trivial or is all of ∧3H/H. With more work, one can extend the result of Proposition 2.10 from Ig toKg =Ig(2); see [ F a3].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09952 | AMR-109-0260

### Question 3 - What can be said about the dimensions of Dλ?

O / open | AMR 问题清单汇编

What can be said about the dimensions of Dλ?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09953 | AMR-109-0022

### Question 3.1 - (Fast word problem).

P / partially_solved | AMR 问题清单汇编

(Fast word problem). Is there a sub-quadratic time algorithm to solve the word problem in Modg? One might guess that n logn is possible here, as there is such an algorithm for certain relatively (strongly) hyperbolic groups (see [ F a3]), and mapping class groups are at least weakly hyperbolic, as proven by Masur-Minsky (Theorem 1.3 of [ MM1]). The conjugacy problem for Mod g is harder. The original algorithm of Hemion [ He] seems to give no reasonable (even exponential) time bound. One refinement of the problem would be to prove that Mod g is biautomatic. However, even a biautomatic structure gives only an exponential time algorithm to solve the conjugacy problem. Another approach to solving the conjugacy problem is the following.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09954 | AMR-109-0281

### Question 3.1 - If n≥ 4 and g≥ 1, does every homomorphism from Aut(Fn) to Mod±(Sg) have finite image?

O / open | AMR 问题清单汇编

If n≥ 4 and g≥ 1, does every homomorphism from Aut(Fn) to Mod±(Sg) have finite image? By [ 21], one cannot obtain homomorphisms with infinite image unless Mod ±(Sg) contains the symmetric group Σ n+1. For large enough genus, you can realize any symmetric group; but the order of a finite group of symmetries is at most 84g-6, so here one needs 84 g- 6≥ (n + 1)!. 326 M. Bridson and K. Vogtmann There are no injective maps from Aut( Fn) to mapping class groups. This follows from the result of Brendle and Hamidi-Tehrani that we quoted earlier. For certain g one can construct homomorphisms Aut(F3)→Mod±(Sg) with infinite image, but we do not know the minimal such g.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09955 | AMR-109-0032

### Question 3.11 - (Isomorphism problem for subgroups).

S / solved | AMR 问题清单汇编

(Isomorphism problem for subgroups). Is the isomorphism problem for the collection of finitely presented subgroups of Modg solvable? Note that the isomorphism problem is not solvable for the collection of all finitely generated linear groups, nor is it even solvable for the collection of finitely generated subgroups of GL( n, Z). There are many other algorithmic questions one can ask; we mention just one more.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09956 | AMR-103-0062

### Question 3.12 - (C.

O / open | AMR 问题清单汇编

(C. Lescop) Is the Kontsevich integral of a (zero-framed) knot equal to the Chern-Simons series of configuration space integrals of the same knot (with Gauss integral 0)?

背景链接：https://msp.org/gtm/2002/04/gtm-2002-04-024s.pdf

09957 | AMR-109-0033

### Question 3.12 - Is there an algorithm to decide whether or not a given subgroup H <Modg is freely indecomposable?

O / open | AMR 问题清单汇编

Is there an algorithm to decide whether or not a given subgroup H <Modg is freely indecomposable? Whether or not H splits over Z? 28 B. Farb

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09958 | AMR-109-0034

### Question 3.13 - (Rational growth).

O / open | AMR 问题清单汇编

(Rational growth). Does Modg have rational growth function with respect to some set of generators? with respect to every set of generators? Of course one can also ask the same question for any finitely generated subgroup of Mod g, for examplelg. Note that the existence of an automatic structure is not known to imply rationality of growth (even for one generating set); one needs in addition the property that the automatic structure consist of geodesics. Unfortunately Mosher’s automatic structure does not satisfy this stronger condition. It is natural to ask for other recursive patterns in the Cayley graph of Mod g. To be more precise, let P denote a property that elements of Mod g might or might not have. For example, P might be the property of being finite order, of lying in a fixed subgroup H <Modg, or of being pseudo-Anosov. Now let cP (r) = # {BModg (r)∩{x∈ Modg:x has P}} We now define the growth series for the property P, with respect to a fixed generating set for Γ, to be the power series fP (z) = ∞∑ i=0 cP (i)zi

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09959 | AMR-109-0035

### Question 3.14 - (Rational growth for properties).

O / open | AMR 问题清单汇编

(Rational growth for properties). For which properties P is the function fP is rational? Densities. For any subset S⊂ Modg, it is natural to ask how common elements of S are in Mod g. There are various ways to interpret this question, and the answer likely depends in a strong way 2. Some problems on mapping class groups and moduli space 29 on the choice of interpretation 4. One way to formalize this is via the density d(S) of S in Mod g, where d(S) = lim r→∞ #[B(r)∩S] #B(r) (9) where B(r) is the number of elements of Mod g in the ball of radius r, with respect to a fixed set of generators. While for subgroups H <Modg the number d(H) itself may depend on the choice of generating sets for H and Mod g, it is not hard to see that the (non)positivity of d(H) does not depend on the choices of generating sets. As the denominator and (typically) the numerator in (9) are exponential, one expects that d(S) = 0 for most S. Thus is it natural to replace both the numerator and denominator of (9) with their logarithms; we denote the corresponding limit as in (9) by dlog(S), and we call this the logarithmic density of S in Mod g. The following is one interpretation of a folklore conjecture.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09960 | AMR-109-0125

### Question 3.2 - .

P / partially_solved | AMR 问题清单汇编

. Is κ2i = 0 in H ∗(BT∞; Q)? Very little is known about the cohomology of the Torelli group past dimension 1, and as far as I know it might be possible (although hard to believe) that BT∞→ hofiber(BΓ+ ∞→B Sp(Z)+) (5) is identically zero on cohomology. The calculations in [ 4] show that H ∗(BΓ∞; Z) contains a wealth of torsion classes and that there are torsion classes of any order. Some of these classes might be of interest in other areas of mathematics. But the difficulty is that the description of torsion classes from [ 4] is indirect and complicated, and very hard to communicate to non-specialists.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09961 | AMR-109-0245

### Question 3.2 - Are there are only finitely many Γg-conjugacy classes of purely pseudo-Anosov surface subgroups of any fixed genus?

S / solved | AMR 问题清单汇编

Are there are only finitely many Γg-conjugacy classes of purely pseudo-Anosov surface subgroups of any fixed genus? Of course given that Question 1.1 is open, the answer to Question 3.2 could be zero.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09962 | AMR-109-0282

### Question 3.2 - Let Γ be an irreducible lattice in a semisimple Lie group of R-rank at least 2.

S / solved | AMR 问题清单汇编

Let Γ be an irreducible lattice in a semisimple Lie group of R-rank at least 2. Does every homomorphism from Γ to Out(Fn) have finite image? This is known for non-uniform lattices (see [ 16]; it follows easily from the Kazdhan-Margulis finiteness theorem and the fact that solvable subgroups of Out( Fn) are virtually abelian [ 5]). Farb and Masur provided a positive answer to the analogous question for maps to mapping class groups [32]. The proof of their theorem was based on results of Kaimanovich and Masur [ 56] concerning random walks on Teichm¨ uller space. (See [ 54] and, for an alternative approach, [ 6].)

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09963 | AMR-109-0283

### Question 3.3 - Is there a theory of random walks on Outer space similar to that of Kaimanovich and Masur for Teichm¨ uller space?

S / solved | AMR 问题清单汇编

Is there a theory of random walks on Outer space similar to that of Kaimanovich and Masur for Teichm¨ uller space? Perhaps the most promising approach to Question 17 is via bounded cohomology, following the template of Bestvina and Fujiwara’s work on subgroups of the mapping class group [ 6].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09964 | AMR-109-0246

### Question 3.3 - LetG~=π1(S2g)→ Γg be the injection given by Theorem 3.1.

O / open | AMR 问题清单汇编

LetG~=π1(S2g)→ Γg be the injection given by Theorem 3.1. Consider ∂∞(G) which can be canonically identified with the circle at infinity of the universal cover ˜S2g~= H2 of S2g. Does there exist a continuous G-equivariant map ∂∞(G)→ PML0(Σ)? 3.5. In the context of 3-manifold topology, given an essential surface Σ in a finite volume hyperbolic 3-manifold M, a natural question is whether there is a finite cover of M to which Σ lifts to an embedded surface. A group theoretic property that is closely related to this question is LERF, which now define. If Γ is a group, and H a subgroup of Γ, then Γ is called H-separable if for every g∈G\H, there is a subgroup K of finite index in Γ such that H⊂K but g /∈K. Γ is called LERF or subgroup separable if Γ is H-separable for all finitely generated subgroups H. This has been widely studied in the setting of low-dimensional topology (see [ 1] and [ 36] for example). Indeed, it is often the case that one does not need the full power of LERF for applications to hyperbolic manifolds, separating geometrically finite subgroups often suffices; this led to the property of GFERF, that is separable on all geometrically finite subgroups. 280 A.W. Reid Now it is known that the groups Γ g are not LERF whenever g≥ 2 since they contain F2×F2, however in analogy with the case for hyperbolic manifolds we pose (for g = 1 it is known that Γ 1 is LERF since it is virtually free):

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09965 | AMR-109-0025

### Question 3.4 - (Almost convexity).

O / open | AMR 问题清单汇编

(Almost convexity). Does there exist a finite generating set for Modg for which it is almost convex? One would also like to know the answer to this question for various subgroups of Mod g. Here is a related, but different, basic question about the geometry of Teichm¨ uller space.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09966 | AMR-109-0127

### Question 3.4 - .

S / solved | AMR 问题清单汇编

. Is Z×BAut(F∞)+ homotopy equivalent to Ω ∞S∞? 4. The interplay between homotopy theory and the mapping class group is inspired by the study of conformal field theories, seen in [ 18] as representations of the surface category. The space of morphisms in Segal’s surface category is the moduli space of Riemann surfaces with parametrized boundary circles. Teichm¨ uller theory implies that this morphism space is homotopy equivalent to ⊔BΓg,b, g≥ 0,b≥ 0. Under suitable stability conditions, in our case Harer’s stability theorem, there is a close rela- tionship between the loop space of the classifying space of a category and its space of morphisms. 98 I. Madsen This was used by Tillmann to show that BΓ+ ∞ is an infinite loop space, [ 19], and in [ 13] to show that the map α of Theorem 2.1 is an infinite loop map. The situation is similar to that of algebraic K-theory, see e.g. [ 20]. The surface category is a special case of the cobordism category Cd that exists in all dimen- sions d≥ 1. The objects of Cd are oriented closed ( d- 1)-dimensional submanifolds of R∞ and morphisms are oriented compact d-dimensional submanifolds of [0,a ]× R∞, 0 <a, that meet the walls{0,a}× R∞ transversely, see [ 6] for details. For d = 2, Cd is homotopy equivalent to Segal’s surface category, i.e. they have homotopy equivalent classifying spaces, [ 13].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09967 | AMR-109-0284

### Question 3.4 - If a subgroup G⊂Out(Fn) is not virtually abelian, then is H 2 b (G; R) infinite dimensional?

P / partially_solved | AMR 问题清单汇编

If a subgroup G⊂Out(Fn) is not virtually abelian, then is H 2 b (G; R) infinite dimensional? Ifm≥n then there are obvious embeddings GL( n, Z)→ GL(m, Z) and Aut( Fn)→ Aut(Fm), but there are no obvious embeddings Out( Fn)→ Out(Fm). Bogopolski and Puga [ 10] have shown that, for m = 1 + ( n- 1)kn, where k is an arbitrary natural number coprime to n- 1, there is in fact an embedding, by restricting automorphisms to a suitable characteristic subgroup of Fm.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09968 | AMR-109-0247

### Question 3.4 - Is Γg GFERF for g≥ 2?

O / open | AMR 问题清单汇编

Is Γg GFERF for g≥ 2?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09969 | AMR-109-0285

### Question 3.5 - For which values of m does Out(Fn) embed in Out(Fm)?

P / partially_solved | AMR 问题清单汇编

For which values of m does Out(Fn) embed in Out(Fm)? What is the minimal such m, and is it true for all sufficiently large m? It has been shown that when n is sufficiently large with respect to i, the homology group Hi(Out(Fn), Z) is independent of n [50, 51 ].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09970 | AMR-109-0026

### Question 3.5 - Is Teich(Σg), endowed with the Teichm¨ uller metric, almost convex?

O / open | AMR 问题清单汇编

Is Teich(Σg), endowed with the Teichm¨ uller metric, almost convex? Note that Cannon proves in [ Ca] that fundamental groups of closed, negatively curved man- ifolds are almost convex with respect to any generating set. He conjectures that this should generalize both to the finite volume case and to the nonpositively curved case.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09971 | AMR-109-0248

### Question 3.5 - LetH be a convex cocompact subgroup of Γg.

O / open | AMR 问题清单汇编

LetH be a convex cocompact subgroup of Γg. Is Γg H-separable? Just focusing on surface subgroups, we can ask:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09972 | AMR-109-0286

### Question 3.6 - Is there a map Out(Fn)→ Out(Fm) that induces an isomorphism on homology in the stable range?

O / open | AMR 问题清单汇编

Is there a map Out(Fn)→ Out(Fm) that induces an isomorphism on homology in the stable range? A number of the questions in this section and (2.2) ask whether certain quotients of Out( Fn) or Aut( Fn) are necessarily finite. The following quotients arise naturally in this setting: define Q(n,m ) to be the quotient of Aut( Fn) by the normal closure of λm, where λ is the Nielsen move defined on a basis {a1,...,a n} bya1↦→a2a1. (All such Nielsen moves are conjugate in Aut( Fn), so the choice of basis does not alter the quotient.) The image of a Nielsen move in GL( n, Z) is an elementary matrix and the quotient of GL( n, Z) by the normal subgroup generated by the m-th powers of the elementary matrices is the finite group GL( n, Z/m). But Bridson and Vogtmann [ 21] showed that if m is sufficiently large then 20. Automorphism groups of free, surface and free abelian groups 327 Q(n,m ) is infinite because it has a quotient that contains a copy of the free Burnside group B(n- 1,m ). Some further information can be gained by replacing B(n- 1,m ) with the quotients of Fn considered in subsection 39.3 of A.Yu. Ol’shanskii’s book [ 73]. But we know very little about the groups Q(n,m ). For example:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09973 | AMR-109-0249

### Question 3.6 - LetH be a surface subgroup of Γg.

O / open | AMR 问题清单汇编

LetH be a surface subgroup of Γg. Is Γg H-separable? For recent progress on various classes of subgroups of Γ g that are separable, we refer the reader to [ 24].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09974 | AMR-109-0287

### Question 3.7 - For which values of n and m is Q(n,m ) infinite?

P / partially_solved | AMR 问题清单汇编

For which values of n and m is Q(n,m ) infinite? Is Q(3, 5) infinite?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09975 | AMR-109-0288

### Question 3.8 - Can Q(n,m ) have infinitely many finite quotients?

P / partially_solved | AMR 问题清单汇编

Can Q(n,m ) have infinitely many finite quotients? Is it residually finite?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09976 | AMR-109-0030

### Question 3.9 - Does every finitely presented subgroup H < Modg have solvable conjugacy problem?

S / solved | AMR 问题清单汇编

Does every finitely presented subgroup H < Modg have solvable conjugacy problem? is it combable? automatic? Note that every finitely-generated subgroup of a group with solvable word problem has solvable word problem. The same is not true for the conjugacy problem: there are subgroups of GL( n, Z) with unsolvable conjugacy problem; see [ Mi]. It is not hard to see that Mod g, like GL( n, Z), has finitely many conjugacy classes of finite subgroups. However, we pose the following.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09977 | AMR-109-0261

### Question 4 - How much of this paper can be generalized to the Birman-Wenzl-Murakami algebra?

O / open | AMR 问题清单汇编

How much of this paper can be generalized to the Birman-Wenzl-Murakami algebra? In this direction, John Enyang [ Eny04] has shown that the Birman-Murakami-Wenzl algebra is a cellular algebra, and used this to give a definition of its irreducible representations similar to the approach in Section 7. Next we would like to generalize [ Big04] to the Birman-Wenzl-Murakami algebra.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09978 | AMR-109-0289

### Question 4.1 - Can one detect the growth of a surface or free-group homomorphism by its action on the homology of a characteristic s…

P / partially_solved | AMR 问题清单汇编

Can one detect the growth of a surface or free-group homomorphism by its action on the homology of a characteristic subgroup of finite index? Notice that one has to pass to a subgroup of finite index in order to have any hope because automorphisms of exponential growth can act trivially on homology. A. Piggott [ 74] has an- swered the above question for free-group automorphisms of polynomial growth, and linear-growth automorphisms of surfaces are easily dealt with, but the exponential case remains open in both settings. 328 M. Bridson and K. Vogtmann Finer questions concerning growth are addressed in the on-going work of Handel and Mosher [43]. They explore, for example, the implications of the following contrast in behaviour between surface automorphisms and free-group automorphisms: in the surface case the exponential growth rate of a pseudo-Anosov automorphism is the same as that of its inverse, but this is not the case for iwip free-group automorphisms. For mapping tori of automorphisms of free abelian groups G = Zn ⋊φ Z, the following con- ditions are equivalent (see [ 17]): G is automatic; G is a CAT(0) group 3; G satisfies a quadratic isoperimetric inequality. In the case of mapping tori of surface automorphisms, all mapping tori satisfy the first and last of these conditions and one understands exactly which Sg ⋊ Z are CAT(0) groups. Brady, Bridson and Reeves [ 12] show that there exist mapping tori of free-group automor- phismsF ⋊ Z that are not automatic, and Gersten showed that some are not CAT(0) groups [ 36]. On the other hand, many such groups do have these properties, and they all satisfy a quadratic isoperimetric inequality [ 18].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09979 | AMR-109-0250

### Question 4.1 - Does there exist a closed hyperbolic 4-manifold X that is the total space of a smooth fiber bundle Σg→X→ Σh?

O / open | AMR 问题清单汇编

Does there exist a closed hyperbolic 4-manifold X that is the total space of a smooth fiber bundle Σg→X→ Σh? We will call such an X a surface bundle over a surface. The following is well-known.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09980 | AMR-109-0256

### Question 4.10 - Let Γ be a lattice in SO(m, 1), m≥ 3 or SU(q, 1), q≥ 2 which is admissable for Γg.

P / partially_solved | AMR 问题清单汇编

Let Γ be a lattice in SO(m, 1), m≥ 3 or SU(q, 1), q≥ 2 which is admissable for Γg. Does Γ inject in Γg? Can there be purely pseudo-Anosov representations? If such an injection exists does there exist a continuous Γ-equivariant map ∂∞(Γ)→ PML0(Σ)? More generally, for a fixed Σ and hence fixed vcd, a further natural generalization of the discussion here is: 284 A.W. Reid

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09981 | AMR-109-0257

### Question 4.11 - Which 1-ended admissable word hyperbolic groups G inject in Γg (as purely pseudo-Anosov subgroups)?

P / partially_solved | AMR 问题清单汇编

Which 1-ended admissable word hyperbolic groups G inject in Γg (as purely pseudo-Anosov subgroups)? If such an injection exists does there exist a continuous G-equivariant map ∂∞(G)→ PML0(Σ)?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09982 | AMR-109-0290

### Question 4.2 - Classify those φ∈ Aut(Fn) for which Fn ⋊φ Z is automatic and those for which it is CAT(0).

P / partially_solved | AMR 问题清单汇编

Classify those φ∈ Aut(Fn) for which Fn ⋊φ Z is automatic and those for which it is CAT(0). Of central importance in trying to understand mapping tori is:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09983 | AMR-109-0129

### Question 4.3 - .

O / open | AMR 问题清单汇编

. Is there a geometric map from BC2 into topological cyclic homology of a point? A recent manuscript by Kevin Costello, [ 3], relates conformal field theories to (the linear) Hochschild homology, and this might well be the place to start. In the same paper Costello announces a con- struction of the Deligne-Mumford compactification of the moduli space of Riemann surface in 6. The mapping class groups and homotopy theory 99 terms of the open moduli space. There is also preliminary work of Søren Galatius and Yasha Eliashberg which gives a homotopical description of a partial compactification of the stable mod- uli space.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09984 | AMR-109-0251

### Question 4.3 - Does there exist a closed hyperbolic 4-manifold that is a surface bundle over a surface where the genus of the fiber…

O / open | AMR 问题清单汇编

Does there exist a closed hyperbolic 4-manifold that is a surface bundle over a surface where the genus of the fiber and base is 2? 4.2. Some evidence for a negative answer to Question 4.1 is given in [ 21] where the following conjecture is stated (this is a special case of a more general conjecture on vanishing of Seiberg- Witten invariants). We refer to [ 37] and [ 14] for definitions.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09985 | AMR-109-0291

### Question 4.3 - Is there an alogrithm to decide isomorphism among groups of the form F ⋊ Z.

O / open | AMR 问题清单汇编

Is there an alogrithm to decide isomorphism among groups of the form F ⋊ Z. In the purest form of this question one is given the groups as finite presentations, so one has to address issues of how to find the decomposition F ⋊ Z and one has to combat the fact that this decomposition may not be unique. But the heart of any solution should be an answer to:

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09986 | AMR-109-0042

### Question 4.3 - LetY be any finite cover of Mg, and let f:Y →Y be a finite order homeo- morphism.

O / open | AMR 问题清单汇编

LetY be any finite cover of Mg, and let f:Y →Y be a finite order homeo- morphism. If f is homotopic to the identity, must f equal the identity?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09987 | AMR-109-0130

### Question 4.4 - .

O / open | AMR 问题清单汇编

. Can one generalize Theorem 2.1 to the Deligne-Mumford compactification of the moduli space of Riemann surfaces?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09988 | AMR-109-0292

### Question 4.4 - Is the conjugacy problem solvable in Out(Fn)?

P / partially_solved | AMR 问题清单汇编

Is the conjugacy problem solvable in Out(Fn)? Martin Lustig posted a detailed outline of a solution to this problem on his web page some years ago [ 65], but neither this proof nor any other has been accepted for publication. This problem is of central importance to the field and a clear, compelling solution would be of great interest. The conjugacy problem for mapping class groups was shown to be solvable by Hemion [52], and an effective algorithm for determining conjugacy, at least for pseudo-Anosov mapping classes, was given by Mosher [ 70]. The isomorphism problem for groups of the form Sg ⋊ Z can be viewed as a particular case of the solution to the isomorphism problem for fundamental groups of geometrizable 3-manifolds [ 76]. The solvability of the conjugacy problem for GL( n, Z) is due to Grunewald [ 39]

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09989 | AMR-109-0253

### Question 4.6 - Does there exist a closed hyperbolic 4-manifold X for which no finite cover admits a symplectic structure?

O / open | AMR 问题清单汇编

Does there exist a closed hyperbolic 4-manifold X for which no finite cover admits a symplectic structure? (ie X is not virtually symplectic.) We note that it is easy to construct non-symplectic hyperbolic 4-manifolds. For example it is easy to see that the Davis manifold D (see [ 9]) admits no symplectic structure using the following simple parity rule (see [ 14] Corollary 10.1.10): Suppose that (M,ω ) is a closed symplectic 4-manifold. Then 1-b1(M ) + b+ 2 (M ) is even. For the Davis manifold we have from [ 35] that b1(D) = 24 and b2(D) = 72. Since the signature of D is zero, b+ 2 (D) = 36, and so the parity condition fails. 4.3. A natural weakening of Question 4.1 is the following.

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09990 | AMR-109-0254

### Question 4.7 - Forg,h≥ 2, does there exist a short exact sequence: 1→π1(Σg)→ Γ→π1(Σh)→ 1 for which Γ is a word hyperbolic group?

O / open | AMR 问题清单汇编

Forg,h≥ 2, does there exist a short exact sequence: 1→π1(Σg)→ Γ→π1(Σh)→ 1 for which Γ is a word hyperbolic group? Arguing as in the proof of Theorem 4.2, it follows that such an extension defines a purely pseudo-Anosov surface subgroup of Γ g. However, even in this case, little is known. We make two comments in this regard. 1. One result is that Γ cannot be the fundamental group of a closed complex hyperbolic surface. As described in [ 20], this follows from [ 25], on showing that if Γ is as decribed, then there is a non-singular holomorphic fibration X = H2 C/Γ→ Σh that induces the short exact sequence (see also [ 17]). 2. The following idea to construct purely pseudo-Anosov surface subgroups was described to me by Ian Agol. In [ 40], Thurston proves the following result (see [ 40] for terminology and further details). 17. Surface subgroups of mapping class groups 283

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09991 | AMR-109-0255

### Question 4.9 - Does there exist a cocompact Fuchsian subgroup of Δ5 that misses the com- pactification locus?

O / open | AMR 问题清单汇编

Does there exist a cocompact Fuchsian subgroup of Δ5 that misses the com- pactification locus? Given such a Fuchsian subgroup F <Δ5, Agol produces a purely pseudo-Anosov subgroup in some Γ g using a branched cover construction. 4.4. One generalization of looking for surface subgroups of Γ g is to look for injections of (cocompact) lattices in Lie groups into Γ g. If the lattices are superrigid the image of such a lattice in Γ g is necessarily finite (see [ 12] and [ 43]), and so it follows (cf. Theorem 2 of [ 43]), that the only lattices that can admit a faithful representation (or even an infinite representation) into Γ g are lattices in SO( m, 1), m≥ 2 or SU( q, 1), q≥ 1. Indeed, since solvable subgroups of Γ g are virtually abelian [ 4], this observation also excludes non-cocompact lattices of SU( q, 1) for q≥ 2 from injecting. In addition, there is a simple obstruction to injecting certain of these lattices, or indeed for any group. Namely if a finitely generated group G admits an injection into Γ g, then vcd(G)≤ vcd(Γg) (see [ 5]). If the vcd of a group G satisfies the above inequality, then we call G admissable. The vcd’s of the groups Γ g are known to be 4 g- 5 when g≥ 2 [ 16]. Motivated by this discussion we pose.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09992 | AMR-109-0262

### Question 5 - Is there a homological definition of representations of the Birman-Wenzl- Murakami algebra?

O / open | AMR 问题清单汇编

Is there a homological definition of representations of the Birman-Wenzl- Murakami algebra? I believe the answer to this is yes. Furthermore, the homological construction suggests a new algebra Zn, which would further generalize the Iwahori-Hecke and Birman-Wenzl-Murakami algebras. I will conclude this paper with a definition of Zn and some related open questions. I hope these might be amenable to some combinatorial computations, even without the homological motivation, which is currently unclear and unpublished. We use the notation σi1...ik as shorthand for σi1...σ ik, and ¯ σi1...ik for σ-1 i1...ik. Define the following elements of RBn. X2 = q¯σ1 + 1-q-σ1 X3 = ( q2¯σ21-σ12)X2 X4 = ( q3¯σ321-σ123)X3... Xn = ( qn-1¯σ(n-1)...1-σ1...(n-1))Xn-1 Then Zn is the algebra RBn modulo the following relations. (q¯σ2 + 1-q-σ2)X2 = ( q¯σ21-σ12)X2, (q2¯σ32-σ23)X3 = ( q2¯σ321-σ123)X3, (q3¯σ432-σ234)X4 = ( q3¯σ4321-σ1234)X4,... (qn-1¯σn...2-σ2...n)Xn = ( qn-1¯σn...1-σ1...n)Xn. 316 S. Bigelow Note that Hn(1,-q) is the quotient of Zn by the relation X2 = 0. Also the Birman-Wenzl- Murakami algebra is the quotient of Zn by the relations X3 = 0 and σ1X2 =tX2. The following basically asks if Zn is bigger than the Birman-Wenzl-Murakami algebra.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09993 | AMR-109-0293

### Question 5.1 - Where precisely does the rational homology of Aut(Fn) stabilize?

O / open | AMR 问题清单汇编

Where precisely does the rational homology of Aut(Fn) stabilize? And for Out(Fn)? There are only two known non-trivial classes in the (unstable) rational homology of Out( Fn) [49, 26 ]. However, Morita [ 69] has defined an infinite series of cycles, using work of Kontsevich which identifies the homology of Out( Fn) with the cohomology of a certain infinite-dimensional Lie algebra. The first of these cycles is the generator of H4(Out(F4); Q) ∼= Q, and Conant and Vogtmann showed that the second also gives a non-trivial class, in H8(Out(F6); Q) [ 26]. Both Morita and Conant-Vogtmann also defined more general cycles, parametrized by odd-valent graphs.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09994 | AMR-109-0062

### Question 5.18 - For which k≥ 1 is it true that Aut(Ig(k)) = Mod ± g?

S / solved | AMR 问题清单汇编

For which k≥ 1 is it true that Aut(Ig(k)) = Mod ± g? that Comm(Ig(k)) = Mod± g? Theorem 5.17 answers the question for k = 1, 2. It would be remarkable if all of Mod g could be reconstructed from subgroups deeper down in its lower central series.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09995 | AMR-109-0050

### Question 5.2 - (Morita).

P / partially_solved | AMR 问题清单汇编

(Morita). Is H1(Kg, Z) finitely generated for g≥ 3? Note that Birman-Craggs-Johnson (see, e.g., [ BC, Jo1 ]) and Morita [ Mo2] have found large abelian quotients of Kg. The proof in [ BF] of Theorem 5.1, suggests an approach to answering to Question 5.2. Let me briefly describe the idea. Following the the outline in [ McM], we first find an action of Kg on the 2. Some problems on mapping class groups and moduli space 37 first homology of a certain abelian cover Y of Σ g; this action respects the structure of H1(Y, Z) as a module over the Galois group of the cover. The crucial piece is that we are able to reduce this to a representation ρ:Kg→ SL2(Z[t,t -1]) on the special linear group over the ring of integral laurent polynomials in one variable. This group acts on an associated Bruhat-Tits-Serre tree, and one can then analyze this action using combinatorial group theory. One might now try to answer Question 5.2 in the negative by systematically computing more elements in the image of ρ, and then analyzing more closely the action on the tree for SL 2. One potentially useful ingredient is a theorem of Grunewald-Mennike-Vaserstein [ GMV] which gives free quotients of arbitrarily high rank for the group SL 2(Z[t]) and the group SL 2(K[s,t ]), where K is an arbitrary finite field. Since we know for g ≥ 3 that Ig is finitely generated and Kg is not, it is natural to ask about the subgroups interpolating between these two. To be precise, consider the exact sequence (11). Corresponding to each subgroup L <∧3H/H is its pullback π-1(L). The lattice of such subgroups L can be thought of as a kind of interpolation between Ig andKg.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09996 | AMR-109-0294

### Question 5.2 - Are Morita’s original cycles non-trivial in homology?

P / partially_solved | AMR 问题清单汇编

Are Morita’s original cycles non-trivial in homology? Are the generalizations due to Morita and to Conant and Vogtmann non-trivial in homology? No other classes have been found to date in the homology of Out( Fn), leading naturally to the question of whether these give all of the rational homology.

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09997 | AMR-109-0066

### Question 5.22 - (Lie algebra for the Johnson filtration).

O / open | AMR 问题清单汇编

(Lie algebra for the Johnson filtration). Is hg a finitely presented Lie algebra? If so, give an explicit finite presentation for it.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09998 | AMR-109-0295

### Question 5.3 - Do the Morita classes generate all of the rational homology of Out(Fn)?

P / partially_solved | AMR 问题清单汇编

Do the Morita classes generate all of the rational homology of Out(Fn)? The maximum dimension of a Morita class is about 4 n/3. Morita's cycles lift naturally to Aut(Fn), and again the first two are non-trivial in homology. By Galatius' result, all of these cycles must eventually disappear under the stabilization map Aut( Fn)→ Aut(Fn+1). Conant and Vogtmann show that in fact they disappear immediately after they appear, i.e. one application of the stabilization map kills them [ 25]. If it is true that the Morita classes generate all of the rational homology of Out( Fn) then this implies that the stable range is significantly lower than the current bound. We note that Morita has identified several conjectural relationships between his cycles and various other interesting objects, including the image of the Johnson homomorphism, the group of homology cobordism classes of homology cylinders, and the motivic Lie algebra associated to the algebraic mapping class group (see Morita's article in this volume). Since the stable rational homology of Out( Fn) is trivial, the natural maps from mapping class groups to Out( Fn) and from Out( Fn) to GL( n, Z) are of course zero. However, the unstable homology of all three classes of groups remains largely unkown and in the unstable range these maps might well be nontrivial. In particular, we note that H8(GL(6, Z); Q)~= Q [30]; this leads naturally to the question 330 M. Bridson and K. Vogtmann

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

09999 | AMR-109-0296

### Question 5.4 - Is the image of the second Morita class in H8(GL(6, Z); Q)) non-trivial?

O / open | AMR 问题清单汇编

Is the image of the second Morita class in H8(GL(6, Z); Q)) non-trivial? For further discussion of the cohomology of Aut( Fn) and Out( Fn) we refer to [ 81].

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

10000 | AMR-109-0263

### Question 6 - DoesX3 equal 0 in Zn?

P / partially_solved | AMR 问题清单汇编

DoesX3 equal 0 in Zn? Presumably some extra relations should be added to Zn, such as σ1X2 = tX2, or something more general.

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10001 | AMR-109-0297

### Question 6.1 - Is there a set of simple Steinberg-type relations for the mapping class group?

O / open | AMR 问题清单汇编

Is there a set of simple Steinberg-type relations for the mapping class group? There is also a presentation of Aut( Fn) coming from the action of Aut( Fn) on the subcomplex of Auter space spanned by graphs of degree at most 2. This is simply-connected by [ 48], so Brown's method [ 22] can be used to write down a presentation. The vertex groups are stabilizers of marked graphs, and the edge groups are the stabilizers of pairs consisting of a marked graph and a forest in the graph. The quotient of the subcomplex modulo Aut( Fn) can be computed explicitly, and one finds that Aut( Fn) is generated by the (finite) stabilizers of seven specific marked graphs. In addition, all of the relations except two come from the natural inclusions of edge stabilizers into vertex stabilizers, i.e. either including the stabilizer of a pair (graph, forest) into the stabilizer of the graph, or into the stabilizer of the quotient of the graph modulo the forest. Thus the whole group is almost (but not quite) a pushout of these finite subgroups. In the terminology of Haefliger (see [ 19], II.12), the complex of groups is not simple.

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10002 | AMR-109-0071

### Question 6.2 - (Rigidity of the Mod g,1 action on S1).

P / partially_solved | AMR 问题清单汇编

(Rigidity of the Mod g,1 action on S1). Is any faithful action ρ: Mod g,1→ Homeo+(S1) conjugate in Homeo+(S1) to the standard action, given in (19)? What about the same question for finite index subgroups of Modg,1? Perhaps there is a vastly stronger, topological dynamics characterization of Mod g,1 inside Homeo+(S1), in the style of the Convergence Groups Conjecture (theorem of Tukia, Casson- Jungreis and Gabai), with "asymptotically source - sink" being replaced here by "asymptotically source - sink - ⋯ - source - sink", or some refinement/variation of this. Now, the group of lifts of elements of Homeo +(S1) to homeomorphisms of R gives a central extension 1→ Z→ ˜Homeo(S1)→ Homeo+(S1)→ 1 which restricts via (19) to a central extension 1→ Z→ ˜Modg,1→ Modg,1→ 1 (20) Note that Mod g,1 has torsion. Since ˜Homeo(S1)⊂ Homeo+(R) which clearly has no torsion, it follows that (20) does not split. In particular the extension (20) gives a nonvanishing class ξ∈ H 2(Modg,1, Z). Actually, it is not hard to see that ξ is simply the "euler cocycle", which assigns to any pointed map Σ h→M g the euler class of the pullback bundle of the "universal circle bundle" over Mg. The torsion in Mod g,1 and in Mod g preclude each from having a left-ordering, or acting faithfully on R. As far as we know this is the only obstruction; it disappears when one passes to appropriate finite index subgroups.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

10003 | AMR-109-0298

### Question 6.2 - Can Out(Fn) and Mod±(Sg) be obtained as a pushout of a finite subsystem of their finite subgroups, i.e.

O / open | AMR 问题清单汇编

Can Out(Fn) and Mod±(Sg) be obtained as a pushout of a finite subsystem of their finite subgroups, i.e. is either the fundamental group of a developable simple complex of finite groups on a 1-connected base?

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10004 | AMR-109-0072

### Question 6.3 - (orderability).

O / open | AMR 问题清单汇编

(orderability). Does Modg,g ≥ 2 have some finite index subgroup which acts faithfully by homeomorphisms on S1? Does either Modg or Modg,1 have a finite index subgroup which acts faithfully by homeomorphisms on R? Note that Thurston proved that braid groups are orderable. Since Ig andIg,1 are residually torsion-free nilpotent, they are isomorphic to a subgroup of Homeo +(R); in fact one can show 8One can see this by averaging any Riemannian metric on S1 by the group action. 48 B. Farb that (20) splits when restricted to Ig,1. On the other hand, Witte [ Wi] proved that no finite index subgroup of Sp(2 g, Z) acts faithfully by homoeomorphisms on S1 or on R. Non-residual finiteness of the universal central extension. The Lie group Sp(2 g, R) has infinite cyclic fundamental group. Its universal cover ˜Sp(2g, R) gives a central extension 1→ Z→ ˜Sp(2g, R)→ Sp(2g, R)→ 1 which restricts to a central extension 1→ Z→ ˜Sp(2g, Z)→ Sp(2g, Z)→ 1 (21) The cocycle ζ∈H 2(Sp(2g, Z), Z) defining the extension (21) is nontrivial and bounded; this comes from the fact that it is proportional to the Kahler class on the corresponding locally symmetric quotient (which is a K(π, 1) space). Deligne proved in [ De] that ˜Sp(2g, Z) is not residually finite. Since there is an obvious surjection of exact sequences from (19) to (21), and since both central extensions give a bounded cocycle, one begins to wonder about the following.

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10005 | AMR-109-0299

### Question 6.3 - Establish finiteness properties of the kernel IA(n) of the map from Out(Fn) to GL(n, Z).

O / open | AMR 问题清单汇编

Establish finiteness properties of the kernel IA(n) of the map from Out(Fn) to GL(n, Z). In particular, determine whether IA(n) is finitely presentable for n> 3. The subgroup IA( n) is analogous to the Torelli subgroup of the mapping class group of a surface, which also remains quite mysterious in spite of having been extensively studied.

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10006 | AMR-109-0073

### Question 6.4 - ((Non)residual finiteness).

S / solved | AMR 问题清单汇编

((Non)residual finiteness). Is the (universal) central extension ˜Modg,1 of Modg,1 residually finite, or not? Note that an old result of Grossman states that Mod g and Mod g,1 are both residually finite. The group Sp(2 g, Z) is easily seen to be residually finite; indeed the intersection of all congruence subgroups of Sp(2 g, Z) is trivial.

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10007 | AMR-109-0075

### Question 6.6 - (Sections over finite index subgroups).

O / open | AMR 问题清单汇编

(Sections over finite index subgroups). Does the natural map Homeo+(Σg)→ Modg have a section over a finite index subgroup of Modg, or not? Of course the ideas in [ Mar] are likely to be pertinent. Answers to Problem 6.5 even for specific subgroups (e.g. for Ig or more generally Ig(k)) would be interesting. It also seems reasonable to believe that the existence of sections is affected greatly by the degree of smoothness one requires. Instead of asking for sections in the above questions, one can ask more generally whether there are any actions of Mod g on Σ g.

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10008 | AMR-109-0076

### Question 6.7 - Does Modg or any of its finite index subgroups have any faithful action by homeomorphisms on Σg?

O / open | AMR 问题清单汇编

Does Modg or any of its finite index subgroups have any faithful action by homeomorphisms on Σg?

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10009 | AMR-109-0264

### Question 7 - What extra relations should be added to Zn to make it finite-dimensional?

O / open | AMR 问题清单汇编

What extra relations should be added to Zn to make it finite-dimensional?

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10010 | AMR-109-0077

### Question 7.1 - Does limg→∞gL(Modg) exist?

O / open | AMR 问题清单汇编

Does limg→∞gL(Modg) exist? Another basic open question is the following.

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10011 | AMR-109-0300

### Question 7.1 - What are the Dehn functions of Aut(Fn) and Out(Fn) for n> 3?

S / solved | AMR 问题清单汇编

What are the Dehn functions of Aut(Fn) and Out(Fn) for n> 3?

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10012 | AMR-109-0085

### Question 7.11 - Give upper and lower bounds for L(Ig(k)) for all k≥ 2 which are of the same order of magnitude.

O / open | AMR 问题清单汇编

Give upper and lower bounds for L(Ig(k)) for all k≥ 2 which are of the same order of magnitude. In [ FLM] bounds on L(H) are given for various special classes of subgroups H < Modg. It seems like there is much more to explore in this direction. One can also combine these types of questions with problems such as Problem 7.5.

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10013 | AMR-109-0086

### Question 7.12 - For various subgroups H <Modg, compute the density of spec(H) in spec(Modg) and the density of H∩Pg inPg.

O / open | AMR 问题清单汇编

For various subgroups H <Modg, compute the density of spec(H) in spec(Modg) and the density of H∩Pg inPg. In particular, what is the density of spec(Modg[L]) in spec(Modg)? What about H =Ig(k) with k≥ 1?

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10014 | AMR-109-0078

### Question 7.2 - Is the sequence {L(Modg)} monotone decreasing?

O / open | AMR 问题清单汇编

Is the sequence {L(Modg)} monotone decreasing? strictly so? Explicit values of L(Modg) are known only when g = 1. In this case one is simply asking for the minimum value of the largest root of a polynomial as one varies over all integral polynomials x2-bx + 1 with b≥ 3. This is easily seen to occur when b = 3. For g = 2 Zhirov [ Zh] found the smallest dilatation for pseudo-Anosovs with orientable foliation. It is not clear if this should equal L(Mod2) or not.

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10015 | AMR-109-0301

### Question 7.2 - What are the higher-dimensional isoperimetric functions of GL(n, Z), Aut(Fn)and Out(Fn)?

P / partially_solved | AMR 问题清单汇编

What are the higher-dimensional isoperimetric functions of GL(n, Z), Aut(Fn)and Out(Fn)?

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10016 | AMR-109-0302

### Question 7.3 - Is Aut(Fn) automatic for n> 3?

S / solved | AMR 问题清单汇编

Is Aut(Fn) automatic for n> 3?

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10017 | AMR-109-0080

### Question 7.4 - Is there a unique (up to conjugacy) minimal dilation pseudo-Anosov in Modg?

O / open | AMR 问题清单汇编

Is there a unique (up to conjugacy) minimal dilation pseudo-Anosov in Modg? Note that this is true for g = 1; the unique minimum is realized by the conjugacy class of the matrix ( 2 1 1 1 ). Here is a natural refinement of the problem of finding L(Modg). Fix a genus g. Fix a possible r-tuple ( k1,...,k r) of singularity data for Σ g. By this we mean to consider possible foliations withr singularities with k1,...,k r prongs, respectively. For a fixed g, there are only finitely many possible tuples, as governed by the Poincare-Hopf index theorem. Masur-Smillie [ MS] proved that, for every admissible tuple, there is some pseudo-Anosov on Σ g with stable foliation having the given singularity data. Hence the following makes sense. 2. Some problems on mapping class groups and moduli space 51

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10018 | AMR-109-0082

### Question 7.6 - Does spec(Ig(k)) have bounded multiplicity for k≥ 3?

O / open | AMR 问题清单汇编

Does spec(Ig(k)) have bounded multiplicity for k≥ 3? One way to get around unbounded multiplicities is to look at the simple length spectrum, which is the subset of spec(Mod g) coming from pseudo-Anosovs represented by simple (i.e. non- self-intersecting) geodesic loops in Mg.

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10019 | AMR-109-0083

### Question 7.7 - (Simple length spectrum).

O / open | AMR 问题清单汇编

(Simple length spectrum). Does the simple length spectrum of Mg, endowed with the Teichm¨ uller metric, have bounded multiplicity? If so, how does the bound depend on g? Of course this question contains the corresponding question for (many) hyperbolic surfaces, which itself is still open. These questions also inspire the following.

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10020 | AMR-109-0265

### Question 8 - How much of this paper can be generalized to Zn?

O / open | AMR 问题清单汇编

How much of this paper can be generalized to Zn? It might be easier to first study these questions for the quotient of Zn by the relation X4 = 0.

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10021 | AMR-109-0097

### Question I - s it possible that all nontrivial (i.e., ⁄= 1 ) elements of a normal subgroup of ModS are pseudo-Anosov?

S / solved | AMR 问题清单汇编

s it possible that all nontrivial (i.e.⁄= 1 ) elements of a normal subgroup of ModS are pseudo-Anosov? To the best of my knowledge, this question was posed independently by D. D. Long, J. D. McCarthy, and R. C. Penner in early eighties. I learned it from R. C. Penner in 1984. A moderate progress in the direction of the solution this problem is due to K. Whittlesey [Whi], who constructed examples of such subgroups in the case of spheres with 5 or more punctures and for closed surfaces of genus 2 (in the latter case the mapping class group is well-known to be 76 N. Ivanov intimately connected with the mapping class group of the sphere with 6 punctures). Unfortunately, there is, apparently, no hope to extend the method of proof to the other cases, especially to the closed surfaces of higher genus (which were the focus of all previous attempts at this problem, I suspect).

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10022 | AMR-109-0096

### Question I - s it true that any normal subgroup is commensurable with such a subgroup?

O / open | AMR 问题清单汇编

s it true that any normal subgroup is commensurable with such a subgroup? Recall that two subgroups Γ 1, Γ2 of a group G are commensurable if the intersection Γ 1∩ Γ2 has finite index in both Γ 1 and Γ 2. This problem was suggested by a discussion with H. Bass.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

10023 | AMR-109-0100

### Question I - s it true that H 1(Γ) = 0 for any subgroup Γ of finite index in ModS?

P / partially_solved | AMR 问题清单汇编

s it true that H 1(Γ) = 0 for any subgroup Γ of finite index in ModS? It is well known that H 1(ModS) = 0. In his 1999 MSU Ph.D. thesis F. Taherkhani carried out some extensive computer calculation aiming at finding a subgroup of finite index with non-zero first cohomology group. For genus 2 he found several subgroups Γ with H 1(Γ⁄= 0, but in genus 3 all examined subgroups Γ turned out to have H 1(Γ) = 0. The higher genus cases apparently were well beyond the available at the time computer resources. See [ T ah]. J. D. McCarthy [ McC2] proved that if S is a closed surface of genus ≥ 3 and Γ is a subgroup of finite index in Mod S containing the Torelli subgroup, then the first cohomology group H 1(Γ) is trivial. His methods are based on D. Johnson results about the Torelli subgroup, the solution of the congruence subgroups problem for Sp2g(Z),g≥ 3, and the Kazhdan property (T) of Sp2g(Z), g≥ 3.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

10024 | AMR-109-0105

### Question I - s the subgroup of ModS generated by the N -th powers of all elements of ModS of infinite index in ModS for sufficiently…

O / open | AMR 问题清单汇编

s the subgroup of ModS generated by the N -th powers of all elements of ModS of infinite index in ModS for sufficiently big N? Notice that such a subgroup is obviously normal and if it is of infinite index, then the quotient group is an infinite Burnside group.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

10025 | AMR-109-0099

### Question I - s there a constant NS, depending only on S, such that the following holds?

O / open | AMR 问题清单汇编

s there a constant NS, depending only on S, such that the following holds? Let f∈ ModS and let t =t±1 α1◦t±1 α2◦⋯◦ t±1 αn is a Dehn multi-twist. If A ={fm(αi): 1 ≤i≤n,m∈ Z} fills S (i.e. for any nontrivial circle γ there exist an α∈ A such that i(γ,α*j*= 0 ), then only a finite number of elements of the form tj◦f, j∈ Z are not pseudo-Anosov, and they are among NS consecutive elements of this family. 4. Fifteen problems about the mapping class groups 77 By a theorem of A. Fathi [ F ath], if t is a Dehn twist, then this is true for NS = 7. A weaker version of this question is still interesting: under the same conditions, is it true that no more than NS elements among tj◦f are not pseudo-Anosov? The initial motivation for this question was that the positive answer would allow to prove the Conjecture 4 in some nontrivial cases. This motivation is now (completely?) obsolete, but I still consider this question as interesting. I believe that Fathi’s paper [ F ath] is one of the deepest and the most underappreciated works in the theory of the mapping class groups. In fact, I am aware of only one application of his results: the author’s theorem [ I2] to the effect that the mapping class groups have rank 1 in an appropriate sense; see [ I7], Section 9.4 for a discussion. The above question may be considered as a test of our understanding of the Fathi’s ideas.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

10026 | AMR-109-0149

### Question - Are the relations (1) - (7) the only relations which express the twist along the boundary as a product of positive tw…

P / partially_solved | AMR 问题清单汇编

Are the relations (1) - (7) the only relations which express the twist along the boundary as a product of positive twists, up to the positive equivalence in the Artin group? The motivation for this question comes from symplectic geometry. If we are given a Sym- plectic Lefschetz Fibration p: M → D of a 4-manifold M over a disk D we can describe it by its monodromy product of positive Dehn twists of a fiber over the base point p on ∂D. These are twists along vanishing cycles α1,α 2,...,α k corresponding to geometric basis of π1(D- {x1,x 2,...,x k},p ), where xi’s are the critical values of the fibration. We consider the situation where the generic fiber has one boundary component and the monodromy along ∂D is equal to the twist along the boundary of the fiber. I have proven in [ 12] that in the case of a generic Lefschetz pencil of plane curves of any fixed degree we can choose a geometric basis of π1(CP 2-{x1,...,x k}) in such a way that the corresponding vanishing cycles αi form a bouquet. In particular each pair of curves intersects in 0 (when tangent) or 1 point. I do not know if a similar result is true for a generic pencil of curves on an algebraic surface or even more generally for any Symplectic Lefschetz Fibration without reducible fibers but it is reasonable to ask the Smith question under these restricted conditions: we assume that the twists in our product are along curves which intersect pairwise in 0 or 1 point. In particular they lie in the image of a small Artin group by a geometric homomorphism.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

10027 | AMR-109-0104

### Question - Are there any other relations between N -th powers Tγ = tN γ of Dehn twists for sufficiently high N?

S / solved | AMR 问题清单汇编

Are there any other relations between N -th powers Tγ = tN γ of Dehn twists for sufficiently high N? In other words, do the above relations provide a presentation of the group generated by the N -th powers of Dehn twists? If not, what are the additional relations?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

10028 | AMR-109-0151

### Question - Consider the Artin group A5 (the braid group on six strings) divided by the relation (a1a2a3a4)5 = a5a4a3a2a2 1a2a3a4a5.

O / open | AMR 问题清单汇编

Consider the Artin group A5 (the braid group on six strings) divided by the relation (a1a2a3a4)5 = a5a4a3a2a2 1a2a3a4a5. Is it true that every positive word in this group which is equal to Δ 4 4 = (a1a2a3a4)10 must have either 40 or 30 or 20 letters and in fact must be Hurwitz equivalent (see [ 1]) to Δ 4 4 or to Δ 2 5 = (a1a2a3a4a5)6 or to h2 = (a5a4a3a2a2 1a2a3a4a5)2? 8. Relations in the mapping class group 127 A similar question on a closed surface has an affirmative answer. Siebert and Tian proved in their recent paper (see [ 1]) that any genus two Symplectic Lefschetz Fibration without reducible fibers and with transitive monodromy is holomorphic. Therefore it corresponds to a monodromy factorization which is a product of conjugates of factorizations which are Hurwitz equivalent to Δ2 5 or to h2 (Δ4 4 does not have transitive monodromy). On a surface with boundary each such factor is equal to the twist along the boundary so there may be only one factor if the product is equal to a single twist, but the equivalence on the closed surface is weaker, has bigger equivalence classes so the answer on the surface with boundary may be different.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

10029 | AMR-109-0101

### Question - Does ModS has the Kazhdan property (T)?

O / open | AMR 问题清单汇编

Does ModS has the Kazhdan property (T)? A positive answer would imply the positive answer to the previous question, but this problems seems to be much more difficult. Most of the known proofs of the property (T) for discrete groups are eventually based on the relations of these discrete groups with Lie groups and on the representation theory of Lie groups. Such an approach is not available for Mod S. New approaches to the Kazhdan Property (T) (see, for example Y. Shalom [ Shal], A. ˙Zuk [ Z] and the N. Bourbaki Seminar report of A. Valette [ V]) hold a better promise for our problem, but, to the best of my knowledge no serious work in this direction was done. The problem remains completely open.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

10030 | AMR-109-0147

### Question - Does there exist a set of at least three pseudo-Anosov homeomorpisms such that every pair satisfies a braid relation.

S / solved | AMR 问题清单汇编

Does there exist a set of at least three pseudo-Anosov homeomorpisms such that every pair satisfies a braid relation. The referee to this paper observed that one can easily find a pair of such pseudo-Anosov homeomorphisms satisfying braid relation of length 3. We choose elements x and y of M (S) of order 2 and 3 respectively and let g = xy and h = yx. Then g and h satisfy the relation ghg =hgh. For suitable x andy the homeomorphisms g andh are pseudo-Anosov. If S is a torus then M (S) = SL2(Z). We may require x2 =-1 and y3 = 1 and almost every choice of such matrices x and y will produce Anosov matrices g and h. For a higher genus a choice of elements x and y for which g and h are pseudo-Anosov is a little more difficult.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

10031 | AMR-109-0102

### Question - What is the growth rate of dW (tn, 1)?

S / solved | AMR 问题清单汇编

What is the growth rate of dW (tn, 1)? One would expect that either the growth is linear, or dW (tn, 1) = O(logn). In the arithmetic groups case, the logarithmic growth corresponds to virtually unipotent elements of arithmetic groups of rank ≥ 2, according to a theorem of A. Lubotzky, S. Moses and M.S. Raghunathan [LMR]. B. Farb, A. Lubotzky and Y. Minsky [ FLM] proved that the growth is linear, so the problem is solved completely. 4. Fifteen problems about the mapping class groups 79

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

10032 | AMR-109-0145

### Question - Which Artin groups admit non-geometric embeddings into M (S)?

O / open | AMR 问题清单汇编

Which Artin groups admit non-geometric embeddings into M (S)? If we do not require the homomorphism to be geometric we do not need to restrict the question to small Artin groups. In view of the third part of lemma 2 it is easy to embed some Artin groups with some mi,j =∞ so we should restrict the question to Artin groups with finite exponents mi,j. One example is very simple. The group Bn similar to An with the first edge of the Dynkin diagram having the label 4, maps into An if we send the first generator of Bn onto the square of the first generator of An. The twists T 2 α1 and Tα2 satisfy T 2 α1Tα2T 2 α1Tα2 =Tα2T 2 α1Tα2T 2 α1. Any other Artin group of finite type maps into E8 in a similar way. The generators map onto products of powers of small number of twists. We can call such homomorphisms almost geometric. These are the only almost geometric homomorphisms which I know.

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

10033 | AMR-109-0329

### Question - Which properties of the braid groups can be extended to the mapping class groups?

O / open | AMR 问题清单汇编

Which properties of the braid groups can be extended to the mapping class groups?

背景链接：https://www.math.uchicago.edu/~farb/papers/mcgbook.pdf

10034 | AIM-TOPOLOGY-0262

### Quotient Ricci flow as a coupled and weighted super-Ricci flow

P / partially_solved | AIM 研讨会问题汇编

**Problem 49.** Suppose \(G\) acts freely and isometrically on \(M\). What kind of flows on \(M/G\) are induced by the Ricci flow on \(M\)?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10035 | OWR-3477-017

### Rank Conjecture for Free Actions on Products of Spheres

P / partially_solved | Oberwolfach 报告问题汇编

If $(\mathbb Z/p)^r$ acts freely on $S^{n_1}\times\cdots\times S^{n_k}$ for a prime $p$, must $r\leq k$?

来源字段：https://doi.org/10.4171/owr/2009/27

10036 | OWR-14299915-009

### Rank Conjecture for Free Elementary-Abelian Actions

P / partially_solved | Oberwolfach 报告问题汇编

If $G=(\mathbb Z/p)^r$ acts freely on a finite-dimensional CW complex homotopy equivalent to a product of $k$ spheres, prove that $r\leq k$. In particular, settle the remaining equidimensional cases $p=2$ with sphere dimension $3$ or $7$.

来源字段：https://doi.org/10.4171/owr/2026/9

10037 | OWR-4428-006

### Rank Versus Heegaard Genus of Hyperbolic Three-Manifolds

S / solved | Oberwolfach 报告问题汇编

Does every closed hyperbolic $3$-manifold $M$ satisfy

$$d(\pi_1(M))=g(M),$$

where $d(\pi_1(M))$ is the rank and $g(M)$ is the Heegaard genus?

来源字段：https://doi.org/10.4171/owr/2010/40

10038 | OPG-1783

### Rank vs. Genus

P / partially_solved | Open Problem Garden

Question Is there a hyperbolic 3-manifold whose fundamental group rank is strictly less than its Heegaard genus? How much can the two differ by?

背景链接：http://www.openproblemgarden.org/op/rank_vs_genus

10039 | AIM-TOPOLOGY-0143

### Rank-one intersection-growth signatures for a big mapping-class family

P / partially_solved | AIM 研讨会问题汇编

For a mapping class of an infinite-type surface that preserves no discrete system of disjoint essential curves and proper arcs, describe the dynamics of curve iterates and, in particular, the asymptotics of $I_n(\alpha,\beta)$.

来源字段：http://aimpl.org/genusinfinity/1/

10040 | AIM-TOPOLOGY-0258

### Rank-one no-new theorem for quasi-positive Geroch quotients

P / partially_solved | AIM 研讨会问题汇编

**Problem 45.** Suppose \(G\) is a compact Lie group with a left-invariant metric. Are there any new examples \(H\backslash G\) with quasi-positive curvature?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10041 | OWR-3477-004

### Rank-One-Isotropy Actions of $S_{5}$ on Spheres

O / open | Oberwolfach 报告问题汇编

Does the symmetric group $S_5$ act smoothly on a sphere with rank-one isotropy?

来源字段：https://doi.org/10.4171/owr/2009/27

10042 | AIM-TOPOLOGY-0253

### Rank-six bundles over S^3 x S^3: classification, stabilization, and soul rigidity

P / partially_solved | AIM 研讨会问题汇编

Is there a metric with \(\sec\geq 0\) on \(\mathbb R^6\)-bundles over \(S^3\times S^3\) with non-trivial Euler class? Wilking has shown that the answer is “No” if the soul is \(S^3\times S^3\) with the product metric.

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10043 | OWR-17127-008

### Ranks of Reeb Action Spectra

O / open | Oberwolfach 报告问题汇编

Which additive ranks can occur for the action spectrum of a contact form on a closed $3$-manifold? If the rank is $2$, must the Reeb flow have exactly two geometrically distinct periodic orbits?

来源字段：https://doi.org/10.4171/owr/2019/31

10044 | AIM-TOPOLOGY-0252

### Rational ellipticity for exceptional polar actions

P / partially_solved | AIM 研讨会问题汇编

**Problem 39.** Suppose \(\Sigma\subset M^n\) is the section of a polar action. If \(\Sigma\) is rationally elliptic, must \(M^n\) be rationally elliptic?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10045 | OWR-15581-006

### Rational Formula for the Knot Invariant $\rho_1$

O / open | Oberwolfach 报告问题汇编

For a knot $K \subset S^3$, let $P^{(1)}(t)$ be the $1$-loop polynomial (perturbed Alexander invariant) arising in the asymptotic expansion of the colored Jones polynomial, and let $\rho_1(t)$ be Rozansky's rational $1$-loop knot invariant. Does the conjectured identity $$ \rho_1(t^2) = \frac{t^2}{(1 - t^2)^2} P^{(1)}(t) $$ hold for every knot $K$?

来源字段：https://doi.org/10.4171/owr/2017/38

10046 | OPG-37131

### Realisation problem for the space of knots in the 3-sphere

P / partially_solved | Open Problem Garden

Problem Given a link $L$ in $S^3$, let the symmetry group of $L$ be denoted $Sym(L) = \pi_0 Diff(S^3,L)$ ie: isotopy classes of diffeomorphisms of $S^3$ which preserve $L$, where the isotopies are also required to preserve $L$.

Now let $L$ be a hyperbolic link. Assume $L$ has the further `Brunnian' property that there exists a component $L_0$ of $L$ such that $L \setminus L_0$ is the unlink. Let $A_L$ be the subgroup of $Sym(L)$ consisting of diffeomorphisms of $S^3$ which preserve $L_0$ together with its orientation, and which preserve the orientation of $S^3$.

There is a representation $A_L \to \pi_0 Diff(L \setminus L_0)$ given by restricting the diffeomorphism to the $L \setminus L_0$. It's known that $A_L$ is always a cyclic group. And $\pi_0 Diff(L \setminus L_0)$ is a signed symmetric group -- the wreath product of a symmetric group with $\mathbb Z_2$.

Problem: What representations can be obtained?

背景链接：http://www.openproblemgarden.org/op/realisation_problem_for_the_space_of_knots_in_the_3_sphere

10047 | OWR-17132-005

### Realizable Type Functions of Equivariant Spectra

P / partially_solved | Oberwolfach 报告问题汇编

Which functions $$\operatorname{Sub}(G)\longrightarrow\mathbb N\cup\{\infty\}$$ occur as the chromatic type function of a finite $p$-local $G$-spectrum?

来源字段：https://doi.org/10.4171/owr/2019/36

10048 | OWR-2089-009

### Realizable Unstable Modules over the Steenrod Algebra

O / open | Oberwolfach 报告问题汇编

Let $\mathcal U$ be the category of unstable modules over the Steenrod algebra. For which objects $M$ does there exist a space $X$ with $H^*(X;\mathbb F_2)\cong M$ in $\mathcal U$? Equivalently, which objects of $\mathcal U$ are of the form $H^*(X;\mathbb F_2)$?

来源字段：https://doi.org/10.4171/owr/2008/43

10049 | OWR-14298592-004

### Realization and Classification of Poincaré Duality Four-Complexes

O / open | Oberwolfach 报告问题汇编

Given a fundamental triple $(T,w,b)$, determine whether it is realized by a Poincaré-duality $4$-complex or by a manifold. For a realizable triple, determine which classes $b'\in H_4(T;\mathbb Z)$ are realized, and identify the additional data needed beyond the symmetric $2$-type to recover the homotopy type.

来源字段：https://doi.org/10.4171/owr/2024/43

10050 | OWR-1728-002

### Realization of Oriented Dimension Functions by Equivariant Spheres

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a finite group and $k$ a field of characteristic $p$. Is every oriented dimension function realizable by a $kO_p(G)^{\mathrm{op}}$-sphere if and only if it satisfies the Borel-Smith conditions?

来源字段：https://doi.org/10.4171/owr/2007/45

10051 | OWR-2040-003

### Realization of Persistence Pairings by Filtration Functions

O / open | Oberwolfach 报告问题汇编

Let $P$ be a persistence pairing on a filtered simplicial complex $K$, let $V_P$ be its associated discrete gradient vector field, and let $P_{V_P,f}$ be the persistence pairing induced by an integer-valued filtration function $f$. Is there a choice of $f$ for which $P_{V_P,f}=P$?

来源字段：https://doi.org/10.4171/owr/2008/29

10052 | OWR-17132-007

### Realizing Chromatic Types at Nested Fixed Points

P / partially_solved | Oberwolfach 报告问题汇编

Let $H<G$ be subgroups of a finite $p$-group. For which pairs $(m,n)$ does there exist a finite $G$-spectrum $X$ such that $\Phi^H X$ has chromatic type $m$ and $\Phi^G X$ has chromatic type $n$?

来源字段：https://doi.org/10.4171/owr/2019/36

10053 | OWR-14299915-018

### Recovering Fixed-Point Homology in Smith Theory

P / partially_solved | Oberwolfach 报告问题汇编

For a finite $\mathbb Z/p$-CW complex $X$, determine how much of the homology of the fixed-point space $X^{\mathbb Z/p}$ can be reconstructed from the homological and equivariant-cohomological data of $X$.

来源字段：https://doi.org/10.4171/owr/2026/9

10054 | OWR-12582-006

### Recovering Three-Manifold Topology from Contact Dynamics

P / partially_solved | Oberwolfach 报告问题汇编

Can the diffeomorphism type of a closed connected 3-manifold be determined from dynamical properties of a flow defined on it, viewed as a contact-type energy level, together with its induced contact structure?

来源字段：https://doi.org/10.4171/owr/2013/34

10055 | OWR-15987-020

### Recovering Toric Arrangement Posets from Complement Cohomology

O / open | Oberwolfach 报告问题汇编

For a central toric arrangement $A$ with layer poset $L$ and complement $M(A)$, does the cohomology algebra $H^\bullet(M(A))$ determine $L$?

来源字段：https://doi.org/10.4171/owr/2018/2

10056 | OWR-4416-016

### Recurrence Polynomials and the Knot A-Polynomial

P / partially_solved | Oberwolfach 报告问题汇编

For every knot $K$, does specializing the recurrence polynomial $\alpha_K$ at $q^{1/2}=1$ give the $A$-polynomial up to a factor depending only on $M$, with the $A$-polynomial having no $M$-factor?

来源字段：https://doi.org/10.4171/owr/2010/35

10057 | OWR-1967-011

### Recursive Determination of Quantum Knot Invariants

O / open | Oberwolfach 报告问题汇编

For a complex reductive Lie algebra $\mathfrak g$ and a knot $K$, is the Witten-Reshetikhin-Turaev function $J_{\mathfrak g,K}$ on the weight lattice uniquely determined among Weyl-equivariant functions by finitely many values together with the relations in the Weyl-invariant recursive ideal $I^W_{\mathfrak g,K}$?

来源字段：https://doi.org/10.4171/owr/2008/22

10058 | OWR-1967-010

### Recursive Ideals and Knot Character Varieties

O / open | Oberwolfach 报告问题汇编

For a knot $K\subset S^3$ and a complex reductive group $G$ with Lie algebra $\mathfrak g$, is the zero set of $\varepsilon(I^W_{\mathfrak g,K})$ equal to the Zariski closure of the image of $$X_G(S^3\setminus K)\longrightarrow X_G(T)?$$ Equivalently, is $\sqrt{\varepsilon(I^W_{\mathfrak g,K})}=A_{G,K}$?

来源字段：https://doi.org/10.4171/owr/2008/22

10059 | AIM-TOPOLOGY-0100

### Reduced Euler characteristic and a stable K1-summand

P / partially_solved | AIM 研讨会问题汇编

Based on the work of Carranza-Doherty-Kapulkin-Opie-Sarazola-Wong, there is a cofibration category structure on the category of digraphs whose weak equivalences are digraph maps that induce isomorphisms on all path homology groups. One can use it to define a Waldhausen category on (pointed) digraphs, which can then be used to take algebraic K-theory.

Compute $K_0(*)$ and check if it agrees with what one might expect from spaces, i.e., an isomorphism $K_0(*)\cong\mathbb Z$, given by taking a class $[X]\in K_0(*)$ to the Euler characteristic of $X$.

Can other $K$-groups be computed?

来源字段：http://aimpl.org/combhomotop/3/

10060 | OWR-5155-008

### Reducibility and Manifold Realization of Finite Poincaré Complexes

P / partially_solved | Oberwolfach 报告问题汇编

Let $\Gamma$ be a virtual Poincaré-duality group of cohomological dimension $k$ whose Farrell cohomology is periodic of period $n+1$, and let $X$ be the resulting finite Poincaré complex of dimension $n+k$ with $\pi_1(X)\cong\Gamma$ and $\widetilde X\simeq S^n$. Is the Spivak normal fibration of $X$ reducible? Is $X$ homotopy equivalent to a closed manifold?

来源字段：https://doi.org/10.4171/owr/2011/27

10061 | OWR-12697686-003

### Reducibility of Trisections with Nonseparating Three Spheres

O / open | Oberwolfach 报告问题汇编

If a smooth $4$-manifold $X$ contains a nonseparating $3$-sphere, is every trisection of $X$ reducible?

来源字段：https://doi.org/10.4171/owr/2023/3

10062 | AIM-TOPOLOGY-0137

### Reduction systems must remember wandering domains

P / partially_solved | AIM 研讨会问题汇编

**Give a Nielsen--Thurston classification type theorem for big mapping classes.**

Suppose a big mapping class is reducible if it preserves a (possibly infinite) discrete collection of pairwise disjoint simple closed curves and simple proper arcs (where discrete means that the collection “does not accumulate anywhere inside the surface”), and irreducible otherwise.

来源字段：http://aimpl.org/genusinfinity/1/

10063 | AIM-TOPOLOGY-0238

### Reduction, holonomy, and symplectic-pencil obstructions

P / partially_solved | AIM 研讨会问题汇编

**Problem 25.** Can one reduce the structure group of a fat principal \(G\)-bundle?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10064 | AMR-102-0009

### Reeb components - Question 5.1

O / open | AMR 问题清单汇编

How many Reeb components must a foliation of an open $3$-manifold contain?

背景链接：https://arxiv.org/abs/math/0209081

10065 | AMR-102-0010

### Reeb components - Question 5.2

O / open | AMR 问题清单汇编

What generalizations of the notion of taut foliation make sense on an open $3$-manifold?

背景链接：https://arxiv.org/abs/math/0209081

10066 | AIM-TOPOLOGY-0250

### Reflection obstruction and a one-parameter Einstein shooting problem

P / partially_solved | AIM 研讨会问题汇编

Study the existence of Einstein metrics on manifolds of cohomogeneity-one.

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10067 | AIM-TOPOLOGY-0002

### Regularity threshold for Weinstein-convex approximation

S / solved | AIM 研讨会问题汇编

Can hypersurfaces be $c^\infty$-approximated by Weinstein convex hypersurfaces?

来源字段：http://aimpl.org/highdimcontacttop/2/

10068 | OWR-12984-006

### Relations Among Algebraic PL and Fourier Knot Degrees

O / open | Oberwolfach 报告问题汇编

For a knot $K$, establish inequalities in both directions among its algebraic, piecewise-linear, and Fourier degrees. Determine the extrema of their three pairwise quotients and whether those quotients have limits along sequences of knots with crossing number tending to infinity.

来源字段：https://doi.org/10.4171/owr/2014/26

10069 | AIM-TOPOLOGY-0022

### Relative Hamiltonian isotopy and the standard contact sphere filling

P / partially_solved | AIM 研讨会问题汇编

Classify symplectic fillings of contact submanifolds of $(S^5,\xi_{std})$ in its standard $D^6$ filling. E.g. Show that the standard $S^3\subset S^5$ has a unique filling up to Hamiltonian isotopy.

来源字段：http://aimpl.org/highdimcontacttop/9/

10070 | OWR-15987-025

### Representation Stability and Compactifications for Braid Milnor Fibers

P / partially_solved | Oberwolfach 报告问题汇编

For the Milnor fiber $F_n=\{(x_1,\ldots,x_n):\prod_{i<j}(x_i-x_j)^2=1\}$, determine $H^*(F_n,\mathbb Q)$ as an $(S_n\times\mu_{n(n-1)})$-module; determine whether a category analogous to FI makes $H^i(F_n,\mathbb Q)$ a finitely generated module; and determine whether $M_{d,n}$ has a natural compactification.

来源字段：https://doi.org/10.4171/owr/2018/2

10071 | AIM-TOPOLOGY-0109

### Reversal defects in the bipolar filtration

P / partially_solved | AIM 研讨会问题汇编

**Reversal and the bipolar filtration.** For each $n\in\mathbb N$, find an $n$-bipolar topologically slice knot $K$ such that $K\#-K^r$ is not smoothly slice.

来源字段：http://aimpl.org/concordsliceknot/1/

10072 | OWR-15208-023

### Ricci Flow of Anosov Metrics on Surfaces

O / open | Oberwolfach 报告问题汇编

For a closed surface of genus at least 2, does Ricci flow eventually carry every metric with Anosov geodesic flow to a negatively curved metric?

来源字段：https://doi.org/10.4171/owr/2017/3

10073 | OWR-14297743-001

### Rigid Subgraphs of the Curve Complex

P / partially_solved | Oberwolfach 报告问题汇编

Which subgraphs of the curve complex are rigid?

来源字段：https://doi.org/10.4171/owr/2023/33

10074 | AMR-102-0005

### Rigidity and moduli - Question 3.1

P / partially_solved | AMR 问题清单汇编

Let $M$ be atoroidal. Is there a natural refinement of the polyhedral structure of the unit ball of the Thurston norm to a polyhedron $\mathscr{P}_\mathscr{F}$ whose faces parameterize the set of taut of foliations of $M$ in the following sense:

- Each open face $c$ of $\mathscr{P}_\mathscr{F}$ corresponds to a pseudo-Anosov flow $X_c$ or equivalence classes of pseudo-Anosov flows. The taut foliations ``parameterized'' by $\mathscr{P}_\mathscr{F}$ can be isotoped to be transverse (or almost transverse) to $X_c$, or some sublamination is monotone equivalent into the stable or unstable singular foliation of $X_c$.

- Every geometric limit of taut foliations $\mathscr{F}_i$ associated to a cell $c$ should be associated to some cell of the closure $\overline{c}$.

- There should be a natural polyhedral map from $\mathscr{P}_\mathscr{F}$ to (some polyhedral subdivision of) the unit ball of the Thurston norm.

背景链接：https://arxiv.org/abs/math/0209081

10075 | AMR-102-0006

### Rigidity and moduli - Question 3.2

O / open | AMR 问题清单汇编

Generalize the Teichmüller polynomial from the fibered faces of the Thurston norm ball to the other faces (of some possibly generalized polyhedron, perhaps the polyhedron sought in question 3.1).

背景链接：https://arxiv.org/abs/math/0209081

10076 | OWR-12485-008

### Ropelength, Menger Energy, and Loop-Number Problems

P / partially_solved | Oberwolfach 报告问题汇编

Can a ropelength-minimizing or ropelength-critical knot be described explicitly, and is the minimizing trefoil piecewise $C^2$ or piecewise smooth? Which geometric quantities or topological invariants are bounded by higher-dimensional integral Menger energies $M_p(\Sigma)$? For an unknot diagram $D$ with writhe $w(D)$, is its loop number at least $|w(D)|$, and how is loop number related to other knot invariants?

来源字段：https://doi.org/10.4171/owr/2013/22

10077 | OWR-1381-011

### Rosenberg Indices and Codimension-Two Curvature Obstructions

P / partially_solved | Oberwolfach 报告问题汇编

For a closed smooth spin manifold $M$, what is the relation between the Rosenberg indices $\alpha_{\mathrm{red}}(M)$ and $\alpha_{\mathrm{max}}(M)$ and the codimension-two positive-scalar-curvature obstructions of Gromov and Lawson?

来源字段：https://doi.org/10.4171/owr/2006/43

10078 | AIM-TOPOLOGY-0150

### Rosendal geometry of the ladder and blooming 3-pod

P / partially_solved | AIM 研讨会问题汇编

Give a complete quasi-isometry classification (in the sense of Rosendal) of big mapping class groups. For instance, let $S$ be the ladder surface and let $S'$ be the $3$--pod blooming surface. Is $\operatorname{Map}(S)$ quasi-isometric to $\operatorname{Map}(S')$?

来源字段：http://aimpl.org/genusinfinity/3/

10079 | OPG-751

### S(S(f)) = S(f) for reloids

P / partially_solved | Open Problem Garden

Question $S(S(f)) = S(f)$ for every endo-reloid $f$?

背景链接：http://www.openproblemgarden.org/op/s_s_f_s_f_for_reloids

10080 | AIM-TOPOLOGY-0089

### Same-shape tableau traces and a pointed Morita obstruction

O / open | AIM 研讨会问题汇编

**Likely intended question.** Take $P,Q\in SYT(\lambda)$. How do $\operatorname{tr}(P)$ and $\operatorname{tr}(Q)$ relate?

来源字段：http://aimpl.org/agclinkhom/9/

10081 | OWR-15987-022

### Schwarz Genus of the Flex-Point Cover

O / open | Oberwolfach 报告问题汇编

Let $X$ be the space of smooth complex homogeneous cubic polynomials in three variables modulo $\mathbb C^\times$, and let $Y=\{(F,p):p\in\mathbb{CP}^2\text{ is a flex of }F=0\}$. The map $Y\to X$ is a degree-nine covering; determine its Schwarz genus.

来源字段：https://doi.org/10.4171/owr/2018/2

10082 | AIM-TOPOLOGY-0024

### Sectorial and cotangent models for convex corners

P / partially_solved | AIM 研讨会问题汇编

In higher dimensions, do intersections of convex hypersurfaces admit a standard model? Can we describe edge rounding?

来源字段：http://aimpl.org/highdimcontacttop/10/

10083 | OWR-5146-007

### Semifree-Module Form of the Toral-Rank Conjecture

S / solved | Oberwolfach 报告问题汇编

If $(M,d)=(\bigwedge(x_1,\ldots,x_r)\otimes V,D)$ is a semifree module with finite-dimensional cohomology, must $\dim V\geq2^r$?

来源字段：https://doi.org/10.4171/owr/2011/18

10084 | OWR-1321-005

### Separable Subgroups of Haken Three-Manifold Groups

S / solved | Oberwolfach 报告问题汇编

Which subgroups of the fundamental group of a Haken 3-manifold are separable?

来源字段：https://doi.org/10.4171/owr/2006/35

10085 | OPG-60043

### Several ways to apply a (multivalued) multiargument function to a family of filters

P / partially_solved | Open Problem Garden

Problem Let $\mathcal{X}$ be an indexed family of filters on sets. Which of the below items are always pairwise equal?

1. The funcoid corresponding to this function (considered as a single argument function on indexed families) applied to the reloidal product of filters $\mathcal{X}$.

2. The funcoid corresponding to this function (considered as a single argument function on indexed families) applied to the starred reloidal product of filters $\mathcal{X}$.

3. $\bigcap_{F\in\operatorname{up}^{\mathrm{FCD}}}\prod^{\mathrm{Strd}}\mathcal{X}}\langle f \rangle F$.

背景链接：http://www.openproblemgarden.org/op/several_ways_to_apply_a_multivalued_multiargument_function_to_a_family_of_filters

10086 | OWR-2089-012

### Sharp Bounds for Exceptional Dehn-Filling Slopes

P / partially_solved | Oberwolfach 报告问题汇编

For every hyperbolic knot manifold $M$, let $E(M)$ be its exceptional filling slopes and $\Delta(E(M))$ their maximum pairwise distance. Is $\#E(M)\leq10$ and $\Delta(E(M))\leq8$? Moreover, unless $M\in\{M_1,M_2,M_3,M_4\}$, is $\#E(M)\leq7$ and $\Delta(E(M))\leq5$?

来源字段：https://doi.org/10.4171/owr/2008/43

10087 | OWR-1703876-006

### Sharp Constrained-Drawing Bounds for Closure Operators

O / open | Oberwolfach 报告问题汇编

Let $\operatorname{cl}$ be a closure operator on $\mathbb R^2$ such that $\operatorname{cl}(S)$ has at most $b$ path-connected components for every finite $S\subset\mathbb R^2$. Are $kb+1$ points always sufficient to obtain a constrained drawing of the star $K_{1,k}$? What sharper bounds hold for constrained drawings of $K_n$, and can constrained-drawing arguments be extended to nontrivial homology or homotopy in higher dimensions?

来源字段：https://doi.org/10.4171/owr/2020/30

10088 | OWR-2654833-002

### Sharp Relations Between Knot Ropelength and Crossing Number

P / partially_solved | Oberwolfach 报告问题汇编

How is the ropelength of a knot $K$ related to its crossing number $\operatorname{Cr}(K)$?

来源字段：https://doi.org/10.4171/owr/2020/37

10089 | OWR-12172-003

### Sharp Syzygy Bounds for Torus Actions

S / solved | Oberwolfach 报告问题汇编

For a rational Poincaré duality space with torus rank $r$, if its equivariant cohomology is not free, the syzygy bound is at most $\lfloor(r-1)/2\rfloor$. Is this bound sharp for $r\geq5$?

来源字段：https://doi.org/10.4171/owr/2012/49

10090 | AIM-TOPOLOGY-0111

### Signed-crossing certificates and a strongly quasipositive characterization at the zeroth bipolar level

P / partially_solved | AIM 研讨会问题汇编

**Characterization of 0-bipolar knots.** Can one characterize 0-positive, 0-negative, or 0-bipolar knots, either via the vanishing of invariants or geometrically?

来源字段：http://aimpl.org/concordsliceknot/1/

10091 | OWR-2089-001

### Simon’s Epimorphism Conjecture for Knot Groups

S / solved | Oberwolfach 报告问题汇编

For every nontrivial knot $K\subset S^3$, does the knot group $\pi_1(S^3\setminus K)$ surject onto only finitely many distinct knot groups?

来源字段：https://doi.org/10.4171/owr/2008/43

10092 | OWR-2089-002

### Simplicial Volume and Rational Rank of Locally Symmetric Spaces

P / partially_solved | Oberwolfach 报告问题汇编

For a finite-volume locally symmetric space of noncompact type, is its simplicial volume nonzero when its $\mathbb Q$-rank is 1? Does its simplicial volume vanish when its $\mathbb Q$-rank is 2?

来源字段：https://doi.org/10.4171/owr/2008/43

10093 | OWR-5158-009

### Simplicial Volume Bounds for Euler Characteristics

O / open | Oberwolfach 报告问题汇编

For every aspherical manifold $M$, does $\lVert M\rVert=0$ imply $\chi(M)=0$? More strongly, does there exist a constant $C(n)$ depending only on the dimension such that

$$|\chi(M)|\leq C(n)\lVert M\rVert$$

for every $n$-dimensional aspherical manifold $M$?

来源字段：https://doi.org/10.4171/owr/2011/30

10094 | OWR-15177-002

### Simplicial Volumes of Surface Bundles

O / open | Oberwolfach 报告问题汇编

For surface bundles with fixed closed fiber $F$ and base $B$, can the simplicial volume be bounded above in terms of the Euler characteristics, and are there finitely or infinitely many possible simplicial-volume values?

来源字段：https://doi.org/10.4171/owr/2016/56

10095 | OWR-5158-013

### Simplicial-Volume-Generated Seminorms in Third Homology

O / open | Oberwolfach 报告问题汇编

Does the functorial seminorm on $H_3(\mathord{\cdot};\mathbb R)$ generated by simplicial volume coincide with the $\ell^1$-seminorm?

来源字段：https://doi.org/10.4171/owr/2011/30

10096 | AIM-TOPOLOGY-0104

### Singular homology of cyclic graph powers

S / solved | AIM 研讨会问题汇编

Singular homology of graphs

Every simple graph can be viewed as a closure space as follows: given a graph $X = (V, E)$, we define a closure space by taking its underlying set to be $V$ and $c(A) = \bigcup_{v \in A} c(v)$, where $c(v) = \{ w \in V \ | \ \{ v, w \} \in E\}$. For $n , k \in \mathbb{Z}$, define the graph $(\mathbb{Z}/n, c_k)$ to have the set of vertices $\mathbb{Z}/n = \{ 0, 1, \ldots, n-1\}$ and an edge between $i$ and $j$ whenever $i$ and $j$ are no more than $k$ away.

Compute $H^{sing}_*(\mathbb{Z}/n, c_k)$. Is it isomorphic to the homomology of the clique complex of $(\mathbb{Z}/n, c_k)$?

Three substantive repairs are necessary for a literal well-posed reading.

1. A Čech closure must satisfy \(A\subseteq c(A)\). For a loopless simple graph the displayed \(c(v)\) omits \(v\). The standard intended graph closure is the **closed-neighborhood closure** \[ c_G(A)=A\cup\bigcup_{v\in A}N_G(v), \] equivalently the graph relation is made reflexive. 2. “No more than \(k\) away” is taken in cyclic distance \[ \delta_n(i,j)=\min\{|i-j|,n-|i-j|\}. \] Thus the underlying simple graph is the \(k\)-th power \(C_n^k\) of the \(n\)-cycle, and its reflexive relation is \(iR_kj\iff\delta_n(i,j)\le k\). 3. The notation \(H_*^{\mathrm{sing}}\) is ambiguous because several singular theories for closure spaces now exist. The paper attached to this AIM question by Milićević uses ordinary topological simplices: a singular \(q\)-simplex is a continuous map \(|\Delta^q|\to X\). This is the \(I\)- or topological-simplex singular theory computed here, with integral coefficients; no \(J\)-interval theory is silently identified with it.

The recovered problem is therefore: for \(n\ge3\), \(k\ge0\), compute the \(I\)-singular homology of the finite closure space associated with the reflexive symmetric graph \(C_n^k\), and compare it with \(H_*(\operatorname{Cl}(C_n^k);\mathbb Z)\). The degenerate cases \(n=1,2\) are recorded separately below. Negative \(k\) does not produce an extensive closure by this distance prescription.

来源字段：http://aimpl.org/combhomotop/4/

10097 | OWR-17471-019

### Slice Candidates and Shared Trace Boundaries of Knots

P / partially_solved | Oberwolfach 报告问题汇编

Is the positive Whitehead double of the left-handed trefoil slice? Is the $(2,1)$-cable of the figure-eight knot slice? For which knots $K$ does there exist a distinct knot $J$ with $\partial X(J)\cong\partial X(K)$, or with the two trace boundaries merely integer-homology cobordant? Give an algorithm for producing such knots $J$.

来源字段：https://doi.org/10.4171/owr/2020/8

10098 | OWR-12697687-001

### Slice-Genus Bounds for Homogeneous Braid Quasimorphisms

P / partially_solved | Oberwolfach 报告问题汇编

For every homogeneous quasimorphism $f:B_n\to\mathbb R$, do constants $A,C\in\mathbb R$ exist such that $|f(\beta)|\leq A\chi_4(\widehat\beta)+C$ for every braid $\beta\in B_n$?

来源字段：https://doi.org/10.4171/owr/2023/4

10099 | OPG-37129

### Slice-ribbon problem

P / partially_solved | Open Problem Garden

Conjecture Given a knot in $S^3$ which is slice, is it a ribbon knot?

背景链接：http://www.openproblemgarden.org/op/slice_ribbon_problem_0

10100 | OWR-17132-008

### Small-Genus Cohomology of High-Dimensional Diffeomorphism Groups

P / partially_solved | Oberwolfach 报告问题汇编

For $$W_{g,1}=D^{2n}\mathbin{\#}_g(S^n\times S^n),\qquad 2n\geq6,$$ determine the rational cohomology ring $$H^*(B\operatorname{Diff}_\partial(W_{g,1});\mathbb Q)$$ at small genus, particularly $g=1$.

来源字段：https://doi.org/10.4171/owr/2019/36

10101 | OWR-13346-007

### Smeariness, Stability, and Strong Consistency on Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Determine whether a non-positively curved manifold can exhibit only non-positive $k$-smeariness; determine how manifold stability and stickiness are related on a non-Hausdorff Lie-group quotient; and determine whether the inclusion in Ziezold's strong consistency result, $\bigcup_{n=1}^{\infty}\bigcap_{k=n}^{\infty}\widehat E_\rho^{(k)}\subset E_\rho$, can be strict.

来源字段：https://doi.org/10.4171/owr/2014/44

10102 | OPG-37125

### Smooth 4-dimensional Poincare conjecture

P / partially_solved | Open Problem Garden

Conjecture If a $4$-manifold has the homotopy type of the $4$-sphere $S^4$, is it diffeomorphic to $S^4$?

背景链接：http://www.openproblemgarden.org/op/smooth_4_dimensional_poincare_conjecture

10103 | TOP-001

### Smooth 4-Dimensional Poincaré Conjecture

O / open | 独立 / 未分配集合

Is every smooth homotopy 4-sphere diffeomorphic to the standard 4-sphere $S^4$?

背景链接：https://doi.org/10.4310/jdg/1214437136

10104 | OPG-37123

### Smooth 4-dimensional Schoenflies problem

P / partially_solved | Open Problem Garden

Problem Let $M$ be a $3$-dimensional smooth submanifold of $S^4$, $M$ diffeomorphic to $S^3$. By the Jordan-Brouwer separation theorem, $M$ separates $S^4$ into the union of two compact connected $4$-manifolds which share $M$ as a common boundary. The Schoenflies problem asks, are these $4$-manifolds diffeomorphic to $D^4$? ie: is $M$ unknotted?

背景链接：http://www.openproblemgarden.org/op/smooth_4_dimensional_schoenflies_problem

10105 | AIM-TOPOLOGY-0003

### Smooth convex-suspension approximation of irrational contact rotations

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** Let \(\phi\) be the time-one contactomorphism of an ergodic contact flow. For which \(k\) can \(\phi\) be approximated in the \(C^k\) topology by contactomorphisms \(\phi_j\) whose contact-Hamiltonian suspensions (mapping-torus hypersurfaces) are convex?

来源字段：http://aimpl.org/highdimcontacttop/2/

10106 | OWR-17471-018

### Smooth Embeddings of Knot Traces in Homology Four-Spheres

O / open | Oberwolfach 报告问题汇编

For the Conway knot $K$, does the complement $S^4\setminus\nu(f(X(K)))$ of an embedded trace admit a smooth structure? Does some smooth integer-homology $4$-sphere contain the trace $X(K)$ of a nonslice knot smoothly? Finally, do there exist homotopy-equivalent closed oriented simply connected smooth $4$-manifolds $X$ and $W$ such that $X(K)$ embeds smoothly in $X$ but not in $W$?

来源字段：https://doi.org/10.4171/owr/2020/8

10107 | OWR-832-006

### Smooth Extensions of Multiplicative Cohomology Transformations

P / partially_solved | Oberwolfach 报告问题汇编

Given a natural multiplicative transformation $r:h\to H\mathbb{R}$ from a generalized cohomology theory $h$ to ordinary real cohomology, determine whether a smooth extension of $r$ exists and when it is unique.

来源字段：https://doi.org/10.4171/owr/2005/27

10108 | OWR-12697686-012

### Smooth Isotopy of Boundary Twists on Spin Four Manifolds

O / open | Oberwolfach 报告问题汇编

For which closed spin simply connected smooth $4$-manifolds $X$ is the boundary twist $t_X$ smoothly isotopic to the identity?

来源字段：https://doi.org/10.4171/owr/2023/3

10109 | OWR-16415-015

### Smooth Lifts of the Dehn-Twist Braid Relation

O / open | Oberwolfach 报告问题汇编

Let $a$ and $b$ be right Dehn twists about essential simple closed curves meeting once, so that $aba=bab$ in the mapping class group. Can they be lifted to homeomorphisms $A$ and $B$ satisfying $ABA=BAB$? Can such lifts be chosen of class $C^1$, or more generally $C^r$?

来源字段：https://doi.org/10.4171/owr/2018/40

10110 | AIM-TOPOLOGY-0255

### Soul reduction and rigidity in a doubly warped class

P / partially_solved | AIM 研讨会问题汇编

**Problem 42.** Classify metrics with \(\sec\geq0\) on \(S^2\times\mathbb R^4\) (or, more generally, on \(S^n\times\mathbb R^k\)).

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10111 | OWR-12485-001

### Spectra of Multi-Crossing Knot Numbers

P / partially_solved | Oberwolfach 报告问题汇编

For the spectrum of multi-crossing numbers of a knot, what is its behavior, which knot families arise naturally from it, what braids arise from petal projections, and how do these invariants relate to traditional knot invariants?

来源字段：https://doi.org/10.4171/owr/2013/22

10112 | OWR-14297741-003

### Spectral Moduli Interpretations for Stable Module Blocks

P / partially_solved | Oberwolfach 报告问题汇编

Do the spectral schemes associated with principal blocks of stable module categories admit natural spectral moduli-theoretic interpretations beyond the case of elementary abelian groups?

来源字段：https://doi.org/10.4171/owr/2023/40

10113 | AIM-TOPOLOGY-0135

### Split-union closure for round-handle sliceness

P / partially_solved | AIM 研讨会问题汇编

**Round Handle Problem.** Given a linking number \(0\) link \(L\), construct a \(4\)-manifold \(W\) by attaching a round handle to \(B^4\) along each component of \(L\). Must \(L\) be topologically slice in \(W\)?

来源字段：http://aimpl.org/concordsliceknot/5/

10114 | OWR-14299521-010

### Stability of Noncollapsing Manifold Submetries

O / open | Oberwolfach 报告问题汇编

Let $P_i:M_i\to Y_i$ be manifold submetries, with compact $M_i$ of uniformly bounded geometry, converging without collapse to a submetry $P:M\to Y$. Are $P_i$ and $P_j$, for all sufficiently large $i,j$, equivalent up to homeomorphisms of their source and target spaces?

来源字段：https://doi.org/10.4171/owr/2025/30

10115 | OWR-3477-015

### Stable Cohomology of Higher-Dimensional Diffeomorphism Groups

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a homotopy-commutative monoid $M$, constructed from classifying spaces of diffeomorphism groups of oriented $d$-manifolds with boundary and with multiplication induced by gluing, such that $$H^*(\Omega_0^\infty MTSO(d);\mathbb Q)\longrightarrow H^*(M;\mathbb Q)^{\pi_0(M)}$$ is surjective and has zero kernel when $d$ is even, while for odd $d$ its kernel is the ideal generated by $s\,\mathrm{th}(L_k)$ for $4k\geq d$, with $L_k$ the degree-$4k$ component of the Hirzebruch $L$-class?

来源字段：https://doi.org/10.4171/owr/2009/27

10116 | AIM-TOPOLOGY-0068

### Stable envelopes versus fixed-slope objects in Hecke cocenters

P / partially_solved | AIM 研讨会问题汇编

Fixing a slope, what corresponds to a stable basis in \(K\)-theory of Hilbert schemes in the cocenter of the Hecke algebra?

来源字段：http://aimpl.org/agclinkhom/4/

10117 | OWR-1381-010

### Stable K-Area and Nonvanishing Rosenberg Indices

P / partially_solved | Oberwolfach 报告问题汇编

For a closed smooth spin manifold $M$ with infinite stable $K$-area, must $\alpha_{\mathrm{max}}(M)\ne0$? Does an analogous conclusion for $\alpha_{\mathrm{red}}(M)$ hold, at least in special cases such as area-enlargeability?

来源字段：https://doi.org/10.4171/owr/2006/43

10118 | AIM-TOPOLOGY-0260

### Stable quasipositive towers: exact reconstruction, model cases, and a stable-flag reduction

P / partially_solved | AIM 研讨会问题汇编

**Problem 47.** Fix $k\in\mathbb N$. Is there an $n_0=n_0(k)$ such that for any quasipositively curved manifold $(M^n,g)$ with $n\ge n_0$ and \(\operatorname{cohom}(M^n,g)\le k\), there exists a chain \[ M_0=M^n\subset M_1^{\,n+k}\subset M_2^{\,n+2k}\subset\cdots \] such that all inclusions are totally geodesic, the manifolds $M_i$ are quasipositively curved, \(\bigcup_iM_i\) is the classifying space of a Lie group, and \(M_i/\operatorname{Iso}(M_i,g)\) is isometric to \(M/\operatorname{Iso}(M,g)\)?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10119 | OWR-832-004

### Stable Superpolynomials for Khovanov-Rozansky Homology

P / partially_solved | Oberwolfach 报告问题汇编

For every knot $K$, do there exist finite polynomials $\overline{\mathcal{P}}_K\in\mathbb{Z}[a^{\pm1},q^{\pm1},t^{\pm1}]$ and $\mathcal{P}_K\in\mathbb{Z}_{\geq0}[a^{\pm1},q^{\pm1},t^{\pm1}]$ such that, for all sufficiently large $N$, the unreduced and reduced $\mathfrak{sl}(N)$ Khovanov-Rozansky Poincaré polynomials satisfy the following identities? $$\overline{\operatorname{KhR}}_N(K;q,t)=\frac{\overline{\mathcal{P}}_K(q^N,q,t)}{q-q^{-1}}, \qquad \operatorname{KhR}_N(K;q,t)=\mathcal{P}_K(q^N,q,t)$$.

来源字段：https://doi.org/10.4171/owr/2005/27

10120 | AIM-TOPOLOGY-0226

### Stable tangential markings on a noncollapsed limit

P / partially_solved | AIM 研讨会问题汇编

**Problem 13. Given a non-collapsing Gromov-Hausdorff convergence \(M_i\to X\), can one find a “tangent bundle” structure on \(X\) that is sensitive to the diffeomorphism class of the \(M_i\)?**

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10121 | AIM-TOPOLOGY-0110

### Stable-genus dichotomy inside the bipolar filtration

P / partially_solved | AIM 研讨会问题汇编

4-genus and the bipolar filtration

For $n \in \mathbb{N}$, are there knots in $\mathcal{T}_n$ with arbitrarily large smooth 4-genus?

来源字段：http://aimpl.org/concordsliceknot/1/

10122 | OWR-16415-006

### Stable-Range Comparison of Torelli Lie Algebras

O / open | Oberwolfach 报告问题汇编

In the stable genus range, is $\mathfrak t_g(k)\cong\mathfrak m_g(k)$ for every degree $k\neq2$, where $\mathfrak t_g$ and $\mathfrak m_g$ denote the Torelli and symplectic-derivation graded Lie algebras?

来源字段：https://doi.org/10.4171/owr/2018/40

10123 | OPG-37293

### Sticky Cantor sets

S / solved | Open Problem Garden

Conjecture Let $C$ be a Cantor set embedded in $\mathbb{R}^n$. Is there a self-homeomorphism $f$ of $\mathbb{R}^n$ for every $\epsilon$ greater than $0$ so that $f$ moves every point by less than $\epsilon$ and $f(C)$ does not intersect $C$? Such an embedded Cantor set for which no such $f$ exists (for some $\epsilon$ ) is called "sticky". For what dimensions $n$ do sticky Cantor sets exist?

背景链接：http://www.openproblemgarden.org/op/sticky_cantor_sets

10124 | AIM-TOPOLOGY-0183

### Stretch lines and subsurface active intervals

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.5 (K. Rafi).** Assume that \(G(\lambda)\) is a stretch line directed by a maximal geodesic lamination \(\lambda\), then how does the geodesic \(G(\lambda)\) look like? Moreover, if the projection of \(G(\lambda)\) to the arc and curve graph of a subsurface \(Y\) is large, can we expect that an interval of time where each boundary component \(\beta\) of \(Y\) has bounded length and \(\beta\) is close to a geodesic on \(Y\)?

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

10125 | OPG-37339

### Strict inequalities for products of filters

S / solved | Open Problem Garden

Conjecture $\mathcal{A} \times^{\mathsf{\ensuremath{\operatorname{RLD}}}}_F \mathcal{B} \subset \mathcal{A} \ltimes \mathcal{B} \subset \mathcal{A} \times^{\mathsf{\ensuremath{\operatorname{RLD}}}} \mathcal{B}$ for some filter objects $\mathcal{A}$, $\mathcal{B}$. Particularly, is this formula true for $\mathcal{A} = \mathcal{B} = \Delta \cap \uparrow^{\mathbb{R}} \left( 0; + \infty \right)$?

A weaker conjecture:

Conjecture $\mathcal{A} \times^{\mathsf{\ensuremath{\operatorname{RLD}}}}_F \mathcal{B} \subset \mathcal{A} \ltimes \mathcal{B}$ for some filter objects $\mathcal{A}$, $\mathcal{B}$.

背景链接：http://www.openproblemgarden.org/op/strict_inequalities_for_products_of_filters

10126 | OWR-8415349-004

### String Diagram Models of Moduli Spaces

O / open | Oberwolfach 报告问题汇编

Is the string-diagram cell complex $SD$ homotopy equivalent to the moduli space of Riemann surfaces with boundary?

来源字段：https://doi.org/10.4171/owr/2021/46

10127 | AMR-106-0001

### String topology and smooth structures on 4-manifolds

P / partially_solved | AMR 问题清单汇编

Is string topology sensitive to smooth structures on 4-manifolds?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2000-2014/14-07/

10128 | OWR-14298580-008

### Strong Geography for Rational Homology Spheres

O / open | Oberwolfach 报告问题汇编

别名：Strong Geography Restrictions in Heegaard Floer Homology [OWR-14298580-006]

For every rational homology sphere $Y$, does $HF^-(Y)$ contain a direct summand $\mathbb F[U]/U^\ell\oplus\mathbb F[U]/U^{\ell-1}\oplus\cdots\oplus\mathbb F[U]/U$, where $\ell$ is the least integer satisfying $U^\ell HF^-_{\mathrm{red}}(Y)=0$?

来源字段：https://doi.org/10.4171/owr/2024/34

10129 | AIM-TOPOLOGY-0140

### Strong track stabilizers and locally finite transition matrices

P / partially_solved | AIM 研讨会问题汇编

Describe all big mapping classes that preserve a train track on the surface.

来源字段：http://aimpl.org/genusinfinity/1/

10130 | AMR-102-0011

### Sublaminations and superlaminations - Question 6.1

O / open | AMR 问题清单汇编

Characterize those essential laminations which contain genuine sublaminations.

背景链接：https://arxiv.org/abs/math/0209081

10131 | AMR-102-0012

### Sublaminations and superlaminations - Question 6.2

O / open | AMR 问题清单汇编

Suppose $\Lambda$ is a full genuine lamination; i.e. it has some complementary region which is an ideal polygon bundle over a circle. Suppose $M$ is hyperbolic. Is the core circle of this region isotopic to a geodesic? Does it have a noncoalescable insulator family?

背景链接：https://arxiv.org/abs/math/0209081

10132 | AMR-102-0013

### Sublaminations and superlaminations - Question 6.3

O / open | AMR 问题清单汇编

Suppose $\Lambda$ is a genuine lamination. When can $\Lambda$ be ``filled in'' to a very full lamination $\Lambda'$? Does it help for $M$ to be hyperbolic?

背景链接：https://arxiv.org/abs/math/0209081

10133 | AMR-102-0014

### Sublaminations and superlaminations - Question 6.5

O / open | AMR 问题清单汇编

Are loosesse laminations good for anything? Are leaves of the universal cover of a loosesse lamination properly embedded? If $M$ contains a loosesse lamination, does $\widetilde{M} = \mathbb{R}^3$?

背景链接：https://arxiv.org/abs/math/0209081

10134 | AMR-102-0015

### Sublaminations and superlaminations - Question 6.6

O / open | AMR 问题清单汇编

Give an example of a lamination in an atoroidal manifold -- perhaps loosesse -- which can never be realized by minimal surfaces for any metric, but which certifies some useful topological property of $M$.

背景链接：https://arxiv.org/abs/math/0209081

10135 | AIM-TOPOLOGY-0184

### Subsurface activity and a hereditary stretch-defect lemma

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.6 (K. Rafi).** Can we understand the behavior of Thurston's geodesics inductively, going from surfaces to subsurfaces?

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

10136 | OWR-14297741-011

### Support-Filtration Classification of Compactly Generated t-Structures

O / open | Oberwolfach 报告问题汇编

Let $A$ be a connective bounded commutative differential graded ring, let $\mathcal D=D(A)$ be its derived category, and set $X=\operatorname{Spec}(\operatorname{End}_{\mathcal D}(\mathbf 1))$. Are the compactly generated $t$-structures on $\mathcal D$ whose nonpositive aisle is a tensor ideal classified by support filtrations $f:\mathbb Z\to 2^X$?

来源字段：https://doi.org/10.4171/owr/2023/40

10137 | AIM-TOPOLOGY-0240

### Surface quotient, bundle normal forms, and a WNN escape condition

P / partially_solved | AIM 研讨会问题汇编

Is there a positively curved 5-manifold with a free isometric $S^3$ or $SO(3)$ action?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10138 | OWR-10252937-011

### Surgery Decomposition of Positive Scalar Curvature Manifolds

O / open | Oberwolfach 报告问题汇编

Can every topologically positive-scalar-curvature $n$-manifold be built from simple topologically positive-scalar-curvature $n$-manifolds by surgeries of codimension at least $3$?

来源字段：https://doi.org/10.4171/owr/2022/28

10139 | AIM-TOPOLOGY-0178

### Symplectic duality and a complex-linearity obstruction for the Thurston metric

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.5 (D. Dumas).** Is there any sense in which the Thurston metric is compatible with the complex structure of Teichmüller space? We may ask the same question for symplectic structure.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

10140 | OPG-37245

### The 4x5 chessboard complex is the complement of a link, which link?

P / partially_solved | Open Problem Garden

Problem Ian Agol and Matthias Goerner observed that the 4x5 chessboard complex is the complement of many distinct links in the 3-sphere. Their observation is non-constructive, as it uses the resolution of the Poincare Conjecture. Find specific links that have the 4x5 chessboard complex as their complement.

背景链接：http://www.openproblemgarden.org/op/the_4x5_chessboard_complex_is_the_complement_of_a_link_which_link

10141 | AIM-TOPOLOGY-0227

### The Alexandrov boundary conjecture and a polyhedral Gauss formula

P / partially_solved | AIM 研讨会问题汇编

**Problem 14.** Is there a Gauss formula for Alexandrov spaces, i.e. must a convex hypersurface $Y$ of an Alexandrov space $X$ have $\sec Y\ge \sec X$?

If $X$ is a complete finite-dimensional Alexandrov space with curvature $\ge\kappa$ and $Y$ is a convex hypersurface, is $Y$, with its induced intrinsic length metric, an Alexandrov space with curvature $\ge\kappa$?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10142 | AIM-TOPOLOGY-0198

### The arc metric is geodesic and doubling preserves its geodesics

S / solved | AIM 研讨会问题汇编

**Problem 3.7.** Is the metric space $(\mathcal T(S),d_A)$ a length space? Given $X,Y\in\mathcal T(S)$, is there a geodesic $\Gamma$ of $(\mathcal T(S^d),d_{\rm Th})$ connecting $X$ and $Y$ such that $\Gamma\subset\mathcal T(S)$?

The immediately preceding text fixes the notation:

- $S$ is a finite-type surface with nonempty boundary; - $\mathcal T(S)$ is its reduced Teichmüller space of marked bordered hyperbolic structures; - $d_A$ is the arc metric; - $S^d$ is the double of $S$ along its geodesic boundary; and - the doubling map $j:X\mapsto X^d$ is used to identify $\mathcal T(S)$ with its image in $\mathcal T(S^d)$.

With the typography restored, the question is exactly:

In the second sentence, $X,Y$ mean their doubles after the stated identification.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

10143 | OWR-17471-003

### The Berge Conjecture for Lens-Space Surgeries

P / partially_solved | Oberwolfach 报告问题汇编

If a nontrivial knot $K\subset S^3$ admits a Dehn surgery yielding a lens space, must $K$ be doubly primitive? Analogously, if $K\subset S^2\times S^1$ admits a nontrivial lens-space Dehn surgery, must $K$ be doubly primitive?

来源字段：https://doi.org/10.4171/owr/2020/8

10144 | OWR-11137-009

### The Bismut-Lott Transfer Index Conjecture

O / open | Oberwolfach 报告问题汇编

Let $\pi:W\to B$ be a proper submersion with Riemannian structure $g$, and let $(V,h^V)$ be a local system of finitely generated projective modules over a number ring, equipped with metrics. If $\operatorname{index}^{\mathrm{top}}(V,h^V)$ is the differential Becker-Gottlieb transfer index and $\operatorname{index}^{\mathrm{an}}(V,h^V)$ is the analytic index corrected by the Bismut-Lott torsion form, is

$$\delta=\operatorname{index}^{\mathrm{top}}(V,h^V)-\operatorname{index}^{\mathrm{an}}(V,h^V)=0?$$

Equivalently, do the two index classes agree?

来源字段：https://doi.org/10.4171/owr/2011/45

10145 | OWR-17470-002

### The Boundary Singer Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $B\Gamma$ be a finite $d$-dimensional complex and let $\Gamma$ be a residually finite CAT(0) or hyperbolic group. If the action of $\Gamma$ on $H^{\mathrm{st}}_{d-1}(\partial\Gamma;\mathbb F_2)$ has positive entropy, must the boundary $\partial\Gamma$ fail to embed in $S^{2d-2}$?

来源字段：https://doi.org/10.4171/owr/2020/7

10146 | OWR-13108-010

### The Bounded-Height Conjecture for Hyperbolic Dehn Fillings

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Hyperbolic Dehn-Filling Bounded-Height Question [OWR-13108-011]

For every $k$-cusped hyperbolic $3$-manifold $M$, are the heights of all Dehn fillings of $M$ uniformly bounded?

来源字段：https://doi.org/10.4171/owr/2014/38

10147 | AIM-TOPOLOGY-0163

### The classical Tits alternative fails for standard big mapping class groups

S / solved | AIM 研讨会问题汇编

No big mapping class groups satisfies the strong Tits alternative, but it is not known about the classical Tits Alternatives.

Are there any big mapping class groups that satisfy the classical Tits alternative?

来源字段：http://aimpl.org/genusinfinity/4/

10148 | AIM-TOPOLOGY-0275

### The degree-four quaternionic Toledo invariant

P / partially_solved | AIM 研讨会问题汇编

**Question 4.2 (Oscar García-Prada).** Consider a real group $G$ with a symmetric space $X=G/K$ of quaternion-Kähler type. Let $\Omega$ be a 4-form on $X$, $M$ a 4-manifold and $\rho:\pi_1(M)\to G$. Study the Toledo invariant $$\langle f^*\Omega,[M]\rangle.$$ An interesting special case is when $M$ is a Kähler surface.

来源字段：https://aimath.org/WWN/surfacegroups/surfacegroups.pdf

10149 | AIM-TOPOLOGY-0215

### The differentiable diameter sphere problem: a three-gate obstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.** If $\sec M\geq 1$ and $\operatorname{diam}(M)>\pi/2$, must $M$ be diffeomorphic to $S^n$?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10150 | AIM-TOPOLOGY-0219

### The dual Wilking bound and singular join tests

P / partially_solved | AIM 研讨会问题汇编

**Problem 6.** Extend the Wilking Connectivity Theorem to Alexandrov spaces, i.e. if $X$ is a positively curved Alexandrov space and $Y\subset X$ a totally geodesic subspace of codimension $k$, is it true that $X-Y$ has homology only up to dimension $2k-2$?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10151 | AIM-TOPOLOGY-0076

### The Euler pairing, not the raw Hom series

P / partially_solved | AIM 研讨会问题汇编

Let $P\in SYT(\lambda)$, $Q\in SYT(\mu)$ relate $Hom(tr(P),tr(Q))$ to the Macdonald inner product.

来源字段：http://aimpl.org/agclinkhom/5/

10152 | AIM-TOPOLOGY-0028

### The even-color Volume Conjecture needs a logarithm

S / solved | AIM 研讨会问题汇编

**The Colored Jones Polynomial Volume Conjecture for even root of unity**

Let $L\in S^3$ be a hyperbolic link. For n even, let $J_n(L)\in\mathbb Z[t^{\pm}]$ be the n-th normalized Colored Jones Polynomial. Then, as $n\to\infty$, $$J_n(L,t=e^{2\pi i/n})\sim \exp\!\left[\frac{n}{2\pi}\bigl(\operatorname{vol}(S^3-L)+i\,CS(S^3-L)\bigr)\right].\tag{1.1}$$

来源字段：http://aimpl.org/quantumlowdimtop/1/

10153 | AIM-TOPOLOGY-0193

### The forced length-spectrum max norm and a compatible-geodesic criterion

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.2 (A. Papadopoulos).** Is the length-spectrum metric Finsler? If yes, give a formula for the infinitesimal norm of a vector on Teichmüller space with respect to this Finsler structure.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

10154 | AIM-TOPOLOGY-0070

### The integer-slope trace--Hilbert--Cherednik bridge

P / partially_solved | AIM 研讨会问题汇编

Relate \(tr(T(m,n))\) to rings of quasi-invariants and to A. Wilson's conjecture. Describe \(H^*(\text{Hilb}(x^{nd}=y^n))\) as a module over rational Double Affine Hecke algebra (DAHA) and relate it to the above.

来源字段：http://aimpl.org/agclinkhom/4/

10155 | AIM-TOPOLOGY-0277

### The K2 avatar of the higher Weil-Petersson form

P / partially_solved | AIM 研讨会问题汇编

Following on from the previous question, there is also a central extension for \(\mathrm{SL}(n)\)

\[ 1\longrightarrow K_2(A)\longrightarrow E(n,A)\longrightarrow \mathrm{SL}(n,A)\longrightarrow 1. \] Let \(H_n\) be the Hitchin component, and let \(A=\mathbb Q(H_n)\). Given \(\rho:\pi_1(S)\to\mathrm{SL}(n,A)\), we obtain \(q\in K_2(A)\). There is \(d\log:K_2(A)\to\Omega^2(H_n)\). Is \(d\log(q)\) the Weil--Petersson form?

来源字段：https://aimath.org/WWN/surfacegroups/surfacegroups.pdf

10156 | OWR-13108-002

### The L-Space and Left-Orderability Conjecture

O / open | Oberwolfach 报告问题汇编

An irreducible rational homology $3$-sphere $W$ is not an L-space if and only if $\pi_1(W)$ is left-orderable.

来源字段：https://doi.org/10.4171/owr/2014/38

10157 | OWR-14298592-008

### The L-Space and Left-Orderability Conjecture

O / open | Oberwolfach 报告问题汇编

For every irreducible rational homology $3$-sphere $M$, are the following equivalent: $M$ is not an L-space; $M$ admits a coorientable taut foliation; and $\pi_1(M)$ is left-orderable?

来源字段：https://doi.org/10.4171/owr/2024/43

10158 | AIM-TOPOLOGY-0251

### The literal polar-rigidity question is false, with a sharp repaired theorem

S / solved | AIM 研讨会问题汇编

Suppose \(M\) is a polar manifold with \(\sec>0\). Must \(M\) be diffeomorphic to a compact rank-one symmetric space?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10159 | AIM-TOPOLOGY-0243

### The maximal symmetry-rank conjecture and a sharp product defect formula

P / partially_solved | AIM 研讨会问题汇编

**Problem 30.** Suppose \((M^n,g)\) is simply-connected and \(\sec M\geq0\). Is \[ \operatorname{rank}(\operatorname{Iso}(M^n,g))\leq \frac23 n? \]

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10160 | OWR-14299512-006

### The N-or-Infinity Conjecture for Reeb Pseudo-Rotations

O / open | Oberwolfach 报告问题汇编

Let $M$ be the smooth boundary of a star-shaped domain in $\mathbb R^{2n}$. If its Reeb flow has only finitely many prime closed orbits, must it have exactly $n$?

来源字段：https://doi.org/10.4171/owr/2025/31

10161 | OWR-1703864-004

### The Novikov Higher-Signature Conjecture

O / open | Oberwolfach 报告问题汇编

Novikov conjecture: the higher signatures of closed oriented smooth manifolds are invariant under orientation-preserving homotopy equivalences.

来源字段：https://doi.org/10.4171/owr/2020/17

10162 | AIM-TOPOLOGY-0202

### The one-sided length-spectrum obstruction and a regularized asymmetric metric

P / partially_solved | AIM 研讨会问题汇编

**Recovered Problem 4.2.** Can one define the Thurston metric on the quasiconformal Teichmüller space \(\mathcal T_{qc}(\Gamma _0)\)?

The PDF context is essential. Section 4 begins by declaring that \(\Gamma _0\) is an infinitely generated Fuchsian group of the first kind. Immediately preceding Problem 4.2, Problem 4.1 asks whether \(\Gamma _0\) can be quasiconformally deformed so that the lengths of all simple closed curves do not increase.

Here \(X_0=\mathbb H^2/\Gamma _0\) is a complete borderless hyperbolic surface of infinite topological type, and \(\mathcal T_{qc}(\Gamma _0)\) consists of marked hyperbolic structures quasiconformally equivalent to \(X_0\). “Of the first kind” means that the limit set is the full ideal circle; it does not mean that \(X_0\) is compact. The workshop report separately mentions the *symmetrized* Thurston/length-spectrum metric, so the substantive question is the one-sided analogue, not merely the already-established symmetric metric.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

10163 | OWR-8415345-007

### The Q-Form Classification Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Q-form conjecture: let $M$ and $M'$ be simply connected $2q$-manifolds with normal $(q-1)$-smoothings $\bar\nu:M\to B$ and $\bar\nu':M'\to B$, where $q$ is even. If $\bar\nu$ and $\bar\nu'$ are bordant and their Q-forms $E_q(M,\bar\nu)$ and $E_q(M',\bar\nu')$ are isomorphic, then $M$ and $M'$ are h-cobordant over $B$.

来源字段：https://doi.org/10.4171/owr/2021/42

10164 | OWR-5146-003

### The Rational Toral-Rank Conjecture

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Rational Toral-Rank Conjecture [OWR-5146-004]

If a nilpotent finite CW complex $X$ admits an almost free action of the torus $T^k$, must $\dim H^*(X;\mathbb Q)\geq2^k$?

来源字段：https://doi.org/10.4171/owr/2011/18

10165 | OWR-2089-011

### The Refined Unstable-Module Realization Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a space for which the unstable module $H^*(X;\mathbb F_2)$ does not belong to the locally finite subcategory $\mathcal U_0$. Does the unbounded strong realization conjecture hold, and is $S(H^*(X))\leq N(H^*(X))$, where $S$ is the first nilpotence-filtration level whose quotient belongs to no $\mathcal U_n$ and $N$ is the invariant determined by Steenrod operations?

来源字段：https://doi.org/10.4171/owr/2008/43

10166 | OWR-1063-005

### The Segal Conjecture for Infinite Groups

S / solved | Oberwolfach 报告问题汇编

If a group $G$ has a cocompact model for $\underline EG$, is $\pi^0(BG)$ isomorphic to the $I$-adic completion of $A_{\mathrm{ho}}(G)$, where $I$ is the augmentation ideal?

来源字段：https://doi.org/10.4171/owr/2005/42

10167 | AIM-TOPOLOGY-0263

### The solved smooth Hsiang-Kleiner classification and the limits of a direct Ricci-flow proof

S / solved | AIM 研讨会问题汇编

**Problem 50.** Can one improve the Hsiang--Kleiner theorem on positively curved $4$-manifolds with symmetry from homeomorphism to diffeomorphism by using the Ricci flow?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10168 | OWR-15581-004

### The Strong Slope Conjecture for Colored Jones Polynomials

P / partially_solved | Oberwolfach 报告问题汇编

Does the asymptotic degree growth of the colored Jones polynomial of every knot determine the boundary slopes, Euler characteristics, and numbers of sheets of essential Jones surfaces, as predicted by the Strong Slope Conjecture?

来源字段：https://doi.org/10.4171/owr/2017/38

10169 | AIM-TOPOLOGY-0056

### The superspace quotient is a noncompact trace module

P / partially_solved | AIM 研讨会问题汇编

Does the superspace coinvariant ring defined as \[ \mathbb{C}[x_1,\ldots,x_n,\theta_1,\ldots,\theta_n]\big/ \big(\mathbb{C}[x_1,\ldots,x_n,\theta_1,\ldots,\theta_n]^{S_n}_+\big) \] relate to link homology? What object does \[ \mathbb{C}[x_1,\ldots,x_n,\theta_1,\ldots,\theta_n]\big/ \big(\mathbb{C}[x_1,\ldots,x_n,\theta_1,\ldots,\theta_n]^{S_n}_+\big) \] define in the trace of Hecke category?

来源字段：http://aimpl.org/agclinkhom/1/

10170 | AIM-TOPOLOGY-0209

### The Thurston metric for unit-volume flat n-tori

S / solved | AIM 研讨会问题汇编

**Problem 5.3.** Define and study the Thurston metric on the space of flat $n$-tori $SL_n(\mathbb R)/SL_n(\mathbb Z)$.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

10171 | AIM-TOPOLOGY-0207

### The Thurston metric on the punctured-torus plane

P / partially_solved | AIM 研讨会问题汇编

**Problem 5.1.** What is Thurston metric on $\mathbb H^2$, viewed as the Teichmüller space of punctured tori?

Belkhirat, Papadopoulos and Troyanov have studied the Thurston metric on Teichmüller space of flat tori.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

10172 | TOP-002

### The Volume Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Volume Conjecture [TOP-011]; Volume Conjecture [TOP-015]; Volume Conjecture for Knot Polynomials [OWR-832-008]

For a hyperbolic knot $K$, the limit of normalized colored Jones polynomials equals the hyperbolic volume of the knot complement.

背景链接：https://doi.org/10.1007/BF02392716

10173 | OWR-8415345-003

### The Weak Gordon-Luecke Conjecture

O / open | Oberwolfach 报告问题汇编

Let $K_1$ and $K_2$ be knots in a closed oriented irreducible $3$-manifold $Y$, neither a rational unknot in a lens space. Prove that $Y\setminus K_1$ and $Y\setminus K_2$ are orientation-preservingly homeomorphic if and only if an orientation-preserving self-homeomorphism of $Y$ takes $K_1$ to $K_2$.

来源字段：https://doi.org/10.4171/owr/2021/42

10174 | OWR-4411-001

### The Weinstein Periodic-Orbit Conjecture

O / open | Oberwolfach 报告问题汇编

Does every Reeb vector field on a closed three-manifold have a periodic orbit?

来源字段：https://doi.org/10.4171/owr/2010/30

10175 | TOP-003

### The Whitehead Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Whitehead Conjecture [TOP-016]

Is every aspherical closed manifold whose fundamental group has no non-trivial perfect normal subgroups a $K(\pi, 1)$ space?

背景链接：https://pi.math.cornell.edu/~hatcher/AT/AT.pdf

10176 | OWR-16939-002

### Theta Function Rings and Symplectic Cohomology

P / partially_solved | Oberwolfach 报告问题汇编

Let $(M,D)$ be a positive maximally degenerate pair, let $X=M\setminus D$, and let $A$ be the ring generated by its Gross-Siebert theta functions. Beyond the known additive identification, is $A$ isomorphic as a ring to the symplectic cohomology $SH^*(X)$?

来源字段：https://doi.org/10.4171/owr/2019/27

10177 | OWR-13855-004

### Three Closed Reeb Orbits Beyond Lens Spaces

S / solved | Oberwolfach 报告问题汇编

Does every Reeb vector field on a three-manifold that is not a lens space have at least three closed Reeb orbits?

来源字段：https://doi.org/10.4171/owr/2015/34

10178 | OWR-13940-008

### Three-Dimensional Coloring Number of the Sphere

S / solved | Oberwolfach 报告问题汇编

What is the $3$-dimensional coloring number $\chi_3(S^3)$ for triangulations of the $3$-sphere, where monochromatic triangles are allowed but monochromatic tetrahedra are forbidden; in particular, is it finite or is there an upper bound?

来源字段：https://doi.org/10.4171/owr/2015/45

10179 | AIM-TOPOLOGY-0180

### Thurston geodesic envelopes: solved surface classes and a directed interval order

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.2 (F. Guéritau).** Given $X,Y\in\mathcal T(S)$, describe the set $$\operatorname{Env}(X,Y):=\bigcup\{G\},$$ where $G$ denotes a Thurston geodesic connecting $X$ to $Y$.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

10180 | AIM-TOPOLOGY-0009

### Tight realization through almost Weinstein fillings

P / partially_solved | AIM 研讨会问题汇编

**Must every almost contact class admit a tight structure?**

If \(M^{2n-1}\) is a closed cooriented almost contact manifold of dimension at least five, does every homotopy class \(a\) of almost contact structures on \(M\) contain a Borman--Eliashberg--Murphy-tight contact structure?

来源字段：http://aimpl.org/highdimcontacttop/4/

10181 | OWR-11454-006

### Topological $\widehat A$-Classes on Witt Spaces

O / open | Oberwolfach 报告问题汇编

For a Witt space $X$, under what hypotheses can the homology $\widehat A$-class be defined topologically, how does it depend on a positive-scalar-curvature metric on the links, and is it the Chern character of a suitable $K$-homology class defined by the spin Dirac operator?

来源字段：https://doi.org/10.4171/owr/2011/56

10182 | OWR-16406-011

### Topological Characterization of Complex Algebraic Links

O / open | Oberwolfach 报告问题汇编

Which link types arise as $L=V\cap S^3$, where $V\subset\mathbb C^2$ is an algebraic curve and $S^3$ is the unit sphere, and what topological obstructions or characterizations detect such C-links?

来源字段：https://doi.org/10.4171/owr/2018/31

10183 | OWR-1596-011

### Topological Complexity of Fibrations and Nonorientable Surfaces

S / solved | Oberwolfach 报告问题汇编

If $F\to E\to B$ is a locally trivial fibration, is $\operatorname{TC}(E)\leq\operatorname{TC}(F)\operatorname{TC}(B)$? Also, if $N_g$ is the nonorientable surface of genus $g$, what is $\operatorname{TC}(N_g)$?

来源字段：https://doi.org/10.4171/owr/2007/40

10184 | OWR-1596-012

### Topological Complexity of Polygon Spaces and Three-Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

For a generic length vector $\ell$, what is the topological complexity $\operatorname{TC}(M_\ell)$ of the moduli space of closed planar $n$-gons with side lengths $\ell$, modulo rotations and translations? More generally, for a closed 3-manifold $M$, does $\operatorname{TC}(M)$ depend only on $\pi_1(M)$; in particular, is $\operatorname{TC}(L(p,q))=\operatorname{TC}(L(p,1))$?

来源字段：https://doi.org/10.4171/owr/2007/40

10185 | OWR-11454-004

### Topological Isolation of Singularities in Witt Bordism

O / open | Oberwolfach 报告问题汇编

Can every $\mathbb Q$-Witt space be shown, by a purely topological construction, to be $\mathbb Q$-Witt bordant to a $\mathbb Q$-Witt space with at worst isolated singularities?

来源字段：https://doi.org/10.4171/owr/2011/56

10186 | OWR-17132-011

### Topological Modular Forms for Cyclic Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a general positive integer $n$, calculate the spectrum $\operatorname{TMF}^{C_n}$ or determine its homotopy groups.

来源字段：https://doi.org/10.4171/owr/2019/36

10187 | OWR-14297744-008

### Topological Modular Forms over Generalized Elliptic Curves

O / open | Oberwolfach 报告问题汇编

Extend topological modular-form operations to the moduli stack of generalized elliptic curves; identify $\lim_{\mathrm{Isog}}\mathcal O^{\mathrm{top}}$; and determine whether its map from the TMF-local sphere yields integral or near-integral resolutions of that sphere.

来源字段：https://doi.org/10.4171/owr/2023/34

10188 | OWR-13681-012

### Topological Period-Index Conjecture

S / solved | Oberwolfach 报告问题汇编

For a $2d$-dimensional finite CW complex $X$ and $\alpha\in \operatorname{Br}(X)$, the topological period-index conjecture predicts $\operatorname{ind}_{\mathrm{topo}}(\alpha)\mid \operatorname{per}_{\mathrm{topo}}(\alpha)^{d-1}$, except possibly when $\operatorname{per}(\alpha)\equiv2\pmod 4$ and $d=3$.

来源字段：https://doi.org/10.4171/owr/2015/24

10189 | OWR-4534-003

### Topological Repair of Non-Hausdorff Quotient Spaces

O / open | Oberwolfach 报告问题汇编

Let $X$ be a non-Hausdorff space and let $S\subseteq X$ be a closed crack, meaning that any distinct $x,y\in X$ not both in $S$ have disjoint neighborhoods. Put $R=X\setminus S$. How should one choose $L\subseteq2^S$ and topologize $\widetilde X=L\sqcup R$ so as to repair the non-Hausdorff quotient?

来源字段：https://doi.org/10.4171/owr/2010/52

10190 | OWR-12697686-015

### Topological Sliceness of the Figure-Eight Cable

O / open | Oberwolfach 报告问题汇编

Is the $(2,1)$-cable of the figure-eight knot topologically slice?

来源字段：https://doi.org/10.4171/owr/2023/3

10191 | OWR-8415362-004

### Topology of Affine and Doubly Extended Associahedra

O / open | Oberwolfach 报告问题汇编

Is the affine generalized associahedron of rank $n+1$ homeomorphic to $S^n$; is the cluster complex of a doubly extended cluster algebra of rank $n+2$ homotopy equivalent to $S^{n-1}$; and does the stated product-of-spheres description hold for the associated doubly extended associahedron?

来源字段：https://doi.org/10.4171/owr/2021/57

10192 | OWR-15208-026

### Topology of Complete Noncompact Metric Moduli Spaces

O / open | Oberwolfach 报告问题汇编

For a noncompact manifold $V$, what is the appropriate topology on the space of complete smooth metrics with nonnegative sectional curvature and on its quotient by the diffeomorphism group of $V$?

来源字段：https://doi.org/10.4171/owr/2017/3

10193 | OWR-12007-022

### Topology of Strict-to-General Contactomorphism Inclusions

P / partially_solved | Oberwolfach 报告问题汇编

What is the topology of the natural inclusion

$$j:\operatorname{Cont}_0(M,\alpha)\hookrightarrow\operatorname{Cont}_0(M,\xi)$$

for a contact manifold?

来源字段：https://doi.org/10.4171/owr/2012/35

10194 | OWR-12014-011

### Torsion Growth and Hyperbolic Volume in Finite Covers

O / open | Oberwolfach 报告问题汇编

Let $X$ be an irreducible orientable $3$-manifold with empty or torus boundary, let $\operatorname{Vol}(X)$ be the sum of hyperbolic-piece volumes, and let $t(X,G)=|\operatorname{Tor}H_1(X_G,\mathbb Z)|$ for finite-index normal subgroups $G\triangleleft\pi_1(X)$. Is $\limsup_{G\to\infty}t(X,G)^{1/[\pi_1(X):G]}=\exp(\operatorname{Vol}(X)/(6\pi))$?

来源字段：https://doi.org/10.4171/owr/2012/42

10195 | OWR-11101915-005

### Torsion in the Integral Homology-Cobordism Group

O / open | Oberwolfach 报告问题汇编

Does the integer homology-cobordism group $\Theta^3_{\mathbb Z}$ contain torsion?

来源字段：https://doi.org/10.4171/owr/2022/34

10196 | AIM-TOPOLOGY-0059

### Torus-link vacuum vectors from one-box Catalanimals

P / partially_solved | AIM 研讨会问题汇编

**Do Catalanimal operators appear naturally in link homology?**

来源字段：http://aimpl.org/agclinkhom/1/

10197 | OWR-4416-012

### Trace Fields of Hyperbolic 3-Manifolds

O / open | Oberwolfach 报告问题汇编

Is every pair $(K,\sigma)$ with $\sigma(K)\not\subset\mathbb{R}$ the trace field of some hyperbolic $3$-manifold?

来源字段：https://doi.org/10.4171/owr/2010/35

10198 | OWR-11787-006

### Transfer Index Conjecture in Differential Algebraic $K$-Theory

O / open | Oberwolfach 报告问题汇编

For a proper submersion $\pi:W\to B$ equipped with a Riemannian structure and a geometrized local system $[V,h^V]$, does the topological differential-algebraic $K$-theory index $\operatorname{ind}_{\mathrm{top}}[V,h^V]$ equal the analytic index $\operatorname{ind}_{\mathrm{an}}[V,h^V]$?

来源字段：https://doi.org/10.4171/owr/2012/25

10199 | OWR-12007-012

### Translated-Point Bounds for Small Contact Isotopies

P / partially_solved | Oberwolfach 报告问题汇编

For a $C^0$-small contact isotopy $\{\phi_t\}$ on a contact manifold $(M,\xi)$, does every $\phi_t$ have at least as many translated points as the minimum number of critical points of a smooth function on $M$?

来源字段：https://doi.org/10.4171/owr/2012/35

10200 | OWR-11786-027

### Triangulations of 3-Manifolds with Edge Valences Five and Six

O / open | Oberwolfach 报告问题汇编

Can every orientable 3-manifold be triangulated using only edge valences 5 and 6, improving the known construction with valences in the set 4,5,6?

来源字段：https://doi.org/10.4171/owr/2012/24

10201 | OWR-12697686-002

### Trisection Distance Bounds from Nonseparating Hypersurfaces

O / open | Oberwolfach 报告问题汇编

Suppose a smooth $4$-manifold $X$ contains a nonseparating $3$-manifold $Y$, and $\Sigma$ is the central surface of a trisection of $X$. Can the topology of $Y$ and the induced loop in the three disk sets be used to bound the trisection distance $d(\Sigma)$ from above?

来源字段：https://doi.org/10.4171/owr/2023/3

10202 | OWR-14298580-009

### Triviality of the Parametrized-Surgery Class on $S^4$

O / open | Oberwolfach 报告问题汇编

Is the at-most-two-torsion class $\operatorname{ps}\circ r([c])=\operatorname{wat}(\Theta)=\operatorname{impl}(yx)=G(\nu T_c)$ in $\pi_0\operatorname{Diff}(S^4)$ trivial?

来源字段：https://doi.org/10.4171/owr/2024/34

10203 | OWR-13490-021

### Tverberg Intersections Under Good Cover Conditions

P / partially_solved | Oberwolfach 报告问题汇编

For continuous maps in the topological Tverberg setting: does the Tverberg conclusion hold when images of faces form a good cover or are contractible; is there a common point in all images of faces of dimension m-r; and does the disjoint-face intersection assertion hold for affine maps from an m-polytope?

来源字段：https://doi.org/10.4171/owr/2015/5

10204 | OWR-1195-004

### Twisted Dirac Kernels Under Homotopy Equivalence

O / open | Oberwolfach 报告问题汇编

Given homotopy-equivalent closed Riemannian manifolds $f:M'\to M$, is there $\delta>0$ such that every vector bundle $(E,\nabla)$ on $M$ with curvature norm at most $\delta$ has $\ker D_E\cong\ker D_{f^*E}$?

来源字段：https://doi.org/10.4171/owr/2006/13

10205 | OWR-14607-001

### Twisted Field Theories, Factorization Algebras, and TMF Periodicity

P / partially_solved | Oberwolfach 报告问题汇编

How does the theory of twisted functorial field theories relate to factorization algebras, and can this relationship explain the 576-fold periodicity of TMF?

来源字段：https://doi.org/10.4171/owr/2016/25

10206 | OWR-14299512-007

### Two-Point Connectivity by Fixed-Energy Magnetic Geodesics

O / open | Oberwolfach 报告问题汇编

Given two points on a closed manifold and a fixed positive energy, does there always exist a magnetic geodesic of that energy joining them, and how does the answer depend on the Mañé critical values?

来源字段：https://doi.org/10.4171/owr/2025/31

10207 | AIM-TOPOLOGY-0153

### Two-sided curve compactness and a multitwist separation from coarse boundedness

P / partially_solved | AIM 研讨会问题汇编

Describe the compact subsets/subgroups of $\operatorname{Map}(S)$. Also describe all bounded subsets/subgroups of $\operatorname{Map}(S)$.

来源字段：http://aimpl.org/genusinfinity/4/

10208 | AIM-TOPOLOGY-0081

### Two-strand equivariant cohomology of Coxeter-power braid varieties

P / partially_solved | AIM 研讨会问题汇编

Compute $H^*_T(X(\sigma_1\dots\sigma_{n-1})^m\Delta)$.

来源字段：http://aimpl.org/agclinkhom/6/

10209 | AIM-TOPOLOGY-0112

### Two-torsion and the zeroth bipolar level

P / partially_solved | AIM 研讨会问题汇编

**Torsion and the bipolar filtration.** Are all 2-torsion knots 0-bipolar?

来源字段：http://aimpl.org/concordsliceknot/1/

10210 | OWR-15177-008

### Unbounded Simplicial Volume in Surface-Bundle Families

O / open | Oberwolfach 报告问题汇编

For the surface bundles $E_R$ over $\Sigma_2$ constructed with pullback simplicial volumes at least $R$, does the simplicial volume of $E_R$ tend to infinity as $R\to\infty$?

来源字段：https://doi.org/10.4171/owr/2016/56

10211 | AIM-TOPOLOGY-0206

### Uniform critical-exponent control under quasiconformal deformation

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.5 (F. Guéritaud).** Can we have a quasiconformal deformation of \(\Gamma _0\) which bend a variation of \(\delta(\Gamma _n)\) independent of \(n\)?

The printed source says, verbatim (apart from typographical spacing),

The string “5. The intended mathematical verb cannot be recovered with confidence. The immediately preceding text supplies the setting: \(\Gamma _0\) is an infinitely generated Fuchsian group of the first kind, and \(\Gamma _n\) is the subgroup carried by the \(n\)-ball in \(\mathbb H^2/\Gamma _0\). Problem 4.4 asks whether \(\delta(\Gamma _n)\to\delta(\Gamma _0)\).

I therefore keep the source wording visible and separate three plausible readings:

1. **uniform-bound reading:** “bend” is a typo for “bound,” and one asks for a bound on the finite change \(\delta(\Gamma_n')-\delta(\Gamma_n)\) whose constant is independent of \(n\); 2. **infinitesimal reading:** one asks for first-variation bounds for \(\frac d{dt}\delta(\Gamma_n(t))\) independent of \(n\); 3. **exact-equality reading:** one asks for a deformation for which \(\delta(\Gamma_n(t))-\delta(\Gamma_n(0))\) is literally the same function of \(t\) for every \(n\).

The first reading admits an explicit answer, and a natural asymptotic form of “independent of \(n\)” admits a positive answer along compact deformation families. The second has a conditional almost-everywhere answer. The third is much stronger and is obstructed by invariant elementary stages. Because the original intended reading remains unresolved, the result is classified as conditional rather than as a solution of the printed problem.

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

10212 | OWR-17471-009

### Uniform Perfectness of Four-Manifold Diffeomorphism Groups

O / open | Oberwolfach 报告问题汇编

Are there closed orientable smooth $4$-manifolds $M$ for which the identity component $\operatorname{Diff}_0(M)$ is not uniformly perfect?

来源字段：https://doi.org/10.4171/owr/2020/8

10213 | AIM-TOPOLOGY-0268

### Uniform pinching via the best-pinching functional

P / partially_solved | AIM 研讨会问题汇编

**Problem 55.** Is there a \(\delta(n)>0\) such that any \(n\)-dimensional, positively curved manifold carries a \(\delta(n)\)-pinched metric?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10214 | OWR-17471-014

### Universal Additive Vassiliev Invariants from Embedding Calculus

P / partially_solved | Oberwolfach 报告问题汇编

For every $n\geq1$, is the evaluation map $\pi_0\operatorname{ev}_n:K(I^3)\to\pi_0T_n(I,I^3)$ a universal additive Vassiliev invariant of type at most $n-1$ over $\mathbb Z$? Equivalently, is it surjective with kernel consisting exactly of knots that are $n$-equivalent to the unknot?

来源字段：https://doi.org/10.4171/owr/2020/8

10215 | AIM-TOPOLOGY-0152

### Universal countable subgroups and a finite-genus obstruction

P / partially_solved | AIM 研讨会问题汇编

For every $S$ does there exist a countable group $G$ such that $G$ is not a subgroup of $\operatorname{Map}(S)$?

Aougab-Patel-Vlamis have shown that when $S$ is the Loch Ness monster surface, there is no such group $G$. What about other surfaces?

来源字段：http://aimpl.org/genusinfinity/4/

10216 | OWR-13940-006

### Universal Sequences of Plane Curves for Knot Diagrams

O / open | Oberwolfach 报告问题汇编

For a sequence of generic plane immersions of the circle that eventually carries every knot after resolving double points, what characterizes such a universal sequence of plane curves?

来源字段：https://doi.org/10.4171/owr/2015/45

10217 | TOP-004

### Unknotting Problem

P / partially_solved | 独立 / 未分配集合

别名：Unknotting Problem [TOP-014]; Slice-Ribbon Conjecture [OWR-12697686-013]

Can unknots be recognized in polynomial time?

背景链接：https://doi.org/10.1145/301970.301971

10218 | OPG-37237

### Unsolvability of word problem for 2-knot complements

O / open | Open Problem Garden

Problem Does there exist a smooth/PL embedding of $S^2$ in $S^4$ such that the fundamental group of the complement has an unsolvable word problem?

背景链接：http://www.openproblemgarden.org/op/unsolvability_of_word_problem_for_2_knot_complements

10219 | AMR-110-0060

### Unstable homotopy theory 1 - The Johnson question.

O / open | AMR 问题清单汇编

The Johnson question. This says that if X is a space, and x is in BP_n (X), then x is not v_n torsion. My guess is that one should consider this question as a test case for whether there can be a really powerful analog of the Lannes theory of unstable algebras over the Steenrod algebra using BP instead. Such a theory has been constructed by Boardman, Johnson, and Wilson, but so far it has been unable to resolve this question.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

10220 | AMR-110-0061

### Unstable homotopy theory 2 - Determine the v_1 -exponents for the spheres.

S / solved | AMR 问题清单汇编

Determine the v_1 -exponents for the spheres. Recall that Cohen, Moore, and Neisendorfer showed that the p-torsion in the homotopy of S^2n+1 is all killed by p^n, but not p^n-1, for odd p. Determine the analogous v_1-exponent. I am not quite clear on the statement of this problem; I think we want to look at maps from the Moore space into X, so that the Adams map actually acts. There is probably a conjecture out there about what the exponent should be, but again I do not know it. Hopefully somebody will let me know and I can clean up this problem.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

10221 | AMR-110-0062

### Unstable homotopy theory 3 - Suppose X is a simply connected finite complex.

P / partially_solved | AMR 问题清单汇编

Suppose X is a simply connected finite complex. Do the Steenrod reduced powers P^t act trivially on the mod p cohomology of the loop space of X when p is sufficiently large? This question is apparently due to Wilkerson, and is taken from Chuck McGibbon's problem list on phantom maps, in Stable and unstable homotopy, Fields Institute Communications 19 (a book published by the AMS), where there are other questions as well.

背景链接：https://www-users.cse.umn.edu/~tlawson/hovey/

10222 | OPG-37385

### Upgrading a completary multifuncoid

S / solved | Open Problem Garden

Let $\mho$ be a set, $\mathfrak{F}$ be the set of filters on $\mho$ ordered reverse to set-theoretic inclusion, $\mathfrak{P}$ be the set of principal filters on $\mho$, let $n$ be an index set. Consider the filtrator $\left( \mathfrak{F}^n; \mathfrak{P}^n \right)$.

Conjecture If $f$ is a completary multifuncoid of the form $\mathfrak{P}^n$, then $E^{\ast} f$ is a completary multifuncoid of the form $\mathfrak{F}^n$.

See below for definition of all concepts and symbols used to in this conjecture.

Refer to this Web site for the theory which I now attempt to generalize.

背景链接：http://www.openproblemgarden.org/op/upgrading_a_completary_multifuncoid

10223 | OWR-14299915-021

### Vanishing for Non-$p$-Nilpotent Compact Lie Groups

O / open | Oberwolfach 报告问题汇编

Let $G$ be a positive-dimensional compact Lie group such that $\pi_0(G)$ is not $p$-nilpotent. Is the map $\pi_*(\Gamma_k C_*(\Omega BG^\wedge_p))\to\pi_*(C_*(\Omega BG^\wedge_p))$ zero?

来源字段：https://doi.org/10.4171/owr/2026/9

10224 | OWR-17470-010

### Vanishing Normalized Homology Under Local Manifold Convergence

P / partially_solved | Oberwolfach 报告问题汇编

Let $d\geq3$, and let $(M_n)$ be triangulated compact $d$-manifolds whose triangulation $1$-skeleta have uniformly bounded vertex degrees and which Benjamini-Schramm converge to a triangulation of $\mathbb R^d$. Is

$$\lim_{n\to\infty}\frac{\dim H_1(M_n)}{|M_n|}=0$$?

来源字段：https://doi.org/10.4171/owr/2020/7

10225 | AIM-TOPOLOGY-0170

### Veech groups on Jacob's ladder

P / partially_solved | AIM 研讨会问题汇编

Which Veech groups arise from translation structures on the ladder surface?

来源字段：http://aimpl.org/genusinfinity/5/

10226 | OWR-13490-013

### Vertex-Symmetry Criteria for Mod-Two Tightness

O / open | Oberwolfach 报告问题汇编

If X is a 2-dimensional simplicial complex such that $c_Y(y_1,y_2)=c_Y(y_2,y_1)$ for every induced subcomplex Y and every pair of distinct vertices $y_1,y_2$ of Y, must X be $\mathbb{Z}_2$-tight?

来源字段：https://doi.org/10.4171/owr/2015/5

10227 | OWR-4416-018

### Virtual Geometric Structures of 3-Manifolds

O / open | Oberwolfach 报告问题汇编

Which hyperbolic 3-manifolds are secretly Seifert-fibered, fibered, or have a specified JSJ decomposition, and which 3-manifolds secretly have torsion in their fundamental groups?

来源字段：https://doi.org/10.4171/owr/2010/35

10228 | AIM-TOPOLOGY-0225

### Virtual rational ellipticity and the nilmanifold-fiber reduction

P / partially_solved | AIM 研讨会问题汇编

**Problem 12.** Give an Alexandrov analogue of rational ellipticity. In particular, are manifolds with almost non-negative sectional curvature rationally elliptic?

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10229 | OWR-13687-008

### Virtual Surjectivity and Generalized Torelli Kernels

O / open | Oberwolfach 报告问题汇编

For a finite quotient $p:\pi_1(\Sigma_g)\twoheadrightarrow H$, is the induced Prym representation $\rho_{H,p}$ virtually surjective onto every component of the centralizer of $H$ in the relevant symplectic group? Are the kernels of these representations, viewed as generalized Torelli groups, finitely generated?

来源字段：https://doi.org/10.4171/owr/2015/30

10230 | AMR-105-0001

### Virtual-knot problem 1 - Recognising the Kishino Knot

P / partially_solved | AMR 问题清单汇编

Recognising the Kishino Knot: There have been invented many ways to recognize the Kishino virtual knot (from the unknot): The $3$-strand Jones polynomial, i.e. the Jones polynomial of the $3$-strand cabling of the knot , the $\Xi$-polynomial, see , the quaternionic biquandle , and the surface bracket polynomial (Dye and Kauffman ). In Kadokami proves the knot is non-trivial by examining the immersion class of a shadow curve in genus two. One can use the Manturov parity bracket to show that essentially the flat Kishino diagram is its own invariant, exhibiting the non-triviality of this knot. See for an exposition of this proof.

Are we done with this knot? Perhaps not. Other proofs of its non-triviality may be illuminating. The fact that the Kishino diagram is non-trivial and yet a connected sum of trivial virtual knots suggests the question: Classify when a non-trivial virtual knot can be the connected sum of two trivial virtual knots. A key point here is that the connected sum of closed virtual knots is not well defined and hence the different choices give some interesting effects. With long virtual knots, the connected sum is ordered but well-defined, and the last question is closely related to the question of classifying the different long virtual knots whose closures are equivalent to the unknot.

背景链接：https://arxiv.org/abs/1409.2823

10231 | AMR-105-0010

### Virtual-knot problem 10 - The Fundamental Biquandle

P / partially_solved | AMR 问题清单汇编

The Fundamental Biquandle: Does the fundamental biquandle, see classify virtual links up to mirror images? (We know that the biquandle has the same value on the orientation reversed mirror image where the mirror stands perpendicular to the plane (see ).

Are there good examples of weak biquandles which are not strong? This problem is solved in .

We would like to know more about the algebra with 2 generators $A,\,B$ and one relation $[B,(A-1)(A,B)]=0$ (see ). It is associated to the linear case.

背景链接：https://arxiv.org/abs/1409.2823

10232 | AMR-105-0011

### Virtual-knot problem 11 - Virtualization and Unit Jones Polynomial

O / open | AMR 问题清单汇编

Virtualization and Unit Jones Polynomial: Suppose the knot $K$ is classical and not trivial. Suppose that ${\tilde K}$ (obtained from $K$ by virtualizing a subset of its crossings) is not trivial and has a unit Jones polynomial, $V(\tilde K)=1$. Is it possible that ${\tilde K}$ is classical (i.e. isotopic through virtual equivalence to a classical knot)?

Suppose $K$ is a virtual knot diagram with unit Jones polynomial. Is $K$ equivalent to a classical diagram via virtual equivalence plus crossing virtualization? (Recall that by crossing virtualization, we mean flanking a classical crossing by two virtual crossings. This operation does not affect the value of the Jones polynomial.)

Given two classical knots $K$ and $K',$ if $K$ can be obtained from $K'$ by a combination of crossing virtualization and virtual Reidemeister moves, then is $K$ classically equivalent to $K'?$

If the above two questions have affirmative answers, then the only classical knot with unit Jones polynomial is the unknot.

背景链接：https://arxiv.org/abs/1409.2823

10233 | AMR-105-0012

### Virtual-knot problem 12 - Virtual Quandle Homology

O / open | AMR 问题清单汇编

Virtual Quandle Homology: Study virtual quandle homology in analogy to quandle homology .

背景链接：https://arxiv.org/abs/1409.2823

10234 | AMR-105-0013

### Virtual-knot problem 13 - Khovanov Homology

P / partially_solved | AMR 问题清单汇编

Khovanov Homology: Construct a generalization of the Khovanov complex for the case of virtual knots that will work for arbitrary virtual diagrams. Investigate the Khovanov homology constructed in . The main construction in this approach uses an orientable atom condition to give a Khovanov homology over the integers for large classes of virtual links. The import of our question, is to investigate this structure and to possibly find a way to do Khovanov homology for all virtual knots over the ring of integers. Similar questions can be raised for the presently evolving new classes of Khovanov homology theories related to other quantum invariants (cf. ).

By a K-full virtual knot we mean a knot for which there exists a diagram such that the leading (the lowest, or both) term comes from the $B$-state. Analogously, one defines the Kho-full knot relative to the Khovanov invariant. Call such diagrams optimal diagrams. (It is easy to find knots which are neither K-full nor Kho-full.)

Classify all K-full (Kho-full) knots.

Are optimal diagrams always minimal with respect to the number of classical crossings?

Classify all diagram moves that preserve optimality.

Is it true that if a classical knot $K$ has minimal classical diagram with $n$ crossings then any virtual diagram of $K$ has at least $n$ classical crossings?

Can any virtual knot have torsion in the $B$-state of the Khovanov homology (the genuine leading term of some diagram)? Here we use the formulation of Khovanov homology given in .

The behaviour of the lowest and the leading term of the Kauffman bracket for virtual knots was studied in and and .

背景链接：https://arxiv.org/abs/1409.2823

10235 | AMR-105-0014

### Virtual-knot problem 14 - Brauer algebra

P / partially_solved | AMR 问题清单汇编

Brauer algebra: The appropriate domain for the virtual recoupling theory is to place the Jones-Wenzl projectors in the Brauer algebra. That is, when we add virtual crossings to the Temperley-Lieb Algebra to obtain “Virtual Temperley-Lieb Algebra” the result is the Brauer algebra of all connections from $n$ points to $n$ points (see ). What is the structure of the projectors in this context? Can a useful algebraic generalization of the classical recoupling theory be formulated?

背景链接：https://arxiv.org/abs/1409.2823

10236 | AMR-105-0015

### Virtual-knot problem 15 - Virtual Alternating Knots

P / partially_solved | AMR 问题清单汇编

Virtual Alternating Knots: Define and classify alternating virtual knots.

Find an analogue of the Tait flyping conjecture and prove it. Compare .

Classify all alternating weaves on surfaces (without stabilization).

背景链接：https://arxiv.org/abs/1409.2823

10237 | AMR-105-0016

### Virtual-knot problem 16 - Crossing Number

P / partially_solved | AMR 问题清单汇编

Crossing Number: One of the most important problems in knot theory is the problem of finding the minimal crossing number for a given knot. It can be estimated by means of various invariants, the simplest one, perhaps, being the Kauffman bracket estimate: $$ \mathrm{span}\langle K\rangle \leqslant 4n-4g, $$ Here $\mathrm{span}$ stands for the difference between the leading degree and the lowest degree non-zero terms in the Laurent polynomial defined by the Kauffman bracket, $n$ is the crossing number, $g$ is the atom genus for the knot $K$ (also called the Turaev genus).

In the case $g=0$, we get the celebrated Murasugi-Kauffman-Thistlethwaite's theorem on minimality of alternating knot diagrams.

Is there is a way to prove that the atom genus $g$ is minimal for a concrete knot diagram? Then this together with a sharp estimate for the span of the Kauffman bracket guarantees the minimality of the diagram.

The estimate $4n-4g$ is exact for adequate diagrams (after Thistlethwaite). It is exact for many diagrams which are not adequate however: there are lots of minimal diagrams where the span of the Kauffman bracket “drops down”, for example, for torus $(p,q)$ knots for $p>q>2$.

Why is the span of the Kauffman bracket smaller than expected for such classes of knots?

One of possible explanations may be that the projection surface is wrongly chosen. For a torus knot, it is more convenient to consider it as being projected to the torus with no crossing points rather than to the plane with many crossing points. Thus the following problem arises: How do we use the Kauffman bracket, Thistlethwaite spanning tree, atoms and other invariants to get estimates for the number of crossings when projecting the knot diagram not to the plane but rather to some surfaces of higher genera?

Certainly, when applying smoothing on the plane we get the final expansion of the Kauffman bracket as a linear combination of the bracket for the unknot; we shall get expression in terms of the bracket of some “basic” torus knots, which makes the problem harder.

Can we get estimates from Thistlethwaite's spanning tree and/or from the Khovanov homology?

Which minimality theorems can be proved in this direction?

Is it possible to get an estimate for the atom genus when the atom is considered not in the neighborhood of the plane but rather in a neighborhood of the surface the knot is projected to and the thickness of the Khovanov homology does not exceed $2+g$?

背景链接：https://arxiv.org/abs/1409.2823

10238 | AMR-105-0017

### Virtual-knot problem 17 - Crossing number problems

P / partially_solved | AMR 问题清单汇编

Crossing number problems: For each virtual link $L$, there are three crossing numbers: the minimal number $C$ of classical crossings, the minimal number $V$ of virtual crossings, and the minimal total number $T$ of crossings for representatives of $L$. There are also a number of unknotting numbers: The classical unknotting number is the number of crossing switches needed to unknot the knot (using any diagram for the knot). The virtual unknotting number is the number of crossings one needs to convert from classical to virtual (by direct flattening) in order to unknot the knot (using any virtual diagram for the knot). Very little is known. Find out more about the virtual unknotting number.

What is the relationship between the least number of virtual crossings and the least genus in a surface representation of the virtual knot.

Is it true that $T=V+L$?

Is there any algorithm for finding $V$ for some class of virtual knots. For $T$, this is partially done for two classes of links: quasialternating and some other, see . For classical links and alternating diagrams see .

Are there some (non-trivial) upper and lower bounds for $T,\,V,\,L$ coming from virtual knot polynomials (see )?

背景链接：https://arxiv.org/abs/1409.2823

10239 | AMR-105-0018

### Virtual-knot problem 18 - Wild Virtuals

O / open | AMR 问题清单汇编

Wild Virtuals: Create the category of “wild virtual knots” and establish its axiomatics. In particular, one needs a theorem that states when a wild equivalence of tame virtual links implies a tame equivalence of these links.

背景链接：https://arxiv.org/abs/1409.2823

10240 | AMR-105-0019

### Virtual-knot problem 19 - Vassiliev Invariants

P / partially_solved | AMR 问题清单汇编

Vassiliev Invariants: Understand the connection between virtual knot polynomials and the Vassiliev knot invariants of virtual knots (in Kauffman's sense). Some of that was done in .

The key question about this collection of invariants is this: Does every Vassiliev invariant of finite type, for classical knots extend to an invariant of finite type for long virtual knots? Here we mean the problem in the sense of the formulation given in . In it was pointed out that there is a natural notion of Vassiliev invariants for virtual knots that has a different notion of finite type from that given in . This alternate formulation needs further investigation.

背景链接：https://arxiv.org/abs/1409.2823

10241 | AMR-105-0002

### Virtual-knot problem 2 - Flat Virtuals

P / partially_solved | AMR 问题清单汇编

Flat Virtuals: Flat virtual knots, also known as virtual strings , are difficult to classify. Find new combinatorial invariants of flat virtual knots.

We would like to know more about the flat biquandle algebra. This algebra is isomorphic to the Weyl algebra and has no (non-trivial) finite dimensional representations. An example that goes beyond the usual restrictions is the (very simple) affine biquandle used in to construct the Affine Index polynomial invariant of virtual knots. One can make small examples of the flat biquandle algebra, that detect some flat linking beyond mod $2$ linking numbers, but the absence of other finite dimensional representations presents a problem.

背景链接：https://arxiv.org/abs/1409.2823

10242 | AMR-105-0020

### Virtual-knot problem 20 - Embeddings of Surfaces

P / partially_solved | AMR 问题清单汇编

Embeddings of Surfaces: Given a non-trivial virtual knot $K$. Prove that there exists a minimal realization of $K$ in $N=S_{g}\times I$ and an unknotted embedding of $N\subset\mathbb{R}^{3}$ such that the obtained classical knot in $\mathbb{R}^{3}$ is not trivial. (This problem is partially solved by Dye in .

背景链接：https://arxiv.org/abs/1409.2823

10243 | AMR-105-0021

### Virtual-knot problem 21 - Non-Commutativity and Long Knots

P / partially_solved | AMR 问题清单汇编

Non-Commutativity and Long Knots: It is known that any classical long knot commutes with any long knot. This is definitely not the case in the virtual category, see paper by Kamadas, Fenn et al. However the commutativity of classical long knots may be incorporated into the commutativity of virtual knots if the following question is true. Is it true that if $K$ and $K'$ are long knots and $K{\#} K'$ is isotopic to $K'{\#} K$ then there exists a virtual long knot $L$, classical long knots $Q,\,Q'$, and non-negative integer numbers $m,n$ such that $$ K=L^{m} {\#} Q, \quad K'=L^{n} {\#} Q', $$ where by $L^{m}$ we mean the connected sum of $m$ copies of the same knot?

背景链接：https://arxiv.org/abs/1409.2823

10244 | AMR-105-0022

### Virtual-knot problem 22 - The Rack Space

P / partially_solved | AMR 问题清单汇编

The Rack Space: The rack space was invented by Fenn, Rourke and Sanderson . The homology of the rack space has been considered by the above authors and Carter, Kamada, Saito . For low dimensions, the homology has the following interesting interpretations. Two dimensional cycles are represented by virtual link diagrams consistently colored by the rack, and three dimensional cycles by the same but with the regions also colored. See the thesis of Greene . So virtual links can give, in this way, information about classical knots! For the second homology of the dihedral rack, the results are given in Greene's thesis. We now know that for a prime $p$ the third homology has a factor $\mathbb{Z}_p$, see .

Another line of enquiry is to look at properties of the birack space and associated homology.

Beyond problem number the cited item we list new problems that are added to this revised problem list.

背景链接：https://arxiv.org/abs/1409.2823

10245 | AMR-105-0023

### Virtual-knot problem 23 - Find new geometric/topological interpretations for the Jones polynomial and for Khovanov homology.

O / open | AMR 问题清单汇编

Find new geometric/topological interpretations for the Jones polynomial and for Khovanov homology.

背景链接：https://arxiv.org/abs/1409.2823

10246 | AMR-105-0024

### Virtual-knot problem 24 - Does it follow that $K$ and $K'$ are equivalent as classical knots?

O / open | AMR 问题清单汇编

Let $VKT/Z$ denote virtual knot theory modulo $Z$-equivalence, as defined in the section above on virtual knot theory. Recall that two virtual diagrams that are $Z$-equivalent have the same Jones polynomial. It is known that classical knot theory embeds in virtual knot theory. Does classical knot theory embed in $VKT/Z?$ That is, suppose that $K$ and $K'$ are classical knot diagrams, and suppose that $K$ and $K'$ are equivalent using virtual moves and $Z$-moves. Does it follow that $K$ and $K'$ are equivalent as classical knots?

背景链接：https://arxiv.org/abs/1409.2823

10247 | AMR-105-0025

### Virtual-knot problem 25 - Rotational virtual knot theory introduced in is virtual knot theory without the first virtual move (thus one does not…

P / partially_solved | AMR 问题清单汇编

Rotational virtual knot theory introduced in is virtual knot theory without the first virtual move (thus one does not allow the addition or deletion of a virtual curl). This version of virtual knot theory is significant because all quantum link invariants originally defined for classical links extend to rotational virtual knot theory. This theory has begun to be explored and deserves further exploration. We have formulated a version of the bracket polynomial for rotational virtuals that assigns variables according to the absolute value of the Whitney degree of state curves. See Figure the cited item for examples of non-trivial rotational virtual links.

The rotational bracket can be generalised in the fashion of the Arrow Polynomial. Beyond this there are all the quantum link invariants and how they behave on rotational virtuals. We are in the process of writing a new paper on rotational virtual knot theory. A natural class of invariants for rotational virtuals arises as quantum invariants associated with finite dimensional quasi-triangular Hopf algebras. This harks back to an early project that needs further articulation. In that work with Radford we articulate invariants that immediately generalize to invariants of rotational virtual knots. These invariants are defined via integrals on finite dimensional Hopf algebras and should be studied for their own sake. We would also like to know about the possibility of categorification of finite dimensional Hopf algebras and, in particular, the meaning of categorifying a right integral.

背景链接：https://arxiv.org/abs/1409.2823

10248 | AMR-105-0026

### Virtual-knot problem 26 - surface arrow invariant

P / partially_solved | AMR 问题清单汇编

The arrow polynomial generalizes to a more powerful invariant of virtual knots by defining a version of it for knots in specific thickened surfaces. The arrow polynomial for a knot in a thickened surface has state variables that take into account the isotopy class of the state curve and its arrow number as well. This gives a very strong invariant of knots in thickened surfaces and it can be applied to virtual knots by using a minimal surface representative for the virtual knot. Investigating this surface arrow invariant generalizes previous work on the surface bracket polynomial .

背景链接：https://arxiv.org/abs/1409.2823

10249 | AMR-105-0027

### Virtual-knot problem 27 - Study concordance and cobordism invariants of virtual knots.

P / partially_solved | AMR 问题清单汇编

Study concordance and cobordism invariants of virtual knots. In particular, solve the question of virtual knots up to pass-equivalence (taking a direct generalization of classical pass equivalence that gives the Arf invariant of knots). Understanding this specific problem will promote our understanding of concordance of virtual knots.

背景链接：https://arxiv.org/abs/1409.2823

10250 | AMR-105-0028

### Virtual-knot problem 28 - Biquandles

O / open | AMR 问题清单汇编

Biquandles: The first two problems are old chestnuts.

Give a descriptive representation of the free biquandle.

Give a topological explanation of the fundamental biquandle of a knot.

Is there such a thing as a free partial biquandle: probably not.

背景链接：https://arxiv.org/abs/1409.2823

10251 | AMR-105-0029

### Virtual-knot problem 29 - Khovanov homology for virtual knots and links works directly with mod-$2$ coefficients.

P / partially_solved | AMR 问题清单汇编

Khovanov homology for virtual knots and links works directly with mod-$2$ coefficients. With mod-2 coefficients there are no technical difficulties associated with the fact that a virtual state curve can resmooth to a single curve. Over the integers, this is a problem that has to be dealt with. An integral Khovanov homology for virtual knots has been constructed by Manturov . It is also the case that the work of Khovanov-Rozansky categorifiying infinitely many specializations of the Homflypt polynomial is an integral invariant for virtual knots. In principle, the Khovanov-Rozansky work solves the problem of integral Khovanov homology for virtuals, but to see this in terms of Khovanov's original definition is a very good technical problem. Solve this problem to clarify the Manturov construction.

背景链接：https://arxiv.org/abs/1409.2823

10252 | AMR-105-0003

### Virtual-knot problem 3 - The Flat Hierarchy

O / open | AMR 问题清单汇编

The Flat Hierarchy: The flat hierarchy is constructed for any ordinal $\alpha$. We label flat crossings with members of this ordinal. In a flat third Reidemeister move, a line with two $a$ labels can slide across a crossing labeled $b$ only if $a$ is greater than $b.$ This generalizes the usual to theory of flat virtual diagrams to a system with arbitrarily many different types of flat crossings. Classify the diagrams in this hierarchy. This concept is due to Kauffman (unpublished). A first step in working with the flat hierarchy can be found in .

背景链接：https://arxiv.org/abs/1409.2823

10253 | AMR-105-0030

### Virtual-knot problem 30 - In we have studied a mod-2 categorification of the arrow polynomial and discovered many baffling examples of pairs of…

P / partially_solved | AMR 问题清单汇编

In we have studied a mod-2 categorification of the arrow polynomial and discovered many baffling examples of pairs of virtual knots that are discriminated by this link homology and not distinguished by Khovanov homology mod-2 or the arrow polynomial. We want to understand this new link homology and to that end will do more computations and will try to understand the structure of this homology theory for virtual links.

背景链接：https://arxiv.org/abs/1409.2823

10254 | AMR-105-0031

### Virtual-knot problem 31 - Make a systematic study of Vassiliev invariants for virtual knots and links.

P / partially_solved | AMR 问题清单汇编

Make a systematic study of Vassiliev invariants for virtual knots and links. There is ongoing work on this problem.

背景链接：https://arxiv.org/abs/1409.2823

10255 | AMR-105-0032

### Virtual-knot problem 32 - Generalize virtual knot theory to virtual 2-spheres in 4-space.

P / partially_solved | AMR 问题清单汇编

Generalize virtual knot theory to virtual 2-spheres in 4-space.

背景链接：https://arxiv.org/abs/1409.2823

10256 | AMR-105-0033

### Virtual-knot problem 33 - Find a way to effectively compute the Kauffman-Radford-Hennings (KRH) invariants for three-manifolds via right integr…

P / partially_solved | AMR 问题清单汇编

Find a way to effectively compute the Kauffman-Radford-Hennings (KRH) invariants for three-manifolds via right integrals on finite dimensional Hopf algebras . Find a way to categorify the Kauffman-Radford-Hennings invariants for all finite dimensional quasitriangular Hopf algebras. The KRH invariants are actually invariants of rotational virtual links that are also, after normalization, invariant under the moves of the Kirby calculus. In this way, they give invariants of three manifolds and virtual three manifolds. The formulation of the KRH invariants in terms of integrals for finite dimensional Hopf algebras is very elegant, but computation is very difficult. Thus these invariants form a challenge for both virtual and classical knot theory. It is worth pointing out again, that the proper domain for all quantum link invariants is rotational virtual knot theory. Thus, this problem about the KRH invariants is a test case for the structure of quantum link invariants as a whole.

背景链接：https://arxiv.org/abs/1409.2823

10257 | AMR-105-0034

### Virtual-knot problem 34 - Create a combinatorial homotopy theory for Khovanov homology so that the Khovanov homology of a knot or link is equiv…

P / partially_solved | AMR 问题清单汇编

Create a combinatorial homotopy theory for Khovanov homology so that the Khovanov homology of a knot or link is equivalent to the homotopy type of an abstract complex (or category) associated with the knot of link. This problem is meant in the sprit of Bar-Natan's reformulation of Khovanov homology as an abstract chain homotopy class of a categorical chain complex associated with the knot or link. Recent work of Lifshitz and Sarkar goes very far in this direction. Of course we would like a deeper connection, and we would like to have the theory working for virtual knots and links.

背景链接：https://arxiv.org/abs/1409.2823

10258 | AMR-105-0035

### Virtual-knot problem 35 - In we give a formula for the Kauffman polynomial, due to Jaeger, that expresses this invariant as a state sum over or…

P / partially_solved | AMR 问题清单汇编

In we give a formula for the Kauffman polynomial, due to Jaeger, that expresses this invariant as a state sum over oriented, partially smoothed links associated with the given link. these states are evaluated by using the Homfly polynomial. Generalize the Jaeger point of view to obtain a categorification of the Kauffman polynomial by using chain complexes for these states that are derived from the Khovanov-Rozansky categorification of the Homflypt polynomial.

背景链接：https://arxiv.org/abs/1409.2823

10259 | AMR-105-0036

### Virtual-knot problem 36 - electrical

O / open | AMR 问题清单汇编

In we show how, by translating between knots and planar graphs (the checkerboard and medial constructions) one can associate a signed graph to a classical knot and that, with a choice of nodes and an interpretation of the signs as generalized electrical conductance, the conductance between two nodes of the graph is an invariant of isotopy of the knot restricted to move in the complement of these nodes. The moves on the graphs involve replacements of pedant loops and edges, series and parallel replacements and a star-triangle exchange move. These moves can be applied to abstract (possibly non-planar) graphs. As a result we obtain a generalization of knot theory (similar in spirit but quite different from virtual knot theory) by taking the electrical equivalence classes of signed graphs. Call this theory $EG_{\pm}.$ The conductance remains an invariant of these Electric-Knots and shows that the theory is highly non-trivial. This formulation of Electric Knots gives rise to many problems. Find new invariants of Graph-Knots. Note that if we take a planar immersion of a signed graph, then the induced cyclic orders of edges at the nodes of the planar embedding give a natural way (analogous to the medial graph) to associate a virtual knot diagram to the immersion. Explore this relationship of Electric Knots and virtual knots.

背景链接：https://arxiv.org/abs/1409.2823

10260 | AMR-105-0037

### Virtual-knot problem 37 - It has been pointed out that the chain complex for Knot Floer Homology (categorifying the Alexander-Conway polynomial…

P / partially_solved | AMR 问题清单汇编

It has been pointed out that the chain complex for Knot Floer Homology (categorifying the Alexander-Conway polynomial) is generated by the states described in Formal Knot Theory . These states involve an assignment of pointers to each region of the diagram (with two adjacent regions omitted) such that each pointer marks a unique crossing in the diagram. The key problem about this complex is that while the chain groups are described via knot diagram combinatorics, the differential in the complex seem to require high dimensional contact geometry. Even though there is now a combinatorial definition of Knot Floer homology via arc diagrams, this is an unsatisfactory state of affairs. The problem is to find a combinatorial definition of the differential on the complex generated by the Formal Knot Theory states.

背景链接：https://arxiv.org/abs/1409.2823

10261 | AMR-105-0038

### Virtual-knot problem 38 - The Kauffman bracket polynomial of a virtual diagram can have the leading term, i.e., the term having the highest pos…

O / open | AMR 问题清单汇编

The Kauffman bracket polynomial of a virtual diagram can have the leading term, i.e., the term having the highest possible degree, equal to zero.

Assume that we have a diagram for which the Kauffman bracket polynomial has the nonzero leading term. Classify moves (compositions of the generalized Reidemeister moves) establishing the equivalence in the class of diagrams with nonzero leading terms.

背景链接：https://arxiv.org/abs/1409.2823

10262 | AMR-105-0039

### Virtual-knot problem 39 - We still do not know whether the free knot whose Gauss diagram is a heptagon (i.e.

S / solved | AMR 问题清单汇编

We still do not know whether the free knot whose Gauss diagram is a heptagon (i.e. consists of 7 chords each of which is linked with precisely two adjacent ones) is trivial.

背景链接：https://arxiv.org/abs/1409.2823

10263 | AMR-105-0004

### Virtual-knot problem 4 - Virtuals and the Theory of Doodles

P / partially_solved | AMR 问题清单汇编

Virtuals and the Theory of Doodles: Compare flat theories of virtual knots with theories of doodles. A doodle is represented by a flat diagram in the plane. Reidemeister moves of type I and II are allowed but not type III. So a triple point must not be allowed. Khovanov has associated a group to doodles . Commutator identities can also be associated to a doodle. Cobordism of doodles is defined by an immersed surface without triple points. Cobordism classes represent elements of a free abelian group. There is undoubtably a rich seam of results which could be found by investigating doodles. For example there is no reason not to have virtual crossings as well as flat crossings. This would give doodles on a surface of higher genus, see .

背景链接：https://arxiv.org/abs/1409.2823

10264 | AMR-105-0040

### Virtual-knot problem 40 - Given a chord diagram $D$ we can construct the following formal chain complex.

P / partially_solved | AMR 问题清单汇编

Given a chord diagram $D$ we can construct the following formal chain complex. Formally speaking, this complex will look like a simplicial complex in all dimensions except zero, but one can get rid of this by using duality arguments.

In dimension $-1$ there is just one chain denoted by $\emptyset$. In dimension $0$, the chains are in one-to-one correspondence with unoriented chords of the diagram $D$; in dimension $1$ the chains are in one-to-one correspondence with pairs of unlinked chords, for dimension $2$ we take triples of pairwise unlinked chords, etc.

The differential acts on the $k$-tuple, $\{a_{1},\dots, a_{k}\}$ of pairwise unlinked chords to the $(k+1)$-tuple $\{a_{1},\dots, a_{k},a_{l}\}$ obtained by the addition of a chord unlinked with all the previous ones. The sign is taken in a standard way to make the complex well-defined with integral coefficients. Such a complex is naturally defined by the intersection graph of the chord diagram.

Example. Let the chord diagram consist of $2n$ chords, where the the following pairs of chords are linked: $(1,2)$, $(3,4)$, $(5,6)$, \dots,(2n-1,2n)$. Then the corresponding simplicial complex is the $(n-1)$-sphere (the $n$-th join of $S^{0}$.

Can we get any other simplicial complexes this way other than just bouquets of spheres? An affirmative answer to this question may lead to some homological calculations which are very easy for computer implementation. This problem was motivated several years ago by an attempt to construct the spectrification of the Khovanov homology. It turns out that if some (virtual) knot in the $A$-state has exactly one circle then its Khovanov homology in the lowest quantum grading looks exactly as described above. Thus, this approach may be useful for understanding the Khovanov homology of chord diagrams as well as some other approaches to the Khovanov homology spectrification obtained by recent work with Lipshitz.

On the other hand, this problem is motivated the construction of graph-link theory. Indeed, every triangulated manifold admits a shallow triangulation such that together with any 1-frame of a $k$-simplex, it contains the whole simplex.

Considering the one-dimensional frame of such a triangulation as the non-intersection graph of some chord diagrams, we get exactly the chord diagram, whose lower term Khovanov homology looks as described above. The problem is that not every graph is an intersection graph (neither any graph is a non-intersection graph) of any chord diagram. This leads to the graph-link theory for which all possible manifolds appear in the lower degree homology theory. This leads to some lower homotopy of graph links. The next problem can be formulated as follows.

Given two diagrams of (virtual) knots or links with exactly one circle in the $A$-state. Which combinations of Reidemeister moves on the diagrams have to be be considered in such a way that whenever two diagrams generate equivalent knots, there exists a chain of moves from one diagram to the other with all intermediate diagrams having single-circle $A$-state and such that chord diagram homology (and homotopy) corresponding to those intermediate diagrams do not change under such moves?

It is worth comparing these thoughts with the recent work of Lipschitz on the spectrification of Khovanov homology. Possibly, the Bloom-Nikonov approach to Khovanov homology can be used here.

背景链接：https://arxiv.org/abs/1409.2823

10265 | AMR-105-0041

### Virtual-knot problem 41 - Which quantum invariants extend to virtual knots themselves without restrictions?

P / partially_solved | AMR 问题清单汇编

Which quantum invariants extend to virtual knots themselves without restrictions? Certainly, there are such ones, e.g., the Kauffman bracket and the Jones polynomial.

背景链接：https://arxiv.org/abs/1409.2823

10266 | AMR-105-0042

### Virtual-knot problem 42 - One can consider braids with even numbers of strands.

O / open | AMR 问题清单汇编

One can consider braids with even numbers of strands. Markov's moves change the parity of the number of strands. Can one reformulate Markov's theorem in such a way that only braids with even numbers of strands take part in this formulation?

背景链接：https://arxiv.org/abs/1409.2823

10267 | AMR-105-0043

### Virtual-knot problem 43 - In his wonderful paper , which firstly had the title “Knot Floer Homotopy”, Sarkar constructs a cell complex.

P / partially_solved | AMR 问题清单汇编

In his wonderful paper , which firstly had the title “Knot Floer Homotopy”, Sarkar constructs a cell complex. The homology of this complex coincides with the Floer homology. In this work the following main ingredients are used: Manolescu-Ozsváth-Sarkar-Szab\'o-Thurston approach allowing one to define Heegaard-Floer homology by using grid diagrams and shellability. Roughly speaking, if complexes look like order complexes, then they can be replaced with cell (simplicial) complexes homotopically equivalent to them.

Try to do the same things for Khovanov homology. In the initial Khovanov definition this problem seems to be very difficult, since the set of chains and states can hard be ordered. However, taking into account Bloom's construction this problem looks more optimistic.

背景链接：https://arxiv.org/abs/1409.2823

10268 | AMR-105-0044

### Virtual-knot problem 44 - A complete invariant is used for proving that one theory is a part of another theory.

O / open | AMR 问题清单汇编

A complete invariant is used for proving that one theory is a part of another theory. For example, the fact that the set of classical knots is a part of the set of virtual knots, is true for the same reason as a quandle with peripheral structure can be extended from classical knots into virtual knots. Analogous statements can be done for classical and virtual braids (the extension of Hurwitz's action).

So, we get an actual problem: how can one construct a complete invariant for virtual knots?

背景链接：https://arxiv.org/abs/1409.2823

10269 | AMR-105-0045

### Virtual-knot problem 45 - To prove that the invariant $\mathcal{F}$ constructed by V.

P / partially_solved | AMR 问题清单汇编

To prove that the invariant $\mathcal{F}$ constructed by V. O. Manturov for virtual braids, is complete for virtual braids with more than two strands.

背景链接：https://arxiv.org/abs/1409.2823

10270 | AMR-105-0046

### Virtual-knot problem 46 - In it was showed how one could determine non-invertibility of long virtual knots and non-commutativity of long virtua…

P / partially_solved | AMR 问题清单汇编

In it was showed how one could determine non-invertibility of long virtual knots and non-commutativity of long virtual knots. This method used two type crossings: an early undercrossing and an early overcrossing. Further, for two types we have different operations, these operations are similar but not coincide (these operations are related to quandles). Regretfully, this method gives nearly nothing for classical knots. But it is universal: instead of the construction of quandle with two operations one can consider other ways of constructing knot invariants. In these ways we partition the set of crossings into two subsets, and for each subset we use an operation. Moreover, two operations are different enough to determine non-invertibility, and in the same time are similar to give an invariant structure. In the level of the Kauffman bracket polynomial this construction does not work. We get two problems.

Try to do the same for other objects than quandles, for example: nonlinear quandles, quandles with rings having non-unique decomposition of multiples.

Try to apply it on a “categorified level”: use the usual chain space for Khovanov homology and in it two different differentials but commute with each other (these differentials correspond to circle and star operations). These two different differentials could arise from usual and odd Khovanov homologies in the case when we have an identification (but maybe not canonical) between chain spaces of these complexes. It is expected that these invariants could recognize non-invertibility of knots.

背景链接：https://arxiv.org/abs/1409.2823

10271 | AMR-105-0047

### Virtual-knot problem 47 - It is well known that flat virtual knots are easily algorithmically recognizable, see .

P / partially_solved | AMR 问题清单汇编

It is well known that flat virtual knots are easily algorithmically recognizable, see . The absence of a geometric approach to the definition of free knots leads to the lack of the methods for solving the following three problems.

Is it true that (any) connected sum of free knots is trivial?

Are free knots algorithmically recognizable?

Prove that free knots do not commute in the general case.

In the first case we suggest that the answer is affirmative, and the proof should be a certain modification of the analogous proof for virtual knots (see, e.g., )

In the second case we suggest that the answer is negative. It seems to us that free knots are a very complicated object, where one can construct models for many logical constructions.

The third problem seems to be solvable by using (and, possibly, extending) standard methods in the free knot theory.

背景链接：https://arxiv.org/abs/1409.2823

10272 | AMR-105-0048

### Virtual-knot problem 48 - The functorial mapping was constructed, by means of it we constructed the map from the set of virtual knots to the se…

S / solved | AMR 问题清单汇编

The functorial mapping was constructed, by means of it we constructed the map from the set of virtual knots to the set of virtual knots with orientable atoms.

Can one construct an analogous map which projects the set of virtual knots to the set of classical knots? For example, one can try to find an index by using characteristic classes. Here only one condition which arises under the third Reidemeister move should hold.

Note that in the case of homologically trivial knots we can consider a two-branched covering like a projection.

背景链接：https://arxiv.org/abs/1409.2823

10273 | AMR-105-0049

### Virtual-knot problem 49 - Turaev constructed the map from the set of long flat knots to the set of long virtual knots.

O / open | AMR 问题清单汇编

Turaev constructed the map from the set of long flat knots to the set of long virtual knots. Can one construct any map from the set of long free knots to the set of long virtual knots? This question can be extended and one can try to construct any map which “enlarge” the structure.

背景链接：https://arxiv.org/abs/1409.2823

10274 | AMR-105-0005

### Virtual-knot problem 5 - Virtual Three Manifolds

O / open | AMR 问题清单汇编

Virtual Three Manifolds: There is a theory of virtual $3$-manifolds constructed as formal equivalence classes of virtual diagrams modulo generalized Kirby moves, see . From this point of view, there are two equivalences for ordinary $3$-manifolds: homeomorphisms and virtual equivalence. Do these equivalences coincide? That is, given two ordinary three manifolds, presented by surgery on framed links $K$ and $L$, suppose that $K$ and $L$ are equivalent through the virtual Kirby calculus. Does this imply that they are equivalent through the classical Kirby calculus?

What is a virtual $3$-manifold?: That is, give an interpretation of these equivalence classes in the domain of geometric topology.

Construct another theory of virtual $3$-manifolds by performing surgery on links in thickened surfaces $S_{g}\times \mathbb{R}$ considered up to stabilization. Will this theory coincide to that proposed by Kauffman and Dye ?

背景链接：https://arxiv.org/abs/1409.2823

10275 | AMR-105-0050

### Virtual-knot problem 50 - The problem about cobordisms in sections: Let us have a free knot and its cobordism.

O / open | AMR 问题清单汇编

The problem about cobordisms in sections: Let us have a free knot and its cobordism. Construct a parity on the given free knot, which is defined by using only this cobordism and respect only moves inside this cobordism.

背景链接：https://arxiv.org/abs/1409.2823

10276 | AMR-105-0051

### Virtual-knot problem 51 - Prove or disprove the conjecture about the non-uniqueness of minimal representative of a free link, i.e.

O / open | AMR 问题清单汇编

Prove or disprove the conjecture about the non-uniqueness of minimal representative of a free link, i.e. there exists a free link having several minimal representatives.

背景链接：https://arxiv.org/abs/1409.2823

10277 | AMR-105-0052

### Virtual-knot problem 52 - If $X$ is the free rack, is $\Gamma X$ a cat(0) space?

O / open | AMR 问题清单汇编

If $X$ is the free rack, is $\Gamma X$ a cat(0) space? A positive answer would imply that all its higher homotopy groups are trivial.

背景链接：https://arxiv.org/abs/1409.2823

10278 | AMR-105-0053

### Virtual-knot problem 53 - Is there any algorithm for recognition whether two graph-links are equivalent or not?

P / partially_solved | AMR 问题清单汇编

Is there any algorithm for recognition whether two graph-links are equivalent or not? Our conjecture is “no”. The idea behind that is that graph-links are complicated enough, and possibly, they contain sufficiently many degrees of freedom to include something like Turing machines. All mathematics is roughly split into the recognizable one (low-dimensional topology, hyperbolic groups, decidability) and the non-recognizable one (topology of dimension $4$ and higher, arbitrary finitely presentable groups, Turing machines etc). The usual argument for undecidability for finitely presented group allows one to construct some universal (semi)groups which includes the apparatus of the Turing machines. We think that graph-links are enough complicated to include similar things: they contain arbitrary graphs, and Reidemeister moves, in principle, could play the role of “rules for formal languages” or “group relations”. It seems unlikely that Reidemeister moves collapse graph-links to anything simple enough because of the parity considerations: there are “irreducibly odd” graph-links which are stable in the sense that every equivalent graph contains the initial graph as a subgraph.

背景链接：https://arxiv.org/abs/1409.2823

10279 | AMR-105-0054

### Virtual-knot problem 54 - Can one construct a projection from the set of graph-links to the set of realizable graph-links?

O / open | AMR 问题清单汇编

Can one construct a projection from the set of graph-links to the set of realizable graph-links?

背景链接：https://arxiv.org/abs/1409.2823

10280 | AMR-105-0055

### Virtual-knot problem 55 - Construct a parity on graph-links by using Bouchet's criterion about the realizability of a graph.

O / open | AMR 问题清单汇编

Construct a parity on graph-links by using Bouchet's criterion about the realizability of a graph. Try to find a parity which is responsible for the cyclically 6-edge connection, see .

背景链接：https://arxiv.org/abs/1409.2823

10281 | AMR-105-0056

### Virtual-knot problem 56 - Construct generalizations of the Frobenius extension and the Rasmussen for “rigid” graph-links with orientable atoms.

O / open | AMR 问题清单汇编

Construct generalizations of the Frobenius extension and the Rasmussen for “rigid” graph-links with orientable atoms.

背景链接：https://arxiv.org/abs/1409.2823

10282 | AMR-105-0057

### Virtual-knot problem 57 - Is it true that two equivalent realizable graph-links are equivalent in the class of realizable graph-links?

O / open | AMR 问题清单汇编

Is it true that two equivalent realizable graph-links are equivalent in the class of realizable graph-links? If it is not true, then construct an example.

背景链接：https://arxiv.org/abs/1409.2823

10283 | AMR-105-0058

### Virtual-knot problem 58 - Construct a group for graph-links which is analogous to the group from .

O / open | AMR 问题清单汇编

Construct a group for graph-links which is analogous to the group from .

背景链接：https://arxiv.org/abs/1409.2823

10284 | AMR-105-0059

### Virtual-knot problem 59 - Construct “graph-braids”.

O / open | AMR 问题清单汇编

Construct “graph-braids”.

背景链接：https://arxiv.org/abs/1409.2823

10285 | AMR-105-0006

### Virtual-knot problem 6 - Welded Knots

P / partially_solved | AMR 问题清单汇编

Welded Knots: We would like to understand welded knots . It is well known that if we admit forbidden moves to the virtual link diagrams, each virtual knot can be transformed to the unknot. If we allow only one forbidden move (e.g. the upper one), then there are lots of different equivalence classes of knots. In fact the fundamental group and the quandle of the virtual diagram are invariant under the upper forbidden move. The resulting equivalence classes are called welded knots. Similarly, welded braids were studied in , and every welded knot is the closure of a welded braid. The question is to construct good invariants of welded knots and, if possible, to classify them. In a mapping is constructed from welded knots to ambient isotopy classes of embeddings of tori (ribbon tori to be exact) in four dimensional space, and it is proved that this mapping is an isomorphism from the combinatorial fundamental group (in fact the quandle) of the welded knot to the fundamental group of the complement of the corresponding torus embedding in four-space. Is this Satoh mapping faithful from equivalence classes of welded knots (links) to ambient isotopy classes of ribbon torus embeddings in four-space?

Another interesting question is the following: If a welded knot has trivial fundamental group, does this imply that the knot is trivial as a welded knot?

背景链接：https://arxiv.org/abs/1409.2823

10286 | AMR-105-0060

## Virtual-knot problem 60 - Problems on Free Knot Cobordism

P / partially_solved | AMR 问题清单汇编

Problems on Free Knot Cobordism: The methods used for proving the fact that the invariant $L$ gives an obstruction to the sliceness are not immediately generalized for obtaining lower estimates on the slice genus of free knots. Two reasons are as follows. First, when we define a parity and justified parity for double lines on $\mathcal{D}$, we chose an arbitrary curve connecting the two preimages of the point on the double curve. We assert that any two curves connecting these two preimages (and behaving correctly in neighborhoods of the ends) are homotopic. It is true in the case of cobordism of genus zero, but in the case of a surface of an arbitrary genus $h$ it is, indeed, not true.

Thus, in order to define a parity for double lines we have to impose some restrictions on the spanning surface: we have to require that the cohomology class dual to the graph $\Psi$ was $\mathbb{Z}_{2}$-homologically trivial. The significance of the property for even-valent graphs to be $\mathbb{Z}_{2}$-homologically trivial is closely connected with atoms (for details see Sec. the cited item and ).

Another problem is that the Reeb graph of an arbitrary Morse function (not necessarily corresponding to the disc) is not necessarily a disc. Consider Fig. the cited item.

Thus, starting from a free knot $K$ for which, we say, $L(K)=8$, we (in principle) can turn it by a Morse bifurcation into free two-component link consisting of two free knots $K_{1}$ and $K_{2}$, for which $L(K_{i})=4$, and then by another Morse bifurcation we can reconstruct this trivial link into the unknot. The invariant $L$ is not an obstruction to this, since the sum of $4$ and $-4$ is zero.

In some cases we can overcome these two difficulties for cobordisms (of arbitrary genus).

Let $\mathcal{D}_{g}$ be a surface with boundary $S^1$. Obviously, the collection of double lines of $\mathcal{D}_{g}$ defines a relative $\mathbb{Z}_{2}$-homology class $\kappa\in H_1(\mathcal{D}_g,S^1;\mathbb{Z}_2)$. This homology class is an obstruction for the surface to be checkerboard-colorable; also, this is an obstruction for well-definedness of even/odd double lines.

Namely, if we look at the definition of an even/odd double line: we see that there is an ambiguity in the choice of path connecting two preimages of a generic point on the double line. For the case of a disc cobordism, the parity of double lines is well defined, because all such curves are homotopic. For $\mathcal{D}_{g}$ the unique obstruction to this well-definedness is the class $\kappa$.

We call a cobordism of genus $g$ checkerboard (or atomic\/) if the corresponding class $\kappa$ vanishes.

The next task (after detecting which $1$-stratum is even and which one is odd) is to distinguish between $b$ and $b'$. To this end, one should do the same for preimages of points lying on odd $1$-strata, connect them by a generic curve, and count the intersection with even double lines. So, we see that the only obstruction is the relative $\mathbb{Z}_{2}$-homology class $\kappa'\in H_1(\mathcal{D}_g,S^1;\mathbb{Z}_2)$ generated by even double lines.

We say that a checkerboard cobordism is $2$-atomic if $\kappa'$ vanishes.

The following theorem holds.

[see ] Assume for a $1$-component framed $4$-graph $K$ we have $L(K)\neq 0$. Then there is no $2$-atomic cobordism spanning the knot $K$ of any genus.

In the papers the fourth author constructed a strengthening $G_{m}$ of the group $G$ given in the present work (in the notation our group $G$ is $G_{1}$) and the invariants of free knots with values in the classes of conjugate elements from $G_{m}$. The idea is as follows. Even chords are further partitioned into chords of different types, it leads to more accurate definition of generators and relations in the group; these constructions are closely connected with iterated parities and the map deleting odd crossings. It seems that all invariants related to the groups $G_{m}$ also give an obstruction to the sliceness of a knot. Moreover, in the paper the author constructed invariants of virtual knots in which the over/undercrossing structure was taken into account besides the parity of chords. We devote a separate paper to the investigation of a connection of these invariants with cobordisms.

背景链接：https://arxiv.org/abs/1409.2823

10287 | AMR-105-0007

## Virtual-knot problem 7 - Long Knots and Long Flat Knots

P / partially_solved | AMR 问题清单汇编

Long Knots and Long Flat Knots: Enlarge the long knot invariant structure proposed in . Can one get new classical knot invariants from the approach in this paper? Bring together the ideas from with the biquandle construction from to obtain more powerful invariants of long knots. Long flat virtual knots can be studied via a powerful remark due to Turaev (in conversation) to the effect that one can associate to a given long flat virtual knot diagram $F$ a descending diagram $D(F)$ (by always going over before going under in resolving the flat (non-virtual) crossings in the diagram). The long virtual knot type of $D(F)$ is an invariant of the long flat knot $F.$ This means that one can apply any other invariant $I$ of virtual knots that one likes to $D(F)$ and $I[D(F)]$ will be an invariant of the long flat $F.$ It is quite interesting to do sample calculations of such invariants and this situation underlines the deeper problem of finding a full classification of long flat knots.

Bartholomew, Fenn, S. Kamada and N. Kamada have applied quaternion invariants to long virtual knots, see .

See also Sec. the cited item for more questions about long virtual knots and long flat virtual knots.

背景链接：https://arxiv.org/abs/1409.2823

10288 | AMR-105-0008

### Virtual-knot problem 8 - Virtual Biquandle

O / open | AMR 问题清单汇编

Virtual Biquandle: Construct presentations of the virtual biquandle with the a linear (non-commutative) representation at classical crossings and some interesting structure at virtual crossings. A start has been made by Bartholomew and Fenn . In this paper various biquandle decorations are made at classical and virtual crossings which were found by a computer search.

背景链接：https://arxiv.org/abs/1409.2823

10289 | AMR-105-0009

### Virtual-knot problem 9 - Virtual braids

P / partially_solved | AMR 问题清单汇编

Virtual braids: Is there a birack such that its action on virtual braids is faithful?

Is the invariant of virtual braids in , see also , faithful?

The action defined by linear biquandles is not faithful. This almost certainly means that the corresponding linear invariants of virtual knots and links are not faithful (see also ).

背景链接：https://arxiv.org/abs/1409.2823

10290 | AIM-TOPOLOGY-0114

### Visibility and fixed-pattern obstructions beyond two-solvability

P / partially_solved | AIM 研讨会问题汇编

**Highly solvable knots with large 4-genera.** For arbitrary $n>2$ and $g>0$ prove that there exist $n$-solvable knots $K$ such that the topological 4-genus of $K$ is strictly more than $g$.

来源字段：http://aimpl.org/concordsliceknot/1/

10291 | OWR-15581-009

### Volume Bounds from Mahler Measures of Biperiodic Links

O / open | Oberwolfach 报告问题汇编

Let $L$ be a semi-regular biperiodic alternating link, let $\Lambda$ be its toroidally alternating quotient, and let $p(z,w)$ be the characteristic polynomial of the associated toroidal dimer model. Is

$$\operatorname{vol}\!\left(T^2\times(-1,1)\setminus L\right)\leq2\pi m(p(z,w))$$

hold?

来源字段：https://doi.org/10.4171/owr/2017/38

10292 | AIM-TOPOLOGY-0030

### Volume-scale asymptotics and a dominant relative RT-channel reduction

P / partially_solved | AIM 研讨会问题汇编

**Asymptotics of Reshetikhin-Turaev and Turaev-Viro invariants for hyperbolic 3-manifolds.** Let \(M\) be a hyperbolic \(3\)-manifold. What are the asymptotics of the Turaev-Viro and Reshetikhin-Turaev invariants for this manifold?

来源字段：http://aimpl.org/quantumlowdimtop/1/

10293 | OWR-11790-001

### Weak Injectivity of the Whitehead-Double Concordance Operator

O / open | Oberwolfach 报告问题汇编

Is the Whitehead-double satellite operator weakly injective on the smooth knot concordance group; equivalently, does $\operatorname{Wh}(K)=\operatorname{Wh}(U)$ imply $K=U$?

来源字段：https://doi.org/10.4171/owr/2012/28

10294 | AIM-TOPOLOGY-0075

### Weak Macdonald categorification and a three-box algebra-lift obstruction

P / partially_solved | AIM 研讨会问题汇编

Categorify the modified Macdonald polynomials $\widetilde H_\mu$.

来源字段：http://aimpl.org/agclinkhom/5/

10295 | OWR-12697686-008

### Weeks Manifold Realization of the Margulis Constant

O / open | Oberwolfach 报告问题汇编

Is the three-dimensional Margulis constant $\varepsilon_3$ uniquely realized by the Weeks manifold $M_W$, with $\varepsilon_3=\mu(M_W)\approx 0.774$?

来源字段：https://doi.org/10.4171/owr/2023/3

10296 | AIM-TOPOLOGY-0234

### Weighted infinite tori as quantitative phase spaces

P / partially_solved | AIM 研讨会问题汇编

Find an application where infinite-dimensional Alexandrov spaces appear as limits of manifolds of increasing dimension.

3. Group Actions and Submersions

来源字段：https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf

10297 | OPG-59973

### What are hyperfuncoids isomorphic to?

P / partially_solved | Open Problem Garden

Let $\mathfrak{A}$ be an indexed family of sets.

Products are $\prod A$ for $A \in \prod \mathfrak{A}$.

Hyperfuncoids are filters $\mathfrak{F} \Gamma$ on the lattice $\Gamma$ of all finite unions of products.

Problem Is $\bigcap^{\mathsf{\tmop{FCD}}}$ a bijection from hyperfuncoids $\mathfrak{F} \Gamma$ to:

- prestaroids on $\mathfrak{A}$; - staroids on $\mathfrak{A}$; - completary staroids on $\mathfrak{A}$?

If yes, is $\operatorname{up}^{\Gamma}$ defining the inverse bijection? If not, characterize the image of the function $\bigcap^{\mathsf{\tmop{FCD}}}$ defined on $\mathfrak{F} \Gamma$.

Consider also the variant of this problem with the set $\Gamma$ replaced with the set $\Gamma^{\ast}$ of complements of elements of the set $\Gamma$.

背景链接：http://www.openproblemgarden.org/op/what_are_hyperfuncoids_isomorphic_to

10298 | OPG-37159

### What is the homotopy type of the group of diffeomorphisms of the 4-sphere?

P / partially_solved | Open Problem Garden

Problem $Diff(S^4)$ has the homotopy-type of a product space $Diff(S^4) \simeq \mathbb O_5 \times Diff(D^4)$ where $Diff(D^4)$ is the group of diffeomorphisms of the 4-ball which restrict to the identity on the boundary. Determine some (any?) homotopy or homology groups of $Diff(D^4)$.

背景链接：http://www.openproblemgarden.org/op/what_is_the_homotopy_type_of_the_group_of_diffeomorphisms_of_the_4_sphere

10299 | AIM-TOPOLOGY-0173

### What the infinitesimal Thurston norm determines

S / solved | AIM 研讨会问题汇编

Problem 1.1 (D. Dumas, K. Rafi). Does the Thurston norm at X ∈ T (S)determines X?

Does the Thurston infinitesimal norm on \(T_X\mathcal T(S)\), \(X\in\mathcal T(S)\), determine \(X\)?

来源字段：https://aimath.org/pastworkshops/lipschitzteichproblems.pdf

10300 | OPG-37154

### Which compact boundaryless 3-manifolds embed smoothly in the 4-sphere?

P / partially_solved | Open Problem Garden

Problem Determine a computable set of invariants that allow one to determine, given a compact boundaryless 3-manifold, whether or not it embeds smoothly in the 4-sphere. This should include a constructive procedure to find an embedding if the manifold is embeddable.

背景链接：http://www.openproblemgarden.org/op/which_compact_boundaryless_3_manifolds_embed_smoothly_in_the_4_sphere

10301 | OPG-37145

### Which homology 3-spheres bound homology 4-balls?

P / partially_solved | Open Problem Garden

Problem Is there a complete and computable set of invariants that can determine which (rational) homology $3$-spheres bound (rational) homology $4$-balls?

背景链接：http://www.openproblemgarden.org/op/which_homology_3_spheres_bound_homology_4_balls

10302 | TOP-010

### Zeeman Conjecture

O / open | 独立 / 未分配集合

Is $K \times [0,1]$ collapsible for every finite contractible 2-dimensional CW complex K?

背景链接：https://en.wikipedia.org/wiki/Zeeman%27s_conjecture

# 资料汇编 07 | 分析 / Analysis

1,379 条记录。O=开放；P=部分解决；S=已解决。状态与英文/LaTeX 表述均沿用来源。

10303 | OWR-1966-002

### $\Gamma$-Limits for Vector-Valued Dislocation Phase Fields

P / partially_solved | Oberwolfach 报告问题汇编

Beyond kernels of the form $\Gamma(z)=|z|^{-3}\gamma(z/|z|)$ with bounded uniformly elliptic angular part, is the $\Gamma$-limit of the vector-valued dislocation phase-field energy the lower-semicontinuous envelope in $BV$ of $$F_{\mathrm{flat}}(u)=\int\gamma_0(\nu,[u])\,d\mathcal H^1$$?

来源字段：https://doi.org/10.4171/owr/2008/21

10304 | OWR-17130-008

### $C^1$ Regularity of Planar Convex Variational Minimizers

O / open | Oberwolfach 报告问题汇编

If $F$ is strictly convex, is every Lipschitz minimizer of

$$\int_{B_1}F(\nabla u)\,dx$$

in dimension $n=2$ of class $C^1$?

来源字段：https://doi.org/10.4171/owr/2019/34

10305 | OPG-725

### $C^r$ Stability Conjecture

P / partially_solved | Open Problem Garden

Conjecture Any $C^r$ structurally stable diffeomorphism is hyperbolic.

背景链接：http://www.openproblemgarden.org/op/c_r_stability_conjecture

10306 | OWR-14299906-009

### $L^p$ Approximation for Higher-Rank Group Factors

O / open | Oberwolfach 报告问题汇编

For $d\geq 2$, does $L^p(L(\operatorname{PSL}_{2d-1}(\mathbb Z)))$ have the completely bounded approximation property for some $p\neq 2$?

来源字段：https://doi.org/10.4171/owr/2026/4

10307 | OWR-15577-002

### $L^p$ Bounds for the Triangular Hilbert Transform

O / open | Oberwolfach 报告问题汇编

Determine the range of exponents for which the triangular Hilbert transform, the trilinear singular integral with kernel $(x+y+z)^{-1}$, is bounded on the corresponding $L^p$ spaces.

来源字段：https://doi.org/10.4171/owr/2017/34

10308 | OWR-720-005

### $L^p$ Extensions for Shift-Invariant Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Extend the $L^2$ theorem on generic linear combinations of generators of finitely generated shift-invariant spaces to $L^p(\mathbb{R}^d)$ for $p\ne 2$.

来源字段：https://doi.org/10.4171/owr/2004/10

10309 | OWR-13750336-011

### $L^p$ Ranges for Radial Projection Estimates

P / partially_solved | Oberwolfach 报告问题汇编

Determine the optimal exponent range in mixed $L^p$ estimates for radial projections of Frostman measures, and find geometric applications in which that range improves the conclusion.

来源字段：https://doi.org/10.4171/owr/2023/25

10310 | AIM-FUNCTIONAL_ANALYSIS-0041

### A Borel-categorical framework and an isotropy-loss lemma

O / open | AIM 研讨会问题汇编

What is the right set-theoretic framework to deal with functorial classification?

Which measurable/categorical objects and comparison maps simultaneously record (i) the descriptive complexity of assigning invariants to coded objects and (ii) the action of an invariant on isomorphisms, or more general morphisms?

来源字段：http://aimpl.org/settheorycstar/7/

10311 | AIM-ANALYSIS-0034

### A closure obstruction for the proposed oval flow

P / partially_solved | AIM 研讨会问题汇编

**Conjecture (The ovals of Benguria and Loss).** Let $\Gamma \subset \mathbb{R}^2$ be a closed smooth curve of length $2\pi$, let $\kappa(s)$ be its curvature as a function of arclength, and let $$ H_\Gamma = -\frac{d^2}{ds^2} + \kappa(s)^2 $$ act on periodic functions in $W^{2,2}(\Gamma)$. Then its lowest eigenvalue satisfies $\lambda_1(H_\Gamma) \geq 1$.

来源字段：http://aimpl.org/shapesurface/6/

10312 | AIM-SEVERAL_COMPLEX_VARIABLES-0039

### A cohomological screen for Fefferman Pontryagin forms

P / partially_solved | AIM 研讨会问题汇编

**Problem (S. Dragomir).** Which Pontryagin forms of the Fefferman metric of a strictly pseudoconvex abstract CR hypersurface \(M\) are obstructions to global embeddability of \(M\)?

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10313 | AIM-SEVERAL_COMPLEX_VARIABLES-0032

### A conditional Bezout bound and a quadratic node family

P / partially_solved | AIM 研讨会问题汇编

**3. (S. Ji)** Let \(M\subset \mathbb C^n\) be a compact CR manifold defined by polynomials, with \(n\geq 2\) and CR codimension greater or equal to 1. The Harvey--Lawson variety may have isolated singularities. What is the relationship between the degree of the defining polynomials and the number of singularities?

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10314 | AIM-ANALYSIS-0162

### A constant-intensity obstruction to negatively correlated GAF zeros

P / partially_solved | AIM 研讨会问题汇编

Find a nondegenerate complex Gaussian entire function having infinitely many random zeros and satisfying \(\rho_2(z,w)\leq \rho_1(z)\rho_1(w)\) for every pair of distinct points.

来源字段：https://aimath.org/WWN/randomzeros/arcc1.pdf

10315 | AIM-FUNCTIONAL_ANALYSIS-0018

### A corrected relative-commutant corner preorder for maximal O2-ultrapower embeddings

P / partially_solved | AIM 研讨会问题汇编

If \(\phi_j\colon A\to \prod_{\mathcal U}\mathcal O_2\) are *-homomorphisms, write \(\phi_1\leq\phi_2\) if there is a partial isometry \(u\) in \(A\) such that \(u^*\phi_2u=\phi_1\).

Assume \(A\) is separable and \(A\) embeds into \(\prod_{\mathcal U}\mathcal O_2\). Is there a \(\leq\)-maximal embedding \(\phi\) of \(A\) into \(\prod_{\mathcal U}\mathcal O_2\)?

来源字段：http://aimpl.org/settheorycstar/1/

10316 | AIM-ANALYSIS-0160

### A CUE critical-point sum rule and the small-exponent Jensen bridge

P / partially_solved | AIM 研讨会问题汇编

**Question 3 (Ashkan Nikeghbali).** What can one tell about the distribution of the zeros of the derivative of characteristic polynomial of random unitary matrix, especially near the boundary of the unit circle. Also what can one tell about \(E[|f'|^s]\) as \(s\) tends to zero?

来源字段：https://aimath.org/WWN/randomzeros/arcc1.pdf

10317 | AIM-ANALYSIS-0111

### A current-valued obstruction to torsion

P / partially_solved | AIM 研讨会问题汇编

**Lipschitz homotopy groups of the Heisenberg group II.** Does $\pi_n^{\mathrm{Lip}}(\mathbb H^n)$ contain torsion elements?

来源字段：http://aimpl.org/mappingmetric/1/

10318 | AIM-ANALYSIS-0078

### A deterministic dense unequal-norm Weaver partition algorithm

P / partially_solved | AIM 研讨会问题汇编

Find an algorithm for Weaver's conjecture/equivalence/theorem (or call it the vector partitioning problem), similarly for Paving Conjecture

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10319 | AIM-ANALYSIS-0070

### A diagonal-trace reduction for a bilinear polynomial Carleson operator

P / partially_solved | AIM 研讨会问题汇编

With $m=\widehat K$,

\[ |\partial^\alpha m(\xi)|\lesssim |\xi|^{-|\alpha|}, \qquad \xi\in\mathbb R^2\setminus\{0\}, \tag{0.13} \]

for sufficiently many multi-indices $\alpha$. It then asks for a multilinear analogue of the Stein--Wainger polynomial Carleson theorem, giving as its example

\[ \left\|\sup_{\lambda\in\mathbb R} \left|\int_{\mathbb R^2}f(x-t)g(x-s)K(t,s) e^{i\lambda s^2t^2}\,dt\,ds\right|\right\|_{L^2_x} \lesssim \|f\|_{L^4}\|g\|_{L^4}. \tag{0.14} \]

Thus the phase is unambiguously the mixed quartic monomial $s^2t^2$, not a quadratic or a sum of one-variable monomials.

The PDF next cites Li and Muscalu's translated-multiplier theorem. Comparison with Li--Muscalu's primary statement recovers the intended symbol as

\[ m(\xi-N)=m(\xi_1-N_1,\xi_2-N_2), \]

and the corresponding bilinear maximal operator is

\[ C_m(f_1,f_2)(x)=\sup_{N\in\mathbb R^2} \left|\int_{\mathbb R^2}m(\xi-N)\widehat f_1(\xi_1) \widehat f_2(\xi_2)e^{2\pi ix(\xi_1+\xi_2)}\,d\xi\right|. \]

That theorem concerns linear translations of a multiplier in frequency; it does not contain the physical-variable quartic phase in (0.14).

For a genuinely singular $K$, (0.14) must be interpreted by principal values or compatible smooth truncations. The workshop question does not specify a truncation convention. Below, $K$ is given the usual translation-invariant Calderón--Zygmund realization associated with the Mikhlin symbol $m$. If one does not assume enough regularity in (0.13) to obtain a standard Hölder Calderón--Zygmund kernel, the all-scale Sobolev theorem below should instead be read under the explicit ambient maximal hypothesis in Theorem 5.1.

来源字段：https://aimath.org/pastworkshops/multilinopsproblems.pdf

10320 | AIM-SEVERAL_COMPLEX_VARIABLES-0022

### A dimension-free boundary-regularity threshold for proper ball maps

P / partially_solved | AIM 研讨会问题汇编

Does there exist a universal constant \(t\) such that a proper holomorphic map between \(B^n\) and \(B^N\), with \(1<n<N\), which is \(C^t\) up to the boundary is actually a rational map?

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10321 | AIM-SEVERAL_COMPLEX_VARIABLES-0045

### A divisor-trace criterion for removability of meromorphic sphere maps

P / partially_solved | AIM 研讨会问题汇编

Let \((M,p)\subset \mathbb C^n\) be a generic real-analytic submanifold, and suppose \[ f:M\longrightarrow \partial\mathbb B^N,\qquad n\leq N, \] is a continuous CR mapping which extends meromorphically to a neighborhood of \(p\). Under what conditions does \(f\) extend holomorphically near \(p\)? In particular, is minimality of \(M\) sufficient?

来源字段：https://aimath.org/WWN/complexitycrmap/complexitycrmap.pdf

10322 | AMR-036-0016

### A fifth-root functional equation

O / open | AMR 问题清单汇编

For $\omega=e^{2\pi i/5}$, is an entire solution of $$f(\omega z)f(\omega^{-1}z)=f(z)-1$$ unique up to rotation of the variable $z$?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10323 | AIM-FUNCTIONAL_ANALYSIS-0036

### A finite-cardinal counterexample and the intended infinite problem

S / solved | AIM 研讨会问题汇编

Assume \(A\) is a tensor product of algebras of the form \(\mathbb M_n(\mathbb C)\), for \(n\in\mathbb N\), \(\kappa<\kappa'\) are cardinals, and \(D_2(\kappa')\) unitally embeds into \(A\otimes D_2(\kappa)\). Can we conclude that there is a unital embedding of \(D_2(\kappa)\) into \(A\)?

来源字段：http://aimpl.org/settheorycstar/6/

10324 | AIM-ANALYSIS-0038

### A finite-degree moment obstruction for finite-dimensional Bergman spaces

P / partially_solved | AIM 研讨会问题汇编

Is there a pseudoconvex domain $\Omega \subset \mathbb{C}^n$ with $0 < {\rm dim} \, A^2(\Omega) < \infty$?

来源字段：http://aimpl.org/scvproblems/1/

10325 | AIM-ANALYSIS-0001

### A finite-part BOCG identity for Fisher--Hartwig symbols

P / partially_solved | AIM 研讨会问题汇编

The Borodin-Okounkov-Case-Geronimo identity is given by \(\text{det}(T_N(e^f))=(e^{N\hat f(0)+\sum k\hat f(k)\hat f(-k)}) \text{det}(1-k)\). Find a useful adjustment to the right-hand side so that an identity holds in the Fisher-Hartwig case.

来源字段：http://aimpl.org/riemhilberttoeplitz/1/

10326 | AIM-ANALYSIS-0003

### A finite-section criterion for ratios of Toeplitz determinants

P / partially_solved | AIM 研讨会问题汇编

Study merging singularities or double scaling limits for Toeplitz determinants with two symbols. Compare \(\text{det}(T_n(a_{\lambda}b_{\lambda}))\) and \(\text{det}T_n(a_{\lambda})T_n(b_{\lambda})\). Under what conditions does the ratio converge?

来源字段：http://aimpl.org/riemhilberttoeplitz/1/

10327 | AIM-ANALYSIS-0126

### A five-state area-preserving bi-Lipschitz counterexample to product rigidity

P / partially_solved | AIM 研讨会问题汇编

**Product quasiconformal mappings of the plane.** Let $f$ be a quasiconformal map of the plane. Suppose at a.e. point, the differential of $f$ either preserves both the $x$-direction and the $y$-direction, or switches the two directions. Does this imply $f$ is a product map?

来源字段：http://aimpl.org/mappingmetric/5/

10328 | AIM-FUNCTIONAL_ANALYSIS-0026

### A forced ideal skeleton and quotient-norm reduction

P / partially_solved | AIM 研讨会问题汇编

What is the ideal structure of \(\mathcal B(H)\otimes_{\min}\mathcal Q(H)\)? Does it depend on the model of set theory?

来源字段：http://aimpl.org/settheorycstar/3/

10329 | AIM-SEVERAL_COMPLEX_VARIABLES-0015

### A formal-orbit obstruction to convergent CR normal forms

P / partially_solved | AIM 研讨会问题汇编

Let \(C\) be a class of hypersurfaces defined by a finite order condition. A normal form is a subclass \(C_0\) where normal representatives are determined up to a finite dimensional group. For example, for Levi non-degenerate hypersurfaces there is the Chern-Moser normal form and for finite type hypersurfaces in \(\mathbb C^2\) Kollar presented a normal form. Can you find a class where it can be proved that there is no convergent normal form?

来源字段：http://aimpl.org/crscv/5/

10330 | AIM-FUNCTIONAL_ANALYSIS-0010

### A formulation audit and ball-calibration obstruction for space-form quermassintegral problems

O / open | AIM 研讨会问题汇编

Isoperimetric type problems for quermassintegrals in hyperbolic or spherical space.

It is item 1.5 in the “Inequalities” section of the 2019 AIM workshop *Symmetry and convexity in geometric inequalities*. The record contains no remarks or literature field.

This sentence names a research direction, not a proposition with a truth value: it specifies neither dimension, normalization of quermassintegrals, pair of indices, admissible sets, convexity condition, inequality direction, nor equality case. I therefore classify this record as `context_only`, not as a solved or open theorem.

A defensible modern decomposition is:

1. the classical volume/surface-area isoperimetric inequality in a space form; 2. pairwise Alexandrov--Fenchel inequalities between genuine quermassintegrals, with equality at geodesic balls; 3. analogous inequalities for raw curvature integrals; 4. variants under convexity, horospherical convexity, star-shapedness, or curvature-cone hypotheses; 5. stability estimates measuring distance from a ball by the inequality deficit.

These are different problems. Results for one class or normalization must not be reported as resolving all the others.

来源字段：http://aimpl.org/symconvgeomineq/1/

10331 | AIM-SPECIAL_FUNCTIONS-0014

### A Garsia--Haiman filtration theorem and a small-k obstruction for diagonal harmonics

P / partially_solved | AIM 研讨会问题汇编

(14) Are there any nice relationships between $k$-Schur functions and diagonal harmonics modules or Garsia-Haiman modules, whose Frobenius series are given by $\nabla(e_n)$ and $\widetilde{H}_\mu$, respectively?

来源字段：https://aimath.org/WWN/kostka/prob2.pdf

10332 | AIM-SPECIAL_FUNCTIONS-0009

### A generic clique-block basis of individual LLT polynomials

P / partially_solved | AIM 研讨会问题汇编

**(9)** The Schur basis of $\Lambda$ is contained in the spanning set consisting of all LLT polynomials. Are there other interesting bases that have some simple relation to LLT polynomials? Lam remarked that all Hall-Littlewood polynomials are LLT polynomials, as are all skew Schur functions. Morse remarked that not every $k$-Schur function is an LLT polynomial, for example $s^{(3)}_{(2,1,1)}$. Modified Macdonald polynomials and, conjecturally, $\nabla(e_n)$ can be expressed as weighted sums of LLT polynomials indexed by tuples of ribbons and tuples of shifted columns, respectively.

来源字段：https://aimath.org/WWN/kostka/prob2.pdf

10333 | AIM-SEVERAL_COMPLEX_VARIABLES-0007

## A global-local counterexample to type detection by Bergman restriction

S / solved | AIM 研讨会问题汇编

Let \(\Omega_2\subset\Omega_1\) be two smooth bounded pseudoconvex domains that share a boundary point \(p\). If the restriction map \(A^2(\Omega_1)\to A^2(\Omega_2)\) is not compact, is the D'Angelo type of \(p\) the same with respect to \(\Omega_1\) and \(\Omega_2\)?

When the inside domain \(\Omega_2\) is a ball, is \(p\) strictly pseudoconvex with respect to \(\Omega_1\)?

来源字段：http://aimpl.org/crscv/2/

10334 | AIM-ANALYSIS-0119

## A good-scale area-efficiency reduction for integrable lower metric distortion

P / partially_solved | AIM 研讨会问题汇编

The lower metric dilatation of a homeomorphism \(f:X\to Y\) of metric spaces is \[ h_f(x)=\liminf_{r\to0} \frac{\sup\{d(f(x),f(y)):d(x,y)\le r\}} {\inf\{d(f(x),f(z)):d(x,z)\ge r\}}. \] Let \(f\) be a homeomorphism of planar domains with \(h_f\in L^1_{\mathrm{loc}}\) and \(h_f\) finite off a set of \(\sigma\)-finite length. Does \(f\) have partial derivatives almost everywhere?

来源字段：http://aimpl.org/mappingmetric/4/

10335 | AIM-ANALYSIS-0087

## A half-measure Fourier frame with no two-Riesz partition

P / partially_solved | AIM 研讨会问题汇编

Does there exist a measurable \(E\subset\mathbb T=\mathbb R/\mathbb Z\), with \(|E|=1/2\), such that for every partition \(\mathbb Z=A\sqcup B\), at least one of \(\mathcal E_E(A)\) and \(\mathcal E_E(B)\) fails to be a Riesz sequence in \(L^2(E)\)?

It appears under Darrin Speegle's name.

The preceding lines say that \(E\subset[0,1]\) is measurable and that the displayed family is an exponential frame. The standard notation in Speegle's work and the surrounding sentence recover the intended family unambiguously as

\[ \mathcal E_E(\mathbb Z) =\{e_n\chi_E:n\in\mathbb Z\}, \qquad e_n(x)=e^{2\pi i n x}, \]

viewed either in \(L^2(\mathbb T)\) with support in \(E\), or canonically in \(L^2(E)\). Thus the question is:

This report answers **yes**. The set can be chosen compact and contained strictly inside one representative interval \([0,1)\).

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10336 | AIM-ANALYSIS-0064

## A hybrid spectral-complexity reduction for the maximal directional Hilbert transform

P / partially_solved | AIM 研讨会问题汇编

2 of the AIM PDF. The authoritative display is

\[ \left\|\sup_{j\in\{1,2,\ldots,N\}}|H_{v_j}f|\right\|_{L^{2,\infty}(\mathbb R^2)} \lesssim \sqrt{\log N}\,\|f\|_2, \tag{0.4} \]

for a collection \(\{v_j\in S^1:1\leq j\leq N\}\), where the PDF prints

\[ H_{v_j}f(x):=\int_{\mathbb R}f(x-tv_j)\,\frac{dt}{t}. \tag{0.5} \]

Three conventions or repairs are necessary.

1. The singular integral in (0.5) must be interpreted as a Cauchy principal value. This is the convention in all cited primary literature and is required even for Schwartz functions. 2. The weak norm used here is \[ \|g\|_{2,\infty}^{*}:=\sup_{\lambda>0}\lambda |\{x:|g(x)|>\lambda\}|^{1/2}. \] The superscript \({}^{*}\) is suppressed below when no ambiguity arises. Thus (0.4) is equivalent, up to squaring its constant, to \[ |\{H_Vf>\lambda\}|\lesssim \frac{\log N}{\lambda^2}\|f\|_2^2, \qquad H_Vf:=\sup_{v\in V}|H_vf|. \] 3. The literal statement is false for \(N=1\), since \(\sqrt{\log 1}=0\) but \(H_v\) is nonzero. The intended asymptotic statement is for \(N\geq2\), or uniformly with \(\sqrt{\log(1+N)}\). Repetitions and antipodal directions do not change the operator, so the most intrinsic parameter is the number \(M\) of distinct projective directions \([v]=\{v,-v\}\), and the clean formulation uses \(\sqrt{\log(1+M)}\).

The symbol \(\lesssim\) in the question means a constant independent of \(N\), the directions, and \(f\) (after fixing the normalization of the Hilbert transform).

来源字段：https://aimath.org/pastworkshops/multilinopsproblems.pdf

10337 | AIM-SPECIAL_FUNCTIONS-0007

### A Lam--Kato bridge for the geometric and graded meanings of k-Schur functions

P / partially_solved | AIM 研讨会问题汇编

“(7) Elucidate the representation-theoretical significance of the \(k\)-Schur functions. The answer may involve either representation theory of symmetric groups or representation theory of Lie algebras.”

The exact AIM record is:

It is Problem 7 in the 2005 AIM workshop list on Generalized Kostka Polynomials, <https://aimath.org/WWN/kostka/prob2.pdf>. The surrounding workshop notes use the one-parameter recursive \(k\)-Schur functions \(s_\lambda^{(k)}[X;t]\). The problem is deliberately open-ended: it asks for a representation-theoretic model, not for a yes/no assertion.

There are now two answers which must not be conflated:

1. the parameter-forgotten specialization \(s_\lambda^{(k)}[X;1]\) represents an affine-Grassmannian homology Schubert class; 2. the full graded modern \(k\)-Schur function is the graded character of a finite-dimensional module for a current algebra, equivalently for a skew group algebra containing \(\mathbb C\mathfrak S_m\).

The first answer is geometric and was published by Lam. The second is the main subject of Kato's current arXiv preprint. The parameter is called \(t\) in the older literature and \(q\) in Kato's convention. Below it is denoted by \(u\) whenever the name is immaterial.

来源字段：https://aimath.org/WWN/kostka/prob2.pdf

10338 | AIM-ANALYSIS-0144

### A lattice criterion and a curvature screen for log-concave mixtures

P / partially_solved | AIM 研讨会问题汇编

Let $\{\phi_k(x)\}_{k=0}^\infty$ and $\{f_k\}_{k=0}^\infty$ be log-concave. When is $\sum f_k\phi_k(x)$ log-concave ?

来源字段：http://aimpl.org/hyperbolicpoly/3/

10339 | AIM-ANALYSIS-0145

### A literal counterexample and an exact symmetric-quartic curvature classification

S / solved | AIM 研讨会问题汇编

**Conjecture (Edwards--Gordon).** Let $p \in \mathbb{R}[x]$ be a real polynomial of degree $n > 2$, and let $\kappa(x)$ denote the signed curvature of the graph $y = p(x)$ in $\mathbb{R}^2$. Does $\kappa(x)$ have at most $n - 1$ points of extreme curvature (or more strongly, does $\kappa'(x) = 0$ have at most $n - 1$ distinct real solutions)? (Note: the original AIM problem list entry wrote 'at least $n - 1$', reversing the inequality of Edwards and Gordon, 2004, MR2104693.)

来源字段：http://aimpl.org/hyperbolicpoly/5/

10340 | AIM-ANALYSIS-0014

### A matrix Fisher-Hartwig atlas and a partial-index obstruction

P / partially_solved | AIM 研讨会问题汇编

Formulate and then analyze general finite matrix symbol with the analog of zeros and jumps.

来源字段：http://aimpl.org/riemhilberttoeplitz/2/

10341 | AIM-FUNCTIONAL_ANALYSIS-0030

## A mixed local approximation obstruction for the nuclear UCT problem

P / partially_solved | AIM 研讨会问题汇编

**UCT problem.** Does every **separable nuclear** C\(^*\)-algebra satisfy the Rosenberg--Schochet UCT?

The source omits the word “separable.” In the Rosenberg--Schochet framework, and in modern statements of the UCT problem, the intended question is:

The separability qualification is material: classical Kasparov \(KK\)-theory and the bootstrap category used here have separable C\(^*\)-algebras as objects. There are nonseparable variants of bivariant theory, but the literal all-cardinalities reading would require an additional choice not made in the AIM record. This report treats the standard separable problem.

For separable C\(^*\)-algebras \(A,D\), let \[ \gamma_{A,D}:KK_*(A,D)\longrightarrow \operatorname{Hom}_{\mathbb Z}^{*}(K_*(A),K_*(D)) \] be induced by Kasparov product. The algebra \(A\) satisfies the UCT if, for every separable \(D\), there is a natural short exact sequence of \(\mathbb Z/2\)-graded groups \[ 0\longrightarrow \operatorname{Ext}_{\mathbb Z}^{1,*}(K_*(A),K_{*+1}(D)) \longrightarrow KK_*(A,D) \xrightarrow{\gamma_{A,D}} \operatorname{Hom}_{\mathbb Z}^{*}(K_*(A),K_*(D)) \longrightarrow0. \tag{1} \] The shift records that the \(\operatorname{Ext}^1\) map has odd degree. The sequence splits, but not naturally. For separable nuclear \(A\), Rosenberg--Schochet theory gives the equivalent conditions:

1. (1) holds for every separable \(D\); 2. \(A\) is \(KK\)-equivalent to a commutative C\(^*\)-algebra; 3. \(A\) lies in the Large Bootstrap Class \(\mathcal N\), the smallest class of separable nuclear algebras containing \(\mathbb C\), closed under \(KK\)-equivalence and the two-out-of-three property for semisplit extensions.

The preceding AIM record, Problem 4.2, is the **Small Bootstrap Problem**. In Blackadar’s terminology, the Small Bootstrap Class \(\mathcal B\) is generated constructively from finite-dimensional and separable commutative algebras using two-out-of-three for extensions, finite tensor products, inductive limits, stable isomorphism, and crossed products by \(\mathbb Z\). The Large Bootstrap/UCT Class \(\mathcal N\) additionally closes from \(\mathbb C\) under \(KK\)-equivalence. One has \[ \mathcal B\subseteq\mathcal N. \] Thus a positive answer to Problem 4.2 implies a positive answer to Problem 4.3, but the UCT problem is potentially strictly weaker. The two AIM records should not be collapsed into one question.

来源字段：http://aimpl.org/settheorycstar/4/

10342 | AIM-ANALYSIS-0094

## A mixed-discriminant reconstruction of the five-way spanning problem

O / open | AIM 研讨会问题汇编

There's an algorithm: start with k vectors in n-dimnl space. Can you partition them into 5 sets so that set spans the space. Take each vector, repeat it 5 times and augment those matrices by identity to get dimension k. You'll get k semi-def-pos. matrices. Now you check each matrix for some property xxxx. This is an algorithm to carry out Rado-Horn?

* Mirko Visontai

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10343 | AIM-SEVERAL_COMPLEX_VARIABLES-0031

## A moment counterexample and a Levi wall for sphere links

P / partially_solved | AIM 研讨会问题汇编

Let \(n\geq 2\) and \(M\subseteq\mathbb C^n\) be a compact CR manifold which is maximally complex (so the CR codimension is \(1\)). Suppose in the class of maximally complex CR manifolds, \(M\) is homotopic to a sphere in the complex \(m\)-plane, where \(m-1\) is the CR dimension. Must \(M\) be the boundary of a smooth analytic submanifold?

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10344 | AIM-ANALYSIS-0019

## A Neumann-point incenter-moment expansion for the discrete Bareket conjecture

P / partially_solved | AIM 研讨会问题汇编

In the same spirit as Bareket's conjecture one can ask what polygon maximizes \(\lambda_1(\Omega,\alpha)\), \(\alpha<0\), among all polygons of the same area and having no more than \(N\) sides. Similarly, one can ask the corresponding question when \(\alpha>0\) in which case the problem is to minimize the eigenvalue. The following conjecture concerns the case \(N=3\).

**Discrete Bareket conjecture for triangles.** Among all triangles \(T\) of a given area, the lowest Robin eigenvalue \(\lambda_1(T,\alpha)\) is

来源字段：http://aimpl.org/shapesurface/1/

10345 | AIM-FUNCTIONAL_ANALYSIS-0034

## A nonseparable W*-algebra not isomorphic to its opposite

S / solved | AIM 研讨会问题汇编

Is there a nonseparable AF C*-algebra or W*-algebra not isomorphic to its opposite algebra?

来源字段：http://aimpl.org/settheorycstar/6/

10346 | AIM-ANALYSIS-0133

### A null-padding counterexample to the literal compact-deformation formulation

S / solved | AIM 研讨会问题汇编

An \(m\)-rectifiable closed set \(S \subset V=\ell^\infty_n\) is called *minimizing* if for every Lipschitz map \(f:V\to V\) such that \(f\) is the identity off a cube \(C\) and \(f(C)\subset C\), \[ \mathcal H^m(C\cap S)\leq \mathcal H^m(f(C\cap S)). \] Let \(S\) be minimizing. Is it true that for \(\mathcal H^m\)-almost every \(x\in S\), there exists a neighborhood \(U\) of \(x\) such that \(S\cap U\) is a Lipschitz graph over an \(m\)-dimensional subspace \(\Pi\) of \(V\)?

来源字段：http://aimpl.org/mappingmetric/6/

10347 | AIM-ANALYSIS-0123

### A one-dimensional obstruction and C3 rigidity in Hilbert dimension at least three

P / partially_solved | AIM 研讨会问题汇编

Are all $1$-quasiconformal homeomorphisms of Hilbert space similarities?

Let $f$ be a homeomorphism between Hilbert spaces with $H_f=1$ everywhere. Is $f$ a similarity?

来源字段：http://aimpl.org/mappingmetric/5/

10348 | AIM-ANALYSIS-0103

### A one-sided endpoint criterion for largest-root selection

P / partially_solved | AIM 研讨会问题汇编

Using interlacing polynomials but if you only want the maximum root, you don't need all these other roots to exist, can we trim our conditions so that the machinery is more usable? Is there some weaker form of interlacing if you don't care about all the roots.

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10349 | AIM-ANALYSIS-0143

### A paired-tail reduction and coefficientwise positivity through n=5

P / partially_solved | AIM 研讨会问题汇编

Let \[ Q_n^{(\alpha,\beta)}(x):=\sum_{k=0}^n\binom nk f_kf_{n-k} \left((x+\alpha)_k(x+\beta)_{n-k}-(x+\alpha+\beta)_k(x)_{n-k}\right), \] where \(\alpha,\beta>0\), \(f_k^2\ge f_{k-1}f_{k+1}\), and \((x)_k=x(x+1)\cdots(x+k-1)\). Then the Maclaurin coefficients of \(Q^{(\alpha,\beta)}\) are nonnegative.

Karp, and remarks that the case \(\alpha=\beta=1\) was proved in arXiv:1203.1482.

The cited paper removes several ambiguities in the abbreviated AIM text:

* \(n\) is a nonnegative integer; the substantive conjecture is for \(n\ge3\), while the determinant generating series begins at \(n=2\). * \((x)_0=1\). * The sequence is intended to be **nonnegative and log-concave without internal zeros**. The local inequalities alone are not the intended definition. * The final \(Q^{(\alpha,\beta)}\) on the AIM page means \(Q_n^{(\alpha,\beta)}\); omission of \(n\) is a typographical abbreviation, as is clear from the defining equation and from Karp’s equation (3). * Karp’s original Conjecture 1 assumes strict log-concavity and asks for positive coefficients for \(n\ge3\). The AIM statement is its natural weak/nonnegative version.

This report proves that weak version for every \(\alpha,\beta>0\) and every \(0\le n\le5\), gives a general tail-kernel reduction, and shows exactly why nonnegativity and the no-internal-zero condition cannot be omitted.

来源字段：http://aimpl.org/hyperbolicpoly/3/

10350 | AIM-ANALYSIS-0029

### A parity reduction and gradient-cone test for directional hot spots

P / partially_solved | AIM 研讨会问题汇编

For every nonzero real eigenfunction $u$ in the first positive Neumann eigenspace $E_1(\Omega)$, does there exist a nonzero vector $e=e(u)$ such that $e\cdot\nabla u(x)>0$ for every $x\in\Omega$?

The phrase "the eigenfunction" is ambiguous when the first positive eigenvalue is multiple. This matters for the disk and the square. The intended source, Jerison--Nadirashvili's Conjecture 8.4, says **any** Neumann eigenfunction with the lowest nonzero eigenvalue and lets the direction depend on that eigenfunction. Accordingly, the recovered formulation used here is:

Multiplying $e$ by a positive scalar changes nothing, so $e$ may be required to be a unit vector. Replacing $u$ by $-u$ replaces a good direction by $-e$. The inequality is only in the open domain. It generally degenerates at boundary support points because of the Neumann condition, so it must not be silently strengthened to $\overline\Omega$.

For a nonsmooth convex domain, the Neumann problem is understood weakly. Eigenfunctions are analytic in the interior, which is enough to interpret the displayed inequality there. The new parity theorem below is deliberately stated for a smooth boundary because its proof differentiates the boundary condition.

来源字段：http://aimpl.org/shapesurface/3/

10351 | AIM-ANALYSIS-0097

### A phase-shift dictionary for Kadison-Singer and a free-phase obstruction

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** Recast the interlacing-family and mixed-characteristic-polynomial proof of Kadison--Singer in terms of perturbation determinants, rational Herglotz--Nevanlinna functions, spectral phase shifts, and resolvents. Can this dictionary provide a genuinely analytic proof, and what quantitative resolvent statement must replace the multivariate polynomial barrier argument?

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10352 | AIM-SEVERAL_COMPLEX_VARIABLES-0021

### A polar-edge uniqueness criterion for removable meromorphic singularities

P / partially_solved | AIM 研讨会问题汇编

Assume \(M\subset \mathbb C^n\) is a real-analytic generic submanifold, \(f=p/q:\mathbb C^n\to\mathbb C^N\) is the germ of a meromorphic map at \(z\in M\), and \(f\) is holomorphic in a one-sided wedge \(W\) attached to \(M\) near \(z\). Assume that \(|f(w)|^2\) increases to 1 as \(w\to M\) in \(W\). Does \(f\) extend to a full neighborhood of \(z\) in \(\mathbb C^n\)?

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10353 | AIM-ANALYSIS-0146

### A polynomial/full-Laguerre-Polya boundary for zero-spacing under higher derivatives

P / partially_solved | AIM 研讨会问题汇编

What are the zero spacing effects of higher order differential operators on functions which belong to $\mathcal{L}\text{-}\mathcal{P}$?

来源字段：http://aimpl.org/hyperbolicpoly/5/

10354 | AIM-ANALYSIS-0033

### A quantitative defect criterion upgrading honeycomb energy asymptotics to empirical cell convergence

P / partially_solved | AIM 研讨会问题汇编

**Asymptotic partitioning problems.** In several problems where one wants to find an in some sense optimal partition of a planar domain $\Omega$ into $N$ subdomains it has been observed that as $N$ becomes large the partition converges to the hexagonal one.

Find conditions for problems involving optimal $N$-partitions of a planar domain such that one observes in the limit $N\to\infty$ a packing of hexagons.

来源字段：http://aimpl.org/shapesurface/6/

10355 | AIM-SPECIAL_FUNCTIONS-0017

### A rectangle-data obstruction and a precise k-Schur boundary test

P / partially_solved | AIM 研讨会问题汇编

**(17)** Find a good definition of level-restricted $q,t$-Kostka numbers. Does this make sense for root systems other than type $A$? Is there any connection to $k$-$q,t$-Kostka numbers?

来源字段：https://aimath.org/WWN/kostka/prob2.pdf

10356 | AIM-SEVERAL_COMPLEX_VARIABLES-0028

### A reducible obstruction and the quadratic irreducible case

P / partially_solved | AIM 研讨会问题汇编

**Problem 1 (B. Stensønes).** Let \(p(z,\bar z,w,\bar w)\) be a plurisubharmonic, homogeneous, degree \(2k\) polynomial on \(\mathbb C^2\). Let \(\Sigma\) be an algebraic curve not containing the origin. If \(p|_\Sigma\equiv0\), then is \(\Sigma\) the level set of a homogeneous holomorphic polynomial?

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10357 | AIM-ANALYSIS-0072

## A resonant obstruction to the displayed physical-space maximal estimate

S / solved | AIM 研讨会问题汇编

With the Fourier convention

\[ \widehat f(\xi)=\int_{\mathbb R^n}f(x)e^{-2\pi i x\cdot\xi}\,dx, \]

Question 13 (Carbery) first records the higher-dimensional spherical Carleson problem

\[ \left\|\sup_R\left|\int_{|\xi|\leq R} \widehat f(\xi)e^{2\pi i x\cdot\xi}\,d\xi\right|\right\|_2 \lesssim \|f\|_2. \tag{0.18} \]

It then asks, literally,

\[ \left\|\sup_R\left|f*\left( \frac{e^{i|x|}}{|x|^{(n+1)/2}}\, \chi_{\{|x|\leq R\}}\right)\right|\right\|_2 \lesssim \|f\|_2? \tag{0.19} \]

The context makes the supremum an unbounded supremum over positive radii. In particular, neither a square on the phase nor a conjugate oscillatory branch is present in (0.19). This was also checked directly in the decompressed PDF content stream. The PDF cites Carbery--Gorges--Marletta--Thiele, although its bibliography misspells Marletta as “Marietta.”

For every finite \(R\), the kernel in (0.19) is locally integrable exactly when \(n\geq2\). Thus the literal formula needs an additional prescription in dimension one; all results below concern \(n\geq2\), the range relevant to the higher-dimensional question.

来源字段：https://aimath.org/pastworkshops/multilinopsproblems.pdf

10358 | AIM-SPECIAL_FUNCTIONS-0012

## A rigging-refined stable-level fusion polynomial

P / partially_solved | AIM 研讨会问题汇编

(12) Consider the product $B=B^{r_\ell s_\ell}\otimes_k\cdots\otimes_k B^{r_1 s_1}$. In this case, where the factors are products of rectangles in type $A$, there are known combinatorial interpretations and fermionic formulas for the fusion coefficients via rigged configurations (Schilling et al.). There exist conjectural formulas in other types (Hatayama et al.). The problem of generalizing the grading of the tensor product to the two-variable $(q,t)$ case is open.

来源字段：https://aimath.org/WWN/kostka/prob2.pdf

10359 | AIM-SEVERAL_COMPLEX_VARIABLES-0001

## A scale obstruction to the coefficient-one formulation

S / solved | AIM 研讨会问题汇编

Let \(\Omega\subset\mathbb C^n\) be pseudoconvex and contain \(0\), let \(H\) be a hyperplane through \(0\), and let \(\phi\in L^1_{\mathrm{loc}}(\Omega)\). Assuming \[ i\partial\bar\partial\phi\geq -C\,i\partial\bar\partial\log B_\Omega(z,z), \] how large may \(C\) be while every holomorphic \(f\) on \(H\cap\Omega\) of finite weighted \(L^2\)-norm has an extension \(F\in\mathcal O(\Omega)\) satisfying the coefficient-one estimate \[ \int_\Omega |F|^2e^{-\phi}\,dV_n \leq \int_{H\cap\Omega}|f|^2e^{-\phi}\,dV_{n-1}? \tag{1} \] It also asks whether \(C\) can be universal. The source says that \(C=0\) is Ohsawa--Takegoshi and that, for strictly pseudoconvex \(\Omega\), ``\(C\geq 1/(n+1)\) works.''

There are two problems with this literal wording.

1. The coefficient \(1\) in (1) has units of length squared and is not preserved by dilation. Classical Euclidean Ohsawa--Takegoshi extension has a constant depending on a transverse size of the domain (in a common elementary statement, on its diameter), unless a transverse defining function and its normalization are included. 2. Since \(\omega_{B,\Omega}:=i\partial\bar\partial\log B_\Omega(z,z)\geq0\), the class of allowed weights grows as \(C\) grows. Thus a theorem at \(C_0\) implies the same theorem for \(0\leq C\leq C_0\), not for \(C\geq C_0\). The bullet can coherently mean either ``\(C=1/(n+1)\) works'' or ``the supremum of admissible \(C\)'s is at least \(1/(n+1)\).''

来源字段：http://aimpl.org/crscv/1/

10360 | AIM-SEVERAL_COMPLEX_VARIABLES-0017

## A scale-correct winding bound for the essential spectral threshold

P / partially_solved | AIM 研讨会问题汇编

Relate the infimum of the essential spectrum of the $\overline{\partial}$-Neumann Laplacian on the worm domain to the winding of the domain.

来源字段：http://aimpl.org/crscv/6/

10361 | AIM-SEVERAL_COMPLEX_VARIABLES-0002

## A sign-corrected, defining-function-sensitive reduction of the worm Dirichlet problem

P / partially_solved | AIM 研讨会问题汇编

Fix a defining function \(\rho<0\) for which \(\Phi_\rho=-\log(-\rho)\) is strictly plurisubharmonic. Let \(\omega_\rho=i\partial\bar\partial\Phi_\rho\) and let \(\Delta_\rho\) be its Laplace--Beltrami operator. For \(f\in C^\infty(\partial\mathcal W)\), determine the boundary regularity of the solution of \(\Delta_\rho u=0\) in \(\mathcal W\), \(u=f\) on \(\partial\mathcal W\), after fixing or normalizing the defining function.

The supplied AIM record asks for boundary regularity on the smooth Diederich--Fornæss worm \(\mathcal W\subset\mathbb C^2\) for the Laplace--Beltrami operator associated with the displayed form \[ i\partial\bar\partial\log(-\rho), \] where \(\rho<0\) in \(\mathcal W\), and prints the boundary condition as `\(\nu=f\)`.

There are two formulation problems.

1. Since the problem explicitly calls itself Dirichlet, `\(\nu=f\)` is reconstructed as \[ u=f\quad\hbox{on }\partial\mathcal W. \] This is a semantic reconstruction of an apparent extraction/OCR error. The current AIM source URL was unavailable during this run, so it was not independently verified against the original rendered page. 2. With the standard positivity convention for a Kähler form and \(\rho<0\), the displayed potential has the wrong sign. Indeed, \[ i\partial\bar\partial\log(-\rho) =-i\partial\bar\partial[-\log(-\rho)]. \] Thus whenever the usual logarithmic metric is positive, the displayed form is negative.

The mathematically coherent version studied here is therefore:

The qualification on \(\rho\) is essential: an arbitrary defining function need not make \(-\log(-\rho)\) strictly plurisubharmonic.

来源字段：http://aimpl.org/crscv/1/

10362 | AIM-ANALYSIS-0137

## A spectral-projector Hermitian form and the singular Bezoutian obstruction

P / partially_solved | AIM 研讨会问题汇编

Given $p\in\mathbb{C}[x]$, $\deg(p)=n$, find an $n\times n$ Hermitian matrix whose inertia $(n_+,n_-,n_0)$ counts the zeros of $p$ in $H_+,H_-,H_0=\mathbb R$.

来源字段：http://aimpl.org/hyperbolicpoly/2/

10363 | AIM-ANALYSIS-0012

## A spherical log gas on a smooth loop

P / partially_solved | AIM 研讨会问题汇编

The AIM record is Problem 2.7 from the 2024 workshop *Riemann-Hilbert problems, Toeplitz matrices, and applications*. Its exact stored text is

\[ Z_N=\int_{\Gamma} \cdots \int_{\Gamma} \prod_{i \leq j < k \leq N}|z_k-z_j|^{2\beta} \prod_{j=1}^N\left(\frac{1}{1+|z_j|^2}\right)^{(N-1)\beta+1} d|z_1|\cdots d|z_N|, \]

Two conservative repairs are necessary before the displayed quantity defines the customary symmetric log gas.

1. The unused index \(i\) in \(\prod_{i\leq j<k\leq N}\) would repeat the pair \((j,k)\) exactly \(j\) times and destroy permutation symmetry. We therefore read it as the standard Vandermonde product \(\prod_{1\leq j<k\leq N}\). 2. We read \(d|z|\) as Euclidean arclength \(|dz|\). Read literally as the differential of the radial coordinate, it vanishes on a circle centered at the origin and is not a positive measure on a general loop.

The reconstructed problem studied below is consequently

\[ \boxed{ Z_N(\Gamma,\beta)=\int_{\Gamma^N} \prod_{1\leq j<k\leq N}|z_j-z_k|^{2\beta} \prod_{j=1}^N(1+|z_j|^2)^{-[(N-1)\beta+1]}|dz_j|.} \tag{1.1} \]

Here \(\Gamma\) is assumed to be an embedded \(C^2\) loop, and the particles are labelled: there is no factor \(1/N!\). The main large-\(N\) result is for fixed \(\beta>0\). For \(\beta=0\) the integral is elementary. For a fixed negative real \(\beta\), the integral ceases to exist once \(N\) is large enough, so no unrenormalized large-\(N\) question remains.

来源字段：http://aimpl.org/riemhilberttoeplitz/2/

10364 | AIM-SEVERAL_COMPLEX_VARIABLES-0033

## A stratified finite-jet criterion for smooth infinitesimal CR homogeneity

P / partially_solved | AIM 研讨会问题汇编

**Problem 4 (D. Zaitsev).** Let \(M\) be a locally homogeneous CR manifold. That is, suppose for any two points \(p,q\in M\), there exists a CR diffeomorphism \[ h:(U,p)\longrightarrow (V,q) \] for some neighborhoods \(U\) of \(p\) and \(V\) of \(q\). Does it follow that the Lie algebra of infinitesimal automorphisms spans \(T_pM\) for one (and hence all) \(p\in M\)?

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10365 | AIM-ANALYSIS-0115

### A strict critical-dimension obstruction for Heisenberg snowflakes

P / partially_solved | AIM 研讨会问题汇编

**Snowflake embeddings of the Heisenberg group into low-dimensional Euclidean spaces.** Let $d_{cc}$ denote the Carnot--Carathéodory metric. Does there exist $\epsilon<1$ so that the first Heisenberg group $\mathbb H^1$, equipped with $d_{cc}^{\epsilon}$, bi-Lipschitz embeds into $\mathbb R^5$?

来源字段：http://aimpl.org/mappingmetric/2/

10366 | AIM-ANALYSIS-0099

### A support-discrepancy formulation for zonotope generator partitions

P / partially_solved | AIM 研讨会问题汇编

Is there an analog of KS for zonotopes? Let \(z_1,\ldots,z_k\in\mathbb R^n\), \(k>n\), and let \(\bar z_i=\{\alpha z_i:0\leq\alpha\leq1\}\). Consider \(\mathrm{Zon}=\alpha_1\bar z_1+\cdots+\alpha_k\bar z_k\).

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10367 | AIM-ANALYSIS-0118

### A symmetry criterion for bi-Lipschitz metric spheres

P / partially_solved | AIM 研讨会问题汇编

**Bi-Lipschitz parameterization of metric $2$-spheres.** Give a geometric characterization of metric $2$-spheres which are equivalent to the standard $2$-sphere via a bi-Lipschitz map.

来源字段：http://aimpl.org/mappingmetric/3/

10368 | AIM-ANALYSIS-0031

### A target-specific q_Omega criterion for strict disk comparison

P / partially_solved | AIM 研讨会问题汇编

Prove or disprove that among all planar domains of a given area, the disk is the unique minimizer for the smallest positive eigenvalue \(\nu _1(\Omega)\) of the Dirac problem with infinite mass boundary conditions

\[ \begin{cases} -\mathsf i(\partial _1v-\mathsf i\partial _2v)=\nu u&\text{in }\Omega,\\ -\mathsf i(\partial _1u+\mathsf i\partial _2u)=\nu v&\text{in }\Omega,\\ \mathsf i(n_1+\mathsf i n_2)u=v&\text{on }\partial\Omega, \end{cases} \] where \(n=(n_1,n_2)^\top\) is the outer unit normal.

The phrase “all planar domains” needs a regularity convention. The 2021 formulation used bounded, simply connected \(C^\infty\) domains. The current results used below establish self-adjointness, discreteness, and the required variational characterization for bounded planar domains with \(C^2\) boundary, without a simple-connectivity hypothesis. Accordingly, the rigorous partial theorem in this report is for bounded connected \(C^2\) domains. No claim is made for arbitrary rough domains, where even the operator realization in the displayed form requires care.

Writing the complex normal as \(\mathbf n=n_1+\mathsf i n_2\), the boundary condition is \(v=\mathsf i\mathbf n u\). This is exactly the massless infinite-mass member \((\theta,m)=(0,0)\) of the quantum-dot Dirac family used in the 2026 literature: its convention \(\overline{\mathbf n}v=\mathsf i\vartheta(0)u\), with \(\vartheta(0)=1\), becomes the same condition after multiplication by \(\mathbf n\).

来源字段：http://aimpl.org/shapesurface/5/

10369 | AIM-ANALYSIS-0142

## A three-nonzero-root theorem for Fisk's 3x3 Toeplitz-minor transform

P / partially_solved | AIM 研讨会问题汇编

It asks the following. Given

\[ p(x)=\sum_{k=0}^{n}a_kx^k\in\mathcal L\!-\!\mathcal P^+, \]

\[ \mathcal T_3p(x):=\sum_{k=0}^{n}D_k^{(3)}x^k\in\mathcal L\!-\!\mathcal P^+, \qquad D_k^{(3)}:=\det[a_{k+j-i}]_{i,j=0}^{2}? \tag{1} \]

Here the determinant is

\[ \det\!\begin{pmatrix} a_k&a_{k+1}&a_{k+2}\\ a_{k-1}&a_k&a_{k+1}\\ a_{k-2}&a_{k-1}&a_k \end{pmatrix}. \]

The AIM record only declares \(a_{-1}=a_{n+1}=0\) immediately before its \(2\times2\) formula. The \(3\times3\) display also uses \(a_{-2}\) and \(a_{n+2}\), so it is not literally defined without an added convention. Fisk's paper and Yoshida's formulation use the natural zero extension

\[ a_j=0\quad\text{for every }j\notin\{0,\ldots,n\}. \tag{2} \]

All results below use (2). For a nonzero finite polynomial, membership in \(\mathcal L\!-\!\mathcal P^+\) is equivalent to having nonnegative coefficients and only real, nonpositive zeros. If the zero polynomial is admitted, it maps to zero and is a separate trivial boundary case.

来源字段：http://aimpl.org/hyperbolicpoly/3/

10370 | AIM-ANALYSIS-0066

## A transfer principle and the full Banach range for lacunary rotation-coupled circle averages

P / partially_solved | AIM 研讨会问题汇编

On the plane \(\mathbb R^2\), let \(R:S^1\to S^1\) be rotation by \(\pi/3\). Prove, for certain \(p,q,r\), \[ \left\|\sup_{t>0}\left|\int_{S^1}f(x-tw)g(x-tR w)\,d\sigma(w)\right|\right\|_{L^r(\mathbb R^2)} \lesssim \|f\|_{L^p(\mathbb R^2)}\|g\|_{L^q(\mathbb R^2)}. \tag{0.7} \] During the workshop, this was shown to be equivalent to proving \[ \left\|\sup_{t>0}\left|\int_{S^1}f(x-tw)g(x+tw)\,d\sigma(w)\right|\right\|_{L^r(\mathbb R^2)} \lesssim \|f\|_{L^p(\mathbb R^2)}\|g\|_{L^q(\mathbb R^2)}. \tag{0.8} \]

来源字段：https://aimath.org/pastworkshops/multilinopsproblems.pdf

10371 | AIM-FUNCTIONAL_ANALYSIS-0044

## A triangular AF reduction and the broad-versus-matricial complexity split

P / partially_solved | AIM 研讨会问题汇编

What is the complexity of isometric isomorphism of direct limits of not necessarily self-adjoint subalgebras of finite dimensional C*-algebras? How does it compare to the complexity of isomorphism of AF algebras?

来源字段：http://aimpl.org/settheorycstar/7/

10372 | AIM-SEVERAL_COMPLEX_VARIABLES-0034

## A two-stage orbit decomposition and finite-jet groupoid criterion

P / partially_solved | AIM 研讨会问题汇编

**5. (D. Zaitsev)** Given \(M\) a CR manifold and \(p\in M\), let \(O(p)\) be the set of points \(q\in M\) for which there exists a CR automorphism \(h:(U,p)\to(V,q)\) for some neighborhoods \(U\) of \(p\) and \(V\) of \(q\). Is \(O(p)\) always a manifold? In that case, is the span of the Lie algebra of infinitesimal automorphisms of \(M\) equal to \(T_qO(p)\) if \(q\in O(p)\)?

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10373 | AIM-ANALYSIS-0042

## A weighted Bergman space on the disk cannot have finite positive dimension

S / solved | AIM 研讨会问题汇编

Let $\Delta$ be the unit disk in $\mathbb{C}$ and let $\omega > 0$ be an upper semicontinuous function on $\Delta$. Consider the weightd Bergman space $A^2(\Delta, \omega)$. Is it possible that $0 < {\rm dim} \, A^2(\Delta, \omega) < \infty$?

来源字段：http://aimpl.org/scvproblems/1/

10374 | AIM-ANALYSIS-0047

## A weighted Kähler characteristic and a toric logarithmic Hardy model

P / partially_solved | AIM 研讨会问题汇编

Given a Kähler manifold $(M, \omega)$. Define $H^2_\omega:=\{f: \Delta \rightarrow M : \int_\Delta \log |z|^2 f^* \omega > -\infty \}$. Investigate these spaces, for example: completeness, radial limits, etc... A special case to consider is when $M$ being toric.

For holomorphic maps $f:\Delta\to M$, analyze the maps for which $\int_\Delta\log|z|^2 f^*\omega$ is finite rather than $-\infty$; specify a topology or metric before asking about completeness; and determine meaningful forms of boundary convergence, especially for toric targets.

来源字段：http://aimpl.org/scvproblems/1/

10375 | AIM-SEVERAL_COMPLEX_VARIABLES-0004

## A weighted planar reduction for the logarithmic test function on the smooth worm

P / partially_solved | AIM 研讨会问题汇编

Let \(\mathcal W\) be the smooth Diederich--Fornæss worm domain in \(\mathbb C^2\), and let \(\mathbf B_{\mathcal W}\) be its Bergman projection. Prove or disprove that \[ \mathbf B_{\mathcal W}(\log \overline{z_1})\in C^\infty(\overline{\mathcal W}). \] If it is not smooth up to the boundary, this gives an alternative proof that Condition R fails on \(\mathcal W\).

来源字段：http://aimpl.org/crscv/1/

10376 | AIM-FUNCTIONAL_ANALYSIS-0039

### A ZFC counterexample and the projection-poset obstruction

S / solved | AIM 研讨会问题汇编

Suppose a C*-algebra has an approximate identity consisting of projections. Does it have an increasing approximate identity (on some index set, possibly different) consisting of projections?

来源字段：http://aimpl.org/settheorycstar/6/

10377 | AIM-FUNCTIONAL_ANALYSIS-0033

### A ZFC non-diagonalizable pure state and a uniform diagonal-gap criterion

S / solved | AIM 研讨会问题汇编

Consider the following statement: every pure state on \(\mathcal{B}\left( H\right) \) is diagonalizable. Is it consistent with with ZFC?

Is it consistent with ZFC that every pure state on \(\mathcal B(H)\) is diagonalizable?

来源字段：http://aimpl.org/settheorycstar/5/

10378 | OWR-15952-003

### Absolute Continuity and Densities of Bernoulli Convolutions

P / partially_solved | Oberwolfach 报告问题汇编

For which parameters $\lambda\in(1/2,1)$ is the Bernoulli convolution $\nu_\lambda$ absolutely continuous, and what regularity properties can its density have?

来源字段：https://doi.org/10.4171/owr/2017/47

10379 | AMR-036-0002

### Accessory parameters of the Heun equation

P / partially_solved | AMR 问题清单汇编

For the Heun equation $$y''+\left(\sum_{j=0}^2\frac{1-\alpha_j}{z-a_j}\right)y'+\frac{Az-\lambda}{(z-a_0)(z-a_1)(z-a_2)}y=0,$$ where $\alpha_j>0$, $A=\alpha'\alpha''$, and $\sum_{j=0}^2\alpha_j+\alpha'+\alpha''=2$, describe the accessory parameters $\lambda$ for which projective monodromy is conjugate into $\operatorname{PSU}(2)$. Treat also real $a_j,\lambda$; characterize nonemptiness; and give an explicit upper bound for the number of such parameters. Finiteness itself is known.

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10380 | OWR-12005-004

### Additive Eigenvalue Bounds for Domain Boundaries

O / open | Oberwolfach 报告问题汇编

For every dimension $m$, can the eigenvalue estimate for a smooth bounded domain $\Omega\subset\mathbb{R}^{m+1}$ with boundary $\Sigma$ be improved to $\lambda_k(\Sigma)\operatorname{Vol}(\Sigma)^{2/m}\leq A_m I(\Omega)^{1+2/m}+B_m k^{2/m}$, with constants $A_m,B_m$ depending only on $m$?

来源字段：https://doi.org/10.4171/owr/2012/33

10381 | AIM-SEVERAL_COMPLEX_VARIABLES-0025

### Affine hyperplane rank bounds and an explicit Veronese benchmark

S / solved | AIM 研讨会问题汇编

For \(n\geq 2\), define the rank of a real polynomial \(p(z,\bar z)\) on \(\mathbb C^n\) to be the smallest integer \(N\) such that \[ p(z,\bar z)=\sum_{j=1}^N \pm |p_j(z)|^2. \] Given a real-analytic \(p:\mathbb C^n\to\mathbb R\), suppose that for every complex hyperplane \(H\), the restriction \(p|_H\) has rank at most \(k\). Does there exist \(c(n,k)\) such that \(\operatorname{rank}p\leq c(n,k)\)?

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10382 | AIM-ANALYSIS-0041

### Affine lineality in the pseudoconvex case and a rate-sensitive nonconvex separation

P / partially_solved | AIM 研讨会问题汇编

What can we say about the geometry of a complete Reinhardt domain with trivial Bergman space?

来源字段：http://aimpl.org/scvproblems/1/

10383 | AIM-SEVERAL_COMPLEX_VARIABLES-0026

### Affine zero obstructions for powers that are squared norms

P / partially_solved | AIM 研讨会问题汇编

Find necessary and sufficient conditions on a real-valued polynomial \(R(z,\bar z)\) such that there exists \(N\) with \[ R(z,\bar z)^N=\sum_{j=1}^{k}|p_j(z)|^2. \]

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10384 | AIM-ANALYSIS-0155

### Agler-denominator characterization and a monomial-pullback product certificate

P / partially_solved | AIM 研讨会问题汇编

For which polynomials \(p\) with no zeros on \(\overline{\mathbb D}^{3}\) does \[ p(T)p(T)^*\succeq \widetilde p(T)\widetilde p(T)^* \tag{1.2} \] hold for every commuting triple \(T=(T_1,T_2,T_3)\) of strict contraction matrices, of every finite size?

The source does not spell out the multivariate reflection. The standard and degree-consistent recovery is the following. If \(p\in\mathbb C[z_1,\ldots,z_d]\) has **coordinate multidegree** \[ \mathbf n=(n_1,\ldots,n_d),\qquad n_j=\deg_{z_j}p, \] then \[ \widetilde p(z) =z^{\mathbf n}\overline{p(1/\bar z_1,\ldots,1/\bar z_d)}, \qquad z^{\mathbf n}=\prod_{j=1}^d z_j^{n_j}. \tag{1.1} \] This is not reflection with respect to total degree. The recovered tridisk question is:

The closed-polydisk hypothesis is retained exactly; it is stronger than the “no zeros in the open polydisk” convention used in some papers.

来源字段：http://aimpl.org/hyperbolicpoly/9/

10385 | OWR-14298003-001

### Algebraic Framework for Stochastic Gauge Symmetries

O / open | Oberwolfach 报告问题汇编

Is there a common algebraic framework for the renormalization and gauge symmetries arising in geometric stochastic heat equations, two-dimensional Yang-Mills Langevin dynamics, and three-dimensional stochastic Yang-Mills-Higgs theory?

来源字段：https://doi.org/10.4171/owr/2023/54

10386 | AIM-ANALYSIS-0093

### Algorithmic Kadison-Singer beyond weighted BSS sparsification

P / partially_solved | AIM 研讨会问题汇编

Algorithmic version of KS: try the Batson-S-S approach to KS? BUT, they had convinced themselves that their approach wouldn't work. Perhaps it is worth to revisit this?

* Leonid Gurvits

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10387 | AIM-FUNCTIONAL_ANALYSIS-0043

### All seven classes have the universal Polish-orbit degree

S / solved | AIM 研讨会问题汇编

Is the isomorphism relation in the following classes of separable C*-algebras:

- all

- nuclear

- exact

- simple

- simple exact

- nuclear \(\mathcal Z\)-stable

- simple nuclear

complete analytic? below a group actions? above all group actions?

来源字段：http://aimpl.org/settheorycstar/7/

10388 | AIM-ANALYSIS-0018

### All-coupling reverse Robin inequality for tangential domains

P / partially_solved | AIM 研讨会问题汇编

“...with a boundary parameter $\alpha0$ centred at the origin.”

This record is item 1.2, “Bareket's conjecture,” in the AIM list from the workshop *Shape optimization with surface interactions*.

The enumerated conjecture that follows is intact.

1. If $d=2$, $\alpha<0$, and $\Omega$ is bounded and simply connected, then the disk $B_R$ with $|B_R|=|\Omega|$ satisfies \[ \lambda_1(\Omega,\alpha)\leq \lambda_1(B_R,\alpha). \] 2. If $d\geq3$, $\alpha<0$, and $\Omega$ is bounded and convex, then the ball $B_R$ with $|B_R|=|\Omega|$ satisfies the same inequality.

Here and below the sign convention is \[ -\Delta u=\lambda u\quad\hbox{in }\Omega, \qquad \partial_\nu u+\alpha u=0\quad\hbox{on }\partial\Omega, \qquad \alpha<0. \] Only the damaged prose before the displayed list has been reconstructed. No mathematical symbol missing from the two-part conjecture is needed below.

来源字段：http://aimpl.org/shapesurface/1/

10389 | AIM-ANALYSIS-0020

### All-coupling Robin bound states for compact graph deformations in dimensions two and three

P / partially_solved | AIM 研讨会问题汇编

Consider the Robin Laplacian with a negative boundary parameter \(\alpha\) on a locally perturbed half-space (its boundary is a hyperplane away from a bounded region). The essential spectrum is expected to be \([-\alpha^2,\infty)\). Prove or disprove that there are always negative eigenvalues below \(-\alpha^2\).

Its mathematical content is:

The source does not specify the ambient dimension, regularity, whether the zero deformation is excluded, or whether “always” means every attractive coupling and every nontrivial deformation. These distinctions matter:

- the undeformed half-space itself has no eigenvalue below its threshold; - in dimension two every sufficiently regular nontrivial deformation with the same straight line on both tails is already known to bind for every attractive coupling; - in higher dimensions the answer depends on the class of deformation, and recurrence of the tangential directions stops after ambient dimension three.

Accordingly, the intended version is taken to ask for at least one discrete eigenvalue for every attractive parameter and every **nontrivial** sufficiently regular local deformation. The result below proves this for compactly supported graph deformations in ambient dimensions two and three and gives a capacity criterion in every dimension. It does not assert a full answer for all higher-dimensional or non-graph perturbations.

To avoid a sign collision with the literature, set \(\beta=-\alpha>0\). Thus \[ \partial_\nu u=\beta u, \qquad q_\beta[u]=\int_\Omega |\nabla u|^2\,dz -\beta\int_{\partial\Omega}|u|^2\,dS, \] and the flat threshold is \(-\beta^2=-\alpha^2\).

来源字段：http://aimpl.org/shapesurface/1/

10390 | OWR-12866-007

### Almost-Everywhere Fourier Convergence in $L^{1}$

O / open | Oberwolfach 报告问题汇编

Characterize the functions $f\in L^1(\mathbb T)$, for $\mathbb T=\mathbb R/\mathbb Z$, whose Fourier series converges almost everywhere.

来源字段：https://doi.org/10.4171/owr/2014/06

10391 | AIM-ANALYSIS-0120

### An axis-trace reduction and an obstruction to the canonical fat-Cantor trace

P / partially_solved | AIM 研讨会问题汇编

**Mapping properties of planar maps of exponentially integrable distortion.** Does there exist a homeomorphism \(f\) of the plane with exponentially integrable distortion such that \(f\) fixes the real axis and sends a set of positive length onto the \(\tfrac13\) Cantor set?

来源字段：http://aimpl.org/mappingmetric/4/

10392 | AIM-ANALYSIS-0007

### An exact cyclic r-star reduction for higher-size Angelesco equilibrium problems

P / partially_solved | AIM 研讨会问题汇编

Study potential theory for \(k\times k\) Riemann--Hilbert problems for \(k>2\) when multiple orthogonal polynomials are involved.

**Recovered research program.** For RHPs of size \(k=r+1>2\) arising from MOPs, identify the correct vector logarithmic-energy problem (masses, interaction coefficients, support and upper constraints, and external fields), prove existence and uniqueness or identify critical alternatives, derive its Euler--Lagrange and spectral-curve data, and use those data to construct the \(g\)-functions and sheet structure required for nonlinear steepest descent. The official report emphasizes the external-source \(4\times4\) case, but the canonical wording also includes Angelesco, Nikishin, star, two-matrix, and other MOP systems.

来源字段：http://aimpl.org/riemhilberttoeplitz/2/

10393 | AIM-ANALYSIS-0056

### An exact petal criterion for a diagonal family with a degenerate direction

P / partially_solved | AIM 研讨会问题汇编

Let \(F\) be a germ of automorphism of \(\mathbb C^n\) (\(n>1\)) tangent to the identity with a degenerate characteristic direction \(v\). When is there a domain of attraction tangent to the origin along \(v\)?

来源字段：http://aimpl.org/scvproblems/2/

10394 | AIM-OPTIMIZATION-0001

### An explicit lifted-LMI cone that is not hyperbolic

P / partially_solved | AIM 研讨会问题汇编

**Definition 1.10.1.** \(G\subseteq\mathbb R^d\) admits a lifted-LMI representation if there is a linear map \[ L:\mathbb R^d\oplus\mathbb R^m\longrightarrow\mathbb S^n \] such that \[ x\in\operatorname{int}(G) \quad\Longleftrightarrow\quad L(x,u)\succ0\quad\text{for some }u\in\mathbb R^m. \]

**Open Problem 1.10.1.** Characterize all \(d\)-dimensional, pointed, closed, convex cones in \(\mathbb R^d\) which admit a lifted-LMI representation. I believe that this set of cones strictly contain the hyperbolic cones.

来源字段：https://aimath.org/WWN/matrixineq/matrixineq.pdf

10395 | AIM-FUNCTIONAL_ANALYSIS-0017

### An ultrafilter-free local criterion for O2-ultrapower embeddability

P / partially_solved | AIM 研讨会问题汇编

Does every separable C*-algebra embed into an ultrapower of \(\mathcal O_2\) with respect to an ultrafilter on \(\mathbb N\)?

来源字段：http://aimpl.org/settheorycstar/1/

10396 | AIM-ANALYSIS-0063

### An unequal-cardinality trilinear Kakeya estimate in four dimensions

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** Suppose that we are in \(\mathbb R^4\), and let \(\varepsilon>0\). Suppose that \(\mathbb T_1,\mathbb T_2,\mathbb T_3\) are three transversal families of \(\delta\)-tubes (three short sides of length \(\delta\) and one long side of length \(1\)) such that, for each \(j\in\{1,2,3\}\), \[ \{e(T):T\in\mathbb T_j\} \] is a \(\delta\)-separated subset of \(S^3\). If \[ q\geq \frac43, \qquad \frac1p+\frac3q\leq 3, \] then is there a constant \(C_\varepsilon>0\) such that \[ \left\| \prod_{j=1}^{3} \left(\sum_{T\in\mathbb T_j}\chi_T\right) \right\|_{L^{q/3}(\mathbb R^4)} \leq C_\varepsilon \prod_{j=1}^{3} \left[ \delta^{\,4/q-3/p'-\varepsilon} (\#\mathbb T_j)^{1/p} \right]? \tag{AIM} \] Here \(e(T)\in S^3\) is the direction of the long side of \(T\), and \(p'\) is the conjugate exponent.

来源字段：https://aimath.org/pastworkshops/multilinopsproblems.pdf

10397 | AMR-036-0013

### Analytic germs with prescribed convex barriers

O / open | AMR 问题清单汇编

Let $A$ be a set of vectors $a=(a_1,\ldots,a_n)\in\mathbb C^n$ such that every $n$ are linearly independent, and let $K_a$ be plane convex compact sets. Characterize when analytic germs $F_1,\ldots,F_n$ at infinity, with $F_j(\infty)=0$, can be chosen so that $a_1F_1+\cdots+a_nF_n$ continues analytically to $\mathbb C\setminus K_a$ but to no larger domain $\mathbb C\setminus K$ with $K$ convex compact.

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10398 | OWR-12697711-002

### Analytic Hilbert Seventeenth Problem

P / partially_solved | Oberwolfach 报告问题汇编

Is every positive-semidefinite real-analytic function $f:\mathbb R^n\to\mathbb R$ a sum of squares of real-analytic meromorphic functions?

来源字段：https://doi.org/10.4171/owr/2023/15

10399 | OWR-13750336-008

### Analytic Maps of Complex Cantor Sets

O / open | Oberwolfach 报告问题汇编

For Cantor sets $X,Y\subset\mathbb C$, is the set of analytic maps from a neighborhood of $X$ into $\mathbb C$ that send $X$ into $Y$ at most countable?

来源字段：https://doi.org/10.4171/owr/2023/25

10400 | OWR-9790366-004

### Analytic Proofs of Dimension-Free Nonlocal Regularity

O / open | Oberwolfach 报告问题汇编

Give a purely analytic proof that solutions $w$ of the perturbed Kolmogorov equation satisfy $\|D_x^2w\|_{L^p((0,T)\times\mathbb R^2)}\leq M_p\|f\|_{L^p((0,T)\times\mathbb R^2)}$, ideally with the sharp constant $M_p=C_p$.

来源字段：https://doi.org/10.4171/owr/2022/15

10401 | OWR-13750336-007

### Analytic Self-Maps and Automorphism Groups of Cantor Sets

O / open | Oberwolfach 报告问题汇编

For a Cantor set $X\subset\mathbb R$, characterize its real-analytic self-maps and the groups arising as subgroups of $\operatorname{Aut}_\omega(X)$. When is $\operatorname{Aut}_\omega(X)$ finitely generated? If $X\subset\mathbb R^d$ and every real-analytic submanifold $M$ has $M\cap X$ with empty interior in $X$, is the set of real-analytic maps $X\to X$ at most countable?

来源字段：https://doi.org/10.4171/owr/2023/25

10402 | OWR-4799-009

### Analytic Torsion under Holomorphic Degeneration

P / partially_solved | Oberwolfach 报告问题汇编

For a proper holomorphic degeneration $\pi:X\to S$ with a possibly singular central fiber $X_{0}$, what is the asymptotic behavior as $s\to0$ of the analytic torsion $\tau(X_{s},\omega_{X_{s}}(\xi_{s}))$?

来源字段：https://doi.org/10.4171/owr/2011/09

10403 | OWR-15992-002

### Antibiotic Resistance Persistence in Companion Animal Models

P / partially_solved | Oberwolfach 报告问题汇编

For a within-host model containing wild-type, mutation-resistant, and plasmid-resistant bacteria in a companion animal, determine whether the animal can become a source of antibiotic resistance and characterize parameter regimes with persistence or coexistence of the strains.

来源字段：https://doi.org/10.4171/owr/2018/7

10404 | OWR-16163-008

### Applications of CPH Semigroups and Dilations

O / open | Oberwolfach 报告问题汇编

Determine useful applications of CPH-maps, CPH-semigroups, and CPH-dilations, including whether these structures yield significant new dilation or operator-theoretic results.

来源字段：https://doi.org/10.4171/owr/2018/22

10405 | OWR-720-008

### Approximation by Convolution Idempotents

O / open | Oberwolfach 报告问题汇编

How close can an $L^1$ function be to a convolution idempotent?

来源字段：https://doi.org/10.4171/owr/2004/10

10406 | OWR-14298366-001

### Asymptotic Growth of a Matrix Cone Norm Constant

O / open | Oberwolfach 报告问题汇编

Determine the asymptotic growth of $C_n=\sup\{\gamma_+(A)/\lVert A\rVert_{(e,1)}:A\in\mathcal{C}^{n\times n},\ A\succeq0,\ A\neq0\}$.

来源字段：https://doi.org/10.4171/owr/2024/21

10407 | OWR-12866-014

### Averaging Times for $H^p$ Dirichlet Polynomials

O / open | Oberwolfach 报告问题汇编

For $C>1$, let $T_p(N,C)$ be the least $T$ such that $$\frac1T\int_0^T|f(it)|^p\,dt\leq C\|f\|_p^p$$ for every Dirichlet polynomial $f(s)=\sum_{n=1}^N a_n n^{-s}$ of degree at most $N$. How does $T_p(N,C)$ depend on $p$? In particular, is $T_p(N,C)=O(N)$ for $1\leq p<2$?

来源字段：https://doi.org/10.4171/owr/2014/06

10408 | AMR-036-0010

### Backward uniqueness for the heat equation

P / partially_solved | AMR 问题清单汇编

Let $D\subset\mathbb R^n$ have regular boundary for the Dirichlet problem. Prove or disprove that the following are equivalent: (PI) there is a nonzero bounded solution of $u_t=\Delta u$ on $D\times[0,1]$ with $u(x,1)=0$; (PII) there is a nonzero harmonic function $v$ on $D$ satisfying $v(x)=O(e^{-|x|^2})$ as $|x|\to\infty$.

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10409 | AIM-ANALYSIS-0091

### Balanced uncentered sparsification by a one-coordinate lift

S / solved | AIM 研讨会问题汇编

Find \(S\subset[m]\), \(|S|\approx d\), so that \[ C\sum_{j\in S}x_jx_j^T\approx I \] with a balancing condition \(\left\|\sum x_j\right\|\leq B\).

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10410 | AMR-138-0001

### Banach Spaces and Ramsey Theory - Problem 1

O / open | AMR 问题清单汇编

Recall that the *Banach-Mazur distance* between two isomorphic Banach spaces $X$ and $Y$ (not necessarily infinite-dimensional) is defined by \[ d_{BM}(X,Y)=\inf\big\{\|T\|\cdot\|T^{-1}\|: T:X\to Y \text{ is an isomorphism}\big\}.\] For every $n\in\mathbb{N}$ define \[ R^n_\infty=\max\big\{ d_{BM}(X,\ell^n_\infty): \mathrm{dim}(X)=n\big\}.\]

\emph{(Distance from the cube -- A. Pe{\l}czy\'{n}ski [Pe2])} Determine the asymptotic behavior of $R^n_\infty$ as $n$ tends to infinity.

背景链接：https://arxiv.org/abs/1006.2668

10411 | AMR-138-0010

### Banach Spaces and Ramsey Theory - Problem 10

O / open | AMR 问题清单汇编

*(Fixing properties of operators on $C[0,1]$ -- H. P. Rosenthal [Ro2])* Let $T:C[0,1]\to C[0,1]$ be a bounded linear operator and $X$ be a separable Banach space not containing a copy $c_0$. Assume that the operator $T$ fixes a copy of $X$. Is it true that $T$ fixes a copy of $C[0,1]$? Equivalently, is it true that the dual operator $T^*$ of $T$ has non-separable range?

背景链接：https://arxiv.org/abs/1006.2668

10412 | AMR-138-0011

### Banach Spaces and Ramsey Theory - Problem 11

O / open | AMR 问题清单汇编

The *Distortion Problem* is discussed in detail in [OS-new]. A surprising phenomenon observed so far is that ``most'' distortable Banach spaces are arbitrarily distortable.

*(Bounded distortion -- see [OS-new])* Does there exist a boundedly distortable Banach space?

背景链接：https://arxiv.org/abs/1006.2668

10413 | AMR-138-0012

### Banach Spaces and Ramsey Theory - Problem 13

O / open | AMR 问题清单汇编

*(Converse Aharoni problem -- see [BL,Kalton])* Let $X$ be a separable Banach space with the property that every separable Banach space is Lipschitz isomorphic to a subset of $X$. Is it true that $X$ contains a copy of $c_0$?

背景链接：https://arxiv.org/abs/1006.2668

10414 | AMR-138-0013

### Banach Spaces and Ramsey Theory - Problem 14

O / open | AMR 问题清单汇编

Recall the very old result due to S. Banach and S. Mazur saying that every infinite-dimensional Banach space has an infinite basic sequence. It turns out that the dual version of this *Basic Sequence Problem* is still widely open.

\emph{(S. Banach [B] -- A. Pe{\l}czy\'{n}ski [Pe0])} Does every infinite-dimensional Banach space $X$ has an infinite-dimensional quotient with a Schauder basis?

背景链接：https://arxiv.org/abs/1006.2668

10415 | AMR-138-0014

### Banach Spaces and Ramsey Theory - Problem 15

O / open | AMR 问题清单汇编

So the problem of S. Banach and A. Pe{\l}czy\'{n}ski now becomes the following.

*(Separable quotient problem)* Does every infinite-dimensional Banach space have a non-trivial infinite-dimensional separable quotient?

背景链接：https://arxiv.org/abs/1006.2668

10416 | AMR-138-0015

### Banach Spaces and Ramsey Theory - Problem 16

O / open | AMR 问题清单汇编

Are the following equivalent assuming the P-ideal dichotomy?

(1) Every non-separable Banach space has an uncountable biorthogonal system.

(2) $\mathfrak{b}=\aleph_2$.

背景链接：https://arxiv.org/abs/1006.2668

10417 | AMR-138-0016

### Banach Spaces and Ramsey Theory - Problem 17

O / open | AMR 问题清单汇编

Recall that a *support set* in a Banach space $X$ is a nonempty convex set $C$ with the property that every point $x$ of $C$ is its support point, i.e. there is a functional $f$ of $X$ such that \[ f(x)=\inf\{f(y): y\in C\}<\sup\{f(y): y\in C\}. \] The existence of a support set in every non-separable Banach space is an old problem of S. Rolewicz [Rol] who showed that separable Banach spaces do not admit support sets and who noticed that many of the non-separable Banach spaces do admit support sets.

*(S. Rolewicz [Rol])* Which Banach spaces admit support sets? Do all non-separable Banach spaces have support sets?

背景链接：https://arxiv.org/abs/1006.2668

10418 | AMR-138-0017

### Banach Spaces and Ramsey Theory - Problem 18

O / open | AMR 问题清单汇编

This leaves however open the following interesting variant of Rolewicz's problem.

*(S. Todorcevic [To1])* Does every Banach space of density $>\aleph_1$ have a support set?

背景链接：https://arxiv.org/abs/1006.2668

10419 | AMR-138-0018

### Banach Spaces and Ramsey Theory - Problem 19

O / open | AMR 问题清单汇编

*(Unconditional basic sequences in non-separable spaces)* Does every Banach space $X$ of density $>\mathfrak{c}$ have an infinite unconditional basic sequence? Does additional properties, like reflexivity, make a difference?

背景链接：https://arxiv.org/abs/1006.2668

10420 | AMR-138-0002

### Banach Spaces and Ramsey Theory - Problem 2

O / open | AMR 问题清单汇编

\emph{(Elton unconditionality constant -- S. J. Dilworth, E. Odell, Th. Schlumprecht and A. Zs\'{a}k [DOSZ])} Does there exist $M\geqslant 1$ such that $K(\delta)\leqslant M$ for every $0<\delta\leqslant 1$?

背景链接：https://arxiv.org/abs/1006.2668

10421 | AMR-138-0019

### Banach Spaces and Ramsey Theory - Problem 20

O / open | AMR 问题清单汇编

On the other hand no consistent bounds, like the one of [DLT] mentioned above, is known in this context. So the following problem is wide open.

Does every (respectively, every reflexive) Banach space of density $>\mathfrak{c}$ (respectively, of density $>\aleph_1$) has an infinite sub-symmetric basic sequence?

背景链接：https://arxiv.org/abs/1006.2668

10422 | AMR-138-0020

### Banach Spaces and Ramsey Theory - Problem 22

O / open | AMR 问题清单汇编

*(D. H. Fremlin [Fr2])* Is there a compact hereditary family of finite subsets of $\omega_1$ containing the singletons and having the property that every finite subset of $\omega_1$ has a subset half of its size belonging to the family $\mathcal{F}$?

背景链接：https://arxiv.org/abs/1006.2668

10423 | AMR-138-0003

### Banach Spaces and Ramsey Theory - Problem 3

O / open | AMR 问题清单汇编

*(Rosenthal basis problem -- H. P. Rosenthal [FPR])* Is it true that every Rosenthal basis $(x_n)$ is equivalent to the standard unit vector basis of $c_0$ or $\ell_p$ for some $1\leqslant p<+\infty$?

背景链接：https://arxiv.org/abs/1006.2668

10424 | AMR-138-0004

### Banach Spaces and Ramsey Theory - Problem 4

O / open | AMR 问题清单汇编

*(Unique spreading model problem -- S. A. Argyros [AOST])* Let $X$ be a reflexive Banach space such that $|\mathrm{SP_w}(X)/\sim|=1$ (i.e. all normalized weakly null sequences in $X$ generate a unique spreading model up to equivalence). Is it true that the unique spreading model is equivalent to the standard unit vector basis of $c_0$ or $\ell_p$ for some $1\leqslant p<+\infty$?

背景链接：https://arxiv.org/abs/1006.2668

10425 | AMR-138-0005

### Banach Spaces and Ramsey Theory - Problem 5

O / open | AMR 问题清单汇编

*(Countable spreading models problem -- S. J. Dilworth, E. Odell and B. Sari [DOS])* Let $X$ be a reflexive Banach space such that $|\mathrm{SP_w}(X)/\sim|\leqslant \aleph_0$. Does there exist a normalized weakly null sequence in $X$ generating a spreading model equivalent to the standard unit vector basis of $c_0$ or $\ell_p$ for some $1\leqslant p<+\infty$?

背景链接：https://arxiv.org/abs/1006.2668

10426 | AMR-138-0006

### Banach Spaces and Ramsey Theory - Problem 6

O / open | AMR 问题清单汇编

*(Block homogeneous basis problem -- see [FR])* Let $X$ be a Banach space with a normalized block homogeneous Schauder basis. Is it true that $X$ is isomorphic to $c_0$ or $\ell_p$ for some $1\leqslant p<+\infty$?

背景链接：https://arxiv.org/abs/1006.2668

10427 | AMR-138-0007

### Banach Spaces and Ramsey Theory - Problem 7

O / open | AMR 问题清单汇编

Although it is known that every separable non-Hilbertian Banach space contains two non-isomorphic subspaces, we point out that there are several related open problems. The following one is probably the most natural.

*(G. Godefroy [Go])* Let $X$ be a separable non-Hilbertian Banach space. Is it true that there exists a sequence $(X_n)$ of subspaces of $X$ which are pairwise non-isomorphic?

背景链接：https://arxiv.org/abs/1006.2668

10428 | AMR-138-0008

### Banach Spaces and Ramsey Theory - Problem 8

O / open | AMR 问题清单汇编

Let $X$ be a separable Banach space which fails to have type $2- $ (or fails to have co-type $2+ $) for some $ >0$. Is it true that $X$ contains continuum many pairwise non-isomorphic subspaces?

背景链接：https://arxiv.org/abs/1006.2668

10429 | AMR-138-0009

### Banach Spaces and Ramsey Theory - Problem 9

O / open | AMR 问题清单汇编

*(Complemented Subspace Problem for $C[0,1]$ -- see [BP-new,LW-new,Pe1])* Is it true that every complemented subspace of $C[0,1]$ is isomorphic to a $C(K)$ space for some closed subset $K$ of $[0,1]$.

背景链接：https://arxiv.org/abs/1006.2668

10430 | AIM-ANALYSIS-0024

### Beck's solution of van den Berg's conjecture and an exact prism refinement

S / solved | AIM 研讨会问题汇编

If \(\Omega\subset\mathbb R^n\) is a convex domain, \(\rho\) and \(D\) are its inradius and diameter, and \(u\) is a first Dirichlet eigenfunction, then there should be a dimensional constant \(C_n\) such that

\[ \|u\|_{L^\infty(\Omega)} \le C_n\rho^{-n/2}\left(\frac{\rho}{D}\right)^{1/6} \|u\|_{L^2(\Omega)}. \tag{1} \]

The record first recalls Chiti's estimate, which is (1) without the factor \((\rho/D)^{1/6}\), and says that Georgiev--Mukherjee--Steinerberger proved (1) for \(n=2\). It also says that the exponent \(1/6\) is suggested by a thin cone and asks even for a weaker positive power of \(\rho/D\).

The intended domain is **open, bounded, full-dimensional, and convex**. Van den Berg's original Conjecture 3 explicitly says “open, bounded convex set.” Boundedness is also needed for finite \(D\) and for the usual discrete first Dirichlet eigenvalue. A convex open set is connected. The eigenfunction may be multiplied by any nonzero scalar; (1) is invariant under that choice. We take the positive ground state.

来源字段：http://aimpl.org/shapesurface/2/

10431 | AMR-036-0027

### Belgian Chocolate constant

P / partially_solved | AMR 问题清单汇编

Let $f$ be real and holomorphic in the unit disk, with one simple zero at $0$ and two simple $1$-points at $\pm ia$. Determine the minimum possible value of $a$.

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10432 | OWR-13750335-004

### Bergman-Kernel Asymptotics at Curvature-Vanishing Points

O / open | Oberwolfach 报告问题汇编

Establish a full pointwise asymptotic expansion of the Bergman kernel at curvature-vanishing points for semipositive line bundles in complex dimension greater than one.

来源字段：https://doi.org/10.4171/owr/2023/24

10433 | OWR-14297736-010

### Bessel-Function Solutions of Inhomogeneous Laplace Equations

O / open | Oberwolfach 报告问题汇编

For every admissible tuple $(r,a,b)$, prove existence of Fourier-mode solutions expressed through modified Bessel functions, find closed formulas for their Laurent-polynomial coefficients, and relate the solution space to differential Galois theory.

来源字段：https://doi.org/10.4171/owr/2023/36

10434 | AMR-036-0030

### Better estimates for Littlewood constants

P / partially_solved | AMR 问题清单汇编

Obtain better rigorous estimates for the Littlewood exponents $\alpha$, $\beta$, and for $\sup_c P_c$, where $P_c$ is the pressure for the hyperbolic quadratic polynomial $p_c(z)=z^2+c$ corresponding to the potential $|(p_c^n)'|^{-1}$.

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10435 | OWR-14298587-021

### Bianchi-Egnell Optimizers

P / partially_solved | Oberwolfach 报告问题汇编

Characterize the optimizers of the sharp Bianchi-Egnell stability constant $c_{\mathrm{BE}}$. In particular, determine whether optimizers are radial and whether they can change sign.

来源字段：https://doi.org/10.4171/owr/2024/37

10436 | OWR-8415348-001

### Bilinear Wave-Schrödinger Restriction Without Transversality

O / open | Oberwolfach 报告问题汇编

Determine the admissible exponents and sharp scale dependence in bilinear restriction estimates for wave-Schrödinger interactions when the transversality assumption is removed.

来源字段：https://doi.org/10.4171/owr/2021/45

10437 | OWR-1787-007

### Blaschke Approximation of Inner-Function Arguments

O / open | Oberwolfach 报告问题汇编

For an inner function $B$ on the unit disk $D$, does there exist an interpolating Blaschke product $I$ such that $\operatorname{Arg}B-\operatorname{Arg}I=\widetilde{v}$ for some $v\in L^\infty(\partial D)$? Does there exist such an $I$ with $\operatorname{Arg}B-\operatorname{Arg}I=u+\widetilde{v}$ for $u,v\in L^\infty(\partial D)$ and $\lVert u\rVert_\infty<\pi/2$?

来源字段：https://doi.org/10.4171/owr/2008/06

10438 | OWR-738-007

### Bloch Spectra and Hilbert-Space Characterizations

P / partially_solved | Oberwolfach 报告问题汇编

If $\Lambda_{\mathrm{Bloch}},\ell^1}(X)=\ell^{(2,1)}$ for a complex Banach space $X$, must $X$ be isomorphic to a Hilbert space?

来源字段：https://doi.org/10.4171/owr/2004/36

10439 | AIM-FUNCTIONAL_ANALYSIS-0013

### Block-line sections and a covariance criterion for diagonal ell_p dilations

P / partially_solved | AIM 研讨会问题汇编

Let \(p>1\), let \(H\subset\mathbb R^n\) be a fixed \(k\)-dimensional linear subspace, and put

\[
D_t=\operatorname{diag}(e^{t_1},\ldots,e^{t_n}),\qquad B_p^n=\left\{x:\sum_{i=1}^n|x_i|^p\leq1\right\}. \]

\[ f_H(t)=\operatorname{vol}_k(D_tB_p^n\cap H),\qquad t\in\mathbb R^n, \tag{1.1} \]

jointly log-concave?

Here \(\operatorname{vol}_k\) is Euclidean Lebesgue measure induced on the **fixed** subspace \(H\). The source remark says that \(p=1,2\) are known from Nayar--Tkocz and that the \(p=\infty\) assertion is equivalent to the logarithmic Brunn--Minkowski conjecture. Neighboring AIM records introduce no additional convention.

Two possible misreadings must be excluded:

- this is section volume, not the volume of the orthogonal projection onto \(H\); - in general \(D_tB_p^n\cap H\neq D_t(B_p^n\cap H)\), because \(D_t\) need not preserve \(H\).

For finite \(t\), the section contains a relative neighborhood of \(0\) in \(H\), so \(f_H(t)>0\) when \(k\geq1\). There is therefore no zero-volume convention hidden in log-concavity.

来源字段：http://aimpl.org/symconvgeomineq/2/

10440 | AIM-ANALYSIS-0139

### BMV after Stahl: reducible measures and the first three-state phase obstruction

S / solved | AIM 研讨会问题汇编

Find a simple proof of BMV conjecture for $3\times 3$ matrices (and an explicit formula for the BMV measure for $3\times 3$ matrices; the existence of the measure is due to recent work of Stahl).

来源字段：http://aimpl.org/hyperbolicpoly/2/

10441 | OWR-12866-016

### Bohr-Radius Characterization of $\ell^1$

O / open | Oberwolfach 报告问题汇编

Let $X$ be a Banach sequence space such that $$\inf_n K(B_{X_n})>0,$$ where $K(B_{X_n})$ is the Bohr radius of the unit ball of its first $n$ coordinates. Must $X=\ell^1$ as sets?

来源字段：https://doi.org/10.4171/owr/2014/06

10442 | AIM-SEVERAL_COMPLEX_VARIABLES-0050

### Boundary and Milnor obstructions to deformation into a ball

P / partially_solved | AIM 研讨会问题汇编

Let \(n\geq 3\), and let \(V\subset\mathbb C^n\) be a complex analytic subvariety of codimension one. Assume that \(0\in V\) is an isolated singularity and that \(\partial V\) is compact and smooth. Can \(V\) be “deformed” into \(\mathbb B^{\,n-1}\subset\mathbb C^{n-1}\)? A smooth deformation \((V_t,\partial V_t)\), for \(t\) in an interval \(I\), is required to have the following local descriptions. Near every \((p_0,t_0)\) with \(p_0\in V_{t_0}\), there is a function \(F(p,t)\), smooth in \((p,t)\) and holomorphic in \(p\), such that \(V_t=\{F(\cdot,t)=0\}\). Near every boundary point there is also a smooth real function \(r(p,t)\) such that locally \[ V_t=\{F(\cdot,t)=0,\ r(\cdot,t)<0\}, \] and the indicated \(p\)-derivatives of \(F\) and \(r\) are linearly independent.

来源字段：https://aimath.org/WWN/complexitycrmap/complexitycrmap.pdf

10443 | OWR-1703869-003

### Boundary Branch Singularities of Area-Minimizing Currents

O / open | Oberwolfach 报告问题汇编

Let $T$ be an area-minimizing two-dimensional current with analytic prescribed boundary curve $\Gamma\subset\mathbb R^{2+n}$. Is the set of branched singular points of $T$ at the boundary isolated, with $T$ having finite topology locally at each such point?

来源字段：https://doi.org/10.4171/owr/2020/22

10444 | OWR-14299519-004

### Boundary Branching of Area-Minimizing Currents

O / open | Oberwolfach 报告问题汇编

For an area-minimizing current with analytic boundary, what is the structure and maximal dimension of the boundary branching set?

来源字段：https://doi.org/10.4171/owr/2025/33

10445 | OWR-3389-018

### Boundary Corrections to Domain Lieb-Thirring Constants

P / partially_solved | Oberwolfach 报告问题汇编

Can one obtain improved Lieb-Thirring constants for a Schrödinger operator on a domain $\Omega$ rather than on all of $\mathbb R^n$, for example by proving a boundary correction term?

来源字段：https://doi.org/10.4171/owr/2009/06

10446 | OWR-3389-015

### Boundary Corrections to Li-Yau Eigenvalue Bounds

O / open | Oberwolfach 报告问题汇编

For Dirichlet Laplacian eigenvalues, can the Li-Yau lower bound for sums of eigenvalues be strengthened by a correction term involving boundary surface area? Do analogous boundary-measure corrections hold for sums of Neumann eigenvalues and for Berezin-type bounds?

来源字段：https://doi.org/10.4171/owr/2009/06

10447 | AIM-ANALYSIS-0049

### Boundary multipliers and Levi resonance for autonomous attracting basins

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** Let $F\in\operatorname{Aut}(\mathbb C^2)$ be one holomorphic automorphism, and let $p$ be an attracting fixed point (or an attracting periodic cycle). Can the proper full basin of attraction of $p$ have nonempty boundary which is globally an embedded $C^1$ real hypersurface of $\mathbb R^4$?

来源字段：http://aimpl.org/scvproblems/1/

10448 | OWR-13750336-014

### Boundary Sparse Bounds for Planar Bochner-Riesz Means

O / open | Oberwolfach 报告问题汇编

别名：Endpoint Sparse Bounds for Planar Averaging Operators [OWR-13750336-013]

For $0<\lambda<1/2$, do the planar Bochner-Riesz means $R_t^\lambda$ satisfy every sparse bound $\operatorname{Sp}(p,q)$ whose reciprocal-exponent pair $(1/p,1/q)$ lies on the boundary segment $P_2P_3$ of the necessary-condition trapezoid $T_2(\lambda)$?

来源字段：https://doi.org/10.4171/owr/2023/25

10449 | OWR-1057-004

### Boundary-Supported Interpolating Measures for Bergman Spaces

O / open | Oberwolfach 报告问题汇编

Does there exist a bounded planar region $G$ for which multiplication by $z$ on the Bergman space $L_a^2(G)$ has a continuous interpolating measure supported on the boundary of $G$?

来源字段：https://doi.org/10.4171/owr/2005/36

10450 | AIM-SEVERAL_COMPLEX_VARIABLES-0014

### Boundary-to-filling equivalence and a local affirmative answer

P / partially_solved | AIM 研讨会问题汇编

It was proven by Eliashberg that any embeddable CR-structure on $\mathbb{S}^3$ bounds a Stein manifold $X\simeq \mathbb{B}^4$ with a strictly plurisubharmonic exhaustion function with a single critical point. Can $X$ always be embedded into $\mathbb{C}^2$?

Does the compact Stein filling $\overline X$, with its given smooth strictly pseudoconvex boundary, admit a holomorphic embedding into $\mathbb C^2$ smooth up to the boundary, so that its image is a bounded strictly pseudoconvex domain and the boundary CR structure is preserved?

来源字段：http://aimpl.org/crscv/3/

10451 | OWR-11101912-001

### Boundedness of Multilinear Hilbert Transforms

O / open | Oberwolfach 报告问题汇编

For $n\geq3$, establish the conjectured $L^p$ boundedness of the $n$-linear Hilbert transform; more generally, establish boundedness of the simplex Hilbert transform.

来源字段：https://doi.org/10.4171/owr/2022/31

10452 | OWR-15577-003

### Bounds for Multilinear and Simplex Hilbert Transforms

O / open | Oberwolfach 报告问题汇编

Establish bounds in the appropriate $L^p$ ranges for multilinear and simplex Hilbert transforms, including the higher-dimensional singular forms arising from the corners problem.

来源字段：https://doi.org/10.4171/owr/2017/34

10453 | OWR-1703869-005

### Branching Complexity in the Branched Transportation Mailing Problem

O / open | Oberwolfach 报告问题汇编

For an optimizer of the mailing problem in branched transportation, does it have finite graph structure in every ball outside the supports of the initial and target measures, and is the number of branches at a branching point uniformly bounded?

来源字段：https://doi.org/10.4171/owr/2020/22

10454 | OWR-4531-015

### Brennan-Type Integrability for Crescent-Domain Conformal Maps

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be the bounded complementary component of a crescent domain and let $\phi:G\to\mathbb D$ be conformal. For which $\tau$ is $$\int_G|\phi'(z)|^{3+\tau}\,dA(z)<\infty$$ independently of $G$?

来源字段：https://doi.org/10.4171/owr/2010/49

10455 | AIM-ANALYSIS-0132

### Calibrated faces in ell_p^n and the solved dimensional ranges

P / partially_solved | AIM 研讨会问题汇编

Let $\mathcal H^m$ denote the Hausdorff $m$-measure corresponding to the norm in $\ell_p^n$. Let $P$ be a polyhedral $m$-cycle in $\ell_p^n$, $p\ne2$, with faces $F_1,\ldots,F_k$. Is \[ \mathcal H^m(F_1)\le \sum_{j=2}^k\mathcal H^m(F_j)? \]

来源字段：http://aimpl.org/mappingmetric/6/

10456 | OWR-13103-004

### Calibrations for Unoriented Minimum Matching

O / open | Oberwolfach 报告问题汇编

How much of the calibration and filling-minimization procedure survives for the unoriented minimum-matching problem after discarding the linear duality between currents and differential forms?

来源字段：https://doi.org/10.4171/owr/2014/33

10457 | OWR-13486-006

### Canonical Endpoint Normalizations for Singular Sturm-Liouville Problems

O / open | Oberwolfach 报告问题汇编

Formulate satisfactory normalizations of a fundamental system of solutions near endpoints of limit-point or limit-circle type that are not regular singular points, for example for the Lennard-Jones potential $q(x)=4\epsilon((\sigma/x)^{12}-(\sigma/x)^6)$, with $\epsilon>0$ and $\sigma>0$.

来源字段：https://doi.org/10.4171/owr/2015/1

10458 | AIM-FUNCTIONAL_ANALYSIS-0025

### Cardinality bounds and forcing localization for tensor norms

P / partially_solved | AIM 研讨会问题汇编

Does the number of C*-norms on $A\otimes _{alg}B$ for

- $A=B=\mathcal{B}\left( H\right) $

- $A=B=C^{\ast }\left( \mathbb{F}_{\infty }\right) $, where $\mathbb{F}_{\infty }$ is the free group on countably many generators

- $A=B\left( H\right) $ and $B=\mathcal{Q}\left( H\right) $ where $\mathcal{Q}\left( H\right) $ is the Calkin algebra

depend on the model of set theory?

来源字段：http://aimpl.org/settheorycstar/3/

10459 | AMR-036-0033

### Carleson-Jones quarter conjecture

O / open | AMR 问题清单汇编

For a regular connected compact plane set $E$, let $\beta_E=\limsup_{\varepsilon\to0}\log l(\varepsilon)/(-\log\varepsilon)$, where $l(\varepsilon)$ is the length of the Green-function level curve $G=\varepsilon$, and let $\beta=\sup_E\beta_E$. Is $\beta=1/4$?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10460 | AIM-ANALYSIS-0017

### Cartesian-product closure and endpoint tests for Robin-gap monotonicity

P / partially_solved | AIM 研讨会问题汇编

**Monotonicity of the spectral gap.** For a bounded convex domain \(\Omega\subset\mathbb R^n\), the spectral gap \(\lambda _2(\Omega,\alpha)-\lambda _1(\Omega,\alpha)\) is strictly increasing as a function of \(\alpha>0\). In particular, the Neumann gap provides a lower bound for the Dirichlet gap: \[ \mu _1(\Omega)<\lambda _2(\Omega)-\lambda _1(\Omega). \]

来源字段：http://aimpl.org/shapesurface/1/

10461 | AIM-ANALYSIS-0077

### Certified constructive descent for interlacing proofs

P / partially_solved | AIM 研讨会问题汇编

There are existence proofs for KS, how to make them constructive? Likes stable polynomials and interlacers. How to make constructive proofs?

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10462 | AIM-SPECIAL_FUNCTIONS-0015

### Charge on LS galleries and paths

P / partially_solved | AIM 研讨会问题汇编

Define a charge statistic on galleries. Recall that the tableaux form a subset of the galleries under a natural injection, so the new statistic should restrict to the usual charge statistic on tableaux. Galleries, in turn, can be viewed as Littelmann paths, so one could even ask for a charge statistic on the latter objects.

来源字段：https://aimath.org/WWN/kostka/prob2.pdf

10463 | AIM-SPECIAL_FUNCTIONS-0016

### Charge, the Brylinski--Kostant filtration, and canonical bases

P / partially_solved | AIM 研讨会问题汇编

(16) Give a non-miraculous representation-theoretical explanation of charge. There exists a well-defined parametrizing set for a basis of the representation for each weight space and filtration. Yet the canonical Lusztig-Kashiwara bases are not compatible. How can these be related?

(16) Give a “non-miraculous” representation-theoretical explanation of charge. Note that there exists a well-defined parametrizing set for a basis of the representation for each weight space and filtration. Yet the canonical Lusztig-Kashiwara bases are not compatible - how can these be related?

来源字段：https://aimath.org/WWN/kostka/prob2.pdf

10464 | OWR-14298585-006

### Cheeger-Space Criteria for Metric-Measure Spaces

O / open | Oberwolfach 报告问题汇编

Characterize when a metric-measure space $(X,d,m)$ is a Cheeger space, and determine when the resulting metric-measure construction gives the appropriate $p$-energy on $X$.

来源字段：https://doi.org/10.4171/owr/2024/42

10465 | OWR-12979-008

### Choquet Boundaries and $C^*$-Envelopes of Commuting Tuples

O / open | Oberwolfach 报告问题汇编

For a commuting operator tuple $A=(A_1,\ldots,A_d)$, can one compute its noncommutative Choquet boundary and hence its $C^*$-envelope?

来源字段：https://doi.org/10.4171/owr/2014/21

10466 | OWR-14748-002

### Classification Beyond $\mathcal Z$-Stable Simple $C^*$-Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Can separable, unital, simple, nuclear $C^*$-algebras in the non-$\mathcal Z$-stable case be classified by the Elliott invariant together with the Cuntz semigroup? What classification is possible in the nonsimple case?

来源字段：https://doi.org/10.4171/owr/2016/40

10467 | OWR-3470-001

### Classification of Analytic Hilbert Modules on Strictly Convex Domains

O / open | Oberwolfach 报告问题汇编

Let $\Omega\subset\mathbb C^d$ be a strictly convex domain with smooth boundary, and let $A$ be a matrix with entries in the disk algebra $\mathcal A(\Omega)$ whose rank is constant on $\partial\Omega$. Classify, within the framework of quasi-free Hilbert modules, the privileged analytic Hilbert modules $$F=\operatorname{coker}\bigl(A: L_a^2(\Omega)^m\longrightarrow L_a^2(\Omega)^n\bigr),$$ and describe their parameter space.

来源字段：https://doi.org/10.4171/owr/2009/20

10468 | OWR-14748-009

### Classification of General Free Araki-Woods Factors

P / partially_solved | Oberwolfach 报告问题汇编

Classify the free Araki-Woods factors $\Gamma(\mu,m)''$ beyond the almost-periodic case. In particular, what complete invariant should replace the currently known partial invariants?

来源字段：https://doi.org/10.4171/owr/2016/40

10469 | OWR-10252928-004

### Closed Forms for Multivariate Polynomial Approximation Kernels

O / open | Oberwolfach 报告问题汇编

Can the coefficients of the optimal multivariate polynomial-approximation kernel be expressed in closed form?

来源字段：https://doi.org/10.4171/owr/2022/20

10470 | OWR-15181-004

### Closure Properties of Logarithmic Hardy Fields

O / open | Oberwolfach 报告问题汇编

For the quasianalytic asymptotic field $F=\bigcup_k F_k$ built from Ilyashenko algebras, what closure properties does $F$ have beyond being a Hardy field and being closed under logarithmic composition?

来源字段：https://doi.org/10.4171/owr/2016/60

10471 | OWR-4211-008

### Coercivity of Acoustic Boundary Integral Operators

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Acoustic Boundary-Operator Coercivity Question [OWR-4211-010]

Determine the conditions on the wavenumber, coupling parameter, obstacle geometry, and Lipschitz boundary under which the acoustic-scattering boundary integral operator

$$A_{k,\eta}=I+K'-i\eta S$$

is coercive.

来源字段：https://doi.org/10.4171/owr/2010/10

10472 | OWR-13948-006

### Collapse of Multivalued Minimizers with Single-Valued Boundary Data

O / open | Oberwolfach 报告问题汇编

If u minimizes a quasiconvex functional and $u|_{\partial B}=Q\llbracket\phi\rrbracket$, must $u=Q\llbracket\Phi\rrbracket$ for some single-valued minimizer $\Phi$?

来源字段：https://doi.org/10.4171/owr/2015/53

10473 | OWR-16160-019

### Collected Coarse-Geometric Embedding Problems

P / partially_solved | Oberwolfach 报告问题汇编

Does $L_p([0,1])$ embed coarsely into $\ell_p$? Does there exist a free measure-preserving action of a free group on a manifold whose warped cone admits a wall structure? Can one construct a slowly growing box space with bounded successive indices? For finite groups $\operatorname{SL}(n_m,\mathbb F_{p_m})$, do fixed-cardinality generating sets exist whose Cayley graphs form a superexpander family?

来源字段：https://doi.org/10.4171/owr/2018/19

10474 | OWR-16163-013

## Collected Problems in Operator Algebras and Quantum Information

O / open | Oberwolfach 报告问题汇编

Resolve the following problems.

1. Find deterministic unitaries $U_1,\ldots,U_k\in M_n(\mathbb C)$ and constants $C>0$ and $k>\exp(2C^2)$ such that

$$\left\lVert\sum_{i\ne j}a_{ij}U_iU_j^*\right\rVert\leq C\left(\sum_{i\ne j}\lvert a_{ij}\rvert^2\right)^{1/2}$$

for every coefficient matrix $(a_{ij})$.

2. For a compact-quantum-group algebra $A$, is the property that every $\pi$-$\varepsilon$ cocycle extends to a Schürmann triple equivalent to $H^2(A,{}_{\varepsilon}\mathbb C_{\varepsilon})=0$?

3. Extend the Haagerup reduction theorem to type III Orlicz spaces.

4. For positive states $\varphi,\psi$, set $\delta(\varphi,\psi)=H((\varphi+\psi)/2)-(H(\varphi)+H(\psi))/2$. Is

$$\lvert\varphi-\psi\rvert(A)\leq\left(8\delta(\varphi,\psi)(\varphi+\psi)(A^2)\right)^{1/2}$$

for every positive operator $A$? For every completely positive trace-preserving map $T$ and density matrix $\varphi$, is

$$1-\lVert T(\varphi^{1/2})\rVert_2^2\leq4\bigl(H(T\varphi)-H(\varphi)\bigr)?$$

5. In $\operatorname{VN}(F_2)$, is an automorphism able to map the generator masa $\{a\}''$ to the radial masa $\{a+a^{-1}+b+b^{-1}\}''$? Does a masa free from the radial masa exist?

来源字段：https://doi.org/10.4171/owr/2018/22

10475 | OWR-10252938-003

## Combinatorial Models of Discrete Vector Bundle Geometry

P / partially_solved | Oberwolfach 报告问题汇编

What might a combinatorial model of discrete vector bundles with connection and curvature look like?

来源字段：https://doi.org/10.4171/owr/2022/29

10476 | OWR-13357-002

## Commutants and Factor Types of Brownian Energy Representations

P / partially_solved | Oberwolfach 报告问题汇编

For the left and right Brownian energy representations, determine whether the von Neumann algebras they generate are mutual commutants; if so, determine the type of the resulting factor.

来源字段：https://doi.org/10.4171/owr/2014/55

10477 | OWR-1787-004

## Commutants of Volterra Operators on $L^p$ Spaces

P / partially_solved | Oberwolfach 报告问题汇编

For the Volterra operator $V$ on $L^p[0,1]$, defined by $(Vf)(x)=\int_0^xf(s)\,ds$, characterize its commutant $[V]_p$ when $p\notin\{1,2,\infty\}$. The commutant is known for $p\in\{1,2,\infty\}$ but remains open for all other values of $p$.

来源字段：https://doi.org/10.4171/owr/2008/06

10478 | OWR-1054-003

## Commutator Derivatives of Convex Functions on Carnot Groups

O / open | Oberwolfach 报告问题汇编

For every $H$-convex function $u$ on an arbitrary Carnot group, are the commutator derivatives $[X_i,X_j]u$ Radon measures?

来源字段：https://doi.org/10.4171/owr/2005/33

10479 | AIM-SEVERAL_COMPLEX_VARIABLES-0008

## Compactness versus Stein neighborhoods: a two-axis criterion and a converse counterexample

P / partially_solved | AIM 研讨会问题汇编

Describe the relation between compactness of the \(\overline\partial\)-Neumann operator and existence of a Stein neighborhood basis. In particular, does the geometric sufficient condition for compactness in Straube (*Ann. Inst. Fourier* 54, 2004, MR2097419, on geometric vector-field conditions implying compactness of the $\bar{\partial}$-Neumann operator) imply existence of a Stein neighborhood basis?

来源字段：http://aimpl.org/crscv/2/

10480 | AIM-FUNCTIONAL_ANALYSIS-0005

## Comparable truncators and a radial reduction for the localized Santalo conjecture

P / partially_solved | AIM 研讨会问题汇编

**Conjecture due to D. Cordero-Erausquin.** If \(K\) and \(L\) symmetric convex bodies in \(\mathbb R^n\), then

\[ |K\cap L|\,|K^\circ\cap L| \le |B_2^n\cap L|\,|B_2^n\cap L|, \]

where \[ K^\circ=\{y\in\mathbb R^n:\ \langle x,y\rangle\le1 \text{ for every }x\in K\} \] is the polar body of \(K\).

来源字段：http://aimpl.org/symconvgeomineq/1/

10481 | OWR-14216-001

### Comparative Accuracy of Spline and Finite-Element Bases

O / open | Oberwolfach 报告问题汇编

For numerical approximation, how accurate and robust are maximally continuous splines, minimally continuous splines, divergence-conforming splines, and curl-conforming splines; how do splines compare with classical finite elements; how do the answers depend on dimension; which polynomial and Fourier bases are best for multidimensional approximation; and when can those preferences be quantified?

来源字段：https://doi.org/10.4171/owr/2016/8

10482 | OWR-13678-003

### Complementarity Form of the Schumaker Strang Conjecture

O / open | Oberwolfach 报告问题汇编

For every $q\in\mathbb{R}^n$, does there exist $x\in\mathbb{R}^n$ such that $x\geq 0$, $Mx+q\geq 0$, and $\langle Mx+q,x\rangle=0$?

来源字段：https://doi.org/10.4171/owr/2015/21

10483 | OWR-4211-006

### Complete Radiation Conditions for Curved and Elastic Boundaries

O / open | Oberwolfach 报告问题汇编

Can the complete radiation boundary condition method be adapted to CRBC corner treatments, extended from corners to smooth boundaries, and applied to more complicated hyperbolic systems such as elasticity, with a rigorous analysis?

来源字段：https://doi.org/10.4171/owr/2010/10

10484 | AMR-036-0014

### Composite periodic entire functions

O / open | AMR 问题清单汇编

Classify entire functions $f,g$ for which $f\circ g$ is periodic. Prove that, up to the natural equivalences, the possibilities are exhausted by: $g$ periodic; $g(z+T)=g(z)+K$ with $f$ $K$-periodic; $g$ quadratic; or $g=P\circ h$ with $P$ quadratic, $h(z+T)=h(z)+K$, and $f\circ P$ $K$-periodic.

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10485 | OWR-13750332-003

### Compton Transform Inversion with Spatially Unresolved Detectors

O / open | Oberwolfach 报告问题汇编

Can the three-stage Compton transform be inverted when the final detector records energy but not the spatial location of the detected photon?

来源字段：https://doi.org/10.4171/owr/2023/21

10486 | OWR-13486-008

### Concise Inverse Spectral Theory for Indefinite Strings

P / partially_solved | Oberwolfach 报告问题汇编

Find an equally concise analogue of the solution of the inverse spectral problem for indefinite strings.

来源字段：https://doi.org/10.4171/owr/2015/1

10487 | AIM-ANALYSIS-0127

### Conditional cell-mass concentration and a radial no-go theorem for the standard Sierpinski carpet

P / partially_solved | AIM 研讨会问题汇编

Is the usual \(\tfrac13\) Sierpiński carpet \(S_3\) quasisymmetrically equivalent to a Loewner space?

来源字段：http://aimpl.org/mappingmetric/5/

10488 | AIM-ANALYSIS-0159

### Conditional-Poisson covariance proxy and Jensen-gap reduction for Ginibre permanental roots

P / partially_solved | AIM 研讨会问题汇编

**Question 2: Yan Fyodorov** What is the mean density of permanental polynomials? This is unknown for random matrices of size greater than 5 × 5.

来源字段：https://aimath.org/WWN/randomzeros/arcc1.pdf

10489 | AIM-SEVERAL_COMPLEX_VARIABLES-0009

### Confinement of compact-disc candidates and a flat local survivor in C^3

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** Is there a smooth bounded pseudoconvex domain \(\Omega\subset\mathbb C^n\) (\(n\geq3\)) whose boundary contains a nonconstant analytic disc and for which the \(\bar\partial\)-Neumann operator \(N_1\) on \((0,1)\)-forms is compact? For \(n=2\) this is impossible.

来源字段：http://aimpl.org/crscv/2/

10490 | AIM-ANALYSIS-0150

### Confluent Slater quotients and the first kurtosis obstruction

P / partially_solved | AIM 研讨会问题汇编

Let $p_k(z)$ be the polynomials orthogonal with respect to $\Phi$ on $(-\infty,\infty)$. Define

\[ P_n(z):=\frac{\det[p_i(z_j)]_{i,j=1}^n}{\det[z_i^{j-1}]_{i,j=1}^n},\qquad Q_n(z):=\frac{\det[p_i(z_j)]_{i,j=2}^n}{\det[z_i^{j-2}]_{i,j=2}^n}. \] Investigate the zeros of $P_n$ and $Q_n$.

来源字段：http://aimpl.org/hyperbolicpoly/7/

10491 | OWR-13486-005

### Congruence from Complete Robin and Dirichlet Spectra

O / open | Oberwolfach 报告问题汇编

Let $\Omega_1$ and $\Omega_2$ be Lipschitz domains whose Robin Laplacians have the same eigenvalue sequence for every $\beta \geq 0$ and for Dirichlet boundary conditions. Must $\Omega_1$ and $\Omega_2$ be congruent?

来源字段：https://doi.org/10.4171/owr/2015/1

10492 | AMR-035-0002

### Conjectural Large Genus Asymptotics of Area Siegel-Veech Constants

P / partially_solved | AMR 问题清单汇编

Let $\boldsymbol{d}=(d_1,\ldots,d_n)$ be an unordered partition of $4g-4$ with $d_i\in\{-1,0,1,2,\ldots\}$, and let $\widehat\Pi_{4g-4}$ be the set of such partitions having at most $\log g$ entries equal to $-1$. For non-hyperelliptic components $\mathcal{Q}$ of all strata $\mathcal{Q}(\boldsymbol{d})$ of meromorphic quadratic differentials with at most simple poles, where $\boldsymbol{d}\in\widehat\Pi_{4g-4}$ and $g\geq6$, $$c_{\mathrm{area}}(\mathcal{Q})=\frac14\bigl(1+\varepsilon_2(\boldsymbol{d})\bigr),$$ where $$\lim_{g\to\infty}\max_{\boldsymbol{d}\in\widehat\Pi_{4g-4}}|\varepsilon_2(\boldsymbol{d})|=0.$$

背景链接：https://arxiv.org/abs/1912.11702

10493 | AMR-035-0001

### Conjectural Large Genus Asymptotics of Masur-Veech Volumes

P / partially_solved | AMR 问题清单汇编

Let $\boldsymbol{d}=(d_1,\ldots,d_n)$ be an unordered partition of $4g-4$ with $d_i\in\{-1,0,1,2,\ldots\}$, and let $\widehat\Pi_{4g-4}$ be the set of such partitions having at most $\log g$ entries equal to $-1$. For every $\boldsymbol{d}\in\widehat\Pi_{4g-4}$, $$\operatorname{Vol}\mathcal{Q}(d_1,\ldots,d_n)=\frac{4}{\pi}\prod_{i=1}^n\frac{2^{d_i+2}}{d_i+2}\bigl(1+\varepsilon_1(\boldsymbol{d})\bigr),$$ where $$\lim_{g\to\infty}\max_{\boldsymbol{d}\in\widehat\Pi_{4g-4}}|\varepsilon_1(\boldsymbol{d})|=0.$$

背景链接：https://arxiv.org/abs/1912.11702

10494 | AIM-FUNCTIONAL_ANALYSIS-0046

### Conjugacy, inner cosets, and a turbulence phase diagram

P / partially_solved | AIM 研讨会问题汇编

For a separable C*-algebra $A$, what is the complexity of orbit equivalence relations associated to the action $\mathrm{Aut}\left( A\right) $ on itself by conjugacy and to the action of $\mathrm{Inn}\left( A\right) $ on $\mathrm{\mathrm{Au}t}\left( A\right) $ by left translation? Are these action turbulent? How are they related to structural properties of $A$?

来源字段：http://aimpl.org/settheorycstar/7/

10495 | AMR-036-0045

### Conjugate one-points in the unit disk

P / partially_solved | AMR 问题清单汇编

Let $f$ be holomorphic in the unit disk with a simple zero at $0$, exactly two simple $1$-points at $a$ and $\overline a$, and no other zeros or $1$-points. Determine the minimum possible $|a|$ and the extremal function; in particular, is the minimum the conjectured covering constant $c\approx0.0505468$?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10496 | AMR-036-0034

### Connectedness of extremal Green level sets

O / open | AMR 问题清单汇编

If the definition of $\sup_E\beta_E$ is extended from connected regular compact plane sets to all regular compact sets, is the supremum attained on connected sets?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10497 | OWR-717-003

### Connectivity Detection by Planar-Domain Functions

O / open | Oberwolfach 报告问题汇编

For the monotone domain functions constructed from Green and kernel functions of planar domains, which expressions require the compared domains to be homotopic, and can these functions detect connectivity?

来源字段：https://doi.org/10.4171/owr/2004/07

10498 | AIM-SEVERAL_COMPLEX_VARIABLES-0024

### Constant-degree denominator removal and stable polynomial endpoints

P / partially_solved | AIM 研讨会问题汇编

Let $f:\mathbb B^n\to\mathbb B^N$ be a proper rational map. Is $f$ homotopic to a polynomial map $p$ of the same degree? That is, for every $t\in[0,1]$, each $f_t:\mathbb B^n\to\mathbb B^N$ is proper, with $f_0=f$ and $f_1=p$.

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10499 | AIM-ANALYSIS-0148

### Constraint pivots, canonical orthogonal bases, and a dual-lacunary Padé equivalence

P / partially_solved | AIM 研讨会问题汇编

Consider a subspace $S$ of $\mathbb C[x]$ of codimension $k$ (defined by $k$ linear constraints, explicitly given). Given a scalar product on $\mathbb C[x]$, find an orthogonal basis of $S$ (ordered by degree); some degrees will be missing. 1. Which degrees get skipped? 2. Characterize these orthogonal polynomials. 3. Does this have anything to do with lacunary Padé approximants?

来源字段：http://aimpl.org/hyperbolicpoly/6/

10500 | OWR-12595-002

### Constraint-Preserving Methods for Maxwell-Coupled Systems

P / partially_solved | Oberwolfach 报告问题汇编

Can high-resolution numerical methodology for involution-constrained systems be applied to equations coupled with Maxwell's equations, such as elastodynamics?

来源字段：https://doi.org/10.4171/owr/2013/47

10501 | OWR-14298585-005

### Construction of p-Energies without Classical Gradients

P / partially_solved | Oberwolfach 报告问题汇编

How can one construct a canonical $p$-energy on a space that has no classical gradient? Determine hypotheses under which graph-limit, upper-gradient, and Korevaar-Schoen constructions exist and yield the same energy.

来源字段：https://doi.org/10.4171/owr/2024/42

10502 | OWR-13859-007

### Constructive Algorithms for Nonlinear Sparse Approximation

O / open | Oberwolfach 报告问题汇编

How can one devise effective constructive algorithms for nonlinear sparse approximation, including m-term approximation with respect to bases and redundant dictionaries?

来源字段：https://doi.org/10.4171/owr/2015/38

10503 | AIM-ANALYSIS-0088

### Constructive syndetic partitions via restricted DFT matroids and irrational rotations

P / partially_solved | AIM 研讨会问题汇编

**Recovered constructive problem.** Given a measurable set \(S\subset\mathbb T\) of positive measure, construct a finite coloring \(\mathbb Z=\bigsqcup_{j=1}^r\Lambda_j\) such that every \(\Lambda_j\) is syndetic and \(\{e^{2\pi i n x}:n\in\Lambda_j\}\) is a Riesz sequence in \(L^2(S)\). Ideally, give effective membership rules, a gap bound, and verifiable Riesz bounds.

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10504 | OWR-9790366-003

### Continuous Minimizers for a Regional Fractional-Laplacian Functional

O / open | Oberwolfach 报告问题汇编

For the Rayleigh-Faber-Krahn variational problem associated with the regional fractional Laplacian, does there exist a minimizer continuous on all of $\mathbb R^n$?

来源字段：https://doi.org/10.4171/owr/2022/15

10505 | OWR-15959-004

### Continuum Stability of Gradient Flows on Geometric Graphs

P / partially_solved | Oberwolfach 报告问题汇编

Under which hypotheses do gradient-flow evolution equations on geometric graphs converge to a continuum evolution as the number of sample points $n\to\infty$ and the connectivity scale $\varepsilon\to0$?

来源字段：https://doi.org/10.4171/owr/2017/54

10506 | OWR-1058-001

### Convergence and Optimality of Adaptive Finite Elements

P / partially_solved | Oberwolfach 报告问题汇编

For which adaptive finite-element problems and methods are convergence and error reduction guaranteed; when are adaptive algorithms optimal; is coarsening necessary; and do bulk-criterion convergence proofs extend to other marking criteria?

来源字段：https://doi.org/10.4171/owr/2005/37

10507 | OWR-7155452-001

### Convergence of Far-Field Refractor Schemes

O / open | Oberwolfach 报告问题汇编

For the numerical scheme discretizing the general Monge-Ampère equation in the far-field refractor setting, do the solutions of the scheme converge?

来源字段：https://doi.org/10.4171/owr/2021/31

10508 | OWR-2654833-004

### Convergence of Self-Contact Penalty Forces

P / partially_solved | Oberwolfach 报告问题汇编

For elastic models without higher-order regularizing terms, can analogous convergence results be proved for the nonlocal self-contact penalty, and do the functional derivatives of the penalty energy converge to the forces caused by self-contact in the constrained limit problem?

来源字段：https://doi.org/10.4171/owr/2020/37

10509 | OWR-17468-002

### Convergence of Standard Cantor Prefractal Boundary Approximations

O / open | Oberwolfach 报告问题汇编

For the Cantor-set boundary-element approximation with $i(j)=j$, $\delta=0$, and $C_j$ the standard prefractal sequence for the Cantor set $C$, prove that the approximations $\phi_j^h$ converge to $\phi$.

来源字段：https://doi.org/10.4171/owr/2020/5

10510 | OWR-14299093-005

### Converse Bessel-Pair Criterion for Grushin Operators

O / open | Oberwolfach 报告问题汇编

For Grushin vector fields, prove or disprove the converse Bessel-pair criterion: if the associated abstract Hardy or Hardy-Rellich inequalities hold for a pair of weights $(V,W)$, must $(V,W)$ be a Bessel pair?

来源字段：https://doi.org/10.4171/owr/2025/14

10511 | OWR-14298373-007

### Coulomb-Gas Asymptotics Beyond Weil-Petersson Quasicircles

P / partially_solved | Oberwolfach 报告问题汇编

What are the Coulomb-gas asymptotics when the boundary curve $\eta$ is not a Weil-Petersson quasicircle?

来源字段：https://doi.org/10.4171/owr/2024/25

10512 | AIM-ANALYSIS-0164

### Covariance, diagonal, and normalized-kernel criteria for Gaussian entire functions

P / partially_solved | AIM 研讨会问题汇编

Is there a necessary and sufficient condition for a given correlation function to be the correlation function of an enire Gaussian function?

来源字段：https://aimath.org/WWN/randomzeros/arcc1.pdf

10513 | OPG-36928

### Criterion for boundedness of power series

P / partially_solved | Open Problem Garden

Question Give a necessary and sufficient criterion for the sequence $(a_n)$ so that the power series $\sum_{n=0}^{\infty} a_n x^n$ is bounded for all $x \in \mathbb{R}$.

背景链接：http://www.openproblemgarden.org/op/criterion_for_boundedness_of_power_series

10514 | OWR-11101916-007

### Critical Cwikel Estimates for Noncompact Spectral Triples

O / open | Oberwolfach 报告问题汇编

Prove critical Cwikel-type estimates at $p=2$ for noncompact spectral triples, including estimates for $M_f(1-\Delta)^{-d/4}$.

来源字段：https://doi.org/10.4171/owr/2022/35

10515 | OWR-3389-020

### Critical Fractional and Hardy-Lieb-Thirring Inequalities

P / partially_solved | Oberwolfach 报告问题汇编

Can one prove a critical Lieb-Thirring inequality for arbitrary, including noninteger, powers of the Laplacian? For $0<\theta\leq1$, what is the sharp constant $C_\theta$ in the Hardy-Lieb-Thirring inequality on the half-line with weight $V(r)r^{1-\theta}$?

来源字段：https://doi.org/10.4171/owr/2009/06

10516 | OWR-14741-006

### Critical Sampling Density Beyond Beurling Density

O / open | Oberwolfach 报告问题汇编

For settings such as Bergman spaces and wavelet frames where Beurling density is not the correct notion, does a critical density separating sampling from interpolation exist?

来源字段：https://doi.org/10.4171/owr/2016/33

10517 | OWR-732-003

### Crystallization from Many-Particle Interaction Potentials

O / open | Oberwolfach 报告问题汇编

Under what conditions on an interaction potential do energy-minimizing configurations of many particles form crystals?

来源字段：https://doi.org/10.4171/owr/2004/30

10518 | OWR-16632-001

### Crystallization under Elastic Interaction Potentials

O / open | Oberwolfach 报告问题汇编

Can the crystallization and limiting results for the particle system be extended to more general interaction potentials that allow elastic deformations?

来源字段：https://doi.org/10.4171/owr/2018/49

10519 | OWR-13678-002

### Cubic Spline Interpolation on General Triangulations

P / partially_solved | Oberwolfach 报告问题汇编

Given a planar triangulation, is arbitrary interpolation of function values at its vertices always possible by a spline in $S_3^1$? In the corresponding quartic $C^1$ setting, is simultaneous interpolation of function and gradient values at the vertices impossible for some triangulations?

来源字段：https://doi.org/10.4171/owr/2015/21

10520 | OWR-16163-005

### Cutoff Parameters for Quantum-Group Random Walks

O / open | Oberwolfach 报告问题汇编

On the quantum permutation group $S_N^+$, let $\varphi$ define the quantum random walk by $$\varphi(u_{ij}^n)=\delta_{ij}\left(\frac{N-1}{N}\frac{T_{2n}(\sqrt{N-2})}{T_{2n}(\sqrt N)}+\frac1N\right),$$ where $T_0(X)=1$, $T_1(X)=X$, and $XT_n(X)=T_{n+1}(X)+T_{n-1}(X)$. Prove or disprove that the completely bounded norm of $P_{\varphi}=(\operatorname{id}\otimes\varphi)\circ\Delta$ gives the correct cutoff parameter.

来源字段：https://doi.org/10.4171/owr/2018/22

10521 | OWR-4531-002

### Cyclic Right Translates in Weighted $\ell^p$ Spaces

O / open | Oberwolfach 报告问题汇编

For $p\geq1$ and a positive weight $\omega$ on $\mathbb{Z}$ with bounded successive ratios, characterize the weights for which some $x\in\ell^{p}_{\omega}(\mathbb{Z})$ has complete right translates $\{S^{n}x:n\geq0\}$.

来源字段：https://doi.org/10.4171/owr/2010/49

10522 | AIM-FUNCTIONAL_ANALYSIS-0006

### Dar's conjecture: homogeneous repair and Cartesian-product closure

P / partially_solved | AIM 研讨会问题汇编

**Dar's conjecture.** Let \(\mu(K,L)\) is defined by \[ \mu(K,L)=\max_{x\in\mathbb R^n}|K\cap(x+L)|. \] Then for convex bodies \(K,L\subset\mathbb R^n\), \[ |K+L|^{1/n}\geq \mu(K,L)+ \frac{|K|^{1/n}|L|^{1/n}}{\mu(K,L)}. \tag{AIM} \]

来源字段：http://aimpl.org/symconvgeomineq/1/

10523 | AMR-036-0022

### Decay of separated subharmonic level components

P / partially_solved | AMR 问题清单汇编

Let $u$ be subharmonic on $1<|z|<2$, and let pairwise disjoint open sets $D_k$ be unions of components of $\{u<0\}$. Suppose that for every $r\in(1,2)$, $$\int_{\{\theta:re^{i\theta}\in D_k\}}u(re^{i\theta})\,d\theta\le-\delta_k.$$ Determine the possible decay rate of $\delta_k$, closing the gap between known exponentially decaying examples and the universal bound $\sum_k\delta_k^\alpha<\infty$ for some $\alpha<1/3$.

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10524 | AMR-036-0040

### Defect relation for points in $\mathbb P^2$

O / open | AMR 问题清单汇编

Let $f:\mathbb C\to\mathbb P^2$ be linearly nondegenerate and let $\delta(a,f)$ be the Nevanlinna deficiency of a point $a\in\mathbb P^2$. Prove that for every system of points in general position, $$\sum_a\delta(a,f)\le1.$$

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10525 | AIM-SEVERAL_COMPLEX_VARIABLES-0018

### Degree audit and a smooth convex spectral trichotomy

P / partially_solved | AIM 研讨会问题汇编

Is the spectrum of the $\overline{\partial}$-Neumann Laplacian on a bounded smooth pseudoconvex domain in $\mathbb{C}^n$ always discrete? Is the infimum of the spectrum always a point eigenvalue (ground state energy)?

来源字段：http://aimpl.org/crscv/6/

10526 | AIM-SEVERAL_COMPLEX_VARIABLES-0016

### Degree-by-degree spectral reduction for the Hartogs triangle and worm

P / partially_solved | AIM 研讨会问题汇编

Compute the spectrum of the $\overline{\partial}$-Neumann Laplacian on the Hartogs triangle and on the Diederich-Fornaess worm domain. In particular, determine whether or not the corresponding spectra are discrete.

来源字段：http://aimpl.org/crscv/6/

10527 | OWR-720-013

### Density and Balian-Low Theory for Gabor Schauder Bases

P / partially_solved | Oberwolfach 报告问题汇编

If a Gabor system is a Schauder basis, must its lower and upper densities both equal $1$? Is there an analogue of the Balian-Low theorem for Gabor Schauder bases?

来源字段：https://doi.org/10.4171/owr/2004/10

10528 | OWR-11130-006

### Density Conditions for Optimal-Transport Regularity

O / open | Oberwolfach 报告问题汇编

Can conditions on the source and target densities guarantee regularity of the optimal transport map when standard geometric conditions such as the MTW condition fail?

来源字段：https://doi.org/10.4171/owr/2011/38

10529 | AIM-FUNCTIONAL_ANALYSIS-0037

### Density-controlled nuclear hulls and a coherent local-to-global criterion

P / partially_solved | AIM 研讨会问题汇编

Is every exact C*-algebra a subalgebra of a nuclear C*-algebra?

The answer is positive for separable algebras since every separable exact C*-algebra is a subalgebra of $\mathcal O_2$.

来源字段：http://aimpl.org/settheorycstar/6/

10530 | OWR-1964-001

### Deployment Barriers for Time-Domain Boundary Element Methods

O / open | Oberwolfach 报告问题汇编

Why has time-domain boundary element methodology (BEM) not achieved the operational deployment expected a decade ago, and what mathematical and computational developments are needed to overcome the stability, precision, and cost barriers for time-domain integral equations?

来源字段：https://doi.org/10.4171/owr/2008/19

10531 | OWR-14299583-003

### Detecting Geometric Scales in Fractal Measurements

O / open | Oberwolfach 报告问题汇编

Given a space with different microscopic and macroscopic geometries, develop an intrinsic procedure for detecting the correct scale at which to measure volume, perimeter, and isoperimetric behavior.

来源字段：https://doi.org/10.4171/owr/2025/54

10532 | OWR-11101912-011

### Determining Improving Regions from Assouad Spectra

P / partially_solved | Oberwolfach 报告问题汇编

For a given set $E$, can the improving region $T_E$ be determined from the Assouad spectrum of $E$ and its subsets?

来源字段：https://doi.org/10.4171/owr/2022/31

10533 | AIM-ANALYSIS-0112

### Deterministic Assouad embeddings and a finite-scale stopping certificate

S / solved | AIM 研讨会问题汇编

If \((X,d)\) is doubling and \(1/2<\alpha<1\), then \((X,d^\alpha)\) admits a bi-Lipschitz embedding into \(\mathbb R^N\), where \(N\) depends on the doubling constant but is independent of \(\alpha\).

This is Problem 2.1 in the “Embeddings” section of the AIM problem list from the 2012 workshop *Mapping theory in metric spaces*.

The record writes \(\epsilon\) for the snowflake exponent. Naor--Neiman instead write the exponent as \(1-\varepsilon\), with \(0<\varepsilon<1/2\). To avoid reversing limits, this report uses \(\alpha\) for the exponent and \[ s=1-\alpha \] for the loss of exponent. Thus “\(\alpha\) near one” means “\(s\) near zero.”

The source remark correctly identifies the solution by Guy David and Marie Snipes. Their arXiv preprint is titled *A constructive proof of the Assouad embedding theorem with bounds on the dimension*; the published title is *A non-probabilistic proof of the Assouad embedding theorem with bounds on the dimension* [DS]. No reconstruction of the mathematical statement was needed.

来源字段：http://aimpl.org/mappingmetric/2/

10534 | AIM-ANALYSIS-0106

### Deterministic sharp-order Lee--Naor extension and finite gentle-partition aggregation

S / solved | AIM 研讨会问题汇编

**Deterministic proof of the Lee--Naor Lipschitz extension theorem.** The Lee--Naor theorem states that, given a doubling subset \(A\) of a metric space \(X\) and an \(L\)-Lipschitz map from \(A\) to a Banach space \(Y\), there exists a \(C(\log m)L\)-Lipschitz extension \(X\to Y\). Here \(m\) is the doubling constant of \(A\). Is there a deterministic proof of the Lee--Naor Lipschitz extension theorem giving the sharp dependence on the doubling constant?

来源字段：http://aimpl.org/mappingmetric/1/

10535 | OWR-15212-005

### Deterministic Transport Under Mutual Multiphase Domination

O / open | Oberwolfach 报告问题汇编

Let $(X,\sigma)$ and $(Y,\eta)$ be atomless vector-valued measure spaces that mutually dominate each other in the multiphase-transport order. Must there be measurable maps $T:X\to Y$ and $S:Y\to X$ such that $T_\#\sigma_i=\eta_i$ and $S_\#\eta_i=\sigma_i$ for every component $i$?

来源字段：https://doi.org/10.4171/owr/2017/7

10536 | AIM-ANALYSIS-0131

### Dilatation energies and diameter-constrained obstacles

P / partially_solved | AIM 研讨会问题汇编

**Variational problems for dilatation functionals.** Study extremal variational problems for dilatation functionals of mappings. For instance, find the extremal domain (and possibly also the extremal mappings) associated to the configuration of a doubly connected planar domain including an obstacle of prescribed diameter.

来源字段：http://aimpl.org/mappingmetric/6/

10537 | AIM-FUNCTIONAL_ANALYSIS-0012

### Dimension repair and a sharp ellipsoid stability formula

P / partially_solved | AIM 研讨会问题汇编

**Question by Makai and Martini.** It is known that if for any direction \(u\in S^{n-1}\), a parallel section function \(A_{K,u}(t)\) has maximum at \(t=0\), then body \(K\) is symmetric.

What if we pose the same question but replacing volumes by surface areas, that is \[ \widetilde A_{K,u}(t)=\left|\partial\left(K\cap\{tu+u^\perp\}\right)\right|_{n-1}, \] is body \(K\) still symmetric?

来源字段：http://aimpl.org/symconvgeomineq/2/

10538 | OWR-14299582-004

### Dimension-Free Bounds and Optimal Ornstein-Uhlenbeck Calculus Angles

P / partially_solved | Oberwolfach 报告问题汇编

If $X$ is $K$-convex, are the bounds $\sup_{d\geq1}M_{p,d,\theta}^X$ finite for every $1<p<\infty$ and every $0<\theta<\theta_p$, possibly even at $\theta=\theta_p$? If $X$ is UMD, does the Ornstein-Uhlenbeck operator $-L_X$ have a bounded $H^\infty(\Sigma_\theta)$ functional calculus for every $0<\theta<\theta_p$?

来源字段：https://doi.org/10.4171/owr/2025/53

10539 | OWR-14299582-003

### Dimension-Independent Analyticity Bounds for K-Convex Spaces

O / open | Oberwolfach 报告问题汇编

Let $X$ be a $K$-convex Banach space. For every $1<p<\infty$, do there exist an angle $\theta>0$ and a bound independent of the dimension $d$ such that $\sup_{d\geq1}M_{p,d,\theta}^X<\infty$?

来源字段：https://doi.org/10.4171/owr/2025/53

10540 | OWR-17469-003

### Dimension-Uniform Lower Bounds for Twice-Smooth Integration

O / open | Oberwolfach 报告问题汇编

For a bounded convex domain $D_d\subset\mathbb R^d$, define

$$\overline C^2(D_d)=\left\{f\in C^2(D_d):\|f\|_\infty\leq1,\ \operatorname{Lip}(f)\leq d^{-1/2},\ \operatorname{Lip}(D_\theta f)\leq d^{-1}\right\}.$$

Is there an absolute constant $C>0$, independent of $d$ and $D_d$, such that

$$e_n(\overline C^2(D_d),\operatorname{INT})\geq Cn^{-2/d}$$?

来源字段：https://doi.org/10.4171/owr/2020/6

10541 | OWR-17290-004

### Dimensional Thresholds for Pointed Sub-Laplacian Self-Adjointness

P / partially_solved | Oberwolfach 报告问题汇编

For the pointed sub-Laplacian on a 3-dimensional equiregular sub-Riemannian structure modeled locally on the Heisenberg group, determine whether the topological dimension 3, the horizontal dimension 2, or the homogeneous dimension 4 governs essential self-adjointness.

来源字段：https://doi.org/10.4171/owr/2019/47

10542 | OWR-16767-003

### Direct Proofs of Hilbert-Space Symplectic Nonsqueezing

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a direct proof of symplectic non-squeezing in Hilbert space?

来源字段：https://doi.org/10.4171/owr/2019/5

10543 | AIM-ANALYSIS-0080

### Directional hyperbolic families and the obstruction to a uniform leaf

P / partially_solved | AIM 研讨会问题汇编

Extend the theory of interlacing families to several variables

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10544 | OWR-12723-007

### Discrepancy Lower Bounds for Line Unions in Three Dimensions

O / open | Oberwolfach 报告问题汇编

For

$$A=\bigcup_{i=1}^N(\ell_i\cap[0,1]^3),\qquad \ell_i\subset\mathbb R^3,$$

and

$$D_N(A)=\sup_Q\left|\frac{H^1(A\cap Q)}{H^1(A)}-H^3(Q)\right|,$$

does there exist an absolute constant $c>0$ such that

$$D_N(A)\geq c\frac{\log H^1(A)}{H^1(A)}$$

for every such union of lines?

来源字段：https://doi.org/10.4171/owr/2013/49

10545 | AIM-ANALYSIS-0013

### Discrete Bessel gap probabilities, Riemann-Hilbert problems, and the Toda-KdV limit

S / solved | AIM 研讨会问题汇编

Express the largest-particle Fredholm determinant of the (possibly finite-temperature) discrete Bessel process through a discrete Riemann--Hilbert problem, recover cylindrical Toda from its Lax pair, and identify the Riemann--Hilbert object and equation obtained under the soft-edge Toda-to-KdV scaling.

来源字段：http://aimpl.org/riemhilberttoeplitz/2/

10546 | OWR-16163-011

### Distributions of Self-Adjoint Bimonotone Limit Operators

O / open | Oberwolfach 报告问题汇编

Give a concrete description of the probability measures that are the distributions of the self-adjoint bimonotone limit operators $\lambda_t^*+\lambda_t+\rho_t^*+\rho_t$.

来源字段：https://doi.org/10.4171/owr/2018/22

10547 | OWR-3470-003

### Dixmier Traces of Commutators on Bidisk Quotient Modules

P / partially_solved | Oberwolfach 报告问题汇编

Let $M\subset H^2(\mathbb D^2)$ be an invariant subspace generated by homogeneous polynomials, and let $S_f$ denote compression of multiplication by $f$ to the quotient module $M^\perp$. Classify the quotient modules for which all cross-commutators among $S_z,S_w,S_z^*,S_w^*$ belong to the trace class, the Macaev class, or a weak Schatten class, and compute the Dixmier traces of powers of $[S_f^*,S_f]$ whenever such powers lie in the weak trace class.

来源字段：https://doi.org/10.4171/owr/2009/20

10548 | OWR-16766-003

### Dual Recovery of Binary Tomography Solution Intersections

O / open | Oberwolfach 报告问题汇编

别名：Exact Recovery by Lagrangian-Dual Binary Tomography [OWR-16766-002]; Binary Tomography Recovery Conjecture [OWR-16766-004]

Let $A\in\mathbb R^{m\times n}$, let $x\in\{-1,1\}^n$ produce noiseless data $y=Ax$, and let $\mu^*$ minimize $$ \frac12\lVert\mu-y\rVert_2^2+\lVert A^T\mu\rVert_1. $$ With $\widehat x=\operatorname{sign}(A^T\mu^*)$, must $\widehat x$ recover $x$ whenever the binary tomography problem has a unique solution? When it has multiple solutions, does the dual solution recover the coordinates common to all of them?

来源字段：https://doi.org/10.4171/owr/2019/4

10549 | OWR-720-002

### Dual Wavelet Frames and Tight Gabor Frames

P / partially_solved | Oberwolfach 报告问题汇编

Characterize when the canonical dual of a wavelet frame has wavelet structure; extend descriptions of dual wavelet frames to subspaces; and construct an explicit tight Gabor frame with exponentially decaying generator and Fourier transform.

来源字段：https://doi.org/10.4171/owr/2004/10

10550 | OWR-12339-007

### Dual-Frame Completion of Bessel Wavelet Systems

O / open | Oberwolfach 报告问题汇编

Given two Bessel dyadic wavelet systems generated by $\psi_1$ and $\widetilde\psi_1$, can $\psi_2$ and $\widetilde\psi_2$ always be chosen so that the two two-generator systems are dual frames? Equivalently, does Han's conjectured tight-frame completion hold for every sufficiently large bound $K$?

来源字段：https://doi.org/10.4171/owr/2013/11

10551 | OWR-766-001

### Edge-Recovery Accuracy from Noisy Projection Data

P / partially_solved | Oberwolfach 报告问题汇编

What is the best attainable accuracy for recovering image edges from noisy observations of projections?

来源字段：https://doi.org/10.4171/owr/2004/52

10552 | OWR-5150-002

### Effective Toughness of Periodically Defective Materials

O / open | Oberwolfach 报告问题汇编

For a fracture material with many periodically distributed defects, rigorously determine the homogenized material and its effective toughness governing the limiting dynamic evolution.

来源字段：https://doi.org/10.4171/owr/2011/22

10553 | OWR-14298803-001

### Efficient Nonlinear Balanced-Truncation Model Reduction

O / open | Oberwolfach 报告问题汇编

Develop more efficient balanced reduced-order models for nonlinear systems and compare them systematically with other nonlinear model-reduction methods. Use scalable approximations of the associated Hamilton-Jacobi-Bellman equations to enable real-time output-feedback control. Determine whether iterative solvers and alternative basis functions can replace direct polynomial-coefficient solvers while reducing runtime and memory enough to reach substantially higher state-space dimensions.

来源字段：https://doi.org/10.4171/owr/2024/57

10554 | OWR-14226-002

### Efficient Numerical Transport between Smooth Densities

P / partially_solved | Oberwolfach 报告问题汇编

How can the density transport problem, namely finding a diffeomorphism that transports one smooth density to another, be solved numerically in an efficient way?

来源字段：https://doi.org/10.4171/owr/2016/18

10555 | OWR-9790364-002

### Elementary Proofs of Laguerre-Pólya Total Positivity

O / open | Oberwolfach 报告问题汇编

Can the complex-analytic proof of the total-positivity characterization of the Laguerre-Pólya class be replaced by an elementary argument?

来源字段：https://doi.org/10.4171/owr/2022/13

10556 | AIM-FUNCTIONAL_ANALYSIS-0042

### Embedding-certificate and range-map audit for Borel O2 selection

S / solved | AIM 研讨会问题汇编

Is there a constructive Borel proof of the $\mathcal O_2$ embedding theorem?

Is there a Borel map $F:\Gamma_{\mathrm{ex}}\to\mathcal O_2^{\mathbb N}$ such that $\gamma_i\mapsto F(\gamma)_i$ extends to a *-monomorphism $A_\gamma\to\mathcal O_2$ for every $\gamma$?

来源字段：http://aimpl.org/settheorycstar/7/

10557 | AIM-OPTIMIZATION-0003

### Encoding and convexity obstructions in the stated hyperbolic-feasibility problem

O / open | AIM 研讨会问题汇编

Are all Hyperbolic Feasibility Problems polynomial-time equivalent to LMI problems?

来源字段：https://aimath.org/WWN/matrixineq/matrixineq.pdf

10558 | OWR-742-002

### Endpoint Compactness in Complex Interpolation

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Complex-Interpolation Compactness Question [OWR-742-001]

If $T:A_j\to B_j$ is compact for $j=0,1$, must $T:[A_0,A_1]_\theta\to[B_0,B_1]_\theta$ be compact for every $0<\theta<1$?

来源字段：https://doi.org/10.4171/owr/2004/40

10559 | AIM-ANALYSIS-0044

### Endpoint d-bar estimates on polydisks and bounded symmetric domains

P / partially_solved | AIM 研讨会问题汇编

Given a \(\overline{\partial}\)-closed \((0,1)\)-form \(f \in L^\infty\) on a polydisk or bounded symmetric domain, is there a \(u\) solving \(\overline{\partial}u = f\) with the estimate \(\lVert u\rVert_\infty \leq c \lVert f\rVert_\infty\)?

来源字段：http://aimpl.org/scvproblems/1/

10560 | OWR-15954-006

### Endpoint Lieb-Thirring Bounds for Infinite-Gap Jacobi Matrices

O / open | Oberwolfach 报告问题汇编

For perturbations of Jacobi matrices in infinite-gap isospectral tori, does the Lieb-Thirring inequality hold at the endpoint $p=1$? More generally, which infinite-gap spectral sets admit a Lieb-Thirring bound?

来源字段：https://doi.org/10.4171/owr/2017/49

10561 | AIM-FUNCTIONAL_ANALYSIS-0011

### Endpoint parity and root-order obstructions for intrinsic-volume section polynomials

P / partially_solved | AIM 研讨会问题汇编

Replacing volume of the sections by the surface area (or any intrinsic volume), is it true that the only bodies in even dimensions with such property are ellipsoids?

来源字段：http://aimpl.org/symconvgeomineq/2/

10562 | AIM-FUNCTIONAL_ANALYSIS-0008

### Endpoint reductions and strict one-direction local concavity at the ball

P / partially_solved | AIM 研讨会问题汇编

**Brunn-Minkowski inequality for \(L_p\) surface area.** Show that for convex bodies \(K\) and \(L\) in \(\mathbb R^n\) and for \(0\le p\le1\), \[ S_p(K+L)^{1/(n-p)}\ge S_p(K)^{1/(n-p)}+S_p(L)^{1/(n-p)}, \] where \[ S_p(K)=\int_{S^{n-1}}h_K(u)^{1-p}\,dS_K(u) \] is the \(L_p\) surface area.

来源字段：http://aimpl.org/symconvgeomineq/1/

10563 | OWR-11101912-013

### Endpoint Spectral-Cluster Improvements on Product Manifolds

O / open | Oberwolfach 报告问题汇编

For an arbitrary $d$-dimensional product manifold $M=X\times Y$, do the logarithmically improved spectral-cluster bounds with window width $\varepsilon(\lambda)=(\log\lambda)^{-1}$ extend to the endpoint $q_d=2(d+1)/(d-1)$ and to the range $2<q<q_d$?

来源字段：https://doi.org/10.4171/owr/2022/31

10564 | OWR-5150-001

### Energy Release for General Crack Geometries

O / open | Oberwolfach 报告问题汇编

Characterize the energy release rate for the very general crack geometries occurring in the Francfort-Marigo fracture model.

来源字段：https://doi.org/10.4171/owr/2011/22

10565 | AIM-ANALYSIS-0130

### Energy-jump rigidity for planar Hopf-Laplace maps

P / partially_solved | AIM 研讨会问题汇编

The exact AIM record asks about a planar Sobolev map \(f\) satisfying

\[ \partial_{\bar z}\!\left(f_z\overline{f_{\bar z}}\right)=0, \qquad f\in W^{1,2}_{\mathrm{loc}},\qquad J_f\geq 0, \]

There is an important notational defect in that text. The displayed second-order equation contains the conjugate, but the next sentence says that it reduces locally to \(f_zf_{\bar z}=1\), without a conjugate. The mathematically consistent reduction is

\[ F_w\overline{F_{\bar w}}=1. \tag{1.1} \]

Indeed, \(f_z\overline{f_{\bar z}}\in L^1_{\mathrm{loc}}\), and Weyl's lemma for \(\bar\partial\) gives

\[ f_z\overline{f_{\bar z}}=\varphi(z) \tag{1.2} \]

for a holomorphic function \(\varphi\). Near a point where \(\varphi\ne0\), choose a branch of \(\sqrt\varphi\) and the natural coordinate

\[ w=\int^z\sqrt{\varphi(\zeta)}\,d\zeta. \]

For \(F(w)=f(z(w))\), the chain rule gives

\[ F_w\overline{F_{\bar w}} =f_z\overline{f_{\bar z}}(z_w)^2 =\varphi\,\frac1\varphi=1. \]

Thus (1.1), with the conjugate retained, is the local differential inclusion relevant to the question.

There is also a scope ambiguity.
Cristina--Iwaniec--Kovalev--Onninen formulate the originating open question as: “Suppose that \(h\) is an energy-minimal deformation. Is \(\lvert Dh\rvert\) continuous?” The AIM record foregrounds only finite energy and \(J_f\ge0\). This report distinguishes:

1. the **broad Hopf-solution reading**, namely (1.2), \(f\in W^{1,2}_{\mathrm{loc}}\), and \(J_f\ge0\); and 2. the **energy-minimal-deformation reading**, which adds the global deformation/minimality conditions from the source paper.

Since \(Df\) is initially defined only almost everywhere, “continuous” is understood to mean that its Frobenius norm has a continuous representative.

来源字段：http://aimpl.org/mappingmetric/6/

10566 | OWR-4337-004

### Energy-Slope Bounds for Thin-Film Coarsening

O / open | Oberwolfach 报告问题汇编

For thin liquid-film evolution with superlinear mobility, can one estimate the steepness of the energy along solution paths well enough to obtain optimal rigorous upper bounds on the coarsening rate?

来源字段：https://doi.org/10.4171/owr/2010/24

10567 | AMR-036-0003

### Entire solutions of higher-order Briot-Bouquet equations

S / solved | AMR 问题清单汇编

Classify the entire solutions of $F(y^{(k)},y)=0$ when $F$ is irreducible and its highest-degree homogeneous part has a single distinct linear factor, equivalently equations of the remaining form $$(y^{(k)}-ay)^d+Q_{d-1}(y^{(k)},y)=0,\qquad \deg Q_{d-1}\le d-1.$$ Are all such entire solutions exponential polynomials?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10568 | OWR-742-005

### Entropy Numbers of Vector-Valued Sequence Operators

P / partially_solved | Oberwolfach 报告问题汇编

Let $T : X \to Y$ be a bounded linear operator between Banach spaces, and let $\widetilde{T} : \ell_p^N(X) \to \ell_q^N(Y)$ be its coordinatewise lift $(x_1, \ldots, x_N) \mapsto (Tx_1, \ldots, Tx_N)$. How do the entropy numbers $e_k(\widetilde{T})$ of $\widetilde{T} : \ell_p^N(X) \to \ell_q^N(Y)$ relate quantitatively to the entropy numbers $e_k(T)$ of $T : X \to Y$?

来源字段：https://doi.org/10.4171/owr/2004/40

10569 | AIM-ANALYSIS-0086

### Equal-measure sets can have different exponential-frame partition data

P / partially_solved | AIM 研讨会问题汇编

Whether the number of partitions depends on \(|E|\) for exponential frames or any other properties of \(E\)? Are there any examples of two sets with same measure such that the number of partitions is different (or the constants are different)?

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10570 | AIM-SEVERAL_COMPLEX_VARIABLES-0042

### Equality rigidity and low-rank structure for roots of polynomial squared norms

P / partially_solved | AIM 研讨会问题汇编

**The radical of the set of squared norms of polynomial mappings** (proposed by John D'Angelo and Dror Varolin). Let $\mathcal P$ denote the set of Hermitian symmetric polynomials $R(z,\bar w)$, and let $\mathcal S\subset\mathcal P$ denote the set of squared norms of holomorphic polynomial mappings. For $R\in\mathcal P$, give a necessary and sufficient condition for the existence of a positive integer $N$ such that $R^N\in\mathcal S$. In other words, when do there exist $N$ and holomorphic polynomials $p_j$ such that \[ R(z,\bar z)^N=\sum_{j=1}^k |p_j(z)|^2? \tag{1.1} \] It is known that $R^N\in\mathcal S$ implies \[ |R(z,\bar w)|^2\le R(z,\bar z)R(w,\bar w). \tag{1.2} \] This inequality can hold without $R$ being a squared norm. Also, $R\in\mathcal S$ if and only if every matrix $[R(z_i,\bar z_j)]_{i,j=1}^k$ is nonnegative semidefinite. Equivalently, under hereditary functional calculus, $R([A],[A^*])\succeq0$ for all finite commuting square-matrix tuples $[A]=(A_1,\ldots,A_n)$.

来源字段：https://aimath.org/WWN/complexitycrmap/complexitycrmap.pdf

10571 | AMR-036-0023

### Equilibrium for infinitely many positive masses

P / partially_solved | AMR 问题清单汇编

Let positive masses $a_k$ be placed at a discrete set $x_k\in\mathbb R^n$ and suppose $\sum_k a_k/|x_k|^{n-1}<\infty$, so that $$F(x)=\sum_k\frac{a_k(x_k-x)}{|x-x_k|^n}$$ converges. Must $F$ vanish somewhere?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10572 | AMR-099-0003

### Equilibrium point configurations on the line

P / partially_solved | AMR 问题清单汇编

Let $(a_n)_{n\in\mathbb{Z}}$ be a locally finite configuration of points on $\mathbb{R}$. For the force law $F(x,y)=|x-y|^{-2}$, call the configuration an equilibrium if $$\sum_{i\ne n}|a_i-a_n|^{-2}\operatorname{sgn}(a_i-a_n)=0$$ for every $n$. Must there exist $\alpha,\beta\in\mathbb{R}$ such that $a_n=\alpha n+\beta$ for all $n$?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/equilibrium.pdf

10573 | OWR-14298168-001

### Equilibrium Solutions in Nonlinear Elastostatics

P / partially_solved | Oberwolfach 报告问题汇编

In nonlinear elastostatics, under what conditions do equilibrium solutions exist as minimizers of the elastic energy $\int_\Omega W(x,Du(x))\,dx$ for deformations $u:\Omega\to\mathbb{R}^n$?

来源字段：https://doi.org/10.4171/owr/2024/1

10574 | AIM-SEVERAL_COMPLEX_VARIABLES-0037

### Equivariant hyperquadric embeddings and a standard-parabolic obstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 1 (M. Kolář).** Assume \(M\subset \mathbb C^2\) is a real analytic hypersurface that admits a nonlinearizable automorphism \(H\). Does there exist a hyperquadric \(Q\subset \mathbb C^n\) such that \(M\hookrightarrow Q\) nontrivially and \(H\) is induced by an automorphism of \(Q\)? Can a bound on \(n\) be found?

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10575 | EP-1028

### Erdős Problem #1028: Graph Theory, Discrepancy

S / solved | Erdős 问题

Let\[H(n)=\min_f \max_{X\subseteq \{1,\ldots,n\}} \left\lvert \sum_{x\neq y\in X} f(x,y)\right\rvert,\]where $f$ ranges over all functions $f:X^2\to \{-1,1\}$. Estimate $H(n)$.

来源字段：https://www.erdosproblems.com/1028

10576 | EP-1041

### Erdős Problem #1041: with $\lvert z_i\rvert < 1$ for all $i$

O / open | Erdős 问题

Let $f(z)=\prod_{i=1}^n(z-z_i)\in \mathbb{C}[z]$ with $\lvert z_i\rvert < 1$ for all $i$.

Must there always exist a path of length less than $2$ in\[\{z: \lvert f(z)\rvert < 1\}\]which connects two of the roots of $f$?

来源字段：https://www.erdosproblems.com/1041

10577 | EP-1042

### Erdős Problem #1042:

S / solved | Erdős 问题

Let $F\subset\mathbb{C}$ be a closed set of transfinite diameter $1$ which is not contained in any closed disc of radius $1$.

If $f(z)=\prod_{i=1}^n(z-z_i)\in\mathbb{C}[x]$ with all $z_i\in F$ then can\[\{ z: \lvert f(z)\rvert < 1\}\]have $n$ connected components?

If the transfinite diameter of $F$ is $<1$ then must this set only have at most $(1-c)n$ connected components, where $c>0$ depends only on $F$ (or just the transfinite diameter of $F$)?

来源字段：https://www.erdosproblems.com/1042

10578 | EP-1043

### Erdős Problem #1043: Analysis

S / solved | Erdős 问题

Let $f\in \mathbb{C}[x]$ be a monic non-constant polynomial. Must there exist a straight line $\ell$ such that the projection of\[\{ z: \lvert f(z)\rvert\leq 1\}\]onto $\ell$ has measure at most $2$?

来源字段：https://www.erdosproblems.com/1043

10579 | EP-1044

### Erdős Problem #1044: Analysis

S / solved | Erdős 问题

Let $f(z)=\prod_{i=1}^n(z-z_i)\in\mathbb{C}[x]$ where $\lvert z_i\rvert\leq 1$ for all $i$. If $\Lambda(f)$ is the maximum of the lengths of the boundaries of the connected components of\[\{ z: \lvert f(z)\rvert<1\}\]then determine the infimum of $\Lambda(f)$.

来源字段：https://www.erdosproblems.com/1044

10580 | EP-1045

### Erdős Problem #1045: Analysis

O / open | Erdős 问题

Let $z_1,\ldots,z_n\in \mathbb{C}$ with $\lvert z_i-z_j\rvert\leq 2$ for all $i,j$, and\[\Delta(z_1,\ldots,z_n)=\prod_{i\neq j}\lvert z_i-z_j\rvert.\]What is the maximum possible value of $\Delta$? Is it maximised by taking the $z_i$ to be the vertices of a regular polygon?

来源字段：https://www.erdosproblems.com/1045

10581 | EP-1046

### Erdős Problem #1046: \]If $E$ is connected then is $E$ contained in a disc of radius $2$?

S / solved | Erdős 问题

Let $f\in \mathbb{C}[x]$ be a monic polynomial and\[E=\{ z: \lvert f(z)\rvert <1\}.\]If $E$ is connected then is $E$ contained in a disc of radius $2$?

来源字段：https://www.erdosproblems.com/1046

10582 | EP-1047

### Erdős Problem #1047: Analysis

S / solved | Erdős 问题

Let $f\in \mathbb{C}[x]$ be a monic polynomial with $m$ distinct roots, and let $c>0$ be a constant small enough such that $\{ z: \lvert f(z)\rvert\leq c\}$ has $m$ distinct connected components.

Must all these components be convex?

来源字段：https://www.erdosproblems.com/1047

10583 | EP-1048

### Erdős Problem #1048: Analysis

S / solved | Erdős 问题

If $f\in \mathbb{C}[x]$ is a monic polynomial with all roots satisfying $\lvert z\rvert \leq r$ for some $r<2$, then must\[\{ z: \lvert f(z)\rvert <1\}\]have a connected component with diameter $>2-r$?

来源字段：https://www.erdosproblems.com/1048

10584 | EP-1114

### Erdős Problem #1114:

S / solved | Erdős 问题

Let $f(x)\in \mathbb{R}[x]$ be a polynomial of degree $n$ whose roots $\{a_0<\cdots<a_n\}$ are all real and form an arithmetic progression.

The differences between consecutive zeros of $f'(x)$, beginning from the midpoint of $(a_0,a_m)$ towards the endpoints, are monotonically increasing.

来源字段：https://www.erdosproblems.com/1114

10585 | EP-1115

### Erdős Problem #1115: Analysis

S / solved | Erdős 问题

Let $f(z)$ be an entire function of finite order, and let $\Gamma$ be a rectifiable path on which $f(z)\to \infty$. Let $\ell(r)$ be the length of $\Gamma$ in the disc $\lvert z\rvert<r$.

Find a path for which $\ell(r)$ grows as slowly as possible, and estimate $\ell(r)$ in terms of $M(r)=\max_{\lvert z\rvert=r}\lvert f(z)\rvert$.

In particular, can such a path $\Gamma$ be found for which $\ell(r)\ll r$?

来源字段：https://www.erdosproblems.com/1115

10586 | EP-1116

### Erdős Problem #1116: Analysis

S / solved | Erdős 问题

For a meromorphic function $f$ let $n(r,a)$ count the number of roots of $f(z)=a$ in the disc $\lvert z\rvert <r$. Does there exist a meromorphic (or entire) $f$ such that for every $a\neq b$\[\limsup_{r\to \infty}\frac{n(r,a)}{n(r,b)}=\infty?\]

来源字段：https://www.erdosproblems.com/1116

10587 | EP-1118

### Erdős Problem #1118: for some $c$, the set $E(c)=\{ z: \lvert f(z)\rvert >c\}$ has finite measure

S / solved | Erdős 问题

Let $f(z)$ be a non-constant entire function such that, for some $c$, the set $E(c)=\{ z: \lvert f(z)\rvert >c\}$ has finite measure.

What is the minimum growth rate of $f(z)$?

If $E(c)$ has finite measure then must there exist $c'<c$ such that $E(c')$ has finite measure?

来源字段：https://www.erdosproblems.com/1118

10588 | EP-1119

### Erdős Problem #1119: Analysis, Set Theory

S / solved | Erdős 问题

Let $\mathfrak{m}$ be an infinite cardinal with $\aleph_0<\mathfrak{m}<\mathfrak{c}=2^{\aleph_0}$. Let $\{f_\alpha\}$ be a family of entire functions such that, for every $z_0\in \mathbb{C}$, there are at most $\mathfrak{m}$ distinct values of $f_\alpha(z_0)$. Must $\{f_\alpha\}$ have cardinality at most $\mathfrak{m}$?

来源字段：https://www.erdosproblems.com/1119

10589 | EP-1125

### Erdős Problem #1125: Must $f$ be monotonic?

S / solved | Erdős 问题

Let $f:\mathbb{R}\to \mathbb{R}$ be such that\[2f(x) \leq f(x+h)+f(x+2h)\]for every $x\in \mathbb{R}$ and $h>0$. Must $f$ be monotonic?

来源字段：https://www.erdosproblems.com/1125

10590 | EP-1126

### Erdős Problem #1126: Analysis

S / solved | Erdős 问题

If\[f(x+y)=f(x)+f(y)\]for almost all $x,y\in \mathbb{R}$ then there exists a function $g$ such that\[g(x+y)=g(x)+g(y)\]for all $x,y\in\mathbb{R}$ such that $f(x)=g(x)$ for almost all $x$.

来源字段：https://www.erdosproblems.com/1126

10591 | EP-1150

### Erdős Problem #1150: Analysis, Polynomials

O / open | Erdős 问题

Does there exist a constant $c>0$ such that, for all large $n$ and all polynomials $P$ of degree $n$ with coefficients $\pm 1$,\[\max_{\lvert z\rvert=1}\lvert P(z)\rvert > (1+c)\sqrt{n}?\]

来源字段：https://www.erdosproblems.com/1150

10592 | EP-1151

### Erdős Problem #1151: Analysis, Polynomials

O / open | Erdős 问题

Given $a_1,\ldots,a_n\in [-1,1]$ let\[\mathcal{L}^nf(x) = \sum_{1\leq i\leq n}f(a_i)\ell_i(x)\]be the unique polynomial of degree $n-1$ which agrees with $f$ on $a_i$ for $1\leq i\leq n$ (that is, the Lagrange interpolation polynomial).

Let $a_i$ be the set of Chebyshev nodes. Prove that, for any closed $A\subseteq [-1,1]$, there exists a continuous function $f$ such that $A$ is the set of limit points of $\mathcal{L}^nf(x)$.

来源字段：https://www.erdosproblems.com/1151

10593 | EP-1152

### Erdős Problem #1152: Analysis, Polynomials

O / open | Erdős 问题

For $n\geq 1$ fix some sequence of $n$ distinct numbers $x_{1n},\ldots,x_{nn}\in [-1,1]$. Let $\epsilon=\epsilon(n)\to 0$.

Does there always exist a continuous function $f:[-1,1]\to \mathbb{R}$ such that if $p_n$ is a sequence of polynomials, with degrees $\deg p_n<(1+\epsilon(n))n$, such that $p_n(x_{kn})=f(x_{kn})$ for all $1\leq k\leq n$, then $p_n(x)\not\to f(x)$ for almost all $x\in [-1,1]$?

来源字段：https://www.erdosproblems.com/1152

10594 | EP-1153

### Erdős Problem #1153: Analysis, Polynomials

S / solved | Erdős 问题

For $x_1,\ldots,x_n\in [-1,1]$ let\[l_k(x)=\frac{\prod_{i\neq k}(x-x_i)}{\prod_{i\neq k}(x_k-x_i)},\]which are such that $l_k(x_k)=1$ and $l_k(x_i)=0$ for $i\neq k$.

Let\[\lambda(x)=\sum_k \lvert l_k(x)\rvert.\]Is it true that, for any fixed $-1\leq a< b\leq 1$,\[\max_{x\in [a,b]}\lambda(x)> \left(\frac{2}{\pi}-o(1)\right)\log n?\]

来源字段：https://www.erdosproblems.com/1153

10595 | EP-1154

### Erdős Problem #1154: Analysis

P / partially_solved | Erdős 问题

Does there exist, for every $\alpha \in [0,1]$, a ring or field in $\mathbb{R}$ with Hausdorff dimension $\alpha$?

来源字段：https://www.erdosproblems.com/1154

10596 | EP-115

### Erdős Problem #115: Polynomials, Analysis

S / solved | Erdős 问题

If $p(z)$ is a polynomial of degree $n$ such that $\{z: \lvert p(z)\rvert\leq 1\}$ is connected then is it true that\[\max_{\substack{z\in\mathbb{C}\\ \lvert p(z)\rvert\leq 1}} \lvert p'(z)\rvert \leq (\tfrac{1}{2}+o(1))n^2?\]

来源字段：https://www.erdosproblems.com/115

10597 | EP-116

### Erdős Problem #116: Polynomials, Analysis

S / solved | Erdős 问题

Let $p(z)=\prod_{i=1}^n (z-z_i)$ for $\lvert z_i\rvert \leq 1$. Is it true that\[\lvert\{ z: \lvert p(z)\rvert <1\}\rvert>n^{-O(1)}\](or perhaps even $>(\log n)^{-O(1)}$)?

来源字段：https://www.erdosproblems.com/116

10598 | EP-1197

### Erdős Problem #1197: Analysis

S / solved | Erdős 问题

Let $E\subset (0,\infty)$ be a set of positive measure. Is it true that, for almost all $x>0$, for all sufficiently large (depending on $x$) integers $n$ there exists an integer $r\geq 1$ such that $nx\in r\cdot E$?

来源字段：https://www.erdosproblems.com/1197

10599 | EP-1215

### Erdős Problem #1215: Analysis

S / solved | Erdős 问题

Does there exist a constant $C$ such that for every polynomial $P$ with $P(0)=1$, all of whose roots are on the unit circle, there exists a path in\[\{ z: \lvert P(z)\rvert < 1\}\]which connects $0$ to the unit circle of length at most $C$?

来源字段：https://www.erdosproblems.com/1215

10600 | EP-1221

### Erdős Problem #1221: Analysis, Distribution

O / open | Erdős 问题

Let $a=(a_1,\ldots,a_n)$ be points, ordered consecutively, around the circle (normalised to have circumference $1$). Let\[M_r(a)=\max_r \lvert a_{i+r}-a_i\rvert\]and\[m_r(a)=\min_r \lvert a_{i+r}-a_i\rvert,\]where the indices are interpreted in a cyclic fashion and the distance is around the circle (in other words, the maximum and minimum distance between $r$-consecutive points).

Let\[\Lambda_r=\inf_a \limsup_n n M_r(a_1,\ldots,a_n),\]\[\lambda_r = \sup_a \liminf_n nm_r(a_1,\ldots,a_n),\]and\[\mu_r=\inf_a \limsup_n \frac{M_r(a)}{m_r(a)},\]where the outer $\inf/\sup$ is over all infinite sequences $a=(a_1,\ldots)$.

Is it true that $r(\Lambda_r-1)$, $r(1-\lambda_r)$, and $r(\mu_r-1)$ tend to infinity as $r\to \infty$?

来源字段：https://www.erdosproblems.com/1221

10601 | EP-178

### Erdős Problem #178: Discrepancy

S / solved | Erdős 问题

Let $A_1,A_2,\ldots$ be an infinite collection of infinite sets of integers, say $A_i=\{a_{i1}<a_{i2}<\cdots\}$. Does there exist some $f:\mathbb{N}\to\{-1,1\}$ such that\[\max_{m, 1\leq i\leq d} \left\lvert \sum_{1\leq j\leq m} f(a_{ij})\right\rvert \ll_d 1\]for all $d\geq 1$?

来源字段：https://www.erdosproblems.com/178

10602 | EP-225

### Erdős Problem #225: Analysis

S / solved | Erdős 问题

Let\[ f(\theta) = \sum_{0\leq k\leq n}c_k e^{ik\theta}\]be a trigonometric polynomial all of whose roots are real, such that $\max_{\theta\in [0,2\pi]}\lvert f(\theta)\rvert=1$. Then\[\int_0^{2\pi}\lvert f(\theta)\rvert \mathrm{d}\theta \leq 4.\]

来源字段：https://www.erdosproblems.com/225

10603 | EP-226

### Erdős Problem #226: Analysis

S / solved | Erdős 问题

Is there an entire non-linear function $f$ such that, for all $x\in\mathbb{R}$, $x$ is rational if and only if $f(x)$ is?

来源字段：https://www.erdosproblems.com/226

10604 | EP-227

### Erdős Problem #227: Analysis

S / solved | Erdős 问题

Let $f=\sum_{n=0}^\infty a_nz^n$ be an entire function which is not a polynomial. Is it true that if\[\lim_{r\to \infty} \frac{\max_n\lvert a_nr^n\rvert}{\max_{\lvert z\rvert=r}\lvert f(z)\rvert}\]exists then it must be $0$?

来源字段：https://www.erdosproblems.com/227

10605 | EP-228

### Erdős Problem #228: Analysis, Polynomials

S / solved | Erdős 问题

Does there exist, for all large $n$, a polynomial $P$ of degree $n$, with coefficients $\pm 1$, such that\[\sqrt{n} \ll \lvert P(z) \rvert \ll \sqrt{n}\]for all $\lvert z\rvert =1$, with the implied constants independent of $z$ and $n$?

来源字段：https://www.erdosproblems.com/228

10606 | EP-229

### Erdős Problem #229: Analysis, Iterated Functions

S / solved | Erdős 问题

Let $(S_n)_{n\geq 1}$ be a sequence of sets of complex numbers, none of which have a finite limit point. Does there exist an entire transcendental function $f(z)$ such that, for all $n\geq 1$, there exists some $k_n\geq 0$ such that\[f^{(k_n)}(z) = 0\textrm{ for all }z\in S_n?\]

来源字段：https://www.erdosproblems.com/229

10607 | EP-230

### Erdős Problem #230: Analysis, Polynomials

S / solved | Erdős 问题

Let $P(z)=\sum_{1\leq k\leq n}a_kz^k$ for some $a_k\in \mathbb{C}$ with $\lvert a_k\rvert=1$ for $1\leq k\leq n$. Does there exist a constant $c>0$ such that, for $n\geq 2$, we have\[\max_{\lvert z\rvert=1}\lvert P(z)\rvert \geq (1+c)\sqrt{n}?\]

来源字段：https://www.erdosproblems.com/230

10608 | EP-255

### Erdős Problem #255: Discrepancy

S / solved | Erdős 问题

Let $z_1,z_2,\ldots \in [0,1]$ be an infinite sequence, and define the discrepancy\[D_N(I) = \#\{ n\leq N: z_n\in I\} - N\lvert I\rvert.\]Must there exist some interval $I\subseteq [0,1]$ such that\[\limsup_{N\to \infty}\lvert D_N(I)\rvert =\infty?\]

来源字段：https://www.erdosproblems.com/255

10609 | EP-395

### Erdős Problem #395: Analysis

S / solved | Erdős 问题

If $z_1,\ldots,z_n\in \mathbb{C}$ with $\lvert z_i\rvert=1$ then is it true that the probability that\[\lvert \epsilon_1z_1+\cdots+\epsilon_nz_n\rvert \leq \sqrt{2},\]where $\epsilon_i\in \{-1,1\}$ uniformly at random, is $\gg 1/n$?

来源字段：https://www.erdosproblems.com/395

10610 | EP-485

### Erdős Problem #485: Analysis, Polynomials

S / solved | Erdős 问题

Let $f(k)$ be the minimum number of terms in $P(x)^2$, where $P\in \mathbb{Q}[x]$ ranges over all polynomials with exactly $k$ non-zero terms. Is it true that $f(k)\to\infty$ as $k\to \infty$?

来源字段：https://www.erdosproblems.com/485

10611 | EP-494

### Erdős Problem #494: Analysis, Additive Combinatorics

S / solved | Erdős 问题

If $A\subset \mathbb{C}$ is a finite set and $k\geq 1$ then let\[A_k = \{ z_1+\cdots+z_k: z_i\in A\textrm{ distinct}\}.\]For $k>2$ does the multiset $A_k$ (together with the size of $A$) uniquely determine the set $A$?

来源字段：https://www.erdosproblems.com/494

10612 | EP-498

### Erdős Problem #498: Combinatorics, Analysis

S / solved | Erdős 问题

Let $z_1,\ldots,z_n\in\mathbb{C}$ with $1\leq \lvert z_i\rvert$ for $1\leq i\leq n$. Let $D$ be an arbitrary disc of radius $1$. Is it true that the number of sums of the shape\[\sum_{i=1}^n\epsilon_iz_i \textrm{ for }\epsilon_i\in \{-1,1\}\]which lie in $D$ is at most $\binom{n}{\lfloor n/2\rfloor}$?

来源字段：https://www.erdosproblems.com/498

10613 | EP-511

### Erdős Problem #511: Analysis

S / solved | Erdős 问题

Let $f(z)\in \mathbb{C}[z]$ be a monic polynomial of degree $n$. Is it true that, for every $c>1$, the set\[\{ z\in \mathbb{C}: \lvert f(z)\rvert< 1\}\]has at most $O_c(1)$ many connected components of diameter $>c$ (where the implied constant is in particular independent of $n$)?

来源字段：https://www.erdosproblems.com/511

10614 | EP-512

### Erdős Problem #512: Analysis

S / solved | Erdős 问题

Is it true that, if $A\subset \mathbb{Z}$ is a finite set of size $N$, then\[\int_0^1 \left\lvert \sum_{n\in A}e(n\theta)\right\rvert \mathrm{d}\theta \gg \log N,\]where $e(x)=e^{2\pi ix }$?

来源字段：https://www.erdosproblems.com/512

10615 | EP-515

### Erdős Problem #515: Analysis

S / solved | Erdős 问题

Let $f(z)$ be an entire function, not a polynomial. Does there exist a locally rectifiable path $C$ tending to infinity such that, for every $\lambda>0$, the integral\[\int_C \lvert f(z)\rvert^{-\lambda} \mathrm{d}z\]is finite?

来源字段：https://www.erdosproblems.com/515

10616 | EP-516

### Erdős Problem #516: Analysis

S / solved | Erdős 问题

Let $f(z)=\sum_{k\geq 1}a_k z^{n_k}$ be an entire function of finite order such that $\lim n_k/k=\infty$. Let $M(r)=\max_{\lvert z\rvert=r}\lvert f(z)\rvert$ and $m(r)=\min_{\lvert z\rvert=r}\lvert f(z)\rvert$. Is it true that\[\limsup\frac{\log m(r)}{\log M(r)}=1?\]

来源字段：https://www.erdosproblems.com/516

10617 | EP-519

### Erdős Problem #519: Analysis

S / solved | Erdős 问题

Let $z_1,\ldots,z_n\in \mathbb{C}$ with $z_1=1$. Must there exist an absolute constant $c>0$ such that\[\max_{1\leq k\leq n}\left\lvert \sum_{i}z_i^k\right\rvert>c?\]

来源字段：https://www.erdosproblems.com/519

10618 | EP-523

### Erdős Problem #523: where $\epsilon_k\in \{-1,1\}$ independently uniformly at random for $0\leq k\leq n$

S / solved | Erdős 问题

Let $f(z)=\sum_{0\leq k\leq n} \epsilon_k z^k$ be a random polynomial, where $\epsilon_k\in \{-1,1\}$ independently uniformly at random for $0\leq k\leq n$.

Does there exist some constant $C>0$ such that, almost surely,\[\max_{\lvert z\rvert=1}\left\lvert \sum_{k\leq n}\epsilon_k(t)z^k\right\rvert=(C+o(1))\sqrt{n\log n}?\]

来源字段：https://www.erdosproblems.com/523

10619 | EP-525

### Erdős Problem #525: Analysis, Probability

S / solved | Erdős 问题

Is it true that all except at most $o(2^n)$ many degree $n$ polynomials with $\pm 1$-valued coefficients $f(z)$ have $\lvert f(z)\rvert <1$ for some $\lvert z\rvert=1$? What is the behaviour of\[m(f)=\min_{\lvert z\rvert=1}\lvert f(z)\rvert?\]

来源字段：https://www.erdosproblems.com/525

10620 | EP-527

### Erdős Problem #527: Analysis, Probability

S / solved | Erdős 问题

Let $a_n\in \mathbb{R}$ be such that $\sum_n \lvert a_n\rvert^2=\infty$ and $\lvert a_n\rvert=o(1/\sqrt{n})$. Is it true that, for almost all $\epsilon_n=\pm 1$, there exists some $z$ with $\lvert z\rvert=1$ (depending on the choice of signs) such that\[\sum_n \epsilon_n a_n z^n\]converges?

来源字段：https://www.erdosproblems.com/527

10621 | EP-67

### Erdős Problem #67: Discrepancy

S / solved | Erdős 问题

If $f:\mathbb{N}\to \{-1,+1\}$ then is it true that for every $C>0$ there exist $d,m\geq 1$ such that\[\left\lvert \sum_{1\leq k\leq m}f(kd)\right\rvert > C?\]

来源字段：https://www.erdosproblems.com/67

10622 | EP-907

### Erdős Problem #907: Analysis

S / solved | Erdős 问题

Let $f:\mathbb{R}\to \mathbb{R}$ be such that $f(x+h)-f(x)$ is continuous for every $h>0$. Is it true that\[f=g+h\]for some continuous $g$ and additive $h$ (i.e. $h(x+y)=h(x)+h(y)$)?

来源字段：https://www.erdosproblems.com/907

10623 | EP-908

### Erdős Problem #908: Analysis

S / solved | Erdős 问题

Let $f:\mathbb{R}\to \mathbb{R}$ be such that $f(x+h)-f(x)$ is measurable for every $h>0$. Is it true that\[f=g+h+r\]where $g$ is continuous, $h$ is additive (so $h(x+y)=h(x)+h(y)$), and $r(x+h)-r(x)=0$ for every $h$ and almost all (depending on $h$) $x$?

来源字段：https://www.erdosproblems.com/908

10624 | EP-909

### Erdős Problem #909: Analysis, Topology

S / solved | Erdős 问题

Let $n\geq 2$. Is there a space $S$ of dimension $n$ such that $S^2$ also has dimension $n$?

来源字段：https://www.erdosproblems.com/909

10625 | EP-974

### Erdős Problem #974: Analysis

S / solved | Erdős 问题

Let $z_1,\ldots,z_n\in \mathbb{C}$ be a sequence such that $z_1=1$. Suppose that the sequence of\[s_k=\sum_{1\leq i\leq n}z_i^k\]contains infinitely many $(n-1)$-tuples of consecutive values of $s_k$ which are all $0$. Then (essentially)\[z_j=e(j/n),\]where $e(x)=e^{2\pi ix}$.

来源字段：https://www.erdosproblems.com/974

10626 | EP-988

### Erdős Problem #988: Discrepancy

S / solved | Erdős 问题

If $P\subseteq S^2$ is a subset of the unit sphere then define the discrepancy\[D(P) = \max_C \lvert \lvert C\cap P\rvert - \alpha_C \lvert P\rvert \rvert,\]where the maximum is taken over all spherical caps $C$, and $\alpha_C$ is the appropriately normalised measure of $C$.

Is it true that\[\min_{\lvert P\rvert=n}D(P)\to \infty\]as $n\to \infty$?

来源字段：https://www.erdosproblems.com/988

10627 | EP-989

### Erdős Problem #989: Discrepancy

S / solved | Erdős 问题

If $A=\{z_1,z_2,\ldots\}\in \mathbb{R}^2$ is an infinite sequence then let\[f(r)=\max_C \left\lvert \lvert A\cap C\rvert-\pi r^2\right\rvert,\]where the maximum is taken over all circles $C$ of radius $r$.

Is $f(r)$ unbounded for every $A$? How fast does $f(r)$ grow?

来源字段：https://www.erdosproblems.com/989

10628 | EP-991

### Erdős Problem #991: Discrepancy

S / solved | Erdős 问题

Suppose $A=\{w_1,\ldots,w_n\}\subset S^2$ maximises\[\prod_{i<j}\lvert w_i-w_j\rvert\]over all possible sets of size $n$.

Is it true that\[\max_C\lvert \lvert A\cap C\rvert - \alpha_C n\rvert =o(n),\]where the maximum is taken over all spherical caps $C$ and $\alpha_C$ is the area of $C$ (normalised so that the entire sphere has area $1$)?

来源字段：https://www.erdosproblems.com/991

10629 | EP-994

### Erdős Problem #994: Analysis, Discrepancy

S / solved | Erdős 问题

Let $E\subseteq (0,1)$ be a meaurable subset with Lebesgue measure $\lambda(E)$. Is it true that, for almost all $\alpha$,\[\lim_{n\to \infty}\frac{1}{n}\sum_{1\leq k\leq n}1_{\{k\alpha \}\in E}=\lambda(E)\]for all $E$?

来源字段：https://www.erdosproblems.com/994

10630 | EP-998

### Erdős Problem #998: Analysis, Diophantine Approximation

S / solved | Erdős 问题

Let $\alpha$ be an irrational number. Is it true that if, for all large $n$,\[\#\{ 1\leq m\leq n: \{ \alpha m\} \in [u,v)\} = n(v-u)+O(1)\]then $u=\{\alpha k\}$ and $v=\{\alpha \ell\}$ for some integers $k$ and $\ell$?

来源字段：https://www.erdosproblems.com/998

10631 | OWR-14299906-004

### Ergodic Limiting Quantum Isometry Actions

O / open | Oberwolfach 报告问题汇编

Establish ergodicity of the limiting quantum isometry action on the path space $\Sigma_A$ for matrices $A$ beyond the full-shift matrix whose entries are all one.

来源字段：https://doi.org/10.4171/owr/2026/4

10632 | OWR-12979-007

### Essential Normality for Homogeneous Varieties

P / partially_solved | Oberwolfach 报告问题汇编

For a homogeneous variety $V\subset\mathbb{C}^d$ and $M=\{f\in H_d^2:f(z)=0\text{ for all }z\in V\cap\mathbb B_d\}$, do the compressed coordinate commutators satisfy $[S_j,S_k^*]\in\mathcal S^q$ for every $q>\dim_{\mathbb C}V$?

来源字段：https://doi.org/10.4171/owr/2014/21

10633 | OWR-17290-001

### Essential Self-Adjointness on Almost-Riemannian Regular Regions

P / partially_solved | Oberwolfach 报告问题汇编

Let $R$ be the regular region of an almost-Riemannian manifold. Is the Laplace-Beltrami operator $\Delta_g$, initially defined on $C_c^\infty(R)$, essentially self-adjoint on $L^2(R,\mu_g)$? In particular, is this true for every real-analytic two-dimensional almost-Riemannian structure without tangency points?

来源字段：https://doi.org/10.4171/owr/2019/47

10634 | OWR-17290-007

### Essential Spectrum of Sign-Indefinite Transmission Laplacians

O / open | Oberwolfach 报告问题汇编

Let $L=-\nabla\cdot(h\nabla)$ on a bounded domain $\Omega$, where $h=1$ on $\Omega_+$ and $h=-\mu$ on $\Omega_-$, with interface $\Sigma$. When $\mu=1$, can the essential spectrum of $L$ contain more than one point or a nonempty interval? If it is a singleton, how do eigenvalues accumulate? Can boundary-triple methods be developed without assuming $\Omega_-\subseteq\Omega$, and for nonsmooth $\Sigma$?

来源字段：https://doi.org/10.4171/owr/2019/47

10635 | OWR-14298168-002

### Eulerian Formulation of Unidirectional Damage

O / open | Oberwolfach 报告问题汇编

Develop a well-posed analytical Eulerian formulation of unidirectional damage.

来源字段：https://doi.org/10.4171/owr/2024/1

10636 | OWR-4538-002

### Evolution-Semigroup Generators for Nonautonomous Operator Families

P / partially_solved | Oberwolfach 报告问题汇编

For a family of generators $A(t)$ on a Banach space $X$, consider $X_0=C_0(\mathbb R,X)$ and

$$(\mathcal Af)(s)=-f'(s)+A(s)f(s)$$

on an appropriate domain. Under what conditions on $A(\cdot)$, and for which domain, is $\mathcal A$ the generator of a $C_0$-semigroup?

来源字段：https://doi.org/10.4171/owr/2010/56

10637 | OWR-763-002

### Evolutionary Effects of Vaccines and Antibiotics

O / open | Oberwolfach 报告问题汇编

How do bacterial populations respond to imperfect strain-specific vaccines and to antibiotic selection pressures, and how do these responses change over time?

来源字段：https://doi.org/10.4171/owr/2004/49

10638 | OWR-14297736-008

### Exact Bases with Exponential-Polyhomogeneous Asymptotics

O / open | Oberwolfach 报告问题汇编

For generalized semiclassical ordinary differential operators $P_h=\sum_{k=0}^mC_k(x,h)\partial_x^k$, prove that $\ker P_h$ has a basis of exact solutions with the same exponential-polyhomogeneous asymptotics as the corresponding quasimodes.

来源字段：https://doi.org/10.4171/owr/2023/36

10639 | AIM-ANALYSIS-0149

### Exact degree-two Jacobi interlacing and its physical interpretation

P / partially_solved | AIM 研讨会问题汇编

Let \(P_n^{\;\alpha,\,\beta}\) denote the Jacobi polynomial of \(\deg n\) with parameters \(\alpha,\beta>-1\). Then \(P_n^{\;\alpha',\,\beta'}\) and \(P_n^{\;\alpha,\,\beta}\) interlace for \(\alpha\approx\alpha'\), and \(\beta\approx\beta'\). Does the failure of interlacing correspond to some physical law being “violated”?

来源字段：http://aimpl.org/hyperbolicpoly/6/

10640 | OWR-13678-010

### Exact Dimensions of Trivariate Spline Spaces

O / open | Oberwolfach 报告问题汇编

In the remaining cases where the upper and lower dimension bounds differ, does the dimension equal the upper bound?

来源字段：https://doi.org/10.4171/owr/2015/21

10641 | AIM-ANALYSIS-0085

### Exact finite Feichtinger partition count and a sharp ceiling obstruction

P / partially_solved | AIM 研讨会问题汇编

Let \((x_n)_{n\in\mathbb N}\) be a Parseval frame with \(\|x_n\|\geq \epsilon>0\) for every \(n\). Let \(N(\epsilon)\) be the least universal number of classes needed to partition the index set so that every class indexes a Riesz sequence. Determine \(N(\epsilon)\), especially its asymptotic behavior as \(\epsilon\downarrow0\).

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10642 | AIM-ANALYSIS-0108

### Exact finite localization and summable Hilbert-block extensions for the L2-to-L1 problem

P / partially_solved | AIM 研讨会问题汇编

**Banach space pairs with the Lipschitz extension property.** Does the pair \((L^2,L^1)\) have the Lipschitz extension property?

来源字段：http://aimpl.org/mappingmetric/1/

10643 | AIM-ANALYSIS-0058

### Exact Green functions and zero leafwise critical measure on the Cayley fiber

P / partially_solved | AIM 研讨会问题汇编

Let \(C\) be the cubic surface defined by \[ x^2+y^2+z^2-xyz=D, \] where \(D\) is a complex parameter. Let \(s_x:C\to C\) be defined by \[ s_x(x,y,z)=(yz-x,y,z). \] The maps \(s_y,s_z\) are defined similarly.

Consider, for example, the map \(S:=s_x\circ s_y\circ s_z\). It is known that \(S\) has non-zero entropy. Adapt the study of critical measure, etc. from the study of Hénon maps to that of \(F\).

来源字段：http://aimpl.org/scvproblems/2/

10644 | AIM-ANALYSIS-0166

### Exact GUE-zero insertion amplitudes and a coefficient-space obstruction

P / partially_solved | AIM 研讨会问题汇编

Question 9: Ashkan Nikeghbali What are the natural physical examples of random functions with GUE zeros on the real line?

**Question 9: Ashkan Nikeghbali What are the natural physical examples of random functions with GUE zeros on the real line?**

*Partially supported by NSF grants DMS-0211458, CAREER DMS-0349309, and AIM.

来源字段：https://aimath.org/WWN/randomzeros/arcc1.pdf

10645 | AIM-ANALYSIS-0082

### Exact lower witnesses in the complete-edge Weaver family

P / partially_solved | AIM 研讨会问题汇编

For the random MSS, there's a witness for the best bound, is there a conjecture for Weaver's conjecture? Weaver has a construction that \(\sqrt\eps\) is necessary but MSS has no constructive lower bound. Pete C. *does* have a constructive lower bound.

Determine or conjecture the sharp quantitative two-color Weaver discrepancy, and seek explicit exact-Parseval families proving that a \(\sqrt\alpha\) term is necessary, while distinguishing this task from mixed-polynomial root optimality and from projection-paving/Riesz lower examples.

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10646 | AIM-SEVERAL_COMPLEX_VARIABLES-0048

### Exact monomial degree spectra and an obstruction to failed degree lowering

P / partially_solved | AIM 研讨会问题汇编

Let \(R:\mathbb B^n\to\mathbb B^N\), \(n\ge3\), be a rational proper mapping, \[ R(z)=\frac{p(z)}{q(z)}. \] Let \(d=d_0(R)\) be the degree of \(R\).

**Question 8a.** Is \[ d\le \frac{N-1}{n-1}? \] **Question 8b.** Is this inequality true if one also assumes \[ n\ge 2d^2+2d? \] Characterize all \(R\) for which equality holds.

For monomial maps the answer to Question 8b is yes [DLP]. For \(n=2\), a proper monomial map has \(d\le2N-3\), sharply [DKR].

A further question asks: if \(f:\mathbb B^n\to\mathbb B^N\) is proper, is rational, polynomial, or monomial, and \(d=d_0(f)>1\), must there be a proper \(g:\mathbb B^n\to\mathbb B^N\) in the corresponding class with \(d_0(g)=d_0(f)-1\)?

来源字段：https://aimath.org/WWN/complexitycrmap/complexitycrmap.pdf

10647 | AIM-ANALYSIS-0079

### Exact one-way communication bounds for a frame-partition surrogate

P / partially_solved | AIM 研讨会问题汇编

Chuck started to describe a cooperative two-player game with partitioning the sets --> could we get an impossibility result via communication complexity?

* Greg Knese

Can one formulate a cooperative communication problem whose outputs are Kadison--Singer/Weaver-type partitions, and can a communication lower bound give a valid obstruction to constructing such partitions efficiently?

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10648 | AIM-ANALYSIS-0165

### Exact planar probabilities and sphere witnesses for random Viro diagrams

P / partially_solved | AIM 研讨会问题汇编

**Question 8: Maurice Rojas** What is the probability that a random Viro diagram contains no sphere?

Investigate the connections between random sparse polygons and Newton polytopes. This should be extended to the case of random Viro diagrams.

A sphere is an entire connected component of the Viro hypersurface that is PL-homeomorphic to $S^{d-1}$, where $d$ is the dimension of the ambient triangulated space.

来源字段：https://aimath.org/WWN/randomzeros/arcc1.pdf

10649 | AIM-ANALYSIS-0025

### Exact tensorization of optimal Dirichlet eigenfunction power-concavity

P / partially_solved | AIM 研讨会问题汇编

Let $\Omega\subset\mathbb R^n$ be a bounded convex domain and let $u$ be the principal eigenfunction of the Dirichlet Laplacian on $\Omega$. Brascamp and Lieb proved that $u$ is log-concave. It is also known that $u^\alpha$ is concave for certain $\alpha>0$. Define $$ \alpha(\Omega)=\sup\{\alpha\geq 0:u^\alpha\text{ is concave}\}. $$ Lindqvist conjectured that $\alpha(\Omega)$ is maximized by the ball.

The source URL is <http://aimpl.org/shapesurface/2/>.

The eigenfunction solves $$ -\Delta u=\lambda_1(\Omega)u\quad\text{in }\Omega, \qquad u>0\quad\text{in }\Omega, \qquad u=0\quad\text{on }\partial\Omega. $$ Its positive normalization is immaterial: multiplying $u$ by a constant multiplies $u^\alpha$ by a positive constant and hence does not change concavity. Translations, rotations, and uniform dilations of $\Omega$ also leave $\alpha(\Omega)$ unchanged. A general affine map does not, because it changes the Laplacian into an anisotropic operator.

来源字段：http://aimpl.org/shapesurface/2/

10650 | AIM-ANALYSIS-0026

### Exact thin-box counterexamples and an unweighted flat-collapse obstruction

S / solved | AIM 研讨会问题汇编

**Optimizers for Neumann eigenvalues among convex sets of a fixed perimeter.** Prove the existence of optimizers for $$ \sup\{\mu_k(\Omega):\Omega\subset\mathbb R^n\text{ convex}, P(\Omega)=P_0\} $$ and $$ \inf\{\mu_k(\Omega):\Omega\subset\mathbb R^n\text{ convex}, P(\Omega)=P_0\}. $$

来源字段：http://aimpl.org/shapesurface/3/

10651 | AIM-ANALYSIS-0138

### Exact zeros and stability obstructions for BMV trace polynomials

P / partially_solved | AIM 研讨会问题汇编

Study the zeros of Bessis-Moussa-Villani polynomials $t\to\text{tr}(A+tB)^m$.

来源字段：http://aimpl.org/hyperbolicpoly/2/

10652 | AIM-FUNCTIONAL_ANALYSIS-0048

### Exactness-independent coarse bounds and explicit stabilization of Mackey Borel relations

P / partially_solved | AIM 研讨会问题汇编

Does the complexity of the Mackey Borel structure of a simple separable C*-algebra increase as one goes from nuclear C*-algebras to exact ones to ones that are not even exact?

来源字段：http://aimpl.org/settheorycstar/7/

10653 | AMR-036-0018

### Exceptional directions in Gross's theorem

P / partially_solved | AMR 问题清单汇编

For a local inverse germ $\phi_z$ of a meromorphic function $f$ at a noncritical value $w=f(z)$, Gross's theorem gives analytic continuation along almost every ray from $w$. Sharpen 'almost every': what is the smallest possible exceptional set theorem, in capacity, dimension, or another natural sense?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10654 | OWR-13486-001

### Existence and Uniqueness of Sobolev Traces on Continuous Domains

P / partially_solved | Oberwolfach 报告问题汇编

If $\Omega$ has continuous boundary, does every element of $H^1(\Omega)$ have at most one trace, and does existence of a trace imply uniqueness? In particular, if every element of $\widetilde{H}^1(\Omega)$ has a trace, does every element of $H^1(\Omega)$ have at most one trace?

来源字段：https://doi.org/10.4171/owr/2015/1

10655 | OWR-723-005

### Expected Discrepancy of Generalized Lattice Point Sets

O / open | Oberwolfach 报告问题汇编

For even $p$, let $M_n^{z,\Delta}=\{jz+\Delta\pmod 1:0\leq j<n\}$ with $z,\Delta\in[0,1]^d$. Is $\mathbb E_{z,\Delta}[\operatorname{disc}_p^p(M_n^{z,\Delta})]\leq\mathbb E[\operatorname{disc}_p^p(M_n)]$ for $n$ independent uniformly distributed points $M_n$?

来源字段：https://doi.org/10.4171/owr/2004/13

10656 | AIM-SEVERAL_COMPLEX_VARIABLES-0049

### Explicit interior jet bounds for proper rational ball maps

P / partially_solved | AIM 研讨会问题汇编

**Problem 9 (Bernhard Lamel).** Suppose that \(R,\widetilde R:\mathbb B^n\to\mathbb B^N\) are proper rational mappings. Find the smallest integer \(r\) such that \[ j_0^rR=j_0^r\widetilde R \quad\Longrightarrow\quad R=\widetilde R. \] Ask the same question when \(R,\widetilde R\) are merely proper, rather than rational.

来源字段：https://aimath.org/WWN/complexitycrmap/complexitycrmap.pdf

10657 | AIM-ANALYSIS-0129

### Explicit positive-area carpets with non-isometric quasisymmetries

P / partially_solved | AIM 研讨会问题汇编

A set $X\subset\mathbb R^n$ is quasisymmetrically rigid if every quasisymmetric map of $X$ onto itself is the restriction to $X$ of an isometry of $\mathbb R^n$. Bonk and Merenkov showed that the usual $1/3$ Sierpiński carpet is quasisymmetrically rigid. Which positive-area Sierpiński carpets are rigid, and which are nonrigid?

来源字段：http://aimpl.org/mappingmetric/5/

10658 | OWR-4531-017

### Exponential Boundary Growth in the Dirichlet Weak Product

O / open | Oberwolfach 报告问题汇编

别名：Duplicate: Boundary Behavior in the Dirichlet Weak Product [OWR-4531-018]

Let $D_0$ be the Dirichlet space without constants, with orthonormal basis $n^{-1/2}z^n$, and let $D_0^{\odot2}$ consist of sums $F=\sum_jt_jf_jg_j$ with bounded $t_j\geq0$ and orthonormal bases $(f_j)$ and $(g_j)$. What boundary behavior do functions in $D_0^{\odot2}$ have? In particular, if $\|F\|<1$, how fast can $$\int_{|z|=r}\exp\!\left(\frac{|F(z)|^2}{\log(1/(1-r^2))}\right)$$ grow as $r\to1^-$?

来源字段：https://doi.org/10.4171/owr/2010/49

10659 | OWR-720-012

### Exponential Frames for Measurable Sets

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a subset $\Gamma\subset\mathbb{R}$ and $\eta$ in a Hilbert space $H$ such that the $\Gamma$-translates of $\eta$ form a frame, or even a tight frame? Equivalently, does $L^2(A)$ admit a frame of exponentials?

来源字段：https://doi.org/10.4171/owr/2004/10

10660 | OWR-11101912-007

### Exponential Maximizers for Cone Adjoint Restriction

P / partially_solved | Oberwolfach 报告问题汇编

For each $d\geq4$, is $f_\star(\tau,\xi)=e^{-|\tau|}$ a global maximizer of the $p=2$ Fourier adjoint-restriction inequality on the one-sheeted cone $K_d^+=\{(\tau,\xi):\tau=|\xi|\}$?

来源字段：https://doi.org/10.4171/owr/2022/31

10661 | OWR-3470-005

### Extremal Tuples of Commuting Contractions

O / open | Oberwolfach 报告问题汇编

For $d>2$, identify the extremal $d$-tuples $T$ of commuting contractions, where $T$ is extremal if every extension $R$ within the family reduces the original Hilbert space and is therefore only a direct-sum extension.

来源字段：https://doi.org/10.4171/owr/2009/20

10662 | AMR-036-0031

### Extremality of iterated quadratic polynomials

P / partially_solved | AMR 问题清单汇编

Are the iterates $p_c^n$ of hyperbolic quadratic polynomials extremal, or nearly extremal, for the Littlewood exponent $\alpha$ governing mean spherical derivatives of polynomials?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10663 | OWR-4531-016

### Extreme Points and Fermat Equations in Star-Invariant Spaces

O / open | Oberwolfach 报告问题汇编

For an inner function $\theta$ and $K^{p}_{\theta}=H^{p}\cap\theta H^{p}_{0}$, characterize the extreme points of the unit ball of $K^{\infty}_{\theta}$; and prove or disprove the Fermat equation $f^{n}+g^{n}=h^{n}$ in $K^{p}_{\theta}$ for $n\geq3$, except when $f,g,h$ span a one-dimensional subspace.

来源字段：https://doi.org/10.4171/owr/2010/49

10664 | AIM-ANALYSIS-0084

### Facewise strongly-Rayleigh completion and a sharp marginal packing threshold

P / partially_solved | AIM 研讨会问题汇编

Consequences to matroid theory: can we prove a continuous version of matroid theory using MSS theorem and what consequences to matroid theory does MSS have?

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10665 | OWR-13750336-004

### Favard Length and Lipschitz Graph Structure

P / partially_solved | Oberwolfach 报告问题汇编

Let $E\subset[0,1]^2$ be $1$-Ahlfors regular and define $\operatorname{Fav}(E)=\int_{S^1}\mathcal H^1(\pi_\theta(E))\,d\theta$. If $\operatorname{Fav}(E)\geq C>0$, must there exist a Lipschitz graph $\Gamma$ such that $\operatorname{Lip}(\Gamma)\lesssim_C1$ and $\mathcal H^1(E\cap\Gamma)\gtrsim_C1$?

来源字段：https://doi.org/10.4171/owr/2023/25

10666 | AMR-036-0043

### Few inflection points of holomorphic curves

P / partially_solved | AMR 问题清单汇编

Let $f=(f_0,\ldots,f_n)$ be a linearly nondegenerate holomorphic curve, let $T(r,f)$ have finite lower order $\lambda$, and let $N_1(r)$ be the averaged counting function of zeros of its Wronskian. If $N_1(r)=o(T(r,f))$, prove that $\lambda$ is rational and that $\lim_{r\to\infty}\log T(r,f)/\log r$ exists.

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10667 | OWR-4211-002

### Finite Elements for Sign-Changing Maxwell Interfaces

P / partially_solved | Oberwolfach 报告问题汇编

For time-harmonic Maxwell equations with sign-changing material coefficients across a general Lipschitz interface, establish convergence of the standard finite element schemes.

来源字段：https://doi.org/10.4171/owr/2010/10

10668 | AIM-ANALYSIS-0128

### Finite-cell obstructions for proper self-embeddings of the standard Sierpinski carpet

P / partially_solved | AIM 研讨会问题汇编

**Bi-Lipschitz embeddings of the Sierpiński carpet into itself.** Is every bi-Lipschitz embedding of the \(\tfrac13\) Sierpiński carpet \(S_3\) into itself the restriction of an affine mapping?

来源字段：http://aimpl.org/mappingmetric/5/

10669 | AIM-ANALYSIS-0006

### Finite-cover droplets and an exact soft-Riemann--Hilbert sector reduction

P / partially_solved | AIM 研讨会问题汇编

Given \(Q\), produce \(Q(z^d)\) with multiple disjoint support sets for the droplet. Study the soft Riemann-Hilbert problem for orthogonal polynomials.

来源字段：http://aimpl.org/riemhilberttoeplitz/2/

10670 | OWR-742-004

### Finite-Dimensional Approximation of Bounded Analytic Functions

O / open | Oberwolfach 报告问题汇编

Let $f_1,\ldots,f_k\in H^\infty(D,\mathbb C^N)$ and $M(z)=\operatorname{span}\{f_1(z),\ldots,f_k(z)\}$. For every $\varepsilon>0$, is there a subspace $S\subset\mathbb C^N$ of dimension depending only on $k,\varepsilon$ such that every $g\in H^\infty(D,\mathbb C^N)$ with $\|g(z)\|_p\leq1$ on $D$ and $\operatorname{dist}_{\ell_p}(g(z),M(z))\leq\varepsilon$ almost everywhere on $\partial D$ satisfies $\operatorname{dist}_{\ell_p}(g(0),S)\leq\psi(\varepsilon)$ for some fixed $\psi$ with $\psi(\varepsilon)\to0$?

来源字段：https://doi.org/10.4171/owr/2004/40

10671 | AIM-SEVERAL_COMPLEX_VARIABLES-0036

### Finite-jet status and a complex-locus similarity sieve

P / partially_solved | AIM 研讨会问题汇编

**7. (D. Zaitsev)** Let \(M\) be a real submanifold of \(\mathbb C^n\) which is generically minimal and holomorphically nondegenerate. What can you say about \(\operatorname{Aut}(M,p)\) at nonminimal points?

The assigned AIM record is Problem 7 of the workshop list *Emerging applications of complexity for CR mappings*. Its text is:

The line **“CR embeddings”** that follows is a new section heading. It is not part of Problem 7.

Two genuine ambiguities remain in the source.

1. It says “real submanifold,” not “real-analytic generic CR submanifold.” Holomorphic nondegeneracy and the notation used in the surrounding problems strongly suggest the real-analytic embedded CR category, but this is not stated. 2. In the standard usage of the cited literature, \[ \operatorname{Aut}(M,p) =\{\text{germs at }p\text{ of biholomorphisms }H:(\mathbb C^n,p)\to(\mathbb C^n,p), \ H(M)\subset M\}. \] Thus it is a stability group: every element fixes \(p\). It is not the collection of all local arrows taking \(p\) to varying points. This reading is reinforced by the immediately preceding AIM problems, which introduce separate notation \(\mathcal O(p)\) for the orbit under such arrows.

This report therefore gives three answers: the established real-analytic hypersurface answer, the still-open higher-codimension answer, and the limitations of a merely smooth interpretation.

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10672 | AIM-SEVERAL_COMPLEX_VARIABLES-0023

### Finite-kernel reduction for cocompact invariant ball embeddings

P / partially_solved | AIM 研讨会问题汇编

**Problem (X. Huang).** Assume \(\Gamma\subset\operatorname{Aut}(\mathbb B^N)\) is a discrete subgroup and \(f:\mathbb B^n\to\mathbb B^N\) is a proper holomorphic embedding such that \(\Gamma(f(\mathbb B^n))\subset f(\mathbb B^n)\) and \(f(\mathbb B^n)/\Gamma\) is compact. Is \(f\) linear?

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10673 | OWR-1113-002

### Finite-Order Laminates for Rank-One Convex Envelopes

P / partially_solved | Oberwolfach 报告问题汇编

For an energy density $W$, find readily checkable conditions ensuring that its rank-one convex envelope $W^{\mathrm{rc}}$ is attained by a laminate of finite order $N$; also bound $N$ in terms of $W$ and confine all supporting matrices to a ball of finite prescribed radius.

来源字段：https://doi.org/10.4171/owr/2005/52

10674 | AIM-ANALYSIS-0060

### Finite-range reduction and measurable affine patching for a one-variable vector-field maximal operator

P / partially_solved | AIM 研讨会问题汇编

Let \(u:\mathbb R\to\mathbb R\) be a measurable function. Define the maximal operator along the planar vector field \((1,u)\) by \[ M_u f(x,y):=\sup_{\epsilon>0}\left| \frac{1}{2\epsilon}\int_{-\epsilon}^{\epsilon} f(x-t,y-u(x)t)\,dt \right|. \] Does \(M_u\) satisfy any \(L^p\) bound for certain \(p<\infty\)?

来源字段：https://aimath.org/pastworkshops/multilinopsproblems.pdf

10675 | OWR-14298803-004

### Finite-Sample Error Bounds for EDMD Time-Average Optimization

O / open | Oberwolfach 报告问题汇编

Derive explicit bounds for the gap between the exact auxiliary-function optimum and its EDMD-based approximation as a function of the number $N$ of data points, the dictionary degree $m$, and the sampling step $\tau$.

来源字段：https://doi.org/10.4171/owr/2024/57

10676 | AMR-036-0035

### Finiteness of Newtonian equilibrium points

P / partially_solved | AMR 问题清单汇编

For finitely many positive charges $a_k$ at points $x_k\in\mathbb R^3$, is the critical set of $u(x)=\sum_{k=1}^n a_k/|x-x_k|$ always finite?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10677 | AIM-ANALYSIS-0122

### First-order scaling and the elementary-proof bottleneck

P / partially_solved | AIM 研讨会问题汇编

Is there an elementary proof for the fact that every entire \(n\)-harmonic function in \(\mathbb R^n\) with linear growth is affine?

来源字段：http://aimpl.org/mappingmetric/5/

10678 | AIM-SEVERAL_COMPLEX_VARIABLES-0027

### Fixed harmonic parts: exact low-degree breakpoints and a deformation criterion

P / partially_solved | AIM 研讨会问题汇编

Let \(S(z,\bar z)\) be a harmonic Hermitian polynomial of bidegree \((d,d)\) on \(\mathbb C^n\). For a given \(k\), how many Hermitian polynomials \(A(z,\bar z)\) of bidegree \((d-1,d-1)\) exist such that \[ S(z,\bar z)+A(z,\bar z)\|z\|^2 =\sum_{j=1}^{k}|f_j(z)|^2? \] Which properties of \(S\) determine this number?

If \(S=0\) and \(k<n\), the only solution is \(A=0\), by a lemma of Huang. If \(S\) is arbitrary and \(k<n/2\), there is at most one solution.

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10679 | OWR-12005-011

### Fixed Volume Magnetic Eigenvalue Minimizers in Three Dimensions

P / partially_solved | Oberwolfach 报告问题汇编

For a real magnetic parameter $\beta$ and a bounded open set $\Omega\subset\mathbb{R}^3$, can one find a fixed-volume $\Omega$ that minimizes the first Dirichlet eigenvalue $E_1$ of $\left(i\nabla+\frac{\beta}{2}(-x_2,x_1,0)\right)^2$?

来源字段：https://doi.org/10.4171/owr/2012/33

10680 | AIM-ANALYSIS-0098

### Fixed-diagonal geometry and an exact 4x4 two-paving profile

P / partially_solved | AIM 研讨会问题汇编

Wants to geometrize this work! Paving projections is especially interesting. Fix r. Find the conditions on the diagonal of a projection so it can be r-paved? Conditions should be independent of dimension.

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10681 | AIM-FUNCTIONAL_ANALYSIS-0020

### Flatness exactly characterizes a nontrivial B(H) ultrapower commutant

S / solved | AIM 研讨会问题汇编

An ultrafilter \(\mathcal U\) on \(\mathbb N\) is *flat* if there are nonincreasing functions \(h_n:\mathbb N\to[0,1]\) such that \[ h_n(0)=1,\qquad \lim_{j\to\infty}h_n(j)=0, \] and, for every \(f:\mathbb N\nearrow\mathbb N\), \[ \lim_{n\to\mathcal U}\sup_{j\in\mathbb N} |h_n(j)-h_n(f(j))|=0. \] Is a nonprincipal ultrafilter satisfying \[ \mathcal B(H)'\cap\mathcal B(H)^{\mathcal U}\ne\mathbb C I \] flat?

来源字段：http://aimpl.org/settheorycstar/1/

10682 | OWR-1592-003

### Flexible Geometric Signal-Reproduction Systems

O / open | Oberwolfach 报告问题汇编

Beyond curvelets and shearlets, are there other constructions based on operations on the plane that guarantee reproduction of a signal while offering more flexible analytic and geometric properties?

来源字段：https://doi.org/10.4171/owr/2007/36

10683 | AMR-023-0013

### Flint Hills series

O / open | AMR 问题清单汇编

Does the Flint Hills series $\sum_{n=1}^{\infty} 1/(n^3\sin^2 n)$ converge?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767

10684 | AIM-FUNCTIONAL_ANALYSIS-0027

### Forcing absoluteness of minimal and separable maximal tensor products

P / partially_solved | AIM 研讨会问题汇编

Does \(\otimes_{\min}\) or \(\otimes_{\max}\) commute with forcing?

The record names no pair of C\(^*\)-algebras and gives no remarks or literature. The two preceding records ask whether the number of C\(^*\)-norms on certain algebraic tensor products, and the ideal structure of \(\mathcal B(H)\otimes_{\min}\mathcal Q(H)\), can depend on the model of set theory. This supports a model-comparison reading, but it does **not** justify silently inserting \(\mathcal B(H)\), \(\mathcal Q(H)\), or any other particular pair into Problem 3.3.

There is also a real foundational ambiguity. If \(M\subseteq N=M[G]\) is a forcing extension and \(A\in M\) is complete according to \(M\), then \(N\) may contain new Cauchy sequences in the old metric space. The convention used in later operator-algebra forcing work is to identify a ground-model C\(^*\)-algebra with its metric completion in the extension [FKV]. We adopt precisely that convention.

For a ground-model C\(^*\)-algebra \(A\), write \[ A^N=\widehat{A^M}^{\,N}. \] For \(\alpha\in\{\min,\max\}\), the canonical formulation studied here is whether \[ \widehat{(A\otimes_\alpha B)^M}^{\,N} \quad\hbox{and}\quad A^N\otimes_\alpha^N B^N \tag{1} \] are canonically isomorphic by the identity on the old algebraic tensor product.

This is different from recomputing a definition in \(N\). For example, the completion of the ground-model Calkin algebra need not be the Calkin algebra computed in the extension [FKV, §2]. It is also different from asking whether forcing preserves nuclearity or exactness, and from developing a Boolean-valued tensor-product functor. Those are plausible broader questions, not recovered content of the one-line AIM record.

来源字段：http://aimpl.org/settheorycstar/3/

10685 | AIM-SEVERAL_COMPLEX_VARIABLES-0035

### Formal and biholomorphic CR equivalence diverge

S / solved | AIM 研讨会问题汇编

**Problem 6 (D. Zaitsev).** Let \((M,p),(M',p')\) be germs of CR submanifolds in \(\mathbb C^n\) which are formally equivalent. Are they biholomorphically equivalent?

**Remark.** The non-CR case doesn't necessarily have this property. The answer in the CR case is yes in many situations but the general question is open.

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10686 | OWR-14211-010

## Formal Duality of Energy Minimizing Periodic Point Sets

O / open | Oberwolfach 报告问题汇编

Why do energy-minimizing periodic point sets appear to have the rare formal-duality property, namely equality of their averaged pair sums with the density-scaled Fourier-dual sums for every Schwartz function?

来源字段：https://doi.org/10.4171/owr/2016/3

10687 | OWR-11101912-006

## Foschi Maximizers for Wave Strichartz Inequalities

P / partially_solved | Oberwolfach 报告问题汇编

For every spatial dimension $d\geq2$, does the wave-equation initial datum

$$(u_0,\dot u_0)=\bigl((1+|x|^2)^{-(d-1)/2},0\bigr)$$

maximize the sharp Strichartz inequality?

来源字段：https://doi.org/10.4171/owr/2022/31

10688 | OWR-14299906-012

## Foundations for Quantum Measurable Spaces

O / open | Oberwolfach 报告问题汇编

Define a category of quantum measurable spaces and quantum channels whose commutative subcategory is equivalent to suitable measurable spaces and Markov kernels, and in which positive operator-valued measurements are equivalent to quantum-to-classical channels.

来源字段：https://doi.org/10.4171/owr/2026/4

10689 | AIM-ANALYSIS-0032

## Four sharp constants and a conical affine lower bound

P / partially_solved | AIM 研讨会问题汇编

Let \(\Omega\subset\mathbb R^n\) be a convex domain and let \(f:\Omega\to[0,\infty)\) be either convex or subharmonic. Find the smallest constants in \[ \frac1{|\Omega|}\int_\Omega f\,dx \leq \frac{c_n}{|\partial\Omega|}\int_{\partial\Omega}f\,d\sigma \tag{HH1} \] and \[ \int_\Omega f\,dx \leq \widetilde c_n|\Omega|^{1/n} \int_{\partial\Omega}f\,d\sigma . \tag{HH2} \]

It asks the following.

Here and below, \(|\partial\Omega|=\mathcal H^{n-1}(\partial\Omega)\). The source uses the same symbols for two different function classes. This is a material ambiguity: there are actually four universal sharp constants, and one of the four is now known exactly while the other three are not all known.

The source also says that (HH2) follows from (HH1) “with \(\widetilde c_n=c_n/\sqrt n\).” Read as a choice of a sufficient, nonsharp constant, this is valid. Read as an equality between the two optimal constants, it is not justified. Section 3 gives the exact normalization relation.

The standard admissible class in the cited higher-dimensional papers is that of bounded convex domains, without a smooth-boundary restriction. Thus the conical domains used below are directly admissible. A smooth approximation argument is nevertheless included in Section 7.3.

来源字段：http://aimpl.org/shapesurface/6/

10690 | OWR-11128-005

## Four-Point Configurations on Three Lines

P / partially_solved | Oberwolfach 报告问题汇编

Can a nonzero Schwartz function satisfy a recurrence of the form $f(x+1)=C_1f(x-1)+C_2f(x)e(\alpha x)+C_3f(x)e(\beta x)$, where $e(t)=e^{2\pi i t}$? Equivalently, can the associated two-dimensional matrix recurrence have a nonzero solution that decays rapidly for a positive-measure set of initial values $x$?

来源字段：https://doi.org/10.4171/owr/2011/36

10691 | OWR-14297736-009

## Fourier Coefficients of Inhomogeneous Automorphic Laplace Equations

O / open | Oberwolfach 报告问题汇编

For solutions of $(\Delta-r(r+1))f=E_aE_b$ on $\operatorname{PSL}_2(\mathbb Z)\backslash\mathbb H$, compute all Fourier coefficients and evaluate the divisor-correlation sums occurring in them.

来源字段：https://doi.org/10.4171/owr/2023/36

10692 | AIM-ANALYSIS-0061

### Fourier norms of indicator functions and a two-interval splitting benchmark

P / partially_solved | AIM 研讨会问题汇编

**Question 2 (Christ).** On $\mathbb R^d$, let $E\subset\mathbb R^d$ with $|E|=1$. Given $q>2$, describe the set $E$ that maximises the quantity $\|\widehat{\mathbf 1_E}\|_q$.

This record is Question 2 from the AIM workshop problem list *Carleson theorems and multilinear operators*. The PDF states:

The displayed expression in the PDF is unambiguously \[ \|\widehat{\mathbf 1_E}\|_{L^q(\mathbb R^d)}. \] The accompanying 2015 status paragraph reports existence for all $q>2$ and several perturbative classification results. Its symbol $2N$ means an even integer $2N\ge4$, rather than multiplication by a fixed but unspecified parameter.

来源字段：https://aimath.org/pastworkshops/multilinopsproblems.pdf

10693 | OWR-11101912-012

### Fourier Restriction Conjecture in Higher Dimensions

O / open | Oberwolfach 报告问题汇编

Does the Fourier restriction conjecture hold in every dimension $d\geq 3$?

来源字段：https://doi.org/10.4171/owr/2022/31

10694 | OWR-15956-009

### Fourier Uniqueness from Spherical Zero Sets

O / open | Oberwolfach 报告问题汇编

Let $f\in\mathcal S(\mathbb R^n)$. If $f$ vanishes on every sphere $|x|=r_j$ and its Fourier transform $\widehat f$ vanishes on every sphere $|\xi|=\rho_j$, where $r_j,\rho_j\to\infty$, must $f\equiv0$?

来源字段：https://doi.org/10.4171/owr/2017/51

10695 | OWR-1703872-019

### Fourier Uniqueness from Vanishing on General Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

For Fourier-uniqueness problems in which a test function and its Fourier transform vanish on prescribed spheres, do analogous uniqueness or integration-by-parts results hold when the vanishing sets are replaced by other surfaces?

来源字段：https://doi.org/10.4171/owr/2020/26

10696 | OWR-14299513-006

### Fractal Local Smoothing for General Time Sets

O / open | Oberwolfach 报告问题汇编

For an arbitrary time set $E\subset[1,2]$, does fractal local smoothing hold throughout the range $p>2$ and $s>p^{-1}\nu_E^{\#}(ps_p)$?

来源字段：https://doi.org/10.4171/owr/2025/32

10697 | OWR-11101912-002

### Fractal Schrödinger Convergence Above Half Dimension

O / open | Oberwolfach 报告问题汇编

For $d/2<\alpha\leq d$, determine the sharp exponent $s(\alpha)$ such that every $f\in H^s(\mathbb R^d)$ satisfies $e^{it\Delta}f(x)\to f(x)$ as $t\to0$ for $\mathcal H^\alpha$-almost every $x$.

来源字段：https://doi.org/10.4171/owr/2022/31

10698 | OWR-14299513-004

### Fractal-Time Local Smoothing

P / partially_solved | Oberwolfach 报告问题汇编

For a fractal time set $E\subset[1,2]$ and a maximal $2^{-j}$-separated subset $E_j$, establish sharp local smoothing: $\left(\sum_{t\in E_j}\lVert e^{it\sqrt{-\Delta}}P_jf\rVert_{L^p(\mathbb{R}^d)}^p\right)^{1/p}\leq C2^{js}\lVert f\rVert_{L^p(\mathbb{R}^d)}$.

来源字段：https://doi.org/10.4171/owr/2025/32

10699 | OWR-14299513-007

### Fractal-Time Local Smoothing Between Lebesgue Spaces

O / open | Oberwolfach 报告问题汇编

Let $1<p\leq q<\infty$, and let $p'=p/(p-1)$ be the Hölder conjugate of $p$. If $q>p'$, does $\left(\sum_{t\in E_j}\lVert e^{it\sqrt{-\Delta}}P_jf\rVert_{L^q(\mathbb{R}^d)}^q\right)^{1/q}\leq C2^{js}\lVert f\rVert_{L^p(\mathbb{R}^d)}$ hold whenever $s>\frac{d+1}{2}(\frac1p-\frac1q)+\frac1q\nu_E^{\#}(\frac{q(d-1)}2(1-\frac1p-\frac1q))$?

来源字段：https://doi.org/10.4171/owr/2025/32

10700 | OWR-12339-003

### Frame Property of the Gaussian Gabor System on $\Lambda_3$

O / open | Oberwolfach 报告问题汇编

Let $g$ be the normalized Gaussian on $\mathbb R^2$, let

$$A_3=\begin{pmatrix}1&2/5\\0&\sqrt{21}/5\end{pmatrix},\qquad \Lambda_3=A_3(\mathbb Z+i\mathbb Z)^2,$$

and let $G(g,\Lambda_3)$ be the associated Gabor system. Is $G(g,\Lambda_3)$ a frame for $L^2(\mathbb R^2)$?

来源字段：https://doi.org/10.4171/owr/2013/11

10701 | OWR-11791-005

### Frame Sets of General Gabor Windows

O / open | Oberwolfach 报告问题汇编

For which triples $(g,\alpha,\beta)$, with $g\in L^2(\mathbb R)$ and $\alpha,\beta>0$, is the Gabor system $\{e^{2\pi i\beta lt}g(t-\alpha k):k,l\in\mathbb Z\}$ a frame for $L^2(\mathbb R)$?

来源字段：https://doi.org/10.4171/owr/2012/29

10702 | OWR-14298372-008

### Free Polar Decomposition Under a Nondegeneracy Condition

O / open | Oberwolfach 报告问题汇编

Let $z$ be a $B$-valued circular element with a free bipolar decomposition, and suppose $\operatorname{span}\{E((z^*z)^k):k\geq 0\}$ is dense in $B$. Must $z$ admit a free polar decomposition?

来源字段：https://doi.org/10.4171/owr/2024/22

10703 | OWR-1057-009

### Frequently Hypercyclic Weighted Backward Shifts

S / solved | Oberwolfach 报告问题汇编

Characterize frequently hypercyclic weighted backward shifts; and determine whether the set of frequently hypercyclic vectors of an operator can be of second category.

来源字段：https://doi.org/10.4171/owr/2005/36

10704 | AMR-036-0001

### Fuchsian equations with unitary monodromy

P / partially_solved | AMR 问题清单汇编

Fix singularities $a_1,\ldots,a_n$ and real exponent differences $\alpha_1,\ldots,\alpha_n$ for second-order Fuchsian equations on the Riemann sphere. Let $E(a_1,\ldots,a_n,\alpha_1,\ldots,\alpha_n)$ be the set of accessory parameters for which the projective monodromy is contained in $\operatorname{PSU}(2)$. Is $E$ always discrete? Is it always finite?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10705 | AMR-023-0002

### Fuglede's conjecture in dimensions one and two

P / partially_solved | AMR 问题清单汇编

For nonconvex subsets of $\mathbb{R}$ and $\mathbb{R}^2$, is a set spectral if and only if it tiles by translations?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767

10706 | AIM-FUNCTIONAL_ANALYSIS-0014

### Gauge integrals, cube sections, and profile universality

O / open | AIM 研讨会问题汇编

Take $a_1, \dots, a_n \in \mathbb R$ and $v_1, \dots, v_n \in \mathbb R^n$ and define $$ f(t)=\int_{\mathbb R^k}e^{-\max_{1\leq i\leq n}e^{a_it}|\left |}\,dx. $$ Then $f(t)$ is log-concave is equivalent to log-Brunn-Minkowski. What is possible $f$ that we can use for Brunn-Minkowski?

来源字段：http://aimpl.org/symconvgeomineq/2/

10707 | OWR-9790358-011

### Gaussian Cubature for Three-Variable Product Chebyshev Weights

O / open | Oberwolfach 报告问题汇编

Does a Gaussian cubature rule of the prescribed degree exist for the product Chebyshev weight in three variables?

来源字段：https://doi.org/10.4171/owr/2022/7

10708 | OWR-15577-012

### Gaussian Extremizers for Strichartz Inequalities

P / partially_solved | Oberwolfach 报告问题汇编

Do Gaussian functions extremize the Strichartz inequality $$ \lVert u\rVert_{L^{2+4/d}(\mathbb R^{d+1})}\leq C\lVert f\rVert_{L^2(\mathbb R^d)} $$ for solutions $u$ of the free Schrödinger equation in every dimension?

来源字段：https://doi.org/10.4171/owr/2017/34

10709 | AIM-ANALYSIS-0036

### Gaussian Poincaré equality is equivalent to splitting off a line

S / solved | AIM 研讨会问题汇编

Consider the operator $-\gamma^{-1}\nabla \gamma \nabla$ on the weighted space $L^2(\Omega, \gamma)$ with Neumann boundary conditions, $\Omega\subset \mathbb{R}^n$ and where $\gamma$ is a Gaussian weight. It is known that the first non-trivial eigenvalue of this operator satisfies $\mu_1(\Omega, \gamma)\geq 1$ (this is the Poincaré-Wirtinger inequality).

Prove that the equality in the Poincaré-Wirtinger inequality is attained only for infinite strips.

来源字段：http://aimpl.org/shapesurface/6/

10710 | OWR-2493-006

### Gauss-Green Boundaries of Infinite Electrical Networks

O / open | Oberwolfach 报告问题汇编

For an infinite electrical network $G$ with energy space $\mathcal H_E$ and finite-energy harmonic subspace $\operatorname{Harm}$, what is the appropriate boundary $\partial G$, including its space and measure, whose integral represents the boundary term in the discrete Gauss-Green formula?

来源字段：https://doi.org/10.4171/owr/2008/55

10711 | OWR-13750336-006

### General Kakeya Dimension Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $E\subset\mathbb R^n$, and let $D$ be the set of directions in which $E$ contains a unit line segment. If $D\neq\varnothing$, must $\dim_{\mathrm H}E\geq\dim_{\mathrm H}D+1$?

来源字段：https://doi.org/10.4171/owr/2023/25

10712 | OWR-14216-002

### General Quadrature for Isogeometric Galerkin Methods

O / open | Oberwolfach 报告问题汇编

Can a general and effective quadrature solution be developed for Galerkin-based isogeometric analysis methods?

来源字段：https://doi.org/10.4171/owr/2016/8

10713 | AIM-ANALYSIS-0096

### Generating interlacing families by real-rootedness preservers

P / partially_solved | AIM 研讨会问题汇编

**Can we generate new interlacing families?**

Find reusable operations which turn already-certified interlacing families into new ones, and identify hypotheses under which those operations preserve every conditional common-interlacing relation in the family tree.

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10714 | OWR-12339-001

### Generator Bounds for Multivariate Tight Wavelet Frames

P / partially_solved | Oberwolfach 报告问题汇编

Obtain estimates for the number of generators needed by multivariate tight wavelet frames.

来源字段：https://doi.org/10.4171/owr/2013/11

10715 | OWR-1115-002

### Geometric Characterization of Active Robin Boundaries

O / open | Oberwolfach 报告问题汇编

别名：Redundant Extraction - Active Robin Boundaries [OWR-1115-003]

Let $D\subset\mathbb{R}^d$ be a bounded domain containing a closed ball $B^*$. Let $u$ solve $\Delta u=0$ on $D\setminus B^*$, with $u=1$ on $\partial B^*$ and $\partial u/\partial n=u$ on $\partial D$. Give necessary and sufficient geometric conditions for $\inf_{x\in D\setminus B^*}u(x)>0$, equivalently, for the whole boundary of $D$ to be active.

来源字段：https://doi.org/10.4171/owr/2005/54

10716 | OWR-4429-005

### Geometry and Order of Caustics

O / open | Oberwolfach 报告问题汇编

What are caustics? In the oscillatory-integral setting, they are envelopes of optical rays or degenerate critical points of the phase where uniform $L^\infty$ bounds fail; determine their geometry and order.

来源字段：https://doi.org/10.4171/owr/2010/41

10717 | OWR-12979-001

### Geometry and Spectral Theory of Generalized Radial-Part Algebras

O / open | Oberwolfach 报告问题汇编

Let $D\subset\mathbb C^n$ be a Cartan domain of rank $r$, let $G=\operatorname{Aut}(D)$, and form the continuous-parameter algebra $\mathcal R$ obtained from the $K$-radial parts of the $G$-invariant differential operators by allowing the dimension parameters to vary. Does $\mathcal R$ have a natural geometric realization? Is it commutative and normal with respect to the generalized radial measure? Does it admit sufficiently many joint eigenfunctions to define a Harish-Chandra transform, and are its elements canonical functions of a distinguished generating set?

来源字段：https://doi.org/10.4171/owr/2014/21

10718 | OWR-13859-006

### Global Optimality and Initialization in Sparse Dictionary Learning

O / open | Oberwolfach 报告问题汇编

In sparse dictionary learning, is the generating dictionary a global optimum of the K-SVD/ITKsM criteria; are there efficient initialization strategies when sparsity $S=O(\mu^{-2})$; and can the convergence radii of ITKsM and ITKrM be extended to $O(1)$?

来源字段：https://doi.org/10.4171/owr/2015/38

10719 | AIM-ANALYSIS-0125

### Global permutation rigidity for repeated Carnot factors

P / partially_solved | AIM 研讨会问题汇编

Let $H$ be an irreducible Carnot group, $G$ a product of several copies of $H$, and $f:G\to G$ a quasisymmetric map. Must $f$ be a product map (after permutation of the factors)?

A Carnot group is irreducible if it is not the real line and is not the direct sum of two Carnot groups.

来源字段：http://aimpl.org/mappingmetric/5/

10720 | OWR-4214-009

### Gluing Primitive Ideal Spaces of Amenable Algebras

O / open | Oberwolfach 报告问题汇编

If $X$ is a locally compact sober space with an open subset $U$ such that both $U$ and $X\setminus U$ are homeomorphic to primitive ideal spaces of amenable $C^*$-algebras, is $X$ itself homeomorphic to the primitive ideal space of an amenable $C^*$-algebra?

来源字段：https://doi.org/10.4171/owr/2010/13

10721 | AMR-036-0024

### Goldberg's constant

P / partially_solved | AMR 问题清单汇编

Let $f$ be holomorphic in the unit disk with exactly one simple zero $z_0$ and two $1$-points $z_1,z_2$, counted with multiplicity. Determine the largest universal lower bound $A_2$ for the radius of the smallest hyperbolic disk containing $z_0,z_1,z_2$, and determine whether an extremal function must have a double $1$-point.

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10722 | OWR-11128-007

### Gowers Inverse Theorems for Oscillatory Integral Decay

O / open | Oberwolfach 报告问题汇编

For a multilinear oscillatory integral with nondegenerate polynomial phase, repeated Cauchy-Schwarz estimates can bound the integral by a high-order Gowers norm of one input. If the expected decay does not follow directly, the Green-Tao-Ziegler inverse theorem supplies algebraic structure for that input. Does this structural information suffice to prove power decay of the oscillatory integral?

来源字段：https://doi.org/10.4171/owr/2011/36

10723 | OWR-15959-003

### Gradient Flow of Wasserstein-Quantization Supports

P / partially_solved | Oberwolfach 报告问题汇编

For $N$-point Wasserstein quantization of a density supported on a compact subset of $\mathbb R^d$, how does the support set evolve under steepest descent, and how does this discrete evolution compare with the Wasserstein gradient flow of the limiting energy?

来源字段：https://doi.org/10.4171/owr/2017/54

10724 | OWR-4990379-008

### Gradient Flows for Network Transport Metrics

O / open | Oberwolfach 报告问题汇编

Analyze gradient flows for the network optimal-transport metrics $W_\kappa^2$ and their limiting Wasserstein and Fisher-Rao metrics.

来源字段：https://doi.org/10.4171/owr/2021/10

10725 | AMR-036-0019

### Gross property of implicit functions

P / partially_solved | AMR 问题清单汇编

Let $F$ be entire in two variables and let a holomorphic germ $\phi$ satisfy $F(z,\phi(z))=0$ near a nonsingular point. Must $\phi$ admit analytic continuation along almost every ray from its base point, as inverse germs of entire functions do?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10726 | OWR-4341-005

### Harmonic Forms on Manifolds with Long Necks

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a compact Riemannian manifold partitioned along a separating hypersurface $Y$, and let $X(r)$ denote the family of Riemannian manifolds obtained by stretching a cylindrical collar $[-r, r] \times Y$ to length $2r$. How does the space of $L^2$ harmonic differential forms $\mathcal{H}^*(X(r))$ behave asymptotically in the adiabatic/neck-stretching limit as $r \to \infty$?

来源字段：https://doi.org/10.4171/owr/2010/28

10727 | OWR-12005-008

### Harmonic Test Inequalities Above the First Eigenvalue

O / open | Oberwolfach 报告问题汇编

For every bounded sufficiently smooth domain $\Omega\subset\mathbb{R}^m$ and every $\lambda>\lambda_1(\Omega)$, does there exist a harmonic function $h$ such that $\sum_{j=1}^{\infty}\frac{\lambda_j}{\lambda_j-\lambda}|\langle h,u_j\rangle|^2<0$, where the $u_j$ are Dirichlet eigenfunctions?

来源字段：https://doi.org/10.4171/owr/2012/33

10728 | OWR-11787-002

### Heat-Trace Asymptotics on Singular and Noncompact Manifolds

O / open | Oberwolfach 报告问题汇编

For an elliptic operator $P$ on a possibly noncompact or singular manifold, give criteria ensuring that $e^{-tP}$ is trace class and that its heat trace has an asymptotic expansion $\sum_{\alpha,k}a_{\alpha,k}t^\alpha(\log t)^k$ as $t\to0$.

来源字段：https://doi.org/10.4171/owr/2012/25

10729 | AIM-SEVERAL_COMPLEX_VARIABLES-0019

### Hessian growth and the weighted dbar-Neumann spectrum

P / partially_solved | AIM 研讨会问题汇编

Let \(\phi:\mathbb C^n\to\mathbb R\) be a \(C^2\) plurisubharmonic function. Determine the relation between the eigenvalues of the complex Hessian of \(\phi\) and the compactness of the \(\bar\partial\)-Neumann operator on \(L^2_{(0,q)}(\mathbb C^n,e^{-\phi})\). See [Haslinger 2012] for \(n=1\). Determine the relation between the spectrum of the complex Laplacian \(\Box_\phi\) and the weight function \(\phi\). In particular, understand the bottom (infimum) of the essential spectrum.

来源字段：http://aimpl.org/crscv/6/

10730 | OWR-12005-002

### Hexagonal Asymptotics for Spectral Minimal Partitions

O / open | Oberwolfach 报告问题汇编

For a bounded planar domain $\Omega$, does the minimal $k$-partition energy satisfy the hexagonal asymptotic $\lim_{k\to\infty}L_k(\Omega)/k=\lambda(\mathrm{Hexa}_1)A(\Omega)^{-1}$, where $\mathrm{Hexa}_1$ is the regular unit-area hexagon?

来源字段：https://doi.org/10.4171/owr/2012/33

10731 | OWR-4211-004

### High-Order Impedance Conditions Near Boundary Corners

O / open | Oberwolfach 报告问题汇编

For thin-coating scattering problems whose boundary contains an angle, can one derive high-order generalized impedance boundary conditions, and can the resulting asymptotic models be given a rigorous analysis in time-domain and more general settings?

来源字段：https://doi.org/10.4171/owr/2010/10

10732 | OWR-7155438-002

### High-Order Variational Integrators for Degenerate Lagrangians

O / open | Oberwolfach 报告问题汇编

For general degenerate Lagrangians, can degenerate variational integrators be generalized to arbitrary order using Galerkin or Runge-Kutta discretizations; can they and symplectic Runge-Kutta methods be obtained without assuming any component of the symplectic potential vanishes; and do symplectic Runge-Kutta methods follow from a discrete variational principle?

来源字段：https://doi.org/10.4171/owr/2021/17

10733 | AIM-ANALYSIS-0116

### Higher-dimensional quantitative bi-Lipschitz parameterization and a scalar-curvature obstruction

P / partially_solved | AIM 研讨会问题汇编

**Bi-Lipschitz parameterization by Euclidean spaces.** Are there higher dimensional analogs of the Bonk--Lang parameterization theorem which give quantitative control on bi-Lipschitz constants?

来源字段：http://aimpl.org/mappingmetric/3/

10734 | OWR-10252938-001

### Hilbert Complexes in Numerical Partial Differential Equations

O / open | Oberwolfach 报告问题汇编

How did Hilbert complexes come to play a major role in numerical partial differential equations?

来源字段：https://doi.org/10.4171/owr/2022/29

10735 | OWR-14610-002

### Hodge Spectral Gaps Under Ricci Bounded Convergence

O / open | Oberwolfach 报告问题汇编

For a noncollapsed Gromov-Hausdorff convergent sequence with $|\operatorname{Ric}_{X_i}|\leq K$, is there a uniform spectral gap for the Hodge Laplacian on 1-forms and, equivalently, does $b_1(X_i)\to b_1(X)$, with the harmonic 1-form Betti number of the limit equal to the ordinary $b_1(X)$?

来源字段：https://doi.org/10.4171/owr/2016/28

10736 | AMR-036-0041

### Holomorphic curves with bounded spherical derivative

O / open | AMR 问题清单汇编

Let $f:\mathbb C\to\mathbb P^n$ be holomorphic with spherical derivative $\|f'\|(z)=O(|z|^\sigma)$ for some $\sigma>-1$, and let $a_1,\ldots,a_q$ be hyperplanes in general position not covering $f(\mathbb C)$. Prove $$\sum_{j=1}^qN(r,a_j,f)\ge(q+1-n)T(r,f)+O(r^{\sigma+1}).$$

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10737 | AMR-135-0001

### Holomorphic Foliation Theory - Problem 1

O / open | AMR 问题清单汇编

Instead of considering the weighted current $\tau^a_r$, we can consider an un-weighted current (*Ahlfors current*) $$\sigma_r^a:={1\over A(r)} (\phi_a)_*[\mathbb{D}(r)] \qquad \text{with} \qquad A(r):=\int_{\mathbb{D}(r)} \phi_a^*(\omega).$$

Do we have $\sigma_r^a\to T$ as $r$ tends to $1$ for $T$-almost every $a$ ?

背景链接：https://arxiv.org/abs/1807.05640

10738 | AMR-135-0010

### Holomorphic Foliation Theory - Problem 10

O / open | AMR 问题清单汇编

Let $Z$ be a holomorphic vector field in $\mathbb{C}^k$ which is not the product of a polynomial vector field with a holomorphic function. Let $\phi$ be the uniformization of a leaf $L$. The source could be $\mathbb{C}$ or the unit disc. Consider the Nevanlinna currents $$\tau^\phi_r:={1\over T(r)} \phi_*\Big(\log^+{r\over |\zeta|}\Big).$$ Are the cluster points of $\tau_r^\phi$, when $r$ tends to $\infty$ for the case of $\mathbb{C}$ and $r$ tends to $1$ for the case of the unit disc, supported necessarily by the hyperplane at infinity ?

背景链接：https://arxiv.org/abs/1807.05640

10739 | AMR-135-0011

### Holomorphic Foliation Theory - Problem 11

O / open | AMR 问题清单汇编

Find the minimal growth of the Nevanlinna characteristic function $T_h$ for $h\in\mathscr{U}$, where $$T_h(r):=\int_0^r {dt \over t} \int_{\mathbb{D}(t)} h^*(\omega)$$ and $\omega$ is the Fubini-Study form on $\mathbb{P}^1$.

背景链接：https://arxiv.org/abs/1807.05640

10740 | AMR-135-0012

### Holomorphic Foliation Theory - Problem 12

O / open | AMR 问题清单汇编

Recently, Benjamin Weiss has proved the existence of non-trivial probability measures on $\mathscr{H}$ which are invariant by the above translation operator, see [Weiss]. These measures are studied in particular by Buhovsky, Gl\"ucksam, Logunov and Sodin [BGLS]. A measure is *non-trivial* if it gives zero measure to the set of constant maps. Using standard results from ergodic theory, one gets the existence of non-trivial ergodic measures, i.e. measures such that the only invariant measurable subsets are of zero or full measure.

Given such a measure $\mu$ on $\mathscr{H}$ as above, what we can say about $\mu(\mathscr{U})$ ?

背景链接：https://arxiv.org/abs/1807.05640

10741 | AMR-135-0013

### Holomorphic Foliation Theory - Problem 13

O / open | AMR 问题清单汇编

Consider a complex manifold $M$ of dimension $n$ which is homogeneous under the action of a Lie group $G$. Let $\mathscr{M}$ denote the space of holomorphic maps from $M$ to $\mathbb{P}^n$, endowed with the topology of uniform convergence on compact sets. The group $G$ acts on this space.

(i) Under which conditions this action admits a dense orbit ?

(ii) Extend the construction by B. Weiss to this context.

背景链接：https://arxiv.org/abs/1807.05640

10742 | AMR-135-0014

### Holomorphic Foliation Theory - Problem 14

O / open | AMR 问题清单汇编

Explore the above properties for the following classes of maps in $\mathbb{C}^n$ : biholomorphisms of $\mathbb{C}^n$ and volume preserving (resp. symplectic) automorphisms of $\mathbb{C}^n$ (resp. $\mathbb{C}^n$ with $n$ even).

背景链接：https://arxiv.org/abs/1807.05640

10743 | AMR-135-0015

### Holomorphic Foliation Theory - Problem 15

O / open | AMR 问题清单汇编

In order to study foliations with leaves of dimension larger than $1$, what is the right notion which substitutes directed ${dd^c}$-closed currents in dimension $1$ that should be considered ? Such objects should permit to develop the ergodic theory for these foliations.

背景链接：https://arxiv.org/abs/1807.05640

10744 | AMR-135-0016

### Holomorphic Foliation Theory - Problem 16

O / open | AMR 问题清单汇编

Construct non-elementary foliations $\mathscr{F}$ in $\mathbb{P}^k$ with leaves of dimension larger than $1$ and such that every leaf does not contain a compact curve.

背景链接：https://arxiv.org/abs/1807.05640

10745 | AMR-135-0017

### Holomorphic Foliation Theory - Problem 17

O / open | AMR 问题清单汇编

Is it always possible to approximate the foliation $\mathscr{F}$ by foliations in $\mathbb{C}^k$ or\break $\mathbb{P}^k$ ? More precisely, is it always possible to approximate the forms $\alpha_j$, uniformly on compact sets of $\mathbb{D}^k$, by entire forms (resp. polynomial forms) $\alpha_{j,n}$ which still satisfy the above Frobenius condition ?

背景链接：https://arxiv.org/abs/1807.05640

10746 | AMR-135-0002

### Holomorphic Foliation Theory - Problem 2

O / open | AMR 问题清单汇编

Assume that $\mathscr{F}$ is a foliation of degree $d\geq 2$ with non-hyperbolic singularities. Describe the ${dd^c}$-closed positive currents, directed by the foliation. In which case diffuse positive closed currents directed by the foliation do exist ?

背景链接：https://arxiv.org/abs/1807.05640

10747 | AMR-135-0003

### Holomorphic Foliation Theory - Problem 3

O / open | AMR 问题清单汇编

Let $\mathscr{H}_d$ denote the set of foliations of degree $d$ with only hyperbolic singularities. This is an open real Zariski dense set in $\mathscr{B}_d$. It is easy to show that $\mathscr{B}_d^{a}\cap\mathscr{H}_d$ is an analytic set [FS3] because one can bound the degree of the curve, when all singularities are hyperbolic. Let $\mathscr{B}_d^{c}$ denote the set of foliations which admit directed positive closed currents. It is clearly a closed set in $\mathscr{B}_d$. Part of Theorem 3.1 can be stated as $ \mathscr{B}_d^{c}\cap\mathscr{H}_d=\mathscr{B}_d^{a}\cap\mathscr{H}_d$.

Is the space $\mathscr{B}_d^{c}$ a countable union of proper analytic subsets of $\mathscr{B}_d$, when $d\geq2$ ?

背景链接：https://arxiv.org/abs/1807.05640

10748 | AMR-135-0004

### Holomorphic Foliation Theory - Problem 4 (Anosov)

O / open | AMR 问题清单汇编

Consider foliations on $\mathbb{P}^2$. Recall that $\mathscr{B}_d$ denote the space of holomorphic foliations of geometric degree $d$ on $\mathbb{P}^2$. Let $\mathscr{A}_d$ denote the space of foliations given in an affine coordinate chart $\mathbb{C}^2$ in $\mathbb{P}^2$ by a vector field $$Z:=P(z,w) {\partial \over \partial z} + Q(z,w) {\partial \over \partial w}$$ with $\max(\deg P, \deg Q)=d$. This notion of degree is not invariant under automorphisms of $\mathbb{P}^2.$ Generically in $\mathscr{B}_d$, the foliations have no compact leaf, and generically in $\mathscr{A}_d$, the line at infinity is invariant.

For a generic foliation in $\mathscr{A}_d$ (resp. $\mathscr{B}_d$), are generic leaves biholomorphic to the unit disc ?

背景链接：https://arxiv.org/abs/1807.05640

10749 | AMR-135-0005

### Holomorphic Foliation Theory - Problem 5

O / open | AMR 问题清单汇编

Let $\mathscr{F}$ be a holomorphic foliation on $\mathbb{P}^2$. Denote by $\mathrm{sing}(\mathscr{F})$ the singular set of $\mathscr{F}$ which is finite.

Is it true that for every leaf $L$ of $\mathscr{F}$, we have $\overline L\cap \mathrm{Sing}(\mathscr{F})\not=\varnothing$ ?

背景链接：https://arxiv.org/abs/1807.05640

10750 | AMR-135-0006

### Holomorphic Foliation Theory - Problem 6

O / open | AMR 问题清单汇编

The question is quite old and is due to Camacho-Lins Neto-Sad [CLS]. The problem is also discussed in [FS3]. The answer is positive if $\mathscr{F}$ admits a directed positive closed $(1,1)$-current $T$, in particular, when $\mathscr{F}$ has an invariant algebraic curve. The support of this current $T$ should contain a singularity since otherwise the cohomology class of $T$ satisfies $\{T\}^2=0$ which is impossible in $\mathbb{P}^2$. Using the leaf $L$, we can construct a directed positive ${dd^c}$-closed current $S$ supported by $\overline L$. Again, for cohomological reason, one can show that $S$ and $T$ have to intersect and the intersection should be at the singularities as both currents are directed by the foliation. For this purpose, one can use the theory of densities for currents, see also [DS1].

Does there exist a Levi flat smooth or real analytic hypersurface in $\mathbb{P}^2$ ?

背景链接：https://arxiv.org/abs/1807.05640

10751 | AMR-135-0007

### Holomorphic Foliation Theory - Problem 7

O / open | AMR 问题清单汇编

(i) Assume that $\mathscr{F}$ is a foliation of geometric degree $d\geq 2$ on $\mathbb{P}^2$ with the line at infinity invariant and having only hyperbolic singularities. Do the leaves, except for the compact curves, have Green functions ?

(ii) More generally, assume that $\mathscr{F}$ is in $\mathscr{B}_d$ with $d\geq 2.$ What is the function theory of leaves (in terms of Green functions), when the foliation admits a non-zero, positive closed current directed by $\mathscr{F}$ ?

背景链接：https://arxiv.org/abs/1807.05640

10752 | AMR-135-0008

### Holomorphic Foliation Theory - Problem 8

O / open | AMR 问题清单汇编

Construct holomorphic minimal foliation (resp. lamination) on a compact complex manifold with uncountably many extremal ${dd^c}$-closed, non-closed, currents directed by the foliation (resp. lamination).

背景链接：https://arxiv.org/abs/1807.05640

10753 | AMR-135-0009

### Holomorphic Foliation Theory - Problem 9

O / open | AMR 问题清单汇编

Let $Z$ be a holomorphic vector field in $\mathbb{C}^2$.

(i) Is it possible to have a set of positive measure of leaves covered by $\mathbb{C}$ and another set of positive measure of leaves covered by the unit disc ?

(ii) If a non-pluripolar set of leaves are parabolic, are all leaves parabolic ?

(iii) Find conditions which imply that all leaves are covered by the unit disc.

背景链接：https://arxiv.org/abs/1807.05640

10754 | AIM-ANALYSIS-0053

### Holomorphic Neumann data as Hardy traces with period constraints

P / partially_solved | AIM 研讨会问题汇编

Characterize the boundary data space for Neumann boundary problem for holomorphic functions on planar domain. Are there any representation formula for the solution?

来源字段：http://aimpl.org/scvproblems/1/

10755 | AIM-OPTIMIZATION-0002

### Homogenization, exact PSD lift size, and hyperbolic cone obstructions

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.10.2.** Characterize all \(d\)-dimensional, pointed, closed, convex cones in \(\mathbb R^d\) which admit a poly-time, lifted-LMI representation. I believe that this set of cones strictly contain the hyperbolic cones.

来源字段：https://aimath.org/WWN/matrixineq/matrixineq.pdf

10756 | AIM-ANALYSIS-0028

### Hot spots: high-dimensional refutation, product rules, and heat-flow quantifiers

S / solved | AIM 研讨会问题汇编

Hot spots conjecture

This conjecture is often attributed to J. Rauch, who has formulated it in terms of large-time behaviuor of the heat semigroup generated by the Neumann Laplacian. On the qualitative level, this conjecture manifests as follows: for an insulated flat piece of metal with an arbitrary initial temperature distribution, given enough time, the hottest point on the metal will lie on its boundary.

The first non-trivial Neumann eigenfunction does not have global extrema in the interior of \(\Omega\) if

1. \(\Omega\subset\mathbb R^2\) is simply connected. 2. \(\Omega\subset\mathbb R^n\) (\(n\geq 3\)) is convex.

来源字段：http://aimpl.org/shapesurface/3/

10757 | AIM-ANALYSIS-0141

### Hurwitz minors as controlled Toeplitz minors and staircase Schur functions

P / partially_solved | AIM 研讨会问题汇编

Investigate the signs of Hurwitz minors for Pólya frequency sequences.

来源字段：http://aimpl.org/hyperbolicpoly/2/

10758 | OWR-14298796-004

### Improved Error Rates for QMC Slicing

O / open | Oberwolfach 报告问题汇编

Prove the improved convergence rates observed numerically when quasi-Monte Carlo designs replace independent random directions in sliced approximations of kernel sums.

来源字段：https://doi.org/10.4171/owr/2024/48

10759 | OWR-13491-001

### Inclusions and Approximability in Operator Ideals

O / open | Oberwolfach 报告问题汇编

For operator ideals generated by $s$-numbers, is $L_q^{\mathrm{hilb}}$ contained in $L_p^{\mathrm{app}}$ whenever $1/q$ is at least $1/p+1$, and for which $p$ are all operators in $L_p^{\mathrm{ent}}$ approximable?

来源字段：https://doi.org/10.4171/owr/2015/6

10760 | AIM-SEVERAL_COMPLEX_VARIABLES-0005

### Index one, global regularity, and Stein neighborhoods

P / partially_solved | AIM 研讨会问题汇编

Let \(\Omega\) be a smooth bounded pseudoconvex domain such that for any \(0<\eta<1\), there exists a defining function \(\rho_\eta\) such that \(-(-\rho_\eta)^\eta\) is plurisubharmonic on \(\Omega\). Note that such an \(\eta\) always exists [MR0430315]; however, it could be small. Determine minimal assumptions on \(\Omega\) to show that the \(\bar\partial\)-Neumann operator \(N\) (or the Bergman projection \(\mathbf B\)) is globally regular. Determine minimal assumptions on \(\Omega\) to show existence of a Stein neighborhood basis for the closure. Are there domains such that \(\mathbf B\) is globally regular but no such family of defining functions exists?

来源字段：http://aimpl.org/crscv/1/

10761 | AIM-ANALYSIS-0152

### Index tests and quantitative boundary-root pushing for zero-free Mergelyan approximation

P / partially_solved | AIM 研讨会问题汇编

If \(\mathbb C\setminus K\) is connected and \(f\in A(K)\) has no zeros in \(K^\circ\), must \(f\) be a uniform limit on \(K\) of polynomials having no zeros on \(K\)?

The live page was checked on 2026-07-28. It attributes the problem to P. Gauthier and reads:

The phrase “Let be \(K\) be” is a source typo; the intended opening is “Let \(K\subset\mathbb C\) be compact.” The page does not specify a topology for \(p_n\to f\). The intended meaning is **uniform convergence on \(K\)**:

\[ \|p_n-f\|_K:=\max_{z\in K}|p_n(z)-f(z)|\longrightarrow0. \]

This reconstruction is verified by the precise formulations in Andersson, Gauthier--Knese, Khrushchev, and Danielyan. It is also confirmed internally by adjacent AIM Conjecture 9.2, which uses \(\|\cdot\|_{\infty,K}\) and says that Andersson proved its equivalence to Problem 9.1.

\[ A(K)=C(K)\cap\mathcal O(K^\circ). \]

The recovered question is therefore:

来源字段：http://aimpl.org/hyperbolicpoly/9/

10762 | AMR-036-0012

### Indicators of linear combinations

O / open | AMR 问题清单汇编

Let $A$ be a set of vectors $a=(a_1,\ldots,a_n)\in\mathbb C^n$ such that every $n$ of them are linearly independent, and assign to every $a\in A$ a $\rho$-trigonometrically convex function $h_a$. Characterize when there exist entire functions $f_1,\ldots,f_n$ of order $\rho$ and normal type such that $a_1f_1+\cdots+a_nf_n$ has indicator $h_a$ for every $a$. Treat also the completely-regular-growth version.

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10763 | AIM-ANALYSIS-0089

### Induced ell-p paving endpoints and simultaneous all-p conference paving

P / partially_solved | AIM 研讨会问题汇编

For every \(\varepsilon>0\), is there an integer \(k=k(\varepsilon,p)\), independent of \(n\), such that every zero-diagonal \(n\times n\) matrix \(A\) with \(\|A\|_p=1\) has a partition \([n]=\sigma_1\sqcup\cdots\sqcup\sigma_k\) satisfying \[ \left\|\sum_{j=1}^kP_{\sigma_j}AP_{\sigma_j}\right\|_p =\max_j\|P_{\sigma_j}AP_{\sigma_j}\|_p\leq\varepsilon? \]

The record is Bill Johnson's question from the AIM workshop *Beyond Kadison--Singer: paving and consequences*. It asks, for \(1\leq p<\infty\):

The source has the typographical fragment `\|\dot\|_`; its following prose explicitly says that this is the **operator norm on \(\ell_p^n\)**. Accordingly, throughout this report \[ \|A\|_{p\to p}:=\sup_{x\ne0}\frac{\|Ax\|_p}{\|x\|_p}. \] This is not the Schatten norm \((\operatorname{tr}|A|^p)^{1/p}\). We write \(R_p(\varepsilon)\) for the least uniform number of blocks, when it exists, and \(p'=p/(p-1)\), with \(1'=\infty\).

The displayed equality in the question is correct, but it needs a Banach-space justification rather than a Hilbert-space orthogonality argument. If \(B_j=P_{\sigma_j}AP_{\sigma_j}\), then \(\ell_p^n=(\bigoplus_j\ell_p(\sigma_j))_p\) and \(\sum_jB_j=\bigoplus_jB_j\). Hence, for \(1\leq p<\infty\), \[ \left\|\bigoplus_jB_jx_j\right\|_p^p \leq \bigl(\max_j\|B_j\|_{p\to p}\bigr)^p\sum_j\|x_j\|_p^p. \] The reverse inequality follows by taking a norming (or approximately norming) vector supported in one maximizing block. Thus \[ \left\|\sum_jP_{\sigma_j}AP_{\sigma_j}\right\|_{p\to p} =\max_j\|P_{\sigma_j}AP_{\sigma_j}\|_{p\to p}. \] The same equality holds for \(p=\infty\).

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10764 | OPG-41335

### Inequality for square summable complex series

S / solved | Open Problem Garden

Conjecture For all $\alpha=(\alpha_1,\alpha_2,\ldots)\in l_2(\cal{C})$ the following inequality holds $$\sum_{n\geq 1}|\alpha_n|^2\geq \frac{6}{\pi^2}\sum_{k\geq0}\bigg|\sum_{l\geq0}\frac{1}{l+1}\alpha_{2^k(2l+1)}\bigg|^2$$

背景链接：http://www.openproblemgarden.org/op/inequality_for_square_summable_complex_series

10765 | AMR-023-0015

### Infinitely many Lehmer pairs

O / open | AMR 问题清单汇编

Are there infinitely many Lehmer pairs of zeros in the sense used in the theory of the de Bruijn-Newman constant?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767

10766 | OWR-17469-001

### Information Complexity of Multivariate Integration and Approximation

O / open | Oberwolfach 报告问题汇编

For a function class $F$ on a bounded domain $D_d\subset\mathbb R^d$, how many function values $f(x_i)$ are needed to approximate either the integral $\operatorname{INT}(f)$ or the function $\operatorname{APP}(f)$ with error at most $\varepsilon$?

来源字段：https://doi.org/10.4171/owr/2020/6

10767 | OWR-17468-003

### Inf-Sup Stability for Nonconforming Multiple Traces

O / open | Oberwolfach 报告问题汇编

For the spectral nonconforming Petrov-Galerkin multiple-traces discretization, prove an inf-sup condition for the variational problem of finding $\lambda\in X_L:=X_L^0\times\cdots\times X_L^M$ such that $\langle F(s)\lambda,\phi\rangle_\times=\langle g,\phi\rangle_\times$ for every $\phi\in Y_L:=Y_L^0\times\cdots\times Y_L^M$. Such a condition would establish existence and uniqueness of the discrete solution.

来源字段：https://doi.org/10.4171/owr/2020/5

10768 | OWR-1320-002

### Injectivity and Inversion of Restricted Spherical Radon Transforms

O / open | Oberwolfach 报告问题汇编

For the spherical Radon transform restricted to centers on a hypersurface $S$, determine injectivity, inversion formulas and their stability, the effect of incomplete data, and the range of $R_S$. In particular, does $R_Sf=0$ imply $f=0$ for compactly supported $f$?

来源字段：https://doi.org/10.4171/owr/2006/34

10769 | OWR-13103-005

### Integral Representatives of Normal Mod-Two Currents

O / open | Oberwolfach 报告问题汇编

别名：Cross-Report Duplicate of the Normal Mod-Two Current Question [OWR-13852-005]

Is every normal mod-2 current equivalent to a normal integral current?

来源字段：https://doi.org/10.4171/owr/2014/33

10770 | OWR-14298003-002

### Integration by Parts and Itô Formulas for Rough Paths

P / partially_solved | Oberwolfach 报告问题汇编

Which additional classes of rough paths satisfy integration-by-parts identities and therefore an Itô formula?

来源字段：https://doi.org/10.4171/owr/2023/54

10771 | OWR-1057-012

### Intermediate Bergman Backward-Shift Invariant Subspaces

O / open | Oberwolfach 报告问题汇编

For the backward shift $B$ on a Bergman space $A^p$, if $M_1\subsetneq M_2$, does there exist a $B$-invariant subspace $M$ strictly between them?

来源字段：https://doi.org/10.4171/owr/2005/36

10772 | OWR-12866-011

### Intermediate BMO Spaces for Dirichlet Series

O / open | Oberwolfach 报告问题汇编

Does there exist a BMO-type space $\mathcal B$ for Dirichlet series, intermediate between $H^\infty$ and the candidate Dirichlet-series BMO spaces, such that: for every $f\in\mathcal B$, $\sigma>0$, and $p<\infty$, $$\limsup_{T\to\infty}\frac1{2T}\int_{-T}^{T}|f(\sigma+it)|^p\,dt<\infty;$$ and, for every $\sigma>1/2$, the corresponding horizontal shift of the alternating zeta function belongs to $\mathcal B$?

来源字段：https://doi.org/10.4171/owr/2014/06

10773 | OWR-1057-002

### Intermediate Invariant Subspaces Between Nested Subspaces

O / open | Oberwolfach 报告问题汇编

For an operator $T$, if $M_1\subset M_2$ in $\operatorname{Lat}T$ and $\dim(M_2/M_1)>1$, does there exist $M\in\operatorname{Lat}T$ strictly between $M_1$ and $M_2$?

来源字段：https://doi.org/10.4171/owr/2005/36

10774 | AIM-ANALYSIS-0046

### Intermediate pseudoconvex domains above Fatou-Bieberbach domains

P / partially_solved | AIM 研讨会问题汇编

Given a Fatou-Bieberbach domain $\Omega$, must there exist a pseudoconvex domain $\tilde{\Omega}$ such that $\tilde{\Omega}$ with $\Omega \subsetneq \tilde{\Omega} \subsetneq \mathbb{C}^n$.

**Recovered problem.** For \(n\geq2\), given a Fatou--Bieberbach domain \(\Omega\subsetneq\mathbb C^n\), must there exist a connected pseudoconvex domain \(D\) such that \[ \Omega\subsetneq D\subsetneq\mathbb C^n? \]

来源字段：http://aimpl.org/scvproblems/1/

10775 | OWR-4209-006

### Interpolation of Sobolev Extension Domains and Jacobian Signs

P / partially_solved | Oberwolfach 报告问题汇编

If $\Omega\subset\mathbb R^n$ is both a $W^{1,p}$- and a $W^{1,q}$-extension domain with $1\leq p<q\leq\infty$, must it be a $W^{1,r}$-extension domain for every $p<r<q$? Also, can a Sobolev homeomorphism $f:\Omega_1\to\Omega_2$ have $J_f>0$ on a set of positive measure and $J_f<0$ on a set of positive measure?

来源字段：https://doi.org/10.4171/owr/2010/08

10776 | OWR-15582-009

### Intrinsic $L^2$ Extension Beyond Nilpotent Sections

O / open | Oberwolfach 报告问题汇编

Let $0=m_0<m_1<\cdots$ be the jumping numbers of a multiplier-ideal filtration $\mathcal I(m\psi)$, let $J=\mathcal I(m_p\psi):\mathcal I(m_{p-1}\psi)$, and let $Y=V(J)$. Without requiring a section $f$ on $Y$ to be nilpotent and associated with $\mathcal I(m_{p-1}\psi)/\mathcal I(m_p\psi)$, is there an intrinsic $L^2$ norm on $f$ and a corresponding intrinsic $L^2$ extension estimate?

来源字段：https://doi.org/10.4171/owr/2017/39

10777 | OPG-36697

### Invariant subspace problem

O / open | Open Problem Garden

Problem Does every bounded linear operator on an infinite-dimensional separable Hilbert space have a non-trivial closed invariant subspace?

背景链接：http://www.openproblemgarden.org/op/invariant_subspace_problem

10778 | OWR-1057-003

### Invariant Subspaces of Weighted Translation Operators

O / open | Oberwolfach 报告问题汇编

Let $\omega=(\omega(n))_{n\in\mathbb Z}$ be positive and satisfy $c^{-1}\leq\omega(n+1)/\omega(n)\leq c$ for all $n$. Must the translation-invariant Hilbert space $\ell_\omega^2(\mathbb Z)$ have a nontrivial translation-invariant subspace?

来源字段：https://doi.org/10.4171/owr/2005/36

10779 | OWR-4797-002

### Inverse Identification of Nonseparated Inclusions

O / open | Oberwolfach 报告问题汇编

Can unknown non-separated inclusions be identified from the available inverse-problem data?

来源字段：https://doi.org/10.4171/owr/2011/05

10780 | OWR-12595-003

### Involution Structure and Stability of MHD Solvers

P / partially_solved | Oberwolfach 报告问题汇编

What is the role of the involution structure in numerical calculations, and can it influence the stability of Riemann solvers for ideal magnetohydrodynamics?

来源字段：https://doi.org/10.4171/owr/2013/47

10781 | OWR-12595-001

### Involutions and Accuracy in Constrained PDE Methods

O / open | Oberwolfach 报告问题汇编

For involution-constrained PDEs in magnetohydrodynamics and nonlinear elasticity, how can the proper involutions be identified; what is the relation between enforcing involutions and using higher-order methods; and are lower-order methods with involution constraints sufficient?

来源字段：https://doi.org/10.4171/owr/2013/47

10782 | OWR-13103-001

### Involutivity Obstructions for Normal Currents

P / partially_solved | Oberwolfach 报告问题汇编

Let $X(x,y,z)=(1,0,-x/2)$ and $Y(x,y,z)=(0,1,y/2)$ be the horizontal vector fields on the Heisenberg group $\mathbb H^1\cong\mathbb R^3$, and put $\xi=X\wedge Y$. If $T=\xi\wedge\mu$ is a normal $2$-current in $\mathbb R^3$, must $\mu$ be absolutely continuous with respect to Lebesgue measure? More generally, can normal currents that fail the Frobenius-type involutivity conclusion for integral currents be classified, and what remains true when the vector field has less than $C^1$ regularity or involutivity is defined without Lie brackets?

来源字段：https://doi.org/10.4171/owr/2014/33

10783 | OWR-4807-012

### Isomorphism of Free Group Factors

O / open | Oberwolfach 报告问题汇编

For free groups $F_n$ and $F_m$ with $n,m>1$ and $n\neq m$, are the corresponding free-group von Neumann factors isomorphic? Equivalently, for a nontrivial projection $P\in\lambda(F_n)'$, is $\lambda(F_n)'$ $*$-isomorphic to $P\lambda(F_n)'P$?

来源字段：https://doi.org/10.4171/owr/2011/17

10784 | OWR-14298372-001

### Isomorphism of Free Group Factors

O / open | Oberwolfach 报告问题汇编

Are the free group factors $L(\mathbb F_2)$ and $L(\mathbb F_3)$ isomorphic?

来源字段：https://doi.org/10.4171/owr/2024/22

10785 | OWR-1277-002

### Joint $\Gamma$-Convergence for Evolutionary Variational Problems

O / open | Oberwolfach 报告问题汇编

Let $(\mathcal E_k,\mathcal R_k)$, or $(\mathcal E_k,\mathcal K_k)$ in the Hamiltonian case, be pairs of time-dependent functionals whose two components $\Gamma$-converge independently to $(\mathcal E,\mathcal R)$, respectively $(\mathcal E,\mathcal K)$. If associated solutions $q_k$ converge pointwise to $q$, what joint $\Gamma$-convergence conditions ensure that $q$ solves the evolution problem determined by the pair $(\mathcal E,\mathcal R)$, respectively $(\mathcal E,\mathcal K)$?

来源字段：https://doi.org/10.4171/owr/2006/28

10786 | OWR-1115-007

### Jump Processes from Families of Lévy Measures

P / partially_solved | Oberwolfach 报告问题汇编

Given a family of Lévy measures $\mu(x,dh)$ on $\mathbb{R}^d$, determine when it defines corresponding jump processes; obtain their heat-kernel properties; determine regularity of their harmonic functions; and determine whether the generator extends to that of a Feller semigroup.

来源字段：https://doi.org/10.4171/owr/2005/54

10787 | AIM-SPECIAL_FUNCTIONS-0005

### k-Schur functions as affine Grassmannian Schubert bases

S / solved | AIM 研讨会问题汇编

**(5)** Prove that the dual $k$-Schur functions are the symmetric component of affine Schubert polynomials. Prove also that the $k$-Schur functions are a Schubert basis of the homology of the affine Grassmannian, and that the dual $k$-Schur functions give a Schubert basis of the cohomology of the affine Grassmannian.

来源字段：https://aimath.org/WWN/kostka/prob2.pdf

10788 | AIM-FUNCTIONAL_ANALYSIS-0021

### K1 sign normal form for the Calkin shift and independence of the commutative analogue

P / partially_solved | AIM 研讨会问题汇编

Is there a model of set theory in which the Calkin algebra has an automorphism sending the unilateral shift \(S\) to \(S^\ast\)? Is the analogous fact for \(\ell^\infty\left/c_0\right.\) true?

来源字段：http://aimpl.org/settheorycstar/2/

10789 | AIM-FUNCTIONAL_ANALYSIS-0031

### Kadison-Singer via cardinal-uniform paving

S / solved | AIM 研讨会问题汇编

Does every pure state of the atomic masa in \(\mathcal{B}\left(H\right)\) extend uniquely to a pure state of \(\mathcal{B}\left(H\right)\)?

来源字段：http://aimpl.org/settheorycstar/5/

10790 | OWR-14299906-010

### Kakeya Conjecture in Dimensions Four and Higher

P / partially_solved | Oberwolfach 报告问题汇编

For every integer $d\geq 4$, prove that every Kakeya set in $\mathbb R^d$ has Hausdorff dimension $d$.

来源字段：https://doi.org/10.4171/owr/2026/4

10791 | OWR-14299906-001

### KMS-Symmetric Quantum Markov Semigroups on von Neumann Algebras

O / open | Oberwolfach 报告问题汇编

Extend the derivation-correspondence characterization of bounded generators of KMS-symmetric quantum Markov semigroups from matrix algebras to arbitrary von Neumann algebras.

来源字段：https://doi.org/10.4171/owr/2026/4

10792 | AMR-023-0005

### Kung-Traub conjecture

O / open | AMR 问题清单汇编

For an iteration without memory that uses $n$ evaluations of a function or its derivatives per step, is its convergence order always at most $2^{n-1}$?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767

10793 | AIM-ANALYSIS-0114

### Lacunary orthogonal layers admit nonlinear dimension reduction

P / partially_solved | AIM 研讨会问题汇编

**Bi-Lipschitz embeddings: Hilbert spaces vs. finite-dimensional Euclidean spaces.** Suppose that $X$ is a doubling space which admits a bi-Lipschitz embedding into Hilbert space. Does $X$ necessarily admit a bi-Lipschitz embedding into some finite-dimensional Euclidean space?

Does every doubling subset of Hilbert space admit a bi-Lipschitz embedding into $\mathbb R^N$ for some finite $N$?

来源字段：http://aimpl.org/mappingmetric/2/

10794 | OWR-12012-005

### Laplacians on Nonsymmetric Self-Similar Sets

O / open | Oberwolfach 报告问题汇编

Does a Laplacian exist on every connected nonsymmetric self-similar set?

来源字段：https://doi.org/10.4171/owr/2012/40

10795 | AIM-ANALYSIS-0151

## Large-power zero-free disks for powers of the Riemann xi kernel

P / partially_solved | AIM 研讨会问题汇编

Are there any non-real zeros of \[ \int_0^\infty \Phi^\alpha(t)\cos(zt)\,dt \qquad\text{for}\qquad \alpha>0? \]

In its section “7. Riemann \(\Xi\)-function,” it defines \[ \Xi_{\mathrm{AIM}}(z)=\int_{-\infty}^{\infty}\Phi(t)\cos(zt)\,dt, \qquad \Phi(t)=\sum_{n=1}^{\infty} \bigl(2n^4\pi^2e^{9t}-3n^2\pi e^{5t}\bigr) e^{-n^2\pi e^{4t}}, \tag{1} \] and displays the question as **Problem 7.2**, attributed to J. Haglund.

Csordas's Theorems 4.1 and 4.5 verify that the function in (1) extends evenly to the real line, is \(C^\infty\), strictly positive, strictly decreasing on \((0,\infty)\), superexponentially decreasing, and strictly log-concave. Thus, for every real \(\alpha>0\), the notation \(\Phi(t)^\alpha\) has the unambiguous meaning \[ \Phi(t)^\alpha=\exp\!\bigl(\alpha\log\Phi(t)\bigr)>0. \] Set \[ F_\alpha(z):=\int_0^\infty \Phi(t)^\alpha\cos(zt)\,dt. \tag{2} \]

来源字段：http://aimpl.org/hyperbolicpoly/7/

10796 | OWR-4531-020

## Law-of-Iterated-Logarithm Bounds for Dyadic Martingales

S / solved | Oberwolfach 报告问题汇编

For dyadic intervals $I_k(x)$ of length $2^{-k}$, let $f_I$ be the interval mean and $Q_n(x)=\sum_{k\geq n}(f_{I_{k+1}(x)}-f_{I_k(x)})^{2}$. Is it true that $\limsup_{n\to\infty}\frac{|f_{I_n(x)}-f(x)|}{\sqrt{Q_n(x)|\log|\log Q_n(x)||}}<\infty$ for almost every $x\in\mathbb{R}$?

来源字段：https://doi.org/10.4171/owr/2010/49

10797 | OWR-14299513-005

## Legendre-Assouad Bounds for Fractal Local Smoothing

O / open | Oberwolfach 报告问题汇编

For $p>2$, does $\left(\sum_{t\in E_j}\lVert e^{it\sqrt{-\Delta}}P_jf\rVert_{L^p(\mathbb{R}^d)}^p\right)^{1/p}\leq C2^{js}\lVert f\rVert_{L^p(\mathbb{R}^d)}$ hold whenever $s>p^{-1}\nu_E^{\#}(ps_p)$, where $s_p=(d-1)(1/2-1/p)$ and $\nu_E^{\#}$ is the Legendre-Assouad function of $E$?

来源字段：https://doi.org/10.4171/owr/2025/32

10798 | AIM-ANALYSIS-0048

## Levi-flat reduction and a global first-integral obstruction for real-analytic Fatou-Bieberbach boundaries

P / partially_solved | AIM 研讨会问题汇编

Is there a Fatou-Bieberbach domain with real analytic boundary?

来源字段：http://aimpl.org/scvproblems/1/

10799 | AMR-036-0017

## Levin's derivative-zero problem

O / open | AMR 问题清单汇编

Let $f$ be entire and suppose every zero of every derivative $f^{(n)}$, $n\ge0$, lies in the closed lower half-plane. Must $f$ lie in the compact-open closure of polynomials whose zeros lie in that half-plane, or have one of the forms $ce^{az}$ and $c(e^{ibz}-e^{id})$, where $c,a\in\mathbb C$ and $b,d\in\mathbb R$?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10800 | OWR-3389-017

## Lieb-Thirring Refinements and the Ovals Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $E_1<E_2\leq\cdots\leq0$ be the negative eigenvalues of $-\Delta-V$ on $L^2(\mathbb R^n)$, where $V\geq0$. When a power of $V$ is integrable strongly enough to make the left-hand side finite, does $\sum_j|E_j|^\gamma\leq C|E_1|^\gamma$ hold with $C$ depending only on $n$, $\gamma$, and that integrability data? In one dimension with $\gamma=1$, can the Eden-Foias result be proved directly or sharpened by correction terms, and can the eigenvalue sum be estimated without the Birman-Schwinger transformation? Finally, among smooth closed curves of length $2\pi$ in $\mathbb R^3$, is the principal eigenvalue of $-d^2/ds^2+\kappa(s)^2$ minimized by the circle?

来源字段：https://doi.org/10.4171/owr/2009/06

10801 | OWR-13750332-002

## Limits of Phase Retrieval and Tomographic Imaging

O / open | Oberwolfach 报告问题汇编

How much improvement in phase retrieval and tomographic imaging can be obtained by fully exploiting indirect measurements, sophisticated analysis, and high-performance algorithms?

来源字段：https://doi.org/10.4171/owr/2023/21

10802 | AIM-ANALYSIS-0104

## Linear bi-Lipschitz extension constants and affine perturbative certificates

P / partially_solved | AIM 研讨会问题汇编

**Linear dependence of constants in bi-Lipschitz extension theorems.** Do bi-Lipschitz extension theorems admit linear dependence on constants?

来源字段：http://aimpl.org/mappingmetric/1/

10803 | OWR-14298796-002

### Linear Error Growth for Large Operator Perturbations

O / open | Oberwolfach 报告问题汇编

Prove that the approximation-error bound for the adjoint-free operator-learning method continues to grow linearly with the perturbation size beyond the small-perturbation regime.

来源字段：https://doi.org/10.4171/owr/2024/48

10804 | OWR-720-003

### Linear Independence of Four Time-Frequency Shifts

P / partially_solved | Oberwolfach 报告问题汇编

For every nonzero $g\in L^2(\mathbb{R})$, are the four functions $g(x)$, $g(x-1)$, $g(x-2)$, and $e^{2\pi i x}g(x)$ linearly independent?

来源字段：https://doi.org/10.4171/owr/2004/10

10805 | AIM-ANALYSIS-0110

### Lipschitz homotopy of Heisenberg groups and an infinite-order Hopf class

P / partially_solved | AIM 研讨会问题汇编

Let \(\mathbb H^n\) be the \(n\)-th Heisenberg group with its Carnot--Carathéodory metric, and let \(\pi_k^{\mathrm{Lip}}(X)\) denote the \(k\)-th Lipschitz homotopy group. If \(\pi_k(\mathbb S^n)\) is nontrivial, is \(\pi_k^{\mathrm{Lip}}(\mathbb H^n)\) nontrivial? In particular, is \(\pi_3^{\mathrm{Lip}}(\mathbb H^2)\ne0\)?

来源字段：http://aimpl.org/mappingmetric/1/

10806 | AIM-SEVERAL_COMPLEX_VARIABLES-0041

### Literal and boundary counterexamples for Hermitian sums of squares

S / solved | AIM 研讨会问题汇编

**1. Sums of squares of polynomials. (Proposed by John D'Angelo)** Let \(\Omega\subset\mathbb C^n\) be a strongly pseudoconvex domain with compact algebraic boundary. Let \(R(z,\overline z)\) be a real polynomial which is positive on the boundary of \(\Omega\). Does there exist a positive integer \(k\) and polynomials \(p_1(z),p_2(z),\ldots,p_k(z)\) such that \[ R(z,\overline z)=\sum_{j=1}^k |p_j(z)|^2 \] on \(\Omega\)? The answer is known to be yes if \(\Omega\) is the unit ball in \(\mathbb C^n\). (See [CD].)

[CD] David W. Catlin and John P. D'Angelo, “A stabilization theorem for Hermitian forms and applications to holomorphic mappings,” *Mathematical Research Letters* 3, no. 2 (1996), 149--166.

来源字段：https://aimath.org/WWN/complexitycrmap/complexitycrmap.pdf

10807 | AIM-FUNCTIONAL_ANALYSIS-0001

### Literal equality correction and direct-sum tensorization for log-Brunn--Minkowski

P / partially_solved | AIM 研讨会问题汇编

**Log-Brunn-Minkowski Inequality.** For convex bodies \(K\) and \(L\) in \(\mathbb R^n\) with support functions \(h_K\) and \(h_L\), the logarithmic sum \(tK+_\circ(1-t)L\) is defined as

\[ tK+_\circ(1-t)L= \bigcap_{u\in S^{n-1}} \{x\in\mathbb R^n:\langle x,u\rangle \le h_K(u)^t h_L(u)^{1-t}\}, \]

where \(h_K(u)=\max_{x\in K}\langle x,u\rangle\). Show that if \(K\) and \(L\) symmetric convex bodies in \(\mathbb R^n\), then for all \(0\le t\le1\) the following **hods**

\[ |tK+_\circ(1-t)L|\ge |K|^t|L|^{1-t} \]

with equality if and only if \(K=L\) or \(K\) and \(L\) are parallelograms with parallel sides.

来源字段：http://aimpl.org/symconvgeomineq/1/

10808 | AMR-036-0029

### Littlewood constants $lpha$ and $eta$

P / partially_solved | AMR 问题清单汇编

For $\phi(n)=\sup_{\deg p=n}\int_{|z|<1}|p'|/(1+|p|^2)\,dm$, let $\alpha=\limsup\log\phi(n)/\log n$. For a regular compact set $E$, define $\beta_E$ from the growth of lengths of Green-function level curves, and let $\beta=\sup_E\beta_E$ over connected $E$. Is there a connection between $\alpha$ and $\beta$, and in particular is $\alpha=\beta$?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10809 | AIM-SPECIAL_FUNCTIONS-0008

### LLT expansions in graded k-Schur functions: status, explicit families, and a normalization obstruction

P / partially_solved | AIM 研讨会问题汇编

**(8)** Describe the expansions of LLT polynomials in terms of \(k\)-Schur functions. This problem may serve as a stepping-stone toward the problem of computing Schur expansions of LLT polynomials. For instance, LLT polynomials are \(k\)-Schur functions in certain special cases.

来源字段：https://aimath.org/WWN/kostka/prob2.pdf

10810 | OWR-4533-001

### Local Compactness and Nuclear-Space Extensions of Character Spectra

O / open | Oberwolfach 报告问题汇编

Let $\mathcal L\cong C_0(X)$ be a commutative $C^*$-algebra, and let $X_0\subseteq X$ correspond to the characters in $\operatorname{Rep}_0(\mathcal L,\mathbb C)$. Is $X_0$ locally compact in the relative topology? Does an analogous character-spectrum construction exist for separable nuclear spaces, extending the case of $\mathbb R^{\mathbb N}$?

来源字段：https://doi.org/10.4171/owr/2010/51

10811 | OWR-12979-004

### Local Geometry and Algebraic Classification of Multiplier Varieties

O / open | Oberwolfach 报告问题汇编

For multiplier varieties, is there an intrinsic local definition of a variety, and are the classical irreducible components varieties in that sense? Are multiplier-biholomorphic irreducible varieties in a finite-dimensional ball associated with isomorphic multiplier algebras, and is multiplier biholomorphism an equivalence relation? Finally, if $f:\mathbb D\to\mathbb B_\infty$ is a proper embedding with $f(0)=0$, is evaluation at $0$ the only character that vanishes on every coordinate multiplier?

来源字段：https://doi.org/10.4171/owr/2014/21

10812 | OWR-8415348-002

### Local Smoothing for the Euclidean Wave Equation

P / partially_solved | Oberwolfach 报告问题汇编

For the Euclidean half-wave propagator on $\mathbb{R}^n$, prove the local smoothing estimate $\left(\int_0^1\lVert e^{it\sqrt{-\Delta}}f\rVert_{L^p(\mathbb{R}^n)}^p\,dt\right)^{1/p}\leq C_\varepsilon\lVert f\rVert_{W^{\sigma(p)+\varepsilon,p}(\mathbb{R}^n)}$ for every $\varepsilon>0$ in the conjectured range, where $\sigma(p)=0$ for $2<p\leq 2n/(n-1)$ and $\sigma(p)=(n-1)(1/2-1/p)-1/p$ for $p>2n/(n-1)$.

来源字段：https://doi.org/10.4171/owr/2021/45

10813 | AIM-FUNCTIONAL_ANALYSIS-0028

### Local type I approximation and the UCT barrier

P / partially_solved | AIM 研讨会问题汇编

Is every stably finite nuclear separable C*-algebra locally approximable by type I C*-algebras?

来源字段：http://aimpl.org/settheorycstar/4/

10814 | OWR-4807-011

### Localization Principles for Lie Algebras

P / partially_solved | Oberwolfach 报告问题汇编

Which notions of localization should replace the classical uncertainty principle for Lie algebras?

来源字段：https://doi.org/10.4171/owr/2011/17

10815 | AMR-036-0020

### Locally constant logarithmic potentials

P / partially_solved | AMR 问题清单汇编

Let $\mu$ be a positive plane measure with $\mu(\{|z|\le r\})\le cr^\alpha$ for some $0<\alpha<1/2$. Can its logarithmic potential $$u(z)=\int\log\left|1-\frac z\zeta\right|\,d\mu(\zeta)$$ be locally constant on an open set meeting every circle centered at the origin?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10816 | AIM-ANALYSIS-0135

### Log-curvature obstructions and exact interpolation for n at most 2

P / partially_solved | AIM 研讨会问题汇编

Let \(\{c_k\}_{k=0}^n\subset\mathbb R\) be given. Does there exist a real rooted polynomial, \(p(x)\), with zeros outside \([0,n]\) such that \(c_k=p(k)\), \(k=0,1,\ldots,n\)?

来源字段：http://aimpl.org/hyperbolicpoly/1/

10817 | AIM-ANALYSIS-0050

### Logarithmic lattice-packing reduction and necessary character constraints

P / partially_solved | AIM 研讨会问题汇编

Is there a Fatou-Bieberbach domain contained in $\mathbb{C}^*\times\mathbb{C}^*$?

来源字段：http://aimpl.org/scvproblems/1/

10818 | OWR-4337-005

### Logarithmic Sobolev Inequalities for Nonconvex Spin Systems

O / open | Oberwolfach 报告问题汇编

For spin systems with a nonconvex Hamiltonian, what conditions imply a logarithmic Sobolev inequality and the associated hydrodynamic-limit estimates?

来源字段：https://doi.org/10.4171/owr/2010/24

10819 | OWR-4806-002

### Long-Time Action Preservation for Fourier Splitting of NLS

O / open | Oberwolfach 报告问题汇编

For a fully discrete Fourier splitting method for the one-dimensional cubic nonlinear Schrödinger equation, let $I_j^{K,n}=|\xi_j^n|^2$ denote the numerical actions, with $K$ prime. A bound $|I_j^{K,n}-I_j^{K,0}|\leq C\varepsilon$ is known for $n\tau\leq T/\varepsilon$. Does a comparable action-preservation bound hold up to times of order $\varepsilon^{-r}$ for large $r$?

来源字段：https://doi.org/10.4171/owr/2011/16

10820 | OWR-4990379-004

### Long-Time Relations between Mean-Field Games and Aggregation Equations

O / open | Oberwolfach 报告问题汇编

What is the relationship, if any, between the long-time behavior of mean-field-game systems and that of the associated aggregation or kinetic equation?

来源字段：https://doi.org/10.4171/owr/2021/10

10821 | OWR-723-004

### Low-Randomness Constructions of Low-Discrepancy Point Sets

P / partially_solved | Oberwolfach 报告问题汇编

Can one construct, in time polynomial in $n$ and $d$, an $n$-point set $M_n\subset[0,1]^d$ with star discrepancy $O(\sqrt{d/n})$, or at least find a small subset of $[0,1]^{dn}$ guaranteed to contain such a set?

来源字段：https://doi.org/10.4171/owr/2004/13

10822 | OWR-1058-003

### Low-Regularity Right-Hand-Side Approximation in Adaptive Methods

O / open | Oberwolfach 报告问题汇编

If the solution $u$ belongs to the approximation class $\mathcal A^s$ for $0<s\leq1/2$, does the corresponding right-hand side, for every $n$, have a piecewise-constant approximation on an $n$-element partition with error $O(n^{-s})$ in the dual norm?

来源字段：https://doi.org/10.4171/owr/2005/37

10823 | OWR-720-006

### Lower Density of Wave-Packet Frames

O / open | Oberwolfach 报告问题汇编

If a wave-packet system generated by $\psi\in L^2(\mathbb{R})$ and a discrete parameter set $M$ has an upper frame bound, then the lower dimension of $M$ should equal $0$.

来源字段：https://doi.org/10.4171/owr/2004/10

10824 | OWR-17130-009

### Martinez-Maure Support Functions as Singular Minimizers

O / open | Oberwolfach 报告问题汇编

Is the support function $R$ of the Martínez-Maure example a Lipschitz singular minimizer of

$$\int_{B_1}F(\nabla u)\,dx$$

for some integrand $F$ whose singular set $D_F$ consists exactly of the four noncoplanar cusps?

来源字段：https://doi.org/10.4171/owr/2019/34

10825 | OWR-13674-017

### Matrix Spencer Signing Conjecture

O / open | Oberwolfach 报告问题汇编

Given $n\times n$ matrices $A_1,\ldots,A_n$ with $\lVert A_i\rVert\leq 1$, does there exist a signing $s_i\in\{\pm1\}$ such that $\left\lVert\sum_{i=1}^n s_iA_i\right\rVert=O(\sqrt n)$? More generally, for $m>n$, is the bound $O\left(\sqrt{n\log(m/n)+1}\right)$ valid?

来源字段：https://doi.org/10.4171/owr/2015/17

10826 | OWR-12005-005

### Maximal Normalized Steklov Eigenvalues

O / open | Oberwolfach 报告问题汇编

For a fixed $k$, how large can the $k$th Steklov eigenvalue $\sigma_k$ be when boundary measure is normalized; equivalently, what is the maximum of $\sigma_k(g)|\Sigma|_g^{1/n}$ over smooth Riemannian metrics?

来源字段：https://doi.org/10.4171/owr/2012/33

10827 | AMR-036-0032

### Maximizing quadratic pressure

P / partially_solved | AMR 问题清单汇编

For which parameters $c$ is the pressure $P_c$ of the hyperbolic quadratic polynomial $p_c(z)=z^2+c$, for the potential $|(p_c^n)'|^{-1}$, close to or equal to its supremum?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10828 | AMR-036-0037

### Maximum length of a polynomial lemniscate

P / partially_solved | AMR 问题清单汇编

For a monic polynomial $p$ of degree $d$, determine the maximum length of the lemniscate $E(p)=\{z:|p(z)|=1\}$. Is the extremal asymptotically $p(z)=z^d+1$, giving maximum length $2d+o(1)$ as $d\to\infty$?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10829 | OWR-15577-004

### Maximum Principle for Intermediate Riesz Kernels

P / partially_solved | Oberwolfach 报告问题汇编

For $s\in(1,d-1)$ and $K(x)=x/|x|^{s+1}$, does there exist $C>0$ such that every nonnegative smooth compactly supported function $f$ satisfies $$ \sup_{x\in\mathbb R^d}|K*f(x)|\leq C\sup_{x\in\operatorname{supp}f}|K*f(x)| $$?

来源字段：https://doi.org/10.4171/owr/2017/34

10830 | AMR-023-0007

### Mean value problem for polynomial critical points

O / open | AMR 问题清单汇编

Given a complex polynomial $f$ of degree $d\geq2$ and $z\in\mathbb{C}$, must there be a critical point $c$ of $f$ such that $|f(z)-f(c)|\leq |f'(z)|\,|z-c|$?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767

10831 | OWR-16632-004

### Mean-Field Limits with Finite-Volume Effects

O / open | Oberwolfach 报告问题汇编

In the presence of finite-volume effects, can one rigorously justify the limiting passage from the microscopic particle system to the macroscopic mean-field system?

来源字段：https://doi.org/10.4171/owr/2018/49

10832 | OWR-1062-009

### Measure-Topology Continuity and the Algebra $M^A$

O / open | Oberwolfach 报告问题汇编

Let $A$ be a weakly dense $*$-subalgebra of a finite von Neumann algebra $(M,\tau)$, and let $M^A$ be the smallest subalgebra of $M$ closed under taking weak closures of abelian $*$-subalgebras. When is $M^A=M$? Is every derivation $\Delta:M\to U(M\mathbin{\overline\otimes}M^{\mathrm{op}})$ automatically continuous from bounded convergence in measure to convergence in measure?

来源字段：https://doi.org/10.4171/owr/2005/41

10833 | AMR-036-0039

### Median inequality for three subharmonic functions

O / open | AMR 问题清单汇编

Let $u_1,u_2,u_3$ be subharmonic in the plane with $u_j(0)=0$, and let $v_1\le v_2\le v_3$ be their pointwise increasing rearrangement. Put $I(r,v)=\int_0^{2\pi}v(re^{i\theta})\,d\theta$ and $B(r)=\max_{|z|=r}v_3(z)$. Prove $$\sup_{r>0}\frac{I(r,v_2)}{B(r)}\ge0,$$ or the stronger assertion with $\limsup_{r\to0}$.

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10834 | AIM-ANALYSIS-0136

### Meromorphic sampling, gauge freedom, and exponential-rational rigidity

P / partially_solved | AIM 研讨会问题汇编

Let $f(z)$ be a meromorphic function. When is the sequence $\{f(k)\}_{k=0}^{\infty}$ a multiplier sequence?

来源字段：http://aimpl.org/hyperbolicpoly/1/

10835 | OWR-3474-002

### Metric Diameters of Measures with Prescribed Interval Masses

S / solved | Oberwolfach 报告问题汇编

Given $h>0$ and $a_n\geq0$, let $A=\{\mu\in M_+(\mathbb{R}_+):\mu([nh,(n+1)h])=a_n\text{ for every }n\geq1\}$. Determine the diameters of $A$ in the flat metric $\rho_F$ and the Wasserstein-type metric $\rho_W$, and describe their dependence on $h$.

来源字段：https://doi.org/10.4171/owr/2009/24

10836 | OWR-14297736-031

### Metric-Independent Bounds for Bakry-Émery Eigenvalues

O / open | Oberwolfach 报告问题汇编

For the normalized eigenvalues $\overline\lambda_i(M,g,e^{-f})$ of the Bakry-Émery Laplacian on a closed surface, find an upper bound independent of the metric $g$ and weight $f$.

来源字段：https://doi.org/10.4171/owr/2023/36

10837 | OWR-15219-015

### Minimal Flat Extension Orders for Truncated Moments

P / partially_solved | Oberwolfach 报告问题汇编

Given a truncated moment sequence $\beta^{(2d)}$, estimate the smallest $k\geq0$ for which the moment matrix $M_{d+k}$ has a flat extension $M_{d+k+1}$.

来源字段：https://doi.org/10.4171/owr/2017/14

10838 | OWR-11791-001

### Minimizers for Banach-Space Subspace Segmentation

O / open | Oberwolfach 报告问题汇编

For the subspace-segmentation problem of minimizing the approximation error over $M$-tuples of closed subspaces of a Banach space, what are necessary and sufficient conditions for a minimizer to exist when the exponent $p\ne2$?

来源字段：https://doi.org/10.4171/owr/2012/29

10839 | OWR-3470-004

### Models for Three or More Commuting Contractions

O / open | Oberwolfach 报告问题汇编

Find an effective model, analogous to Ando's model for two commuting contractions, for three or more commuting contraction operators on a Hilbert space.

来源字段：https://doi.org/10.4171/owr/2009/20

10840 | AIM-SPECIAL_FUNCTIONS-0004

### Modern k-(q,t)-Kostka positivity and a change-of-basis obstruction

P / partially_solved | AIM 研讨会问题汇编

(4) Prove that the \(k\)-\(q,t\)-Kostka polynomials lie in \(\mathbb N[q,t]\).

来源字段：https://aimath.org/WWN/kostka/prob2.pdf

10841 | AMR-036-0042

### Modified Cartan conjecture

P / partially_solved | AMR 问题清单汇编

For $p\ge3$, let $V(D)$ consist of zero-free holomorphic vectors $(f_1,\ldots,f_p)$ on $D$ with $\sum f_j=0$, and use the source's definition of a $C$-class for an infinite sequence. Prove that every infinite sequence in $V(D(1))$ has a subsequence for which $\{1,\ldots,p\}$ is a union of disjoint $C$-classes on $D(R_p)$ for some $R_p>0$ depending only on $p$. Can $R_p$ be chosen independently of $p$, and what is the geometric interpretation?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10842 | OWR-4990379-005

### Monge Minimizers for Multimarginal Coulomb Transport

O / open | Oberwolfach 报告问题汇编

Do Monge-type minimizers always exist for the multimarginal optimal-transport problem with Coulomb cost?

来源字段：https://doi.org/10.4171/owr/2021/10

10843 | AIM-OPTIMIZATION-0004

### Monotone Galerkin approximation of finite Schur complements

P / partially_solved | AIM 研讨会问题汇编

A problem analogous to Problem 1.10.3 was solved in [5] by showing that Second Order Cone Programming is poly.-time equivalent to Linear Programming.

**1.11 Hugo Woerdeman.** One of the questions I am interested in is how to approximate numerically the Schur complement of a positive definite operator supported on a finite subspace. Of course there is the formula that for a block matrix \[ \begin{pmatrix}A&B\\ C&D\end{pmatrix} \] the Schur complement is \(A-BD^{-1}C\), but this requires determining the inverse of the infinite operator \(D\). The way this question arose is through attempts to develop multivariable analogs of the Gohberg--Semencul formula. One way to prove the Gohberg--Semencul formula is by determining the Schur complement of the inverse of a Toeplitz operator whose symbol is the reciprocal of a positive trigonometric polynomial. In several variables the analogous attempt runs into difficulties. In order to at least get a decent numerical approximation of the inverse of a doubly Toeplitz matrix, one may try to find reasonable numerical methods to determine finitely based Schur complements of infinite operators.

来源字段：https://aimath.org/WWN/matrixineq/matrixineq.pdf

10844 | OWR-3389-022

### Monotonicity of Weighted Yamabe Heat Traces

O / open | Oberwolfach 报告问题汇编

Fix $n\geq4$. For each smooth positive weight $W$ on $S^n$, normalized by $\int_{S^n}W^{n/2}=\operatorname{vol}(S^n)$, form the weighted Yamabe operator $Y_W=W^{-1/2}YW^{-1/2}$ from the round conformal Laplacian $Y$. Must the scaled heat trace $t\longmapsto t^{n/2}\operatorname{Tr}(e^{-tY_W})$ be decreasing on $(0,\infty)$?

来源字段：https://doi.org/10.4171/owr/2009/06

10845 | AIM-ANALYSIS-0163

### Morse reduction and exact Fubini-Study minimization in degrees one and two

P / partially_solved | AIM 研讨会问题汇编

**Question 6: Bernard Shiffman.** Does the Fubini-Study metric on P1 minimize the expected number of critical points? There are reasons to conjecture that the answer is Yes.

来源字段：https://aimath.org/WWN/randomzeros/arcc1.pdf

10846 | OWR-12339-009

### Muckenhoupt Thresholds for Integer-Translate Schauder Bases

O / open | Oberwolfach 报告问题汇编

Let $\psi\in L^2(\mathbb R)$, let $B_\psi=\{\psi(\cdot-k):k\in\mathbb Z\}$ be a Schauder basis for its closed linear span, and set

$$p_\psi(\xi)=\sum_{k\in\mathbb Z}|\widehat\psi(\xi+k)|^2,\qquad q_\psi=\inf\{p>1:p_\psi\text{ is an }A_p\text{-weight}\}.$$

For each $q$ with $1\leq q<2$, which properties of $B_\psi$ characterize the equality $q_\psi=q$?

来源字段：https://doi.org/10.4171/owr/2013/11

10847 | AMR-035-0003

### Multiplicity-one support of area Siegel-Veech constants

O / open | AMR 问题清单汇编

Let $\boldsymbol{d}=(d_1,\ldots,d_n)$ be an unordered partition of $4g-4$ with $d_i\in\{-1,0,1,2,\ldots\}$, and let $\widehat\Pi_{4g-4}$ be the set of such partitions having at most $\log g$ entries equal to $-1$. For every stratum $\mathcal Q(d_1,\ldots,d_n)$ with $\boldsymbol d\in\widehat\Pi_{4g-4}$, prove that as $g\to\infty$ its area Siegel-Veech constant is asymptotically supported only on the multiplicity-one configurations $\mathcal C_{b,\mathrm I}(d_i,d_j)$ and $\mathcal C_{b,\mathrm{II}}(a_1,a_2)$ of homologous saddle connections, where $i\ne j$, $d_i,d_j\ge1$, $a_1,a_2\ge0$, $a_1+a_2\ge3$, and $a_1+a_2+2$ is an entry of $\boldsymbol d$.

背景链接：https://arxiv.org/abs/1912.11702

10848 | OWR-15222-006

### Multivariate Smoothness Classes and Their Approximation

O / open | Oberwolfach 报告问题汇编

What is a natural multivariate analogue of univariate smoothness classes, and how should functions in those classes be approximated?

来源字段：https://doi.org/10.4171/owr/2017/17

10849 | OWR-4214-010

### Naimark's Problem for Nonseparable $C^*$-Algebras

O / open | Oberwolfach 报告问题汇编

别名：Repeated Extraction of Naimark's Problem [OWR-4214-003]

In the nonseparable case, does every $C^*$-algebra whose irreducible representations are all equivalent have to be an algebra of compact operators?

来源字段：https://doi.org/10.4171/owr/2010/13

10850 | AIM-ANALYSIS-0069

### Near-sharp trilinear convolution range on transverse patches of the 3-sphere

P / partially_solved | AIM 研讨会问题汇编

**Question 10 (Bez).** Let \(S_1,S_2,S_3\) be transversal patches of the unit sphere \(\mathbb S^3\) in \(\mathbb R^4\). Let \(\sigma_1,\sigma_2,\sigma_3\) be the surface measures, respectively. Determine the full range of exponents \(p,q>0\) such that \[ \|g_1d\sigma_1*g_2d\sigma_2*g_3d\sigma_3\|_q \lesssim \prod_{j=1}^3\|g_j\|_p. \tag{0.12} \]

来源字段：https://aimath.org/pastworkshops/multilinopsproblems.pdf

10851 | OWR-720-010

### Nearest Normalized Tight Frames

P / partially_solved | Oberwolfach 报告问题汇编

Find metrics on the space of frames for which the normalized tight frame associated to a given frame is a closest normalized tight frame, and determine the geometry of all closest-frame subspaces.

来源字段：https://doi.org/10.4171/owr/2004/10

10852 | AIM-SEVERAL_COMPLEX_VARIABLES-0029

### Neighboring Newton edges and the corner-allocation obstruction

S / solved | AIM 研讨会问题汇编

**Problem 2 (B. Stensønes).** Let \[ p(z,\bar z,w,\bar w)=\sum p_{\alpha\beta\gamma\delta} z^\alpha\bar z^\beta w^\gamma\bar w^\delta \] be a plurisubharmonic real-valued polynomial on \(\mathbb C^2\). Write \[ N(p)=\{(a,b)\in\mathbb N^2:a=\alpha+\beta, b=\gamma+\delta,\ p_{\alpha\beta\gamma\delta}\ne0\}. \] Let \(\Gamma_1,\Gamma_2\) be extreme edges of \(N(p)\) such that \(\Gamma_1\cap\Gamma_2\ne\varnothing\), where an extreme edge is a line in \(N(p)\) with no points of \(N(p)\) below. Can \[ q=\sum_{(\alpha+\beta,\gamma+\delta)\in\Gamma_1\cup\Gamma_2} p_{\alpha\beta\gamma\delta}z^\alpha\bar z^\beta w^\gamma\bar w^\delta \] be written as the sum of two plurisubharmonic weighted-homogeneous polynomials?

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10853 | AIM-ANALYSIS-0158

### No ensemble-free extreme-root asymptotic and a Kac small-ball transfer theorem

P / partially_solved | AIM 研讨会问题汇编

Question 1: Wenbo Li Consider random polynomials in one variable (real or complex). Find the asymptotics of the norm of the largest zero as the degree of the random polynomials tends to infinity.

来源字段：https://aimath.org/WWN/randomzeros/arcc1.pdf

10854 | OWR-12005-017

### Nodal Counts for Fractional Schrödinger Eigenfunctions

O / open | Oberwolfach 报告问题汇编

Let $0<s<1$, let $(-\Delta)^s+V$ be a lower-semibounded fractional Schrödinger operator on $L^2(\mathbb R^m)$ with real $V$, and let $\psi_k$ be a continuous real eigenfunction associated with the $k$th eigenvalue below the essential spectrum. Bound the number of connected components of $\{\psi_k\ne0\}$ in terms of $k$.

来源字段：https://doi.org/10.4171/owr/2012/33

10855 | OWR-12174-003

### Noise Conditions for Quasioptimal Total-Variation Regularization

P / partially_solved | Oberwolfach 报告问题汇编

For total-variation regularization with generalized quasioptimality and Hanke-Raus parameter-choice rules, what noise-restriction set $N$ ensures convergence of the combined method?

来源字段：https://doi.org/10.4171/owr/2012/51

10856 | OWR-12593-007

### Noncommutative Geometry of Automorphic Function Spaces

O / open | Oberwolfach 报告问题汇编

Can noncommutative geometry clarify the analytic spaces of automorphic cusp forms and Eisenstein series, possibly through an appropriate noncommutative boundary?

来源字段：https://doi.org/10.4171/owr/2013/45

10857 | AIM-SEVERAL_COMPLEX_VARIABLES-0003

### Noncompact automorphism groups and anisotropic orbit rates

P / partially_solved | AIM 研讨会问题汇编

Let \(\Omega\) be a smooth bounded domain in \(\mathbb C^n\) such that the automorphism group of \(\Omega\) is non-compact. What can be said about \(\Omega\)? See Isaev--Krantz (*Adv. Math.* 146, 1999, MR1706680, survey on domains with non-compact automorphism group).

来源字段：http://aimpl.org/crscv/1/

10858 | OWR-10252925-009

### Nondegeneracy in the Spectral Edges Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For a sufficiently general periodic discrete Schrödinger operator $L$, prove that every extremum of the dispersion function $\lambda$ on the real Bloch variety is nondegenerate.

来源字段：https://doi.org/10.4171/owr/2022/17

10859 | AIM-ANALYSIS-0011

### Nonperiodic 2-by-2 products: a literal factorization diagnostic, exact commuting asymptotics, and a no-limit obstruction

P / partially_solved | AIM 研讨会问题汇编

Consider \[ \prod_{i=1}^N \begin{bmatrix}1&a_i z\\ b_i&1\end{bmatrix}, \] for constants \(a_i,b_i\). Suppose Wiener-Hopf factorization has no partial indices. Study the behavior as \(N\to\infty\). The asymptotics are known if the \(a_i\)'s and \(b_i\)'s are periodic. What happens in the more general case?

来源字段：http://aimpl.org/riemhilberttoeplitz/2/

10860 | OWR-718-002

### Nonsimple Nonlinear Eigenvalues and Isolated Spectra

P / partially_solved | Oberwolfach 报告问题汇编

Describe the nonlinear spectral behavior near a nonsimple eigenvalue $\lambda_0$. Also characterize a class of homogeneous maps on $\mathbb{R}^N$ all of whose eigenvalues are isolated.

来源字段：https://doi.org/10.4171/owr/2004/08

10861 | OWR-8415363-001

### Nonsymmetric Extensions of Dirichlet-Form Trace Constructions

O / open | Oberwolfach 报告问题汇编

Determine whether the trace construction for symmetric Dirichlet forms extends to nonsymmetric forms.

来源字段：https://doi.org/10.4171/owr/2021/58

10862 | AIM-ANALYSIS-0040

### Nontrivial Bergman space versus the plurisubharmonic Liouville property

P / partially_solved | AIM 研讨会问题汇编

Is there a pseudoconvex domain $\Omega$ with ${\rm dim}\,A^2(\Omega)\neq 0$ such that there is no bounded plurisubharmonic function $\phi$ on $\Omega$.

**Recovered question.** Does there exist a connected pseudoconvex domain $\Omega\subset\mathbb C^n$ such that $A^2(\Omega)\neq\{0\}$ and every bounded plurisubharmonic function on $\Omega$ is constant?

来源字段：http://aimpl.org/scvproblems/1/

10863 | OWR-15960-004

### Nontriviality of Local Algebras from Deformed Borchers Triples

O / open | Oberwolfach 报告问题汇编

Are the local von Neumann algebras generated by the deformed Borchers triples nontrivial?

来源字段：https://doi.org/10.4171/owr/2017/55

10864 | AIM-ANALYSIS-0073

## Normalization and the constant-one multilinear Kakeya question

P / partially_solved | AIM 研讨会问题汇编

The official TeX says that, for collections \(\mathcal T_i\) of doubly infinite tubes of width one in \(\mathbb R^n\),

\[ \int \left[ \left(\sum_{T_1\in\mathcal T_1}\chi_{T_1}\right)\cdots \left(\sum_{T_n\in\mathcal T_n}\chi_{T_n}\right) w(T_1)\wedge\cdots\wedge w(T_n) \right]^{1/(n-1)} \le C_n\prod_{i=1}^n(\#\mathcal T_i)^{1/(n-1)}, \tag{0.20} \]

and asks whether one can take \(C_n=1\).

As typeset, the symbols \(T_i\) in the wedge factor are free: they occur outside the displayed sums. The primary theorem from which the display comes, Carbery--Valdimarsson [CV13, Theorem 1], and the later explicit display in Bennett--Iliopoulou [BI18, p. 1095], show that the intended left side is the following \(n\)-fold weighted sum. Put \(p=1/(n-1)\) and

\[ F_{\boldsymbol{\mathcal T}}(x) :=\sum_{(T_1,\ldots,T_n)\in\mathcal T_1\times\cdots\times\mathcal T_n} \left|\det\bigl(w(T_1),\ldots,w(T_n)\bigr)\right| \prod_{i=1}^n\mathbf 1_{T_i}(x). \tag{1} \]

The recovered question is therefore

\[ \mathcal K_n(\boldsymbol{\mathcal T}) :=\int_{\mathbb R^n}F_{\boldsymbol{\mathcal T}}(x)^p\,dx \stackrel{?}{\le} \prod_{i=1}^n(\#\mathcal T_i)^p. \tag{2} \]

Here the wedge is the **unsigned** volume of the parallelotope, as in [CV13]. Without the absolute value, a fractional power need not even be real.

There is a second source ambiguity that is decisive for the number \(1\): neither the problem TeX nor the PDF defines “width one.” In [CV13, pp. 1644--1645], a \(1\)-tube means the \(1\)-neighbourhood of a line, hence a round tube of radius one and total transverse width two. Other formulations normalize a tube's \((n-1)\)-dimensional cross-sectional volume to one; [BI18, p. 1094] explicitly uses that convention. These conventions differ by a dimensional factor that cannot be ignored in a best-constant question.

来源字段：https://aimath.org/pastworkshops/multilinopsproblems.pdf

10865 | AIM-SPECIAL_FUNCTIONS-0003

## Normalization obstruction, membership reduction, and stable-range tableau atoms

P / partially_solved | AIM 研讨会问题汇编

(3) Prove that the $k$-atoms (see the first definition of $k$-Schur functions in Morse's lecture notes) form a basis for the linear span of $\{J_\mu:\mu_1\le k\}$. Prove also that the expansion coefficients of the $J_\lambda$'s in terms of the $k$-atoms refine the $q,t$-Kostka numbers.

来源字段：https://aimath.org/WWN/kostka/prob2.pdf

10866 | AMR-036-0036

## Number of Newtonian equilibrium points

P / partially_solved | AMR 问题清单汇编

If the critical set of $u(x)=\sum_{k=1}^n a_k/|x-x_k|$ for positive point charges in $\mathbb R^3$ is finite, how many points can it contain? In particular, is Maxwell's bound $(n-1)^2$ valid?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10867 | AIM-ANALYSIS-0021

## One-dimensional counterexample and classification for optimal Dirichlet eigenvalue multiplicity

P / partially_solved | AIM 研讨会问题汇编

Let $\Omega_k^*$ be a minimizer of the shape optimization problem \[ \min\{\lambda_k(\Omega):\Omega\subset\mathbb R^n, |\Omega|=V_0\}. \] Prove that $\lambda_{k-1}(\Omega_k^*)=\lambda_k(\Omega_k^*)$.

来源字段：http://aimpl.org/shapesurface/2/

10868 | AMR-147-0001

## Open Problems in Pluripotential Theory - Question 1

O / open | AMR 问题清单汇编

[Bedford] Given $\varphi$ an arbitrary plurisubharmonic function in some domain $\Omega \subset {\mathbb{C}}^n$, what can be said about the set of cluster points of $(dd^c \varphi_j)^n$, where $\varphi_j$ are bounded plurisubharmonic functions which decrease to $\varphi$ ?

背景链接：https://arxiv.org/abs/1511.00705

10869 | AMR-147-0010

## Open Problems in Pluripotential Theory - Question 10

O / open | AMR 问题清单汇编

A subtle point is to understand precisely when a function $\varphi \in DMA(X,\omega)$ belongs to ${\mathcal{E}}(X,\omega)$. This has been investigated in [CGZ08] with poor success:

[Coman-Guedj-Zeriahi] Assume $\varphi \in DMA(X,\omega)$ is such that $MA(\varphi)$ does not charge $(\varphi=-\infty)$. Does it imply that $\varphi$ belongs to ${\mathcal{E}}(X,\omega)$ ?

背景链接：https://arxiv.org/abs/1511.00705

10870 | AMR-147-0011

### Open Problems in Pluripotential Theory - Question 11

O / open | AMR 问题清单汇编

[Phong] Let $X \stackrel{\pi}{\rightarrow} {\mathbb{C}}\mathbb{P}^2$ be the blow up of the complex projective plane at one point. This is a DelPezzo surface, hence its first Chern class is positive. Pick $\omega \in c_1(X)$ a K\"ahler form, $D$ a smooth anticanonical divisor, and let $\mu_D=\omega \wedge [D] /V$ denote a normalized Lebesgue measure along $D$. Can one find $\varphi_D \in DMA(X,\omega)$ such that $$ MA(\varphi_D)=\mu_D \; \; ? $$

背景链接：https://arxiv.org/abs/1511.00705

10871 | AMR-147-0012

### Open Problems in Pluripotential Theory - Question 12

O / open | AMR 问题清单汇编

We now focus on the case of *bounded* and *continuous* plurisubharmonic functions. Although it is the setting of the original breakthrough of Bedford and Taylor [BT76,BT82], the range of the complex Monge-Amp\`ere operator acting on bounded/continuous (quasi-)psh functions is yet not characterized (despite rather sharp sufficient conditions [Kol98]). We recall the following question from [Bed93, p. 70]:

[Bedford] Is the range of the complex Monge-Amp\`ere operator acting on $PSH \cap L_{loc}^{\infty}$ characterized by a uniform integrability condition ?

背景链接：https://arxiv.org/abs/1511.00705

10872 | AMR-147-0013

### Open Problems in Pluripotential Theory - Question 13

O / open | AMR 问题清单汇编

[Ko\l{}odziej] Assume there is a continuous subsolution to the Dirichlet problem, i.e. there is a continuous psh function $u$ in $\Omega$ such that $u_{|\partial \Omega} =0$ and $\mu \leq (dd^c u)^n$ in $\Omega$. Does it imply that there is a continuous solution, i.e. a continuous psh function $\varphi$ such that $\varphi_{|\partial \Omega} =0$ and $\mu = (dd^c \varphi)^n$ in $\Omega$ ?

背景链接：https://arxiv.org/abs/1511.00705

10873 | AMR-147-0014

### Open Problems in Pluripotential Theory - Question 14

O / open | AMR 问题清单汇编

[Ko\l{}odziej] Let $(X,\omega)$ be a compact K\"ahler manifold. Let $\mu$ be a probability measure which is locally dominated by the Monge-Amp\`ere measures of bounded plurisubharmonic functions. Can one find $\varphi \in PSH(X,\omega) \cap L^{\infty}(X)$ such that $\mu=MA(\varphi)$ ?

背景链接：https://arxiv.org/abs/1511.00705

10874 | AMR-147-0015

### Open Problems in Pluripotential Theory - Question 15

O / open | AMR 问题清单汇编

[DDGHKZ] Are the following properties equivalent ?

(i) There exists $\varepsilon_{\mu}>0$ such that $\exp(-\varepsilon_\mu PSH(X,\omega) ) \subset L^1(\mu)$.

(ii) There exists $\varphi \in {\mathcal{H}ol}(X,\omega)$ sucht that $\mu=MA(\varphi)$.

背景链接：https://arxiv.org/abs/1511.00705

10875 | AMR-147-0016

### Open Problems in Pluripotential Theory - Question 16

O / open | AMR 问题清单汇编

[Zeriahi] Let $\mu$ be a probability measure in a bounded pseudoconvex domain $\Omega \subset {\mathbb{C}}^n$. Assume there is a H\"older continuous subsolution to the Dirichlet problem, i.e. there is a H\"older continuous psh function $u$ in $\Omega$ such that $u_{|\partial \Omega} =0$ and $\mu \leq (dd^c u)^n$ in $\Omega$. Is there a H\"older continuous psh function $\varphi$ such that $\varphi_{|\partial \Omega} =0$ and $\mu = (dd^c \varphi)^n$ in $\Omega$ ?

背景链接：https://arxiv.org/abs/1511.00705

10876 | AMR-147-0017

### Open Problems in Pluripotential Theory - Question 17

O / open | AMR 问题清单汇编

It is interesting that no PDE proof of the above fact is known.

[Blocki-Ko\l{}odziej] Find a proof of the above estimate that relies on PDE techniques.

背景链接：https://arxiv.org/abs/1511.00705

10877 | AMR-147-0018

### Open Problems in Pluripotential Theory - Question 18

O / open | AMR 问题清单汇编

If the form $\omega$ is merely semi-positive and big (i.e. $\int_X\omega^n>0$) then solutions to the Monge-Amp\`ere equations with $L^p$- right hand sides are continuous as shown in [EGZ11] by using viscosity techniques (solutions are then unique up to an additive constant). Further regularity is then largely open. In particular the following is an important problem:

[Eyssidieux-Guedj-Zeriahi] Assume $\omega$ is merely semi-positive and big and $$ (\omega+dd^c \varphi)^n=f dV,\ f\in L^p,\ p>1. $$

Is $\varphi$ H\"older continuous on $X$ ?

背景链接：https://arxiv.org/abs/1511.00705

10878 | AMR-147-0019

### Open Problems in Pluripotential Theory - Question 19 (Berman)

O / open | AMR 问题清单汇编

Let $\mu_S$ denote the normalized $n$-dimensional Hausdorff measure on a smooth totally real manifold $S \subset X$ of maximal real dimension $n$. It has been shown by Sadullaev in [Sad] that $S$ is non pluripolar, hence $\mu_S$ is the Monge-Amp\`ere measure $\mu_S=(\omega+dd^c \varphi_S)^n$ of a unique function $\varphi_S \in {\mathcal{E}}(X,\omega)$ normalized by $\sup_X \varphi_S=0$.

What is the optimal regularity of $\varphi_S$ ?

背景链接：https://arxiv.org/abs/1511.00705

10879 | AMR-147-0002

### Open Problems in Pluripotential Theory - Question 2

O / open | AMR 问题清单汇编

[Bedford] Given $\varphi$ a bounded plurisubharmonic function in some domain $\Omega \subset {\mathbb{C}}^n$, what can be said about the set of cluster points of $(dd^c \varphi_j)^n$, where $\varphi_j$ are bounded plurisubharmonic functions which converge to $\varphi$ in $L_{loc}^1$ ?

背景链接：https://arxiv.org/abs/1511.00705

10880 | AMR-147-0020

### Open Problems in Pluripotential Theory - Question 20

O / open | AMR 问题清单汇编

Let $\alpha \in H^{1,1}(X,{\mathbb{R}})$ be a big cohomology class, $\mu$ a smooth volume form and let $T$ be the unique finite energy current in $\alpha$ such that $\langle T \rangle^n=\mu$. Is $T$ smooth in $Amp(\alpha)$ ?

背景链接：https://arxiv.org/abs/1511.00705

10881 | AMR-147-0021

### Open Problems in Pluripotential Theory - Question 21 (DiNezza-Lu)

O / open | AMR 问题清单汇编

Let $(X,\omega)$ be a compact $n$-dimensional K\"ahler manifold. Assume $\varphi \in {\mathcal{E}}(X,\omega)$ solves the equation $(\omega+dd^c \varphi)^n=f\omega^n$, with $f\in L^1$ being smooth off a divisor $D\subset X$. Is the function $\varphi$ smooth on $X\setminus D$?

背景链接：https://arxiv.org/abs/1511.00705

10882 | AMR-147-0022

### Open Problems in Pluripotential Theory - Question 22

O / open | AMR 问题清单汇编

[Blocki] Let $\Omega \subset {\mathbb{C}}^n$ be a bounded hyperconvex domain. Let $u$ be the unique continuous plurisubharmonic function in $\Omega$ such that $u_{| \partial \Omega}=0$ and $$ (dd^c u)^n=f dV, $$ with $f$ smooth and bounded away from zero $f \geq \varepsilon >0$. Is $u$ automatically smooth ?

背景链接：https://arxiv.org/abs/1511.00705

10883 | AMR-147-0023

### Open Problems in Pluripotential Theory - Question 23

O / open | AMR 问题清单汇编

[Blocki-Dinew] Let $u$ be a plurisubharmonic function in a smooth strictly pseudoconvex bounded domain $\Omega \subset {\mathbb{C}}^n$. Assume $u \in {\mathcal{C}}^{1,\alpha}(\Omega),\alpha>1-\frac2n$ and $$ (dd^c u)^n=f dV, $$ with $f$ smooth and bounded away from zero $f \geq \varepsilon >0$. Is $u$ automatically smooth ?

背景链接：https://arxiv.org/abs/1511.00705

10884 | AMR-147-0024

### Open Problems in Pluripotential Theory - Question 24

O / open | AMR 问题清单汇编

[Demailly] Fix $\Omega \subset {\mathbb{C}}^n$ a smooth strictly pseudoconvex bounded domain, and $\phi \in {\mathcal{C}}^{1,1}(\partial \Omega)$. Let $u$ be the unique plurisubharmonic function in $\Omega$, continuous up to the boundary, such that $u=\phi$ on $\partial \Omega$ and $$ (dd^c u)^n=0 \; \; \text{ in } \; \; \Omega. $$

Show, by using complex analytic methods, that $u$ is ${\mathcal{C}}^{1,1}$-smooth in $\Omega$.

背景链接：https://arxiv.org/abs/1511.00705

10885 | AMR-147-0025

### Open Problems in Pluripotential Theory - Question 25

O / open | AMR 问题清单汇编

Bedford and Fornaess [BeFo] have shown that $u$ need not belong to $\mathcal{C}^2(\bar{\Omega})$ under the assumptions above. Their construction relies on the boundary behavior of the $\mathcal{C}^2$ norm.

Under what minimial assumptions does $u$ belong to $\mathcal{C}^2(\Omega)$ ?

背景链接：https://arxiv.org/abs/1511.00705

10886 | AMR-147-0026

### Open Problems in Pluripotential Theory - Question 26 (Blocki)

O / open | AMR 问题清单汇编

It is easy to prove that for a hyperconvex domain $\Omega$ and a fixed pole $w$ one has $$\lim_{z\rightarrow\partial\Omega}g_{\Omega}(z,w)=0.$$

Is it true that for a bounded hyperconvex domain $g_{\Omega}(z,w)$ converges locally uniformly to zero, as the pole $w$ converges to the boundary of $\Omega$?

背景链接 : https://arxiv.org/abs/1511.00705

10887 | AMR-147-0027

### Open Problems in Pluripotential Theory - Question 27 (Blocki)

O / open | AMR 问题清单汇编

Let $\Omega$ be smoothly bounded strictly pseudoconvex domain. It is known that $g_{\Omega}\in\mathcal{C}^{1,1}(\Omega\setminus\lbrace w\rbrace)$. Is there an example where $g$ fails to be in $\mathcal{C}^2(\Omega\setminus\lbrace w\rbrace)$?

背景链接 : https://arxiv.org/abs/1511.00705

10888 | AMR-147-0028

### Open Problems in Pluripotential Theory - Question 28 (Dinew-Ko\l{}odziej)

O / open | AMR 问题清单汇编

Let $(X,\omega)$ be a compact Hermitian manifold of dimension greater or equal to $3$. Characterize the forms $\omega$ for which $\int_X(\omega+dd^c\varphi)^n=\int_X\omega^n$ for any bounded $\omega$-plurisubharmonic function $\varphi$.

背景链接 : https://arxiv.org/abs/1511.00705

10889 | AMR-147-0029

### Open Problems in Pluripotential Theory - Question 29 (Tosatti)

O / open | AMR 问题清单汇编

Let $(X,\omega)$ be a compact Hermitian manifold. Define $T(X,\omega)$ by $$ T(X,\omega)=\inf \left\lbrace\int_X(\omega+dd^c\varphi)^n|\ \varphi\in\mathcal{C}^{\infty}(X)\cap PSH(X,\omega) \right\rbrace. $$ Is it possible that $T(X,\omega)=0$ for some manifold?

背景链接 : https://arxiv.org/abs/1511.00705

10890 | AMR-147-0003

### Open Problems in Pluripotential Theory - Question 3

O / open | AMR 问题清单汇编

When $n=1$, maximal plurisubharmonic function are precisely harmonic ones, in particular they are smooth, but in higher dimension they may be quite irregular: a plurisubharmonic function that only depends on $n-1$ variables is maximal. When $\varphi \in DMA(\Omega)$ then $$ \varphi \text{ is maximal } \Longleftrightarrow MA(\varphi)=0, $$ as explained by Blocki in [Blo04]. In particular maximality is a local notion in $DMA(\Omega)$.

[Blocki] Is maximality a local notion for arbitrary plurisubharmonic functions?

背景链接 : https://arxiv.org/abs/1511.00705

10891 | AMR-147-0030

### Open Problems in Pluripotential Theory - Question 30 (Dinew-Ko\l{}odziej)

O / open | AMR 问题清单汇编

Let $(X,\omega)$ be a compact Hermitian manifold. Define $cap(X,\omega)$ by $$ cap(X,\omega)=\inf \left\lbrace\int_X(\omega+dd^c\varphi)^n|\ \varphi\in PSH(X,\omega),\ 0\leq\varphi\leq 1\right\rbrace. $$ Relate the quantity $cap(X,\omega)$ to the geometry of $(X,\omega)$. In particular is this quantity computable in terms of geometric data? How does it behave under deformations?

背景链接 : https://arxiv.org/abs/1511.00705

10892 | AMR-147-0031

### Open Problems in Pluripotential Theory - Question 31 (Tosatti-Weinkove)

O / open | AMR 问题清单汇编

Let $z_1,\cdots,z_k$ be a collection of points on a Hermitian manifold $X$ and let $\tau_1,\cdots,\tau_k$ be a collection of positive numbers satisfying $\sum_{j=1}^k\tau_j^n<\int_X\omega^n$. Construct an $\omega$-plurisubharmonic function, so that $\phi(z)\leq \tau_j \log(||z-z_j||)+O(1)$ near $z_j$.

背景链接 : https://arxiv.org/abs/1511.00705

10893 | AMR-147-0032

### Open Problems in Pluripotential Theory - Question 32

O / open | AMR 问题清单汇编

When $n=\dim_{\mathbb{C}} \leq 2$, it follows from the work of Blel-Mimouni [BM05], the second author [G05b] and Favre-Jonsson [FJ05] that $\varphi$ can be further decomposed as follows: for any $\varepsilon>0$, there exists $\pi_\varepsilon:X_\varepsilon \rightarrow X$ a finite composition of blow-ups such that $$ \pi_\varepsilon^* (\omega+dd^c \varphi)=[D_\varepsilon]+R_\varepsilon, $$ where $D_\varepsilon$ is a simple normal crossing divisor with coefficients in ${\mathbb{R}}^+$ and $R_\varepsilon$ is a positive current with Lelong numbers smaller than $\varepsilon$ at all points.

[Favre-Guedj-Jonsson] Can one similarly desingularize quasi-plurisubharmonic functions in any dimension $n \geq 3$ ?

背景链接 : https://arxiv.org/abs/1511.00705

10894 | AMR-147-0033

### Open Problems in Pluripotential Theory - Question 33 (ACHKZ)

O / open | AMR 问题清单汇编

While giving the optimal exponential integrability the result is still not sharp because of the polynomial term in front.

Is it true that (for a different constant $c_n$) $$ \mathrm{Vol}(\lbrace\varphi<-s\rbrace)\leq c_n\delta_{\Omega}^{2n}exp(-2ns\mu^{-1})? $$

背景链接 : https://arxiv.org/abs/1511.00705

10895 | AMR-147-0034

### Open Problems in Pluripotential Theory - Question 34

O / open | AMR 问题清单汇编

[Berman-Boucksom-Eyssidieux-Guedj-Zeriahi] Assume $\varphi \in {\mathcal{E}}(X,\omega)$ is a $\omega$-psh function with finite energy. Is it true that $\varphi$ has zero Lelong number at all points ?

背景链接 : https://arxiv.org/abs/1511.00705

10896 | AMR-147-0035

### Open Problems in Pluripotential Theory - Question 35

O / open | AMR 问题清单汇编

Let $(X,\omega)$ be a compact K\"ahler manifold and $Y \subset X$ a complex submanifold. It is natural to wonder whether all $\omega$-plurisubharmonic functions on $Y$ are induced by global $\omega$-plurisubharmonic functions:

[Coman-Guedj-Zeriahi] Is it true that $$ PSH \left(Y,\omega_{|Y} \right)=PSH(X,\omega)_{|Y} \; \; ? $$

背景链接 : https://arxiv.org/abs/1511.00705

10897 | AMR-147-0036

### Open Problems in Pluripotential Theory - Question 36

O / open | AMR 问题清单汇编

[P\u aun] Can one find $h=e^{-\varphi}$ a hermitian metric of ${\mathcal{L}}$ on ${\mathcal{X}}$ which is a semi-positive extension of $\varphi_0$, i.e. such that $dd^c \varphi \geq 0$ and $\varphi_{|X_0}=\varphi_0$ ?

背景链接 : https://arxiv.org/abs/1511.00705

10898 | AMR-147-0037

### Open Problems in Pluripotential Theory - Question 37

O / open | AMR 问题清单汇编

The point in the example above is that there is no differential test from above for $\varphi$ along the line $(w=0)$. A simpler example in any dimension is provided by $\varphi(z)=||z||$: there is no differential test from above at the origin; the function is strictly plurisubharmonic in the viscosity sense, but not in the classical sense.

[Guedj-Zeriahi] Let $\varphi$ be a plurisubharmonic function. What can be said about the set of points at which $\varphi$ does not admit a differential test from above ?

背景链接 : https://arxiv.org/abs/1511.00705

10899 | AMR-147-0038

### Open Problems in Pluripotential Theory - Question 38

O / open | AMR 问题清单汇编

[Eyssidieux-Guedj-Zeriahi] Let $v$ be a continuous viscosity supersolution of the equation $$ -(dd^c V)^n+fdV=0. $$

Is $P(v)$ a pluripotential supersolution of the same equation ?

背景链接 : https://arxiv.org/abs/1511.00705

10900 | AMR-147-0039

### Open Problems in Pluripotential Theory - Question 39

O / open | AMR 问题清单汇编

[Eyssidieux-Guedj-Zeriahi] Let $\omega$ be a smooth semi-positive and big form on a compact K\"ahler manifold $X$. Is any function $\varphi \in PSH(X,\omega)$ the decreasing limit of smooth $\omega$-psh functions ?

背景链接 : https://arxiv.org/abs/1511.00705

10901 | AMR-147-0004

### Open Problems in Pluripotential Theory - Question 4

O / open | AMR 问题清单汇编

[Dujardin] Assume $0 \leq \alpha <1$ and $n=2$. Assume $\varphi$ is ${\mathcal{C}}^{2,\alpha}$-smooth and maximal. Is the suppport of $dd^c \varphi$ foliated (or laminated) by holomorphic disc along which $\varphi$ is harmonic ?

背景链接 : https://arxiv.org/abs/1511.00705

10902 | AMR-147-0040

### Open Problems in Pluripotential Theory - Question 41

O / open | AMR 问题清单汇编

When $S_0$ has some non-zero Lelong numbers, DiNezza and Lu construct a maximal family of positive currents $(\omega_t)_{0<t<T_{max}}$ which are honest K\"ahler forms on a time-increasing family of Zariski open subsets $\Omega_t$ where they satisfy the K\"ahler-Ricci flow equation. By analogy with Demailly regularization results [Dem92,Dem93], it is natural to expect that this family has analytic singularities.

[DiNezza-Lu] Does the K\"ahler-Ricci flow produce approximants with analytic singularities ?

背景链接：https://arxiv.org/abs/1511.00705

10903 | AMR-147-0041

### Open Problems in Pluripotential Theory - Question 42

O / open | AMR 问题清单汇编

Do actually deals with more general flows which contain, as a particular case, the local version of the K\"ahler-Ricci flow. These flows can thus be run from a bounded plurisubharmonic initial data. The following question is natural, in view of the corresponding global results from [GZ13,DiNL14b]:

[Guedj] Can one run the above complex Monge-Amp\`ere flow from an arbitrary plurisubharmonic function ?

背景链接：https://arxiv.org/abs/1511.00705

10904 | AMR-147-0042

### Open Problems in Pluripotential Theory - Question 43

O / open | AMR 问题清单汇编

[Eyssidieux-Guedj-Zeriahi] Assume $u$ (resp. $v$) is a viscosity subsolution (resp. supersolution) to the above degenerate complex Monge-Amp\`ere flow such that $u \leq v$ on the parabolic boundary $\{0\} \times X$. Is it true that that $u \leq v$ on $[0,T] \times X$ ?

背景链接：https://arxiv.org/abs/1511.00705

10905 | AMR-147-0043

### Open Problems in Pluripotential Theory - Question 44 (Dinh)

O / open | AMR 问题清单汇编

Lelong numbers are well-defined not only for positive *closed* currents- in particular it is enough to assume that the current is positive and *harmonic* i.e. $dd^c T=0$. The next question deals with Siu- type problem for harmonic currents:

Let $X$ be a compact K\"ahler manifold and $T$ be a positive harmonic current on $X$. Is it true that the sets $$E_c(T)=\lbrace z\in X|\ v(T,z)\geq c\rbrace$$ are analytic?

背景链接：https://arxiv.org/abs/1511.00705

10906 | AMR-147-0044

### Open Problems in Pluripotential Theory - Question 45 (Bloom-Leveberg)

O / open | AMR 问题清单汇编

The supremum is achieved by the relative extremal function $$h_K(z)=\limsup_{\zeta\rightarrow z}sup\lbrace u(\zeta)|\ u\in {\mathrm{PSH}}(\mathbb{B}^n),\ u\leq 1,\ u|_K\leq 0\rbrace.$$

The set $K$ is called *regular* if the relative extremal function is continuous.

Given any subset $K$ of positive capacity and $\varepsilon>0$ is there a regular set $V\subset K$ such that $$cap(V)\geq cap(K)-\varepsilon?$$

背景链接：https://arxiv.org/abs/1511.00705

10907 | AMR-147-0005

### Open Problems in Pluripotential Theory - Question 5

O / open | AMR 问题清单汇编

The (global) domain of definition $DMA(X,\omega)$ of the complex Monge-Amp\`ere operator $\varphi \mapsto (\omega+dd^c \varphi)^n$ is different than its local counterpart $DMA_{loc}(X,\omega)$. The construction of Blocki [Blo04] breaks down here as the total mass of the Monge-Amp\`ere measures of any approximating sequence is fixed, equal to the volume $Vol_\omega(X)=\int_X \omega^n$. It turns out indeed that $DMA(X,\omega)$ contains functions which do not have gradient in $L^2$.

[Coman-Guedj-Zeriahi] Characterize $DMA(X,\omega)$.

背景链接：https://arxiv.org/abs/1511.00705

10908 | AMR-147-0006

### Open Problems in Pluripotential Theory - Question 6

O / open | AMR 问题清单汇编

Let $\varphi$ be a plurisubharmonic function in the unit ball ${\mathbb{B}}$ of ${\mathbb{C}}^n$. Assume $\varphi$ has a well-defined complex Monge-Amp\`ere measure $(dd^c \varphi)^n$, we could for instance assume that $\varphi$ is locally bounded outside the origin.

[Guedj, Rashkovskii] Assume that $(dd^c \varphi)^n $ has a Dirac mass at the origin. Does it imply that $\varphi$ has a positive Lelong number at $0 \in {\mathbb{C}}^n$ ?

背景链接：https://arxiv.org/abs/1511.00705

10909 | AMR-147-0007

### Open Problems in Pluripotential Theory - Question 7

O / open | AMR 问题清单汇编

As observed by Demailly in [Dem93], it follows from the Ohsawa-Takegoshi theorem that the $\varphi_j$'s approximate $\varphi$ in a very accurate manner, hence motivating the following:

[Demailly] Does one have $(dd^c \varphi_j)^n(0) \longrightarrow (dd^c \varphi)^n(0)$ ?

背景链接：https://arxiv.org/abs/1511.00705

10910 | AMR-147-0008

### Open Problems in Pluripotential Theory - Question 8

O / open | AMR 问题清单汇编

[Coman-Guedj] Assume $(X,\omega)$ is the complex projective space ${\mathbb{C}}\mathbb{P}^n$ equipped with the Fubini-Study K\"ahler form. Can one find examples of a $\omega$-psh function $G_a \in DMA(X,\omega)$, whose Monge-Amp\`ere measure is concentrated at some point $a \in {\mathbb{C}}\mathbb{P}^n$ and having an irrational Lelong number at $a$ ?

背景链接：https://arxiv.org/abs/1511.00705

10911 | AMR-147-0009

### Open Problems in Pluripotential Theory - Question 9

O / open | AMR 问题清单汇编

It has been shown in [BEGZ10, Theorem 1.16] that two cohomologous positive closed currents $T \sim S$ satisfy $$ \langle T^n \rangle \leq \langle S^n \rangle $$ if $T$ is more singular than $S$ and both currents have *small unbounded locus*, i.e. their local potentials are locally bounded outside a closed complete pluripolar set. This latter condition seems purely technical and the question makes sense in full generality:

[Boucksom-Eyssidieux-Guedj-Zeriahi] Assume $T \sim S$ with $T$ more singular than $S$. Is it true that $$ \langle T^n \rangle \leq \langle S^n \rangle \; \; ? $$

背景链接：https://arxiv.org/abs/1511.00705

10912 | AMR-155-0002

### Open Problems in Wavelet Theory - Conjecture 2

O / open | AMR 问题清单汇编

Let $\mathscr D$ be any wavelet dimension function, i.e., $\mathscr D$ satisfies (D1)--(D4). Then, the collection of o.n. wavelets with the same dimension function \[ \{\psi \in L^2(\mathbb{R}): \psi \text{ is an o.n. wavelet and } \mathscr D_\psi=\mathscr D\}. \] is a path connected subset of $L^2(\mathbb{R})$.

背景链接：https://arxiv.org/abs/1905.08230

10913 | AMR-155-0001

### Open Problems in Wavelet Theory - Problem 1

O / open | AMR 问题清单汇编

One of the fundamental areas in the theory of wavelets is the investigation of properties of the collection of all wavelets as a subset of $L^2(\mathbb{R})$. The most prominent problem in this area was formulated independently by D. Larson and G. Weiss around the year 1995.

Is the collection of all orthonormal wavelets (as a subset of the unit sphere in $L^2(\mathbb{R})$) path connected in $L^2(\mathbb{R})$ norm?

背景链接：https://arxiv.org/abs/1905.08230

10914 | AMR-155-0010

### Open Problems in Wavelet Theory - Problem 10

O / open | AMR 问题清单汇编

Does Calder\'on's formula \begin{equation} \sum_{j\in{\mathbb{Z}}} |\hat \psi((A^T)^j \xi)|^2=1 \qquad\text{for a.e. }\xi \in \mathbb{R}^n \end{equation} hold for any orthonormal (or Parseval) wavelet $\psi$ associated with $(A,\Gamma)$?

背景链接：https://arxiv.org/abs/1905.08230

10915 | AMR-155-0011

### Open Problems in Wavelet Theory - Problem 11

O / open | AMR 问题清单汇编

A variant of Problem 3.1 asks for a characterization of dilations for which well-localized o.n. wavelets exist. We say that a function $\psi \in L^2(\mathbb{R}^n)$ is *well-localized* if both $\psi$ and $\hat \psi$ have polynomial decay. That is, for some large $N>0$, we have \[ \psi(x) = O(|x|^{-N}) \text{ as } |x| \to \infty \qquad\text{and}\qquad \hat \psi(\xi) = O(|\xi|^{-N}) \text{ as } |\xi| \to \infty. \]

Let $\Gamma={\mathbb{Z}}^n$ be the lattice of translates. For what expansive dilations $A$ do there exist well-localized wavelets (possibly with multiple generators)?

背景链接：https://arxiv.org/abs/1905.08230

10916 | AMR-155-0012

### Open Problems in Wavelet Theory - Problem 12

O / open | AMR 问题清单汇编

Schwartz class is defined as a collection of all $\infty$-regular functions on $\mathbb{R}^n$.

Do Schwartz class wavelets exist for integer expansive dilations $A$ and lattice $\Gamma ={\mathbb{Z}}^n$?

背景链接：https://arxiv.org/abs/1905.08230

10917 | AMR-155-0013

### Open Problems in Wavelet Theory - Problem 13

O / open | AMR 问题清单汇编

We end by stating not that serious, yet curious problem. The only known construction of wavelets in the Schwartz class is a Meyer wavelet which is a band-limited function. Hence, it is natural ask the following question.

Suppose $\psi$ is an orthonormal wavelet such that $\psi$ belongs to the Schwartz class. Is $\hat\psi$ necessarily compactly supported?

背景链接：https://arxiv.org/abs/1905.08230

10918 | AMR-155-0003

### Open Problems in Wavelet Theory - Problem 3

O / open | AMR 问题清单汇编

This leaves open the problem of existence of Riesz wavelets which was posed by Seip [Seip]. We say that $\psi$ is a Riesz wavelet for $\mathcal{H} =H^2(\mathbb{R})$ or $L^2(\mathbb{R})$ if the wavelet system is a Riesz basis of $\mathcal{H}$. A Riesz basis in a Hilbert space $\mathcal{H}$ can be defined as an image of an orthonormal basis under an invertible operator on $\mathcal{H}$. Every Riesz basis has a dual Riesz basis. However, the dual of Riesz wavelet system might not be a wavelet system. If it is, then we say that $\psi$ is a biorthogonal (Riesz) wavelet.

Does there exist a Riesz wavelet $\psi$ in $H^2(\mathbb{R})$ such that $\psi$ belongs to the Schwartz class?

背景链接：https://arxiv.org/abs/1905.08230

10919 | AMR-155-0004

### Open Problems in Wavelet Theory - Problem 4

O / open | AMR 问题清单汇编

Larson [La2] has posed an interesting problem about frequency supports of wavelets. Must the support of the Fourier transform of a wavelet contain a wavelet set? This problem stems from the observation that there are two ways of describing minimality of frequency support. The first one is that $\operatorname{supp} \hat \psi$ has the smallest possible Lebesgue measure (equal to 1), which is used in the actual definition of an MSF wavelet. The second possibility is to insist that the support is minimal with respect to the inclusion partial order. It is not known whether these two natural definitions of minimality of frequency supports are the same. This is the essence of the following problem posed by Larson in late 1990's although its official formulation appeared only in [La2].

Is it true that for any orthonormal wavelet $\psi \in L^2(\mathbb{R})$, there exists a wavelet set $W$ such that such that $W \subset \operatorname{supp} \hat\psi$?

背景链接：https://arxiv.org/abs/1905.08230

10920 | AMR-155-0005

### Open Problems in Wavelet Theory - Problem 5

O / open | AMR 问题清单汇编

Another fundamental problem posed by Larson [La2] asks about density of Riesz wavelets.

Is the collection of all Riesz wavelets dense in $L^2(\mathbb{R})$?

背景链接：https://arxiv.org/abs/1905.08230

10921 | AMR-155-0006

### Open Problems in Wavelet Theory - Problem 6

O / open | AMR 问题清单汇编

Suppose that $\psi \in L^2(\mathbb{R}^n)$ is a Parseval wavelet. Then, we can define spaces $$ V_j = D^j(V(\psi)) \qquad j\in {\mathbb{Z}}, $$ where $Df(x)=\sqrt{2} f(x)$ is a dilation operator. Baggett has shown that a sequence $\{V_j:j\in{\mathbb{Z}}\}$ satisfies all properties of GMRA (M1), (M2), (M4), and (M5') possibly with the exception of (M3). Hence, it is natural to ask the following question.

Let $\psi$ be a Parseval wavelet with the space of negative dilates $V=V(\psi)$. Is it true that $$ \bigcap_{j\in{\mathbb{Z}}} D^j(V(\psi))=\{0\}? $$

背景链接：https://arxiv.org/abs/1905.08230

10922 | AMR-155-0007

### Open Problems in Wavelet Theory - Problem 7

O / open | AMR 问题清单汇编

A more recent problem was proposed by Christensen and his collaborators [CKK1, CKK2].

Suppose $\psi$ is Bessel wavelet with bound $<1$. Does there exist $\psi'$ such that the combined wavelet system $$ \{\psi_{j,k}: j,k \in {\mathbb{Z}}\} \cup \{\psi'_{j,k}: j,k \in {\mathbb{Z}}\}. $$ generated by $\psi$ and $\psi'$ is a Parseval frame?

背景链接：https://arxiv.org/abs/1905.08230

10923 | AMR-155-0008

### Open Problems in Wavelet Theory - Problem 8

O / open | AMR 问题清单汇编

The last problem illustrates the difficulty of determining whether a function is a frame wavelet or not. The following problem was proposed by Weber and the first author [BW].

For $0<b<1$ define $\psi_b \in L^2(\mathbb{R})$ by $\hat \psi_b = \mathbf 1_{(-1,-b) \cup (b, 1)}$. For what values of $1/8< b \le 1/6$, is $\psi_b$ a frame wavelet?

背景链接：https://arxiv.org/abs/1905.08230

10924 | AMR-155-0009

### Open Problems in Wavelet Theory - Problem 9

O / open | AMR 问题清单汇编

One of the most fundamental problems in wavelet theory asks for a characterization of dilations for which o.n. wavelets exist. Although this problem has been explicitly stated by Speegle [S03] and Wang [Wa], it has been studied earlier in late 1990's.

For what dilations $A \in GL_n(\mathbb{R})$ and lattices $\Gamma \subset \mathbb{R}^n$, there exist an orthonormal wavelet associated with $(A,\Gamma)$?

背景链接：https://arxiv.org/abs/1905.08230

10925 | AIM-SEVERAL_COMPLEX_VARIABLES-0030

### Openness criteria from maximum modulus and fiber geometry

P / partially_solved | AIM 研讨会问题汇编

**Problem 1 (S. Berhanu).** Let \(M^{2n-1}\subseteq\mathbb C^n\) be a real hypersurface satisfying the maximum principle for CR functions in the strong sense (continuous CR functions attaining a (weak) local maximum must be constant). Is it necessary that any local continuous CR function is open?

**Remark.** In \(\mathbb C^2\) the hypothesis cannot hold.

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10926 | OWR-12979-002

### Operator Models for Higher-Rank Cowen-Douglas Flag Classes

P / partially_solved | Oberwolfach 报告问题汇编

For $n>2$, develop a natural higher-rank Cowen-Douglas flag class extending

$$FB_2(\Omega)=\left\{\begin{pmatrix}T_0&S\\0&T_1\end{pmatrix}:T_i\in B_1(\Omega),\ T_0S=ST_1\right\},$$

and determine its unitary invariants and reproducing-kernel model.

来源字段：https://doi.org/10.4171/owr/2014/21

10927 | OWR-14298161-001

### Optimal Eulerian Liftings of Second-Order Functionals

O / open | Oberwolfach 报告问题汇编

For the second-order functional $E(u)=\int_\Omega H(\nabla^2u(x))\,dx$, what is the largest convex, lower-semicontinuous, subadditive lifting of $E$?

来源字段：https://doi.org/10.4171/owr/2024/7

10928 | OWR-16767-004

### Optimal Fourier Restriction Estimates on Submanifolds

O / open | Oberwolfach 报告问题汇编

For the Fourier restriction operator $R$ to a submanifold $C\subset\mathbb R^{1+n}$, determine the optimal exponents $p,q$ for which

$$\lVert Rf\rVert_{L^p(C)}\lesssim\lVert f\rVert_{L^q(\mathbb R^{1+n})}$$

holds.

来源字段：https://doi.org/10.4171/owr/2019/5

10929 | OWR-4531-003

### Optimal Inverse-Growth Conditions for Bicyclic Operator Cyclicity

O / open | Oberwolfach 报告问题汇编

For a bicyclic operator $T$, is $\log\lVert T^{-n}\rVert/\sqrt n\to0$ an optimal sufficient growth hypothesis for cyclicity? Specifically, can cyclicity fail under the weaker assumption $\log\lVert T^{-n}\rVert=O(\sqrt n)$?

来源字段：https://doi.org/10.4171/owr/2010/49

10930 | OWR-12723-002

### Optimal Order of Multidimensional Point-Set Discrepancy

O / open | Oberwolfach 报告问题汇编

For $s\geq3$, is the sharp order of the minimum $L^\infty$ discrepancy of $N$-point sets in the unit half-open $s$-cube equal to $(\log N)^{s-1}/N$, and do the Chen-Skriganov and digital-net constructions discussed in the source attain that order?

来源字段：https://doi.org/10.4171/owr/2013/49

10931 | OWR-17136-001

### Optimal Regularity of $p$-Harmonic Functions

O / open | Oberwolfach 报告问题汇编

Determine the optimal regularity of $p$-harmonic functions in dimensions $d\geq3$.

来源字段：https://doi.org/10.4171/owr/2019/40

10932 | OWR-14299582-005

### Optimal Sector Angles for UMD Ornstein-Uhlenbeck Calculus

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a UMD Banach space. For every $1<p<\infty$, does the Ornstein-Uhlenbeck operator $-L_X$ admit a bounded $H^\infty(\Sigma_\theta)$ functional calculus for every sector angle $0<\theta<\theta_p$?

来源字段：https://doi.org/10.4171/owr/2025/53

10933 | OWR-12331-001

### Optimal SVD-Based Space-Time Focusing Signals

O / open | Oberwolfach 报告问题汇编

For a two-dimensional acoustic medium containing unknown point scatterers and an array of transducers, let $\phi_{\mathrm{sym}}(\omega)$ be the phase selected by a symmetric singular-value decomposition of the frequency-domain scattering operator for a simple singular value. Does the associated superposition of time-harmonic input signals optimally concentrate acoustic energy in space and time near the selected scatterer?

来源字段：https://doi.org/10.4171/owr/2013/03

10934 | OWR-14298585-004

### p-Energies on Nonsmooth Metric-Measure Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Develop an intrinsic notion of $p$-energy for nonsmooth metric-measure spaces and characterize the geometric hypotheses under which the competing Sobolev, Dirichlet-form, upper-gradient, and discrete-approximation constructions agree.

来源字段：https://doi.org/10.4171/owr/2024/42

10935 | AIM-FUNCTIONAL_ANALYSIS-0038

### Padding and catalogue bounds for universal nuclear and exact C*-algebras

P / partially_solved | AIM 研讨会问题汇编

Is there a universal nuclear C*-algebra of character density $\aleph_1$? More generally, for which cardinals $\kappa$ is there a universal nuclear C*-algebra of character density $\kappa$? Similar question can be asked for exact algebras.

来源字段：http://aimpl.org/settheorycstar/6/

10936 | AIM-ANALYSIS-0002

### Painleve tau functions as limits of genuine block Toeplitz determinants

P / partially_solved | AIM 研讨会问题汇编

Investigate the connection between the \(\tau\) function from Painlevé equations and block Toeplitz determinants. Seek explicit examples in which \(\tau\) function asymptotics is known. Find the block Toeplitz determinant. What is it and what do we learn? Use Riemann-Hilbert to connect this to Fuchsian Riemann-Hilbert problems.

来源字段：http://aimpl.org/riemhilberttoeplitz/1/

10937 | OWR-15208-013

### Parallel Harmonic Spinors on Closed Ricci-Flat Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

If $(M,g)$ is closed and Ricci-flat, does it admit a globally parallel harmonic spinor?

来源字段：https://doi.org/10.4171/owr/2017/3

10938 | OWR-4792-003

### Parametrizations of Separable $C^*$-Algebras

O / open | Oberwolfach 报告问题汇编

Are all reasonable parametrizations of separable $C^*$-algebras by elements of a standard Borel space equivalent?

来源字段：https://doi.org/10.4171/owr/2011/02

10939 | OWR-13948-005

### Partial Regularity of Multivalued Quasiconvex Minimizers

P / partially_solved | Oberwolfach 报告问题汇编

Do minimizers of quasiconvex functionals on Q-valued functions have partial regularity, and for which subclasses of functionals do regularity results hold?

来源字段：https://doi.org/10.4171/owr/2015/53

10940 | AIM-ANALYSIS-0005

### Perturbative solvability beyond the reciprocal-symbol symmetry

P / partially_solved | AIM 研讨会问题汇编

“The determinant of Toeplitz \(+\) Handle is expressible in terms of the \(4 \times 4\) Riemann-Hilbert problem for Szego type symbols. The obstacle to asymptotic analysis is the existence of a solution to model problems. If \(d \tilde{d}=1\) on \(\{|z|=1\}\), where \(d=\frac{\phi(z)}{w(z)}\) and \(\tilde{d}=\frac{\phi(1/z)}{w(1/z)}\), then it is solvable. What if this doesn't hold?”

For Szegő-type nonvanishing symbols, what can be said about existence of the model Riemann--Hilbert problem when the ratio \(d=w/\phi\) does not obey \(d(z)d(z^{-1})=1\)?

来源字段：http://aimpl.org/riemhilberttoeplitz/1/

10941 | AIM-ANALYSIS-0030

### Planar nonattainment and a nodal-puncture obstruction for simple first Steklov eigenvalues

P / partially_solved | AIM 研讨会问题汇编

**Non-existence of a maximizer for the first Steklov eigenvalue with perimeter constraint.** Let \(\sigma _1(\Omega)\) denote the first non-trivial Steklov eigenvalue for a bounded domain \(\Omega\subset\mathbb R^n\). Prove that there is no \(\Omega\subset\mathbb R^n\) which maximizes the quantity \[ P(\Omega)^{1/(n-1)}\sigma _1(\Omega). \]

来源字段：http://aimpl.org/shapesurface/4/

10942 | OWR-12005-012

### Planar Steklov Isospectral Domains

O / open | Oberwolfach 报告问题汇编

Do there exist non-isometric planar domains that are isospectral for the Steklov eigenvalue problem?

来源字段：https://doi.org/10.4171/owr/2012/33

10943 | OWR-1535-002

### Poincaré-Function Limits Characterizing Exceptional Polynomials

O / open | Oberwolfach 报告问题汇编

Let $p$ be a real polynomial of degree $d>1$ with $p'(0)=\lambda>1$, and let $f$ be the entire solution of $f(\lambda z)=p(f(z))$, $f(0)=0$, $f'(0)=1$. With $N_f(x)=\#\{\xi:f(\xi)=0,\ |\xi|<x\}$, $\mu$ the associated harmonic measure, and $\rho=\log_\lambda d$, does the existence of either limit $\lim_{x\to\infty}x^{-\rho}N_f(x)$ or $\lim_{t\to0}t^{-\rho}\mu(B(0,t))$ characterize polynomials linearly conjugate to a Chebyshev polynomial or a pure power?

来源字段：https://doi.org/10.4171/owr/2007/19

10944 | OWR-1196-006

### Poisson Midpoint Methods for Nonconstant Structure Matrices

O / open | Oberwolfach 报告问题汇编

For a Poisson system $y'=J(y)\nabla H(y)$, does a nonstandard generalization of the implicit midpoint rule exist that is conjugate to a Poisson map for a perturbed $J(y)$?

来源字段：https://doi.org/10.4171/owr/2006/14

10945 | AIM-ANALYSIS-0055

### Pole-orbit stability for rational Henon recurrences

P / partially_solved | AIM 研讨会问题汇编

Investigate the dynamics of maps of the form \[ (x,y)\longmapsto (p(x)-\delta y,x), \] where \(p(x)\) is a degree \(2\) rational map.

来源字段：http://aimpl.org/scvproblems/2/

10946 | AIM-ANALYSIS-0023

### Polygonal Faber-Krahn status and an exact affine-gauge reduction

P / partially_solved | AIM 研讨会问题汇编

**Discrete Faber--Krahn inequality.** Among all polygons of a given area and having not more than \(N\) edges, the lowest Dirichlet eigenvalue \(\lambda_1(\Omega)\) is minimized by the regular \(N\)-gon.

\[ \lambda _1(\Omega)=\min_{0\ne v\in H^1_0(\Omega)} \frac{\int_\Omega |\nabla v|^2\,dx}{\int_\Omega v^2\,dx}. \]

The record also asks about five subsidiary directions: local minimality of the regular polygon, convexity of a minimizer, monotonicity of the regular-polygon eigenvalues in the number of sides, the analogous first nonzero Neumann problem, and perimeter-constrained analogues. The exact AIM record says that a minimizer exists and has exactly \(N\) sides, citing Henrot, Theorem 3.3.1.

No reconstruction of mathematical notation was needed.

来源字段：http://aimpl.org/shapesurface/2/

10947 | AIM-SEVERAL_COMPLEX_VARIABLES-0040

### Polynomial approximation of integrable holomorphic one-forms

P / partially_solved | AIM 研讨会问题汇编

**Problem 1 (J. E. Fornæss).** Let \(\omega\) be a holomorphic \(1\)-form on a polydisc in \(\mathbb C^3\) with \(\omega\wedge d\omega=0\). Can \(\omega\) be approximated on compact subsets by polynomial \(1\)-forms \(h\) satisfying \(h\wedge dh=0\)?

**Remark.** A positive answer would lead to a nontrivial foliation of \(\mathbb{CP}^3\) by hypersurfaces.

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10948 | OWR-4531-014

### Polynomial Density on Crescent Domains

P / partially_solved | Oberwolfach 报告问题汇编

Let $\Omega$ be a crescent domain with bounded complementary component $G$ and unbounded complementary component $\Omega_\infty$, put $\delta(z)=\operatorname{dist}(z,\Omega_\infty)$, and let $\omega$ be harmonic measure on $\partial G$. If $$\int_{\partial G}\log\delta(z)\,d\omega(z)=-\infty,$$ must $H^p(\Omega,dA)=L_a^p(\Omega,dA)$ hold for every $1\leq p<\infty$?

来源字段：https://doi.org/10.4171/owr/2010/49

10949 | AIM-ANALYSIS-0043

### Polynomial density, multivariable defect kernels, and boundary models for de Branges--Rovnyak spaces

P / partially_solved | AIM 研讨会问题汇编

Let \(b\in H^\infty(\Delta)\) with \(\|b\|_{H^\infty}\leq 1\). Define \(H(b):=(I-T_bT_{\overline b})^{1/2}H^2(\Delta)\), where \(T_b\) denotes multiplication by \(b\).

1. When are polynomials dense in \(H(b)\)? (known in affirmative for non-extremal \(b\) or inner \(b\)) 2. How to generate the \(H(b)\)s as reproducing kernel Hilbert spaces to the polydisk or the ball in \(\mathbb C^n\)? 3. Find integral representations of these spaces in settings beyond the unit disk \(\Delta\).

来源字段：http://aimpl.org/scvproblems/1/

10950 | OWR-1787-006

### Polynomial Stability of Hypercyclic Entire Functions

O / open | Oberwolfach 报告问题汇编

Let $HC(D)$ denote the entire functions whose sequence of derivatives is dense in $H(\mathbb{C})$, equivalently the hypercyclic vectors for differentiation. If $f\in HC(D)$ and $p$ is a nonconstant polynomial, is $pf\in HC(D)$?

来源字段：https://doi.org/10.4171/owr/2008/06

10951 | OWR-1787-005

### Polynomial Stability of Universal Taylor Series

O / open | Oberwolfach 报告问题汇编

Let $U(D,0)$ denote universal Taylor series in the unit disk. If $f\in U(D,0)$ and $a\in\mathbb{C}$ with $|a|\geq1$, is $(z-a)f(z)$ again in $U(D,0)$? More generally, for a nonconstant polynomial $p$, is $pf\in U(D,0)$? Is $f'\in U(D,0)$? Does there exist an inner function $f\in U_1(D,0)$?

来源字段：https://doi.org/10.4171/owr/2008/06

10952 | OWR-4990379-006

### Polynomial-Time Point Optimization and Abrikosov Crystallization

O / open | Oberwolfach 报告问题汇编

How can one construct, in polynomial time, $N$-point configurations on a prescribed submanifold of $\mathbb R^d$ that are optimal to high accuracy, and how can large-$N$ Abrikosov crystallization, especially in two dimensions, be understood mathematically?

来源字段：https://doi.org/10.4171/owr/2021/10

10953 | AMR-023-0008

### Pompeiu problem

P / partially_solved | AMR 问题清单汇编

Characterize the domains for which there exists a nonzero function whose integral vanishes over every congruent copy of the domain.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767

10954 | AIM-FUNCTIONAL_ANALYSIS-0035

### Positive consistency of Naimark's problem and a local hereditary reduction

P / partially_solved | AIM 研讨会问题汇编

Is the following relatively consistent with ZFC?

Every C*-algebra \(A\) that has a unique irreducible representation up to the unitary equivalence is isomorphic to the algebra of compact operators on some Hilbert space.

来源字段：http://aimpl.org/settheorycstar/6/

10955 | AIM-ANALYSIS-0051

### Positive logarithmic capacity does not force a nonconstant H^2 function

S / solved | AIM 研讨会问题汇编

For a domain \(\Omega\subset\mathbb C\), define \[ H^2(\Omega):=\{f\text{ holomorphic on }\Omega: |f|^2\text{ has a harmonic majorant}\}. \] Does \(\operatorname{cap}(\mathbb C\setminus\Omega)>0\) imply \(\dim H^2(\Omega)>1\)?

来源字段：http://aimpl.org/scvproblems/1/

10956 | OWR-15219-020

### Positive Radon Measures for Truncated Moment Data

O / open | Oberwolfach 报告问题汇编

Let $X$ be locally compact, let $V\subseteq C(X,\mathbb R)$ have basis $a_1,\ldots,a_m$, and let $s_1,\ldots,s_m\in\mathbb R$. Characterize when there is a positive Radon measure $\mu$ on $X$ such that $$s_j=\int_X a_j(x)\,d\mu(x)\qquad(1\leq j\leq m)$$.

来源字段：https://doi.org/10.4171/owr/2017/14

10957 | OWR-4136-007

### Positive-Definite Functions of Minkowski Functionals

P / partially_solved | Oberwolfach 报告问题汇编

Let $Y$ be a random variable. Call $X=(X_1,\ldots,X_n)$ an $n$-dimensional version of $Y$ if there is a positive function $\gamma$ on $\mathbb{R}^n\setminus\{0\}$ such that $\sum_i a_iX_i$ and $\gamma(a)Y$ have the same distribution for every $a\neq0$. Characterize all such versions and all possible functions $\gamma$. Equivalently, for an origin-symmetric star body $K$, characterize the continuous real-valued functions $f$ on the nonnegative real axis for which $f(\|\cdot\|_K)$ is positive definite, and characterize those $K$ for which the resulting function class is nontrivial.

来源字段：https://doi.org/10.4171/owr/2009/53

10958 | OWR-15219-017

### Positivity and Closure of Flat Moment Matrices

P / partially_solved | Oberwolfach 报告问题汇编

Give concrete conditions on a truncated moment sequence $\beta^{(m)}$ that ensure positivity of its Riesz functional $L_\beta$, and characterize the closure $F_{n,d}$ of the cone of flat positive $n$-dimensional moment matrices of degree $2d$.

来源字段：https://doi.org/10.4171/owr/2017/14

10959 | OWR-16163-006

### Positivity of Hermitian Forms on Symmetric Tensor Powers

P / partially_solved | Oberwolfach 报告问题汇编

Let $W$ be a Hermitian operator on the $k$th symmetric power of $\mathbb C^d$, and define

$$p_W(y)=\langle y^{\otimes k},Wy^{\otimes k}\rangle.$$

Characterize when $p_W(y)\geq0$ for every $y\in\mathbb C^d$.

来源字段：https://doi.org/10.4171/owr/2018/22

10960 | OWR-11128-006

### Power Decay for Multilinear Oscillatory Integrals

O / open | Oberwolfach 报告问题汇编

For the multilinear oscillatory integral $I_\lambda$ with a nondegenerate polynomial phase, does there exist $\delta>0$ such that $|I_\lambda(f_1,\ldots,f_n)|\leq C|\lambda|^{-\delta}\prod_j\|f_j\|_{L^\infty}$ as $|\lambda|\to\infty$?

来源字段：https://doi.org/10.4171/owr/2011/36

10961 | OWR-764-001

### Power Decay for Polynomial Oscillatory Integrals

O / open | Oberwolfach 报告问题汇编

Does every nondegenerate polynomial phase function $P$ yield power decay for the associated oscillatory integral?

来源字段：https://doi.org/10.4171/owr/2004/50

10962 | AIM-SEVERAL_COMPLEX_VARIABLES-0013

### Product branching and the missing irreducibility hypothesis

S / solved | AIM 研讨会问题汇编

Let $D_1$ and $D_2$ be two bounded symmetric domains such that neither is a ball. Let $F:D_1\to D_2$ be a proper holomorphic map. Is $F$ a *trivial map*, that is $F(z)=(z,g(z))$, in suitable coordinates, where $g(z)$ is a vector valued holomorphic function? By a result of Tsai, the answer is yes when the rank (as a bounded symmetric domain) of $D_2$ is not greater than the rank of $D_1$.

来源字段：http://aimpl.org/crscv/3/

10963 | OWR-1062-012

### Pukánszky Invariants of Masas in $L(F_{2})$

O / open | Oberwolfach 报告问题汇编

Must every maximal abelian self-adjoint subalgebra of the free group factor $L(\mathbb F_2)$ have $\infty$ in its Pukánszky invariant?

来源字段：https://doi.org/10.4171/owr/2005/41

10964 | AIM-ANALYSIS-0075

### Pure-state extension from a MASA of a factor

P / partially_solved | AIM 研讨会问题汇编

**QUESTION 2.** Given a MASA \(A\) of a factor \(M\), do the pure states of \(A\) have unique state extension to \(M\)?

The compilation is dated November 30, 2014.

The question is:

The source then records: a positive answer for discrete MASAs of \(B(H)\) when \(H\) is separable (by Marcus--Spielman--Srivastava); a negative answer when the MASA has a continuous part; a negative answer for all MASAs in type II factors on separable \(H\); mixed behavior for nonseparable type II factors; and “no results at all” for type III factors.

The quantifier in the question is important. It asks whether **every** pure state of \(A\), including singular pure states, has exactly one state extension to \(M\). It does not ask whether a particular pure state has a unique extension, and it does not ask only about normal states.

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10965 | AIM-ANALYSIS-0076

### Pure-state extensions across a split ideal

P / partially_solved | AIM 研讨会问题汇编

**Question 3.** Let a C*-algebra \(B\) be the direct sum of an ideal and a MASA, \(B=I\oplus A\). Under what circumstances do the pure states of \(A\) have unique extension to states of \(B\)?

The mathematical question is

Those bullets are not part of Question 3. Moreover, the extractor enqueues those bullets again as subsequent AIM records. Only the quoted sentence is analyzed here.

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10966 | AIM-ANALYSIS-0095

### Quantitative counts of Weaver-good bipartitions

P / partially_solved | AIM 研讨会问题汇编

Understand limitations of the proof, we don't have any idea of how many good partitions there are? come up with more quantitative versions of the proof.

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

10967 | AIM-SEVERAL_COMPLEX_VARIABLES-0020

### Quantitative interior jet bounds and boundary determination regimes

P / partially_solved | AIM 研讨会问题汇编

Let \(n\geq 2\).

- Given a proper rational map \(f:\mathbb B^n\to\mathbb B^N\), find the smallest number \(k(n,N)\) such that \(f\) is determined by its \(k\)-jet at the origin. - Let \(\Omega\subset\mathbb C^n\) and \(\Omega'\subset\mathbb C^N\) be strongly pseudoconvex domains. If \(f:\Omega\to\Omega'\) is a proper holomorphic map which extends smoothly to \(\partial\Omega\), does there exist a \(k\) such that \(f\) is determined by its \(k\)-jet at a point?

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

10968 | AIM-ANALYSIS-0109

### Quantitative local obstruction and conditional Nagata upper bound for Hadamard ALRs

P / partially_solved | AIM 研讨会问题汇编

**Absolute Lipschitz retracts** Are all Hadamard manifolds absolute Lipschitz retracts?

来源字段：http://aimpl.org/mappingmetric/1/

10969 | OWR-13674-019

### Quantitative Real-Rootedness and Two-Eigenvalue Interlacing

O / open | Oberwolfach 报告问题汇编

Can one define a quantitative distance from real-rootedness for which inequalities implied by real-rootedness persist with a distance-dependent penalty? Can interlacing-polynomial methods be extended to control two eigenvalues simultaneously?

来源字段：https://doi.org/10.4171/owr/2015/17

10970 | AIM-ANALYSIS-0037

### Quantitative transfer of ordered nodal slabs

P / partially_solved | AIM 研讨会问题汇编

Consider the nodal domains of the $k$-th Neumann eigenfunction on a sequence of domains collapsing to a lower dimensional object.

Under what assumptions can one prove that the nodal domains are in a sense ordered?

来源字段：http://aimpl.org/shapesurface/6/

10971 | OWR-17468-001

### Quasi-Optimality for Nonconforming Galerkin Spaces

O / open | Oberwolfach 报告问题汇编

For nonconforming Galerkin spaces $V_{j}^{h}\not\subset V$, what error estimate should replace the standard Céa quasi-optimality bound $\|\phi-\phi_j^h\|\leq C\inf_{\psi\in V_j^h}\|\phi-\psi\|$?

来源字段：https://doi.org/10.4171/owr/2020/5

10972 | OWR-13948-004

### Quasiconvexity of Multiple-Valued Integrands

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Quasiconvexity Extraction [OWR-13948-003]

Let $f:B\times\operatorname{Hom}(\mathbb{R}^m,TM)\to\mathbb{R}_{\geq0}$ be quasiconvex and define $e_M(x,\nu_1,\ldots,\nu_Q,N_1,\ldots,N_Q)=\sum_{i=1}^Qf(x,\nu_i,N_i)$. Is $e_M$ quasiconvex on the space of $Q$-valued maps into $M$?

来源字段：https://doi.org/10.4171/owr/2015/53

10973 | AIM-ANALYSIS-0117

### Quasisymmetric uniformization via weak metric doubling measures

S / solved | AIM 研讨会问题汇编

**Quasisymmetric uniformization of metric \(2\)-spheres**

Give a geometric characterization of metric \(2\)-spheres which are quasisymmetrically equivalent to the standard \(2\)-sphere.

来源字段：http://aimpl.org/mappingmetric/3/

10974 | AIM-SEVERAL_COMPLEX_VARIABLES-0044

### Ramification obstructions and the compact-potential inconsistency

P / partially_solved | AIM 研讨会问题汇编

Let \(X,Y\) be compact Riemann surfaces and let \(a_j\in X\) be points with positive finite multiplicities \(m_j\). Does there exist a holomorphic map \(f:X\to Y\) such that, in local coordinates at \(a_j\) and \(f(a_j)\), \[ f(z)=z^{m_j+1}, \] and \(f\) has no other critical points?

来源字段：https://aimath.org/WWN/complexitycrmap/complexitycrmap.pdf

10975 | OWR-11101912-010

### Random Set Methods for Harmonic Analysis Estimates

O / open | Oberwolfach 报告问题汇编

Let $B_p(\Gamma)$ be the least correction factor in the weak $L^p([0,1]^d)$ majorant inequality for frequencies in $\Gamma\subset[1,N]^d$. For which sets $\Gamma$ does $B_p(\Gamma)\ll N^\varepsilon$ hold for every $\varepsilon>0$, and can random-set methods be adapted to prove such estimates?

来源字段：https://doi.org/10.4171/owr/2022/31

10976 | AIM-ANALYSIS-0161

### Random supports, Newton polytopes, and local Viro complexity

P / partially_solved | AIM 研讨会问题汇编

Investigate the connections between random sparse polygons and Newton polytopes. This should be extended to the case of random Viro diagrams.

来源字段：https://aimath.org/WWN/randomzeros/arcc1.pdf

10977 | OWR-723-006

### Random-Order Discrepancy Bounds for Lattice Point Sets

O / open | Oberwolfach 报告问题汇编

Prove for shifted lattice point sets $M_n$ a star-discrepancy bound $\operatorname{disc}_\infty(M_n)\leq C\sqrt{d/n}$, or a comparably sharp bound of the same order as the probabilistic bound for unrestricted $n$-point sets.

来源字段：https://doi.org/10.4171/owr/2004/13

10978 | OWR-5154-006

### Rank Completions of Crossed-Product Rings

O / open | Oberwolfach 报告问题汇编

For a measure-preserving action $\alpha$ of a countable group $\Gamma$, which invariants of the action are determined by the rank completion $A_\alpha$ of the regular closure of its crossed-product ring? For which nonamenable groups can this completion be independent of the action?

来源字段：https://doi.org/10.4171/owr/2011/26

10979 | AIM-SEVERAL_COMPLEX_VARIABLES-0012

### Rank spectra for Hermitian squared norms divisible by norm powers

P / partially_solved | AIM 研讨会问题汇编

Let \(R(z,\bar z)\) be a Hermitian polynomial which is a sum of squares \[ R(z,\bar z)=\sum_{j=1}^{m}|p_j(z)|^2, \] where the holomorphic polynomials \(p_j\) are linearly independent. If \(R=\|z\|^2A\), what are the possible values of \(m\)? Huang's lemma implies that \(m=0\) or \(m\ge n\). Consider the same question with \(\|z\|^{2d}\).

The two MathSciNet citations were recovered exactly.

- MR1703603 is Xiaojun Huang, “On a linearity problem for proper holomorphic maps between balls in complex spaces of different dimensions,” *Journal of Differential Geometry* 51 (1999), 13--33. The lemma used by AIM is the rank obstruction for a nonzero expression divisible by \(\|z\|^2\). - MR2869101 is John P. D'Angelo and Jiří Lebl, “Pfister's theorem fails in the Hermitian case,” *Proceedings of the American Mathematical Society* 140 (2012), 1151--1157. Its Theorem 2.1 is the requested higher-power generalization.

The canonical record itself is internally coherent and the primary papers remove any ambiguity about \(d\): \(d\) is a positive integer and \[ \|z\|^{2d}=\left(\sum_{i=1}^n |z_i|^2\right)^d. \] No homogeneity of \(R\), \(A\), or the \(p_j\) is stated. Importantly, the D'Angelo--Lebl bound applies even to real-analytic, nonhomogeneous multiples.

来源字段：http://aimpl.org/crscv/3/

10980 | OWR-8415338-011

### Rank-One Convex Energies Excluding Nontrivial Young Measures

P / partially_solved | Oberwolfach 报告问题汇编

Find rank-one convex functions whose differential inclusion for critical points under outer variations only supports exclusively trivial Young measures.

来源字段：https://doi.org/10.4171/owr/2021/35

10981 | OWR-4337-003

### Rates and Self-Similarity in Energy-Driven Coarsening

O / open | Oberwolfach 报告问题汇编

For energy-driven coarsening systems, what is the rate of coarsening and can it be established rigorously, what features do the evolving patterns have, and are their statistics self-similar in time?

来源字段：https://doi.org/10.4171/owr/2010/24

10982 | AIM-ANALYSIS-0147

### Rational counterexamples and a finite gap-two parameter theorem

S / solved | AIM 研讨会问题汇编

Let \((L_n^\alpha)\) denote the generalized Laguerre polynomials. It is known that \(L_m^\alpha\) and \(L_n^\alpha\) can have common zeros if \(|m-n|>1\). Is the number of common zeros linked to arithmetic properties of \(\alpha\)?

There are no common zeros between \(L_m^\alpha\) and \(L_n^\alpha\), \(|m-n|>1\), for \(\alpha\) rational.

来源字段：http://aimpl.org/hyperbolicpoly/6/

10983 | AMR-036-0028

### Rational Goldberg extremals

O / open | AMR 问题清单汇编

For rational functions of each fixed degree, determine the analogues of Goldberg's constant and the unit-disk zero/one-point extremal quantities, and characterize their extremal functions.

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10984 | AMR-036-0026

### Real two-one-point extremals

P / partially_solved | AMR 问题清单汇编

Solve the two-one-point extremal problem for real holomorphic $f$: determine the minimum of $\max(|z_1|,|z_2|)$ and maximum of $|f'(0)|$ when $f(0)=0$, $0$ is its only zero, and $z_1,z_2$ are its two $1$-points.

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10985 | AIM-ANALYSIS-0009

### Real-linear small-norm criteria for conjugate dbar and additive jump problems

P / partially_solved | AIM 研讨会问题汇编

For planar orthogonal polynomials, we know that \[ \overline{\partial}Y=\overline Y \begin{bmatrix}0&w\\0&0\end{bmatrix} \] and \[ \int p_j\overline{p_k}w\,dA=h_j\delta_{j,k}. \] Is there an analogue of small norm theory of such \(\overline\partial\)-problems (i.e. \(\overline\partial A=\overline A W+AV\)?) Or small norm theory for \(A_+-A_-=\overline{A_-}V_C+A_-V_n\)?

来源字段：http://aimpl.org/riemhilberttoeplitz/2/

10986 | OWR-12979-005

### Realization of Matricial Families by Tensor-Algebra Elements

P / partially_solved | Oberwolfach 报告问题汇编

Let $M$ be a $W^*$-algebra, let $E$ be a $W^*$-correspondence over $M$, and let $Z$ be an admissible sequence of weights defining the weighted tensor algebra $\mathcal T_+(E,Z)$. Every $F\in\mathcal T_+(E,Z)$ has a Berezin transform that forms a matricial family on the associated weighted intertwiner domains. Conversely, does every such matricial family arise as the Berezin transform of some $F\in\mathcal T_+(E,Z)$?

来源字段：https://doi.org/10.4171/owr/2014/21

10987 | OWR-4214-008

### Realization of Sober Spaces as Primitive Ideal Spaces

O / open | Oberwolfach 报告问题汇编

Is every second-countable coherent, locally compact, sober $T_0$ space $X$ homeomorphic to $\operatorname{Prim}(A)$ for some amenable $C^*$-algebra $A$?

来源字段：https://doi.org/10.4171/owr/2010/13

10988 | OWR-12866-013

### Rectangular Factorization in the Disk Algebra

O / open | Oberwolfach 报告问题汇编

Does every $f$ in the disk algebra $A(\mathbb D)$ admit a rectangular factorization $f=g*h$ with $g,h\in A(\mathbb D)$?

来源字段：https://doi.org/10.4171/owr/2014/06

10989 | AMR-036-0038

### Rectangular-lattice Landau extremal

S / solved | AMR 问题清单汇编

For the rectangular lattice $\Lambda=\{an+ibm:n,m\in\mathbb Z\}$ with $a^2+b^2=1$ and $a\in(0,1)$, let $f_a:\mathbb D\to\mathbb C\setminus\Lambda$ be the universal cover normalized by $f_a(0)=(a+ib)/2$ and $f_a'(0)>0$. Maximize $f_a'(0)$ over $a\in(0,1)$.

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

10990 | OWR-14297739-001

### Recurrence-Coefficient Characterization of Bernstein-Szegő Measures

O / open | Oberwolfach 报告问题汇编

Characterize Bernstein-Szegő measures completely through recurrence coefficients of their associated multivariate orthogonal spaces, analogously to the known spectral characterization on $\mathbb T^2$.

来源字段：https://doi.org/10.4171/owr/2023/43

10991 | AIM-ANALYSIS-0004

### Refutation status and a uniformly sectorial endpoint theorem

P / partially_solved | AIM 研讨会问题汇编

Please see C.A. Berger and L.A. Coburn's 1994 article “Heat Flow and Berlin-Toeplitz Estimates” for background on this conjecture.

(Berger-Coburr 1994): On the Fock/Bergman space, define the projection \(\prod:L^2(\mathbb C,e^{-|z|^2}dA)\to\mathscr F_{\mathbb C}\). Take \(p:\mathbb C\to\mathbb C\) and define \(T_p(f)=\prod(p(z)f(z))\). \(T_p\) is bounded on the Fock space iff \(u_t=\Delta u,\ u|_{t=0}=p\), and \(u(1/4,x,y)\) are bounded.

**Recovered conjecture (Berger--Coburn, original normalization).** Let \(g:\mathbb C^n\to\mathbb C\) be measurable and suppose \(gk_a\in L^2(d\mu_{1/2})\) for every normalized coherent state \(k_a\). Then the Toeplitz operator \[ T_gf=P_{1/2}(gf), \] initially defined on the span of coherent states, extends boundedly on \(F^2_{1/2}\) if and only if \(H_{1/4}g\) is bounded.

来源字段：http://aimpl.org/riemhilberttoeplitz/1/

10992 | AIM-ANALYSIS-0008

### Regular and obstructed matrix Riemann--Hilbert formulations for KdV

P / partially_solved | AIM 研讨会问题汇编

**In general, can you pose matrix Riemann Hilbert problems for KdV?**

来源字段：http://aimpl.org/riemhilberttoeplitz/2/

10993 | OWR-15960-006

### Relative Commutants and Möbius Nets for Finite-Group Inclusions

O / open | Oberwolfach 报告问题汇编

Let $(N_0\subset M_0,\Omega_0)$ be a standard half-sided modular inclusion, and form the free-product inclusion $(N\subset M,\Omega)$ from finitely many isomorphic copies indexed by a finite set $K$. Determine the relative commutant $N'\cap M$. If it is nontrivial, which Möbius-covariant net does the inclusion define?

来源字段：https://doi.org/10.4171/owr/2017/55

10994 | OWR-1113-001

### Relaxation and Microstructure Evolution in Finite Plasticity

P / partially_solved | Oberwolfach 报告问题汇编

For rate-independent finite plasticity, determine the relaxation of the first incremental variational problem, including the quasiconvex envelope of the effective energy density, and formulate a mathematically appropriate time-continuous evolution of microstructures and pattern formation.

来源字段：https://doi.org/10.4171/owr/2005/52

10995 | OWR-13750332-006

### Reliability of Bone Mineral Density Imaging

P / partially_solved | Oberwolfach 报告问题汇编

Are bone-mineral-density measurements obtained by DXA or CT reliable?

来源字段：https://doi.org/10.4171/owr/2023/21

10996 | OWR-15219-014

### Representing Measures for Finite-Dimensional Function Spaces

O / open | Oberwolfach 报告问题汇编

Let $S$ be a topological space, let $V$ be a finite-dimensional vector space of continuous real-valued functions on $S$, and let $L\in V^*$. Characterize when there exists a Borel measure $\mu$ on $S$ such that $$L(f)=\int_S f\,d\mu\qquad\text{for every }f\in V$$.

来源字段：https://doi.org/10.4171/owr/2017/14

10997 | OWR-1592-004

### Reproducing Subgroups of Metaplectic-Heisenberg Representations

O / open | Oberwolfach 报告问题汇编

For the metaplectic-Heisenberg representation on $L^2(\mathbb{R}^d)$, which subgroups $H$ of the ambient group $G$ admit an analyzing window $\varphi\in L^2(\mathbb{R}^d)$ such that $f=\int_H\langle f,\widetilde{\mu}_e(h)\varphi\rangle\widetilde{\mu}_e(h)\varphi\,dh$ for every $f\in L^2(\mathbb{R}^d)$?

来源字段：https://doi.org/10.4171/owr/2007/36

10998 | AMR-022-1010

### Research Problems in Function Theory - Problem 1.10

O / open | AMR 问题清单汇编

If $f(z)$ is a meromorphic function of finite order with more than two deficient values, is it true that if $\sigma>1$, then \[\limsup_{r\to\infty}\frac{T(\sigma r)}{T(r)}<+\infty.\]

背景链接：https://arxiv.org/abs/1809.07200

10999 | AMR-022-1015

### Research Problems in Function Theory - Problem 1.15

S / solved | AMR 问题清单汇编

(Edrei's spread conjecture) If $f(z)$ is meromorphic in the plane and of lower order $\lambda$, and if $\delta=\delta(a,f)>0$, is it true that, for a sequence $r=r_\nu\to\infty$, $f(z)$ is close to $a$ on a part of the circle $|z|=r_\nu$ having angular measure at least \[\frac{4}{\lambda}\sin^{-1}\sqrt{\left(\frac{\delta}{2}\right)}+o(1)?\] (For a result in this direction, see Edrei .)

背景链接：https://arxiv.org/abs/1809.07200

11000 | AMR-022-1016

### Research Problems in Function Theory - Problem 1.16

P / partially_solved | AMR 问题清单汇编

For any function $f(z)$ in the plane, let $n(r)=\sup_a n(r,a)$ be the maximum number of roots of the equation $f(z)=a$ in $|z|<r$, and \[A(r) = \frac{1}{\pi}\int\int_{|z|<r}\frac{|f'(z)|^2}{\{1+|f(z)|^2\}^2}\,dx\,dy = \frac{1}{\pi}\int\int_{|a|<\infty}\frac{n(r,a)\,|da|^2}{(1+|a|^2)^2}.\] Then $\pi A(r)$ is the area, with due count of multiplicity, of the image on the Riemann sphere of the disc $|z|<r$ under $f$, and $A(r)$ is the average value of $n(r,a)$ as $a$ moves over the Riemann sphere. It is known (see Hayman ) that \[1\leq\liminf_{r\to\infty}\frac{n(r)}{A(r)}\leq e.\] Can $e$ be replaced by any smaller quantity, and in particular, by $1$?

背景链接：https://arxiv.org/abs/1809.07200

11001 | AMR-022-1002

### Research Problems in Function Theory - Problem 1.2

P / partially_solved | AMR 问题清单汇编

How big can the set of Valiron deficiencies be for functions in the plane? It is known that $$ N(r,a)=T(r,f)+O\big(T(r,f)^{\frac{1}{2}+\varepsilon}\big) $$ as $r\to\infty$, for all $a$ outside a set of capacity zero (see Nevanlinna . In the case $R<+\infty$ this is more or less best-possible, but in the plane we only know from an example of Valiron that the corresponding set of $a$ can be non-countably infinite. It is also not known whether ([source label: 1.3]) can be sharpened.

背景链接：https://arxiv.org/abs/1809.07200

11002 | AMR-022-1023

### Research Problems in Function Theory - Problem 1.23

P / partially_solved | AMR 问题清单汇编

Under what circumstances does $f(z_0+z)$ have the same deficiencies as $f(z)$? It was shown by Dugu{\'e} that this need not be the case for meromorphic functions, and by Hayman that it is not necessarily true for entire functions of infinite order. The case of functions of finite order remains open. Valiron notes that a sufficient condition is \[\frac{T(r+1,f)}{T(r,f)}\to 1, \hspace{1cm}\text{ as }r\to\infty,\] and this is the case in particular if $\rho-\lambda<1$. Since for entire functions of lower order $\lambda$, $\lambda\leq\frac{1}{2}$ there are no deficiencies anyway, it follows that the result is true at any rate, for entire functions of order $\rho<\frac{3}{2}$ and, since $\lambda\geq0$ always, for meromorphic functions of order less than one.

背景链接：https://arxiv.org/abs/1809.07200

11003 | AMR-022-1026

### Research Problems in Function Theory - Problem 1.26

P / partially_solved | AMR 问题清单汇编

The analogue of Problem 1.7 may be asked for meromorphic functions. The proposers conjecture that in this case \[\sum\delta(a,f)\leq\max\{\Lambda_1(\rho),\Lambda_2(\rho)\},\] where for $\rho\geq1, q=[2\rho]$ we have \[\Lambda_1(\rho)=2-\frac{2\sin\big(\frac{1}{2}\pi(2\rho-q)\big)}{q+2\sin\big(\frac{1}{2}\pi(2\rho-q)\big)},\] \[\Lambda_2(\rho)=2-\frac{2\cos\big(\frac{1}{2}\pi(2\rho-q)\big)}{q+1}.\] Weitsman shows that this result would be sharp. The correct bound is known for $0\leq\rho\leq1$. (D. Drasin and A. Weitsman)

背景链接：https://arxiv.org/abs/1809.07200

11004 | AMR-022-1027

### Research Problems in Function Theory - Problem 1.27

P / partially_solved | AMR 问题清单汇编

Let $E$ be the set for which $m(r,a)\to\infty$ as $r\to\infty$. How large can $E$ be if: [(a)] ; $f$ is entire and of order $\frac{1}{2}$ mean type, ; $f$ is meromorphic of order $\rho$, where $0\leq\rho\leq\frac{1}{2}$. The proposers settled this problem in all other cases (see Update 2.1 for more details). (D. Drasin and A. Weitsman)

背景链接：https://arxiv.org/abs/1809.07200

11005 | AMR-022-1003

### Research Problems in Function Theory - Problem 1.3

S / solved | AMR 问题清单汇编

If $f(z)$ is meromorphic of finite order $\rho$ and $\sum\delta(a,f)=2$, it is conjectured that $\rho=n/2$, where $n$ is an integer and $n\geq 2$, and all the deficiencies are rational. F. Nevanlinna has proved this result on the condition that $f(z)$ has no multiple values, so that $n(r,a)=\overline{n}(r,a)$ for every $a$ (see also R. Nevanlinna ).

背景链接：https://arxiv.org/abs/1809.07200

11006 | AMR-022-1030

### Research Problems in Function Theory - Problem 1.30

O / open | AMR 问题清单汇编

Can one establish an upper bound on the number of finite asymptotic values of a meromorphic function $f(z)$ in $\mathbb{C}$, taking into account both the order of $f$, and the angular measure of its tracts? (W. Al-Katifi)

背景链接：https://arxiv.org/abs/1809.07200

11007 | AMR-022-1031

### Research Problems in Function Theory - Problem 1.31

O / open | AMR 问题清单汇编

Let the function $f$ be meromorphic in the plane, and not rational, and satisfy the condition $$ \frac{T(r,f)}{(\log r)^3}\to\infty,\hspace{1cm}\text{ as }r\to\infty, $$ where $T(r,f)$ is the Nevanlinna characteristic. A theorem of Yang Lo states that then there exists a direction $\theta_0\in[0,2\pi)$ such that for every positive $\varepsilon$, either $f$ attains every finite value infinitely often in $D_\varepsilon=\{z:|\arg z - \theta_0|<\varepsilon\}$, or else $f^{(k)}$ attains every value, except possibly zero, infinitely often in $D_\varepsilon$ for all positive integers $k$. Can the condition ([source label: star]) be dropped completely? Or, possibly, can it be replaced be the `more usual' condition \[\frac{T(r,f)}{(\log r)^2}\to\infty,\hspace{1cm}\text{ as }r\to\infty\,?\] One cannot expect any more from Yang Lo's method of finding $\theta_0$ through the use of `filling discs'. Rossi has shown that ([source label: star]) cannot be improved if $\theta_0$ is sought in this way. (D. Drasin; communicated by J. Rossi)

背景链接：https://arxiv.org/abs/1809.07200

11008 | AMR-022-1032

### Research Problems in Function Theory - Problem 1.32

P / partially_solved | AMR 问题清单汇编

Let $f$ be meromorphic in $\mathbb{C}$, and let $f^{-1}$ denote any element of the inverse function that is analytic in a neighbourhood of a point $w$. A well-known theorem of Gross , states that $f^{-1}$ may be continued analytically along almost all rays beginning at $w$. Is it possible to refine the exceptional set in this theorem? (A. Eremenko)

背景链接：https://arxiv.org/abs/1809.07200

11009 | AMR-022-1035

### Research Problems in Function Theory - Problem 1.35

P / partially_solved | AMR 问题清单汇编

Determine the upper and lower estimates for the growth of entire and meromorphic solutions of algebraic ordinary differential equations (AODE). (This is a classical problem.) For AODEs of first order, it is known that the meromorphic solutions $f$ must have finite order (see Gol'dberg ) and that $(\log r)^2=O(T(r,f))$ (see Eremenko , ). (The latter two references contain a general account of first order AODEs, including modern proofs of some classical results.) The order of entire solutions of first order AODEs is an integral multiple of $\frac{1}{2}$ (see Malmquist ). For AODEs of second order, it is known that the order of entire solutions is positive, see Zimogljad . There is no upper estimate valid for all entire or meromorphic solutions of AODEs of order grreater than one, but there is an old conjecture that $\log |f(z)|\leq exp_n(|z|)$ for entire solutions $f$ of an AODE of order $n$. (A. Eremenko)

背景链接：https://arxiv.org/abs/1809.07200

11010 | AMR-022-1036

### Research Problems in Function Theory - Problem 1.36

P / partially_solved | AMR 问题清单汇编

Let $F$ be a polynomial in two variables, and let $y$ be a meromorphic solution of the algebraic ordinary differential equation $F(y^{(n)},y)=0$. Is it true that $y$ must be an elliptic function, or a rational function of exponentials, or a rational function? This is known in the following cases: [(a)] ; $n=1$: a classical result, probably due to Abel; ; $n=2$: an old result of Picard , and independently, Bank and Kaufman ; ; $n$ is odd and $y$ has at least one pole, Eremenko ; and ; the genus of the curve $F(x_1,x_2)=0$ is at least equal to one, Eremenko . (A. Eremenko)

背景链接：https://arxiv.org/abs/1809.07200

11011 | AMR-022-1037

### Research Problems in Function Theory - Problem 1.37

O / open | AMR 问题清单汇编

Find criteria for and/or give explicit methods for the construction of meromorphic functions $f$ in $\mathbb{C}$ with the following properties: [(a)] ; all poles of $f$ are of odd multiplicity; ; all zeros of $f$ are of even multiplicity. (Here `explicit methods' means that all computations must be practicable.) The background of this problem lies in the question of meromorphic solutions of the differential equation $y''+A(z)y=0$ in the whole plane. (J. Winkler)

背景链接：https://arxiv.org/abs/1809.07200

11012 | AMR-022-1038

### Research Problems in Function Theory - Problem 1.38

O / open | AMR 问题清单汇编

[(a)] ; Let $f$ be non-constant and meromorphic in the open unit disc $\mathbb{D}$, with $\alpha<+\infty$, and define $$ \alpha = \limsup_{r\to1}\frac{T(r, f)}{-\log (1-r)}, $$ and \[\Psi=(f)^{m_0}(f')^{m_1}\ldots(f^{(k)})^{m_k}.\] It is known that $\Psi$ assumes all finite values, except possibly zero, infinitely often, provided that $m_0\geq3$ and $\alpha>2/(m_0-2)$, (or $m_0\geq2$ and \mbox{$\alpha>2/(m_0-1)$}, if $f$ is analytic). For which smaller values of $\alpha$ does the same conclusion hold? ; Let $f$ be non-constant and meromorphic in $\mathbb{D}$, with $\alpha<+\infty$ in ([source label: alphadef]); assume also that $f$ has only finitely many zeros and poles in $\mathbb{D}$. Let $l$ be a positive integer, and write $\Psi=\sum^l_{\nu=0}a_\nu f^{(\nu)}$, where the $a_\nu$ are functions in $\mathbb{D}$ for which $T(r,a_\nu)=O(T(r,f))$ as $r\to1$ (for each $\nu$). It is known that if $\Psi$ is non-constant, then $\Psi$ assumes every finite value, except possibly zero, infinitely often, provided that $\alpha>\frac{1}{2}l(l+1)+1$. For which smaller values of $\alpha$ does the same conclusion hold? (L. R. Sons)

背景链接：https://arxiv.org/abs/1809.07200

11013 | AMR-022-1039

### Research Problems in Function Theory - Problem 1.39

O / open | AMR 问题清单汇编

Let $f$ be a function meromorphic in $\mathbb{D}$, for which $\alpha<+\infty$ in ([source label: alphadef]). [(a)] ; Shea and Sons have shown that if $f(z)\neq0,\infty$ and $f'(z)\neq1$ in $\mathbb{D}$, then $\alpha\leq 2$. Is $2$ best possible? ; Shea and Sons have shown that, if $f(z)\neq0$ and $f'(z)\neq1$ in $\mathbb{D}$, then $\alpha\leq7$. What is the best possible $\alpha$ in this case? (L. R. Sons)

背景链接：https://arxiv.org/abs/1809.07200

11014 | AMR-022-1040

### Research Problems in Function Theory - Problem 1.40

O / open | AMR 问题清单汇编

Let $f$ be a function meromorphic in $\mathbb{D}$ of finite order $\rho$. Shea and Sons have shown that \[\sum_{a\neq \infty}\delta(a,f)\leq\delta(0,f')\big(1+k(f)\big)+\frac{2}{\lambda}(\rho+1),\] where \[k(f)=\limsup_{r\to1}\frac{\overline{N}(r,\infty,f)}{T(r,f)+1}\hspace{1cm}\text{ and }\hspace{1cm}\lambda(f)=\liminf_{r\to\infty}\frac{T(r,f)}{\log(1/(1-r))}.\] Can the factor $2$ be eliminated? (If so, the result is then best possible). (L. R. Sons)

背景链接：https://arxiv.org/abs/1809.07200

11015 | AMR-022-1041

### Research Problems in Function Theory - Problem 1.41

O / open | AMR 问题清单汇编

Let $f$ be a function meromorphic in $\mathbb{D}$, for which $\alpha=+\infty$ in ([source label: alphadef]). Then it is known that \[\sum_{a\in\mathbb{C}\cup\{\infty\}}\delta(a,f)\leq2.\] Are there functions which have an `arbitrary' assignment of deficiencies at an arbitrary sequence of complex numbers, subject only to these conditions? For analytic functions, Girynk has a result; whereas for arbitrary meromorphic functions, there is a result of Krutin . (L. R. Sons)

背景链接：https://arxiv.org/abs/1809.07200

11016 | AMR-022-1043

### Research Problems in Function Theory - Problem 1.43

O / open | AMR 问题清单汇编

Let $f$ be a meromorphic function of lower order $\lambda$. Let \[m_0(r,f)=\inf\{|f(z)|:|z|=r\}\] and \[M(r,f)=\sup\{|f(z)|:|z|=r\}\] and suppose that \[\log r=o(\log M(r,f)),\hspace{1cm}\text{ as }r\to\infty.\] Gol'dberg and Ostrovskii proved that if $0<\lambda<\frac{1}{2}$, then \[\limsup_{r\to\infty}\frac{\log m_0(r,f)}{\log M(r,f)}+\pi\lambda\sin(\pi\lambda)\limsup_{r\to\infty}\frac{N(r,f)}{\log M(r,f)}\geq\cos(\pi\lambda).\] Does this inequality remain valid for $\frac{1}{2}\leq\lambda<1$? See also Gol'dberg and Ostrovskii . (A. A. Gol'dberg and I. V. Ostrovskii)

背景链接：https://arxiv.org/abs/1809.07200

11017 | AMR-022-1005

### Research Problems in Function Theory - Problem 1.5

O / open | AMR 问题清单汇编

Under what conditions can $\sum\delta(a,f)$ be nearly $2$ for an entire function of finite order $\rho$? Pfluger proved that if $\sum\delta(a,f)=2$, then (see Hayman ) $\rho$ is a positive integer $q$, the lower order $\lambda$ is such that $\lambda=\rho$ and all the deficiencies are integral multiplicities of $1/q$. If further \[\sum\delta(a,f)>2-\varepsilon(\lambda),\] where $\varepsilon(\lambda)$ is a positive quantity depending on $\lambda$, then Edrei and Fuchs (, ) proved that these results remain true `nearly', in the sense that there exist `large' deficiencies which are nearly positive integral multiplicities of $1/q$, and whose sum of deficiencies is `nearly' $2$. Can there be a finite or infinite number of small deficiences as well in this case?

背景链接：https://arxiv.org/abs/1809.07200

11018 | AMR-022-1007

### Research Problems in Function Theory - Problem 1.7

O / open | AMR 问题清单汇编

If $f(z)$ is an entire function of finite order $\rho$ which is not an integer, it is known that (see Pfluger and Hayman ), \[\sum \delta(a,f)\leq 2-K(\rho)\] where $K(\rho)$ is a positive quantity depending on $\rho$. What is the best possible value for $K(\rho)$? Edrei and Fuchs conjectured (see also Hayman ) that if $q$ is the integral part of $\rho$, and if $q\geq1$, then \[K(\rho)=\frac{|\sin(\pi\rho)|}{q+|\sin(\pi\rho)|},\hspace{1cm}q\leq\rho<q+\frac{1}{2},\] \[K(\rho)=\frac{|\sin(\pi\rho)|}{q+1},\hspace{1cm}q+\frac{1}{2}\leq\rho<q+1.\] This result would be sharp. If $\rho\leq\frac{1}{2}$, there are no deficient values, so that $K(\rho)=1$. If $\frac{1}{2}<\rho<1$, Pfluger proved that $K(\rho)=\sin(\pi\rho)$. See also Hayman .

背景链接：https://arxiv.org/abs/1809.07200

11019 | AMR-022-1008

### Research Problems in Function Theory - Problem 1.8

O / open | AMR 问题清单汇编

Following the notation in Problem 1.7, if $f(z)$ is meromorphic in the plane of order $\rho$, it is conjectured by Pfluger , that for $a\neq b$ \[\limsup_{r\to\infty}\frac{N(r,a)+N(r,b)}{T(r,f)}\geq K(\rho).\] This is known to be true for $0<\rho\leq1$. If equality holds in the above inequality, it is conjectured that $f(z)$ has regular growth, i.e. $\rho=\lambda$.

背景链接：https://arxiv.org/abs/1809.07200

11020 | AMR-022-2011

### Research Problems in Function Theory - Problem 2.11

P / partially_solved | AMR 问题清单汇编

If $f(z)=\sum a_nz^{\lambda_n}$ is an entire function, and $\sum(1/\lambda_n)$ converges, is it true that: [(a)] ; $f(z)$ has no finite asymptotic value, ; $\limsup_{r\to\infty}\frac{\log m_0(r,f)}{\log M(r,f)}=1$, where $m_0(r,f)=\inf_{|z|=r}|f(z)|$ is the minimum modulus of $f(z)$? $(a)$ is known for $\lambda_n>n(\log n)^{1+\varepsilon}$ (see K\"ovari ); and $(b)$ is known for $\lambda_n>n(\log n)^2$ (see K\"ovari ). It is also known that $f(z)$ has no finite radial asymptotic value if $\sum(1/\lambda_n)$ converges, and that here this hypothesis cannot be replaced by any weaker condition (see Macintyre ).

背景链接：https://arxiv.org/abs/1809.07200

11021 | AMR-022-2012

### Research Problems in Function Theory - Problem 2.12

P / partially_solved | AMR 问题清单汇编

If the entire function $f(z)$ has finite order $\rho$, and the maximal density of non-zero coefficients is $\Delta$, is it true that if $\rho\Delta<\frac{1}{2}$, $f(z)$ cannot have a finite deficient value with deficiency one?

背景链接：https://arxiv.org/abs/1809.07200

11022 | AMR-022-2512

### Research Problems in Function Theory - Problem 2.12a

P / partially_solved | AMR 问题清单汇编

Under the same conditions as in Problem 2.12, is it true that if $\rho\Delta<1$, $f(z)$ cannot have a finite asymptotic value? This is known if $\rho\Delta<1/\pi^2$, see K\"ovari . Is it true that, if $m_0(r,f)$ is the minimum modulus \[\limsup_{r\to\infty}\frac{\log m_0(r,f)}{\log M(r,f)}\geq\cos(\pi\rho\Delta)?\] See K\"ovari .

背景链接：https://arxiv.org/abs/1809.07200

11023 | AMR-022-2013

### Research Problems in Function Theory - Problem 2.13

P / partially_solved | AMR 问题清单汇编

If $f(z)=\sum a_n z^{\lambda_n}$ is an entire function, and $\lambda_n/n\to\infty$, is it true that $f(z)$ has [(a)] ; no Picard value, ; no Borel exceptional value, ; no deficient value? All this is known for functions of finite order (see Fuchs ). If the answer is `no', are $(b)$ and $(c)$ true for $\sum(1/\lambda_n)<\infty$? Certainly by a theorem of Biernacki , $(a)$ is true in this case. (T.K\"ovari)

背景链接：https://arxiv.org/abs/1809.07200

11024 | AMR-022-2014

### Research Problems in Function Theory - Problem 2.14

P / partially_solved | AMR 问题清单汇编

[(a)] ; Let $f(z)=\sum a_n z^n$ be entire and $m(r)=\max_n |a_n|r^n$. If $C>\frac{1}{2}$ then does there exist an entire $f$ with \[m(r)/M(r,f)\to C ?\] Any value of $C$ such that $0<C\leq\frac{1}{2}$ is possible. ; If $f(z)\neq0$, then \[\liminf_{r\to\infty}\frac{m(r)}{M(r,f)}=0;\] Is \[\lim_{r\to\infty}\frac{m(r)}{M(r,f)}=0\,?\] ; What is the exact upper bound $\beta$ of \[\beta_f=\liminf_{r\to\infty}\frac{m(r)}{M(r,f)}\,?\] It is known that $\frac{4}{7}<\beta<2/\pi$. See Clunie and Hayman .

背景链接：https://arxiv.org/abs/1809.07200

11025 | AMR-022-2015

### Research Problems in Function Theory - Problem 2.15

O / open | AMR 问题清单汇编

(Blumenthal's conjecture) Let $w=f_1(z), f_2(z)$ be entire functions. Is it true that if \[M(r,f_1)=M(r,f_2),\hspace{1cm}0<r<\infty,\] then $f_1(z), f_2(z)$ are equivalent, apart from rotations and reflections in the $z$ and $w$ planes? The corresponding problem for polynomials (of degree higher than about $6$) is also open. The functions $(1-z)e^z$ and $e^{\frac{1}{2}z^2}$ have the same value of $M(r,f)$ for $0<r<2$, and $e^{z-z^2}$ and $e^{z^2+\frac{1}{2}}$ for $r\geq\frac{1}{4}$.

背景链接：https://arxiv.org/abs/1809.07200

11026 | AMR-022-2016

### Research Problems in Function Theory - Problem 2.16

P / partially_solved | AMR 问题清单汇编

Let $\nu(r)$ be the number of points on $|z|=r$, such that \mbox{$|f(z)|=M(r,f)$}. Can we have [(a)] ; $\limsup_{r\to\infty}\nu(r)=\infty$\,? ; $\liminf_{r\to\infty}\nu(r)=\infty$\,? (P. Erd\"os)

背景链接：https://arxiv.org/abs/1809.07200

11027 | AMR-022-2017

### Research Problems in Function Theory - Problem 2.17

P / partially_solved | AMR 问题清单汇编

If $f(z)$ is a non-constant entire function and \[b(r)=\left(r\frac{d}{dr}\right)^2\log M(r,f),\] then $$ \limsup_{r\to\infty} b(r)\geq A $$ where $A$ is an absolute constant, such that $0.18<A\leq\frac{1}{4}$, see Hayman . What is the best value of $A$? It seems fair to conjecture that the correct constant in ([source label: 2.3]) is in fact $\frac{1}{4}$.

背景链接：https://arxiv.org/abs/1809.07200

11028 | AMR-022-2019

### Research Problems in Function Theory - Problem 2.19

P / partially_solved | AMR 问题清单汇编

If $f(z)$ is an entire function of exponential type, i.e. satisfying \mbox{$|f(z)|\leq Me^{K|z|}$} for some constants $M$, $K$, and if, further, $|f(x)|\leq A$ for negative $x$, and $|f(x)|\leq B$ for positive $x$, what is the sharp bound for $|f(z)|$? If $A=B$, it is known that \[|f(z)|\leq Ae^{Ky}\] is true and sharp.

背景链接：https://arxiv.org/abs/1809.07200

11029 | AMR-022-2026

### Research Problems in Function Theory - Problem 2.26

O / open | AMR 问题清单汇编

What is the least integer $k=k(N)$, such that every entire function $f(z)$ can be written as \[f(z)=\sum^k_{\nu=1}[f_\nu(z)]^N,\] where $f(z)$ and $f_\nu(z)$ are entire functions? It is enough to solve the problem for $f(z)=z$, since then one can substitute $f(z)$ for $z$. The equation \[z=\frac{1}{N^2}\sum^N_{\nu=1}\frac{(1+\omega_\nu z)^N}{\omega_\nu},\] where $\omega_\nu$ are the distinct $N$-th roots of unity, shows that $k(N)\leq N$. On the other hand, for $N=1, 2, 3$ we have $k(N)=N$. To see e.g. that $k(3)\geq 3$, suppose that \[z=f^3+g^3=(f+g)(f+\omega g)(f+\omega^2 g),\] where $\omega=exp(2\pi i/3)$. It follows that the meromorphic function $\phi(z)=f(z)/g(z)$ satisfies $\phi(z)\neq-1, -\omega, -\omega^2$, except possibly at $z=0$. Thus, by Picard's Theorem, $\phi(z)$ must be rational, and so $\phi(z)$ assumes at least two of the three values $-1, -\omega, -\omega^2$. This gives a contradiction. (H.A. Heilbronn)

背景链接：https://arxiv.org/abs/1809.07200

11030 | AMR-022-2027

### Research Problems in Function Theory - Problem 2.27

O / open | AMR 问题清单汇编

Let $\phi_1, \ldots, \phi_n$ denote entire functions of the form $$ \phi(z)=\sum e^{f_\nu(z)}/\sum e^{g_\nu(z)} $$ where $f_\nu(z), g_\nu(z)$ are entire functions. Does there exist an entire function $f(z)$, not of the form $\phi(z)$, but satisfying an algebraic equation of the form \mbox{$f^n+\phi_1f^{n-1}+\ldots+\phi_n=0$}? The special cases $n=2$, or when the $f_\nu(z)$ are linear polynomials, may be easier to settle. Note that \[f(z)=\frac{\sin \pi z^2}{\sin \pi z}\] is not of the form $\sum e^{f_\nu(z)}$, although it is a ratio of such functions.

背景链接：https://arxiv.org/abs/1809.07200

11031 | AMR-022-2003

### Research Problems in Function Theory - Problem 2.3

P / partially_solved | AMR 问题清单汇编

If $\phi(z)$ is an entire function growing slowly compared with the function $f(z)$, we can consider $\phi(z)$ to be an asymptotic function of $f(z)$, if $f(z)-\phi(z)\to0$ as $z\to\infty$ along a path $\Gamma$. Is it true that an entire function of order $\rho$ can have almost $2\rho$ distinct asymptotic functions of order less than $\frac{1}{2}$? (If $\phi=\phi_1(z)-\phi_2(z)$ and the minimum modulus of $\phi$ tends to zero, then $\phi$ has lower order at least $\frac{1}{2}$ mean type (See Hayman ). A positive result in this direction is due to Denjoy , but only when the paths are straight lines. The result when the $\phi_\lambda(z)$ are polynomials is true (and is a trivial consequence of Ahlfors' theorem for asymptotic values ).

背景链接：https://arxiv.org/abs/1809.07200

11032 | AMR-022-2033

### Research Problems in Function Theory - Problem 2.33

O / open | AMR 问题清单汇编

Is it possible to obtain the exact value of $C_\infty$, or the asymptotic behaviour of $\frac{C_\lambda}{\log\lambda}$ as $\lambda\to\infty$? The question seems related to the number of zeros a function can have in a small disc centred on a point of $|z|=r$, see Hayman .

背景链接：https://arxiv.org/abs/1809.07200

11033 | AMR-022-2034

### Research Problems in Function Theory - Problem 2.34

P / partially_solved | AMR 问题清单汇编

Is it possible to say something more precise about $C(\lambda)$ when $\lambda$ is just greater than $1$? In particular, is it true that $C(\lambda)=-1$ for such $\lambda$, or alternatively, is $C(\lambda)$ a strictly decreasing function of $\lambda$?

背景链接：https://arxiv.org/abs/1809.07200

11034 | AMR-022-2036

### Research Problems in Function Theory - Problem 2.36

O / open | AMR 问题清单汇编

Suppose that $0<\rho<\alpha\leq1$, where $\rho$ is the order of an entire function $f$. Let $E_\alpha$ be the set of $r$ for which $\log m_0(r,f)>\cos (\pi\alpha) \log M(r,f)$. Besicovitch showed that the upper density of $E_\alpha$ is at least $1-\rho/\alpha$, and Barry proved the stronger result, that the same is true of the lower logarithmic density of $E_\alpha$. Examples given by Hayman , show that Barry's theorem is sharp; in these examples, the logarithmic density exists, but the upper density is larger. This suggests that Besicovitch's theorem may be sharpened.

背景链接：https://arxiv.org/abs/1809.07200

11035 | AMR-022-2039

## Research Problems in Function Theory - Problem 2.39

P / partially_solved | AMR 问题清单汇编

We can also compare $m_0(r,f)$ with the characteristic $T(r)$. We have \[\limsup_{r\to\infty}\frac{\log m_0(r,f)}{T(r)}\geq D(\lambda)\] and ask for the best constant $D(\lambda)$. In view of Petrenko's solution of Problem 1.17, we certainly have $D(\lambda)\geq-\pi\lambda$ with $1\leq\lambda<\infty$. Also, Ess\'en and Shea show that $D(\lambda)\leq\frac{\pi\lambda}{1+|\sin(\pi\lambda)|}$ for $1<\lambda<\frac{3}{2}$, and $D(\lambda)\leq\frac{-\pi\lambda}{2}$ for $\frac{3}{2}<\lambda<\infty$. Further, it follows from results of Valiron , and Edrei and Fuchs ( and ) that \[D(\lambda)= \begin{cases} \pi\lambda\cot(\pi\lambda) & \text{if }0\leq\lambda<\frac{1}{2} \pi\lambda\cos(\pi\lambda) & \text{if }\frac{1}{2}\leq\lambda<1. \end{cases} \] (D. Shea)

背景链接：https://arxiv.org/abs/1809.07200

11036 | AMR-022-2043

## Research Problems in Function Theory - Problem 2.43

P / partially_solved | AMR 问题清单汇编

Let $f(z)$ be a transcendental entire function which permutes the integers, i.e. gives an injective mapping of the integers onto themselves. Is it true that $f(z)$ is at least of order $1$, type $\pi$? We can also ask the corresponding question for a function permuting the positive integers with the same conjectured answer. Note that $f(z)=z+\sin z$ satisfies both conditions and is of order $1$, type $\pi$. If $f(z)$ assumes integer values on the positive integers, then Hardy and P\'olya proved (see ) that $f(z)$ is at least of order $1$, type $2$; and if $f(z)$ assumes integer values on all the integers, then Buck has shown that $f(z)$ is at least of order $1$, type $\log(\frac{3+\sqrt{5}}{2})=0.962\ldots$. (L. A. Rubel)

背景链接：https://arxiv.org/abs/1809.07200

11037 | AMR-022-2045

## Research Problems in Function Theory - Problem 2.45

O / open | AMR 问题清单汇编

Let $J_0(z)$ be the Bessel function of order zero. Is it true that the equation $J_0(z)=1$ has at most one solution on each ray from the origin? An affirmative answer would show that the exceptional set in a theorem of Delsarte and Lions is, in fact, void. Asymptotic estimates show that there can be at most a finite number of solutions on any ray, and yield even stronger information. (L. Zalcman)

背景链接：https://arxiv.org/abs/1809.07200

11038 | AMR-022-2046

## Research Problems in Function Theory - Problem 2.46

P / partially_solved | AMR 问题清单汇编

Let $\{f_\alpha(z)\}$ be a family of entire functions, and assume that for every $z_0$, there are only denumerably many distinct values of $f_\alpha(z_0)$. Then if $c=2^{\mathfrak{N}_0}>\mathfrak{N}_1$, the family $\{f_\alpha(z)\}$ is itself denumerable. The above result was proved by Erd\"os . If $c=\mathfrak{N}_1$, he constructed a counter-example. Suppose now that $m$ is an infinite cardinal, $\mathfrak{N}_0<m<c$. Assume that for every $z_0$, there are at most $m$ distinct values $f_\alpha(z_0)$. Does it then follow that the family $\{f_\alpha(z)\}$ has at most power $m$? If $m^+<c$, where $m^+$ is the successor of $m$, it is easy to see that the answer is `yes'. However, if $c=m^+$, the counter-example fails. It is possible the question is undecidable. (P. Erd\"os)

背景链接：https://arxiv.org/abs/1809.07200

11039 | AMR-022-2047

## Research Problems in Function Theory - Problem 2.47

O / open | AMR 问题清单汇编

Let $E_\rho$ be the linear space of entire functions $f$ such that \mbox{$|f(z)|\leq B\exp(A|z|^\rho)$} for some positive $A$ and $B$. Let $K_\rho$ be the family of functions $k(z)$ positive and continuous on $\mathbb{C}$, with $\exp(A|z|^\rho)=o(k(z))$ as $|z|\to\infty$, for all positive $A$. Let $S$ be a subset of $\mathbb{C}$, and $\|\cdot\|_{k,S},\|\cdot\|_k$ the semi-norms defined for $f\in E_\rho, k\in K_\rho$ by \[\|f\|_{k,S}=\sup_S\Big\{\frac{|f(z)|}{k(z)}\Big\},\] \[\|f\|_k=\sup_\mathbb{C}\Big\{\frac{|f(z)|}{k(z)}\Big\}.\] We say that $S$ is a sufficient set for $E_\rho$ if the topologies defined by the semi-norms $\{\|\cdot\|_{k,k\in K_\rho\},\{\|\cdot\|_{k,S},k\in K_\rho\}$ coincide, see Ehrenpreis . [(a)] ; Suppose that, whenever $f\in E_\rho$, $f$ is bounded on $S$ if and only if $f$ is bounded on $\mathbb{C}$. Does it follow that $S$ is a sufficient set for $E_\rho$? ; Suppose that $S$ is a sufficient set for $E_\rho$. Then, if $f\in E_\rho$ and $f$ is bounded on $S$, does it follow that $f$ is bounded on $\mathbb{C}$, or (maybe) of small growth? Some recent work of Oliver suggests the latter, at least, is likely. (D. A. Brannan)

背景链接：https://arxiv.org/abs/1809.07200

11040 | AMR-022-2048

### Research Problems in Function Theory - Problem 2.48

O / open | AMR 问题清单汇编

If $A, B$ are countable dense subsets of $\mathbb{R}$, $\mathbb{C}$ respectively, does there necessarily exist a transcendental entire function that maps $A$ onto $B$, and $\mathbb{R} \setminus A$ into $\mathbb{C} \setminus B$? Suppose that $E, F$ are countable dense subsets of $\mathbb{R}$, and that there exists an entire function $f$, monotonic on $\mathbb{R}$, that maps $E$ onto $F$, and $\mathbb{R}\setminus E$ onto $\mathbb{R}\setminus F$. Find interesting conditions which imply that $f$ is trivial. For example, if $E, F$ are real rationals, under what conditions is $f$ necessarily linear with rational coefficients? One could also investigate this question in the case of real-valued harmonic or subharmonic functions in $\mathbb{R}^n$, $n\geq2$. If $A, B$ are two countable dense subsets of $\mathbb{R}$, Barth and Schneider have shown that there exists a transcendental entire function, monotonic on $\mathbb{R}$, that maps $A$ onto $B$ and $\mathbb{R}\setminus A$ onto $\mathbb{R}\setminus B$; also, if $A,B$ are countable dense subsets of $\mathbb{C}$, see Barth and Schneider . (K. F. Barth)

背景链接：https://arxiv.org/abs/1809.07200

11041 | AMR-022-2049

### Research Problems in Function Theory - Problem 2.49

O / open | AMR 问题清单汇编

If $f(z)$ is a transcendental entire function, we define \[M=\{z:|f(z)|=M(|z|,f)\}.\] Tyler has shown that $M$ can have isolated points, and that, given any $N$, we can have $\nu(r)>N$ for infinitely many $r$, where $\nu(r)$ is the number of points in $M \cap \{|z|=r\}$. Herzog and Piranian have shown that $\limsup_{r\to\infty}\nu(r)$ can be infinite; is it true that $\liminf_{r\to\infty}\nu(r)<\infty$ for all entire $f$? (J. G. Clunie)

背景链接：https://arxiv.org/abs/1809.07200

11042 | AMR-022-2050

### Research Problems in Function Theory - Problem 2.50

O / open | AMR 问题清单汇编

Characterise those entire functions having at least one continuous maximum modulus path going from $0$ to $\infty$. (W. Al-Katifi)

背景链接：https://arxiv.org/abs/1809.07200

11043 | AMR-022-2051

### Research Problems in Function Theory - Problem 2.51

O / open | AMR 问题清单汇编

Suppose that an entire function $f$ has exactly one curve $\Gamma$ of maximum modulus (that is, $\Gamma$ is connected, joins $0$ to $\infty$, and $f$ has no other maximum modulus points). What can be said about the minimum rate of growth of $M(r,f)$, if one is given information about the geometry of the curve $\Gamma$, for example, that $\Gamma$ is a given infinitely-spiralling spiral? If $\Gamma$ is a radial line, clearly nothing much can be said. (In a sense, this is the opposite of a Phragm\'en-Lindel\"of principle). (D. A. Brannan)

背景链接：https://arxiv.org/abs/1809.07200

11044 | AMR-022-2053

### Research Problems in Function Theory - Problem 2.53

O / open | AMR 问题清单汇编

For entire or, more generally, meromorphic functions $f$ and $g$, let `$f\leq g$' mean that, for any sequence $\{z_n\}^\infty_1$ for which $|f(z_n)|\to\infty$, then $|g(z_n)|\to\infty$. For entire functions, it can be proved that, if $f'\leq f$, then $f$ is of exponential type; what can be said if $f''\leq f$? Does $f''\leq f$ imply that $f$ is normal (that is, that $|f'(z)|(1+|f(z)|^2)^{-1}$ is bounded)? (An analogue in the case of the unit disc $\mathbb{D}$ has been proved by Pommerenke ). In the above ordering, does there exist $f\wedge g$ and $f\vee g$ for any two entire functions $f$ and $g$? That is, given $f$ and $g$, does there exist an $h$ such that $h\leq f$ and $h\leq g$, and so that if $k\leq f$ and $k\leq g$, then $k\leq h$? Similarly, for $f\vee g$. Finally, if $f$ is meromorphic and $f'\leq f$, does this imply a growth restriction on $f$, e.g. is the order of $f$ at most two? Note: Any constant $c$ satisfies $c\leq f$ for all entire $f$, and any non-constant polynomial $p$ satisfies $f\leq p$ for any entire $f$. Observe also that $e^{cz}$ are all equivalent if $c>0$. (L. A. Rubel and J. M. Anderson)

背景链接：https://arxiv.org/abs/1809.07200

11045 | AMR-022-2054

### Research Problems in Function Theory - Problem 2.54

O / open | AMR 问题清单汇编

Let $E$ be a closed set in $\mathbb{C}$, with the following properties: $(1)$ there exists a transcendental entire function $f(z)$ that is bounded on $E$; and $(2)$, there exists a transcendental entire function $g(z)$ that is bounded away from $0$ on the complement of $E$. For each such set $E$, must there exist one transcendental entire function that is simultaneously bounded on $E$ and bounded away from $0$ on the complement of $E$? (L. A. Rubel)

背景链接：https://arxiv.org/abs/1809.07200

11046 | AMR-022-2055

### Research Problems in Function Theory - Problem 2.55

O / open | AMR 问题清单汇编

Let $f_i(z)$, $i=1, 2, 3$ be non-constant entire functions of one complex variable, and \[V=\{z:z=(z_1,z_2,z_3)\in\mathbb{C}^3,f_1(z_1)+f_2(z_2)+f_3(z_3)=0\}.\] If $F$ is an entire function of $z=(z_1,z_2,z_3)$ that is bounded on $V$, is $F$ necessarily constant on $V$? (J. M. Anderson and L. A. Rubel)

背景链接：https://arxiv.org/abs/1809.07200

11047 | AMR-022-2056

### Research Problems in Function Theory - Problem 2.56

O / open | AMR 问题清单汇编

Prove or disprove the conjecture that an entire function $f$ of $n$ complex variables is an $L$-atom (where this is defined in a way analogous to the definition for $n=1$; see Problem 5.55 with $\mathbb{D}$ replaced by $\mathbb{C}$); if and only if there are entire functions $f_2,f_3,\ldots,f_n$ of the $n$ variables such that $(f, f_2,f_3,\ldots,f_n)$ is an analytic automorphism of $\mathbb{C}^n$, that is, an injective biholomorphic map of $\mathbb{C}^n$ onto $\mathbb{C}^n$. Rubel can prove the result in the case $n=1$. (L. A. Rubel)

背景链接：https://arxiv.org/abs/1809.07200

11048 | AMR-022-2059

### Research Problems in Function Theory - Problem 2.59

P / partially_solved | AMR 问题清单汇编

(A width conjecture) Given a power series $\sum^\infty_{k=0}a_kz^k$, suppose that there is a non-negative $\rho$ such that all of the partial sums $S_n(z)=\sum^n_{k=0}a_kz^k$, $n=1, 2, 3, \ldots$ are non-zero in the region \[S\rho=\{z=x+iy:|y|<Kx^{1-(\rho/2)},x>0\}.\] We conjecture that $f(z)$ must be entire of order at most $\rho$. When $\rho=0$, $S_\rho$ is a sector with vertex at $z=0$, and the conjecture is a consequence of results of Carlson , which were later generalised by Rosenbloom , Ganelius , and Korevaar and McCoy . Remark: If $f(z)=e^z$, which is of order $1$, Saff and Varga have shown that the partial sums $\sum^n_{k=0}z^k/k!$ are in fact zero-free in the parabolic region \[\{z=z+iy:y^2\leq4(x+1), x>-1\}.\] (E. B. Saff and R. S. Varga)

背景链接：https://arxiv.org/abs/1809.07200

11049 | AMR-022-2006

### Research Problems in Function Theory - Problem 2.6

P / partially_solved | AMR 问题清单汇编

Let $f(z)$ be an entire function. Then Boas (unpublished) proved that there exists a path $\Gamma_\infty$ such that, for every $n$, $$ \left|\frac{f(z)}{z^n}\right|\to\infty, \hspace{1cm}\text{ as }z\to\infty \text{ along } \Gamma. $$ Can we improve this result if something is known about the lower growth of $M(r,f)$? Hayman has shown that there exist functions of infinite order and, in fact, growing arbitrarily rapidly, such that, on every path $\Gamma$ on which $f(z)\to\infty$, we have \[\log \log |f(z)|=O(\log |z|),\] i.e. $f(z)$ has finite order on $\Gamma$.

背景链接：https://arxiv.org/abs/1809.07200

11050 | AMR-022-2060

### Research Problems in Function Theory - Problem 2.60

O / open | AMR 问题清单汇编

Let $\sum^\infty_{k=0}a_kz^k$ be a non-vanishing entire function, and let \mbox{$S_n(z)=\sum^n_{k=0}a_kz^k$}. Given $\varepsilon>0$, must there exist a $z_0$ and an $n$ such that $S_n(z_0)=0$ and $|f(z_0)|<\varepsilon$? (The Hurwitz theorem shows that the result is true if $f$ has a zero.) (D. J. Newman and A. Abian)

背景链接：https://arxiv.org/abs/1809.07200

11051 | AMR-022-2061

### Research Problems in Function Theory - Problem 2.61

P / partially_solved | AMR 问题清单汇编

Let $\Gamma$ be a rectifiable curve. Suppose $f$ is a continuous function on the plane satisfying \[\int_{\sigma(\Gamma)}f(z)dz=0\hspace{1cm}\text{ for all rigid motions }\sigma.\] Does this imply that $f$ is an entire function? The answer is `yes' for some choices of $\Gamma$, and `no' for others. For example, the answer is `no' for the circle ($f(x+iy)=\sin(ax)$ is a counter-example for a suitable choice of $a$); `yes' for an ellipse; `yes' for any polygonal Jordan curve; and `yes' for the boundary of any convex set with at least one corner. Prove that the circle is the only closed rectifiable Jordan curve (or the only curve among the class of curves which are boundaries of bounded convex sets) for which the answer is `no'. This problem is related to the `Pompeiu Problem' discussed by Brown, Schreiber and Taylor . (L. Brown)

背景链接：https://arxiv.org/abs/1809.07200

11052 | AMR-022-2062

### Research Problems in Function Theory - Problem 2.62

S / solved | AMR 问题清单汇编

Let $f$ denote a rational or entire function of a complex variable, and $f^n, n=1, 2, \ldots$, the $n$-th iterate of $f$, so that $f^1=f, f^{n+1}=f\circ f^n=f^n\circ f$. Provided that $f$ is not rational of degree $0$ or $1$, the set $C$ of those points where $\{f^n\}$ forms a normal family is a proper open subset of the plane, and is invariant under the map $z \mapsto f(z)$. A component $G$ of $C$ is a wandering domain of $f$ if $f^k(G)\cap f^n(G)=\emptyset$ for all $\{k,n\,|\,k\geq1,n\geq1,k\neq n\}$. Jakobson has asked whether it is possible for a rational function $f$ to have a wandering domain. Baker gave a transcendental entire function which does have such domains. (I. N. Baker)

背景链接：https://arxiv.org/abs/1809.07200

11053 | AMR-022-2063

### Research Problems in Function Theory - Problem 2.63

P / partially_solved | AMR 问题清单汇编

Let $f$ be a rational function and $C$ be as in Problem 2.62. We say that $g$ is a limit function for $f$ if $g$ is defined in some component $G$ of $C$ and is the limit of some subsequence of $(f^n)$ in $G$. In the simplest examples, the number of limit functions is finite, which implies that each has a constant $\alpha$, say, such that $f^k(\alpha)=\alpha$ for some positive integer $k$. If, in addition, $|(f^k)'(\alpha)|<1$ for each of the limit functions, we say that the function $f$ belongs to the class $N$. [(a)] ; Does there exist a rational $f$ which has an infinity of constant limit functions? ; Is the property of belonging to $N$ `generic' in some sense for rational functions? Some account of the older established results can be found in Fatou (, , ) and a sketch from a more modern point of view is given by Guckenheimer . (I. N. Baker)

背景链接：https://arxiv.org/abs/1809.07200

11054 | AMR-022-2067

### Research Problems in Function Theory - Problem 2.67

O / open | AMR 问题清单汇编

Let $f$ be an entire function, and let $D$ be a component of the set in $\mathbb{C}$ where the family of iterates $\{f_n\}$ is normal. Can this family have an infinite bounded set of constant limit functions? Eremenko and Lyubich have shown that the set of constant limit functions may be infinite. (A. Eremenko)

背景链接：https://arxiv.org/abs/1809.07200

11055 | AMR-022-2068

### Research Problems in Function Theory - Problem 2.68

P / partially_solved | AMR 问题清单汇编

Let $f$ be an entire function satisfying the condition \[\log M(r,f)\leq(1+o(1))r^\rho,\hspace{1cm}\text{ as }r\to\infty.\] Suppose that there exists a curve $\Gamma$ tending to $\infty$ such that on $\Gamma$ \[\log|f(z)|\leq(\alpha+o(1))r^\rho,\hspace{1cm}\text{ as }r=|z|\to\infty,\] for some $\alpha$ in $[-1,1)$; and denote by $E(r,\varepsilon)$ the angular measure of the set \[\{re^{i\theta}:\log|f(re^{i\theta})|\leq(1-\varepsilon)r^\rho\}.\] Eremenko conjectures that \[\limsup_{\varepsilon\to0,\, r\to\infty}E(r,\varepsilon)\geq\frac{2}{\rho}\arccos\alpha.\] Jaenisch has proved some related results. (A. Eremenko)

背景链接：https://arxiv.org/abs/1809.07200

11056 | AMR-022-2069

### Research Problems in Function Theory - Problem 2.69

O / open | AMR 问题清单汇编

Hayman has shown that $$ \liminf_{r\to\infty}\frac{T(r,f)}{T(r,f')}\leq1 $$ for transcendental entire functions $f$ of lower order zero. Toppila has shown that there exists an entire function of order one which does not satisfy ([source label: asterix]). Does there exist a constant $d > 0$ such that ([source label: asterix]) holds for all transcendental entire functions $f$ of order less than $d$? (S. Toppila)

背景链接：https://arxiv.org/abs/1809.07200

11057 | AMR-022-2071

### Research Problems in Function Theory - Problem 2.71

P / partially_solved | AMR 问题清单汇编

It is shown by Hellerstein and Rossi , and Gundersen that if $f_1$ and $f_2$ are two linearly independent solutions to the differential equation $w'' + Hw = 0$, where $H$ is a polynomial and $f_1$ and $f_2$ have only finitely many non-real zeros, then $H$ is a non-negative constant. It is also shown that, if $H(z) = az + b$, for $a,b\in\mathbb{R}$, then the differential equation admits a solution with only real zeros (and infinitely many of them). Furthermore, as pointed out by Gundersen , Titchmarsh () showed that, if $H(z) = z^4-\beta$ for special choices of $\beta$, then the differential equation also admits solutions with only real zeros (and infinitely many of them). Characterise all non-constant polynomials $H$ such that the differential equation admits a solution with only real zeros (and infinitely many of them). (S. Hellerstein and J. Rossi)

背景链接：https://arxiv.org/abs/1809.07200

11058 | AMR-022-2073

### Research Problems in Function Theory - Problem 2.73

O / open | AMR 问题清单汇编

Let $F(z, a, b)$ be an entire function of three complex variables, and suppose that $F$ is not of the form $$ F(z,a,b) = G(z,H(a,b)) $$ for any entire functions $G$ and $H$ of two complex variables. Can \[\{F(z, a,b):a,b\in\mathbb{C}\}\] constitute a normal family of entire functions of $z$? (Put rather loosely, does there exist a two-parameter normal family of entire functions?) Notice that if $F$ does have the form ([source label: 2.73]) then $F_bF_{a,z} = F_aF_{b,z }$, where subscripts denote partial differentiation. The purpose of ruling out the form ([source label: 2.73]) is to ensure that there are two honest parameters of the family. Otherwise we could have, say, \[F(z, a, b) = z + 5a^2 + \sin b,\] which is really a one-parameter family in disguise. (L. A. Rubel)

背景链接：https://arxiv.org/abs/1809.07200

11059 | AMR-022-2074

### Research Problems in Function Theory - Problem 2.74

O / open | AMR 问题清单汇编

Suppose that $f(z) = 1 + a_1z + a_2 z^2 +\ldots \in U_{2p}$. If $p = 0$ (so that $f\in U_0)$ and if $f$ is not a polynomial, it is well-known that $f$ cannot have two consecutive Taylor coefficients equal to zero. If $p > 0$, can an analogous assertion be made? Is it true, for example, that the Taylor series of $f$ in $U_{2p}$ cannot have $2p + 2$ consecutive coefficients equal to zero? (J. Williamson)

背景链接：https://arxiv.org/abs/1809.07200

11060 | AMR-022-2075

### Research Problems in Function Theory - Problem 2.75

P / partially_solved | AMR 问题清单汇编

Suppose that $f$ is entire of proximate order $\rho(r)$, and that $f$ has a representation as a Dirichlet series \[f( z ) = \sum^\infty_{n=1}a_ne^{\lambda_n z},\hspace{1cm} 0\leq\lambda_1<\lambda_2<\ldots\to\infty, \hspace{1cm}a_n>0.\] Can one give a complete characterisation of the indicator \[h(\theta,f)=\limsup_{r\to\infty}r^{-\rho(r)}\log|f(re^{i\theta})|\] of such functions? If $f$ has a representation \[f(z)=\int^\infty_0 e^{iz}\,dF(t)\] where $F$ is positive and increasing, Gol'dberg and Ostrovskii gave such a characterisation. (A. A. Gol'dberg and I. V. Ostrovskii)

背景链接：https://arxiv.org/abs/1809.07200

11061 | AMR-022-2077

### Research Problems in Function Theory - Problem 2.77

O / open | AMR 问题清单汇编

Let $\Omega$ be a component of the normal set of an entire function $f$ (under iteration). Do there exist such an $f$ and such an $\Omega$ with the following properties: [(a)] ; $f^n(\Omega)$ is uniformly bounded, for $n = 0,1,2,\ldots$; ; $f^n(\Omega)\cap f^m(\Omega)=\emptyset$ for $n\neq m$. (I. N. Baker, R. Herman and I. Kra; communicated by D. Hamilton)

背景链接：https://arxiv.org/abs/1809.07200

11062 | AMR-022-2078

### Research Problems in Function Theory - Problem 2.78

P / partially_solved | AMR 问题清单汇编

(Fatou's conjecture) Show that the subset $U$ of functions $g$ in $R_d$, such that all the critical points of $g$ are in the basins of attraction of periodic sinks, is dense in $R_d$. The property `$g\in U$' is also sometimes called Axiom A. See also Fatou . (P. Fatou; communicated by M. R. Herman)

背景链接：https://arxiv.org/abs/1809.07200

11063 | AMR-022-2079

### Research Problems in Function Theory - Problem 2.79

P / partially_solved | AMR 问题清单汇编

[(a)] ; Show that, if a function $g$ in $R_d$ has the property that its Julia set $J(g) \neq \hat{\mathbb{C}}$, then $g$ does not leave invariant a non-trivial Beltrami form on $J(g)$. Here a Beltrami form $\mu$ means that $\mu\in L^{\infty}( \hat{\mathbb{C}})$ and $\|\mu\|_{L^\infty}<1$; and trivial on $J(g)$ means that $\mu(x)=0$ for almost all $x$ in $J(g)$. ; More generally, is the Lebesgue measure of $J(g)$ in $\hat{\mathbb{C}}$ equal to zero? (This is the analogous conjecture to the Ahlfors conjecture for finitely-generated Kleinian groups.) Negative answers to both $(a)$ and $(b)$ have been proved (by McMullen (uncited) and Eremenko and Lyubich ) for the class of transcendental entire functions $g$. Douady has conjectured that the answer is negative, and that a counter-example is the function $P_\lambda(z) = \lambda(z + z^2)$, for some $\lambda$ of modulus one. See also Douady , Lyubich and Ma\ n\'e . (D. Sullivan; communicated by M. R. Herman)

背景链接：https://arxiv.org/abs/1809.07200

11064 | AMR-022-2008

### Research Problems in Function Theory - Problem 2.8

O / open | AMR 问题清单汇编

Does ([source label: 2.1]) remain true if the number $n(r)$ of poles of $f(z)$ in $|z|<r$ satisfies $n(r)=O(r^k)$, where $k<\frac{1}{2}<\lambda$, and $\lambda$ is the lower order of $f(z)$? Gol'dberg and Ostrovskii have shown that ([source label: 2.1]) can be false if $\frac{1}{2}<k<\lambda$.

背景链接：https://arxiv.org/abs/1809.07200

11065 | AMR-022-2080

### Research Problems in Function Theory - Problem 2.80

O / open | AMR 问题清单汇编

Let the function $g$ in $R_d$ have the property that its Julia set $J(g) = \hat{\mathbb{C}}$. Is the dimension $k$ of the space of Beltrami forms on $\hat{\mathbb{C}}$, invariant under $g$, at most one? Also, are certain of the Latt\`es examples the only rational functions such that $k\neq0$? For further information, see . (D. Sullivan; slightly modified and communicated by M. R. Herman)

背景链接：https://arxiv.org/abs/1809.07200

11066 | AMR-022-2081

### Research Problems in Function Theory - Problem 2.81

O / open | AMR 问题清单汇编

Let the function $g$ in $R_d$ have the property that its Julia set $J(g) = \hat{\mathbb{C}}$. Is $g$ ergodic for Lebesgue measure? In other words, if $B\subset\hat{\mathbb{C}}$ is a Borel-invariant set under $g$ (that is, $g^{-1}(B) = B$), does it follow that either $B$ or $\hat{\mathbb{C}} \setminus B$ has Lebesgue measure zero? For further information, see . (D. Sullivan; communicated by M. R. Herman)

背景链接：https://arxiv.org/abs/1809.07200

11067 | AMR-022-2082

### Research Problems in Function Theory - Problem 2.82

O / open | AMR 问题清单汇编

Let $L_d$ denote the class of those functions $g\in R_d$ such that every critical point of $g$ is preperiodic but not periodic. Show that, if the function $g$ in $R_d$ has the property that $J(g) = \hat{\mathbb{C}}$, then $g$ belongs to the closure of $L_d$ in $R_d$. (M. R. Herman)

背景链接：https://arxiv.org/abs/1809.07200

11068 | AMR-022-2083

### Research Problems in Function Theory - Problem 2.83

P / partially_solved | AMR 问题清单汇编

Let a function $f$ in $R_d$ have the property that \[f(z)=\lambda_\alpha z+O(z^2)\hspace{1cm}\text{ as }z\to0,\] where $\lambda_\alpha=e^{2\pi i\alpha}$ and $\alpha\in\mathbb{R}\setminus\mathbb{Q}$. Assume also that $f$ is linearisable at $0$, and denote by $S$ its Siegel singular disc. [(a)] ; Is $\alpha$ necessarily a Brjuno number? In other words, is it true that \[\sum^\infty_{n=0}(\log q_{n+1})/q_n<+\infty,\] where $\{p_n/q_n\}^\infty_{n=0}$ are the convergents of the continued fraction expansion of $\alpha$? Also, what is the situation here when $f$ is a non-linear entire function? ; Is $f$ necessarily injective on the boundary $\partial S$ of $S$ in $\hat{\mathbb{C}}$? ; Is it true that $f$ has no periodic points on $\partial S$? Both $(b)$ and $(c)$ are open even under the additional hypothesis that $f$ has no critical point on $\partial S$. A positive answer to $(b)$, under this additional hypothesis, would imply that when $\alpha$ satisfies a diophantine condition (that is, there exists $\beta$, $\gamma$, $\gamma > 0$ and $\beta\geq2$ such that, for every number $p/q$ in $\mathbb{Q}$, we have $|\alpha-(p/q)|\geq\gamma q^{-\beta})$ then $f$ has a critical point on $\partial S$. For further information, see Herman and Przytycki . (J.-C. Yoccoz and M. R. Herman)

背景链接：https://arxiv.org/abs/1809.07200

11069 | AMR-022-2084

### Research Problems in Function Theory - Problem 2.84

O / open | AMR 问题清单汇编

Does there exist a number $\lambda$ of modulus one that is not a root of unity, such that the positive orbit of $-\frac{1}{2}$ under $P_ \lambda(z) = \lambda(z + z^2)$ is dense in $J(P_ \lambda)$? (M. R. Herman)

背景链接：https://arxiv.org/abs/1809.07200

11070 | AMR-022-2085

### Research Problems in Function Theory - Problem 2.85

O / open | AMR 问题清单汇编

Suppose that $\lambda$ is of modulus one and not a root of unity, let $P_ \lambda(z) = \lambda(z + z^2)$ and \[h_\lambda(z)=z+O(z^2)\] is the unique formal power series such that $P_ \lambda(h_\lambda(z)) = h_\lambda(\lambda z)$. Denote by $R(\lambda)$ the radius of convergence of $h_\lambda$. [(a)] ; Calculate (or, at least, estimate up to $\pm10^{-10}$) the value of $m = \sup_\lambda R(\lambda)$. ; Prove that $m$ is realised by a function $h_\lambda$, where $\lambda = e^{2\pi i\alpha}$ and $\alpha$ is a real algebraic number of degree $2$. ; If $R(\lambda) = 0$, is the following true: for every positive $\varepsilon$, the function $P_\lambda$ has a repelling periodic cycle included in $\{|z| < \varepsilon\}$? This property is known to hold for a dense $G_\delta$-set of numbers $\lambda$ of modulus one, see Cremer . (M. R. Herman and J. -C. Yoccoz)

背景链接：https://arxiv.org/abs/1809.07200

11071 | AMR-022-2087

### Research Problems in Function Theory - Problem 2.87

O / open | AMR 问题清单汇编

Does there exist a non-linear entire function $g$ with wandering domain $W$ such that $\bigcup_{n\geq0}g^n(W)$ is bounded in $\mathbb{C}$? It has been conjectured by Lyubich that if $g$ and $W$ exist, then $g^n(W)$ cannot converge as $n\to\infty$ to a fixed point of $g$. (M. R. Herman; A. Eremenko and M. Yu. Lyubich)

背景链接：https://arxiv.org/abs/1809.07200

11072 | AMR-022-2088

### Research Problems in Function Theory - Problem 2.88

P / partially_solved | AMR 问题清单汇编

Let $B$ denote the boundary of the Mandelbrot set (or, equivalently, the topological bifurcation set of the family $z\mapsto z^2+c$, $c\in\mathbb{C}$. [(a)] ; Is $B$ locally-connected? ; Prove that $B$ has Hausdorff dimension $2$. ; Does $B$ have Lebesgue measure zero? For further information, see Douady and Lyubich . (A. Douady and J. H. Hubbard; N. Sibony; M. Rees; M. R. Herman)

背景链接：https://arxiv.org/abs/1809.07200

11073 | AMR-022-2089

### Research Problems in Function Theory - Problem 2.89

O / open | AMR 问题清单汇编

Let the function $f_0$ in $R_d$ have an invariant Herman singular ring $A_f$ of rotation number $\alpha$, where $\alpha$ satisfies a diophantine condition. Denote by $H_{d,\alpha}$ the class of all functions $f_1$ in $R_d$ such that $f_1$ can be joined to $f_0$ by a continuous path $f_t$, $0\leq t\leq1$, in $R_d$, where each $f_t$ has a Herman singular ring $A_{f_t}$ of rotation $\alpha$, and the annuli $A_f$ vary continuously with $f$ (in the sense of Carath\'eodory). [(a)] ; Is $H_{d,\alpha}$ locally closed in $R_d$? ; Is the boundary of $H_{d,\alpha}$ in its closure in $R_d$ a topological manifold? Both $(a)$ and $(b)$ are related to the investigation of rational functions with an invariant Herman singular ring, when the moduli of their invariant rings tend to zero. See Herman . (M. R. Herman)

背景链接：https://arxiv.org/abs/1809.07200

11074 | AMR-022-2009

### Research Problems in Function Theory - Problem 2.9

P / partially_solved | AMR 问题清单汇编

We ask the analogues of Problems 2.6, 2.7 and 2.8 if, in addition, $f(z)$ has another finite Picard value, e.g. $f(z)\neq0$. In this case, if $\infty$ has deficiency one, in the sense of Nevanlinna, ([source label: 2.1]) remains true for functions of finite order (see Edrei and Fuchs ), but not necessarily for functions of infinite order, see Gol'dberg and Ostrovskii (, ).

背景链接：https://arxiv.org/abs/1809.07200

11075 | AMR-022-2090

### Research Problems in Function Theory - Problem 2.90

S / solved | AMR 问题清单汇编

Does there exist a number $\alpha$ in $\mathbb{R}\setminus\mathbb{Q}$ that does not satisfy a diophantine condition, such that every $\mathbb{R}$-analytic orientation-preserving diffeomorphism of the circle with rotation number $\alpha$ is $\mathbb{R}$-analytically conjugated to a rotation? For related questions, see Douady . If $\alpha$ satisfies a diophantine condition, the global analytical conjugacy theorem has been proved. See Herman and Yoccoz . (M. R. Herman)

背景链接：https://arxiv.org/abs/1809.07200

11076 | AMR-022-3010

### Research Problems in Function Theory - Problem 3.10

P / partially_solved | AMR 问题清单汇编

Suppose that $u(X)$ is harmonic on the unit ball $|X|<1$, and remains continuous with partial derivatives of all orders on $|X|=1$, where $X$ is a point $(x_1, x_2, x_3)$ in space, and \[|X|^2=x^2_1+x^2_2+x^2_3.\] Suppose further that there is a set $E$ of positive area on $|X|=1$, such that both $u$ and its normal derivative vanish on $E$. Is it true that $u\equiv0$? Here the two-dimensional analogue is almost trivial, since if $u$ is harmonic in $\mathbb{D}$, and $u$ and its partial derivatives remain continuous on the unit cirlce $\mathbb{T}$, we may consider \[f(z)=z\Big(\frac{\partial u}{\partial x}-i\frac{\partial u}{\partial y}\Big), \hspace{1cm}\text{ where }z=x+iy.\] If $u(z)$ and its normal derivatives both vanish on a set $E$, then $f(z)$ vanishes at all limit points of $E$. Now the Poisson-Jensen formula shows at once that $\log|f(z)|=-\infty,f(z)=0,$ identically in $\mathbb{D}$, provided that the closure of $E$ has positive $1$-dimensional measure. (L. Bers)

背景链接：https://arxiv.org/abs/1809.07200

11077 | AMR-022-3012

### Research Problems in Function Theory - Problem 3.12

P / partially_solved | AMR 问题清单汇编

Consider a domain of infinite connectivity in $\mathbb{R}^3$ whose complement $E$ lies in the plane $P : x_3 =0$. Suppose further than any disc of positive radius $R$ in $P$ contains a subset of $E$ having area at least $\varepsilon$, where $\varepsilon, R$ are fixed positive constants. If $u$ is positive and harmonic in $D$, continuous in $\mathbb{R}^3$ and zero on $E$, is it true that \[u = cx_3 + \phi(x), \hspace{1cm}x_3 > 0,\] where $c$ is a constant and $\phi(x)$ is uniformly bounded? One can also ask the analogue of this result for $\mathbb{R}^m$ when $m>3$. It is true in $\mathbb{R}^2$ (but Kjellberg cannot remember who proved it). (B. Kjellberg)

背景链接：https://arxiv.org/abs/1809.07200

11078 | AMR-022-3013

### Research Problems in Function Theory - Problem 3.13

P / partially_solved | AMR 问题清单汇编

Let $u(x)$ be subharmonic in $\mathbb{R}^m$. One can define the quantities $n(r, 0), N(r, 0), T(r)$ as in Nevanlinna theory in the plane, taking the analogue of the case $u(z) = \log|f(z) |$, where $f(z)$ is an entire function (see, for example Hayman ). Define \[\delta(u) = 1 - \limsup_{r\to\infty}\frac{N(r,u)}{T(r)}.\] If the order $\rho$ of $u$ is less than $1$, it is possible to obtain the sharp upper bound for $\delta(u)$ in terms of $\rho$ and $m$. The bound is attained when $u(x)$ has all its mass on a ray (see Hayman and Kennedy ). One can ask the corresponding question for $\rho > 1$. One can ask whether a lower bound $A(\rho)$ can be obtained for $\delta(u)$ if $\rho > 1$ and all the mass of $u(x)$ lies on a ray, or more generally on some suitable lower dimensional subspace $S$ of $\mathbb{R}^m$ and $\rho>\rho_0(S)$. For fixed $S$ we may conjecture by analogy with the case $m = 2$, that $A(\rho)\to1$ as $\rho\to\infty$. This is proved by Hellerstein and Shea in the case $m = 2$. (D. Shea)

背景链接：https://arxiv.org/abs/1809.07200

11079 | AMR-022-3014

### Research Problems in Function Theory - Problem 3.14

O / open | AMR 问题清单汇编

Let there be given an integrable function $F$ on $\mathbb{T}$ and a point $z_0$ in $\mathbb{D}$. The problem is to maximise $u(z_0)$, where $u$ runs through all functions which are subharmonic in $\mathbb{D}$, equal to $F$ on $\mathbb{T}$, and which satisfy \[\inf u(re^{i\theta})\leq 0, \hspace{1cm}0 < r < 1.\] (A. Baernstein)

背景链接：https://arxiv.org/abs/1809.07200

11080 | AMR-022-3015

### Research Problems in Function Theory - Problem 3.15

O / open | AMR 问题清单汇编

Let $D$ be a doubly-connected domain with boundary curves $\alpha$ and $\beta$ and let $z_0, z_1$ be points of $D$. Let $A, B$ be given real numbers. The problem is to maximise $u(z_0)$, where $u$ runs through all functions which are subharmonic in $D$, take the values $A$ and $B$ on $\alpha$ and $\beta$ respectively and are non-positive on some curve connecting $z_1$ to $\alpha$. (A. Baernstein)

背景链接：https://arxiv.org/abs/1809.07200

11081 | AMR-022-3016

### Research Problems in Function Theory - Problem 3.16

O / open | AMR 问题清单汇编

A compact set $E$ in $\mathbb{R}^n$, $n\geq3$ is said to be thin at $P_0$ if $$ \int^1_0\frac{c(P_0,r)}{r^{n-1}}\,dr<\infty, $$ where $c(P_0,r)= \text{cap}\big[E\cap\{P:|P-P_0| \leq r\}\big]$; this is the integrated form of the Wiener criterion. It follows from Kellogg's theorem that the points of $E$ where ([source label: B3.1]) holds, form a polar set. Can one give a direct proof of this fact, which shows perhaps that ([source label: B3.1]) is best possible? (P. J. Rippon)

背景链接：https://arxiv.org/abs/1809.07200

11082 | AMR-022-3018

### Research Problems in Function Theory - Problem 3.18

S / solved | AMR 问题清单汇编

It is known that the set $E$ of least capacity $C$ and given volume is a ball. If $E$ displays some measure of asymmetry (for instance, if every ball with the same volume as $E$ in space contains a minimum proportion $\delta$ in the complement of $E$), can one obtain a lower bound for the capacity of $E$ which exceeds $C$ by some positive function of $\delta$? (E. Fraenkel; communicated by W. K. Hayman)

背景链接：https://arxiv.org/abs/1809.07200

11083 | AMR-022-3020

### Research Problems in Function Theory - Problem 3.20

O / open | AMR 问题清单汇编

Suppose that you have a continuous real function $u(x)$ on $\mathbb{R}^n$, and you want to know whether a homeomorphism $\phi:\mathbb{R}^n\to\mathbb{R}^n$ and a harmonic function $v$ on $\mathbb{R}^n$ exist, such that \[v(x)=u(\phi(x)).\] Is it necessary and sufficient that there should exist mappings $\mu_1, \mu_2, \ldots, \mu_n$ so that \[U=(u, \mu_1, \mu_2, \ldots, \mu_n)\] is a light open mapping of $\mathbb{R}^n$ into $\mathbb{R}^n$? The case $n=2$ is a result of Stoilow which is in , for example. (L. A. Rubel, communicated by D. A. Brannan)

背景链接：https://arxiv.org/abs/1809.07200

11084 | AMR-022-3022

### Research Problems in Function Theory - Problem 3.22

P / partially_solved | AMR 问题清单汇编

Let $D$ be a domain containing the origin whose `outer boundary' is $\mathbb{T}$ and whose `inner boundary' is a closed set $E$ in $\mathbb{D}$. If every radius of the unit disc meets $E$, determine the supremum of the harmonic measure at $0$ of $\mathbb{T}$ with respect to $D$. (W. H. J. Fuchs)

背景链接：https://arxiv.org/abs/1809.07200

11085 | AMR-022-3028

### Research Problems in Function Theory - Problem 3.28

S / solved | AMR 问题清单汇编

Determine all domains $\Omega$ in $\mathbb{R}^n$, $n\geq2$, satisfying the identity $\int_\Omega h(x)\,dx = 0$ for every function $h$ harmonic and integrable on $\Omega$. In the case $n = 2$, the answer is given by Sakai in . (M. Sakai)

背景链接：https://arxiv.org/abs/1809.07200

11086 | AMR-022-3029

### Research Problems in Function Theory - Problem 3.29

S / solved | AMR 问题清单汇编

It is known that the Newtonian potential of a uniform mass distribution spread over an ellipsoid $K$ in $\mathbb{R}^n$, $n\geq2$ is a quadratic function of the coordinates of $x = (x_1,\ldots,x_n)$ for $x\in K$. Nikliborc (, ) and Dive (, ) independently proved that for $n = 2$ and $n = 3$, the ellipsoid is the only body with this property. Prove this converse assertion for $n > 3$ (preferably by a new method, since Nikliborc and Dive both use methods involving highly non-trivial calculations). (H. S. Shapiro)

背景链接：https://arxiv.org/abs/1809.07200

11087 | AMR-022-3030

### Research Problems in Function Theory - Problem 3.30

O / open | AMR 问题清单汇编

Let $K(z, z')$ denote the kernel of the double layer potential occurring in Fredholm's theory where $z, z' \in \Gamma$, $\Gamma$ being a smooth Jordan curve. (Recall that $K(z,z')=\frac{\cos \phi}{|z-z'|}$, where $\phi$ is the angle between the inward normal to $\Gamma$ at $z$ and the line $(z,z')$.) When $\Gamma$ is a circle, the function $z \mapsto K(z, z')$ is a constant (i.e. the same for each choice of $z'$); and consequently the integral operator \[T_\Gamma:f \mapsto \int_\Gamma f(z)K(z,z')\,ds_z\] is of rank one (as an operator from $C(\Gamma)$ to $C(\Gamma)$). Are there any other $\Gamma$ for which the rank of $T_\Gamma$ is finite? (H. S. Shapiro)

背景链接：https://arxiv.org/abs/1809.07200

11088 | AMR-022-3031

## Research Problems in Function Theory - Problem 3.31

O / open | AMR 问题清单汇编

Let $D$ be an unbounded domain in $\mathbb{R}^n$, $n\geq 2$. Points in $\mathbb{R}^n$ will be denoted by $x = (x_1,x_2,\ldots,x_n)$, and $|x|$ will denote the Euclidean norm of $x$. The following result was given by Ess\'en . \begin{theorem} Assume that the least harmonic majorant $\Psi$ of $|x_1|$ in $D$ is such that $\Psi(x)=O(|x|)$ as $x\to\infty$ in $D$. If $|x|$ has a harmonic majorant in $D$, then $|x_1|\log^+|x_1|$ has a harmonic majorant in $D$. \end{theorem} In the plane, $|x|^p$ (if $p > 0$) has a harmonic majorant in $D$, if and only if $F\in H^p$, where $F:\{|x| < 1\} \to D$ is a universal covering map with $F(0) = 0$. There are similar statements for $|x_1|\log^+|x_1|$ and $\text{Re}\, F\in L\log L$. Thus, when $n = 2$, Theorem 1 is closely related to the following. \begin{theorem} Suppose that $n = 2$, and that $F\in H^1(\mathbb{D})$ (where $\mathbb{D}$ is the unit disc). Then $\text{Re}\, F\in L\log L$ if and only if $$ \int^\infty_{-\infty}N(1,iv,F)\log^+|v|\,dv<\infty, $$ where $N(1,\omega,F)$ is the Nevanlinna counting function (see ). \end{theorem} Theorem 2 is an extension of a well-known result of Zygmund. Note also that, in the case $n = 2$, there are functions $F$ in $H^1(\mathbb{D})$ such that $\text{Re}\, F\notin L\log L$, see . [(a)] ; In the case $n=2$, what is the relation between the condition on $\Psi$ in Theorem 1 and condition ([source label: Jstar]) in Theorem 2? ; Suppose now that $n\geq2$. The assumption on $\Psi$ in Theorem 1 was introduced in the proof for purely technical reasons. Is this the correct condition needed in Theorem 1 (we assume always that $|x|$ has a harmonic majorant in $D$)? Does there exist a domain $D$ such that $|x|$ has a harmonic majorant in $D$ while $|x_1|\log^+|x_1|$ does not have a harmonic majorant in $D$? (M. Ess\'en)

背景链接：https://arxiv.org/abs/1809.07200

11089 | AMR-022-3033

## Research Problems in Function Theory - Problem 3.33

O / open | AMR 问题清单汇编

Let $\Omega$ be a bounded open subset of $\mathbb{R}^n$, $n\geq 2$, and suppose that $y\in\partial\Omega$. Denote the open ball of centre $y$ and radius $r$ by $B(r)$. A point $y$ is said to be $B$-regular for $\Omega$ if, for each resolutive function $f$ on $\partial\Omega$ that is bounded in $B(R)\cap\partial\Omega$ for some positive $R$, the Perron-Wiener-Brelot solution $H^\Omega$ of the Dirichlet problem is bounded in $B(r)\cap\Omega$ for some positive $r$. Also, a point $y$ is said to be $lB$-regular for $\Omega$ if there exists a positive null sequence $\{r_m\}^\infty_1$, such that $y$ is $B$-regular for $B(r_m)\cap\Omega$ for all $m$. If $y$ is $lB$-regular for $\Omega$, then it is $B$-regular for $\Omega$, see Sadi . Is the converse true? (D. H. Armitage and A. Sadi)

背景链接：https://arxiv.org/abs/1809.07200

11090 | AMR-022-3034

## Research Problems in Function Theory - Problem 3.34

P / partially_solved | AMR 问题清单汇编

Let $\Omega$ be a bounded domain in $\mathbb{R}^n$, $n\geq 2$, with the property that there exists $\alpha$ in $(0,\pi]$ such that for every point $y$ in $\partial\Omega$ there is an open truncated cone with vertex $y$ and angle $\alpha$ contained in $\Omega$. Is there some positive number $p=p(\alpha)$ such that every positive superharmonic function in $\Omega$ belongs to $L^p(\Omega)$? If so, can such a $p$ be characterised in terms of $\alpha$? (D. H. Armitage)

背景链接：https://arxiv.org/abs/1809.07200

11091 | AMR-022-3035

## Research Problems in Function Theory - Problem 3.35

O / open | AMR 问题清单汇编

For $r_1<r_2$, we will call the set $\{x\in\mathbb{R}^n:n\geq3,r_1<\|x\|<r_2\}$ an annulus and its closure a closed annulus. Let $\Omega$ be a non-empty subset of $\mathbb{R}^n$, $n\geq3$, such that $\lambda(\overline{\Omega})<+\infty$, where $\lambda$ denotes $n$-dimensional Lebesgue measure. Then if, for each point $x\in\mathbb{R}^n\setminus\overline{\Omega}$ we have that \[\frac{1}{\lambda(\overline{\Omega})}\int_{\overline{\Omega}}\|x-y\|^{2-n}\,d\lambda(y)\] equals the mean-value of the function $y\mapsto\|x-y\|^{2-n}$ over the unit sphere of $\mathbb{R}^m$, it can be shown that $\overline{\Omega}$ is a closed annulus. If, throughout the hypotheses, we replace $\overline{\Omega}$ by $\Omega$, can we conclude that $\Omega$ is an annulus? For details of closely-related work, see and . (D. H. Armitage and M. Goldstein)

背景链接：https://arxiv.org/abs/1809.07200

11092 | AMR-022-3004

## Research Problems in Function Theory - Problem 3.4

O / open | AMR 问题清单汇编

Consider the class of functions subharmonic in the unit disc $\mathbb{D}$, and satisfying $u(z)\leq0$ there. Suppose also that $A(r,u)\leq-1$, for $r$ lying on a set $E$ consisting of a finite number of straight line segments. Then it is known (see e.g. Nevanlinna ) that $B(r,u)$ is maximal, when $u(z)=u_E(z)$, where $u_E(z)$ is harmonic in $\mathbb{D}$, except on a set $E$ of the positive real axis, and $u(z)$ assumes boundary values $0$ on $|z|=1$, and $-1$ on $E$. This is the solution to the so-called Carleman-Milloux problem . For $0<r<1$, $0<K<1$, let $C(r,K)$ be the set of all $\theta$ such that $u(re^{i\theta})<-K$. Is it true that $C(r,K)$ has minimal length only if $u(z)=u_E(z)$? The special case where $E$ consists of the whole interval $[0,1]$ has particular interest. (T. K\"ovari)

背景链接：https://arxiv.org/abs/1809.07200

11093 | AMR-022-3005

### Research Problems in Function Theory - Problem 3.5

P / partially_solved | AMR 问题清单汇编

Suppose that $u(z)$ is positive and subharmonic in $\mathbb{D}$, and that there exists a series of arcs $\gamma_n$ tending to the arc $\alpha\leq\theta\leq\beta$ of $|z|=1$, such that $$ u(z)\leq M,\hspace{1cm} z\text{ on }\gamma_n,\hspace{1cm} n=1,2. $$ If in addition, $$ \int^1_0(1-r)u(re^{i\theta})\,d\theta<+\infty, $$ for a set $E$ on $\theta$, which is dense in the interval $(\alpha, \beta)$, then Maclane proved that $u(re^{i\theta})$ is uniformly bounded for \[\alpha+\delta\leq\theta\leq\beta-\delta, \hspace{1cm}0\leq r<1,\] and any fixed positive $\delta$. These conclusions thus hold in particular, if $$ \int^1_0(1-r)B(r,u)\,dr<+\infty. $$ Can the growth conditions ([source label: 3.3]) and ([source label: 3.4]) be weakened without weakening the conclusions?

背景链接：https://arxiv.org/abs/1809.07200

11094 | AMR-022-3009

### Research Problems in Function Theory - Problem 3.9

O / open | AMR 问题清单汇编

If $D$ is a convex domain in space of $3$ or more dimensions, can we assert any inequalities for the Green's function $g(P,Q)$ of $D$ which generalise the results of $2$ dimensions, that follow from the classical inequalities for convex univalent functions? Gabriel proved that the level surfaces $G(P,Q)=\lambda>0$ are convex, but the proof is long. It would be interesting to find a simpler proof, and also to obtain definite inequalities for the curvatures. It may be conjectured that half-space gives the extreme case. (G. E. H. Reuter)

背景链接：https://arxiv.org/abs/1809.07200

11095 | AMR-022-4010

### Research Problems in Function Theory - Problem 4.10

P / partially_solved | AMR 问题清单汇编

Is it true that the length of the curve $|f_n(z)|=1$ is maximal for $f_n(z)=z^n-1$? (P. Erd\"os)

背景链接：https://arxiv.org/abs/1809.07200

11096 | AMR-022-4011

### Research Problems in Function Theory - Problem 4.11

O / open | AMR 问题清单汇编

If $|z_i|\leq1$, estimate from below, the area of $E^{(n)}_f$. Erd\"os, Herzog and Piranian prove that, given positive $\varepsilon$, the area of $E^{(n)}_f$ can be made less that $\varepsilon$, if $n>n_0(\varepsilon)$.

背景链接：https://arxiv.org/abs/1809.07200

11097 | AMR-022-4013

### Research Problems in Function Theory - Problem 4.13

O / open | AMR 问题清单汇编

It is known that there exists a polynomial $P(z)$ \[P(z)=\sum^n_{k=1}\varepsilon_k z^k, \hspace{1cm}\varepsilon_k=\mp1\] for which $$ \max_{|z|=1}|P(z)|<C_1\sqrt{n}. $$ (See Clunie ). Is it necessarily true that $C_1>1+A$ if ([source label: 4.1]) holds, where $A$ is a positive absolute constant?

背景链接：https://arxiv.org/abs/1809.07200

11098 | AMR-022-4014

### Research Problems in Function Theory - Problem 4.14

S / solved | AMR 问题清单汇编

Does there exist a polynomial of the type in Problem 4.13, for which $$ \min_{|z|=1}|P(z)|>C_2\sqrt{n} $$ for every $n$? More generally, does there exist such a polynomial satisfying both ([source label: 4.1]) and ([source label: 4.2])?

背景链接：https://arxiv.org/abs/1809.07200

11099 | AMR-022-4015

### Research Problems in Function Theory - Problem 4.15

P / partially_solved | AMR 问题清单汇编

If again $\varepsilon_k=\mp1$, is it true that, for large $n$, all but $o(2^n)$ polynomials $P(z)=\sum^n_{k=1}\varepsilon_k z^k$ have just $n/2+o(n)$ roots in $\mathbb{D}$?

背景链接：https://arxiv.org/abs/1809.07200

11100 | AMR-022-4016

### Research Problems in Function Theory - Problem 4.16

S / solved | AMR 问题清单汇编

Is it true that for all but $o(2^n)$ polynomials $P(z)$ \[\min_{|z|=1}|P(z)|<1,\] or, if not, what is the corresponding correct result?

背景链接：https://arxiv.org/abs/1809.07200

11101 | AMR-022-4018

### Research Problems in Function Theory - Problem 4.18

P / partially_solved | AMR 问题清单汇编

If $f$ is any polynomial or rational function of degree $N$, find the least upper bound $\phi(N)$ of \[\frac{1}{r}\int^r_0dt\int^\pi_{-\pi}\frac{|f'(re^{i\theta})|}{1+|f|^2}\,d\theta,\] for varying $r$ and $f$. It is known only that $\phi(N)=O(N^{\frac{1}{2}})$, $\phi(N)\neq O(\log N)^{\frac{1}{2}-\varepsilon}$, as $N\to\infty$. The upper bound for varying $f$ of \[\int^\infty_0dt\int^\pi_{-\pi}\frac{|f'|}{1+|f|^2}\,dr\,d\theta\] should not be much larger than $\phi(N)$, but there is no proof that it is not $+\infty$. The special cases $N=2, 3, \ldots$ would be of interest. See Littlewood .

背景链接：https://arxiv.org/abs/1809.07200

11102 | AMR-022-4021

### Research Problems in Function Theory - Problem 4.21

O / open | AMR 问题清单汇编

If $a_k=\mp1, k=0,\ldots,n$ and \[b_k=a_na_{n-k}+a_{n-1}a_{n-k-1}+\ldots+a_ka_0,\] is it true that \[\sum^n_1|b_k|^2>An^2,\] where $A$ is an absolute constant? If $p(z)=a_0+a_1z+\ldots+a_nz^n$, then \[|p(z)|^2=n+2\sum^n_1b_k\cos(k\theta),\] so that the truth of the above inequality would imply \[\frac{1}{2\pi}\int^{2\pi}_0\big|p(e^{i\theta})\big|^4\,d\theta\geq n^2(1+A).\] (F. R. Keogh)

背景链接：https://arxiv.org/abs/1809.07200

11103 | AMR-022-4022

### Research Problems in Function Theory - Problem 4.22

O / open | AMR 问题清单汇编

Using the notation of Problem 4.7, if $|z_i|\leq1$, Clunie and Netanyahu (personal communication) showed that a path exists joining the origin to $| z| = 1$ in $E^{(n)}_f$. What is the shortest length $L^{(n)}_f$ of such a path? Presumably $L^{(n)}_f$ tends to infinity with $n$, but not too fast. (P. Erd\"os)

背景链接：https://arxiv.org/abs/1809.07200

11104 | AMR-022-4023

### Research Problems in Function Theory - Problem 4.23

O / open | AMR 问题清单汇编

Some of the Problems 4.7 to 4.12 extend naturally to the space of higher dimensions. Let $x_i$ be a set of $n$ points in $\mathbb{R}^m$ and let $E^{(m)}_n$ be the set of points for which \[\prod^n_{i=1}|x-x_i|\leq1.\] When is the maximum volume of $E^{(m)}_n$ attained and how large can it be? Piranian observed that the ball is not extreme for $m = 3$, $n = 2$. If $E^{(m)}_n$ is connected, can it be covered by a ball of radius $2$? For $m = 2$, this was proved by Pommerenke . (P. Erd\"os)

背景链接：https://arxiv.org/abs/1809.07200

11105 | AMR-022-4024

### Research Problems in Function Theory - Problem 4.24

O / open | AMR 问题清单汇编

Let \[P(z)=\sum^n_0a_kz^k\] be a self-inversive polynomial, i.e. if $\zeta$ is a zero of $P(\zeta)$ with multiplicity $m$, then $1/\zeta$ is also a zero with multiplicity $m$. Is it true that $w = P(z)$ maps $\mathbb{D}$ onto a domain containing a disc of radius $A= \max_{0\leq k\leq n} |a_k|$? (T. Sheil-Small)

背景链接：https://arxiv.org/abs/1809.07200

11106 | AMR-022-4025

### Research Problems in Function Theory - Problem 4.25

O / open | AMR 问题清单汇编

Determine \[\inf\int^\pi_{-\pi}\big|1-e^{i\theta}\big|^{2\lambda}\big|P(e^{i\theta})\big|^2\,d\theta,\hspace{1cm}\lambda>0,\] where $P(z)$ ranges over all polynomials with integer coefficients and leading coefficient unity. (The solution would have number-theoretic applications.) (W. H. J. Fuchs)

背景链接：https://arxiv.org/abs/1809.07200

11107 | AMR-022-4026

### Research Problems in Function Theory - Problem 4.26

P / partially_solved | AMR 问题清单汇编

Let $P_n$ denote the class of polynomials $p(z)$, $p(0) = 1$, of degree at most $n$ and of positive real part in $\mathbb{D}$. Find \[\max_{p\in P_n}\int^{2\pi}_0|p(e^{i\theta})|^2\,d\theta.\] (F. Holland)

背景链接：https://arxiv.org/abs/1809.07200

11108 | AMR-022-4027

### Research Problems in Function Theory - Problem 4.27

O / open | AMR 问题清单汇编

Let $p(x)$ be a real polynomial of degree $n$ in the real variable $x$ such that $p(x) = 0$ has $n$ distinct (real) rational roots. Does there necessarily exist a (real) non-zero number $t$ such that $p(x)- t = 0$ has $n$ distinct (real) rational roots? (I can prove this for $n = 1,2, 3$.) (L. A. Rubel)

背景链接：https://arxiv.org/abs/1809.07200

11109 | AMR-022-4003

### Research Problems in Function Theory - Problem 4.3

O / open | AMR 问题清单汇编

Let $P_N(z)$ be a polynomial with $N$ terms, satisfying $|P_N(z)|\leq1$ on $|z|=1$. How large can $P_n(z)$ be if $P_n(z)$ is a partial sum of $P_N(z)$? (P. Erd\"os)

背景链接：https://arxiv.org/abs/1809.07200

11110 | AMR-022-4030

## Research Problems in Function Theory - Problem 4.30

P / partially_solved | AMR 问题清单汇编

Let $\mathcal{P}$ denote the set of all polynomials of the form \[p(z)=\prod^n_{\nu=1}(z-\zeta_\nu),\] where $n\geq2$ and $|\zeta_\nu|\leq1$, $\nu=1, 2, \ldots, n$. The Sendov conjecture (see Problem 4.5) states: if $p(z)\in\mathcal{P}$ then each disc \[\{z:|z-\zeta_\nu|\leq1\},\hspace{1cm} \nu=1, 2, \ldots, n,\] contains at least one zero of $p'(z)$. Schmeisser proved this conjecture for certain subsets of $\mathcal{P}$. In all but two of these special cases, the proof show that the following stronger result is true: \textit{If $\zeta$ is an arbitrary point of the convex hull of the zeros of $p(z)$, then the disc \mbox{$\{z:|z-\zeta|\leq1\text{ contains at least one zero of }p'(z)\}$}}. We ask: [(a)] ; is this stronger result true for all $p(z)$ in $\mathcal{P}_1$, where $\mathcal{P}_1$ is the subset of all polynomials in $\mathcal{P}$ which vanish at $0$? ; is this stronger result true for all $p(z)$ in $\mathcal{P}_2$, where $\mathcal{P}_2$ is the subset of all polynomials in $\mathcal{P}$ which are of the form $p(z)=z^n+a_{n-1}z^{n-1}+\ldots+a_0$, $a_\nu\leq0$, $\nu=0, 1, \ldots, n-1$? (G. Schmeisser)

背景链接：https://arxiv.org/abs/1809.07200

11111 | AMR-022-4031

## Research Problems in Function Theory - Problem 4.31

P / partially_solved | AMR 问题清单汇编

Erd\"os and Newman conjectured that if $$ f(z)=\sum^n_{k=0}a_kz^k,\hspace{1cm} |a_k|=1, \hspace{1cm}0\leq k\leq n, $$ then there is an absolute constant $c$ such that $$ \max_{|z|=1}|f(z)|>(1+c)n^{1/2}. $$ The weaker form of our conjecture stated that ([source label: D2]) holds if we assume \linebreak \mbox{$a_k=\pm1$} (Problem 4.13). A stronger form would be that ([source label: D2]) holds even if \mbox{$f(z)=\sum^n_{k=1}a_{n_k}z^{n_k}$}, $n_k$ natural numbers, $|a_{n_k}|=1$. However, ([source label: D2]) was disproved by Kahane (no citation). In fact, he showed that, given that $\varepsilon>0$, there are polynomials of the form ([source label: D1]) for which $$ \max_{|z|=1}|f(z)|<n^{1/2}+O(n^{(3/10)+\varepsilon})\hspace{1cm}\text{ as }n\to\infty. $$ Show that $n^{(3/10)+\varepsilon}$ cannot be replaced by $n^\varepsilon$ in ([source label: D3]). Is there any $n$-th degree polynomial of the form ([source label: D1]) for which $$ \min_{|z|=1}|f(z)|>(1-\varepsilon)n^{1/2} $$ for every positive $\varepsilon$ if $n>n_0(\varepsilon)$? Perhaps there is an $n$-th degree polynomial of the form ([source label: D1]), for which, for all $z$, $|z|=1$, $$ (1-\varepsilon)n^{1/2}<|f(z)|<(1+\varepsilon)n^{1/2}. $$ In fact, ([source label: D5]) could possibly hold with $n^{1/2}+O(1)$ on the left and right. It would be worthwhile to determine if ([source label: D2]) holds for $a_k=\pm1$ and for $a_{n_k}=\pm1$. (P. Erd\"os and D. Newman)

背景链接：https://arxiv.org/abs/1809.07200

11112 | AMR-022-4004

## Research Problems in Function Theory - Problem 4.4

S / solved | AMR 问题清单汇编

Is there a function $f(k)$ of the positive integer $k$, so that the square of every polynomial having at least $f(k)$ terms has a least $k$ terms? Erd\"os proved that at any rate, $f(k)>k^{1+c}$ for a positive constant $c$. (P. Erd\"os)

背景链接：https://arxiv.org/abs/1809.07200

11113 | AMR-022-4005

## Research Problems in Function Theory - Problem 4.5

P / partially_solved | AMR 问题清单汇编

Let $P(z)$ be a polynomial whose zeros $z_1, z_2, \ldots, z_n$ lie in $|z|\leq1$. Is it true that $P'(z)$ always has a zero in $|z-z_1|\leq1$? (Bl. Sendov)

背景链接：https://arxiv.org/abs/1809.07200

11114 | AMR-022-4006

## Research Problems in Function Theory - Problem 4.6

O / open | AMR 问题清单汇编

If $H_\nu(z)$ is the $\nu$-th Hermite polynomial, so that \[H_\nu(z)e^{-z^2}=(-1)^\nu\big(\frac{d}{dz}\big)^\nu e^{-z^2},\] is it true that the equation \[1+H_1(z)+aH_n(z)+bH_m(z)=0,\] where $2\leq n<m$, and $a, b$ are complex, has at least one zero in the strip $|\text{Im}\, z|\leq c$, where $c$ is an absolute constant? (This is true if $b=0$, see Makai and Tur\'an .) (P. Tur\'an)

背景链接：https://arxiv.org/abs/1809.07200

11115 | AMR-022-4007

## Research Problems in Function Theory - Problem 4.7

P / partially_solved | AMR 问题清单汇编

Let $f(z)=z^n+a_1z^{n-1}+\ldots+a_n$ be a polynomial of degree $n$. Cartan proved that the set $|f(z)|\leq1$, which we call $E^{(n)}_f$ can always be covered by discs, the sum of whose radii is at most $2e$. It seems likely that $2e$ can be replaced by $2$. If $E^{(n)}_f$ is connected, this was proved by Pommerenke , who also proved the general result, with $2.59$ instead of $2$.

背景链接：https://arxiv.org/abs/1809.07200

11116 | AMR-022-5010

### Research Problems in Function Theory - Problem 5.10

O / open | AMR 问题清单汇编

If, in addition, $f(z)$ is univalent in $\mathbb{D}$, the conclusions of Problem 5.8 and Problem 5.9 follow with a constant $S$, $S\geq L$, known as the schlicht Bloch's constant. What is the value of $S$? We may also ask the same question when, in addition, $f(z)$ is star-like, thus obtaining a still larger constant $S_1$. If $f(z)$ is convex, the correct value of the constant is $\pi/4$, attained for $f(z)=\frac{1}{2}\log\frac{1+z}{1-z}$, which maps $\mathbb{D}$ onto the strip $|\text{Im}\, z|<\pi/4$.

背景链接：https://arxiv.org/abs/1809.07200

11117 | AMR-022-5016

### Research Problems in Function Theory - Problem 5.16

P / partially_solved | AMR 问题清单汇编

Do corresponding results to Problem 5.15(a) apply to the means \[I_\lambda(r,f)=\Big\{\frac{1}{2\pi}\int^{2\pi}_0\big|f(re^{i\theta})\big|^\lambda \,d\theta\Big\}^{1/\lambda}, \hspace{1cm}0<\lambda<\infty\] or the Nevanlinna characteristic \[T(r,f)=\frac{1}{2\pi}\int^{2\pi}_0\log^+\big|f(re^{i\theta})\big|\,d\theta?\] This is known to be the case when $D$ is already symmetrical (but possibly multiply-connected) so that $D=D^*$, as a consequence of the theory of subordination (see e.g Littlewood ).

背景链接：https://arxiv.org/abs/1809.07200

11118 | AMR-022-5017

### Research Problems in Function Theory - Problem 5.17

O / open | AMR 问题清单汇编

Let $D=D_0$ be a domain, $g(z,a_0)$ be the Green's function of $D$ with respect to a point $a_0$ on the positive real axis, and let $D_\lambda$ be the part of $D$ where $g>\lambda$ for $0<\lambda<\infty$. Is it true, at least in some simple cases, that $(D_\lambda)^*\subset(D^*_\lambda)$? The cases where $D^*$ consists of the plane or the unit disc cut along the negative real axis are of particular interest. A positive answer to this problem for simply-connected domains $D$ would lead to a positive answer of Problem 5.16 for the same class of domains, using a formula of Hardy-Stein-Spencer (see Hayman ). With the general notation of the introduction above, when can we assert that $|a_n|\leq|a_n^*|$ for general $n$? This is true, for instance, when $D^*$ is the plane cut along the negative real axis, so that we obtain the Littlewood conjecture that $|a_n|\leq4|a_0|n$ for non-zero univalent $f$. In fact, $|a_1|\leq4|a_0|$ in this case, by symmetrisation and subordination, and $|a_n|\leq n|a_1|$ by De Branges' proof of the Bieberbach conjecture (see Update 6.1). If $D$ is convex and $D=D^*$, the exact bound for $|a_n|$ is $|a_1|$, but $|a_n^*|\leq|a_1|$ in general (see Hayman ).

背景链接：https://arxiv.org/abs/1809.07200

11119 | AMR-022-5018

### Research Problems in Function Theory - Problem 5.18

O / open | AMR 问题清单汇编

Let $f(z)=\lambda+a_1z+\ldots$ be analytic in $\mathbb{D}$, where $0<\lambda<1$. Find the best constant $B(\lambda)$ such that if \[F(r)=\lambda+|a_1|r+|a_2|r^2+\ldots,\] then \[F[B(\lambda)]\leq1\] for all $f$. It is known that (see Bombieri ): \[B(\lambda)= \begin{cases} (1+2\lambda)^{-1} &\text{if }\frac{1}{2}\leq\lambda<1, 1/\sqrt{2}&\text{if }\lambda=0, \end{cases}\] and that \[B(\lambda)>[(1-\lambda)/2]^{\frac{1}{2}},\hspace{1cm}\text{if }0<\lambda<\frac{1}{2}.\]

背景链接：https://arxiv.org/abs/1809.07200

11120 | AMR-022-5019

### Research Problems in Function Theory - Problem 5.19

S / solved | AMR 问题清单汇编

A function meromorphic in $\mathbb{D}$ which has no asymptotic value, assumes every value infinitely often in the disc. Every point of the circumference $\mathbb{T}$ is a Picard point for such a function, i.e. a point such that all except perhaps two values are taken in every neighbourhood. Functions with no exceptional values in the global sense are known. Can locally exceptional values occur? (See Cartwright and Collingwood , ). (E. F. Collingwood)

背景链接：https://arxiv.org/abs/1809.07200

11121 | AMR-022-5002

### Research Problems in Function Theory - Problem 5.2

O / open | AMR 问题清单汇编

Is it true that ([source label: 5.3]) implies that $$ I_1(r,f)=O(1-r)^{-1} $$ and $$ |a_n|=O(n)? $$

背景链接：https://arxiv.org/abs/1809.07200

11122 | AMR-022-5020

### Research Problems in Function Theory - Problem 5.20

O / open | AMR 问题清单汇编

Plessner proves after Privaloff that if $f$ is analytic in $\mathbb{D}$, almost all points $P$ of the boundary are of two kinds. Either [(a)] ; $f$ tends to a finite limit, as $z\to P$ in any Stolz angle $S$, lying in $\mathbb{D}$, or ; as $z\to P$ in any $S$, $f$ takes (infinitely) often all values of a dense set. Can `dense set' be replaced by anything bigger here; e.g. the complement of a set of measure zero? (E. F. Collingwood)

背景链接：https://arxiv.org/abs/1809.07200

11123 | AMR-022-5021

### Research Problems in Function Theory - Problem 5.21

O / open | AMR 问题清单汇编

Corresponding to each function $f$ analytic in $\mathbb{D}$, and each value $w$, with $(|w|<1)$, write \[ f_w(z)=f\Big(\frac{z-w}{1-wz}\Big)=\sum^\infty_{n=0}a_n^{(w)}z^n,\] \[\|f_w\|=\sum^\infty_{n=0}|a^{(w)}_n|\] and let $W_f$ be the set of all values $w$, with $(|w|<1)$ for which \[\|f_w\|<\infty,\] Since $a^{(w)}_n$ is a continuous function of $w$, it follows that $W_f$ is a set of type $F_\sigma$. What more can be said? For example, if $W_f$ is everywhere dense in $D$ (or uncountable, or of positive measure), is $W_f$ the unit disc? It is known that $W_f$ may be a proper non-empty subset of $D$. Is $W_f$ either empty, or all of $D$ if $f$ is univalent?

背景链接：https://arxiv.org/abs/1809.07200

11124 | AMR-022-5022

### Research Problems in Function Theory - Problem 5.22

O / open | AMR 问题清单汇编

Let $H^p$ be the space of functions $f(z)=\sum^\infty_{n=0}a_nz^n$ analytic in $\mathbb{D}$, and such that \[\int^{2\pi}_0\big|f(re^{i\theta})\big|^p\,d\theta\] remains bounded as $r\to1$. We define $H^\infty$ to be the class of bounded functions in $\mathbb{D}$. For $0 < p < 1$ describe the coefficient multipliers from $H^p$ to $H^p$. That is, for each such $p$ describe the sequences $\lambda_n$ such that \[\sum\lambda_n a_n z^n\in H^p\hspace{1cm}\text{whenever }\sum a_nz^n\in H^p.\] (P. L. Duren)

背景链接：https://arxiv.org/abs/1809.07200

11125 | AMR-022-5023

### Research Problems in Function Theory - Problem 5.23

P / partially_solved | AMR 问题清单汇编

Describe similarly the coefficient multipliers from $S$ to $S$, where $S$ is the class of functions $\sum^\infty_{n=1}a_nz^n$ univalent in $\mathbb{D}$ either [(a)] ; with the normalisation $a_1 = 1$, or\ ; generally. ; What are the multipliers of the space of close-to-convex functions into itself? ; What are the multipliers of $S$ into the class $C$ of convex functions? ; What are the multipliers from the class $N$ of functions of bounded characteristic into itself? The analogous problem for the class $N^+$ may be more tractable. (The definition of $N^+$ is too lengthy for this work, but the reader is directed to Duren for more details). Ruscheweyh and Sheil-Small in solving Problem 6.9 have shown that $(\lambda_n)$ is a multiplier sequence from $C$ into itself if and only if $\sum\lambda_nz^n\in C$. In general, one can obtain only some sufficient conditions. Thus in most cases $f(z) = \sum a_n z^n$ belongs to a class $A$ if $a_n$ is sufficiently small, and conversely if $f\in A$, then $a_n$ cannot be too big. For example, $\sum^\infty_2n|a_n|\leq1$ is a sufficient condition for $f(z)$ in $S$, and $|a_n|\leq n\sqrt{7/6}$ is a necessary condition (see Fitzgerald ). Similarly, if $\sum^\infty_1|a_n|<\infty$, then $f(z)$ is continuous in $\overline{\mathbb{D}}$, and so belongs to $H^p$ for every positive $p$ and to $N$, whilst if $f\in N$, then $|a_n|\leq\exp(cn^\frac{1}{2})$ for some constant $c$. Again, if $f(z)$ belongs to one of the above classes, then so does $\frac{1}{t}f(tz)$ for $0 < t < 1$, so that the sequence $(t^{n-1})$ is a multiplier sequence. In other cases, negative results are known. Thus Frostman showed that $(n)$ is not a multiplier sequence from $N$ to $N$, and Duren showed that $\big(\frac{1}{n+1}\big)$ is not such a sequence either. (P. L. Duren, except for $(e)$ which is due to A. Shields)

背景链接：https://arxiv.org/abs/1809.07200

11126 | AMR-022-5025

### Research Problems in Function Theory - Problem 5.25

O / open | AMR 问题清单汇编

Let $W^+$ be the Banach algebra of power series $f(z)=\sum^\infty_{n=0}a_nz^n$ absolutely convergent in $|z|\leq1$, with $\|f\|=\sum^\infty_{n=0}|a_n|$. Which functions generate $W^+$? More precisely, for which functions $f$ is it true that the polynomials in $f$ are dense in $W^+$? It is clear that a necessary condition is that $f$ be univalent in $\overline{\mathbb{D}}$. Newman has shown that if in addition, $f'\in H^1$ then $f$ generates $W^+$. Hedberg and Lisin have shown (independently) that if $f$ is univalent and $\sum n|a_n|^2(\log n)^{1+\varepsilon}<\infty$ for some positive $\varepsilon$, then $f$ generates $W^+$. Neither Newman's condition nor Hedberg-Lisin's condition implies the other. Is univalentness enough? (L. Zalcman)

背景链接：https://arxiv.org/abs/1809.07200

11127 | AMR-022-5026

### Research Problems in Function Theory - Problem 5.26

P / partially_solved | AMR 问题清单汇编

Let $B$ be the Bergman space of square integrable functions in $\mathbb{D}$, that is, those functions $f(z) = \sum^\infty_{n=0} a_nz^n$ for which $\sum^\infty_{n=1} n^{-1}|a_n|^2<\infty$. A subspace $S$ is said to be invariant if $zf \in S$ whenever $f \in S$. What are the invariant subspaces of $B$? The corresponding problem for $H^2$ was solved by Beurling and uses the inner-outer factorisation of $H^p$ functions, a tool unavailable in the present context. (L. Zalcman)

背景链接：https://arxiv.org/abs/1809.07200

11128 | AMR-022-5027

### Research Problems in Function Theory - Problem 5.27

P / partially_solved | AMR 问题清单汇编

(The corona conjecture) Let $D$ be an arbitrary domain in the plane that supports non-constant bounded analytic functions. Suppose that $f_1(z),\ldots, f_n(z)$ are bounded and analytic in $D$ and satisfy \[\sum^n_{\nu=1}|f_\nu(z)|\geq\delta>0\] in $D$. Can one find bounded analytic functions $g_\nu(z)$ in $D$ such that \[\sum^n_{\nu=1}f_\nu(z)g_\nu(z)\equiv1\] in $D$? When $D$ is a disc, Carleson proved that the answer is `yes', and the result extends to finitely connected domains. The result is also known to be true for certain infinitely connected domains (see Behrens , Gamelin ), but false for general Riemann surfaces of infinite genus (Cole, unpublished). Presumably the answer for the general plane domain is negative. Proofs of all positive results depend on Carleson's theorem. (L. Zalcman)

背景链接：https://arxiv.org/abs/1809.07200

11129 | AMR-022-5030

### Research Problems in Function Theory - Problem 5.30

P / partially_solved | AMR 问题清单汇编

Let $\mathcal{B}$ be the space of Bloch functions, that is the space of functions analytic in $\mathbb{D}$ with \[\|f\|_\mathcal{B}=|f(0)|+\sup_\mathbb{D}(1-|z|^2)|f'(z)|<\infty.\] Let $\mathcal{B}_S$ be the space of functions of the form $$ f(z) = \log g'(z),\hspace{1cm} g \in S $$ where $S$ is the class of functions as in Problem 5.23(a). Let $\mathcal{B}_Q$ be the space of functions $g$ in $S$ that have a quasi-conformal extension to the closed plan, see Anderson, Clunie and Pommerenke . [(a)] ; Is $\mathcal{B}_S$ connected in the norm topology? ; Is $\mathcal{B}_Q$ dense in $\mathcal{B}_S$ in the norm topology? (L. Bers)

背景链接：https://arxiv.org/abs/1809.07200

11130 | AMR-022-5031

### Research Problems in Function Theory - Problem 5.31

P / partially_solved | AMR 问题清单汇编

It was shown by Becker that \[\big\{f:\|f\|_\mathcal{B}<1\big\}\subset \mathcal{B}_Q.\] Is the radius 1 best possible? Is it true that for $f \in \mathcal{B}_S$, \[ \limsup_{|z|\to1}\,(1 - |z|^2)|f'(z)| < 1\hspace{0.5cm}\implies\hspace{0.5cm} f \in\mathcal{B}.\] (Ch. Pommerenke)

背景链接：https://arxiv.org/abs/1809.07200

11131 | AMR-022-5032

### Research Problems in Function Theory - Problem 5.32

O / open | AMR 问题清单汇编

Suppose that $f_n\in\mathcal{B}_S$. What does $\|f_n-f\|_\mathcal{B}\to0$ as $n\to\infty$ mean geometrically for the functions $g_n$ related to $f_n$ by ([source label: B5.11])? (Ch. Pommerenke)

背景链接：https://arxiv.org/abs/1809.07200

11132 | AMR-022-5034

### Research Problems in Function Theory - Problem 5.34

O / open | AMR 问题清单汇编

It was proved by Hall that every Bloch function has (possibly infinite) angular limits on an uncountably dense subset of $|z | = 1$. Do there always exist angular limits on a set of positive measure relative to some fixed Hausdorff measure, such as logarithmic measure for example? (J. E. McMillan, Ch. Pommerenke)

背景链接：https://arxiv.org/abs/1809.07200

11133 | AMR-022-5035

### Research Problems in Function Theory - Problem 5.35

P / partially_solved | AMR 问题清单汇编

Let $F$ be any discontinuous group of M\"obius transformations of $\mathbb{D}$. Does there always exist a meromorphic function automorphic with respect to $\Gamma$ and normal, i. e. such that \[(1-|z|^2)\frac{|f'(z)|}{1+|f(z)|^2}\] is bounded in $\mathbb{D}$? (Ch. Pommerenke)

背景链接：https://arxiv.org/abs/1809.07200

11134 | AMR-022-5036

### Research Problems in Function Theory - Problem 5.36

P / partially_solved | AMR 问题清单汇编

Let $(n_k)$ be a sequence of positive integers such that $$ n_{k+1}>\lambda n_k,\text{ where }\lambda>1, $$ and suppose that $$ f(z)=\sum^\infty_{k=0}a_kz^{n_k} $$ is analytic in $\mathbb{D}$. Is it true that if $\sum^\infty_{k=0}|a_k|=\infty$, then $f(z)$ assumes every finite value [(a)] ; at least once; ; infinitely often; ; in every angle $\alpha<\arg z<\beta$ of $| z | < 1$? See Weiss and Weiss . (J. P. Kahane)

背景链接：https://arxiv.org/abs/1809.07200

11135 | AMR-022-5037

### Research Problems in Function Theory - Problem 5.37

O / open | AMR 问题清单汇编

Suppose that $f(z)$ is a function as in ([source label: B5.13]) and define \[\mu=\limsup_{r\to1}\frac{\log\log M(r,f)}{-\log(1-r)},\] where $M(r,f)$ is the maximum modulus of $f(z)$ on $|z| = r$. We do not now assume ([source label: B5.12]), but let $N^0(t)$ be the number of $n_k$ not greater than $t$. If $N(r, a)$ is the function of Nevanlinna theory (see Chapter 1) it is known that \[\limsup_{r\to1}\frac{N(r,0)}{\log M(r,f)}=1\] provided that either [(a)] ; $\mu > 0$ and \[\liminf_{k\to\infty}\frac{\log(n_{k+1}-n_k)}{\log n_k}>\frac{1}{2}\Big(\frac{2+\mu}{1+\mu}\Big),\] (this is implicit in Wiman, see Sunyer and Balaguer ), or ; $\mu>\frac{1-\beta}{\beta}$ and $N^0(t)=O(t^{1-\beta})$ as $t\to\infty$, where $0<\beta<1$, see Sons . If $0<\mu<\frac{1-\beta}{\beta}$ with $N^0(t)=O(t^{1-\beta})$ as $t\to\infty$ we ask $(a), (b)$ and $(c)$ of the preceding problem, at least for those cases not covered above. In particular we may consider the cases $n_k = [k^\alpha]$, where $1 < \alpha < \frac{3}{2}$. (L. R. Sons)

背景链接：https://arxiv.org/abs/1809.07200

11136 | AMR-022-5038

### Research Problems in Function Theory - Problem 5.38

O / open | AMR 问题清单汇编

Tao-Shing Shah has shown that if $g(z)\prec f(z)$ in $\mathbb{D}$, $g'(0)/f'(0)$ is real, and $f$ in $S$ then $$ |g(z)|\leq|f(z)|\text{ for }|z|\leq\frac{1}{2}(3-\sqrt{5}), $$ $$ |g'(z)|\leq|f'(z)|\text{ for }|z|\leq3-\sqrt{8}. $$ Both constants are `best-possible'. Shah's proofs are technically very involved and it would be nice to have simpler proofs. Goluzin gave simpler proofs but with worse constants in each case. His methods appear to be incapable of yielding ([source label: C5.1]) and ([source label: C5.2]). (P. L. Duren)

背景链接：https://arxiv.org/abs/1809.07200

11137 | AMR-022-5039

### Research Problems in Function Theory - Problem 5.39

O / open | AMR 问题清单汇编

Goluzin has shown that, if $g(z)\prec f(z)$ in $\mathbb{D}$, then \[M_2(r,g')\leq M_2(r,f'),\hspace{1cm}0\leq r\leq\frac{1}{2}.\] Here, for $p > 0$, \[M_p(r,h)=\Big(\frac{1}{2\pi}\int^{2\pi}_0|h(re^{i\theta})|^p\,d\theta\Big)^{1/p}.\] The result is not necessarily true if $r > \frac{1}{2}$, as $f(z) = z, g(z) = z^2$ shows, though it follows from a theorem of Littlewood that, for all $p$, \[M_p(r,g)\leq M_p(r,f),\hspace{1cm} 0 < r < 1.\] Find the largest number $r_p, 0 < r_p < 1$, independent of $f$ and $g$ so that \[M_p(r,g')\leq M_p(r,f'),\hspace{1cm} 0<r<r_p,\] if $g\prec f$. Note: If $g(z) \prec f(z)$ in $\mathbb{D}$, then $g(z) = f(\phi(z))$ so that \[|g'(z)|\leq |f'(\phi(z))|,\hspace{1cm}|z|\leq\sqrt{2}-1,\] (see Carath\'eodory ). Thus if $h(z) = f'(\phi(z))$ then $h\prec f'$ so that, by Littlewood's theorem, $M_p(r,g') \leq M_p(r,h) \leq M_p(r,f')$ for $p > 0, r \leq\sqrt{2}-1$. This improves what one gets if one applied ([source label: C5.2]) to the $p$-th means since $\sqrt{2}-1>3-\sqrt{8}$; but it may not be best possible. (P. L. Duren)

背景链接：https://arxiv.org/abs/1809.07200

11138 | AMR-022-5004

### Research Problems in Function Theory - Problem 5.4

P / partially_solved | AMR 问题清单汇编

If the sequence $w_n$ satisfies $$ \arg w_n=O\big(|w_n|^{\frac{1}{2}}\big) $$ and $$ |w_{n+1}-w_n|=O(|w_n|^{\frac{1}{2}}) $$ then it is known (see Hayman ) that ([source label: 5.5]) and hence ([source label: 5.6]) hold. It is interesting to ask whether the method will yield the same conclusions under somewhat weaker hypotheses, such as for instance, replacing the index $\frac{1}{2}$ in ([source label: 5.7]) and ([source label: 5.8]) by a smaller positive number.

背景链接：https://arxiv.org/abs/1809.07200

11139 | AMR-022-5042

### Research Problems in Function Theory - Problem 5.42

S / solved | AMR 问题清单汇编

Suppose that \[f(z)=\sum^\infty_{n=0}a_nz^n\] is analytic in $\mathbb{D}$, with \[\sum^\infty_{n=0}|a_n|=1,\hspace{1cm} |f(z)|\geq\delta>0\text{ in }\mathbb{D},\hspace{1cm}\text{ and}\hspace{1cm}\frac{1}{f(z)}=\sum^\infty_{n=0}b_nz^n.\] The following facts are known about $M=\sum^\infty_{n=0}|b_n|$: [(a)] ; $M < +\infty$; ; if $\delta<\frac{1}{2}$, $M$ cannot be bounded in terms of $\delta$ (Katznelson (no citation)). ; if $\delta>2^{-\frac{1}{2}}$, $M$ can be bounded in terms of $\delta$ (Katznelson (no citation), Newman (no citation)). What is the infimum of those $\delta$ such that $M$ can be bounded in terms of $\delta$? (A likely guess is $\frac{1}{2}$). (H. S. Shapiro)

背景链接：https://arxiv.org/abs/1809.07200

11140 | AMR-022-5043

### Research Problems in Function Theory - Problem 5.43

O / open | AMR 问题清单汇编

Determine the Laurent coefficient bodies for analytic functions taking values of modulus at most unity in a given annulus \[A_r = \{z : r < |z| < 1\}.\] Determine the extremal functions. (M. Heins)

背景链接：https://arxiv.org/abs/1809.07200

11141 | AMR-022-5044

### Research Problems in Function Theory - Problem 5.44

S / solved | AMR 问题清单汇编

Suppose that $0 < \alpha < 1$ and \[\frac{(1+xz)^\alpha}{1-z}=\sum^\infty_{n=0}A_n(x)z^n,\hspace{1cm} A_0(x)=1,\hspace{1cm} |x|=1.\] Is it true that \[|A_{2n+1}(x)|\leq|A_{2n+1}(1)|,\hspace{1cm}n\geq2,\hspace{1cm}|x|=1?\] The above is true for $n = 1$; the corresponding result is false for all \mbox{$A_{2m}(x)$} with $m \geq 1$, see Brannan . More generally, one can ask the same question for the coefficients of \[(1+xz)^\alpha(1-z)^{-\beta}, \hspace{1cm}|x|=1, \hspace{1cm}\alpha>0,\hspace{1cm}\beta>0.\] Here it is not even known if $|A_3(x)| \leq A_3(1)$. (D. A. Brannan)

背景链接：https://arxiv.org/abs/1809.07200

11142 | AMR-022-5045

### Research Problems in Function Theory - Problem 5.45

O / open | AMR 问题清单汇编

If $A$ is any analytic subset of the Riemann sphere it was shown by Kierst that $A$ is (exactly) the set of asymptotic values of a function meromorphic in $\mathbb{D}$. If $\infty\in A$, Kierst also proved that $A$ is the set of asymptotic values of a function $f$ analytic in $\mathbb{D}$. However, there exist analytic sets which are not the set of asymptotic values of a function analytic in $\mathbb{D}$. Ryan has characterised those subsets of the Riemann sphere which are the set of asymptotic values of a function analytic in $\mathbb{D}$. Can one find a simpler characterisation? (K. F. Barth)

背景链接：https://arxiv.org/abs/1809.07200

11143 | AMR-022-5046

### Research Problems in Function Theory - Problem 5.46

O / open | AMR 问题清单汇编

A non-constant function $f$ analytic in $\mathbb{D}$, is said to be in the MacLane class $\mathcal{A}$ if the set of points of the unit circle $\mathbb{T}$ at which $f$ has an asymptotic value is a dense subset of $\mathbb{T}$ (see MacLane ). A function $f$ is said to have an arc tract if there exists a sequence $\{\gamma_n\}$ of arcs, $\gamma_n\subset \mathbb{D}$, and a non-degenerate subarc $\gamma$ of $\mathbb{T}$ such that $\gamma_n\to\gamma$ (in the obvious fashion) and $\min\{|f(z)| :z\in\gamma_n\}\to\infty$ as $n\to\infty$. Does there exist a function $f\in\mathcal{A}$ with an arc tract and with non-zero derivative (see MacLane )? (K. F. Barth)

背景链接：https://arxiv.org/abs/1809.07200

11144 | AMR-022-5047

### Research Problems in Function Theory - Problem 5.47

P / partially_solved | AMR 问题清单汇编

Let \[f(z)=\sum^\infty_{n=0}a_nz^{\lambda_n}\] be analytic in $\mathbb{D}$ and have Hadamard gaps, i.e. $\lambda_{n+1}/\lambda_n\geq q>1$. Need $f$ have any radial limits (finite or infinite)? If not, need $f$ have any asymptotic value on a path ending at a single point? For the case $q\geq3$, see MacLane . (J. M. Anderson and R. Hornblower)

背景链接：https://arxiv.org/abs/1809.07200

11145 | AMR-022-5048

### Research Problems in Function Theory - Problem 5.48

O / open | AMR 问题清单汇编

Let \[f(z)=\sum^\infty_{k=1}a_{n_k}z^{n_k}\] be analytic in $\mathbb{D}$ and have Hadamard gaps. Characterise sets $S$ in $\mathbb{D}$ with the property that if $f$ is bounded on $S$ then $f(z)$ is bounded for $z\in\mathbb{D}$. (K. G. Binmore)

背景链接：https://arxiv.org/abs/1809.07200

11146 | AMR-022-5049

### Research Problems in Function Theory - Problem 5.49

P / partially_solved | AMR 问题清单汇编

What kind of gaps can the Taylor expansion of a non-constant automorphic function have? For example, can it have Hadamard gaps? (Presumably the sharp answer would depend on the group concerned.) This is closely related to a theorem of R\'enyi (no citation) that a non-constant periodic entire function cannot have more than half of its coefficients zero. (L. A. Rubel)

背景链接：https://arxiv.org/abs/1809.07200

11147 | AMR-022-5051

### Research Problems in Function Theory - Problem 5.51

O / open | AMR 问题清单汇编

It can be shown that there exists a Blaschke product $B(z)$ with \mbox{$B(0) = 0$} such that \[\frac{1+B(z)}{1-B(z)}\] is a Bloch function. Give an explicit construction for such a product. (F. Holland)

背景链接：https://arxiv.org/abs/1809.07200

11148 | AMR-022-5052

### Research Problems in Function Theory - Problem 5.52

O / open | AMR 问题清单汇编

Let F be a Fuchsian group in $\mathbb{D}$, and let $B$ be a set on $\mathbb{T}$ such that $B\cap \gamma(B)=\emptyset$ for $\gamma\in\Gamma,\gamma\neq I$. (This is the case, for instance, if \[B=(\partial F\cap\mathbb{T})\setminus C\hspace{1cm}(C\text{ countable})\] where $F$ is a normal fundamental domain.) If cap $B > 0$, does it follow that $F$ is of convergence type? (Conversely, it is known that, for every group $F$ of convergence type, there exists such a set $B$ with cap $B > 0$). (Ch. Pommerenke)

背景链接：https://arxiv.org/abs/1809.07200

11149 | AMR-022-5053

### Research Problems in Function Theory - Problem 5.53

O / open | AMR 问题清单汇编

The ratio $R$ of two Blaschke products $B(z,a_n), B(z,b_n)$ is of bounded characteristic, but need not be a normal meromorphic function. That is, if $a_n$ is `close' to $b_n$ for infinitely many $n$ we can arrange that the spherical derivative of $R$ is too large for $R$ to be normal. When is $R$ a normal function? Cima and Colwell have shown that if $\{a_n\}$ and $\{b_n\}$ are both interpolating sequences, then $R$ is normal if and only if $\{a_n\} \cup \{b_n\}$ is also an interpolating sequence. What happens if the sequences are not interpolating sequences? (J. M. Anderson)

背景链接：https://arxiv.org/abs/1809.07200

11150 | AMR-022-5054

### Research Problems in Function Theory - Problem 5.54

O / open | AMR 问题清单汇编

Suppose that \[f(z)=\sum^\infty_{n=0}a_nz^n\] is convergent in $\mathbb{D}$, that $|z_0| = 1$, and that neither of the series \[\sum^\infty_{n=0}(\text{Re}\, a_n)z_0^n,\hspace{1cm}\sum^\infty_{n=0}(\text{Im}\, a_n)z_0^n,\] is absolutely convergent. Let $S$ be the set of complex values assumed by the series \[\sum^\infty_{n=0}\varepsilon_n a_n z_0^n,\] where $\varepsilon=\pm1$, and is allowed to vary over all possible choices. Is it true that $S = \mathbb{C}$? (A. C. Offord; communicated by J. G. Clunie)

背景链接：https://arxiv.org/abs/1809.07200

11151 | AMR-022-5055

### Research Problems in Function Theory - Problem 5.55

O / open | AMR 问题清单汇编

Let the function $h(z)$ be analytic in $\mathbb{D}$, $G_h = \{h(\mathbb{D})\}$, $K_h$ a compact subset of $G_h$. The function $h(z)$ will be said to have the $L$-property (`$h(z_n)$ leaves the range of $h$') on the sequence $\{z_n\}^\infty_1$ with $\lim |z_n|=1$, if only finitely many points of $\{h(z_n)\}^\infty_1$ lie in $K_h$ for every $K_h$. We will define two functions $f,g$ analytic in $\mathbb{D}$ to be an ordered $L$-pair if, on any sequence for which $f$ has the $L$-property, $g$ also has the $L$-property. A non-constant function $\alpha(z)$ analytic in $\mathbb{D}$ will be called an $L$-atom if, corresponding to any function $f$ such that $f, \alpha$ is an ordered $L$-pair, there exists a function $\phi_f$ analytic on $G_\alpha$ such that $f=\phi_f\circ\alpha$. Prove or disprove the conjecture that $\alpha$ is an $L$-atom if and only if it is univalent. What happens if $\mathbb{D}$ is replaced by more general domains? (L. A. Rubel)

背景链接：https://arxiv.org/abs/1809.07200

11152 | AMR-022-5056

### Research Problems in Function Theory - Problem 5.56

O / open | AMR 问题清单汇编

By a first-order property we shall mean a ring-theoretic property in the first-order predicate calculus. For functions $f$ analytic in $\mathbb{D}$, are the following first-order properties: [(a)] ; $f$ is constant? ; $f$ is bounded ? ; $f$ is admissible, i.e. $T(r,f) \neq O(\log(1-r)^{-1})?$ What is the situation in more general domains? (L. A. Rubel)

背景链接：https://arxiv.org/abs/1809.07200

11153 | AMR-022-5057

### Research Problems in Function Theory - Problem 5.57

O / open | AMR 问题清单汇编

Suppose that $f$ is analytic in $\mathbb{D}$. Plessner's theorem asserts that at almost all points $e^{i\theta}$ on the unit circle, either $f$ has an angular limit, or else the image $f(S)$ of every Stolz angle $S$ with vertex at $e^{i\theta}$ is dense in $\mathbb{C}$. How much can `dense' be improved? In particular, is it true that at almost all $e^{i\theta}$ either $f$ has an angular limit, or else $f(S)$ is all of $\mathbb{C}$, except perhaps for a set of zero logarithmic capacity? This result would be best possible, since if $E$ is any closed set of capacity zero then the universal covering map of the disc onto $\mathbb{C} \setminus E$ has angular limits almost nowhere. (A. Baernstein II)

背景链接：https://arxiv.org/abs/1809.07200

11154 | AMR-022-5058

### Research Problems in Function Theory - Problem 5.58

O / open | AMR 问题清单汇编

Suppose that $f$ is univalent and zero-free in $\mathbb{D}$. It has been shown by Baernstein that, for each $p\in(0,\frac{1}{2})$, $f$ admits a factorisation $f=B_p(F_p)^{1/p}$, where $B_p\in H^\infty, 1/B_p\in H^\infty$, and $\text{Re}\, F_p > 0$. Is it possible to pass to the limit $p = \frac{1}{2}$, and thus factor $f$ into a bounded function times a function subordinate to a map onto a slit plane? (A. Baernstein II)

背景链接：https://arxiv.org/abs/1809.07200

11155 | AMR-022-5059

### Research Problems in Function Theory - Problem 5.59

O / open | AMR 问题清单汇编

(Subordination and extreme point problem) Let $g(z)=\sum^\infty_{n=0}B_nz^n$ be analytic in $\mathbb{D}$. Denote by $S_g$ the family of functions $f(z)$ subordinate to $g$. Find general conditions on $g$ so that the only extreme points of the closed convex hull of $S_g$ are the functions $g(ze^{it})(0\leq t<2\pi)$. This is known for certain functions $g$, see Clunie , for example: [(a)] ; $g(z)=[(1+cz)/(1-z)]^\alpha$ where $|c|\leq1,\alpha\geq1$; ; $g(z)=\exp[(1+z)/(1-z)]$. Sheil-Small can prove it for a functions of the form \linebreak\mbox{$g(z)=h((1+z)/(1-z))$}, where $h(w)$ is a univalent quadratic polynomial in $\text{Re}\, w>0$. One might expect that the conclusion would hold for a functions $g$ with positive coefficients increasing in a suitably regular manner. (T. Sheil-Small)

背景链接：https://arxiv.org/abs/1809.07200

11156 | AMR-022-5006

### Research Problems in Function Theory - Problem 5.6

S / solved | AMR 问题清单汇编

It is known that there exist functions which fail to take any of the values $2\pi ik$, $-\infty<k<+\infty$ and which do not satisfy ([source label: 5.9]), and in fact $|a_n|\leq \log\log n$ (see Littlewood , and Hayman ). However ([source label: 5.9]) holds if $f(z)$ omits all but a finite interval of the imaginary axis (again by subordination). This suggests that ([source label: 5.9]) might still hold if the omitted values $w_n$ cluster near $\infty$ sufficiently close to the imaginary axis.

背景链接：https://arxiv.org/abs/1809.07200

11157 | AMR-022-5060

### Research Problems in Function Theory - Problem 5.60

O / open | AMR 问题清单汇编

(Hadamard convolutions) Suppose that $\alpha\geq1, \beta\geq1$ and that $\phi$ is analytic in $\mathbb{D}$, and satisfies \[\phi(z)\ast\frac{(1+xz)^\alpha}{(1-z)^\beta}\neq0,\hspace{1cm}|x|=1, |z|<1.\] Is it true that \[\phi(z)\ast\frac{(1+xz)^{\alpha-1}}{(1-z)^\beta}\neq0,\hspace{1cm}|x|=1, |z|<1\,?\] This is true when $\alpha$ is a natural number. For the proof and further background, see Sheil-Small . (T. Sheil-Small)

背景链接：https://arxiv.org/abs/1809.07200

11158 | AMR-022-5062

### Research Problems in Function Theory - Problem 5.62

O / open | AMR 问题清单汇编

Let $u$ be a continuous real-valued function on the unit circle $\mathbb{T}$. Give a necessary and sufficient condition on $u$ such that $u$ is the real part of a function $f$ in the disc algebra $A(\overline{\mathbb{D}})$. Remarks: [(a)] ; A solution would have applications in the algebraic ideal theory of $A(\overline{\mathbb{D}})$. ; An answer to the analogous problem for $L^p(T), H^p(\mathbb{D})$ is the Burkholder-Gundy-Silverstein Theorem (see Peterson ). (M. von Renteln)

背景链接：https://arxiv.org/abs/1809.07200

11159 | AMR-022-5063

### Research Problems in Function Theory - Problem 5.63

O / open | AMR 问题清单汇编

One of the many equivalent norms on BMOA on $\mathbb{D}$ is defined by \[\|f\|_h=\inf_q\sup_{z\in\mathbb{D}}|f(z)+\overline{q(z)}|,\] the infimum being taken over all functions analytic in $\mathbb{D}$. Given $f$ in BMOA, does there exist a $q$ such that \[|f(z)+\overline{q(z)}|\equiv\|f\|_h\text{ a.e. on }|z|=1?\] The answer is `yes' when $f$ is a rational function. (J. A. Hempel)

背景链接：https://arxiv.org/abs/1809.07200

11160 | AMR-022-5064

### Research Problems in Function Theory - Problem 5.64

O / open | AMR 问题清单汇编

Let $f$ be analytic in $\mathbb{D}$ with $$ |f(z)|=O((1-|z|)^{-k}),\hspace{1cm}k\geq0. $$ Then $f$ induces a distribution on $C^\infty(T)$, as follows. For $\phi\in C^\infty(T)$ \[\lim_{r\to1}\Big(\frac{1}{2\pi}\int f(re^{i\theta})\phi(e^{-i\theta})\,d\theta\Big)=\Lambda_f(\phi).\] What can be said about the order of the distributions satisfying ([source label: D5.64.1]) and \[\lim_{|z|\to1}|f(z)|(1-|z|)^k=0\] (J. A. Cima)

背景链接：https://arxiv.org/abs/1809.07200

11161 | AMR-022-5066

### Research Problems in Function Theory - Problem 5.66

S / solved | AMR 问题清单汇编

Let $B$ be an infinite Blaschke product in $\mathbb{D}$. Does there exist a positive $\delta$, depending on $B$, such that, for every $w$, $|w|<\delta$, the set $B^{-1}(\{w\})$ is infinite? Stephenson has obtained some related results. (K. Stephenson)

背景链接：https://arxiv.org/abs/1809.07200

11162 | AMR-022-5067

### Research Problems in Function Theory - Problem 5.67

O / open | AMR 问题清单汇编

Let the function $f$ in $\mathbb{D}$ be given by \[f(z)=\sum^\infty_{k=0}a_kz^{n_k},\hspace{1cm}\frac{n_{k+1}}{n_k}\geq\lambda>1,\hspace{1cm} k\geq0,\] with \[m_0(r,f)\equiv \max_{k\geq0}|a_k|r^{n_k}\to\infty\hspace{1cm}\text{ as }r\to 1^-.\] Define \[E=\Big\{\theta:\liminf_{r\to1^-}\frac{|f(re^{i\theta})|}{m_0(r,f)}>0\Big\}.\] Is it true that $E$ has measure $0$? This has been proved for the case that $\mu(\frac{1}{2}(1+r))/m_0(r,f)\leq$ constant. (D. Gnuschke and Ch. Pommerenke)

背景链接：https://arxiv.org/abs/1809.07200

11163 | AMR-022-5068

### Research Problems in Function Theory - Problem 5.68

O / open | AMR 问题清单汇编

Let the function $f$ where \[f(z)=1+\sum^\infty_{n=1}a_nz^n,\hspace{1cm}|z|\leq1,\] be a Bloch function with positive real part in $\mathbb{D}$. Determine the rate of growth (as $N\to\infty$) of the sequence $\big\{\sum^N_{n=1}|a_n|^2\big\}$. (F. Holland)

背景链接：https://arxiv.org/abs/1809.07200

11164 | AMR-022-5069

### Research Problems in Function Theory - Problem 5.69

O / open | AMR 问题清单汇编

Let the function $f$ where \[f(z)=1+\sum^\infty_{n=1}a_nz^n,\hspace{1cm}|z|\leq1,\] be a Bloch function with positive real part in $\mathbb{D}$ and such that each $a_n\geq0$. Does it follow that $\sum^\infty_{n=1}a^2_n<\infty$? An equivalent formulation of the problem is the following: let $\mu$ be a probability measure in Zygmund's class $\Lambda_*$ on the circle, and let \[\hat{\mu}(n)=\int^{2\pi}_0e^{-inx}\,d\mu(x)\geq0,\hspace{1cm}n\in\mathbb{Z}.\] Is it true that $\hat{\mu}\in l_2$? A counter-example, if one exists, cannot be constructed using Riesz products (see Duren , and Holland and Twomey ). An affirmative answer would mean that, if $f(z) = 1 + \sum^\infty_{n=1}a_nz^n$ is a Bloch function in $\mathbb{D}$ with positive real part, then $\sum^\infty_{n=1}|a_n|^4<\infty$. Even if this last inequality is false, perhaps it is still true in the general case that there exists $p > 4$, such that $\sum^\infty_{n=1}|a_n|^p<\infty$. (F. Holland)

背景链接：https://arxiv.org/abs/1809.07200

11165 | AMR-022-5070

### Research Problems in Function Theory - Problem 5.70

O / open | AMR 问题清单汇编

Barth and Clunie have constructed a bounded analytic function in $\mathbb{D}$ with a level set component of infinite length; this component is highly branched. Can one construct a bounded analytic function with an unbranched level set component of infinite length? (K. F. Barth and J. G. Clunie)

背景链接：https://arxiv.org/abs/1809.07200

11166 | AMR-022-5071

### Research Problems in Function Theory - Problem 5.71

O / open | AMR 问题清单汇编

Suppose that \[f(z) = \sum^\infty_{k=1}a_kz^{n_k},\hspace{1cm}n_{k+1}/n_k\geq q>1,\] is an analytic function in $\mathbb{D}$ with Hadamard gaps, such that $T(r,f)\to\infty$ as $r\to1$. Does $\delta(w,f)=0$ hold for every (finite) complex number $w$? (T. Murai)

背景链接：https://arxiv.org/abs/1809.07200

11167 | AMR-022-5072

### Research Problems in Function Theory - Problem 5.72

O / open | AMR 问题清单汇编

Let the function $f$ have the power series $f(z)=\sum^\infty_{n=0}a_nz^n$ of radius of convergence $1$; let $E$ be the singular set on $\mathbb{T}$, and suppose that \[\sup_{\xi\in E}\, \sup_{N\geq0}\Big|\sum^N_{n=0}a_n\xi\Big|<\infty.\] It has been shown by Allan, O'Farrell and Ransford that, if $E$ has measure zero, then $\sum^\infty_{n=0}a_nz^n$ converges to $f(z)$ at each point $z$ in $\mathbb{T}\setminus E$. Does the conclusion remain true if $E$ has positive measure? (Nothing appears to be known either way.) (T. J. Ransford)

背景链接：https://arxiv.org/abs/1809.07200

11168 | AMR-022-5073

### Research Problems in Function Theory - Problem 5.73

O / open | AMR 问题清单汇编

Let $0<\alpha<1$ and let $R_\alpha$ denote the set of all Riesz potentials $p(x)$ of finite positive Borel measures $\mu$ on $\mathbb{R}$: \[p(x)=\int^\infty_{-\infty}|x-t|^{-\alpha}\,d\mu(t).\] Characterise those non-negative measurable functions $f(x)$ on $\mathbb{R}$ that are dominated by some function $p$ in $R_\alpha$. (B. Korenblum)

背景链接：https://arxiv.org/abs/1809.07200

11169 | AMR-022-5074

### Research Problems in Function Theory - Problem 5.74

O / open | AMR 问题清单汇编

Characterise those non-negative measurable functions $f$ on the unit circle that are dominated almost everywhere by moduli of the boundary values of an analytic function in the unit disc with positive real part. (Problem 5.73 might be considered a step towards solving this problem.) (B. Korenblum)

背景链接：https://arxiv.org/abs/1809.07200

11170 | AMR-022-5075

### Research Problems in Function Theory - Problem 5.75

S / solved | AMR 问题清单汇编

Does there exist a bounded analytic function in $\mathbb{D}$ such that the image of every radius has infinite length? See, for example, Anderson , and Rudin . (W. Rudin; communicated by K. F. Barth)

背景链接：https://arxiv.org/abs/1809.07200

11171 | AMR-022-5076

### Research Problems in Function Theory - Problem 5.76

P / partially_solved | AMR 问题清单汇编

Let $\gamma$ be a non-tangential arc that lies in $\mathbb{D}$ except for one endpoint at $z = 1$, and define \[\gamma_\theta=e^{i\theta\gamma},\hspace{1cm} \theta\in[0,2\pi).\] Does there exist a function $g\in H^\infty$ such that \[\lim_{z\to e^{i\theta},\,z\in\gamma_\theta}g(z)\] exists for no value of $\theta$? (See Rudin .) If $\gamma$ is tangential, the answer is `yes', see Collingwood and Lohwater . (W. Rudin; communicated by K. F. Barth)

背景链接：https://arxiv.org/abs/1809.07200

11172 | AMR-022-5077

### Research Problems in Function Theory - Problem 5.77

O / open | AMR 问题清单汇编

In general, the radial behaviour of the derivative of a bounded analytic function in $\mathbb{D}$ can be pretty arbitrary; in fact, even under much stronger restrictions than just bounded, it can still be quite arbitrary. Is it true that, given any measurable function $m(\theta)$ on $[0,2\pi)$, there exists a function $f(z)$, continuous on $\overline{\mathbb{D}}$ and univalent on $\mathbb{D}$, such that \[\lim_{r\to1}f'(re^{i\theta})=m(\theta)\] for almost all $\theta$? (In terms of known results, the conjecture seems quite plausible. Ortel and Schneider showed that the conjecture is true under slightly strengthened hypotheses; and Lohwater, Piranian and Rudin showed that the conjecture is true with a slightly weakened conclusion.) (W. J. Schneider)

背景链接：https://arxiv.org/abs/1809.07200

11173 | AMR-022-5078

### Research Problems in Function Theory - Problem 5.78

O / open | AMR 问题清单汇编

Let $H^1$ denote Hausdorff one-dimensional measure on $\mathbb{C}$, and $\mathbb{T}$; let $g:\mathbb{T}\to[-\infty,\infty]$ denote an arbitrary Borel function. Does there exist a corresponding function $f$, analytic in $\mathbb{D}$ and with bounded Taylor coefficients, such that \[\lim_{r\to1^-}f(rz)=g(z)\] for $H^1$-almost all $z$ in $\mathbb{T}$? For work on related questions, see Ortel and Schneider . (M. Ortel and W. J. Schneider)

背景链接：https://arxiv.org/abs/1809.07200

11174 | AMR-022-5079

### Research Problems in Function Theory - Problem 5.79

O / open | AMR 问题清单汇编

[(a)] ; Let $f$ be a non-constant analytic function in $\mathbb{D}$, $m$ be a positive integer, and define $\psi = (f)^mf'$. Then it is shown by Sons () that, when $f$ and $\psi$ both belong to MacLane's class $A$ , either [(i)] ; $f$ has finite asymptotic values on a dense subset of $\mathbb{T}$, or ; $\psi$ assumes every finite value infinitely often. What replacements can be found for $(i)$ (to give another correct theorem)? Can $\psi$ be taken to be of the form
\[\psi=(f)^{m_0}(f')^{m_1}\ldots(f^{(k)})^{m_k},\]
whenever $k,m_0,m_1,\ldots,m_k\in\mathbb{N}\cup\{0\}$? ; It is known that the conclusion in part $(a)$ is true when $\psi$ is replaced by
\[\psi=f^{(l)}+\sum^{i-1}_{\nu=0}a_\nu f^{(\nu)},\]
where $l\in\mathbb{N}$ and the $a_\nu$ are analytic functions in $\{|z| < 1 +r\}$, for some positive $r$. What replacements can be found in this case for $(i)$ to give another correct theorem? Can the $a_\nu$ be taken as functions analytic in $\mathbb{D}$ with $T(r,a_\nu) = o(T(r,f))$ as $r\to1$? (For both parts, compare Problem 1.38.) (L. R. Sons)

背景链接 : https://arxiv.org/abs/1809.07200

11175 | AMR-022-5008

### Research Problems in Function Theory - Problem 5.8

O / open | AMR 问题清单汇编

Suppose that $f(z)=z+a_2z^2+\ldots$ is analytic in $\mathbb{D}$. Then $f(z)$ maps some sub-domain of $\mathbb{D}$ univalently into a disc of radius at least $B$, where $B$ is Bloch's constant. What is the value of $B$? The best results known are $B\geq\sqrt{3}/4>0.433$, due to Ahlfors , and $B<0.472$ due to Ahlfors and Grunsky . The upper bound is conjectured to be the right one. Heins has shown that $B>\sqrt{3}/4$.

背景链接 : https://arxiv.org/abs/1809.07200

11176 | AMR-022-5009

### Research Problems in Function Theory - Problem 5.9

O / open | AMR 问题清单汇编

With the hypotheses of Problem 5.8, it follows that $f(z)$ assumes all values in some disc of radius $L$, $L\geq B$. What is the value of $L$? The best lower bound for the Landau constant $L$ is $L\geq\frac{1}{2}$, due to Ahlfors .

背景链接 : https://arxiv.org/abs/1809.07200

11177 | AMR-022-6001

### Research Problems in Function Theory - Problem 6.1

S / solved | AMR 问题清单汇编

The Bieberbach conjecture Is it true that $|a_n|\leq n$ for $f$ in $S$ with equality only for $f(z)\equiv f_\theta(z)$? The result is known to be true for $n=2$ (see Bieberbach ), $n=3$ (see L\"owner ), and $n=4$ (see Garabedian and Schiffer , Charzynski and Schiffer ).

背景链接 : https://arxiv.org/abs/1809.07200

11178 | AMR-022-6010

### Research Problems in Function Theory - Problem 6.10

P / partially_solved | AMR 问题清单汇编

If $F(z)$, $G(z)$ are convex functions in $\Sigma$, it is known that for $0<\lambda<1$, \[H(z)=\lambda F(z)+(1-\lambda)G(z)\in\Sigma,\] see Pommerenke . Is it true that $H(z)$ is also convex? (Ch. Pommerenke)

背景链接 : https://arxiv.org/abs/1809.07200

11179 | AMR-022-6100

### Research Problems in Function Theory - Problem 6.100

O / open | AMR 问题清单汇编

Given two functions $f$, $g$ in the (usual) class $S$, we can form the new functions (arithmetic and geometric mean functions) \[F(z)=\alpha f(z)+\beta g(z)\hspace{1cm}\text{ and }\hspace{1cm} G(z)=z\Big(\frac{f(z)}{z}\Big)^\alpha\Big(\frac{g(z)}{z}\Big)^\beta,\] where $\alpha$, $\beta\in(0,1)$ and $\alpha+\beta=1$. It is known that, if \[0.042\simeq\frac{1}{1+e^\pi}<\alpha,\hspace{1cm}\beta<\frac{e^\pi}{1+e^\pi}\simeq0.988,\] then there are functions $f$, $g$ in $S$ such that $F$ and $G$ have valence infinity in $\mathbb{D}$. What can be said about the `fringes' of the interval $(0,1)$? Is there some bound on the valence of $F$ and $G$ that is a function of $\alpha$ for $0 < \alpha\leq 1/(1+e^\pi)$? (A. W. Goodman)

背景链接 : https://arxiv.org/abs/1809.07200

11180 | AMR-022-6101

### Research Problems in Function Theory - Problem 6.101

O / open | AMR 问题清单汇编

Let $K$ be a closed set of points in $\mathbb{C}$, and let $F(K)$ denote the family of functions $f$ of the form \[f(z)=\sum^n_{k=1}\frac{A_k}{z-a_k},\] where $A_k > 0$ and $a_k\in K$, $k=1, 2,\ldots, n$. Find a maximal domain of $p$-valence for the class $F(K)$. For $p=1$, this problem was completely solved by Distler ; however, for $p> 1$, we do not even have a good conjecture. (See also Goodman .) (A. W. Goodman)

背景链接 : https://arxiv.org/abs/1809.07200

11181 | AMR-022-6102

### Research Problems in Function Theory - Problem 6.102

O / open | AMR 问题清单汇编

Let $\{v_n\}^\infty_1$ be a sequence of positive integers (which may include $\infty$); the sequence is called a valence sequence if there is a function $f(z)$, analytic in $\mathbb{D}$, such that $f^{(n)}(z)$ has valence $v_n$ in $\mathbb{D}$, for $n = 1,2,\ldots$. Find interesting necessary conditions for $\{v_n\}$ to be a valence sequence. Also, find sufficient conditions for $\{v_n\}$ to be a valence sequence. (For some results of this type, see Goodman .) (A. W. Goodman)

背景链接：https://arxiv.org/abs/1809.07200

11182 | AMR-022-6103

### Research Problems in Function Theory - Problem 6.103

O / open | AMR 问题清单汇编

The function \[k(z)=2\text{Re }\Big(\frac{z+\frac{1}{3}z^3}{(1-z)^3}\Big)=\sum^\infty_{n=1}\frac{1}{3}(2n^2+1)r^n(e^{in\theta}+e^{-in\theta}),\] where $z = re^{i\theta}$, lies in the closure of $S_H$. Prove that $k$ is extremal for the coefficient bounds in $S_H$. It is known by Clunie and Sheil-Small that $|a_n| <\frac{1}{3}(2n^2+1)$ for functions in $S_H$ with real coefficients, and that $|a_n| \leq\frac{1}{3}(2n^2+1)$ for functions in $S_H$ for which the domain $f(\mathbb{D})$ is close-to-convex. (T. Sheil-Small)

背景链接：https://arxiv.org/abs/1809.07200

11183 | AMR-022-6104

### Research Problems in Function Theory - Problem 6.104

O / open | AMR 问题清单汇编

It is known that, for functions $f$ in $S^0_H$, $\{|w| < \frac{1}{16}\}\subset f(\mathbb{D})$. Prove that the correct value $d$, of the Koebe constant for the class $S^0_H$ is $\frac{1}{6}$. Note that the function \[k_0(z)=\text{Re }\Big(\frac{z+\frac{1}{3}z^3}{(1-z)^3}\Big)+i\text{Im }\Big(\frac{z}{(1-z)^2}\Big)\] belongs to $S^0_H$ and maps $\mathbb{D}$ onto the plane cut along the real axis from $-\frac{1}{6}$ to $-\infty$, and that $\frac{1}{6}$ is the correct constant for those functions $f$ in $S^0_H$ for which $f(\mathbb{D})$ is close-to-convex. Also, determine the number \[\alpha=\sup\{|a_2|:f\in S_H\}.\] The best known estimate for $\alpha$ is $\alpha < 57.05$, see Sheil-Small . It is also known that $d\geq 1/(2\alpha)$. (T. Sheil-Small)

背景链接：https://arxiv.org/abs/1809.07200

11184 | AMR-022-6105

### Research Problems in Function Theory - Problem 6.105

P / partially_solved | AMR 问题清单汇编

What are the convolution multipliers $\phi^*:K_H\to K_H$, where $K_H$ is the subclass of functions $f$ in $S_H$ with convex images $f(\mathbb{D})$? A particularly interesting case is the radius of convexity problem: for which values of $r$ in $(0,1)$ is the function $z\mapsto f(rz)$ convex in $\mathbb{D}$, when $f$ is convex in $\mathbb{D}$? (It is known that $r\leq\sqrt(2)-1$, see Clunie and Sheil-Small .) (T. Sheil-Small)

背景链接：https://arxiv.org/abs/1809.07200

11185 | AMR-022-6106

### Research Problems in Function Theory - Problem 6.106

O / open | AMR 问题清单汇编

Let $J$ be a Jordan curve in $\mathbb{C}$ bounding a domain $D$. Suppose that $f:e^{it}\mapsto f(e^{it})$ is a sense-preserving homeomorphism of the unit cirlce $\mathbb{T}$ onto $J$, and that the harmonic extension of $f$ to $\mathbb{D}$ satisfies the relation $f(\mathbb{D})\subset D$. Prove that $f$ is a homeomorphism of $\mathbb{D}$ onto $D$. This is known to be true if $J$ is convex, when the hypothesis $f(\mathbb{D})\subset D$ is automatically satisfied because of the positivity of the Poisson kernel by the Kneser-Rado-Choquet theorem, see Duren . The result is also known to be true when $\partial f/\partial t$ is continuous and non-zero (on $\mathbb{T}$), and when the co-analytic and analytic parts of $f$ have continuous derivatives on $\overline{\mathbb{D}}$. (T. Sheil-Small)

背景链接：https://arxiv.org/abs/1809.07200

11186 | AMR-022-6107

### Research Problems in Function Theory - Problem 6.107

O / open | AMR 问题清单汇编

Prove that, for $f\in S^0_H$, $$ \big||a_n|-|a_{-n}|\big|\leq n,\hspace{1cm} n=2,3,4,\ldots. $$ (This is a generalisation of the Bieberbach conjecture for $S$.) It is known that ([source label: J6.107]) holds in the following cases: [(a)] ; when $f$ has real coefficients, see Clunie and Sheil-Small ; ; when $f(\mathbb{D})$ is star-like with respect to the origin, see Sheil-Small ; ; when $f(\mathbb{D})$ is convex in one direction, see Sheil-Small . (T. Sheil-Small)

背景链接：https://arxiv.org/abs/1809.07200

11187 | AMR-022-6108

## Research Problems in Function Theory - Problem 6.108

O / open | AMR 问题清单汇编

Let $f$ be analytic univalent in $\mathbb{D}$, and consider \[I_\lambda(r,f')=\Big(\frac{1}{2\pi}\int^{2\pi}_0\big|f'(re^{i\theta})\big|^\lambda\,d\theta\Big)^{1/\lambda}\] where $\lambda > 0$. ; What is the maximal order of magnitude of $I_\lambda(r,f')$ as $r\to1$, where $0\leq \lambda\leq\frac{2}{5}$? If $\lambda>\frac{2}{5}$, it is known that $I_\lambda(r,f') = O\big(1/(1-r)^{3-1/\lambda}\big)$ with equality when $f$ is the Koebe function. The case $\lambda >\frac{1}{2}$ follows easily from classical facts, while the case $\frac{2}{5}\leq \lambda \leq\frac{1}{2}$ is due to Feng and MacGregor . (It was formerly conjectured that the Koebe function would still be extremal for $\lambda>\frac{1}{3}$; but an example of Makarov has shown that this is not the case for $\lambda\leq\frac{1}{3}+\varepsilon$, for some positive $\varepsilon$.) ; Now normalise the function $f$ to belong to the usual class $S$. For $\lambda>\frac{2}{5}$, we know that \[I_\lambda(r,f')\leq C_\lambda I_\lambda(r,k'),\] where $k$ is the Koebe function. For which $\lambda$ is the best constant $C_\lambda$ equal to $1$? (It follows from de Branges' theorem that $C_\lambda = 1$ for \mbox{$\lambda = 2,4,6,\ldots$}. Presumably $C_\lambda=1$ for $\lambda\geq2$, but the proposer knows of no proof of this.) (A. Baernstein)

背景链接：https://arxiv.org/abs/1809.07200

11188 | AMR-022-6109

## Research Problems in Function Theory - Problem 6.109

O / open | AMR 问题清单汇编

Let $f$ be analytic univalent in $\mathbb{D}$, and consider \[I_{-\lambda}(r,f')=\Big(\frac{1}{2\pi}\int^{2\pi}_0\big|f'(re^{i\theta})\big|^{-\lambda}\,d\theta\Big)^{1/\lambda},\] where $\lambda>0$. Except for the elementary case $\lambda=\infty$, the maximal order of magnitude as $r\to1$ is not known for any positive $\lambda$. A particularly interesting case is when $\lambda=2$. It seems possible that $$ I_{-2}(r,f')=O\big(I_{-2}(r,k')\big)=O\big((1-r)^{-1/2}\big) $$ might be true, where $k$ is the Koebe function. This equation ([source label: J6.109]) is slightly stronger than an earlier conjecture of Brennan (see Problem 6.94). The best known bounds are due to Pommerenke . Related information appears in . (A. Baernstein)

背景链接：https://arxiv.org/abs/1809.07200

11189 | AMR-022-6110

## Research Problems in Function Theory - Problem 6.110

P / partially_solved | AMR 问题清单汇编

Let $\Omega$ be a simply-connected domain in the finite plane whose complement contains $n$ disjoint closed balls with centres on the interval $[0,1]$ and common radius $\varepsilon$, $\varepsilon<1/n$, (such domains are the so-called `ball and chain domains'). Let $z_0$ in $\Omega$ be a point at a distance at least $1$ from each ball, and let $w(z_0,\Omega)$ denote the harmonic measure at $z_0$ of the union of the balls, relative to $\Omega$. Is it true that, for every positive $\delta$, there is an estimate $$ w(z_0,\Omega)\leq C_\delta(n\varepsilon)^{\frac{1}{2}-\delta}, $$ where $C_\delta$ depends only on $\delta$. An affirmative answer would imply the $L^p$ extension, for $1 < p < 2$, of the Hayman-Wu theorem mentioned in Problem 6.93. In the case $n = 1$, the Beurling-Nevanlinna projection theorem shows that ([source label: J1.110]) is true with exponent $\frac{1}{2}$; however, for large $n$ there are examples that show that this is then false. (A. Baernstein)

背景链接：https://arxiv.org/abs/1809.07200

11190 | AMR-022-6111

## Research Problems in Function Theory - Problem 6.111

O / open | AMR 问题清单汇编

Let $A$ denote the class of functions $f(z) = z + a_2z^2 +\ldots$ analytic in $\mathbb{D}$. For $\delta\geq0$ and $T= \{T_k\}^\infty_2$ a sequence of non-negative real numbers, define a $T$-$\delta$-neighbourhood of $f\in A$ by \[TN_\delta(f)=\Big\{g:g(z)=z+b_2z^2+\ldots\in A,\sum^\infty_{k=2}T_k|a_k-b_k|\leq\delta\Big\}.\] When $T = \{k\}^\infty_2$, we call $TN_\delta(f) = N_\delta(f)$ a $\delta$-neighbourhood of $f$ ($\delta$-neighbourhoods were introduced by Ruscheweyh , who used them to generalise the result that $N_1(z)\subset St$, the class of normalised star-like functions in $\mathbb{D}$.) Now let $K[A,B]$ denote the class of univalent functions \[\{f:f\in S,1+zf''(z)/f'(z)\prec(1+Az)/(1+Bz),z\in \mathbb{D}\},\] where $-1\leq B<A\leq1$, introduced by Janowski . See also Pommerenke for discussion on $\prec$ and subordination. The proposers have shown that if $f\in K[A,B]$, and either \[(a)\hspace{1cm}-\frac{1}{4}(\sqrt{3}+2)\leq B<A\leq1\hspace{1cm}\text{ or }\hspace{1cm} (b)\hspace{1cm} -1\leq-A\leq B<A\leq1,\] then $N_\delta(f)\subset St$, where \[\delta= \begin{cases} (1-B)^{(A-B)/B},&\hspace{1cm} B\neq0, \\ e^{-A},&\hspace{1cm} B=0, \end{cases}\] and that this value of $\delta$ is best possible. Is this conclusion still valid without the hypotheses $(a)$ and $(b)$? (T. Sheil-Small and E. M. Silvia)

背景链接：https://arxiv.org/abs/1809.07200

11191 | AMR-022-6112

### Research Problems in Function Theory - Problem 6.112

O / open | AMR 问题清单汇编

If $f$ in $A$ and $\delta > 0$, define a $\Sigma_\delta(f)$ neighbourhood of $f$ to be \[\Big\{g:g\in A,\big|(g'(z)-f'(z))-\frac{1}{z}(g(z)-f(z))\big|+\big|(g'(z)-f'(z))+\frac{1}{z}|g(z)-f(z))\big|<2\delta\Big\}.\] Clearly, $N_\delta(f)\subset\sum_\delta(f)$. Then it is known by Sheil-Small and Silvia that $\sum_1(z)\subset St$, and \mbox{$\sum_\frac{1}{4}(f)\subset St$} for convex functions $f$, with the notation in Problem 6.111. Given a normal family $\mathcal{F}$ in $A$, the dual, $\mathcal{F}^*$, of $\mathcal{F}$ is the set \[\big\{f:f\in A,f\ast g\neq0\text{ for all }g\in\mathcal{F},0<|z|<1\big\},\] where $\ast$ denotes the Hadamard product, see Ruscheweyh . The Bieberbach conjecture (see de Branges ) is equivalent to the statement that $N_1(z)\subset S^*$ (where $S^*$ is the dual of $S$). It seems reasonable therefore to ask whether $\sum_1(z)\subset S^*$. (Recall that $S^*\subset St$, see Ruscheweyh .) (T. Sheil-Small and E. M. Silvia)

背景链接：https://arxiv.org/abs/1809.07200

11192 | AMR-022-6113

### Research Problems in Function Theory - Problem 6.113

O / open | AMR 问题清单汇编

Following the notation of Problem 6.111 and 6.112, it is known that, if $|x|\leq\rho\leq1$ and $\gamma=1/(1+\rho)^2$, then \[N_\gamma\Big(\frac{z}{1-xz}\Big)\subset S^*.\] Is it true that \[\Sigma_\gamma\Big(\frac{z}{1-xz}\Big)\subset S^*?\] Notice that, since not all convex functions belong to $S^*$ (see Sheil-Small and Silvia ), it follows that we cannot replace in the above, $N_\gamma(z/(1-xz))$ by $N_\gamma(g)$, where $g$ is an arbitrary convex function. (T. Sheil-Small and E. M. Silvia)

背景链接：https://arxiv.org/abs/1809.07200

11193 | AMR-022-6114

### Research Problems in Function Theory - Problem 6.114

S / solved | AMR 问题清单汇编

Let $\Gamma$ be a regular curve and $f$ an analytic and conformal function in the open unit disc. Does $f^{-1}(\Gamma)$ necessarily have finite length? (A curve $\Gamma$ is said to be regular if the intersection of $\Gamma$ with a disc of radius $r$ has one-dimensional measure at most $Cr$, where $C$ is a constant independent of $r$.) (J. L. Fernandez and D. Hamilton)

背景链接：https://arxiv.org/abs/1809.07200

11194 | AMR-022-6115

### Research Problems in Function Theory - Problem 6.115

S / solved | AMR 问题清单汇编

Let $\Gamma$ be a rectifiable curve, and let $E$ be a subset of $\Gamma$ having zero length. If $\Omega$ is any simply-connected domain and $z\in\Omega$, is it true that the harmonic measure satisfies the equation $\omega(z,E,\Omega)=0$? \textit{(B. \O{}ksendal; R. Kaufman and J.-M. G. Wu; communicated by D. Hamilton)}

背景链接：https://arxiv.org/abs/1809.07200

11195 | AMR-022-6116

### Research Problems in Function Theory - Problem 6.116

O / open | AMR 问题清单汇编

Let $D$ be a domain in $\mathbb{C}$ containing the origin $0$; for $t>0$, let $\Omega_t$ be the component of $D \cap\{|z|\leq t\}$ containing $0$. In the usual notation for harmonic measure, define the function $\omega = \omega_D:[0, \infty)\to[0,1]$ by the formula \[\omega(t)=\omega\big(0,\partial\Omega_t\cap\{|z|=t\},\Omega_t\big).\] What can be said about $\omega$ and about its relations with $D$? For instance: [(a)] ; What are necessary and sufficient conditions on a function to be $\omega_D$ for some $D$? ; If $\omega_{D_1 }\equiv \omega_{D_2}$, are $D_1$ and $D_2$ essentially the same? (That is, will $D_1$ be a rotation or reflection of $D_2$? Or will $D_1$ and $D_2$ differ only on sets of capacity zero?) What happens if $\omega_{D_1 }\equiv \omega_{D_2}$ only on some subinterval of $[0,1]$? ; Given a function $\omega_D$, can one `reconstruct' $D$? ; Can one infer such properties as connectivity of $D$ fromthe behaviour of $\omega_D$? One might start by dealing with domains $D$ that are circularly symmetric. (K. Stephenson)

背景链接：https://arxiv.org/abs/1809.07200

11196 | AMR-022-6117

## Research Problems in Function Theory - Problem 6.117

O / open | AMR 问题清单汇编

Let $G$ be a domain in $\mathbb{C}$ that contains the origin $0$ and is axially-symmetric with respect to the real axis, that is, if a point $z\in G$ then the line segment with endpoints $z$ and $\bar{z}$ also lies in $G$. (In particular it follows that $G$ is simply-connected.) For $t>0$, let $G_t$ and $G_t'$ be the components of the domains \[\{z:z\in G,\text{Im }z< t\}\hspace{1cm}\text{ and }\hspace{1cm}\{z:z\in G,\text{Re }z< t\}\] containing $0$. Let $\omega_t$ and $\omega_t'$ denote the harmonic measure at $0$ of the sets \[\partial G_t\cap\{\text{Im }z=t\}\hspace{1cm}\text{ and }\hspace{1cm}\partial G_t'\cap\{\text{Re }z=t\}\] with respect to the domains $G_t$ and $G_t'$ respectively. Now let $D_1$ and $D_2$ be two domains in $\mathbb{C}$ of this type, and use for each the notation just described. Then: [(a)] ; If $\omega_t(D_1)\equiv \omega_t(D_2)$ for each $t>0$, is it true that $D_1 = D_2$? ; If $\omega_t'(D_1)\equiv \omega_t'(D_2)$ for each $t>0$, is it true that $D_1 = D_2$? ; If both $(a)$ and $(b)$ fail, is it true that $D_1 = D_2$ if $\omega_t(D_1)\equiv \omega_t(D_2)$ for each $t>0$ and $\omega_t'(D_1)\equiv \omega_t'(D_2)$ for each $t>0$? If the answer to $(a)$ or $(b)$ is `yes', in that case, how few $t$'s does one need for the conclusion to hold? (For example, infinitely many $t$'s such that $\{\text{Im }z = t\}$ or $\{\text{Re }z = t\}$ meet the domains? Or do the $t$'s need to be dense?) If the answer to $(a)$, $(b)$ or $(c)$ is `yes', in that case can one replace `axially-symmetric' by `simply-connected', or perhaps drop this requirement completely? (D. A. Brannan)

背景链接：https://arxiv.org/abs/1809.07200

11197 | AMR-022-6012

## Research Problems in Function Theory - Problem 6.12

O / open | AMR 问题清单汇编

If $f(z)=z+\sum^\infty_{k=2}a_{n_k}z^{n_k}\in S$, and \[\liminf_{k\to\infty}\frac{n_{k+1}}{n_k}>1,\] then Pommerenke has proved that $$ a_n=o\Big(\frac{1}{n}\Big) $$ and this is sharp. If \[\liminf_{k\to\infty}(n_{k+1}-n_k)>4,\] then Hayman has shown that $$ a_n=o(n^{-\frac{1}{2}}). $$ Are there intermediate gap conditions which allow us to interpolate between ([source label: 6.3]) and ([source label: 6.4])?

背景链接：https://arxiv.org/abs/1809.07200

11198 | AMR-022-6013

## Research Problems in Function Theory - Problem 6.13

O / open | AMR 问题清单汇编

Suppose that $f(z)$ in $S$, and that positive integers $k, m, n,$ are given. It is known that there exist complex numbers $c_0, c_1,\ldots,c_m,$ depending on $k, m, n$ and $f$, such that $c_0=1$, $|c_m|\geq4^{-m}$ and $$ |c_0a_{n+j}+c_1a_{n+j+1}+\ldots+c_ma_{n+j+m}|\leq Kn^{\alpha_m},\hspace{1cm}0\leq j\leq k, $$ where $K$ is a constant depending on $k, m$ and $f$, and $\alpha_m=8/\sqrt{m-\frac{1}{2}}$. What are the best possible values for the $\alpha_m$? See Pommerenke . It is known that $\alpha_1=0$, and it may be conjectured that $\alpha_2=-\frac{1}{3}$, $\alpha_3=-\frac{1}{2}+\varepsilon$, and $\alpha_m<-\frac{1}{2}$ for $m>4$. If, in addition, $f(z)$ is star-like, then Pommerenke has shown that ([source label: 6.5]) holds with $\alpha_m=-1+\frac{2}{m+1}$. The result is sharp.

背景链接：https://arxiv.org/abs/1809.07200

11199 | AMR-022-6813

## Research Problems in Function Theory - Problem 6.13′

O / open | AMR 问题清单汇编

The results of Pommerenke were proved in fact for mean $p$-valent functions, and if $f(z)=\sum^\infty_{n=0}a_nz^n$ is mean $p$-valent with $p>\frac{1}{4}$, then ([source label: 6.5]) holds with $\alpha_m=-\frac{1}{2}+8p^\frac{3}{2}/\sqrt{m}$. This result is probably far from the best possible, though clearly $\alpha_m\geq-\frac{1}{2}$ in all cases. Lucas has shown that $\alpha_1=2p-2$ if $p\geq1$, and $\alpha_1\leq 2p-2\sqrt{p}$ for $\frac{1}{4}<p<1$. It is fair to conjecture that the correct value of $\alpha_1$ is $p-1$ for $\frac{1}{2}<p<1$ and $-\frac{1}{2}$ if $p<\frac{1}{2}$. It might also be conjectured that $\alpha_m=-\frac{1}{2}$ for $m+1>4p$. The functions \[f(z)=\Big(\frac{1+z^{m+1}}{1-z^{m+1}}\Big)^{2p/(m+1)}=1+\sum^\infty_{n=2}a_nz^n\] provide counter-examples. These functions are mean $p$-valent, and $a_n=0$ except when $(m+1)$ divides $n$, and for the remaining values of $n$ we have, as $n\to\infty$, \[|a_n|\sim Cn^{2p/(m+1)-1},\hspace{1cm}\text{ where }C\text{ is a constant}.\] Thus ([source label: 6.5]) cannot hold with $\alpha_m<2p/(m+1)-1$.

背景链接：https://arxiv.org/abs/1809.07200

11200 | AMR-022-6014

### Research Problems in Function Theory - Problem 6.14

O / open | AMR 问题清单汇编

If $f(z)$ in $S$, set \[A^{(k)}_n= \begin{vmatrix} a_n,&a_{n+1},&\ldots,&a_{n+k-1} \hdotsfor{4} a_{n+k-1},&a_{n+k},&\ldots,&a_{n+2k-2} \end{vmatrix}\] Using ([source label: 6.5]), Pommerenke proved that if $k$ is fixed $$ |A^{(k)}_n|^{1/k}=O(n^{j_k}),\hspace{1cm}\text{ as }n\to\infty, $$ with $j_k=-\frac{1}{2}+16/\sqrt{k}$. Here the conjecture is that $j_k<-\frac{1}{2}$ for large $k$. It can be shown that for $k=2$, ([source label: 6.6]) is false with $j_2=0$, and holds with $j_2=\frac{1}{4}$, see Hayman . If $f(z)$ is in addition star-like, then Pommerenke showed that the best possible values of $j_k$ are $j_k=-1+2/k$.

背景链接：https://arxiv.org/abs/1809.07200

11201 | AMR-022-6814

### Research Problems in Function Theory - Problem 6.14′

O / open | AMR 问题清单汇编

Here again the main conclusions extend to mean $p$-valent functions. In this case ([source label: 6.6]) holds with $j_k=-\frac{1}{2}+16(p^3/k)^{\frac{1}{2}}$. This is still unlikely to be best possible.

背景链接：https://arxiv.org/abs/1809.07200

11202 | AMR-022-6015

### Research Problems in Function Theory - Problem 6.15

P / partially_solved | AMR 问题清单汇编

If $f(z)$ in $S$, write \[f_\alpha(z)=\int^z_0f'(\zeta)^\alpha\, d\zeta.\] For what values of $\alpha$, is it true that $f_\alpha(z)\in S$? The result is known to hold for $\alpha\leq(\sqrt{5}-2)/3$ (see Duren, Shapiro and Shields ), but not for $\alpha>\frac{1}{3}$, see Royster . If for $z$ in $\mathbb{D}$, $f(z)$ is analytic and $|f''(z)/f'(z)|\leq c/(1-|z|^2)$, then $f(z)$ is univalent if $$ c\leq2(\sqrt{5}-2), $$ see Duren, Shapiro and Shields ; but not if $c>2$, see Hille . What is the best value of $c$? (P. L. Duren)

背景链接：https://arxiv.org/abs/1809.07200

11203 | AMR-022-6017

### Research Problems in Function Theory - Problem 6.17

O / open | AMR 问题清单汇编

If $f(z)=z+\sum^\infty_{n=2}a_nz^n$ in $S$, then \[ A=\pi\sum^\infty_{n=1}n|a_n|^2\] is the area of the image domain. What is the minimum value of $A$ when $a_2$ is given? Clearly $A\geq\pi(1+2|a_2|^2)$ always, but this bound is not sharp if $|a_2|>\frac{1}{2}$, since in this case $f(z)=z+a_2z^2$ is not univalent in $\mathbb{D}$. (H. S. Shapiro)

背景链接：https://arxiv.org/abs/1809.07200

11204 | AMR-022-6018

### Research Problems in Function Theory - Problem 6.18

O / open | AMR 问题清单汇编

If $F(z)=z+\sum^\infty_{n=1}b_nz^{-n}$ in $\Sigma$, then \[A(F)=\pi-\pi\sum^\infty_{n=1}n|b_n|^2\] is the area of the set of values not assumed by $F(z)$ in $\mathbb{D}$. If $F_n(z)\in\Sigma$, and \[F_n(z)\to F(z),\hspace{1cm}\text{ as }n\to\infty,\] for $|z|>1$, under what additional hypotheses is it true that $$ A(F_n)\to A(F),\hspace{1cm}\text{ as }n\to\infty\,? $$ It is suggested that ([source label: 6.7]) might be true under some hypotheses on $(1-|z|^2)^2\{F(z),z\}$ where \[\{F(z),z\}=\Big(\frac{F''}{F'}\Big)^\prime-\frac{1}{2}\Big(\frac{F''}{F'}\Big)^2\] is the Schwarzian derivative of $F(z)$. (L. Bers)

背景链接：https://arxiv.org/abs/1809.07200

11205 | AMR-022-6019

### Research Problems in Function Theory - Problem 6.19

O / open | AMR 问题清单汇编

If $f(z)=\sum^\infty_{n=1}a_nz^n$ is analytic in $\mathbb{D}$ and $\sum^\infty_{n=1}|a_n|<+\infty$, can $f(z)$ map the unit circle $\mathbb{T}$ onto a curve of positive two-dimensional measure if [(a)] ; $f(z)$ in $S$, ; more generally, $f'(z)\neq0$ in $\mathbb{D}$?

背景链接：https://arxiv.org/abs/1809.07200

11206 | AMR-022-6002

### Research Problems in Function Theory - Problem 6.2

S / solved | AMR 问题清单汇编

Define $A_n=\sup_{f\in S}|a_n|$. It is shown by Hayman that \[\frac{A_n}{n}\to K_0,\hspace{1cm}\text{ as }n\to\infty.\] Is it true that $K_0=1$? The best known result so far is $K_0<1.243$ due to Milin .

背景链接：https://arxiv.org/abs/1809.07200

11207 | AMR-022-6020

### Research Problems in Function Theory - Problem 6.20

O / open | AMR 问题清单汇编

Let $C$ be a closed curve inside the unit circle $\mathbb{T}$. Under what conditions on $C$ does there exist a univalent function $f$ in $\mathbb{D}$ such that $f(C)$ and $f(\mathbb{T})$ are both convex?

背景链接：https://arxiv.org/abs/1809.07200

11208 | AMR-022-6021

### Research Problems in Function Theory - Problem 6.21

O / open | AMR 问题清单汇编

A function $f(z)$ analytic in $\mathbb{D}$ is said to be typically real if $f(z)$ is real, when and only when $z$ is real, see Rogosinski . If $f(z)=z+\sum^\infty_{n=2}a_nz^n$ is typically real in $\mathbb{D}$, then \[f(z)=\frac{z}{1-z^2}P(z),\] where $P(0)=0$, $\text{Re}\, P(z)>0$ in $\mathbb{D}$. What other conditions must $P(z)$ satisfy to make $f(z)$ univalent in $\mathbb{D}$?

背景链接：https://arxiv.org/abs/1809.07200

11209 | AMR-022-6022

### Research Problems in Function Theory - Problem 6.22

S / solved | AMR 问题清单汇编

If $f(z)=z+\sum^\infty_{n=2}a_nz^n$ is univalent and star-like of order $\frac{1}{2}$ in $\mathbb{D}$, i.e. \[\text{Re}\,\frac{zf'(z)}{f(z)}\geq\frac{1}{2},\] find the radius of the largest disc $|z|<r$ in which $f(z)$ is convex. In other words, when is \[\min_{|z|=r}\text{Re}\,\Big\{\frac{P+1}{2}+z\frac{P'(z)}{P+1}\Big\}>0,\] where $\text{Re}\, P(z)>0$ in $\mathbb{D}$ and $P(0)=1$?

背景链接：https://arxiv.org/abs/1809.07200

11210 | AMR-022-6023

### Research Problems in Function Theory - Problem 6.23

O / open | AMR 问题清单汇编

A related problem concerns upper bounds for $|a_{n+1}|-|a_n|$ when $f(z)$ is mean $p$-valent. Lucas has proved that \[\big||a_{n+1}|-|a_n|\big|=O(n^{j_d}),\] where $j_d=2p-2$, if $p\geq1$; $j_d\leq2p-2\sqrt{p}$ if $\frac{1}{4}<p<1$; and $j_d=-\frac{1}{2}$, if $p<\frac{1}{4}$. The result for $\frac{1}{4}<p<1$ is probably not sharp. A similar question may be asked for symmetric mean $p$-valent functions of the type \[f(z)=\sum^\infty_{n=0}a_nz^{an+b}.\] The coefficients of such functions behave rather like those of functions $\sum a_nz^n$ which are mean $(p/a)$-valent. In particular, if $f(z)=z+\sum a_nz^{2n+1}$ is mean univalent, then Lucas proved that \[\big||a_{n+1}|-|a_n|\big|=O(n^{1-\sqrt{2}}).\]

背景链接：https://arxiv.org/abs/1809.07200

11211 | AMR-022-6024

### Research Problems in Function Theory - Problem 6.24

O / open | AMR 问题清单汇编

If $f(z)=z+\sum^\infty_{n=2}a_nz^n\in S(1)$, prove that on $|z|=r$, \[|f(z)|\leq\frac{r}{(1-r)^2}.\] It is shown by Garabedian and Royden that $f(z)$ assumes in $\mathbb{D}$ each value $w$ such that $|w|<\frac{1}{4}$ and hence that \[|f(z)|\geq\frac{r}{(1+r)^2},\hspace{1cm}|z|=r.\] The question of sharp bounds for $|f'(z)|$ and $|f'(z)|/|f(z)|$ is also open. The corresponding results for $S$ are elementary, see e.g. Hayman .

背景链接：https://arxiv.org/abs/1809.07200

11212 | AMR-022-6025

### Research Problems in Function Theory - Problem 6.25

O / open | AMR 问题清单汇编

Suppose that $p$ is an integer and $f(z)=\sum^\infty_{n=0}a_nz^n$ is $p$-valent in $\mathbb{D}$. It is conjectured by Goodman that \[|a_n|\leq\sum^p_{k=1}|a_k|D(p,k,n),\] where \[D(p,k,n)=\frac{2kn\prod^p_{\alpha=1}(n^2-\alpha^2)}{(p+k)!(p-k)!(n^2-k^2)},\hspace{1cm}1\leq k\leq p<n.\] This result, containing the Bieberbach conjecture as a special case, is likely to be extremely difficult. The inequality if true would be sharp in all cases. No counter-examples are known and the conjecture has been proved only if $a_k=0$ for $k=1, 2, \ldots, (p-1)$ and $n=p+1$, see Spencer ; or $n=p+2$, see Jenkins . The conjecture is true and sharp for certain classes of $p$-valent functions, namely those which are typically real of order $p$, see Goodman and Robertson . We recall that the conjecture is definitely false for areally mean $p$-valent functions, if $p=1$ and $n=3$ by the example of Spencer , though for circumferentially mean $p$-valent functions it remains true in this case, see Jenkins .

背景链接：https://arxiv.org/abs/1809.07200

11213 | AMR-022-6026

## Research Problems in Function Theory - Problem 6.26

P / partially_solved | AMR 问题清单汇编

Suppose that $f(z)=\sum^\infty_{n=0}a_nz^n$ is circumferentially mean $p$-valent and $f(z)\neq0$ in $\mathbb{D}$. (This latter condition is a consequence of mean $p$-valency if $p<1$). It is conjectured that in this case, at least if $p\geq 1$, we have $$ |a_n|\leq A_{n,p} $$ where $$ F(z)=a_0\Big(\frac{1+z}{1-z}\Big)^{2p}=\sum^\infty_{n=0}A_{n,p}z^{2n}. $$ The conjecture ([source label: 6.10]) is known to be true for $n=1$ and all $p$; and for $n=2, 3$ if $p=1$, by the results quoted in Problem 6.25. It is certainly false for small positive $p$, and large $n$, since it would imply $a_n=O(n^{\varepsilon-1})$ for every positive $\varepsilon$ as $n\to\infty$, for a bounded univalent function. For if $g(z)$ is bounded and univalent, and $\varepsilon>0$, then if $K$ is a sufficiently large positive constant $g(z)+K$ is circumferentially mean $\varepsilon$-valent. On the other hand, if $p>\frac{1}{4}$ and $f(z)$ is fixed, ([source label: 6.10]) holds for all sufficiently large $n$, see Hayman . In the special case when $p=1$ and $f(z)$ is univalent, ([source label: 6.10]) reduces to the Littlewood conjecture $|a_n|\leq4|a_0|n$. This conjecture is somewhat weaker than the Bieberbach conjecture. It was shown by Nehari (see also Bombieri ) that we have at any rate $|a_n|\leq 4K_0|a_0|n$ if $f(z)$ is univalent, where $K_0$ is the constant of Problem 6.2. Thus, Littlewood's conjecture holds since $K_0=1$, as was pointed out in Update 6.2. It also seems likely that for $\lambda p>1$ \[I_\lambda(r,f)\leq I_\lambda(r,F)\] if $f(z)$ satisfies the above hypotheses, and $F(z)$ is given by ([source label: 6.11]).

背景链接：https://arxiv.org/abs/1809.07200

11214 | AMR-022-6027

## Research Problems in Function Theory - Problem 6.27

O / open | AMR 问题清单汇编

Suppose that \[g(z) = z + b_0 + b_1z^{-1} + \ldots\] is univalent in $|z|>1$. Is it true that for each positive $\varepsilon$ we have \[n|b_n|=O(n^\varepsilon)\big\{\max_{0<|\nu-n|<\frac{n}{2}}(\nu(|b_\nu|+1)\big\},\hspace{1cm}\text{ as }n\to\infty\,?\] This is suggested by some results of Clunie and Pommerenke. (J. Clunie, Ch. Pommerenke)

背景链接：https://arxiv.org/abs/1809.07200

11215 | AMR-022-6028

## Research Problems in Function Theory - Problem 6.28

S / solved | AMR 问题清单汇编

Suppose that $f(z) = z+\sum^\infty_{n=2}a_nz^n$ in $S$ and that \mbox{$P(z) = \sum^n_{k=0}b_kz^k$} is a polynomial of degree at most $n$. Is it true that \[\max_{|z|=1} |P(z)\ast f(z)| \leq n \max_{|z|=1} |P(z )|?\] Here $P \ast f = \sum^n_{k=0}a_kb_kz^k$. The above result would imply Rogosinski's generalised Bieberbach conjecture but is weaker than Robertson's conjecture, see Sheil-Small . (T. Sheil-Small)

背景链接：https://arxiv.org/abs/1809.07200

11216 | AMR-022-6029

## Research Problems in Function Theory - Problem 6.29

O / open | AMR 问题清单汇编

With the above notation $f (z)$ in $S$ if and only if for each pair of numbers $\xi_1, \xi_2$ satisfying $|\xi_1|\leq1$, $|\xi_2|\leq1$, we have \[f(z)\ast\frac{z}{(1-\xi_1z)(1-\xi_2z)}\neq0,\hspace{1cm}0<|z|<1.\] On the other hand it is true that if $F(z) \ast f(z) \neq 0$, $0 <|z| < 1$ whenever $f$ in $S$, then $F(z)$ is star-like. What is the complete class of star-like functions having this property? $F(z) = z + z^n/n$ has the property since the Bieberbach conjecture holds. (T. Sheil-Small)

背景链接：https://arxiv.org/abs/1809.07200

11217 | AMR-022-6802

## Research Problems in Function Theory - Problem 6.2′

O / open | AMR 问题清单汇编

If $A^{(p)}_n=\sup_{f\in S_p}|a_n|$ is it true that \[\frac{A^{(p)}_n}{n^{2p-1}}\to K_p,\hspace{1cm}\text{ as }n\to\infty,\] and if so, what is $K_p$? It is known that for a fixed $f$ in $S(p)$, the limit \[\alpha_f=\lim_{n\to\infty}\frac{|a_n|}{n^{2p-1}}\] exists if $p>\frac{1}{4}$, see Hayman and Eke .

背景链接：https://arxiv.org/abs/1809.07200

11218 | AMR-022-6003

## Research Problems in Function Theory - Problem 6.3

O / open | AMR 问题清单汇编

If $f(z)$ in $S$ it is shown by Bombieri , that there exist constants $c_n$ such that for $f(z)$ in $S$ \[|\text{Re}\,(n-a_n)|\leq c_n\text{Re}\,(2-a_2).\] What is it the exact size of the constants $c_n$? Is it true that there exists $d_n$ such that \[\big|n-|a_n|\big|\leq d_n\big(2-|a_2|\big)\,?\] (E. Bombieri)

背景链接：https://arxiv.org/abs/1809.07200

11219 | AMR-022-6030

### Research Problems in Function Theory - Problem 6.30

P / partially_solved | AMR 问题清单汇编

If $f$ in $S$, Baernstein has shown that \[\int^{2\pi}_0|f(re^{i\theta})|^p\,d\theta\leq\int^{2\pi}_0|k(re^{i\theta})|^p\,d\theta,\hspace{1cm}0<r<1,\hspace{1cm}0<p<\infty,\] where $k(z)$ is the Koebe function. Does the corresponding inequality hold for integral means of the derivatives at least for certain values of $p$? The best we can hope for is that it holds for $p\geq\frac{1}{3}$ because $k'(z)\in H^p$ for $p<\frac{1}{3}$, and there exist functions $f$ in $S$ for which $f'(z)$ does not belong to any $H^p$. (An example is due to Lohwater, Piranian and Rudin ). For close-to-convex functions it was proved by MacGregor that the result holds for $p\geq1$, and in fact the corresponding inequality holds for derivatives of all orders. (A. Baernstein)

背景链接：https://arxiv.org/abs/1809.07200

11220 | AMR-022-6031

### Research Problems in Function Theory - Problem 6.31

O / open | AMR 问题清单汇编

Duren has shown that if $f(z) = \sum^\infty_{n=0}a_nz^n$ in $S$ and if \[(1 - r )^2f( r ) = \lambda + O\big(( 1 - r )^\delta\big),\hspace{1cm}\text{ as }r\to1-,\] for some $\lambda$, $\delta$, where $\lambda\neq0$, $\delta>0$, then \[\frac{a_n}{n}=\lambda+O\Big(\frac{1}{\log n}\Big),\hspace{1cm}\text{ as }n\to\infty.\] To what extent can this estimate be improved? (P. L. Duren)

背景链接：https://arxiv.org/abs/1809.07200

11221 | AMR-022-6032

### Research Problems in Function Theory - Problem 6.32

O / open | AMR 问题清单汇编

Let $S_\alpha$, $0 < \alpha \leq 1$ be the subclass of $S$ of functions $f$ such that $\mathbb{C}\setminus f(\mathbb{D})$ is a single piecewise analytic slit from some finite point $\omega_0$ to $\infty$ that makes an angle at most $\alpha \pi/2$ with the radii vectors. What can be said about the Taylor coefficients of functions in $S_\alpha$? If $f(-1) = \infty$ and $f(e^{i\phi_f}) = \omega_0$, $-\pi< \phi < \pi$, find $\sup_{f\in S_\alpha} |\phi_f|$. (K. W. Lucas)

背景链接：https://arxiv.org/abs/1809.07200

11222 | AMR-022-6033

### Research Problems in Function Theory - Problem 6.33

O / open | AMR 问题清单汇编

The same questions as in Problem 6.32 can be asked under the alternative hypothesis that $\mathbb{C}\setminus\{f(\mathbb{D})\}$ is a single piecewise analytic slit from some finite point ($\omega_0$, say) to $\infty$ lying in an infinite sector with opening $\alpha\pi$ $(0<\alpha\leq2)$ and vertex $\omega_0$. (K. W. Lucas and D. A. Brannan)

背景链接：https://arxiv.org/abs/1809.07200

11223 | AMR-022-6034

### Research Problems in Function Theory - Problem 6.34

O / open | AMR 问题清单汇编

A function $f(z) = z + a_2z^2 +\ldots$ analytic in $\mathbb{D}$ is said to belong to Ruscheweyh's class $M$ if the $*$ (i.e. Hadamard) convolution of $f$ with every (normalised) convex function is univalent. All close-to-convex functions lie in $M$. Suppose that $g(z) = z + b_2z^2 +\ldots$ is analytic in $\mathbb{D}$ and satisfies the condition $$ \text{Re}\,\Big\{\frac{\phi*(gF)}{\phi*g}\Big\}>0\hspace{1cm}|z|<1, $$ for all normalised convex functions $\phi$, and all normalised functions $F$ of positive real part in $\mathbb{D}$. If $g$ is star-like, then ([source label: C6.1]) certainly holds; is ([source label: C6.1]) true for any other $g$, or (maybe) for some significant larger family of $g$? If $g$ satisfies ([source label: 6.2]) and the condition \[\text{Re}\, \Big(\frac{zf'}{g}\Big) > 0,\hspace{1cm} |z| < 1,\] then $f\in M$; does this classify $M$? (T. Sheil-Small)

背景链接：https://arxiv.org/abs/1809.07200

11224 | AMR-022-6035

### Research Problems in Function Theory - Problem 6.35

O / open | AMR 问题清单汇编

Let $\mathbb{O}$ be a subset of $\mathbb{D}=\{|\omega|< 1\}$. Find a characterisation of those $\mathbb{O}$ that are of the form $(\mathbb{C}\setminus f(\mathbb{D}))\cap\mathbb{D}$ for some $f$ in $S$. What is the maximum area of $\mathbb{O}$ ? Given that $\omega_1, \omega_2\in\{|\omega|<1\}$, how does one tell whether there exists such an $f$ with $\omega_1, \omega_2 \in\mathbb{O}_f$? The same question could be asked for $\omega_1, \omega_2, \omega_3$ etc. (A. W. Goodman)

背景链接：https://arxiv.org/abs/1809.07200

11225 | AMR-022-6037

### Research Problems in Function Theory - Problem 6.37

O / open | AMR 问题清单汇编

Suppose that $f(z)=z + c_3z^3 + c_5z^5 +\ldots$ is an odd univalent function in $\mathbb{D}$, and let $d_n = |c_{2n+1}|-|c_{2n-1}|$. It is known that $d_n\to0$, the best known estimate being $d_n =O(n^{1-\sqrt{2}})$; can this be improved to $$ d_n=O(n^{-\frac{1}{2}})\,? $$ (Nothing better is possible, as is shown by the fourth-root transform of the Koebe-function). Milin (see ; ; and ) proves that $d_n\leq K(\alpha)n^{-\frac{1}{2}}$ for functions $f$ such that $g(z) = [f(z^{\frac{1}{2}})]^2 = z + a_2z^2 +\ldots$ is univalent in $\mathbb{D}$ and has positive Hayman number $\alpha = \lim_{n\to\infty} n^{-1}|a_n|$; but $K(\alpha)\to\infty$ as $\alpha\to0$. Levin showed that $d_n=O(n^{-\frac{1}{2}}\log n)$ if $c_n$ vanishes for $n\not\equiv1\pmod 4$. (P. L. Duren)

背景链接：https://arxiv.org/abs/1809.07200

11226 | AMR-022-6039

### Research Problems in Function Theory - Problem 6.39

S / solved | AMR 问题清单汇编

Suppose $f$ in $S$ and define $h(z) = \{f(z^2)\}^{\frac{1}{2}} = z + c_3z^3 + c_5z^5 +\ldots$ Robertson's conjecture (see Sheil-Small ) asserts that \[1+|c_3|^2+|c_5|^2+\ldots+|c_{2n-1}|^2\leq n.\] This is known to be true if $f$ is star-like; is it true if $f$ has real coefficients, or if $f$ is close-to-convex? (P. L. Duren)

背景链接：https://arxiv.org/abs/1809.07200

11227 | AMR-022-6040

### Research Problems in Function Theory - Problem 6.40

O / open | AMR 问题清单汇编

If $f(z)$ in $S$ and if the $a_n$ are real, then $$ 1+a_3+\ldots+a_{2n-1}\geq a_n^2,\hspace{1cm}n\geq1. $$ The Bieberbach conjecture for such functions follows easily, see e.g. Fitzgerald . It was pointed out by Clunie and Robertson that ([source label: C6.2]) holds for normalised typically-real functions; the inequality is clear from the representation formula for these functions. The Bieberbach conjecture for $S$ would follow analogously if we could prove that if $f$ in $S$, $$ 1+|a_3|+\ldots+|a_{2n-1}|\geq|a_n|^2,\hspace{1cm}n\geq1. $$ Bshouty (unpublished) has shown that if $f$ in $S$, then there exists an $N(f)$ such that ([source label: C6.3]) holds for $n > N(f)$. (C. Fitzgerald)

背景链接：https://arxiv.org/abs/1809.07200

11228 | AMR-022-6042

### Research Problems in Function Theory - Problem 6.42

S / solved | AMR 问题清单汇编

If $f$ in $S$, write \[\log[f(z)/z]=2\sum^\infty_{k=1}\gamma_kz^k.\] If $f$ is star-like then $|\gamma_k| \leq1/k$; this is false in general, even in order of magnitude. Milin (see ) has shown that \[\sum^n_{k=1}k|\gamma_k|^2\leq\sum^n_{k=1}\frac{1}{k} +\delta\] where $\delta<0.312$, and that $\delta$ cannot be reduced to $0$; Milin conjectured that $$ \sum^N_{n=1}\sum^n_{k=1}k|\gamma_k|^2\leq\sum^N_{n=1}\sum^n_{k=1}\frac{1}{k}, $$ which would imply Robertson's conjecture (see Problem 6.39). Inequality ([source label: C6.4]) is known to be true for $N = 1, 2, 3$ (see Grin\v span ). Is it true in general if $f$ has real coefficients, or if $f$ is close-to-convex? (P. L. Duren)

背景链接：https://arxiv.org/abs/1809.07200

11229 | AMR-022-6043

### Research Problems in Function Theory - Problem 6.43

O / open | AMR 问题清单汇编

Using the notation of Problem 6.42, it is well-known that \[\Big|\sum^\infty_{k=1}k\gamma_kz^k\Big|=O\Big(\frac{1}{1-r}\Big),\hspace{1cm}r\to1-,\] for $|z| = r < 1$; is it true that \[\sum^\infty_{k=1}k|\gamma_k|r^k=O\Big(\frac{1}{1-r}\Big),\hspace{1cm}r\to1-\,?\] (D. Aharonov)

背景链接：https://arxiv.org/abs/1809.07200

11230 | AMR-022-6047

### Research Problems in Function Theory - Problem 6.47

O / open | AMR 问题清单汇编

If $f$ in $S$ and $f'$ is also univalent in $\mathbb{D}$, what can be said about $\max|a_n|$, $n \geq 2$? The function $z(1-z)^{-1}$ shows that $\max|a_n|\geq1$ whenever $n\geq2$. (J. G. Clunie)

背景链接：https://arxiv.org/abs/1809.07200

11231 | AMR-022-6048

## Research Problems in Function Theory - Problem 6.48

O / open | AMR 问题清单汇编

Suppose that $f$ in $S$. The coefficient problem, except in certain cases, remains open for each of the following subclasses of univalent functions. (We limit ourselves to one-parameter families). [(a)] ; $B(\alpha)$, the class of Basilevi\^c functions comprising functions $f$ such that \[f(z)=\Big[\int^z_0 p(t)s^\alpha(t)t^{-1}\,dt\Big]^{1/\alpha},\] where $\alpha> 0$, $p(t) = 1+p_1t+\ldots$ is analytic and of positive real part in $|t|<1$ and $s(t) = t + s_2t^2 +\ldots$ is star-like in $|t| < 1$. ; $M(\alpha)$, the class of Mocanu-Reade functions comprising functions $f$ such that \[\text{Re}\,\Big[\alpha\Big(1+\frac{zf''}{f'}\Big)+(1-\alpha)\frac{zf'}{f}\Big]>0,\] where $0 <\alpha < 1$. ; $S^*(\alpha)$, the class of strongly-star-like functions, comprising functions $f$ such that \[\Big|\arg\Big(\frac{zf'}{f}\Big)\Big|<\frac{\alpha\pi}{2},\] where $0 < \alpha < 1$. (D. A. Brannan)

背景链接：https://arxiv.org/abs/1809.07200

11232 | AMR-022-6049

## Research Problems in Function Theory - Problem 6.49

O / open | AMR 问题清单汇编

What are the extreme points of the following classes of functions? [(a)] ; Basilevi\^c functions (see Problem 6.48). ; $S^*(\alpha)$ (see Problem 6.48). ; Close-to-convex functions of order $\alpha$, $0 < \alpha < 1$. ; Functions of boundary rotation $k\pi$, $2 < k < 4$. (D. A. Brannan)

背景链接：https://arxiv.org/abs/1809.07200

11233 | AMR-022-6005

## Research Problems in Function Theory - Problem 6.5

O / open | AMR 问题清单汇编

If it proves too difficult to obtain sharp bounds for all of the coefficients in Problem 6.4, we ask for the orders of magnitude. An area principle shows that \[\sum^\infty_{n=1}n|b_n|^2\leq1\] and hence \[b_n=o(n^{-\frac{1}{2}}).\] Clunie and Pommerenke have shown that \[|b_n|=O(n^{-\frac{1}{2}-\frac{1}{300}}).\] In the opposite direction, examples due to Clunie show that \[|b_n|>n^{-1+\delta}\] is possible for indefinitely many $n$ and a fixed $F(z)$ in $\Sigma$, where $\delta$ is an absolute constant.

背景链接：https://arxiv.org/abs/1809.07200

11234 | AMR-022-6050

## Research Problems in Function Theory - Problem 6.50

O / open | AMR 问题清单汇编

If $0 \le \alpha \le 1$, and $f(z)$, $g(z)\in \Sigma$, and if we define $F(z)$ by \begin{eqnarray} F(z)&=&f(z)^{1-\alpha}g(z)^\alpha, \hspace{1cm}|z|>1; &=&z+\sum^\infty_{n=0}A_nz^{-n}, \hspace{1cm} |z|\text{ sufficiently large}; \end{eqnarray} is it true that \[\sum^\infty_{n=1}n|A_n|^2\le 1?\] See Thomas . (D. K. Thomas)

背景链接：https://arxiv.org/abs/1809.07200

11235 | AMR-022-6051

## Research Problems in Function Theory - Problem 6.51

O / open | AMR 问题清单汇编

Let $D$ be a domain in $\mathbb{C}$ (containing the origin) of connectivity $n$, and let $S(D)$ be the class of analytic univalent functions in $D$ with $f(0) = 0$, $f'(0) = 1$. Find the functions $f$ in $S(D)$ that minimise \[\int_D\int|f'(z)|^2\,d\sigma_z.\] (D. Aharonov)

背景链接：https://arxiv.org/abs/1809.07200

11236 | AMR-022-6052

## Research Problems in Function Theory - Problem 6.52

O / open | AMR 问题清单汇编

Suppose that $f(z)$ is analytic in $\mathbb{D}$, and has the whole complex plane as its range. Does there necessarily exist a bounded univalent function $g(z)$ in $\mathbb{D}$ such that $f(z)+g(z)$ has the whole complex plane as its range? (L. A. Rubel)

背景链接：https://arxiv.org/abs/1809.07200

11237 | AMR-022-6054

## Research Problems in Function Theory - Problem 6.54

O / open | AMR 问题清单汇编

Let $D$ be a Jordan domain with boundary $C$, $\{F_n(z)\}^\infty_1$ the sequence of Faber polynomials for $D$, and $S(D)$ the class of univalent functions \[f(z)=F_1(z)+\sum^\infty_{n=2}a_nF_n(z)\] in $D$. Does the coefficient region of at least one $a_n$ have the same shape as $C$, or is it at least in the subclass of star-like functions? Royster has shown that, if $f$ is star-like in an ellipse, then there exists a direction $\theta_f$ and sequences \[\{\lambda_n\}^\infty_2,\hspace{1cm}\{\mu_n\}^\infty_2,\hspace{1cm}\lambda_n>\mu_n>0,\] such that each coefficient $a_n$ lies in an ellipse of centre $0$, major axis $\lambda_n$, minor axis $\mu_n$, inclined at an angle $\theta_f$ to the real axis. (The $\lambda_n, \mu_n$ are known.) (K. W. Lucas)

背景链接：https://arxiv.org/abs/1809.07200

11238 | AMR-022-6055

### Research Problems in Function Theory - Problem 6.55

O / open | AMR 问题清单汇编

Let $f(z)$ be a normalised bounded star-like function in $\mathbb{D}$, and set \[f(\xi)=\lim_{r\to1-}f(r\xi),\] where $|\xi|=1$, $\xi\in E$, $E\subset\{|z|=1\}$. Is it true that $f(E) = \{f(\xi):\xi\in E\}$ has zero linear (= one-dimensional Hausdorff) measure if $\text{cap } E = 0$, or at least if $E$ has zero logarithmic measure? (It is known that $\text{meas} \{\arg f(\xi):\xi\in E\}=0$ if $\text{cap } E=0$.) (Ch. Pommerenke)

背景链接：https://arxiv.org/abs/1809.07200

11239 | AMR-022-6056

### Research Problems in Function Theory - Problem 6.56

O / open | AMR 问题清单汇编

Let $S_R(q)$ be the class of normalised univalent functions in $\mathbb{D}$ with real coefficients that admit a quasi-conformal extension to the whole plane with complex dilatation bounded (in modulus) almost everywhere by $q$, $q < 1$. Following the notation in Problem 6.44, prove (or disprove) that, if $f\in S_R(p)$ and $g\in S_R(q)$, then $f\otimes g\in S_R(r)$. (J. G. Krzyz)

背景链接：https://arxiv.org/abs/1809.07200

11240 | AMR-022-6058

### Research Problems in Function Theory - Problem 6.58

O / open | AMR 问题清单汇编

Following the notation in Problem 6.57, the well-known Golusin inequality for functions $f$ in $\Sigma(q)$ (defined in Problem 6.57) is: $$ \Big|\log\frac{f'(z)f'(\zeta)(z-\zeta)^2}{[f(z)-f(\zeta)]^2}\Big|\leq q\log\frac{|z\overline{\xi}-1|^2}{(|z|^2-1)(|\zeta|^2-1)}. $$ Prove (or disprove) that ([source label: C6.6]) is also sufficient for $f$ to have a quasi-conformal extension to the whole plane, possibly for $q$ sufficiently small. (J. G. Krzyz)

背景链接：https://arxiv.org/abs/1809.07200

11241 | AMR-022-6006

### Research Problems in Function Theory - Problem 6.6

O / open | AMR 问题清单汇编

What are the orders of magnitude of the $c_n$ in Problem 6.4? Springer obtained the estimate \[|c_n|\leq\frac{2^n}{n}\] and also showed that, given $\varepsilon>0$, \[|c_{2n-1}|>(1-\varepsilon)2^{2n-2}e/(\pi n^3)^{\frac{1}{2}}\] is possible for all sufficiently large $n$.

背景链接：https://arxiv.org/abs/1809.07200

11242 | AMR-022-6060

### Research Problems in Function Theory - Problem 6.60

O / open | AMR 问题清单汇编

Let $C$ be a closed Jordan curve. Then if $f(z) = z + a_2z^2 +\ldots$, $g(z) =z^{-1}+b_0 +b_1z + \ldots$ map $\mathbb{D}$ onto the inside and outside of $C$ respectively, the area principle shows that $C$ is the unit circumference $\mathbb{T}$. If we remove the normalisation on $g$ and replace $g$ by $g_1(z) = b_{-1}z^{-1} + b_0 + b_1z + \ldots$, what can be said about the connection between $f$ and $g_1$? For example, what about the asymptotic behaviour of their coefficients, or their Lipschitz behaviour on $\mathbb{T}$, or about prime ends (if we make $C$ slightly non-Jordan)? (D. A. Brannan)

背景链接：https://arxiv.org/abs/1809.07200

11243 | AMR-022-6061

### Research Problems in Function Theory - Problem 6.61

O / open | AMR 问题清单汇编

Let $D_1, D_2$ be Jordan domains bounded by rectifiable curves $C_1, C_2$ of equal length. Suppose that an isometric sewing of $C_1$ and $C_2$ is everywhere conformally admissible, thus generating a Riemann surface equivalent to a sphere $S$. Is the curve $C$ on $S$ which corresponds to $C_1$ (and $C_2$) necessarily rectifiable? (A. Huber)

背景链接：https://arxiv.org/abs/1809.07200

11244 | AMR-022-6062

### Research Problems in Function Theory - Problem 6.62

O / open | AMR 问题清单汇编

Let $D_1$ and $D_2$ be bounded Jordan domains, bounded by curves $C_1$ and $C_2$ of bounded boundary rotation (in the sense of Paatero, see e.g. Noonan ); then $C_1$ and $C_2$ are rectifiable, and we shall assume that they have the same length. It is known that in this case, every isometric sewing is conformally admissible and generates a Riemann surface which is equivalent to a sphere $S$. It follows from results of Aleksandrov (no citation) and Reshetnjak (no citation) that the curve $C$ on $S$ which corresponds to $C_1$ (and $C_2$) is of bounded boundary rotation. Can one find a function-theoretic proof of this? (A. Huber)

背景链接：https://arxiv.org/abs/1809.07200

11245 | AMR-022-6063

## Research Problems in Function Theory - Problem 6.63

O / open | AMR 问题清单汇编

Let $\alpha$ be a homeomorphic mapping of $(0, \infty)$ onto $(\alpha(0), \infty)$, $\alpha(0)\geq 0$, such that $x\to\alpha(x)+i$ defines a conformal sewing of the half-strip \[H = \big\{z:\text{Re}\, z > 0, 0 \leq \text{Im}\,(z) \leq 1\big\}.\] If $\alpha$ is hyperbolic (that is, if the generated Riemann surface has a hyperbolic end at $\infty$), does there exist a positive continuous function $\varepsilon: (0,\infty)\to\mathbb{R}$ with the property that each conformal sewing $x\to\beta(x)+i$ of $H$ satisfying the inequality \[|\beta(x)-\alpha(x)|<\varepsilon(x) ,\hspace{1cm}0<x<\infty,\] is also hyperbolic? (C. Constantinescu; communicated by A. Huber)

背景链接：https://arxiv.org/abs/1809.07200

11246 | AMR-022-6064

## Research Problems in Function Theory - Problem 6.64

O / open | AMR 问题清单汇编

Let $\alpha$ be real and suppose that $f(z)=z+\sum^\infty_{n=2}a_nz^n$ is analytic in $\mathbb{D}$ with $f(z)f'(z)/z\neq0$. We say $f$ is in $M_\alpha$, the class of alpha-convex functions, if \[\text{Re}\,\Big[(1-\alpha)\frac{zf'}{f}+\alpha\Big(1+\frac{zf''}{f'}\Big)\Big]>0\] for $|z|<1$. Note that $M_0=S^*$, the class of star-like functions and $M_1=K$, the class of convex functions. It is known that $M_\alpha\subset S^*$ for all $\alpha$. Clunie and Keogh have shown the following: [(a)] ; If $\sum^\infty_{n=2}n|a_n|<1$ then $f\in M_0$ ; If $\sum^\infty_{n=2}n^2|a_n|<1$ then $f\in M_1$. What generalisation of these conditions implies that $f$ belongs to $M_\alpha$? (S. Miller)

背景链接：https://arxiv.org/abs/1809.07200

11247 | AMR-022-6065

## Research Problems in Function Theory - Problem 6.65

O / open | AMR 问题清单汇编

Given $M$, $1<M<\infty$, let $S^*(M)$ be the class of star-like univalent functions $f$ in $\mathbb{D}$ with $f(0)=0$, $f'(0)=1$, and $|f(z)|\leq M$ for $|z|<1$. If $f(z)=z+\sum^\infty_{n=2}a_nz^n$, find $\sup\{|a_3|:f\in S^*(M)\}$ for $e<M<5$. Barnard and Lewis have found the supremum for other values of $M$. (J. Lewis)

背景链接：https://arxiv.org/abs/1809.07200

11248 | AMR-022-6068

## Research Problems in Function Theory - Problem 6.68

O / open | AMR 问题清单汇编

Let $\Sigma$ be the class of univalent functions in $\{|z|>1\}$ with the usual normalisation $f(z)=z+\sum^\infty_{n=0}b_nz^{-n}$. Let $S_f$ denote the Schwarzian derivative of $f$. Does \[\sup_{|z|>1}(|z|^2-1)^2\big|S_{f_n}(z)-S_f(z)\big|\to0\hspace{1cm}\text{ as }n\to\infty\] imply that \[\sup_{|z|>1}(|z|^2-1)\Big|\frac{f_n''(z)}{f_n'(z)}-\frac{f''(z)}{f'(z)}\Big|\to0\hspace{1cm}\text{ as }n\to\infty,\] if $f, f_n\in\Sigma$ $(n=1, 2, \ldots)$? This is known to be true if $f$, $f_n$ have quasiconformal extensions onto the plane. (J. Becker)

背景链接：https://arxiv.org/abs/1809.07200

11249 | AMR-022-6007

## Research Problems in Function Theory - Problem 6.7

O / open | AMR 问题清单汇编

If $f(z)$ in $S$ and is bounded, i.e. satisfies $|f(z)|<M$ for $z\in\mathbb{D}$, we again ask for the order of magnitude of the coefficients $a_n$. Since the area of the image of $\mathbb{D}$ by $f(z)$ is at most $\pi M^2$, we deduce that $\sum n|a_n|^2\leq M^2$, so that again \[|a_n|=o(n^{-\frac{1}{2}}),\hspace{1cm}\text{ as }n\to\infty.\] Here also Clunie and Pommerenke have shown that \[|a_n|=O(n^{-\frac{1}{2}-\frac{1}{300}}).\] Examples in the opposite direction due to Littlewood , show again that for a sufficiently small positive $\delta$, we can have \[|a_n|>n^{-1+\delta}\] for infinitely many $n$ and a fixed $f(z)$. The problems for this class of functions seem very analogous to the corresponding problems for $\Sigma$, see Problem 6.5.

背景链接：https://arxiv.org/abs/1809.07200

11250 | AMR-022-6071

## Research Problems in Function Theory - Problem 6.71

O / open | AMR 问题清单汇编

For each $f$ in $S$, it can be shown that \[\int^{2\pi}_0\Big|\frac{f'(Re^{i\theta})}{f(Re^{i\theta})}\Big|^2d\theta=O\Big(\frac{1}{1-R}\log\frac{1}{1-R}\Big)\] as $R\to1$. Hayman has constructed an example showing that `$O$' cannot be replaced by `$o$'. For certain subclasses of $S$, such as the star-like functions and the functions with positive Hayman index (including all support points of $S$), the estimate can be improved to $O(1/(1-R))$. Can the same improvement be made for the extreme points of $S$?; for close-to-convex functions? (P. L. Duren)

背景链接：https://arxiv.org/abs/1809.07200

11251 | AMR-022-6072

### Research Problems in Function Theory - Problem 6.72

O / open | AMR 问题清单汇编

Let $\Gamma$ be the analytic arc omitted by a support point of $S$. Must $\Gamma$ have monotonic argument? Must the angle between the radius and tangent vectors be monotonic on $\Gamma$? (The second property implies the first. Brown has shown that the support points associated with point-evaluation functions have both properties.) (P. L. Duren)

背景链接：https://arxiv.org/abs/1809.07200

11252 | AMR-022-6074

### Research Problems in Function Theory - Problem 6.74

O / open | AMR 问题清单汇编

Suppose $f(z)=z+\sum^\infty_{n=2}a_nz^n$ is univalent and bounded by $M$ in $\mathbb{D}$. Find \[\sup_t \max_{0\leq t\leq 2\pi}|s_n(e^{it})|,\] where \[s_n(z)=z+\sum^n_{k=2}a_kz^k.\] (F. Holland)

背景链接：https://arxiv.org/abs/1809.07200

11253 | AMR-022-6075

### Research Problems in Function Theory - Problem 6.75

O / open | AMR 问题清单汇编

Let $\mathcal{P}_n$ be the class of polynomials \[P_n(z)=z+a_2z^2+\ldots+a_nz^n\] univalent in $\mathbb{D}$, and let \[A_m(n)=\max_{\mathcal{P}_n}|a_m|.\] If $n$ is fixed, is it true that as $m$ increases, the quantity $A_m(n)$ increases strictly up to some index $n_0$, and then decreases strictly? Again, for fixed (but arbitrary) $n$ determine the least $n_1$ such that \[A_m(n)\leq1,\hspace{1cm}n_1\leq m\leq n.\] (D. A. Brannan)

背景链接：https://arxiv.org/abs/1809.07200

11254 | AMR-022-6076

### Research Problems in Function Theory - Problem 6.76

O / open | AMR 问题清单汇编

Let $\mathcal{V}_n$ denote the class of polynomials \[P_n(z)=z+a_2z^2+\ldots+a_nz^n\] analytic and bi-univalent in $\mathbb{D}$ (that is, $P_n$ and $P^{-1}_n$ are both univalent in the unit disc). Determine $\max_{\mathcal{V}_n}|a_2|$ and $\max_{\mathcal{V}_n}|a_n|$. (D. A. Brannan)

背景链接：https://arxiv.org/abs/1809.07200

11255 | AMR-022-6077

### Research Problems in Function Theory - Problem 6.77

O / open | AMR 问题清单汇编

Let $\mathcal{P}_n$ be the class of polynomials \[p_n(z)=z+a_2z^2+\ldots+a_nz^n\] univalent in $\mathbb{D}$. Determine \[\max_{p\in\mathcal{P}_n}\int^{2\pi}_0|p(e^{it})|^q\,dt,\hspace{1cm}0<q<\infty.\] (F. Holland)

背景链接：https://arxiv.org/abs/1809.07200

11256 | AMR-022-6078

### Research Problems in Function Theory - Problem 6.78

S / solved | AMR 问题清单汇编

Suppose that $f$ in $S$. Consider the region $\mathbb{D}(f)$ on the Riemann sphere which is the stereographic projection of the image of the unit disc under $f$. We can associate with each $f$ in $S$ the spherical area of $\mathbb{D}(f)$. What is \[\min_{f\in S}\{\text{area of }\mathbb{D}(f)\},\] and what is the extremal function? (Y. Avci)

背景链接：https://arxiv.org/abs/1809.07200

11257 | AMR-022-6807

### Research Problems in Function Theory - Problem 6.7′

O / open | AMR 问题清单汇编

Here our counter-example shows that $|a_n|=o(n^{-\frac{1}{2}})$ is best possible for bounded $f(z)$ in $S(p)$.

背景链接：https://arxiv.org/abs/1809.07200

11258 | AMR-022-6080

### Research Problems in Function Theory - Problem 6.80

O / open | AMR 问题清单汇编

If $f$ is univalent analytic in $\mathbb{D}$, then it is well known (see Pommerenke ) that both $f$ and its first derivative $f'$ must be normal, while the higher derivatives $f^{(n)}$ need not be normal if $n\geq2$. Setting $f^{(-1)}(z)=\int^z_0f(t)\,dt$ and $f^{(-n-1)}(z)=\int^z_0f^{(-n)}(t)\,dt$, it is easy to verify that $f^{(-2)}$ is Bloch; while if $n\geq3$, $f^{(-n)}$ is bounded (hence Bloch, and therefore normal). Thus, if $f$ is univalent analytic in $\mathbb{D}$, then the functions $f^{(n)}$ must be normal for $n=1, 0, -2, -3,\ldots,$ and need not be normal for $n=2, 3, \ldots$. Must $f^{(-1)}(z)=\int^z_0f(t)\,dt$ be normal if $f(z)$ is univalent analytic in $\mathbb{D}$? (D. Campbell)

背景链接：https://arxiv.org/abs/1809.07200

11259 | AMR-022-6081

### Research Problems in Function Theory - Problem 6.81

O / open | AMR 问题清单汇编

Let $G$ be the set of functions analytic and not univalent in $\mathbb{D}$. Set, for $f\in G$, \[M_f=\sup\{|f'(z)|:|z|<1\},\hspace{1cm}m_f=\inf\{|f'(z)|:|z|<1\}\] and put \[\gamma=\inf\Big\{\frac{M_f}{m_f}:f\in G\Big\}.\] Find $\gamma$. John has proved that $\gamma\geq e^{\pi/2}\approx4.7$ and Yamashita has shown that $\gamma\leq e^\pi\approx 23.1$. (G. M., J. M. and T. M. Rassias)

背景链接：https://arxiv.org/abs/1809.07200

11260 | AMR-022-6082

### Research Problems in Function Theory - Problem 6.82

O / open | AMR 问题清单汇编

The above definition of a bi-univalent function is difficult to understand. What is also meant is that the inverse function $f^{-1}$ has an analytic continuation to the unit disc where it is univalent. Let $\sigma$ denote the class of bi-univalent functions, namely the class of functions $f(z)=z+a_2z^2+\ldots$ analytic and univalent in $\mathbb{D}$, such that their inverses $f^{-1}$ are also analytic univalent in the $\mathbb{D}$. Brannan conjectures that \[a^*_2\equiv\sup_\sigma|a_2|=\sqrt{2}.\] It is known that $a^*_2<1.51$ (see Lewin ) and that $a^*_2>\frac{4}{3}+0.02$ (see Styer and Wright ). Prove or disprove the statement that, if $f\in\sigma$, then \[d_n\equiv\big||a_{n+1}|-|a_n|\big|\leq A\] for some absolute constant $A$ with $0<A<1$. Alternatively, what is $\sup_\sigma d_n$ for $n\geq2$? (D. A. Brannan)

背景链接：https://arxiv.org/abs/1809.07200

11261 | AMR-022-6083

### Research Problems in Function Theory - Problem 6.83

O / open | AMR 问题清单汇编

Let $S$ be the usual class of normalised univalent functions in the unit disc $\mathbb{D}$. Characterise those sequences $\{z_n\}$ of points in $\mathbb{D}$ for which $f(z_n) = g(z_n)$ for two different functions $f, g$ in $S$. Notice that a necessary condition is that $\sum^\infty_{n=1}(1-|z_n|) < \infty$, because $(f-g)\in H^p$ for all $p <\frac{1}{2}$. (P. L. Duren)

背景链接：https://arxiv.org/abs/1809.07200

11262 | AMR-022-6084

### Research Problems in Function Theory - Problem 6.84

O / open | AMR 问题清单汇编

If $f(z)$ in $S$, write \[\log\frac{f(z)}{z}=2\sum^\infty_{n=1}\gamma_nz^n\] and \[f(z^p)^{1/p}=z+\sum^\infty_{n=1}c^{(p)}_nz^{pn+1}\hspace{1cm}(p=1,2,\ldots).\] Szeg\"o's conjecture asserts that $c^{(p)}_n=O(n^{2/p-1})$ as $n\to\infty$. This has been proved for $p = 1,2$ and $3$, but Pommerenke has shown that it is false for $p\geq 12$; his example also has $\gamma_n\neq O(1/n)$. Milin has shown that, if a function $f$ in $S$ has the property that $\gamma_n=O(1/n)$, then $c^{(p)}_n=O(n^{2/p-1})$ for every $p$. Is the converse true? (P. L. Duren)

背景链接：https://arxiv.org/abs/1809.07200

11263 | AMR-022-6086

### Research Problems in Function Theory - Problem 6.86

O / open | AMR 问题清单汇编

Sundberg notes that it is well known fact (see Hayman ) that, for each fixed $z_0$ in $\mathbb{D}$, \[\Big|z_0\frac{f''(z_0)}{f'(z_0)}-\frac{2\rho^2}{1-\rho^2}\Big|\leq\frac{4\rho}{1-\rho^2},\] and asks if this can be improved, if we have that $f$ is real on the real axis, or equivalently, if $f$ has real coefficients? (C. Sundberg; communicated by P. L. Duren)

背景链接：https://arxiv.org/abs/1809.07200

11264 | AMR-022-6087

### Research Problems in Function Theory - Problem 6.87

O / open | AMR 问题清单汇编

Let $L_1$, $L_2$ be two complex-valued continuous linear functionals on $H(\mathbb{D})$, the space of all analytic functions on the unit disc $\mathbb{D}$, that are not constant on the set $S$ of all normalised univalent functions on $\mathbb{D}$. Assume, in addition, that $L_1 \neq tL_2$ for any positive $t$. If a function $f$ in $S$ maximises both $\text{Re}\, \{L_1\}$ and $\text{Re}\, \{L_2\}$ in $S$, must $f$ be a rotation of the Koebe function? (P. L. Duren)

背景链接：https://arxiv.org/abs/1809.07200

11265 | AMR-022-6088

## Research Problems in Function Theory - Problem 6.88

O / open | AMR 问题清单汇编

Let the function $f = z + a_2z^2 + \ldots$ in $S$ map $\mathbb{D}$ onto a domain with finite area $A$. Then Bieberbach's inequality $|a_2|\leq2$ can be sharpened to the following: $$ |a_2|\leq2-cA^{-1/2} $$ where $c$ is an absolute constant. What is the best value of $c$? Aharonov and Shapiro have shown that ([source label: J6.88]) holds for some $c$, and have a conjecture concerning the sharp constant $c$ and the extremal function for ([source label: J6.88]). They also conjecture that $|a_2|\leq 2-c_1l^{-1}$, where $l$ is the length of $\partial f(\mathbb{D})$. For some background information see Aharonov and Shapiro ( and ) and Abarov, Shapiro and Solynin ( and ). (H. S. Shapiro)

背景链接：https://arxiv.org/abs/1809.07200

11266 | AMR-022-6089

## Research Problems in Function Theory - Problem 6.89

O / open | AMR 问题清单汇编

Let $S^*(\frac{1}{2})$ denote the class of functions $g$ analytic in $\mathbb{D}$ and such that $\text{Re}\, (zg'/g) > \frac{1}{2}$ in $\mathbb{D}$. Is it true that, if $f\in S^*(\frac{1}{2})$, $r\in (0,1)$ and $\theta\in[0,2\pi)$, then $$ \frac{1}{|f(re^{-i\theta})|}\int^r_0|f'(te^{i\theta})|\,dt\leq\frac{\arcsin r}{r}? $$ (The left-hand-side of ([source label: J6.89]) is the ratio of the length of the image of a radius and the distance between the endpoints of that image (as the crow flies).) Inequality ([source label: J6.89]) is true when $f(z) = z/(1-z)$; in addition, the left-hand-side of ([source label: J6.89]) never exceeds $\pi/2$ for any $r$ or $f$. Let $z_1,z_2$ lie in $\mathbb{D}$. Then, in the smaller class of convex functions $f$, what can be said about \[\frac{1}{|f(z_1)-f(z_2)|}\int^{z_2}_{z_1}|f'(t)|\,|dt|?\] (R. R. Hall)

背景链接：https://arxiv.org/abs/1809.07200

11267 | AMR-022-6808

## Research Problems in Function Theory - Problem 6.8′

O / open | AMR 问题清单汇编

If we ask the analogous problems to those of Problem 6.8 for the class $S(p)$, the correct orders of magnitude are again known in many cases, but not the exact bounds. If $f$ in $S(p)$ and $f$ is bounded, then \[I_1(r,f')=o(1-r)^{-\frac{1}{2}}\] and this is sharp for the class, by our introductory example.

背景链接：https://arxiv.org/abs/1809.07200

11268 | AMR-022-6009

## Research Problems in Function Theory - Problem 6.9

S / solved | AMR 问题清单汇编

(Schoenberg's conjecture) If $f(z)=\sum^\infty_{n=1}a_nz^n$ and $g(z)=\sum^\infty_{n=1}b_nz^n$ are convex, and $f$, $g$ belong to $S$, is it true that \[f\ast g=\sum^\infty_{n=1}a_nb_nz^n\] is also a convex function in $S$? See P\'olya and Schoenberg .

背景链接：https://arxiv.org/abs/1809.07200

11269 | AMR-022-6090

## Research Problems in Function Theory - Problem 6.90

O / open | AMR 问题清单汇编

Let $E$ be a set of positive logarithmic capacity on the unit circle $\mathbb{T}$. Is $E$ necessarily a set of uniqueness for functions univalent in the unit disc $\mathbb{D}$? Carleson has shown that this is false for functions $f$ analytic in $\mathbb{D}$ for which ${\int\int}_{|z|<1}|f'(z)|^2\,dx\,dy<\infty$. Beurling has shown that univalent functions cannot have constant boundary values on a set of positive capacity. (D. H. Hamilton)

背景链接：https://arxiv.org/abs/1809.07200

11270 | AMR-022-6091

## Research Problems in Function Theory - Problem 6.91

O / open | AMR 问题清单汇编

Let $\Omega$ be an arbitrary domain in $\mathbb{C}$. Does there necessarily exist a set $E$ in $\partial\Omega$, of full harmonic measure, with the following property: for each $z$ in $E$, there exist circular arcs $C_r$ in $\Omega$, of radius $r$ (where $r$ is small) and centred at $z$, for which \[\lim_{R\to0}\Big(\frac{1}{\pi R^2}\int^R_0\theta(C_r)r\,dr\Big)=\frac{1}{2},\] where $\theta(\,\cdot\,)$ denotes angular measure? For simply-connected domains $\Omega$, this is a theorem of McMillan . For general domains $\Omega$, at least there is a sequence $\{r_n\}$ decreasing to $0$ for which $\theta(C_{r_n})$ approaches $\pi/4$. (D. Stegenga and K. Stephenson)

背景链接：https://arxiv.org/abs/1809.07200

11271 | AMR-022-6092

### Research Problems in Function Theory - Problem 6.92

O / open | AMR 问题清单汇编

If $\mathbb{R}^2_+=\{(x,y)\in\mathbb{R}^2:y>0\}$, suppose that $E\subset\mathbb{R}^2_+$, and let $f:\mathbb{R}^2_+\to B^2$ be analytic and conformal where $B^2 = \{(x, y)\in\mathbb{R}^2_+: x^2 + y^2 < 1\}$; assume also that \[\lim_{x\to0,\,x\in E}f(x)=\alpha.\] Lindel\"of's theorem shows that $\alpha$ is an angular limit of $f$ at $0$ if $E$ is a curve terminating at $0$. Vuorinen has shown that the same conclusion holds under much weaker hypotheses on $E$; for instance, the condition that \[\liminf_{r\to0}m(A\cap(0,r))/r>0,\] where $A = \{|x|: x \in E\}$, is sufficient (see Vuorinen ). This problem concerns a converse result. Denote by $\mathcal{F}$ the class of all analytic and conformal maps of $\mathbb{R}^2_+$ into $B^2$. Let $K$ be a subset of $\mathbb{R}^2_+$ with the property that, whenever \[\lim_{x\to0,\,x\in E}f(x)=\alpha\hspace{1cm}\text{ and }\hspace{1cm}f\in\mathcal{F},\] then necessarily $f$ has an angular limit $\alpha$ at $0$. Since $f$ is conformal it follows (see Vuorinen ) that \[\lim_{x\to0,\,x\in K_1}f(x)=\alpha,\] where $K_1 = \{(x,y)\in\mathbb{R}^2_+ : \rho((x,y), K) < l\}$ and $\rho$ is the Poincar\'e metric of $\mathbb{R}^2_+$. What can be said about the thickness of $K_1$ at $0$? Is it true that $$ \limsup_{r\to0}m\big(B^\prime\cap(0,r)\big)/r>0, $$ where $B^\prime = \{(x^2 + y^2)^{1/2} : (x,y)\in K_1\}$? It is easy to see that ([source label: J6.92]) must hold if $K$ is contained in an angle with vertex at $0$ whose closure lies inside $\mathbb{R}^2_+$. (M. Vuorinen)

背景链接：https://arxiv.org/abs/1809.07200

11272 | AMR-022-6093

### Research Problems in Function Theory - Problem 6.93

O / open | AMR 问题清单汇编

Let the function $f(z) = z + a_2z^2 + \ldots$ map $\mathbb{D}$ univalently onto a domain $\Omega$, and let $F: \Omega\to\mathbb{D}$ denote the inverse of $f$. Is it true that \[\int_{\Omega\cap \mathbb{R}}|F'(x)|^p\,dx<\infty\hspace{1cm}\text{ for }1\leq p<2?\] Hayman and Wu and Garnett, Gehring and Jones have shown that the answer is `yes' for $p = 1$. For $p = 2$, it is possible that $\int_{\Omega\cap\mathbb{R}}|F'(x)|^2\,dx = \infty$; for example, when $\Omega = \{w: |w| < R, R > 1\}\setminus L$ where, for a suitably chosen $R_1$, $L = \{(u,0): -R\leq u \leq-R_1 < 0\}$. (A. Baernstein II)

背景链接：https://arxiv.org/abs/1809.07200

11273 | AMR-022-6094

### Research Problems in Function Theory - Problem 6.94

O / open | AMR 问题清单汇编

Let $\Omega$ be a simply-connected domain in $\mathbb{C}$ with at least two boundary points, and let the function $\phi$ map $\Omega$ analytically and conformally onto $\mathbb{D}$. For which values of $p$ is it true that $$ {\int\int}_\Omega|\phi'|^p\,dx\,dy<\infty? $$ If $\Omega$ is star-like or close-to-convex, ([source label: J6.94]) holds for $\frac{4}{3} < p < 4$ and this is sharp. More generally, it is known that there is a universal constant $\tau$, independent of $\Omega$, with $0 < \tau < 1$, such that ([source label: J6.94]) holds whenever $\frac{4}{3} < p < 3 + \tau$. Is $\frac{4}{3}< p < 4$ the correct range for general types of $\Omega$? (For background material and additional information, see .) (J. E. Brennan)

背景链接：https://arxiv.org/abs/1809.07200

11274 | AMR-022-6095

### Research Problems in Function Theory - Problem 6.95

O / open | AMR 问题清单汇编

Determine an intrinsic characterisation for the class $\mathcal{H}$ of functions $h$ analytic in $\mathbb{D}$ that admit a decomposition of the form $2h = f + f^{-1}$ for some function $f$ in the class $S$ of normalised univalent functions in $\mathbb{D}$. (Here $f^{-1}$ denotes the function inverse to $f$ and it is assumed that $f^{-1}$ has an analytic continuation to $\mathbb{D}$.) Notice that the functions $z\mapsto z$ and $z\mapsto z/(1-z^2)$ both belong to $\mathcal{H}$. In particular, does the function $z \mapsto z + a \sin (2\pi z)$ belong to $\mathcal{H}$ for any non-zero constant $a$? (F. Holland)

背景链接：https://arxiv.org/abs/1809.07200

11275 | AMR-022-6096

### Research Problems in Function Theory - Problem 6.96

O / open | AMR 问题清单汇编

For $-\infty<p<+\infty$ let \[B(p):=\sup\{\beta_f(p):f\text{ conformal map of }\mathbb{D}\text{ into }\mathbb{D}\}\] where \[\beta_f(p)=\limsup_{r\to1}\Big(\int_{|\zeta|=r}|f'(r\zeta)|^p\,|d\zeta|\Big)/\log\frac{1}{1-r}.\] The BCJK-conjecture states that \[ B(p)= \begin{cases} -p-1 & \text{for } p\leq-2, p^2/4 & \text{for } -2\leq p\leq2, p-1 & \text{for } p\geq2. \end{cases} \] The claim that $B(p)=|p|-1$ is the famous Brennan conjecture. Carleson and Jones proved that $B(p)=p-1+O((p-2)^2)$ as $p\to2$, $p>2$. Based on extensive computer experiments Kraetzer conjectured that $B(p)=p^2/4$ for $|p|\leq2$. See Garnett and Marshall . (Ch. Pommerenke)

背景链接：https://arxiv.org/abs/1809.07200

11276 | AMR-022-6097

### Research Problems in Function Theory - Problem 6.97

O / open | AMR 问题清单汇编

Goodman conjectured that if $f(z) = \sum^\infty_{n=1}a_nz^n$ is $p$-valent in $\mathbb{D}$, then for each $n> p$, we have \[|a_n|\leq\sum^p_{k=1}\frac{2k(n+p)!}{(p+k)!(p-k)!(n-p-1)!(n^2-k^2)}|a_k|.\] If the bound is true, then it is sharp in all of the variables $p$, $n$, $a_1,\ldots, a_p$. The conjecture has been proved for large subclasses (, , ), but is still open in general. The simplest case, namely for all $2$-valent functions, is still open; but Watson has made important contributions towards proving this inequality. (A. W. Goodman)

背景链接：https://arxiv.org/abs/1809.07200

11277 | AMR-022-6098

### Research Problems in Function Theory - Problem 6.98

O / open | AMR 问题清单汇编

The coefficients of a $p$-valent function are bounded by some function of its zeros. In particular, let the function \[f(z)=z^q+\sum^\infty_{n=q+1}a_nz^n\] be $p$-valent in $\mathbb{D}$ and have $s$ zeros $\beta_k$, $k= 1,2, \ldots, s$, where $0 < |\beta_k|<1$. Goodman conjectured that, under these hypotheses, $|a_n|\leq |A_n|$ where $A_n$ is defined by the identity \[F(z)=\frac{z^q}{(1-z)^{2q+2s}}\Big(\frac{1+z}{1-z}\Big)^{2t}\prod^s_{k=1}\Big(1+\frac{z}{|\beta_k|}\Big)\big(1+|\beta_k|z\big)=z+\sum^\infty_{n=2}A_nz^n,\] where $t = p-q-s\geq0$. The conjecture has been proved if $t = 0$ and $f(z)$ is $p$-valent star-like with respect to the origin . However, it is still open in general. (A. W. Goodman)

背景链接：https://arxiv.org/abs/1809.07200

11278 | AMR-022-6099

### Research Problems in Function Theory - Problem 6.99

O / open | AMR 问题清单汇编

A function $f(z) = z + a_2 z^2 +\ldots$ is said to belong to the class $CV(R_1,R_2)$ if it is univalent and convex in $\mathbb{D}$, and if on $f(\{|z|=1\})$ the curvature $\rho$ satisfies the inequalities $R_1\leq\rho\leq R_2$. So far, little progress has been made on the study of this class of functions (see Goodman and ); we do not even know the sharp bound for $|a_2|$ in $CV(R_1,R_2)$. (A. W. Goodman)

背景链接：https://arxiv.org/abs/1809.07200

11279 | AMR-022-7001

### Research Problems in Function Theory - Problem 7.1

O / open | AMR 问题清单汇编

Let $E$ be the compact plane set of transfinite diameter ($=$capacity) $d(E)=1$ and let \[d_n(E)^{n(n-1)/2}=\max_{w_\nu\in E}\prod_{1\leq\mu<\nu\leq n}|w_\mu-w_\nu|.\] It is known that $d_n(E)$ decreases with $n$ and $d_n(E)\to d(E)$ as $n\to\infty$, so that $d_n(E)\geq1$ is trivial. It is known by Pommerenke , that $d_n(E)\geq n^{2/(n-1)}$ if $E$ is connected. Is this inequality true in general? Further, is it true that \[d_n(E)^{(n-1)/2}\leq Kn\] if $E$ is connected, where $K$ is some absolute constant? It is known that \[d_n(E)^(n-1)/2}\leq\Big(\frac{4}{e}\log n +4\Big)n,\] in this case. (Ch. Pommerenke)

背景链接：https://arxiv.org/abs/1809.07200

11280 | AMR-022-7010

### Research Problems in Function Theory - Problem 7.10

S / solved | AMR 问题清单汇编

It was proved by Boyarski\u\i\, that the partial derivatives of a plane $K$-quasiconformal mapping are locally $L$-integrable for $2\leq p < 2+c$, where $c = c(K) > 0$. Show that this is true with $c =\frac{2}{K-1}$. The example $f(z) = |z|^{\frac{1}{K}-1}.z$ shows that such a result would be sharp. (F. W. Gehring)

背景链接：https://arxiv.org/abs/1809.07200

11281 | AMR-022-7013

### Research Problems in Function Theory - Problem 7.13

O / open | AMR 问题清单汇编

One part of Nevanlinna theory is devoted to the following problem. How does the geometric structure of a simply connected Riemann covering surface of the sphere influence the value distribution of the meromorphic function generating the surface? It is suggested that one also consider halfsheets among the constituent pieces of such surfaces, which have an infinite number of branch points on their boundary. A typical example is the covering surface generated by $e^z - z$. It contains a right half-plane with a second order branch point at $2\pi in$ for each integer $n$. (F. Huckeman)

背景链接：https://arxiv.org/abs/1809.07200

11282 | AMR-022-7015

## Research Problems in Function Theory - Problem 7.15

O / open | AMR 问题清单汇编

Let $D$ be a domain in the extended complex plane. A finite point $z$ on the boundary $\partial D$ of $D$ is called angular (relative to $D$) if there exists $\varepsilon > 0$ such that every component domain of $D\cap\{|z - z_0| < \varepsilon\}$ which has $z_0$ as a boundary point, is contained in an angle less than $\pi$ with vertex at $z_0$. Angularity at $\infty$ is similarly defined. Let $A = A(D)$ be the set of angular points of $\partial D$ relative to $D$. Obviously $A$ not empty implies that $\partial D$ has positive capacity. The set $A(D)$ can have positive linear measure. E.g. let $C$ be a Cantor set on $|z|=1$ and let $D$ consist of the open unit disc from which have been deleted all points $rz$ with $z \in C$ and $\frac{1}{2}\leq r < 1$. Then $A(D) = C$ which can, of course, have positive linear measure. Yet the following holds for arbitrary domains: $A(D)$ is either empty or its harmonic measure, relative to any point of $D$, is zero. This result follows easily from an unpublished theorem on Brownian paths $\omega(t)$ in the complex plane. This says that almost all such paths have the property that for every real $t_0$ and every $\varepsilon > 0$ the set of numbers $\Big|\frac{\omega(t)-\omega(t_0)}{\omega(t)-\omega(t_0)}\Big|$ with $t_0<t<t_0+\varepsilon$ fills at least an open arc of length $\pi$ on the unit circle. This theorem is not easy and it would be desirable to give a direct proof of the above result on $A(D)$. Moreover, the Brownian paths approach will certainly not yield a similar result for the set $B_\alpha(D)$ of $\partial D$ whose points are defined by replacing the angles less than $\pi$ with translates of $\{x+iy : 0<x< |y|^\alpha\}$ for a given $\alpha$ with $\frac{1}{2}< \alpha < 1$. (For $\alpha=\frac{1}{2}$ the result is false as can be seen by taking $D$ to be a disc.) Problem: For which $\alpha\in(\frac{1}{2},1)$ is the harmonic measure of the set $B_\alpha(D)$ always zero? (Of course we may assume that the capacity of $\partial D$ is positive.) A much more difficult problem would be to characterise the monotone functions $f(y)$ having the property that the set obtained on replacing the angles by translates of $\{x+iy : 0 < x < f(|y|)\}$ has necessarily harmonic measure zero. Similar questions can be asked for Riemann surfaces and for $n$-dimensional space. (A. Dvoretzky)

背景链接：https://arxiv.org/abs/1809.07200

11283 | AMR-022-7016

## Research Problems in Function Theory - Problem 7.16

P / partially_solved | AMR 问题清单汇编

Let $\gamma$ be a Jordan arc, $d\mu$ a measure on $\gamma$. Does the Laplace transform $$ f(z) = \int_\gamma e^{z\zeta}\,d\mu(\zeta) $$ always have `asymptotically analytic' growth as $|z|\to\infty$. The answer might be `no'. However it could be true anyway that the zeros of $f(z)$ have `measurable distribution' in the sense of A. Pfluger . (J. Korevaar)

背景链接：https://arxiv.org/abs/1809.07200

11284 | AMR-022-7018

## Research Problems in Function Theory - Problem 7.18

O / open | AMR 问题清单汇编

Let $\Gamma$ be a Jordan curve and suppose that $z = 0$ lies inside it. Wermer showed that when $\Gamma$ has infinite length, the powers $z^n$, $n\neq0$ span all of $C(\Gamma)$. One can indicate conditions on $\Gamma$ under which the powers $z^n$, $n\neq n_1,\ldots, n_k$, form a spanning set (see Korevaar and Pfluger, ). Under what conditions on $\Gamma$ can one omit an infinite set of powers, and still have a spanning set? (J. Korevaar)

背景链接：https://arxiv.org/abs/1809.07200

11285 | AMR-022-7019

## Research Problems in Function Theory - Problem 7.19

O / open | AMR 问题清单汇编

For what sets $\Omega$ of lattice points $(m_k, n_k)$ do the monomials $x^{m_k}y^{n_k}$ span $L^2$ or $C_0$ on the unit square $0\leq x\leq1$, $0\leq y\leq1$? It is conjectured that the condition $\sum\frac{1}{m_kn_k}=\infty$ is sufficient for sets $\Omega$ in an angle $\varepsilon x \leq y \leq x/\varepsilon$, $\varepsilon>0$. Hellerstein has shown that the condition is not necessary. (J. Korevaar)

背景链接：https://arxiv.org/abs/1809.07200

11286 | AMR-022-7002

## Research Problems in Function Theory - Problem 7.2

O / open | AMR 问题清单汇编

Let $f(z)$ be analytic in a simply-connected domain $D$. It is known that $f(z)$ can be expanded in a series of Faber polynomials \[f(z)=\sum^\infty_{n=0}a_nP_n(z).\] Find the domain of variability $V$ of $a_n$ as $f(z)$ runs through all functions analytic in $D$, and having positive real part there. It is known that if $D$ is a circle, $V$ is a circle; and if $D$ is an ellipse, then $V$ is an ellipse, see Royster . (W. C. Royster)

背景链接：https://arxiv.org/abs/1809.07200

11287 | AMR-022-7020

## Research Problems in Function Theory - Problem 7.20

P / partially_solved | AMR 问题清单汇编

(Two constant theorems for the polydisc) Let $F(z_1, z_2)$ be defined for $|z_1\leq1, |z_2|\leq2$ except when $z_1 = z_2$ and $|z_1|=|z_2|=1$. Suppose that $F$ is plurisubharmonic in $|z_1|<1, |z_2|<1$ and that $F(z_1,z_2)\leq\log\frac{1}{|z_1-z_2|}$, whenever the left hand side is defined. Further, suppose that $F(z_1, z_2) \leq 0$ for $\{|z_1| = |z_2| = 1, z_1 \neq z_2\}$. Does it follow that $F(z_1,z_2)\leq0$ for $|z_1|<1$, $|z_2|<1$? (L. A. Rubel, A. Shields)

背景链接：https://arxiv.org/abs/1809.07200

11288 | AMR-022-7021

### Research Problems in Function Theory - Problem 7.21

P / partially_solved | AMR 问题清单汇编

Suppose that $|z_k | = 1$ $(1 \leq k < \infty)$. Put \[A_l=\limsup_{m\to\infty}\big|\sum^m_{k=1}z_k^l\big|.\] It is easy to see that there is a sequence $z_k,$ for which $A_l<cl$ for all $l$, and Clunie proved $A_l>cl^{\frac{1}{2}}$ for infinitely many $l$. Is there a sequence for which $A_l = o(l)$ as $l\to\infty$? (P. Erd\"os)

背景链接：https://arxiv.org/abs/1809.07200

11289 | AMR-022-7023

### Research Problems in Function Theory - Problem 7.23

O / open | AMR 问题清单汇编

The expression $u(z, \zeta)$ of Problem 6.57 is closely related to the Schwarzian derivative $\{f(z), z\}$, e.g. it is invariant under compositions with M\"obius transformations and \[\lim_{\zeta\to z}u(z,\zeta)=\frac{1}{6}\{f(z),z\}\] K\"uhnau (no citation) showed that if $f(z)$ is analytic in $\mathbb{D}$ and has a quasiconformal extension to the whole plane with \[|f_{\overline{z}}/f_z|\leq q<1\hspace{1cm}\text{ a.e.}\] then $$ |u(z,\zeta)|\leq q(1-|z|^2)^{-1}(1-|\zeta|^2)^{-1}. $$ Show that ([source label: C7.1]) is also sufficient for $f$ to have a quasi-conformal extension to the whole plane (possibly with $\frac{1}{3}\leq q < 1$). (J. G. Krzyz)

背景链接：https://arxiv.org/abs/1809.07200

11290 | AMR-022-7024

### Research Problems in Function Theory - Problem 7.24

O / open | AMR 问题清单汇编

Let $E = \{|z|< 1\}$, let $0$, $a > 0$, $b = |b|e^{i\beta}(-\pi < \beta\leq\pi)$ be distinct points of $E$; and let $\mathcal{K}$ be the family of continua $K$ with the properties that $\{a, b\} \subset K \subset E\setminus\{0\}$ and $E\setminus K$ is connected. On any $K\in\mathcal{K}$ there is a continuous function $\arg z$; we thus subdivide $\mathcal{K}$ into homotopy classes $\{\mathcal{K}_n\}$ according to the value of \[V(K)=\arg b-\arg a\in\{\beta+2n\pi,n\in\mathbb{Z}\}.\] Let us call $K\in\mathcal{K}$ a natural continuum if $K$ is a trajectory of a quadratic differential $a$ with the following properties: [(a)] ; $0, a, b$ are simple poles of $\sigma$, and there is no other pole of $\sigma$ in $E$; ; $\sigma$ is real on $\partial E$. There are many problems that then arise, for example: [(i)] ; Do all homotopy classes contain natural continua? (There are many in $\mathcal{K}_n$ when $|\beta+2n\pi| < 2\pi$.) ; Find all natural continua in $\mathcal{K}$. ; How does the modulus of $E\setminus K$ vary when $K$ runs through the natural continua in $\mathcal{K}_n$? (F. Huckemann)

背景链接：https://arxiv.org/abs/1809.07200

11291 | AMR-022-7025

### Research Problems in Function Theory - Problem 7.25

O / open | AMR 问题清单汇编

Let $K$ be a compact set of positive measure in $\mathbb{C}$. Does there necessarily exist a non-constant analytic function in $\mathbb{C}\setminus K$ with $f(\infty) = 0$ such that $[f(z)-f(\zeta)]/[z-\zeta]\neq\pm1$ for any $z, \zeta\in\mathbb{C}\setminus K$? Conceivably, the hypotheses even imply the existence of non-linear analytic functions $f$ with $|f(z) -f(\zeta)| / |z - \zeta|$ bounded, but this is a well-known unsolved problem. Of course one can pose more general problems again such as requiring that the difference quotient omit all values in some pre-assigned plane set. (These problems arise in the use of variational methods.) (D. Aharonov and H. S. Shapiro)

背景链接：https://arxiv.org/abs/1809.07200

11292 | AMR-022-7026

### Research Problems in Function Theory - Problem 7.26

O / open | AMR 问题清单汇编

Is there a homeomorphism of the open unit ball in $\mathbb{R}^3$ onto $\mathbb{R}^3$, whose coordinate functions are harmonic? In other words, do there exist $u_1,u_2,u_3$ harmonic in $|x|<1$, $x = (x_1, x_2, x_3)$, such that \[ (x_1, x_2, x_3)\to (u_1, u_2, u_3)\] is a homeomorphism of $|x| < 1$ onto all of $\mathbb{R}^3$? The analogous problem in $\mathbb{R}^2$ is answered negatively; the result is due to T. Rado (no citation), and is an important lemma in the theory of minimal surfaces. (H. S. Shapiro)

背景链接：https://arxiv.org/abs/1809.07200

11293 | AMR-022-7027

### Research Problems in Function Theory - Problem 7.27

O / open | AMR 问题清单汇编

For a domain $D$ in $\mathbb{C}$, define \[\rho(x,y) = \sup\{|f(x)-f(y)| : x,y\in D; f \text{ analytic in }D; |f'|\leq1\text{ in }D\}.\] If $D$ is convex, then $\rho(x,y) = |x-y|$, but not otherwise. Clearly $\rho(x,y)\leq L(x,y)$, the infimum of the lengths of paths in $G$ that join $x$ to $y$. What can be said about $\rho$ for general $D$, in terms of the geometry of $D$? (L. A. Rubel)

背景链接：https://arxiv.org/abs/1809.07200

11294 | AMR-022-7028

### Research Problems in Function Theory - Problem 7.28

O / open | AMR 问题清单汇编

Suppose that $f(z)$ is continuous in a domain $D$ and that either [(i)] ; $\int_{|\zeta-z|=r}f(\zeta)\,d\zeta=0$ for all $z\in D$ and $0<r\leq r(z)$, or (weaker), [(ii)] ; $\lim_{r\to0}\Big[r^{-2}\int_{|\zeta-z|=r}f(\zeta)\,d\zeta\Big]=0$ for all $z\in D.$ Does it follow that $f(z)$ is analytic in $D$? (see e.g. , ). (D. Gaier and L. Zalcman)

背景链接：https://arxiv.org/abs/1809.07200

11295 | AMR-022-7029

### Research Problems in Function Theory - Problem 7.29

P / partially_solved | AMR 问题清单汇编

Let $f(z)$ be continuous on $\mathbb{D}$, and let $\alpha$ be a fixed number with $0 < \alpha \leq 1$. If, for each $z$ in $\{|z|< 1\}$, \[\int_{|\zeta-z|<\alpha(1-|z|)}f(\zeta)\,d\zeta=0,\] is $f$ necessarily analytic in $\mathbb{D}$? What happens if we are given that $f$ is continuous only in $\mathbb{D}$? (L. Zalcman)

背景链接：https://arxiv.org/abs/1809.07200

11296 | AMR-022-7003

### Research Problems in Function Theory - Problem 7.3

O / open | AMR 问题清单汇编

Let $z_i$, $1\leq i\leq n$ be a finite sequence of complex numbers such that $|z_i|\leq1$. Set \[S_k=\sum^n_{i=1}z^k_i.\] Can we have $$ \max_{2\leq k\leq n+1}|S_k|<A^{-n}, $$ where $A$ is an absolute constant greater than one? If we assume $z_1=1$, $|z_i|\leq1$, $2\leq i\leq n$, then ([source label: 7.1]) can be satisfied. See Tur\'an . (P. Erd\"os)

背景链接：https://arxiv.org/abs/1809.07200

11297 | AMR-022-7031

### Research Problems in Function Theory - Problem 7.31

O / open | AMR 问题清单汇编

Suppose that \[a_1>0,\hspace{1cm}0\leq a_n\leq n,\hspace{1cm} n\geq1,\hspace{1cm} b_n=\sum^n_{\nu=1}a_\nu,\hspace{1cm} c_n=\sum^n_{\nu=1}b_\nu.\] Then $\sum(a_n/c_n)^\alpha<\infty$ if $\alpha > \frac{1}{2}$. For what other functions $f(t)$ is it true that $\sum f(c_n/a_n)<\infty$? Is it true, for instance, that (under some smoothness condition on $f$) $\sum f(c_n/a_n)$ converges with $\sum f(n^2)$? The analogous result for $c_n/b_n$ was obtained by Borwein , that if $xf(x)$ is positive and non-increasing for $x\geq a>0$ and $\sum f(n)<\infty$, then $\sum f(c_n/b_n)<\infty$. (W. K. Hayman)

背景链接：https://arxiv.org/abs/1809.07200

11298 | AMR-022-7033

### Research Problems in Function Theory - Problem 7.33

O / open | AMR 问题清单汇编

Let $P(\theta)=\sum^N_{n=1}e^{i\lambda_n\theta}$ be a finite Dirichlet series with exponents $\gamma_m\neq\gamma_n$ for $m\neq n$. What can be said about $\mu\equiv\inf|P(\theta)|$? A trivial argument shows that $\mu\leq (N-1)^{1/2}$. In fact, $|P|^2=N+2\sum_{m\neq n}\cos(\lambda_m-\lambda_n)\theta$. If $w(\theta)=|P|^2-N\geq-c$, then let $h(\theta)=c+w(\theta)=2\sum b_n\cos(\delta_n\theta)$, where the $b_n$ are positive integers. Then \[b_n=\lim_{T\to\infty}\frac{1}{2T}\int^T_{-T}h(t)\cos(\delta_nt)\,dt\leq\lim_{T\to\infty}\frac{1}{2T}\int^T_{-T}h(t)\,dt=c.\] Thus $c\geq1$. The problem arises in prediction theory where $\mu\leq(N-1)^{1/2}$ is adequate. If the $\lambda_n$ are rational, then the problem reduces to a problem on polynomials with coefficients $0$ and $1$. (S. Rudolfer and W. K. Hayman)

背景链接：https://arxiv.org/abs/1809.07200

11299 | AMR-022-7034

### Research Problems in Function Theory - Problem 7.34

O / open | AMR 问题清单汇编

Let $\beta_j\in\mathbb{R}^+$ and $\zeta_j\in\mathbb{C}$, and for suitable small $z$ define $$ f(z)=\prod^n_{j=1}(1-\zeta_jz)^{\beta_j}=1+a_1z+a_2z^2+\ldots $$ Suppose further that the $a_k$ are all real. Then there exists $N=N(\beta_1, \beta_2, \beta_3, \ldots, \beta_n)<\infty$, independent of the $\zeta_j$, such that $\min(a_1, a_2, \ldots, a_N)\leq0$. Find a sharp, or good, upper bound for $N$ as a function of the $\beta$'s. This would be significant for Tur\'an's power sum method. To see that $N<\infty$: if the $\beta_j$'s are all integers, then $f$ is a polynomial and $N=\beta_1+\beta_2+\beta_3+\ldots+\beta_n+1$. (The sort of estimate wanted in the general case.) If not, assume that $|\zeta_1|\leq|\zeta_2|\leq\ldots\leq|\zeta_n|\leq1$, and let $m=\max\{j:\beta_j\in\mathbb{Z}\}$. The radius of convergence of ([source label: 7.341]) is $|\zeta_m|^{-1}$ and if the $a_k$ were all positive (or even greater than or equal to $0$). then by Pringshein's theorem (see Titchmarsh ), $|\zeta_m|^{-1}$ would be a singularity of $f$. Since $f$ only has the singularities $\{\zeta^{-1}_j:\beta_j\notin\mathbb{Z}\}$ we deduce that $\zeta_m>0$. We multiply through by $(1-\zeta_m)^{-\beta_m}$ which has positive coefficients, and repeat the arguments. After at most $n$ steps, we obtain a contradiction. This gives a finite $N$ for any particular set of $\zeta_j$ and the uniformity is straightforward. (R. Hall)

背景链接：https://arxiv.org/abs/1809.07200

11300 | AMR-022-7035

### Research Problems in Function Theory - Problem 7.35

O / open | AMR 问题清单汇编

According to Fefferman's theorem (no citation), a real function $u$ on the unit circle which has bounded mean oscillation, can be decomposed as $u=b_1+\tilde{b_2}$ where $b_1$ and $b_2$ are bounded functions and $\tilde{b_2}$ denotes the conjugate of $b_2$. Given $u$, what is the smallest possible $\|b_2\|_\infty$? This problem is discussed by Baernstein (Section 10). An affirmative answer to would prove the conjecture about factoring non-zero univalent functions that Baernstein made (see Problem 5.58). (A. Baernstein)

背景链接：https://arxiv.org/abs/1809.07200

11301 | AMR-022-7036

### Research Problems in Function Theory - Problem 7.36

S / solved | AMR 问题清单汇编

This problem is equivalent to Problem 7.9 due to Gehring and Reich about best bounds for area distribution under quasiconformal mapping. Let $E$ denote a measurable subset of the unit disc $\mathbb{D}$, $m$ denote $2$-dimensional measure, and define \[f_E(z)=-\frac{1}{\pi}\int\int_E\frac{dm(w)}{(w-z)^2},\hspace{1cm}z\in\mathbb{C}\setminus E.\] Thus, $f_E$ is the $2$-dimensional Hilbert transform of the characteristic function of $E$. It follows from the Calder\'on-Zygmund theory of singular integrals that there are constants $a$ and $b$ such that \[\int\int_{\mathbb{D}\setminus E}|f_E|\,dm\leq am(E)\log\frac{\pi}{m(E)}+bm(E)\] for every $E$. The problem is to find the smallest possible $a$ (for which there exists some $b$ such that the inequality holds for every $E\subset\mathbb{D}$). Consideration of $E=\{z:|z|<\delta\}$ for small $\delta$ shows that $a=1$ would be best possible, and this is conjectured by Gehring and Reich. An analogous sharp inequality for sets $E\subset[-1,1]$ and $1$-dimensional Hilbert transforms \[f_E(x)=-\frac{1}{\pi}\int_E\frac{dt}{x-t}\] is known, and can be proved either by a subordination argument or by use of a theorem of Stein and Weiss. (A. Baernstein)

背景链接：https://arxiv.org/abs/1809.07200

11302 | AMR-022-7037

### Research Problems in Function Theory - Problem 7.37

O / open | AMR 问题清单汇编

Let $T$ denote the class of all rational functions $g$ of the form \[g(z)=\sum^n_{j=1}\frac{\lambda_j}{(z-z_j)^2},\] where the constants $\lambda_j$ satisfy $\lambda_j>0$ and $\sum^n_{j=1}\lambda_j=1$. Prove (or disprove): There is a constant $C$ with the property that for each $g$ in $T$ we can find a set $S=S(g)$ with $m(S)=\pi$ such that \[\int\int_{\Delta(R)\setminus S}|g|\,dx\,dy\leq2\pi\log R+C\] for every $R$ in $(1,\infty)$. Here $\Delta(R)=\{z:|z|<R\}$. This assertion, if true, would imply that the inequality of Problem 7.36 holds with $a=1$, and thus solve the area problem of Gehring and Reich. Problems of this sort have been considered by Fuchs and MacIntyre . (A. Baernstein)

背景链接：https://arxiv.org/abs/1809.07200

11303 | AMR-022-7038

### Research Problems in Function Theory - Problem 7.38

O / open | AMR 问题清单汇编

The Hankel matrices of a function $f$ having a Taylor expansion \[f(z)=a_0+a_1z+\ldots\] are defined by \[H^{(n)}_p=(a_{ij});\hspace{1cm}a_{ij}=a_{n+1+j-2};\hspace{1cm}1\leq i,\, j\leq p+1.\] If $f$ belongs to the Pick-Nevanlinna class ($\text{Det } H^{(n)}_p\geq0$, all $n, p$), then all the poles of $f$ are simple and they lie on the positive real axis. Denote by \[\varepsilon^{(n)}_1\geq\varepsilon^{(n)}_2\geq\ldots\geq\varepsilon^{(n)}_p\geq0\] the eigenvalues of $H^{(n)}_p$. Then \[\limsup_{n\to\infty}(\varepsilon^{(n)}_j)^{1/n}=\frac{1}{\lambda_j},\] where $\lambda_j$ is the $j$-th pole, where the poles are numbered in order of increasing modulus. What can be said about the eigenvalues under less restrictive conditions? (R. Bouteiller)

背景链接：https://arxiv.org/abs/1809.07200

11304 | AMR-022-7039

### Research Problems in Function Theory - Problem 7.39

S / solved | AMR 问题清单汇编

(Subadditivity problem for analytic capacity) Prove or disprove the existence of a constant $M$ such that \[\gamma(K_1\cup K_2)\leq M\{\gamma(K_1)+\gamma(K_2)\}\] for all compact sets $K_1, K_2$ in $\mathbb{C}$. Even the case where $\gamma(K_2)=0$ is open. For background, see Garnett . (J. Korevaar)

背景链接：https://arxiv.org/abs/1809.07200

11305 | AMR-022-7004

### Research Problems in Function Theory - Problem 7.4

O / open | AMR 问题清单汇编

If $z_1=1$, and the $z_i$ are arbitrary complex numbers for $2\leq i\leq n$, then Atkinson proved that \[\max_{1\leq k\leq n}|S_k|>c\] with $c=\frac{1}{3}$. What is the best value for the constant $c$?

背景链接：https://arxiv.org/abs/1809.07200

11306 | AMR-022-7042

### Research Problems in Function Theory - Problem 7.42

O / open | AMR 问题清单汇编

Define the Harnack function $H_{z_0}(z)$ for a Green domain $D$ relative to $z_0\in D$ to be the supremum of all positive harmonic functions $h$ on $D$ to be the supremum of all positive harmonic functions $h$ on $D$ which satisfy $h(z_0)\leq1$. If $K_\zeta(z)$ is the Martin kernel for $D$ relative to $z_0$, $K(\zeta_0)=1$ for all $\zeta$ in the Martin boundary $\Delta_1$, then \[H_{z_0}(z)=\sup\{H_\zeta(z):z\in D, \zeta\in\Delta_1\}.\] In particular for $D$ the unit disc, the boundary $\Delta_1$ may be identified with the unit circle $\mathbb{T}$ and, if $z_0=0$, then $K_\zeta(z)$ is the Poisson kernel with pole at $\zeta\in T$. In this case, $K_\zeta=H_0$ along the radius to $\zeta$. Using the Riemann mapping functions, this property may be described for simply connected $D$ as follows: for each $\zeta\in \Delta_1$, $K_\zeta$ touches $H_{z_0}$ along a Green line for $D$ issuing from $z_0$. Does this property continue to hold for $D$ multiply connected? (M. Arsove and G. Johnson)

背景链接：https://arxiv.org/abs/1809.07200

11307 | AMR-022-7047

### Research Problems in Function Theory - Problem 7.47

O / open | AMR 问题清单汇编

Let $K\subset\mathbb{C}$ be compact and let $x_0\in K$ be a non-peak point for $R(K)$, the uniform limits on $K$ of rational functions with poles outside $K$. Will there always exist a continuous curve $\Gamma$ in $K$ terminating at $x_0$? It is easy to see that if $\sum^\infty_{n=1}2^n M_1(A_n(x_0)\setminus K)<\infty$, where $A_n(x_0)=\{z:2^{-n-1}\leq|z-x_0|\leq2^{-n}\}$, and $M_1$ denotes $1$-dimensional Hausdorff content, then $\Gamma$ can be chosen to be a straight line segment. In this case, \O ksendal showed that the integrated Brownian motion starting at $x_0$ stays inside $K$ for a positive period of time, almost surely. There is an example of a compact set $K$ and a non-peak point $x_0\in K$ such that no straight line segment terminating at $x_0$ is included in $K$. (B. \O ksendal)

背景链接：https://arxiv.org/abs/1809.07200

11308 | AMR-022-7048

### Research Problems in Function Theory - Problem 7.48

O / open | AMR 问题清单汇编

A domain $D\subseteq \mathbb{R}^n$ is said to be linearly accessible, if each point in the complement of $D$ can be joined to $\infty$ by a ray which does not meet $D$. Let $g(\,\cdot\,,x_0)$ be the Green's function for $D$ with pole at $x_0$ in $D$. Is $\{x:g(x,x_0)>t\}$ linearly accessible for $0<t<\infty$. This conclusion is valid in $\mathbb{R}^2$. (J. Lewis)

背景链接：https://arxiv.org/abs/1809.07200

11309 | AMR-022-7049

### Research Problems in Function Theory - Problem 7.49

P / partially_solved | AMR 问题清单汇编

Let $x=(x_1, x_2, \ldots, x_n)$ be a point in Euclidean $n$-space, $n\geq3$, with $|x|=\Big(\sum^n_{i=1}x^2_i\Big)^{1/2}$. A function $u$ on $\mathbb{R}^n$ is said to be homogeneous of degree $m$ if $u(\lambda x)=\lambda^m u(x)$ for all $\lambda>0$. If $u$ is differentiable, put $\triangledown u=(u_{x_1}, u_{x_2},\ldots, u_{x_n})$. Prove there are no homogeneous polynomials $u$ with real coefficients, $m\geq2$, such that $$ \triangledown u\cdot(|\triangledown u|^{p-2}\triangledown u)\equiv0,\hspace{1cm}p\text{ fixed},\hspace{1cm}p\neq2,\hspace{1cm}1<p<\infty. $$ In $\mathbb{R}^2$ there are no polynomial solutions. A proof of the above would imply that if $f$ is any solution to ([source label: 7.49.1]) on a domain $D\subseteq\mathbb{R}^n$, then $f$ is real analytic in $D$ if and only if $\triangledown f$ does not vanish in $D$. (J. Lewis)

背景链接：https://arxiv.org/abs/1809.07200

11310 | AMR-022-7050

### Research Problems in Function Theory - Problem 7.50

O / open | AMR 问题清单汇编

Let $U$ be a connected open set in $\mathbb{R}^n$. Brelot and Choquet showed that the set of points on the boundary of $U$ which are accessible from the interior by (finite length) rectifiable paths supports harmonic measure. It is natural, in view of polygonal path connectedness of finely open sets, to ask if the same is true for finely open sets and the Keldysh measure. (T. Lyons and B. \O ksendal)

背景链接：https://arxiv.org/abs/1809.07200

11311 | AMR-022-7051

## Research Problems in Function Theory - Problem 7.51

O / open | AMR 问题清单汇编

Let $D$ be a bounded strictly pseudo-convex domain in $\mathbb{C}^n, n>1$, with smooth boundary. Denote by $A^\infty(D)$ the class of functions analytic in $D$, continuous on $\overline{D}$, all of whose derivatives are continuous on $\overline{D}$. Let $E$ be a closed subset of the boundary of $D$ which is not a set of uniqueness for $A^\infty(D)$, i.e. there exists a function $f\not\equiv0$, which belongs to $A^\infty(D)$ such that $f$ vanishes exactly on $E$ and all of the derivatives of $f$ vanish on $E$. Is every closed subset of $E$ a set of non-uniqueness for $A^\infty(D)$? This is true in the case of the unit disc in $\mathbb{C}$. (A. -M. Chollet)

背景链接：https://arxiv.org/abs/1809.07200

11312 | AMR-022-7052

## Research Problems in Function Theory - Problem 7.52

O / open | AMR 问题清单汇编

Let $K\subset\mathbb{C}^n$ be a compact set and let $P_0(K)$ be the set of all polynomials on $K$. The $P$-hull of $K$, the polynomial convex hull of $K$, is defined by \[\text{$P$-hull } K=\{z\in\mathbb{C}^n:|p(z)|\leq\sup_k|P(z)|\text{ for all }p\in P_0(K)\}.\] Let $P(K)$ be the uniform closure of $P_0(K)$ in $C(K)$, the continuous function on $K$. Let \v{S}ilov Bd $(P(K))$ denote the \v{S}ilov boundary of the uniform algebra $P(K)$. Determine all compact sets $K\subset\mathbb{C}^n (n>1)$ such that \v{S}ilov Bd $(P(K))=\text{Boundary }(\text{$P$-hull }K)$. For $n=1$, every compact set $K\subset\mathbb{C}$ has this property. For $n>2$, examples of $K$ are compact sets and closed spheres. (S. Kilambi)

背景链接：https://arxiv.org/abs/1809.07200

11313 | AMR-022-7053

## Research Problems in Function Theory - Problem 7.53

O / open | AMR 问题清单汇编

[(i)] ; (One-dimensional version.) Let $E$ be a compact set in $\mathbb{R}$; and, for each $x \in E$, let $\delta_x > 0$ be given. Let $I_x = (x-\delta_x, x + \delta_x)$. For what values of $c$ can one always find a disjoint collection of such intervals, $\{I_{x_j}\}$ say, such that $\sum_j|I_{x_j}|\geq c|E|$? It is known that this is possible for $c = \frac{1}{2}$, but is impossible in general for $c > \frac{2}{3}$. ; ($n$-dimensional version.) Let $E$ be a compact set in $\mathbb{R}^n$; and let $K$ be an open bounded symmetric convex set in $\mathbb{R}^n$. For each $x \in E$, let $\delta_x > 0$ be given; and let $K_x = x + \delta_xK$, the dilation of $K$ by a factor $\delta_x$, centred at $x$. For what values of $c$ can one always find a disjoint collection of such sets, $\{K_x\}$ say, such that $\sum_j|K_{x_j}|\geq c|E|$? If $c(K)$ denotes the best value, it is known that $2^{-n}\leq c(K)<1$. The facts in $(i)$ and $(ii)$, together with some sketchy information on $c(Q_n)$ (where $Q_n$ is the $n$-cube) and $c(S_n)$ (where $S_n$ is the $n$-sphere), are given in , but there is no information on the correct asymptotic behaviour of $c(Q_n)$ or $c(S_n)$. The problem has applications to the best constants in results concerning the Hardy-Littlewood maximal function. (P. L. Walker)

背景链接：https://arxiv.org/abs/1809.07200

11314 | AMR-022-7054

## Research Problems in Function Theory - Problem 7.54

O / open | AMR 问题清单汇编

Let $\phi_t(z)= e^{tz}-1$, $\phi^1_t=\phi_t$ and $\phi^{n+1}_t=\phi_t\circ\phi^n_t$ for $n\geq1$; it follows that \[\phi^1_t(-1)=e^{-t}-1 = -t+\frac{t}{2!}-\ldots,\] \[\phi^2_t(-1)=e^{t(e^{-t}-1)}-1 = -t^2+\ldots\text{ etc.}\] Are the coefficients in these formal power series for $\{\phi^n_t(-1)\}^\infty_{n=1}$ uniformly bounded by $1$ in modulus? (P. J. Rippon)

背景链接：https://arxiv.org/abs/1809.07200

11315 | AMR-022-7055

## Research Problems in Function Theory - Problem 7.55

O / open | AMR 问题清单汇编

It is known that any quasi-conformal homeomorphism of $B^n = \{x : x \in\mathbb{R}^n,|x|<1\}$ onto a Jordan domain $D$ in $\mathbb{R}^n$ can be extended to a homeomorphism of $B^n$ onto $D$. If $\partial D$ is rectifiable (in the sense that $\Lambda^{n-1}(\partial D)<\infty$), is $f|_{\partial B^n}$ absolutely continuous (in the sense that $\Lambda^{n-1}\big(f(E)\big)=0$ for every set $E$ in $\partial B^n$ with $|E|=0$)? One can also ask the analogous question about $f^{-1}$. When $n = 2$, the answer to both questions is `yes' for conformal mappings, but `no' for quasi-conformal mappings. When $n = 3$, Gehring has proved that, if in addition the function $f$ has a quasi-conformal extension to $\mathbb{R}^n$, then $f|_{\partial B^n}$ is absolutely continuous; but, even in this special case, it is not known if $f^{-1}|_{\partial D}$ is absolutely continuous. (A. Baernstein II)

背景链接：https://arxiv.org/abs/1809.07200

11316 | AMR-022-7057

## Research Problems in Function Theory - Problem 7.57

S / solved | AMR 问题清单汇编

For $n\geq2$, let \[\mathbb{R}^n_+=\big\{(x_1,\ldots,x_n)\in\mathbb{R}^n:x_n>0\big\}\] and \[\mathbb{R}^n_-=\big\{(x_1,\ldots,x_n)\in\mathbb{R}^n:x_n<0\big\};\] let $E$, $F$ be non-empty compact subsets of $\mathbb{R}^n_+$, $\mathbb{R}^n_-$ (respectively), and let $F^*$ denote the symmetric image of $F$ in $\partial \mathbb{R}^n_+$. Denote by $\Delta(E,F)$, $\Delta(E,F^*)$ the families of all curves in $\overline{\mathbb{R}^n}$ joining $E$ and $F$, $E$ and $F^*$ (respectively). Is it true that $$ M\big(\Delta(E,F)\big)\leq M\big(\Delta(E,F^*)\big), $$ where $M$ denotes the $n$-modulus of a curve family? It is easy to show that strict inequality holds in ([source label: J7.57]) if $E$ and $F$ are balls. Also, if $\Delta(E,F)$ is obtained from $\Delta(E, F^*)$ as a result of symmetrisation, then ([source label: J7.57]) holds. Note too that it follows from the symmetry principle for the modulus that $\frac{1}{2}M\big(\Delta(E,F)\big)\leq M\big(\Delta(E,F^*)\big)$, at least if $E\cap f^*=\emptyset$. (M. Vuorinen)

背景链接：https://arxiv.org/abs/1809.07200

11317 | AMR-022-7058

## Research Problems in Function Theory - Problem 7.58

O / open | AMR 问题清单汇编

Let $E\subset[0,1]$ be a compact set on the positive real axis in $\mathbb{R}^2$; and let $E$ have positive conformal $2$-capacity, that is $M\big(\Delta(E, \partial B^2(2);\mathbb{R}^2)\big)> 0$ where $\Delta(E,\partial B^2(2);\mathbb{R}^2)$ is the family of all curves joining $E$ to $\partial B^2(2)$ and $M(\Delta)$ is the $2$-modulus of $\Delta$. Is it true that $M\big(\Delta(E, F; \mathbb{R}^2)\big) = \infty$, where $F = \mathbb{R}\setminus E$? (A. A. Gon\^car; communicated by M. Vuorinen)

背景链接：https://arxiv.org/abs/1809.07200

11318 | AMR-022-7061

## Research Problems in Function Theory - Problem 7.61

O / open | AMR 问题清单汇编

Is it true that, for any polynomial $P$ with complex coefficients, the mapping $f\mapsto P^*(D)(Pf)$, where $P^*(z) = \overline{P(\overline{z})}$, is a bijection of $\mathcal{E}$ onto $\mathcal{E}$? The proposer can prove that this is true when $P$ is a homogeneous polynomial; and D. J. Newman told the proposer that he could prove the injectivity half of the conjecture, but the proposer has seen no details of the proof. The conjecture would follow if one could show that the partial differential equation $P^*(D)(Pf) = z^\alpha$ where $P^*(z) = \overline{P(\overline{z})}$, has a solution $f$ that is entire and of exponential type for every multi-index $\alpha$. (H. S. Shapiro)

背景链接：https://arxiv.org/abs/1809.07200

11319 | AMR-022-7062

## Research Problems in Function Theory - Problem 7.62

O / open | AMR 问题清单汇编

Given a countable number of convergent series with positive decreasing terms, one can find such a series converging more slowly than any of these. Without making any assumption about the Continuum Hypothesis, can one associate with every countable ordinal number $\alpha$ a convergent series $\sum^\infty_{n=1}x_{n,\alpha}$ with $0\leq x_{n+1,\alpha}\leq x_{n,\alpha}$ such that [(a)] ; if $\alpha<\beta$, then $x_{n,\alpha}/x_{n,\beta}\to0$ as $n\to\infty$, and ; if $x_n > 0$ and $\sum^\infty_{n=1}x_n<\infty$, then there exists $\alpha$ such that $x_n/x_{n,\alpha}\to0$ as $n\to\infty$. [See also Problem 2.66.] (A. Hinkkanen)

背景链接：https://arxiv.org/abs/1809.07200

11320 | AMR-022-7064

### Research Problems in Function Theory - Problem 7.64

S / solved | AMR 问题清单汇编

Let $\Gamma$ be a Fuchsian group in $\mathbb{D}$, and let $i(z) \equiv z$. Is it true that \[\sum_{\gamma\in\Gamma}|\gamma'(0)|\geq\prod_{\gamma\in\Gamma,\,\gamma\neq i}|\gamma(0)|^2\,?\] Note that this is an equivalent formulation of a problem connecting the Bergman kernel function and the capacity function of a Riemann surface. (N. Suita; communicated by Ch. Pommerenke)

背景链接：https://arxiv.org/abs/1809.07200

11321 | AMR-022-7065

### Research Problems in Function Theory - Problem 7.65

P / partially_solved | AMR 问题清单汇编

Let $\mathcal{W}$ be a hyperbolic Riemann surface, $G_\omega$ the Green's function with pole $\omega\in\mathcal{W}$, and $\Gamma=\{[\gamma_n]\}$ the fundamental group. Let $\tilde{\Gamma}$ be the subgroup of equivalence classes $[\gamma]$ such that, for every $\omega\in\mathcal{W}$, the harmonic conjugate $G^*_\omega$ of $G_\omega$ changes by an integral multiple of $2\pi$ on a representative path in $[\gamma]$. Certainly $[\Gamma,\Gamma]\triangleleft\tilde{\Gamma}\triangleleft\Gamma$. Is it true that, if $[\Gamma, \Gamma] \neq\tilde{\Gamma}$, then $\mathcal{W}$ is necessarily of the form $\mathcal{V}\setminus A$ where $\mathcal{V}$ is a hyperbolic surface and $A\neq\emptyset$ is a relatively closed subset of zero logarithmic capacity? The conjecture is true if $\tilde{\Gamma}=\Gamma$ and $\mathcal{V}$ is the unit disc; and it is false if $[\Gamma,\Gamma] = \tilde{\Gamma}$. (The conjecture arises from work in .) (K. Stephenson)

背景链接：https://arxiv.org/abs/1809.07200

11322 | AMR-022-7066

### Research Problems in Function Theory - Problem 7.66

O / open | AMR 问题清单汇编

A continuous mapping $f:B^n\to\mathbb{R}^n$, where $B^n = \{x : x\in\mathbb{R}^n,|x|<1\}$, and $n\geq2$, is said to be proper if $f^{-1}(K)$ is a compact subset of $B^n$ whenever $K$ is a proper subset of $f(B^n)$. Let \[B_f = \{z : z \in B^n, f\text{ is not a local homeomorphism at }z\}.\] Is it true that, if [(i)] ; $n\geq3$, ; $f:B^n\to\mathbb{R}^n$ is proper and quasi-analytic, and [(iii)] ; $B_f$ is compact then $f$ is necessarily injective? Note that the mapping $f(z) = z^2$, where $z \in B^2$, shows that the conjecture is false when $n = 2$. The conjecture is known to be true in the special case that $f(B^n) = B^n$, $n\geq3$. The conjecture is a special case of a more general open problem in . (M. Vuorinen)

背景链接：https://arxiv.org/abs/1809.07200

11323 | AMR-022-7067

### Research Problems in Function Theory - Problem 7.67

P / partially_solved | AMR 问题清单汇编

Let $V$ be the zero set of some analytic function in a strictly pseudo-convex domain $\Omega$ in $\mathbb{C}^2$. If $V$ has finite area inside $\Omega$, is it necessarily true that $V$ is the zero set of some bounded analytic function on $\Omega$? Bo Berndttson (no citation) has shown that the answer is `yes' when $\Omega$ is the ball; and easy examples show that the answer is `no' for strictly pseudo-convex domains in $\mathbb{C}$ when $n > 2$. Skoda (, ) and independently Henkin have shown (under some cohomology condition on $\Omega$) for all $n$ that $V$ is the zero set of a function of Nevanlinna class only if $V$ satisfies the Blaschke condition. (R. Zeinstra)

背景链接：https://arxiv.org/abs/1809.07200

11324 | AMR-022-7068

### Research Problems in Function Theory - Problem 7.68

O / open | AMR 问题清单汇编

For a Hadamard gap sequence $\{n_k\}^\infty_{k=1}$, $n_{k+1}/n_k\geq q>1$, is it true that the measure in $[0,2\pi[$ of the set of those points $x$ for which \[\liminf_{m\to\infty}\Big|\sum^m_{k=1}\cos(n_kx)-\xi\Big|=0,\hspace{1cm}\text{ for all }\xi\in\mathbb{R},\] equals $2\pi$? (T. Murai)

背景链接：https://arxiv.org/abs/1809.07200

11325 | AMR-022-7069

### Research Problems in Function Theory - Problem 7.69

O / open | AMR 问题清单汇编

Given an associative algebra $A$, with identity $1$ and countable basis, then for a finite-dimensional subspace $V$ spanned by the vectors $\{e_j\}^k_{j=1}$ we have the differential operator \[\sum^k_{j=1}e_j\frac{\partial}{\partial x_j}\] acting on differentiable functions defined over domains in $V$ and taking their values in $A$. We call a function $f: U\subseteq V\to A$ a left analytic function if \[\sum^k_{j=1}e_j\frac{\partial f}{\partial x_j}(x)=0\] for all $x\in U$. Ryan has shown that there is a generalised Cauchy integral formula \[f(x_0)=\int_{\partial M}G(x,x_0)\sum^k_{j=1}(-1)^je_j\,d\hat{x}_jf(x),\] with real-analytic kernel $G(x,x_0)$, where $M$ is an arbitrary, compact real $n$-dimensional manifold lying in $U$; and $x_0\in \mathring{M}$ if and only if there are elements $\{p_j\}^k_{j=1}\subseteq A$ satisfying the relation \[p_je_l+p_le_j=2\delta_{jl}.\] [(a)] ; Is the result still valid if we only assume $G(x,x_0)$ to be a $C^1$ function? ; What analogous result holds if we assume the algebra to be non-associative? (J. Ryan)

背景链接：https://arxiv.org/abs/1809.07200

11326 | AMR-022-7070

### Research Problems in Function Theory - Problem 7.70

O / open | AMR 问题清单汇编

Using arguments due to Ahlfors (see, for example ) any M\"obius transformation in $\mathbb{R}^n$ can be written in the form $(ax + b)(cx+d)^{-1}$, where $x\in\mathbb{R}^n$ and $a, b, c, d$ are elements of a Clifford algebra $A_n$ that satisfies certain constraints. It can be shown that the linear differential equations whose solution spaces are conformally invariant are of the type \[D^kf_k((ax+b)(cx+d)^{-1})=0,\hspace{1cm} k\in\mathbb{N},\] where $D$ is the Euclidean Dirac operator, and the associated conformal weight is \[J_k(cx+d)= \begin{cases} (cx+d)\ast |cx+d|^{-n-1+k}&\text{for }k=2p-1, |cx+d|^{-n+k}&\text{for }k=2p, \end{cases}\] where $\ast$ is the involution described in . ; What are the non-linear differential equations whose solution spaces are conformally invariant? ; Can their conformal weights also be expressed in terms of $cx + d$, and what relationship do these solutions have to the linear conformally-invariant differential equations? (J. Ryan)

背景链接：https://arxiv.org/abs/1809.07200

11327 | AMR-022-7071

### Research Problems in Function Theory - Problem 7.71

O / open | AMR 问题清单汇编

Given a domain of holomorphy $\Omega\subseteq\mathbb{C}^n$, $n\geq2$, what conditions are required on $\Omega$ to admit an analytic function $p:\Omega\to\mathbb{C}$ which cannot be analytically extended beyond the boundary of $\Omega$, and satisfies the complex version \[\sum^n_{k=1}\frac{\partial^2}{\partial z^2_j}p(z)=0\] of Laplace's equation? (J. Ryan)

背景链接：https://arxiv.org/abs/1809.07200

11328 | AMR-022-7072

### Research Problems in Function Theory - Problem 7.72

O / open | AMR 问题清单汇编

Let $N$ denote the class of complex-valued $L^\infty$-functions $v$ on the unit disc $U$ such that $\int_U v\phi\,dx\,dy=0$ whenever $\phi$ is analytic in $U$ with $\int_U|\phi(x+iy)|\,dx\,dy<\infty$. The Cauchy principal value of \[(Bv)(z)=\frac{-1}{\pi}\int_U\frac{v(\zeta)}{(z-\zeta)^2}\,d\xi\,d\eta,\hspace{1cm}\zeta=\xi+i\eta,\] defines the Beurling transform $Bv$ of $v$. Is it true that $Bv\in L^\infty$ and, furthermore, that $$ \|Bv\|_\infty\leq C\|v\|_\infty, $$ for some $C < \infty$, whenever $v\in N$. A weaker question is whether this holds for all $v\in N\cap P$, where $P$ is the class of all polynomials in $z$ and $\overline{z}$. (The inequality ([source label: K7.72]) is true at least for certain subclasses of $N\cap P$.) (A. Hinkkanen)

背景链接：https://arxiv.org/abs/1809.07200

11329 | AMR-022-7073

### Research Problems in Function Theory - Problem 7.73

P / partially_solved | AMR 问题清单汇编

Let $D_1$, $D_2$ be domains in $\{|z|< R\}$, and let $\lambda_1(z)\,|dz|$ and $\lambda_2(z)\,|dz|$ be their hyperbolic metrics. What is the least number $A = A(R)$ such that the hyperbolic metric $\lambda(z)\,|dz|$ of $D_1 \cap D_2$ satisfies the inequality \[\lambda(z)< A(\lambda_1(z)+\lambda_2(z))?\] (W. H. J. Fuchs)

背景链接：https://arxiv.org/abs/1809.07200

11330 | AMR-022-7074

### Research Problems in Function Theory - Problem 7.74

P / partially_solved | AMR 问题清单汇编

In their famous Acta paper , Hardy and Littlewood introduced the celebrated Hardy-Littlewood maximal function in connection with complex function theory. Since then it has proved an invaluable tool in real analysis. Here we ask some questions about the dependence of constants on dimension. Let $B$ be a convex compact symmetric body in $\mathbb{R}^n$, normalised to have Euclidean volume $1$. Let the Hardy-Littlewood maximal functions be \[Mf(x)= \sup_{t>0}\big(\frac{1}{t^n}\int_{tB}|f(x+y)|\,dy\Big)\] and \[\tilde{M}f(x)=\sup_{k\in Z}\big(\frac{1}{2^{kn}}\int_{2^kB}|f(x+y)|\,dy\Big).\] [(a)] ; If $B$ is the Euclidean ball in $\mathbb{R}^n$, does there exist a constant $C$ such that $\text{meas}\{x:\tilde{M}f(x)>\lambda\}\leq C\lambda^{-1}\|f\|_1$ for all $\lambda> 0$, with $C$ independent of $n$? ; If so, what is the answer to the same question for $Mf$? ; In the case that $n = 1$, a conjecture of F. Sonia and the proposer is that the best constant in the inequality $\text{meas}\{x:Mf(x)>\lambda\}\leq C\lambda^{-1}\|f\|_1$ is $C = \frac{3}{2}$. Prove this. ; Let $1<p\leq\frac{3}{2}$. Can the best constant in the inequality \[\|Mf\|_p\leq C_p\|f\|_p\] be taken to be independent of $n$ and the body $B$? Even if $B = [-\frac{1}{2},\frac{1}{2}]^n$, can the constant be chosen independent of $n$? The following relevant facts are known. For $(a)$ and $(b)$, the best known constants are found in . For $(c)$, Sonia and the proposer have shown that the answer is `yes' if $\frac{3}{2}< p \leq\infty$; for $B$ with suitably-curved boundary, the answer is `yes' for $1< p\leq\infty$; for the sphere, see . (A. Carbery)

背景链接：https://arxiv.org/abs/1809.07200

11331 | AMR-022-7075

### Research Problems in Function Theory - Problem 7.75

P / partially_solved | AMR 问题清单汇编

Prove or disprove the following statements about analytic capacity $\gamma$. [(a)] ; If $E\subseteq\mathbb{C}$ is compact and $\phi$ is a $C^1$-diffeomorphism of $\mathbb{C}$ onto $\mathbb{C}$, then $\gamma(E) = 0$ if and only if $\gamma(\phi(E))=0$. The statement is false if $\phi$ is a homeomorphism or a quasi-conformal mapping. ; If $E\subseteq\mathbb{C}$ is compact and $\phi\in \text{GL}(2,\mathbb{R})$, then $\gamma(E) = 0$ if and only if $\gamma(\phi(E))=0$. ; If $E$, $F$ are compact subsets of $\mathbb{C}$, then there exists a constant $K > 0$ (independent of the choice of $E$ and $F$) such that \[\gamma(E\cup F)\leq K(\gamma(E)+\gamma(F)).\] Perhaps one can take $K = 1$? See for related results. ; If $K$ is a compact subset of $\mathbb{C}$ with $\gamma(K) = 0$, then \[\gamma(E\setminus K) = \gamma(E)\] for all compact subsets $E$ of $\mathbb{C}$. Here $\gamma(E\setminus K)$ means the inner capacity \[\sup\{\gamma(L):L\subseteq E\setminus K, L\text{ compact}\}.\] An interesting special case would be that when $K$ is the `corner quarters square Cantor set' (the so-called Garnett set). (A. G. O'Farrell)

背景链接：https://arxiv.org/abs/1809.07200

11332 | AMR-022-7076

### Research Problems in Function Theory - Problem 7.76

O / open | AMR 问题清单汇编

Prove or disprove the following statement: If $K$ is a compact subset of $\mathbb{C}$ with continuous analytic capacity $\alpha(K)= 0$, then \[\alpha(E\setminus K)=\alpha(E)\] for all compact subsets $E$ of $\mathbb{C}$. An interesting special case is when $K$ is a $C^1$-arc. The case when $K$ is a $C^{1+\varepsilon}$-arc has already been settled. (A. G. O'Farrell)

背景链接：https://arxiv.org/abs/1809.07200

11333 | AMR-022-7077

### Research Problems in Function Theory - Problem 7.77

O / open | AMR 问题清单汇编

We will say that $g(x)$ is a rearrangement of $f(x)$ if \[m\{x:g(x) < y\} = m\{x:f(x) < y\}\hspace{1cm}\text{ for all } y\in\mathbb{R},\] where $m$ is Lebesgue measure and $f$ and $g$ are defined on some finite interval $I$. What are those functions $f(x)$ for which $f'(x)$ is a rearrangement $f(x)$? Obvious examples of such functions are $f(x)=ke^x$ on any interval and $f(x) = k\sin x$ on $[0,\frac{1}{2}\pi]$. What others are there? How about other `differential rearrangements' than $f'(z)\sim f(x)$? (L. A. Rubel)

背景链接：https://arxiv.org/abs/1809.07200

11334 | AMR-022-7079

### Research Problems in Function Theory - Problem 7.79

O / open | AMR 问题清单汇编

Let $f$ be analytic on a domain $G$ in $\mathbb{C}$. We will say that a point $z_0\in G$ is a MacLane point of $f$ if there exists some neighbourhood $N$ of $z_0$ such that the restrictions to $N$ of the successive derivatives of $f$, \[\{f^{(n)}|_N:n\in\mathbb{N}\},\] form a normal family of functions on $N$. Let $M(f)$ denote the set of MacLane points of $f$. What can be said about the set $M(f)$, besides the fact that it is open? Must $M(f)$ be connected? If $G$ is simply-connected, must $M(f)$ be simply-connected? Or can $M(f)$ be an arbitrarily prescribed open subset of $G$? Similar questions can be asked about functions meromorphic in $G$. Perhaps it is more natural, also, to ask such questions about$\{f^{(n)}(z)/n!\}$ rather than simply $\{f^{(n)}(z)\}$? Some relevant facts can be found in and . (L. A. Rubel)

背景链接：https://arxiv.org/abs/1809.07200

11335 | AMR-022-7008

### Research Problems in Function Theory - Problem 7.8

O / open | AMR 问题清单汇编

Is it possible to express each $K$-quasiconformal map in $3$-space as the composition of two quasiconformal maps with maximal dilatation less than $K$? The corresponding plane result is true. (F. W. Gehring)

背景链接：https://arxiv.org/abs/1809.07200

11336 | AMR-022-7080

### Research Problems in Function Theory - Problem 7.80

S / solved | AMR 问题清单汇编

Let $f$ be an inner function, with $f(0) = 0$; then $f$ induces an ergodic (Lebesgue-) measure-preserving map of the circle onto itself. What is the entropy, $h(f)$, of $f$? It is conjectured that $h(f) < \infty$ if and only if $f'$ belongs to the Nevanlinna class; and that, in that case, then \[h(f)=\frac{1}{2\pi}\int^{2\pi}_0\log\big|f'(e^{i\theta})\big|\,d\theta .\] (See and .) (J. L. Fernandez)

背景链接：https://arxiv.org/abs/1809.07200

11337 | AMR-022-7081

### Research Problems in Function Theory - Problem 7.81

O / open | AMR 问题清单汇编

Let $I$ denote the class of all inner functions. Then $I$, as a subset of $H^\infty$, enjoys some of the properties that the collection of unimodular functions have as a subset of $L^\infty_\mathbb{R}$ (the real-valued functions in $L^\infty(\Pi)$ ). Is it true that if $\{\Lambda_n\} \subset (H^\infty)^*$ and if for each function $\phi\in I$ one has $|\Lambda_n(\phi)|\leq C(\phi)$, then \[\sup_n\|\Lambda_n\|_{(H^\infty)^*}<\infty?\] This is known to hold if $\Lambda_n\in L^1\setminus H^1_0(\subset(H^\infty)^*)$ (, , ); the corresponding real-variable result is also known . (J. L. Fernandez)

背景链接：https://arxiv.org/abs/1809.07200

11338 | AMR-022-7083

### Research Problems in Function Theory - Problem 7.83

O / open | AMR 问题清单汇编

Let $G$ be a finitely-generated Fuchsian group of the first kind. [(a)] ; If $F:\Pi \to \Pi$ is a $G$-invariant quasi-symmetric function, is $F$ totally singular? ; Is the Teichmuller space $T_G$ dense in the space $S_G$ (Schwarzians of $G$-invariant univalent functions)? (O. Lehto; L. Bers; I. Kra; communicated by D. Hamilton)

背景链接：https://arxiv.org/abs/1809.07200

11339 | AMR-022-8001

### Research Problems in Function Theory - Problem 8.1

O / open | AMR 问题清单汇编

Let $L = \{L\}^\infty_0$ be a non-negative increasing sequence such that \[\sum^\infty_{k=0}L_kr^k<\infty\hspace{1cm}(0<r<1).\] If $f(z)$ is analytic in $|z|< 1$, we will say that $f\in\mathcal{P}_L$ if there exists a constant $A$ such that, for each integer $n\geq0$ and each polynomial $P_n$ of degree $n$, \[\|P_n\ast f\|_\infty\leq AL_n\|P\|_\infty,\] where $\ast$ denotes the Hadamard product. The infimum of such $A$ for a given $f\in\mathcal{P}_L$ defines a norm on $\mathcal{P}_L$ which then becomes a Banach space. A variety of linear operators (such as subordination) have the property that they are norm-decreasing on $\mathcal{P}_L$. This enables one to obtain sharp coefficient inequalities for subordinate functions etc., once a function has been shown to lie in $\mathcal{P}_L$. The spaces $\mathcal{P}_L$ are `large-growth' spaces; for example, the case $L_k = 1$ $(k\geq 0)$ is the space of Cauchy-Stieltjes transforms. Convex sequences $\{L_k\}^\infty_0$ (such as $L_k = k$) have the property that \[\sum^\infty_{k=0}L_kz^k\] lies in the unit ball of $\mathcal{P}_L$. Can one relate known spaces to these spaces, for example $H^p$, for $0 < p < 1$? (T. Sheil-Small)

背景链接：https://arxiv.org/abs/1809.07200

11340 | AMR-022-8010

### Research Problems in Function Theory - Problem 8.10

S / solved | AMR 问题清单汇编

If $f$ and $1/f$ belong to the Bergman space $A^2$, does it follow that $\mathcal{P}f$ is dense in $A^2$? Here $\mathcal{P}f$ denotes the set of all polynomial multiples of $f$, i.e. \[\mathcal{P}f = \{pf: p\text{ a polynomial}\}.\] More generally, if $f\in A^2$ and $|f(z)|\geq c(1-|z|)^a$ for some $a$, $c > 0$, then does it follow that $\mathcal{P}f$ is dense in $A^2$? (For partial results in this direction, see ). (A. L. Shields)

背景链接：https://arxiv.org/abs/1809.07200

11341 | AMR-022-8011

### Research Problems in Function Theory - Problem 8.11

S / solved | AMR 问题清单汇编

Let $g$ be a function in the space $D$ of functions analytic in $|z|< 1$ with finite Dirichlet integral, i.e., \[{\int\int}_{|z|<1}|h'(z)|^2\,dx\,dy<\infty.\] If $\mathcal{P}g$, as defined in Problem 8.10, is dense in $D$ and if, for some $f\in D$, $|f(z)|\geq|g(z)|$ for all $|z|<1$, is it necessarily true that $\mathcal{P}f$ is dense in $D$? The analogous result is true in $H^2$ and in $A^2$. Shields solved the special case $g(z)\equiv1$. (A. L. Shields)

背景链接：https://arxiv.org/abs/1809.07200

11342 | AMR-022-8012

### Research Problems in Function Theory - Problem 8.12

O / open | AMR 问题清单汇编

Let $A$ denote the set of functions continuous on $|z|= 1$ which extend continuously to analytic functions on $|z|< 1$ (the disc algebra). Let $\tilde{A}$ denote the set of functions of the form $f\circ\phi$ where $f$ ranges over $A$ and $\phi$ ranges over the set of sense-preserving homeomorphisms of $|z| = 1$. Find a `good' characterisation of $\tilde{A}$. Is there a function in $\tilde{A}$ which coincides with \[\sum^\infty_{n=1}2^{-n}\exp(-i2^n\theta)\] on some subset of $|z|= 1$ having positive measure? The proposer conjectures not. (This problem is related to generalised analytic continuation.) (H. S. Shapiro)

背景链接：https://arxiv.org/abs/1809.07200

11343 | AMR-022-8013

### Research Problems in Function Theory - Problem 8.13

O / open | AMR 问题清单汇编

Does there exist a singular measure in Zygmund's class $A^*$ all of whose Fourier-Stieltjes coefficients are non-negative? (F. Holland)

背景链接：https://arxiv.org/abs/1809.07200

11344 | AMR-022-8014

### Research Problems in Function Theory - Problem 8.14

O / open | AMR 问题清单汇编

Which functions in $L^\infty$ on the unit circle generate positive Hankel operators? (F. Holland)

背景链接：https://arxiv.org/abs/1809.07200

11345 | AMR-022-8015

### Research Problems in Function Theory - Problem 8.15

S / solved | AMR 问题清单汇编

Characterise the Hankel operators on the Hardy space $H^2$ on the circle that are of trace class. (F. Holland)

背景链接：https://arxiv.org/abs/1809.07200

11346 | AMR-022-8017

### Research Problems in Function Theory - Problem 8.17

O / open | AMR 问题清单汇编

In the ring of bounded analytic functions on the unit ball or the polydisc in $n$ variables, is the intersection of two finitely-generated ideals again finitely generated? This was proved for $n = 1$ by McVoy and Rubel . (L. A. Rubel)

背景链接：https://arxiv.org/abs/1809.07200

11347 | AMR-022-8018

### Research Problems in Function Theory - Problem 8.18

O / open | AMR 问题清单汇编

For which simply-connected domains $D$ (with $0 \in D$) is it true that there is a constant $K = K(D)$ such that $$ {\int\int}_D|f|^2\,dx\,dy\leq K{\int\int}_D|f'|^2\,dx\,dy $$ for all functions $f$ analytic on $D$ with $f(0) = 0$? This inequality ([source label: J8.18]) is known as the analytic Poincar\'e inequality. Courant and Hilbert have given a Jordan domain for which ([source label: J8.18]) is false; and Hummel has given an example of a spiral domain $D$ for which ${\int\int}_D|f|^2\,dx\,dy = \infty$ and ${\int\int}_D|f'|^2\,dx\,dy < \infty$. (D. H. Hamilton)

背景链接：https://arxiv.org/abs/1809.07200

11348 | AMR-022-8019

### Research Problems in Function Theory - Problem 8.19

O / open | AMR 问题清单汇编

Let $\mu\geq1$ be a singular measure on the boundary of the unit disc $\mathbb{D}$; and let $S_\mu$ be the corresponding singular inner function \[S_\mu(z)=\exp\Big\{\int^{2\pi}_0\frac{z+e^{i\theta}}{z-e^{i\theta}}\,d\mu(\theta)\Big\},\hspace{1cm}z\in \mathbb{D}.\] The function $S_\mu$ is said to be discrete if $\mu$ is discrete, and continuous if it has no discrete part. If $S_\mu$ is discrete, does there exist some $\delta > 0$ such that $\|S_\mu-s_\nu\|_\infty>\delta$ for all continuous functions $S_\nu$? The answer is `no' if `discrete' and `continuous' are interchanged . (K. Stephenson)

背景链接：https://arxiv.org/abs/1809.07200

11349 | AMR-022-8021

### Research Problems in Function Theory - Problem 8.21

O / open | AMR 问题清单汇编

Let $w\geq0$ be an integrable function on the circle or on the line. Then $w$ is said to belong to the class $A_2$ if \[\sup_I\Big(\frac{1}{|I|}\int_Iw\,dx\Big)\Big(\frac{1}{|I|}\int_I\frac{1}{w}\,dx\Big)<\infty\] over all (appropriate) intervals $I$ ; and $w$ is said to satisfy the Helson-Szeg\"o condition if $\log w = u + \tilde{v}$ where $u\in L^\infty,\|v\|_\infty<\frac{1}{2}\pi$ and $\tilde{v}$ is the conjugate function of $v$. It is known that the Helson-Szeg\"o condition is equivalent to $A_2$, since both conditions are necessary and sufficient for $$ \int_I|\tilde{f}|^2w\,dx\leq\text{ const.}\int_I|f|^2w\,dx. $$ The problem is to prove that the Helson-Szeg\"o condition and $A_2$ are equivalent without using ([source label: J8.21]). For information on the case $\|v\|_\infty<\pi$, see . (J. B. Garnett and P. W. Jones)

背景链接：https://arxiv.org/abs/1809.07200

11350 | AMR-022-8023

### Research Problems in Function Theory - Problem 8.23

O / open | AMR 问题清单汇编

In Pompeiu's formula, $f(z)$ and $\int_\Gamma f(\zeta)/(\zeta-z)\,d\zeta$ make sense for merely continuous functions, but \[\int\int_\Omega\frac{\partial f}{\partial\overline{\zeta}}\frac{1}{\zeta-z}\,d\zeta\,d\eta\] needs at least some weak differentiability of $f$. It would be useful to extend the validity of the formula, for example to cover the case of functions $f\in \text{Lip }\alpha$, $\alpha> 0$. This prompts the following question. For which Borel sets $\Omega$ is the inequality \[\Big|\left<\frac{\partial\chi_\Omega}{\partial\overline{z}},f\right>\Big|\leq C_\Omega\|f\|_{\text{Lip }\alpha}\] valid? Putting it another way, for which Borel sets $\Omega$ does $\chi_\Omega$ act on the Besov space $B^{\alpha-1}_{\infty,\infty}$? An interesting special case would be that when $\Omega$ is a `Swiss cheese'. (A. G. O'Farrell)

背景链接：https://arxiv.org/abs/1809.07200

11351 | AMR-022-8024

### Research Problems in Function Theory - Problem 8.24

O / open | AMR 问题清单汇编

Let the sequence $\{M_k\}^\infty_0$ of positive numbers be such that \[M_0=1\hspace{1cm}\text{ and }\hspace{1cm}\frac{M_{k+j}}{M_kM_j}\geq\begin{pmatrix}k+j\\j\end{pmatrix}.\] Assume also the non-quasianalyticity condition that $\sum_k\big(M_k/M_{k+1}\big)<\infty$. Consider those functions $f$ on a compact set $X\subset\mathbb{C}$ that are limits of rational functions with poles of $X$ in the norm \[g\mapsto\sum^\infty_{k=0}\frac{1}{M_k}\sup_{X}|g^{(k)}|;\] these functions form a Banach algebra. Is $X$ its maximal ideal space? (The answer is `yes' if $X$ is the unit disc or the unit interval.) (A. G. O'Farrell)

背景链接：https://arxiv.org/abs/1809.07200

11352 | AMR-022-8025

### Research Problems in Function Theory - Problem 8.25

O / open | AMR 问题清单汇编

Let $\psi:S^1\to S^1$ be a direction-reversing homeomorphism, and let $A_\psi$ denote the set of functions $f:S^1\to\mathbb{C}$ such that both $f$ and $f\circ\psi$ belong to the disc algebra. When does $A_\psi$ contain only constant functions? (A. G. O'Farrell)

背景链接：https://arxiv.org/abs/1809.07200

11353 | AMR-022-8026

### Research Problems in Function Theory - Problem 8.26

P / partially_solved | AMR 问题清单汇编

Let $\psi:S^1\to S^1$ be a homeomorphism. When is it true that $\text{Re }A = \text{Re }A \circ \psi$? That is, when is each function $f$ in the real part of the disc algebra also of the form $g\circ\psi$, for some function $g$ in the disc algebra? O'Connell has shown that it is necessary that $\psi$ is absolutely continuous, and sufficient that $\psi$ is $C^{1+\varepsilon}$. (A. G. O'Farrell)

背景链接：https://arxiv.org/abs/1809.07200

11354 | AMR-022-8003

### Research Problems in Function Theory - Problem 8.3

O / open | AMR 问题清单汇编

For a bounded plane domain $D$, denote by $N(D)$ the class of functions analytic on $D$ of bounded characteristic (i.e., all quotients of functions in $H^\infty(D)$ with nonvanishing denominator); let $f_1,\ldots,f_N\in N(D)$ have no common zeros in $D$. Find necessary and sufficient conditions on $f_1,\ldots,f_N$ in order that they generate the full ring $N(D)$. Equivalently, if $g_1,\ldots,g_N\in H^\infty(D)$, when does the ideal generated by $g_1,\ldots,g_N$ in $H^\infty(D)$, or in $N(D)$, contain a non-vanishing function? For $D = \mathbb{D}$, for example, the zeros of $g_1,\ldots,g_N$ should not get too close together as one approaches $\partial D$, and one would like to obtain a corona-type theorem for this problem - i.e., a lower estimate for \[|g_1(z)|+\ldots|g_N(z)|\] in $\mathbb{D}$. (J. J. Kelleher)

背景链接：https://arxiv.org/abs/1809.07200

11355 | AMR-022-8004

### Research Problems in Function Theory - Problem 8.4

O / open | AMR 问题清单汇编

Let $D$ be a simply-connected domain in the complex plane $\mathbb{C}$ and $A(D)$ the ring of functions $f: D\to\mathbb{C}$ analytic in $D$. Bers (no citation) has shown that (for domains of arbitrary connectivity) the algebraic structure of $A(D)$ determines the conformal structure of $D$. Can $A(D)$ be the direct sum of two non-trivial subrings of itself? (This would represent a generalisation of Taylor's Theorem.) Is there a more general result for multiply-connected domains $D$? (J. J. Kelleher)

背景链接：https://arxiv.org/abs/1809.07200

11356 | AMR-022-8005

### Research Problems in Function Theory - Problem 8.5

O / open | AMR 问题清单汇编

Let $G$ be a domain in $\mathbb{C}$, and $H(G)$ the ring of functions analytic on $G$. It is known that, for two domains $G_1$ and $G_2$, $H(G_1)$ is isomorphic to $H(G_2)$ if and only if $G_1$ and $G_2$ are conformally (or anticonformally) equivalent. What can be said under only the hypothesis that $H(G_1)$ and $H(G_2)$ are elementarily equivalent in the sense of model theory? For a large class of domains the corresponding problem has been solved for $H_\mathbb{C}(G)$, the algebra of functions analytic on $G$, by Henson and Rubel (see and ). (L. A. Rubel)

背景链接：https://arxiv.org/abs/1809.07200

11357 | AMR-022-8007

### Research Problems in Function Theory - Problem 8.7

O / open | AMR 问题清单汇编

We use the notation of Problem 8.6. Let $\{z_n\}^\infty_1$ be a sequence of points in $\mathbb{D}$ such that the kernel functions $k_n(z) = (1-z_nz)^{-2}$ do not span the space $A^2$, and let $\{f_j\}^\infty_1$ be finite linear combinations of the $k_n$ which converge (in norm) to some function $f$ in $A^2$. Prove that, if the sequence $\{f_j\}$ converges uniformly to $0$ on some disc $\Delta$ in $\{|z| > 1\}$, then $f\equiv0$. Similar problems can of course be stated for the functions $(1- z_nz)^{-1}$, and for spaces other than $A^2$. This slightly `weird' problem arises in the theory of generalised analytic continuation; it is known to be true if the closure of $\{z_n\}^\infty_1$ does not contain all of $\{|z| = 1\}$; it is also known that the analogous result for $H^2$ is true. (H. S. Shapiro)

背景链接：https://arxiv.org/abs/1809.07200

11358 | AMR-022-8008

### Research Problems in Function Theory - Problem 8.8

O / open | AMR 问题清单汇编

Suppose that $F$ is a relatively-closed subset of $\mathbb{D}$, and let \[\|f\|_F=\sup\{|f(z)|;z\in F\}\] for functions $f$ in the Bergman space $A^2$. Describe geometrically the set \[\{z: |z|< 1, |f(z)|\leq\|f\|_F\text{ for all }f\in A^2\}.\] (A. Stray)

背景链接：https://arxiv.org/abs/1809.07200

11359 | AMR-022-8009

### Research Problems in Function Theory - Problem 8.9

O / open | AMR 问题清单汇编

Determine \[\|\Lambda\|=\sup\Big|{\int\int}_{|z|<1}f(z)\phi(z)\,d\sigma_z\Big|\] over those $f\in A^1$ with $\|f\|_1\leq1$, where \[\phi(z)=\text{sgn}(\text{Re } z).\] It is shown by Reich and Strebel that $\|\Lambda\|<1$, and that there exists an extremal function for the problem. See also Harrington and Ortel . Problems of this type are of interest in connection with the theorem of Hamilton in quasi-conformal mapping. (K. Strebel; communicated by M. Ortel)

背景链接：https://arxiv.org/abs/1809.07200

11360 | AMR-022-9001

### Research Problems in Function Theory - Problem 9.1

O / open | AMR 问题清单汇编

A sequence $\{z_n\}^\infty_1$ in $|z|< 1$ is interpolating for bounded analytic (harmonic) functions if, for each bounded sequence $\{\alpha_n\}^\infty_1$ there exists a function $u(z)$ bounded and analytic (harmonic) in $|z|< 1$ with $u(z_n) = \alpha_n (n\geq 1)$. It is known that a sequence is interpolating for bounded harmonic functions if and only if it is interpolating for bounded analytic functions (, , ); however all such proofs require knowledge of conditions implying that a sequence interpolates for bounded analytic functions. Find a simple proof of this equivalence which does not rely on knowledge of such conditions. (L. Zalcman)

背景链接：https://arxiv.org/abs/1809.07200

11361 | AMR-022-9010

### Research Problems in Function Theory - Problem 9.10

O / open | AMR 问题清单汇编

Let $F$ be a closed subest of $\mathbb{R}^n$, $n\geq2$. Call $F$ a set of harmonic approximation if every function continuous on $F$ and harmonic in the interior of $F$ can be uniformly approximated there by a harmonic function in $\mathbb{R}^n$. Give necessary and sufficient conditions that $F$ be a set of harmonic approximation. If $F$ is nowhere dense, \v{S}aginjan has done this. If $F$ is the closure of its interior, Gauthier, Ow and Goldstein have given necessary conditions and sufficient conditions when $n=2$, but not necessary and sufficient conditions. This has applications to Rubel's Problem 9.7. (M. Goldstein)

背景链接：https://arxiv.org/abs/1809.07200

11362 | AMR-022-9011

### Research Problems in Function Theory - Problem 9.11

O / open | AMR 问题清单汇编

Let $D$ be a planar domain. A sequence $\{z_j\}$ of points in $D$ is said to be an interpolating sequence if whenever $\{\alpha_j\}\in\ell^\infty$ there is a function $F\in H^\infty(D)$ such that $F(z_j) = \alpha_j$, for all $j$. Suppose that the sequence $\{z_j\}$ has the property that for each $j$ there is a function $F_j \in H^\infty(D)$ such that $F_j(z_k)=0$ if $k\neq j$, $F_j(z_j)=1$ and $\|F\|_\infty\leq C$. Is $\{z_j\}$ necessarily an interpolating sequence? (P. W. Jones)

背景链接：https://arxiv.org/abs/1809.07200

11363 | AMR-022-9012

### Research Problems in Function Theory - Problem 9.12

O / open | AMR 问题清单汇编

Let $\Gamma\subset\mathbb{C}$ be a Jordan curve of logarithmic capacity $1$, and let $\phi$ be a conformal map from the exterior of $\Gamma$ to the exterior of the unit circle such that $\phi(\infty)=\infty$. We consider charge distributions on $\Gamma$ consisting of $n$ point charges $1/n$ at $n$th order Fekete points $z_1,\ldots,z_n$ on $\Gamma$, $n\in\mathbb{N}$. If $\Gamma$ is smooth enough, the corresponding potentials \[\frac{1}{n}\sum^n_{k=1}\log|z-z_k|\] give approximations to $\log|\phi(z)|$ (outside $\Gamma$) and to $0$ (inside $\Gamma$) which are $O(1/n)$ away from $\Gamma$ (see , , , ). Prove a similar result for the case where $\Gamma$ is a square. It does hold in the degenerate case $\Gamma = [-2,2]$. (J. Korevaar)

背景链接：https://arxiv.org/abs/1809.07200

11364 | AMR-022-9013

### Research Problems in Function Theory - Problem 9.13

O / open | AMR 问题清单汇编

Let $K$ be a compact subset of $\mathbb{R}^n$, $n\geq3$. For $\phi\in\mathcal{D}$, let $D(\phi)$ be a least-diameter disc containing $\text{spt }\phi$; let $d(\phi) = \text{diam }(\text{spt }\phi)$, and let \[\|\phi\|_\ast=\|\phi\|_\infty+d(\phi)\cdot\|\triangledown\phi\|_\infty.\] Are the following conditions equivalent for continuous functions $f:\mathbb{R}^n\to\mathbb{R}$? [(1)] ; There exists a sequence $\{f_n\}^\infty_1$ of functions harmonic near $K$, such that $f_n\to f$ uniformly on $K$. ; There exists a function $n$ such that $n(\delta)$ decreases to $0$ as $\delta$ decreases to $0$, for which \[\Big|\int_{\mathbb{R}^n}f\delta\phi\,dx\Big|\leq\eta(d(\phi))\|\phi\|_\ast C(D(\phi)-X).\] Here $C$ denotes the harmonic capacity of $\mathbb{R}^n$ obtained from the kernel $r^{-n+2}$. Note that $(1)$ implies $(2)$, and that $(2)$ implies $(1)$ if $f$ is a $C^2$-function. The condition $(2)$ is formally analogous to one that occurs in rational approximation theory. (A. G. O'Farrell)

背景链接：https://arxiv.org/abs/1809.07200

11365 | AMR-022-9014

### Research Problems in Function Theory - Problem 9.14

O / open | AMR 问题清单汇编

Let $f$ be continuous on a compact subset $K$ of $\mathbb{C}$. If there exists a sequence $\{f_n\}^\infty_1$ of functions analytic near $K$ for which $g_n\to f^2$ uniformly on $K$, does there necessarily exist a sequence $\{h_n\}^\infty_1$ of functions analytic near $K$ for which $h_n\to f$ uniformly on $K$? Paramanov has proved this under the stronger hypothesis that $f\in\text{Lip }(\frac{1}{2})$; it is also true under the hypothesis that $f\in W^{1, p}$ $(p> 2)$. (A. G. O'Farrell)

背景链接：https://arxiv.org/abs/1809.07200

11366 | AMR-022-9015

### Research Problems in Function Theory - Problem 9.15

S / solved | AMR 问题清单汇编

Let $f_1$ and $f_2\in H^\infty(\text{unit disc}) = H^\infty$; and let the function $g\in H^\infty$ satisfy the inequality \[|g(z)|\leq|f_1(z)|+|f_2(z)|.\] Do there necessarily exist functions $g_1$ and $g_2$ in $H^\infty$ such that \[g^2=f_1g_1+f_2g_2\,?\] In other words, is it true that $g^2\in I(f_1, f_2)$ (the ideal generated by $f_1$ and $f_2$)? Wolff has proved that $g^3\in I(f_1,f_2)$. Also Rao has given an example of a function $g\notin I(f_1,f_2)$. For related results by Tolokonnikov, see . (J. Garnett)

背景链接：https://arxiv.org/abs/1809.07200

11367 | AMR-022-9016

### Research Problems in Function Theory - Problem 9.16

P / partially_solved | AMR 问题清单汇编

Let $\Gamma$ be a curve of the form \[\{x+iA(x):-\infty<x<\infty\}\] with \[|A(x_1)-A(x_2)|\leq M|x_1-x_2|.\] Let $E$ be a compact subset of $\Gamma$, $\Delta_1(t)> 0$, and let \[\Omega=\mathbb{C}^*\setminus E,\hspace{1cm}\text{ where }\mathbb{C}^*=\mathbb{C}\cup\{\infty\}.\] Prove the corona theorem for $\Omega$. (J. Garnett)

背景链接：https://arxiv.org/abs/1809.07200

11368 | AMR-022-9017

### Research Problems in Function Theory - Problem 9.17

O / open | AMR 问题清单汇编

Let $K$ denote the $\frac{1}{3}$-Cantor set on $\mathbb{R}$; let $E = K\times K$, and let $\Omega=\mathbb{C}^*\setminus E$. Prove the corona theorem for $\Omega$. (J. Garnett)

背景链接：https://arxiv.org/abs/1809.07200

11369 | AMR-022-9002

### Research Problems in Function Theory - Problem 9.2

O / open | AMR 问题清单汇编

Let $f(z)\in H^\infty$, and let $\{z_n\}$ be a Blaschke sequence \[\sum^\infty_{n=1}(\-|z_n|)<\infty.\] [(a)] ; Does there always exist a Blaschke product $B(z)$ of norm not necessarily equal to $1$, such that $$ B(z_n)=f(z_n) $$ for $n\geq1$? This is certainly the case if the Blaschke sequence is uniformly separated, that is \[\inf_n\prod_{m\neq n}\big|\frac{z_m-z_n}{1-\overline{z_m}z_n}\big|>0.\] See, for example, Earl . ; Is the unique $H^\infty$ function of minimal norm assuming the values $\{f(z_n)\}$ at $\{z_n\}$ a constant multiple of a Blaschke product? The answer is `yes' if the sequence $\{z_n\}$ is finite, see Earl . What happens if we know, in addition, that $f(z_n)\to0$ as $n\to\infty$? The answer is again `yes' if $z_n\to1$ non-tangentially as $n\to\infty$. ; One can ask questions $(a)$ and $(b)$ for sequences $\{z_n\}$ that are weakly separated, viz \[\inf_{m\neq n}\big|\frac{z_m-z_n}{1-\overline{z_m}z_n}\big|>0.\] ; Also, one can ask questions $(a)$ and $(c)$ for inner functions instead of for Blaschke products. (J. P. Earl and A. Stray)

背景链接：https://arxiv.org/abs/1809.07200

11370 | AMR-022-9004

### Research Problems in Function Theory - Problem 9.4

O / open | AMR 问题清单汇编

For each pair of $f,g\in H^\infty$, does there necessarily exist another pair of functions $a,b\in H^\infty$ such that $$ af+gb\neq0,\hspace{1cm}|z|<1\,? $$ It is easy to see that a necessary condition for this is that $\log(|f|-|g|)$ have a harmonic minorant. Is this condition also sufficient for ([source label: C9.1])? This problem is closely related to Problem 9.3. (B. A. Taylor; communicated by L. A. Rubel)

背景链接：https://arxiv.org/abs/1809.07200

11371 | AMR-022-9005

### Research Problems in Function Theory - Problem 9.5

O / open | AMR 问题清单汇编

Let $K_1, K_2, K_3$ be disjoint closed sets in the extended complex plane, and $C_1, C_2, C_3$ constants. Let $\rho_n(f)$ be the best rational approximation to the function $f$ which equals $C_i$ on $K_i$ $(i = 1, 2, 3)$; i.e. \[\rho_n(f)=\inf_{g\in R_n}\max_{z\in\cup_i K_i}|f(z)-g(z)|,\] where $R_n$ is the class of rational functions $f$ order at most $n$. Find a geometric characterisation of $\lim_{n\to\infty}\rho_n^{1/n}$. For the case of two sets, see Gonchar . (T. Ganelius)

背景链接：https://arxiv.org/abs/1809.07200

11372 | AMR-022-9006

### Research Problems in Function Theory - Problem 9.6

O / open | AMR 问题清单汇编

Let $D$ be an open subset of the extended complex plane with non-empty boundary $\partial D$, and let $F$ be a relatively-closed subset of $D$. Let $f$ be a function given on $F$, and $\{f_n\}^\infty_1$ a sequence of functions analytic on $D$ such that $f_n\to f$ uniformly on $F$. If $E\subset(\partial F_n\cap\partial D)$ and if $f$ extends continuously to $F\cup E$, can each $f_n$ be extended continuously to $F\cup E$? The answer is `yes' if $D$ is the unit disc, or if $E$ is compact. (A. Stray)

背景链接：https://arxiv.org/abs/1809.07200

11373 | AMR-022-9007

### Research Problems in Function Theory - Problem 9.7

O / open | AMR 问题清单汇编

Let us call a closed set $E$ in $\mathbb{C}$ a weak Arakelian set if, corresponding to each function $g(z)$ continuous on $E$ and analytic in the interior of $E$, there exists an entire function $g(z)$ such that, for any sequence $\{z_n\}^\infty_1$ in $E$, $|f(z_n)|\to\infty$ if and only if $|g(z_n)|\to\infty$. Find a geometric characterisation of the weak Arakelian sets. (L. A. Rubel)

背景链接：https://arxiv.org/abs/1809.07200

11374 | AMR-022-9008

### Research Problems in Function Theory - Problem 9.8

O / open | AMR 问题清单汇编

Let $\gamma$ be a Jordan arc in $\mathbb{C}^n$, $n\geq2$ such that the projections $\gamma_j$ on the complex coordinate planes $j=1,\ldots,n$ have area zero. Then $R(\gamma)=C(\gamma)$. Is it true that $P(\gamma)=C(\gamma)$? See Korevaar and Wermer . (J. Korevaar)

背景链接：https://arxiv.org/abs/1809.07200

11375 | AMR-022-9009

### Research Problems in Function Theory - Problem 9.9

O / open | AMR 问题清单汇编

Does the condition $\sum 1/p_n<\infty$ for positive integers $p_n$ guarantee that the sequences of powers $\{z^{p_n}\}$ fails to span $C(\gamma)$ for every Jordan arc $\gamma$? Korevaar and Dixon have shown that for arcs of locally limited rotation (for example $C^1$ arcs), the condition \[p_n\geq nL(n),\hspace{1cm}0<L(n)\uparrow,\hspace{1cm}\sum 1/nL(n)<\infty\] assues a non-spanning sequence $\{z^{p_n}\}$. (J. Korevaar)

背景链接：https://arxiv.org/abs/1809.07200

11376 | AIM-SEVERAL_COMPLEX_VARIABLES-0047

### Residual obstruction to Artin approximation of formal CR maps

P / partially_solved | AIM 研讨会问题汇编

Let \[ 0\in M\subset \mathbb C^n,\qquad 0\in M'\subset \mathbb C^N \] be real-analytic CR manifolds, both of Bloom--Graham--Kohn finite type, and let \[ f:(\mathbb C^n,0)\longrightarrow(\mathbb C^N,0) \] be a formal holomorphic map that sends (M) into (M') formally. Is it true that, for every integer \(\ell>0\), there is a convergent holomorphic germ \[ f^\ell:(\mathbb C^n,0)\longrightarrow(\mathbb C^N,0) \] such that (f^\ell(M)\subset M') and \[ j_0^\ell f^\ell=j_0^\ell f? \]

If \(\rho\) and \(\rho'\) are defining functions, formal containment means, invariantly, that substitution by \((f,\bar f)\) sends the defining ideal of \(M'\) into the completed defining ideal of \(M\). In the hypersurface case this can be written \[ \rho'(f(z),\bar f(\bar z))=a(z,\bar z)\rho(z,\bar z). \] For higher codimension, \(\rho,\rho'\) are vector-valued and \(a\) is a matrix of formal series.

1. the approximating map is printed with codomain \(\mathbb C^n\); it must have codomain \(\mathbb C^N\); 2. the final informal “complexity” sentence subtracts \(N\) from the dimension of an analytic set containing the graph. A graph over \(\mathbb C^n\) has dimension \(n\), so the natural excess dimension is \(\dim A_f-n\), not \(\dim A_f-N\). Moreover, the analytic set must be specified canonically; an arbitrary containing set gives no invariant.

来源字段：https://aimath.org/WWN/complexitycrmap/complexitycrmap.pdf

11377 | OWR-11791-006

### Restricted Isometries for Random Scattering Matrices

O / open | Oberwolfach 报告问题汇编

Does the random scattering matrix used for sparse recovery satisfy a restricted isometry property in a parameter regime strong enough to yield uniform recovery of all sparse signals?

来源字段：https://doi.org/10.4171/owr/2012/29

11378 | OWR-14299906-002

### Reverse Approximation Implications for Traces

O / open | Oberwolfach 报告问题汇编

Is every hyperlinear trace an MF trace? Is every amenable trace quasidiagonal?

来源字段：https://doi.org/10.4171/owr/2026/4

11379 | OWR-13489-004

### Riemannian Control of Bergman Geometry

P / partially_solved | Oberwolfach 报告问题汇编

Let $M$ be an open Riemann surface whose complex structure is induced by a complete Riemannian metric $ds^2$. How does the Riemannian geometry of $ds^2$ control the Bergman kernel and Bergman metric of $M$?

来源字段：https://doi.org/10.4171/owr/2015/4

11380 | OWR-15577-013

### Riesz Transform Boundedness and $\beta$-Number Carleson Conditions

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mu$ be a Borel measure on $\mathbb R^{n+1}$ satisfying $\mu(B(x,r))\leq c_0r^n$. Is the codimension-one Riesz transform bounded on $L^2(\mu)$ if and only if every ball $B$ satisfies $$ \int_B\int_0^{r(B)}\beta_{\mu,2}(x,r)^2\Theta_\mu(x,r)\,\frac{dr}{r}\,d\mu(x)\leq C\mu(B), $$ where $\Theta_\mu(x,r)=\mu(B(x,r))/r^n$ and $\beta_{\mu,2}$ is the Jones $n$-plane coefficient?

来源字段：https://doi.org/10.4171/owr/2017/34

11381 | AIM-SPECIAL_FUNCTIONS-0006

### Root-sensitive k-Schur functions and an exponent-profile obstruction

P / partially_solved | AIM 研讨会问题汇编

**(6) Generalize \(k\)-Schur functions to root systems other than type \(A\).**

来源字段：https://aimath.org/WWN/kostka/prob2.pdf

11382 | OWR-12005-009

### Round-Sphere Maximization of the First Boundary Eigenvalue

O / open | Oberwolfach 报告问题汇编

For every integer $m\geq3$, does the round sphere maximize $\lambda_1(\Sigma)|\Sigma|^{2/m}$ among convex hypersurfaces $\Sigma^m\subset\mathbb R^{m+1}$, where $\lambda_1(\Sigma)$ is the first nontrivial Laplace-Beltrami eigenvalue?

来源字段：https://doi.org/10.4171/owr/2012/33

11383 | OWR-12723-006

### Sample-Based Approximation of Ridge Functions

P / partially_solved | Oberwolfach 报告问题汇编

What algorithm can exploit the ridge structure of functions in $R_d^{\alpha,p}$ to approximate them efficiently from samples?

来源字段：https://doi.org/10.4171/owr/2013/49

11384 | AIM-ANALYSIS-0010

### Scalar dbar normalization and planar Hermite multipoles for the elliptic Gaussian

P / partially_solved | AIM 研讨会问题汇编

Analyze the condition given by \[ \overline{\partial}\phi=\overline{H_n}e^{-NQ}, \] where \(Q=|z|^2+2t\operatorname{Re}(z^2)\) and where \(\overline{H_n}\) is Hermite.

来源字段：http://aimpl.org/riemhilberttoeplitz/2/

11385 | AIM-SEVERAL_COMPLEX_VARIABLES-0038

### Scalar-parabolic rank obstructions for equivariant hyperquadric embeddings

P / partially_solved | AIM 研讨会问题汇编

**1. (M. Kolar)** Assume $M\subseteq\mathbb C^2$ is a real analytic hypersurface that admits a nonlinearizable automorphism $H$. Does there exist a hyperquadric $Q\subseteq\mathbb C^n$ such that $M\hookrightarrow Q$ nontrivially and $H$ is induced by an automorphism of $Q$? Can a bound on $n$ be found?

**2. (D. Zaitsev)** Assume $M\subseteq\mathbb C^2$ admits a nontrivial map tangent to the identity of order 2. Does $M$ have an embedding into a hyperquadric $Q$ as in the preceding problem?

来源字段：https://aimath.org/WWN/crmappings/crmappings.pdf

11386 | AIM-FUNCTIONAL_ANALYSIS-0003

### Scale invariance and the fibers of normalized cone-volume measure

S / solved | AIM 研讨会问题汇编

**Cone-volume measure.** If \(K\) and \(L\) symmetric convex bodies in \(\mathbb R^n\) such that \(\bar V_K=\bar V_L\), then \(K=L\) or \(K\) and \(L\) are parallelograms with parallel sides.

Moreover, Problem 1.15 on that same page explicitly calls \(\bar V_L\) “the cone-volume probability measure of \(L\).” Thus the bar denotes normalization, and the page does not state equal volume, dimension two, or equality up to dilation.

The statement as written cannot be correct. It has at least two independent defects:

1. normalized cone-volume measure is invariant under dilation, so it cannot distinguish \(K\) from \(cK\); 2. in dimensions \(n\geq 3\), distinct equal-volume parallelotopes with the same facet-normal directions have the same cone-volume measure, but are not parallelograms.

No silent correction is made below. Instead, the literal assertion is refuted, the known planar repair is identified, and its higher-dimensional polyhedral obstruction is classified exactly.

来源字段：http://aimpl.org/symconvgeomineq/1/

11387 | OWR-14298585-002

### Scaling Constants for Discrete Energies on Metric Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Characterize the compact metric spaces that admit a proper scaling constant $\sigma_p$ for graph approximations whose renormalized discrete $p$-energies converge to a meaningful continuum energy. Give further examples beyond the known $p$-conductively homogeneous spaces.

来源字段：https://doi.org/10.4171/owr/2024/42

11388 | OWR-12979-006

### Schatten-Class Commutators for Homogeneous Polynomial Modules

P / partially_solved | Oberwolfach 报告问题汇编

Let $M$ be the submodule of the Drury-Arveson space $H_d^2$ generated by homogeneous polynomials $p_1,\ldots,p_m\in\mathbb C[z_1,\ldots,z_d]$, let $S_j$ be the compression of multiplication by $z_j$ to $M^\perp$, and let $Z(p)$ be the common zero set of the generators. Are all commutators $[S_j,S_k^*]$ in the Schatten class $\mathcal S^q$ for every $q>\dim Z(p)$?

来源字段：https://doi.org/10.4171/owr/2014/21

11389 | OWR-14748-006

### Scope of the Generalized Tracial-Rank-One Class

P / partially_solved | Oberwolfach 报告问题汇编

Define $\mathcal N_1$ to consist of separable, unital, simple, amenable, $\mathcal Z$-stable $C^*$-algebras $A$ satisfying the UCT and $\operatorname{gTR}(A\otimes U)\leq 1$ for some infinite-type UHF algebra $U$. How large is $\mathcal N_1$, and which $C^*$-algebras belong to it?

来源字段：https://doi.org/10.4171/owr/2016/40

11390 | OWR-17290-003

### Self-Adjointness Across Almost-Riemannian Tangency Points

O / open | Oberwolfach 报告问题汇编

When an almost-Riemannian singular set has tangency points, is the Laplace-Beltrami operator on the regular region essentially self-adjoint? In particular, does this hold on $\mathbb R^2$ for

$$X_1=\partial_x,\qquad X_2=(z-x^2)\partial_z,$$

whose singular set is $z=x^2$?

来源字段：https://doi.org/10.4171/owr/2019/47

11391 | OWR-17290-005

### Self-Adjointness and Extensions of Infinite Metric-Graph Laplacians

O / open | Oberwolfach 报告问题汇编

For a metric graph with infinitely many edges and Kirchhoff Laplacian $H$, find geometric or combinatorial criteria guaranteeing self-adjointness. When self-adjointness fails, determine the deficiency indices and self-adjoint extensions of $H$.

来源字段：https://doi.org/10.4171/owr/2019/47

11392 | OWR-17290-002

### Self-Adjointness for Degenerate No-Tangency Metrics

P / partially_solved | Oberwolfach 报告问题汇编

For the vector fields

$$X_1=\partial_x,\qquad X_2=x(x^{2\ell}+z^2)\partial_z$$

on $\mathbb R^2$, is the Laplace-Beltrami operator on the regular region essentially self-adjoint for every $\ell\geq1$?

来源字段：https://doi.org/10.4171/owr/2019/47

11393 | OWR-4538-001

### Semigroup Generation by Sums of Unbounded Operators

O / open | Oberwolfach 报告问题汇编

For two unbounded linear operators $A$ and $B$ on a Banach space, under what conditions, and for which choice of domains, is $A+B$ the generator of a $C_0$-semigroup?

来源字段：https://doi.org/10.4171/owr/2010/56

11394 | OWR-13350-005

### Semigroup Uniqueness for Dirichlet Operators

O / open | Oberwolfach 报告问题汇编

Is the Friedrichs extension the only closed extension of $L_0$ generating a $C_0$-semigroup on $L^p(\mu)$ for $p\geq1$, equivalently, is $L_0$ essentially self-adjoint when $p=2$?

来源字段：https://doi.org/10.4171/owr/2014/48

11395 | AMR-119-0001

### Several Complex Variables Problem List - Problem 1

O / open | AMR 问题清单汇编

{\bf (New) (See Section 7.2.) Let $V$ be a germ at $0$ of an irreducible complex variety in $\mathbb{C}^n.$ For any small enough $\epsilon > 0$, let $E= E_\epsilon:= V \cap \mathbb{R}^n \cap \{|z|< \epsilon \}.$ Define the extremal function $U_E$ on $\{ |z|< \epsilon \}$: U_E(z):= \sup \{u(z); u \leq 0\; \mbox{on}\; E,\; 0 \leq u \leq 1 \; \mbox{on} \; \{|z|< \epsilon \}$ The problem is to classify those $V$ for which we have a constant $A>0$ for which $(\alpha) \; U_E(z) \leq A | \Im z |,\; |z|< \epsilon/2.$}

背景链接：https://arxiv.org/abs/math/9509202

11396 | AMR-119-0010

### Several Complex Variables Problem List - Problem 10

O / open | AMR 问题清单汇编

Let $X$ be a complex subvariety of the unit ball $\mathbb{B}$ with an isolated singularity at the origin. Suppose $\Omega_t$ is a continuously increasing family of Stein open subsets of $X$. Also, suppose that $0 \in \partial \Omega_{1/2} , 0\in \Omega_t, t>1/2.$ Is $\Omega_s$ Runge in $\Omega_t$ $\forall s<t$?

背景链接：https://arxiv.org/abs/math/9509202

11397 | AMR-119-0011

### Several Complex Variables Problem List - Problem 11

O / open | AMR 问题清单汇编

Let $K^{compact} \subset X \backslash \lbrace 0 \rbrace$, $X$ as above, $0 \in \hat{K}$, the polynomially convex hull of $K$. Suppose that $L \subset X$, $L$ contains a neighborhood of $K$ in $X$. Does $\hat{L}$ contain a neighborhood of $0$ in $X$? (This is equivalent to the previous problem.)\\

背景链接：https://arxiv.org/abs/math/9509202

11398 | AMR-119-0012

### Several Complex Variables Problem List - Problem 12

O / open | AMR 问题清单汇编

Let $\lbrace U_n \rbrace$ be an increasing sequence of Stein open subsets of a Stein Space $X$. Is $\bigcup_{n=1}^{\infty} U_n$ Stein? (If X is a complex manifold, not necessarily Stein, the answer is in general no.)\\

背景链接：https://arxiv.org/abs/math/9509202

11399 | AMR-119-0013

### Several Complex Variables Problem List - Problem 13

O / open | AMR 问题清单汇编

Suppose $M$ is a complex manifold of dimension 2, and suppose $\forall K^{compact} \subset M$ there exists an open subset $U$, $K \subset U \subset M$ such that $U$ is biholomorphic to the unit ball. Does this imply that M is equivalent to the unit ball $\mathbb{B}$, $\Delta \times \mathbb{C}$ or $\mathbb{C}^2$? (This is false in $\mathbb{C}^3$).\\

背景链接：https://arxiv.org/abs/math/9509202

11400 | AMR-119-0014

### Several Complex Variables Problem List - Problem 14

O / open | AMR 问题清单汇编

Is ``long'' $\mathbb{C}^2$ biholomorphic to $\mathbb{C}^2$? ( A complex manifold is a long $\mathbb{C}^2$ if it is the union of proper subsets that are biholomorphic to $\mathbb{C}^2$.)\\

背景链接：https://arxiv.org/abs/math/9509202

11401 | AMR-119-0015

### Several Complex Variables Problem List - Problem 15

O / open | AMR 问题清单汇编

Let $\Omega \subset \subset \mathbb{C}^n$ be a smooth pseudoconvex domain. Let $A^k(\bar\Omega)= H(\Omega)\cap C^k(\bar\Omega)$ with $C^k$ topology, $1 \leq k \leq \infty$. Is $A^{\infty}(\bar\Omega)$ dense in $A^k(\bar\Omega)$?\\ -N. Sibony, sibony@anh.matups.fr

背景链接：https://arxiv.org/abs/math/9509202

11402 | AMR-119-0016

### Several Complex Variables Problem List - Problem 16

O / open | AMR 问题清单汇编

Let $\Omega \subset \subset \mathbb{C}^n$ be a smooth pseudoconvex domain. Assume that $\bar\Omega$ has a Stein neighborhood basis. Is every function in $A^{\infty}(\bar\Omega)$ uniformly approximable by holomorphic functions in a neighborhood of $\bar\Omega$?\\ -N. Sibony, sibony@anh.matups.fr

背景链接：https://arxiv.org/abs/math/9509202

11403 | AMR-119-0017

### Several Complex Variables Problem List - Problem 17

O / open | AMR 问题清单汇编

Let $f_1, ..., f_k$ be bounded holomorphic functions on $\Omega.$ Suppose $\sum |f_i|> c> 0.$ The Corona problem is whether there exist bounded holomorphic funcions $g_1, ..., g_k$ on $\Omega$ such that $\sum f_i g_i \equiv 1.$ Can the Corona problem be solved if $\Omega$ is the unit ball or the unit polydisc?

背景链接：https://arxiv.org/abs/math/9509202

11404 | AMR-119-0018

### Several Complex Variables Problem List - Problem 18

O / open | AMR 问题清单汇编

\ \ \ Let $\Phi =\{ \Phi^0, \Phi^1, \Phi^2, \Phi^3\}$ be a subset of the body $(=skew-field) \ \mathbb{H}$ of the real quaternions with the conditions $$ \Phi^p \bar\Phi^q + \Phi^q \bar\Phi^p = 2 \delta_{p,q}, $$ $\Omega$ is a domain in $\mathbb{R}^4=\mathbb{C}^2$. On the $\mathbb{H}$-bimodule $C^1(\Omega ; \mathbb{H})$ the operator $D_\Phi = \sum^3_{k=0} \Phi^k \frac \partial {\partial x_k} $ defines the set of the $(left-\Phi)$-hyper-holomorphic functions $M_\Phi (\Omega ; \mathbb{H}) = Ker D_\Phi$. Let $\hat \Phi =\{ 1, i, j, -k\}$ with i, j, k being the ordinary imaginary units in $\mathbb{H}$. It is known that $$ Hol (\Omega ; \mathbb{C}^2) \subset M_{\hat \Phi} (\Omega ; \mathbb{H}), $$ where $Hol (\Omega ; \mathbb{C}^2)$ is the subset of all holomorphic (in the usual sense) mappings. 1) Describe the set of all hyper-holomorphic but non-holomorphic functions in $\Omega$ (it is not empty). 2) Describe "locations" of the sets $Hol$ and $M_\Phi Hol$ inside $M_\Phi$. 3) How do they depend on $\Omega$?

-H. Shapiro and N. Vasilievsky, matemat@cinvesmx.bitnet

背景链接：https://arxiv.org/abs/math/9509202

11405 | AMR-119-0019

### Several Complex Variables Problem List - Problem 19

O / open | AMR 问题清单汇编

Is every boundary point of a bounded pseudoconvex domain $D$ in $\mathbb{C}^n$ of finite type a peak point for $A(D)$?

背景链接：https://arxiv.org/abs/math/9509202

11406 | AMR-119-0002

### Several Complex Variables Problem List - Problem 2

O / open | AMR 问题清单汇编

{\bf (New) (See Section 8.2.) What is the lowest degree of a polynomial mapping $P: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ for which the (real) Jacobian conjecture fails, i.e. the Jacobian of $P$ near vanishes while $p$ fails to be invertible.}

背景链接：https://arxiv.org/abs/math/9509202

11407 | AMR-119-0020

### Several Complex Variables Problem List - Problem 20

O / open | AMR 问题清单汇编

Describe peak sets on weakly pseudoconvex domains of finite type.

背景链接：https://arxiv.org/abs/math/9509202

11408 | AMR-119-0021

### Several Complex Variables Problem List - Problem 21

O / open | AMR 问题清单汇编

Characterize the representing measures for the origin in the unit ball in $\mathbb{C}^n.$

背景链接：https://arxiv.org/abs/math/9509202

11409 | AMR-119-0022

### Several Complex Variables Problem List - Problem 22

O / open | AMR 问题清单汇编

Let $X \subset \mathbb{C}^n$ be compact and totally disconnected. Assume that $X$ has finite one-dimensional Hausdorff measure. Then is $X$ polynomially convex?\\ -H. Alexander, U22330@UICVM.bitnet

背景链接：https://arxiv.org/abs/math/9509202

11410 | AMR-119-0023

### Several Complex Variables Problem List - Problem 23

O / open | AMR 问题清单汇编

Let $\Omega \subset \mathbb{C}^n$ be a smoothly bounded, pseudoconvex domain. Let $V \subseteq \Omega$ be a divisor which is bounded away from the Silov boundary in $\partial \Omega.$ Prove that there is a bounded holomorphic function on $\Omega$ that vanishes on $V.$ (This is known for the polydisc---a result of Rudin.)\\ -S. Krantz, sk @math.wustl.edu

背景链接：https://arxiv.org/abs/math/9509202

11411 | AMR-119-0024

### Several Complex Variables Problem List - Problem 24

O / open | AMR 问题清单汇编

Give a geometric characterization of zero sets of $H^p$ functions on strongly pseudoconvex domains.

背景链接：https://arxiv.org/abs/math/9509202

11412 | AMR-119-0025

### Several Complex Variables Problem List - Problem 26

O / open | AMR 问题清单汇编

Solve $\bar\partial \; Neumann$ in $L^2 (d\mu)$, $d\mu$ a measure, even on the unit disc.

背景链接：https://arxiv.org/abs/math/9509202

11413 | AMR-119-0026

### Several Complex Variables Problem List - Problem 27

O / open | AMR 问题清单汇编

Let $\Omega$ be a smooth bounded planar domain and let $T:L^2(\Omega)\mapsto L^2(\Omega)$ be the Kohn solution operator for $\partial/\partial\bar z$. For $h\in L^\infty(\Omega)$ we define the compact non-self-adjoint operator $S_h:L^2(\Omega)\mapsto L^2(\Omega), f\mapsto \bar hTf$. Under what circumstance will $S_h$ have eigenvalues? Consider in particular the case where $h$ is holomorphic. If $h$ has a single-valued holomorphic primitive $H$ then it is not hard to see that $S_h$ has no eigenvalues. (Just note that $e^{-\bar H/\lambda}Tf$ is holomorphic and orthogonal to holomorphic functions.) Does the converse hold? That is: do periods of $H$ guarantee the existence of eigenvalues? (For annuli, the answer is``yes.")\\ -D. Barrett, barrett@math.lsa.umich.edu

背景链接：https://arxiv.org/abs/math/9509202

11414 | AMR-119-0027

### Several Complex Variables Problem List - Problem 28

O / open | AMR 问题清单汇编

If D is a real analytically bounded pseudoconvex domain in $\mathbb{C}^n$, is the $\bar\partial$ Neumann problem globally real analytic hypoelliptic up to the boundary?\\ - D. S. Tartakoff, U22393@UICVM.bitnet

背景链接：https://arxiv.org/abs/math/9509202

11415 | AMR-119-0028

### Several Complex Variables Problem List - Problem 29

O / open | AMR 问题清单汇编

If M is a compact real analytic pseudoconvex CR manifold of real dimension at least 5 is $\Box_b$ globally real analytic hypoelliptic (say on (0,1) forms) on M?\\ - D. S. Tartakoff, U22393@UICVM.bitnet

背景链接：https://arxiv.org/abs/math/9509202

11416 | AMR-119-0003

### Several Complex Variables Problem List - Problem 3

O / open | AMR 问题清单汇编

**(New) (See Section 11.3) (Liouville type assertion on CR 3-manifolds) On a CR 3-manifold one can always define a Carnot-Caratheodory metric. A homeomorphism between two strongly pseudoconvex CR 3-manifolds is called conformal if it maps infinitesimal spheres with respect to a Carnot-Caratheodory metric to infinitesimal spheres. Is a conformal homeomorphism CR? -Puqi Tang, tang@math.purdue.edu**

背景链接：https://arxiv.org/abs/math/9509202

11417 | AMR-119-0029

### Several Complex Variables Problem List - Problem 30

O / open | AMR 问题清单汇编

Can one solve $\bar\partial$ with H\"{o}lder estimates on any real analytic bounded pseudoconvex domain in $\mathbb{C}^n$?\\

背景链接：https://arxiv.org/abs/math/9509202

11418 | AMR-119-0030

### Several Complex Variables Problem List - Problem 31

O / open | AMR 问题清单汇编

Solve $\bar\partial$ in $L^p$ in bounded pseudoconvex domains, without necessarily smooth boundary, $1 < p < 2$.

背景链接：https://arxiv.org/abs/math/9509202

11419 | AMR-119-0031

### Several Complex Variables Problem List - Problem 32

O / open | AMR 问题清单汇编

Solve $\bar\partial$ with $L^p$ estimates, $1< p< \infty$ for smooth convex domains in $\mathbb{C}^n$.

背景链接：https://arxiv.org/abs/math/9509202

11420 | AMR-119-0032

### Several Complex Variables Problem List - Problem 33

O / open | AMR 问题清单汇编

Can one solve $\bar \partial u = f$ in sup norm on smooth bounded convex domains in $\mathbb{C}^2$?\\

背景链接：https://arxiv.org/abs/math/9509202

11421 | AMR-119-0033

### Several Complex Variables Problem List - Problem 34

O / open | AMR 问题清单汇编

Which geometric properties of the domain do uniform estimates depend on? $\lbrack e.g. L^2$ estimates depend on the diameter and the dimension $\rbrack$.

背景链接：https://arxiv.org/abs/math/9509202

11422 | AMR-119-0034

### Several Complex Variables Problem List - Problem 35

O / open | AMR 问题清单汇编

If $\Omega \subset \subset \mathbb{C}^n$ is pseudoconvex with smooth boundary, does $\bar \partial: \Lambda^{\alpha}(\bar \Omega) \mapsto \Lambda_{(0, 1)}^{\beta} (\bar \Omega)$ have closed range if $\alpha < \beta$?

背景链接：https://arxiv.org/abs/math/9509202

11423 | AMR-119-0035

### Several Complex Variables Problem List - Problem 36

O / open | AMR 问题清单汇编

Let $\Omega$ in $\mathbb{C}^n$ be a bounded biholomorphic image of the unit ball in $\mathbb{C}^n$. Does the $\bar\partial$ satisfy uniform estimates on $\Omega$? The problem is of interest mainly when $\Omega$ has very rough boundary.\\ -B. Berndtsson, bob@math.chalmers.se

背景链接：https://arxiv.org/abs/math/9509202

11424 | AMR-119-0036

### Several Complex Variables Problem List - Problem 37

O / open | AMR 问题清单汇编

Does there exist a pseudoconvex domain in $\mathbb{C}^2$ with the same homotopy type as $S^2$?\\ -F. Forstneric, forstner@math.wisc.edu

背景链接：https://arxiv.org/abs/math/9509202

11425 | AMR-119-0037

### Several Complex Variables Problem List - Problem 38

O / open | AMR 问题清单汇编

Is B complete on bounded pseudoconvex domains with less than $C^1$ boundary?\\ -J. McNeal, mcneal@math.princeton.edu

背景链接：https://arxiv.org/abs/math/9509202

11426 | AMR-119-0038

### Several Complex Variables Problem List - Problem 39

O / open | AMR 问题清单汇编

For which domains is the Bergman kernal function $K(z, w)$ nonvanishing everywhere? This is related to Lu Qi-Keng conjecture.

背景链接：https://arxiv.org/abs/math/9509202

11427 | AMR-119-0004

### Several Complex Variables Problem List - Problem 4

O / open | AMR 问题清单汇编

**(New) (See Section 11.3) A diffeomorphism between two strongly pseudoconvex CR $(2n+1)$-manifolds is called quasiconformal if its differential preserves the underlying contact structures and distorts the CR structures boundedly. When Hermitian metrics on the contact bundles are fixed, this distortion can be measured by checking how spheres in the contact space is mapped to ellipsoids. However, this measurement depends on the choices of the Hermitian metrics if $n>1$. Fixing a quasiconformal diffeomorphism, can we choose Hermitian metrics so that the distortion is minimal? - Puqi Tang, tang@math.purdue.edu**

背景链接：https://arxiv.org/abs/math/9509202

11428 | AMR-119-0039

### Several Complex Variables Problem List - Problem 41

O / open | AMR 问题清单汇编

Is there a smoothly bounded pseudoconvex domain in $\mathbb{C}^n$ on which the Kobayashi and Caratheodory metrics are not comparable?

-S. Krantz, SK@math.wustl.edu

背景链接：https://arxiv.org/abs/math/9509202

11429 | AMR-119-0040

### Several Complex Variables Problem List - Problem 42

O / open | AMR 问题清单汇编

Is the regular order of contact of a smooth pseudoconvex domain the same as the commutator type, one vector field at a time.\\ -J. McNeal, mcneal@math.princeton.edu

背景链接：https://arxiv.org/abs/math/9509202

11430 | AMR-119-0041

### Several Complex Variables Problem List - Problem 43

O / open | AMR 问题清单汇编

Are finite type pseudoconvex domains in $\mathbb{C}^2$ locally convexifiable with biholomorphic maps with continuous boundary values?

背景链接：https://arxiv.org/abs/math/9509202

11431 | AMR-119-0042

### Several Complex Variables Problem List - Problem 45

O / open | AMR 问题清单汇编

What's the asymptotic behavior of k on finite type domains in $\mathbb{C}^n$?

背景链接：https://arxiv.org/abs/math/9509202

11432 | AMR-119-0043

### Several Complex Variables Problem List - Problem 46

O / open | AMR 问题清单汇编

Let $D = \lbrace z \in \mathbb{C}^n ; h(z) < 1 \rbrace$ be a bounded pseudoconvex complete$-$circular domain with Minkowski function h. Characterize the Caratheodory, the Bergman or the Kobayashi completeness of D via properties of h. [The following is known: a) If h is continuous, then D is Bergman complete. b) If $n\geq 3$, continuity of h does not imply the Kobayashi completeness of D.]\\ -P. Pflug, pflugvec@dosuni1.bitnet

背景链接：https://arxiv.org/abs/math/9509202

11433 | AMR-119-0044

### Several Complex Variables Problem List - Problem 47

O / open | AMR 问题清单汇编

Given a compact complex hypersurface X in $\mathbb{P}^n$ with degree at least $n+2$ and only normal crossing singularities. Show that the infinitesimal Kobayashi pseudometric on $\mathbb{P}^n \setminus X$ is degenerate at most on a complex hypersurface.\\ -N. Sibony, sibony@anh.matups.fr

背景链接：https://arxiv.org/abs/math/9509202

11434 | AMR-119-0045

### Several Complex Variables Problem List - Problem 48

O / open | AMR 问题清单汇编

Find (or prove it does not exist) a complete hyperbolic manifold not biholomorphic to a convex domain where the Kobayashi and Carath\'eodory distances agree everywhere.\\ -M. Abate, abate@vaxsns.infn.it

背景链接：https://arxiv.org/abs/math/9509202

11435 | AMR-119-0046

### Several Complex Variables Problem List - Problem 49

O / open | AMR 问题清单汇编

Can you recover certain derivatives of the Levi form at a weakly pseudoconvex point in terms of the weighted boundary behavior of the Kobayashi metric?\\ -S. Krantz, sk@math.wustl.edu

背景链接：https://arxiv.org/abs/math/9509202

11436 | AMR-119-0005

### Several Complex Variables Problem List - Problem 5

O / open | AMR 问题清单汇编

Let M be a smooth real hypersurface in $\mathbb{C}^n$ with defining function r. Let $p_0$ be a point in M, and let U be a neighborhood in $\mathbb{C}^n$ of $p_0$. When do C-R functions on $M\cap U$ extend to $\lbrace r > 0 \rbrace$?\\ -B. Stensones, berit@math.lsa.umich.edu

背景链接：https://arxiv.org/abs/math/9509202

11437 | AMR-119-0047

### Several Complex Variables Problem List - Problem 50

O / open | AMR 问题清单汇编

Let $\Omega \subset \subset \mathbb{C}^n$ be pseudoconvex with smooth boundary. For $K \subset \bar\Omega$ compact, let $\hat K:= \lbrace z \in \bar \Omega ; \phi(z) \leq sup_K \phi, \phi$ continuous on $\bar \Omega, \phi$ plurisubharmonic on $\Omega \rbrace$. Assume $\hat K \cap \partial \Omega = K \cap \partial \Omega$. Then $\bar \Omega$ has a Stein neighborhood basis.\\ -N. Sibony, sibony@anh.matups.fr

背景链接：https://arxiv.org/abs/math/9509202

11438 | AMR-119-0048

### Several Complex Variables Problem List - Problem 51

O / open | AMR 问题清单汇编

Find conditions insuring that if a sequence of plurisubharmonic functions $u_m \rightarrow u$ in $\mathbb{C}^n$ then $(dd^c u_m)^n \rightarrow (dd^c u)^n$.

背景链接：https://arxiv.org/abs/math/9509202

11439 | AMR-119-0049

### Several Complex Variables Problem List - Problem 52

O / open | AMR 问题清单汇编

{\bf (New) Let $V$ be a germ at $0$ of an irreducible complex variety in $\mathbb{C}^n.$ For any small enough $\epsilon > 0$, let $E= E_\epsilon:= V \cap \mathbb{R}^n \cap \{|z|< \epsilon \}.$ Define the extremal function $U_E$ on $\{ |z|< \epsilon \}: U_E(z):= \sup \{u(z); u \leq 0\; \mbox{on}\; E,\; 0 \leq u \leq 1 \; \mbox{on} \; \{|z|< \epsilon \}$ The problem is to classify those $V$ for which we have a constant $A>0$ for which $(\alpha) \; U_E(z) \leq A | \Im z |,\; |z|< \epsilon/2.$}

背景链接：https://arxiv.org/abs/math/9509202

11440 | AMR-119-0050

### Several Complex Variables Problem List - Problem 53

O / open | AMR 问题清单汇编

Which smooth domains have non compact automorphism groups?\\

背景链接：https://arxiv.org/abs/math/9509202

11441 | AMR-119-0051

### Several Complex Variables Problem List - Problem 54

O / open | AMR 问题清单汇编

If P: $\mathbb{C}^2 \rightarrow \mathbb{C}^2$ is a holomorphic polynomial map with nonvanishing Jacobian, then P is biholomorphic.

背景链接：https://arxiv.org/abs/math/9509202

11442 | AMR-119-0052

### Several Complex Variables Problem List - Problem 55

O / open | AMR 问题清单汇编

{\bf (New) What is the lowest degree of a polynomial mapping $P: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ for which the (real) Jacobian conjecture fails, i.e. the Jacobian of $P$ never vanishes while $p$ fails to be invertible.}

背景链接：https://arxiv.org/abs/math/9509202

11443 | AMR-119-0053

### Several Complex Variables Problem List - Problem 56

O / open | AMR 问题清单汇编

Classify Proper Holomorphic Correspondences from the unit disc to itself.

背景链接：https://arxiv.org/abs/math/9509202

11444 | AMR-119-0054

### Several Complex Variables Problem List - Problem 57

O / open | AMR 问题清单汇编

Is every proper holomorphic self-mapping of a smooth bounded domain in $\mathbb{C}^n (n>1)$ biholomorphic?

背景链接：https://arxiv.org/abs/math/9509202

11445 | AMR-119-0055

### Several Complex Variables Problem List - Problem 58

O / open | AMR 问题清单汇编

Does f have a continuous extension to $\bar U$? (This is true if U, V are pseudoconvex.)

背景链接：https://arxiv.org/abs/math/9509202

11446 | AMR-119-0056

### Several Complex Variables Problem List - Problem 59

O / open | AMR 问题清单汇编

If M is an n-dimensional parallellizable Stein manifold, can M be holomorphically immersed in $\mathbb{C}^n$? [ True for n=1 (Gunning and Narasimhan); also such an M can be immersed in $\mathbb{C}^{n+1}$ (Eliashberg-Gromov)]. \\ -R. Narasimhan, Dept. of Math., Univ. of Chicago, Chicago, IL 60637

背景链接：https://arxiv.org/abs/math/9509202

11447 | AMR-119-0006

### Several Complex Variables Problem List - Problem 6

O / open | AMR 问题清单汇编

Let D be a bounded domain in $\mathbb{C}^n$ with smooth boundary, with envelope of holomorphy $D^{\sim}$. A)Is $D^{\sim}$ finitely sheeted over $\mathbb{C}^n$? B) Does $D^{\sim}$ have finite volume?\\ -B. Stensones, berit@math.lsa.umich.edu.

背景链接：https://arxiv.org/abs/math/9509202

11448 | AMR-119-0057

### Several Complex Variables Problem List - Problem 60

O / open | AMR 问题清单汇编

Can $SL(n, \mathbb{C})$ be holomorphically immersed in $\mathbb{C}^{n^2-1}$. [Cannot be done algebraically, true if n=2]\\ -R. Narasimhan, Dept. of Math., Univ. of Chicago, Chicago, IL 60637

背景链接：https://arxiv.org/abs/math/9509202

11449 | AMR-119-0058

### Several Complex Variables Problem List - Problem 61

O / open | AMR 问题清单汇编

Can the algebraic hypersurface $$ w^3 + w g(w_1, ....., w_n) + f(w_1, ....., w_n) = 1, $$ where g is homogeneous of degree 2 and f homogeneous of degree 3, be holomorphically immersed in $\mathbb{C}^n$ [ Not true for algebraic immersion]\\ -R. Narasimhan, Dept. of Math., Univ. of Chicago, Chicago, IL 60637

背景链接：https://arxiv.org/abs/math/9509202

11450 | AMR-119-0059

### Several Complex Variables Problem List - Problem 62

O / open | AMR 问题清单汇编

Suppose that the polynomial equation $$F(w_0, w_1, ....., w_n)=1$$ defines a smooth affine algebraic hypersurface (these are always parallellizable (Murthy, Swan, n=2; Suslin for general n). Can they be holomorphically immersed in $\mathbb{C}^n$?\\ -R. Narasimhan, Dept. of Math., Univ. of Chicago, Chicago, IL 60637

背景链接：https://arxiv.org/abs/math/9509202

11451 | AMR-119-0060

### Several Complex Variables Problem List - Problem 63

O / open | AMR 问题清单汇编

Do there exist Fatou--Bieberbach domains which are not Runge? A Fatou--Bieberbach domain is a proper subdomain of $\mathbb{C}^2$ which is biholomorphic to $\mathbb{C}^2$.\\ -F. Forstneric, forstner@math.wisc.edu

背景链接：https://arxiv.org/abs/math/9509202

11452 | AMR-119-0061

### Several Complex Variables Problem List - Problem 64

O / open | AMR 问题清单汇编

Do there exist wandering domains for complex Henon maps?

背景链接：https://arxiv.org/abs/math/9509202

11453 | AMR-119-0062

### Several Complex Variables Problem List - Problem 65

O / open | AMR 问题清单汇编

The Fatou Set of a Polynomial Automorphism of $\mathbb{C}^2$. Let $f$ be a polynomial automorphism of $\mathbb{C}^2$, and suppose that $f$ is not conjugate to an elementary map. Let $\Omega$ be a connected component of the set of points where the forward iterates $\lbrace f^n:n=1,2,3,...\rbrace $ are locally bounded, which is the Fatou set of $f$. Let us suppose that $\Omega$ is periodic, i.e. $f^n\Omega=\Omega$ for some $n\ne0$. Let us first consider the case in which $\Omega$ is recurrent, i.e. $\Omega$ contains a point whose orbit does not converge to $\partial\Omega$. In this case $\Omega$ is either (a) a basin of attraction of a sink orbit, (b) the basin of attractin of a Siegel disk, or (c) the basin of a Herman ring. It is not hard to construct examples of (a) and (b).\\ (i) Can case (c) occur? Nothing at all seems to be known about periodic domains which are not recurrent.\\ (ii) If $\Omega$ is not recurrent, does there exist $P\in\partial\Omega$ such that $\lim_{k\to\infty}f^{nk}(Q)=P$ for all $Q\in\Omega$?\\ (iii) If $\Omega$ is not recurrent, does there exist $P\in\partial\Omega$ such that $f^nP=P$, and one of the eigenvalues of $DF(P)$ is $e^{2\pi ip/q}$?\\ -E. Bedford, bedford@iubacs.bitnet or bedford@ucs.indiana.edu.

背景链接：https://arxiv.org/abs/math/9509202

11454 | AMR-119-0063

### Several Complex Variables Problem List - Problem 66

O / open | AMR 问题清单汇编

Let $ F_c : (z,w) \rightarrow (z^2 +c - w, z)$, $c \in \mathbb{C}$ a symplectic automorphism of $\mathbb{C}^2$, i.e. $F_c$ is biholomorphic and $ F^*_c(dz \wedge dw) = dz \wedge dw$. Can there exist a $c \in \mathbb{C}$ and a periodic orbit $\{z_i\}^k_{i=0}$ for $F_c$ such that $z_0$ belongs to a Siegel domain ?\\ -M. Herman

背景链接：https://arxiv.org/abs/math/9509202

11455 | AMR-119-0064

### Several Complex Variables Problem List - Problem 67

O / open | AMR 问题清单汇编

Suppose $f$ is a holomorphic map on $\mathbb{P}^2$ of degree $d \geq 2$. Must $f$ have a repelling fixed point.\\

背景链接：https://arxiv.org/abs/math/9509202

11456 | AMR-119-0065

### Several Complex Variables Problem List - Problem 68

O / open | AMR 问题清单汇编

Does every holomorphic map $F:\mathbb{P}^k \rightarrow \mathbb{P}^k$ of degree at least two have a repelling periodic point? Do meromorphic maps have periodic orbits? -sibony@anh.matups.fr

背景链接：https://arxiv.org/abs/math/9509202

11457 | AMR-119-0066

### Several Complex Variables Problem List - Problem 69

O / open | AMR 问题清单汇编

Is the support of $\mu$ equal to the closure of the repelling periodic orbits?

背景链接：https://arxiv.org/abs/math/9509202

11458 | AMR-119-0007

### Several Complex Variables Problem List - Problem 7

O / open | AMR 问题清单汇编

If D is a strictly pseudoconvex domain in $\mathbb{C}^2$ diffeomorphic to $\mathbb{R}^4$ and A is an embedded analytic disc outside D with $bA \subset bD$, then there is an embedded smooth disc M in D, with bM=bA, and such that M is totally real except at one elliptic point. With such a configuration (M,A) in $\mathbb{C}^2$, does it follow that the envelope of holomorphy of M contains A ?\\ -F. Forstneric, forstner@math.wisc.edu

背景链接：https://arxiv.org/abs/math/9509202

11459 | AMR-119-0067

### Several Complex Variables Problem List - Problem 70

O / open | AMR 问题清单汇编

Does there exist a holomorphic map $F:\mathbb{P}^k \rightarrow \mathbb{P}^k$ with a wandering Fatou component $U$, i.e. $F^n(U) \bigcap F^m(U) = \emptyset $ for all $n \neq m$?

背景链接：https://arxiv.org/abs/math/9509202

11460 | AMR-119-0068

### Several Complex Variables Problem List - Problem 71

O / open | AMR 问题清单汇编

Classify the dynamics around a fixed point of a holomorphic map on $\mathbb{P}^k$ or even just defined in a neighborhood of p. -sibony@anh.matups.fr

背景链接：https://arxiv.org/abs/math/9509202

11461 | AMR-119-0069

### Several Complex Variables Problem List - Problem 72

O / open | AMR 问题清单汇编

Let $f:\mathbb{P}^2 \rightarrow \mathbb{P}^2$ be a holomorphic map of degree $d \geq 2$. Assume K is a totally invariant set. Let C denote the critical set of $f$. Assume $\overline{\bigcup_{n=1}^{\infty} f^n(C)} \bigcap K = \emptyset$. Is f hyperbolic on K?

背景链接：https://arxiv.org/abs/math/9509202

11462 | AMR-119-0070

### Several Complex Variables Problem List - Problem 73

O / open | AMR 问题清单汇编

Let $H_d$ denote the space of holomorphic maps $f:\mathbb{P}^2 \rightarrow \mathbb{P}^2$ of degree d. This is a finite dimensional space parametrized by the coefficients. Does the set of $ f\in H_d$ with infinitely many attractive basins have measure zero?

背景链接：https://arxiv.org/abs/math/9509202

11463 | AMR-119-0071

### Several Complex Variables Problem List - Problem 74

O / open | AMR 问题清单汇编

Let $f:\mathbb{P}^2 \rightarrow \mathbb{P}^2$ be a holomorphic map of degree $d \geq 2$ and let $\lambda_1 \leq \lambda_2$ be the Lyapunov exponents for the ergodic measure $\mu = T \wedge T$. Is $\lambda_1 >0$?

背景链接：https://arxiv.org/abs/math/9509202

11464 | AMR-119-0072

### Several Complex Variables Problem List - Problem 76

O / open | AMR 问题清单汇编

Is every compact reduced analytic space biholomorphically equivalent to a subvariety of a complex manifold? If so, can the manifold be chosen to be compact?\\ -E. L. Stout, stout@math.washington.edu

背景链接：https://arxiv.org/abs/math/9509202

11465 | AMR-119-0073

### Several Complex Variables Problem List - Problem 77

O / open | AMR 问题清单汇编

Let C be a smooth cubic in $\mathbb{P}^2$. Then $\mathbb{P}^2\setminus C$ is algebraically parallellizable, but cannot be algebraically immersed in $\mathbb{C}^2$. Can it be holomorhically immersed in $\mathbb{C}^2$? [ There are smooth cubics for which it is true . For related problems see the section called ``Stein manifolds''.]\\ -R. Narasimhan, Dept. of Math., Univ. of Chicago, Chicago, IL 60637

背景链接：https://arxiv.org/abs/math/9509202

11466 | AMR-119-0074

### Several Complex Variables Problem List - Problem 78

O / open | AMR 问题清单汇编

Consider the following statement: `` Let Y be a compact subset of $\mathbb{C}^2$ which lies over the circle $|\lambda|= 1$, and let U be an open neighborhood of K, the polynomially convex hull of Y. Then there exists a finitely sheeted Riemann surface S lying over $|\lambda| < 1$ as a branched covering such that $S\subset U$.'' Is this statement true for every choice of Y and U?\\ -H. Alexander, u22330@uicvm.bitnet and J. Wermer, Brown University.

背景链接：https://arxiv.org/abs/math/9509202

11467 | AMR-119-0075

### Several Complex Variables Problem List - Problem 79

O / open | AMR 问题清单汇编

Let M be a K$\ddot{a}$hler manifold. Assume that all curvatures are between $-a^2$ and $-b^2$. Construct nontrivial bounded holomorphic functions on M.\\ -N. Sibony, sibony@anh.matups.fr

背景链接：https://arxiv.org/abs/math/9509202

11468 | AMR-119-0008

### Several Complex Variables Problem List - Problem 8

O / open | AMR 问题清单汇编

Let $U$ be an open subset of a Stein space. If $U$ is locally Stein, is $U$ Stein?\\

背景链接：https://arxiv.org/abs/math/9509202

11469 | AMR-119-0076

### Several Complex Variables Problem List - Problem 80

O / open | AMR 问题清单汇编

Are strongly pseudoconvex CR surfaces of real dimension 5 locally embeddable?

背景链接：https://arxiv.org/abs/math/9509202

11470 | AMR-119-0077

### Several Complex Variables Problem List - Problem 81

O / open | AMR 问题清单汇编

Let $M \subset P^n$ be a compact odd dimensional CR manifold. Find conditions on M which ensure that M bounds a complex variety.(This is well understood in $\mathbb{C}^n$.)\\ -N.Sibony, sibony@anh.matups.fr

背景链接：https://arxiv.org/abs/math/9509202

11471 | AMR-119-0078

### Several Complex Variables Problem List - Problem 82

O / open | AMR 问题清单汇编

**(New) (Liouville type assertion on CR 3-manifolds) On a CR 3-manifold one can always define a Carnot-Caratheodory metric. A homeomorphism between two strongly pseudoconvex CR 3-manifolds is called conformal if it maps infinitesimal spheres with respect to a Carnot-Caratheodory metric to infinitesimal spheres. Is a conformal homeomorphism CR? \\ -Puqi Tang, tang@math.purdue.edu**

背景链接：https://arxiv.org/abs/math/9509202

11472 | AMR-119-0079

### Several Complex Variables Problem List - Problem 83

O / open | AMR 问题清单汇编

**(New) A diffeomorphism between two strongly pseudoconvex CR $(2n+1)$-manifolds is called quasiconformal if its differential preserves the underlying contact structures and distorts the CR structures boundedly. When Hermitian metrics on the contact bundles are fixed, this distortion can be measured by checking how spheres in the contact space is mapped to ellipsoids. However, this measurement depends on the choices of the Hermitian metrics if $n>1$. Fixing a quasiconformal diffeomorphism, can we choose Hermitian metrics so that the distortion is minimal?\\ - Puqi Tang, tang@math.purdue.edu**

背景链接：https://arxiv.org/abs/math/9509202

11473 | AMR-119-0080

### Several Complex Variables Problem List - Problem 84

O / open | AMR 问题清单汇编

Let $\Omega \subseteq \mathbb{C}^n$ be a bounded domain with complete Kobayashi metric. Is there a constant $C = C(p) > 0$ such that if $f$ is holomorphic and $p > 0$ then for all $z \in \Omega$ and all $r > 0$ small it holds that $$ |f(z)|^p \leq C \frac{1}{|B(z,r)|} \int_{B(z,r)} |f(\zeta)|^p dV(\zeta) . $$ Here $f$ is any holomorphic function and $B(z,R)$ is the Kobayashi metric ball. The measure being used is up for grabs. It can be Euclidean or one of the canonical measures associated with the Kobayashi metric construction (such as that due to Eisenman or Bun Wong). In case $\Omega$ is finite type in $\mathbb{C}^2$ or strongly pseudoconvex in any dimension then the results is true just because metric balls are comparable to polydiscs and then classical arguments of Stein (or see my SCV book) will suffice. In general I think that something like this should be true but I have no idea how to prove it.\\ -S. Krantz, sk@math.wustl.edu

背景链接：https://arxiv.org/abs/math/9509202

11474 | AMR-119-0081

### Several Complex Variables Problem List - Problem 85

O / open | AMR 问题清单汇编

Prove that if $\Omega$ is strongly pseudoconvex then there is an absolute constant $K > 0$ such that the integrated Kobayashi distance between any two points can be realized by a Kobayashi chain of $K$ discs. [For instance, on a convex domain, $K = 1.$] \\ Prize \$50 -S. Krantz. sk @math.wustl.edu

背景链接：https://arxiv.org/abs/math/9509202

11475 | AMR-119-0082

### Several Complex Variables Problem List - Problem 86

O / open | AMR 问题清单汇编

Prove that if $\Omega$ is a smooth bounded pseudoconvex domain in $\mathbb{C}^n$ which is finite type in the sense of D'Angelo, then it is of finite type in the sense of Kohn (ideal type).\\ Prize \$50 -J. McNeal, mcneal@math.princeton.edu

背景链接：https://arxiv.org/abs/math/9509202

11476 | AMR-119-0009

### Several Complex Variables Problem List - Problem 9

O / open | AMR 问题清单汇编

Let M be a compact complex manifold. Find some appropriate conditions on M such that if U is an open subset of M which is locally Stein, then U is Stein.\\ -N. Sibony, sibony@anh.matups.fr

背景链接：https://arxiv.org/abs/math/9509202

11477 | OWR-720-009

### Sharp $L^1$ Bounds for Approximate Convolution Idempotents

O / open | Oberwolfach 报告问题汇编

For symmetric $g\in L^1(\mathbb{R})$ with $\lVert g\rVert_1=1$, determine the infimum of $\lVert g*g-g\rVert_1$, and decide whether it is attained.

来源字段：https://doi.org/10.4171/owr/2004/10

11478 | AIM-SEVERAL_COMPLEX_VARIABLES-0046

### Sharp 2-jet determination and an accumulation obstruction

P / partially_solved | AIM 研讨会问题汇编

Let \(M\) be a connected generic real-analytic manifold in \(\mathbb C^n\), holomorphically nondegenerate and of finite type. For \(p\in M\), let \(k(p)\) be the least \(k\) for which \[ \operatorname{Aut}_M(p)\longrightarrow G^k(\mathbb C^n), \qquad f\longmapsto j_p^k f \tag{1.1} \] is injective. Does there exist such an \(M\subset\mathbb C^2\) and a sequence \(p_j\in M\) with \(k(p_j)\to\infty\)?

Comment: such an \(M\) exists in higher dimensions.

If \(M\) is only generically of finite type, does there exist an \(M\) as above with \(p_j\) converging to a point of \(M\)?

It contains the running header “AIM WORKSHOP ON COMPLEXITY OF MAPPINGS IN CR GEOMETRY 3” inside the phrase “of finite type” and flattens subscripts and superscripts in the jet map.

Here \(\operatorname{Aut}_M(p)\) is the stability group of germs at \(p\) of local biholomorphisms of the ambient space fixing \(p\) and preserving the germ \((M,p)\); \(G^k(\mathbb C^n)\) is the group of invertible \(k\)-jets after centering \(p\) at the origin. For real-analytic finite-type hypersurfaces, this agrees with the corresponding group of germs of real-analytic CR automorphisms.

The PDF really says “generic real analytic manifold,” not “hypersurface.” In \(\mathbb C^2\), however, a positive-dimensional generic submanifold of positive CR dimension and finite Bloom--Graham type must have real codimension one, hence is a hypersurface. A codimension-two generic submanifold has CR dimension zero and cannot generate its tangent bundle by CR brackets.

来源字段：https://aimath.org/WWN/complexitycrmap/complexitycrmap.pdf

11479 | OWR-14298587-020

### Sharp Bianchi-Egnell Stability Constant

O / open | Oberwolfach 报告问题汇编

Determine the optimal constant $c_{\mathrm{BE}}$ in the Bianchi-Egnell quantitative stability inequality for the Sobolev inequality.

来源字段：https://doi.org/10.4171/owr/2024/37

11480 | AIM-ANALYSIS-0065

### Sharp bounds and an elliptic-radial nullspace for the determinantal trilinear form

S / solved | AIM 研讨会问题汇编

Prove \[ \left|\int_{\mathbb R^2}\int_{\mathbb R^2} f(x)g(y)h(x+y)\frac{1}{\det(x,y)}\,dx\,dy\right| \lesssim \|f\|_p\|g\|_q\|h\|_r, \] for certain \(p,q,r\).

This question has a quite satisfactory answer. See Gressman et al.

来源字段：https://aimath.org/pastworkshops/multilinopsproblems.pdf

11481 | OWR-12723-004

### Sharp Bounds for Halton-Kronecker Sequence Discrepancy

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Halton-Kronecker Discrepancy Question [OWR-12723-003]

Does the discrepancy $D_N$ of the two-dimensional Halton-Kronecker sequence

$$z_n=(\phi_2(n),\{n\sqrt2\}),\qquad n\geq0,$$

where $\phi_2$ is the base-$2$ van der Corput sequence, satisfy $D_N=O((\log N)^{2+\varepsilon}/N)$ for every $\varepsilon>0$?

来源字段：https://doi.org/10.4171/owr/2013/49

11482 | OWR-17469-002

### Sharp Complexity Constants for Smooth Multivariate Integration

O / open | Oberwolfach 报告问题汇编

For integration on $C^k(D_d)$, does

$$\limsup_{n\to\infty} e_n(C^k(D_d),\operatorname{INT})\,n^{k/d}$$

depend on the bounded domain $D_d\subset\mathbb R^d$? Is the limsup in fact a limit?

来源字段：https://doi.org/10.4171/owr/2020/6

11483 | AIM-ANALYSIS-0062

### Sharp decay for bounded-degree chirps along null directions

P / partially_solved | AIM 研讨会问题汇编

\[ e^{i\lambda x_3^2}, \]

the exponent on the right is a positive \(\epsilon\), and the isolated `1` before “So far” is the page number. Thus the displayed question is

\[ \left|\int_B e^{i\lambda x_3^2} \prod_{j=1}^N f_j(x\cdot v_j)\,dx\right| \lesssim \lambda^{-\epsilon}\prod_{j=1}^N\|f_j\|_\infty, \tag{1.1} \]

where \(B\) is the unit ball in \(\mathbb R^3\).

This cannot be the intended CLTT datum, since CLTT use \(V_j\) for linear subspaces. The mathematically coherent reconstruction is corroborated by the exact model in Oliveira e Silva's exposition of the CLTT problem:

\[ \Gamma=\{v\in\mathbb R^3\setminus\{0\}:v_3^2=v_1^2+v_2^2\}, \qquad v_j\in\Gamma,\qquad V_j=\operatorname{span}(v_j), \tag{1.2} \]

with the lines \(V_j\) distinct and \(\pi_j(x)=x\cdot v_j\). Unit normalization of the \(v_j\) is harmless because it can be absorbed into \(f_j\). This reconstruction also exactly explains the source's statement that (1.1) was known for \(N\leq5\). All results below use (1.2), and no silent claim is made for the literal, undefined “different light cones” wording.

来源字段：https://aimath.org/pastworkshops/multilinopsproblems.pdf

11484 | OWR-15954-007

### Sharp Eigenvalue Bounds for Nonselfadjoint Jacobi Perturbations

O / open | Oberwolfach 报告问题汇编

Let $E\subset\mathbb R$ be a finite-gap set, let $J_0\in\mathcal T_E$, and let $J=J_0+\delta J$ be a compact nonselfadjoint Jacobi perturbation. Does the bound

$$\sum_{z\in\sigma_d(J)}\frac{\operatorname{dist}(z,E)^{p+\varepsilon}(1+\lvert z\rvert)^{1/2-2\varepsilon}}{\operatorname{dist}(z,\partial E)^{1/2}}\leq L_{\varepsilon,p,E}\sum_{n=-\infty}^{\infty}\left(\lvert\delta a_n\rvert^p+\lvert\delta b_n\rvert^p+\lvert\delta c_n\rvert^p\right)$$

remain valid for $\varepsilon=0$? Can the denominator $\operatorname{dist}(z,\partial E)^{1/2}$ be replaced by $\operatorname{dist}(z,E)^{1/2}$?

来源字段：https://doi.org/10.4171/owr/2017/49

11485 | OWR-15577-011

### Sharp Exponents in the Fourier Restriction Conjecture

O / open | Oberwolfach 报告问题汇编

Determine the sharp exponent range in the Fourier restriction inequality $$ \lVert\widehat f\rVert_{L^q(S^{d-1})}\leq C\lVert f\rVert_{L^p(\mathbb R^d)}, $$ especially when $q<2$.

来源字段：https://doi.org/10.4171/owr/2017/34

11486 | OWR-11695863-002

### Sharp Fractional Inradius Constants with Fixed Connectivity

O / open | Oberwolfach 报告问题汇编

For $0<s<1$ and $k\geq2$, determine the sharp constant $$ T_{s,k}=\inf\{\lambda_1^s(\Omega)r_\Omega^{2s}:\Omega\subset\mathbb R^2\text{ has connectivity }k\}. $$.

来源字段：https://doi.org/10.4171/owr/2022/53

11487 | OWR-14299582-002

### Sharp Hilbert-Transform Bounds for UMD Spaces

O / open | Oberwolfach 报告问题汇编

For a UMD Banach space $X$, let $h_p(X)$ be the $L^p(X)$ operator norm of the Hilbert transform and let $m_p(X)$ be the UMD-$p$ constant. Determine the sharp dependence of $h_p(X)$ on $m_p(X)$; in particular, decide whether $h_p(X)$ and $m_p(X)$ are comparable up to universal linear factors.

来源字段：https://doi.org/10.4171/owr/2025/53

11488 | AIM-ANALYSIS-0045

### Sharp Holder duality and a homogeneous-sector reduction for scaled Fock spaces

P / partially_solved | AIM 研讨会问题汇编

Define $\mathcal H_{p,\alpha}:=\{f\ \text{entire}: (\alpha p/(2\pi))^n(\int_{\mathbb C^n}|f|^p e^{-\alpha p\|z\|^2/2}\,dm)^{1/p}<\infty\}$. What is the best constant for the H\"older inequality between $\mathcal H_{p,\alpha}$ and $\mathcal H_{q,\alpha}$?

来源字段：http://aimpl.org/scvproblems/1/

11489 | OWR-13750335-005

### Sharp Homogeneous Weights in Fractional Hardy Inequalities

O / open | Oberwolfach 报告问题汇编

In the sharp Hardy inequality for the fractional sub-Laplacian $\mathcal L_s$ on $\mathbb H^n$, replace the homogeneous weight $w_s(z,a)$ by $|(z,a)|^{2s}$ while retaining the sharp constant.

来源字段：https://doi.org/10.4171/owr/2023/24

11490 | AIM-FUNCTIONAL_ANALYSIS-0009

### Sharp Kähler-angle classification for two-dimensional unitary projection averages

P / partially_solved | AIM 研讨会问题汇编

Consider a convex body in $\mathbb{C}^n$ and fix a real subspace $E$ of $\mathbb{C}^n$. Define $V_E(K) = \int_{U(n)} |K|\phi(E)|\,d\phi$, where $U(n)$ is a unitary group. Does Brunn--Minkowski type inequality hold for such quantities: \[ V_E(K+L)^{1/i}\geq V_E(K)^{1/i}+V_E(L)^{1/i}? \] Also, for what type of subspaces $E$ this will work?

来源字段：http://aimpl.org/symconvgeomineq/1/

11491 | OWR-4209-007

### Sharp Korn-Sobolev Constants and Shape-Preserving Approximation

P / partially_solved | Oberwolfach 报告问题汇编

For the endpoint Korn-Sobolev inequality $$\left(\int_{\mathbb R^n}|u|^{n/(n-1)}\,dx\right)^{(n-1)/n}\leq C\int_{\mathbb R^n}|\varepsilon(u)|\,dx$$ for compactly supported vector fields, determine the best constant and an isoperimetric interpretation. Also determine whether a $C^2$ approximation to a convex, respectively subharmonic, function can itself be chosen convex, respectively subharmonic.

来源字段：https://doi.org/10.4171/owr/2010/08

11492 | AIM-FUNCTIONAL_ANALYSIS-0016

### Sharp log-concavity threshold for Fubini--Study geodesic balls

P / partially_solved | AIM 研讨会问题汇编

For a smooth convex domain \(K\) in \(\mathbb R^n\), Laplace equation with Dirichlet boundary conditions is \[ \begin{cases} \Delta u=\lambda u&\text{ on }K,\\ u\equiv0&\text{ on }\partial K. \end{cases} \] Choose \(u_1>0\) in \(\operatorname{int}K\), then \[ \operatorname{Hess}\log u_1\leq0? \] True for \(K=B_2\). Question is whether it is true for complex projective space \(\mathbb CP^n\) or for other symmetric \(K\)'s.

来源字段：http://aimpl.org/symconvgeomineq/2/

11493 | OWR-12723-001

### Sharp Multidimensional Star-Discrepancy Asymptotics

O / open | Oberwolfach 报告问题汇编

Determine the sharp asymptotic order, as $N\to\infty$, of the minimum star discrepancy, equivalently the minimum $L^\infty$ discrepancy, of $N$-point sets in the unit half-open $s$-cube for $s\geq3$, and clarify its relation to the small-ball conjecture and Kolmogorov entropy.

来源字段：https://doi.org/10.4171/owr/2013/49

11494 | AIM-ANALYSIS-0027

### Sharp Neumann eigenvalue ratios and an all-index ordered-simplex family

P / partially_solved | AIM 研讨会问题汇编

Let \(\Omega\subset\mathbb R^n\) be a nonempty bounded open convex set, and list the Neumann eigenvalues, with multiplicity, as \[ 0=\mu _0(\Omega)<\mu _1(\Omega)\leq\mu _2(\Omega)\leq\cdots. \] Then \[ \tag{a}\frac{\mu _2(\Omega)}{\mu _1(\Omega)}\leq4, \qquad \tag{b}\frac{\mu _k(\Omega)}{\mu _1(\Omega)}\leq k^2. \]

来源字段：http://aimpl.org/shapesurface/3/

11495 | OWR-16168-004

### Sharp Normalized Steklov Supremum in Higher Dimensions

O / open | Oberwolfach 报告问题汇编

Among smooth bounded domains $\Omega\subset\mathbb R^n$ with $n\geq3$, determine the supremum of $$\sigma_1(\Omega)|\partial\Omega|^{1/(n-1)}$$ and characterize the domain, or degenerating sequence of domains, on which it is attained.

来源字段：https://doi.org/10.4171/owr/2018/27

11496 | AIM-ANALYSIS-0081

### Sharp root motion for a first-order stability preserver

P / partially_solved | AIM 研讨会问题汇编

How do operators that preserve stability affect locations of roots (i.e., quantitative versions of MSS)

* Petter Branden

**Question.** How do operators that preserve stability affect locations of roots (i.e. quantitative versions of Marcus--Spielman--Srivastava)?

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

11497 | AIM-ANALYSIS-0105

### Sharp scalar parameter stability and a finite square-root obstruction

P / partially_solved | AIM 研讨会问题汇编

**Dependence on parameters**

Do canonical Lipschitz extensions exist which depend nicely on parameters?

来源字段：http://aimpl.org/mappingmetric/1/

11498 | OWR-12005-007

### Sharp Shape-Dependent Bounds for Dirichlet Eigenvalues

O / open | Oberwolfach 报告问题汇编

How does the shape of a bounded planar starlike domain affect the eigenvalues of its Dirichlet Laplacian, and what geometrically sharp shape-dependent upper bounds hold?

来源字段：https://doi.org/10.4171/owr/2012/33

11499 | OWR-15577-006

### Sharp Weighted Bounds for Rough Singular Integrals

P / partially_solved | Oberwolfach 报告问题汇编

For a rough homogeneous singular integral $T_\Omega$ with $\Omega\in L^\infty(S^{n-1})$ and mean zero, can the weighted estimate $\lVert T_\Omega\rVert_{L^2(w)\to L^2(w)}\lesssim[w]_{A_2}^2$ be improved to linear dependence on $[w]_{A_2}$? Is the maximal singular integral $T_\Omega^\star$ of weak type $(1,1)$?

来源字段：https://doi.org/10.4171/owr/2017/34

11500 | OWR-15219-016

### Shortest Moment Extensions and Minimal Atomic Measures

O / open | Oberwolfach 报告问题汇编

Does a shortest convergent extension sequence for a truncated moment sequence $\beta^{(2d)}$ always produce a representing measure with the minimum possible number of atoms?

来源字段：https://doi.org/10.4171/owr/2017/14

11501 | OWR-736-001

### Shrinkage and Democracy of Warped Wavelet Bases

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Warped Wavelet-Basis Problem [OWR-736-002]; Duplicate Warped Wavelet-Basis Extraction [OWR-736-003]

For wavelet bases warped by a sufficiently regular function $G$, under what hypotheses are the shrinkage (unconditionality) and $p$-democracy properties of compactly supported wavelet bases preserved?

来源字段：https://doi.org/10.4171/owr/2004/34

11502 | AIM-ANALYSIS-0121

### Simultaneous Sobolev approximation for radial planar homeomorphisms

P / partially_solved | AIM 研讨会问题汇编

**Approximation of Sobolev homeomorphisms by diffeomorphisms.** Let \(f\) be a homeomorphism of the plane such that \(f\) and \(f^{-1}\) lie in some Sobolev class, e.g. \(W^{1,2}\). Can one approximate \(f\) by diffeomorphisms \(f_j\) so that \(f_j\) and \(f_j^{-1}\) converge in the Sobolev norm?

来源字段：http://aimpl.org/mappingmetric/4/

11503 | AIM-FUNCTIONAL_ANALYSIS-0050

### Single generation beyond Z-stability

P / partially_solved | AIM 研讨会问题汇编

A C*-algebra is called singly generated if it contains an element that is not contained in any proper sub-C*-algebra. A C*-algebra is called \(\mathcal{Z}\)-stable if it absorbs the Jiang-Su algebra tensorially.

Which separable, unital, simple C*-algebras are singly generated? Is there a simple, nuclear C*-algebra that is singly generated, yet is not \(\mathcal{Z}\)-stable? Is single generation connected to the regularity properties coming up in the classification of nuclear, simple C*-algebras?

来源字段：http://aimpl.org/settheorycstar/8/

11504 | OWR-12005-001

### Singular Coalescence Limits in Plasmonic Problems

P / partially_solved | Oberwolfach 报告问题汇编

As two domains in a plasmonic boundary-value problem coalesce with separation parameter $a\to0$, do solutions exist at $a=0$, how should the equations be modified there, and do qualitatively different branches such as localized or hybridized solutions occur?

来源字段：https://doi.org/10.4171/owr/2012/33

11505 | AMR-036-0021

### Small components of subharmonic level sets

P / partially_solved | AMR 问题清单汇编

Let subharmonic functions $u_k$ on the unit square converge uniformly to $u(x,y)=x$. If $D_k=\{u_k<0\}$ and $D_k^*$ is the component containing $-1/2$, can $D_k\setminus D_k^*$ meet every horizontal segment $[-1+it,1+it]$, $-1<t<1$?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

11506 | AIM-SEVERAL_COMPLEX_VARIABLES-0043

### Smooth reduced pullbacks under finite holomorphic maps

P / partially_solved | AIM 研讨会问题汇编

Let \(f:(\mathbb C^n,0)\to(\mathbb C^n,0)\) be a finite holomorphic map germ and let \(V\subset(\mathbb C^n,0)\) be a reduced complex-analytic variety germ. If the reduced inverse-image germ \(f^{-1}(V)_{\mathrm{red}}\) is a complex manifold, must \(V\) be a complex manifold?

The AIM PDF asks the following question, proposed by Linda Rothschild.

来源字段：https://aimath.org/WWN/complexitycrmap/complexitycrmap.pdf

11507 | OWR-15430-007

### Sobolev Sheaves on Subanalytic Sites

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a Sobolev sheaf complex $H^s$ on the subanalytic Grothendieck topology of a manifold $M$ such that, on every relatively compact Lipschitz open set $U$, $H^s(U)$ is concentrated in degree zero and equals the Sobolev space $H^s(U)$, particularly for $s\geq1/2$?

来源字段：https://doi.org/10.4171/owr/2017/22

11508 | AIM-FUNCTIONAL_ANALYSIS-0007

### Solved dual-quermassintegral inequality and a sharp translation threshold

S / solved | AIM 研讨会问题汇编

If \(K,L\subset\mathbb R^n\) are origin-symmetric convex bodies containing the origin in their interiors and \(i=2,\ldots,n-1\), prove \[ \widetilde V_i(K+L)^{1/i} \geq \widetilde V_i(K)^{1/i}+\widetilde V_i(L)^{1/i}. \tag{1.2} \]

来源字段：http://aimpl.org/symconvgeomineq/1/

11509 | AIM-ANALYSIS-0124

### Solved status and the null G_delta structure of completely nonrectifiable fibers

S / solved | AIM 研讨会问题汇编

**Quasiconformal mappings of the plane which destroy the rectifiability of uncountably many disjoint lines.** Is there a quasiconformal map \(f\) of the plane and an uncountable set \(E\subset\mathbb R\) so that \(f(E\times\mathbb R)\) contains no rectifiable curves?

来源字段：http://aimpl.org/mappingmetric/5/

11510 | OPG-37185

### Something like Picard for 1-forms

P / partially_solved | Open Problem Garden

Conjecture Let $D$ be the open unit disk in the complex plane and let $U_1,\dots,U_n$ be open sets such that $\bigcup_{j=1}^nU_j=D\setminus\{0\}$. Suppose there are injective holomorphic functions $f_j: U_j \to \mathbb{C},$ $j=1,\ldots,n,$ such that for the differentials we have ${\rm d}f_j={\rm d}f_k$ on any intersection $U_j\cap U_k$. Then those differentials glue together to a meromorphic 1-form on $D$.

背景链接：http://www.openproblemgarden.org/op/something_like_picard_for_1_forms

11511 | AIM-FUNCTIONAL_ANALYSIS-0004

## Source repair and exact Cartesian factorization for L_p-Wulff combinations

P / partially_solved | AIM 研讨会问题汇编

It reads, verbatim in the relevant place, \[ \bigcap_{u\in S^{n-1}}\left\{x\in\mathbb R^n:\ \left\leq \left(t h_K(u)^p+(1-t)h_L(u)^p\right)^{1/p}\right\}.

The intended expression is nevertheless unambiguous from the primary source of this formulation, Böröczky--Lutwak--Yang--Zhang (2012), equation (1.8): it is \[ \langle x,u\rangle\leq \left(t h_K(u)^p+(1-t)h_L(u)^p\right)^{1/p}. \tag{1.1} \] Thus, for \(p>0\), define the Wulff body \[ W_p(t;K,L):=\bigcap_{u\in S^{n-1}} \left\{x\in\mathbb R^n: \langle x,u\rangle\leq M_{p,t}(h_K(u),h_L(u))\right\}, \tag{1.2} \] where \[ M_{p,t}(a,b)=\left(t a^p+(1-t)b^p\right)^{1/p}. \] For \(p=0\), the displayed \(1/p\) expression is undefined. For \(0<t<1\), its limiting reconstruction on the positive quadrant is \[ M_{0,t}(a,b)=a^t b^{1-t}, \qquad W_0(t;K,L)=\bigcap_{u\in S^{n-1}} \{x:\langle x,u\rangle\leq h_K(u)^t h_L(u)^{1-t}\}. \tag{1.3} \] Indeed, for \(a,b>0\), taking logarithms and expanding \(a^p,b^p\) at \(p=0\) gives \(M_{p,t}(a,b)\to a^t b^{1-t}\). We extend this mean continuously to \([0,\infty)^2\): for \(0<t<1\), \(M_{0,t}(a,b)=0\) if \(ab=0\), including \(M_{0,t}(0,0)=0\). For every \(p\geq0\), the endpoint weights are defined separately by \[ M_{p,0}(a,b)=b,\qquad M_{p,1}(a,b)=a. \] This convention avoids every \(0^0\) ambiguity. For \(p>0\), the displayed power-mean formula already extends continuously to the closed nonnegative quadrant.

The recovered AIM conjecture is \[ |W_p(t;K,L)|\geq |K|^t|L|^{1-t}, \qquad 0\leq t\leq1,\quad p\geq0, \tag{1.4} \] for **origin-symmetric** convex bodies \(K=-K\), \(L=-L\). “Symmetric” cannot mean symmetry about arbitrary centers because the construction is tied to the origin.

For \(0<p<1\), the function in the right side of (1.1) need not itself be a support function; the intersection/Wulff operation is essential. For \(p\geq1\), it is a support function.

The AIM display (1.4) is pointwise weaker than the standard homogeneous \(L_p\)-Brunn--Minkowski inequality \[ |W_p(t;K,L)|^{p/n} \geq t|K|^{p/n}+(1-t)|L|^{p/n}, \qquad p>0, \tag{1.5} \] because weighted arithmetic--geometric mean applied to \(|K|^{p/n}\) and \(|L|^{p/n}\) turns (1.5) into (1.4). They should not be silently identified for one fixed triple \((K,L,t)\): their equality cases can differ. However, Proposition 5.1 below shows that their **universal** versions (all bodies and all weights, at a fixed \(p>0\)) are equivalent by homogeneity and volume normalization. At \(p=0\), (1.5) tends exactly to (1.4).

来源字段：http://aimpl.org/symconvgeomineq/1/

11512 | AIM-ANALYSIS-0068

## Sparse-degree discrete restriction and a finite random-phase theorem

P / partially_solved | AIM 研讨会问题汇编

**Question 9 (Li).** Let \(d\geq 3\). For \(p\geq 2(d+1)\), prove \[ \left\|\sum_{n=1}^{N}a_n e^{2\pi i n^d t}e^{2\pi i n\cdot x}\right\|_{L^p(\mathbb T^2)} \lesssim N^{\frac12-\frac{d+1}{p}+\varepsilon} \left(\sum_{n=1}^{N}|a_n|^2\right)^{1/2}. \tag{0.11} \] This is related to Waring's problem.

来源字段：https://aimath.org/pastworkshops/multilinopsproblems.pdf

11513 | OWR-15437-002

## Spectra of Nonlinear Problems with Energy-Dependent Boundary Conditions

O / open | Oberwolfach 报告问题汇编

Are the observed spectral features-nontrivial discrete spectrum and, in irrational geometry, existence of a frequency limit point-general for nonlinear boundary-value problems with energy-dependent boundary conditions?

来源字段：https://doi.org/10.4171/owr/2017/29

11514 | OWR-14297746-012

## Spectra of Operators from Heisenberg Substitutions

P / partially_solved | Oberwolfach 报告问题汇编

For a nonperiodic configuration $\omega$ arising from a symbolic substitution system over the discrete Heisenberg group, determine the spectrum $\sigma(H_\omega)$ and its spectral distribution function.

来源字段：https://doi.org/10.4171/owr/2023/37

11515 | AIM-ANALYSIS-0140

### Spectral arbitrariness of the tridiagonal T_n sign pattern

S / solved | AIM 研讨会问题汇编

**Normalized statement.** For every \(n\geq2\) and every monic \(q\in\mathbb R[x]\) of degree \(n\), there is \(A\in Q(\mathcal S_n)\) such that \(\chi_A=q\).

Tyaglov and supplies no further status or hypotheses. In sign-pattern terminology, every \(+\) means a strictly positive real entry, every \(-\) a strictly negative real entry, and every \(0\) an exact zero. Denote this order-\(n\) sign pattern by \(\mathcal S_n\), and its qualitative class by \(Q(\mathcal S_n)\).

Two corrections are required to make the sentence literally true.

1. For every real \(n\times n\) matrix, \[ \det(A-\lambda I)=(-1)^n\lambda^n+\text{lower-degree terms}. \] Thus exact equality is possible only for degree-\(n\) polynomials with leading coefficient \((-1)^n\). Equivalently, the characteristic polynomial \(\chi_A(x)=\det(xI-A)\) can be any **monic** real polynomial of degree \(n\). An arbitrary nonzero real polynomial is a scalar multiple of such a determinant, not necessarily the determinant itself. 2. The fixed pattern starts at \(n=2\). At \(n=1\), its sole diagonal entry would have to be both positive and negative.

The recovered assertion is therefore:

This is the statement that \(\mathcal S_n\) is **spectrally arbitrary**.

来源字段：http://aimpl.org/hyperbolicpoly/2/

11516 | OWR-12005-003

### Spectral Determination of Toric Moment Polytopes

P / partially_solved | Oberwolfach 报告问题汇编

For a compact symplectic toric manifold $(M,g)$ with toric metric, does the Laplace spectrum determine its moment (Delzant) polytope and hence its symplectomorphism type?

来源字段：https://doi.org/10.4171/owr/2012/33

11517 | OWR-12584-002

### Spectral Measures of Perturbed Isospectral Tori

O / open | Oberwolfach 报告问题汇编

Characterize the spectral measures of all $\ell^2$ perturbations of points in the isospectral torus $T_E$. Do such perturbations preserve the absolutely continuous spectrum?

来源字段：https://doi.org/10.4171/owr/2013/36

11518 | OWR-710-003

### Spectral Methods for First-Kind Equations on Nonspherical Geometries

P / partially_solved | Oberwolfach 报告问题汇编

Prove stability and convergence of fully discrete spectral methods for first-kind equations on geometries other than the sphere.

来源字段：https://doi.org/10.4171/owr/2004/27

11519 | OWR-4990370-001

### Spectral Shapes of Group-Based Graph Operators

O / open | Oberwolfach 报告问题汇编

What shapes can occur as spectra of weighted Laplace or Markov operators on graphs arising from groups? Can a Cayley graph have Cantor spectrum? Can a Cayley graph of a torsion-free group have a spectral gap?

来源字段：https://doi.org/10.4171/owr/2021/2

11520 | OWR-738-004

### Spectral Subspaces of Measures on Infinite Groups

O / open | Oberwolfach 报告问题汇编

Let $G$ be an infinite discrete group, let $\mu \in M_0(G) \setminus \{0\}$ be a nonzero finitely supported (or $c_0$) measure on $G$, and let $m_\mu$ be the associated convolution operator on $\ell_p(G)$ ($1 < p < \infty$). Does the operator space $N_{|\mu|}^{(p)}$ coincide with the space $L_1^{(p)}(m_\mu)$?

来源字段：https://doi.org/10.4171/owr/2004/36

11521 | OWR-8415353-006

### Spectral-Gap Conjecture for Compact Simple Lie Groups

O / open | Oberwolfach 报告问题汇编

Let $\mu$ be a symmetric probability measure on a compact connected simple Lie group. Does $\mu$ have a spectral gap if and only if it is nondegenerate?

来源字段：https://doi.org/10.4171/owr/2021/50

11522 | OWR-17290-006

### Spectrum of Boundary Problems with Noncompact Multiplier Weights

O / open | Oberwolfach 报告问题汇编

Let $P$ be an $n$-term self-similar function of zero spectral order such that its generalized derivative $P'$ is a noncompact Sobolev multiplier. For the singular Sturm-Liouville problem

$$-y''=\lambda P'y,\qquad y(0)=y(1)=0,$$

does the spectrum consist of $n-1$ possibly overlapping intervals of continuous spectrum, with at most one eigenvalue in each gap and at most $n-2$ eigenvalues in total?

来源字段：https://doi.org/10.4171/owr/2019/47

11523 | OWR-14297746-006

### Spectrum of Randomly Perturbed Fibonacci Hamiltonians

O / open | Oberwolfach 报告问题汇编

Let $H$ be the Fibonacci Hamiltonian and let $W_\omega$ be a random perturbation. Determine the topological structure of the almost-sure spectrum of $H+W_\omega$.

来源字段：https://doi.org/10.4171/owr/2023/37

11524 | AIM-SEVERAL_COMPLEX_VARIABLES-0053

### Spherical CR boundaries and Veronese quotient singularities

P / partially_solved | AIM 研讨会问题汇编

Let \(M\subset\mathbb C^N\) be a compact real-analytic spherical CR manifold of hypersurface type, and suppose that \(M\) is the boundary of an analytic variety \(V\). What kinds of singularities can \(V\) have?

Problem 13 in the AIM workshop list *Complexity of mappings in CR geometry*, proposed by Xiaojun Huang, asks:

The source adds that in the algebraic case the filling has at most one isolated singular point, citing [HJ] and Huang's survey [H]. Direct inspection of the PDF verifies the intended reconstruction: \[ f:\mathbb B^2\longrightarrow\mathbb B^3,\qquad f(z_1,z_2)=(z_1^2,\sqrt2\,z_1z_2,z_2^2), \] and the polynomial extension \(F:\mathbb C^2\to\mathbb C^3\) has image \[ X=\{w\in\mathbb C^3:Q(w)=w_2^2-2w_1w_3=0\}. \] The restricted image and its relative boundary are \[ V=f(\mathbb B^2)=X\cap\mathbb B^3,\qquad M=f(S^3)=X\cap S^5. \] Indeed, \[ \|f(z)\|^2 =|z_1|^4+2|z_1z_2|^2+|z_2|^4 =\bigl(|z_1|^2+|z_2|^2\bigr)^2. \]

1. The image of \(f:\mathbb B^2\to\mathbb B^3\) is not the entire unbounded cone \(X\); the polynomial extension \(F:\mathbb C^2\to\mathbb C^3\) has image \(X\), while \(f(\mathbb B^2)=X\cap\mathbb B^3\). 2. The PDF says that \(M\) has one singularity, then computes \[ \nabla Q=(-2w_3,2w_2,-2w_1). \] This proves that the affine cone \(X\), and hence the filling \(V\), has its unique singularity at \(0\). It does **not** prove that \(M\) is singular. In fact \(0\notin M\), because \(|f(z)|=1\) on \(S^3\), and \(M\) is smooth.

Here “hypersurface type” means CR codimension one: if \(V\) has complex dimension \(d\), then \(M\) has real dimension \(2d-1\) and CR dimension \(d-1\). It need not be a real hypersurface of the larger ambient \(\mathbb C^N\). “Boundary” is the boundary relative to the analytic filling (or its closure), not the ambient topological boundary in \(\mathbb C^N\).

来源字段：https://aimath.org/WWN/complexitycrmap/complexitycrmap.pdf

11525 | OWR-2491-007

### Spherical Plancherel Decomposition of $p$-Adic Symmetric Varieties

P / partially_solved | Oberwolfach 报告问题汇编

For a p-adic symmetric variety $X_v=H_v\backslash G_v$ with fixed invariant measure and maximal compact subgroup $K_v$, decompose $L^2(X_v)^{K_v}$ as a direct integral of irreducible representations and determine the representations and Plancherel density occurring in the decomposition.

来源字段：https://doi.org/10.4171/owr/2008/53

11526 | AIM-SPECIAL_FUNCTIONS-0001

### Spin-preserving type A crystals for two-cell ribbon quotients

P / partially_solved | AIM 研讨会问题汇编

**(1)** Find a crystal structure on ribbon tableaux compatible with the spin statistic. This has been solved for type \(A\) domino tableaux.

**Recovered formulation.** For each fixed ribbon length \(r\), ribbon-tileable skew shape \(\lambda/\rho\), and alphabet \([m]\), construct a regular classical type \(A_{m-1}\) crystal on the semistandard \(r\)-ribbon tableaux of that shape, with weight equal to content, such that spin (equivalently cospin for a fixed shape) is constant on every classical crystal component.

来源字段：https://aimath.org/WWN/kostka/prob2.pdf

11527 | OWR-13104-004

### Square-Function Criterion for Higher-Dimensional Rectifiability

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mu$ be a Radon measure on $\mathbb R^d$ such that $0<\Theta_{n,*}(x,\mu)<\infty$ for $\mu$-almost every $x$ and

$$\int_0^1\left(\frac{\mu(B(x,r))}{r^n}-\frac{\mu(B(x,2r))}{(2r)^n}\right)^2\frac{dr}{r}<\infty.$$

Must $\mu$ be $n$-rectifiable? In particular, if $\mu=\mathcal H^n\!\restriction E$, must $E$ be $n$-rectifiable?

来源字段：https://doi.org/10.4171/owr/2014/34

11528 | AIM-SEVERAL_COMPLEX_VARIABLES-0011

### Squared-norm powers: exact Gram test and two complete special classifications

P / partially_solved | AIM 研讨会问题汇编

Let \(R(z,\overline z)\) be a bihomogeneous polynomial. Find necessary and sufficient conditions such that there exists an integer \(N\) with \[ (R(z,\overline z))^N=\sum_{j=1}^m |p_j(z)|^2. \] Here the \(p_j(z)\)'s are linearly independent holomorphic polynomials. See MR1682713 and MR2770459.

来源字段：http://aimpl.org/crscv/3/

11529 | AIM-ANALYSIS-0113

### SRA curvature and finite Carnot-tangent obstructions

P / partially_solved | AIM 研讨会问题汇编

**Curvature conditions and bi-Lipschitz embeddings.** Let $X$ be a doubling metric space. Can one formulate a curvature-type condition which, if satisfied by $X$, ensures that $X$ admits a bi-Lipschitz embedding into some finite-dimensional Euclidean space?

来源字段：http://aimpl.org/mappingmetric/2/

11530 | AIM-ANALYSIS-0154

### Stability in logarithmic moment coordinates

S / solved | AIM 研讨会问题汇编

Let \(\mu\) be a probability measure on \(\{0,1\}^n\). Consider \[ f(\vec z)=\int \vec z^{\,\vec\alpha}\,d\mu(\vec\alpha) \] and \[ g(\vec\lambda)=\int e^{\vec\lambda\cdot\vec\alpha} \,d\mu(\vec\alpha). \] What property of \(g\) is equivalent to the stability of \(f\)?

来源字段：http://aimpl.org/hyperbolicpoly/9/

11531 | OWR-4136-004

### Stability of the Prékopa-Leindler Inequality

P / partially_solved | Oberwolfach 报告问题汇编

Let $\lambda\in(0,1)$, and let $f,g,m$ be nonnegative integrable functions on $\mathbb{R}^n$ such that $f$ and $g$ are probability densities with zero mean and $m(\lambda x+(1-\lambda)y)\geq f(x)^\lambda g(y)^{1-\lambda}$ for all $x,y\in\mathbb{R}^n$. With $\delta_1(f,g)=\min\{1,\int_{\mathbb{R}^n}|f-g|\}$, is there a dimension-dependent constant $\gamma>0$ such that $\int_{\mathbb{R}^n}m\geq1+\gamma\lambda\delta_1(f,g)^2$?

来源字段：https://doi.org/10.4171/owr/2009/53

11532 | AIM-ANALYSIS-0156

### Stability-preserving Markov semigroups on three bits

P / partially_solved | AIM 研讨会问题汇编

**Recovered statement.** Characterize continuous-time Markov generators $\psi$ on the coefficient vectors of multiaffine generating polynomials for probability measures on $\{0,1\}^3$ such that \[ T_t=e^{t\psi}:\mathbb R_{MA}[x,y,z]\longrightarrow\mathbb R_{MA}[x,y,z] \] preserves real stability for every $t\ge 0$.

来源字段：http://aimpl.org/hyperbolicpoly/9/

11533 | OWR-14298808-017

### Stable Banach Isometric Subspace Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $V$ be a real or complex $n$-dimensional normed space and $2\leq k<n$. If every pair of $k$-dimensional subspaces $E,E'\subset V$ satisfies $d_{\mathrm{BM}}(E,E')<1+\delta$, prove that $d_{\mathrm{BM}}(V,\ell_2^n)<1+\varepsilon(n,k,\delta)$ with $\varepsilon(n,k,\delta)\to 0$ as $\delta\to 0$.

来源字段：https://doi.org/10.4171/owr/2024/58

11534 | OWR-12697711-012

### Stable Division and Arveson-Douglas Essential Normality

O / open | Oberwolfach 报告问题汇编

If a homogeneous ideal $I\subset\mathbb C[z_1,\ldots,z_d]$ gives an essentially normal compressed coordinate tuple, must $I$ have the stable division property?

来源字段：https://doi.org/10.4171/owr/2023/15

11535 | OWR-11791-002

### Stable Oversampling Approximations of Linear Systems

O / open | Oberwolfach 报告问题汇编

Which linear time-invariant systems can be stably approximated by approximation processes that use oversampling?

来源字段：https://doi.org/10.4171/owr/2012/29

11536 | OWR-15996-002

### Stable PDE-Informed Neural Networks for Inverse Problems

O / open | Oberwolfach 报告问题汇编

For convolutional neural networks used in inverse problems, determine whether suitably discretized PDE-motivated networks enforce stability, and how explicit prior knowledge and constraints should be incorporated.

来源字段：https://doi.org/10.4171/owr/2018/11

11537 | OWR-15959-005

### Stable Wasserstein Gradient Flow Algorithms for Data Analysis

P / partially_solved | Oberwolfach 报告问题汇编

Which clustering or classification algorithms can be formulated as Wasserstein gradient flows, and can such algorithms be designed to remain stable as the sample size $n\to\infty$?

来源字段：https://doi.org/10.4171/owr/2017/54

11538 | OWR-12866-012

### Stationarity of Perfect Squares for the Pisier Norm

O / open | Oberwolfach 报告问题汇编

Is the set of perfect squares stationary for the Pisier norm?

来源字段：https://doi.org/10.4171/owr/2014/06

11539 | AIM-ANALYSIS-0168

### Stationary zero-process obstruction to reconstruction from real zeros

P / partially_solved | AIM 研讨会问题汇编

**Recovered inverse problem.** Let \(F\) be a random real-entire function of order \(1\), meaning \(F(\bar z)=\overline{F(z)}\) almost surely. If the law of its real-zero divisor is given, is the law of its complete complex-zero divisor determined, and if so how?

来源字段：https://aimath.org/WWN/randomzeros/arcc1.pdf

11540 | OWR-11101912-009

### Strict Majorant Properties at Fixed Noneven Exponents

P / partially_solved | Oberwolfach 报告问题汇编

For $p\notin 2\mathbb{N}$, which sets $\Gamma\subset\mathbb{Z}^d$ satisfy the strict majorant property on $L^p([0,1]^d)$?

来源字段：https://doi.org/10.4171/owr/2022/31

11541 | OWR-5154-007

### Strong Approximation of $L^p$ Convolution Operators

O / open | Oberwolfach 报告问题汇编

Is every bounded left convolution operator $\lambda_p(f)$ on $L^p(G)$ in the strong-operator closure of the finitely supported convolution operators $\lambda_p(\mathbb C\Gamma)$?

来源字段：https://doi.org/10.4171/owr/2011/26

11542 | OWR-16766-005

### Strong Duality and Multilevel Discrete Tomography

O / open | Oberwolfach 报告问题汇编

Does strong duality hold between the binary-tomography primal problem and its Lagrangian dual? Can an analogous dual formulation be developed for discrete tomography with more than two gray levels?

来源字段：https://doi.org/10.4171/owr/2019/4

11543 | AIM-ANALYSIS-0134

### Strong relative isoperimetry from the measure-theoretic boundary

S / solved | AIM 研讨会问题汇编

Can the perimeter measure \(P\) on the right hand side of the relative isoperimetric inequality \[ \min \{\mu(E\cap B),\mu(B\setminus E)\}\leq C r P(E,\lambda B), \] where \(E\subset X\) is a Borel set and \(B\subset X\) is a ball of radius \(r\), be replaced with the codimension one Hausdorff measure of the part of the measure-theoretic boundary of \(E\) inside \(B\) even without knowing ahead of time whether \(E\) is of finite perimeter?

来源字段：http://aimpl.org/mappingmetric/6/

11544 | OWR-11791-004

### Super-Resolution Accuracy beyond the Cutoff Scale

O / open | Oberwolfach 报告问题汇编

Given noisy low-pass measurements of a signal $x$, how accurately can super-resolution recover features below the cutoff scale?

来源字段：https://doi.org/10.4171/owr/2012/29

11545 | OWR-16160-018

### Superexpanders and Separation of L-p from Hilbert Embeddability

O / open | Oberwolfach 报告问题汇编

Does there exist a bounded-geometry metric space that embeds into $\ell_p$ for some $2<p<\infty$ but does not embed into a Hilbert space?

来源字段：https://doi.org/10.4171/owr/2018/19

11546 | OWR-12976-002

### Support Uncertainty for Finite Cyclic Short-Time Fourier Transforms

O / open | Oberwolfach 报告问题汇编

Let $G$ be a finite cyclic group. Does there exist a nonzero window $\varphi\in\mathbb C^G$ such that the attainable support-size pairs

$$\bigl(\lVert x\rVert_0,\lVert V_\varphi x\rVert_0\bigr)$$

are exactly

$$\bigl(\lVert x\rVert_0,\lvert G\rvert^2-\lvert G\rvert+\lVert\widehat x\rVert_0\bigr),\qquad x\in\mathbb C^G$$

as $x$ ranges over $\mathbb C^G$?

来源字段：https://doi.org/10.4171/owr/2014/18

11547 | AIM-ANALYSIS-0090

### Support-coloring cases and a recursive paving bottleneck

P / partially_solved | AIM 研讨会问题汇编

Take A = n \(\times\) n matrix that's a commutator and write it as a commutator of B,C with B diagonal of norm 1 and C with bounded norm, with some minimization. Bill's recent PNAS paper with Ozawa and Schechtmann showed something that needed KS with optimal estimates to improve bound from \(n^\alpha\) (which uses only restricted invertibility) to poly(n), and that the existence of a bound independent of n implies KS. Can you improve the result using KS to get a bound independent of n?

Is there an absolute \(K\) such that every zero-diagonal \(A\in M_n(\mathbb C)\) has \(A=[B,C]\), where \(B\) is diagonal, \(\|B\|\leq 1\), and \(\|C\|\leq K\|A\|\), independently of \(n\)?

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

11548 | OWR-15181-022

### Surreal Integration Beyond Globally Subanalytic Functions

P / partially_solved | Oberwolfach 报告问题汇编

Can the surreal-number measure and integration theory be extended beyond semialgebraic and globally subanalytic sets and functions, in particular to restricted analytic functions and exponentiation?

来源字段：https://doi.org/10.4171/owr/2016/60

11549 | AMR-036-0046

### Symmetric one-points in the unit disk

P / partially_solved | AMR 问题清单汇编

Let $f$ be holomorphic in the unit disk with a simple zero at $0$, exactly two simple $1$-points at $b$ and $-b$, and no other zeros or $1$-points. Determine the minimum $b_0$ of $|b|$ and the extremal function. Is $b_0>c$, where $c\approx0.0505468$ is the conjectured constant in the conjugate-one-point problem?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

11550 | OWR-16767-002

### Symplectic Nonsqueezing in Infinite-Dimensional Hilbert Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Does the finite-dimensional symplectic non-squeezing theorem extend to symplectomorphisms of infinite-dimensional Hilbert symplectic spaces?

来源字段：https://doi.org/10.4171/owr/2019/5

11551 | OWR-4534-002

### Tempered Embeddings of Unimodular Quasi-Affine Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be an algebraic group, let $H\subseteq G$ be algebraic, and suppose $Z=G/H$ is unimodular and quasi-affine. Does there exist a rational $G$-module $V$ and a $G$-equivariant embedding $Z\hookrightarrow V$ such that the pushforward of a $G$-invariant measure $\mu_Z$ is a tempered distribution on $V$?

来源字段：https://doi.org/10.4171/owr/2010/52

11552 | OWR-4990370-005

### Tensor Stability of Quantum Markov Gradient Estimates

O / open | Oberwolfach 报告问题汇编

Is the quantum Markov gradient estimate $\operatorname{GE}(K,\infty)$ stable under tensor products of quantum Markov semigroups?

来源字段：https://doi.org/10.4171/owr/2021/2

11553 | AIM-FUNCTIONAL_ANALYSIS-0002

### Tensorization and local product families for the log-Minkowski deficit

P / partially_solved | AIM 研讨会问题汇编

**Log-Minkowski inequality.** Show that if \(K\) and \(L\) symmetric convex bodies in \(\mathbb R^n\), then \[ \int_{S^{n-1}}\log\frac{h_K(u)}{h_L(u)}\,d\bar V_L(u) \geq \frac1n\log\frac{|K|}{|L|}, \] where \(\bar V_L\) is the cone-volume probability measure of \(L\).

来源字段：http://aimpl.org/symconvgeomineq/1/

11554 | OWR-1057-014

### The Corona Problem for Plane-Domain Function Algebras

O / open | Oberwolfach 报告问题汇编

Let $\Omega\subseteq\mathbb C$ be a domain such that $A=H^\infty(\Omega)$ contains a nonconstant function. If $f_1,\ldots,f_N\in A$ satisfy $\sum_{j=1}^N|f_j(z)|\geq\delta>0$ on $\Omega$, must there exist $g_1,\ldots,g_N\in A$ such that $\sum_{j=1}^Ng_jf_j=1$?

来源字段：https://doi.org/10.4171/owr/2005/36

11555 | AIM-ANALYSIS-0071

### The critical logarithmic endpoint model

P / partially_solved | AIM 研讨会问题汇编

**Question 12 (Guo).** Prove that there is a universal constant $C>0$ such that, for every $0<\varepsilon<1/2$, \[ \left\|\sup_{\lambda\in\mathbb R} \left|\operatorname{p.v.}\!\int_{\mathbb R} f(x-t)e^{i\lambda |t|^\varepsilon}\,\frac{dt}{t}\right| \right\|_{L^2(\mathbb R)} \le C\|f\|_{L^2(\mathbb R)}. \tag{Q12} \]

来源字段：https://aimath.org/pastworkshops/multilinopsproblems.pdf

11556 | AIM-SEVERAL_COMPLEX_VARIABLES-0010

### The Faran homotopy question and a rank-five barrier

P / partially_solved | AIM 研讨会问题汇编

Given proper holomorphic maps $f,g:\mathbb B^n\to\mathbb B^N$, how does one show that $f$ and $g$ are not homotopic? Find homotopy invariants for proper holomorphic maps between balls.

Special case (Lebl): Is the Faran map from $\mathbb B^2$ to $\mathbb B^4$ given by $$ (z,w)\longmapsto (z^3,\sqrt3zw,w^3,0) $$ homotopic to $(z,w)\longmapsto(z,w,0,0)$?

来源字段：http://aimpl.org/crscv/3/

11557 | AIM-ANALYSIS-0015

### The Hilbert matrix on power-weighted ell-squared spaces

P / partially_solved | AIM 研讨会问题汇编

The \(\mathscr l^2\)-spectrum for the Hilbert matrix \(H_{mn}=1/(m+n+1)\) is known. Find the spectrum on spaces of the form \(\mathscr l_2(\alpha)=\{\sum |a_n|^2(n+1)^\alpha<\infty\}\).

来源字段：http://aimpl.org/riemhilberttoeplitz/3/

11558 | AIM-FUNCTIONAL_ANALYSIS-0022

### The index-trivial layer of Calkin automorphisms

P / partially_solved | AIM 研讨会问题汇编

Is there a model of ZFC in which the Calkin algebra has an automorphism that is not approximately inner, in the point-norm net sense, and induces the identity on K-theory?

来源字段：http://aimpl.org/settheorycstar/2/

11559 | OWR-16163-004

### The Lévy-Khintchine Property for Quantum Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a unital $*$-bialgebra $A$, does every generating functional admit a Lévy-Khintchine-type decomposition? More generally, classify the $*$-bialgebras with this property.

来源字段：https://doi.org/10.4171/owr/2018/22

11560 | AIM-FUNCTIONAL_ANALYSIS-0032

### The liftable-Calkin-masa and diffuse-witness reductions

P / partially_solved | AIM 研讨会问题汇编

Consider the following statement: for every pure state \(\phi\) of \(\mathcal B(H)\) there is a masa \(\mathcal A\) such that \(\phi\restriction\mathcal A\) is multiplicative (i.e., pure). Is it consistent with ZFC?

Its negation is known to follow from the Continuum Hypothesis (Akemann--Weaver).

来源字段：http://aimpl.org/settheorycstar/5/

11561 | AIM-SPECIAL_FUNCTIONS-0013

### The LLT/generalized-Kostka bridge and its limits

P / partially_solved | AIM 研讨会问题汇编

Suspicions were voiced at the workshop that the two gradings of products of Schur functions - one from LLT polynomials and one from Morse and Lapointe's \(t\)-statistic - may be related. Can this relationship be stated precisely? What role does the affine Hecke algebra play? What is the connection between \(q\)-Littlewood-Richardson coefficients, Macdonald polynomials, and the definition of \(k\)-Schur functions via generalized Kostka polynomials?

来源字段：https://aimath.org/WWN/kostka/prob2.pdf

11562 | AIM-FUNCTIONAL_ANALYSIS-0023

### The nonseparable Calkin consistency problem and two transplantation obstructions

P / partially_solved | AIM 研讨会问题汇编

**Readability-normalized question.** Is it relatively consistent with ZFC that there are a nonseparable complex Hilbert space \(H\) and an outer automorphism of \[ \mathcal Q(H):=\mathcal B(H)/\mathcal K(H), \] where \(\mathcal K(H)\) is the ideal of compact operators?

来源字段：http://aimpl.org/settheorycstar/2/

11563 | AIM-FUNCTIONAL_ANALYSIS-0040

### The one-Cartan-mark gap for conjugacy of masas

P / partially_solved | AIM 研讨会问题汇编

Is conjugacy by automorphism of masas of the hyperfinite II\(_1\) factor classifiable by countable structures?

来源字段：http://aimpl.org/settheorycstar/7/

11564 | OWR-12723-005

### The Optimal Constant in One-Dimensional Sequence Discrepancy

P / partially_solved | Oberwolfach 报告问题汇编

Let $I=\{x\in\mathbb R:0\leq x<1\}$. Determine the optimal constant $c^*=\sup\{c:\text{every sequence }(x_n)\subset I\text{ satisfies }D_N\geq c(\log N)/N\text{ for infinitely many }N\}$.

来源字段：https://doi.org/10.4171/owr/2013/49

11565 | AIM-ANALYSIS-0035

### The Pólya-Szegő buckling conjecture and a radial projection gap on balls

P / partially_solved | AIM 研讨会问题汇编

Let \(\Omega\subset\mathbb R^2\) be a bounded domain and let \(\Lambda _1(\Omega)\) be the lowest eigenvalue of \[ \Delta^2u+\Lambda _1(\Omega)\Delta u=0\quad\hbox{in }\Omega, \qquad u=\partial_\nu u=0\quad\hbox{on }\partial\Omega. \] Prove that \(\Lambda _1(\Omega)\) is minimized by the disk among all simply connected sets of equal area.

来源字段：http://aimpl.org/shapesurface/6/

11566 | AIM-SPECIAL_FUNCTIONS-0010

## The quotient and forced-lift dictionary for k-Schur and level-restricted structure constants

S / solved | AIM 研讨会问题汇编

(10) What is the relationship between \(k\)-Schur functions and \(k\)-level-restricted Schur functions? These may be essentially the same: it was asserted that some of the structure constants for \(k\)-Schur functions, the \(k\)-Littlewood-Richardson coefficients, give all of the structure constants for level-restricted Schur functions. A paper by Goodman and Wenzl on Iwahori-Hecke algebras of type \(A\) at roots of unity (J. Algebra 215 (1999), 694-734) may be relevant in this context.

来源字段：https://aimath.org/WWN/kostka/prob2.pdf

11567 | AIM-ANALYSIS-0102

## The scalar-and-zero mixed-characteristic extremizer in dimension two

P / partially_solved | AIM 研讨会问题汇编

"There's a conjecture about repeated copies of Identity with zero padding that this gives you the worse bounds on roots?"

The original workshop document places the sentence under a heading for Adam Marcus. The next bullets discuss using interlacing polynomials when only the maximum root is wanted and using rank-two rather than rank-one perturbations. Thus “worse” is almost certainly “worst,” and “roots” should not be read as coordinatewise extremality of every root.

There is an essentially exact match in Marcus--Spielman--Srivastava (MSS), Section 7.2, page 346. For positive semidefinite Hermitian matrices \(A_1,\ldots,A_m\in\mathbb C^{d\times d}\), define

\[ \mu[A_1,\ldots,A_m](x) =\left.\prod_{i=1}^m(1-\partial_{z_i}) \det\!\left(xI_d+\sum_{i=1}^m z_iA_i\right) \right|_{z_1=\cdots=z_m=0}. \]

The precise recovered conjecture is:

**MSS scalar-and-zero extremizer conjecture.** Fix \(d,m\) and \(\epsilon>0\), and suppose the feasible set

\[ \mathcal F_{d,m,\epsilon} =\left\{(A_i)_{i=1}^m:A_i\succeq0, \ \sum_iA_i=I_d,\ \operatorname{tr}A_i\leq\epsilon\right\} \]

is nonempty. The largest root of \(\mu[A_1,\ldots,A_m]\) is maximized by taking as many matrices as possible equal to \(\epsilon I_d/d\), one further smaller scalar multiple of \(I_d\) if needed, and all remaining matrices zero. Explicitly, with

\[ q=\left\lfloor\frac d\epsilon\right\rfloor, \qquad s=d-q\epsilon\in[0,\epsilon), \]

the proposed family consists of \(q\) copies of \((\epsilon/d)I_d\), one copy of \((s/d)I_d\) when \(s>0\), and zero matrices filling the remaining slots. Feasibility implies \(m\geq\lceil d/\epsilon\rceil\), so this family fits.

Two kinds of “padding” occur in the source material and probably became compressed in the workshop shorthand. The conjectured family is padded by zero matrices. In the paragraph immediately before the conjecture, MSS also define a partial mixed discriminant by padding its arguments with identities:

\[ D(A_1,\ldots,A_k) =\frac{D(A_1,\ldots,A_k,I_d,\ldots,I_d)}{(d-k)!}. \]

The recovered conjecture concerns **only the largest root**. Section 8 below gives a counterexample to the stronger all-roots reading.

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

11568 | AIM-FUNCTIONAL_ANALYSIS-0029

### The Small Bootstrap Problem and a separable-hull reduction

P / partially_solved | AIM 研讨会问题汇编

All presently known nuclear C*-algebras are obtained from $\mathbb C$ by taking closure under simple operations such as tensoring with finite-dimensional matrix algebras and taking inductive limits. They belong to the so-called bootstrap class.

Does every nuclear C*-algebra belong to the bootstrap class?

**Small Bootstrap Problem.** Is every separable nuclear C*-algebra in Blackadar's Small Bootstrap Class $\mathsf B$?

来源字段：http://aimpl.org/settheorycstar/4/

11569 | AIM-ANALYSIS-0067

### The solved discrete quadratic Carleson problem and a sharp-scale finite sampling reduction

S / solved | AIM 研讨会问题汇编

3 of the authoritative AIM PDF. The PDF asks for the discrete analogue of Stein--Wainger's polynomial Carleson theorem:

\[ \left\| \sup_{\lambda} \left| \sum_{m\in\mathbb Z} f(n-m)\frac{e^{i\lambda m^2}}{m} \right| \right\|_{\ell^2_n} \lesssim \|f\|_{\ell^2}. \tag{0.9} \]

It then takes \(\Lambda\subset[0,1]\), replaces the unrestricted supremum by \(\sup_{\lambda\in\Lambda}\), and cites Krause--Lacey for a sufficient condition for boundedness of that restricted operator.

Two repairs/conventions are forced by the display and by the cited paper.

1. The term \(m=0\) is undefined and must be omitted. The infinite sum is the symmetric principal value \[ \lim_{N\to\infty}\sum_{0<|m|\leq N} f(n-m)\frac{e^{i\lambda m^2}}{m}. \] Krause--Lacey explicitly write \(m\ne0\). 2. The source uses \(e^{i\lambda m^2}\), while the later literature normally writes \(e(\alpha m^2)\), where \(e(t)=e^{2\pi i t}\). The change \(\alpha=\lambda/(2\pi)\) is harmless for the unrestricted supremum. We use the latter convention, for which the parameter belongs naturally to \(\mathbb T=\mathbb R/\mathbb Z\).

Thus the clean recovered operator is

\[ \mathcal C_2 f(n) :=\sup_{\lambda\in\mathbb T} \left|\operatorname{p.v.}\sum_{m\ne0} f(n-m)\frac{e(\lambda m^2)}{m}\right|. \tag{1.1} \]

The norm in (0.9) is the sequence norm in the free variable \(n\in\mathbb Z\).

来源字段：https://aimath.org/pastworkshops/multilinopsproblems.pdf

11570 | AIM-SPECIAL_FUNCTIONS-0011

### The solved two-step flag puzzle conjecture and projection-lift support tests

S / solved | AIM 研讨会问题汇编

**(11)** Prove Buch's weakened version of Knutson's false conjecture for the Schubert structure constants that arise in Schubert calculus on flag manifolds. The original conjecture turns out to be false for full flags, but seems to be OK for two-step flags \(V^a\subseteq V^b\subseteq \mathbb C^n\). A paper by Buch, Kresch, Tamvakis, and Yong (Duke Math. J. 122 (2004), 125--143) reduces \(q\)-Schubert calculus on Grassmannians to this two-step case.

来源字段：https://aimath.org/WWN/kostka/prob2.pdf

11571 | AIM-ANALYSIS-0022

### The spectral Mahler functional on the planar cube orbit

P / partially_solved | AIM 研讨会问题汇编

A hypercube in \(\mathbb R^n\) is an attainer for \[ \sup_{K\in\mathcal K^n_\star}\ \inf_{T\in\mathsf{GL}_n} \lambda _1(TK)\lambda _1((TK)^\circ). \]

Harrell and its status note.

Let \(\mathcal K^n_\star\) denote the centrally symmetric bounded convex domains in \(\mathbb R^n\), and for \(K\in\mathcal K^n_\star\) let \[ K^\circ=\{x\in\mathbb R^n:x\mathbin\cdot y\leq 1\text{ for every }y\in K\}. \] Writing \(\lambda _1(\Omega)\) for the first Dirichlet eigenvalue of \(-\Delta\) on the interior of \(\Omega\), the conjecture is:

Central symmetry forces the center to be the origin in this formulation. We harmlessly identify a bounded convex domain with its closure when taking the polar; its Dirichlet eigenvalue is that of its interior. No corruption of the mathematical statement was found.

来源字段：http://aimpl.org/shapesurface/2/

11572 | OWR-14748-012

### Thom’s Conjecture for the Hyperfinite Factor’s Unitary Group

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Thom Conjecture for the Hyperfinite Factor [OWR-14748-011]

Does the unitary group of the hyperfinite $\mathrm{II}_1$ factor have no uniformly discrete free nonabelian subgroup, where uniform discreteness is measured with respect to the $2$-norm?

来源字段：https://doi.org/10.4171/owr/2016/40

11573 | AIM-ANALYSIS-0016

### Three atomic singular inner functions: source diagnosis, pair selection, and point reconstruction

O / open | AIM 研讨会问题汇编

Define \[ u_{\xi,\alpha(z)}=e^{-\alpha\left(\frac{\xi+z}{\xi-z}\right)}, \] for \(\alpha>0\) and \(|\xi|=1\). Given \(\xi_1,\xi_2,\xi_3\) and \(\alpha_1,\alpha_2,\alpha_3\), can we approximate \[ \sum_{m=0}^{\infty}c_m^{(1)}u_1^m+c_m^{(2)}u_2^m+c_m^{(3)}u_3^m? \]

来源字段：http://aimpl.org/riemhilberttoeplitz/3/

11574 | OWR-759-005

### Three-Cocycles in the Hyperfinite Type $\mathrm{II}_1$ Factor

P / partially_solved | Oberwolfach 报告问题汇编

For a locally compact group $G$, can every class in $H^3_{\mathrm{Borel}}(G,\mathbb T)$ be realized as the Connes-Jones invariant of an automorphic representation $G\to\operatorname{Out}(R)$ on the hyperfinite type $\mathrm{II}_1$ factor $R$?

来源字段：https://doi.org/10.4171/owr/2004/45

11575 | AIM-ANALYSIS-0052

### Toeplitz and mode reductions for weighted Szegő projections on the smooth worm

P / partially_solved | AIM 研讨会问题汇编

Can we choose continuous multiples of surface area on the boundary of the Diederich-Fornaess worm (for example, powers of Levi-form) mitigating the irregularity of the Szeg\"o projection (defined using surface area/modified measure?)

来源字段：http://aimpl.org/scvproblems/1/

11576 | OWR-13750336-003

### Tomas-Stein Extremizers on the Circle

P / partially_solved | Oberwolfach 报告问题汇编

Are constant functions extremizers for the Tomas-Stein extension inequality $\lVert\widehat{f\sigma}\rVert_{L^6(\mathbb R^2)}\leq C\lVert f\rVert_{L^2(\sigma)}$ on the unit circle?

来源字段：https://doi.org/10.4171/owr/2023/25

11577 | OWR-16766-001

### Tomographic Reconstruction near Discontinuities

P / partially_solved | Oberwolfach 报告问题汇编

For reconstruction of functions with discontinuities from discrete tomographic data, what is the reconstruction behavior near singularities, what is the effect of remote singularities, and why can discontinuous functions be reconstructed accurately?

来源字段：https://doi.org/10.4171/owr/2019/4

11578 | OWR-4214-007

### Toms-Winter Conjecture for Nuclear $C^*$-Algebras

O / open | Oberwolfach 报告问题汇编

别名：Repeated Extraction of the Toms-Winter Conjecture [OWR-4214-005]

For a simple, unital, nuclear $C^*$-algebra $A$, are the following equivalent: (1) $A$ has finite nuclear dimension; (2) $A\cong A\otimes\mathcal{Z}$, where $\mathcal{Z}$ is the Jiang-Su algebra; and (3) $A$ has strict comparison?

来源字段：https://doi.org/10.4171/owr/2010/13

11579 | AIM-SEVERAL_COMPLEX_VARIABLES-0051

### Top-degree W1 solvability and a finite-order open-mapping reduction

P / partially_solved | AIM 研讨会问题汇编

**Problem 11 (proposed by Mei-Chi Shaw).** Let \(\Omega\subset\mathbb {CP}^{n}\) be pseudoconvex with \(C^\infty\) boundary. Let us solve \(\bar\partial u=f\) in \(\Omega\). If \(f\) is a \((p,q)\)-form such that \(\bar\partial f=0\) and \(f\in C^\infty(\overline\Omega)\), does there exist \(u\in W^1(\overline\Omega)\) such that \(\bar\partial u=f\)? It is known that there is such a \(u\) in \(L^2(\overline\Omega)\), and also one in \(W^\varepsilon(\overline\Omega)\) for some \(\varepsilon>0\), where \(\varepsilon\) depends on \(\Omega\). See [CSW]. The answer was not known even when \(\partial\Omega\) was real analytic.

来源字段：https://aimath.org/WWN/complexitycrmap/complexitycrmap.pdf

11580 | OWR-15954-001

### Totik-Widom Bounds Beyond Parreau-Widom Sets

O / open | Oberwolfach 报告问题汇编

Does there exist a compact set $E\subset\mathbb R$ that is not Parreau-Widom but whose Chebyshev constants satisfy a Totik-Widom bound?

来源字段：https://doi.org/10.4171/owr/2017/49

11581 | AIM-ANALYSIS-0074

### Trace-norm matrix signing: positive case solved with sharp exponent

P / partially_solved | AIM 研讨会问题汇编

Let \(\{a_j:j=1,\ldots,m\}\) be “(self-adjoint?, positive?)” \(n\times n\) matrices, each with trace norm less than \(\epsilon\ll1\), and put \(a=\sum_j a_j\), with \(\|a\|\le1\). “For starters” take every \(a_j>0\) and \(a=1\).

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

11582 | OWR-4536-005

### Transferring Hypocoercivity to Larger Banach Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Let $L$ be an operator on a Hilbert space $H$ such that, for some $a<0$, $\Sigma(L)\cap\{z:\operatorname{Re}z>a\}=\{0\}$, $\ker L$ is finite-dimensional, and $\|e^{tL}f-\Pi f\|_H\leq C_ae^{at}\|f\|_H$. For an extension of $L$ to a Banach space $E\supset H$, under what hypotheses do the same spectral-gap and semigroup-decay conclusions hold on $E$?

来源字段：https://doi.org/10.4171/owr/2010/54

11583 | OWR-15954-002

### Transitional Asymptotics for Toda Shock Waves

O / open | Oberwolfach 报告问题汇编

For the Toda shock problem, what are the long-time asymptotics in the transitional regions near the critical rays $n/t=\xi_{\rm cr}$ and $n/t=\xi_{\rm cr,1}$, including possible asymptotic solitons?

来源字段：https://doi.org/10.4171/owr/2017/49

11584 | OWR-720-001

### Translation Invariance of Finitely Generated Shift-Invariant Spaces

O / open | Oberwolfach 报告问题汇编

Find a general characterization of finitely generated shift-invariant spaces $V$ satisfying invariance under translation by $1/n$.

来源字段：https://doi.org/10.4171/owr/2004/10

11585 | AIM-FUNCTIONAL_ANALYSIS-0015

### Transport-adapted and finite-scale variable-curvature Caffarelli estimates

P / partially_solved | AIM 研讨会问题汇编

Let \(T=\nabla\Phi:\mu\to\nu\) be a Brenier map, where \(\mu\) and \(\nu\) have densities \(e^{-V(x)}\) and \(e^{-W(x)}\). If \[ \nabla^2W(x)\ge K\,\mathrm{Id}, \] then, for \(e\in S^{n-1}\), \[ \sup_{x\in\mathbb{R}^n}\Phi_{ee}^2 \le \frac1K\sup_{x\in\mathbb{R}^n}V_{ee}. \] Can one obtain better results if \(\nabla^2W(x)\ge F(x)\,\mathrm{Id}\)?

来源字段：http://aimpl.org/symconvgeomineq/2/

11586 | AIM-SEVERAL_COMPLEX_VARIABLES-0052

### Transversality criteria and a classified quadratic counterexample family

P / partially_solved | AIM 研讨会问题汇编

**Problem 12 (Peter Ebenfelt).** Let \(0\in M\subset\mathbb C^{n+1}\) and \(0\in M'\subset\mathbb C^{N+1}\), where \(M,M'\) are \(C^\infty\) or real-analytic hypersurfaces and \(n\le N\). Let \(H:(\mathbb C^{n+1},0)\to(\mathbb C^{N+1},0)\) be holomorphic, with \(H(M)\subset M'\), and suppose that \(H\) does not map a full neighborhood of zero into \(M'\). Give conditions on \(M,M'\) ensuring that every such \(H\) is transversal to \(M'\) at zero. Here transversality at \(p\) means \[ d_pH(T_p\mathbb C^{n+1})+T_{H(p)}M'=T_{H(p)}\mathbb C^{N+1}. \]

来源字段：https://aimath.org/WWN/complexitycrmap/complexitycrmap.pdf

11587 | OWR-12339-008

### Trigonometric Factorization of Positive Matrix Polynomials

O / open | Oberwolfach 报告问题汇编

For every positive-semidefinite symmetric $m\times m$ trigonometric matrix polynomial $M$ on the two-dimensional torus, is $M=S^{\mathsf T}S$ for some trigonometric matrix polynomial $S$? The case $d=2$ and $m\geq2$ is open.

来源字段：https://doi.org/10.4171/owr/2013/11

11588 | AIM-ANALYSIS-0039

### Trivial Bergman space and quantitative A^p collapse

S / solved | AIM 研讨会问题汇编

What is the dimension of the Bergman space of a short $\mathbb{C}^2$? (We may assume vanishing Kobayashi metric)

As an example of short $\mathbb{C}^2$, consider the sequence of mappings $f_n(z,w)=(a_nw+z^2,a_nz)$, with the coefficients $a_n\searrow0$ rapidly. Then let $\Omega=\bigcup_{n\geq1}(\text{basin of attraction of }f_n\text{ at }0)$.

来源字段：http://aimpl.org/scvproblems/1/

11589 | AIM-FUNCTIONAL_ANALYSIS-0019

### Trivial relative commutants are exactly non-flat ultrafilters

P / partially_solved | AIM 研讨会问题汇编

Can one prove in ZFC that for some free ultrafilter \(\mathcal U\) on \(\mathbb N\) we have \[ \mathcal B(H)'\cap \prod_{\mathcal U}\mathcal B(H)=\mathbb C I\ ? \]

来源字段：http://aimpl.org/settheorycstar/1/

11590 | AMR-036-0025

### Two one-points in the unit disk

P / partially_solved | AMR 问题清单汇编

Let $f$ be holomorphic in the unit disk, with $f(0)=0$, $f'(0)\ne0$, no other zeros, and exactly two solutions $z_1,z_2$ of $f(z)=1$, counted with multiplicity. Determine the minimum of $\max(|z_1|,|z_2|)$ and the maximum of $|f'(0)|$.

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

11591 | AIM-SPECIAL_FUNCTIONS-0002

### Two remaining comparison edges and a hook-stable equivalence for the four k-Schur candidates

P / partially_solved | AIM 研讨会问题汇编

**(2)** Prove that the four definitions of \(k\)-Schur functions (see Morse's lecture notes) are indeed equivalent.

来源字段：https://aimath.org/WWN/kostka/prob2.pdf

11592 | AIM-ANALYSIS-0057

### Two-neutral hedgehogs and a rigorous product model

P / partially_solved | AIM 研讨会问题汇编

**Recovered statement.** Study hedgehogs for **germs** of automorphisms of $\mathbb{C}^2$ with both elliptic eigenvalues.

来源字段：http://aimpl.org/scvproblems/2/

11593 | OWR-13674-020

### Two-Sided Paving Bounds

S / solved | Oberwolfach 报告问题汇编

Can the reduction from partitioning to paving be improved so that a single partition controls both the top and bottom eigenvalues, reducing the known $O(\varepsilon^{-4})$ block bound toward the lower bound $\Omega(\varepsilon^{-2})$?

来源字段：https://doi.org/10.4171/owr/2015/17

11594 | OWR-8415336-012

### Two-Term Weyl Asymptotics and Ivrii's Billiard Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For every smooth planar domain $\Omega$, does the eigenvalue counting function satisfy $$ N(\lambda)=\frac{\operatorname{Area}(\Omega)}{4\pi}\lambda+C\ell(\partial\Omega)\lambda^{1/2}+o(\lambda^{1/2})? $$ Also, does every smooth planar billiard domain have periodic trajectories of measure zero?

来源字段：https://doi.org/10.4171/owr/2021/33

11595 | OWR-14748-004

### Type-I Rigidity of Boundary-Transitive Tree Groups

P / partially_solved | Oberwolfach 报告问题汇编

Let $T$ be a locally finite tree and let $G\leq\operatorname{Aut}(T)$ be a closed subgroup acting transitively on the boundary $\partial T$. Must $G$ be a type I group?

来源字段：https://doi.org/10.4171/owr/2016/40

11596 | AMR-153-0001

### Ultraproduct II_1 Factors - Problem 1

O / open | AMR 问题清单汇编

Classify elementary equivalence classes of $\mathbb{M}^{\mathcal{U}}$ where $\mathcal{U}$ ranges over ultrafilters on $\mathbb{N}$. What about elementary equivalence classes of $\mathbb{M}_{\mathcal{U}}$?

背景链接：https://arxiv.org/abs/2511.20377

11597 | AMR-153-0010

### Ultraproduct II_1 Factors - Problem 10

O / open | AMR 问题清单汇编

Let $M$ be a separable II$_1$ factor that contains an existential copy of $N$ where $N$ is a nonamenable biexact II$_1$ factor. Then is $M$ prime?

背景链接：https://arxiv.org/abs/2511.20377

11598 | AMR-153-0011

### Ultraproduct II_1 Factors - Problem 11

O / open | AMR 问题清单汇编

For the definition of bi-exactness see the work [ding2023biexactvonneumannalgebras]. The above behavior was observed in $L(\mathbb{F}_2)$, see [Sridiagonal].

Let $N$ be a non amenable II$_1$ factor with Haagerup property, and $N\subset M$ existentially. Then can $M$ have property (T)?

背景链接：https://arxiv.org/abs/2511.20377

11599 | AMR-153-0012

### Ultraproduct II_1 Factors - Problem 12

O / open | AMR 问题清单汇编

Let $N$ be a II$_1$ factor with Haagerup's property, and $B\subset N^\mathcal{U}$, with Property (T). Then are there finite dimensional subalgebras $M_n\subset N$ such that $B\subset \prod_{n\to \mathcal{U}}M_n$?

背景链接：https://arxiv.org/abs/2511.20377

11600 | AMR-153-0013

### Ultraproduct II_1 Factors - Problem 13

O / open | AMR 问题清单汇编

Progress and evidence towards the above has been obtained in [peterson2026ultraproductapproximationspropertyt].

Does there exist an existential inclusion of $L(\mathbb{F}_2)$ into $\mathbb{M}^\mathcal{U}$? Is the random embedding existential, in the sense of Haar-sampling?

背景链接：https://arxiv.org/abs/2511.20377

11601 | AMR-153-0014

### Ultraproduct II_1 Factors - Problem 14

O / open | AMR 问题清单汇编

Recall that a II$_1$ factor $N$ is said to be existentially closed if whenever it embeds into a II$_1$ factor $M$, it can embed existentially into $M$. Such factors exist by [ECfactors], also see [IOANA2024110264, GKEPTConj] for concrete constructions.

Does there exist a countable group $G$ such that $L(G)$ is an existentially closed II$_1$ factor? Does there exist an existentially closed II$_1$ factor which is not isomorphic to a group von Neumann algebra?

背景链接：https://arxiv.org/abs/2511.20377

11602 | AMR-153-0015

### Ultraproduct II_1 Factors - Problem 15

O / open | AMR 问题清单汇编

Does there exist a sequence of separable non Gamma factors $N_i$ such that their ultraproduct $\prod_{i\to \mathcal{U}}N_i$ has Property Gamma?

背景链接：https://arxiv.org/abs/2511.20377

11603 | AMR-153-0016

### Ultraproduct II_1 Factors - Problem 16

O / open | AMR 问题清单汇编

[Solved by J. Peterson [peterson2026ultraproductapproximationspropertyt]] Does there exist a pair of Connes embeddable non Gamma factors that are not elementarily equivalent? Does there exist continuum many non Gamma factors that are pairwise not elementarily equivalent?

背景链接：https://arxiv.org/abs/2511.20377

11604 | AMR-153-0017

### Ultraproduct II_1 Factors - Problem 17

O / open | AMR 问题清单汇编

[Solved by J. Peterson [peterson2026ultraproductapproximationspropertyt]] Does there exist an icc non inner amenable group $G$ such that $L(G)$ is not elementarily equivalent to $L(\mathbb{F}_2)$?

背景链接：https://arxiv.org/abs/2511.20377

11605 | AMR-153-0018

### Ultraproduct II_1 Factors - Problem 18

O / open | AMR 问题清单汇编

[Solved by J. Peterson [peterson2026ultraproductapproximationspropertyt]] Are there two non elementarily equivalent Connes embeddable super McDuff II$_1$ factors $N,M$? Are there continuum many?

背景链接：https://arxiv.org/abs/2511.20377

11606 | AMR-153-0019

### Ultraproduct II_1 Factors - Problem 19

O / open | AMR 问题清单汇编

If $N$ is a super McDuff II$_1$ factor, and $N$ is elementarily equivalent to $M$ then is $M$ super McDuff? In particular, does there exist a super McDuff II$_1$ factor which is not uniformly super McDuff?

背景链接：https://arxiv.org/abs/2511.20377

11607 | AMR-153-0002

### Ultraproduct II_1 Factors - Problem 2

O / open | AMR 问题清单汇编

For every Connes embeddable non Gamma II$_1$ factor $N$ does there exist an embedding $\pi: N\hookrightarrow \mathbb{M}^{\mathcal{U}}$ such that $N'\cap \mathbb{M}^{\mathcal{U}}=\mathbb{C}$?

背景链接：https://arxiv.org/abs/2511.20377

11608 | AMR-153-0020

### Ultraproduct II_1 Factors - Problem 20

O / open | AMR 问题清单汇编

[Solved by J. Peterson [peterson2026ultraproductapproximationspropertyt]] Does there exist two factors with property Gamma and are not McDuff, that are not elementarily equivalent? Are there continuum many such examples?

背景链接：https://arxiv.org/abs/2511.20377

11609 | AMR-153-0021

### Ultraproduct II_1 Factors - Problem 21

O / open | AMR 问题清单汇编

The above is complementary to [boutonnetchifanioana], which established many non elementarily equivalent McDuff factors.

Is $L(F)$ super McDuff, where $F$ is the well known Thompson's group?

背景链接：https://arxiv.org/abs/2511.20377

11610 | AMR-153-0022

### Ultraproduct II_1 Factors - Problem 22

O / open | AMR 问题清单汇编

See [Jolissaint1998] where it was proved that these group von Neumann algebras are McDuff.

If $M\cong N_1\overline{\otimes} N_2$, and $M$ is McDuff, does it imply that either $N_1$ or $N_2$ are McDuff?

背景链接：https://arxiv.org/abs/2511.20377

11611 | AMR-153-0023

### Ultraproduct II_1 Factors - Problem 23

O / open | AMR 问题清单汇编

For the next few problems, recall the equivalence relation of sequantial commutation [elayavalli2023sequential] on the haar unitaries in a II$_1$ factor. For $u,v\in \mathcal{H}(N)$ haar unitaries, say that $u\sim_{N^\mathcal{U}} v$ if there exists a finite set $w_1, \cdots, w_n$ of haar unitaries in $N^{\mathcal{U}}$, such that $[u, w_1]=[w_1,w_2]= \cdots = [w_{n-1}, w_n]= [w_n,v]=0$. In the case that there is a unique orbit among haar unitaries, additionally recall the commutation diameter $\mathfrak{O}(M) := \sup_{u,v\in \mathcal{H}(N)} p_N(u,v)$, where $p_N(u,v)$ denotes the length of the shortest path of sequentially commuting haar unitaries in the ultrapower connecting $u$ and $v$.

Does there exist a non Gamma factor with unique sequential commutation orbit, and has sequential diameter equal to $n$, $n\in \mathbb{N}$?

背景链接：https://arxiv.org/abs/2511.20377

11612 | AMR-153-0024

### Ultraproduct II_1 Factors - Problem 24

O / open | AMR 问题清单汇编

Does there exist a property (T) II$_1$ factor with unique sequential commutation orbit?

背景链接：https://arxiv.org/abs/2511.20377

11613 | AMR-153-0025

### Ultraproduct II_1 Factors - Problem 25

O / open | AMR 问题清单汇编

Does there exist a II$_1$ factor $N$ with no diffuse relative $(T)$ subalgebras and with unique sequential commutation orbit? Is there such an $N$ with additionally the Haagerup property?

背景链接：https://arxiv.org/abs/2511.20377

11614 | AMR-153-0026

### Ultraproduct II_1 Factors - Problem 26

O / open | AMR 问题清单汇编

Does there exist a II$_1$ factor $N$ satisfying $h(N)=0$ but $N$ has continuum many sequential commutation orbits? Secondly does there exist a II$_1$ factor $N$ such that $h(N)=0$, and $N$ is not generated as a von Neumann algebra by a single sequential commutation orbit?

背景链接：https://arxiv.org/abs/2511.20377

11615 | AMR-153-0027

### Ultraproduct II_1 Factors - Problem 27

O / open | AMR 问题清单汇编

Let $M= N_1*_B N_2$, and $u_i\in \mathcal{H}(N_i)$ such that $E_B(u_i^k)=0$ for all $k\in \mathbb{N}$. Suppose $v_i\in \mathcal{H}(M^\mathcal{U})$ such that $u_i\sim_{M^\mathcal{U}} v_i$ and $E_{B^\mathcal{U}}(v_i^k)=0$ for all $k \in \mathbb{N}$. Then is $v_1$ freely independent from $v_2$?

背景链接：https://arxiv.org/abs/2511.20377

11616 | AMR-153-0028

### Ultraproduct II_1 Factors - Problem 28

O / open | AMR 问题清单汇编

See [Houdayerloana2024] where this is proven for commutants.

Let $N$ be a non-prime II$_1$ factor. Does it have a unique sequential commutation orbit?

背景链接：https://arxiv.org/abs/2511.20377

11617 | AMR-153-0029

### Ultraproduct II_1 Factors - Problem 29

O / open | AMR 问题清单汇编

Let $N$ be a separable II$_1$ factor. Then do $N$ and $N^t$ have the same number of sequential commutation orbits?

背景链接：https://arxiv.org/abs/2511.20377

11618 | AMR-153-0003

### Ultraproduct II_1 Factors - Problem 3

O / open | AMR 问题清单汇编

[Solved by J. Peterson [peterson2026ultraproductapproximationspropertyt]] A non Gamma factor $N$ is said to be pseudocompact if there exists an ultrafilter $\mathcal{U}$ such that $N$ is elementarily equivalent to $\mathbb{M}^{\mathcal{U}}$. Is $L(\mathbb{F}_n)$ pseudocompact for $n\geq 2$?

背景链接：https://arxiv.org/abs/2511.20377

11619 | AMR-153-0030

### Ultraproduct II_1 Factors - Problem 30

O / open | AMR 问题清单汇编

Recall $A,B\subset M$ are $k$-independent if for all alternating centered words $w$ of length $k$ in $A,B$, one has $\tau(w)=0$. Usage of plain independent refers to $2$-independent. Additionally, $u,v\in \mathcal{U}(M)$ are $k$-independent means that the von Neumann algebras they generate are $k$-independent.

Suppose $u,v\in N^\mathcal{U}$ independent unitaries, then can they be lifted to sequences of pairwise independent unitaries in $N$?

背景链接：https://arxiv.org/abs/2511.20377

11620 | AMR-153-0031

### Ultraproduct II_1 Factors - Problem 31 (Solved by G. Patchell and A. Shiner, paper in preparation)

O / open | AMR 问题清单汇编

This is done in some particular cases in [exoticCIKE].

Suppose $u,v\in N^\mathcal{U}$ are freely independent unitaries, then can they be lifted to sequences of pairwise $k$-independent unitaries in $N$ for all $k$?

背景链接：https://arxiv.org/abs/2511.20377

11621 | AMR-153-0032

### Ultraproduct II_1 Factors - Problem 32

O / open | AMR 问题清单汇编

Suppose $M$ is a separable subfactor admitting two embeddings $\pi_i:M\to N^\mathcal{U}$ and further suppose $\varphi_n:N\to N$ are a sequence of unital completely positive maps, such that $\Phi=(\varphi_n)_{\mathcal{U}}$ conjugates $\pi_1$ to $\pi_2$. Then does there exist a unitary $u\in N^\mathcal{U}$ that conjugates $\pi_1$ and $\pi_2$?

背景链接：https://arxiv.org/abs/2511.20377

11622 | AMR-153-0033

### Ultraproduct II_1 Factors - Problem 33

O / open | AMR 问题清单汇编

Is there a non amenable separable II$_1$ factor $N$ such that any pair of embeddings of $N$ into $N^{\mathcal{U}}$ are conjugate by a unitary in $N^{\mathcal{U}}$?

背景链接：https://arxiv.org/abs/2511.20377

11623 | AMR-153-0034

### Ultraproduct II_1 Factors - Problem 34

O / open | AMR 问题清单汇编

If a Connes embeddable separable II$_1$ factor $N$ satisfies that any pair of embeddings of $N$ into $R^{\mathcal{U}}$ are conjugate by an automorphism of $R^{\mathcal{U}}$, then is $N\cong R$?

背景链接：https://arxiv.org/abs/2511.20377

11624 | AMR-153-0035

### Ultraproduct II_1 Factors - Problem 35

O / open | AMR 问题清单汇编

See [ScottSri2019ultraproduct, AGKE] for a detailed discussion of such Jung-type properties ([JungTubularity]) and what is known and not known.

Is the pentagon right angled Artin group II$_1$-factor stable?

背景链接：https://arxiv.org/abs/2511.20377

11625 | AMR-153-0036

### Ultraproduct II_1 Factors - Problem 36 (II$_1$ factor Tarski's problem)

O / open | AMR 问题清单汇编

Are the factors $L(\mathbb{F}_n)$ elementarily equivalent for different values of $n$?

背景链接：https://arxiv.org/abs/2511.20377

11626 | AMR-153-0037

### Ultraproduct II_1 Factors - Problem 37

O / open | AMR 问题清单汇编

The reduced $C^*$-algebra version of this problem has been settled in the negative in the recent work [elayavalli2025negativeresolutioncalgebraictarski].

The problem below is due to S. Popa:

Let $N$ be a Connes embeddable separable II$_1$ factor. Does it admit an embedding into $R^\mathcal{U}$ with factorial commutant?

背景链接：https://arxiv.org/abs/2511.20377

11627 | AMR-153-0038

### Ultraproduct II_1 Factors - Problem 38

O / open | AMR 问题清单汇编

Let $N_0$ be an index $2$ subfactor of an ultrapower $N^\mathcal{U}$. Then is $N_0\cong \prod_{i\to \mathcal{U}} N_i$ where $N_i\subset N$ are index $2$ subfactors?

背景链接：https://arxiv.org/abs/2511.20377

11628 | AMR-153-0039

### Ultraproduct II_1 Factors - Problem 39

O / open | AMR 问题清单汇编

Is $N^{\mathcal{U}}$ ever isomorphic to a group von Neumann algebra?

背景链接：https://arxiv.org/abs/2511.20377

11629 | AMR-153-0004

### Ultraproduct II_1 Factors - Problem 4

O / open | AMR 问题清单汇编

Does there exist a group von Neumann algebra $L(G)$ that is pseudocompact? Does there exist an icc non inner amenable group von Neumann algebra $L(G)$ that is not pseudocompact? (Second part solved by J. Peterson [peterson2026ultraproductapproximationspropertyt])

背景链接：https://arxiv.org/abs/2511.20377

11630 | AMR-153-0040

### Ultraproduct II_1 Factors - Problem 40

O / open | AMR 问题清单汇编

It follows from arguments of S. Popa [PopaOrthoPairs] that ultrapowers can never be non trivial tensor products, crossed products, amalgamated free products, or have diffuse regular abelian subalgebras.

Does there exist a II$_1$ factor $N$ such that $h(N^{\mathcal{U}})=\infty$ and $h(N)=0$, where $h$ denotes 1-bounded entropy?

背景链接：https://arxiv.org/abs/2511.20377

11631 | AMR-153-0041

### Ultraproduct II_1 Factors - Problem 42

O / open | AMR 问题清单汇编

The above ``*ultrapower-bicommutant* problem'' is credited to S. Popa, see [Popaindep].

If $M$ is a II$_1$ factor, then does $\mathbb{M}_2(\mathbb{C}) \otimes M$ always embed into $M^\mathcal{U}$? (Note this is immediate if $M$ is Connes embeddable.)

背景链接：https://arxiv.org/abs/2511.20377

11632 | AMR-153-0042

### Ultraproduct II_1 Factors - Problem 43

O / open | AMR 问题清单汇编

Recall that a II$_1$ factor $N$ is said to be solid if any subalgebra $A\subset N$ with diffuse commutant in $N$ is amenable. Recall that a II$_1$ factor $N$ is said to be $\mathcal{U}$-solid if any subalgebra $A\subset N$ with diffuse commutant in $N^\mathcal{U}$ is amenable.

Is solidity equivalent to $\mathcal{U}$-solidity in II$_1$ factors?

背景链接：https://arxiv.org/abs/2511.20377

11633 | AMR-153-0043

### Ultraproduct II_1 Factors - Problem 44

O / open | AMR 问题清单汇编

See Proposition 7 in [OzawaSolidActa] (see also [HURigid]). The problem is still open.

Is there a pair of II$_1$ factors with the same 2-quantifier theory but are not elementarily equivalent?

背景链接：https://arxiv.org/abs/2511.20377

11634 | AMR-153-0044

### Ultraproduct II_1 Factors - Problem 45

O / open | AMR 问题清单汇编

For the definition of 2-quantifier theory, please consult [quantifiertheory].

Does there exist II$_1$ factors $N,M$ that are elementarily equivalent, but not as as $C^*$-algebras?

背景链接：https://arxiv.org/abs/2511.20377

11635 | AMR-153-0045

### Ultraproduct II_1 Factors - Problem 46

O / open | AMR 问题清单汇编

Is every seperable II$_1$ factor selfless as a $C^*$-algebra?

背景链接：https://arxiv.org/abs/2511.20377

11636 | AMR-153-0046

### Ultraproduct II_1 Factors - Problem 47

O / open | AMR 问题清单汇编

For a definition of selfless see [robertselfless]. This is known for all ultraproduct II$_1$ factors, and some select other examples. Also recall that it is known that strict comparison holds for all II$_1$ factors viewed as $C^*$-algebras.

Does there exist a II$_1$ factor $N$ such that $N^\mathcal{U}$ has non-full fundamental group?

背景链接：https://arxiv.org/abs/2511.20377

11637 | AMR-153-0047

### Ultraproduct II_1 Factors - Problem 48

O / open | AMR 问题清单汇编

Suppose $N_1, N_2, M_1, M_2$ are separable II$_1$ factors such that $N_1^\mathcal{U}\overline{\otimes} N_2^\mathcal{U}\cong M_1^\mathcal{U}\overline{\otimes} M_2^\mathcal{U}$, then is $N_i$ elementarily equivalent to $M_i$ up to permutation?

背景链接：https://arxiv.org/abs/2511.20377

11638 | AMR-153-0048

### Ultraproduct II_1 Factors - Problem 49

O / open | AMR 问题清单汇编

Suppose $I,J$ are finite sets, and $\{N_i\}_{i\in I}, \{M_j\}_{j\in J}$ are separable II$_1$ factors such that $*_{i\in I}N_i^\mathcal{U}\cong *_{j\in J}M_j^\mathcal{U}$, then is $I\cong J$ and the factors are mutually elementarily equivalent up to permutation?

背景链接：https://arxiv.org/abs/2511.20377

11639 | AMR-153-0005

### Ultraproduct II_1 Factors - Problem 5

O / open | AMR 问题清单汇编

Let $N$ be a finite index subfactor of the matrix ultraproduct $\mathbb{M}^\mathcal{U}$. Then is $N$ also isomorphic to a matrix ultraproduct?

背景链接：https://arxiv.org/abs/2511.20377

11640 | AMR-153-0006

### Ultraproduct II_1 Factors - Problem 6

O / open | AMR 问题清单汇编

Denote by $\mathbb{S}^{\mathcal{U}}=\prod_{i\to \mathcal{U}}\mathbb{S}_n$, the ultraproduct of the symmetric groups with the normalized Hamming distance. Clearly, $\mathbb{S}^{\mathcal{U}}\subset \mathbb{M}^\mathcal{U}$ in a trace preserving way (where the trace on $\mathbb{S}^{\mathcal{U}}$ is given by one minus the ultralimit of the Hamming distance to the identity elements). Denote by $W^*(\mathbb{S}^{\mathcal{U}})$ the von Neumann subalgebra of $\mathbb{M}^\mathcal{U}$ generated by $\mathbb{S}^{\mathcal{U}}$. Does $W^*(\mathbb{S}^{\mathcal{U}})$ contain a copy of a matrix ultraproduct?

背景链接：https://arxiv.org/abs/2511.20377

11641 | AMR-153-0007

### Ultraproduct II_1 Factors - Problem 7

O / open | AMR 问题清单汇编

Recall that an embedding of $\iota: N\hookrightarrow M$ is existential if there is an embedding of $M\subset N^{\mathcal{U}}$ that restricts to the diagonal embedding under $\iota$. One can define this notion similarly for inclusions of groups.

Does there exist a separable $N$ admitting an existential embedding $L(\mathbb{F}_2)\subset N$ and moreover satisfies $N$ is not isomorphic to a non trivial free product?

背景链接：https://arxiv.org/abs/2511.20377

11642 | AMR-153-0008

### Ultraproduct II_1 Factors - Problem 8

O / open | AMR 问题清单汇编

For some mild evidence for the above in the negative, see [jekel2024upgradedfreeindependencephenomena] wherein a general result on free complementation in $L(\mathbb{F}_2)^\mathcal{U}$ is proved, generalizing previous forms of absorption results in the literature, for instance [Popa1983, HoudayerIoana2024, CyrilAOP, FreePinsker].

Does there exist a nonamenable strongly solid $N$ and $M$ containing an existential copy of $N$ such that $M$ admits a Cartan subalgebra?

背景链接：https://arxiv.org/abs/2511.20377

11643 | AMR-153-0009

### Ultraproduct II_1 Factors - Problem 9

O / open | AMR 问题清单汇编

Does there exist a group $G$ that contains an existential inclusion of $\mathbb{F}_2$ such that $L(G)\ncong L(\mathbb{F}_n)$ for some $n\in \mathbb{N}$? Specifically, does there exist a non elementary surface group $G$ such that $L(G)\ncong L(\mathbb{F}_n)$?

背景链接：https://arxiv.org/abs/2511.20377

11644 | OWR-13678-001

### Unbounded Sizes of Sensitive Triangulations

O / open | Oberwolfach 报告问题汇编

Are the sizes of sensitive triangulations unbounded?

来源字段：https://doi.org/10.4171/owr/2015/21

11645 | OWR-4531-001

### Unconditional Reproducing-Kernel Bases in Model Spaces

O / open | Oberwolfach 报告问题汇编

For a model space $K_{\Theta}=H^{2}\ominus\Theta H^{2}$ associated with an inner function $\Theta$, does $K_{\Theta}$ possess an unconditional basis of reproducing kernels?

来源字段：https://doi.org/10.4171/owr/2010/49

11646 | OWR-11101912-003

### Uniform Bounds for Two-Dimensional Bilinear Hilbert Transforms

O / open | Oberwolfach 报告问题汇编

Can one obtain uniform bounds for the forms associated with the entire family of two-dimensional bilinear Hilbert transforms?

来源字段：https://doi.org/10.4171/owr/2022/31

11647 | OWR-14298372-006

### Uniform Determinant Integrals for Row Contractions

O / open | Oberwolfach 报告问题汇编

For every $0\leq r<1$ and integer $k\geq 1$, prove that there is $C(r,k)$ such that, for every $d$-tuple $A=(A_1,\ldots,A_d)$ of $k\times k$ matrices with $\|A\|_{\mathrm{row}}\leq r$, one has $\sup_N\int_{U(N)^d}|\det(I-A\otimes U)|^2\,dU\leq C(r,k)$.

来源字段：https://doi.org/10.4171/owr/2024/22

11648 | OWR-1058-002

### Uniform Difference-Quotient Estimates on Adaptive Meshes

O / open | Oberwolfach 报告问题汇编

For every $u\in C^2(\Omega)$, does its finite-element Ritz projection $u_h$ satisfy $\sum_{E\subset\partial K}h_E^{-3}\|[\partial_nu_h]\|_{L^2(E)}^2\leq C(u)$ for every cell $K\in\mathcal T_h$, with $C(u)$ independent of $h$?

来源字段：https://doi.org/10.4171/owr/2005/37

11649 | AIM-FUNCTIONAL_ANALYSIS-0047

### Uniform features and the missing isomorphism step for nonstandard simple spectra

P / partially_solved | AIM 研讨会问题汇编

Is the Mackey Borel structure on the spectrum of a simple separable C*-algebra always the same when it is not standard?

All nonstandard spectra of AF algebras are isomorphic.

来源字段：http://aimpl.org/settheorycstar/7/

11650 | OWR-11101912-004

### Uniform Intermediate-Norm Square-Function Estimates

O / open | Oberwolfach 报告问题汇编

For the moment curve in dimension two, let $E$ be the Fourier extension operator and let $\mathcal I(\delta)$ partition the parameter interval into intervals of length $\delta$. Does there exist some $2<p<4$ and a constant $C_p$, independent of $\delta$, such that

$$\|Ef\|_{L^p(B_{\delta^{-2}})}\leq C_p\left\|\left(\sum_{I\in\mathcal I(\delta)}|E_If|^2\right)^{1/2}\right\|_{L^p(w_{B_{\delta^{-2}}})}?$$

Equivalently, can the logarithmic losses in the known estimates be removed?

来源字段：https://doi.org/10.4171/owr/2022/31

11651 | OWR-11128-002

### Uniform Rectifiability from Bounded Riesz Transforms

P / partially_solved | Oberwolfach 报告问题汇编

别名：Higher-Dimensional Riesz Transforms and Uniform Rectifiability [OWR-11128-010]

Is an $n$-dimensional AD-regular measure $\mu$ uniformly $n$-rectifiable if and only if the maximal Riesz transform $R_\mu^*$ is bounded on $L^2(\mu)$?

来源字段：https://doi.org/10.4171/owr/2011/36

11652 | AIM-FUNCTIONAL_ANALYSIS-0045

### Uniformization obstruction to Borel inverse functors

P / partially_solved | AIM 研讨会问题汇编

**Reconstruction.** Let $F:\mathcal C\to\mathcal D$ be a classification functor in Elliott's sense, where objects and arrows of the two categories have standard Borel codings and $F$ is Borel on both. Suppose an inverse functor $G:\mathcal D\to\mathcal C$ exists in the appropriate one-sided or equivalence sense. Must some such $G$ be Borel?

来源字段：http://aimpl.org/settheorycstar/7/

11653 | OWR-4211-005

### Uniformly Bounded Cochain Projections for p-Finite Elements

O / open | Oberwolfach 报告问题汇编

Do $p$-uniformly $L^2$-bounded cochain projections exist on suitable discrete subcomplexes of the de Rham complex for the $p$-version finite-element method, or can their nonexistence be proved in general?

来源字段：https://doi.org/10.4171/owr/2010/10

11654 | OWR-4807-001

### Unique Pure State Extensions from Diagonal Operators

S / solved | Oberwolfach 报告问题汇编

Does every pure state on the diagonal operators on an infinite-dimensional Hilbert space extend uniquely to a pure state on the algebra of all bounded operators on that Hilbert space?

来源字段：https://doi.org/10.4171/owr/2011/17

11655 | OWR-13750332-004

### Unique Recovery of Continuous Motions

O / open | Oberwolfach 报告问题汇编

Classify the functions for which a continuous motion can be uniquely recovered from the tomographic data.

来源字段：https://doi.org/10.4171/owr/2023/21

11656 | OWR-1057-006

### Uniqueness Subsets in Bergman and Korenblum Spaces

O / open | Oberwolfach 报告问题汇编

Extend the quantitative minorant criteria for separated uniqueness subsets-where decay $\log|f(z)|\leq-M(|z|)$ forces $f=0$-to Bergman spaces and to the growth spaces $A_r^s$ with $0<s<1$.

来源字段：https://doi.org/10.4171/owr/2005/36

11657 | OWR-14298372-005

### Unitary Integral Identity for Noncommutative Polynomials

O / open | Oberwolfach 报告问题汇编

Let $p$ and $q$ be stable polynomials in noncommuting variables with $p(0)=q(0)=1$ and determinantal representations $\det p(X)=\det(I-A\otimes X)$ and $\det q(X)=\det(I-B\otimes X)$. Prove $\lim_{N\to\infty}\int_{U(N)^d}\det(p(U))\overline{\det(q(U))}\,dU=\det((I-A\otimes\overline B)^{-1})$.

来源字段：https://doi.org/10.4171/owr/2024/22

11658 | OWR-1787-002

### Universal Faber Series Versus Universal Taylor Series

O / open | Oberwolfach 报告问题汇编

Let $\Omega=\mathbb{C}\setminus K$, where $K$ is compact and connected and $\Omega$ is connected, and let $Y$ be the family of Faber-expansion sets $\Gamma\subset\Omega$. Is the inclusion $\bigcap_{\Gamma\in Y}U(\Omega,\Gamma)\subseteq\bigcap_{\zeta\in\Omega}U(\Omega,\zeta)$ strict? Equivalently, does there exist a function universal for every Taylor-series center $\zeta\in\Omega$ but not for every Faber-expansion set $\Gamma\in Y$?

来源字段：https://doi.org/10.4171/owr/2008/06

11659 | OWR-14741-005

### Universal Sampling and Interpolation Density in Kernel Spaces

P / partially_solved | Oberwolfach 报告问题汇编

For a reproducing-kernel Hilbert space $H$, how many samples are needed to recover a function completely, and is there a universal density theorem or a critical density separating sampling sets from interpolation sets?

来源字段：https://doi.org/10.4171/owr/2016/33

11660 | OWR-720-015

### Universal Wavelet Generators for Dense Affine Sets

O / open | Oberwolfach 报告问题汇编

Does there exist $\psi\in L^2(\mathbb{R})$ such that every affine parameter set $\Gamma$ with lower density greater than $1$ and finite upper density yields a wavelet frame $W(\psi,\Gamma)$ for $L^2(\mathbb{R})$?

来源字段：https://doi.org/10.4171/owr/2004/10

11661 | OWR-1787-008

### Universality Under Composition of Holomorphic Self-Maps

O / open | Oberwolfach 报告问题汇编

Suppose $(\varphi_n)$ and $(\vartheta_n)$ are sequences of holomorphic self-maps of the unit disk, or of a domain in $\mathbb{C}$ or $\mathbb{C}^n$, and each sequence admits a universal function relative to $H^\infty$. Under what natural conditions does the composed sequence $(\varphi_n\circ\vartheta_n)$ also admit a universal function?

来源字段：https://doi.org/10.4171/owr/2008/06

11662 | AIM-ANALYSIS-0107

### Unordered tuples of metric trees: qualitative ALR versus injectivity

P / partially_solved | AIM 研讨会问题汇编

\({\mathbb Q}_Q(Y)\) denotes the space of unordered \(Q\)-tuples \(y=[y_1,\ldots,y_Q]\) of elements of \(Y\) equipped with the metric \[ d(y,z)=\min_{\sigma\in S_Q}\max_{1\leq j\leq Q} d(y_j,z_{\sigma(j)}). \] If \(Y\) is an absolute Lipschitz retract, is the same true of the space \({\mathbb Q}_Q(Y)\)?

来源字段：http://aimpl.org/mappingmetric/1/

11663 | AIM-ANALYSIS-0153

### Upper-density universality and zero-free approximation

P / partially_solved | AIM 研讨会问题汇编

For compact \(K\) with connected complement and \(f\in A(K)\) nonzero on \(K^\circ\), can \(f\) be uniformly approximated by polynomials having no zeros on \(K\)?

**Recovered AIM conjecture.** Let \(K\subset\mathbb C\) be compact, let \(\mathbb C\setminus K\) be connected, and let \(f\) be continuous on \(K\), holomorphic and nonzero on \(K^\circ\). If

来源字段：http://aimpl.org/hyperbolicpoly/9/

11664 | OWR-11578-002

### Vanishing-Regularization Convergence for Nonlinear Iterations

O / open | Oberwolfach 报告问题汇编

In iterative regularization for nonlinear inverse PDE problems $R(u) = g$, let $B$ be a nonlinear forward/reconstruction operator with local approximations $B_k$, and let $S_n = N_{\rho, \alpha_n} \circ R$ be the regularized inverse step using Tikhonov regularization with parameter $\alpha_n \to 0$. Can the convergence analysis of the iteration $$ g_{n+1} = g_k + B(S(g_n)) - B_k(S(g_n)) $$ be extended when the exact inverse $S$ is replaced by $S_n = N_{\rho, \alpha_n} \circ R$ as $\alpha_n \to 0$?

来源字段：https://doi.org/10.4171/owr/2012/11

11665 | OWR-13859-004

### Variable Bandwidth for Locally Oscillatory Signals

P / partially_solved | Oberwolfach 报告问题汇编

What is a rigorous notion of variable bandwidth for signals $f\in L^2(\mathbb R)$ that captures local frequency variation while retaining appropriate sampling, density, and analyticity properties?

来源字段：https://doi.org/10.4171/owr/2015/38

11666 | OWR-14748-016

### Von Neumann Generation of the Algebra T-Psi

O / open | Oberwolfach 报告问题汇编

Let $A$ be a unital $C^*$-algebra and let $\psi$ be a symmetric state on the infinite free product $*_{j=1}^\infty A_j$. In the GNS representation $\pi_\psi$, let

$$T_\psi=\bigcap_{n\geq1}W^*\!\left(\bigcup_{j\geq n}\pi_\psi(A_j)\right)$$

be the tail von Neumann algebra, and let $D_\psi\subseteq T_\psi$ be the associated tail $C^*$-algebra obtained from the canonical state-preserving conditional expectation onto $T_\psi$. Is $T_\psi$ generated as a von Neumann algebra by $D_\psi$?

来源字段：https://doi.org/10.4171/owr/2016/40

11667 | OWR-720-011

### Wavelet Structure of Canonical Tight Frames

O / open | Oberwolfach 报告问题汇编

Does the canonical normalized tight frame associated with every wavelet frame always have wavelet structure?

来源字段：https://doi.org/10.4171/owr/2004/10

11668 | OWR-12012-007

### Weighted Condenser-Energy Minimizers

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathbf A=(A_i)_{i\in I}$ be a countable condenser of pairwise disjoint nonempty closed subsets of $\mathbb R^n$, let $\mathbf f=(f_i)$ be an admissible external field, let $\mathbf a=(a_i)$ satisfy $a_i>0$ and $\sum_i a_i<\infty$, and let $g\in C(\mathbb R^n)$ satisfy $0<c_1<g<c_2<\infty$. If $G_{\mathbf f}(\mathbf A,\mathbf a,g)<\infty$, does there exist $\boldsymbol\lambda\in\mathcal E_{\mathbf f}^+(\mathbf A,\mathbf a,g)$ with $G_{\mathbf f}(\boldsymbol\lambda)=G_{\mathbf f}(\mathbf A,\mathbf a,g)$?

来源字段：https://doi.org/10.4171/owr/2012/40

11669 | OWR-11128-008

### Weighted Riesz Transform Inequalities for General Laguerre Expansions

O / open | Oberwolfach 报告问题汇编

Let $d\mu_m(r)=\prod_{j=1}^n r_j^{2m_j+1}\,dr_j$. Does the weighted estimate $\int_{\mathbb{R}_+^n}|R_{j,m}f(r)|^p\prod_{j=1}^n r_j^{m_j(p-2)}\,d\mu_m(r)\leq C_p\int_{\mathbb{R}_+^n}|f(r)|^p\prod_{j=1}^n r_j^{m_j(p-2)}\,d\mu_m(r)$ hold for $1<p<\infty$, with $C_p$ independent of $n$ and $m$, for Laguerre expansions of arbitrary type?

来源字段：https://doi.org/10.4171/owr/2011/36

11670 | OWR-8415343-001

### Weighted Sobolev Stability Under Nonlinear Composition

O / open | Oberwolfach 报告问题汇编

Let $G$ be smooth with a first-order zero at the origin, and let $s>1$. Does there exist $C_2=C_2(G,s)$ such that every coefficient family $\{\phi_{\widetilde{\ell}}(t)\}_{\widetilde{\ell}\in\mathbb{N}^2}$ on $[t_-,t_+]$ satisfying $\sum_{\widetilde{\ell}}(1+|\widetilde{\ell}|)^{2s}\sup_{t\in[t_-,t_+]}|\phi_{\widetilde{\ell}}(t)|^2\leq1$ also satisfies $\sum_{\widetilde{\ell}}(1+|\widetilde{\ell}|)^{2s}\sup_{t\in[t_-,t_+]}|(G\circ\phi)_{\widetilde{\ell}}(t)|^2\leq C_2$?

来源字段：https://doi.org/10.4171/owr/2021/40

11671 | OWR-3389-021

### Weighted Yamabe Heat-Trace Comparison

O / open | Oberwolfach 报告问题汇编

Let $Y$ be the conformal Laplacian on the round sphere $S^n$, let $W>0$ be smooth with $\int_{S^n}W^{n/2}$ equal to the round-sphere volume, and set $Y_W=W^{-1/2}YW^{-1/2}$. For $n\geq3$, is $\max_{t>0}t^{n/2}\operatorname{Tr}(e^{-tY_W})\leq\max_{t>0}t^{n/2}\operatorname{Tr}(e^{-tY})$?

来源字段：https://doi.org/10.4171/owr/2009/06

11672 | AIM-ANALYSIS-0092

### What a Kadison-Singer partition would have to glue

P / partially_solved | AIM 研讨会问题汇编

KS says that we can partition K into subspaces like this. Can we improve the Banach-Mazur distance of a convex body to the cube using this partition? The best bound currently is n^{5/6}.

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

11673 | AIM-FUNCTIONAL_ANALYSIS-0049

### What pure-state reducibility remembers

P / partially_solved | AIM 研讨会问题汇编

If A is a C*-algebra, denote by $E_A$ the relation of unitary equivalence of pure states of A.

Assume $A$ and $B$ are C*-algebras and $E_A$ is Borel-reducible to $E_B$. What does this fact imply about the relation between $A$ and $B$?

来源字段：http://aimpl.org/settheorycstar/7/

11674 | AIM-ANALYSIS-0100

### What rank-one perturbation can and cannot eliminate

P / partially_solved | AIM 研讨会问题汇编

**Bodmann's question.** Can the explicit mixed/multivariate characteristic-polynomial machinery used after the Kadison--Singer theorem be replaced by a Bourgain--Tzafriri-style rank-one perturbation argument? More specifically, can one reproduce the relevant multivariate argument using resolvents and rank-one updates instead of manipulating the polynomials themselves?

来源字段：https://docs.google.com/document/d/1XRw0WyG5PshqboCp99tovCRD1GCK0jXZiNLH_nw618c/edit

11675 | AIM-SEVERAL_COMPLEX_VARIABLES-0006

### What scalar Bergman commutators detect

P / partially_solved | AIM 研讨会问题汇编

Let $\Omega$ be a smooth bounded pseudoconvex domain, let $\mathbf B_\Omega$ denote the Bergman projection operator and $M_\psi$ denote multiplication by $\psi$. Suppose that $[\mathbf B_\Omega,M_\psi]$ is compact on $L^2(\Omega)$ for all $\psi\in C(\overline\Omega)$. Is $N_1$ compact on $L^2_{(0,1)}(\Omega)$?

来源字段：http://aimpl.org/crscv/2/

11676 | AIM-ANALYSIS-0167

### Which web is meant by a Gaussian free-field zero set?

P / partially_solved | AIM 研讨会问题汇编

Question 10: Scott Sheffield Consider a function who Fourier transform is white noise on the unit circle. We aim to understand the web like appearance, that is the zero level lines of the Gaussian free field. Zelditch and Schramm make the above question more precise.

Direct inspection of the two-page source PDF gives the exact text:

There is no additional definition or nearby clarification after Question 10.

The two sentences do not specify one mathematical object. Three readings must be separated.

1. **White noise supported on the frequency circle.** In the conventional random-wave reading, the inverse Fourier transform of Gaussian white noise on $S^1\subset\mathbb R^2$ is the monochromatic Gaussian plane wave. It is a real-analytic random function satisfying the Helmholtz equation, so its pointwise zero set is a genuine collection of curves. 2. **White noise as a boundary distribution.** White noise on the spatial circle has independent Fourier coefficients and is not a pointwise function. Its Poisson extension is a smooth harmonic function in the open disk, completely determined by the boundary distribution. Neither the boundary noise nor this harmonic extension is the two-dimensional zero-boundary GFF. 3. **The two-dimensional GFF.** The continuum GFF has Fourier multiplier $|\xi|^{-1}$ applied to bulk white noise, or covariance multiplier $|\xi|^{-2}$. It uses all radial frequencies and is a random distribution, not a function. Consequently, a raw set $\{z:h(z)=0\}$ is undefined. Canonical “level lines” require boundary data and a local-set/Markov characterization, or a specifically proved approximation scheme.

Zelditch later explicitly distinguished the almost-fixed-frequency random wave ensemble, whose proposed percolation-type scaling is associated conjecturally with $\mathrm{SLE}_6$, from cutoff GFF ensembles associated with $\mathrm{SLE}_4$ [3]. This supports treating the AIM text as a compact juxtaposition of two web questions, not as an identity between the plane wave and the GFF.

The reconstructed research tasks are therefore:

- determine what is rigorously known about the monochromatic plane-wave nodal web; - state the now-canonical meaning of GFF level lines and loop webs; - determine what, if anything, can be inferred for raw smooth or Fourier truncation zero sets.

来源字段：https://aimath.org/WWN/randomzeros/arcc1.pdf

11677 | AIM-ANALYSIS-0054

### Wild realization and a constant-Jacobian packing obstruction

P / partially_solved | AIM 研讨会问题汇编

Let \(F\) be a non-polynomial automorphism of \(\mathbb C^2\). What are the possible biholomorphic types of Fatou components of \(F\)?

来源字段：http://aimpl.org/scvproblems/2/

11678 | OWR-2043-005

### Wolff Cone-Decomposition and Local-Smoothing Conjectures

P / partially_solved | Oberwolfach 报告问题汇编

For functions $f_k$ whose Fourier supports lie in $\delta$-plates of the truncated light cone in $\mathbb{R}^{d+1}$, does $\lVert\sum_k f_k\rVert_p\leq C_\varepsilon\delta^{-\alpha(p)-\varepsilon}(\sum_k\lVert f_k\rVert_p^p)^{1/p}$ hold for $p>2(d+1)/(d-1)$, where $\alpha(p)=d(1/2-1/p)-1/2$? Does the stronger mixed-norm version with $(\sum_k\lVert f_k\rVert_p^2)^{1/2}$ and exponent $\beta(p)=(d-1)/4-(d+1)/(2p)$ hold for $p>2+4/(d-1)$? Do the associated Sogge local-smoothing and cone-multiplier estimates hold in their conjectured ranges?

来源字段：https://doi.org/10.4171/owr/2008/32

11679 | OWR-1062-007

### Z-Stability of Infinite Tensor Powers

P / partially_solved | Oberwolfach 报告问题汇编

Let $A$ be a separable, unital, nuclear, non-elementary $C^*$-algebra, and let $\mathcal{Z}$ denote the Jiang--Su algebra. Is the infinite minimal (spatial) tensor power $A^{\otimes \infty}$ always $\mathcal{Z}$-stable (i.e., $A^{\otimes \infty} \cong A^{\otimes \infty} \otimes \mathcal{Z}$)?

来源字段：https://doi.org/10.4171/owr/2005/41

11680 | AMR-036-0015

### Zeros and one-points on three rays

O / open | AMR 问题清单汇编

Does there exist an entire function whose zeros lie on the positive ray and whose $1$-points lie on two rays making angles $\pm\alpha$ with it, for some $\alpha\in(\pi/3,\pi/2)\setminus\{2\pi/5\}$?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

11681 | AIM-FUNCTIONAL_ANALYSIS-0024

### ZFC nonliftable projection-generated Calkin masas and separable local lifting

P / partially_solved | AIM 研讨会问题汇编

Does every masa in the Calkin algebra that is generated by its projections lift to a masa in $\mathcal{B}\left( H\right) $?

来源字段：http://aimpl.org/settheorycstar/2/

# 资料汇编 08 | 偏微分方程 / Partial Differential Equations

438 条记录。O=开放；P=部分解决；S=已解决。状态与英文/LaTeX 表述均沿用来源。

11682 | OWR-4339-008

### $\varepsilon$-Regularity Estimate for Donaldson's Equation

O / open | Oberwolfach 报告问题汇编

别名：Duplicate: $\varepsilon$-Regularity for Donaldson's Equation [OWR-4339-007]

For Donaldson's almost-complex Calabi-Yau equation, do uniform constants $\varepsilon,C>0$ exist such that $$r^{-2}\int_{B_g(p,r)}\operatorname{tr}_g(\widetilde g)\,dV_g\leq\varepsilon$$ implies $$\sup_{B_g(p,r/2)}\operatorname{tr}_g(\widetilde g)\leq Cr^{-4}\int_{B_g(p,r)}\operatorname{tr}_g(\widetilde g)\,dV_g$$?

来源字段：https://doi.org/10.4171/owr/2010/26

11683 | OWR-2047-001

### $p$-Version Convergence of Plane-Wave Discontinuous Galerkin Methods

P / partially_solved | Oberwolfach 报告问题汇编

For plane-wave discontinuous Galerkin methods for the homogeneous Helmholtz equation, develop and analyze the $p$-version in which the approximation order or number of plane-wave directions is increased, including its convergence rates, error estimates, and pollution behavior.

来源字段：https://doi.org/10.4171/owr/2008/36

11684 | DARPA-004

### 21st Century Fluids

P / partially_solved | DARPA 挑战

Extend classical fluid dynamics to handle complex substances like foams, suspensions, gels, and liquid crystals.

背景链接：https://web.math.utk.edu/~vasili/refs/darpa07.MathChallenges.html

11685 | AIM-PDES-0016

### A co-periodic constrained-positivity certificate for generalized KdV waves

S / solved | AIM 研讨会问题汇编

Prove that the spectrum for spatially periodic solutions to generalised KdV lies entirely on the imaginary axis.

来源字段：http://aimpl.org/compproofstability/6/

11686 | AIM-PDES-0006

### A coherent-stretching diagnostic for exponential material-line growth

P / partially_solved | AIM 研讨会问题汇编

Exponential growth of the length of Lagrangian flow map for 2D Euler.

Find a solution $u$ of 2D Euler with $$ \sup_t \|u\|_{C^1(\mathbb T^2)}<\infty, $$ and a curve, $l$, of finite length such that the length of $\Phi_t(l)$ grows exponentially in time. Here, $\Phi_t$ is the Lagrangian flow map of $u$.

**Recovered working statement.** Find a global classical solution of two-dimensional incompressible Euler on $\mathbb T^2$, a regular embedded $C^1$ curve $\gamma:[0,L_0]\to\mathbb T^2$, and constants $c,\lambda>0$ such that $$ \sup_{t\geq0}\|u(t)\|_{C^1}<\infty, \qquad \mathcal L(t):=\operatorname{Length}(\Phi_t\circ\gamma) \geq c e^{\lambda t}\quad(t\geq0). $$ Length means material arclength, with the initial parametrization carried by the flow. It does **not** mean an operator norm of $\Phi_t$. Because $\Phi_t$ is a diffeomorphism, an initially embedded curve stays embedded.

来源字段：http://aimpl.org/smallscalefluid/6/

11687 | AIM-PDES-0094

### A conservation-aware calibration of PDE Arnold diffusion

P / partially_solved | AIM 研讨会问题汇编

5. (Paul Rabinowitz) Is there a reasonable PDE analogue of Arnold's diffusion? It seems that a version of finite dimensional transition chains is not the right mech-anism for PDE. There are results of Kuksin which prove "diffusion" for PDE with respect to high order Sobolev norms. The same problem for infinite lattices like Fermi-Pasta-Ulam system. 2

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11688 | AIM-PDES-0082

### A dense analytic rational skeleton for an analytic Aubry foliation

P / partially_solved | AIM 研讨会问题汇编

Let $f:\mathbb T\times\mathbb R\to\mathbb T\times\mathbb R$ be an analytic twist map. Suppose that $\mathbb T\times\mathbb R$ is foliated by invariant curves $\Gamma_c$ (which are Aubry--Mather sets). Are these curves $\Gamma_c$ necessarily analytic?

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11689 | AIM-PDES-0019

### A determinant-line phase certificate for higher-dimensional stability counts

P / partially_solved | AIM 研讨会问题汇编

Develop a geometric approach to stability that generalizes to higher dimensions.

来源字段：http://aimpl.org/compproofstability/9/

11690 | AIM-PDES-0020

### A full-line Maslov certificate without pulse reversibility

P / partially_solved | AIM 研讨会问题汇编

Given a possibly non-reversible homoclinic pulse whose full-line spectral problem is Hamiltonian in space, can its relevant real eigenvalues be counted by a Maslov index without using reflection symmetry, and what additional data replace the symmetry reduction?

来源字段：http://aimpl.org/compproofstability/10/

11691 | AIM-PDES-0084

### A full-support certificate for excluding Denjoy dynamics on invariant twist-map curves

P / partially_solved | AIM 研讨会问题汇编

Does there exist a \(C^3\) twist map \[ f:\mathbb T\times\mathbb R\longrightarrow\mathbb T\times\mathbb R \] with an invariant essential curve \(\Gamma\), having irrational rotation number, such that \(f|_\Gamma\) is not topologically conjugate to the corresponding rigid rotation?

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11692 | AIM-PDES-0014

### A mode-complete certificate for two-dimensional Evans computations

P / partially_solved | AIM 研讨会问题汇编

Develop rigorous computer assisted methods for computing the Evans function in two dimensions.

来源字段：http://aimpl.org/compproofstability/4/

11693 | AIM-PDES-0010

### A necessary pulled-wake compatibility filter for cubic complex Ginzburg-Landau fronts

P / partially_solved | AIM 研讨会问题汇编

Prove existence and stability of pulled fronts for the complex Ginzburg-Landau equation.

来源字段：http://aimpl.org/compproofstability/2/

11694 | AIM-PDES-0021

### A normalized a posteriori center-graph and chart-exit certificate

P / partially_solved | AIM 研讨会问题汇编

Develop rigorous computer assisted methods for handling center manifolds.

来源字段：http://aimpl.org/compproofstability/11/

11695 | AIM-PDES-0052

### A positive four-velocity kinetic profile and its transverse Evans determinant

P / partially_solved | AIM 研讨会问题汇编

P. Szmolyan: (Kinetic and Boltzmann Equations)

but then includes Y. Li's water-wave heading and setup. The original AIM PDF was inspected directly. On page 2, immediately before this question, it has

\[ f_t+v\mathbin{\cdot}f_x=Q(f,f),\qquad f=f(x,v,t),\quad x\in\mathbb R^d, \tag{1.1} \] followed by

\[ (v-c)f'=Q(f,f). \tag{1.2} \] The PDF text stream itself drops the equality sign between \(v\cdot f_x\) and \(Q(f,f)\); (1.1) is the unambiguous reconstruction from the displayed kinetic equation and its profile equation. The line break “pertur-bations” is only line-end hyphenation. After the parenthetical suggestion about discrete velocities, the PDF starts a new heading, “Y.

There is one additional notational compression. For a planar wave normal to \(x_1\), with \(z=x_1-ct\), the multidimensional kinetic equation gives

\[ (v_1-c)\bar f'(z)=Q(\bar f(z),\bar f(z)), \qquad \bar f(\pm\infty)=f_\pm. \tag{1.3} \] Thus the scalar \(v-c\) in (1.2) is read as the normal velocity \(v_1-c\). The recovered problem is:

1. under what collision, end-state, and speed conditions does (1.3) have a positive heteroclinic profile; 2. when is that planar profile stable to perturbations depending on transverse spatial variables; and 3. how should an Evans function be constructed for the resulting kinetic spectral problem?

来源字段：https://aimath.org/WWN/multidimwaves/multidimwaves.pdf

11696 | AIM-PDES-0018

### A precise one-dimensional Swift-Hohenberg quasipattern reduction

O / open | AIM 研讨会问题汇编

A starting point would be the following problem.

Prove existence of quasiperiodic patters for the 1d Swift-Hogenberg.

来源字段：http://aimpl.org/compproofstability/8/

11697 | OWR-1591-001

### A Priori Estimates for Nonlinear Hessian Equations

O / open | Oberwolfach 报告问题汇编

For the Hessian equation $\sigma_2(D^2u)=f(x,u,Du)$, what conditions on $f$ ensure estimates analogous to those for $\sigma_2(D^2u)=1$?

来源字段：https://doi.org/10.4171/owr/2007/35

11698 | AIM-PDES-0102

### A quantitative primal-dual certificate for stochastic Mather measures

P / partially_solved | AIM 研讨会问题汇编

Investigate possible extensions of the techniques used in the Aubry--Mather theory and Monge--Kantorovich problems to study linear programming problems in infinite dimensions. An example of such extensions are the stochastic Mather measures which can be used to analyze second order Hamilton--Jacobi equations.

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11699 | AIM-PDES-0034

### A radial phase diagram for a singular complex Monge-Ampere density

P / partially_solved | AIM 研讨会问题汇编

Let \(\det(u_{\overline k j})=f(z)/|z|^\alpha\) for \(f(z)\) smooth. What is the optimal regularity of \(u\)?

来源字段：http://aimpl.org/nonlinpdegeom/1/

11700 | AIM-PDES-0037

### A source-defect diagnosis, forced Aubin--Mabuchi zero mode, and cscK linearization

O / open | AIM 研讨会问题汇编

(Question of Donaldson) Find smooth solutions to the equation \[ \ddot{\varphi}- \frac 12 |\nabla \dot \varphi|^2_{\omega_\varphi}=-\lambda R_{\omega_\varphi} \] for $\lambda>0$ on $M \times [0,1]$ with $\varphi|_{M\times\{0\}}=0$, $\varphi|_{M\times\{0\}}=\varphi_0$.

来源字段：http://aimpl.org/nonlinpdegeom/2/

11701 | AIM-PDES-0064

### A symmetry obstruction and reversible two-mode diagnostic for vortex-pattern return

P / partially_solved | AIM 研讨会问题汇编

**T. Kapitula: (Transient dynamics for vortex patterns).** Setup: Start with the Gross--Pitaevskii equation \[ i q_t+\Delta q\mathbin{\pm}|q|^2q=V(x)q. \] Numerical investigations show that if one starts with a ``nice'' vortex pattern with a lot of symmetry, it evolves to an ``ugly'' vortex pattern with little symmetry or structure, and then evolves further to a ``nice'' pattern again. **Problem:** By what mechanism does this happen and how can one capture the general dynamics?

来源字段：https://aimath.org/WWN/multidimwaves/multidimwaves.pdf

11702 | AIM-PDES-0013

### A two-level validated certificate for Maslov indices of planar elliptic problems

P / partially_solved | AIM 研讨会问题汇编

For self-adjoint elliptic spectral problems on a bounded physical domain \(\Omega\subset\mathbb R^2\), develop validated-numerics certificates for the Maslov index of a path of Cauchy-data Lagrangian subspaces in the boundary trace space.

来源字段：http://aimpl.org/compproofstability/4/

11703 | AIM-PDES-0012

### A two-radius core-tail certificate for infinite-dimensional spatial dynamics

P / partially_solved | AIM 研讨会问题汇编

Use computer assisted methods to extend spatial dynamics in infinite dimensions from the perturbative regime.

来源字段：http://aimpl.org/compproofstability/3/

11704 | AIM-PDES-0002

### A viscosity-uniform drift-shear family

P / partially_solved | AIM 研讨会问题汇编

Consider the 2D stationary Navier-Stokes equations on \(\mathbb{T}^2\) with forcing \(f\) and viscosity \(\nu>0\). Find a fixed (\(\nu\)-independent) \(f\) such that \(\sup\|u^\nu\|_{L^2}<\infty\). This is a problem suggested by Peter Constantin.

来源字段：http://aimpl.org/smallscalefluid/2/

11705 | OWR-699-001

### Adaptive Diffusion Parameters for Discretization Accuracy

P / partially_solved | Oberwolfach 报告问题汇编

Can the accuracy of a discretization be enhanced by locally adapting diffusion parameters?

来源字段：https://doi.org/10.4171/owr/2004/16

11706 | OWR-2494-002

### Adaptive Error Control for Navier-Stokes Equations

O / open | Oberwolfach 报告问题汇编

Develop reliable a posteriori error control and associated adaptive mesh algorithms for numerical approximations of the Navier-Stokes equations.

来源字段：https://doi.org/10.4171/owr/2008/56

11707 | OWR-2654835-001

### Additional Measurements for Semilinear Fractional Inverse Sources

P / partially_solved | Oberwolfach 报告问题汇编

For an inverse source problem in a semilinear fractional PDE, what additional measurement is sufficient to recover the unknown source term?

来源字段：https://doi.org/10.4171/owr/2020/39

11708 | OWR-1792-001

### Admissibility and Kinetic Relations for Undercompressive Shocks

O / open | Oberwolfach 报告问题汇编

For the singularly perturbed conservation law $u_t^\varepsilon+f(u^\varepsilon)_x=R(\varepsilon u_x^\varepsilon,\varepsilon^2u_{xx}^\varepsilon,\ldots)_x$ and its limit $u_t+f(u)_x=0$, determine which discontinuities are admissible and the corresponding Riemann solution. In particular, characterize undercompressive shocks and the kinetic relation needed for uniqueness.

来源字段：https://doi.org/10.4171/owr/2008/11

11709 | OWR-4990380-001

### Admissibility and Prediction for Multidimensional Hyperbolic Balance Laws

O / open | Oberwolfach 报告问题汇编

Find suitable admissibility criteria for physically reasonable solutions of multidimensional hyperbolic balance laws, especially the Euler system, determine how these criteria affect multiphase-flow models, random systems, and structure-preserving numerical schemes, and assess whether such laws can predict pandemic evolution.

来源字段：https://doi.org/10.4171/owr/2021/11

11710 | AIM-PDES-0027

### Affine-normal arrival-time models and sharp center regularity

P / partially_solved | AIM 研讨会问题汇编

For \(u\) strictly convex, determine the existence and regularity of solutions to \[ \operatorname{det}(D^2u)\,u^{ij}u_i u_j=1. \]

来源字段：http://aimpl.org/nonlinpdegeom/1/

11711 | AIM-PDES-0097

### Aggregate singular sets of minimizing laminations and the codimension-one stable norm

P / partially_solved | AIM 研讨会问题汇编

8. (Franz Auer) Conjecture: If $T\in C_{n-1}M$ is a minimizing closed current, then the Hausdorff dimension of the corresponding singularity set is at most $n-7$.

**Problem 28 (Franz Auer).** If $M^n$ is a compact Riemannian manifold and $T\in C_{n-1}M$ is a minimizing closed real current, conjecturally the Hausdorff dimension of the corresponding singularity set is at most $n-7$.

**Problem 29.** Define the stable norm on $H_q(M,\mathbb R)$ by \[ \|h\|=\inf\{\mathbf M(T):[T]=h\}. \] This is an analogue of Mather's \(\beta\)-function. Study convexity and differentiability properties of \[ B=\{h\in H_q(M,\mathbb R):\|h\|\leq 1\}. \]

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11712 | OWR-12589-001

### All-Time Soliton Resolution for Equivariant Wave Maps

S / solved | Oberwolfach 报告问题汇编

For the finite-energy equivariant wave-map equation, can soliton resolution be proved for all times rather than only along a sequence of times? Can one construct a solution with at least two bubbles?

来源字段：https://doi.org/10.4171/owr/2013/41

11713 | OWR-16410-008

### Almost Everywhere Regularity of Parametrized Stationary Varifolds

O / open | Oberwolfach 报告问题汇编

For a general parametrized integer stationary $n$-varifold, does the singular set have zero $n$-dimensional Hausdorff measure without additional hypotheses?

来源字段：https://doi.org/10.4171/owr/2018/35

11714 | AIM-PDES-0065

### An angular-isotypic stability filter for higher-dimensional encapsulated vortices

P / partially_solved | AIM 研讨会问题汇编

**Open problem.** Under what conditions are the encapsulated-vortex solutions \(u(x,t)=e^{\mu Kt}\widehat\psi(\widehat x)w(r)\) of NLS or NLKG with \(N\ge3\) stable or unstable?

来源字段：https://aimath.org/WWN/multidimwaves/multidimwaves.pdf

11715 | AIM-PDES-0009

### An exact annular inflow-outflow boundary layer

P / partially_solved | AIM 研讨会问题汇编

Vanishing viscosity limit in a setting like Temam--Wang.

Study the vanishing viscosity limit from Navier-Stokes to Euler on the annulus in a setting like the work of Temam and Wang, with in-flow and out-flow conditions.

**Recovered working statement.** Let \[ A_{r_-,r_+}=\{x\in\mathbb R^2:r_-<|x|<r_+\}, \qquad 0<r_-<r_+. \] For smooth, compatible inflow/outflow data whose normal component is nonzero on both boundary components, prove that solutions \(u^\nu\) of incompressible Navier--Stokes converge as \(\nu\downarrow0\) to a smooth solution \(u^0\) of incompressible Euler, identify the outflow boundary layer, and quantify convergence.

来源字段：http://aimpl.org/smallscalefluid/9/

11716 | AIM-PDES-0056

### An explicit two-family viscous corner layer

P / partially_solved | AIM 研讨会问题汇编

Williams, “Two Interacting Shocks in 1D,” in the AIM workshop list *Open Problems in Multidimensional Stability of Waves and Patterns* (2005). The setup is

\[ u_t+f(u)_x=0 \tag{A} \]

and its identity-viscosity regularization

\[ u_t^\varepsilon+f(u^\varepsilon)_x =\varepsilon u^\varepsilon_{xx}, \qquad u\in\mathbb R^2,\quad x\in\mathbb R. \tag{B} \]

The recovered problem asks whether one can construct explicit solutions of (B), “say of the form \(u^\varepsilon(t,x,t/\varepsilon,x/\varepsilon)\),” converging reasonably to solutions of (A), while explicitly resolving the dynamics at the corner where \(U_l\) and \(U_r\) meet in the \(xt\)-plane.

The original PDF was checked directly. Thus the displayed target is a forward two-shock Riemann fan, normally involving two distinct characteristic families, not the merger of two scalar same-family shocks. The source sentence says “two-shock solution to the viscous conservation law,” although the sharp sector diagram is naturally the inviscid pattern; this apparent wording slip is recorded rather than silently altered.

Humphreys, K. Promislow, and B. Sandstede entries. Those entries are not part of AIM-PDES-0056 and are excluded here.

来源字段：https://aimath.org/WWN/multidimwaves/multidimwaves.pdf

11717 | OWR-17130-012

### Ancient Ovals as Post-Singularity Limit Flows

O / open | Oberwolfach 报告问题汇编

Can ancient ovals arise as limit flows at singularities of a mean curvature flow after the first singular time?

来源字段：https://doi.org/10.4171/owr/2019/34

11718 | OWR-14299521-004

### Angle Dependence of One-Phase Bernoulli Singularities

O / open | Oberwolfach 报告问题汇编

Can the optimal singular-set dimension for one-phase Bernoulli free boundaries depend on the contact angle?

来源字段：https://doi.org/10.4171/owr/2025/30

11719 | OWR-16936-002

### Anomalous Dissipation in Stochastic Navier-Stokes Equations

O / open | Oberwolfach 报告问题汇编

Do there exist stationary martingale solutions of the three-dimensional stochastic Navier-Stokes equations on $\mathbb{T}^3$ satisfying weak anomalous dissipation, $\lim_{\nu\to0}\nu\mathbb{E}\lVert u\rVert_{L^2_x}^2=0$?

来源字段：https://doi.org/10.4171/owr/2019/24

11720 | OWR-829-001

### Approximation Classes for Adaptive Finite Elements

P / partially_solved | Oberwolfach 报告问题汇编

For the two-dimensional Poisson equation with linear finite elements, suppose $0<s\leq\frac12$ and the solution $u$ belongs to the approximation class $\mathcal{A}^s$, meaning that its best energy-norm error on suitable $n$-element partitions is $O(n^{-s})$. Must the right-hand side $f$ admit piecewise-constant approximations on $n$-element partitions whose error in the dual norm is also $O(n^{-s})$?

来源字段：https://doi.org/10.4171/owr/2005/24

11721 | AIM-PDES-0003

### Arbitrary-large one-phase solutions for 2D biharmonic Burgers

P / partially_solved | AIM 研讨会问题汇编

Large data global wellposedness for 2D Burgers with hyper-dissipation

Prove the global existence of (strong) solutions to the 2D Burgers' equation with hyperviscosity; namely, \(\partial_tu+u\cdot\nabla u=-\nu\Delta^2u\).

来源字段：http://aimpl.org/smallscalefluid/3/

11722 | AIM-PDES-0059

### Arbitrary-period stability exchange on the local Zaitsev branch

P / partially_solved | AIM 研讨会问题汇编

Consider the KP-I equation \[ (u_t-u_{xxx}+c u_x+u u_x)_x+u_{yy}=0. \] The KdV line soliton has no \(y\)-variation and is unstable to transverse perturbations. A family of \(y\)-periodic waves bifurcates from it and connects to the lump solution, which was then presumed stable.

**What is the stability of the KP-I soliton solution with periodic structure?**

来源字段：https://aimath.org/WWN/multidimwaves/multidimwaves.pdf

11723 | AIM-PDES-0093

### Arnold diffusion, radial genericity, and separatrix-map actions

P / partially_solved | AIM 研讨会问题汇编

**Problem 23 (Arnold diffusion).** Consider a \(C^r\), \(3\le r\le\omega\), Hamiltonian \[ H_\varepsilon:\mathbb T^n\times\mathbb R^n\times\mathbb T\to \mathbb R,\qquad H_\varepsilon(x,p,t)=H_0(p)+\varepsilon H_1(x,p,t), \] where \(H_0\) is positive definite and superlinear. Suppose \(n\ge2\). Given open sets \(U_0,U_1\subset\mathbb R^n\), is it true that for a typical \(H_1\) in the \(C^r\) topology there is \(\varepsilon_0>0\) such that, for **every** \(\varepsilon\in(0,\varepsilon_0)\), some trajectory satisfies \(p(t_0)\in U_0\) and \(p(t_1)\in U_1\)? What is the right meaning of “typical”? Does Mather's \(n=2\) cusp-residual theorem hold within trigonometric polynomials of sufficiently high order \(N\)?

**Problem 24.** Consider an *a priori* unstable Hamiltonian \[ H(\theta,I,x,y)=H_0(I)+F(x,y)+\varepsilon H_1(\theta,I,x,y) \] on \(\mathbb T^n\times\mathbb R^n\times\mathbb R^2\), where \(H_0\) is convex and superlinear, \(F\) has a separatrix loop, and \(H_1\) is generic. Understand the relation between Mather--Xia variational methods and the separatrix-map method; they seem similar in spirit.

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11724 | OWR-1789-001

### Asymptotic Stability of Orbitally Stable NLS Ground States

P / partially_solved | Oberwolfach 报告问题汇编

For the nonlinear Schrödinger equation $iu_t+\Delta u+\beta(|u|^2)u=0$ on $\mathbb{R}\times\mathbb{R}^3$, with standing waves $e^{it\omega}\varphi_\omega$, is every orbitally stable ground state asymptotically stable? Specifically, for every nearby finite-energy solution, do there exist $\omega_0>0$, $\theta\in C^1(\mathbb{R}_+,\mathbb{R})$, and $h_0\in H^1$ such that $\|u(t)-e^{i\theta(t)}\varphi_{\omega_0}-e^{it\Delta}h_0\|_{H^1}\to0$ as $t\to\infty$?

来源字段：https://doi.org/10.4171/owr/2008/08

11725 | OWR-10252939-004

### Asymptotic Stability of the $\phi^4$ Kink

P / partially_solved | Oberwolfach 报告问题汇编

Establish full asymptotic stability of the $\phi^4$ kink under odd perturbations, including control of the quadratic source term created by the internal mode.

来源字段：https://doi.org/10.4171/owr/2022/30

11726 | OWR-11788-007

### Asymptotic Symmetry of Geometric PDE Singularities

O / open | Oberwolfach 报告问题汇编

How should the folklore conjecture that finite-time singularities of parabolic geometric PDE become asymptotically as symmetric as their topology permits be formulated and interpreted?

来源字段：https://doi.org/10.4171/owr/2012/26

11727 | OWR-14298011-002

### Automatic Convexity of Optimal Spiral Strategies

O / open | Oberwolfach 报告问题汇编

Must every optimal spiral-like fire-containment strategy be locally convex and satisfy the initial-angle condition (A1)?

来源字段：https://doi.org/10.4171/owr/2023/52

11728 | AIM-PDES-0004

### Backward self-similarity in a half-space and a parasitic-profile audit

P / partially_solved | AIM 研讨会问题汇编

Existence/nonexistence of self-similar solutions for 3D Navier-Stokes

Establish the existence/nonexistence of self-similar solutions for 3D Navier-Stokes in half space.

来源字段：http://aimpl.org/smallscalefluid/4/

11729 | OWR-12988-003

### Basis Design for Numerical Homogenization

P / partially_solved | Oberwolfach 报告问题汇编

Is there a general method for constructing accurate finite-dimensional basis functions for numerical homogenization of partial differential equations with arbitrary rough or multiscale coefficients?

来源字段：https://doi.org/10.4171/owr/2014/30

11730 | DYN-004

### Birkhoff Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Birkhoff Conjecture for Integrable Convex Billiards [OWR-4330-002]

If a billiard table is strictly convex and integrable, must its boundary be an ellipse?

背景链接：https://arxiv.org/abs/2008.03566

11731 | OWR-10252939-001

### Blowup and Weak Soliton Resolution for Calogero-Moser DNLS

S / solved | Oberwolfach 报告问题汇编

For general solutions of the Calogero-Moser derivative nonlinear Schrödinger equation, determine whether finite-time blowup can occur and whether weak soliton resolution holds.

来源字段：https://doi.org/10.4171/owr/2022/30

11732 | OWR-14299912-002

### Blowup Residues, Stability, and Rates for Mass-Critical gKdV

O / open | Oberwolfach 报告问题汇编

For the mass-critical generalized Korteweg-de Vries equation, determine whether every finite-time, finite-point blowup solution with rate exponent $\nu\in(1/3,1/2)$ has an $H^1$ blowup residue, and whether the finite-time, finite-point blowup family constructed for $\nu\in(3/7,1/2)$ is stable. Determine whether the only $H^1$ power blowup rates are $\nu=2/5$ and $\nu>3/7$; if so, construct the remaining rates $\nu\in(1/3,3/7)\setminus\{2/5\}$ for infinite-energy solutions in a suitable $H^\kappa$, with $0<\kappa<1$.

来源字段：https://doi.org/10.4171/owr/2026/10

11733 | OWR-13858-004

### BMO Blowup Criteria for Incompressible Flow Potentials

O / open | Oberwolfach 报告问题汇编

For incompressible Navier-Stokes flow, does blowup at a finite time $t_*$ imply $\lVert A\rVert_{\mathrm{BMO}}\to\infty$ in three dimensions and $\lVert\psi\rVert_{\mathrm{BMO}}\to\infty$ in two dimensions, where $A$ is the three-dimensional vector potential and $\psi$ is the two-dimensional stream function?

来源字段：https://doi.org/10.4171/owr/2015/37

11734 | OWR-12011-001

### Borderline Sobolev Blowup Criteria for Hyperdissipative Flow

O / open | Oberwolfach 报告问题汇编

For a possible finite-time blowup of the $d$-dimensional hyperdissipative flow, does the borderline conjecture

$$\lVert u(t)\rVert_{H^{(d+2)/6}(\mathbb R^d)}^2\to\infty\quad\text{as }t\to t_*$$

hold? Equivalently, is finiteness of this norm on $[0,T]$ sufficient for regularity?

来源字段：https://doi.org/10.4171/owr/2012/39

11735 | OWR-4429-002

### Borderline Soliton-Potential Interactions

O / open | Oberwolfach 报告问题汇编

Consider $$u_t+\bigl(u_{xx}-\lambda u+a(\varepsilon x)u^m\bigr)_x=0$$ on $\mathbb R_t\times\mathbb R_x$, where $m\in\{2,3,4\}$, $\varepsilon>0$, $0\leq\lambda<1$, and $a$ is smooth and positive with $a(-\infty)=1$ and $a(+\infty)=2$. For the soliton-like solution asymptotic to $Q(\cdot-(1-\lambda)t)$ as $t\to-\infty$, what is its long-time behavior at the threshold $\lambda=\widetilde\lambda$ separating refraction from reflection?

来源字段：https://doi.org/10.4171/owr/2010/41

11736 | OWR-767-004

### Bound-State Moments and Embedded Singular Spectrum

O / open | Oberwolfach 报告问题汇编

For Schrödinger operators in two dimensions, does finiteness of the bound-state moments imply absence of embedded singular spectrum?

来源字段：https://doi.org/10.4171/owr/2004/53

11737 | OWR-4209-004

### Boundary Conditions Preserving Optimal Hardy-Sobolev Constants

O / open | Oberwolfach 报告问题汇编

Which boundary conditions, other than Dirichlet and Navier conditions, yield the same optimal constants $H(p,m,n)$ and $S(p,m,n)$ in the Hardy-Sobolev embeddings for $W^{m,p}(\Omega)$?

来源字段：https://doi.org/10.4171/owr/2010/08

11738 | OWR-9790366-002

### Boundary Curvature and Stickiness of Nonlocal Minimal Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

For nonlocal minimal surfaces beyond planar graphs, determine whether the nonlocal mean-curvature equation holds up to the boundary and characterize how generally boundary stickiness occurs.

来源字段：https://doi.org/10.4171/owr/2022/15

11739 | OWR-14298587-002

### Boundary Frequency Gap for Optimal Partitions

O / open | Oberwolfach 报告问题汇编

For a boundary free-interface point in a spectral optimal-partition problem in dimension $d>2$, determine the exact frequency gap $\delta_d$ above frequency $2$. Prove or disprove that $\delta_d=5/2$, attained by three domains forming a collinear triple junction of $3/2$-points that meets the boundary orthogonally.

来源字段：https://doi.org/10.4171/owr/2024/37

11740 | OWR-4412-004

### Boundary Gradient Attainment in the Affine Plateau Problem

O / open | Oberwolfach 报告问题汇编

For solutions of the affine Plateau problem, what conditions on the boundary data and domain ensure that the gradient reaches the boundary data, or equivalently that the Gauss map has the appropriate boundary behavior?

来源字段：https://doi.org/10.4171/owr/2010/31

11741 | AIM-PDES-0083

### Boundary integrability and the explicit confocal-caustic dictionary

P / partially_solved | AIM 研讨会问题汇编

Consider a billiard inside a region bounded by a closed convex curve \(\Gamma\subset\mathbb R^2\). Suppose that a neighborhood of \(\Gamma\) is foliated by caustics for the billiard. Is it true that then \(\Gamma\) is an ellipse and hence the caustics are analytic curves? The billiard in \(\Gamma\) is described by a twist map \(f\), and caustics correspond to invariant curves of \(f\). Hence this question is a particular case of the previous one.

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11742 | OWR-11101918-006

### Boundary Unique Continuation on General Lipschitz Domains

O / open | Oberwolfach 报告问题汇编

Let $\Omega\subset\mathbb R^n$ be an arbitrary Lipschitz domain, let $B$ be centered on $\partial\Omega$, and let $u\in C(\overline\Omega)$ be harmonic in $\Omega$. If $u=0$ on $\Sigma=B\cap\partial\Omega$ and

$$\mathcal H^{n-1}\bigl(\{x\in\Sigma:\nabla u(x)=0\}\bigr)>0,$$

must $u\equiv0$ in $\Omega$, without any smallness assumption on the Lipschitz constant?

来源字段：https://doi.org/10.4171/owr/2022/37

11743 | OWR-14298587-019

### Boundary Uniqueness on Lipschitz Domains

O / open | Oberwolfach 报告问题汇编

Let $\Omega\subset\mathbb R^n$ be an arbitrary Lipschitz domain, let $u\in C(\overline\Omega)$ be harmonic in $\Omega$, and suppose that $u$ vanishes on a relatively open boundary patch $\Sigma$. If $\mathcal H^{n-1}(\{x\in\Sigma:\nabla u(x)=0\})>0$, must $u\equiv 0$?

来源字段：https://doi.org/10.4171/owr/2024/37

11744 | AIM-PDES-0046

### Boundary-audited toric Calabi dissipation and a global product family

P / partially_solved | AIM 研讨会问题汇编

What is the behavior of $g_{j\overline k}(t)$ solving the Calabi flow, \[ \frac{\partial}{\partial t}g_{j\overline k} =\partial_j\partial_{\overline k}R? \] Can long time existence be established for toric surfaces?

**Recovered statement.** Let $(X^n,J,\omega_0)$ be a compact Kähler manifold of complex dimension $n$, let $[\omega_0]$ be fixed, and let \[ \omega_{\varphi(t)}=\omega_0+\sqrt{-1}\,\partial\bar\partial\varphi(t)>0 \] start from a smooth Kähler metric. With $S(\omega_\varphi)$ denoting scalar curvature and $\bar S$ its class-dependent average, use the convention \[ \partial_t\varphi=S-\bar S, \qquad \partial_tg_{j\bar k}=\partial_j\partial_{\bar k}S. \tag{1.1} \] If $X$ is a compact toric surface and the initial metric is invariant under its compact torus, does the unique smooth solution exist for all $t\geq0$? What can be said about convergence, possibly after the extremal automorphism and hence for the modified Calabi flow?

来源字段：http://aimpl.org/nonlinpdegeom/4/

11745 | OWR-14298011-006

### Bounded-Variation Solutions for Continuum Traffic Models

O / open | Oberwolfach 报告问题汇编

Does the hysteretic traffic-flow model admit a canonical Riemann solver and solutions of the initial-value problem for general bounded-variation data, at least for sufficiently small initial data?

来源字段：https://doi.org/10.4171/owr/2023/52

11746 | AIM-PDES-0057

### Branch-point resonances and parabolic time decay

P / partially_solved | AIM 研讨会问题汇编

For which more general parabolic systems can a zero of an Evans function at a branch point of the spatial dispersion relation be related rigorously to the leading large-time decay of the linear Green function and, through nonlinear estimates, to the decay of perturbations of a coherent structure?

In the original AIM PDF, the entry is headed by B. Sandstede and begins with the observation that an Evans function can often be extended to branch points of the linear dispersion relation. It then says that Murata showed that the temporal decay rates for scalar linear heat equations depend strongly on the presence of roots at those branch points. The open problem is, verbatim,

The singular “equation” appears in the source and is probably a grammatical error; it has not been silently changed. Here “this” means the dependence of large-time Green-function decay on a threshold/branch-point root or resonance. The text beginning “S. Malham: (Biscale chaos)” belongs to the following AIM record, AIM-PDES-0058, and is not part of the present problem.

Thus the recovered mathematical question is:

来源字段：https://aimath.org/WWN/multidimwaves/multidimwaves.pdf

11747 | OWR-14298797-001

### Brownian Loop Measure for Stochastic Geometric Heat Flow

O / open | Oberwolfach 报告问题汇编

For the renormalized stochastic geometric heat equation in the full subcritical regime, prove the expected connection between its invariant measure and Brownian loop measure.

来源字段：https://doi.org/10.4171/owr/2024/49

11748 | OWR-1050-018

### Category Bounds for Positive Nonlinear Schrödinger Solutions

O / open | Oberwolfach 报告问题汇编

Let $-\Delta u+(\lambda^2a(x)+1)u=|u|^{p-1}u$ on $\mathbb{R}^N$, where the zero-potential region is the disjoint union $\Omega_1\cup\Omega_2$. For sufficiently large $\lambda$, are there at least $\operatorname{cat}(\Omega_1)+\operatorname{cat}(\Omega_2)+\operatorname{cat}(\Omega_1)\operatorname{cat}(\Omega_2)$ positive solutions?

来源字段：https://doi.org/10.4171/owr/2005/29

11749 | OWR-16414-005

### Cell Segregation by Differential Attraction and Repulsion

P / partially_solved | Oberwolfach 报告问题汇编

For a two-species model with differential attraction, repulsion, and diffusion, can these mechanisms lead to cell sorting and segregation?

来源字段：https://doi.org/10.4171/owr/2018/39

11750 | AIM-PDES-0017

### Certified mass bounds for basins in finite-dimensional gradient flows

P / partially_solved | AIM 研讨会问题汇编

Estimate size of the basin of attraction for each equilibria.

来源字段：http://aimpl.org/compproofstability/7/

11751 | OWR-16414-004

### Chemotactic Cell Sorting in Keller-Segel Models

P / partially_solved | Oberwolfach 报告问题汇编

In the two-dimensional Keller-Segel chemotaxis model, can chemotaxis be the principal mechanism of cell sorting, and what occurs in the regime $\chi_2\ll1$?

来源字段：https://doi.org/10.4171/owr/2018/39

11752 | AIM-PDES-0098

### Chern and slope obstructions for pseudoholomorphic plane foliations on the four-torus

P / partially_solved | AIM 研讨会问题汇编

**Problem 30 (Victor Bangert).** Let $M^{2n}$ be a manifold with an almost complex structure $J$. A pseudo-holomorphic line is a map $f:\mathbb R^2\to M$ satisfying $$ f_y=J(f)f_x. $$ Moser proved that, for an almost complex structure $J$ on $\mathbb T^{2n}$ which is close to a standard complex structure $J_0$, and any sufficiently irrational vector $v\in\mathbb R^{2n}$, there exists a foliation of $\mathbb T^{2n}$ by pseudo-holomorphic curves which is conjugate to a linear foliation by real two-planes containing $v$. Develop a global (non-perturbative) theory of such foliations or laminations. The theory is expected to be particularly rich for $\mathbb T^4$, due to positivity of the intersection number of pseudoholomorphic curves, analogously to the intersection number for geodesics on $\mathbb T^2$ underlying Hedlund's work.

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11753 | AIM-PDES-0045

### Chern--Ricci flow on the standard Hopf surface

P / partially_solved | AIM 研讨会问题汇编

The *Chern--Ricci curvature* of a Hermitian metric $g$ is \[ R^{\mathcal{Ch}}_{\bar k j}=-\partial_j\partial_{\bar k}\log\det(g), \] and the Chern--Ricci flow is \[ \frac{\partial}{\partial t}g_{\bar k j}=-R^{\mathcal{Ch}}_{\bar k j}. \] What is its behavior on the simplest Hopf surface, \[ \mathbb (\mathbb C^2\setminus\{0\})/\big((z_1,z_2)\sim(2z_1,2z_2)\big)? \]

来源字段：http://aimpl.org/nonlinpdegeom/4/

11754 | OWR-10252937-009

### Classification of Ancient Noncollapsed Mean Curvature Flows

O / open | Oberwolfach 报告问题汇编

Classify ancient noncollapsed mean-curvature flows in higher dimensions without assuming two-convexity.

来源字段：https://doi.org/10.4171/owr/2022/28

11755 | OWR-13750335-002

### Classification of Critical CR Yamabe Solutions

P / partially_solved | Oberwolfach 报告问题汇编

Classify positive solutions of the critical CR Yamabe equation on the Heisenberg group without assuming finite volume; obtain analogous classifications for fourth-order CR $Q$-curvature equations and for the $p$-Laplace equation on $\mathbb H^n$.

来源字段：https://doi.org/10.4171/owr/2023/24

11756 | OWR-14297735-006

### Classification of Noncollapsed Four-Dimensional Steady Solitons

O / open | Oberwolfach 报告问题汇编

Prove that the only noncollapsed four-dimensional steady gradient Ricci solitons with nonnegative curvature operator are the Bryant soliton and the $\mathbb Z_2\times O(3)$-symmetric family, and that every soliton in the latter family has a ray as its blow-down.

来源字段：https://doi.org/10.4171/owr/2023/32

11757 | OWR-9790366-001

### Classification of Singular Radial Fractional-Laplacian Solutions

P / partially_solved | Oberwolfach 报告问题汇编

Classify the radial solutions of $(-\Delta)^\gamma u=A u^p$ on $\mathbb R^n\setminus\{0\}$ with an isolated singularity, equivalently the solutions of the autonomous nonlocal equation obtained by the cylindrical change of variables.

来源字段：https://doi.org/10.4171/owr/2022/15

11758 | OWR-14298587-014

### Classification of Stable Allen-Cahn Configurations

O / open | Oberwolfach 报告问题汇编

Classify the global stable critical points of the Allen-Cahn energy in $\mathbb R^3$. In particular, determine whether all stable configurations have the macroscopic regularity and minimal-surface behavior known for absolute minimizers.

来源字段：https://doi.org/10.4171/owr/2024/37

11759 | OWR-15578-006

### Closed Magnetic Geodesics and Multiplicity-One Minimal Hypersurfaces

P / partially_solved | Oberwolfach 报告问题汇编

Does every topological two-sphere in a magnetic field admit a closed embedded curve of each prescribed constant geodesic curvature? Separately, does the min-max construction of minimal hypersurfaces always produce a hypersurface with multiplicity one?

来源字段：https://doi.org/10.4171/owr/2017/35

11760 | OWR-10252935-002

### Codimension-One Stability of Lorentzian Catenoids

P / partially_solved | Oberwolfach 报告问题汇编

For $2\leq n\leq4$, is the $2n$-parameter family of Lorentzian catenoids codimension-one stable under arbitrary nonsymmetric perturbations in the appropriate weighted $L^2$ topology?

来源字段：https://doi.org/10.4171/owr/2022/26

11761 | OWR-12336-001

### Coincidence of Supports and Maxima in Infinity-Eigenvalue Limits

O / open | Oberwolfach 报告问题汇编

For the limiting infinity-eigenvalue and optimal-transport problem, is $\operatorname{spt}(f_\infty)=\operatorname*{argmax}u_\infty=\operatorname*{argmax}d_\Omega$, rather than merely $\operatorname{spt}(f_\infty)\subseteq\operatorname*{argmax}u_\infty\subseteq\operatorname*{argmax}d_\Omega$?

来源字段：https://doi.org/10.4171/owr/2013/08

11762 | OWR-16157-002

## Complementarity Conditions in Nonlinear PDE Control

O / open | Oberwolfach 报告问题汇编

Let $\Omega$ be the spatial domain and $I=(0,T)$. Consider the rate-independent control system

$$0\in-\Delta z-g-\sigma\Delta\dot z+\partial\lvert\dot z\rvert\quad\text{in }H^{-1}(\Omega),\qquad z(0)=0,$$

with objective

$$j_1(z)+j_2(z(T))+\frac12\lVert g\rVert_{H^1(I;L^2(\Omega))}^2.$$

At a locally optimal pair, write $\bar y=\dot{\bar z}$, choose $\bar q\in\partial\lvert\bar y\rvert$, and let $p$ be the adjoint variable. Is

$$\langle p,1-\lvert\bar q\rvert\rangle=0$$

a necessary optimality condition?

来源字段：https://doi.org/10.4171/owr/2018/16

11763 | AIM-PDES-0025

## Complex Monge-Ampere symmetry: a literal counterexample and a proper repair

S / solved | AIM 研讨会问题汇编

Let $u$ plurisubharmonic satisfy \[ \mathrm{det}(u_{j\overline k})=f(u) \] on $B_1(0)$ with Dirichlet boundary conditions, then $u$ is rotationally symmetric.

It gives the same text verbatim and attributes Problem 1.05 to Xu-Jia Wang. Its database object has empty `intro` and `status` fields and no remarks. A Wayback CDX query found a 2019 snapshot of the same section.

The source does **not** specify:

- whether $B_1(0)$ lies in $\mathbb C^n$ or what $n$ is (the complex indices strongly indicate $B_1\subset\mathbb C^n$); - the actual Dirichlet boundary value, in particular whether it is constant (usually normalized to zero) or arbitrary; - the regularity of $u$ or whether it is strictly plurisubharmonic; - the sign, regularity, or monotonicity of $f$; - whether the determinant is classical or Bedford--Taylor/pluripotential; - whether “rotationally symmetric” means $U(n)$-invariant or invariant under all of $O(2n)$.

These are essential mathematical data. In particular, “with Dirichlet boundary conditions” ordinarily means that some boundary trace is prescribed, not that the trace is automatically constant. Under that literal reading the statement is false (Proposition 5.1 below). A natural repaired reading is the classical problem on $B_1\subset\mathbb C^n$ with constant boundary value, positive nondecreasing $f$, and a $C^2$ strictly plurisubharmonic solution. That version follows from uniqueness and unitary covariance (Theorem 5.2).

No source text is silently changed in this report. The literal claim and the repaired theorem are kept separate.

来源字段：http://aimpl.org/nonlinpdegeom/1/

11764 | OWR-1050-006

## Connected Concentration Sets and Solution Counts

O / open | Oberwolfach 报告问题汇编

Suppose $E_\varepsilon[u_\varepsilon]\leq C\varepsilon^{N-m}$ and the concentration set is connected. Must its limiting concentration set have dimension $m$? What is the optimal asymptotic bound on the number of solutions as $\varepsilon\to0$?

来源字段：https://doi.org/10.4171/owr/2005/29

11765 | OWR-11576-002

## Constraint Functions in $Q_2$-$P_1$ Incompressible Elasticity

O / open | Oberwolfach 报告问题汇编

For a $Q_2$-$P_1$ discretization of incompressible finite elasticity, how does the choice of constraint function $\Theta$ affect stability and the numerical solution as the mesh is refined?

来源字段：https://doi.org/10.4171/owr/2012/09

11766 | OWR-14298587-010

## Continuity of Gradient Distance for Monotone Equations

O / open | Oberwolfach 报告问题汇编

Let $G\in C^0(\mathbb R^2;\mathbb R^2)$ be strictly monotone, and let $u$ be a Lipschitz solution of $\operatorname{div}G(\nabla u)=0$. If $\mathcal B$ is the set where ellipticity degenerates simultaneously from below and above, is $x\mapsto\operatorname{dist}(\nabla u(x),\mathcal B)$ continuous?

来源字段：https://doi.org/10.4171/owr/2024/37

11767 | OWR-14298587-016

## Continuum Laws for Dislocation Motion

P / partially_solved | Oberwolfach 报告问题汇编

Derive continuum evolution laws for dislocation-density fields and their associated elastoplastic effects from the physical laws governing individual dislocations.

来源字段：https://doi.org/10.4171/owr/2024/37

11768 | OWR-14750-001

## Convergence of Adaptive Hybrid Finite Element Methods

O / open | Oberwolfach 报告问题汇编

Does the adaptive hybrid finite-element procedure driven by the two-step a posteriori flux-error estimator converge?

来源字段：https://doi.org/10.4171/owr/2016/42

11769 | OWR-15212-003

### Convergence of Complex-Balanced Systems with Boundary Equilibria

O / open | Oberwolfach 报告问题汇编

For general complex-balanced reaction-diffusion systems with boundary equilibria, prove that renormalized solutions converge to equilibrium, or establish an entropy-dissipation estimate that implies such convergence.

来源字段：https://doi.org/10.4171/owr/2017/7

11770 | OWR-16414-001

### Convergence of Dynamic Monge-Kantorovich Systems

O / open | Oberwolfach 报告问题汇编

For the dynamic Monge-Kantorovich system, does the solution pair $(\mu(t),u(t))$ converge as $t\to\infty$ to the optimal-transport density and potential $(\mu^*,u^*)$ solving the Monge-Kantorovich equations?

来源字段：https://doi.org/10.4171/owr/2018/39

11771 | OWR-4990380-002

### Convergence of Fully Discrete Hyperbolic Balance-Law Schemes

P / partially_solved | Oberwolfach 报告问题汇编

Prove convergence of fully discrete approximations of hyperbolic balance laws, including Lax-Friedrichs and Godunov schemes.

来源字段：https://doi.org/10.4171/owr/2021/11

11772 | OWR-15432-001

### Convergence of Spectral Estimators for Diffusion Tensors

O / open | Oberwolfach 报告问题汇编

Let $(\nu_k,u_k)$ be noisy observations of eigenpairs and let $\mu$ be an invariant density satisfying $$\operatorname{div}(S\nabla u_k)=\nu_k\mu u_k$$ with Neumann boundary conditions. Prove convergence of the spectral reconstruction estimator obtained by fitting these equations to recover the matrix-valued coefficient $S$.

来源字段：https://doi.org/10.4171/owr/2017/24

11773 | OWR-4990380-003

### Convergence Without Continuous or TVD Numerical Fluxes

O / open | Oberwolfach 报告问题汇编

Determine whether a finite-volume flux that satisfies neither the usual continuity condition nor a TVD condition can nevertheless yield a convergent scheme, in particular for VOF-machine-learning solvers.

来源字段：https://doi.org/10.4171/owr/2021/11

11774 | OWR-1050-012

### Critical Exponents for Positive Superlinear Parabolic Solutions

S / solved | Oberwolfach 报告问题汇编

Does the ancient-time equation $u_t-\Delta u=u^p$ on $\mathbb{R}^N\times\mathbb{R}$ have a positive classical solution if and only if $p(N-2)\geq N+2$?

来源字段：https://doi.org/10.4171/owr/2005/29

11775 | OWR-16767-008

### Critical Self-Similar Profiles for Focusing Cubic Wave Blowup

O / open | Oberwolfach 报告问题汇编

For the focusing cubic wave equation in dimensions $d\geq5$, is $f^*$ a critical self-similar profile describing threshold solutions and their intermediate-time behavior between stable blowup and global existence?

来源字段：https://doi.org/10.4171/owr/2019/5

11776 | OWR-764-002

### Critical Well-Posedness for Derivative Nonlinear Schrödinger Equations

O / open | Oberwolfach 报告问题汇编

Is the one-dimensional derivative nonlinear Schrödinger Cauchy problem $iu_t+u_{xx}=i(|u|^2u)_x$, $u(0)=u_0$, locally well-posed for initial data $u_0\in\widehat H^{1/2}_1(\mathbb R)$ at the critical Fourier-Lebesgue endpoint?

来源字段：https://doi.org/10.4171/owr/2004/50

11777 | OWR-1050-001

### Critical-Exponent Polyharmonic Dirichlet Problems

O / open | Oberwolfach 报告问题汇编

Let $\Omega\subset\mathbb{R}^n$ be bounded and conformally contractible, with $n>2m$. Does the small-$\lambda$ uniqueness result for $(-\Delta)^m u=\lambda(1+|u|^{p-1}u)$ with boundary conditions $u=\nabla u=\cdots=D^{m-1}u=0$ extend from $p>(n+2m)/(n-2m)$ to the critical exponent $p=(n+2m)/(n-2m)$?

来源字段：https://doi.org/10.4171/owr/2005/29

11778 | AIM-PDES-0086

### Critical-level surgery for nonconvex Aubry problems

P / partially_solved | AIM 研讨会问题汇编

**Problem 11 (Albert Fathi).** For a Hamiltonian \[ H:\mathbb T^n\times\mathbb R^n\longrightarrow\mathbb R \] that is superlinear but not necessarily convex in momentum, ask:

**Problem 12 (attributed to Sergey Bolotin in [GMT18]).** Let \(c_0\) be the infimum of the constants \(c\) for which \[ H(z,Du(z))=c \] has a global viscosity subsolution on \(\mathbb T^n\). Let \(\mathcal{SS}\) be the critical subsolutions and define \[ S(z,y)=\sup_{u\in\mathcal{SS}}\{u(z)-u(y)\}. \] For a fixed vertex \(y\), the standard formulation says that \(z\mapsto S(z,y)\) is a critical subsolution on \(\mathbb T^n\) and a solution on \(\mathbb T^n\setminus\{y\}\). Does some \(y\) make it a solution on the whole torus when \(H\) is not necessarily convex?

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11779 | OWR-767-001

### Curves Minimizing $\lambda_{1}(g,C)$

O / open | Oberwolfach 报告问题汇编

For a closed planar curve $C \subset \mathbb{R}^2$ of fixed length and coupling parameter $g \in \mathbb{R}$, let $\lambda_1(g, C)$ denote the lowest eigenvalue of the two-dimensional Schrödinger operator with a curvature/boundary interaction of strength $g$ supported on $C$. Determine the closed curve $C$ of given length that minimizes $\lambda_1(g, C)$ in the parameter ranges $1/4 < g \leq 1$ and $g < 0$ with $g \neq -1$.

来源字段：https://doi.org/10.4171/owr/2004/53

11780 | OWR-12008-004

### De Giorgi Conjecture for Allen-Cahn

O / open | Oberwolfach 报告问题汇编

If a bounded solution of $$\Delta u+(1-u^2)u=0\quad\text{in }\mathbb R^N$$ satisfies $\partial_{x_N}u>0$, must all level sets of $u$ be hyperplanes when $N\leq8$?

来源字段：https://doi.org/10.4171/owr/2012/36

11781 | OWR-3476-002

### De Giorgi Conjecture for the Allen-Cahn Equation

O / open | Oberwolfach 报告问题汇编

For a bounded solution $u$ of the Allen-Cahn equation $\Delta u+(1-u^2)u=0$ in $\mathbb{R}^N$ satisfying $\partial_{x_N}u>0$, are all level sets hyperplanes for $N\leq8$? Equivalently, is $u(x)=\tanh((a\cdot x-b)/\sqrt{2})$ for some $|a|=1$ and $b\in\mathbb{R}$?

来源字段：https://doi.org/10.4171/owr/2009/26

11782 | OWR-830-001

### Decay Rates for Nonlinear Waves on Black-Hole Backgrounds

P / partially_solved | Oberwolfach 报告问题汇编

For the zeroth spherical harmonic of a linear wave $\phi$ on a Schwarzschild or Reissner-Nordström exterior, compactly supported Cauchy data yield $$|r\phi(u,\infty)|\leq Cu^{-2},\qquad |\partial_u(r\phi)(u,\infty)|\leq C_\varepsilon u^{-3+\varepsilon},$$ and, when $r(u,v)\leq R$, $$|\phi(u,v)|\leq C_{R,\varepsilon}v^{-3+\varepsilon},\qquad |\partial_v\phi(u,v)|\leq C_{R,\varepsilon}v^{-3+\varepsilon}.$$ Are these decay rates sufficient to treat a nonlinearity perturbatively? In particular, determine the small-data behavior of $\Box_g\phi=|\phi|^p$ in the remaining range $1+\sqrt2\leq p\leq4$.

来源字段：https://doi.org/10.4171/owr/2005/25

11783 | AIM-PDES-0008

### Deformation and invariant barriers for low-Sobolev 2D Euler norm inflation

P / partially_solved | AIM 研讨会问题汇编

2D Euler norm inflation

Consider the 2D Euler equation with initial velocity \(u_0\in H^s\), \(s\in(0,1)\). Prove norm inflation. Note that the \(L^2\) and \(H^1\) norms are conserved. Same question for \(L^p\) norms with \(p\ne2\).

来源字段：http://aimpl.org/smallscalefluid/8/

11784 | OWR-14297732-013

### Degrees of Asymptotically Conical Expanders Under Connected Sum

O / open | Oberwolfach 报告问题汇编

How are $\deg_{\exp}(X_1)$, $\deg_{\exp}(X_2)$, and $\deg_{\exp}(X_1\#_\partial X_2)$ related, and what is $\deg_{\exp}(D^3\times S^1)$?

来源字段：https://doi.org/10.4171/owr/2023/29

11785 | OWR-16168-008

### Detection and Uniqueness of Expanding Harmonic-Map Flows

P / partially_solved | Oberwolfach 报告问题汇编

For harmonic-map-flow solutions emerging from a $0$-homogeneous map $u_0:S^{n-1}\to N$, characterize which solutions are expanders and formulate and prove an appropriate uniqueness theorem for expanding solutions.

来源字段：https://doi.org/10.4171/owr/2018/27

11786 | OWR-13486-002

### Determination of Domains by Diffusion

P / partially_solved | Oberwolfach 报告问题汇编

Does diffusion determine a domain? Equivalently, under suitable hypotheses, if two domains have the same diffusion dynamics or heat semigroup, must they be congruent?

来源字段：https://doi.org/10.4171/owr/2015/1

11787 | OWR-13858-003

### Determining Wavenumbers Near Three-Dimensional Turbulence

O / open | Oberwolfach 报告问题汇编

For three-dimensional Navier-Stokes dynamics with intermittency dimension $d$ near $3$, is the time-averaged determining wavenumber $\langle\Lambda\rangle$ bounded by a constant multiple of the Kolmogorov dissipation wavenumber $\kappa_d$, as predicted by Kolmogorov turbulence theory?

来源字段：https://doi.org/10.4171/owr/2015/37

11788 | AIM-PDES-0054

### Dimensional and spectral resolution map for planar Euler--Korteweg phase boundaries

P / partially_solved | AIM 研讨会问题汇编

The system admits a planar profile \((\bar\rho,\bar u)(x_1-ct)\), and the spectrum of the linearization about it lies on the imaginary axis. Are there eigenvalues on the imaginary axis? Is the planar profile spectrally stable or unstable? Does the linear information imply full nonlinear stability?

来源字段：https://aimath.org/WWN/multidimwaves/multidimwaves.pdf

11789 | OWR-1050-005

### Dimensions of Energy-Concentration Sets

O / open | Oberwolfach 报告问题汇编

For positive solutions $u_\varepsilon$ of $\varepsilon^2\Delta u-u+u^p=0$ in a bounded smooth domain $\Omega$, with $\partial u/\partial\nu=0$ on $\partial\Omega$ and $E_\varepsilon[u_\varepsilon]\leq C\varepsilon^{N-m}$, must the energy-concentration set have dimension $m$?

来源字段：https://doi.org/10.4171/owr/2005/29

11790 | OWR-8415338-004

### Dirac-Mass Tangent Cones in Monge-Ampère Equations

O / open | Oberwolfach 报告问题汇编

For the singular convex solutions of $$\det D^2u=1+\sum_i a_i\delta_{p_i},$$ determine the precise tangent cones at the Dirac masses $p_i$.

来源字段：https://doi.org/10.4171/owr/2021/35

11791 | OWR-1050-014

### Discrete Interaction-Matrix Inequalities in Three Dimensions

O / open | Oberwolfach 报告问题汇编

For distinct points $a_1,\ldots,a_p\in\mathbb{R}^3$, let $A=(\alpha_{ij})$ be given by $\alpha_{ii}=0$ and $\alpha_{ij}=|a_i-a_j|^{-1}$ for $i\neq j$. Prove that there exists $c(p)>0$ such that, for every $u\in\mathbb{R}^p$, $\sup_{i,k}|u^{\mathsf T}(\partial A/\partial a_i^k)u|+|Au|^2\geq c(p)\sum_{i\neq j}u_i^2/|a_i-a_j|^2$.

来源字段：https://doi.org/10.4171/owr/2005/29

11792 | OWR-12336-003

### Discreteness of the Dirichlet $p$-Laplacian Spectrum

O / open | Oberwolfach 报告问题汇编

Is the spectrum of the Dirichlet $p$-Laplacian on a bounded domain discrete? In particular, do higher eigenvalues have finite multiplicity, and is there at least one number above the second eigenvalue that is not an eigenvalue?

来源字段：https://doi.org/10.4171/owr/2013/08

11793 | OWR-715-001

### Dissipation and Decay for Nonlinear Wave Equations

P / partially_solved | Oberwolfach 报告问题汇编

In three spatial dimensions, how does effective or ineffective damping alter $L^p$-$L^q$ decay estimates for nonlinear wave equations, in relation to the behavior near zero of $f(u_t,\nabla u,\nabla u_t,\nabla^2u)$?

来源字段：https://doi.org/10.4171/owr/2004/05

11794 | AMR-024-0004

### Dissipation bounds for flow past an obstacle

P / partially_solved | AMR 问题清单汇编

For viscous incompressible flow in $\mathbb{R}^3\setminus B$ past a fixed obstacle $B$, with velocity approaching a nonzero constant vector at infinity, obtain realistic rigorous upper bounds on the energy-dissipation rate.

背景链接：https://web.math.princeton.edu/~const/2k.pdf

11795 | OWR-11453-001

### Dissipative Noether Theory and Quantum Fluid Energies

O / open | Oberwolfach 报告问题汇编

For the two energy functionals arising from the dissipative variational formulation of quantum fluid dynamics, are there other transformations yielding new energy functionals, how can a dissipative Noether theory be developed on the space of measures, can the theory be applied to classical fluid dynamics and identify models with several energies, and can it lead to an optimal-transport formulation of quantum mechanics?

来源字段：https://doi.org/10.4171/owr/2011/55

11796 | OWR-16414-002

### Dynamic Limits for Congested and Branched Transport

O / open | Oberwolfach 报告问题汇编

For the dynamic optimal-transport system with exponent $\beta>0$, does $(\mu(t),u(t))$ converge as $t\to\infty$ to an equilibrium associated with congested transport when $0<\beta<1$ and with branched transport when $\beta>1$?

来源字段：https://doi.org/10.4171/owr/2018/39

11797 | OWR-10252937-005

### Dynamical Alekseevskii Conjecture for Homogeneous Ricci Flows

O / open | Oberwolfach 报告问题汇编

Let $M^n$ be a homogeneous simply connected manifold that is not contractible. Does every homogeneous Ricci-flow solution on $M^n$ have finite extinction time?

来源字段：https://doi.org/10.4171/owr/2022/28

11798 | OWR-4085-003

### Efficient Computation of Reduced Free Energy

P / partially_solved | Oberwolfach 报告问题汇编

Can the reduced free energy per removed particle, $A(X_r)/N_c$, be computed efficiently in the limit $N_c\to\infty$?

来源字段：https://doi.org/10.4171/owr/2009/42

11799 | OWR-12338-004

### Einstein-Type Dynamics for Concentrated Euler Flows

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Einstein-Type Fluid-Dynamics Question [OWR-12338-003]

Derive an analogue of Einstein's formula for the dynamics of localized objects in a fluid-mechanics framework, for example through concentrating solutions of the Euler equations.

来源字段：https://doi.org/10.4171/owr/2013/10

11800 | AIM-PDES-0015

### Endpoint correction and sharp acoustic conditioning for planar Gross-Pitaevskii waves

O / open | AIM 研讨会问题汇编

Prove existence of a traveling wave for Gross-Pitaevskii equation in \(\mathbb{R}^2\) for all wave speeds \(c \in [0, \sqrt{2}]\).

来源字段：http://aimpl.org/compproofstability/5/

11801 | OWR-12336-007

### Endpoint Regularity for $p$-Laplacian Gradients

O / open | Oberwolfach 报告问题汇编

For $p$-harmonic functions in dimensions $n\geq3$, can one construct higher-dimensional analogues of the planar least-regular examples and determine their optimal gradient regularity? In particular, does $V(\nabla u)=|\nabla u|^{(p-2)/2}\nabla u$ belong locally to $C^{1/2}$ in every dimension?

来源字段：https://doi.org/10.4171/owr/2013/08

11802 | OWR-1544-001

### Energetic Limits for General Complete-Damage Energies

O / open | Oberwolfach 报告问题汇编

For the complete-damage model with stored energy $$\widehat{\mathcal E}(u,z)=\int_\Omega W(x,e(u),z)+\frac{\kappa}{r}|\nabla z|^r\,dx,$$ extend the existence and convergence theorem for energetic solutions from quadratic to general nonquadratic densities $W$ and from $r>d$ to the range $1\leq r\leq d$.

来源字段：https://doi.org/10.4171/owr/2007/28

11803 | OWR-4429-003

### Energy Thresholds for Landau-Lifshitz Dynamics

P / partially_solved | Oberwolfach 报告问题汇编

Does the heat-flow and wave-map blow-up/global-existence dichotomy extend to the full Landau-Lifshitz family $\partial_tu=a_1(\Delta u+|\nabla u|^2u)+a_2u\times\Delta u$, including the Schrödinger-map case $a_1=0$; namely, are all data with $E(u_0)\leq4\pi$ global while finite-time blow-up occurs for some data at every $E(u_0)>4\pi$?

来源字段：https://doi.org/10.4171/owr/2010/41

11804 | OWR-1703869-006

### Energy-Angular Momentum Relations for Rotating Skyrmions

O / open | Oberwolfach 报告问题汇编

How do the minimal energies of skyrmion solutions depend on the magnitude of angular momentum, and can this dependence be used to estimate the rotation frequency $\omega$ and prove that the resulting Landau-Lifshitz solution is non-static?

来源字段：https://doi.org/10.4171/owr/2020/22

11805 | OWR-1598-005

### Entropy Selection for Conservation Laws with Discontinuous Flux

P / partially_solved | Oberwolfach 报告问题汇编

For scalar conservation laws with discontinuous flux arising as hydrodynamic limits of interacting particle systems, which entropy solution is selected by the hydrodynamic limit, so as to identify the physically relevant entropy notion?

来源字段：https://doi.org/10.4171/owr/2007/42

11806 | AIM-PDES-0058

### Evans diagnostics for primary and secondary cubic-autocatalysis front instability

P / partially_solved | AIM 研讨会问题汇编

**S. Malham (Biscale chaos).** Consider the cubic-autocatalysis system \[ u_t=\delta\Delta u-uv^2, \qquad v_t=\Delta v+uv^2. \] Here \(u\) is the fuel/reactant, \(v\) is the autocatalyst, and \(\delta\) is the ratio of their molecular diffusivities. When \(\delta\sim8\), small perturbations of a planar front evolve into a complex wrinkled front. Malevanets, Careta, and Kapral, *Biscale chaos in propagating fronts*, Phys. Rev. E 52 (1995), 4724--4735, suggested that its spatiotemporal chaos has two length scales.

**Open problem.** What are the stability properties of the planar front? Can we understand the secondary instability by getting an Evans function for the front that arises from the [primary] instability?

来源字段：https://aimath.org/WWN/multidimwaves/multidimwaves.pdf

11807 | AIM-PDES-0062

### Evans functions at an algebraic threshold

P / partially_solved | AIM 研讨会问题汇编

Open Problem: Can one construct an Evans function for this problem? If so, how does it behave? Note that for this problem the absolute spectrum touches the essential spectrum so there is no exponential dichotomy that we can exploit.

M. Wechselberger: (Loss of hyperbolicity, algebraic decay and the Evans function). For certain values of \(p\), the generalized KdV equation \(u_t-u_{xxx}+u_x^p=0\) admits standing waves \(\phi(x)\) which decay exponentially to zero as \(|x|\to\infty\). However, the eigenfunctions of the operator obtained by linearizing about \(\phi\) need not decay exponentially and depending on the power \(p\) (say \(p=5\)) may only decay algebraically.

来源字段：https://aimath.org/WWN/multidimwaves/multidimwaves.pdf

11808 | AIM-PDES-0078

### Exact flat-torus Mather quotient and current status

P / partially_solved | AIM 研讨会问题汇编

Inspection of page 3 of the original AIM PDF (PDF page 2), version 15 August 2003, supports the following recovery.

- The symbol after \(TM\times\) is \(\mathbb T=\mathbb R/\mathbb Z\), the time circle. Thus \(L:TM\times\mathbb T\to\mathbb R\) is a one-periodic time-dependent Tonelli Lagrangian. - The class is \(c\in H^1(M;\mathbb R)\), not homology \(H_1\). The PDF's superscript was flattened in extraction, and the following instruction to choose a closed one-form confirms the cohomological reading. - Since the PDF writes \(\mathcal A_c\subset M\), while a time-periodic projected Aubry set naturally lives in \(M\times\mathbb T\), \(\mathcal A_c\) is the projected time-zero section. This is also forced by paths running from time \(0\) to the integer time \(n\). - The broken strings hnc, ∫ n > 0, and lim inf > n→∞ recover as \(h_c^n\), \(\int_0^n\), and \(\liminf_{n\to\infty}\). The word probablility is misspelled in the PDF itself and is editorially recovered as “probability.”

Precisely, choose a closed one-form \(\eta_c\) with \([\eta_c]=c\), and set

\[ L_c(x,v,t)=L(x,v,t)-\eta_c(x)[v]+\alpha(c), \qquad \min_{\mu\ {\rm invariant}}\int_{TM\times\mathbb T}L_c\,d\mu=0. \tag{1.1} \]

For \(x,y\in M\) and \(n\in\mathbb N\),

\[ h_c^n(x,y)=\inf_{\substack{\gamma(0)=x\\\gamma(n)=y}} \int_0^n L_c(\gamma(t),\dot\gamma(t),t)\,dt, \tag{1.2} \]

where the infimum is over absolutely continuous curves. On the time-zero projected Aubry set \(\mathcal A_c\), define

\[ \rho_c(x,y)=\liminf_{n\to\infty}h_c^n(x,y) +\liminf_{n\to\infty}h_c^n(y,x). \tag{1.3} \]

The metric quotient by \(\rho_c=0\) is \(\overline{\mathcal A}_c\). Mather asks whether it is always totally disconnected and whether its Hausdorff dimension in the induced Mather metric is zero.

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11809 | AIM-PDES-0011

### Exact pushed-front benchmark and pulled spectral obstruction

P / partially_solved | AIM 研讨会问题汇编

A starting point for the above problem would be

Prove existence and stability for pushed or pulled fronts for a toy model using a computer assisted proof.

It also supplies the context omitted by the words "the above problem": Problem 2.1, attributed to Arnd Scheel, asks for existence and stability of pulled fronts for the complex Ginzburg--Landau equation. Problem 2.2 proposes a toy model as a starting point.

The intended phrase is almost surely "existence and marginal stability," but this reconstruction is not silently substituted for the source.

The 2023 AIM workshop announcement emphasizes two complementary rigorous-numerics tasks: validation of heteroclinic or homoclinic connections and computation of quantities controlling spectral stability of one-dimensional waves. This report accordingly chooses a scalar reaction--diffusion toy model and treats both profile existence and weighted spectral stability. It does **not** claim to solve the preceding complex Ginzburg--Landau problem.

来源字段：http://aimpl.org/compproofstability/2/

11810 | AIM-PDES-0063

### Exact state chamber and integrable interaction for a separating thin-film shock pair

P / partially_solved | AIM 研讨会问题汇编

The thin-film equation \[ u_t+(u^2-u^3)_x=-\varepsilon (u^3u_{xxx})_x \] supports solutions comprising a Lax shock moving left with speed \(s_1\) and an undercompressive shock moving right with speed \(s_2\), as in Figure 2. Can one prove stability or instability of this structure?

来源字段：https://aimath.org/WWN/multidimwaves/multidimwaves.pdf

11811 | AIM-PDES-0089

### Exact-energy reachability at the universal Mañé critical value

P / partially_solved | AIM 研讨会问题汇编

The source file prints:

\[ c_u=\inf\{h\in\mathbb R\mid \exists x\in\widetilde M\ \forall y\in M\ \exists\text{ trajectory with energy }h\text{ joining }x\text{ to }y\}. \tag{1} \]

Here \(M\) is compact, \(L:TM\to\mathbb R\) is a convex superlinear (Tonelli) Lagrangian, \(\widetilde L:T\widetilde M\to\mathbb R\) is its lift to the universal cover, and \(c_u\) is the Mañé critical value of the lift.

More importantly, the mixed quantifiers \(x\in\widetilde M\) and \(y\in M\) occur in both the official PDF and HTML.

Contreras later restated the question in the following unambiguous all-pairs form:

\[ c_u(L)=\inf\{k\in\mathbb R\mid \forall x,y\in\widetilde M\ \exists\text{ an Euler--Lagrange orbit from }x\text{ to }y \text{ with energy }k\}\,? \tag{2} \]

This later version changes both parts of the problematic quantifier: \(y\) lies in \(\widetilde M\), and the base point is universally rather than existentially quantified. In the same paper, a connecting orbit is a positive-time curve \(\gamma:[0,T]\to\widetilde M\), \(T>0\), satisfying the Euler--Lagrange equation and the endpoint conditions.

There is still a diagonal convention hidden in (2). If positive time is required and \(x=y\) is included, the formula is false even for the free particle on a circle. The mathematically viable version studied below is therefore:

\[ \mathcal R_{\ne}(h):\qquad \text{every ordered pair }x,y\in\widetilde M,\ x\ne y, \text{ is joined in positive time by an orbit of exact energy }h. \tag{3} \]

Equivalently, one may keep \(x=y\) but declare the diagonal condition vacuous. This is not a silent rewriting of the source: (1), (2), the projected-endpoint reading of (1), and (3) are recorded as distinct formulations.

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11812 | OWR-731-002

### Existence and Convergence in Irreversible Fracture Evolution

P / partially_solved | Oberwolfach 报告问题汇编

For the time-discrete irreversible-fracture evolution minimizing a Mumford-Shah-type elastic energy with increasing jump set, does each minimization problem have a solution, and is there a limiting evolution as the time step tends to zero?

来源字段：https://doi.org/10.4171/owr/2004/29

11813 | OWR-12180-001

### Existence of Optimal Two-Phase Eigenvalue Configurations

S / solved | Oberwolfach 报告问题汇编

Let $\Omega\subset\mathbb R^d$ be bounded, let $0<m<|\Omega|$, and let $A=\Omega\setminus B$ and $\sigma=\alpha\chi_A+\beta\chi_B$ with $0<\alpha<\beta$ and $|B|=m$. Does minimizing the first Dirichlet eigenvalue $\lambda(B)$ over measurable $B\subset\Omega$ with $|B|=m$ always have a solution?

来源字段：https://doi.org/10.4171/owr/2012/57

11814 | OWR-17130-006

### Explicit Cheeger Sets and Constants of Bounded Domains

P / partially_solved | Oberwolfach 报告问题汇编

Given a bounded domain $\Omega\subset\mathbb R^n$, explicitly describe its Cheeger sets $E\subseteq\Omega$, with $\lvert E\rvert>0$, minimizing $P(E)/\lvert E\rvert$, and compute

$$h(\Omega)=\inf_{E\subseteq\Omega}\frac{P(E)}{\lvert E\rvert}.$$

Can the minimizing sets and the value of this infimum be determined explicitly?

来源字段：https://doi.org/10.4171/owr/2019/34

11815 | OWR-14297737-004

### External-Field Stabilization of Unstable Kinetic Equilibria

P / partially_solved | Oberwolfach 报告问题汇编

For an unstable kinetic equilibrium, determine whether an imposed external electric field can stabilize the dynamics and characterize an optimal stabilizing field.

来源字段：https://doi.org/10.4171/owr/2023/38

11816 | OWR-3390-001

### Extreme Gravity-Capillary Waves with General Vorticity

P / partially_solved | Oberwolfach 报告问题汇编

For periodic traveling water waves with general vorticity, does a sequence of almost-extreme symmetric monotone waves with $\max \psi_Y^j\to0$ converge to an extreme wave satisfying the Stokes $120^\circ$ corner condition at stagnation points, rather than to a smooth wave with a bottom stagnation point?

来源字段：https://doi.org/10.4171/owr/2009/07

11817 | OWR-721-002

### Far-Field Solvability in Electromagnetic Inverse Scattering

P / partially_solved | Oberwolfach 报告问题汇编

In electromagnetic inverse scattering, is the factorization-method far-field equation solvable if and only if the sampling point lies in the scatterer $D$?

来源字段：https://doi.org/10.4171/owr/2004/11

11818 | AMR-024-0003

### Finite-dimensional dynamics for two-dimensional Navier-Stokes

O / open | AMR 问题清单汇编

Is the global attractor of the periodically forced two-dimensional Navier-Stokes equations conjugate to a smooth finite-dimensional dynamical system? Can its transient dynamics be described or controlled in finite-dimensional terms?

背景链接：https://web.math.princeton.edu/~const/2k.pdf

11819 | OWR-14609-001

### Finite-Time Blowup for Three-Dimensional Zakharov-Kuznetsov

O / open | Oberwolfach 报告问题汇编

Does the three-dimensional Zakharov-Kuznetsov equation admit a finite-time blow-up solution, as expected in the $L^2$-critical regime?

来源字段：https://doi.org/10.4171/owr/2016/27

11820 | AIM-PDES-0048

### Finite-time Kähler-Ricci diameter: status trichotomy and an entropy obstruction

P / partially_solved | AIM 研讨会问题汇编

Is the diameter bounded at finite time singularities of the K\"ahler-Ricci flow on compact K\"ahler manifolds?

来源字段：http://aimpl.org/nonlinpdegeom/4/

11821 | OWR-12338-001

### Finite-Time Perfect Mixing Under Power Constraints

S / solved | Oberwolfach 报告问题汇编

Determine whether finite-time perfect mixing can occur for a two-dimensional incompressible flow subject to a finite power constraint.

来源字段：https://doi.org/10.4171/owr/2013/10

11822 | OWR-12697707-001

### Finite-Time Singularities in Free-Boundary Fluid Models

O / open | Oberwolfach 报告问题汇编

Determine whether finite-time singularities form in vortex patches, water waves, porous-media interface flows, Hele-Shaw cells, and atmospheric front dynamics.

来源字段：https://doi.org/10.4171/owr/2023/11

11823 | OWR-13486-007

### First-Singularity Profiles in Nonlocal Transport Models

O / open | Oberwolfach 报告问题汇编

Consider the transport equation $\theta_t+u\theta_x=0$. Compare the Córdoba-Córdoba-Fontelos velocity

$$u(x,t)=\operatorname{PV}\int_{\mathbb R}\frac{\theta(y,t)}{y-x}\,dy$$

with the less singular velocity

$$u(x,t)=\int_{\mathbb R}\lvert y-x\rvert^{-\gamma}\theta(y,t)\,dy,\qquad 0<\gamma<1.$$

For symmetric nonnegative initial data maximized at the origin, must the first singularity always be a cusp, or can one construct data producing square-root behavior? Do singularities for the second velocity occur only where $\theta'\neq0$, while those for the first occur only where $\theta'=0$? Can one construct a blowup solution for the second velocity whose first singularity is an infinite slope at the origin while $\theta$ remains smooth elsewhere?

来源字段：https://doi.org/10.4171/owr/2015/1

11824 | AIM-PDES-0090

### Five Mane-critical questions and stability of a full Aubry graph

P / partially_solved | AIM 研讨会问题汇编

1. **Problem 16 (aperiodic levels).** For a compact manifold \(M\) and a Tonelli Lagrangian \(L:TM\to\mathbb R\), the horocycle-type example at the universal-cover Mañé critical value \(c_u\) was described as the only known energy level \(\{H=c_u\}\) having neither periodic orbits nor singularities. Does such a regular aperiodic level occur at another energy? 2. **Problem 17 (contact in the lower gap).** For a Tonelli Hamiltonian \(H:T^*M\to\mathbb R\), put \(\Sigma_h=H^{-1}(h)\), let \(c_u\) be the universal-cover critical value, and \[ e_0=\min\{h\in\mathbb R:\pi(\Sigma_h)=M\}. \] If \(M\ne\mathbb T^2\), is every \(\Sigma_h\) with \(e_0<h<c_u\) not of contact type? 3. **Problem 18 (noncompact weak KAM boundary).** For a complete noncompact Tonelli system, understand the geometry of Contreras's compactification of \(M/d_c\) by an extended Aubry/static set represented by Busemann weak KAM solutions. 4. **Problem 19 (capacity of a lifted sublevel).** For a \(\mathbb Z^n\)-periodic Tonelli Hamiltonian on \(\mathbb T^n\times\mathbb R^n\), with \(c_0=\alpha(0)\), does the open lifted sublevel \[ \{(x,p)\in\mathbb R^{2n}:H(x,p)<h\} \] have finite symplectic capacity for every \(h<c_0\)? 5. **Problem 20 (cohomological gradient stability).** Suppose \(H:\mathbb T^n\times\mathbb R^n\to\mathbb R\) is Tonelli, \(\pi(\mathcal A_0)=\mathbb T^n\), and \(H(x,Du)=\alpha(0)\) has a unique viscosity solution modulo constants. Define \[ H_c(x,p)=H(x,p-c),\qquad H_c(x,Dv_c)=\alpha(c). \] Is \[ \lim_{|c|\to0}\|Dv_c-Du\|_{\operatorname{Lip}(\pi(\mathcal A_c))}=0? \]

The source itself uses \(\operatorname{Lip}(\pi(\mathcal A_c))\), but does not define whether this is the full Lipschitz norm or only the uniform fiber norm of the Aubry graph. Both readings are distinguished below.

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11825 | AMR-026-0002

### Five Open Problems - Compensated compactness for symmetric matrices

P / partially_solved | AMR 问题清单汇编

Develop a compensated-compactness calculus for symmetric matrices when compensated compactness yields only inequalities. As a first application, prove complete continuity of the semigroup for the multidimensional scalar-conservation-law Cauchy problem.

背景链接：https://perso.ens-lyon.fr/denis.serre/DPF/Ouverts.pdf

11826 | AMR-026-0003

### Five Open Problems - Compressible Navier-Stokes near vacuum

P / partially_solved | AMR 问题清单汇编

For compressible Navier-Stokes equations with constant viscosities near vacuum, does the unphysical one-dimensional consequence identified by Hoff and Serre have a multidimensional counterpart?

背景链接：https://perso.ens-lyon.fr/denis.serre/DPF/Ouverts.pdf

11827 | AMR-026-0004

### Five Open Problems - Eternal finite-energy compressible Euler flow

P / partially_solved | AMR 问题清单汇编

Does the compressible Euler system in odd spatial dimension admit a nontrivial smooth eternal solution having finite nonzero mass and energy?

背景链接：https://perso.ens-lyon.fr/denis.serre/DPF/Ouverts.pdf

11828 | AMR-026-0001

### Five Open Problems - Global Cauchy theory for one-dimensional Euler-Fourier flow

P / partially_solved | AMR 问题清单汇编

Develop a global-in-time theory of the Cauchy problem for the one-dimensional Euler-Fourier system for initial data constrained only by finite energy and entropy, possibly with local boundedness assumptions on density, reciprocal density, velocity, temperature, and reciprocal temperature.

背景链接：https://perso.ens-lyon.fr/denis.serre/DPF/Ouverts.pdf

11829 | AMR-026-0005

### Five Open Problems - Regular reflection without irrotationality

P / partially_solved | AMR 问题清单汇编

Prove existence of a regular reflection for compressible flow against a wedge without assuming that the flow is irrotational.

背景链接：https://perso.ens-lyon.fr/denis.serre/DPF/Ouverts.pdf

11830 | AIM-PDES-0104

### Flat-germ obstruction to large minimal periods and genuine quasi-periodicity

P / partially_solved | AIM 研讨会问题汇编

**36. (Massimiliano Berti)** For a nonlinear wave equation \[ u_{tt}-u_{xx}=f(u),\qquad f(0)=f'(0)=0, \] with Dirichlet boundary conditions and general nonlinearity there exist a large number of small-amplitude periodic orbits with fixed period. Could one find small-amplitude periodic orbits of large period and quasi-periodic solutions for the wave equation?

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11831 | OWR-14612-002

### Flux Stability for Multidimensional Conservation Laws

O / open | Oberwolfach 报告问题汇编

Can stability of entropy solutions with respect to the flux be established for multidimensional conservation laws, or even for one-dimensional laws with space-dependent flux, in initial-boundary-value problems?

来源字段：https://doi.org/10.4171/owr/2016/30

11832 | AIM-PDES-0051

### Focusing spherical shocks: similarity norms and gauge modes

P / partially_solved | AIM 研讨会问题汇编

**Open Problem:** A classical and apparently very hard problem is the analysis of non stationary spherical waves, e.g. a focusing inviscid shock (c.f. R. Courant & K.O. Friedrichs, *Supersonic Flow and Shock Waves*, Interscience Publishers, New York, (1948) or L.D. Landau & E.M. Lifshitz *Fluid Mechanics*, Pergamon Press, Oxford (1959)) Are such waves stable?

The original first page contains the following complete problem under S. Benzoni-Gavage, H. K. Jenssen, and M. Williams:

Szmolyan: (Kinetic and Boltzmann Equations)” and the kinetic equation that follow are the next presenter’s setup on page 2. They are not part of AIM-PDES-0051. The recovered statement studied here is therefore only the focusing-inviscid-shock question above. No mathematical symbols are missing from that question.

The phrase “such waves” is broad. To obtain a precise object, most of the literature studies the spherical, strong-shock, ideal-perfect-gas, self-similar Guderley solution. Results for that model do not automatically answer the question for finite-strength shocks, nonideal equations of state, nonsimilar focusing waves, or viscous flows.

来源字段：https://aimath.org/WWN/multidimwaves/multidimwaves.pdf

11833 | OWR-13950-001

### Formation of Singular Cytoskeleton Profiles from Smooth Data

O / open | Oberwolfach 报告问题汇编

For the spatially one-dimensional hyperbolic-parabolic cytoskeleton model, can the explicitly constructed Dirac peaks, moving profiles, and shocks arise dynamically from a suitable class of smooth initial data?

来源字段：https://doi.org/10.4171/owr/2015/55

11834 | AIM-PDES-0087

### Frequency arithmetic does not determine Mather-set regularity

P / partially_solved | AIM 研讨会问题汇编

Consider a positive definite Lagrangian \(L:TM\times\mathbb T\to\mathbb R\). Let \(\alpha:H^1(M,\mathbb R)\to\mathbb R\) be Mather's \(\alpha\)-function. Relate the regularity of the Mather set \(\mathcal M_c\) and the arithmetic properties of the frequency \(\omega=D\alpha(c)\in H_1(M,\mathbb R)\).

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11835 | AIM-PDES-0103

### From formal graph and density expansions to Mather-measure error bounds

P / partially_solved | AIM 研讨会问题汇编

**Problem 35 (Diogo Gomes).** For a small perturbation of a completely integrable Hamiltonian system a viscosity solution corresponding to a non-resonant unperturbed torus can be uniformly approximated by formal expansions. Similar expansions can be constructed for the density of the Mather measures. However, it is not known how well Mather measures themselves are approximated by these formal expansions.

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11836 | AIM-PDES-0050

### Frozen Lopatinski symbols and exact spherical capillary mode corrections

P / partially_solved | AIM 研讨会问题汇编

Open Problem:Determine the relationship between the Lopatinski determinant for stability and recent work on stability of spherical waves using spherical harmonics (c.f. C.C. Wu & P. H. Roberts, Bubble shape instability and sonoluminescence, Phys. Lett. A 250, 131 (1998)).

来源字段：https://aimath.org/WWN/multidimwaves/multidimwaves.pdf

11837 | AIM-PDES-0081

### Full-projection Aubry sets under a regular Mather alpha function

P / partially_solved | AIM 研讨会问题汇编

6. (Gonzalo Contreras) Is it true that under the same condition on αL, each Aubry set

Ac covers the whole configuration space Tn?

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11838 | OWR-1317-002

### Full-Range Calderón-Zygmund Estimates for the $p$-Laplacian

O / open | Oberwolfach 报告问题汇编

For the $p$-Laplacian equation $\operatorname{div}(|Du|^{p-2}Du)=\operatorname{div}(|F|^{p-2}F)$, does $F\in L_{\mathrm{loc}}^q$ imply $Du\in L_{\mathrm{loc}}^q$ throughout the full range $q\geq p-1$?

来源字段：https://doi.org/10.4171/owr/2006/31

11839 | OWR-12588-006

### Gauss-Bonnet-Chern Positive Mass and Penrose Inequalities

O / open | Oberwolfach 报告问题汇编

Do the Positive Mass Theorem and the Penrose Inequality for the Gauss-Bonnet-Chern mass hold for asymptotically flat manifolds?

来源字段：https://doi.org/10.4171/owr/2013/40

11840 | OWR-4077-001

### General-Flux Existence for Stationary Three-Dimensional Navier-Stokes

P / partially_solved | Oberwolfach 报告问题汇编

For the stationary three-dimensional Navier-Stokes equations in a bounded multiply connected domain, does the existence theorem under restricted flux extend to arbitrary flux satisfying zero total flux on each boundary component?

来源字段：https://doi.org/10.4171/owr/2009/34

11841 | AIM-PDES-0091

### Generic Mather measures, periodic support, and rational locking flats

P / partially_solved | AIM 研讨会问题汇编

Mañé proved that for a generic positive definite Lagrangian \(L:TM\times\mathbb T\to\mathbb R\) and generic \(c\in H^1(M,\mathbb R)\), the minimizing measure in the Mather set \(\mathcal M_c\) is unique. For generic \(L\) and \(c\), is this unique minimizing measure supported on a periodic orbit? Jeff Xia conjectured that, for generic \(L\) and every \(c\in H^1(M,\mathbb R)\), the number of ergodic invariant measures in \(\mathcal M_c\) is finite.

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11842 | OWR-14612-001

### Generic Regularity of Two-by-Two Conservation Laws

O / open | Oberwolfach 报告问题汇编

For $2\times 2$ systems of conservation laws such as isentropic gas dynamics, is generic $C^3$ data smooth outside a locally finite family of shock curves, and can one prove uniqueness and continuous dependence of the resulting general solutions?

来源字段：https://doi.org/10.4171/owr/2016/30

11843 | OWR-8415338-001

### Generic Singularities of Mean Curvature Flow

P / partially_solved | Oberwolfach 报告问题汇编

Huisken conjectured that spherical and generalized cylindrical singularities are generic singularity models for mean curvature flow.

来源字段：https://doi.org/10.4171/owr/2021/35

11844 | AMR-143-0001

### Geometric Hydrodynamics in Open Problems - Problem 1

O / open | AMR 问题清单汇编

Find a variational (preferably geodesic) formulation describing the motion of an ideal fluid in a fixed domain $M$ containing sources and sinks. What is the correct formulation of the variational problem: should one take into account the exterior forces and/or the ``memory" of the fluid?

背景链接：https://arxiv.org/abs/2205.01143

11845 | AMR-143-0010

### Geometric Hydrodynamics in Open Problems - Problem 10

O / open | AMR 问题清单汇编

Are particle trajectories analytic in time in any dimension? For fluids on manifolds with boundary how does this analyticity depends on whether the boundary is analytic or not?

背景链接：https://arxiv.org/abs/2205.01143

11846 | AMR-143-0011

### Geometric Hydrodynamics in Open Problems - Problem 11

O / open | AMR 问题清单汇编

Prove that for any two-dimensional compact analytic Riemannian manifold $M$ with analytic boundary the set $\mathscr{GM}(M)$ of generalized minimal flows is a nonempty and proper subset of the Yudovich space $YU(M)$.

背景链接：https://arxiv.org/abs/2205.01143

11847 | AMR-143-0012

### Geometric Hydrodynamics in Open Problems - Problem 12

O / open | AMR 问题清单汇编

Prove that for any GM-flow $u(x,t)$ the level lines of vorticity $\mathrm{curl} u(x,t)=\text{const}$ are real-analytic.

背景链接：https://arxiv.org/abs/2205.01143

11848 | AMR-143-0013

### Geometric Hydrodynamics in Open Problems - Problem 13

O / open | AMR 问题清单汇编

Define anisotropic spaces more appropriate for hydrodynamics with larger smoothness in the direction of flow lines. Prove existence and uniqueness theorems for ideal hydrodynamics in those function spaces in any dimension.

背景链接：https://arxiv.org/abs/2205.01143

11849 | AMR-143-0014

### Geometric Hydrodynamics in Open Problems - Problem 14

O / open | AMR 问题清单汇编

The concept of GM-flows appears closely related to the ``ancient flows" for the Navier--Stokes equations but, in fact, it is different. Both concepts express the general idea that, in reality, we observe flows that have existed since long time ago (one may imagine a river flow). In this regard, it is not natural to consider flows starting at some definite time $t_0$ with some (arbitrary) initial velocity $u_0$. GM-flows have natural, intrinsic structure (analytic level lines of vorticity) and some natural, finite regularity in the transversal direction. The question of which functional classes are suitable for their study is an intriguing open problem.

Determine the true regularity of GM-flows in the framework of anisotropic function spaces.

背景链接：https://arxiv.org/abs/2205.01143

11850 | AMR-143-0015

### Geometric Hydrodynamics in Open Problems - Problem 15

O / open | AMR 问题清单汇编

Thus the structure of singularities of the exponential map in 2D hydrodynamics resembles that of a smooth map between finite-dimensional manifolds. This leads to the following natural question.

Quantify the failure of the Fredholm property of the $L^2$ exponential map in $\mathscr{D}^s_\mu(M)$ for a compact Riemannian 3-manifold $M$.

背景链接：https://arxiv.org/abs/2205.01143

11851 | AMR-143-0016

### Geometric Hydrodynamics in Open Problems - Problem 16

O / open | AMR 问题清单汇编

- []

(a) Determine the order of the first conjugate point along any $L^2$ geodesic starting from the identity in $\mathscr{D}_\mu^s(M)$.

(b) Is there a relation between existence of conjugate points in $\mathscr{D}_\mu^s(M)$ and Arnold stability criterion for stationary flows in $M$?

背景链接：https://arxiv.org/abs/2205.01143

11852 | AMR-143-0017

### Geometric Hydrodynamics in Open Problems - Problem 17

O / open | AMR 问题清单汇编

Show that for any geodesic $\gamma(t)$ in $\mathscr{D}_\mu^s(M)$ its absolute complexity is exponentially bounded above: $$ \mathcal{C}(\gamma(t)) \lesssim e^{t\|\dot{\gamma}_0\|_{H^s}} . $$

背景链接：https://arxiv.org/abs/2205.01143

11853 | AMR-143-0018

### Geometric Hydrodynamics in Open Problems - Problem 18 (Aging Problem)

O / open | AMR 问题清单汇编

Consider an arbitrary solution $u=u(t,x)$ of the Euler equations in $M$. Given any two time instants can one decide which velocity field $u(t_1,x)$ or $u(t_2,x)$ corresponds to an earlier time instant? In other words, is it possible to determine the aging of the fluid from its velocity field? Numerical experiments suggest that fluids ``age" with time: starting with a smooth initial velocity $u_0(x)$ the corresponding solution becomes ``wrinkled" in that its derivatives generally grow. (V. Yudovich referred to this phenomenon as ``regularity deterioration".)

Is it possible to quantify the ``aging" property of the fluid?

背景链接：https://arxiv.org/abs/2205.01143

11854 | AMR-143-0019

### Geometric Hydrodynamics in Open Problems - Problem 19

O / open | AMR 问题清单汇编

Consider an ideal fluid in a periodic channel $M=\mathbb{T} \times (0,a)$. Assume that its initial velocity $u_0$ is $C^1$ close to that of a plane-parallel flow whose velocity profile $v=v(x_2)$ satisfies $v'>0$ and $v''>0$ in $(0,a)$. Suppose that the level lines of $\omega_0 = \mathrm{curl} u_0$ satisfy $\nabla\omega_0 \neq 0$ in $M$ and that one of the lines $\omega_0 = \text{const}$ has a ``kink".

Show that the ``kink" does not disappear as the fluid evolves in time.

背景链接：https://arxiv.org/abs/2205.01143

11855 | AMR-143-0002

### Geometric Hydrodynamics in Open Problems - Problem 2

O / open | AMR 问题清单汇编

It turned out that in all of the above cases the corresponding configuration space has a natural groupoid structure. Using the corresponding Lie groupoids of multiphase diffeomorphisms instead of the Lie group of volume-preserving transformations in Arnold's setting one can describe the corresponding Lie algebroids and obtain geometric and Hamiltonian interpretations for the motion of the corresponding multiphase fluids, ``homogenized" vortex sheets, and generalized flows. Solutions of the above Euler equations were proved to be precisely the geodesics of an $L^2$-type right-invariant (source-wise) metric on the corresponding Lie groupoids of multiphase volume-preserving diffeomorphisms [IK18, IK22]. Another interesting domain for applications of Lie groupoids is provided by elasticity theory, cf. [mar-hugh, Kupf, JimLeon].

Extend the geodesic and Hamiltonian descriptions of the Euler-Arnold equations on Lie groupoids to problems of elasticity theory.

背景链接：https://arxiv.org/abs/2205.01143

11856 | AMR-143-0020

### Geometric Hydrodynamics in Open Problems - Problem 20

O / open | AMR 问题清单汇编

Investigate properties of the entropy functions $H_\epsilon, H_{\epsilon,\delta}, \tilde{H}_\epsilon$ and $\tilde{H}_{\epsilon,\delta}$ in this section and explain relations between them.

背景链接：https://arxiv.org/abs/2205.01143

11857 | AMR-143-0021

### Geometric Hydrodynamics in Open Problems - Problem 21

O / open | AMR 问题清单汇编

In the two-dimensional case, besides the kinetic energy, this system has infinitely many enstrophy invariants. These invariants are Casimir functionals: they do not depend on the metric on $M$, but specify a coadjoint orbit (i.e. a particular set of isovortical vector fields) on which the Euler evolution takes place but say nothing about the dynamics on the orbit itself.

Prove the non-integrability of the 2D Euler equation.

背景链接：https://arxiv.org/abs/2205.01143

11858 | AMR-143-0022

### Geometric Hydrodynamics in Open Problems - Problem 22

O / open | AMR 问题清单汇编

Justify the phenomenon observed in {\rm [ModViv19, ModViv21]} that integrable cases of point vortices seem to be attractors for 2D Euler flows with generic smooth initial vorticity in 2D (for small numerical viscosity). Design a model of dissipation in the 2D Euler equation which would produce such an integrable dynamics at large times.

背景链接：https://arxiv.org/abs/2205.01143

11859 | AMR-143-0023

### Geometric Hydrodynamics in Open Problems - Problem 23

O / open | AMR 问题清单汇编

Study in detail the motion and bifurcations of a small number of vortices on various manifolds: half- and quarter-plane, hemi- and quarter-sphere, disk, torus, cylinder and half-cylinder, etc.

背景链接：https://arxiv.org/abs/2205.01143

11860 | AMR-143-0024

### Geometric Hydrodynamics in Open Problems - Problem 24

O / open | AMR 问题清单汇编

Study the first non-integrable cases for a small number of point vortices on non-orientable surfaces. Describe the full set of Casimirs and finite-dimensional approximations for the group of area-preserving diffeomorphisms of non-orientable surfaces, analogous to the ones in the orientable cases.

背景链接：https://arxiv.org/abs/2205.01143

11861 | AMR-143-0025

### Geometric Hydrodynamics in Open Problems - Problem 25

O / open | AMR 问题清单汇编

This is not the case for a general infinite-dimensional dynamical system and, in particular, for the Eulerian hydrodynamics. Nadirashvili showed that the Euler equation of a 2D fluid has wandering solutions: there is an initial condition of a fluid in a 2D annulus whose neighbourhood never returns sufficiently closely to the initial condition after a certain time [Nadir]. A 3D analogue of that result is unknown.

Prove that the 3D Euler equation has wandering solutions.

背景链接：https://arxiv.org/abs/2205.01143

11862 | AMR-143-0026

### Geometric Hydrodynamics in Open Problems - Problem 26

O / open | AMR 问题清单汇编

Relax the restrictions on the smoothness index $k$ of the $C^k$ spaces (related to application of the KAM) and prove non-transitivity and non-mixing properties of the 3D Euler in full generality.

背景链接：https://arxiv.org/abs/2205.01143

11863 | AMR-143-0027

### Geometric Hydrodynamics in Open Problems - Problem 27

O / open | AMR 问题清单汇编

Consider an integrable system on a compact $2n$-dimensional manifold, which has $n$ first integrals in involution, functionally independent almost everywhere. How wild could such a system be on a singular submanifold, where the integrals become dependent? What are constraints dictated by the integrability? Could one observe a chaos on some tiny submanifold of an integrable system?

背景链接：https://arxiv.org/abs/2205.01143

11864 | AMR-143-0028

### Geometric Hydrodynamics in Open Problems - Problem 28

O / open | AMR 问题清单汇编

Arnold pointed out the remarkable topology of steady 3D fields: for an analytic steady field, not everywhere collinear with its curl, the flow domain is almost everywhere fibered into invariant tori and annuli, see [arn66, ak]. If the steady field is everywhere collinear with its vorticity but the proportionality coefficient is a generic function, then the domain is still fibered in a similar way. If the steady field is Beltrami, i.e., it is an eigenfield for the curl operator, $\mathrm{curl} v=\lambda v$, then its topology can be very intricate. Hence a paradox arises: a generic steady field has a very regular topology, while a sufficiently chaotic field must necessarily be an eigenfield for the curl operator.

Explain this paradox: what is a typical steady field and what genericity notion is natural for steady fields?

背景链接：https://arxiv.org/abs/2205.01143

11865 | AMR-143-0029

### Geometric Hydrodynamics in Open Problems - Problem 29 ([ak])

O / open | AMR 问题清单汇编

Find a sequence of higher helicity invariants for vector fields so that, given a field, if all the previous invariants are equal to zero for it, then the first nonzero invariant bounds the field's energy from below.

背景链接：https://arxiv.org/abs/2205.01143

11866 | AMR-143-0003

### Geometric Hydrodynamics in Open Problems - Problem 3

O / open | AMR 问题清单汇编

Describe the differential geometry (including computations of sectional and Ricci curvatures, conjugate points, etc.) for the right-invariant (source-wise) $L^2$-metric on the Lie groupoid of multiphase or generalized volume-preserving diffeomorphisms (analogous to Arnold's description of the differential geometry of the group $\mathscr{D}_\mu(M)$).

背景链接：https://arxiv.org/abs/2205.01143

11867 | AMR-143-0030

### Geometric Hydrodynamics in Open Problems - Problem 30

O / open | AMR 问题清单汇编

Find a direct link from 3D NLS equation to 1D NLS equation as a reduction to singular solutions, rather than going through the LIA procedure. A similar question arises for the compressible Euler equations as a reduction from 3D to singular solutions supported on 1D submanifolds.

背景链接：https://arxiv.org/abs/2205.01143

11868 | AMR-143-0031

### Geometric Hydrodynamics in Open Problems - Problem 31

O / open | AMR 问题清单汇编

Geometric description of vortex sheets leads to an interesting non-local metric of hydrodynamical pedigree on shape spaces: it is an $H^{-1/2}$-metric on closed hypersurfaces bounding the same volume (or equivalently, on constant densities inside those hypersurfaces), see [IK18]. Such a metric is constructed with the help of the Neumann-to-Dirichlet operators and it is always nondegenerate since it is bounded below by the Kantorovich--Wasserstein distance. Geodesics with respect to this metric describe motions of potential fluid flows with vortex sheets and fluids with free dynamic boundary, cf. [IK18, Marsden, Loe].

Describe the differential geometry of shape spaces equipped with such $H^{-1/2}$-metrics obtained as metrics on dynamic boundaries or vortex sheets.

背景链接：https://arxiv.org/abs/2205.01143

11869 | AMR-143-0032

### Geometric Hydrodynamics in Open Problems - Problem 32

O / open | AMR 问题清单汇编

Extend the above results on symplectomorphism and the K\"ahler map to the wave functions with zeros on $M$. Explain the quantization condition controversy {\rm [Wallstrom, Fritsche]} in the language of the momentum map for the above semi-direct product group.

背景链接：https://arxiv.org/abs/2205.01143

11870 | AMR-143-0033

### Geometric Hydrodynamics in Open Problems - Problem 33

O / open | AMR 问题清单汇编

The connection between equations of quantum mechanics and hydrodynamics might shed some light on the hydrodynamic quantum analogues studied, e.g., in [Couder, Bush]: the motion of bouncing droplets in certain vibrating liquids manifests many properties of quantum mechanical particles.

Explain the droplet-quantum particle correspondence by a combination of averaging and the Madelung transform.

背景链接：https://arxiv.org/abs/2205.01143

11871 | AMR-143-0034

### Geometric Hydrodynamics in Open Problems - Problem 34

O / open | AMR 问题清单汇编

Now, the next natural question concerns what happens with the energy that has escaped from the system? One can introduce the ``limit absorption principle'' for our system. Namely, one can introduce a friction for the balls of the family $F_N$ which absorbs the energy and then let $N$ go to infinity. But this solution is not completely legitimate as here the energy sink is included from the beginning and it is only moved farther and farther away. So, this explanation of the energy escape is circular.

Is there a more natural way to introduce energy dissipation without explicit introduction of a friction mechanism?

背景链接：https://arxiv.org/abs/2205.01143

11872 | AMR-143-0035

### Geometric Hydrodynamics in Open Problems - Problem 35

O / open | AMR 问题清单汇编

Is there a similar mechanism which describes a cascade in real hydrodynamics? Is it possible to define the parameters $n_i, m_i,$ and $ r_i$ in such a way that the above apparatus works as intended?

背景链接：https://arxiv.org/abs/2205.01143

11873 | AMR-143-0004

### Geometric Hydrodynamics in Open Problems - Problem 4

O / open | AMR 问题清单汇编

Is it possible to extend Arnold's geodesic framework from the group of diffeomorphisms $\mathscr{D}(M)$ to the semigroup of maps ${\mathcal{M}ap}(M)$ that would capture both smooth solutions and their continuations beyond emergence of shock waves for Burgers-like and compressible fluid equations?

背景链接：https://arxiv.org/abs/2205.01143

11874 | AMR-143-0005

### Geometric Hydrodynamics in Open Problems - Problem 5

O / open | AMR 问题清单汇编

Find a variational description of the system of material particles, moving in $\mathbb{R}^d$ with $ d>1$ and sticking upon collision, for both potential and non-potential fluid flows. This formulation should be sufficiently flexible to describe the formation and development of shock waves in the system described by the multi-dimensional Burgers equation.

背景链接：https://arxiv.org/abs/2205.01143

11875 | AMR-143-0006

### Geometric Hydrodynamics in Open Problems - Problem 6

O / open | AMR 问题清单汇编

In all other cases the metric is neither left- nor right-invariant. It would be interesting to investigate this system for generic $S$ and $\rho(\omega)$. It would also be of interest to define and study a model system where the configuration space is $SL(2)$, since this group looks more ``liquid-like". A similar question about possible fluid models is very intriguing.

Describe a model for an $L^2$ \mathrm{left-invariant metric} on the group $\mathscr{D}_\mu(M)$. Are there any interesting physical systems demonstrating that type of invariance?

背景链接：https://arxiv.org/abs/2205.01143

11876 | AMR-143-0007

### Geometric Hydrodynamics in Open Problems - Problem 7 (Two-point boundary value problem)

O / open | AMR 问题清单汇编

Let $M$ be a compact two-dimensional Riemannian manifold and let $\varphi$ be a volume-preserving diffeomorphism of $M$ of Sobolev class $H^s$.

(i) \mathrm{(Surjectivity Problem)} Find a divergence-free vector field $v \in H^s(M)$ such that $\exp_e{v} = \varphi$.

(ii) \mathrm{(Variational Problem)} Find a curve $\gamma(t)$ in $\mathscr{D}_\mu^s(M)$ from $e$ to $\varphi$ which minimizes the $L^2$ energy functional $ \mathcal{E}(\gamma) = \frac{1}{2} \int_0^1 \| \dot{\gamma}(t)\|^2_{L^2} dt. $

背景链接：https://arxiv.org/abs/2205.01143

11877 | AMR-143-0008

### Geometric Hydrodynamics in Open Problems - Problem 8

O / open | AMR 问题清单汇编

Another interesting question is related to closed geodesics. Suppose that $M$ is a compact surface (possibly with boundary) of genus at least $2$ or that $M$ is a multi-connected bounded domain in $\mathbb{R}^2$ with at least two holes.

Does there exist a closed geodesic in $\mathscr{D}^s_\mu(M)$?

背景链接：https://arxiv.org/abs/2205.01143

11878 | AMR-143-0009

### Geometric Hydrodynamics in Open Problems - Problem 9

O / open | AMR 问题清单汇编

How ``far" is $\mathscr{D}_\mu(M)$ from being an infinite-dimensional (locally) symmetric Riemannian space?

背景链接：https://arxiv.org/abs/2205.01143

11879 | OWR-8415338-003

### Geometry of Singular Monge-Ampère Solution Loci

P / partially_solved | Oberwolfach 报告问题汇编

Let $n\geq5$ and consider convex solutions of $$\det D^2u=1+\sum_i a_i\delta_{p_i}.$$ Determine whether singular solutions analogous to the known lower-dimensional examples exist; whether their singularities persist under small perturbations of the locations $p_i$ and masses $a_i$; and the dimension, regularity, connectivity, and other geometry of the singular-solution locus, including algebro-geometric conditions that exclude singular structures.

来源字段：https://doi.org/10.4171/owr/2021/35

11880 | OWR-1972-005

### Global Existence and Singularities for the Euler-Nordström System

O / open | Oberwolfach 报告问题汇编

For the Euler-Nordström system and initial data in a function or distribution space $X$, does a unique solution exist locally in time, remain in $X$, and depend continuously on the initial data? Does it exist for all time, or develop a finite-time singularity whose nature can be determined?

来源字段：https://doi.org/10.4171/owr/2008/27

11881 | OWR-11788-002

### Global Regularity and Uniqueness of Three-Dimensional Navier-Stokes

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Three-Dimensional Navier-Stokes Problem [OWR-11788-009]

For the three-dimensional Navier-Stokes equations on $\mathbb R^3$, are Leray-Hopf solutions globally regular and unique?

来源字段：https://doi.org/10.4171/owr/2012/26

11882 | OWR-13858-001

### Global Regularity of Three-Dimensional Euler and Navier-Stokes Flows

O / open | Oberwolfach 报告问题汇编

Are smooth solutions of the three-dimensional incompressible Euler equations or the three-dimensional Navier-Stokes equations globally regular, or can they develop singularities in finite time?

来源字段：https://doi.org/10.4171/owr/2015/37

11883 | OWR-9790360-002

### Global Regularity of Three-Dimensional Navier-Stokes Flow

O / open | Oberwolfach 报告问题汇编

Do the three-dimensional incompressible Navier-Stokes equations admit a global-in-time smooth solution for every smooth divergence-free initial velocity field?

来源字段：https://doi.org/10.4171/owr/2022/9

11884 | OWR-11101912-015

### Global Scattering for Cubic Defocusing Dispersive Flows

P / partially_solved | Oberwolfach 报告问题汇编

Do one-dimensional dispersive flows with cubic defocusing nonlinearities and small, not necessarily localized, initial data have global-in-time scattering solutions?

来源字段：https://doi.org/10.4171/owr/2022/31

11885 | OWR-769-001

### Global Solutions and Sharp-Interface Limits in Three Dimensions

O / open | Oberwolfach 报告问题汇编

Let $\Omega\subset\mathbb{R}^3$ and consider the phase-field system $$-\operatorname{div}_xT=b,\qquad T=D\bigl(\varepsilon(\nabla_xu)-\bar\varepsilon S\bigr),$$ $$S_t=-c\bigl(\psi_S(\varepsilon(\nabla_xu),S)-\nu\Delta_xS\bigr)|\nabla_xS|,$$ with $\nu>0$. Does the associated initial-boundary-value problem have global-in-time solutions in three dimensions, and do these solutions converge as $\nu\to0$ to solutions of the corresponding sharp-interface model?

来源字段：https://doi.org/10.4171/owr/2004/55

11886 | OWR-1050-019

### Global Stability Thresholds in Two-Species Migration Models

O / open | Oberwolfach 报告问题汇编

For the two-species Lotka-Volterra competition system with nonnegative heterogeneous growth rate $m(x)$, let $c\in(c^*,1)$ and suppose $(\mu,\nu)\in\Sigma_c$, so that the semitrivial steady state $(\theta,0)$ is linearly stable. Is $(\theta,0)$ globally asymptotically stable for every $0<b<1$?

来源字段：https://doi.org/10.4171/owr/2005/29

11887 | OWR-14298164-002

### Global Weak Solutions for Perfect-Gas Navier-Stokes-Fourier Flow

O / open | Oberwolfach 报告问题汇编

Prove global-in-time existence of weak solutions to the compressible Navier-Stokes-Fourier system with the perfect-gas law $p=\rho\vartheta$ without the additional structural restrictions required by existing results.

来源字段：https://doi.org/10.4171/owr/2024/10

11888 | OWR-2494-003

### Global Weak Solutions from Transport Discretizations of Euler Flow

O / open | Oberwolfach 报告问题汇编

For the isentropic Euler equations, does there exist a global-in-time weak solution, and can the proposed optimal-transport time discretization be shown to converge to such a solution?

来源字段：https://doi.org/10.4171/owr/2008/56

11889 | OWR-2654836-002

### Global Well-Posedness of Three-Dimensional Navier-Stokes

O / open | Oberwolfach 报告问题汇编

Can one establish global well-posedness for the three-dimensional Navier-Stokes equations?

来源字段：https://doi.org/10.4171/owr/2020/40

11890 | AIM-PDES-0036

### Goldberg conjecture: corrected statement and counterexample sieve

P / partially_solved | AIM 研讨会问题汇编

Let \((M^{2n},g,J,\omega)\) be compact and almost Kähler, meaning that \(J^2=-\mathrm{Id}\), \(g(JX,JY)=g(X,Y)\), and the fundamental form \(\omega(X,Y)=g(JX,Y)\) is closed. If \(g\) is Einstein, must \(J\) be integrable (and hence the structure Kähler)?

来源字段：http://aimpl.org/nonlinpdegeom/2/

11891 | OWR-13113-001

### Gradient Flows for $N$-Species Reactive Cahn-Hilliard Systems

P / partially_solved | Oberwolfach 报告问题汇编

Extend the generalized gradient-flow formulation, well-posedness theory, and energy-dissipation theory from the two-component reactive Cahn-Hilliard model to reactive Cahn-Hilliard systems with an arbitrary number $N$ of species.

来源字段：https://doi.org/10.4171/owr/2014/43

11892 | OWR-14298587-009

### Gradient-Constrained Ginzburg-Landau Minimizers

P / partially_solved | Oberwolfach 报告问题汇编

For $2\leq N\leq 6$, let $u_\varepsilon(x)=f_\varepsilon(|x|)x/|x|$ be the radial vortex solution on $B^N$. Is $u_\varepsilon$ the unique minimizer of the Ginzburg-Landau energy $G_\varepsilon$ over the gradient fields $\mathcal V=\{u=\nabla\phi:\phi\in H^2(B^N),\ \nabla\phi=\operatorname{Id}\text{ on }\partial B^N\}$?

来源字段：https://doi.org/10.4171/owr/2024/37

11893 | AIM-PDES-0085

### Hamilton-Jacobi graph regularity versus KAM conjugacy

P / partially_solved | AIM 研讨会问题汇编

The existence of KAM invariant tori of a Hamiltonian system on \(\mathbb T^n\times\mathbb R^n\) is equivalent to the existence of regular solutions of the Hamilton--Jacobi equation \[ H(x,c+Du(x))=\alpha(c). \] Hence KAM theory can be regarded as regularity result for viscosity solutions. Question: Is it possible to obtain proofs of KAM results using this connection?

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11894 | OWR-17130-011

### Hausdorff Absolute Continuity of PDE-Constrained Measures

O / open | Oberwolfach 报告问题汇编

Let

$$A=\partial_{x_1}^3+\partial_{x_2}^3+\partial_{x_3}^3.$$

If a real Radon measure $\mu$ on $\mathbb R^3$ satisfies $A\mu=0$ distributionally, must $\mu\ll\mathcal H^2$?

来源字段：https://doi.org/10.4171/owr/2019/34

11895 | OWR-15576-002

### High-Frequency Coercivity on Nonconvex Nontrapping Domains

O / open | Oberwolfach 报告问题汇编

For high-frequency scattering, does coercivity of the combined boundary-integral operator $A_k$ persist for strongly nontrapping, nonconvex domains such as smooth star-shaped domains, possibly with a coercivity constant $\gamma(k)$ that degenerates mildly as $k\to\infty$?

来源字段：https://doi.org/10.4171/owr/2017/33

11896 | OWR-3476-009

### Higher Eigensolutions of the $1$-Laplacian

P / partially_solved | Oberwolfach 报告问题汇编

For the variational eigenproblem of the nonsmooth $1$-Laplace operator on $BV(\Omega)$, how should higher eigensolutions be defined and characterized beyond the first minimizers, given the ambiguity between single and multiple eigenvalue equations?

来源字段：https://doi.org/10.4171/owr/2009/26

11897 | OWR-14299912-003

### Higher Sobolev Norm Growth for Half-Wave Maps

O / open | Oberwolfach 报告问题汇编

For nonrational initial data for the half-wave maps equation on the circle, determine the behavior of the higher Sobolev norms $H^s$ for $s>1/2$. Can an $H^s$ norm blow up in finite or infinite time?

来源字段：https://doi.org/10.4171/owr/2026/10

11898 | OWR-17128-001

### Higher-Dimensional Frenkel-Kontorova Traveling Waves

O / open | Oberwolfach 报告问题汇编

Extend the one-dimensional Frenkel-Kontorova existence theorem for traveling heteroclinic waves in smooth anharmonic two-well potentials to higher spatial dimensions.

来源字段：https://doi.org/10.4171/owr/2019/32

11899 | OWR-4990380-005

### Higher-Order Entropy-Conservative Finite-Volume Methods

O / open | Oberwolfach 报告问题汇编

Generalize the entropy-conservative finite-volume discretization to higher-order methods.

来源字段：https://doi.org/10.4171/owr/2021/11

11900 | OWR-4339-009

### Holomorphic-Curve Structure of Calabi-Yau Blow-Up Sets

O / open | Oberwolfach 报告问题汇编

Assume the $\varepsilon$-regularity estimate for Donaldson's almost-complex Calabi-Yau equation. If the a priori estimates in Donaldson's conjecture fail, must the resulting blow-up set be a union of $J$-holomorphic curves?

来源字段：https://doi.org/10.4171/owr/2010/26

11901 | AIM-PDES-0001

### Horizontal enstrophy budgets and a Couette orientation obstruction

P / partially_solved | AIM 研讨会问题汇编

Consider the 2D Euler equation with horizontal viscosity on $\mathbb T\times[0,1]$: $$ \partial_t\omega+u\cdot\nabla\omega=\nu\partial_{xx}\omega, \qquad u=\nabla^\perp\Delta^{-1}\omega. $$ How large can $\nu\int_\Omega|\partial_x\omega|^2$ get as $\nu\to0$ and $t\to\infty$?

来源字段：http://aimpl.org/smallscalefluid/1/

11902 | OWR-14299287-003

### Inertial Manifolds for Navier-Stokes Equations

O / open | Oberwolfach 报告问题汇编

Do the two-dimensional Navier-Stokes equations admit an inertial manifold?

来源字段：https://doi.org/10.4171/owr/2025/24

11903 | OWR-11101924-004

### Infinite Tails of Multiple-Scattering Series

O / open | Oberwolfach 报告问题汇编

Develop a collective method for computing exponentially many single-scattering returns and control the infinite tail of the multiple-scattering series even when that series diverges.

来源字段：https://doi.org/10.4171/owr/2022/43

11904 | OWR-17130-013

### Instantaneously Complete Yamabe Flow in Higher Dimensions

P / partially_solved | Oberwolfach 报告问题汇编

Does the theory of instantaneously complete Yamabe flows on noncompact surfaces extend to higher-dimensional noncompact manifolds? For conformally hyperbolic Yamabe flows, is

$$g(t)\geq m(m-1)t\,g_H$$

equivalent to instantaneous completeness?

来源字段：https://doi.org/10.4171/owr/2019/34

11905 | OWR-4077-003

### Integrability of Free-Surface Potential Flow

P / partially_solved | Oberwolfach 报告问题汇编

Is the free-surface potential-flow problem in hydrodynamics integrable?

来源字段：https://doi.org/10.4171/owr/2009/34

11906 | OWR-4085-004

### Interaction Potentials for Single and Double Peak States

P / partially_solved | Oberwolfach 报告问题汇编

What conditions on the interaction potential $V$ allow a steady state to have one peak or two equal-sized peaks?

来源字段：https://doi.org/10.4171/owr/2009/42

11907 | OWR-1050-004

### Interface-Layer Solutions on Perforated Domains

O / open | Oberwolfach 报告问题汇编

Let $\Omega\subset\mathbb R^N$, with $N\geq3$, be a domain with one hole. For sufficiently large $p$, does $-\Delta u=u^p$ admit a solution that converges away from a suitable $(N-1)$-dimensional manifold $M$ to a harmonic function and tends to $1$ on $M$?

来源字段：https://doi.org/10.4171/owr/2005/29

11908 | OWR-1050-003

### Interior-Layer Peaks in Singularly Perturbed Elliptic Problems

O / open | Oberwolfach 报告问题汇编

For $-\varepsilon^2\Delta u=u(u-a(x))(1-u)$ in a bounded smooth domain $\Omega$, with explosive boundary condition $u|_{\partial\Omega}=+\infty$, construct peak solutions superimposed on solutions that possess interior transition layers.

来源字段：https://doi.org/10.4171/owr/2005/29

11909 | OWR-13948-001

### Intermediate Regularity Regime for Monge-Ampère Solutions

P / partially_solved | Oberwolfach 报告问题汇编

What is the qualitative behavior of weak and very weak solutions of the Monge-Ampère equation $\operatorname{Det}\nabla^2v=-\tfrac12\operatorname{curl}\operatorname{curl}(\nabla v\otimes\nabla v)=f$ in the regularity regime $1/3<\alpha<2/3$?

来源字段：https://doi.org/10.4171/owr/2015/53

11910 | AIM-PDES-0096

### Intermediate-dimensional minimizing currents and a flat-torus obstruction

P / partially_solved | AIM 研讨会问题汇编

Let $M^n$ be a compact Riemannian manifold and let $C_qM$ be the set of closed $q$-currents $[ C_qM=\{T\in(\Omega^q(M))^*:T(d\alpha)=0\text{ for all }\alpha\in\Omega^{q-1}(M)\}. ]$ A closed current determines $[T]\in H_q(M,\mathbb R)$ and has a mass. Every real homology class has a mass-minimizing representative. For $q=1$, supports of minimizers consist of minimal geodesics; for $q=n-1$, minimizers are measured laminations by minimizing hypersurfaces, possibly with singularities. What can be said for $1<q<n-1$?

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11911 | OWR-12013-001

### Intrinsic Viscosity Solutions for Rough Hamilton-Jacobi Equations

O / open | Oberwolfach 报告问题汇编

For rough Hamilton-Jacobi-Bellman equations driven by a rough path, can solutions defined as limits of viscosity solutions for smooth approximations be characterized intrinsically, for example by touching test functions as in classical viscosity theory?

来源字段：https://doi.org/10.4171/owr/2012/41

11912 | OWR-11130-005

### Intrinsic-Flat Stability of the Positive Mass Theorem

P / partially_solved | Oberwolfach 报告问题汇编

If a sequence of asymptotically flat three-manifolds has nonnegative scalar curvature, no interior closed minimal surfaces, and ADM mass tending to zero, must it converge in the pointed intrinsic-flat sense to Euclidean space when the base points lie on constant-mean-curvature surfaces of fixed area?

来源字段：https://doi.org/10.4171/owr/2011/38

11913 | OWR-4080-002

### Inverse Spectral Recovery of Vibrating-String Damping

P / partially_solved | Oberwolfach 报告问题汇编

Can the damping coefficient $a(x)$ in the viscously damped vibrating-string equation $$u_{tt}=u_{xx}-2a(x)u_t,\qquad 0<x<1,$$ with homogeneous Dirichlet boundary conditions be recovered from the spectrum?

来源字段：https://doi.org/10.4171/owr/2009/37

11914 | OWR-13852-008

### Kähler-Ricci Flow from Currents with Positive Lelong Numbers

P / partially_solved | Oberwolfach 报告问题汇编

What happens to the Kähler-Ricci flow when its initial positive current $T_0$ has positive Lelong numbers?

来源字段：https://doi.org/10.4171/owr/2015/31

11915 | OWR-17138-004

### KdV Evolution from General Ergodic Random Fields

P / partially_solved | Oberwolfach 报告问题汇编

For the Korteweg-de Vries equation on $\mathbb R$, is the evolution almost surely well posed when the initial condition is a general ergodic random field, beyond the known white-noise case?

来源字段：https://doi.org/10.4171/owr/2019/42

11916 | OWR-4990379-003

### Lagrangian Schemes for Irregular Porous-Medium Solutions

O / open | Oberwolfach 报告问题汇编

Does convergence of the Lagrangian discretization scheme remain valid for less regular porous-medium solutions, especially when $-\nabla U'(\rho)$ is nonzero on the boundary of the support of $\rho$, as for the Barenblatt solution?

来源字段：https://doi.org/10.4171/owr/2021/10

11917 | OWR-14299093-002

### Landis Unique-Continuation Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let $V\in L^\infty(\mathbb R^N)$ satisfy $\|V\|_\infty\leq 1$, and let $u$ solve $(-\Delta+V)u=0$ on $\mathbb R^N$. If $u$ decays at least as fast as $e^{-k|x|}$ for some $k>1$, must $u\equiv 0$ without assuming positivity of the Schrödinger operator?

来源字段：https://doi.org/10.4171/owr/2025/14

11918 | OWR-11788-010

### Large-Data Scattering for Energy-Critical Schrödinger Maps

P / partially_solved | Oberwolfach 报告问题汇编

For Schrödinger maps from $\mathbb R^{2+1}$ to $S^2$ with zero homotopy class and energy below $2E(Q_1)=8\pi$, do global well-posedness and scattering hold?

来源字段：https://doi.org/10.4171/owr/2012/26

11919 | OWR-767-002

### Liouville Theorems for Periodic Elliptic Equations

P / partially_solved | Oberwolfach 报告问题汇编

Does the Liouville property for periodic second-order scalar elliptic equations extend beyond divergence form, to higher-order equations and matrix-valued systems? Can one decide effectively whether a given periodic elliptic equation has the Liouville property? Can analogous criteria be established for operators on abelian coverings?

来源字段：https://doi.org/10.4171/owr/2004/53

11920 | AIM-PDES-0030

### Literal torus obstruction and background-Hessian reconstruction

O / open | AIM 研讨会问题汇编

Let \(u\) be a solution to \[ u_{tt}+\operatorname{det}(u_{ij})=0 \] on \(T^n\times[0,1]\) with convex boundary data. Prove a \(C^2\) estimate for \(u(\cdot,t)\) for \(t\) fixed.

A solution would imply the maximal rank conjecture for complex tori.

来源字段：http://aimpl.org/nonlinpdegeom/1/

11921 | AMR-064-0007

### Localized Degenerate Control of Navier-Stokes

P / partially_solved | AMR 问题清单汇编

For incompressible Navier-Stokes on $\mathbb{T}^d$, $d=2,3$, is the system approximately controllable and/or controllable in finite-dimensional projections by a localized degenerate forcing, where the control space $E$ is a finite-dimensional subspace of $\{u\in V:\operatorname{supp}u\subset\overline{\mathcal D}\}$? Construct such an $E$ independently of the viscosity $\nu$.

背景链接：https://arxiv.org/abs/1304.2590

11922 | OWR-15212-006

### Long-Time Behavior of Degenerate Keller-Segel Equations

P / partially_solved | Oberwolfach 报告问题汇编

For the degenerate-diffusion Keller-Segel equation $$\partial_t\rho=\Delta(\rho^m)-\nabla\!\cdot\!\bigl(\rho\nabla(N*\rho)\bigr)$$ in dimension $d>2$, determine the long-time behavior of solutions, including whether they converge to stationary states or lose mass at infinity.

来源字段：https://doi.org/10.4171/owr/2017/7

11923 | OWR-16159-001

### Long-Time Behavior of Random-Data Nonlinear Schrödinger Equations

P / partially_solved | Oberwolfach 报告问题汇编

Determine the long-time behavior of the almost-sure global solution to the one-dimensional nonlinear Schrödinger equation $$i\partial_sU+\partial_y^2U=|U|^{p-1}U,$$ where $1<p<5$ and the initial data are Gaussian and of low regularity.

来源字段：https://doi.org/10.4171/owr/2018/18

11924 | OWR-17128-004

### Long-Time Dynamics of Fractional Dispersive Equations

O / open | Oberwolfach 报告问题汇编

For the fractional dispersive equation $$\partial_tu=\partial_x\!\left(|\partial_x|^\alpha u-u^p\right)\quad\text{on }\mathbb R,$$ determine how the parameters $\alpha$ and $p$ govern global behavior, singularity formation, and the stability of solitary waves.

来源字段：https://doi.org/10.4171/owr/2019/32

11925 | OWR-16767-006

### Long-Time Dynamics of Randomized Mass-Subcritical NLS

O / open | Oberwolfach 报告问题汇编

Let $d\geq1$, $2/d<p<4/d$, and $\phi\in L^2(\mathbb R^d)$. For

$$i\partial_tu-\Delta u=\lvert u\rvert^pu,\qquad u(0)=\phi^\omega,$$

what is the long-time asymptotic behavior beyond free scattering profiles?

来源字段：https://doi.org/10.4171/owr/2019/5

11926 | OWR-12338-002

### Long-Time Navier-Stokes Flow in Exterior Domains

O / open | Oberwolfach 报告问题汇编

Clarify the large-time behavior of solutions of the Navier-Stokes initial-value problem in a two-dimensional exterior domain, including the associated Oseen evolution.

来源字段：https://doi.org/10.4171/owr/2013/10

11927 | OWR-14609-003

### Long-Time Resonant Dynamics on Large Domains

P / partially_solved | Oberwolfach 报告问题汇编

For nonlinear dispersive equations on domains of very large size $L$, what happens after the Euclidean approximation breaks down, and can the effective dynamics still be described on longer resonant time scales?

来源字段：https://doi.org/10.4171/owr/2016/27

11928 | OWR-14226-001

### Low-Rank Splitting Error Analysis for Discretized PDEs

O / open | Oberwolfach 报告问题汇编

Extend the error analysis of the dynamical low-rank splitting scheme to spatial discretizations of partial differential equations.

来源字段：https://doi.org/10.4171/owr/2016/18

11929 | OWR-12730-002

### Low-Rank Tensor Methods for High-Dimensional Eigenproblems

O / open | Oberwolfach 报告问题汇编

Can low-rank tensor methods overcome the curse of dimensionality when computing the $p$ smallest eigenvalues of $-\Delta u+V(x)u=\lambda u$ on $(a,b)^d$ with Dirichlet boundary conditions, including for very large $d$?

来源字段：https://doi.org/10.4171/owr/2013/56

11930 | AIM-PDES-0044

### Low-regularity flexibility and smooth instability for embedded torus metrics

S / solved | AIM 研讨会问题汇编

(Problem originally proposed by Fang-Hua Lin) Let \((T^2,g_0) \subseteq (\mathbb R^3,g_{\mathrm{eucl}})\) be an isometric embedding and \(g\) be a metric on \(T^2\) which is \(C^0\) close to \(g_0\). Is there an isometric embedding of \((T^2,g)\) in \(\mathbb R^3\)?

来源字段：http://aimpl.org/nonlinpdegeom/3/

11931 | OWR-12331-002

### Lyapunov Functions for Three-Dimensional Acoustic PML Systems

O / open | Oberwolfach 报告问题汇编

For the three-dimensional acoustic perfectly matched layer system with spatially varying damping coefficients, construct a Lyapunov function that is nonincreasing and uniformly controls the acoustic energy; in particular, control the terms created by spatial variation of the damping.

来源字段：https://doi.org/10.4171/owr/2013/03

11932 | AIM-PDES-0038

### Mabuchi distances in a big class and their homothety calibration

P / partially_solved | AIM 研讨会问题汇编

Let \(X\) be a compact Kähler manifold of complex dimension \(n\), let \(\theta\) be a smooth closed real \((1,1)\)-form whose class is big, and let \(p\geq1\). Does the full non-pluripolar Monge--Ampère mass finite-energy space \(\mathcal E^p(X,\theta)\) carry a complete metric \(d_p\), agreeing with the Mabuchi--Darvas \(L^p\) metric when the class is Kähler, such that weak pluripotential Mabuchi geodesics are constant-speed metric geodesics?

来源字段：http://aimpl.org/nonlinpdegeom/2/

11933 | OWR-731-004

### Macroscopic Limits of Surfactant Energy Minimizers

O / open | Oberwolfach 报告问题汇编

If pairs $(u_\varepsilon,\rho_\varepsilon)$ minimize the surfactant energy $G_\varepsilon$, do they converge to a macroscopic state, and is the limit characterized as a minimizer of a relaxed macroscopic energy?

来源字段：https://doi.org/10.4171/owr/2004/29

11934 | OWR-14299286-002

### Maximal Computable Dissipative Solutions of Euler Equations

O / open | Oberwolfach 报告问题汇编

If a numerical scheme for the Euler system converges to a weak solution, is its limit a maximal computable dissipative solution? If the same scheme is oscillatory, does a maximal computable dissipative solution fail to exist?

来源字段：https://doi.org/10.4171/owr/2025/22

11935 | OWR-17130-003

### Mean-Curvature Blowup Limits of Wings and Pitchfork Translators

O / open | Oberwolfach 报告问题汇编

Do the finite-entropy $\Delta-wings$ and pitchfork translators of mean curvature flow arise as blow-up limits of mean curvature flows?

来源字段：https://doi.org/10.4171/owr/2019/34

11936 | OWR-14299907-001

### Mesh Structures Behind Even-Odd Superconvergence

O / open | Oberwolfach 报告问题汇编

Consider the surface finite-element discretization of the surface Stokes equations on a smooth two-dimensional surface $\Gamma\subset\mathbb{R}^3$, using discrete geometries $\Gamma_h^k=\Phi_h^k(\widehat\Gamma_h)$ obtained from a uniformly refined reference triangulation $\widehat\Gamma_h$ by a piecewise polynomial parametrization $\Phi_h^k$ of degree $k$. Characterize the internal mesh structures that cause the observed superconvergence for even $k$ but not odd $k$. In particular, are mirrored pairs of triangles the essential mechanism, and how can the same parity effect occur when no such pairing is visible?

来源字段：https://doi.org/10.4171/owr/2026/5

11937 | OWR-4085-001

### Microstructure Control of Martensitic Phase Transformations

O / open | Oberwolfach 报告问题汇编

For mechanically stabilized martensitic materials, determine the relation between temperature shift and initial microstructure, the thermodynamic principles governing interfacial microstructures, the mechanism of austenite nucleation, and whether complete suppression of austenite nucleation can be achieved by tuning microstructure, orientation, and specimen dimensions.

来源字段：https://doi.org/10.4171/owr/2009/42

11938 | AIM-PDES-0029

### Missing quantifiers and exact Sobolev thresholds for a divergence-form model

O / open | AIM 研讨会问题汇编

If $u\in W^{1,2}_{\mathrm{loc}}(\Omega)$ solves $\operatorname{div}(A(x)\nabla u)=0$, where $A$ is bounded measurable and uniformly elliptic with fixed constants, does $\nabla u$ belong locally to $L^{2+\varepsilon}$ for some $\varepsilon>0$ depending only on dimension and ellipticity?

来源字段：http://aimpl.org/nonlinpdegeom/1/

11939 | DYN-003

### MLC Conjecture

P / partially_solved | 独立 / 未分配集合

Is the Mandelbrot set locally connected?

背景链接：https://en.wikipedia.org/wiki/Mandelbrot_set#Local_connectivity

11940 | OWR-8415338-005

### Monge-Ampère Singularities for Nonpolyhedral Obstacles

O / open | Oberwolfach 报告问题汇编

Develop singular solutions for Monge-Ampère equations with obstacles whose graphs need not be convex polytopes.

来源字段：https://doi.org/10.4171/owr/2021/35

11941 | OWR-14298011-001

### Multidimensional Null Controllability for Bilinear Evolution Equations

O / open | Oberwolfach 报告问题汇编

For bilinear evolution equations $\dot u+Au+p(t)Bu=0$, prove $j$-null controllability in two and three dimensions without the moment method. Establish controllability for the Fokker-Planck equation with perfectly reflecting boundary conditions.

来源字段：https://doi.org/10.4171/owr/2023/52

11942 | OWR-732-002

### Multiphase Models with Multiple Conserved Quantities

O / open | Oberwolfach 报告问题汇编

Extend second-order sharp-interface approximations of diffuse-interface phase-field models to systems with multiple phases and an arbitrary number of conserved quantities.

来源字段：https://doi.org/10.4171/owr/2004/30

11943 | OWR-1050-011

### Multiplicity and Attainment for Critical Hardy p-Laplacian Problems

O / open | Oberwolfach 报告问题汇编

Describe the multiplicity of solutions of the $p$-Laplacian equation with critical Hardy potential, for example when $\Omega$ is a ball, and prove or disprove the converse attainment criterion for $\lambda_p(\Omega)$.

来源字段：https://doi.org/10.4171/owr/2005/29

11944 | OWR-3476-003

### Multiplicity of Blow-Up Points in the Sinh-Gordon Equation

P / partially_solved | Oberwolfach 报告问题汇编

In the sinh-Gordon blow-up problem, what is the possible simple or multiple character of the blow-up points; in particular, when can multiple blow-up points occur?

来源字段：https://doi.org/10.4171/owr/2009/26

11945 | PDE-001

### Navier-Stokes Existence and Smoothness

O / open | Hilbert 问题

Do smooth solutions to Navier-Stokes equations exist globally in 3D? Or do finite-time singularities occur?

背景链接：https://www.claymath.org/wp-content/uploads/2022/02/MPPc.pdf

11946 | MPP-004

### Navier-Stokes Existence and Smoothness

O / open | 千禧年问题

别名：Navier-Stokes Regularity [ANA-006]

Prove or give a counterexample: Do solutions to the Navier-Stokes equations in three dimensions always exist and remain smooth for all time?

背景链接：https://www.claymath.org/millennium-problems/

11947 | AIM-PDES-0053

### Near-critical solitary gravity waves: recovered status and a two-term KP-II dispersion audit

P / partially_solved | AIM 研讨会问题汇编

**Recovered statement.** Determine the stability of the finite-depth, pure-gravity travelling solitary waves \((\eta_c,\Phi_c)(x-ct)\) of the full water-wave equations as \(c\) approaches the critical long-wave speed \(c_*\). In the multidimensional workshop context, this includes stability of the line solitary wave under perturbations depending on a transverse horizontal coordinate.

来源字段：https://aimath.org/WWN/multidimwaves/multidimwaves.pdf

11948 | OWR-15437-003

### NLS Approximation on Periodic Nonlinear Wave Graphs

O / open | Oberwolfach 报告问题汇编

Can the nonlinear Schrödinger (NLS) approximation be justified for original nonlinear wave systems on periodic graphs with quadratic nonlinearities and Kirchhoff vertex conditions?

来源字段：https://doi.org/10.4171/owr/2017/29

11949 | OWR-4209-005

### Nodal Geometry of Second $p$-Laplacian Eigenfunctions

P / partially_solved | Oberwolfach 报告问题汇编

Determine the second eigenfunction of the $p$-Laplacian on a bounded domain, including its nodal set; in particular, prove or disprove that the nodal line is a diameter for a disk for every $p\in(1,\infty)$, and is vertical or horizontal on a square for $1<p<2$ but diagonal for $p>2$.

来源字段：https://doi.org/10.4171/owr/2010/08

11950 | OWR-13347-001

### Non-Kähler Ricci Solitons and Broader Existence Methods

O / open | Oberwolfach 报告问题汇编

Can Hermitian non-Kähler Ricci solitons be constructed using methods successful in the Kähler setting? Can partial differential equation methods establish existence on more general Fano varieties? Are there other special geometries that yield nontrivial Ricci solitons?

来源字段：https://doi.org/10.4171/owr/2014/45

11951 | OWR-14299519-001

### Nonflat Graphical Special Lagrangian Cones in Higher Dimensions

O / open | Oberwolfach 报告问题汇编

For every $n\geq5$, do there exist nonflat graphical special Lagrangian cones in $\mathbb C^n$?

来源字段：https://doi.org/10.4171/owr/2025/33

11952 | OWR-17134-002

### Nonlinear Stability of Columnar Euler Vortices

O / open | Oberwolfach 报告问题汇编

What conditions, if any, ensure nonlinear stability of columnar vortices for the three-dimensional Euler equations?

来源字段：https://doi.org/10.4171/owr/2019/38

11953 | OWR-2495-005

### Nonlinear Stability of Periodic Reaction-Diffusion Waves

P / partially_solved | Oberwolfach 报告问题汇编

Let $u^*(t)$ be a spectrally stable periodic solution of a reaction-diffusion PDE. Does spectral stability suffice for nonlinear stability under a larger class of perturbations, such as bounded rather than spatially localized perturbations?

来源字段：https://doi.org/10.4171/owr/2008/57

11954 | OWR-17128-003

### Nonlocal Water Waves with Generalized Vorticity

O / open | Oberwolfach 报告问题汇编

Extend the nonlocal interface formulation of the water-wave problem from constant vorticity to general vorticity, and determine whether the extension yields additional conservation laws.

来源字段：https://doi.org/10.4171/owr/2019/32

11955 | OWR-15212-004

### Numerical Convergence for General Monge-Ampère Boundary Problems

P / partially_solved | Oberwolfach 报告问题汇编

Prove convergence of numerical schemes to viscosity solutions for general Monge-Ampère equations and boundary conditions, including the second boundary-value problem.

来源字段：https://doi.org/10.4171/owr/2017/7

11956 | OWR-14298011-005

### Occupational Measures for Tonelli Hamilton-Jacobi Equations

O / open | Oberwolfach 报告问题汇编

Extend the theory relating critical sets, singular sets, and limiting occupational measures from $\frac12|Du+c|^2+V(x)=\alpha[c]$ on $\mathbb T^d$ to general Tonelli Hamilton-Jacobi equations $H(x,Du+c)=\alpha[c]$.

来源字段：https://doi.org/10.4171/owr/2023/52

11957 | OWR-757-005

### Oleinik Entropy for the Leroux System

O / open | Oberwolfach 报告问题汇编

Establish the Oleinik entropy condition for the Riemann invariants arising in the hydrodynamic limit of the Leroux system.

来源字段：https://doi.org/10.4171/owr/2004/43

11958 | OWR-11695861-001

### Onsager Thresholds for Hydrostatic Euler Energy Conservation

O / open | Oberwolfach 报告问题汇编

Determine the critical anisotropic regularity thresholds above which weak solutions of the hydrostatic Euler equations conserve horizontal kinetic energy and below which anomalous dissipation can occur.

来源字段：https://doi.org/10.4171/owr/2022/51

11959 | AMR-151-0001

### Open Problems for Painleve Equations - Problem 1

O / open | AMR 问题清单汇编

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\itemsep=0pt

- The Runge--Kutta method, and variants, are standard ODE solvers. Can we do better for integrable equations such as the Painlev\'e equations?

- Painlev\'e equations are ``integrable" and solvable by the isomonodromy method through an associated Riemann--Hilbert problem. How can we use this in the development of software for studying the Painlev\'e equations numerically?

- It is well known that there are {discrete Painlev\'e equations}, which are integrable discrete equations that tend to the associated Painlev\'e equations in an appropriate continuum limit. Should we use a ``integrable discretization" of the Painlev\'e equations?

背景链接：https://arxiv.org/abs/1901.10122

11960 | AMR-151-0002

### Open Problems for Painleve Equations - Problem 2

O / open | AMR 问题清单汇编

The Painlev\'e equations are known to have a cornucopia of properties such as: a Hamiltonian representation; exact solutions (rational solutions, algebraic solutions, solutions in terms of classical special solutions); B\"acklund transformations (which relate two solutions of a Painlev\'e equation); associated isomonodromy problems (which are Lax pairs that express the Painlev\'e equation as the compatibility of two linear systems); and asymptotic approximations in the complex plane, with associated connection formulae relating the asymptotics. For details see [refPAC05review,refConte,refFIKN,refGLS,refIKSY,refKNY,refNoumi] and the references therein.

A complete classification and a unifying structure for these properties is needed as the presently known results are rather fragmentary and non-systematic.

背景链接：https://arxiv.org/abs/1901.10122

11961 | OWR-13487-001

### Optimal Adaptive Approximation of Transport-Dominated Equations

O / open | Oberwolfach 报告问题汇编

Does adaptive finite-element approximation of an elliptic or parabolic transport-dominated diffusion equation attain the convergence rate of the best partition generated by a refinement method such as newest-vertex bisection?

来源字段：https://doi.org/10.4171/owr/2015/2

11962 | OWR-1592-005

### Optimal Adaptive Wavelet Schemes for Nonlinear Boundary Problems

O / open | Oberwolfach 报告问题汇编

For the nonlinear boundary-value problem $-\Delta u=F(u)$ in $\Omega$, with $u=0$ on $\partial\Omega$ and nonlinear $F:H_0^1(\Omega)\to H^{-1}(\Omega)$, can semiorthogonal spline wavelets be used to construct an adaptive scheme that evaluates the nonlinear terms and solves the equation with asymptotically optimal complexity?

来源字段：https://doi.org/10.4171/owr/2007/36

11963 | OWR-4209-002

### Optimal Domain Conditions for Positivity of Hinged Plates

O / open | Oberwolfach 报告问题汇编

For the hinged biharmonic plate problem $\Delta^2v=f$ with $v=0$ and $\Delta v-(1-\sigma)\kappa v_n=0$ on $\partial\Omega$, determine whether the convexity and $C^{2,1}$-boundary assumptions ensuring $f\geq0\Rightarrow v\geq0$ are optimal, or can be weakened.

来源字段：https://doi.org/10.4171/owr/2010/08

11964 | OWR-699-002

### Optimal Mesh Selection with A Posteriori Error Estimates

P / partially_solved | Oberwolfach 报告问题汇编

Develop adaptive error control and optimal mesh-selection theories that use a posteriori estimates allowing cancellation of elementwise errors while preserving relevant stability properties.

来源字段：https://doi.org/10.4171/owr/2004/16

11965 | OWR-13858-002

### Optimal Mixing Rates Under Palinstrophy Constraints

S / solved | Oberwolfach 报告问题汇编

For passive-scalar transport with a velocity field constrained by a uniform palinstrophy, or $H^2$, budget, what is the optimal long-time decay rate of the mix-norm, and is it polynomial or exponential?

来源字段：https://doi.org/10.4171/owr/2015/37

11966 | OWR-14298011-004

### Optimal Set Motions from Nonconvex Initial Regions

P / partially_solved | Oberwolfach 报告问题汇编

For the optimal set-motion problem, characterize optimal strategies when the initial set $\Omega_0$ is nonconvex and determine when all $\Omega(t)$ remain connected. Which planar results extend to $\mathbb R^n$ or Riemannian manifolds, and in particular do optimal set motions remain convex in $\mathbb R^3$?

来源字段：https://doi.org/10.4171/owr/2023/52

11967 | OWR-9790355-002

### Optimization and Multiscale Simulation in Additive Manufacturing

O / open | Oberwolfach 报告问题汇编

Develop methods for variable-density structural optimization in a fixed design domain, for optimizing deposition paths in fiber-oriented additive manufacturing, and for immersed two-level space-time simulation of strongly nonlinear local problems.

来源字段：https://doi.org/10.4171/owr/2022/4

11968 | OWR-17130-010

### Persistence of Euler Vortex Filaments Under Binormal Flow

O / open | Oberwolfach 报告问题汇编

For the three-dimensional Euler equations with initial vorticity concentrated along a smooth spatial curve, does the vorticity remain concentrated around a curve for finite times, and can it remain close for a significant time to a filament evolving by binormal flow?

来源字段：https://doi.org/10.4171/owr/2019/34

11969 | OWR-15991-001

### Perturbed Ergodic Constants in Hamilton-Jacobi Equations

P / partially_solved | Oberwolfach 报告问题汇编

For randomly perturbed periodic coercive Hamilton-Jacobi equations, determine the behavior of the perturbed ergodic constant $\overline H^\eta$ in general settings, especially the asymptotics of $\eta^{-1}(\overline H^\eta-\overline H)$ as $\eta\to0$.

来源字段：https://doi.org/10.4171/owr/2018/6

11970 | OWR-4077-002

### Physically Admissible Weak Solutions of Incompressible Euler

O / open | Oberwolfach 报告问题汇编

What is a weak solution of the incompressible Euler equation, particularly one that is physically reasonable?

来源字段：https://doi.org/10.4171/owr/2009/34

11971 | OWR-4140-001

### Pinning and Depinning of Random-Obstacle Interfaces

P / partially_solved | Oberwolfach 报告问题汇编

For the random-obstacle interface equation, how does the forcing $F$ determine whether the solution satisfies finite pinning, $\lim_{t\to\infty}u(x,t)<\infty$, or depinning, $\lim_{t\to\infty}u(x,t)=\infty$?

来源字段：https://doi.org/10.4171/owr/2009/57

11972 | OWR-4085-005

### Pinning Thresholds for Forced Interfaces

P / partially_solved | Oberwolfach 报告问题汇编

For which forcing strengths $F$ is the interface pinned, meaning $\lim_{t\to\infty}u(x,t)<\infty$, and for which is there no pinning, meaning $\lim_{t\to\infty}u(x,t)=\infty$?

来源字段：https://doi.org/10.4171/owr/2009/42

11973 | OWR-14226-003

### Pluri-Lagrangian Integrability of Discrete Variational Systems

P / partially_solved | Oberwolfach 报告问题汇编

Can integrability of discrete variational systems be characterized by the pluri-Lagrangian property that a field on a multi-time space is critical for the action of a $d$-form on every $d$-dimensional multi-time manifold?

来源字段：https://doi.org/10.4171/owr/2016/18

11974 | OWR-1277-001

### Point-Load Asymptotics for Thin Shells

O / open | Oberwolfach 报告问题汇编

Determine the small-thickness asymptotic behavior near the load point of the solution of the normal point-load problem for a shell, especially for hyperbolic and parabolic shell geometries.

来源字段：https://doi.org/10.4171/owr/2006/28

11975 | OWR-14298587-012

### Pointwise Quadratic Bounds for Extremal Solutions

O / open | Oberwolfach 报告问题汇编

For an extremal solution $u^\star$ of a semilinear elliptic equation near an isolated singularity, prove or disprove the pointwise estimate $\limsup_{x\to 0} f'(u^\star(x))|x|^2<\infty$.

来源字段：https://doi.org/10.4171/owr/2024/37

11976 | OWR-14299287-005

### Predicting Topographically Pinned Vortices

O / open | Oberwolfach 报告问题汇编

Extend selective-decay theory so that it predicts the strength and location of vortices pinned to topography.

来源字段：https://doi.org/10.4171/owr/2025/24

11977 | OWR-4990378-001

### Quadratic Growth without Quadratic Control Regularization

O / open | Oberwolfach 报告问题汇编

For distributed optimal control of the $H^1$-critical defocusing wave equation on a bounded domain in $\mathbb R^3$, do the strong quadratic-growth property and the resulting stability estimates remain valid when $\beta_2=0$, so that the objective has no quadratic $L^2$ control penalty?

来源字段：https://doi.org/10.4171/owr/2021/9

11978 | AIM-PDES-0077

## Quadratic-chain characterization of the Aubry set

P / partially_solved | AIM 研讨会问题汇编

The primary AIM PDF was checked directly. With typographical layout normalized, it asks about \[ L_f(x,v)=\frac12\lVert v-f(x)\rVert_x^2 \] for a smooth vector field \(f\) on a compact Riemannian manifold \(M\), the relation between its zero-class Aubry set and recurrence of the flow \(\phi^t\) of \(f\), and uniqueness of the zero solution of a stationary Hamilton--Jacobi equation. The PDF literally says that the zero section in \(TM\) is invariant and literally prints \[ H_-(x,p)=\frac12\lVert p\rVert_x^2-p(f(x)). \]

There are three independent formulation defects.

1. In standard tangent-bundle coordinates the invariant copy of the \(f\)-flow is \(\operatorname{Graph}(f)=\{(x,f(x)):x\in M\}\), not the zero section. It is this graph that the Legendre transform sends to the zero section of \(T^*M\). 2. The Legendre transform of the displayed \(L_f\) is \[ H_+(x,p)=\frac12\lVert p\rVert_x^2+p(f(x)), \] since \(p=(v-f)^\flat\). The printed \(H_-\) is instead dual to \(L_{-f}(x,v)=\frac12\lVert v+f(x)\rVert^2\), the Mañé Lagrangian for the reversed vector field. 3. A stationary Hamilton--Jacobi equation is invariant under \(u\mapsto u+C\). Thus \(u=0\) is never literally the unique solution on a nonempty \(M\). On connected \(M\), the meaningful question is uniqueness modulo constants, or uniqueness after imposing \(u(x_*)=0\).

The PDF's \(H1(M,\mathbb R)\) should also be read as \(H^1(M,\mathbb R)\). None of these corrections is silently substituted for the source: below, \(L_f,H_+\) are the consistent formulation, and the printed minus-sign equation is handled separately by time reversal.

I assume henceforth that \(M\) is compact, connected, without boundary, and Riemannian, and that \(f\) is smooth. Write \(\mathcal A_f\subset M\) for the **projected** zero-class Aubry set and \(\widetilde{\mathcal A}_f\subset TM\) for the tangent-bundle Aubry set.

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11979 | AIM-PDES-0032

## Quadratic-growth complex Monge-Ampere rigidity with a completeness gap

P / partially_solved | AIM 研讨会问题汇编

Let \(u\) be plurisubharmonic solution to \[ \det(u_{\bar k j})=1\quad\hbox{on }\mathbb C^n \] satisfying \[ C^{-1}(|z|^2+1)\leq u\leq C(|z|^2+1), \] then \(u\) is quadratic.

The reversal \(u_{\bar k j}\) versus the more usual \(u_{j\bar k}\) is harmless. There are, however, three source-level ambiguities or omissions:

1. no regularity or solution concept is specified (classical, viscosity, or pluripotential); 2. it is not said whether the displayed inequalities hold everywhere or only outside a compact set; 3. completeness of the Kähler metric defined by the complex Hessian is not assumed.

The AIM record also says that An-Min Li and Li Sheng solved the problem. Their theorem has extra smoothness and completeness hypotheses, so that remark is correct only for the strengthened reading recorded in Section 2 below.

Throughout the elementary arguments here, \[ \partial_{z_j}=\tfrac12(\partial_{x_j}-i\partial_{y_j}),\qquad \partial_{\bar z_j}=\tfrac12(\partial_{x_j}+i\partial_{y_j}). \] Thus \(u_{z\bar z}=\frac14\Delta u\) in complex dimension one.

来源字段：http://aimpl.org/nonlinpdegeom/1/

11980 | OWR-11101918-010

## Quantitative Rigidity for Aviles-Giga Differential Inclusions

O / open | Oberwolfach 报告问题汇编

Do there exist constants $C>0$ and $\alpha\in(0,1)$ such that every $w\in H^1(B_1;\mathbb R^2)$ satisfies

$$\inf_{A\in K}\int_{B_{1/2}}|Dw-A|^2\,dx\leq C\left(\int_{B_1}\operatorname{dist}^2(Dw,K)\,dx\right)^\alpha?$$

This asks for a quantitative rigidity estimate for the differential inclusion.

来源字段：https://doi.org/10.4171/owr/2022/37

11981 | OWR-4079-001

## Quantization of Concentration Mass for Higher-Order Equations

O / open | Oberwolfach 报告问题汇编

For solutions of the higher-order critical equation $(-\Delta)^m u=\lambda u\exp(m u)$ with bounded concentration energy, must the concentration mass equal $L\Lambda_1$ for some positive integer $L$ when $m>2$, as it does for $m=1,2$?

来源字段：https://doi.org/10.4171/owr/2009/36

11982 | AIM-PDES-0099

### Quantizing Mather measures by equilibrium packets and smooth-graph WKB states

P / partially_solved | AIM 研讨会问题汇编

Consider the Hamiltonian \[ H(x,p)=\frac12|p|^2+V(x), \] where \(V\) is periodic on \(\mathbb T^n\). The corresponding quantum Hamiltonian is \[ -\frac{h^2}{2}\Delta+V(x). \] Do Mather sets play a role in the semiclassical limit? Conversely, can one “quantize” Mather sets by building corresponding quasimodes with good error bounds? Would a Diophantine condition be useful?

It is Craig Evans's problem from the AIM workshop *New connections between dynamical systems and PDE's*.

The mathematical text recovered from the original AIM page is:

Immediately after identifying the operator as \(-h^2\Delta/2+V\), the source displays the eigenproblem with \(+h^2\Delta u/2+Vu=Eu\). With the analytic convention \(\Delta=\sum_j\partial_{x_j}^2\), the quantization of the stated symbol \(|p|^2/2+V\) is \[ P_h=-\frac{h^2}{2}\Delta+V. \] That is the operator treated below. The plus-sign display would instead have principal symbol \(-|p|^2/2+V\), which is not the stated Tonelli Hamiltonian.

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11983 | AIM-PDES-0101

### Quenched Brownian large deviations in stationary ergodic potentials without mixing

P / partially_solved | AIM 研讨会问题汇编

**33. (Elena Kosygina)** Let \(\Omega\) be a probability space with probability measure \(P\) ergodic under a shift transformation group \(\tau_x:\Omega\to\Omega\), \(x\in\mathbb R^n\). Consider the stochastic Hamilton--Jacobi equation \[ u_t+H(D_xu,\tau_x\omega)=0,\qquad (x,t)\in\mathbb R^n\times(0,\infty), \] where \(H(p,\omega)\) is convex in \(p\) and satisfies some regularity assumptions, for example \[ H(p,\omega)=\frac12|p|^2-V(\omega). \] A homogenization result for this problem was obtained by Lions and Souganidis. For the Hamiltonian above homogenization is equivalent to a large deviation result for Brownian motion among random obstacles (Snitzman). Problem: Derive a large deviation result for general stationary ergodic setting without any independence or mixing assumptions on \(P\) under the shifts.

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11984 | OWR-1050-017

### Radiality and Liouville Properties for Critical Elliptic Operators

O / open | Oberwolfach 报告问题汇编

For Pucci's maximal uniformly elliptic operator $\mathcal{M}^+_{\lambda,\Lambda}$ and its critical exponent $p^*$, is every nonnegative $C^2$ solution of $\mathcal{M}^+_{\lambda,\Lambda}(D^2u)+u^{p^*}=0$ on $\mathbb{R}^N$ radial? For $1<p<p^*$, is every nonnegative entire solution identically zero?

来源字段：https://doi.org/10.4171/owr/2005/29

11985 | OWR-1317-001

### Rank-One Convexity Versus Quasiconvexity in Two Dimensions

O / open | Oberwolfach 报告问题汇编

Does rank-one convexity imply quasiconvexity for functions on $\mathbb R^{2\times2}$?

来源字段：https://doi.org/10.4171/owr/2006/31

11986 | OWR-823-002

### Reachable States of Time-Dependent Schrödinger Systems

O / open | Oberwolfach 报告问题汇编

For the bilinear controlled Schrödinger equation $i\partial_t\Psi=(H_0-\varepsilon(t)\mu)\Psi$ with $\Psi(0)=\Psi_0$, characterize the attainable final states in infinite dimension and give implementable criteria for the appropriate notions of controllability.

来源字段：https://doi.org/10.4171/owr/2005/18

11987 | AIM-PDES-0047

### Recovered HCF+ and a complete curve-case continuation and entropy theorem

P / partially_solved | AIM 研讨会问题汇编

Can the Streets--Tian continuation criterion for \(HCF_+\) be reduced to control of fewer geometric quantities, and does this exact flow possess Perelman-type \(\mathcal F\) and \(\mathcal W\) functionals?

about a flow of inverse Hermitian metrics and displays \[ \frac{\partial}{\partial t}g^{j\bar k} =g^{m\bar n}\partial_m\partial_{\bar n}g^{j\bar k} -\partial_m g^{j\bar k}\,\partial_{\bar n}g^{m\bar k}. \tag{bad} \] It then states the known finite-time obstruction \[ |\mathrm{Rm}^{\mathcal{Ch}}|+|T|+|\nabla T|\longrightarrow\infty \] and asks whether this can be improved and whether analogues of Perelman's \(\mathcal F\) and \(\mathcal W\) functionals exist.

Equation (bad) is not a tensor equation under the Einstein convention. In its quadratic term, \(\bar n\) occurs only once while the supposedly free index \(\bar k\) occurs twice. Therefore the literal displayed equation cannot be analyzed as a geometric flow.

The intended equation is independently verified by Proposition 3.9 and Corollary 3.10 of Ustinovskiy's 2018 Princeton thesis. The recovered inverse-metric equation is \[ \boxed{\quad \frac{\partial}{\partial t}g^{i\bar j} =g^{m\bar n}\partial_m\partial_{\bar n}g^{i\bar j} -\bigl(\partial_m g^{i\bar n}\bigr) \bigl(\partial_{\bar n}g^{m\bar j}\bigr). \quad} \tag{HCF\(_+^{-1}\)} \] Thus the first \(g^{j\bar k}\) in the bad quadratic term must be \(g^{j\bar n}\). All contracted and free indices then occur correctly.

Invariantly, this is Ustinovskiy's distinguished Hermitian curvature flow, now often called the **positive Hermitian curvature flow** \(HCF_+\): \[ \frac{\partial}{\partial t}g_{i\bar j} =-\Theta_{i\bar j},\qquad \Theta_{i\bar j}=S^{(2)}_{i\bar j}+\frac12Q^2_{i\bar j}, \tag{HCF\(_+\)} \] where \[ S^{(2)}_{i\bar j}=g^{m\bar n}\Omega_{m\bar n i\bar j},\qquad Q^2_{i\bar j} =g^{m\bar n}g^{p\bar s}T_{pm\bar j}T_{\bar s\bar n i}. \] Here \(\Omega=\mathrm{Rm}^{\mathcal{Ch}}\) and \(T\) are respectively the curvature and torsion of the Chern connection. The recovered problem is:

来源字段：http://aimpl.org/nonlinpdegeom/4/

11988 | OWR-16411-007

### Reduction of Order for Perturbatively Corrected Equations

O / open | Oberwolfach 报告问题汇编

别名：Well Posed Perturbative Einstein Field Equations [OWR-16411-008]

For perturbatively corrected partial differential equations, is there a systematic reduction-of-order procedure that remains accurate to leading perturbative order without introducing spurious degrees of freedom, coordinate dependence, runaway solutions, or ill-posedness?

来源字段：https://doi.org/10.4171/owr/2018/36

11989 | AIM-PDES-0080

### Regular Mather functions, convex duality, and integrability

P / partially_solved | AIM 研讨会问题汇编

**5. (Patrick Bernard)** Consider an analytic positive definite Lagrangian \[ L:\mathbb T^n\times\mathbb R^n\times\mathbb T\longrightarrow\mathbb R, \qquad L=L(x,v,t), \] satisfying Mather's conditions. Let \[ \alpha_L(c)=-\inf_\mu\int\bigl(L(x,v,t)-c\cdot v\bigr)\,d\mu \] be Mather's function \(\alpha_L:\mathbb R^n\to\mathbb R\). Is it true that if \(\alpha_L\) is analytic with positive definite second derivative, then the Lagrangian system is completely integrable?

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

11990 | OWR-767-005

### Regularity and Monotonicity of Atomic Ground-State Densities

O / open | Oberwolfach 报告问题汇编

Determine the regularity of the spherically averaged ground-state electron density $\rho_e$ near the origin. Is $\rho_e(r)$ monotonically decreasing?

来源字段：https://doi.org/10.4171/owr/2004/53

11991 | OWR-1534-001

### Regularity and Symmetry in Laplacian Shape Optimization

P / partially_solved | Oberwolfach 报告问题汇编

Let $D\subset\mathbb{R}^d$ be bounded and open, fix a volume $m$, and let $\widehat{\Omega}\subset D$ be an optimal quasi-open set of volume $m$ for the $k$-th Dirichlet eigenvalue. For $k=1$, obtain sharper bounds for the size of the singular part of $\partial\widehat{\Omega}$. For $k=2$, when $d=2$ and $D$ is a rectangle, must $\widehat{\Omega}$ be symmetric? For $k\geq3$ and connected $D$, are the associated eigenfunctions Lipschitz continuous?

来源字段：https://doi.org/10.4171/owr/2007/18

11992 | OWR-14298587-003

### Regularity of Boundary Free Interfaces

P / partially_solved | Oberwolfach 报告问题汇编

For the boundary free interface $\mathcal F_{\partial D}$ of a spectral optimal partition, let $\operatorname{Reg}$ be the set of points whose limiting frequency is an integer. Is $\operatorname{Reg}$ a $C^1$ submanifold of codimension one in $\partial D$? Does $\mathcal F_{\partial D}\setminus\operatorname{Reg}$ have codimension two in $\partial D$?

来源字段：https://doi.org/10.4171/owr/2024/37

11993 | OWR-14298587-006

### Regularity of Fractional Multiphase Interfaces

O / open | Oberwolfach 报告问题汇编

Develop a regularity theory for fractional multiphase minimal partitions with general surface-tension coefficients, whether or not those coefficients satisfy the triangle inequality.

来源字段：https://doi.org/10.4171/owr/2024/37

11994 | OWR-14298587-005

### Regularity of Stationary Constraint Maps and Their Heat Flow

O / open | Oberwolfach 报告问题汇编

Extend the regularity theory for energy-minimizing maps constrained to a target domain to stationary constraint maps, and develop the corresponding regularity theory for their parabolic heat flow.

来源字段：https://doi.org/10.4171/owr/2024/37

11995 | OWR-8415338-010

### Regularity of Varifolds for Anisotropic Energies

P / partially_solved | Oberwolfach 报告问题汇编

Does the regularity theorem for rectifiable varifolds with generalized mean curvature bounded in $L^p$, where $p$ is greater than the varifold dimension, extend to anisotropic energies $\Sigma_\Psi(V)=\int\Psi(T_z\Gamma)\theta(z)\,d\mathcal{H}^m(z)$, under the corresponding density-ratio hypothesis?

来源字段：https://doi.org/10.4171/owr/2021/35

11996 | AIM-PDES-0042

### Regularity of weak geodesics on singular Kähler varieties

P / partially_solved | AIM 研讨会问题汇编

Let \(X\) be a compact singular Kähler variety, let \(\omega_0,\omega_1\) be cohomologous Kähler metrics on \(X\), and define their weak Mabuchi geodesic by the pluripotential Perron envelope on a resolution. Is that envelope \(C^{1,1}_{\mathrm{loc}}(X_{\mathrm{reg}}\times A)\), where \(A\) is the annulus encoding geodesic time?

来源字段：http://aimpl.org/nonlinpdegeom/2/

11997 | OWR-14297735-001

### Regularity of Weak Hamiltonian-Minimal Legendrian Surfaces

O / open | Oberwolfach 报告问题汇编

Prove that every weak solution of the Hamiltonian-minimal equation arising from Legendrian area variation is a smooth branched Legendrian immersion away from isolated Schoen-Wolfson conical singularities of nonzero Maslov class.

来源字段：https://doi.org/10.4171/owr/2023/32

11998 | OWR-15582-005

### Relative Kähler-Ricci Flow on Projective-Space Fibrations

O / open | Oberwolfach 报告问题汇编

For a projective-space fibration $f:X\to S$ arising from a rank-$r$ bundle $E$, can the relative Kähler-Ricci flow be developed directly on $\mathcal O_E(r)$, using

$$K_{X/S}^{-1}\cong\mathcal O_E(r)\otimes f^*(\det E)^{-1},$$

rather than on $K_{X/S}$?

来源字段：https://doi.org/10.4171/owr/2017/39

11999 | OWR-12730-001

### Reliability of Numerically Computed PDE Eigenvalues

P / partially_solved | Oberwolfach 报告问题汇编

How many numerical eigenvalues from finite-difference or finite-element approximations of PDE eigenvalue problems are reliable?

来源字段：https://doi.org/10.4171/owr/2013/56

12000 | OWR-12336-004

### Representation and Limits of Fractional Infinity Eigenfunctions

O / open | Oberwolfach 报告问题汇编

For the fractional infinity-eigenvalue problem, does $u(x)=\delta(x)^\alpha/(\delta(x)^\alpha+\varrho_1(x)^\alpha)$ produce every first eigenfunction when $\varrho_1(x)=\operatorname{dist}(x,\Gamma_1)$ and $\Gamma_1\subseteq\Gamma$? Is the maximal solution, with $\Gamma_1=\Gamma$, the limit of the fractional $p$-eigenfunctions as $p\to\infty$?

来源字段：https://doi.org/10.4171/owr/2013/08

12001 | OWR-14297732-009

### Ricci Flow Uniqueness from Metric Space Initial Data

P / partially_solved | Oberwolfach 报告问题汇编

For two Ricci flows converging locally to the same metric space as $t\to0$, what regularity assumptions on the limiting metric ensure a common gauge in which the flows can be effectively compared and remain close for small time?

来源字段：https://doi.org/10.4171/owr/2023/29

12002 | OWR-11101918-012

### Rigidity of Higher Index Differential Inclusion Curves

S / solved | Oberwolfach 报告问题汇编

For the nonelliptic differential-inclusion sets $\Gamma_n$ with $n\geq 3$, does sufficiently local membership $Dw(x)\in\Gamma_n$ force $Dw$ to be rigid?

来源字段：https://doi.org/10.4171/owr/2022/37

12003 | OWR-14298587-013

### Rigidity of Stable One-Phase Solutions

P / partially_solved | Oberwolfach 报告问题汇编

Let $u$ be a global stable solution of the one-phase free-boundary problem in $\mathbb R^n$. Prove that $u$ is one-dimensional when $n\leq 6$.

来源字段：https://doi.org/10.4171/owr/2024/37

12004 | OWR-12008-005

### Rigidity of Symmetric-Gradient Differential Inclusions

P / partially_solved | Oberwolfach 报告问题汇编

What structure can be proved for sufficiently regular $u$ satisfying the differential inclusion $Eu\in\{\lambda(a\mathbin{\odot}b):\lambda\in\mathbb R\}$?

来源字段：https://doi.org/10.4171/owr/2012/36

12005 | OWR-1703869-001

### Rigidity, Regularity, and Vectorial Nucleation in Nonlinear Inclusions

O / open | Oberwolfach 报告问题汇编

For geometrically nonlinear differential inclusions, is there a critical threshold separating rigid and flexible regimes, and can one bound the maximal regularity of irregular solutions? Can scalar nucleation models for shape-memory alloys be extended to vectorial models with compatibility conditions, related to convex integration, and analyzed through their associated random processes?

来源字段：https://doi.org/10.4171/owr/2020/22

12006 | OWR-11101924-003

### Rigorous Boundary-Layer Density Asymptotics

O / open | Oberwolfach 报告问题汇编

Prove that the asymptotic rate-of-convergence formulas derived for the slow densities $\eta_m$ along periodic multiple-scattering orbits describe the actual densities.

来源字段：https://doi.org/10.4171/owr/2022/43

12007 | AIM-PDES-0100

### Robust failure in nonconvex stochastic Hamilton-Jacobi homogenization

P / partially_solved | AIM 研讨会问题汇编

**32. (Takis Souganidis)** Homogenization problem for random Hamilton-Jacobi equation \[ F(D^2u,Du,u,x/\varepsilon,x,\omega)=0 \] with non-convex stationary ergodic in \(x/\varepsilon\) Hamiltonian \(F\).

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

12008 | OWR-2495-004

### Robustness of Liesegang Precipitation Patterns

P / partially_solved | Oberwolfach 报告问题汇编

For reaction-diffusion systems exhibiting Liesegang precipitation patterns with characteristic spatial scaling, how robust are the patterns under perturbations, and for which open sets of kinetic and diffusion constants and boundary conditions do they persist?

来源字段：https://doi.org/10.4171/owr/2008/57

12009 | OWR-11101912-014

### Scattering for General Cubic Defocusing Dispersive Flows

P / partially_solved | Oberwolfach 报告问题汇编

Do one-dimensional dispersive flows with cubic defocusing nonlinearities and small initial data have global-in-time scattering solutions?

来源字段：https://doi.org/10.4171/owr/2022/31

12010 | OWR-10252939-003

### Scattering for One-Dimensional Cubic Defocusing Flows

P / partially_solved | Oberwolfach 报告问题汇编

Prove that every one-dimensional dispersive flow with cubic defocusing nonlinearity and sufficiently small, not necessarily localized, initial data has a global scattering solution.

来源字段：https://doi.org/10.4171/owr/2022/30

12011 | OWR-11788-004

### Scattering for the Quintic Resonant System

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Quintic Resonant-System Scattering Problem [OWR-11788-005]

For a sequence $\vec u=(u_j)_{j\in\mathbb Z^2}$ governed by the quintic resonant system

$$i\partial_tu_j+\partial_{xx}u_j=\sum_{(j_1,\ldots,j_5)\in\mathcal R(j)}u_{j_1}\overline{u}_{j_2}u_{j_3}\overline{u}_{j_4}u_{j_5},$$

where

$$\mathcal R(j)=\left\{(j_1,\ldots,j_5)\in(\mathbb Z^2)^5:\ j_1-j_2+j_3-j_4+j_5=j,\ |j_1|^2-|j_2|^2+|j_3|^2-|j_4|^2+|j_5|^2=|j|^2\right\},$$

define

$$E_{\mathrm{ls}}(\vec u)=\frac12\sum_{j\in\mathbb Z^2}\langle j\rangle^2\lVert u_j\rVert_{L^2(\mathbb R)}^2.$$

For every $E>0$ and every smooth initial datum with $E_{\mathrm{ls}}(\vec u_0)\leq E$, does there exist a global solution conserving $E_{\mathrm{ls}}$ that scatters as $t\to\pm\infty$?

来源字段：https://doi.org/10.4171/owr/2012/26

12012 | OWR-14299287-002

### Selecting Among Nonunique Fluid-Mechanics Solutions

O / open | Oberwolfach 报告问题汇编

Find a satisfactory solution concept for fluid-mechanics PDEs that resolves or selects among nonunique admissible weak solutions.

来源字段：https://doi.org/10.4171/owr/2025/24

12013 | OWR-1110-002

### Selection in Finite-Volume LSW Coarsening Models

P / partially_solved | Oberwolfach 报告问题汇编

For finite-volume corrections to the Lifshitz-Slyozov-Wagner coarsening model, determine whether the self-consistent theory closed at the two-particle distribution level resolves the weak selection problem for self-similar asymptotic states; determine also the role of particle collisions in selection.

来源字段：https://doi.org/10.4171/owr/2005/49

12014 | OWR-14299287-004

### Selective Decay in Forced-Dissipative Flows

P / partially_solved | Oberwolfach 报告问题汇编

Can selective-decay variational principles be extended to forced-dissipative flows? Do their predictions remain valid for complex topography, what causes topographically pinned vortices, and can the strengths and locations of those vortices be predicted?

来源字段：https://doi.org/10.4171/owr/2025/24

12015 | OWR-2494-001

### Self-Similar Shock Diffraction by Wedges

P / partially_solved | Oberwolfach 报告问题汇编

Given $\gamma>1$ and $\rho_0>0$, let $\Lambda$ be the upper half-plane outside a wedge and let $\phi$ solve $\operatorname{div}(\rho(|D\phi|^2,\phi)D\phi)+2\rho(|D\phi|^2,\phi)=0$, where $\rho=(\rho_0-(\gamma-1)(\phi+|D\phi|^2/2))^{1/(\gamma-1)}$. Seek a self-similar weak solution satisfying $D\phi\cdot\nu=0$ on the wedge boundary and tending at infinity to the two incident constant-state potentials separated by the incident shock.

来源字段：https://doi.org/10.4171/owr/2008/56

12016 | OWR-14298161-002

### Semigeostrophic Equations on the Whole Rotating Sphere

O / open | Oberwolfach 报告问题汇编

Can local well-posedness for the semigeostrophic equation on a closed spherical subdomain contained strictly in one hemisphere be extended to the whole rotating sphere, despite the degeneracy of the Coriolis parameter at the equator?

来源字段：https://doi.org/10.4171/owr/2024/7

12017 | AIM-PDES-0033

### Separated-variable counterexamples and sharp boundary blow-up

P / partially_solved | AIM 研讨会问题汇编

Let $u:\mathbb R^2\to\mathbb R$ be a solution of $$ u_{11}u_{22}=1. $$ If $u(x)\leq C(1+|x|^2)$, then $u$ must be quadratic.

Can the growth condition be removed?

If one considers instead solutions $u:\Omega\to\mathbb R$ for $\Omega\subseteq\mathbb R^2$ convex, and bounded, and $u$ with Dirichlet boundary conditions, can one obtain a $C^2$ boundary estimate?

来源字段：http://aimpl.org/nonlinpdegeom/1/

12018 | AIM-PDES-0039

### Sharp block-power obstruction for complex Monge-Ampere regularity

P / partially_solved | AIM 研讨会问题汇编

Let $u$ be a plurisubharmonic function satisfying $$ (i\partial\overline\partial u)^n=\psi $$ with $\psi>0$ and smooth. Assume $u$ is $C^{1,\alpha}$ for $\alpha>1-\frac2n$, then $u\in C^\infty$.

**Recovered statement (conjecture).** Let $\Omega\subset\mathbb C^n$ be a domain, $n\ge1$, and let $$ u\in \operatorname{PSH}(\Omega)\cap C_{\mathrm{loc}}^{1,\beta}(\Omega), \qquad 0<\beta\le1. $$ Suppose, in the Bedford-Taylor sense, $$ (i\partial\bar\partial u)^n=\Psi, \tag{1.1} $$ where $\Psi$ is a smooth strictly positive $(n,n)$-form. If $\beta>1-2/n$, prove that $u\in C^\infty_{\mathrm{loc}}(\Omega)$.

来源字段：http://aimpl.org/nonlinpdegeom/2/

12019 | OWR-11453-002

### Sharp Free-Energy Decay for Fast Diffusion

O / open | Oberwolfach 报告问题汇编

For the fast-diffusion solution $u(t,\cdot)$ and its free-energy functional $\mathcal F_\sigma$, can one obtain sharp estimates for the decay of $\mathcal F_\sigma[u(t,\cdot)]$ for every $t\geq0$?

来源字段：https://doi.org/10.4171/owr/2011/55

12020 | AIM-PDES-0026

### Sharp radial regularity and the ambiguity of smooth cone decomposition

P / partially_solved | AIM 研讨会问题汇编

Let $u$ strictly convex with Dirichlet boundary conditions solve $$ \det(u_{ij})=1+\delta_{x_0}, $$ then $u=\varphi+w$ for $\varphi$ smooth uniformly convex cone, and $w$ smooth.

来源字段：http://aimpl.org/nonlinpdegeom/1/

12021 | OWR-1703875-003

### Sharp Waiting-Time Estimates for Variational PDE Discretizations

O / open | Oberwolfach 报告问题汇编

Can the variational/Lagrangian approach for the porous-medium and thin-film equations be sharpened to give mesh-independent, quantitatively sharper estimates for the waiting-time phenomenon and to match the observed behavior of adapted discretization meshes?

来源字段：https://doi.org/10.4171/owr/2020/29

12022 | OWR-4990383-001

### Sharp-Interface Limits of Periodic Phase-Transition Equations

P / partially_solved | Oberwolfach 报告问题汇编

Let $a$ be a $\mathbb Z^d$-periodic uniformly elliptic matrix field, let $W$ be a double-well potential with global minima at $\pm1$, and let $u_\varepsilon$ solve $$ \partial_tu_\varepsilon-\operatorname{div}\!\bigl(a(\varepsilon^{-1}x)Du_\varepsilon\bigr)+\varepsilon^{-2}W'(u_\varepsilon)=0 $$ on $\mathbb R^d\times(0,\infty)$ with initial datum $u_0$. What is the limiting behavior of $u_\varepsilon$ as $\varepsilon\to0^+$?

来源字段：https://doi.org/10.4171/owr/2021/14

12023 | OWR-1050-013

### Sign-Changing Solutions of the Critical Equation

O / open | Oberwolfach 报告问题汇编

For $-\Delta u=|u|^{4/(N-2)}u$ with $u\in D^{1,2}(\mathbb{R}^N)$, what is the optimal energy-gap constant $\varepsilon_0$, and is the minimum energy among sign-changing solutions attained? Are there sign-changing solutions that, modulo conformal transformations, do not have $O(2)$ symmetry? Is every bounded sign-changing solution on $\mathbb{R}^N$ contained in $D^{1,2}(\mathbb{R}^N)$?

来源字段：https://doi.org/10.4171/owr/2005/29

12024 | OWR-7155448-001

### Simpler Quantization of Weil-Petersson Iterated Fibrations

O / open | Oberwolfach 报告问题汇编

Is there a simpler construction and quantization of the iterated fibration structure used for the Weil-Petersson metric and its associated pseudodifferential calculus?

来源字段：https://doi.org/10.4171/owr/2021/27

12025 | OWR-1050-016

### Simultaneous Control of Nodal Blow-Up Values and Positions

O / open | Oberwolfach 报告问题汇编

For ancient solutions of $u_t=u_{xx}+\lambda u+|u|^{p-1}u$ on $(0,\pi)$ with Dirichlet boundary conditions, can one prescribe simultaneously the alternating extrema $u_1,\ldots,u_m$ and their blow-up positions $0<x_1<\cdots<x_m<\pi$, subject to the compatibility conditions appearing in the separate existence theorems?

来源字段：https://doi.org/10.4171/owr/2005/29

12026 | OWR-16414-003

### Singular Branched-Transport Structures as Lyapunov Minima

P / partially_solved | Oberwolfach 报告问题汇编

For $\beta>1$, determine whether the singular structures observed in long-time numerical solutions are related to local minima of the candidate Lyapunov functional $L_{\beta}$.

来源字段：https://doi.org/10.4171/owr/2018/39

12027 | OWR-11101918-004

### Singular Free Boundaries in the Critical Dimension

O / open | Oberwolfach 报告问题汇编

Determine whether the critical dimension $d_*$ for singular free boundaries in the one-phase problem is $5$, $6$, or $7$; describe the singular set in the critical dimension; remove the symmetry assumption from the two-dimensional branch-point regularity result; and find a frequency-function proof that may extend to higher dimensions.

来源字段：https://doi.org/10.4171/owr/2022/37

12028 | AIM-PDES-0095

### Smooth periodic bounded-gradient minimizers need not be Birkhoff

S / solved | AIM 研讨会问题汇编

If \(u\) is minimal and \(\sup_{\mathbb R^n}|Du|<\infty\), must the projection of \(\operatorname{graph}u\) to \(\mathbb T^{n+1}\) have no self-intersections?

Let \[ L:\mathbb T^n\times\mathbb T\times\mathbb R^n\longrightarrow\mathbb R, \qquad L=L(x,u,p), \] be smooth and satisfy the usual convexity assumptions in \(p\). A model is \[ L(x,u,p)=\frac12|p|^2+F(x,u), \] where \(F:\mathbb T^{n+1}\to\mathbb R\) is periodic in all variables. A function \(u:\mathbb R^n\to\mathbb R\) is *minimal* if it minimizes the action against every compactly supported variation. Is there an analytic condition guaranteeing that a minimal graph has no self-intersections on the torus? More concretely:

The phrase “without self-intersections” is not the tautological assertion that an ordinary graph in \(\mathbb R^{n+1}\) is embedded. For \(k=(m,\ell)\in\mathbb Z^n\times\mathbb Z\), set \[ (T_k u)(x)=u(x+m)+\ell . \] Bangert's definition is that the orbit \(\{T_k u\}\) is totally ordered pointwise: for every \(k\), either \(T_k u\ge u\) everywhere or \(T_k u\le u\) everywhere (with the strong maximum principle refining this to strict inequality or equality). Equivalently, integer translates of the graph are disjoint or identical. This is also called the Birkhoff property.

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

12029 | OWR-1054-004

### Smoothness of Phase-Separation Limit Supports

S / solved | Oberwolfach 报告问题汇编

For limit measures arising from stable critical points of the phase-separation energy, prove that the support is smooth in dimensions $n\leq7$.

来源字段：https://doi.org/10.4171/owr/2005/33

12030 | OWR-8415340-005

### Solitary-Wave Stability for the Lugiato-Lefever Equation

O / open | Oberwolfach 报告问题汇编

For the Lugiato-Lefever equation $$\partial_t\psi=-i\beta\,\partial_x^2\psi-(1+i\alpha)\psi+i|\psi|^2\psi+F,$$ determine the stability or instability of its solitary-wave solutions.

来源字段：https://doi.org/10.4171/owr/2021/37

12031 | OWR-4429-001

### Soliton Resolution and Blow-Up near Soliton Families

O / open | Oberwolfach 报告问题汇编

What can be said about solutions near solitons, and can one describe the interaction between the soliton's modal parameters and the wave part of the solution, including blow-up along the soliton family?

来源字段：https://doi.org/10.4171/owr/2010/41

12032 | OWR-16767-007

### Soliton Resolution for Randomized Mass-Subcritical NLS

O / open | Oberwolfach 报告问题汇编

Let $2/d<p<4/d$ and $\phi\in L^2(\mathbb{R}^d)$. For almost every $\omega$, does the solution of $i\partial_tu-\Delta u=|u|^pu$, $u(0)=\phi^\omega$, admit solitons $u_j(t,x)=e^{it a_j+i x\cdot k_j}\phi_j(x-c_jt)$ and a free profile $e^{-it\Delta}\phi$ such that $\left\|u(t)-\sum_j u_j(t)-e^{-it\Delta}\phi\right\|_{L^2(\mathbb{R}^d)}\to0$ as $t\to\infty$?

来源字段：https://doi.org/10.4171/owr/2019/5

12033 | OWR-14299287-001

### Solution Concepts for Fluid-Mechanics Equations

O / open | Oberwolfach 报告问题汇编

What solution concepts are appropriate for the partial differential equations of fluid mechanics?

来源字段：https://doi.org/10.4171/owr/2025/24

12034 | AMR-019-0006

### Some Open Problems in Elasticity - A positive Jacobian bound

O / open | AMR 问题清单汇编

Prove or disprove that, under reasonable growth conditions on $W$, an energy-minimizing deformation satisfies $\det Dy^*(x)\ge\varepsilon>0$.

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12035 | AMR-019-0015

### Some Open Problems in Elasticity - Atomistic foundations

O / open | AMR 问题清单汇编

Establish the status of elasticity theory with respect to atomistic models.

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12036 | AMR-019-0010

### Some Open Problems in Elasticity - Bifurcation theory

P / partially_solved | AMR 问题清单汇编

Develop local and global bifurcation theories for nonlinear elastostatics with mixed displacement-traction boundary conditions.

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12037 | AMR-019-0018

### Some Open Problems in Elasticity - Dimension reduction

P / partially_solved | AMR 问题清单汇编

Give a rigorous derivation of models of rods, plates, and shells from three-dimensional elasticity as thickness tends to zero.

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12038 | AMR-019-0014

### Some Open Problems in Elasticity - Dynamic stability

P / partially_solved | AMR 问题清单汇编

Develop criteria for dynamic stability and instability of equilibria in nonlinear elasticity.

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12039 | AMR-019-0017

### Some Open Problems in Elasticity - Elastic-crystal free energies

O / open | AMR 问题清单汇编

For free-energy functions $\psi(A,\theta)$ of elastic crystals, determine boundary conditions under which the minimum is attained and conditions under which it is not.

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12040 | AMR-019-0001

### Some Open Problems in Elasticity - Existence of minimizers

P / partially_solved | AMR 问题清单汇编

Prove the existence of energy minimizers for elastostatics for quasiconvex stored-energy functions $W$ satisfying $W(A)\to\infty$ as $\det A\to0^+$ .

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12041 | AMR-019-0012

### Some Open Problems in Elasticity - Global dynamics

P / partially_solved | AMR 问题清单汇编

Prove global existence and uniqueness for suitable initial-boundary-value problems in dynamic nonlinear elasticity.

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12042 | AMR-019-0004

### Some Open Problems in Elasticity - Lavrentiev phenomena

P / partially_solved | AMR 问题清单汇编

Can the Lavrentiev phenomenon occur for elastostatics under growth conditions ensuring that all finite-energy deformations are continuous?

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12043 | AMR-019-0009

### Some Open Problems in Elasticity - Nonglobal local minimizers

O / open | AMR 问题清单汇编

Devise general methods for proving the existence of local but nonglobal minimizers and other weak equilibria in nonlinear elastostatics.

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12044 | AMR-019-0013

### Some Open Problems in Elasticity - Qualitative dynamics

O / open | AMR 问题清单汇编

Develop a qualitative dynamics for dynamic theories of elasticity.

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12045 | AMR-019-0016

### Some Open Problems in Elasticity - Quasiconvexification of energy wells

P / partially_solved | AMR 问题清单汇编

For the set of energy-minimizing gradients $K(\theta)$ defined in the source, determine its quasiconvex hull $K(\theta)^{qc}$ for $\theta\le\theta_c$.

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12046 | AMR-019-0003

### Some Open Problems in Elasticity - Regularity of minimizers

O / open | AMR 问题清单汇编

Determine when the minimizer $y^*$ in Theorem 2.1 of the source is smooth.

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12047 | AMR-019-0007

### Some Open Problems in Elasticity - Smooth self-contact

P / partially_solved | AMR 问题清单汇编

Justify the Ciarlet-Nečas minimization problem, or an appropriate modification, in situations involving smooth self-contact.

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12048 | AMR-019-0002

### Some Open Problems in Elasticity - Testing convexity conditions

O / open | AMR 问题清单汇编

Find useful ways of verifying polyconvexity and quasiconvexity for stored-energy functions arising in anisotropic nonlinear elasticity.

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12049 | AMR-019-0008

### Some Open Problems in Elasticity - Uniqueness of equilibrium

P / partially_solved | AMR 问题清单汇编

Prove or disprove uniqueness of sufficiently smooth equilibrium solutions for pure-displacement problems in homogeneous bodies homeomorphic to a ball when $W$ is strictly polyconvex.

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12050 | AMR-019-0011

### Some Open Problems in Elasticity - Variational fracture models

P / partially_solved | AMR 问题清单汇编

Clarify the status of models based on the fracture energy functional (2.31) in the source relative to classical fracture and nonlinear elastostatics.

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12051 | AMR-019-0005

### Some Open Problems in Elasticity - Weak Euler-Lagrange equations

P / partially_solved | AMR 问题清单汇编

Prove or disprove that, under reasonable growth conditions on $W$, energy minimizers satisfy the weak Euler-Lagrange equations.

背景链接：https://people.maths.ox.ac.uk/ball/Articles%20in%20Conference%20Proceedings%20and%20Books/JMB%202002%20re%20Marsden%2060th.pdf

12052 | OWR-11101924-008

### Spectral Approximation of Discrete-Dipole Operators

O / open | Oberwolfach 报告问题汇编

For discrete-dipole matrices $T_{\kappa h}$, characterize and prove the observed convergence as $h\to0$ of isolated nonreal eigenvalues outside the real accumulation interval $[\Lambda_-,\Lambda_+]$.

来源字段：https://doi.org/10.4171/owr/2022/43

12053 | OWR-1050-002

### Spike Counts in GM1 Reaction-Diffusion Steady States

P / partially_solved | Oberwolfach 报告问题汇编

Determine the number of spikes in steady states of the GM1 reaction-diffusion system in general.

来源字段：https://doi.org/10.4171/owr/2005/29

12054 | OWR-14298374-005

### Splitting Schemes for Rough Stochastic Heat Equations

O / open | Oberwolfach 报告问题汇编

For the fully discrete Lie-Trotter scheme applied to $\partial_tu=\partial_{xx}u+g(u)\dot W$ on $[0,1]$ with homogeneous Dirichlet data, extend positivity and mean-square convergence of order $1/4$ to rougher nonlinearities $g$, and determine a weak convergence rate.

来源字段：https://doi.org/10.4171/owr/2024/26

12055 | OWR-14298161-003

### Stability and Alternative Costs in Spherical Optimal Transport

O / open | Oberwolfach 报告问题汇编

Can stability of spherical semigeostrophic solutions be proved as the curvature $\kappa\to0$? Is there another cost $c(x,y)$ on $S^2$ for which the transport map $c\text{-}\exp_x(\nabla p_t(x))$ yields a continuity equation for the transported density that does not involve the velocity $u_t$?

来源字段：https://doi.org/10.4171/owr/2024/7

12056 | OWR-735-001

### Stability and Complexity of Helmholtz Boundary Elements

P / partially_solved | Oberwolfach 报告问题汇编

For Helmholtz Galerkin boundary-element discretizations, how do the stability threshold and error depend on the wave number and mesh width, and can the stabilization be implemented efficiently, with what computational complexity?

来源字段：https://doi.org/10.4171/owr/2004/33

12057 | OWR-11576-001

### Stability of Finite-Element Methods for Incompressible Elasticity

O / open | Oberwolfach 报告问题汇编

For two benchmark problems in two-dimensional incompressible finite elasticity, what are the stability properties of the continuum solutions and of their displacement, mixed, and mixed-enhanced finite-element discretizations?

来源字段：https://doi.org/10.4171/owr/2012/09

12058 | OWR-11101918-011

### Stability of Nonelliptic Differential Inclusions

O / open | Oberwolfach 报告问题汇编

Do there exist constants $C>0$ and $\alpha\in(0,1)$ giving the proposed stability estimate for approximate differential inclusions into the nonelliptic set $K$?

来源字段：https://doi.org/10.4171/owr/2022/37

12059 | OWR-17128-002

### Stability of Peaked Reduced Ostrovsky Waves

P / partially_solved | Oberwolfach 报告问题汇编

For the reduced Ostrovsky equation, are the peaked periodic traveling waves $$U_*(z)=\frac{3z^2-\pi^2}{18},$$ extended periodically, nonlinearly stable or unstable?

来源字段：https://doi.org/10.4171/owr/2019/32

12060 | OWR-1196-005

### Stable High-Order Regularization of Momentum Sheets

O / open | Oberwolfach 报告问题汇编

For the momentum-sheet application, find a high-order regularization of singular Green functions without temporal instability, preserve potential vorticity, integrate normal motion of straight sheets, and relate integrator symplecticity directly to optimization behavior.

来源字段：https://doi.org/10.4171/owr/2006/14

12061 | OWR-11101918-009

### Stable Wulff Shapes for Crystalline Nonlocal Energies

O / open | Oberwolfach 报告问题汇编

Classify the crystalline anisotropies whose Wulff shapes remain global minimizers of the nonlocal energy $E_\gamma$ for all sufficiently small $\gamma>0$.

来源字段：https://doi.org/10.4171/owr/2022/37

12062 | AIM-PDES-0079

### Static-class criterion for generic Hamilton-Jacobi uniqueness

P / partially_solved | AIM 研讨会问题汇编

4. (Albert Fathi) Is it true that for generic \(L\in C^\infty\) there exists only a one-parameter family of viscosity solutions of the corresponding Hamilton-Jacobi equation?

The sentence is not self-contained. The preceding item supplies the intended weak-KAM setting: \(M\) is compact; \(L:\mathbb T\times TM\to\mathbb R\) is a smooth time-periodic Lagrangian satisfying the usual Mather assumptions (strict fiberwise convexity, superlinearity, and completeness); and one works at a cohomology class \(c\). The autonomous convention \(\mathbb T=\{0\}\) is included in the same formalism. If \(\eta_c\) is a closed one-form representing \(c\), the critical equation is

\[ u_t+H(t,x,\eta_c(x)+D_xu)=\alpha_L(c) \tag{HJ}_{L,c} \]

on \(\mathbb R/\mathbb Z\times M\); in the autonomous case it is

\[ H(x,\eta_c(x)+Du)=\alpha_L(c). \]

Adding a constant to a solution produces another solution. Thus “only a one-parameter family” is recovered as **uniqueness modulo additive constants**, or equivalently uniqueness after a normalization such as \(u(z_0)=0\).

The source does not say whether “generic” means a residual subset of the full space of Tonelli Lagrangians or Mañé genericity under smooth potential perturbations, and it does not quantify over \(c\). These distinctions matter. The standard theorem recovered below uses: a fixed strong Tonelli Lagrangian \(L_0\), a fixed class \(c\), and a residual set of potentials \(f\in C^\infty(\mathbb T\times M)\) in the \(C^\infty\) Fréchet topology, applied to \(L_0-f\). No conclusion for every uncountably many \(c\) is silently inserted.

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

12063 | AIM-PDES-0049

### Stationary spherical shocks: annular existence and center compatibility

P / partially_solved | AIM 研讨会问题汇编

The record comes from the problem list *Open Problems in Multidimensional Stability of Waves and Patterns*, compiled by N. Costanzino after the AIM workshop of May 16--20, 2005. In space dimension \(d=2\) or \(3\), put \(m=d-1\), write the density as \(\rho(r,t)\), and take the velocity vector to be \(u(r,t)x/r\). The displayed barotropic system is

\[ \rho_t+(\rho u)_r+\frac{m\rho u}{r}=s(r,t), \] \[ (\rho u)_t+(\rho u^2+P(\rho))_r+\frac{m\rho u^2}{r} =\nu\left(u_r+\frac{mu}{r}\right)_r+F(r,t). \tag{1} \]

Here \(s\) and \(F\) are the mass and momentum sources. The displayed setup uses uppercase \(F\), while the question later calls the same momentum source lowercase \(f\). This report uses \(F\); it is a notation inconsistency in the source, not a mathematical distinction.

The recovered question asks whether (1) has stationary spherical viscous or inviscid shocks, how \(s\) and \(F\) must be chosen, whether either may vanish, how curvature affects stability and nearby nonspherical fronts, and whether stability can change when the radius crosses a critical value \(r_*\).

来源字段：https://aimath.org/WWN/multidimwaves/multidimwaves.pdf

12064 | OWR-1050-007

### Stationary-Curve Skeletons in Uniformly Parabolic Equations

O / open | Oberwolfach 报告问题汇编

Given a skeleton of stationary curves for a uniform parabolic PDE, determine how many and which additional stationary curves are forced to exist.

来源字段：https://doi.org/10.4171/owr/2005/29

12065 | OWR-15431-001

### Stochastic Regularization of Two-Dimensional Euler Flows

P / partially_solved | Oberwolfach 报告问题汇编

For the two-dimensional incompressible Euler equations on the torus, can transport-type stochastic noise restore uniqueness, prevent singular behavior, or otherwise improve well-posedness?

来源字段：https://doi.org/10.4171/owr/2017/23

12066 | OWR-1794-002

### Stokes Eigenfunctions with Constant Boundary Pressure

O / open | Oberwolfach 报告问题汇编

For the Stokes eigenvalue problem $-\Delta u+\nabla p=\lambda u$, $\operatorname{div}u=0$ in $\Omega$, and $u=0$ on $\partial\Omega$, does constancy of $p$ on a boundary portion imply that $u=0$ and $p$ is constant in $\Omega$? Is the disk the only domain admitting a nonzero eigenfunction with $p=0$? Establish the related unique-continuation statement for nonsmooth boundaries.

来源字段：https://doi.org/10.4171/owr/2008/13

12067 | OWR-1277-003

### Stokes Phases and Multipliers for Nonlinear PDEs

O / open | Oberwolfach 报告问题汇编

In a Stokes-phenomenon calculation for a $(1+1)$-dimensional PDE, determine the possible exponential phases, the regions where each Stokes line is active, and the multiplier switched on when an active Stokes line is crossed.

来源字段：https://doi.org/10.4171/owr/2006/28

12068 | OWR-11101918-005

### Strong Onsager Conjecture for Euler Flows

P / partially_solved | Oberwolfach 报告问题汇编

Do there exist Hölder-continuous incompressible three-dimensional Euler flows with total kinetic-energy dissipation that satisfy the local energy inequality?

来源字段：https://doi.org/10.4171/owr/2022/37

12069 | OWR-1591-003

### Strong Quantization for Fourth-Order Navier Problems

O / open | Oberwolfach 报告问题汇编

Let $u_k>0$ solve $\Delta^2u_k=\lambda_k u_k e^{2u_k^2}$ on a bounded domain $\Omega\subset\mathbb{R}^4$, with Navier conditions $u_k=\Delta u_k=0$ on $\partial\Omega$, $\lambda_k\to0$, and $\int_\Omega|\Delta u_k|^2\,dx=\lambda_k\int_\Omega u_k^2e^{2u_k^2}\,dx\to\Lambda>0$. If $\Lambda=L\Lambda_1$ and the sequence has $I$ blow-up points, must $L=I$; equivalently, is every blow-up point simple?

来源字段：https://doi.org/10.4171/owr/2007/35

12070 | AIM-PDES-0061

### Strong viscous shocks and an amplitude-independent scalar transverse certificate

P / partially_solved | AIM 研讨会问题汇编

**Open Problem.** Determine the stability properties of strong shocks for viscous conservation laws in one or several space dimensions.

来源字段：https://aimath.org/WWN/multidimwaves/multidimwaves.pdf

12071 | OWR-14612-003

### Structure-Preserving Variational Closures of Kinetic Equations

O / open | Oberwolfach 报告问题汇编

For a kinetic equation reduced by a closure $f^{\star}(W(x,t),c)$, what test functions $\phi_{\alpha}(x,t,c)$ should be used in the variational formulation so that the reduced equations retain the physical conservation laws, and what can be said about the resulting evolution equations for $W(x,t)$, in particular whether they are symmetric hyperbolic balance laws with an entropy?

来源字段：https://doi.org/10.4171/owr/2016/30

12072 | OWR-3476-007

### Sublinear Lane-Emden Ground-State Asymptotics

P / partially_solved | Oberwolfach 报告问题汇编

For the Lane-Emden problem $-\Delta_pu=\lambda|u|^{q-2}u$ in a bounded domain with $u=0$ on the boundary, do the asymptotic results for ground states and least-energy nodal solutions as $q\to p$ remain valid when $1<q<p$?

来源字段：https://doi.org/10.4171/owr/2009/26

12073 | OWR-2654836-003

### Support of Renormalized Singular Stochastic Equations

P / partially_solved | Oberwolfach 报告问题汇编

For a singular SPDE with renormalisation, can the topological support of the solution law $\mathcal{L}_T(u)$ be characterized in terms of solutions to the associated renormalised controlled equations?

来源字段：https://doi.org/10.4171/owr/2020/40

12074 | AIM-PDES-0092

### Tensorization of white-noise hyperbolic minimizers

P / partially_solved | AIM 研讨会问题汇编

\[ L(x,v,t)=L_0(x,v)+\sum_{i=1}^{N}F_i(x)\dot w_i(t) \quad\text{on }\mathbb T^n\times\mathbb R^n, \tag{1} \]

where the \(w_i\) are independent Brownian motions. It assumes that a forcing map is an embedding, states that almost surely there is a unique global minimizer, and conjectures that this minimizer is a hyperbolic trajectory of the Lagrangian flow. It notes that E, Khanin, Mazel, and Sinai proved the assertion for \(n=1\).

1. The PDF prints \(F=(F_1,\ldots,F_n):\mathbb T^n\to\mathbb R^N\), although (1) contains \(N\) functions. The type-correct and standard reading is \[ F=(F_1,\ldots,F_N):\mathbb T^n\longrightarrow\mathbb R^N. \tag{2} \] This is the embedding condition used in the later random Hamilton--Jacobi literature. 2. It prints \(\gamma:\mathbb R\to M\), but no \(M\) is introduced. The configuration space in the problem is \(\mathbb T^n\), so the intended codomain is \(\mathbb T^n\).

The words “positive definite” are understood in the usual Tonelli sense of positive-definite velocity Hessian, not as pointwise positivity of the value of \(L_0\).

Because \(\dot w_i\) is distribution-valued, (1) is not an ordinary time-dependent function. For an absolutely continuous competitor \(\gamma\) on \([s,t]\), the stochastic action is read pathwise as

\[ A^\omega_{s,t}(\gamma) =\int_s^t L_0(\gamma,\dot\gamma)\,du +\sum_i\int_s^t F_i(\gamma(u))\,dw_i(u), \tag{3} \]

with the last integral defined by integration by parts:

\[ \int_s^t F_i(\gamma)\,dw_i =F_i(\gamma(t))w_i(t)-F_i(\gamma(s))w_i(s) -\int_s^t w_i(u)\,d(F_i\circ\gamma)(u). \tag{4} \]

This is pathwise well defined because \(F_i\circ\gamma\) is absolutely continuous. A two-sided global minimizer is a curve every compact segment of which minimizes (3) among curves with the same endpoints.

Here “hyperbolic” is taken in the random-dynamical sense used in the later work: the derivative cocycle along the global minimizer has no zero Lyapunov exponent, equivalently in this symplectic setting it has \(n\) stable and \(n\) unstable directions with exponential rates in the tempered/nonuniform sense. Where uniform exponential-dichotomy estimates are available, the product result below preserves those as well.

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

12075 | AIM-PDES-0060

## The differentiated Benjamin--Ono spectral problem

P / partially_solved | AIM 研讨会问题汇编

**Setup.** Consider the Benjamin--Ono equation \[ u_t+u u_x-Ku_x=0, \qquad \widehat{Ku}(k)=|k|\widehat u(k). \] This equation supports travelling waves of the form \[ \phi(z)=\frac{4}{1+z^2}. \] The resulting eigenvalue problem is \[ (cv+Kv-\phi v)_z=\lambda v. \tag{A} \]

**Open Problem.** What is the spectrum of (A) and how does it relate to the spectrum of \[ cv+Kv-\phi v=\lambda v, \] which is well known. A possible approach would be to treat the eigenvalue equation as a two dimensional problem.

The PDF confirms the coefficient \(4/(1+z^2)\), the multiplier \(|k|\), and the derivative outside the full parenthesis in (A). The K. Zumbrun material on strong viscous shocks starts the next presenter and is not part of this problem.

It is useful to restore the speed parameter:

\[ \phi_c(x)=\frac{4c}{1+c^2x^2},\qquad c>0. \]

At \(c=1\) this is the profile printed in the AIM source. Throughout this report \(D=|D|\) denotes the Fourier multiplier \(|\xi|\), and

\[ L_c=D+c-\phi_c,\qquad A_c=\partial_xL_c. \tag{1.1} \]

The spectral parameter for \(A_c\) is denoted by \(\lambda\); a selfadjoint spectral parameter for \(L_c\) is denoted by \(\mu\). This prevents the two different eigenvalue problems from being conflated.

来源字段：https://aimath.org/WWN/multidimwaves/multidimwaves.pdf

12076 | OWR-14299519-006

## The Hölder Threshold for Isometric-Immersion Rigidity

O / open | Oberwolfach 报告问题汇编

Does there exist a critical exponent $\alpha_0$ such that $C^{1,\alpha}$ isometric immersions are rigid for $\alpha>\alpha_0$, while the Nash-Kuiper flexibility theorem extends to $C^{1,\alpha}$ for $\alpha<\alpha_0$?

来源字段：https://doi.org/10.4171/owr/2025/33

12077 | AIM-PDES-0028

## The refuted C1 conjecture and radial rigidity

S / solved | AIM 研讨会问题汇编

A function \(u:\Omega\to\mathbb R\) satisfies the special Lagrangian equation if \[ \sum_i\arctan(\lambda_i)=c, \] where \(c\) is constant and the \(\lambda_i\) are the eigenvalues of \(D^2u\). (Conjecture of Nadirashvili--Vlăduț) Any viscosity solution on the unit ball is \(C^1(B)\) for sufficiently smooth boundary data.

The original paper states the conjecture as follows: any *Harvey--Lawson solution* of \(\mathrm{SLE}_\theta\) on a ball \(B\) lies in \(C^1(B)\), provided the boundary value \(\varphi\) is sufficiently smooth [NV10, p. 1180]. The original statement does not specify a Hölder/Sobolev threshold for “sufficiently smooth.” It is a branchwise Dirichlet statement; the middle branch in dimension three was the motivating nonconcave case.

For the analysis below, the recovered equation is \[ F(D^2u):=\operatorname{tr}\arctan(D^2u) =\sum_{j=1}^n\arctan\lambda_j(D^2u)=c, \qquad c\in(-n\pi/2,n\pi/2), \tag{SLE} \] with every arctangent on its principal branch. “\(C^1(B)\)” is read as interior \(C^1\) regularity.

来源字段：http://aimpl.org/nonlinpdegeom/1/

12078 | PDE-002

## The Regularity Problem for Euler Equations

P / partially_solved | 独立 / 未分配集合

Do solutions to the 3D Euler equations for incompressible fluid flow remain smooth for all time, given smooth initial data?

背景链接：https://doi.org/10.1137/23M1580395

12079 | AIM-PDES-0041

### The solved J-equation criterion and a sharp chamber on the one-point blow-up

P / partially_solved | AIM 研讨会问题汇编

The displayed mathematics agrees with Lejmi--Székelyhidi's original formulation.

The recovered statement is the following. Let $M$ be a compact Kähler manifold of complex dimension $n$, let $\omega,\chi$ be Kähler forms, and set

$$ \omega_\varphi=\omega+\sqrt{-1}\,\partial\bar\partial\varphi>0, \qquad c=n\frac{[\omega]^{n-1}\cdot[\chi]}{[\omega]^n}. $$

Determine whether

$$ \operatorname{tr}_{\omega_\varphi}\chi=c $$

has a smooth solution exactly when, for every irreducible proper analytic subvariety $V\subset M$ of complex dimension $p$, where $1\le p\le n-1$,

$$ \int_V\left(c\,\omega^p-p\,\omega^{p-1}\wedge\chi\right)>0. \tag{LS} $$

For singular $V$, the integral means pairing the closed form with the integration current $[V]$, equivalently integration after resolution. It depends only on the Kähler classes. Reducible subvarieties add no conditions beyond their irreducible components. The source phrase “all proper subvarieties” is conventionally restricted to positive dimensions $1,\ldots,n-1$; zero-dimensional sets do not supply a J-positivity obstruction.

来源字段：http://aimpl.org/nonlinpdegeom/2/

12080 | OWR-4806-001

### Thermodynamic Reservoirs in Grid-Based Reduced Models

O / open | Oberwolfach 报告问题汇编

In extending thermostat-based reduced modeling to grid-based PDE methods, can the thermodynamic reservoir be taken to be the unresolved unknowns on a fine grid, with the reduced model represented by a subset of those unknowns? Can the full trajectory then be approximated weakly by the resulting reduced dynamics?

来源字段：https://doi.org/10.4171/owr/2011/16

12081 | OWR-11101918-008

### Thomas Yau Singularity Picture for Lagrangian Mean Curvature Flow

P / partially_solved | Oberwolfach 报告问题汇编

For generic initial Lagrangian surfaces, are the only tangent flows at singularities two multiplicity-one planes meeting either at a point or along a line, as suggested by the Thomas-Yau singularity picture?

来源字段：https://doi.org/10.4171/owr/2022/37

12082 | OWR-1972-003

### Three-Dimensional Navier-Stokes Regularity

O / open | Oberwolfach 报告问题汇编

For the three-dimensional incompressible Navier-Stokes equations with given initial data, do global-in-time solutions exist, and are they unique and regular?

来源字段：https://doi.org/10.4171/owr/2008/27

12083 | OWR-1703869-002

### Three-Dimensional Nucleation Structures and Stochastic Microstructure Models

O / open | Oberwolfach 报告问题汇编

Can analogous nucleation structures be produced in three dimensions, and how do such structures scale with a prescribed disclination angle? Can stochastic partial differential equations describing the resulting microstructure be derived?

来源字段：https://doi.org/10.4171/owr/2020/22

12084 | OWR-11101918-013

### Threshold Regularity for the Eikonal Equation

P / partially_solved | Oberwolfach 报告问题汇编

At what Besov regularity threshold do divergence-free solutions $m:\Omega\to S^1$ of the eikonal equation become rigid, and is the threshold exactly $B^{1/3}_{3,\mathrm{loc},\infty}$?

来源字段：https://doi.org/10.4171/owr/2022/37

12085 | OWR-14742-001

### Thresholds and Traces for Supercritical Lane-Emden Blowup

O / open | Oberwolfach 报告问题汇编

For the supercritical Lane-Emden heat flow $u_t-\Delta u=|u|^{p-2}u$ on a smooth bounded domain $\Omega\subset\mathbb R^n$, $p>2^*=2n/(n-2)$, can the complete-blow-up threshold be chosen as $M=\alpha(n-2-\alpha)$, $\alpha=2/(p-2)$, for sufficiently small $p>2^*$; and does every finite-time blow-up solution with bounded energy have a trace $u(T)\in L^{p,\lambda}_0(\Omega)$?

来源字段：https://doi.org/10.4171/owr/2016/34

12086 | OWR-14298011-007

### Time-Varying Traffic Control and Lyapunov Learning

P / partially_solved | Oberwolfach 报告问题汇编

For the coupled traffic PDE-ODE system with an automated-vehicle control, prove existence for time-varying controls and solve the stabilization problem. Develop reliable neural methods for constructing Lyapunov functions for nonlinear systems.

来源字段：https://doi.org/10.4171/owr/2023/52

12087 | AIM-PDES-0055

### Toda one-soliton spectrum, sharp weight threshold, and Floquet convention audit

S / solved | AIM 研讨会问题汇编

**R. Pego (Stability of Toda Lattice Solitons).** The Toda lattice \[ \ddot q_k=\exp(q_{k+1}-q_k)-\exp(q_k-q_{k-1}) \] has the well-known solutions \[ q_k(t)=\log\!\left( \frac{\cosh(\beta(k-ct+1))}{\cosh(\beta(k-ct))} \right), \qquad c=\frac{\sinh\beta}{\beta}. \] What are their spectral stability properties?

来源字段：https://aimath.org/WWN/multidimwaves/multidimwaves.pdf

12088 | OWR-1703869-004

### Topology and Morse Theory of Allen-Cahn Moduli Spaces

O / open | Oberwolfach 报告问题汇编

For the moduli space $M_{2m}$ of $2m$-ended solutions of the planar Allen-Cahn equation $\Delta u+u-u^3=0$, is $M_{2m}$ connected, what is its dimension, and what is the Morse index of its solutions?

来源字段：https://doi.org/10.4171/owr/2020/22

12089 | OWR-1050-015

### Topology of Yamabe Asymptotic Sets

O / open | Oberwolfach 报告问题汇编

Determine the topology of $S=\{u\in\mathbb R^p\setminus\{0\}:(Au,u)\leq0\}$ arising from the sign-changing Yamabe asymptotes.

来源字段：https://doi.org/10.4171/owr/2005/29

12090 | AIM-PDES-0035

### Two quantitative gates for toric Chen-Weber compactness

P / partially_solved | AIM 研讨会问题汇编

Find a toric proof of the toric case of the orbifold compactness theorem proved by Chen-Weber.

来源字段：http://aimpl.org/nonlinpdegeom/2/

12091 | OWR-14298587-018

### Two-Dimensionality of Low-Energy Abelian Higgs Fields

P / partially_solved | Oberwolfach 报告问题汇编

For the abelian Higgs energy in dimension $n\geq 5$, prove that every critical pair in the low-energy, nearly flat regime is gauge-equivalent to a two-dimensional solution.

来源字段：https://doi.org/10.4171/owr/2024/37

12092 | AIM-PDES-0088

### Two-scale fate of minimax periodic orbits near an irrational cantorus

P / partially_solved | AIM 研讨会问题汇编

(Walter Craig) This question concerns the minimax Birkhoff orbits in an area-preserving twist map of an annulus. It is well known that (i) there exist at least two periodic orbits with a given rational rotation number \(\alpha=p/q\), namely a minimal orbit and a minimax orbit; (ii) as \(\alpha_j\to\omega\in\mathbb R\setminus\mathbb Q\), the minimal Birkhoff orbits converge to the Mather set \(\Pi_\omega\); and (iii) if \(\Pi_\omega\) is an invariant circle, the minimax orbits also converge to \(\Pi_\omega\). What is the fate of the minimax orbits when the Mather set is a Cantor set?

来源字段：https://aimath.org/WWN/dynpde/dynpde.pdf

12093 | OWR-12874-003

### Unconditional Uniqueness for Critical Equivariant Wave Maps

O / open | Oberwolfach 报告问题汇编

Let $M^n$ and $N^\ell$ be rotationally symmetric manifolds with metrics $dr^2+h(r)^2d\omega^2$ and $d\phi^2+g(\phi)^2d\chi^2$, where $n\geq 3$, $k\geq 1$, and $\ell=k(k+n-2)$. For an admissible $M$, consider the equivariant wave-map equation

$$\phi_{tt}-\phi_{rr}-(n-1)\frac{h'(r)}{h(r)}\phi_r+\frac{\ell}{h(r)^2}g(\phi)g'(\phi)=0.$$

Does its Cauchy problem have unconditional uniqueness in the critical space $H^{n/2}\times H^{n/2-1}$, without imposing an additional Strichartz-space condition?

来源字段：https://doi.org/10.4171/owr/2014/14

12094 | OWR-7155445-001

### Unified Analysis of Higher-Dimensional Evolution Equations

O / open | Oberwolfach 报告问题汇编

Can there be a unifying theory for the analytical and numerical treatment of evolution equations on higher-dimensional domains?

来源字段：https://doi.org/10.4171/owr/2021/24

12095 | AIM-PDES-0031

### Unified radial C2 criterion and thin-shell obstruction

P / partially_solved | AIM 研讨会问题汇编

Find conditions that allow for \(C^2\) regularity for degenerate Monge-Ampère equation (both real and complex).

来源字段：http://aimpl.org/nonlinpdegeom/1/

12096 | OWR-8415340-004

### Uniform Low-Regularity Schemes for Coupled Klein-Gordon Systems

P / partially_solved | Oberwolfach 报告问题汇编

Extend uniformly accurate low-regularity numerical approximations from the Klein-Gordon equation to systems such as coupled Klein-Gordon equations and Klein-Gordon-Schrödinger systems.

来源字段：https://doi.org/10.4171/owr/2021/37

12097 | OWR-16157-001

### Uniform Minimal Control Time for Advection-Diffusion

O / open | Oberwolfach 报告问题汇编

Fix $L,T>0$ and $M\neq0$. For $0<\varepsilon\ll1$, consider

$$\partial_t y^\varepsilon-\varepsilon\partial_{xx}y^\varepsilon+M\partial_xy^\varepsilon=0$$

on $(0,L)\times(0,T)$, with $y^\varepsilon(0,t)=v^\varepsilon(t)$, $y^\varepsilon(L,t)=0$, and $y^\varepsilon(\cdot,0)=y_0^\varepsilon\in H^{-1}(0,L)$. Determine the minimal time $T_M$ for which null controls $v^\varepsilon\in L^2(0,T)$ can be chosen with cost uniformly bounded as $\varepsilon\to0$.

来源字段：https://doi.org/10.4171/owr/2018/16

12098 | OWR-14609-007

### Unique Nonequilibrium Invariant Measures for Resonant NLS

O / open | Oberwolfach 报告问题汇编

For the nonequilibrium resonant NLS toy model coupled to heat baths with $T_1<T_n$, does there exist a unique smooth ergodic nonequilibrium invariant measure?

来源字段：https://doi.org/10.4171/owr/2016/27

12099 | OWR-14299093-004

### Uniqueness and Fundamental Solutions for Fractional Diffusion

O / open | Oberwolfach 报告问题汇编

Let $M$ be a complete connected noncompact $N$-dimensional Riemannian manifold with $\operatorname{Ric}(M)\geq-(N-1)k$ and $\lambda_1(\Omega)\geq c\,m(\Omega)^{-2/N}$ for every relatively compact open $\Omega$. For $u_t=-(-\Delta_M)^s(u^m)$, where $s\in(0,1)$ and $m>1$, prove uniqueness of weak dual solutions, construct fundamental solutions with Dirac-delta initial data, and establish pointwise bounds for general solutions.

来源字段：https://doi.org/10.4171/owr/2025/14

12100 | OWR-11453-003

### Uniqueness for Periodic Semigeostrophic Systems

O / open | Oberwolfach 报告问题汇编

Are distributional solutions unique for the two-dimensional periodic semi-geostrophic system and for its dual system?

来源字段：https://doi.org/10.4171/owr/2011/55

12101 | OWR-701-001

### Uniqueness for Spatially Dependent Two-Phase Diffusion

P / partially_solved | Oberwolfach 报告问题汇编

Can Kruzkov-type techniques prove uniqueness, without strong regularity assumptions, for multidimensional two-phase-flow equations whose diffusion function depends on the spatial variables?

来源字段：https://doi.org/10.4171/owr/2004/18

12102 | OWR-13856-004

### Uniqueness of Condensation-Model Parameters

O / open | Oberwolfach 报告问题汇编

别名：Expository Metastability Heading for a Nonlocal Dirichlet Problem [OWR-13856-003]

Fix $\beta>1$, let $m_\beta>0$ solve $m_\beta=\tanh(\beta m_\beta)$, and put $m^*=\sqrt{1-\beta^{-1}}$. For $u_+\in(m^*,m_\beta)$, consider a profile $q:\mathbb R_{\leq0}\to\mathbb R$ satisfying

$$q(x)=\tanh\!\left(\beta(J*q)(x)+\beta h_*\right),\qquad \lim_{x\to-\infty}q(x)=m_-,\qquad q(0)=u_+.$$

Are the parameters $h_*$ and $m_-$ uniquely determined by $u_+$?

来源字段：https://doi.org/10.4171/owr/2015/35

12103 | OWR-15961-001

### Uniqueness of Higher-Dimensional Critical Minimizers

O / open | Oberwolfach 报告问题汇编

In spatial dimension $N>1$, for the fair-competition aggregation-diffusion model at critical interaction strength $\chi=\chi_c$, is the global minimizer of the free-energy functional unique up to dilations? In the subcritical rescaled problem, are stationary states global minimizers and are those minimizers unique?

来源字段：https://doi.org/10.4171/owr/2017/56

12104 | OWR-16936-001

### Uniqueness of Rough Continuity Equations

P / partially_solved | Oberwolfach 报告问题汇编

For the continuity equation $\partial_tu+\operatorname{div}(bu)=0$ in $d$ dimensions, with bounded divergence-free $b\in W^{1,p}$ and $u\in L^q$, are distributional solutions unique and Lagrangian throughout the regime $1<1/p+1/q<1+1/d$?

来源字段：https://doi.org/10.4171/owr/2019/24

12105 | OWR-11788-008

### Uniqueness of Tangent Flows in Mean Curvature Flow

P / partially_solved | Oberwolfach 报告问题汇编

Do mean-curvature-flow singularities have unique tangent flows, and more generally can the full asymptotic-symmetry conjecture for finite-time geometric PDE singularities be proved?

来源字段：https://doi.org/10.4171/owr/2012/26

12106 | OWR-16410-007

### Uniqueness of Tangents for Multivalued Energy Minimizers

P / partially_solved | Oberwolfach 报告问题汇编

Let $u$ be a Dirichlet-energy-minimizing $q$-valued function and let $Y$ be a singular point. Is the tangent function obtained by blow-up at $Y$ independent of the sequence of scales $\rho_j\to0^+$?

来源字段：https://doi.org/10.4171/owr/2018/35

12107 | OWR-14298164-001

### Vanishing Viscosity and Propagating Terraces

O / open | Oberwolfach 报告问题汇编

For the $n$-tube incompressible porous-media model, establish well-posedness when $\nu=0$ and characterize the vanishing-viscosity limit as $\nu\to 0$. For the two-tube model, prove existence and study stability of the propagating terrace for every tube separation $l>0$.

来源字段：https://doi.org/10.4171/owr/2024/10

12108 | OWR-13750332-005

### Vanishing-Viscosity Limits for Transport Equations

O / open | Oberwolfach 报告问题汇编

Let $u_\varepsilon$ be the unique weak solution of the viscosity-regularized transport equation $$-\varepsilon\Delta u_\varepsilon+Hu_\varepsilon+\alpha u_\varepsilon=\langle f,\xi^m\rangle.$$ Does $u_\varepsilon$ converge as $\varepsilon\to0$ in a suitable topology, and if so, does its limit solve the corresponding unregularized transport equation?

来源字段：https://doi.org/10.4171/owr/2023/21

12109 | OWR-11578-001

### Variational Eddy Currents with Degenerate Conductivity

O / open | Oberwolfach 报告问题汇编

For general nonnegative $\sigma\in L^\infty(\mathbb R^3)$, can the mapping $A\mapsto\phi_A$ be defined in the variational formulation of the eddy-current equation?

来源字段：https://doi.org/10.4171/owr/2012/11

12110 | OWR-14297735-008

### Vectorial Allen-Cahn Limits Without Energy Convergence

O / open | Oberwolfach 报告问题汇编

Prove convergence of the vectorial Allen-Cahn equation to a De Giorgi solution of multiphase mean-curvature flow without assuming convergence of the energies.

来源字段：https://doi.org/10.4171/owr/2023/32

12111 | AMR-023-0014

### Vlasov-Maxwell regularity

P / partially_solved | AMR 问题清单汇编

Establish global regularity, or exhibit breakdown, for solutions of the Vlasov-Maxwell equations from appropriate smooth initial data.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767

12112 | OWR-4208-001

### Weak Solutions of Zero-Mobility Diffuse and Sharp Interfaces

O / open | Oberwolfach 报告问题汇编

For the diffuse-interface system, prove existence of weak solutions when the interface thickness $\varepsilon>0$ and mobility $m=0$; and for the limiting sharp-interface system, prove existence of global-in-time weak solutions when the limiting mobility $m_0=0$.

来源字段：https://doi.org/10.4171/owr/2010/07

12113 | OWR-16418-001

### Weak Tumor-Invasion Solutions Without Growth Compatibility

O / open | Oberwolfach 报告问题汇编

Let $n_1,n_2$ satisfy

$$\partial_tn_1-\operatorname{div}(n_1\nabla p)=n_1F_1(p)+n_2G_1(p),$$
$$\partial_tn_2-\operatorname{div}(n_2\nabla p)=n_1F_2(p)+n_2G_2(p)$$

on $\mathbb R^d$ for $t\geq0$, where $n=n_1+n_2$, $p=n^\gamma$, $\gamma>1$, $F=F_1+F_2$, $G=G_1+G_2$, and $F(p),G(p)\leq0$ for $p\geq P_H$. Can weak solutions be constructed without assuming

$$\sup_{0\leq p\leq P_H}\frac{\lvert F(p)-G(p)\rvert^2}{p^{1/\gamma}}\leq C_H$$

Can weak solutions be constructed when no such uniform bound is imposed?

来源字段：https://doi.org/10.4171/owr/2018/43

12114 | AIM-PDES-0040

### Weak-closure reduction for the maximal-rank counterexample problem

P / partially_solved | AIM 研讨会问题汇编

Find a counterexample to the maximal rank conjecture.

**Conjecture (Maximal Rank Conjecture).** Let $A\subseteq\mathbb C$ be the annulus $$ A=\{1<|w|<e\mid w\in\mathbb C\}. $$ Given an $n$-dimensional compact Kähler manifold $(M,\omega)$, and potentials $\varphi _0,\varphi _1$, the geodesic on $M\times A$ is obtained as the weak limit of solutions to $$ \begin{cases} (\pi^*\omega+i\partial\bar\partial\Phi_\delta)^{n+1} =\delta(\pi^*\omega+i\,dw\wedge d\bar w)^{n+1},\\ \Phi_\delta|_{M\times\{0\}}=\varphi _0,\\ \Phi_\delta|_{M\times\{1\}}=\varphi _1. \end{cases} $$ Then $$ (\pi^*\omega+i\partial\bar\partial\Phi_\delta)|_{M\times\{\mathrm{pt.}\}} \geq \varepsilon\omega $$ for some $\varepsilon>0$ independent of $\delta$.

来源字段：http://aimpl.org/nonlinpdegeom/2/

12115 | OWR-823-001

### Weak-Solution Uniqueness for Coupled Thermomechanical Systems

O / open | Oberwolfach 报告问题汇编

Are weak solutions unique for the coupled thermomechanical model of steel heat treatment comprising quasistatic force balance, phase-fraction evolution, transformation strain, and an energy balance with a quadratic stress source?

来源字段：https://doi.org/10.4171/owr/2005/18

12116 | AIM-PDES-0007

### Weighted convection cancellation for topographic geostrophic profiles

P / partially_solved | AIM 研讨会问题汇编

**The QG equation with Ekman layer with topography**

It gives no equation, domain, quantifiers, remarks, or bibliography. Consequently there is no uniquely verified mathematical statement to solve. In particular, “QG” may mean the three-dimensional quasi-geostrophic system with Ekman pumping, surface QG, a barotropic potential-vorticity model, or a quasi-geostrophic limit of a rotating primitive model. These are not interchangeable.

The following is therefore a **closely matching modern formulation**, not a claimed transcription of the intended AIM problem. Chemin, Fanelli, and Gallagher study a rapidly rotating incompressible fluid in

$$ \Omega_H=\{(x_h,z):x_h\in O,\ -H(x_h)<z<0\}, \qquad H(x_h)=\varphi(\rho(x_h)), $$

where $\rho$ is the distance from a smooth convex island (after a shore cutoff), $|\nabla_h\rho|=1$, and the isobaths are parallel to the coast. Their Navier--Stokes--Coriolis system is

$$ \partial_tu_\varepsilon-\varepsilon\beta\Delta u_\varepsilon +\frac1\varepsilon e_3\times u_\varepsilon +(u_\varepsilon\cdot\nabla)u_\varepsilon =-\frac1\varepsilon\nabla p_\varepsilon, \qquad \nabla\cdot u_\varepsilon=0, \qquad u_\varepsilon|_{\partial\Omega_H}=0. \tag{NSC} $$

Here $\beta>0$ is fixed and the Rossby number $\varepsilon\downarrow0$. This model has precisely the three ingredients visible in the AIM title-geostrophic balance, Ekman boundary layers, and non-flat topography-but the available evidence does not prove that it is the intended model.

Sign convention: throughout this report,

$$ \nabla^\perp h=(-\partial_2h,\partial_1h), \qquad e_3\times(u_1,u_2,0)=(-u_2,u_1,0). $$

Reversing the orientation changes both tangential directions but not the orthogonality lemma or the positive damping rate below.

来源字段：http://aimpl.org/smallscalefluid/7/

12117 | OWR-14298011-003

### Well-Defined Trajectories in Controlled Set Motions

O / open | Oberwolfach 报告问题汇编

For an optimal planar set motion $t\mapsto\Omega(t)$ minimizing terminal area under a bounded boundary-control effort, what regularity do the sets $\Omega(t)$ have, and can trajectories orthogonal to $\partial\Omega(t)$ always be defined?

来源字段：https://doi.org/10.4171/owr/2023/52

12118 | OWR-15431-002

### Well-Posedness Beyond Known Noisy Euler Regimes

O / open | Oberwolfach 报告问题汇编

For the stochastic two-dimensional Euler vorticity equation on the torus, can existence be extended below the treated $L^p$ classes, uniqueness from $L^\infty$ to finite $L^p$, and noncollapse of the stochastic flow be proved?

来源字段：https://doi.org/10.4171/owr/2017/23

12119 | AIM-PDES-0043

### Zero-distance conjugate Kähler-Einstein manifolds need not be deformation equivalent

S / solved | AIM 研讨会问题汇编

Let \((M,\omega)\) and \((M',\omega')\) be compact Kähler-Einstein manifolds which are \(\varepsilon\)-close in Gromov-Hausdorff distance. Is there a holomorphic family \(M_t\) with \(M_0=M\) and \(M_1=M'\)?

来源字段：http://aimpl.org/nonlinpdegeom/2/

# 资料汇编 09 | 动力系统 / Dynamical Systems

888 条记录。O=开放；P=部分解决；S=已解决。状态与英文/LaTeX 表述均沿用来源。

12120 | OWR-14298367-006

### $\Gamma$-Supercyclicity of Composition Operators and Weighted Shifts

S / solved | Oberwolfach 报告问题汇编

Let $T_f$ be a dissipative composition operator of bounded distortion and let $B_w$ be its associated weighted shift. For which subsets $\Gamma\subseteq\mathbb C$ is $T_f$ $\Gamma$-supercyclic if and only if $B_w$ is $\Gamma$-supercyclic? In particular, does this equivalence hold for every $\Gamma\subseteq\mathbb C$?

来源字段：https://doi.org/10.4171/owr/2024/19

12121 | OWR-8415336-019

### $\gamma$-Topology Limits of Hamiltonians with Singular Sets

P / partially_solved | Oberwolfach 报告问题汇编

Let $H$ be a continuous Hamiltonian on $M\setminus V$, approximated uniformly on compact subsets by smooth Hamiltonians $H_k$ with time-one maps $\phi_k$. Does $\phi_k$ converge in the $\gamma$-topology to an element of the Humilière completion of $\operatorname{Ham}(M,\omega)$?

来源字段：https://doi.org/10.4171/owr/2021/33

12122 | OWR-12007-002

### $C^0$ Hamiltonian Dynamics and Continuous Symplectic Structures

O / open | Oberwolfach 报告问题汇编

Can one define $C^0$ counterparts of the classical smooth symplectic objects, in particular a meaningful $C^0$ Hamiltonian dynamics?

来源字段：https://doi.org/10.4171/owr/2012/35

12123 | AMR-108-0045

### 6.1 (I. Kapovitch) - Random walks and generic pseudo-Anosov singularities

S / solved | AMR 问题清单汇编

Show that a random walk on the mapping class group gives a pseudo-Anosov element whose invariant foliations have generic trivalent singularities with probability tending to $1$ as the walk length tends to infinity.

背景链接：https://arxiv.org/pdf/1512.04620#page=5

12124 | AMR-108-0048

### 6.4 (Maher) - Generic mapping-class orbit points in Teichmüller balls

S / solved | AMR 问题清单汇编

For the orbit of a point $x$ in Teichmüller space under the mapping class group, show that as $r\to\infty$: (1) the proportion of orbit points in the Teichmüller ball of radius $r$ that are pseudo-Anosov with invariant foliations having generic trivalent singularities tends to $1$; and (2) the proportion that yield hyperbolic manifolds when used as Heegaard-splitting gluing maps tends to $1$.

背景链接：https://arxiv.org/pdf/1512.04620#page=5

12125 | AMR-046-0012

### A class of Hamiltonian systems

O / open | AMR 问题清单汇编

Consider a Hamiltonian system with a center at the origin and Hamiltonian $$H(x,y)=H_{2n}(x,y)+H_m(x,y),\qquad m>2n,$$ where $H_{2n}$ and $H_m$ are homogeneous polynomials of degrees $2n$ and $m$. Does the period annulus of the origin have at most one critical period?

背景链接：https://arxiv.org/abs/2012.02524

12126 | AIM-DYNAMICAL_SYSTEMS-0065

### A covariance-profile criterion for moving-billiard functional limit laws

P / partially_solved | AIM 研讨会问题汇编

**Problem 22.** What kind of limit theorems can be found for random billards with moving/deforming scatterers?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12127 | AIM-DYNAMICAL_SYSTEMS-0037

### A cyclotomic ramification clock in a regular arboreal tower

P / partially_solved | AIM 研讨会问题汇编

Say something interesting about wild ramification for arboreal representations. e.g. study Galois representations over local fields of residue characteristic dividing degree of \(f\) (or even less than or equal to the degree). e.g. An arboreal version of Sen's theorem.

来源字段：http://aimpl.org/galarithdyn/12/

12128 | AIM-DYNAMICAL_SYSTEMS-0027

### A cyclotomic--Chebyshev obstruction

S / solved | AIM 研讨会问题汇编

If \(K_fK_g\) is finite over both \(K_f\) and \(K_g\), are \(f\) and \(g\) conjugate, an iterate, or conjugate to something that commutes?

来源字段：http://aimpl.org/galarithdyn/7/

12129 | AIM-DYNAMICAL_SYSTEMS-0021

### A degree-prime dichotomy for critical portrait level structures

P / partially_solved | AIM 研讨会问题汇编

1. Analogy between dynatomic and modular curves (See Silverman, ADS Section 4.1 and 4.2 for dynatomic polynomials). 2. What are other proxies for level structure, e.g. marked critical points, for which bad primes are more natural?

来源字段：http://aimpl.org/galarithdyn/4/

12130 | AIM-DYNAMICAL_SYSTEMS-0092

### A flexible PCF family on P^2 beyond symmetric products

S / solved | AIM 研讨会问题汇编

**Flexible Family of PCF maps.** Is there a flexible family of PCF maps that does not come from \(\mathbb P^1\)?

来源字段：http://aimpl.org/finitedynamics/5/

12131 | AIM-DYNAMICAL_SYSTEMS-0034

### A full toric Kummer image on projective space with an exact cyclotomic correlation

P / partially_solved | AIM 研讨会问题汇编

Generalizing: Can you find an example in higher dimensions?

Find classes of examples of polynomials or rational maps \(d\geq3\) for which we can compute explicitly the arboreal Galois representation up to finite index over various fields.

This sentence is not self-contained.

It also suggests trying Stoll's techniques for \(x^p+c\). The archived section has no further introduction, so “generalizing” must at least carry forward the request for computable arboreal images. Since Problem 10.1 already varies the **degree** of a map of \(\mathbb P^1\), “higher dimensions” most naturally means a phase space of dimension greater than one. This is consistent with the official workshop report, where Joseph Silverman describes maps of \(\mathbb P^2\) as candidates for higher-dimensional arboreal representations.

Several choices nevertheless remain unstated:

* the phase space could be \(\mathbb P^r\), a product, an abelian variety, or a general quasiprojective variety; * a “preimage tree” is still a one-dimensional rooted graph, even when the phase space has dimension \(r>1\); * for an endomorphism of \(\mathbb P^r\), its algebraic degree and generic number of preimages differ: degree \(d\) gives topological degree \(d^r\); * “the expected group” could mean the full automorphism group of the regular preimage tree or the smaller group forced by algebraic structure.

We preserve these distinctions and give an honest endomorphism of \(\mathbb P^r\), not merely a self-map written on a Cartesian product. Its image is computed exactly. It is maximal in the natural toric Kummer comparison group, although it has infinite index in the full tree automorphism group.

来源字段：http://aimpl.org/galarithdyn/10/

12132 | AIM-DYNAMICAL_SYSTEMS-0052

### A gap-optimized Poisson approximation bound with a clustering obstruction

P / partially_solved | AIM 研讨会问题汇编

Problem 9. Can one either determine or improve any estimates on the rate of convergence to the Poisson limit law in slowly mixing systems? For instance, from say a decay rate of 1/n 4 to 1/n.

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12133 | AIM-DYNAMICAL_SYSTEMS-0036

### A gcd law for dynatomic-field intersections

P / partially_solved | AIM 研讨会问题汇编

“Describe $K_n\cap K_{n'}$. Prove that this is small.”

It is problem 11.2 in the section “Formal period $n$” of the 2016 AIM workshop *The Galois theory of orbits in arithmetic dynamics*. The preceding record asks, “When is the image maximal?”, but neither record defines $K_n$. The workshop report confirms that dynatomic Galois groups were intended: Jones discusses Galois groups of fields generated by periodic points, and Krumm writes $\Phi_n(c,x)$ for the $n$th dynatomic polynomial of $x^2+c$ and studies its splitting field over $\mathbb Q(c)$ [AIM16]. Still, the fragment does not say whether the map is generic or specialized, whether the base is a number field or function field, or whether $K_n$ means the exact/formal-period field or the field of all points whose period divides $n$.

We use the following explicit reconstruction. Let $K$ be a characteristic-zero field and $f\in K[x]$ have degree at least two. Define the dynatomic polynomial \[ \Phi^*_{f,n}(x) =\prod_{r\mid n}\bigl(f^r(x)-x\bigr)^{\mu(n/r)}, \tag{1.1} \] which is a polynomial despite its quotient presentation. Let \[ K_n(f/K)=\text{the splitting field over $K$ of }\Phi^*_{f,n}(x). \tag{1.2} \] When repeated roots occur, “splitting field” means the field generated by the distinct roots. Roots can have formal period $n$ but smaller exact period in parabolic cases [MP94, MS95]; this distinction is kept explicit. The question is meaningful for distinct $n,n'$. If $n=n'$, the intersection is tautologically $K_n$.

The proved slice below takes $f(x)=x^d$ over $\mathbb Q$, with $d$ even. In this family all relevant polynomials are separable and formal period equals exact period. Moreover, the splitting field of $f^n(x)-x$ is the same field as (1.2), so the answer is insensitive to the second ambiguity above.

来源字段：http://aimpl.org/galarithdyn/11/

12134 | AMR-046-0019

### A Markus-Yamabe problem for differential equations

P / partially_solved | AMR 问题清单汇编

Are there smooth vector fields in $\mathbb{R}^3$ satisfying the hypotheses of the Markus-Yamabe conjecture and having periodic orbits?

背景链接：https://arxiv.org/abs/2012.02524

12135 | AMR-046-0020

### A Markus-Yamabe/La Salle problem for discrete dynamical systems

O / open | AMR 问题清单汇编

Let $F:\mathbb{R}^2\to\mathbb{R}^2$ be smooth, have a fixed point, and satisfy $$\rho\bigl(|DF(x)|\bigr)<1\quad\text{for every }x\in\mathbb{R}^2.$$ Is the fixed point globally asymptotically stable?

背景链接：https://arxiv.org/abs/2012.02524

12136 | AIM-DYNAMICAL_SYSTEMS-0011

### A maximal dyadic-pair stabilizer in Thompson's group T

P / partially_solved | AIM 研讨会问题汇编

**Maximal subgroups in Thompson groups**

Find new maximal subgroups of infinite index in Thompson groups

来源字段：http://aimpl.org/groupdynamorigin/3/

12137 | AIM-DYNAMICAL_SYSTEMS-0095

### A mod-2 criterion for rigid/flexible Lattes portrait overlap

P / partially_solved | AIM 研讨会问题汇编

**portraits of rigid Latt\`es**

What portraits occur for the rigid Latt\`es maps? In particular, are any of them the flexible Latt\`es portraits?

来源字段：http://aimpl.org/finitedynamics/6/

12138 | AIM-DYNAMICAL_SYSTEMS-0044

### A modewise rotating-AR analogue for discrete-time stochastic reaction-diffusion systems

P / partially_solved | AIM 研讨会问题汇编

Its damaged displays recover as

\[ \dot x=f(x,y),\qquad \dot y=g(x,y), \]

and, after linearization at the stable fixed point,

\[ \begin{bmatrix}dx\\dy\end{bmatrix} =A\begin{bmatrix}x\\y\end{bmatrix}dt +\begin{bmatrix}\sigma _1\,dW_1\\ \sigma _2\,dW_2\end{bmatrix}. \]

On page 2 the same question is repeated as Problem 13 and followed by a separate fast/slow-systems question.

“A discrete time reaction diffusion equation” is not specified further: it could mean a coupled map lattice, a time discretization of an SPDE, an integrodifference equation, or a microscopic discrete-time population process. The result below therefore treats a clearly stated coupled-map lattice model. Other interpretations require changing the multiplier and noise covariance, but the modewise argument survives whenever a spatial eigenbasis diagonalizes both.

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12139 | AMR-046-0008

### A new Hilbert sixteenth-type problem

P / partially_solved | AMR 问题清单汇编

Let $\mathcal M_m$ be the family of planar polynomial vector fields that are linear combinations of $m$ distinct monomial vector fields $(x^{n_j}y^{k_j},0)$ or $(0,x^{n_j}y^{k_j})$, and let $\mathcal H^M[m]$ be their maximum possible number of limit cycles. (i) Find upper and lower bounds for $\mathcal H^M[m]$. (ii) Find the least $m$ for which a planar polynomial differential system with $m$ monomials has at least $m+1$ limit cycles.

背景链接：https://arxiv.org/abs/2012.02524

12140 | AMR-046-0018

### A piecewise-linear Hilbert sixteenth-type problem

P / partially_solved | AMR 问题清单汇编

Let $\mathcal H(n)$ be the maximum number of limit cycles of degree-$n$ planar polynomial systems, and let $\mathcal L(n)$ be the maximum number of crossing limit cycles of planar piecewise-linear systems whose two zones are separated by a branch of a degree-$n$ algebraic curve. Improve, if possible, the known lower bounds $$\mathcal H(2)\ge4,\ \mathcal H(3)\ge13,\ \mathcal H(n)\ge Kn^2\log n$$ and $$\mathcal L(1)\ge3,\ \mathcal L(2)\ge4,\ \mathcal L(n)\ge\lfloor n/2\rfloor.$$

背景链接：https://arxiv.org/abs/2012.02524

12141 | AIM-DYNAMICAL_SYSTEMS-0041

### A precise inverse dynatomic Galois problem and two complete realization cases

P / partially_solved | AIM 研讨会问题汇编

Variation of problem 14.1: ask again only in a dynatomic setting. Formulate and solve.

来源字段：http://aimpl.org/galarithdyn/14/

12142 | AIM-DYNAMICAL_SYSTEMS-0066

### A regenerative stable--Gaussian trichotomy

P / partially_solved | AIM 研讨会问题汇编

**Problem 23.** What type of limiting theorems/statistical properties can be found for slowly mixing systems.

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12143 | AMR-046-0009

### A second-order differential equation

O / open | AMR 问题清单汇编

Let $f$ be a continuous, nonzero, $T$-periodic function and let $p>0$. Find necessary and sufficient conditions on $f$ for the existence of positive $T$-periodic solutions of $x^p(t)x''(t)=f(t)$.

背景链接：https://arxiv.org/abs/2012.02524

12144 | AIM-DYNAMICAL_SYSTEMS-0074

### A sharp sequential Markov spectral certificate

P / partially_solved | AIM 研讨会问题汇编

**Problem 31.** What can be said about spectral properties/Lyapunov spectrum in sequential or nonstationary settings?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12145 | AIM-DYNAMICAL_SYSTEMS-0094

### A tame positive-characteristic form of Milnor's four-point criterion

P / partially_solved | AIM 研讨会问题汇编

**Milnor's characterization of Lattès maps with four postcritical points.**

- no critical point is postcritical; - all critical points are simple.

Does this hold over positive characteristic? What does one get from the dual of Frobenius in the inseparable case? What happens in characteristics \(2\) and \(3\), where \(\operatorname{Aut}(E)\) can be nonabelian?

来源字段：http://aimpl.org/finitedynamics/6/

12146 | AIM-DYNAMICAL_SYSTEMS-0072

### A Wasserstein rate dictionary for local-update generators

P / partially_solved | AIM 研讨会问题汇编

Problem 29. How can one bound rates of convergence, using the Wasserstein distance, for Markov generators of the type,

`LA (x1,..., x n) = ∑`

`> i`

`∫`

`ωi(x, dy )[ A(y) - A(x)],`

which involve only a finite number of x′s and y′s near i.

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12147 | OWR-8415336-013

### Absence of Absolutely Periodic Euclidean Billiard Orbits

P / partially_solved | Oberwolfach 报告问题汇编

Do Euclidean billiards have no absolutely periodic orbits?

来源字段：https://doi.org/10.4171/owr/2021/33

12148 | OWR-3396-011

### Absence of Eigenvalues for Pisot Substitution Operators

O / open | Oberwolfach 报告问题汇编

Let $\sigma$ be a primitive substitution with hull $\Omega$ and, for a one-to-one $f:A\to\mathbb{R}$, define $[H_\omega\psi](n)=\psi(n+1)+\psi(n-1)+f(\omega_n)\psi(n)$ on $\ell^2(\mathbb{Z})$. If $\sigma$ is Pisot, must every $H_\omega$ have no eigenvalues; equivalently, must $\Omega_c=\{\omega\in\Omega:H_\omega\text{ has no eigenvalues}\}=\Omega$?

来源字段：https://doi.org/10.4171/owr/2009/13

12149 | AMR-052-0054

### Absolute continuity at full dimension

P / partially_solved | AMR 问题清单汇编

For a positive-entropy invariant measure $m$ on a holomorphic quasi-repeller $\Lambda$, is $m$ absolutely continuous with respect to Hausdorff measure in dimension $\dim_Hm$ if and only if $\dim_Hm=\dim_H\overline\Lambda$?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12150 | OWR-1319-030

### Absolute Continuity for Symplectic $\mathbb Z^2$-Actions

O / open | Oberwolfach 报告问题汇编

Let $\alpha_0$ be the action on $\mathbb T^4$ of a real-diagonalizable maximal abelian subgroup of $\operatorname{Sp}(4,\mathbb Z)$, and let $\alpha$ be a $\mathbb Z^2$-action whose elements are homotopic to the corresponding elements of $\alpha_0$. If an $\alpha$-invariant Borel probability measure $\mu$ satisfies $h_*\mu=\lambda$ for the semiconjugacy $h$ and Lebesgue measure $\lambda$, prove that $\mu$ is absolutely continuous and $\mu=\omega\wedge\omega$ for a Lebesgue-measurable exterior two-form $\omega$ that is closed in an appropriate sense.

来源字段：https://doi.org/10.4171/owr/2006/33

12151 | OWR-14297733-002

### Abundance of Zoll Flows in Hamiltonian Systems

P / partially_solved | Oberwolfach 报告问题汇编

Characterize how abundant Zoll flows are within natural classes of Hamiltonian systems, including geodesic and magnetic flows on the two-torus.

来源字段：https://doi.org/10.4171/owr/2023/30

12152 | AMR-052-0048

### Accessibility of basin-boundary periodic points

P / partially_solved | AMR 问题清单汇编

Let $f:U\to f(U)$ be a proper holomorphic map of degree at least two on a simply connected attracting basin $U$, and suppose $f$ extends holomorphically across $\overline U$. Is every periodic point of $\partial U$ accessible from $U$?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12153 | AMR-052-0049

### Accessibility of positive-exponent boundary points

O / open | AMR 问题清单汇编

In the setting of Przytycki Problem 1.1, is every $x\in\partial U$ with $\liminf_{n\to\infty}n^{-1}\log|(f^n)'(x)|>0$ accessible from $U$?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12154 | AMR-052-0030

### Accessibility of the critical point in a Cremer Julia set

S / solved | AMR 问题清单汇编

Can the critical point of a Cremer polynomial be accessible from the complement of its Julia set?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12155 | OWR-8415336-005

### Accumulation Points of Convex Billiard Length Spectra

P / partially_solved | Oberwolfach 报告问题汇编

For a real-analytic convex billiard domain $\Omega$, are the multiples of the perimeter $|\partial\Omega|$ the only accumulation points of its length spectrum?

来源字段：https://doi.org/10.4171/owr/2021/33

12156 | AMR-045-0049

### Adler's renewal question

P / partially_solved | AMR 问题清单汇编

Is every irreducible sofic shift topologically conjugate to a renewal shift?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12157 | OWR-4410-001

### Affine Invariant Submanifolds in Higher-Genus Strata

O / open | Oberwolfach 报告问题汇编

Classify the affine invariant submanifolds in strata of abelian differentials in genus at least three.

来源字段：https://doi.org/10.4171/owr/2010/29

12158 | AMR-039-0019

### Alexandrov spaces

P / partially_solved | AMR 问题清单汇编

Do spaces with positive sectional curvature in the sense of Alexandrov have positive coarse Ricci curvature for a natural choice of Markov kernels? Can this be proved by manifold approximation or parallel transport in Alexandrov spaces?

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12159 | AMR-046-0017

### Algebraic limit cycles and related questions

P / partially_solved | AMR 问题清单汇编

Determine the entries currently marked unknown in the following comparison between quadratic systems and planar piecewise-linear systems with a straight separation line: maximum limit cycles ($4?$ versus $3?$); algebraic limit cycles ($1?$ versus $2?$); non-hyperbolic algebraic limit cycles ($0?$ versus $1?$); coexistence of algebraic and non-algebraic limit cycles (unknown for quadratic systems and impossible in the piecewise-linear case); and critical periods ($2?$ versus unknown).

背景链接：https://arxiv.org/abs/2012.02524

12160 | AIM-DYNAMICAL_SYSTEMS-0002

### Algebraic path shifts for the power-map tower

P / partially_solved | AIM 研讨会问题汇编

Let $k$ be a field, let $f\in k(z)$ have degree at least $2$, and let $t$ be transcendental. Put \[ k_\infty=k\Bigl(\bigcup_{n\geq0}f^{-n}(t)\Bigr),\qquad L=k_\infty\cap\overline{k}, \] and $G_\infty=\operatorname{Gal}(k_\infty/k(t))$, $G_\infty^{\rm geom}=\operatorname{Gal}(k_\infty/L(t))$. Can one algebraically construct “nice paths” which give explicit wreath recursions for $G_\infty^{\rm geom}$? More precisely, for $z\in f^{-1}(t)$, can one find explicit, perhaps recursively described, endomorphisms of the tower sending $t$ to $z$?

Its question is:

Four formulation points matter.

1. The symbol $t$ should be an indeterminate over $k$, and all fields should be placed in a fixed algebraic closure of $k(t)$. 2. For a rational map, $k_\infty$ is best read as the directed union of the splitting fields of $f^{\circ n}(X)=t$. 3. The final $K_\infty$ in the source is evidently the same tower previously called $k_\infty$. 4. Every member of $\operatorname{Gal}(k_\infty/L(t))$ fixes $t$. Therefore a map sending $t$ to a nontrivial first preimage cannot be a deck transformation. The algebraic counterpart of a path must instead be an isomorphism of pointed towers or an automorphism of the abstract $L$-field which changes the distinguished subfield $L(t)$ to $L(z)$.

Separability is also necessary for a regular rooted preimage tree. The theorem below assumes $\operatorname{char}L\nmid d$.

来源字段：http://aimpl.org/groupdynamorigin/1/

12161 | OWR-8415361-002

### Algebraic Rigidity of Higher-Rank Anosov Actions

O / open | Oberwolfach 报告问题汇编

别名：Unspecified Higher-Rank Dynamics Conjecture [OWR-8415361-001]

Is every $C^\infty$ Anosov action of $\mathbb R^k$ or $\mathbb Z^k$, with $k\geq2$, on a compact manifold and without $C^\infty$ rank-one factors $C^\infty$-conjugate, after passage to finite covers, to an algebraic action?

来源字段：https://doi.org/10.4171/owr/2021/56

12162 | OWR-3394-010

### Almost Input-to-State Stability of the Damped Pendulum

S / solved | Oberwolfach 报告问题汇编

For the damped pendulum on $S^1\times\mathbb{R}$ given by $\dot\theta=\omega$ and $\dot\omega=-\sin\theta-\omega+d$, is the system almost input-to-state stable? That is, does there exist a class-$\mathcal{K}$ function $\gamma$ such that, for every essentially bounded input $d$, there is a measure-zero exceptional set $B_d$ and every initial state outside $B_d$ satisfies $\limsup_{t\to\infty}|x(t)|\leq\gamma(\lVert d\rVert_\infty)$?

来源字段：https://doi.org/10.4171/owr/2009/11

12163 | AMR-045-0037

### Amenable Cantor actions

O / open | AMR 问题清单汇编

Is every minimal action of a countable amenable group on the Cantor set topologically orbit equivalent to a $\mathbb Z$ action?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12164 | AIM-DYNAMICAL_SYSTEMS-0086

### An algebraic period-three test case for infinitesimal Thurston rigidity

P / partially_solved | AIM 研讨会问题汇编

Algebraic proof of Thurston rigidity

Algebraic proof of infinitesimal Thurston Rigidity, i.e. if $f$ is PCF non-Latt\`es and degree at least 2, then $1$ is not an eigenvalue of $f^{\ast}:H^1(\mathbb{P}^1,\Theta_{P_f}) \to H^1(\mathbb{P}^1,\Theta_{P_f})$ where $P_f$ is the post-critical set and $\Theta$ is the sheaf of holomorphic vector fields with zeros along the post-critical set.

Its problem text is:

Here \[ P_f=\bigcup_{n\geq 1} f^n(C_f) \] is understood as a reduced finite marked set. No reconstruction of its mathematical content is needed, but the notation $f^*$ requires an important clarification made in Section 3.

来源字段：http://aimpl.org/finitedynamics/3/

12165 | AIM-DYNAMICAL_SYSTEMS-0033

### An exact affine family in every odd prime degree

P / partially_solved | AIM 研讨会问题汇编

Find classes of examples of polynomials or rational maps $d\geq 3$ for which we can compute explicitly the arboreal Galois representation up to finite index over various fields.”

“Try Stoll's techniques for $x^p+c$ for $p$ prime.

来源字段：http://aimpl.org/galarithdyn/10/

12166 | AIM-DYNAMICAL_SYSTEMS-0099

### An explicit graph transform on the state-dependent-delay solution manifold

P / partially_solved | AIM 研讨会问题汇编

(9) Alternative proofs of the existence of infinite dimensional local invariant manifolds at equilibria for differential equations with state-dependent delay

For instance, there is no result using the graph transform or Lyapunov-Perron technique to prove the existence of local stable manifolds at equilibria for differential equations with state-dependent delay.

来源字段：https://aimath.org/WWN/variabletimelag/variabletimelag.pdf

12167 | AIM-DYNAMICAL_SYSTEMS-0022

### An intrinsic level-4 obstruction and a discriminant sieve

P / partially_solved | AIM 研讨会问题汇编

1. For $p\mid N$, $X^{\mathrm{dyn}}(N)$ has bad reduction at $p$ for the standard model, but evidence suggests there exists a different model with good reduction. Is this true? If not, why not? Is there a ``reason'' for this to be true? 2. (Silverman) In general, what are the primes of bad reduction and why?

来源字段：http://aimpl.org/galarithdyn/4/

12168 | AIM-DYNAMICAL_SYSTEMS-0040

### An odometer obstruction to geometric polynomial realization

P / partially_solved | AIM 研讨会问题汇编

Work over $\mathbb{C}(x)$ for a start. Given a subgroup $G\subseteq \operatorname{Aut}$(regularly rooted $d$-ary tree). Does there exist a polynomial $f(z)\in K[z]$ with $G$ as its image of Galois representation? Interpret and solve.

来源字段：http://aimpl.org/galarithdyn/14/

12169 | AMR-052-0069

### An orbit converging to an irrationally indifferent fixed point

S / solved | AMR 问题清单汇编

Under the hypotheses of Eremenko-Lyubich Question 2, can even a single orbit converge to $z_0$?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12170 | AMR-042-0003

### Analogues of Pesin theory

O / open | AMR 问题清单汇编

Is there an analogue of Pesin theory for suitably defined smooth maps of the objects that arise naturally in algebraic dynamical systems-compact sets locally resembling a manifold times a Cantor set, or totally disconnected compact sets? Can any algebraic $\mathbb{Z}^d$-systems with $d>1$ be perturbed in a meaningful way?

背景链接：https://www.imath.kiev.ua/~skolyada/kevin.pdf

12171 | AMR-036-0007

### Analytic degenerate Herman rings

P / partially_solved | AMR 问题清单汇编

Does there exist a rational function having an analytic invariant Jordan curve on which it is topologically conjugate to an irrational rotation, where the curve is neither a circle nor a level curve of a linearizer in a rotation domain? Equivalently, does an analytic degenerate Herman ring exist?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

12172 | AMR-052-0023

### Angle and renormalization at the golden-mean Siegel critical point

P / partially_solved | AMR 问题清单汇编

For the quadratic Siegel polynomial with rotation angle $\theta_0=(\sqrt5-1)/2$, prove that the Siegel-disk boundary has the experimentally observed opening angle of about $120^\circ$ at the critical point, and construct the expected renormalization there.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12173 | OWR-5147-004

### Annular Recurrent Fatou Components of Hénon Maps

O / open | Oberwolfach 报告问题汇编

Can a Hénon map have a recurrent Fatou component that is a one-dimensional rotation domain conformally equivalent to an annulus?

来源字段：https://doi.org/10.4171/owr/2011/19

12174 | OWR-14299512-003

### Aperiodic Stable Transitivity Near Anosov Flows

O / open | Oberwolfach 报告问题汇编

Does there exist a stably transitive diffeomorphism with no periodic points? More specifically, can the time-one map of a geodesic flow on a negatively curved surface be stably transitive?

来源字段：https://doi.org/10.4171/owr/2025/31

12175 | OWR-1191-003

### Aperiodicity from Cohomological Trace Density in Delone Hulls

O / open | Oberwolfach 报告问题汇编

Let $P$ be a repetitive finite-local-complexity Delone set whose hull translation action is uniquely ergodic with ergodic measure $\mu$. Is $P$ aperiodic if and only if $\tau_\mu^*(\check H^1(\Omega_P,\mathbb Z))$ is dense in $\mathbb R^d$?

来源字段：https://doi.org/10.4171/owr/2006/09

12176 | AMR-052-0058

### Approximating quasi-repeller dimension by measures

P / partially_solved | AMR 问题清单汇编

For a holomorphic quasi-repeller $\Lambda$, is $\sup_{m\in\mathcal M^+(\Lambda)}\dim_Hm=\dim_H\overline\Lambda$? Does allowing all invariant ergodic measures change the answer?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12177 | AMR-052-0026

### Arc in a Cremer Julia set

O / open | AMR 问题清单汇编

For $P_\alpha(z)=z^2+e^{2\pi i\alpha}z$ with a Cremer fixed point at $0$, is there an arc in its Julia set joining $0$ to its preimage $-e^{2\pi i\alpha}$?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12178 | AMR-052-0022

### Arithmetic criterion for Jordan Siegel disks

P / partially_solved | AMR 问题清单汇编

For a quadratic Siegel polynomial with rotation angle $\theta$, find the arithmetic condition on $\theta$ that makes the Siegel-disk boundary a Jordan curve. In particular, is there $\gamma_0>2$ such that a Diophantine bound $|\theta-p/q|>C/q^\gamma$ forces a Jordan domain for $\gamma<\gamma_0$ but not for $\gamma>\gamma_0$?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12179 | AMR-048-0001

### Arnold and Arnold-Givental conjectures

P / partially_solved | AMR 问题清单汇编

For a Hamiltonian diffeomorphism of a closed symplectic manifold, prove the Arnold lower bound on its number of fixed points in terms of Morse-theoretic data. In the Lagrangian Arnold-Givental form, prove the corresponding lower bound for intersections of a Lagrangian submanifold with its Hamiltonian image under the standard hypotheses.

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Dynamical_systems

12180 | AIM-DYNAMICAL_SYSTEMS-0051

### ASIP rates as strong averaging rates

P / partially_solved | AIM 研讨会问题汇编

**Problem 8.** Is there an annealed almost sure invariant principle (ASIP) with rate \(n^{1/4}\log n\) instead of \(n^{1/4+\delta}\)?

Suppose \(\mu\) is an invariant measure of the system \(y_{n+1}=Ty_n\). For \(\epsilon>0\), how does the dynamics of the system \[ x_{n+1}^{\epsilon,y}=x_n^{\epsilon,y}+\epsilon f(x_n^{\epsilon,y},y_n) \] compare with the dynamics of the system \[ \bar x_{n+1}=\bar x_n+\epsilon\int f(\bar x_n,y_n)\,d\mu(y)? \] Moreover, what can be said about \(\sup_{n<1/\epsilon}|\bar x_n-x_n^{\epsilon,y}|\), specifically with respect to limit theorems?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12181 | OWR-4330-004

### Asymptotic Counts of Periodic Rational Polygon Billiards

O / open | Oberwolfach 报告问题汇编

For every rational polygon $P$, does $\lim_{L\to\infty}N(P,L)/L^2$ exist, where $N(P,L)$ counts combinatorially distinct periodic billiard orbits of length at most $L$?

来源字段：https://doi.org/10.4171/owr/2010/17

12182 | OWR-4075-005

### Asymptotics of Dispersing Billiards with Holes

P / partially_solved | Oberwolfach 报告问题汇编

For a finite-horizon dispersing billiard map with a hole $H$, if an initial distribution $\mu_0$ is evolved forward to time $n$ to give $\mu_n$, how does $\mu_n$ behave?

来源字段：https://doi.org/10.4171/owr/2009/32

12183 | AMR-045-0035

### Automorphism groups of full shifts

O / open | AMR 问题清单汇编

Are the groups $\operatorname{Aut}(\sigma_2)$ and $\operatorname{Aut}(\sigma_3)$ isomorphic?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12184 | AMR-048-0003

### Banach's simple Lebesgue spectrum problem

P / partially_solved | AMR 问题清单汇编

Does there exist an ergodic measure-preserving transformation whose Koopman operator has simple Lebesgue spectrum?

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Dynamical_systems

12185 | AMR-045-0025

### Bernoulli factors of group shifts

O / open | AMR 问题清单汇编

Does every nonabelian $\mathbb Z^d$ group shift factor algebraically onto a Bernoulli group shift?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12186 | AMR-048-0002

### Berry-Tabor conjecture

P / partially_solved | AMR 问题清单汇编

For a generic quantum system whose classical counterpart is integrable, prove that the unfolded high-energy level spacings have Poisson statistics.

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Dynamical_systems

12187 | AMR-045-0011

### Beta functions

P / partially_solved | AMR 问题清单汇编

Characterize the functions that occur as beta functions of mixing Markov shifts.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12188 | OWR-1383-006

### Birkhoff Integrable-Billiards Conjecture

O / open | Oberwolfach 报告问题汇编

The only integrable planar inner billiards should be those inside ellipses.

来源字段：https://doi.org/10.4171/owr/2006/45

12189 | OWR-17127-001

### Birkhoff Rigidity Conjecture for Convex Billiards

P / partially_solved | Oberwolfach 报告问题汇编

Are the completely integrable billiards inside smooth strictly convex planar domains exactly the billiards inside ellipses?

来源字段：https://doi.org/10.4171/owr/2019/31

12190 | OWR-14297733-003

### Birkhoff Sections for Three-Dimensional Reeb Flows

O / open | Oberwolfach 报告问题汇编

Does every three-dimensional Reeb vector field admit a Birkhoff section?

来源字段：https://doi.org/10.4171/owr/2023/30

12191 | AMR-039-0012

### Bishop-Gromov theorem

P / partially_solved | AMR 问题清单汇编

Is there an analogue, for positive coarse Ricci curvature, of the Bishop-Gromov theorem or the isoperimetric form of the Gromov-Lévy theorem? Identify suitable comparison spaces or formulate a version that also captures discrete examples such as the cube.

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12192 | OWR-15995-004

### Bitrate Requirements for Lossless Dynamical System Digitalization

P / partially_solved | Oberwolfach 报告问题汇编

For lossless digitalization of a discrete-time topological dynamical system, determine the minimum channel capacity or signal bitrate required for a receiver to identify the initial state with error tending to zero.

来源字段：https://doi.org/10.4171/owr/2018/10

12193 | OWR-9790354-006

### Borel Extensions of Hyper-Aperiodic Coloring Theorems

O / open | Oberwolfach 报告问题汇编

Let $S\subseteq q^\Gamma$ be a subshift of finite type and let $f:\operatorname{Free}(k^\Gamma)\to q$ be an $S$-coloring whose coding map has no nonidentity element of $\Gamma$ fixing its entire image. Determine additional hypotheses ensuring that, when $f$ is Borel rather than continuous, every free Borel $\Gamma$-action still admits a Borel hyper-aperiodic $S$-coloring.

来源字段：https://doi.org/10.4171/owr/2022/3

12194 | OWR-2053-010

### Borel Structure and Generic Triviality of Flow Self-Similarities

O / open | Oberwolfach 报告问题汇编

For any measurable measure-preserving flow $T$, is its self-similarity group $I(T)$ a Borel group? Is the absence of self-similarity generic among measure-preserving flows?

来源字段：https://doi.org/10.4171/owr/2008/42

12195 | OWR-1460-009

### Boundary Behavior under Asymptotic Schwarz-Pick Equality

P / partially_solved | Oberwolfach 报告问题汇编

Let $f:\mathbb{D}\to\mathbb{D}$ be holomorphic and suppose $\lim_{z\to1}(1-|z|^2)|f'(z)|/(1-|f(z)|^2)=1$. What can be said about the boundary behavior of $f$ at $z=1$?

来源字段：https://doi.org/10.4171/owr/2007/09

12196 | AMR-052-0050

### Boundary entropy of an attracting basin

P / partially_solved | AMR 问题清单汇编

In the setting of Przytycki Problem 1.1, is $h_{\mathrm{top}}(f|_{\partial U})=\log\deg(f|_U)$?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12197 | AMR-052-0080

### Boundary fixed points in rank-zero Hénon components

P / partially_solved | AMR 问题清单汇编

In the rank-zero case, if the limiting map on an invariant stable component is constant with value $x_0\in\partial U$, prove that one eigenvalue at $x_0$ equals $1$.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12198 | AMR-052-0013

### Boundary of the principal hyperbolic component

P / partially_solved | AMR 问题清单汇编

Let $B(z^n)$ be the set of degree-$n$ polynomials with an attracting fixed point whose immediate basin contains every critical point. Describe the boundary of $B(z^n)$ in the space of degree-$n$ polynomials.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12199 | OWR-17127-007

### Boundary of Zero-Entropy Dissipative Systems

P / partially_solved | Oberwolfach 报告问题汇编

Within the space of mildly dissipative disk diffeomorphisms, is the boundary of the zero-topological-entropy locus exactly the set of infinitely renormalizable zero-entropy systems?

来源字段：https://doi.org/10.4171/owr/2019/31

12200 | AMR-052-0015

### Boundary quotient independent of base Blaschke product

O / open | AMR 问题清单汇编

Quotient the boundary of $B(A)$ by quasiconformal conjugacy, writing the quotient as $\partial(A)$. Prove that the natural isomorphism $F:B(z^n)\to B(A)$ extends to a homeomorphism $\partial(z^n)\to\partial(A)$.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12201 | OWR-1460-010

### Boundary Regularity for Complete Conformal Metrics

O / open | Oberwolfach 报告问题汇编

Let $\Omega\subseteq\mathbb C$ be a domain and $\Gamma\subseteq\partial\Omega$ a boundary set. For a regular conformal metric $\lambda(z)|dz|$ on $\Omega$ with curvature bounded below by $-4$, when are local completeness near $\Gamma$, meaning $d_\lambda(z,z_0)\to\infty$ as $z$ approaches every point of $\Gamma$, and divergence $\lambda(z)\to\infty$ at every point of $\Gamma$ equivalent? In particular, what minimal regularity of $\Gamma$ makes them equivalent?

来源字段：https://doi.org/10.4171/owr/2007/09

12202 | AMR-052-0057

### Boundary theorems for geometric coding trees

O / open | AMR 问题清单汇编

Which theorems about boundary behavior of Riemann maps have analogues for geometric coding trees?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12203 | AMR-052-0056

### Bounded Jacobian cocycles and absolute continuity

O / open | AMR 问题清单汇编

For which positive-entropy invariant measures $m$ does uniform $L^2(m)$ boundedness of the sums of $\log\operatorname{Jac}_m f-\kappa\log|f'|$, where $\kappa=\dim_Hm$, imply $m\ll H^\kappa$?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12204 | AIM-DYNAMICAL_SYSTEMS-0073

### Bounded observables with Gaussian, borderline, and stable limits

P / partially_solved | AIM 研讨会问题汇编

**Problem 30.** What type of limiting theorems/statistical properties can be found for slowly mixing systems?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12205 | AMR-052-0067

### Bounded orbit of a wandering domain

P / partially_solved | AMR 问题清单汇编

Does there exist an entire function with a wandering Fatou component whose orbit of components is bounded?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12206 | AMR-036-0004

### Bounded wandering domains of entire functions

P / partially_solved | AMR 问题清单汇编

Let $f$ be a nonlinear entire function and let $D$ be a Fatou component on which all limit functions of the iterates $f^n$ are constant. Can the set of those constant limit functions be both infinite and bounded? Equivalently, can a subdomain of a wandering domain wander within a bounded subset of the plane?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

12207 | OWR-11570-001

### Bowen's Formula for Nonirreducible Dynamical Systems

O / open | Oberwolfach 报告问题汇编

For which large classes of non-irreducible systems does Bowen's formula hold, and is there a variation of Bowen's formula encompassing an even broader class of systems?

来源字段：https://doi.org/10.4171/owr/2012/03

12208 | OWR-12864-002

### Burnstile Lobe Sequences and Loss of Primary Intersections

O / open | Oberwolfach 报告问题汇编

In burnstile theory for propagating fronts, characterize the sequence of modified entrance and exit lobes: is there a unique locally smallest lobe, what transport questions are answered by the other lobes, and can continuous-time treatment improve efficiency? What are the physical consequences when the stable and unstable burning invariant manifolds lose a primary intersection point, particularly for ballistic front propagation, and can the theory be extended continuously through that loss?

来源字段：https://doi.org/10.4171/owr/2014/04

12209 | AMR-041-0013

### Calogero-Moser-Vlasov

P / partially_solved | AMR 问题清单汇编

For the infinite-dimensional Calogero-Moser system, in which particles on the real line interact through the inverse-square potential, does the dynamics exist? Is it integrable in the sense that every invariant measure gives rise to almost-periodic dynamics?

背景链接：https://people.math.harvard.edu/~knill/seminars/intr/

12210 | OWR-15436-005

### Canonical Equations for Spatial Adaptive Dynamics

O / open | Oberwolfach 报告问题汇编

In the spatial adaptive-dynamics limit described by the jump process of monomorphic equilibria, can one derive a canonical equation governing evolution under asymptotically small mutation steps?

来源字段：https://doi.org/10.4171/owr/2017/28

12211 | OWR-1323-013

### Chaos of Individual Operators in Chaotic Semigroups

S / solved | Oberwolfach 报告问题汇编

If $(T_t)_{t\geq0}$ is a chaotic $C_0$-semigroup on a separable complex Banach space, is every $T_t$ with $t>0$ chaotic? Does at least one $T_{t_0}$, with $t_0>0$, have to be chaotic?

来源字段：https://doi.org/10.4171/owr/2006/37

12212 | OWR-1323-010

### Chaotic Operators on Nuclear Fréchet Spaces

O / open | Oberwolfach 报告问题汇编

Does every infinite-dimensional nuclear Fréchet space admit a chaotic operator?

来源字段：https://doi.org/10.4171/owr/2006/37

12213 | AIM-DYNAMICAL_SYSTEMS-0049

### Chronological Lasota--Yorke bounds for nonstationary compositions

P / partially_solved | AIM 研讨会问题汇编

Problem 6. Can we find Lasota-York type inequalities for a composition of oper-ators?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12214 | OWR-15582-002

### Classification of Codimension-One Projective Foliations

O / open | Oberwolfach 报告问题汇编

别名：Theorem on Orbicurve Factorization of Holonomy Representations [OWR-15582-003]

Is every codimension-one holomorphic foliation $\mathcal F$ on a projective manifold $X$ either transversely projective or the pullback $\mathcal F=\pi^*\mathcal G$ of a one-dimensional foliation $\mathcal G$ on a projective surface under a rational map $\pi:X\dashrightarrow Y$?

来源字段：https://doi.org/10.4171/owr/2017/39

12215 | AMR-045-0027

### Classify group shifts

P / partially_solved | AMR 问题清单汇编

Classify group shifts up to topological conjugacy, especially abelian group shifts with completely positive entropy.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12216 | AMR-045-0033

### Classify one-sided sofic shifts

O / open | AMR 问题清单汇编

Classify one-sided sofic shifts up to topological conjugacy.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12217 | AMR-045-0002

### Classify shifts of finite type

O / open | AMR 问题清单汇编

Classify shifts of finite type up to topological conjugacy; in particular, give a decision procedure determining whether two nonnegative integer matrices define conjugate shifts.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12218 | AMR-045-0050

### Classify sofic shifts

P / partially_solved | AMR 问题清单汇编

Classify two-sided sofic shifts up to topological conjugacy.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12219 | OWR-14748-015

### Classifying Mixing Smale Spaces by Homoclinic $C^*$-Algebras

O / open | Oberwolfach 报告问题汇编

For mixing Smale spaces, can the computable Elliott invariant of the homoclinic $C^*$-algebra be used to establish an equivalence of the underlying Smale spaces?

来源字段：https://doi.org/10.4171/owr/2016/40

12220 | OWR-14748-008

### Clopen Strict Comparison for Amenable Cantor Actions

P / partially_solved | Oberwolfach 报告问题汇编

Does every minimal action of a countably infinite amenable group $G$ on the Cantor set $X$ have clopen strict comparison? Explicitly, whenever clopen sets $A,B\subseteq X$ satisfy $\mu(A)<\mu(B)$ for every $G$-invariant Borel probability measure $\mu$, can $A$ be partitioned into finitely many clopen sets that can be translated by elements of $G$ to pairwise disjoint subsets of $B$?

来源字段：https://doi.org/10.4171/owr/2016/40

12221 | OWR-1323-009

### Closed Common Hypercyclic Subspaces

O / open | Oberwolfach 报告问题汇编

For a countable family of hypercyclic operators, when can a common hypercyclic vector subspace be chosen to be a closed common hypercyclic subspace?

来源字段：https://doi.org/10.4171/owr/2006/37

12222 | AMR-049-0001

### Closed-versus-preclosed trajectory lengths

O / open | AMR 问题清单汇编

In a regular $n$-gon, call a trajectory preclosed when its endpoints divide their boundary edges into equal-length parts and meet those oriented edges at equal angles. Closed and preclosed trajectories are strongly parallel when they have the same sequence of reflection edges. After canceling all common factors from the ratios of lengths of a family of parallel closed trajectories, prove that the remaining integer factors relating closed-trajectory lengths to the corresponding preclosed-trajectory lengths are never equal to $n$.

背景链接：https://amj.math.stonybrook.edu/pdf-Springer-final/020-0170.pdf

12223 | OWR-4214-002

### Cocycle Superrigidity for Bernoulli Actions

O / open | Oberwolfach 报告问题汇编

If a countable group $\Gamma$ has vanishing first $\ell^2$-Betti number, is its Bernoulli action $\mathrm{U}_{\mathrm{fin}}$-cocycle superrigid, in particular for $\Gamma=\mathbb{Z}\wr F_2$?

来源字段：https://doi.org/10.4171/owr/2010/13

12224 | AMR-052-0016

### Combinatorial boundary of Blaschke-product space

O / open | AMR 问题清单汇编

Give a combinatorial description, possibly by laminations, of the quotient boundary space $\partial(z^n)$ obtained from the boundary of $B(z^n)$ by identifying quasiconformally conjugate maps.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12225 | AMR-052-0087

### Combinatorial theory for geometrically finite maps

P / partially_solved | AMR 问题清单汇编

Extend Thurston's finite combinatorial classification from critically finite rational maps to all geometrically finite rational maps: give finite topological data classifying the map relative to the closure of its postcritical set and characterize the data realized by rational maps.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12226 | OWR-4132-003

### Common Basin Boundaries in a Transfer-Operator System

O / open | Oberwolfach 报告问题汇编

For the transfer-operator system in the regime $6<B\leq16$, with fixed points $u_{-r_*},u_0,u_{r_*}$, is the basin of $u_0$ the common boundary of the basins of $u_{-r_*}$ and $u_{r_*}$?

来源字段：https://doi.org/10.4171/owr/2009/49

12227 | OWR-13497-003

### Commutation of Variational Discretization and Optimal Control

O / open | Oberwolfach 报告问题汇编

For general classes of variational integrators used to discretize controlled Euler-Lagrange systems and their optimal-control problems, determine whether discretization and optimization commute, i.e. whether the discrete adjoint and optimality conditions have the same approximation order as the variational state discretization.

来源字段：https://doi.org/10.4171/owr/2015/12

12228 | AMR-045-0014

### Commuting expansive automorphisms

O / open | AMR 问题清单汇编

If $S$ is an expansive automorphism of an irreducible shift of finite type, must $S$ itself be a shift of finite type?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12229 | AMR-045-0016

### Commuting powers conjecture

O / open | AMR 问题清单汇编

If $S$ and $T$ are mixing shifts of finite type, prove that $S^i$ and $T^j$ can commute for all sufficiently large integers $i,j$.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12230 | OWR-13750333-010

### Compatible Geodesic and Horocycle Symbolic Codings

O / open | Oberwolfach 报告问题汇编

Construct a symbolic coding of the geodesic flow on $\operatorname{PSL}_2(\mathbb Z)\backslash\mathbb H$ that recovers the gap statistics obtained from the corresponding horocycle-flow coding.

来源字段：https://doi.org/10.4171/owr/2023/22

12231 | OWR-4137-010

### Complete Invariance of Singular-Value Basins

O / open | Oberwolfach 报告问题汇编

If $f$ is entire and $D$ is an immediate basin containing all critical and asymptotic values of $f$, must $D$ be completely invariant? More generally, if a region $D$ contains all such values, must $f^{-1}(D)$ be connected?

来源字段：https://doi.org/10.4171/owr/2009/54

12232 | AMR-036-0006

### Completely invariant Fatou components

P / partially_solved | AMR 问题清单汇编

How many completely invariant components can the Fatou set of a transcendental entire function have? In particular, can there be more than one?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

12233 | OWR-1460-011

### Completely Invariant Julia Sets of Rational Semigroups

O / open | Oberwolfach 报告问题汇编

For a rational semigroup $G$, let $J(G)$ be its Julia set and let $E(G)$ be the smallest closed completely invariant subset of the Riemann sphere containing at least three points. If $G$ contains $f$ and $g$ with $J_f\neq J_g$ and $E(G)$ is not the whole sphere, must $E(G)$ be Möbius-equivalent to a line segment or circle? If $G$ is a polynomial semigroup, $J_h=J(G)$ for some $h\in G$, and $J(G)$ is not a line segment, must $J_f=J_g$ for all $f,g\in G$, and hence must $G$ be nearly abelian?

来源字段：https://doi.org/10.4171/owr/2007/09

12234 | AMR-048-0019

### Completeness of Lyapunov's second method

O / open | AMR 问题清单汇编

For which classes of ordinary differential equations do the classical and canonically generalized forms of Lyapunov's second method give necessary as well as sufficient conditions for stability or asymptotic stability of motion?

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Dynamical_systems

12235 | AMR-052-0031

### Components after removing a Cremer fixed point

P / partially_solved | AMR 问题清单汇编

For a quadratic Cremer polynomial $P_\alpha$, how many connected components does $J(P_\alpha)\setminus\{0\}$ have? In particular, is the number countably infinite?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12236 | AMR-052-0028

### Computer picture of a Cremer Julia set

O / open | AMR 问题清单汇编

Produce a reliable computer picture of the Julia set of a Cremer polynomial.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12237 | OWR-2094-005

### Conformal Radius Bounds for Polynomial Siegel Disks

O / open | Oberwolfach 报告问题汇编

For every $d\geq3$, does there exist $C=C(d)$ such that every degree-$d$ polynomial $f$ with an indifferent fixed point at the origin satisfies $\log\operatorname{rad}(f)\leq-B(\theta)/(d-1)+\log(\min_i|c_i|)+C$, where $\theta$ is the rotation number, $B(\theta)$ is the Brjuno sum, the $c_i$ are the critical points, and $\operatorname{rad}(f)$ is the Siegel-disk conformal radius?

来源字段：https://doi.org/10.4171/owr/2008/48

12238 | OWR-12875-002

### Congruence Deficiency of Veech Groups in $\mathcal H(4)$

P / partially_solved | Oberwolfach 报告问题汇编

How far are the Veech groups of square-tiled surfaces in $\mathcal H(4)$, especially those in the Prym locus, from being congruence groups?

来源字段：https://doi.org/10.4171/owr/2014/15

12239 | AMR-041-0006

### Conjugacy

P / partially_solved | AMR 问题清单汇编

If two Birkhoff billiard maps $T_1$ and $T_2$ satisfy $T_1=ST_2S^{-1}$ for a homeomorphism $S$, must their tables be similar? Relatedly, can one hear the shape of a real-analytic convex drum?

背景链接：https://people.math.harvard.edu/~knill/seminars/intr/

12240 | OWR-4214-001

### Conjugacy Rigidity from Group-Measure-Space Factors

P / partially_solved | Oberwolfach 报告问题汇编

Let $\Gamma\curvearrowright(X,\mu)$ be a Bernoulli action of a countable ICC group with an infinite normal subgroup whose inclusion has relative property (T). If a free ergodic action $\Lambda\curvearrowright(Y,\nu)$ has an isomorphic group-measure-space von Neumann algebra, must $\Lambda\curvearrowright Y$ be conjugate to $\Gamma\curvearrowright X$?

来源字段：https://doi.org/10.4171/owr/2010/13

12241 | OWR-4137-011

### Connected Escaping Sets of Postsingularly Finite Entire Functions

O / open | Oberwolfach 报告问题汇编

Let $f$ be a transcendental entire function whose postsingular set is finite and contained in the Julia set. Under which hypotheses is the escaping set $I(f)$ connected?

来源字段：https://doi.org/10.4171/owr/2009/54

12242 | AMR-052-0042

### Conservativity when the Julia set is the sphere

S / solved | AMR 问题清单汇编

If a rational map $f$ has $J(f)=\widehat{\mathbb C}$, is $\omega(z)=\widehat{\mathbb C}$ for almost every $z$, and is $f$ conservative with respect to Lebesgue measure?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12243 | OWR-1962-001

### Continuity of Julia Sets Under Polynomial Degeneration

P / partially_solved | Oberwolfach 报告问题汇编

Given complex polynomials $f_n$ of fixed degree $d$ with $f_n\to f$ coefficientwise, determine in which sense $K(f_n)\to K(f)$ and $J(f_n)\to J(f)$, for example via upper or lower semicontinuity in the compact-set topology.

来源字段：https://doi.org/10.4171/owr/2008/17

12244 | AMR-052-0011

### Continuity of polynomial intertwining

P / partially_solved | AMR 问题清单汇编

For a fixed first polynomial $P_1$, does the polynomial obtained by intertwining $P_1$ with $P_2$ vary continuously with $P_2$?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12245 | AMR-052-0003

### Continuity of polynomial mating

P / partially_solved | AMR 问题清单汇编

When one or both input polynomials in a mating vary continuously, does the resulting rational function vary continuously?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12246 | AMR-052-0007

### Continuity of tuning in the host polynomial

O / open | AMR 问题清单汇编

Among polynomials $P_1$ of degree greater than two with a superstable orbit of fixed period, does the tuning with a fixed $P_2$ vary continuously with $P_1$?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12247 | AMR-052-0006

### Continuity of tuning in the inserted polynomial

O / open | AMR 问题清单汇编

For a fixed polynomial $P_1$, does the polynomial obtained by tuning $P_1$ with $P_2$ vary continuously with $P_2$?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12248 | AMR-052-0074

### Continuous extension of external-ray rotation number

P / partially_solved | AMR 问题清单汇编

For monic polynomials $z^n+a_{n-1}z^{n-1}+\cdots+a_1z$ with $|a_1|\ge1$, external rays landing at the fixed point $0$ have a rotation number. Does this rotation number extend uniquely and continuously to the whole parameter space, agreeing with $\theta$ when $a_1=e^{2\pi i\theta}$?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12249 | AMR-039-0004

### Continuous-time

P / partially_solved | AMR 问题清单汇编

For a continuous-time Markov semigroup $(m_x^t)$ define $$\kappa(x,y)=\liminf_{t\to0^+}\frac1t\frac{d(x,y)-T_1(m_x^t,m_y^t)}{d(x,y)}.$$ Under a natural assumption such as non-explosion, does this definition give the standard elementary consequences of positive coarse Ricci curvature for both diffusions and jump processes, even with an unbounded generator? Is positivity of $\kappa$ enough to imply non-explosion?

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12250 | OWR-12173-002

### Convergence Criteria for Hardy-Field Multiple Ergodic Averages

P / partially_solved | Oberwolfach 报告问题汇编

Let $T$ be a measure-preserving transformation, let $f_1,\ldots,f_k\in L^\infty$, and let $a_1,\ldots,a_k$ be functions in a common Hardy field with polynomial growth. Consider $$\frac1N\sum_{n=1}^N T^{\lfloor a_1(n)\rfloor}f_1\cdots T^{\lfloor a_k(n)\rfloor}f_k.$$ Do these averages converge if and only if every nonzero $a\in\operatorname{span}(a_1,\ldots,a_k)$ satisfies at least one of the following: $|a(t)-cp(t)|\succ\log t$ for every $c\in\mathbb R$ and $p\in\mathbb Z[t]$; $a(t)-cp(t)\to d$ for some $c,d\in\mathbb R$ and $p\in\mathbb Z[t]$; or $|a(t)-t/m|\ll\log t$ for some $m\in\mathbb Z$?

来源字段：https://doi.org/10.4171/owr/2012/50

12251 | AMR-052-0075

### Convergence of the real Thurston algorithm

P / partially_solved | AMR 问题清单汇编

For a piecewise monotone interval map, iteratively replace its critical values by those of a polynomial with the same ordered critical data and conjugate back. Formulate and prove precise hypotheses under which the resulting maps converge, including beyond the postcritically finite case.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12252 | OWR-8415336-009

### Copreservation of Even and Odd Periodic Billiard Families

P / partially_solved | Oberwolfach 报告问题汇编

Can one-parameter families of $2$-periodic and $(2m+1)$-periodic billiard trajectories be preserved simultaneously under deformations of the billiard table?

来源字段：https://doi.org/10.4171/owr/2021/33

12253 | OWR-8415336-010

### Counting Rational Caustics in Convex Billiards

O / open | Oberwolfach 报告问题汇编

How many $(m,n)$-rational caustics should a sufficiently smooth strictly convex billiard domain have?

来源字段：https://doi.org/10.4171/owr/2021/33

12254 | OWR-4990373-007

### Critical Fixation Threshold for Zero-Temperature Glauber Dynamics

P / partially_solved | Oberwolfach 报告问题汇编

For zero-temperature Glauber dynamics on $\mathbb{Z}^d$, started from i.i.d. Bernoulli-$p$ opinions with fair-coin tie breaking, is the critical threshold $p_c(d)$ separating eventual fixation from infinite opinion changes equal to $1/2$ for every $d\geq 2$, in particular for $\mathbb{Z}^2$?

来源字段：https://doi.org/10.4171/owr/2021/4

12255 | AIM-DYNAMICAL_SYSTEMS-0079

### Critical orbit relations and simultaneous PCF quadratic parameters

P / partially_solved | AIM 研讨会问题汇编

An algebraic subvariety in $M_d$ has a Zariski dense subset of PCF maps if and only if that subvariety is defined by critical orbit relations.

Let $X \subset \mathbb{A}^2$ be a curve. Is it true that you have infinitely many points $(c_1,c_2) \in X$ such that $z^2+c_1$ and $z^2+c_2$ are both PCF if and only if it is vertical line, a horizontal line, or the diagonal?

来源字段：http://aimpl.org/finitedynamics/2/

12256 | AMR-052-0089

### Critically finite maps with hyperbolic postcritical complement

P / partially_solved | AMR 问题清单汇编

For $n>1$, do there exist nontrivial critically finite rational maps $f:\mathbb P^n\to\mathbb P^n$ whose postcritical hypersurface $V$ has Kobayashi-hyperbolic complement $\mathbb P^n\setminus V$? If so, describe their dynamics.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12257 | AIM-DYNAMICAL_SYSTEMS-0023

### Cubic GIT boundary and a characteristic-two Frobenius stratum

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** Describe the boundary $\overline M_d\setminus M_d$ of the GIT compactification algebraically over fields of arbitrary characteristic, identify the degenerate maps represented there, and describe the behavior of iteration on those boundary maps; begin with $d=3$ and account for inseparability in small characteristic.

来源字段：http://aimpl.org/galarithdyn/5/

12258 | OWR-4330-006

### Cutting Sequences on Regular Polygon Translation Surfaces

O / open | Oberwolfach 报告问题汇编

Which bi-infinite sequences over $A=\{a_1,\ldots,a_k\}$ occur as cutting sequences of geodesics on translation surfaces from regular $2k$-gons, and can the geodesic be recovered from its cutting sequence?

来源字段：https://doi.org/10.4171/owr/2010/17

12259 | OWR-14298803-003

### Data-Driven Estimation of Maximal Time Averages

P / partially_solved | Oberwolfach 报告问题汇编

Given sampled trajectory data but no explicit governing vector field, estimate the maximal long-time average of an observable by approximating the Koopman generator. Establish convergence beyond polynomial ODEs and characterize the required data, dictionary, and sampling assumptions.

来源字段：https://doi.org/10.4171/owr/2024/57

12260 | AMR-045-0030

### Decidability problems in symbolic dynamics

P / partially_solved | AMR 问题清单汇编

Determine whether algorithms exist to: decide conjugacy of two SFTs; decide conjugacy of two two-sided or one-sided sofic shifts; compute the expansive component generated by an expansive SFT automorphism; and decide, for a surjective block code of a full shift, whether it has a continuous right inverse.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12261 | AMR-052-0070

### Degenerate-flow limits of bad Newton polynomials

O / open | AMR 问题清单汇编

Call a polynomial bad if its Newton map has an attracting cycle that is not a root. Prove that every bad degree-$d$ polynomial $f_1$ belongs to a one-parameter family $f_h$, $0<h\le1$, bad for the relaxed Newton maps $N_{h,f_h}(z)=z-hf_h(z)/f_h'(z)$, such that as $h\to0$ the Newton flows tend to a degenerate flow.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12262 | OWR-4137-007

### Degree Bounds for Degenerate Herman Rings

O / open | Oberwolfach 报告问题汇编

Can the number of degenerate Herman rings of a rational function $f$ be estimated in terms of $\deg f$?

来源字段：https://doi.org/10.4171/owr/2009/54

12263 | OWR-4075-001

### Dense Hamiltonian Trajectories Near Integrable Systems

P / partially_solved | Oberwolfach 报告问题汇编

Can one find a $C^\infty$ Hamiltonian $H$ in a $C^r$-small neighborhood of $H_0(I)=\frac12\sum_i I_i^2$ such that its unit-energy surface contains a dense trajectory?

来源字段：https://doi.org/10.4171/owr/2009/32

12264 | AMR-052-0018

### Density of cusps in a cubic parameter boundary

S / solved | AMR 问题清单汇编

For $f_\lambda(z)=\lambda z^2+z^3$, let $U$ be the parameter component where both finite critical points lie in the immediate basin of zero. Prove that parameters with a parabolic periodic cycle are dense in $\partial U$.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12265 | AMR-052-0083

### Density of hyperbolic rational maps

O / open | AMR 问题清单汇编

For every degree $d$, prove that expanding (hyperbolic, Axiom A) maps are dense in the spaces $\operatorname{Rat}_d$ of rational maps and $\operatorname{Poly}_d$ of polynomials.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12266 | AMR-052-0040

### Diameter of Mandelbrot limbs

P / partially_solved | AMR 问题清单汇编

For the Mandelbrot limb $M(p/q)$ of internal angle $p/q$, is $\operatorname{diam}M(p/q)<K/q^2$ for an absolute constant $K$? If not, is it at least bounded by $K\log(q)/q^2$?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12267 | OWR-14297746-003

### Diffraction Scaling and Mahler-Function Asymptotics

O / open | Oberwolfach 报告问题汇编

Relate the order of scaling near zero of a substitution system's diffraction measure to the radial asymptotics of its associated Mahler function.

来源字段：https://doi.org/10.4171/owr/2023/37

12268 | AMR-052-0084

### Dimension and ergodicity of geometrically finite Julia sets

P / partially_solved | AMR 问题清单汇编

For a geometrically finite rational map $f$, prove that either its Julia set is the whole sphere and $f$ is ergodic there, or its Julia set has Hausdorff dimension $\delta<2$; in the latter case determine its $\delta$-dimensional measure and the dynamics in that measure class.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12269 | AMR-052-0032

### Dimension and measure of Cremer Julia sets

P / partially_solved | AMR 问题清单汇编

Does every Cremer polynomial have Julia set of Hausdorff dimension two? Does every Cremer Julia set have Lebesgue measure zero?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12270 | OWR-5153-002

### Direction-Dimension Dichotomies on Nonarithmetic Veech Surfaces

O / open | Oberwolfach 报告问题汇编

Does the Hausdorff-dimension dichotomy for non-ergodic or divergent directions persist when the square torus is replaced by a non-arithmetic Veech surface, and what is the analogue of Perez Marco's condition?

来源字段：https://doi.org/10.4171/owr/2011/25

12271 | AMR-039-0014

### Discrete Ricci flow

P / partially_solved | AMR 问题清单汇编

Let the metric of a Markov space evolve by $$\frac{d}{dt}d(x,y)=-\kappa(x,y)d(x,y),$$ where $\kappa$ is computed from the current metric, with either a fixed or suitably evolving transition kernel. What can be said about the existence, behavior, and limiting geometry of this discrete Ricci flow?

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12272 | AIM-DYNAMICAL_SYSTEMS-0056

### Discrete rotating modes and their fast/slow reduction

P / partially_solved | AIM 研讨会问题汇编

\[ \dot x=f(x,y),\qquad \dot y=g(x,y), \tag{1.1} \]

and, after linearization at the stable fixed point,

\[ \begin{bmatrix}dx\\dy\end{bmatrix} =A\begin{bmatrix}x\\y\end{bmatrix}dt +\begin{bmatrix}\sigma _1\,dW_1\\ \sigma _2\,dW_2\end{bmatrix}. \tag{1.2} \]

The source says that this process can be approximated by “a constant times a rotation times pair of standard indepen-dent Ornstein-Uhlenbeck processes” and asks for the analogue in a discrete-time reaction--diffusion equation and its connection to more general fast/slow systems. The PDF breaks “independent” at the end of a line, so `indepen-dent` is a typesetting line-break artifact.

The same equations and discrete-time question occur as Problem 1 on page 1. Problem 13 repeats them and adds, verbatim, “Also, what is the connection of this system to more general fast/slow systems?” That additional clause is the main focus of the new contribution below.

The source does not define the discrete-time reaction--diffusion model, the approximation parameter, or “fast/slow.” We therefore make three explicit choices:

- a discrete reaction--diffusion linearization is a noisy coupled map on a finite undirected graph; - its discrete Ornstein--Uhlenbeck analogue is a stable vector AR(1), or VAR(1), process; - a stable spatial mode can serve as a fast subsystem coupled to variables whose increments are of order \(\varepsilon\).

These are controlled interpretations, not claims that the AIM question has a unique model-independent answer.

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12273 | AMR-039-0017

### Discrete scalar curvature

O / open | AMR 问题清单汇编

Define a scalar-curvature candidate by $S(x)=\int\kappa(x,y)\,dm_x(y)$, possibly with a distance-dependent weight. Does this quantity have useful geometric or probabilistic properties, such as controlling volume growth?

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12274 | AMR-039-0016

### Discrete sectional curvature

P / partially_solved | AMR 问题清单汇编

Replace the $T_1$ distance in the coarse-Ricci definition by $L^\infty$ transport, requiring a coupling that moves every point by at most $d(x,y)$. Does this define a useful discrete sectional curvature, can it be assigned a numerical value, and is it related to Alexandrov sectional curvature?

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12275 | OWR-1323-003

### Disjoint-Hypercyclic Operators on Banach Spaces

S / solved | Oberwolfach 报告问题汇编

Does every separable infinite-dimensional Banach space support a pair, or a larger family, of disjoint-hypercyclic operators?

来源字段：https://doi.org/10.4171/owr/2006/37

12276 | OWR-12724-008

### Distal Actions with Negative Infinite Sofic Entropy

O / open | Oberwolfach 报告问题汇编

Does there exist a distal action whose sofic topological entropy is $-\infty$?

来源字段：https://doi.org/10.4171/owr/2013/50

12277 | AIM-DYNAMICAL_SYSTEMS-0010

### Domain corrections and internal width bounds

P / partially_solved | AIM 研讨会问题汇编

Let \(G\) be \(F,T\) or \(V\) or a topological full group. Let \(\rho _1(g)\) be the commutator width of \(g\) and \(\rho _2(g)\) the minimum number of involutions in a factorization of \(g\).

1. Prove that \(\rho _1\) and \(\rho _2\) are bounded. 2. Prove that \(\rho _1=1\) and \(\rho _2=3\). 3. Is there a finitely presented simple group with commutator width greater or equal to \(2\)?

来源字段：http://aimpl.org/groupdynamorigin/3/

12278 | AMR-052-0017

### Domains of holomorphy for expanding-map components

O / open | AMR 问题清单汇编

Is $B(z^n)$ a domain of holomorphy? More generally, is every component of the space of expanding rational maps, or of expanding polynomials, a domain of holomorphy?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12279 | OWR-2094-004

### Douady Conjecture on Polynomial Linearizability

O / open | Oberwolfach 报告问题汇编

For a polynomial with an irrationally indifferent fixed point at the origin and multiplier $e^{2\pi i\theta}$, when is the fixed point linearizable, and what is the conformal radius of its Siegel disk? Is the Brjuno condition necessary for linearizability of every polynomial of degree $d\geq3$?

来源字段：https://doi.org/10.4171/owr/2008/48

12280 | OWR-9790354-001

### Dynamical Comparison for Amenable Minimal Nonamenable-Group Actions

P / partially_solved | Oberwolfach 报告问题汇编

Conjecture that every amenable minimal action of a nonamenable group on a compact metric space has dynamical comparison.

来源字段：https://doi.org/10.4171/owr/2022/3

12281 | OWR-16160-013

### Dynamical Consequences of Weak Warped-Cone Equivalence

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Weak Warped-Cone Equivalence Question [OWR-16160-014]

If two sequences of warped-cone levels are weak-equivalent, meaning that subsequences coarsely embed into one another in both directions, how much does this equivalence determine about the dynamics of the underlying group actions?

来源字段：https://doi.org/10.4171/owr/2018/19

12282 | OWR-744-007

### Dynamics and Algebraic Dimension of K3 and Hyperkähler Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Does a positive-entropy automorphism of a projective K3 surface have a dense orbit? For a hyperkähler manifold $M$, is $a(M)\in\{0,\dim M/2,\dim M\}$?

来源字段：https://doi.org/10.4171/owr/2004/42

12283 | AMR-052-0060

### Dynamics of exponential-trigonometric entire maps

P / partially_solved | AMR 问题清单汇编

Describe the dynamics of the entire maps $z\mapsto\lambda e^z\sin z$ and $z\mapsto\lambda e^z\cos z$.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12284 | OWR-12582-008

### Dynamics of Large-Coupling Quasiperiodic Schrödinger Cocycles

O / open | Oberwolfach 报告问题汇编

For the quasiperiodic Schrödinger cocycle with $f:\mathbb T\to\mathbb R$ of class $C^2$, exactly two nondegenerate critical points, Diophantine frequency $\omega$, and sufficiently large coupling $\lambda$, describe the dynamics of the induced projective map $\Phi_E:\mathbb T^2\to\mathbb T^2$ for every $E\in\mathbb R$ under more general assumptions on $f$.

来源字段：https://doi.org/10.4171/owr/2013/34

12285 | AMR-052-0051

### Dynamics on a Siegel-disk boundary

P / partially_solved | AMR 问题清单汇编

Can the boundary of a Siegel disk contain periodic points or points with positive Lyapunov exponent? Must the topological entropy of the boundary dynamics be zero?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12286 | OWR-15208-021

### Ebin Analysis for Anosov Diffeomorphism Moduli

O / open | Oberwolfach 报告问题汇编

Can a smooth-conjugacy class of Anosov diffeomorphisms have Hausdorff dimension greater than 2, and can Ebin's infinite-dimensional analysis of Riemannian metrics modulo diffeomorphisms be adapted to Anosov diffeomorphisms and their moduli?

来源字段：https://doi.org/10.4171/owr/2017/3

12287 | AMR-048-0006

### Eden's conjecture on local Lyapunov dimension

O / open | AMR 问题清单汇编

For a smooth dissipative dynamical system with a global attractor, is the supremum of the local Lyapunov dimension on the attractor attained at an equilibrium or at an unstable periodic orbit contained in the attractor?

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Dynamical_systems

12288 | AMR-052-0091

### Effective computation of entropy for surface diffeomorphisms

P / partially_solved | AMR 问题清单汇编

Given an explicitly specified smooth orientation-preserving diffeomorphism $F$ of the $2$-sphere, is its topological entropy Turing-computable to arbitrary prescribed error, and can it be computed in useful time? Resolve this in particular for Hénon and standard-map families; in the area-preserving cases, ask the analogous question for measure-theoretic entropy.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12289 | AMR-050-0001

### Elliptic-billiard invariant k_{107}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Here $\theta_i$ is the angle of $P$ at $P_i$, $L$ is its perimeter, and $l_i=|P''_i-P_i|$, $r_i=|P_{i+1}-P''_i|$. Prove that, for $N\equiv0\pmod4$, the quantity $(A'/A)\prod_i\sin(\theta_i/2)$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{107}.

背景链接：https://arxiv.org/abs/2004.12497

12290 | AMR-050-0002

### Elliptic-billiard invariant k_{108}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Here $\theta_i$ is the angle of $P$ at $P_i$, $L$ is its perimeter, and $l_i=|P''_i-P_i|$, $r_i=|P_{i+1}-P''_i|$. Prove that, for $N\equiv2\pmod4$, the quantity $(A'/A)/\prod_i\sin(\theta_i/2)$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{108}.

背景链接：https://arxiv.org/abs/2004.12497

12291 | AMR-050-0003

### Elliptic-billiard invariant k_{109}

S / solved | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Here $\theta_i$ is the angle of $P$ at $P_i$, $L$ is its perimeter, and $l_i=|P''_i-P_i|$, $r_i=|P_{i+1}-P''_i|$. Prove that, for odd $N$, the quantity $A/A''$ (and prove it equals the invariant $A'/A$) is constant as $P$ varies through the family. The source labels this assertion invariant code k_{109}.

背景链接：https://arxiv.org/abs/2004.12497

12292 | AMR-050-0004

### Elliptic-billiard invariant k_{110}

P / partially_solved | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Here $\theta_i$ is the angle of $P$ at $P_i$, $L$ is its perimeter, and $l_i=|P''_i-P_i|$, $r_i=|P_{i+1}-P''_i|$. Prove that, for even $N$, the quantity $AA''$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{110}.

背景链接：https://arxiv.org/abs/2004.12497

12293 | AMR-050-0005

### Elliptic-billiard invariant k_{111}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Here $\theta_i$ is the angle of $P$ at $P_i$, $L$ is its perimeter, and $l_i=|P''_i-P_i|$, $r_i=|P_{i+1}-P''_i|$. Prove that, for even $N$, the quantity $A'A''$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{111}.

背景链接：https://arxiv.org/abs/2004.12497

12294 | AMR-050-0006

### Elliptic-billiard invariant k_{114}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Here $\theta_i$ is the angle of $P$ at $P_i$, $L$ is its perimeter, and $l_i=|P''_i-P_i|$, $r_i=|P_{i+1}-P''_i|$. Prove that, for $N\equiv2\pmod4$, the quantity $\prod_i|P_i-f_1|$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{114}.

背景链接：https://arxiv.org/abs/2004.12497

12295 | AMR-050-0007

### Elliptic-billiard invariant k_{115}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Here $\theta_i$ is the angle of $P$ at $P_i$, $L$ is its perimeter, and $l_i=|P''_i-P_i|$, $r_i=|P_{i+1}-P''_i|$. Prove that, for $N\equiv0\pmod4$, the quantity $\prod_i|P'_i-f_1|$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{115}.

背景链接：https://arxiv.org/abs/2004.12497

12296 | AMR-050-0008

### Elliptic-billiard invariant k_{117}

P / partially_solved | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Here $\theta_i$ is the angle of $P$ at $P_i$, $L$ is its perimeter, and $l_i=|P''_i-P_i|$, $r_i=|P_{i+1}-P''_i|$. Prove that, for even $N$, the quantity each of $\prod_i l_i$ and $\prod_i r_i$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{117}.

背景链接：https://arxiv.org/abs/2004.12497

12297 | AMR-050-0009

### Elliptic-billiard invariant k_{118}

P / partially_solved | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Here $\theta_i$ is the angle of $P$ at $P_i$, $L$ is its perimeter, and $l_i=|P''_i-P_i|$, $r_i=|P_{i+1}-P''_i|$. Prove that, for odd $N$, the quantity each of $\sum_i l_i$ and $\sum_i r_i$, with value $L/2$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{118}.

背景链接：https://arxiv.org/abs/2004.12497

12298 | AMR-050-0010

### Elliptic-billiard invariant k_{120}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Here $\theta_i$ is the angle of $P$ at $P_i$, $L$ is its perimeter, and $l_i=|P''_i-P_i|$, $r_i=|P_{i+1}-P''_i|$. Prove that, for all $N$, the quantity $\sum_i\cos\angle P_i f_1 P_{i+1}$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{120}.

背景链接：https://arxiv.org/abs/2004.12497

12299 | AMR-050-0011

### Elliptic-billiard invariant k_{203,a}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Drop perpendiculars from $M$ to the sides of $P$ and let $A_M$ be the area of the resulting pedal polygon. Prove that, for $N\equiv0\pmod4$, every $M$, the quantity $AA_M$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{203,a}.

背景链接：https://arxiv.org/abs/2004.12497

12300 | AMR-050-0012

### Elliptic-billiard invariant k_{203,b}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Drop perpendiculars from $M$ to the sides of $P$ and let $A_M$ be the area of the resulting pedal polygon. Prove that, for $N\not\equiv2\pmod4$, $M=O$, the quantity $AA_M$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{203,b}.

背景链接：https://arxiv.org/abs/2004.12497

12301 | AMR-050-0013

### Elliptic-billiard invariant k_{204}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Drop perpendiculars from $M$ to the sides of $P$ and let $A_M$ be the area of the resulting pedal polygon. Prove that, for $N\equiv2\pmod4$, every $M$, the quantity $A/A_M$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{204}.

背景链接：https://arxiv.org/abs/2004.12497

12302 | AMR-050-0014

### Elliptic-billiard invariant k_{303,a}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Drop perpendiculars from $M$ to the sides of $P'$; let $A'_M$ and $C'_2$ be the area and signed area centroid of the resulting pedal polygon. Prove that, for $N\equiv2\pmod4$, every $M$, the quantity $A'A'_M$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{303,a}.

背景链接：https://arxiv.org/abs/2004.12497

12303 | AMR-050-0015

### Elliptic-billiard invariant k_{303,b}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Drop perpendiculars from $M$ to the sides of $P'$; let $A'_M$ and $C'_2$ be the area and signed area centroid of the resulting pedal polygon. Prove that, for $N\not\equiv0\pmod4$, $M=O$, the quantity $A'A'_M$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{303,b}.

背景链接：https://arxiv.org/abs/2004.12497

12304 | AMR-050-0016

### Elliptic-billiard invariant k_{304}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Drop perpendiculars from $M$ to the sides of $P'$; let $A'_M$ and $C'_2$ be the area and signed area centroid of the resulting pedal polygon. Prove that, for $N\equiv0\pmod4$, every $M$, the quantity $A'/A'_M$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{304}.

背景链接：https://arxiv.org/abs/2004.12497

12305 | AMR-050-0017

### Elliptic-billiard invariant k_{307}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Drop perpendiculars from $M$ to the sides of $P'$; let $A'_M$ and $C'_2$ be the area and signed area centroid of the resulting pedal polygon. Prove that, for even $N$, every $M$, the quantity the signed area centroid $C'_2$ of the pedal polygon is constant as $P$ varies through the family. The source labels this assertion invariant code k_{307}.

背景链接：https://arxiv.org/abs/2004.12497

12306 | AMR-050-0018

### Elliptic-billiard invariant k_{401}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. The antipedal polygon of a polygon $R$ with respect to $M$ has sides through its vertices perpendicular to the corresponding $R_i-M$; $A_M^*,C_0^*,C_2^*$ denote its area and centroids, with primes for $R=P'$. Prove that, for $N\equiv2\pmod4$, every $M$, the quantity $A'A_M^*$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{401}.

背景链接：https://arxiv.org/abs/2004.12497

12307 | AMR-050-0019

### Elliptic-billiard invariant k_{402}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. The antipedal polygon of a polygon $R$ with respect to $M$ has sides through its vertices perpendicular to the corresponding $R_i-M$; $A_M^*,C_0^*,C_2^*$ denote its area and centroids, with primes for $R=P'$. Prove that, for $N\equiv0\pmod4$, every $M$, the quantity $A'/A_M^*$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{402}.

背景链接：https://arxiv.org/abs/2004.12497

12308 | AMR-050-0020

### Elliptic-billiard invariant k_{403,a}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. The antipedal polygon of a polygon $R$ with respect to $M$ has sides through its vertices perpendicular to the corresponding $R_i-M$; $A_M^*,C_0^*,C_2^*$ denote its area and centroids, with primes for $R=P'$. Prove that, for odd $N$, $M=O$, the quantity $A_MA_M^*$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{403,a}.

背景链接：https://arxiv.org/abs/2004.12497

12309 | AMR-050-0021

### Elliptic-billiard invariant k_{403,b}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. The antipedal polygon of a polygon $R$ with respect to $M$ has sides through its vertices perpendicular to the corresponding $R_i-M$; $A_M^*,C_0^*,C_2^*$ denote its area and centroids, with primes for $R=P'$. Prove that, for $N\equiv0\pmod4$, $M=f_1$ or $f_2$, the quantity $A_MA_M^*$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{403,b}.

背景链接：https://arxiv.org/abs/2004.12497

12310 | AMR-050-0022

### Elliptic-billiard invariant k_{404}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. The antipedal polygon of a polygon $R$ with respect to $M$ has sides through its vertices perpendicular to the corresponding $R_i-M$; $A_M^*,C_0^*,C_2^*$ denote its area and centroids, with primes for $R=P'$. Prove that, for $N\equiv2\pmod4$, $M=f_1$ or $f_2$, the quantity $A_M^*/A_M$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{404}.

背景链接：https://arxiv.org/abs/2004.12497

12311 | AMR-050-0023

### Elliptic-billiard invariant k_{405}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. The antipedal polygon of a polygon $R$ with respect to $M$ has sides through its vertices perpendicular to the corresponding $R_i-M$; $A_M^*,C_0^*,C_2^*$ denote its area and centroids, with primes for $R=P'$. Prove that, for even $N$, $M=O,f_1,$ or $f_2$, the quantity the vertex centroid $C_0^*$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{405}.

背景链接：https://arxiv.org/abs/2004.12497

12312 | AMR-050-0024

### Elliptic-billiard invariant k_{406,a}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. The antipedal polygon of a polygon $R$ with respect to $M$ has sides through its vertices perpendicular to the corresponding $R_i-M$; $A_M^*,C_0^*,C_2^*$ denote its area and centroids, with primes for $R=P'$. Prove that, for even $N$, $M=O$, the quantity the vertex and signed area centroids ${C'_0}^*,{C'_2}^*$ (both equal $O$) is constant as $P$ varies through the family. The source labels this assertion invariant code k_{406,a}.

背景链接：https://arxiv.org/abs/2004.12497

12313 | AMR-050-0025

### Elliptic-billiard invariant k_{406,b}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. The antipedal polygon of a polygon $R$ with respect to $M$ has sides through its vertices perpendicular to the corresponding $R_i-M$; $A_M^*,C_0^*,C_2^*$ denote its area and centroids, with primes for $R=P'$. Prove that, for $N=4$, $M=f_1$ or $f_2$, the quantity the centroids ${C'_0}^*,{C'_2}^*$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{406,b}.

背景链接：https://arxiv.org/abs/2004.12497

12314 | AMR-050-0026

### Elliptic-billiard invariant k_{407}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. The antipedal polygon of a polygon $R$ with respect to $M$ has sides through its vertices perpendicular to the corresponding $R_i-M$; $A_M^*,C_0^*,C_2^*$ denote its area and centroids, with primes for $R=P'$. Prove that, for even $N$, $M=f_1$ or $f_2$, the quantity the vertex centroid ${C'_0}^*$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{407}.

背景链接：https://arxiv.org/abs/2004.12497

12315 | AMR-050-0027

### Elliptic-billiard invariant k_{501}

P / partially_solved | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Let $K,K',K''$ be the Steiner centroids of curvature of $P,P',P''$, and let $A_K,A'_{K'},A''_{K''}$ be the areas of their pedal polygons with respect to those centroids. Prove that, for odd $N$, the quantity $A/A_K$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{501}.

背景链接：https://arxiv.org/abs/2004.12497

12316 | AMR-050-0028

### Elliptic-billiard invariant k_{502}

P / partially_solved | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Let $K,K',K''$ be the Steiner centroids of curvature of $P,P',P''$, and let $A_K,A'_{K'},A''_{K''}$ be the areas of their pedal polygons with respect to those centroids. Prove that, for odd $N$, the quantity $A'/A'_{K'}$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{502}.

背景链接：https://arxiv.org/abs/2004.12497

12317 | AMR-050-0029

### Elliptic-billiard invariant k_{503}

P / partially_solved | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. Let $K,K',K''$ be the Steiner centroids of curvature of $P,P',P''$, and let $A_K,A'_{K'},A''_{K''}$ be the areas of their pedal polygons with respect to those centroids. Prove that, for odd $N$, the quantity $A''/A''_{K''}$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{503}.

背景链接：https://arxiv.org/abs/2004.12497

12318 | AMR-050-0030

### Elliptic-billiard invariant k_{601}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j=1,2$, let $q_{j,i}$ and $q^*_{j,i}$ be the distances from $f_j$ to the corresponding pedal and antipedal vertices. Let $\bar A_j,\bar A'_j,\bar A''_j$ be pedal areas of $P,P',P''$ with respect to $f_j$, and use a star for antipedal areas. Prove that, for odd $N$, the quantity $(\sum_iq_{1,i})(\sum_iq_{2,i})$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{601}.

背景链接：https://arxiv.org/abs/2004.12497

12319 | AMR-050-0031

### Elliptic-billiard invariant k_{602}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j=1,2$, let $q_{j,i}$ and $q^*_{j,i}$ be the distances from $f_j$ to the corresponding pedal and antipedal vertices. Let $\bar A_j,\bar A'_j,\bar A''_j$ be pedal areas of $P,P',P''$ with respect to $f_j$, and use a star for antipedal areas. Prove that, for all $N$, the quantity $(\prod_iq_{1,i})(\prod_iq_{2,i})$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{602}.

背景链接：https://arxiv.org/abs/2004.12497

12320 | AMR-050-0032

### Elliptic-billiard invariant k_{603}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j=1,2$, let $q_{j,i}$ and $q^*_{j,i}$ be the distances from $f_j$ to the corresponding pedal and antipedal vertices. Let $\bar A_j,\bar A'_j,\bar A''_j$ be pedal areas of $P,P',P''$ with respect to $f_j$, and use a star for antipedal areas. Prove that, for all $N$, the quantity $(\sum_iq^*_{1,i})/(\sum_iq^*_{2,i})$ (equal to $1$) is constant as $P$ varies through the family. The source labels this assertion invariant code k_{603}.

背景链接：https://arxiv.org/abs/2004.12497

12321 | AMR-050-0033

### Elliptic-billiard invariant k_{605,a}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j=1,2$, let $q_{j,i}$ and $q^*_{j,i}$ be the distances from $f_j$ to the corresponding pedal and antipedal vertices. Let $\bar A_j,\bar A'_j,\bar A''_j$ be pedal areas of $P,P',P''$ with respect to $f_j$, and use a star for antipedal areas. Prove that, for odd $N$, the quantity $\bar A'_1\bar A'_2$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{605,a}.

背景链接：https://arxiv.org/abs/2004.12497

12322 | AMR-050-0034

### Elliptic-billiard invariant k_{606}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j=1,2$, let $q_{j,i}$ and $q^*_{j,i}$ be the distances from $f_j$ to the corresponding pedal and antipedal vertices. Let $\bar A_j,\bar A'_j,\bar A''_j$ be pedal areas of $P,P',P''$ with respect to $f_j$, and use a star for antipedal areas. Prove that, for all $N$, the quantity $\bar A_1/\bar A_2=\bar A'_1/\bar A'_2$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{606}.

背景链接：https://arxiv.org/abs/2004.12497

12323 | AMR-050-0035

### Elliptic-billiard invariant k_{607}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j=1,2$, let $q_{j,i}$ and $q^*_{j,i}$ be the distances from $f_j$ to the corresponding pedal and antipedal vertices. Let $\bar A_j,\bar A'_j,\bar A''_j$ be pedal areas of $P,P',P''$ with respect to $f_j$, and use a star for antipedal areas. Prove that, for $N\equiv0\pmod4$, the quantity $\bar A^*_1/\bar A^*_2$ (equal to $1$) is constant as $P$ varies through the family. The source labels this assertion invariant code k_{607}.

背景链接：https://arxiv.org/abs/2004.12497

12324 | AMR-050-0036

### Elliptic-billiard invariant k_{608}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j=1,2$, let $q_{j,i}$ and $q^*_{j,i}$ be the distances from $f_j$ to the corresponding pedal and antipedal vertices. Let $\bar A_j,\bar A'_j,\bar A''_j$ be pedal areas of $P,P',P''$ with respect to $f_j$, and use a star for antipedal areas. Prove that, for even $N$, the quantity ${\bar A'_1}^*/{\bar A'_2}^*$ (equal to $1$) is constant as $P$ varies through the family. The source labels this assertion invariant code k_{608}.

背景链接：https://arxiv.org/abs/2004.12497

12325 | AMR-050-0037

### Elliptic-billiard invariant k_{609}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j=1,2$, let $q_{j,i}$ and $q^*_{j,i}$ be the distances from $f_j$ to the corresponding pedal and antipedal vertices. Let $\bar A_j,\bar A'_j,\bar A''_j$ be pedal areas of $P,P',P''$ with respect to $f_j$, and use a star for antipedal areas. Prove that, for even $N$, the quantity $\bar A''_1/\bar A''_2$ (equal to $1$) is constant as $P$ varies through the family. The source labels this assertion invariant code k_{609}.

背景链接：https://arxiv.org/abs/2004.12497

12326 | AMR-050-0038

### Elliptic-billiard invariant k_{610}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j=1,2$, let $q_{j,i}$ and $q^*_{j,i}$ be the distances from $f_j$ to the corresponding pedal and antipedal vertices. Let $\bar A_j,\bar A'_j,\bar A''_j$ be pedal areas of $P,P',P''$ with respect to $f_j$, and use a star for antipedal areas. Prove that, for even $N$, the quantity ${\bar A''_1}^*/{\bar A''_2}^*$ (equal to $1$) is constant as $P$ varies through the family. The source labels this assertion invariant code k_{610}.

背景链接：https://arxiv.org/abs/2004.12497

12327 | AMR-050-0039

### Elliptic-billiard invariant k_{701}

P / partially_solved | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. The evolute polygon of $R$ has vertices at successive intersections of perpendicular bisectors of its sides; $A_{ev},A'_{ev},A''_{ev}$ are the areas for $P,P',P''$. Prove that, for $N>4$, the quantity $A/A_{ev}$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{701}.

背景链接：https://arxiv.org/abs/2004.12497

12328 | AMR-050-0040

### Elliptic-billiard invariant k_{702}

P / partially_solved | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. The evolute polygon of $R$ has vertices at successive intersections of perpendicular bisectors of its sides; $A_{ev},A'_{ev},A''_{ev}$ are the areas for $P,P',P''$. Prove that, for $N>4$, the quantity $A'/A'_{ev}$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{702}.

背景链接：https://arxiv.org/abs/2004.12497

12329 | AMR-050-0041

### Elliptic-billiard invariant k_{703}

P / partially_solved | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. The evolute polygon of $R$ has vertices at successive intersections of perpendicular bisectors of its sides; $A_{ev},A'_{ev},A''_{ev}$ are the areas for $P,P',P''$. Prove that, for $N>4$, the quantity $A''/A''_{ev}$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{703}.

背景链接：https://arxiv.org/abs/2004.12497

12330 | AMR-050-0042

### Elliptic-billiard invariant k_{802}

S / solved | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for all $N$, the quantity $L_j^\dagger$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{802}.

背景链接：https://arxiv.org/abs/2004.12497

12331 | AMR-050-0043

### Elliptic-billiard invariant k_{803}

P / partially_solved | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for $N\ne4$, the quantity $\sum_i\cos\theta_{j,i}^\dagger$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{803}.

背景链接：https://arxiv.org/abs/2004.12497

12332 | AMR-050-0044

### Elliptic-billiard invariant k_{804,a}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for $N\equiv0\pmod4$, the quantity $AA_j^\dagger$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{804,a}.

背景链接：https://arxiv.org/abs/2004.12497

12333 | AMR-050-0045

### Elliptic-billiard invariant k_{804,b}

P / partially_solved | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for $N=4$, the quantity $AA_j^\dagger=4$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{804,b}.

背景链接：https://arxiv.org/abs/2004.12497

12334 | AMR-050-0046

### Elliptic-billiard invariant k_{805}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for $N\equiv2\pmod4$, the quantity $A/A_j^\dagger$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{805}.

背景链接：https://arxiv.org/abs/2004.12497

12335 | AMR-050-0047

### Elliptic-billiard invariant k_{806,a}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for all $N$, the quantity ${A'_j}^\dagger/A_j^\dagger$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{806,a}.

背景链接：https://arxiv.org/abs/2004.12497

12336 | AMR-050-0048

### Elliptic-billiard invariant k_{806,b}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for $N=4$, the quantity ${A'_j}^\dagger/A_j^\dagger=2$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{806,b}.

背景链接：https://arxiv.org/abs/2004.12497

12337 | AMR-050-0049

### Elliptic-billiard invariant k_{807}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for even $N$, the quantity $AA^\otimes$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{807}.

背景链接：https://arxiv.org/abs/2004.12497

12338 | AMR-050-0050

### Elliptic-billiard invariant k_{808}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for odd $N$, the quantity $A/A^\otimes$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{808}.

背景链接：https://arxiv.org/abs/2004.12497

12339 | AMR-050-0051

### Elliptic-billiard invariant k_{809}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for even $N$, the quantity $A'{A'}^\ominus$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{809}.

背景链接：https://arxiv.org/abs/2004.12497

12340 | AMR-050-0052

### Elliptic-billiard invariant k_{810}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for odd $N$, the quantity $A'/{A'}^\ominus$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{810}.

背景链接：https://arxiv.org/abs/2004.12497

12341 | AMR-050-0053

### Elliptic-billiard invariant k_{811}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for all $N$, the quantity $\sum_iw_i^2$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{811}.

背景链接：https://arxiv.org/abs/2004.12497

12342 | AMR-050-0054

### Elliptic-billiard invariant k_{812,a}

P / partially_solved | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for all $N$, the quantity $\sum_i\cos\psi_{1,i}$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{812,a}.

背景链接：https://arxiv.org/abs/2004.12497

12343 | AMR-050-0055

### Elliptic-billiard invariant k_{812,b}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for $N=4$, the quantity $\sum_i\cos\psi_{1,i}=0$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{812,b}.

背景链接：https://arxiv.org/abs/2004.12497

12344 | AMR-050-0056

### Elliptic-billiard invariant k_{813}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for all $N$, the quantity $A_{j,pol}/A_j^\dagger$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{813}.

背景链接：https://arxiv.org/abs/2004.12497

12345 | AMR-050-0057

### Elliptic-billiard invariant k_{814}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for all $N$, the quantity $A_{j,pol}/A_{j,dual}$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{814}.

背景链接：https://arxiv.org/abs/2004.12497

12346 | AMR-050-0058

### Elliptic-billiard invariant k_{815}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for odd $N$, the quantity $A_{j,ped}^\dagger A_{j,dual}$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{815}.

背景链接：https://arxiv.org/abs/2004.12497

12347 | AMR-050-0059

### Elliptic-billiard invariant k_{816}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for even $N$, the quantity $A_{j,ped}^\dagger/A_{j,dual}$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{816}.

背景链接：https://arxiv.org/abs/2004.12497

12348 | AMR-050-0060

### Elliptic-billiard invariant k_{817}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for $N\equiv0\pmod4$, the quantity $A_j^\dagger A_{j,ant}$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{817}.

背景链接：https://arxiv.org/abs/2004.12497

12349 | AMR-050-0061

### Elliptic-billiard invariant k_{818}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Prove that, for $N\equiv2\pmod4$, the quantity $A_j^\dagger/A_{j,ant}$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{818}.

背景链接：https://arxiv.org/abs/2004.12497

12350 | AMR-050-0062

### Elliptic-billiard invariant k_{903,a}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Subscripts $1,2$ distinguish the two foci; double primes refer to $P''$, and $A_j'^\ddagger$ is the area obtained by inverting $P'$ about a focus of its own elliptic vertex locus. Prove that, for odd $N$, the quantity $A_1^\dagger A_2^\dagger$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{903,a}.

背景链接：https://arxiv.org/abs/2004.12497

12351 | AMR-050-0063

### Elliptic-billiard invariant k_{904,a}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Subscripts $1,2$ distinguish the two foci; double primes refer to $P''$, and $A_j'^\ddagger$ is the area obtained by inverting $P'$ about a focus of its own elliptic vertex locus. Prove that, for odd $N$, the quantity ${A'_1}^\dagger{A'_2}^\dagger$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{904,a}.

背景链接：https://arxiv.org/abs/2004.12497

12352 | AMR-050-0064

### Elliptic-billiard invariant k_{905}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Subscripts $1,2$ distinguish the two foci; double primes refer to $P''$, and $A_j'^\ddagger$ is the area obtained by inverting $P'$ about a focus of its own elliptic vertex locus. Prove that, for even $N$, the quantity ${A''_1}^\dagger/{A''_2}^\dagger=1$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{905}.

背景链接：https://arxiv.org/abs/2004.12497

12353 | AMR-050-0065

### Elliptic-billiard invariant k_{906}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Subscripts $1,2$ distinguish the two foci; double primes refer to $P''$, and $A_j'^\ddagger$ is the area obtained by inverting $P'$ about a focus of its own elliptic vertex locus. Prove that, for even $N$, the quantity ${A'_1}^\ddagger/{A'_2}^\ddagger=1$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{906}.

背景链接：https://arxiv.org/abs/2004.12497

12354 | AMR-050-0066

### Elliptic-billiard invariant k_{907,a}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Subscripts $1,2$ distinguish the two foci; double primes refer to $P''$, and $A_j'^\ddagger$ is the area obtained by inverting $P'$ about a focus of its own elliptic vertex locus. Prove that, for odd $N$, the quantity $A_{1,dual}^\dagger A_{2,dual}$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{907,a}.

背景链接：https://arxiv.org/abs/2004.12497

12355 | AMR-050-0067

### Elliptic-billiard invariant k_{907,b}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Subscripts $1,2$ distinguish the two foci; double primes refer to $P''$, and $A_j'^\ddagger$ is the area obtained by inverting $P'$ about a focus of its own elliptic vertex locus. Prove that, for even $N$, the quantity $A_{1,dual}^\dagger/A_{2,dual}^\dagger=1$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{907,b}.

背景链接：https://arxiv.org/abs/2004.12497

12356 | AMR-050-0068

### Elliptic-billiard invariant k_{908,a}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Subscripts $1,2$ distinguish the two foci; double primes refer to $P''$, and $A_j'^\ddagger$ is the area obtained by inverting $P'$ about a focus of its own elliptic vertex locus. Prove that, for even $N$, the quantity $A_{1,ped}^\dagger/A_{2,ped}^\dagger=1$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{908,a}.

背景链接：https://arxiv.org/abs/2004.12497

12357 | AMR-050-0069

### Elliptic-billiard invariant k_{908,b}

O / open | AMR 问题清单汇编

Fix an ellipse with center $O$ and foci $f_1,f_2$, and let $P=(P_i)_{i=1}^N$ range over its Poncelet family of $N$-periodic billiard orbits tangent to a fixed confocal caustic. Let $P'$ be the outer polygon formed by consecutive tangent intersections and $P''$ the inner polygon of caustic tangency points; write $A,A',A''$ for their areas. For $j\in\{1,2\}$ invert the vertices of $P$ in the unit circle centered at $f_j$; denote the resulting polygon by $P_j^\dagger$, with perimeter $L_j^\dagger$, area $A_j^\dagger$, and angles $\theta_{j,i}^\dagger$. Primes refer to $P'$. Let $A^\otimes$ and ${A'}^\ominus$ be the areas after elliptic inversion of $P$ in its caustic and of $P'$ in the billiard ellipse. Let $A_{j,pol},A_{j,dual},A_{j,ant}$ be focal polar, dual, and antipedal areas, and $w_i,\psi_{j,i}$ the dual side lengths and polar angles. Subscripts $1,2$ distinguish the two foci; double primes refer to $P''$, and $A_j'^\ddagger$ is the area obtained by inverting $P'$ about a focus of its own elliptic vertex locus. Prove that, for $N=3$, the quantity $A_{1,ped}^\dagger/A_{2,ped}^\dagger=1$ is constant as $P$ varies through the family. The source labels this assertion invariant code k_{908,b}.

背景链接：https://arxiv.org/abs/2004.12497

12358 | AMR-045-0031

### Embedding one-sided subshifts

P / partially_solved | AMR 问题清单汇编

Let $T$ be a mixing one-sided SFT and $S$ a subshift with $h(S)<h(T)$. Give useful necessary and sufficient conditions for $S$ to be isomorphic to a subshift of $T$.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12359 | AMR-045-0032

### Embedding under a preimage bound

O / open | AMR 问题清单汇编

Let $S$ be a one-sided subshift and $T$ the full one-sided shift on $N$ symbols, with $h(S)<\log N$, and suppose no point of $S$ has more than $N$ preimages. Must $S$ be isomorphic to a subshift of $T$?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12360 | OWR-1382-001

### Emergence of Irreversible Macroscopic Dynamics

O / open | Oberwolfach 报告问题汇编

What additional nonmechanical, probabilistic information or assumption is needed to derive irreversible thermodynamic, macroscopic, or kinetic behavior from time-reversal-invariant microscopic Newtonian or Schrödinger dynamics?

来源字段：https://doi.org/10.4171/owr/2006/44

12361 | OWR-16767-001

### Energy Transfer Among Modes in Resonant Evolution

O / open | Oberwolfach 报告问题汇编

For the cubic resonant system

$$i\frac{d\alpha_n}{d\tau}=\sum_{n+j=k+l}S_{njkl}\overline{\alpha_j}\alpha_k\alpha_l,$$

how is the conserved energy $E=\sum_{n\geq0}E_n\lvert\alpha_n\rvert^2$ distributed among modes during evolution? In particular, under what conditions does energy flow from low to high modes?

来源字段：https://doi.org/10.4171/owr/2019/5

12362 | OWR-11579-001

### Ensemble Controllability from Individual Controllability

P / partially_solved | Oberwolfach 报告问题汇编

If every individual system in a parameterized ensemble is controllable, under what additional conditions is the ensemble jointly controllable?

来源字段：https://doi.org/10.4171/owr/2012/12

12363 | AMR-042-0006

### Entropy and Deligne periods

P / partially_solved | AMR 问题清单汇编

Let $\log_p:\mathbb{C}_p^*\to\mathbb{C}_p$ be the branch of the $p$-adic logarithm with $\log_p(p)=0$, and let $T_\lambda:x\mapsto\lambda x$ on $\mathbb{Q}_p$. Is there a meaningful entropy-like invariant $h_p$, for example invariant under topological conjugacy, such that $h_p(T_\lambda)=\log_p\lambda$?

背景链接：https://www.imath.kiev.ua/~skolyada/kevin.pdf

12364 | OWR-1319-012

### Entropy and Geometric Growth of Induced Endomorphisms

O / open | Oberwolfach 报告问题汇编

What is an analogue of Manning's theorem relating the topological entropy of a continuous map $f:M\to M$ to the geometric exponential growth rate of the induced endomorphism $f_*:\pi_1(M)\to\pi_1(M)$?

来源字段：https://doi.org/10.4171/owr/2006/33

12365 | OWR-9790354-005

### Entropy and Nondiagonal Asymptotic Pairs

P / partially_solved | Oberwolfach 报告问题汇编

Let an amenable countably infinite discrete group $\Gamma$ act expansively and algebraically on a compact metrizable space $X$. Is $h(\Gamma\curvearrowright X)>0$ equivalent to the existence of a nondiagonal asymptotic pair?

来源字段：https://doi.org/10.4171/owr/2022/3

12366 | AMR-039-0013

### Entropy decay

P / partially_solved | AMR 问题清单汇编

Does positive coarse Ricci curvature imply a useful exponential entropy-decay statement analogous to that obtained from logarithmic Sobolev inequalities, while correctly treating examples such as binomial distributions on the cube?

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12367 | OWR-2053-023

### Entropy of Globally Minimal Geodesic Flows

P / partially_solved | Oberwolfach 报告问题汇编

Let $M$ be a compact manifold of hyperbolic type, let $g$ be any Riemannian metric on $M$, let $\varphi_t$ be its geodesic flow on $SM$, and let $SM_{\min}$ consist of unit vectors whose lifts are globally minimizing geodesics. Is $h_{\mathrm{top}}(\varphi_t|_{SM_{\min}})=h_{\mathrm{Vol}}(g)$? If $M$ is a higher-genus surface, is the Liouville-measure entropy $h_{\mu_L}(\varphi_t)>0$ for every metric $g$?

来源字段：https://doi.org/10.4171/owr/2008/42

12368 | OWR-1319-001

### Entropy of Hamiltonian Flows on Restricted Contact-Type Hypersurfaces

O / open | Oberwolfach 报告问题汇编

Does positive topological entropy for the Hamiltonian flow on a fiberwise star-shaped hypersurface extend to every hypersurface of restricted contact type?

来源字段：https://doi.org/10.4171/owr/2006/33

12369 | OWR-11570-002

### Entropy Production on Folded Hyperbolic Fractals

P / partially_solved | Oberwolfach 报告问题汇编

Find other folded fractals $\Lambda$ that are hyperbolic for noninvertible maps $f$, together with families of $f$-invariant measures $\mu$ on $\Lambda$ whose entropy production is negative or positive.

来源字段：https://doi.org/10.4171/owr/2012/03

12370 | OWR-1319-029

### Entropy Spectra of Maximal-Rank Abelian Actions

O / open | Oberwolfach 报告问题汇编

Let $\alpha$ be a $C^{1+\varepsilon}$ action of $\mathbb Z^k$ on a $(k+1)$-manifold with an ergodic invariant measure, no proportional Lyapunov exponents, and at least one element of positive entropy. What values can the entropies of elements of $\alpha$ take? Are they algebraic integers of degree at most $k+1$?

来源字段：https://doi.org/10.4171/owr/2006/33

12371 | AMR-045-0040

### Entropy structures

P / partially_solved | AMR 问题清单汇编

For each $1\le r\le\infty$, characterize the entropy structures that can occur for $C^r$ diffeomorphisms of compact Riemannian manifolds.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12372 | AMR-045-0007

### Equal-entropy factors conjecture

O / open | AMR 问题清单汇编

Let $A,B$ be irreducible integer matrices of the same spectral radius. Suppose $\operatorname{tr}(A^n)>0$ implies $\operatorname{tr}(B^n)>0$ for every $n$, and the dimension module of $B$ is a quotient of a closed submodule of that of $A$. Prove that the shift $S_B$ is a factor of $S_A$.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12373 | AMR-045-0020

### Equal-entropy SFT covers

O / open | AMR 问题清单汇编

For $d>1$, must every $\mathbb Z^d$ sofic shift be a factor of a $\mathbb Z^d$ shift of finite type having the same entropy?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12374 | AMR-045-0021

### Equal-entropy subcovers

O / open | AMR 问题清单汇编

For $d>1$, if a continuous factor map sends a $\mathbb Z^d$ SFT $X$ onto a sofic shift $Y$, must $X$ contain a sofic subshift $W$ with $h(W)=h(Y)$ and image $Y$?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12375 | OWR-14749-001

### Equality of Rokhlin and Sofic Entropies

O / open | Oberwolfach 报告问题汇编

For a free ergodic probability-measure-preserving action of a sofic group $G$, does Rokhlin entropy equal sofic entropy whenever sofic entropy is not $-\infty$?

来源字段：https://doi.org/10.4171/owr/2016/41

12376 | AMR-045-0034

### Equivalence of canonical covers

O / open | AMR 问题清单汇编

Give a procedure deciding whether two canonical left-resolving irreducible covers of a one-sided irreducible SFT are related by an automorphism carrying one quotient relation to the other.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12377 | OWR-3396-007

### Equivalence of Dual Pisot Substitution Constructions

O / open | Oberwolfach 报告问题汇编

For primitive Pisot substitutions beyond the known two-letter unimodular cases, are the star-dual, Galois-dual, and geometric dual-substitution constructions equivalent, and does the resulting dual tiling become a proper tiling rather than a multiple covering, thereby proving the Pisot substitution conjecture?

来源字段：https://doi.org/10.4171/owr/2009/13

12378 | OWR-2044-001

### Error Propagation in Asteroid Dynamics Models

O / open | Oberwolfach 报告问题汇编

For an uncertain dynamical model of an asteroid moving mainly in the gravitational fields of the Sun and Jupiter, how much error is introduced by neglecting Saturn, how does that compare with errors in the initial conditions, and how do error balls propagate under the dynamics?

来源字段：https://doi.org/10.4171/owr/2008/33

12379 | AIM-DYNAMICAL_SYSTEMS-0054

### Escape and conditional limits after eclipse pruning

P / partially_solved | AIM 研讨会问题汇编

Problem 11. Is there some limiting distribution and/or escape rate in the case that can be found in the case of eclipsing scatterers?

Open Problems and Questions: Tuesday

I inspected the text operators in the original AIM PDF. Two source issues are now resolved.

Two minimal grammatical repairs have the same apparent meaning:

- “Is there some limiting distribution and/or escape rate that can be found in the case of eclipsing scatterers?” - “Is there some limiting distribution and/or escape rate in the case of eclipsing scatterers?”

The source does not define the initial ensemble, the time variable (physical time or collision count), the random variable whose distribution should converge, or the admissible geometry. I use the standard open-systems interpretation: survival mass, a collision-count escape rate, and the distribution conditioned on survival.

For pairwise disjoint convex obstacles \(K_1,\ldots,K_q\subset\mathbb R^d\), the **no-eclipse condition** is \[ \operatorname{conv}(K_i\cup K_j)\cap K_k=\varnothing \quad\text{for all distinct }i,j,k. \] “Eclipsing scatterers” means this condition fails: a third obstacle obstructs part, or sometimes all, of the line of sight between two others. Failure of no eclipse is not itself a dynamical model; different failures can cause symbolic pruning, grazing trapped rays, loss of compactness of the nongrazing trapped set, or none of these severe effects.

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12380 | OWR-4075-006

### Escape Dynamics Under Varying Billiard Holes

O / open | Oberwolfach 报告问题汇编

For a dispersing billiard map, how do the escape behavior, surviving mass, and asymptotics of $\mu_n$ change when the hole $H$ or the initial distribution $\mu_0$ varies?

来源字段：https://doi.org/10.4171/owr/2009/32

12381 | OWR-4137-004

### Escaping Boundary Points of Baker Domains

O / open | Oberwolfach 报告问题汇编

Let $f : \mathbb{C} \to \mathbb{C}$ be a transcendental entire function, and let $I(f) = \{z \in \mathbb{C} : f^n(z) \to \infty \text{ as } n \to \infty\}$ denote its escaping set. If $U \subset \mathbb{C}$ is a periodic escaping Fatou component (Baker domain) of $f$, must its boundary contain at least one escaping point, i.e., $\partial U \cap I(f) \neq \varnothing$?

来源字段：https://doi.org/10.4171/owr/2009/54

12382 | AIM-DYNAMICAL_SYSTEMS-0063

### Exact diffusion and chiral area transport in a four-direction collision model

P / partially_solved | AIM 研讨会问题汇编

**Problem 20.** Calculate transport coefficients or related physical quantities for a variety of billard systems.

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12383 | AIM-DYNAMICAL_SYSTEMS-0019

### Exact infinite-index Chebyshev specializations

S / solved | AIM 研讨会问题汇编

Is there any example where $G$ does not have finite index in $G'?$

Take $f(z)\in\mathbb Q[z]$, assume $0$ is **not** periodic. Consider two groups \[ G:=\lim_{n\to\infty}\operatorname{Gal}(f^n),\qquad G':=\lim\operatorname{Gal}(f^n(z)-t). \]

来源字段：http://aimpl.org/galarithdyn/3/

12384 | AIM-DYNAMICAL_SYSTEMS-0098

### Exact ODE realization by a flow-defect delay equation

P / partially_solved | AIM 研讨会问题汇编

**(5)** Global bifurcations in differential equations with state-dependent delay.

Some related issues are

- continuation for homoclinic or heteroclinic solutions; - continuation of (rapidly oscillating) periodic solutions for all relevant parameters.

**(6)** Analyticity of solutions of analytic delay differential equation (especially, with respect to the delay).

In particular, the “Whiskey Problem” suggested by Roger.

**(7)** Study of differential equations with unbounded finite delays as for instance arising in echo control.

Some particular aspects are

- equations with unbounded state-dependent delay; - equation with delay given by an infinite integral.

**(8)** How can a given ODE dynamics be realized in a class of differential equations with delay?

来源字段：https://aimath.org/WWN/variabletimelag/variabletimelag.pdf

12385 | AIM-DYNAMICAL_SYSTEMS-0016

### Exact prime-support endpoints over F_q(T)

P / partially_solved | AIM 研讨会问题汇编

Let $K=\mathbb F_q(T)$, let $f\in K(z)$ have degree at least two (or $f\in K[z]$ in the polynomial version), and let $\alpha\in K$. Replace rational primes by finite places of $K$. How large is the set of places dividing a nonzero numerator of some $f^n(\alpha)$? What analogues hold over other fields?

Taken literally, $f(z)\in\mathbb F_q(T)$ omits the variable in which $f$ is iterated. The two preceding records remove the ambiguity. Problem 1.1 defines \[ N(x)=\#\{p\le x:p\mid f^n(\alpha)\text{ for some }n\} \] and asks for its growth; Problem 1.2 asks how that growth differs for PCF and non-PCF maps. Accordingly, the minimal reconstruction of Problem 1.3 is:

For $K=\mathbb F_q(T)$ and a monic irreducible $P\in\mathbb F_q[T]$, write $v_P$ for its normalized valuation. We use the degree cutoff \[ \mathcal P_f(\alpha)=\{P:v_P(f^n(\alpha))>0 \text{ for some }n\ge0\text{ with }f^n(\alpha)\ne0\}, \qquad N_{f,\alpha}(D)=\#\{P\in\mathcal P_f(\alpha):\deg P\le D\}. \] The corresponding norm cutoff is $N_{f,\alpha}^{\rm norm}(X)= N_{f,\alpha}(\lfloor\log_qX\rfloor)$. Divisors of denominators instead measure visits to $\infty$, so numerator and denominator conventions must not be conflated.

来源字段：http://aimpl.org/galarithdyn/1/

12386 | AIM-DYNAMICAL_SYSTEMS-0069

### Exact transport in an equilibrium Pareto random flight

P / partially_solved | AIM 研讨会问题汇编

**Problem 26.** Calculate transport coefficients or related physical quantities for a variety of billard systems.

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12387 | OWR-744-004

### Exceptional Minimal Sets for Foliations of $\mathbb{CP}^2$

O / open | Oberwolfach 报告问题汇编

Every leaf of a holomorphic foliation of $\mathbb{CP}^2$ accumulates on a singularity.

来源字段：https://doi.org/10.4171/owr/2004/42

12388 | OWR-4137-014

### Existence of Ahlfors Islands Maps Between Compact Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

For compact Riemann surfaces $X,Y$ with $X$ hyperbolic and $W\subset Y$, when does an Ahlfors islands map $f:W\to X$ exist, and does the answer depend on $X$?

来源字段：https://doi.org/10.4171/owr/2009/54

12389 | OWR-4137-006

### Existence of Noncircular Degenerate Herman Rings

P / partially_solved | Oberwolfach 报告问题汇编

Let $f$ be rational and let $\gamma$ be an analytic $f$-periodic Jordan curve on which an iterate acts with irrational rotation number, and suppose $\gamma$ is not in the closure of a Herman ring. Can $\gamma$ be a degenerate Herman ring that is not a circle?

来源字段：https://doi.org/10.4171/owr/2009/54

12390 | AMR-039-0020

### Expanders

S / solved | AMR 问题清单汇编

Does there exist a family of bounded-degree expander graphs, with spectral gap bounded away from zero and diameter tending to infinity, whose coarse Ricci curvature is non-negative?

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12391 | AMR-052-0073

### Expanding conformal metric for nonrecurrent quadratics

P / partially_solved | AMR 问题清单汇编

If the quadratic polynomial $P_c(z)=z^2+c$ is nonrecurrent, does there exist a conformal metric $\rho(z)|dz|$ with integrable singularities in which $P_c$ is expanding on its Julia set?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12392 | AMR-045-0013

### Expansive components

O / open | AMR 问题清单汇编

Suppose a $\mathbb Z^2$ action $\alpha$ has $\alpha^{\boldsymbol n}$ an SFT for some $\boldsymbol n$. Can $\alpha$ have infinitely many expansive components? Can an expansive component have a boundary on a line of irrational slope?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12393 | AMR-045-0012

### Expansive directions of two-dimensional SFTs

P / partially_solved | AMR 问题清单汇编

For a $\mathbb Z^2$ shift of finite type $\alpha$, characterize the possible sets $E_1(\alpha)$ of expansive directions, especially under the assumption that $\alpha^{\boldsymbol n}$ is an SFT for some $\boldsymbol n$. The unrestricted first subquestion has since been solved; retain the constrained classification problem.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12394 | AMR-052-0045

### Explicit full-dimensional Julia set

P / partially_solved | AMR 问题清单汇编

Find an explicit rational map whose Julia set has Hausdorff dimension two. When such a Julia set has zero Lebesgue measure, identify a natural geometric measure on it.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12395 | OWR-11570-003

### Explicit Invariant Densities for Even Continued Fractions

O / open | Oberwolfach 报告问题汇编

For the set $E_{\mathrm{even}}$ of irrational numbers in $[0,1]$ with all continued-fraction entries even, whose Gauss-map invariant measure $\mu$ is equivalent to its Hausdorff-dimension packing measure $P_h$, find a closed explicit formula for the real-analytic extension of $d\mu/dP_h$ to $[0,1]$, or prove that no such formula exists.

来源字段：https://doi.org/10.4171/owr/2012/03

12396 | AIM-DYNAMICAL_SYSTEMS-0038

### Explicit size and wild ramification bounds for the 0-rooted tree of z^2+1 over Q_2

P / partially_solved | AIM 研讨会问题汇编

Let \(f(z)=z^2+1\) over \(\mathbb Q_2\). What is the size, etc., of the arboreal representation?

来源字段：http://aimpl.org/galarithdyn/12/

12397 | AIM-DYNAMICAL_SYSTEMS-0070

### Explicit symbolic martingales for a sequential hyperbolic class

P / partially_solved | AIM 研讨会问题汇编

Problem 27. Can one use function spaces to get martingale decompositions and concatenations for some class of hyperbolic maps?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12398 | AIM-DYNAMICAL_SYSTEMS-0029

### Exploding dynatomic index for cyclotomic exceptional maps

P / partially_solved | AIM 研讨会问题汇编

Begin to classify those \(f\) for which \(\operatorname{Gal}(K_f/K)\) does not have finite index in the thing it's supposed to. Ask this question again for \[ K_f'=K(\text{all periodic points of }f). \]

来源字段：http://aimpl.org/galarithdyn/7/

12399 | OWR-1460-014

### Exponential Auxiliary Systems for Four-Value-Sharing Meromorphic Functions

O / open | Oberwolfach 报告问题汇编

Let $f$ and $g$ be meromorphic functions sharing $a_1,a_2,a_3,a_4=\infty$, and define $$\psi_1=\frac{f'(f-g)}{\prod_{\nu=1}^3(f-a_\nu)},\qquad \psi_2=\frac{g'(f-g)}{\prod_{\nu=1}^3(g-a_\nu)}.$$ Classify the solutions of $\{\psi_1=e^z,\psi_2=e^{-z}\}$ and $\{\psi_1=e^{z^2},\psi_2=e^{-z^2}\}$, up to replacing $(f,g)$ by $(T\circ f(az+b),T\circ g(az+b))$, where $T$ is a Möbius transformation and $a\neq0$.

来源字段：https://doi.org/10.4171/owr/2007/09

12400 | AMR-045-0023

### Extension of a block code I

O / open | AMR 问题清单汇编

Let $T$ be a mixing sofic shift with a receptive fixed point. When is there a block code $f:T\to T$ and an SFT $T'\supset T$ such that $f(T')\subset T$?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12401 | AMR-045-0024

### Extension of a block code II

O / open | AMR 问题清单汇编

Let $f$ be a surjective block code from a mixing sofic shift $T$ to itself. When does there exist an SFT $T'\supset T$ such that $f(T')\subset T$?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12402 | OWR-1050-008

### Extensions of the Braid Conley Index

O / open | Oberwolfach 报告问题汇编

For the braid Conley index, which braid-group data suffice to compute the index? What is the smallest discretization for which a relative braid class has a single component? When does the index contain more dynamical information than the braid class alone? Does the theory extend to general fourth-order conservative equations?

来源字段：https://doi.org/10.4171/owr/2005/29

12403 | AMR-052-0029

### External rays landing at a Cremer point

P / partially_solved | AMR 问题清单汇编

Can any external ray land at a Cremer periodic point?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12404 | OWR-4137-018

### Extremal Hausdorff Measures for Hyperbolic Exponential Maps

O / open | Oberwolfach 报告问题汇编

For a hyperbolic exponential function $f(z)=\lambda e^z$, is the finite Hausdorff measure $H^h$ on each finite-width horizontal strip extremal in the sense of Kleinbock, Lindenstrauss, and Weiss?

来源字段：https://doi.org/10.4171/owr/2009/54

12405 | AMR-045-0008

### Factor maps between sofic shifts

P / partially_solved | AMR 问题清单汇编

For sofic shifts $S,T$ with $h(S)\ge h(T)$, give necessary and sufficient conditions for a factor map from $S$ onto $T$. The most fundamental unequal-entropy transitive case has been solved, but the general problem remains.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12406 | AIM-DYNAMICAL_SYSTEMS-0042

### Field intersections and an exact Kummer independence criterion

P / partially_solved | AIM 研讨会问题汇编

To what extent and when are the actions of Galois on preimages of $\alpha$ independent from the action of the preimages of $\beta?$.

来源字段：http://aimpl.org/galarithdyn/15/

12407 | AMR-045-0029

### Finitary images of IID processes

O / open | AMR 问题清单汇编

If a $\mathbb Z^d$ SFT has a unique measure of maximal entropy and that measure is Bernoulli, must an i.i.d. process map finitarily onto it?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12408 | OWR-3396-006

### Finite Correlation Determination of Primitive Substitutions

O / open | Oberwolfach 报告问题汇编

Is a Pisot substitution determined by its first three correlations? More generally, is every primitive substitution determined by finitely many correlations?

来源字段：https://doi.org/10.4171/owr/2009/13

12409 | OWR-13497-010

### Finite Fourier Feedback for Driftless-System Stabilization

P / partially_solved | Oberwolfach 报告问题汇编

For a driftless control-affine system $\dot x=\sum_{j=1}^{m}u_jf_j(x)$ on $D\subset\mathbb R^n$ satisfying $\operatorname{Lie}_{x=0}\{f_j:1\leq j\leq m\}=\mathbb R^n$, can one construct a finite trigonometric time-varying feedback $u_j=\sum_{k=-N}^{N}v_{kj}(x)e^{i\omega kt}$ whose closed-loop trivial solution is asymptotically stable, with piecewise-smooth coefficients $v_{kj}(x)=v_{-k,j}(x)\to0$ as $x\to0$?

来源字段：https://doi.org/10.4171/owr/2015/12

12410 | AIM-DYNAMICAL_SYSTEMS-0087

### Finite ramification, critical-height envelopes, and a tame-tower obstruction

P / partially_solved | AIM 研讨会问题汇编

Ramfication in pre-image towers

Let \(X\) be a projective variety, \(f:X/K \to X/K\) PCF. Let \(\alpha \in X(K)\). Look at the tower of fields from adding the pull-backs of \(\alpha\) \[ K_n=K(f^{-n}(\alpha)). \]

There exists a finite set of primes \(S\) such that all \(K_n\) are unramified outside of \(S\).

(Known for \(\mathbb P^1\)).

1. If not PCF and \(X=\mathbb P^1\) let \(S_n=\{\text{primes of \(K\) where \(K_n/K\) is ramified}\}\). Is the growth rate of \(\#S_n\) related to some height of \(f\), specifically \(\hat h_{\mathrm{crit}}\)? 2. Assume conjecture (\(X=\mathbb P^1\)). Can you arrange it so that all of the ramification in the tower is tame? (even for one example).

来源字段：http://aimpl.org/finitedynamics/4/

12411 | AIM-DYNAMICAL_SYSTEMS-0043

### Finite-depth globalization and all-primes obstructions for local arboreal representations

P / partially_solved | AIM 研讨会问题汇编

Given an arboreal Galois representations over $\mathbf Q_p$ (for all or some $p$) what are some nonobvious, sufficient, or necessary conditions under which all of these come from some global representation attached to some $f?$

来源字段：http://aimpl.org/galarithdyn/16/

12412 | OWR-705-005

### Finite-Field Detection of Area-Preserving Dynamics

P / partially_solved | Oberwolfach 报告问题汇编

Can reduction of an area-preserving map modulo a finite field detect whether the original map has dynamical signatures of order, chaos, or time-reversal symmetry?

来源字段：https://doi.org/10.4171/owr/2004/22

12413 | AIM-DYNAMICAL_SYSTEMS-0008

## Finite-kernel entropy rescaling and obstructions to intermediate SFT spectra

P / partially_solved | AIM 研讨会问题汇编

Let \(G\) be a finitely generated infinite ameanable group, \(A\) be a finite alphabet and define \[ \mathcal{F} =\left\{f:B\rightarrow A:B\subseteq G\text{ finite}\right\} \] a finite set of forbidden patterns and \[ S_{G,\mathcal{F},A} =\left\{x\in A^G:x\text{ contains no forbidden subpatterns}\right\}. 

Choose \(\{F_n\}_{n\in\mathbb N}\) a Følner sequence and define \[ \mathcal H_{G,\mathcal F,A} =\lim_{n\to+\infty} \frac{\log\#\{f:F_n\to A:f\text{ appears in some } x\in S_{G,\mathcal F,A}\}}{n}. \]

Define \[ E_G=\{\mathcal H_{G,\mathcal F,A}: \mathcal F\text{ is a finite set of forbidden subwords}\}. \]

1. Given a recursively presented and finitely generated infinite amenable group \(G\), what is \(E_G\)? 2. Does there exist a recursively presented and finitely generated infinite amenable group \(G\) such that \(E_{\mathbb Z}\subsetneq E_G\subsetneq E_{\mathbb Z^2}\)?

来源字段：http://aimpl.org/groupdynamorigin/2/

12414 | AIM-DYNAMICAL_SYSTEMS-0007

## Finite-relator detection of the free-subgroup cover property

P / partially_solved | AIM 研讨会问题汇编

Let $A$ be a finite alphabet and $\Omega$ be a 2-sided subshift, i.e., a closed subset of $A^\mathbb{Z}$ that is invariant under the shift map $\sigma$. An element $g \in Homeo(\Omega)$ belong to the full group of the shift ($F(\Omega)$) if each point in $\Omega$ has an open neighborhood $U$ such that $g|_U = (\sigma^n)|_U$ for some $n \in \mathbb{Z}$ depending on $U$ is equal to resembles a power of the shift map.

Is it true that for all finitely generated subgroup $H \leq F(\Omega)$, $H$ is either elementary amenable or for any $K$ finitely presented subgroup such that $K \twoheadrightarrow H$, $K$ contains a free subgroup?

**Recovered question.** Let $[[\sigma]]$ be the topological full group of $(\Omega,\sigma)$. Is every finitely generated subgroup $H\leq[[\sigma]]$ either elementary amenable, or does it have the property \[ \mathcal C(H):\quad \text{every finitely presented group $K$ admitting an epimorphism $K\twoheadrightarrow H$ contains a nonabelian free subgroup $F_2$?} \]

来源字段：http://aimpl.org/groupdynamorigin/2/

12415 | OWR-12864-005

## Finite-Time Invariant Manifolds and Unsteady Transport

O / open | Oberwolfach 报告问题汇编

Develop a theory of finite-time invariant manifolds that genuinely captures the time dependence of aperiodic unsteady flows. In particular, quantify transport created by intersecting or nonintersecting stable and unstable manifold segments and relate those manifolds to practical flow-barrier diagnostics.

来源字段：https://doi.org/10.4171/owr/2014/04

12416 | AIM-DYNAMICAL_SYSTEMS-0061

## Finite-time Lyapunov diagnostics for nonstationary matrix products

P / partially_solved | AIM 研讨会问题汇编

Problem 18. What can be said about spectral properties/Lyapunov spectrum in sequential or nonstationary settings?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12417 | AMR-039-0002

## Finsler manifolds

P / partially_solved | AMR 问题清单汇编

The space $\mathbb{R}^N$ equipped with an $L^p$ norm has zero coarse Ricci curvature. Does this observation yield useful results for Finsler manifolds?

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12418 | OWR-4990373-004

## Fixation Times and Consensus in Majority Dynamics

O / open | Oberwolfach 报告问题汇编

For majority dynamics on a given graph, if fixation occurs, how quickly is the final configuration reached, and is unanimity reached or do both opinions survive?

来源字段：https://doi.org/10.4171/owr/2021/4

12419 | AIM-DYNAMICAL_SYSTEMS-0001

## Fixed-point proportions as boundary measure and a family approaching full Hausdorff dimension

P / partially_solved | AIM 研讨会问题汇编

It asks the following five broad questions after defining

\[ \operatorname{FPP}(G)=\lim_{n\to\infty} \frac{\#\{g\in G_n:g\text{ fixes a vertex on level }n\}}{\#G_n}: \]

1. What can be said for groups acting on rooted trees? 2. On a fixed tree, can one find groups with nonzero fixed-point proportion and Hausdorff dimensions converging to (1)? 3. What can be said for iterated Galois groups? 4. What can be said for iterated monodromy groups of complex rational functions? 5. What can be said for complex polynomials that are not dynamically exceptional?

Three repairs are necessary before doing mathematics.

* The canonical JSON repeats the paragraph defining (k_n,G_n,G_\infty); the PDF has it only once. * The simultaneous phrases “(t\in k),” (G_n=\operatorname{Gal}(k_n/k(t))), and “(k=\mathbb C(t)) with (t) transcendental over (\mathbb C)” are inconsistent. In the generic formulation, (k) is the constant field, (t) is transcendental over (k), (K=k(t)), and (K_n=K(f^{-n}(t))) is the splitting field of (f^n(x)-t) (with the analogous numerator-minus-(t)-times-denominator polynomial for a rational map). Then (G_n=\operatorname{Gal}(K_n/K)). In the specialized formulation one instead chooses (a\in k), takes (K_n=k(f^{-n}(a))), and uses (\operatorname{Gal}(K_n/k)). * The cited identifier MR2162164 is Nekrashevych’s book [Nek05]. Proposition 6.4.2 says, for a **postcritically finite** complex rational function, that its profinite iterated monodromy group is the Galois group of the union of the generic splitting fields. The record’s citation arXiv:1204.2843 is Jones’s paper [Jon15], whose main dynamical theorem concerns postcritically finite **polynomials**, not arbitrary rational functions. A later rational-function theorem is [BJKL22], under additional hypotheses. Thus the record overstates both the scope of the book proposition and the Jones result.

The repaired generic setting will be used below. We assume the map is separable. Then a generic fiber of (f^n) has (d^n) distinct points, the preimage tree is the regular (d)-ary rooted tree, and the Galois action on all roots is faithful because those roots generate the splitting field. In positive characteristic these qualifications cannot be omitted: inseparability can destroy the regular tree.

来源字段：http://aimpl.org/groupdynamorigin/1/

12420 | AMR-045-0052

## Flow equivalence of subshifts

P / partially_solved | AMR 问题清单汇编

Classify subshifts up to flow equivalence.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12421 | AIM-DYNAMICAL_SYSTEMS-0068

## Flux normalization and clustering tests for billiard strip targets

P / partially_solved | AIM 研讨会问题汇编

**Problem 25.** What can be said about hitting times/EVT to shrinking targets in billiards where the target is not a ball but a strip induced by a set in configuration space?

Problem 25. What can be said about hitting times/EVT to shrinking targets in billards where the target is not a ball but a strip induced by a set in configuration space?

Open Problems and Questions: Thursday

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12422 | OWR-13497-007

## Formal Languages for Hybrid-System Control

O / open | Oberwolfach 报告问题汇编

Specify suitable formal languages, analogous to finite-state languages defined by accepting states, for controlling hybrid systems whose inputs combine a continuous control function $u$ on a finite interval with a symbol string $v$.

来源字段：https://doi.org/10.4171/owr/2015/12

12423 | AIM-DYNAMICAL_SYSTEMS-0083

### Formal-period covers and the phase-quotient ramification budget

P / partially_solved | AIM 研讨会问题汇编

Fix $d$. If $n >>_d 1$, then is $M_d(n)$ of general type, where $M_d(n)$ marks periodic points of formal period $n$.

Known for abelian varieties and for $d=2$, $n=1,\ldots,5$ $M_d(n)$ is rational and $M_2(6)$ is general type (Blanc-Canci)

Its exact wording is:

The record adds, verbatim:

The notation $n\gg_d1$ is read as “all sufficiently large $n$, with threshold allowed to depend on $d$.”

There are two bibliographic corrections to the literature note. First, the small-period paper has three authors: Jérémy Blanc, Jung Kyu Canci, and Noam D. Elkies. Second, its theorem is stated for a point of **exact** period $n$, whereas Michelle Manes's $M_d(n)$ is constructed using **formal** period. Blanc--Canci--Elkies explicitly say that their exact-period variety is a $\operatorname{PGL}_2$-invariant open subset of Manes's formal-period variety. Thus the birational assertions agree on the relevant component, but the moduli functors are not literally identical.

The abelian-variety analogy is broadly correct but too imprecise to be a citation. A precise modern result of Abramovich and Várilly-Alvarado says that for a locally closed $X\subset\mathcal A_g$, all irreducible components of its pullback to sufficiently high full-level moduli are of general type. Full abelian level structure is much richer than one marked dynamical periodic point, so this is an analogy rather than a reduction.

来源字段：http://aimpl.org/finitedynamics/2/

12424 | OWR-1460-013

### Four-Value-Sharing Meromorphic Functions with $\psi=1$

O / open | Oberwolfach 报告问题汇编

Let $f$ and $g$ be meromorphic functions sharing the four values $a_1,a_2,a_3,a_4=\infty$. Define $\psi=f'g'(f-g)^2/\prod_{\nu=1}^3((f-a_\nu)(g-a_\nu))$, $\psi_1=f'(f-g)/\prod_{\nu=1}^3(f-a_\nu)$, and $\psi_2=g'(f-g)/\prod_{\nu=1}^3(g-a_\nu)$. Determine all tuples $(a_1,a_2,a_3,f,g)$ satisfying $\psi=1$.

来源字段：https://doi.org/10.4171/owr/2007/09

12425 | AMR-041-0005

### Fractal caustics

O / open | AMR 问题清单汇编

Are there geodesic flows or Birkhoff billiards with fractal caustics? More specifically, for every $1\leq s<2$, is there a caustic of a convex billiard with Hausdorff dimension $s$; and for every $s\geq1$, is there a Riemannian manifold $M$ and a point whose caustic has Hausdorff dimension $s$?

背景链接：https://people.math.harvard.edu/~knill/seminars/intr/

12426 | AMR-041-0008

### Free gas in a moving container

P / partially_solved | AMR 问题清单汇编

Does a free gas coupled to a convex rigid container by conservation of momentum converge weakly to equilibrium, with the container-which moves only by translation-coming to rest?

背景链接：https://people.math.harvard.edu/~knill/seminars/intr/

12427 | OWR-11126-005

### Frequency Images from Nonadapted Hermitian Structures

S / solved | Oberwolfach 报告问题汇编

别名：Frequency Maps of Central N-Body Equilibria [OWR-11126-006]

For the frequency map of angular momenta of relative equilibria, do non-adapted Hermitian structures contribute nothing to its image; equivalently, is $\operatorname{Im}F=A_{P_0}$?

来源字段：https://doi.org/10.4171/owr/2011/34

12428 | AIM-DYNAMICAL_SYSTEMS-0003

### Full-group criterion and a nonamenability obstruction to IMG orbit cyclicization

P / partially_solved | AIM 研讨会问题汇编

Suppose that \(\Gamma\) is a group acting on the boundary of a rooted tree, e.g. the iterated monodromy group of a continuous map. Is there a \(\mathbb Z\)-action with the same orbits? Find an explicit homeomorphism implementing the orbit equivalence.

**Intended problem.** Given the boundary action of an iterated monodromy group \(G=IMG(f)\), does there exist \(\varphi\in\operatorname{Homeo}(\partial T)\) such that \(E_\varphi=E_G\), where \(E_\varphi\) is the orbit relation of the cyclic group \(\langle\varphi\rangle\)? Can \(\varphi\), or an orbit-equivalence conjugacy producing it, be made explicit?

来源字段：http://aimpl.org/groupdynamorigin/1/

12429 | OWR-13855-002

### Full-Measure Families of Ergodic Irrational Polygons

O / open | Oberwolfach 报告问题汇编

Does there exist a set of full Lebesgue measure consisting of ergodic non-rational polygons with any prescribed number of edges?

来源字段：https://doi.org/10.4171/owr/2015/34

12430 | AMR-052-0061

### Full-plane Julia sets in the exponential family

P / partially_solved | AMR 问题清单汇编

For $E_\lambda(z)=\lambda e^z$, characterize completely the parameters $\lambda$ for which $J(E_\lambda)=\mathbb C$.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12431 | AMR-045-0018

### Full-support invariant measures

P / partially_solved | AMR 问题清单汇编

For the symbolic system $X$ constructed in Section 14 of the source from a commuting pair of full-shift endomorphisms, is Haar measure its only invariant ergodic Borel probability measure with full support?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12432 | AMR-039-0010

### Functional inequalities

P / partially_solved | AMR 问题清单汇编

Can concentration consequences of positive coarse Ricci curvature be formulated as transportation or other functional inequalities? In a coarse setting, determine a suitable version that permits non-Gaussian tails at small measures or scales, perhaps using a quadratic-then-linear transport cost.

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12433 | AIM-DYNAMICAL_SYSTEMS-0039

### Functorial operations on arboreal representations

P / partially_solved | AIM 研讨会问题汇编

Define a category containing the rooted inverse-image Galois actions attached to pointed finite dynamical systems $(X,f,\alpha)$. Determine which standard operations on these actions are induced by coordinate conjugacy, semiconjugacy, iteration, extension of the ground field, and products of dynamical systems; identify exact obstructions to treating these operations as unrestricted operations on abstract tree representations.

来源字段：http://aimpl.org/galarithdyn/13/

12434 | OWR-12593-003

### Furstenberg Classification of Times-$p$ and Times-$q$ Invariant Measures

O / open | Oberwolfach 报告问题汇编

Let $p,q\geq 2$ be multiplicatively independent. Are the probability measures on $\mathbb R/\mathbb Z$ invariant under both $T_p(x)=px\pmod 1$ and $T_q(x)=qx\pmod 1$ precisely the finitely supported invariant measures, Lebesgue measure, and their convex combinations?

来源字段：https://doi.org/10.4171/owr/2013/45

12435 | AMR-045-0017

### Furstenberg times-p-times-q problem

P / partially_solved | AMR 问题清单汇编

For multiplicatively independent integers $p,q>1$, can there exist a nonatomic Borel probability measure on the circle, other than Haar measure, that is jointly invariant under $x\mapsto px\pmod1$ and $x\mapsto qx\pmod1$?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12436 | OWR-14299091-002

### Generalized Fried Conjecture for Anosov Flows

P / partially_solved | Oberwolfach 报告问题汇编

Identify the dynamical counterpart of the Ray-Singer metric, formulate the corresponding generalized Fried conjecture, and prove the predicted equality between the Ray-Singer norm and the dynamical invariant for general Anosov flows.

来源字段：https://doi.org/10.4171/owr/2025/13

12437 | AMR-045-0006

### Generalized spectral conjecture

S / solved | AMR 问题清单汇编

Let $S\subset\mathbb R$ be a unital subring and let $A$ be a square matrix over $S$ whose nonzero spectrum satisfies the spectral-conjecture conditions. Prove that $A$ is strong shift equivalent over $S$ to a primitive matrix.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12438 | OWR-1589-003

### Generic Finite Complexity in Multidimensional Dispersing Billiards

P / partially_solved | Oberwolfach 报告问题汇编

For finite-horizon dispersing billiard configurations in a $d$-dimensional flat torus, is the subset satisfying the finite-complexity condition generic in the $C^3$ topology? Equivalently, does a typical configuration have no phase point whose trajectory undergoes tangential collisions more than $2d-2$ times?

来源字段：https://doi.org/10.4171/owr/2007/33

12439 | AMR-036-0044

### Generic static-output stabilizability

O / open | AMR 问题清单汇编

For real matrices $A\in\operatorname{Mat}_{n\times n}$, $B\in\operatorname{Mat}_{n\times p}$, and $C\in\operatorname{Mat}_{m\times n}$ with $n=mp$, determine for which pairs $(m,p)$ a generic system $\dot x=Ax+Bu$, $y=Cx$ admits a real static output feedback $u=Ky$ such that every eigenvalue of $A+BKC$ lies in the open left half-plane.

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

12440 | OWR-4330-008

### Geodesic Flows on Koch Snowflake Fractal Surfaces

O / open | Oberwolfach 报告问题汇编

Does the inverse limit of the Veech groups of the prefractal Koch-snowflake surfaces define a dynamically relevant object yielding a well-defined geodesic flow on a fractal flat surface dynamically equivalent to the Koch-snowflake billiard flow?

来源字段：https://doi.org/10.4171/owr/2010/17

12441 | AIM-DYNAMICAL_SYSTEMS-0085

### Gleason irreducibility and exact critical portraits

P / partially_solved | AIM 研讨会问题汇编

Irreducibility of Gleason polynomials

Define the Gleason polynomials as the polynomials in \(c\) such that \(f_c(x)=x^2+c\) is post-critically finite.

Are the Gleason polynomials irreducible?

Is \(G_n\) irreducible over \(\mathbb Q\) for every \(n\geq1\)?

来源字段：http://aimpl.org/finitedynamics/3/

12442 | OWR-12582-004

### Global Rigidity of Diophantine Pseudo-Rotations

P / partially_solved | Oberwolfach 报告问题汇编

Are Diophantine pseudo-rotations always globally conjugate to a rotation?

来源字段：https://doi.org/10.4171/owr/2013/34

12443 | AIM-DYNAMICAL_SYSTEMS-0064

### Gluing blockwise martingale decompositions across seams

P / partially_solved | AIM 研讨会问题汇编

Problem 21. Can one use function spaces to get martingale decompositions and concatenations for some class of hyperbolic maps?

Can one use generalized function spaces to get martingales for concatenations of random hyperbolic maps?

No mathematical symbols are missing.

This is a repeated workshop question. On the preceding day, Problem 15 asked:

Problem 21 replaces “generalized function spaces” by “function spaces” and “martingales for concatenations” by “martingale decompositions and concatenations,” but the research intent is the same. Problem 27 on the following day repeats Problem 21 verbatim. The repetition is useful evidence that this was a persistent workshop theme rather than three unrelated questions.

The completed analysis of Problem 15 (AIM-DYNAMICAL_SYSTEMS-0058) derived a global sequential Gordin recursion from summable transfer-cocycle memory. When do the local martingale-coboundary decompositions glue globally, and what precisely is paid at the seams?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12444 | AMR-045-0009

### Good finitary conjecture

O / open | AMR 问题清单汇编

Prove that two mixing Markov shifts admit a magic-word isomorphism exactly when they have the same beta function, the same ratio group $\Delta$, and the same canonical generator of the weights quotient group $\Gamma/\Delta$.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12445 | OWR-12724-004

### Gottschalk Surjunctivity Problem

O / open | Oberwolfach 报告问题汇编

Is every countable discrete group $\Gamma$ surjunctive; that is, for every finite alphabet $A$, is every continuous injective $\Gamma$-equivariant map $A^\Gamma\to A^\Gamma$ necessarily surjective?

来源字段：https://doi.org/10.4171/owr/2013/50

12446 | AIM-DYNAMICAL_SYSTEMS-0055

### Green--Kubo reduction and an exact persistent-flight transport diagnostic

P / partially_solved | AIM 研讨会问题汇编

Develop methods to calculate diffusion, conductivity, mobility, and related transport quantities across multiple deterministic or random billiard models.

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12447 | AMR-120-0001

### Group Actions on 1-Manifolds - Question 1

O / open | AMR 问题清单汇编

Does there exist a lattice in a higher-rank simple Lie group admitting a nontrivial action by orientation-preserving homeomorphisms of the interval ?

背景链接：https://arxiv.org/abs/1712.06462

12448 | AMR-120-0010

### Group Actions on 1-Manifolds - Question 11

O / open | AMR 问题清单汇编

Does the group of piecewise-affine circle homeomorphisms contain distorted elements ?

背景链接：https://arxiv.org/abs/1712.06462

12449 | AMR-120-0011

### Group Actions on 1-Manifolds - Question 12

O / open | AMR 问题清单汇编

What are the groups of piecewise-projective homeomorphisms of the interval/circle that are topologically conjugate to groups of $C^1$ diffeomorphisms ?

背景链接：https://arxiv.org/abs/1712.06462

12450 | AMR-120-0012

### Group Actions on 1-Manifolds - Question 13

O / open | AMR 问题清单汇编

Given $0 < \alpha < \beta < 1$, does there exist a finitely-generated group of $C^{1+\alpha}$ diffeomorphisms of the circle/interval that does not embed into the group of $C^{1+\beta}$ diffeomorphisms ?

背景链接 : https://arxiv.org/abs/1712.06462

12451 | AMR-120-0013

### Group Actions on 1-Manifolds - Question 14

O / open | AMR 问题清单汇编

Let $G$ be a finitely-generated group of $C^2$ circle diffeomorphisms. Assume that $G$ admits an exceptional minimal set $\Lambda$. Is the Lebesgue measure of $\Lambda$ equal to zero ?

背景链接 : https://arxiv.org/abs/1712.06462

12452 | AMR-120-0014

### Group Actions on 1-Manifolds - Question 17

O / open | AMR 问题清单汇编

Let $G$ be a finitely-generated group of $C^2$ circle diffeomorphisms. If the action of $G$ is minimal, is it necessarily ergodic with respect to the Lebesgue measure ?

背景链接 : https://arxiv.org/abs/1712.06462

12453 | AMR-120-0015

### Group Actions on 1-Manifolds - Question 18

O / open | AMR 问题清单汇编

Let $G$ be a finitely-generated group of $C^2$ orientation-preserving circle diffeomorphisms. Does there exist a non-degenerate probability distribution on $G$ for which the stationary measure is absolutely continuous with respect to the Lebesgue measure ?

背景链接 : https://arxiv.org/abs/1712.06462

12454 | AMR-120-0016

### Group Actions on 1-Manifolds - Question 19

O / open | AMR 问题清单汇编

Given a symmetric, finitely-supported, non-degenerate probability distribution on Thompson's group $T$, does the Poisson-Furstenberg boundary of $T$ with respect to it identifies with the circle endowed with the corresponding stationary measure ?

背景链接 : https://arxiv.org/abs/1712.06462

12455 | AMR-120-0002

### Group Actions on 1-Manifolds - Question 2

O / open | AMR 问题清单汇编

Does there exist an infinite, finitely-generated Kazhdan group of circle homeomorphisms ?

背景链接 : https://arxiv.org/abs/1712.06462

12456 | AMR-120-0017

### Group Actions on 1-Manifolds - Question 20

O / open | AMR 问题清单汇编

Does there exist a faithful action of $\mathbb{Z}^2$ by $C^{\infty}$ orientation-preserving circle diffeomorphisms that is $C^{\infty}$ structurally stable ?

背景链接 : https://arxiv.org/abs/1712.06462

12457 | AMR-120-0018

### Group Actions on 1-Manifolds - Question 21

O / open | AMR 问题清单汇编

Given $r \geq 1$, is the subset of $\mathrm{Diff}_+^{r}(\mathrm{S}^1)^2$ consisting of pairs of commuting diffeomorphisms locally connected ?

背景链接 : https://arxiv.org/abs/1712.06462

12458 | AMR-120-0019

### Group Actions on 1-Manifolds - Question 22

O / open | AMR 问题清单汇编

Given a faithful action $\phi_0$ of a finitely generated group $G$ by $C^{\infty}$ circle diffeomorphisms admitting an exceptional minimal set, does there exist a path $\phi_t$ of faithful actions of $G$ that is continuous in the $C^{\infty}$ topology and starts with $\phi_0 = \phi$ so that each $\phi_t$ admits an exceptional minimal set for $t < 1$ and $\phi_1$ is minimal ?

背景链接 : https://arxiv.org/abs/1712.06462

12459 | AMR-120-0020

### Group Actions on 1-Manifolds - Question 23

O / open | AMR 问题清单汇编

Let $G$ be a finitely-generated group of circle homeomorphisms whose action is $C^0$ structurally stable. Suppose that $G$ admits infinitely many non semiconjugate actions on the circle. Does $G$ contain a free subgroup in two generators ?

背景链接 : https://arxiv.org/abs/1712.06462

12460 | AMR-120-0021

### Group Actions on 1-Manifolds - Question 24

O / open | AMR 问题清单汇编

Let $G$ be a finitely-generated group of $C^1$ circle diffeomorphisms. Suppose that $G$ has no resilient pairs. Given $\varepsilon > 0$, can $G$ be conjugated (by a homeomorphism) into a group of Lipschitz homeomorphisms for which the Lipschitz constants of the generators are all $\leq e^{\varepsilon}$ ?

背景链接 : https://arxiv.org/abs/1712.06462

12461 | AMR-120-0022

### Group Actions on 1-Manifolds - Question 25

O / open | AMR 问题清单汇编

Let $f$ be a $C^2$ circle diffeomorphism of irrational rotation number $\rho$. Does the set of $C^2$ conjugates of $f$ contain the rotation $R_{\rho}$ in its $C^2$-closure ?

背景链接：https://arxiv.org/abs/1712.06462

12462 | AMR-120-0023

### Group Actions on 1-Manifolds - Question 26

O / open | AMR 问题清单汇编

For which values of $r > 1$ there exists $s \geq r$ such that for every $C^s$ circle diffeomorphism $f$ of irrational rotation number $\rho$, the sequence $f^{q_n}$ converges to the identity in the $C^r$ topology, where $(q_n)$ is the sequence of denominators of the rational approximations of $\rho$ ?

背景链接：https://arxiv.org/abs/1712.06462

12463 | AMR-120-0024

### Group Actions on 1-Manifolds - Question 27

O / open | AMR 问题清单汇编

Let $f$ be a $C^2$ circle diffeomorphism of irrational rotation number $\rho$. Given $\varepsilon > 0$, let $M_{\varepsilon}$ be the *mapping torus* of $f$ over $\mathrm{S}^1 \times [0,\varepsilon]$, that is, the surface obtained by identifying $(x,0) \sim (f(x), \varepsilon)$. Let $\rho (\varepsilon) \in \mathbb{R} / \mathbb{Z}$ be such that $M_{\varepsilon}$ corresponds to the elliptic curve $\mathbb{C} / ( \mathbb{Z} + i \rho(\varepsilon) \mathbb{Z})$. Does $\rho(\varepsilon)$ converges to $\rho$ as $\varepsilon \to 0$ ?

背景链接：https://arxiv.org/abs/1712.06462

12464 | AMR-120-0025

### Group Actions on 1-Manifolds - Question 28

O / open | AMR 问题清单汇编

Is the (restriction of the) Godbillon-Vey class invariant under topological conjugacy for groups of $C^2$ diffeomorphisms ?

背景链接：https://arxiv.org/abs/1712.06462

12465 | AMR-120-0026

### Group Actions on 1-Manifolds - Question 29

O / open | AMR 问题清单汇编

What are the groups of $C^2$ circle diffeomorphisms acting minimally for which the normalizer inside the group of absolutely continuous homeomorphisms coincides with that inside the group of diffeomorphisms ?

背景链接：https://arxiv.org/abs/1712.06462

12466 | AMR-120-0003

### Group Actions on 1-Manifolds - Question 3

O / open | AMR 问题清单汇编

Does there exist a nontrivial (hence infinite) Kazhdan group of orientation-preserving homeomorphisms of the interval ?

背景链接：https://arxiv.org/abs/1712.06462

12467 | AMR-120-0027

### Group Actions on 1-Manifolds - Question 30

O / open | AMR 问题清单汇编

Does Denjoy's theorem hold for circle homeomorphisms whose graphs are real-analytic ?

背景链接：https://arxiv.org/abs/1712.06462

12468 | AMR-120-0028

### Group Actions on 1-Manifolds - Question 31

O / open | AMR 问题清单汇编

Is the continuous cohomology of the group $\mathrm{Diff}^{\omega}_+(\mathrm{S}^1)$ of orientation-preserving, real-analytic circle diffeomorphisms generated by the Euler and the Godbillon--Vey classes ?

背景链接：https://arxiv.org/abs/1712.06462

12469 | AMR-120-0029

### Group Actions on 1-Manifolds - Question 32

O / open | AMR 问题清单汇编

Is the group of orientation-preserving $C^2$ circle diffeomorphisms simple ?

背景链接：https://arxiv.org/abs/1712.06462

12470 | AMR-120-0030

### Group Actions on 1-Manifolds - Question 33

O / open | AMR 问题清单汇编

Does the Tits alternative hold in $\mathrm{Diff}_+^{\omega} (\mathrm{S}^1)$ ? More precisely, does every non-metabelian subgroup of this group contain a free subgroup ?

背景链接：https://arxiv.org/abs/1712.06462

12471 | AMR-120-0004

### Group Actions on 1-Manifolds - Question 4

O / open | AMR 问题清单汇编

Does there exist an infinite, finitely-generated group that acts on the real line all of whose actions by orientation-preserving homeomorphisms of the line without global fixed points are of type (iii) ?

背景链接：https://arxiv.org/abs/1712.06462

12472 | AMR-120-0005

### Group Actions on 1-Manifolds - Question 5

O / open | AMR 问题清单汇编

Does there exist a finitely-generated, amenable, left-orderable group having an isolated order inside an infinite space of left orders ?

背景链接：https://arxiv.org/abs/1712.06462

12473 | AMR-120-0006

### Group Actions on 1-Manifolds - Question 6

O / open | AMR 问题清单汇编

Does there exist a finitely-generated, left-orderable group for which the conjugacy action on its space of left orders is minimal (that is, all the orbits are dense) ?

背景链接：https://arxiv.org/abs/1712.06462

12474 | AMR-120-0007

### Group Actions on 1-Manifolds - Question 7

O / open | AMR 问题清单汇编

Does there exist a finitely-generated, left-orderable group without free subgroups and admitting no nontrivial homomorphism into $\mathbb{Z}$ ?

背景链接：https://arxiv.org/abs/1712.06462

12475 | AMR-120-0008

### Group Actions on 1-Manifolds - Question 8

O / open | AMR 问题清单汇编

Does there exist a finitely-generated, left-orderable group $G$ with no nontrivial homomorphism into $\mathbb{Z}$ and trivial group of bounded cohomology $\mathrm{H}^2_b(G, \mathbb{R})$ ?

背景链接：https://arxiv.org/abs/1712.06462

12476 | AMR-120-0009

### Group Actions on 1-Manifolds - Question 9

O / open | AMR 问题清单汇编

Is the group of orientation-preserving homeomorphisms of the 2-disk that are the identity at the boundary left orderable ?

背景链接：https://arxiv.org/abs/1712.06462

12477 | OWR-1460-001

### Growth and Order of Entire Functions with Newton Baker Domains

P / partially_solved | Oberwolfach 报告问题汇编

Let $f$ be an entire transcendental function and $N_f(z)=z-f(z)/f'(z)$ its Newton function. Suppose $N_f$ has an invariant Baker domain $U$, meaning $N_f(U)\subseteq U$ and its iterates tend to infinity. Must there be a constant $K>0$ and a curve $\gamma$ tending to infinity such that $|f(z)|\leq|z|^K$ for $z\in\gamma$? Must the order of $f$ be at least $1/2$?

来源字段：https://doi.org/10.4171/owr/2007/09

12478 | AMR-045-0043

### Growth of jointly periodic points I

O / open | AMR 问题清单汇编

For a surjective one-dimensional cellular automaton $f$ on the full $N$-shift, let $\nu(f,S_N)$ be the limsup exponential growth rate of points that are spatially fixed by a power of the shift and periodic under $f$. Must $\nu(f,S_N)>1$?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12479 | AMR-045-0044

### Growth of jointly periodic points II

O / open | AMR 问题清单汇编

With $\nu(f,S_N)$ defined as in Question 25.3, must every surjective one-dimensional cellular automaton satisfy $\nu(f,S_N)\ge\sqrt N$?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12480 | AMR-052-0086

### Haken-type decomposition for rational maps

O / open | AMR 问题清单汇编

Develop an analogue of the Haken decomposition for geometrically finite rational maps. In particular, if the Julia set is disconnected, can the map be constructed by surgery from rational maps with connected Julia sets?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12481 | OWR-15951-003

### Half-Line Spectra for Fibonacci Schrödinger Operators

O / open | Oberwolfach 报告问题汇编

Let $V_\omega$ be a continuum Fibonacci potential on $\mathbb R$, and define the separable Schrödinger operator on $\mathbb R^2$ by $$ L^{(2)}_{\lambda_1V_\omega,\lambda_2V_\omega}=-\Delta+\lambda_1V_\omega(x)+\lambda_2V_\omega(y). $$ Does its spectrum contain a half-line for every $\lambda_1,\lambda_2>0$?

来源字段：https://doi.org/10.4171/owr/2017/46

12482 | OWR-1388-001

### Hamiltonian Dynamics Near Infinite-Dimensional Saddle-Center Bifurcations

O / open | Oberwolfach 报告问题汇编

Determine the dynamics of an infinite-dimensional Hamiltonian system near a saddle-center bifurcation and the nature of the induced homoclinic bifurcation.

来源字段：https://doi.org/10.4171/owr/2006/50

12483 | OWR-2495-003

### Hamiltonian Meandering Transitions in Infinite Dimensions

P / partially_solved | Oberwolfach 报告问题汇编

Can results on the Hamiltonian meandering transition-from relative equilibria to relative periodic orbits-be extended to infinite-dimensional Hamiltonian systems, such as PDE models of vortex dynamics?

来源字段：https://doi.org/10.4171/owr/2008/57

12484 | OWR-1961-002

### Hausdorff Dimension of Critical Almost Mathieu Spectra

P / partially_solved | Oberwolfach 报告问题汇编

For the almost Mathieu operator on $\ell^2(\mathbb{Z})$, defined by $(H_{\alpha,\lambda,\theta}\psi)(n)=\psi(n+1)+\psi(n-1)+\lambda\cos(2\pi\alpha n+\theta)\psi(n)$, let $S(\alpha,\lambda)$ be its spectrum, independent of $\theta$ when $\alpha$ is irrational. Is $\dim_HS(\alpha,2)<1/2$ for every irrational $\alpha$?

来源字段：https://doi.org/10.4171/owr/2008/16

12485 | OWR-4137-017

### Hausdorff Gauges for Conformal Measures of Elliptic Maps

O / open | Oberwolfach 报告问题汇编

For an elliptic function $f$ whose critical points are attracted to attracting or parabolic periodic points and with at least one parabolic point, does there exist a gauge function $g$ such that the atomless $h$-conformal measure on $J(f)$ equals the Hausdorff measure $\mathcal H^g$ on $J(f)$?

来源字段：https://doi.org/10.4171/owr/2009/54

12486 | AMR-152-0024

### Henon Maps: Open Problems - Conjecture 24

O / open | AMR 问题清单汇编

where $(c, b)\in\mathbb{C}^2$ is a parameter. (We include the case $b=0$ to simplify the presentation.) Let $J_{c, b}$ be the Julia set of $f_{c, b}$: by definition this is the intersection between the boundaries of the sets of points having bounded forward (resp. backward) orbits. By extension, we let $J_{c, 0}$ be the Julia set of $p(z)=z^2+c$. The *connectedness locus* of $f_{c, b}$ is defined as

\[\mathcal{M}=\big\{(c, b)\in\mathbb{C}^2 : J_{c, b} \mbox{ is connected}\big\}.\]

$\mathcal{M}$ is disconnected.

背景链接：https://arxiv.org/abs/2312.03907

12487 | AMR-152-0047

### Henon Maps: Open Problems - Conjecture 47

O / open | AMR 问题清单汇编

Let $K$ be a number field, let $B\geq 1$, and let $d\geq 2$. Then as $f$ varies over generalized H\'{e}non maps of degree $d$ over $K$, the quantity \[\#\{P\in \operatorname{Per}(f):[K(P):K]\leq B\}\] is bounded just in terms of $[K:\mathbb{Q}]$, $d$, and $B$.

背景链接：https://arxiv.org/abs/2312.03907

12488 | AMR-152-0048

### Henon Maps: Open Problems - Conjecture 48 (see [I])

O / open | AMR 问题清单汇编

We have already seen why this bound must depend on $d$, and adjoining periodic points of $f$ to the base field shows why the bound must depend on $[K:\mathbb{Q}]$. As a starting point for further exploration, we mention two more readily falsifiable conjectures.

Over $\mathbb{Q}$, $(x, y)\mapsto (y, y^2+c+x)$ has no point of period $N$, other than $N\in \{1, 2, 3, 4, 6, 8\}$.

背景链接：https://arxiv.org/abs/2312.03907

12489 | AMR-152-0049

### Henon Maps: Open Problems - Conjecture 49

O / open | AMR 问题清单汇编

For all but finitely many $\delta\in \mathbb{Q}$, the $\mathbb{Q}$-rational periodic points of any $f(x, y)=(y, y^2+c-\delta x)$ with $c\in \mathbb{Q}$ have period dividing 2.

背景链接：https://arxiv.org/abs/2312.03907

12490 | AMR-152-0050

### Henon Maps: Open Problems - Conjecture 50

O / open | AMR 问题清单汇编

Let $K$ be a number field, let $B\geq 1$, and let $d\geq 2$. Then, there exist an $\epsilon>0$ and constant $C$ (depending on these data) such that, as $f$ varies over generalized H\'{e}non maps of degree $d$ over $K$, the quantity \[\#\{P\in (\overline{K})^2 :[K(P):K]\leq B\text{ and }\hat{h}_f(P)<\epsilon h(f)+C\}\] is bounded uniformly, where $h(f)$ is the height of the tuple of coefficients of $f$.

背景链接：https://arxiv.org/abs/2312.03907

12491 | AMR-152-0012

### Henon Maps: Open Problems - Problem 12

O / open | AMR 问题清单汇编

Very recently, in [BSt], the authors developed a kneading theory for the H\'enon maps $H_{a,b}$ within a set of parameters $\mathcal{WY}$ for which Wang and Young proved the existence of a strange attractor. This set has positive measure and consist of maps which are strongly dissipative. We refer to [BSt] and [WY] for details.

Describe the set of kneading sequences $\mathfrak{K}$ of the H\'enon map $H_{a,b}$, with $(a,b) \in \mathcal{WY}$.

背景链接：https://arxiv.org/abs/2312.03907

12492 | AMR-152-0033

### Henon Maps: Open Problems - Problem 33

O / open | AMR 问题清单汇编

Prove the existence and the uniqueness of the equilibrium state $\mu_\phi$ associated to any sufficiently regular weight $\phi$ (for instance, every H\"older continuous $\phi$, and possibly with some bound on $\max \phi - \min \phi$).

背景链接：https://arxiv.org/abs/2312.03907

12493 | AMR-152-0035

### Henon Maps: Open Problems - Problem 35

O / open | AMR 问题清单汇编

Show that $\mu_\phi$ satisfies the CLT for H\"older continuous observables. Namely, prove that, for any H\"older continuous observable $g$, there exists $\sigma \geq 0$ such that for any interval $I \subset \mathbb{R}$, we have \[ \lim_{n\to \infty} \mu_\phi \left(\left\{ \frac{S_n (g) - n\langle \mu_\phi,g\rangle }{\sqrt{n} } \in I \right\}\right) = \begin{cases} 1 \mbox{ when } I \mbox{ is of the form } I=(-\delta,\delta) & \mbox{ if } \sigma^2=0,\\ \frac{1}{\sqrt{2\pi\sigma^2}}\displaystyle\int_I e^{-t^2 / (2\sigma^2)} \mathrm{d}t & \mbox{ if } \sigma^2 >0. \end{cases} \]

背景链接：https://arxiv.org/abs/2312.03907

12494 | AMR-152-0036

### Henon Maps: Open Problems - Problem 36

O / open | AMR 问题清单汇编

Show that $\mu_\phi$ satisfies the Large Deviation Principle (LDP) for H\"older continuous observables. Namely, prove that, for any H\"older continuous observable $g$ with $\langle \mu_\phi, g\rangle=0$ and which is not a coboundary, there exists a non-negative, strictly convex function $c$ which is defined on a neighborhood of $0 \in \mathbb{R}$, vanishes only at $0$, and such that, for all $\epsilon > 0$ sufficiently small, \[ \lim_{n\to \infty} \frac{1}{n} \log \mu_\phi \left(\left\{ x \in X\colon \frac{S_n(g)(x) }{n} >\epsilon \right\}\right) = -c(\epsilon).\]

背景链接：https://arxiv.org/abs/2312.03907

12495 | AMR-152-0044

### Henon Maps: Open Problems - Problem 44

O / open | AMR 问题清单汇编

Let $f\in \mathrm{Aut}[\mathbb{C}^d]$ be any regular polynomial automorphism.

- Prove that one can find a compact complex manifold $\bar{S}(f)$ and an open immersion $S(f) \subset \bar{S}(f)$ such that the complement $\bar{S}(f)\setminus S(f)$ is a divisor.

- Prove that $\bar{S}(f)$ is unique up to bimeromorphism.

- Describe complex objects on $\bar{S}(f)$ (analytic subvarieties, vector fields, holomorphic foliations, positive closed currents,...). Compute its deformation space.

背景链接：https://arxiv.org/abs/2312.03907

12496 | AMR-152-0001

### Henon Maps: Open Problems - Question 1

O / open | AMR 问题清单汇编

Can we find a H\'enon map (of some degree) which admits a wild attractor, i.e., a Cantor set which attracts a set of Lebesgue positive measure and which is strictly included in a transitive set ?

背景链接：https://arxiv.org/abs/2312.03907

12497 | AMR-152-0010

### Henon Maps: Open Problems - Question 10

O / open | AMR 问题清单汇编

As mentioned previously, for mildly dissipative real H\'enon maps the set of periodic points is dense in the union of the supports of the invariant probability measures [CP]. The same property holds for any complex H\'enon maps [MR4142445].

For any real H\'enon map, does the closure of the set of periodic orbits support all the invariant probability measures?

背景链接：https://arxiv.org/abs/2312.03907

12498 | AMR-152-0011

### Henon Maps: Open Problems - Question 11

O / open | AMR 问题清单汇编

For any map $f_{c,b}$ and any infinite set of periodic saddles $(p_n)$ which are pairwise not homoclinically related, do we have $\min(|\lambda^-(p_n)|,|\lambda^+(p_n)|)\underset{n\to \infty}\longrightarrow 0$?

背景链接：https://arxiv.org/abs/2312.03907

12499 | AMR-152-0013

### Henon Maps: Open Problems - Question 13

O / open | AMR 问题清单汇编

Is it true that any two distinct kneading sequences of the H\'enon map $H_{a,b}$, with $(a,b) \in \mathcal{WY}$, determine a unique pair of parameters $(a,b)$, and in that way govern all the other kneading sequences in $\mathfrak{K}$ of $H_{a,b}$?

背景链接：https://arxiv.org/abs/2312.03907

12500 | AMR-152-0014

### Henon Maps: Open Problems - Question 14

O / open | AMR 问题清单汇编

We say that $H_2$ has small cycles if for any neighborhood $U$ of the origin $\mathbf{s}ymbol{0}$ in $\mathbb{C}^2$, there exists a cycle of $H_2$ which is entirely contained in $U\smallsetminus \{\mathbf{s}ymbol{0}\}$.

Does $H_2$ have small cycles?

背景链接：https://arxiv.org/abs/2312.03907

12501 | AMR-152-0015

### Henon Maps: Open Problems - Question 15

O / open | AMR 问题清单汇编

Does $H_2$ have both small saddle cycles and small elliptic cycles?

背景链接：https://arxiv.org/abs/2312.03907

12502 | AMR-152-0016

### Henon Maps: Open Problems - Question 16

O / open | AMR 问题清单汇编

We say that $H_2$ has a Herman ring with rotation number $\theta\in (\mathbb{R}\smallsetminus \mathbb{Q})/\mathbb{Z}$ if there exists an annulus $V= \bigl\{z\in \mathbb{C}/\mathbb{Z} : \mathrm{Im}(z)<h\bigr\}$ with $h>0$, a holomorphic map $\phi\colon V\to \mathbb{C}^2$, and an integer $n\geq 2$ such that \[\forall z\in V,\quad H_2^{\circ n}\circ \phi(z) = \phi(z+\theta).\]

Does $H_2$ have a Herman ring?

背景链接：https://arxiv.org/abs/2312.03907

12503 | AMR-152-0017

### Henon Maps: Open Problems - Question 17

O / open | AMR 问题清单汇编

If the answer is yes, we may consider the set $\Theta\subset (\mathbb{R}-\mathbb{Q})/\mathbb{Z}$ of rotation numbers $\theta$ such that $H_2$ has a Herman ring with rotation number $\theta$.

Does $\Theta$ have positive Lebesgue measure? More precisely, is $0$ a Lebesgue density point of $\Theta$?

背景链接：https://arxiv.org/abs/2312.03907

12504 | AMR-152-0018

### Henon Maps: Open Problems - Question 18

O / open | AMR 问题清单汇编

Assume $H_2$ has a Herman ring with bounded type rotation number $\theta$. Is it possible to find parameters $a\in \mathbb{D}\smallsetminus \{0\}$ arbitrarily close to $1$ such that the dissipative H\'enon map $H\colon\mathbb{C}^2\to \mathbb{C}^2$ defined by \[H(x,y) = \left(ay,x+y^2\right).\] has a Herman ring with rotation number $\theta$?

背景链接：https://arxiv.org/abs/2312.03907

12505 | AMR-152-0019

### Henon Maps: Open Problems - Question 19

O / open | AMR 问题清单汇编

Recall the following standard definitions (see [BS]). Let $K^\pm$ denote the set of points with bounded forward orbits for $f^\pm$. We denote $J^\pm:= \partial K^\pm$. We also define $J:= J^+ \cap J^-$ and $J^*$ to be the closure of ${\mathcal{S}}$ (hence $J^* \subset J$).

If $f$ is quasi-hyperbolic, then is $\mathrm{int}(K^+)$ the union of a finite number of basins of sink orbits?

背景链接：https://arxiv.org/abs/2312.03907

12506 | AMR-152-0002

### Henon Maps: Open Problems - Question 2

O / open | AMR 问题清单汇编

Returning to stochastic attractors, observe that the existence of a parameter $c\in \mathbb{R}$ for which the quadratic map $x \mapsto x^2+c$ displays an absolutely continuous measure is easy to ensure. It suffices to pick a parameter $c$ such that the post-critical orbit is finite but not periodic. In dimension $2$, the existence of a positive measure set of parameters of H\'enon maps displaying an attractor supporting an invariant SRB measure (i.e., a measure whose conditional measures along unstable curves are absolutely continuous.) is a fundamental result, whose proof still remain difficult and lengthy ([BC,Berger,Ta,WY]).

Is there a quick proof for the existence of SRB for some parameters of the H\'enon maps?

背景链接：https://arxiv.org/abs/2312.03907

12507 | AMR-152-0020

### Henon Maps: Open Problems - Question 20

O / open | AMR 问题清单汇编

If $f$ is quasi-hyperbolic, then is there no wandering Fatou component?

背景链接：https://arxiv.org/abs/2312.03907

12508 | AMR-152-0021

### Henon Maps: Open Problems - Question 21

O / open | AMR 问题清单汇编

If $f$ is quasi-hyperbolic, do the unstable slices satisfy a John-type condition (as in [BS7])?

背景链接：https://arxiv.org/abs/2312.03907

12509 | AMR-152-0022

### Henon Maps: Open Problems - Question 22

O / open | AMR 问题清单汇编

If $f$ is quasi-hyperbolic and dissipative, and if $J$ is connected, do the external rays land at $J$? Is $J$ a finite quotient of the real solenoid?

背景链接：https://arxiv.org/abs/2312.03907

12510 | AMR-152-0023

### Henon Maps: Open Problems - Question 23

O / open | AMR 问题清单汇编

Let us now suppose that $f$ is a real surface automorphism. In other words, we suppose $X$ to be projective and defined by real polynomial equations, and $f$ to be also defined over the real numbers. We may thus consider the restriction map $f_\mathbb{R}$ to the set of real points $X_\mathbb{R}$. Observe that $h_{\mathrm top}(f_\mathbb{R})\le \log \lambda$. It was proved in [BS8] that for generalized H\'enon maps $h_{\mathrm top}(f_\mathbb{R})= \log \lambda$ implies $f$ to be quasi-hyperbolic.

If $f$ is a real surface automorphism such that the entropy of $f_\mathbb{R}$ is the same as the entropy of $f$, does it follow that $f$ is quasi-hyperbolic?

背景链接：https://arxiv.org/abs/2312.03907

12511 | AMR-152-0025

### Henon Maps: Open Problems - Question 25

O / open | AMR 问题清单汇编

of the fundamental group of $\mathcal{H}_{\mathbb{C}}$ (with the base-point at $(c, b)=(-4, 0)$) to the group of shift-commuting automorphisms of $\{0, 1\}^{\mathbb{Z}}$ is surjective (see, e.g., [A,BS1]).

Is the locus $\mathcal{H}_{\mathbb{C}}$ connected?

背景链接：https://arxiv.org/abs/2312.03907

12512 | AMR-152-0026

### Henon Maps: Open Problems - Question 26 (van Strien [vanstrien])

O / open | AMR 问题清单汇编

In a topological term, monotonicity of the topological entropy of the real H\'enon map $f_{c, b}|_{\mathbb{R}^2}$ can be formulated as

Is the isentrope $\mathcal{E}_{\alpha}$ connected for any $0\leq \alpha \leq \log 2$ ?

背景链接：https://arxiv.org/abs/2312.03907

12513 | AMR-152-0027

### Henon Maps: Open Problems - Question 27

O / open | AMR 问题清单汇编

Several articles attempt at giving lower bounds for topological entropy of real H\'enon maps, e.g., [NBGM,NP]. Among others, the paper [NBGM] rigorously showed that $h_{\mathrm{top}}(f_{c, b}|_{\mathbb{R}^2})>0.46469$ for the classical H\'enon's parameter, and this bound is believed to be close to optimal. For upper bounds, the paper [BLS] has shown that $h_{\mathrm{top}}(f_{c, b}|_{\mathbb{R}^2})<\log 2$ if and only if the Julia set of $f_{c, b}$ (as a complex dynamical system) is not contained in $\mathbb{R}^2$. However, there is no algorithm which provides rigorous (non-trivial) upper bounds.

We thus propose the following problem.

Construct an algorithm to compute a rigorous upper bound for the topological entropy of a real H\'enon map $f_{c, b}|_{\mathbb{R}^2}$.

背景链接：https://arxiv.org/abs/2312.03907

12514 | AMR-152-0028

### Henon Maps: Open Problems - Question 28

O / open | AMR 问题清单汇编

Give an asymptotic expansion of the Hausdorff dimension of $J^+_{a,c}$ as $(a,c)$ tends to $(0, 0)$. In particular is there a uniform lower bound of $\dim(J^+_{a,c})$ of the form $\dim(J^+_{a,c})\geq 3+h(a)$ with $h(a)>0$ in the neighborhood of $c=0$?

背景链接：https://arxiv.org/abs/2312.03907

12515 | AMR-152-0029

### Henon Maps: Open Problems - Question 29

O / open | AMR 问题清单汇编

Is it possible to find a generalized complex H\'enon map $f$, an open set $U\subset \mathbb{C}^2$ intersecting $J^+$ and a holomorphic foliation of $U$ such that $J^+\cap U$ is a union of leaves of this foliation?

背景链接：https://arxiv.org/abs/2312.03907

12516 | AMR-152-0003

### Henon Maps: Open Problems - Question 3

O / open | AMR 问题清单汇编

Do there exist $\varepsilon>0$ and a locally dense set (i.e., whose closure has non-empty interior.) $E$ of quadratic H\'enon maps for which every point starting in some set of Lebesgue measure at least $\varepsilon$ has non-statistical behavior?

背景链接：https://arxiv.org/abs/2312.03907

12517 | AMR-152-0030

### Henon Maps: Open Problems - Question 30

O / open | AMR 问题清单汇编

Under which circumstances is it possible that there exist saddle points $p$ and $q$, and relative open subsets $U\subset W^u(p)$ and $V\subset W^u(q)$ such that $U\cap K^+$ is biholomorphic to $V\cap K^+$?

背景链接：https://arxiv.org/abs/2312.03907

12518 | AMR-152-0031

### Henon Maps: Open Problems - Question 31

O / open | AMR 问题清单汇编

Let $f_1$, $f_2$ be two generalized H\'enon maps. Under which circumstances is it possible that some local unstable slice of $f_1$ (i.e., a set of the form $K^+\cap U$, where $U$ is a relatively open subset of an unstable manifold) is biholomorphically equivalent to an unstable slice of $f_2$?

背景链接：https://arxiv.org/abs/2312.03907

12519 | AMR-152-0032

### Henon Maps: Open Problems - Question 32

O / open | AMR 问题清单汇编

Since a local unstable slice of a generalized H\'enon maps contains essentially a complete information about unstable multipliers, the previous question is reminiscent of the classical ``spectral rigidity'' problem:

To which extent is a generalized H\'enon map determined by the list of its unstable multipliers (resp. by the list of moduli of its unstable multipliers)?

背景链接：https://arxiv.org/abs/2312.03907

12520 | AMR-152-0034

### Henon Maps: Open Problems - Question 34

O / open | AMR 问题清单汇编

Suppose $\phi$ is sufficiently regular, so that the equilibrium state $\mu_\phi$ exists and is unique. Is it true that \begin{equation} \frac{1}{e^{n P(\phi)}} \sum_{x \in SP_n} e^{S_n (\phi)} \delta_x \to \mu_\phi \quad ? \end{equation}

背景链接：https://arxiv.org/abs/2312.03907

12521 | AMR-152-0037

### Henon Maps: Open Problems - Question 37

O / open | AMR 问题清单汇编

Do there exist norms for functions on the Julia set which are bounded on H\"older continuous functions, contract (on the complement of an invariant line) under the action of $f_*$ (or, more generally, of $f_* (e^{\phi-P(\phi)} \cdot)$), and such that the contraction is stable by perturbation of $\phi$?

背景链接：https://arxiv.org/abs/2312.03907

12522 | AMR-152-0038

### Henon Maps: Open Problems - Question 38

O / open | AMR 问题清单汇编

The case of cubic automorphisms on $\mathbb{C}^3$, and the case of automorphisms obtained as a composition of an affine transformation and a triangular one were considered by Blanc and Van Santen [blanc-V1,blanc-V2]. Their computations lead them to formulate the following intriguing problem. A weak Perron number is an algebraic integer $\lambda \ge1$ such that all its Galois conjugates satisfy $|\mu|\le \lambda$.

Is the dynamical degree of any polynomial automorphism of $\mathbb{C}^d$ equal to a weak Perron number of degree $\le d-1$?

背景链接：https://arxiv.org/abs/2312.03907

12523 | AMR-152-0039

### Henon Maps: Open Problems - Question 39

O / open | AMR 问题清单汇编

Suppose $f$ is a polynomial automorphism of $\mathbb{C}^d$ satisfying $\lambda(f)=1$. Is it true that $\deg(f^n) \asymp n^k$ for some $k \in \mathbb{N}$? Moreover, if $k\ge 1$, does $f$ preserve a rational fibration?

背景链接：https://arxiv.org/abs/2312.03907

12524 | AMR-152-0004

### Henon Maps: Open Problems - Question 4

O / open | AMR 问题清单汇编

It is natural to ask whether the third condition is superfluous. The recent result [OC23] suggests that this may be the case.

Is there a real, conservative, polynomial H\'enon map with an annular rotation domain? Is there an open set of such real, conservative, polynomial H\'enon maps?

背景链接：https://arxiv.org/abs/2312.03907

12525 | AMR-152-0040

### Henon Maps: Open Problems - Question 40

O / open | AMR 问题清单汇编

Urech proved that $\deg(f^n)$ tends to infinity whenever it is unbounded, see [urech]. His result was made stronger by Cantat and Xie in [CX]: there exists a universal function $\sigma\colon \mathbb{N}\to\mathbb{N}$ such that $\limsup \sigma = \infty$ and $\deg(f^n) \ge \sigma(n)$. They raised the following weaker form of the previous problem.

Suppose that $\lambda(f)=1$, and $\deg(f^n)$ is unbounded for some polynomial automorphism $f$ of $\mathbb{C}^d$. Does there exist $C>0$ such that $\deg(f^n) \ge Cn$?

背景链接：https://arxiv.org/abs/2312.03907

12526 | AMR-152-0041

### Henon Maps: Open Problems - Question 41

O / open | AMR 问题清单汇编

Let $\mathrm{Tame}(3)$ be the subgroup of polynomial automorphisms of $\mathbb{C}^3$ which is generated by affine and triangular transformations. A theorem of Shestakov and Umirbaev [SU] states that $\mathrm{Tame}(3)$ is a strict subgroup of the full group of polynomial automorphisms of $\mathbb{C}^3$ (as opposed to the $2$-dimensional situation). A decisive progress on the structure of $\mathrm{Tame}(3)$ was recently made by Lamy and Przytycky [LP], who constructed a $\mathrm{CAT}(0)$-complex $\mathcal{C}$ over which $\mathrm{Tame}(3)$ acts by isometries.

Is it possible to characterize those $f\in \mathrm{Tame}(3)$ for which $\lambda(f)=1$ in terms of their action on $\mathcal{C}$?

背景链接：https://arxiv.org/abs/2312.03907

12527 | AMR-152-0042

### Henon Maps: Open Problems - Question 42

O / open | AMR 问题清单汇编

Let $f$ be any generalized H\'enon map. Describe the set of all generalized H\'enon maps $g$ such that $\bar{S}(f)$ is biholomorphic to $\bar{S}(g)$.

背景链接：https://arxiv.org/abs/2312.03907

12528 | AMR-152-0043

### Henon Maps: Open Problems - Question 43

O / open | AMR 问题清单汇编

A complex manifold which admits a holomorphic atlas whose transition maps are restriction of birational maps of $\mathbb{P}^d_\mathbb{C}$ is said to carry a birational structure.

The following problem is extracted from [dloussky].

Does any non-K\"ahler compact complex surface admit a birational structure?

背景链接：https://arxiv.org/abs/2312.03907

12529 | AMR-152-0045

### Henon Maps: Open Problems - Question 45

O / open | AMR 问题清单汇编

It is unclear how to extend this construction to a larger class of polynomial automorphisms of $\mathbb{C}^3$. However when $\lambda(f)^2>\lambda(f^{-1})$ an invariant valuation on the ring of polynomial functions in three variables (One way to interpret geometrically such an object is to say that it picks an irreducible subvariety in *any* algebraic compactification of $\mathbb{C}^3$ in a compatible way.) is known to exist by [DF20], which suggests the following question.

Let $f\in \mathrm{Aut}[\mathbb{C}^3]$ be any polynomial automorphism such that $\lambda(f)^2>\lambda(f^{-1})$. Prove the existence of a projective compactification $X$ of $\mathbb{C}^3$ such that the induced birational map $f\colon X \dashrightarrow X$ admits a super-attracting fixed point $p$ on the divisor at infinity.

背景链接：https://arxiv.org/abs/2312.03907

12530 | AMR-152-0046

### Henon Maps: Open Problems - Question 46

O / open | AMR 问题清单汇编

Over a number field $K$, is it possible to construct infinite families of generalized H\'{e}non maps of algebraic degree $d$, and $K$-rational cycles of length at least $d+3$? Can one construct maps with $N_d$ periodic points, where $N_d-d\to \infty$, or even $N_d/d\to \infty$ as $d\to\infty$?

背景链接：https://arxiv.org/abs/2312.03907

12531 | AMR-152-0005

### Henon Maps: Open Problems - Question 5

O / open | AMR 问题清单汇编

An annular rotation domain is said to be trivial when the whole dynamics is conjugate to a rotation.

Is there an entire (i.e., an analytic map which extends to a holomorphic map of $\mathbb{C}^2$.) symplectomorphism of $\mathbb{R}^2$ with a non trivial annular rotation domain?

背景链接：https://arxiv.org/abs/2312.03907

12532 | AMR-152-0051

### Henon Maps: Open Problems - Question 51

O / open | AMR 问题清单汇编

Under what circumstance is it true that \[\log\operatorname{Norm}(\mathfrak{a}_n)=o(d^n) \] as $n\to \infty$? Under what circumstance does there exist an ideal $\mathfrak{b} \subset \mathcal{O}_{K, S}$ such that $\mathfrak{a}_n | \mathfrak{b}$ for infinitely many $n$? Note that, since $\mathfrak{a}_1 | \mathfrak{a}_n$ for all $n$, $\mathfrak{b} =\mathfrak{a}_1$ is a reasonable candidate.

背景链接：https://arxiv.org/abs/2312.03907

12533 | AMR-152-0006

### Henon Maps: Open Problems - Question 6

O / open | AMR 问题清单汇编

Under this assumption, and in restriction to any open topological disc $D\subset \mathbb{R}^2$ that is compactly mapped inside itself, the dynamics is semi-conjugated to a (non-trivial) continuous map of a real tree. Other strong consequences can be derived (e.g., a closing lemma, or a description of zero entropy dynamics, see below). Using Wiman's theorem (in the same spirit like in [DL, MR3213832]), one can prove that H\'enon maps are mildly dissipative once $|b|<1/4$. In this way we obtain dynamical informations for *all* H\'enon maps having their Jacobian in $(-1/4,1/4)$ and not only for those satisfying $|b|\ll 1$. One expects that this property extends beyond the bound obtained through Wiman's theorem.

Which real H\'enon maps $f_{c,b}$ are mildly dissipative? Is this property satisfied by all H\'enon maps with $|b|<1$?

背景链接：https://arxiv.org/abs/2312.03907

12534 | AMR-152-0007

### Henon Maps: Open Problems - Question 7

O / open | AMR 问题清单汇编

When $|b|$ is larger but smaller than $1/4$ (so that it is mildly dissipative), [CPT] describes the dynamics of $f_{c,b}\in \mathcal{E}_0$. In particular, all maps $f_{c,b}\in \partial \mathcal{E}_0$ are infinitely renormalizable (with renormalization periods eventually equal to $2$), solving a conjecture by Tresser (which is still open when we don't assume the mild dissipation). One may wonder if the converse holds.

Let us consider any infinitely renormalizable mildly dissipative map $f_{c,b}\in\mathcal{E}_0$. Is it the limit of maps with positive entropy? Is it the limit of Morse-Smale maps?

背景链接：https://arxiv.org/abs/2312.03907

12535 | AMR-152-0008

### Henon Maps: Open Problems - Question 8

O / open | AMR 问题清单汇编

It is also natural to try to describe the boundary $\partial\mathcal{E}_0$: is it a (piecewise smooth) arc? One would like to implement the strategy developed in [dCLM] for $b$ close to $0$:

Let us consider any infinitely renormalizable mildly dissipative map in $\mathcal{E}_0$. Does the sequence of renormalizations converge?

背景链接：https://arxiv.org/abs/2312.03907

12536 | AMR-152-0009

### Henon Maps: Open Problems - Question 9

O / open | AMR 问题清单汇编

We also don't know if different combinatorics of the renormalizations may occur.

Does there exist a mildly dissipative map $f_{c,b}\in \mathcal{E}_0$ which admits arbitrarily deep renormalizations with odd periods?

背景链接：https://arxiv.org/abs/2312.03907

12537 | AMR-052-0081

### Herman-ring retracts for Hénon maps

O / open | AMR 问题清单汇编

Can the subsequential limit map on an invariant stable component of a polynomial diffeomorphism of $\mathbb C^2$ be a retraction onto a Herman ring or a punctured Siegel disk?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12538 | OWR-15995-003

### Higher Dimensional Feedback Stabilization by Sampling

O / open | Oberwolfach 报告问题汇编

Extend sampling-based feedback-stabilization procedures from systems with two unstable eigenvalues to higher-dimensional systems while preserving their stabilization guarantees.

来源字段：https://doi.org/10.4171/owr/2018/10

12539 | OWR-15993-009

### Higher Multistationarity in T-Cell Activation Models

O / open | Oberwolfach 报告问题汇编

For the T-cell activation ODE model with antigen amount and receptor-unbinding rate as parameters, do there exist parameter values for which more than three steady states occur?

来源字段：https://doi.org/10.4171/owr/2018/8

12540 | AMR-052-0063

### Homeomorphism type of exponential Knaster continua

P / partially_solved | AMR 问题清单汇编

For parameters $\lambda,\mu>1/e$, are the Knaster-like continua arising in the dynamics of $E_\lambda(z)=\lambda e^z$ and $E_\mu(z)=\mu e^z$ homeomorphic?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12541 | OWR-9790354-007

### Hyper-Aperiodic Colorings from Aperiodic SFT Colorings

O / open | Oberwolfach 报告问题汇编

Let $S\subseteq q^\Gamma$ be a subshift of finite type and let $f:\operatorname{Free}(k^\Gamma)\to q$ be a Borel aperiodic $S$-coloring. Must $\operatorname{Free}(k^\Gamma)$ also admit a Borel hyper-aperiodic $S$-coloring?

来源字段：https://doi.org/10.4171/owr/2022/3

12542 | OWR-2053-026

### Hyperbolic Minimal Geodesics and Measurable Rotation Models

O / open | Oberwolfach 报告问题汇编

Let $M$ be a compact manifold of hyperbolic type with a Riemannian metric $g$, and let $SM_{\min}^g$ consist of velocities of globally $g$-minimizing geodesics. Does $SM_{\min}^g$ contain a closed hyperbolic geodesic? Is there an ergodic hyperbolic invariant measure supported on $SM_{\min}^g$? Separately, if $f$ is an area-preserving diffeomorphism of a surface, can $f$ be measurably conjugated to a rotation of $S^1$ by a Diophantine angle $\beta$?

来源字段：https://doi.org/10.4171/owr/2008/42

12543 | AMR-036-0009

### Hypotheses for analytic invariant curves

P / partially_solved | AMR 问题清单汇编

Let $C$ be an analytic invariant curve of a rational function $f$, suppose $f:C\to C$ is not a homeomorphism, and assume $C$ contains a repelling fixed point, lies in $J(f)$, and contains neither critical points nor rational fixed points. A theorem concludes that either $f$ is a Lattès function or $C$ is algebraic. Which of these hypotheses can be removed?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

12544 | AIM-DYNAMICAL_SYSTEMS-0005

### Identical maximal iterated Galois images with incompatible p-adic Julia sets

P / partially_solved | AIM 研讨会问题汇编

**Reconstructing the Julia set of a function from the iterated Galois group**

Let $k$ be a finite field extension of $\mathbb{Q}_p$, $f \in k(z)$ of degree at least $2$, and $t$ be a transcendental number. Define \[ k_\infty=k\left(\bigcup_{n\geq0}f^{-n}(t)\right),\qquad L=k_\infty\cap\overline{k}, \] \[ G_\infty=\operatorname{Gal}(k_\infty/k(t)),\qquad G_\infty^{\mathrm{geom}}=\operatorname{Gal}(k_\infty/L(t)). \]

Can we reconstruct the $p$-adic Julia set of $f$ from $G_\infty$ or $G_\infty^{\mathrm{geom}}$?

来源字段：http://aimpl.org/groupdynamorigin/1/

12545 | OWR-1460-004

### IM-Sharing Variants of Theorem H for Entire Functions

O / open | Oberwolfach 报告问题汇编

Let $f$ be a nonconstant entire function, let $k\geq2$, and let $a,b\in\mathbb{C}$ with $b\neq0$ and $b\neq a$. Suppose that $f$ and $f'$ share the value $a$ ignoring multiplicity, and that $f^{(k)}(z)=b$ whenever $f'(z)=b$. Must $f(z)=de^{cz}+c^{-1}a$ for nonzero constants $c,d$ satisfying $c^{k-1}=1$?

来源字段：https://doi.org/10.4171/owr/2007/09

12546 | AMR-052-0047

### Image of a geometric coding tree

P / partially_solved | AMR 问题清单汇编

For a geometric coding tree of inverse branches of a holomorphic map, let $z_\infty:D(z_\infty)\to\overline U$ map each convergent symbolic branch to its limit. How large can the image $z_\infty(D(z_\infty))$ be?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12547 | AMR-052-0039

### Infinite intersections of small Mandelbrot sets

P / partially_solved | AMR 问题清单汇编

Does every nested intersection $\bigcap_k H_1*\cdots*H_k*M$ of tuned copies of the Mandelbrot set consist of one point? Equivalently, are infinitely renormalizable parameters totally disconnected; do they have measure zero or small Hausdorff dimension?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12548 | AMR-045-0039

### Infinite symbolic-extension entropy

P / partially_solved | AMR 问题清单汇编

Can a $C^r$ diffeomorphism of a compact Riemannian manifold have infinite symbolic-extension entropy?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12549 | AIM-DYNAMICAL_SYSTEMS-0031

### Infinitesimal rigidity of the discrete tree action and visible root displacement

P / partially_solved | AIM 研讨会问题汇编

Is there a deformation theory for arboreal Galois representations?

Consider $x^2+t$ where $t^2=0$ over Artinian ring $K[t]/t^2$ or $K[t]/t^n$ for $n\geq 1$.

Can we parametrize arboreal Galois representations as we do p-adic Galois representations? This is different from the direction of the other remark.

来源字段：http://aimpl.org/galarithdyn/9/

12550 | AIM-DYNAMICAL_SYSTEMS-0077

### Ingram's critical-height theorem and an explicit unicritical height estimate

S / solved | AIM 研讨会问题汇编

**Critical height conjecture.** Let \(f:\mathbb P^1\to\mathbb P^1\), not Lattès. Define \[ \widehat h_{\mathrm{crit}}(f)= \sum_{c\in\mathrm{crit}(f)}\widehat h(c). \] Do there exist \(c_1,c_2>0\), depending on \(d,h_M\), such that for all \(f\in M_d(\overline{\mathbb Q})-\{\text{Lattès}\}\), \[ c_1h_M(f)-c_2\leq \widehat h_{\mathrm{crit}}(f)? \]

来源字段：http://aimpl.org/finitedynamics/1/

12551 | AMR-052-0088

### Injectivity radius from the number of generators

P / partially_solved | AMR 问题清单汇编

If a complete hyperbolic $3$-manifold $N$ has fundamental group generated by $n$ elements, is there a bound $R_n$, depending only on $n$, on the radius of an embedded ball contained entirely in its convex core?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12552 | OWR-3474-001

### Instability of Multi-Peaked Orientational Steady States

O / open | Oberwolfach 报告问题汇编

For the noisy orientational aggregation model on $S^1$, under the hypotheses giving $N$-peaked steady states for sufficiently small noise and $N\geq3$, are all such steady states unstable?

来源字段：https://doi.org/10.4171/owr/2009/24

12553 | OWR-4806-003

### Integrability Under Hirota-Kimura Discretization

O / open | Oberwolfach 报告问题汇编

Does every algebraically completely integrable system with a quadratic vector field remain algebraically completely integrable after applying its Hirota-Kimura discretization?

来源字段：https://doi.org/10.4171/owr/2011/16

12554 | OWR-5153-001

### Intermediate Dimensions of Nonergodic Directions on Translation Surfaces

O / open | Oberwolfach 报告问题汇编

For translation surfaces, can the Hausdorff dimension of the set of non-ergodic directions or divergent directions take a value strictly between $0$ and $1/2$?

来源字段：https://doi.org/10.4171/owr/2011/25

12555 | AIM-DYNAMICAL_SYSTEMS-0026

### Intrinsic versus modular bad reduction for quadratic COR curves

P / partially_solved | AIM 研讨会问题汇编

Bad reduction for models of critical orbit relation curves. e.g. $M_2$.

**Recovered question.** For COR curves in $M_2$, determine bad primes separately for the abstract normalized curve, its punctures, and its critical/portrait marking; explain which bad primes are intrinsic and which are artifacts or necessary degenerations of a chosen modular model.

来源字段：http://aimpl.org/galarithdyn/6/

12556 | OWR-4137-005

### Invariant Analytic Jordan Curves of Rational Maps

P / partially_solved | Oberwolfach 报告问题汇编

Which analytic Jordan curves in the Riemann sphere can be invariant under rational functions? In particular, are there examples not arising from rotation domains or Lattès functions?

来源字段：https://doi.org/10.4171/owr/2009/54

12557 | AMR-052-0044

### Invariant line fields on Julia sets

O / open | AMR 问题清单汇编

Are Lattès maps the only rational maps having measurable invariant line fields on their Julia sets?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12558 | AMR-045-0019

### Invariant measures and subsystems

O / open | AMR 问题清单汇编

Determine all shift-invariant Borel probability measures and all subsystems of the symbolic system $X$ constructed in Section 14 of the source from the Furstenberg commuting-endomorphism problem.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12559 | OWR-717-006

### Invariant Quasicircles for Rational Maps

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a quasicircle $C\subset\mathbb C$ containing the postcritical set of a rational map $R$ and satisfying $C\subseteq R^{-1}(C)$?

来源字段：https://doi.org/10.4171/owr/2004/07

12560 | AMR-039-0008

### Isoperimetric profile and curvature at infinity

O / open | AMR 问题清单汇编

Suppose the global infimum of coarse Ricci curvature is zero, while its infimum on every finite-radius ball about an origin is positive. Is there a systematic relation between the rate at which curvature tends to zero at infinity and the isoperimetric profile? Analyze, for example, the $M/M/k$ queue.

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12561 | AMR-052-0025

### John domains at general Misiurewicz points

S / solved | AMR 问题清单汇编

Analyze Julia and Fatou geometry at a general Misiurewicz parameter whose critical point never returns close to itself. To what extent does the real-quadratic equivalence between this nonrecurrence condition and the Fatou set being a John domain remain valid?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12562 | AMR-045-0042

### Jointly periodic points in one dimension

O / open | AMR 问题清单汇编

Prove that the jointly periodic points of every surjective one-dimensional cellular automaton are dense.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12563 | OWR-2094-003

### Jordan Boundaries of Bounded Polynomial Fatou Components

P / partially_solved | Oberwolfach 报告问题汇编

Is every bounded Fatou component of a polynomial a Jordan domain, meaning a domain whose boundary is a Jordan curve?

来源字段：https://doi.org/10.4171/owr/2008/48

12564 | AMR-052-0019

### Jordan boundary of a cubic parameter component

S / solved | AMR 问题清单汇编

For $f_\lambda(z)=\lambda z^2+z^3$, let $U$ be the parameter component where both finite critical points lie in the immediate basin of zero. Prove that $\partial U$ is a Jordan curve.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12565 | AMR-045-0051

### K-groups of canonical matrix systems

O / open | AMR 问题清单汇编

Which pairs of abelian groups occur as $K_0(M,I)$ and $K_1(M,I)$ for a canonical matrix system of a subshift?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12566 | AMR-048-0009

### Kaplan-Yorke dimension conjecture

P / partially_solved | AMR 问题清单汇编

Under the hypotheses in which the Lyapunov (Kaplan-Yorke) dimension is defined from the ordered Lyapunov exponents, prove that it equals the appropriate dimension of the attractor or invariant measure.

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Dynamical_systems

12567 | OWR-13348-003

### Kaplan-Yorke Equality for Coupled Baker's Maps

O / open | Oberwolfach 报告问题汇编

For bidirectionally coupled skinny baker's maps, does the Kaplan-Yorke equality hold for a prevalent set of coupling functions?

来源字段：https://doi.org/10.4171/owr/2014/46

12568 | AMR-041-0009

### Kolmogorov mixing-torus problem

S / solved | AMR 问题清单汇编

Does there exist a Hamiltonian system with a smooth invariant torus on which the induced dynamics is mixing? No such mixing can occur on a two-dimensional torus; the question asks especially for examples in higher dimensions.

背景链接：https://people.math.harvard.edu/~knill/seminars/intr/

12569 | OWR-14215-002

### Koopman Operators for Switched Infinite-Dimensional Systems

O / open | Oberwolfach 报告问题汇编

Can the joint spectral quantities and stabilizability radius of switched linear systems be generalized to infinite-dimensional linear operators such as Koopman operators, and can a canonical Koopman operator be defined for a given switching system?

来源字段：https://doi.org/10.4171/owr/2016/7

12570 | AMR-039-0018

### L2 Bonnet-Myers and dimension

O / open | AMR 问题清单汇编

Under the strengthened transport estimate $$T_1(m_x^{*t},m_{x'}^{*t'})\leq e^{-\kappa\min(t,t')}d(x,x')+C\frac{(\sqrt t-\sqrt{t'})^2}{2d(x,x')},$$ the diameter is at most $\pi\sqrt{C/(2\kappa)}$. Is $C$ intrinsically related to a dimension, in particular to the parameter $n$ in the Bakry-Émery condition $CD(K,n)$?

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12571 | AIM-DYNAMICAL_SYSTEMS-0009

### Labeled subgroups of subshift full groups

O / open | AIM 研讨会问题汇编

Let \(\Omega\subseteq B^{\mathbb Z}\) be a two-sided subshift with shift \(\sigma\), and let \(\{\delta_a:a\in A\}\) be a specified finite family in the topological full group \([[\sigma]]\). Study the subgroup \(H=\langle\delta_a:a\in A\rangle\).

That page gives Problem 2.4 no introduction, proposer, status, remarks, or bibliography. In particular, it does not define:

- the space or action whose topological full group is intended; - the alphabet or index set \(A\); - the homeomorphisms \(\delta_a\); or - the probability measure and convention meant by “Liouville property.”

The preceding Problem 2.2 on the same page defines a two-sided subshift \(\Omega\subseteq A^{\mathbb Z}\), the shift \(\sigma\), and its topological full group \(F(\Omega)\). It is plausible that Problem 2.4 was intended to inherit that setting, but the page contains no cross-reference. Problem 2.1 uses different named generators \(\varphi_a\), not \(\delta_a\), and gives no identification between them.

The official 2024 AIM workshop report has a section on the structure of topological full groups and reproduces Problems 2.1 and 2.2 as Questions 0.3 and 0.4. It then moves to the entropy problem as Question 0.5 and does not reproduce Problem 2.4. Therefore neither the official report nor the archived source recovers \(\delta_a\).

The literal prompt has no determinate truth value. The conservative reconstruction analyzed below is explicitly conditional:

The symbol \(B\) is used for the subshift alphabet and \(A\) only for generator labels, since the source does not establish that they coincide.

来源字段：http://aimpl.org/groupdynamorigin/2/

12572 | OWR-11101923-002

### Land Heterogeneity and Monin-Obukhov Similarity Uncertainty

O / open | Oberwolfach 报告问题汇编

How much uncertainty in Monin-Obukhov similarity theory can be systematically attributed to local heterogeneity of the land surface?

来源字段：https://doi.org/10.4171/owr/2022/42

12573 | OWR-5153-006

### Laurent Coefficients and Rotational Components of Singular One-Forms

O / open | Oberwolfach 报告问题汇编

Let $\omega=f(z)\,dz$ be a holomorphic one-form on the punctured unit disk, with an essential singularity at $z=0$ and Laurent expansion $f(z)=\sum_{k=-\infty}^{\infty}a_kz^k$. Can the coefficients $(a_k)$ determine the rotational components of the linear-approach space $L(0)$?

来源字段：https://doi.org/10.4171/owr/2011/25

12574 | AMR-052-0043

### Lebesgue ergodicity on a spherical Julia set

O / open | AMR 问题清单汇编

If $J(f)=\widehat{\mathbb C}$, is $f$ ergodic for Lebesgue measure? At least, does it have at most $2\deg f-2$ ergodic components?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12575 | AMR-049-0006

### Length ratios of parallel short trajectories

O / open | AMR 问题清单汇编

Unfold a billiard trajectory in a regular $n$-gon by reflecting the polygon across each encountered side. A short trajectory joins two vertices and meets no vertex in between; a reachable point is an endpoint vertex occurring in such an unfolding. For an oriented short trajectory, let $\alpha$ and $\beta$ be its source and target endpoint angles as in Definition 2.1 of the source, and write $A_k$ for the resulting one of $n-2$ trajectory types (indices modulo $n-2$); in particular $A_0$ is characterized by $\beta-\alpha=2\pi/n$. Prove that the length ratio of parallel short trajectories of types $A_k$ and $A_\ell$ is $$\sin\frac{(k+1)\pi}{n}:\sin\frac{(\ell+1)\pi}{n}.$$

背景链接：https://amj.math.stonybrook.edu/pdf-Springer-final/020-0170.pdf

12576 | AMR-052-0078

### Lift-family criterion for arbitrary kneading data

O / open | AMR 问题清单汇编

Find a general property of a lifting family that guarantees convergence of the real Thurston algorithm for arbitrary kneading sequences.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12577 | AMR-052-0077

### Lift-family criterion for finite kneading data

O / open | AMR 问题清单汇编

Find a general property of a lifting family that guarantees convergence of the real Thurston algorithm for every periodic or preperiodic kneading sequence.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12578 | AMR-052-0052

### Lifting invariant measures through coding trees

P / partially_solved | AMR 问题清单汇编

For a holomorphic quasi-repeller $\Lambda$, is every invariant ergodic measure on $\overline\Lambda$ the image of a measure on a one-sided shift under a nearby geometric coding tree? What if the measure has positive entropy, and what is the answer for measures on Julia sets of rational maps?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12579 | AIM-DYNAMICAL_SYSTEMS-0053

### Light cones, return events, and ASIP truncations for coupled map lattices

P / partially_solved | AIM 研讨会问题汇编

What can be said about return times for coupled map lattices on an infinite lattice? What can be said about almost sure invariance principles for such maps? How fast does a signal propagate to spatial infinity? Also, what happens for finite-range versus infinite-range coupling?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12580 | OWR-4539-004

### Limit Cycles and Zeros of Abelian Integrals

O / open | Oberwolfach 报告问题汇编

For the polynomial perturbation of a planar Hamiltonian system with ovals $\{\gamma_h\}$ and Abelian integral $I(h)=\oint_{\gamma_h}(f(x,y)\,dy-g(x,y)\,dx)$, how many periodic orbits of the unperturbed system persist or bifurcate as periodic orbits of the perturbed system for small $\varepsilon$?

来源字段：https://doi.org/10.4171/owr/2010/57

12581 | AIM-DYNAMICAL_SYSTEMS-0059

### Limit laws and a deterministic variance-switching obstruction for moving billiards

P / partially_solved | AIM 研讨会问题汇编

**Problem 16.** What kind of limit theorems can be found for random billiards with moving/deforming scatterers?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12582 | AMR-052-0053

### Limit laws on holomorphic quasi-repellers

P / partially_solved | AMR 问题清单汇编

Characterize the positive-entropy invariant measures $m$ on a holomorphic quasi-repeller for which the almost-sure invariance principle, law of the iterated logarithm, and central limit theorem hold for Birkhoff sums of every reasonable observable with positive variance.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12583 | AMR-052-0008

### Limit of tunings along growing periods

O / open | AMR 问题清单汇编

Let $P_{1,k}$ have a superstable orbit whose period tends to infinity and suppose $P_{1,k}\to P_{1,\infty}$. Do the tunings with a fixed polynomial $P_2$ also converge to $P_{1,\infty}$?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12584 | OWR-1323-008

### Linear Subspaces of Hereditarily Hypercyclic Vectors

O / open | Oberwolfach 报告问题汇编

For a hereditarily hypercyclic operator $T$, does $HC(T)$ contain the range of an injective bounded operator $S$? If so, is every linearly independent $n$-tuple in $\operatorname{Im}(S)$ hypercyclic for the direct sum $T\oplus\cdots\oplus T$?

来源字段：https://doi.org/10.4171/owr/2006/37

12585 | AMR-045-0001

### Little shift equivalence conjecture

O / open | AMR 问题清单汇编

If a nonnegative integer matrix $A$ has a unique, simple, nonzero eigenvalue $n$, prove that $A$ is strong shift equivalent over $\mathbb Z_+$ to the $1\times1$ matrix $[n]$.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12586 | AMR-039-0009

### Local assumptions for concentration

P / partially_solved | AMR 问题清单汇编

Can the bounded-local-variance hypothesis used for concentration under positive coarse Ricci curvature be relaxed while retaining estimates governed by the actual local variance, including estimates that remain bounded in a continuous-time limit when transition probabilities become small?

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12587 | OWR-17127-002

### Local Birkhoff Rigidity Near Elliptical Billiards

P / partially_solved | Oberwolfach 报告问题汇编

Let a smooth strictly convex planar billiard have invariant caustics with every rotation number in $(0,\delta)$ for some $\delta>0$. Under what hypotheses does this boundary foliation force the billiard table to be an ellipse?

来源字段：https://doi.org/10.4171/owr/2019/31

12588 | AMR-052-0037

### Local connectivity for bounded-type renormalization

S / solved | AMR 问题清单汇编

If a quadratic polynomial $f_c$ is infinitely renormalizable of bounded type, must $J(f_c)$ be locally connected? In particular, is the Julia set of the quadratic Feigenbaum map locally connected?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12589 | AMR-052-0085

### Local connectivity of geometrically finite Julia components

P / partially_solved | AMR 问题清单汇编

Prove that every connected component of the Julia set of a geometrically finite rational map is locally connected.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12590 | AMR-052-0072

### Local connectivity of quadratic Julia sets

O / open | AMR 问题清单汇编

For $P_c(z)=z^2+c$ with connected Julia set, characterize the parameters $c$ for which $J(P_c)$ is locally connected.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12591 | AMR-052-0038

### Local connectivity of real quadratic Julia sets

S / solved | AMR 问题清单汇编

For every real $c\in[-2,1/4]$, is the Julia set of $f_c(z)=z^2+c$ locally connected?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12592 | AMR-052-0012

### Local connectivity of the Mandelbrot set

O / open | AMR 问题清单汇编

Is the Mandelbrot set locally connected? Equivalently, for the quadratic family $z\mapsto z^2+\lambda$, is the boundary of the unbounded component of the structurally stable parameter set locally connected?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12593 | AMR-141-0006

### Local Discrete Holomorphic Dynamics - Open Problem OP1

O / open | AMR 问题清单汇编

\emph{Find families $\{f_{\lambda, a}(z)=\lambda z+z^2 g_a(z)\}_{a\in M}\subset\mathop\mathrm{End}\nolimits(\mathbb{C},O)$ of elliptic discrete holomorphic dynamical systems such that $f_{\lambda,a}$ is holomorphically linearizable if and only if $\lambda$ is a Brjuno number.} For instance, *is it true that an elliptic polynomial is holomorphically linearizable if and only if its multiplier is a Brjuno number?* Or, even more specifically, *is it true that an elliptic cubic polynomial is holomorphically linearizable if and only if its multiplier is a Brjuno number?*

In this context, it might be useful to remember the following dichotomy due to Ilyashenko and Perez-Marco:

\begin{theorem}[Ilyashenko, 1979 [I1], Perez-Marco, 2001 [P9]]

Given $g\in\mathop\mathrm{End}\nolimits(\mathbb{C},O)$ and $\lambda\in \mathbb{C}^*$ not a root of unity, put \[ f_{\lambda,a}=\lambda z+azg(z) \] for all $a\in\mathbb{C}$. Then:

\mathrm{(i)} either the origin is a Siegel point of $f_{\lambda,a}$ for all $a\in\mathbb{C}$, or

\mathrm{(ii)} the origin is a Cremer point of $f_{\lambda,a}$ for all $a\in\mathbb{C}\setminus K$, where $K\subset\subset\mathbb{C}$ is a bounded exceptional set of capacity (and hence Lebesgue measure) zero.

\end{theorem}

背景链接：https://arxiv.org/abs/1107.4876

12594 | AMR-141-0014

### Local Discrete Holomorphic Dynamics - Open Problem OP10

O / open | AMR 问题清单汇编

\emph{Classify with respect to formal conjugation the superattracting germs $f\in\mathop\mathrm{End}\nolimits(\mathbb{C}^n,O)$ of order 2 when $n\ge 3$, or of order $3$ when $n=2$.}

背景链接：https://arxiv.org/abs/1107.4876

12595 | AMR-141-0015

### Local Discrete Holomorphic Dynamics - Open Problem OP11

O / open | AMR 问题清单汇编

Of course, one would like to extend these results at least to dimension 3, and thus we can add two more open problems to the list:

*Classify $3$-dimensional non-invertible dominant rigid germs.*

背景链接：https://arxiv.org/abs/1107.4876

12596 | AMR-141-0016

### Local Discrete Holomorphic Dynamics - Open Problem OP15

O / open | AMR 问题清单汇编

*Find formal normal forms and invariants for germs tangent to the identity having only degenerate characteristic directions.*

Partial results on this problem can be find in [BM], [AT2] and [AR].

背景链接：https://arxiv.org/abs/1107.4876

12597 | AMR-141-0007

### Local Discrete Holomorphic Dynamics - Open Problem OP2

O / open | AMR 问题清单汇编

The Brjuno function is clearly quite an irregular function, diverging at a $G_\delta$-dense set of irrational numbers . A surprising fact is that, on the contrary, numerical experiments (see, e.g., [Car] and references therein) as well as theoretical results (see [MMY]) suggest that $\log r+B$ is more regular, even better than continuous. More specifically, Marmi, Mattei and Yoccoz proposed the following problem:

*Does $\log r+B$ admit a $1/2$-H\"older continuous extension to $[0,1)$?*

背景链接：https://arxiv.org/abs/1107.4876

12598 | AMR-141-0008

### Local Discrete Holomorphic Dynamics - Open Problem OP3

O / open | AMR 问题清单汇编

\emph{Are the multiplier and the homeomorphism class of the stable set a complete set of topological invariants for discrete holomorphic local dynamical systems in $\mathop\mathrm{End}\nolimits(\mathbb{C},O)$ having a Cremer point at the origin?}

背景链接：https://arxiv.org/abs/1107.4876

12599 | AMR-141-0009

### Local Discrete Holomorphic Dynamics - Open Problem OP4

O / open | AMR 问题清单汇编

*Give an effective procedure for deciding whether two germs tangent to the identity are holomorphically locally conjugated.*

Clearly, a similar question can be asked for parabolic germs; in that case another (topological, formal and holomorphic) invariant is the multiplier, and one has to replace the multiplicity by a suitably defined parabolic multiplicity, and the index by a suitably defined parabolic index (or, better yet, by \'Ecalle's iterative residue). Then the multiplier and the parabolic multiplicity are a complete set of topological invariants, the multiplier, the parabolic multiplicity and the iterative residue are a complete set of formal invariants, and adding suitably adjusted \'Ecalle-Voronin functional invariants one obtains a complete set of holomorphic invariants. Again, most of the times the computation of the functional invariants is hopeless, and thus we have the following generalization of the previous question:

背景链接：https://arxiv.org/abs/1107.4876

12600 | AMR-141-0010

### Local Discrete Holomorphic Dynamics - Open Problem OP5

O / open | AMR 问题清单汇编

\emph{Let $f\in\mathop\mathrm{End}\nolimits(\mathbb{C}^n,O)$ be non-invertible and dominant. Is it true that $c_\infty=c_\infty(f)$ is an algebraic integer of order at most $n$, i.e., there are integers $a_0,\ldots, a_{n-1}\in\mathbb{Z}$ such that $c^n_\infty+a_{k-1} c_\infty^{k-1}+\cdots+a_0=0$? Moreover, does there exist $\delta\in(0, 1]$ such that $\delta c^k_\infty\le c(f^k)\le c^k_\infty$ for all $k\ge 1$?}

Roughly speaking, the order of a germ may be thought of as a sort of analog of the degree of a polynomial map: the former somewhat measure the rate of attraction of the origin, while the latter measure the rate of attraction of infinity. For the sequence of the degree of the iterates of a polynomial map, Favre and Jonsson [FJ2] have proved the following

\begin{theorem}[Favre-Jonsson, 2011 [FJ2]]

Let $F\colon\mathbb{C}^2\to\mathbb{C}^2$ be a polynomial map. Then the sequence $\{\deg F^j\}_{j\in\mathbb{N}}$ satisfies a linear recursion formula with integer coefficients. \end{theorem}

We can then wonder whether a similar property holds for the sequence of orders of the iterates of a germ:

背景链接：https://arxiv.org/abs/1107.4876

12601 | AMR-141-0011

### Local Discrete Holomorphic Dynamics - Open Problem OP6

O / open | AMR 问题清单汇编

\emph{Let $f\in\mathop\mathrm{End}\nolimits(\mathbb{C}^n,O)$ be non-invertible and dominant. Is it true that the sequence $\{c(f^k)\}_{k\in\mathbb{N}}$ satisfies, at least for $k$ large enough, a linear recursion formula with integer coefficients?}

背景链接：https://arxiv.org/abs/1107.4876

12602 | AMR-141-0012

### Local Discrete Holomorphic Dynamics - Open Problem OP7

O / open | AMR 问题清单汇编

\emph{Let $f\in\mathop\mathrm{End}\nolimits(\mathbb{C}^n,O)$ be a superattracting germ, and let $H$ be the first non-zero term in the homogeneous expansion of $f$. Assume that $H$ is degenerate, that is there exists $v\in\mathbb{C}^n\setminus\{O\}$ such that $H(v)=O$. Is it still true that if the postcritical set of $f$ is an analytic cone then $f$ is holomorphically locally conjugated to $H$?}

背景链接：https://arxiv.org/abs/1107.4876

12603 | AMR-141-0013

### Local Discrete Holomorphic Dynamics - Open Problem OP9

O / open | AMR 问题清单汇编

\emph{Let $f$, $g\in\mathop\mathrm{End}\nolimits(\mathbb{C}^n,O)$ be two superattracting germs, with $c(f)=c(g)$. Is it true that $f$ and $g$ are topologically (respectively, holomorphically) locally conjugated if and only if their postcritical sets are topologically (respectively, holomorphically) conjugated, that is, there is a local homeomorphism (respectively, biholomorphism) sending the postcritical set of $f$ onto the postcritical set of $g$?}

Concerning instead the classification problem with respect to the formal conjugacy, [AR] contains a formal classification of superattracting germs $f\in\mathop\mathrm{End}\nolimits(\mathbb{C}^2,O)$ of order 2. The methods given there can in principle be used to attack the general formal classification problem; here we limit ourselves to a more specific question:

背景链接：https://arxiv.org/abs/1107.4876

12604 | AMR-141-0001

### Local Discrete Holomorphic Dynamics - Open Problem Q1

O / open | AMR 问题清单汇编

*What is the topological structure of (the germ at $p$ of) $K_f$?*

For instance, does $K_f$ have non-empty interior? Is it locally connected at $p$? Is $K_f\setminus\{p\}$ connected? What is the topological (homological, cohomological) structure of $U\setminus K_f$? And so on.

\begin{remark} Both the definition of stable set and Question 1 are topological in character; we might also state them for local dynamical systems which are continuous only. However, the *answers* might (and usually will) strongly depend on the holomorphicity of the dynamical system. \end{remark}

Clearly, the stable set $K_f$ is completely $f$-invariant, and thus the pair $(K_f,f)$ is the promised discrete global dynamical system, canonically associated with the given discrete local dynamical system. In particular, the second natural question in discrete holomorphic local dynamics is

背景链接：https://arxiv.org/abs/1107.4876

12605 | AMR-141-0002

## Local Discrete Holomorphic Dynamics - Open Problem Q2

O / open | AMR 问题清单汇编

*What is the dynamical structure of $(K_f,f)$?*

For instance, what is the asymptotic behavior of the orbits? Do they converge to $p$, or have they a chaotic behavior? Is there a dense orbit? Do there exist proper $f$-invariant subsets, that is sets $L\subset K_f$ such that $f(L)\subseteq L$? If they do exist, what is the dynamics on them?

To answer all these questions, one of the most efficient ways is to replace $f$ by a dynamically equivalent but simpler (e.g., linear) system $g$. In our context, ``dynamically equivalent" means ``locally conjugated"; and we have different kinds of conjugacy to consider.

\begin{definition} Let $f_1\colon U_1\to M_1$ and $f_2\colon U_2\to M_2$ be two holomorphic local dynamical systems at $p_1\in M_1$ and $p_2\in M_2$ respectively. We shall say that $f_1$ and $f_2$ are *holomorphically* (respectively, *smoothly,* $C^k$ with $k\in\mathbb{N}^*$, or *topologically*) *locally conjugated* if there are open neighbourhoods $W_1\subseteq U_1$ of $p_1$, $W_2\subseteq U_2$ of $p_2$, and a biholomorphism (respectively, a $C^\infty$ diffeomorphism, a $C^k$ diffeomorphism, or a homeomorphism) $\varphi\colon W_1\to W_2$ with $\varphi(p_1)=p_2$ such that $$ f_1=\varphi^{-1}\circ f_2\circ\varphi $$ on $\varphi^{-1}\bigl( W_2\cap f_2^{-1}(W_2)\bigr)=W_1\cap f_1^{-1}(W_1)$. \end{definition}

If $f_1\colon U_1\to M_1$ and $f_2\colon U_2\to M_2$ are locally conjugated we clearly have $$ K_{f_2|_{W_2}}=\varphi(K_{f_1|_{W_1}})\;; $$ so the local dynamics of $f_1$ about $p_1$ is to all purposes equivalent (up to the order of smoothness of the local conjugation) to the local dynamics of $f_2$ about $p_2$.

In particular, using local coordinates centered at $p\in M$ it is easy to show that any holomorphic local dynamical system at $p$ is holomorphically locally conjugated to a holomorphic local dynamical system at $O\in\mathbb{C}^n$, where $n=\dim M$; so from now on we shall mostly work only with $\mathop\mathrm{End}\nolimits(\mathbb{C}^n,O)$.

Whenever we have an equivalence relation in a class of objects, classification problems come out. So the third natural question in local holomorphic dynamics is

背景链接：https://arxiv.org/abs/1107.4876

12606 | AMR-141-0003

## Local Discrete Holomorphic Dynamics - Open Problem Q3

O / open | AMR 问题清单汇编

\emph{Find a (possibly small) class $\mathcal{F}$ of holomorphic local dynamical systems at $O\in\mathbb{C}^n$ such that every holomorphic local dynamical system $f$ at a point in an $n$-dimensional complex manifold is holomorphically (respectively, smoothly, $C^k$, or topologically) locally conjugated to a (possibly) unique element of $\mathcal{F}$, called *holomorphic* (respectively, *smooth,* $C^k$ or *topological*) *normal form* of $f$.}

Unfortunately, the holomorphic classification is often too complicated to be practical; the family $\mathcal{F}$ of normal forms might be uncountable. A possible replacement is looking for invariants instead of normal forms:

背景链接：https://arxiv.org/abs/1107.4876

12607 | AMR-141-0004

## Local Discrete Holomorphic Dynamics - Open Problem Q4

O / open | AMR 问题清单汇编

\emph{Find a way to associate a (possibly small) class of (possibly computable) objects, called *invariants,* to any holomorphic local dynamical system $f$ at $O\in\mathbb{C}^n$ so that two holomorphic local dynamical systems at $O$ can be holomorphically (respectively, smoothly, $C^k$, or topologically) locally conjugated only if they have the same invariants. The class of invariants is *complete* if two holomorphic local dynamical systems at $O$ are holomorphically (respectively, smoothly, $C^k$, or topologically) locally conjugated if and only if they have the same invariants.}

Contrarily to the previous ones, our final general question makes sense only for *holomorphic* local dynamical systems. Indeed, a discrete holomorphic local dynamical system at $O\in\mathbb{C}^n$ is given by an element of $\mathbb{C}_0\{z_1,\ldots,z_n\}^n$, the space of $n$-tuples of converging power series in $z_1,\ldots,z_n$ without constant terms, which is a subspace of the space $\mathbb{C}_0[\![z_1,\ldots,z_n]\!]^n$ of $n$-tuples of formal power series without constant terms. It is well known that $\mathbb{C}_0[\![z_1,\ldots,z_n]\!]^n$ is closed under composition of power series, and that an element $\Phi\in \mathbb{C}_0[\![z_1,\ldots,z_n]\!]^n$ has an inverse (with respect to composition) still belonging to $\mathbb{C}_0[\![z_1,\ldots,z_n]\!]^n$ if and only if its linear part is a linear automorphism of $\mathbb{C}^n$.

\begin{definition} We say that $f_1$, $f_2\in\mathbb{C}_0[\ negthinspace[z_1,\ldots,z_n]\negthinspace ]^n$ are *formally conjugated* if there is an invertible $\Phi\in\mathbb{C}_0[\![z_1,\ldots,z_n]\!]^n$ such that $f_1=\Phi^{-1}\circ f_2\circ\Phi$ in $\mathbb{C}_0[\![z_1,\ldots,z_n]\!]^n$. \end{definition}

Clearly, two holomorphically locally conjugated holomorphic local dynamical systems are both formally and topologically locally conjugated too. On the other hand, there are examples of holomorphic local dynamical systems that are topologically locally conjugated without being neither formally nor holomorphically locally conjugated, and examples of holomorphic local dynamical systems that are formally conjugated without being neither holomorphically nor topologically locally conjugated. So the last natural general question in local holomorphic dynamics is

背景链接：https://arxiv.org/abs/1107.4876

12608 | AMR-141-0005

## Local Discrete Holomorphic Dynamics - Open Problem Q5

O / open | AMR 问题清单汇编

\emph{Find normal forms and invariants with respect to the relation of formal conjugacy for holomorphic local dynamical systems at $O\in\mathbb{C}^n$.}

In the rest of this paper we shall describe a few specific open problems, both in one and in several complex variables; we refer to [A5,A6] and [Br2] for surveys on the theory of discrete holomorphic local dynamical systems, and for more details on the known answers to the previous questions and on background of the open problems we selected.

背景链接：https://arxiv.org/abs/1107.4876

12609 | AIM-DYNAMICAL_SYSTEMS-0089

## Local inertia and Frobenius in PCF preimage towers

P / partially_solved | AIM 研讨会问题汇编

Inertia groups for preimage towers

- What can you say about inertia subgroups of Galois groups in towers of fields generated by $f^{-n}(\alpha)$ for $f$ PCF? - What happens to decomposition groups?

来源字段：http://aimpl.org/finitedynamics/4/

12610 | AIM-DYNAMICAL_SYSTEMS-0062

## Local return laws and ASIP transfer in infinite coupled map lattices

P / partially_solved | AIM 研讨会问题汇编

**Problem 19.** What can be said about return times in coupled map lattices (CML) in the infinite case? What about related problems regarding almost sure invariance principles?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12611 | AMR-048-0020

## Local reversibility of reversible cellular automata

O / open | AMR 问题清单汇编

In every dimension at least three, is each reversible cellular automaton locally reversible?

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Dynamical_systems

12612 | OWR-8415336-016

## Local Typicality of the Newhouse Phenomenon

P / partially_solved | Oberwolfach 报告问题汇编

Is the Newhouse phenomenon locally typical in the space of smooth dynamical systems?

来源字段：https://doi.org/10.4171/owr/2021/33

12613 | AMR-052-0034

### Locally connected Siegel Julia sets

P / partially_solved | AMR 问题清单汇编

Give an example of a Siegel polynomial whose Julia set is provably locally connected. Is the Julia set locally connected for Lebesgue-almost every Siegel rotation angle, and what can be said about its Hausdorff dimension?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12614 | OWR-4137-009

### Locally Omitted Values in Completely Invariant Fatou Components

P / partially_solved | Oberwolfach 报告问题汇编

Let $f$ be an entire function with a completely invariant Fatou component $D$. Must every locally omitted value of $f$ belong to $D$, where $a$ is locally omitted if some component $G$ of $\{z:|f(z)-a|<\varepsilon\}$ contains no solution of $f(z)=a$?

来源字段：https://doi.org/10.4171/owr/2009/54

12615 | AMR-046-0025

### Loewner's conjecture

P / partially_solved | AMR 问题清单汇编

Let $f$ be real analytic near the origin, with $f(0,0)=0$, and let $n>1$. Suppose the origin is an isolated equilibrium of $$\dot x=2^n\operatorname{Re}\!\left(\frac{\partial^n f}{\partial\bar z^n}\right),\qquad \dot y=2^n\operatorname{Im}\!\left(\frac{\partial^n f}{\partial\bar z^n}\right).$$ Prove that the index of this vector field at the origin is at most $n$.

背景链接：https://arxiv.org/abs/2012.02524

12616 | AMR-039-0001

### Log-concave measures

P / partially_solved | AMR 问题清单汇编

For Ollivier's coarse Ricci curvature, smooth uniformly strictly log-concave measures on $\mathbb{R}^N$ have positive curvature. What can be said for a general log-concave measure? In particular, analyze a convex body equipped with Brownian motion conditioned not to leave it.

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12617 | AMR-046-0001

### Low degree rigid systems

O / open | AMR 问题清单汇编

Consider the planar cubic rigid systems $$\begin{cases}\dot x=-y+x(a+bx+cy+dx^2+exy),\\ \dot y=x+y(a+bx+cy+dx^2+exy).\end{cases}$$ Is $2$ the maximum number of limit cycles?

背景链接：https://arxiv.org/abs/2012.02524

12618 | AMR-046-0004

### Low-degree classical Liénard systems

P / partially_solved | AMR 问题清单汇编

For the classical polynomial Liénard system $$\dot x=y-F(x),\qquad \dot y=-x,$$ let $\operatorname{Lie}(n)$ be the maximum number of limit cycles when $F$ has degree $n$. Find proofs using Dulac functions that $\operatorname{Lie}(3)=1$ and $\operatorname{Lie}(4)=1$.

背景链接：https://arxiv.org/abs/2012.02524

12619 | AIM-DYNAMICAL_SYSTEMS-0096

### Low-period multiplier-locus conspiracies

P / partially_solved | AIM 研讨会问题汇编

Let \(\mathop{Per}_n(\lambda)\) be the locus in the moduli space of formal \(n\)-cycles with multiplier \(\lambda\). \(\mathop{Per}_n(\lambda)\) is in general an irreducible cubic, \(\\mathop{Per}_3(1)\) factors. Let \(\mathop{Per}^{\ast}_n(\lambda)\) be the same but with actual \(n\)-cycles instead of formal \(n\)-cycles. \(\mathop{Per}^{\ast}_3(1)=\mathop{Per}_2(-3)\) are the same line in \(\mathbb A^2\) (Milnor). \(\mathop{Per}_n(\lambda)\cap\mathop{Per}_m(\lambda')\) should be zero dimensional (need at least one multiplier outside the unit circle). Are there any other examples \(n,m,\lambda,\lambda'\) where the intersection is not zero dimensional?

来源字段：http://aimpl.org/finitedynamics/6/

12620 | AIM-DYNAMICAL_SYSTEMS-0067

### Lyapunov points versus uniform dichotomy intervals

P / partially_solved | AIM 研讨会问题汇编

Problem 24. What can be said about spectral properties/Lyapunov spectrum in sequential or nonstationary settings?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12621 | OWR-1112-004

### Lyapunov Positivity for Smooth Quasiperiodic Schrödinger Operators

O / open | Oberwolfach 报告问题汇编

If a smooth potential $v$ on the circle satisfies the transversality condition and $\omega$ satisfies a weak Diophantine condition, does there exist $\lambda_0$ such that for every $|\lambda|>\lambda_0$ and every energy $E$, the Lyapunov exponent of the quasiperiodic Schrödinger operator is positive, indeed at least $\tfrac14\log|\lambda|$?

来源字段：https://doi.org/10.4171/owr/2005/51

12622 | OWR-5157-003

### Macroscopic Pivotals Without Infinite-Cluster Exceptional Times

O / open | Oberwolfach 报告问题汇编

Does there exist a regime $q\in(q^*,4)$, equivalently $\kappa\in(4,\kappa^*)$ in the $\mathrm{SLE}_\kappa$ setting, for critical heat-bath dynamics in which macroscopic pivotals exist but no exceptional time has an infinite cluster?

来源字段：https://doi.org/10.4171/owr/2011/29

12623 | AMR-036-0005

### Makienko conjecture

P / partially_solved | AMR 问题清单汇编

Let $f:\widehat{\mathbb C}\to\widehat{\mathbb C}$ be rational with Julia set $J$, and suppose that a component $D$ of $\widehat{\mathbb C}\setminus J$ satisfies $\partial D=J$. Must $D$ be completely invariant under $f^2$, that is, $f^{-2}(D)=D$?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

12624 | AMR-048-0010

### Margulis measure-classification conjecture

P / partially_solved | AMR 问题清单汇编

Classify invariant ergodic probability measures for higher-rank diagonalizable group actions on homogeneous spaces; in particular, prove that the measures satisfying the usual nondegeneracy hypotheses are algebraic (homogeneous).

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Dynamical_systems

12625 | AMR-045-0028

### Markov random fields and Bernoulli shifts

O / open | AMR 问题清单汇编

If a translation-invariant Markov random field $\mu$ on a shift is the unique Markov random field, even without assuming translation invariance, with its conditional probabilities, must the measured shift be isomorphic to a Bernoulli shift?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12626 | AIM-DYNAMICAL_SYSTEMS-0058

### Martingales from sequential transfer contraction

P / partially_solved | AIM 研讨会问题汇编

**Problem 15.** Can one use generalized function spaces to get martingales for concatenations of random hyperbolic maps?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12627 | AMR-041-0014

### Mather theory near integrable systems

O / open | AMR 问题清单汇编

Are there quasiperiodic global minimals for metrics on the torus that are close to a flat three-dimensional torus? Here a geodesic is a global minimal if the segment between any two of its points is a minimizing geodesic.

背景链接：https://people.math.harvard.edu/~knill/seminars/intr/

12628 | AIM-DYNAMICAL_SYSTEMS-0035

### Maximal dynatomic images: a one-cycle criterion and low-period classification

P / partially_solved | AIM 研讨会问题汇编

Formal period \(n\), Problem 11.1: “When is the image maximal?”

来源字段：http://aimpl.org/galarithdyn/11/

12629 | AIM-DYNAMICAL_SYSTEMS-0071

### Maximal entropy and periodic orbits for dispersing billiards

P / partially_solved | AIM 研讨会问题汇编

**Problem 28.** Can one prove existence and/or physical properties of a measure of maximal entropy for dispersing billards? Connections to periodic orbits?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12630 | AIM-DYNAMICAL_SYSTEMS-0047

### Maximal entropy for finite-horizon Lorentz collision maps

P / partially_solved | AIM 研讨会问题汇编

Problem 4. For a discrete-time Lorentz gas with a finite horizon, is there always a measure of maximal entropy?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12631 | OWR-1319-028

### Maximal-Rank Smooth Abelian Actions

O / open | Oberwolfach 报告问题汇编

Which compact $(k+1)$-manifolds carry $C^{1+\varepsilon}$ $\mathbb Z^k$-actions with no proportional Lyapunov exponents and with some element of positive entropy?

来源字段：https://doi.org/10.4171/owr/2006/33

12632 | AMR-046-0014

### Maximum number of critical periods

O / open | AMR 问题清单汇编

Let $\mathcal T(n)$ be the maximum number of critical periods that a planar polynomial differential system of degree $n$ can have. Is there a constant $C>0$ such that $\mathcal T(n)\ge Cn^2\log n$?

背景链接：https://arxiv.org/abs/2012.02524

12633 | AMR-052-0065

### Measure and dimension of transcendental parameter hairs

P / partially_solved | AMR 问题清单汇编

Determine the measure and Hausdorff dimension of the parameter hairs in the exponential, sine, and cosine families.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12634 | OWR-4132-002

### Meromorphic Ruelle Zeta Functions for Piecewise Hyperbolic Systems

P / partially_solved | Oberwolfach 报告问题汇编

For piecewise-affine dynamical systems preserving complementary stable and unstable cone fields, construct meromorphic extensions of their Ruelle zeta functions and determine their domains of continuation.

来源字段：https://doi.org/10.4171/owr/2009/49

12635 | OWR-12175-012

### Metric Mean Dimension for Amenable Group Actions

P / partially_solved | Oberwolfach 报告问题汇编

For a continuous action of a countable amenable group on a compact metrizable space, is mean topological dimension equal to the minimum metric mean dimension over all compatible metrics?

来源字段：https://doi.org/10.4171/owr/2012/52

12636 | OWR-13687-004

### Minimal Dilatations of Pseudo-Anosov Maps

O / open | Oberwolfach 报告问题汇编

What are the exact values and qualitative properties of pseudo-Anosov maps realizing the minimal length $\ell_g$, or more generally with $\log\lambda(f)\asymp g^{-1}$?

来源字段：https://doi.org/10.4171/owr/2015/30

12637 | OWR-1112-001

### Minimal Sets and Lyapunov Positivity for Quasiperiodic Schrödinger Cocycles

P / partially_solved | Oberwolfach 报告问题汇编

For a quasiperiodic Schrödinger cocycle at the lowest spectral energy $E_0$ with positive Lyapunov exponent, determine the structure of its minimal set under sufficient regularity of the potential and rational independence of the frequency; in particular, can the fibers be Cantor sets? Also determine whether sufficiently large coupling forces a positive Lyapunov exponent for all energies when the potential is only finitely smooth.

来源字段：https://doi.org/10.4171/owr/2005/51

12638 | OWR-1460-003

### Minimal Unique-Range Sets for Entire Functions

P / partially_solved | Oberwolfach 报告问题汇编

For a set $S\subset\mathbb C$, let $E(S,f)$ be the multiset of points at which $f$ takes a value in $S$, counted with multiplicity. What is the minimum cardinality of a finite $S$ such that $E(S,f)=E(S,g)$ for nonconstant entire functions $f$ and $g$ implies $f=g$?

来源字段：https://doi.org/10.4171/owr/2007/09

12639 | OWR-11131-001

### Minimum-Energy $N$-Body Configurations at Fixed Angular Momentum

P / partially_solved | Oberwolfach 报告问题汇编

What is the minimum-energy configuration of an $N$-body system at a fixed angular-momentum level?

来源字段：https://doi.org/10.4171/owr/2011/39

12640 | OWR-12864-001

### Mixing and Transport from Finite-Time Noisy Data

O / open | Oberwolfach 报告问题汇编

Extend the theory of mixing and transport from periodic, quasiperiodic, and single-frequency-dominated models to realistic finite-time noisy observations, including stochastic effects.

来源字段：https://doi.org/10.4171/owr/2014/04

12641 | AMR-042-0002

### Mixing of all orders

P / partially_solved | AMR 问题清单汇编

For $\mathbb{Z}^d$-actions by automorphisms of a connected group, mixing actions are mixing of all orders. Can this result be proved using simpler ideas from dynamics, without the known Diophantine estimates?

背景链接：https://www.imath.kiev.ua/~skolyada/kevin.pdf

12642 | OWR-12007-014

### Mixing Zero-Entropy Area-Preserving Disk Diffeomorphisms

O / open | Oberwolfach 报告问题汇编

别名：Answered Survey of Zero-Entropy Disk Dynamics [OWR-12007-013]; Duplicate Mixing Zero-Entropy Disk Question [OWR-12007-015]

Does there exist a mixing area-preserving $C^\infty$-diffeomorphism of the closed $2$-disk with zero topological entropy?

来源字段：https://doi.org/10.4171/owr/2012/35

12643 | OWR-4990373-005

### Monotonicity of Opinion Distributions in Median Dynamics

O / open | Oberwolfach 报告问题汇编

In median dynamics on a graph $G=(V,E)$, each vertex starts with an independent uniform opinion in $[0,1]$ and updates at rate one to the median of its neighbors' opinions, including its own opinion when the degree is even. If $\mu_t(x)$ is the marginal opinion measure at vertex $x$ at time $t$, is $t\mapsto\mu_t(x)$ monotone non-decreasing for every $x\in V$ and every $\alpha>1/2$?

来源字段：https://doi.org/10.4171/owr/2021/4

12644 | AMR-047-0001

### Multiple ergodic averages - Problem 1

O / open | AMR 问题清单汇编

Determine the structure of the multiple correlation sequences $(\mathcal{C}(n_1,\ldots,n_\ell))$ defined by (source reference E:MultCor). Is it true that any such sequence is an (approximate) integral combination of generalized $\ell$-step nilsequences in $\ell$-variables?

背景链接：https://arxiv.org/abs/1103.3808

12645 | AMR-047-0010

### Multiple ergodic averages - Problem 10

P / partially_solved | AMR 问题清单汇编

Suppose that the sequence of $\ell$-tuples of polynomials $(p_{1,N},\ldots, p_{\ell,N})$ is good. Show that for every ergodic system $(X,\mathcal X,\mu,T)$ and functions $f_1,\ldots,f_\ell\in L^\infty(\mu)$ we have $$ \lim_{N\to\infty}\frac{1}{N}\sum_{n=1}^N\, T^{[p_{1,N}(n)]}f_1\cdots T^{[p_{\ell,N}(n)]}f_\ell=\int f_1\, d\mu\, \cdots \int f_\ell\, d\mu $$ where convergence takes place in $L^2(\mu)$.

背景链接：https://arxiv.org/abs/1103.3808

12646 | AMR-047-0011

### Multiple ergodic averages - Problem 11

P / partially_solved | AMR 问题清单汇编

Let $(X,\mathcal X,\mu,T)$ be a system and $f, g, h\in L^\infty(\mu)$ be functions. Show that the averages $$ \frac{1}{N}\sum_{n=1}^N f(T^nx)\cdot g(T^{2n}x)\cdot h(T^{3n}x) \quad \text{and the averages} \quad \frac{1}{N}\sum_{n=1}^N f(T^nx)\cdot g(T^{n^2}x) $$ converge pointwise almost everywhere.

背景链接：https://arxiv.org/abs/1103.3808

12647 | AMR-047-0012

### Multiple ergodic averages - Problem 12

P / partially_solved | AMR 问题清单汇编

Let $(X,\mathcal X,\mu,T)$ be a system and $f,g\in L^\infty(\mu)$ be functions. If $\Lambda$ is the von Mangoldt function and $\phi$ is a multiplicative function that takes values on the complex unit disc and has convergent means, then show that the averages $$ \frac{1}{N}\sum_{n=1}^N \Lambda(n)\, f(T^nx)\cdot g(T^{2n}x) \quad \text{and the averages} \quad \frac{1}{N}\sum_{n=1}^N \phi(n)\, f(T^nx)\cdot g(T^{2n}x) $$ converge pointwise almost everywhere.

背景链接：https://arxiv.org/abs/1103.3808

12648 | AMR-047-0013

### Multiple ergodic averages - Problem 13

S / solved | AMR 问题清单汇编

Let $(a(n))$ be the sequence of integers $(p_n)$, where $p_n$ is the $n$-th prime, or $([n^c])$ where $c>0$, or $(2^n)$. Is it true that for every ergodic system $(X,\mathcal X,\mu,T)$ and all functions $f_0,\ldots, f_\ell\in L^\infty(\mu)$, one has a decomposition $$ \int f_0 \cdot T^{a(n)} f_1 \cdot \ldots \cdot T^{\ell a(n)} f_\ell \ d\mu= \psi(a(n))+e(n), $$ where $(\psi(n))$ is an ($\ell$-step) nilsequence and $\lim_{N\to\infty}\frac{1}{N}\sum_{n=1}^N|e(n)|=0$?

背景链接：https://arxiv.org/abs/1103.3808

12649 | AMR-047-0014

### Multiple ergodic averages - Problem 14

O / open | AMR 问题清单汇编

Let $p_1,\ldots, p_\ell$ be integer valued generalized polynomials. Show that the averages $$ \frac{1}{N}\sum_{n=1}^{N} T_1^{p_1(n)}f_1\cdots T_\ell^{p_\ell(n)}f_\ell $$ converge in the mean as $N\to\infty$.

背景链接：https://arxiv.org/abs/1103.3808

12650 | AMR-047-0015

### Multiple ergodic averages - Problem 15

S / solved | AMR 问题清单汇编

Suppose that the polynomials $p_1,\ldots, p_\ell\in \mathbb Z[t]$ are pairwise independent. Show that there exists $d\in \mathbb N$ such that the factors $\mathcal Z_{d,T_1}, \ldots, \mathcal Z_{d,T_\ell}$ are characteristic factors for the averages (source reference E:Multies).

背景链接：https://arxiv.org/abs/1103.3808

12651 | AMR-047-0016

### Multiple ergodic averages - Problem 16

S / solved | AMR 问题清单汇编

Suppose that the polynomials $p_1,\ldots, p_\ell\in \mathbb Z[t]$ are rationally independent. Show that $\mathcal{K}_{rat}(T_1),\ldots, \mathcal{K}_{rat}(T_\ell)$ are characteristic factors for the averages (source reference E:Multies).

背景链接：https://arxiv.org/abs/1103.3808

12652 | AMR-047-0017

### Multiple ergodic averages - Problem 17

S / solved | AMR 问题清单汇编

Suppose that the polynomials $p_1,\ldots,p_\ell\in \mathbb Z[t]$ are rationally independent and have zero constant term. Show that for every $A\in \mathcal X$ and every $\varepsilon>0$, there exists $n\in \mathbb N$ such that $$ \mu(A\cap T_1^{p_1(n)}A\cap \cdots\cap T_\ell^{p_\ell(n)}A)\geq \mu(A)^{\ell+1}-\varepsilon. $$

背景链接：https://arxiv.org/abs/1103.3808

12653 | AMR-047-0018

### Multiple ergodic averages - Problem 18

P / partially_solved | AMR 问题清单汇编

Let $(X,\mathcal X,\mu,T_1,\ldots, T_\ell)$ be a system and $\{p_1,\ldots, p_\ell\}$ be a family of intersective integer polynomials. Show that for every set $A\in \mathcal X$ with $\mu(A)>0$ one has $$ \mu(A\cap T_1^{p_1(n)}A\cap \cdots\cap T_\ell^{p_\ell(n)}A)>0 $$ for some $n\in\mathbb N$.

背景链接：https://arxiv.org/abs/1103.3808

12654 | AMR-047-0019

### Multiple ergodic averages - Problem 19

P / partially_solved | AMR 问题清单汇编

Let $(X,\mathcal X,\mu, T,S)$ be a system and $f,g\in L^\infty(\mu)$ be functions. Show that the averages $$ \frac1N \sum_{n=1}^N f(T^nx)\cdot g(S^nx) $$ converge pointwise almost everywhere.

背景链接：https://arxiv.org/abs/1103.3808

12655 | AMR-047-0002

### Multiple ergodic averages - Problem 2

P / partially_solved | AMR 问题清单汇编

Let $\mathcal C_{T,S}$ be the set of sequences $(\int f\,T^ng\,S^nh\,d\mu)_{n\ge1}$ over probability-preserving systems with commuting $T,S$ and bounded $f,g,h$, and let $\mathcal C_T$ be the subclass in which $T$ and $S$ are powers of one transformation. Show that $\mathcal{C}_{T,S}=\mathcal{C}_{T}$.

背景链接：https://arxiv.org/abs/1103.3808

12656 | AMR-047-0020

### Multiple ergodic averages - Problem 20

P / partially_solved | AMR 问题清单汇编

Is it true that one always has a decomposition $$ \int f_0 \cdot T_1^n f_1 \cdot \ldots \cdot T_\ell^n f_\ell \ d\mu= \psi(n)+e(n) $$ where $(\psi(n))$ is an $\ell$-step nilsequence and $\lim_{N\to\infty}\frac{1}{N}\sum_{n=1}^N|e(n)|=0$?

背景链接：https://arxiv.org/abs/1103.3808

12657 | AMR-047-0021

### Multiple ergodic averages - Problem 21

O / open | AMR 问题清单汇编

Let $(X,\mathcal X,\mu)$ be a probability space, $T_1,\ldots, T_\ell \colon X\to X$ be invertible measure preserving transformations, and $p_1,\ldots,p_\ell$ be distinct polynomials with zero constant term. Show that for every $A\in \mathcal X$ with $\mu(A)>0$ we have $$ \mu(A\cap T_1^{m+p_1(n)}A\cap\cdots\cap T_\ell^{m+p_\ell(n)}A)>0 $$ for some $m,n\in \mathbb N$.

背景链接：https://arxiv.org/abs/1103.3808

12658 | AMR-047-0022

### Multiple ergodic averages - Problem 22

P / partially_solved | AMR 问题清单汇编

Here $\mathcal F=\{a_1,\ldots,a_\ell\}$ is a family of functions of polynomial growth in one Hardy field, and $\operatorname{span}^*(\mathcal F)$ denotes its nonzero linear combinations. . Show that the family of sequences $\{([a_1(n)]),\ldots,$ $([a_\ell(n)])\}$ is good for $\ell$-convergence of a single transformation if and only if every function $a\in\text{span}^*(\mathcal{F})$ satisfies one of the following conditions: • $|a(t)-cp(t)|/ \log t\to \infty$ for every $c\in\mathbb R$ and $p\in \mathbb Z[t]$; \text{ or } • $a(t)-cp(t)\to d$ for some $c,d\in\mathbb R$; \text{ or } • $|a(t)-t/m|\leq C \log{t}$ for some non-zero $m\in\mathbb Z$ and $C>0$.

背景链接：https://arxiv.org/abs/1103.3808

12659 | AMR-047-0023

### Multiple ergodic averages - Problem 23

S / solved | AMR 问题清单汇编

Here $\mathcal F=\{a_1,\ldots,a_\ell\}$ is a family of functions of polynomial growth in one Hardy field, and $\operatorname{span}^*(\mathcal F)$ denotes its nonzero linear combinations. and suppose that for every function $a\in \text{span}^*(\mathcal{F})$ we have $|a(t)-cp(t)|/ \log{t}\to\infty$ for every $c\in \mathbb R$ and $p\in \mathbb Z[t]$. Show that for every ergodic system $(X,\mathcal{B},\mu,T)$ and functions $f_1,\ldots,f_\ell\in L^\infty(\mu)$ we have $$ \lim_{N\to\infty} \frac1N \sum_{n=1}^N T^{[a_1(n)]}f_1\cdots T^{[a_\ell(n)]}f_\ell=\int f_1\, d\mu\, \cdots \, \int f_\ell\, d\mu $$ where the convergence takes place in $L^2(\mu)$.

背景链接：https://arxiv.org/abs/1103.3808

12660 | AMR-047-0024

### Multiple ergodic averages - Problem 24

P / partially_solved | AMR 问题清单汇编

Let $a,b$ be distinct positive non-integers. Show that for every ergodic system $(X,\mathcal{X},\mu,T)$ and functions $f, g \in L^\infty(\mu)$, we have $$ \lim_{N\to\infty} \frac1N \sum_{n=1}^N f(T^{[n^a]}x) \cdot g(T^{[n^b]}x)=\int f \ d\mu \cdot \int g\ d\mu $$ for almost every $x\in X$.

背景链接：https://arxiv.org/abs/1103.3808

12661 | AMR-047-0025

### Multiple ergodic averages - Problem 25

S / solved | AMR 问题清单汇编

Here $\mathcal F=\{a_1,\ldots,a_\ell\}$ is a family of functions of polynomial growth in one Hardy field, and $\operatorname{span}^*(\mathcal F)$ denotes its nonzero linear combinations. and suppose that for every function $a\in \text{span}^*(\mathcal{F})$ we have $|a(t)-cp(t)|\to \infty$ for every $c\in \mathbb R$ and $p\in \mathbb Z[t]$. Show that the collection of sequences $\{([a_1(n)]),\ldots,([a_\ell(n)])\}$ is good for $\ell$-recurrence of a single transformation.

背景链接：https://arxiv.org/abs/1103.3808

12662 | AMR-047-0026

### Multiple ergodic averages - Problem 26

O / open | AMR 问题清单汇编

Find an example of a function $a\in \mathcal H$ that grows faster than polynomials, meaning, $a(t)/t^k \to\infty$ for every $k\in \mathbb N$, such that the sequence $[a(n)]$ is good for multiple recurrence and convergence of powers.

背景链接：https://arxiv.org/abs/1103.3808

12663 | AMR-047-0027

### Multiple ergodic averages - Problem 27

S / solved | AMR 问题清单汇编

Let $c$ be a positive non-integer. Show that the sequence $([p_n^c])$ is good for multiple recurrence and convergence of powers.

背景链接：https://arxiv.org/abs/1103.3808

12664 | AMR-047-0028

### Multiple ergodic averages - Problem 28

P / partially_solved | AMR 问题清单汇编

Show that the sequence $([n \sin n])$ is good for multiple recurrence and convergence of powers.

背景链接：https://arxiv.org/abs/1103.3808

12665 | AMR-047-0029

### Multiple ergodic averages - Problem 29

P / partially_solved | AMR 问题清单汇编

Show that if $c>1$ is not an integer, then the sequence $([n^c])$ is good for multiple recurrence and convergence of commuting transformations. Moreover, show that if $(X,\mathcal X,\mu, T_1,\ldots,T_\ell)$ is a system and $f_1,\ldots,f_\ell\in L^\infty(\mu)$, then the $L^2(\mu)$-limit $$ \lim_{N\to\infty} \frac1N \sum_{n=1}^N T^{[n^c]}_1f_1 \cdots T^{[n^c]}_\ell f_\ell $$ is equal to the $L^2(\mu)$-limit $\lim_{N\to\infty} \frac1N \sum_{n=1}^N T^{n}_1 f_1\cdots T^{n}_\ell f_\ell $.

背景链接：https://arxiv.org/abs/1103.3808

12666 | AMR-047-0003

### Multiple ergodic averages - Problem 3

P / partially_solved | AMR 问题清单汇编

If $(a_1(n)),\ldots, (a_\ell(n))$ are sequences of integers, then show that the following %%three statements are equivalent: • The sequences $(a_1(n)),\ldots, (a_\ell(n))$ are good for $\ell$-convergence of commuting transformations. • The sequences $(a_1(n)),\ldots, (a_\ell(n))$ are good for $\ell$-convergence of $\ell$-step nilsystems. • The sequence $\big( \frac{1}{N}\sum_{n=1}^N \psi(a_1(n),\ldots,a_\ell(n)) \big)$ converges for every basic generalized $\ell$-step nilsequence $\psi$ in $\ell$-variables.

背景链接：https://arxiv.org/abs/1103.3808

12667 | AMR-047-0030

### Multiple ergodic averages - Problem 30

P / partially_solved | AMR 问题清单汇编

Let $\ell \in \mathbb N$ and $c,c_1,\ldots, c_\ell$ be positive real numbers. Show that the prime numbers contain patterns of the form $$ \{m,m+[n^{c}],m+2[n^{c}], \ldots, m+\ell[n^c]\} \quad \text{ and } \quad \{m,m+[n^{c_1}],\ldots,m+[n^{c_\ell}]\} $$ for infinitely many $n\in\mathbb N$.

背景链接：https://arxiv.org/abs/1103.3808

12668 | AMR-047-0031

### Multiple ergodic averages - Problem 31

O / open | AMR 问题清单汇编

Suppose that $n\sigma_n\to\infty$. Show that almost surely the sequence $(a_n(\omega))$ is good for multiple recurrence and convergence of commuting transformations. Moreover, show that almost surely the following holds: For every system $(X,\mathcal X,\mu, T_1,\ldots,T_\ell)$ and functions $f_1,\ldots, f_\ell \in L^\infty(\mu)$, the averages %%$$ $$ \frac{1}{N}\sum_{n=1}^N T_1^{a_n(\omega)}f_1\cdots T_\ell^{ a_n(\omega)}f_\ell $$ converge in $L^2(\mu)$ and their limit equals the limit of the averages $\frac{1}{N}\sum_{n=1}^N T_1^{n}f_1\cdots T_\ell^nf_\ell$.

背景链接：https://arxiv.org/abs/1103.3808

12669 | AMR-047-0032

### Multiple ergodic averages - Problem 32

P / partially_solved | AMR 问题清单汇编

Suppose that $n\sigma_n\to\infty$. Show that almost surely the following holds: For every system $(X,\mathcal X,\mu, T,S)$ and functions $f, g \in L^\infty(\mu)$, we have $$ \lim_{N\to\infty} \frac{1}{N}\sum_{n=1}^N T^nf\cdot S^{a_n(\omega)}g= \mathbb E(f|\mathcal{I}_T)\cdot \mathbb E(g|\mathcal{I}_S) $$ where the limit is taken in $L^2(\mu)$. Furthermore, if $\sigma_n=n^{-a}$ for some $a\in (0,1)$, show that the convergence also holds pointwise almost everywhere.

背景链接：https://arxiv.org/abs/1103.3808

12670 | AMR-047-0033

### Multiple ergodic averages - Problem 33

O / open | AMR 问题清单汇编

Suppose that $a,b\in(0,1)$ and $a\neq b$. Show that almost surely the following holds: For every system $(X,\mathcal X,\mu,T,S)$ and functions $f, g \in L^\infty(\mu)$ we have $$ \lim_{N\to\infty} \frac{1}{N}\sum_{n=1}^N T^{a_n(\omega)}f\cdot S^{b_n(\omega)}g= \mathbb E(f|\mathcal{I}_T)\cdot \mathbb E(g|\mathcal{I}_S) $$ where the limit is taken in $L^2(\mu)$ or pointwise.

背景链接：https://arxiv.org/abs/1103.3808

12671 | AMR-047-0034

### Multiple ergodic averages - Problem 34

S / solved | AMR 问题清单汇编

Let $(X,\mathcal X,\mu, T_n)$ be a measure preserving system with multiplicative structure and $A\in \mathcal X$ with $\mu(A)>0$. Is it true that there exist $m,n\in \mathbb N$, $m> n$, such that $$ \mu\bigl(T_{2mn} A\cap T_{(m-n)(m+n)} A \bigr)>0\, ? $$

背景链接：https://arxiv.org/abs/1103.3808

12672 | AMR-047-0035

### Multiple ergodic averages - Problem 35

O / open | AMR 问题清单汇编

Let $(X,\mathcal X,\mu, T_n)$ be a measure preserving system with multiplicative structure and $A\in \mathcal X$ with $\mu(A)>0$. Is it true that there exist $m,n\in \mathbb N$ such that $$ \mu\bigl(T_{m(m+n)} A\cap T_{(m+2n)(m+3n)}\cap T_{(m+4n)(m+5n)} A \bigr)>0\, ? $$

背景链接：https://arxiv.org/abs/1103.3808

12673 | AMR-047-0004

### Multiple ergodic averages - Problem 4

P / partially_solved | AMR 问题清单汇编

Let $(a(n))$ be a sequence that satisfies: • for every connected $\ell$-step nilmanifold $X$ and every irrational nilrotation $b$ in $X$ the sequence $(b^{a(n)}\Gamma)_{n \in\mathbb N}$ is equidistributed in $X$, and • the set $\{n\in\mathbb N \colon r| a(n)\}$ has positive upper density for every $r\in \mathbb N$. Show that the sequence $(a(n))$ is good for $\ell$-recurrence of commuting transformations.

背景链接：https://arxiv.org/abs/1103.3808

12674 | AMR-047-0005

### Multiple ergodic averages - Problem 5

S / solved | AMR 问题清单汇编

If $(a(n))$ is good for $\ell$-recurrence of powers, is then $(a(n)^k)$ good for $1$-recurrence for $k=1,\ldots, \ell$?

背景链接：https://arxiv.org/abs/1103.3808

12675 | AMR-047-0006

### Multiple ergodic averages - Problem 6

P / partially_solved | AMR 问题清单汇编

If a sequence is good for $2$-convergence of powers, then show that it is good for $2$-convergence of commuting transformations.

背景链接：https://arxiv.org/abs/1103.3808

12676 | AMR-047-0007

### Multiple ergodic averages - Problem 7

O / open | AMR 问题清单汇编

Is there a sequence that is good for $2$-recurrence of powers but is not good for $2$-recurrence of commuting transformations?

背景链接：https://arxiv.org/abs/1103.3808

12677 | AMR-047-0008

### Multiple ergodic averages - Problem 8

O / open | AMR 问题清单汇编

Give an explicit example of a fast growing sequence that is good for multiple recurrence and convergence of powers and commuting transformations.

背景链接：https://arxiv.org/abs/1103.3808

12678 | AMR-047-0009

### Multiple ergodic averages - Problem 9

O / open | AMR 问题清单汇编

Let $\mathcal P$ be an essentially distinct family of integer polynomials, and let $d_{\min}(\mathcal P)$ be the least $d$ for which the Host-Kra factor $\mathcal Z_{d,T}$ is characteristic for the associated polynomial multiple averages in every system. If $|\mathcal P|\geq 2$, show that $ d_{min}(\mathcal P)\leq |\mathcal P|-1. $

背景链接：https://arxiv.org/abs/1103.3808

12679 | OWR-14601-002

### Multistationarity and Permanence in MESSI Reaction Networks

P / partially_solved | Oberwolfach 报告问题汇编

For MESSI mass-action biochemical systems, characterize when a stoichiometric compatibility class can contain more than one positive steady state with the same conserved quantities, and determine the related permanence and invariant-subspace properties.

来源字段：https://doi.org/10.4171/owr/2016/19

12680 | OWR-15436-004

### Multistationarity in Kinetic-Proofreading Networks

O / open | Oberwolfach 报告问题汇编

For the kinetic-proofreading module in the T-cell activation model, which has deficiency one under the strongest simplifying assumptions, does the dynamical system admit multiple positive steady states, and what are the asymptotics of its solutions?

来源字段：https://doi.org/10.4171/owr/2017/28

12681 | AMR-041-0002

### N-body problem

P / partially_solved | AMR 问题清单汇编

What is the measure of the set of initial conditions of the Newtonian $N$-body problem that lead to global solutions? The complementary set of singularities splits into collision and non-collision singularities.

背景链接：https://people.math.harvard.edu/~knill/seminars/intr/

12682 | OWR-4429-004

### Near-Soliton Dynamics of Equivariant Schrödinger Maps

P / partially_solved | Oberwolfach 报告问题汇编

For equivariant Schrödinger maps in $\mathbb{R}^2$ with initial data close in $\dot H^1$ to a harmonic-map soliton family $Q_m$, characterize the global dynamics, including whether finite-time blow-up can occur while the solution remains close to the noncompact soliton family.

来源字段：https://doi.org/10.4171/owr/2010/41

12683 | OWR-12870-003

### New Rational Lyapunov Exponents on Hilbert Modular Surfaces

O / open | Oberwolfach 报告问题汇编

Can one construct Kobayashi geodesics on Hilbert modular surfaces with rational Lyapunov exponents other than 1, 1/2, 1/3, 1/5, and 1/7?

来源字段：https://doi.org/10.4171/owr/2014/10

12684 | AMR-052-0066

### Newton dynamics for entire functions

O / open | AMR 问题清单汇编

Describe the dynamics of Newton's method when applied to broad natural classes of transcendental entire functions.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12685 | AMR-039-0003

### Nilpotent groups

P / partially_solved | AMR 问题清单汇编

What is the coarse Ricci curvature of discrete or continuous nilpotent groups? In particular, for the natural random walk generated by $a,b$ on the discrete Heisenberg group $\langle a,b,c\mid ac=ca,bc=cb,[a,b]=c\rangle$, does the negative small-scale curvature tend to zero at larger scales?

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12686 | AMR-045-0054

### Nivat's conjecture

P / partially_solved | AMR 问题清单汇编

Let $x\in A^{\mathbb Z^2}$ and let $N_x(n_1,n_2)$ be the number of distinct $n_1\times n_2$ rectangular patterns in $x$. If $N_x(n_1,n_2)\le n_1n_2$ for some positive $n_1,n_2$, prove that $x$ is periodic.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12687 | AMR-039-0007

### Non-constant curvature

P / partially_solved | AMR 问题清单汇编

Can estimates based on a uniform lower bound for coarse Ricci curvature be extended to spaces where curvature has only a controlled number of negative or zero values? Can curvature of iterated kernels be related to an average curvature along random-walk trajectories, perhaps by large deviations?

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12688 | AMR-052-0014

### Non-equivalent compactifications of Blaschke-product space

O / open | AMR 问题清单汇编

For a degree-$n$ Blaschke product $A$, let $B(A)$ be the rational maps obtained by mating $A$ with a varying Blaschke product, and let $F:B(z^n)\to B(A)$ be the natural biholomorphism. Prove that if $n>2$ and $A\ne z^n$, then $F$ does not extend to a homeomorphism of their actual boundaries.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12689 | AMR-052-0035

### Non-Jordan Siegel-disk boundary

S / solved | AMR 问题清单汇编

Can a Siegel disk have a boundary that is not a Jordan curve?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12690 | AMR-039-0005

### Non-reversible spectral gap

P / partially_solved | AMR 问题清单汇编

Positive coarse Ricci curvature gives a spectral-gap bound for reversible random walks and on finite spaces. What spectral-radius, operator-norm, or Poincaré-inequality bounds hold in the non-reversible case? Can finite-space approximation be used?

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12691 | OWR-1703875-001

### Nondissipative Dynamics from Antisymmetric Large Deviations

O / open | Oberwolfach 报告问题汇编

Does the dynamics generated by the time-antisymmetric large-deviation rate function $L_F^{\mathrm{asym}}$ provide a counterpart of gradient flow that describes purely nondissipative dynamics?

来源字段：https://doi.org/10.4171/owr/2020/29

12692 | AMR-046-0010

### Number of centers

P / partially_solved | AMR 问题清单汇编

Determine the maximum number $\mathcal{C}_n$ of centers for planar polynomial differential systems of degree $n\geq4$.

背景链接：https://arxiv.org/abs/2012.02524

12693 | AMR-036-0008

### Number of degenerate Herman rings

O / open | AMR 问题清单汇编

Is the number of degenerate Herman rings of a rational function finite, and can it be bounded in terms of the degree of the rational function?

背景链接：https://www.math.purdue.edu/~eremenko/uns1.html

12694 | OWR-14215-005

### Observable Spaces Capturing Koopman Dynamics

O / open | Oberwolfach 报告问题汇编

What space of observables $F$ should be chosen for a Koopman operator so that its spectral properties meaningfully encode the dynamics?

来源字段：https://doi.org/10.4171/owr/2016/7

12695 | AMR-045-0015

### One-sided full-shift automorphisms

S / solved | AMR 问题清单汇编

Prove that every expansive automorphism of a one-sided full shift is topologically conjugate to a two-sided full shift.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12696 | AMR-126-0011

### Open Problems in Arithmetic Dynamics - Conjecture 11

O / open | AMR 问题清单汇编

[Dynamical Mordell--Lang Conjecture: Bell, 2006, [MR2225492]; Denis, 1994, [MR1259107]; Ghioca--Tucker, 2009, [arxiv0805.1560]]

Let $X/\mathbb{C}$ be a quasi-projective variety, let $f:X\to X$ be a morphism, let $P\in X$ be a point, and let $Y\subseteq X$ be a subvariety of $X$. Then \[ \{ n\ge 0 : f^n(P)\in Y \} \] is a finite union of arithmetic progressions.

背景链接：https://arxiv.org/abs/1806.04980

12697 | AMR-126-0012

### Open Problems in Arithmetic Dynamics - Conjecture 12

O / open | AMR 问题清单汇编

[Amerik--Bogomolov--Rovinsky, 2011, [MR2862064]; Medvedev--Scanlon, 2009, [arxiv0901.2352]]

Let $X$ be a quasi-projective variety defined over an algebraically closed field $K$ of characteristic $0$. Let $f:X\to X$ be an endomorphism defined over $K$, and assume that $f$ is not a fibered map. \begin{parts}

(a) \mathrm{(Weak form)} There exists a point $P\in X(K)$ whose orbit $\mathcal{O}_f(P)$ is Zariski dense in $X$.

(b) \mathrm{(Strong form)} The set \[ \bigl\{ P\in X(K) : \text{$\mathcal{O}_f(P)$ is Zariski dense in $X$} \bigr\} \] is Zariski dense in $X$. \end{parts}

背景链接：https://arxiv.org/abs/1806.04980

12698 | AMR-126-0013

### Open Problems in Arithmetic Dynamics - Conjecture 13

O / open | AMR 问题清单汇编

[Ghioca--Tucker--Zhang, 2011, {[MR2854724, Conjecture 2.4]}]

Let $X/\mathbb{C}$ be a projective variety, let $f: X \to X$ be a polarizable endomorphism, and let $Y$ be a subvariety of $X$ which has no component contained entirely within the singular locus of $X$. Then the following are equivalent\textup: \begin{parts}

(a) $Y$ is preperiodic under $f$.

(b) The intersection $Y\cap \operatorname{PrePer}_{f}(X)$ contains a Zariski dense subset of smooth points such that for every $x$ in the subset, the tangent subspace of $Y$ at $x$ is preperiodic under the induced action of $f$ on the Grassmanian $\operatorname{Gr}_{\dim(Y)}\left(T_{X,x}\right)$, where we write $T_{X,x}$ for the tangent space of $X$ at the point $x$. \end{parts}

背景链接：https://arxiv.org/abs/1806.04980

12699 | AMR-126-0017

### Open Problems in Arithmetic Dynamics - Conjecture 17 (Dynamical Andr{\'e}--Oort Conjecture)

O / open | AMR 问题清单汇编

\textup{(Baker--DeMarco, ${}\approx2011$, [arxiv1211.0255,MR2884382])} Let $X\subseteq\mathcal{M}_d^{\mathrm{crit}}$ be an algebraic subvariety such that the PCF maps in $X$ are Zariski dense in $X$. Then $X$ is cut out by ``*critical orbit relations*.''

背景链接：https://arxiv.org/abs/1806.04980

12700 | AMR-126-0018

### Open Problems in Arithmetic Dynamics - Conjecture 18 (Dynamical Andr{\'e}--Oort Conjecture for 1-Dimensional Families)

O / open | AMR 问题清单汇编

\textup{(Baker--DeMarco, 2013, [arxiv1211.0255])} Let $T$ be a non-singular algebraic curve, and let \[ T \longrightarrow\mathcal{M}_d^{\mathrm{crit}},\quad t \longmapsto (f_t,P_{1,t},\ldots,P_{2d-2,t}) \] be a family of rational maps with marked critical points. Then the following are equivalent\textup: \begin{parts}

(a) There are infinitely many $t\in T$ such that $f_t$ is PCF.

(b) For every $i$ and $j$, the maps $P_i:T\to\mathbb{P}^1$ and $P_j:T\to\mathbb{P}^1$ are $f$-dynamically related. \end{parts}

背景链接：https://arxiv.org/abs/1806.04980

12701 | AMR-126-0002

### Open Problems in Arithmetic Dynamics - Conjecture 2

O / open | AMR 问题清单汇编

In view of Northcott's theorem, it is natural to inquire as to the possible size of the finite set $\operatorname{PrePer}\bigl(f,\mathbb{P}^N(K)\bigr)$, which leads to the following fundamental conjecture.

Fix integers $N\ge1$, $d\ge2$, and $D\ge1$. There is a constant $C(N,d,D)$ such that for all degree $d$ morphisms $f:\mathbb{P}^N\to\mathbb{P}^N$ defined over a number field $K$ of degree $[K:\mathbb{Q}]=D$, the number of $K$-rational preperiodic points is uniformly bounded, \[ \#\operatorname{PrePer}\bigl(f,\mathbb{P}^N(K)\bigr) \le C(N,d,D). \]

背景链接：https://arxiv.org/abs/1806.04980

12702 | AMR-126-0023

### Open Problems in Arithmetic Dynamics - Conjecture 23 (Orbit Density Conjecture for $\mathbb{A}^N$)

O / open | AMR 问题清单汇编

\textup{(Medvedev--Scanlon, 2009, {[arxiv0901.2352, Conjecture 5.10]}; Amerik--Bogomolov--Rovinsky, 2011, {[MR2862064, Conjecture 1.2]})}

Let $f:\mathbb{A}^N\to\mathbb{A}^N$ be a morphism defined over ${\bar{\mathbb{Q}}}$, i.e., $f=(f_1,\ldots,f_N)$, where the $f_i\in{\bar{\mathbb{Q}}}[x_1,\ldots,x_N]$ are polynomials with ${\bar{\mathbb{Q}}}$-coefficients. Assume that there is no rational function $g\in{\bar{\mathbb{Q}}}(x_1,\ldots,x_N)$ satisfying $g\circ f=g$. Then there exists a point $P\in\mathbb{P}^N({\bar{\mathbb{Q}}})$ whose orbit $\mathcal{O}_f(P)$ is Zariski dense in $\mathbb{P}^N$.

背景链接：https://arxiv.org/abs/1806.04980

12703 | AMR-126-0024

### Open Problems in Arithmetic Dynamics - Conjecture 24 (Orbit Density Conjecture)

O / open | AMR 问题清单汇编

\textup{(Medvedev and Scanlon, 2009, {[arxiv0901.2352, Conjecture 5.10]}; Amerik, Bogomolov and Rovinsky, 2011, {[MR2862064, Conjecture 1.2]})}

Let ${\bar K}$ be an algebraically closed field of characteristic $0$, let $X/K$ be a quasi-projective variety, and let $f:X\to X$ be a dominant morphism. Assume that there does not exist a non-constant rational function $g\in{\bar K}(X)$ satisfying $g\circ f=g$. Then there exists a point $P\in X({\bar K})$ whose orbit $\mathcal{O}_f(P)$ is Zariski dense in $X$.

背景链接：https://arxiv.org/abs/1806.04980

12704 | AMR-126-0026

### Open Problems in Arithmetic Dynamics - Conjecture 26 (Integrality Conjecture)

O / open | AMR 问题清单汇编

\textup{(Bellon--Viallet, [MR1704282], 1999)} Let $f:\mathbb{P}^N\dashrightarrow\mathbb{P}^N$ be a dominant rational map. Then the dynamical degree $\delta_f$ of $f$ is an algebraic integer, i.e., $\delta_f$ is the root of a monic polynomial having integer coefficients.

背景链接：https://arxiv.org/abs/1806.04980

12705 | AMR-126-0003

### Open Problems in Arithmetic Dynamics - Conjecture 3

O / open | AMR 问题清单汇编

\textup{(Poonen, 1998, [poonen:uniformboundrefined])} For all $c\in\mathbb{Q}$, the set of periodic points \text{$\operatorname{Per}(x^2+c,\mathbb{Q})$} contains no points of period strictly greater than $3$.

背景链接：https://arxiv.org/abs/1806.04980

12706 | AMR-126-0031

### Open Problems in Arithmetic Dynamics - Conjecture 31 (Dynamical Lang Height Conjecture)

O / open | AMR 问题清单汇编

Let $N\ge1$, $D\ge1$, and $d\ge2$. There are positive constants $c_1(D,N,d)$ and $c_2(D,N,d)$ such that for all number fields $K/\mathbb{Q}$ with $[K:\mathbb{Q}]\le D$, all degree $d$ morphisms $f:\mathbb{P}^N\to\mathbb{P}^N$ defined over $K$, and all points $P\in\mathbb{P}^N(K)$ whose orbit $\mathcal{O}_f(P)$ is Zariski dense in $\mathbb{P}^N$, we have \[ {\hat h}_f(P) \ge c_1(D,N,d) h_\mathcal{M}(f) + c_2(D,N,d). \]

背景链接：https://arxiv.org/abs/1806.04980

12707 | AMR-126-0034

### Open Problems in Arithmetic Dynamics - Conjecture 34

O / open | AMR 问题清单汇编

Let $f\in\mathbb{C}_v(z)$ be a rational function of degree at least $2$, and let ${\mathcal{J}}_f\subset\mathsf{P}_{\mathrm{Berk}}^1$ be the Julia set of $f$. Then ${\mathcal{J}}_f\cap\mathbb{P}^1(\mathbb{C}_v)$ is equal to the closure in $\mathbb{P}^1(\mathbb{C}_v)$ of the set of Type I repelling periodic points of $f$.

背景链接：https://arxiv.org/abs/1806.04980

12708 | AMR-126-0036

### Open Problems in Arithmetic Dynamics - Conjecture 36

O / open | AMR 问题清单汇编

[[MR1781331]] Let $K$ be a locally compact subfield of $\mathbb{C}_v$, and suppose that the algebraic closure $\bar{K}$ of $K$ is dense in $\mathbb{C}_v$. Let $f\in{K}(z)$ be a rational function of degree at least $2$. Then $f$ has no wandering domains.

背景链接：https://arxiv.org/abs/1806.04980

12709 | AMR-126-0037

### Open Problems in Arithmetic Dynamics - Conjecture 37

O / open | AMR 问题清单汇编

Let $K\subset\mathbb{C}_v$ be a discretely valued subfield whose algebraic closure $\bar{K}$ is dense in $\mathbb{C}_v$. Let $p\geq 0$ be the residue characteristic of $\mathbb{C}_v$, and let $f\in\mathbb{C}_v(z)$ be a rational function of degree at least $2$. Suppose that there are no points $\zeta\in\mathsf{P}_{\mathrm{Berk}}^1$ at which $f$ has local degree divisible by $p$. Then every wandering domain of $f$ lies in the attracting basin of a Type II Julia periodic point.

背景链接：https://arxiv.org/abs/1806.04980

12710 | AMR-126-0039

### Open Problems in Arithmetic Dynamics - Conjecture 39

O / open | AMR 问题清单汇编

\begin{parts}

(a) Let $f:\mathbb{P}^N\to\mathbb{P}^N$ be a morphism of degree $d\ge2$ defined over $\mathbb{Q}$, and let $P\in\mathbb{P}^N(\mathbb{Q})$ be a point whose $f$-orbit $\mathcal{O}_f(P)$ is Zariski dense in $\mathbb{P}^N$. Then for every $\epsilon>0$, the set of primes \[ \bigl\{ p : \#\mathcal{O}_{\tilde f}(\tilde P\bmod p) \ge p^{(1/2-\epsilon)N} \bigr\} \] is a set of density $1$.

(b) Let $f:\mathbb{A}^N\to\mathbb{A}^N$ be a polynomial automorphism defined over $\mathbb{Q}$, i.e., the map $f$ is defined by polynomials, and it has an inverse defined by polynomials. Let $P\in\mathbb{A}^N(\mathbb{Q})$ be a point whose $f$-orbit $\mathcal{O}_f(P)$ is Zariski dense in $\mathbb{A}^N$. Then for every $\epsilon>0$, the set of primes \[ \bigl\{ p : \#\mathcal{O}_{\tilde f}(\tilde P\bmod p) \ge p^{(1-\epsilon)N} \bigr\} \] is a set of density $1$.
\end{parts}

背景链接：https://arxiv.org/abs/1806.04980

12711 | AMR-126-0004

### Open Problems in Arithmetic Dynamics - Conjecture 4

O / open | AMR 问题清单汇编

[Manes, 2008, [MR2407816]]

A $\mathbb{Q}$-rational periodic point of a map of the form $bx+cx^{-1}$ with $b,c\in\mathbb{Q}^*$ has period $1$, $2$, or $4$. Further, \[ \sup_{b,c\in\mathbb{Q}^*} \#\operatorname{PrePer}\bigl(bx+cx^{-1},\mathbb{P}^1(\mathbb{Q})\bigr) = 12. \]

背景链接：https://arxiv.org/abs/1806.04980

12712 | AMR-126-0044

### Open Problems in Arithmetic Dynamics - Conjecture 44

O / open | AMR 问题清单汇编

Let $f:\mathbb{P}^N\dashrightarrow\mathbb{P}^N$ be a dominant rational map defined over $\mathbb{F}_q$. \begin{parts}

(a) $\mathbb{P}^N(\bar\mathbb{F}_q)_f$ is Zariski dense in $\mathbb{P}^N$.

(b) $\displaystyle \lim_{k\to\infty} \#\mathbb{P}^N(\mathbb{F}_{q^k})_f/\#\mathbb{P}^N(\mathbb{F}_{q^k}) = 1. $ \end{parts}

背景链接：https://arxiv.org/abs/1806.04980

12713 | AMR-126-0045

### Open Problems in Arithmetic Dynamics - Conjecture 45

O / open | AMR 问题清单汇编

\textup{(Jones--Levy, 2017, [MR3704363, Conjecture 1.2])} Let $K$ be a number field, let $f(x)\in K(x)$ be a rational function of degree at least $2$, and let $\alpha\in K$ be a point that is not periodic for $f$. Then $(f,\alpha)$ is eventually stable over $K$.

背景链接：https://arxiv.org/abs/1806.04980

12714 | AMR-126-0046

### Open Problems in Arithmetic Dynamics - Conjecture 46

O / open | AMR 问题清单汇编

Let $K$ be a number field, let $f:\mathbb{P}^N\to\mathbb{P}^N$ be an endomorphism of degree at least $2$ defined over $K$, and let $P\in\mathbb{P}^N(K)$ be a point that is not periodic for $f$. Then $(f,P)$ is eventually stable over $K$.

背景链接：https://arxiv.org/abs/1806.04980

12715 | AMR-126-0049

### Open Problems in Arithmetic Dynamics - Conjecture 49

O / open | AMR 问题清单汇编

Let $f(x)\in\mathbb{Z}[x]$ be a polynomial of degree at least $2$, and let $\alpha\in\mathbb{Z}$ be a wandering point for $f$. Then $\mathcal{O}_f(\alpha)$ contains only finitely many primes.

背景链接：https://arxiv.org/abs/1806.04980

12716 | AMR-126-0005

### Open Problems in Arithmetic Dynamics - Conjecture 5

O / open | AMR 问题清单汇编

[Benedetto, Chen, Hyde, Kovacheva, White, 2014, {[MR3277939]}; Blanc--Canci--Elkies, 2015, [MR3431627]]

Let $f\in\operatorname{End}_2^1(\mathbb{Q})$ be a rational map of degree $2$ defined over $\mathbb{Q}$. \begin{parts}

(a) Let $P\in\mathbb{P}^1(\mathbb{Q})$ be periodic for $f$. Then $P$ has period at most $7$.

(b) Let $P\in\mathbb{P}^1(\mathbb{Q})$ be preperiodic for $f$. Then $\#\mathcal{O}_f(P)\le9$.

(c) The full set of rational preperiodic points satisfies $\#\operatorname{PrePer}\bigl(f,\mathbb{P}^1(\mathbb{Q})\bigr) \le 14$. \end{parts}

背景链接：https://arxiv.org/abs/1806.04980

12717 | AMR-126-0050

### Open Problems in Arithmetic Dynamics - Conjecture 50

O / open | AMR 问题清单汇编

Let $K$ be a number field, let $f(x)\in K(x)$ be a rational function of degree at least $2$, let $\alpha\in K$ be a wandering point for $f$, let $\beta\in K$, and let $c$ be a constant. Then the set \[ \left\{ n\ge0 : \sum_{{\mathfrak{p}}} \operatorname{ord}_{\mathfrak{p}}^{\scriptscriptstyle+}\bigl(f^n(\alpha)-\beta\bigr) \le c \right\} \quad\text{is finite.} \]

背景链接：https://arxiv.org/abs/1806.04980

12718 | AMR-126-0051

### Open Problems in Arithmetic Dynamics - Conjecture 51

O / open | AMR 问题清单汇编

Let $f(x)\in\mathbb{Q}(x)$ be a rational function of degree at least $2$, and let $\alpha,\beta\in\mathbb{Q}$ be $f$-wandering points. Then the Zsigmondy set of ${\mathcal{A}}(f,\alpha,\beta)$ is finite.

背景链接：https://arxiv.org/abs/1806.04980

12719 | AMR-126-0052

### Open Problems in Arithmetic Dynamics - Conjecture 52 (Dynamical Lang--Siegel Conjecture for $\mathbb{P}^N$)

O / open | AMR 问题清单汇编

Let $f:\mathbb{P}^N\to\mathbb{P}^N$ be a morphism of degree $d\ge2$ defined over $\mathbb{Q}$, and assume that for all $1\le n\le N+1$, the first coordinate function of $f^n$ is not a constant multiple of $X_0^{d^n}$. Let $P\in\mathbb{P}^N(\mathbb{Q})$. Then $\mathcal{O}_f(P)$ contains only finitely points that are integeral relative to the hyperplane $x_0=0$. More precisely, assume that $P$ is wandering, and for each $n\ge1$, write \begin{multline*} f^n(P) = \bigl[A_0(n),A_1(n),\ldots,A_N(n)\bigr] * \quad\text{with $A_0(n),\ldots,A_N(n)\in\mathbb{Z}$ and $\gcd\bigl(A_0(n),\ldots,A_N(n)\bigr)=1$.} \end{multline*} Then \[ \lim_{n\to\infty} \frac{\log\bigl|A_0(n)\bigr|} {\log\max\bigl\{ \bigl|A_0(n)\bigr|,\ldots,\bigl|A_N(n)\bigr| \bigr\} } = 1. \]

背景链接：https://arxiv.org/abs/1806.04980

12720 | AMR-126-0054

### Open Problems in Arithmetic Dynamics - Conjecture 54 (Dynamical Brauer--Manin Conjecture)

O / open | AMR 问题清单汇编

\textup{(Hsia--Silverman, 2009, [MR2537714])} Let $K$ be a number field, let $X/K$ be a variety, let $f:X\to X$ be a morphism defined over $K$, let $Y\subset X$ be a subvariety defined over $K$, and let $P\in X(K)$. Let ${\mathcal{A}}_K$ be the ring of adeles of $K$, and for any set $S$, let ${\mathcal{C}}(S)$ denote the adelic closure of $S$. Then \[ {\mathcal{C}}\bigl(Y\cap\mathcal{O}_f(P)\bigr) = Y({\mathcal{A}}_K) \cap {\mathcal{C}}\bigl(\mathcal{O}_f(P)\bigr). \]

背景链接：https://arxiv.org/abs/1806.04980

12721 | AMR-126-0006

### Open Problems in Arithmetic Dynamics - Conjecture 6

O / open | AMR 问题清单汇编

[Poonen, 2012, {[arxiv1206.7104, Question 3.2.]}]

Let $K$ be a number field, let $X$ and $S$ be varieties defined over $K$, let $\pi:X\to S$ be a morphism, and let $f:X\to X$ be an $S$-morphism, which means that $\pi\circ f=\pi$. For each point $s\in S({\bar K})$, the map $f$ restricted to the fiber $X_s:=\pi^{-1}(s)$ gives a morphism $f_s:X_s\to X_s$, and we view $f:X\to X$ as a family of maps parameterized by the points of $S$. Then for all $D\ge1$ we have \[ \sup_{\substack{L/K\\ [L:K]=D\\}} \sup_{s\in S(L)} \#\operatorname{PrePer}\bigl(f_s,X_s(L)\bigr) < \infty. \]

背景链接：https://arxiv.org/abs/1806.04980

12722 | AMR-126-0008

### Open Problems in Arithmetic Dynamics - Conjecture 8 (Odoni, 1985, [MR805714])

O / open | AMR 问题清单汇编

Let $K$ be a Hilbertian field, (A field $K$ is *Hilbertian* if it is not possible to cover $\mathbb{P}^1(K)$ by a union of finitely manys sets of the form $F\bigl(C(K)\bigr)$, where $C/K$ is a smooth curve and $F:C\to\mathbb{P}^1$ is a finite map of degree at least $2$. For example, all number fields are Hilbertian, but $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{Q}_p$ are not.) and let $d\ge2$. There exists a monic polynomial $f(X)\in K[X]$ of degree $d$ and a point $P\in K$ such that $\iota_K(f,P)=1$. \textup(Addendum: [MR3220023, Jones, 2015] If $K$ is a number field, then one can take the coefficients of $f(X)$ to lie in the ring of integers of $K$.\textup)

背景链接：https://arxiv.org/abs/1806.04980

12723 | AMR-126-0009

### Open Problems in Arithmetic Dynamics - Conjecture 9

O / open | AMR 问题清单汇编

Let $K$ be a field of characteristic $0$, and let $f:\mathbb{P}^N\to\mathbb{P}^N$ be a morphism of degree $d\ge2$ defined over $K$. Then for all sufficiently large $n$ we have (The function $\mu$ is the M\"obius function, which we recall is defined as follows: If $n$ is a product of $k$ distinct primes, then $\mu(n)=(-1)^k$, otherwise $\mu(n)=0$.) \[ \#\operatorname{Per}_n^*\bigl(f,\mathbb{P}^N({\bar K})\bigr)

:= \sum_{m\mid n} \mu\left(\frac{n}{m}\right) (d^m+1)^N. \]

背景链接：https://arxiv.org/abs/1806.04980

12724 | AMR-126-0001

### Open Problems in Arithmetic Dynamics - Question 1

O / open | AMR 问题清单汇编

Let $S$ be a set, and let $T\subseteq S$ be a subset. Classify those maps $f:S\to S$ with the property that for every $\alpha\in S$, the set \[ \bigl\{ n \ge 0 : f^n(\alpha) \in T \bigr\} \] has some nice structure, for example, it is the union of finitely many one-sided arithmetic progressions.

背景链接：https://arxiv.org/abs/1806.04980

12725 | AMR-126-0010

## Open Problems in Arithmetic Dynamics - Question 10

O / open | AMR 问题清单汇编

Let $K$ be a field of characteristic $0$, let $f:\mathbb{P}^N\to\mathbb{P}^N$ be a morphism of degree $d\ge2$ defined over $K$, and for $n\ge1$, let $\nu=\nu(n,f)=\#\operatorname{Per}_n^*(f)$. Find general necessary and sufficient conditions on $f$ for the following statements to hold\textup: \begin{parts}

(a) The index of the dynatomic representation $\rho_{n,f}^{\mathrm{dyn}}$ is bounded, i.e., \[ \sup_{n\ge1}\; \bigl( \operatorname{Aut}({\mathcal{P}}_{n,\nu}^*) : \rho_{n,f}^{\mathrm{dyn}}(G_K) \bigr) < \infty. \]

(b) For all sufficiently large $n$, the dynatomic representation $\rho_{n,f}^{\mathrm{dyn}}$ is as surjective as possible, i.e., \[ \rho_{n,f}^{\mathrm{dyn}}(G_K) = \operatorname{Aut}({\mathcal{P}}_{n,\nu}^*)^{\operatorname{Aut}_K(f)}. \] \end{parts}

背景链接：https://arxiv.org/abs/1806.04980

12726 | AMR-126-0014

## Open Problems in Arithmetic Dynamics - Question 14

O / open | AMR 问题清单汇编

Let $G_{d,m,n}(c)\in\mathbb{Z}[c]$ be the Gleason polynomial of $x^d+c$. \begin{parts}

(a) Is it true that $G_{2,m,n}$ is irreducible in $\mathbb{Q}[c]$ for all $m\ge0$ and $n\ge1$?

(b) For which $(d,m,n)$ is $G_{d,m,n}$ irreducible in $\mathbb{Q}[c]$?

(c) More generally, what is the complete factorization of $G_{d,m,n}$ in $\mathbb{Q}[c]$? \end{parts}

背景链接：https://arxiv.org/abs/1806.04980

12727 | AMR-126-0015

## Open Problems in Arithmetic Dynamics - Question 15

O / open | AMR 问题清单汇编

We collect some of the natural questions that one might ask about the moduli spaces $\mathcal{M}_d^N$ and about automorphism groups. \begin{parts}

(a) For $N\ge2$, are all \textup(some, none\textup) of the moduli spaces $\mathcal{M}_d^N$ rational varieties?

(b) For each $(N,d)$, describe the possible groups $\operatorname{Aut}(f)$ for $f\in\mathcal{M}_d^N$, or at least give a good upper bound for \[ U(N,d):=\sup_{f\in\mathcal{M}_d^N}\#\operatorname{Aut}(f) \] in terms of $N$ and $d$. It is known that $U(N,d)$ is finite [MR2741188], that $U(1,d)\le\max\{60,2d+2\}$ [MR2884382, Example 2.54], and that $U(2,d)\le60d^6$ [arxiv1509.06670, Theorem 6.2].

(c) Describe the subvarieties of $\mathcal{M}_d^N$ corresponding to maps $f$ whose automorphism group $\operatorname{Aut}(f)$ is non-trivial. See [MR3709645] for a detailed analysis for $N=1$ and [arxiv1607.05772] for some partial results for $N=d=2$. For example, \[ \{f\in\mathcal{M}_2^1\cong\mathbb{A}^2:\operatorname{Aut}(f)\ne1\}=\text{cuspidal cubic curve in $\mathbb{A}^2$,} \] with $\operatorname{Aut}(f)\cong\mathbb{Z}/2\mathbb{Z}$ except for the map $f$ at the cusp, for which $\operatorname{Aut}(f)\cong{\mathcal{S}}_3$.

(d) Most of the spaces $\mathcal{M}_d^N$ are singular. Describe the singularity locus. The case $N=1$ is analyzed in [MR3709645]. \end{parts}

背景链接：https://arxiv.org/abs/1806.04980

12728 | AMR-126-0016

## Open Problems in Arithmetic Dynamics - Question 16

O / open | AMR 问题清单汇编

The fact that the bound for $\operatorname{FOM/FOD}(1,d)$ does not depend on $d$ suggests the following question.

For a given $N\ge1$, is \[ \sup_{d\ge2} \operatorname{FOM/FOD}(N,d) < \infty? \] If so, what is its value as a function of $N$?

背景链接：https://arxiv.org/abs/1806.04980

12729 | AMR-126-0019

## Open Problems in Arithmetic Dynamics - Question 19

O / open | AMR 问题清单汇编

Find a ``nice'' function $C(N,d,D,s)$ so that every degree $d$ morphism $f:\mathbb{P}^N\to\mathbb{P}^N$ defined over a field $K$ of degree at most $D$ and having at most $s$ primes of bad reduction satisfies \[ \#\operatorname{PrePer}\bigl(f,\mathbb{P}^N(K)\bigr) \le C(N,d,D,s). \]

背景链接：https://arxiv.org/abs/1806.04980

12730 | AMR-126-0020

### Open Problems in Arithmetic Dynamics - Question 20

O / open | AMR 问题清单汇编

What is the exact value of $\operatorname{ShafDim}_d^N\mathrm{?}$ In particular, is it true that \[ \operatorname{ShafDim}_d^1=d+1\quad\text{for all $d\ge3$}\mathrm{?}} \]

背景链接：https://arxiv.org/abs/1806.04980

12731 | AMR-126-0021

### Open Problems in Arithmetic Dynamics - Question 21

O / open | AMR 问题清单汇编

Give criteria, in terms of an integer $d\ge2$ and the geometry of a portrait ${\mathcal{P}}$, that distinguish between the following two situations\textup: \begin{parts}

(1) $\operatorname{\mathcal{G\!R}}_d^1[{\mathcal{P}}](K,S)/\operatorname{PGL}_2(R_S)$ is finite for all number fields $K$ and all finite sets of primes $S$.

(2) $\operatorname{\mathcal{G\!R}}_d^1[{\mathcal{P}}](K,S)/\operatorname{PGL}_2(R_S)$ is infinite for some number fields $K$ and some finite sets of primes $S$. \end{parts}

背景链接：https://arxiv.org/abs/1806.04980

12732 | AMR-126-0022

### Open Problems in Arithmetic Dynamics - Question 22

O / open | AMR 问题清单汇编

Let $Y_0^{\mathrm{dyn}}(n)$ and $Y_1^{\mathrm{dyn}}(n)$ be the dynamical modular curves classifying points, respectively cycles, of formal period $n$ attached to the family of maps $x^2+c$, or $x^m+c$, or some other natural $1$-parameter family of rational maps on $\mathbb{P}^1$ such as $bx+x^{-1}$. \begin{parts}

(a) Describe the primes of bad reduction for $Y_0^{\mathrm{dyn}}(n)$ and $Y_1^{\mathrm{dyn}}(n)$.

(b) Give a natural geometric/dynamical explanation for the ``sporadic'' primes of bad reduction, which we imprecisely define as primes not dividing $n$ and not dividing a finite set of bad primes associated to the family $f$.

(c) Describe the primes $p$ for which the reductions of $Y_0^{\mathrm{dyn}}(n)$ and/or $Y_1^{\mathrm{dyn}}(n)$ modulo $p$ are geometrically irreducible, e.g., for the latter, this is asking for the primes $p$ such that the polynomial $\Phi_n(x,z)$ does not factor in the ring $\bar\mathbb{F}_p[x,z]$. \end{parts}

背景链接：https://arxiv.org/abs/1806.04980

12733 | AMR-126-0025

### Open Problems in Arithmetic Dynamics - Question 25

O / open | AMR 问题清单汇编

Let $f:\mathbb{P}^N\dashrightarrow\mathbb{P}^N$ be a dominant rational map. Do there exist a finite collection of constant coefficient linear recursions \[ L_n^{(1)},L_n^{(2)},\ldots,L_n^{(r)} \] and an index function $\iota:\mathbb{N}\to\{1,2,\ldots,r\}$ so that for all $n\ge1$ we have \[

\deg(f^n) = L_n^{(\iota_n)} ? \]

背景链接：https://arxiv.org/abs/1806.04980

12734 | AMR-126-0027

### Open Problems in Arithmetic Dynamics - Question 27

O / open | AMR 问题清单汇编

Let $f:\mathbb{P}^N\dashrightarrow\mathbb{P}^N$ be a dominant rational map. Is it true that there exists an integer $\ell_f$ satisfying $0\le\ell_f\le N$ such that (We recall that the notation $F(n)\asymp G(n)$ means that there are positive constants $c_1,c_2,c_3$, depending on $F$ and $G$, so that $c_1F(n) \le G(n) \le c_2F(n)$ for all $n\ge c_3$.) \[ \deg(f^n) \asymp n^{\ell_f} \delta_f^n, \] where the implied constants may depend on $f$, but not on $n$?

背景链接：https://arxiv.org/abs/1806.04980

12735 | AMR-126-0028

### Open Problems in Arithmetic Dynamics - Question 28

O / open | AMR 问题清单汇编

Let $X$ be a smooth projective variety of dimension $N$, and let $f:X\dashrightarrow X$ be a dominant rational map. Is it true that the higher order dynamical degrees $\delta_f^{(i)}$ are algebraic integers for all $1\le i\le N$\mathrm{?}

背景链接：https://arxiv.org/abs/1806.04980

12736 | AMR-126-0029

### Open Problems in Arithmetic Dynamics - Question 29

O / open | AMR 问题清单汇编

Let $f:\mathbb{P}^N\dashrightarrow\mathbb{P}^N$ be a dominant rational map defined over ${\bar{\mathbb{Q}}}$, and let $P\in\mathbb{P}^N({\bar{\mathbb{Q}}})_f$ be a point whose orbit $\mathcal{O}_f(P)$ is Zariski dense in $\mathbb{P}^N$. Is it true that there exist integers $k_f$ and $\ell_f$ with $0\le\ell_f\le N$ such that \begin{equation}

h\bigl(f^n(P)\bigr) \asymp \delta_f^n\cdot n^{\ell_f}\cdot (\log n)^{k_f}, \end{equation} where the implied constants depend on $f$ and $P$, but not on $n$? Further, if $\delta_f>1$, is it true that $k_f=0$?

背景链接：https://arxiv.org/abs/1806.04980

12737 | AMR-126-0030

### Open Problems in Arithmetic Dynamics - Question 30

O / open | AMR 问题清单汇编

It is known that $\overline\mu_d(\mathbb{P}^N)\le d^{-(N-1)}$ for all $N\ge2$ and all $d\ge2$; see [arxiv0908.3835, Proposition 10] and [MR2975153]. This leads to the obvious question.

What is the exact value of $\overline\mu_d(\mathbb{P}^N)$?

背景链接：https://arxiv.org/abs/1806.04980

12738 | AMR-126-0032

### Open Problems in Arithmetic Dynamics - Question 32

O / open | AMR 问题清单汇编

Given a family of maps $f:\mathbb{P}^N\to\mathbb{P}^N$ and a family of points $P$, parametrized by $B$, describe the map \[ B(K_v)\longrightarrow\mathbb{R}\cup\{\infty\},\quad t \longmapsto \hat\lambda_{f_t,v}(P_t). \] In particular, to what extent is it a Weil function associated to some divisor in $\operatorname{Div}(B)\otimes\mathbb{R}$?

背景链接：https://arxiv.org/abs/1806.04980

12739 | AMR-126-0033

### Open Problems in Arithmetic Dynamics - Question 33

O / open | AMR 问题清单汇编

Let $K(B)$ be the function field of an algebraic curve, let $f:\mathbb{P}^N\to\mathbb{P}^N$ be a morphism of degree $d\ge2$ defined over the field $K(B)$, and let $P\in\mathbb{P}^N\bigl(K(B)\bigr)$. Is the canonical height ${\hat h}_f(P)$ always in $\mathbb{Q}$? If not, give necessary and sufficient conditions for ${\hat h}_f(P)\in\mathbb{Q}$.

背景链接：https://arxiv.org/abs/1806.04980

12740 | AMR-126-0035

### Open Problems in Arithmetic Dynamics - Question 35

O / open | AMR 问题清单汇编

Let $f\in\mathbb{C}_v(z)$ with $\deg f = d\geq 2$. Can $f$ have infinitely many periodic cycles of indifferent components that are not disks? If not, is there a bound in terms of $d$ for the number of such cycles? Do the answers depend on the field $\mathbb{C}_v$?

背景链接：https://arxiv.org/abs/1806.04980

12741 | AMR-126-0038

### Open Problems in Arithmetic Dynamics - Question 38 (Vague Motivating Question)

O / open | AMR 问题清单汇编

If we let $f$ range over $\operatorname{End}_{\mathsf{Set}}(S)$ or $\operatorname{Aut}_{\mathsf{Set}}(S)$ and let $\#S\to\infty$, then there are known statistics for $\#\operatorname{Per}(f,S)$ and for various natural quantities associated to the graph $\Gamma_f$ such as its number of components, its largest component, its largest cycle, and many others. See for example [MR1083961,MR0119227,MR0062973]. In particular, Flajolet and Odlyzko [MR1083961] analyze a large number of properties for average random maps, and one may ask whether similar statistics hold if one restricts to maps defined by polynomials.

To what extent do polynomial maps behave like random set maps?

背景链接：https://arxiv.org/abs/1806.04980

12742 | AMR-126-0040

### Open Problems in Arithmetic Dynamics - Question 40

O / open | AMR 问题清单汇编

Let $f:\mathbb{P}^N\to\mathbb{P}^N$ be a morphism of degree $d\ge2$ defined over $\mathbb{Q}$, and let $P\in\mathbb{P}^N(\mathbb{Q})$ be a point whose $f$-orbit $\mathcal{O}_f(P)$ is Zariski dense in $\mathbb{P}^N$. Under what conditions on $f$ is it true that for every $\epsilon>0$, the set of primes \[ \bigl\{ p : \#\mathcal{O}_{\tilde f}(\tilde P\bmod p) \le p^{(1/2+\epsilon)N} \bigr\} \] is a set of density $1$?

背景链接：https://arxiv.org/abs/1806.04980

12743 | AMR-126-0041

### Open Problems in Arithmetic Dynamics - Question 41

O / open | AMR 问题清单汇编

Fix an integer $N\ge1$ and a prime power $q$. Is it true that \[ \frac{1}{\#\operatorname{End}_d^N(\mathbb{F}_q)} \sum_{f\in\operatorname{End}_d^N(\mathbb{F}_q)} \operatorname{Comp}(\Gamma_f) = \operatorname{Comp}_{\mathrm{avg}}\bigl(\#\mathbb{P}^N(\mathbb{F}_q)\bigr) + O(1) \quad\text{as $d\to\infty$?} \]

背景链接：https://arxiv.org/abs/1806.04980

12744 | AMR-126-0042

### Open Problems in Arithmetic Dynamics - Question 42

O / open | AMR 问题清单汇编

Let $f:\mathbb{P}^N\to\mathbb{P}^N$ be a morphism of degree $d\ge2$ defined over $\mathbb{Q}$. Is it true that for every $\epsilon>0$, the set of primes \[ \left\{ p : \operatorname{Comp}(\Gamma_{\tilde f\bmod p}) \le \operatorname{Comp}_{\mathrm{avg}}(p^{(1-\epsilon)N})\right\} \] is a set of density $0$? Similarly with $\mathbb{A}^N$ in place of $\mathbb{P}^N$? For which maps $f$ can one obtain stronger results, e.g., $\operatorname{Comp}(\Gamma_{\tilde{f}\bmod p})\asymp\operatorname{Comp}_{\mathrm{avg}}(p^N)$ as $p\to\infty$?

背景链接：https://arxiv.org/abs/1806.04980

12745 | AMR-126-0043

### Open Problems in Arithmetic Dynamics - Question 43

O / open | AMR 问题清单汇编

Let $f(x)\in\mathbb{F}_q[x]$ be an \textup(irreducible\textup) polynomial, let $g(x)\in\mathbb{F}_q[x]$, and for each $n\ge1$, factor $g\bigl(f^n(x)\bigr)$ in $\mathbb{F}_q[x]$ as \[ g\bigl(f^n(x)\bigr) = p_1(x)^{e_1} p_2(x)^{e_2} \cdots p_r(x)^{e_r} \] with $p_i(x)$ irreducible in $\mathbb{F}_q[x]$. \begin{parts}

(a) Describe the behavior, as $n\to\infty$, of quantities such as \[ r,\quad \sum_{i=1}^r e_i,\quad \max_{1\le i\le r} e_i,\quad \max_{1\le i\le r}\deg p_i(x). \]

(b) Describe or predict the factorization of $g\bigl(f^{n+1}(x)\bigr)$ from a given factorization of $g\bigl(f^n(x)\bigr)$.

(c) Answer the analogous questions for the iterates of a rational function $f(x)\in\mathbb{F}_q(x)$. Note that the exponents $e_1,\ldots,e_r$ may now be both positive and negative, and hence it may be more natural to study $\sum |e_i|$ instead of $\sum e_i$, and $\max |e_i|$ instead of $\max e_i$. \end{parts}

背景链接：https://arxiv.org/abs/1806.04980

12746 | AMR-126-0047

### Open Problems in Arithmetic Dynamics - Question 47

O / open | AMR 问题清单汇编

Let $K$ be a number field, let $f:\mathbb{P}^N\to\mathbb{P}^N$ be an endomorphism of degree at least $2$ defined over $K$, and let $X\subsetneq\mathbb{P}^N$ be a subvariety that is defined over $K$ and irreducible over $K$. \begin{parts}

(a) If the iterated forward orbit $\bigcup_{n\ge1} f^n(X)$ of $X$ is Zariski dense in $\mathbb{P}^N$, is it true that $(f,X)$ is eventually stable over $K$?

(b) If the iterated forward orbit is Zariski dense and $\dim(X)\ge1$, is it true that $(f,X)$ is geometrically eventually stable?

(c) If $\dim(X)\ge1$, is eventual stability over $K$ equivalent to geometric eventual stability? \end{parts}

背景链接：https://arxiv.org/abs/1806.04980

12747 | AMR-126-0048

### Open Problems in Arithmetic Dynamics - Question 48

O / open | AMR 问题清单汇编

We recall that $\operatorname{End}_d^1(K)$ denotes the set of degree $d$ rational functions $f\in K(x)$, and we write $\operatorname{Poly}_d^1(K)$ for the set of degree $d$ polynomials $f\in K[x]$. We define \begin{align*} \operatorname{Poly}_d^1(\mathbb{F}_q)^\mathrm{stab} &:= \bigl\{ f\in\operatorname{Poly}_d^1(\mathbb{F}_q) : \text{$(f,0)$ is stable over $\mathbb{F}_q$} \bigr\}, \\ \operatorname{Rat}_d^1(\mathbb{F}_q)^\mathrm{stab} &:= \bigl\{ f\in\operatorname{Rat}_d^1(\mathbb{F}_q) : \text{$(f,0)$ is stable over $\mathbb{F}_q$} \bigr\}, \end{align*} to be the indicated sets of $\mathbb{F}_q$-stable maps. For fixed $d\ge2$, how does the size of the sets $\operatorname{Poly}_d^1(\mathbb{F}_q)^\mathrm{stab}$ and $\operatorname{Rat}_d^1(\mathbb{F}_q)^\mathrm{stab}$ grow as $q\to\infty$? Analogous questions apply for the sets of eventually stable maps, and for maps of $\mathbb{A}^N$ or $\mathbb{P}^N$ with $N\ge2$.

背景链接：https://arxiv.org/abs/1806.04980

12748 | AMR-126-0053

### Open Problems in Arithmetic Dynamics - Question 53

O / open | AMR 问题清单汇编

Let $f:\mathbb{P}^N\to\mathbb{P}^N$ be a morphism of degree $d$ defined over $\mathbb{Q}$. \begin{parts}

(a) For which $n\ge1$ is it true that $f$ has a point of exact period $n$ in $\mathbb{P}^N(\mathbb{Q})$ if and only if $f$ has a point of exact period $n$ in $\mathbb{P}^N(\mathbb{R})$ and in $\mathbb{P}^N(\mathbb{Q}_p)$ for all primes $p\/$\mathrm{?}

(b) For which $n\ge1$ is it true that there are infinitely many primes $p$ such that the map $f$ has no points of exact period $n$ in $\mathbb{P}^N(\mathbb{Q}_p)$\mathrm{?} Is this true for all $n\ge n_0(N,d)$\mathrm{?} \end{parts} More generally, we may ask the same questions for preperiodic points of type $(m,n)$, i.e., points with tails of exact length $m$ and periods of exact length $n$.

背景链接：https://arxiv.org/abs/1806.04980

12749 | AMR-126-0007

### Open Problems in Arithmetic Dynamics - Question 7

O / open | AMR 问题清单汇编

\begin{parts}

(a) For which choices of fields $K$, maps $f$, and basepoints $P$ is the Odoni index $\iota_K(f,P)$ equal to $1$, i.e., when is the arboreal representation $\rho^{\mathrm{arb}}_{f,P}$ surjective?

(b) For which choices of fields $K$, maps $f$, and basepoints $P$ is the Odoni index $\iota_K(f,P)$ finite, i.e., when does the image of the arboreal representation $\rho^{\mathrm{arb}}_{f,P}$ have finite index in the full tree automorphism group? \end{parts}

背景链接：https://arxiv.org/abs/1806.04980

12750 | AMR-020-0504

### Open Problems in Integrable Systems - Argument shift method and Jordan-Kronecker invariants of Lie algebras

P / partially_solved | AMR 问题清单汇编

Compute the Jordan--Kronecker invariants for the most interesting classes of Lie algebras, and particularly for the following {\rm(}a{\rm)} semidirect sums $\mathfrak{g}+_{\rho}V$, where $\rho: \mathfrak{g} \rightarrow End(V)$ is a representation of a semisimple Lie algebra $\mathfrak{g}$ and $V$ is assumed to be commutative; {\rm(}b{\rm)} Borel subalgebras of simple Lie algebras; {\rm(}c{\rm)} parabolic subalgebras of simple Lie algebras; {\rm(}d{\rm)} the centralisers of singular elements $a\in\mathfrak{g}$, where $\mathfrak{g}$ is simple; {\rm(}e{\rm)} Lie algebras of small dimensions.

背景链接：https://arxiv.org/abs/1804.03737

12751 | AMR-020-0505

### Open Problems in Integrable Systems - Argument shift method and Jordan-Kronecker invariants of Lie algebras

P / partially_solved | AMR 问题清单汇编

Are there any restrictions on the algebraic type of the pencils $\mathcal{A}_{x+\lambda a}$? Which algebraic types can be realised by means of an appropriately chosen Lie algebra{\/\rm?}

背景链接：https://arxiv.org/abs/1804.03737

12752 | AMR-020-0506

### Open Problems in Integrable Systems - Argument shift method and Jordan-Kronecker invariants of Lie algebras

O / open | AMR 问题清单汇编

Study examples of ``quadratic $+$ linear'' Poisson pencils. Compute their algebraic types and construct complete families of polynomials in bi-involution for such pencils.

背景链接：https://arxiv.org/abs/1804.03737

12753 | AMR-020-0507

### Open Problems in Integrable Systems - Argument shift method and Jordan-Kronecker invariants of Lie algebras

O / open | AMR 问题清单汇编

Is it true that for any quadratic Poisson bracket (defined on a vector space), there exists a polynomial integrable system?

背景链接：https://arxiv.org/abs/1804.03737

12754 | AMR-020-0401

### Open Problems in Integrable Systems - Around the Birkhoff conjecture

P / partially_solved | AMR 问题清单汇编

Assume that the exterior of an outer billiard table is foliated by invariant curves. Prove that the table is an ellipse.

背景链接：https://arxiv.org/abs/1804.03737

12755 | AMR-020-0402

### Open Problems in Integrable Systems - Around the Birkhoff conjecture

O / open | AMR 问题清单汇编

Prove the algebraic version of Birkhoff conjecture for outer billiards in a non-Euclidean surface of constant curvature.

背景链接：https://arxiv.org/abs/1804.03737

12756 | AMR-020-0403

### Open Problems in Integrable Systems - Around the Birkhoff conjecture

P / partially_solved | AMR 问题清单汇编

Prove Conjecture [source label: Descon] it the case that the foliation admits (i) a rational, (ii) an algebraic first integral.

背景链接：https://arxiv.org/abs/1804.03737

12757 | AMR-020-0404

### Open Problems in Integrable Systems - Around the Birkhoff conjecture

O / open | AMR 问题清单汇编

Is it possible to choose the domain so that the dynamics of the corresponding billiard map are locally (near the 2-periodic orbit) conjugated to the dynamics of the rigid rotation through some angle $\alpha$?

背景链接：https://arxiv.org/abs/1804.03737

12758 | AMR-020-0501

### Open Problems in Integrable Systems - Bi-Poisson vector spaces

O / open | AMR 问题清单汇编

Consider the action of $ Aut (V,J)$ on $V$. Describe the partition of $V$ into $ Aut (V,J)$-orbits. More generally, describe the action of $ Aut (V,J)$ on the set of all $k$-dimensional subspaces $U\subset V$ (orbits, invariants, fixed points). Notice that fixed points of this action (i.e., $ Aut (V,J)$-invariant subspaces) are important, as they correspond to well-defined (co)distributions in the context of Poisson pencils.

背景链接：https://arxiv.org/abs/1804.03737

12759 | AMR-020-0502

### Open Problems in Integrable Systems - Bi-Poisson vector spaces

O / open | AMR 问题清单汇编

Find necessary and sufficient conditions for the bi-Lagrangian Grassmannian $LG(V,J)$ to be a smooth algebraic variety. Describe the partition of $LG(V,J)$ into $Aut (V,J)$-orbits.

背景链接：https://arxiv.org/abs/1804.03737

12760 | AMR-020-0503

### Open Problems in Integrable Systems - Bi-Poisson vector spaces

O / open | AMR 问题清单汇编

Do bi-integrable systems exist for each algebraic type?

背景链接：https://arxiv.org/abs/1804.03737

12761 | AMR-020-0405

### Open Problems in Integrable Systems - Geometry of caustics, invariant surfaces, and commuting billiard maps

O / open | AMR 问题清单汇编

Are there plane billiards, other than ellipses, that possess rational caustics with two different values of the rotation numbers? Same question for outer billiards.

背景链接：https://arxiv.org/abs/1804.03737

12762 | AMR-020-0406

### Open Problems in Integrable Systems - Geometry of caustics, invariant surfaces, and commuting billiard maps

P / partially_solved | AMR 问题清单汇编

Suppose that a billiard table has a sequence of convex caustics with rotation numbers converging to some number $\omega \in (0,1/2)$. Does it imply that the table is an ellipse?

背景链接：https://arxiv.org/abs/1804.03737

12763 | AMR-020-0407

### Open Problems in Integrable Systems - Geometry of caustics, invariant surfaces, and commuting billiard maps

P / partially_solved | AMR 问题清单汇编

{\rm (M. Bialy)} Are there multi-dimensional convex billiards, other than ellipsoids, having invariant hypersurfaces in the phase space? Same question for multi-dimensional outer billiards.

背景链接：https://arxiv.org/abs/1804.03737

12764 | AMR-020-0408

### Open Problems in Integrable Systems - Geometry of caustics, invariant surfaces, and commuting billiard maps

S / solved | AMR 问题清单汇编

{\rm (A. Glutsyuk, S. Tabachnikov )} Given two nested closed convex hypersurfaces, assume that the billiard transformations on the set of oriented lines that intersect both hypersurfaces commute. Prove that the hypersurfaces are confocal ellipsoids.

背景链接：https://arxiv.org/abs/1804.03737

12765 | AMR-020-0701

### Open Problems in Integrable Systems - Integrability and Quantisation

P / partially_solved | AMR 问题清单汇编

Assume that a classical integrable system $(p_1,\dots,p_n)$ is given on $M$. Does there exist a quantum integrable system $(P_1,\dots,P_n)$ such that $p_j=\sigma(P_j)$?

背景链接：https://arxiv.org/abs/1804.03737

12766 | AMR-020-0702

### Open Problems in Integrable Systems - Integrability and Quantisation

O / open | AMR 问题清单汇编

[S. V\ u Ng{\d o}c] Given a set of semiclassical operators that verify conditions [source label: item:self-adjoint ] and [source label: item:commute] above, can one detect the independence axiom [source label: item:independent] in a purely spectral way? More precisely, can one tell from asymptotics of joint eigenvalues or eigenfunctions of $(P_1,\dots,P_n)$ that the principal symbols are almost everywhere independent?

背景链接：https://arxiv.org/abs/1804.03737

12767 | AMR-020-0703

### Open Problems in Integrable Systems - Integrability and Quantisation

P / partially_solved | AMR 问题清单汇编

[S. V\ u Ng{\d o}c] Write Bohr-Sommerfeld rules for focus-focus singularities in Berezin-Toeplitz quantisation.

背景链接：https://arxiv.org/abs/1804.03737

12768 | AMR-020-0704

### Open Problems in Integrable Systems - Integrability and Quantisation

O / open | AMR 问题清单汇编

[S. V\ u Ng{\d o}c] Define (and detect) the quantum Chern class.

背景链接：https://arxiv.org/abs/1804.03737

12769 | AMR-020-0705

### Open Problems in Integrable Systems - Integrability and Quantisation

P / partially_solved | AMR 问题清单汇编

[S. V\ u Ng{\d o}c] Compute the Taylor series invariant of semitoric systems directly from the spectrum ``at'' the focus-focus critical value.

背景链接：https://arxiv.org/abs/1804.03737

12770 | AMR-020-0706

### Open Problems in Integrable Systems - Integrability and Quantisation

P / partially_solved | AMR 问题清单汇编

[S. V\ u Ng{\d o}c] Is the singular Bohr-Sommerfeld formal power series in $\hbar$ a spectral invariant?

背景链接：https://arxiv.org/abs/1804.03737

12771 | AMR-020-0707

### Open Problems in Integrable Systems - Integrability and Quantisation

P / partially_solved | AMR 问题清单汇编

[\'{A}. Pelayo] What information about the principal symbols $f_1,\ldots, f_n$ of a quantum integrable system $T_{1,\hbar},\ldots, T_{n,\hbar}$ can be detected from their semiclasical joint spectrum? For instance, suppose that we know that a certain object -- say, an integer $z$, or a matrix $A$ -- is a symplectic invariant of $(M, \omega, f_1,\ldots, f_n)$. Can we compute this object from the semiclassical joint spectrum?

背景链接：https://arxiv.org/abs/1804.03737

12772 | AMR-020-0708

### Open Problems in Integrable Systems - Integrability and Quantisation

P / partially_solved | AMR 问题清单汇编

[\'{A}. Pelayo] Can one detect from the joint spectrum of a quantum integrable system $T_{1,\hbar},\ldots, T_{n,\hbar}$ that a singularity is degenerate?

背景链接：https://arxiv.org/abs/1804.03737

12773 | AMR-020-0709

### Open Problems in Integrable Systems - Integrability and Quantisation

O / open | AMR 问题清单汇编

[\'{A}. Pelayo] Can one make progress in counting the number of fixed points by studying the spectrum of the quantisation of $\mu \colon M \to S^1$?

背景链接：https://arxiv.org/abs/1804.03737

12774 | AMR-020-0712

### Open Problems in Integrable Systems - Integrable systems and geometric quantisation

O / open | AMR 问题清单汇编

[Miranda-Presas-Solha ] Modify this scheme to get finite dimensional representation spaces for focus-focus and hyperbolic singularities that still capture the invariants of the corresponding integrable systems and their singularities.

背景链接：https://arxiv.org/abs/1804.03737

12775 | AMR-020-0508

### Open Problems in Integrable Systems - Nijenhuis operators: singular points and global properties

P / partially_solved | AMR 问题清单汇编

Describe closed manifolds $M$ which admit Nijenhuis operators $L(x)$ that are algebraically regular at each point $x\in M$.

背景链接：https://arxiv.org/abs/1804.03737

12776 | AMR-020-0509

### Open Problems in Integrable Systems - Nijenhuis operators: singular points and global properties

O / open | AMR 问题清单汇编

Let us fix a certain algebraic type of a linear operator, i.e., its Segre characteristic (see above). Does there exist a Nijenhuis operator $L$ in $\mathbb{R}^n$ with the following two properties: ; $L$ has compact support; ; in a certain domain $U\subset \mathbb{R}^n$, the algebraic type of $L$ does not change and coincides with the given one? More generally, find necessary and sufficient conditions, in terms of the Segre characteristic, for the existence of a Nijenhuis operator in $\mathbb{R}^n$ with the above properties.

背景链接：https://arxiv.org/abs/1804.03737

12777 | AMR-020-0510

### Open Problems in Integrable Systems - Nijenhuis operators: singular points and global properties

S / solved | AMR 问题清单汇编

Construct real analytic examples of Nijenhuis operators on closed two-dimensional surfaces whose eigenvalues are real and generically distinct.

背景链接：https://arxiv.org/abs/1804.03737

12778 | AMR-020-0511

### Open Problems in Integrable Systems - Nijenhuis operators: singular points and global properties

P / partially_solved | AMR 问题清单汇编

Describe all the functions $f(x,y)$ of two variables defined in a neighbourhood of $(0,0)\in\mathbb{R}^2$ such that ; $f_y(0,0)\not\equiv 0$; ; $f_y(0,0)=0$; ; $\frac{f_x(x-f_x) - f}{f_y} $ is locally smooth (or, more precisely, extends up to a locally smooth function).

背景链接：https://arxiv.org/abs/1804.03737

12779 | AMR-020-0512

### Open Problems in Integrable Systems - Nijenhuis operators: singular points and global properties

P / partially_solved | AMR 问题清单汇编

Consider a smooth map $\Phi=(\sigma_1,\dots,\sigma_n): U(0) \to \mathbb{R}^n$, where $U(0)$ is a neighbourhood of the origin $0\in \mathbb{R}^n$. We assume that $0$ is a singular point of $\Phi$, that is, $\operatorname{rank} d\Phi(0) < n$, but almost all points $x\in U(0)$ are regular. Describe those maps $\Phi$ for which all the components of the Nijenhuis operator $L(x)$ defined by [source label: eq10.1] are smooth functions (notice that $L$ is well defined and smooth at regular points of $\Phi$, and we are interested in necessary and sufficient conditions for $L$ to be smoothly extendable onto the whole neighbourhood $U(0)$).

背景链接：https://arxiv.org/abs/1804.03737

12780 | AMR-020-0513

### Open Problems in Integrable Systems - Nijenhuis operators: singular points and global properties

P / partially_solved | AMR 问题清单汇编

Describe/classify the collections of algebraically independent homogeneous polynomials $\sigma_1, \dots, \sigma_n$, $\deg \sigma_k = k$, in $n$ variables $x_1,\dots, x_n$ with the required property.

背景链接：https://arxiv.org/abs/1804.03737

12781 | AMR-020-0514

### Open Problems in Integrable Systems - Nijenhuis operators: singular points and global properties

O / open | AMR 问题清单汇编

Describe the structure of singularities of $\Phi=(\sigma_1,\dots,\sigma_n): M \to \mathbb{R}^n$ in terms of the singular points of the recursion operator $R$, their linearisations, and the corresponding left-symmetric algebras. What are sufficient and/or necessary conditions for such singularities to be non-degenerate (in the sense of Eliasson )? Notice that, in this case, the algebraic type of $R$ is rather special: at a generic point, $R$ is semisimple, but each of its eigenvalues has multiplicity 2.

背景链接：https://arxiv.org/abs/1804.03737

12782 | AMR-020-0409

### Open Problems in Integrable Systems - Noncommutative integrable maps

P / partially_solved | AMR 问题清单汇编

{\rm (V. Retakh) Establish complete integrability of the noncommutative version of the leapfrog map. Define noncommutative versions of the pentagram map and its higher-dimensional analogs and establish their complete integrability. }

背景链接：https://arxiv.org/abs/1804.03737

12783 | AMR-020-0601

### Open Problems in Integrable Systems - Poisson geometry and action-angle variables

P / partially_solved | AMR 问题清单汇编

Extend the action-angle theorem in a neighbourhood of a Liouville torus at singular points of the Poisson structure.

背景链接：https://arxiv.org/abs/1804.03737

12784 | AMR-020-0602

### Open Problems in Integrable Systems - Poisson geometry and action-angle variables

P / partially_solved | AMR 问题清单汇编

Extend the action-angle theorem in a neighbourhood of a Liouville torus at singular points of the Poisson structure for splittable integrable systems.

背景链接：https://arxiv.org/abs/1804.03737

12785 | AMR-020-0603

### Open Problems in Integrable Systems - Poisson geometry and action-angle variables

P / partially_solved | AMR 问题清单汇编

Determine obstructions to the global existence of action-angle coordinates on regular Poisson manifolds.

背景链接：https://arxiv.org/abs/1804.03737

12786 | AMR-020-0604

### Open Problems in Integrable Systems - Poisson geometry and action-angle variables

O / open | AMR 问题清单汇编

Which foliations with affine leaves can be described as the image of the moment map?

背景链接：https://arxiv.org/abs/1804.03737

12787 | AMR-020-0605

### Open Problems in Integrable Systems - Poisson geometry and action-angle variables

O / open | AMR 问题清单汇编

Consider a Poisson manifold $M$ of even dimension such that it is symplectic on a dense set $U\subset M$. Assume that $M$ is endowed with a toric action which is Hamiltonian on $U$. Is there an analog of Delzant theorem for the image of the moment map?

背景链接：https://arxiv.org/abs/1804.03737

12788 | AMR-020-0606

### Open Problems in Integrable Systems - Poisson geometry and action-angle variables

P / partially_solved | AMR 问题清单汇编

[S. V\ u Ng{\d o}c] Find natural examples of integrable systems in classical mechanics with non-trivial Duistermaat--Chern classes.

背景链接：https://arxiv.org/abs/1804.03737

12789 | AMR-020-0303

### Open Problems in Integrable Systems - Polynomial in momenta integrals in higher dimensions.

P / partially_solved | AMR 问题清单汇编

Construct new examples of natural Hamiltonian systems on higher dimensional manifolds which are integrable in the class of integrals polynomial in momenta.

背景链接：https://arxiv.org/abs/1804.03737

12790 | AMR-020-0304

### Open Problems in Integrable Systems - Polynomial in momenta integrals in higher dimensions.

O / open | AMR 问题清单汇编

Construct stationary axially symmetric 4-dimensional Einstein metrics admitting Killing tensors of higher order.

背景链接：https://arxiv.org/abs/1804.03737

12791 | AMR-020-0305

### Open Problems in Integrable Systems - Polynomial in momenta integrals in higher dimensions.

O / open | AMR 问题清单汇编

[Gilkey ] In the Riemannian case, is every $1$-homogeneous manifold locally homogeneous?

背景链接：https://arxiv.org/abs/1804.03737

12792 | AMR-020-0306

### Open Problems in Integrable Systems - Polynomial in momenta integrals in higher dimensions.

S / solved | AMR 问题清单汇编

In a symmetric space, is every Killing tensor a sum of symmetric products of Killing vectors? Equivalently, is it true that the algebra of all polynomial integrals of the geodesic flow in a symmetric space is generated by linear integrals?

背景链接：https://arxiv.org/abs/1804.03737

12793 | AMR-020-0710

### Open Problems in Integrable Systems - Quantum integrability for polynomial in momenta integrals

O / open | AMR 问题清单汇编

What are necessary and/or sufficient conditions on a metric $g$ such that every polynomial integral of its geodesic flow is quantisable?

背景链接：https://arxiv.org/abs/1804.03737

12794 | AMR-020-0711

### Open Problems in Integrable Systems - Quantum integrability for polynomial in momenta integrals

P / partially_solved | AMR 问题清单汇编

Quantise the Mischenko-Fomenko algebra $\mathcal F_a \subset \mathcal P(\goth g)$ for an arbitrary finite-dimensional Lie algebra $\mathfrak g$ {\/\rm(}or find an obstruction to quantisation{\/\rm)}.

背景链接：https://arxiv.org/abs/1804.03737

12795 | AMR-020-0307

### Open Problems in Integrable Systems - Superintegrable systems

O / open | AMR 问题清单汇编

{\it Construct a natural Hamiltonian system on the 2-sphere with a nonconstant potential which is superintegrable by integrals of degree $\ge 3$ and admits no nontrivial integral of degree one and two. }

背景链接：https://arxiv.org/abs/1804.03737

12796 | AMR-020-0308

### Open Problems in Integrable Systems - Superintegrable systems

P / partially_solved | AMR 问题清单汇编

Does there exist a non-conformally flat metric on the sphere $S^n$, $n>2$, whose geodesic flow is maximally superintegrable (in the class of integrals which are polynomial in momenta)?

背景链接：https://arxiv.org/abs/1804.03737

12797 | AMR-020-0201

### Open Problems in Integrable Systems - Topology of integrable systems, Lagrangian fibrations, and their invariants

O / open | AMR 问题清单汇编

Under which additional assumptions does this principle become a rigorous theorem?

背景链接：https://arxiv.org/abs/1804.03737

12798 | AMR-020-0202

### Open Problems in Integrable Systems - Topology of integrable systems, Lagrangian fibrations, and their invariants

P / partially_solved | AMR 问题清单汇编

Consider two Lagrangian fibrations $\phi: M^4 \to B$ and $\phi': {M'}^4 \to B'$. Assume that $B$ and $B'$ are affinely equivalent in the sense that there exists an affine diffeomorphism $\psi : B \to B'$. Is it true that under these assumptions the corresponding Lagrangian fibrations are symplectomorphic?

背景链接：https://arxiv.org/abs/1804.03737

12799 | AMR-020-0203

### Open Problems in Integrable Systems - Topology of integrable systems, Lagrangian fibrations, and their invariants

P / partially_solved | AMR 问题清单汇编

Do local symplectic invariants exist for diffeomorphic degenerate singularities? How many and of what kind are they? This question makes sense even in the simplest case of one-degree-of-freedom systems.

背景链接：https://arxiv.org/abs/1804.03737

12800 | AMR-020-0204

### Open Problems in Integrable Systems - Topology of integrable systems, Lagrangian fibrations, and their invariants

P / partially_solved | AMR 问题清单汇编

Describe the symplectic invariants of stable rank-one singularities described by V. V. Kalashnikov. For such singularities, one of the action variables, say $I_1$, is smooth, and the other $I_2$ is singular. Is it true that $I_1$ and $I_2$ are sufficient for the symplectic classification? Here is an almost equivalent version of this question. Assume that we have two functions $H$ and $F$ that commute with respect to two different symplectic structures $\omega_1$ and $\omega_2$ and define (in both cases) a stable rank-one singularity described by V. V. Kalashnikov. Assume that the action variables $I_1$ and $I_2$ are the same for $\omega_1$ and $\omega_2$. This condition is equivalent to the relation $$ \oint_\gamma \alpha_1 = \oint_\gamma \alpha_2 $$ for any cycle $\gamma$ on any regular fiber $\mathcal L_{f,h}=\{ F=f, H=h\}$ and appropriately chosen $1$-forms $\alpha_i$ such that $d\alpha_i = \omega_i$, $i=1,2$. Is it true that, under these conditions, there exists a smooth map $\psi$ that preserves the functions $F$ and $H$ and such that $\psi^*(\omega_2)=\omega_1$? The source notes that the simplest case is known.

背景链接：https://arxiv.org/abs/1804.03737

12801 | AMR-020-0205

### Open Problems in Integrable Systems - Topology of integrable systems, Lagrangian fibrations, and their invariants

P / partially_solved | AMR 问题清单汇编

Assume that we know explicit formulas for the action variables $I_1,\dots, I_n$ so that we are able to analyse their asymptotic behaviour in a neighbourhood of a singular fiber. Can we recover the topology of this singularity from the asymptotic behaviour (or at least to distinguish between different types of singularities)? For example, we know that, in the case of non-degenerate hyperbolic singularities, the singular part $I_{sing}$ is of the form $h\ln h + \dots$ . Is this property characteristic for non-degenerate hyperbolic singularities? The case of one degree of freedom is understood in .

背景链接：https://arxiv.org/abs/1804.03737

12802 | AMR-020-0206

### Open Problems in Integrable Systems - Topology of integrable systems, Lagrangian fibrations, and their invariants

P / partially_solved | AMR 问题清单汇编

Describe all topological types of singularities that may appear in algebraically integrable systems with a small ($\leq 3$) number of degrees of freedom, and describe the symplectic invariants of such singularities.

背景链接：https://arxiv.org/abs/1804.03737

12803 | AMR-020-0207

### Open Problems in Integrable Systems - Topology of integrable systems, Lagrangian fibrations, and their invariants

P / partially_solved | AMR 问题清单汇编

[\'{A}. Pelayo] Extend the classification of semitoric systems $F=(J,H)$ in to allow for $F$ having non-degenerate singularities with hyperbolic blocks (a version of this problem was mentioned in various forms in ).

背景链接：https://arxiv.org/abs/1804.03737

12804 | AMR-020-0208

### Open Problems in Integrable Systems - Topology of integrable systems, Lagrangian fibrations, and their invariants

P / partially_solved | AMR 问题清单汇编

[\'{A}. Pelayo] Consider a compact connected $2n$-dimensional symplectic manifold $M$, endowed with a Hamiltonian $(S^1)^{n-1}$-action; these are called complexity one spaces. Consider an integrable system $f_1,\ldots, f_{n}$ on $M$ where $(f_1,\ldots, f_{n-1}) \colon M \to \mathbb{R}^{n-1}$ is the momentum map of the Hamiltonian $(S^1)^{n-1}$-action. Suppose that the singularities of the integrable system are non-degenerate and also they do not contain hyperbolic blocks. Study how the invariants of the complexity one space are related to the invariants of the semitoric system.

背景链接：https://arxiv.org/abs/1804.03737

12805 | AMR-020-0209

### Open Problems in Integrable Systems - Topology of integrable systems, Lagrangian fibrations, and their invariants

O / open | AMR 问题清单汇编

[N. T. Zung] Study the topology and geometry of these singular fibers and their small neighbourhoods.

背景链接：https://arxiv.org/abs/1804.03737

12806 | AMR-020-0210

### Open Problems in Integrable Systems - Topology of integrable systems, Lagrangian fibrations, and their invariants

O / open | AMR 问题清单汇编

[N. T. Zung] Give a clear description of these special singular fibers.

背景链接：https://arxiv.org/abs/1804.03737

12807 | AMR-020-0301

### Open Problems in Integrable Systems - Two-dimensional case.

O / open | AMR 问题清单汇编

Complete the above table: {\rm (1)} construct new examples of natural Hamiltonian systems on closed two-dimensional surfaces admitting polynomial integrals, describe and, if possible, classify them; {\rm (2)} prove, if possible, nonexistence of such integrals (perhaps under some additional assumptions).

背景链接：https://arxiv.org/abs/1804.03737

12808 | AMR-020-0302

### Open Problems in Integrable Systems - Two-dimensional case.

P / partially_solved | AMR 问题清单汇编

Construct a natural Hamiltonian system with a nonconstant potential on $S^2$ that admits a nontrivial polynomial integral of degree $5$ and does not admit any nontrivial integrals of smaller degrees.

背景链接：https://arxiv.org/abs/1804.03737

12809 | AMR-051-0001

### Open Problems on Billiards and Geometric Optics

P / partially_solved | AMR 问题清单汇编

Let $\gamma$ be a smooth closed strictly convex plane curve, parametrized by arc length $s$, and let $d>0$. Form a sphere-like surface by gluing the two ends of the height-$d$ cylinder over $\gamma$ to two copies of its enclosed domain. Its geodesic return map on the phase cylinder is $$T(s,\alpha)=(s_1+d\cot\alpha_1,\alpha_1),$$ where $(s_1,\alpha_1)$ is the ordinary billiard image of $(s,\alpha)$. (1) Does $T$ have invariant curves, for example KAM curves near the boundary? (2) Other than circles, for which $\gamma$ is $T$ integrable? (3) Can $T$ be ergodic?

背景链接：https://arxiv.org/abs/2110.10750

12810 | AMR-051-0002

### Open Problems on Billiards and Geometric Optics

P / partially_solved | AMR 问题清单汇编

Let $\gamma$ be a smooth closed strictly convex plane curve. The outer billiard map $T$ sends a point $A$ near $\gamma$ to the point $T(A)$ for which $[A,T(A)]$ is tangent to $\gamma$ at its midpoint. Outer billiards about ellipses are integrable, with phase space foliated by homothetic invariant ellipses. Are there other integrable outer billiards?

背景链接：https://arxiv.org/abs/2110.10750

12811 | AMR-051-0003

### Open Problems on Billiards and Geometric Optics

P / partially_solved | AMR 问题清单汇编

Let $\gamma$ be a smooth closed strictly convex curve and $\delta\in(0,\pi/2)$. Say that $\gamma$ has the $\delta$-Gutkin property if the curve of incoming oriented lines meeting $\gamma$ at the constant angle $\delta$ is invariant under the Birkhoff billiard. What are the Gutkin billiards on the sphere and on the hyperbolic plane?

背景链接：https://arxiv.org/abs/2110.10750

12812 | AMR-051-0004

### Open Problems on Billiards and Geometric Optics

O / open | AMR 问题清单汇编

For a Birkhoff billiard inside a closed smooth strictly convex hypersurface $S\subset\mathbb{R}^d$, let $T$ act on the space $\mathbb{A}$ of oriented lines intersecting $S$. Find a non-ellipsoidal $S$ for which $T$ leaves invariant a smooth hypersurface $\Sigma\subset\mathbb{A}$, and determine the geometric and dynamical properties of such invariant hypersurfaces.

背景链接：https://arxiv.org/abs/2110.10750

12813 | AMR-051-0005

### Open Problems on Billiards and Geometric Optics

O / open | AMR 问题清单汇编

Let $\gamma$ be a smooth convex plane billiard table symmetric about an axis $l$, and let $C$ be a convex caustic. Must $C$ be symmetric about $l$? Prove this or give a counterexample.

背景链接：https://arxiv.org/abs/2110.10750

12814 | AMR-051-0006

### Open Problems on Billiards and Geometric Optics

P / partially_solved | AMR 问题清单汇编

A planar projective billiard is a bounded domain $\Omega$ whose piecewise-smooth boundary carries a transverse line field $L$. At $p\in\partial\Omega$, an incident line $\ell$ reflects to $\ell'$ when $\ell,\ell',L(p),T_p\partial\Omega$ form a harmonic quadruple. Call the billiard $k$-reflective if its billiard map has an open set of $k$-periodic points. (1) Construct a $k$-reflective projective billiard for some odd $k\ge5$. (2) For fixed $k\ge4$, classify $k$-reflective projective billiards within natural boundary-smoothness classes (polygonal, piecewise algebraic, analytic, and so on)-the projective Ivrii conjecture.

背景链接：https://arxiv.org/abs/2110.10750

12815 | AMR-051-0007

### Open Problems on Billiards and Geometric Optics

O / open | AMR 问题清单汇编

Suppose a bounded strictly convex planar billiard has two nested closed caustics such that the smaller caustic is itself a caustic for the billiard in the larger caustic. Must the billiard boundary be an ellipse?

背景链接：https://arxiv.org/abs/2110.10750

12816 | AMR-051-0008

### Open Problems on Billiards and Geometric Optics

P / partially_solved | AMR 问题清单汇编

Let $\gamma\subset\mathbb{R}^n$ be a closed strictly convex hypersurface, and let $\Pi$ be the phase cylinder of oriented lines meeting $\gamma$ transversely twice. For every $\varepsilon>0$ and $k\in\mathbb{N}$, is every $C^\infty$ Hamiltonian symplectomorphism $\Pi\to\Pi$ a $C^\infty$ limit of compositions of reflections from $\gamma$ and from hypersurfaces $\varepsilon$-close to $\gamma$ in the $C^k$ topology?

背景链接：https://arxiv.org/abs/2110.10750

12817 | AMR-051-0009

### Open Problems on Billiards and Geometric Optics

O / open | AMR 问题清单汇编

Let $C\subset\mathbb{R}^2$ be a curve and translate it through an $\varepsilon$-square lattice, recording a click whenever it meets a lattice point. Can the shape of $C$ be recovered from the distribution of click parameters, either for $C$ alone or for all its rotations? Is there a Fourier-type transform that extracts curvature patterns from this click distribution? Analyze the cases of a segment, a polygon, and a circle.

背景链接：https://arxiv.org/abs/2110.10750

12818 | AMR-051-0010

### Open Problems on Billiards and Geometric Optics

P / partially_solved | AMR 问题清单汇编

Let $g^t:\mathbb{R}^2\to\mathbb{R}^2$ be a Lebesgue-measure-preserving flow or cascade. A point is trapped if its positive semiorbit is bounded and its negative semiorbit is unbounded; let $T_g$ be the trapped set. What is the maximum Hausdorff dimension of $T_g$? More specifically, for $C^m$ systems determine the maximal dimension $d(m)$, between $1$ and $2$.

背景链接：https://arxiv.org/abs/2110.10750

12819 | AMR-051-0011

### Open Problems on Billiards and Geometric Optics

P / partially_solved | AMR 问题清单汇编

A uniformly massive planar body $B$ moves through a uniform medium of initially stationary infinitesimal particles, which reflect elastically from $\partial B$. For simple noncircular shapes such as an ellipse, triangle, or rod, describe the translational and rotational motion for $t\geq0$. In particular, if a rod or centrally symmetric body starts rotating about its center without translation, is the total number of turns finite; if not, does angular velocity tend to zero, and with what asymptotic behavior?

背景链接：https://arxiv.org/abs/2110.10750

12820 | AMR-051-0012

### Open Problems on Billiards and Geometric Optics

P / partially_solved | AMR 问题清单汇编

A body moves freely in a rarefied medium in $\mathbb{R}^n$, $n\geq1$, under Newtonian aerodynamics. Determine the equations of motion and prove existence and uniqueness. In the one-dimensional case, formulate the motion using a measure $\mu_t$ on particle phase space together with the massive particle's position $X(t)$ and velocity $P(t)=X'(t)$.

背景链接：https://arxiv.org/abs/2110.10750

12821 | AMR-051-0013

### Open Problems on Billiards and Geometric Optics

P / partially_solved | AMR 问题清单汇编

In $\mathbb{R}^n$, the space $\mathcal{L}$ of oriented lines has dimension $2n-2$ and a natural symplectic structure. Normal families of rays form Lagrangian submanifolds and can be trapped by mirrors. What is the greatest dimension of a family of rays that can be trapped? In $\mathbb{R}^3$, can a non-normal two-parameter family of rays be trapped?

背景链接：https://arxiv.org/abs/2110.10750

12822 | AMR-051-0014

### Open Problems on Billiards and Geometric Optics

P / partially_solved | AMR 问题清单汇编

For a planar oval $\gamma$, alternately follow chords in two fixed directions to obtain a circle map $F:\gamma\to\gamma$. If $F$ is conjugate to a rotation for every pair of directions, must $\gamma$ be an ellipse? In the projective version, use pencils through points $P,Q$; if for every $P,Q$ with $PQ$ meeting $\gamma$ the resulting $F$ is conjugate to a Möbius transformation, must $\gamma$ be an ellipse?

背景链接：https://arxiv.org/abs/2110.10750

12823 | AMR-051-0015

### Open Problems on Billiards and Geometric Optics

P / partially_solved | AMR 问题清单汇编

In a planar symplectic billiard on an oval, the chord $xy$ reflects to $yz$ when the tangent at $y$ is parallel to $xz$; define polygonal symplectic billiards similarly. (1) Classify polygons for which every symplectic billiard orbit is periodic. (2) Does every polygon have a periodic orbit? (3) Is the symplectic billiard in a stadium chaotic?

背景链接：https://arxiv.org/abs/2110.10750

12824 | AMR-051-0016

### Open Problems on Billiards and Geometric Optics

P / partially_solved | AMR 问题清单汇编

For an oval $\gamma$ and a light source inside it, call the envelope of rays after $n$ reflections the $n$th caustic by reflection. Is every generic caustic by reflection in an ellipse a curve with exactly four cusps? Does this four-cusp property characterize ellipses?

背景链接：https://arxiv.org/abs/2110.10750

12825 | AMR-051-0017

### Open Problems on Billiards and Geometric Optics

O / open | AMR 问题清单汇编

For an oval $\gamma$, the area spectrum of its outer billiard is the set of areas of the circumscribed polygons formed by periodic outer-billiard trajectories. Is this area spectrum related to the spectrum of a differential operator?

背景链接：https://arxiv.org/abs/2110.10750

12826 | OWR-4410-010

### Orbit Closures of Full Cartan Torus Actions

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be an abelian subgroup of $\operatorname{SL}_d(\mathbb Z)$ of rank at least two that contains a totally irreducible toral automorphism and is maximal in rank among abelian subgroups containing it. Classify the orbit closures of the diagonal $G$-action on $\mathbb T^d\times\mathbb T^d$.

来源字段：https://doi.org/10.4171/owr/2010/29

12827 | AMR-045-0038

### Orbit equivalence and ordered cohomology

P / partially_solved | AMR 问题清单汇编

Classify irreducible SFTs up to topological orbit equivalence; characterize and classify their unital ordered cohomology groups; determine when winding and standard orders agree; and decide whether every subshift orbit-equivalent to an irreducible SFT is itself an SFT.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12828 | AMR-042-0001

### Order of mixing

P / partially_solved | AMR 问题清单汇编

Let $p$ be a prime for which $$f(u_1,u_2)=1+u_1u_2+u_1^2u_2+u_1^3u_2+u_1^4+u_2^2+u_1^4u_2^2$$ is irreducible, and consider the algebraic $\mathbb{Z}^2$-action associated to $\mathbb{Z}[u_1^{\pm1},u_2^{\pm1}]/\langle p,f\rangle$. Its exact order $M$ of mixing satisfies $3\leq M<7$. What is $M$?

背景链接：https://www.imath.kiev.ua/~skolyada/kevin.pdf

12829 | AIM-DYNAMICAL_SYSTEMS-0004

### Outside-subtree field criteria and affine wreath recursions for arboreal Galois groups

P / partially_solved | AIM 研讨会问题汇编

**Wreath recursion for Galois groups**

1. Contruct explicit wreath recursions for Galois groups

2. Under what conditions is a group coming from a Galois arboreal representation a branch group?

来源字段：http://aimpl.org/groupdynamorigin/1/

12830 | AMR-052-0064

### Parameter spaces of cosine and sine families

P / partially_solved | AMR 问题清单汇编

Describe the parameter-space structure for the entire families $C_\lambda(z)=\lambda\cos z$ and $S_\lambda(z)=\lambda\sin z$.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12831 | AIM-DYNAMICAL_SYSTEMS-0082

### PCF descent and an odd postcritical-divisor criterion

P / partially_solved | AIM 研讨会问题汇编

Field of definition vs field of moduli

Are all PCF maps defined over their field of moduli?

来源字段：http://aimpl.org/finitedynamics/2/

12832 | AIM-DYNAMICAL_SYSTEMS-0090

### PCF descent, higher-dimensional sparsity, and a finite-stratum certificate

P / partially_solved | AIM 研讨会问题汇编

PCF if the orbit of the ramification locus \(C\), \(V=\bigcup_{n\geq0}f^n(C)\), is a proper algebraic subvariety; “PCF all the way down” if the restriction of \(f\) to every periodic component of the post-critical component \(V\) is PCF.

It asks, for a morphism \[ f:\mathbb P^N\longrightarrow\mathbb P^N, \] whether ordinary PCF implies “PCF all the way down,” whether either class is Zariski dense in \(M_d^N\), and whether there is a less restrictive notion that is abundant, dense in a reasonable topology, and has an André-Oort property. The two definitions in the record are:

The live source page, http://aimpl.org/finitedynamics/5/, was checked on 2 August 2026. On the live page the latter is a broken link to `#it1`; it probably means definition (1), but this cannot be recovered with certainty and is not silently corrected here.

Three conventions are needed to make the question precise.

1. Work over \(\mathbb C\), take \(d\geq2\), and use reduced supports. Write \[ P(f)=\overline{\bigcup_{n\geq1}f^n(\operatorname{Crit}(f))}^{\rm Zar}. \] Including \(\operatorname{Crit}(f)\) itself, as the source does in \(V\), does not change whether its orbit is contained in a proper algebraic set. 2. A periodic component means an irreducible component \(L\) with \(f^r(L)=L\) set-theoretically for some \(r>0\). If components form a cycle, the relevant map is the first return \(f^r|_L\), not \(f|_L\). 3. The source definition is one-step. Astorg's later definition is recursive: take the union of periodic irreducible components, pass to a common return iterate, take its intrinsic postcritical set, and repeat. “All the way down” below refers to this modern recursive version; the literal source version follows already at the first step.

If a periodic component is singular, “ramification locus of the restriction” has no automatic smooth-manifold meaning. This is a real mathematical obstruction, not merely notation; Sections 8 and 9 explain it.

来源字段：http://aimpl.org/finitedynamics/5/

12833 | AIM-DYNAMICAL_SYSTEMS-0078

### PCF equidistribution requires a sampling law

P / partially_solved | AIM 研讨会问题汇编

**Equidistribution Problems**

1. \(f:\mathbb P^1\to\mathbb P^1\). Is the set PCF equidistributed on \(M_d\) with respect to the bifurcation measure. 2. Same question for \(f:\mathbb P^N\to\mathbb P^N\) with respect to *what* measure?

来源字段：http://aimpl.org/finitedynamics/2/

12834 | AIM-DYNAMICAL_SYSTEMS-0020

### PCFness, specialization, and the Basilica boundary case

P / partially_solved | AIM 研讨会问题汇编

Take $f(z)\in\mathbb Q[z]$, assume $0$ is not periodic. Consider two groups \[ G:=\lim_{n\to\infty}\operatorname{Gal}(f^n),\qquad G':=\lim\operatorname{Gal}(f^n(z)-t). \]

**Recovered question.** How does postcritical finiteness of $f$ affect the finite-index comparison between the specialized iterated Galois group at the target $0$ and the generic iterated Galois group? How does this comparison control the prime-divisor question in Problem 1.1? Analyze the Basilica polynomial $f(z)=z^2-1$ first.

来源字段：http://aimpl.org/galarithdyn/3/

12835 | AMR-046-0013

### Period functions for systems with homogeneous components

O / open | AMR 问题清单汇编

For $$\dot x=P_{2k+1}(x,y),\qquad \dot y=Q_{2\ell+1}(x,y),$$ where $P_{2k+1}$ and $Q_{2\ell+1}$ are homogeneous polynomials of the indicated odd degrees, (i) characterize all centers and (ii) determine the maximum number of oscillations of the period function among these centers.

背景链接：https://arxiv.org/abs/2012.02524

12836 | OWR-17127-006

### Period Structure of Dissipative Disk Diffeomorphisms

P / partially_solved | Oberwolfach 报告问题汇编

For every dissipative diffeomorphism of the disk, does there exist $n_0$ such that every period belongs to $$\{n2^k:n\leq n_0,\ k\in\mathbb N\},$$ and, if there are infinitely many periods, must they contain $\{n2^k:k\in\mathbb N\}$ for some $n$?

来源字段：https://doi.org/10.4171/owr/2019/31

12837 | AMR-045-0046

### Periodic and finite configurations

P / partially_solved | AMR 问题清单汇编

For a $d$-dimensional cellular automaton with $d\ge2$, let $f_P$ and $f_F$ be its restrictions to spatially periodic and finite configurations. Does injectivity of $f_P$ imply surjectivity of $f_F$? Does surjectivity of $f_F$ imply surjectivity of $f_P$? Does surjectivity of $f$ imply surjectivity of $f_P$?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12838 | OWR-8415336-003

### Periodic Billiards and the Wave Trace of Ellipses

P / partially_solved | Oberwolfach 报告问题汇编

Can a set of periodic billiard orbits, for example a positive-measure Cantor set of a fixed length, produce the wave-trace expansion of an ellipse?

来源字段：https://doi.org/10.4171/owr/2021/33

12839 | OWR-13750333-009

### Periodic Minimizers in Compact Linear Domino Games

O / open | Oberwolfach 报告问题汇编

For a linear domino game $(\Sigma,T,s)$ with infinite compact topological symbol space $\Sigma$ and continuous size function $s$, must an average-size minimizer be periodic? Equivalently, construct a game for which no periodic sequence minimizes average size, or prove none exists.

来源字段：https://doi.org/10.4171/owr/2023/22

12840 | AMR-041-0007

### Periodic orbits

P / partially_solved | AMR 问题清单汇编

(a) Is the set of $n$-periodic orbits of a smooth strictly convex Birkhoff billiard nowhere dense for every $n$? (b) Does every polygonal Birkhoff billiard have a periodic orbit?

背景链接：https://people.math.harvard.edu/~knill/seminars/intr/

12841 | AMR-052-0033

### Periodic orbits near a Cremer point

P / partially_solved | AMR 问题清单汇编

For a Cremer point of an arbitrary rational map, does every neighborhood contain infinitely many periodic orbits?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12842 | AMR-052-0036

### Periodic point on a Siegel-disk boundary

P / partially_solved | AMR 问题清单汇编

Does any rational function have a Siegel disk with a periodic point on its boundary?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12843 | AMR-045-0041

### Periodic points of cellular automata

P / partially_solved | AMR 问题清单汇编

Must every surjective $d$-dimensional cellular automaton have dense periodic points? Must its jointly spatially and temporally periodic points be dense? Are its minimal subsystems dense?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12844 | AMR-046-0011

### Periodic rational difference equations

O / open | AMR 问题清单汇编

Consider $$x_{n+k}=\frac{A_0+A_1x_n+\cdots+A_kx_{n+k-1}}{B_0+B_1x_n+\cdots+B_kx_{n+k-1}},$$ where the coefficients are nonnegative, $\sum A_i,\sum B_i>0$, and $A_1^2+B_1^2\ne0$. Call it $p$-periodic if every positive initial condition generates a sequence of least common period $p$. Apart from rescaling and index-dilation equivalents of $$x_{n+1}=x_n,\quad x_{n+1}=1/x_n,\quad x_{n+2}=x_{n+1}/x_n,\quad x_{n+2}=(1+x_{n+1})/x_n,$$ and $$x_{n+3}=(1+x_{n+1}+x_{n+2})/x_n,$$ are there any such periodic equations?

背景链接：https://arxiv.org/abs/2012.02524

12845 | AMR-046-0005

### Periodic Riccati differential equations

P / partially_solved | AMR 问题清单汇编

For a general $T$-periodic Riccati equation $$\frac{dx}{dt}=A_2(t)x^2+A_1(t)x+A_0(t),$$ give effective criteria determining whether it has a continuum of periodic solutions or exactly $2$, $1$, or $0$ limit cycles.

背景链接：https://arxiv.org/abs/2012.02524

12846 | OWR-1383-005

### Periodic Trajectories and Invariant Curves in Billiards

O / open | Oberwolfach 报告问题汇编

For a plane billiard table, the set of periodic billiard trajectories should have measure zero with respect to the canonical invariant measure. Also, are convex billiard tables with an invariant curve consisting of periodic trajectories rare or common, and how can such tables be constructed systematically? The same questions may be asked for outer billiards.

来源字段：https://doi.org/10.4171/owr/2006/45

12847 | AIM-DYNAMICAL_SYSTEMS-0088

### Periodic-point fields and a cyclotomic obstruction to ramification detection

P / partially_solved | AIM 研讨会问题汇编

For maps of a fixed degree (up to an explicitly chosen notion of conjugacy), is there a property of the labeled periodic splitting-field system, or of specified Galois/ramification data extracted from it, that is equivalent to PCF?

来源字段：http://aimpl.org/finitedynamics/4/

12848 | AMR-044-0001

### Periods of Pseudo-Integrable Billiards

O / open | AMR 问题清单汇编

Consider billiard tables formed by two concentric semicircles joined by two line segments. Consider trajectories with a fixed circle, concentric with the boundary semicircles, as caustic, and with rotation numbers $\rho_1$ and $\rho_2$ relative to the two semicircles. Are these trajectories periodic, and what periods are possible for given $\rho_1$ and $\rho_2$?

背景链接：https://armj.math.stonybrook.edu/html-articles/Files-2015-2024/14-04

12849 | AMR-039-0021

### Permutation groups

O / open | AMR 问题清单汇编

For permutation groups with the transposition random walk, coarse Ricci curvature is positive but gives concentration of the wrong order. Can this discrepancy be explained by hyperbolic-like properties of permutation groups or by a mixture of positive and negative curvature behavior?

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12850 | OWR-15436-003

### Persistence and Permanence of Weakly Reversible Reaction Networks

O / open | Oberwolfach 报告问题汇编

For mass-action dynamical systems arising from weakly reversible chemical reaction networks, do all solutions have persistent dynamics with a positive lower bound and permanent dynamics with uniform positive lower and finite upper bounds for all sufficiently large time?

来源字段：https://doi.org/10.4171/owr/2017/28

12851 | OWR-12582-001

### Persistence of KdV Solutions Under Quasilinear Perturbations

P / partially_solved | Oberwolfach 报告问题汇编

Let $N_4(x,u,u_x,u_{xx},u_{xxx})$ be a Hamiltonian quasilinear term vanishing to fourth order at the origin. For the perturbed equations on $\mathbb T$

$$u_t+u_{xxx}-\partial_x(u^2)+N_4(x,u,u_x,u_{xx},u_{xxx})=0$$

and

$$u_t+u_{xxx}\pm\partial_x(u^3)+N_4(x,u,u_x,u_{xx},u_{xxx})=0,$$

do the periodic, quasiperiodic, or almost-periodic solutions of the underlying KdV and modified KdV equations persist under sufficiently small perturbations?

来源字段：https://doi.org/10.4171/owr/2013/34

12852 | OWR-2495-001

### Persistence of Singular Solutions in State-Dependent Delay Equations

P / partially_solved | Oberwolfach 报告问题汇编

For $\varepsilon\dot x(t)=g(x(t-r_1),\ldots,x(t-r_m))$ with $r_i=r_i(x(t))$, if a singular solution composed of inner and outer solutions exists at $\varepsilon=0$, does a genuine solution exist for every sufficiently small $\varepsilon>0$?

来源字段：https://doi.org/10.4171/owr/2008/57

12853 | AMR-043-0007

### Pingree open problems - Boyle problem 1

O / open | AMR 问题清单汇编

Characterize mixing shifts of finite type up to topological orbit equivalence.

背景链接：https://math.huji.ac.il/~mhochman/open-problems/pingree-open-problems.pdf

12854 | AMR-043-0008

### Pingree open problems - Boyle problem 2

O / open | AMR 问题清单汇编

Let $S$ and $T$ be subshifts. If $S$ is a mixing shift of finite type and $T$ is topologically orbit equivalent to $S$, must $T$ also be a mixing shift of finite type?

背景链接：https://math.huji.ac.il/~mhochman/open-problems/pingree-open-problems.pdf

12855 | AMR-043-0001

### Pingree open problems - Hochman problem 1

O / open | AMR 问题清单汇编

Let $X=\{0,1\}^{\mathbb{Z}}$ and $Y=\{y\in\{0,1,2\}^{\mathbb{Z}}:y_i\neq y_{i+1}\}$. Both are mixing shifts of finite type with entropy $\log2$, but they are not isomorphic. If $\operatorname{Per}(X)$ and $\operatorname{Per}(Y)$ denote their periodic points, are $X\setminus\operatorname{Per}(X)$ and $Y\setminus\operatorname{Per}(Y)$ topologically conjugate?

背景链接：https://math.huji.ac.il/~mhochman/open-problems/pingree-open-problems.pdf

12856 | AMR-043-0002

### Pingree open problems - Hochman problem 2

O / open | AMR 问题清单汇编

Let $T:[0,1)\to[0,1)$ be the doubling map $x\mapsto 2x \pmod 1$, and let $\mu$ be an ergodic measure for $T$ with $0<h(\mu)<1$. Call $f:\mathbb{R}\to\mathbb{R}$ non-singular for $\mu$ if there is a Borel set $A$ such that $\mu|_A$ and $(f\mu)|_A$ are not mutually singular. Which affine maps $f(x)=ax+b$ are non-singular for $\mu$?

背景链接：https://math.huji.ac.il/~mhochman/open-problems/pingree-open-problems.pdf

12857 | AMR-043-0005

### Pingree open problems - Ledrappier problem 1

S / solved | AMR 问题清单汇编

Let $M$ be a compact Riemannian manifold of constant negative curvature and $(g_t)_{t\in\mathbb{R}}$ its geodesic flow. Does there exist a probability measure $\mu$ invariant under the time-one map $g_1$ but not under the full flow $(g_t)_{t\in\mathbb{R}}$?

背景链接：https://math.huji.ac.il/~mhochman/open-problems/pingree-open-problems.pdf

12858 | AMR-043-0003

### Pingree open problems - Petersen tail-field problem 1

O / open | AMR 问题清单汇编

Let $A=\{0,1,\ldots,d-1\}$ and let $\sigma$ be the shift on $A^{\mathbb{Z}}$. Define $(v_n(x))_i=\#\{0\leq j\leq n:x_j=i\}$ and $(w_n(x))_i=\#\{0\leq j\leq n:x_{-j}=i\}$. Put $\mathcal{F}_n^+=\bigcap_{k\geq n}\mathcal{B}(v_n,v_{n+1},\ldots)$, $\mathcal{F}^+=\bigcap_{n\geq0}\mathcal{F}_n^+$, and define $\mathcal{F}^-$ analogously using the $w_n$. Is $\mathcal{F}^+=\mathcal{F}^-$?

背景链接：https://math.huji.ac.il/~mhochman/open-problems/pingree-open-problems.pdf

12859 | AMR-043-0004

### Pingree open problems - Petersen tail-field problem 2

O / open | AMR 问题清单汇编

With $\mathcal{F}^+$ and $\mathcal{F}^-$ defined from the forward and backward symbol-count tail fields on the full shift $A^{\mathbb{Z}}$, if $\mathcal{F}^+\neq\mathcal{F}^-$, is it nevertheless true that $\mathcal{F}^+$ is trivial if and only if $\mathcal{F}^-$ is trivial?

背景链接：https://math.huji.ac.il/~mhochman/open-problems/pingree-open-problems.pdf

12860 | AMR-043-0006

### Pingree open problems - Thouvenot problem

S / solved | AMR 问题清单汇编

Let $(M,T)$ be a smooth map on a manifold with a good symbolic cover: a mixing shift of finite type factors onto $(M,T)$ and is injective on a set of full measure for every invariant probability measure. Let $f\in C^1(M)$ (or smoother), and let $(X,(T_t)_{t\in\mathbb{R}})$ be the flow under $f$. Can every ergodic measure-preserving system with entropy less than $h(X,T_1)$ be realized as an invariant measure for $(X,T_1)$?

背景链接：https://math.huji.ac.il/~mhochman/open-problems/pingree-open-problems.pdf

12861 | AIM-DYNAMICAL_SYSTEMS-0017

### Pink generating functions for affine Frobenius tree actions

P / partially_solved | AIM 研讨会问题汇编

Is Pink's generating function $\Phi_w$ always rational for every $\rho(\text{Frob}_p)=w?$

Suggestion: Try matching data to degree $1$ rational functions.

来源字段：http://aimpl.org/galarithdyn/2/

12862 | AMR-045-0053

### Pisot substitution conjecture

P / partially_solved | AMR 问题清单汇编

Prove that the substitutive shift of every irreducible Pisot substitution has pure discrete spectrum.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12863 | OWR-3396-002

### Pisot Substitution Conjecture for Pure Point Spectrum

O / open | Oberwolfach 报告问题汇编

Does every primitive substitution on a finite alphabet whose inflation multiplier is a Pisot-Vijayaraghavan number, under the usual irreducibility assumptions, produce a continuous dynamical system with pure point dynamical spectrum? Can this be proved by showing that its geometric point-set realization is a regular model set?

来源字段：https://doi.org/10.4171/owr/2009/13

12864 | OWR-14215-006

### Pointwise Convergence of Koopman Laplace Averages

O / open | Oberwolfach 报告问题汇编

For generalized Laplace averages used to construct Koopman eigenfunctions, does the known norm convergence also hold pointwise? For which other observable spaces, especially spaces with readily computable norms, does the same eigenfunction construction remain valid?

来源字段：https://doi.org/10.4171/owr/2016/7

12865 | OWR-13750336-012

### Pointwise Convergence of Multiple Ergodic Averages

O / open | Oberwolfach 报告问题汇编

For commuting invertible measure-preserving transformations $T_1,\ldots,T_d$ and bounded functions $f_1,\ldots,f_d$, do the averages $\frac1N\sum_{n=1}^N\prod_{k=1}^d f_k(T_k^n x)$ converge pointwise almost everywhere for every $d\geq2$?

来源字段：https://doi.org/10.4171/owr/2023/25

12866 | OWR-14299577-004

### Pointwise Convergence of Polynomial Ergodic Averages

P / partially_solved | Oberwolfach 报告问题汇编

Let $(X,\mathcal{B},\mu)$ be a probability space with commuting invertible measure-preserving transformations $T_1,\ldots,T_m$, and let $P_1,\ldots,P_m\in\mathbb{Z}[y]$. For $f_1,\ldots,f_m\in L^\infty(X)$, do the averages $\frac{1}{N}\sum_{n=1}^N\prod_{i=1}^m f_i(T_i^{P_i(n)}x)$ converge pointwise almost everywhere as $N\to\infty$?

来源字段：https://doi.org/10.4171/owr/2025/51

12867 | OWR-17298-003

### Pointwise Equidistribution Under Toral Endomorphisms with Fewer Hypotheses

O / open | Oberwolfach 报告问题汇编

Let $A$ and $B$ be nonsingular integer matrices and let $\mu$ be an $A$-invariant ergodic probability measure on $\mathbb{T}^d$ with positive entropy, satisfying irreducibility of every characteristic polynomial of $B^n$ and with no power of $B$ conjugate to a power of $A$. Are additional hypotheses such as (3a)-(3c) unnecessary for almost-everywhere pointwise equidistribution under $B$?

来源字段：https://doi.org/10.4171/owr/2019/55

12868 | AMR-052-0001

### Polynomial matings that are rational

P / partially_solved | AMR 问题清单汇编

Given two monic polynomials of the same degree with connected filled Julia sets, form their topological mating by identifying their circles at infinity with opposite angles and collapsing external rays. Which matings are conjugate to rational functions?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12869 | AMR-052-0009

### Polynomial realization of intertwining

P / partially_solved | AMR 问题清单汇编

When does the topological intertwining construction for two polynomial dynamical planes yield a branched map conjugate to a polynomial?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12870 | AMR-052-0004

### Polynomial realization of tuning

P / partially_solved | AMR 问题清单汇编

For polynomials $P_1,P_2$ satisfying the tuning construction's connectedness and critical-basin hypotheses, is the resulting topological branched map always conjugate to a polynomial?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12871 | AMR-039-0015

### Positive curvature up to delta

O / open | AMR 问题清单汇编

Define curvature up to $\delta$ by $$T_1(m_x,m_y)\leq(1-\kappa(x,y))d(x,y)+\delta.$$ Which theorems for positive coarse Ricci curvature extend to this setting? Can one choose a discrete subset and redefine its random walk naturally so that it has genuinely positive Ricci curvature?

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12872 | AMR-045-0004

### Positive rational shift equivalence

S / solved | AMR 问题清单汇编

If positive square matrices $A,B$ are shift equivalent over $\mathbb Q_+$, prove that they are strong shift equivalent over $\mathbb Q_+$.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12873 | AMR-052-0041

### Positive-area nowhere-dense Julia sets

S / solved | AMR 问题清单汇编

Can a nowhere-dense Julia set of a rational map have positive Lebesgue measure?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12874 | AIM-DYNAMICAL_SYSTEMS-0014

### Positive-density PCF obstruction and a quantitative Chebyshev bound

P / partially_solved | AIM 研讨会问题汇编

Describe \[ N(x)=\#\{p\leq x : p\mid f^n(\alpha) \text{ for some } n\} \] for \(p\) prime as \(x\to\infty\).

来源字段：http://aimpl.org/galarithdyn/1/

12875 | AIM-DYNAMICAL_SYSTEMS-0028

### Power twists obstruct arithmetic reconstruction from the preperiodic-point field

P / partially_solved | AIM 研讨会问题汇编

What do you need to know about \(K_f\) to figure out what \(f\) was (or how close to \(f\) can you get)?

The archived AIM page supplies the missing definition verbatim: for a number field \(K\) and \(f(z)\in K(z)\) of degree at least two, \[ K_f=K(\text{all preperiodic points of }f). \] The next AIM item separately introduces \(K_f'=K(\text{all periodic points of }f)\), confirming that \(K_f\) in this problem means the field generated by **preperiodic**, not merely periodic, points. The official workshop summary likewise distinguishes dynatomic fields generated by periodic points from arboreal fields generated by backward orbits, but it does not discuss Problem 7.2 specifically.

There is a second ambiguity: “conjugate” may mean conjugate by \(\operatorname{PGL}_2(K)\) or by \(\operatorname{PGL}_2(\overline K)\). The distinction is essential below. The neighboring Problem 7.1 explicitly suggests the equivalence types “conjugate, an iterate, or conjugate to something that commutes,” and asks only for commensurability of \(K_f\) and \(K_g\). We therefore analyze exact equality and commensurability of embedded extensions, and distinguish arithmetic from geometric conjugacy.

来源字段：http://aimpl.org/galarithdyn/7/

12876 | OWR-15174-006

### Preperiods and Cycles of Quadratic Polynomial Iteration

P / partially_solved | Oberwolfach 报告问题汇编

For a quadratic polynomial $f\in\mathbb F_q[x]$ and $m\in\mathbb F_q$, determine the preperiod and eventual period of the orbit $m,f(m),f^{\circ2}(m),\ldots$.

来源字段：https://doi.org/10.4171/owr/2016/53

12877 | AIM-DYNAMICAL_SYSTEMS-0081

### Prime-uniform p-adic PCF accumulation on flexible Lattes loci

P / partially_solved | AIM 研讨会问题汇编

Is $M_d(\mathbb Q_p)\cap PCF$ (or polynomials $P_d(\mathbb Q_p)\cap PCF$) infinite for all $p$? (maybe no for $p>d$, but what about $p<d$?) What are the accumulation points?

来源字段：http://aimpl.org/finitedynamics/2/

12878 | AMR-052-0082

## Products involving Herman rings as stable components

O / open | AMR 问题清单汇编

In the rank-two case for a polynomial diffeomorphism of $\mathbb C^2$, can an invariant stable component be a product of two Herman rings, or a product of a Herman ring and a Siegel disk?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12879 | AIM-DYNAMICAL_SYSTEMS-0097

## Quantitative causal mollification and structural obstructions for state-dependent delay

P / partially_solved | AIM 研讨会问题汇编

This is record 1 from the AIM workshop *Low dimensional structures in dynamical systems with variable time lags* (June 2010). Its four subquestions, with line wrapping repaired but mathematical signs preserved, are:

1. **Existence and smoothness of local invariant manifolds at equilibria for neutral differential equations with state-dependent delay.** A particular class is \[ x'(t)=A\bigl(x'(t-r(x(t)))\bigr)+f(x(t)), \] with a linear operator \(A\), a delay function \(r\), and a smooth nonlinearity \(f\). 2. **Extension of “lap-number” techniques to the widest possible class of equations with delay.** The PDF lists equations with non-monotonic delay, implicitly defined delay, and “multiple delay,” and suggests numerical case studies as input to analysis. 3. **How does a state-dependent delay change the dynamics in contrast to a constant delay?** Suggested test cases are scalar Wright or Mackey--Glass equations, scalar equations with two or more delays, and higher-order equations with one or more delays. 4. **Approximation of delay dependence** \[ x(t+r(x(t))) \] **by distributed/integral terms to improve smoothness properties.**

来源字段：https://aimath.org/WWN/variabletimelag/variabletimelag.pdf

12880 | AIM-DYNAMICAL_SYSTEMS-0015

## Quantitative prime support in polynomial orbits

P / partially_solved | AIM 研讨会问题汇编

What is the growth rate of $N(x)$ for PCF and non PCF maps?

Assume $f(z)\in\mathbb Q[z]$ is PCF (orbit of critical points is finite) and some reasonable starting point $\alpha\in\mathbb Q$.

The definition of $N(x)$ is omitted from this record. The January 13, 2017 capture gives the section introduction

Problem 1.1 then defines

\[ N(x)=\#\{p\le x:p\mid f^n(\alpha)\text{ for some }n\}. \]

Thus the intended setting is a polynomial $f\in\mathbb Q[z]$, a rational starting point $\alpha$, and rational primes $p$. The page does not define “reasonable,” nor specify whether $n=0$ is included. We use $n\ge1$; including $n=0$ changes the count by finitely many primes unless $\alpha=0$, itself a degenerate case.

Write each nonzero orbit value in lowest terms as

\[ f^n(\alpha)=A_n/B_n,\qquad A_n\in\mathbb Z,\quad B_n\in\mathbb Z_{>0},\quad \gcd(A_n,B_n)=1. \]

We interpret $p\mid f^n(\alpha)$ as $v_p(f^n(\alpha))>0$, equivalently $p\mid A_n$. Primes dividing coefficient or starting-point denominators may be put in a fixed exceptional set, changing $N(x)$ by $O_{f,\alpha}(1)$. A polynomial is postcritically finite (PCF) when every critical point has finite forward orbit.

The source ambiguity is mathematically important: without nondegeneracy conditions on $\alpha$, neither the PCF nor the non-PCF class has one growth rate.

来源字段：http://aimpl.org/galarithdyn/1/

12881 | AIM-DYNAMICAL_SYSTEMS-0030

### Quantitative root-degree obstructions for preimage extensions

P / partially_solved | AIM 研讨会问题汇编

“The previous section's problem appropriately phrased for preimage extensions.”

It is problem 8.1 in the section “Arboreal side” of the 2016 AIM workshop *The Galois theory of orbits in arithmetic dynamics*.

The fragment does not specify a map, a root, the tree, or the comparison group. The workshop report makes the intended arboreal reading substantially clearer. Tucker considers a number field $K$, a polynomial $f$, the inverse limit $G$ of the Galois groups of the splitting fields of $f^n(x)$, and the inverse limit $G'$ for $f^n(x)-t$ over $K(t)$; he asks, under the assumption that $0$ is not periodic, for an example in which $G$ has infinite index in $G'$. Jones's contribution to the same report asks for classes of higher-degree maps for which finite index in the preimage-tree automorphism group should fail.

Accordingly, the following is the **reconstructed question** used here.

Let $K$ be a number field, let $f\in K(z)$ have degree $d\geq2$, and let $\alpha\in\mathbb P^1(K)$. Assume that $\alpha$ is not in the postcritical set of $f$, so that the level-labelled preimage tree \[ T(f,\alpha)=\coprod_{n\geq0}f^{-n}(\alpha) \] is a regular rooted $d$-ary tree. Put \[ K_n=K\bigl(f^{-n}(\alpha)\bigr),\qquad G_n=\operatorname{Gal}(K_n/K),\qquad G_\infty=\varprojlim G_n. \] Let $E_n$ be the arithmetic generic monodromy group of $f^n(x)-t$ over $K(t)$, and $E_\infty=\varprojlim E_n$. After choosing a compatible specialization embedding, $G_\infty\subseteq E_\infty\subseteq\operatorname{Aut}(T_d)$. Classify pairs $(f,\alpha)$ for which \[ [E_\infty:G_\infty]=\infty. \tag{1.1} \] When $E_\infty=\operatorname{Aut}(T_d)$, this is the usual full-tree finite-index problem. When critical relations or symmetries force $E_\infty$ to be smaller, $E_\infty$ (or a still smaller subgroup encoding root-dependent structure) is “the thing it is supposed to” be.

There remains genuine reconstruction uncertainty. The short AIM text may have intended only polynomials rooted at $0$, as in Tucker's formulation, or the now-standard pair $(f,\alpha)$. It also does not say whether the expected group is always $\operatorname{Aut}(T_d)$ or the generic monodromy group of the fixed map. Both readings are kept explicit below.

来源字段：http://aimpl.org/galarithdyn/8/

12882 | AMR-048-0014

### Quantum unique ergodicity

P / partially_solved | AMR 问题清单汇编

Let $M$ be a compact negatively curved Riemannian manifold. Do the probability measures $|\varphi_j|^2\,d\operatorname{vol}$ associated with every orthonormal sequence of Laplace eigenfunctions equidistribute to normalized volume as the eigenvalues tend to infinity?

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Dynamical_systems

12883 | AMR-052-0010

### Quasiconformal construction of intertwinings

P / partially_solved | AMR 问题清单汇编

Can polynomial intertwinings be constructed by quasiconformal surgery?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12884 | AMR-052-0002

### Quasiconformal construction of matings

P / partially_solved | AMR 问题清单汇编

Can polynomial matings, including cases with infinite critical orbits, be constructed directly by quasiconformal cut-and-paste surgery?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12885 | AMR-052-0005

### Quasiconformal construction of tunings

O / open | AMR 问题清单汇编

Can polynomial tunings be constructed by quasiconformal surgery?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12886 | AIM-DYNAMICAL_SYSTEMS-0046

## Quenched CLTs, centering, and explicit tests for random expanding maps

P / partially_solved | AIM 研讨会问题汇编

Given an observable \(\phi:X\to\mathbb R\), can one show in the annealed dynamics that, for almost every \(P\in X\times\Omega\), the quenched system has a central limit theorem?

Its notation is

\[ T_{\beta_i}:X\longrightarrow X,\qquad i=1,2,\qquad X=[0,1], \]

not an unsubscripted product \(T\beta i\). The source says that random compositions have a central limit theorem and then asks:

This wording is verified from page 1. It is mathematically ambiguous in several ways.

- The capital \(P\in X\times\Omega\) appears to denote a point, not a probability. We reserve \(\mathbb P\) for the law on environments. - An annealed CLT is a statement under a joint law on \(X\times\Omega\). A quenched CLT fixes \(\omega\) and takes distribution only over \(x\) under a fiber measure. It is therefore normally asserted for \(\mathbb P\)-almost every environment, not for almost every fixed \(x\). - No map-selection law, stationary measure, centering, normalization, observable class, or uniform expansion/mixing hypothesis is specified. - A statement for almost every point \((x,\omega)\) would more naturally be an almost-sure CLT, which is different from an ordinary quenched CLT.

Nearby context matters. Page 2 contains Problem 14: “Quenched central limit theorem (CTL): Are the normalizing constants and variance almost surely the same?” This confirms that equality of centering and variance between annealed and quenched laws was an intended issue.

The controlled reconstruction used below is: \(\Omega=\{1,2\}^{\mathbb Z}\) has a stationary ergodic law \(\mathbb P\), \(\sigma\) is the shift, and \(T_\omega=T_{\beta_{\omega_0}}\). The one-sided iid model in the question is obtained by projection; the two-sided natural extension is used when invertibility of the base is convenient.

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12887 | AIM-DYNAMICAL_SYSTEMS-0076

## Quotient height versus arithmetic orbit height

P / partially_solved | AIM 研讨会问题汇编

**Moduli height.** Let \(X\) be a quasi-projective variety and \(G\) a group action on \(X\). Let \(Y=X//G\) be a geometric quotient of \(X\) by \(G\). Is \[ h_Y(x)\asymp \min_{g\in G}h_X(gx)? \] (ample heights)

In particular, fix an embedding \(M_d\subset\mathbb P^M\). Then define a height \(h_M:M_d\to\mathbb R\) using the embedding. Is this height function comparable to the minimum height of the maps in the class in \(\operatorname{Rat}_d\)?

来源字段：http://aimpl.org/finitedynamics/1/

12888 | OWR-11570-007

## Radial Julia Dimensions of Exponential Maps

O / open | Oberwolfach 报告问题汇编

For $f_\lambda(z)=\lambda e^z$ with hyperbolic parameter $\lambda$, determine the supremum of $\dim_H(J_r(f_\lambda))$ over all hyperbolic parameters, and determine whether the function $\lambda\mapsto\dim_H(J_r(f_\lambda))$ is increasing on $(0,1/e)$.

来源字段：https://doi.org/10.4171/owr/2012/03

12889 | AMR-046-0021

## Random linear differential equations

O / open | AMR 问题清单汇编

Let $A_0,\ldots,A_n$ be independent $N(0,1)$ random variables and let $p_n$ be the probability that the zero solution of $$A_nx^{(n)}+A_{n-1}x^{(n-1)}+\cdots+A_1x'+A_0x=0$$ is globally asymptotically stable, equivalently that every root of the characteristic polynomial has negative real part. Find the asymptotic expansion of $p_n$ as $n\to\infty$. Is $(p_n)$ strictly decreasing?

背景链接：https://arxiv.org/abs/2012.02524

12890 | AMR-045-0003

## Range of the dimension representation

O / open | AMR 问题清单汇编

Given a mixing shift of finite type $S_A$, determine the range of the dimension representation $\operatorname{Aut}(S_A)\to\operatorname{Aut}(G_A)$.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12891 | OWR-3396-004

## Rauzy Fractals from Multidimensional Continued Fractions

O / open | Oberwolfach 报告问题汇编

Given a Brun expansion of a multidimensional continued-fraction algorithm, how can one associate a Rauzy fractal to the expansion, and, assuming almost-everywhere exponential convergence to that fractal, can one prove that its subtiles are disjoint in measure and determine when they form a tiling?

来源字段：https://doi.org/10.4171/owr/2009/13

12892 | AMR-049-0003

### Reachable points lie on reachable polygons

O / open | AMR 问题清单汇编

Unfold a billiard trajectory in a regular $n$-gon by reflecting the polygon across each encountered side. A short trajectory joins two vertices and meets no vertex in between; a reachable point is an endpoint vertex occurring in such an unfolding. For an oriented short trajectory, let $\alpha$ and $\beta$ be its source and target endpoint angles as in Definition 2.1 of the source, and write $A_k$ for the resulting one of $n-2$ trajectory types (indices modulo $n-2$); in particular $A_0$ is characterized by $\beta-\alpha=2\pi/n$. Prove that every reachable point is a vertex of infinitely many reachable $n$-gons.

背景链接：https://amj.math.stonybrook.edu/pdf-Springer-final/020-0170.pdf

12893 | AMR-049-0004

### Reachable points on lines through a unitary pair

O / open | AMR 问题清单汇编

Unfold a billiard trajectory in a regular $n$-gon by reflecting the polygon across each encountered side. A short trajectory joins two vertices and meets no vertex in between; a reachable point is an endpoint vertex occurring in such an unfolding. For an oriented short trajectory, let $\alpha$ and $\beta$ be its source and target endpoint angles as in Definition 2.1 of the source, and write $A_k$ for the resulting one of $n-2$ trajectory types (indices modulo $n-2$); in particular $A_0$ is characterized by $\beta-\alpha=2\pi/n$. For $n\ge5$, call reachable points $u,v$ of type $A_0$ a unitary pair if $\det(u,v)=\sin((n-2)\pi/n)$, and put $\lambda=2\cos(\pi/n)$. Prove: (a) $u_m=u+m(\lambda+1)v$ is reachable of type $A_0$ for every integer $m$ for which it lies in the upper half-plane; (b) $w_m=u+(\lambda+m(\lambda+1))v$ is reachable of type $A_1$ whenever it lies there; (c) there are no other reachable points on $u+\mathbb Rv$; and (d) the analogous description on $v+\mathbb Ru$ yields only types $A_0$ and $A_{n-3}$.

背景链接：https://amj.math.stonybrook.edu/pdf-Springer-final/020-0170.pdf

12894 | OWR-17475-003

### Real Rational Maps with Exclusively Real Periodic Points

P / partially_solved | Oberwolfach 报告问题汇编

For $d\geq2$, let $\mathcal R_d$ be the set of real morphisms $\phi:\mathbb P^1\to\mathbb P^1$ of degree $d$ whose periodic points are all real. The existence question has an affirmative answer via Chebyshev maps; does $\mathcal R_d$ have full dimension, equivalently nonempty interior, in the space of real degree-$d$ morphisms?

来源字段：https://doi.org/10.4171/owr/2020/12

12895 | OWR-7155444-005

### Reduced Variables Preserving Dominant Dynamical Timescales

P / partially_solved | Oberwolfach 报告问题汇编

Given only a high-dimensional description of a dynamical system, how can one construct a low-dimensional collective variable that reproduces its dominant dynamical timescales?

来源字段：https://doi.org/10.4171/owr/2021/23

12896 | OWR-1112-002

### Reducibility Dichotomies for Liouville Quasiperiodic Cocycles

P / partially_solved | Oberwolfach 报告问题汇编

Let $\alpha$ be Liouville and $A$ a smooth $SL(2,\mathbb R)$-valued cocycle over rotation by $\alpha$. Is it true that for Lebesgue-almost every $\theta$, the cocycle $(\alpha,R_\theta A)$ is either nonuniformly hyperbolic or rotations-reducible? More generally, does the corresponding almost-rotations-reducibility dichotomy hold for every irrational $\alpha$?

来源字段：https://doi.org/10.4171/owr/2005/51

12897 | OWR-2488-002

### Reflectionless Potentials and Global Camassa-Holm Solutions

P / partially_solved | Oberwolfach 报告问题汇编

Is every Segal-Wilson potential for the Sturm-Liouville hierarchy reflectionless? Does the Camassa-Holm equation admit a global solution when its initial datum is derived from a reflectionless potential $a=(1,1,y)$?

来源字段：https://doi.org/10.4171/owr/2008/50

12898 | OWR-3396-005

### Regular Model Sets from Primitive Pisot Substitutions

O / open | Oberwolfach 报告问题汇编

For a primitive Pisot substitution with geometric point set $\Lambda=\bigsqcup_i\Lambda_i$, tile intervals $A_i$, and cut-and-project windows $W_i$, is each $\Lambda_i$ a regular model set with window $W_i$ up to boundary points? Equivalently, does the model-set construction tile the physical line, or does the corresponding product of intervals and windows form a fundamental domain of the cut-and-project lattice?

来源字段：https://doi.org/10.4171/owr/2009/13

12899 | OWR-12864-004

### Relations and Accuracy Among Unsteady Flow-Barrier Definitions

O / open | Oberwolfach 报告问题汇编

How are the competing definitions of unsteady flow barriers related, for example curves of extremal deformation versus transfer-operator eigenfunctions or mesohyperbolic trajectories versus extremal flux? How accurately does each definition identify barriers, which properties are essential, and must a valid definition be frame invariant?

来源字段：https://doi.org/10.4171/owr/2014/04

12900 | AIM-DYNAMICAL_SYSTEMS-0084

### Relative PCF rigidity, the necessary quotient, and a unicritical slice

P / partially_solved | AIM 研讨会问题汇编

Fix \(d,N\). \(f:\mathbb{P}^N \to \mathbb{P}^N\) is a regular polynomial endomorphism if it leaves a hyperplane invariant, call the space of them \(R_d^N\). If we restrict each \(f\) to its hyperplane it gives a projection \(\pi: R_d^N \to M_d^{N-1}\). In any given fiber of \(\pi\) is there a curve of PCF maps? (Known: Ingram-There are horizontal families, i.e. intersect every fiber in finitely many places. There are no PCF maps above the power map).

Alternate formulation: If you have a map from a curve \(C \to R_d^N \to M_d^{N-1}\) where the map \(C \to M_d^{N-1}\) is constant and \(C\) lands in the PCF locus of \(R_d^N\). Is the map \(C \to R_d^N\) already constant?

Can a nonconstant curve contained in the PCF locus lie in one fiber of \(\pi:\mathsf{RPE}_d^N\to\mathsf M_d^{N-1}\)?

来源字段：http://aimpl.org/finitedynamics/2/

12901 | OWR-11779-002

### Renormalization Basins of Quasiperiodic Schrödinger Cocycles

O / open | Oberwolfach 报告问题汇编

For a one-frequency analytic quasiperiodic Schrödinger cocycle $(\alpha,A)$ over the rotation $x\mapsto x+\alpha$, what is the basin of the renormalization attractor?

来源字段：https://doi.org/10.4171/owr/2012/17

12902 | AMR-052-0059

### Representative transcendental entire dynamics

O / open | AMR 问题清单汇编

Find a collection of representative examples of transcendental entire maps whose dynamics may serve as models for general phenomena.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12903 | OWR-1741-001

### Representativeness of Symmetry-Reduced Molecular Dynamics

O / open | Oberwolfach 报告问题汇编

How representative are the symmetry-reduced, group-generated solutions of the equations of molecular dynamics? In particular, do simulations on increasingly large fundamental domains with increasingly small corresponding subgroups approximate, in an average sense, generic solutions with suitably restricted initial conditions?

来源字段：https://doi.org/10.4171/owr/2007/58

12904 | AIM-DYNAMICAL_SYSTEMS-0091

### Residual collisions and quotient-safe Lattes exceptions

P / partially_solved | AIM 研讨会问题汇编

**Thurston Rigidity Exceptions**

What are the ``obvious'' exception to Thurston Rigidity in $M_d^N$?

来源字段：http://aimpl.org/finitedynamics/5/

12905 | OWR-5153-004

### Residual Sets Supporting Interval-Exchange Ergodic Measures

O / open | Oberwolfach 报告问题汇编

Can any residual set carry an ergodic measure of a minimal non-uniquely ergodic interval-exchange transformation?

来源字段：https://doi.org/10.4171/owr/2011/25

12906 | AMR-046-0016

### Reversible equivariant planar differential systems

S / solved | AMR 问题清单汇编

Is the period function associated with the period annulus of the origin for $$\dot z=iz+(z\bar z)^n z^{k+1},$$ where $n$ and $k$ are positive integers, monotonically decreasing?

背景链接：https://arxiv.org/abs/2012.02524

12907 | AMR-046-0015

### Reversible quadratic systems

P / partially_solved | AMR 问题清单汇编

For the family of reversible quadratic centers $$\begin{cases}\dot x=-y+xy,\\ \dot y=x+Dx^2+Fy^2,\end{cases}$$ is $2$ the maximum number of critical periods?

背景链接：https://arxiv.org/abs/2012.02524

12908 | AMR-048-0015

### Rokhlin multiple-mixing problem

O / open | AMR 问题清单汇编

Is every strongly mixing measure-preserving transformation strongly mixing of order three?

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Dynamical_systems

12909 | OWR-2094-006

### Rotation Domains and Solenoid Models in Higher-Dimensional Dynamics

O / open | Oberwolfach 报告问题汇编

Does there exist a Siegel ball whose boundary is homeomorphic to the 3-sphere? Does there exist a rotation domain analytically equivalent to the product of an annulus and a complex plane? What does the domain of convergence of the linearizing map for a two-dimensional rotation domain look like? For a complex Hénon map with connected Julia set $J$, when is the natural map from $J^-\setminus K$ to the complex solenoid one-to-one; when is $J$ a quotient of a solenoid, and which equivalence relations occur? Find a good characterization of connectedness of the Julia set for Hénon-like maps.

来源字段：https://doi.org/10.4171/owr/2008/48

12910 | OWR-12173-003

### Roth Recurrence in Groups of Intermediate Growth

O / open | Oberwolfach 报告问题汇编

Can there exist a positive-measure set $B$ in a measure-preserving action of a group of intermediate growth such that $\mu(g_1^nB\cap g_2^nB)=0$ for every $n\ne0$?

来源字段：https://doi.org/10.4171/owr/2012/50

12911 | OWR-8415336-007

### Ruelle Zeta Vanishing for Generic Negatively Curved Metrics

P / partially_solved | Oberwolfach 报告问题汇编

For a generic negatively curved metric on a closed connected oriented three-manifold $\Sigma$, does the order $m(0)$ of vanishing at zero of its Ruelle zeta function equal $4-b_1(\Sigma)$?

来源字段：https://doi.org/10.4171/owr/2021/33

12912 | OWR-13497-006

### Saddle Structure of Incorrect Formation Equilibria

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Four-Agent Formation-Control Heading [OWR-13497-001]; Inferred Stronger Four-Agent Formation Claim [OWR-13497-004]; Duplicate General Four-Agent Formation Question [OWR-13497-005]

Let four point agents $p_i\in\mathbb R^2$ obey

$$\dot p_i=\sum_{j\neq i}e_{ij}(p_j-p_i),\qquad e_{ij}=\lVert p_i-p_j\rVert^2-\bar d_{ij}^{2}.$$

When the desired shape is a general quadrilateral, or a triangle with the fourth agent in its interior, is every equilibrium with some $e_{ij}\neq0$ a saddle point?

来源字段：https://doi.org/10.4171/owr/2015/12

12913 | AMR-045-0048

### Salem beta-transformations

O / open | AMR 问题清单汇编

If $\beta$ is a Salem number, prove that the periodic points of the beta-transformation $x\mapsto\beta x\pmod1$ are exactly $\mathbb Q\cap[0,1)$.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12914 | OWR-12173-005

### Sarnak's Möbius Disjointness Conjecture

O / open | Oberwolfach 报告问题汇编

别名：Duplicate: Sarnak's Möbius Disjointness Conjecture [OWR-12173-007]

For every sequence $a_n=f(T^n x)$ generated by a zero-topological-entropy dynamical system, is $$\sum_{n\leq N}\mu(n)a_n=o(N)$$?

来源字段：https://doi.org/10.4171/owr/2012/50

12915 | AIM-DYNAMICAL_SYSTEMS-0075

### Scalar no-loss homogenization rate transfer and a multidimensional area obstruction

P / partially_solved | AIM 研讨会问题汇编

\[ \dot x_\varepsilon=a(x_\varepsilon,y_\varepsilon) +\varepsilon^{-1}b(x_\varepsilon)v(y_\varepsilon), \qquad \dot y_\varepsilon=\varepsilon^{-2}g(y_\varepsilon), \qquad \int v(y)\,d\mu(y)=0. \]

It then asks, assuming decay of multiple correlations at a specified rate for the invariant measure \(\mu\), whether one can show

\[ x_\varepsilon\longrightarrow_w X, \qquad dX=a(X)\,dt+b(X)*d\omega . \]

Here \(\longrightarrow_w\) is weak convergence; this repair is also corroborated by later papers treating exactly this fast--slow scaling. Three mathematical ambiguities remain in the printed question:

1. the topology of weak convergence is not stated; 2. the covariance of \(\omega\) and the meaning of the star integral are not stated; 3. because the original drift is \(a(x,y)\), the limiting \(a(x)\) must mean the \(\mu\)-average unless \(a\) is independent of \(y\).

The natural topology below is the uniform topology on \(C([0,T])\).

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12916 | OWR-1323-011

### Scalar-Stable Hypercyclic Operators on Nuclear Fréchet Spaces

O / open | Oberwolfach 报告问题汇编

Does every nuclear Fréchet space $E$ support a hypercyclic operator $T$ such that $\lambda T$ is hypercyclic for every nonzero scalar $\lambda$?

来源字段：https://doi.org/10.4171/owr/2006/37

12917 | OWR-14298367-003

### Shadowing Without Bounded Distortion

O / open | Oberwolfach 报告问题汇编

For dissipative composition operators without bounded distortion, is generalized hyperbolicity equivalent to the shadowing property?

来源字段：https://doi.org/10.4171/owr/2024/19

12918 | AMR-039-0006

### Sharp Lichnerowicz theorem

P / partially_solved | AMR 问题清单汇编

For the $\varepsilon$-step random walk on an $N$-dimensional Riemannian manifold, the coarse-curvature argument gives the lower spectral-gap estimate $\inf\operatorname{Ric}$, whereas the sharp Lichnerowicz estimate is $\frac{N}{N-1}\inf\operatorname{Ric}$. Can the missing directional information be incorporated, for example through reflection couplings, to recover the sharp theorem?

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12919 | OWR-1323-012

### Simultaneous Orbit Vanishing and Scalar Hypercyclicity

P / partially_solved | Oberwolfach 报告问题汇编

If $T$ is an operator on a Hilbert space with $\inf_n\lVert T^nx\rVert=0$ for every $x$, must $\inf_n(\lVert T^nu\rVert+\lVert T^nv\rVert)=0$ for all $u,v$? If both $T$ and $3T$ are hypercyclic, is $2T$ hypercyclic? Does the first condition imply that $T$ is hypercyclic?

来源字段：https://doi.org/10.4171/owr/2006/37

12920 | AMR-052-0046

### Size of the instability locus

P / partially_solved | AMR 问题清单汇编

For an analytic family $\mathcal A$ of rational maps, let $Q\subset\mathcal A$ be the $J$-unstable locus. What is the Lebesgue measure of $Q$, and is $\dim_H Q=\dim\mathcal A$?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12921 | AIM-DYNAMICAL_SYSTEMS-0060

### Slow mixing, fractional scaling, and a zero-frequency filter dichotomy

P / partially_solved | AIM 研讨会问题汇编

**Problem 17.** What new statistical properties can be found for slowly mixing systems.

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12922 | OWR-8415336-006

### Smooth Rigidity of Diophantine Step-Two Affine Maps

P / partially_solved | Oberwolfach 报告问题汇编

For a Diophantine step-two affine map $a$, does every smooth perturbation that is $C^1$-conjugate to $a$ admit a smooth conjugacy to $a$?

来源字段：https://doi.org/10.4171/owr/2021/33

12923 | AMR-052-0062

### Smoothness of exponential-family parameter hairs

P / partially_solved | AMR 问题清单汇编

Many parameters with $J(E_\lambda)=\mathbb C$ lie on parameter curves or hairs. Are these hairs $C^\infty$? Are they analytic?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12924 | AMR-045-0045

### Sparse jointly periodic points

O / open | AMR 问题清单汇编

Prove that for some $N>1$ there is a surjective one-dimensional cellular automaton $f$ with $\nu(f,S_N)<N$.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12925 | AIM-DYNAMICAL_SYSTEMS-0080

### Sparsity of the PCF locus and a quotient tangent criterion for critical portraits

P / partially_solved | AIM 研讨会问题汇编

PCF locus

1. How big is the PCF locus in \(M_d^N\)? 2. Given a “critical portrait” when is the set of PCF maps with that portrait zero dimensional? (what are the obvious counter examples?)

来源字段：http://aimpl.org/finitedynamics/2/

12926 | OWR-14297733-005

### Spatially Periodic Reaction-Diffusion Patterns and Homoclinic Entropy

O / open | Oberwolfach 报告问题汇编

For reaction-diffusion systems on $\mathbb R^m$ with $m>1$, can the complexity theorem for stationary, asymptotically spatially periodic solutions with localized defects be extended from one spatial dimension? Do symmetric homoclinic tangles involving multi-parameter families of periodic orbits always have positive topological entropy?

来源字段：https://doi.org/10.4171/owr/2023/30

12927 | OWR-14215-004

### Spectra of Koopman Operators for Linear Maps

P / partially_solved | Oberwolfach 报告问题汇编

For a matrix $A$ acting on the unit ball $U\subseteq\mathbb C^k$, how is the spectrum, and especially the point spectrum, of the Koopman operator $T_A$ on $C(U)$ determined by the spectrum of $A$?

来源字段：https://doi.org/10.4171/owr/2016/7

12928 | OWR-1323-005

### Spectral and Finite-Dimensional Problems for Hypercyclic Tuples

P / partially_solved | Oberwolfach 报告问题汇编

If an operator tuple is hypercyclic, must it commute with a cyclic operator $B$, and can $B$ satisfy $\sigma_p(B^*)=\varnothing$? Does there exist an $n$-tuple with a somewhere-dense nondense orbit, a hypercyclic $n$-tuple of matrices on $\mathbb{C}^n$, or a hypercyclic $n$-tuple of hyponormal operators on an infinite-dimensional Hilbert space?

来源字段：https://doi.org/10.4171/owr/2006/37

12929 | OWR-12864-003

### Spectral Bifurcations and Entropy Changes

O / open | Oberwolfach 报告问题汇编

Does the observed correspondence between bifurcations in the eigenspectrum of the Perron-Frobenius operator and changes in a dynamical system’s topological entropy hold beyond the studied lid-driven-cavity flows?

来源字段：https://doi.org/10.4171/owr/2014/04

12930 | AMR-045-0005

### Spectral conjecture

S / solved | AMR 问题清单汇编

Let $S\subset\mathbb R$ be a unital subring. Prove that a tuple $\Lambda=(\lambda_1,\ldots,\lambda_k)$ is the nonzero spectrum of a primitive matrix over $S$ exactly when it satisfies the Perron, Galois, and nonnegative net-trace conditions stated in the source.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12931 | OWR-1323-007

### Spectral Criteria for Supercyclic Subspaces

O / open | Oberwolfach 报告问题汇编

Are there natural classes of operators for which the spectral condition associated with the Supercyclicity Criterion is necessary for the existence of a supercyclic subspace?

来源字段：https://doi.org/10.4171/owr/2006/37

12932 | OWR-4793-001

### Spectral Thickness of Fibonacci Hamiltonians

P / partially_solved | Oberwolfach 报告问题汇编

What can careful computations reveal about the spectral structure of Fibonacci Hamiltonians? In particular, is the thickness of the spectrum a monotone decreasing function of $\lambda$ that behaves like $1/\lambda$ as $\lambda\to0$?

来源字段：https://doi.org/10.4171/owr/2011/03

12933 | OWR-4132-001

### SRB Measures for Piecewise-Affine Hyperbolic Systems

P / partially_solved | Oberwolfach 报告问题汇编

Do all piecewise-affine systems preserving complementary stable and unstable cone fields have a Sinai-Ruelle-Bowen measure, and does this hold for most such systems or for an open dense set in a natural topology?

来源字段：https://doi.org/10.4171/owr/2009/49

12934 | OWR-792-004

### Stability of Nonlinear Multicascade Systems

O / open | Oberwolfach 报告问题汇编

Establish stability criteria for general nonlinear multicascade systems.

来源字段：https://doi.org/10.4171/owr/2005/11

12935 | OWR-15995-002

### Stabilization Limits for Scalar Multiplicative Observation Channels

P / partially_solved | Oberwolfach 报告问题汇编

别名：Optimal Control with Multiplicative Measurement Noise [OWR-15995-001]

Consider the scalar system $$X_{t+1}=aX_t+U_t,\qquad Y_t=C_tX_t,$$ where the $C_t$ are independent and identically distributed continuous random variables and $U_t$ may be any causal function of $Y_0,\ldots,Y_t$. What is the optimal causal control law for second-moment stabilization, what is the largest stabilizable gain $a$, and can a tight impossibility bound be proved?

来源字段：https://doi.org/10.4171/owr/2018/10

12936 | OWR-8415361-003

### Stable Hyperbolicity of Closed Geodesics for Riemannian Metrics

O / open | Oberwolfach 报告问题汇编

For a closed manifold $M$, let $\mathcal R_{\mathrm{hyp}}(M)$ be the set of Riemannian metrics for which every closed geodesic is hyperbolic. Is the $C^2$ interior of $\mathcal R_{\mathrm{hyp}}(M)$ exactly the set of metrics whose geodesic flows are Anosov, in every dimension?

来源字段：https://doi.org/10.4171/owr/2021/56

12937 | AIM-DYNAMICAL_SYSTEMS-0050

### Stable laws and diffusion clocks for nonstationary sums

P / partially_solved | AIM 研讨会问题汇编

Problem 7. Do stable laws exist for a non-stationary process? Also, is it possible to find a diffusion rate if there are is no limiting law?

Do stable laws exist for a non-stationary process? Also, is it possible to find a diffusion rate if there is no limiting law?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12938 | OWR-7155444-006

### State Reconstruction and Reduced Coordinates from Sparse Sensors

P / partially_solved | Oberwolfach 报告问题汇编

What are the mathematical limits and possibilities of sensor placement, measurement-map discovery, time-delay methods, and multimodal data fusion for reconstructing the full state of a complex dynamical system and producing useful reduced coordinates?

来源字段：https://doi.org/10.4171/owr/2021/23

12939 | OWR-14215-003

### State-Dependent Koopman Operators for Switching Systems

O / open | Oberwolfach 报告问题汇编

Let $\mathcal M=\{A_0,A_1\}$, let $\widehat{\mathcal M}=\mathcal M/\rho(\mathcal M)$, and suppose $\widehat{\mathcal M}^{*}$ is bounded. Define $$ f(x)=\begin{cases} A_0x,&\sup_{A\in\widehat{\mathcal M}^{*}}\lVert Ax\rVert_2=\sup_{A\in\widehat{\mathcal M}^{*}}\lVert AA_0x\rVert_2,\\ A_1x,&\text{otherwise}. \end{cases} $$ Is the Koopman operator induced by $f$ an appropriate operator for the switching system, and what information about the system does it encode?

来源字段：https://doi.org/10.4171/owr/2016/7

12940 | AMR-045-0010

### Stochastic zeta functions

O / open | AMR 问题清单汇编

Characterize the functions that occur as stochastic zeta functions of mixing Markov shifts.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12941 | OWR-2094-002

### Structural Problems in One-Dimensional Complex Dynamics

P / partially_solved | Oberwolfach 报告问题汇编

Find a combinatorial description of Branner-Hubbard slices of cubic polynomials; determine whether every Cantor Julia set of a rational map has zero spherical area; decide whether the Brjuno condition is optimal for rational maps, entire transcendental maps, and relatively compact maximal Siegel disks; decide whether $\operatorname{area}(K_\lambda)$ for $P_\lambda(z)=\lambda z+z^2$ is bounded below by a constant $C>0$ when $|\lambda|\leq1$; determine whether every Siegel-disk boundary of a polynomial or rational map is a Jordan curve; determine whether Zakeri sets are Jordan arcs for cubic polynomials and for quadratic rational or more general bicritical maps, and quasi-arcs for bounded-type rotation numbers; decide whether generalized tuning can replace the closure of a $z\mapsto z^d$ basin of a rational map $R$ by the filled Julia set of any degree-$d$ polynomial $P$, and whether local connectivity transfers; determine whether almost every bifurcation-locus parameter in harmonic measure has a critical orbit dense in its Julia set; characterize when a topological mating is a sphere and a faithful model, and whether formal matings with arbitrarily long ray connections exist; determine whether the closure of the cubic rigidity locus is the Misiurewicz locus; decide whether the cubic connectedness loci satisfy $\mathcal{C}_3=\mathcal{C}_3^\circ$; and determine whether every wandering connected component of a rational map eventually becomes a point or a topological circle.

来源字段：https://doi.org/10.4171/owr/2008/48

12942 | AMR-039-0011

### Sturm-Lott-Villani definition

P / partially_solved | AMR 问题清单汇编

What is the relationship, if any, between Ollivier coarse Ricci curvature and the Sturm-Lott-Villani displacement-convexity notion, including its $CD(K,N)$ form, especially for discrete spaces?

背景链接：https://www.yann-ollivier.org/rech/publs/problems_curvmarkov.pdf

12943 | AMR-045-0047

### Surjunctive groups

P / partially_solved | AMR 问题清单汇编

Characterize the countable groups $G$ for which every injective $G$-equivariant cellular automaton on $A^G$, for every finite alphabet $A$, is surjective.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12944 | OWR-12183-001

### Switching Ratios in Discretized Hysteresis Dynamics

P / partially_solved | Oberwolfach 报告问题汇编

For the discretized hysteresis equation $\dot u_n=(u_{n-1}-2u_n+u_{n+1})/h^2+\mathcal H(u_n)$ with $u_n(0)=-c(hn)^2$, $c>0$, and thresholds $H_1>0>H_{-1}$, is $\#\{\text{switched points}\}/\#\{\text{nonswitched points}\}\approx|H_1/H_{-1}|$?

来源字段：https://doi.org/10.4171/owr/2012/60

12945 | AIM-DYNAMICAL_SYSTEMS-0025

### Synchronous critical collisions force parity deficits, while trailing relations need not

P / partially_solved | AIM 研讨会问题汇编

What is the influence of critical relations on the image of Galois over a number field (Pink condition)? More specifically which sorts of critical orbit relations force infinite index for the image of Galois?

来源字段：http://aimpl.org/galarithdyn/6/

12946 | AMR-046-0002

### Systems with homogeneous components I

O / open | AMR 问题清单汇编

Is $(n+m)/2$ the maximum number of limit cycles of $$\dot x=P_n(x,y),\qquad \dot y=Q_m(x,y),$$ where $n\neq m$ and $P_n,Q_m$ are homogeneous polynomials of odd degrees $n,m$, respectively?

背景链接：https://arxiv.org/abs/2012.02524

12947 | AMR-046-0003

### Systems with homogeneous components II

P / partially_solved | AMR 问题清单汇编

(i) For the cubic family $$\begin{cases}\dot x=ax+by,\\ \dot y=cx^3+dx^2y+exy^2+fy^3,\end{cases}$$ is $2$ the maximum number of limit cycles? (ii) If it is true, give a simple proof of uniqueness of the limit cycle for $$\begin{cases}\dot x=y,\\ \dot y=-x^3+dx^2y+y^3.\end{cases}$$

背景链接：https://arxiv.org/abs/2012.02524

12948 | OWR-15951-005

### Tame Dynamics of Nonamenable Group Actions

O / open | Oberwolfach 报告问题汇编

What happens for tame minimal dynamical systems when the acting group is not assumed to be commutative or even amenable?

来源字段：https://doi.org/10.4171/owr/2017/46

12949 | AMR-052-0024

### Taylor-coefficient regularity of a Siegel conjugacy

O / open | AMR 问题清单汇编

For $P_\rho'(z)=\lambda(1-z)^\rho$, $P_\rho(0)=0$, let $h$ linearize the Siegel disk and write $h'(\zeta)/(1-h(\zeta))=\sum_{\nu\ge0}a_\nu\zeta^\nu$. Make rigorous the observed approximation of these coefficients by those of the simplified equation, at least for small $\rho$; in particular for $\theta=(\sqrt5-1)/2$ and $\rho=1$, prove a uniform bound such as $|a_\nu-2/3|<0.1$.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12950 | AMR-048-0018

### Termination of juggler sequences

O / open | AMR 问题清单汇编

Starting from a positive integer $a_0$, define $a_{n+1}=\lfloor a_n^{1/2}\rfloor$ when $a_n$ is even and $a_{n+1}=\lfloor a_n^{3/2}\rfloor$ when $a_n$ is odd. Does every such juggler sequence eventually reach $1$?

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Dynamical_systems

12951 | AMR-041-0010

### The good, the bad, and the ugly

O / open | AMR 问题清单汇编

For a Hamiltonian system, call the good set the maximal invariant subset on which the invariant Liouville measure is almost periodic; call the bad set the Pesin set on which the invariant measure has a positive Lyapunov exponent; and call the ugly set the complement of their union. Is there an example in which the ugly set has positive measure?

背景链接：https://people.math.harvard.edu/~knill/seminars/intr/

12952 | OWR-12582-002

### The Hofer-Zehnder Periodic-Orbit Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Hofer-Zehnder conjecture: A Hamiltonian diffeomorphism with more than the necessary number of fixed points-in particular, more than $n+1$ on $\mathbb{CP}^n$ in the expected case-has infinitely many periodic orbits.

来源字段：https://doi.org/10.4171/owr/2013/34

12953 | DYN-002

### The Painlevé Conjecture

S / solved | 独立 / 未分配集合

In the $n$-body problem with $n \geq 4$, can non-collision singularities occur in finite time?

背景链接：https://archive.ymsc.tsinghua.edu.cn/pacm_paperurl/20220228202444113958904

12954 | AIM-DYNAMICAL_SYSTEMS-0013

### The solved nucleus criterion for simplicity

S / solved | AIM 研讨会问题汇编

For a faithful contracting self-similar action $(G,X)$ on a finite alphabet, characterize simplicity of the universal Nekrashevych algebra $\mathcal O(G,X)$.

来源字段：http://aimpl.org/groupdynamorigin/4/

12955 | AIM-DYNAMICAL_SYSTEMS-0024

### The surviving arboreal tree at a degree-drop boundary

P / partially_solved | AIM 研讨会问题汇编

1. (Want to look at a curve in \(M_2\)). For \[ f(z)=\frac{z^2-1}{z^2-c}, \] there is a collection of PCF maps which approach \(c=1\) on the curve. Compare the arboreal representation for \(f\) to the representation of the limit map. 2. More generally, how do these representations behave as you deform to the boundary?

来源字段：http://aimpl.org/galarithdyn/5/

12956 | OWR-12582-009

### The Tits Alternative for Analytic Surface Symplectomorphisms

O / open | Oberwolfach 报告问题汇编

Tits-alternative conjecture: For every compact surface $M$, every finitely generated subgroup of the real-analytic area-preserving diffeomorphism group $\operatorname{Symp}^\omega_\mu(M)$ is either virtually solvable or contains a nonabelian free group.

来源字段：https://doi.org/10.4171/owr/2013/34

12957 | DYN-001

### The Weinstein Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Weinstein Conjecture [DYN-005]; Weinstein Conjecture for Contact-Type Energy Hypersurfaces [OWR-1052-001]

Does every Reeb vector field on a closed contact manifold have at least one periodic orbit?

背景链接：https://arxiv.org/abs/math/0611007

12958 | AMR-052-0076

### Thurston algorithm for power-law lift families

P / partially_solved | AMR 问题清单汇编

For the lift family $x\mapsto k-k|2x-1|^\alpha$, $\alpha>1$, does the real Thurston algorithm converge whenever the initial interval map has a periodic or preperiodic kneading sequence?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12959 | AMR-052-0020

### Thurston's algorithm without critical finiteness

P / partially_solved | AMR 问题清单汇编

Starting with an orientation-preserving branched covering $f_0:S^2\to S^2$ and three marked base points, iteratively conjugate it as in Thurston's pullback algorithm to obtain rational maps $r_n$ and coordinate maps $\phi_n$. Under what conditions does $r_n$ converge uniformly to a rational map $r_\infty$, and under what conditions and on what subset of $S^2$ does $\phi_n$ converge uniformly?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12960 | OWR-14299512-001

### Topological Invariance of Helicity

P / partially_solved | Oberwolfach 报告问题汇编

Is helicity invariant under topological conjugacy of exact volume-preserving vector fields?

来源字段：https://doi.org/10.4171/owr/2025/31

12961 | AMR-052-0027

### Topological model for a Cremer Julia set

O / open | AMR 问题清单汇编

Give a plausible topological model for the Julia set of a Cremer polynomial.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12962 | OWR-5153-007

### Topological Types of Linear-Approach Spaces

O / open | Oberwolfach 报告问题汇编

For a translation surface $X$ and a point $x$ in its metric completion, define $L(x)$ as the germs of unit-speed geodesic trajectories in $X$ approaching $x$. Which topological spaces can arise as $L(x)$?

来源字段：https://doi.org/10.4171/owr/2011/25

12963 | OWR-15951-001

### Topology and Symmetries of Aperiodic Tiling Spaces

O / open | Oberwolfach 报告问题汇编

How can aperiodic tilings be reconstructed as limits of growing coronae? Which homeomorphisms can occur between tiling spaces, how many such homeomorphisms can there be, and which extended symmetries can subshifts possess?

来源字段：https://doi.org/10.4171/owr/2017/46

12964 | AMR-052-0090

### Topology of hyperbolic attractors in dimension three

P / partially_solved | AMR 问题清单汇编

Let $A$ be a hyperbolic attractor of a diffeomorphism of a compact $3$-manifold. Beyond the known Anosov, laminated, Williams, and invariant-torus cases, can another topology occur? In particular, can the transversal structure of the unstable lamination be a Sierpiński carpet?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12965 | OWR-2053-022

### Torus Realizations of Adding Machines and Bernoulli Shifts

O / open | Oberwolfach 报告问题汇编

Can the adding machine and the Bernoulli shift each be realized on the two-torus $T^2$ as a minimal uniquely ergodic topological system $(T^2,\lambda,h)$, where $\lambda$ is an $h$-invariant ergodic measure of full support, in the sense of measurable conjugacy?

来源字段：https://doi.org/10.4171/owr/2008/42

12966 | OWR-15993-008

### Trait Selection in Fluctuating Environments

P / partially_solved | Oberwolfach 报告问题汇编

For phenotypically structured populations in a time-periodic or fluctuating environment, determine survival or extinction, population-size dynamics, trait selection, and the effect of environmental variation on the phenotype distribution.

来源字段：https://doi.org/10.4171/owr/2018/8

12967 | OWR-15430-010

### Trajectory Geometry Near Analytic Vector-Field Singularities

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a real analytic vector field with a singular point at $0\in\mathbb R^n$. What qualitative dynamics near $0$ can be described from geometric properties of trajectories converging to $0$, including their finiteness properties with respect to analytic sets and their behavior relative to neighboring trajectories?

来源字段：https://doi.org/10.4171/owr/2017/22

12968 | OWR-737-002

### Trajectory Trichotomy for Linear Vector Fields

O / open | Oberwolfach 报告问题汇编

Let $M$ be a finite set of real linear vector fields. There is a finite set $W\subseteq(\mathbb{R}\setminus\mathbb{Q})\cup i(\mathbb{R}\setminus\{0\})$ such that, for every family $G$ of trajectories of members of $M$, some $W'\subseteq W$ makes $G$ interdefinable over the real field with at least one of $e^x$, $\mathbb{Z}$, or $(x^w)_{w\in W'}$, where $x^{a+ib}$ denotes $t\mapsto t^a(\cos(b\log t),\sin(b\log t))$ on $(0,\infty)$. Are these three alternatives a trichotomy?

来源字段：https://doi.org/10.4171/owr/2004/35

12969 | OWR-12977-006

### Transitions and Transport Among Cancer-Cell Migration Strategies

P / partially_solved | Oberwolfach 报告问题汇编

For amoeboid, mesenchymal, and collective cancer-cell movement, which strategy is advantageous for invasion; what triggers transitions between strategies; what dynamics occurs when all are present; and what are their transport properties such as invasion speeds?

来源字段：https://doi.org/10.4171/owr/2014/19

12970 | AIM-DYNAMICAL_SYSTEMS-0045

### Transport and equipartition in a two-mass toral billiard

P / partially_solved | AIM 研讨会问题汇编

Given a Saini billard on a torus with a finite horizon and two masses \(m_1\) and \(m_2\), is there a way to find the system's viscosity, i.e. transfer of momentum. Additionally, is there a way to define heat in this model and can one check the Einstein relation? Lastly, if \(m_1\ne m_2\) then, in the limit, do the masses have the same kinetic energy?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12971 | OWR-17127-010

### Transversal Foliations and Global Sections for Reeb Flows

P / partially_solved | Oberwolfach 报告问题汇编

If a Reeb flow on a closed $3$-manifold admits a transverse foliation and has zero topological entropy, must it admit a global surface of section after removing finitely many invariant tori?

来源字段：https://doi.org/10.4171/owr/2019/31

12972 | AMR-046-0022

### Triangular billiards

P / partially_solved | AMR 问题清单汇编

Does every triangular billiard have a periodic trajectory?

背景链接：https://arxiv.org/abs/2012.02524

12973 | AMR-046-0006

### Trigonometric Abel differential equations I

S / solved | AMR 问题清单汇编

For $$\frac{dx}{dt}=(a_0+a_1\sin t+a_2\cos t)x^3+(b_0+b_1\sin t+b_2\cos t)x^2,$$ is $3$ the maximum number of $2\pi$-periodic limit cycles?

背景链接：https://arxiv.org/abs/2012.02524

12974 | AMR-046-0007

### Trigonometric Abel differential equations II

O / open | AMR 问题清单汇编

Given integers $p>q\geq2$ and $m,n\in\mathbb{N}$, find the maximum number of $2\pi$-periodic limit cycles of $$\frac{dx}{dt}=A_m(t)x^p+B_n(t)x^q,$$ where $A_m$ and $B_n$ are $2\pi$-trigonometric polynomials of degrees $m$ and $n$.

背景链接：https://arxiv.org/abs/2012.02524

12975 | OWR-13497-011

### Trigonometric-Polynomial Feedback for Nonlinear Stabilization

O / open | Oberwolfach 报告问题汇编

Let $\dot x=f(x,u)$ be smooth near $(x,u)=(0,0)$ with $f(0,0)=0$. Assume the Lie algebra rank condition holds at the origin, the origin is locally continuously reachable in arbitrarily small time, and the state dimension is not $2$ or $3$. Can one construct a stabilizing feedback $u=k(x,t)$ that is a trigonometric polynomial in $t$ with coefficients depending on $x$?

来源字段：https://doi.org/10.4171/owr/2015/12

12976 | OWR-13497-008

### Turnpike Properties as Criteria for Strict Dissipativity

P / partially_solved | Oberwolfach 报告问题汇编

For the discrete-time optimal-control problem $x(k+1)=f(x(k),u(k))$ with stage cost $\ell(x,u)$, under which assumptions does the turnpike property imply strict dissipativity? Here strict dissipativity means that for some equilibrium $(x_e,u_e)$, storage function $\lambda$, and $\rho\in\mathcal K_\infty$, one has $\ell(x,u)-\ell(x_e,u_e)+\lambda(x)-\lambda(f(x,u))\geq\rho(\lVert x-x_e\rVert)$. The known implication is strict dissipativity $\Rightarrow$ turnpike property.

来源字段：https://doi.org/10.4171/owr/2015/12

12977 | AIM-DYNAMICAL_SYSTEMS-0032

### Two square-class tests for finite-level agreement of quadratic arboreal representations

P / partially_solved | AIM 研讨会问题汇编

To what level can the representation associated to two quadratic polynomials agree?

来源字段：http://aimpl.org/galarithdyn/9/

12978 | AMR-049-0005

### Types of parallel short trajectories

O / open | AMR 问题清单汇编

Unfold a billiard trajectory in a regular $n$-gon by reflecting the polygon across each encountered side. A short trajectory joins two vertices and meets no vertex in between; a reachable point is an endpoint vertex occurring in such an unfolding. For an oriented short trajectory, let $\alpha$ and $\beta$ be its source and target endpoint angles as in Definition 2.1 of the source, and write $A_k$ for the resulting one of $n-2$ trajectory types (indices modulo $n-2$); in particular $A_0$ is characterized by $\beta-\alpha=2\pi/n$. If a short trajectory emanating from $O$ with slope angle $\alpha$ has type $A_k$, prove that the parallel short trajectory from $O$ with slope angle $\ell\pi/n+\varepsilon\alpha$, where $\varepsilon\in\{-1,1\}$, has type $A_{\varepsilon k-\ell}$. When $n$ is even, restrict $\ell$ to even integers.

背景链接：https://amj.math.stonybrook.edu/pdf-Springer-final/020-0170.pdf

12979 | AMR-049-0002

### Types of vertices of reachable polygons

O / open | AMR 问题清单汇编

Unfold a billiard trajectory in a regular $n$-gon by reflecting the polygon across each encountered side. A short trajectory joins two vertices and meets no vertex in between; a reachable point is an endpoint vertex occurring in such an unfolding. For an oriented short trajectory, let $\alpha$ and $\beta$ be its source and target endpoint angles as in Definition 2.1 of the source, and write $A_k$ for the resulting one of $n-2$ trajectory types (indices modulo $n-2$); in particular $A_0$ is characterized by $\beta-\alpha=2\pi/n$. Call an $n$-gon reachable if it is an $SL(2,\mathbb R)$ image of the original regular $n$-gon and all its vertices except the distinguished vertex $O$ are reachable points. If $n\ge5$ and its clockwise vertices are $v_0=O,v_1,\ldots,v_{n-1}$, prove that $v_1,\ldots,v_{n-1}$ have types $A_0,A_1,\ldots,A_{n-3},A_0$, respectively.

背景链接：https://amj.math.stonybrook.edu/pdf-Springer-final/020-0170.pdf

12980 | AMR-052-0055

### Unbounded Jacobian cocycles and singularity

O / open | AMR 问题清单汇编

For which positive-entropy invariant measures $m$ does failure of uniform $L^2(m)$ boundedness of the sums of $\log\operatorname{Jac}_m f-\kappa\log|f'|$, where $\kappa=\dim_Hm$, imply $m\perp H^\kappa$ and $\dim_Hm<\dim_H\overline\Lambda$?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12981 | AMR-048-0013

### Unbounded outer-billiard orbits for almost every polygon

P / partially_solved | AMR 问题清单汇编

Prove that the outer billiard about almost every convex polygon has an unbounded orbit.

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Dynamical_systems

12982 | AMR-052-0071

### Uniform access to roots for relaxed Newton maps

P / partially_solved | AMR 问题清单汇编

Let all roots of a degree-$d$ polynomial $f$ lie in the unit disk, let $\alpha$ be a root of multiplicity $m$, and let $A^*_{h}(\alpha)$ be its immediate basin for $N_{h,f}$. Prove that $\bigcap_{0<h\le m}A^*_{h}(\alpha)$ meets every circle of radius $R\ge3$ in arcs of total length at least $2\pi R/(cd)$ for an absolute constant $c$.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12983 | AMR-052-0068

### Uniform convergence to an irrationally indifferent fixed point

P / partially_solved | AMR 问题清单汇编

Let $\varphi$ be a holomorphic germ fixing $z_0$ with multiplier $e^{2\pi i\alpha}$ for irrational $\alpha$. Can $\varphi^n(z)\to z_0$ uniformly on some domain?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12984 | AMR-052-0021

### Uniform geometry in complex renormalization

O / open | AMR 问题清单汇编

Let $f_i(z)=z^2+c_i$ range over finitely many critically periodic quadratic polynomials, let $g_n$ be the iterated tuning $f_1\vdash\cdots\vdash f_n$, and write $n_k=\prod_{i\le k}m_i$ for the products of critical periods. Prove that every set $\{g_n^{n_k\ell+i}(0):0\le\ell<m_{k+1}\}$ has geometry bounded uniformly for all $i\le n_k$, $k<n$, and $n$.

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

12985 | OWR-3396-010

### Uniform Recurrence Rates for Minimal Interval Exchanges

O / open | Oberwolfach 报告问题汇编

For a minimal interval-exchange transformation $T$ on $J=\{x:0\leq x<1\}$, define $\phi_1(x,y)=\liminf_{n\to\infty}n\lvert T^n(x)-y\rvert$. Can $\phi_1(x,y)=0$ hold for every $x,y\in J$, rather than only for almost every pair?

来源字段：https://doi.org/10.4171/owr/2009/13

12986 | OWR-1112-003

### Uniformity of $SL(2,\mathbb R)$ Cocycles over Minimal Subshifts

P / partially_solved | Oberwolfach 报告问题汇编

Does uniformity of every locally constant $SL(2,\mathbb R)$ cocycle hold for every uniquely ergodic minimal subshift? Does every uniquely ergodic minimal dynamical system admit a nonuniform continuous $SL(2,\mathbb R)$ cocycle, or a nonuniform continuous Schrödinger cocycle?

来源字段：https://doi.org/10.4171/owr/2005/51

12987 | OWR-2053-025

### Unique Integrability of Center Bundles

O / open | Oberwolfach 报告问题汇编

For a partially hyperbolic diffeomorphism with splitting $TM=E^s\oplus E^c\oplus E^u$, where $E^s$ and $E^u$ are uniquely integrable to $C^0$ foliations with smooth leaves, is the center bundle $E^c$ always uniquely integrable?

来源字段：https://doi.org/10.4171/owr/2008/42

12988 | OWR-8415336-004

### Uniqueness of Fixed-Bounce Geodesic Loops in Billiards

P / partially_solved | Oberwolfach 报告问题汇编

For a billiard domain $\Omega$ and a boundary point $x$, is there a unique geodesic loop based at $x$ with winding number $1$ and $q$ reflections, so that the loop-length function $\psi_q$ is well defined?

来源字段：https://doi.org/10.4171/owr/2021/33

12989 | OWR-2094-007

### Uniqueness of Maximal Entropy Measures for Surface Maps

O / open | Oberwolfach 报告问题汇编

For a meromorphic self-map of a projective compact complex surface with small topological degree, is the invariant measure $\mu$ constructed under finite dynamical energy the unique measure of maximal entropy?

来源字段：https://doi.org/10.4171/owr/2008/48

12990 | AIM-DYNAMICAL_SYSTEMS-0012

### Universal bounds and an exact uniformly recurrent family

P / partially_solved | AIM 研讨会问题汇编

**Growths functions of repetitive graphs.** A graph $\Gamma$ is called **repetitive** if for all $r>0$ there exists $d>0$ such that for all $x,y\in\Gamma$, there exists an isomorphic copy of $B_r(x)$ (the ball of radius $r$ centered at $x$) at distance less or equal to $d$ from $y$.

Consider two kind of growths functions for $\Gamma$: pick $x_0\in\Gamma$ and define $f_{x_0}(n)=\#B_n(x_0)$ and $f(x)=\max\{f_x(n):x\in\Gamma\}$.

What functions can be realized as growths functions of repetitive graphs?

来源字段：http://aimpl.org/groupdynamorigin/4/

12991 | OWR-1460-007

### Universal Holomorphic Functions under Composition Operators

P / partially_solved | Oberwolfach 报告问题汇编

Let $\Omega\subseteq\mathbb{C}$ be a domain, let $X$ be either $H(\Omega)$, the Fréchet space of holomorphic functions, or its unit ball, and let $(\varphi_n)$ be a sequence of holomorphic self-maps of $\Omega$. Call $f$ $X$-universal if $\{f\circ\varphi_n:n\geq1\}$ is locally uniformly dense in $X$. Give necessary and sufficient conditions on $(\varphi_n)$ for $X$-universal functions to exist on an arbitrary domain. In particular, does an annulus admit $X$-universal functions?

来源字段：https://doi.org/10.4171/owr/2007/09

12992 | OWR-12007-010

### Universal Rotation-Zero Words and Analytic Extremizers

O / open | Oberwolfach 报告问题汇编

Does there exist a nontrivial word $w$ in the rank-two free group such that the lifted rotation number of $\rho(w)$ is zero for every representation $\rho$ into the universal cover of $\operatorname{Homeo}^+(S^1)$? For arbitrary $w$, does there exist a real-analytic representation $\rho$ with $\operatorname{rot}(\rho(w))=R(w,r,s)$?

来源字段：https://doi.org/10.4171/owr/2012/35

12993 | OWR-13348-002

### Unstable Invariance of Multiple-Correlation Limits

O / open | Oberwolfach 报告问题汇编

Are all weak-$L^2$ accumulation points of the multiple-correlation sequence $\varphi_1\circ f^{n_1}\,\varphi_2\circ f^{n_1+n_2}\cdots\varphi_k\circ f^{n_1+\cdots+n_k}$, as all $n_i\to\infty$, also $W^u$-invariant, given that they are $W^s$-invariant?

来源字段：https://doi.org/10.4171/owr/2014/46

12994 | OWR-1323-006

### Vanishing Subsequences for Unbounded Operators

O / open | Oberwolfach 报告问题汇编

Can an unbounded operator $T$ have the property that, for every infinite-dimensional closed subspace $E_0$ and every sequence $(n_k)$, some $x_0\in E_0$ satisfies $T^{n_k}x_0\to0$?

来源字段：https://doi.org/10.4171/owr/2006/37

12995 | AIM-DYNAMICAL_SYSTEMS-0048

### Variable-block coupling bounds for nonstationary statistics

P / partially_solved | AIM 研讨会问题汇编

Problem 5. For a time dependent non-stationary process, is it possible to use coupling to study the statistical properties of this process?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

12996 | AMR-045-0036

### Virtual FOG conjecture

O / open | AMR 问题清单汇编

For a mixing SFT $S$, let $\operatorname{Aut}_0(S)$ be the inert automorphisms and $F_0(S)$ its finite-order-generated subgroup. Prove that $\operatorname{Aut}_0(S)/F_0(S)$ is finite.

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

12997 | OWR-4137-012

### Wandering Domains of Polynomial-Type Quasiregular Maps

P / partially_solved | Oberwolfach 报告问题汇编

If $p:\mathbb R^2\to\mathbb R^2$ is $K$-quasiregular of polynomial type with $K<\deg p$, can $p$ have a wandering Fatou component?

来源字段：https://doi.org/10.4171/owr/2009/54

12998 | OWR-2094-008

### Wandering Domains, Hyperbolicity, and Thurston Algorithms

O / open | Oberwolfach 报告问题汇编

For a transcendental entire function $f$ with wandering domain $U$, can the limit functions of $\{f^n|_U\}$ be bounded? If its postsingular set is bounded, can $f$ have wandering domains, including when all critical points are nonrecurrent? For $f_{\lambda,k}(z)=\lambda ze^z+k$ with $\lambda=e^{2\pi i\theta}$ and the required non-recurrence hypotheses, can $f_{\lambda,k}$ have a Siegel disk? Is the complement of the hyperbolicity locus precompact in $\operatorname{Rat}_d$ modulo Möbius conjugacy? For $f_\lambda(z)=\lambda e^z$, characterize the parameters for which the nonescaping set, or that set together with $\{\infty\}$, is connected. Can Thurston's algorithm be implemented in $O((\deg(f)+|\Omega_f|)^n)$ iterations, producing either a point sufficiently close to a fixed point of the pullback map or a Thurston obstruction?

来源字段：https://doi.org/10.4171/owr/2008/48

12999 | AMR-052-0079

### Wandering stable components for complex Hénon maps

P / partially_solved | AMR 问题清单汇编

Let $f$ be a polynomial diffeomorphism of $\mathbb C^2$ with Jacobian determinant $\delta$, let $U$ be a component of the interior of the bounded-forward-orbit set, and let a subsequence of iterates converge on $U$ to $g$. Can $U$ wander? If so, can $g$ have rank zero or one, and can $U$ be bounded, unbounded of finite volume, or unbounded of infinite volume?

背景链接：https://www.math.stonybrook.edu/open-problems-dynamical-systems

13000 | OWR-14297734-001

### Wave Amplification of Upper Ocean Currents

O / open | Oberwolfach 报告问题汇编

Can waves amplify upper-ocean currents?

来源字段：https://doi.org/10.4171/owr/2023/31

13001 | OWR-11101923-001

### Wave-Driven Surface Ocean Currents

O / open | Oberwolfach 报告问题汇编

How does wave-field evolution drive surface ocean currents?

来源字段：https://doi.org/10.4171/owr/2022/42

13002 | AMR-045-0026

### Weak algebraic equivalence

O / open | AMR 问题清单汇编

Is every nonabelian $\mathbb Z^d$ group shift weakly algebraically equivalent to a Bernoulli group shift?

背景链接：https://www.math.umd.edu/~mboyle/papers/openfinalsub3nov2007.pdf

13003 | AIM-DYNAMICAL_SYSTEMS-0093

### Weak-Perron entropy exponents and reachable-component edge growth

P / partially_solved | AIM 研讨会问题汇编

**Entropy**

1. Let \(f(z)=z^2+c\), \(c\in\mathbb R\) and PCF with invariant \([a,b]\) with fixed point at one end. Which algebraic numbers arise as \(e^s\), where \(s\) is the entropy of \(f\)? 2. More generally, for \(z^2+c\) on its Hubbard tree (or higher degree). 3. Is there an algebraic analogue of entropy (like dynamical degree vs. arithmetic degree in higher dimensions)?

来源字段：http://aimpl.org/finitedynamics/6/

13004 | AIM-DYNAMICAL_SYSTEMS-0057

### What is deterministic in a quenched CLT?

P / partially_solved | AIM 研讨会问题汇编

**Problem 14.** Quenched central limit theorem (CTL): Are the normalizing con-stants and variance almost surely the same?

来源字段：https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

13005 | OWR-12593-004

### Zero Entropy Measures for Higher Rank Toral Actions

P / partially_solved | Oberwolfach 报告问题汇编

Can invariant measures for higher-rank toral actions, including those in the Furstenberg-Rudolph setting, be partially classified under a hypothesis weaker than positive entropy, particularly in the zero-entropy case?

来源字段：https://doi.org/10.4171/owr/2013/45

13006 | OWR-11126-009

### Zero Entropy Systems as Limits of Integrable Systems

O / open | Oberwolfach 报告问题汇编

In low dimensions, are all conservative dynamical systems with zero topological entropy limits of integrable systems; in particular, can the rational-rotation approximations of every smooth area-preserving disk pseudo-rotation be chosen to preserve the standard area form?

来源字段：https://doi.org/10.4171/owr/2011/34

13007 | OWR-17127-005

### Zero-Entropy Surface Dynamics and Their Bifurcations

O / open | Oberwolfach 报告问题汇编

Classify surface diffeomorphisms with zero topological entropy and describe the mechanisms by which they bifurcate to positive-entropy dynamics.

来源字段：https://doi.org/10.4171/owr/2019/31

# 资料汇编 10 | 组合数学 / Combinatorics

1,032 条记录。O=开放；P=部分解决；S=已解决。状态与英文/LaTeX 表述均沿用来源。

13008 | AMR-030-0045

### "1/3 - 2/3 Conjecture" For every poset that is not a chain, there is some pair of elements x and y so that x appears abo

P / partially_solved | AMR 问题清单汇编

Kislitsyn: "1/3 - 2/3 Conjecture" For every poset that is not a chain, there is some pair of elements x and y so that x appears above y in a random linear extension of the poset at least 1/3 of the time and at most 2/3 of the time. See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13009 | AMR-030-0021

### "Seymour's Second Neighborhood Conjecture" Any oriented graph has a vertex whose outdegree is at most its second outdegr

P / partially_solved | AMR 问题清单汇编

Seymour: "Seymour's Second Neighborhood Conjecture" Any oriented graph has a vertex whose outdegree is at most its second outdegree (vertices at directed distance 2). See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13010 | AMR-054-0007

### $k$-sets

O / open | AMR 问题清单汇编

What is the maximum number of $k$-sets? (Equivalently, what is the maximum complexity of a $k$-level in an arrangement of hyperplanes?)

背景链接：https://topp.openproblem.net/p7

13011 | OWR-16775-007

### $q$-Analogues of Quadrilaterals

O / open | Oberwolfach 报告问题汇编

What should be regarded as a $q$-analogue of a quadrilateral?

来源字段：https://doi.org/10.4171/owr/2019/13

13012 | AMR-030-0043

### ("Diamond-Free Posets Problem") What is the size of the largest subset of the Boolean lattice B_(n) which includes no B_

P / partially_solved | AMR 问题清单汇编

Griggs, Lu: ("Diamond-Free Posets Problem") What is the size of the largest subset of the Boolean lattice B_(n) which includes no B_(2) as a subposet? (What is the maximum number of subsets of [n] so that there are no A, B, C, D with A < B, A < C, B < D, and C < D?)

背景链接：https://people.math.sc.edu/cooper/combprob.html

13013 | COMB-001

### 1/3-2/3 Conjecture

P / partially_solved | 独立 / 未分配集合

别名：1/3-2/3 Conjecture [COMB-018]

Does every non-totally-ordered finite poset have two elements with probability between 1/3 and 2/3 in random linear extensions?

背景链接：https://epubs.siam.org/doi/10.1137/0405037

13014 | OPG-1797

### 2-accessibility of primes

P / partially_solved | Open Problem Garden

Question Is the set of prime numbers 2-accessible?

背景链接：http://www.openproblemgarden.org/op/2_accessibility_of_primes

13015 | OPG-1825

### 3-accessibility of Fibonacci numbers

P / partially_solved | Open Problem Garden

Question Is the set of Fibonacci numbers 3-accessible?

背景链接：http://www.openproblemgarden.org/op/3_accessibility_of_fibonacci_numbers

13016 | GREEN-013

### 4-term APs in Fourier Uniform Sets

P / partially_solved | Ben Green 问题

Suppose that $A \subset \mathbb{Z}/N\mathbb{Z}$ has density $\alpha$ and is Fourier uniform (all Fourier coefficients of $1_A - \alpha$ are $o(N)$). Does $A$ contain at least $\gg \alpha^{100}N^2$ 4-term arithmetic progressions?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13017 | AIM-COMBINATORICS-0271

### A 21-vertex counterexample via the complement of the WBGKSF

S / solved | AIM 研讨会问题汇编

**Question 1.** Is it true that every Berge graph is either basic or has a 2-join or has an even-pair skew-partition?

来源字段：https://aimath.org/WWN/perfectgraph/perfectgraph.pdf

13018 | AIM-COMBINATORICS-0155

## A 4/9 quadratic upper bound for the first Brown--Erdos--Sos case

P / partially_solved | AIM 研讨会问题汇编

For any $r\geq 3$ and $s\geq 4$ we have $$ f^r(n,s(r-2)+3,s)=o(n^2). $$

To remove a convention ambiguity, throughout this report $$ f_r(n,v,e)=\max\{|E(H)|:H\text{ is a simple }r\text{-uniform hypergraph on }n\text{ vertices and no }e\text{ distinct edges have union of size }\leq v\}. $$ For $n\geq v$, this is equivalently the largest number of edges in an $r$-graph in which every $v$-vertex set spans at most $e-1$ edges. Some sources write $f^{(r)}(n;v,e)$, and the original Sárközy--Selkow paper defines the least edge number which forces the configuration; the latter differs from the extremal maximum by one. Neither distinction changes an $o(n^2)$ assertion. Thus the AIM conjecture is $$ f_r(n,e(r-2)+3,e)=o(n^2) \tag{BES} $$ for each fixed $r\geq3$ and $e\geq4$.

There is a substantive transcription error in the first AIM remark. The cited Sárközy--Selkow theorem is $$ f_r\bigl(n,e(r-2)+2+\lfloor\log_2e\rfloor,e\bigr)=o(n^2), \tag{1} $$ not the displayed formula in the record, which omits the $+2$. This is confirmed both by the primary paper and by later papers quoting it. For $(r,e)=(3,4)$, (1) gives $f_3(n,8,4)=o(n^2)$, leaving $f_3(n,7,4)$ open, exactly as the next sentence of the AIM record says. The uncorrected formula would instead assert $f_3(n,6,4)=o(n^2)$, which is false: Glock, Joos, Kim, Kühn, Lichev and Pikhurko proved $$ f_3(n,6,4)=\left(\frac7{36}+o(1)\right)n^2. $$ This report preserves the source statement and corrects only the cited remark.

The second AIM remark also makes the roles of its parameters difficult to parse. A precise repeated-edge version is defined in Section 3 below.

来源字段：http://aimpl.org/hypergraphturan/4/

13019 | AIM-COMBINATORICS-0105

## A boundary-flux reduction for the mean square-grid sandpile identity

P / partially_solved | AIM 研讨会问题汇编

Jeremy Martin: “What is the average value of the scaling limit of the identity element of the sandpile group on $\mathbb Z^2$?”

The AIM record asks:

Wesley Pegden adds that an explicit construction for the square lattice might compute both that average and the side length of the central square of constant height $2$, but that this was not then known.

Taken literally, there is no finite sandpile group “on $\mathbb Z^2$” with a distinguished recurrent identity. The standard reading, confirmed by the cited pictures and subsequent literature, is the following finite-volume problem. Let

$$ Q_L=\{1,\ldots,L\}^2\subset\mathbb Z^2 $$

and wire every edge leaving $Q_L$ to one sink. Thus every nonsink vertex retains lattice degree four. Let $e_L:Q_L\to\{0,1,2,3\}$ be the recurrent representative of zero in the sandpile group. Rescale $Q_L$ to the unit square and ask for

$$ A=\lim_{L\to\infty}A_L, \qquad A_L:=\frac1{L^2}\sum_{x\in Q_L}e_L(x). $$

This reconstruction matters. A $1,2,3,4$ height convention adds exactly $1$ to the answer. Changing the exhaustion, aspect ratio, or dissipation rule changes the identities and may change the limit. Even and odd squares also have visibly different central sets (an odd square has a thin central cross), although that cross has vanishing area.

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13020 | AIM-COMBINATORICS-0260

## A chain-deficit theorem and exact multidimensional constructions for restricted sums

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.4 (G. Freiman).** Let $A\subseteq\mathbb Z$ with $|A|=n$ and let $G$ be a subset of $A\times A$. Write $$ G_1=\{a_i+a_j:(a_i,a_j)\in G\} $$ and $$ G_2=\{a_i-a_j:(a_i,a_j)\in G\}. $$ Given $|G_1|$, estimate $|G_2|$ from above and describe those sets $A$ with largest possible $|G_2|$.

Examples: 1. If $|G_1|=1$, then $|G_2|=n$. 2. If $|G_1|=2$, then $|G_2|=2n-1$, and $A$ is an arithmetic progression. 3. If $|G_1|=4$, then $|G_2|=4n-c\sqrt n$, and $A$ is isomorphic to the set of interior points of some convex set. 4. If $|G_1|=8$, then $|G_2|=8n-cn^{2/3}$, and $A$ is near a three-dimensional convex body.

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13021 | AIM-COMBINATORICS-0183

### A collision budget for Eulerian second neighborhoods

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.15 (Seymour and/or Jackson).** If $D$ is an Eulerian digraph with no loops or digons, then $$ \sum_{v\in V(D)} |N_2^+(v)| \;\geq\; \sum_{v\in V(D)} |N^+(v)|. \tag{1} $$

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13022 | AIM-COMBINATORICS-0190

### A convention-sensitive counterexample and a maximal-family obstruction certificate

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.20 (Hoàng--Reed).** If $G$ is a digraph with minimum outdegree $r$, then there are directed cycles $C_1,\ldots,C_r$ such that, for every $\ell$, $$ \left|V(C_\ell)\cap\left(\bigcup_{i=1}^{\ell-1}V(C_i)\right)\right|\leq 1. $$

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13023 | AIM-COMBINATORICS-0191

### A convention-sensitive counterexample to Hamidoune's reverse local-connectivity conjecture

S / solved | AIM 研讨会问题汇编

**Conjecture (Hamidoune, 1981).** Let $D$ be a digraph with $\delta_D^+\ge r$, $\delta_D^-\ge r$, and $r\ge 1$. Then there is an edge $(x,y)$ such that $\kappa_D(y,x)\ge r$.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13024 | AIM-COMBINATORICS-0111

### A coordinate-free form of simplicial multidegree and an orientation obstruction

P / partially_solved | AIM 研讨会问题汇编

Many participants: "How can we generalize Baker and Norine's graphical Riemann-Roch theorem [7] from graphs to cell complexes? Solving this problem may require a good combinatorial definition of critical configurations or superstables in arbitrary dimension, and/or understanding higher-dimensional generalizations of the Riemann-Roch theorem in algebraic geometry."

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13025 | AIM-COMBINATORICS-0164

### A corrected colored reduction and a Sullivan-2 counterexample normal form

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.4.** Any simple digraph with no loops or digons has a vertex $v$ such that $$ |N_2^+(v)|+|N^+(v)|\geq 2|N^-(v)|. $$

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13026 | AIM-COMBINATORICS-0193

### A counterexample to the unqualified weak-weight formulation

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.23 (Bollobás & Scott).** Let $p:E(G)\to[0,1]$. If $$ \sum_{v\in N_G^+(u)}p(uv)\ge 1 \quad\text{and}\quad \sum_{v\in N_G^-(u)}p(vu)\ge 1 $$ for all $u\in V(G)$, there is a directed cycle in $G$ of total weight at least $1$.

**Note.** There is a nice proof that there is a dipath of total weight at least $1$.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13027 | AMR-030-0073

### A covering code of radius R is a set of binary n-words so that every binary n-word can be reached from one of the codewo

P / partially_solved | AMR 问题清单汇编

A covering code of radius R is a set of binary n-words so that every binary n-word can be reached from one of the codewords by changing at most R bits. How big is the smallest covering code of radius R with n bits? It is known to be a constant c(R) times 2^(R )/ n^(R), but it's not known what c(R) is. Some believe that c(R) = R!.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13028 | AIM-COMBINATORICS-0083

### A cut-cone certificate for binary transitive hosts

P / partially_solved | AIM 研讨会问题汇编

**Conjecture (Leader--Russell--Walters).** A finite set is Ramsey if and only if it is a subset of some set with a transitive symmetry group.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13029 | AMR-030-0037

### A d-dimensional permutation of order n is an n-by-n-by

P / partially_solved | AMR 问题清单汇编

Linal/Luria: A d-dimensional permutation of order n is an n-by-n-by-...-by-n (d+1)-dimensional array of zeroes and ones, with the property that every row, column, etc., has exactly one 1 in it. (More precisely, if the array entries are A(j_(1)j_(2)j_(3)...j_(d+1)), for each k in [d+1] and t in [n], there is exactly one choice of all coordinates j_(i) except i=k so that setting j_(i)=k makes A(j_(1)j_(2)j_(3)...j_(d+1))=1.) Let P(n,d) be the number of d-dimensional permutations of order n. Linial and Luria showed that P(n,d) < ((1+o(1))n/e^(d))^n^(d). Is this tight, i.e., is P(n,d) > ((1+o(1))n/e^(d))^n^(d )also true? If so, this would be a very satisfying generalization of Stirling's formula and well-known bounds on the number of Latin squares (which are 2-dimensional permutations in disguise). See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13030 | AMR-030-0076

### A de Bruijn covering code of radius R is a binary string so that the set of words appearing as n consecutive symbols (wi

P / partially_solved | AMR 问题清单汇编

Chung/: A de Bruijn covering code of radius R is a binary string so that the set of words appearing as n consecutive symbols (with wrap-around) is a covering code of radius R. What is the smallest de Bruijn covering code with parameters R and n? It is known to be between c2^(n)/n^(R) and c2^(n) log n/n^(R).

背景链接：https://people.math.sc.edu/cooper/combprob.html

13031 | AIM-COMBINATORICS-0067

### A defect-graph certificate for large crossing families

P / partially_solved | AIM 研讨会问题汇编

If $K_n$ is drawn in the plane with straight line edges, can we always find $\Omega(n)$ pairwise crossing edges?

The record also cites the then-known $\Omega(\sqrt n)$ result of Aronov, Erdős, Kleitman, Pach, and collaborators, and a polynomial result for curves with bounded pairwise intersection multiplicity.

The standard precise reading is as follows. Let $P$ be a set of $n$ points in the plane in general position (no three collinear), and draw every edge of the complete graph as the straight segment between its endpoints. Two edges cross when their relative interiors intersect. A crossing family is a set of edges every two of which cross. Endpoint contacts therefore do not count: the edges in a crossing family have pairwise disjoint endpoints, so its size is at most $\lfloor n/2\rfloor$. Interior crossings of three or more edges at the same point are allowed by this definition. Write $$ T(P)=\max\{|\mathcal F|:\mathcal F\text{ is a crossing family spanned by }P\}, \qquad T(n)=\min_{|P|=n}T(P). $$ The recovered question is whether $T(n)=\Omega(n)$.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13032 | AIM-COMBINATORICS-0108

### A degree and orientation obstruction for a higher Merino theorem

P / partially_solved | AIM 研讨会问题汇编

Caroline Klivans: “How can we extend to higher dimensions Merino's result [25] that the generating function of the critical configurations (by degrees) is equal to an evaluation of the Tutte polynomial of $G$ (in particular, is equal to $T(1,y)$))?”

David Perkinson asks whether there is a corresponding external-activity statistic. Klivans observes that the cellular matroid is available but that higher critical groups are counted by a torsion-weighted sum of spanning trees, suggesting an arithmetic rather than ordinary Tutte polynomial. Igor Pak suggests that degree might be constant on each Laplacian coset, permitting a formal generating function without first choosing critical representatives.

Reference [25] in the workshop list is Criel Merino López, *Chip firing and the Tutte polynomial* (1997). With a connected undirected graph $G=(V,E)$, sink $q$, and recurrent stable configurations $c$ on $V\setminus\{q\}$, the normalization is

$$ \operatorname{level}_G(c) =\sum_{v\ne q}c(v)-|E|+\deg_G(q), \qquad T_G(1,y)=\sum_{c\in\operatorname{Rec}(G,q)}y^{\operatorname{level}_G(c)}. \tag{1.1} $$

This is a statistic of the distinguished recurrent representative of a reduced-Laplacian coset. It is not the value of an integer homomorphism on the sandpile group.

The higher-dimensional setting intended by the workshop is that of Duval--Klivans--Martin. For a finite simplicial complex $X$, their $i$-critical group is

$$ K_i(X)=\ker \partial_i/\operatorname{im}(\partial_{i+1}\partial_{i+1}^{*}). \tag{1.2} $$

Under their acyclicity and torsion-free-root hypotheses it is finite, is a cokernel of a reduced up-down Laplacian, and its order is a spanning-tree sum weighted by squared torsion.

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13033 | AIM-COMBINATORICS-0121

### A diagonal-lattice reduction for the tropical Picard group of a product of graphs

P / partially_solved | AIM 研讨会问题汇编

Dustin Cartwright: “What is the (tropical) Picard group (in the sense of [13]) of the product of graphs? I have a guess if one of the graphs is a tree.”

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13034 | AIM-COMBINATORICS-0215

### A diameter-free Property B bound for arbitrary van der Waerden sets

P / partially_solved | AIM 研讨会问题汇编

Define $W^*(k)$ to be the minimum cardinality of a finite set $X\subseteq\mathbb Z$ such that every red--blue coloring of $X$ contains a monochromatic nonconstant $k$-term arithmetic progression. Then $W^*(k)\le W(k)$. Is $W(k)-W^*(k)$ unbounded as $k\to\infty$? Is $$ \lim_{k\to\infty}\frac{W^*(k)}{W(k)}=1? $$

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13035 | AIM-COMBINATORICS-0112

### A dissipative ridge-routing sandpile from a fitting orientation

P / partially_solved | AIM 研讨会问题汇编

Caroline Klivans: “Can we develop a chip-firing model for directed rooted forests (as defined, for example, by Olivier Bernardi [10]) in higher dimension?”

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13036 | AIM-COMBINATORICS-0244

### A divisor-profile lower bound and a polynomial-height near-Sidon theorem for squares

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.7 (Brought by B. Green and T. Tao).** Take a finite subset of the squares of integers, \(|A|=N\). What lower bound can you get on \(A+A\)? One can get better than \(cN\) via Freiman's theorem. B. Green comments: I am not sure who posed this. It is certainly implicit in a paper of Chang on Rudin's problem (are the squares a \(\Lambda(p)\) set?).

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13037 | AIM-COMBINATORICS-0058

### A downgrade criterion for clique-order monotonicity

P / partially_solved | AIM 研讨会问题汇编

Let \(G\xrightarrow r K_k\) be minimal, i.e. \(G-e\not\xrightarrow r K_3\) for any \(e\in E(G)\).

The displayed source text contains

The occurrence of \(K_3\) is an internal typo: the parameter that follows varies with \(k\), and the standard definition in Fox--Grinshpun--Liebenau--Person--Szabó is minimality for \(K_k\). The unambiguous recovered statement is therefore as follows.

For integers \(r\geq 2\) and \(k\geq 3\), write \[ G\longrightarrow (K_k)_r \] if every \(r\)-edge-colouring of \(G\) contains a monochromatic \(K_k\). Let \(\mathcal M_r(K_k)\) be the family of graphs that are \(r\)-Ramsey-minimal for \(K_k\): \[ G\longrightarrow(K_k)_r, \qquad G-e\not\longrightarrow(K_k)_r\quad(e\in E(G)). \] Because the Ramsey property is monotone under taking supergraphs, minimality under every one-edge deletion is equivalent to having no proper Ramsey subgraph. Define \[ s_r(K_k)=\min\{\delta(G):G\in\mathcal M_r(K_k)\}. \] The conjecture is \[ \boxed{s_r(K_k)\leq s_r(K_{k+1})\quad\text{for all }r\geq2, k\geq3.} \] The workshop writes \(s_r(k)\) for \(s_r(K_k)\).

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13038 | AIM-COMBINATORICS-0159

### A feedback-set bound for second neighborhoods

P / partially_solved | AIM 研讨会问题汇编

**Seymour's Second Neighborhood Conjecture.** Every finite oriented graph has a vertex \(v\) such that \[ |N_2^+(v)|\ge |N_1^+(v)|. \]

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13039 | AIM-COMBINATORICS-0084

### A finite certificate for three-stripe triangle avoidance

P / partially_solved | AIM 研讨会问题汇编

**Conjecture.** Let \(T\) be a set of three points in \(\mathbb E^2\). Then there exists a 3-coloring of \(\mathbb E^2\) with no monochromatic copy of \(T\).

The quantifiers intended by the source are \[ \forall T\subset\mathbb E^2\ (|T|=3),\quad \exists c_T:\mathbb E^2\to\{0,1,2\},\quad \forall\text{ Euclidean isometries }g, \quad c_T\text{ is not constant on }g(T). \tag{1.1} \] Thus the coloring may depend on \(T\). This is an arbitrary point-coloring; no measurability or regularity is assumed. A “set of three points” has three distinct elements. Repeated-point multisets are therefore not intended.

The preceding record ends “For 3-point sets”, while the following record conjectures that two colors force every three-point configuration, with a possible equilateral exception. Graham and Tressler's published formulation says “for any triangle \(T\)” and identifies a triangle with its three vertices. We use the standard meaning of congruent copy: translation followed by an orthogonal map, including a reflection. This convention also agrees with the 2019 paper of Aichholzer and Perz, which explicitly includes the reflected triangle. The constructions below in fact handle reflections automatically.

This fixed-plane question must not be confused with the higher-dimensional Ramsey property. A nondegenerate triangle is a simplex and hence is Ramsey in some sufficiently high-dimensional Euclidean space for every fixed number of colors. Conjecture (1.1) instead asks whether three colors can avoid it when the ambient space is kept equal to \(\mathbb E^2\).

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13040 | AIM-COMBINATORICS-0002

### A finite entropic coupling calculus with forest and even-cycle certificates

P / partially_solved | AIM 研讨会问题汇编

**Entropy proof of Sidorenko**

There is a proof system for graph homomorphism inequalities using flag algebras.

Define a proof system for graph homomorphism inequalities using entropy.

来源字段：http://aimpl.org/highdimdiscrete/1/

13041 | AIM-COMBINATORICS-0097

### A fixed Laplacian quotient and hereditary firing staircases

P / partially_solved | AIM 研讨会问题汇编

Acyclic orientations of a graph \(G\) are the regions of its graphic arrangement. The equivalence relation generated by converting a source into a sink encodes chip-firing on a special class of divisors, and its classes are represented geometrically by passing to the toric graphic arrangement. Is there a similar geometric model for chip-firing equivalence on larger classes of divisors? Can multi-toppling, or hereditary chip-firing, provide it, and when does that dynamics have the abelian property?

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13042 | AIM-COMBINATORICS-0081

### A four-point Hamming barrier in Euclidean Ramsey theory

P / partially_solved | AIM 研讨会问题汇编

**Conjecture (\$1000).** Every finite spherical set is Ramsey.

Its exact mathematical claim is:

Here a finite set \(X\subseteq\mathbb E^n\) is Ramsey if, for every positive integer \(r\), there is an \(N=N(X,r)\) such that every arbitrary \(r\)-coloring of all points of \(\mathbb E^N\) contains a monochromatic set congruent to \(X\). A set is spherical if all its points lie on one Euclidean sphere.

The source record ends with the fragment “A `warm up':”. This is not an additional hypothesis or an incomplete part of Conjecture 34. Graham's original published problem list confirms the intended prose: after the \(\$1000\) problem it says that, as a warm-up, one might prove that any four-point subset of a circle is Ramsey. Thus the warm-up belongs semantically to record 0082.

A congruent copy in \(\mathbb E^N\) is a distance-preserving image of the finite metric on \(X\). Equivalently, after restricting to \(\operatorname{aff}(X)\), it is the image under an affine isometric embedding \[ x\longmapsto Ux+t, \] where \(U\) preserves inner products. Its affine subspace, orientation, and translation in \(\mathbb E^N\) are arbitrary; it is not required to lie in the standard copy of \(\mathbb E^n\). Reflections make no substantive difference (and can be extended to rotations after adding one unused dimension). Both pairwise distances and circumradius are preserved.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13043 | AIM-COMBINATORICS-0258

### A global density theorem below the 3k-4 threshold

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.2 (V. Sós).** Let \(A\) be a strictly increasing infinite sequence of integers, and denote by \(A_n\) the initial \(n\)-element segment of \(A\). Suppose that \[ |A_n+A_n|<Cn. \] What can be said about the structure of \(A\)?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13044 | AMR-030-0026

### A graph G is said to be uniquely colorable it has only one optimal coloring up to permutation of the colors

O / open | AMR 问题清单汇编

: A graph G is said to be uniquely colorable it has only one optimal coloring up to permutation of the colors. (That is, there is only one partition into a minimum number of independent sets.) Is it false that G(n,1/2) is uniquely colorable (aas)?

背景链接：https://people.math.sc.edu/cooper/combprob.html

13045 | AMR-030-0025

### A graph G is said to be uniquely H-saturated if it contains no H, but adding any edge to G creates exactly one copy of H

O / open | AMR 问题清单汇编

: A graph G is said to be uniquely H-saturated if it contains no H, but adding any edge to G creates exactly one copy of H (up to isomorphism). Clearly, if G is uniquely K_(r)-saturated, then joining a single vertex to G creates a uniquely K_(r)_(+1)-saturated graph. Call a uniquely K_(r)-saturated graph sporadic if it has no dominating vertex. Is it true that, for each r, there are finitely many sporadic uniquely K_(r)-saturated graphs?

背景链接：https://people.math.sc.edu/cooper/combprob.html

13046 | AIM-COMBINATORICS-0039

### A high-probability lower bound for random permutation discrepancy

P / partially_solved | AIM 研讨会问题汇编

**Discrepancy of permutations**

a) discrepancy of k-permutations

For $k=3$, $[NNN'12]$ showed a lower bound of $\Omega(\log n)$ where $n$ is the number of elements in the ground set. For general $k$, we know that the answer lies in $[\sqrt{k}+\log n,\sqrt{k}\log n]$. An open question is to clsoe this gap.

b) discrepancy of k-random permutations

A lower bound of $\Omega(\sqrt{k})$ is known.

来源字段：http://aimpl.org/hereddiscrep/1/

13047 | AIM-COMBINATORICS-0014

### A holonomy reduction to equiangular lines

P / partially_solved | AIM 研讨会问题汇编

Given \(k\)-dimensional linear subspaces \(U,V\leq\mathbb R^d\), say that \(U,V\) are equi-isoclinic with angle \(\alpha\) if every unit vector in \(U\) projects onto \(V\) as a vector of length \(\alpha\).

Let \(N_\alpha^k(d)\) be the maximum number of \(k\)-dimensional subspaces of \(\mathbb R^d\), every pair of which is equi-isoclinic with angle \(\alpha\). Then \[ \text{(1) }N_\alpha^k(d)=O_{\alpha,k}(d), \qquad \text{(2) }N_\alpha^k(d)=O_k(d)+o_{\alpha,k}(d). \]

The AIM record, problem 4.1 from the workshop *High-dimensional phenomena in discrete analysis*, states:

The word “angle” is potentially misleading. In the displayed definition, \(\alpha\) is the projection length, hence the cosine of every principal angle. The common principal angle itself is \[ \theta=\arccos\alpha. \] Some papers instead call \(\lambda=\alpha^2\) the parameter of isoclinism. This report always uses the AIM convention: \(\alpha\) is the projection length.

The meaningful range is \(0\leq\alpha\leq1\), and the family is a set of distinct subspaces. Without distinctness, repetitions make the case \(\alpha=1\) unbounded. I assume \(d\geq k\).

Statement (2) needs a quantifier audit. Its nonredundant natural reading is: \[ \exists C_k<\infty\ \ \forall\text{ fixed }\alpha\in(0,1)\ \exists r_{\alpha,k}(d)=o(d) \quad N_\alpha^k(d)\leq C_kd+r_{\alpha,k}(d). \tag{1.1} \] Thus the leading constant is independent of \(\alpha\), while the threshold and lower-order term may depend on \(\alpha,k\). If both terms were allowed unrestricted \(\alpha\)-dependence, (2) would say no more than (1).

来源字段：http://aimpl.org/highdimdiscrete/4/

13048 | AIM-COMBINATORICS-0138

### A hybrid distance-to-tree and sparse separating families

P / partially_solved | AIM 研讨会问题汇编

Find (different) notions of distance on the space of graphs, or perhaps measures on this space. In particular: how far is a given graph from being a tree?

Tree-width is not good enough. There is a relation to tree-length.

来源字段：http://aimpl.org/largenetworks/1/

13049 | AIM-COMBINATORICS-0257

### A linear unit-distance bound for algebraically concentrated p-point sets

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.1 (T. Tao).** Suppose that \(S\subseteq\mathbb F_p^2\) with \(|S|=p\). How many pairs of points of distance one apart can there be? That is, how large can \[ \#\{(x,y)\in S\times S:(x_1-x_2)^2+(y_1-y_2)^2=1\} \] be?

**Remark.** The trivial upper bound is \(p^{3/2}\). The best lower-bound example is of size \(p\).

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13050 | AIM-COMBINATORICS-0061

### A linear worst-case lower bound for Ramsey numbers of rectilinear knots

P / partially_solved | AIM 研讨会问题汇编

Fix a knot \(K\). For all sufficiently large \(n\), every linear spatial embedding of \(K_n\) contains \(K\). Here the vertices lie in \(\mathbb R^3\) and every edge is a straight segment. Containment can mean either (i) equivalence by a homeomorphism, which allows reflection, or (ii) ambient isotopy, which distinguishes the two hands of a chiral knot. If \(K\) has \(k\) crossings, how large must \(n=n(k)\) be?

The record attributes a five-fold exponential upper bound to Negami and a bound \(n(k)\leq 2^{2^{ck}}\) to Conlon--Fox. It also records that every rectilinear \(K_7\) contains a trefoil up to reflection.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13051 | AIM-COMBINATORICS-0285

### A literal NP class for the P4-structure of proper 2-joins

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.** Find a class \(\mathcal C_1\) of \(4\)-uniform hypergraphs such that

- the \(P_4\)-structure of every graph with a \(2\)-join belongs to \(\mathcal C_1\); - no odd ring belongs to \(\mathcal C_1\); - \(\mathcal C_1\) belongs to NP.

If \(G\) is a graph on \(V\), then its **\(P_4\)-structure** \[ \mathsf P_4(G)=(V,\mathcal E_4(G)) \] is the \(4\)-uniform hypergraph whose hyperedges are precisely the four-element sets inducing a chordless four-vertex path in \(G\).

For odd \(k\geq5\), the **odd ring** \(\mathcal R_k\) has vertices \(u_0,\ldots,u_{k-1}\) and the \(k\) hyperedges \[ \{u_{i+1},u_{i+2},u_{i+3},u_{i+4}\}, \qquad i\in\mathbb Z_k. \] Thus \(\mathcal R_k=\mathsf P_4(C_k)\).

The phrase “no odd ring belongs” is weaker than the phrase used immediately above it for the basic classes, namely “no \(4\)-uniform hypergraph in \(\mathcal C\) contains an induced odd ring.” This difference matters and is audited below.

The only explicit \(2\)-join definition in the source PDF appears in subsection F.4. This report follows that source convention. A split is \[ (V_1,V_2;A_1,B_1,A_2,B_2), \] where \(V_1,V_2\) partition \(V(G)\), and \(A_i,B_i\) are disjoint nonempty subsets of \(V_i\). Across the split, all edges between \(A_1,A_2\), and all edges between \(B_1,B_2\), are present, and there are no other cross edges. For \(i=1,2\), let \(\mathcal P_i\) be the chordless paths in \(G[V_i]\) having one end in \(A_i\), the other in \(B_i\), and no internal vertex in \(A_i\cup B_i\). The source additionally requires \[ \mathcal P_i\ne\varnothing \quad\text{and}\quad G[V_i]\text{ is not itself a path belonging to }\mathcal P_i. \tag{1.1} \] This is a connected, non-path, or “proper” \(2\)-join convention. The non-path condition is essential: if path \(2\)-joins are allowed, every sufficiently long odd hole has such a split, making the first two bullets inconsistent.

来源字段：https://aimath.org/WWN/perfectgraph/perfectgraph.pdf

13052 | AIM-COMBINATORICS-0184

### A literal parallel-arc counterexample and a collision obstruction for the simple repair

S / solved | AIM 研讨会问题汇编

**Conjecture 6.16 (Thomassé & Král).** Let \(G\) be an Eulerian digraph on \(n\) vertices, so that \(|E(G)|\ge n^2/3\). Then \(G\) has a directed cycle of length at most three.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13053 | AIM-COMBINATORICS-0227

### A local mirrored-shadow law for sum representation functions

P / partially_solved | AIM 研讨会问题汇编

What general properties, other than \((*)\), are shared by all functions \(\nu=1_A*1_B\) for finite \(A,B\subseteq\mathbb Z\)?

The raw corpus record defines, for finite integer sets \(A,B\), \[ \nu(n):=\#\{(a,b)\in A\times B:a+b=n\},\qquad n\in\mathbb Z, \] then sorts the positive values of \(\nu\) as column heights \(r_1\ge r_2\ge\cdots\) of a Ferrers diagram.

The official AIM PDF, Problem 2.2, page 5 of the PDF (printed pages 5--6 across the break), shows that the intended formula is \[ r_k^2\le r_k+r_{k+1}+r_{k+2}+\cdots \qquad(k\ge1). \tag{*} \] Here nonexistent later columns contribute zero. The recovered problem asks:

If \[ N_t:=\#\{n\in\mathbb Z:\nu(n)\ge t\}, \] then the source also records Pollard's inequality \[ N_1+\cdots+N_t\ge t(|A|+|B|-t), \qquad 1\le t\le\min\{|A|,|B|\}, \tag{P} \] and notes that (P) follows from \((*)\).

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13054 | AIM-COMBINATORICS-0145

### A manifold scale needs calibration and identifiability

P / partially_solved | AIM 研讨会问题汇编

Is there a scale at which the structure of a (real) network “looks like” a manifold? (e.g., Krioukov's result)

来源字段：http://aimpl.org/largenetworks/1/

13055 | AIM-COMBINATORICS-0284

### A merged record and the corrected balanced-circulant classification

O / open | AIM 研讨会问题汇编

**CONJECTURE. Every non-empty balanced circulant is isomorphic to a \(G(k,m)\).**

The first coherent problem in the record can nevertheless be recovered from the immediately preceding K.1 source text. For integers \(k,m\geq 1\), put \(n=4km\), take \(V=\mathbb Z_n\), and define \(G(k,m)\) by \[ xy\in E(G(k,m)) \quad\Longleftrightarrow\quad x-y\equiv \pm1\pmod {4m}. \] Equivalently, the source writes \(x-y+4mt\equiv\pm1\pmod n\) for some integer \(t\). The conjecture appearing verbatim in this record is:

The preceding sentences also assert that the earlier decomposition conjecture holds for this family: when \(k>1\), \[ S=\{0\}\cup\{4mi+1,4mi-1:1\leq i\leq k\} \] is a star cutset, while for \(k=1\) the graph is a basic \(4m\)-cycle.

First, it omits the definition of \(G(k,m)\), which occurs in the preceding record. Second, its claim that \(4mj\) is isolated in \(G[V\setminus S]\) for every \(1\leq j\leq k\) cannot include \(j=k\): modulo \(n\), \(4mk=0\), and vertex \(0\) belongs to \(S\). The correct range is \(1\leq j\leq k-1\).

来源字段：https://aimath.org/WWN/perfectgraph/perfectgraph.pdf

13056 | AIM-COMBINATORICS-0211

### A missing threshold and an explicit Freiman-model repair

S / solved | AIM 研讨会问题汇编

**Problem 1.5 (N. Katz).** Fix an integer \(k\geq3\) and a real number \(C\geq2\). Given that \(A\) is a finite set of integers with \[ |A+A|<C|A|, \] is it true that the number of \(k\)-term arithmetic progressions in \(A\) is at least \(c|A|^2\), with a positive constant \(c\) depending on \(k\) and \(C\) only?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13057 | AIM-COMBINATORICS-0189

### A multigraph counterexample and a low-exposure reduction

S / solved | AIM 研讨会问题汇编

**Conjecture 6.19 (Bermond--Thomassen).** In a digraph $D$ with $\delta_D^+\ge 2k-1$, there are $k$ vertex-disjoint cycles.

Sullivan's report from the 2006 AIM workshop on the Caccetta--Haggkvist conjecture. After repairing only the PDF line breaks, its statement is

The source note says that the conjecture was open for $k\ge3$ and that Thomassen's proof for $k=2$ was not intuitive. 1 9”; no mathematical symbol is missing from the recovered sentence.

There is a consequential convention problem. At the start of the same report, a digraph is defined by vertex and edge sets with head and tail incidence relations, and the report explicitly says that this definition allows multiple edges from $u$ to $v$. A *simple digraph* is then defined separately as having at most one such edge. Conjecture 6.19 says “digraph,” not “simple digraph.” Its definition of an edge as going from a vertex to “another vertex” excludes loops, but digons are not excluded. The report defines outdegree without saying whether parallel arcs are counted with multiplicity.

Under the usual incidence-counted degree of a directed multigraph, the exact printed statement is false. Under the alternative convention in which outdegree means the number of distinct out-neighbors, parallel copies can be deleted and the statement is exactly the standard simple-digraph conjecture. Both readings are treated below. The note about the known $k=2$ case and all modern formulations checked show that the latter was the intended problem.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13058 | AIM-COMBINATORICS-0019

## A normalization counterexample for analytic-rank tensorization

P / partially_solved | AIM 研讨会问题汇编

The following is the `problem` field reproduced verbatim, including the printed symbol `\phi\neq S\subset[d]` and all original conventions:

The record's `literature` field says verbatim:

The exact statement needs the following reconstruction before it has a definite truth value.

1. The printed `\(\phi\neq S\subset[d]\)` is almost certainly an extraction/OCR error for the standard condition \(\varnothing\ne S\subsetneq[d]\). Both factors in a partition-rank-one decomposition must use nonempty complementary sets of modes. 2. \(\omega\) must be a **nontrivial** additive character, i.e. a primitive \(p\)-th root in the prime-field notation. If \(\omega=1\), every bias is one. 3. Standard analytic rank is \[ \operatorname{arank}_p(T)=-\log_p\operatorname{bias}(T). \] The missing logarithm base matters critically when a numerical rank is raised to the \(m\)-th power. Base \(p\) is also the normalization for which matrix analytic rank equals matrix rank and Lovett's inequality \(\operatorname{arank}\le\operatorname{prank}\) has constant one. 4. The identity must be the order-\(d\), side-\(n\) diagonal tensor \[ I_{n,d}=\sum_{i=1}^n e_i^*\otimes\cdots\otimes e_i^*. \] 5. Tensor power must mean the **Kronecker power**, denoted below by \(A^{\boxtimes m}\), which groups corresponding modes and has order \(d\) and side length \(n^m\). The ungrouped external tensor product has order \(dm\) and is a different operation. 6. The only nonvacuous quantification is: there exists \(c=c(A,p,d,n)\in(0,1)\), independent of \(m\), such that the inequality holds for every sufficiently large \(m\) (or, more strongly, every \(m\ge1\)). If \(c\) may depend on \(m\), the question does not express exponential amplification.

Under these standard choices, the analytic-rank branch of the displayed question is false. The partition-rank branch retains meaningful content in order at least four.

There is also a type distinction in the cap-set paragraph. The array \(C(x,y,z)\), indexed by group elements, is a coefficient tensor. Its analytic rank is the bias of its multilinear extension on three copies of \(\mathbb F_3^{3^k}\), not the average of \(\omega^{C(x,y,z)}\) over the three group indices.

来源字段：http://aimpl.org/addcombapp/1/

13059 | OPG-148

## A nowhere-zero point in a linear mapping

P / partially_solved | Open Problem Garden

Conjecture If ${\mathbb F}$ is a finite field with at least 4 elements and $A$ is an invertible $n \times n$ matrix with entries in ${\mathbb F}$, then there are column vectors $x,y \in {\mathbb F}^n$ which have no coordinates equal to zero such that $Ax=y$.

背景链接：http://www.openproblemgarden.org/op/a_nowhere_zero_point_in_a_linear_mapping

13060 | AIM-COMBINATORICS-0195

## A parametric audit of a refuted logarithmic outdegree condition

S / solved | AIM 研讨会问题汇编

**Conjecture 6.25.** If \(D\) is a digraph on \(n\) vertices with \[ \sum_{v\in V(D)} \log\left(1+\frac{1}{\delta_D^+(v)}\right) \geq n\log\left(1+\frac1r\right), \] then \(D\) has a directed cycle of length at most \[ \left\lceil\frac nr\right\rceil. \]

Take a transitive tournament of size \(n-1\), and replace the edge from the vertex of outdegree \(n-2\) to the vertex of outdegree zero with a path of length \(2\) in the opposite direction, thereby increasing the number of vertices to \(n\).

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13061 | AIM-COMBINATORICS-0236

## A parity-enhanced shell inequality for the Erdős-Turán additive-basis problem

P / partially_solved | AIM 研讨会问题汇编

Suppose \(A\subseteq\mathbb N\) is an additive basis of order \(n\).

This is not a meaningful parameter in context. The same sentence then defines \(r(m)\) using pairs \((a_1,a_2)\), and the problem is explicitly attributed to Erdős and Turán. The intended order is therefore **2**. This is confirmed by the original conjecture and every primary modern restatement checked.

The recovered problem is:

Let \(A\subseteq\mathbb N_0=\{0,1,2,\ldots\}\) be an additive basis of order \(2\), so every sufficiently large integer belongs to \(A+A\). Define the ordered representation function

\[ r_A(m)=\#\{(a_1,a_2)\in A^2:a_1+a_2=m\}. \]

\[ \limsup_{m\to\infty}r_A(m)=\infty? \]

Some sources reserve “basis” for coverage of every nonnegative integer and “asymptotic basis” for eventual coverage. The result below applies to the eventual version. The choice \(\mathbb N=\mathbb N_0\) versus \(\{1,2,\ldots\}\), and any finite change to \(A\), do not affect whether the limsup is infinite.

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13062 | AIM-COMBINATORICS-0117

## A planar analytic certificate for metric chip-firing

P / partially_solved | AIM 研讨会问题汇编

**(27) Adrien Kassel:** “How does chip-firing on planar metric graphs relate to questions in analysis?”

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13063 | AIM-COMBINATORICS-0123

### A positive-density obstruction for the prop-agator limit

P / partially_solved | AIM 研讨会问题汇编

(33) Lionel Levine: "What is the infinite volume limit of Andrea's prop-agator model [12] with a finite density of propagators?"

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13064 | AIM-COMBINATORICS-0093

### A q-Eulerian last-site law for cycle IDLA

P / partially_solved | AIM 研讨会问题汇编

Problem (3), attributed to Persi Diaconis, asks about the “explorers game” of reference [16] on a cycle: repeatedly send an explorer from vertex \(0\), and study the vertex occupied last. It reports an Eulerian-number law and suggests studying higher-dimensional sets and circulant graphs. The recorded comments mention Jim Propp's earlier observation, possible connections with Pólya's and Bernard Friedman's urns, the difficulty of \(\mathbb Z^2\), and the possibility that weights in one dimension yield a new \(q\)-analogue. Footnote 2 merely points to a definition of Eulerian numbers; it is not an exponent.

Reference [16] is Diaconis and Fulton, *A growth model, a game, an algebra, Lagrange inversion, and characteristic classes*. Its explorer process is now usually called sequential internal diffusion-limited aggregation (IDLA):

1. initially the source \(0\) is occupied (equivalently, the first explorer starts with the empty aggregate and settles immediately at \(0\)); 2. each subsequent explorer is an independent copy of the specified Markov chain, started at \(0\); 3. it stops on first visiting an unoccupied vertex, which is then added to the aggregate.

This is not one random walk continued until its cover time.

There is an indexing inconsistency in the recorded cycle formula. With \(n\) vertices labelled \(0,\ldots,n-1\), vertex \(0\) is occupied first and cannot be last when \(n>1\). Direct derivation below gives \[ \mathbb P(L=j) =\frac{\left\langle{\,n-1\atop j-1\,}\right\rangle}{(n-1)!}, \qquad 1\le j\le n-1, \] for simple symmetric walk, where the bracket is an ordinary Eulerian number. The PDF's \(A(n,j)/n!\) becomes correct after the corresponding one-unit change of cycle size and Eulerian index, for example on a cycle with \(n+1\) vertices with \(j\) replaced by \(j-1\). The discrepancy is consistent with the PDF's explicit warning that its quotations are informal summaries.

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13065 | AIM-COMBINATORICS-0094

### A quotient-root formula for every parking/toppling Betti number

P / partially_solved | AIM 研讨会问题汇编

**Sam Hopkins.** “This question concerns a recent preprint of Venkatesh and Viswanath [34], which shows that for a finite graph \(G\) on \(n\) vertices, the dimension of the root space associated to the root that is the sum of all simple roots in the Kac-Moody algebra having Cartan matrix \(2I_n-A(G)\) is the number of maximal \(G\)-parking functions. (Here \(A(G)\) is the adjacency matrix of \(G\).) Can this be exploited further? Are all \(G\)-parking functions present somewhere? Does the sandpile group act on some such root space?”

(a) **Sam Hopkins.** “I should also mention that there is a natural labeling of a basis of this root space by maximal parking functions.”

(b) **Anton Dochtermann.** “The number of maximal parking functions is also the number of last syzygies in a minimal free resolution of the toppling ideal, and also the number of chambers of a certain slice of the graphic arrangement. Can we connect the Lie theory to commutative algebra or hyperplane arrangements?”

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13066 | AIM-COMBINATORICS-0077

## A reciprocity bound for long directed paths

P / partially_solved | AIM 研讨会问题汇编

**Directed paths in Eulerian digraphs** (Jacques Verstraete). Does every Eulerian digraph of average degree \(d\) have a directed path of length \(\Omega(d)\)? Getting \(\Omega(\sqrt d)\) is not too hard.

The record describes this as a special case of a conjecture of Bollobás and Scott associated with their paper *A Proof of a Conjecture of Bondy Concerning Paths in Weighted Digraphs*.

The conventions must be made explicit. In the modern literature on this problem, a digraph is finite, loopless, and simple: each ordered pair of distinct vertices supports at most one arc, though both opposite arcs may be present. A directed path is vertex-simple, and its length is its number of arcs. A digraph is Eulerian when \[ d^+(v)=d^-(v)\qquad(v\in V(D)). \] Writing \(n=|V(D)|\) and \(m=|A(D)|\), the average degree in the question is the average out-degree \[ d=\frac{m}{n}, \] not the average total degree \(2m/n\). These are the conventions explicitly used by Janzer--Sudakov--Tomon and Knierim--Larcher--Martinsson.

These conventions are logically necessary. If parallel arcs are allowed, two vertices with \(M\) arcs in each direction form an Eulerian multidigraph of average out-degree \(M\), while its longest vertex-simple directed path has length one. If “path” means a trail or walk, an Euler tour makes the connected case immediate. Thus neither of those readings is the intended problem.

Connectivity causes no difficulty. Every weak component of an Eulerian digraph is itself Eulerian, and every nontrivial weak component is strongly connected. Since the average degree of the whole digraph is the vertex-weighted average of the component averages, one component has average degree at least \(d\).

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13067 | AIM-COMBINATORICS-0004

## A reduced-degree masking counterexample to nonlinear Roth

S / solved | AIM 研讨会问题汇编

Let \(P\) be any (non-linear) polynomial, \(A\subset \mathbb F_p\) with \(|A|>p^{0.99}\). Does there always exist distinct \(x,y\) such that \[ x,y,y+P(x)-P(y)\in A? \]

来源字段：http://aimpl.org/highdimdiscrete/2/

13068 | AIM-COMBINATORICS-0103

## A residual-diamond model for abelian dynamics with shared memory

P / partially_solved | AIM 研讨会问题汇编

(13) Lionel Levine: "What is the right definition of an abelian network with shared memory?" (a) Lionel Levine: "I have unified some subset of the class of models that behave like chip-firing does, but a lot of models with an abelian property still fall outside this definition and seem to re-quire some kind of shared memory. An example of shared mem-ory is cluster-firing, as investigated by Spencer Backman [6], who in particular studied when the abelian property is pre-served while allowing firing of multiple vertices at once." (b) Jordan Ellenberg: "This seems formally similar to Persi Diaco-nis's question on non-overlapping discs."

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13069 | AIM-COMBINATORICS-0007

## A root-system reduction for the spherical triangle form

P / partially_solved | AIM 研讨会问题汇编

Denote by $S^n\subset\mathbb R^{n+1}$ the unit sphere. For a function $f:S^n\to\mathbb R$, define \[ Tf=\mathbb E_{x,y,z\in S^n,\ x+y+z=0}f(x)f(y)f(z). \] If $f$ is 1-bounded and $|Tf|>\delta$, does $f$ correlate with a function that depends on $O_\delta(1)$ coordinates (with respect to some basis)?

来源字段：http://aimpl.org/highdimdiscrete/2/

13070 | AIM-COMBINATORICS-0051

## A saturated-list kernel for one-point extensions of four-color Ramsey colorings

P / partially_solved | AIM 研讨会问题汇编

Find \(r(3,3,3,3)\). Is it 51?

It records the historical bounds

\[ 51\le r(3,3,3,3)\le 62. \]

\[ R_4(3):=r(3,3,3,3) \]

is the **least** integer \(N\) such that every coloring of \(E(K_N)\) with four colors contains a monochromatic triangle. I call a coloring with no monochromatic triangle **good**. With this convention:

- \(R_4(3)\ge51\) means that a good coloring of \(K_{50}\) exists. - \(R_4(3)\le62\) means that no good coloring of \(K_{62}\) exists. - The two statements do **not** assert that a good \(K_{51}\) or \(K_{61}\) exists. - The conjectural equality \(R_4(3)=51\) says precisely that 50 is the maximum possible order of a good four-coloring.

This off-by-one distinction is maintained throughout.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13071 | AIM-COMBINATORICS-0287

## A scope trichotomy for the balanced-skew-partition class problem

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.** Find a class \(\mathcal C_3\) of 4-uniform hypergraphs such that

1. the \(P_4\)-structure of every graph with a balanced skew partition belongs to \(\mathcal C_3\); 2. no odd ring belongs to \(\mathcal C_3\); 3. \(\mathcal C_3\) belongs to NP.

That is a distinct, unnumbered research question: it changes the invariant and is not one of the three requirements on \(\mathcal C_3\). The official HTML likewise places it after Problem 3.

There is a crucial wording difference in the source. Immediately before the three numbered problems, the desired basic-graph class is required to have **no member containing an induced odd ring**. Problem 3 itself merely says that **no odd ring belongs**. These are not equivalent without a heredity requirement. The distinction controls the answer.

来源字段：https://aimath.org/WWN/perfectgraph/perfectgraph.pdf

13072 | AMR-030-0002

## A set of points S is Euclidean Ramsey if, for every k, there exists an N so that every k-coloring of Euclidean N-space c

O / open | AMR 问题清单汇编

Graham: A set of points S is Euclidean Ramsey if, for every k, there exists an N so that every k-coloring of Euclidean N-space contains a monochromatic copy of S. Show that, if S can be embedded in some d-dimensional sphere, then it is Euclidean Ramsey. This problem is open even for four points on a circle, although it is known to be true for triangles. (Boris Bukh points out that there is good evidence against this now: here and here, for example.)

背景链接：https://people.math.sc.edu/cooper/combprob.html

13073 | AIM-COMBINATORICS-0200

## A sharp lower-bound certificate for the first open transitive tournament case

P / partially_solved | AIM 研讨会问题汇编

**Conjecture (Mader, 1985).** For every positive integer $k$, there is a positive integer $r$ such that every $r$-out-regular digraph contains a subdivision of the transitive tournament on $k$ vertices.

The source notes that the assertion is known for $k=3$, with $r=2$, and for $k=4$, with $r=3$ by Mader (1996). It says that even the existence of an $r$ for $k=5$ is unknown and that $r=6$ is conjectured.

This report uses finite simple digraphs: loops and parallel arcs are excluded, but a digon (both $xy$ and $yx$) is allowed. This convention is material; Section 8 gives a counterexample if parallel arcs count with multiplicity.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13074 | AIM-COMBINATORICS-0156

## A sharp three-edge-color counterexample to the literal conjecture

P / partially_solved | AIM 研讨会问题汇编

The exact AIM record is Conjecture 10 in the 2011 workshop list, in the section “Ruzsa-Szemerédi Theorem and Relatives.” It takes three disjoint sets \(A,B,C\), each of size \(n\), and matchings \(M_1,\ldots,M_\ell\) of triples with one vertex in each part. The forbidden configuration consists of three edges

\[ abc,\qquad a'b'c',\qquad a''b''c'' \]

in one \(M_i\), together with the crossing edge \(ab'c''\) in “some other” \(M_j\). The union \(M=\bigcup_i M_i\) is required to be a simple linear \(3\)-graph. The displayed conjecture is

\[ |M|=o(n^2). \]

It does not state a bound on \(\ell\), a lower bound on \(|M_i|\), maximality or perfection of the matchings, or that the \(M_i\) form a partition.

The following quantifier audit is therefore necessary.

- “Some other \(M_j\)” naturally means \(j\ne i\). - A matching has its standard meaning: its edges are pairwise vertex-disjoint. It may have size \(0,1,\) or \(2\). - Simple means no repeated hyperedge; linear means that two distinct hyperedges meet in at most one vertex. - The set-theoretic union being simple does not by itself imply that an edge belongs to only one indexed \(M_i\). The counterexample below satisfies the stronger condition that the \(M_i\) are pairwise edge-disjoint and partition \(M\), so this ambiguity cannot rescue the statement.

Under these standard literal quantifiers, the conjecture is false. The source document itself warns elsewhere that some workshop questions may be “rather easy (or even not well-defined),” which is material here.

来源字段：http://aimpl.org/hypergraphturan/4/

13075 | AIM-COMBINATORICS-0167

## A sharp two-label frontier for Seymour's load conjecture

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.5 (Seymour).** Under the hypotheses of Conjecture 6.1, if \(|V|=d\) and \[ \sum_{i=1}^{k}\delta^+_{G_i}(v)\geq d \] for all \(v\), then \(G\) has a rainbow directed cycle.

**Note.** This conjecture is false if \(|V|=d\) is replaced by \(|V|=d+1\).

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13076 | AIM-COMBINATORICS-0254

### A singleton obstruction to the Katz critical-set question

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.17 (N. Katz).** Does there exist a finite subset \(A\subset\mathbb Z\), \(|2A|=k|A|\), so that for the function \(f(n,k)\) defined in the previous problem every proper \(A'\subset A\) satisfies \[ |nA'|\geq f(n,k)|A'|? \]

Suppose \(A\subseteq\mathbb Z\). If \(|2A|<k|A|\), describe the sumset \(|nA|\) as \(n\) tends to infinity. Find \(f(n,k)\), which is the smallest number such that \[ |nA|\leq f(n,k)|A| \] for all \(A\) such that \(|2A|<k|A|\).

For a fixed integer \(n\geq2\) and real \(k>1\), does there exist a nonempty finite \(A\subset\mathbb Z\), preferably \(|A|\geq2\), with \(|2A|=k|A|\) and the stated property for every proper subset?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13077 | AIM-COMBINATORICS-0075

### A small Fano-good tight path and the connectedness convention

P / partially_solved | AIM 研讨会问题汇编

Which connected 3-uniform hypergraphs \(H\) satisfy \(R(H,F)=2v(H)-1\)?

The AIM record, attributed to David Conlon, asks which 3-uniform hypergraphs are Ramsey-good with respect to the Fano plane. The record defines \[ r(H,F)=2(v(H)-1)+1 \] as the equality of interest and describes a two-part coloring intended to prove the corresponding lower bound.

I use the asymmetric convention now standard in the literature: \(R(H,F)\) is the least \(N\) such that every red/blue coloring of \(\binom{[N]}3\) contains either a **red** copy of \(H\) or a **blue** copy of the Fano plane \(F\). The Fano plane is the 3-graph on seven vertices whose seven edges form the unique Steiner triple system \(S(2,3,7)\). One convenient labeling is \[ 123,145,167,246,257,347,356. \tag{1} \]

Here hypergraph chromatic number means the weak vertex chromatic number: no edge may be monochromatic. The Fano plane has \(\chi(F)=3\). It has a proper 3-coloring with a singleton class (color \(1\) alone, and color \(\{2,3,4\}\) and \(\{5,6,7\}\) separately in (1)), so \(\sigma(F)=1\). Thus the graph-style Burr expression becomes \(2(v(H)-1)+1\).

There is an important implicit hypothesis. It is not a valid construction for an arbitrary disconnected \(H\): different components can be placed in the two red parts. The literature on Ramsey goodness makes the connectedness convention explicit. Accordingly, the substantive classification question recovered here is:

For completeness, disconnected hypergraphs can still be tested against the numerical equality, but then that equality need not arise from the claimed construction and can have accidental small-order causes.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13078 | AIM-COMBINATORICS-0028

### A sparse interval obstruction and corrected product-factor tradeoff

S / solved | AIM 研讨会问题汇编

There exists \(n,\varepsilon,\delta\) for which the following holds. Let \(A\subset\mathbb F_p\) and suppose \(|A+A|\le |A|^{1+\varepsilon}\). Then \(nA-nA\) contains a product set \(XY\) where \(|X|,|Y|\ge |A|^{1+\varepsilon}\). If \(X\) is polynomially large, how big can you make \(Y\)?

来源字段：http://aimpl.org/addcombapp/2/

13079 | AIM-COMBINATORICS-0262

### A two-sided rich core in an extremal point-line arrangement

P / partially_solved | AIM 研讨会问题汇编

Given \(n\) distinct points and \(n\) distinct lines in \(\mathbb R^2\), suppose their number of incidences is of order \(n^{4/3}\). What structural conclusions follow?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13080 | AIM-COMBINATORICS-0212

### A two-sided transfer principle for the least doubling of 3-AP-free sets

P / partially_solved | AIM 研讨会问题汇编

Given that $A$ is an $n$-element integer set free of three-term arithmetic progressions, how small can $|2A|$ be?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13081 | AIM-COMBINATORICS-0182

### A two-vertex digon counterexample and an oriented local descent theorem

S / solved | AIM 研讨会问题汇编

For any \(k\) and any digraph \(G\), every vertex of out-degree \(k\) is at directed distance at most \(k\) from a vertex in \(G^*\).

The source list declares at the outset that digraphs are finite unless explicitly stated otherwise, and defines \(N_j^+(v)\) to be the vertices at directed distance exactly \(j\) from \(v\). With \[ G^*=\{v\in V(G): |N_2^+(v)|\geq |N^+(v)|\}, \] the displayed conjecture says:

This is a conclusive counterexample to the literal statement. The surrounding problem list nevertheless suggests an intended repair to simple digraphs without loops or digons, and the motivating out-degree-one argument requires that repair. The substantive positive analysis below therefore treats finite **oriented graphs**: loopless digraphs with at most one of \(uv,vu\) present.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13082 | AIM-COMBINATORICS-0140

### A typed and scale-aware definition protocol for network curvature

P / partially_solved | AIM 研讨会问题汇编

Given a finite network, what additional structure, observation scale, output location, comparison principle, and normalization must be declared before the word *curvature* is mathematically well-defined? Can any one graph-only definition serve local and global purposes?

来源字段：http://aimpl.org/largenetworks/1/

13083 | AIM-COMBINATORICS-0024

### A variational energy core for stronger BSG

P / partially_solved | AIM 研讨会问题汇编

For a finite nonempty set \(A\) in an abelian group, put \[ E(A)=|\{(a,b,c,d)\in A^4:a+b=c+d\}|, \qquad e(A)=\frac{E(A)}{|A|^3}. \] If \(e(A)\ge 1/K\), must there be \(A'\subseteq A\) such that \[ |A'|\ge \frac{|A|}{K^{100}},\qquad |A'+A'|\le K^{100}|A'|,\qquad e(A')\ge e(A)? \] The stronger requested conclusion is \[ e(A')\ge e(A)\left(\frac{|A|}{|A'|}\right)^{1/100}. \tag{1} \] Here and below, “subset” is allowed to include the whole set. The ambient group is taken to be abelian, as in the cited BSG literature.

来源字段：http://aimpl.org/addcombapp/2/

13084 | OWR-741-008

### A Weakened Wills Inequality for Convex Bodies

O / open | Oberwolfach 报告问题汇编

Does there exist $c(n)$, depending only on $n$, such that $G(K)\leq V_n(K)+V_{n-1}(K)+c(n)\sum_{i=0}^{n-2}V_i(K)$ for every convex body $K\subset\mathbb R^n$?

来源字段：https://doi.org/10.4171/owr/2004/39

13085 | AIM-COMBINATORICS-0269

### A weighted product of chordal distances

S / solved | AIM 研讨会问题汇编

Suppose that \(r\geq 1\), that \(z_1,\ldots,z_r\) are points on the unit circle, and that \(p_1,\ldots,p_r\geq0\) satisfy \(p_1+\cdots+p_r=r\). Prove that there is a point \(z\) on the unit circle such that \[ \prod_{j=1}^{r}|z-z_j|^{p_j}\geq2. \]

Its mathematical statement, with the typography restored but without changing its content, is:

The PDF calls this a conjecture, says that \(2\) is best possible, and records the cases of equal weights and equally spaced points. It also mentions a polynomial reformulation. The source does not say that the \(z_j\) are distinct, nor does it impose a distinctness condition on the new point \(z\). Here \(r\) is necessarily a positive integer because it is the number of indexed points. In the rest of this report it is renamed \(n\).

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13086 | AIM-COMBINATORICS-0102

### Abelian hard-object growth and a greedy obstruction

P / partially_solved | AIM 研讨会问题汇编

**(12) Persi Diaconis:** “Is there an abelian growth model for non-overlapping discs growing in the plane?”

**(a) James Propp:** “There is such an abelian model in the case of non-nesting discs; firing means if you have a disc that is totally contained in another one, you move it in some fixed way.”

**(b) Anton Dochtermann:** “We could even ask if there is an abelian model of the one-dimensional case of non-overlapping intervals on a line.”

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13087 | OWR-734-012

### Abstract Graded Structures for Macaulay Inequalities

O / open | Oberwolfach 报告问题汇编

Define an abstract graded structure with a possibly ambiguous product that implies Macaulay's inequalities, and the Kruskal-Katona inequalities in exterior or squarefree cases.

来源字段：https://doi.org/10.4171/owr/2004/32

13088 | AMR-029-0001

### Achieve global rigidity by pinning nodes

O / open | AMR 问题清单汇编

Given a graph $G(V,E)$, find a minimum cardinality set $S \subset V$ of nodes such that adding a complete graph on $S$ renders the graph $G+K_S$ globally rigid in 2-dimensional space.

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13089 | AMR-029-0002

### Acyclic orientation with connectivity prescriptions

O / open | AMR 问题清单汇编

Problem 1. Given an undirected graph $\displaystyle G=(V,E)$ and $\displaystyle s,t\in V,\;\; k\in N$, decide whether the graph has an acyclic orientation, so that for every vertex $\displaystyle v\in V\setminus\{s,t\}$, there are $\displaystyle k$ pairwise edge-disjoint directed paths from $\displaystyle s$ to $\displaystyle v$, and also $\displaystyle k$ pairwise edge-disjoint directed paths from $\displaystyle v$ to $\displaystyle t$. Problem 2. Given an undirected graph $\displaystyle G=(V,E)$ and $\displaystyle s, t_1, t_2\in V,\;\; k\in N$, decide whether the graph has an acyclic orientation, so that there are $\displaystyle k$ pairwise edge-disjoint directed paths from $\displaystyle s$ to $\displaystyle t_1$, and also $\displaystyle k$ pairwise edge-disjoint directed paths from $\displaystyle s$ to $\displaystyle t_2$.

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13090 | AMR-029-0003

### Acyclic orientation with parity constraints

P / partially_solved | AMR 问题清单汇编

Problem 1. Find a good characterization for undirected graphs having an acyclic orientation so that the in-degree of every node is even. Problem 2. Find a good characterization for undirected graphs $\displaystyle G=(V,E)$ and $\displaystyle T\subseteq V$, having an acyclic orientation so that the in-degree of a node is odd iff it is in $\displaystyle T.$ Problem 3. Find a good characterization for undirected graphs $\displaystyle G=(V,E)$ which, for every possible $\displaystyle T\subsetneq V$ satisfying $\displaystyle |T|+|E|$ even, have an acyclic orientation so that the in-degree of a node is odd iff it is in $\displaystyle T.$

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13091 | GREEN-025

### Additive Complements and Cosets

P / partially_solved | Ben Green 问题

Suppose that $A \subset \mathbb{F}_2^n$ has an additive complement of size $K$. Does $2A$ contain a coset of codimension $O_K(1)$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13092 | OWR-2090-025

### Additive Coverage by Boundedly Many Bases over $\mathbb F_p$

P / partially_solved | Oberwolfach 报告问题汇编

For a prime $p$, does a constant $r=r(p)$ exist such that, for every finite-dimensional vector space $V$ over $\mathbb F_p$ and every collection of $r$ bases $B_1,\ldots,B_r$ of $V$, each vector in $V$ is a $\{0,1\}$-linear combination of vectors from those bases?

来源字段：https://doi.org/10.4171/owr/2008/44

13093 | OWR-12697684-026

### Additive Dimension of Transpositions

P / partially_solved | Oberwolfach 报告问题汇编

Let $M_n$ be the least integer such that every family of $M_n$ transpositions in $S_n$ contains pairwise distinct transpositions that can be ordered to have product equal to the identity. Determine $M_n$.

来源字段：https://doi.org/10.4171/owr/2023/1

13094 | OWR-12173-004

### Additive Sets Preserving Piecewise Syndeticity

O / open | Oberwolfach 报告问题汇编

Let $A\subset\mathbb Z$ have the property that $A+B$ is piecewise syndetic, or alternatively piecewise Bohr, whenever $B\subset\mathbb Z$ has positive upper Banach density. What can be said about $A$?

来源字段：https://doi.org/10.4171/owr/2012/50

13095 | GREEN-067

### Affine Translates of {0,1,3}

O / open | Ben Green 问题

If $A$ is a set of $n$ integers, what is the maximum number of affine translates of the set $\{0, 1, 3\}$ that $A$ can contain?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13096 | OPG-382

### Aharoni-Berger conjecture

O / open | Open Problem Garden

Conjecture If $M_1,\ldots,M_k$ are matroids on $E$ and $\sum_{i=1}^k rk_{M_i}(X_i) \ge \ell (k-1)$ for every partition $\{X_1,\ldots,X_k\}$ of $E$, then there exists $X \subseteq E$ with $|X| = \ell$ which is independent in every $M_i$.

背景链接：http://www.openproblemgarden.org/op/aharoni_berger_conjecture

13097 | OWR-734-007

### Algebraic Shifting and Homological VC Dimension

O / open | Oberwolfach 报告问题汇编

Does algebraic shifting by generic initial ideals preserve HVC dimension: if a simplicial complex $K$ has HVC dimension $d$, must its algebraic shifting also have HVC dimension $d$?

来源字段：https://doi.org/10.4171/owr/2004/32

13098 | OWR-14298158-003

### Algebraicity of Labelled Well-Quasi-Ordered Permutation Classes

O / open | Oberwolfach 报告问题汇编

Does every labelled well-quasi-ordered permutation class have an algebraic generating function?

来源字段：https://doi.org/10.4171/owr/2024/6

13099 | GREEN-056

### Almost Invariant Sets Under Affine Maps

P / partially_solved | Ben Green 问题

Suppose $A \subset \mathbb{Z}/p\mathbb{Z}$ has density $\frac{1}{2}$. Under what conditions on $K$ can $A$ be almost invariant under all maps $\phi(x) = ax + b$ with $|a|, |b| \leq K$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13100 | GREEN-008

### Almost Sum-Free Sets

P / partially_solved | Ben Green 问题

Suppose that $A \subset [N]$ has no more than $\varepsilon N^2$ solutions to $x + y = z$. Can one remove $\varepsilon' N$ elements to leave a sum-free set, where $\varepsilon' \to 0$ as $\varepsilon \to 0$, with a reasonable bound?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13101 | OWR-14298163-025

### Almost-Disjoint Geometric Triangles in Three-Space

O / open | Oberwolfach 报告问题汇编

Let $t(n)$ be the maximum number of geometric triangles in $\mathbb R^3$ on an $n$-point vertex set such that two triangles are either disjoint or meet in one common vertex. Prove or disprove that $t(n)=o(n^2)$.

来源字段：https://doi.org/10.4171/owr/2024/3

13102 | AIM-COMBINATORICS-0147

### An alternating flat family for the complete 3-graph lower bound

P / partially_solved | AIM 研讨会问题汇编

Its exact mathematical assertion is

$$ \pi(K_m^3)=1-\left(\frac{2}{m-1}\right)^2. \tag{1} $$

The record remarks that Sidorenko gives a construction achieving the lower bound and that Mubayi and Keevash found another construction via digraphs.

For a 3-uniform hypergraph $F$, let $\operatorname{ex}(n,F)$ be the maximum number of triples in an $F$-free 3-graph and set

$$ \pi(F)=\lim_{n\to\infty}\frac{\operatorname{ex}(n,F)}{\binom n3}. $$

Sidorenko uses the complementary covering density $t(m,3)$, the limiting minimum fraction of triples required to meet every $m$-vertex set. Thus $\pi(K_m^3)=1-t(m,3)$, and (1) is equivalent to

$$ t(m,3)=\frac{4}{(m-1)^2}. $$

Sidorenko writes this as $t(\alpha+1,3)=(2/\alpha)^2$, while Keevash writes $\pi(K_{t+1}^3)=1-(2/t)^2$. The AIM record omits a quantifier on $m$. The nontrivial intended reading is: **for every integer $m\geq4$**. The case $m=3$ is tautologically compatible but is not the subject of this complete-hypergraph conjecture.

来源字段：http://aimpl.org/hypergraphturan/1/

13103 | AIM-COMBINATORICS-0229

### An anchor-container reduction for imbalanced restricted sumsets

P / partially_solved | AIM 研讨会问题汇编

Suppose that \(A\) and \(B\) are finite sets of integers with \(|A|=m\) and \(|B|=n\), where \(m>n\), and suppose that \(G\subseteq A\times B\) has size at least \(\delta mn\). Further suppose that \[ \bigl|\{a+b:a\in A,\ b\in B,\ (a,b)\in G\}\bigr|<Cm. \] Must there exist \(A'\subseteq A\), \(B'\subseteq B\), with \(|A'|\ge cm\), \(|B'|\ge cn\), such that \(|A'+B'|\le Km\), where \(c,K\) depend only on \(\delta,C\)?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13104 | AIM-COMBINATORICS-0171

### An arc--nonedge defect identity for Sullivan-1

P / partially_solved | AIM 研讨会问题汇编

Conjecture 6.6. ("Compromise Conjecture") Any simple digraph with no loops or digons has a vertex v such that |N +2 (v)| ≥ | N -(v)|.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13105 | AIM-COMBINATORICS-0124

### An arrival-gated interpolation from IDLA to the abelian sandpile

P / partially_solved | AIM 研讨会问题汇编

What does a stochastic sandpile model that interpolates between DLA [16] and the sandpile model look like?

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13106 | AMR-030-0074

### An asymmetric covering code of radius R is a set of binary n-words so that every binary n-word can be reached from one o

O / open | AMR 问题清单汇编

/Ellis/Kahng: An asymmetric covering code of radius R is a set of binary n-words so that every binary n-word can be reached from one of the codewords by changing at most R zeroes to ones. How big is the smallest asymmetric covering code of radius R with n bits? It is known to be a constant c(R) times 2^(R )/ n^(R), but it's not known what c(R) is. Some believe that c(R) = 2^(R) R!.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13107 | AMR-030-0075

### An asymmetric packing code of radius R is a set of binary n-words so that no binary n-word can be reached from more than

O / open | AMR 问题清单汇编

/Ellis/Kahng: An asymmetric packing code of radius R is a set of binary n-words so that no binary n-word can be reached from more than one of the codewords by changing at most R zeroes to ones. How big is the largest asymmetric covering code of radius R with n bits?

背景链接：https://people.math.sc.edu/cooper/combprob.html

13108 | AIM-COMBINATORICS-0286

### An existential NP class for P4-structures of M-join graphs

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.** Find a class \(\mathbf C_2\) of 4-uniform hypergraphs such that (i) the \(P_4\)-structure of every graph with an M-join belongs to \(\mathbf C_2\), (ii) no odd ring belongs to \(\mathbf C_2\), and (iii) \(\mathbf C_2\) belongs to NP.

来源字段：https://aimath.org/WWN/perfectgraph/perfectgraph.pdf

13109 | AIM-COMBINATORICS-0087

### An explicit 94-vertex cubic-residue candidate and certificate reduction

P / partially_solved | AIM 研讨会问题汇编

A Folkman graph is a $K_4$-free graph such that every 2-edge-coloring contains a monochromatic triangle. Challenge ($100): find a Folkman graph with fewer than 100 vertices. Candidate construction (Exoo): take $\mathbb Z_{127}$ as vertices, join $i$ to $j$ when $i-j$ is a cubic residue modulo $127$, and delete 33 vertices (three independent sets of size 11).

The phrase “three independent sets of size 11” does not itself specify the deleted vertices. Below it is reconstructed canonically from the triangle $\{0,1,2\}$: delete the three common-neighbor sets belonging to its three edges. This is an explicit realization consistent with the source wording, but I did not verify from the inaccessible original Overleaf history that these were Exoo’s intended three sets.

Write \[ H=\{x^3:x\in\mathbb F_{127}^{\times}\},\qquad i\sim j\quad\Longleftrightarrow\quad i-j\in H, \] and call the resulting Cayley graph $G_{127}$. Define \[ D=H\cap(1+H)=\{t\in H:t-1\in H\}. \] The explicit deletion set considered here is \[ U=D\cup 2D\cup(1+D),\qquad X=G_{127}[\mathbb F_{127}\setminus U]. \]

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13110 | AIM-COMBINATORICS-0238

### An explicit finite-field two-source expander

S / solved | AIM 研讨会问题汇编

**Problem 3.1 (J. Bourgain).** Find explicitly a function \(f:\mathbb F_p\times\mathbb F_p\to\mathbb F_p\) such that for every \(A,B\subseteq\mathbb F_p\) with \(|A|,|B|\sim p^{1/2}\) we have \[ |f(A\times B)|\geq p^{1/2+\epsilon}. \]

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13111 | AIM-COMBINATORICS-0214

### An explicit local-lemma certificate and bounded high-overlap neighbourhood for van der Waerden hypergraphs

P / partially_solved | AIM 研讨会问题汇编

Let \(W(k)\) be the least \(N\) such that every red/blue colouring of \([N]=\{1,\ldots,N\}\) has a monochromatic arithmetic progression of length \(k\). What is the true order of growth of \(W(k)\)? Graham offered \(\$1000\) for a proof that \[ W(k)\leq 2^{k^2}. \]

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13112 | AIM-COMBINATORICS-0265

### An incomplete sums-differences prompt and an exact slope-height obstruction

O / open | AIM 研讨会问题汇编

\(SD(r_1,\ldots,r_n;\alpha)\), for some \(r_i\in\mathbb R\).

For how small an exponent \(\alpha\) can one find finitely many slopes \(r_1,\ldots,r_n\) such that \(SD(r_1,\ldots,r_n;\alpha)\) holds?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13113 | AIM-COMBINATORICS-0224

### An odd-order five-coset counterexample above the one-third threshold

P / partially_solved | AIM 研讨会问题汇编

1

1, d 2 = a′

2

2

1

2, and it follows that d1 - d2 = a′

1

2

<pre> Problem 1.18 (V. Lev). Let A be a finite non-empty subset of an abelian group G, and write D:= A-A. If any d ∈ D has strictly more than |A|/2 representations of the form

d = a′ - a″ with a′, a ″ ∈ A, then D is a subgroup: indeed, by the pigeonhole principle for any d1, d 2 ∈ D there exists a pair of representations d1 = a′

∈ D.Assume now that any d ∈ D is only guaranteed to have at least |A|/2 representations as d = a′ -a″ with a′, a ″ ∈ A. In this case the argument above doesn't work, and in fact, the conclusion is not true either. To see this, consider the set A:= H ∪ (g + H), where PALO ALTO PROBLEMS 5

H < G is a finite subgroup and g ∈ G is so chosen that the order of g in the quotient group G/H is at least five. Then D = ( -g + H) ∪ H ∪ (g + H) is not a subgroup, but a union of three cosets. At the same time, it is easily seen that any d ∈ D has at least

|H| = |A|/2 representations of the form d = a′ - a″.Is this example unique? In other words, given that any d ∈ D has at least |A|/2representations as d = a′ - a″, is it necessarily true that D is either a subgroup or a union of three cosets? For practical applications one should go somewhat beyond the |A|/2 bound. Problem: assuming that any d ∈ D:= A - A has more than |A|/3representations of the form d = a′ - a″ with a′, a ″ ∈ A, is it necessarily true that D is either a subgroup or a union of three cosets? </pre>

Writing \[ r_A(d):=\bigl|\{(a',a'')\in A^2:a'-a''=d\}\bigr| =|A\cap(A+d)|, \] the final question is whether \[ \min_{d\in A-A}r_A(d)>\frac{|A|}{3} \tag{1.1} \] forces \(A-A\) to be a subgroup or a union of at most three cosets of a common subgroup. The answer is no, even in a cyclic group of odd order.

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13114 | AIM-COMBINATORICS-0201

### An order-statistic form of Shen's deficit conjecture

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.31 (Shen).** For a digraph \(G\) on \(n\) vertices of girth \(g\), define \[ t(G,r)=\sum_{u:\,\delta_G^+(u)<r} \bigl(r-\delta_G^+(u)\bigr). \] If \(\delta_G^+\ge1\), then \[ n\ge r(g-1)+1-t(G,r). \]

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13115 | AIM-COMBINATORICS-0109

### An orientation-double-cover sandpile across all cell dimensions

P / partially_solved | AIM 研讨会问题汇编

Andrea Sportiello and Jordan Ellenberg: "Can we develop a chip-firing model for cell complexes that includes chips on cells of different dimensions and features interaction between dimensions? Probably, chip configurations and firing in lower dimensions influence what can happen in higher dimensions." (a) Gregg Musiker: "Is there a Chow ring lurking here?"

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13116 | AMR-029-0004

### Are t-perfect graphs strongly t-perfect?

O / open | AMR 问题清单汇编

Is it true that every t-perfect graph is strongly t-perfect?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13117 | AMR-029-0005

### Are there deletion-contraction formulas for the polymatroid Tutte polynomial?

S / solved | AMR 问题清单汇编

Are there deletion-contraction formulas for the polymatroid Tutte polynomial?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13118 | AIM-COMBINATORICS-0020

### Arithmetic removal in cyclic groups: corrected quantifiers, matching distances, and exact CRT-compatible lifts

P / partially_solved | AIM 研讨会问题汇编

For every \(\varepsilon>0\), find the best order of a number \(\delta(\varepsilon)>0\), independent of \(N\), such that whenever \(G=\mathbb Z/N\mathbb Z\) and \(A,B,C\subseteq G\) span at most \(\delta(\varepsilon)N^2\) ordered zero-sum triples, one can delete at most \(\varepsilon N\) memberships from **each colored set** and leave no zero-sum triple.

来源字段：http://aimpl.org/addcombapp/1/

13119 | OWR-17472-019

### Arm-Length Refinements for Complementary Partitions

O / open | Oberwolfach 报告问题汇编

Find a representation-theoretic interpretation of the stronger arm-length-refined multiset identity for complementary partitions in a rectangle.

来源字段：https://doi.org/10.4171/owr/2020/9

13120 | OWR-14604-016

### Asymptotic Enumeration of Arithmetic-Progression-Free Sets

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Progression-Free-Set Enumeration Question [OWR-14604-015]

For fixed $k\geq3$, is the number of $k$-term-AP-free subsets of $[n]$ equal to $2^{(1+o(1))r_k(n)}$ as $n\to\infty$, where $r_k(n)$ is the maximum size of a $k$-term-AP-free subset of $[n]$?

来源字段：https://doi.org/10.4171/owr/2016/22

13121 | OWR-11136-010

### Asymptotic Enumeration of Higher-Dimensional Permutations

P / partially_solved | Oberwolfach 报告问题汇编

别名：Lower Bounds for Higher-Dimensional Permutation Counts [OWR-11136-011]

For fixed d, let $S_{d,n}$ be the d-dimensional permutations of order n. Is $|S_{d,n}|=\left(n/e^d+o(n)\right)^{n^d}$ as $n\to\infty$?

来源字段：https://doi.org/10.4171/owr/2011/44

13122 | OWR-14298163-014

### Asymptotics of Higher-Dimensional General-Position Sets

O / open | Oberwolfach 报告问题汇编

For fixed $d$ and $k$, determine the asymptotic order of $\alpha_{d,k}(n)$, the maximum size of a subset of $[n]^d$ containing no $k+2$ points in a $k$-dimensional affine subspace. In particular, determine $\alpha_{4,2}(n)$.

来源字段：https://doi.org/10.4171/owr/2024/3

13123 | OWR-11136-007

### Asymptotics of Uniform Union-Free Families

O / open | Oberwolfach 报告问题汇编

Given $r\geq t\geq1$, determine the asymptotic behavior of $\operatorname{UF}_t(n,r)$, the maximum size of a $t$-union-free $r$-uniform family on an $n$-element ground set.

来源字段：https://doi.org/10.4171/owr/2011/44

13124 | AIM-COMBINATORICS-0222

### Balanced power-sum obstructions and the one-sign Behrend family

P / partially_solved | AIM 研讨会问题汇编

Wooley), asks whether Behrend's construction can be generalized to give large sets \(S\subset [N]\) having no solutions of \[ \sum_{i=1}^{s}a_i x_i=0, \qquad \sum_{i=1}^{s}a_i=0, \qquad |a_i|<A, \] and asks the analogous question for \[ \sum_{i=1}^{t}a_i x_i^2=0. \] The PDF then records N. Alon's two negative examples: coefficients \(1,1,-1,-1\) force a Sidon-type \(O(\sqrt N)\) bound, while fifty \(+1\)'s and fifty \(-1\)'s in the quadratic equation force \(O(N^{1/25})\). It also records that the linear construction works when only one coefficient has one sign, and points to Ruzsa's Acta Arithmetica papers.

The following source issues matter mathematically.

Thus this report treats the prompt as a family/classification question rather than a single yes/no proposition.

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13125 | OWR-17464-021

### Balanced Three-Colorings of Six-Uniform Hypergraphs

O / open | Oberwolfach 报告问题汇编

Does every 6-regular 6-uniform hypergraph have a 3-coloring of its vertices such that no hyperedge contains four vertices of the same color?

来源字段：https://doi.org/10.4171/owr/2020/1

13126 | OWR-1455-006

### Ball Conjecture for Twisted Bruhat Intervals

O / open | Oberwolfach 报告问题汇编

Let $(W,S)$ be a Coxeter system and let $\theta$ be an involutive automorphism such that $s\theta(s)$ never has odd order for $s\in S$ unless $\theta(s)=s$. Is every interval in $\operatorname{Br}(\iota(\theta))$ either a PL sphere or a ball?

来源字段：https://doi.org/10.4171/owr/2007/04

13127 | OPG-369

### Bases of many weights

O / open | Open Problem Garden

Let $G$ be an (additive) abelian group, and for every $S \subseteq G$ let ${\mathit stab}(S) = \{ g \in G: g + S = S \}$.

Conjecture Let $M$ be a matroid on $E$, let $w: E \rightarrow G$ be a map, put $S = \{ \sum_{b \in B} w(b): B \mbox{ is a base} \}$ and $H = {\mathit stab}(S)$. Then $$|S| \ge |H| \left( 1 - rk(M) + \sum_{Q \in G/H} rk(w^{-1}(Q)) \right).$$

背景链接：http://www.openproblemgarden.org/op/bases_of_many_weights

13128 | OPG-37181

### Beneš Conjecture

P / partially_solved | Open Problem Garden

别名：Beneš Conjecture (graph-theoretic form) [OPG-37210]

Let $E$ be a non-empty finite set. Given a partition $\bf h$ of $E$, the stabilizer of $\bf h$, denoted $S(\bf h)$, is the group formed by all permutations of $E$ preserving each block of $\mathbf h$.

Problem ( $\star$ ) Find a sufficient condition for a sequence of partitions ${\bf h}_1, \dots, {\bf h}_\ell$ of $E$ to be complete, i.e. such that the product of their stabilizers $S({\bf h}_1) S({\bf h}_2) \dots S({\bf h}_\ell)$ is equal to the whole symmetric group $\frak S(E)$ on $E$. In particular, what about completeness of the sequence $\bf h,\delta(\bf h),\dots,\delta^{\ell-1}(\bf h)$, given a partition $\bf h$ of $E$ and a permutation $\delta$ of $E$?

Conjecture (Beneš) Let $\bf u$ be a uniform partition of $E$ and $\varphi$ be a permutation of $E$ such that $\bf u\wedge\varphi(\bf u)=\bf 0$. Suppose that the set $\big(\varphi S({\bf u})\big)^{n}$ is transitive, for some integer $n\ge2$. Then $$\frak S(E) = \big(\varphi S({\bf u})\big)^{2n-1}.$$

背景链接：http://www.openproblemgarden.org/op/bene_conjecture

13129 | AMR-029-0006

### Berge's conjecture on path partitions

P / partially_solved | AMR 问题清单汇编

Let D be a digraph without loops and k a positive integer. For a partition $\Pi$ of V(D) into directed paths (a path partition) let $|\Pi|_k=\sum_{P \in \Pi}\min\{|P|,k\},$ where $|P|$ is the number of nodes of the path (which can be 1). A partial k-colouring is the union of k stable sets. Is it true that for every path partition $\Pi$ minimizing $|\Pi|_k$, there exists a partial k-colouring which meets each path $P\in \Pi$ in $\min\{|P|,k\}$ stable sets?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13130 | OWR-734-009

### Betti Bounds and Nerve HVC Dimension

O / open | Oberwolfach 报告问题汇编

For fixed $A>0$, if every intersection $L$ of $m$ members of a family $\mathcal M$ of subsets of $\mathbb R^d$ satisfies $b(L)\leq A\binom{m}{d}$, must the nerve of $\mathcal M$ have HVC dimension $d$?

来源字段：https://doi.org/10.4171/owr/2004/32

13131 | AMR-028-0001

### Betti Posets and the Stanley Depth

O / open | AMR 问题清单汇编

The Betti poset of a monomial ideal $I$ determines the Stanley projective dimension of $S/I$ and $I$. More precisely, if $I\subseteq S$ and $I'\subseteq S'$ are monomial ideals in polynomial rings $S$ and $S'$ with $\mathcal{B}(I)\cong\mathcal{B}(I')$, then $\operatorname{spdim}_{S}(S/I)=\operatorname{spdim}_{S'}(S'/I')$ and $\operatorname{spdim}_{S}I=\operatorname{spdim}_{S'}I'$.

背景链接：https://arxiv.org/abs/1509.08275

13132 | OWR-12697684-028

### Biased Sub-Balanced Slices of Boolean Functions

O / open | Oberwolfach 报告问题汇编

For every indicator function $f$ on the balanced slice with $\mathbb E f=1/2$, must there be a sub-balanced slice of dimension $\log^*n$, or at least of dimension tending to infinity, on which $\mathbb E f>0.51$?

来源字段：https://doi.org/10.4171/owr/2023/1

13133 | OWR-1536-001

### Biased Weak-Win Games on Sparse Hypergraphs

O / open | Oberwolfach 报告问题汇编

Can Beck's weak-win Meta-Conjecture for generalized tic-tac-toe on finite hypergraphs be extended to biased games? When the underlying hypergraph is sparse, what positive results remain possible after complete occupation is replaced by an appropriate majority or surplus objective?

来源字段：https://doi.org/10.4171/owr/2007/20

13134 | OWR-16164-009

### Bijection Between Eulerian Orientations and Bicolored Trees

O / open | Oberwolfach 报告问题汇编

Find a bijective proof that rooted planar Eulerian orientations are equinumerous with the corresponding class of bicolored trees.

来源字段：https://doi.org/10.4171/owr/2018/23

13135 | OWR-11695867-014

### Bijection Between Symmetric Walk Families

O / open | Oberwolfach 报告问题汇编

Find a bijection between the symmetric walk families $X(\widehat S)$ and $Y(\widehat S)$ for every multiset $\widehat S$ of small steps.

来源字段：https://doi.org/10.4171/owr/2022/57

13136 | OWR-11695867-010

### Bijective Proof of a Hurwitz Number Identity

O / open | Oberwolfach 报告问题汇编

Let $H_{n,\ell}$ count ordered $\ell$-tuples of transpositions in $S_n$ whose product is the identity. For $\lambda\vdash n$, let $f_\lambda$ count standard Young tableaux and let $C_\lambda=\sum_{(i,j)\in\lambda}(j-i)$. Give a bijective proof, for all $\ell\geq 0$, of $n!H_{n,\ell}=\sum_{\lambda\vdash n}f_\lambda^2 C_\lambda^\ell$.

来源字段：https://doi.org/10.4171/owr/2022/57

13137 | AMR-029-0007

### Binary matroid representation of cyclic families

P / partially_solved | AMR 问题清单汇编

Let $B=\{b_1,b_2,\ldots ,b_k\}\subset\{0,1,\ldots ,n-1\}$, and let $B_i=\{b_1+i, b_2+i,\ldots, b_k+i\}$ where addition is modulo n. That is, ${\mathcal B}=\{B_1,B_2,\ldots B_N\}$ is the collection of cyclic translates of B. Prove that there exists a binary matroid M such that every $B_i$ is a basis of M.

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13138 | AIM-COMBINATORICS-0277

### Bisimplicial vertices and protected elimination in even-hole-free graphs

S / solved | AIM 研讨会问题汇编

**Recovered Hayward--Reed conjecture.** Every finite simple non-null even-hole-free graph has a vertex \(v\) such that \(N(v)\) is the disjoint union of at most two cliques.

来源字段：https://aimath.org/WWN/perfectgraph/perfectgraph.pdf

13139 | OWR-11695867-008

### Borwein Sign Conjectures

P / partially_solved | Oberwolfach 报告问题汇编

Do the Second and Cubic Borwein conjectures hold: for every positive integer $n$, do the coefficients of $P_n(q)^2$ and $P_n(q)^3$ have the repeating sign pattern $+,-,-$?

来源字段：https://doi.org/10.4171/owr/2022/57

13140 | AMR-029-0008

### Bounded degree matroid basis

O / open | AMR 问题清单汇编

Let M be a matroid on ground set V, let H=(V,E) be a hypergraph with maximum degree $\Delta$, let c(v) be the cost of node v, and let $l(e) \leq u(e)$ ($e \in E$) be lower and upper bounds on the hyperedges. Let OPT denote the minimum cost of a basis B of M which satisfies $l(e) \leq |B \cap e| \leq u(e)$ for every $e \in E$. Is there a polynomial algorithm to find a basis B such that $c(B) \leq OPT$ and $l(e)-\Delta+1 \leq |B \cap e| \leq u(e)+\Delta-1$?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13141 | AIM-COMBINATORICS-0225

### Bourgain's arbitrary-modulus theorem answers Chang's sum-product question

S / solved | AIM 研讨会问题汇编

**Problem 1.19 (M.-C. Chang).** Is it true that for any \(\varepsilon>0\) there exists \(\varepsilon'>0\) with the following property: if \(A\subseteq\mathbb Z/q\mathbb Z\) (with a sufficiently large integer \(q\)) satisfies \[ |A+A|+|A\cdot A|<q^\varepsilon, \] then either \(|A|>q^{1-\varepsilon'}\), or there exists \(q_1\mid q\), \(q_1>1\), such that the canonical image of \(A\) in \(\mathbb Z/q_1\mathbb Z\) has at most \(q^{\varepsilon'}\) elements?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13142 | AIM-COMBINATORICS-0249

### Bourgain's three-dimensional rich-line problem

S / solved | AIM 研讨会问题汇编

**Problem 3.12 (J. Bourgain).** Consider the Szemerédi--Trotter theorem in \(\mathbb R^3\) with \(n^2\) lines, each containing \(n\) points from a given set \(S\). Assume no \(n\) lines are coplanar. Find a lower bound for \(S\) (e.g. \(|S|\ge n^{3-\varepsilon}\)).

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13143 | OWR-15206-010

### Box Discrepancy of Higher-Dimensional Permutations

O / open | Oberwolfach 报告问题汇编

For every fixed $d\geq1$, do there exist $d$-dimensional permutations $X\subseteq[n]^{d+1}$ such that every axis-parallel box $B\subseteq[n]^{d+1}$ satisfies $$ \left||X\cap B|-\frac{\operatorname{vol}(B)}{n}\right|=O\!\left(\sqrt{\operatorname{vol}(B)}\right) $$?

来源字段：https://doi.org/10.4171/owr/2017/1

13144 | AIM-COMBINATORICS-0069

### Burr--Erdos, skeletal degeneracy, and a sharp hereditary edge envelope

S / solved | AIM 研讨会问题汇编

Conjecture: (Burr--Erdős) $r(G,G) \le c(d)n$ if $G$ is an $n$-vertex $d$-degenerate graph.

Known: (Kostochka--Sudakov) $r(G,G) \le n^{1+o_d(1)}$.

If $G$ is a 3-uniform $d$-degenerate hypergraph, the statement can be false. There is a 1-degenerate 4-uniform $G$ such that $r_4(G,G) \ge 2^{\Omega(n^{1/3})}$.

Perhaps we can redefine degeneracy for $k$-uniform hypergraphs: call a $k$-uniform hypergraph $d$-degenerate if the 2-graph obtained by replacing every edge by a graph clique $K_k^{(2)}$ is $d$-degenerate as a graph.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13145 | AIM-COMBINATORICS-0135

### Calibrating curvature measurements on empirical networks

P / partially_solved | AIM 研讨会问题汇编

**11.05.** “How do we measure the curvature of real networks?”

来源字段：http://aimpl.org/largenetworks/1/

13146 | AIM-COMBINATORICS-0240

### Cancellation thresholds and a prime-repunit obstruction for multiplicative subgroups

P / partially_solved | AIM 研讨会问题汇编

Let \(H\leq \mathbb F_p^*\) be a multiplicative subgroup. How large must \(H\) be in order that

\[ \left|\sum_{x\in H}e_p(ax)\right|=o(|H|) \tag{1} \]

hold for all \(a\neq0\)?

Here the standard notation is

\[ e_p(t)=\exp(2\pi i t/p). \]

The natural uniform interpretation of (1), and the one used in the literature, is that along a family of primes \(p\to\infty\) and eligible subgroups \(H=H_p\),

\[ M(H):=\max_{a\in\mathbb F_p^*} \left|\sum_{x\in H}e_p(ax)\right|=o(|H|). \tag{2} \]

The notation \(H\leq\mathbb F_p^*\), rather than \(H\subseteq\mathbb F_p^*\), confirms that \(H\) is a subgroup. No change to the mathematical statement beyond restoring the PDF notation is made here.

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13147 | AIM-COMBINATORICS-0261

### Cantor-direction Kakeya sets and a Hölder graph obstruction

P / partially_solved | AIM 研讨会问题汇编

Let \(C\) be a triadic Cantor set. Does there exist \(E\subset \mathbb R^2\) with the following properties: 1) \(E\) is the union of a 1-D family of unit line segments whose slopes are in \(C\); 2) the Lebesgue measure of \(E\) is 0. 3) the union of the doubles of the above line segments has positive Lebesgue measure.

Katz, in the AIM workshop list *Recent trends in additive combinatorics*. The official PDF reads:

The source does not define “triadic Cantor set,” “1-D family,” or “double.” The later paper of Bateman and Katz [BK] uses almost exactly the phrase “a ‘1 dimensional’ family of unit line segments” and explicitly calls that description loose.

For the precise results below, let

\[ C=\left\{\sum_{k=1}^{\infty}\frac{2\varepsilon_k}{3^k}:\varepsilon_k\in\{0,1\}\right\}\subset[0,1] \]

be the standard middle-thirds Cantor set and let

\[ u(m)=\frac{(1,m)}{\sqrt{1+m^2}} \]

be the Euclidean unit direction of slope \(m\). If \(I\) is a segment with midpoint \(c_I\), its centered \(\lambda\)-dilate is

\[ \lambda I=\{c_I+\lambda(x-c_I):x\in I\}. \]

In particular, \(2I\) has twice the length and the same midpoint and direction. Kroc--Pramanik [KP, Definition 1.1] use precisely the analogous tube convention: same center, orientation, and cross-sectional radius, but multiplied axial length.

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13148 | GREEN-080

### Cap Sets in F_7^n

P / partially_solved | Ben Green 问题

What is the largest subset $A \subset \mathbb{F}_7^n$ for which $A - A$ intersects $\{-1, 0, 1\}^n$ only at 0?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13149 | AIM-COMBINATORICS-0213

### Cap-set status and a sharp quadratic-graph dimension barrier

P / partially_solved | AIM 研讨会问题汇编

For a positive integer \(r\), what is the largest possible size of a subset \(A\subseteq\mathbb F_3^r\) containing no three points on a line?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13150 | AMR-029-0009

### Capacitated packing of k-arborescences

O / open | AMR 问题清单汇编

Let D=(V,A) be a digraph with arc-capacities $c : A \to \mathbb{N}$ and a root node $r_0\in V$. A k-arborescence is the arc-disjoint union of k spanning arborescences rooted at $r_0$. Is it true that if $c(a)\leq l$ for every $a \in A$ and there is a capacity-obeying packing of kl spanning arborescences, then there is a capacity-obeying packing of l k-arborescences?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13151 | AIM-COMBINATORICS-0126

### Cartwright local matrices as link walks with potential

P / partially_solved | AIM 研讨会问题汇编

(36) Lionel Levine: “The graph Laplacian is the generator of a random walk on a graph. Does the Laplacian of a simplicial complex generate some kind of stochastic process on this higher-dimensional object?” (a) Caroline Klivans: “Look up the work of Russel Lyons [24].” (b) Lionel Levine: “I have and it is interesting but not what I want.” (c) Andrea Sportiello: “Dustin Cartwright's model [13] looks more like a random walk with the way it assigns weights.” (d) Matt Baker: “So the concrete question is: does Dustin's model generate some stochastic process?”

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13152 | OWR-16164-020

### Catalan-Power Divisibility of a Set-Partition Transition Determinant

O / open | Oberwolfach 报告问题汇编

Let $A=(a_{ij})_{1\leq i,j\leq n}$. For set partitions $P$ of $X=\{x_1,\ldots,x_n\}$ and $Q$ of $Y=\{y_1,\ldots,y_n\}$, call a set $S$ of edges $x_i y_j$ a transition from $P$ to $Q$ if adjoining $S$ to spanning trees on the blocks of $P$ produces a spanning forest of $X\cup Y$ in which every component meets $Y$ and the induced partition of $Y$ is $Q$. Give $S$ weight $\prod_{x_i y_j\in S}a_{ij}$, and define the Bell-number-sized matrix $T$ by letting $T_{P,Q}$ be the sum of the weights of all transitions from $P$ to $Q$. Is $(\det A)^{C_n}$ always a factor of $\det T$, where $$C_n=\frac{1}{n+1}\binom{2n}{n}$$ is the $n$-th Catalan number?

来源字段：https://doi.org/10.4171/owr/2018/23

13153 | AIM-COMBINATORICS-0217

### Ceiling-aware bounds and a carry-free density Hales-Jewett construction

P / partially_solved | AIM 研讨会问题汇编

{ "id": "AIM-COMBINATORICS-0217", "source": "AIM Problem Lists", "source_type": "PDF", "domain": "Combinatorics", "domain_slug": "combinatorics", "workshop": "Recent trends in additive combinatorics", "workshop_slug": "additivecomb", "section": "", "number": "1.11", "tag": "problem", "problem": "Problem 1.11 (R. Graham).

It does **not** explicitly state an ambient set for \(S_{\alpha,k}\). Problem 1.9 on the same page explicitly writes a set as a subset of \(\mathbb Z\), and Problems 1.8, 1.10, and 1.12 also use integer lattices or \(\mathbb Z\). I therefore take finite subsets of \(\mathbb Z\) as the intended reading, but label this as a reconstruction rather than source text.

The results below are robust to this ambiguity. Lower bounds are proved for arbitrary finite subsets of \(\mathbb R\), while all upper constructions lie in \(\mathbb Z_{>0}\). A nontrivial \(j\)-term arithmetic progression means \[ \{a,a+d,\ldots,a+(j-1)d\},\qquad d>0, \] with distinct terms.

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13154 | AIM-COMBINATORICS-0031

### Centered affine-fibre density increments for almost containment in A+A

P / partially_solved | AIM 研讨会问题汇编

Let \(A\subset \mathbb F_2^n\) be a set of density \(\alpha\). Then there is a subspace \(V\) of codimension \(\log^{O(1)}(\varepsilon^{-1}\alpha^{-1})\) so that \[ |V\cap(A+A)|\geq (1-\varepsilon)|V|. \]

Although written declaratively, comparison with O'Donnell's 2012 open-problem list and Green's updated open-problem list shows that this is intended as a conjecture/problem.

We take \(0<\alpha\leq 1\) and \(0<\varepsilon<1\). The cases \(\varepsilon\geq1\) and \(\alpha=1\) are immediate. The case \(\alpha=0\) is outside the natural formulation, since its claimed bound involves \(\alpha^{-1}\).

来源字段：http://aimpl.org/addcombapp/2/

13155 | OWR-17472-014

### Chamber Structures of Schur-Positive Multiplicity Functions

O / open | Oberwolfach 报告问题汇编

Characterize the multiplicity functions $m_F$ arising from Schur-positive symmetric functions, including their piecewise-quasipolynomial chamber structures and the behavior of $m_F(\omega_0+k\omega)$.

来源字段：https://doi.org/10.4171/owr/2020/9

13156 | AMR-029-0010

### Changing conservative weightings in bipartite graphs

O / open | AMR 问题清单汇编

Let G=(A,B;E) be a bipartite graph, and $w:E \to \{1,-1\}$ a conservative weighting. Can we determine in polynomial time the maximum number of positive edges that can be negated so that the weighting remains conservative?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13157 | AIM-COMBINATORICS-0188

### Charbit's spectral short-cycle conjecture: the sharp digon case and a Perron-period obstruction

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.18 (Charbit).** Let \(A\) be the adjacency matrix of a digraph \(G\), with zero diagonal and \(a_{ij}=1\) if and only if \((i,j)\in E(G)\). If the spectral radius of \(A\) is at least \(n/k\), then \(G\) has a cycle of length at most \(k\).

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13158 | AMR-029-0011

### Chromatic number of t-perfect graphs

O / open | AMR 问题清单汇编

Is every t-perfect graph 4-colourable?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13159 | OWR-12861-022

### Chromatic Uniqueness of Nonrepresentable Matroids

O / open | Oberwolfach 报告问题汇编

Does there exist a matroid whose chromatic polynomial is not the chromatic polynomial of any matroid representable over a field; does there exist a matroid not representable over any field that is chromatically unique; and how small can such a matroid be?

来源字段：https://doi.org/10.4171/owr/2014/01

13160 | OWR-16633-021

### Clique-Sequence Rank and Generic Rigidity Matroids

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Clique-Sequence Submodularity Conjecture [OWR-16633-022]

For $F\subseteq E(K_n)$, define

$$r_t(F)=\min_{(C_1,\ldots,C_k)}\left(\left\lvert F\cup\bigcup_{i=1}^kC_i\right\rvert-k\right),$$

where the minimum ranges over proper $K_t$-sequences in $K_n$. Is $r_t$ submodular, so that $(E(K_n),r_t)$ is a matroid? In particular, is $(E(K_n),r_5)$ the generic $3$-rigidity matroid?

来源字段：https://doi.org/10.4171/owr/2018/50

13161 | OWR-15987-007

### Closed Formulas for Braid-Matroid Kazhdan-Lusztig Polynomials

P / partially_solved | Oberwolfach 报告问题汇编

Find a simple closed formula for the Kazhdan-Lusztig polynomials of braid matroids.

来源字段：https://doi.org/10.4171/owr/2018/2

13162 | OWR-11695867-013

### Closed Sets Supporting Permutons

P / partially_solved | Oberwolfach 报告问题汇编

Which closed subsets of $[0,1]^2$ can occur as the support of a permuton?

来源字段：https://doi.org/10.4171/owr/2022/57

13163 | OWR-1455-007

### Clustered Colorings of Triangulated Grids

P / partially_solved | Oberwolfach 报告问题汇编

For a $d$-dimensional triangulated grid $T_n^d$, let $\xi_k(G)$ be the least $m$ for which the vertices of $G$ can be colored with $k$ colors so that every monochromatic connected subgraph has at most $m$ vertices, and let $L_n^d(i)=\{x\in[n]^d:\sum_jx_j=i\}$. Is the worst-case value of $\xi_2(T_n^d)$ equal to $|L_n^d(i_{1/2})|$, making the diagonal-layer coloring optimal? For fixed $d$ and $3\leq k\leq d-1$, is $\xi_k(T_n^d)$ of order $n^{d-k+1}$?

来源字段：https://doi.org/10.4171/owr/2007/04

13164 | OWR-17135-011

### Cocircuit Graph Diameter Conjecture for Oriented Matroids

O / open | Oberwolfach 报告问题汇编

If $M$ is an oriented matroid of rank $r$ on $n$ elements, must the diameter of its cocircuit graph satisfy $$\operatorname{diam}(G^*(M))\leq n-r+2$$?

来源字段：https://doi.org/10.4171/owr/2019/39

13165 | OWR-1782-009

### Codegree Thresholds for Tight Hypergraph Hamilton Cycles

P / partially_solved | Oberwolfach 报告问题汇编

别名：Codegree Thresholds for Hypergraph Hamilton Cycles [OWR-1386-013]

Let $H$ be a $k$-uniform hypergraph on $n$ vertices with $n\geq k+1\geq4$. If every $(k-1)$-element vertex set is contained in at least $\lfloor(n-k+3)/2\rfloor$ edges, must $H$ contain a tight Hamiltonian cycle?

来源字段：https://doi.org/10.4171/owr/2008/01

13166 | OWR-15987-008

### Coefficient Complexity of Braid-Matroid Kazhdan-Lusztig Polynomials

P / partially_solved | Oberwolfach 报告问题汇编

Can the known flag-Whitney-number and Stirling-number formulas for coefficients of braid-matroid Kazhdan-Lusztig polynomials be converted into effective complexity bounds and computations of the remaining coefficients of $P(B_n,t)$?

来源字段：https://doi.org/10.4171/owr/2018/2

13167 | OWR-14298163-015

### Collinear-Triple-Free Sets in Random Grids

O / open | Oberwolfach 报告问题汇编

Characterize $\alpha([n]^2,p)$, the maximum size of a collinear-triple-free subset of a random subset of $[n]^2$ in which each grid point is retained independently with probability $p$.

来源字段：https://doi.org/10.4171/owr/2024/3

13168 | OWR-14299518-019

### Color Patterns and the Erdős-Rogers Function

P / partially_solved | Oberwolfach 报告问题汇编

For every fixed $k\geq 3$, is the color-pattern parameter $P_r(k)$ asymptotic in order to $r\,f_{k,k+1}(r)^2$, that is, is $P_r(k)=\Theta\!\left(rf_{k,k+1}(r)^2\right)$?

来源字段：https://doi.org/10.4171/owr/2025/42

13169 | OWR-4798-026

### Colored Realizations of Gorenstein* $cd$-Indices

O / open | Oberwolfach 报告问题汇编

Can the entire $cd$-index of every Gorenstein* poset be realized as the flag $f$-vector of a colored simplicial complex? More precisely, for a vertex coloring of a simplicial complex $\Delta$ with $\dim(\Delta)+1$ colors, where $f_S(\Delta)$ counts faces whose set of vertex colors is $S$, can the coefficients of the $cd$-index always be represented by such numbers $f_S(\Delta)$?

来源字段：https://doi.org/10.4171/owr/2011/08

13170 | OWR-3473-002

### Coloring $F$-Free Uniform Hypergraphs

O / open | Oberwolfach 报告问题汇编

For every fixed $k$-uniform hypergraph $F$, does a constant $c_F$ exist such that every $F$-free $k$-uniform hypergraph of maximum degree $\Delta$ has chromatic number at most $$c_F\left(\frac{\Delta}{\log\Delta}\right)^{1/(k-1)}$$?

来源字段：https://doi.org/10.4171/owr/2009/23

13171 | OPG-762

### Combinatorial covering designs

P / partially_solved | Open Problem Garden

A $(v, k, t)$ covering design, or covering, is a family of $k$-subsets, called blocks, chosen from a $v$-set, such that each $t$-subset is contained in at least one of the blocks. The number of blocks is the covering's size, and the minimum size of such a covering is denoted by $C(v, k, t)$.

Problem Find a closed form, recurrence, or better bounds for $C(v,k,t)$. Find a procedure for constructing minimal coverings.

背景链接：http://www.openproblemgarden.org/op/covering_designs

13172 | OWR-14299523-002

### Combinatorial Formulas for Graph Zeta Functions

O / open | Oberwolfach 报告问题汇编

For every graph $\Gamma$ on a fixed vertex set, find a meaningful combinatorial formula for its zeta function $W_\Gamma^-(X,T)$ that extends the known hypergraph formula for cographs.

来源字段：https://doi.org/10.4171/owr/2025/38

13173 | OWR-12872-017

### Combinatorial Interpretation of an $A_{2n}^{(2)}$ Rogers-Ramanujan Identity

P / partially_solved | Oberwolfach 报告问题汇编

For positive integers $m,n$, set $\kappa=2m+2n+1$. Find a combinatorial interpretation of the Hall-Littlewood identity

$$\sum_{\lambda_1\leq m}q^{|\lambda|}P_{2\lambda}(1,q,q^2,\ldots;q^{2n-1}) =\frac{(q^\kappa;q^\kappa)_\infty^n}{(q;q)_\infty^n} \prod_{i=1}^n\theta(q^{i+m};q^\kappa) \prod_{1\leq i<j\leq n}\theta(q^{j-i},q^{i+j-1};q^\kappa),$$

where $P_{2\lambda}$ is a Hall-Littlewood polynomial, $(a;q)_\infty=\prod_{r\geq 0}(1-aq^r)$, and $\theta(a_1,\ldots,a_k;q)=\prod_{s=1}^k(a_s;q)_\infty(q/a_s;q)_\infty$.

来源字段：https://doi.org/10.4171/owr/2014/12

13174 | AMR-031-0011

### Combinatorial interpretation of Kronecker coefficients

O / open | AMR 问题清单汇编

For partitions $\lambda,\mu,\nu$ of $n$, the Kronecker coefficient $g_{\mu\nu}^{\lambda}$ is defined by $$V_\mu\otimes V_\nu\cong\bigoplus_\lambda g_{\mu\nu}^{\lambda}V_\lambda$$ for irreducible symmetric-group modules $V_\lambda$. Give a combinatorial interpretation of $g_{\mu\nu}^{\lambda}$.

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Combinatorics

13175 | OWR-17472-013

### Combinatorial Interpretation of Stable Kronecker Coefficients

O / open | Oberwolfach 报告问题汇编

Find a combinatorial interpretation of the stable (reduced) Kronecker coefficients, analogous to the Littlewood-Richardson rule.

来源字段：https://doi.org/10.4171/owr/2020/9

13176 | OWR-14298158-017

### Combinatorial Interpretations of Negative k-Arrangements

O / open | Oberwolfach 报告问题汇编

Let $A(n,k)$ denote the number of permutations of $[n]$ whose fixed points are colored with one of $k$ colors, extended polynomially from nonnegative integers $k$. Does $A(n,k)$ have a combinatorial interpretation for every negative integer $k$? More generally, does $A(2n,k)$ have a combinatorial interpretation for every $k\in\mathbb Z$?

来源字段：https://doi.org/10.4171/owr/2024/6

13177 | OWR-11783-010

### Combinatorial Nonnegativity for Polytope $S$-Polynomials

O / open | Oberwolfach 报告问题汇编

Does every lattice polytope $P$ have a combinatorial proof that its $S$-polynomial has nonnegative coefficients?

来源字段：https://doi.org/10.4171/owr/2012/21

13178 | OWR-12872-015

### Combinatorial Proof of the Two-Thirds Leaf Limit

S / solved | Oberwolfach 报告问题汇编

Let $T_n$ range over rooted plane trees whose child labels decrease, and choose uniformly both a tree and one of its vertices. Give a direct combinatorial proof, avoiding generating functions and induction, that the probability $p_n$ that the chosen vertex is a leaf satisfies $p_n\to 2/3$ as $n\to\infty$.

来源字段：https://doi.org/10.4171/owr/2014/12

13179 | OWR-16164-005

### Combinatorics of Signed Adjacency Polytopes

P / partially_solved | Oberwolfach 报告问题汇编

For a sign sequence $\varepsilon_1,\ldots,\varepsilon_{n-1}\in\{-,+\}$, consider the polytope in $[0,1]^n$ defined, for each $i$, by $x_i+x_{i+1}\leq1$ when $\varepsilon_i=-$ and $x_i+x_{i+1}\geq1$ when $\varepsilon_i=+$. Give a combinatorial interpretation of the $h^*$-polynomial of this polytope.

来源字段：https://doi.org/10.4171/owr/2018/23

13180 | OWR-17472-008

### Combinatorics of Sylow Branching Coefficients

P / partially_solved | Oberwolfach 报告问题汇编

Find a combinatorial description of the Sylow branching-coefficient map $Z^{\mathbf1_{P_n}}:\mathcal P(n)\to\mathbb N_0$, $\lambda\mapsto Z^{\mathbf1_{P_n}}_\lambda$.

来源字段：https://doi.org/10.4171/owr/2020/9

13181 | OWR-4425-011

### Common Parse Words for Rooted Tree Pairs

P / partially_solved | Oberwolfach 报告问题汇编

For a pair of rooted trees that is not mutually crooked, meaning that some consecutive leaves have an uncle-nephew relationship in both trees, does there exist a common parse word in which those uncle and nephew leaves receive the same label?

来源字段：https://doi.org/10.4171/owr/2010/37

13182 | AIM-COMBINATORICS-0047

### Compact coloring lists: a quantitative disjoint-block counterexample

S / solved | AIM 研讨会问题汇编

**Polysize list of colorings.** Is there a list of colorings of polynomial size such that for any subset of the columns, one of these colorings gives it a discrepancy of $O(h)$, where $h$ is the hereditary discrepancy?

来源字段：http://aimpl.org/hereddiscrep/1/

13183 | AIM-COMBINATORICS-0256

### Compact Freiman models and relation peeling

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.19 (brought by V. Lev, originally stated by S. Konyagin and V. Lev).** For any integer $n\geq2$, the set $$D_n=\{0,1,2,4,\ldots,2^{\,n-2}\}$$ is “linear” (has Freiman rank one) and is not contained in an arithmetic progression with difference larger than one; hence it is not isomorphic to a set of integers of length smaller than $2^{\,n-2}$. Is this extremal? Is it true that every Freiman-isomorphism class of $n$-element integer sets has a representative of length at most $2^{\,n-2}$?

**Conjecture.** For $n\geq7$, every $n$-element set of integers is Freiman-isomorphic to a subset of $$[0,2^{\,n-2}].$$

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13184 | AMR-029-0012

### Compactness of Kőnig-property

O / open | AMR 问题清单汇编

A hypergraph $H=(V,E)$ has the Kőnig-property if there is a set $\mathcal{D}\subseteq E$ of pairwise disjoint hyperedges such that there is a vertex cover consisting of one vertex from each hyperedge in $\mathcal{D}$. (Using this terminology, Kőnig's theorem says that every finite bipartite graph has the Kőnig-property). R. Aharoni and N. Bowler conjectured independently the following. If $H=(V,E)$ is a hypergraph such that all of its hyperedges are finite and for all finite $E'\subseteq E$ the hypergraph $(V,E')$ has the Kőnig-property, then $H$ has the Kőnig-property as well ( p. 19. Problem 6.7).

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13185 | GREEN-054

### Comparable Elements in Integer Lattices

O / open | Ben Green 问题

Consider a set $S \subset [N]^3$ with the property that any two distinct elements $s, s'$ of $S$ are comparable (in the coordinatewise partial order). Is $|S| \leq N^{2-\delta}$ for some $\delta > 0$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13186 | AMR-029-0013

### Compatible Euler-tours

O / open | AMR 问题清单汇编

If G is an undirected graph with even degrees then call two closed Eulerian walks compatible if no pair of incident edges occurs consecutively in both. Is it true that the maximum number of pairwise compatible Euler walks in an Eulerian graph of minimum degree 2d is either 2d-2 or 2d-1?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13187 | AIM-COMBINATORICS-0268

### Complete slope classification for the four-corner Cantor projection

S / solved | AIM 研讨会问题汇编

For which real numbers \(\alpha\) the set \(\mathcal C_4+\alpha\mathcal C_4\) has zero Lebesgue measure?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13188 | AIM-COMBINATORICS-0099

### Complexity and an exact two-vertex formula for divisor rank

P / partially_solved | AIM 研讨会问题汇编

**Problem (Lionel Levine).** Is there an efficient way to compute the rank (in the sense of Baker and Norine [7]) of a divisor?

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13189 | AMR-121-0001

### Complexity in Enumerative Combinatorics - Conjecture 1 (Wilf)

O / open | AMR 问题清单汇编

Now, for a *combinatorial sequence* $\{a_n\}$ one can ask if there is a Wilfian formula. In the original paper [Wilf] an explicit example is given:

Let $a_n$ be the number of unlabeled graphs on $n$ vertices. Then $\{a_n\}$ has no Wilfian formula of type \textup{\small $($\textrm{W1}$)$}.

背景链接：https://arxiv.org/abs/1803.06636

13190 | AMR-121-0013

### Complexity in Enumerative Combinatorics - Conjecture 13 (cf. [Ges-sem])

O / open | AMR 问题清单汇编

In [Ges-sem], the author defined the following *Gessel sequence*: $$ b_n := \ 2\cdot 5^n \. - \. (3+4i)^{n} \. - \. (3-4i)^{n}, \quad \text{where} \ \ i \. = \. \sqrt{-1} $$ see also [OEIS, A250102 (https://oeis.org/A250102)]. Note that $b_n\in \mathbb{Z}$ since $$b_n = 2\cdot 5^n \. - \. 2 {2r} \. 3^{n-2r} $$ and that $b_i \ge 0$ since $|3\pm 4i|=5$. Note also that $\{b_n\}$ is C-recursive since $$ B(t) \. := \sum_{n=0}^\infty \. b_n {(1-5t) $$ This is an example of a C-recursive nonnegative sequence without an easy combinatorial interpretation.

Sequence $ of type $\textup{\small $($\textrm{C1}$)$}$.

背景链接：https://arxiv.org/abs/1803.06636

13191 | AMR-121-0002

### Complexity in Enumerative Combinatorics - Conjecture 2

O / open | AMR 问题清单汇编

Let $a_n$ be the number of unlabeled plane triangulations with $n$ vertices, and let $b_n$ be the number of $3$-connected planar graphs with $n$ vertices. Then $\{a_n\}$ and $\{b_n\}$ can be computed in \mathrm{poly}$(n)$ time.

背景链接：https://arxiv.org/abs/1803.06636

13192 | AMR-121-0003

### Complexity in Enumerative Combinatorics - Conjecture 3

O / open | AMR 问题清单汇编

It was shown in [Ges] that similar polynomial recurrences exist for the number of $k$-regular labeled graphs, for every fixed $k\ge 1$.

Fix $k\ge 1$ and let $a_n$ be the number of unlabeled $k$-regular graphs with $n$ vertices. Then $\{a_n\}$ can be computed in \textup{\textrm{poly}}$(n)$ time.

背景链接：https://arxiv.org/abs/1803.06636

13193 | AMR-121-0004

### Complexity in Enumerative Combinatorics - Conjecture 4

O / open | AMR 问题清单汇编

For a fixed ${\text **T** } $, let $g(m,n)$ denote the number of tilings of $[m\times n]$ with ${\text **T** } $. Is $g(m,n)$ computable in polynomial time? The following conjecture suggests otherwise.

There exists a finite set of tiles ${\text **T** } $ such that counting the number of tilings of $[n\times n]$ } $ is $\textup{\textsf{\#EXP}}$-complete.

背景链接：https://arxiv.org/abs/1803.06636

13194 | AMR-121-0006

### Complexity in Enumerative Combinatorics - Conjecture 6

O / open | AMR 问题清单汇编

Before we proceed to more combinatorial examples, let us mention that D-transcendental GFs are the subject of *Differential Galois Theory*, which goes back to Liouville, Lie, Picard and Vessiot in the 19th century (see e.g. [Ritt]), and continues to be developed [Put-S]. Some natural GFs are known to be D-transcendental, e.g.\ $\Gamma(z)$, $\zeta(z)$, etc., but there are too few methods to prove this in most cases of interest. Here are some of our favorite open problems along these lines, unapproachable with existing tools.

$\sum_{n\ge 1}$ are D-transcendental.

背景链接：https://arxiv.org/abs/1803.06636

13195 | AMR-121-0007

### Complexity in Enumerative Combinatorics - Conjecture 7

O / open | AMR 问题清单汇编

Here $p_n$ is the $n$-th prime, and $\pi(n)$ is the number of primes $\le n$, as above. Both GFs are known to be non-D-finite, as shown by Flajolet, Gerhold and Salvy in [FGS] by asymptotic arguments. The authors quip: ``*Almost anything is non-holonomic unless it is holonomic by design*''. Well, maybe so. But the same applies for D-transcendence where the gap between what we believe and what we can prove is much wider. The reader should think of such open problems as the irrationality of $e+\pi$ and $\zeta(5)$, and imagine a similar phenomenon in this case (cf. [KoZ]).

}}$ is D-transcendental.

背景链接：https://arxiv.org/abs/1803.06636

13196 | AMR-121-0008

### Complexity in Enumerative Combinatorics - Conjecture 8 (cf. [Zei])

O / open | AMR 问题清单汇编

Let $a_n$ denotes the number of self-avoiding walks $O\to O$ in $\mathbb{Z}^2$ of length $n$. Then the sequence $\{a_n\}$ has no Wilfian formula of type \textup{\small $($\textrm{W1}$)$}.

背景链接：https://arxiv.org/abs/1803.06636

13197 | AMR-121-0010

### Complexity in Enumerative Combinatorics - Problem 10

O / open | AMR 问题清单汇编

Can }}_n(1324)\}$ $(n)$ time? (In 2005, Doron Zeilberger expressed doubts that $\text{{\rm \textsf{A}}}}_{1000}(1324)$ This sentiment has been roundly criticized on both mathematical and theological grounds (see [Ste]). ) More generally, can one find a single permutation $\pi$ such that }}_n(\pi)\}$ $(n)$ time? Is the computation of }}_n(\pi)\}$ or harder for *random* permutations $\pi \in S_k$?

背景链接：https://arxiv.org/abs/1803.06636

13198 | AMR-121-0011

### Complexity in Enumerative Combinatorics - Problem 11

O / open | AMR 问题清单汇编

One especially notable case of a combinatorial interpretation is the problem of computing the *Kronecker coefficients* of the symmetric group, which are defined by $$\chi^\lambda \cdot \chi^\mu = \sum_{\nu\vdash n} g(\lambda,\mu,\nu) \. \chi^\nu, \quad \text{where} \ \ \lambda,\mu \vdash n $$ and $\chi^\lambda, of $S_n$.

Find a combinatorial interpretation for the Kronecker coefficients $$\bigl\{ .$$

背景链接：https://arxiv.org/abs/1803.06636

13199 | AMR-121-0012

### Complexity in Enumerative Combinatorics - Problem 12

O / open | AMR 问题清单汇编

The original proof is an elegant double counting argument modulo $2$. This suggests the following open problem: (This problem was suggested to us by Peter Shor https://tinyurl.com/y7yafneh (https://cstheory.stackexchange.com/questions/33508).)

Find a combinatorial interpretation for $N_e(G)/2$.

背景链接：https://arxiv.org/abs/1803.06636

13200 | AMR-121-0005

### Complexity in Enumerative Combinatorics - Problem 5 (Pak--Yeliussizov)

O / open | AMR 问题清单汇编

This also implies that $y(t)$ is not D-finite. Indeed, observe by definition, that if a sequence $\{a_n\}$ is P-recursive, then so is $\{n! $, which implies the result by taking $a_n=B_n/n!$ (cf. [LR]). Of course, there is a more direct way to prove that $y(t)$ is not $D$-finite by repeated differentiation or via the asymptotics, see below. This suggests the following advanced generalization of Klazar's theorem.

Suppose $\{a_n/n!\}$ is D-algebraic but not P-recursive. Does this imply that $\{a_n\}$ is D-transcendental?

背景链接：https://arxiv.org/abs/1803.06636

13201 | AMR-121-0009

### Complexity in Enumerative Combinatorics - Problem 9

O / open | AMR 问题清单汇编

In a forthcoming paper [GP-ADE] we construct an explicit but highly artificial non-symmetric set $S\subset F_k\times F_\ell$ In [KP2] we use the tools in [KP1] to prove that groups have an uncountable set of *spectral radii* $$ \rho(G,S) := \lim_{n\to \infty} a_n(G,S)^{1/n} $$ Since the set of D-algebraic sequence is countable, this implies the existence of D-transcendental Cayley graphs with symmetric $S$, but such a proof is nonconstructive.

Find an explicit construction of $\Gamma(G,S)$ when $S$ is symmetric, and $

背景链接：https://arxiv.org/abs/1803.06636

13202 | AMR-029-0014

### Complexity of computing a v-reduced divisor in multigraphs

O / open | AMR 问题清单汇编

Is there a polynomial algorithm for computing a $v_0$-reduced divisor equivalent to a given divisor of an undirected multigraph?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13203 | AMR-029-0015

### Complexity of computing the rotor-router action

O / open | AMR 问题清单汇编

Let $G$ be an undirected graph. What is the complexity of computing the rotor-router action of the sandpile group of $G$ on the spanning trees of $G$?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13204 | AMR-029-0016

### Complexity of the chip-firing reachability problem for general digraphs

O / open | AMR 问题清单汇编

Is the chip-firing reachability problem co-NP-hard for general digraphs?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13205 | AMR-029-0017

### Complexity of the halting problem for Eulerian multigraphs

O / open | AMR 问题清单汇编

Is the chip-firing halting problem in P for Eulerian digraphs with multiple edges?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13206 | AMR-029-0018

### Complexity of the halting problem for simple digraphs

O / open | AMR 问题清单汇编

Is it true that the chip-firing halting problem for simple digraphs is NP-complete?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13207 | AIM-COMBINATORICS-0133

### Composition-resolved LL fibers and a finite-flatness obstruction

P / partially_solved | AIM 研讨会问题汇编

The supplied source URL is <http://aimpl.org/rationalcatalan/1/>. In particular, the misspelling “Lyashoko-Looijenga” is in the source; the standard spelling is “Lyashko--Looijenga.”

Let $T$ be the full set of pseudo-reflections, let $\ell_T$ be reflection length, let $c$ be a Coxeter element, and let

\[ NC(W,c)=[1,c]_T=\{w:\ell_T(w)+\ell_T(w^{-1}c)=n\}. \]

For an integer $p\geq 0$, count weak $p$-tuples

\[ w_1\leq_T\cdots\leq_Tw_p\leq_Tc. \]

The assertion is that their number is

\[ C_W(p):=\prod_{i=1}^n\frac{ph+d_i}{d_i}, \]

equivalently that

\[ C_W(p)=\sum_{k=1}^n\binom{p+1}{k}F_k, \qquad F_k:=\operatorname{Fact}_k(c). \tag{1} \]

The factors in $F_k$ are ordered, nonidentity, and length-additive. The $p+1$, rather than $p$, comes from the $p+1$ increments between $1,w_1,\ldots,w_p,c$.

来源字段：http://aimpl.org/rationalcatalan/1/

13208 | OPG-404

### Concavity of van der Waerden numbers

O / open | Open Problem Garden

For $k$ and $\ell$ positive integers, the (mixed) van der Waerden number $w(k,\ell)$ is the least positive integer $n$ such that every (red-blue)-coloring of $[1,n]$ admits either a $k$-term red arithmetic progression or an $\ell$-term blue arithmetic progression.

Conjecture For all $k$ and $\ell$ with $k \geq \ell$, $w(k,\ell) \geq w(k+1,\ell-1)$.

背景链接：http://www.openproblemgarden.org/op/concavity_of_van_der_waerden_numbers

13209 | AIM-COMBINATORICS-0130

### Cone-stack moduli and stabilizer criteria for harmonic metric-graph quotients

P / partially_solved | AIM 研讨会问题汇编

Jordan Ellenberg: “Is there a Deligne-Mumford stack on graphs?” Matt Baker: “We have a paper on metric graphs that accom-plishes some of this.” Jordan: “But there is still a question of what are harmonic group actions on metric graphs?”

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13210 | AMR-029-0019

### Conforti-Cornuéjols conjecture on the MFMC property

O / open | AMR 问题清单汇编

Is it true that a clutter has the MFMC property if and only if it has the packing property?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13211 | AMR-046-0031

### Conjecture of multiplicative persistence

P / partially_solved | AMR 问题清单汇编

For $n\in\mathbb{N}$, let $\Pi(n)$ be the product of its decimal digits, and let $\operatorname{Pm}(n)$ be the least positive integer such that $\Pi^{\operatorname{Pm}(n)}(n)=\Pi^{\operatorname{Pm}(n)+1}(n)$. Is $\operatorname{Pm}(n)\leq11$ for every $n$?

背景链接：https://arxiv.org/abs/2012.02524

13212 | OWR-2090-021

### Connectivity of Nonnegative Flats in Weighted Matroids

P / partially_solved | Oberwolfach 报告问题汇编

Let $M$ be a finite simple rank-$3$ matroid on $n$ points with real weights $a_i$ of nonnegative total sum. Is the subconfiguration of nonnegative-weight points connected by the lines of nonnegative weight? More generally, for a rank-$r$ matroid, is the order complex of the nonnegative proper flats $(r-3)$-connected, and is it shellable?

来源字段：https://doi.org/10.4171/owr/2008/44

13213 | AMR-030-0082

### Consider p(v, t), the probability that a walk beginning from the origin ends at the point v on the d-dimensional integer

O / open | AMR 问题清单汇编

/Spencer: Consider p(v, t), the probability that a walk beginning from the origin ends at the point v on the d-dimensional integer lattice in time t. Conjecture : Ignoring the obvious "parity issue", this function is unimodal in t >= 0.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13214 | AIM-COMBINATORICS-0036

### Constructive Beck-Fiala status and an exact pseudoforest incidence theorem

P / partially_solved | AIM 研讨会问题汇编

Given a set system \((V,\mathcal S)\) with \(V=[n]\), \(\mathcal S=S_1,\ldots,S_m\). Each \(i\in V\) occurs in at most \(t\) sets from \(\mathcal S\). Then there is a coloring to the elements of \(V\) achieving a discrepancy of \(O(\sqrt t)\). Komlós conjecture implies this by treating the incidence vector of each \(i\in V\) as a vector and scaling it down by \(\sqrt t\).

来源字段：http://aimpl.org/hereddiscrep/1/

13215 | AMR-029-0020

### Constructive characterization of dumpy graphs

O / open | AMR 问题清单汇编

Find a constructive characterization of k-dumpy graphs.

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13216 | AIM-COMBINATORICS-0035

### Constructive Komlos and deterministic balancing with bounded direction overlap

P / partially_solved | AIM 研讨会问题汇编

**Constructive Komlós.** Find a polynomial time algorithm to find the signs above.

来源字段：http://aimpl.org/hereddiscrep/1/

13217 | OWR-1265-006

### Containment of Simultaneous Core Partitions

S / solved | Oberwolfach 报告问题汇编

For coprime positive integers $p$ and $q$, let $\kappa_{p,q}$ denote the unique simultaneous $(p,q)$-core partition of maximum size $(p^2-1)(q^2-1)/24$ (constructed by Olsson and Stanton). Is the Young diagram of every simultaneous $p$-core and $q$-core partition contained in the Young diagram of $\kappa_{p,q}$?

来源字段：https://doi.org/10.4171/owr/2006/15

13218 | AIM-COMBINATORICS-0205

## Convention obstructions and an oriented-pseudoforest theorem

S / solved | AIM 研讨会问题汇编

**Conjecture 6.35 (Thomassé).** In a digraph \(D\), \[ \sum_{v\in V(D)} \left|\delta_D^+(v)-\delta_D^-(v)\right| + \left|\{(u,v):d(u,v)\leq2\}\right| \geq 2|E(D)| + \left|\{v:\delta_D^+(v)>\delta_D^-(v)\}\right|. \tag{1.1} \]

**Note.** This is exact for transitive tournaments.

**Repaired oriented version.** Let \(D\) be a finite simple oriented graph (no loops, parallel arcs, or digons), and count ordered pairs \((u,v)\) with \(u\ne v\). Then (1.1) holds.

The notation section of the same PDF is important:

- a digraph is finite unless stated otherwise; - its incidence-relation definition explicitly allows multiple edges from \(u\) to \(v\); - a *simple* digraph has at most one edge from \(u\) to \(v\); - the text separately uses “no loops or digons” when those properties are required; - \(\delta_D^+(v)\) and \(\delta_D^-(v)\) are the outdegree and indegree of the vertex \(v\), while \(\delta_D^+\) and \(\delta_D^-\) without an argument denote minimum degrees; - \(d(u,v)\) is the length of a shortest directed path from \(u\) to \(v\).

The displayed set must consist of **ordered pairs of distinct vertices**. Indeed, if the diagonal pairs \((v,v)\), whose usual distance is zero, were included, its size would increase by \(n\). The note that (1.1) is exact for every transitive tournament would then be false. Unreachable ordered pairs are understood to have distance \(+\infty\) and are not counted.

The source does not say “simple,” “oriented,” or “digon-free” in Conjecture 6.35. Under its own general definition of digraph, the statement is false:

- a two-vertex digon is a simple counterexample; - even after forbidding digons, the directed triangle with every arc doubled is a counterexample.

The definition's phrase “from \(u\) to another vertex \(v\)” suggests that loops were not intended, although later statements sometimes exclude loops explicitly. If a loop is permitted, the one-vertex one-loop digraph is another counterexample: its two degrees are equal, it has no distinct ordered vertex pair, and (1.1) reads \(0\geq2\).

Thus both simplicity and exclusion of digons are necessary for the most evident repair. The likely intended version is:

This report distinguishes the refuted literal statement from this plausible repaired conjecture.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13219 | OWR-741-007

## Convex-Body Bounds for Lattice-Point Enumerators

O / open | Oberwolfach 报告问题汇编

Find effective upper bounds for the lattice-point enumerator $G(K)=\#(K\cap\mathbb Z^n)$ in terms of functionals of a convex body $K$.

来源字段：https://doi.org/10.4171/owr/2004/39

13220 | GREEN-018

## Corner Problem in Product Sets

P / partially_solved | Ben Green 问题

Suppose $G$ is a finite group, and let $A \subset G \times G$ be a subset of density $\alpha$. Are there $\gg_\alpha |G|^3$ triples $x, y, g$ such that $(x, y), (gx, y), (x, gy)$ all lie in $A$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13221 | AIM-COMBINATORICS-0042

## Corrected volumetric characterization of vector balancing

S / solved | AIM 研讨会问题汇编

Let $X,Y$ be norms on $\mathbb R^d$, with unit balls $C=B_X$ and $K=B_Y$. For $v_1,\ldots,v_N\in C$, the source defines \[ \operatorname{disc}_Y(v_1,\ldots,v_N) =\min_{\varepsilon\in\{-1,1\}^N} \left\|\sum_{i=1}^N\varepsilon_i v_i\right\|_Y, \] then takes the maximum over all such $N$-tuples to obtain $\alpha_N(X,Y,d)$ and finally $\alpha(X,Y,d)=\sup_N\alpha_N(X,Y,d)$.

The main text attributes to Banaszczyk the lower bound \[ \alpha(X,Y,d)\gtrsim \sqrt d\left(\frac{\operatorname{vol}_d(B_X)} {\operatorname{vol}_d(B_Y)}\right)^{1/d}, \tag{1.1} \] and asks if it is also an upper bound up to logarithmic factors. A universal constant is suppressed in the AIM display; Banaszczyk's 1993 paper [Ban93] gives an explicit constant.

The two database remarks must not be treated as two competing formulas. The first says, in part, \[ \min_{B_Y\cap W\subseteq E}\sqrt d\, \frac{\operatorname{vol}(E)^{1/d}} {\operatorname{vol}(B_X\cap W)^{1/d}}, \tag{1.2} \] whereas the second explicitly says “The formula above should be” \[ \max_{1\le k\le d}\max_{\dim W=k} \min_{E:\,B_X\cap W\subseteq E} \sqrt{k}\left( \frac{\operatorname{vol}(E)}{\operatorname{vol}(B_Y\cap W)} \right)^{1/k}. \tag{1.3} \] The official 2016 AIM workshop report [AIM16, equation (3)] confirms (1.3) verbatim in substance and specifies that the volumes are Lebesgue measures inside $W$. Thus the reconstruction is:

* $E$ contains the **input** section $B_X\cap W$; * the denominator is the **output** section $B_Y\cap W$; * both volumes are intrinsic $k$-dimensional volumes on $W$; * the roots are $1/k$ and the prefactor is $\sqrt{k}$; * the second remark is a correction of the first.

Formula (1.2) also fails a basic scaling test: enlarging the admissible input ball should make balancing harder, but (1.2) decreases. This independently rules it out as a functional for $\alpha(X,Y,d)$.

来源字段：http://aimpl.org/hereddiscrep/1/

13222 | AMR-054-0037

### Counting Polyominoes

O / open | AMR 问题清单汇编

How many polyominoes on $n$ squares are there? A polyomino is a connected interior-disjoint union of axis-aligned unit squares joined edge-to-edge, in other words, an edge-connected union of cells in the planar square lattice. The order of a polyomino is the number of unit squares forming it. The problem asks for the number of polyominoes of order $n$. The key constraint here is that polyominoes must be edge-connected. There are three variations on the problem, depending on whether two polyominoes are considered equivalent by factoring out just translations (fixed polyominoes), rotations and translations (chiral polyominoes), or reflections, rotations, and translations (free polyominoes).

背景链接：https://topp.openproblem.net/p37

13223 | OWR-14298158-006

### Counting-Sequence Ratios for Length-Six Principal Classes

O / open | Oberwolfach 报告问题汇编

Let $A=\operatorname{Av}(132456)$ and $B=\operatorname{Av}(124356)$. How different are their counting sequences? Is their ratio constant or polynomial, and how close can the counting sequences of distinct principal permutation classes be?

来源字段：https://doi.org/10.4171/owr/2024/6

13224 | AMR-029-0021

### Covering a crossing supermodular function with graph edges

O / open | AMR 问题清单汇编

Given a crossing supermodular function $p:2^V\to \mathbb{Z}$ satisfying $p(\emptyset)=p(V)=0$, what is the minimum number of edges of an undirected graph G covering p (that is, with the property that $d_G(X)\ge p(X)$ holds for every $X\subseteq V$)?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13225 | AMR-029-0022

### Covering a crossing supermodular function with pairwise non-parallel arcs

O / open | AMR 问题清单汇编

Given a crossing supermodular function $p:2^V\to \mathbb{Z}$ satisfying $p(\emptyset)=p(V)=0$, what is the minimum number of pairwise non-parallel arcs (i.e. only one copy of the arc uv is allowed for any pair $u,v\in V$, but opposite pairs are allowed) covering p, where a digraph D=(V,A) is said to cover p if $\varrho_D(X)\ge p(X)$ holds for every $X\subseteq V$?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13226 | AMR-029-0023

### Covering a symmetric crossing supermodular function with hyperedges of prescribed size

O / open | AMR 问题清单汇编

Given a symmetric crossing supermodular function $p:2^V\to \mathbb{R}$ and positive integers $n_1,n_2,\dots,n_k$, does there exist a hypergraph H=(V,E) covering p and having exactly k hyperedges of sizes $n_1,n_2,\dots,n_k$?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13227 | GREEN-081

### Covering by Random Translates

O / open | Ben Green 问题

If $A \subset \mathbb{Z}/p\mathbb{Z}$ is random with $|A| = \sqrt{p}$, can we almost surely cover $\mathbb{Z}/p\mathbb{Z}$ with $100\sqrt{p}$ translates of $A$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13228 | AMR-125-0013

### Coxeter Groups and Unimodality - Conjecture 13

O / open | AMR 问题清单汇编

Let $(W,S)$ be a Coxeter system. It is of interest, and difficult, to obtain properties of the rank generating function of Bruhat intervals $[u,v]$, $u,v \in W$ (see, e.g., [BjoEke]). In this respect, I feel that the following holds.

Let $W$ be a Weyl group, and $u,v \in W$. Then $[u,v]$ is rank log-concave.

背景链接：https://arxiv.org/abs/2410.09897

13229 | AMR-125-0014

### Coxeter Groups and Unimodality - Conjecture 14

O / open | AMR 问题清单汇编

Many matrices arising in combinatorics are known to be TP (see, e.g., [BreMem], [BreCombTP], and [BreTPComb]). It is therefore surprising that for a fundamental and old combinatorial matrix such as the one consisting of the Eulerian numbers this property has not yet been settled. For $n \in \mathbb{P}$ and $k \in \mathbb{N}$ let $A(n,k)$ be the corresponding *Eulerian number* (so, $A(n,k)$ is the number of permutations in $S_n$ that have $k$ descents). The following conjecture was first put forward in [BreCombTP, Conj. 6.10].

The matrix \[ A:=( A(n+1,k))_{n,k \in \mathbb{N}} \] is totally positive.

背景链接：https://arxiv.org/abs/2410.09897

13230 | AMR-125-0015

### Coxeter Groups and Unimodality - Conjecture 15

O / open | AMR 问题清单汇编

For all $i,j,r \in \mathbb{N}$ the determinant of the submatrix determined by the rows indexed by $i,i+1, \ldots , i+r$ and columns indexed by $j,j+1, \ldots , j+r$

is a monotonically increasing function of $i \in \mathbb{N}$.

背景链接：https://arxiv.org/abs/2410.09897

13231 | AMR-125-0002

### Coxeter Groups and Unimodality - Conjecture 2

O / open | AMR 问题清单汇编

Let $(W_1,S_1)$, $(W_2,S_2)$ be two Coxeter systems, and $u,v \in W_1$, $w,z \in W_2$ be such that $[u,v] \simeq [w,z]$ (isomorphic as posets). Then \begin{equation}

P_{u,v}(q)=P_{w,z}(q). \end{equation}

背景链接：https://arxiv.org/abs/2410.09897

13232 | AMR-125-0003

### Coxeter Groups and Unimodality - Conjecture 3

O / open | AMR 问题清单汇编

Let $(W_1,S_1)$, $(W_2,S_2)$ be two Coxeter systems, $J_1 \subseteq S_1$, $J_2 \subseteq S_2$, $u,v \in W_1^{J_1}$, $w,z \in W_2^{J_2}$ and $f: [u,v] \rightarrow [w,z]$ be a poset isomorphism such that $f([u,v]^{J_1})=[w,z]^{J_2}$. Then \begin{equation}

P_{u,v}^{J_1,q}(q)=P_{w,z}^{J_2,q}(q). \end{equation}

背景链接：https://arxiv.org/abs/2410.09897

13233 | AMR-125-0004

### Coxeter Groups and Unimodality - Conjecture 4

O / open | AMR 问题清单汇编

Let $W$ be a Weyl group of type $A$ and $u,v \in W$, $u \leq v$. Then \begin{equation}

P_{u,v}(q) = \widetilde{P}_{u,v,z}(q) \end{equation} for any hypercube decomposition $[u,z]$ of $[u,v]$.

背景链接：https://arxiv.org/abs/2410.09897

13234 | AMR-125-0005

### Coxeter Groups and Unimodality - Conjecture 5

O / open | AMR 问题清单汇编

Let $(W,S)$ be a Coxeter system, and $u,v \in W$, $u \leq v$. Then \[ \widetilde{\Phi}_{u,v} \in \mathbb{N} \langle c,d \rangle. \]

背景链接：https://arxiv.org/abs/2410.09897

13235 | AMR-125-0008

### Coxeter Groups and Unimodality - Conjecture 8

O / open | AMR 问题清单汇编

For $\sigma \in S_n$ let \[ L(\sigma) := | \{ (i,j) \in [n]^2 : i<j, \sigma(i)>\sigma(j), \; i \not \equiv j \pmod{2} \} |. \] The statistic $L$ is known as the *odd length* or *odd inversion number* of $\sigma$. The statistic was introduced in [KloVol] in relation to formed spaces and has been further studied in [BreCav] and [BreCa2]. The following conjecture appears in [BreCa2, Conj. 6.2].

Let $n \in \mathbb{P}$, $n \geq 5$. Then \[ L_n(x) := \sum_{\sigma \in S_n} x^{L(\sigma)} \] is symmetric and unimodal.

背景链接：https://arxiv.org/abs/2410.09897

13236 | AMR-125-0009

### Coxeter Groups and Unimodality - Conjecture 9

O / open | AMR 问题清单汇编

So, knowledge of the $ \widetilde{R}$-polynomials is equivalent to knowledge of the $R$-polynomials. Combinatorial interpretations of the polynomials $\{ \widetilde{R}_{u,v}(t) \}_{u,v \in W}$ have been given in [DeoInv] and [Dye93], see [BB, Thms 5.3.4 and 5.3.7]. It is easy to see that, if $u \leq v$, then $deg( \widetilde{R}_{u,v})=\ell(u,v)$ and the powers appearing in $ \widetilde{R}_{u,v}(t)$ are all of the same parity. More precisely, there is a polynomial $Q_{u,v}(t) \in \mathbb{N}[t]$ such that $ \widetilde{R}_{u,v}(t)= Q_{u,v}(t^2)$ if $\ell(u,v) \equiv 0 \pmod{2}$ and $ \widetilde{R}_{u,v}(t)= t Q_{u,v}(t^2)$ if $\ell(u,v) \equiv 1 \pmod{2}$. It is of interest to characterize the $R$-polynomials. In this respect I propose the following conjecture which is a generalization of [BreDM, Conj. 7.1].

Let $(W,S)$ be a finite Coxeter system and $u,v \in W$, $u \leq v$. Then the polynomial $Q_{u,v}(t)$ is log-concave.

背景链接：https://arxiv.org/abs/2410.09897

13237 | AMR-125-0001

### Coxeter Groups and Unimodality - Problem 1

O / open | AMR 问题清单汇编

It is well known (see, e.g., [Hum, Prop. 5.12] or [BB, Cor. 7.1.4]) that if $(W,S)$ is a Coxeter system then $\sum_{u \in W} x^{\ell(u)}$ is a rational generating function. The following problem was proposed by Stembridge in [Ste97].

Let $(W,S)$ be a Coxeter system. Is it true that then \[ \sum_{t \in T} x^{\ell(t)} \] is a rational generating function?

背景链接：https://arxiv.org/abs/2410.09897

13238 | AMR-125-0010

### Coxeter Groups and Unimodality - Problem 10

O / open | AMR 问题清单汇编

The following problem was raised in [BreChro, Prob. 7.1].

Does $\tau(G;x)$ have only real roots for all graphs $G$?

背景链接：https://arxiv.org/abs/2410.09897

13239 | AMR-125-0011

### Coxeter Groups and Unimodality - Problem 11

O / open | AMR 问题清单汇编

Let $\sum_{i=0}^{d} h_i t^i \in \mathbb{N}[t]$ be a symmetric, monic polynomial with only real roots. Is there a simplicial convex polytope whose $h$-vector is $(h_0, \ldots, h_d)$?

背景链接：https://arxiv.org/abs/2410.09897

13240 | AMR-125-0012

### Coxeter Groups and Unimodality - Problem 12

O / open | AMR 问题清单汇编

Let $\sum_{i=0}^{d} h_i t^i \in \mathbb{N}[t]$ be a polynomial with only real roots such that $h_0=1$ and $h_0 < h_1 < \cdots < h_k$ for some $0 \leq k \leq d$. Is there a simplicial complex whose $f$-vector is $(h_0, h_1-h_0, \ldots, h_k-h_{k-1})$?

背景链接：https://arxiv.org/abs/2410.09897

13241 | AMR-125-0006

### Coxeter Groups and Unimodality - Problem 6

O / open | AMR 问题清单汇编

It is known ([Polo], see also [CasRep]) that if $P(q) \in \mathbb{N}[q]$ is such that $P(0)=1$ then there are $n \in \mathbb{P}$ and $u,v \in S_n$ such that $P(q)=P_{u,v}(q)$. The following related problem was posed by Bj\"{o}rner ([BjoPC]).

Let $(W,S)$ be a Coxeter system and $u,v \in W$, $u \leq v$. Is it true that then there are a Coxeter system $(W',S')$ and $w \in W'$ such that \[ P_{u,v}(q)=P_{e,w}(q) \; ? \]

背景链接：https://arxiv.org/abs/2410.09897

13242 | AMR-125-0007

### Coxeter Groups and Unimodality - Problem 7

O / open | AMR 问题清单汇编

The proof of the celebrated nonnegativity conjecture for Kazhdan-Lusztig polynomials ([EW]) makes the following problem even more compelling.

Find a combinatorial interpretation for Kazhdan-Lusztig polynomials.

背景链接：https://arxiv.org/abs/2410.09897

13243 | OWR-11695867-009

### Cubic and Higher Modulus Borwein Conjectures

P / partially_solved | Oberwolfach 报告问题汇编

Can the remaining cubic and higher-modulus Borwein sign conjectures be proved, including the modulus-$5$ and modulus-$7$ cases?

来源字段：https://doi.org/10.4171/owr/2022/57

13244 | AIM-COMBINATORICS-0139

### Curvature versus clustering in regular networks

P / partially_solved | AIM 研讨会问题汇编

**Problem.** “Is curvature related to clustering?”

**Remark.** “What is meant by clustering?”

For a simple unweighted graph with hop distance and uniform neighbor measures, compare non-lazy Ollivier edge curvature, and its vertex average, with the Watts--Strogatz local triangle coefficient. What quantitative relation holds, and does either quantity determine the other?

来源字段：http://aimpl.org/largenetworks/1/

13245 | AIM-COMBINATORICS-0143

### Curvature, cores, capacity, and the price of detours

P / partially_solved | AIM 研讨会问题汇编

**What precise notions of negative curvature imply congestion?**

What price are you willing to pay to avoid congestion?

*Where* does congestion occur? (something analogous to convex core?)

来源字段：http://aimpl.org/largenetworks/1/

13246 | AIM-COMBINATORICS-0192

### Cut covers and small cases of Mader's local-connectivity conjecture

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.22 (Mader).** If a digraph $D$ has $\delta_D^+\ge r$, then there are vertices $x\ne y$ such that $\lambda(x,y)\ge r$. Also, there is an edge $(x,y)$ so that $\lambda(x,y)\ge r$.

The first part of this conjecture was proven by Mader [24] for $\lambda(x,y)\ge r-1$.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13247 | AIM-COMBINATORICS-0071

### Cycle--clique Ramsey numbers and an endpoint-fan certificate

P / partially_solved | AIM 研讨会问题汇编

Cycle versus cliques

Jozef Skokan

Easy: $r(C_m,K_n)\ge (m-1)(n-1)+1$.

Conjecture: (Erdős) $r(C_m,K_n)=(m-1)(n-1)+1$ if $m\ge n$.

Known: (Bondy--Erdős 1973) True if $m\ge n^2$. (Nikiforov) True if $m\ge4n+6$.

Why should $m\ge n$?

Easy: $r(P_m,K_n)=(m-1)(n-1)+1$ for all $m,n$.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13248 | AMR-029-0024

### Cyclic orderings of matroids

O / open | AMR 问题清单汇编

Let M be a matroid on ground set S, and suppose that S can be partitioned into k bases. Is it true that there is a cyclic ordering of the elements of S such that any $|S|/k$ consecutive elements in this ordering form a basis?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13249 | OWR-11781-002

### Decidability of Hypergraph Turán Densities

O / open | Oberwolfach 报告问题汇编

Given a finite family $F$ of $r$-uniform hypergraphs and $\alpha\in\mathbb{Q}$, is it decidable whether the Turán density $\pi(F)$ satisfies $\pi(F)\leq\alpha$?

来源字段：https://doi.org/10.4171/owr/2012/19

13250 | AMR-029-0025

### Deciding kernel-perfectness

O / open | AMR 问题清单汇编

What is the complexity of deciding kernel-perfectness in various classes of digraphs?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13251 | AMR-029-0026

### Deciding the validity of the score sequence of a soccer tournament

P / partially_solved | AMR 问题清单汇编

In a soccer tournament of n teams, every pair of teams plays one match. The winner gets 3 points, the loser gets 0, while both teams receive 1 point in case of a draw. Is there a polynomial algorithm to decide whether a given score sequence (a score for each team) can be the end result of a tournament?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13252 | OWR-11786-019

### Decomposability Thresholds for Simplicial Polytopes

P / partially_solved | Oberwolfach 报告问题汇编

What is the least $k=k(d)$ such that every simplicial $d$-polytope is weakly $k$-decomposable, respectively $k$-decomposable, while some simplicial $d$-polytope is not weakly $(k-1)$-decomposable, respectively $(k-1)$-decomposable?

来源字段：https://doi.org/10.4171/owr/2012/24

13253 | AMR-029-0027

### Decomposing rooted (k,l)-connected graphs into rooted k-connected parts

O / open | AMR 问题清单汇编

Let G=(V,E) be an undirected graph, and $r \in V$ a root node. G is called rooted (k,l)-connected if G-X is $(k-\vert X\vert)l$-edge-connected for any $X \subseteq V-r$. Is it true that every rooted (k,l)-connected graph contains l edge-disjoint spanning rooted k-connected subgraphs?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13254 | AMR-029-0028

### Decomposition of oriented k-partition-connected digraphs

O / open | AMR 问题清单汇编

Let D=(V,A) be a digraph whose underlying graph is k-partition-connected, and let $r_0 \in V$ be a node of in-degree 0. Suppose that the in-degree of every other node is at least k. Is it true that D can be decomposed into k weakly connected spanning subgraphs, so that every node $v \in V-r_0$ has positive in-degree in each subgraph?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13255 | OWR-16633-025

### Decomposition of Representable Polymatroids into Matroid Ranks

P / partially_solved | Oberwolfach 报告问题汇编

Let $f$ be a polymatroid function on a ground set $V$ that is linearly representable over a field $F$: there are subspaces $T_v$ such that

$$f(S)=\operatorname{rk}\!\left(\bigcup_{v\in S}T_v\right).$$

Is $f$ always a nonnegative linear combination of linear-matroid rank functions over $F$?

来源字段：https://doi.org/10.4171/owr/2018/50

13256 | AIM-COMBINATORICS-0203

### Defect feedback sets and a fractional obstruction for four-order majority digraphs

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.33 (Thomassé and Charbit).** Caccetta--Häggkvist holds for \(3/4\)-digraphs.

**Recovered four-order conjecture.** Every \(n\)-vertex \(3/4\)-digraph has a vertex \(v\) with \[ d^+(v)<\frac n3. \]

The definition is in the immediately preceding source paragraph. Given integers \(\alpha,\beta\) and \(\beta\) linear orders \(\sigma_1,\ldots,\sigma_\beta\) on a finite vertex set \(V\), the associated \(\alpha/\beta\)-digraph \(D\) has

\[ xy\in A(D)\quad\Longleftrightarrow\quad x<_{\sigma_i}y\text{ in at least }\alpha\text{ of the }\beta\text{ orders}. \]

Thus a \(3/4\)-digraph is represented by **exactly four** linear orders and has \(xy\) precisely when at least three of the four put \(x\) before \(y\). A \(2\)-\(2\) tie produces no arc in either direction. Repetition among the four orders is not forbidden.

The short AIM sentence is potentially ambiguous about how much of the full Caccetta--Häggkvist girth conjecture is intended. Its own note, and the primary source in Charbit's 2005 thesis, make the concrete question unambiguous:

Because outdegree is integral, this is \(\delta^+(D)\leq\lfloor(n-1)/3\rfloor\). In particular, when \(3\mid n\) the conjecture requires \(\delta^+(D)\leq n/3-1\), not merely \(\delta^+(D)\leq n/3\).

The AIM paragraph also calls an \(\alpha/\beta\)-digraph a “majority digraph” when \(\alpha/\beta>2/3\). This is a supermajority convention, not the unrelated notions called majority digraph or majority coloring in other literatures. For four orders, “strict majority of four voters” in social choice means the same threshold, namely at least three.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13257 | AMR-030-0016

### Define the "crossing number" of a graph to be the minimum number of (topological) crossings of edges in any straight-lin

O / open | AMR 问题清单汇编

For a finite graph G, let cr(G) be the minimum number of crossings over topological drawings in the plane, and let pcr(G) be the minimum number of pairs of edges that cross, over such drawings. Must cr(G)=pcr(G) for every G?

背景链接：https://people.math.sc.edu/cooper/combprob.html

13258 | AMR-030-0017

### Define the discrepancy of a graph to be the largest value of D(S,T) = | |S||T|/2 - e(S,T) |, over all disjoint vertex se

P / partially_solved | AMR 问题清单汇编

Chung, Graham: Define the discrepancy of a graph to be the largest value of D(S,T) = | |S||T|/2 - e(S,T) |, over all disjoint vertex sets S and T. Suppose that G has no induced copy of some graph H. How large must the discrepancy be?

背景链接：https://people.math.sc.edu/cooper/combprob.html

13259 | AIM-COMBINATORICS-0064

### Degree-sensitive semi-algebraic Ramsey bounds

P / partially_solved | AIM 研讨会问题汇编

Question (Bukh--Matoušek): Can this be improved to a constant number of logs?

Known: In dimension 1, two logs suffice, and this is tight.

(Conlon--Fox--Pach--Sudakov--Suk) \(r_f(n)\geq c\log\cdots\log n\), with \(k-2\) logarithms.

If the dimension is high enough, this is sharp (in \(\mathbb R^{c(k)}\)).

Variant: the unbalanced \(R_f^{(k)}(s,n)\).

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13260 | AIM-COMBINATORICS-0232

### Dense nonsumsets and cyclic-interval square roots

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.7 (B. Green).** What is the size of the largest subset of \(\mathbb F_p\) which is not a sumset \(B+B\)?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13261 | AMR-029-0029

### Destroying rigidity

O / open | AMR 问题清单汇编

Let G be a graph that is rigid in two-dimensional space. Can we determine in polynomial time the minimum number of edges whose deletion from G results in a graph which is not rigid?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13262 | OWR-4425-001

### Detection of Highly Fluctuating Sequences

O / open | Oberwolfach 报告问题汇编

How can sequences with large variations or fluctuations be detected and defined?

来源字段：https://doi.org/10.4171/owr/2010/37

13263 | OWR-14298158-015

### Diagonal Permuton Limits for Skew-Sum Avoidance

P / partially_solved | Oberwolfach 报告问题汇编

别名：Point-Probability Criterion Commentary [OWR-14298158-016]

For every nonempty permutation $\alpha$, is the permuton limit of $\operatorname{Av}(\alpha\ominus1)$ the diagonal permuton?

来源字段：https://doi.org/10.4171/owr/2024/6

13264 | OPG-351

### Diagonal Ramsey numbers

O / open | Open Problem Garden

Let $R(k,k)$ denote the $k^{th}$ diagonal Ramsey number.

Conjecture $\lim_{k \rightarrow \infty} R(k,k) ^{\frac{1}{k}}$ exists.

Problem Determine the limit in the above conjecture (assuming it exists).

背景链接：http://www.openproblemgarden.org/op/diagonal_ramsey_numbers

13265 | OWR-11786-008

### Diameter Bounds for Connected Layer Families

P / partially_solved | Oberwolfach 报告问题汇编

For connected layer families on $d$-element multisets from $n$ symbols, is $\widetilde f(d,n)=d(n-1)$? In particular, can this be proved or refuted for $d=3$? For layer families arising from triangulated surfaces, are the upper bounds $2n$ with boundary and $n$ without boundary tight?

来源字段：https://doi.org/10.4171/owr/2012/24

13266 | OWR-14299577-018

### Difference Sets of Signed Representatives

P / partially_solved | Oberwolfach 报告问题汇编

Let $n\ge1$ and let $A\subseteq\{-n,\ldots,n\}\setminus\{0\}$ contain exactly one of $k$ and $-k$ for each $1\le k\le n$. Must $|(A-A)\cap[1,n]|=(1-o(1))n$? More strongly, for each $\varepsilon>0$ and every $B\subseteq[1,n]$ with $|B|\ge\varepsilon n$, must $\sum_{a,b\in A}\mathbf1_B(a-b)\gg_\varepsilon n^2$?

来源字段：https://doi.org/10.4171/owr/2025/51

13267 | OWR-4425-016

### Digit Sum Bias and Nonvanishing Hankel Determinants

P / partially_solved | Oberwolfach 报告问题汇编

With $\gamma_\infty(t)=\lim_{k\to\infty}2^{-k}|\{0\leq n<2^k:s_2(n+t)\geq s_2(n)\}|$, is $\gamma_\infty(t)>1/2$ for every integer $t\geq1$, what is $\liminf_{t\to\infty}\gamma_\infty(t)$, and does there exist an infinite sequence $a$ over a finite subset of $\mathbb{Z}$ for which every Hankel determinant $\det(a_{k+i+j})_{0\leq i,j<n}$ is nonzero for all $n\geq1$ and $k\geq0$?

来源字段：https://doi.org/10.4171/owr/2010/37

13268 | OPG-696

### Ding's tau_r vs. tau conjecture

P / partially_solved | Open Problem Garden

Conjecture Let $r \ge 2$ be an integer and let $H$ be a minor minimal clutter with $\frac{1}{r}\tau_r(H) < \tau(H)$. Then either $H$ has a $J_k$ minor for some $k \ge 2$ or $H$ has Lehman's property.

背景链接：http://www.openproblemgarden.org/op/dings_tau_r_vs_tau_conjecture

13269 | AIM-COMBINATORICS-0246

### Direction-complete finite-field sets and quadratic secant lifting

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.9 (A. Granville).** Given a finite field \(F\) and an integer \(n\ge 1\), find the smallest size of a subset \(E\subset F^n\) which determines all directions in \(F^n\).

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13270 | AMR-029-0030

### Disjoint spanning in- and out-arborescences

O / open | AMR 问题清单汇编

Does there exist a value k so that in every k-arc-connected directed graph D=(V,A) and for every node $v\in V$, there is a spanning in-arborescence and a disjoint spanning out-arborescence rooted in v?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13271 | GREEN-079

### Disjoint Sumsets Construction

O / open | Ben Green 问题

For arbitrarily large $n$, does there exist an abelian group $H$ with $|H| = n^{2+o(1)}$ and subsets $A_1, \dots, A_n, B_1, \dots, B_n$ satisfying $|A_i||B_i| \geq n^{2-o(1)}$, $|A_i + B_i| = |A_i||B_i|$, such that $A_i + B_i$ are pairwise disjoint from $A_j + B_k$ ($j \neq k$)?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13272 | OWR-16763-024

### Disjoint Transversal Bases from a Family of Matroid Bases

O / open | Oberwolfach 报告问题汇编

Given a rank-$n$ matroid and bases $B_1,\ldots,B_n$, do there exist $n$ pairwise disjoint transversal bases, each choosing one representative from every $B_i$?

来源字段：https://doi.org/10.4171/owr/2019/1

13273 | AMR-054-0039

### Distances among Point Sets in $\mathbb{R}^2$ and $\mathbb{R}^3$

P / partially_solved | AMR 问题清单汇编

For a point set $P$ in $\mathbb{R}^d$, let $f_d(P)$ be the number of unit-distance point pairs: $$f_d(P) = \left| \{ (u,v) \mid u, v \in P, \, \|u-v\| = 1 \} \right| \;;$$ and let $f_d(n)$ be the maximum over all sets of $n$ points: $$f_d(n) = \max_{|P| = n} f_d(P) \;.$$ Further, let $g_d(P)$ denote the number of distinct distances induced by a set of points $P$: $$g_d(P) = \left| \{ \|u-v\| \mid u, v \in P \} \right| \;;$$ and let $g_d(n)$ be the minimum over all sets of $n$ points: $$g_d(n) = \min_{|P|=n} g_d(P) \;.$$ Give upper and lower bounds on $f_d(n)$ and $g_d(n)$, particularly for $d=2$ and $d=3$.

背景链接：https://topp.openproblem.net/p39

13274 | AMR-031-0002

### Dittert-Hajek conjecture

P / partially_solved | AMR 问题清单汇编

Let $A=(a_{ij})$ be an $n\times n$ matrix with nonnegative entries and total entry sum $n$. Define $$\phi(A)=\prod_{i=1}^n\sum_{j=1}^n a_{ij}+\prod_{j=1}^n\sum_{i=1}^n a_{ij}-\operatorname{per}(A).$$ Is $\phi$ uniquely maximized by the matrix $A=J_n/n$, where every entry of $J_n$ is $1$?

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Combinatorics

13275 | OPG-37222

### Dividing up the unrestricted partitions

O / open | Open Problem Garden

Begin with the generating function for unrestricted partitions:

(1+x+x^2+...)(1+x^2+x^4+...)(1+x^3+x^6+...)...

Now change some of the plus signs to minus signs. The resulting series will have coefficients congruent, mod 2, to the coefficients of the generating series for unrestricted partitions. I conjecture that the signs may be chosen such that all the coefficients of the series are either 1, -1, or zero.

背景链接：http://www.openproblemgarden.org/op/dividing_up_the_unrestricted_partitions

13276 | AIM-COMBINATORICS-0017

### Divisor-energy reduction and quantifier obstruction for square polynomials

P / partially_solved | AIM 研讨会问题汇编

**Rudin's problem for $\mathbb F_p[x]$, $p>2$.** Suppose $A\subset \mathbb F_p[x]$ is of finite size, and every element of $A$ is a square. Show that $|A+A|\geq |A|^{1+\epsilon}$ for some constant $\epsilon$.

来源字段：http://aimpl.org/addcombapp/1/

13277 | AMR-030-0008

### Does every thrackle have average degree at most 2

P / partially_solved | AMR 问题清单汇编

Conway : Does every thrackle have average degree at most 2? A thrackle is a drawing of a graph in the plane so that every two edges share exactly one point (whether it is an endpoint of two incident edges or a transverse crossing -- no tangencies allowed).

背景链接：https://people.math.sc.edu/cooper/combprob.html

13278 | AMR-030-0014

### Does lim R(k,k)^(1/k) exist

O / open | AMR 问题清单汇编

Erdős: Does lim R(k,k)^(1/k) exist? What is it? (If it exists, it's between sqrt(2) and 4.) See "Small Ramsey Numbers" by Stanislaw Radziszowski.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13279 | AIM-COMBINATORICS-0120

### Drift splits generalized IDLA scaling

P / partially_solved | AIM 研讨会问题汇编

(30) Persi Diaconis: “What does a limiting shape theory for generalized internal DLA (see [22], [4], and [21]) look like?”

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13280 | OPG-2359

### Edge-antipodal colorings of cubes

P / partially_solved | Open Problem Garden

We let $Q_d$ denote the $d$-dimensional cube graph. A map $\phi: E(Q_d) \rightarrow \{0,1\}$ is called edge-antipodal if $\phi(e) \neq \phi(e')$ whenever $e,e'$ are antipodal edges.

Conjecture If $d \ge 2$ and $\phi: E(Q_d) \rightarrow \{0,1\}$ is edge-antipodal, then there exist a pair of antipodal vertices $v,v' \in V(Q_d)$ which are joined by a monochromatic path.

背景链接：http://www.openproblemgarden.org/op/edge_antipodal_colorings_of_cubes

13281 | AMR-029-0031

### Edge-covering number of 2-polymatroids

O / open | AMR 问题清单汇编

Let f be a 2-polymatroid function on S that has a matroid representation $M=(S \times \{1,2\},r)$ with the following property: $|C\cap \{(e,1),(e,2)\}| \leq 1$ for every $e \in S$ and every circuit C of M. Is it true that $\beta(f) \leq \beta^*(f)+1$, i.e. the minimum cover by matchings is at most one more than the minimum fractional cover by matchings?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13282 | AMR-029-0032

### Edge-independent spanning trees

O / open | AMR 问题清单汇编

In a graph G=(V,E) with a root node r, two spanning trees $T_1$ and $T_2$ are called edge-independent if for any node x in V-r, the unique paths between r and x in $T_1$ and $T_2$ are edge-disjoint. Is it true that if G is k-edge-connected then there exist k edge-independent spanning trees for arbitrary r?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13283 | OWR-15215-002

### Endpoint Discrepancy of Finite Point Sets

O / open | Oberwolfach 报告问题汇编

For finite point sets $P_n\subset[0,1]^d$, determine the sharp lower and upper asymptotic orders of the $L^p$-discrepancy for $p=1$ and $p=\infty$.

来源字段：https://doi.org/10.4171/owr/2017/10

13284 | AIM-COMBINATORICS-0053

### Endpoint-deck gluing for one-edge deletion at a 3-uniform Ramsey threshold

P / partially_solved | AIM 研讨会问题汇编

**Problem (Dudek--Rödl).** Let \(N=r_3(n,n)\). Is it true that every red--blue edge-coloring of \(K_N^{(3)}-e\) contains a monochromatic \(K_n^{(3)}\)? Here \(K_N^{(3)}-e\) denotes the complete 3-uniform hypergraph on \(N\) vertices with one edge deleted.

The source record (AIM problem list, workshop *Graph Ramsey theory*, section *Hypergraph size Ramsey numbers*) asks:

The record also notes the lower bound \(\widehat r_3(n,n)\geq \operatorname{poly}(n)r_3(n,n)^2\), and asks specifically about \(n=4\), where \(r_3(4,4)=13\).

All choices of the deleted triple are equivalent, since the symmetric group on the vertices of \(K_N^{(3)}\) is transitive on triples. Thus the word "every" does not create different deletion types.

There is one boundary issue. Literally, the question is false for \(n=3\): \(K_3^{(3)}\) consists of one edge, so \(r_3(3,3)=3\), whereas deleting that edge leaves no copy of \(K_3^{(3)}\). The explicit emphasis on \(n=4\) and the published formulation for nontrivial clique Ramsey problems indicate that the intended range is \(n\geq4\). We analyze that range below.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13285 | OWR-15206-009

### Enumeration and Structure of Higher-Dimensional Permutations

O / open | Oberwolfach 报告问题汇编

A $d$-dimensional permutation of order $n$ is a $\{0,1\}$-valued array on $[n]^{d+1}$ with exactly one entry equal to $1$ on every axis-parallel line. For each fixed $d\geq3$, is the number of such arrays $$ \left((1+o(1))\frac{n}{e^d}\right)^{n^d} $$ as $n\to\infty$?

来源字段：https://doi.org/10.4171/owr/2017/1

13286 | OWR-12872-014

### Enumeration of Graphs with Prescribed Degree Sequences

P / partially_solved | Oberwolfach 报告问题汇编

Determine, exactly or asymptotically, the number of simple labelled graphs with a prescribed nonnegative degree sequence $\mathbf d=(d_1,\ldots,d_n)$.

来源字段：https://doi.org/10.4171/owr/2014/12

13287 | AMR-122-0001

### Enumeration of Matchings - Problem 1

O / open | AMR 问题清单汇编

Show that in the $2n-1,2n,2n-1$ semiregular hexagon, the central location (consisting of the two innermost triangles) is covered by a lozenge in exactly one-third of the tilings.

(Equivalently: Show that if one chooses a random perfect matching of the dual graph, the probability that the central edge is contained in the matching is exactly 1/3.)

背景链接：https://arxiv.org/abs/math/9801060

13288 | AMR-122-0010

### Enumeration of Matchings - Problem 10

O / open | AMR 问题清单汇编

What about $(2n-1)$-by-$2n$ rectangles? For these regions, removing the central square does not make the region tilable. However, if one removes any one of the four squares adjacent to the middle square, one obtains a region that is tilable, and moreover, for this region the number of tilings appears to be a nice round number.

Find a formula for the number of domino tilings of a $(2n-1)$-by-$2n$ Aztec rectangle with a square adjoining the central square removed.

背景链接：https://arxiv.org/abs/math/9801060

13289 | AMR-122-0011

### Enumeration of Matchings - Problem 11

O / open | AMR 问题清单汇编

Perhaps this is a special case of a two-parameter fact that says that you can take an intrusion of length $m$ in a $2n$-by-$2n$ square and the number of tilings of the resulting region will always be a square or twice a square.

What is going on with ``intruded Aztec diamonds''? In particular, why is the number of tilings so square-ish?

背景链接：https://arxiv.org/abs/math/9801060

13290 | AMR-122-0012

### Enumeration of Matchings - Problem 12

O / open | AMR 问题清单汇编

Consider the Kasteleyn matrix $K_n$ for the Aztec diamond of order $n$, in which every other vertical domino has its sign flipped (that is, the corresponding 1's in the bipartite adjacency matrix are replaced by $-1$'s).

Show that the sum of the entries of the matrix inverse of $K_n$ is $(n-1)(n+3)/2 - 2^{n-1} + 2$.

(This formula works for $n=1$ through $n=8$.)

I should mention in this connection that Greg Kuperberg has some high-tech ruminations on the inverses of Kasteleyn matrices, and there is a chance that representation-theory methods will be useful here.

背景链接：https://arxiv.org/abs/math/9801060

13291 | AMR-122-0013

### Enumeration of Matchings - Problem 13

O / open | AMR 问题清单汇编

Find a general formula for the number of domino tilings of even pillows.

背景链接：https://arxiv.org/abs/math/9801060

13292 | AMR-122-0014

### Enumeration of Matchings - Problem 14

O / open | AMR 问题清单汇编

Find a general formula for the number of domino tilings of Aztec windows.

Even an argument explaining why the number of tilings for windows of inner order $x$ and outer order $x+w$ should be given by a polynomial in $x$ (for each fixed $w$) would constitute progress!

背景链接：https://arxiv.org/abs/math/9801060

13293 | AMR-122-0015

### Enumeration of Matchings - Problem 15

O / open | AMR 问题清单汇编

One case that has not yet been settled is the case that arises from a rather symmetric dissection of the plane into equilateral triangles, squares, and regular hexagons, with 4 polygons meeting at each vertex. Empirically, one finds that the number of diform tilings is $2^{n(n+1)}$.

Prove that for the Aztec-type regions in the dissection of the plane into triangles, squares, and hexagons, the number of tilings of the region of order $n$ is $2^{n(n+1)}$.

(See {\tt http://www-math.mit.edu/$\sim$propp/dragon.ps} for a picture of one of these tilings.)

背景链接：https://arxiv.org/abs/math/9801060

13294 | AMR-122-0016

## Enumeration of Matchings - Problem 16

O / open | AMR 问题清单汇编

For instance, we have the roughly hexagonal region shown at the top of the next page (a union of square and triangular pieces, with certain boundary vertices marked with a ``.''\ so as to bring out the large-scale 2,3,2,2,3,2 hexagonal structure more clearly); it has 17920 tilings, where 17920 is $2^9 \cdot 5 \cdot 7$. More generally, if one takes an $a,b,c$ quasi-hexagon, one finds that one gets a large power of 2 times a product of powers of odd primes in which all the primes are fairly small (and their exponents are too).

Find a formula for the number of diform tilings in the $a,b,c$ quasi-hexagon in the dissection of the plane that arises from slicing the dissection into squares along every third upward-sloping diagonal.

I should mention that one reason for my special interest in Problem 16 is that it seems to be a genuine hybrid of domino tilings of Aztec diamonds and lozenge tilings of hexagons. I should also mention that I think the problem will yield to some generalization of the method of subgraph substitution that has already been of such great use in enumeration of matchings of graphs.

\begin{verbatim} ._____ /| | / | | ./____|_____|_____._____ /| | | /| | / | | | / | | ./____|_____|_____|/____|_____|_____. /| | | /| | | /| / | | | / | | | / | ./____|_____|_____|/____|_____|_____|/____|_____ | | | /| | | /| | | | | | / | | | / | | | |_____|_____|/____|_____|_____|/____|_____|_____. | /| | | /| | | /| | / | | | / | | | / | ./____|_____|_____|/____|_____|_____|/____|_____ | | | /| | | /| | | | | | / | | | / | | | |_____|_____|/____|_____|_____|/____|_____|_____. | /| | | /| | | / | / | | | / | | | / ./____|_____|_____|/____|_____|_____./ | | /| | | / | | | / | | | / |_____./____|_____|_____./ | | / | | / |_____./ \end{verbatim}

I will not go into great detail here on this method (the interested reader can examine {\tt http://www-math.mit.edu/$\sim$propp/} {\tt fpsac96.ps.gz}). Here is an overview: One studies not graphs but weighted graphs, with weights assigned to edges, and one does weighted enumeration of perfect matchings, where the weight of a matching is the product of the weights of the constituent edges. One then looks at local substitutions with a graph that preserve the sum of the weights of the matchings, or more generally, multiply the sum of the weights of the matchings by some predictable factor. Then the problem of weight-enumerating matchings of one graph reduces to the problem of weight-enumerating matchings of another (hopefully simpler) graph. Iterating this procedure, one can often eventually reduce the graph to something one already understands. I am confident that this will apply to Problem 16, yielding a reduction to the standard $a,b,c$ semiregular hexagon.

背景链接：https://arxiv.org/abs/math/9801060

13295 | AMR-122-0017

## Enumeration of Matchings - Problem 17

O / open | AMR 问题清单汇编

Characterize those local substitutions that have a predictable effect on the weighted sum of matchings of a graph.

The most useful local substitution so far has been \begin{verbatim}

q q | C / \ D a u b / \ / \ / \ p---t v---r ----> p r \ / \ / d w c \ / | B \ / A s s \end{verbatim} (where unmarked edges have weight 1 and where $A,B,C,D$ are obtained from $a,b,c,d$ by dividing by $ac+bd$), but Rick Kenyon's substitution \begin{verbatim} 3/2 p---q p---q | | | | 1/2 r---s ----> r---s | | 2 | | t---u t u \end{verbatim} has also been of use (where vertices $p,q,t,u$ may be attached elsewhere in the graph, but not the vertices $r,s$).

背景链接：https://arxiv.org/abs/math/9801060

13296 | AMR-122-0018

## Enumeration of Matchings - Problem 18

O / open | AMR 问题清单汇编

How many perfect matchings does the triangle graph of order $n$ have?

背景链接：https://arxiv.org/abs/math/9801060

13297 | AMR-122-0019

## Enumeration of Matchings - Problem 19

O / open | AMR 问题清单汇编

One can also look at graphs that are bipartite but not planar. A natural example is the $n$-cube (that is, the $n$-dimensional cube with all sides of length 2). It has been shown that the number of perfect matchings of the $n$-cube goes like $1$, $2$, $9=3^2$, $272=16 \cdot 17$, $589185=3^2 \cdot 5 \cdot 13093$, $\dots$.

Find a formula for the number of perfect matchings of the $n$-cube.

(This may be intractable; after all, the graph has exponentially many vertices.)

背景链接：https://arxiv.org/abs/math/9801060

13298 | AMR-122-0002

## Enumeration of Matchings - Problem 2

O / open | AMR 问题清单汇编

\end{verbatim} These are similar to the numbers one gets from counting lozenge-tilings of an $n,n,n,n,n,n$ hexagon, in that the largest prime factor seems to be bounded by a linear function of $n$, and it ought to be possible to come up with a conjectural exact formula. What might be harder is proving it.

Enumerate the lozenge-tilings of the region obtained from the $n,n+1,n,n+1,n,n+1$ hexagon by removing the central triangle.

背景链接：https://arxiv.org/abs/math/9801060

13299 | AMR-122-0003

### Enumeration of Matchings - Problem 3

O / open | AMR 问题清单汇编

10 3 8 2 4 2 (2) (3) (5) (13) (17) (19)

2 2 2 3 4 4 8 4 (2) (5) (7) (11) (13) (17) (19) (23) \end{verbatim}

Enumerate the lozenge-tilings of the region obtained from the $2n,2n+3,2n,2n+3,2n,2n+3$ hexagon by removing a triangle from the middle of each of its long sides.

背景链接：https://arxiv.org/abs/math/9801060

13300 | AMR-122-0004

### Enumeration of Matchings - Problem 4

O / open | AMR 问题清单汇编

Determine the number of lozenge-tilings of a regular hexagon from which two of its innermost unit triangles (one upward-pointing and one downward-pointing) have been removed.

背景链接：https://arxiv.org/abs/math/9801060

13301 | AMR-122-0005

### Enumeration of Matchings - Problem 5

O / open | AMR 问题清单汇编

Determine the cokernel of the Carlitz matrices, or equivalently of the Kasteleyn matrices of $a,b,c$ hexagons, and if possible find a way to interpret the cokernel in terms of the tilings.

He points out that in the case $a=b=c=2$, one gets a non-cyclic group as the cokernel.

背景链接：https://arxiv.org/abs/math/9801060

13302 | AMR-122-0006

### Enumeration of Matchings - Problem 6

O / open | AMR 问题清单汇编

Digressing for a moment from the topic of lozenge tilings, I point out that in general, the Kasteleyn matrix $K$ is not canonically defined, in the sense that there may be many ways of modifying the signs of certain entries of the bipartite adjacency matrix of a graph so that all non-zero contributions to the determinant have the same sign. Thus, one should not expect the eigenvalues of $K$ to possess combinatorial significance. However, the spectrum of $K$ times its adjoint $K^*$ is independent of which Kasteleyn matrix $K$ one chooses (independently shown by David Wilson and Horst Sachs). Thus, it is natural to ask:

What is the significance of the spectrum of $K K^*$, where $K$ is any Kasteleyn matrix associated with a bipartite planar graph?

背景链接：https://arxiv.org/abs/math/9801060

13303 | AMR-122-0007

### Enumeration of Matchings - Problem 7

O / open | AMR 问题清单汇编

Find the ``vertical moments of inertia'' for the mass on edges arising from edge-probabilities for random matchings of the $a,b,c$ honeycomb graph.

背景链接：https://arxiv.org/abs/math/9801060

13304 | AMR-122-0008

### Enumeration of Matchings - Problem 8

O / open | AMR 问题清单汇编

47 2 (2) (3) (5) \end{verbatim} Note that only the presence of the large prime factor 73 makes one doubt that there is a general formula; the other prime factors are reassuringly small. Further data might make it clear what that 73 is doing there.

Count the domino tilings of an Aztec diamond from which two close-to-central squares, related by a knight's move, have been deleted.

背景链接：https://arxiv.org/abs/math/9801060

13305 | AMR-122-0009

### Enumeration of Matchings - Problem 9

O / open | AMR 问题清单汇编

One can also look at ``Aztec rectangles'' from which squares have been removed so as to restore the balance between black and white squares (a necessary condition for tilability). For instance, one can remove the central square from an $a$-by-$b$ Aztec rectangle in which $a$ and $b$ differ by 1, with the larger of $a,b$ odd: \begin{verbatim} XX XXXX XXXXXX XXXXXXXX XXXX XXXX XXXXXXXX XXXXXX XXXX XX \end{verbatim}

Find a formula for the number of domino tilings of a $2n$-by-$(2n+1)$ Aztec rectangle with its central square removed.

背景链接：https://arxiv.org/abs/math/9801060

13306 | OWR-1536-014

### Equalities Among Critical Biases in Positional Games

O / open | Oberwolfach 报告问题汇编

In $(1:b)$ biased Maker--Breaker positional games played on the edges of the complete graph $K_n$, let $b_H$, $b_{D_k}$, and $b_{C_k}$ denote respectively the critical threshold biases for the Hamiltonicity game, the minimum-degree-$k$ game, and the $k$-vertex-connectivity game. Do the asymptotic equalities $b_H = b_{D_2}$ and $b_{C_k} = b_{D_k}$ hold?

来源字段：https://doi.org/10.4171/owr/2007/20

13307 | OWR-13490-009

### Equality Cases in the Balanced Lower Bound Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For every balanced simplicial d-polytope P, the normalized h-numbers satisfy $\frac{h_0(P)}{\binom{d}{0}}\leq\cdots\leq\frac{h_{\lfloor d/2\rfloor}(P)}{\binom{d}{\lfloor d/2\rfloor}}$; equality at the relevant adjacent terms should characterize the balanced stacked property.

来源字段：https://doi.org/10.4171/owr/2015/5

13308 | OPG-692

### Equality in a matroidal circumference bound

P / partially_solved | Open Problem Garden

Question Is the binary affine cube $AG(3,2)$ the only 3-connected matroid for which equality holds in the bound $$E(M) \leq c(M) c(M^*) / 2$$where$c(M)$is the circumference (i.e. largest circuit size) of$M$?

背景链接：http://www.openproblemgarden.org/op/equality_in_a_matroidal_circumference_bound

13309 | OWR-15206-004

### Equiangular Lines and Few-Angle Spherical Codes

P / partially_solved | Oberwolfach 报告问题汇编

For $r\geq2$, is the maximum number $N_{1/(2r-1)}(n)$ of equiangular lines in $\mathbb{R}^n$ equal to $r(n-1)/(r-1)+O(1)$ as $n\to\infty$? For fixed $\alpha$, does every spherical code whose inner products belong to $\{x:-1\leq x<0\}\cup\{\alpha\}$ have size $O(n^2)$?

来源字段：https://doi.org/10.4171/owr/2017/1

13310 | AMR-029-0033

### Equitable list colouring

O / open | AMR 问题清单汇编

Is it true that every graph G is equitably k-list-colourable for any $k \geq \Delta(G)+1$?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13311 | OWR-17135-035

### Equivariant Log-Concavity of Orlik-Solomon Algebras

O / open | Oberwolfach 报告问题汇编

Let $M$ be a matroid, let $W$ be a finite symmetry group of $M$, and regard its Orlik-Solomon algebra $OS^*(M)$ as a graded $W$-representation. For every $i$, does $OS^i(M)\otimes OS^i(M)$ contain $OS^{i-1}(M)\otimes OS^{i+1}(M)$ as a subrepresentation?

来源字段：https://doi.org/10.4171/owr/2019/39

13312 | EP-1002

### Erdős Problem #1002

O / open | Erdős 问题

For any $0<\alpha<1$, let\[f(\alpha,n)=\frac{1}{\log n}\sum_{1\leq k\leq n}(\tfrac{1}{2}-\{ \alpha k\}).\]Does $f(\alpha,n)$ have an asymptotic distribution function?

In other words, is there a non-decreasing function $g$ such that $g(-\infty)=0$, $g(\infty)=1$, and\[\lim_{n\to \infty}\lvert \{ \alpha\in (0,1): f(\alpha,n)\leq c\}\rvert=g(c)?\]

来源字段：https://www.erdosproblems.com/1002

13313 | EP-101

### Erdős Problem #101

O / open | Erdős 问题

Given $n$ points in $\mathbb{R}^2$, no five of which are on a line, the number of lines containing four points is $o(n^2)$.

来源字段：https://www.erdosproblems.com/101

13314 | EP-102

### Erdős Problem #102

O / open | Erdős 问题

Let $c>0$ and $h_c(n)$ be such that for any $n$ points in $\mathbb{R}^2$ such that there are $\geq cn^2$ lines each containing more than three points, there must be some line containing $h_c(n)$ many points. Estimate $h_c(n)$. Is it true that, for fixed $c>0$, we have $h_c(n)\to \infty$?

来源字段：https://www.erdosproblems.com/102

13315 | EP-1023

### Erdős Problem #1023: Combinatorics

S / solved | Erdős 问题

Let $F(n)$ be the maximal size of a family of subsets of $\{1,\ldots,n\}$ such that no set in this family is the union of other members of the family. Is it true that there is a constant $c>0$ such that\[F(n)\sim c \frac{2^n}{n^{1/2}}?\]

来源字段：https://www.erdosproblems.com/1023

13316 | EP-1025

### Erdős Problem #1025: Combinatorics

S / solved | Erdős 问题

Let $f$ be a function from all pairs of elements in $\{1,\ldots,n\}$ to $\{1,\ldots,n\}$ such that $f(x,y)\neq x$ and $\neq y$ for all $x,y$. We call $X\subseteq \{1,\ldots,n\}$ independent if whenever $x,y\in X$ we have $f(x,y)\not\in X$.

Let $g(n)$ be such that, in every function $f$, there is an independent set of size at least $g(n)$. Estimate $g(n)$.

来源字段：https://www.erdosproblems.com/1025

13317 | EP-1026

### Erdős Problem #1026: Combinatorics

S / solved | Erdős 问题

Let $x_1,\ldots,x_n$ be a sequence of distinct real numbers. Determine\[\max\left(\sum x_{i_r}\right),\]where the maximum is taken over all monotonic subsequences.

来源字段：https://www.erdosproblems.com/1026

13318 | EP-1027

### Erdős Problem #1027: Combinatorics

S / solved | Erdős 问题

Let $c>0$, and let $n$ be sufficiently large depending on $c$. Suppose that $\mathcal{F}$ is a family of at most $c2^n$ many finite sets of size $n$. Let $X=\cup_{A\in \mathcal{F}}A$.

Must there exist $\gg_c 2^{\lvert X\rvert}$ many sets $B\subset X$ which intersect every set in $\mathcal{F}$, yet contain none of them?

来源字段：https://www.erdosproblems.com/1027

13319 | EP-1030

### Erdős Problem #1030

O / open | Erdős 问题

Let $R(k,l)$ be the usual Ramsey number: the smallest $n$ such that if the edges of $K_n$ are coloured red and blue then there exists either a red $K_k$ or a blue $K_l$.

Prove the existence of some $c>0$ such that\[\lim_{k\to \infty}\frac{R(k+1,k)}{R(k,k)}> 1+c.\]

来源字段：https://www.erdosproblems.com/1030

13320 | EP-1038

### Erdős Problem #1038

P / partially_solved | Erdős 问题

Determine the infimum and supremum of\[\lvert \{ x\in \mathbb{R}: \lvert f(x)\rvert < 1\}\rvert\]as $f\in \mathbb{R}[x]$ ranges over all non-constant monic polynomials, all of whose roots are real and in the interval $[-1,1]$.

来源字段：https://www.erdosproblems.com/1038

13321 | EP-1039

### Erdős Problem #1039

P / partially_solved | Erdős 问题

Let $f(z)=\prod_{i=1}^n(z-z_i)\in \mathbb{C}[z]$ with $\lvert z_i\rvert \leq 1$ for all $i$. Let $\rho(f)$ be the radius of the largest disc which is contained in $\{z: \lvert f(z)\rvert< 1\}$.

Determine the behaviour of $\rho(f)$. In particular, is it always true that $\rho(f)\gg 1/n$?

来源字段：https://www.erdosproblems.com/1039

13322 | EP-1040

### Erdős Problem #1040

P / partially_solved | Erdős 问题

Let $F\subseteq \mathbb{C}$ be a closed infinite set, and let $\mu(F)$ be the infimum of\[\lvert \{ z: \lvert f(z)\rvert < 1\}\rvert,\]as $f$ ranges over all polynomials of the shape $\prod (z-z_i)$ with $z_i\in F$.

Is $\mu(F)$ determined by the transfinite diameter of $F$? In particular, is $\mu(F)=0$ whenever the transfinite diameter of $F$ is $\geq 1$?

来源字段：https://www.erdosproblems.com/1040

13323 | EP-1062

### Erdős Problem #1062

P / partially_solved | Erdős 问题

Let $f(n)$ be the size of the largest subset $A\subseteq \{1,\ldots,n\}$ such that there are no three distinct elements $a,b,c\in A$ such that $a\mid b$ and $a\mid c$. How large can $f(n)$ be? Is $\lim f(n)/n$ irrational?

来源字段：https://www.erdosproblems.com/1062

13324 | EP-1063

### Erdős Problem #1063

P / partially_solved | Erdős 问题

Let $k\geq 2$ and define $n_k\geq 2k$ to be the least value of $n$ such that $n-i$ divides $\binom{n}{k}$ for all but one $0\leq i<k$. Estimate $n_k$.

来源字段：https://www.erdosproblems.com/1063

13325 | EP-1071

### Erdős Problem #1071

S / solved | Erdős 问题

Is there a finite set of unit line segments (rotated and translated copies of $(0,1)$) in the unit square, no two of which intersect, which are maximal with respect to this property?

Is there a region $R$ with a maximal set of disjoint unit line segments that is countably infinite?

来源字段：https://www.erdosproblems.com/1071

13326 | EP-1076

### Erdős Problem #1076: Hypergraphs

S / solved | Erdős 问题

Let $k\geq 5$ and let $\mathcal{F}_k$ be the family of all $3$-uniform hypergraphs with $k$ vertices and $k-2$ edges. Is it true that\[\mathrm{ex}_3(n,\mathcal{F}_k)\sim \frac{n^2}{6}?\]

来源字段：https://www.erdosproblems.com/1076

13327 | EP-1083

### Erdős Problem #1083

P / partially_solved | Erdős 问题

Let $d\geq 3$, and let $f_d(n)$ be the minimal $m$ such that every set of $n$ points in $\mathbb{R}^d$ determines at least $m$ distinct distances. Estimate $f_d(n)$ - in particular, is it true that\[f_d(n)=n^{\frac{2}{d}-o(1)}?\]

来源字段：https://www.erdosproblems.com/1083

13328 | EP-1087

### Erdős Problem #1087

P / partially_solved | Erdős 问题

Let $f(n)$ be minimal such that every set of $n$ points in $\mathbb{R}^2$ contains at most $f(n)$ many sets of four points which are 'degenerate' in the sense that some pair are the same distance apart. Estimate $f(n)$ - in particular, is it true that $f(n)\leq n^{3+o(1)}$?

来源字段：https://www.erdosproblems.com/1087

13329 | EP-1088

### Erdős Problem #1088

P / partially_solved | Erdős 问题

Let $f_d(n)$ be the minimal $m$ such that any set of $m$ points in $\mathbb{R}^d$ contains a set of $n$ points such that any two determined distances are distinct. Estimate $f_d(n)$. In particular, is it true that, for fixed $n\geq 3$,\[f_d(n)=2^{o(d)}?\]

来源字段：https://www.erdosproblems.com/1088

13330 | EP-1096

### Erdős Problem #1096

S / solved | Erdős 问题

Let $1<q<1+\epsilon$ and consider the set of numbers of the shape $\sum_{i\in S}q^i$ (for all finite $S$), ordered by size as $0=x_1<x_2<\cdots$.

Is it true that, provided $\epsilon>0$ is sufficiently small, $x_{k+1}-x_k \to 0$?

来源字段：https://www.erdosproblems.com/1096

13331 | EP-109

### Erdős Problem #109: Additive Combinatorics

S / solved | Erdős 问题

Any $A\subseteq \mathbb{N}$ of positive upper density contains a sumset $B+C$ where both $B$ and $C$ are infinite.

来源字段：https://www.erdosproblems.com/109

13332 | EP-1108

### Erdős Problem #1108

P / partially_solved | Erdős 问题

Let\[A = \left\{ \sum_{n\in S}n!: S\subset \mathbb{N}\textrm{ finite}\right\}.\]If $k\geq 2$, then does $A$ contain only finitely many $k$th powers? Does it contain only finitely many powerful numbers?

来源字段：https://www.erdosproblems.com/1108

13333 | EP-1117

### Erdős Problem #1117

P / partially_solved | Erdős 问题

Let $f(z)$ be an entire function which is not a monomial. Let $\nu(r)$ count the number of $z$ with $\lvert z\rvert=r$ such that $\lvert f(z)\rvert=\max_{\lvert z\rvert=r}\lvert f(z)\rvert$. (This is a finite quantity if $f$ is not a monomial.)

Is it possible for\[\limsup \nu(r)=\infty?\]Is it possible for\[\liminf \nu(r)=\infty?\]

来源字段：https://www.erdosproblems.com/1117

13334 | EP-1122

### Erdős Problem #1122

P / partially_solved | Erdős 问题

Let $f:\mathbb{N}\to \mathbb{R}$ be an additive function (i.e. $f(ab)=f(a)+f(b)$ whenever $(a,b)=1$). Let\[A=\{ n \geq 1: f(n+1)< f(n)\}.\]If $\lvert A\cap [1,X]\rvert =o(X)$ then must $f(n)=c\log n$ for some $c\in \mathbb{R}$?

来源字段：https://www.erdosproblems.com/1122

13335 | EP-1130

### Erdős Problem #1130

S / solved | Erdős 问题

For $x_1,\ldots,x_n\in [-1,1]$ let\[l_k(x)=\frac{\prod_{i\neq k}(x-x_i)}{\prod_{i\neq k}(x_k-x_i)},\]which are such that $l_k(x_k)=1$ and $l_k(x_i)=0$ for $i\neq k$.

Let $x_0=-1$ and $x_{n+1}=1$ and\[\Upsilon(x_1,\ldots,x_n)=\min_{0\leq i\leq n}\max_{x\in[x_i,x_{i+1}]} \sum_k \lvert l_k(x)\rvert.\]Is it true that\[\Upsilon(x_1,\ldots,x_n)\ll \log n?\]Describe which choice of $x_i$ maximise $\Upsilon(x_1,\ldots,x_n)$.

来源字段：https://www.erdosproblems.com/1130

13336 | EP-1131

### Erdős Problem #1131

P / partially_solved | Erdős 问题

For $x_1,\ldots,x_n\in [-1,1]$ let\[l_k(x)=\frac{\prod_{i\neq k}(x-x_i)}{\prod_{i\neq k}(x_k-x_i)},\]which are such that $l_k(x_k)=1$ and $l_k(x_i)=0$ for $i\neq k$.

What is the minimal value of\[I(x_1,\ldots,x_n)=\int_{-1}^1 \sum_k \lvert l_k(x)\rvert^2\mathrm{d}x?\]In particular, is it true that\[\min I =2-(1+o(1))\frac{1}{n}?\]

来源字段：https://www.erdosproblems.com/1131

13337 | EP-1132

### Erdős Problem #1132

P / partially_solved | Erdős 问题

For $x_1,\ldots,x_n\in [-1,1]$ let\[l_k(x)=\frac{\prod_{i\neq k}(x-x_i)}{\prod_{i\neq k}(x_k-x_i)},\]which are such that $l_k(x_k)=1$ and $l_k(x_i)=0$ for $i\neq k$.

Let $x_1,x_2,\ldots\in [-1,1]$ be an infinite sequence, and let\[L_n(x) = \sum_{1\leq k\leq n}\lvert l_k(x)\rvert,\]where each $l_k(x)$ is defined above with respect to $x_1,\ldots,x_n$.

Must there exist $x\in (-1,1)$ such that\[L_n(x) >\frac{2}{\pi}\log n-O(1)\]for infinitely many $n$?

Is it true that\[\limsup_{n\to \infty}\frac{L_n(x)}{\log n}\geq \frac{2}{\pi}\]for almost all $x\in (-1,1)$?

来源字段：https://www.erdosproblems.com/1132

13338 | EP-1145

### Erdős Problem #1145: and $B=\{1\leq b_1<b_2<\cdots\}$ be sets of integers with $a_n/b_n\to 1$

O / open | Erdős 问题

Let $A=\{1\leq a_1<a_2<\cdots\}$ and $B=\{1\leq b_1<b_2<\cdots\}$ be sets of integers with $a_n/b_n\to 1$.

If $A+B$ contains all sufficiently large positive integers then is it true that $\limsup 1_A\ast 1_B(n)=\infty$?

来源字段：https://www.erdosproblems.com/1145

13339 | EP-1157

### Erdős Problem #1157: Hypergraphs, Turan Number

O / open | Erdős 问题

Let $t,k,r\geq 2$. Let $\mathcal{F}$ be the family of all $r$-uniform hypergraphs with $k$ vertices and $s$ edges. Determine\[\mathrm{ex}_r(n,\mathcal{F}).\]

来源字段：https://www.erdosproblems.com/1157

13340 | EP-1158

### Erdős Problem #1158:

O / open | Erdős 问题

Let $K_{t}(r)$ be the complete $t$-partite $t$-uniform hypergraph with $r$ vertices in each class.

Is it true that\[\mathrm{ex}_t(n,K_t(r)) \geq n^{t-r^{1-t}-o(1)}\]for all $t,r$?

来源字段：https://www.erdosproblems.com/1158

13341 | EP-1159

### Erdős Problem #1159: Determine whether there exists a constant $C>1$ such that the following holds

O / open | Erdős 问题

Determine whether there exists a constant $C>1$ such that the following holds.

Let $P$ be a finite projective plane. Must there exist a set of points $S$ such that $1\leq \lvert S\cap \ell\rvert \leq C$ for all lines $\ell$?

来源字段：https://www.erdosproblems.com/1159

13342 | EP-117

### Erdős Problem #117

O / open | Erdős 问题

Let $h(n)$ be minimal such that any group $G$ with the property that any subset of $>n$ elements contains some $x\neq y$ such that $xy=yx$ can be covered by at most $h(n)$ many Abelian subgroups.

Estimate $h(n)$ as well as possible.

来源字段：https://www.erdosproblems.com/117

13343 | EP-1179

### Erdős Problem #1179: Additive Combinatorics, Probability

S / solved | Erdős 问题

Let $0<\epsilon<1$ and let $g_\epsilon(N)$ be the minimal $k$ such that if $G$ is an abelian group of size $N$ and $A\subseteq G$ is a uniformly random subset of size $k$, and\[F_A(g) = \#\left\{ S\subseteq A: g = \sum_{x\in S}x\right\},\]then, with probability $\to 1$ as $N\to \infty$,\[\left\lvert F_A(g)-\frac{2^k}{N}\right\rvert \leq \epsilon \frac{2^k}{N}\]for all $g\in G$.

Estimate $g_\epsilon(N)$ - in particular, is it true that for all $\epsilon>0$\[g_\epsilon(N)=(1+o_\epsilon(1))\log_2N?\]

来源字段：https://www.erdosproblems.com/1179

13344 | EP-1183

### Erdős Problem #1183: Combinatorics, Ramsey Theory

O / open | Erdős 问题

Let $f(n)$ be maximal such that in any $2$-colouring of the subsets of $\{1,\ldots,n\}$ there is always a monochromatic family of at least $f(n)$ sets which is closed under taking unions and intersections. Estimate $f(n)$.

Let $F(n)$ be defined similarly, except that we only require the family be closed under taking unions. Estimate $F(n)$. In particular, is it true that $F(n)\geq n^{\omega(n)}$ for some $\omega(n)\to \infty$ as $n\to \infty$, and $F(n)<(1+o(1))^n$?

来源字段：https://www.erdosproblems.com/1183

13345 | EP-1185

### Erdős Problem #1185: Additive Combinatorics, Arithmetic Progressions

S / solved | Erdős 问题

Let $\delta>0$ and $k\geq 3$. Is it true that there exists $m\geq 1$ (depending only on $\delta$ and $k$) such that, for all large $N$, if $A,B\subseteq \{1,\ldots,N\}$ with $\lvert A\rvert \geq \delta N$ and $\lvert B\rvert \geq m$ then there is a non-trivial $k$-term arithmetic progression in $A$ whose common difference is in $B-B$?

来源字段：https://www.erdosproblems.com/1185

13346 | EP-1186

### Erdős Problem #1186: Additive Combinatorics, Arithmetic Progressions

O / open | Erdős 问题

Let $\delta_k$ be such that in any $2$-colouring of $\{1,\ldots,n\}$ there exist at least $(\delta_k+o(1))n^2$ many monochromatic $k$-term arithmetic progressions. Give reasonable bounds (or even an asymptotic formula) for $\delta_k$.

来源字段：https://www.erdosproblems.com/1186

13347 | EP-119

### Erdős Problem #119

S / solved | Erdős 问题

Let $z_i$ be an infinite sequence of complex numbers such that $\lvert z_i\rvert=1$ for all $i\geq 1$, and for $n\geq 1$ let\[p_n(z)=\prod_{i\leq n} (z-z_i).\]Let $M_n=\max_{\lvert z\rvert=1}\lvert p_n(z)\rvert$.

Is it true that $\limsup M_n=\infty$?

Is it true that there exists $c>0$ such that for infinitely many $n$ we have $M_n > n^c$?

Is it true that there exists $c>0$ such that, for all large $n$,\[\sum_{k\leq n}M_k > n^{1+c}?\]

来源字段：https://www.erdosproblems.com/119

13348 | EP-1191

### Erdős Problem #1191: Additive Combinatorics, Sidon Sets

O / open | Erdős 问题

Let $A\subset\mathbb{N}$ be an infinite Sidon set. Is it true that\[\liminf_{x\to \infty} \frac{\lvert A\cap [1,x]\rvert}{x^{1/2}}(\log x)^{1/2}=0?\]Does there exist an infinite Sidon set $A$ such that\[\liminf_{x\to \infty} \frac{\lvert A\cap [1,x]\rvert}{x^{1/2}}(\log x)^c>0\]for some $c>0$?

来源字段：https://www.erdosproblems.com/1191

13349 | EP-1192

### Erdős Problem #1192: Additive Combinatorics, Additive Basis

O / open | Erdős 问题

For $A\subset \mathbb{N}$ let $f_r(n)$ count the number of solutions to $n=a_1+\cdots+a_r$ with $a_i\in A$.

Does there exist, for all $r\geq 2$, a basis $A$ of order $r$ (so that $f_r(n)>0$ for all large $n$) such that\[\sum_{n\leq x}f_r(n)^2 \ll x\]for all $x$?

来源字段：https://www.erdosproblems.com/1192

13350 | EP-1193

### Erdős Problem #1193: and let $g(n)$ be a non-decreasing function of $n$ which is always $>0$

S / solved | Erdős 问题

Let $A\subset \mathbb{N}$ and let $g(n)$ be a non-decreasing function of $n$ which is always $>0$.

Is the lower density of\[\{ n: 1_A\ast 1_A(n)=g(n)\}\]always $0$? Is the upper density always $<c$ for some constant $c<1$?

来源字段：https://www.erdosproblems.com/1193

13351 | EP-1194

### Erdős Problem #1194: b_n\in A$. How fast must $a_n/n$ increase?

O / open | Erdős 问题

Let $A\subset\mathbb{N}$ be such that every integer $n\geq 1$ can be written uniquely as $a_n-b_n$ for some $a_n,b_n\in A$. How fast must $a_n/n$ increase?

来源字段：https://www.erdosproblems.com/1194

13352 | EP-1198

### Erdős Problem #1198: Additive Combinatorics, Ramsey Theory

S / solved | Erdős 问题

If $\mathbb{N}$ is $2$-coloured then must there exist an infinite set $A=\{a_1<\cdots\}$ such that all expressions of the shape\[\prod_{i\in S_1}a_i+\cdots+\prod_{i\in S_k}a_i,\]for disjoint $S_1,\ldots,S_k$ (excluding the trivial expressions $a_i$) are the same colour?

来源字段：https://www.erdosproblems.com/1198

13353 | EP-1199

### Erdős Problem #1199: Additive Combinatorics, Ramsey Theory

O / open | Erdős 问题

Is it true that in any $2$-colouring of $\mathbb{N}$ there exists an infinite set $A$ such that all elements of $A+A$ are the same colour?

来源字段：https://www.erdosproblems.com/1199

13354 | EP-120

### Erdős Problem #120

O / open | Erdős 问题

Let $A\subseteq\mathbb{R}$ be an infinite set. Must there be a set $E\subset \mathbb{R}$ of positive measure which does not contain any set of the shape $aA+b$ for some $a,b\in\mathbb{R}$ and $a\neq 0$?

来源字段：https://www.erdosproblems.com/120

13355 | EP-1213

### Erdős Problem #1213: Additive Combinatorics

S / solved | Erdős 问题

Let $a,K\geq 1$. Does there exist $f(a,K)$ such that if\[a=a_1<\cdots <a_s\]is a sequence of integers with $a_s> f(a,K)$ and with bounded gaps $a_{i+1}-a_i\leq K$ then there are two distinct intervals $I$ and $J$ such that\[\sum_{i\in I}a_i=\sum_{j\in J}a_j?\]

来源字段：https://www.erdosproblems.com/1213

13356 | EP-131

### Erdős Problem #131

P / partially_solved | Erdős 问题

Let $F(N)$ be the maximal size of $A\subseteq\{1,\ldots,N\}$ such that no $a\in A$ divides the sum of any distinct elements of $A\backslash\{a\}$. Estimate $F(N)$. In particular, is it true that\[F(N) > N^{1/2-o(1)}?\]

来源字段：https://www.erdosproblems.com/131

13357 | EP-139

### Erdős Problem #139: Additive Combinatorics, Arithmetic Progressions

S / solved | Erdős 问题

Let $r_k(N)$ be the size of the largest subset of $\{1,\ldots,N\}$ which does not contain a non-trivial $k$-term arithmetic progression. Prove that $r_k(N)=o(N)$.

来源字段：https://www.erdosproblems.com/139

13358 | EP-140

### Erdős Problem #140: Additive Combinatorics, Arithmetic Progressions

S / solved | Erdős 问题

Let $r_3(N)$ be the size of the largest subset of $\{1,\ldots,N\}$ which does not contain a non-trivial $3$-term arithmetic progression. Prove that $r_3(N)\ll N/(\log N)^C$ for every $C>0$.

来源字段：https://www.erdosproblems.com/140

13359 | EP-142

### Erdős Problem #142

O / open | Erdős 问题

Let $r_k(N)$ be the largest possible size of a subset of $\{1,\ldots,N\}$ that does not contain any non-trivial $k$-term arithmetic progression. Prove an asymptotic formula for $r_k(N)$.

来源字段：https://www.erdosproblems.com/142

13360 | EP-145

### Erdős Problem #145

O / open | Erdős 问题

Let $s_1<s_2<\cdots$ be the sequence of squarefree numbers. Is it true that, for any $\alpha \geq 0$,\[\lim_{x\to \infty}\frac{1}{x}\sum_{s_n\leq x}(s_{n+1}-s_n)^\alpha\]exists?

来源字段：https://www.erdosproblems.com/145

13361 | EP-152

### Erdős Problem #152

S / solved | Erdős 问题

For any $M\geq 1$, if $A\subset \mathbb{N}$ is a sufficiently large finite Sidon set then there are at least $M$ many $a\in A+A$ such that $a+1,a-1\not\in A+A$.

来源字段：https://www.erdosproblems.com/152

13362 | EP-153

### Erdős Problem #153

O / open | Erdős 问题

Let $A$ be a finite Sidon set and $A+A=\{s_1<\cdots<s_t\}$. Is it true that\[\frac{1}{t}\sum_{1\leq i<t}(s_{i+1}-s_i)^2 \to \infty\]as $\lvert A\rvert\to \infty$?

来源字段：https://www.erdosproblems.com/153

13363 | EP-154

### Erdős Problem #154: Sidon Sets

S / solved | Erdős 问题

Let $A\subset \{1,\ldots,N\}$ be a Sidon set with $\lvert A\rvert\sim N^{1/2}$. Must $A+A$ be well-distributed over all small moduli? In particular, must about half the elements of $A+A$ be even and half odd?

来源字段：https://www.erdosproblems.com/154

13364 | EP-155

### Erdős Problem #155

O / open | Erdős 问题

Let $F(N)$ be the size of the largest Sidon subset of $\{1,\ldots,N\}$. Is it true that for every $k\geq 1$ we have\[F(N+k)\leq F(N)+1\]for all sufficiently large $N$?

来源字段：https://www.erdosproblems.com/155

13365 | EP-156

### Erdős Problem #156

O / open | Erdős 问题

Does there exist a maximal Sidon set $A\subset \{1,\ldots,N\}$ of size $O(N^{1/3})$?

来源字段：https://www.erdosproblems.com/156

13366 | EP-157

### Erdős Problem #157: Does there exist an infinite Sidon set which is an asymptotic basis of order 3?

S / solved | Erdős 问题

Does there exist an infinite Sidon set which is an asymptotic basis of order 3?

来源字段：https://www.erdosproblems.com/157

13367 | EP-158

### Erdős Problem #158

O / open | Erdős 问题

Let $A\subset \mathbb{N}$ be an infinite set such that, for any $n$, there are most $2$ solutions to $a+b=n$ with $a\leq b$. Must\[\liminf_{N\to\infty}\frac{\lvert A\cap \{1,\ldots,N\}\rvert}{N^{1/2}}=0?\]

来源字段：https://www.erdosproblems.com/158

13368 | EP-160

### Erdős Problem #160

O / open | Erdős 问题

Let $h(N)$ be the smallest $k$ such that $\{1,\ldots,N\}$ can be coloured with $k$ colours so that every four-term arithmetic progression must contain at least three distinct colours. Estimate $h(N)$.

来源字段：https://www.erdosproblems.com/160

13369 | EP-168

### Erdős Problem #168

P / partially_solved | Erdős 问题

Let $F(N)$ be the size of the largest subset of $\{1,\ldots,N\}$ which does not contain any set of the form $\{n,2n,3n\}$. What is\[ \lim_{N\to \infty}\frac{F(N)}{N}?\]Is this limit irrational?

来源字段：https://www.erdosproblems.com/168

13370 | EP-170

### Erdős Problem #170

O / open | Erdős 问题

Let $F(N)$ be the smallest possible size of $A\subset \{0,1,\ldots,N\}$ such that $\{0,1,\ldots,N\}\subset A-A$. Find the value of\[\lim_{N\to \infty}\frac{F(N)}{N^{1/2}}.\]

来源字段：https://www.erdosproblems.com/170

13371 | EP-171

### Erdős Problem #171: Additive Combinatorics, Combinatorics

S / solved | Erdős 问题

Is it true that for every $\epsilon>0$ and integer $t\geq 1$, if $N$ is sufficiently large and $A$ is a subset of $[t]^N$ of size at least $\epsilon t^N$ then $A$ must contain a combinatorial line $P$ (a set $P=\{p_1,\ldots,p_t\}$ where for each coordinate $1\leq j\leq t$ the $j$th coordinate of $p_i$ is either $i$ or constant).

来源字段：https://www.erdosproblems.com/171

13372 | EP-176

### Erdős Problem #176

O / open | Erdős 问题

Let $N(k,\ell)$ be the minimal $N$ such that for any $f:\{1,\ldots,N\}\to\{-1,1\}$ there must exist a $k$-term arithmetic progression $P$ such that\[ \left\lvert \sum_{n\in P}f(n)\right\rvert\geq \ell.\]Find good upper bounds for $N(k,\ell)$. Is it true that for any $c>0$ there exists some $C>1$ such that\[N(k,ck)\leq C^k?\]What about\[N(k,2)\leq C^k\]or\[N(k,\sqrt{k})\leq C^k?\]

来源字段：https://www.erdosproblems.com/176

13373 | EP-179

### Erdős Problem #179: Additive Combinatorics, Arithmetic Progressions

S / solved | Erdős 问题

Let $1\leq k<\ell$ be integers and define $F_k(N,\ell)$ to be minimal such that every set $A\subset \mathbb{N}$ of size $N$ which contains at least $F_k(N,\ell)$ many $k$-term arithmetic progressions must contain an $\ell$-term arithmetic progression. Find good upper bounds for $F_k(N,\ell)$. Is it true that\[F_3(N,4)=o(N^2)?\]Is it true that for every $\ell>3$\[\lim_{N\to \infty}\frac{\log F_3(N,\ell)}{\log N}=2?\]

来源字段：https://www.erdosproblems.com/179

13374 | EP-185

### Erdős Problem #185: 1,2\}^n$ which contains no three points on a line. Is it true that $f_3(n)=o(3^n)$?

S / solved | Erdős 问题

Let $f_3(n)$ be the maximal size of a subset of $\{0,1,2\}^n$ which contains no three points on a line. Is it true that $f_3(n)=o(3^n)$?

来源字段：https://www.erdosproblems.com/185

13375 | EP-186

### Erdős Problem #186: Additive Combinatorics

S / solved | Erdős 问题

Let $F(N)$ be the maximal size of $A\subseteq \{1,\ldots,N\}$ which is 'non-averaging', so that no $n\in A$ is the arithmetic mean of at least two elements in $A$. What is the order of growth of $F(N)$?

来源字段：https://www.erdosproblems.com/186

13376 | EP-191

### Erdős Problem #191: Combinatorics, Ramsey Theory

S / solved | Erdős 问题

Let $C>0$ be arbitrary. Is it true that, if $n$ is sufficiently large depending on $C$, then in any $2$-colouring of $\binom{\{2,\ldots,n\}}{2}$ there exists some $X\subseteq \{2,\ldots,n\}$ such that $\binom{X}{2}$ is monochromatic and\[\sum_{x\in X}\frac{1}{\log x}\geq C?\]

来源字段：https://www.erdosproblems.com/191

13377 | EP-192

### Erdős Problem #192: Arithmetic Progressions, Combinatorics

S / solved | Erdős 问题

Let $A=\{a_1,a_2,\ldots\}\subset \mathbb{R}^d$ be an infinite sequence such that $a_{i+1}-a_i$ is a positive unit vector (i.e. is of the form $(0,0,\ldots,1,0,\ldots,0)$). For which $d$ must $A$ contain a three-term arithmetic progression?

来源字段：https://www.erdosproblems.com/192

13378 | EP-193

### Erdős Problem #193

S / solved | Erdős 问题

Let $S\subseteq \mathbb{Z}^3$ be a finite set and let $A=\{a_1,a_2,\ldots,\}\subset \mathbb{Z}^3$ be an infinite $S$-walk, so that $a_{i+1}-a_i\in S$ for all $i$. Must $A$ contain three collinear points?

来源字段：https://www.erdosproblems.com/193

13379 | EP-194

### Erdős Problem #194: Arithmetic Progressions

S / solved | Erdős 问题

Let $k\geq 3$. Must any ordering of $\mathbb{R}$ contain a monotone $k$-term arithmetic progression, that is, some $x_1<\cdots<x_k$ which forms an increasing or decreasing $k$-term arithmetic progression?

来源字段：https://www.erdosproblems.com/194

13380 | EP-196

### Erdős Problem #196

O / open | Erdős 问题

Must every permutation of $\mathbb{N}$ contain a monotone 4-term arithmetic progression? In other words, given a permutation $x$ of $\mathbb{N}$ must there be indices with either $i<j<k<l$ or $i>j>k>l$ such that $x_i,x_j,x_k,x_l$ are an arithmetic progression?

来源字段：https://www.erdosproblems.com/196

13381 | EP-198

### Erdős Problem #198: Additive Combinatorics, Sidon Sets

S / solved | Erdős 问题

If $A\subset \mathbb{N}$ is a Sidon set then must the complement of $A$ contain an infinite arithmetic progression?

来源字段：https://www.erdosproblems.com/198

13382 | EP-199

### Erdős Problem #199: Arithmetic Progressions

S / solved | Erdős 问题

If $A\subset \mathbb{R}$ does not contain a 3-term arithmetic progression then must $\mathbb{R}\backslash A$ contain an infinite arithmetic progression?

来源字段：https://www.erdosproblems.com/199

13383 | EP-20

### Erdős Problem #20

O / open | Erdős 问题

别名：The Sunflower Conjecture [COMB-015]; Sunflower Conjecture [COMB-006]; Sunflower Conjecture [COMB-020]

Let $f(n,k)$ be minimal such that every family $\mathcal{F}$ of $n$-uniform sets with $\lvert \mathcal{F}\rvert \geq f(n,k)$ contains a $k$-sunflower. Is it true that\[f(n,k) < c_k^n\]for some constant $c_k>0$?

来源字段：https://www.erdosproblems.com/20

13384 | EP-207

### Erdős Problem #207: Combinatorics, Hypergraphs

S / solved | Erdős 问题

For any $g\geq 2$, if $n$ is sufficiently large and $\equiv 1,3\pmod{6}$ then there exists a 3-uniform hypergraph on $n$ vertices such that

- every pair of vertices is contained in exactly one edge (i.e. the graph is a Steiner triple system) and

- for any $2\leq j\leq g$ any collection of $j$ edges contains at least $j+3$ vertices.

来源字段：https://www.erdosproblems.com/207

13385 | EP-208

### Erdős Problem #208

O / open | Erdős 问题

Let $s_1<s_2<\cdots$ be the sequence of squarefree numbers. Is it true that, for any $\epsilon>0$ and large $n$,\[s_{n+1}-s_n \ll_\epsilon s_n^{\epsilon}?\]Is it true that\[s_{n+1}-s_n \leq (1+o(1))\frac{\pi^2}{6}\frac{\log s_n}{\log\log s_n}?\]

来源字段：https://www.erdosproblems.com/208

13386 | EP-212

### Erdős Problem #212

O / open | Erdős 问题

Is there a dense subset of $\mathbb{R}^2$ such that all pairwise distances are rational?

来源字段：https://www.erdosproblems.com/212

13387 | EP-217

### Erdős Problem #217

O / open | Erdős 问题

For which $n$ are there $n$ points in $\mathbb{R}^2$, no three on a line and no four on a circle, which determine $n-1$ distinct distances and so that (in some ordering of the distances) the $i$th distance occurs $i$ times?

来源字段：https://www.erdosproblems.com/217

13388 | EP-21

### Erdős Problem #21: Combinatorics, Intersecting Family

S / solved | Erdős 问题

Let $f(n)$ be minimal such that there is an intersecting family $\mathcal{F}$ of sets of size $n$ (so $A\cap B\neq\emptyset$ for all $A,B\in \mathcal{F}$) with $\lvert \mathcal{F}\rvert=f(n)$ such that any set $S$ with $\lvert S\rvert \leq n-1$ is disjoint from at least one $A\in \mathcal{F}$.

Is it true that\[f(n) \ll n?\]

来源字段：https://www.erdosproblems.com/21

13389 | EP-231

### Erdős Problem #231: Combinatorics

S / solved | Erdős 问题

Let $S$ be a string of length $2^k-1$ formed from an alphabet of $k$ characters. Must $S$ contain an abelian square: two consecutive blocks $x$ and $y$ such that $y$ is a permutation of $x$?

来源字段：https://www.erdosproblems.com/231

13390 | EP-241

### Erdős Problem #241

O / open | Erdős 问题

Let $f(N)$ be the maximum size of $A\subseteq \{1,\ldots,N\}$ such that the sums $a+b+c$ with $a,b,c\in A$ are all distinct (aside from the trivial coincidences). Is it true that\[ f(N)\sim N^{1/3}?\]

来源字段：https://www.erdosproblems.com/241

13391 | EP-245

### Erdős Problem #245: Additive Combinatorics

S / solved | Erdős 问题

Let $A\subseteq \mathbb{N}$ be an infinite set such that $\lvert A\cap \{1,\ldots,N\}\rvert=o(N)$. Is it true that\[\limsup_{N\to \infty}\frac{\lvert (A+A)\cap \{1,\ldots,N\}\rvert}{\lvert A\cap \{1,\ldots,N\}\rvert}\geq 3?\]

来源字段：https://www.erdosproblems.com/245

13392 | EP-260

### Erdős Problem #260

O / open | Erdős 问题

Let $a_1<a_2<\cdots$ be an increasing sequence such that $a_n/n\to \infty$. Is the sum\[\sum_n \frac{a_n}{2^{a_n}}\]irrational?

来源字段：https://www.erdosproblems.com/260

13393 | EP-288

### Erdős Problem #288

O / open | Erdős 问题

Is it true that there are only finitely many pairs of intervals $I_1,I_2$ such that\[\sum_{n_1\in I_1}\frac{1}{n_1}+\sum_{n_2\in I_2}\frac{1}{n_2}\in \mathbb{N}?\]

来源字段：https://www.erdosproblems.com/288

13394 | EP-295

### Erdős Problem #295

O / open | Erdős 问题

Let $N\geq 1$ and let $k(N)$ denote the smallest $k$ such that there exist $N\leq n_1<\cdots <n_k$ with\[1=\frac{1}{n_1}+\cdots+\frac{1}{n_k}.\]Is it true that\[\lim_{N\to \infty} k(N)-(e-1)N=\infty?\]

来源字段：https://www.erdosproblems.com/295

13395 | EP-317

### Erdős Problem #317

O / open | Erdős 问题

Is there some constant $c>0$ such that for every $n\geq 1$ there exists some $\delta_k\in \{-1,0,1\}$ for $1\leq k\leq n$ with\[0< \left\lvert \sum_{1\leq k\leq n}\frac{\delta_k}{k}\right\rvert < \frac{c}{2^n}?\]Is it true that for sufficiently large $n$, for any $\delta_k\in \{-1,0,1\}$,\[\left\lvert \sum_{1\leq k\leq n}\frac{\delta_k}{k}\right\rvert > \frac{1}{[1,\ldots,n]}\]whenever the left-hand side is not zero?

来源字段：https://www.erdosproblems.com/317

13396 | EP-320

### Erdős Problem #320

S / solved | Erdős 问题

Let $S(N)$ count the number of distinct sums of the form $\sum_{n\in A}\frac{1}{n}$ for $A\subseteq \{1,\ldots,N\}$. Estimate $S(N)$.

来源字段：https://www.erdosproblems.com/320

13397 | EP-321

### Erdős Problem #321

S / solved | Erdős 问题

What is the size of the largest $A\subseteq \{1,\ldots,N\}$ such that all sums $\sum_{n\in S}\frac{1}{n}$ are distinct for $S\subseteq A$?

来源字段：https://www.erdosproblems.com/321

13398 | EP-326

### Erdős Problem #326

P / partially_solved | Erdős 问题

Does there exist $A=\{a_1<a_2<\cdots\}\subset \mathbb{N}$ which is a minimal basis of order $2$ (i.e. every large integer is the sum of $2$ elements from $A$, and no proper subset of $A$ has this property), such that\[\lim_{k\to \infty}\frac{a_k}{k^2}=c\]for some $c\neq 0$?

来源字段：https://www.erdosproblems.com/326

13399 | EP-327

### Erdős Problem #327

O / open | Erdős 问题

Suppose $A\subseteq \{1,\ldots,N\}$ is such that if $a,b\in A$ and $a\neq b$ then $a+b\nmid ab$. Can $A$ be 'substantially more' than the odd numbers?

What if $a,b\in A$ with $a\neq b$ implies $a+b\nmid 2ab$? Must $\lvert A\rvert=o(N)$?

来源字段：https://www.erdosproblems.com/327

13400 | EP-335

### Erdős Problem #335

P / partially_solved | Erdős 问题

Let $d(A)$ denote the density of $A\subseteq \mathbb{N}$. Characterise those $A,B\subseteq \mathbb{N}$ with positive density such that\[d(A+B)=d(A)+d(B).\]

来源字段：https://www.erdosproblems.com/335

13401 | EP-388

### Erdős Problem #388

O / open | Erdős 问题

Can one classify all solutions of\[\prod_{1\leq i\leq k_1}(m_1+i)=\prod_{1\leq j\leq k_2}(m_2+j)\]where $k_1,k_2>3$ and $m_1+k_1\leq m_2$? Are there only finitely many solutions?

来源字段：https://www.erdosproblems.com/388

13402 | EP-389

### Erdős Problem #389

O / open | Erdős 问题

Is it true that for every $n\geq 1$ there is a $k$ such that\[n(n+1)\cdots(n+k-1)\mid (n+k)\cdots (n+2k-1)?\]

来源字段：https://www.erdosproblems.com/389

13403 | EP-39

### Erdős Problem #39

O / open | Erdős 问题

Is there an infinite Sidon set $A\subset \mathbb{N}$ such that\[\lvert A\cap \{1\ldots,N\}\rvert \gg_\epsilon N^{1/2-\epsilon}\]for all $\epsilon>0$?

来源字段：https://www.erdosproblems.com/39

13404 | EP-390

### Erdős Problem #390

P / partially_solved | Erdős 问题

Let $f(n)$ be the minimal $m$ such that\[n! = a_1\cdots a_k\]with $n< a_1<\cdots <a_k=m$. Is there (and what is it) a constant $c$ such that\[f(n)-2n \sim c\frac{n}{\log n}?\]

来源字段：https://www.erdosproblems.com/390

13405 | EP-40

### Erdős Problem #40

O / open | Erdős 问题

For what functions $g(N)\to \infty$ is it true that\[\lvert A\cap \{1,\ldots,N\}\rvert \gg \frac{N^{1/2}}{g(N)}\]implies $\limsup 1_A\ast 1_A(n)=\infty$?

来源字段：https://www.erdosproblems.com/40

13406 | EP-41

### Erdős Problem #41

O / open | Erdős 问题

Let $A\subset\mathbb{N}$ be an infinite set such that the triple sums $a+b+c$ are all distinct for $a,b,c\in A$ (aside from the trivial coincidences). Is it true that\[\liminf \frac{\lvert A\cap \{1,\ldots,N\}\rvert}{N^{1/3}}=0?\]

来源字段：https://www.erdosproblems.com/41

13407 | EP-417

### Erdős Problem #417

O / open | Erdős 问题

Let\[V'(x)=\#\{\phi(m): 1\leq m\leq x\}\]and\[V(x)=\#\{\phi(m) \leq x: 1\leq m\}.\]Does $\lim V(x)/V'(x)$ exist? Is it $>1$?

来源字段：https://www.erdosproblems.com/417

13408 | EP-42

### Erdős Problem #42

S / solved | Erdős 问题

Let $M\geq 1$ and $N$ be sufficiently large in terms of $M$. Is it true that for every Sidon set $A\subset \{1,\ldots,N\}$ there is another Sidon set $B\subset \{1,\ldots,N\}$ of size $M$ such that $(A-A)\cap(B-B)=\{0\}$?

来源字段：https://www.erdosproblems.com/42

13409 | EP-421

### Erdős Problem #421

S / solved | Erdős 问题

Is there a sequence $1\leq d_1<d_2<\cdots$ with density $1$ such that all products $\prod_{u\leq i\leq v}d_i$ are distinct?

来源字段：https://www.erdosproblems.com/421

13410 | EP-43

### Erdős Problem #43

S / solved | Erdős 问题

If $A,B\subset \{1,\ldots,N\}$ are two Sidon sets such that $(A-A)\cap(B-B)=\{0\}$ then is it true that\[ \binom{\lvert A\rvert}{2}+\binom{\lvert B\rvert}{2}\leq\binom{f(N)}{2}+O(1),\]where $f(N)$ is the maximum possible size of a Sidon set in $\{1,\ldots,N\}$? If $\lvert A\rvert=\lvert B\rvert$ then can this bound be improved to\[\binom{\lvert A\rvert}{2}+\binom{\lvert B\rvert}{2}\leq (1-c+o(1))\binom{f(N)}{2}\]for some constant $c>0$?

来源字段：https://www.erdosproblems.com/43

13411 | EP-44

### Erdős Problem #44

O / open | Erdős 问题

Let $N\geq 1$ and $A\subset \{1,\ldots,N\}$ be a Sidon set. Is it true that, for any $\epsilon>0$, there exist $M$ and $B\subset \{N+1,\ldots,M\}$ (which may depend on $N,A,\epsilon$) such that $A\cup B\subset \{1,\ldots,M\}$ is a Sidon set of size at least $(1-\epsilon)M^{1/2}$?

来源字段：https://www.erdosproblems.com/44

13412 | EP-447

### Erdős Problem #447: Combinatorics

S / solved | Erdős 问题

How large can a union-free collection $\mathcal{F}$ of subsets of $[n]$ be? By union-free we mean there are no solutions to $A\cup B=C$ with distinct $A,B,C\in \mathcal{F}$. Must $\lvert \mathcal{F}\rvert =o(2^n)$? Perhaps even\[\lvert \mathcal{F}\rvert <(1+o(1))\binom{n}{\lfloor n/2\rfloor}?\]

来源字段：https://www.erdosproblems.com/447

13413 | EP-479

### Erdős Problem #479

O / open | Erdős 问题

Is it true that, for all $k\neq 1$, there are infinitely many $n$ such that $2^n\equiv k\pmod{n}$?

来源字段：https://www.erdosproblems.com/479

13414 | EP-497

### Erdős Problem #497: Combinatorics

S / solved | Erdős 问题

How many antichains in $[n]$ are there? That is, how many families of subsets of $[n]$ are there such that, if $\mathcal{F}$ is such a family and $A,B\in \mathcal{F}$, then $A\not\subseteq B$?

来源字段：https://www.erdosproblems.com/497

13415 | EP-499

### Erdős Problem #499: Combinatorics

S / solved | Erdős 问题

Let $M=(a_{ij})$ be a real $n\times n$ doubly stochastic matrix (i.e. the entries are non-negative and each column and row sums to $1$). Does there exist some $\sigma\in S_n$ such that\[\prod_{1\leq i\leq n}a_{i\sigma(i)}\geq n^{-n}?\]

来源字段：https://www.erdosproblems.com/499

13416 | EP-50

### Erdős Problem #50

O / open | Erdős 问题

Schoenberg proved that for every $c\in [0,1]$ the density of\[\{ n\in \mathbb{N}: \phi(n)<cn\}\]exists. Let this density be denoted by $f(c)$. Is it true that there are no $x$ such that $f'(x)$ exists and is positive?

来源字段：https://www.erdosproblems.com/50

13417 | EP-501

### Erdős Problem #501

P / partially_solved | Erdős 问题

For every $x\in\mathbb{R}$ let $A_x\subset \mathbb{R}$ be a bounded set with outer measure $<1$.

Must there exist an infinite independent set, that is, some infinite $X\subseteq \mathbb{R}$ such that $x\not\in A_y$ for all $x\neq y\in X$?

If the sets $A_x$ are closed and have measure $<1$, then must there exist an independent set of size $3$?

来源字段：https://www.erdosproblems.com/501

13418 | EP-503

### Erdős Problem #503

P / partially_solved | Erdős 问题

What is the size of the largest $A\subseteq \mathbb{R}^d$ such that every three points from $A$ determine an isosceles triangle? That is, for any three points $x,y,z$ from $A$, at least two of the distances $\lvert x-y\rvert,\lvert y-z\rvert,\lvert x-z\rvert$ are equal.

来源字段：https://www.erdosproblems.com/503

13419 | EP-509

### Erdős Problem #509

P / partially_solved | Erdős 问题

Let $f(z)\in\mathbb{C}[z]$ be a monic non-constant polynomial. Can the set\[\{ z\in \mathbb{C}: \lvert f(z)\rvert \leq 1\}\]be covered by a set of circles the sum of whose radii is $\leq 2$?

来源字段：https://www.erdosproblems.com/509

13420 | EP-510

### Erdős Problem #510

P / partially_solved | Erdős 问题

别名：Chowla's cosine problem [OPG-739]

If $A\subset \mathbb{Z}$ is a finite set of size $N$ then is there some absolute constant $c>0$ and $\theta$ such that\[\sum_{n\in A}\cos(n\theta) < -cN^{1/2}?\]

来源字段：https://www.erdosproblems.com/510

13421 | EP-513

### Erdős Problem #513

P / partially_solved | Erdős 问题

Let $f=\sum_{n=0}^\infty a_nz^n$ be a transcendental entire function. What is the greatest possible value of\[\liminf_{r\to \infty} \frac{\max_n\lvert a_nr^n\rvert}{\max_{\lvert z\rvert=r}\lvert f(z)\rvert}?\]

来源字段：https://www.erdosproblems.com/513

13422 | EP-517

### Erdős Problem #517

P / partially_solved | Erdős 问题

Let $f(z)=\sum_{k=1}^\infty a_kz^{n_k}$ be an entire function (with $a_k\neq 0$ for all $k\geq 1$). Is it true that if $n_k/k\to \infty$ then $f(z)$ assumes every value infinitely often?

来源字段：https://www.erdosproblems.com/517

13423 | EP-524

### Erdős Problem #524

O / open | Erdős 问题

For any $t\in (0,1)$ let $t=\sum_{k=1}^\infty \epsilon_k(t)2^{-k}$ (where $\epsilon_k(t)\in \{0,1\}$). What is the correct order of magnitude (for almost all $t\in(0,1)$) for\[M_n(t)=\max_{x\in [-1,1]}\left\lvert \sum_{k\leq n}(-1)^{\epsilon_k(t)}x^k\right\rvert?\]

来源字段：https://www.erdosproblems.com/524

13424 | EP-528

### Erdős Problem #528

O / open | Erdős 问题

Let $f(n,k)$ count the number of self-avoiding walks of $n$ steps (beginning at the origin) in $\mathbb{Z}^k$ (i.e. those walks which do not intersect themselves). Determine\[C_k=\lim_{n\to\infty}f(n,k)^{1/n}.\]

来源字段：https://www.erdosproblems.com/528

13425 | EP-536

### Erdős Problem #536

O / open | Erdős 问题

Let $f(N)$ be the largest size of $A\subseteq \{1,\ldots,N\}$ with the property that there are no distinct $a,b,c\in A$ such that\[[a,b]=[b,c]=[a,c],\]where $[a,b]$ denotes the least common multiple.

Estimate $f(N)$ - in particular, is it true that $f(N)=o(N)$?

来源字段：https://www.erdosproblems.com/536

13426 | EP-539

### Erdős Problem #539

P / partially_solved | Erdős 问题

Let $h(n)$ be such that, for any set $A\subseteq \mathbb{N}$ of size $n$, the set\[\left\{ \frac{a}{(a,b)}: a,b\in A\right\}\]has size at least $h(n)$. Estimate $h(n)$.

来源字段：https://www.erdosproblems.com/539

13427 | EP-543

### Erdős Problem #543

S / solved | Erdős 问题

Define $f(N)$ be the minimal $k$ such that the following holds: if $G$ is an abelian group of size $N$ and $A\subseteq G$ is a random set of size $k$ then, with probability $\geq 1/2$, all elements of $G$ can be written as $\sum_{x\in S}x$ for some $S\subseteq A$. Is\[f(N) \leq \log_2 N+o(\log\log N)?\]

来源字段：https://www.erdosproblems.com/543

13428 | EP-588

### Erdős Problem #588

P / partially_solved | Erdős 问题

Let $f_k(n)$ be minimal such that if $n$ points in $\mathbb{R}^2$ have no $k+1$ points on a line then there must be at most $f_k(n)$ many lines containing at least $k$ points. Is it true that\[f_k(n)=o(n^2)\]for $k\geq 4$?

来源字段：https://www.erdosproblems.com/588

13429 | EP-598

### Erdős Problem #598

P / partially_solved | Erdős 问题

Let $m$ be an infinite cardinal and $\kappa$ be the successor cardinal of $2^{\aleph_0}$. Can one colour the countable subsets of $m$ using $\kappa$ many colours so that every $X\subseteq m$ with $\lvert X\rvert=\kappa$ contains subsets of all possible colours?

来源字段：https://www.erdosproblems.com/598

13430 | EP-623

### Erdős Problem #623

P / partially_solved | Erdős 问题

Let $X$ be a set of cardinality $\aleph_\omega$ and $f$ be a function from the finite subsets of $X$ to $X$ such that $f(A)\not\in A$ for all $A$. Must there exist an infinite $Y\subseteq X$ that is independent - that is, for all finite $B\subset Y$ we have $f(B)\not\in Y$?

来源字段：https://www.erdosproblems.com/623

13431 | EP-624

### Erdős Problem #624

P / partially_solved | Erdős 问题

Let $X$ be a finite set of size $n$ and $H(n)$ be such that there is a function $f:\{A: A\subseteq X\}\to X$ so that for every $Y\subseteq X$ with $\lvert Y\rvert \geq H(n)$ we have\[\{ f(A): A\subseteq Y\}=X.\]Prove that\[H(n)-\log_2 n \to \infty.\]

来源字段：https://www.erdosproblems.com/624

13432 | EP-635

### Erdős Problem #635

P / partially_solved | Erdős 问题

Let $t\geq 1$ and $A\subseteq \{1,\ldots,N\}$ be such that whenever $a,b\in A$ with $b-a\geq t$ we have $b-a\nmid b$. How large can $\lvert A\rvert$ be? Is it true that\[\lvert A\rvert \leq \left(\frac{1}{2}+o_t(1)\right)N?\]

来源字段：https://www.erdosproblems.com/635

13433 | EP-654

### Erdős Problem #654

O / open | Erdős 问题

Let $f(n)$ be such that, given any $x_1,\ldots,x_n\in \mathbb{R}^2$ with no four points on a circle, there exists some $x_i$ with at least $f(n)$ many distinct distances to other $x_j$. Estimate $f(n)$ - in particular, is it true that\[f(n)>(1-o(1))n?\]Or at least\[f(n) > (1/3+c)n\]for some $c>0$, for all large $n$?

来源字段：https://www.erdosproblems.com/654

13434 | EP-657

### Erdős Problem #657

P / partially_solved | Erdős 问题

Is it true that if $A\subset \mathbb{R}^2$ is a set of $n$ points such that every subset of $3$ points determines $3$ distinct distances (i.e. $A$ has no isosceles triangles) then $A$ must determine at least $f(n)n$ distinct distances, for some $f(n)\to \infty$?

来源字段：https://www.erdosproblems.com/657

13435 | EP-658

### Erdős Problem #658: Additive Combinatorics

S / solved | Erdős 问题

Let $\delta>0$ and $N$ be sufficiently large depending on $\delta$. Is it true that if $A\subseteq \{1,\ldots,N\}^2$ has $\lvert A\rvert \geq \delta N^2$ then $A$ must contain the vertices of a square?

来源字段：https://www.erdosproblems.com/658

13436 | EP-66

### Erdős Problem #66

O / open | Erdős 问题

Is there $A\subseteq \mathbb{N}$ such that\[\lim_{n\to \infty}\frac{1_A\ast 1_A(n)}{\log n}\]exists and is $\neq 0$?

来源字段：https://www.erdosproblems.com/66

13437 | EP-664

### Erdős Problem #664: Combinatorics

S / solved | Erdős 问题

Let $c<1$ be some constant and $A_1,\ldots,A_m\subseteq \{1,\ldots,n\}$ be such that $\lvert A_i\rvert >c\sqrt{n}$ for all $i$ and $\lvert A_i\cap A_j\rvert\leq 1$ for all $i\neq j$.

Must there exist some set $B$ such that $B\cap A_i\neq \emptyset$ and $\lvert B\cap A_i\rvert \ll_c 1$ for all $i$?

来源字段：https://www.erdosproblems.com/664

13438 | EP-670

### Erdős Problem #670

P / partially_solved | Erdős 问题

Let $A\subseteq \mathbb{R}^d$ be a set of $n$ points such that all pairwise distances differ by at least $1$. Is the diameter of $A$ at least $(1+o(1))n^2$?

来源字段：https://www.erdosproblems.com/670

13439 | EP-702

### Erdős Problem #702: Combinatorics

S / solved | Erdős 问题

Let $k\geq 4$. If $\mathcal{F}$ is a family of subsets of $\{1,\ldots,n\}$ with $\lvert A\rvert=k$ for all $A\in \mathcal{F}$ and $\lvert \mathcal{F}\rvert >\binom{n-2}{k-2}$ then there are $A,B\in\mathcal{F}$ such that $\lvert A\cap B\rvert=1$.

来源字段：https://www.erdosproblems.com/702

13440 | EP-703

### Erdős Problem #703: Combinatorics

S / solved | Erdős 问题

Let $r\geq 1$ and define $T(n,r)$ to be maximal such that there exists a family $\mathcal{F}$ of subsets of $\{1,\ldots,n\}$ of size $T(n,r)$ such that $\lvert A\cap B\rvert\neq r$ for all $A,B\in \mathcal{F}$.

Estimate $T(n,r)$ for $r\geq 2$. In particular, is it true that for every $\epsilon>0$ there exists $\delta>0$ such that for all $\epsilon n<r<(1/2-\epsilon) n$ we have\[T(n,r)<(2-\delta)^n?\]

来源字段：https://www.erdosproblems.com/703

13441 | EP-707

### Erdős Problem #707: Additive Combinatorics, Sidon Sets

S / solved | Erdős 问题

Let $A\subset \mathbb{N}$ be a finite Sidon set. Is there some set $B$ with $A\subseteq B$ which is perfect difference set modulo $p^2+p+1$ for some prime $p$?

来源字段：https://www.erdosproblems.com/707

13442 | EP-722

### Erdős Problem #722: Combinatorics

S / solved | Erdős 问题

Let $k>r$ and $n$ be sufficiently large in terms of $k$ and $r$. Does there always exist a block $r-(n,k,1)$ design (or Steiner system with parameters $(n,k,r)$), provided the trivial necessary divisibility conditions $\binom{k-i}{r-i}\mid \binom{n-i}{r-i}$ are satisfied for every $0\leq i<r$?

That is, can one find a family of $\binom{n}{k}\binom{k}{r}^{-1}$ many subsets of $\{1,\ldots,n\}$, all of size $k$, such that any $A\subseteq \{1,\ldots,n\}$ of size $r$ is contained in exactly one set in the family?

来源字段：https://www.erdosproblems.com/722

13443 | EP-723

### Erdős Problem #723: If there is a finite projective plane of order $n$ then must $n$ be a prime power?

O / open | Erdős 问题

If there is a finite projective plane of order $n$ then must $n$ be a prime power?

A finite projective plane of order $n$ is a collection of subsets of $\{1,\ldots,n^2+n+1\}$ of size $n+1$ such that every pair of elements is contained in exactly one set.

来源字段：https://www.erdosproblems.com/723

13444 | EP-725

### Erdős Problem #725

P / partially_solved | Erdős 问题

Give an asymptotic formula for the number of $k\times n$ Latin rectangles.

来源字段：https://www.erdosproblems.com/725

13445 | EP-732

### Erdős Problem #732: Combinatorics

S / solved | Erdős 问题

Call a sequence $1< X_1\leq \cdots \leq X_m\leq n$ block-compatible if there is a pairwise balanced block design $A_1,\ldots,A_m\subseteq \{1,\ldots,n\}$ such that $\lvert A_i\rvert=X_i$ for $1\leq i\leq m$. (A pairwise block design means that every pair in $\{1,\ldots,n\}$ is contained in exactly one of the $A_i$.)

Are there necessary and sufficient conditions for $(X_i)$ to be block-compatible?

Is there some constant $c>0$ such that for all large $n$ there are\[\geq \exp(c n^{1/2}\log n)\]many block-compatible sequences for $\{1,\ldots,n\}$?

来源字段：https://www.erdosproblems.com/732

13446 | EP-734

### Erdős Problem #734

P / partially_solved | Erdős 问题

Find, for all large $n$, a non-trivial pairwise balanced block design $A_1,\ldots,A_m\subseteq \{1,\ldots,n\}$ such that, for all $t$, there are $O(n^{1/2})$ many $i$ such that $\lvert A_i\rvert=t$.

来源字段：https://www.erdosproblems.com/734

13447 | EP-741

### Erdős Problem #741

S / solved | Erdős 问题

Let $A\subseteq \mathbb{N}$ be such that $A+A$ has positive (upper) density. Can one always decompose $A=A_1\sqcup A_2$ such that $A_1+A_1$ and $A_2+A_2$ both have positive (upper) density?

Is there a basis $A$ of order $2$ such that if $A=A_1\sqcup A_2$ then $A_1+A_1$ and $A_2+A_2$ cannot both have bounded gaps?

来源字段：https://www.erdosproblems.com/741

13448 | EP-747

### Erdős Problem #747: Combinatorics, Hypergraphs

S / solved | Erdős 问题

How large should $\ell(n)$ be such that, almost surely, a random $3$-uniform hypergraph on $3n$ vertices with $\ell(n)$ edges must contain $n$ vertex-disjoint edges?

来源字段：https://www.erdosproblems.com/747

13449 | EP-749

### Erdős Problem #749

P / partially_solved | Erdős 问题

Let $\epsilon>0$. Does there exist $A\subseteq \mathbb{N}$ such that the lower density of $A+A$ is at least $1-\epsilon$ and yet $1_A\ast 1_A(n) \ll_\epsilon 1$ for all $n$?

来源字段：https://www.erdosproblems.com/749

13450 | EP-776

### Erdős Problem #776

P / partially_solved | Erdős 问题

Let $r\geq 2$ and $A_1,\ldots,A_m\subseteq \{1,\ldots,n\}$ be such that $A_i\not\subseteq A_j$ for all $i\neq j$ and for any $t$ if there exists some $i$ with $\lvert A_i\rvert=t$ then there must exist at least $r$ sets of that size.

How large must $n$ be (as a function of $r$) to ensure that there is such a family which achieves $n-3$ distinct sizes of sets?

来源字段：https://www.erdosproblems.com/776

13451 | EP-777

### Erdős Problem #777: Graph Theory, Combinatorics

S / solved | Erdős 问题

If $\mathcal{F}$ is a family of subsets of $\{1,\ldots,n\}$ then we write $G_{\mathcal{F}}$ for the graph on $\mathcal{F}$ where $A\sim B$ if $A$ and $B$ are comparable - that is, $A\subseteq B$ or vice versa.

Is it true that, if $\epsilon>0$ and $n$ is sufficiently large, whenever $m\leq (2-\epsilon)2^{n/2}$ the graph $G_\mathcal{F}$ has $<2^{n}$ many edges?

Is it true that if $G_{\mathcal{F}}$ has $\geq cm^2$ edges then $m\ll_c 2^{n/2}$?

Is it true that, for any $\epsilon>0$, there exists some $\delta>0$ such that if there are $>m^{2-\delta}$ edges then $m<(2+\epsilon)^{n/2}$?

来源字段：https://www.erdosproblems.com/777

13452 | EP-780

### Erdős Problem #780: Combinatorics, Hypergraphs

S / solved | Erdős 问题

Suppose $n\geq kr+(t-1)(k-1)$ and the edges of the complete $r$-uniform hypergraph on $n$ vertices are $t$-coloured. Prove that some colour class must contain $k$ pairwise disjoint edges.

来源字段：https://www.erdosproblems.com/780

13453 | EP-781

### Erdős Problem #781: Additive Combinatorics

S / solved | Erdős 问题

Let $f(k)$ be the minimal $n$ such that any $2$-colouring of $\{1,\ldots,n\}$ contains a monochromatic $k$-term descending wave: a sequence $x_1<\cdots <x_k$ such that, for $1<j<k$,\[x_j \geq \frac{x_{j+1}+x_{j-1}}{2}.\]Estimate $f(k)$. In particular is it true that $f(k)=k^2-k+1$ for all $k$?

来源字段：https://www.erdosproblems.com/781

13454 | EP-782

### Erdős Problem #782

P / partially_solved | Erdős 问题

Do the squares contain arbitrarily long quasi-progressions? That is, does there exist some constant $C>0$ such that, for any $k$, the squares contain a sequence $x_1,\ldots,x_k$ where, for some $d$ and all $1\leq i<k$,\[x_i+d\leq x_{i+1}\leq x_i+d+C.\]Do the squares contain arbitrarily large cubes\[a+\left\{ \sum_i \epsilon_ib_i: \epsilon_i\in \{0,1\}\right\}?\]

来源字段：https://www.erdosproblems.com/782

13455 | EP-785

### Erdős Problem #785: Additive Combinatorics

S / solved | Erdős 问题

Let $A,B\subseteq \mathbb{N}$ be infinite sets such that $A+B$ contains all large integers. Let $A(x)=\lvert A\cap [1,x]\rvert$ and similarly for $B(x)$. Is it true that if $A(x)B(x)\sim x$ then\[A(x)B(x)-x\to \infty\]as $x\to \infty$?

来源字段：https://www.erdosproblems.com/785

13456 | EP-787

### Erdős Problem #787

P / partially_solved | Erdős 问题

Let $g(n)$ be maximal such that given any set $A\subset \mathbb{R}$ with $\lvert A\rvert=n$ there exists some $B\subseteq A$ of size $\lvert B\rvert\geq g(n)$ such that $b_1+b_2\not\in A$ for all $b_1\neq b_2\in B$.

Estimate $g(n)$.

来源字段：https://www.erdosproblems.com/787

13457 | EP-788

### Erdős Problem #788

P / partially_solved | Erdős 问题

Let $f(n)$ be maximal such that if $B\subset (2n,4n)\cap \mathbb{N}$ there exists some $C\subset (n,2n)\cap \mathbb{N}$ such that $c_1+c_2\not\in B$ for all $c_1\neq c_2\in C$ and $\lvert C\rvert+\lvert B\rvert \geq f(n)$.

Estimate $f(n)$. In particular is it true that $f(n)\leq n^{1/2+o(1)}$?

来源字段：https://www.erdosproblems.com/788

13458 | EP-789

### Erdős Problem #789

P / partially_solved | Erdős 问题

Let $h(n)$ be maximal such that if $A\subseteq \mathbb{Z}$ with $\lvert A\rvert=n$ then there is $B\subseteq A$ with $\lvert B\rvert \geq h(n)$ such that if $a_1+\cdots+a_r=b_1+\cdots+b_s$ with $a_i,b_i\in B$ then $r=s$.

Estimate $h(n)$.

来源字段：https://www.erdosproblems.com/789

13459 | EP-790

### Erdős Problem #790

P / partially_solved | Erdős 问题

Let $l(n)$ be maximal such that if $A\subset\mathbb{Z}$ with $\lvert A\rvert=n$ then there exists a sum-free $B\subseteq A$ with $\lvert B\rvert \geq l(n)$ - that is, $B$ is such that there are no solutions to\[a_1=a_2+\cdots+a_r\]with $a_i\in B$ all distinct.

Estimate $l(n)$. In particular, is it true that $l(n)n^{-1/2}\to \infty$? Is it true that $l(n)< n^{1-c}$ for some $c>0$?

来源字段：https://www.erdosproblems.com/790

13460 | EP-791

### Erdős Problem #791

P / partially_solved | Erdős 问题

Let $g(n)$ be minimal such that there exists $A\subseteq \{0,\ldots,n\}$ of size $g(n)$ with $\{0,\ldots,n\}\subseteq A+A$. Estimate $g(n)$. In particular is it true that $g(n)\sim 2n^{1/2}$?

来源字段：https://www.erdosproblems.com/791

13461 | EP-806

### Erdős Problem #806: Additive Combinatorics

S / solved | Erdős 问题

Let $A\subseteq \{1,\ldots,n\}$ with $\lvert A\rvert \leq n^{1/2}$. Must there exist some $B\subset\mathbb{Z}$ with $\lvert B\rvert=o(n^{1/2})$ such that $A\subseteq B+B$?

来源字段：https://www.erdosproblems.com/806

13462 | EP-808

### Erdős Problem #808: Additive Combinatorics, Graph Theory

S / solved | Erdős 问题

Let $c,\epsilon>0$ and $n$ be sufficiently large. If $A\subset \mathbb{N}$ has $\lvert A\rvert=n$ and $G$ is any graph on $A$ with at least $n^{1+c}$ edges then\[\max(\lvert A+_GA\rvert,\lvert A\cdot_G A\rvert) \geq \lvert A\rvert^{1+c-\epsilon},\]where\[A+_GA = \{ a+b: (a,b)\in G\}\]and similarly for $A\cdot_GA$.

来源字段：https://www.erdosproblems.com/808

13463 | EP-812

### Erdős Problem #812

O / open | Erdős 问题

Is it true that\[\frac{R(n+1)}{R(n)}\geq 1+c\]for some constant $c>0$, for all large $n$? Is it true that\[R(n+1)-R(n)\gg n^2?\]

来源字段：https://www.erdosproblems.com/812

13464 | EP-817

### Erdős Problem #817

O / open | Erdős 问题

Let $k\geq 3$ and define $g_k(n)$ to be the minimal $N$ such that $\{1,\ldots,N\}$ contains some $A$ of size $\lvert A\rvert=n$ such that\[\langle A\rangle = \left\{\sum_{a\in A}\epsilon_aa: \epsilon_a\in \{0,1\}\right\}\]contains no non-trivial $k$-term arithmetic progression. Estimate $g_k(n)$. In particular, is it true that\[g_3(n) \gg 3^n?\]

来源字段：https://www.erdosproblems.com/817

13465 | EP-818

### Erdős Problem #818: Additive Combinatorics

S / solved | Erdős 问题

Let $A$ be a finite set of integers such that $\lvert A+A\rvert \ll \lvert A\rvert$. Is it true that\[\lvert AA\rvert \gg \frac{\lvert A\rvert^2}{(\log \lvert A\rvert)^C}\]for some constant $C>0$?

来源字段：https://www.erdosproblems.com/818

13466 | EP-819

### Erdős Problem #819

P / partially_solved | Erdős 问题

Let $f(N)$ be maximal such that there exists $A\subseteq \{1,\ldots,N\}$ with $\lvert A\rvert=\lfloor N^{1/2}\rfloor$ such that $\lvert (A+A)\cap [1,N]\rvert=f(N)$. Estimate $f(N)$.

来源字段：https://www.erdosproblems.com/819

13467 | EP-826

### Erdős Problem #826

O / open | Erdős 问题

Are there infinitely many $n$ such that, for all $k\geq 1$,\[\tau(n+k)\ll k?\]

来源字段：https://www.erdosproblems.com/826

13468 | EP-827

### Erdős Problem #827

O / open | Erdős 问题

Let $n_k$ be minimal such that if $n_k$ points in $\mathbb{R}^2$ are in general position then there exists a subset of $k$ points such that all $\binom{k}{3}$ triples determine circles of different radii.

Determine $n_k$.

来源字段：https://www.erdosproblems.com/827

13469 | EP-829

### Erdős Problem #829

O / open | Erdős 问题

Let $A\subset\mathbb{N}$ be the set of cubes. Is it true that\[1_A\ast 1_A(n) \ll (\log n)^{O(1)}?\]

来源字段：https://www.erdosproblems.com/829

13470 | EP-830

### Erdős Problem #830

O / open | Erdős 问题

We say that $a,b\in \mathbb{N}$ are an amicable pair if $\sigma(a)=\sigma(b)=a+b$. Are there infinitely many amicable pairs? If $A(x)$ counts the number of amicable $1\leq a\leq b\leq x$ then is it true that\[A(x)>x^{1-o(1)}?\]

来源字段：https://www.erdosproblems.com/830

13471 | EP-838

### Erdős Problem #838

O / open | Erdős 问题

Let $f(n)$ be maximal such that any $n$ points in $\mathbb{R}^2$, with no three on a line, determine at least $f(n)$ different convex subsets. Estimate $f(n)$ - in particular, does there exist a constant $c$ such that\[\lim \frac{\log f(n)}{(\log n)^2}=c?\]

来源字段：https://www.erdosproblems.com/838

13472 | EP-83

### Erdős Problem #83: Combinatorics

S / solved | Erdős 问题

Suppose that we have a family $\mathcal{F}$ of subsets of $[4n]$ such that $\lvert A\rvert=2n$ for all $A\in\mathcal{F}$ and for every $A,B\in \mathcal{F}$ we have $\lvert A\cap B\rvert \geq 2$. Then\[\lvert \mathcal{F}\rvert \leq \frac{1}{2}\left(\binom{4n}{2n}-\binom{2n}{n}^2\right).\]

来源字段：https://www.erdosproblems.com/83

13473 | EP-846

### Erdős Problem #846

S / solved | Erdős 问题

Let $A\subset \mathbb{R}^2$ be an infinite set for which there exists some $\epsilon>0$ such that in any subset of $A$ of size $n$ there are always at least $\epsilon n$ with no three on a line.

Is it true that $A$ is the union of a finite number of sets where no three are on a line?

来源字段：https://www.erdosproblems.com/846

13474 | EP-847

### Erdős Problem #847

S / solved | Erdős 问题

Let $A\subset \mathbb{N}$ be an infinite set for which there exists some $\epsilon>0$ such that in any subset of $A$ of size $n$ there is a subset of size at least $\epsilon n$ which contains no three-term arithmetic progression.

Is it true that $A$ is the union of a finite number of sets which contain no three-term arithmetic progression?

来源字段：https://www.erdosproblems.com/847

13475 | EP-857

### Erdős Problem #857

P / partially_solved | Erdős 问题

Let $m=m(n,k)$ be minimal such that in any collection of sets $A_1,\ldots,A_m\subseteq \{1,\ldots,n\}$ there must exist a sunflower of size $k$ - that is, some collection of $k$ of the $A_i$ which pairwise have the same intersection.

Estimate $m(n,k)$, or even better, give an asymptotic formula.

来源字段：https://www.erdosproblems.com/857

13476 | EP-863

### Erdős Problem #863

S / solved | Erdős 问题

Let $r\geq 2$ and let $A\subseteq \{1,\ldots,N\}$ be a set of maximal size such that there are at most $r$ solutions to $n=a+b$ with $a\leq b$ for any $n$. (That is, $A$ is a $B_2[r]$ set.)

Similarly, let $B\subseteq \{1,\ldots,N\}$ be a set of maximal size such that there are at most $r$ solutions to $n=a-b$ for any $n\geq 1$.

If $\lvert A\rvert\sim c_rN^{1/2}$ as $N\to \infty$ and $\lvert B\rvert \sim c_r'N^{1/2}$ as $N\to \infty$ then is it true that $c_r\neq c_r'$ for $r\geq 2$? Is it true that $c_r'<c_r$?

来源字段：https://www.erdosproblems.com/863

13477 | EP-864

### Erdős Problem #864

P / partially_solved | Erdős 问题

Let $A\subseteq \{1,\ldots N\}$ be a set such that there exists at most one $n$ with more than one solution to $n=a+b$ (with $a\leq b\in A$). Estimate the maximal possible size of $\lvert A\rvert$ - in particular, is it true that\[\lvert A\rvert \leq (1+o(1))\frac{2}{\sqrt{3}}N^{1/2}?\]

来源字段：https://www.erdosproblems.com/864

13478 | EP-867

### Erdős Problem #867: Additive Combinatorics

S / solved | Erdős 问题

Is it true that if $A=\{a_1<\cdots <a_t\}\subseteq \{1,\ldots,N\}$ has no solutions to\[a_i+a_{i+1}+\cdots+a_j\in A\]then\[\lvert A\rvert \leq \frac{N}{2}+O(1)?\]

来源字段：https://www.erdosproblems.com/867

13479 | EP-873

### Erdős Problem #873

P / partially_solved | Erdős 问题

Let $A=\{a_1<a_2<\cdots\}\subseteq \mathbb{N}$ and let $F(A,X,k)$ count the number of $i$ such that\[[a_i,a_{i+1},\ldots,a_{i+k-1}] < X,\]where the left-hand side is the least common multiple. Is it true that, for every $\epsilon >0$, there exists some $k$ such that\[F(A,X,k)<X^\epsilon?\]

来源字段：https://www.erdosproblems.com/873

13480 | EP-875

### Erdős Problem #875

P / partially_solved | Erdős 问题

Let $A=\{a_1<a_2<\cdots\}\subset \mathbb{N}$ be an infinite set such that the sets\[S_r = \{ a_1+\cdots +a_r: a_1<\cdots<a_r\in A\}\]are disjoint for distinct $r\geq 1$. How fast can such a sequence grow? How small can $a_{n+1}-a_n$ be? In particular, for which $c$ is it possible that $a_{n+1}-a_n\leq n^{c}$?

来源字段：https://www.erdosproblems.com/875

13481 | EP-876

### Erdős Problem #876

P / partially_solved | Erdős 问题

Let $A=\{a_1<a_2<\cdots\}\subset \mathbb{N}$ be an infinite sum-free set - that is, there are no solutions to\[a=b_1+\cdots+b_r\]with $b_1<\cdots<b_r<a\in A$. How small can $a_{n+1}-a_n$ be? Is it possible that $a_{n+1}-a_n<n$?

来源字段：https://www.erdosproblems.com/876

13482 | EP-877

### Erdős Problem #877: Additive Combinatorics

S / solved | Erdős 问题

Let $f_m(n)$ count the number of maximal sum-free subsets $A\subseteq\{1,\ldots,n\}$ - that is, there are no solutions to $a=b+c$ in $A$ and $A$ is maximal with this property. Estimate $f(n)$ - is it true that $f_m(n)=o(2^{n/2})$?

来源字段：https://www.erdosproblems.com/877

13483 | EP-881

### Erdős Problem #881

P / partially_solved | Erdős 问题

Let $A\subset\mathbb{N}$ be an additive basis of order $k$ which is minimal, in the sense that if $B\subset A$ is any infinite set then $A\backslash B$ is not a basis of order $k$.

Must there exist an infinite $B\subset A$ such that $A\backslash B$ is a basis of order $k+1$?

来源字段：https://www.erdosproblems.com/881

13484 | EP-895

### Erdős Problem #895: Additive Combinatorics, Graph Theory

S / solved | Erdős 问题

Is it true that, for all sufficiently large $n$, if $G$ is a triangle-free graph on $\{1,\ldots,n\}$ then there must exist three independent points $a,b,a+b$?

来源字段：https://www.erdosproblems.com/895

13485 | EP-896

### Erdős Problem #896

S / solved | Erdős 问题

Estimate the maximum of $F(A,B)$ as $A,B$ range over all subsets of $\{1,\ldots,N\}$, where $F(A,B)$ counts the number of $m$ such that $m=ab$ has exactly one solution (with $a\in A$ and $b\in B$).

来源字段：https://www.erdosproblems.com/896

13486 | EP-899

### Erdős Problem #899: Additive Combinatorics

S / solved | Erdős 问题

Let $A\subseteq \mathbb{N}$ be an infinite set such that $\lvert A\cap \{1,\ldots,N\}\rvert=o(N)$. Is it true that\[\limsup_{N\to \infty}\frac{\lvert (A-A)\cap \{1,\ldots,N\}\rvert}{\lvert A\cap \{1,\ldots,N\}\rvert}=\infty?\]

来源字段：https://www.erdosproblems.com/899

13487 | EP-903

### Erdős Problem #903: Combinatorics

S / solved | Erdős 问题

Let $n=p^2+p+1$ for some prime power $p$, and let $A_1,\ldots,A_t\subseteq \{1,\ldots,n\}$ be a block design (so that every pair $x,y\in \{1,\ldots,n\}$ is contained in exactly one $A_i$).

Is it true that if $t>n$ then $t\geq n+p$?

来源字段：https://www.erdosproblems.com/903

13488 | EP-906

### Erdős Problem #906

P / partially_solved | Erdős 问题

Is there an entire non-zero function $f:\mathbb{C}\to \mathbb{C}$ such that, for any infinite sequence $n_1<n_2<\cdots$, the set\[\{ z: f^{(n_k)}(z)=0 \textrm{ for some }k\geq 1\}\]is everywhere dense?

来源字段：https://www.erdosproblems.com/906

13489 | EP-936

### Erdős Problem #936

P / partially_solved | Erdős 问题

Are\[2^n\pm 1\]and\[n!\pm 1\]powerful (i.e. if $p\mid m$ then $p^2\mid m$) for only finitely many $n$?

来源字段：https://www.erdosproblems.com/936

13490 | EP-938

### Erdős Problem #938

P / partially_solved | Erdős 问题

Let $A=\{n_1<n_2<\cdots\}$ be the sequence of powerful numbers (if $p\mid n$ then $p^2\mid n$).

Are there only finitely many three-term progressions of consecutive terms $n_k,n_{k+1},n_{k+2}$?

来源字段：https://www.erdosproblems.com/938

13491 | EP-943

### Erdős Problem #943

P / partially_solved | Erdős 问题

Let $A$ be the set of powerful numbers (if $p\mid n$ then $p^2\mid n$). Is it true that\[1_A\ast 1_A(n)=n^{o(1)}\]for every $n$?

来源字段：https://www.erdosproblems.com/943

13492 | EP-949

### Erdős Problem #949

P / partially_solved | Erdős 问题

Let $S\subset \mathbb{R}$ be a set containing no solutions to $a+b=c$. Must there be a set $A\subseteq \mathbb{R}\backslash S$ of cardinality continuum such that $A+A\subseteq \mathbb{R}\backslash S$?

来源字段：https://www.erdosproblems.com/949

13493 | EP-956

### Erdős Problem #956

P / partially_solved | Erdős 问题

If $C,D\subseteq \mathbb{R}^2$ then the distance between $C$ and $D$ is defined by\[\delta(C,D)=\inf_{\substack{c\in C\\ d\in D}}\| c-d\|.\]Let $h(n)$ be the maximal number of unit distances between disjoint convex translates. That is, the maximal $m$ such that there is a compact convex set $C\subset \mathbb{R}^2$ and a set $X$ of size $n$ such that all $(C+x)_{x\in X}$ are disjoint and there are $m$ pairs $x_1,x_2\in X$ such that\[\delta(C+x_1,C+x_2)=1.\]Determine $h(n)$ - in particular, prove that there exists a constant $c>0$ such that $h(n)>n^{1+c}$ for all large $n$.

来源字段：https://www.erdosproblems.com/956

13494 | EP-959

### Erdős Problem #959

P / partially_solved | Erdős 问题

Let $A\subset \mathbb{R}^2$ be a set of size $n$ and let $\{d_1,\ldots,d_k\}$ be the set of distinct distances determined by $A$. Let $f(d)$ be the number of times the distance $d$ is determined, and suppose the $d_i$ are ordered such that\[f(d_1)\geq f(d_2)\geq \cdots \geq f(d_k).\]Estimate\[\max (f(d_1)-f(d_2)),\]where the maximum is taken over all $A$ of size $n$.

来源字段：https://www.erdosproblems.com/959

13495 | EP-960

### Erdős Problem #960

S / solved | Erdős 问题

Let $r,k\geq 2$ be fixed. Let $A\subset \mathbb{R}^2$ be a set of $n$ points with no $k$ points on a line. Determine the threshold $f_{r,k}(n)$ such that if there are at least $f_{r,k}(n)$ many ordinary lines (lines containing exactly two points) then there is a set $A'\subseteq A$ of $r$ points such that all $\binom{r}{2}$ many lines determined by $A'$ are ordinary.

Is it true that $f_{r,k}(n)=o(n^2)$, or perhaps even $\ll n$?

来源字段：https://www.erdosproblems.com/960

13496 | EP-963

### Erdős Problem #963

P / partially_solved | Erdős 问题

Let $f(n)$ be the maximal $k$ such that in any set $A\subset \mathbb{R}$ of size $n$ there is a subset $B\subseteq A$ of size $\lvert B\rvert\geq k$ which is dissociated that is, the sums $\sum_{b\in S}b$ are distinct for all $S\subseteq B$. Estimate $f(n)$ - in particular, is it true that\[f(n)\geq \lfloor \log_2 n\rfloor?\]

来源字段：https://www.erdosproblems.com/963

13497 | EP-965

### Erdős Problem #965: Ramsey Theory

S / solved | Erdős 问题

Is it true that, for any $2$-colouring of $\mathbb{R}$, there is a set $A\subseteq \mathbb{R}$ of cardinality $\aleph_1$ such that all sums $a+b$ with $a\neq b$ and $a,b\in A$ are the same colour?

来源字段：https://www.erdosproblems.com/965

13498 | EP-973

### Erdős Problem #973

P / partially_solved | Erdős 问题

Does there exist a constant $C>1$ such that, for every $n\geq 2$, there exists a sequence $z_i\in \mathbb{C}$ with $z_1=1$ and $\lvert z_i\rvert \geq 1$ for all $1\leq i\leq n$ with\[\max_{2\leq k\leq n+1}\left\lvert \sum_{1\leq i\leq n}z_i^k\right\rvert < C^{-n}?\]

来源字段：https://www.erdosproblems.com/973

13499 | EP-984

### Erdős Problem #984: Arithmetic Progressions, Additive Combinatorics

S / solved | Erdős 问题

Can $\mathbb{N}$ be $2$-coloured such that if\[\{a,a+d,\ldots,a+(k-1)d\}\]is a $k$-term monochromatic arithmetic progression then $k\ll_\epsilon a^\epsilon$ for all $\epsilon>0$?

来源字段：https://www.erdosproblems.com/984

13500 | EP-990

### Erdős Problem #990

S / solved | Erdős 问题

Let $f=a_0+\cdots+a_dx^d\in \mathbb{C}[x]$ be a polynomial. Is it true that, if $f$ has roots $z_1,\ldots,z_d$ with corresponding arguments $\theta_1,\ldots,\theta_d\in [0,2\pi]$, then for all intervals $I\subseteq [0,2\pi]$\[\left\lvert (\# \theta_i \in I) - \frac{\lvert I\rvert}{2\pi}d\right\rvert \ll \left(n\log M\right)^{1/2},\]where $n$ is the number of non-zero coefficients of $f$ and\[M=\frac{\lvert a_0\rvert+\cdots +\lvert a_d\rvert}{(\lvert a_0\rvert\lvert a_d\rvert)^{1/2}}.\]

来源字段：https://www.erdosproblems.com/990

13501 | OWR-2090-024

### Essential Hyperplane Covers of the Boolean Cube

O / open | Oberwolfach 报告问题汇编

Let $f(n)$ be the minimum number of affine hyperplanes needed to cover $\{0,1\}^n$ so that every hyperplane is essential and every variable has nonzero coefficient in some defining equation. Determine $f(n)$, or close the gap $\Omega(\sqrt n)\leq f(n)\leq n/2+1$.

来源字段：https://doi.org/10.4171/owr/2008/44

13502 | OWR-12173-001

### Euclidean Ramsey Configurations and Sphericity

O / open | Oberwolfach 报告问题汇编

Which finite point configurations $K\subset\mathbb R^n$ are Ramsey for Euclidean $r$-colorings? In particular, are the Ramsey configurations exactly the spherical configurations?

来源字段：https://doi.org/10.4171/owr/2012/50

13503 | OWR-17464-004

### Euclidean Red-Blue Ramsey Sets

O / open | Oberwolfach 报告问题汇编

For which subsets $K$ of Euclidean space $\mathbb E^n$ does every red-blue coloring of $\mathbb E^n$ contain either a red pair at distance $1$ or a blue isometric copy of $K$?

来源字段：https://doi.org/10.4171/owr/2020/1

13504 | OPG-636

### Even vs. odd latin squares

P / partially_solved | Open Problem Garden

A latin square is even if the product of the signs of all of the row and column permutations is 1 and is odd otherwise.

Conjecture For every positive even integer $n$, the number of even latin squares of order $n$ and the number of odd latin squares of order $n$ are different.

背景链接：http://www.openproblemgarden.org/op/even_vs_odd_latin_squares

13505 | AIM-COMBINATORICS-0272

### Even-pair skew partitions and a natural two-block localization theorem

O / open | AIM 研讨会问题汇编

**E.8 Even-Pair Skew-Partition.** An even-pair skew-partition is a partition of the vertex set of a graph \(G\) into four sets \(A,B,C,D\) such that \(A\) is complete to \(B\), \(C\) is anti-complete to \(D\), and any two non-adjacent vertices in \(A\) or \(B\) are an even pair. **Question 2.** Is even-pair skew-partition a composition? Contributed by Bruce Reed.

来源字段：https://aimath.org/WWN/perfectgraph/perfectgraph.pdf

13506 | AIM-COMBINATORICS-0160

### Exact anatomy of the BCW circular construction

P / partially_solved | AIM 研讨会问题汇编

**Behzad--Chartrand--Wall conjecture.** For integers \(r\ge1\) and \(g\ge2\), the minimum order of an \(r\)-regular digraph of directed girth \(g\) is \[ r(g-1)+1. \]

Sullivan's 2006 AIM workshop survey *A Summary of Results and Problems Related to the Caccetta--Häggkvist Conjecture*. Section 4 defines an \(r\)-regular digraph by \[ d^+(v)=d^-(v)=r\qquad\text{for every vertex }v.

Equivalently, every \(r\)-regular digraph of order \(n\) has directed girth \[ g(D)\le \left\lceil\frac nr\right\rceil. \] The construction giving the upper bound has vertex set \(\mathbb Z_{r(g-1)+1}\) and arcs \[ i\longrightarrow i+s\pmod {r(g-1)+1}, \qquad 1\le s\le r. \tag{1} \] This is the precise meaning of placing the vertices on a circle and joining each vertex to the next \(r\) clockwise vertices [1,2,10].

There is a convention subtlety. The AIM survey allows parallel arcs in its ambient definition of “digraph,” but the original cage problem and later precise formulations use at most one arc for each ordered pair. Shen and Gregory explicitly allow loops but no multiple arcs [7]. In the cage range \(g\ge2\), loops are automatically absent. Digons are allowed when \(g=2\), while \(g\ge3\) automatically excludes them. Thus the construction is loopless for all \(r\ge1,g\ge2\), is the complete bidirected digraph when \(g=2\), and is oriented when \(g\ge3\).

The raw record continues through Sections 5 and 6 of the source. Those passages on connectivity, additive combinatorics, Plünnecke graphs, and rainbow conjectures are nearby source context, not part of Conjecture 4.1.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13507 | AIM-COMBINATORICS-0095

### Exact block-product sampling of maximal G-parking functions

P / partially_solved | AIM 研讨会问题汇编

**(5) Lionel Levine:** “How can we uniformly and efficiently sample from the set of maximal \(G\)-parking functions?” **(a) Persi Diacons:** “There may be a connection to my paper with Christos Athanasiadis [5].”

Christos A. Athanasiadis and Persi Diaconis, “Functions of random walks on hyperplane arrangements,” *Advances in Applied Mathematics* **45** (2010), 410--437.

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13508 | AIM-COMBINATORICS-0114

### Exact cellular sandpile dynamics on simplex-boundary spheres

P / partially_solved | AIM 研讨会问题汇编

Art Duval: “Can we figure out the abelian sandpile dynamics in higher dimensions?” Matt Baker asks, “Do we have pretty three-dimensional pictures?” Caroline Klivans replies, “I have code for this.”

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13509 | AMR-031-0013

### Exact Dedekind numbers

O / open | AMR 问题清单汇编

Let $M(n)$ be the number of monotone Boolean functions of $n$ variables, equivalently the number of antichains of subsets of an $n$-element set. Determine the exact values $M(n)$ for $n\geq10$.

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Combinatorics

13510 | AIM-COMBINATORICS-0153

### Exact deficit identity for recursive tight-cycle constructions

P / partially_solved | AIM 研讨会问题汇编

It defines \[ C_5^3=\{123,234,345,451,512\} \] and asks for the classical edge-density Turán value. Its conjecture is \[ \boxed{\pi(C_5^3)=2\sqrt3-3.} \] Here \[ \pi(F)=\lim_{n\to\infty}\frac{\operatorname{ex}(n,F)}{\binom n3}, \] where copies are ordinary (not necessarily induced) injective copies of a 3-uniform hypergraph.

The displayed five edges are all cyclically consecutive triples on five distinct vertices, so this is the **full tight cycle**. It is not:

- the four-edge hypergraph \(C_5^{3-}\), obtained by deleting one edge; - a loose or linear cycle; - the expansion of an ordinary graph cycle (which introduces private new vertices for graph edges); - a minimum-codegree, covering, tiling, or uniformly-dense Turán problem.

These distinctions matter. In particular, the 2024--2025 results proving \(\pi(C_5^{3-})=1/4\), and codegree results about tight cycles, do not answer this record.

The source statement is mathematically coherent. Its only historical compression is that the currently public numerical upper certificate is usually quoted more accurately as \(0.468287\), while the record rounds it to \(<0.4683\).

来源字段：http://aimpl.org/hypergraphturan/3/

13511 | AIM-COMBINATORICS-0141

### Exact degree-sequence obstruction to hyperbolicity indicators

P / partially_solved | AIM 研讨会问题汇编

For random graphs, is there an idea for a degree of heterogeneity that could indicate hyperbolicity

Its exact problem text is:

The accompanying remarks ask for conditions ensuring hyperbolicity of an ensemble of random graphs, meaning that hyperbolic graphs have probability one, and emphasize that randomness is essential.

来源字段：http://aimpl.org/largenetworks/1/

13512 | AIM-COMBINATORICS-0197

### Exact feedback sets in cyclic tournament substitutions

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.27 (Chudnovsky--Seymour--Sullivan).** Let \(G\) be a simple digraph with \(k\) non-edges, where a non-edge is an unordered pair \(\{u,v\}\) for which neither \(uv\) nor \(vu\) is an arc. If \(G\) has no directed cycle of length at most \(3\), then one can delete at most \(k/2\) edges from \(G\) and obtain an acyclic digraph.

Sullivan's 2006 AIM survey.

For a digraph \(G\), write \[ \gamma(G):= \bigl|\{\{u,v\}:u\ne v,\ uv,vu\notin E(G)\}\bigr| \] and let \(\beta(G)\) be the minimum cardinality of an **arc** set \(F\subseteq E(G)\) such that \(G-F\) is acyclic. The source conjecture is therefore \[ \beta(G)\leq \frac{\gamma(G)}2. \tag{CSS} \] Because \(\beta(G)\) is integral, the literal meaning of “at most \(k/2\) edges” is \[ \beta(G)\leq \left\lfloor\frac{k}{2}\right\rfloor. \] In particular, equality in the integer formulation can occur for odd \(k\), even though equality in the real-valued display cannot.

The conventions need care.

- Some authors use “simple digraph” while still allowing a pair of opposite arcs. This ambiguity is harmless here: opposite arcs form a directed \(2\)-cycle and are excluded by the hypothesis. Thus every admissible digraph is oriented. - The deletion is of arcs, not vertices. A feedback vertex set is a different parameter and is not addressed by the conjecture. - “No directed cycle of length at most \(3\)” excludes loops, digons, and directed triangles. Simplicity already excludes loops in the usual convention. - The source note says that tight examples include transitive tournaments, \(C_4\), and “products,” but it does not define the product operation. The primary Chudnovsky--Seymour--Sullivan paper displays a balanced four-bag substitution: each vertex of a directed \(4\)-cycle is replaced by a transitive tournament, and consecutive bags are joined completely in the cyclic direction. This is the lexicographic product \(C_4\circ T_t\). Section 5 treats this operation precisely.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13513 | AIM-COMBINATORICS-0259

### Exact finite-parameter optimization of the largest product-free families in alternating groups

S / solved | AIM 研讨会问题汇编

**Problem 4.3 (V. Sós).** Let \(P\) be a product-free subset of an abelian group \(G\) with \(|G|=n\). How large can \(P\) be? For abelian groups one has \(|P|\ge 2n/7\) (Alon--Kleitman), which is sharp. The case \(G=A_n\) is already interesting, and Green conjectures that in this case \(|P|=o(|A_n|)\).

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13514 | AIM-COMBINATORICS-0255

### Exact horizontal Plancherel identities for the diagonal Diffie--Hellman sum

P / partially_solved | AIM 研讨会问题汇编

Problem 3.18 (J. Bourgain). Find a good upper bound for the absolute value of the exponential sum ... where \(\theta\) is a generator for \(\mathbb F_p^*\).

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13515 | AIM-COMBINATORICS-0202

### Exact identities and rounding loss for induced directed paths

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.32 (Thomassé).** If \(G\) is a loopless digon-free digraph, the maximum number of induced directed 2-edge paths is \[ \frac{n^3}{15}+O(n^2). \]

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13516 | AMR-029-0034

### Exact matching in red-blue bipartite graphs

P / partially_solved | AMR 问题清单汇编

Give an algorithm and/or a good characterization to decide if a red-blue edge-coloured bipartite graph contains a perfect matching with exactly k red edges.

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13517 | AIM-COMBINATORICS-0221

### Exact no-carry constructions for sparse two-fold sumsets

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.15 (E. Croot).** Let \(f(S)\) denote the length of the longest arithmetic progression in a set of integers \(S\). Given a real \(\theta\in(0,1]\) and an integer \(N\ge1\), among all subsets \(A\subseteq[N]\) satisfying \(|A|\ge N^\theta\), how small can \(f(A+A)\) be?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13518 | AIM-COMBINATORICS-0152

### Exact profiles for the odd book construction

P / partially_solved | AIM 研讨会问题汇编

The AIM record, from the workshop list “Hypergraph Turán problem,” section “Turán functions for books,” conjectures

\[ \pi(B_{5,5})=\frac{40}{81},\qquad \pi(B_{6,6})=\frac12. \]

The notation is important. Following Bohman-Frieze-Mubayi-Pikhurko, for \(2\leq m\leq k\), the \(k\)-uniform book \(B_{k,m}\) has pages

\[ [k-1]\cup\{k+i-1\}\quad (i=1,\ldots,m) \]

and binding edge \(\{k,k+1,\ldots,2k-1\}\). Thus \(B_{k,k}\) consists of a spine \(S\) of size \(k-1\), a set \(T\) of \(k\) tips, all pages \(S\cup\{t\}\) for \(t\in T\), and the binding edge \(T\). This is not the ordinary graph-theoretic book convention.

For a \(k\)-graph \(H\), define

\[ N_H(S)=\{v\notin S:S\cup\{v\}\in H\},\qquad |S|=k-1. \]

Then \(H\) is \(B_{k,k}\)-free if and only if every \(N_H(S)\) is an independent set in \(H\): an edge \(T\subseteq N_H(S)\) is exactly the binding of a copy with spine \(S\). This is the “independent neighborhoods” formulation used in the primary paper.

来源字段：http://aimpl.org/hypergraphturan/2/

13519 | AIM-COMBINATORICS-0210

### Exact quadratic-residue half-set counts and a quantitative Varnavides transfer

P / partially_solved | AIM 研讨会问题汇编

For a prime \(p\), what is the least number of three-term arithmetic progressions that a subset \(A\subset \mathbb F_p\) with \(|A|=(p-1)/2\) can have? What happens if \(|A|=\delta p\), where \(\delta<1/2\)?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13520 | AIM-COMBINATORICS-0150

### Exact recursion and local profile stability for the broken tetrahedron construction

P / partially_solved | AIM 研讨会问题汇编

**\(K_4^3\) minus an edge.** Let \(K_4^-\) be obtained from \(K_4^3\) by removing one edge. Conjecture that \[ \pi(K_4^-)=\frac27. \]

来源字段：http://aimpl.org/hypergraphturan/1/

13521 | AIM-COMBINATORICS-0280

### Exact structure and coloring of the proposed even-hole-free circulants

O / open | AIM 研讨会问题汇编

Every non-empty even-hole-free circulant is isomorphic to a $G(k,m)$.

It begins in the middle of the proof of the implications among Conjectures 2, 3, and 4 in Section G.4, contains all of Section G.5, and then absorbs G.6, Chapters H, I, and J, and the beginning of Chapter K.

The embedded items can be delimited as follows.

1. carried-over proof prose from G.4 on even-hole-free graphs; 2. G.5, the classification conjecture for even-hole-free circulants; 3. G.6, the question asking for a characterization of beta-perfect graphs; 4. H.1--H.4, several unrelated questions about partitionable graphs; 5. Chapter I, a collection of imperfection-ratio bounds and conjectures; 6. Chapter J, two integer-programming questions; and 7. the opening definitions of Chapter K on balanced graphs.

Only item 2 is studied here. No claim below addresses any other embedded record.

For integers $k\geq 1$ and $m\geq 0$, put

\[ L=2m+1,\qquad n=kL, \]

and take $V=\mathbb Z_n$. The source includes loops and declares $(i,j)$ to be an edge when, for some integer $t$,

\[ i-j+tL\equiv 0,\ +1,\text{ or }-1\pmod n. \tag{1} \]

We write $G(k,m)$ for the **underlying simple graph**, obtained by deleting the source's convenient loops. The displayed conjecture is:

The source additionally prints

\[ \omega(G(k,m))=2k, \qquad 2k+\left\lceil\frac{k}{m}\right\rceil \leq \chi(G(k,m)) \leq 2k+\left\lceil\frac{k}{m}\right\rceil+1. \tag{2} \]

There are two necessary repairs.

* Formula (2) is meaningful only for $m\geq1$, and its clique assertion is correct only for $m\geq2$. * In their subsequent 2005 technical report, Andrade, Boros, and Gurvich say that they always restrict to connected circulants. Without this convention, the displayed classification is false; a counterexample is proved below.

Accordingly, the intended unresolved question is recovered as the classification of **connected**, nonempty, even-hole-free simple circulants. The family itself retains the source range $k\geq1,m\geq0$.

来源字段：https://aimath.org/WWN/perfectgraph/perfectgraph.pdf

13522 | AIM-COMBINATORICS-0032

### Exact structure inside a two-fold sumset

P / partially_solved | AIM 研讨会问题汇编

Let \(A\subset \mathbb F_2^n\) be a subset of density \(0.01\). Then there is a subset \(B\subset \mathbb F_2^n\) with density absolute constant, so that \[ B+B\subset A+A \] and \[ B=S_1+\cdots+S_\ell \] for some sets \(S_i\)'s with \(|S_i|\leq \operatorname{poly}(n)\) and \(\ell\leq \operatorname{poly}(n)\).

The source says:

The live page was checked on 29 July 2026. It has the same text, the attribution above, and no status or explanatory remark. Its placement and `problem` tag show that the declarative sentence is a proposed assertion to prove, not a theorem reported with a proof.

There are two necessary quantifier repairs.

1. **Literal density \(0.01\) is impossible.** A subset of \(\mathbb F_2^n\) has density \(m/2^n\). Equality \(m/2^n=1/100\) would imply \(100m=2^n\), impossible because \(25\nmid 2^n\). Thus the literal statement has no instances and is vacuously true. In additive-combinatorics usage, “density \(0.01\)” here must mean “density at least \(0.01\)” (or a fixed positive density, with harmless rounding). 2. The words “absolute constant” and “\(\operatorname{poly}(n)\)” must be uniform. A precise intended reading is: there exist constants \(c>0\), \(C,D<\infty\), and \(n_0\), independent of \(n\) and \(A\), such that every \(n\geq n_0\) and every \(A\subset\mathbb F_2^n\) with \(|A|\geq .01\,2^n\) admit nonempty \(S_1,\dots,S_\ell\) and \(B\) satisfying \[ |B|\geq c2^n,\qquad B+B\subset A+A, \qquad B=S_1+\cdots+S_\ell, \] \[ \ell\leq n^D,\qquad |S_i|\leq n^C. \] Without uniform exponents, one could take \(\ell=1\), \(S_1=B=A\), and choose an exponent separately for each finite instance; that is not a meaningful conjecture.

All mathematical analysis below concerns this repaired, uniform version.

来源字段：http://aimpl.org/addcombapp/2/

13523 | AIM-COMBINATORICS-0263

### Exact subgroup self-sumsets and a robust quotient-character leakage bound

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.7 (A. Granville and T. Tao).** Suppose that \(G\) is a proper subgroup of the multiplicative group of \(\mathbb F_p\). Let \(A\subset \mathbb F_p\) such that \(A+A=G\) or \(A+A\) is slightly larger than \(G\). Do such \(A\) exist, and do they necessarily have structure?

Another version of this question is as follows: given that \(|A|>p^\varepsilon\), can \(A+A\) be contained in a proper subgroup of \(\mathbb F_p^*\)?

**Remark(s).** Probably a very hard question, at least for small \(\varepsilon>0\). Negative answer would imply Vinogradov's conjecture that the least quadratic non-residue modulo \(p\) is smaller than \(p^\varepsilon\), for \(p\) sufficiently large. This problem is very close to Problem 3.6.

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13524 | OWR-12697684-021

### Exact Thresholds for Rainbow Matchings in Hypergraphs

P / partially_solved | Oberwolfach 报告问题汇编

How large must $N$ be in terms of $t$ and $r$ to guarantee a rainbow matching of size $t$?

来源字段：https://doi.org/10.4171/owr/2023/1

13525 | AIM-COMBINATORICS-0158

### Exact trace certificates and a regular counterexample normal form for Caccetta--Haggkvist

P / partially_solved | AIM 研讨会问题汇编

**Caccetta--Häggkvist conjecture.** Every simple \(n\)-vertex digraph with minimum outdegree at least \(r\) has a directed cycle of length at most \[ \left\lceil\frac nr\right\rceil . \]

Sullivan's 14 April 2006 AIM workshop survey, *A Summary of Results and Problems Related to the Caccetta--Häggkvist Conjecture*. Inspection of the original PDF recovers Conjecture 2.1 as follows.

Here \(r\) is a positive integer. We take the digraph to be loopless, with at most one arc in each ordered direction. The **directed girth** \(g(D)\) is the least number of arcs in a directed cycle, and is infinity for an acyclic digraph. The ceiling is essential: for example \(n=7,r=2\) predicts a cycle of length at most four, not three.

There is a convention mismatch inside the source which must be made explicit. Its definition of “simple digraph” excludes parallel copies of an arc but does not exclude having both \(u\to v\) and \(v\to u\). Such a pair is a **digon**. The matrix restatement, however, requires \(a_{ij}=1\Rightarrow a_{ji}=0\) for \(i\ne j\), which encodes an **oriented graph** (no digons). This causes no loss: a digon is already a directed 2-cycle, and \(\lceil n/r\rceil\geq2\) whenever a loopless \(n\)-vertex digraph can have outdegree \(r\leq n-1\). Thus every counterexample would be oriented.

For the matrix formulation, let \(B\) be the ordinary adjacency matrix: \(b_{ii}=0\), and \(b_{ij}=1\) exactly when \(i\to j\) is an actual arc. The source's matrix is \[ A=I+B. \tag{1} \] Its diagonal ones are artificial “wait” moves, not loops of the original digraph. Hence a row sum at least \(r+1\) means actual outdegree at least \(r\). With \(L=\lceil n/r\rceil\), the source asserts \[ \operatorname{tr}(A^L)>n. \tag{2} \] Theorem 1 below proves the exact equivalence, including why the inequality is strict and why a closed walk suffices to obtain a simple directed cycle.

The historical bullets in the record describe the state in 2006; they are not treated as present-day claims.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13526 | AMR-031-0015

### Exact van der Waerden numbers

P / partially_solved | AMR 问题清单汇编

Let $W(r,k)$ be the least $N$ such that every coloring of $\{1,\ldots,N\}$ with $r$ colors contains a monochromatic arithmetic progression of length $k$. Determine the unknown exact values of $W(r,k)$.

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Combinatorics

13527 | OWR-11136-022

### Exceptional Points in Favourite Distance Digraphs

O / open | Oberwolfach 报告问题汇编

Given a finite point set $S \subset \mathbb{R}^3$ and a assignment of a 'favourite distance' $d(p) > 0$ to each point $p \in S$, form the directed graph whose edges are $(p, q)$ with $|p - q| = d(p)$. In the extremal characterization of maximum-outdegree favourite-distance digraphs in $\mathbb{R}^3$ (Swanepoel), let $T \subset S$ be the exceptional set of at most two vertices not lying on the canonical concentric spheres/circles. Is the exceptional set $T$ always empty?

来源字段：https://doi.org/10.4171/owr/2011/44

13528 | OWR-11129-001

### Existence of an Extremal Doubly Even $[72,36,16]$ Code

O / open | Oberwolfach 报告问题汇编

Does an extremal binary self-dual doubly even code with parameters $[72,36,16]$ exist?

来源字段：https://doi.org/10.4171/owr/2011/37

13529 | OWR-12861-001

### Existence of Combinatorial Designs at Admissible Orders

P / partially_solved | Oberwolfach 报告问题汇编

For integers $q>r\geq1$ and $\lambda\geq1$, an $(n,q,r,\lambda)$-design is a collection of $q$-subsets of an $n$-element set in which every $r$-subset occurs in exactly $\lambda$ blocks. Characterize the orders $n$ for which such a design exists. The necessary divisibility conditions are $$\binom{q-i}{r-i}\mid\lambda\binom{n-i}{r-i}\qquad(0\leq i<r).$$ For fixed $q$, $r$, and $\lambda$, these conditions are known to be sufficient for all sufficiently large $n$; the remaining question is to determine the finite exceptional admissible orders. When $\lambda=1$, the designs are Steiner systems.

来源字段：https://doi.org/10.4171/owr/2014/01

13530 | OWR-11136-006

### Existence of Sparse Steiner Triple Systems

P / partially_solved | Oberwolfach 报告问题汇编

For which $s$ and admissible $v$ does there exist an $s$-sparse Steiner triple system $\operatorname{STS}(v)$, containing no $i$ blocks on $i+2$ points for any $4\leq i\leq s$?

来源字段：https://doi.org/10.4171/owr/2011/44

13531 | AIM-COMBINATORICS-0011

### Expansion can hide only global Sidorenko obstructions

P / partially_solved | AIM 研讨会问题汇编

Given a \(k\)-graph \(F\), define the \(r\)-expansion of \(F\) to be the \(r\)-graph obtained by enlarging each \(k\)-edge of \(F\) with a set of \(r-k\) vertices of degree one.

Does there exist a non-Sidorenko (hyper)graph \(H\) whose \(r\)-expansion is Sidorenko?

来源字段：http://aimpl.org/highdimdiscrete/3/

13532 | OWR-15215-003

### Explicit Low-Discrepancy Sets and Efficient Dispersion

P / partially_solved | Oberwolfach 报告问题汇编

Let $$ \operatorname{disc}^*(n,d)=\inf_{|P|=n}\sup_{x\in[0,1]^d}\left|\frac{|P\cap[0,x]|}{n}-x_1\cdots x_d\right|. $$ Can the gap between the bounds $c\,d/n\leq\operatorname{disc}^*(n,d)\leq C\sqrt{d/n}$ be closed? Can point sets satisfying the upper bound be constructed explicitly, or with less randomness than independent uniform sampling? For dispersion, what is the optimal order of the largest empty axis-parallel box, and what is the complexity of finding one?

来源字段：https://doi.org/10.4171/owr/2017/10

13533 | AIM-COMBINATORICS-0151

### Explicit residue formulas for the solved generalized-triangle problem

S / solved | AIM 研讨会问题汇编

Frankl and Füredi determined $\pi(B_{k,2})$ for $k=5,6$; in both cases the lower bound comes from blowing up a small design. Determine $\operatorname{ex}(n,B_{5,2})$ and $\operatorname{ex}(n,B_{6,2})$ exactly for all large $n$.

The record adds that the missing stability property was the main difficulty.

The 2011 workshop report defines the $k$-uniform book $B_{k,m}$ as follows. There are $m$ page edges sharing a common set $X$ of $k-1$ vertices:

$$ X\cup\{y_1\},\ldots,X\cup\{y_m\}, $$

and one further $k$-edge contains the $m$ vertices $y_1,\ldots,y_m$ and is otherwise disjoint from the page edges. In particular, $B_{k,2}$ has vertex set

$$ X\mathbin{\dot\cup}\{y_1,y_2\}\mathbin{\dot\cup}Z, \qquad |X|=k-1,\quad |Z|=k-2, $$

$$ X\cup\{y_1\},\qquad X\cup\{y_2\},\qquad \{y_1,y_2\}\cup Z. \tag{1} $$

Thus $B_{k,2}$ is exactly the generalized $k$-uniform triangle $\mathcal T_k$ used by Norin and Yepremyan. This identification is important: the word “book” alone does not specify the uniformity or the closing edge.

来源字段：http://aimpl.org/hypergraphturan/2/

13534 | AIM-COMBINATORICS-0115

### Explorer aggregation on oriented codimension-one faces

P / partially_solved | AIM 研讨会问题汇编

“Is this model more amenable to generalization **than** the abelian sandpile model?”

Persi Diaconis and William Fulton, “A growth model, a game, an algebra, Lagrange inversion, and characteristic classes,” *Rendiconti del Seminario Matematico della Università e Politecnico di Torino* **49** (1991), 95--119.

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13535 | AIM-COMBINATORICS-0239

### Exponential upper and lower bounds for the additive basis order of multiplicative subgroups

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.2 (J. Bourgain).** Given \(\delta\in(0,1)\), what is the smallest integer \(k\geq 1\) such that, for every prime \(p\) and every multiplicative subgroup \(H\leq \mathbb F_p^\times\) with \(|H|>p^\delta\), one has \[ kH:=\underbrace{H+\cdots+H}_{k\text{ summands}}=\mathbb F_p? \] It was known that this holds if \(\log k>(1/\delta)^C\). Can one do better?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13536 | AMR-054-0034

### Extending Pseudosegment Arrangements by Subdivision

O / open | AMR 问题清单汇编

How many intersections among an arrangement of pseudosegments in the plane must be added as vertices to allow the pseudosegment arrangment to be extended to a pseudoline arrangement? An arrangement of pseudosegments in the plane is a family of finite-length planar curves such that every two curves intersect in at most one point. An arrangement of pseudolines in the plane is a family of planar curves that extend to infinity on both ends such that every two curves intersect in at most one point. Only some pseudosegment arrangements can be extended to pseudoline arrangements. However, if we allow turning intersection points into vertices of the arrangement, thereby subdividing the segments, then it is always possible to make a pseudosegment arrangement extendible. The question is how many such vertices must be added in the worst-case in terms of the number $n$ of pseudosegments.

背景链接：https://topp.openproblem.net/p34

13537 | OPG-60003

### Extremal $4$-Neighbour Bootstrap Percolation in the Hypercube

P / partially_solved | Open Problem Garden

Problem Determine the smallest percolating set for the $4$-neighbour bootstrap process in the hypercube.

背景链接：http://www.openproblemgarden.org/op/extremal_4_neighbour_bootstrap_percolation_in_the_hypercube

13538 | OWR-1386-011

### Extremal Bounds for Intersecting $K(d)$-Families

S / solved | Oberwolfach 报告问题汇编

Fix $k\geq d\geq3$. A family $\mathcal G$ of $k$-subsets is a $K(d)$-family if any distinct $A_1,\ldots,A_d\in\mathcal G$ with $|\bigcup_iA_i|\leq2k$ have $\bigcap_iA_i\neq\varnothing$. Is $|\mathcal G|\leq\binom{n-1}{k-1}$ for every $K(d)$-family on $[n]$ when $n\geq dk/(d-1)$, with equality only for a common-point family?

来源字段：https://doi.org/10.4171/owr/2006/48

13539 | OWR-4425-005

### Extremal Bounds Relating Word Period Parameters

P / partially_solved | Oberwolfach 报告问题汇编

Does $|w|>\frac{7}{3}\tau(w)-4$ imply $\tau(w)=\pi(w)$? What is the infimum of $d$ for which a counterexample satisfies $|w|\geq\frac{7}{3}\tau(w)-d$? Must such extremal words be Assous-Pouzet words, and, for fixed $\delta$, how can all words satisfying $\tau(w)<\pi(w)$ and $|w|\geq\frac{7}{3}\tau(w)-\delta$ be described?

来源字段：https://doi.org/10.4171/owr/2010/37

13540 | OWR-10252930-020

### Extremal Conjecture for Partially Directed Uniform Hypergraphs

P / partially_solved | Oberwolfach 报告问题汇编

For fixed $k\geq 2$ and all sufficiently large $n$, does every $n$-vertex $k$-PDG without $\widetilde T_k$ as a subgraph satisfy $\alpha+(\log_2 3)\beta\leq 1$?

来源字段：https://doi.org/10.4171/owr/2022/22

13541 | OWR-1587-006

### Extremal Densities of Triangles and Uniform Hypergraphs

P / partially_solved | Oberwolfach 报告问题汇编

Determine the minimum triangle density above the critical edge density and the Turán density of the complete $3$-uniform hypergraph $K_4^3$, together with the analogous questions for $K_r$ when $r>3$.

来源字段：https://doi.org/10.4171/owr/2007/31

13542 | OWR-1782-008

### Extremal Independent-Neighborhood Hypergraphs of Uniformity Five and Six

O / open | Oberwolfach 报告问题汇编

Let $f(n,k)$ be the maximum number of edges in an $n$-vertex $k$-uniform hypergraph whose every neighborhood is independent. Let $b(n,k)$ be the maximum number of edges in an odd $n$-vertex $k$-uniform hypergraph, where odd means that some bipartition $X\cup Y$ meets every edge in an odd number of vertices of $Y$. Conjecture: for $k\in\{5,6\}$ and all sufficiently large $n$, $f(n,k)=b(n,k)$.

来源字段：https://doi.org/10.4171/owr/2008/01

13543 | OWR-12861-018

### Extremal Numbers of Tight Hypergraph Paths and Cycles

O / open | Oberwolfach 报告问题汇编

For a $k$-uniform tight path $P_{\ell}^{k}$ and tight cycle $C_{\ell}^{k}$, what are the extremal numbers $\operatorname{ex}(n;P_{\ell}^{k})$ and $\operatorname{ex}(n;C_{\ell}^{k})$ when $k$ divides $\ell$?

来源字段：https://doi.org/10.4171/owr/2014/01

13544 | OWR-1703876-016

## Extremal Questions in Discrete Geometry and Topological Combinatorics

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Discrete-Geometry Problem Collection [OWR-1703876-017]

For a simplicial $4$-polytope $P$ with $n$ vertices in which every three vertices contain an edge, must its graph contain a clique of size at least $n/10$? Given two continuous probability measures on $S^2$ and $0<\alpha<1/2$, must some spherical cap have measure $\alpha$ for both? Must every simple red-blue pseudoline arrangement contain a bounded triangular face incident to both colors? For a hypergraph $H$, does $m_2^*=3$ imply that $m_3^*$ is finite? Finally, what is the minimum number of partial triangulations of $n$ points in general position in the plane, and is it $C_{n-2}=\frac{1}{n-1}\binom{2n-4}{n-2}$?

来源字段：https://doi.org/10.4171/owr/2020/30

13545 | AIM-COMBINATORICS-0025

## Extremal sumset examples: Hamming balls and sharp no-carry digit cubes

P / partially_solved | AIM 研讨会问题汇编

The AIM record, under the heading “Finding examples,” contains three requests:

1. “Find counter examples to the polynomial Bogolyubov Ruzsa theorem.” 2. “Find interesting examples of subset \(A\subset\mathbb F_2^n\) so that \(A+A\) is far from all of \(\mathbb F_2^n\).” 3. “Find interesting examples of \(A\subset[N]\) of density \(\alpha\) so that \(4A\) does not contain long APs.”

It did not assert that the conjecture was known false. The first request also does not specify a theorem statement, a summand count, or a parameter to falsify. Source-era terminology distinguished at least two assertions:

* A **dense polynomial Bogolyubov statement** asks whether every density-\(\alpha\) set \(A\subseteq\mathbb F_2^n\) has a linear subspace of codimension \(O(\log(1/\alpha))\) inside \(4A\). * A **small-doubling polynomial Bogolyubov--Ruzsa statement** asks whether \(|A+A|\leq K|A|\) forces a subspace \(V\subseteq tA\), for some fixed absolute \(t\), with \(|V|\geq K^{-O(1)}|A|\).

These are stronger fixed-sumset statements than merely covering \(A\) by polynomially many cosets of a subspace. The latter is Marton's polynomial Freiman--Ruzsa (PFR) conjecture, now a theorem in \(\mathbb F_2^n\). Thus “find counterexamples to the theorem” is obsolete if “theorem” meant PFR, while the stronger fixed-\(t\) polynomial Bogolyubov conjecture remains a viable open problem in general. In special structured settings polynomial Bogolyubov-type results have also been proved. No counterexample to a proved theorem can be sought within its hypotheses.

We therefore recover the mathematical task as follows: construct and exactly analyze extremal families which test the best possible polynomial dependence, are far from affine-subspace models while making \(A+A\) quantitatively far from the ambient vector space, or keep arithmetic progressions in an integer fourfold sumset short.

Conventions are essential:

* In \(\mathbb F_2^n\), addition is coordinatewise modulo 2 and \(jA=\{a_1+\cdots+a_j:a_i\in A\}\). Repetition is allowed. Because the characteristic is 2, subtraction and addition agree. * In \(\mathbb Z\), \(jA\) is the ordinary Minkowski sum with repetition. There is no modular wraparound. * An arithmetic progression is nonconstant and has positive integer common difference. * For the second request, we report two explicit metrics. The missing density of the sumset is \[ 1-\frac{|A+A|}{2^n}. \] For comparison with affine-subspace models, we use normalized symmetric difference \[ \frac{|A\mathbin\triangle U|}{|A|+|U|} \] for affine subspaces \(U\) on the same cardinality scale as \(A\). The source itself does not prescribe a unique metric or scale.

来源字段：http://aimpl.org/addcombapp/2/

13546 | AMR-029-0035

### Extreme direction Sperner for square 0-1 matrix

O / open | AMR 问题清单汇编

Let A be an $n \times n$ 0-1 matrix, and suppose that the facets of the polyhedron $P=\{x: A x \leq {\mathbf 1},\ x \leq {\mathbf 1}\}$ are coloured by $n$ colours such that a facet with extreme direction $-e_i$ does not get colour $i$, and every colour appears twice. Can we find in polynomial time a vertex that is incident to facets of every colour (a panchromatic vertex)?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13547 | OWR-4798-019

### Face Vectors of Flag Simplicial Complexes

O / open | Oberwolfach 报告问题汇编

Which integer vectors occur as the face vectors of flag simplicial complexes, and can one give a complete characterization of them?

来源字段：https://doi.org/10.4171/owr/2011/08

13548 | OWR-702-003

### Facet Preservation in Recursive Hoffman-Cut Decompositions

O / open | Oberwolfach 报告问题汇编

In the recursive decomposition of a non-facet Hoffman cut into Hoffman cuts on smaller node sets, every resulting cut should already define a facet.

来源字段：https://doi.org/10.4171/owr/2004/19

13549 | OWR-13355-005

### Failure Rates in Sequential Aperiodic Tiling

O / open | Oberwolfach 报告问题汇编

For a finite aperiodic tile set, determine how often a finite patch that cannot extend to a tiling of the whole plane causes an attempt to assemble a tiling one tile at a time to fail.

来源字段：https://doi.org/10.4171/owr/2014/53

13550 | OWR-16633-023

### Fast Submodular Minimization for Represented Matroid Ranks

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Represented-Matroid Minimization Problem [OWR-16633-024]

Let $r$ be the rank function of a matroid on ground set $V$, supplied by an explicit linear representation over a field, and let $z:V\to\mathbb R$. Can

$$\min_{S\subseteq V}\bigl(r(S)-z(S)\bigr)$$

be computed more efficiently by exploiting the representation?

来源字段：https://doi.org/10.4171/owr/2018/50

13551 | AIM-COMBINATORICS-0267

### Favard length and an exact tube formula for the four-corner Cantor set

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.11 (Brought by B. Green, implicit in a paper of Solomyak and Peres).** Give an estimate for the size of the \(\delta\)-thickened Kahane Besicovich set \(C_4\times C_4\), where \[ C_4=\left\{\sum_{n=0}^{\infty}\frac{a_n}{4^n}: a_n\in\{0,1\}\right\}. \]

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13552 | AMR-029-0036

### Finding kernels in special digraphs

O / open | AMR 问题清单汇编

In which classes of digraphs can we decide if a kernel exists and find one in polynomial time?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13553 | OWR-4425-007

### Fine-Wilf Bounds for Antimorphic Periods

O / open | Oberwolfach 报告问题汇编

Let $\theta$ be an antimorphic involution on $A^*$ and let $p$ and $q$ be alternating $\theta$-periods of $w$. Is it true that $|w|\geq p+q-\gcd(p,q)$ implies that $\gcd(p,q)$ is a $\theta$-period of $w$?

来源字段：https://doi.org/10.4171/owr/2010/37

13554 | AIM-COMBINATORICS-0220

### Finite Beatty blocks and balanced gap words

O / open | AIM 研讨会问题汇编

Problem 1.14 (R. Graham). Instead of k-term arithmetic progressions, one could consider a more flexible structure, namely weak k-term arithmetic progressions, which are sets of the form

{b nα + βc: 1 ≤ n ≤ k} (for some α ≥ 1 and β).

Since there are substantially more weak k-term arithmetic progressions than k-term arithmetic progressions, some of the standard problems and results might be easier to attack.

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13555 | AIM-COMBINATORICS-0106

### Finite certificates and a small-area obstruction for symmetric hexagonal polyomino tilings

P / partially_solved | AIM 研讨会问题汇编

(16) Wesley Pegden: "Are all the ways to hexagonally tile the plane with polyominoes that are 90 degrees rotationally symmetric given by the tilings (up to dilations, say) considered in my recent paper with Lionel Levine and Charles Smart [23] on the Apollonian circle-packing patterns that emerge in the scaling limit of the sandpile model? We believe this conjecture (and in fact state a stronger conjecture in the paper)."

**Conjecture 1.4 [LPS].** If $\mathcal T$ is a primitive, regular, hexagonal tiling of $\mathbb Z^2$ by $90$-degree symmetric tiles, then $\mathcal T=T_C+L_C+v$ for some circle $C$ in the band Apollonian packing $\mathcal B$ and some $v\in\mathbb Z^2$.

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13556 | AIM-COMBINATORICS-0179

### Finite-boundary transfer for infinite second neighborhoods

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.11.** Is Seymour's Second Neighborhood true for locally finite digraphs? What if we just require the outdegrees to be finite?

Any simple digraph with no loops or digons has a vertex \(v\) with \(\lvert N_2^+(v)\rvert\geq \lvert N^+(v)\rvert\).

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13557 | AIM-COMBINATORICS-0247

### Finite-field point-line incidences and a projective no-wrap criterion

P / partially_solved | AIM 研讨会问题汇编

Find an analogue of the Szemerédi--Trotter theorem for \(\mathbb F_p\times\mathbb F_p\). More precisely, suppose that there are \(n\) points and \(l\) lines in \(\mathbb F_p^2\). Does the number \(i\) of point--line incidences necessarily satisfy \[ i\ll (nl)^{2/3}+n+l? \] In particular, if both \(n\) and \(l\) are about \(\log p\), is \(i=O((nl)^{2/3})\)?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13558 | AIM-COMBINATORICS-0065

### Fixed predicates, bounded complexity, and a universal parameter lift

P / partially_solved | AIM 研讨会问题汇编

**Problem (Conlon--Fox--Pach--Sudakov--Suk).** Is \(R_f^{(3)}(4,n)\) polynomial in \(n\)?

The preceding source records supply the intended convention. A single polynomial \[ f:(\mathbb R^2)^3\longrightarrow \mathbb R \] is fixed. For each ordered finite set \(P=(v_1,\ldots,v_N)\subset\mathbb R^2\), form the ordered 3-uniform hypergraph \(H_f(P)\) in which, for \(i<j<k\), \[ \{v_i,v_j,v_k\}\in E(H_f(P)) \quad\Longleftrightarrow\quad f(v_i,v_j,v_k)\geq 0. \] Then \(R_f^{(3)}(4,n)\) is the least \(N\) such that every such ordered \(N\)-point set gives either a 4-clique or an independent set of size \(n\). Thus the literal question fixes all of the following across all \(n\): the plane as ambient space, one polynomial, its coefficients, and the edge rule \(f\geq0\).

This is not the same quantifier order as the now-standard bounded-complexity number \(R_r^{\mathbf t}(s,n)\), where \(\mathbf t=(d,D,m)\) is fixed but the \(m\) defining polynomials and the Boolean function of their sign pattern may vary with the instance and hence with \(n\).

The source also records a connection between \(R_f^{(4)}(5,n)\) and the Erdős--Szekeres convex-polygon problem. Section 7 below makes that connection exact under the usual general-position convention.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13559 | AIM-COMBINATORICS-0129

### Fixed-graph recognition and a rotation certificate for maximal harmonic actions

P / partially_solved | AIM 研讨会问题汇编

Gregg Musiker: “Can we characterize which genus \(g\) graphs have a maximal harmonic group (in the sense of Corry [15]) of size \(6(g-1)\)?” Scott Corry notes that a maximal graph group can be used to construct a graph by a Cayley-graph variation; Musiker asks for the converse, fixed-graph question: “But if I hand you a graph, can you tell me if it has a group of size \(6(g-1)\) that acts harmonically on it?”

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13560 | OWR-11695867-006

### Fixed-Occurrence Permutation Pattern Asymptotics

O / open | Oberwolfach 报告问题汇编

For permutations with exactly $r$ occurrences of a fixed pattern, is $\psi_r(n)\sim C_r n^r\psi_0(n)$, with the conjectured algebraic form for the generating function $\Psi_r(x)$?

来源字段：https://doi.org/10.4171/owr/2022/57

13561 | AIM-COMBINATORICS-0063

### Fixed-polynomial semi-algebraic Ramsey bounds and unbounded-degree universality

P / partially_solved | AIM 研讨会问题汇编

The source asks the following question, attributed to Andrew Suk. For each \(n\), let \(H_n\) be a \(k\)-uniform hypergraph whose vertices are labelled points \[ P=(v_1,\ldots,v_n)\in(\mathbb R^2)^n, \] and, for \(i_1<\cdots<i_k\), put \[ \{v_{i_1},\ldots,v_{i_k}\}\in E(H_n) \quad\Longleftrightarrow\quad f(v_{i_1},\ldots,v_{i_k})\geq 0, \] where \(f\) is one polynomial in the \(2k\) coordinates. What order of magnitude can be guaranteed for the largest clique or independent set? The source denotes the worst guarantee by \[ r_f(n)=\min_{H_n}\max\{\alpha(H_n),\omega(H_n)\}. \]

There are two quantifier issues in that notation.

1. I interpret \(f\) as fixed while \(n\) and the point sequence vary. More explicitly, for an ordered sequence \(P\) of distinct points let \(H_f(P)\) be the hypergraph above, and define \[ r_f(n)=\min_{P\in(\mathbb R^2)^n\text{ distinct}} \max\{\alpha(H_f(P)),\omega(H_f(P))\}. \] This is the reading for which the subscript \(f\) is meaningful. 2. The relation need not be invariant under permutations of its \(k\) arguments. Thus the labels, or equivalently the linear order on the vertices, are part of the structure. This agrees with the ordered-set formulation in Adibelli--Tomon. If the intended object was an unordered geometric relation, one should additionally assume that the sign of \(f\) is permutation-invariant on distinct inputs.

Define the associated Ramsey threshold \[ R_f(m)=\min\{N:\text{ every ordered }N\text{-point configuration has an }f\text{-homogeneous }m\text{-set}\}. \] Then, with integer endpoints understood, \[ r_f(N)\geq m\quad\Longleftrightarrow\quad R_f(m)\leq N. \tag{1} \] This elementary inverse relation is useful because the literature is mostly stated in terms of \(R_f\).

The source records the ordinary hypergraph-Ramsey guarantee \(c\log^{(k-1)}n\), where \(\log^{(j)}\) denotes \(j\)-fold iteration, not a power. Below, all logarithms are base two; changing the base only changes constants.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13562 | AMR-030-0044

### For any poset P, define ex(n,P) to be the size of the largest subset of the Boolean lattice B_(n) which includes no (inj

P / partially_solved | AMR 问题清单汇编

Griggs, Lu: For any poset P, define ex(n,P) to be the size of the largest subset of the Boolean lattice B_(n) which includes no (injective) copy of P as a (not-necessarily induced) subposet. Show that lim_(n→∞) ex(n,P) n^(1/2) 2^( -n) exists.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13563 | AMR-030-0078

### For each k and every sufficiently large n with k dividing ((n-1) choose (k-1)), there is a universal cycle for the k-sub

S / solved | AMR 问题清单汇编

Chung/Diaconis/Graham: For each k and every sufficiently large n with k dividing ((n-1) choose (k-1)), there is a universal cycle for the k-subsets of an n set.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13564 | AMR-030-0003

### For every non-equilateral triangle T, show that it is possible to color the plane with three colors so that there is no

O / open | AMR 问题清单汇编

Graham: For every non-equilateral triangle T, show that it is possible to color the plane with three colors so that there is no monochromatic (congruent) copy of T.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13565 | OWR-14299904-013

### Forest Analogues of Schubert-Polynomial Properties

O / open | Oberwolfach 报告问题汇编

Find forest counterparts of the structural and positivity properties of Schubert polynomials. In particular, for an indexed forest $F$, give explicit coefficients $a^G_{i,F}$ in $(x_1+\cdots+x_i)P_F=\sum_Ga^G_{i,F}P_G$.

来源字段：https://doi.org/10.4171/owr/2026/2

13566 | AIM-COMBINATORICS-0082

### Four concyclic points and bounded Euclidean Ramsey witnesses

P / partially_solved | AIM 研讨会问题汇编

Conjecture. ($100) Every 4-point subset of a circle is Ramsey.

A rival conjecture:

**Recovered conjecture.** Let \[ X=\{x_1,x_2,x_3,x_4\} \] be four distinct points on a circle of positive radius in a Euclidean plane. For every positive integer \(r\), does there exist \(N\) such that every map \[ \chi:\mathbb R^N\longrightarrow [r] \] is constant on some set \(\{y_1,y_2,y_3,y_4\}\) satisfying \[ \|y_i-y_j\|=\|x_i-x_j\|\qquad(1\le i,j\le4)? \tag{1.1} \]

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13567 | AIM-COMBINATORICS-0091

### Fourier completion of a roots-of-unity hypergraph Laplacian

P / partially_solved | AIM 研讨会问题汇编

The mathematical question is whether chip-firing on graphs has a useful extension to a \(k\)-uniform hypergraph equipped with a cyclic order on every edge, with the \(k\)th roots of unity replacing the signs \(+1,-1\). Spencer Backman reported a complex positive-semidefinite Laplacian, independence from an auxiliary total order on the vertices, and asked for its kernel and for analogues of nonnegative, reduced, and recurrent configurations. Andrea Sportiello suggested replacing roots of unity by arbitrary edge vectors whose coordinates sum to zero.

1. The isolated “1” after “workshop's website” is a footnote marker; the footnote gives `aimath.org/WWN/chipfiring/`. 2. The last displayed phrase in the PDF is “between \(\omega\) and \(\omega^{n-1}\).” The problem otherwise uses \(k\), not \(n\). Thus \(n\) is undefined there and may be a notational slip. It is not silently changed below.

The old workshop landing page remains accessible, but its participant-list endpoint (which was supposed to contain Backman's statement of interest) was not recoverable in this run. Consequently, the exact unpublished firing convention could not be verified. It realizes exactly the advertised positivity and order-independence, specializes to the ordinary graph Laplacian when \(k=2\), and is the same general operator later studied by Mulas and Reff. Claims about Backman's unpublished setup are therefore phrased conditionally on this reconstruction.

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13568 | OWR-734-008

### Fractional Helly Theorems for Bounded HVC Dimension

O / open | Oberwolfach 报告问题汇编

Let $\mathcal K$ be a strongly hereditary class of simplicial complexes such that every $n$-vertex $K\in\mathcal K$ satisfies $\sum_i b_i(K)<An^d$. Must $\mathcal K$ have the fractional Helly property of order $d$: for every $t>0$, does there exist $s(t)>0$ such that $f_d(K)\geq t\binom{n}{d+1}$ implies $\dim K\geq s(t)n$?

来源字段：https://doi.org/10.4171/owr/2004/32

13569 | OWR-4425-014

### Fragile and Robust Squarefree Infinite Words

P / partially_solved | Oberwolfach 报告问题汇编

Do there exist infinite words over $\{0,1,2\}$ that are fragile for squarefreeness, meaning every nonempty finite set of symbol flips destroys squarefreeness, and do there exist such words that are robust, meaning infinitely many disjoint finite flip sets preserve squarefreeness?

来源字段：https://doi.org/10.4171/owr/2010/37

13570 | COMB-005

### Frankl's Union-Closed Sets Conjecture

P / partially_solved | 独立 / 未分配集合

别名：The Union-Closed Sets Conjecture [COMB-013]; Union-Closed Sets Conjecture [COMB-017]; Union-Closed Sets Conjecture [COMB-021]; Frankl's union-closed sets conjecture [OPG-2108]

For every finite union-closed family of sets (other than the empty family), there exists an element that belongs to at least half of the sets.

背景链接：https://arxiv.org/abs/2211.09055

13571 | AIM-COMBINATORICS-0078

### Free-cut decomposition for ordered matchings versus triangles

P / partially_solved | AIM 研讨会问题汇编

Does there exist \(\epsilon>0\) such that \[ r_<(K_3,M)=O(n^{2-\epsilon}) \] for every ordered matching \(M\) on \(n\) vertices?

The source also records the uniform bound \(r_<(K_3,M)\le r_<(K_3,K_n)=\Theta(n^2/\log n)\) and an existential lower bound with exponent \(4/3+o(1)\).

Here and in the cited primary papers, \(n\) is the **number of vertices** of the ordered matching, not the number of matching edges. Thus a perfect matching has \(n/2\) edges and \(n\) is even. This matters when comparing the nested matching \(NM_q\), which is conventionally indexed by its \(q\) edges and has \(2q\) vertices.

There are competing argument/color conventions in the literature. To remove the ambiguity, throughout this report define \[ R(M):=\min\{N:\text{every red/blue coloring of }K_N^< \text{ has a red }K_3\text{ or a blue ordered }M\}. \] This is the quantity written \(r_<(K_3,M)\) in the AIM question. Papers that write \(r_<(M,K_3)\) with “blue \(M\) or red \(K_3\)” are studying exactly the same quantity. More generally, swapping both arguments and color names does not change an off-diagonal Ramsey number.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13572 | AIM-COMBINATORICS-0181

### Frontier Hall obstructions for exact distance shells

P / partially_solved | AIM 研讨会问题汇编

Conjecture 6.13. Any simple digraph with no directed cycles of length at most k has a vertex v such that \(|N^+_{>k}(v)|\geq |N^+_{>k-1}(v)|\).

**Conjecture 6.13 (Burckel, recovered).** Let \(k\geq2\). Any finite simple digraph with no directed cycle of length at most \(k\) has a vertex \(v\) such that \[ |N_k^+(v)|\geq |N_{k-1}^+(v)|. \tag{1} \]

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13573 | OWR-16763-030

### Full Rainbow Matchings in Densely Colored Bipartite Multigraphs

P / partially_solved | Oberwolfach 报告问题汇编

别名：Composite Duplicate of Two Rainbow-Matching Conjectures [OWR-16763-028]; Aharoni-Berger Rainbow Matching Conjecture [OWR-9790352-023]

If a bipartite multigraph is properly edge-colored with $n$ colors and has at least $n+1$ edges of each color, must it have a rainbow matching using every color?

来源字段：https://doi.org/10.4171/owr/2019/1

13574 | OWR-4791-015

### Functional Inequality on the Boolean Cube

O / open | Oberwolfach 报告问题汇编

别名：Functional Inequality on the Boolean Cube [OWR-4791-016]

For every nonempty $A\subseteq\{0,1\}^n$ and every $f:\{0,1\}^n\to\mathbb{R}$, conjecture $E(f,f)+4\sum_x f(x)^2\geq2\left(\frac{\log_2(2^n/|A|)}{|A|}\sum_{x\in A}|f(x)|\right)^2$, where $E(f,f)=\sum_{x\sim y}(f(x)-f(y))^2$.

来源字段：https://doi.org/10.4171/owr/2011/01

13575 | OWR-4798-025

### Gamma Vectors of Flag Homology Spheres

O / open | Oberwolfach 报告问题汇编

What are the possible face numbers of flag simplicial spheres? In particular, if $\Delta$ is a flag homology sphere, is its $\gamma$-vector nonnegative, and is it the $f$-vector of a balanced simplicial complex?

来源字段：https://doi.org/10.4171/owr/2011/08

13576 | OWR-16164-008

### Generalized Alternating-Sign Matrices and Shifted Plane Partitions

O / open | Oberwolfach 报告问题汇编

Find a natural definition of $(n,k)$-alternating-sign matrices such that $(n-1,2)$-alternating-sign matrices coincide with, or are in a simple bijection with, ordinary $n$-alternating-sign matrices, and such that their number equals the number of $(n,k)$ column-strict shifted plane partitions.

来源字段：https://doi.org/10.4171/owr/2018/23

13577 | OWR-11695867-015

### Generating Functions for Parity Restricted 321 Avoiding Permutations

O / open | Oberwolfach 报告问题汇编

Is the generating function $\sum_{n\geq 1}P_{321}(n)q^n$ algebraic, D-finite, or D-algebraic, and what are the asymptotics of $P_{321}(n)$?

来源字段：https://doi.org/10.4171/owr/2022/57

13578 | OWR-17135-002

### Generating Gross Substitutes Valuations from Matroid Ranks

P / partially_solved | Oberwolfach 报告问题汇编

For every $n$, does there exist a finite $m\geq n$ such that every gross-substitutes valuation on $[n]$ can be generated from weighted matroid-rank valuations on $[m]$?

来源字段：https://doi.org/10.4171/owr/2019/39

13579 | OWR-12867-006

### Generating Series for Immersed Plane Curves

O / open | Oberwolfach 报告问题汇编

Let $N(g)$ be the number, up to isotopy, of immersed curves in $\mathbb R^2$ with only transverse self-intersections and exactly $g$ such intersections. Find a combinatorial or geometric interpretation of the generating series $\sum_{g\geq 0}N(g)z^g$.

来源字段：https://doi.org/10.4171/owr/2014/07

13580 | OWR-14298158-012

### Generating-Function Field for Separable Permutation Subclasses

O / open | Oberwolfach 报告问题汇编

Does every proper subclass of the separable permutations have a generating function in $$\mathbb Q\!\left(z,\sqrt{1-4z},\sqrt{1-6z+5z^2}\right)?$$ Determine whether this field contains all such generating functions.

来源字段：https://doi.org/10.4171/owr/2024/6

13581 | OWR-14298007-007

### Generic Anisotropy of Simplicial Spheres

P / partially_solved | Oberwolfach 报告问题汇编

Is every simplicial sphere generically anisotropic over every field?

来源字段：https://doi.org/10.4171/owr/2023/58

13582 | AMR-029-0037

### Generic global rigidity in three dimensions

P / partially_solved | AMR 问题清单汇编

Decide whether a graph is globally rigid in three-dimensional space.

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13583 | AMR-029-0038

### Generic rigidity in three dimensions

P / partially_solved | AMR 问题清单汇编

Can we decide in polynomial time whether a given graph is rigid in 3-dimensional space?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13584 | AMR-030-0032

### Given a permutation σ, what is the maximum number of copies of σ that a permutation on n symbols may contain

O / open | AMR 问题清单汇编

Given a permutation σ, what is the maximum number of copies of σ that a permutation on n symbols may contain?

背景链接：https://people.math.sc.edu/cooper/combprob.html

13585 | AMR-030-0033

### Given two permutations σ and τ, what is the expected number of copies of σ in a permutation chosen uniformly at random f

P / partially_solved | AMR 问题清单汇编

: Given two permutations σ and τ, what is the expected number of copies of σ in a permutation chosen uniformly at random from those permutations on n symbols which avoid τ? Newsflash! Miklós Bóna has a nice paper addressing this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13586 | AMR-029-0039

### Goddyn's conjecture on thin spanning trees

P / partially_solved | AMR 问题清单汇编

A spanning tree of a graph G is called $\epsilon$-thin if it contains at most an $\epsilon$ fraction of the edges of each cut. Is there a function $f: (0,1)\to {\mathbb Z}_+$ such that every $f(\epsilon)$-edge-connected graph has an $\epsilon$-thin spanning tree?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13587 | AMR-029-0040

### Gonality and edge subdivisions

O / open | AMR 问题清单汇编

How does gonality change if each edge of the graph is subdivided $k$ times?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13588 | AIM-COMBINATORICS-0027

### Good Freiman models from polynomial coset covers

P / partially_solved | AIM 研讨会问题汇编

If \(A\) is a finite subset of an abelian group \(G\) and

\[ |A+A|\leq K|A|, \]

must there be \(A'\subseteq A\), with \(|A'|\geq K^{-O(1)}|A|\), and a finite group \(H\), with \(|H|\leq K^{O(1)}|A|\), such that \(A'\) is Freiman-isomorphic to a subset of \(H\)? Here a Freiman isomorphism means a map \(\phi\) such that, for every \(a,b,c,d\) in its domain,

\[ a+b=c+d \quad\Longleftrightarrow\quad \phi(a)+\phi(b)=\phi(c)+\phi(d). \]

The 2018 workshop report states the target slightly more strongly as \(|H|\leq C K^C|A'|\), with an absolute \(C\). The results below keep these normalizations separate and, in the bounded-torsion cases, satisfy both.

The input sentence “this is false for arbitrary groups” is read as saying that the preceding *whole-set* polynomial modeling assertion is false in that generality; it does not answer the subsequent large-subset question.

来源字段：http://aimpl.org/addcombapp/2/

13589 | GREEN-028

### Gowers Box Norms over Finite Fields

O / open | Ben Green 问题

Let $p$ be an odd prime and suppose $f : \mathbb{F}_p^n \times \mathbb{F}_p^n \to \mathbb{C}$ is bounded pointwise by 1. Suppose $\mathbb{E}_h \|\Delta_{(h,h)}f\|_\square^4 \geq \delta$. Does $f$ correlate with a function of the form $a(x)b(y)c(x+y)(-1)^{q(x,y)}$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13590 | OWR-11695867-003

### Graded Analogues of Parking-Space Enumeration

O / open | Oberwolfach 报告问题汇编

Find a graded, or $q$-, analogue of the parking-space enumeration theorem, for example via Cherednik-algebra shift functors, Lie-theoretic $q$-Kreweras numbers, or the freeness conjecture.

来源字段：https://doi.org/10.4171/owr/2022/57

13591 | OWR-1455-004

### Grading of Twisted-Involution Bruhat Posets

O / open | Oberwolfach 报告问题汇编

Let $(W,S)$ be a Coxeter system, let $\theta$ be an involutive automorphism of $W$, and let $\mathcal{I}(\theta)=\{w\in W:\theta(w)=w^{-1}\}$ be the set of twisted involutions. For which $(W,S)$ and $\theta$ is the induced Bruhat-order subposet $\operatorname{Br}(\mathcal{I}(\theta))$ graded?

来源字段：https://doi.org/10.4171/owr/2007/04

13592 | AIM-COMBINATORICS-0216

### Graham's reciprocal-mass square conjecture: a dyadic extremal reduction

P / partially_solved | AIM 研讨会问题汇编

Let \(A\subseteq\mathbb Z\times\mathbb Z\) satisfy \[ \sum_{(x,y)\in A}\frac1{x^2+y^2}=\infty. \] Conjecture (\(\$1000\)): \(A\) contains the four vertices of a square, i.e. four points of the form \((x,y),(x+a,y),(x,y+a),(x+a,y+a)\). More generally, it should be true that \(A\) contains a \(k\times k\) square grid.

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13593 | OWR-12872-018

### Growth Rate of Quadrant Loops

O / open | Oberwolfach 报告问题汇编

Let $Q_n(a)$ count length-$2n$ square-lattice loops confined to the nonnegative quadrant, weighted by $a$ for each northwest or east-south corner, and let $Q(a,u)=\sum_{n\geq 0}Q_n(a)u^n$. For every $a\geq -1$, is the exponential growth rate of $Q_n(a)$ equal to that of the corresponding unconfined loops?

来源字段：https://doi.org/10.4171/owr/2014/12

13594 | GAME-009

### Grundy's Game Periodicity

O / open | 独立 / 未分配集合

Is the nim-sequence of Grundy's game eventually periodic?

背景链接：https://wwwhomes.uni-bielefeld.de/achim/grundy.html

13595 | AIM-COMBINATORICS-0068

### Hall ratio three: current coloring bound and a nested critical-core certificate

P / partially_solved | AIM 研讨会问题汇编

Is there an $\epsilon>0$ such that $\chi(G)\le d^{1-\epsilon}$ for every graph $G$ of maximum degree $d$ such that $\alpha(H)\ge |V(H)|/3$ for every subgraph $H\subseteq G$?

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13596 | GREEN-082

### Hamming Ball Covering Growth

P / partially_solved | Ben Green 问题

Let $r$ be fixed and let $H(r)$ be the Hamming ball of radius $r$ in $\mathbb{F}_2^n$. Let $f(r)$ be the smallest constant such that there exist infinitely many $n$ with subspaces $V_n \leq \mathbb{F}_2^n$ satisfying $V_n + H(r) = \mathbb{F}_2^n$ and $|V_n| = (f(r) + o(1)) \frac{2^n}{|H(r)|}$. Does $f(r) \to \infty$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13597 | AMR-029-0041

### Head-disjoint strongly connected orientations

O / open | AMR 问题清单汇编

An orientation of a hypergraph is a directed hypergraph obtained by choosing a single head-node in each hyperedge. We call a set of orientations of a hypergraph head-disjoint if none of the hyperedges have the same head-node in two of them. When does a 3-uniform hypergraph admit three head-disjoint strongly connected orientations?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13598 | OWR-14299905-014

### Height Bounds for Ideals in Finite Posets

O / open | Oberwolfach 报告问题汇编

For every ideal $A$ of a finite poset, is

$$\max_{x\in A}h(x)\geq |A|-|\max(A)|+1?$$

More generally, is $|A|-\max_{x\in A}h(x)$ bounded by a function of $|\max(A)|$ alone?

来源字段：https://doi.org/10.4171/owr/2026/1

13599 | OWR-12481-012

### Hierarchical Refinement of Symmetric-Group Coset Partitions

P / partially_solved | Oberwolfach 报告问题汇编

If a $t$-coset is a coset of the stabilizer of $t$ points in the symmetric group $S_n$, is every partition of $S_n$ into $t$-cosets a refinement of a partition into $(t-1)$-cosets?

来源字段：https://doi.org/10.4171/owr/2013/18

13600 | AIM-COMBINATORICS-0033

### Higher additive energy and quantitative stability of the Holder defect

P / partially_solved | AIM 研讨会问题汇编

**Recovered conjecture.** For every integer \(n\ge2\) and every \(0<\varepsilon<1\), there exist \(c=c(n,\varepsilon)>0\) and \(c'=c'(n,\varepsilon)>0\) such that the following holds uniformly for every finite abelian group \(G\), every nonempty \(A\subseteq G\), and every \(K>1\). If \[ \rho_{2n+2}(A)^{(n-1)/n}\le K\rho_{2n}(A), \] then there is a subset \(H\subseteq(n-1)A\) such that \[ |H|\ge \rho_{2n}(A)^{1/(n-1)}K^{-c}, \qquad E_4(H)\ge \frac{|H|^3}{|A|^\varepsilon K^{c'}}. \]

来源字段：http://aimpl.org/addcombapp/2/

13601 | AIM-COMBINATORICS-0107

### Higher-dimensional critical configurations and an orientation obstruction

P / partially_solved | AIM 研讨会问题汇编

What are higher dimensional (in the sense of [18] and [17]) critical configurations?" Sam Hopkins asks whether the desired objects should specifically generalize stable recurrent configurations, or also parking functions and superstable elements. Klivans replies: "We would like to generalize any of these notions from the graph case to the simplicial complex case.

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13602 | OWR-2090-006

### Higher-Dimensional Enumeration and Tensor-Rank Problems

P / partially_solved | Oberwolfach 报告问题汇编

Improve the asymptotic bounds for the number $L_n$ of $n\times n$ Latin squares. Formulate and solve higher-dimensional analogues of this enumeration problem. Are there higher-dimensional analogues of the gamma function? What is the maximum rank of a real $n\times n\times n$ tensor, and is it $(1+o(1))n^2/2$?

来源字段：https://doi.org/10.4171/owr/2008/44

13603 | AMR-029-0042

### Highly element-connected orientation

O / open | AMR 问题清单汇编

Is it true that if an undirected graph G with terminal set T is 2k-element-connected, then it has a k-element-connected orientation?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13604 | OWR-14299910-004

### Homological Interpretation of Semidistributive Rowmotion

O / open | Oberwolfach 报告问题汇编

For a finite semidistributive lattice $L$, find a homological interpretation of its rowmotion bijection extending the Auslander-Reiten interpretation available for distributive lattices.

来源字段：https://doi.org/10.4171/owr/2026/6

13605 | OWR-822-005

### Homology-Sphere Factors in Simplicial Decompositions

O / open | Oberwolfach 报告问题汇编

Let $K$ be a $d$-dimensional doubly Cohen-Macaulay simplicial complex over a field, with $d>1$. Can $K$ be decomposed as $K=\bigcup_{i=1}^m S_i$, where every $S_i$ is a homology sphere and, after choosing any $S_i$ first, each subsequent $S_i$ intersects the preceding union in a $d$-face?

来源字段：https://doi.org/10.4171/owr/2005/17

13606 | AMR-030-0001

### How many colors is it necessary to use so that, if you paint every single point of the two-dimensional plane some color

P / partially_solved | AMR 问题清单汇编

Erdős: How many colors is it necessary to use so that, if you paint every single point of the two-dimensional plane some color, no two points which are a distance one from each other are the same color? (That is, what is the chromatic number of the unit distance graph in the plane?) It's not hard to show that the number is between 4 and 7 -- but nobody has a clue where it falls in between. Recently, De Grey showed it's at least 5. See this. A related problem, due to Chris Dillard: Consider B_(r), the Ball of radius r about 0 in R^(2). What is the chromatic number of the unit distance graph on B_(r)? We know that, for r < 1/2, it is 1; for r = 1/2, it is 2; for 1/2 < r <= sqrt(3)/3, it's 3. How about for r > sqrt(3)/3? At some r_(0), the chromatic number must become 4. It is easy to see that r_(0 )is no more than 3/sqrt(11) : see the figure below. In fact, it is possible to improve this to sqrt(3)/2, but that still leaves a sizable gap.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13607 | AMR-030-0041

### How many comparisons are needed to determine a linear order of the Boolean poset

O / open | AMR 问题清单汇编

Fishburn, Pekec, Reeds: How many comparisons are needed to determine a linear order of the Boolean poset? That is, what is the fewest number of questions of the form "Is S < T?" must one ask in order to find out a linear ordering of all subsets of an n-set? Conjecture: n-1.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13608 | AMR-030-0010

### How many congruent regular tetrahedra can touch at a point

O / open | AMR 问题清单汇编

How many congruent regular tetrahedra can touch at a point? Easy to show it's at least 20, and at most 22. Apparently, this has been open a long time.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13609 | AMR-030-0087

### If A is an invertible n x n matrix, is there always an n x n submatrix B of [A A] so that perm(B) is nonzero

O / open | AMR 问题清单汇编

Kahn: If A is an invertible n x n matrix, is there always an n x n submatrix B of [A A] so that perm(B) is nonzero. The notation perm(B) means the permanent of B, which is the "determinant without the signs". This implies Jaeger's conjecture above. See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13610 | AIM-COMBINATORICS-0016

### Incidence-faithful projection and a fixed-axis obstruction for Kupitz discrepancy

P / partially_solved | AIM 研讨会问题汇编

**Kupitz conjecture.** Given \(n\) points in \(\mathbb R^2\), is there always a line through at least two points such that the numbers of points on the two sides differ by \(O(1)\)?

Given \(n\) points in \(\mathbb R^3\), is there always a plane through at least two points such that the numbers of points on the two sides differ by \(O(1)\)?

来源字段：http://aimpl.org/highdimdiscrete/4/

13611 | AMR-029-0043

### Incomplete splitting-off in digraphs

O / open | AMR 问题清单汇编

Given a digraph D=(V+s,A) which is k-arc-connected in V, what is the maximum number of (disjoint) pairs of arcs, consisting of entering and leaving arcs at $s$, whose (simultaneous) splitting off does not destroy the k-arc-connectivity of D in V?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13612 | AIM-COMBINATORICS-0076

### Increasing paths and two certificates for the Chung-Graham sum question

P / partially_solved | AIM 研讨会问题汇编

The AIM record, attributed to Ron Graham, gives every edge of a graph a distinct label. For the complete graph it asks whether there is an absolute constant \(c>0\) such that \[ \max_{v\in V(K_n)}t(v)\ge cn, \] where \(t(v)\) is the maximum length of an increasing path starting at \(v\). It then asks, for every edge-ordered graph \(G\), whether \[ \sum_{v\in V(G)}t(v)\ge |E(G)|. \tag{1.1} \]

The conventions are recoverable from the primary literature. A path is **simple**: its vertices are distinct. Its length is its number of edges. An increasing path \(v_0v_1\cdots v_\ell\) satisfies \[ \lambda(v_0v_1)<\lambda(v_1v_2)<\cdots<\lambda(v_{\ell-1}v_\ell). \] A trail may revisit vertices and is treated as a different, easier problem. This distinction is explicit in Milans and in Bucić--Kwan--Pokrovskiy--Sudakov--Tran--Wagner. Isolated vertices have \(t(v)=0\).

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13613 | OWR-13490-020

### Independence Complexes and Alternating-Sign Matrices

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be a graph whose independence complex has one cohomology generator, let $e$ be an edge, and form $G_k$ by replacing $e$ with a path of $3k+1$ edges. Find a closed formula and a natural generating function for the largest weight in a cohomology representative of $G_k$. Also, find independence complexes that realize Kuperberg's symmetry classes of alternating-sign matrices.

来源字段：https://doi.org/10.4171/owr/2015/5

13614 | AMR-029-0044

### Independent arborescences in acyclic digraphs

O / open | AMR 问题清单汇编

Let D=(V,A) be an acyclic digraph with designated root-nodes $r_1,...,r_k\in V$. Let $U_1,...,U_k$ be convex node sets with $r_i\in U_i$. Is it true that there exist independent $r_i$-arborescences $F_i$ with $V(F_i)=U_i$ if and only if there exist openly node-disjoint $r_i$-v$ paths $P_i$ ($v\in U_i$) for each $v\in V$? (We call the $r_i$-v$ and $r_j$-v$ paths $P_i$ and $P_j$ openly node-disjoint if they are edge-disjoint and $V(P_i)\cap V(P_j)=\{v\}\cup(\{r_i\}\cap\{r_j\})$.)

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13615 | AMR-029-0045

### Independent trees

P / partially_solved | AMR 问题清单汇编

In a graph G=(V,E) with a root node r, two spanning trees $T_1$ and $T_2$ are called r-independent if for any node x in V-r, the unique paths between r and x in $T_1$ and $T_2$ are internally node disjoint. Is it true that if G is k-connected then there exist k r-independent trees for arbitrary r?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13616 | OWR-4425-006

### Independent Word Equations with Nonperiodic Solutions

O / open | Oberwolfach 报告问题汇编

For word equations over a free monoid, does there exist an independent system of three equations in three unknowns that has a nonperiodic solution? Here independent means that no proper subsystem has the same solution set.

来源字段：https://doi.org/10.4171/owr/2010/37

13617 | AIM-COMBINATORICS-0012

### Induced packing versus dynamic editing for C5

P / partially_solved | AIM 研讨会问题汇编

Let \(H\) be a fixed graph, \(G\) an \(n\)-vertex graph. If the minimum number of edges needed to add or remove from \(G\) in order to remove all induced copies of \(H\) is \(o(n^2)\), then the regularity lemma implies that the maximum number of induced copies of \(H\), with each pair intersecting in at most one vertex, is \(o(n^2)\).

Can we get better bounds, even for the case \(H=C_5\)?

来源字段：http://aimpl.org/highdimdiscrete/3/

13618 | AMR-029-0046

### Infinite Lucchesi-Younger

O / open | AMR 问题清单汇编

For a digraph $D=(V,A)$, we call a nonempty $C\subseteq A$ a dicut if there is some $X\subseteq V$ such that no edge enters $X$ and $C$ consists of the outgoing edges of $X$. The conjecture states that for any $D$, there exists a system $\mathcal{C}$ of pairwise disjoint finite dicuts and an edge set $F\subseteq A$ consisting of one edge from each element of $\mathcal{C}$, such that $F$ intersects every finite dicut of $D$.

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13619 | OWR-1183-011

### Infinite Transversals from Points to Spanned Hyperplanes

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Point-to-Hyperplane Matching Conjecture [OWR-1183-012]

For every $d$ and every $P\subseteq\mathbb R^d$ whose affine span is $\mathbb R^d$, let $H$ be the set of hyperplanes spanned by $P$. Does there exist an injective map $f:P\to H$ with $p\in f(p)$ for every $p\in P$?

来源字段：https://doi.org/10.4171/owr/2006/01

13620 | OWR-4425-018

### Infinite Words Avoiding Equal-Length, Equal-Sum Factors

O / open | Oberwolfach 报告问题汇编

别名：Repeated Extraction of the Equal-Sum-Factor Problem [OWR-4425-019]

Does there exist an infinite sequence over a finite subset of $\mathbb{Z}$ with no factor $x_1x_2$ such that $|x_1|=|x_2|$ and the sums of the letters in $x_1$ and $x_2$ are equal?

来源字段：https://doi.org/10.4171/owr/2010/37

13621 | AMR-029-0047

### Integer decomposition of smooth polytopes

P / partially_solved | AMR 问题清单汇编

Is it true that every smooth polytope has the integer decomposition property?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13622 | OWR-741-001

### Integrality of Constrained Polytope Families

O / open | Oberwolfach 报告问题汇编

Fix $r>0$ and let $B$ and $R$ be the integral matrices defining type-$A_r$ hive polytopes. Must every nonempty polytope $\{h\in\mathbb R^{(r+1)(r+2)/2}:Bh=b,\ Rh\leq c\}$ with integral $b$ and $c$ contain an integral point?

来源字段：https://doi.org/10.4171/owr/2004/39

13623 | GREEN-029

### Inverse Theorem for Gowers Norms

P / partially_solved | Ben Green 问题

Determine bounds for the inverse theorem for Gowers norms.

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13624 | OWR-14298158-004

### Inversion Monotonicity for 1324-Avoiding Permutations

P / partially_solved | Oberwolfach 报告问题汇编

Let $\operatorname{Av}_{n,k}(\tau)$ denote length-n permutations avoiding $\tau$ with exactly k inversions. For every n and k, is $|\operatorname{Av}_{n,k}(1324)|\le|\operatorname{Av}_{n+1,k}(1324)|$? Does the analogous inequality hold for $\operatorname{Av}_{n,k}(1324,231)$?

来源字段：https://doi.org/10.4171/owr/2024/6

13625 | OWR-14298158-005

### Inversion Monotonicity in the Domino Subclass

P / partially_solved | Oberwolfach 报告问题汇编

Can the inversion-monotonicity conjecture for $1324$-avoiding permutations be proved on the domino subclass $[\operatorname{Av}(132)\mid\operatorname{Av}(231)]$?

来源字段：https://doi.org/10.4171/owr/2024/6

13626 | AMR-030-0046

### Is enumeration of pressing sequences of bicolored graphs (aka simple pseudographs) #P-hard

O / open | AMR 问题清单汇编

: Is enumeration of pressing sequences of bicolored graphs (aka simple pseudographs) #P-hard? Is there an FPRAS for sampling them?

背景链接：https://people.math.sc.edu/cooper/combprob.html

13627 | AMR-030-0011

### Is every polygonal room in the plane illuminable from some point

O / open | AMR 问题清单汇编

Is every polygonal room in the plane with mirrored walls illuminable from some interior point, allowing light rays to undergo specular reflections at the walls?

背景链接：https://people.math.sc.edu/cooper/combprob.html

13628 | AMR-030-0034

### Is it possible for a permutation on n symbols to contain exactly n

O / open | AMR 问题清单汇编

: Is it possible for a permutation on n symbols to contain exactly n!/(m!^(2)(n - m)!) copies of each permutation on m symbols? (Yes for m=1,2,3. Unknown for m>3.) Do infinitely many such perfectly m-symmetric permutations exist?

背景链接：https://people.math.sc.edu/cooper/combprob.html

13629 | AMR-030-0012

### Is it true that every graph whose vertices have odd degree greater than one contains a cycle of length 2^(n) for some n

P / partially_solved | AMR 问题清单汇编

Erdős-Gyárfás: Is it true that every graph whose vertices have odd degree greater than one contains a cycle of length 2^(n) for some n? This one has kept me (and apparently, lots of others) up many a night trying fruitlessly to construct a counterexample. See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13630 | AMR-030-0031

### Is it true that the sum of the k largest Laplacian eigenvalues of a graph with m edges is at most k(k+1)/2+m

P / partially_solved | AMR 问题清单汇编

Brouwer : Is it true that the sum of the k largest Laplacian eigenvalues of a graph with m edges is at most k(k+1)/2+m?

背景链接：https://people.math.sc.edu/cooper/combprob.html

13631 | AMR-030-0080

### Is it true that, for some k, if all (K-1)-words are encountered by a t-ary word at the same rate as a uniform random t-a

O / open | AMR 问题清单汇编

/Rorabaugh: Is it true that, for some k, if all (K-1)-words are encountered by a t-ary word at the same rate as a uniform random t-ary word, then this is also true for K-words, when K>k? That is, do substitution instance counts exhibit quasirandom threshold-like behavior?

背景链接：https://people.math.sc.edu/cooper/combprob.html

13632 | AMR-030-0047

### Is the 1/3-2/3 Conjecture for Pressing Sequences true

O / open | AMR 问题清单汇编

: Is the 1/3-2/3 Conjecture for Pressing Sequences true? That is, if a graph G is not uniquely pressable, is it true that there much be two vertices x and y so that the probability that x comes before y in a uniform random pressing sequence of G is between 1/3 and 2/3? See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13633 | AMR-030-0086

### Is the exponent of matrix multiplication 2

P / partially_solved | AMR 问题清单汇编

Is the exponent of matrix multiplication 2? In other words, can two n b n matrices be multiplied in O(n^(2+)^(l')) steps? See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13634 | AMR-030-0040

### Is the poset of integer partitions ordered by refinement Sperner

P / partially_solved | AMR 问题清单汇编

Is the poset of integer partitions ordered by refinement Sperner?

背景链接：https://people.math.sc.edu/cooper/combprob.html

13635 | AMR-030-0039

### Is the weak order on S_(n) (the "inversion" poset) Sperner

S / solved | AMR 问题清单汇编

Is the weak order on S_(n) (the "inversion" poset) Sperner?

背景链接：https://people.math.sc.edu/cooper/combprob.html

13636 | AMR-030-0077

### Is there a word which is unavoidable over a k letter alphabet, but not a (k-1) letter alphabet, for each integer k > 1

O / open | AMR 问题清单汇编

For each integer k>1, is there a pattern avoidable over a k-letter alphabet but not avoidable over a (k-1)-letter alphabet (that is, of avoidability index exactly k)?

背景链接：https://people.math.sc.edu/cooper/combprob.html

13637 | OWR-11695867-011

### Joint Genus and Degree Asymptotics for Connected Hurwitz Numbers

O / open | Oberwolfach 报告问题汇编

Determine asymptotic formulas for connected Hurwitz numbers when both genus and degree tend to infinity.

来源字段：https://doi.org/10.4171/owr/2022/57

13638 | AIM-COMBINATORICS-0251

### Joints of lines over finite fields

P / partially_solved | AIM 研讨会问题汇编

Take \(n\) lines in \(\mathbb R^3\). Define a joint to be a point with three lines passing through it that are not coplanar. How many joints can there be?

Let \(\mathcal L\) be a set of \(L\) distinct affine lines in \(\mathbb F_q^3\). A point is a **joint** if it lies on three lines of \(\mathcal L\) whose direction vectors are linearly independent. Bound, and if possible determine, the maximum number \(J\) of joints in terms of \(L\) (and the field size \(q\)).

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13639 | OWR-16763-025

### Kahn’s Matroidal Rota Basis Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Given a rank-$n$ matroid and bases $B_{i,j}$ for $1\leq i,j\leq n$, can representatives $b_{i,j}\in B_{i,j}$ be chosen so that every column $\{b_{1,j},\ldots,b_{n,j}\}$ and every row $\{b_{i,1},\ldots,b_{i,n}\}$ is a basis?

来源字段：https://doi.org/10.4171/owr/2019/1

13640 | OWR-1455-009

### Kalai-Sarkaria Algebraic-Shifting Conjecture

O / open | Oberwolfach 报告问题汇编

Let $K$ be a simplicial complex on $n$ vertices that embeds in the $(d-1)$-sphere. Its symmetric or exterior algebraic shifting should satisfy $\Delta(K)\subseteq\Delta(d,n)$, where $\Delta(d,n)$ is the symmetric shifting of the cyclic $d$-polytope on $n$ vertices.

来源字段：https://doi.org/10.4171/owr/2007/04

13641 | AIM-COMBINATORICS-0034

### Kernel saturation and rank-sensitive Komlos discrepancy

P / partially_solved | AIM 研讨会问题汇编

There is a universal constant \(c>0\) such that for every \(v_1,\ldots,v_n\in \mathbb B_2^d\), there are signs \(\varepsilon_1,\ldots,\varepsilon_n\in\{-1,1\}\) for which \[ \left\|\sum_{i=1}^n\varepsilon_i v_i\right\|_\infty\leq c. \]

Here \(\mathbb B_2^d=\{v\in\mathbb R^d:\|v\|_2\leq1\}\). Equivalently, if \(A=[v_1\ \cdots\ v_n]\in\mathbb R^{d\times n}\) has column Euclidean norms at most one, the conjecture says \[ \operatorname{disc}(A):=\min_{\varepsilon\in\{-1,1\}^n} \|A\varepsilon\|_\infty=O(1), \] with an absolute implied constant independent of \(d,n\), and of \(A\).

The source literature field says that \(O(\sqrt{\log\min\{d,n\}})\) was given by “Banaszczyk'98” and “Naor'05.” Banaszczyk's published 1998 theorem was verified. The record gives no title for the Naor shorthand, and a matching primary 2005 publication was not located, so no bibliographic metadata is invented here.

来源字段：http://aimpl.org/hereddiscrep/1/

13642 | OWR-734-013

### Khovanskii Bounds Beyond Simplicial Polytopes

O / open | Oberwolfach 报告问题汇编

Extend Khovanskii's upper-bound theorem to simplicial spheres, determine the face numbers of a section $P$ from those of $Q$, and obtain analogous bounds for general polytopes.

来源字段：https://doi.org/10.4171/owr/2004/32

13643 | OWR-14299088-011

### Komlos Discrepancy Conjecture and Unsatisfiable Formulas

O / open | Oberwolfach 报告问题汇编

Determine whether discrepancy instances constructed from unsatisfiable Boolean formulas can have lower bound greater than $1+\sqrt{2}$. More broadly, prove the Komlós conjecture that vectors of Euclidean norm at most one admit signs with uniformly bounded coordinate discrepancy.

来源字段：https://doi.org/10.4171/owr/2025/9

13644 | OWR-13490-022

### Labeled Floor Diagrams in Three Dimensions

P / partially_solved | Oberwolfach 报告问题汇编

Determine the combinatorics and enumeration of labeled floor diagrams for tropical curves in three-dimensional projective space.

来源字段：https://doi.org/10.4171/owr/2015/5

13645 | AIM-COMBINATORICS-0230

### Lacunary counterexamples to the asymmetric Freiman container question

P / partially_solved | AIM 研讨会问题汇编

Tao. Let \(A,B\subseteq\mathbb Z\) be finite, with \[ |A|=m>|B|=n, \qquad |A+B|<Km. \] Must there be a generalized arithmetic progression \(P\) of rank \(c(K)\), containing \(B\), and a set \(X\subseteq\mathbb Z\) such that \[ A\subseteq P+X, \qquad |P+X|\le c(K)|A|? \tag{Q} \]

Its missing symbols are \[ |P+P|\le c(K,\varepsilon)m^{\varepsilon}|P|. \tag{R} \] The complete recovered remark says that Plünnecke's inequality gives the desired type of covering if the requirement that \(P\) be a progression is weakened to (R); in that version \(P\) is merely a set with somewhat small sumset. The source also notes that \(m=n\) is the case of Freiman's theorem.

Croot and Lev later restated the question as “Freiman's theorem for distinct set summands,” using a constant \(C=C(K)\) to bound both the rank and \(|X+P|/|A|\). They explicitly note that the comparable-size regime \(|A|\asymp|B|\) follows from Freiman's theorem and Ruzsa covering.

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13646 | GREEN-023

### Large Cosets in Iterated Sumsets

P / partially_solved | Ben Green 问题

Suppose that $A \subset \mathbb{F}_2^n$ has density $\alpha$. Does $10A$ contain a coset of some subspace of dimension at least $n - O(\log(1/\alpha))$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13647 | GREEN-075

### Large Gaps in Dilates

O / open | Ben Green 问题

Let $p$ be a prime and let $A \subset \mathbb{Z}/p\mathbb{Z}$ be a set of size $\sqrt{p}$. Is there a dilate of $A$ with a gap of length $100\sqrt{p}$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13648 | GREEN-001

### Large Sum-Free Sets

S / solved | Ben Green 问题

Let $A$ be a set of $n$ positive integers. Does $A$ contain a sum-free set of size at least $n/3 + \Omega(n)$, where $\Omega(n) \to \infty$ as $n \to \infty$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13649 | OWR-1782-016

### Large Triple-Product-Free Subsets of Finite Groups

O / open | Oberwolfach 报告问题汇编

Do there exist infinitely many groups $G_n$ of order $n$ containing subsets $S_n$ such that $ab^{-1}cd^{-1}ef^{-1}=1$, with $a,b,c,d,e,f\in S_n$, has only trivial solutions and $|S_n|\gg n^{1/3}$?

来源字段：https://doi.org/10.4171/owr/2008/01

13650 | GREEN-024

### Largest Coset in 2A

O / open | Ben Green 问题

Suppose that $A \subset \mathbb{F}_2^n$ has density $\alpha$. What is the largest size of coset guaranteed to be contained in $2A$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13651 | OWR-11695867-001

### Lee-Rains-Warnaar Macdonald Integral Identities

P / partially_solved | Oberwolfach 报告问题汇编

Do the Lee-Rains-Warnaar vanishing-integral identities for Macdonald polynomials hold beyond the known Schur specialization $q=t$?

来源字段：https://doi.org/10.4171/owr/2022/57

13652 | OWR-734-011

### Lefschetz Isomorphisms for Gorenstein* Complexes

S / solved | Oberwolfach 报告问题汇编

别名：Duplicate Lefschetz Property Conjecture [OWR-734-010]

For every $(d-1)$-dimensional Gorenstein* complex $K$, does multiplication by a generic linear form induce isomorphisms $\omega^{d-2i}:H_i(K)\to H_{d-i}(K)$?

来源字段：https://doi.org/10.4171/owr/2004/32

13653 | OWR-2654827-010

### Lefschetz Properties of Tropical Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Can one find other combinatorially interesting families of tropical varieties with the Lefschetz property?

来源字段：https://doi.org/10.4171/owr/2020/31

13654 | OPG-37416

### Length of surreal product

P / partially_solved | Open Problem Garden

Conjecture Every surreal number has a unique sign expansion, i.e. function $f: o\rightarrow \{-, +\}$, where $o$ is some ordinal. This $o$ is the length of given sign expansion and also the birthday of the corresponding surreal number. Let us denote this length of $s$ as $\ell(s)$.

It is easy to prove that

$$\ell(s+t) \leq \ell(s)+\ell(t)$$

What about

$$\ell(s\times t) \leq \ell(s)\times\ell(t)$$?

背景链接：http://www.openproblemgarden.org/op/length_of_surreal_product

13655 | AMR-030-0085

### Let f(p;n,k) = C(n,k) p^(k) (1-p)^(n-k)

O / open | AMR 问题清单汇编

Galvin: Let f(p;n,k) = C(n,k) p^(k) (1-p)^(n-k). If p is not 0, 1/2, or 1, is it possible for f(p;n,k) = f(p;n,l) and f(p;n,k') = f(p;n,l') for distinct k, k', l, l'? This is connected with the question of which permutations can arise as the pattern of independence polynomials of trees.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13656 | AMR-030-0089

### Let G be a bicolored graph, and let H be the graph whose vertices are the valid pressing sequences of G and whose edges

O / open | AMR 问题清单汇编

For a bicoloured graph G, form a graph on its successful pressing sequences, all of common length L. Join two sequences when their longest common subsequence has length at least L-4. Must this graph be connected?

背景链接：https://people.math.sc.edu/cooper/combprob.html

13657 | AMR-030-0088

### Let S_(n) be a subset of 2^(n), interpreted as a family of truth assignments to x_(1)

O / open | AMR 问题清单汇编

: Let S_(n) be a subset of 2^(n), interpreted as a family of truth assignments to x_(1),...,x_(n). Let S_(n)-SAT be the problem of determining satisfiability over the possible assignments in S_(n). Call S_(n) "exponential" if S_(n) > ą^(n) for some ą > 1 and all sufficiently large n. Is S_(n)-SAT NP-Hard? See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13658 | AIM-COMBINATORICS-0044

### Linear Beck--Fiala systems and an incidence-forest reduction

P / partially_solved | AIM 研讨会问题汇编

Consider a matrix where both columns are rows are t-sparse. LLL gives a discrepancy bound of $O(\sqrt{t\log t})$ here. If we assume the stronger property that every pair of rows intersect in at most one element, can we get a $O(\sqrt t)$?

来源字段：http://aimpl.org/hereddiscrep/1/

13659 | GREEN-016

### Linear Equation x + 3y = 2z + 2w

O / open | Ben Green 问题

What is the largest subset of $[N]$ with no solution to $x + 3y = 2z + 2w$ in distinct integers $x, y, z, w$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13660 | OWR-14299905-013

### Linear-Extension Wins and Poset Variance

O / open | Oberwolfach 报告问题汇编

Let $P$ be an $n$-element poset and let $\operatorname{win}(x)$ and $\sigma^2(x)$ denote the win statistic and rank variance of an element $x$. Is $\operatorname{win}(x)=\Omega(\sigma^2(x)/n)$ for every $P$ and $x$?

来源字段：https://doi.org/10.4171/owr/2026/1

13661 | AMR-054-0061

### Lines Tangent to Four Unit Balls

O / open | AMR 问题清单汇编

Given a set of $n$ unit-radius balls in $\mathbb{R}^3$, what is the number of lines that are tangent to four of the balls in the set, and miss all the others? (The balls are not necessarily disjoint.)

背景链接：https://topp.openproblem.net/p61

13662 | AIM-COMBINATORICS-0048

### Link structure and sharp edge bounds for three-edge tight-path-free triple systems

P / partially_solved | AIM 研讨会问题汇编

\[ P_s=\{123,234,345,\ldots,s(s+1)(s+2)\}. \]

Thus the object asked about is the unordered simple 3-uniform hypergraph

\[ P_3=\bigl\{\{1,2,3\},\{2,3,4\},\{3,4,5\}\bigr\}. \]

The word “unordered” matters: the displayed labels specify an isomorphism type, not an order that an embedding must preserve. The problem is to improve

\[ \Omega\!\left(\frac{t^2}{\log t}\right) \le r_3(P_3,K_t^{(3)})\le O(t^2), \]

and in particular to decide whether the upper bound is \(o(t^2)\), perhaps even of order \(t^2/\log t\).

There is a convention shift in the literature. Mubayi--Suk write \(TP_m\) for the unordered tight 3-path on \(m\) vertices, with \(m-2\) edges. Consequently, the AIM object \(P_3\) is their \(TP_5\), while AIM \(P_2\) is their \(TP_4\). Ordered tight-path Ramsey numbers are a different problem and are not used here.

Equivalently, if \(H\) is the hypergraph of red triples in a red/blue coloring of \(K_N^{(3)}\), a blue \(K_t^{(3)}\) is exactly an independent \(t\)-set of \(H\). Hence \(r_3(P_3,K_t^{(3)})\) is the least \(N\) such that every \(N\)-vertex \(P_3\)-free 3-graph has independence number at least \(t\).

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13663 | GREEN-015

### Lipschitz AP-Free Graphs

O / open | Ben Green 问题

Does there exist a Lipschitz function $f : \mathbb{N} \to \mathbb{Z}$ whose graph $\Gamma = \{(n, f(n)) : n \in \mathbb{Z}\} \subset \mathbb{Z}^2$ is free of 3-term progressions?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13664 | AIM-COMBINATORICS-0276

### List coloring and equality structure for even-hole-free graphs

S / solved | AIM 研讨会问题汇编

**Conjecture 3 (Hoàng).** If \(G\) is an even-hole-free graph, then \(\chi(G)\leq2\omega(G)-1\).

来源字段：https://aimath.org/WWN/perfectgraph/perfectgraph.pdf

13665 | AMR-029-0048

### List colouring of two matroids

P / partially_solved | AMR 问题清单汇编

Given some matroids on the same ground set $S$, a colouring of $S$ is called proper if each monochromatic set is independent in each matroid. Let $M_1$ and $M_2$ be two matroids on ground set $S$, and suppose that there is a proper colouring by $k$ colours. Is it true that $M_1$ and $M_2$ can be k-list-coloured, i.e. for arbitrary lists $L_s$ of k colours for each $s \in S$, there is a proper colouring that uses colours from these lists?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13666 | AIM-COMBINATORICS-0015

### Literal counterexamples and exact bisector-defect bookkeeping

S / solved | AIM 研讨会问题汇编

(1) Let $S\subset\mathbb R^2$ with no three points collinear. Then $S$ determines at least $|S|/2$ distinct distances.

(2) Given $n$ points in $\mathbb R^2$ with no three collinear, one can remove $O(1)$ points so that the number of isosceles triangles is at most $1.99\binom n2$.

(3) Given $n$ points in $\mathbb R^2$ with no three collinear, $S$ determines $\Omega(n^2)$ perpendicular bisectors containing at most one point of $S$, unless $S$ is the vertices of a regular polygon together with its center.

来源字段：http://aimpl.org/highdimdiscrete/4/

13667 | AIM-COMBINATORICS-0252

### Literal normalization obstruction and the solved polynomial Freiman-Ruzsa theorem

S / solved | AIM 研讨会问题汇编

**Problem 3.15 (Brought by T. Tao, originally stated by I. Ruzsa).** Choose \(A\subseteq\mathbb F_2^n\) such that \[ |A+A|\le k|A|. \] Does there exist a subspace \(V\subseteq\mathbb F_2^n\) such that \[ |V|<k^c|A|,\qquad |A\cap V|\ge k^{-c}|A|? \]

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13668 | AIM-COMBINATORICS-0196

### Local average outdegree and short directed cycles

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.26 (Shen [32]).** Let \(G\) be a digraph with \(n\) vertices and minimum outdegree at least one. If \[ d_G^+(v)+d_G^+(u)\ge 2r \] for every edge \((u,v)\) in \(G\), then the girth \(g\) of \(G\) is at most \[ \left\lceil\frac nr\right\rceil. \]

Jian Shen, “Short Cycles in Digraphs with Local Average Outdegree at Least Two,” *The Electronic Journal of Combinatorics* **10** (2003), no. 1, Research Paper R26, 11 pages, DOI 10.37236/1719.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13669 | AIM-COMBINATORICS-0180

### Local edge budgets for Thomassé's nonneighbor conjecture

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.12 (Thomassé).** Let \(G\) be a digraph with no directed cycle of length at most three. Then there is a vertex \(v\in V(G)\) with \(\delta^+(v)\) at most the number of non-neighbors of \(v\).

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13670 | AMR-029-0049

### Local edge-connectivity augmentation of a hypergraph with fixed rank

O / open | AMR 问题清单汇编

We are given a hypergraph $G_0=(V,\mathcal{E}_0)$ of rank at most $k$ (where $k$ is fixed, not part of the input) and a symmetric function $r:V\times V\to \mathbb{Z}_+$. Can we find in polynomial time a graph $G=(V,{E})$ with a minimum number of edges such that $\lambda_{G_0+G}(x,y)\ge r(x,y)$ for every $x,y\in V$?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13671 | AIM-COMBINATORICS-0026

### Local Hilbertian variance and obstructions to p-tightness

P / partially_solved | AIM 研讨会问题汇编

Let \(2\le p<\infty\), let \(A\subseteq B_p^d=\{x\in\mathbb R^d:\|x\|_p\le1\}\) be finite, and let \(a\in\operatorname{conv}(A)\). For every \(\varepsilon>0\), there are at most \(O(p\varepsilon^{-2})\) distinct points \(x_1,\ldots,x_s\in A\) and nonnegative weights \(\theta_i\) with \(\sum_i\theta_i=1\) such that \[ \left\|a-\sum_{i=1}^s\theta_i x_i\right\|_p\le\varepsilon. \]

In particular, this preserves the phrases “unit Hamming ball,” the omitted approximation norm, and the trailing word `status`.

The phrase “the $L^p$ unit Hamming ball” is not standard. A literal unit Hamming ball is a subset of a discrete alphabet such as \(\{0,1\}^d\), not a ball in \(\mathbb R^d\), and it has no parameter \(p\). By contrast, Barman's theorem and every subsequent primary source checked use the unit \(\ell_p\) ball and measure the approximation error in \(\ell_p\). The following is therefore a **strongly supported reconstruction**, not a verbatim recovery:

If “Hamming ball” were instead read literally as \(\{0,e_1,\ldots,e_d\}\), the problem would become a special simplex-approximation question and the asserted linear dependence on \(p\) would not describe its true behavior; Proposition 5.5 below quantifies this.

来源字段：http://aimpl.org/addcombapp/2/

13672 | AIM-COMBINATORICS-0057

### Local trees of triangles force quadratic order

S / solved | AIM 研讨会问题汇编

Question Does there exist any finite graph where every edge is in at least two triangles and the graph is locally a tree of triangles? (Solved by J. Verstraete)

(Nešetřil--Rödl) Partial results

**Recovered problem.** For every positive integer \(\ell\), find a finite nontrivial graph \(G\), of order \(n\geq\ell\), such that:

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13673 | AIM-COMBINATORICS-0056

### Local triangle trees and K4-minor support girth

S / solved | AIM 研讨会问题汇编

For every positive integer \(\ell\), is there a finite graph \(G\) such that \(G\xrightarrow{2}K_3\) and, for every set \(X\subseteq V(G)\) of at most \(\ell\) vertices, the induced graph \(G[X]\) is a subgraph of a tree of triangles?

The record comes from the January 2015 AIM workshop *Graph Ramsey theory* and attributes the question to Vojtěch Rödl. In normalized notation it asks:

Here \(G\xrightarrow{2}K_3\) means that every red/blue coloring of \(E(G)\) has a monochromatic triangle.

The record also states the motivating converse phenomenon: for each fixed triangle tree \(T\), there is a number of colors \(k\) such that every \(G\xrightarrow{k}K_3\) contains \(T\). That assertion is background rather than an ingredient in the affirmative solution below.

The supplied record says that a tree of triangles starts with a triangle and is enlarged by adding two edges that make one new triangle with an existing edge. The official workshop summary makes the intended recursion fully explicit: at each step one adds **one fresh vertex** \(x\) and the two edges \(xa,xb\), where \(ab\) is an existing edge. Thus these graphs are exactly the graphs usually called **2-trees** (under the convention that the initial graph is \(K_3\)).

There are two quantifier and containment points to settle.

1. The phrase “on every \(\ell\) vertices” is interpreted locally, as “on every set of at most \(\ell\) vertices.” If it meant only sets of exactly \(\ell\) vertices with no requirement \(|V(G)|\ge \ell\), the assertion would become vacuous whenever the witness had fewer than \(\ell\) vertices. If \(|V(G)|\ge\ell\), the “exactly \(\ell\)” and “at most \(\ell\)” versions are equivalent because the class of subgraphs of 2-trees is hereditary. 2. The word “induces” applies to \(G[X]\): all edges of \(G\) within \(X\) are retained. The subsequent containment in a triangle tree is ordinary subgraph containment, not induced-subgraph containment. This is exactly the wording of the source and is the notion addressed below.

Isolated vertices and disconnected induced graphs cause no problem: a “subgraph” may delete vertices and edges from the ambient triangle tree.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13674 | OWR-11695867-016

### Log-Concavity of q-Stirling Polynomials

O / open | Oberwolfach 报告问题汇编

Are the $q$-Stirling polynomials $S_q[n,k]$ log-concave for all $n$ and $k$?

来源字段：https://doi.org/10.4171/owr/2022/57

13675 | OWR-14299089-007

### Log-Concavity under Hadamard Products

S / solved | Oberwolfach 报告问题汇编

For a polynomial p, define $W(p)$ by $\sum_{n\ge0}p(n)x^n=W(p)(x)/(1-x)^{\deg p+1}$. If $W(p)$ and $W(q)$ have nonnegative, log-concave coefficient sequences with no internal zeros, must $W(pq)$ have the same property?

来源字段：https://doi.org/10.4171/owr/2025/12

13676 | OPG-426

### Long rainbow arithmetic progressions

P / partially_solved | Open Problem Garden

For $k\in \mathbb{N}$ let $T_k$ denote the minimal number $t\in \mathbb{N}$ such that there is a rainbow $AP(k)$ in every equinumerous $t$-coloring of $\{ 1,2,\ldots,tn\}$ for every $n\in \mathbb{N}$

Conjecture For all $k\geq 3$, $T_k=\Theta (k^2)$.

背景链接：http://www.openproblemgarden.org/op/long_rainbow_arithemtic_progressions

13677 | OWR-10252930-034

### Long Twin Subsequences in Ternary Words

O / open | Oberwolfach 报告问题汇编

For ternary words, is the maximum twin-subsequence length asymptotically $n/2-o(n)$, both for a uniformly random word and in the worst case?

来源字段：https://doi.org/10.4171/owr/2022/22

13678 | AIM-COMBINATORICS-0074

### Loose hypergraph triangles versus cliques: an edge-rooted component certificate

P / partially_solved | AIM 研讨会问题汇编

Known: (Kim, Ajtai--Komlós--Szemerédi) $r(3,t)=\Theta(t^2/\log t)$.

Let $\triangle$ denote the 3-uniform hypergraph with vertex set $[6]$ and edges $\{123,345,561\}$.

Known: $\Omega\!\left(t^{3/2}/(\log t)^{3/4}\right)=r_3(\triangle,K_t^{(3)})=O(t^{3/2})$.

Conjecture (Kostochka--Mubayi--Verstraete): $r_3(\triangle,K_t^{(3)})=o(t^{3/2})$.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13679 | AIM-COMBINATORICS-0185

### Low temporal repetition forces excess trace

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.17, item 1 (Seymour).** Let \[ t=\left\lceil\frac nr\right\rceil . \] Let \(A_1,A_2,\ldots,A_t\) be \(n\times n\) zero-one matrices, not necessarily distinct, such that \[ (A_k)_{ii}=1,\qquad (A_k)_{ij}=1\Longrightarrow (A_k)_{ji}\ne1\quad(i\ne j), \tag{1} \] and every row sum of every \(A_k\) is at least \(r+1\). Is \[ \operatorname{tr}(A_1A_2\cdots A_t)>n? \tag{2} \] The sentence split from the canonical record ends: “This is a special case of Conjecture 6.5.”

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13680 | AIM-COMBINATORICS-0198

### LP obstructions and cyclic blow-ups for DeVos's distribution conjecture

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.28 (DeVos).** For any digraph \(G\) with no directed cycles of length at most three, there is a probability distribution \(p\) on \(V(G)\) such that, at every vertex \(v\), \[ p(N^+(v))\geq p(N^-(v)) \quad\text{and}\quad p(N^-_2(v))\geq p(N^-(v)), \] where \(p(S)=\sum_{x\in S}p(x)\).

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13681 | AMR-029-0050

### Making the union of two directed spanning trees strongly connected

O / open | AMR 问题清单汇编

Let D=(V,E) be a directed graph that is the union of two disjoint directed spanning trees. Can we characterize when does D have a directed spanning tree $T \subseteq E$ such that E-T is also a directed spanning tree and reversing the orientation of each edge of T results in a strongly connected digraph?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13682 | AIM-COMBINATORICS-0040

### Matrix Spencer: exact dilation bookkeeping and a sharp Frobenius-energy case

P / partially_solved | AIM 研讨会问题汇编

**Matrix Spencer.** Given matrices $A_1,\ldots,A_n\in\mathbb R^{n\times n}$ satisfying $\|A_i\|_{\mathrm{op}}\le1$, do there exist signs $\varepsilon_1,\ldots,\varepsilon_n\in\{-1,1\}$ such that $\|\sum_i\varepsilon_iA_i\|=O(\sqrt n)$?

来源字段：http://aimpl.org/hereddiscrep/1/

13683 | OWR-17135-010

### Matroid Structure and Hyperpaths in Collapsible Hypertrees

O / open | Oberwolfach 报告问题汇编

Do collapsible $d$-dimensional hypertrees admit a useful matroidal structure, and is there a satisfactory higher-dimensional analogue of a path for them?

来源字段：https://doi.org/10.4171/owr/2019/39

13684 | OWR-4791-022

### Maximum Density of Directed Simplices in Tournaments

P / partially_solved | Oberwolfach 报告问题汇编

For a $d$-tournament on $n$ vertices, obtained by assigning a compatible orientation to every $d$-set, what is the greatest possible number of directed $(d+1)$-sets? Equivalently, how does the limiting density $c_d$ of directed $(d+1)$-sets behave for fixed $d$ and as $d$ grows?

来源字段：https://doi.org/10.4171/owr/2011/01

13685 | OWR-1183-010

### Maximum Families of Pairwise Colliding Permutations

O / open | Oberwolfach 报告问题汇编

Let $\rho(n)$ be the maximum size of a family of pairwise colliding permutations of $[n]$, where two permutations collide if they map some element to consecutive integers. Is $\limsup_{n\to\infty}n^{-1}\log_2\rho(n)=1$?

来源字段：https://doi.org/10.4171/owr/2006/01

13686 | GAME-003

### Maximum Givens in Minimal Sudoku

P / partially_solved | 独立 / 未分配集合

What is the maximum number of givens for a minimal Sudoku puzzle?

背景链接：https://sudokutheory.com/wiki/index.php?title=Snipes

13687 | OWR-17464-019

### Maximum Independent Sets in Regular Uniform Hypergraphs

O / open | Oberwolfach 报告问题汇编

For $r\geq2$ and $d\geq1$, determine the maximum number of independent sets in a $d$-regular $r$-uniform hypergraph $G$ on $n$ vertices. In particular, does the canonical $d$-regular hypergraph $H_d^r$ satisfy the conjectured extremal bound

$$\operatorname{ind}(G)\leq \operatorname{ind}(H_d^r)^{n/(rd)}$$?

来源字段：https://doi.org/10.4171/owr/2020/1

13688 | OWR-1386-012

### Maximum Intersection Order for Asymptotic $K(d)$-Families

O / open | Oberwolfach 报告问题汇编

For fixed $k$, determine the largest $d=d(k)$ for which the asymptotic $K(d)$-family bound $|\mathcal G|\leq(1+o(1))\binom{n-1}{k-1}$ holds, and extend that bound to the exact extremal result.

来源字段：https://doi.org/10.4171/owr/2006/48

13689 | OWR-14299518-028

### Maximum Sizes of Cover-Free Families

O / open | Oberwolfach 报告问题汇编

Determine the maximum possible size of an $r$-cover-free family.

来源字段：https://doi.org/10.4171/owr/2025/42

13690 | AMR-029-0051

### Maximum square-free 2-matching

P / partially_solved | AMR 问题清单汇编

Given an undirected graph G=(V,E), find a maximum cardinality 2-matching containing no cycles of length 4 in polynomial time.

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13691 | AMR-029-0052

### Maximum weakly stable matchings in graphs without odd preference cycles

O / open | AMR 问题清单汇编

Let (G,<) be a preference system with ties where in every odd cycle there is a node that prefers its clockwise neighbour to its other neighbour, and there is another node that prefers its anti-clockwise neighbour to its other neighbour. How well can we approximate the maximum size weakly stable matching?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13692 | AMR-029-0053

### Maximum weight bounded fractional matching

P / partially_solved | AMR 问题清单汇编

Given a graph G=(V,E), and weight and capacity functions $w,u: E \to {\mathbb R_+}$ defined on the edge set, is there a combinatorial, strongly polynomial algorithm to find a vector x maximizing wx subject to the constraints $x \in P(G)$ and $x \leq u$, where P(G) is the convex hull of the incidence vectors of perfect matchings of G?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13693 | AMR-029-0054

### Maximum weight k-element subsets of perfect matchings

P / partially_solved | AMR 问题清单汇编

Given a bipartite graph G with edge weights, can we find in polynomial time a maximum weight k-element matching in G that can be extended to a perfect matching?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13694 | AMR-029-0055

### Min-sum two edge-disjoint paths

P / partially_solved | AMR 问题清单汇编

Let G=(V,E) be an undirected graph and let $(s_1,t_1), (s_2,t_2)$ be two node pairs. Give a combinatorial, polynomial-time algorithm to find edge-disjoint $s_1-t_1$ and $s_2-t_2$ paths with minimum total length.

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13695 | AIM-COMBINATORICS-0199

### Minimal feedback arc sets and long paths

S / solved | AIM 研讨会问题汇编

**Conjecture 6.29 (Lichiardopol's Conjecture).** Every digraph \(D\) has some minimal feedback arc set \(F\) which contains a path of length \(\delta_D^+\).

a feedback arc set immediately before the conjecture: if \(D=(V,E)\), then \(F\subseteq E\) is a feedback arc set (FAS) when \((V,E\setminus F)\) has no directed cycle. The exact statement is:

The source's note says that this conjecture implies the Hoàng--Reed, Caccetta--Häggkvist, Bermond--Thomassen, and Thomassé path conjectures.

The AIM text itself does not define “minimal.” The subsequent primary work of Schneider and Botler on this exact conjecture resolves the ambiguity:

- **minimal means inclusion-minimal**, not minimum cardinality: there is no FAS \(F'\subsetneq F\); - the path lies in the spanning arc-subdigraph \(D[F]=(V(D),F)\), so every arc of the path belongs to \(F\); this is not a vertex-induced subdigraph; - a path has distinct vertices \(v_0,\ldots,v_\ell\) and arcs \(v_iv_{i+1}\), and its length is its number \(\ell\) of arcs.

Their formulation says “length at least \(\delta^+(D)\).” This is equivalent to the AIM wording because a longer directed path has a subpath of exactly the required length.

There is a convention mismatch in the 2006 PDF. Its opening definition allows parallel directed edges, while later work on Conjecture 6.29 uses an arc set \(A\subseteq V\times V\), hence no parallel copies. Proposition 4 below shows that the literal multiplicity-degree multidigraph reading is already false in a particularly elementary way. More importantly, Theorem 3 gives a simple, loopless, digon-free counterexample, so the present-day refutation does not depend on this ambiguity.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13696 | AIM-COMBINATORICS-0235

### Minimal order-two bases and quantitative thinning

S / solved | AIM 研讨会问题汇编

Suppose \(A\) is a subset of the naturals, and is an additive basis of \(\mathbb N\) of order 2. Does there exist a proper subset \(B\) of \(A\), where \(B\) is an additive basis of the naturals of order 2? What is the slowest growing \[ B(x):=|\{b\in B:b\leq x\}|? \]

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13697 | AIM-COMBINATORICS-0231

### Minimum difference sets in fixed dimension

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.6 (Y. Stanchescu).** Suppose that \(A\) is a finite subset of \(\mathbb Z^d\), not contained in a hyperplane of dimension smaller than \(d\). Determine the smallest possible value of \(|A-A|\) as a function of \(|A|\).

(d + 1)|A| - 1 > 2 d(d + 1)

2d - 2 + 1 / d - 1

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13698 | AMR-029-0056

### Minimum k-way cut in a hypergraph

S / solved | AMR 问题清单汇编

Can we find a minimum k-way cut in a capacitated hypergraph in polynomial time, if k is fixed?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13699 | AMR-031-0005

### Minimum length of a superpermutation

P / partially_solved | AMR 问题清单汇编

A superpermutation on $n$ symbols is a string containing every permutation of the $n$ symbols as a contiguous substring. Determine the minimum possible length of a superpermutation for every $n>5$.

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Combinatorics

13700 | AMR-029-0057

### Minimum polychromatic number for plane graphs with fixed girth

P / partially_solved | AMR 问题清单汇编

For a plane graph $G$, let $g(G)$ denote the length of the shortest face in $G$. For a (not necessarily proper) $k$-coloring of $V(G)$ we say that a face is polychromatic if all $k$ colors appear on its vertices. A k-coloring of $V(G)$ is called polychromatic if every face of $G$ is polychromatic. The polychromatic number of $G$, denoted by $p(G)$, is the largest number $k$ such that there is a polychromatic $k$-coloring of $V(G)$. Define $p(g)=\min\{p(G)|g(G)=g\}$. Determine $p(g)$ exactly for every positive integer $g$. The first open case is $g=5$ where we know that $2 \leq p(5) \leq 4$.

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13701 | OWR-14298158-014

### Moment Sequences from Matching Pattern Avoidance

P / partially_solved | Oberwolfach 报告问题汇编

For which matchings $\tau$ is $(M_n(\tau))_{n\geq0}$ a Hamburger moment sequence, equivalently $(M_{2n}(\tau))_{n\geq0}$ a Stieltjes moment sequence?

来源字段：https://doi.org/10.4171/owr/2024/6

13702 | OWR-4791-031

### Monochromatic Concatenations of Permutation Words

O / open | Oberwolfach 报告问题汇编

View $S_n$ as the permutations of $1,\ldots,n$, written as words with concatenation $xy$. Given $k$, does there exist $n$ such that every $k$-colouring of $S_n$ has words $x,y,z$ whose six concatenations $xyz,xzy,yxz,yzx,zxy,zyx$ are permutations in $S_n$ and all have the same colour?

来源字段：https://doi.org/10.4171/owr/2011/01

13703 | OWR-14298163-027

### Monochromatic Empty Triangles

O / open | Oberwolfach 报告问题汇编

Let $f(n)$ be the largest integer such that every set of $n$ red and $n$ blue points in general position in the plane contains at least $f(n)$ monochromatic empty triangles. Prove or disprove that $f(n)=o(n^2)$.

来源字段：https://doi.org/10.4171/owr/2024/3

13704 | OWR-11781-006

### Monochromatic Structures below Random Density

O / open | Oberwolfach 报告问题汇编

Does an analogue of the quadratic-structure phenomenon for two-colorings with fewer than the random density of monochromatic $K_4$'s hold for graph colorings, and what is the asymptotically minimal number of monochromatic $3$-term arithmetic progressions in a two-coloring of $\{1,\ldots,n\}$?

来源字段：https://doi.org/10.4171/owr/2012/19

13705 | OWR-12697684-031

### Monochromatic Subspaces in Binary Vector Spaces

O / open | Oberwolfach 报告问题汇编

Let $f(n)$ be the largest number such that every two-coloring of $\mathbb F_2^n\setminus\{0\}$ contains a monochromatic set of $f(n)$ nonzero vectors forming a linear subspace together with $0$. Is $f(n)=O(n)$?

来源字段：https://doi.org/10.4171/owr/2023/1

13706 | AMR-054-0058

### Monochromatic Triangles

P / partially_solved | AMR 问题清单汇编

For any (planar) triangle $T$, is there is a $3$-coloring of the (infinite) plane with no monochromatic copy of $T$? We imagine congruent copies of $T$ moved around the plane via rigid motions, and seek a spot where $T$ is monochromatic. $T$ is monochromatic if its three vertices are painted the same color, by virtue of lying on points of the plane painted that color. Note that the coloring in the question may depend on the given triangle $T$.

背景链接：https://topp.openproblem.net/p58

13707 | GREEN-066

### Monochromatic x+y and xy

P / partially_solved | Ben Green 问题

If $\{1, \dots, N\}$ is $r$-coloured, then for $N \geq N_0(r)$ there exist integers $x, y \geq 3$ such that $x+y$ and $xy$ have the same colour. Find reasonable bounds for $N_0(r)$.

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13708 | OPG-618

### Monotone 4-term Arithmetic Progressions

P / partially_solved | Open Problem Garden

Question Is it true that every permutation of positive integers must contain monotone 4-term arithmetic progressions?

背景链接：http://www.openproblemgarden.org/op/monotone_4_term_arithmetic_progressions

13709 | OWR-14298158-018

### Monotone k-Shuffles in Long Permutations

O / open | Oberwolfach 报告问题汇编

For each fixed $k\in\mathbb{N}$, does every sufficiently long permutation contain a consecutive factor that is a $k$-shuffle of $12$ or $21$?

来源字段：https://doi.org/10.4171/owr/2024/6

13710 | OWR-14299904-009

### Monotonicity of Hook Counts in Symmetric Partitions

P / partially_solved | Oberwolfach 报告问题汇编

For each fixed positive integer $t$, let $\operatorname{sc}^{\star}_t(n)$ and $\operatorname{dd}^{\star}_t(n)$ count the $t$-hooks in self-conjugate and doubled-distinct partitions of $n$. Is $\operatorname{sc}^{\star}_t(n)\geq\operatorname{sc}^{\star}_{t+1}(n)$ and $\operatorname{dd}^{\star}_t(n)\geq\operatorname{dd}^{\star}_{t+1}(n)$ for all sufficiently large $n$, with a threshold bounded uniformly in $t$?

来源字段：https://doi.org/10.4171/owr/2026/2

13711 | GREEN-020

### Multidimensional Szemerédi Theorem Bounds

P / partially_solved | Ben Green 问题

Find reasonable bounds for instances of the multidimensional Szemerédi theorem.

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13712 | AIM-COMBINATORICS-0218

### Multiplicity and an exact weighted-cut identity for three-term progressions

P / partially_solved | AIM 研讨会问题汇编

Inspection of the official PDF confirms that the lost superscripts and fractions are \[ \frac{n^2(1+o(1))}{22}\qquad\hbox{and}\qquad \frac{n^2}{16}. \] The words beginning “COLLECTED BY” are a page footer, not part of the problem. The source also says “Roberts-Zeilberger”; the cited paper is by Aaron Robertson and Doron Zeilberger.

Given a homogeneous linear system $S$ that is partition regular for $r$ colors, determine the least possible number of nontrivial monochromatic solutions in an $r$-coloring of $[n]=\{1,\ldots,n\}$. The list notes that for two colors and $x+y=z$ the answer is $n^2(1+o(1))/22$, whereas a random coloring gives asymptotically $n^2/16$. It asks in particular for $x+y=2z$, and whether random two-colorings are optimal.

For $x+y=z$, count triples with $x\le y$, so reversal of the summands is identified. There are \[ U(n)=\sum_{z=2}^n\left\lfloor\frac z2\right\rfloor =\left\lfloor\frac{n^2}{4}\right\rfloor \] such triples. The $\lfloor n/2\rfloor$ cases $x=y$ have two distinct values and are monochromatic with probability $1/2$; all others have three distinct values and are monochromatic with probability $1/4$. Thus a random two-coloring has expectation \[ \frac{U(n)-\lfloor n/2\rfloor}{4} +\frac{\lfloor n/2\rfloor}{2} =\frac{n^2}{16}+O(n). \] This is exactly the normalization in the AIM question. Counting ordered pairs $(x,y)$ instead doubles the leading constants to $n^2/8$ randomly and $n^2/11$ optimally. Robertson--Zeilberger and Schoen proved the reversal-identified minimum $n^2/22+O(n)$.

For $x+y=2z$, this report counts each nonconstant three-term progression \[ (a,a+d,a+2d),\qquad d>0,\quad a+2d\le n, \] once. In the equation, it corresponds to $(x,y,z)=(a,a+2d,a+d)$, with the reversal of $x,y$ identified. Ordered endpoint solutions have twice the count. The diagonal $x=y=z$ is excluded as trivial; including it adds exactly $n$ monochromatic solutions and changes no leading constant.

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13713 | GREEN-061

### N Queens Problem Asymptotics

P / partially_solved | Ben Green 问题

别名：N-Queens Problem Asymptotics [GREEN-097]

In how many ways (asymptotically) $Q(n)$ may $n$ non-attacking queens be placed on an $n \times n$ chessboard?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13714 | AIM-COMBINATORICS-0098

### Naturality as an affine automorphism action

P / partially_solved | AIM 研讨会问题汇编

By the Matrix--Tree theorem, the number of spanning trees of a graph \(G\) equals the size of the critical group of \(G\). When are the spanning trees a torsor for the critical group? Making \(G\) into a ribbon graph, and given choice of sink, we do get a torsor structure from the rotor-router model [HLM+08]. But under what conditions can we get a natural torsor action without this additional data?

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13715 | OWR-14298163-026

### Nearly Right-Angled Point Sets

O / open | Oberwolfach 报告问题汇编

Let $X\subseteq\mathbb R^n$ be finite and suppose no three points of $X$ determine an angle greater than $\frac\pi2(1-10^{-10000})$. Prove or disprove that $|X|\leq 1.99^n$.

来源字段：https://doi.org/10.4171/owr/2024/3

13716 | COMB-011

### No-Three-in-Line Problem

P / partially_solved | 独立 / 未分配集合

别名：No-Three-in-Line Problem [OWR-14298163-013]

What is the maximum number of points in an $n \times n$ grid with no three collinear?

背景链接：https://doi.org/10.1016/j.ejc.2023.103796

13717 | OWR-784-004

### Nonbinary Matroid Certificates

O / open | Oberwolfach 报告问题汇编

What certificates prove that a matroid is not binary?

来源字段：https://doi.org/10.4171/owr/2005/03

13718 | OWR-14299904-014

### Nonnegative Coefficients for Cylindric Order Polynomials

O / open | Oberwolfach 报告问题汇编

For every cylindric shape $\lambda/\mu/d$, is $\operatorname{CPP}_{\lambda/\mu/d}(n)$ a polynomial in $n$ with nonnegative coefficients?

来源字段：https://doi.org/10.4171/owr/2026/2

13719 | OWR-17135-031

### Nonnegative Square Decompositions of Bipartite Determinant Differences

P / partially_solved | Oberwolfach 报告问题汇编

别名：Nonnegative Square Decompositions of Bipartite Determinant Differences [OWR-17135-032]

Let $G$ be bipartite with parts $X=[n]$ and $Y$, set $X_S=|\{y\in Y:N(y)=S\}|$ for nonempty $S\subseteq[n]$, and define $M$ by $$M_{ii}=\sum_{S:i\in S}X_S,\qquad M_{ij}=\sum_{S:i\in S,\,j\notin S}\frac{X_S}{|S|}\quad(i\neq j)$$. Is there an identity $$\det(\operatorname{diag}M)-\det M=\sum_\mu c_\mu x^\mu+\sum_{\mu,\nu}c_{\mu\nu}(x^\mu-x^\nu)^2$$ in which $x^\mu$ and $x^\nu$ are monomials in the variables $X_S$ and all coefficients $c_\mu,c_{\mu\nu}$ are nonnegative?

来源字段：https://doi.org/10.4171/owr/2019/39

13720 | OWR-1455-001

### Nonnegativity in Complete Indices of Bruhat Intervals

O / open | Oberwolfach 报告问题汇编

For every Bruhat interval $[u,v]$, are all coefficients of the lower-degree part $\widetilde{\Psi}_{u,v}$ of its complete cd-index nonnegative?

来源字段：https://doi.org/10.4171/owr/2007/04

13721 | OWR-4425-017

### Nonvanishing Hankel Determinants of Morphic Sequences

O / open | Oberwolfach 报告问题汇编

Let $\varphi$ be the morphism $1\mapsto12$, $2\mapsto23$, $3\mapsto14$, $4\mapsto32$ and let $a=\varphi^\omega(1)$. Can one prove that every Hankel determinant of $a$ is nonzero? More generally, is it true that for every integer $m\geq2$ no infinite sequence over a finite subset of $\mathbb{Z}$ has all Hankel determinants nonzero modulo $m$?

来源字段：https://doi.org/10.4171/owr/2010/37

13722 | AIM-COMBINATORICS-0045

### Normalization-safe hardness below constructive Komlos bounds

P / partially_solved | AIM 研讨会问题汇编

Can we prove some hardness statement about finding a coloring achieving Komlos guarantee or the Banaszczyk discrepancy guarantee?

来源字段：http://aimpl.org/hereddiscrep/1/

13723 | GAME-008

### Octal Games Periodicity

O / open | 独立 / 未分配集合

Are the nim-sequences of all finite octal games eventually periodic?

背景链接：https://wwwhomes.uni-bielefeld.de/achim/octal.html

13724 | AIM-COMBINATORICS-0080

### Off-diagonal convex geometric Ramsey bounds

P / partially_solved | AIM 研讨会问题汇编

Problem: Improve these bounds.

**Recovered problem.** For integers \(s,n\ge2\), improve the lower and upper bounds for the least integer \(g(s,n)\) such that every red/blue coloring of all pairs of every \(g(s,n)\)-point set in the plane in general position contains either

Its stored section, “Ordered Ramsey numbers,” is misleading because a section boundary was absorbed into the preceding record.

The AIM workshop summary independently describes the diagonal parameter: the least size of a general-position planar point set whose every red/blue pair-coloring contains a monochromatic convex \(p\)-set. More decisively, Mubayi and Suk's 2014 paper defines exactly this convex geometric Ramsey problem and gives the diagonal bound \[ 2^{\Omega(n)}<g(n,n)\le 2^{O(n^2\log n)}. \] Its concluding remarks state the same off-diagonal problem and the same direct Ramsey/Erdős--Szekeres upper-bound mechanism. The original Overleaf project was not directly retrievable, but the repository adjacency, the AIM workshop summary, and this primary paper agree. Thus the recovered problem is:

- \(s\) points in convex position whose every pair is red, or - \(n\) points in convex position whose every pair is blue.

Here “general position” means no three points are collinear. The order of the colors is immaterial, so \(g(s,n)=g(n,s)\).

The powers displayed in the source are coarse asymptotic bounds and should not be read as literal inequalities for every small \(n\). In particular, the standard Erdős--Szekeres lower bound is \(2^{k-2}+1\le ES(k)\), where \(ES(k)\) is defined below.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13725 | AIM-COMBINATORICS-0023

### Off-diagonal van der Waerden numbers: refuted quadratic conjecture and a difference-sensitive transversal criterion

P / partially_solved | AIM 研讨会问题汇编

Let \(W(k,\ell)\) be the minimum \(n\) so that any red/blue coloring of \(\{1,\cdots,n\}\) has either a red \(k\)-AP or a blue \(\ell\)-AP. The following conjecture is supported by numerical evidence: \[ W(3,\ell)=\ell^{2+o(1)}. \]

来源字段：http://aimpl.org/addcombapp/2/

13726 | AIM-COMBINATORICS-0172

### One-sided lifting and polarized minimal counterexamples for Sullivan-3

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.7 (later called Sullivan-3).** Every finite simple digraph with no loops or digons has a vertex $v$ such that \[ |N_2^+(v)|+|N^+(v)| \geq2\min\{|N^-(v)|,|N^+(v)|\}. \tag{1.1} \]

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13727 | AIM-COMBINATORICS-0137

### Operational meaning and identifiability limits of network curvature

P / partially_solved | AIM 研讨会问题汇编

Why measure it? (What are the real-world applications?)

Wasn't that your result, that the internet has curvature \(-1\) and dimension \(2\)?

So, yes, we need more examples like that, showing the significance of curvature.

来源字段：http://aimpl.org/largenetworks/1/

13728 | AMR-029-0058

### Opposite vertices of base polyhedra

O / open | AMR 问题清单汇编

Is it true that if all vertices of a base polytope B are in $\{0,1,-1\}^n$ and $0 \in B$, then B has a vertex v such that -v is also a vertex?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13729 | OWR-4791-026

### Optimal Cofilling Profiles for Uniform Set Systems

O / open | Oberwolfach 报告问题汇编

For the cofilling profile $\phi_d(\alpha)$ defined from minimal d-uniform systems and their coboundaries, is the upper bound $\phi_d(\alpha)\leq\frac{d}{d+1}\alpha$ optimal, and is the corresponding extremal example essentially unique?

来源字段：https://doi.org/10.4171/owr/2011/01

13730 | OWR-2090-008

### Optimal Dilation Thresholds for Ehrhart $\delta$-Polynomials

O / open | Oberwolfach 报告问题汇编

Fix $d>0$. Given $h(t)=h_0+\cdots+h_dt^d$ with nonnegative integer coefficients and $h_0\geq1$, write $h(t)/(1-t)^{d+1}=\sum_{m\geq0}g(m)t^m$ and define $U_nh(t)$ by $U_nh(t)/(1-t)^{d+1}=\sum_{m\geq0}g(nm)t^m$. Determine the optimal positive integers $m_d,n_d$ such that, for every such $h$ and $n\geq n_d$, the polynomial $U_nh(t)=\sum_i h_i(n)t^i$ has $d$ negative real roots converging to the roots of the Eulerian polynomial $A_d(t)$, has positive strictly log-concave coefficients, and satisfies $h_i(n)<m_dh_d(n)$ for $0\leq i\leq d$. Also determine the optimal choices in the Ehrhart special case $h(t)=\delta_P(t)$, where $P$ is a $d$-dimensional lattice polytope and the coefficients of $\delta_{nP}(t)$ satisfy $1=\delta_0(n)$, $$\delta_i(n)<\delta_{d-i}(n)<\delta_{i+1}(n)\quad\text{for }0\leq i\leq\lfloor d/2\rfloor-1,$$ and $\delta_{\lfloor(d+1)/2\rfloor}(n)<m_d\delta_d(n)$.

来源字段：https://doi.org/10.4171/owr/2008/44

13731 | OWR-15215-004

### Optimal Minimal Dispersion in High Dimensions

P / partially_solved | Oberwolfach 报告问题汇编

For the minimal dispersion $\operatorname{disp}^*(n,d)$ of $n$ points in $[0,1]^d$, can one improve $$ \operatorname{disp}^*(n,d)\leq\min\left\{\frac{c_1d}{n},\frac{c_2d\log(n/d)}{n}\right\}, $$ and in particular prove $\operatorname{disp}^*(n,d)\leq cd/n$ for $n\geq d$? Can one also improve $$ \operatorname{disp}^*(n,d)\geq\frac{\log_2d}{4(n+\log_2d)} $$?

来源字段：https://doi.org/10.4171/owr/2017/10

13732 | GREEN-076

### Optimal Sidon Bases

O / open | Ben Green 问题

Are there infinitely many $q$ for which there is a set $A \subset \mathbb{Z}/q\mathbb{Z}$ with $|A| = (\sqrt{2} + o(1))q^{1/2}$ and $A + A = \mathbb{Z}/q\mathbb{Z}$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13733 | OWR-14299904-015

### Optimization over Nonzero Littlewood-Richardson Coefficients

O / open | Oberwolfach 报告问题汇编

For a partition $\lambda$ of $n$ with length $d$, define $\eta(\lambda)=n^2+\frac{d(d-1)(2d-1)}6-\sum_{i=1}^d(\lambda_i-(i-1))^2$ and $\Delta(\lambda,\mu,\nu)=\eta(\lambda)-\eta(\mu)-\eta(\nu)$. Given $n,k,d$ with $k\leq n/2$, maximize $\Delta(\lambda,\mu,\nu)$ over triples with $|\lambda|=n$, $|\mu|=k$, $|\nu|=n-k$, and $c^\lambda_{\mu,\nu}\neq0$.

来源字段：https://doi.org/10.4171/owr/2026/2

13734 | AIM-COMBINATORICS-0233

### Order lifting for sparse additive bases of polynomial values

P / partially_solved | AIM 研讨会问题汇编

This is Problem 2.8, attributed to T. Wooley, in the AIM workshop list *Recent trends in additive combinatorics*.

For a set $A\subseteq\mathbb N$, write

\[ hA=\{a_1+\cdots+a_h:a_i\in A\},\qquad A(x)=|A\cap[0,x]|. \]

The question is: if $f\in\mathbb Z[t]$ has degree $d\ge 2$, takes natural-number values, and

\[ \{f(1),f(2),\ldots\}\subseteq hA, \]

must $A(x)$ satisfy a substantially stronger lower bound than the counting bound $A(x)\gg x^{1/(hd)}$? As is customary for an asymptotic question, finitely many initial values of $f$ are irrelevant. We therefore assume below that $f$ is positive and strictly increasing on the range in use; a polynomial of positive leading coefficient has this property eventually.

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13735 | AIM-COMBINATORICS-0079

### Order-robust ordered Ramsey numbers of regular graphs

S / solved | AIM 研讨会问题汇编

Does there exist a constant \(c>0\) such that for every sufficiently large \(n\) there exists a 100-regular graph \(H\) on \(n\) vertices such that \(r_<(H)\ge n^{1+c}\) for every ordering of \(H\)?

The remainder of the extracted record starts with the literal command \[ \texttt{\string\section\{Ramsey meets Erd\H{o}s--Szekeres\}} \] and introduces Dhruv Mubayi's function \(g(s,n)\). This is a source-boundary error, not part of Question 32. Thus this job treats only the 100-regular ordered-Ramsey question.

For an underlying graph \(H\) and a total order \(\prec\) on \(V(H)\), write \((H,\prec)\) for the resulting ordered graph. Its ordered Ramsey number \(r_<((H,\prec))\) is the least \(N\) such that every red-blue coloring of the edges of the naturally ordered complete graph on \([N]\) contains a monochromatic copy of \(H\) whose vertex embedding preserves \(\prec\).

The phrase "for every ordering" is inside the choice of \(H\). Equivalently, with \[ \rho(H):=\min_{\prec}r_<((H,\prec)), \tag{1.1} \] the question is \[ \exists c>0\ \exists n_0\ \forall n\ge n_0\ \exists\text{ a 100-regular }H_n\text{ on }n\text{ vertices with }\rho(H_n)\ge n^{1+c}. \tag{1.2} \]

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13736 | AMR-029-0059

### Orientation conjecture of Nash-Williams

P / partially_solved | AMR 问题清单汇编

Any $2k$-edge-connected (possibly infinite) multigraph admits a $k$-edge-connected orientation.

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13737 | AMR-029-0060

### Orientation of nonideal clutters

O / open | AMR 问题清单汇编

Let $\mathcal{C}$ be a clutter on ground set V, and let $\mathcal{B}$ be its blocker. Is it true that $\mathcal{C}$ is nonideal if and only if there exist $p:{\mathcal C} \to V$ and $q: {\mathcal B} \to V$ such that $p(X)\in X$ for every $X \in {\mathcal C}$, $p(Y)\in Y$ for every $Y \in {\mathcal B}$, if $p(X)=q(Y)$, then $|X \cap Y|>1$?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13738 | AMR-029-0061

### Orientation with shortest round trip

O / open | AMR 问题清单汇编

Let G=(V,E) be a mixed graph with non-negative edge-lengths and let $s,t \in V$. Can we find in polynomial time an orientation where the sum of the lengths of the shortest s-t directed path and the shortest t-s directed path is minimal?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13739 | AMR-029-0062

### Orientation-compatible w-vertex cover

O / open | AMR 问题清单汇编

Given a digraph D=(V,A) and non-negative even-valued arc weights $w_a\ (a \in A)$, can we find in polynomial time a w-vertex cover $x$ of the underlying undirected graph with the additional property that for every node v with $x_v>0$ there is an arc $uv\in A$ with $x_u+x_v=w_a$?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13740 | OWR-14210-005

### Packing and Covering for Infinite Matroids

O / open | Oberwolfach 报告问题汇编

Let $M$ and $N$ be matroids on a common ground set $E$. Does there exist a partition $E=P\mathbin{\dot\cup}C$ such that the restrictions $M|P$ and $N|P$ have disjoint spanning sets, while the contractions $M.C$ and $N.C$ have independent sets whose union is $C$?

来源字段：https://doi.org/10.4171/owr/2016/2

13741 | AIM-COMBINATORICS-0072

### Packing equal-distance witnesses under the local (5,9) condition

P / partially_solved | AIM 研讨会问题汇编

(Erdős 1986) What is the least possible number of distinct distances among \(n\) points in the plane if every 5 of them span at least 9 distinct distances? Conjecture: \(\Omega(n^2)\). Best known: \(\Omega(n)\). Related to \(F(r,s,9)\).

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13742 | AIM-COMBINATORICS-0176

### Palette transversals and a rooted Hall obstruction

P / partially_solved | AIM 研讨会问题汇编

Conjecture 6.1, if ∑ti=1 δ+ Gi(v) ≥ |V| for all vertices v, there is a rainbow cycle in G.

Conjecture 6.9. Under the hypotheses of Conjecture 6.1, if \(\sum_{i=1}^{t}\delta^+_{G_i}(v)\geq |V|\) for all vertices \(v\), there is a rainbow cycle in \(G\).

**Conjecture 6.9 (recovered).** If \(n=|V|\) and \[ \lambda(v):=\sum_{i=1}^{k}d^+_{G_i}(v) =\sum_{e\in E_G^+(v)}|S_e| \geq n \qquad\text{for every }v\in V, \tag{2} \] then \(G\) has a rainbow directed cycle.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13743 | OWR-4425-004

### Palindromic Recurrent Words Outside Morphic Class $P$

O / open | Oberwolfach 报告问题汇编

Are there uniformly recurrent infinite words containing arbitrarily long palindromes that arise from primitive morphisms, none of which belongs to class $P$?

来源字段：https://doi.org/10.4171/owr/2010/37

13744 | AMR-029-0063

### Parity constrained strongly connected orientations

P / partially_solved | AMR 问题清单汇编

Find a good characterization for undirected graphs having a strongly connected (more generally k-edge-connected) orientation so that the in-degree of every node is odd.

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13745 | OWR-11695867-017

### Parity Log-Concavity of Type-B q-Stirling Polynomials

O / open | Oberwolfach 报告问题汇编

Are the type-$B$ $q$-Stirling polynomials $S_q^B[n,k]$ parity log-concave for all $n$ and $k$?

来源字段：https://doi.org/10.4171/owr/2022/57

13746 | AIM-COMBINATORICS-0021

### Parity, polynomial rank, and an exact prime-distance case

P / partially_solved | AIM 研讨会问题汇编

There is an absolute constant $C$ such that, for $0<\alpha<1$, if $k$ is an even integer satisfying \[ C\sqrt{n\log(1/\alpha)}\le k\le n-C\sqrt{n\log(1/\alpha)}, \] and $A\subseteq\mathbb F_2^n$ has no two distinct elements at Hamming distance $k$, then $|A|/2^n\le\alpha$.

This preserves the missing parity hypothesis and the ambiguous expression under the square root.

The `literature` field is empty.

The literal statement is not correct: $k$ must be even. The standard parity subspace

\[ E_n=\{x\in\mathbb F_2^n:|x|\equiv0\pmod2\} \]

has density $1/2$, and every element of $E_n+E_n=E_n$ has even Hamming weight. Thus it avoids every odd $k$.

There is a second, genuinely consequential ambiguity. The TeX string

\[ \sqrt{\log \alpha^{-1}n} \]

could mean either $\sqrt{n\log(1/\alpha)}$ or $\sqrt{\log(n/\alpha)}$. The first reading is strongly supported by the AIM workshop summary: for forbidden distance $k=(1-\gamma)n$, it records the Frankl--Rödl bound

\[ |A|\le (2-\Omega(\gamma^2))^n. \]

After division by $2^n$, this is an $\exp(-\Omega(\gamma^2n))$ density bound, and solving for the endpoint gap $\gamma n=n-k$ gives the scale $\sqrt{n\log(1/\alpha)}$. It does not give $\sqrt{\log(n/\alpha)}$.

Accordingly, the following is the **strongly supported reconstruction**, not a claim that the missing typography has been recovered verbatim:

Here $A+A=\{x+y:x,y\in A\}$, addition is in $\mathbb F_2^n$, and $|z|$ is ordinary integer Hamming weight. Since $x+y=0$ exactly when $x=y$, the fact that $0\in A+A$ for every nonempty $A$ is harmless when $k>0$. Density means $|A|/2^n$. The endpoint inequality makes the assertion vacuous when $\alpha$ is so small that the interval contains no integer.

来源字段：http://aimpl.org/addcombapp/1/

13747 | AMR-029-0064

### Partition median problem

O / open | AMR 问题清单汇编

Let P be the set of partitions of a ground set S. We allow two operations on P: (1) splitting a class into two arbitrary classes and (2) joining two classes into one. For two partitions X and Y, let us define the distance d(X,Y) as the minimum number of such operations transforming X to Y. Given partitions $X_1,X_2,\ldots,X_k\in P$, find a partition $Y\in P$ in polyinomial time minimizing the total distance, $\sum_{i=1}^k d(X_i,Y).$

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13748 | AMR-029-0065

### Partitioning a bipartite graph into proportional factors

O / open | AMR 问题清单汇编

Let G=(V,E) be a bipartite graph, and $c_1,\dots c_k$ positive reals whose sum is 1. Can E always be partitioned into k parts $E_1,\dots,E_k,$ so that for every $v \in V$ and every $i \in \{1,\dots,k\}$ we have $\lfloor c_i d_E(v) \rfloor \leq d_{E_i}(v) \leq \lceil c_i d_E(v) \rceil$?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13749 | GREEN-026

### Partitions and Large Cosets

O / open | Ben Green 问题

Suppose that $\mathbb{F}_2^n$ is partitioned into sets $A_1, \dots, A_K$. Does $2A_i$ contain a coset of codimension $O_K(1)$ for some $i$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13750 | OWR-4425-012

### Pattern Characterization of Fractional Powers in Words

O / open | Oberwolfach 报告问题汇编

For a noninteger real $\alpha>1$, does there exist a finite or infinite set $P$ of word patterns such that a word $x$ contains an occurrence of an $\alpha$-power if and only if some factor of $x$ matches a pattern in $P$ under a nonerasing morphism?

来源字段：https://doi.org/10.4171/owr/2010/37

13751 | OWR-16164-019

### Pattern-Avoidance Inequalities on Digitally Convex Shapes

P / partially_solved | Oberwolfach 报告问题汇编

Let $\lambda\subset\mathbb Z^2$ be a digitally convex shape of side $n$, and let $S_\lambda$ consist of the permutations $\sigma\in S_n$ whose graph lies in $\lambda$. Let $A_\lambda$ count the permutations in $S_\lambda$ that avoid every increasing triple $i_1<i_2<i_3$ for which both $(i_1,\sigma_{i_3})$ and $(i_3,\sigma_{i_1})$ lie in $\lambda$. Let $B_\lambda$ count the red-and-blue colorings of permutations in $S_\lambda$ that avoid both a red increasing pair $i_1<i_2$ with $(i_2,\sigma_{i_1})\in\lambda$ and a blue increasing pair $i_1<i_2$ with $(i_1,\sigma_{i_2})\in\lambda$. Is $B_\lambda\geq A_\lambda$ for every digitally convex shape $\lambda$?

来源字段：https://doi.org/10.4171/owr/2018/23

13752 | OPG-37196

### Perfect 2-error-correcting codes over arbitrary finite alphabets.

P / partially_solved | Open Problem Garden

Conjecture Does there exist a nontrivial perfect 2-error-correcting code over any finite alphabet, other than the ternary Golay code?

背景链接：http://www.openproblemgarden.org/op/perfect_2_error_correcting_codes_over_arbitrary_finite_alphabets

13753 | GAME-005

### Perfect Chess

O / open | 独立 / 未分配集合

What is the outcome of a perfectly played game of chess?

背景链接：https://en.wikipedia.org/wiki/Combinatorial_game_theory

13754 | AIM-COMBINATORICS-0273

### Perfect contractility and 1-amalgam blocks

P / partially_solved | AIM 研讨会问题汇编

**Conjecture.** 1-Amalgam decomposition is \(\mathcal C^{pc}\)-preserving.

来源字段：https://aimath.org/WWN/perfectgraph/perfectgraph.pdf

13755 | GAME-006

### Perfect Komi in Go

O / open | 独立 / 未分配集合

What is the perfect value of komi (compensation points) in Go?

背景链接：https://en.wikipedia.org/wiki/Go_(game

13756 | OWR-4425-002

### Perfection of Pattern-Avoiding Infinite-Word Spaces

O / open | Oberwolfach 报告问题汇编

Let $L$ be the set of infinite words over $\Sigma$ avoiding a pattern $p$. Is $L$ perfect?

来源字段：https://doi.org/10.4171/owr/2010/37

13757 | OWR-741-004

### Period Collapse in Ehrhart Quasipolynomials

O / open | Oberwolfach 报告问题汇编

Give necessary or sufficient conditions for the period of the Ehrhart quasipolynomial of a rational polytope to differ from its denominator $D$.

来源字段：https://doi.org/10.4171/owr/2004/39

13758 | AIM-COMBINATORICS-0101

### Periodic Laplacian certificates for robustness and explosion

P / partially_solved | AIM 研讨会问题汇编

Lionel Levine: “Suppose I give you a stable, periodic sandpile configuration \(c\) on the square grid \(\mathbb Z^2\). Is it decidable whether there is an integer \(n\) such that \(c+n\delta_0\) (i.e. the addition of \(n\) grains of sand to the origin) is not stabilizable?”

The original AIM PDF, *Problems from the AIM Chip-Firing Workshop*, was checked directly. Problem 11 reads:

Matt Baker then asks what the set of such \(n\) looks like. Levine says that, if it is nonempty, it has a minimum. Andrea Sportiello asks whether that minimum must depend on the period, and Levine observes that a suitable period-dependent bound would make the problem decidable.

No mathematical text is missing.

The intended finite encoding is taken to be two positive periods \(P,Q\), followed by the \(PQ\) heights \(c(i,j)\in\{0,1,2,3\}\) in a rectangular fundamental cell. The declared periods need not be minimal. The decision question is uniform in this finite input.

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13759 | OWR-2090-026

### Permanent Bounds for Diffuse Doubly Stochastic Matrices

O / open | Oberwolfach 报告问题汇编

Fix $\gamma\geq1$. If $A$ is an $n\times n$ doubly stochastic matrix with $\sum_{i,j}a_{ij}^2\leq\gamma$, is $\operatorname{per}(A)=n^{O(1)}e^{-n}$?

来源字段：https://doi.org/10.4171/owr/2008/44

13760 | OWR-14298158-008

### Permuton Limits for Consecutive Avoiders and Uniquely Sorted Permutations

P / partially_solved | Oberwolfach 报告问题汇编

What are the permuton limits of consecutive-pattern avoiders? What are the permuton limits of uniquely sorted permutations, and of uniquely sorted permutations that avoid a fixed pattern?

来源字段：https://doi.org/10.4171/owr/2024/6

13761 | OWR-1536-011

### Picker-Chooser and Maker-Breaker Strategy Transfers

P / partially_solved | Oberwolfach 报告问题汇编

For a hypergraph $(V,F)$, does Picker win the Picker-Chooser game whenever Maker, playing second, wins the corresponding Maker-Breaker game? Likewise, does Picker win the Chooser-Picker game whenever Breaker, playing second, wins the corresponding Maker-Breaker game?

来源字段：https://doi.org/10.4171/owr/2007/20

13762 | AMR-029-0066

### Polyhedral description of kernels

O / open | AMR 问题清单汇编

For which classes of digraphs can we explicitly give a linear description of the convex hull of kernels?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13763 | OWR-13490-014

### Polyhedrality of Cones of Nonnegative Valuations

O / open | Oberwolfach 报告问题汇编

For the cone $\overline{V}_+$ of lattice-invariant $h^*$-nonnegative valuations, is $\overline{V}_+$ a polyhedral cone, and in the stronger form is it linearly isomorphic to $\mathbb{R}_{\geq 0}^{d+1}$?

来源字段：https://doi.org/10.4171/owr/2015/5

13764 | OWR-4798-005

### Polynomial Bounds for Polytope Diameters

O / open | Oberwolfach 报告问题汇编

别名：Duplicate: Polynomial Bounds for Polytope Diameters [OWR-4798-022]

For $d$-polytopes with $n$ facets, is the maximum graph diameter bounded by a polynomial in $n$? Can a family have diameter growing superlinearly in its number of facets? Does every $d$-polytope with $n$ facets have diameter at most $d(n-d)$?

来源字段：https://doi.org/10.4171/owr/2011/08

13765 | AIM-COMBINATORICS-0062

### Polynomial bounds for rectilinear knot Ramsey numbers

P / partially_solved | AIM 研讨会问题汇编

Question: Is it polynomial in $k$?

Lower bound: $n(K) \geq \sqrt{\nu(K)}$, where $\nu(K)$ is the crossing number of a knot.

We can also ask the same question for links. For the simplest nontrivial link of two unknots (i.e., Hopf link), the answer is 6.

The preceding record supplies the missing antecedents. A **linear** (or rectilinear) spatial embedding of $K_n$ sends its vertices to points of $\mathbb R^3$ and every edge to the straight segment joining its endpoints, with distinct edges meeting only at common endpoints. For a knot or link type $L$, let $R(L)$ be the least $n$ such that every linear embedding of $K_n$ contains a cycle, or a vertex-disjoint union of cycles, equivalent to $L$. This is the quantity denoted $n(K)$ in the question. Negami's original equivalence permits an arbitrary homeomorphism of $\mathbb R^3$, and hence does not distinguish a chiral link from its mirror. He separately proves finiteness of the ambient-isotopy version $R^+(L)$.

The phrase “polynomial in $k$” needs a uniform quantifier. For knots define \[ N_{=}(k)=\max_{c(K)=k}R(K),\qquad N_{\leq}(k)=\max_{c(K)\leq k}R(K), \] where $c(K)$ is crossing number. These maxima are finite because there are only finitely many knot types with bounded crossing number and Negami proved every $R(K)$ finite. The natural question is whether constants $C,d$, independent of $K$, satisfy \[ R(K)\leq C(1+c(K))^d, \] equivalently whether $N_{\leq}(k)$ is polynomially bounded. Since \[ N_{\leq}(k)=\max_{0\leq j\leq k}N_{=}(j), \] a uniform polynomial bound for either formulation yields one for the other (after replacing a bound by its monotone envelope).

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13766 | OWR-2043-007

### Polynomial Freiman-Ruzsa Conjecture over Binary Vector Spaces

S / solved | Oberwolfach 报告问题汇编

If $A\subset\mathbb{F}_2^N$ is finite and $|A+A|\leq K|A|$, does there exist a subspace $H\subset\mathbb{F}_2^N$ such that $|H|\leq K^C|A|$ and $|H\cap A|\geq K^{-C}|A|$ for an absolute constant $C$?

来源字段：https://doi.org/10.4171/owr/2008/32

13767 | AIM-COMBINATORICS-0090

### Polynomial growth is false for both diagonal and off-diagonal readings

S / solved | AIM 研讨会问题汇编

For an integer $k \geq 1$, let $W(k)$ denote the minimum $n$ (the van der Waerden number $W(3,k)$) such that every $2$-coloring of $\{1, 2, \ldots, n\}$ contains either a monochromatic $3$-term arithmetic progression in the first color or a monochromatic $k$-term arithmetic progression in the second color.

**Question.** Does $W(k)$ grow polynomially in $k$?

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13768 | OWR-10252930-023

### Polynomial-Density Hales-Jewett Problem for Graph Families

O / open | Oberwolfach 报告问题汇编

If $\mathcal{F}$ is a family of graphs on $[n]$ such that $F_1\triangle F_2$ is not a clique for any distinct $F_1,F_2\in\mathcal{F}$, must $|\mathcal{F}|=o(2^{\binom n2})$?

来源字段：https://doi.org/10.4171/owr/2022/22

13769 | AIM-COMBINATORICS-0009

## Popular square differences: a radix obstruction and a sharp characteristic-two no-wrap model

P / partially_solved | AIM 研讨会问题汇编

**Popular Furstenberg--Sárközy.** For all $\epsilon>0$, there exists $N_0=N_0(\epsilon)$ such that for all $N\ge N_0$, if $A\subseteq[N]$ and $|A|\ge\delta N$, then there exists $y\ne0$ such that $$ |\{x:x,x+y^2\in A\}|\ge(\delta^2-\epsilon)N, $$ with $N_0(\epsilon)\le e^{1/\epsilon^c}$.

Its main paragraph reads:

It then asks for superpolynomial lower bounds on $N_0(\epsilon)$, or even $N_0(\epsilon)>\epsilon^{-10}$, and for upper or lower bounds over $\mathbb F_p[t]_{<n}$.

The displayed statement omits a quantifier on $\delta$. The natural uniform reading is $$ \forall\epsilon>0\ \exists N_0(\epsilon)\ \forall N\ge N_0\ \forall\delta\in[0,1]\ \forall A\subseteq[N]\quad |A|\ge\delta N\Longrightarrow\cdots . \tag{1} $$ This is equivalent to replacing $\delta$ in the conclusion by the actual density $\alpha=|A|/N$: one direction takes $\delta=\alpha$, and the other uses $\delta\le\alpha$. This formulation makes it clear why the threshold can depend only on $\epsilon$. The range $\delta^2\le\epsilon$ is trivial, since the requested lower bound is nonpositive.

Here $y$ is an integer. Since $y$ and $-y$ give the same shift, and since a positive count forces $y^2\le N-1$, the meaningful range is $$ 1\le y\le\lfloor\sqrt{N-1}\rfloor. \tag{2} $$ The variable $x$ can be taken over $\mathbb Z$; membership of both terms in $A\subseteq[N]$ automatically imposes $1\le x\le N-y^2$. This report uses the non-strict inequalities in the AIM record. The known literature often proves a strict inequality, which is stronger. Our counterexamples also have a strict gap, so endpoint conventions do not affect them.

An arbitrary admissible $N_0$ can always be enlarged, so a lower-bound question is meaningful only for the least threshold. Define $$ \mathcal N_{\mathbb Z}(\epsilon) =\min\{N_*:\text{the actual-density version holds for every }N\ge N_*\}. \tag{3} $$ Existence follows from known optimal polynomial-recurrence results.

来源字段：http://aimpl.org/highdimdiscrete/2/

13770 | AIM-COMBINATORICS-0022

## Popular-direction planarization for no-five-coplanar sumsets

P / partially_solved | AIM 研讨会问题汇编

Doubling of generic set of points

There is a result of Stancescu as follows: If $A\subset \mathbb{R}^2$ is a finite set with no three points on a line, then $$ |A+A|\geq |A|(\log |A|)^c $$ for some $c>0$. Also we know that the lower bound of $|A|^{1-\varepsilon}$ is false because of the Behrend construction. We would like to generalize this statement to higher dimensions.

Let $A\subset \mathbb{R}^3$ be a finite set with no five points on a plane. What is the best inequality $$ |A+A|\geq |A|f(A) $$ that we can have?

The exact AIM record says:

The source status says that $f(A)\geq\exp(\log^c|A|)$ is known, and that the Behrend construction does not imply $f(A)\leq|A|^c$.

The named author is Yonutz V. Stanchescu, not “Stancescu.” His 2002 paper proves, in particular, that if $X$ is an $m$-point planar set with no three collinear, then $$ |X+X|\geq \frac12m \left(\frac{m}{r_3(m)}\right)^{1/4}, \tag{1.1} $$ where $r_3(N)$ is the largest size of a subset of $\{1,\ldots,N\}$ with no nontrivial three-term arithmetic progression.

The sentence about a false $|A|^{1-\varepsilon}$ lower bound can only refer to the factor $f(A)$. Read literally as a lower bound for $|A+A|$, it is weaker than the elementary torsion-free inequality $|A+A|\geq2|A|-1$. The intended assertion is that one cannot hope in the planar problem for $$ f(A)\geq |A|^{1-\varepsilon}, $$ or equivalently $|A+A|\geq|A|^{2-\varepsilon}$, because Stanchescu's Behrend-type construction has $$ |A+A|\leq |A|\exp(C\sqrt{\log |A|}) =|A|^{1+o(1)} $$ for arbitrarily large sizes.

In this report, “plane” means an affine plane and all points in a forbidden five-tuple are distinct. The condition allows four coplanar points and three collinear points. For $|A|\geq5$, however, it forbids four collinear points: a line containing four points, together with any fifth point, lies in an affine plane.

来源字段：http://aimpl.org/addcombapp/2/

13771 | AIM-COMBINATORICS-0142

## Positive graph curvature requires a scale audit

P / partially_solved | AIM 研讨会问题汇编

What would be meant by a graph having *positive* curvature? (Grids don't have it; trees don't; how about spheres?)

来源字段：http://aimpl.org/largenetworks/1/

13772 | OWR-16164-014

### Positive Polynomial Comparisons for Unitriangular Conjugacy Classes

O / open | Oberwolfach 报告问题汇编

Put $T=q-1$ and define $A_n(T)$ by $$\sum_{n\geq0}A_n(T)\frac{x^n}{n!}=(1-\sin(Tx))^{-1/T}.$$ For an infinite family of prime powers $q$ for which the number $k(U_n(\mathbb F_q))$ of conjugacy classes in $U_n(\mathbb F_q)$ is polynomial in $q$, is $$k(U_n(\mathbb F_{T+1}))-A_n(T)\in\mathbb N[T]$$?

来源字段：https://doi.org/10.4171/owr/2018/23

13773 | OWR-14298010-005

### Positive Weighted $h^*$-Polynomials beyond Product Weights

O / open | Oberwolfach 报告问题汇编

For a lattice polytope $P$, characterize homogeneous polynomial weights beyond $R_P$, particularly weights in $S_P$, for which the weighted $h^*$-polynomial has nonnegative coefficients and is coefficientwise monotone under inclusions $P\subseteq Q$.

来源字段：https://doi.org/10.4171/owr/2023/51

13774 | OWR-14298158-013

### Positivity Under Pattern Avoidance in Matchings

P / partially_solved | Oberwolfach 报告问题汇编

For which matching patterns does avoidance preserve positivity, and what can be deduced for general avoided patterns?

来源字段：https://doi.org/10.4171/owr/2024/6

13775 | OWR-723-002

### Power-Law Fixed-Radius Circle Discrepancy

O / open | Oberwolfach 报告问题汇编

For every globally balanced red-blue coloring of the $n\times n$ square lattice viewed as a torus, and every $2<R<n/2$, must some translated circle of radius $R$ have color imbalance at least $cR^\alpha$ for absolute constants $c,\alpha>0$?

来源字段：https://doi.org/10.4171/owr/2004/13

13776 | OWR-16164-010

### Prevalence of Wilf Equivalence and Exponential Collapse

O / open | Oberwolfach 报告问题汇编

In natural families of finite structures, how common are Wilf equivalences? In particular, is exponential Wilf collapse typical, and what structural mechanism explains the prevalence of Wilf-equivalent objects?

来源字段：https://doi.org/10.4171/owr/2018/23

13777 | OWR-14299905-012

### Product-Free Sets in Symmetric and Alternating Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a permutation group $G$ and an integer $k$, determine the maximum density $m_k(G)$ of a $k$-product-free subset. In particular, determine $m_k(\operatorname{Sym}(n))$ and $m_k(\operatorname{Alt}(n))$ in the unresolved ranges of $k$.

来源字段：https://doi.org/10.4171/owr/2026/1

13778 | GREEN-017

### Progressions in F_3^n with Boolean Common Differences

P / partially_solved | Ben Green 问题

Suppose that $A \subset \mathbb{F}_3^n$ is a set of density $\alpha$. Under what conditions on $\alpha$ is $A$ guaranteed to contain a 3-term progression with nonzero common difference in $\{0, 1\}^n$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13779 | GREEN-009

### Progressions in Subsets of Z/NZ

P / partially_solved | Ben Green 问题

Is $r_5(N) \ll N(\log N)^{-c}$? Is $r_4(\mathbb{F}_5^n) \ll N^{1-c}$ where $N = 5^n$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13780 | AIM-COMBINATORICS-0226

### Pure-diagonal deficits in Lev's restricted Scherk conjecture

P / partially_solved | AIM 研讨会问题汇编

Thus the recovered problem is:

**Lev's restricted Scherk conjecture.** Let \(A,B\) be finite subsets of an abelian group \(G\), with \[ A\cap(-B)=\{0\}. \] For \[ A\mathbin{\dot+}B:=\{a+b:a\in A,\ b\in B,\ a\ne b\}, \] prove or disprove \[ |A\mathbin{\dot+}B|\ge |A|+|B|-3. \tag{1} \]

The hypothesis says that \(0\in A\cap B\) and that \(0=0+0\) is the unique representation of zero from \(A\times B\).

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13781 | OWR-4425-020

### PVHH-Cube Avoidance in a Morphic Fixed Point

S / solved | Oberwolfach 报告问题汇编

别名：Repeated Extraction of the PVHH-Cube Conjecture [OWR-4425-021]

Does the fixed point beginning at $0$ of the morphism $0\mapsto03$, $1\mapsto43$, $3\mapsto1$, $4\mapsto01$ avoid every factor $x_1x_2x_3$ with $|x_1|=|x_2|=|x_3|$ and equal sums of letters in $x_1$, $x_2$, and $x_3$?

来源字段：https://doi.org/10.4171/owr/2010/37

13782 | OWR-14604-008

## Quadratic-Length Universal Random Permutations

O / open | Oberwolfach 报告问题汇编

Let $f(k)$ be the minimum length for which a uniformly random sequence of reals is $k$-universal with high probability, meaning it contains every permutation in $S_k$ as a subsequence pattern. Is $f(k)\leq 1000k^2$?

来源字段：https://doi.org/10.4171/owr/2016/22

13783 | AIM-COMBINATORICS-0066

## Quantifier-resolved off-diagonal semi-algebraic graph Ramsey numbers

P / partially_solved | AIM 研讨会问题汇编

Question: (Fox) Is \(R_f^{(2)}(3,n)=n^{1+o(1)}\)?

The preceding records introduce \(R_f^{(k)}(s,n)\) as the unbalanced version of a Ramsey problem for a semi-algebraic relation, with the superscript recording uniformity. They do not say whether \(f\) is fixed once and for all or ranges over all descriptions of bounded complexity. That distinction is decisive here.

For a fixed symmetric predicate \(f\) on \(\mathbb R^d\), define \[ R_2^f(3,n)=\min\{N:\text{every }N\text{-point realization of }f\text{ contains a }K_3\text{ or an independent }n\text{-set}\}. \tag{1} \] If \(f\) is not symmetric, the nearby source convention evaluates \(f(p_i,p_j)\) for \(i<j\); then an ordering of the points is part of the structure. All new predicates used below are symmetric, so this issue does not affect the proofs.

Modern papers more often fix only a complexity tuple \(\mathbf t=(d,D,m)\). A graph has complexity \(\mathbf t\) if its vertices are points of \(\mathbb R^d\) and adjacency is a fixed Boolean combination of the signs of \(m\) polynomials, each of degree at most \(D\). Then \[ R_2^{\mathbf t}(3,n) \tag{2} \] is the least \(N\) forcing the Ramsey alternative in **every** graph of that complexity. The polynomials and Boolean formula may vary between graphs, whereas \(d,D,m\) do not vary with \(n\). A third notion, used explicitly by Jin--Tomon, fixes the entire description \(\Lambda=(f_1,\ldots,f_m;\Phi)\) and writes \(R_2^\Lambda(3,n)\).

Thus (1), (2), and the fixed-description version are not interchangeable. A uniform theorem for (2) implies the corresponding upper bound for each fixed description in that class. A single lower-bound construction for (2) says nothing about another specified \(f\), unless its description is identified.

Finally, \(R(3,n)=n^{1+o(1)}\) is a two-sided assertion. For a class that contains the empty graph on \(n-1\) vertices, \(R(3,n)\geq n\) is automatic, so a bound \(R(3,n)\leq n(\log n)^C\) establishes it. For one fixed predicate, the lower bound \(n\) is not universal: if \(f\equiv1\), then \(R_2^f(3,n)=3\) for every \(n\geq3\).

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13784 | AIM-COMBINATORICS-0013

## Quantifier-sensitive random barriers for dense linear tripartite patterns

P / partially_solved | AIM 研讨会问题汇编

**Turán exponents of hypergraphs.** There exists \(c>0\) such that the following holds. If \(H\) is an \(n\)-vertex, 3-uniform hypergraph with at least \(n^{3-c/d}\) edges, then \(H\) contains some/every 3-uniform, 3-partite linear hypergraph on \(d\) vertices with at least \(d^2/100\) edges.

The AIM record reads:

The slash in "some/every" cannot be silently resolved: it separates two different extremal problems. To avoid using \(H\) both for a host and a pattern, write \(\mathcal G\) for the host, put \[ k_d=\left\lceil\frac{d^2}{100}\right\rceil, \] and let \(\mathcal L_d\) be the finite family of isomorphism types of 3-uniform, 3-partite, linear hypergraphs on exactly \(d\) vertices with at least \(k_d\) edges. Isolated vertices are allowed under the usual meaning of "on \(d\) vertices"; none of the asymptotic results below depends on this convention.

With \(\operatorname{ex}_3(n,\mathcal F)\) denoting the maximum number of edges in an \(\mathcal F\)-free 3-graph, the two readings are:

* **Some:** \(\operatorname{ex}_3(n,\mathcal L_d)<n^{3-c/d}\). * **Every:** \(\max_{F\in\mathcal L_d}\operatorname{ex}_3(n,F)<n^{3-c/d}\).

Here "every" means that the host contains at least one copy of every isomorphism type \(F\); it does not mean that every labeled injection of \(F\) is present. The first extremal number is at most the minimum of the individual extremal numbers, whereas the second assertion is governed by their maximum.

The natural order of quantifiers is \[ \exists c>0\ \forall d\geq d_0\ \exists n_0(d)\ \forall n\geq n_0(d) \ \forall\mathcal G. \tag{1} \] In particular, \(d\) is fixed before \(n\to\infty\), and \(c\) is absolute. Letting \(d>n\) without such an order makes containment impossible. Small \(d\) can be handled separately, so the choice of a fixed \(d_0\) is immaterial to the intended asymptotic question.

来源字段：http://aimpl.org/highdimdiscrete/3/

13785 | AIM-COMBINATORICS-0219

## Quantitative Hales-Jewett bounds and exact binary calibration

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.13 (R. Graham).** Obtain "reasonable" bounds for the Hales--Jewett theorem and for the density version of it.

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13786 | AIM-COMBINATORICS-0030

### Quantitative transfer from PFR covers to dense Bogolyubov containment

P / partially_solved | AIM 研讨会问题汇编

Polynomial Freiman Ruzsa conjecture implies Polynomial Bogolyubov Ruzsa conjecture.

Prove that polynomial Freiman Ruzsa conjecture implies Polynomial Bogolyubov Ruzsa conjecture.

来源字段：http://aimpl.org/addcombapp/2/

13787 | AMR-029-0067

### Quasi-kernels and quasi-sinks

O / open | AMR 问题清单汇编

A quasi-kernel of a digraph $D$ is an independent vertex set $K$ sucht that every vertex is reachable from $K$ in $D$ by a path of length at most two. A quasi-sink of $D$ is a quasi-kernel of the digraph that we obtain by changing the direction of the edges in $D$. Is it true that for any infinite digraph $D=(V,A)$ there is a partition $\{V_1, V_2\}$ of $V$ such that $D[V_1]$ admits a quasi-kernel and $D[V_2]$ admits a quasi-sink?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13788 | GREEN-065

### Rado's Boundedness Conjecture

P / partially_solved | Ben Green 问题

Suppose $a_1, \dots, a_k$ are integers which do not satisfy Rado's condition. Is $c(a_1, \dots, a_k)$ bounded in terms of $k$ only?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13789 | OWR-14299905-005

### Radon Numbers for Unions of Convex Sets

P / partially_solved | Oberwolfach 报告问题汇编

What is the least integer $f(d,s,t)$ such that every set $P$ of $f(d,s,t)$ points in $\mathbb R^d$ admits a partition $P=A\sqcup B$ for which every $s$-convex set containing $A$ intersects every $t$-convex set containing $B$? Is $f(d,s,t)$ polynomial in its parameters?

来源字段：https://doi.org/10.4171/owr/2026/1

13790 | OPG-478

### Rainbow AP(4) in an almost equinumerous coloring

O / open | Open Problem Garden

Problem Do 4-colorings of $\mathbb{Z}_{p}$, for $p$ a large prime, always contain a rainbow $AP(4)$ if each of the color classes is of size of either $\lfloor p/4\rfloor$ or $\lceil p/4\rceil$?

背景链接：http://www.openproblemgarden.org/op/rainbow_ap_4_in_an_almost_equinumerous_coloring

13791 | AMR-029-0068

### Rainbow matchings in bipartite graphs

O / open | AMR 问题清单汇编

Given k disjoint matchings in a bipartite graph, a rainbow matching is a matching that contains one edge from each of them. Is it true that any family of k disjoint matchings of size k+1 has a rainbow matching?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13792 | AIM-COMBINATORICS-0161

### Rainbow reachability and the multi-label core obstruction

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.1 (Seymour, Sullivan).** Let $G$ be a simple digraph on vertex set $V$, and let $E_1,\ldots,E_k\subseteq E(G)$. Give an arc $e$ the label set $S_e\subseteq\{1,\ldots,k\}$, where $i\in S_e$ if and only if $e\in E_i$, and put $G_i=(V,E_i)$. Then either

1. $G$ contains a rainbow directed cycle; or 2. there is a vertex $v$ such that \[ \bigl|\{w:\text{there is a rainbow directed path of positive length from }v\text{ to }w\}\bigr| \geq \sum_{i=1}^k\delta^+_{G_i}(v). \]

The record comes from Blair D.

If each arc $e$ has a label set $S_e\subseteq[k]$, a directed path or directed cycle is **rainbow** when one can choose a label $\ell(e)\in S_e$ for every one of its arcs so that the chosen labels are pairwise distinct. Thus labels are assigned to arcs of the structure; an arc may carry several available labels.

With the PDF's typography restored, the statement is:

The notation is nonstandard but unambiguous in the PDF: $\delta^+_G$ without an argument is minimum outdegree, while $\delta^+_G(v)$ is explicitly defined to be the outdegree of the particular vertex $v$. Thus the right side is $\sum_i d^+_{G_i}(v)$, not a sum of global minimum degrees.

The positive-length convention, and hence exclusion of $v$ from its own reachability set, is forced by the document's stated implications. With $E_1=E_2=E(H)$, Conjecture 6.1 is claimed to imply Seymour's second-neighborhood conjecture exactly; including the trivial path would yield only $|N_2^+(v)|\geq |N^+(v)|-1$. Likewise, the document's deduction of Caccetta--Häggkvist uses the fact that the reachable set has size at most $|V|-1$.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13793 | COMB-003

### Ramsey Number R(5,5)

P / partially_solved | 独立 / 未分配集合

别名：Ramsey Number $R(5,5)$ [COMB-016]; Ramsey Number R(5,5) [COMB-022]

What is the exact value of $R(5,5)$, the smallest number $n$ such that any 2-coloring of the edges of $K_n$ contains a monochromatic $K_5$?

背景链接：https://arxiv.org/abs/2409.15709

13794 | OWR-1782-006

### Ramsey Sets and Finite Transitive Embeddings

O / open | Oberwolfach 报告问题汇编

A finite set $S$ in Euclidean space is Ramsey if every finite coloring of some sufficiently large Euclidean space contains a monochromatic isometric copy of $S$. Is $S$ Ramsey if and only if it embeds in a finite set whose isometry group acts transitively?

来源字段：https://doi.org/10.4171/owr/2008/01

13795 | AIM-COMBINATORICS-0037

### Random Beck-Fiala systems and the hereditary saturation obstruction

P / partially_solved | AIM 研讨会问题汇编

**Beck-Fiala for random set system.** Each element $i\in[n]$ lies in $t$ randomly selected sets from $\mathcal S$.

Its literature field says that when $m\ge n$ the hereditary discrepancy is $O(\sqrt{t\log t})$ with high probability, that when $n\ge m^t$ the hereditary discrepancy is $O(1)$, and asks about the remaining range $n^{1/t}<m<n$.

Their precise model is as follows. Let $X=[n]$ be the elements and let $\Sigma=\{S_1,\ldots,S_m\}$ be the sets, with $1\le t\le m$. Independently for every $x\in X$, choose a subset

$$ T_x\in\binom{[m]}t $$

uniformly among the $N=\binom mt$ possibilities, and put $x\in S_j$ exactly when $j\in T_x$. Equivalently, the $m\times n$ incidence matrix has independent columns, each uniform among the $0$--$1$ vectors of Hamming weight exactly $t$.

Thus there is **no replacement within a column**: an element belongs to exactly $t$ distinct sets. Columns are sampled **with replacement across elements**, and those column choices are independent. This is not the Bernoulli-entry model studied by Hoberg and Rothvoss [HR], where memberships are mutually independent with a prescribed probability.

Ezra--Lovett's first theorem is indeed hereditary. Their dense-regime theorem as actually stated in the paper is about ordinary discrepancy. The paper's abstract, and apparently the AIM synopsis copied from it, use “hereditary discrepancy” also for the dense regime, but Theorem 1.4/4 and its proof state only $\operatorname{disc}$. The distinction is mathematically essential and is resolved below by an explicit obstruction.

来源字段：http://aimpl.org/hereddiscrep/1/

13796 | GREEN-053

### Random Permutations Fixing k-Sets

P / partially_solved | Ben Green 问题

Let $p(k)$ be the limit as $n \to \infty$ of the probability that a random permutation on $[n]$ preserves some set of size $k$. Is $p(k)$ a decreasing function of $k$? Is $p(k) = (C + o(1))k^{-\alpha}(\log k)^{-3/2}$ for some absolute constant $C$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13797 | AIM-COMBINATORICS-0038

### Random sub-neighborhood discrepancy on the Boolean cube

P / partially_solved | AIM 研讨会问题汇编

**Hypercube set system.** Given a set system where the elements are the vertices of the hypercube \(\{0,1\}^n\). Corresponding to each vertex \(v\), there is a set \(S_v\) which is generated by picking a random subset of the neighbours of \(v\) in the hypercube. What is the discrepancy of this set system?

来源字段：http://aimpl.org/hereddiscrep/1/

13798 | AMR-029-0069

### Rank-respecting augmentation of hypergraphs with negamodular constraints

O / open | AMR 问题清单汇编

Given a crossing negamodular function $R:2^V\to \mathbb{Z}$ such that $R(X)\ne 1$ for every $X\subseteq V$ and a hypergraph $G_0=(V,\mathcal{E}_0)$, find a hypergraph $G=(V,\mathcal{E})$ of minimum total size such that $d_G(X)\ge R(X)-d_{G_0}(X)$ holds for every $X\subseteq V$ and the rank of $G$ does not exceed the rank of $G_0$.

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13799 | OWR-14298158-007

### Rationality Below Growth Rate Four

P / partially_solved | Oberwolfach 报告问题汇编

Is every finitely based permutation class of growth rate less than $4$ rational?

来源字段：https://doi.org/10.4171/owr/2024/6

13800 | OWR-14298158-002

### Rationality of Separable Permutation Subclasses

P / partially_solved | Oberwolfach 报告问题汇编

别名：Separable-Permutation Enumeration Introduction [OWR-14298158-010]

If a subclass of the separable permutations contains all of $\operatorname{Av}(231)$ or one of its symmetries, must its generating function be nonrational?

来源字段：https://doi.org/10.4171/owr/2024/6

13801 | OWR-4425-003

### Reachability and Power Avoidance in Infinite Words

P / partially_solved | Oberwolfach 报告问题汇编

For power-free words, does the reachability property hold for every prefix $u$ and suffix $v$ occurring infinitely often; does every curling-number sequence eventually reach $1$; and does the lexicographically least infinite word over $\mathbb{N}$ avoiding $5/2$-powers use only three letters?

来源字段：https://doi.org/10.4171/owr/2010/37

13802 | OWR-1782-017

### Real-Rooted Barycentric h-Polynomials and g-Theorem Conditions

P / partially_solved | Oberwolfach 报告问题汇编

If $P$ is a $d$-dimensional polytope, not necessarily simplicial, does the $h$-polynomial of the barycentric subdivision $\operatorname{sd}(\partial P)$ have only real roots? More generally, if $h(t)=h_0+h_1t+\cdots+h_dt^d$ has integer coefficients with $h_0=1$ and $h_i\geq1$, and all its roots are real, must $(h_0,\ldots,h_d)$ satisfy the classical $g$-theorem conditions?

来源字段：https://doi.org/10.4171/owr/2008/01

13803 | OWR-13490-008

### Real-Rootedness of Generalized Chebyshev Polynomials

O / open | Oberwolfach 报告问题汇编

For which triangulations $L$ are all roots of the generalized Tchebyshev polynomial $T_n^L(x)$ real for every $n$?

来源字段：https://doi.org/10.4171/owr/2015/5

13804 | OWR-14298007-014

### Realizability of M-Convex Functions

O / open | Oberwolfach 报告问题汇编

Give a natural definition of a realizable $M$-convex function and of a Plucker vector for a subspace arrangement, extending the corresponding notions for valuated matroids and $M$-convex sets.

来源字段：https://doi.org/10.4171/owr/2023/58

13805 | AIM-COMBINATORICS-0008

### Reciprocal constants and exact entropy-to-set bounds for easier arithmetic Kakeya

P / partially_solved | AIM 研讨会问题汇编

Let $S\subset\mathbb Z^n$ be such that for all $(d_1,\ldots,d_r)\in([N]^n)^r$, $S$ contains, for some translate $a_0$, the generalized arithmetic progression $$\{a_0+i_1d_1+\cdots+i_rd_r: i_1,\ldots,i_r\in\{0,1,\ldots,k-1\}\}.$$ Then $|S|\geq N^{c_kn+o_{k\to\infty}(1)}$.

The supplied AIM record, Problem 2.5 in the “Additive combinatorics” section of *High-dimensional phenomena in discrete analysis*, states:

The record adds that $r=1$ is known with exponent “$0.59n+o_k(1)$,” and calls the following formulation equivalent: if integer-valued random variables $X,Y_1,\ldots,Y_r$ obey $$H(X+a_1Y_1+\cdots+a_rY_r)\leq h \quad\text{for all }a_j\in\{0,1,\ldots,k-1\},$$ then $H(Y_1,\ldots,Y_r)\leq rhc_k$.

Several details are not specified and cannot be silently filled in.

* The displayed $o_{k\to\infty}(1)$ does not say how $N,n,r$ are quantified, is not multiplied by $n$, and uses $k$ both in $c_k$ and as the limiting variable. A precise asymptotic formulation should instead define a normalized exponent by a limit in $N$, and only then discuss its limit in $k$. * If the entropy multiplier is called $c_k$, its direct set consequence has exponent $1/c_k$, not $c_k$. Section 5 gives the exact calculation and a $k=2$ falsification of the same-symbol reading. * The entropy variables must be jointly defined. The standard primary sources assume that they are discrete and finitely supported; independence is not assumed. In the set-to-entropy application, the directions are chosen independently but $X$ is a deterministic function of all of them. * The problem is nontrivial only for $k\geq2$. For $k=1$, every nonempty $S$ works and the entropy hypothesis controls only $H(X)$. * The notation $[N]$ may mean $\{1,\ldots,N\}$ or $\{0,\ldots,N-1\}$. The proofs below use only $|[N]|=N$ and remain valid if zero is allowed. The source does not require the labeled directions to be distinct or linearly independent; neither do the proofs.

We therefore work with integers $k\geq2,r\geq1,n\geq1,N\geq1$, finite $S$, and all labeled direction tuples, including dependent or repeated ones.

来源字段：http://aimpl.org/highdimdiscrete/2/

13806 | AIM-COMBINATORICS-0194

### Reciprocal-weight cycles and a winding-number obstruction

S / solved | AIM 研讨会问题汇编

**Conjecture 6.24 (Zhang).** Let \(G\) be a digraph on \(n\) vertices, and \(f:E(G)\to\{0,1,\ldots\}\). If \[ \sum_{e\in E^+(v)} f(e)\ge \frac nk \qquad\text{for every }v\in V(G), \] then there is a directed cycle \(C\) such that \[ \sum_{e\in C}\frac1{f(e)}\le k. \]

The source immediately prints Charbit's counterexample: take the directed Cayley graph on \(\mathbb Z_8\) with generators \(\{1,2\}\), put weight \(1\) on the arcs \(x\to x+2\), weight \(2\) on the arcs \(x\to x+1\), and take \(k=3\).

Two defects in the printed text require explicit treatment.

1. The codomain includes zero although \(1/f(e)\) is then undefined. The most direct repair is \(f:E(G)\to\mathbb Z_{>0}\). Equivalently, one may set \(1/0=+\infty\); Lemma 1 below explains why zero-weight arcs can then simply be deleted. 2. The source says that the outgoing-weight sum in Charbit's example is \(4\). There is one outgoing 1-arc of weight \(2\) and one outgoing 2-arc of weight \(1\), so the correct sum is \[ 2+1=3, \] not \(4\). This is an arithmetic typo, not a fatal error, since \(3\ge8/3\).

The source does not state the domain of \(k\). The Caccetta--Haggkvist context indicates a positive integer, and all counterexamples below use integers \(k\ge2\). The sharp-threshold calculation also makes sense for real \(k>0\).

Primary source: Blair D. Sullivan, *A Summary of Problems and Results related to the Caccetta--Haggkvist Conjecture* (2006), arXiv:math/0605646, Section 6.6, Conjecture 6.24.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13807 | AMR-029-0070

### Recognition of Seymour graphs

O / open | AMR 问题清单汇编

A graph G is said to be a Seymour graph if for any edge set F that satisfies $|C\cap F|\le |C\setminus F|$ for every circuit C of G, there exist $|F|$ pairwise disjoint cuts each containing exactly one element of F. Can we decide in polynomial time whether a graph is Seymour?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13808 | AIM-COMBINATORICS-0223

### Recognizing self-sumsets over finite fields

S / solved | AIM 研讨会问题汇编

Problem 1.17 (A. Granville). Given a set in a finite field, how to determine (in rea-sonable time) whether it is a sumset of yet another set?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13809 | AIM-COMBINATORICS-0128

### Recognizing undirected reduced Laplacian lattices

P / partially_solved | AIM 研讨会问题汇编

If you have an \(n\times n\) integer matrix of full rank, is that the Laplacian of a graph? Well, of course not in general, but if you are allowed to do row operations you can always get a graph (in fact an infinite number of them) with this property. This is due to an algorithm of Wilmes [29]. But to accomplish this, you need to allow the resulting graph to be directed.”

“When is it the case that a submodule of \(\mathbb Z^n\) with rank \(n\) is the reduced Laplacian lattice of an undirected graph?

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13810 | OWR-1536-020

### Recycled Maker-Breaker Games and Kaplansky Four-in-a-Line

O / open | Oberwolfach 报告问题汇编

In the recycled Maker-Breaker game, if Breaker has a winning strategy, does that imply a winning strategy in the recycled game, in particular when Breaker's win is guaranteed by the Erdős-Selfridge theorem? Also, is the 4-in-a-line Kaplansky game, in which players alternately claim points of $\mathbb Z^2$ and the first to obtain four points on an opponent-free line wins, a draw?

来源字段：https://doi.org/10.4171/owr/2007/20

13811 | AMR-029-0071

### Red-blue cut problem

P / partially_solved | AMR 问题清单汇编

Given a directed graph whose arcs are coloured red and blue and integers r and b, can we decide in polynomial time whether the digraph has a cut with at most r red arcs and at most b blue arcs?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13812 | AIM-COMBINATORICS-0092

### Regular-matroid Jacobian sampling is solved; torsion obstructs the unrestricted simplicial version

P / partially_solved | AIM 研讨会问题汇编

This is Problem 2 in the problem list from the July 8--12, 2013 AIM workshop *Generalizations of chip-firing and the critical group*. The source PDF is the authoritative text [AIM13].

The mathematically precise reconstruction is as follows. Let

\[ D\in \mathbb Z^{n\times m},\qquad \operatorname{rank}D=n, \qquad L=DD^{\mathsf T}, \]

\[ J(D)=\mathbb Z^n/L\mathbb Z^n=\operatorname{coker}(L). \]

Suppose one can choose a uniformly random element of $J(D)$. Can one use it to choose a uniformly random column basis of $D$, efficiently? The discussion proposes regular matroids as the correct first generalization of graphs, observes via Cauchy--Binet that the cardinalities agree in the unimodular case, and asks for an efficient bijection. The final sentence says that it should suffice that the determinants of the maximal minors are $\pm1$.

There is an important literal-versus-intended distinction in that last sentence.

- If **every** maximal minor is literally $\pm1$, then no maximal minor vanishes, every $n$-subset of columns is a basis, and uniform sampling is just uniform sampling of an $n$-subset. - The nontrivial regular-matroid reading is that every **nonzero** maximal minor is $\pm1$, equivalently every maximal minor belongs to $\{0,\pm1\}$. This report calls that condition *maximal-minor unimodularity*.

The source's phrase “the size of a basis” in part (a) is likewise understood as “the number of bases.”

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13813 | GAME-010

### Rendezvous Problem

P / partially_solved | 独立 / 未分配集合

What is the optimal strategy for two agents to meet on a network without communication?

背景链接：https://arxiv.org/abs/1406.2795

13814 | OWR-1455-012

### Representation Models for Toric Cohomology and Symmetric Functions

P / partially_solved | Oberwolfach 报告问题汇编

Construct a direct isomorphism between the symmetric-group representation on the cohomology of the toric variety associated with the type-$A$ Coxeter complex and the sign-twisted representation on the homology of maximal open intervals in $(B_n\setminus\{\widehat{0}\})*C_n$. Also, for a partition $\lambda$ of $n$, is the symmetric function $Q_{\lambda,j}$ the Frobenius characteristic of a representation?

来源字段：https://doi.org/10.4171/owr/2007/04

13815 | AIM-COMBINATORICS-0177

### Resolving the average in DeVos's ordered-label conjecture

P / partially_solved | AIM 研讨会问题汇编

Conjecture 6.10. (Devos) Under the hypotheses of

**Conjecture 6.10 (DeVos).** Under the hypotheses of Conjecture 6.1, if there is no rainbow cycle in $G$ with strictly increasing edge labels, then the average number of vertices reachable from a fixed vertex $v$ by label-increasing (possibly trivial) paths is at least \[ 1+\sum_{i=1}^k\delta^+_{G_i}. \] **Note.** This can be proved when $G_i=\operatorname{Cayley}(\Gamma,A_i)$ for some group $\Gamma$, using Lemma 5.9.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13816 | AIM-COMBINATORICS-0005

### Restricted Roth progressions and their local obstructions

P / partially_solved | AIM 研讨会问题汇编

**Roth with common difference \(p-1\).** For all \(C>0\) and sufficiently large \(N\), if \(A\subset[N]\) with \(|A|\ge N/\log^C N\), then there exist \(x\) and \(p\) prime such that \[ x,x+p-1,x+2(p-1)\in A. \]

Its exact main statement is:

The record asks separately what happens if \(p-1\) is replaced by a positive integer that is a sum of two squares. It also proposes a model: for \[ W=\prod_{q\le w,\ q\ {\rm prime}}q, \qquad S_W=\{y\ge1:(y,W)=1\}, \] find \(x,x+y,x+2y\in A\) with \(y\in S_W\).

The main quantifiers are \[ \forall C>0\ \exists N_0(C)\ \forall N\ge N_0(C)\ \forall A\subset[N]. \] Thus \(C\) is fixed before \(N\to\infty\); the assertion is not uniform for a choice \(C=C(N)\). Containment forces \(1\le p-1\le (N-1)/2\), hence \(2\le p\le (N+1)/2\). The literal statement allows \(p=2\), which asks for three consecutive elements of \(A\). No lower bound tending to infinity is imposed on \(p\).

For the sum-of-two-squares variant, this report uses the standard convention \(y=u^2+v^2>0\) with \(u,v\in\mathbb Z\), not both zero. The periodic result below remains true even if both coordinates are required nonzero.

来源字段：http://aimpl.org/highdimdiscrete/2/

13817 | GREEN-002

### Restricted Sumset Problem

P / partially_solved | Ben Green 问题

Let $A \subset \mathbb{Z}$ be a set of $n$ integers. Is there a subset $S \subset A$ of size $(\log n)^{100}$ such that $S \hat{+} S$ is disjoint from $A$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13818 | GREEN-068

### Restricted Sumsets in Partitions

O / open | Ben Green 问题

For which values of $k$ is the following true: whenever we partition $[N] = A_1 \cup \dots \cup A_k$, we have $|\bigcup_{i=1}^k (A_i \hat{+} A_i)| \geq \frac{1}{10} N$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13819 | AIM-COMBINATORICS-0046

### Reverse Banaszczyk minimization and the ellipsoidal regime

P / partially_solved | AIM 研讨会问题汇编

Say that a convex body $K$ has property B if for any set of vectors of length no more than $c$ (some small constant), there exists a signing of them s.t. this signed sum lies inside $K$. What is the convex body with the smallest gaussian volume which has property $B$?

During the workshop it was shown that if $K$ has property $B$, then $\mathbb E\|G\|_K=(\log n)^{O(1)}$, where $G$ is a standard Gaussian random vector.

来源字段：http://aimpl.org/hereddiscrep/1/

13820 | AIM-COMBINATORICS-0029

### Riemann-profile transference for discrete entropy monotonicity

P / partially_solved | AIM 研讨会问题汇编

Let \(G\) be a torsion free abelian group. Suppose \(X\) is a random variable on \(G\) and \(S_n\) is sum of \(n\) many i.i.d. copies of \(X\). Let \(H(\cdot)\) be the discrete entropy function. We have the following theorem by T. Tao: \[ H(S_2)\geq H(S_1)+\log_2\sqrt 2-o(1), \] where the \(o(1)\) term goes to zero as \(H(X)\) grows to infinity. Show \[ H(S_n)\geq H(S_{n-1})+\log_2\sqrt{\frac n{n-1}}-o(1). \]

来源字段：http://aimpl.org/addcombapp/2/

13821 | AIM-COMBINATORICS-0168

### Rigidity of the deficit-one rainbow-cycle obstruction

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.5 (Seymour).** Under the hypotheses of Conjecture 6.1, if \(|V|=d\) and \[ \sum_{i=1}^{k} d^+_{G_i}(v)\ge d \] for all \(v\), then \(G\) must have a rainbow cycle.

**Note.** This conjecture is false if \(|V|=d\) is replaced by \(|V|=d+1\).

(Seymour) Under the hypotheses of". Reading the two records against page 6 of the source PDF recovers the complete statement:

The searchable PDF text suppresses the superscript \(+\) in the displayed formula for Conjecture 6.5. The intended quantity is nevertheless the outdegree: Conjecture 6.1 uses \(d^+_{G_i}(v)\), the source's opening notation defines \(\delta^+_{G_i}(v)\) as the outdegree of \(v\), and Conjecture 6.9 repeats the total-outdegree condition with an explicit \(+\). We use the modern notation \(d^+\) to avoid confusing a vertex degree with minimum degree.

The hypotheses inherited from Conjecture 6.1 are as follows. A finite simple digraph \(G=(V,E)\) is supplied with edge sets \(E_1,\ldots,E_k\subseteq E\), and \(G_i=(V,E_i)\). Equivalently, every arc \(e\) has a list \[ S_e=\{i\in[k]:e\in E_i\}. \] A directed cycle is *rainbow* if one can choose from the list of each of its arcs a different label. In the sharpness note, the same integer \(d\) remains on the right-hand side while the number of vertices is changed from \(d\) to \(d+1\).

This attempt owns only the continuation and sharpness note in record 0168. It uses record 0167 only to recover the broken sentence and does not edit or claim that record.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13822 | OWR-11695867-020

### Rigorous Asymptotics for Hamiltonian Cycles in Random Maps

O / open | Oberwolfach 报告问题汇编

Can the predicted asymptotics for Hamiltonian cycles in random maps be proved by probabilistic or combinatorial methods?

来源字段：https://doi.org/10.4171/owr/2022/57

13823 | OPG-58213

### Roller Coaster permutations

P / partially_solved | Open Problem Garden

Let $S_n$ denote the set of all permutations of $[n]=\set{1,2,\ldots,n}$. Let $i(\pi)$ and $d(\pi)$ denote respectively the number of increasing and the number of decreasing sequences of contiguous numbers in $\pi$. Let $X(\pi)$ denote the set of subsequences of $\pi$ with length at least three. Let $t(\pi)$ denote $\sum_{\tau\in X(\pi)}(i(\tau)+d(\tau))$.

A permutation $\pi\in S_n$ is called a Roller Coaster permutation if $t(\pi)=\max_{\tau\in S_n}t(\tau)$. Let $RC(n)$ be the set of all Roller Coaster permutations in $S_n$.

Conjecture For $n\geq 3$,

- If $n=2k$, then $|RC(n)|=4$. - If $n=2k+1$, then $|RC(n)|=2^j$ with $j\leq k+1$.

Conjecture (Odd Sum conjecture) Given $\pi\in RC(n)$,

- If $n=2k+1$, then $\pi_j+\pi_{n-j+1}$ is odd for $1\leq j\leq k$. - If $n=2k$, then $\pi_j + \pi_{n-j+1} = 2k+1$ for all $1\leq j\leq k$.

背景链接：http://www.openproblemgarden.org/op/roller_coaster_permutations

13824 | AMR-054-0068

### Rolling a Die over a Labeled Board

O / open | AMR 问题清单汇编

Label the faces of a unit cube with numbers $1$--$6$ as in a die. Place the cube to sit on an integer lattice grid, with one corner at the origin and sides aligned with the axes. Completely label every lattice square of a rectangular ``board'' $R$, whose corner is at the origin, with numbers in $\{1,2,3,4,5,6\}$. The problem is to roll the cube over its edges so that, for each square $s \in B$ labeled $l$, the cube lands on $s$ precisely once, and when it does so, the top face of the cube has label $l$. What is the computational complexity of solving an instance of this problem?

背景链接：https://topp.openproblem.net/p68

13825 | AMR-029-0072

### Rota's conjecture on disjoint bases

P / partially_solved | AMR 问题清单汇编

Let $M$ be a matroid of rank n whose ground set S can be partitioned into n disjoint bases $B_1,\dots,B_n$. Is it true that $B_1,\dots,B_n$ always have n disjoint transversals that are bases of $M$?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13826 | OPG-361

### Rota's unimodal conjecture

P / partially_solved | Open Problem Garden

Let $M$ be a matroid of rank $r$, and for $0 \le i \le r$ let $w_i$ be the number of closed sets of rank $i$.

Conjecture $w_0,w_1,\ldots,w_r$ is unimodal.

Conjecture $w_0,w_1,\ldots,w_r$ is log-concave.

背景链接：http://www.openproblemgarden.org/op/rotas_unimodal_conjecture

13827 | GREEN-010

### Roth's Theorem with Random Common Differences

P / partially_solved | Ben Green 问题

Let $S \subset \mathbb{N}$ be random. Under what conditions is Roth's theorem for progressions of length 3 true with common differences in $S$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13828 | AIM-COMBINATORICS-0122

### Rotor antiparticles and inverse topplings: a common quotient but different full dynamics

P / partially_solved | AIM 研讨会问题汇编

Jim Propp: “Is Andrea Sportiello's model of inverse topplings [12] the same as the particle creation and deletion model?”

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13829 | AMR-029-0073

### Rotor-routing halting problem

P / partially_solved | AMR 问题清单汇编

The rotor-routing halting problem asks the following: Given an initial chip-and-rotor configuration on a digraph, does the rotor-routing game eventually terminate? A more refined version of the problem is the halting configuration problem: If the rotor-routing game terminates, what is the final configuration?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13830 | AMR-031-0008

### Rudin's conjecture on squares in progressions

P / partially_solved | AMR 问题清单汇编

For positive integers $N,q,a$, let $Q(N;q,a)$ be the number of perfect squares among $a,a+q,\ldots,a+(N-1)q$, and let $Q(N)=\max_{q,a\geq1}Q(N;q,a)$. Prove that $Q(N)=O(\sqrt N)$. In the stronger form, prove that $Q(N)=Q(N;24,1)$ for every $N>6$.

背景链接：https://en.wikipedia.org/wiki/List_of_unsolved_problems_in_mathematics#Combinatorics

13831 | OWR-12697684-020

### Ryser-Brualdi-Stein Transversal Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

别名：Ryser-Brualdi-Stein Latin Square Transversal Conjecture [OWR-12697684-001]

Does every Latin square of order $n$ have a transversal of size $n-1$, and a full transversal when $n$ is odd?

来源字段：https://doi.org/10.4171/owr/2023/1

13832 | OWR-16763-027

### Ryser-Brualdi-Stein Transversals in Latin Squares

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Latin-Square Transversal Conjecture [OWR-16763-026]; Composite Duplicate of Two Rainbow-Matching Conjectures [OWR-16763-028]; Ryser-Brualdi-Stein Transversal Conjecture [OWR-10252930-010]

Does every Latin square of order $n$ have a partial transversal of size $n-1$, and a full transversal when $n$ is odd? Equivalently, does every proper $n$-edge-coloring of $K_{n,n}$ contain a rainbow matching of size $n-1$?

来源字段：https://doi.org/10.4171/owr/2019/1

13833 | AMR-029-0074

### S-T edge-connectivity augmentation

P / partially_solved | AMR 问题清单汇编

Given a digraph D=(V,A), two (not necessarily disjoint) subsets $S,T\subseteq V$ and a connectivity requirement k, develop a strongly polynomial time combinatorial algorithm for finding a minimum cardinality arc-set whose addition makes D k-edge-connected between S and T, that is, it contains k edge-disjoint paths from any node in S to any node in T.

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13834 | AMR-029-0075

### Sabidussi's compatibility conjecture

P / partially_solved | AMR 问题清单汇编

Let G=(V,E) be an Eulerian graph with minimum degree at least 4, and let W be a closed Eulerian walk of G. Is it true that G has a cycle decomposition such that no pair of consecutive edges of W appear in the same cycle of the decomposition?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13835 | AIM-COMBINATORICS-0282

### Safe edge deletion, modular unichord certificates, and the 3-core

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 2 (Conforti, Rao).** In every balanced graph, there exists an edge that can be deleted, so that the resulting graph remains balanced.

Hence, \(G(k,m)-e\) is still balanced for any edge \(e\), in agreement with Conjecture 2 from the section "Balanced Graphs".

来源字段：https://aimath.org/WWN/perfectgraph/perfectgraph.pdf

13836 | OPG-59928

### Saturated $k$-Sperner Systems of Minimum Size

P / partially_solved | Open Problem Garden

Question Does there exist a constant $c>1/2$ and a function $n_0(k)$ such that if $|X|\geq n_0(k)$, then every saturated $k$-Sperner system $\mathcal{F}\subseteq \mathcal{P}(X)$ has cardinality at least $2^{(1+o(1))ck}$?

背景链接：http://www.openproblemgarden.org/op/saturated_k_sperner_systems_of_minimum_size

13837 | OPG-60002

### Saturation in the Hypercube

P / partially_solved | Open Problem Garden

Question What is the saturation number of cycles of length $2\ell$ in the $d$-dimensional hypercube?

背景链接：http://www.openproblemgarden.org/op/saturation_in_the_hypercube

13838 | AIM-COMBINATORICS-0127

### Scaled intertwining, time change, and lumpability for harmonic Markov maps

P / partially_solved | AIM 研讨会问题汇编

James Propp: “What is known about harmonic maps (in the sense of Baker and Norine [8], Cory [15], etc.) between Markov chains?” Matt Baker comments that Urakawa's paper [33], cited in the original article, already recognizes a connection with random walks, and that there is also a notion of harmonic maps for metric graphs.

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13839 | AMR-029-0076

### Scrambled Rota conjecture

P / partially_solved | AMR 问题清单汇编

Let $M=(S,r)$ be a loopless matroid of rank k whose ground set can be partitioned into k bases. Is it true that no matter how we partition S into sets of size k, the partition will have k-1 disjoint transversals that are bases?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13840 | OWR-16931-023

### Second-Order Cut Bounds for Uniform Hypergraphs

P / partially_solved | Oberwolfach 报告问题汇编

Fix integers $k\geq r\geq2$, with either $r\geq3$ or $k\geq4$. Does every $k$-uniform hypergraph with $m$ edges have an $r$-cut of size at least $$\alpha_{r,k}m+\Omega(m^{2/3}),\qquad \alpha_{r,k}=\frac{S(k,r)r!}{r^k}$$?

来源字段：https://doi.org/10.4171/owr/2019/19

13841 | AIM-COMBINATORICS-0100

### Semisimple character partitions and a rational cycle obstruction

P / partially_solved | AIM 研讨会问题汇编

Lionel Levine: “This question is inspired by Criel Merino's work [25] relating the Tutte polynomial to chip-firing. The Tutte polynomial is the most general invariant of a graph that has a deletion-contraction recurrence. Is there a deletion-contraction recurrence of some sort for critical groups? It would have to be at the level of group algebras rather than groups themselves.” Jeremy Martin suggested starting with a cycle graph. Levine also asked whether the deletion--restriction exact sequence for hyperplane-arrangement algebras has a critical-group analogue.

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13842 | OPG-37226

### Sequence defined on multisets

S / solved | Open Problem Garden

Conjecture Define a $2 \times n$ array of positive integers where the first row consists of some distinct positive integers arranged in increasing order, and the second row consists of any positive integers in any order. Create a new array where the first row consists of all the integers that occur in the first array, arranged in increasing order, and the second row consists of their multiplicities. Repeat the process. For example, starting with the array $[1; 1]$, the sequence is: $[1; 1]$-> $[1; 2]$-> $[1, 2; 1, 1]$-> $[1, 2; 3, 1]$-> $[1, 2, 3; 2, 1, 1]$-> $[1, 2, 3; 3, 2, 1]$-> $[1, 2, 3; 2, 2, 2]$-> $[1, 2, 3; 1, 4, 1]$-> $[1, 2, 3, 4; 3, 1, 1, 1]$-> $[1, 2, 3, 4; 4, 1, 2, 1]$-> $[1, 2, 3, 4; 3, 2, 1, 2]$-> $[1, 2, 3, 4; 2, 3, 2, 1]$, and we now have a fixed point (loop of one array).

The process always results in a loop of 1, 2, or 3 arrays.

背景链接：http://www.openproblemgarden.org/op/a_sequence_defined_on_multisets

13843 | AMR-029-0077

### Serial symmetric exchanges

O / open | AMR 问题清单汇编

Let M be a matroid, and let A and B be two bases of M. A subset X of A and a subset Y of B, both of size k, form a serial symmetric exchange with respect to A and B if there is an ordering $x_1,\dots,x_k$ of X and an ordering $y_1,\dots,y_k$ of Y such that both $A \setminus \{x_1,\dots,x_i\} \cup \{y_1,\dots,y_i\}$ and $B \setminus \{y_1,\dots,y_i\} \cup \{x_1,\dots,x_i\}$ are bases for every $i \in \{1,\dots,k\}$. Is it true that for any matroid M, any two bases A and B, and any $X \subseteq A$, there exists $Y \subseteq B$ such that X and Y form a serial symmetric exchange with respect to A and B?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13844 | GREEN-070

### Sets with No Unique Sum Representations

P / partially_solved | Ben Green 问题

What is the size of the smallest set $A \subset \mathbb{Z}/p\mathbb{Z}$ (with at least two elements) for which no element in the sumset $A + A$ has a unique representation?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13845 | AIM-COMBINATORICS-0234

### Seven-corner completion sets

S / solved | AIM 研讨会问题汇编

**B. Green comments:** The answer to this is “no” as it stands. For example \(S\) could be a dissociated set. Tim, Terry and I [Green] tried to formulate a decent question along these lines but couldn't come up with anything we liked.

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13846 | OWR-11451-009

### Seymour One-Flowing Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Does Seymour’s $1$-flowing conjecture hold: does the max-flow min-cut relation hold for every binary matroid with no minor isomorphic to $\operatorname{AG}(3,2)$, $T_{11}$, or $T_{11}^*$?

来源字段：https://doi.org/10.4171/owr/2011/53

13847 | OWR-4425-008

### Sharp Bounds for Alternating Involution Periods

O / open | Oberwolfach 报告问题汇编

For a morphic involution $\theta$ on $A^*$, with $\tau_\theta(w)$ the maximum length of a $\theta$-unbordered factor and $\pi_\theta^{\mathrm{alt}}(w)$ the shortest alternating $\theta$-period, what is the sharp relation between these quantities, analogous to the ordinary unbordered-factor and period relation?

来源字段：https://doi.org/10.4171/owr/2010/37

13848 | AIM-COMBINATORICS-0059

### Sharp lower bound from the resolved Erdős--Rogers scale

P / partially_solved | AIM 研讨会问题汇编

Conjecture: (Fox--Grinshpun--Liebenau--Person--Szabó) \(s_r(k)=\Theta(r\,f_{k,k+1}(r)^2)\), where \(f_{k,k+1}\) is the Erdős--Rogers function.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13849 | AIM-COMBINATORICS-0001

### Sharp lower bounds for entropic subspace approximation

P / partially_solved | AIM 研讨会问题汇编

**Theorem.** Let \(X\) be any \(\mathbb F_2^n\)-valued random variable such that \(d[X,X]\leq \log K\), where \(d\) is the entropic Ruzsa distance. Then there is a subspace \(H\subset \mathbb F_2^n\) such that \(d[X,U_H]\leq 11\log K\), where \(U_H\) is uniform on \(H\).

Can we give lower bounds for \(d[X,U_H]\)?

来源字段：http://aimpl.org/highdimdiscrete/1/

13850 | AIM-COMBINATORICS-0010

### Sharp Sidorenko and finite embedding bounds for 3-uniform loose forests

P / partially_solved | AIM 研讨会问题汇编

**How badly can Sidorenko fail for hypergraphs.** Let \(H\) be a 3-uniform, 3-partite hypergraph. Then any \(n\)-vertex 3-uniform hypergraph with \(pn^3\) edges contains at least \(p^{C e(H)}n^{v(H)}\) copies of \(H\).

The AIM record asks:

The accompanying note says that, when \(H\) is contained in \(K_{t,t,t}^{(3)}\), one has the exponent \(t^3\), and compares this with the graph exponent \(e(H)+v(H)/2\).

There are three necessary clarifications.

1. The quantity \(p=e(G)/n^3\) is not the usual ordered edge density. For a simple 3-graph \(G\), put \[ \rho=t_{K_3^{(3)}}(G)=\frac{6e(G)}{n^3}=6p. \] If \(q=e(G)/\binom n3\) is the usual unordered density, then \(\rho=q(n)_3/n^3\). In particular the source's \(p\) is at most \(1/6\), up to the finite-\(n\) correction. 2. A universal assertion is meaningful only if \(C\) is an absolute constant, independent of \(H\). If \(C\) may depend on \(H\), the complete-partite envelope already gives such a bound. 3. The word "copies" cannot literally mean injective copies for every positive \(p\). More precisely, for every fixed simple \(H\) with at least two edges, every finite \(C\), and every \(n\geq v(H)\), let \(G\) consist of one edge and \(n-3\) isolated vertices. Then \(p=n^{-3}>0\), the claimed right side \(p^{Ce(H)}n^{v(H)}\) is strictly positive, but \(\operatorname{inj}(H,G)=0\). The density formulation naturally counts labeled homomorphisms; injective copies follow only above a finite-size threshold, with a collision error.

Accordingly, the normalized form of the open question is whether an absolute \(C\) satisfies \[ t_H(G):=\frac{\operatorname{hom}(H,G)}{n^{v(H)}} \geq \rho^{C e(H)} \tag{1} \] for every 3-partite 3-uniform \(H\) and every finite simple 3-graph \(G\).

来源字段：http://aimpl.org/highdimdiscrete/3/

13851 | AIM-COMBINATORICS-0207

### Sharp threshold and finite torsion certificates for torus difference sets

S / solved | AIM 研讨会问题汇编

Let \(a>0\). Is there a \(d_0=d_0(a)\) such that, if \(d>d_0\) and \(A\subset \mathbb T^d\) is closed with \(\mu(A)>a\), then \(A-A\) contains a subgroup?

Here \(\mathbb T^d=(\mathbb R/\mathbb Z)^d\) and \(\mu\) is normalized Haar measure. Literally, every difference set contains the trivial subgroup \(\{0\}\), so the only nonvacuous reading is “contains an infinite subgroup.” This is also the reading forced by the accompanying AIM remark, which says that Bourgain's examples have difference sets containing no *infinite* subgroup. Green's later formulation asks whether \(A-A\) contains a one-dimensional subgroup.

For a closed \(A\), these formulations are equivalent: \(A-A\) is compact, and if it contains an infinite abstract subgroup \(H\), it contains \(\overline H\). An infinite closed subgroup of a torus has positive-dimensional identity component and hence contains a circle subgroup \[ C_v:=\{tv\pmod {\mathbb Z^d}:t\in\mathbb R/\mathbb Z\} \] for some primitive \(v\in\mathbb Z^d\setminus\{0\}\). Conversely a circle is infinite. Finite nontrivial subgroups are not what the recovered problem asks about.

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13852 | OWR-12006-004

### Shellability of Barycentric Subdivisions of Convex Complexes

O / open | Oberwolfach 报告问题汇编

If $C$ is a convex simplicial complex, is its barycentric subdivision shellable?

来源字段：https://doi.org/10.4171/owr/2012/34

13853 | AMR-030-0018

### Show that every (1/2+ł)|E(ő_(n))| edges of the n-cube ő_(n) contains a C_(4) when n is sufficiently large

O / open | AMR 问题清单汇编

Erdős: Show that every (1/2+ł)|E(ő_(n))| edges of the n-cube ő_(n) contains a C_(4) when n is sufficiently large. (The best known value of ł is around .19 due to Chung.)

背景链接：https://people.math.sc.edu/cooper/combprob.html

13854 | AMR-030-0013

### Show that the discrepancy of any hypergraph H is at most c|E(H)|^(1/2)

S / solved | AMR 问题清单汇编

Beck: Show that the discrepancy of any hypergraph H is at most c|E(H)|^(1/2)

背景链接：https://people.math.sc.edu/cooper/combprob.html

13855 | AMR-030-0035

### Show that the inversion permutation, i

P / partially_solved | AMR 问题清单汇编

Propp: Show that the inversion permutation, i.e., the one which takes s to 1/s mod p, has longest increasing subsequence of length 2√ p(1+o(1)), i.e., the length it would have if the permutation were truly random.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13856 | AMR-030-0042

### Show that the jump number of a random linear extension of a grid poset (i

O / open | AMR 问题清单汇编

: Show that the jump number of a random linear extension of a grid poset (i.e., a product of chains) is close to the maximum w.h.p. (For the "symmetric grid" poset [m]^n, this is known for n = exp(o(log m)).)

背景链接：https://people.math.sc.edu/cooper/combprob.html

13857 | OPG-37167

### Shuffle-Exchange Conjecture

P / partially_solved | Open Problem Garden

别名：Shuffle-Exchange Conjecture (graph-theoretic form) [OPG-37089]

Given integers $k,n\ge2$, let $d(k,n)$ be the smallest integer $d\ge2$ such that the symmetric group $\frak S$ on the set of all words of length $n$ over a $k$-letter alphabet can be generated as $\frak S = (\sigma \frak G)^d:=\sigma\frak G \sigma\frak G \dots \sigma\frak G$ ( $d$ times), where $\sigma\in \frak S$ is the shuffle permutation defined by $\sigma(x_1 x_2 \dots x_{n}) = x_2 \dots x_{n} x_1$, and $\frak G$ is the exchange group consisting of all permutations in $\frak S$ preserving the first $n-1$ letters in the words.

Problem (SE) Evaluate $d(k,n)$.

Conjecture (SE) $d(k,n)=2n-1$, for all $k,n\ge2$.

背景链接：http://www.openproblemgarden.org/op/shuffle_exchange_conjecture

13858 | GREEN-074

### Sidon Set Size Bounds

P / partially_solved | Ben Green 问题

Write $F(N)$ for the largest Sidon subset of $[N]$. Improve, at least for infinitely many $N$, the bounds $N^{1/2} + O(1) \leq F(N) \leq N^{1/2} + N^{1/4} + O(1)$.

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13859 | AIM-COMBINATORICS-0073

### Simplex Ramsey numbers and a robust boundary-extension lemma

P / partially_solved | AIM 研讨会问题汇编

Estimate $r_k(k+1):=r_k(k+1,k+1)$.

For positive integers $s,t>k$, $r_k(s,t)$ is the least $N$ such that every red-blue coloring of $\binom{[N]}k$ contains either an $s$-set all of whose $k$-subsets are red or a $t$-set all of whose $k$-subsets are blue. Thus $$ r_k(k+1):=r_k(k+1,k+1) $$ is the least $N$ for which every two-coloring of the $k$-edges of $K_N^{(k)}$ contains a monochromatic $K_{k+1}^{(k)}$.

The complete $k$-graph $K_{k+1}^{(k)}$ consists of the $k+1$ facets of an abstract $k$-simplex, explaining the section title. There is no notation ambiguity in the record. The parameter tending to infinity is the uniformity $k$, while the desired monochromatic set has the minimum nontrivial size $k+1$.

The source’s stated lower bound $2^{\Omega(k)}$ was correct when recorded but is no longer current.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13860 | AIM-COMBINATORICS-0237

### Simultaneously avoiding bounded weighted averages

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.12 (Y. Stanchescu).** Fix an integer $t\geq1$ and suppose that $A\subseteq[N]$ is a set such that none of the $t^2$ equations $$ mx+ny=(m+n)z,\qquad 1\leq m,n\leq t, $$ has a non-trivial solution in the variables $x,y,z\in A$. How large can $A$ be under this assumption?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13861 | AIM-COMBINATORICS-0070

### Skeletal degeneracy and an asymmetric Ramsey transfer

P / partially_solved | AIM 研讨会问题汇编

Call a $k$-uniform hypergraph $d$-degenerate if the 2-graph obtained by replacing every edge by a graph clique $K_k^{(2)}$ is $d$-degenerate as a graph.

For fixed $k,d$, does every $n$-vertex $k$-uniform hypergraph $H$, whose pair shadow is $d$-degenerate, satisfy $r(H;2)=O_{k,d}(n)$?

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13862 | AMR-029-0078

### Skew-supermodular colouring with two class sizes

O / open | AMR 问题清单汇编

Let $p_1$ and $p_2$ be integer skew-supermodular set functions on ground set S such that $\max\{p_1(X),p_2(X)\}\leq \min\{|X|,k\}$ for every $X \subseteq S$, and let $m_1,m_2,n_1,n_2$ be positive integers such that $m_1 n_1+m_2 n_2=|S|$. Can we decide in polynomial time if there is a partition ${\mathcal P}$ of S with $m_1$ classes of size $n_1$ and $m_2$ classes of size $n_2$, such that $|\{Y \in {\mathcal P}: Y \cap X \neq \emptyset\}| \geq \max\{p_1(X), p_2(X)\}$ for every $X \subseteq S$?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13863 | AIM-COMBINATORICS-0209

### Sliding progression-free windows via gap words

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.3 (G. Freiman).** Fix an integer $s\geq3$ and suppose that $A=(a_1,a_2,\ldots)$ is a strictly increasing sequence of integers such that no segment $$ (a_{i+1},a_{i+2},\ldots,a_{i+s})\qquad(i=0,1,\ldots) $$ contains a three-term arithmetic progression. How large can the density of $A$ be? Describe all extremal sets.

It gives two examples:

1. for $s=4$, the nonnegative integers in the residue classes $0,1,3,4\pmod 6$, of density $2/3$; 2. for $s=8$, the nonnegative integers in the residue classes $0,1,3,4,9,10,12,13\pmod {18}$, of density $4/9$.

The source fixes several conventions implicitly rather than explicitly.

- The examples are subsets of the nonnegative integers, but the statement says only “integers.” Since an increasing one-sided sequence is bounded below by $a_1$, translating it makes it nonnegative and does not change any density or arithmetic-progression condition. - A three-term progression is necessarily **nontrivial**, with three distinct terms $x<y<z$ satisfying $x+z=2y$. Otherwise every nonempty block would contain a trivial progression. - Every sliding block of **exactly** $s$ consecutive sequence terms is meant, including the first block because $i=0$. A shorter collection of consecutive terms is automatically covered by extending it to an $s$-term block. - “Density” is not defined. For an arbitrary set the natural density need not exist. This report therefore optimizes the upper asymptotic density $$ \overline d(A)=\limsup_{N\to\infty} \frac{|A\cap[0,N]|}{N+1}. \tag{1.1} $$ When the natural density exists, it agrees with this value. Both examples are periodic and have natural density.

The two displayed moduli are redundant but correct: $$ \{0,1,3,4\}\pmod6=\{0,1\}\pmod3, $$ and $$ \{0,1,3,4,9,10,12,13\}\pmod {18} =\{0,1,3,4\}\pmod9. $$

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13864 | AIM-COMBINATORICS-0242

### Small doubling, tripling, and balanced quotients below two

S / solved | AIM 研讨会问题汇编

**Problem 3.5 (T. Tao).** Let \(A\) be a finite subset of a (not necessarily abelian) group. Given that \(A\cdot A\) is small, is \(A\cdot A\cdot A\) necessarily small, too?

It then records that Ben Green and Mei-Chu Chang pointed out the counterexample template \(A=H\cup\{x\}\), with \(H\) non-normal and \(x\) outside its normalizer. The source proposes two replacements:

1. Does \(A\) contain a large subset \(A'\) for which \((A')^3\) is small? 2. What inverse information follows from \(|A^2|<2|A|\)?

Here and below, “small” is interpreted uniformly: \(|A^2|\le K|A|\) for a fixed \(K\), and the desired bounds depend only on \(K\).

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13865 | AMR-029-0079

### Small quasi-kernels in directed graphs

P / partially_solved | AMR 问题清单汇编

Let D=(V,A) be a finite digraph in which every vertex has positive outdegree. A quasi-kernel is an independent set K such that every vertex of D can reach K by a directed path of length at most two. Must D have a quasi-kernel of size at most |V|/2?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13866 | AIM-COMBINATORICS-0248

### Small sums and products near the square-root scale

S / solved | AIM 研讨会问题汇编

Do there exist sets $A,B,C\subseteq \mathbb F_p$ with

\[ |A|,|B|,|C|\asymp \sqrt p \]

and $|A+B|+|AC|<p$?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13867 | AIM-COMBINATORICS-0228

### Small sumsets with no three collinear

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.3 (G. Freiman).** Suppose that \(A\subseteq \mathbb Z^2\) is a finite set, no three points of which are on a line. What is the smallest size of \(2A=A+A\), given that \(|A|=n\)?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13868 | AIM-COMBINATORICS-0125

### Small-density i.i.d. sandpiles stabilize

S / solved | AIM 研讨会问题汇编

(35) Leonardo Rolla: “If you start with an i.i.d. number of chips with some small density \(\mu\) in \(\mathbb Z^d\), is this configuration stabilizable with high probability?”

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13869 | AIM-COMBINATORICS-0060

### Solved by semi-random triangle-free graph packings

S / solved | AIM 研讨会问题汇编

Conjecture: \(s_r(3)=\Theta(r^2\log r)\).

Remark: This would follow from the extension of the triangle-free process to \(r\) colors.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13870 | AIM-COMBINATORICS-0054

### Solved diagonal online-Ramsey lower bound and a common-core obstruction to greedy Painter

S / solved | AIM 研讨会问题汇编

**Problem.** Show that \[ \widetilde r(n,n)\ge (\sqrt2+\epsilon)^n \] for some \(\epsilon>0\).

`\section{Erd\H{o}s--Hajnal for tournaments}`

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13871 | AIM-COMBINATORICS-0208

## Sparse relative hypergraph regularity and its necessary hypotheses

P / partially_solved | AIM 研讨会问题汇编

This is Problem 1.2, attributed to T. Tao, in the AIM workshop list *Recent trends in additive combinatorics* (collected by Ernie Croot and Vsevolod F. Lev). It asks whether a subset of large relative density in a pseudorandom sparse hypergraph satisfies a hypergraph regularity lemma, and observes that such a result could reprove the existence of arbitrarily long arithmetic progressions in the primes.

\[ |A|=|B|=N,\qquad G_0\subseteq A\times B, \]

$G_0$ is “sparsely $c(\varepsilon,\delta)$-quasirandom,” and $G\subseteq G_0$ has $|G|>\delta|G_0|$, then there are equitable partitions

\[ A=A_1\cup\cdots\cup A_s,\qquad B=B_1\cup\cdots\cup B_t, \]

with $s,t<C(\varepsilon,\delta)$, such that $G$ is $\varepsilon$-regular relative to $G_0$ on at least $(1-\varepsilon)st$ pairs $A_i\times B_j$.

More importantly, the source does not formally define “sparsely quasirandom” or the desired hypergraph regularity. The literal question is therefore not one fully quantified proposition: the answer depends on the pseudorandomness hypothesis and on whether “regularity” is required merely to decompose an edge function or also to support a counting/removal lemma.

The presenter remarks that, despite the difficulty of dense hypergraph regularity, the correct sparse induction might be cleaner.

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13872 | AIM-COMBINATORICS-0089

## Sparse witnesses and off-diagonal van der Waerden numbers

P / partially_solved | AIM 研讨会问题汇编

Question: Does $W(n)-W^*(n)\to\infty$?

Off diagonal van der Waerden numbers $W(k,3)$

From computed values, $W(k,3)$ seems to grow quadratically in $k$.

Known: $k^{2-o(1)}<W(k)<e^{O(k\log^5 k)}$.

Here $W(n)$ is the least $N$ such that every two-colouring of $[N]$ has a monochromatic $n$-term arithmetic progression, while $W^*(n)$ is the least cardinality of an arbitrary finite set $S\subset\mathbb Z$ such that every two-colouring of $S$ has such a progression. It also records $W^*(3)=W(3)=9$ and $W^*(4)\le 27<35=W(4)$.

1. Is the saving from allowing an arbitrary set unbounded, i.e. $W(n)-W^*(n)\to\infty$? 2. Does the off-diagonal number $W(k,3)$ have quadratic order?

In the last displayed bound of the record, the occurrence of $W(k)$ is almost certainly a transcription error for $W(k,3)$. All conclusions below use this explicitly labelled reconstruction.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13873 | AMR-029-0081

## Sparsifier subgraphs

P / partially_solved | AMR 问题清单汇编

Devise combinatorial polynomial-time algorithms for the following two problems. Given a graph G, find a subgraph H with $O(n)$ edges such that $d_H(X) \geq \Omega(\frac{n}{m}) d_G(X)$ for every $X \subseteq V$ Given a graph G, find a subgraph H with $\Omega(n)$ edges such that $d_H(X) \leq O(\frac{n}{m}) d_G(X)$ for every $X \subseteq V$

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13874 | OWR-710-004

## Spherical Designs with Square Cardinality

O / open | Oberwolfach 报告问题汇编

Do there exist spherical designs with $(n+1)^2$ points that are exact for all polynomials of degree $n$?

来源字段：https://doi.org/10.4171/owr/2004/27

13875 | OPG-37228

## Square achievement game on an n x n grid

P / partially_solved | Open Problem Garden

Problem Two players alternately write O's (first player) and X's (second player) in the unoccupied cells of an $n \times n$ grid. The first player (if any) to occupy four cells at the vertices of a square with horizontal and vertical sides is the winner. What is the outcome of the game given optimal play? Note: Roland Bacher and Shalom Eliahou proved that every 15 x 15 binary matrix contains four equal entries (all 0's or all 1's) at the vertices of a square with horizontal and vertical sides. So the game must result in a winner (the first player) when n=15.

背景链接：http://www.openproblemgarden.org/op/a_game_on_an_n_x_n_grid

13876 | AIM-COMBINATORICS-0243

## Square-difference sets, Paley cliques, and an exact external-defect variance identity

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.6 (brought by V. Lev).** Given a prime \(p\), how large can a set \(A\subseteq\mathbb F_p\) be, given that the difference between any two elements of \(A\) is a quadratic residue modulo \(p\)?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13877 | OWR-14604-001

## Square-Root Box Discrepancy for Multidimensional Permutations

O / open | Oberwolfach 报告问题汇编

For every $d\geq2$, do there exist arbitrarily large $d$-dimensional permutations $A$ such that for every box $T\subseteq[n]^{d+1}$, $|A(T)-\operatorname{vol}(T)/n|=O(\sqrt{\operatorname{vol}(T)})$?

来源字段：https://doi.org/10.4171/owr/2016/22

13878 | GREEN-055

### Stable Density on Subspaces

P / partially_solved | Ben Green 问题

Let $A \subset \mathbb{F}_2^n$. If $V$ is a subspace, write $\alpha(V)$ for the density of $A$ on $V$. Is there some $V$ of moderately small codimension on which $\alpha$ is stable?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13879 | OWR-16164-013

### Stanley Rank-Size Bounds for Differential Posets

O / open | Oberwolfach 报告问题汇编

For every $r$-differential poset with rank sizes $p_n$, improve the bound $$p_n\leq1+r\sum_{k=0}^{n-2}p_k+(r-1)p_{n-1}$$ toward Stanley's conjectured inequality $p_n\leq rp_{n-1}+p_{n-2}$.

来源字段：https://doi.org/10.4171/owr/2018/23

13880 | AMR-156-0009

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 10: Combinatorial interpretation of Kronecker product coefficients of symmetric group characters

O / open | AMR 问题清单汇编

Let $\chi^\lambda$ and $\chi^\mu$ be irreducible characters of the symmetric group $S_n$ indexed by partitions $\lambda,\mu\vdash n$, and let $g_{\lambda\mu\nu} = \langle \chi^\lambda \chi^\mu, \chi^\nu\rangle$ be the Kronecker product coefficients, equivalently defined by $\prod_{i,j,k}(1-x_i y_j z_k)^{-1} = \sum_{\lambda,\mu,\nu} g_{\lambda\mu\nu} s_\lambda(x)s_\mu(y)s_\nu(z)$. Find a positive combinatorial interpretation of the Kronecker coefficients $g_{\lambda\mu\nu}$, thereby combinatorially reproving that they are nonnegative.

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13881 | AMR-156-0010

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 11: Combinatorial interpretation of Schubert polynomial structure constants

O / open | AMR 问题清单汇编

Let $\mathfrak{S}_w$ denote the Schubert polynomial indexed by a permutation $w\in S_\infty$. In the polynomial ring $\mathbb{Z}[x_1,x_2,\dots]$, the product of two Schubert polynomials expands as $\mathfrak{S}_u \mathfrak{S}_v = \sum_w c_{uv}^w \mathfrak{S}_w$, where the structure constants $c_{uv}^w$ are known to be nonnegative integers via intersection theory on the flag variety $\operatorname{GL}(n,\mathbb{C})/B$. Find a positive combinatorial interpretation of the Schubert intersection coefficients $c_{uv}^w$, thereby combinatorially reproving their nonnegativity.

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13882 | AMR-156-0011

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 12: Combinatorial interpretation of the row sums of the character table of S_n

O / open | AMR 问题清单汇编

Let $S_n$ act on itself by conjugation, with character $\psi_{S_n}$. The multiplicity of an irreducible character $\chi^\lambda$ of $S_n$ in $\psi_{S_n}$ is equal to the row sum $\sum_K \chi^\lambda(K)$ over all conjugacy classes $K$ of $S_n$, which is therefore a nonnegative integer. Give a combinatorial interpretation of the row sums $\sum_K \chi^\lambda(K)$ of the character table of $S_n$, thereby combinatorially reproving that they are nonnegative.

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13883 | AMR-156-0012

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 13: Positivity of the (q,t)-Kostka polynomials

S / solved | AMR 问题清单汇编

Let $K_{\lambda\mu}(q,t)$ denote the $(q,t)$-Kostka polynomials defined via the integral form Macdonald symmetric functions $J_\mu(x;q,t) = \sum_\lambda K_{\lambda\mu}(q,t) s_\lambda[X(1-t)]$. Show that $K_{\lambda\mu}(q,t) \in \mathbb{N}[q,t]$, i.e., that $K_{\lambda\mu}(q,t)$ has nonnegative integer coefficients.

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13884 | AMR-156-0013

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 14: Combinatorial symmetry and Schur-positivity of LLT ribbon polynomials

P / partially_solved | AMR 问题清单汇编

For a skew shape $\lambda/\mu$ tileable by $k$-ribbons, let $H^{(k)}_\lambda(x;q) = \sum_T q^{s(T)} x^T$ be the Lascoux-Leclerc-Thibon (LLT) $k$-ribbon generating polynomial summed over $k$-ribbon tableaux $T$ of shape $\lambda$ with spin $s(T)$.

(a) Give a combinatorial proof that the polynomials $H^{(k)}_\lambda(x;q)$ are symmetric functions.

(b) When $H^{(k)}_\lambda(x;q)$ is expanded in terms of Schur functions, are the coefficients polynomials in $q$ with nonnegative integer coefficients?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13885 | AMR-156-0014

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 15: Kazhdan-Lusztig polynomial nonnegativity for arbitrary Coxeter groups

S / solved | AMR 问题清单汇编

Let $W$ be an arbitrary Coxeter group and let $u \le v$ be elements in the Bruhat order of $W$. Are the coefficients of the Kazhdan-Lusztig polynomial $P_{u,v}(q)$ nonnegative integers?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13886 | AMR-156-0015

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 16: Combinatorial interpretation of Kazhdan-Lusztig polynomial coefficients for S_n

O / open | AMR 问题清单汇编

Give a direct positive combinatorial interpretation of the coefficients of the Kazhdan-Lusztig polynomial $P_{u,v}(q)$ when $W$ is a finite or affine Weyl group, especially for the symmetric group $W = S_n$, thereby combinatorially reproving that they are nonnegative.

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13887 | AMR-156-0017

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 17': h-positivity of the permanent-like generating symmetric function of a totally nonnegative matrix

O / open | AMR 问题清单汇编

Let $A=(a_{ij})$ be a totally nonnegative $n\times n$ real matrix, and define the symmetric function $$F_A = \sum_{w\in S_n} a_{1,w(1)}a_{2,w(2)}\cdots a_{n,w(n)} p_{\rho(w)},$$ where $p_{\rho(w)}$ is the power sum symmetric function indexed by the cycle type of $w$. Is $F_A$ $h$-positive (i.e., does its expansion in the basis $\{h_\lambda\}$ of complete homogeneous symmetric functions have nonnegative coefficients)?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13888 | AMR-156-0016

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 17: Stembridge's monomial immanant positivity conjecture for totally nonnegative matrices

O / open | AMR 问题清单汇编

For a partition $\lambda \vdash n$, let $\phi^\lambda$ denote the unique class function on $S_n$ whose Frobenius characteristic $\operatorname{ch}\phi^\lambda = \frac{1}{n!}\sum_{w\in S_n}\phi^\lambda(w)p_{\rho(w)}$ is the monomial symmetric function $m_\lambda$, and define the immanant $\operatorname{Imm}_{\phi^\lambda}(A) = \sum_{w\in S_n} \phi^\lambda(w) a_{1,w(1)}\cdots a_{n,w(n)}$. If $A$ is a totally nonnegative $n\times n$ real matrix (every minor of $A$ is nonnegative) and $\lambda \vdash n$, is $\operatorname{Imm}_{\phi^\lambda}(A) \ge 0$?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13889 | AMR-156-0018

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 18: Stembridge's Schur-positivity conjecture for monomial immanants of Jacobi-Trudi matrices

O / open | AMR 问题清单汇编

Let $\mu$ and $\nu$ be partitions with $\nu \subseteq \mu$ and $\ell(\mu) \le n$, and let $H_{\mu/\nu} = [h_{\mu_i - \nu_j + j - i}]_{i,j=1}^n$ be the Jacobi-Trudi matrix. For any partition $\lambda \vdash n$, is the monomial immanant $\operatorname{Imm}_{\phi^\lambda}(H_{\mu/\nu})$ Schur-positive ($s$-positive)?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13890 | AMR-156-0019

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 19: Haiman's unimodality and positivity conjecture for Hecke algebra monomial characters on Kazhdan-Lusztig basis elements

O / open | AMR 问题清单汇编

Let $H_n(q)$ be the Iwahori-Hecke algebra of type $A_{n-1}$, let $\phi^\lambda$ be the monomial virtual character of $H_n(q)$ corresponding to $\lambda \vdash n$, and let $C'_w$ be the Kazhdan-Lusztig basis element indexed by $w\in S_n$. Is it true that $\phi^\lambda(q^{\ell(w)/2} C'_w)$ is a polynomial in $q$ with nonnegative integer coefficients, and moreover that these coefficients are unimodal and symmetric about $q^{\ell(w)/2}$?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13891 | AMR-156-0001

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 1: Generalized Lower Bound Theorem and g-theorem for Gorenstein* complexes

S / solved | AMR 问题清单汇编

Let $\Delta$ be a $(d-1)$-dimensional Gorenstein* simplicial complex (i.e., for every face $F\in\Delta$, including $F=\varnothing$, the reduced simplicial homology $\tilde{H}_i(\operatorname{link}(F);\mathbb{Q})$ is $\mathbb{Q}$ if $i=\dim\operatorname{link}(F)$ and $0$ otherwise), with $f$-vector $(f_0,\dots,f_{d-1})$ and $h$-vector $(h_0,h_1,\dots,h_d)$ defined by $\sum_{i=0}^d f_{i-1}(x-1)^{d-i}=\sum_{i=0}^d h_i x^{d-i}$. Does the Generalized Lower Bound Theorem $h_0\le h_1\le \cdots \le h_{\lfloor d/2\rfloor}$ (or more generally McMullen's $g$-theorem) hold for Gorenstein* complexes (and in particular for triangulations of spheres, PL-spheres, and complete simplicial fans)?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13892 | AMR-156-0021

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 20': Real-rootedness of the chain polynomial of a finite distributive lattice

S / solved | AMR 问题清单汇编

For a finite poset $P$, let $c_k(P)$ denote the number of $k$-element chains in $P$ (with $c_0=1$), and define the chain polynomial $C(P,t) = \sum_{k\ge 0} c_k(P) t^k$. When $L$ is a finite distributive lattice, is every zero of $C(L,t)$ real?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13893 | AMR-156-0020

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 20: Neggers-Stanley Poset Conjecture on real zeros of the P-Eulerian polynomial W(P,x)

S / solved | AMR 问题清单汇编

Let $P$ be a finite poset on $n$ elements, let $\mathcal{L}(P)$ denote the set of linear extensions of $P$ (for a natural labeling, or via $P$-partitions in general), and let $W(P,x) = \sum_{w\in\mathcal{L}(P)} x^{d(w)}$ be the order polynomial generating the descent statistic $d(w)$. Is every zero of $W(P,x)$ real?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13894 | AMR-156-0022

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 21: Stanley-Stembridge e-positivity conjecture for chromatic symmetric functions of (3+1)-free posets

O / open | AMR 问题清单汇编

Let $P$ be a finite $(3+1)$-free poset (i.e., $P$ contains no induced subposet isomorphic to the disjoint union of a $3$-element chain and a $1$-element chain), and let $G = \operatorname{inc}(P)$ be its incomparability graph. Is the chromatic symmetric function $X_G(x) = \sum_\kappa x_1^{a_1(\kappa)} x_2^{a_2(\kappa)}\cdots$ of $G$ $e$-positive (i.e., a nonnegative linear combination of elementary symmetric functions $e_\lambda$)?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13895 | AMR-156-0023

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 22: Gasharov's Schur-positivity conjecture for chromatic symmetric functions of claw-free graphs

O / open | AMR 问题清单汇编

Let $G$ be a finite claw-free graph (i.e., $G$ has no induced subgraph isomorphic to $K_{1,3}$). Is the chromatic symmetric function $X_G(x)$ of $G$ Schur-positive ($s$-positive)?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13896 | AMR-156-0024

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 23: Real-rootedness of the independence (stable set) polynomial of a claw-free graph

S / solved | AMR 问题清单汇编

For a finite graph $G$, let $c_k$ be the number of stable (independent) $k$-element subsets of vertices of $G$, and define the stable set (independence) polynomial $D(G,t) = \sum_{k\ge 0} c_k t^k$. If $G$ is a claw-free graph, does $D(G,t)$ have only real zeros?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13897 | AMR-156-0025

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 24: Haglund-Ono-Wagner real-rootedness conjecture for per(A + xJ)

S / solved | AMR 问题清单汇编

Let $A$ be an $n\times n$ real matrix for which every column is weakly increasing (and nonnegative), and let $J$ denote the $n\times n$ matrix of all $1$'s. Does the rook/permanent polynomial $\operatorname{per}(A + xJ)$ have only real zeros?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13898 | AMR-156-0026

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 25: Unimodality and log-concavity of matroid and geometric lattice sequences (Read-Hoggar-Welsh-Mason conjectures)

S / solved | AMR 问题清单汇编

Are the following sequences unimodal or log-concave for every finite graph or matroid?

(a) The absolute values of the coefficients of the chromatic polynomial of a graph, or more generally, the characteristic polynomial of a matroid.

(b) The number of $i$-edge spanning forests of a graph, or more generally, the number of $i$-element independent sets of a matroid.

(c) The number of elements of rank $i$ of a geometric lattice (the Whitney numbers of the second kind).

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13899 | AMR-156-0002

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 2: GLBT and nonnegativity for the toric h-vector of convex polytopes and Gorenstein* lattices

P / partially_solved | AMR 问题清单汇编

Let $\mathcal{P}$ be an arbitrary convex polytope (not necessarily rational or simplicial), or more generally a Gorenstein* poset that is a lattice with bottom element $\hat{0}$ and top element $\hat{1}$ removed. Does the toric $h$-vector (generalized $h$-vector) $h(\mathcal{P})=(h_0,h_1,\dots,h_d)$ of $\mathcal{P}$ satisfy the Generalized Lower Bound Theorem $h_0\le h_1\le \cdots \le h_{\lfloor d/2\rfloor}$, or at least $h_i\ge 0$ for all $i$?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13900 | AMR-156-0003

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 4: Charney-Davis conjecture for Gorenstein* flag complexes

O / open | AMR 问题清单汇编

Let $\Delta$ be a $(2e-1)$-dimensional Gorenstein* flag simplicial complex (i.e., a Gorenstein* complex in which every minimal non-face has two vertices) with $h$-vector $(h_0,h_1,\dots,h_{2e})$. Is it true that $$(-1)^e(h_0 - h_1 + h_2 - \cdots + h_{2e}) \ge 0?$$

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13901 | AMR-156-0004

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 5: Interval partitioning of simplicial complexes with acyclic low-dimensional links

S / solved | AMR 问题清单汇编

Suppose that $\Delta$ is a finite abstract simplicial complex such that the link of every face $F$ of $\Delta$ of dimension at most $j$ (including $\operatorname{link}(\varnothing)=\Delta$) is acyclic over $\mathbb{Q}$. Can $\Delta$ (regarded as a poset under set inclusion) be partitioned into intervals $[F,F']$ such that (a) $\dim F' - \dim F = j+2$, and (b) the bottom elements $F$ of these intervals form a subcomplex of $\Delta$?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13902 | AMR-156-0005

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 6: Partitionability of Cohen-Macaulay simplicial complexes

S / solved | AMR 问题清单汇编

Let $\Delta$ be a pure $(d-1)$-dimensional simplicial complex that is Cohen-Macaulay over $\mathbb{Q}$ (i.e., $\tilde{H}_i(\operatorname{link}(F);\mathbb{Q})=0$ for all $F\in\Delta$ and $i\ne \dim\operatorname{link}(F)$). Is every Cohen-Macaulay simplicial complex partitionable into intervals $[F,F']$ where each $F'$ is a facet (maximal face) of $\Delta$?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13903 | AMR-156-0006

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 7: Nonnegativity of the cd-index of a Gorenstein* poset

S / solved | AMR 问题清单汇编

Let $P$ be a Gorenstein* graded poset of rank $n-1$, and let $\Phi_P(c,d)$ be its $cd$-index in noncommuting variables $c$ and $d$ (where $\Psi_P(a,b)=\Phi_P(a+b,ab+ba)$ is the flag $h$-vector generating polynomial). Are all coefficients of the $cd$-index $\Phi_P(c,d)$ nonnegative?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13904 | AMR-156-0007

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 8: Adin's cubical h-vector nonnegativity and GLBT for cubical complexes

O / open | AMR 问题清单汇编

Let $L$ be a pure cubical complex of rank $d$ (dimension $d-1$), and let $h(L)=(h_0(L),h_1(L),\dots,h_d(L))$ be Adin's cubical $h$-vector defined by $\sum_{i=0}^d h_i(L)x^i = \frac{1}{1+x}\left(2^{d-1} + x\sum_s h(\Delta_s,x) + (-2)^{d-1}\widetilde{\chi}(L)x^{d+1}\right)$, where $\Delta_s=\operatorname{link}(s)$ for each vertex $s$ of $L$.

(a) If $L$ is a Cohen-Macaulay poset, is $h_i(L)\ge 0$ for all $i$?

(b) If $L$ is in addition a Gorenstein* poset, is it true that $h_0(L)\le h_1(L)\le \cdots \le h_{\lfloor d/2\rfloor}(L)$?

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13905 | AMR-156-0008

### Stanley's Positivity Problems in Algebraic Combinatorics - Problem 9: Combinatorial interpretation of plethysm coefficients

O / open | AMR 问题清单汇编

Let $s_\lambda$ denote the Schur function indexed by a partition $\lambda$, and let $s_m[s_n]$ (or more generally $s_\lambda[s_\mu]$) denote the plethysm of Schur functions. Find a positive combinatorial interpretation of the plethysm coefficients $\langle s_m[s_n], s_\lambda\rangle$, thereby combinatorially reproving that they are nonnegative.

背景链接：https://math.mit.edu/~rstan/papers/problems.pdf

13906 | AMR-030-0081

### start at (0,0), at each point in time, we take a step from (x, y) to (x+1, y), (x-1, y), (x, y+1), or (x, y-1) with prob

O / open | AMR 问题清单汇编

Consider the following walk: start at (0,0), at each point in time, we take a step from (x, y) to (x+1, y), (x-1, y), (x, y+1), or (x, y-1) with probability proportional to one plus the number of steps we have already taken from (x, y) to the destination point in the past. What is the probability of returning to the origin at some point? Even if the öweightö of an edge (i.e., one plus the number of steps taken between two points) is never incremented beyond 2, this problem is open.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13907 | AMR-029-0082

### Strong colouring of matroid-graph pairs

O / open | AMR 问题清单汇编

Let G=(V,E) be a graph with maximum degree $\Delta \geq 2$, and let M=(V,r) be a matroid that has $2 \Delta$ disjoint bases. Is it true that M has $2 \Delta$ disjoint bases that are all independent in G?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13908 | OWR-730-001

### Strongly Balanced Almost-Uniform Vector Permutations

O / open | Oberwolfach 报告问题汇编

Does every finite set of $m$-dimensional vectors of Euclidean norm at most $1$, whose vector sum has norm at most $1$, admit a permutation whose every prefix sum has norm at most $\sqrt{2.7m}$ and whose block sums are uniformly bounded by an absolute constant?

来源字段：https://doi.org/10.4171/owr/2004/28

13909 | AMR-029-0083

### Strongly maximal H-free spanning subgraph

P / partially_solved | AMR 问题清单汇编

Let the graphs $G=(V,E)$ and $H$ be fixed. An edge set $F\subseteq E$ is called $H$-free if $(V,F)$ does not contain $H$ as a subgraph. We say that $F$ is strongly maximal if for any $H$-free edge set $I\subseteq E$ we have $\left|I \setminus F\right| \leq \left|F \setminus I\right|$. If $H$ is the path of length two, then we talk about strongly maximal matchings, and any $G$ admits such a matching ( p. 16. Theorem 5.6). For what other graphs $H$ can we guarantee the existence of a strongly maximal $H$-free $F$ for any $G$?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13910 | AMR-029-0084

### Strongly maximal matchings

S / solved | AMR 问题清单汇编

Is it true that if all the hyperedges of a hypergraph $H$ have size at most $k$ for some $k\in \mathbb{N}$, then $H$ admits a strongly maximal matching?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13911 | AMR-029-0085

### Strongly minimal edge cover

P / partially_solved | AMR 问题清单汇编

Is it true that if the hypergraph $H$ has no isolated vertices and all of its hyperedges are finite, then $H$ admits a strongly minimal edge cover?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

13912 | OWR-14298158-011

### Structural Decomposition of Separable Permutation Subclasses

P / partially_solved | Oberwolfach 报告问题汇编

For a subclass $C$ of the separable permutations, must $C$ be closed under direct sums or skew sums, a finite union of such classes, or contained in an inflation $X[U]$ by a proper subclass $U$ as conjectured?

来源字段：https://doi.org/10.4171/owr/2024/6

13913 | OWR-768-003

### Structural Thresholds for Sum-Free Sets in $\mathbb{Z}/p\mathbb{Z}$

P / partially_solved | Oberwolfach 报告问题汇编

Determine the sharp structural threshold between densities $0.2p$ and $0.33p$ for sum-free subsets of $\mathbb Z/p\mathbb Z$ to be contained in a dilation of an interval $[n,p-n]$.

来源字段：https://doi.org/10.4171/owr/2004/54

13914 | GREEN-077

### Structure of Sets with Bounded Representation

P / partially_solved | Ben Green 问题

Suppose $A \subset [N]$ has size $\geq c\sqrt{N}$ and representation function $r_A(n) \leq r$ for all $n$. What can be said about the structure of $A$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13915 | GREEN-073

### Structured Subsets with Bounded Doubling

P / partially_solved | Ben Green 问题

Given a set $A \subset \mathbb{Z}$ with $D(A) \leq K$, find a large structured subset $A'$ which "obviously" has $D(A') \leq K + \varepsilon$.

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13916 | GAME-002

### Sudoku: Minimal Puzzles Count

P / partially_solved | 独立 / 未分配集合

How many Sudoku puzzles with exactly one solution are minimal (removing any clue creates multiple solutions)?

背景链接：https://en.wikipedia.org/wiki/Sudoku

13917 | GAME-001

### Sudoku: Unique Solution Puzzles

P / partially_solved | 独立 / 未分配集合

How many Sudoku puzzles have exactly one solution?

背景链接：https://en.wikipedia.org/wiki/Sudoku

13918 | GREEN-069

### Sum of Cubes in F_3^n

S / solved | Ben Green 问题

Let $A_1, \dots, A_{100}$ be "cubes" in $\mathbb{F}_3^n$ (images of $\{0, 1\}^n$ under linear automorphisms). Is $A_1 + \dots + A_{100} = \mathbb{F}_3^n$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13919 | GREEN-006

### Sum-Free Subsets of [N]^d

P / partially_solved | Ben Green 问题

Fix an integer $d$. What is the largest sum-free subset of $[N]^d$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13920 | OWR-11781-001

### Supersaturation Blow-Ups in Uniform Hypergraphs

O / open | Oberwolfach 报告问题汇编

For all positive integers $\ell$ and $r$ and every $c>0$, does there exist $\gamma>0$ such that every $r$-uniform hypergraph $H$ with at least $cn^\ell$ copies of the complete $r$-uniform hypergraph $K^r_\ell$ contains a complete $\ell$-partite $r$-uniform hypergraph with equal part size $t\geq\gamma(\log n)^{1/(r-1)}$?

来源字段：https://doi.org/10.4171/owr/2012/19

13921 | AMR-030-0007

### Suppose a family V of n points are chosen in R^(3) and L is a collection of lines so that every triangle spanned by thre

O / open | AMR 问题清单汇编

Solymosi : Suppose a family V of n points are chosen in R^(3) and L is a collection of lines so that every triangle spanned by three points of V is pierced by some element of L. Show that |L| must grow at least linearly in n. (Best known: n^(1/2)!)

背景链接：https://people.math.sc.edu/cooper/combprob.html

13922 | AMR-030-0004

### Suppose a geometric graph has no pairwise k-crossing lines

O / open | AMR 问题清单汇编

Pach : Suppose a geometric graph has no pairwise k-crossing lines. That is, no k edges all cross each other. Must the graph have O_(k)(n) edges?

背景链接：https://people.math.sc.edu/cooper/combprob.html

13923 | AMR-030-0015

### Suppose G has n vertices and no induced copy of H

P / partially_solved | AMR 问题清单汇编

For each finite graph H, does there exist $\varepsilon(H)>0$ such that every n-vertex graph with no induced copy of H contains a clique or independent set of size at least $n^{\varepsilon(H)}$?

背景链接：https://people.math.sc.edu/cooper/combprob.html

13924 | AMR-030-0006

### Suppose H is a linear 3-uniform hypergraph, i

P / partially_solved | AMR 问题清单汇编

Kalai : Suppose H is a linear 3-uniform hypergraph, i.e., a subset of the set of all triples of n points with the property that no two edges intersect in more than one point. Further suppose that H is embedded in R^(3) faithfully, i.e., the vertices are placed in such a way that, for any two edges e_(1) and e_(2) of H, the planar triangles spanned by their respective vertices intersect if and only if e_(1) and e_(2) intersect in H, and, if they do intersect, they do so in precisely their common vertex. Show that the number of edges in H is o(n^(2)).

背景链接：https://people.math.sc.edu/cooper/combprob.html

13925 | AMR-030-0022

### Suppose that G is a tree

O / open | AMR 问题清单汇编

Graham: Suppose that G is a tree. Denote by L(G) the line graph of G. Is the sequence |G|, |L(G)|, |L(L(G))|, |L(L(L(G)))| ... unique to G? That is, can a tree be reconstructed from the sequence of sizes of its iterated line graphs? See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13926 | AMR-030-0005

### Suppose we begin with a set of points S in the plane

O / open | AMR 问题清单汇编

: Suppose we begin with a set of points S in the plane. Let T(S) be the set of points one gets by taking all lines through pairs of points in S, and taking all intersections of those lines. If one iterates T(.), how quickly does the cardinality of the set grow? See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13927 | OWR-17135-030

### Symmetric Function Bases for Bipartite Determinant Inequalities

P / partially_solved | Oberwolfach 报告问题汇编

Let $G$ be bipartite with parts $X=[n]$ and $Y$, and set $X_S=|\{y\in Y:N(y)=S\}|$ for every nonempty $S\subseteq[n]$. Define the $n\times n$ matrix $M$ by $$M_{ii}=\sum_{S:i\in S}X_S,\qquad M_{ij}=\sum_{S:i\in S,\,j\notin S}\frac{X_S}{|S|}\quad(i\neq j)$$. Since $\det M$ and $\det(\operatorname{diag}M)$ are invariant under relabeling $X$, is there a natural basis from symmetric-function theory in which to express these polynomials in the variables $X_S$?

来源字段：https://doi.org/10.4171/owr/2019/39

13928 | OWR-17464-012

### Symmetric Intersecting Families in Discrete Product Spaces

P / partially_solved | Oberwolfach 报告问题汇编

For intersecting families $A\subseteq[k]^n$ with $k\geq3$, what bounds follow when highly asymmetric, single-coordinate extremal examples are forbidden? In particular, how small must $A$ be when its automorphism group acts transitively on the coordinates?

来源字段：https://doi.org/10.4171/owr/2020/1

13929 | OWR-15987-016

### Symmetric-Group Characters Determined by Reflection Length

O / open | Oberwolfach 报告问题汇编

Give a simple description of the characters $\chi(g)$ of $S_n$ that depend only on the reflection length $\ell(g)$, equivalently only on the number of cycles of $g$.

来源字段：https://doi.org/10.4171/owr/2018/2

13930 | AIM-COMBINATORICS-0118

### Tate defects and character splitting for critical groups of abelian graph covers

P / partially_solved | AIM 研讨会问题汇编

We know results about the homomorphism between critical groups for two graphs that are related by a Galois covering map (see, e.g., [31]). How do these results relate to class field theory, when the Galois group of the covering is abelian?

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13931 | AIM-COMBINATORICS-0186

### Temporal trace semantics and a bounded-repetition special case

P / partially_solved | AIM 研讨会问题汇编

The matrices \(A_1,\ldots,A_t\) are \(n\times n\) \(0\)-\(1\) matrices, \(a^{(i)}_{uv}=1\) implies \(a^{(i)}_{vu}\ne1\) for \(u\ne v\), and all diagonal entries equal \(1\). They need not be distinct. If every row sum of \(A_i\) is at least \(r_i+1\) and \[ \sum_{i=1}^t r_i\ge n, \] must ordinary matrix multiplication satisfy \[ \operatorname{tr}(A_1A_2\cdots A_t)>n? \]

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13932 | AIM-COMBINATORICS-0006

### Tensor growth rate and exact one-dimensional obstructions for constrained differences

P / partially_solved | AIM 研讨会问题汇编

For which \(0\in D\subseteq \mathbb F_p\) is it true that any \(A\subseteq\mathbb F_p^n\) avoiding \[ x,x+y,x+2y\in A\qquad\text{with }y\in D^n\setminus\{0\} \] has size at most \(C^n\) for some \(C<p\)?

来源字段：http://aimpl.org/highdimdiscrete/2/

13933 | OWR-14298007-015

### Tensor Products of Matroids

P / partially_solved | Oberwolfach 报告问题汇编

Study the pairs of matroids that admit a tensor product: construct further examples, describe tensor products combinatorially for particular matroid classes, and characterize obstructions by forbidden minors.

来源字段：https://doi.org/10.4171/owr/2023/58

13934 | AMR-030-0019

### The "cycle double cover conjecture" states that every bridgeless graph contains a set of cycles which cover each edge of

P / partially_solved | AMR 问题清单汇编

Seymour/Szekeres: The "cycle double cover conjecture" states that every bridgeless graph contains a set of cycles which cover each edge of the graph exactly twice. See this and this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13935 | AMR-046-0032

### The 196 conjecture

O / open | AMR 问题清单汇编

Define $f:\mathbb{N}\to\mathbb{N}$ by $f(n)=n+\operatorname{rev}(n)$, where $\operatorname{rev}$ reverses the decimal digits. Are there infinitely many $n$ for which no iterate $f^k(n)$, $k>0$, is a palindrome? Is the least such $n$ equal to $196$?

背景链接：https://arxiv.org/abs/2012.02524

13936 | OPG-150

### The additive basis conjecture

P / partially_solved | Open Problem Garden

Conjecture For every prime $p$, there is a constant $c(p)$ (possibly $c(p)=p$ ) so that the union (as multisets) of any $c(p)$ bases of the vector space $({\mathbb Z}_p)^n$ contains an additive basis.

背景链接：http://www.openproblemgarden.org/op/the_additive_basis_conjecture

13937 | OPG-152

### The Alon-Tarsi basis conjecture

O / open | Open Problem Garden

Conjecture If $B_1,B_2,\ldots B_p$ are invertible $n \times n$ matrices with entries in ${\mathbb Z}_p$ for a prime $p$, then there is a $n \times (p-1)n$ submatrix $A$ of $[B_1 B_2 \ldots B_p]$ so that $A$ is an AT-base.

背景链接：http://www.openproblemgarden.org/op/the_alon_tarsi_basis_conjecture

13938 | OWR-14299518-016

### The Alon-Tarsi Conjecture for Latin Squares

O / open | Oberwolfach 报告问题汇编

For every even integer $n$, is the difference between the numbers of even and odd Latin squares of order $n$ nonzero?

来源字段：https://doi.org/10.4171/owr/2025/42

13939 | AIM-COMBINATORICS-0266

### The arithmetic Kakeya entropy constant and an exact bounded-support case

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.10 (T. Tao).** What is the best \(\varepsilon\) for which there exist real numbers \(r_1,\ldots,r_n\) with the following property: given any two random values \(x,y\) taking finitely many real values, and obeying the entropy bound \(H(x+r_jy)<\log N\) for all \(j=1,\ldots,N\), one necessarily has \(H(x-y)<(1+\varepsilon)\log N\).

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13940 | AIM-COMBINATORICS-0281

### The balanced-graph trichotomy for C4-free and subcubic graphs

P / partially_solved | AIM 研讨会问题汇编

Conjecture 1 (Conforti, Cornuéjols, Rao). Every balanced graph is either basic or has a 2-join or a skew-partition.

- A graph is **balanced** if every induced cycle has length divisible by \(4\). Such a graph is bipartite. - A balanced graph is **basic** if either: 1. it admits a bipartition with every vertex on one side of degree at most \(2\); or 2. it has a hole \(H\) such that \(G\setminus V(H)\) induces a complete bipartite graph. - A **2-join** has a partition \(V(G)=X_1\dot\cup X_2\), with disjoint nonempty \(A_i,B_i\subseteq X_i\) for \(i=1,2\), such that all \(A_1A_2\) and all \(B_1B_2\) edges are present and there are no other edges between \(X_1\) and \(X_2\). Each \(G[X_i]\) contains an \(A_i\)-to-\(B_i\) path. Under the corrected standard convention, if \(|A_i|=|B_i|=1\), then \(G[X_i]\) is not itself a chordless path between them. This is the proper/non-path convention intended by the source; later authors note and repair a minor wording error in an older printed definition. - A **skew partition** is a partition into four nonempty sets \((A,B,C,D)\) such that \(A\) is anticomplete to \(B\) and \(C\) is complete to \(D\). Nonemptiness is necessary: allowing empty pieces would make the decomposition statement vacuous.

Here and below, graphs are finite and simple. A \(C_4\) means an induced four-cycle.

来源字段：https://aimath.org/WWN/perfectgraph/perfectgraph.pdf

13941 | COMB-010

### The Cap Set Problem

P / partially_solved | 独立 / 未分配集合

别名：Cap Set Problem [GAME-007]

What is the maximum size of a cap set in $\mathbb{F}_3^n$?

背景链接：https://arxiv.org/abs/1605.09223

13942 | AIM-COMBINATORICS-0104

### The DFS-burning solution and a chordwise Hamilton-path refinement

P / partially_solved | AIM 研讨会问题汇编

Is there a natural bijection \(\phi\) between labeled trees \(T\) on \(\{0,1,2,\ldots,n\}\), rooted at \(0\), and parking functions of length \(n\), such that \[ \deg(\phi(T))=g-\operatorname{inv}(T), \] where \(g\) is the genus of the complete graph \(K_{n+1}\) and \(\operatorname{inv}(T)\) is the number of inversions of \(T\)? Perkinson further suggested that an arbitrary-graph version should replace inversions by Gessel's \(\kappa\)-statistic.

Comparison with the linked AIM PDF recovers the question as follows:

Here \[ g=|E(K_{n+1})|-|V(K_{n+1})|+1=\binom n2. \] The recorder's note says that the problem was resolved in reference [30]. In the bibliography of the AIM document, [30] is David Perkinson, Qiaoyu Yang, and Kuai Yu, *G-parking functions and tree inversions*, arXiv:1309.2201, subsequently published in *Combinatorica* 37 (2017), 269--282.

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13943 | OPG-60000

### The Double Cap Conjecture

P / partially_solved | Open Problem Garden

Conjecture The largest measure of a Lebesgue measurable subset of the unit sphere of $\mathbb{R}^n$ containing no pair of orthogonal vectors is attained by two open caps of geodesic radius $\pi/4$ around the north and south poles.

背景链接：http://www.openproblemgarden.org/op/the_double_cap_conjecture

13944 | AIM-COMBINATORICS-0162

### The doubled-label slice is exactly Seymour's second-neighborhood conjecture

P / partially_solved | AIM 研讨会问题汇编

Conjecture 6.2. Seymour's Second Neighborhood Conjecture Notes: To see this, for a digraph H, let k = 2, G = H, and E1 = E2 = E(H).

Seymour's Second Neighborhood Conjecture. Notes: To see this, for a digraph \(H\), let \(k=2\), \(G=H\), and \(E_1=E_2=E(H)\).

Any simple digraph with no loops or digons has a vertex \(v\) whose second neighborhood is at least as big as its first neighborhood, i.e. \(|N_2^+(v)|\geq |N^+(v)|\).

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13945 | AIM-COMBINATORICS-0055

### The Erdős-Hajnal property for the Paley seven-vertex tournament with one vertex removed

S / solved | AIM 研讨会问题汇编

Does the six-vertex tournament with backward arcs \((4,1),(6,3),(6,1),(5,2)\) have the tournament Erdős--Hajnal property?

Its historical remark says that Berger--Choromanski--Chudnovsky had reduced the six-vertex case to this one tournament.

The extraction split the definition of \(T^*\) into the preceding record and also incorrectly retained the preceding section title, "Online Ramsey numbers." The nearby source text reconstructs the intended section as **Erdős--Hajnal for tournaments**, attributed to Maria Chudnovsky, and defines \(T^*\) on \(\{1,\ldots,6\}\) by the four exceptional arcs \[ 6\to1,\qquad 6\to3,\qquad 5\to2,\qquad 4\to1, \] with every remaining pair oriented \(i\to j\) when \(i<j\).

Thus the exact recovered question is:

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13946 | OWR-8415359-002

### The Erdős-Rado Sunflower Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For every integer $r\geq3$, does there exist a constant $c(r)$ such that every family $\mathcal{F}$ of sets of size at most $w$ contains an $r$-sunflower whenever $|\mathcal{F}|>c(r)^w$?

来源字段：https://doi.org/10.4171/owr/2021/54

13947 | OWR-14298371-004

### The Erdős-Rado Sunflower Conjecture

O / open | Oberwolfach 报告问题汇编

Does the Erdős-Rado sunflower conjecture hold with the conjectured exponential bound on the size of sunflower-free uniform set systems?

来源字段：https://doi.org/10.4171/owr/2024/27

13948 | AIM-COMBINATORICS-0041

### The Euclidean Steinitz conjecture and a sharp ell_1 obstruction

P / partially_solved | AIM 研讨会问题汇编

The AIM record is Problem 1.4, “Steinitz conjecture,” from the workshop list *Hereditary discrepancy and factorization norms*. Its displayed text says that, for zero-sum vectors \(v_1,\ldots,v_n\in\mathbb R^d\) of norm at most one in a norm \(X\), \[ \min_{\pi}\max_{1\le k\le n} \left\|\sum_{i=1}^k v_{\pi(i)}\right\|_X =O(\sqrt d), \] after taking a supremum over \(n\). As written, this is not a consistent definition or conjecture:

1. the displayed \(S_n(X,d)\) omits the supremum over admissible vector families, so it still depends on the particular family just introduced; 2. \(S^n\) should be the symmetric group on \([n]\), conventionally denoted \(\mathfrak S_n\); 3. an \(O(\sqrt d)\) estimate uniform over all norms is false; and 4. the record itself reports a linear lower bound for \(\ell_p\), \(1<p<2\), which would contradict the displayed conjecture.

For a finite indexed family \(V=(v_1,\ldots,v_n)\), put \[ c_X(V):=\min_{\pi\in\mathfrak S_n}\max_{1\le k\le n} \left\|\sum_{i=1}^k v_{\pi(i)}\right\|_X, \] and \[ S_{X,n}(d):= \sup_{\substack{v_1,\ldots,v_n\in\mathbb R^d\\ \sum_i v_i=0,\ \|v_i\|_X\le1}} c_X(V), \qquad S_X(d):=\sup_{n\ge1}S_{X,n}(d). \] For \(X=\ell_p^d\), write \(S_p(d)\).

The **recovered intended conjecture** is \[ S_2(d)=O(\sqrt d), \tag{EC} \] with a universal implied constant. This is the Euclidean Steinitz problem, also attributed to Bergström in the modern literature. Current primary sources explicitly formulate the conjecture for \(\ell_2\), while describing the \(O(d)\) bound as the result for arbitrary norms. Thus the word “a norm \(X\)” in the AIM formulation has almost certainly been carried over from the definition when the conjectural sentence should have specialized to \(X=\ell_2\).

The AIM line “if \(X=\ell_p\) for \(1<p<2\), \(S(X,d)=\Omega(d)\)” is also corrupt under the definition above. The most plausible repair is \(\Omega(d^{1/p})\), but the record supplies no citation for that line, so this report treats the exponent repair as a reconstruction rather than as a verified bibliographic fact. The theorem below proves the \(\Omega(d^{1/p})\) lower bound directly.

来源字段：http://aimpl.org/hereddiscrep/1/

13949 | AIM-COMBINATORICS-0174

## The exact two-label boundary for directed rainbow cycles

P / partially_solved | AIM 研讨会问题汇编

Conjecture 6.1, if $\delta^+_{G_i}\geq r_i$, and $\sum_{i=1}^{t}r_i\geq |V|$, there is a rainbow cycle in $G$.

**Conjecture 6.8 (recovered).** If numbers $r_1,\ldots,r_k$ satisfy \[ \delta^+(G_i)\geq r_i\quad(i\in[k]), \qquad \sum_{i=1}^k r_i\geq n:=|V|, \tag{1.3} \] then $G$ contains a rainbow directed cycle.

Its literal text is

The preceding record contains “Conjecture 6.8. Under the hypotheses of”. The original AIM PDF and arXiv source verify that the intended statement is therefore Conjecture 6.8, continuing the setup of Conjecture 6.1. The printed source uses $k$ layers in Conjecture 6.1 but sums to $t$ in Conjecture 6.8. The coherent reconstruction treats this as a dummy-index mismatch and sets $t=k$.

Let $G=(V,E)$ be a finite simple digraph, let $E_1,\ldots,E_k\subseteq E(G)$, and put \[ S_e=\{i\in[k]:e\in E_i\},\qquad G_i=(V,E_i). \tag{1.1} \] The source defines “simple” to mean that there is at most one arc from $u$ to $v$; it does not build looplessness or the absence of digons into that word. A rainbow directed path or cycle is one whose arcs admit a labeling \[ \ell:E(H)\longrightarrow[k],\qquad \ell(e)\in S_e, \tag{1.2} \] such that $\ell$ is injective. Thus a multiply labeled arc uses one chosen admissible label, and the two arcs of a rainbow digon must receive different labels.

For completeness, Conjecture 6.1 says that either there is a rainbow directed cycle or some vertex $v$ can reach at least $\sum_i d^+_{G_i}(v)$ vertices by rainbow paths. The intended paths there have positive length: the source's proof of an implication uses the fact that, without a rainbow cycle, a vertex cannot reach all $n$ vertices in this sense, while Conjecture 6.10 later explicitly says “possibly trivial” when it wishes to include length zero. Accordingly, the start vertex is **not** included in the Conjecture 6.1 reachable set merely via the trivial path. If one instead allows the trivial path, it has no arcs, is vacuously rainbow, and includes its start; that alternative convention does not affect Conjecture 6.8 or any theorem below.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13950 | AIM-COMBINATORICS-0050

## The exceptional three-color path problem: star-forest and triangle-density reductions

P / partially_solved | AIM 研讨会问题汇编

Is it true that every three-edge-coloring of every graph of chromatic number at least six contains a monochromatic path of length three?

that \(P_k\) means a path of **length** \(k\), hence with \(k\) edges. It records \[ r(P_3,P_3,P_3)=6 \] and asks:

To avoid a clash with the more usual convention in the literature, this report writes \[ Q=v_0v_1v_2v_3 \] for the path with three edges and four vertices. Thus the source's \(P_3\) is the graph called \(P_4\) by Garrison and most of the cited papers.

The intended setting is finite simple graphs and ordinary, not necessarily induced, copies of the path. Garrison explicitly works with finite simple graphs, and both the Ramsey statement and the use of the Turán number are for ordinary subgraphs. A three-edge-coloring may leave a color unused. The host may be disconnected, but its chromatic number is the maximum of the chromatic numbers of its components, so this causes no extra case.

Equivalently, if \(\chi(Q,3)\) denotes the smallest \(m\) such that every three-edge-coloring of every \(m\)-chromatic graph contains a monochromatic \(Q\), the question asks whether \[ \chi(Q,3)=R_3(Q)=6. \]

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13951 | AIM-COMBINATORICS-0110

## The F-lattice feasible set and elementary apex representatives

S / solved | AIM 研讨会问题汇编

Let \(\Gamma\) be the set of \(2\times2\) real symmetric matrices \(A\) for which there is a function \(u:\mathbb Z^2\to\mathbb Z\) satisfying \[ u(x)=\frac12x\mathbin{\cdot}Ax+o(1+|x|^2), \qquad \Delta u(x)\leq0\quad(x\in\mathbb Z^2). \] For the usual square-lattice Laplacian, \(\Gamma\) exhibits an Apollonian circle-packing phenomenon. For the F-lattice, which has two incoming and two outgoing edges at every vertex, computations appear to give a very simple set. Is this really the case?

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13952 | AIM-COMBINATORICS-0245

## The finite-field Kakeya problem in dimension three

S / solved | AIM 研讨会问题汇编

**Problem 3.8 (T. Tao).** Take \(F\) to be a finite field, and suppose that \(E\subseteq F\times F\times F\), where \(E\) is a Besicovitch set; i.e. \(E\) contains a line in every direction. It is known from the work of Wolff that \(|E|\gtrsim |F|^{5/2}\); prove the better lower bound \(|E|\gtrsim |F|^{5/2+\varepsilon}\).

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13953 | AIM-COMBINATORICS-0113

### The full toppling Gröbner fan in the tree case

P / partially_solved | AIM 研讨会问题汇编

For a connected undirected graph (or multigraph) \(G\), describe all cones of the Gröbner fan of the homogeneous lattice ideal \(I_G\) attached to the image of the graph Laplacian, not only the distinguished cone whose monomial initial ideal is the \(G\)-parking-function ideal.

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13954 | AIM-COMBINATORICS-0119

### The full Tutte polynomial from chip-firing data

S / solved | AIM 研讨会问题汇编

Matt Baker: “Is there a relation between the full Tutte polynomial (as opposed to just \(T_G(1,y)\) from Merino's theorem [25]) and chip-firing? We made to use some additional geometric structure.”

“We **may need** to use some additional geometric structure.”

This is Problem 29 in the 2013 AIM list *Problems from the workshop Generalizations of chip-firing and the critical group*.

The last sentence is ungrammatical. A highly plausible reconstruction is

but this emendation is not treated as verified source text. The mathematical question in the preceding sentence is unambiguous.

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13955 | OWR-16775-005

### The Generalized Hamada p-Rank Conjecture

O / open | Oberwolfach 报告问题汇编

Among all combinatorial designs with the same parameters as a subspace design

$$t\text{-}(v,k,\lambda)_q,$$

does the induced design with parameters

$$2\text{-}\left(\binom{v}{1}_q,\binom{k}{1}_q,\lambda\right)$$

have minimum $p$-rank?

来源字段：https://doi.org/10.4171/owr/2019/13

13956 | OWR-14298371-006

### The Graphic Kahn-Kalai Conjecture

O / open | Oberwolfach 报告问题汇编

Is there a universal $K>0$ such that, for every $H\subseteq K_n$, $p\geq Kp_E(H)\log v(H)$ implies $\Pr(G_{n,p}\supseteq H)\geq1/2$?

来源字段：https://doi.org/10.4171/owr/2024/27

13957 | AIM-COMBINATORICS-0052

### The hypergraph size-Ramsey equality is false, with critical-edge structure for minimal witnesses

S / solved | AIM 研讨会问题汇编

The size Ramsey number $\hat r(n,n)$ is the minimum number $m$ for which there exists a graph with $m$ edges such that every 2-edge-coloring of it produces a monochromatic $K_n$. Similarly define $\hat r_3(n,n)$ for the corresponding 3-uniform size Ramsey number.

For graphs, $\hat r(n,n)=\binom{r(n,n)}2$.

Problem (Dudek--Rödl): Is \[ \hat r_3(n,n)=\binom{r_3(n,n)}3? \]

In particular,

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13958 | AIM-COMBINATORICS-0250

### The joints theorem and fixed-direction stability

S / solved | AIM 研讨会问题汇编

Take \(n\) lines in \(\mathbb R^3\). Define a joint to be a point with three lines passing through it that are not coplanar. How many joints can there be?

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13959 | OPG-373

### The large sets conjecture

O / open | Open Problem Garden

Conjecture If $A$ is 2-large, then $A$ is large.

背景链接：http://www.openproblemgarden.org/op/the_large_sets_conjecture

13960 | COMB-004

### The Lonely Runner Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Lonely Runner Conjecture [COMB-002]; Lonely Runner Conjecture [COMB-019]; Lonely runner conjecture [OPG-416]

For any $n$ runners on a circular track with distinct constant speeds, each runner is "lonely" (distance at least $1/n$ from all others) at some time.

背景链接：https://arxiv.org/abs/2604.23906

13961 | AIM-COMBINATORICS-0049

### The minimal empty-core tight path and the 2026 uniformity-four resolution

P / partially_solved | AIM 研讨会问题汇编

Is it true that for fixed $k$, and $s$ sufficiently large (in terms of $k$), one has \[ r_k(P_s^{(k)},K_t^{(k)})=\Theta(t^{k-1})? \]

1. $P_s^{(k)}$ has **$s$ edges**, not $s$ vertices. Explicitly, \[ V(P_s^{(k)})=[s+k-1],\qquad E_i=\{i,i+1,\ldots,i+k-1\}\quad(1\le i\le s). \] 2. The Ramsey problem is **unordered**. Thus a copy is an ordinary hypergraph copy; its vertices need not respect an ambient linear order.

For fixed integers $k\ge2$ and $s\ge1$, $r_k(P_s^{(k)},K_t^{(k)})$ is the least $N$ such that every red/blue coloring of $\binom{[N]}k$ contains either a red ordinary copy of $P_s^{(k)}$ or a blue $K_t^{(k)}$. Equivalently, if the red edges form a $k$-graph $H$, it is the least $N$ for which every $P_s^{(k)}$-free $H$ on $N$ vertices satisfies $\alpha(H)\ge t$. All asymptotics below are for $t\to\infty$, with constants allowed to depend on the fixed pair $(k,s)$.

This differs from the ordered tight-path notation in Mubayi--Suk: their $P_s^{(k)}$ has $s$ ordered vertices and $s-k+1$ edges, and embeddings preserve the order. Their tower-type ordered results do not answer this AIM question.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13962 | AIM-COMBINATORICS-0132

### The pentagon lift and a periodic obstruction on general graphs

P / partially_solved | AIM 研讨会问题汇编

Matt Baker: "Peter Winkler [35] gives an interesting solution to a famous math Olympiad problem involving numbers on a pentagon in his book on math puzzles. His solution takes infinite covers of the pentagon. I have not tried to generalize this proof to the usual chip-firing game; Farbod and I [9] have a solution that uses potential theory. But maybe we could apply Winkler's method to general graphs?" Matt Macauley asks whether this is the numbers game for Coxeter groups, and Baker replies that it is a special case, while the numbers game and graph chip-firing are different generalizations away from a cycle.

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

13963 | OPG-151

### The permanent conjecture

O / open | Open Problem Garden

Conjecture If $A$ is an invertible $n \times n$ matrix, then there is an $n \times n$ submatrix $B$ of $[A A]$ so that $perm(B)$ is nonzero.

背景链接：http://www.openproblemgarden.org/op/the_permanent_conjecture

13964 | AIM-COMBINATORICS-0206

### The prime maximal Z15 boundary counterexample

S / solved | AIM 研讨会问题汇编

**Conjecture 6.36 (Thomassé).** Let \(G\) be a digraph on \(n\) vertices with minimum outdegree at least \(4n/15\) so that \(G\) is maximal with no cycles of length at most three, and has no homogeneous set (in other words, \(G\) cannot be obtained by substitution). Then \(G\) is a Cayley graph on \(3k+1\) vertices with \(1,\ldots,k\) as generators for some \(k\) (i.e. the circular interval graph with everyone joined to the next \(k\) clockwise).

Sullivan's 2006 AIM report. It prints the following statement:

It immediately adds:

The heading “Acknowledgements” begins on the next printed line.

Under the standard conventions made explicit below, the note and the conjecture are in direct tension. This report proves that the noted \(\mathbb Z_{15}\) digraph satisfies **every** printed hypothesis, including arc-maximality and absence of a nontrivial homogeneous set. Since \(15\neq3k+1\) for every integer \(k\), it refutes the literal statement. The likely repair is to replace “at least” by “strictly greater than,” but no claim is made here that the repaired conjecture is true.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13965 | AIM-COMBINATORICS-0149

### The refuted de Caen clique-covering conjecture and an exact rational recursion

S / solved | AIM 研讨会问题汇编

Does \(k(1-\pi(K_{k+1}^k))\) tend to \(\infty\) as \(k\to\infty\)?

The AIM record asks:

Here \(K_{k+1}^k\) is the complete \(k\)-uniform hypergraph on \(k+1\) vertices and \[ \pi(K_{k+1}^k) =\lim_{n\to\infty} \frac{\operatorname{ex}(n,K_{k+1}^k)}{\binom nk}. \] The statement is uncorrupted. Its order of limits is essential: for each fixed \(k\), first let the number \(n\) of vertices tend to infinity in the definition of \(\pi\); only afterwards let \(k\to\infty\).

来源字段：http://aimpl.org/hypergraphturan/1/

13966 | AIM-COMBINATORICS-0003

### The Renyi-Shannon bottleneck in entropic inverse arguments

P / partially_solved | AIM 研讨会问题汇编

Neighbor 1.1 states the entropic Polynomial Freiman-Ruzsa theorem with constant 11, confirming that Problem 1.3 asks for a more direct entropic derivation of inverse consequences, not for a proof of PFR itself.

The source asks about two assertions for \[ f:\mathbb F_2^n\longrightarrow\mathbb F_2^m. \]

First, if \[ \mathbb P_{x+y=z+w}\bigl(f(x)+f(y)=f(z)+f(w)\bigr)\geq\epsilon, \tag{1} \] PFR implies correlation with a linear function. Second, with \[ D_hf(x)=f(x+h)-f(x), \] if \[ \mathbb P_{x,h_1,h_2,h_3} \bigl(D_{h_1}D_{h_2}D_{h_3}f(x)=0\bigr)\geq\epsilon, \tag{2} \] PFR implies correlation with a quadratic function. It asks for direct entropic proofs with polynomial bounds.

Several conventions are absent and must be made explicit.

1. In (1), this report interprets the subscript as the uniform distribution on ordered quadruples satisfying \(x+y=z+w\). Equivalently, choose \(x,y,z\) independently and uniformly and put \(w=x+y+z\). 2. In (2), \(x,h_1,h_2,h_3\) are independent and uniform. 3. In characteristic two, subtraction equals addition: \[ D_hf(x)=f(x+h)+f(x). \] Thus the third derivative is the sum of the eight cube values, and derivative order is immaterial. 4. The source never defines “correlates.” For vector-valued functions, two inequivalent meanings occur in the literature: - **agreement:** \(\mathbb P_x(f(x)=q(x))\) is at least a polynomial in the test parameter; - **scalar phase correlation:** for some nonzero output character \(\lambda\), the phase \((-1)^{\langle\lambda,f(x)\rangle}\) has nontrivial inner product with a linear or quadratic phase.

The PFR/approximate-homomorphism formulation naturally gives agreement with an affine map on a large set, while the standard \(U^3\) inverse theorem gives scalar phase correlation. These should not be conflated. “Linear” is also naturally affine in (1), since its test is invariant under adding a constant. A quadratic map here means a map of combinatorial degree at most two, equivalently one whose third additive derivatives vanish identically.

来源字段：http://aimpl.org/highdimdiscrete/1/

13967 | AIM-COMBINATORICS-0163

### The repeated-label reduction and its sharp threshold

P / partially_solved | AIM 研讨会问题汇编

For a digraph \(H\), take \(G=H\), \(k=\lceil n/\delta_H^+\rceil\), and \(E_1=\cdots=E_k=E(H)\). We must get a rainbow cycle because the sum of the outdegrees at each vertex is at least \(n\). This corresponds to a directed cycle of length at most \(\lceil n/\delta_H^+\rceil\) in \(H\), as desired.

Here \(H\) is a finite simple oriented graph on \(n\) vertices and \(\delta_H^+\geq 1\).

For completeness, the preceding rainbow conjecture gives every arc \(e\) a label list \(S_e\subseteq [k]\), equivalently \(E_i=\{e:i\in S_e\}\), and says that either:

1. there is a directed cycle whose arcs can be assigned pairwise distinct admissible labels; or 2. some vertex \(v\) can reach at least \(\sum_i d^+_{G_i}(v)\) vertices by rainbow paths.

The source's deduction requires “reachable” in the second alternative to mean reachable by a positive-length rainbow path (or, equivalently for this argument, to exclude \(v\) itself when no rainbow cycle exists). Section 8 records what changes under the convention that the trivial path counts.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13968 | AIM-COMBINATORICS-0157

### The sparse-removal conjecture fails literally and at the true extremal exponent

S / solved | AIM 研讨会问题汇编

Let $F$ be a graph and $\alpha>1$ be such that $\operatorname{ex}(n,F)=\Omega(n^\alpha)$. Then for any $\epsilon>0$ there is $n_0$ so that if $n>n_0$ and a graph $H$ is the edge-disjoint union of $m=\lceil\epsilon n^\alpha\rceil$ copies of $F$, then $H$ contains another copy of $F$ (i.e. has at least $m+1$ copies of $F$).

The AIM record (Hypergraph Turan problem workshop, Section “Ruzsa--Szemerédi Theorem and Relatives,” item 44.3) reads:

The stored source has one evident TeX error: `\Omega(n^{\alpha}$` is missing the closing brace after $\alpha$. The display above repairs only that brace. In what follows:

- $F$ and $H$ are finite simple graphs, $F$ is fixed, and $H$ has exactly $n$ vertices (isolated vertices of $H$ are permitted); - “copy” means an unlabelled, not necessarily induced, subgraph isomorphic to $F$; - the hypothesis supplies a specified family of $m$ pairwise edge-disjoint copies whose edge sets partition $E(H)$; and - “another” means a copy distinct from those specified $m$ copies. It need not be edge-disjoint from them.

Counting labelled embeddings would make the numerical phrase “at least $m+1$ copies” inappropriate, since one image of $F$ generally has several labelled embeddings. The unlabelled-image convention is the standard one here.

来源字段：http://aimpl.org/hypergraphturan/4/

13969 | OWR-1183-006

### The Two-Dimensional Time-Bomb Conjecture

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Two-Dimensional Time-Bomb Conjecture [OWR-1183-007]

In the Time-Bomb game on $\mathbb Z^2$, the Angel moves at most one square per turn and the Devil names one position-time pair $(p,t)$ per turn; the Angel loses if its $t$-th move lands at a previously named $p$. Can the Angel survive forever?

来源字段：https://doi.org/10.4171/owr/2006/01

13970 | AMR-030-0079

### There is (essentially) a unique sequence over {1,2} which is its own run-length encoding

P / partially_solved | AMR 问题清单汇编

Kolakoski: There is (essentially) a unique sequence over {1,2} which is its own run-length encoding. Is the density of 1's in this sequence 1/2? See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

13971 | AIM-COMBINATORICS-0204

### Thomasse's path-girth conjecture: refutation and exact cyclic blow-ups

S / solved | AIM 研讨会问题汇编

**Conjecture 6.34 (Thomassé).** Every digraph $D$ has a path of length $\delta_D^+(g-1)$, where $g$ is the girth of $D$.

**Note.** This is open for $g=3$, and implies Caccetta--Häggkvist.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13972 | AIM-COMBINATORICS-0275

### Three-way divisibility and a list strengthening for even-hole-free graphs

P / partially_solved | AIM 研讨会问题汇编

A \(k\)-division of a graph \(G\) is a partition of \(V(G)\) into \(V_1,\ldots,V_k\) such that no \(V_i\) contains a largest clique of \(G\). A graph is \(k\)-divisible if each of its induced subgraphs with at least one edge has a \(k\)-division.

**Conjecture 2 (Hoàng).** Every even-hole-free graph is 3-divisible.

来源字段：https://aimath.org/WWN/perfectgraph/perfectgraph.pdf

13973 | GAME-004

### Tic-Tac-Toe Winning Dimension

P / partially_solved | 独立 / 未分配集合

Given the width of a tic-tac-toe board, what is the smallest dimension guaranteeing X has a winning strategy?

背景链接：https://www.unsolvedmath.com/problems/1361

13974 | AIM-COMBINATORICS-0154

### Tight 5-cycle minus an edge: solved density and a recursive deficit identity

P / partially_solved | AIM 研讨会问题汇编

**Conjecture (Tight $5$-cycle minus an edge).** Let the $3$-uniform hypergraph $C_5^-$ be obtained from the tight $5$-cycle $C_5^3$ by removing one edge. An example of a $C_5^-$-free $3$-graph can be obtained by taking a complete $3$-partite $3$-graph and repeating this construction recursively within each of the three parts, giving edge density $1/4$ in the limit. Prove that the Turán density $\pi(C_5^-) = \lim_{n \to \infty} \operatorname{ex}(n, C_5^-) / \binom{n}{3}$ satisfies $\pi(C_5^-) = 1/4$.

来源字段：http://aimpl.org/hypergraphturan/3/

13975 | OWR-1455-005

### Topology of Twisted-Involution Bruhat Intervals

O / open | Oberwolfach 报告问题汇编

Let $(W,S)$ be a Coxeter system, let $\theta$ be an involutive automorphism of $W$, and let $\mathcal{I}(\theta)=\{w\in W:\theta(w)=w^{-1}\}$ be the set of twisted involutions. What is the topology of the intervals in the induced Bruhat-order subposet $\operatorname{Br}(\mathcal{I}(\theta))$?

来源字段：https://doi.org/10.4171/owr/2007/04

13976 | AIM-COMBINATORICS-0187

### Trace equivalence and a triple-overlap obstruction

P / partially_solved | AIM 研讨会问题汇编

Conjecture 6.17. The following were all presented by Seymour, with no other attributions given. For the following questions, matrices are assumed to be \(n\times n\) \(0\)-\(1\) matrices such that \(a_{ij}=1\) implies \(a_{ji}\ne1\), and all diagonal elements equal \(1\).

The original AIM PDF verifies that the fragment is actually **items 3 and 4 of Conjecture 6.17**, in Section 6.3, “Matrices.” The stray definition of spectral radius and terminal “8” belong to the transition to Conjecture 6.18, not to this problem.

The following source text is verified directly from page 8 of the PDF.

Item 2, immediately preceding the assigned fragment, supplies the shared quantitative hypotheses: \[ A_1,\ldots,A_t,\qquad \text{row sums of }A_i\text{ at least }r_i+1,\qquad \sum_{i=1}^t r_i\ge n, \tag{1.1} \] and asks whether \[ \operatorname{tr}(A_1A_2\cdots A_t)>n. \tag{1.2} \] The source explicitly says that this is equivalent to Conjecture 6.5.

Item 3 forms \(t+1\) copies of the common vertex set \(V\), puts horizontal edges between equal vertices in consecutive copies, and puts the \(A_k\)-edges between copies \(k\) and \(k+1\). It asks for a nonhorizontal path from the first copy of some \(u\) to the last copy of the same \(u\). Item 4 squashes the layers onto \(V\), colors an edge from layer \(k\) by \(k\), and asks for a nontrivial rainbow directed cycle whose colors appear in increasing order.

The diagonal \(1\)'s are the horizontal waiting steps. Put \[ B_i=A_i-I,\qquad E_i=\{(u,v):u\ne v,\ (B_i)_{uv}=1\}. \tag{1.3} \] Then \(B_i\) is a zero-diagonal \(0\)-\(1\) matrix, every \(E_i\) is an oriented edge set, and \[ \delta^+(E_i)\ge r_i. \tag{1.4} \] The wording of item 3 can be read as adding a matrix diagonal edge on top of the already specified horizontal edge. Whether those coincident edges are viewed as one edge or two does not affect the existence question. The matrix model counts the diagonal choice once, and that is the convention used here.

An “increasing” colored cycle means that one can choose a cyclic starting vertex so that the distinct colors encountered along the directed traversal are \[ i_1<i_2<\cdots<i_s. \tag{1.5} \] This is exactly the convention forced by the layered construction.

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

13977 | GREEN-057

### Trace Reconstruction

P / partially_solved | Ben Green 问题

Given a string $x \in \{0, 1\}^n$, let $\tilde{x}$ be obtained by deleting bits independently at random with probability $\frac{1}{2}$. How many independent traces $\tilde{x}_1, \dots, \tilde{x}_m$ are needed to reconstruct $x$ with probability 0.9?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13978 | AIM-COMBINATORICS-0264

### Transcendental dilate sumsets and an exact digit-box construction

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.8 (I. Łaba).** Suppose that \(\alpha\) is transcendental, and \(|A|=n\). What is the best lower bound for \(|A+\alpha A|\)?

**Remark(s).** Konyagin and Łaba have shown that this cardinality is \[ \gg \frac{n\log n}{\log\log n}. \] The best example (lowest known cardinality) is \[ n e^{c\sqrt{\log n}}. \]

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

13979 | OPG-37230

### Transversal achievement game on a square grid

P / partially_solved | Open Problem Garden

Problem Two players alternately write O's (first player) and X's (second player) in the unoccupied cells of an $n \times n$ grid. The first player (if any) to occupy a set of $n$ cells having no two cells in the same row or column is the winner. What is the outcome of the game given optimal play?

背景链接：http://www.openproblemgarden.org/op/a_transversal_achievement_game_on_a_square_grid

13980 | OWR-17135-033

### Triangle Equation Hyperplane Arrangements

O / open | Oberwolfach 报告问题汇编

Let $\mathcal H$ be the arrangement in $\mathbb R^{\binom n2}$ consisting, for every unordered triple $\{i,j,k\}\subseteq[n]$, of the hyperplane $$H_{i,j,k}=\{x:x_{i,j}+x_{j,k}=x_{i,k}\}$$. Determine the intersection lattice, characteristic polynomial, regions, and other combinatorial invariants of $\mathcal H$.

来源字段：https://doi.org/10.4171/owr/2019/39

13981 | AMR-123-0001

### Trimer Covers in the Triangular Grid - Problem 1

O / open | AMR 问题清单汇编

Find an exact formula for $T_{003}(k(3k-1)/2,k(3k+1)/2)$.

Of course, even short of an exact formula, any method of determining the number of tilings that is more efficient than brute-force enumeration (e.g., a recurrence relation) would be of interest.

See [KiPr] for more discussion of no-stones tilings of benzels.

背景链接：https://arxiv.org/abs/2206.06472

13982 | AMR-123-0004

### Trimer Covers in the Triangular Grid - Problem 11

O / open | AMR 问题清单汇编

With $T(n) = T_{112}(3n-1,3n)$ with $n \geq 1$, it appears that $T(n)$ has no prime factor greater than or equal to $4n$.

Find a formula governing $T_{112}(3n-1,3n)$. See OEIS entry A364483 (https://oeis.org/A364483).

We now switch from central diagonals to peripheral diagonals.

背景链接：https://arxiv.org/abs/2206.06472

13983 | AMR-123-0005

### Trimer Covers in the Triangular Grid - Problem 12

O / open | AMR 问题清单汇编

Is it true that $T_{112}(n+2,2n+1)$ is the ``$n$th large Schr\"oder number'' (see OEIS entry A006318 (https://oeis.org/A006318)) for all $n \geq 1$? Equality has been verified for $1 \leq n \leq 15$.

背景链接：https://arxiv.org/abs/2206.06472

13984 | AMR-123-0006

### Trimer Covers in the Triangular Grid - Problem 13

O / open | AMR 问题清单汇编

Is it true that $T_{112}(n+2,2n)$ is the number of ``royal paths in a lattice of order $n$'' (see OEIS entry A006319 (https://oeis.org/A006319)) for all $n \geq 1$? Equality has been verified for $1 \leq n \leq 15$.

背景链接：https://arxiv.org/abs/2206.06472

13985 | AMR-123-0007

### Trimer Covers in the Triangular Grid - Problem 15

O / open | AMR 问题清单汇编

Is $T_{113}(n,2n-4)$ 2-adically continuous as a function of $n \geq 5$?

Comment: The sequence appears to be constant mod 2, constant mod 4, 2-periodic mod 8, and 8-periodic mod 16 (with repeating pattern 4, 4, 4, 4, 4, 12, 4, 12). The case $n = 4$ breaks the pattern. See OEIS entry A364416 (https://oeis.org/A364416).

背景链接：https://arxiv.org/abs/2206.06472

13986 | AMR-123-0008

### Trimer Covers in the Triangular Grid - Problem 16

O / open | AMR 问题清单汇编

Is $T_{113}(n,2n-3)$ 2-adically continuous as a function of $n \geq 4$?

Comment: The sequence appears to be constant mod 2, constant mod 4, 2-periodic mod 8, and 8-periodic mod 16 (with repeating pattern 2, 14, 2, 14, 10, 6, 10, 6). The case $n = 3$ breaks the pattern. See OEIS entry A364417 (https://oeis.org/A364417).

At the OPAC 2022 meeting, David Speyer suggested that one might use a different definition of a trimer, namely, a path consisting of three vertices and two edges. Thus, each stone would correspond to three different trimers according to which 2 of the 3 possible edges one used. Equivalently, one could use the original definition of a trimer, with the proviso that each stone will count with weight 3. The new numbers do not satisfy any nice patterns, with one exception: in the case where all prototiles are allowed, using stones of weight 3 seems to give rise to the same 2-adic continuity phenomenon we saw in Problems 11 and 12.

Let $T_{ijk}(a,b;3)$ denote the weighted sum of the $i,j,k$ tilings of the $(a,b)$-benzel as defined near the end of section 1,

where a tiling with $m$ stones has weight $3^m$.

背景链接：https://arxiv.org/abs/2206.06472

13987 | AMR-123-0009

### Trimer Covers in the Triangular Grid - Problem 17

O / open | AMR 问题清单汇编

Is $T_{113}(n,2n-4;3)$ 2-adically continuous as a function of $n \geq 5$?

Comment: The sequence appears to be constant mod 2, constant mod 4, constant mod 8, and 8-periodic mod 16 (with repeating pattern 4, 4, 12, 12, 4, 12, 12, 4). The case $n = 4$ breaks the pattern. See OEIS entry A364418 (https://oeis.org/A364418).

背景链接：https://arxiv.org/abs/2206.06472

13988 | AMR-123-0010

### Trimer Covers in the Triangular Grid - Problem 18

O / open | AMR 问题清单汇编

Comment: The sequence appears to be constant mod 2, constant mod 4, constant mod 8, and 8-periodic mod 16 (with repeating pattern 4, 4, 12, 12, 4, 12, 12, 4). The case $n = 4$ breaks the pattern. See OEIS entry A364418 (https://oeis.org/A364418).

Is $T_{113}(n,2n-3;3)$ 2-adically continuous as a function of $n \geq 4$?

Comment: The sequence appears to be constant mod 2, constant mod 4, 2-periodic mod 8, and 8-periodic mod 16 (with repeating pattern 14, 6, 10, 2, 6, 14, 2, 10). The case $n = 3$ breaks the pattern. See OEIS entry A364438 (https://oeis.org/A364438).

Comment: Given that the number of right stones minus the number of left stones in a tiling of a benzel is the same for all tilings of that benzel, a different way to describe the weighting assigned to tilings in Problems 17 and 18 (equivalent up to a power of 3) is to give weight 1 to left stones and weight 9 to right stones. During the final stages of preparing this article for publication, I realized that one could more generally give weight 1 to left stones and weight $m$ to right stones. The resulting enumerations appear to exhibit $p$-adic continuity when $m+1$ is a power of the prime $p$. Other forms of congruential consistency appear in the data as well; for instance, the data for $m=9$ appear to satisfy 5-adic continuity. I hope to address such patterns in future work.

背景链接：https://arxiv.org/abs/2206.06472

13989 | AMR-123-0011

### Trimer Covers in the Triangular Grid - Problem 20

O / open | AMR 问题清单汇编

The question we ask is, how many such tiling are there? OEIS entry A334875 (https://oeis.org/A334875) gives the answers for small values of $n$. If we look at the prime factorizations of the answers, we notice that the exponent of the prime 2 is creeping upward. Specifically, the multiplicity of the prime 2 in the factorizations of the nonzero terms in this sequence goes 0, 0, 1, 3, 2, 3, 4, 3, 4, 3, 5, 8, 6, 8, \dots. This is a priori surprising, since the probability that a ``random'' positive integer is divisible by $2^m$ decreases exponentially as $m$ increases.

As $n$ goes to infinity within the set of natural numbers congruent to 0, 2, 9, or 11 (mod 12), does the number of tiling of $T_n$ by stones converge 2-adically to 0?

背景链接：https://arxiv.org/abs/2206.06472

13990 | AMR-123-0002

### Trimer Covers in the Triangular Grid - Problem 4

O / open | AMR 问题清单汇编

When $a$ and $b$ are such that the Conway-Lagarias invariant is strictly positive, the $(a,b)$-benzel cannot be tiled by bones and left stones; the corresponding entries in the table must be zero. On the other hand, when $a$ and $b$ are such that the Conway-Lagarias invariant is negative or zero, the entries in the table are observed to be positive, though I see no reason for concluding that they are.

Is it true that when the Conway-Lagarias invariant associated with the $(a,b)$-benzel is negative or zero, tilings of type 013 exist?

背景链接：https://arxiv.org/abs/2206.06472

13991 | AMR-123-0003

### Trimer Covers in the Triangular Grid - Problem 7

O / open | AMR 问题清单汇编

In parallel with the observations that preceded Problem 4, note that when the Conway-Lagarias invariant of a benzel is strictly negative, the benzel cannot be tiled by bones and right-pointing stones; the corresponding entries in the table must be zero. On the other hand, when $a$ and $b$ are such that the Conway-Lagarias invariant is positive or zero, the entries in the table are observed to be positive, though I see no reason for concluding that they are.

Is it true that when the Conway-Lagarias invariant associated with the $(a,b)$-benzel is positive or zero, tilings of type 103 exist?

背景链接：https://arxiv.org/abs/2206.06472

13992 | GREEN-012

### Tuples in Dense Sets

P / partially_solved | Ben Green 问题

Let $G$ be an abelian group of size $N$, and suppose that $A \subset G$ has density $\alpha$. Are there at least $\alpha^{15}N^{10}$ tuples $(x_1, \ldots, x_5, y_1, \ldots, y_5) \in G^{10}$ such that $x_i + y_j \in A$ whenever $j \in \{i, i+1, i+2\}$?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

13993 | OWR-11781-004

### Turán Densities of Hypergraph Daisies

P / partially_solved | Oberwolfach 报告问题汇编

Let $D_r$ be the $r$-daisy consisting of the six $r$-sets $A$ with $P\subset A\subset P\cup Q$ for an $(r-2)$-set $P$ and a disjoint $4$-set $Q$. How large can a family $\mathcal A$ of $r$-sets from an $n$-set be if it contains no copy of $D_r$?

来源字段：https://doi.org/10.4171/owr/2012/19

13994 | OPG-60006

### Turán Problem for $10$-Cycles in the Hypercube

P / partially_solved | Open Problem Garden

Problem Bound the extremal number of $C_{10}$ in the hypercube.

背景链接：http://www.openproblemgarden.org/op/turan_problem_for_10_cycles_in_the_hypercube

13995 | AIM-COMBINATORICS-0043

### Tusnady's problem: canonical models and monotone-chain certificates

P / partially_solved | AIM 研讨会问题汇编

**Tusnady's problem.** Let $d_n$ be the discrepancy of $n$ points in $\mathbb R^2$ wrt all alix-aligned rectangles i.e. the set system comprises of all axis-aligned rectangles in $\mathbb R^2$ as sets, maximised over all choices of $n$ points. Tusnady's problem then asks for the exact value of $d_n$. We can also ask for the same question in $\mathbb R^d$.

来源字段：http://aimpl.org/hereddiscrep/1/

13996 | OWR-14299905-006

### Tverberg Numbers for Unions of Convex Sets

P / partially_solved | Oberwolfach 报告问题汇编

What is the least integer $f_r(d,s_1,\ldots,s_r)$ such that every set $P$ of that many points in $\mathbb R^d$ has a partition $P=P_1\sqcup\cdots\sqcup P_r$ with the following property: whenever each $C_i$ is $s_i$-convex and contains $P_i$, the intersection $\bigcap_{i=1}^r C_i$ is nonempty?

来源字段：https://doi.org/10.4171/owr/2026/1

13997 | OWR-11695867-002

### Twenty-Vertex Models and Pachter-Triangle Tilings

P / partially_solved | Oberwolfach 报告问题汇编

Is the number of configurations of the twenty-vertex model with DWBC3 on an $n\times n$ grid equal to the number of domino tilings of Pachter's triangle of size $2n$, and does the analogous equality hold for the extended $(n+k)\times n$ model and enhanced triangle for every $k\geq 0$?

来源字段：https://doi.org/10.4171/owr/2022/57

13998 | AIM-COMBINATORICS-0085

### Two-colored planes and the exact stripe spectrum

P / partially_solved | AIM 研讨会问题汇编

Conjecture. In any 2-coloring of E^2, every 3-point set X occurs monochromatically (as some congruent copy), except possibly for the three points that form some specific equilateral triangle.

(Example coloring: half-open horizontal stripes of height sqrt(3)/2 in alternating colors will stop the three vertices of a unit equilateral triangle from occurring monochromatically).

Known: If T is a set of 3 collinear points, then there exists a 16-coloring of E^N with no monochromatic copy of T.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

13999 | AIM-COMBINATORICS-0178

### Two-label extremality and the meaning of average reachability

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 6.10 (DeVos).** Let $G$ be a directed graph whose edge set is partitioned into $k$ color classes $E_1, \ldots, E_k$, and let $G_i = (V(G), E_i)$. If $G$ has no rainbow directed cycle (a directed cycle with at most one edge from each $E_i$), must there exist a vertex $v \in V(G)$ such that $\sum_{i=1}^k d_{G_i}^+(v) \cdot d_{G_i}^-(v) \geq \sum_{i \neq j} d_{G_i}^+(v) \cdot d_{G_j}^-(v)$?

来源字段：https://aimath.org/WWN/caccetta/caccetta.pdf

14000 | AIM-COMBINATORICS-0241

### Two-step product growth in SL_2(F_p)

S / solved | AIM 研讨会问题汇编

**Problem 3.4 (Brought by B. Green, originally stated by A. Venkatesh).** Let \(A\subseteq SL_2(\mathbb F_p)\) satisfy \(|A|\sim p^{5/2}\). Does it follow that \[ |A\cdot A|>p^{5/2+\delta}\,? \]

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

14001 | AIM-COMBINATORICS-0144

### Typed identifiability and minimal data for congestion

P / partially_solved | AIM 研讨会问题汇编

What are the data required to define congestion?

Do weights on the network or some such affect this?

来源字段：http://aimpl.org/largenetworks/1/

14002 | OWR-14298007-013

### Ultra-Log-Concavity from a Matrix Recursion

O / open | Oberwolfach 报告问题汇编

For $n\geq 4$, define $(T_{n+1},Q_{n+1},C_{n+1})^T=M(T_n,Q_n,C_n)^T$, where $M=\begin{pmatrix}z&1+z&1+z\\0&1+z&z\\z+z^2&0&1+z\end{pmatrix}$ and $(T_4,Q_4,C_4)^T=(z^4+2z^3,z^4,2z^4+2z^3)^T$. Writing $V_n=T_n+Q_n+C_n=\sum_kv_{n,k}z^k$, prove that $(v_{n,k})_k$ is ultra-log-concave. More generally, develop criteria for unimodality and ultra-log-concavity of polynomial vectors generated by nonnegative polynomial-matrix recursions.

来源字段：https://doi.org/10.4171/owr/2023/58

14003 | OWR-4425-009

### Unbordered Factors and Alternating Involution Periods

O / open | Oberwolfach 报告问题汇编

Let $\theta$ be a morphic involution on $A^*$ and $w$ an $A$-word. Is it true that $|w|\geq3\tau_\theta(w)$ implies $\tau_\theta(w)=\pi_\theta^{\mathrm{alt}}(w)$?

来源字段：https://doi.org/10.4171/owr/2010/37

14004 | OWR-10252930-032

### Uncommon Pairs of Linear Equations

P / partially_solved | Oberwolfach 报告问题汇编

For even $k\geq 6$, if the common solution space of $\sum_{i=1}^k a_ix_i=0$ and $\sum_{i=1}^k b_ix_i=0$ has dimension $k-2$, is the pair of equations uncommon over every sufficiently large finite field?

来源字段：https://doi.org/10.4171/owr/2022/22

14005 | AIM-COMBINATORICS-0086

### Uniform color bounds for avoiding three-point collinear configurations

P / partially_solved | AIM 研讨会问题汇编

Known: If \(T\) is a set of 3 collinear points, then there exists a 16-coloring of \(\mathbb E^N\) with no monochromatic copy of \(T\).

**Recovered question.** Determine \(q_*\). In particular, can the uniform upper bound \(q_*\leq16\) be reduced?

The immediately preceding record, AIM-COMBINATORICS-0085, ends with:

The next record begins the unrelated section on Folkman graphs. Thus “above” refers to the constant 16, not to the preceding conjectures about arbitrary plane triples. This reading is independently confirmed by the historical formulation of Erdős and Graham and by Graham--Tressler: for every three-point collinear set and in every dimension there is a 16-coloring avoiding a monochromatic congruent copy, and the question is whether 16 is the best universal constant.

Here is a precise formulation. For a nondegenerate three-point collinear set \(T\), let \[ \chi_n(T)=\min\{q:\mathbb E^n\text{ has a coloring with at most }q\text{ colors and no monochromatic congruent copy of }T\}. \] Define \[ q_*:=\sup_{T}\sup_{n\geq 1}\chi_n(T), \] where \(T\) ranges over triples of distinct collinear points. The recovered question is:

The coloring may depend on \(T\) and \(n\); only its number of colors must be uniform. No measurability or other regularity is imposed. A “copy” means a congruent copy, so its scale is fixed. Degenerate triples with repeated points are excluded.

The source Overleaf page was not available for direct inspection in this run.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

14006 | AIM-COMBINATORICS-0253

### Uniform iterated-sumset growth at the first doubling threshold

P / partially_solved | AIM 研讨会问题汇编

Suppose \(A\subseteq\mathbb Z\). If \(|2A|<k|A|\), describe \(|nA|\) as \(n\) tends to infinity. Find \(f(n,k)\), the smallest number such that \[ |nA|\le f(n,k)|A| \] for all \(A\) such that \(|2A|<k|A|\).

来源字段：https://aimath.org/WWN/additivecomb/additivecomb.pdf

14007 | GREEN-071

### Uniform Random Variables with Uniform Sum

P / partially_solved | Ben Green 问题

Suppose $X, Y$ are finitely-supported independent random variables taking integer values such that $X + Y$ is uniformly distributed on its range. Are $X$ and $Y$ themselves uniformly distributed on their ranges?

来源字段：https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

14008 | AIM-COMBINATORICS-0116

### Uniform versus torsion-squared higher spanning trees

P / partially_solved | AIM 研讨会问题汇编

Caroline Klivans: “Is there an efficient algorithm to generate a higher dimensional spanning tree uniformly at random?”

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

14009 | AIM-COMBINATORICS-0018

### Uniform-prime counterexample and a fixed-characteristic polynomial reduction

S / solved | AIM 研讨会问题汇编

3APs in sets of small growth

Obtain a proof of the following theorem using the polynomial method.

Theorem. There is an absolute constant \(c>0\) that the following holds. If \(A\subset \mathbb F_p^n\) and \(|A+A|\leq |A|^{1+c}\) then \(A\) contains a 3AP.

来源字段：http://aimpl.org/addcombapp/1/

14010 | OWR-741-003

### Unimodality of Gorenstein Polytope h-Star Vectors

S / solved | Oberwolfach 报告问题汇编

If the semigroup ring $R_P$ of an integral polytope $P$ is standard and Gorenstein, must $h^*(P)$ be unimodal? More generally, if $h_i^*(P)=h_{d-i}^*(P)$ for all $i$, must $h^*(P)$ be unimodal?

来源字段：https://doi.org/10.4171/owr/2004/39

14011 | AMR-054-0004

### Union of Fat Objects in 3D

O / open | AMR 问题清单汇编

What is the complexity of the union of ``fat'' objects in $\mathbb{R}^3$?

背景链接：https://topp.openproblem.net/p4

14012 | OWR-12697684-014

### Unions of Linear Bases as Additive Bases

P / partially_solved | Oberwolfach 报告问题汇编

For every prime $p$, does there exist $k(p)$ such that every union of $k(p)$ linear bases of $\mathbb F_p^n$ is an additive basis of $\mathbb F_p^n$?

来源字段：https://doi.org/10.4171/owr/2023/1

14013 | OWR-2654827-004

### Universal Lefschetz Parameters for Neighborly Spheres

O / open | Oberwolfach 报告问题汇编

For neighborly spheres, does every linear system of parameters have the Lefschetz property?

来源字段：https://doi.org/10.4171/owr/2020/31

14014 | OWR-14298158-009

### Universal Permuton Convergence and Algebraic Functional Equations

O / open | Oberwolfach 报告问题汇编

For a permutation class $C$, does a uniformly random permutation of size $n$ in $C$ always converge to a limiting permuton, and what is the analogue of this question for graphons? Also, which polynomial functional equations of the form $$P\bigl(F(t,u),F(t,0),\partial_uF(t,0),t,u\bigr)=0$$ force the formal power series $F(t,u)$ to be algebraic under natural hypotheses? More generally, when a series $F(t,u,v)$ satisfies a polynomial relation involving $F(t,u,v)$, auxiliary series $G_i(t,u)$, and series $H_j(t)$, what natural hypotheses guarantee that $F$ is algebraic?

来源字段：https://doi.org/10.4171/owr/2024/6

14015 | OWR-741-006

### Universality of Real Table Polytopes

S / solved | Oberwolfach 报告问题汇编

Is every real polytope representable as a table polytope, namely a polytope of nonnegative arrays with prescribed line sums?

来源字段：https://doi.org/10.4171/owr/2004/39

14016 | AMR-029-0086

### Upper bound on common independent set cover

O / open | AMR 问题清单汇编

For a loopless matroid $M=(S,r)$, let $\Delta(M)=\max_{X\subseteq S} |X|/r(X)$. Let $M_1=(S,r_1)$ and $M_2=(S,r_2)$ be two arbitrary loopless matroids on S. Is it true that S can be partitioned into $\lceil\max\{\Delta(M_1),\Delta(M_2)\}\rceil+1$ common independent sets?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

14017 | AMR-029-0087

### Upper bound on the divisorial gonality of a graph

P / partially_solved | AMR 问题清单汇编

$\rm{gon}(G) \leq \frac{|E(G)|-|V(G)|}{2} + 2$, where $\rm{gon}(G)$ the denotes the divisorial gonality of graph $G$.

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

14018 | OWR-1183-008

### Upper Bounds for Low-Level Hemisphere Arrangements

P / partially_solved | Oberwolfach 报告问题汇编

If $\mathcal A$ is an arrangement of $n$ great hemispheres in $S^d$ and $0\leq\ell<(n-d)/2$, is the number of vertices at level at most $\ell$ at most $v(\ell,n,d)$, the corresponding number for the arrangement polar-dual to a cyclic $d$-polytope on $n$ vertices? A vertex's level is the number of hemispheres not containing it.

来源字段：https://doi.org/10.4171/owr/2006/01

14019 | OWR-14299904-002

### Upward Closure of Grothendieck Schubert Supports

P / partially_solved | Oberwolfach 报告问题汇编

别名：Upper Elements in Grothendieck Schubert Supports [OWR-14299904-003]

Let $G_w$ be a Grothendieck polynomial. If $u\in\operatorname{Supp}_S(G_w)$ and $\ell(u)<\deg(G_w)$, must there exist $v\in\operatorname{Supp}_S(G_w)$ with $u<_L v$?

来源字段：https://doi.org/10.4171/owr/2026/2

14020 | AIM-COMBINATORICS-0088

### van der Waerden numbers and sparse forcing sets

P / partially_solved | AIM 研讨会问题汇编

The source record, attributed to Ron Graham, defines \(W(n,m)\) to be the least \(N\) such that every red/blue coloring of \([N]=\{1,\ldots,N\}\) contains either a red \(n\)-term arithmetic progression or a blue \(m\)-term arithmetic progression, and puts \(W(n)=W(n,n)\). It records

\[ W(2)=3,\quad W(3)=9,\quad W(4)=35,\quad W(5)=178,\quad W(6)=1132, \]

Berlekamp's lower bound \(W(p+1)\ge p2^p\) for prime \(p\), Gowers's upper bound

\[ W(n)\le 2^{2^{2^{2^{2^{n+9}}}}}, \]

and Graham's prize conjecture

\[ \boxed{W(n)\le 2^{n^2}}. \]

It also defines \(W^*(n)\) as the minimum cardinality of a finite set \(S\subset\mathbb N\) such that every two-coloring of \(S\) contains a monochromatic \(n\)-term arithmetic progression. It records \(W^*(3)=W(3)=9\) and \(W^*(4)\le27<W(4)=35\).

Throughout, an arithmetic progression is \(a,a+d,\ldots,a+(n-1)d\) with \(d>0\), so its terms are distinct. The source's damaged tabular markup was reconstructed only as the displayed list of exact values above. Translating a finite subset of \(\mathbb Z\) into \(\mathbb N\) preserves its progressions, so the variants using \(S\subset\mathbb Z\) and \(S\subset\mathbb N\) have the same parameter.

来源字段：https://www.overleaf.com/read/mnvcscjjysvg

14021 | AIM-COMBINATORICS-0136

### Variable-curvature distortion bounds for embedded neighborhood graphs

P / partially_solved | AIM 研讨会问题汇编

Need ways to compute geodesics on spaces where curvature is variable (using embedded graphs).

来源字段：http://aimpl.org/largenetworks/1/

14022 | AMR-054-0019

### Vertical Decompositions in $\mathbb{R}^d$

O / open | AMR 问题清单汇编

What is the complexity of the vertical decomposition of $n$ surfaces in $\mathbb{R}^d$, $d \ge 5$?

背景链接：https://topp.openproblem.net/p19

14023 | AIM-COMBINATORICS-0096

### Volume and resistance crossover for the proposed parabolic IDLA domains

P / partially_solved | AIM 研讨会问题汇编

For internal diffusion-limited aggregation (IDLA), repeatedly launch explorers from the origin. On \(\mathbb Z^d\) the limiting shape is round, with radius of order \(n^{1/d}\) after \(n\) explorers. The quoted fluctuation scales are \(\sqrt{\log n}\) for \(d\geq 3\), \(\log n\) for \(d=2\), and \(\sqrt n\) for \(d=1\). Are there growth models with intermediate fluctuation scales? In particular, what happens on a wedge in \(\mathbb Z^2\), or on the induced lattice region \[ |y|\leq x^{1/\gamma}, \] and, more generally, on \(|y|\leq f(x)\)?

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

14024 | OWR-14299904-005

### Weak-Order Convexity of Grothendieck Schubert Supports

P / partially_solved | Oberwolfach 报告问题汇编

For every $w\in S_n$, is $\operatorname{Supp}_S(G_w)$ convex in left weak order? Equivalently, if $u,u^{\prime}\in\operatorname{Supp}_S(G_w)$, is every $v$ with $u\leq_L v\leq_L u^{\prime}$ also in the support?

来源字段：https://doi.org/10.4171/owr/2026/2

14025 | OWR-14299904-004

### Weak-Order Growth of Grothendieck Schubert Supports

P / partially_solved | Oberwolfach 报告问题汇编

Let $w\in S_n$. If $u\in\operatorname{Supp}_S(G_w)$ and $\ell(u)<\deg(G_w)$, must there exist $k$ such that $u<_L s_k u$ and $s_k u\in\operatorname{Supp}_S(G_w)$?

来源字段：https://doi.org/10.4171/owr/2026/2

14026 | AMR-029-0088

### Weighted bipartite edge colouring

P / partially_solved | AMR 问题清单汇编

Let G=(S,T;E) be a bipartite graph, with weights $w:E \to [0,1]$. A proper weighted edge colouring is a colouring of the edges such that at each vertex, the sum of weights of edges of the same colour is at most 1. A lower bound for the number of colours is the minimum number of unit bins needed to pack the weights incident to any vertex, denoted by b. Is there always a proper weighted edge colouring using $2b-1$ colours?

背景链接：https://oldlemon.cs.elte.hu/egres/open/List_of_open_problems

14027 | AIM-COMBINATORICS-0274

### What a 2-join block must remember for perfect contractility

O / open | AIM 研讨会问题汇编

**Conjecture.** 2-Join decomposition is \(\mathcal C^{pc}\)-preserving.

Here \(\mathcal C^{pc}\) is the class of perfectly contractile graphs. The preceding text in the AIM source defines a 2-join as a partition \(V(G)=V_1\dot\cup V_2\) with pairwise disjoint nonempty sets \(A_i,B_i\subseteq V_i\) such that:

1. all edges between the two sides are precisely the edges of the complete joins \(A_1\)--\(A_2\) and \(B_1\)--\(B_2\); and 2. if \(\mathcal P_i\) is the family of chordless \(A_i\)--\(B_i\) paths in \(G[V_i]\) whose internal vertices avoid \(A_i\cup B_i\), then \(\mathcal P_i\ne\varnothing\), and \(G[V_i]\) is not isomorphic to a member of \(\mathcal P_i\).

The last condition excludes a path side, in terminology used by later 2-join literature.

* **Extraction defect.** After the displayed 2-join conjecture it absorbs a different star-cutset conjecture, Sections F.5 and F.6, and Sections G.1--G.4. The next canonical record starts with a conjecture from G.4. Those later paragraphs are not qualifications of the 2-join conjecture. * **Statement defect.** The source explains abstractly that blocks are formed by adding vertices and edges, but never specifies the blocks for this 2-join. Thus “the 2-join decomposition” is not a defined operation in this record. The opening conjecture has a clear research intent but no literal truth value until a block convention is fixed.

The mathematical work below concerns only the exact opening conjecture. It does not treat the unrelated material accidentally merged into the record.

来源字段：https://aimath.org/WWN/perfectgraph/perfectgraph.pdf

14028 | AMR-030-0029

### What are the (homogeneous adjacency) spectra of the ultracube and the complete hypergraph

O / open | AMR 问题清单汇编

/Dutle : What are the (homogeneous adjacency) spectra of the ultracube and the complete hypergraph? (Ultracube = cartesian power of a hyperedge.)

背景链接：https://people.math.sc.edu/cooper/combprob.html

14029 | AMR-030-0038

### What are the Whitney numbers of the (lattice of contractions of the) n-cube

O / open | AMR 问题清单汇编

: What are the Whitney numbers of the (lattice of contractions of the) n-cube? What if contractions equivalent under symmetries of the cube are identified? How many contractions are there up to isomorphism? See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

14030 | AMR-030-0030

### What is the (homogeneous adjacency) spectrum of the Fano plane

O / open | AMR 问题清单汇编

/Clark : What is the (homogeneous adjacency) spectrum of the Fano plane?

背景链接：https://people.math.sc.edu/cooper/combprob.html

14031 | AMR-030-0036

### What is the length of the shortest sequence in [n]* containing, as a (consecutive) subword, each permutation of [n]

P / partially_solved | AMR 问题清单汇编

What is the length of the shortest sequence in [n]* containing, as a (consecutive) subword, each permutation of [n]? See this, this, this, this, and this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

14032 | AMR-030-0024

### What is the list-chromatic number of Sudoku

O / open | AMR 问题清单汇编

: What is the list-chromatic number of Sudoku? That is, suppose one places k symbols (aka colors) in each cell of a Sudoku board -- not necessarily all the same lists of symbols. What is the least k so that it is always possible to choose a symbol/color from each list for its cell that the resulting choices have no conflicts in rows, columns, or blocks (the 3X3 submatrices that cannot have conflicts in ordinary Sudoku)? This is the Sudoku analogue of the famous Dinitz Conjecture / Galvin Theorem. Even for k=4, the question is open.

背景链接：https://people.math.sc.edu/cooper/combprob.html

14033 | AMR-030-0027

### What is the meaning of the multiplicity of zero as a root of a hypergraph's (or graph's) characteristic polynomial

O / open | AMR 问题清单汇编

Nikiforov: What is the meaning of the multiplicity of zero as a root of a hypergraph's (or graph's) characteristic polynomial?

背景链接：https://people.math.sc.edu/cooper/combprob.html

14034 | AMR-030-0009

### What is the minimum number of n-simplexes needed to triangulate the n-cube

O / open | AMR 问题清单汇编

What is the minimum number of n-simplexes needed to triangulate the n-cube? See this.

背景链接：https://people.math.sc.edu/cooper/combprob.html

14035 | AMR-030-0084

### What is the probability that a random nXn matrix over Z_(p) has zero permanent as n goes to infinity

P / partially_solved | AMR 问题清单汇编

Tao: What is the probability that a random nXn matrix over Z_(p) has zero permanent as n goes to infinity? (Surely 1/p... as long as p is not 2.)

背景链接：https://people.math.sc.edu/cooper/combprob.html

14036 | AMR-030-0083

### What is the threshold function n = f(k) for the event that a random permutation on n symbols contains all patterns on k

P / partially_solved | AMR 问题清单汇编

Alon: What is the threshold function n = f(k) for the event that a random permutation on n symbols contains all patterns on k symbols? Conjecture: f(k) = k^(2)/2 (1+o(1)).

背景链接：https://people.math.sc.edu/cooper/combprob.html

14037 | AIM-COMBINATORICS-0131

### What the critical group sees of a harmonic action

O / open | AIM 研讨会问题汇编

“If you have a graph with a harmonic action, then its critical group comes with some extra structure, namely a group of automorphisms of the graph. That extra object might be worth studying.”

This is item (41), attributed to Lionel Levine, in the AIM workshop list *Generalizations of chip-firing and the critical group*:

There is no missing formula and no question mark to reconstruct. The item is a research suggestion rather than a proposition or a question with fixed quantifiers. The precise object suggested by the remark is the finite \(\mathbb Z[G]\)-module \(\operatorname{Jac}(X)\), where \(G\) acts harmonically on \(X\) and hence acts on degree-zero divisors and their quotient by principal divisors.

There is one important correction to a literal reading of “a group of automorphisms”: a faithful action on the graph need not induce a faithful action on its critical group. Thus the extra datum is a representation \[ \rho_X:G\longrightarrow \operatorname{Aut}(\operatorname{Jac}(X)), \] together with its kernel, not automatically an embedded copy of \(G\).

来源字段：https://aimath.org/pastworkshops/chipfiringproblems.pdf

14038 | OPG-2063

### Wide partition conjecture

P / partially_solved | Open Problem Garden

Conjecture An integer partition is wide if and only if it is Latin.

背景链接：http://www.openproblemgarden.org/op/wide_partition_conjecture

14039 | OWR-11695867-007

### Wilf Class Asymptotics with Fixed Pattern Occurrences

O / open | Oberwolfach 报告问题汇编

For each Wilf class with exactly $r$ occurrences of the pattern, is $\psi_r(n)\sim C_r n^r\psi_0(n)$, and what moment-sequence properties do the corresponding occurrence generating functions have?

来源字段：https://doi.org/10.4171/owr/2022/57

# 资料汇编 11 | 图论 / Graph Theory

1,181 条记录。O=开放；P=部分解决；S=已解决。状态与英文/LaTeX 表述均沿用来源。

14040 | OPG-139

### (m,n)-cycle covers

P / partially_solved | Open Problem Garden

Conjecture Every bridgeless graph has a (5,2)-cycle-cover.

背景链接：http://www.openproblemgarden.org/op/m_n_cycle_covers

14041 | GRAPH-008

### 1-Factorization Conjecture

P / partially_solved | 独立 / 未分配集合

Does every k-regular graph on 2n vertices admit a 1-factorization when k ≥ n (or k ≥ n-1 for even n)?

背景链接：https://arxiv.org/abs/1401.4159

14042 | AMR-027-0203

### 10 Lectures and 42 Open Problems - The planted clique problem

O / open | AMR 问题清单汇编

Is there a polynomial time algorithm that is able to find the largest clique of $G$ (with high probability) for $\omega \ll \sqrt{n}$ ? For example, for $\omega \approx \frac{\sqrt{n}}{\log n}$ ? Is there a polynomial time algorithm that is able to distinguish, with high probability, $G$ from a draw of $G\left( n, \frac12\right)$ for $\omega \ll \sqrt{n}$ ? For example, for $\omega \approx \frac{\sqrt{n}}{\log n}$ ? Is there a quasi-linear time algorithm able to find the largest clique of $G$ (with high probability) for $\omega\leq \left(\frac1{\sqrt{e}}-\epsilon \right)\sqrt{n}$ , for some $\epsilon>0$ ? (This question was posed here )

背景链接：https://afonsobandeira.wordpress.com/2015/09/30/10l42pgraphs/

14043 | OPG-56230

### 2-colouring a graph without a monochromatic maximum clique

P / partially_solved | Open Problem Garden

Conjecture If $G$ is a non-empty graph containing no induced odd cycle of length at least $5$, then there is a $2$-vertex colouring of $G$ in which no maximum clique is monochromatic.

背景链接：http://www.openproblemgarden.org/op/2_colouring_a_graph_without_a_monochromatic_maximum_clique

14044 | OPG-34915

### 3-Colourability of Arrangements of Great Circles

P / partially_solved | Open Problem Garden

Consider a set $S$ of great circles on a sphere with no three circles meeting at a point. The arrangement graph of $S$ has a vertex for each intersection point, and an edge for each arc directly connecting two intersection points. So this arrangement graph is 4-regular and planar.

Conjecture Every arrangement graph of a set of great circles is $3$-colourable.

背景链接：http://www.openproblemgarden.org/op/3_colourability_of_arrangements_of_great_circles

14045 | OPG-60027

### 3-Decomposition Conjecture

P / partially_solved | Open Problem Garden

Conjecture (3-Decomposition Conjecture) Every connected cubic graph $G$ has a decomposition into a spanning tree, a family of cycles and a matching.

背景链接：http://www.openproblemgarden.org/op/3_decomposition_conjecture

14046 | OPG-60046

### 3-Edge-Coloring Conjecture

P / partially_solved | Open Problem Garden

Conjecture Suppose $G$ with $|V(G)|>2$ is a connected cubic graph admitting a $3$-edge coloring. Then there is an edge $e \in E(G)$ such that the cubic graph homeomorphic to $G-e$ has a $3$-edge coloring.

背景链接：http://www.openproblemgarden.org/op/3_edge_coloring_conjecture

14047 | OPG-128

### 3-flow conjecture

O / open | Open Problem Garden

Conjecture Every 4-edge-connected graph has a nowhere-zero 3-flow.

背景链接：http://www.openproblemgarden.org/op/3_flow_conjecture

14048 | AMR-054-0046

### 3D Minimum-Bend Orthogonal Graph Drawings

O / open | AMR 问题清单汇编

Does every simple graph with maximum vertex degree $\Delta \leq 6$ have a 3D orthogonal point-drawing with no more than two bends per edge? A 3D orthogonal point-drawing of a graph maps each vertex to a unique point of the 3D cubic lattice, and maps each edge to a lattice path between the endpoints; these paths can only intersect at common endpoints. In this problem, each path must have at most two bends, that is, consist of at most three orthogonal line segments (links).

背景链接：https://topp.openproblem.net/p46

14049 | OPG-47294

### 4-connected graphs are not uniquely hamiltonian

O / open | Open Problem Garden

Conjecture Every $4$-connected graph with a Hamilton cycle has a second Hamilton cycle.

背景链接：http://www.openproblemgarden.org/op/4_connected_graphs_are_not_uniquely_hamiltonian

14050 | OPG-127

### 4-flow conjecture

O / open | Open Problem Garden

Conjecture Every bridgeless graph with no Petersen minor has a nowhere-zero 4-flow.

背景链接：http://www.openproblemgarden.org/op/4_flow_conjecture

14051 | OPG-830

### 4-regular 4-chromatic graphs of high girth

O / open | Open Problem Garden

Problem Do there exist 4-regular 4-chromatic graphs of arbitrarily high girth?

背景链接：http://www.openproblemgarden.org/op/high_girth_low_degree_4_chromatic_graphs

14052 | OPG-126

### 5-flow conjecture

O / open | Open Problem Garden

Conjecture Every bridgeless graph has a nowhere-zero 5-flow.

背景链接：http://www.openproblemgarden.org/op/5_flow_conjecture

14053 | OPG-411

### 5-local-tensions

O / open | Open Problem Garden

Conjecture There exists a fixed constant $c$ (probably $c=4$ suffices) so that every embedded (loopless) graph with edge-width $\ge c$ has a 5-local-tension.

背景链接：http://www.openproblemgarden.org/op/5_local_tensions

14054 | OPG-160

### 57-regular Moore graph?

O / open | Open Problem Garden

Question Does there exist a 57-regular graph with diameter 2 and girth 5?

背景链接：http://www.openproblemgarden.org/op/57_regular_moore_graph

14055 | OPG-182

### A generalization of Vizing's Theorem?

O / open | Open Problem Garden

Conjecture Let $H$ be a simple $d$-uniform hypergraph, and assume that every set of $d-1$ points is contained in at most $r$ edges. Then there exists an $r+d-1$-edge-coloring so that any two edges which share $d-1$ vertices have distinct colors.

背景链接：http://www.openproblemgarden.org/op/a_generalization_of_vizings_theorem

14056 | OPG-134

### A homomorphism problem for flows

O / open | Open Problem Garden

Conjecture Let $M,M'$ be abelian groups and let $B \subseteq M$ and $B' \subseteq M'$ satisfy $B=-B$ and $B' = -B'$. If there is a homomorphism from $Cayley(M,B)$ to $Cayley(M',B')$, then every graph with a B-flow has a B'-flow.

背景链接：http://www.openproblemgarden.org/op/a_homomorphism_problem_for_flows

14057 | AMR-084-0001

### Absolute bounds for rational Diophantine tuples

O / open | AMR 问题清单汇编

Is there an absolute upper bound for the size of a rational Diophantine $m$-tuple, a set of nonzero rationals for which the product of every two distinct elements plus $1$ is a rational square?

背景链接：https://web.math.pmf.unizg.hr/~duje/dtuples.html

14058 | OPG-145

### Acyclic edge-colouring

P / partially_solved | Open Problem Garden

Conjecture Every simple graph with maximum degree $\Delta$ has a proper $(\Delta+2)$-edge-colouring so that every cycle contains edges of at least three distinct colours.

背景链接：http://www.openproblemgarden.org/op/acyclic_edge_coloring

14059 | OPG-46940

### Acyclic list colouring of planar graphs.

P / partially_solved | Open Problem Garden

Conjecture Every planar graph is acyclically 5-choosable.

背景链接：http://www.openproblemgarden.org/op/acyclic_list_colouring_of_planar_graphs

14060 | AMR-099-0005

### Additive-error graph model of the Euclidean plane

O / open | AMR 问题清单汇编

Does there exist a graph $G$ and a map $f:V(G)\to\mathbb{R}^2$ such that $|\|f(x)-f(y)\|_2-d_G(x,y)|<C$ for every $x,y\in V(G)$ and some constant $C<\infty$? Can such a graph have bounded degree?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=10

14061 | GRAPH-015

### Albertson Conjecture

P / partially_solved | 独立 / 未分配集合

Can the crossing number of a graph be lower-bounded by the crossing number of a complete graph with the same chromatic number?

背景链接：https://arxiv.org/abs/2512.08020

14062 | AMR-087-0074

### Algebraic algorithms for ideal-lattice problems

P / partially_solved | AMR 问题清单汇编

Use algebraic tools to solve computational problems on ideal lattices efficiently.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14063 | OWR-14298007-006

### Algebraic Connectivity of Complete Graphs

P / partially_solved | Oberwolfach 报告问题汇编

For $d\geq 3$ and $n\geq d+1$, prove the conjectured upper bound $a_d(K_n)\leq n/d$ for the $d$-dimensional algebraic connectivity of the complete graph.

来源字段：https://doi.org/10.4171/owr/2023/58

14064 | AMR-087-0026

### Algebraic independence in multivariate elimination

O / open | AMR 问题清单汇编

Give conditions or constructions that ensure algebraic independence in multivariate elimination for small-root attacks.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14065 | OPG-37232

### Algorithm for graph homomorphisms

P / partially_solved | Open Problem Garden

Question

Is there an algorithm that decides, for input graphs $G$ and $H$, whether there exists a homomorphism from $G$ to $H$ in time $O(c^{|V(G)|+|V(H)|})$ for some constant $c$?

背景链接：http://www.openproblemgarden.org/op/algorithm_for_graph_homomorphisms

14066 | OPG-56028

### Almost all non-Hamiltonian 3-regular graphs are 1-connected

O / open | Open Problem Garden

Conjecture Denote by $NH(n)$ the number of non-Hamiltonian 3-regular graphs of size $2n$, and similarly denote by $NHB(n)$ the number of non-Hamiltonian 3-regular 1-connected graphs of size $2n$.

Is it true that $\lim\limits_{n \rightarrow \infty} \displaystyle\frac{NHB(n)}{NH(n)} = 1$?

背景链接：http://www.openproblemgarden.org/op/almost_all_non_hamiltonian_3_regular_graphs_are_1_connected

14067 | AMR-089-0004

### Alternative and arithmetic proofs of the pi identities

P / partially_solved | AMR 问题清单汇编

Give another proof of the paper's identities (Ж) and (ж) using the methods for identity (1). Can $f(a,b,c,d)$ be interpreted as a residue at $(a+b)+(c+d)i$, or otherwise related to the Gaussian integers?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/17-75/index.html

14068 | AMR-087-0069

### An algebraic LLL algorithm

P / partially_solved | AMR 问题清单汇编

Develop an algebraic analogue of the LLL lattice-reduction algorithm that exploits ideal-lattice structure.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14069 | OPG-323

### Antichains in the cycle continuous order

S / solved | Open Problem Garden

If $G$, $H$ are graphs, a function $f: E(G) \rightarrow E(H)$ is called cycle-continuous if the pre-image of every element of the (binary) cycle space of $H$ is a member of the cycle space of $G$.

Problem Does there exist an infinite set of graphs $\{G_1,G_2,\ldots \}$ so that there is no cycle continuous mapping between $G_i$ and $G_j$ whenever $i \neq j$?

背景链接：http://www.openproblemgarden.org/op/antichains_in_the_cycle_continuous_order

14070 | OPG-46279

### Antidirected trees in digraphs

P / partially_solved | Open Problem Garden

An antidirected tree is an orientation of a tree in which every vertex has either indegree 0 or outdergree 0.

Conjecture Let $D$ be a digraph. If $|A(D)| > (k-2) |V(D)|$, then $D$ contains every antidirected tree of order $k$.

背景链接：http://www.openproblemgarden.org/op/antidirected_trees_in_digraphs

14071 | OWR-16931-021

### Antiferromagnetic Homomorphism Bounds Without Triangle-Freeness

O / open | Oberwolfach 报告问题汇编

Let $H$ be a weighted graph whose edge-weight matrix has at most one positive eigenvalue. Can the triangle-free assumption on a $d$-regular graph $G$ be removed from the reverse-Sidorenko homomorphism inequality for $\operatorname{hom}(G,H)$?

来源字段：https://doi.org/10.4171/owr/2019/19

14072 | OWR-730-010

### Approximation Complexity of Group Connector

O / open | Oberwolfach 报告问题汇编

Determine the approximation complexity of the Group Connector problem: given a weighted graph and disjoint terminal sets with requirements, find a minimum-cost forest satisfying each requirement.

来源字段：https://doi.org/10.4171/owr/2004/28

14073 | OPG-37217

### Approximation ratio for k-outerplanar graphs

S / solved | Open Problem Garden

Conjecture Is the approximation ratio for the Maximum Edge Disjoint Paths (MaxEDP) or the Maximum Integer Multiflow problem (MaxIMF) bounded by a constant in $k$-outerplanar graphs or tree-width graphs?

背景链接：http://www.openproblemgarden.org/op/approximation_ratio_for_k_outerplanar_graphs

14074 | OPG-37211

### Approximation Ratio for Maximum Edge Disjoint Paths problem

P / partially_solved | Open Problem Garden

Conjecture Can the approximation ratio $O(\sqrt{n})$ be improved for the Maximum Edge Disjoint Paths problem (MaxEDP) in planar graphs or can an inapproximability result stronger than $\mathcal{APX}$-hardness?

背景链接：http://www.openproblemgarden.org/op/approximation_ratio_for_maximum_edge_disjoint_paths_problem

14075 | OPG-49573

### Arc-disjoint directed cycles in regular directed graphs

O / open | Open Problem Garden

Conjecture If $G$ is a $k$-regular directed graph with no parallel arcs, then $G$ contains a collection of ${k+1 \choose 2}$ arc-disjoint directed cycles.

背景链接：http://www.openproblemgarden.org/op/arc_disjoint_directed_cycles_in_regular_directed_graphs

14076 | OPG-46495

### Arc-disjoint out-branching and in-branching

P / partially_solved | Open Problem Garden

Conjecture There exists an integer $k$ such that every $k$-arc-strong digraph $D$ with specified vertices $u$ and $v$ contains an out-branching rooted at $u$ and an in-branching rooted at $v$ which are arc-disjoint.

背景链接：http://www.openproblemgarden.org/op/arc_disjoint_out_branching_and_in_branching

14077 | OPG-46496

### Arc-disjoint strongly connected spanning subdigraphs

P / partially_solved | Open Problem Garden

Conjecture There exists an ineteger $k$ so that every $k$-arc-connected digraph contains a pair of arc-disjoint strongly connected spanning subdigraphs?

背景链接：http://www.openproblemgarden.org/op/arc_disjoint_strongly_connected_spanning_subdigraphs

14078 | OPG-48368

### Are almost all graphs determined by their spectrum?

P / partially_solved | Open Problem Garden

Problem Are almost all graphs uniquely determined by the spectrum of their adjacency matrix?

背景链接：http://www.openproblemgarden.org/op/are_almost_all_graphs_determined_by_their_spectrum

14079 | OPG-37117

### Are different notions of the crossing number the same?

P / partially_solved | Open Problem Garden

Problem Does the following equality hold for every graph $G$? $$ \text{pair-cr}(G) = \text{cr}(G) $$

The crossing number $\text{cr}(G)$ of a graph $G$ is the minimum number of edge crossings in any drawing of $G$ in the plane. In the pairwise crossing number $\text{pair-cr}(G)$, we minimize the number of pairs of edges that cross.

背景链接：http://www.openproblemgarden.org/op/are_different_notions_of_the_crossing_number_the_same

14080 | OPG-36939

### Are vertex minor closed classes chi-bounded?

P / partially_solved | Open Problem Garden

Question Is every proper vertex-minor closed class of graphs chi-bounded?

背景链接：http://www.openproblemgarden.org/op/vertex_minor_closed_classes_are_chi_bounded

14081 | AMR-090-0003

### Arnold sequence as an f-transform

O / open | AMR 问题清单汇编

Is Arnold's sequence $(u_k)_{k\ge1}$ the $f$-transform of the quadruple $(2,4,4,4)$?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/18-79/index.html

14082 | OPG-36933

### Asymptotic Distribution of Form of Polyhedra

S / solved | Open Problem Garden

Problem Consider the set of all topologically inequivalent polyhedra with $k$ edges. Define a form parameter for a polyhedron as $\beta:= v/(k+2)$ where $v$ is the number of vertices. What is the distribution of $\beta$ for $k \to \infty$?

背景链接：http://www.openproblemgarden.org/op/asymptotic_distribution_of_form_of_polyhedra

14083 | OWR-16763-019

### Asymptotic Enumeration of Rank-Three Matroids

O / open | Oberwolfach 报告问题汇编

Let $p(n,3)$ count rank-$3$ matroids and $p'(n,3)$ count rank-$3$ paving matroids with no hyperplane larger than $4$. Is there a constant $c>-2$ such that

$$\log p(n,3)\sim\log p'(n,3)=\frac1{n-2}\binom n3\bigl(\log(n-2)+c\bigr)$$

as $n\to\infty$?

来源字段：https://doi.org/10.4171/owr/2019/1

14084 | OWR-1386-007

### Asymptotic Size-Ramsey Numbers of Paths

S / solved | Oberwolfach 报告问题汇编

Let $P_n$ be the path on $n$ vertices and let $r_e(H)$ be the size-Ramsey number of $H$. Is it true that $r_e(P_n)/n\to\infty$ and $r_e(P_n)/n^2\to0$?

来源字段：https://doi.org/10.4171/owr/2006/48

14085 | OWR-14299911-027

### Asymptotic-Cone Characterization of Hierarchically Hyperbolic Spaces

O / open | Oberwolfach 报告问题汇编

Characterize, through their asymptotic cones, the coarse median groups that are quasi-isometric to hierarchically hyperbolic spaces. Specifically, is a coarse median group quasi-isometric to a hierarchically hyperbolic space if and only if each nonprincipal asymptotic cone is bi-Lipschitz equivalent to a universal real cubing?

来源字段：https://doi.org/10.4171/owr/2026/8

14086 | OPG-46432

### Ádám's Conjecture

O / open | Open Problem Garden

Conjecture Every digraph with at least one directed cycle has an arc whose reversal reduces the number of directed cycles.

背景链接：http://www.openproblemgarden.org/op/adams_conjecture

14087 | AMR-073-0001

### Babai's problem

O / open | AMR 问题清单汇编

Babai's problem: which groups are Babai invariant groups?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Graph_theory

14088 | GRAPH-027

### Barnette's Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Barnette's Conjecture [OPG-385]

Does every cubic bipartite three-connected planar graph have a Hamiltonian cycle?

背景链接：https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.GD.2025.6

14089 | AMR-087-0084

### Better trapdoor pairings

O / open | AMR 问题清单汇编

Construct improved practical trapdoor pairings.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14090 | OWR-1183-009

### Bipartizing Triangle-Free Graphs with Few Edge Deletions

P / partially_solved | Oberwolfach 报告问题汇编

Can every triangle-free graph on $n$ vertices be made bipartite by deleting at most $n^2/25$ edges?

来源字段：https://doi.org/10.4171/owr/2006/01

14091 | OWR-9790352-029

### Blow-Ups of Clique Subdivisions in Dense Graphs

P / partially_solved | Oberwolfach 报告问题汇编

For fixed integers $m,r\geq2$, does every $n$-vertex graph with $\Omega(n^{2-1/r})$ edges contain the $r$-blow-up of a subdivision of $K_m$?

来源字段：https://doi.org/10.4171/owr/2022/1

14092 | OWR-9790352-031

### Bollobás-Erdős-Tuza Independent-Set Hitting Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

If a sequence of graphs $G$ satisfies $\alpha(G)=\Omega(|V(G)|)$, must the minimum order $h(G)$ of a vertex set meeting every maximum independent set satisfy $h(G)=o(|V(G)|)$?

来源字段：https://doi.org/10.4171/owr/2022/1

14093 | OPG-34908

### Book Thickness of Subdivisions

P / partially_solved | Open Problem Garden

Let $G$ be a finite undirected simple graph.

A $k$-page book embedding of $G$ consists of a linear order $\preceq$ of $V(G)$ and a (non-proper) $k$-colouring of $E(G)$ such that edges with the same colour do not cross with respect to $\preceq$. That is, if $v\prec x\prec w\prec y$ for some edges $vw,xy\in E(G)$, then $vw$ and $xy$ receive distinct colours.

One can think that the vertices are placed along the spine of a book, and the edges are drawn without crossings on the pages of the book.

The book thickness of $G$, denoted by bt $(G)$ is the minimum integer $k$ for which there is a $k$-page book embedding of $G$.

Let $G'$ be the graph obtained by subdividing each edge of $G$ exactly once.

Conjecture There is a function $f$ such that for every graph $G$, $$\text{bt}(G) \leq f( \text{bt}(G') )\enspace.$$

背景链接：http://www.openproblemgarden.org/op/book_thickness_of_subdivisions

14094 | OPG-131

### Bouchet's 6-flow conjecture

O / open | Open Problem Garden

Conjecture Every bidirected graph with a nowhere-zero $k$-flow for some $k$, has a nowhere-zero $6$-flow.

背景链接：http://www.openproblemgarden.org/op/bouchets_6_flow_conjecture

14095 | OWR-4212-008

### Boundaries of Random Hyperbolic Level Graphs

O / open | Oberwolfach 报告问题汇编

For infinite random hyperbolic graphs formed from random finite horizontal levels joined by a perpendicular spanning tree, what is the structure of the random boundary, and is there a threshold for its path-connectedness related to the threshold for connected horizontal levels?

来源字段：https://doi.org/10.4171/owr/2010/11

14096 | OWR-14298163-021

### Bounded Pairwise Crossings in Hypercube Drawings

O / open | Oberwolfach 报告问题汇编

Does there exist an absolute constant $c\in\mathbb N$ such that every hypercube graph $Q_d$ has a topological drawing in the plane with no collection of $c$ pairwise crossing edges?

来源字段：https://doi.org/10.4171/owr/2024/3

14097 | AMR-088-0013

### Bounded prime-sum criterion for rational points

O / open | AMR 问题清单汇编

For an elliptic curve $E/\mathbb{Q}$, let $N_p$ be its number of solutions modulo $p$ plus one and put $f(X)=\sum_{p\le X}\log(N_p/p)$. Is $f(X)$ bounded if and only if $E(\mathbb{Q})$ is finite?

背景链接：https://www.maths.nottingham.ac.uk/plp/pmzcw/download/open.pdf

14098 | OWR-11101922-007

### Bounded-Degree Sensitivity of Mixing Times

O / open | Oberwolfach 报告问题汇编

Can one construct uniformly quasi-isometric sequences of finite Cayley graphs $G_n,H_n$, both of uniformly bounded degree, such that

$$t_{\mathrm{mix}}(G_n)\geq\log\log\log|G_n|\,t_{\mathrm{mix}}(H_n)?$$

Can the factor $\log\log\log|G_n|$ be improved?

来源字段：https://doi.org/10.4171/owr/2022/41

14099 | OWR-14299905-026

### Bounded-Diameter Monochromatic Covers

O / open | Oberwolfach 报告问题汇编

For every integer $r\geq2$, does there exist $D(r)$ such that every $r$-edge-colored complete graph can be covered by $r-1$ monochromatic connected subgraphs, each of diameter at most $D(r)$?

来源字段：https://doi.org/10.4171/owr/2026/1

14100 | OWR-16627-001

### Bounded-Length Strong Ramsey Games

O / open | Oberwolfach 报告问题汇编

For every integer $t$, is there a constant $c_t$ such that the length $L(K_t,n)$ of the strong Ramsey game on $K_n$ is at most $c_t$ for every $n$?

来源字段：https://doi.org/10.4171/owr/2018/44

14101 | OWR-15219-009

### Bounding Generic Completion Rank by Maximum-Likelihood Threshold

O / open | Oberwolfach 报告问题汇编

Does there exist a function $f:\mathbb N\to\mathbb N$ such that, for every finite simple graph $G$, the generic completion rank of $G$ is at most $f(\operatorname{mlt}(G))$?

来源字段：https://doi.org/10.4171/owr/2017/14

14102 | OPG-36907

### Bounding the chromatic number of triangle-free graphs with fixed maximum degree

P / partially_solved | Open Problem Garden

Conjecture A triangle-free graph with maximum degree $\Delta$ has chromatic number at most $\ceil{\frac{\Delta}{2}}+2$.

背景链接：http://www.openproblemgarden.org/op/bounding_the_chromatic_number_of_triangle_free_graphs_with_fixed_maximum_degree

14103 | OPG-48583

### Bounding the on-line choice number in terms of the choice number

S / solved | Open Problem Garden

Question Are there graphs for which $\text{ch}^{\text{OL}}-\text{ch}$ is arbitrarily large?

背景链接：http://www.openproblemgarden.org/op/bounding_the_on_line_choice_number_in_terms_of_the_choice_number

14104 | OWR-1536-006

### Breaking Points in Diameter Positional Games

O / open | Oberwolfach 报告问题汇编

For the biased diameter games $D_d(a:b)$ on $E(K_n)$, is the breaking point, up to a polylogarithmic factor, asymptotic to the known biases at which Breaker wins? In particular, is the breaking point for $D_3(3:b)$ of order $n^{2/3}\operatorname{polylog}(n)$, and what is it for $D_3(2:b)$?

来源字段：https://doi.org/10.4171/owr/2007/20

14105 | AMR-087-0056

### Breaking the pairing system

O / open | AMR 问题清单汇编

Find an attack that breaks the pairing-based cryptographic system discussed in the slides, or establish its resistance to known attacks.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14106 | AMR-087-0057

### Breaking weaker pairing assumptions

O / open | AMR 问题清单汇编

Break, or determine the true hardness of, the weaker security assumptions used in pairing-based cryptography.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14107 | GRAPH-001

### Brouwer's Conjecture on Graph Laplacians

P / partially_solved | 独立 / 未分配集合

Can the sum of eigenvalues of the Laplacian matrix of a graph be bounded by the number of edges?

背景链接：https://www.sciencedirect.com/science/article/abs/pii/S0166218X26002593

14108 | OPG-46385

### Caccetta-Häggkvist Conjecture

O / open | Open Problem Garden

Conjecture Every simple digraph of order $n$ with minimum outdegree at least $r$ has a cycle with length at most $\lceil n/r\rceil$

背景链接：http://www.openproblemgarden.org/op/caccetta_haggkvist_conjecture

14109 | OWR-1532-003

### Caccetta-Häggkvist Conjecture for Directed Triangles

O / open | Oberwolfach 报告问题汇编

If $G$ is a digraph with $n$ vertices and no directed cycle of length at most $3$, must some vertex of $G$ have outdegree less than $n/3$?

来源字段：https://doi.org/10.4171/owr/2007/16

14110 | OWR-14299518-024

### Cayley-Sum Constructions for Triangle Ramsey Numbers

O / open | Oberwolfach 报告问题汇编

Can a Cayley-sum graph generated by the sum-free process on $\mathbb{F}_2^d$ attain $R(3,k)\geq\left(\frac12-o(1)\right)k^2/\log k$ along an infinite sequence?

来源字段：https://doi.org/10.4171/owr/2025/42

14111 | GT-008

### Cereceda's Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Cereceda's Conjecture [GRAPH-061]

For any $k$-chromatic graph, can its $k$-colorings be transformed into each other by recoloring one vertex at a time, staying within $k$ colors, in polynomial time in the number of vertices?

背景链接：https://doi.org/10.1016/j.disc.2007.07.028

14112 | GRAPH-044

### Characterization of Word-Representable Planar Graphs

O / open | 独立 / 未分配集合

Characterize which planar graphs are word-representable.

背景链接：https://arxiv.org/abs/1501.07108

14113 | OPG-1750

### Characterizing (aleph_0,aleph_1)-graphs

P / partially_solved | Open Problem Garden

Call a graph an $(\aleph_0,\aleph_1)$-graph if it has a bipartition $(A,B)$ so that every vertex in $A$ has degree $\aleph_0$ and every vertex in $B$ has degree $\aleph_1$.

Problem Characterize the $(\aleph_0,\aleph_1)$-graphs.

背景链接：http://www.openproblemgarden.org/op/characterizing_aleph_0_aleph_1_graphs

14114 | AMR-099-0011

### Cheeger constants of nets in transitive graphs

O / open | AMR 问题清单汇编

Let $G$ be vertex-transitive and let $G_k$ be a $k$-net graph of $G$. Must $h(G_k)\ge h(G)$ whenever $G_k$ is not a single vertex?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=15

14115 | OPG-44879

### Choice Number of k-Chromatic Graphs of Bounded Order

P / partially_solved | Open Problem Garden

Conjecture If $G$ is a $k$-chromatic graph on at most $mk$ vertices, then $\text{ch}(G)\leq \text{ch}(K_{m*k})$.

背景链接：http://www.openproblemgarden.org/op/choice_number_of_k_chromatic_graphs_of_bounded_order

14116 | OPG-53019

### Choosability of Graph Powers

P / partially_solved | Open Problem Garden

Question (Noel, 2013) Does there exist a function $f(k)=o(k^2)$ such that for every graph $G$, $$ \text{ch}\left(G^2\right)\leq f\left(\chi\left(G^2\right)\right)? $$

背景链接：http://www.openproblemgarden.org/op/choosability_of_graph_powers

14117 | AMR-087-0045

### Choosing a field for an elliptic curve of prescribed order

P / partially_solved | AMR 问题清单汇编

Given $n$, efficiently choose a prime power $q$ and construct an elliptic curve $E/\mathbb{F}_q$ with $\#E(\mathbb{F}_q)=n$.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14118 | OPG-700

### Chords of longest cycles

P / partially_solved | Open Problem Garden

Conjecture If $G$ is a 3-connected graph, every longest cycle in $G$ has a chord.

背景链接：http://www.openproblemgarden.org/op/chords_of_longest_cycles

14119 | OWR-12697684-024

### Chromatic Alternatives from Girth and Local Covers

O / open | Oberwolfach 报告问题汇编

Does a function $f:\mathbb N^2\to\mathbb N$ exist such that every finite graph $G$ with $\chi(G)\geq f(r,k)$ either has a subgraph of chromatic number at least $k$ and girth greater than $r$, or has an $r$-local cover of chromatic number at least $k$?

来源字段：https://doi.org/10.4171/owr/2023/1

14120 | OWR-4791-029

### Chromatic Boundedness of Wheel-Free Graphs

P / partially_solved | Oberwolfach 报告问题汇编

For a graph $G$, let $\omega(G)$ be its clique number and $\chi(G)$ its chromatic number. A wheel is a pair $(C,v)$ where $C$ is an induced cycle of length at least $4$ and $v\notin V(C)$ has at least three neighbors on $C$. Let $\mathcal{C}$ consist of graphs with no wheel. Does there exist $f:\mathbb{N}\to\mathbb{N}$ such that $\chi(G)\leq f(\omega(G))$ for every $G\in\mathcal{C}$?

来源字段：https://doi.org/10.4171/owr/2011/01

14121 | OPG-60055

### Chromatic number of $\frac{3}{3}$-power of graph

P / partially_solved | Open Problem Garden

Let $G$ be a graph and $m,n\in \mathbb{N}$. The graph $G^{\frac{m}{n}}$ is defined to be the $m$-power of the $n$-subdivision of $G$. In other words, $G^{\frac{m}{n}}=(G^{\frac{1}{n}})^m$.

Conjecture Let $G$ be a graph with $\Delta(G)\geq 2$. Then $\chi(G^{\frac{3}{3}})\leq 2\Delta(G)+1$.

背景链接：http://www.openproblemgarden.org/op/chromatic_number_of_frac_3_3_power_of_graph

14122 | OWR-9790352-037

### Chromatic Number of Almost-Linear Graph Unions

P / partially_solved | Oberwolfach 报告问题汇编

Let $G_1,\ldots,G_n$ be graphs with $\chi(G_i)\leq n-1$ such that every two have at most one vertex in common. How large can $\chi(G_1\cup\cdots\cup G_n)$ be?

来源字段：https://doi.org/10.4171/owr/2022/1

14123 | OPG-59952

### Chromatic Number of Common Graphs

P / partially_solved | Open Problem Garden

Question Do common graphs have bounded chromatic number?

背景链接：http://www.openproblemgarden.org/op/chromatic_number_of_common_graphs

14124 | OWR-14298163-004

### Chromatic Number of Overlapping Unit-Circle Constellations

O / open | Oberwolfach 报告问题汇编

Determine the maximum chromatic number of the tangency graph of a possibly overlapping constellation of unit circles. Equivalently, determine the chromatic number of the plane in the Hadwiger-Nelson problem.

来源字段：https://doi.org/10.4171/owr/2024/3

14125 | OPG-37656

### Chromatic number of random lifts of complete graphs

P / partially_solved | Open Problem Garden

Question Is the chromatic number of a random lift of $K_5$ concentrated on a single value?

背景链接：http://www.openproblemgarden.org/op/chromatic_number_of_random_lifts_of_complete_graphs

14126 | GRAPH-028

### Chvátal's Toughness Conjecture

P / partially_solved | 独立 / 未分配集合

Is there a constant t such that every t-tough graph is Hamiltonian?

背景链接：https://doi.org/10.1016/j.disc.2023.114082

14127 | OWR-13346-003

### Circuits and Thresholds in Rank-Two Matrix Completion

O / open | Oberwolfach 报告问题汇编

In rank-$2$ low-rank matrix completion, characterize circuits and circuit polynomials; moreover, if $N=n$ and the observed positions are the edges of a uniformly random $2r$-regular graph, determine whether all entries are completable with high probability as $n=N\to\infty$.

来源字段：https://doi.org/10.4171/owr/2014/44

14128 | OPG-37619

### Circular choosability of planar graphs

P / partially_solved | Open Problem Garden

Let $G = (V, E)$ be a graph. If $p$ and $q$ are two integers, a $(p,q)$-colouring of $G$ is a function $c$ from $V$ to $\{0,\dots,p-1\}$ such that $q \le |c(u)-c(v)| \le p-q$ for each edge $uv\in E$. Given a list assignment $L$ of $G$, i.e.~a mapping that assigns to every vertex $v$ a set of non-negative integers, an $L$-colouring of $G$ is a mapping $c: V \to N$ such that $c(v)\in L(v)$ for every $v\in V$. A list assignment $L$ is a $t$- $(p,q)$-list-assignment if $L(v) \subseteq \{0,\dots,p-1\}$ and $|L(v)| \ge tq$ for each vertex $v \in V$. Given such a list assignment $L$, the graph G is $(p,q)$- $L$-colourable if there exists a $(p,q)$-$L$-colouring $c$, i.e. $c$ is both a $(p,q)$-colouring and an $L$-colouring. For any real number $t \ge 1$, the graph $G$ is $t$- $(p,q)$-choosable if it is $(p,q)$- $L$-colourable for every $t$- $(p,q)$-list-assignment $L$. Last, $G$ is circularly $t$-choosable if it is $t$- $(p,q)$-choosable for any $p$, $q$. The circular choosability (or circular list chromatic number or circular choice number) of G is $$cch(G):= \inf\{t \ge 1: G \text{ is circularly$t$-choosable}\}.$$

Problem What is the best upper bound on circular choosability for planar graphs?

背景链接：http://www.openproblemgarden.org/op/circular_choosability_of_planar_graphs

14129 | OPG-401

### Circular coloring triangle-free subcubic planar graphs

O / open | Open Problem Garden

Problem Does every triangle-free planar graph of maximum degree three have circular chromatic number at most $\frac{20}{7}$?

背景链接：http://www.openproblemgarden.org/op/circular_chromatic_number_of_triangle_free_planar_graphs

14130 | OPG-2040

### Circular colouring the orthogonality graph

P / partially_solved | Open Problem Garden

Let ${\mathcal O}$ denote the graph with vertex set consisting of all lines through the origin in ${\mathbb R}^3$ and two vertices adjacent in ${\mathcal O}$ if they are perpendicular.

Problem Is $\chi_c({\mathcal O}) = 4$?

背景链接：http://www.openproblemgarden.org/op/circular_colouring_the_orthogonality_graph

14131 | OPG-59994

### Circular flow number of regular class 1 graphs

P / partially_solved | Open Problem Garden

A nowhere-zero $r$-flow $(D(G),\phi)$ on $G$ is an orientation $D$ of $G$ together with a function $\phi$ from the edge set of $G$ into the real numbers such that $1 \leq |\phi(e)| \leq r-1$, for all $e \in E(G)$, and $\sum_{e \in E^+(v)}\phi(e) = \sum_{e \in E^-(v)}\phi(e), \textrm{ for all } v \in V(G)$. The circular flow number of $G$ is inf $\{ r | G$ has a nowhere-zero $r$-flow $\}$, and it is denoted by $F_c(G)$.

A graph with maximum vertex degree $k$ is a class 1 graph if its edge chromatic number is $k$.

Conjecture Let $t \geq 1$ be an integer and $G$ a $(2t+1)$-regular graph. If $G$ is a class 1 graph, then $F_c(G) \leq 2 + \frac{2}{t}$.

背景链接：http://www.openproblemgarden.org/op/circular_flow_number_of_regular_class_1_graphs

14132 | OPG-59997

### Circular flow numbers of $r$-graphs

P / partially_solved | Open Problem Garden

A nowhere-zero $r$-flow $(D(G),\phi)$ on $G$ is an orientation $D$ of $G$ together with a function $\phi$ from the edge set of $G$ into the real numbers such that $1 \leq |\phi(e)| \leq r-1$, for all $e \in E(G)$, and $\sum_{e \in E^+(v)}\phi(e) = \sum_{e \in E^-(v)}\phi(e), \textrm{ for all } v \in V(G)$.

A $(2t+1)$-regular graph $G$ is a $(2t+1)$-graph if $|\partial_G(X)| \geq 2t+1$ for every $X \subseteq V(G)$ with $|X|$ odd.

Conjecture Let $t > 1$ be an integer. If $G$ is a $(2t+1)$-graph, then $F_c(G) \leq 2 + \frac{2}{t}$.

背景链接：http://www.openproblemgarden.org/op/circular_flow_numbers_of_r_graphs

14133 | OWR-14299911-010

### Classification of Strongly Prime Smooth Graphs

O / open | Oberwolfach 报告问题汇编

Characterize and classify the strongly prime smooth graphs.

来源字段：https://doi.org/10.4171/owr/2026/8

14134 | OWR-9790352-033

### Clique Minors in Graphs Excluding a Fixed Subgraph

P / partially_solved | Oberwolfach 报告问题汇编

For which graphs $F$ does there exist $c(F)>0$ such that every graph $G$ with no subgraph isomorphic to $F$ contains a clique minor of order at least $\chi(G)^{1+c(F)}$?

来源字段：https://doi.org/10.4171/owr/2022/1

14135 | OWR-16763-011

### Clustered Coloring Without Topological or Odd Clique Minors

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Clustered Hadwiger-Type Conjecture [OWR-16763-012]

For every positive integer $t$, do the $K_{t+1}$-topological-minor-free graphs and the odd-$K_{t+1}$-minor-free graphs admit $t$-colorings whose monochromatic components have uniformly bounded size?

来源字段：https://doi.org/10.4171/owr/2019/1

14136 | OWR-14299911-016

### Coarse Buneman Graphs for Partial Data

O / open | Oberwolfach 报告问题汇编

When half-space membership data are missing or noisy, construct a coarse analogue of the Buneman graph that represents the partial data and permits computation of medians up to a controlled error. Determine when the resulting metric space retains quantitatively controlled median geometry.

来源字段：https://doi.org/10.4171/owr/2026/8

14137 | OWR-14299911-015

### Coarse Buneman Models for Countable Preference Data

O / open | Oberwolfach 报告问题汇编

Extend the Buneman-graph model of single-peaked preference data to a countably infinite set of alternatives. Construct a coarse Buneman graph that accommodates missing or uncertain half-space assignments and computes medians with controlled error.

来源字段：https://doi.org/10.4171/owr/2026/8

14138 | OWR-1532-012

### Colin de Verdière Invariants and Four-Dimensional Panellings

P / partially_solved | Oberwolfach 报告问题汇编

For every undirected graph $G$, are the following equivalent: $\mu(G)\leq5$; the even-mapping invariant satisfies $\sigma(G)\leq5$; and the panelling $\widetilde G$, obtained by attaching a disk to each circuit of $G$, embeds in $\mathbb{R}^4$?

来源字段：https://doi.org/10.4171/owr/2007/16

14139 | OWR-15427-019

### Collinear Grid Triples and Rigid Bipartite Unit Distance Graphs

P / partially_solved | Oberwolfach 报告问题汇编

What is the minimum possible number of collinear triples in a subset $S\subseteq\{1,\ldots,n\}^3$ of cardinality $|S|=n^{3-s}$?

来源字段：https://doi.org/10.4171/owr/2017/19

14140 | OWR-12330-014

### Color Bounds for Weight-Modified Reed Fractional Coloring

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Weight-Modified Reed Coloring Question [OWR-12330-015]

Does the weight-modified Reed fractional-coloring algorithm always use at most $\max_{v\in V(G)}\gamma_\ell(v)$ colors, and can one prove better bounds?

来源字段：https://doi.org/10.4171/owr/2013/02

14141 | OWR-14210-001

### Color Savings and Strong Chromatic Index Bounds

O / open | Oberwolfach 报告问题汇编

For a graph with maximum degree $\Delta$ and clique number $\omega$, how many colors can be saved in ordinary, list, or correspondence coloring relative to bounds from $\Delta$, $\omega$, and maximum average degree; in particular, is the strong chromatic index at most $1.25\Delta^2$?

来源字段：https://doi.org/10.4171/owr/2016/2

14142 | OPG-550

### Coloring and immersion

P / partially_solved | Open Problem Garden

Conjecture For every positive integer $t$, every (loopless) graph $G$ with $\chi(G) \ge t$ immerses $K_t$.

背景链接：http://www.openproblemgarden.org/op/coloring_and_immersion

14143 | OWR-711-009

### Coloring Dense Triangle-Free Graphs and Hajós' Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a bound $f(1/3)$ on the chromatic number of every triangle-free $n$-vertex graph of minimum degree at least $n/3$; in particular, is $f(1/3)=4$? Is Hajós' conjecture true for triangle-free graphs?

来源字段：https://doi.org/10.4171/owr/2004/01

14144 | OPG-827

### Coloring random subgraphs

P / partially_solved | Open Problem Garden

If $G$ is a graph and $p \in [0,1]$, we let $G_p$ denote a subgraph of $G$ where each edge of $G$ appears in $G_p$ with independently with probability $p$.

Problem Does there exist a constant $c$ so that ${\mathbb E}(\chi(G_{1/2})) > c \frac{\chi(G)}{\log \chi(G)}$?

背景链接：http://www.openproblemgarden.org/op/coloring_random_subgraphs

14145 | OPG-616

### Coloring the Odd Distance Graph

P / partially_solved | Open Problem Garden

The Odd Distance Graph, denoted ${\mathcal O}$, is the graph with vertex set ${\mathbb R}^2$ and two points adjacent if the distance between them is an odd integer.

Question Is $\chi({\mathcal O}) = \infty$?

背景链接：http://www.openproblemgarden.org/op/coloring_the_odd_distance_graph

14146 | OPG-46533

### Coloring the union of degenerate graphs

O / open | Open Problem Garden

Conjecture The union of a $1$-degenerate graph (a forest) and a $2$-degenerate graph is $5$-colourable.

背景链接：http://www.openproblemgarden.org/op/coloring_the_union_of_degenerate_graphs

14147 | OWR-14299905-038

### Colorings Without Disjoint Color-Isomorphic Triangles

O / open | Oberwolfach 报告问题汇编

Let $g(n)$ be the fewest colors needed to edge-color $K_n$ with no two vertex-disjoint color-isomorphic triangles. Determine the order of growth of $g(n)$; in particular, is $g(n)=\Theta(n)$, or even $g(n)=n-O(1)$?

来源字段：https://doi.org/10.4171/owr/2026/1

14148 | OPG-47470

### Colouring the square of a planar graph

P / partially_solved | Open Problem Garden

Conjecture Let $G$ be a planar graph of maximum degree $\Delta$. The chromatic number of its square is

- at most $7$ if $\Delta =3$, - at most $\Delta+5$ if $4\leq\Delta\leq 7$, - at most $\left\lfloor\frac32\,\Delta\right\rfloor+1$ if $\Delta\ge8$.

背景链接：http://www.openproblemgarden.org/op/colouring_the_square_of_a_planar_graph

14149 | OWR-15216-004

### Compactness of Chromatic and Coloring Numbers

P / partially_solved | Oberwolfach 报告问题汇编

Is it consistent that some infinite cardinal $\theta$ admits a $(\theta,\theta+2)$-coloring graph? More generally, what is the relationship between compactness of chromatic number and compactness of coloring number?

来源字段：https://doi.org/10.4171/owr/2017/11

14150 | OWR-14299905-032

### Comparing Monotone and Ordinary Crossing Numbers

O / open | Oberwolfach 报告问题汇编

Is there an absolute constant $C$ such that $\operatorname{mon-cr}(G)\leq C\operatorname{cr}(G)$ for every graph $G$?

来源字段：https://doi.org/10.4171/owr/2026/1

14151 | OWR-14299911-018

### Comparing Travelling-Salesman and Travelling-Polytope Metrics

O / open | Oberwolfach 报告问题汇编

For a graph $X$, vertices $x,y$, and a finite vertex set $F$, compare the minimum path length $\operatorname{TS}(x,F,y)$ through every vertex of $F$ with the minimum length $\operatorname{TP}(x,F,y)$ of a path of convex finite-vertex polytopes covering $F$. Determine sharp reverse bounds to $\operatorname{TP}\leq2\operatorname{TS}$ and compute the resulting exponent $\operatorname{TP}(X)$.

来源字段：https://doi.org/10.4171/owr/2026/8

14152 | OWR-14299081-007

### Complete Bipartite Subgraphs and Connected Matchings at $\alpha=2$

O / open | Oberwolfach 报告问题汇编

Is there a constant $c>0$ such that every $n$-vertex graph $G$ with $\alpha(G)=2$ contains both (1) a complete bipartite graph $K_{n^c,n^c}$ and (2) a connected matching of size $cn$?

来源字段：https://doi.org/10.4171/owr/2025/1

14153 | OPG-826

### Complete bipartite subgraphs of perfect graphs

S / solved | Open Problem Garden

Problem Let $G$ be a perfect graph on $n$ vertices. Is it true that either $G$ or $\bar{G}$ contains a complete bipartite subgraph with bipartition $(A,B)$ so that $|A|, |B| \ge n^{1 - o(1)}$?

背景链接：http://www.openproblemgarden.org/op/complete_bipartite_subgraphs_of_perfect_graphs

14154 | OWR-14210-002

### Complete Graph Immersions from Minimum Degree

O / open | Oberwolfach 报告问题汇编

For every integer $n\geq 1$, does every graph of minimum degree at least n contain an immersion of the complete graph $K_n$?

来源字段：https://doi.org/10.4171/owr/2016/2

14155 | OWR-15427-005

### Complete Graph Minors from Homological Graph Dimension

O / open | Oberwolfach 报告问题汇编

For a finite graph $G$, let $\gamma(G)$ be the largest integer $q$ for which some connected family of induced subgraphs of $G$ has a nerve with nonzero $q$-dimensional homology. For every positive integer $d$, is $\gamma(G)\geq d$ equivalent to the existence of a $K_{d+2}$ minor in $G$?

来源字段：https://doi.org/10.4171/owr/2017/19

14156 | AMR-089-0003

### Complex continuation of the associated zeta function

O / open | AMR 问题清单汇编

For $Z(s)=\sum f(a,b,c,d)^s$, which is known to converge for real $s>1/2$, extend $Z$ to complex values of $s$.

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/17-75/index.html

14157 | OWR-9790359-009

### Complexity of Low-Chromatic Compact Free Subshifts

O / open | Oberwolfach 报告问题汇编

Is the collection of compact free subshifts of $2^{\mathbb Z_2*\mathbb Z_2*\mathbb Z_2}$ with Borel chromatic number at most $3$ $\boldsymbol{\Sigma}^1_2$-complete?

来源字段：https://doi.org/10.4171/owr/2022/8

14158 | OWR-1189-014

### Complexity of Maximum Tutte Sets and D-Graph Recognition

O / open | Oberwolfach 报告问题汇编

For MAX TUTTE SET, determine additional polynomial-time graph classes; determine its complexity on $3$-connected planar graphs and on planar $(1-\varepsilon)$-tough graphs; and decide whether level-$1$ D-graphs are recognizable in polynomial time.

来源字段：https://doi.org/10.4171/owr/2006/07

14159 | OPG-46738

### Complexity of the H-factor problem.

P / partially_solved | Open Problem Garden

An $H$-factor in a graph $G$ is a set of vertex-disjoint copies of $H$ covering all vertices of $G$.

Problem Let $c$ be a fixed positive real number and $H$ a fixed graph. Is it NP-hard to determine whether a graph $G$ on $n$ vertices and minimum degree $cn$ contains and $H$-factor?

背景链接：http://www.openproblemgarden.org/op/complexity_of_the_h_factor_problem

14160 | AMR-087-0007

### Computational realization of a second cohomology group

O / open | AMR 问题清单汇编

Turn $H^2(G_K,K_s^*)$ into an explicitly computational group.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14161 | OWR-14299911-017

### Computing Transversality Degrees of Graphs

O / open | Oberwolfach 报告问题汇编

For a graph $G$, let $\operatorname{TD}(G)$ be the minimum degree of a median graph whose transversality graph is isomorphic to $G$. Develop an algorithm to compute $\operatorname{TD}(G)$, and obtain structural or complexity bounds for this invariant.

来源字段：https://doi.org/10.4171/owr/2026/8

14162 | OWR-14299081-005

### Conditions Forcing Long Induced Paths

P / partially_solved | Oberwolfach 报告问题汇编

For a graph class $\mathcal{G}$, let $f_{\mathcal{G}}(n)$ be the maximum integer $k$ such that every member of $\mathcal{G}$ containing an $n$-vertex path also contains an induced $k$-vertex path. What natural conditions on $\mathcal{G}$ ensure that $f_{\mathcal{G}}(n)\to\infty$?

来源字段：https://doi.org/10.4171/owr/2025/1

14163 | GRAPH-023

### Conference Graph Existence

P / partially_solved | 独立 / 未分配集合

Does there exist a conference graph for every number of vertices $v > 1$ where $v \equiv 1 \pmod{4}$ and v is an odd sum of two squares?

背景链接：https://doi.org/10.4153/CJM-1978-029-1

14164 | AMR-088-0008

### Congruent number decision problem

O / open | AMR 问题清单汇编

Given an integer $n$, determine whether there are rational numbers $x,y,z$ satisfying $x^2+y^2=z^2$ and $xy=2n$; equivalently, determine whether $n$ is the area of a right triangle with rational sides.

背景链接：https://www.maths.nottingham.ac.uk/plp/pmzcw/download/open.pdf

14165 | AMR-088-0009

### Congruent numbers in residue classes 5, 6, and 7 modulo 8

O / open | AMR 问题清单汇编

Is every integer $n\equiv5,6,$ or $7\pmod 8$ a congruent number?

背景链接：https://www.maths.nottingham.ac.uk/plp/pmzcw/download/open.pdf

14166 | AMR-099-0001

### Conjecture 0.1 - Is there an infinite expander?

O / open | AMR 问题清单汇编

Call an infinite connected graph $G$ of uniformly bounded degree an infinite expander if there is a constant $c>0$ such that, for every vertex set $S$ and every finite-radius ball $B(r)$ satisfying $|S\cap B(r)|<|B(r)|/2$, one has $|\partial S\cap B(r)|>c|S\cap B(r)|$, where $\partial S$ is the vertex boundary of $S$. Prove that no infinite expander exists.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/infexp.ps

14167 | AMR-086-0002

### Conjecture 1.3 - Pillai

P / partially_solved | AMR 问题清单汇编

Let $k$ be a positive integer. The equation $$ x^p-y^q=k, $$ where the unknowns $x$, $y$, $p$ and $q$ take integer values, all $\ge 2$, has only finitely many solutions $(x,y,p,q)$.

背景链接：https://arxiv.org/abs/math/0312440

14168 | AMR-086-0003

### Conjecture 1.4 - Shorey

O / open | AMR 问题清单汇编

There exists a positive number $C$ which depends only on $L$ and $H$ with the following property. Let $m$, $x$ and $y$ be rational integers with $m\ge 2$ and $|y|>1$ satisfying $$ y^m=f(x). $$ Then either $m\le C$, or else there is a proper subsum in $$ y^m-b_1x^{n_1}-\cdots-b_{L-1}x^{n_{L-1}}-b_L $$ which vanishes.

背景链接：https://arxiv.org/abs/math/0312440

14169 | AMR-086-0004

### Conjecture 1.5

O / open | AMR 问题清单汇编

Let $k\ge 2$ be an integer and $\alpha_1,\ldots,\alpha_n$ be non-zero elements in a field $K$ of zero characteristic, such that no quotient $\alpha_i/\alpha_j$ with $j\not=i$ is a root of unity. Consider the function $$ F(X_1,\ldots,X_k)= \det\pmatrix{ \alpha_1^{X_1}&\cdots&\alpha_k^{X_1}\cr \vdots&\ddots&\vdots\cr \alpha_1^{X_k}&\cdots&\alpha_k^{X_k}\cr }. $$ Then the equation $$ F(0,x_2,\ldots,x_k)=0 $$ has only finitely many solutions $(x_2,\ldots,x_k)\in\mathbb{Z}^{k-1}$ such that in the corresponding determinant, all $(k-1)\times k$ and all $k\times (k-1)$ submatrices have rank $k-1$.

背景链接：https://arxiv.org/abs/math/0312440

14170 | AMR-086-0006

### Conjecture 1.7

O / open | AMR 问题清单汇编

If there is no prime in the interval $[n+1,n+k]$, then the product $(n+1)\cdots(n+k)$ has at least $k$ distinct prime divisors.

背景链接：https://arxiv.org/abs/math/0312440

14171 | AMR-086-0007

### Conjecture 1.8 - Langevin

O / open | AMR 问题清单汇编

Given an increasing sequence $n_1<n_2<\cdots<n_k$ of positive integers such that $n_1,n_2,\ldots,n_k$ are multiplicatively dependent, there exists a prime number in the interval $[n_1,n_k]$.

背景链接：https://arxiv.org/abs/math/0312440

14172 | AMR-086-0008

### Conjecture 1.9

P / partially_solved | AMR 问题清单汇编

Fix a positive integer $m$ for which the equation $$ m^2 + m_1^2 + m_2^2 = 3 mm_1m_2 $$ has a solution in positive integers $(m_1,m_2)$ with $0<m_1\le m_2\le m$. Then such a pair $(m_1,m_2)$ is unique.

背景链接：https://arxiv.org/abs/math/0312440

14173 | AMR-086-0016

### Conjecture 2.12

O / open | AMR 问题清单汇编

Let $\theta$ be real algebraic number of degree at least $3$. Then inequality (2.11) has infinitely many solutions in integers $p$ and $q$ with $q>0$ if and only if the integral $$ \int_1^\infty \psi(x) dx $$ diverges.

背景链接：https://arxiv.org/abs/math/0312440

14174 | AMR-086-0017

### Conjecture 2.14 - Mahler

O / open | AMR 问题清单汇编

There exists an absolute constant $c>0$ such that $$ \Vert \log a\Vert>a^{-c} $$ for all integers $a\ge 2$.

背景链接：https://arxiv.org/abs/math/0312440

14175 | AMR-086-0018

### Conjecture 2.15 - Mahler

P / partially_solved | AMR 问题清单汇编

Let $(\varepsilon_n)_{n\ge 0}$ be a sequence of elements in $\{0,1\}$. Assume that the real number $$ \sum_{n\ge 0}\varepsilon_n 3^{-n} $$ is irrational, then it is transcendental.

背景链接：https://arxiv.org/abs/math/0312440

14176 | AMR-086-0010

### Conjecture 2.2 - Erdős-Woods

P / partially_solved | AMR 问题清单汇编

There exists a positive integer $k$ such that, for $m$ and $n$ positive integers, the conditions $$ R(m+i)=R(n+i)\quad (i=0,\ldots,k-1) $$ imply $m=n$.

背景链接：https://arxiv.org/abs/math/0312440

14177 | AMR-086-0011

### Conjecture 2.3 - Erdős-Dressler

P / partially_solved | AMR 问题清单汇编

If $a$ and $b$ are two positive integers with $a<b$ and $R(a)=R(b)$ then there is a prime $p$ with $a< p< b$.

背景链接：https://arxiv.org/abs/math/0312440

14178 | AMR-086-0012

### Conjecture 2.4 - Philippon

O / open | AMR 问题清单汇编

There exist real numbers $\varepsilon$, $\alpha$ and $\beta$ with $0<\varepsilon<1/2$, $\alpha\ge 1$ and $\beta\ge 0$, and a positive integer $B$, such that for any nonzero rational numbers $x$, $y$ satisfying $xy^B\not=1$, the following is true: if $S$ denotes the set of prime numbers for which $|xy^B+1|_p<1$, then $$ -\sum_{p\in S}\log|xy^B+1|_p\le B\Bigl(\alpha \mathrm{h}(x)+\varepsilon\mathrm{h}(y)+ \bigl(\alpha B+\varepsilon\bigr)\bigl(\beta+\sum_{p\in S}\log p\bigr)\Bigr). $$

背景链接：https://arxiv.org/abs/math/0312440

14179 | AMR-086-0013

### Conjecture 2.5 - Lang-Waldschmidt

O / open | AMR 问题清单汇编

For any $\varepsilon>0$, there exists a constant $C(\varepsilon)>0$ such that, for any nonzero rational integers $a_1,\ldots,a_m$, $b_1,\ldots,b_m$ with $a_1^{b_1}\cdots a_m^{b_m}\not=1$, $$ \left|a_1^{b_1}\cdots a_m^{b_m}-1\right|\ge {C(\varepsilon)^m B\over (|b_1|\cdots |b_m|\cdot |a_1|\cdots |a_m|)^{1+\varepsilon}}, $$ where $B=\max_{1\le i\le m}|b_i|$.

背景链接：https://arxiv.org/abs/math/0312440

14180 | AMR-086-0014

### Conjecture 2.6

O / open | AMR 问题清单汇编

For any $\varepsilon>0$, there is a constant $C(\varepsilon)>0$ such that, for any positive integers $x$, $y$, $p$, $q$ satisfying $x^p\not= y^q$, the inequality $$ |x^p-y^q| \ge C(\varepsilon)\max\{x^p, y^q\}^{1-(1/p)-(1/q)-\varepsilon} $$ holds.

背景链接：https://arxiv.org/abs/math/0312440

14181 | AMR-086-0015

### Conjecture 2.7 - Hall

O / open | AMR 问题清单汇编

If $x$ and $y$ are positive integers with $y^2\not=x^3$, then $$ |y^2-x^3|\ge C\max\{y^2,x^3\}^{1/6}. $$

背景链接：https://arxiv.org/abs/math/0312440

14182 | AMR-086-0028

### Conjecture 3.10 - Gel’fond-Schneider

O / open | AMR 问题清单汇编

The $d$ numbers $$ \log\alpha,\; \alpha^\beta,\; \alpha^{\beta^2},\ldots, \alpha^{\beta^{d-1}} $$ are algebraically independent over $\mathbb{Q}$.

背景链接：https://arxiv.org/abs/math/0312440

14183 | AMR-086-0029

### Conjecture 3.11 - $p$-adic analog of Lindemann-Weierstrass's Theorem

O / open | AMR 问题清单汇编

Let $\beta_1,\ldots,\beta_n$ be $p$-adic algebraic numbers in the domain of convergence of the $p$-adic exponential function $\exp_p$. Then the $n$ numbers $\exp_p\beta_1,\ldots, \exp_p\beta_n$ are algebraically independent over $\mathbb{Q}$.

背景链接：https://arxiv.org/abs/math/0312440

14184 | AMR-086-0030

### Conjecture 3.12 - $p$-adic analog of an algebraic independence result of Gel’fond

O / open | AMR 问题清单汇编

Let $\alpha$ be a non-zero algebraic number in the domain of convergence of the $p$-adic logarithm $\log_p$, and let $\beta$ be a $p$-adic cubic algebraic number, such that $\beta\log_p\alpha$ is in the domain of convergence of the $p$-adic exponential function $\exp_p$. Then the two numbers $$ \alpha^\beta=\exp_p(\beta\log_p\alpha) \quad\text{and}\quad \alpha^{\beta^2}=\exp_p(\beta^2\log_p\alpha) $$ are algebraically independent over $\mathbb{Q}$.

背景链接：https://arxiv.org/abs/math/0312440

14185 | AMR-086-0031

### Conjecture 3.13 - Blum, Cucker, Shub and Smale

O / open | AMR 问题清单汇编

Given an absolute constant $c$ and polynomials $P_1,\ldots,P_m$ with a total of $N$ coefficients and no common complex zeros, there is no program to find, in at most $N^c$ step, the coefficients of polynomials $A_i$ satisfying B\'ezout's relation $$ A_1P_1+\cdots+A_mP_m=1. $$

背景链接：https://arxiv.org/abs/math/0312440

14186 | AMR-086-0032

### Conjecture 3.14

O / open | AMR 问题清单汇编

Let $\Sigma$ be a finite subset of $\mathbb{C}^n$ and $\varepsilon$ a positive number. There exists a positive number $r_0(\Sigma,\varepsilon)$ such that, for any positive integer $t$ and any entire function $f$ in $\mathbb{C}^n$ which vanishes on $\Sigma$ with multiplicity $\ge t$, $$ \Theta_f(r)\ge \omega_t(\Sigma)-t\varepsilon \quad\text{for}\quad r\ge r_0(\Sigma,\varepsilon). $$

背景链接：https://arxiv.org/abs/math/0312440

14187 | AMR-086-0033

### Conjecture 3.15 - Goncharov

P / partially_solved | AMR 问题清单汇编

As a $\mathbb{Q}$-algebra, $\mathfrak{Z}$ is the direct sum of $\mathfrak{Z}_p$ for $p\ge 0$.

背景链接：https://arxiv.org/abs/math/0312440

14188 | AMR-086-0034

### Conjecture 3.16 - Zagier

P / partially_solved | AMR 问题清单汇编

For $p\ge 3$ we have $$ d_p=d_{p-2}+d_{p-3} $$ with $d_0=1$, $d_1=0$, $d_2=1$.

背景链接：https://arxiv.org/abs/math/0312440

14189 | AMR-086-0035

### Conjecture 3.17

O / open | AMR 问题清单汇编

The numbers $\pi$, $\zeta(3),\zeta(5),\ldots,\zeta(2n+1),\ldots$ are algebraically independent over $\mathbb{Q}$.

背景链接：https://arxiv.org/abs/math/0312440

14190 | AMR-086-0036

### Conjecture 3.18

O / open | AMR 问题清单汇编

At least three of the four numbers $$ \pi,\; \Gamma(1/5),\; \Gamma(2/5), \; e^{\pi\sqrt 5} $$ are algebraically independent over $\mathbb{Q}$.

背景链接：https://arxiv.org/abs/math/0312440

14191 | AMR-086-0037

### Conjecture 3.19 - Rohrlich

P / partially_solved | AMR 问题清单汇编

$\overline{G}$ is a universal odd distribution with values in groups where multiplication by $2$ is invertible.

背景链接：https://arxiv.org/abs/math/0312440

14192 | AMR-086-0020

### Conjecture 3.2 - Roy

O / open | AMR 问题清单汇编

Let $k$ be a positive integer, $y_1,\ldots,y_k$ complex numbers which are linearly independent over $\mathbb{Q}$, $\alpha_1,\ldots,\alpha_k$ nonzero complex numbers and $s_0,s_1,t_0,t_1,u$ positive real numbers satisfying $$ \max\{1,t_0,2t_1\} < \min\{s_0,2s_1\} \quad\text{and}\quad \max\{s_0,s_1+t_1\} < u < {1\over 2}(1+t_0+t_1). $$ Assume that, for any sufficiently large positive integer $N$, there exists a nonzero polynomial $P_N\in\mathbb{Z}[X_0,X_1]$ with partial degree $\le N^{t_0}$ in $X_0$, partial degree $\le N^{t_1}$ in $X_1$ and height $\le e^N$ which satisfies $$ \Bigg| \big(\mathcal{D}^kP_N\big) \Big(\sum_{j=1}^k m_jy_j, \prod_{j=1}^k\alpha_j^{m_j} \Big) \Bigg| \le \exp(-N^u) $$ for any nonnegative integers $k$, $m_1,\ldots,m_k$ with $k\le N^{s_0}$ and $\max\{m_1,\ldots,m_k\}\le N^{s_1}$. Then, we have $$ \operatorname{trdeg} \mathbb{Q}(y_1,\ldots,y_k, \alpha_1,\ldots,\alpha_k) \ge k. $$

背景链接：https://arxiv.org/abs/math/0312440

14193 | AMR-086-0038

### Conjecture 3.20 - Nesterenko

P / partially_solved | AMR 问题清单汇编

Let $\tau\in\mathbb{C}$ have positive imaginary part. Assume that $\tau$ is not quadratic. Set $q=e^{2i\pi\tau}$. Then at least $4$ of the $5$ numbers $$ \tau,\; q,\; P(q),\; Q(q),\; R(q) $$ are algebraically independent.

背景链接：https://arxiv.org/abs/math/0312440

14194 | AMR-086-0039

### Conjecture 3.21 - Bertolin

O / open | AMR 问题清单汇编

Let $\mathcal{E}_1,\ldots,\mathcal{E}_n$ be pairwise non isogeneous elliptic curves with modular invariants $j(\mathcal{E}_h)$. For $h=1,\ldots,n$, let $\omega_{1h},\omega_{2h}$ be a pair of fundamental periods of $\wp_h$ with $\eta_{1h},\eta_{2h}$ the associated quasi-periods, $P_{ih}$ points on $\mathcal{E}_h(\mathbb{C})$ and $p_{ih}$ (resp. $d_{ih}$) elliptic integrals of the first (resp. second) kind associated to $P_{ih}$. Define $\kappa_h=[k_h:\mathbb{Q}]$ and let $d_h$ be the dimension of the $k_h$-subspace of $\mathbb{C}/(k_h\omega_{1h}+k_h\omega_{2h})$ spanned by $p_{1h},\ldots,p_{r_hh}$. Then the transcendence degree of the field $$ \mathbb{Q}\Bigl(\bigl\{ j(\mathcal{E}_h), \omega_{1h},\omega_{2h}, \eta_{1h},\eta_{2h}, P_{ih}, p_{ih}, d_{ih} \bigr\}_{1\le i\le r_h \atop 1\le h\le n} \Bigr) $$ is at least $$ 2\sum_{h=1}^n d_h +4\sum_{h=1}^n \kappa_h^{-1}-n+1. $$

背景链接：https://arxiv.org/abs/math/0312440

14195 | AMR-086-0041

### Conjecture 3.23

O / open | AMR 问题清单汇编

Given an elliptic curve with Weierstrass equation $y^2=4x^3-g_2x-g_3$, a nonzero period $\omega$, the associated quasi-period $\eta$ of the zeta function and a complex number $u$ which is not a pole of $\wp$, we have $$ \operatorname{trdeg}\mathbb{Q} \bigl(g_2,g_3,\pi/\omega,\wp(u),\zeta(u)-(\eta/\omega)u\bigr)\ge 2. $$

背景链接：https://arxiv.org/abs/math/0312440

14196 | AMR-086-0042

### Conjecture 3.24 - Bertrand

O / open | AMR 问题清单汇编

Let $q_1,\ldots,q_n$ be nonzero algebraic numbers in the unit open disc such that the $3n$ numbers $$ J(q_i),\; DJ(q_i),\; D^2J(q_i)\qquad (i=1,\ldots,n) $$ are algebraically dependent over $\mathbb{Q}$. Then there exist two indices $i\not = j$ \ ($1\le i\le n$, $1\le j\le n$) such that $q_i$ and $q_j$ are multiplicatively dependent.

背景链接：https://arxiv.org/abs/math/0312440

14197 | AMR-086-0043

### Conjecture 3.25 - Bertrand

O / open | AMR 问题清单汇编

Let $q_1$ and $q_2$ be two nonzero algebraic numbers in the unit open disc. Suppose that there is an irreducible element $P\in\mathbb{Q}[X,Y]$ such that $$ P\bigl(J(q_1),J(q_2)\bigr)=0. $$ Then there exist a constant $c$ and a positive integer $s$ such that $P=c\Phi_s$, where $\Phi_s$ is the modular polynomial of level $s$. Moreover $q_1$ and $q_2$ are multiplicatively dependent.

背景链接：https://arxiv.org/abs/math/0312440

14198 | AMR-086-0044

### Conjecture 3.26

O / open | AMR 问题清单汇编

Is there such a bound depending polynomially on the degree and height of $P$?

背景链接：https://arxiv.org/abs/math/0312440

14199 | AMR-086-0021

### Conjecture 3.3 - Algebraic Independence of Logarithms of Algebraic Numbers

O / open | AMR 问题清单汇编

Let $\lambda_1,\ldots, \lambda_n$ be $\mathbb{Q}$-linearly independent complex numbers. Assume that the numbers $e^{\lambda_1},\ldots,e^{\lambda_n}$ are algebraic. Then the numbers $\lambda_1,\ldots, \lambda_n$ are algebraically independent over $\mathbb{Q}$.

背景链接：https://arxiv.org/abs/math/0312440

14200 | AMR-086-0022

### Conjecture 3.4 - Strong Four Exponentials Conjecture

O / open | AMR 问题清单汇编

Let $x_1,x_2$ be two $\overline{\mathbb{Q}}$-linearly independent complex numbers and $y_1,y_2$ be also two $\overline{\mathbb{Q}}$-linearly independent complex numbers. Then at least one of the four numbers $x_1y_1$, $x_1y_2$, $x_2y_1$, $x_2y_2$ does not belong to $\widetilde{\mathcal{L}}$.

背景链接：https://arxiv.org/abs/math/0312440

14201 | AMR-086-0023

### Conjecture 3.5 - Strong Five Exponentials Conjecture

O / open | AMR 问题清单汇编

Let $x_1, x_2$ be two $\mathbb{Q}$-linearly independent complex numbers and $y_1, y_2$ be also two $\mathbb{Q}$-linearly independent complex numbers. Further let $\beta_{ij}$ \ ($ i=1,2$, $j=1,2$), $\gamma_1$ and $\gamma_2$ be six algebraic numbers with $\gamma_1\not=0$. Assume that the five numbers $$ e^{x_1y_1-\beta_{11}},\; e^{x_1y_2-\beta_{12}},\; e^{x_2y_1-\beta_{21}},\; e^{x_2y_2-\beta_{22}},\; e^{(\gamma_1 x_1/x_2)-\gamma_2} $$ are algebraic. Then all five exponents vanish: $$ x_iy_j=\beta_{ij} \quad (i=1,2, \quad j=1,2) \quad \text{ and }\quad \gamma_1 x_1=\gamma_2 x_2. $$.

背景链接：https://arxiv.org/abs/math/0312440

14202 | AMR-086-0024

### Conjecture 3.6 - Roy

O / open | AMR 问题清单汇编

For any $4\times 4$ skew-symmetric matrix $\mathrm{M}$ with entries in $\mathcal{L}$ and rank $\le 2$, either the rows of $\mathrm{M}$ are linearly dependent over $\mathbb{Q}$, or the column space of $\mathrm{M}$ contains a nonzero element of $\mathbb{Q}^4$.

背景链接：https://arxiv.org/abs/math/0312440

14203 | AMR-086-0026

### Conjecture 3.8 - Gel’fond

O / open | AMR 问题清单汇编

The two numbers $$ \log\alpha\quad\text{and}\quad \alpha^\beta $$ are algebraically independent over $\mathbb{Q}$.

背景链接：https://arxiv.org/abs/math/0312440

14204 | AMR-086-0027

### Conjecture 3.9 - Schneider

O / open | AMR 问题清单汇编

The $d-1$ numbers $$ \alpha^\beta,\; \alpha^{\beta^2},\ldots, \alpha^{\beta^{d-1}} $$ are algebraically independent over $\mathbb{Q}$.

背景链接：https://arxiv.org/abs/math/0312440

14205 | AMR-086-0046

### Conjecture 4.1 - Lehmer's Problem

P / partially_solved | AMR 问题清单汇编

There exists a positive absolute constant $c$ such that, for any nonzero algebraic number $\alpha$ which is not a root of unity, $$ \mathrm{M}(\alpha)\ge 1+c. $$ Equivalently, there exists a positive absolute constant $c$ such that, for any nonzero algebraic number $\alpha$ of degree at most $d$ which is not a root of unity, $$ \mathrm{h}(\alpha)\ge {c\over d}. $$

背景链接：https://arxiv.org/abs/math/0312440

14206 | AMR-086-0053

### Conjecture 4.11 - Wirsing and Schmidt

P / partially_solved | AMR 问题清单汇编

For any positive integer $n$ and any real number $\theta$ which is either transcendental or else is algebraic of degree $>n$, there exists a positive constant $c=c(n,\theta)$ with the following property: there exist infinitely many algebraic numbers $\gamma$ of degree $\le n$ for which $$ 0<|\theta-\gamma|<c\mathrm{H}(\gamma)^{-n-1}. $$

背景链接：https://arxiv.org/abs/math/0312440

14207 | AMR-086-0054

### Conjecture 4.12

P / partially_solved | AMR 问题清单汇编

Let $\underline{\theta}=(\theta_1,\ldots,\theta_m)$ be a $m$-tuple of complex numbers. Define $$ t=\operatorname{trdeg} \mathbb{Q}(\underline{\theta}) $$ and assume $t\ge 1$. There exist positive constants $c_1$ and $c_2$ with the following property. Let $(D_\nu)_{\nu\ge 0}$ and $(\mu_\nu)_{\nu\ge 0}$ be sequences of real numbers satisfying $$ c_1\le D_\nu\le \mu_\nu,\quad D_\nu\le D_{\nu+1}\le 2 D_\nu, \quad \mu_\nu\le \mu_{\nu+1}\le 2 \mu_\nu \qquad (\nu\ge 0). $$ Assume also that the sequence $(\mu_\nu)_{\nu\ge 0}$ is unbounded. Then for infinitely many $\nu$ there exists a $m$-tuple $(\gamma_1,\ldots,\gamma_m)$ of algebraic numbers satisfying $$ [\mathbb{Q}(\underline{\gamma}):\mathbb{Q}]\le D_\nu, \quad \mu(\underline{\gamma})\le \mu_\nu $$ and $$ \max_{1\le i\le m}|\theta_i-\gamma_i|\le \exp\{- c_2 D_\nu^{ 1/t } \mu_\nu\}. $$

背景链接：https://arxiv.org/abs/math/0312440

14208 | AMR-086-0055

### Conjecture 4.13 - Laurent-Roy

O / open | AMR 问题清单汇编

Let $\theta\in\mathbb{C}^m$. There is a positive constant $c$, depending only on $\theta$ and $m$, with the following property. Let $k$ be an integer with $0\le k\le m$. For infinitely many integers $T\ge 1$, there exists an algebraic set $Z\subset\mathbb{C}^m$, defined over $\mathbb{Q}$, of dimension $k$, and a point $\alpha\in Z$, such that $$ t(Z)\le T^{m-k}\quad\text{and}\quad |\theta-\alpha|\le \exp\{-cT^{m+1}\}. $$

背景链接：https://arxiv.org/abs/math/0312440

14209 | AMR-086-0056

## Conjecture 4.14

O / open | AMR 问题清单汇编

There exist two positive absolute constants $c_1$ and $c_2$ with the following property. Let $\lambda_1,\ldots,\lambda_m$ be logarithms of algebraic numbers with $\alpha_i=e^{\lambda_i}$ \ ($1\le i\le m$), let $\beta_0,\ldots, \beta_m$ be algebraic numbers, $D$ the degree of the number field $$ \mathbb{Q}(\alpha_1,\ldots,\alpha_m,\beta_0,\ldots,\beta_m) $$ and finally let $h\ge 1/D$ satisfy $$ h\ge \max_{1\le i\le m} \mathrm{h}(\alpha_i), \quad h\ge {1\over D}\max_{1\le i\le m}|\lambda_i| \quad\text{and}\quad h\ge \max_{0\le j\le m} \mathrm{h}(\beta_j). $$ (1) Assume that the number $$ \Lambda=\beta_0+\beta_1\lambda_1+\cdots+\beta_m\lambda_m $$ is nonzero. Then $$ |\Lambda|\ge \exp\bigl\{-c_1mD^2h \bigr\}. $$ (2) Assume $\lambda_1,\ldots,\lambda_m$ are linearly independent over $\mathbb{Q}$. Then $$ \sum_{i=1}^m|\lambda_i-\beta_i|\ge \exp\bigl\{-c_2mD^{1+(1/m)}h \bigr\}. $$

背景链接：https://arxiv.org/abs/math/0312440

14210 | AMR-086-0057

## Conjecture 4.15

O / open | AMR 问题清单汇编

There exists a positive absolute constant $C$ with the following property. Let $\alpha_1,\ldots,\alpha_n$ be nonzero algebraic numbers and $\log\alpha_1,\ldots,\log\alpha_n$ logarithms of $\alpha_1,\ldots,\alpha_n$ respectively. Assume that the numbers $\log\alpha_1,\ldots,\log\alpha_n$ are $\mathbb{Q}$-linearly independent. Let $\beta_0,\beta_1,\ldots,\beta_n$ be algebraic numbers, not all of which are zero. Denote by $D$ the degree of the number field $$ \mathbb{Q}(\alpha_1,\ldots,\alpha_n,\beta_0,\beta_1,\ldots,\beta_n) $$ over $\mathbb{Q}$. Further, let $A_1,\ldots,A_n$ and $B$ be positive real numbers, each $\ge e$, such that $$ \log A_j\ge\max\left\{\mathrm{h}(\alpha_j),\ {|\log\alpha_j|\over D}, \ {1\over D}\right\} \quad (1\le j\le n), $$ $$ B\ge\max_{1\le j\le n-1} \mathrm{h}(\beta_j). $$ Then the number $$ \Lambda=\beta_0+\beta_1\log\alpha_1+\cdots+\beta_n\log\alpha_n $$ satisfies $$ |\Lambda|> \exp\{-C^n D^{n+2}(\log A_1)\cdots (\log A_n)(\log B+\log D) (\log D)\}. $$

背景链接：https://arxiv.org/abs/math/0312440

14211 | AMR-086-0058

## Conjecture 4.16 - Quantitative Refinement of Schanuel's Conjecture

O / open | AMR 问题清单汇编

Let $x_1,\ldots,x_n$ be $\mathbb{Q}$-linearly independent complex numbers. Assume that for any $\varepsilon>0$, there exists a positive number $H_0$ such that, for any $H\ge H_0$ and $n$-tuple $(h_1,\ldots,h_n)$ of rational integers satisfying $0<\max\{|h_1|,\ldots,|h_n|\}\le H$, the inequality $$ |h_1x_1+\cdots+h_nx_n|\ge \exp\bigl\{-H^\varepsilon\bigr\} $$ holds. Let $d$ be a positive integer. Then there exists a positive number $C=C(x_1,\ldots,x_n,d)$ with the following property: for any integer $H\ge 2$ and any $n+1$ tuple $P_1,\ldots,P_{n+1}$ of polynomials in $\mathbb{Z}[X_1,\ldots,X_n,Y_1,\ldots,Y_n]$ with degrees $\le d$ and usual heights $\le H$, which generate an ideal of $\mathbb{Q}[X_1,\ldots,X_n,Y_1,\ldots,Y_n]$ of rank $n+1$, we have $$ \sum_{j=1}^{n+1}\bigl|P_j(x_1,\ldots,x_n,e^{x_1},\ldots,e^{x_n})\bigr|\ge H^{-C}. $$

背景链接：https://arxiv.org/abs/math/0312440

14212 | AMR-086-0061

## Conjecture 4.19

P / partially_solved | AMR 问题清单汇编

Let $A$ be a simple Abelian variety of dimension $g$ over a number field $K$ embedded in $\mathbb{R}$. Denote by $\ell$ the rank over $\mathbb{Z}$ of the Mordell-Weil group $A(K)$. For any $\varepsilon>0$, there exists $h_0>0$ (which depends only on the Abelian variety $A$, the real number field $K$ and $\varepsilon$) such that, for any $h\ge h_0$ and any $\zeta\in A(\mathbb{R})^0$, there is a point $\gamma\in A(K)$ with N\'eron-Tate height $\le h$ such that $$ \operatorname{dist}(\zeta,\gamma) \le h^{-(\ell/2g)+\varepsilon}. $$

背景链接：https://arxiv.org/abs/math/0312440

14213 | AMR-086-0062

### Conjecture 4.20

O / open | AMR 问题清单汇编

Let $m$, $n$, $k$ be positive integers and $a_{ij\kappa}$ rational integers ($1\le i\le n$, $1\le j\le m$, $1\le\kappa\le k$). For $\underline{x}=(x_1,\ldots,x_k)\in\bigl(\mathbb{R}_+^\times\bigr)^k$ denote by $\Gamma(\underline{x})$ the following finitely generated subgroup of $\bigl(\mathbb{R}_+^\times\bigr)^n$: $$ \Gamma(\underline{x})=\left\{ \left( \prod_{j=1}^m \prod_{\kappa=1}^k x_k^{a_{ij\kappa}s_j}\right)_{1\le i\le n} \; ;\; \underline{s}=(s_1,\ldots,s_m)\in\mathbb{Z}^m\right\}. $$ Assume there exists $\underline{x} \in\bigl(\mathbb{R}_+^\times\bigr)^k$ such that $\Gamma(\underline{x})$ is dense in $\bigl(\mathbb{R}_+^\times\bigr)^n$. Then for any $\underline{\gamma}=(\gamma_1,\ldots,\gamma_k)$ in $\bigl(\mathbb{R}_+^\times\bigr)^k$ with $\gamma_1,\ldots,\gamma_k$ algebraic and multiplicatively independent, the subgroup $\Gamma(\underline{\gamma})$ is dense in $\bigl(\mathbb{R}_+^\times\bigr)^n$.

背景链接：https://arxiv.org/abs/math/0312440

14214 | AMR-086-0063

### Conjecture 4.21

O / open | AMR 问题清单汇编

For any $\varepsilon>0$ there exists $S_0>0$ (depending on $\varepsilon$, $\gamma_1,\ldots,\gamma_m$ and $\mathcal{K}$) such that, for any $S\ge S_0$ and any $\underline{\zeta}\in\mathcal{K}$, there exists $\underline{s}\in\mathbb{Z}^m$ with $|\underline{s}|\le S$ and $$ \max_{1\le i\le n} |\gamma_i(\underline{s})-\zeta_i|\le S^{-1-(1/n)+\varepsilon}. $$

背景链接：https://arxiv.org/abs/math/0312440

14215 | AMR-086-0048

### Conjecture 4.3 - Amoroso-David

O / open | AMR 问题清单汇编

For each positive integer $n\ge 1$ there exists a positive number $c(n)$ having the following property. Let $\alpha_1,\ldots,\alpha_n$ be multiplicatively independent algebraic numbers. Define $D=[\mathbb{Q}(\alpha_1,\ldots,\alpha_n): \mathbb{Q}]$. Then $$ \prod_{i=1}^n\mathrm{h}(\alpha_i)\ge{c(n)\over D}. $$

背景链接：https://arxiv.org/abs/math/0312440

14216 | AMR-086-0049

### Conjecture 4.4 - Amoroso-David

O / open | AMR 问题清单汇编

For each positive integer $n\ge 1$ there exists a positive number $c(n)$ such that, if $\underline{\alpha}=(\alpha_1,\ldots,\alpha_n)$ is a $n$-tuple of multiplicatively independent algebraic numbers, then $$ \mathrm{h}(1\colon \alpha_1\colon \cdots\colon \alpha_n)\ge{c(n)\over \omega(\underline{\alpha})}. $$

背景链接：https://arxiv.org/abs/math/0312440

14217 | AMR-086-0050

### Conjecture 4.5 - Amoroso-David

P / partially_solved | AMR 问题清单汇编

For each integer $n\ge 1$ there exists a positive constant $c(n)$ such that, for any algebraic subvariety $V$ of $\mathbb{G}_m^n$ which is defined over $\mathbb{Q}$, which is $\mathbb{Q}$-irreducible, and which is not a union of translates of algebraic subgroups by torsion points, we have $$ \hat{h}(V)\ge c(n)\deg(V)^{(s-\dim V-1)/(s-\dim V)}, $$ where $s$ is the dimension of the smallest algebraic subgroup of $\mathbb{G}_m^n$ containing $V$.

背景链接：https://arxiv.org/abs/math/0312440

14218 | AMR-086-0052

### Conjecture 4.7 - David-Hindry

P / partially_solved | AMR 问题清单汇编

There exists a positive constant $c$, depending only on $A$ and $\mathcal{L}$, such that for any $P\in A(\overline{\mathbb{Q}})$ which has infinite order modulo any abelian subvariety, we have $$ \hat{h}_{\mathcal{L}}(P)\ge c \delta(P)^{-1}. $$

背景链接：https://arxiv.org/abs/math/0312440

14219 | AMR-086-0066

### Conjecture 5.3

O / open | AMR 问题清单汇编

Let $n$ be a positive integer. For almost all $n$-tuples $(x_1,\ldots,x_n)$, there are positive constants $c$ and $D_0$ (depending on $n$, $x_1,\ldots,x_n$ and $\varepsilon$), with the following property. For any integer $D\ge D_0$, any real number $\mu\ge D$ and any $2n$-tuple $\alpha_1,\ldots,\alpha_n,\beta_1,\ldots,\beta_n$ of algebraic numbers satisfying $$ [\mathbb{Q}( \alpha_1,\ldots,\alpha_n,\beta_1,\ldots,\beta_n):\mathbb{Q}]\le D $$ and $$ [\mathbb{Q}( \alpha_1,\ldots,\alpha_n,\beta_1,\ldots,\beta_n):\mathbb{Q}] \max\left\{\mathrm{h}(\alpha_i),\; \mathrm{h}(\beta_i)\; ; \; 1\le i\le n\right\}\le\mu, $$ we have $$ \max\left\{|x_i-\beta_i|,\; |e^{x_i}-\alpha_i|\; ; \; {1\le i\le n}\right\} \ge \exp\{-c D^{1/(2n)}\mu\}. $$

背景链接：https://arxiv.org/abs/math/0312440

14220 | AMR-086-0067

### Conjecture 5.4 - Loxton and van der Poorten

S / solved | AMR 问题清单汇编

Let $(n_i)_{i\ge 0}$ be an increasing sequence of positive integers. Assume there is a prime number $p$ such that the power series $$ \sum_{i\ge 0}z^{n_i}\in\mathbb{F}_p[[z]] $$ is algebraic over $\mathbb{F}_p(z)$ and irrational (not in $\mathbb{F}_p(z)$). Then the real number $$ \sum_{i\ge 0}10^{-n_i} $$ is transcendental.

背景链接：https://arxiv.org/abs/math/0312440

14221 | OWR-16931-006

### Conlon-Lee Conjecture for Graph Subdivisions

P / partially_solved | Oberwolfach 报告问题汇编

Let $k\geq2$ be even, let $H$ be a graph, and write $H^{k-1}$ for the subdivision obtained by replacing every edge of $H$ with a path having exactly $k-1$ new internal vertices. Does there exist $\delta>0$ such that $$\operatorname{ex}(n,H^{k-1})=O(n^{1+1/k-\delta})$$?

来源字段：https://doi.org/10.4171/owr/2019/19

14222 | OWR-711-002

### Connectivity and Hamilton Circles in Locally Finite Graphs

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a function $f$ such that every locally finite graph whose vertices and ends have degree at least $f(k)$ contains a $k$-connected subgraph? Does every $4$-connected planar locally finite graph, and the square of every $2$-connected locally finite graph, have a Hamilton circle?

来源字段：https://doi.org/10.4171/owr/2004/01

14223 | OWR-1703876-013

### Connectivity of Bistellar Flip Graphs in Dimensions Three and Four

O / open | Oberwolfach 报告问题汇编

Are the bistellar-flip graphs of triangulations of every point set connected in dimensions $3$ and $4$?

来源字段：https://doi.org/10.4171/owr/2020/30

14224 | OWR-12872-013

### Connectivity Under Weakened Bridge Addability

O / open | Oberwolfach 报告问题汇编

Which connectivity bounds and related asymptotic properties of bridge-addable graph classes remain valid when bridge-addability is replaced by a weaker closure condition?

来源字段：https://doi.org/10.4171/owr/2014/12

14225 | OWR-4212-014

### Connectivity-Preserving Paths in Antisymmetric Digraphs

O / open | Oberwolfach 报告问题汇编

For every nonnegative integer $m$, does every $k$-connected antisymmetric digraph $D$ with

$$\delta^+(D)\geq\left\lfloor\frac{3k+1}{2}\right\rfloor+m \quad\text{and}\quad \delta^-(D)\geq\left\lfloor\frac{3k+1}{2}\right\rfloor+m$$

contain a directed path $P$ of length $2m$ such that $\kappa(D-V(P))\geq k$?

来源字段：https://doi.org/10.4171/owr/2010/11

14226 | OPG-172

### Consecutive non-orientable embedding obstructions

P / partially_solved | Open Problem Garden

Conjecture Is there a graph $G$ that is a minor-minimal obstruction for two non-orientable surfaces?

背景链接：http://www.openproblemgarden.org/op/consecutive_non_orientable_embedding_obstructions

14227 | AMR-087-0044

### Constructing an elliptic curve of prescribed order over a fixed field

S / solved | AMR 问题清单汇编

Given integers $n$ and a prime power $q$, construct, when possible, an elliptic curve $E/\mathbb{F}_q$ with $\#E(\mathbb{F}_q)=n$.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14228 | OWR-12697712-004

### Convergence of Expected Graphon Chromatic Numbers

P / partially_solved | Oberwolfach 报告问题汇编

别名：Chromatic Numbers of Exchangeable Graphon Random Graphs [OWR-12697712-003]

For a graphon $W$, does $(\log n/n)\mathbb{E}[\chi(G(n,W))]$ converge as $n\to\infty$?

来源字段：https://doi.org/10.4171/owr/2023/16

14229 | AMR-087-0088

### Converting between divisor-class and infrastructure discrete logarithms

P / partially_solved | AMR 问题清单汇编

Give efficient reductions in both directions between the degree-zero divisor-class-group discrete logarithm problem and the infrastructure discrete logarithm problem.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14230 | GRAPH-024

### Conway's 99-Graph Problem

P / partially_solved | 独立 / 未分配集合

Does there exist a strongly regular graph with parameters (99,14,1,2)?

背景链接：https://doi.org/10.5802/alco.418

14231 | GRAPH-016

### Conway's Thrackle Conjecture

P / partially_solved | 独立 / 未分配集合

Does every thrackle have at most as many edges as vertices?

背景链接：https://arxiv.org/abs/2506.11808

14232 | AMR-089-0001

### Coordinates on convex domains

P / partially_solved | AMR 问题清单汇编

For a compact convex domain $\Omega$, the values of $F_\Omega$ at the vertices of its corner locus $C_\Omega$ give complete coordinates. How are these coordinates for $\Omega$ related to those for the dual domain $\Omega^*$?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/17-75/index.html

14233 | OPG-125

### Cores of Cayley graphs

O / open | Open Problem Garden

For a prime p, is the core of every Cayley graph on a power of $\mathbb Z_p$ itself a Cayley graph on a power of $\mathbb Z_p$?

背景链接：http://www.openproblemgarden.org/op/cores_of_cayley_graphs

14234 | OPG-824

### Cores of strongly regular graphs

P / partially_solved | Open Problem Garden

Question Does every strongly regular graph have either itself or a complete graph as a core?

背景链接：http://www.openproblemgarden.org/op/cores_of_strongly_regular_graphs

14235 | OPG-1764

### Counting 3-colorings of the hex lattice

O / open | Open Problem Garden

Problem Find $\lim_{n \rightarrow \infty} (\chi( H_n, 3)) ^{ 1 / |V(H_n)| }$.

背景链接：http://www.openproblemgarden.org/op/counting_3_colorings_of_the_hex_lattice

14236 | OPG-37325

### Covering powers of cycles with equivalence subgraphs

P / partially_solved | Open Problem Garden

Is there an absolute constant c>0 such that, for every k and all sufficiently large n depending on k, the equivalence covering number of $C_n^k$ is at least ck?

背景链接：http://www.openproblemgarden.org/op/covering_powers_of_cycles_with_equivalence_subgraphs

14237 | OWR-1782-013

### Critical Biases in Positional Graph Games

P / partially_solved | Oberwolfach 报告问题汇编

In biased positional games on the edges of $K_n$, close the remaining gap for the Breaker bias at which Maker can build a Hamilton cycle. Determine the threshold biases for Maker to build a perfect matching or a $k$-connected graph, and determine whether a threshold bias exists for the Avoider-Enforcer Hamiltonicity game.

来源字段：https://doi.org/10.4171/owr/2008/01

14238 | OPG-37068

### Crossing numbers and coloring

P / partially_solved | Open Problem Garden

We let $cr(G)$ denote the crossing number of a graph $G$.

Conjecture Every graph $G$ with $\chi(G) \ge t$ satisfies $cr(G) \ge cr(K_t)$.

背景链接：http://www.openproblemgarden.org/op/crossing_numbers_and_coloring

14239 | OPG-1812

### Crossing sequences

P / partially_solved | Open Problem Garden

Conjecture Let $(a_0,a_1,a_2,\ldots,0)$ be a sequence of nonnegative integers which strictly decreases until $0$.

Then there exists a graph that be drawn on a surface with orientable (nonorientable, resp.) genus $i$ with $a_i$ crossings, but not with less crossings.

背景链接：http://www.openproblemgarden.org/op/crossing_sequences

14240 | GRAPH-048

### Crown Graphs and Longest Word-Representants

O / open | 独立 / 未分配集合

Among bipartite graphs, do crown graphs require the longest word-representants?

背景链接：https://arxiv.org/abs/1609.00674

14241 | AMR-087-0030

### Cryptographic primitives from hard small roots

O / open | AMR 问题清单汇编

Construct additional cryptographic primitives whose security follows from the hardness of finding small roots.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14242 | AMR-087-0002

### Cryptographically useful bilinear structures

P / partially_solved | AMR 问题清单汇编

Find bilinear structures that are useful for cryptographic constructions.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14243 | AMR-087-0076

### Cryptography from worst-case algebraic-number-theory hardness

S / solved | AMR 问题清单汇编

Base cryptographic constructions directly on worst-case hardness assumptions from algebraic number theory.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14244 | GRAPH-035

### Cubic Graph Pathwidth

P / partially_solved | 独立 / 未分配集合

What is the maximum pathwidth of an n-vertex cubic graph?

背景链接：https://doi.org/10.1016/j.ipl.2005.10.010

14245 | AMR-087-0012

### Cup-product pairings in zeta computation

P / partially_solved | AMR 问题清单汇编

Determine whether natural de Rham cup-product pairings can be used to improve zeta-function computations.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14246 | OPG-60029

### Cycle Double Covers Containing Predefined 2-Regular Subgraphs

P / partially_solved | Open Problem Garden

Conjecture Let $G$ be a $2$-connected cubic graph and let $S$ be a $2$-regular subgraph such that $G-E(S)$ is connected. Then $G$ has a cycle double cover which contains $S$ (i.e all cycles of $S$ ).

背景链接：http://www.openproblemgarden.org/op/cycle_double_covers_containing_predefined_2_regular_subgraphs

14247 | OPG-60001

### Cycles in Graphs of Large Chromatic Number

P / partially_solved | Open Problem Garden

For an integer $k\ge3$, if $\chi(G)>k$, then $G$ contains at least $\frac{(k+1)(k-1)!}{2}$ cycles of length $0\bmod k$.

背景链接：http://www.openproblemgarden.org/op/cycles_in_graphs_of_large_chromatic_number

14248 | OPG-50631

### Cyclic spanning subdigraph with small cyclomatic number

P / partially_solved | Open Problem Garden

Conjecture Let $D$ be a digraph all of whose strong components are nontrivial. Then $D$ contains a cyclic spanning subdigraph with cyclomatic number at most $\alpha(D)$.

背景链接：http://www.openproblemgarden.org/op/cyclic_spanning_subdigraph_with_small_cyclomatic_number

14249 | AMR-087-0059

### Decision Linear versus DDH

O / open | AMR 问题清单汇编

Is the Decision Linear problem strictly harder than the decisional Diffie-Hellman problem in the relevant pairing groups?

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14250 | OPG-46583

### Decomposing a connected graph into paths.

P / partially_solved | Open Problem Garden

Conjecture Every simple connected graph on $n$ vertices can be decomposed into at most $\frac{1}{2}(n+1)$ paths.

背景链接：http://www.openproblemgarden.org/op/decomposing_a_connected_graph_into_paths

14251 | OPG-46606

### Decomposing an eulerian graph into cycles with no two consecutives edges on a prescribed eulerian tour.

P / partially_solved | Open Problem Garden

Conjecture Let $G$ be an eulerian graph of minimum degree $4$, and let $W$ be an eulerian tour of $G$. Then $G$ admits a decomposition into cycles none of which contains two consecutive edges of $W$.

背景链接：http://www.openproblemgarden.org/op/decomposing_an_eulerian_graph_into_cycles_with_no_two_consecutives_edges_on_a_prescirbed_eulerian_tour

14252 | OPG-46584

### Decomposing an eulerian graph into cycles.

P / partially_solved | Open Problem Garden

Conjecture Every simple eulerian graph on $n$ vertices can be decomposed into at most $\frac{1}{2}(n-1)$ cycles.

背景链接：http://www.openproblemgarden.org/op/decomposing_an_eulerian_graph_into_cycles

14253 | OPG-46237

### Decomposing an even tournament in directed paths.

P / partially_solved | Open Problem Garden

Conjecture Every tournament $D$ on an even number of vertices can be decomposed into $\sum_{v\in V}\max\{0,d^+(v)-d^-(v)\}$ directed paths.

背景链接：http://www.openproblemgarden.org/op/decomposing_an_even_tournament_in_directed_paths

14254 | OPG-141

### Decomposing eulerian graphs

P / partially_solved | Open Problem Garden

Conjecture If $G$ is a 6-edge-connected Eulerian graph and $P$ is a 2-transition system for $G$, then $(G,P)$ has a compaible decomposition.

背景链接：http://www.openproblemgarden.org/op/decomposing_eulerian_graphs

14255 | OPG-47651

### Decomposing k-arc-strong tournament into k spanning strong digraphs

P / partially_solved | Open Problem Garden

Conjecture Every k-arc-strong tournament decomposes into k spanning strong digraphs.

背景链接：http://www.openproblemgarden.org/op/decomposing_k_arc_strong_tournament_into_k_spanning_strong_digraphs

14256 | OWR-9790354-008

### Definable Analogues of Finite Combinatorial Theorems

O / open | Oberwolfach 报告问题汇编

Which results of finite combinatorics admit counterparts for Borel graphs that are Borel, $\mu$-measurable, or $\tau$-Baire measurable?

来源字段：https://doi.org/10.4171/owr/2022/3

14257 | OPG-798

### Degenerate colorings of planar graphs

P / partially_solved | Open Problem Garden

A graph $G$ is $k$-degenerate if every subgraph of $G$ has a vertex of degree $\le k$.

Conjecture Every simple planar graph has a 5-coloring so that for $1 \le k \le 4$, the union of any $k$ color classes induces a $(k-1)$-degenerate graph.

背景链接：http://www.openproblemgarden.org/op/degenerate_colorings_of_planar_graphs

14258 | GRAPH-025

### Degree Diameter Problem

P / partially_solved | 独立 / 未分配集合

For given maximum degree d and diameter k, what is the largest possible number of vertices in a graph?

背景链接：https://doi.org/10.17632/d75dzbjd4k.11

14259 | AMR-084-0010

### Degree-only bounds for polynomial D(n)-tuples

O / open | AMR 问题清单汇编

Let $P_n$ be the supremum of the sizes of nondegenerate polynomial $D(n)$-tuples over $\mathbb{Z}[X]$. Find an upper bound for $P_n$ depending only on $\deg n$, not on the coefficients of $n$.

背景链接：https://web.math.pmf.unizg.hr/~duje/dtuples.html

14260 | OWR-9790352-027

### Density Threshold for Rainbow Clique Subdivisions

P / partially_solved | Oberwolfach 报告问题汇编

别名：Density Threshold for Rainbow Clique Subdivisions [OWR-9790352-028]

For fixed $m\geq2$, what is the smallest $c$ such that every sufficiently large properly edge-colored graph on $n$ vertices with $\Omega(n(\log n)^c)$ edges contains a rainbow subdivision of $K_m$? In particular, does $c=1$ suffice?

来源字段：https://doi.org/10.4171/owr/2022/1

14261 | AMR-087-0049

### Detecting Jacobians via criteria and Deligne modules

O / open | AMR 问题清单汇编

Combine the Meagher-Ritzenthaler criteria with Deligne modules to detect Jacobians in an ordinary absolutely simple abelian-threefold isogeny class.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14262 | AMR-087-0092

### Deuring lifting for Darmon-Heegner points

O / open | AMR 问题清单汇编

Find an analogue of the Deuring Lifting Theorem for Darmon-Heegner points.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14263 | AMR-087-0028

### Dimension reduction in small-root lattices

O / open | AMR 问题清单汇编

Determine whether the lattice dimension in the stated small-root constructions can be reduced.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14264 | OWR-2654831-004

### Dimer Theory and Lorentz-Minimal Scaling-Limit Surfaces

O / open | Oberwolfach 报告问题汇编

What is the correspondence between the classical Kenyon-Okounkov theory and Lorentz-minimal surfaces, and how does local graph structure lead to conjectural Lorentz-minimality of scaling-limit surfaces?

来源字段：https://doi.org/10.4171/owr/2020/35

14265 | OWR-4212-004

### Directed Disjoint Paths in Restricted Digraph Classes

P / partially_solved | Oberwolfach 报告问题汇编

For which restricted classes of digraphs can the fixed-$k$ edge-disjoint paths problem be solved in polynomial time? Here the input is a digraph $G$ and $k$ terminal pairs, and the task is to decide whether mutually edge-disjoint directed paths connect each pair.

来源字段：https://doi.org/10.4171/owr/2010/11

14266 | OPG-46359

### Directed path of length twice the minimum outdegree

O / open | Open Problem Garden

Conjecture Every oriented graph with minimum outdegree $k$ contains a directed path of length $2k$.

背景链接：http://www.openproblemgarden.org/op/directed_cycle_of_length_twice_the_minimum_outdegree

14267 | OWR-4791-025

### Directed Trees in Slightly Larger Tournaments

O / open | Oberwolfach 报告问题汇编

Let $T$ be a directed tree on $n$ vertices with $k$ leaves. Must every tournament on $n+k-1$ vertices contain a copy of $T$?

来源字段：https://doi.org/10.4171/owr/2011/01

14268 | OWR-12330-007

### Disjoint-Edge Bounds for Simple Topological Graphs

P / partially_solved | Oberwolfach 报告问题汇编

Is it true that every simple topological graph with $n$ vertices and no $k$ pairwise disjoint edges has at most $O(kn)$ edges?

来源字段：https://doi.org/10.4171/owr/2013/02

14269 | OWR-16164-006

### Distances in Decorated Random Planar Maps

O / open | Oberwolfach 报告问题汇编

Determine the order of typical graph distances in decorated random planar-map models, including $O(n)$ loop models and FK percolation on random planar maps.

来源字段：https://doi.org/10.4171/owr/2018/23

14270 | AMR-087-0034

### Distribution of elliptic-curve group structures

P / partially_solved | AMR 问题清单汇编

Study the distribution of group structures $E(\mathbb{F}_q)$ as elliptic curves $E/\mathbb{F}_q$ vary; in particular, determine the correct nonuniform law.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14271 | AMR-087-0043

### Distribution of elliptic-curve pseudorandom sequences

P / partially_solved | AMR 问题清单汇编

Prove the conjecture that the EC-LCG, EC-PG, and EC-NRG sequences defined in the slides are very well distributed.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14272 | OPG-46538

### Do any three longest paths in a connected graph have a vertex in common?

P / partially_solved | Open Problem Garden

Conjecture Do any three longest paths in a connected graph have a vertex in common?

背景链接：http://www.openproblemgarden.org/op/do_any_three_longest_paths_in_a_connected_graph_have_a_vertex_in_common

14273 | OPG-36880

### Does the chromatic symmetric function distinguish between trees?

P / partially_solved | Open Problem Garden

Problem Do there exist non-isomorphic trees which have the same chromatic symmetric function?

背景链接：http://www.openproblemgarden.org/op/does_the_symmetric_chromatic_function_distinguish_trees

14274 | OPG-37038

### Domination in cubic graphs

P / partially_solved | Open Problem Garden

Problem Does every 3-connected cubic graph $G$ satisfy $\gamma(G) \le \lceil |G|/3 \rceil$?

背景链接：http://www.openproblemgarden.org/op/domination_in_cubic_graphs

14275 | OPG-36922

### Domination in plane triangulations

P / partially_solved | Open Problem Garden

Conjecture Every sufficiently large plane triangulation $G$ has a dominating set of size $\le \frac{1}{4} |V(G)|$.

背景链接：http://www.openproblemgarden.org/op/domination_in_plane_triangulations

14276 | OPG-34839

### Double-critical graph conjecture

P / partially_solved | Open Problem Garden

A connected simple graph $G$ is called double-critical, if removing any pair of adjacent vertexes lowers the chromatic number by two.

Conjecture $K_n$ is the only $n$-chromatic double-critical graph

背景链接：http://www.openproblemgarden.org/op/double_critical_graph_conjecture

14277 | OPG-322

### Drawing disconnected graphs on surfaces

O / open | Open Problem Garden

Conjecture Let $G$ be the disjoint union of the graphs $G_1$ and $G_2$ and let $\Sigma$ be a surface. Is it true that every optimal drawing of $G$ on $\Sigma$ has the property that $G_1$ and $G_2$ are disjoint?

背景链接：http://www.openproblemgarden.org/op/drawing_disconnected_graphs_on_surfaces

14278 | OWR-11786-023

### Dual-Diameter Bounds for Triangulated Surfaces

O / open | Oberwolfach 报告问题汇编

If $\overline f(2,n)$ is the maximal dual diameter of a triangulated surface with boundary and $n$ vertices, is $\overline f(2,n)\leq n-3$?

来源字段：https://doi.org/10.4171/owr/2012/24

14279 | GRAPH-010

### Earth-Moon Problem

P / partially_solved | 独立 / 未分配集合

别名：Earth-Moon Problem [OPG-46900]

What is the maximum chromatic number of biplanar graphs?

背景链接：https://mlelarge.github.io/graph-conjectures/op/earth_moon_problem/

14280 | OPG-2110

### Edge list coloring conjecture

P / partially_solved | Open Problem Garden

Conjecture Let $G$ be a loopless multigraph. Then the edge chromatic number of $G$ equals the list edge chromatic number of $G$.

背景链接：http://www.openproblemgarden.org/op/edge_list_coloring_conjecture

14281 | OPG-804

### Edge Reconstruction Conjecture

O / open | Open Problem Garden

Conjecture

Every simple graph with at least 4 edges is reconstructible from it's edge deleted subgraphs

背景链接：http://www.openproblemgarden.org/op/edge_reconstruction_conjecture

14282 | OWR-12330-009

### Edge-Disjoint Connected Spanning Subgraphs

P / partially_solved | Oberwolfach 报告问题汇编

For $a_1,a_2\geq1$, does every $(a_1+a_2+2)$-edge-connected graph $G$ have edge-disjoint spanning subgraphs $G_1,G_2$ such that $G_j$ is $a_j$-edge-connected for $j=1,2$?

来源字段：https://doi.org/10.4171/owr/2013/02

14283 | OPG-47031

### Edge-disjoint Hamilton cycles in highly strongly connected tournaments.

P / partially_solved | Open Problem Garden

Conjecture For every $k\geq 2$, there is an integer $f(k)$ so that every strongly $f(k)$-connected tournament has $k$ edge-disjoint Hamilton cycles.

背景链接：http://www.openproblemgarden.org/op/edge_disjoint_hamilton_cycles

14284 | OWR-4791-008

### Edge-Disjoint Hamilton Cycles in Random Graphs

S / solved | Oberwolfach 报告问题汇编

For any edge probability $p=p(n)$, does $G\sim G(n,p)$ contain $\lfloor\delta(G)/2\rfloor$ edge-disjoint Hamilton cycles with high probability?

来源字段：https://doi.org/10.4171/owr/2011/01

14285 | AMR-087-0048

### Effective representation of principally polarized abelian threefolds

O / open | AMR 问题清单汇编

Give an effective input representation for a principally polarized abelian threefold suitable for deciding whether it is a Jacobian.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14286 | OWR-14299905-023

### Efficient Algorithms for Large Hypergraph Independent Sets

O / open | Oberwolfach 报告问题汇编

For some absolute $c>0$, give a polynomial-time algorithm that finds an independent set of size at least $c\,\mathbb E|I|$ in every large-girth, $d$-regular, $r$-uniform hypergraph, where $I$ is the independent set produced by the random-ordering construction.

来源字段：https://doi.org/10.4171/owr/2026/1

14287 | AMR-087-0003

### Efficient class-group realizations of large cyclic groups

P / partially_solved | AMR 问题清单汇编

Find orders $\mathcal{O}$ whose Picard groups contain $\mathbb{Z}/\ell$, admit compact element representations, and allow group composition in $O(\log\ell)$ operations.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14288 | OWR-14299905-030

### Efficient Recognition of Graph Metrizability

P / partially_solved | Oberwolfach 报告问题汇编

Find a polynomial-time algorithm with practical complexity for deciding whether a finite graph is metrizable.

来源字段：https://doi.org/10.4171/owr/2026/1

14289 | OWR-10252930-021

### Elementary Proof of the Closed Walk Stacking Criterion

O / open | Oberwolfach 报告问题汇编

Find an elementary proof that a closed walk $W$ has a stacking in a graph $G$ if and only if $W\neq kW'$ for every walk $W'$ and every integer $k\geq 2$.

来源字段：https://doi.org/10.4171/owr/2022/22

14290 | AMR-087-0041

### Elliptic curves with smooth group order

P / partially_solved | AMR 问题清单汇编

Prove that sufficiently many elliptic curves $E/\mathbb{F}_p$ have smooth group order $\#E(\mathbb{F}_p)$.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14291 | AMR-087-0042

### Elliptic-curve orders with a large prime divisor

P / partially_solved | AMR 问题清单汇编

Quantify elliptic curves over finite fields whose group order has a large prime divisor.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14292 | OPG-735

### End-Devouring Rays

P / partially_solved | Open Problem Garden

Problem Let $G$ be a graph, $\omega$ a countable end of $G$, and $K$ an infinite set of pairwise disjoint $\omega$-rays in $G$. Prove that there is a set $K'$ of pairwise disjoint $\omega$-rays that devours $\omega$ such that the set of starting vertices of rays in $K'$ equals the set of starting vertices of rays in $K$.

背景链接：http://www.openproblemgarden.org/op/end_devouring_rays

14293 | OWR-1194-001

### Equal-Area Drawings of Plane Triangulations

O / open | Oberwolfach 报告问题汇编

Does every 4-connected triangulated plane graph with a fixed outer face admit a straight-line drawing in which all interior triangles have equal area?

来源字段：https://doi.org/10.4171/owr/2006/12

14294 | OWR-14299081-003

### Equality Cases for Gerzon’s Equiangular-Line Bound

P / partially_solved | Oberwolfach 报告问题汇编

Are there infinitely many dimensions $r$ in which the maximum number of equiangular lines in $\mathbb{R}^r$ attains Gerzon’s bound $\binom{r+1}{2}$?

来源字段：https://doi.org/10.4171/owr/2025/1

14295 | OWR-16931-025

### Equality of List and Ordinary Ramsey Numbers

O / open | Oberwolfach 报告问题汇编

For the list Ramsey number $R_L$, is $$R_L(K_k,2)=R(K_k,2)$$ for every $k$, and is $$R_L(K_3,r)=R(K_3,r)$$ for every $r$?

来源字段：https://doi.org/10.4171/owr/2019/19

14296 | OWR-16763-009

### Equivalence Hierarchies for Connected Graphs

O / open | Oberwolfach 报告问题汇编

If connected graphs $H,H'$ are $3$-equivalent, must they be $2$-equivalent? For every $q\geq2$, do nonisomorphic connected $q$-equivalent graphs exist? Which graphs $H$ make $K_k$ and $K_k+H$ Ramsey equivalent?

来源字段：https://doi.org/10.4171/owr/2019/1

14297 | OWR-16627-004

### Equivalence of Graphs in Strong Ramsey Games

O / open | Oberwolfach 报告问题汇编

Determine the largest $t=t(k)$ such that $K_k$ is 2-equivalent to $K_k+K_{1,t}$, and the largest $t=t(k)$ for which some tree $T$ on $t$ vertices satisfies $K_k\sim_2 K_k+T$; moreover, for each $q\ge 2$, do there exist two non-isomorphic connected $q$-equivalent graphs?

来源字段：https://doi.org/10.4171/owr/2018/44

14298 | EP-1006

### Erdős Problem #1006: Graph Theory, Cycles

S / solved | Erdős 问题

Let $G$ be a graph with girth $>4$ (that is, it contains no cycles of length $3$ or $4$). Can the edges of $G$ always be directed such that there is no directed cycle, and reversing the direction of any edge also creates no directed cycle?

来源字段：https://www.erdosproblems.com/1006

14299 | EP-1007

### Erdős Problem #1007: Graph Theory

S / solved | Erdős 问题

The dimension of a graph $G$ is the minimal $n$ such that $G$ can be embedded in $\mathbb{R}^n$ such that every edge of $G$ is a unit line segment.

What is the smallest number of edges in a graph with dimension $4$?

来源字段：https://www.erdosproblems.com/1007

14300 | EP-1008

### Erdős Problem #1008: Graph Theory, Cycles

S / solved | Erdős 问题

Does every graph with $m$ edges contain a subgraph with $\gg m^{2/3}$ edges which contains no $C_4$?

来源字段：https://www.erdosproblems.com/1008

14301 | EP-1009

### Erdős Problem #1009: Graph Theory

S / solved | Erdős 问题

Is it true that, for every $c>0$, there exists $f(c)$ such that every graph on $n$ vertices with at least $\lfloor n^2/4\rfloor+k$ edges, where $k<c n$, contains at least $k-f(c)$ many edge disjoint triangles?

来源字段：https://www.erdosproblems.com/1009

14302 | EP-1010

### Erdős Problem #1010: Graph Theory

S / solved | Erdős 问题

Let $t<\lfloor n/2\rfloor$. Does every graph on $n$ vertices with $\lfloor n^2/4\rfloor+t$ edges contain at least $t\lfloor n/2\rfloor$ triangles?

来源字段：https://www.erdosproblems.com/1010

14303 | EP-1011

### Erdős Problem #1011

P / partially_solved | Erdős 问题

Let $f_r(n)$ be minimal such that every graph on $n$ vertices with $\geq f_r(n)$ edges and chromatic number $\geq r$ contains a triangle. Determine $f_r(n)$.

来源字段：https://www.erdosproblems.com/1011

14304 | EP-1012

### Erdős Problem #1012: Graph Theory

S / solved | Erdős 问题

Let $k\geq 0$. Let $f(k)$ be such that every graph on $n\geq f(k)$ vertices with at least $\binom{n-k-1}{2}+\binom{k+2}{2}+1$ edges contains a cycle on $n-k$ vertices. Determine or estimate $f(k)$.

来源字段：https://www.erdosproblems.com/1012

14305 | EP-1013

### Erdős Problem #1013

P / partially_solved | Erdős 问题

Let $h_3(k)$ be the minimal $n$ such that there exists a triangle-free graph on $n$ vertices with chromatic number $k$. Find an asymptotic for $h_3(k)$, and also prove\[\lim_{k\to \infty}\frac{h_3(k+1)}{h_3(k)}=1.\]

来源字段：https://www.erdosproblems.com/1013

14306 | EP-1014

### Erdős Problem #1014

S / solved | Erdős 问题

Let $R(k,l)$ be the Ramsey number, so the minimal $n$ such that every graph on at least $n$ vertices contains either a $K_k$ or an independent set on $l$ vertices.

Prove, for fixed $k\geq 3$, that\[\lim_{l\to \infty}\frac{R(k,l+1)}{R(k,l)}=1.\]

来源字段：https://www.erdosproblems.com/1014

14307 | EP-1015

### Erdős Problem #1015: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

Let $f(t)$ be minimal such that, in any two-colouring of the edges of $K_n$, the edges can be partitioned into vertex disjoint monochromatic copies of $K_t$ (not necessarily the same colour) with at most $f(t)$ vertices remaining.

Estimate $f(t)$. In particular, is it true that $f(t)^{1/t}\to 1$? Is it true that $f(t)\ll t$?

来源字段：https://www.erdosproblems.com/1015

14308 | EP-1016

### Erdős Problem #1016

P / partially_solved | Erdős 问题

Let $h(n)$ be minimal such that there is a graph on $n$ vertices with $n+h(n)$ edges which contains a cycle on $k$ vertices, for all $3\leq k\leq n$. Estimate $h(n)$. In particular, is it true that\[h(n) \geq \log_2n+\log_*n-O(1),\]where $\log_*n$ is the iterated logarithmic function?

来源字段：https://www.erdosproblems.com/1016

14309 | EP-1018

### Erdős Problem #1018: Graph Theory, Planar Graphs

S / solved | Erdős 问题

Let $\epsilon>0$. Is there a constant $C_\epsilon$ such that, for all large $n$, every graph on $n$ vertices with at least $n^{1+\epsilon}$ edges must contain a subgraph on at most $C_\epsilon$ vertices which is non-planar?

来源字段：https://www.erdosproblems.com/1018

14310 | EP-1019

### Erdős Problem #1019: Graph Theory, Planar Graphs

S / solved | Erdős 问题

A planar graph on $n$ vertices with $3n-6$ edges (the maximum possible) is called saturated. Does every graph on $n$ vertices with $\lfloor n^2/4\rfloor+\lfloor \frac{n+1}{2}\rfloor$ edges contain a saturated planar graph with $>3$ vertices?

来源字段 : https://www.erdosproblems.com/1019

14311 | EP-1020

### Erdős Problem #1020:

O / open | Erdős 问题

Let $f(n;r,k)$ be the maximal number of edges in an $r$-uniform hypergraph which contains no set of $k$ many independent edges.

For all $r\geq 3$,\[f(n;r,k)=\max\left(\binom{rk-1}{r}, \binom{n}{r}-\binom{n-k+1}{r}\right).\]

来源字段 : https://www.erdosproblems.com/1020

14312 | EP-1021

### Erdős Problem #1021

S / solved | Erdős 问题

Is it true that, for every $k\geq 3$, there is a constant $c_k>0$ such that\[\mathrm{ex}(n,G_k) \ll n^{3/2-c_k},\]where $G_k$ is the bipartite graph between $\{y_1,\ldots,y_k\}$ and $\{z_1,\ldots,z_{\binom{k}{2}}\}$, with each $z_j$ joined to a unique pair of $y_i$?

来源字段 : https://www.erdosproblems.com/1021

14313 | EP-1022

### Erdős Problem #1022

S / solved | Erdős 问题

Is there a constant $c_t$, where $c_t\to \infty$ as $t\to \infty$, such that if $\mathcal{F}$ is a finite family of finite sets, all of size at least $t$, and for every set $X$ there are $<c_t\lvert X\rvert$ many $A\in \mathcal{F}$ with $A\subseteq X$, then $\mathcal{F}$ has chromatic number $2$ (in other words, has property B)?

来源字段 : https://www.erdosproblems.com/1022

14314 | EP-1024

### Erdős Problem #1024: Estimate $f(n)$

S / solved | Erdős 问题

Let $f(n)$ be such that every $3$-uniform linear hypergraph on $n$ vertices contains an independent set on $f(n)$ vertices. Estimate $f(n)$.

来源字段 : https://www.erdosproblems.com/1024

14315 | EP-1029

### Erdős Problem #1029

P / partially_solved | Erdős 问题

If $R(k)$ is the Ramsey number for $K_k$, the minimal $n$ such that every $2$-colouring of the edges of $K_n$ contains a monochromatic copy of $K_k$, then\[\frac{R(k)}{k2^{k/2}}\to \infty.\]

来源字段 : https://www.erdosproblems.com/1029

14316 | EP-1031

### Erdős Problem #1031: Graph Theory

S / solved | Erdős 问题

If $G$ is a graph on $n$ vertices which contains no trivial (empty or complete) subgraph on $\geq 10\log n$ many vertices, then must $G$ contain an induced non-trivial regular subgraph on $\gg \log n$ many vertices?

来源字段 : https://www.erdosproblems.com/1031

14317 | EP-1032

### Erdős Problem #1032

O / open | Erdős 问题

We say that a graph is $4$-chromatic critical if it has chromatic number $4$, and removing any edge decreases the chromatic number to $3$.

Is there, for arbitrarily large $n$, a $4$-chromatic critical graph on $n$ vertices with minimum degree $\gg n$?

来源字段 : https://www.erdosproblems.com/1032

14318 | EP-1033

### Erdős Problem #1033

P / partially_solved | Erdős 问题

Let $h(n)$ be such that every graph on $n$ vertices with $>n^2/4$ many edges contains a triangle whose vertices have degrees summing to at least $h(n)$. Estimate $h(n)$. In particular, is it true that\[h(n)\geq (2(\sqrt{3}-1)-o(1))n?\]

来源字段 : https://www.erdosproblems.com/1033

14319 | EP-1034

### Erdős Problem #1034: Graph Theory

S / solved | Erdős 问题

Let $G$ be a graph on $n$ vertices with $>n^2/4$ many edges. Must there be a triangle $T$ in $G$ and vertices $y_1,\ldots,y_t$, where $t>(\frac{1}{2}-o(1))n$, such that every vertex is joined to at least two vertices of $T$?

来源字段 : https://www.erdosproblems.com/1034

14320 | EP-1035

### Erdős Problem #1035

O / open | Erdős 问题

Is there a constant $c>0$ such that every graph on $2^n$ vertices with minimum degree $>(1-c)2^n$ contains the $n$-dimensional hypercube $Q_n$?

来源字段 : https://www.erdosproblems.com/1035

14321 | EP-1036

### Erdős Problem #1036: Graph Theory

S / solved | Erdős 问题

Let $G$ be a graph on $n$ vertices which does not contain a trivial (empty or complete) graph on more than $c\log n$ vertices. Must $G$ contain at least $2^{\Omega_c(n)}$ many induced subgraphs which are not pairwise isomorphic?

来源字段：https://www.erdosproblems.com/1036

14322 | EP-1037

### Erdős Problem #1037: Graph Theory

S / solved | Erdős 问题

Let $G$ be a graph on $n$ vertices in which every degree occurs at most twice, and the number of distinct degree is $>(\frac{1}{2}+\epsilon)n$. Must $G$ contain a trivial (empty or complete) subgraph of size 'much larger' than $\log n$?

来源字段：https://www.erdosproblems.com/1037

14323 | EP-104

### Erdős Problem #104

O / open | Erdős 问题

Given $n$ points in $\mathbb{R}^2$ the number of distinct unit circles containing at least three points is $o(n^2)$.

来源字段：https://www.erdosproblems.com/104

14324 | EP-1066

### Erdős Problem #1066

P / partially_solved | Erdős 问题

Let $G$ be a graph given by $n$ points in $\mathbb{R}^2$, where any two distinct points are at least distance $1$ apart, and we draw an edge between two points if they are distance $1$ apart.

Let $g(n)$ be maximal such that any such graph always has an independent set on at least $g(n)$ vertices. Estimate $g(n)$, or perhaps $\lim \frac{g(n)}{n}$.

来源字段：https://www.erdosproblems.com/1066

14325 | EP-1067

### Erdős Problem #1067: Graph Theory, Set Theory

S / solved | Erdős 问题

Does every graph with chromatic number $\aleph_1$ contain an infinitely connected subgraph with chromatic number $\aleph_1$?

来源字段：https://www.erdosproblems.com/1067

14326 | EP-1068

### Erdős Problem #1068

P / partially_solved | Erdős 问题

Does every graph with chromatic number $\aleph_1$ contain a countable subgraph which is infinitely vertex-connected?

来源字段：https://www.erdosproblems.com/1068

14327 | EP-1070

### Erdős Problem #1070

P / partially_solved | Erdős 问题

Let $f(n)$ be maximal such that, given any $n$ points in $\mathbb{R}^2$, there exist $f(n)$ points such that no two are distance $1$ apart. Estimate $f(n)$. In particular, is it true that $f(n)\geq n/4$?

来源字段：https://www.erdosproblems.com/1070

14328 | EP-1075

### Erdős Problem #1075

P / partially_solved | Erdős 问题

Let $r\geq 3$. There exists $c_r>r^{-r}$ such that, for any $\epsilon>0$, if $n$ is sufficiently large, the following holds.

Any $r$-uniform hypergraph on $n$ vertices with at least $(1+\epsilon)(n/r)^r$ many edges contains a subgraph on $m$ vertices with at least $c_rm^r$ edges, where $m=m(n)\to \infty$ as $n\to \infty$.

来源字段：https://www.erdosproblems.com/1075

14329 | EP-1077

### Erdős Problem #1077: Graph Theory

S / solved | Erdős 问题

We call a graph $D$-balanced (or $D$-almost-regular) if the maximum degree is at most $D$ times the minimum degree.

Let $\epsilon,\alpha>0$ and $D$ and $n$ be sufficiently large. If $G$ is a graph on $n$ vertices with at least $n^{1+\alpha}$ edges, then must $G$ contain a $D$-balanced subgraph on $m>n^{1-\alpha}$ vertices with at least $\epsilon m^{1+\alpha}$ edges?

来源字段：https://www.erdosproblems.com/1077

14330 | EP-1078

### Erdős Problem #1078: If $G$ has minimum degree $\geq (r-\frac{3}{2}-o(1))n$ then $G$ must contain a $K_r$

S / solved | Erdős 问题

Let $G$ be an $r$-partite graph with $n$ vertices in each part. If $G$ has minimum degree $\geq (r-\frac{3}{2}-o(1))n$ then $G$ must contain a $K_r$.

来源字段：https://www.erdosproblems.com/1078

14331 | EP-1079

### Erdős Problem #1079: Graph Theory, Turan Number

S / solved | Erdős 问题

Let $r\geq 4$. If $G$ is a graph on $n$ vertices with at least $\mathrm{ex}(n;K_r)$ edges then must $G$ contain a vertex with degree $d\gg_r n$ whose neighbourhood contains at least $\mathrm{ex}(d;K_{r-1})$ edges?

来源字段：https://www.erdosproblems.com/1079

14332 | EP-108

### Erdős Problem #108

O / open | Erdős 问题

For every $r\geq 4$ and $k\geq 2$ is there some finite $f(k,r)$ such that every graph of chromatic number $\geq f(k,r)$ contains a subgraph of girth $\geq r$ and chromatic number $\geq k$?

来源字段：https://www.erdosproblems.com/108

14333 | EP-1080

### Erdős Problem #1080: Graph Theory

S / solved | Erdős 问题

Let $G$ be a bipartite graph on $n$ vertices such that one part has $\lfloor n^{2/3}\rfloor$ vertices. Is there a constant $c>0$ such that if $G$ has at least $cn$ edges then $G$ must contain a $C_6$?

来源字段：https://www.erdosproblems.com/1080

14334 | EP-1085

### Erdős Problem #1085

P / partially_solved | Erdős 问题

Let $f_d(n)$ be minimal such that, in any set of $n$ points in $\mathbb{R}^d$, there exist at most $f_d(n)$ pairs of points which distance $1$ apart. Estimate $f_d(n)$.

来源字段：https://www.erdosproblems.com/1085

14335 | EP-1086

### Erdős Problem #1086

P / partially_solved | Erdős 问题

Let $g(n)$ be minimal such that any set of $n$ points in $\mathbb{R}^2$ contains the vertices of at most $g(n)$ many triangles with the same area. Estimate $g(n)$.

来源字段：https://www.erdosproblems.com/1086

14336 | EP-1089

### Erdős Problem #1089

S / solved | Erdős 问题

Let $g_d(n)$ be minimal such that every collection of $g_d(n)$ points in $\mathbb{R}^d$ determines at least $n$ many distinct distances. Estimate $g_d(n)$. In particular, does\[\lim_{d\to \infty}\frac{g_d(n)}{d^{n-1}}\]exist?

来源字段：https://www.erdosproblems.com/1089

14337 | EP-1091

### Erdős Problem #1091

S / solved | Erdős 问题

Let $G$ be a $K_4$-free graph with chromatic number $4$. Must $G$ contain an odd cycle with at least two diagonals?

More generally, is there some $f(r)\to \infty$ such that every graph with chromatic number $4$, in which every subgraph on $\leq r$ vertices has chromatic number $\leq 3$, contains an odd cycle with at least $f(r)$ diagonals?

来源字段：https://www.erdosproblems.com/1091

14338 | EP-1092

### Erdős Problem #1092

S / solved | Erdős 问题

Let $f_r(n)$ be maximal such that, if a graph $G$ has the property that every subgraph $H$ on $m$ vertices is the union of a graph with chromatic number $\leq r$ and a graph with $\leq f_r(m)$ edges, then $G$ has chromatic number $\leq r+1$.

Is it true that $f_2(n) \gg n$? More generally, is $f_r(n)\gg_r n$?

来源字段：https://www.erdosproblems.com/1092

14339 | EP-1104

### Erdős Problem #1104

P / partially_solved | Erdős 问题

Let $f(n)$ be the maximum possible chromatic number of a triangle-free graph on $n$ vertices. Estimate $f(n)$.

来源字段：https://www.erdosproblems.com/1104

14340 | EP-1105

### Erdős Problem #1105

S / solved | Erdős 问题

The anti-Ramsey number $\mathrm{AR}(n,G)$ is the maximum possible number of colours in which the edges of $K_n$ can be coloured without creating a rainbow copy of $G$ (i.e. one in which all edges have different colours).

Let $C_k$ be the cycle on $k$ vertices. Is it true that\[\mathrm{AR}(n,C_k)=\left(\frac{k-2}{2}+\frac{1}{k-1}\right)n+O(1)?\]Let $P_k$ be the path on $k$ vertices and $\ell=\lfloor\frac{k-1}{2}\rfloor$. If $n\geq k\geq 5$ then is $\mathrm{AR}(n,P_k)$ equal to\[\max\left(\binom{k-2}{2}+1, \binom{\ell-1}{2}+(\ell-1)(n-\ell+1)+\epsilon\right)\]where $\epsilon=1$ if $k$ is odd and $\epsilon=2$ otherwise?

来源字段：https://www.erdosproblems.com/1105

14341 | EP-110

### Erdős Problem #110: Graph Theory, Chromatic Number

S / solved | Erdős 问题

Is there some $F(n)$ such that every graph with chromatic number $\aleph_1$ has, for all large $n$, a subgraph with chromatic number $n$ on at most $F(n)$ vertices?

来源字段：https://www.erdosproblems.com/110

14342 | EP-111

### Erdős Problem #111

O / open | Erdős 问题

If $G$ is a graph let $h_G(n)$ be defined such that any subgraph of $G$ on $n$ vertices can be made bipartite after deleting at most $h_G(n)$ edges.

What is the behaviour of $h_G(n)$? Is it true that $h_G(n)/n\to \infty$ for every graph $G$ with chromatic number $\aleph_1$?

来源字段 : https://www.erdosproblems.com/111

14343 | EP-1111

### Erdős Problem #1111

P / partially_solved | Erdős 问题

If $G$ is a finite graph and $A,B$ are disjoint sets of vertices then we call $A,B$ anticomplete if there are no edges between $A$ and $B$.

If $t,c\geq 1$ then there exists $d\geq 1$ such that if $\chi(G)\geq d$ and $\omega(G)<t$ then there are anticomplete sets $A,B$ with $\chi(A)\geq \chi(B)\geq c$.

来源字段 : https://www.erdosproblems.com/1111

14344 | EP-1120

### Erdős Problem #1120

P / partially_solved | Erdős 问题

Let $f\in \mathbb{C}[z]$ be a monic polynomial of degree $n$, all of whose roots satisfy $\lvert z\rvert\leq 1$. Let\[E=\{ z: \lvert f(z)\rvert \leq 1\}.\]What is the shortest length of a path in $E$ joining $z=0$ to $\lvert z\rvert =1$?

来源字段 : https://www.erdosproblems.com/1120

14345 | EP-1133

### Erdős Problem #1133

P / partially_solved | Erdős 问题

Let $C>0$. There exists $\epsilon>0$ such that if $n$ is sufficiently large the following holds.

For any $x_1,\ldots,x_n\in [-1,1]$ there exist $y_1,\ldots,y_n\in [-1,1]$ such that, if $P$ is a polynomial of degree $m<(1+\epsilon)n$ with $P(x_i)=y_i$ for at least $(1-\epsilon)n$ many $1\leq i\leq n$, then\[\max_{x\in [-1,1]}\lvert P(x)\rvert >C.\]

来源字段 : https://www.erdosproblems.com/1133

14346 | EP-113

### Erdős Problem #113: Graph Theory, Turan Number

S / solved | Erdős 问题

If $G$ is bipartite then $\mathrm{ex}(n;G)\ll n^{3/2}$ if and only $G$ is $2$-degenerate, that is, $G$ contains no induced subgraph with minimal degree at least 3.

来源字段 : https://www.erdosproblems.com/113

14347 | EP-114

### Erdős Problem #114

P / partially_solved | Erdős 问题

If $p(z)\in\mathbb{C}[z]$ is a monic polynomial of degree $n$ then is the length of the curve $\{ z\in \mathbb{C}: \lvert p(z)\rvert=1\}$ maximised when $p(z)=z^n-1$?

来源字段 : https://www.erdosproblems.com/114

14348 | EP-1155

### Erdős Problem #1155: Graph Theory

O / open | Erdős 问题

Construct a random graph on $n$ vertices in the following way: begin with the complete graph $K_n$. At each stage, choose uniformly a random triangle in the graph and delete all the edges of this triangle. Repeat until the graph is triangle-free.

Describe the typical parameters and structure of such a graph. In particular, if $f(n)$ is the number of edges remaining, then is it true that\[\mathbb{E}f(n)\asymp n^{3/2}\]and that $f(n) \ll n^{3/2}$ almost surely?

来源字段 : https://www.erdosproblems.com/1155

14349 | EP-1156

### Erdős Problem #1156: in which every edge is included independently with probability $1/2$

O / open | Erdős 问题

Let $G$ be a random graph on $n$ vertices, in which every edge is included independently with probability $1/2$.

Is there some constant $C$ such that that chromatic number $\chi(G)$ is, almost surely, concentrated on at most $C$ values?

Is it true that, if $\omega(n)\to \infty$ sufficiently slowly, then for every function $f(n)$\[\mathbb{P}(\lvert\chi(G)-f(n)\rvert<\omega(n))<1/2\]if $n$ is sufficiently large?

来源字段 : https://www.erdosproblems.com/1156

14350 | EP-1178

### Erdős Problem #1178: Graph Theory, Hypergraphs

O / open | Erdős 问题

For $r\geq 3$ let $d_r(e)$ be the minimal $d$ such that\[\mathrm{ex}_r(n,\mathcal{F})=o(n^2),\]where $\mathcal{F}$ is the family of $r$-uniform hypergraphs on $d$ vertices with $e$ edges.

Prove that\[d_r(e)=(r-2)e+3\]for all $r,e\geq 3$.

来源字段 : https://www.erdosproblems.com/1178

14351 | EP-1182

### Erdős Problem #1182: Graph Theory, Ramsey Theory

O / open | Erdős 问题

Let $f(n)$ be maximal such that there is a connected graph $G$ with $n$ vertices and $f(n)$ edges such that\[R(K_3,G)= 2n-1.\]Let $F(n)$ be maximal such that every connected graph $G$ with $n$ vertices and $\leq F(n)$ edges has\[R(K_3,G)= 2n-1.\]Estimate $f(n)$ and $F(n)$. In particular, is it true that $F(n)/n\to \infty$?

来源字段：https://www.erdosproblems.com/1182

14352 | EP-1216

### Erdős Problem #1216: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

A tournament is a complete directed graph. Let $f(n)$ be such that every tournament on $n$ vertices contains a transitive tournament on $f(n)$ vertices (i.e. one such that if $i\to j\to k$ then $i\to k$).

Is it true that $f(n)=\lfloor \log_2 n\rfloor +1$?

来源字段：https://www.erdosproblems.com/1216

14353 | EP-127

### Erdős Problem #127: Is there an infinite sequence of $m_i$ such that $f(m_i)\to \infty$?

S / solved | Erdős 问题

Let $f(m)$ be maximal such that every graph with $m$ edges must contain a bipartite graph with\[\geq \frac{m}{2}+\frac{\sqrt{8m+1}-1}{8}+f(m)\]edges. Is there an infinite sequence of $m_i$ such that $f(m_i)\to \infty$?

来源字段：https://www.erdosproblems.com/127

14354 | EP-128

### Erdős Problem #128: Must $G$ contain a triangle?

O / open | Erdős 问题

Let $G$ be a graph with $n$ vertices such that every induced subgraph on $\geq \lfloor n/2\rfloor$ vertices has more than $n^2/50$ edges. Must $G$ contain a triangle?

来源字段：https://www.erdosproblems.com/128

14355 | EP-129

### Erdős Problem #129

O / open | Erdős 问题

Let $R(n;k,r)$ be the smallest $N$ such that if the edges of $K_N$ are $r$-coloured then there is a set of $n$ vertices which does not contain a copy of $K_k$ in at least one of the $r$ colours. Prove that there is a constant $C=C(r)>1$ such that\[R(n;3,r) < C^{\sqrt{n}}.\]

来源字段：https://www.erdosproblems.com/129

14356 | EP-132

### Erdős Problem #132

O / open | Erdős 问题

Let $A\subset \mathbb{R}^2$ be a set of $n$ points. Must there be two distances which occur at least once but between at most $n$ pairs of points? Must the number of such distances $\to \infty$ as $n\to \infty$?

来源字段：https://www.erdosproblems.com/132

14357 | EP-133

### Erdős Problem #133:

S / solved | Erdős 问题

Let $f(n)$ be minimal such that every triangle-free graph $G$ with $n$ vertices and diameter $2$ contains a vertex with degree $\geq f(n)$.

What is the order of growth of $f(n)$? Does $f(n)/\sqrt{n}\to \infty$?

来源字段：https://www.erdosproblems.com/133

14358 | EP-134

### Erdős Problem #134: Graph Theory

S / solved | Erdős 问题

Let $\epsilon,\delta>0$ and $n$ be sufficiently large in terms of $\epsilon$ and $\delta$. Let $G$ be a triangle-free graph on $n$ vertices with maximum degree $<n^{1/2-\epsilon}$.

Can $G$ be made into a triangle-free graph with diameter $2$ by adding at most $\delta n^2$ edges?

来源字段：https://www.erdosproblems.com/134

14359 | EP-136

### Erdős Problem #136: Determine the size of $f(n)$

S / solved | Erdős 问题

Let $f(n)$ be the smallest number of colours required to colour the edges of $K_n$ such that every $K_4$ contains at least 5 colours. Determine the size of $f(n)$.

来源字段：https://www.erdosproblems.com/136

14360 | EP-147

### Erdős Problem #147: Graph Theory, Turan Number

S / solved | Erdős 问题

If $H$ is bipartite with minimum degree $r$ then there exists $\epsilon=\epsilon(H)>0$ such that\[\mathrm{ex}(n;H) \gg n^{2-\frac{1}{r-1}+\epsilon}.\]

来源字段：https://www.erdosproblems.com/147

14361 | EP-149

### Erdős Problem #149

O / open | Erdős 问题

The strong chromatic index of a graph $G$, denoted by $\mathrm{sq}(G)$, is the minimum $k$ such that the edges of $G$ can be partitioned into $k$ sets of 'strongly independent' edges, that is, such that the subgraph of $G$ induced by each set is the union of vertex-disjoint edges.

Is it true that, for any graph $G$ with maximum degree $\Delta$,\[\mathrm{sq}(G)\leq\frac{5}{4}\Delta^2?\]

来源字段 : https://www.erdosproblems.com/149

14362 | EP-150

### Erdős Problem #150: Graph Theory

S / solved | Erdős 问题

A minimal cut of a graph is a minimal set of vertices whose removal disconnects the graph. Let $c(n)$ be the maximum number of minimal cuts a graph on $n$ vertices can have.

Does $c(n)^{1/n}\to \alpha$ for some $\alpha <2$?

来源字段 : https://www.erdosproblems.com/150

14363 | EP-151

### Erdős Problem #151

O / open | Erdős 问题

For a graph $G$ let $\tau(G)$ denote the minimal number of vertices that include at least one from each maximal clique of $G$ on at least two vertices (sometimes called the clique transversal number).

Let $H(n)$ be maximal such that every triangle-free graph on $n$ vertices contains an independent set on $H(n)$ vertices.

If $G$ is a graph on $n$ vertices then is\[\tau(G)\leq n-H(n)?\]

来源字段 : https://www.erdosproblems.com/151

14364 | EP-159

### Erdős Problem #159

O / open | Erdős 问题

There exists some constant $c>0$ such that

$$R(C_4,K_n) \ll n^{2-c}.$$

来源字段 : https://www.erdosproblems.com/159

14365 | EP-161

### Erdős Problem #161

P / partially_solved | Erdős 问题

Let $\alpha\in[0,1/2)$ and $n,t\geq 1$. Let $F^{(t)}(n,\alpha)$ be the smallest $m$ such that we can $2$-colour the edges of the complete $t$-uniform hypergraph on $n$ vertices such that if $X\subseteq [n]$ with $\lvert X\rvert \geq m$ then there are at least $\alpha \binom{\lvert X\rvert}{t}$ many $t$-subsets of $X$ of each colour.

For fixed $n,t$ as we change $\alpha$ from $0$ to $1/2$ does $F^{(t)}(n,\alpha)$ increase continuously or are there jumps? Only one jump?

来源字段 : https://www.erdosproblems.com/161

14366 | EP-162

### Erdős Problem #162

P / partially_solved | Erdős 问题

Let $\alpha>0$ and $n\geq 1$. Let $F(n,\alpha)$ be the largest $k$ such that there exists some 2-colouring of the edges of $K_n$ in which any induced subgraph $H$ on at least $k$ vertices contains more than $\alpha\binom{\lvert H\rvert}{2}$ many edges of each colour.

Prove that for every fixed $0\leq \alpha \leq 1/2$, as $n\to\infty$,\[F(n,\alpha)\sim c_\alpha \log n\]for some constant $c_\alpha$.

来源字段 : https://www.erdosproblems.com/162

14367 | EP-163

### Erdős Problem #163: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

For any $d\geq 1$ if $H$ is a graph such that every subgraph contains a vertex of degree at most $d$ then $R(H)\ll_d n$.

来源字段 : https://www.erdosproblems.com/163

14368 | EP-165

### Erdős Problem #165

O / open | Erdős 问题

Give an asymptotic formula for $R(3,k)$.

来源字段 : https://www.erdosproblems.com/165

14369 | EP-166

### Erdős Problem #166: Prove that\[R(4,k) \gg \frac{k^3}{(\log k)^{O(1)}}.\]

S / solved | Erdős 问题

Prove that\[R(4,k) \gg \frac{k^3}{(\log k)^{O(1)}}.\]

来源字段 : https://www.erdosproblems.com/166

14370 | EP-167

### Erdős Problem #167: Graph Theory

O / open | Erdős 问题

If $G$ is a graph with at most $k$ edge disjoint triangles then can $G$ be made triangle-free after removing at most $2k$ edges?

来源字段：https://www.erdosproblems.com/167

14371 | EP-173

### Erdős Problem #173

O / open | Erdős 问题

In any $2$-colouring of $\mathbb{R}^2$, for all but at most one triangle $T$, there is a monochromatic congruent copy of $T$.

来源字段：https://www.erdosproblems.com/173

14372 | EP-180

### Erdős Problem #180

S / solved | Erdős 问题

If $\mathcal{F}$ is a finite set of finite graphs then $\mathrm{ex}(n;\mathcal{F})$ is the maximum number of edges a graph on $n$ vertices can have without containing any subgraphs from $\mathcal{F}$. Note that it is trivial that $\mathrm{ex}(n;\mathcal{F})\leq \mathrm{ex}(n;G)$ for every $G\in\mathcal{F}$.

Is it true that, for every $\mathcal{F}$, there exists $G\in\mathcal{F}$ such that\[\mathrm{ex}(n;G)\ll_{\mathcal{F}}\mathrm{ex}(n;\mathcal{F})?\]

来源字段：https://www.erdosproblems.com/180

14373 | EP-181

### Erdős Problem #181

O / open | Erdős 问题

Let $Q_n$ be the $n$-dimensional hypercube graph (so that $Q_n$ has $2^n$ vertices and $n2^{n-1}$ edges). Prove that\[R(Q_n) \ll 2^n.\]

来源字段：https://www.erdosproblems.com/181

14374 | EP-182

### Erdős Problem #182: Graph Theory

S / solved | Erdős 问题

Let $k\geq 3$. What is the maximum number of edges that a graph on $n$ vertices can contain if it does not have a $k$-regular subgraph? Is it $\ll n^{1+o(1)}$?

来源字段：https://www.erdosproblems.com/182

14375 | EP-184

### Erdős Problem #184

O / open | Erdős 问题

Any graph on $n$ vertices can be decomposed into $O(n)$ many edge-disjoint cycles and edges.

来源字段：https://www.erdosproblems.com/184

14376 | EP-190

### Erdős Problem #190

S / solved | Erdős 问题

Let $H(k)$ be the smallest $N$ such that in any finite colouring of $\{1,\ldots,N\}$ (into any number of colours) there is always either a monochromatic $k$-term arithmetic progression or a rainbow arithmetic progression (i.e. all elements are different colours). Estimate $H(k)$. Is it true that\[H(k)^{1/k}/k \to \infty\]as $k\to\infty$?

来源字段：https://www.erdosproblems.com/190

14377 | EP-19

### Erdős Problem #19: If $G$ is an edge-disjoint union of $n$ copies of $K_n$ then is $\chi(G)=n$?

O / open | Erdős 问题

If $G$ is an edge-disjoint union of $n$ copies of $K_n$ then is $\chi(G)=n$?

来源字段：https://www.erdosproblems.com/19

14378 | EP-22

### Erdős Problem #22: Graph Theory

S / solved | Erdős 问题

Let $\epsilon>0$ and $n$ be sufficiently large depending on $\epsilon$. Is there a graph on $n$ vertices with $\geq n^2/8$ many edges which contains no $K_4$ such that the largest independent set has size at most $\epsilon n$?

来源字段：https://www.erdosproblems.com/22

14379 | EP-23

### Erdős Problem #23: Graph Theory

O / open | Erdős 问题

Can every triangle-free graph on $5n$ vertices be made bipartite by deleting at most $n^2$ edges?

来源字段：https://www.erdosproblems.com/23

14380 | EP-24

### Erdős Problem #24: Graph Theory

S / solved | Erdős 问题

Does every triangle-free graph on $5n$ vertices contain at most $n^5$ copies of $C_5$?

来源字段：https://www.erdosproblems.com/24

14381 | EP-289

### Erdős Problem #289

O / open | Erdős 问题

Is it true that, for all sufficiently large $k$, there exist finite intervals $I_1,\ldots,I_k\subset \mathbb{N}$, distinct, not overlapping or adjacent, with $\lvert I_i\rvert \geq 2$ for $1\leq i\leq k$ such that\[1=\sum_{i=1}^k \sum_{n\in I_i}\frac{1}{n}?\]

来源字段：https://www.erdosproblems.com/289

14382 | EP-311

### Erdős Problem #311

O / open | Erdős 问题

Let $\delta(N)$ be the minimal non-zero value of $\lvert 1-\sum_{n\in A}\frac{1}{n}\rvert$ as $A$ ranges over all subsets of $\{1,\ldots,N\}$. Is it true that\[\delta(N)=e^{-(c+o(1))N}\]for some constant $c\in (0,1)$?

来源字段：https://www.erdosproblems.com/311

14383 | EP-318

### Erdős Problem #318

S / solved | Erdős 问题

Let $A\subseteq \mathbb{N}$ be an infinite arithmetic progression and $f:A\to \{-1,1\}$ be a non-constant function. Must there exist a finite non-empty $S\subset A$ such that\[\sum_{n\in S}\frac{f(n)}{n}=0?\]What about if $A$ is an arbitrary set of positive density? What if $A$ is the set of squares excluding $1$?

来源字段：https://www.erdosproblems.com/318

14384 | EP-352

### Erdős Problem #352

O / open | Erdős 问题

Is there some $c>0$ such that every measurable $A\subseteq \mathbb{R}^2$ of measure $\geq c$ contains the vertices of a triangle of area 1?

来源字段：https://www.erdosproblems.com/352

14385 | EP-426

### Erdős Problem #426: Graph Theory

S / solved | Erdős 问题

We say $H$ is a unique subgraph of $G$ if there is exactly one way to find $H$ as a subgraph (not necessarily induced) of $G$. Is there a graph on $n$ vertices with\[\gg \frac{2^{\binom{n}{2}}}{n!}\]many distinct unique subgraphs?

来源字段：https://www.erdosproblems.com/426

14386 | EP-477

### Erdős Problem #477

S / solved | Erdős 问题

Is there a polynomial $f:\mathbb{Z}\to \mathbb{Z}$ of degree at least $2$ and a set $A\subset \mathbb{Z}$ such that for any $n\in \mathbb{Z}$ there is exactly one $a\in A$ and $b\in \{ f(k): k\in\mathbb{Z}\}$ such that $n=a+b$?

来源字段：https://www.erdosproblems.com/477

14387 | EP-500

### Erdős Problem #500

P / partially_solved | Erdős 问题

What is $\mathrm{ex}_3(n,K_4^3)$? That is, the largest number of $3$-edges which can placed on $n$ vertices so that there exists no $K_4^3$, a set of 4 vertices which is covered by all 4 possible $3$-edges.

来源字段：https://www.erdosproblems.com/500

14388 | EP-508

### Erdős Problem #508

P / partially_solved | Erdős 问题

别名：The Hadwiger-Nelson Problem [COMB-012]; Hadwiger-Nelson Problem [GRAPH-060]

What is the chromatic number of the plane? That is, what is the smallest number of colours required to colour $\mathbb{R}^2$ such that no two points of the same colour are distance $1$ apart?

来源字段：https://www.erdosproblems.com/508

14389 | EP-514

### Erdős Problem #514

P / partially_solved | Erdős 问题

Let $f(z)$ be an entire transcendental function. Does there exist a path $L$ so that, for every $n$,\[\lvert f(z)/z^n\rvert \to \infty\]as $z\to \infty$ along $L$?

Can the length of this path be estimated in terms of $M(r)=\max_{\lvert z\rvert=r}\lvert f(z)\rvert$? Does there exist a path along which $\lvert f(z)\rvert$ tends to $\infty$ faster than a fixed function of $M(r)$ (such that $M(r)^\epsilon$)?

来源字段：https://www.erdosproblems.com/514

14390 | EP-518

### Erdős Problem #518: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

Is it true that, in any two-colouring of the edges of $K_n$, there exist $\sqrt{n}$ monochromatic paths, all of the same colour, which cover all vertices?

来源字段：https://www.erdosproblems.com/518

14391 | EP-521

### Erdős Problem #521

P / partially_solved | Erdős 问题

Let $(\epsilon_k)_{k\geq 0}$ be independently uniformly chosen at random from $\{-1,1\}$. If $R_n$ counts the number of real roots of $f_n(z)=\sum_{0\leq k\leq n}\epsilon_k z^k$ then is it true that, almost surely,\[\lim_{n\to \infty}\frac{R_n}{\log n}=\frac{2}{\pi}?\]

来源字段：https://www.erdosproblems.com/521

14392 | EP-522

### Erdős Problem #522

P / partially_solved | Erdős 问题

Let $f(z)=\sum_{0\leq k\leq n} \epsilon_k z^k$ be a random polynomial, where $\epsilon_k\in \{-1,1\}$ independently uniformly at random for $0\leq k\leq n$.

Is it true that, if $R_n$ is the number of roots of $f(z)$ in $\{ z\in \mathbb{C}: \lvert z\rvert \leq 1\}$, then\[\frac{R_n}{n/2}\to 1\]almost surely?

来源字段：https://www.erdosproblems.com/522

14393 | EP-529

### Erdős Problem #529

O / open | Erdős 问题

Let $d_k(n)$ be the expected distance from the origin after taking $n$ random steps from the origin in $\mathbb{Z}^k$ (conditional on no self intersections) - that is, a self-avoiding walk. Is it true that\[\lim_{n\to \infty}\frac{d_2(n)}{n^{1/2}}= \infty?\]Is it true that\[d_k(n)\ll n^{1/2}\]for $k\geq 3$?

来源字段：https://www.erdosproblems.com/529

14394 | EP-531

### Erdős Problem #531

O / open | Erdős 问题

Let $F(k)$ be the minimal $N$ such that if we two-colour $\{1,\ldots,N\}$ there is a set $A$ of size $k$ such that all subset sums $\sum_{a\in S}a$ (for $\emptyset\neq S\subseteq A$) are monochromatic. Estimate $F(k)$.

来源字段：https://www.erdosproblems.com/531

14395 | EP-533

### Erdős Problem #533

S / solved | Erdős 问题

Let $\delta>0$. If $n$ is sufficiently large and $G$ is a graph on $n$ vertices with no $K_5$ and at least $\delta n^2$ edges then $G$ contains a set of $\gg_\delta n$ vertices containing no triangle.

来源字段：https://www.erdosproblems.com/533

14396 | EP-544

### Erdős Problem #544

O / open | Erdős 问题

Show that\[R(3,k+1)-R(3,k)\to\infty\]as $k\to \infty$. Similarly, prove or disprove that\[R(3,k+1)-R(3,k)=o(k).\]

来源字段：https://www.erdosproblems.com/544

14397 | EP-545

### Erdős Problem #545

O / open | Erdős 问题

Let $G$ be a graph with $m$ edges and no isolated vertices. Is the Ramsey number $R(G)$ maximised when $G$ is 'as complete as possible'? That is, if $m=\binom{n}{2}+t$ edges with $0\leq t<n$ then is\[R(G)\leq R(H),\]where $H$ is the graph formed by connecting a new vertex to $t$ of the vertices of $K_n$?

来源字段：https://www.erdosproblems.com/545

14398 | EP-546

### Erdős Problem #546: Is it true that\[R(G) \leq 2^{O(m^{1/2})}?\]

S / solved | Erdős 问题

Let $G$ be a graph with no isolated vertices and $m$ edges. Is it true that\[R(G) \leq 2^{O(m^{1/2})}?\]

来源字段：https://www.erdosproblems.com/546

14399 | EP-547

### Erdős Problem #547: If $T$ is a tree on $n$ vertices then\[R(T) \leq 2n-2.\]

S / solved | Erdős 问题

If $T$ is a tree on $n$ vertices then\[R(T) \leq 2n-2.\]

来源字段：https://www.erdosproblems.com/547

14400 | EP-548

### Erdős Problem #548: Graph Theory

S / solved | Erdős 问题

Let $n\geq k+1$. Every graph on $n$ vertices with at least $\frac{k-1}{2}n+1$ edges contains every tree on $k+1$ vertices.

来源字段：https://www.erdosproblems.com/548

14401 | EP-549

### Erdős Problem #549: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

If $T$ is a tree which is a bipartite graph with $k$ vertices and $2k$ vertices in the other class then\[R(T)=4k-1.\]

来源字段：https://www.erdosproblems.com/549

14402 | EP-550

### Erdős Problem #550

P / partially_solved | Erdős 问题

Let $m_1\leq\cdots\leq m_k$ and $n$ be sufficiently large. If $T$ is a tree on $n$ vertices and $G$ is the complete multipartite graph with vertex class sizes $m_1,\ldots,m_k$ then prove that\[R(T,G)\leq (\chi(G)-1)(R(T,K_{m_1,m_2})-1)+m_1.\]

来源字段：https://www.erdosproblems.com/550

14403 | EP-551

### Erdős Problem #551: Prove that\[R(C_k,K_n)=(k-1)(n-1)+1\]for $k\geq n\geq 3$ (except when $n=k=3$)

O / open | Erdős 问题

Prove that\[R(C_k,K_n)=(k-1)(n-1)+1\]for $k\geq n\geq 3$ (except when $n=k=3$).

来源字段：https://www.erdosproblems.com/551

14404 | EP-553

### Erdős Problem #553: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

Let $R(3,3,n)$ denote the smallest integer $m$ such that if we $3$-colour the edges of $K_m$ then there is either a monochromatic triangle in one of the first two colours or a monochromatic $K_n$ in the third colour. Define $R(3,n)$ similarly but with two colours. Show that\[\frac{R(3,3,n)}{R(3,n)}\to \infty\]as $n\to \infty$.

来源字段：https://www.erdosproblems.com/553

14405 | EP-554

### Erdős Problem #554

P / partially_solved | Erdős 问题

Let $R_k(G)$ denote the minimal $m$ such that if the edges of $K_m$ are $k$-coloured then there is a monochromatic copy of $G$. Show that\[\lim_{k\to \infty}\frac{R_k(C_{2n+1})}{R_k(K_3)}=0\]for any $n\geq 2$.

来源字段：https://www.erdosproblems.com/554

14406 | EP-556

### Erdős Problem #556: Graph Theory, Ramsey Theory

O / open | Erdős 问题

Let $R_3(G)$ denote the minimal $m$ such that if the edges of $K_m$ are $3$-coloured then there must be a monochromatic copy of $G$. Show that\[R_3(C_n) \leq 4n-3.\]

来源字段：https://www.erdosproblems.com/556

14407 | EP-557

### Erdős Problem #557

S / solved | Erdős 问题

Let $R_k(G)$ denote the minimal $m$ such that if the edges of $K_m$ are $k$-coloured then there is a monochromatic copy of $G$. Is it true that\[R_k(T)\leq kn+O(1)\]for any tree $T$ on $n$ vertices?

来源字段：https://www.erdosproblems.com/557

14408 | EP-558

### Erdős Problem #558

P / partially_solved | Erdős 问题

Let $R_k(G)$ denote the minimal $m$ such that if the edges of $K_m$ are $k$-coloured then there is a monochromatic copy of $G$. Determine\[R_k(K_{s,t})\]where $K_{s,t}$ is the complete bipartite graph with $s$ vertices in one component and $t$ in the other.

来源字段：https://www.erdosproblems.com/558

14409 | EP-559

### Erdős Problem #559: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

Let $\hat{R}(G)$ denote the size Ramsey number, the minimal number of edges $m$ such that there is a graph $H$ with $m$ edges that is Ramsey for $G$.

If $G$ has $n$ vertices and maximum degree $d$ then prove that\[\hat{R}(G)\ll_d n.\]

来源字段：https://www.erdosproblems.com/559

14410 | EP-560

### Erdős Problem #560

P / partially_solved | Erdős 问题

Let $\hat{R}(G)$ denote the size Ramsey number, the minimal number of edges $m$ such that there is a graph $H$ with $m$ edges such that in any $2$-colouring of the edges of $H$ there is a monochromatic copy of $G$.

Determine\[\hat{R}(K_{n,n}),\]where $K_{n,n}$ is the complete bipartite graph with $n$ vertices in each component.

来源字段：https://www.erdosproblems.com/560

14411 | EP-561

### Erdős Problem #561

P / partially_solved | Erdős 问题

Let $\hat{R}(G)$ denote the size Ramsey number, the minimal number of edges $m$ such that there is a graph $H$ with $m$ edges such that in any $2$-colouring of the edges of $H$ there is a monochromatic copy of $G$.

Let $F_1$ and $F_2$ be the union of stars. More precisely, let $F_1=\cup_{i\leq s} K_{1,n_i}$ and $F_2=\cup_{j\leq t} K_{1,m_j}$ with $n_1\geq \cdots \geq n_s\geq 1$ and $m_1\geq \cdots \geq m_t\geq 1$. Prove that\[\hat{R}(F_1,F_2) = \sum_{2\leq k\leq s+t}l_k\]where\[l_k=\max\{n_i+m_j-1: i+j=k\}.\]

来源字段：https://www.erdosproblems.com/561

14412 | EP-563

### Erdős Problem #563

P / partially_solved | Erdős 问题

Let $F(n,\alpha)$ denote the smallest $m$ such that there exists a $2$-colouring of the edges of $K_n$ so that every $X\subseteq [n]$ with $\lvert X\rvert\geq m$ contains more than $\alpha \binom{\lvert X\rvert}{2}$ many edges of each colour.

Prove that, for every $0\leq \alpha< 1/2$,\[F(n,\alpha)\sim c_\alpha\log n\]for some constant $c_\alpha$ depending only on $\alpha$.

来源字段：https://www.erdosproblems.com/563

14413 | EP-565

### Erdős Problem #565:

S / solved | Erdős 问题

Let $R^*(G)$ be the induced Ramsey number: the minimal $m$ such that there is a graph $H$ on $m$ vertices such that any $2$-colouring of the edges of $H$ contains an induced monochromatic copy of $G$.

Is it true that\[R^*(G) \leq 2^{O(n)}\]for any graph $G$ on $n$ vertices?

来源字段：https://www.erdosproblems.com/565

14414 | EP-566

### Erdős Problem #566

O / open | Erdős 问题

Let $G$ be such that any subgraph on $k$ vertices has at most $2k-3$ edges. Is it true that, if $H$ has $m$ edges and no isolated vertices, then\[R(G,H)\ll m?\]

来源字段：https://www.erdosproblems.com/566

14415 | EP-567

### Erdős Problem #567

P / partially_solved | Erdős 问题

Let $G$ be either $Q_3$ or $K_{3,3}$ or $H_5$ (the last formed by adding two vertex-disjoint chords to $C_5$). Is it true that, if $H$ has $m$ edges and no isolated vertices, then\[R(G,H)\ll m?\]

来源字段：https://www.erdosproblems.com/567

14416 | EP-568

### Erdős Problem #568

O / open | Erdős 问题

Let $G$ be a graph such that $R(G,T_n)\ll n$ for any tree $T_n$ on $n$ vertices and $R(G,K_n)\ll n^2$. Is it true that, for any $H$ with $m$ edges and no isolated vertices,\[R(G,H)\ll m?\]

来源字段：https://www.erdosproblems.com/568

14417 | EP-569

### Erdős Problem #569

P / partially_solved | Erdős 问题

Let $k\geq 1$. What is the best possible $c_k$ such that\[R(C_{2k+1},H)\leq c_k m\]for any graph $H$ on $m$ edges without isolated vertices?

来源字段：https://www.erdosproblems.com/569

14418 | EP-570

### Erdős Problem #570: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

Let $k\geq 3$. Is it true that, if $m$ is sufficiently large, for any graph $H$ on $m$ edges without isolated vertices,\[R(C_k,H) \leq 2m+\left\lfloor\frac{k-1}{2}\right\rfloor?\]

来源字段：https://www.erdosproblems.com/570

14419 | EP-571

### Erdős Problem #571

S / solved | Erdős 问题

Show that for any rational $\alpha \in [1,2)$ there exists a bipartite graph $G$ such that\[\mathrm{ex}(n;G)\asymp n^{\alpha}.\]

来源字段：https://www.erdosproblems.com/571

14420 | EP-573

### Erdős Problem #573

P / partially_solved | Erdős 问题

Is it true that\[\mathrm{ex}(n;\{C_3,C_4\})\sim (n/2)^{3/2}?\]

来源字段：https://www.erdosproblems.com/573

14421 | EP-574

### Erdős Problem #574

S / solved | Erdős 问题

Is it true that, for $k\geq 2$,\[\mathrm{ex}(n;\{C_{2k-1},C_{2k}\})=(1+o(1))(n/2)^{1+\frac{1}{k}}.\]

来源字段：https://www.erdosproblems.com/574

14422 | EP-575

### Erdős Problem #575

S / solved | Erdős 问题

If $\mathcal{F}$ is a finite set of finite graphs then $\mathrm{ex}(n;\mathcal{F})$ is the maximum number of edges a graph on $n$ vertices can have without containing any subgraphs from $\mathcal{F}$. Note that it is trivial that $\mathrm{ex}(n;\mathcal{F})\leq \mathrm{ex}(n;G)$ for every $G\in\mathcal{F}$.

Is it true that, for every $\mathcal{F}$, if there is a bipartite graph in $\mathcal{F}$ then there exists some bipartite $G\in\mathcal{F}$ such that\[\mathrm{ex}(n;G)\ll_{\mathcal{F}}\mathrm{ex}(n;\mathcal{F})?\]

来源字段：https://www.erdosproblems.com/575

14423 | EP-576

### Erdős Problem #576

P / partially_solved | Erdős 问题

Let $Q_k$ be the $k$-dimensional hypercube graph (so that $Q_k$ has $2^k$ vertices and $k2^{k-1}$ edges). Determine the behaviour of\[\mathrm{ex}(n;Q_k).\]

来源字段：https://www.erdosproblems.com/576

14424 | EP-577

### Erdős Problem #577: Graph Theory

S / solved | Erdős 问题

If $G$ is a graph with $4k$ vertices and minimum degree at least $2k$ then $G$ contains $k$ vertex-disjoint $4$-cycles.

来源字段：https://www.erdosproblems.com/577

14425 | EP-578

### Erdős Problem #578: Graph Theory

S / solved | Erdős 问题

If $G$ is a random graph on $2^d$ vertices, including each edge with probability $1/2$, then $G$ almost surely contains a copy of $Q_d$ (the $d$-dimensional hypercube with $2^d$ vertices and $d2^{d-1}$ many edges).

来源字段：https://www.erdosproblems.com/578

14426 | EP-579

### Erdős Problem #579

P / partially_solved | Erdős 问题

Let $\delta>0$. If $n$ is sufficiently large and $G$ is a graph on $n$ vertices with no $K_{2,2,2}$ and at least $\delta n^2$ edges then $G$ contains an independent set of size $\gg_\delta n$.

来源字段：https://www.erdosproblems.com/579

14427 | EP-57

### Erdős Problem #57: Graph Theory, Chromatic Number

S / solved | Erdős 问题

If $G$ is a graph with infinite chromatic number and $a_1<a_2<\cdots $ are lengths of the odd cycles of $G$ then $\sum \frac{1}{a_i}=\infty$.

来源字段：https://www.erdosproblems.com/57

14428 | EP-580

### Erdős Problem #580: Must $G$ contain every tree on at most $n/2$ vertices?

O / open | Erdős 问题

Let $G$ be a graph on $n$ vertices such that at least $n/2$ vertices have degree at least $n/2$. Must $G$ contain every tree on at most $n/2$ vertices?

来源字段：https://www.erdosproblems.com/580

14429 | EP-581

### Erdős Problem #581: Determine $f(m)$

S / solved | Erdős 问题

Let $f(m)$ be the maximal $k$ such that a triangle-free graph on $m$ edges must contain a bipartite graph with $k$ edges. Determine $f(m)$.

来源字段：https://www.erdosproblems.com/581

14430 | EP-582

### Erdős Problem #582: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

Does there exist a graph $G$ which contains no $K_4$, and yet any $2$-colouring of the edges produces a monochromatic $K_3$?

来源字段：https://www.erdosproblems.com/582

14431 | EP-583

### Erdős Problem #583: Graph Theory

O / open | Erdős 问题

Every connected graph on $n$ vertices can be partitioned into at most $\lceil n/2\rceil$ edge-disjoint paths.

来源字段：https://www.erdosproblems.com/583

14432 | EP-584

### Erdős Problem #584

P / partially_solved | Erdős 问题

Let $G$ be a graph with $n$ vertices and $\delta n^{2}$ edges. Are there subgraphs $H_1,H_2\subseteq G$ such that

- $H_1$ has $\gg \delta^3n^2$ edges and every two edges in $H_1$ are contained in a cycle of length at most $6$, and furthermore if two edges share a vertex they are on a cycle of length $4$, and

- $H_2$ has $\gg \delta^2n^2$ edges and every two edges in $H_2$ are contained in a cycle of length at most $8$.

来源字段：https://www.erdosproblems.com/584

14433 | EP-585

### Erdős Problem #585

P / partially_solved | Erdős 问题

What is the maximum number of edges that a graph on $n$ vertices can have if it does not contain two edge-disjoint cycles with the same vertex set?

来源字段：https://www.erdosproblems.com/585

14434 | EP-589

### Erdős Problem #589

P / partially_solved | Erdős 问题

Let $g(n)$ be maximal such that in any set of $n$ points in $\mathbb{R}^2$ with no four points on a line there exists a subset on $g(n)$ points with no three points on a line. Estimate $g(n)$.

来源字段：https://www.erdosproblems.com/589

14435 | EP-58

### Erdős Problem #58: Graph Theory, Chromatic Number

S / solved | Erdős 问题

If $G$ is a graph which contains odd cycles of $\leq k$ different lengths then $\chi(G)\leq 2k+2$, with equality if and only if $G$ contains $K_{2k+2}$.

来源字段：https://www.erdosproblems.com/58

14436 | EP-593

### Erdős Problem #593

P / partially_solved | Erdős 问题

Characterize those finite 3-uniform hypergraphs which appear in every 3-uniform hypergraph of chromatic number $>\aleph_0$.

来源字段：https://www.erdosproblems.com/593

14437 | EP-594

### Erdős Problem #594: Graph Theory, Set Theory

S / solved | Erdős 问题

Does every graph $G$ with chromatic number $\geq \aleph_1$ contain all sufficiently large odd cycles?

来源字段：https://www.erdosproblems.com/594

14438 | EP-596

### Erdős Problem #596

P / partially_solved | Erdős 问题

For which graphs $G_1,G_2$ is it true that

- for every $n\geq 1$ there is a graph $H$ without a $G_1$ but if the edges of $H$ are $n$-coloured then there is a monochromatic copy of $G_2$, and yet

- for every graph $H$ without a $G_1$ there is an $\aleph_0$-colouring of the edges of $H$ without a monochromatic $G_2$.

来源字段：https://www.erdosproblems.com/596

14439 | EP-597

### Erdős Problem #597

P / partially_solved | Erdős 问题

Let $G$ be a graph on at most $\aleph_1$ vertices which contains no $K_4$ and no $K_{\aleph_0,\aleph_0}$ (the complete bipartite graph with $\aleph_0$ vertices in each class). Is it true that\[\omega_1^2 \to (\omega_1\omega, G)^2?\]What about finite $G$?

来源字段：https://www.erdosproblems.com/597

14440 | EP-599

### Erdős Problem #599: Graph Theory, Set Theory

S / solved | Erdős 问题

Let $G$ be a (possibly infinite) graph and $A,B$ be disjoint independent sets of vertices. Must there exist a family $P$ of disjoint paths between $A$ and $B$ and a set $S$ which contains exactly one vertex from each path in $P$, and such that every path between $A$ and $B$ contains at least one vertex from $S$?

来源字段：https://www.erdosproblems.com/599

14441 | EP-59

### Erdős Problem #59: Graph Theory, Turan Number

S / solved | Erdős 问题

Is it true that the number of graphs on $n$ vertices which do not contain $G$ is\[\leq 2^{(1+o(1))\mathrm{ex}(n;G)}?\]

来源字段：https://www.erdosproblems.com/59

14442 | EP-600

### Erdős Problem #600

P / partially_solved | Erdős 问题

Let $e(n,r)$ be minimal such that every graph on $n$ vertices with at least $e(n,r)$ edges, each edge contained in at least one triangle, must have an edge contained in at least $r$ triangles. Let $r\geq 2$. Is it true that\[e(n,r+1)-e(n,r)\to \infty\]as $n\to \infty$? Is it true that\[\frac{e(n,r+1)}{e(n,r)}\to 1\]as $n\to \infty$?

来源字段：https://www.erdosproblems.com/600

14443 | EP-602

### Erdős Problem #602

P / partially_solved | Erdős 问题

Let $(A_i)$ be a family of sets with $\lvert A_i\rvert=\aleph_0$ for all $i$, such that for any $i\neq j$ we have $\lvert A_i\cap A_j\rvert$ finite and $\neq 1$. Is there a $2$-colouring of $\cup A_i$ such that no $A_i$ is monochromatic?

来源字段：https://www.erdosproblems.com/602

14444 | EP-603

### Erdős Problem #603

S / solved | Erdős 问题

Let $(A_i)$ be a family of countably infinite sets such that $\lvert A_i\cap A_j\rvert \neq 2$ for all $i\neq j$. Find the smallest cardinal $C$ such that $\cup A_i$ can always be coloured with at most $C$ colours so that no $A_i$ is monochromatic.

来源字段：https://www.erdosproblems.com/603

14445 | EP-608

### Erdős Problem #608: Are there at least $\frac{2}{9}n^2$ edges of $G$ which are contained in a $C_5$?

S / solved | Erdős 问题

Let $G$ be a graph with $n$ vertices and $>n^2/4$ many edges. Are there at least $\frac{2}{9}n^2$ edges of $G$ which are contained in a $C_5$?

来源字段：https://www.erdosproblems.com/608

14446 | EP-609

### Erdős Problem #609

P / partially_solved | Erdős 问题

Let $f(n)$ be the minimal $m$ such that if the edges of $K_{2^n+1}$ are coloured with $n$ colours then there must be a monochromatic odd cycle of length at most $m$. Estimate $f(n)$.

来源字段：https://www.erdosproblems.com/609

14447 | EP-61

### Erdős Problem #61

O / open | Erdős 问题

For any graph $H$ is there some $c=c(H)>0$ such that every graph $G$ on $n$ vertices that does not contain $H$ as an induced subgraph contains either a complete graph or independent set on $\geq n^c$ vertices?

来源字段：https://www.erdosproblems.com/61

14448 | EP-610

### Erdős Problem #610

S / solved | Erdős 问题

For a graph $G$ let $\tau(G)$ denote the minimal number of vertices that include at least one from each maximal clique of $G$ (aside from isolated vertices). This is sometimes called the clique transversal number.

Estimate $\tau(G)$. In particular, is it true that if $G$ has $n$ vertices then\[\tau(G) \leq n-\omega(n)\sqrt{n}\]for some $\omega(n)\to \infty$, or even\[\tau(G) \leq n-c\sqrt{n\log n}\]for some absolute constant $c>0$?

来源字段：https://www.erdosproblems.com/610

14449 | EP-611

### Erdős Problem #611

P / partially_solved | Erdős 问题

For a graph $G$ let $\tau(G)$ denote the minimal number of vertices that include at least one from each maximal clique of $G$ (sometimes called the clique transversal number).

Is it true that if all maximal cliques in $G$ have at least $cn$ vertices then $\tau(G)=o_c(n)$?

Similarly, estimate for $c>0$ the minimal $k_c(n)$ such that if every maximal clique in $G$ has at least $k_c(n)$ vertices then $\tau(G)<(1-c)n$.

来源字段：https://www.erdosproblems.com/611

14450 | EP-612

### Erdős Problem #612

O / open | Erdős 问题

Let $G$ be a connected graph with $n$ vertices, minimum degree $d$, and diameter $D$. Show if that $G$ contains no $K_{2r}$ and $(r-1)(3r+2)\mid d$ then\[D\leq \frac{2(r-1)(3r+2)}{2r^2-1}\frac{n}{d}+O(1),\]and if $G$ contains no $K_{2r+1}$ and $3r-1 \mid d$ then\[D\leq \frac{3r-1}{r}\frac{n}{d}+O(1).\]

来源字段：https://www.erdosproblems.com/612

14451 | EP-613

### Erdős Problem #613: Graph Theory, Ramsey

S / solved | Erdős 问题

Let $n\geq 3$ and $G$ be a graph with $\binom{2n+1}{2}-\binom{n}{2}-1$ edges. Must $G$ be the union of a bipartite graph and a graph with maximum degree less than $n$?

来源字段：https://www.erdosproblems.com/613

14452 | EP-614

### Erdős Problem #614

P / partially_solved | Erdős 问题

Let $f(n,k)$ be minimal such that there is a graph with $n$ vertices and $f(n,k)$ edges where every set of $k+2$ vertices induces a subgraph with maximum degree at least $k$. Determine $f(n,k)$.

来源字段：https://www.erdosproblems.com/614

14453 | EP-615

### Erdős Problem #615: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

Does there exist some constant $c>0$ such that if $G$ is a graph with $n$ vertices and $\geq (1/8-c)n^2$ edges then $G$ must contain either a $K_4$ or an independent set on at least $n/\log n$ vertices?

来源字段：https://www.erdosproblems.com/615

14454 | EP-616

### Erdős Problem #616

P / partially_solved | Erdős 问题

Let $r\geq 3$. For an $r$-uniform hypergraph $G$ let $\tau(G)$ denote the covering number (or transversal number), the minimum size of a set of vertices which includes at least one from each edge in $G$.

Determine the best possible $t$ such that, if $G$ is an $r$-uniform hypergraph $G$ where every subgraph $G'$ on at most $3r-3$ vertices has $\tau(G')\leq 1$, we have $\tau(G)\leq t$.

来源字段：https://www.erdosproblems.com/616

14455 | EP-617

### Erdős Problem #617: Graph Theory

O / open | Erdős 问题

Let $r\geq 3$. If the edges of $K_{r^2+1}$ are $r$-coloured then there exist $r+1$ vertices with at least one colour missing on the edges of the induced $K_{r+1}$.

来源字段：https://www.erdosproblems.com/617

14456 | EP-618

### Erdős Problem #618: Graph Theory

S / solved | Erdős 问题

For a triangle-free graph $G$ let $h_2(G)$ be the smallest number of edges that need to be added to $G$ so that it has diameter $2$ and is still triangle-free. Is it true that if $G$ has maximum degree $o(n^{1/2})$ then $h(G)=o(n^2)$?

来源字段：https://www.erdosproblems.com/618

14457 | EP-619

### Erdős Problem #619

S / solved | Erdős 问题

For a triangle-free graph $G$ let $h_r(G)$ be the smallest number of edges that need to be added to $G$ so that it has diameter $r$ (while preserving the property of being triangle-free).

Is it true that there exists a constant $c>0$ such that if $G$ is a connected graph on $n$ vertices then $h_4(G)<(1-c)n$?

来源字段：https://www.erdosproblems.com/619

14458 | EP-62

### Erdős Problem #62

O / open | Erdős 问题

If $G_1,G_2$ are two graphs with chromatic number $\aleph_1$ then must there exist a graph $G$ whose chromatic number is $4$ (or even $\aleph_0$) which is a subgraph of both $G_1$ and $G_2$?

来源字段：https://www.erdosproblems.com/62

14459 | EP-620

### Erdős Problem #620

P / partially_solved | Erdős 问题

If $G$ is a graph on $n$ vertices without a $K_4$ then how large a triangle-free induced subgraph must $G$ contain?

来源字段：https://www.erdosproblems.com/620

14460 | EP-621

### Erdős Problem #621: Graph Theory

S / solved | Erdős 问题

Let $G$ be a graph on $n$ vertices, $\alpha_1(G)$ be the maximum number of edges that contain at most one edge from every triangle, and $\tau_1(G)$ be the minimum number of edges that contain at least one edge from every triangle.

Is it true that\[\alpha_1(G)+\tau_1(G) \leq \frac{n^2}{4}?\]

来源字段：https://www.erdosproblems.com/621

14461 | EP-622

### Erdős Problem #622: Must $G$ have $\gg 2^{2n}$ subsets that are spanned by a cycle?

S / solved | Erdős 问题

Let $G$ be a regular graph with $2n$ vertices and degree $n+1$. Must $G$ have $\gg 2^{2n}$ subsets that are spanned by a cycle?

来源字段：https://www.erdosproblems.com/622

14462 | EP-625

### Erdős Problem #625

S / solved | Erdős 问题

The cochromatic number of $G$, denoted by $\zeta(G)$, is the minimum number of colours needed to colour the vertices of $G$ such that each colour class induces either a complete graph or empty graph. Let $\chi(G)$ denote the chromatic number.

If $G$ is a random graph with $n$ vertices and each edge included independently with probability $1/2$ then is it true that almost surely\[\chi(G) - \zeta(G) \to \infty\]as $n\to \infty$?

来源字段：https://www.erdosproblems.com/625

14463 | EP-626

### Erdős Problem #626

P / partially_solved | Erdős 问题

Let $k\geq 4$ and $g_k(n)$ denote the largest $m$ such that there is a graph on $n$ vertices with chromatic number $k$ and girth $>m$ (i.e. contains no cycle of length $\leq m$). Does\[\lim_{n\to \infty}\frac{g_k(n)}{\log n}\]exist?

Conversely, if $h^{(m)}(n)$ is the maximal chromatic number of a graph on $n$ vertices with girth $>m$ then does\[\lim_{n\to \infty}\frac{\log h^{(m)}(n)}{\log n}\]exist, and what is its value?

来源字段：https://www.erdosproblems.com/626

14464 | EP-627

### Erdős Problem #627

P / partially_solved | Erdős 问题

Let $\omega(G)$ denote the clique number of $G$ and $\chi(G)$ the chromatic number. If $f(n)$ is the maximum value of $\chi(G)/\omega(G)$, as $G$ ranges over all graphs on $n$ vertices, then does\[\lim_{n\to\infty}\frac{f(n)}{n/(\log_2n)^2}\]exist?

来源字段：https://www.erdosproblems.com/627

14465 | EP-628

### Erdős Problem #628: Graph Theory, Chromatic Number

O / open | Erdős 问题

Let $G$ be a graph with chromatic number $k$ containing no $K_k$. If $a,b\geq 2$ and $a+b=k+1$ then must there exist two disjoint subgraphs of $G$ with chromatic numbers $\geq a$ and $\geq b$ respectively?

来源字段：https://www.erdosproblems.com/628

14466 | EP-629

### Erdős Problem #629

O / open | Erdős 问题

The list chromatic number $\chi_L(G)$ is defined to be the minimal $k$ such that for any assignment of a list of $k$ colours to each vertex of $G$ (perhaps different lists for different vertices) a colouring of each vertex by a colour on its list can be chosen such that adjacent vertices receive distinct colours.

Determine the minimal number of vertices $n(k)$ of a bipartite graph $G$ such that $\chi_L(G)>k$.

来源字段：https://www.erdosproblems.com/629

14467 | EP-630

### Erdős Problem #630: Graph Theory, Chromatic Number

S / solved | Erdős 问题

The list chromatic number $\chi_L(G)$ is defined to be the minimal $k$ such that for any assignment of a list of $k$ colours to each vertex of $G$ (perhaps different lists for different vertices) a colouring of each vertex by a colour on its list can be chosen such that adjacent vertices receive distinct colours.

Does every planar bipartite graph $G$ have $\chi_L(G)\leq 3$?

来源字段：https://www.erdosproblems.com/630

14468 | EP-631

### Erdős Problem #631: Graph Theory, Chromatic Number

S / solved | Erdős 问题

The list chromatic number $\chi_L(G)$ is defined to be the minimal $k$ such that for any assignment of a list of $k$ colours to each vertex of $G$ (perhaps different lists for different vertices) a colouring of each vertex by a colour on its list can be chosen such that adjacent vertices receive distinct colours.

Does every planar graph $G$ have $\chi_L(G)\leq 5$? Is this best possible?

来源字段：https://www.erdosproblems.com/631

14469 | EP-632

### Erdős Problem #632: Graph Theory, Chromatic Number

S / solved | Erdős 问题

A graph is $(a,b)$-choosable if for any assignment of a list of $a$ colours to each of its vertices there is a subset of $b$ colours from each list such that the subsets of adjacent vertices are disjoint.

If $G$ is $(a,b)$-choosable then $G$ is $(am,bm)$-choosable for every integer $m\geq 1$.

来源字段：https://www.erdosproblems.com/632

14470 | EP-636

### Erdős Problem #636: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

Suppose $G$ is a graph on $n$ vertices which contains no complete graph or independent set on $\gg \log n$ many vertices. Must $G$ contain $\gg n^{5/2}$ induced subgraphs which pairwise differ in either the number of vertices or the number of edges?

来源字段：https://www.erdosproblems.com/636

14471 | EP-637

### Erdős Problem #637: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

If $G$ is a graph on $n$ vertices which contains no complete graph or independent set on $\gg \log n$ vertices then $G$ contains an induced subgraph on $\gg n$ vertices which contains $\gg n^{1/2}$ distinct degrees.

来源字段：https://www.erdosproblems.com/637

14472 | EP-638

### Erdős Problem #638

O / open | Erdős 问题

Let $S$ be a family of finite graphs such that for every $n$ there is some $G_n\in S$ such that if the edges of $G_n$ are coloured with $n$ colours then there is a monochromatic triangle.

Is it true that for every infinite cardinal $\aleph$ there is a graph $G$ of which every finite subgraph is in $S$ and if the edges of $G$ are coloured with $\aleph$ many colours then there is a monochromatic triangle.

来源字段：https://www.erdosproblems.com/638

14473 | EP-639

### Erdős Problem #639: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

Is it true that if the edges of $K_n$ are 2-coloured then there are at most $n^2/4$ many edges which do not occur in a monochromatic triangle?

来源字段：https://www.erdosproblems.com/639

14474 | EP-63

### Erdős Problem #63: Graph Theory, Chromatic Number

S / solved | Erdős 问题

Does every graph with infinite chromatic number contain a cycle of length $2^n$ for infinitely many $n$?

来源字段：https://www.erdosproblems.com/63

14475 | EP-640

### Erdős Problem #640

O / open | Erdős 问题

Let $k\geq 3$. Does there exist some $f(k)$ such that if a graph $G$ has chromatic number $\geq f(k)$ then $G$ must contain some odd cycle whose vertices span a graph of chromatic number $\geq k$?

来源字段：https://www.erdosproblems.com/640

14476 | EP-641

### Erdős Problem #641: Graph Theory

S / solved | Erdős 问题

Is there some function $f$ such that for all $k\geq 1$ if a finite graph $G$ has chromatic number $\geq f(k)$ then $G$ has $k$ edge disjoint cycles on the same set of vertices?

来源字段：https://www.erdosproblems.com/641

14477 | EP-642

### Erdős Problem #642

P / partially_solved | Erdős 问题

Let $f(n)$ be the maximal number of edges in a graph on $n$ vertices such that all cycles have more vertices than chords. Is it true that $f(n)\ll n$?

来源字段：https://www.erdosproblems.com/642

14478 | EP-643

### Erdős Problem #643

P / partially_solved | Erdős 问题

Let $f(n;t)$ be minimal such that if a $t$-uniform hypergraph on $n$ vertices contains at least $f(n;t)$ edges then there must be four edges $A,B,C,D$ such that\[A\cup B= C\cup D\]and\[A\cap B=C\cap D=\emptyset.\]Estimate $f(n;t)$ - in particular, is it true that for $t\geq 3$\[f(n;t)=(1+o(1))\binom{n}{t-1}?\]

来源字段：https://www.erdosproblems.com/643

14479 | EP-644

### Erdős Problem #644

P / partially_solved | Erdős 问题

Let $f(k,r)$ be minimal such that if $A_1,A_2,\ldots$ is a family of sets, all of size $k$, such that for every collection of $r$ of the $A_is$ there is some pair $\{x,y\}$ which intersects all of the $A_j$, then there is some set of size $f(k,r)$ which intersects all of the sets $A_i$. Is it true that\[f(k,7)=(1+o(1))\frac{3}{4}k?\]Is it true that for any $r\geq 3$ there exists some constant $c_r$ such that\[f(k,r)=(1+o(1))c_rk?\]

来源字段：https://www.erdosproblems.com/644

14480 | EP-64

### Erdős Problem #64: Graph Theory, Cycles

O / open | Erdős 问题

Does every finite graph with minimum degree at least 3 contain a cycle of length $2^k$ for some $k\geq 2$?

来源字段：https://www.erdosproblems.com/64

14481 | EP-65

### Erdős Problem #65

P / partially_solved | Erdős 问题

Let $G$ be a graph with $n$ vertices and $kn$ edges, and $a_1<a_2<\cdots $ be the lengths of cycles in $G$. Is it true that\[\sum\frac{1}{a_i}\gg \log k?\]Is the sum $\sum\frac{1}{a_i}$ minimised when $G$ is a complete bipartite graph?

来源字段：https://www.erdosproblems.com/65

14482 | EP-660

### Erdős Problem #660

P / partially_solved | Erdős 问题

Let $x_1,\ldots,x_n\in \mathbb{R}^3$ be the vertices of a convex polyhedron. Are there at least\[(1-o(1))\frac{n}{2}\]many distinct distances between the $x_i$?

来源字段：https://www.erdosproblems.com/660

14483 | EP-666

### Erdős Problem #666: Graph Theory

S / solved | Erdős 问题

Let $Q_n$ be the $n$-dimensional hypercube graph (so that $Q_n$ has $2^n$ vertices and $n2^{n-1}$ edges). Is it true that, for every $\epsilon>0$, if $n$ is sufficiently large, every subgraph of $Q_n$ with\[\geq \epsilon n2^{n-1}\]many edges contains a $C_6$?

来源字段：https://www.erdosproblems.com/666

14484 | EP-668

### Erdős Problem #668

P / partially_solved | Erdős 问题

Is it true that the number of incongruent sets of $n$ points in $\mathbb{R}^2$ which maximise the number of unit distances tends to infinity as $n\to\infty$? Is it always $>1$ for $n>3$?

来源字段：https://www.erdosproblems.com/668

14485 | EP-671

### Erdős Problem #671

P / partially_solved | Erdős 问题

Given $a_{i}^n\in [-1,1]$ for all $1\leq i\leq n<\infty$ we define $p_{i}^n$ as the unique polynomial of degree $n-1$ such that $p_{i}^n(a_{i}^n)=1$ and $p_{i}^n(a_{i'}^n)=0$ if $1\leq i'\leq n$ with $i\neq i'$. We similarly define\[\mathcal{L}^nf(x) = \sum_{1\leq i\leq n}f(a_i^n)p_i^n(x),\]the unique polynomial of degree $n-1$ which agrees with $f$ on $a_i^n$ for $1\leq i\leq n$ (that is, the sequence of Lagrange interpolation polynomials).

Is there such a sequence of $a_i^n$ such that for every continuous $f:[-1,1]\to \mathbb{R}$ there exists some $x\in [-1,1]$ where\[\limsup_{n\to \infty} \sum_{1\leq i\leq n}\lvert p_{i}^n(x)\rvert=\infty\]and yet\[\mathcal{L}^nf(x) \to f(x)?\]Is there such a sequence such that\[\limsup_{n\to \infty} \sum_{1\leq i\leq n}\lvert p_{i}^n(x)\rvert=\infty\]for every $x\in [-1,1]$ and yet for every continuous $f:[-1,1]\to \mathbb{R}$ there exists $x\in [-1,1]$ with\[\mathcal{L}^nf(x) \to f(x)?\]

来源字段：https://www.erdosproblems.com/671

14486 | EP-701

### Erdős Problem #701

P / partially_solved | Erdős 问题

Let $\mathcal{F}$ be a family of sets closed under taking subsets (i.e. if $B\subseteq A\in\mathcal{F}$ then $B\in \mathcal{F}$). There exists some element $x$ such that whenever $\mathcal{F}'\subseteq \mathcal{F}$ is an intersecting subfamily we have\[\lvert \mathcal{F}'\rvert \leq \lvert \{ A\in \mathcal{F}: x\in A\}\rvert.\]

来源字段：https://www.erdosproblems.com/701

14487 | EP-704

### Erdős Problem #704

P / partially_solved | Erdős 问题

Let $G_n$ be the unit distance graph in $\mathbb{R}^n$, with two vertices joined by an edge if and only if the distance between them is $1$.

Estimate the chromatic number $\chi(G_n)$. Does it grow exponentially in $n$? Does\[\lim_{n\to \infty}\chi(G_n)^{1/n}\]exist?

来源字段：https://www.erdosproblems.com/704

14488 | EP-705

### Erdős Problem #705

S / solved | Erdős 问题

Let $G$ be a finite unit distance graph in $\mathbb{R}^2$ (i.e. the vertices are a finite collection of points in $\mathbb{R}^2$ and there is an edge between two points if and only if the distance between them is $1$).

Is there some $k$ such that if $G$ has girth $\geq k$ (i.e. $G$ contains no cycles of length $<k$) then $\chi(G)\leq 3$?

来源字段：https://www.erdosproblems.com/705

14489 | EP-706

### Erdős Problem #706

P / partially_solved | Erdős 问题

Let $L(r)$ be such that if $G$ is a graph formed by taking a finite set of points $P$ in $\mathbb{R}^2$ and some set $A\subset (0,\infty)$ of size $r$, where the vertex set is $P$ and there is an edge between two points if and only if their distance is a member of $A$, then $\chi(G)\leq L(r)$.

Estimate $L(r)$. In particular, is it true that $L(r)\leq r^{O(1)}$?

来源字段：https://www.erdosproblems.com/706

14490 | EP-712

### Erdős Problem #712

O / open | Erdős 问题

Determine, for any $k>r>2$, the value of\[\frac{\mathrm{ex}_r(n,K_k^r)}{\binom{n}{r}},\]where $\mathrm{ex}_r(n,K_k^r)$ is the largest number of $r$-edges which can placed on $n$ vertices so that there exists no set of $k$ vertices which is covered by all $\binom{k}{r}$ possible $r$-edges.

来源字段：https://www.erdosproblems.com/712

14491 | EP-714

### Erdős Problem #714

P / partially_solved | Erdős 问题

Is it true that\[\mathrm{ex}(n; K_{r,r}) \gg n^{2-1/r}?\]

来源字段：https://www.erdosproblems.com/714

14492 | EP-715

### Erdős Problem #715: Graph Theory

S / solved | Erdős 问题

Does every regular graph of degree $4$ contain a regular subgraph of degree $3$? Is there any $r$ such that every regular graph of degree $r$ must contain a regular subgraph of degree $3$?

来源字段：https://www.erdosproblems.com/715

14493 | EP-716

### Erdős Problem #716: Is it true that\[\mathrm{ex}_3(n,\mathcal{F})=o(n^2)?\]

S / solved | Erdős 问题

Let $\mathcal{F}$ be the family of all $3$-uniform hypergraphs with $6$ vertices and $3$ $3$-edges. Is it true that\[\mathrm{ex}_3(n,\mathcal{F})=o(n^2)?\]

来源字段：https://www.erdosproblems.com/716

14494 | EP-717

### Erdős Problem #717: Is it true that\[\chi(G) \ll \frac{n^{1/2}}{\log n}\sigma(G)?\]

S / solved | Erdős 问题

Let $G$ be a graph on $n$ vertices with chromatic number $\chi(G)$ and let $\sigma(G)$ be the maximal $k$ such that $G$ contains a subdivision of $K_k$. Is it true that\[\chi(G) \ll \frac{n^{1/2}}{\log n}\sigma(G)?\]

来源字段：https://www.erdosproblems.com/717

14495 | EP-718

### Erdős Problem #718: Graph Theory

S / solved | Erdős 问题

Is there some constant $C>0$ such that any graph on $n$ vertices with $\geq Cr^2n$ edges contains a subdivision of $K_r$?

来源字段：https://www.erdosproblems.com/718

14496 | EP-719

### Erdős Problem #719

O / open | Erdős 问题

Let $\mathrm{ex}_r(n;K_{r+1}^r)$ be the maximum number of $r$-edges that can be placed on $n$ vertices without forming a $K_{r+1}^r$ (the $r$-uniform complete graph on $r+1$ vertices).

Is every $r$-hypergraph $G$ on $n$ vertices the union of at most $\mathrm{ex}_{r}(n;K_{r+1}^r)$ many copies of $K_r^r$ and $K_{r+1}^r$, no two of which share a $K_r^r$?

来源字段：https://www.erdosproblems.com/719

14497 | EP-71

### Erdős Problem #71: Graph Theory, Cycles

S / solved | Erdős 问题

Is it true that for every infinite arithmetic progression $P$ which contains even numbers there is some constant $c=c(P)$ such that every graph with average degree at least $c$ contains a cycle whose length is in $P$?

来源字段：https://www.erdosproblems.com/71

14498 | EP-720

### Erdős Problem #720: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

Let $\hat{R}(G)$ denote the size Ramsey number, the minimal number of edges $m$ such that there is a graph $H$ with $m$ edges such that in any $2$-colouring of the edges of $H$ there is a monochromatic copy of $G$.

Is it true that, if $P_n$ is the path of length $n$, then\[\hat{R}(P_n)/n\to \infty\]and\[\hat{R}(P_n)/n^2 \to 0?\]Is it true that, if $C_n$ is the cycle with $n$ edges, then\[\hat{R}(C_n) =o(n^2)?\]

来源字段：https://www.erdosproblems.com/720

14499 | EP-72

### Erdős Problem #72: Graph Theory, Cycles

S / solved | Erdős 问题

Is there a set $A\subset \mathbb{N}$ of density $0$ and a constant $c>0$ such that every graph on sufficiently many vertices with average degree $\geq c$ contains a cycle whose length is in $A$?

来源字段：https://www.erdosproblems.com/72

14500 | EP-736

### Erdős Problem #736: Graph Theory, Chromatic Number

P / partially_solved | Erdős 问题

Let $G$ be a graph with chromatic number $\aleph_1$. Is there, for every cardinal number $m$, some graph $G_m$ of chromatic number $m$ such that every finite subgraph of $G_m$ is a subgraph of $G$?

来源字段：https://www.erdosproblems.com/736

14501 | EP-737

### Erdős Problem #737: Graph Theory, Chromatic Number

S / solved | Erdős 问题

Let $G$ be a graph with chromatic number $\aleph_1$. Must there exist an edge $e$ such that, for all large $n$, $G$ contains a cycle of length $n$ containing $e$?

来源字段：https://www.erdosproblems.com/737

14502 | EP-738

### Erdős Problem #738

O / open | Erdős 问题

If $G$ has infinite chromatic number and is triangle-free (contains no $K_3$) then must $G$ contain every tree as an induced subgraph?

来源字段：https://www.erdosproblems.com/738

14503 | EP-739

### Erdős Problem #739: Graph Theory, Chromatic Number

P / partially_solved | Erdős 问题

Let $\mathfrak{m}$ be an infinite cardinal and $G$ be a graph with chromatic number $\mathfrak{m}$. Is it true that, for every infinite cardinal $\mathfrak{n}< \mathfrak{m}$, there exists a subgraph of $G$ with chromatic number $\mathfrak{n}$?

来源字段：https://www.erdosproblems.com/739

14504 | EP-73

### Erdős Problem #73: Graph Theory

S / solved | Erdős 问题

Let $k\geq 0$. Let $G$ be a graph such that every subgraph $H$ contains an independent set of size $\geq (n-k)/2$, where $n$ is the number of vertices of $H$. Must $G$ be the union of a bipartite graph and $O_k(1)$ many vertices?

来源字段：https://www.erdosproblems.com/73

14505 | EP-74

### Erdős Problem #74

S / solved | Erdős 问题

Let $f(n)\to \infty$ (possibly very slowly). Is there a graph of infinite chromatic number such that every finite subgraph on $n$ vertices can be made bipartite by deleting at most $f(n)$ edges?

来源字段：https://www.erdosproblems.com/74

14506 | EP-742

### Erdős Problem #742: Graph Theory

O / open | Erdős 问题

Let $G$ be a graph on $n$ vertices with diameter $2$, such that deleting any edge increases the diameter of $G$. Is it true that $G$ has at most $n^2/4$ edges?

来源字段：https://www.erdosproblems.com/742

14507 | EP-743

### Erdős Problem #743: Can we always write $K_n$ as the edge disjoint union of the $T_k$?

O / open | Erdős 问题

Let $T_2,\ldots,T_n$ be a collection of trees such that $T_k$ has $k$ vertices. Can we always write $K_n$ as the edge disjoint union of the $T_k$?

来源字段：https://www.erdosproblems.com/743

14508 | EP-744

### Erdős Problem #744: Graph Theory, Chromatic Number

S / solved | Erdős 问题

Let $k$ be a large fixed constant. Let $f_k(n)$ be the minimal $m$ such that there exists a graph $G$ on $n$ vertices with chromatic number $k$, such that every proper subgraph has chromatic number $<k$, and $G$ can be made bipartite by deleting $m$ edges.

Is it true that $f_k(n)\to \infty$ as $n\to \infty$? In particular, is it true that $f_4(n) \gg \log n$?

来源字段：https://www.erdosproblems.com/744

14509 | EP-745

### Erdős Problem #745: Graph Theory

S / solved | Erdős 问题

Describe the size of the second largest component of the random graph on $n$ vertices, where each edge is included independently with probability $1/n$.

来源字段：https://www.erdosproblems.com/745

14510 | EP-746

### Erdős Problem #746: Graph Theory

S / solved | Erdős 问题

Is it true that, almost surely, a random graph on $n$ vertices with $\geq (\tfrac{1}{2}+\epsilon)n\log n$ edges is Hamiltonian?

来源字段：https://www.erdosproblems.com/746

14511 | EP-75

### Erdős Problem #75

P / partially_solved | Erdős 问题

Is there a graph of chromatic number $\aleph_1$ with $\aleph_1$ vertices such that for all $\epsilon>0$ if $n$ is sufficiently large and $H$ is a subgraph on $n$ vertices then $H$ contains an independent set of size $>n^{1-\epsilon}$?

What about an independent set of size $\gg n$?

来源字段：https://www.erdosproblems.com/75

14512 | EP-750

### Erdős Problem #750

S / solved | Erdős 问题

Let $f(m)$ be some function such that $f(m)\to \infty$ as $m\to \infty$. Does there exist a graph $G$ of infinite chromatic number such that every subgraph on $m$ vertices contains an independent set of size at least $\frac{m}{2}-f(m)$?

来源字段：https://www.erdosproblems.com/750

14513 | EP-751

### Erdős Problem #751: Graph Theory, Chromatic Number

S / solved | Erdős 问题

Let $G$ be a graph with chromatic number $\chi(G)=4$. If $m_1<m_2<\cdots$ are the lengths of the cycles in $G$ then can $\min(m_{i+1}-m_i)$ be arbitrarily large? Can this happen if the girth of $G$ is large?

来源字段：https://www.erdosproblems.com/751

14514 | EP-752

### Erdős Problem #752: Graph Theory, Cycles

S / solved | Erdős 问题

Let $G$ be a graph with minimum degree $k$ and girth $>2s$ (i.e. $G$ contains no cycles of length $\leq 2s$). Must there be $\gg k^s$ many distinct cycle lengths in $G$?

来源字段：https://www.erdosproblems.com/752

14515 | EP-753

### Erdős Problem #753: Graph Theory, Chromatic Number

S / solved | Erdős 问题

The list chromatic number $\chi_L(G)$ is defined to be the minimal $k$ such that for any assignment of a list of $k$ colours to each vertex of $G$ (perhaps different lists for different vertices) a colouring of each vertex by a colour on its list can be chosen such that adjacent vertices receive distinct colours.

Does there exist some constant $c>0$ such that\[\chi_L(G)+\chi_L(G^c)> n^{1/2+c}\]for every graph $G$ on $n$ vertices (where $G^c$ is the complement of $G$)?

来源字段：https://www.erdosproblems.com/753

14516 | EP-758

### Erdős Problem #758: Graph Theory, Chromatic Number

S / solved | Erdős 问题

The cochromatic number of $G$, denoted by $\zeta(G)$, is the minimum number of colours needed to colour the vertices of $G$ such that each colour class induces either a complete graph or empty graph. Let $z(n)$ be the maximum value of $\zeta(G)$ over all graphs $G$ with $n$ vertices.

Determine $z(n)$ for small values of $z(n)$. In particular is it true that $z(12)=4$?

来源字段：https://www.erdosproblems.com/758

14517 | EP-759

### Erdős Problem #759: Graph Theory, Chromatic Number

S / solved | Erdős 问题

The cochromatic number of $G$, denoted by $\zeta(G)$, is the minimum number of colours needed to colour the vertices of $G$ such that each colour class induces either a complete graph or empty graph.

Let $z(S_n)$ be the maximum value of $\zeta(G)$ over all graphs $G$ which can be embedded on $S_n$, the orientable surface of genus $n$. Determine the growth rate of $z(S_n)$.

来源字段：https://www.erdosproblems.com/759

14518 | EP-760

### Erdős Problem #760: Graph Theory, Chromatic Number

S / solved | Erdős 问题

The cochromatic number of $G$, denoted by $\zeta(G)$, is the minimum number of colours needed to colour the vertices of $G$ such that each colour class induces either a complete graph or independent set.

If $G$ is a graph with chromatic number $\chi(G)=m$ then must $G$ contain a subgraph $H$ with\[\zeta(H) \gg \frac{m}{\log m}?\]

来源字段：https://www.erdosproblems.com/760

14519 | EP-761

### Erdős Problem #761

P / partially_solved | Erdős 问题

The cochromatic number of $G$, denoted by $\zeta(G)$, is the minimum number of colours needed to colour the vertices of $G$ such that each colour class induces either a complete graph or empty graph. The dichromatic number of $G$, denoted by $\delta(G)$, is the minimum number $k$ of colours required such that, in any orientation of the edges of $G$, there is a $k$-colouring of the vertices of $G$ such that there are no monochromatic oriented cycles.

Must a graph with large chromatic number have large dichromatic number? Must a graph with large cochromatic number contain a graph with large dichromatic number?

来源字段：https://www.erdosproblems.com/761

14520 | EP-762

### Erdős Problem #762: Graph Theory, Chromatic Number

S / solved | Erdős 问题

The cochromatic number of $G$, denoted by $\zeta(G)$, is the minimum number of colours needed to colour the vertices of $G$ such that each colour class induces either a complete graph or empty graph.

Is it true that if $G$ has no $K_5$ and $\zeta(G)\geq 4$ then $\chi(G) \leq \zeta(G)+2$?

来源字段：https://www.erdosproblems.com/762

14521 | EP-765

### Erdős Problem #765: Give an asymptotic formula for $\mathrm{ex}(n;C_4)$

S / solved | Erdős 问题

Give an asymptotic formula for $\mathrm{ex}(n;C_4)$.

来源字段：https://www.erdosproblems.com/765

14522 | EP-766

### Erdős Problem #766

O / open | Erdős 问题

Let $f(n;k,l)=\min \mathrm{ex}(n;G)$, where $G$ ranges over all graphs with $k$ vertices and $l$ edges.

Give good estimates for $f(n;k,l)$ in the range $k<l\leq k^2/4$. For fixed $k$ and large $n$ is $f(n;k,l)$ a strictly monotone function of $l$?

来源字段：https://www.erdosproblems.com/766

14523 | EP-767

### Erdős Problem #767: Is it true that\[g_k(n)=(k+1)n-(k+1)^2\]for $n$ sufficiently large?

S / solved | Erdős 问题

Let $g_k(n)$ be the maximal number of edges possible on a graph with $n$ vertices which does not contain a cycle with $k$ chords incident to a vertex on the cycle. Is it true that\[g_k(n)=(k+1)n-(k+1)^2\]for $n$ sufficiently large?

来源字段：https://www.erdosproblems.com/767

14524 | EP-76

### Erdős Problem #76: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

Is it true that in any $2$-colouring of the edges of $K_n$ there must exist at least\[(1+o(1))\frac{n^2}{12}\]many edge-disjoint monochromatic triangles?

来源字段：https://www.erdosproblems.com/76

14525 | EP-77

### Erdős Problem #77

O / open | Erdős 问题

If $R(k)$ is the Ramsey number for $K_k$, the minimal $n$ such that every $2$-colouring of the edges of $K_n$ contains a monochromatic copy of $K_k$, then find the value of\[\lim_{k\to \infty}R(k)^{1/k}.\]

来源字段：https://www.erdosproblems.com/77

14526 | EP-775

### Erdős Problem #775: Graph Theory, Hypergraphs

S / solved | Erdős 问题

Is there a $3$-uniform hypergraph on $n$ vertices which contains at least $n-O(1)$ different sizes of cliques (maximal complete subgraphs)

来源字段：https://www.erdosproblems.com/775

14527 | EP-778

### Erdős Problem #778

P / partially_solved | Erdős 问题

Alice and Bob play a game on the edges of $K_n$, alternating colouring edges by red (Alice) and blue (Bob). Alice goes first, and wins if at the end the largest red clique is larger than any of the blue cliques.

Does Bob have a winning strategy for $n\geq 3$? (Erdős believed the answer is yes.)

If we change the game so that Bob colours two edges after each edge that Alice colours, but now require Bob's largest clique to be strictly larger than Alice's, then does Bob have a winning strategy for $n>3$?

Finally, consider the game when Alice wins if the maximum degree of the red subgraph is larger than the maximum degree of the blue subgraph. Who wins?

来源字段：https://www.erdosproblems.com/778

14528 | EP-78

### Erdős Problem #78

O / open | Erdős 问题

Let $R(k)$ be the Ramsey number for $K_k$, the minimal $n$ such that every $2$-colouring of the edges of $K_n$ contains a monochromatic copy of $K_k$.

Give a constructive proof that $R(k)>C^k$ for some constant $C>1$.

来源字段：https://www.erdosproblems.com/78

14529 | EP-794

### Erdős Problem #794: Graph Theory, Hypergraphs

S / solved | Erdős 问题

Is it true that every $3$-uniform hypergraph on $3n$ vertices with at least $n^3+1$ edges must contain either a subgraph on $4$ vertices with $3$ edges or a subgraph on $5$ vertices with $7$ edges?

来源字段：https://www.erdosproblems.com/794

14530 | EP-797

### Erdős Problem #797: Graph Theory, Chromatic Number

S / solved | Erdős 问题

Let $f(d)$ be the maximal acyclic chromatic number of any graph with maximum degree $d$ - that is, the vertices of any graph with maximum degree $d$ can be coloured with $f(d)$ colours such that there is no edge between vertices of the same colour and no cycle containing only two colours.

Estimate $f(d)$. In particular is it true that $f(d)=o(d^2)$?

来源字段：https://www.erdosproblems.com/797

14531 | EP-799

### Erdős Problem #799: Graph Theory, Chromatic Number

S / solved | Erdős 问题

The list chromatic number $\chi_L(G)$ is defined to be the minimal $k$ such that for any assignment of a list of $k$ colours to each vertex of $G$ (perhaps different lists for different vertices) a colouring of each vertex by a colour on its list can be chosen such that adjacent vertices receive distinct colours.

Is it true that $\chi_L(G)=o(n)$ for almost all graphs on $n$ vertices?

来源字段：https://www.erdosproblems.com/799

14532 | EP-79

### Erdős Problem #79: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

We say $G$ is Ramsey size linear if $R(G,H)\ll m$ for all graphs $H$ with $m$ edges and no isolated vertices.

Are there infinitely many graphs $G$ which are not Ramsey size linear but such that all of its subgraphs are?

来源字段：https://www.erdosproblems.com/79

14533 | EP-80

### Erdős Problem #80

P / partially_solved | Erdős 问题

Let $c>0$ and let $f_c(n)$ be the maximal $m$ such that every graph $G$ with $n$ vertices and at least $cn^2$ edges, where each edge is contained in at least one triangle, must contain a book of size $m$, that is, an edge shared by at least $m$ different triangles.

Estimate $f_c(n)$. In particular, is it true that $f_c(n)>n^{\epsilon}$ for some $\epsilon>0$? Or $f_c(n)\gg \log n$?

来源字段：https://www.erdosproblems.com/80

14534 | EP-800

### Erdős Problem #800: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

If $G$ is a graph on $n$ vertices which has no two adjacent vertices of degree $\geq 3$ then\[R(G)\ll n,\]where the implied constant is absolute.

来源字段：https://www.erdosproblems.com/800

14535 | EP-801

### Erdős Problem #801: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

If $G$ is a graph on $n$ vertices containing no independent set on $>n^{1/2}$ vertices then there is a set of $\leq n^{1/2}$ vertices containing $\gg n^{1/2}\log n$ edges.

来源字段：https://www.erdosproblems.com/801

14536 | EP-802

### Erdős Problem #802

P / partially_solved | Erdős 问题

Is it true that any $K_r$-free graph on $n$ vertices with average degree $t$ contains an independent set on\[\gg_r \frac{\log t}{t}n\]many vertices?

来源字段：https://www.erdosproblems.com/802

14537 | EP-803

### Erdős Problem #803: Graph Theory

S / solved | Erdős 问题

We call a graph $H$ $D$-balanced (or $D$-almost-regular) if the maximum degree of $H$ is at most $D$ times the minimum degree of $H$.

Is it true that for every $m\geq 1$, if $n$ is sufficiently large, any graph on $n$ vertices with $\geq n\log n$ edges contains a $O(1)$-balanced subgraph with $m$ vertices and $\gg m\log m$ edges (where the implied constants are absolute)?

来源字段：https://www.erdosproblems.com/803

14538 | EP-804

### Erdős Problem #804:

S / solved | Erdős 问题

Let $f(m,n)$ be maximal such that any graph on $n$ vertices in which every induced subgraph on $m$ vertices has an independent set of size at least $\log n$ must contain an independent set of size at least $f(n)$.

Estimate $f(n)$. In particular, is it true that $f((\log n)^2,n)\geq n^{1/2-o(1)}$? Is it true that $f((\log n)^3,n)\gg (\log n)^3$?

来源字段：https://www.erdosproblems.com/804

14539 | EP-805

### Erdős Problem #805

P / partially_solved | Erdős 问题

For which functions $g(n)$ with $n>g(n)\geq (\log n)^2$ is there a graph on $n$ vertices in which every induced subgraph on $g(n)$ vertices contains a clique of size $\geq \log n$ and an independent set of size $\geq \log n$?

In particular, is there such a graph for $g(n)=(\log n)^3$?

来源字段：https://www.erdosproblems.com/805

14540 | EP-807

### Erdős Problem #807: Graph Theory

S / solved | Erdős 问题

The bipartition number $\tau(G)$ of a graph $G$ is the smallest number of pairwise edge disjoint complete bipartite graphs whose union is $G$. The independence number $\alpha(G)$ is the size of the largest independent subset of $G$.

Is it true that, if $G$ is a random graph on $n$ vertices with edge probability $1/2$, then\[\tau(G)=n-\alpha(G)\]almost surely?

来源字段：https://www.erdosproblems.com/807

14541 | EP-809

### Erdős Problem #809

P / partially_solved | Erdős 问题

Define the anti-Ramsey number $\chi_S(n,e,G)$ as the smallest $r$ such that there is a graph with $n$ vertices and $e$ edges with an $r$-colouring of its edges in which every copy of $G$ has entirely distinct edge colours.

Is it true that, for all $k\geq 3$,\[\chi_S(n, \lfloor n^2/4\rfloor+1,C_{2k+1})\sim n^2/8?\]

来源字段：https://www.erdosproblems.com/809

14542 | EP-810

### Erdős Problem #810

O / open | Erdős 问题

Does there exist some $\epsilon>0$ such that, for all sufficiently large $n$, there exists a graph $G$ on $n$ vertices with at least $\epsilon n^2$ many edges such that the edges can be coloured with $n$ colours so that every $C_4$ receives $4$ distinct colours?

来源字段：https://www.erdosproblems.com/810

14543 | EP-811

### Erdős Problem #811

P / partially_solved | Erdős 问题

Suppose $n\equiv 1\pmod{m}$. We say that an edge-colouring of $K_n$ using $m$ colours is balanced if every vertex sees exactly $\lfloor n/m\rfloor$ many edges of each colours.

For which graphs $G$ is it true that, if $m=e(G)$, for all large $n\equiv 1\pmod{m}$, every balanced edge-colouring of $K_n$ with $m$ colours contains a rainbow copy of $G$? (That is, a subgraph isomorphic to $G$ where each edge receives a different colour.)

来源字段：https://www.erdosproblems.com/811

14544 | EP-813

### Erdős Problem #813

P / partially_solved | Erdős 问题

Let $h(n)$ be minimal such that every graph on $n$ vertices where every set of $7$ vertices contains a triangle (a copy of $K_3$) must contain a clique on at least $h(n)$ vertices. Estimate $h(n)$ - in particular, do there exist constants $c_1,c_2>0$ such that\[n^{1/3+c_1}\ll h(n) \ll n^{1/2-c_2}?\]

来源字段：https://www.erdosproblems.com/813

14545 | EP-814

### Erdős Problem #814: Graph Theory

S / solved | Erdős 问题

Let $k\geq 2$ and $G$ be a graph with $n\geq k-1$ vertices and\[(k-1)(n-k+2)+\binom{k-2}{2}+1\]edges. Does there exist some $c_k>0$ such that $G$ must contain an induced subgraph on at most $(1-c_k)n$ vertices with minimum degree at least $k$?

来源字段：https://www.erdosproblems.com/814

14546 | EP-815

### Erdős Problem #815: Graph Theory

S / solved | Erdős 问题

Let $k\geq 3$ and $n$ be sufficiently large. Is it true that if $G$ is a graph with $n$ vertices and $2n-2$ edges such that every proper induced subgraph has minimum degree $\leq 2$ then $G$ must contain a copy of $C_k$?

来源字段：https://www.erdosproblems.com/815

14547 | EP-816

### Erdős Problem #816: Graph Theory

S / solved | Erdős 问题

Let $G$ be a graph with $2n+1$ vertices and $n^2+n+1$ edges. Must $G$ contain two vertices of the same degree which are joined by a path of length $3$?

来源字段：https://www.erdosproblems.com/816

14548 | EP-82

### Erdős Problem #82

O / open | Erdős 问题

Let $F(n)$ be maximal such that every graph on $n$ vertices contains a regular induced subgraph on at least $F(n)$ vertices. Prove that $F(n)/\log n\to \infty$.

来源字段：https://www.erdosproblems.com/82

14549 | EP-832

### Erdős Problem #832: Graph Theory, Hypergraphs

S / solved | Erdős 问题

Let $r\geq 3$ and $k$ be sufficiently large in terms of $r$. Is it true that every $r$-uniform hypergraph with chromatic number $k$ has at least\[\binom{(r-1)(k-1)+1}{r}\]edges, with equality only for the complete graph on $(r-1)(k-1)+1$ vertices?

来源字段：https://www.erdosproblems.com/832

14550 | EP-833

### Erdős Problem #833: Graph Theory, Hypergraphs

S / solved | Erdős 问题

Does there exist an absolute constant $c>0$ such that, for all $r\geq 2$, in any $r$-uniform hypergraph with chromatic number $3$ there is a vertex contained in at least $(1+c)^r$ many edges?

来源字段：https://www.erdosproblems.com/833

14551 | EP-834

### Erdős Problem #834: Graph Theory, Hypergraphs

S / solved | Erdős 问题

Does there exist a $3$-critical $3$-uniform hypergraph in which every vertex has degree $\geq 7$?

来源字段：https://www.erdosproblems.com/834

14552 | EP-835

### Erdős Problem #835: Graph Theory, Hypergraphs

O / open | Erdős 问题

Does there exist a $k>2$ such that the $k$-sized subsets of $\{1,\ldots,2k\}$ can be coloured with $k+1$ colours such that for every $A\subset \{1,\ldots,2k\}$ with $\lvert A\rvert=k+1$ all $k+1$ colours appear among the $k$-sized subsets of $A$?

来源字段：https://www.erdosproblems.com/835

14553 | EP-837

### Erdős Problem #837

O / open | Erdős 问题

Let $k\geq 2$ and $A_k\subseteq [0,1]$ be the set of $\alpha$ such that there exists some $\beta(\alpha)>\alpha$ with the property that, if $G_1,G_2,\ldots$ is a sequence of $k$-uniform hypergraphs with\[\liminf \frac{e(G_n)}{\binom{\lvert G_n\rvert}{k}} >\alpha\]then there exist subgraphs $H_n\subseteq G_n$ such that $\lvert H_n\rvert \to \infty$ and\[\liminf \frac{e(H_n)}{\binom{\lvert H_n\rvert}{k}} >\beta,\]and further that this property does not necessarily hold if $>\alpha$ is replaced by $\geq \alpha$.

What is $A_3$?

来源字段：https://www.erdosproblems.com/837

14554 | EP-84

### Erdős Problem #84

P / partially_solved | Erdős 问题

The cycle set of a graph $G$ on $n$ vertices is a set $A\subseteq \{3,\ldots,n\}$ such that there is a cycle in $G$ of length $\ell$ if and only if $\ell \in A$. Let $f(n)$ count the number of possible such $A$.

Prove that $f(n)=o(2^n)$.

Prove that $f(n)/2^{n/2}\to \infty$.

来源字段：https://www.erdosproblems.com/84

14555 | EP-842

### Erdős Problem #842: Does $G$ have chromatic number at most $3$?

S / solved | Erdős 问题

Let $G$ be a graph on $3n$ vertices formed by taking $n$ vertex disjoint triangles and adding a Hamiltonian cycle (with all new edges) between these vertices. Does $G$ have chromatic number at most $3$?

来源字段：https://www.erdosproblems.com/842

14556 | EP-85

### Erdős Problem #85: Graph Theory, Ramsey Theory

O / open | Erdős 问题

Let $n\geq 4$ and $f(n)$ be minimal such that every graph on $n$ vertices with minimal degree $\geq f(n)$ contains a $C_4$. Is it true that, for all large $n$, $f(n+1)\geq f(n)$?

来源字段：https://www.erdosproblems.com/85

14557 | EP-86

### Erdős Problem #86

O / open | Erdős 问题

Let $Q_n$ be the $n$-dimensional hypercube graph (so that $Q_n$ has $2^n$ vertices and $n2^{n-1}$ edges). Is it true that every subgraph of $Q_n$ with\[\geq \left(\frac{1}{2}+o(1)\right)n2^{n-1}\]many edges contains a $C_4$?

来源字段：https://www.erdosproblems.com/86

14558 | EP-87

### Erdős Problem #87

O / open | Erdős 问题

Let $\epsilon >0$. Is it true that, if $k$ is sufficiently large, then\[R(G)>(1-\epsilon)^kR(k)\]for every graph $G$ with chromatic number $\chi(G)=k$?

Even stronger, is there some $c>0$ such that, for all large $k$, $R(G)>cR(k)$ for every graph $G$ with chromatic number $\chi(G)=k$?

来源字段：https://www.erdosproblems.com/87

14559 | EP-88

### Erdős Problem #88: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

For any $\epsilon>0$ there exists $\delta=\delta(\epsilon)>0$ such that if $G$ is a graph on $n$ vertices with no independent set or clique of size $\geq \epsilon\log n$ then $G$ contains an induced subgraph with $m$ edges for all $m\leq \delta n^2$.

来源字段：https://www.erdosproblems.com/88

14560 | EP-90

### Erdős Problem #90

S / solved | Erdős 问题

别名：The Unit Distance Problem [GEO-026]; Unit Distance Problem [GEOM-024]; Cube-Simplex conjecture [OPG-778]

Does every set of $n$ distinct points in $\mathbb{R}^2$ contain at most $n^{1+O(1/\log\log n)}$ many pairs which are distance 1 apart?

来源字段：https://www.erdosproblems.com/90

14561 | EP-900

### Erdős Problem #900: Graph Theory

S / solved | Erdős 问题

There is a function $f:(1/2,\infty)\to \mathbb{R}$ such that $f(c)\to 0$ as $c\to 1/2$ and $f(c)\to 1$ as $c\to \infty$ and every random graph with $n$ vertices and $cn$ edges has (with high probability) a path of length at least $f(c)n$.

来源字段：https://www.erdosproblems.com/900

14562 | EP-901

### Erdős Problem #901

P / partially_solved | Erdős 问题

Let $m(n)$ be minimal such that there is an $n$-uniform hypergraph with $m(n)$ edges which is $3$-chromatic. Estimate $m(n)$.

来源字段：https://www.erdosproblems.com/901

14563 | EP-902

### Erdős Problem #902

P / partially_solved | Erdős 问题

Let $f(n)$ be minimal such that there is a tournament (a complete directed graph) on $f(n)$ vertices such that every set of $n$ vertices is dominated by at least one other vertex. Estimate $f(n)$.

来源字段：https://www.erdosproblems.com/902

14564 | EP-904

### Erdős Problem #904: Graph Theory

S / solved | Erdős 问题

Let $r\geq 2$ and let $t_r(n)$ be the Turán number (the maximal number of edges in a graph on $n$ vertices with no $K_{r+1}$).

If $G$ is a graph with $n$ vertices and $m\geq t_r(n)$ edges there exists a clique on $r$ vertices, say $x_1,\ldots,x_r$, such that\[d(x_1)+\cdots+d(x_r)\geq \frac{2rm}{n}.\]

来源字段：https://www.erdosproblems.com/904

14565 | EP-905

### Erdős Problem #905: Graph Theory

S / solved | Erdős 问题

Every graph with $n$ vertices and $>n^2/4$ edges contains an edge which is in at least $n/6$ triangles.

来源字段：https://www.erdosproblems.com/905

14566 | EP-911

### Erdős Problem #911

O / open | Erdős 问题

Let $\hat{R}(G)$ denote the size Ramsey number, the minimal number of edges $m$ such that there is a graph $H$ with $m$ edges that is Ramsey for $G$.

Is there a function $f$ such that $f(x)/x\to \infty$ as $x\to \infty$ such that, for all large $C$, if $G$ is a graph with $n$ vertices and $e\geq Cn$ edges then\[\hat{R}(G) > f(C) e?\]

来源字段：https://www.erdosproblems.com/911

14567 | EP-914

### Erdős Problem #914: Graph Theory

S / solved | Erdős 问题

Let $r\geq 2$ and $m\geq 1$. Every graph with $rm$ vertices and minimum degree at least $m(r-1)$ contains $m$ vertex disjoint copies of $K_r$.

来源字段：https://www.erdosproblems.com/914

14568 | EP-915

### Erdős Problem #915: Must $G$ contain two points which are connected by $m$ disjoint paths?

S / solved | Erdős 问题

Let $G$ be a graph with $1+n(m-1)$ vertices and $1+n\binom{m}{2}$ edges. Must $G$ contain two points which are connected by $m$ disjoint paths?

来源字段：https://www.erdosproblems.com/915

14569 | EP-916

### Erdős Problem #916: Graph Theory, Cycles

S / solved | Erdős 问题

Does every graph with $n$ vertices and $2n-2$ edges contain a cycle and another vertex adjacent to three vertices on the cycle?

来源字段：https://www.erdosproblems.com/916

14570 | EP-917

### Erdős Problem #917

P / partially_solved | Erdős 问题

Let $k\geq 4$ and $f_k(n)$ be the largest number of edges in a graph on $n$ vertices which has chromatic number $k$ and is critical (i.e. deleting any edge reduces the chromatic number).

Is it true that\[f_k(n) \gg_k n^2?\]Is it true that\[f_6(n)\sim n^2/4?\]More generally, is it true that, for $k\geq 6$,\[f_k(n) \sim \frac{1}{2}\left(1-\frac{1}{\lfloor k/3\rfloor}\right)n^2?\]

来源字段：https://www.erdosproblems.com/917

14571 | EP-918

### Erdős Problem #918

P / partially_solved | Erdős 问题

Is there a graph with $\aleph_2$ vertices and chromatic number $\aleph_2$ such that every subgraph on $\aleph_1$ vertices has chromatic number $\leq\aleph_0$?

Is there a graph with $\aleph_{\omega+1}$ vertices and chromatic number $\aleph_1$ such that every subgraph on $\aleph_\omega$ vertices has chromatic number $\leq\aleph_0$?

来源字段：https://www.erdosproblems.com/918

14572 | EP-919

### Erdős Problem #919

P / partially_solved | Erdős 问题

Is there a graph $G$ with vertex set $\omega_2^2$ and chromatic number $\aleph_2$ such that every subgraph whose vertices have a lesser type has chromatic number $\leq \aleph_0$?

What if instead we ask for $G$ to have chromatic number $\aleph_1$?

来源字段：https://www.erdosproblems.com/919

14573 | EP-92

### Erdős Problem #92

S / solved | Erdős 问题

Let $f(n)$ be maximal such that there exists a set $A$ of $n$ points in $\mathbb{R}^2$ in which every $x\in A$ has at least $f(n)$ points in $A$ equidistant from $x$.

Is it true that $f(n)\leq n^{o(1)}$? Or even $f(n) < n^{O(1/\log\log n)}$?

来源字段：https://www.erdosproblems.com/92

14574 | EP-920

### Erdős Problem #920

S / solved | Erdős 问题

Let $f_k(n)$ be the maximum possible chromatic number of a graph with $n$ vertices which contains no $K_k$.

Is it true that, for $k\geq 4$,\[f_k(n) \gg \frac{n^{1-\frac{1}{k-1}}}{(\log n)^{c_k}}\]for some constant $c_k>0$?

来源字段：https://www.erdosproblems.com/920

14575 | EP-921

### Erdős Problem #921: Graph Theory, Chromatic Number

S / solved | Erdős 问题

Let $k\geq 4$ and let $f_k(n)$ be the largest $m$ such that there is a graph on $n$ vertices with chromatic number $k$ in which every odd cycle has length $> m$. Is it true that\[f_k(n) \asymp n^{\frac{1}{k-2}}?\]

来源字段：https://www.erdosproblems.com/921

14576 | EP-922

### Erdős Problem #922: Graph Theory, Chromatic Number

S / solved | Erdős 问题

Let $k\geq 0$. Let $G$ be a graph such that every subgraph $H$ contains an independent set of size $\geq (n-k)/2$, where $n$ is the number of vertices of $H$. Must $G$ have chromatic number at most $k+2$?

来源字段：https://www.erdosproblems.com/922

14577 | EP-923

### Erdős Problem #923: Graph Theory, Chromatic Number

S / solved | Erdős 问题

Is it true that, for every $k$, there is some $f(k)$ such that if $G$ has chromatic number $\geq f(k)$ then $G$ contains a triangle-free subgraph with chromatic number $\geq k$?

来源字段：https://www.erdosproblems.com/923

14578 | EP-924

### Erdős Problem #924: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

Let $k\geq 2$ and $l\geq 3$. Is there a graph $G$ which contains no $K_{l+1}$ such that every $k$-colouring of the edges of $G$ contains a monochromatic copy of $K_l$?

来源字段：https://www.erdosproblems.com/924

14579 | EP-925

### Erdős Problem #925: Graph Theory, Ramsey Theory

S / solved | Erdős 问题

Is there a constant $\delta>0$ such that, for all large $n$, if $G$ is a graph on $n$ vertices which is not Ramsey for $K_3$ (i.e. there exists a 2-colouring of the edges of $G$ with no monochromatic triangle) then $G$ contains an independent set of size $\gg n^{1/3+\delta}$?

来源字段：https://www.erdosproblems.com/925

14580 | EP-926

### Erdős Problem #926: Graph Theory

S / solved | Erdős 问题

Let $k\geq 4$. Is it true that\[\mathrm{ex}(n;H_k) \ll_k n^{3/2},\]where $H_k$ is the graph on vertices $x,y_1,\ldots,y_k,z_1,\ldots,z_{\binom{k}{2}}$, where $x$ is adjacent to all $y_i$ and each pair of $y_i,y_j$ is adjacent to a unique $z_i$.

来源字段：https://www.erdosproblems.com/926

14581 | EP-927

### Erdős Problem #927: Graph Theory

S / solved | Erdős 问题

Let $g(n)$ be the maximum number of different sizes of cliques that can occur in a graph on $n$ vertices. Estimate $g(n)$ - in particular, is it true that\[g(n)=n-\log_2n-\log_*(n)+O(1),\]where $\log_*(n)$ is the number of iterated logarithms such that $\log\cdots \log n <1$.

来源字段：https://www.erdosproblems.com/927

14582 | EP-96

### Erdős Problem #96

O / open | Erdős 问题

If $n$ points in $\mathbb{R}^2$ form a convex polygon then there are $O(n)$ many pairs which are distance $1$ apart.

来源字段：https://www.erdosproblems.com/96

14583 | EP-986

### Erdős Problem #986

S / solved | Erdős 问题

For any fixed $s\geq 3$,\[R(s,k) \gg \frac{k^{s-1}}{(\log k)^c}\]for some constant $c=c(s)>0$.

来源字段：https://www.erdosproblems.com/986

14584 | EP-99

### Erdős Problem #99

O / open | Erdős 问题

Let $A\subseteq\mathbb{R}^2$ be a set of $n$ points with minimum distance equal to 1, chosen to minimise the diameter of $A$. If $n$ is sufficiently large then must there be three points in $A$ which form an equilateral triangle of size 1?

来源字段：https://www.erdosproblems.com/99

14585 | EP-993

### Erdős Problem #993: The independent set sequence of any tree or forest is unimodal

O / open | Erdős 问题

The independent set sequence of any tree or forest is unimodal.

In other words, if $i_k(G)$ counts the number of independent sets of vertices of size $k$ in a graph $G$, and $T$ is any tree or forest, then for some $m\geq 0$ $$i_{0}(T)\leq i_{1}(T)\leq\cdots\leq i_{m}(T)\geq i_{m+1}(T)\geq i_{m+2}(T)\geq\cdots.$$

来源字段：https://www.erdosproblems.com/993

14586 | GRAPH-030

### Erdős-Gyárfás Conjecture

P / partially_solved | 独立 / 未分配集合

Does every graph with minimum degree 3 contain cycles of lengths that are powers of 2?

背景链接：https://arxiv.org/abs/2410.22842

14587 | OPG-52200

### Erdős-Posa property for long directed cycles

P / partially_solved | Open Problem Garden

Conjecture Let $\ell \geq 2$ be an integer. For every integer $n\geq 0$, there exists an integer $t_n=t_n(\ell)$ such that for every digraph $D$, either $D$ has a $n$ pairwise-disjoint directed cycles of length at least $\ell$, or there exists a set $T$ of at most $t_n$ vertices such that $D-T$ has no directed cycles of length at least $\ell$.

背景链接：http://www.openproblemgarden.org/op/erdos_posa_property_for_long_directed_cycles

14588 | OPG-55984

### Erdős-Faber-Lovász conjecture

P / partially_solved | Open Problem Garden

Conjecture If $G$ is a simple graph which is the union of $k$ pairwise edge-disjoint complete graphs, each of which has $k$ vertices, then the chromatic number of $G$ is $k$.

背景链接：http://www.openproblemgarden.org/op/erdos_faber_lovasz_conjecture

14589 | OWR-10252930-004

### Erdős-Gallai Cycle-Decomposition Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Can every $n$-vertex graph be decomposed into $O(n)$ cycles and edges?

来源字段：https://doi.org/10.4171/owr/2022/22

14590 | OWR-1386-017

### Erdős-Gyárfás Power-of-Two Cycle Conjecture

O / open | Oberwolfach 报告问题汇编

Does every graph of minimum degree at least three contain a cycle whose length is a power of two?

来源字段：https://doi.org/10.4171/owr/2006/48

14591 | OWR-12697684-011

### Erdős-Hajnal Clique-or-Independent-Set Conjecture

O / open | Oberwolfach 报告问题汇编

别名：Repeated Erdős-Hajnal Conjecture Context [OWR-12697684-012]

For every fixed graph $H$, does there exist $c(H)>0$ such that every $H$-free graph on $n$ vertices contains a clique or an independent set of size at least $n^{c(H)}$?

来源字段：https://doi.org/10.4171/owr/2023/1

14592 | GRAPH-004

### Erdős-Hajnal Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Erdős-Hajnal Conjecture [GRAPH-031]

For any fixed graph $H$, do $H$-free graphs contain large cliques or independent sets?

背景链接：https://en.wikipedia.org/wiki/Erd%C5%91s%E2%80%93Hajnal_conjecture

14593 | OWR-4791-013

### Erdős-Hajnal Conjecture for Tournaments

O / open | Oberwolfach 报告问题汇编

For every tournament $H$, there exists $\varepsilon<1$ and a constant $c$ such that every $H$-free tournament $G$ satisfies $\chi(G)\leq c|V(G)|^{\varepsilon}$, where $\chi(G)$ is the minimum number of transitive subtournaments partitioning $V(G)$.

来源字段：https://doi.org/10.4171/owr/2011/01

14594 | OWR-16763-007

### Erdős-Pósa Properties of Induced Subdivisions

O / open | Oberwolfach 报告问题汇编

For which graphs $H$ do induced $H$-subdivisions have the Erdős-Pósa property? In particular, do even cycles have it, and what happens for $H=K_4$ or for a fan whose path has length $3$?

来源字段：https://doi.org/10.4171/owr/2019/1

14595 | GRAPH-002

### Eternal Domination vs Domination Number

O / open | 独立 / 未分配集合

Does there exist a graph where the dominating number equals the eternal dominating number and both are less than the clique covering number?

背景链接：https://en.wikipedia.org/wiki/Eternal_dominating_set

14596 | AMR-054-0005

### Euclidean Minimum Spanning Tree

P / partially_solved | AMR 问题清单汇编

Can the Euclidean minimum spanning tree (MST) of $n$ points in $\mathbb{R}^d$ be computed in time close to the lower bound of $\Omega(n \log n)$?

背景链接：https://topp.openproblem.net/p5

14597 | OPG-46943

### Every 4-connected toroidal graph has a Hamilton cycle

O / open | Open Problem Garden

Conjecture Every 4-connected toroidal graph has a Hamilton cycle.

背景链接：http://www.openproblemgarden.org/op/every_4_connected_toroidal_graph_has_a_hamilton_cycle

14598 | OPG-47285

### Every prism over a 3-connected planar graph is hamiltonian.

P / partially_solved | Open Problem Garden

Conjecture If $G$ is a $3$-connected planar graph, then $G\square K_2$ has a Hamilton cycle.

背景链接：http://www.openproblemgarden.org/op/every_prism_over_a_3_connected_planar_graph_is_hamiltonian

14599 | OPG-37229

### Exact colorings of graphs

P / partially_solved | Open Problem Garden

别名：Graphs of exact colorings [OPG-57824]

Conjecture For $c \geq m \geq 1$, let $P(c,m)$ be the statement that given any exact $c$-coloring of the edges of a complete countably infinite graph (that is, a coloring with $c$ colors all of which must be used at least once), there exists an exactly $m$-colored countably infinite complete subgraph. Then $P(c,m)$ is true if and only if $m=1$, $m=2$, or $c=m$.

背景链接：http://www.openproblemgarden.org/op/exact_colorings_of_graphs

14600 | AMR-084-0002

### Exceptional parameters without D(n)-quadruples

O / open | AMR 问题清单汇编

For each $n\in\{-3,3,5,8,12,20\}$, prove that no set of four distinct positive integers has property $D(n)$, meaning that every pairwise product plus $n$ is a square. (The page's original exceptional set also contained $-1$ and $-4$, which subsequent results on the same page settle.)

背景链接：https://web.math.pmf.unizg.hr/~duje/dtuples.html

14601 | OWR-2654831-003

### Existence and Uniqueness of Perfect T-Embeddings

P / partially_solved | Oberwolfach 报告问题汇编

Does every nondegenerate weighted bipartite graph $G$ with sphere topology and a marked face have an augmented dual $G^*$ admitting a perfect t-embedding, equivalently a p-embedding? Is it unique up to Lorentz isometries of $\mathbb R^{2+1}$ preserving the unit hyperboloid $\mathbb H$?

来源字段：https://doi.org/10.4171/owr/2020/35

14602 | AMR-084-0007

### Existence of a rational Diophantine septuple

P / partially_solved | AMR 问题清单汇编

Does there exist a rational Diophantine septuple, that is, seven nonzero rational numbers whose pairwise products plus $1$ are rational squares?

背景链接：https://web.math.pmf.unizg.hr/~duje/dtuples.html

14603 | AMR-084-0009

### Existence of a strong rational Diophantine quadruple

O / open | AMR 问题清单汇编

Does there exist a set of four nonzero rational numbers $\{a_1,a_2,a_3,a_4\}$ such that $a_i a_j+1$ is a rational square for every $1\le i,j\le4$, including $i=j$?

背景链接：https://web.math.pmf.unizg.hr/~duje/dtuples.html

14604 | OWR-10252930-015

### Existence of the Diagonal Ramsey Growth Limit

O / open | Oberwolfach 报告问题汇编

If $R(n)=R(n,n)$ denotes the diagonal Ramsey number, does the limit $$\lim_{n\to\infty}R(n)^{1/n}$$ exist?

来源字段：https://doi.org/10.4171/owr/2022/22

14605 | OWR-12861-023

### Existential Completeness in Triangle-Free Graphs

O / open | Oberwolfach 报告问题汇编

A triangle-free graph is $k$-existentially complete if, for every independent set $A$ and disjoint set $B$ with $|A|+|B|\leq k$, there is a vertex adjacent to every vertex of $A$ and to no vertex of $B$. Is there a constant $r$ such that no triangle-free graph is $r$-existentially complete?

来源字段：https://doi.org/10.4171/owr/2014/01

14606 | OWR-16160-004

### Expanders in Superreflexive Spaces and Uniform Hilbertian Spheres

O / open | Oberwolfach 报告问题汇编

Let $Y$ be a superreflexive Banach space and $(G_n)$ a sequence of expander graphs. Must $(G_n)$ fail to equi-coarsely embed into $Y$? Is the unit sphere of every separable superreflexive Banach space uniformly homeomorphic to the unit sphere of $\ell_2$?

来源字段：https://doi.org/10.4171/owr/2018/19

14607 | AMR-087-0004

### Explicit class-group realization of finite-field discrete logarithms

P / partially_solved | AMR 问题清单汇编

Make the proposed realization of existing finite-field discrete-logarithm systems inside class groups explicit for practical systems.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14608 | AMR-087-0008

### Explicit cocycles and invariants for split local algebras

O / open | AMR 问题清单汇编

Explicitly describe the cocycle $c_u$, equivalently fast-compute invariants of local algebras split by the generalized-dihedral extensions specified in the slides.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14609 | AMR-087-0094

### Explicit test homogeneous spaces of prescribed ramification

O / open | AMR 问题清单汇编

Explicitly construct test elements or principal homogeneous spaces having prescribed ramification and a prescribed large prime order $\ell$.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14610 | OWR-14298163-023

### Extremal Bipartite Containment of B-t

O / open | Oberwolfach 报告问题汇编

Let $B_t$ have bipartition $L\sqcup R$, where $L=\{x_1,\ldots,x_t,y_1,\ldots,y_t,z_1,\ldots,z_t\}$, $R=\{w_{ijk}:1\leq i,j,k\leq t\}$, and $w_{ijk}$ is adjacent to $x_i,y_j,z_k$. Do there exist constants $\sigma<3$ and $C>0$ such that every sufficiently large bipartite graph $G=(A\sqcup B,E)$ with $|A|^\sigma<|B|$ and $|E|>C|B|$ contains $B_t$?

来源字段：https://doi.org/10.4171/owr/2024/3

14611 | OWR-14299525-004

### Extremal Diameters of Symmetric-Group Cayley Graphs

O / open | Oberwolfach 报告问题汇编

Prove the proposed upper bound $\frac{1}{2}n^2+4n$ for diameters of undirected Cayley graphs of $S_n$. Prove or disprove that the explicit square-with-whiskers generating families maximize the diameter for every $n$, extending the computations through $n=15$.

来源字段：https://doi.org/10.4171/owr/2025/43

14612 | OWR-14604-018

### Extremal Graphs Maximizing Non-Triangular Edges

P / partially_solved | Oberwolfach 报告问题汇编

For integers $a,b,c\geq0$, define $G(a,b,c)$ on a vertex partition consisting of a clique $A$ of size $a$ and independent sets $B,C$ of sizes $b,c$, with all edges between $A$ and $B$ and between $B$ and $C$, and no edges between $A$ and $C$. For every $n$ and $e$, can an $n$-vertex graph with at least $e$ edges that maximizes the number of nontriangular edges always be chosen as some $G(a,b,c)$?

来源字段：https://doi.org/10.4171/owr/2016/22

14613 | AMR-084-0005

### Extremal parameters for D(n)-quintuples

P / partially_solved | AMR 问题清单汇编

Determine the least positive integer $n_1$ and the greatest negative integer $n_2$ for which a $D(n_i)$-quintuple exists.

背景链接：https://web.math.pmf.unizg.hr/~duje/dtuples.html

14614 | OPG-37305

### Extremal problem on the number of tree endomorphism

P / partially_solved | Open Problem Garden

Conjecture An endomorphism of a graph is a mapping on the vertex set of the graph which preserves edges. Among all the $n$ vertices' trees, the star with $n$ vertices has the most endomorphisms, while the path with $n$ vertices has the least endomorphisms.

背景链接：http://www.openproblemgarden.org/op/extremal_problem_on_the_number_of_tree_endomorphism

14615 | OWR-9790352-014

### Extremal Subgraph Counts in Sparse Graph Classes

O / open | Oberwolfach 报告问题汇编

For a graph class $\mathcal G$ and a graph $H$, determine $\operatorname{ex}(H,\mathcal G,n)$, the maximum number of subgraphs isomorphic to $H$ in an $n$-vertex member of $\mathcal G$.

来源字段：https://doi.org/10.4171/owr/2022/1

14616 | AMR-087-0019

### Factoring a three-prime integer from fewer known bits

S / solved | AMR 问题清单汇编

Factor $N=pqr$ from fewer known bits of its prime factors.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14617 | AMR-087-0020

### Factoring from nonconsecutive known bits

P / partially_solved | AMR 问题清单汇编

Develop methods to factor an integer when the known bits of its factors are nonconsecutive.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14618 | AMR-087-0018

### Factoring structured semiprimes from fewer known bits

P / partially_solved | AMR 问题清单汇编

Factor $N=p^rq^s$ with $r\approx s$ using fewer known bits of the factors than existing methods require.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14619 | OPG-140

### Faithful cycle covers

P / partially_solved | Open Problem Garden

Conjecture If $G = (V,E)$ is a graph, $p: E \rightarrow {\mathbb Z}$ is admissable, and $p(e)$ is even for every $e \in E(G)$, then $(G,p)$ has a faithful cover.

背景链接：http://www.openproblemgarden.org/op/faithful_cycle_covers

14620 | AMR-087-0033

### Fast construction of five-term geometric progressions for NFS

P / partially_solved | AMR 问题清单汇编

For large $N$, efficiently find the required short five-term geometric progressions modulo $N$ that avoid first- and second-order recurrence, thereby producing two cubic NFS polynomials with a common root and coefficients $O(N^{1/6})$.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14621 | AMR-087-0001

### Fast evaluation of high-degree elliptic-curve isogenies

P / partially_solved | AMR 问题清单汇编

Given an elliptic curve $E/\mathbb{F}_q$ and $P\in E(\mathbb{F}_q)$, characterize maps or isogenies $\psi:E\to E'$ for which $\psi(P)$ can be evaluated in time polynomial in $\log\deg\psi$ and $\log q$, ideally $O(\log\deg\psi\,\log^2q)$.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14622 | OWR-1536-002

### Fast Maker Constructions of Regular Graphs

O / open | Oberwolfach 报告问题汇编

For a fixed $d>0$, are there infinitely many $d$-regular graphs $G$ for which Maker can win $(K_N,G)$ in at most $f(d)|V(G)|$ rounds whenever $N$ is sufficiently large, with $f(d)\leq(1+o(1))d$?

来源字段：https://doi.org/10.4171/owr/2007/20

14623 | AMR-087-0006

### Fast Tate-Lichtenbaum pairing computation

S / solved | AMR 问题清单汇编

Develop a fast algorithm to compute the Tate-Lichtenbaum pairing $T_n$.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14624 | AMR-087-0024

### Faster Coppersmith root methods

O / open | AMR 问题清单汇编

Improve the running time of Coppersmith-type methods for finding small modular or integer roots.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14625 | AMR-087-0032

### Faster cyclic-lattice reduction

O / open | AMR 问题清单汇编

Speed up rotation-augmented reduction algorithms for cyclic or NTRU lattices.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14626 | OWR-1189-010

### Faster Distributed Coloring from Consistent Edge Orientations

O / open | Oberwolfach 报告问题汇编

In the distributed graph-colouring model with globally consistent edge orientations, do the orientations permit faster colouring algorithms?

来源字段：https://doi.org/10.4171/owr/2006/07

14627 | AMR-087-0087

### Faster infrastructure discrete logarithms and point counting

P / partially_solved | AMR 问题清单汇编

Use a baby-step/giant-step infrastructure framework to speed infrastructure discrete logarithms or point counting by a polynomial factor.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14628 | AMR-087-0052

### Faster pairing computation

S / solved | AMR 问题清单汇编

Speed up the computation of cryptographic pairings.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14629 | OPG-37218

### Finding k-edge-outerplanar graph embeddings

P / partially_solved | Open Problem Garden

Conjecture It has been shown that a $k$-outerplanar embedding for which $k$ is minimal can be found in polynomial time. Does a similar result hold for $k$-edge-outerplanar graphs?

背景链接：http://www.openproblemgarden.org/op/finding_k_edge_outerplanar_graph_embeddings

14630 | OWR-4212-011

### Finite Forcibility and Structure of Extremal Graph Limits

P / partially_solved | Oberwolfach 报告问题汇编

别名：Source-Mismatched Extremal-Graph Questions [OWR-4213-001]

What is the possible structure of extremal graphs; in particular, does every extremal graph problem formulated by optimizing subgraph densities subject to fixed densities have an extremal family with a finitely forcible graphon limit, and can extremal graphs be completely characterized?

来源字段：https://doi.org/10.4171/owr/2010/11

14631 | AMR-099-0061

### Finite graphs whose every ball is an expander

O / open | AMR 问题清单汇编

Does there exist a family $(G_n)$ of finite $d$-regular graphs with $|G_n|\to\infty$ and a constant $h>0$ such that every induced metric ball in every $G_n$ has edge-expansion constant at least $h$?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=6

14632 | OWR-16763-015

### Finite Obstruction Sets for Vertex-Minor-Closed Graph Classes

O / open | Oberwolfach 报告问题汇编

Let $\mathcal F$ be a proper graph class closed under isomorphism and vertex minors. Does $\mathcal F$ have only finitely many excluded vertex minors, up to isomorphism?

来源字段：https://doi.org/10.4171/owr/2019/1

14633 | AMR-087-0086

### Finiteness of a Shafarevich-Tate group needed by the lifting method

O / open | AMR 问题清单汇编

Prove finiteness of the Shafarevich-Tate group of the elliptic-curve lift required by the Huang-Raskind method, in the general cases where it is not known.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14634 | AMR-084-0004

### Finiteness of D(n)-quadruples for nonsquare n

O / open | AMR 问题清单汇编

For every nonzero integer $n$ that is not a square, are there only finitely many $D(n)$-quadruples?

背景链接：https://web.math.pmf.unizg.hr/~duje/dtuples.html

14635 | OWR-14298163-019

### Finiteness of Generalized Moore Graphs

O / open | Oberwolfach 报告问题汇编

For a fixed constant $c>0$, are there only finitely many regular graphs satisfying $\operatorname{girth}(G)=(2-c)\operatorname{diam}(G)$, apart from the known complete and odd-cycle families?

来源字段：https://doi.org/10.4171/owr/2024/3

14636 | AMR-084-0003

### Finiteness of parameters admitting at most two D(n)-quadruples

O / open | AMR 问题清单汇编

Let $U$ be the set of integers $n\not\equiv2\pmod4$ for which there are at most two distinct $D(n)$-quadruples. Is $U$ finite?

背景链接：https://web.math.pmf.unizg.hr/~duje/dtuples.html

14637 | OWR-16763-005

### Five-Cycle Near-Triangulation Decomposition Conjecture

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Five-Cycle Near-Triangulation Conjecture [OWR-16763-004]

For every plane near-triangulation $G$ whose outer face is bounded by a $5$-cycle, is $n_G^\star$ a nonnegative linear combination of $n_{G_1},\ldots,n_{G_{11}}$?

来源字段：https://doi.org/10.4171/owr/2019/1

14638 | OWR-4212-006

### Fixed-Minor Testing for Finite-Field-Representable Matroids

P / partially_solved | Oberwolfach 报告问题汇编

For any finite field $F$ and any $F$-representable matroid $N$, is there a polynomial-time algorithm to test whether an $F$-representable matroid contains $N$ as a minor?

来源字段：https://doi.org/10.4171/owr/2010/11

14639 | OWR-11451-008

### Flows in Signed Graphs and Binary Matroids

P / partially_solved | Oberwolfach 报告问题汇编

For a signed graph or binary matroid with nonnegative edge weights satisfying the cut condition, does there exist an integer flow, a fractional flow, and, when the weights are Eulerian, an integer flow?

来源字段：https://doi.org/10.4171/owr/2011/53

14640 | OWR-1532-007

### Forbidden Minors for Knotless Graph Embeddings

O / open | Oberwolfach 报告问题汇编

Find additional forbidden minors for the graphs that admit knotless embeddings in $\mathbb{R}^3$, with the goal of determining the complete minor-obstruction set.

来源字段：https://doi.org/10.4171/owr/2007/16

14641 | OPG-37379

### Forcing a $K_6$-minor

P / partially_solved | Open Problem Garden

Conjecture Every graph with minimum degree at least 7 contains a $K_6$-minor.

Conjecture Every 7-connected graph contains a $K_6$-minor.

背景链接：http://www.openproblemgarden.org/op/forcing_a_k_6_minor

14642 | OPG-59911

### Forcing a 2-regular minor

P / partially_solved | Open Problem Garden

Conjecture Every graph with average degree at least $\frac{4}{3}t-2$ contains every 2-regular graph on $t$ vertices as a minor.

背景链接：http://www.openproblemgarden.org/op/forcing_a_2_regular_minor

14643 | OWR-4791-019

### Forcing Conjecture for Bipartite Cyclic Graphs

O / open | Oberwolfach 报告问题汇编

Forcing conjecture: a graph $H$ is forcing if and only if $H$ is bipartite and contains a cycle.

来源字段：https://doi.org/10.4171/owr/2011/01

14644 | OWR-4791-018

### Forcing Graphs for Quasirandomness

O / open | Oberwolfach 报告问题汇编

For which graphs $H$ does the following implication hold: if a graph sequence $(G_n)$ satisfies $t_{K_2}(G_n)=(1+o(1))p$ and $t_H(G_n)=(1+o(1))p^{|E(H)|}$, then $(G_n)$ is quasirandom with density $p$?

来源字段：https://doi.org/10.4171/owr/2011/01

14645 | OWR-12330-018

### Fork Graphs and Directed Cycle Double Covers

O / open | Oberwolfach 报告问题汇编

How numerous are fork graphs, and is the directed cycle double cover conjecture as hard for fork graphs as for general cubic bridgeless graphs?

来源字段：https://doi.org/10.4171/owr/2013/02

14646 | OWR-16763-006

### Fractional Coloring of Subcubic Triangle-Free Planar Graphs

P / partially_solved | Oberwolfach 报告问题汇编

Does every planar triangle-free graph with maximum degree at most 3 have fractional chromatic number at most 8/3?

来源字段：https://doi.org/10.4171/owr/2019/1

14647 | AMR-087-0036

### Frequency of cyclic elliptic-curve groups

P / partially_solved | AMR 问题清单汇编

How often is the group of a random elliptic curve over $\mathbb{F}_q$ cyclic?

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14648 | OPG-37163

### Friendly partitions

P / partially_solved | Open Problem Garden

A friendly partition of a graph is a partition of the vertices into two sets so that every vertex has at least as many neighbours in its own class as in the other.

Problem Is it true that for every $r$, all but finitely many $r$-regular graphs have friendly partitions?

背景链接：http://www.openproblemgarden.org/op/friendly_partitions

14649 | OWR-9790352-004

### Gallai’s Path-Decomposition Conjecture

O / open | Oberwolfach 报告问题汇编

Can the edge set of every connected graph on $n$ vertices be decomposed into at most $\lceil n/2\rceil$ paths?

来源字段：https://doi.org/10.4171/owr/2022/1

14650 | AMR-087-0055

### Genus-four pairing speed-security tradeoff

O / open | AMR 问题清单汇编

Determine the exact computational-speed and security tradeoff for genus-four curves used in pairing cryptography.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14651 | OPG-500

### Geodesic cycles and Tutte's Theorem

O / open | Open Problem Garden

Problem If $G$ is a $3$-connected finite graph, is there an assignment of lengths $\ell: E(G) \to \mathb R^+$ to the edges of $G$, such that every $\ell$-geodesic cycle is peripheral?

背景链接：http://www.openproblemgarden.org/op/geodesic_cycles_and_tuttes_theorem

14652 | OWR-1536-003

### Global Strategies in Irregular Graph-Construction Games

O / open | Oberwolfach 报告问题汇编

In graph-construction games with highly irregular target graphs, can Maker use a global strategy that postpones deciding which board vertex represents which vertex of the target $G$, and thereby obtain an asymptotically faster construction than local strategies?

来源字段：https://doi.org/10.4171/owr/2007/20

14653 | AMR-087-0009

### Globalizing prescribed local Brauer classes

P / partially_solved | AMR 问题清单汇编

Explicitly construct global algebras or Brauer classes with prescribed local data, especially when the local splitting fields are dihedral.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14654 | GRAPH-017

### GNRS Conjecture

P / partially_solved | 独立 / 未分配集合

Do minor-closed graph families have $\ell_1$ embeddings with bounded distortion?

背景链接：https://doi.org/10.1145/1536414.1536450

14655 | OPG-2242

### Goldberg's conjecture

P / partially_solved | Open Problem Garden

The overfull parameter is defined as follows: $$ w(G) = \max_{H \subseteq G} \left\lceil \frac{ |E(H)| }{ \lfloor \tfrac{1}{2} |V(H)| \rfloor} \right\rceil. $$

Conjecture Every graph $G$ satisfies $\chi'(G) \le \max\{ \Delta(G) + 1, w(G) \}$.

背景链接：http://www.openproblemgarden.org/op/goldbergs_conjecture

14656 | OPG-37323

### Good Edge Labelings

P / partially_solved | Open Problem Garden

Question What is the maximum edge density of a graph which has a good edge labeling?

We say that a graph is good-edge-labeling critical, if it has no good edge labeling, but every proper subgraph has a good edge labeling.

Conjecture For every $c<4$, there is only a finite number of good-edge-labeling critical graphs with average degree less than $c$.

背景链接：http://www.openproblemgarden.org/op/good_edge_labelings

14657 | OPG-164

### Graham's conjecture on tree reconstruction

P / partially_solved | Open Problem Garden

Problem for every graph $G$, we let $L(G)$ denote the line graph of $G$. Given that $G$ is a tree, can we determine it from the integer sequence $|V(G)|, |V(L(G))|, |V(L(L(G)))|, \ldots$?

背景链接：http://www.openproblemgarden.org/op/grahams_conjecture_on_tree_reconstruction

14658 | GRAPH-003

### Graham's Pebbling Conjecture

O / open | 独立 / 未分配集合

Is the pebbling number of the Cartesian product of two graphs at least the product of their pebbling numbers?

背景链接：https://en.wikipedia.org/wiki/Graham%27s_pebbling_conjecture

14659 | OWR-12861-020

### Graph Blocks and Coset Partition Refinements

O / open | Oberwolfach 报告问题汇编

Which properties force a graph $G$ to contain a $k$-block, meaning a maximal set of at least $k$ vertices no two of which can be separated by fewer than $k$ other vertices? How does $$\beta(G)=\max\{k:G\text{ has a }k\text{-block}\}$$ relate to other graph invariants?

来源字段：https://doi.org/10.4171/owr/2014/01

14660 | OWR-1189-004

### Graph Classes with Recursive Chromatic-Polynomial Algorithms

O / open | Oberwolfach 报告问题汇编

Characterize, by simple properties or forbidden minors, the graphs whose chromatic polynomials can be computed efficiently by recursively passing to complements. The desired family includes graphs of bounded treewidth and graphs whose complements have bounded treewidth.

来源字段：https://doi.org/10.4171/owr/2006/07

14661 | GRAPH-007

### Graph Coloring Game Monotonicity

O / open | 独立 / 未分配集合

If Alice has a winning strategy for the vertex coloring game with k colors, does she have one for k+1 colors?

背景链接：https://en.wikipedia.org/wiki/Graph_coloring_game

14662 | OWR-14299088-006

### Graph Distances in Associahedra

P / partially_solved | Oberwolfach 报告问题汇编

Given two vertices of an associahedron, determine their graph distance, equivalently the minimum number of flips or rotations connecting the corresponding triangulations or trees. Develop an exact structural characterization or efficient algorithm beyond the known complexity classification.

来源字段：https://doi.org/10.4171/owr/2025/9

14663 | OWR-14299905-027

### Graph Removal in the $L_infty$ Metric

P / partially_solved | Oberwolfach 报告问题汇编

Does graph removal remain valid when distance from being $H$-free is measured by the $L_\infty$ norm of removed-edge degrees rather than by total edit distance?

来源字段：https://doi.org/10.4171/owr/2026/1

14664 | OPG-36936

### Graphs with a forbidden induced tree are chi-bounded

P / partially_solved | Open Problem Garden

Say that a family ${\mathcal F}$ of graphs is $\chi$-bounded if there exists a function $f: {\mathbb N} \rightarrow {\mathbb N}$ so that every $G \in {\mathcal F}$ satisfies $\chi(G) \le f (\omega(G))$.

Conjecture For every fixed tree $T$, the family of graphs with no induced subgraph isomorphic to $T$ is $\chi$-bounded.

背景链接：http://www.openproblemgarden.org/op/graphs_with_a_forbidden_induced_tree_are_chi_bounded

14665 | AMR-099-0014

### Grid-or-tree embeddings in superlinear Cayley graphs

O / open | AMR 问题清单汇编

Must every Cayley graph of superlinear growth contain, up to rough isometric embedding, either the square grid $\mathbb{Z}^2$ or an infinite binary tree?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=17

14666 | AMR-087-0083

### Groups with infeasible inversion

O / open | AMR 问题清单汇编

Construct groups in which inversion is infeasible under reasonable cryptographic assumptions.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14667 | AMR-087-0093

### Growing-degree improvements to the lifting attack

O / open | AMR 问题清单汇编

Can the lifting attack be improved by allowing the number-field degree $[K:\mathbb{Q}]$ to grow?

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14668 | OWR-1795-006

### Growth Expansion and Mixing in Groups and Graphs

P / partially_solved | Oberwolfach 报告问题汇编

For a graph $\Gamma$, how many vertices can be reached from a given vertex in a given time? For a generating set $A$ of an infinite group $G$, how does $B(t)=|\{\text{products of at most }t\text{ elements of }A\}|$ grow? For the random walk generated by $A$ on a finite group, how quickly does it approach equidistribution, and what lower bounds hold for the gap between the two largest adjacency eigenvalues? For $A$ in an abelian group, how large can $A+A$ be relative to $A$? For $A$ in a general group, how large can $A\cdot A\cdot A$ be relative to $A$?

来源字段：https://doi.org/10.4171/owr/2008/14

14669 | OPG-177

### Grunbaum's Conjecture

S / solved | Open Problem Garden

Conjecture If $G$ is a simple loopless triangulation of an orientable surface, then the dual of $G$ is 3-edge-colorable.

背景链接：http://www.openproblemgarden.org/op/grunbaums_conjecture

14670 | GRAPH-021

### Guy's Crossing Number Conjecture

P / partially_solved | 独立 / 未分配集合

Is the crossing number of the complete graph $K_n$ equal to the value given by Guy's formula?

背景链接：https://doi.org/10.1002/jgt.20249

14671 | OWR-10252930-017

### Gyárfás Tree-Packing Conjecture

O / open | Oberwolfach 报告问题汇编

Can $K_n$ be decomposed into any family of trees $T_1,\ldots,T_n$ in which $T_i$ has $i$ vertices?

来源字段：https://doi.org/10.4171/owr/2022/22

14672 | GRAPH-011

### Gyárfás-Sumner Conjecture

P / partially_solved | 独立 / 未分配集合

Is every graph class defined by excluding one fixed tree as an induced subgraph χ-bounded?

背景链接：https://doi.org/10.1002/jgt.22511

14673 | OWR-15206-005

### Gyárfás-Sumner Conjecture for Forest-Free Graphs

O / open | Oberwolfach 报告问题汇编

For every forest $H$, does there exist a function $f_H$ such that every $H$-free graph $G$ satisfies $\chi(G)\leq f_H(\omega(G))$?

来源字段：https://doi.org/10.4171/owr/2017/1

14674 | GT-001

### Hadwiger Conjecture

P / partially_solved | 独立 / 未分配集合

别名：List Hadwiger Conjecture [OPG-59943]

Every graph with chromatic number $k$ has a $K_k$ minor (where $K_k$ is the complete graph on $k$ vertices).

背景链接：https://mathworld.wolfram.com/HadwigerConjecture.html

14675 | OWR-16763-010

### Hadwiger’s Coloring Conjecture and Its Odd-Minor Variant

O / open | Oberwolfach 报告问题汇编

For every integer $t\geq1$, is every graph with no $K_{t+1}$ minor properly $t$-colorable? Is every graph with no odd $K_{t+1}$ minor properly $t$-colorable?

来源字段：https://doi.org/10.4171/owr/2019/1

14676 | OWR-9790352-018

### Hadwiger’s Graph-Coloring Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Does every graph $G$ with chromatic number $\chi(G)=t$ contain $K_t$ as a minor?

来源字段：https://doi.org/10.4171/owr/2022/1

14677 | OPG-348

### Half-integral flow polynomial values

O / open | Open Problem Garden

Let $\Phi(G,x)$ be the flow polynomial of a graph $G$. So for every positive integer $k$, the value $\Phi(G,k)$ equals the number of nowhere-zero $k$-flows in $G$.

Conjecture $\Phi(G,5.5) > 0$ for every 2-edge-connected graph $G$.

背景链接：http://www.openproblemgarden.org/op/half_integral_flow_polynomial_values

14678 | OWR-1183-005

### Hamilton Circles in Infinite Planar Graphs

O / open | Oberwolfach 报告问题汇编

Does every $4$-connected, planar, locally finite graph have a Hamilton circle in its topological compactification?

来源字段：https://doi.org/10.4171/owr/2006/01

14679 | OWR-1532-005

### Hamilton Circles in Squares and Cubes of Infinite Graphs

O / open | Oberwolfach 报告问题汇编

For a countable two-connected graph $G$, does the Freudenthal compactification $|G^2|$ contain a Hamilton circle? For a countable connected graph $G$, does $|G^3|$ contain a Hamilton circle? Here $G^k$ joins vertices at graph distance at most $k$, and a Hamilton circle contains every vertex and every end.

来源字段：https://doi.org/10.4171/owr/2007/16

14680 | OPG-47028

### Hamilton cycle in small d-diregular graphs

P / partially_solved | Open Problem Garden

An directed graph is $k$-diregular if every vertex has indegree and outdegree at least $k$.

Conjecture For $d >2$, every $d$-diregular oriented graph on at most $4d+1$ vertices has a Hamilton cycle.

背景链接：http://www.openproblemgarden.org/op/hamilton_cycle_in_small_d_diregular_graphs

14681 | OPG-47356

### Hamilton decomposition of prisms over 3-connected cubic planar graphs

P / partially_solved | Open Problem Garden

Conjecture Every prism over a $3$-connected cubic planar graph can be decomposed into two Hamilton cycles.

背景链接：http://www.openproblemgarden.org/op/decomposing_the_prism_of_a_3_connected_cubic_planar_graphs_in_hamilton_cycles

14682 | OPG-485

### Hamiltonian cycles in line graphs

P / partially_solved | Open Problem Garden

Conjecture Every 4-connected line graph is hamiltonian.

背景链接：http://www.openproblemgarden.org/op/hamiltonian_cycles_in_line_graphs

14683 | OPG-488

### Hamiltonian cycles in line graphs of infinite graphs

O / open | Open Problem Garden

Conjecture

- If $G$ is a 4-edge-connected locally finite graph, then its line graph is hamiltonian. - If the line graph $L(G)$ of a locally finite graph $G$ is 4-connected, then $L(G)$ is hamiltonian.

背景链接：http://www.openproblemgarden.org/op/hamiltonian_cycles_in_line_graphs_of_infinite_graphs

14684 | OPG-490

### Hamiltonian cycles in powers of infinite graphs

O / open | Open Problem Garden

Conjecture

- If $G$ is a countable connected graph then its third power is hamiltonian. - If $G$ is a 2-connected countable graph then its square is hamiltonian.

背景链接：http://www.openproblemgarden.org/op/hamiltonian_cycles_in_powers_of_infinite_graphs

14685 | GRAPH-042

### Hamiltonian Decomposition of Hypergraphs

P / partially_solved | 独立 / 未分配集合

Do complete k-uniform hypergraphs admit Hamiltonian decompositions into tight cycles?

背景链接：https://arxiv.org/abs/1403.7932

14686 | OPG-161

### Hamiltonian paths and cycles in vertex transitive graphs

P / partially_solved | Open Problem Garden

Problem Does every connected vertex-transitive graph have a Hamiltonian path?

背景链接：http://www.openproblemgarden.org/op/hamiltonian_paths_and_cycles_in_vertex_transitive_graphs

14687 | OWR-12697684-010

### Hamiltonicity of Cayley and Vertex-Transitive Graphs

P / partially_solved | Oberwolfach 报告问题汇编

Is every connected Cayley graph Hamiltonian? More generally, which connected vertex-transitive graphs fail to have a Hamiltonian path or cycle?

来源字段：https://doi.org/10.4171/owr/2023/1

14688 | OPG-2095

### Hamiltonicity of Cayley graphs

P / partially_solved | Open Problem Garden

Does every finite connected Cayley graph with at least three vertices have a Hamiltonian cycle?

背景链接：http://www.openproblemgarden.org/op/hamiltonicity_of_cayley_graphs

14689 | OWR-14299081-029

### Hamiltonicity of Connected Vertex-Transitive Graphs

O / open | Oberwolfach 报告问题汇编

Does every connected vertex-transitive graph contain a Hamilton path and, apart from the five known exceptions, a Hamilton cycle? In particular, is every connected Cayley graph Hamiltonian, and is a random connected Cayley graph generated by $C\log|G|$ elements asymptotically almost surely Hamiltonian?

来源字段：https://doi.org/10.4171/owr/2025/1

14690 | OWR-14299905-033

### Hamiltonicity of Highly Connected Induced-Star-Free Graphs

O / open | Oberwolfach 报告问题汇编

Does there exist a function $f$ such that every $f(k)$-vertex-connected graph with no induced $K_{1,k}$ is Hamiltonian?

来源字段：https://doi.org/10.4171/owr/2026/1

14691 | OWR-2090-001

### Hanani-Tutte Theorems on Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

The projective-plane Hanani-Tutte statement says that every drawing of a graph not embeddable in the projective plane contains two nonadjacent edges crossing an odd number of times. Does the analogous statement hold on every other surface?

来源字段：https://doi.org/10.4171/owr/2008/44

14692 | GRAPH-018

### Harborth's Conjecture

P / partially_solved | 独立 / 未分配集合

Can every planar graph be drawn with integer edge lengths?

背景链接：https://doi.org/10.4230/LIPIcs.GD.2024.38

14693 | AMR-087-0061

### Hardness of the Pairing Inversion Problem

P / partially_solved | AMR 问题清单汇编

Determine the computational hardness of the Pairing Inversion Problem.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14694 | OWR-12697684-030

### Heavy Cycles in Eulerian Digraphs

O / open | Oberwolfach 报告问题汇编

Let $D$ be an Eulerian weighted digraph in which the total weight of edges leaving each vertex is at least $1$. Must $D$ contain a directed cycle of weight at least a universal positive constant? What stronger bound follows if $D$ is strongly $2$-connected?

来源字段：https://doi.org/10.4171/owr/2023/1

14695 | OPG-806

### Hedetniemi's Conjecture

S / solved | Open Problem Garden

Conjecture If $G,H$ are simple finite graphs, then $\chi(G \times H) = \min \{ \chi(G), \chi(H) \}$.

Here $G \times H$ is the tensor product (also called the direct or categorical product) of $G$ and $H$.

背景链接：http://www.openproblemgarden.org/op/hedetniemis_conjecture

14696 | AMR-087-0091

### Heegner points from nonmaximal orders

O / open | AMR 问题清单汇编

Prove analogues of the stated Heegner-point results for points arising from nonmaximal orders.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14697 | OWR-12697684-033

### High-Chromatic Odd-Cycle-Free Graphs with Large Minimum Degree

O / open | Oberwolfach 报告问题汇编

For fixed $k$, do there exist $n$-vertex graphs containing no odd cycle of length less than $2k+1$, with chromatic number at least four and minimum degree at least $f(k)n$, where $f(k)$ is of order $1/k$?

来源字段：https://doi.org/10.4171/owr/2023/1

14698 | OWR-12481-014

### High-Girth Cubic Graph Homomorphisms to the Five-Cycle

O / open | Oberwolfach 报告问题汇编

Does there exist an integer $\ell$ such that every cubic graph of girth at least $\ell$ admits a graph homomorphism to the cycle $C_5$?

来源字段：https://doi.org/10.4171/owr/2013/18

14699 | OWR-16627-005

### Higher-Color Equivalence of Two-Equivalent Graphs

O / open | Oberwolfach 报告问题汇编

Is every pair of 2-equivalent graphs also 3-equivalent?

来源字段：https://doi.org/10.4171/owr/2018/44

14700 | AMR-089-0002

### Higher-dimensional cropping formula

O / open | AMR 问题清单汇编

Find a higher-dimensional analogue of the paper's cropping and summation argument; in dimension three the expected sum ranges over quadruples $v_1,v_2,v_3,v_4$ for which $\operatorname{ConvHull}(0,v_1,v_2,v_3,v_4)$ contains no lattice points.

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/17-75/index.html

14701 | AMR-087-0015

### Higher-dimensional nondegenerate Frobenius algorithms

P / partially_solved | AMR 问题清单汇编

Develop the higher-dimensional analogue of the nondegenerate-curve Frobenius-lift method.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14702 | OPG-691

### Highly arc transitive two ended digraphs

P / partially_solved | Open Problem Garden

Conjecture If $G$ is a highly arc transitive digraph with two ends, then every tile of $G$ is a disjoint union of complete bipartite graphs.

背景链接：http://www.openproblemgarden.org/op/highly_arc_transitive_two_ended_digraphs

14703 | OWR-4791-033

### Highly Chromatic Subgraphs with Controlled Odd Cycles

O / open | Oberwolfach 报告问题汇编

For all positive integers $k$ and $\ell$, do there exist functions $f(k,\ell)$ and $s(k)$ such that every graph $G$ with $\chi(G)\geq f(k,\ell)$ has a subgraph $G'$ for which either $\chi(G')\geq k$ and $G'$ has no odd cycle of length at most $\ell$, or $G'$ contains a $K_k$ whose edges are each subdivided by at most $s(k)$ vertices?

来源字段：https://doi.org/10.4171/owr/2011/01

14704 | OPG-153

### Highly connected graphs with no K_n minor

P / partially_solved | Open Problem Garden

Problem Is it true for all $n \ge 0$, that every sufficiently large $n$-connected graph without a $K_n$ minor has a set of $n-5$ vertices whose deletion results in a planar graph?

背景链接：http://www.openproblemgarden.org/op/high_connectivity_no_k_n

14705 | OWR-11136-009

### Hill Conjecture for Complete Graph Crossing Numbers

O / open | Oberwolfach 报告问题汇编

Is the crossing number of every drawing of $K_n$ at least $\frac14\lfloor n/2\rfloor\lfloor(n-1)/2\rfloor\lfloor(n-2)/2\rfloor\lfloor(n-3)/2\rfloor$?

来源字段：https://doi.org/10.4171/owr/2011/44

14706 | OWR-9790352-032

### Hitting Maximum Independent Sets in Bounded-Chromatic Graphs

P / partially_solved | Oberwolfach 报告问题汇编

If a sequence of graphs $G$ has $\chi(G)=O(1)$, must the minimum order $h(G)$ of a vertex set meeting every maximum independent set satisfy $h(G)=o(|V(G)|)$?

来源字段：https://doi.org/10.4171/owr/2022/1

14707 | OPG-47282

### Hoàng-Reed Conjecture

P / partially_solved | Open Problem Garden

Conjecture Every digraph in which each vertex has outdegree at least $k$ contains $k$ directed cycles $C_1, \ldots, C_k$ such that $C_j$ meets $\cup_{i=1}^{j-1}C_i$ in at most one vertex, $2 \leq j \leq k$.

背景链接：http://www.openproblemgarden.org/op/hoand_reed_conjecture

14708 | OWR-1106-012

### Hole Projections and Hyperbolicity of Surface Graphs

O / open | Oberwolfach 报告问题汇编

Let $G(S)$ be a graph on a surface $S$ with a mapping-class-group action, and call essential subsurfaces on which subsurface projection is everywhere defined holes. If every pair of holes intersects, must $G(S)$ be Gromov hyperbolic? Is the thresholded sum $\sum_X' d_X(\alpha,\beta)$ over all holes uniformly multiplicatively and additively comparable with the distance between $\alpha$ and $\beta$ in $G(S)$?

来源字段：https://doi.org/10.4171/owr/2005/45

14709 | OWR-708-001

### Hyperbolicity and Imprimitivity in Vertex-Transitive Graphs

O / open | Oberwolfach 报告问题汇编

For every connected locally finite graph $\Gamma$ with a vertex-transitive automorphism group $G$, either $G$ has a finite-block imprimitivity system with finite induced stabilizer, or $\Gamma$ is hyperbolic.

来源字段：https://doi.org/10.4171/owr/2004/25

14710 | AMR-087-0072

### Ideal-lattice digital signatures

S / solved | AMR 问题清单汇编

Construct efficient digital-signature schemes from ideal-lattice problems.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14711 | AMR-087-0071

### Ideal-lattice pseudorandom functions

P / partially_solved | AMR 问题清单汇编

Construct efficient pseudorandom functions from ideal-lattice problems.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14712 | AMR-087-0070

### Ideal-lattice pseudorandom generators

S / solved | AMR 问题清单汇编

Construct efficient pseudorandom generators from ideal-lattice problems.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14713 | AMR-087-0079

### Ideal-lattice Regev cryptosystem

S / solved | AMR 问题清单汇编

Construct an efficient ideal-lattice version of Regev's quantum-SVP-based cryptosystem.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14714 | GRAPH-051

### Imbalance Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Imbalance conjecture [OPG-57613]

If every edge has imbalance ≥1, is the multiset of edge imbalances always graphic?

背景链接：https://arxiv.org/abs/2608.09191

14715 | AMR-087-0095

### Implicit computation with testing characters and homogeneous spaces

O / open | AMR 问题清单汇编

Work efficiently with the testing characters and principal homogeneous spaces without constructing them explicitly.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14716 | GRAPH-052

### Implicit Graph Conjecture

S / solved | 独立 / 未分配集合

Do slowly-growing hereditary graph families admit implicit representations?

背景链接：https://arxiv.org/abs/2111.13198

14717 | AMR-087-0014

### Improved Frobenius lifts for nondegenerate curves

P / partially_solved | AMR 问题清单汇编

Test and analyze whether deleting extra points and fixing the lift $x\mapsto x^p$ improves Frobenius lifts for nondegenerate curves.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14718 | OWR-1189-005

### Independence Bounds for Sparse Subcubic Graphs

O / open | Oberwolfach 报告问题汇编

Determine how the bound $2a(G)\geq n(G)-m(G)$ for $C_3$- and $C_4$-free graphs of maximum degree at most $3$ extends to larger girth or unrestricted maximum degree, and determine the extremal graphs.

来源字段：https://doi.org/10.4171/owr/2006/07

14719 | OWR-14299905-037

### Independence-Number Degeneracy in Induced-Subdivision-Free Graphs

O / open | Oberwolfach 报告问题汇编

For every positive integer $t$, does there exist $C>0$ such that every graph with no induced $K_{t,t}$ and no induced subdivision of $K_t$ in which every edge is subdivided at least once has a vertex $v$ satisfying $\alpha(N(v))\leq C$?

来源字段：https://doi.org/10.4171/owr/2026/1

14720 | OWR-14299905-039

### Independent Sets in Pseudorandom Regular Graphs

O / open | Oberwolfach 报告问题汇编

Does there exist an absolute constant $C>0$ such that every $(n,d,\lambda)$-graph with $\lambda>(\log n)^2$ has at most

$$\left(\frac{4e^2\lambda}{\log n}\right)^t$$

independent sets of size $t=Cn\log n/(d+1)$?

来源字段：https://doi.org/10.4171/owr/2026/1

14721 | OWR-14299905-024

### Independent Sets in Steiner Triple Systems

O / open | Oberwolfach 报告问题汇编

Does every Steiner triple system on $n$ vertices contain an independent set of size at least $(1+o(1))\sqrt{3n\log n}$ as $n\to\infty$?

来源字段：https://doi.org/10.4171/owr/2026/1

14722 | OWR-14299905-022

### Independent Sets in Uniform Hypergraphs

O / open | Oberwolfach 报告问题汇编

Determine the largest possible independence number of a uniform hypergraph with prescribed numbers of vertices and edges.

来源字段：https://doi.org/10.4171/owr/2026/1

14723 | OWR-10252930-028

### Induced Four-Cycle Profiles Above Half Density

P / partially_solved | Oberwolfach 报告问题汇编

Among $n$-vertex graphs with edge density $\xi$, determine asymptotically the maximum number of induced copies of $C_4$. For $\xi>1/2$, is the maximum asymptotically attained by the extremal construction for the triangle-density problem?

来源字段：https://doi.org/10.4171/owr/2022/22

14724 | OWR-10252930-013

### Induced Structures in Highly Chromatic Graphs

O / open | Oberwolfach 报告问题汇编

Which induced subgraphs must occur in graphs of sufficiently large chromatic number?

来源字段：https://doi.org/10.4171/owr/2022/22

14725 | OPG-484

### Infinite uniquely hamiltonian graphs

O / open | Open Problem Garden

Problem Are there any uniquely hamiltonian locally finite 1-ended graphs which are regular of degree $r > 2$?

背景链接：http://www.openproblemgarden.org/op/infinite_uniquely_hamiltonian_graphs

14726 | AMR-088-0012

### Infinitude of rational points on an elliptic curve

O / open | AMR 问题清单汇编

Given an elliptic curve $E:y^2=x^3+Ax+B$ over $\mathbb{Q}$, determine whether $E$ has infinitely many rational points.

背景链接：https://www.maths.nottingham.ac.uk/plp/pmzcw/download/open.pdf

14727 | OWR-17464-020

### Internal Partitions of Regular Graphs

O / open | Oberwolfach 报告问题汇编

An internal partition of a graph is a partition into two nonempty parts in which every vertex has at least as many neighbors in its own part as in the other. For every $d$, are there only finitely many $d$-regular graphs with no internal partition?

来源字段：https://doi.org/10.4171/owr/2020/1

14728 | AMR-087-0046

### Jacobians in abelian-threefold isogeny classes

P / partially_solved | AMR 问题清单汇编

Given the Weil polynomial of an abelian-threefold isogeny class over a finite field, determine whether the class contains a Jacobian.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14729 | OPG-130

### Jaeger's modular orientation conjecture

S / solved | Open Problem Garden

Conjecture Every $4k$-edge-connected graph can be oriented so that ${\mathit indegree}(v) - {\mathit outdegree}(v) \cong 0$ (mod $2k+1$ ) for every vertex $v$.

背景链接：http://www.openproblemgarden.org/op/jaegers_modular_orientation_conjecture

14730 | GRAPH-012

### Jaeger's Petersen Coloring Conjecture

P / partially_solved | 独立 / 未分配集合

Does every bridgeless cubic graph have a cycle-continuous mapping to the Petersen graph?

背景链接：https://doi.org/10.1016/j.dam.2025.07.032

14731 | OPG-638

### Jones' conjecture

P / partially_solved | Open Problem Garden

For a graph $G$, let $cp(G)$ denote the cardinality of a maximum cycle packing (collection of vertex disjoint cycles) and let $cc(G)$ denote the cardinality of a minimum feedback vertex set (set of vertices $X$ so that $G-X$ is acyclic).

Conjecture For every planar graph $G$, $cc(G)\leq 2cp(G)$.

背景链接：http://www.openproblemgarden.org/op/jones_conjecture

14732 | OPG-154

### Jorgensen's Conjecture

P / partially_solved | Open Problem Garden

Conjecture Every 6-connected graph without a $K_6$ minor is apex (planar plus one vertex).

背景链接：http://www.openproblemgarden.org/op/jorgensens_conjecture

14733 | OWR-13674-018

### Kadison-Singer Methods for Ihara Zeta Functions

O / open | Oberwolfach 报告问题汇编

For a connected $k$-regular graph $G$ with adjacency matrix $A$, the Ihara zeta function satisfies $\zeta(u,G)^{-1}=(1-u^2)^{|E|-|V|}\det(I-Au+(k-1)u^2I)$. Can Kadison-Singer and interlacing-polynomial methods yield new information about this zeta function?

来源字段：https://doi.org/10.4171/owr/2015/17

14734 | OPG-56233

### Kriesell's Conjecture

P / partially_solved | Open Problem Garden

Conjecture Let $G$ be a graph and let $T\subseteq V(G)$ such that for any pair $u,v\in T$ there are $2k$ edge-disjoint paths from $u$ to $v$ in $G$. Then $G$ contains $k$ edge-disjoint trees, each of which contains $T$.

背景链接：http://www.openproblemgarden.org/op/kriesells_conjecture

14735 | AMR-088-0010

### L-value criterion for congruent numbers

O / open | AMR 问题清单汇编

For $E_n:y^2=x^3-n^2x$, is $n$ a congruent number if and only if $L(E_n,1)=0$?

背景链接：https://www.maths.nottingham.ac.uk/plp/pmzcw/download/open.pdf

14736 | OPG-407

### Laplacian Degrees of a Graph

S / solved | Open Problem Garden

Conjecture If $G$ is a connected graph on $n$ vertices, then $c_k(G) \ge d_k(G)$ for $k = 1, 2, \dots, n-1$.

背景链接：http://www.openproblemgarden.org/op/laplacian_degrees_of_a_graph

14737 | OWR-13355-006

### Laplacian Spectra of Penrose Tiling Graphs

P / partially_solved | Oberwolfach 报告问题汇编

Determine the spectral properties of the Laplacian on a Penrose-tiling graph or on a graph arising from a higher-dimensional quasicrystal model.

来源字段：https://doi.org/10.4171/owr/2014/53

14738 | OPG-52197

### Large acyclic induced subdigraph in a planar oriented graph.

S / solved | Open Problem Garden

Conjecture Every planar oriented graph $D$ has an acyclic induced subdigraph of order at least $\frac{3}{5} |V(D)|$.

背景链接：http://www.openproblemgarden.org/op/large_acyclic_induced_subdigraph_in_a_planar_oriented_graph

14739 | OWR-12481-011

### Large Anticomplete Sets in Triangle-Free Graphs

O / open | Oberwolfach 报告问题汇编

Does there exist a constant $c>0$ such that every triangle-free graph on more than two vertices has disjoint vertex subsets of orders $cn^c$ and $cn$ with no edges between them?

来源字段：https://doi.org/10.4171/owr/2013/18

14740 | AMR-099-0018

### Large identical neighborhoods and vertex transitivity

O / open | AMR 问题清单汇编

Let $G$ be an $n$-vertex graph whose rooted balls of size $k$ are all isomorphic. If $k>n/2$, or if $k$ is within a fixed constant of $\operatorname{diam}(G)$, must $G$ be vertex-transitive? Separately, for odd $n$, is the degree of a uniformly chosen vertex-transitive graph on $n$ vertices concentrated near $(n-1)/2$?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=23

14741 | OPG-46634

### Large induced forest in a planar graph.

P / partially_solved | Open Problem Garden

Conjecture Every planar graph on $n$ verices has an induced forest with at least $n/2$ vertices.

背景链接：http://www.openproblemgarden.org/op/large_induced_forest_in_a_planar_graph

14742 | OWR-17464-010

### Large-Scale Geometry of Graphs and Manifolds

O / open | Oberwolfach 报告问题汇编

What can be said about the global or large-scale geometry of graphs and its relationship with differential geometry on manifolds?

来源字段：https://doi.org/10.4171/owr/2020/1

14743 | AMR-087-0075

### Lattice reduction for algebraic-number-theory problems

O / open | AMR 问题清单汇编

Use lattice reduction together with average-case problems to solve computational problems in algebraic number theory.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14744 | AMR-087-0022

### Learning from wrong guesses in partial-key factoring

O / open | AMR 问题清单汇编

Extract useful information from incorrect guesses in factoring attacks based on partially known bits.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14745 | GRAPH-049

### Line Graphs of Non-Word-Representable Graphs

P / partially_solved | 独立 / 未分配集合

Is the line graph of a non-word-representable graph always non-word-representable?

背景链接：https://arxiv.org/abs/2509.03339

14746 | GRAPH-032

### Linear Arboricity Conjecture

P / partially_solved | 独立 / 未分配集合

Can every graph with maximum degree Δ be decomposed into at most ⌈(Δ+1)/2⌉ linear forests?

背景链接：https://mathworld.wolfram.com/LinearArboricity.html

14747 | OWR-11136-029

### Linear Bounds for Monotone Crossing Numbers

O / open | Oberwolfach 报告问题汇编

Does there exist a constant $c$ such that $\operatorname{mon-cr}(G)<c\operatorname{cr}(G)$ for every graph $G$?

来源字段：https://doi.org/10.4171/owr/2011/44

14748 | OWR-12861-005

### Linear Cycle-and-Edge Decompositions of Graphs

P / partially_solved | Oberwolfach 报告问题汇编

Does every graph on $n$ vertices decompose into $O(n)$ cycles and edges? This is known for random graphs and for graphs with linear minimum degree.

来源字段：https://doi.org/10.4171/owr/2014/01

14749 | OWR-2090-011

### Linear Edge Bounds for $k$-Quasiplanar Graphs

O / open | Oberwolfach 报告问题汇编

A $k$-quasiplanar topological graph is a graph drawn in the plane with no $k$ pairwise crossing edges. For every fixed integer $k>4$, does every $k$-quasiplanar topological graph with $n$ vertices have $O(n)$ edges?

来源字段：https://doi.org/10.4171/owr/2008/44

14750 | AMR-099-0026

### Linear finite models for locally finite transitive graphs

O / open | AMR 问题清单汇编

If an infinite vertex-transitive graph is $f(r)$-sofic for some function $f$, must it be $cr$-sofic for a constant $c$? More uniformly, for fixed degree $d$, does every radius-$r$ ball that occurs in a finite $d$-regular vertex-transitive graph occur in such a graph of diameter at most $c(d)r$? Determine the optimal $c(d)$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=40

14751 | OPG-596

### Linear Hypergraphs with Dimension 3

P / partially_solved | Open Problem Garden

Conjecture Any linear hypergraph with incidence poset of dimension at most 3 is the intersection hypergraph of a family of triangles and segments in the plane.

背景链接：http://www.openproblemgarden.org/op/linear_hypergraphs_with_dimension_3

14752 | OWR-3473-006

### Linear-Time Algorithms for Deterministic Graphical Games

O / open | Oberwolfach 报告问题汇编

Can finite-state deterministic graphical games of perfect information, given explicitly as graphs, be solved in linear time by a comparison-based algorithm?

来源字段：https://doi.org/10.4171/owr/2009/23

14753 | AMR-054-0051

### Linear-Volume 3D Grid Drawings of Planar Graphs

S / solved | AMR 问题清单汇编

Does every $n$-vertex planar graph have a 3D grid drawing with $O(n)$ volume? A 3D grid drawing of a graph is a placement of the vertices at distinct points with integer coordinates such that the straight line segments representing the edges are pairwise non-crossing. The volume is of the bounding box.

背景链接：https://topp.openproblem.net/p51

14754 | OWR-1320-001

### Linearization of Markov Inverse Problems on Networks

O / open | Oberwolfach 报告问题汇编

Characterize multiterminal directed networks with known topology whose unknown one-step Markov transition probabilities can be recovered from the zeroth and first moments of boundary input-output travel times by analytically reducing the nonlinear inverse equations to a linear system and solving up to the natural gauge freedom.

来源字段：https://doi.org/10.4171/owr/2006/34

14755 | OPG-170

### Linial-Berge path partition duality

P / partially_solved | Open Problem Garden

Conjecture The minimum $k$-norm of a path partition on a directed graph $D$ is no more than the maximal size of an induced $k$-colorable subgraph.

背景链接：http://www.openproblemgarden.org/op/linial_berge_path_partition_duality

14756 | AMR-099-0049

### Liouville extensions by an isometric integer action

O / open | AMR 问题清单汇编

Suppose $\mathbb{Z}$ acts on a graph $G$ by isometries, the quotient $H=G/\mathbb{Z}$ is Liouville, and simple random walk on $G$ visits every translate of a fundamental domain infinitely often almost surely. Must $G$ be Liouville?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=107

14757 | AMR-099-0047

### Liouville property of infinite Ramanujan graphs

O / open | AMR 问题清单汇编

Prove that no infinite connected Ramanujan graph is Liouville; equivalently, every such graph admits a nonconstant bounded harmonic function.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=107

14758 | AMR-099-0048

### Liouville property under rough isometry to nonamenable Cayley graphs

P / partially_solved | AMR 问题清单汇编

Prove that every bounded-degree graph roughly isometric to a nonamenable Cayley graph is non-Liouville.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=107

14759 | OPG-47358

### List chromatic number and maximum degree of bipartite graphs

P / partially_solved | Open Problem Garden

Conjecture There is a constant $c$ such that the list chromatic number of any bipartite graph $G$ of maximum degree $\Delta$ is at most $c \log \Delta$.

背景链接：http://www.openproblemgarden.org/op/list_chromatic_number_and_maximum_degree_of_bipartite_graphs

14760 | GRAPH-013

### List Coloring Conjecture

P / partially_solved | 独立 / 未分配集合

For every graph, does the list chromatic index equal the chromatic index?

背景链接：https://doi.org/10.1006/jctb.1995.1011

14761 | OPG-388

### List colorings of edge-critical graphs

O / open | Open Problem Garden

Conjecture Suppose that $G$ is a $\Delta$-edge-critical graph. Suppose that for each edge $e$ of $G$, there is a list $L(e)$ of $\Delta$ colors. Then $G$ is $L$-edge-colorable unless all lists are equal to each other.

背景链接：http://www.openproblemgarden.org/op/list_colorings_of_edge_critical_graphs

14762 | OPG-59927

### List Colourings of Complete Multipartite Graphs with 2 Big Parts

P / partially_solved | Open Problem Garden

Question Given $a,b\geq2$, what is the smallest integer $t\geq0$ such that $\chi_\ell(K_{a,b}+K_t)= \chi(K_{a,b}+K_t)$?

背景链接：http://www.openproblemgarden.org/op/list_colourings_of_complete_multipartite_graphs_with_2_big_parts

14763 | OPG-56449

### List Total Colouring Conjecture

P / partially_solved | Open Problem Garden

Conjecture If $G$ is the total graph of a multigraph, then $\chi_\ell(G)=\chi(G)$.

背景链接：http://www.openproblemgarden.org/op/list_total_colouring_conjecture

14764 | OWR-1189-008

### List-Labeling Equality for Trees

O / open | Oberwolfach 报告问题汇编

For a graph $G$, let $\chi^{2,1}(G)$ denote the $L(2,1)$-labeling span (where adjacent vertices receive colors at distance at least $2$ and vertices at graph distance $2$ receive distinct colors), and let $\chi^{2,1}_\ell(G)$ denote the corresponding list $L(2,1)$-labeling number. For every tree $T$, does $\chi^{2,1}_\ell(T) = \chi^{2,1}(T)$ hold?

来源字段：https://doi.org/10.4171/owr/2006/07

14765 | OWR-1189-007

### List-Labeling Numbers of Paths

P / partially_solved | Oberwolfach 报告问题汇编

For every path $P_n$, is the $(d,d)$-list-labeling number $\chi^{d,d}_\ell(P_n)=\lfloor3d(1-1/n)\rfloor+1$?

来源字段：https://doi.org/10.4171/owr/2006/07

14766 | OWR-12479-006

### Li-Szegedy Smooth-Retract Conjecture

O / open | Oberwolfach 报告问题汇编

Let $T$ be an induced subtree of a bipartite graph $H$. If $H$ retracts onto $T$, meaning that there is a graph homomorphism $H\to T$ whose restriction to $T$ is the identity, then $T$ is smooth in $H$ in the sense of Li and Szegedy.

来源字段：https://doi.org/10.4171/owr/2013/16

14767 | OWR-12330-012

### Local and Edgewise Reed Bounds

O / open | Oberwolfach 报告问题汇编

Conjecturally, every graph $G$ satisfies both

$$\chi(G)\leq\max_{v\in V(G)}\left\lceil\frac12\bigl(d(v)+1+\omega(G[\widetilde N(v)])\bigr)\right\rceil$$

and

$$\chi(G)\leq\max_{uv\in E(G)}\left\lceil\frac12\bigl(\gamma_\ell(u)+\gamma_\ell(v)\bigr)\right\rceil,$$

where $\gamma_\ell(v)=\frac12\bigl(d(v)+1+\omega(G[\widetilde N(v)])\bigr)$.

来源字段：https://doi.org/10.4171/owr/2013/02

14768 | OWR-12330-011

### Local Clique-Weighted Chromatic Bounds

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a universal constant $\varepsilon>0$ for which

$$\chi(G)\leq\max_{v\in V(G)}\left\lceil(1-\varepsilon)(d(v)+1)+\varepsilon\omega(G[\widetilde N(v)])\right\rceil$$

holds for every graph $G$?

来源字段：https://doi.org/10.4171/owr/2013/02

14769 | OWR-4791-014

### Local Coloring and Transitive Sets in Tournaments

P / partially_solved | Oberwolfach 报告问题汇编

Are the following statements true? (i) For every $k\geq0$ there is $c$ such that any tournament whose every vertex has out-neighbourhood chromatic number at most $k$ has chromatic number at most $c$. (ii) If $H=\Delta(T_2,T_2,T_2)$, then for every $\varepsilon>0$ there is $c>0$ such that every $H$-free tournament has a transitive subset of size at least $c|V(G)|^{1-\varepsilon}$.

来源字段：https://doi.org/10.4171/owr/2011/01

14770 | OWR-12697684-027

### Local Count Thresholds for Bounded-Degree Embeddings

O / open | Oberwolfach 报告问题汇编

Determine the least $f_k(\Delta)$ such that matching all local $k$-uniform homomorphism counts for every graph on at most $f_k(\Delta)$ vertices forces embeddings of all $n$-vertex graphs of maximum degree at most $\Delta$.

来源字段：https://doi.org/10.4171/owr/2023/1

14771 | OWR-793-007

### Local Recognition of Graphs and Geometries

O / open | Oberwolfach 报告问题汇编

For a fixed graph $\Delta$, classify all graphs $\Gamma$ that are locally $\Delta$.

来源字段：https://doi.org/10.4171/owr/2005/12

14772 | OWR-12697712-005

### Local Resilience of Random-Hypergraph Perfect Matchings

P / partially_solved | Oberwolfach 报告问题汇编

For every fixed $k$, do sufficiently large $C$ and sufficiently small $\varepsilon>0$ ensure that, with high probability, deleting at most an $\varepsilon$-fraction of the incident hyperedges at every vertex of $G_k(n,C\log n/n^{k-1})$ still leaves a perfect matching?

来源字段：https://doi.org/10.4171/owr/2023/16

14773 | OWR-16164-002

### Local Weak Limits of Surface-Embedded Random Graphs

O / open | Oberwolfach 报告问题汇编

For the uniform random graph $S_g(n,m)$ embeddable in the orientable surface of genus $g$, determine its Benjamini-Schramm local weak limit in the natural asymptotic regimes of $n$ and $m$.

来源字段：https://doi.org/10.4171/owr/2018/23

14774 | AMR-099-0016

### Local-to-global covering rigidity for Cayley graphs

S / solved | AMR 问题清单汇编

For every Cayley graph $G$, does there exist $r=r(G)$ such that $G$ covers every graph whose radius-$r$ balls are all isomorphic to the radius-$r$ ball in $G$?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=22

14775 | OWR-13110-017

### Long Alternating Paths in Colored Line Arrangements

O / open | Oberwolfach 报告问题汇编

Given n red, n blue, and n green lines in general position in the plane, does the dual graph of the arrangement always contain an alternating path of length $\Theta(n^2)$, where consecutive crossed edges cannot have the same color?

来源字段：https://doi.org/10.4171/owr/2014/40

14776 | OWR-17464-016

### Long Cycles in Induced-Ramsey Graph Constructions

P / partially_solved | Oberwolfach 报告问题汇编

Given a graph $F$ and integers $r,n$, how many and what kinds of cycles longer than $n$ can occur in a graph $H$ with $H\to(F)^r$ and with every induced cycle of length at most $n$ inherited from a copy of $F$?

来源字段：https://doi.org/10.4171/owr/2020/1

14777 | OPG-46460

### Long directed cycles in diregular digraphs

P / partially_solved | Open Problem Garden

Conjecture Every strong oriented graph in which each vertex has indegree and outdegree at least $d$ contains a directed cycle of length at least $2d+1$.

背景链接：http://www.openproblemgarden.org/op/long_directed_cycles_in_digraph_with_minimum_in_and_out_degree

14778 | OWR-16931-024

### Long Increasing Paths in Edge-Ordered Graphs

O / open | Oberwolfach 报告问题汇编

Does there exist $\varepsilon>0$ such that every edge-ordered graph $G$ contains an increasing path of length $$\Omega\!\left((e(G)/|V(G)|)^{1/2+\varepsilon}\right)$$?

来源字段：https://doi.org/10.4171/owr/2019/19

14779 | OWR-1536-017

### Loss Thresholds in Two-Round Random Edge Coloring

O / open | Oberwolfach 报告问题汇编

In the two-round online edge-coloring game in which the edges of $G(n,p)$ are first colored offline and then $N$ additional random edges are colored online, is it true that for every $0\leq\alpha\leq1/6$, if $p=\Omega(n^{-1/2-\alpha})$ and $N=\omega(n^{8\alpha})$, then the player loses?

来源字段：https://doi.org/10.4171/owr/2007/20

14780 | GRAPH-005

### Lovász Conjecture

O / open | 独立 / 未分配集合

别名：Lovász Conjecture [GRAPH-033]

Does every finite connected vertex-transitive graph have a Hamiltonian path?

背景链接：https://en.wikipedia.org/wiki/Lov%C3%A1sz_conjecture

14781 | OPG-46629

### Lovász Path Removal Conjecture

P / partially_solved | Open Problem Garden

Conjecture There is an integer-valued function $f(k)$ such that if $G$ is any $f(k)$-connected graph and $x$ and $y$ are any two vertices of $G$, then there exists an induced path $P$ with ends $x$ and $y$ such that $G-V(P)$ is $k$-connected.

背景链接：http://www.openproblemgarden.org/op/lovasz_path_removal_conjecture

14782 | OWR-1532-014

### Lovász Removable-Path Conjecture for Three Prescribed Vertices

O / open | Oberwolfach 报告问题汇编

Does there exist a function $f_3(k)$ such that, for every $f_3(k)$-connected graph $G$ and every three vertices $s,t,x\in V(G)$, there are a path $P$ from $s$ to $t$ and a $k$-connected subgraph $H$ with $x\in V(H)$ and $V(H)\cap V(P)=\varnothing$?

来源字段：https://doi.org/10.4171/owr/2007/16

14783 | AMR-087-0081

### Malleable RSA modulus generation

O / open | AMR 问题清单汇编

Construct a malleable RSA generator producing publicly related moduli $n,n'$ such that factoring $n'$ makes $n$ easy to factor.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14784 | OPG-412

### Mapping planar graphs to odd cycles

P / partially_solved | Open Problem Garden

Conjecture Every planar graph of girth $\ge 4k$ has a homomorphism to $C_{2k+1}$.

背景链接：http://www.openproblemgarden.org/op/mapping_planar_graphs_to_odd_cycles

14785 | OPG-37364

### Matching cut and girth

P / partially_solved | Open Problem Garden

Question For every $d$ does there exists a $g$ such that every graph with average degree smaller than $d$ and girth at least $g$ has a matching-cut?

背景链接：http://www.openproblemgarden.org/op/matching_cut_and_girth

14786 | OWR-10252930-001

### Matching Thresholds for Pseudorandom Clique Factors

O / open | Oberwolfach 报告问题汇编

For clique factors larger than triangles in sparse pseudorandom graphs, do matching constructions attain the known threshold bounds?

来源字段：https://doi.org/10.4171/owr/2022/22

14787 | OPG-600

### Matchings extend to Hamiltonian cycles in hypercubes

P / partially_solved | Open Problem Garden

Question Does every matching of hypercube extend to a Hamiltonian cycle?

背景链接：http://www.openproblemgarden.org/op/matchings_extends_to_hamilton_cycles_in_hypercubes

14788 | OWR-16763-014

### Maximum Clique in Proper Vertex-Minor-Closed Classes

O / open | Oberwolfach 报告问题汇编

Let $\mathcal F$ be a proper graph class closed under isomorphism and vertex minors. Is there a polynomial-time algorithm that, given $G\in\mathcal F$, computes the clique number of $G$?

来源字段：https://doi.org/10.4171/owr/2019/1

14789 | OWR-14604-011

### Maximum Girth-to-Diameter Ratio

O / open | Oberwolfach 报告问题汇编

Among graphs of minimum vertex degree at least $3$, what is $x=\limsup_{\operatorname{diam}(G)\to\infty}\frac{\operatorname{girth}(G)}{\operatorname{diam}(G)}$? In particular, is $x>1$?

来源字段：https://doi.org/10.4171/owr/2016/22

14790 | OWR-12330-006

### Maximum Simplices in Euclidean Diameter Graphs

P / partially_solved | Oberwolfach 报告问题汇编

Fix $d\neq3$ and $n\geq d+1$, and let $D$ be the diameter graph of $n$ points in $\mathbb R^d$ whose maximum pairwise distance is $1$. Is the maximum possible number of $(d-1)$-dimensional simplices in $D$ equal to $n$?

来源字段：https://doi.org/10.4171/owr/2013/02

14791 | OWR-9790352-030

### Maximum Twin-Width of $n$-Vertex Graphs

P / partially_solved | Oberwolfach 报告问题汇编

What is the maximum twin-width among graphs on $n$ vertices?

来源字段：https://doi.org/10.4171/owr/2022/1

14792 | OWR-14299905-007

### Maximum-Cut Surplus in Clique-Free Graphs

P / partially_solved | Oberwolfach 报告问题汇编

For each $r\geq3$, determine the largest guaranteed surplus over $m/2$ in the maximum cut of an $m$-edge $K_r$-free graph. In particular, does there exist an absolute $\varepsilon>0$ such that every such graph has surplus $\Omega_r(m^{1/2+\varepsilon})$?

来源字段：https://doi.org/10.4171/owr/2026/1

14793 | OWR-15206-013

### Maximum-to-Average Independence Ratios in Triangle-Free Graphs

O / open | Oberwolfach 报告问题汇编

In every triangle-free graph, is the ratio of the maximum independent-set size to the average size of a uniformly random independent set at least $4/3$? In every triangle-free graph of minimum degree $d$, is that ratio at least $2-o_d(1)$?

来源字段：https://doi.org/10.4171/owr/2017/1

14794 | OPG-46575

### Melnikov's valency-variety problem

P / partially_solved | Open Problem Garden

Problem The valency-variety $w(G)$ of a graph $G$ is the number of different degrees in $G$. Is the chromatic number of any graph $G$ with at least two vertices greater than $$\ceil{ \frac{\floor{w(G)/2}}{|V(G)| - w(G)} } ~?$$

背景链接：http://www.openproblemgarden.org/op/melnikovs_valency_variety_problem

14795 | GRAPH-006

### Meyniel's Conjecture on Cop Number

P / partially_solved | 独立 / 未分配集合

Is the cop number of a connected n-vertex graph $O(\sqrt{n})$?

背景链接：https://math.ryerson.ca/~abonato/papers/meyniel_0311.pdf

14796 | OPG-37420

### Minimal graphs with a prescribed number of spanning trees

P / partially_solved | Open Problem Garden

Conjecture Let $n \geq 3$ be an integer and let $\alpha(n)$ denote the least integer $k$ such that there exists a simple graph on $k$ vertices having precisely $n$ spanning trees. Then $\alpha(n) = o(\log{n}).$

背景链接：http://www.openproblemgarden.org/op/minimal_graphs_with_a_prescribed_number_of_spanning_trees

14797 | AMR-099-0017

### Minimum diameter realizing a prescribed local ball

O / open | AMR 问题清单汇编

Fix a rooted radius-$r$ ball $B(o,r)$ that occurs as every radius-$r$ ball of some finite graph. What is the minimum diameter of a finite graph all of whose radius-$r$ balls are isomorphic to $B(o,r)$? Bound this diameter in terms of $r$ and the root degree $d$, and determine whether it can grow faster than linearly in $r$ for fixed $d$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=22

14798 | OWR-12479-004

### Minimum Graph Homomorphism Counts at Fixed Density

P / partially_solved | Oberwolfach 报告问题汇编

For a fixed finite graph $F$ and an $n$-vertex graph $G$ with at least $\rho n^2/2$ edges, determine the minimum possible number of graph homomorphisms $F\to G$, where homomorphisms need not be injective and need not preserve nonedges.

来源字段：https://doi.org/10.4171/owr/2013/16

14799 | OPG-49795

### Minimum number of arc-disjoint transitive subtournaments of order 3 in a tournament

P / partially_solved | Open Problem Garden

Conjecture If $T$ is a tournament of order $n$, then it contains $\left \lceil n(n-1)/6 - n/3\right\rceil$ arc-disjoint transitive subtournaments of order 3.

背景链接：http://www.openproblemgarden.org/op/minimum_number_of_transitive_subtournaments_of_order_3_in_a_tournament

14800 | OWR-14604-013

### Minimum Triangular Edges above Turán Density

P / partially_solved | Oberwolfach 报告问题汇编

For $e>n^2/4$, let $\operatorname{Tr}(n,e)$ be the minimum number of edges lying in triangles among all $n$-vertex graphs with $e$ edges. For integers $a,b,c\geq0$ with $a+b+c=n$, let $\operatorname{Tr}(a,b,c)$ denote the family of graphs whose vertex set is partitioned as $A\cup B\cup C$, where $|A|=a$, $|B|=b$, and $|C|=c$; $B$ and $C$ are independent; every edge between $B$ and $C$ is present; vertices of $C$ have no neighbors in $A$; and the edges inside $A$ and between $A$ and $B$ form an almost-complete graph with more than $\binom{a-1}{2}+ab$ edges. Set $$ g(n,e)=\min\left\{e-bc:a+b+c=n,\ \binom a2+ab+bc\geq e\right\}. $$ Is $\operatorname{Tr}(n,e)=g(n,e)$, and must every extremal graph belong to some family $\operatorname{Tr}(a,b,c)$?

来源字段：https://doi.org/10.4171/owr/2016/22

14801 | OWR-1536-009

### Minimum-Degree Lower Bounds for Game Chromatic Index

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Game-Chromatic-Index Lower-Bound Question [OWR-1536-010]

Does there exist $\varepsilon>0$ and $d_0$ such that every graph $G$ with minimum degree $\delta(G)\geq d_0$ satisfies $\chi'_g(G)\geq(1+\varepsilon)\delta(G)$?

来源字段：https://doi.org/10.4171/owr/2007/20

14802 | OWR-15206-006

### Minimum-Difference Representations of Random Graphs

O / open | Oberwolfach 报告问题汇编

For a graph $G$ on vertices $1,\ldots,n$, let $f_{\min}(G)$ be the least $k$ for which sets $A_1,\ldots,A_n$ can be chosen so that $$ ij\in E(G)\quad\Longleftrightarrow\quad\min\{|A_i\setminus A_j|,|A_j\setminus A_i|\}\geq k. $$ For fixed $0<p<1$, does $G\in G(n,p)$ satisfy $f_{\min}(G)=\Omega(n/\log n)$ with high probability? What is the maximum of $f_{\min}(G)$ over all $n$-vertex graphs, and is $f_{\min}(G)=o(n)$ for every $G\in\mathcal G(n)\cup\mathcal G(n,n)$?

来源字段：https://doi.org/10.4171/owr/2017/1

14803 | OPG-37907

### Mixing Circular Colourings

P / partially_solved | Open Problem Garden

Question Is $\mathfrak{M}_c(G)$ always rational?

背景链接：http://www.openproblemgarden.org/op/mixing_circular_colourings_0

14804 | OWR-3473-010

### Modular and Local Limit Theorems for Subgraph Counts

P / partially_solved | Oberwolfach 报告问题汇编

For every fixed graph $H$, do analogous modular-equidistribution and local-limit theorems hold for the number of copies of $H$ in $G(n,p)$?

来源字段：https://doi.org/10.4171/owr/2009/23

14805 | AMR-089-0005

### Modular extension and analogous lattice series

O / open | AMR 问题清单汇编

Can the function $f$ on $SL(2,\mathbb{Z})$ be extended naturally to $\mathbb{C}/SL(2,\mathbb{Z})$? Can analogous series be constructed for other lattices or tessellations of the plane?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/17-75/index.html

14806 | OPG-60028

### Monochromatic reachability in arc-colored digraphs

O / open | Open Problem Garden

Conjecture For every $k$, there exists an integer $f(k)$ such that if $D$ is a digraph whose arcs are colored with $k$ colors, then $D$ has a $S$ set which is the union of $f(k)$ stables sets so that every vertex has a monochromatic path to some vertex in $S$.

背景链接：http://www.openproblemgarden.org/op/monochromatoc_reachability_in_arc_colored_digraphs

14807 | OPG-1808

### Monochromatic reachability or rainbow triangles

P / partially_solved | Open Problem Garden

In an edge-colored digraph, we say that a subgraph is rainbow if all its edges have distinct colors, and monochromatic if all its edges have the same color.

Problem Let $G$ be a tournament with edges colored from a set of three colors. Is it true that $G$ must have either a rainbow directed cycle of length three or a vertex $v$ so that every other vertex can be reached from $v$ by a monochromatic (directed) path?

背景链接：http://www.openproblemgarden.org/op/monochromatic_reachability_vs_rainbow_triangles

14808 | OPG-60030

### Monochromatic vertex colorings inherited from Perfect Matchings

P / partially_solved | Open Problem Garden

Conjecture For which values of $n$ and $d$ are there bi-colored graphs on $n$ vertices and $d$ different colors with the property that all the $d$ monochromatic colorings have unit weight, and every other coloring cancels out?

背景链接：http://www.openproblemgarden.org/op/monochromatic_vertex_colorings_inherited_from_perfect_matchings

14809 | AMR-087-0050

### Monotonicity of maximal curve point counts in genus

O / open | AMR 问题清单汇编

For fixed $q$, is $N_q(g)=\max_C\#C(\mathbb{F}_q)$ increasing as a function of the genus $g$?

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14810 | GRAPH-026

### Moore Graph Existence

P / partially_solved | 独立 / 未分配集合

Does a Moore graph with girth 5 and degree 57 exist?

背景链接：https://arxiv.org/abs/2210.09577

14811 | AMR-087-0053

### More MNT and pairing-friendly elliptic curves

P / partially_solved | AMR 问题清单汇编

Find more MNT curves, including usable larger embedding degrees, more curve families, and smaller cofactors.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14812 | OWR-1536-019

### Move Complexity and Surplus in Maker Games

O / open | Oberwolfach 报告问题汇编

For Maker's graph-construction game on $K_\infty$, if $R_M(n)$ is the minimum number of moves needed to claim a copy of $K_n$ and $2^{n/2}\leq R_M(n)\leq c2^n$, can the gap be closed or reduced? For a $d$-regular $n$-vertex graph $G$, is there a constant $c>1$ such that $R_M(G)\geq c^dn$? In the Maker-Breaker game on an $n\times n^2$ grid, can the trivial upper bound $n$ on Maker's surplus in a row or column be improved, and is the correct order $\Theta(\sqrt{n\log n})$?

来源字段：https://doi.org/10.4171/owr/2007/20

14813 | OPG-60042

### Multicolour Erdős--Hajnal Conjecture

P / partially_solved | Open Problem Garden

Conjecture For every fixed $k\geq2$ and fixed colouring $\chi$ of $E(K_k)$ with $m$ colours, there exists $\varepsilon>0$ such that every colouring of the edges of $K_n$ contains either $k$ vertices whose edges are coloured according to $\chi$ or $n^\varepsilon$ vertices whose edges are coloured with at most $m-1$ colours.

背景链接：http://www.openproblemgarden.org/op/multicolour_erdos_hajnal_conjecture

14814 | OPG-801

### Nearly spanning regular subgraphs

P / partially_solved | Open Problem Garden

Conjecture For every $\epsilon > 0$ and every positive integer $k$, there exists $r_0 = r_0(\epsilon,k)$ so that every simple $r$-regular graph $G$ with $r \ge r_0$ has a $k$-regular subgraph $H$ with $|V(H)| \ge (1- \epsilon) |V(G)|$.

背景链接：http://www.openproblemgarden.org/op/nearly_spanning_regular_subgraphs

14815 | AMR-087-0090

### Necessity of the no-CM condition for Heegner bounds

O / open | AMR 问题清单汇编

Is the no-complex-multiplication condition in the stated lower bound for Heegner points necessary?

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14816 | AMR-087-0089

### Necessity of the odd-class-number condition for Heegner bounds

O / open | AMR 问题清单汇编

Is the odd-class-number condition in the stated lower bound for Heegner points necessary?

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14817 | GRAPH-019

### Negami's Conjecture

P / partially_solved | 独立 / 未分配集合

Does every graph with a planar cover have a projective-plane embedding?

背景链接：https://arxiv.org/abs/2412.19560

14818 | OPG-1757

### Negative association in uniform forests

P / partially_solved | Open Problem Garden

Conjecture Let $G$ be a finite graph, let $e,f \in E(G)$, and let $F$ be the edge set of a forest chosen uniformly at random from all forests of $G$. Then $$ {\mathbb P}(e \in F \mid f \in F}) \le {\mathbb P}(e \in F) $$

背景链接：http://www.openproblemgarden.org/op/negative_association_in_uniform_forests

14819 | OWR-14299575-001

### Negative Dependence for Random Spanning Forests

O / open | Oberwolfach 报告问题汇编

Let $G=(V,E)$ be a graph and let $\mathcal F$ be its family of spanning forests. Does the generating polynomial $p_{\mathcal F}(x)=\sum_{F\in\mathcal F}x^F$ satisfy the negative-dependence property $R(1,\{1\})$? More broadly, is there a class between real-stable and Lorentzian polynomials for which $R(c,\mathbb R_{\geq0})$ holds for some $c\in[1,2]$?

来源字段：https://doi.org/10.4171/owr/2025/47

14820 | OPG-1793

### Non-edges vs. feedback edge sets in digraphs

P / partially_solved | Open Problem Garden

For any simple digraph $G$, we let $\gamma(G)$ be the number of unordered pairs of nonadjacent vertices (i.e. the number of non-edges), and $\beta(G)$ be the size of the smallest feedback edge set.

Conjecture If $G$ is a simple digraph without directed cycles of length $\le 3$, then $\beta(G) \le \frac{1}{2} \gamma(G)$.

背景链接：http://www.openproblemgarden.org/op/non_edges_vs_feedback_edge_sets_in_digraphs

14821 | AMR-087-0080

### Non-malleability of real RSA key generators

O / open | AMR 问题清单汇编

Use number theory to prove non-malleability properties for real-world RSA key-generation algorithms.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14822 | AMR-099-0006

### Nonamenable subgraphs of transitive graphs with exponential growth

O / open | AMR 问题清单汇编

Must every vertex-transitive graph of exponential growth contain an infinite subgraph $H$ with positive Cheeger constant $h(H)>0$? Can $H$ always be chosen to be a tree?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=13

14823 | AMR-099-0007

### Nonamenable subgraphs under uniform exponential growth

O / open | AMR 问题清单汇编

Must every graph with uniform exponential volume growth contain an infinite subgraph, possibly a tree, having positive Cheeger constant?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=13

14824 | AMR-087-0017

### Nondegenerate surfaces in toric threefolds

P / partially_solved | AMR 问题清单汇编

Work out effective zeta-function computations for nondegenerate surfaces in toric threefolds.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14825 | OWR-11451-014

### Nondeterministic Polynomial Characterization of Seymour Graphs

O / open | Oberwolfach 报告问题汇编

Can one find a nondeterministic polynomial-time characterization of Seymour graphs?

来源字段：https://doi.org/10.4171/owr/2011/53

14826 | OWR-711-007

### Nonprincipal Ramsey-Turán Families

O / open | Oberwolfach 报告问题汇编

Do there exist two graphs $G_1$ and $G_2$ such that the joint Ramsey-Turán density $\rho(\{G_1,G_2\})$ is positive and strictly smaller than both individual densities?

来源字段：https://doi.org/10.4171/owr/2004/01

14827 | AMR-088-0007

### Nonvanishing of the p-adic zeta function at even integers

O / open | AMR 问题清单汇编

Let $\zeta_p:\mathbb{Z}_p\to\mathbb{Q}_p$ be the $p$-adic zeta function. Is $\zeta_p(k)\ne0$ for every even integer $k$?

背景链接：https://www.maths.nottingham.ac.uk/plp/pmzcw/download/open.pdf

14828 | OWR-1189-009

### Normality Probability of Intermediate-Density Random Graphs

O / open | Oberwolfach 报告问题汇编

Let $G\sim G(n,p)$, where the edge probability stays bounded away from both $0$ and $1$ as $n\to\infty$. A graph is normal if it admits a clique cover $\mathcal{Q}$ and a stable-set cover $\mathcal{S}$ such that every member of $\mathcal{Q}$ intersects every member of $\mathcal{S}$. Determine the asymptotic probability that $G$ is normal.

来源字段：https://doi.org/10.4171/owr/2006/07

14829 | AMR-087-0064

### NP-hardness of ideal-lattice SVP

P / partially_solved | AMR 问题清单汇编

Is the shortest vector problem on ideal or cyclic lattices NP-hard, either exactly or under approximation?

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14830 | AMR-087-0065

### NP-hardness of minimum distance for cyclic codes

O / open | AMR 问题清单汇编

Is the minimum-distance problem for cyclic codes NP-hard?

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14831 | OPG-37081

### Number of Cliques in Minor-Closed Classes

P / partially_solved | Open Problem Garden

Question Is there a constant $c$ such that every $n$-vertex $K_t$-minor-free graph has at most $c^tn$ cliques?

背景链接：http://www.openproblemgarden.org/op/number_of_cliques_in_minor_closed_classes

14832 | AMR-083-0011

### O10 - Factoring from roots modulo a composite

P / partially_solved | AMR 问题清单汇编

Let $C_{10}$ find $x$ satisfying $x^e\equiv a\pmod n$ under $\gcd(e,\varphi(n))=\gcd(a,n)=1$. Is complete integer factorization randomized polynomial-time reducible to $C_{10}$?

背景链接：https://mccurley.org/papers/open.ps.gz

14833 | AMR-083-0012

### O11a - Quadratic residuosity modulo a composite

O / open | AMR 问题清单汇编

Can one decide in deterministic polynomial time whether a coprime integer $a$ is a square modulo a composite $n$?

背景链接：https://mccurley.org/papers/open.ps.gz

14834 | AMR-083-0013

### O11b - Factoring from composite quadratic residuosity

O / open | AMR 问题清单汇编

Is complete integer factorization randomized polynomial-time reducible to deciding quadratic residuosity modulo a composite?

背景链接：https://mccurley.org/papers/open.ps.gz

14835 | AMR-083-0014

### O12 - Finding a quadratic nonresidue

O / open | AMR 问题清单汇编

Given a prime $p$, can a quadratic nonresidue modulo $p$ be found in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14836 | AMR-083-0015

### O13 - Realizing a prescribed quadratic signature

O / open | AMR 问题清单汇编

Given a sign vector $\varepsilon\in\{-1,1\}^k$, can the least prime $p$ satisfying $(p_i/p)=\varepsilon_i$ for every $i\le k$ be found in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14837 | AMR-083-0016

### O14 - Square roots modulo a prime

O / open | AMR 问题清单汇编

Given a prime $p$ and a quadratic residue $a$, can a square root $x^2\equiv a\pmod p$ be found in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14838 | AMR-083-0017

### O15 - Polynomial roots modulo a prime

O / open | AMR 问题清单汇编

Given a prime $p$ and $f\in(\mathbb{Z}/p\mathbb{Z})[x]$ known to have a root, can a root be found in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14839 | AMR-083-0018

### O16 - Factoring polynomials modulo a prime

O / open | AMR 问题清单汇编

Given a prime $p$ and $f\in(\mathbb{Z}/p\mathbb{Z})[x]$, can the complete irreducible factorization of $f$ be found in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14840 | AMR-083-0019

### O17 - Constructing irreducible polynomials over finite fields

P / partially_solved | AMR 问题清单汇编

Given a prime $p$ and degree $d$, can an irreducible polynomial of degree $d$ over $\mathbb{F}_p$ be constructed in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14841 | AMR-083-0020

### O18a - Recognizing primitive roots deterministically

O / open | AMR 问题清单汇编

Given a prime $p$ and $b$, can one decide in deterministic polynomial time whether $b$ generates $(\mathbb{Z}/p\mathbb{Z})^*$?

背景链接：https://mccurley.org/papers/open.ps.gz

14842 | AMR-083-0021

### O18b - Recognizing primitive roots randomly

O / open | AMR 问题清单汇编

Is recognition of primitive roots modulo a prime in randomized polynomial time $R$?

背景链接：https://mccurley.org/papers/open.ps.gz

14843 | AMR-083-0022

### O19 - Finding a primitive root modulo a prime

O / open | AMR 问题清单汇编

Given a prime $p$, can a generator of $(\mathbb{Z}/p\mathbb{Z})^*$ be found in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14844 | AMR-083-0023

### O20 - Computing multiplicative orders modulo a prime

O / open | AMR 问题清单汇编

Given a prime $p$ and $a$ coprime to $p$, can $\operatorname{ord}_p(a)$ be computed in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14845 | AMR-083-0024

### O21 - Discrete logarithms modulo a prime

O / open | AMR 问题清单汇编

Given a prime $p$ and elements $g,b$ with $b$ in the subgroup generated by $g$, can an exponent $x$ satisfying $g^x\equiv b\pmod p$ be found in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14846 | AMR-083-0025

### O22a - Discrete logarithms modulo a composite

O / open | AMR 问题清单汇编

Given $g,b,n$ such that $g^x\equiv b\pmod n$ has a solution, can such an exponent $x$ be found in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14847 | AMR-083-0026

### O22b - Factoring from composite discrete logarithms

P / partially_solved | AMR 问题清单汇编

Is complete integer factorization deterministically polynomial-time reducible to discrete logarithms modulo composites?

背景链接：https://mccurley.org/papers/open.ps.gz

14848 | AMR-083-0027

### O23 - Factoring from Euler's totient

P / partially_solved | AMR 问题清单汇编

Is complete integer factorization deterministically polynomial-time reducible to computing $\varphi(n)$?

背景链接：https://mccurley.org/papers/open.ps.gz

14849 | AMR-083-0028

### O24 - Finding a point on an elliptic curve

S / solved | AMR 问题清单汇编

Given $a,b$ and a prime $p$ with nonsingular curve $y^2=x^3+ax+b$, can a point on the curve modulo $p$ be found in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14850 | AMR-083-0029

### O25 - Solving binary quadratic congruences

O / open | AMR 问题清单汇编

Given $k,m,n$ with odd $n$ and $\gcd(km,n)=1$, can integers $x,y$ satisfying $x^2-ky^2\equiv m\pmod n$ be found in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14851 | AMR-083-0030

### O26 - Discrete logarithm versus Diffie-Hellman key distribution

O / open | AMR 问题清单汇编

Is discrete logarithm modulo a prime randomized polynomial-time reducible to computing $g^{xy}$ from $g,g^x,g^y$?

背景链接：https://mccurley.org/papers/open.ps.gz

14852 | AMR-083-0031

### O27 - Elliptic curves of prescribed order

O / open | AMR 问题清单汇编

Given a prime $p$ and $n$, can one construct in deterministic polynomial time an elliptic curve over $\mathbb{F}_p$ having exactly $n$ points whenever one exists?

背景链接：https://mccurley.org/papers/open.ps.gz

14853 | AMR-083-0032

### O28 - Discrete logarithms in elliptic-curve groups

O / open | AMR 问题清单汇编

Given an elliptic curve over $\mathbb{F}_p$ and points $P,Q$ such that $P=nQ$ for some $n$, can such an $n$ be found in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14854 | AMR-083-0033

### O29 - NP-hardness of exact shortest vector

S / solved | AMR 问题清单汇编

For a full-rank integer lattice, is finding a nonzero vector of minimum Euclidean norm NP-hard?

背景链接：https://mccurley.org/papers/open.ps.gz

14855 | AMR-083-0002

### O3 - Finding a prime above a bound

O / open | AMR 问题清单汇编

Given $n\in\mathbb{N}$, can a prime $p>n$ be found in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14856 | AMR-083-0034

### O30 - Polynomial-factor lattice approximation

P / partially_solved | AMR 问题清单汇编

Does there exist a constant $c$ for which one can find in deterministic polynomial time a nonzero lattice vector of length at most $n^c$ times the minimum?

背景链接：https://mccurley.org/papers/open.ps.gz

14857 | AMR-083-0035

### O31 - Order of a polynomial's Galois group

P / partially_solved | AMR 问题清单汇编

Given $f\in\mathbb{Q}[x]$, can the degree of its splitting field, equivalently the order of its Galois group, be computed in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14858 | AMR-083-0036

### O32 - Class numbers of imaginary quadratic orders

P / partially_solved | AMR 问题清单汇编

Given $d\in\mathbb{N}$, can the class number $h(-d)$ of binary quadratic forms of discriminant $-d$ be computed in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14859 | AMR-083-0037

### O33a - NP-hardness of binary quadratic Diophantine solvability

S / solved | AMR 问题清单汇编

Under the promise that $b^2-4ac$ is not a square, is deciding whether $ax^2+bxy+cy^2+dx+ey+f=0$ has an integral solution NP-hard?

背景链接：https://mccurley.org/papers/open.ps.gz

14860 | AMR-083-0038

### O33b - Randomized NP-hardness of binary quadratic Diophantine solvability

S / solved | AMR 问题清单汇编

Is the same binary quadratic Diophantine solvability problem NP-hard under randomized reductions?

背景链接：https://mccurley.org/papers/open.ps.gz

14861 | AMR-083-0039

### O34 - Solvability of the negative Pell equation

O / open | AMR 问题清单汇编

Can one decide in deterministic polynomial time whether $x^2-dy^2=-1$ has an integral solution?

背景链接：https://mccurley.org/papers/open.ps.gz

14862 | AMR-083-0040

### O35 - Greatest common divisors in NC

P / partially_solved | AMR 问题清单汇编

Can $\gcd(a,b)$ be computed in the parallel complexity class $NC$?

背景链接：https://mccurley.org/papers/open.ps.gz

14863 | AMR-083-0041

### O36 - Integer multiplication in linear bit complexity

P / partially_solved | AMR 问题清单汇编

Can two positive integers $a,b$ be multiplied using $O(\log(ab))$ bit operations?

背景链接：https://mccurley.org/papers/open.ps.gz

14864 | AMR-083-0003

### O4 - Finding a prime in an arithmetic progression

O / open | AMR 问题清单汇编

Given coprime $a,n\in\mathbb{N}$, can a prime $p\equiv a\pmod n$ be found in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14865 | AMR-083-0004

### O5a - Deterministic polynomial-time integer factorization

O / open | AMR 问题清单汇编

Is complete integer factorization $C_5$ in deterministic polynomial time $P$?

背景链接：https://mccurley.org/papers/open.ps.gz

14866 | AMR-083-0005

### O5b - Randomized polynomial-time integer factorization

O / open | AMR 问题清单汇编

Is complete integer factorization $C_5$ in randomized polynomial time $R$?

背景链接：https://mccurley.org/papers/open.ps.gz

14867 | AMR-083-0006

### O6 - Factoring a positive-density set of integers

O / open | AMR 问题清单汇编

Does there exist a set $S\subset\mathbb{N}$ of positive lower asymptotic density for which complete factorization of every input $n\in S$ is in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14868 | AMR-083-0007

### O7a - Computing the squarefree part

O / open | AMR 问题清单汇编

Given $n$, can one find $r,s\in\mathbb{N}$ with $n=r^2s$ and $s$ squarefree in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14869 | AMR-083-0008

### O7b - Factoring from a squarefree-part oracle

O / open | AMR 问题清单汇编

Is complete integer factorization randomized polynomial-time reducible to computation of the squarefree part?

背景链接：https://mccurley.org/papers/open.ps.gz

14870 | AMR-083-0009

### O8 - Deterministic polynomial-time squarefreeness testing

P / partially_solved | AMR 问题清单汇编

Can one decide in deterministic polynomial time whether an integer $n$ is squarefree?

背景链接：https://mccurley.org/papers/open.ps.gz

14871 | AMR-083-0010

### O9 - Counting distinct prime factors

O / open | AMR 问题清单汇编

Can $\omega(n)$, the number of distinct prime factors of $n$, be computed in deterministic polynomial time?

背景链接：https://mccurley.org/papers/open.ps.gz

14872 | OWR-16763-017

### Oberwolfach Decompositions into Prescribed Two-Factors

P / partially_solved | Oberwolfach 报告问题汇编

Given an odd integer $n$ and a $2$-regular graph $F$ on $n$ vertices, can $K_n$ be decomposed into edge-disjoint copies of $F$?

来源字段：https://doi.org/10.4171/owr/2019/1

14873 | GRAPH-034

### Oberwolfach Problem

P / partially_solved | 独立 / 未分配集合

For which 2-regular graphs H can the complete graph be decomposed into edge-disjoint copies of H?

背景链接：https://doi.org/10.1016/j.jctb.2021.09.007

14874 | OPG-37357

### Obstacle number of planar graphs

P / partially_solved | Open Problem Garden

Does there exist a planar graph with obstacle number greater than 1? Is there some $k$ such that every planar graph has obstacle number at most $k$?

背景链接：http://www.openproblemgarden.org/op/obstacle_number_of_planar_graphs

14875 | OPG-37182

### Odd cycles and low oddness

S / solved | Open Problem Garden

Conjecture If in a bridgeless cubic graph $G$ the cycles of any $2$-factor are odd, then $\omega(G)\leq 2$, where $\omega(G)$ denotes the oddness of the graph $G$, that is, the minimum number of odd cycles in a $2$-factor of $G$.

背景链接：http://www.openproblemgarden.org/op/odd_cycles_and_low_oddness

14876 | OPG-46824

### Odd-cycle transversal in triangle-free graphs

P / partially_solved | Open Problem Garden

Conjecture If $G$ is a simple triangle-free graph, then there is a set of at most $n^2/25$ edges whose deletion destroys every odd cycle.

背景链接：http://www.openproblemgarden.org/op/odd_cycle_transversal_in_triangle_free_graphs

14877 | AMR-090-0001

### Odd-prime-power periodicity conjecture

P / partially_solved | AMR 问题清单汇编

For every odd prime $p$ and $k\ge1$, is $s(p^k)=k$? For $k\ge2$, is $d(p^k)=p^{k-1}d(p)$?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/18-79/index.html

14878 | OWR-1536-005

### Online Versus Ordinary Size-Ramsey Numbers

O / open | Oberwolfach 报告问题汇编

For every fixed positive integer $c$, is $\lim_{t\to\infty}\mathrm{size\text{-}oRam}_c^2(t)/\mathrm{size\text{-}Ram}_c^2(t)=0$, where the numerator and denominator are the online and ordinary size-Ramsey numbers in the coloring-Ramsey setting?

来源字段：https://doi.org/10.4171/owr/2007/20

14879 | AMR-087-0027

### Optimal polynomial collections for lattice attacks

O / open | AMR 问题清单汇编

Find an optimal collection of polynomials for multivariate lattice-based small-root attacks.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14880 | OWR-9790352-011

### Optimal Product Structure for Planar Graphs

O / open | Oberwolfach 报告问题汇编

What is the smallest integer $k$ such that every planar graph is a subgraph of $H\boxtimes P$ for some path $P$ and some graph $H$ of treewidth at most $k$? Is the optimum between $3$ and $6$?

来源字段：https://doi.org/10.4171/owr/2022/1

14881 | OWR-1183-014

### Ore-Type Conditions for Graph Packing

O / open | Oberwolfach 报告问题汇编

For an $n$-vertex graph $G$, let $\sigma_2(G)=\max_{uv\in E(G)}(\deg_G(u)+\deg_G(v))/2$. If $G_1,G_2$ are $n$-vertex graphs and $(\sigma_2(G_1)+1)(\Delta(G_2)+1)\leq n+1$, must $G_1$ and $G_2$ pack?

来源字段：https://doi.org/10.4171/owr/2006/01

14882 | OPG-494

### Oriented chromatic number of planar graphs

P / partially_solved | Open Problem Garden

An oriented colouring of an oriented graph is assignment $c$ of colours to the vertices such that no two arcs receive ordered pairs of colours $(c_1,c_2)$ and $(c_2,c_1)$. It is equivalent to a homomorphism of the digraph onto some tournament of order $k$.

Problem What is the maximal possible oriented chromatic number of an oriented planar graph?

背景链接：http://www.openproblemgarden.org/op/oriented_chromatic_number_of_planar_graphs

14883 | OPG-46167

### Oriented trees in n-chromatic digraphs

P / partially_solved | Open Problem Garden

Conjecture Every digraph with chromatic number at least $2k-2$ contains every oriented tree of order $k$ as a subdigraph.

背景链接：http://www.openproblemgarden.org/op/oriented_trees_in_n_chromatic_digraphs

14884 | GRAPH-014

### Overfull Conjecture

P / partially_solved | 独立 / 未分配集合

Is a graph with maximum degree Δ(G) ≥ n/3 in class 2 if and only if it has an overfull subgraph with the same maximum degree?

背景链接：https://arxiv.org/abs/2607.02270

14885 | OPG-144

### Packing T-joins

P / partially_solved | Open Problem Garden

Conjecture There exists a fixed constant $c$ (probably $c=1$ suffices) so that every graft with minimum $T$-cut size at least $k$ contains a $T$-join packing of size at least $(2/3)k-c$.

背景链接：http://www.openproblemgarden.org/op/packing_t_joins

14886 | AMR-087-0060

### Pairing signatures without distortion maps

S / solved | AMR 问题清单汇编

Give the cited pairing-based signature constructions and their security proofs without relying on distortion maps.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14887 | AMR-087-0054

### Pairing-friendly hyperelliptic curves

P / partially_solved | AMR 问题清单汇编

Construct pairing-friendly hyperelliptic curves suitable for cryptography.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14888 | AMR-084-0008

### Parameters admitting infinitely many rational D(q)-quintuples

P / partially_solved | AMR 问题清单汇编

For which rational numbers $q$ do there exist infinitely many rational $D(q)$-quintuples?

背景链接：https://web.math.pmf.unizg.hr/~duje/dtuples.html

14889 | OPG-771

### Partial List Coloring

P / partially_solved | Open Problem Garden

别名：Partial List Coloring [OPG-788]

Conjecture Let $G$ be a simple graph with $n$ vertices and list chromatic number $\chi_\ell(G)$. Suppose that $0\leq t\leq \chi_\ell$ and each vertex of $G$ is assigned a list of $t$ colors. Then at least $\frac{tn}{\chi_\ell(G)}$ vertices of $G$ can be colored from these lists.

背景链接：http://www.openproblemgarden.org/op/partial_list_coloring

14890 | OPG-46613

### Partition of a cubic 3-connected graphs into paths of length 2.

O / open | Open Problem Garden

Problem Does every $3$-connected cubic graph on $3k$ vertices admit a partition into $k$ paths of length $2$?

背景链接：http://www.openproblemgarden.org/op/partition_of_a_cubic_3_connected_graphs_into_paths_of_length_2

14891 | OPG-146

### Partitioning edge-connectivity

P / partially_solved | Open Problem Garden

Question Let $G$ be an $(a+b+2)$-edge-connected graph. Does there exist a partition $\{A,B\}$ of $E(G)$ so that $(V,A)$ is $a$-edge-connected and $(V,B)$ is $b$-edge-connected?

背景链接：http://www.openproblemgarden.org/op/partitioning_edge_connectivity

14892 | OPG-47643

### Partitionning a tournament into k-strongly connected subtournaments.

P / partially_solved | Open Problem Garden

Problem Let $k_1, \dots, k_p$ be positve integer Does there exists an integer $g(k_1, \dots, k_p)$ such that every $g(k_1, \dots, k_p)$-strong tournament $T$ admits a partition $(V_1\dots, V_p)$ of its vertex set such that the subtournament induced by $V_i$ is a non-trivial $k_i$-strong for all $1\leq i\leq p$.

背景链接：http://www.openproblemgarden.org/op/partitionning_a_tournament_into_k_strongly_connected_subtournaments

14893 | OWR-15211-001

### PDE-Inspired Graph Functionals and Their Continuum Limits

P / partially_solved | Oberwolfach 报告问题汇编

Given a PDE-inspired functional $f$ on weighted graphs and its continuum counterpart $F$, which structural properties of $F$ have graph analogues, and under what graph-limit regimes does $f$ converge to $F$? Can minimization algorithms for $f$ be used in data analysis or image processing, and which other PDE or variational systems admit useful graph analogues?

来源字段：https://doi.org/10.4171/owr/2017/6

14894 | OPG-586

### Pebbling a cartesian product

P / partially_solved | Open Problem Garden

We let $p(G)$ denote the pebbling number of a graph $G$.

Conjecture $p(G_1 \Box G_2) \le p(G_1) p(G_2)$.

背景链接：http://www.openproblemgarden.org/op/pebbling_a_cartesian_product

14895 | OPG-167

### Pentagon problem

P / partially_solved | Open Problem Garden

Question Let $G$ be a 3-regular graph that contains no cycle of length shorter than $g$. Is it true that for large enough~ $g$ there is a homomorphism $G \to C_5$?

背景链接：http://www.openproblemgarden.org/op/pentagon_problem

14896 | OWR-16930-001

### Percolation Structure in Nearly Optimal Disk Packings

O / open | Oberwolfach 报告问题汇编

For a packing of unit disks in $\mathbb R^2$ whose density is sufficiently close to that of the triangular-lattice packing, join two centers when their distance is at most $2+\varepsilon$. Under what quantitative density assumptions must the resulting graph contain an infinite connected component?

来源字段：https://doi.org/10.4171/owr/2019/18

14897 | GRAPH-009

### Perfect 1-Factorization Conjecture

P / partially_solved | 独立 / 未分配集合

Does every complete graph on an even number of vertices admit a perfect 1-factorization?

背景链接：https://arxiv.org/abs/2607.09459

14898 | OPG-143

### Petersen coloring conjecture

P / partially_solved | Open Problem Garden

Conjecture Let $G$ be a cubic graph with no bridge. Then there is a coloring of the edges of $G$ using the edges of the Petersen graph so that any three mutually adjacent edges of $G$ map to three mutually adjancent edges in the Petersen graph.

背景链接：http://www.openproblemgarden.org/op/petersen_coloring_conjecture

14899 | OWR-10252930-029

### Petersen-Graph Counts in Sparse Pseudorandom Graphs

O / open | Oberwolfach 报告问题汇编

Let $G$ be an $n$-vertex, $C_4$-free graph with edge density $p\gtrsim n^{-1/2}$ such that $$|e_G(A,B)-p|A||B||=o(pn^2)$$ for all $A,B\subseteq[n]$. Is the number of copies of the Petersen graph in $G$, up to a constant factor, at least its expected number in $G(n,p)$?

来源字段：https://doi.org/10.4171/owr/2022/22

14900 | OWR-784-007

### Pfaffian Conditions for Graphs on Orientable Surfaces

O / open | Oberwolfach 报告问题汇编

For a graph $G$ and a nonnegative integer $g$, are the following equivalent: (1) $G$ has a drawing on an orientable surface of genus $g$ such that $\operatorname{cr}(M)$ is even for every perfect matching $M$; (2) $G$ is $4^g$-Pfaffian; and (3) $G$ is $(4^{g+1}-1)$-Pfaffian?

来源字段：https://doi.org/10.4171/owr/2005/03

14901 | OWR-1536-015

### Picker Surplus and Maker-Breaker Monotonicity

O / open | Oberwolfach 报告问题汇编

For the Picker-Chooser game on a hypergraph $H$ consisting of $2^n$ pairwise disjoint sets of size $n$, what happens when $|H|<2^n$, and what if Picker only wants some surplus? For the Maker-Breaker $(1,k)$ connectivity game, with $c(G,k)$ the number of Maker's final edges when Maker wins and $\infty$ otherwise and $C_n(k)=\min_G c(G,k)$ over $n$-vertex graphs, is $C_n(k)\leq C_n(k+1)$, and what analogous functions and questions arise for other Maker-Breaker games?

来源字段：https://doi.org/10.4171/owr/2007/20

14902 | OWR-9790352-034

### Planar Connectivity Models for Cross-Free Societies

O / open | Oberwolfach 报告问题汇编

Which cross-free societies admit a planar connectivity model, and how can such societies be characterized?

来源字段：https://doi.org/10.4171/owr/2022/1

14903 | OWR-1536-018

### Planar Game Chromatic Numbers and Builder Strategies

O / open | Oberwolfach 报告问题汇编

What is the maximum game chromatic number of a planar graph, known to lie between $11$ and $17$? In the online game where Builder adds edges while keeping the chosen-edge graph planar and Painter colors each new edge red or blue, does Builder force a monochromatic copy of a fixed graph $H$ if and only if $H$ is outerplanar?

来源字段：https://doi.org/10.4171/owr/2007/20

14904 | OWR-8415352-012

### Polyhedral Cones of Graph Profile Inequality Exponents

O / open | Oberwolfach 报告问题汇编

For a finite collection $U$ of connected graphs, do the exponents of binomial inequalities valid on the profile $G_U$ form a polyhedral cone, and if so is it necessarily a rational polyhedral cone?

来源字段：https://doi.org/10.4171/owr/2021/49

14905 | OWR-14299905-018

### Polylogarithmic Treewidth in Induced-Subgraph-Free Graph Classes

O / open | Oberwolfach 报告问题汇编

For every positive integer $t$, does there exist an integer $d(t)$ such that every $n$-vertex graph in the hereditary class $\mathcal C_t^*$ has treewidth at most $(\log n)^{d(t)}$?

来源字段：https://doi.org/10.4171/owr/2026/1

14906 | OWR-9790352-010

### Polynomial $chi$-Bounds for Pivot-Minor-Closed Classes

O / open | Oberwolfach 报告问题汇编

For every fixed graph $H$, are graphs with no pivot-minor isomorphic to $H$ polynomially $\chi$-bounded?

来源字段：https://doi.org/10.4171/owr/2022/1

14907 | OWR-11786-017

### Polynomial Hirsch Conjecture for Superstrongly Connected Complexes

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a polynomial $P(n,d)$ that bounds the dual-graph diameter of every superstrongly connected $d$-dimensional simplicial complex on $n$ vertices?

来源字段：https://doi.org/10.4171/owr/2012/24

14908 | OWR-14299081-008

### Polynomial Homogeneous Sets in Induced-Free Graphs

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Erdős-Hajnal Conjecture Passage [OWR-14299081-009]; Duplicate Erdős-Hajnal Conjecture Passage [OWR-14299081-010]; Duplicate Erdős-Hajnal Conjecture Lead-In [OWR-14299081-020]

For every graph $H$, is there a constant $c(H)>0$ such that every $n$-vertex induced-$H$-free graph contains a clique or an independent set of size at least $n^{c(H)}$?

来源字段：https://doi.org/10.4171/owr/2025/1

14909 | OWR-16931-022

### Polynomial Induced Removal Bounds for Graph Pairs

P / partially_solved | Oberwolfach 报告问题汇编

For graphs $H_1,H_2$, let $\delta_{H_1,H_2}(\varepsilon)$ be the largest $\delta$ such that every graph that is $\delta$-close to being induced-$H_i$-free for each $i\in\{1,2\}$ is $\varepsilon$-close to being induced-$\{H_1,H_2\}$-free. Is $\delta_{H_1,H_2}(\varepsilon)$ polynomial in $\varepsilon$ for every pair? For every $a>0$, do some $H_1,H_2$ satisfy $\delta_{H_1,H_2}(\varepsilon)=O(\varepsilon^a)$?

来源字段：https://doi.org/10.4171/owr/2019/19

14910 | AMR-087-0062

### Polynomial-factor hardness of general lattice problems

O / open | AMR 问题清单汇编

Prove that general SVP and SIVP are hard in the worst case to approximate within small polynomial factors.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14911 | AMR-087-0063

### Polynomial-factor hardness of ideal-lattice problems

P / partially_solved | AMR 问题清单汇编

Prove an analogous small-polynomial-factor worst-case hardness result for SVP and SIVP on ideal lattices.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14912 | AMR-087-0025

### Polynomial-shape dependence in small-root algorithms

O / open | AMR 问题清单汇编

Understand and control how the shape of a polynomial affects Coppersmith-type small-root algorithms.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14913 | AMR-087-0011

### Polynomial-time curve zeta computation in genus and field size

O / open | AMR 问题清单汇编

Is computation of a curve's zeta function polynomial simultaneously in the genus $g$ and in $\log q$?

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14914 | OWR-15206-017

### Positive Association in Random Proper Colorings

O / open | Oberwolfach 报告问题汇编

Let $G$ be a finite bipartite graph with parts $A$ and $B$, let $q\geq3$, and let $f$ be a uniformly random proper $q$-coloring of $G$. Does the random set $f^{-1}(0)\cap A$ have positive association? Equivalently, for all increasing functions $X,Y:\{0,1\}^A\to\mathbb R$, is $$ \mathbb E[X(f^{-1}(0)\cap A)Y(f^{-1}(0)\cap A)]\geq\mathbb E[X(f^{-1}(0)\cap A)]\,\mathbb E[Y(f^{-1}(0)\cap A)] $$?

来源字段：https://doi.org/10.4171/owr/2017/1

14915 | OWR-12697684-029

### Positive Discrepancy away from Multipartite Densities

O / open | Oberwolfach 报告问题汇编

For every $\varepsilon>0$, does there exist $a>0$ such that every $n$-vertex graph of density $p$ whose distance from every number $(i-1)/i$ is at least $\varepsilon$ satisfies $\operatorname{disc}^+(G)\geq a\sqrt{p(1-p)}\,n^{3/2}$?

来源字段：https://doi.org/10.4171/owr/2023/1

14916 | OWR-12697711-003

### Positive Weighted Graph Homomorphism Densities

O / open | Oberwolfach 报告问题汇编

Characterize the finite graphs $H$ for which $t(H,G_w)\geq 0$ for every weighted graph $G_w$, allowing real edge weights.

来源字段：https://doi.org/10.4171/owr/2023/15

14917 | AMR-090-0002

### Power-of-two periodicity conjecture

O / open | AMR 问题清单汇编

For every $k\ge1$, is $s(2^k)=u_k$? Is $d(2^k)=2^k$ for $k\ne2$, with $d(4)=2$?

背景链接：https://amj.math.stonybrook.edu/html-articles/Files-2015-2024/18-79/index.html

14918 | AMR-087-0082

### Practical trapdoor discrete-logarithm groups

O / open | AMR 问题清单汇编

Construct practical groups in which discrete logarithms have an effective trapdoor.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14919 | AMR-087-0039

### Prime extension-degree quotients of elliptic-curve orders

O / open | AMR 问题清单汇编

For a fixed $E/\mathbb{F}_q$, prove that $\#E(\mathbb{F}_{q^n})/\#E(\mathbb{F}_q)$ is prime for infinitely many $n$.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14920 | OWR-4791-027

### Prime Labelings of Trees

P / partially_solved | Oberwolfach 报告问题汇编

Call a graph $G$ of order $n$ prime if its vertices can be bijectively labelled by $1,\ldots,n$ so that adjacent vertices receive coprime labels. Is every tree prime?

来源字段：https://doi.org/10.4171/owr/2011/01

14921 | AMR-087-0040

### Prime reductions of elliptic curves over the rationals

O / open | AMR 问题清单汇编

For a torsion-free elliptic curve $E/\mathbb{Q}$, prove that $\#E(\mathbb{F}_p)$ is prime for infinitely many primes $p$.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14922 | AMR-087-0038

### Prime-order curves over every finite field

P / partially_solved | AMR 问题清单汇编

Prove that there are sufficiently many prime-order elliptic curves over every finite field $\mathbb{F}_q$.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14923 | AMR-088-0014

### Prime-sum growth and elliptic-curve rank

O / open | AMR 问题清单汇编

For an elliptic curve $E/\mathbb{Q}$ of rank $r$, does $f(X)=\sum_{p\le X}\log(N_p/p)$ grow asymptotically like $r\log\log X$?

背景链接：https://www.maths.nottingham.ac.uk/plp/pmzcw/download/open.pdf

14924 | AMR-086-0001

### Problem 1.1

P / partially_solved | AMR 问题清单汇编

Let $f\in\mathbb{Z}[X,Y]$ be a polynomial such that the equation $f(x,y)=0$ has only finitely many solutions $(x,y)\in \mathbb{Z}\times\mathbb{Z}$. Give an upper bound for $\max\{|x|,|y|\}$ when $(x,y)$ is such a solution, in terms of the degree of $f$ and of the maximal absolute value of the coefficients of $f$.

背景链接：https://arxiv.org/abs/math/0312440

14925 | AMR-086-0051

### Problem 4.6

P / partially_solved | AMR 问题清单汇编

For $\theta\in(0,\pi)$, define $$ V_\theta=\{re^{it}\; ;\; r>0,\; |t|>\theta\}. $$ Compute $L(V_\theta)$ in terms of $\theta$.

背景链接：https://arxiv.org/abs/math/0312440

14926 | OPG-47646

### PTAS for feedback arc set in tournaments

P / partially_solved | Open Problem Garden

Question Is there a polynomial time approximation scheme for the feedback arc set problem for the class of tournaments?

背景链接：http://www.openproblemgarden.org/op/ptas_for_feedback_arc_set_in_tournaments

14927 | OWR-1194-007

### Quadrisecant-Free Spatial Graph Embeddings

P / partially_solved | Oberwolfach 报告问题汇编

Is a graph embeddable in $\mathbb R^3$ with curved edges and no strong quadrisecant line if and only if it is $1/2$-planar?

来源字段：https://doi.org/10.4171/owr/2006/12

14928 | AMR-087-0031

### Quality of rotation-augmented cyclic-lattice reduction

O / open | AMR 问题清单汇编

Analyze how effective rotation-augmented lattice reduction is on cyclic or NTRU lattices.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14929 | OWR-4212-002

### Quantitative Improvements to Graph Regularity Applications

P / partially_solved | Oberwolfach 报告问题汇编

Can applications of graph regularity be proved with quantitative bounds substantially better than the tower-type growth of $M(\varepsilon)$ for regularity partitions?

来源字段：https://doi.org/10.4171/owr/2010/11

14930 | AMR-087-0078

### Quantum algorithm for ideal-lattice SVP

P / partially_solved | AMR 问题清单汇编

Develop an efficient quantum algorithm for the shortest vector problem on ideal lattices.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14931 | AMR-087-0077

### Quantum algorithm for Smallest Conjugate

O / open | AMR 问题清单汇编

Develop an efficient quantum algorithm for the Smallest Conjugate problem.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14932 | AMR-086-0045

### Question 3.27 - Mahler

P / partially_solved | AMR 问题清单汇编

Are there entire transcendental functions $f(z)$ such that if $x$ is a Liouville number then so is $f(x)$?

背景链接：https://arxiv.org/abs/math/0312440

14933 | AMR-086-0065

### Question 5.2 - Bugeaud

S / solved | AMR 问题清单汇编

Let $n \ge 2$. Denote by $\mathrm{ZH}_n$ the set of real numbers $\xi$ with the following property: there exists $c_1(\xi) >0$ and $c_2(\xi)>0$ such that for algebraic number $\alpha$ of degree $\le n$, $$ |\xi-\alpha|\ge c_2(\xi) \mathrm{H}(\alpha)^{-n-1}, $$ and such that there are infinitely many algebraic numbers $\alpha$ of degree $\le n$ with $$ |\xi-\alpha|\le c_1(\xi) \mathrm{H}(\alpha)^{-n-1}, $$ Do the set $\mathrm{ZH}_n$ strictly contain the set of algebraic numbers of degree $n+1$?

背景链接：https://arxiv.org/abs/math/0312440

14934 | AMR-054-0052

### Queue-Number of Planar Graphs

S / solved | AMR 问题清单汇编

Does every planar graph have $O(1)$ queue-number? A queue layout of a graph consists of a linear order of the vertices and a partition of the edges into non-nested queues. Edge $xy$ is nested inside edge $vw$ if $v<x<y<w$ in the linear order. The queue-number of a graph $G$ is the minimum number of queues in a queue layout of $G$. This question amounts to asking whether every planar graph has a vertex ordering with a constant number of pairwise nested edges (called a rainbow).

背景链接：https://topp.openproblem.net/p52

14935 | OPG-480

### r-regular graphs are not uniquely hamiltonian.

P / partially_solved | Open Problem Garden

Conjecture If $G$ is a finite $r$-regular graph, where $r > 2$, then $G$ is not uniquely hamiltonian.

背景链接：http://www.openproblemgarden.org/op/uniquely_hamiltonian_graphs

14936 | OWR-14299911-008

### Radius-Function Unimodality in Bounded-Dimensional Median Graphs

O / open | Oberwolfach 报告问题汇编

Does there exist an absolute constant $c>0$ such that every median graph of cube dimension $d$ has all of its radius functions $G_{cd}$-unimodal? Extend the theory, if possible, to suitably edge-weighted median graphs and to useful families of partial cubes.

来源字段：https://doi.org/10.4171/owr/2026/8

14937 | OWR-16633-026

### Rainbow Arborescences Across Arc Partitions

P / partially_solved | Oberwolfach 报告问题汇编

Let a directed graph $G=(V,A)$ with $\lvert V\rvert=k+1$ have its arc set partitioned into $k$ arborescences $F_1,\ldots,F_k$, each having $k$ arcs. Must there be an arborescence $F^*\subseteq A$ such that $\lvert F^*\cap F_i\rvert=1$ for every $i$?

来源字段：https://doi.org/10.4171/owr/2018/50

14938 | OWR-17464-003

### Rainbow Matchings Across Families of Matchings

O / open | Oberwolfach 报告问题汇编

Does every collection of $n$ matchings, each of size $n$, in a graph contain a rainbow matching of size $n-1$?

来源字段：https://doi.org/10.4171/owr/2020/1

14939 | OWR-16633-020

### Rainbow Paths in Properly Edge-Colored Regular Graphs

P / partially_solved | Oberwolfach 报告问题汇编

Does every simple $k$-regular graph with a proper $k$-edge-coloring contain a rainbow path of length $k-1$?

来源字段：https://doi.org/10.4171/owr/2018/50

14940 | OWR-9790352-026

### Rainbow Turán Number of Cycles

P / partially_solved | Oberwolfach 报告问题汇编

For the family $\mathcal C$ of all cycles, is the rainbow Turán number $\operatorname{ex}^*(n,\mathcal C)$ bounded by $O(n\log n)$?

来源字段：https://doi.org/10.4171/owr/2022/1

14941 | OWR-16931-027

### Ramsey Bounds for Edge-Ordered Graphs

O / open | Oberwolfach 报告问题汇编

For every edge-ordered graph $H$ on $n$ vertices, is $$r_{\mathrm{edge}}(H)=2^{O(n)}$$? If $H$ is also $d$-degenerate, is $$r_{\mathrm{edge}}(H)=O(C(d)n)$$ for some function $C$ depending only on $d$?

来源字段：https://doi.org/10.4171/owr/2019/19

14942 | OWR-11781-005

### Ramsey Numbers for Loose Triangles versus Complete Hypergraphs

O / open | Oberwolfach 报告问题汇编

For loose $3$-uniform hypergraph triangles $C_3$ and complete $3$-uniform hypergraphs $K^3_t$, does the Ramsey number satisfy $R(C_3,K^3_t)=o(t^{3/2})$?

来源字段：https://doi.org/10.4171/owr/2012/19

14943 | OWR-12861-012

### Ramsey Numbers in Terms of Degeneracy

O / open | Oberwolfach 报告问题汇编

For every graph $H$ on $n$ vertices, is $$\log r(H)=\Theta(d(H)+\log n),$$ where $d(H)$ is the degeneracy of $H$?

来源字段：https://doi.org/10.4171/owr/2014/01

14944 | OPG-372

### Ramsey properties of Cayley graphs

O / open | Open Problem Garden

Conjecture There exists a fixed constant $c$ so that every abelian group $G$ has a subset $S \subseteq G$ with $-S = S$ so that the Cayley graph ${\mathit Cayley}(G,S)$ has no clique or independent set of size $> c \log |G|$.

背景链接：http://www.openproblemgarden.org/op/ramsey_properties_of_cayley_graphs

14945 | OWR-2090-004

### Random Simplicial-Complex Theory

O / open | Oberwolfach 报告问题汇编

Can one develop a theory of random simplicial complexes analogous to random graph theory?

来源字段：https://doi.org/10.4171/owr/2008/44

14946 | OPG-743

### Random stable roommates

P / partially_solved | Open Problem Garden

Conjecture The probability that a random instance of the stable roommates problem on $n \in 2{\mathbb N}$ people admits a solution is $\Theta( n ^{-1/4} )$.

背景链接：http://www.openproblemgarden.org/op/random_stable_roommates

14947 | OWR-16931-005

### Rational Exponents for Extremal Graph Functions

O / open | Oberwolfach 报告问题汇编

For every rational number $r\in(1,2)$, does there exist a graph $H$ such that $$\operatorname{ex}(n,H)=\Theta(n^r)$$?

来源字段：https://doi.org/10.4171/owr/2019/19

14948 | OPG-135

### Real roots of the flow polynomial

S / solved | Open Problem Garden

Conjecture All real roots of nonzero flow polynomials are at most 4.

背景链接：http://www.openproblemgarden.org/op/real_roots_of_the_flow_polynomial

14949 | OWR-12861-010

### Realizability and Consequences of Graph Profiles

O / open | Oberwolfach 报告问题汇编

Which probability distributions on the set $\mathcal H_k$ of isomorphism types of $k$-vertex graphs occur as the $k$-profile $\pi_k(G)$ of a graph $G$? Which global properties of $G$ can be deduced from $\pi_k(G)$?

来源字段：https://doi.org/10.4171/owr/2014/01

14950 | AMR-087-0047

### Recognizing genus-three Jacobians over the base field

P / partially_solved | AMR 问题清单汇编

Decide whether a given principally polarized abelian threefold over a field $k$ is the Jacobian of a curve over $k$.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14951 | OWR-13498-008

### Reconstruction and Response Sets of AC Networks

P / partially_solved | Oberwolfach 报告问题汇编

For connected alternating-current networks with $b$ boundary vertices and complex response matrix $\Lambda$, give an algorithm to reconstruct a network and positive-real edge conductances from $\Lambda$; describe the response sets of series-parallel networks and of planar networks with all boundary vertices on the outer face; and, when every conductance is either $\omega$ or $1/\omega$, describe the possible response matrices as functions of $\omega$.

来源字段：https://doi.org/10.4171/owr/2015/13

14952 | GT-002

### Reconstruction Conjecture

O / open | 独立 / 未分配集合

别名：Reconstruction conjecture [OPG-658]

Every finite simple graph on at least 3 vertices is uniquely determined by its vertex-deleted subgraphs.

背景链接：https://doi.org/10.1016/j.dam.2026.04.023

14953 | OWR-9790352-036

### Reconstruction of Point Sets from $k$-Decks

P / partially_solved | Oberwolfach 报告问题汇编

Does a universal integer $k$ exist such that every finite subset of $\mathbb R^3$ is determined up to isometry by the multiset of its $k$-point subsets, each considered up to isometry?

来源字段：https://doi.org/10.4171/owr/2022/1

14954 | AMR-099-0056

### Recurrence under square-root separation limits

O / open | AMR 问题清单汇编

Let $(G_k)$ be a locally convergent sequence of bounded-degree graphs, each having separation profile of order at most the square root of the subgraph size. Must the local limit be recurrent?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=3

14955 | AMR-087-0066

### Reducing arbitrary lattices to ideal lattices

P / partially_solved | AMR 问题清单汇编

Reduce computational problems on arbitrary lattices to corresponding problems on cyclic or ideal lattices.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14956 | AMR-087-0021

### Reducing guesses in factoring with known bits

O / open | AMR 问题清单汇编

Reduce the number of guesses required by lattice attacks for factoring with partially known bits.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14957 | OWR-1189-001

### Reed's Conjecture on Graph Coloring

O / open | Oberwolfach 报告问题汇编

Every graph $G$ should satisfy $\chi(G)\leq\left\lceil(\Delta(G)+1+\omega(G))/2\right\rceil$.

来源字段：https://doi.org/10.4171/owr/2006/07

14958 | OPG-335

### Reed's omega, delta, and chi conjecture

P / partially_solved | Open Problem Garden

For a graph $G$, we define $\Delta(G)$ to be the maximum degree, $\omega(G)$ to be the size of the largest clique subgraph, and $\chi(G)$ to be the chromatic number of $G$.

Conjecture $\chi(G) \le \ceil{\frac{1}{2}(\Delta(G)+1) + \frac{1}{2}\omega(G)}$ for every graph $G$.

背景链接：http://www.openproblemgarden.org/op/reeds_omega_delta_and_chi_conjecture

14959 | OWR-12330-010

### Reed’s Chromatic Number Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Does every graph $G$ satisfy Reed's conjectured bound? Specifically, must

$$\chi(G)\leq\left\lceil\frac12\bigl(\Delta(G)+1+\omega(G)\bigr)\right\rceil$$

hold?

来源字段：https://doi.org/10.4171/owr/2013/02

14960 | OWR-12697684-019

### Reed’s Chromatic-Number Conjecture

O / open | Oberwolfach 报告问题汇编

Does every graph $G$ satisfy $\chi(G)\leq\left\lceil(\Delta(G)+1+\omega(G))/2\right\rceil$?

来源字段：https://doi.org/10.4171/owr/2023/1

14961 | OWR-16931-008

### Regular Graph Families with Connected Higher-Order Links

P / partially_solved | Oberwolfach 报告问题汇编

For arbitrary positive integers $a,b,c,d,e$, do there exist infinite families of $(a,b,c,d,e)$-regular graphs in which the links of every vertex, edge, triangle, $K_4$, and $K_5$ are connected?

来源字段：https://doi.org/10.4171/owr/2019/19

14962 | OWR-15206-015

### Regular Graphs Beating Clique Homomorphism Densities

O / open | Oberwolfach 报告问题汇编

For graphs $G$ and $F$, let $d_x(G,F)$ be the largest minimum degree of an $F$-free spanning subgraph of $G$. Determine the graphs $F$ for which there exists $\varepsilon>0$ such that, for every sufficiently large $\Delta$, some $\Delta$-regular graph $G_\Delta$ satisfies $$ d_x(G_\Delta,F)<(1-\varepsilon)d_x(K_\Delta,F) $$?

来源字段：https://doi.org/10.4171/owr/2017/1

14963 | AMR-087-0013

### Removing restrictions from hyperelliptic zeta algorithms

P / partially_solved | AMR 问题清单汇编

Remove the imaginary-hyperelliptic and $p\ne2$ restrictions from the complexity bound stated in the slides.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14964 | GRAPH-047

### Representation Number 3 Classification

O / open | 独立 / 未分配集合

Classify graphs with representation number exactly 3.

背景链接：https://arxiv.org/abs/1403.1616

14965 | AMR-099-0059

### Resistance bounds for finite vertex-transitive graphs

P / partially_solved | AMR 问题清单汇编

Prove that there is a universal constant $C$ such that every finite connected vertex-transitive graph $G$ of degree $d$ satisfies $$R_{\mathrm{eff}}(u,v)<Cd+\frac{\operatorname{diam}(G)^2\log|G|}{|G|}$$ for all vertices $u,v$. Also prove that for any sequence with $\operatorname{diam}(G_n)=o(|G_n|)$, the maximum pairwise resistance is $o(\operatorname{diam}(G_n))$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=5

14966 | OWR-14299081-018

### Robertson’s Surface Four-Flow Conjecture

O / open | Oberwolfach 报告问题汇编

Every $2$-edge-connected graph embedded in the projective plane with face-width at least $4$ has a nowhere-zero $4$-flow. More generally, for every surface $S$, does there exist a constant $r(S)$ such that every $2$-edge-connected graph embedded in $S$ with face-width at least $r(S)$ has a nowhere-zero $4$-flow?

来源字段：https://doi.org/10.4171/owr/2025/1

14967 | AMR-087-0023

### Roots of x-squared minus one modulo a composite

O / open | AMR 问题清单汇编

Efficiently solve for, or characterize all relevant roots of, $x^2-1$ modulo a composite integer $N$ in the setting of the slides.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14968 | OWR-1532-006

### Rotational Dimension and Colin de Verdière Invariants

P / partially_solved | Oberwolfach 报告问题汇编

Let $\operatorname{dim}_{\mathrm{rot}}(G)$ be the rotational dimension and $\mu(G)$ the Colin de Verdière invariant. Is $\operatorname{dim}_{\mathrm{rot}}(G)\leq\mu(G)$ for every graph $G$? If $G_1\cap G_2$ is complete, is $\operatorname{dim}_{\mathrm{rot}}(G_1\cup G_2)\leq\max\{\operatorname{dim}_{\mathrm{rot}}(G_1),\operatorname{dim}_{\mathrm{rot}}(G_2),|G_1\cap G_2|+1\}$? Finally, is $\operatorname{dim}_{\mathrm{rot}}(G)\leq3$ exactly when $G$ has no $K_5$ minor?

来源字段：https://doi.org/10.4171/owr/2007/16

14969 | AMR-099-0015

### Roughly transitive graphs versus homogeneous spaces

O / open | AMR 问题清单汇编

If an infinite graph is $C$-roughly transitive for some finite $C$, must it be roughly isometric to a homogeneous metric space? Equivalently, does there exist an infinite roughly transitive graph that is not roughly isometric to any space with a transitive isometry group?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=22

14970 | GRAPH-053

### Ryser's Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Ryser's conjecture [OPG-165]

For r-partite r-uniform hypergraphs, is the vertex cover number at most (r-1) times the matching number?

背景链接：https://doi.org/10.1007/s004930170001

14971 | OWR-9790352-021

### Ryser-Brualdi-Stein Near-Transversal Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Does every Latin square of order $n$ contain a partial transversal of length $n-1$?

来源字段：https://doi.org/10.4171/owr/2022/1

14972 | OWR-4791-009

### Saturation Numbers for Cycles

P / partially_solved | Oberwolfach 报告问题汇编

What is the saturation number $\operatorname{sat}(n,C_k)$, the minimum number of edges in a $C_k$-free graph on $n$ vertices such that adding any missing edge creates a copy of $C_k$?

来源字段：https://doi.org/10.4171/owr/2011/01

14973 | OWR-12872-006

### Scaling Limits of Nonsubcritical Block Stable Random Graphs

O / open | Oberwolfach 报告问题汇编

Determine the distance scaling and scaling limits of random graphs drawn from block-stable classes that are not subcritical, with random planar graphs as the principal case.

来源字段：https://doi.org/10.4171/owr/2014/12

14974 | AMR-087-0010

### Schoof-type zeta computation without bad genus dependence

O / open | AMR 问题清单汇编

Adapt Schoof's method to compute zeta functions of curves without unfavorable dependence on the genus.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14975 | OPG-729

### Seagull problem

P / partially_solved | Open Problem Garden

Conjecture Every $n$ vertex graph with no independent set of size $3$ has a complete graph on $\ge \frac{n}{2}$ vertices as a minor.

背景链接：http://www.openproblemgarden.org/op/seagull_problem

14976 | GRAPH-054

### Second Neighborhood Problem

O / open | 独立 / 未分配集合

Does every oriented graph have a vertex with at least as many vertices at distance 2 as at distance 1?

背景链接：https://doi.org/10.1002/(SICI

14977 | AMR-087-0005

### Security consequences of Weil descent for the class-group realization

P / partially_solved | AMR 问题清单汇编

Determine whether Weil descent compromises the security of the proposed class-group realization of finite-field discrete-logarithm systems.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14978 | AMR-087-0085

### Security of the TGII directed-signature construction

O / open | AMR 问题清单汇编

Prove the simple construction from trapdoor groups with infeasible inversion to directed transitive signatures secure, or repair the construction.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14979 | OWR-13498-012

### Self-Stress Identities for Sparse Graphs

O / open | Oberwolfach 报告问题汇编

For a planar wheel graph $G$ with $n$ vertices and $2n-2$ edges, suppose a self-stress $\omega$ and edge function $f_{ij}$ satisfy $\sum_{ij\in E(G)}\omega_{ij}f_{ij}=1$. Are there other graphs with $n$ vertices and $2n-2$ edges admitting such a self-stress? What general homological or cohomological phenomenon explains the identity?

来源字段：https://doi.org/10.4171/owr/2015/13

14980 | OWR-2489-011

### Semidefinite Bounds Beyond Fractional Chromatic Number

O / open | Oberwolfach 报告问题汇编

For a graph $G$, can one obtain efficiently computable bounds for its chromatic number that improve on the fractional chromatic number, and can such bounds be defined by compact semidefinite programs suitable for practical computation?

来源字段：https://doi.org/10.4171/owr/2008/51

14981 | OWR-1782-005

### Semidegree Thresholds for Directed Cycles

P / partially_solved | Oberwolfach 报告问题汇编

For each fixed positive integer $\ell$ divisible by 3, determine the minimum semidegree that forces every sufficiently large $n$-vertex oriented graph to contain a directed cycle of length $\ell$, without requiring the cycle to contain a prescribed vertex.

来源字段：https://doi.org/10.4171/owr/2008/01

14982 | OWR-4791-010

### Semisaturation Number of the Five-Cycle

O / open | Oberwolfach 报告问题汇编

For the cycle semisaturation number $\operatorname{ssat}(n,C_5)$, is $\operatorname{ssat}(n,C_5)=\frac{11}{8}n+O(1)$?

来源字段：https://doi.org/10.4171/owr/2011/01

14983 | OWR-16763-008

### Separation Between Higher Graph Equivalence Levels

O / open | Oberwolfach 报告问题汇编

If graphs $H,H'$ are $2$-equivalent, must they be $3$-equivalent? More generally, can two graphs be $100$-equivalent but not $101$-equivalent? Why do the known pairs that are $3$-equivalent but not $2$-equivalent include a disconnected graph?

来源字段：https://doi.org/10.4171/owr/2019/1

14984 | OPG-2226

### Seymour's r-graph conjecture

S / solved | Open Problem Garden

An $r$-graph is an $r$-regular graph $G$ with the property that $|\delta(X)| \ge r$ for every $X \subseteq V(G)$ with odd size.

Conjecture $\chi'(G) \le r+1$ for every $r$-graph $G$.

背景链接：http://www.openproblemgarden.org/op/seymours_r_graph_conjecture

14985 | OPG-646

### Seymour's Second Neighbourhood Conjecture

P / partially_solved | Open Problem Garden

Conjecture Any oriented graph has a vertex whose outdegree is at most its second outdegree.

背景链接：http://www.openproblemgarden.org/op/seymours_second_neighbourhood_conjecture

14986 | OPG-333

### Seymour's self-minor conjecture

O / open | Open Problem Garden

Is every countably infinite graph isomorphic to a proper minor of itself?

背景链接：http://www.openproblemgarden.org/op/seymours_self_minor_conjecture

14987 | OWR-711-003

### Shannon Capacity of Odd Cycles

O / open | Oberwolfach 报告问题汇编

Determine the Shannon capacity of odd cycles of length at least $7$.

来源字段：https://doi.org/10.4171/owr/2004/01

14988 | OPG-36879

### Shannon capacity of the seven-cycle

O / open | Open Problem Garden

Problem What is the Shannon capacity of $C_7$?

背景链接：http://www.openproblemgarden.org/op/shannon_capacity_of_the_seven_cycle

14989 | OWR-4212-013

### Sharp Connectivity-Preserving Tree Deletion Bounds

P / partially_solved | Oberwolfach 报告问题汇编

For every positive integer $k$ and every tree $T$, let $g_k(T)$ be the least nonnegative integer such that every $k$-connected graph $G$ with

$$\delta(G)\geq \left\lfloor\frac{3k}{2}\right\rfloor-1+g_k(T)$$

contains a copy $S\cong T$ satisfying $\kappa(G-V(S))\geq k$. Is $g_k(T)=|T|$?

来源字段：https://doi.org/10.4171/owr/2010/11

14990 | OWR-14299905-021

### Sharp Constants for Multicolor Rainbow Triangles

O / open | Oberwolfach 报告问题汇编

For an integer $r$ such that no affine plane of order $r-1$ exists, determine the sharp constant in the maximum possible number of rainbow triangles in a simple $r$-edge-colored graph with $m$ edges. In particular, determine it for $r=7$ and $r=11$.

来源字段：https://doi.org/10.4171/owr/2026/1

14991 | OWR-15962-005

### Sharp Polynomial-Time Upper Bounds for Minimum Bisection

P / partially_solved | Oberwolfach 报告问题汇编

Construct polynomial-time computable bounds $\operatorname{UB}(G)$ for the minimum-bisection objective $\operatorname{OPT}(G)$ such that $\operatorname{OPT}(G)\leq\operatorname{UB}(G)$ for every graph $G$ and $\operatorname{UB}(G)$ is asymptotically sharp on random graphs.

来源字段：https://doi.org/10.4171/owr/2017/57

14992 | OWR-10252930-025

### Sharp Triangle Threshold in Pseudorandom Graphs

P / partially_solved | Oberwolfach 报告问题汇编

Is there a sharp constant $c_0$ such that, for every $\varepsilon>0$, each sufficiently large $(n,d,\lambda)$-graph satisfying $$\lambda<(c_0-\varepsilon)d^2/n$$ contains a triangle, while arbitrarily large triangle-free examples exist with $$\lambda<(c_0+\varepsilon)d^2/n?$$ If so, does the first inequality force many triangles or a proportionally large triangle factor?

来源字段：https://doi.org/10.4171/owr/2022/22

14993 | OWR-13353-009

### Sharper Tour Bounds for the Chinese Postman Problem

O / open | Oberwolfach 报告问题汇编

For the tour bounds in the Chinese postman problem on a $2$-edge-connected graph, especially when the ear parameter $k$ is large and the lower bound is weak, how can the upper and lower bounds be sharpened?

来源字段：https://doi.org/10.4171/owr/2014/51

14994 | OWR-4791-028

### Short Cycles in Highly Dominating Digraphs

O / open | Oberwolfach 报告问题汇编

Call a directed graph $D=(V,E)$ $s$-dominating if every subset $U$ of at most $s$ vertices has a vertex $v$ with $(v,u)\in E$ for every $u\in U$. Does every 100-dominating finite directed graph contain a directed cycle of length at most 100?

来源字段：https://doi.org/10.4171/owr/2011/01

14995 | OWR-14299081-013

### Shortest Paths and Cycles of Prescribed Modular Length

O / open | Oberwolfach 报告问题汇编

Let $\ell,m\geq 0$ with $m\geq 2$. Is there a polynomial-time algorithm that, given a graph and specified vertices $s,t$, finds a shortest $s$-$t$ path whose length is congruent to $\ell\pmod m$, or certifies that none exists? Is there such an algorithm for a shortest cycle whose length is congruent to $\ell\pmod m$?

来源字段：https://doi.org/10.4171/owr/2025/1

14996 | AMR-087-0051

### Shortest vectors in Hermitian lattices

O / open | AMR 问题清单汇编

Find a sharp upper bound for the shortest-vector length in an $n$-dimensional positive-definite Hermitian space of determinant $d$ over an imaginary quadratic principal ideal domain.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

14997 | OPG-60039

### Sidorenko's Conjecture

P / partially_solved | Open Problem Garden

Conjecture For any bipartite graph $H$ and graph $G$, the number of homomorphisms from $H$ to $G$ is at least $\left(\frac{2|E(G)|}{|V(G)|^2}\right)^{|E(H)|}|V(G)|^{|V(H)|}$.

背景链接：http://www.openproblemgarden.org/op/sidorenkos_conjecture

14998 | OWR-14604-021

### Sidorenko's Conjecture for Bipartite Graphons

O / open | Oberwolfach 报告问题汇编

For every bipartite graph $H$ and graphon $W$, is $t_H(W)\geq t_{K_2}(W)^{|E(H)|}$?

来源字段：https://doi.org/10.4171/owr/2016/22

14999 | OWR-12861-004

### Sidorenko's Inequality for Bipartite Graphs

O / open | Oberwolfach 报告问题汇编

For every bipartite graph $H$ and graph $G$, does Sidorenko's inequality $$t_H(G)\geq t_{K_2}(G)^{|E(H)|}$$ hold?

来源字段：https://doi.org/10.4171/owr/2014/01

15000 | OPG-48264

### Signing a graph to have small magnitude eigenvalues

P / partially_solved | Open Problem Garden

Conjecture If $A$ is the adjacency matrix of a $d$-regular graph, then there is a symmetric signing of $A$ (i.e. replace some $+1$ entries by $-1$ ) so that the resulting matrix has all eigenvalues of magnitude at most $2 \sqrt{d-1}$.

背景链接：http://www.openproblemgarden.org/op/signing_a_graph_to_have_small_magnitude_eigenvalues

15001 | OWR-4791-001

### Simonovits-Sós and Sumner Tournament Conjectures

P / partially_solved | Oberwolfach 报告问题汇编

The Simonovits-Sós conjecture asserts that the largest triangle-intersecting families of graphs on a fixed $n$-vertex set are obtained by fixing a triangle and taking all graphs containing it. Sumner's Universal Tournament Conjecture asserts that every tournament on $2n-2$ vertices contains every directed tree on $n$ vertices.

来源字段：https://doi.org/10.4171/owr/2011/01

15002 | OWR-14298163-010

### Simultaneous Embeddings of Random Planar Graphs

O / open | Oberwolfach 报告问题汇编

Let $\mathcal T_n$ be the family of labeled stacked triangulations obtained by starting from $K_4$ on vertices $v_1,\ldots,v_4$ and, for each $i=4,\ldots,n-1$, inserting $v_{i+1}$ into one of the current $2i-4$ triangular faces and joining it to that face's three vertices. If $G_1,G_2$ are sampled independently and uniformly from $\mathcal T_n$, what is the probability, for large $n$, that some $n$-point set in the plane supports crossing-free straight-line embeddings of both graphs?

来源字段：https://doi.org/10.4171/owr/2024/3

15003 | OPG-46817

### Simultaneous partition of hypergraphs

P / partially_solved | Open Problem Garden

Problem Let $H_1$ and $H_2$ be two $r$-uniform hypergraph on the same vertex set $V$. Does there always exist a partition of $V$ into $r$ classes $V_1, \dots, V_r$ such that for both $i=1,2$, at least $r!m_i/r^r -o(m_i)$ hyperedges of $H_i$ meet each of the classes $V_1, \dots, V_r$?

背景链接：http://www.openproblemgarden.org/op/simultaneous_partition_of_hypergraphs

15004 | OWR-16763-020

### Simultaneously Localized List-Coloring Bounds

O / open | Oberwolfach 报告问题汇编

For localized list-coloring theorems, what happens when the local clique number $\omega(v)$ is close to the degree $d(v)$? Is an at-most-exponential range of degrees necessary, and can the bounds be localized simultaneously?

来源字段：https://doi.org/10.4171/owr/2019/1

15005 | OWR-9790359-002

### Six Problems in Borel Graph Combinatorics

P / partially_solved | Oberwolfach 报告问题汇编

Resolve the following questions for Borel graphs: (1) a Borel Lovász Local Lemma for graphs generated by amenable-group actions; (2) uniqueness of Lebesgue measure among finitely additive measures on Borel subsets of $S^n$ for $n\geq2$; (3) existence of unfriendly Borel colorings for every bounded-degree Borel graph; (4) the bound $\chi_B(G)\leq2n+1$ for finitely Borel-colorable graphs generated by $n$ functions; (5) Baire-measurable $(d+1)$-edge-colorings for degree-$d$ Borel graphs; and (6) the measurable independence and chromatic numbers of the Schreier graph of the Bernoulli shift of $F_2$.

来源字段：https://doi.org/10.4171/owr/2022/8

15006 | OWR-10252930-035

### Size Ramsey Numbers of Ramanujan Graphs

O / open | Oberwolfach 报告问题汇编

Let $G$ be an $n$-vertex, $\Delta$-regular Ramanujan graph with $\Delta\geq3$, let $N=10^{10^{10\Delta n}}$, and let $p$ be the probability for which a random graph $G(N,p)$ has probability $1/2$ that every two-coloring contains a monochromatic copy of $G$. Is the size Ramsey number of $G$ equal to $\Theta(pN^2)$?

来源字段：https://doi.org/10.4171/owr/2022/22

15007 | OWR-9790352-012

### Sketchability and Bounded Expansion of Graph Classes

O / open | Oberwolfach 报告问题汇编

If a monotone graph class is $(C,s)$-sketchable for some $C>1$ and some $s$, must it have bounded expansion?

来源字段：https://doi.org/10.4171/owr/2022/1

15008 | OWR-9790352-013

### Small Induced-Universal Graphs for Planar Graphs

O / open | Oberwolfach 报告问题汇编

What is the minimum order of a graph that contains every planar graph on $n$ vertices as an induced subgraph?

来源字段：https://doi.org/10.4171/owr/2022/1

15009 | GRAPH-036

### Snake-in-the-Box Problem

P / partially_solved | 独立 / 未分配集合

What is the longest induced path in an n-dimensional hypercube graph?

背景链接：https://en.wikipedia.org/wiki/Snake-in-the-box

15010 | AMR-011-0011

### Some Questions - Question 11

S / solved | AMR 问题清单汇编

For an infinite $d$-regular Ramanujan graph, does random-walk neighborhood sampling converge to the $d$-regular tree?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15011 | AMR-011-0012

### Some Questions - Question 12

P / partially_solved | AMR 问题清单汇编

For a locally convergent sequence $(G_n)$ of bounded-degree integer-labeled graphs, does the normalized rank modulo $p$ of the adjacency matrix converge?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15012 | AMR-011-0013

### Some Questions - Question 13

O / open | AMR 问题清单汇编

For a graph sequence $(G_n)$ define $e((G_n))=\liminf |E(G_n)|/|V(G_n)|$, and define its combinatorial cost as the infimum of $e((H_n))$ over graph sequences $(H_n)$ on the same vertex sets for which the identity maps have uniformly bounded bi-Lipschitz constants. If $(G_n)$ and $(H_n)$ converge locally to the same limit, must they have the same combinatorial cost?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15013 | AMR-011-0014

### Some Questions - Question 14

O / open | AMR 问题清单汇编

Compactness implies that for every $\varepsilon>0$ there is $K>0$ such that every finite graph can be approximated within error $\varepsilon$ by a finite graph of size $K$. Give an estimate for $K$ in terms of $\varepsilon$.

背景链接：https://www.renyi.hu/~abert/questions.pdf

15014 | AMR-011-0015

### Some Questions - Question 15

S / solved | AMR 问题清单汇编

Let $(G_n)$ be a locally convergent graph sequence and let $\mu_n$ be the probability distribution of the roots of the chromatic polynomial of $G_n$. For every $k$, does $\lim_{n\to\infty}\int_{\mathbb{C}}z^k\,d\mu_n$ exist?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15015 | AMR-011-0016

### Some Questions - Question 16

O / open | AMR 问题清单汇编

Which probability measures can occur as eigenvalue distributions of finite $d$-regular graphs? Find natural restrictions.

背景链接：https://www.renyi.hu/~abert/questions.pdf

15016 | AMR-011-0017

### Some Questions - Question 17

P / partially_solved | AMR 问题清单汇编

Let $G$ be a $d$-regular Cayley graph with spectral measure $\mu$. Is $\mu$ a weak limit of spectral measures of finite $d$-regular graphs?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15017 | AMR-011-0018

### Some Questions - Question 18

P / partially_solved | AMR 问题清单汇编

For each $d\geq3$, does the independence ratio of a uniformly random $d$-regular graph converge in probability as the number of vertices tends to infinity?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15018 | AMR-011-0019

### Some Questions - Question 19

S / solved | AMR 问题清单汇编

Do uniformly random $d$-regular graphs converge, in local-global convergence, to the weak closure of independent identically distributed processes?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15019 | AMR-011-0020

### Some Questions - Question 20

O / open | AMR 问题清单汇编

Is the i.i.d. action of the free group $F_2$ a local-global limit of finite actions of $F_2$?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15020 | AMR-011-0021

### Some Questions - Question 21

P / partially_solved | AMR 问题清单汇编

Let $\Gamma$ have property (T), and let $(G_n)$ be a sofic approximation of a Cayley graph of $\Gamma$. Can $(G_n)$ be changed by an asymptotically vanishing edit distance to a sequence for which every subsequence of connected components is an expander family?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15021 | AMR-011-0022

### Some Questions - Question 22

O / open | AMR 问题清单汇编

Can every ergodic unimodular random network that is almost surely an infinite tree be obtained as the limit of an expander family?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15022 | AMR-011-0023

### Some Questions - Question 23

O / open | AMR 问题清单汇编

Let $G$ be an infinite vertex-transitive graph, let $A$ be a finite vertex set, let $b$ be a vertex, and let $\partial A$ be the set of vertices at distance one from $A$. Prove that $\sum_{x\in\partial A}d(b,x)\geq |A|$.

背景链接：https://www.renyi.hu/~abert/questions.pdf

15023 | AMR-011-0024

### Some Questions - Question 24

O / open | AMR 问题清单汇编

Define the first $L^2$ Betti number of a vertex-transitive graph $G$ from the expected degree of a free spanning forest. Do $G$ and its square $G^2$ have the same first $L^2$ Betti number?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15024 | AMR-011-0025

### Some Questions - Question 25

P / partially_solved | AMR 问题清单汇编

Can free spanning forests be used to prove basic properties of the first $L^2$ Betti number, such as multiplicativity upon passage to a finite-index subgroup?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15025 | AMR-011-0026

### Some Questions - Question 26

S / solved | AMR 问题清单汇编

Let $G$ be an infinite Cayley graph of a group that is not virtually cyclic. Prove that there exists $p<1$ for which Bernoulli $p$-edge percolation on $G$ has an infinite cluster.

背景链接：https://www.renyi.hu/~abert/questions.pdf

15026 | AMR-011-0027

### Some Questions - Question 27

S / solved | AMR 问题清单汇编

**Question 27.** Does every infinite connected Cayley graph admit a $G$-invariant random perfect matching (a random set of edges forming a perfect matching whose law is invariant under the left action of $G$)?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15027 | AMR-011-0028

### Some Questions - Question 28

O / open | AMR 问题清单汇编

Does every infinite Cayley graph, or every infinite vertex-transitive graph, have a spanning tree without leaves?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15028 | AMR-011-0029

### Some Questions - Question 29

O / open | AMR 问题清单汇编

**Question 29.** Let $G$ be an infinite Cayley (vertex-transitive) graph. Does the density of dead ends in $G$ have density 0?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15029 | AMR-011-0030

### Some Questions - Question 30

S / solved | AMR 问题清单汇编

Are factors of i.i.d. on the $3$-regular tree closed in the weak topology? In particular, is the weak limit of majority functions on $n$-balls a factor of i.i.d.?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15030 | AMR-011-0031

### Some Questions - Question 31

P / partially_solved | AMR 问题清单汇编

Does every infinite Cayley graph $G$ admit a $G$-invariant proper coloring with $\chi(G)$ colors?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15031 | AMR-011-0032

### Some Questions - Question 32

P / partially_solved | AMR 问题清单汇编

Let $X$ be the space of $k$-regular Cayley graphs with the local-convergence topology and let $T\subset X$ be the closed subset of transient graphs. Is the Green function, equivalently the expected number of returns to the identity, continuous on $T$?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15032 | AMR-011-0033

### Some Questions - Question 33

S / solved | AMR 问题清单汇编

For every $k>1$, does there exist $C(k)<1$ such that the return probability of every transient $k$-regular Cayley graph is at most $C(k)$?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15033 | AMR-011-0034

### Some Questions - Question 34

S / solved | AMR 问题清单汇编

For a nonamenable group $\Gamma$, does the Bernoulli shift $\{0,1\}^{\Gamma}$ factor onto $\{0,1,2\}^{\Gamma}$?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15034 | AMR-011-0035

### Some Questions - Question 35

S / solved | AMR 问题清单汇编

Can every $d$-regular graphing without multiple edges be properly edge-colored by $d+1$ colors?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15035 | AMR-011-0036

### Some Questions - Question 36

O / open | AMR 问题清单汇编

Let $(G_n)$ and $(H_n)$ converge to the same graph limit, and let $(T_n)$ be a convergent sequence with each $T_n$ a spanning tree of $G_n$. Do there exist spanning trees $P_n$ of $H_n$ converging to the same limit as $(T_n)$?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15036 | AMR-011-0037

### Some Questions - Question 37

O / open | AMR 问题清单汇编

Let $G$ be a bounded-degree expander, or strongly ergodic, graphing that can be properly colored by $C$ colors with arbitrarily small error. Can it be properly $C$-colored?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15037 | AMR-011-0038

### Some Questions - Question 38

P / partially_solved | AMR 问题清单汇编

Let $G$ be a bounded-degree expander, or strongly ergodic, graphing that weakly contains a finite graph $H$. Does $G$ factor onto $H$?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15038 | AMR-011-0039

### Some Questions - Question 39

S / solved | AMR 问题清单汇编

**Question 39.** Does every higher-rank semisimple real lattice (arithmetic lattice in a simple Lie group of real rank $\ge 2$) have rank gradient zero? (i.e. $\mathrm{RG}(\Gamma, (\Gamma_n))=0$ for every chain/sequence $(\Gamma_n)$ of finite-index subgroups with trivial intersection.)

背景链接：https://www.renyi.hu/~abert/questions.pdf

15039 | AMR-011-0040

### Some Questions - Question 40

P / partially_solved | AMR 问题清单汇编

If $A$ and $B$ are countably infinite groups, does $A\times B$ have fixed price $1$?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15040 | AMR-011-0042

### Some Questions - Question 42

P / partially_solved | AMR 问题清单汇编

**Question 42.** Does every residually finite group with Kazhdan's property (T) have rank gradient zero? (i.e. $\mathrm{RG}(\Gamma,(\Gamma_n))=0$ for a Farber sequence $(\Gamma_n)$ of finite-index subgroups.)

背景链接：https://www.renyi.hu/~abert/questions.pdf

15041 | AMR-011-0006

### Some Questions - Question 6

O / open | AMR 问题清单汇编

Can the sequence of metric balls in an infinite Cayley graph form a family of expanders?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15042 | AMR-011-0007

### Some Questions - Question 7

P / partially_solved | AMR 问题清单汇编

Suppose $G$ and $H$ are Cayley, or vertex-transitive, expanders on the same number of vertices and can be matched with asymptotically vanishing edge distance. Must they be isomorphic?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15043 | AMR-011-0008

### Some Questions - Question 8

P / partially_solved | AMR 问题清单汇编

If $G$ is a finite vertex-transitive expander, must every almost automorphism of $G$ be close to an automorphism?

背景链接：https://www.renyi.hu/~abert/questions.pdf

15044 | OWR-12861-021

### Spanning Near-Squares of Odd Cycles

P / partially_solved | Oberwolfach 报告问题汇编

For every odd $n\geq7$, does every $n$-vertex graph with minimum degree at least $\lceil n/2\rceil$ contain a spanning graph formed from the square of an $n$-cycle by deleting alternate peripheral edges until exactly three consecutive vertices have degree $4$ and the remaining $n-3$ vertices have degree $3$?

来源字段：https://doi.org/10.4171/owr/2014/01

15045 | OWR-15581-003

### Spanning-Tree Growth Rates Near One

O / open | Oberwolfach 报告问题汇编

In the graph-complexity reformulation of Lehmer's Mahler-measure problem (Hironaka), for a planar graph $G$ embedded in the punctured plane $\mathbb{R}^2 \setminus \{0\}$, let $G_r$ be its $r$-fold cyclic covering graph and let $\tau(G_r)$ be the number of spanning trees of $G_r$. Given $\varepsilon > 0$, does there exist a graph $G$ in the punctured plane such that $$ 1 < \lim_{r \to \infty} \tau(G_r)^{1/r} < 1 + \varepsilon? $$

来源字段：https://doi.org/10.4171/owr/2017/38

15046 | OWR-10252930-022

### Sparse Graph Sequences with Prescribed Homomorphism Limits

O / open | Oberwolfach 报告问题汇编

For a graph $F$ and a graph $G$ of edge density $p$, define $$t_p(F,G)=p^{-e(F)}|G|^{-|F|}\operatorname{hom}(F,G). $$ Can one construct graphs $G_n$ with $|G_n|\to\infty$ and edge densities $p_n=|G_n|^{-o(1)}$ such that $a_F=\lim_{n\to\infty}t_{p_n}(F,G_n)$ exists for every $F$, while $$a_{C_4}=a_{K_3}=1,\qquad \sup_F a_F^{1/e(F)}<\infty,$$ and $a_F\neq1$ for at least one graph $F$?

来源字段：https://doi.org/10.4171/owr/2022/22

15047 | OWR-4791-034

### Sparse Quadrilateral Removal at the Extremal Threshold

O / open | Oberwolfach 报告问题汇编

For every $\varepsilon>0$, does there exist $n_0=n_0(\varepsilon)$ such that every graph on $n\geq n_0$ vertices that is the edge-disjoint union of at least $\varepsilon n^{3/2}$ quadrilaterals contains another quadrilateral? Moreover, is it true that some $\delta>0$ exists so that such a graph contains at least $\delta n^2$ quadrilaterals?

来源字段：https://doi.org/10.4171/owr/2011/01

15048 | OWR-4791-023

### Sparse Random Graph Convergence Notions

P / partially_solved | Oberwolfach 报告问题汇编

When does a random graph converge under left-convergence, right-convergence, or the Bollobás-Riordan partition metric? In particular, does $G(n,c/n)$ converge with high probability under each notion?

来源字段：https://doi.org/10.4171/owr/2011/01

15049 | OWR-12861-013

### Sparse Random Graphs Avoiding Non-Bipartite Subgraphs

O / open | Oberwolfach 报告问题汇编

For a nonbipartite graph $H$, determine the density thresholds and typical structure of uniformly random $H$-free graphs with $m$ edges.

来源字段：https://doi.org/10.4171/owr/2014/01

15050 | OWR-1965-005

### Spectral and Product Growth in Graphs and Groups

P / partially_solved | Oberwolfach 报告问题汇编

For a graph, bound the gap between its two largest adjacency eigenvalues. For a finite subset $A$ of an abelian group, bound $|A+A|$ in terms of $|A|$. For a finite subset $A$ of an arbitrary group, bound $|AAA|$ in terms of $|A|$.

来源字段：https://doi.org/10.4171/owr/2008/20

15051 | OWR-711-005

### Spectral Bounds for Signed Regular Graphs

P / partially_solved | Oberwolfach 报告问题汇编

Every $d$-regular graph has an edge signing whose signed adjacency matrix has spectral radius at most $2\sqrt{d-1}$.

来源字段：https://doi.org/10.4171/owr/2004/01

15052 | OPG-46456

### Splitting a digraph with minimum outdegree constraints

P / partially_solved | Open Problem Garden

Problem Is there a minimum integer $f(d)$ such that the vertices of any digraph with minimum outdegree at least $f(d)$ can be partitioned into two classes so that the minimum outdegree of the subgraph induced by each class is at least $d$?

背景链接：http://www.openproblemgarden.org/op/splitting_a_digraph_with_minimum_outdegree_constraints

15053 | OWR-12183-002

### Stable Label Patterns on Graphs

O / open | Oberwolfach 报告问题汇编

For a graph-based pattern system with stable vertex labels in $\{+,-,0\}$, which graphs support a stable pattern, and can one graph support several stable patterns not related by the stated trivial symmetries?

来源字段：https://doi.org/10.4171/owr/2012/60

15054 | OPG-46443

### Stable set meeting all longest directed paths.

P / partially_solved | Open Problem Garden

Conjecture Every digraph has a stable set meeting all longest directed paths

背景链接：http://www.openproblemgarden.org/op/stable_set_meeting_all_longest_directed_paths

15055 | OWR-14213-006

### Stanley-Stembridge Chromatic Symmetric Function Conjecture

S / solved | Oberwolfach 报告问题汇编

Stanley-Stembridge conjecture: if $G=G(m)$ is the incomparability graph associated with a Hessenberg function $m$, then its chromatic symmetric function $X_G(x)$ is a nonnegative sum of elementary symmetric functions.

来源字段：https://doi.org/10.4171/owr/2016/5

15056 | OPG-37275

### Star chromatic index of complete graphs

P / partially_solved | Open Problem Garden

Conjecture Is it possible to color edges of the complete graph $K_n$ using $O(n)$ colors, so that the coloring is proper and no 4-cycle and no 4-edge path is using only two colors?

Equivalently: is the star chromatic index of $K_n$ linear in $n$?

背景链接：http://www.openproblemgarden.org/op/star_chromatic_index_of_complete_graphs

15057 | OPG-37271

### Star chromatic index of cubic graphs

P / partially_solved | Open Problem Garden

The star chromatic index $\chi_s'(G)$ of a graph $G$ is the minimum number of colors needed to properly color the edges of the graph so that no path or cycle of length four is bi-colored.

Question Is it true that for every (sub)cubic graph $G$, we have $\chi_s'(G) \le 6$?

背景链接：http://www.openproblemgarden.org/op/star_chromatic_index_of_cubic_graphs

15058 | OWR-12479-008

### Strict Graph-Invariant Bounds and Farber-Chain Rank Gradients

O / open | Oberwolfach 报告问题汇编

Let $\mathcal G=(G_n)$ be a sequence of connected graphs of uniformly bounded degree. For a field $K$, let $C_K(G)$ be the cycle space of $G$, let $C_K^q(G)$ be the subspace generated by cycles of length less than $q$, and define

$$s_K^q(\mathcal G)=\liminf_{n\to\infty}\frac{|E(G_n)|-\dim_K C_K^q(G_n)-|V(G_n)|}{|V(G_n)|},\qquad \beta_K(\mathcal G)=\inf_q s_K^q(\mathcal G).$$

Can either known inequality $\beta_{\mathbb Q}(\mathcal G)\geq\beta_{\mathbb F_p}(\mathcal G)$ or $\beta_K(\mathcal G)+1\leq c(\mathcal G)$ be strict for some graph sequence, where $c(\mathcal G)$ is its combinatorial cost? If $\Gamma$ is finitely generated and residually finite, does the rank gradient associated with a Farber chain of normal finite-index subgroups depend on the chain, or only on $\Gamma$?

来源字段：https://doi.org/10.4171/owr/2013/16

15059 | OPG-37241

### Strong 5-cycle double cover conjecture

P / partially_solved | Open Problem Garden

Conjecture Let $C$ be a circuit in a bridgeless cubic graph $G$. Then there is a five cycle double cover of $G$ such that $C$ is a subgraph of one of these five cycles.

背景链接：http://www.openproblemgarden.org/op/strong_5_cycle_double_cover_conjecture

15060 | OPG-171

### Strong colorability

P / partially_solved | Open Problem Garden

Let $r$ be a positive integer. We say that a graph $G$ is strongly $r$-colorable if for every partition of the vertices to sets of size at most $r$ there is a proper $r$-coloring of $G$ in which the vertices in each set of the partition have distinct colors.

Conjecture If $\Delta$ is the maximal degree of a graph $G$, then $G$ is strongly $2 \Delta$-colorable.

背景链接：http://www.openproblemgarden.org/op/strong_colorability

15061 | OPG-46475

### Strong edge colouring conjecture

P / partially_solved | Open Problem Garden

A strong edge-colouring of a graph $G$ is a edge-colouring in which every colour class is an induced matching; that is, any two vertices belonging to distinct edges with the same colour are not adjacent. The strong chromatic index $s\chi'(G)$ is the minimum number of colours in a strong edge-colouring of $G$.

Conjecture $$s\chi'(G) \leq \frac{5\Delta^2}{4}, \text{if$\Delta$is even,}$$$$s\chi'(G) \leq \frac{5\Delta^2-2\Delta +1}{4},&\text{if$\Delta$is odd.}$$

背景链接：http://www.openproblemgarden.org/op/strong_edge_colouring_conjecture

15062 | OWR-1532-009

### Strong Element-Connected Orientations with Steiner Vertices

O / open | Oberwolfach 报告问题汇编

If the terminal set $T$ is $2k$-element-connected in an undirected graph $G$, does $G$ have a Steiner strongly $k$-element-connected orientation? Here element-connectivity counts paths disjoint in their edges and Steiner vertices, with a specified root in $T$.

来源字段：https://doi.org/10.4171/owr/2007/16

15063 | OWR-16763-023

### Strong Erdős-Hajnal Phenomena in Sparse H-Free Graphs

O / open | Oberwolfach 报告问题汇编

For which graphs $H$ does some $\varepsilon>0$ have the following property? Every $n$-vertex, $\varepsilon$-sparse, $H$-free graph $G$, with $n>1$, contains sets $A,B$ that are complete or anticomplete to each other and satisfy at least one of

$$\lvert A\rvert,\lvert B\rvert\geq\varepsilon n;\qquad \lvert A\rvert\geq\varepsilon n,\ \lvert B\rvert\geq n^\varepsilon;\qquad \lvert A\rvert,\lvert B\rvert\geq n^\varepsilon.$$

Must at least one of these three size conditions always hold?

来源字段：https://doi.org/10.4171/owr/2019/1

15064 | OPG-690

### Strong matchings and covers

P / partially_solved | Open Problem Garden

Let $H$ be a hypergraph. A strongly maximal matching is a matching $F \subseteq E(H)$ so that $|F' \setminus F| \le |F \setminus F'|$ for every matching $F'$. A strongly minimal cover is a (vertex) cover $X \subseteq V(H)$ so that $|X' \setminus X| \ge |X \setminus X'|$ for every cover $X'$.

Conjecture If $H$ is a (possibly infinite) hypergraph in which all edges have size $\le k$ for some integer $k$, then $H$ has a strongly maximal matching and a strongly minimal cover.

背景链接：http://www.openproblemgarden.org/op/strong_matchings_and_covers

15065 | OWR-9790352-019

### Structural and Algorithmic Properties of Burling Graphs

O / open | Oberwolfach 报告问题汇编

别名：Chi-Bounds and Stable Sets in Burling Graphs [OWR-9790352-020]

For a fixed Burling graph $H$, are graphs with no subdivision of $H$ $\chi$-bounded, and do Burling graphs with no induced copy of $H$ have bounded chromatic number? Can Burling graphs be characterized or recognized in polynomial time? Is maximum stable set polynomial-time solvable on Burling graphs? Can the Burling construction be extended to graphs with triangles and unbounded chromatic number?

来源字段：https://doi.org/10.4171/owr/2022/1

15066 | OWR-12861-011

### Structural Determinants of Graph Ramsey Numbers

O / open | Oberwolfach 报告问题汇编

Which structural properties of a graph $H$ approximately determine its Ramsey number?

来源字段：https://doi.org/10.4171/owr/2014/01

15067 | OWR-12330-017

### Structural Invariants for Very Large Graphs

O / open | Oberwolfach 报告问题汇编

What kinds of numerical data or invariants should be extracted from very large graphs, such as protein-interaction graphs, to describe useful structural information?

来源字段：https://doi.org/10.4171/owr/2013/02

15068 | OWR-14299081-004

### Structure and Coloring at Large r-Acyclic Width

O / open | Oberwolfach 报告问题汇编

Fix $r\geq 3$. (1) Is there a function $f:\mathbb{N}\to\mathbb{N}$ such that every graph of $r$-acyclic width at least $f(k)$ contains a subdivided grid of $r$-acyclic width at least $k$? (2) Is there a function $h:\mathbb{N}\to\mathbb{R}$ with $h(r)\to 0$ such that some minimum-width $r$-acyclic $H$-decomposition of every graph $G$ satisfies $\lVert H\rVert=O(|G|^{1+h(r)})$? (3) If an $r$-acyclic decomposition of $G$ has width $w=r\operatorname{-aw}(G)$ and model $H$, let $k$ be the maximum chromatic number among topological minors of $H$ having girth greater than $r$. Can $G$ be colored with $wk$ colors?

来源字段：https://doi.org/10.4171/owr/2025/1

15069 | OWR-1536-007

### Sub-Twice-Maximum-Degree Game Chromatic Index Bounds

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Game-Chromatic-Index Upper-Bound Question [OWR-1536-008]

Does there exist $\varepsilon>0$ such that every nonempty graph $G$ satisfies $\chi'_g(G)\leq(2-\varepsilon)\Delta(G)$, where $\chi'_g(G)$ is the game chromatic index and $\Delta(G)$ is the maximum degree?

来源字段：https://doi.org/10.4171/owr/2007/20

15070 | OPG-46680

### Subdivision of a transitive tournament in digraphs with large outdegree.

P / partially_solved | Open Problem Garden

Conjecture For all $k$ there is an integer $f(k)$ such that every digraph of minimum outdegree at least $f(k)$ contains a subdivision of a transitive tournament of order $k$.

背景链接：http://www.openproblemgarden.org/op/subdivision_of_a_transitive_tournament_in_digraphs_with_large_outdegree

15071 | OPG-46708

### Subgraph of large average degree and large girth.

P / partially_solved | Open Problem Garden

Conjecture For all positive integers $g$ and $k$, there exists an integer $d$ such that every graph of average degree at least $d$ contains a subgraph of average degree at least $k$ and girth greater than $g$.

背景链接：http://www.openproblemgarden.org/op/subgraph_of_large_average_degree_and_large_average_degree

15072 | OWR-9790352-039

### Subgraph Query Complexity for Fixed Pattern Graphs

O / open | Oberwolfach 报告问题汇编

Let $f(H,p)$ be the minimum number of adaptive adjacency queries needed to reveal a copy of a fixed graph $H$ in a hidden countably infinite Erdős-Rényi graph $G(\mathbb N,p)$ with probability at least $1/2$. What is the growth of $f(H,p)$ as $p\to0$, and for which $H$ is it of order $p^{-c}$ for some constant $c$?

来源字段：https://doi.org/10.4171/owr/2022/1

15073 | OWR-12481-016

### Sublinear Bounds for Disjoint Homometric Vertex Sets

O / open | Oberwolfach 报告问题汇编

Let $h(n)$ be the largest integer such that every connected $n$-vertex graph has two disjoint homometric vertex subsets, meaning their multisets of pairwise distances agree. Is $h(n)=o(n)$?

来源字段：https://doi.org/10.4171/owr/2013/18

15074 | OWR-12697712-002

### Successive Edge-Disjoint Spanning Trees in Random Graphs

O / open | Oberwolfach 报告问题汇编

In the successive edge-disjoint minimum-spanning-tree process, do the largest-component profiles satisfy $\rho_k(2k+t)\to\rho_\infty(t)$ as $k\to\infty$, and do the limiting tree weights satisfy $\gamma_k\to2k-1$?

来源字段：https://doi.org/10.4171/owr/2023/16

15075 | GRAPH-037

### Sumner's Conjecture

P / partially_solved | 独立 / 未分配集合

Does every (2n-2)-vertex tournament contain every n-vertex oriented tree?

背景链接：https://arxiv.org/abs/1010.4430

15076 | OWR-784-006

### Surface Chromatic Width and Bounded-Facewidth Snarks

P / partially_solved | Oberwolfach 报告问题汇编

For every $g\geq0$, is $\chi_w(S_g)=4$? More strongly, if $G$ is embedded in $S_g$ and $\operatorname{ew}(G^*)\geq3$, must $\phi_c(G)=4$; equivalently, is every cubic graph embedded in $S_g$ with $\operatorname{fw}(G)\geq3$ three-edge-colorable? At least, for every $g$, does there exist $w$ such that every cubic graph embedded in $S_g$ with $\operatorname{ew}(G)\geq w$ is three-edge-colorable?

来源字段：https://doi.org/10.4171/owr/2005/03

15077 | AMR-087-0067

### SVP-to-CVP reduction within ideal lattices

P / partially_solved | AMR 问题清单汇编

Does SVP reduce to CVP while remaining inside the class of cyclic or ideal lattices?

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

15078 | OPG-46951

### Switching reconstruction conjecture

O / open | Open Problem Garden

Conjecture Every simple graph on five or more vertices is switching-reconstructible.

背景链接：http://www.openproblemgarden.org/op/switching_reconstruction_conjecture

15079 | OPG-46952

### Switching reconstruction of digraphs

P / partially_solved | Open Problem Garden

Question Are there any switching-nonreconstructible digraphs on twelve or more vertices?

背景链接：http://www.openproblemgarden.org/op/switching_reconstruction_of_digraphs

15080 | OWR-1386-018

### Symmetric Relation Coloring and Bounded Monochromatic Components

O / open | Oberwolfach 报告问题汇编

Determine whether $(4,C)$-SymRelCol has a sudden hardness jump, and determine the order of $g(6,n)$, the least $g$ such that every $n$-vertex graph of maximum degree 6 has a two-coloring with every monochromatic component of order at most $g$.

来源字段：https://doi.org/10.4171/owr/2006/48

15081 | OWR-16763-022

### Symmetry of Bidirectional Two-Step Path Density

O / open | Oberwolfach 报告问题汇编

Let $A,B,C$ be disjoint nonempty vertex sets in a graph. Suppose every vertex of $A$ has at least $x\lvert B\rvert$ neighbors in $B$, every vertex of $B$ has at least $x\lvert A\rvert$ neighbors in $A$ and at least $y\lvert C\rvert$ neighbors in $C$, and every vertex of $C$ has at least $y\lvert B\rvert$ neighbors in $B$. Let $\psi(x,y)$ be the largest $z$ such that some vertex of $C$ is joined by two-edge paths to at least $z\lvert A\rvert$ vertices of $A$. Is $\psi(x,y)=\psi(y,x)$ for all $x,y$?

来源字段：https://doi.org/10.4171/owr/2019/1

15082 | AMR-087-0058

### Taxonomy of pairing-related assumptions

O / open | AMR 问题清单汇编

Update Joux's 2002 work by developing a systematic taxonomy of pairing-related computational assumptions.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

15083 | GRAPH-055

### Teschner's Bondage Number Conjecture

P / partially_solved | 独立 / 未分配集合

Is the bondage number of a graph always ≤ 3Δ/2, where Δ is the maximum degree?

背景链接：https://doi.org/10.6084/m9.figshare.33198777

15084 | OPG-142

### The Berge-Fulkerson conjecture

P / partially_solved | Open Problem Garden

Conjecture If $G$ is a bridgeless cubic graph, then there exist 6 perfect matchings $M_1,\ldots,M_6$ of $G$ with the property that every edge of $G$ is contained in exactly two of $M_1,\ldots,M_6$.

背景链接：http://www.openproblemgarden.org/op/the_berge_fulkerson_conjecture

15085 | OPG-611

### The Bermond-Thomassen Conjecture

P / partially_solved | Open Problem Garden

Conjecture For every positive integer $k$, every digraph with minimum out-degree at least $2k-1$ contains $k$ disjoint cycles.

背景链接：http://www.openproblemgarden.org/op/the_bermond_thomassen_conjecture

15086 | OPG-48232

### The Bollobás-Eldridge-Catlin Conjecture on graph packing

P / partially_solved | Open Problem Garden

Conjecture (BEC-conjecture) If $G_1$ and $G_2$ are $n$-vertex graphs and $(\Delta(G_1) + 1) (\Delta(G_2) + 1) < n + 1$, then $G_1$ and $G_2$ pack.

背景链接：http://www.openproblemgarden.org/op/the_bollobas_eldridge_catlin_conjecture_on_graph_packing

15087 | OPG-37670

### The Borodin-Kostochka Conjecture

P / partially_solved | Open Problem Garden

Conjecture Every graph with maximum degree $\Delta \geq 9$ has chromatic number at most $\max\{\Delta-1, \omega\}$.

背景链接：http://www.openproblemgarden.org/op/the_borodin_kostochka_conjecture

15088 | OWR-12330-019

### The Caccetta-Häggkvist Directed-Cycle Conjecture

O / open | Oberwolfach 报告问题汇编

Does every $n$-vertex digraph with minimum outdegree at least $r$ contain a directed cycle of length at most $\lceil n/r\rceil$, as conjectured by Caccetta and Häggkvist?

来源字段：https://doi.org/10.4171/owr/2013/02

15089 | OPG-138

### The circular embedding conjecture

P / partially_solved | Open Problem Garden

Conjecture Every 2-connected graph may be embedded in a surface so that the boundary of each face is a cycle.

背景链接：http://www.openproblemgarden.org/op/the_circular_embedding_conjecture

15090 | OWR-14299905-028

### The Correct Parameter for Local Graph Removal

P / partially_solved | Oberwolfach 报告问题汇编

For every graph $H$, is there an integer $a(H)$ such that an $n$-vertex graph is $o(1)$-close to being $H$-free in $L_\infty$ if and only if every set of $a(H)$ vertices extends to $o(n^{v(H)-a(H)})$ copies of $H$? Is the proposed local-cover parameter the correct value of $a(H)$?

来源字段：https://doi.org/10.4171/owr/2026/1

15091 | OPG-310

### The Crossing Number of the Complete Bipartite Graph

O / open | Open Problem Garden

The crossing number $cr(G)$ of $G$ is the minimum number of crossings in all drawings of $G$ in the plane.

Conjecture $\displaystyle cr(K_{m,n}) = \floor{\frac m2} \floor{\frac {m-1}2} \floor{\frac n2} \floor{\frac {n-1}2}$

背景链接：http://www.openproblemgarden.org/op/the_crossing_number_of_the_complete_bipartite_graph

15092 | OPG-307

### The Crossing Number of the Complete Graph

O / open | Open Problem Garden

The crossing number $cr(G)$ of $G$ is the minimum number of crossings in all drawings of $G$ in the plane.

Conjecture $\displaystyle cr(K_n) = \frac 14 \floor{\frac n2} \floor{\frac{n-1}2} \floor{\frac{n-2}2} \floor{\frac{n-3}2}$

背景链接：http://www.openproblemgarden.org/op/the_crossing_number_of_the_complete_graph

15093 | OPG-313

### The Crossing Number of the Hypercube

O / open | Open Problem Garden

The crossing number $cr(G)$ of $G$ is the minimum number of crossings in all drawings of $G$ in the plane.

The $d$-dimensional (hyper)cube $Q_d$ is the graph whose vertices are all binary sequences of length $d$, and two of the sequences are adjacent in $Q_d$ if they differ in precisely one coordinate.

Conjecture $\displaystyle \lim \frac{cr(Q_d)}{4^d} = \frac{5}{32}$

背景链接：http://www.openproblemgarden.org/op/the_crossing_number_of_the_hypercube

15094 | GT-004

### The Cycle Double Cover Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Cycle Double Cover Conjecture [GRAPH-059]; Cycle Double Cover Conjecture [GRAPH-029]; Cycle double cover conjecture [OPG-137]

Every bridgeless graph has a cycle double cover: a collection of cycles that covers each edge exactly twice.

背景链接：https://arxiv.org/abs/2607.16356

15095 | DARPA-002

### The Dynamics of Networks

P / partially_solved | DARPA 挑战

Develop high-dimensional mathematics to model and predict behavior in large-scale distributed networks.

背景链接：https://web.math.utk.edu/~vasili/refs/darpa07.MathChallenges.html

15096 | OPG-162

### The Erdös-Hajnal Conjecture

P / partially_solved | Open Problem Garden

Conjecture For every fixed graph $H$, there exists a constant $\delta(H)$, so that every graph $G$ without an induced subgraph isomorphic to $H$ contains either a clique or an independent set of size $|V(G)|^{\delta(H)}$.

背景链接：http://www.openproblemgarden.org/op/the_erdos_hajnal_conjecture

15097 | GT-003

### The Graceful Tree Conjecture

O / open | 独立 / 未分配集合

别名：Graceful Tree Conjecture [OPG-439]

Every tree can be gracefully labeled: vertices can be assigned distinct labels from $\{0, 1, \ldots, |E|\}$ such that edge labels (absolute differences) are all distinct.

背景链接：https://mlelarge.github.io/graph-conjectures/op/graceful_tree_conjecture/

15098 | OWR-4212-010

### The Hamiltonicity Conjecture for Four-Connected Line Graphs

O / open | Oberwolfach 报告问题汇编

Is every $4$-connected line graph Hamiltonian?

来源字段：https://doi.org/10.4171/owr/2010/11

15099 | OPG-543

### The intersection of two perfect matchings

O / open | Open Problem Garden

Conjecture Every bridgeless cubic graph has two perfect matchings $M_1$, $M_2$ so that $M_1 \cap M_2$ does not contain an odd edge-cut.

背景链接：http://www.openproblemgarden.org/op/intersecting_two_perfect_matchings

15100 | OWR-14299081-025

### The Log-Log Factor for Rainbow Cycles

O / open | Oberwolfach 报告问题汇编

Is the $\log\log n$ factor in the $O(\log n\log\log n)$ average-degree bound necessary to guarantee a rainbow cycle in a properly edge-colored graph?

来源字段：https://doi.org/10.4171/owr/2025/1

15101 | OWR-14299518-023

### The Sharp Asymptotic for Triangle Ramsey Numbers

O / open | Oberwolfach 报告问题汇编

Is the triangle Ramsey number asymptotic to $R(3,k)=\left(\frac12+o(1)\right)\frac{k^2}{\log k}$?

来源字段：https://doi.org/10.4171/owr/2025/42

15102 | OPG-132

### The three 4-flows conjecture

O / open | Open Problem Garden

Conjecture For every graph $G$ with no bridge, there exist three disjoint sets $A_1,A_2,A_3 \subseteq E(G)$ with $A_1 \cup A_2 \cup A_3 = E(G)$ so that $G \setminus A_i$ has a nowhere-zero 4-flow for $1 \le i \le 3$.

背景链接：http://www.openproblemgarden.org/op/three_4_flows_conjecture

15103 | GT-010

### The Total Coloring Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Total Coloring Conjecture [GRAPH-062]

Can every graph be totally colored with at most $\Delta + 2$ colors, where $\Delta$ is the maximum degree?

背景链接：https://arxiv.org/abs/2507.12737

15104 | OPG-169

### The Two Color Conjecture

P / partially_solved | Open Problem Garden

Conjecture If $G$ is an orientation of a simple planar graph, then there is a partition of $V(G)$ into $\{X_1,X_2\}$ so that the graph induced by $X_i$ is acyclic for $i=1,2$.

背景链接：http://www.openproblemgarden.org/op/partitioning_planar_digraphs

15105 | OWR-822-007

### Theoretical Certification of Crossing-Number Optimality

O / open | Oberwolfach 报告问题汇编

Let $D$ be a touching-free drawing of a simple graph $G$ in which two edges cross at most once and adjacent edges do not cross. For which graphs can one prove theoretically, rather than by exhaustive search, that for every choice of sets $A_e\subseteq V(G)\setminus e$, the number of crossing nonadjacent pairs with $|e\cap A_f|+|f\cap A_e|$ odd does not exceed the corresponding number of noncrossing pairs, and hence that $D$ realizes $\operatorname{cr}(G)$?

来源字段：https://doi.org/10.4171/owr/2005/17

15106 | OWR-14299911-005

### Thiagarajan Grid-Free Decidability Conjecture

S / solved | Oberwolfach 报告问题汇编

For a trace-regular event structure, is its monadic second-order theory decidable if and only if the event structure is grid-free?

来源字段：https://doi.org/10.4171/owr/2026/8

15107 | OWR-14298163-017

### Thick Drawings of Grid Graphs

O / open | Oberwolfach 报告问题汇编

Does there exist an absolute constant $C$ such that every grid graph with $n$ vertices admits a $1$-thick drawing in a square of side length $C\sqrt n$?

来源字段：https://doi.org/10.4171/owr/2024/3

15108 | OWR-1532-004

### Thomassen's Hamiltonicity Conjecture for Infinite Line Graphs

O / open | Oberwolfach 报告问题汇编

Does every infinite four-connected line graph contain a Hamilton circle?

来源字段：https://doi.org/10.4171/owr/2007/16

15109 | OPG-562

### Three-chromatic (0,2)-graphs

P / partially_solved | Open Problem Garden

Question Are there any (0,2)-graphs with chromatic number exactly three?

背景链接：http://www.openproblemgarden.org/op/three_chromatic_0_2_graphs

15110 | OWR-11781-003

### Three-Color Edge-Folkman Numbers for Triangles

O / open | Oberwolfach 报告问题汇编

What are good bounds on the order of a graph $G$ such that every coloring of the edges of $G$ with three colors contains a monochromatic triangle, i.e. $G\longrightarrow (K_3)^3$?

来源字段：https://doi.org/10.4171/owr/2012/19

15111 | OWR-16627-003

### Threshold Bias in the Odd-Cycle Game

O / open | Oberwolfach 报告问题汇编

In the Maker-Breaker odd-cycle game on $K_n$, is the threshold bias $b_{\mathrm{MB}}(OC_n)=(1/2+o(1))n$?

来源字段：https://doi.org/10.4171/owr/2018/44

15112 | OWR-1386-016

### Threshold Constants for Asymmetric Random Ramsey Properties

P / partially_solved | Oberwolfach 报告问题汇编

Let $1<m_2(F_1)\le m_2(F_2)$ and put $m=m_2(F_1,F_2)=\max_{H\subseteq F_2,\,|V(H)|\ge2}|E(H)|/(|V(H)|-2+1/m_2(F_1))$. Does there exist $b>0$ such that for every $\varepsilon>0$, the probability of $G_{n,p}\to(F_1,F_2)$ tends to $0$ when $p\le(1-\varepsilon)bn^{-1/m}$ and to $1$ when $p\ge(1+\varepsilon)bn^{-1/m}$?

来源字段：https://doi.org/10.4171/owr/2006/48

15113 | OWR-1536-016

### Thresholds for Random-Order Online Graph Coloring

O / open | Oberwolfach 报告问题汇编

In the one-player online coloring game where the edges of $K_n$ arrive in random order and are immediately colored with one of $r$ colors while the player avoids a monochromatic copy of a fixed graph $F$, does a threshold $N_0(F,r,n)$ exist such that avoidance is strategically possible for $N\ll N_0$ moves but impossible for $N\gg N_0$? For a tree $F$, is the greedy lower bound $N_0\geq n^{1-1/e_r(F)}$ tight, where $e_r(F)=r(e_F-1)+1$ for a star and $e_r(F)=\frac{kr-1}{k-1}(e_F-1)+1$ otherwise, with $k$ the minimum vertex-cover size of $F$?

来源字段：https://doi.org/10.4171/owr/2007/20

15114 | OWR-12861-009

### Thresholds for Spanning Combs and Bounded-Degree Trees

S / solved | Oberwolfach 报告问题汇编

Is the threshold for a spanning comb $\operatorname{Comb}_n$ in $G(n,p)$ of order $(\log n)/n$? More generally, is this the threshold for every $n$-vertex tree of bounded degree?

来源字段：https://doi.org/10.4171/owr/2014/01

15115 | OWR-1536-012

### Thresholds in Multicolor Online Graph-Coloring Games

O / open | Oberwolfach 报告问题汇编

Can the threshold theorem for two-color online edge-coloring games be extended to every number $r\geq2$ of colors? For trees $F$, does the known lower-bound formula give the correct threshold? What are the thresholds in the balanced variant where Painter receives $r$ vertices or edges at each step and must use each color exactly once?

来源字段：https://doi.org/10.4171/owr/2007/20

15116 | OWR-16633-008

### Topology and Limits in the Asymptotic Spectrum of Graphs

O / open | Oberwolfach 报告问题汇编

For finite graphs, let $\Delta$ be the compact space of graph parameters $f$ satisfying

$$f(G\sqcup H)=f(G)+f(H),\qquad f(G\boxtimes H)=f(G)f(H),$$

monotonicity under cohomomorphisms, and $f(K_1)=1$. Is $\Delta$ connected? Does some $f\in\Delta$ satisfy $f(C_7)=\Theta(C_7)$? As primes $p\to\infty$, do the Haemers bounds satisfy

$$h_{\mathbb F_p}^*\longrightarrow h_{\mathbb Q}^*,\qquad h_{\mathbb F_p}\longrightarrow h_{\mathbb Q}?$$

Finally, if $C_{k,n}$ is the circular graph on the vertices of the $n$-cycle in which two vertices are adjacent exactly when their cyclic distance is less than $k$, are the functions $n/k\mapsto\Theta(C_{k,n})$ and $n/k\mapsto\vartheta(C_{k,n})$ continuous?

来源字段：https://doi.org/10.4171/owr/2018/50

15117 | OPG-815

### Total Colouring Conjecture

P / partially_solved | Open Problem Garden

Conjecture A total coloring of a graph $G = (V,E)$ is an assignment of colors to the vertices and the edges of $G$ such that every pair of adjacent vertices, every pair of adjacent edges and every vertex and incident edge pair, receive different colors. The total chromatic number of a graph $G$, $\chi"(G)$, equals the minimum number of colors needed in a total coloring of $G$. It is an old conjecture of Behzad that for every graph $G$, the total chromatic number equals the maximum degree of a vertex in $G$, $\Delta(G)$ plus one or two. In other words, $$ \chi"(G)=\Delta(G)+1\ \ or\ \ \Delta(G)+2. $$

背景链接：http://www.openproblemgarden.org/op/behzads_conjecture

15118 | AMR-087-0096

### Tractable special cases of the signature problem

O / open | AMR 问题清单汇编

Identify and solve tractable special cases of the signature problem described in the slides.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

15119 | AMR-099-0008

### Transient subtrees of hyperbolic graphs

S / solved | AMR 问题清单汇编

Prove that every bounded-degree transient hyperbolic graph contains a transient subtree.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=14

15120 | GRAPH-050

### Translating Graph Problems to Word Problems

P / partially_solved | 独立 / 未分配集合

Which hard graph problems can be efficiently solved by translating graphs to their word representations?

背景链接：https://arxiv.org/abs/1705.05924

15121 | AMR-087-0097

### Trapdoor-free security from multiple nearby RSA moduli

O / open | AMR 问题清单汇编

For nearby moduli $n_i=n_1+d_i$ and maps $f_i(r)=r^{e_i}\bmod n_i$, prove the conjecture that with sufficiently many components at least one $f_i$ is not polynomial-time invertible, thereby validating the proposed trapdoor-free RSA-like probabilistic function.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

15122 | OWR-14299911-020

### Travelling-Polytope Constants for Products of Trees

O / open | Oberwolfach 报告问题汇编

Compute the travelling-polytope constant $\operatorname{TP}(X)$ when $X$ is a product of regular trees, and more generally determine how this constant behaves under products of median graphs.

来源字段：https://doi.org/10.4171/owr/2026/8

15123 | OWR-12861-025

### Triangle Abundance Forcing Large Clique Blow-Ups

O / open | Oberwolfach 报告问题汇编

For every constant $C>0$, does there exist $\gamma=\gamma(C)>0$ such that every sufficiently large $n$-vertex graph with at least $$\frac{n^{3/2}}{\exp(C\sqrt{\log n})}$$ triangles contains a $t$-blow-up of $K_3$ with $t\geq\gamma\sqrt{\log n}$?

来源字段：https://doi.org/10.4171/owr/2014/01

15124 | OPG-345

### Triangle free strongly regular graphs

O / open | Open Problem Garden

Problem Is there an eighth triangle free strongly regular graph?

背景链接：http://www.openproblemgarden.org/op/triangle_free_strongly_regular_graphs

15125 | OWR-13353-015

### Triangle-Cone Characterization of Graphs

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Triangle-Cone Characterization [OWR-13353-016]

Characterize the graphs for which the all-ones edge vector is a nonnegative combination of triangle incidence vectors, equivalently characterize the linear inequalities defining the cone generated by the triangles of a graph.

来源字段：https://doi.org/10.4171/owr/2014/51

15126 | OPG-46837

### Triangle-packing vs triangle edge-transversal.

P / partially_solved | Open Problem Garden

Conjecture If $G$ has at most $k$ edge-disjoint triangles, then there is a set of $2k$ edges whose deletion destroys every triangle.

背景链接：http://www.openproblemgarden.org/op/triangle_packing_vs_triangle_edge_transversal

15127 | AMR-084-0006

### Triples having property D(n) for several parameters

O / open | AMR 问题清单汇编

Are there infinitely many Diophantine triples that are also $D(n)$-triples for three distinct integers $n\ne1$?

背景链接：https://web.math.pmf.unizg.hr/~duje/dtuples.html

15128 | OPG-46706

### Turán number of a finite family.

P / partially_solved | Open Problem Garden

Given a finite family ${\cal F}$ of graphs and an integer $n$, the Turán number $ex(n,{\cal F})$ of ${\cal F}$ is the largest integer $m$ such that there exists a graph on $n$ vertices with $m$ edges which contains no member of ${\cal F}$ as a subgraph.

Conjecture For every finite family ${\cal F}$ of graphs there exists an $F\in {\cal F}$ such that $ex(n, F ) = O(ex(n, {\cal F}))$.

背景链接：http://www.openproblemgarden.org/op/turan_number_of_a_finite_family

15129 | GRAPH-020

### Turán's Brick Factory Problem

P / partially_solved | 独立 / 未分配集合

What is the minimum crossing number of the complete bipartite graph $K_{m,n}$?

背景链接：https://doi.org/10.1016/j.procs.2023.08.216

15130 | OPG-47343

### Turán's problem for hypergraphs

O / open | Open Problem Garden

Conjecture Every simple $3$-uniform hypergraph on $3n$ vertices which contains no complete $3$-uniform hypergraph on four vertices has at most $\frac12 n^2(5n-3)$ hyperedges.

Conjecture Every simple $3$-uniform hypergraph on $2n$ vertices which contains no complete $3$-uniform hypergraph on five vertices has at most $n^2(n-1)$ hyperedges.

背景链接：http://www.openproblemgarden.org/op/turans_problem_for_hypergraphs

15131 | GRAPH-057

### Tutte's 4-Flow Conjecture for Petersen-Minor-Free Graphs

P / partially_solved | 独立 / 未分配集合

Does every Petersen-minor-free bridgeless graph have a nowhere-zero 4-flow?

背景链接：https://arxiv.org/abs/2607.22267

15132 | GRAPH-056

### Tutte's 5-Flow Conjecture

O / open | 独立 / 未分配集合

Does every bridgeless graph have a nowhere-zero 5-flow?

背景链接：https://doi.org/10.1016/0095-8956(81

15133 | OWR-14299081-015

### Tutte’s Four-Flow Conjecture

O / open | Oberwolfach 报告问题汇编

Does every $2$-connected graph with no Petersen graph minor admit a nowhere-zero $4$-flow?

来源字段：https://doi.org/10.4171/owr/2025/1

15134 | GRAPH-038

### Tuza's Conjecture

O / open | 独立 / 未分配集合

Can the edges of any graph be covered by at most 2ν triangles, where ν is the maximum size of a triangle packing?

背景链接：https://doi.org/10.1016/S0012-365X(98

15135 | OWR-16763-021

### Two-Step Path Density in Tripartite Graphs

O / open | Oberwolfach 报告问题汇编

Let $A,B,C$ be disjoint nonempty vertex sets. Suppose every vertex of $A$ has at least $x\lvert B\rvert$ neighbors in $B$ and every vertex of $B$ has at least $y\lvert C\rvert$ neighbors in $C$. Let $\phi(x,y)$ be the largest $z$ such that some vertex of $C$ is joined by two-edge paths to at least $z\lvert A\rvert$ vertices of $A$. Is $\phi(x,x)\geq1/2$ for $x>1/3$? More generally, do $x+ky>1$ and $kx+y\geq1$ imply $\phi(x,y)\geq1/k$?

来源字段：https://doi.org/10.4171/owr/2019/1

15136 | AMR-087-0037

### Typical arithmetic structure of elliptic-curve orders

P / partially_solved | AMR 问题清单汇编

Characterize the typical arithmetic structure of $\#E(\mathbb{F}_q)$ for elliptic curves over finite fields.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

15137 | AMR-087-0035

### Typical exponent of an elliptic-curve group

P / partially_solved | AMR 问题清单汇编

Is the exponent $e_q(E)$ of $E(\mathbb{F}_q)$ typically close to $q$?

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

15138 | AMR-099-0010

### Unbounded descent through iterated graph nets

O / open | AMR 问题清单汇编

Does there exist a graph for which repeatedly passing to $k$-net graphs, at appropriately chosen scales, produces strictly smaller large-scale graph models unboundedly many times?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=15

15139 | OWR-17464-011

### Unbounded Girth Minus Diameter at Minimum Degree Three

O / open | Oberwolfach 报告问题汇编

Can $\operatorname{girth}(G)-\operatorname{diam}(G)$ be arbitrarily large among graphs $G$ of minimum degree at least $3$?

来源字段：https://doi.org/10.4171/owr/2020/1

15140 | GRAPH-039

### Unfriendly Partition Conjecture

O / open | 独立 / 未分配集合

Does every countable graph admit a partition where every vertex has at least as many neighbors outside its part as inside?

背景链接：https://arxiv.org/abs/2412.14151

15141 | OPG-687

### Unfriendly partitions

P / partially_solved | Open Problem Garden

If $G$ is a graph, we say that a partition of $V(G)$ is unfriendly if every vertex has at least as many neighbors in the other classes as in its own.

Problem Does every countably infinite graph have an unfriendly partition into two sets?

背景链接：http://www.openproblemgarden.org/op/unfriendly_partitions

15142 | OWR-4212-017

### Unfriendly Partitions of Countable Graphs

O / open | Oberwolfach 报告问题汇编

Does every countable graph have an unfriendly partition, i.e. a bipartition in which every vertex has at least as many neighbors in the other class as in its own class?

来源字段：https://doi.org/10.4171/owr/2010/11

15143 | OWR-4135-009

### Uniform Complexity Theory for Definable Graph Polynomials

O / open | Oberwolfach 报告问题汇编

Can the three standard results for graph-polynomial evaluation-almost-everywhere #P-hardness, vertex-exponential evaluation, and fixed-parameter tractability on bounded-treewidth graphs-be proved uniformly for large definable classes of graph polynomials?

来源字段：https://doi.org/10.4171/owr/2009/52

15144 | AMR-099-0012

### Uniform expansion bounds for graph nets

O / open | AMR 问题清单汇编

There is a positive function $f(h,d,k)$ such that every graph $G$ with $h(G)>h>0$ and maximum degree less than $d$ has $h(G_k)>f(h,d,k)$ for each $k$-net graph $G_k$. Is $\inf_k f(1,10,k)>0$? In particular, what is the behavior of this bound for $k$-nets in a regular tree?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=15

15145 | OWR-16627-002

### Uniform Winning Times in Strong Ramsey Games

O / open | Oberwolfach 报告问题汇编

For fixed $k$, does the first player have a winning strategy in the strong Ramsey game $R(K_n,K_k)$ using a number of moves bounded independently of $n$ as $n\to\infty$?

来源字段：https://doi.org/10.4171/owr/2018/44

15146 | OPG-349

### Unions of triangle free graphs

O / open | Open Problem Garden

Problem Does there exist a graph with no subgraph isomorphic to $K_4$ which cannot be expressed as a union of $\aleph_0$ triangle free graphs?

背景链接：http://www.openproblemgarden.org/op/unions_of_triangle_free_graphs

15147 | OPG-136

### Unit vector flows

P / partially_solved | Open Problem Garden

Conjecture For every graph $G$ without a bridge, there is a flow $\phi: E(G) \rightarrow S^2 = \{ x \in {\mathbb R}^3: |x| = 1 \}$.

Conjecture There exists a map $q:S^2 \rightarrow \{-4,-3,-2,-1,1,2,3,4\}$ so that antipodal points of $S^2$ receive opposite values, and so that any three points which are equidistant on a great circle have values which sum to zero.

背景链接：http://www.openproblemgarden.org/op/unit_vector_flows

15148 | OWR-16763-018

### Universal Decomposition of Complete Graphs into Bounded-Degree Factors

O / open | Oberwolfach 报告问题汇编

别名：Decomposing Complete Graphs into Bounded-Degree Regular Factors [OWR-16931-017]

For every $\Delta$, does an $n_0$ exist such that, whenever $n\geq n_0$ and the $n$-vertex graphs $F_1,\ldots,F_t$ are $r_i$-regular with $r_i\leq\Delta$ and $\sum_i r_i=n-1$, the complete graph $K_n$ decomposes into copies of $F_1,\ldots,F_t$?

来源字段：https://doi.org/10.4171/owr/2019/1

15149 | OPG-677

### Universal highly arc transitive digraphs

P / partially_solved | Open Problem Garden

An alternating walk in a digraph is a walk $v_0,e_1,v_1,\ldots,v_m$ so that the vertex $v_i$ is either the head of both $e_i$ and $e_{i+1}$ or the tail of both $e_i$ and $e_{i+1}$ for every $1 \le i \le m-1$. A digraph is universal if for every pair of edges $e,f$, there is an alternating walk containing both $e$ and $f$

Question Does there exist a locally finite highly arc transitive digraph which is universal?

背景链接：http://www.openproblemgarden.org/op/universal_highly_arc_transitive_digraphs

15150 | GRAPH-022

### Universal Point Sets

P / partially_solved | 独立 / 未分配集合

Do planar graphs have universal point sets of subquadratic size?

背景链接：https://doi.org/10.7155/jgaa.00318

15151 | OPG-326

### Universal point sets for planar graphs

P / partially_solved | Open Problem Garden

We say that a set $P \subseteq {\mathbb R}^2$ is $n$-universal if every $n$ vertex planar graph can be drawn in the plane so that each vertex maps to a distinct point in $P$, and all edges are (non-intersecting) straight line segments.

Question Does there exist an $n$-universal set of size $O(n)$?

背景链接：http://www.openproblemgarden.org/op/small_universal_point_sets_for_planar_graphs

15152 | OPG-624

### Universal Steiner triple systems

P / partially_solved | Open Problem Garden

Problem Which Steiner triple systems are universal?

背景链接：http://www.openproblemgarden.org/op/universal_steiner_triple_systems

15153 | OWR-16160-001

### Universality of Superexpansion Among Expander Sequences

O / open | Oberwolfach 报告问题汇编

Are all expander graph sequences superexpanders, meaning that they satisfy a uniform Poincaré inequality for every superreflexive Banach space?

来源字段：https://doi.org/10.4171/owr/2018/19

15154 | AMR-087-0016

### Useful deformations for nondegenerate curves

P / partially_solved | AMR 问题清单汇编

Find useful deformations of nondegenerate curves together with easy starting matrices for Frobenius computation.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

15155 | OWR-14299911-019

### Values of Travelling-Polytope Constants on Median Graphs

O / open | Oberwolfach 报告问题汇编

Which values can the travelling-polytope constant $\operatorname{TP}(X)$ take as $X$ ranges over median graphs? Characterize the graphs realizing each possible value.

来源字段：https://doi.org/10.4171/owr/2026/8

15156 | AMR-088-0006

### Vandiver's conjecture

O / open | AMR 问题清单汇编

For a prime $p$, conjecturally $p$ does not divide the class number of the maximal real subfield $\mathbb{Q}(\zeta_p+\overline{\zeta_p})$ of the $p$th cyclotomic field.

背景链接：https://www.maths.nottingham.ac.uk/plp/pmzcw/download/open.pdf

15157 | OPG-37316

### Vertex Coloring of graph fractional powers

P / partially_solved | Open Problem Garden

Conjecture Let $G$ be a graph and $k$ be a positive integer. The $k-$ power of $G$, denoted by $G^k$, is defined on the vertex set $V(G)$, by connecting any two distinct vertices $x$ and $y$ with distance at most $k$. In other words, $E(G^k)=\{xy:1\leq d_G(x,y)\leq k\}$. Also $k-$ subdivision of $G$, denoted by $G^\frac{1}{k}$, is constructed by replacing each edge $ij$ of $G$ with a path of length $k$. Note that for $k=1$, we have $G^\frac{1}{1}=G^1=G$. Now we can define the fractional power of a graph as follows: Let $G$ be a graph and $m,n\in \mathbb{N}$. The graph $G^{\frac{m}{n}}$ is defined by the $m-$ power of the $n-$ subdivision of $G$. In other words $G^{\frac{m}{n}}\isdef (G^{\frac{1}{n}})^m$. Conjecture. Let $G$ be a connected graph with $\Delta(G)\geq3$ and $m$ be a positive integer greater than 1. Then for any positive integer $n>m$, we have $\chi(G^{\frac{m}{n}})=\omega(G^\frac{m}{n})$. In [1], it was shown that this conjecture is true in some special cases.

背景链接：http://www.openproblemgarden.org/op/vertex_coloring_of_graph_fractional_powers

15158 | OWR-14299905-035

### Vertex Sets Meeting Every Edge Color

O / open | Oberwolfach 报告问题汇编

Let $r\geq3$, and let $H$ be a three-colored $(r-1)$-uniform hypergraph with $R$, $G$, and $B$ edges of the three colors. What is the maximum number of $r$-vertex sets $S$ for which $H[S]$ contains an edge of every color?

来源字段：https://doi.org/10.4171/owr/2026/1

15159 | AMR-099-0009

### Vertex-transitive sub-scale-invariant graphs

O / open | AMR 问题清单汇编

Does there exist a vertex-transitive graph whose multiplicative rough-isometry constants to its $k$-net graphs tend to $1$ as $k\to\infty$?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=14

15160 | AMR-054-0017

### Visibility Graph Recognition

O / open | AMR 问题清单汇编

Given a visibility graph $G$ and a Hamiltonian circuit $C$, determine in polynomial time whether there is a simple polygon whose vertex visibility graph is $G$, and whose boundary corresponds to $C$.

背景链接：https://topp.openproblem.net/p17

15161 | AMR-131-0012

### Visibility Graphs of Points, Segments and Polygons - Conjecture 14

O / open | AMR 问题清单汇编

{ (a) The cycle $C=(p_1, p_{10}, p_9, p_8, p_7, p_5, p_4, p_6, p_3, p_2, p_1)$ is Hamiltonian but is self-intersecting. (b) The cycle $C=(p_1, p_9, p_{10}, p_7, p_8, p_6, p_5, p_4, p_3, p_2, p_1)$ is a Hamiltonian circuit as it forms a boundary of a simple polygon. (c) There is no alternating cycle as both endpoints of the segment $p_9p_{10}$ belong to the convex hull of the segments [rit-cscf-90].

}{fhamilton1.eps}

Every segment visibility graph $G$ contains a Hamiltonian cycle $C$.

背景链接：https://arxiv.org/abs/1012.5187

15162 | AMR-131-0013

### Visibility Graphs of Points, Segments and Polygons - Conjecture 15

O / open | AMR 问题清单汇编

Every segment visibility graph $G$ contains a Hamiltonian cycle $C$ that corresponds to a Hamiltonian circuit in the embedded segment visibility graph $G'$.

背景链接：https://arxiv.org/abs/1012.5187

15163 | AMR-131-0014

### Visibility Graphs of Points, Segments and Polygons - Conjecture 16

O / open | AMR 问题清单汇编

Every segment visibility graph $G$ contains a Hamiltonian cycle $C$ that corresponds to the boundary of a circumscribing polygon in the embedded segment visibility graph $G'$.

背景链接：https://arxiv.org/abs/1012.5187

15164 | AMR-131-0003

### Visibility Graphs of Points, Segments and Polygons - Conjecture 4

O / open | AMR 问题清单汇编

For all integers $k \geq 2$ and $\ell \geq 2$, there is an integer $h(k,\ell)$ such that any point set $P$ of at least $h(k,\ell)$ points in the plane contains $\ell$ collinear points, or $k$ mutually visible points.

背景链接：https://arxiv.org/abs/1012.5187

15165 | AMR-131-0005

### Visibility Graphs of Points, Segments and Polygons - Conjecture 6

O / open | AMR 问题清单汇编

In every $k$-blocked point set, there are at most $k$ points in each colour class.

背景链接：https://arxiv.org/abs/1012.5187

15166 | AMR-131-0001

### Visibility Graphs of Points, Segments and Polygons - Problem 1

O / open | AMR 问题清单汇编

We have stated earlier how to compute the visibility graph $G$ from a given set of points $P$. Consider the opposite problem of determining if there is a set of points $P$ whose visibility graph is the given graph $G$. This problem is called the visibility graph *recognition* problem. Identifying the set of properties satisfied by all visibility graphs is called the visibility graph *characterization* problem. The problem of actually drawing one such set of points $P$ whose visibility graph is the given graph $G$, is called the visibility graph *reconstruction* problem.

Given a graph $G$ in adjacency matrix form, determine whether $G$ is the visibility graph of a set of points $P$ in the plane.

背景链接：https://arxiv.org/abs/1012.5187

15167 | AMR-131-0009

### Visibility Graphs of Points, Segments and Polygons - Problem 10

O / open | AMR 问题清单汇编

\begin{unsolved} For $h>1$, what is the smallest number of vertices of a graph with obstacle number $h$? \end{unsolved}

Does every planar graph have obstacle number $1$?

背景链接：https://arxiv.org/abs/1012.5187

15168 | AMR-131-0010

### Visibility Graphs of Points, Segments and Polygons - Problem 11

O / open | AMR 问题清单汇编

All three above problems are open for segment visibility graphs. Only characterization known is for a sub-class given by Everette et al. [ehkn-cgta2-2000]. They have characterized those segment visibility graphs that do not have $K_5$ (a complete graph of five vertices) as a minor. A graph $M$ is called a *minor* of a graph $G$ if $M$ can be obtained from $G$ by a sequence of vertex deletions, edge deletions and edge contractions. Their characterization gives a straightforward polynomial time algorithm for recognizing this class of graphs.

Given a graph $G$ in adjacency matrix form, determine whether $G$ is the segment visibility graph of a set of disjoint segments $S$ in the plane.

背景链接：https://arxiv.org/abs/1012.5187

15169 | AMR-131-0011

### Visibility Graphs of Points, Segments and Polygons - Problem 12

O / open | AMR 问题清单汇编

\begin{unsolved} Given a graph $G$ in adjacency matrix form, determine whether $G$ is the segment visibility graph of a set of disjoint segments $S$ in the plane. \end{unsolved}

Characterize the segment visibility graphs.

背景链接：https://arxiv.org/abs/1012.5187

15170 | AMR-131-0015

### Visibility Graphs of Points, Segments and Polygons - Problem 17

O / open | AMR 问题清单汇编

Observe that the algorithm of Hoffmann and T\'{o}th [ht-sevg-2003] takes $S$ as an input and then locates a Hamiltonian circuit in the embedded segment visibility graph $G'$. Suppose $S$ is not given as an input but only $G$ is given, then it is not clear how to identify a Hamiltonian cycle in $G$ in polynomial time as there is no known algorithm for segment visibility graph reconstruction problem. So, we have the following problems.

Given a segment visibility graph $G$ in adjacency matrix form, identify a Hamiltonian cycle in $G$ in polynomial time.

背景链接：https://arxiv.org/abs/1012.5187

15171 | AMR-131-0016

### Visibility Graphs of Points, Segments and Polygons - Problem 18

O / open | AMR 问题清单汇编

\begin{unsolved} Given a segment visibility graph $G$ in adjacency matrix form, identify a Hamiltonian cycle in $G$ in polynomial time. \end{unsolved}

Given a segment visibility graph $G$ in adjacency matrix form, identify the edges of $G$ that correspond to segments of $S$.

背景链接：https://arxiv.org/abs/1012.5187

15172 | AMR-131-0017

### Visibility Graphs of Points, Segments and Polygons - Problem 19

O / open | AMR 问题清单汇编

Let us consider variations of bar visibility graphs. While representing vertices of $G$ as bars, there is no restriction on the length of horizontal bars. Suppose, a restriction is imposed that all bars in a visibility representation must have the same length. In that case, we get another type of visibility graphs which are known as *unit bar visibility graphs*. Though there are characterizations for special classes of unit bar visibility graphs [chks-cbvt-2006,dgh-unvl-2004,dv-ubvg-2003], no characterization is known for general graphs. We have the following problems.

Characterize unit bar visibility graphs.

背景链接：https://arxiv.org/abs/1012.5187

15173 | AMR-131-0002

### Visibility Graphs of Points, Segments and Polygons - Problem 2

O / open | AMR 问题清单汇编

\begin{unsolved} Given a graph $G$ in adjacency matrix form, determine whether $G$ is the visibility graph of a set of points $P$ in the plane. \end{unsolved}

Characterize visibility graphs of point sets.

背景链接：https://arxiv.org/abs/1012.5187

15174 | AMR-131-0018

### Visibility Graphs of Points, Segments and Polygons - Problem 21

O / open | AMR 问题清单汇编

Unlike bar visibility graphs, bar $k$-visibility graphs have not been completely characterized. For understanding this class of graphs, Dean et al. [deglst-bvg-2007] considered the problem of bounding the number of edges of a bar $k$-visibility graph $G$. They proved an upper bounds of $(k+1)(3n-7/2 k-5)-1$ edges for $G$ and conjectured an improve edge bound of $(k+1)(3n-4 k-6)$ which was proved later by Hartke et al. [hvw-frbvg-2007]. They also studied $d$-regular $k$-visibility graphs. Dean et al. [deglst-bvg-2007] also proved that $6k+6$ is an upper bound on the chromatic number of $G$. We have the following problems [deglst-bvg-2007,hvw-frbvg-2007].

Characterize bar $k$-visibility graphs.

背景链接：https://arxiv.org/abs/1012.5187

15175 | AMR-131-0019

### Visibility Graphs of Points, Segments and Polygons - Problem 23

O / open | AMR 问题清单汇编

\begin{unsolved} Triangle-free non-planar graphs are forbidden subgraphs of bar $k$-visibility graphs. Is there any other class of forbidden subgraphs of bar $k$-visibility graphs? \end{unsolved}

Are there $(2k+2)$-regular bar $k$-visibility graphs for $k \geq 5$?

背景链接：https://arxiv.org/abs/1012.5187

15176 | AMR-131-0020

### Visibility Graphs of Points, Segments and Polygons - Problem 25

O / open | AMR 问题清单汇编

\begin{unsolved} Are there $(2k+2)$-regular bar $k$-visibility graphs for $k \geq 5$? \end{unsolved}

\begin{unsolved} Are there $d$-regular bar $k$-visibility graphs with $d \geq 2k+3$? \end{unsolved}

Improve the current upper bound of $6k+6$ on the chromatic number of a bar $k$-visibility graph.

背景链接：https://arxiv.org/abs/1012.5187

15177 | AMR-131-0021

### Visibility Graphs of Points, Segments and Polygons - Problem 26

O / open | AMR 问题清单汇编

Dean et al. [deglst-bvg-2007] also studied the thickness of bar $k$-visibility graphs. The *thickness* of a graph $G$ is defined as the minimum number of planar graphs whose union is $G$. Exact values of thickness is know for very few classes of graphs [mos-tgs-1998]. Dean et al. proved an upper bound of $4$ for the thickness of bar $1$-visibility graphs, and conjectured that bar $1$-visibility graphs actually have thickness of at most $2$. The conjecture was disproved by Felsner and Massow [fw-tbvg-2007] by constructing a bar $1$-visibility graph having thickness $3$. For a special class of bar $1$-visibility graphs, Felsner and Massow [fw-tbvg-2007] presented an algorithm for partitioning the edges into two planar graphs showing that the thickness of this special class of graphs is $2$. We have the following problems [deglst-bvg-2007].

It has been shown that the thickness of a bar $k$-visibility graph is bounded by $2k(9k-1)$. Can this upper bound be improved?

背景链接：https://arxiv.org/abs/1012.5187

15178 | AMR-131-0022

### Visibility Graphs of Points, Segments and Polygons - Problem 27

O / open | AMR 问题清单汇编

The crossing number of a graph is the minimum possible number of crossings with which the graph can be drawn in the plane. What is the largest crossing number of a bar $k$-visibility graph?

背景链接：https://arxiv.org/abs/1012.5187

15179 | AMR-131-0023

### Visibility Graphs of Points, Segments and Polygons - Problem 28

O / open | AMR 问题清单汇编

\begin{unsolved} The crossing number of a graph is the minimum possible number of crossings with which the graph can be drawn in the plane. What is the largest crossing number of a bar $k$-visibility graph? \end{unsolved}

The genus of a graph is the minimal integer $g$ such that the graph can be embedded on a surface of genus $g$. What is the largest genus of a bar $k$-visibility graph?

背景链接：https://arxiv.org/abs/1012.5187

15180 | AMR-131-0024

### Visibility Graphs of Points, Segments and Polygons - Problem 29

O / open | AMR 问题清单汇编

Bar visibility graphs have been generalized to rectangle visibility graphs by considering both vertical and horizonal visibility among bars having non-zero thickness [bdhs-orvg-1997,dh-rvrbp-1997,o-opcv-1998,r-drvg-1997,sw-rvgc-2003] . A graph $G$ is called a *rectangle visibility graph* if it can be realized by closed isothetic rectangles in the plane, with pairwise disjoint interiors, with vertices representing rectangles in such a way that two vertices $v_i$ and $v_j$ of $G$ are connected by an edge if and only if their corresponding rectangles are vertically or horizontally visible from each other by a beam of unobstructed visibility of finite width. Unlike bar visibility graphs, no characterization of rectangle visibility graphs are known, and moreover, Shermer [s-orvsiii-1996] has shown that the problem of recognizing them is NP-complete. We have the following problem.

Characterize rectangle visibility graphs.

背景链接：https://arxiv.org/abs/1012.5187

15181 | AMR-131-0025

### Visibility Graphs of Points, Segments and Polygons - Problem 30

O / open | AMR 问题清单汇编

\begin{unsolved} Given a graph $G$ in adjacency matrix form, determine whether $G$ is the visibility graph of a simple polygon $P$. \end{unsolved}

Is the problem of recognizing visibility graphs in $NP$?

背景链接：https://arxiv.org/abs/1012.5187

15182 | AMR-131-0026

### Visibility Graphs of Points, Segments and Polygons - Problem 31

O / open | AMR 问题清单汇编

Given a graph $G$ in adjacency matrix form along with a Hamiltonian cycle $C$ of $G$, determine whether $G$ is the visibility graph of a simple polygon $P$ whose boundary corresponds to the given Hamiltonian cycle $C$.

背景链接：https://arxiv.org/abs/1012.5187

15183 | AMR-131-0027

### Visibility Graphs of Points, Segments and Polygons - Problem 32

O / open | AMR 问题清单汇编

Design an algorithm for testing Necessary Condition 4 in polynomial time.

背景链接：https://arxiv.org/abs/1012.5187

15184 | AMR-131-0028

### Visibility Graphs of Points, Segments and Polygons - Problem 33

O / open | AMR 问题清单汇编

Draw a simple polygon whose visibility graph is the given graph $G$.

背景链接：https://arxiv.org/abs/1012.5187

15185 | AMR-131-0029

### Visibility Graphs of Points, Segments and Polygons - Problem 34

O / open | AMR 问题清单汇编

A *Hamiltonian cycle* is a cycle in an undirected graph which visits each vertex exactly once and also returns to the starting vertex. The Hamiltonian cycle problem is to determine whether a Hamiltonian cycle exists in a given graph $G$. Observe that $G$ may contain several Hamiltonian cycles, and $G$ may be visibility graph for a Hamiltonian cycle and is not a valid visibility graph for another Hamiltonian cycle in $G$.

Given the visibility graph $G$ of a simple polygon $P$, determine the Hamiltonian cycle in $G$ that corresponds to the boundary of $P$.

背景链接：https://arxiv.org/abs/1012.5187

15186 | AMR-131-0030

### Visibility Graphs of Points, Segments and Polygons - Problem 35

O / open | AMR 问题清单汇编

A *dominating set* for a graph $G = (V, E)$ is a subset $D$ of $V$ such that every vertex not in $D$ is joined to at least one member of $D$ by some edge. The minimum dominating set problem in visibility graphs corresponds to the art gallery problem in polygons which has been shown to be NP-hard [ll-ccagp-86, ls-cavg-95]. Following the approximation algorithm for the art gallery problem for polygons given by Ghosh [g-apat-2006,g-apatpp-2010], a minimum dominating set of visibility graph can be computed with an approximation ratio of $O(\log n)$.

Design a constant factor approximation algorithm for computing minimum dominating set of visibility graphs.

背景链接：https://arxiv.org/abs/1012.5187

15187 | AMR-131-0031

### Visibility Graphs of Points, Segments and Polygons - Problem 36

O / open | AMR 问题清单汇编

Given the visibility graph $G$ of a simple polygon $P$ along with the Hamiltonian cycle in $G$ corresponding to the boundary of $P$, determine the maximum hidden set of $G$.

背景链接：https://arxiv.org/abs/1012.5187

15188 | AMR-131-0032

### Visibility Graphs of Points, Segments and Polygons - Problem 37

O / open | AMR 问题清单汇编

\begin{unsolved} Given the visibility graph $G$ of a simple polygon $P$ along with the Hamiltonian cycle in $G$ corresponding to the boundary of $P$, determine the maximum hidden set of $G$. \end{unsolved}

Design an approximation algorithm for computing maximum hidden set of visibility graph.

背景链接：https://arxiv.org/abs/1012.5187

15189 | AMR-131-0033

### Visibility Graphs of Points, Segments and Polygons - Problem 38

O / open | AMR 问题清单汇编

A *clique* in a graph is a set of pairwise adjacent vertices. The problem of computing the maximum clique in the visibility graph is not known to be NP-hard. Observe that the maximum clique in a visibility graph corresponds to the largest empty convex polygon inside the corresponding polygon. Algorithms for computing largest empty convex polygons has been reported by several authors [ar-clec-85, deo-secp-90, es-mcmcpvg-00]. However, for each of these algorithms, the input is either a polygon [es-mcmcpvg-00] or a point set [ar-clec-85, deo-secp-90]. Spinrad [s-grep-03] has discussed possible approaches for computing maximum clique using the notion of *triangle-extendible ordering* which is essentially a transitive orientation of the graph.

Given a visibility graph $G$ in adjacency matrix form, compute a maximum clique of $G$.

背景链接：https://arxiv.org/abs/1012.5187

15190 | AMR-131-0034

### Visibility Graphs of Points, Segments and Polygons - Problem 39

O / open | AMR 问题清单汇编

\begin{unsolved} Given a visibility graph $G$ in adjacency matrix form, compute a maximum clique of $G$. \end{unsolved}

Determine whether a set of vertices of a visibility graph has a triangle-extendible ordering in polynomial time.

背景链接：https://arxiv.org/abs/1012.5187

15191 | AMR-131-0035

### Visibility Graphs of Points, Segments and Polygons - Problem 40

O / open | AMR 问题清单汇编

Two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are *isomorphic* if and only if there is a bijection $f$ that maps vertices of $V_1$ to the vertices of $V_2$ such that an edge $(v, w) \in E_1$ if and only if the edge $(f(v), f(w)) \in E_2$. It has been shown by Ghosh [g-vap-07] that the number of *non-isomorphic* visibility graphs of simple polygons of $n$ vertices is at least $2^{n-4}$. On the other hand, a straightforward application of Warren's theorem [w-lbanm-68] shows that the number of visibility graphs is at most $2^{O(n\log n)}$ [s-grep-03].

Improve the lower and upper bounds on the number of *non-isomorphic* visibility graphs of simple polygons.

背景链接：https://arxiv.org/abs/1012.5187

15192 | AMR-131-0036

### Visibility Graphs of Points, Segments and Polygons - Problem 41

O / open | AMR 问题清单汇编

Although the most natural form of representation for visibility graphs would be to use coordinates of the points, this is not useful if we are looking for a space efficient representation. Lin and Skiena [ls-cavg-95] have proved that visibility graphs require endpoints to have exponential sized integers. However, it is not known whether singly exponential sized integers are sufficient. It is important because if we could guarantee that the number of bits in the integer is polynomial, then visibility graph recognition is in $NP$ [s-grep-03].

Can all endpoints of a visibility graph be assigned integer coordinates such that the integers use a polynomial number of bits?

背景链接：https://arxiv.org/abs/1012.5187

15193 | AMR-131-0037

### Visibility Graphs of Points, Segments and Polygons - Problem 42

O / open | AMR 问题清单汇编

Give a tight bound (with respect to order notation) on the number of bits used in an optimal covering set of a visibility graph.

背景链接：https://arxiv.org/abs/1012.5187

15194 | AMR-131-0038

### Visibility Graphs of Points, Segments and Polygons - Problem 43

O / open | AMR 问题清单汇编

\begin{unsolved} Give a tight bound (with respect to order notation) on the number of bits used in an optimal covering set of a visibility graph. \end{unsolved}

Find a covering set which matches the above bound in polynomial time.

背景链接：https://arxiv.org/abs/1012.5187

15195 | AMR-131-0004

### Visibility Graphs of Points, Segments and Polygons - Problem 5

O / open | AMR 问题清单汇编

Let us discuss upper bounds on $b(n)$. It is obvious that $b(n) \leq {n \choose 2}$. Let $\mu(P)$ denote the size of the set of midpoints of all ${n \choose 2}$ segments between points of $P$. Let $\mu(n)$ denote the minimum of $\mu(P)$ for all $P$ having $n$ points with no three points being collinear. Using Freiman's theorem on set addition [f-fsts-1973], Stanchescu [ys-psc-2002] and Pach [p-msip-2003] have independently shown that $b(n) \leq \mu(n) \leq n2^{c \sqrt{\log n}}$, where $c$ is an absolute constant. This shows that if $ \mu(n)$ is not $O(n)$, it can only be slightly super-linear. Many authors [dpt-nbvp-2009,hhuz-mpp-2001, m-bvpg-2009,pw-ovb-2010,ys-psc-2002] have stated or conjectured that every set of points $P$ in general position requires a super-linear number of blockers as given below.

Prove that as $n \rightarrow \infty$, $\frac{b(n)}{n} \rightarrow \infty$.

背景链接：https://arxiv.org/abs/1012.5187

15196 | AMR-131-0006

### Visibility Graphs of Points, Segments and Polygons - Problem 7

O / open | AMR 问题清单汇编

Consider the problem of obstacle representation of a given graph $G$ of $n$ vertices, which was introduced by Alpet et al. [akl-ong-2010]. Draw every vertex $v_i$ of $G$ as a point $p_i$ in the plane and draw obstacles in such a way that every segment $p_ip_j$ intersects an obstacle if and only if $(v_i,v_j)$ is not an edge in $G$. The *obstacle number* of $G$ is the minimum number of obstacles required in any obstacle representation of $G$. Since an obstacle can be placed to block the visibility between each pair of points, $n \choose 2$ is an upper bound on the obstacle number of $G$. We have the following question from Alpet et al. [akl-ong-2010].

Is the obstacle number of a graph with $n$ vertices bounded above by a linear function of $n$?

背景链接：https://arxiv.org/abs/1012.5187

15197 | AMR-131-0007

### Visibility Graphs of Points, Segments and Polygons - Problem 8

O / open | AMR 问题清单汇编

Regarding the lower bound on obstacle numbers, Alpet et al. [akl-ong-2010] have showed that there exists a graph of $n$ vertices with obstacle number $O(\sqrt{\log n})$, which has been improved to $O(n/\log ^2 n)$ by Mukkamala et al. [mps-glon-2010]. The bound becomes $O(n/\log n)$ if the obstacles are restricted to convex polygons.

Improve the present lower bound $O(n/\log ^2 n)$ of the obstacle number of a graph with $n$ vertices.

背景链接：https://arxiv.org/abs/1012.5187

15198 | AMR-131-0008

### Visibility Graphs of Points, Segments and Polygons - Problem 9

O / open | AMR 问题清单汇编

Regarding the graphs with low obstacle numbers, Alpet et al. [akl-ong-2010] and Mukkamala et al. [mps-glon-2010] have studied graphs with obstacle numbers $1$ and $2$. Mukkamala et al. [mps-glon-2010] showed that for any positive integer $h \geq 3$, there is a graph with obstacle number exactly $h$. The following questions of Alpet et al. [akl-ong-2010] are still open.

For $h>1$, what is the smallest number of vertices of a graph with obstacle number $h$?

背景链接：https://arxiv.org/abs/1012.5187

15199 | GRAPH-041

### Vizing's Conjecture

O / open | 独立 / 未分配集合

For the Cartesian product of graphs $G \square H$, is the domination number at least $\gamma(G) \cdot \gamma(H)$?

背景链接：https://doi.org/10.37236/15

15200 | OWR-16938-003

### Vizing's Conjecture and Algebraic Sum-of-Squares Certificates

O / open | Oberwolfach 报告问题汇编

Prove Vizing's conjecture for arbitrary graphs, and develop symmetry-reduced or Gröbner-basis-free sum-of-squares and semidefinite certificates for its algebraic formulation at larger parameter values.

来源字段：https://doi.org/10.4171/owr/2019/26

15201 | OWR-10252928-002

### Vizing’s Cartesian-Product Domination Conjecture

O / open | Oberwolfach 报告问题汇编

For all finite graphs $G$ and $H$, is $\gamma(G\square H)\geq\gamma(G)\gamma(H)$?

来源字段：https://doi.org/10.4171/owr/2022/20

15202 | OWR-14299911-013

### Wall Structures on Symmetric-Group Cayley Graphs

O / open | Oberwolfach 报告问题汇编

Does there exist a sequence of generating sets $X_n$ for $S_n$, of uniformly bounded size, such that the Cayley graphs $\operatorname{Cay}(S_n,X_n)$ admit wall structures whose wall metrics are uniformly coarsely equivalent to their word metrics?

来源字段：https://doi.org/10.4171/owr/2026/8

15203 | OWR-16763-016

### Weak Coloring Numbers Excluding Complete Bipartite Minors

P / partially_solved | Oberwolfach 报告问题汇编

What is the asymptotically sharp bound on the weak coloring number $\operatorname{wcol}_r(G)$ for graphs $G$ excluding $K_{3,t}$ as a minor?

来源字段：https://doi.org/10.4171/owr/2019/1

15204 | OPG-434

### Weak pentagon problem

P / partially_solved | Open Problem Garden

Conjecture If $G$ is a cubic graph not containing a triangle, then it is possible to color the edges of $G$ by five colors, so that the complement of every color class is a bipartite graph.

背景链接：http://www.openproblemgarden.org/op/weak_pentagon_problem

15205 | OPG-60013

### Weak saturation of the cube in the clique

P / partially_solved | Open Problem Garden

Problem

Determine $\text{wsat}(K_n,Q_3)$.

背景链接：http://www.openproblemgarden.org/op/weak_saturation_of_the_cube_in_the_clique

15206 | OWR-14299911-007

### Weighted Centers on Bounded-Dimensional Median Graphs

O / open | Oberwolfach 报告问题汇编

Is there an almost-linear-time algorithm for the Weighted Center problem on median graphs of bounded cube dimension?

来源字段：https://doi.org/10.4171/owr/2026/8

15207 | OPG-47511

### Weighted colouring of hexagonal graphs.

P / partially_solved | Open Problem Garden

Conjecture There is an absolute constant $c$ such that for every hexagonal graph $G$ and vertex weighting $p:V(G)\rightarrow \mathbb{N}$, $$\chi(G,p) \leq \frac{9}{8}\omega(G,p) + c$$

背景链接：http://www.openproblemgarden.org/op/weighted_colouring_of_hexagonal_graphs

15208 | OWR-4212-016

### Weighted Min-Max Theorems for Disjoint Terminal Paths

O / open | Oberwolfach 报告问题汇编

Given an undirected graph $G=(V,E)$, terminals $S\subseteq V$, and a weight $w:\binom{S}{2}\to\mathbb N$, does the weighted maximum

$$\sum_P w(s(P),t(P))$$

over node-disjoint $S$-paths admit a Mader-type min-max characterization and a polynomial-time algorithm?

来源字段：https://doi.org/10.4171/owr/2010/11

15209 | OPG-157

### What is the largest graph of positive curvature?

S / solved | Open Problem Garden

Problem What is the largest connected planar graph of minimum degree 3 which has everywhere positive combinatorial curvature, but is not a prism or antiprism?

背景链接：http://www.openproblemgarden.org/op/what_is_the_largest_graph_of_positive_curvature

15210 | OPG-567

### What is the smallest number of disjoint spanning trees made a graph Hamiltonian

S / solved | Open Problem Garden

We are given a complete simple undirected weighted graph $G_1=(V,E)$ and its first arbitrary shortest spanning tree $T_1=(V,E_1)$. We define the next graph $G_2=(V,E\setminus E_1)$ and find on $G_2$ the second arbitrary shortest spanning tree $T_2=(V,E_2)$. We continue similarly by finding $T_3=(V,E_3)$ on $G_3=(V,E\setminus \cup_{i=1}^{2}E_i)$, etc. Let k be the smallest number of disjoint shortest spanning trees as defined above and let $T^{k}=(V,\cup_{i=1}^{k}E_i)$ be the graph obtained as union of all $k$ disjoint trees.

Question 1. What is the smallest number of disjoint spanning trees creates a graph $T^{k}$ containing a Hamiltonian path.

Question 2. What is the smallest number of disjoint spanning trees creates a graph $T^{k}$ containing a shortest Hamiltonian path?

Questions 3 and 4. Replace in questions 1 and 2 a shortest spanning tree by a 1-tree. What is the smallest number of disjoint 1-trees creates a Hamiltonian graph? What is the smallest number of disjoint 1-trees creates a graph containing a shortest Hamiltonian cycle?

背景链接：http://www.openproblemgarden.org/op/what_is_the_smallest_number_of_disjoint_spanning_trees_made_a_graph_hamiltonian

15211 | GRAPH-058

### Woodall's Conjecture

P / partially_solved | 独立 / 未分配集合

别名：Woodall's Conjecture [OPG-179]

Is the minimum dicut size equal to the maximum number of disjoint dijoins in a directed graph?

背景链接：https://doi.org/10.1007/s00493-025-00159-x

15212 | GRAPH-045

### Word-Representable Graphs: Forbidden Subgraph Characterization

P / partially_solved | 独立 / 未分配集合

Characterize word-representable graphs in terms of forbidden induced subgraphs.

背景链接：https://arxiv.org/abs/2502.06979

15213 | GRAPH-043

### Word-Representable Graphs: Letter Copies Bound

O / open | 独立 / 未分配集合

Are there graphs on n vertices requiring more than floor(n/2) copies of each letter for word-representation?

背景链接：https://arxiv.org/abs/1501.07108

15214 | GRAPH-046

### Word-Representable Near-Triangulations

P / partially_solved | 独立 / 未分配集合

Characterize word-representable near-triangulations containing $K_4$.

背景链接：https://arxiv.org/abs/2605.25733

15215 | AMR-087-0068

### Worst cases for LLL on ideal lattices

P / partially_solved | AMR 问题清单汇编

Exhibit cyclic or ideal lattices on which LLL achieves its worst-case approximation factor.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

15216 | AMR-087-0073

### Worst-case security of quasi-cyclic cryptosystems

O / open | AMR 问题清单汇编

Prove that quasi-cyclic lattice or code public-key constructions are secure based on worst-case hardness for quasi-cyclic structures.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

15217 | AMR-054-0070

### Yao-Yao Graph a Spanner?

P / partially_solved | AMR 问题清单汇编

Is the Yao-Yao Graph a $t$-spanner for constant $t$? A geometric graph is a $t$-spanner (or just a spanner) if, for every pair of nodes, the shortest distance between the nodes following the edges of the graph is at most $t$ times the Euclidean distance between them. See below for the definition of the Yao-Yao graph.

背景链接：https://topp.openproblem.net/p70

15218 | GRAPH-040

### Zarankiewicz Problem

P / partially_solved | 独立 / 未分配集合

What is the maximum number of edges in a bipartite graph on (m,n) vertices with no complete bipartite subgraph $K_{s,t}$?

背景链接：https://arxiv.org/abs/2410.03702

15219 | AMR-087-0029

### Zero-constant-term Newton-polytope case

O / open | AMR 问题清单汇编

Resolve the zero-constant-term case in the Newton-polytope formulation of multivariate small-root methods.

背景链接：https://www.ipam.ucla.edu/programs/workshops/workshop-i-number-theory-and-cryptography-open-problems/?tab=schedule

15220 | OPG-547

### ¿Are critical k-forests tight?

P / partially_solved | Open Problem Garden

Conjecture

Let $H$ be a $k$-uniform hypergraph. If $H$ is a critical $k$-forest, then it is a $k$-tree.

背景链接：http://www.openproblemgarden.org/op/are_critical_k_forests_tight

# 资料汇编 12 | 概率 / Probability

733 条记录。O=开放；P=部分解决；S=已解决。状态与英文/LaTeX 表述均沿用来源。

15221 | AMR-027-0901

### 10 Lectures and 42 Open Problems - Detection Threshold for SBM for three of more communities

S / solved | AMR 问题清单汇编

What is the partial recovery threshold for the Stochastic Block Model on $k\geq 3$ communities.

背景链接：https://afonsobandeira.wordpress.com/2015/11/28/10l42psbm/

15222 | AMR-027-0406

### 10 Lectures and 42 Open Problems - Feige's conjecture

P / partially_solved | AMR 问题清单汇编

Prove or disprove the following conjecture by Feige : Given $n$ independent random variables $X_1,\dots,X_n$ s.t., for all $i$ , $X_i \geq 0$ and $\mathbb{E} X_i = 1$ we have $\mathrm{Prob}\left( \sum_{i=1}^n X_i \geq n+1 \right) \leq 1 - e^{-1}$ .

背景链接：https://afonsobandeira.wordpress.com/2015/11/29/10l42panextraopenproblem/

15223 | AMR-027-0102

### 10 Lectures and 42 Open Problems - Gaussian singular-value monotonicity

P / partially_solved | AMR 问题清单汇编

For a $d\times d$ real Gaussian matrix $G_{\mathbb{R}}$ and complex Gaussian matrix $G_{\mathbb{C}}$, both normalized to entry variance $1/d$, define $\alpha_{\mathbb{K}}(d)=d^{-1}\mathbb{E}\sum_{k=1}^d\sigma_k(G_{\mathbb{K}})$. Is $\alpha_{\mathbb{R}}(d)$ monotonically increasing in $d$, and is $\alpha_{\mathbb{C}}(d)$ monotonically decreasing in $d$?

背景链接：https://afonsobandeira.wordpress.com/2013/11/01/a-conjecture-on-the-singular-values-of-a-gaussian-matrix/

15224 | AMR-027-0401

### 10 Lectures and 42 Open Problems - Improvement over Non-commutative Khintchine inequality

S / solved | AMR 问题清单汇编

Let $A_1,\dots,A_n\in \mathbb{R}^{d\times d}$ be symmetric matrices and $g_1,\dots,g_n\sim\mathcal{N}(0,1)$ i.i.d.. Does the following hold? $\mathbb{E} \left\| \sum_{k=1}^{n} g_k A_k \right\| \lesssim \sigma + \left(\log d\right)^{\frac12}\sigma_\ast.$

背景链接：https://afonsobandeira.wordpress.com/2015/10/25/10l42concentration/

15225 | AMR-027-0402

### 10 Lectures and 42 Open Problems - Lata $\l$ a-Riemer-Schutt

S / solved | AMR 问题清单汇编

Let $X\in\mathbb{R}^{d\times d}$ be a symmetric matrix with (otherwise) independent gaussian entries. Prove (or disprove): $\mathbb{E} \|X\| \lesssim\mathbb{E} \max_k \|Xe_k\|_2$

背景链接：https://afonsobandeira.wordpress.com/2015/10/25/10l42concentration/

15226 | AMR-027-0101

### 10 Lectures and 42 Open Problems - Mallat-Zeitouni Gaussian-basis problem

P / partially_solved | AMR 问题清单汇编

Let $X$ be a centered Gaussian random vector in $\mathbb{R}^n$ with known covariance matrix, and for an orthonormal basis $B=(b_1,\ldots,b_n)$ let $N_B=\| (\langle X,b_1\rangle,\ldots,\langle X,b_n\rangle)\|_\infty$. Is the Karhunen-Loève basis always optimal for minimizing $\mathbb{E}N_B$ over all orthonormal bases?

背景链接：https://afonsobandeira.wordpress.com/2014/07/02/an-interesting-problem-by-mallat-and-zeitouni/

15227 | AMR-027-0403

### 10 Lectures and 42 Open Problems - Matrix version of 6 deviations suffice

O / open | AMR 问题清单汇编

Prove or disprove: there exists a universal constant $C$ such that, for any choice of $n$ symmetric matrices $H_1,\dots,H_n\in\mathbb{R}^{n\times n}$ satisfying $\|H_k\|\leq 1$ (for all $k=1,\dots,n$ ), there exists $\varepsilon_1,\dots,\varepsilon_n \in \{ \pm1\}$ such that $\left\| \sum_{k=1}^n \varepsilon_k H_k \right\| \leq C \sqrt{n}.$

背景链接：https://afonsobandeira.wordpress.com/2015/10/25/10l42concentration/

15228 | AMR-027-0404

### 10 Lectures and 42 Open Problems - OSNAP

P / partially_solved | AMR 问题清单汇编

Part (3) of the problem: Let $s\leq d\leq m$ and $z_1,\dots,z_m\in \mathbb{R}^d$ i.i.d. random vectors with i.i.d. entries $\left( z_k\right)_j = \left\{ \begin{array}{rcc} -\frac1{\sqrt{s}} & \text{ with probability } & \frac{s}{2m} \\ 0 & \text{ with probability } & 1-\frac{s}{m} \\ \frac1{\sqrt{s}} & \text{ with probability } & \frac{s}{2m} \end{array} \right.$ Note that $\mathbb{E} z_kz_k^T = \frac1m I_{d\times d}.$ The conjecture is that, there exists $c_1$ and $c_2$ positive universal constants such that $\mathrm{Prob}\left\{ \left\| \sum_{k=1}^m \left[z_k z_k^T - \mathbb{E} z_k z_k^T\right] \right\| \geq \varepsilon \right\} < \delta,$ for $m\geq c_1 \frac{d+\log\left( \frac1{\delta} \right)}{\varepsilon^2}$ and $s\geq c_2 \frac{\log\left( \frac{d}{\delta} \right)}{\varepsilon^2}$.

背景链接：https://afonsobandeira.wordpress.com/2015/10/25/10l42concentration/

15229 | AMR-027-0905

### 10 Lectures and 42 Open Problems - Positive PCA tightness

P / partially_solved | AMR 问题清单汇编

In **Positive Principal Component Analysis (PCA)**, the goal is $\max_{x\succ 0,\|x\|=1}x^TWx$ for a symmetric matrix $W$ (equivalently, computing the top eigenvector with a **nonnegativity/sign constraint**). Its **Semidefinite Programming (SDP)** relaxation is $$ \max_{X\succeq0,\,X_{ii}=1}\langle W,X\rangle. $$ **Is this SDP tight with high probability when $W$ is a (Gaussian) Wigner matrix with a planted low-rank (positive) spike / perturbation** - i.e., does the SDP optimum recover the planted positive spike/top eigenvector for signals above a threshold?

背景链接：https://afonsobandeira.wordpress.com/2015/11/28/10l42psbm/

15230 | AMR-027-0405

### 10 Lectures and 42 Open Problems - Random k-lifts of graphs

P / partially_solved | AMR 问题清单汇编

Give a tight upperbound to $\mathbb{E}\left\| A^{\otimes k} -\mathbb{E} A^{\otimes k} \right\|.$

背景链接：https://afonsobandeira.wordpress.com/2015/10/25/10l42concentration/

15231 | AMR-027-0902

### 10 Lectures and 42 Open Problems - Recovery Threshold for SBM for logarithmic many communities

S / solved | AMR 问题清单汇编

In the symmetric SBM with **$k=k(n)$ growing like $O(\log n)$** communities (known in advance), what is the **exact-recovery threshold** - the signal strength above which one can classify *all* vertices correctly (with high probability)? Determine it both **information-theoretically** and **computationally (poly-time)**.

背景链接：https://afonsobandeira.wordpress.com/2015/11/28/10l42psbm/

15232 | AMR-027-0904

### 10 Lectures and 42 Open Problems - Stability conditions for tightness of k-median LP and k-means SDP

P / partially_solved | AMR 问题清单汇编

Give conditions, based on **stability-type properties** of the data (in the sense that the data is "well-explained" by some $k$ clusters), under which the **$k$-median LP** and the **$k$-means SDP** relaxations are **tight (integral)**. I.e., relate integrality of these relaxations to clustering-stability assumptions (e.g., $(c,\varepsilon)$-stable instances of Bilu-Linial type), rather than to a specific generative model.

背景链接：https://afonsobandeira.wordpress.com/2015/11/28/10l42psbm/

15233 | AMR-027-0903

### 10 Lectures and 42 Open Problems - Tightness of k-median LP

P / partially_solved | AMR 问题清单汇编

The **$k$-median** problem admits a natural **Linear Programming (LP)** relaxation (the facility-location-type LP with variables indicating whether a point is opened as a center / assigned to a center). Is the **$k$-median LP tight** - i.e., does its integrality gap equal $1$ and its optimum recover an integral $k$-median solution - even for point clouds sampled from **generative models that do not have a community (cluster) structure**? (The motivation is the SBM-tightness style of analysis: for data coming from mixture models, e.g. over many well-separated clusters $\le k$, the LP/SDP is often tight w.h.p.; here the question is whether tightness extends to non-cluster generative models.)

背景链接：https://afonsobandeira.wordpress.com/2015/11/28/10l42psbm/

15234 | AMR-108-0046

### 6.2 (Maher) - Random tetrahedron-gluing pseudomanifolds

O / open | AMR 问题清单汇编

Start with $n$ tetrahedra and glue their faces together at random. The vertex links need not be spheres but are essentially random triangulated surfaces; call the resulting space a pseudomanifold. Investigate the properties of these pseudomanifolds.

背景链接：https://arxiv.org/pdf/1512.04620#page=5

15235 | AIM-PROBABILITY-0077

### A causal max-plus certificate for nearest-neighbor ballistic deposition

P / partially_solved | AIM 研讨会问题汇编

Expand the universality of the KPZ equation to:

Growth model such as ballistic deposition.

For a standard nearest-neighbor ballistic-deposition height process, prove one or more levels of KPZ universality: KPZ fluctuation exponents, the appropriate geometry-dependent one-point distribution, spatial-field convergence, or convergence as a space-time process to the KPZ fixed point.

来源字段：http://aimpl.org/kpzuniversality/1/

15236 | AIM-PROBABILITY-0151

## A complement correction and a pointed type-B f/h identity

S / solved | AIM 研讨会问题汇编

With the notation normalized but the words unchanged, it asks for elementary explanations of the following two claims.

1. Among centrally symmetric partial $((k+2)$-angulations of a regular $(2kn+2)$-gon having exactly $i$ antipodal orbits of $k$-admissible chords, the proportion containing a diameter is $i/n$. 2. Among centrally symmetric $k$-divisible noncrossing partitions of a $2kn$-gon having exactly $i$ antipodal orbits of nonzero blocks, the proportion containing a zero block is $i/n$.

It also confirms that the second assertion really says **contain a zero block** and really indexes by $i$ orbits of **nonzero** blocks. The recovered remark says that the first and second claims concern, respectively, the $f$- and $h$-numbers of $\Delta^{(k)}_{B_n}$, and that no connection with $NC^{(k)}_{B_n}$ was then known.

Throughout, $n,k\geq 1$. Binomial coefficients outside their usual range are zero.

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15237 | AIM-PROBABILITY-0160

## A conjugacy-class coalgebra and Hopf completion for Cayley intervals

P / partially_solved | AIM 研讨会问题汇编

The record is Problem 4.4, attributed to D. Armstrong, in the AIM workshop list *Braid groups, clusters and free probability*. Let $G$ be generated by a finite set $T$ that is closed under conjugation, let $\ell=\ell_T$, and order $G$ by $$ u\le_T v\quad\Longleftrightarrow\quad \ell(v)=\ell(u)+\ell(u^{-1}v). $$ For $g\in G$, put $P_g=[1,g]$. The problem observes that these intervals are locally self-dual and that conjugate elements give isomorphic intervals. It asks for the structure of the map from conjugacy classes to the Ehrenborg function and, in particular, whether a Hopf structure is preserved.

There are three source-recovery issues.

1. The corpus starts with the duplicated OCR text `Problem 4.\n4.`; the PDF heading is **Problem 4.4**. 2. The displayed sum in the PDF/corpus has the bounds $1\le g_0\le\cdots\le g_k\le g$, while its monomial contains only the jumps $(g_{i-1}^{-1}g_i)$. Taken literally, this omits both boundary jumps and permits arbitrarily many zero jumps, so it is not the Ehrenborg series and does not have finite coefficients. Stanley's precise definition anchors both endpoints and requires the top to occur exactly once. The mathematically coherent recovered formula is therefore $$ F(P_g)=\sum_{k\ge1}\ \sum_{1=g_0\le g_1\le\cdots\le g_{k-1}<g_k=g} x_1^{\ell(g_0^{-1}g_1)}\cdots x_k^{\ell(g_{k-1}^{-1}g_k)}, \tag{1} $$ with $F(P_1)=1$. Equivalently, and without zero-part conventions, $$ F(P_g)=\sum_{1=h_0<h_1<\cdots<h_r=g} M_{(\ell(h_0^{-1}h_1),\ldots,\ell(h_{r-1}^{-1}h_r))}, \tag{2} $$ where $M_\alpha=\sum_{i_1<\cdots<i_r}x_{i_1}^{\alpha_1}\cdots x_{i_r}^{\alpha_r}$. Formula (1) is Stanley's equation (1), specialized to $P_g$; formula (2) is Ehrenborg's strict-chain definition. 3. Everything beginning with the heading `5. Free Probability` is spill from the next section and is not part of Problem 4.4. In particular, reference [47] belongs to that spill. Reference [49] is Stanley's paper cited below.

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15238 | AIM-PROBABILITY-0063

### A crease-balance law for a hat-shaped external field

P / partially_solved | AIM 研讨会问题汇编

`k_2\, \textrm{Re}\, z, & \textrm{Re}\, z0`

Its formula ends with the literal text

immediately followed by the source's closing dollar-sign delimiter,

and has neither a readable inequality in the second branch nor a closing `cases` environment. The linked AIM Problem Lists page returned a 502 error on 2026-08-11, so the missing source text could not be checked directly. The official workshop report does confirm that a group studied a “hat like external field.”

The natural reconstruction analyzed in this report is explicitly labeled as such:

\[ \phi(x+iy)= \begin{cases} -k_1x,&x>0,\\ k_2x,&x<0, \end{cases} \qquad k_1,k_2>0, \tag{1} \]

with \(\phi(iy)=0\) by continuity. Thus \(\phi=-k_1x_+-k_2(-x)_+\), a ridge whose value decreases away from the imaginary axis. The missing condition may have been \(x<0\) or \(x\le0\); these give the same continuous field after defining the value on the axis.

The problem asks whether, for small \(\varepsilon_1,\varepsilon_2>0\), there is an \(S\)-curve joining

\[ z_1=-\varepsilon_1+i, \qquad z_2=\varepsilon_2-i, \tag{2} \]

and asks about the corresponding max--min problem and whether its curve contains a portion of the imaginary axis.

来源字段：http://aimpl.org/vectorequilib/4/

15239 | AIM-PROBABILITY-0084

### A curvature-localization bridge to the GUE Tracy--Widom law

P / partially_solved | AIM 研讨会问题汇编

Provide a variational explanation for the actual form of the GUE Tracy-Widom distribution in terms of the properties of the fixed point. Reduce the exact solvability to purely probabilistic terms.

来源字段：http://aimpl.org/kpzuniversality/1/

15240 | AIM-PROBABILITY-0020

### A definition-aware comparison of classical, complex-amplitude, and quantum Gibbs RBMs

P / partially_solved | AIM 研讨会问题汇编

What are the differences between classical RBMs, Quantum RBMs, and Complex valued RBMs with quantum states?

来源字段：http://aimpl.org/boltzmann/7/

15241 | AIM-PROBABILITY-0029

### A dependence-robust quantitative Anscombe-Renyi bound

P / partially_solved | AIM 研讨会问题汇编

Let \(X_1,X_2,\ldots\) be i.i.d. random variables with mean 0 and variance 1, and let \(S_n=\sum_{i=1}^nX_i\).

For \(b>0\), let \(T_b\) be a sequence of random times. The following theorem is due to Renyi (1961): if \(T_b/n_b\overset p\to1\), where deterministic \(n_b\to\infty\), then \[ \frac{S_{T_b}}{\sqrt{n_b}}\to N(0,1). \] Can we get a bound on \(d(S_{T_b}/n_b,Z)\)?

Some remarks: if \(T_b\) is independent of \(X_1,X_2,\ldots\), then Doebler has some results. Also, see “Randomly Stopped Sums” by Allan Gut.

Bound \(d_K(S_T/\sqrt n,Z)\) using the fixed-sample normal approximation error and a quantitative concentration profile of \(T/n\) near \(1\), allowing arbitrary dependence between \(T\) and all summands.

来源字段：http://aimpl.org/steinhd/1/

15242 | AIM-PROBABILITY-0101

### A diagonal-step log-gamma polymer: Burke obstruction and exact strip moments

P / partially_solved | AIM 研讨会问题汇编

**Exact solvability.** Study solvable variants of the log-Gamma polymer with different types of polymer paths.

来源字段：http://aimpl.org/kpzuniversality/2/

15243 | AIM-PROBABILITY-0143

### A dihedral symmetry barrier for noncrystallographic nonnesting

P / partially_solved | AIM 研讨会问题汇编

Determine how far the nonnesting partitions $NN_W$, the positive-root order, and their enumerative relatives extend from Weyl groups to noncrystallographic finite Coxeter groups. Construct such objects where possible, identify essential barriers where not, and explain what the noncrystallographic $H$-triangle counts.

This is not a standalone Problem 6.5.

In unambiguous paraphrase, the recovered target asks:

That is the part of the larger Problem 1.2 addressed here. “Problem 6.5” in the PDF is instead Zelevinsky's tame/wild classification problem for infinite-type cluster algebras and is not the problem assigned to this attempt.

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15244 | AIM-PROBABILITY-0086

### A direct full-Wick spatial-mollifier construction

P / partially_solved | AIM 研讨会问题汇编

**Stochastic analysis.** Consider the multi-layer extension of the stochastic heat equation introduced by O'Connell and Warren. It is defined via chaos series -- show that it can also be defined by spatially smoothing white-noise and then making sense of the Wick exponential in its formulation as a partition function for non-intersecting Brownian bridges in a space time white-noise environment.

来源字段：http://aimpl.org/kpzuniversality/2/

15245 | AIM-PROBABILITY-0003

### A finite RBM separation from equal-parameter naive Bayes, with an SBN caveat

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** What kind of distributions can be represented by an RBM as opposed to directed models?

Its `problem` field is

The two sentences are identical.

The AIM workshop page does verify the September 2018 workshop and links to that open-problem list.

The phrase “directed models” is mathematically under-specified. A directed acyclic graphical model can have many architectures and conditional families. I analyze two particularly natural one-hidden-layer readings:

1. the binary naïve Bayes model, equivalently a mixture of product distributions; and 2. a one-hidden-layer sigmoid belief network (SBN) with independent binary hidden roots and conditionally independent visible children.

The answers differ. An RBM can require exponentially many naïve Bayes mixture components, but an explicit witness below is also represented by a same-size SBN. Thus “undirected versus directed” alone does not determine a representational comparison.

来源字段：http://aimpl.org/boltzmann/1/

15246 | AIM-PROBABILITY-0087

### A finite-window Burke identity and quenched obstruction for a marked Brownian--Poisson polymer

P / partially_solved | AIM 研讨会问题汇编

**Exact solvability**

Motivated by success in solvable positive temperature polymers, try to study the polynuclear growth model with an underlying Brownian path measure -- is there a Burkes theorem in this setting? Specifically, study the Poissonian last passage model but at positive temperature and with the Poisson points having random weights.

来源字段：http://aimpl.org/kpzuniversality/2/

15247 | AIM-PROBABILITY-0054

### A four-state sign-valued obstruction to fixed-tolerance function mixing bounds

S / solved | AIM 研讨会问题汇编

Let $(X_i)$ be a stationary Markov chain on a finite state space $\Omega$, let $f:\Omega\to[-1,1]$, and put $$ S_N=\frac1N\sum_{i=1}^N f(X_i). $$ Is there an absolute constant $c>0$ such that $$ \Pr\bigl(|S_N-\mathbb E S_N|\geq\varepsilon\bigr) \leq 2\exp\!\left(-\frac{cN\varepsilon^2}{t_f(\delta)}\right) \tag{1} $$ for every $\varepsilon>0$ and $N\geq1$? In particular, can one take the fixed tolerance $\delta=1/4$?

If the braces are read literally as asking for the supremum of all satisfactory times, an ergodic chain normally gives $+\infty$, making (1) vacuous.

The primary source of the terminology, Rabinovich--Ramdas--Jordan--Wainwright, defines instead $$ T_f(\delta):=\min\left\{t\in\{1,2,\ldots\}: \max_{x\in\Omega}\left|P^t f(x)-\pi f\right|\leq\delta\right\}. \tag{2} $$ Here $P^t f(x)=\mathbb E_x f(X_t)$ and $\pi f=\mathbb E_\pi f$. This is the definition used below. The counterexample also works for the stronger persistence version $$ T_f^{\rm pers}(\delta):=\min\left\{t\geq1: \sup_{m\geq t}\max_x|P^m f(x)-\pi f|\leq\delta\right\}, \tag{3} $$ so the conclusion does not depend on the possible distinction between one-time and thereafter mixing.

来源字段：http://aimpl.org/markovmixing/6/

15248 | AIM-PROBABILITY-0108

## A free-Jacobian covariance obstruction to drift-only changes of variables

P / partially_solved | AIM 研讨会问题汇编

The source PDF also drops a visible \(dt\) after each drift and prints a single \(k_t\) where a vector drift is apparently intended. The following separates the questions and records the minimal reconstruction used below.

1. **Process change of variables.** Given self-adjoint variables \(x_1,\ldots,x_m\) that arise at time one from a free stochastic equation \[ dA_i(t)=dS_i(t)+k_{i,t}(A(s):s\leq t)\,dt, \qquad \mu_{A(1)}=\mu_x, \] and variables \(y_j=f_j(x_1,\ldots,x_m)\), is there an operation \(P\) on the drift such that \[ dB_j(t)=dS_j(t)+P(k)_{j,t}(B(s):s\leq t)\,dt, \qquad \mu_{B(1)}=\mu_y? \] Here the inserted \(dt\) and component indices are reconstructions. For a noncommuting tuple, “functional calculus” is not automatic for arbitrary scalar functions, so the rigorous result below takes the \(f_j\) to be self-adjoint noncommutative polynomials. Analytic or sufficiently regular functional calculi would require a separately specified domain.

2. **Microstates free entropy.** Can the \(\limsup\) in the microstates definition of \(\chi\) be replaced by \(\liminf\)? Equivalently, do normalized logarithmic microstate volumes have enough asymptotic regularity that subsequences do not affect the resulting entropy?

3. **Free pressure.** For Hiai’s free pressure \(\pi_R(h)\) and its Legendre-transform entropy \(\eta_R(\tau)\), for which self-adjoint potentials \(h\in *_{i=1}^n C([-R,R])\) is the equilibrium tracial state, satisfying \[ \eta_R(\tau)=\tau(h)+\pi_R(h), \] unique? The source suggests free transportation inequalities as one route.

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15249 | AIM-PROBABILITY-0011

## A full-rank tropical RBM certificate for four visible units

P / partially_solved | AIM 研讨会问题汇编

Is it true that tropical RBMs always have expected dimension?

Is it true that tropical RBMs always have expected dimension, meaning equal to the number of parameters within the model or the dimension of the ambient probability simplex?

来源字段：http://aimpl.org/boltzmann/4/

15250 | AIM-PROBABILITY-0018

## A gated local neural-sampling extension of Boltzmann learning

P / partially_solved | AIM 研讨会问题汇编

Are there biologically plausible extensions of Boltzmann Machines for neural networks?

The wording is deliberately broad. “Biologically plausible” is not a mathematical predicate until one specifies which biological constraints are required, and “extension” could refer to neuron dynamics, learning, architecture, or all three. This report therefore gives a conditional existence answer under explicit criteria, not a claim that a particular mechanism is used by an actual brain.

来源字段：http://aimpl.org/boltzmann/7/

15251 | AIM-PROBABILITY-0079

## A Gaussian-mode audit for weakly nonlinear stochastic Hamilton-Jacobi equations

P / partially_solved | AIM 研讨会问题汇编

Stochastic Hamilton-Jacobi equations $\partial_th=F(\nabla h)+\Delta h+\xi_\epsilon$ with $F$ scaled appropriately.

The exact AIM record is:

This is a sparse research prompt, not a mathematically complete assertion. In particular, the record itself does not specify:

- the spatial dimension or domain; - whether \(\xi_\varepsilon\) is spatially or space-time mollified; - the amplitude and argument scaling of \(F\); - the deterministic height counterterm and possible moving frame; - regularity, symmetry, or growth assumptions on \(F\); or - the topology and solution concept for a limit.

I reconstruct one precise version that is directly covered by the one-dimensional weak-universality literature. On \(\mathbb R\), let \(\xi\) be space-time white noise, choose an even smooth mollifier \(\rho\) with \(\int\rho=1\), set \[ \rho_\varepsilon(x)=\varepsilon^{-1}\rho(x/\varepsilon), \qquad \xi_\varepsilon=\rho_\varepsilon*_{x}\xi, \] and consider \[ \partial_t h_\varepsilon =\partial_x^2h_\varepsilon +\varepsilon^{-1}F\!\left(\sqrt\varepsilon,\partial_xh_\varepsilon\right) +\xi_\varepsilon-C_\varepsilon. \tag{1.1} \] The intended limiting equation, when the effective quadratic coefficient is nonzero, is the Hopf-Cole-renormalized \(1+1\)-dimensional KPZ equation \[ \partial_t h=\partial_x^2h+\lambda(\partial_xh)^2+\xi. \tag{1.2} \] The reconstruction (1.1) is a choice, not a claim that all other readings of the AIM sentence are equivalent. The torus, space-time-mollified noise, discrete smoothing, non-Gaussian noise, fixed-strength nonlinearity, and dimensions \(d\geq2\) require separate statements.

来源字段：http://aimpl.org/kpzuniversality/1/

15252 | AIM-PROBABILITY-0082

### A localization-and-stability route to KPZ fixed-point attraction

P / partially_solved | AIM 研讨会问题汇编

Prove that the fixed point is attractive (i.e., universal in some class of models).

This is problem 11.8 in the section “Big picture questions” from the AIM workshop *The Kardar-Parisi-Zhang equation and universality class*. The record is sparse: it does not name the fixed point, a scaling, a model class, an initial-data space, or a topology.

A model-dependent height function \(H\) should first be centered, tilted, and put under \(1{:}2{:}3\) scaling. Schematically, with nonuniversal constants suppressed,

\[ h_N(t,x)=N^{-1/3}\bigl(H(Nt,vNt+N^{2/3}x)-cNt-\ell N^{2/3}x\bigr). \]

The constants and even the sign convention depend on the model. A precise attraction statement must specify them.

来源字段：http://aimpl.org/kpzuniversality/1/

15253 | AMR-096-0027

### A mathematically natural SIRSN

P / partially_solved | AMR 问题清单汇编

Construct a scale-invariant random spatial network whose law is mathematically natural, for example with an explicit formula for the distribution of $\operatorname{span}(z_1,\ldots,z_k)$.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15254 | AIM-PROBABILITY-0130

### A moment obstruction to the stated freeness and a quantitative diagonal-recovery test

P / partially_solved | AIM 研讨会问题汇编

**Q.** Let \(B_c\) be a band limited operator obtained from \(c\), a circular element, and let \(D\) be the band limited operator obtained from the identity. Then \(D\) is uniformly distributed on \([0,1]\) and \(\star\)-free from \(\{B_c,B_c^*\}\). Is \(D\in W^*(B_c)\)? Or is \(W^*(B_c)=L(\mathbb F_t)\) with \[ t=1+2c\left(1-\frac c2\right)? \]

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15255 | AIM-PROBABILITY-0133

### A moment-free R-transform for an unbounded R-diagonal pair

P / partially_solved | AIM 研讨会问题汇编

**Q:** Can we make sense of the R-transform for \(x_1,x_2\) unbounded (power-series are insufficient to encode all the information)? Easier question is for infinitely divisible unbounded operators.

**Q:** If \(c\) is unbounded R-diagonal, what is the R-transform of \(c,c^*\)?

**0.11 Focus Group on Dirichlet Forms, from Classical to Quantum (day 5)**

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15256 | AIM-PROBABILITY-0037

### A nonreversible stationary-valley reduction for noisy majority

P / partially_solved | AIM 研讨会问题汇编

**Noisy majority model.** The noisy majority model with parameter $\epsilon\in(0,1)$ is a spin system on a graph $G$ that is the stationary distribution of the following Markov chain on $\Omega=\{\pm1\}^{V(G)}$: assign each vertex a rate-$1$ Poisson clock. When the clock at a site rings, the spin at that vertex randomizes according to a $\operatorname{Ber}(1/2)$ with probability $\epsilon$ and chooses the majority of the spins of its neighbors and itself with probability $1-\epsilon$ (flipping a coin in the event of a tie).

Consider the noisy majority model on $\mathbb Z_n^d$ for $d>1$. Show there exists a fixed $\epsilon>0$ such that $t_{\mathrm{mix}}\gtrsim\exp(n^\epsilon)$ (should be true for $\exp(cn)$).

来源字段：http://aimpl.org/markovmixing/1/

15257 | AIM-PROBABILITY-0033

### A one-dimensional counterexample and a bottleneck reduction for spin-glass mixing

P / partially_solved | AIM 研讨会问题汇编

\[ H_n(\sigma)=\sum_{(i,j)\in E_n}J_{ij}\sigma_i\sigma_j, \qquad \mu_{n,J}(\sigma)=Z_{n,J}^{-1}e^{\beta H_n(\sigma)}, \]

where the couplings are described as a “random symmetric matrix with i.i.d. $\pm1$ entries.” After fixing the disorder, it asks whether there are $\beta_0$ and $\varepsilon>0$ such that, for every $\beta\ge\beta_0$, Glauber dynamics has mixing time at least $\exp(n^\varepsilon)$.

The wording omits several quantifiers and conventions.

The exact repository record was therefore preserved, and all repairs above are labeled as reconstructions.

来源字段：http://aimpl.org/markovmixing/1/

15258 | AIM-PROBABILITY-0110

### A one-sided diffuse cyclic amalgam criterion and an entropy-regularity obstruction

P / partially_solved | AIM 研讨会问题汇编

Inspection of pages 1-2 of the original PDF recovers the questions as follows.

1. **Diffuse amalgamation.** Given free group factors $A,B$ with a common diffuse subalgebra $D\subset A,B$, what conditions on $A,B,D$ guarantee that \[ A *_D B \] is a free group factor?

2. **Weak rigidity and dimension.** For a regular weakly rigid (in the sense of Popa) subalgebra of a von Neumann algebra, is the free entropy dimension at most \(1\)?

3. **Betti number versus microstates dimension.** If $\gamma _1,\ldots,\gamma _n\in\Gamma$ generate a group whose first $L^2$-Betti number $\beta _1^{(2)}(\Gamma)$ is large, must the microstates free entropy dimension of the corresponding group generators be large? The notes say that the nonmicrostates version was known through work of Mineyev-Shlyakhtenko.

The symbols printed in the PDF are $A *_D B$ and $\beta _1^{(2)}(\Gamma)$. Thus this record really does contain three questions rather than one question with two numbered subparts.

For the third question this report uses $\delta _0$ for Voiculescu's modified microstates free entropy dimension. For group unitaries this means the dimension of the self-adjoint real-and-imaginary-parts tuple. The symbols $\delta ^*$ and $\delta ^\star$ denote the two nonmicrostates variants; they are not silently identified with $\delta _0$.

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15259 | AIM-PROBABILITY-0135

### A polarization--Fourier criterion for regular GNS cocycle representations

P / partially_solved | AIM 研讨会问题汇编

More generally, given a negative definite function on a group \(\Gamma\) (i.e. a Dirichlet form), we know it gives a representation by affine actions on \(L^2\Gamma\). When is it a multiple of the left regular representation? What conditions on the negative definite function guarantee this?

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15260 | AIM-PROBABILITY-0043

### A portal-gap reduction for a biased small-world walk

P / partially_solved | AIM 研讨会问题汇编

**Cycle + Erdos-Renyi.** Consider the graph \(G=\mathbb Z_n\cup ER(n,p)\) and make the simple random walk on \(G\) nonreversible by adding a counterclockwise drift to the cycle part.

Prove that for \(p=\epsilon/n\), \(\epsilon>0\) fixed, the nonreversible chain has \(t_{\mathrm{mix}}\asymp\log n\).

Prove that when \(p=\epsilon n^{-3/2}\) the nonreversible chain has \(t_{\mathrm{mix}}=\widetilde O(\sqrt n)\).

来源字段：http://aimpl.org/markovmixing/3/

15261 | AIM-PROBABILITY-0071

### A precise high-dimensional Edwards-Wilkinson reduction

P / partially_solved | AIM 研讨会问题汇编

Prove that above 2 spatial dimensions, the KPZ equation is trivial (i.e., discretizations limit to the linearized equation, perhaps with a larger variance in the noise).

来源字段：http://aimpl.org/kpzuniversality/1/

15262 | AIM-PROBABILITY-0159

### A product criterion and a bowtie obstruction for Cayley interval lattices

P / partially_solved | AIM 研讨会问题汇编

Given a group \(G\) generated by a finite conjugacy-closed set \(T\), intervals in the Cayley graph give quasi-Garside structures. Find natural conditions on \((G,T)\) implying that these intervals are lattices, or a natural class in which the presence or absence of the lattice property has a group-theoretic explanation.

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15263 | AIM-PROBABILITY-0022

### A quantitative arcsine bound for permutation-walk occupation time

P / partially_solved | AIM 研讨会问题汇编

It is visibly truncated and is not a syntactically complete mathematical statement.

The lost AIM text was not recovered literally. The intended formulation can nevertheless be identified, up to the harmless choice between a permutation of size \(n\) with \(n-1\) increments and one of size \(n+1\) with \(n\) increments, from the primary paper that arose from this workshop. Fang et al. define a uniform permutation

\[ \pi=(\pi_1,\ldots,\pi_{n+1})\in S_{n+1},\qquad X_k=\begin{cases} +1,&\pi_k<\pi_{k+1},\\ -1,&\pi_k>\pi_{k+1}, \end{cases} \]

and \(S_0=0,\ S_k=\sum_{i=1}^kX_i\). They define the strict-positive occupation count

\[ \Gamma_n=\sum_{k=1}^n \mathbf 1\{S_k>0\}. \]

Their uniform-permutation corollary proves

\[ \frac{\Gamma_n}{n}\ \Rightarrow\ Z, \qquad Z\sim\operatorname{Beta}(1/2,1/2), \]

and the paper then asks for error bounds. This verifies the reconstruction used here; it does not show that the missing AIM characters had exactly this typography.

来源字段：http://aimpl.org/steinhd/1/

15264 | AIM-PROBABILITY-0098

### A quantitative near-equilibrium threshold for KPZ-scale fluctuations

P / partially_solved | AIM 研讨会问题汇编

Universality of initial data for (T)ASEP.

Show \(T^{1/3}\) fluctuations for KPZ equation slightly out of equilibrium by using a variant of the second class particle method for WASEP near equilibrium.

来源字段：http://aimpl.org/kpzuniversality/2/

15265 | AIM-PROBABILITY-0170

### A rank-two obstruction and denominator-growth separator for tame versus wild

P / partially_solved | AIM 研讨会问题汇编

**Problem 6.5 (A. Zelevinsky).** Describe a classification of infinite type cluster algebras as tame or wild. This should generalize the notions of tame/wild Artin groups, tame/wild quivers, etc.

**Remark.** Affine types are certainly tame.

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15266 | AIM-PROBABILITY-0088

### A response formula for the full two-replica overlap law

P / partially_solved | AIM 研讨会问题汇编

**Exact solvability**

Study the overlap for replicated paths of a polymer with respect to the quenched Gibbs measure. What is the distribution of the intersection local time for the continuum polymer? Is there replica symmetry or replica breaking in the strong disorder regime or intermediate disorder regime? Try to use the Macdonald processes contour integral formulas to compute more than just the expected overlap.

来源字段：http://aimpl.org/kpzuniversality/2/

15267 | AIM-PROBABILITY-0065

### A reverse compatibility certificate from spectral curves to equilibrium measures

P / partially_solved | AIM 研讨会问题汇编

The AIM record asks whether string equations, spectral curves, Lax pairs, and WKB asymptotics can be connected explicitly to vector equilibrium problems in two-matrix or external-source models, and whether string equations plus asymptotic spectral-curve data can heuristically recover the equilibrium measure.

This is a broad connection problem, not a single conjecture. The precise issue is logical direction: \[ \text{finite-}N\text{ recurrences/string equation} \Longrightarrow \text{Lax system} \Longrightarrow \text{semiclassical spectral curve}, \] whereas reconstructing an equilibrium measure from the curve requires additional real and variational data.

来源字段：http://aimpl.org/vectorequilib/5/

15268 | AIM-PROBABILITY-0095

### A scaling dictionary and the remaining arbitrary-window polymer problem

P / partially_solved | AIM 研讨会问题汇编

Polymer universality

Consider intermediate scaling for polymer with $\beta=n^{-1/4}c(n)$ where $c(n)$ is any function which tends to infinity with $n$. Show universality of the $F_{GUE}$ fluctuations with a fluctuation exponent which may depend on $c(n)$.

This requires careful estimation of terms in the discrete chaos series which are further and further out as $n$ increases (reminiscent of Soshnikov's approach to edge universality for Wigner matrices).

来源字段：http://aimpl.org/kpzuniversality/2/

15269 | AIM-PROBABILITY-0121

### A sharp extension ladder for unbounded Haagerup--Schultz projections

P / partially_solved | AIM 研讨会问题汇编

**Question 19 (recovered).** Does the main result of Haagerup and Schultz hold for \(T\in L^p(M)\) (for some or all \(p\))? For \(T\in M^\Delta\)? For \(T\in\widetilde M\)?

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15270 | AIM-PROBABILITY-0070

### A sharp Galerkin GFF cancellation and an ultraviolet obstruction for 2D AKPZ

P / partially_solved | AIM 研讨会问题汇编

**Higher dimensions.** Make rigorous sense of the anisotropic KPZ equation \[ \partial_t h=(\partial_xh)^2-(\partial_yh)^2+\Delta h+\xi \] and show that the Gaussian free field is invariant. Is there a way of getting this equation out of the 2d Schur process dynamics of Borodin--Ferrari? Or perhaps out of the 2d q-Whittaker process dynamics of Borodin--Corwin? Is this related to \(2d\) quantum Toda chain?

来源字段：http://aimpl.org/kpzuniversality/1/

15271 | AIM-PROBABILITY-0126

### A slice-Bochner criterion for joint logarithmic potentials

P / partially_solved | AIM 研讨会问题汇编

If \(M\) is a \(\mathrm{II}_1\) factor, \(T_1,\ldots,T_n\in M\), \([T_i,T_j]=0\), then we have the “Brown Measure” defined as the unique measure on \(\mathbb C^n\) such that \[ (*)\qquad \log\Delta\!\left(1-\sum_i\alpha_iT_i\right) =\int\log\!\left(1-\sum_i\alpha_i\zeta_i\right) d\mu_{T_1,\ldots,T_n}(\zeta_1,\ldots,\zeta_n). \]

**Q:** Which functions on \(\mathbb C^n\) have an integral representation as in \((*)\)?

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15272 | AIM-PROBABILITY-0036

### A two-site Ising counterexample to extremal-pair TV maximization

P / partially_solved | AIM 研讨会问题汇编

We seek diagnostic tests for knowing that a general spin system is mixed on a general graph \(G\).

1. Consider the ferromagnetic Ising model on a general graph \(G\). Which two \(x,y\in\Omega\) maximize \(\lVert P^t(x,\cdot)-P^t(y,\cdot)\rVert_{TV}\)? More generally, is it true that in any monotone reversible chain, the maximal and minimal initial configurations maximize the total variation distance between two chains?

2. Can one use a diagnostic to estimate \(t_{\mathrm{mix}}\) up to \(O(1)\)? Of course, answering question (1) would answer this as well.

3. Consider the \(3\)-Potts model for a general graph \(G\). Can one devise a diagnostic to differentiate between \(t_{\mathrm{mix}}=O(n\log n)\) and \(\exp(\Omega(n))\)?

来源字段：http://aimpl.org/markovmixing/1/

15273 | AIM-PROBABILITY-0031

### A uniform-in-q product-chain cutoff and the Potts lower-bound gap

P / partially_solved | AIM 研讨会问题汇编

**Universal lower bound for Potts.** Consider the \(q\)-Potts model or uniform \(q\) proper colorings on a general graph \(G\). Is there a universal \(\asymp n\log n\) lower bound on the mixing time uniform in the temperature? Is there always an \(n\log n/2\) lower bound?

来源字段：http://aimpl.org/markovmixing/1/

15274 | AMR-096-0028

### A visually realistic SIRSN

P / partially_solved | AMR 问题清单汇编

Construct a scale-invariant random spatial network that is visually realistic, in the sense of not looking very different from a real-world road network.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15275 | AMR-099-0073

### Absence of bigeodesics in first-passage percolation

P / partially_solved | AMR 问题清单汇编

Prove that natural i.i.d. first-passage-percolation models on $\mathbb{Z}^d$, including exponential edge lengths, almost surely contain no two-sided infinite geodesic.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=9

15276 | OWR-14603-012

### Absolute Continuity Parameters for Bernoulli Convolutions

O / open | Oberwolfach 报告问题汇编

For $\lambda\in(0,1)$, let $\mu_\lambda$ be the distribution of $\sum_{n\geq0}A_n\lambda^n$, where the $A_n$ are independent unbiased random variables taking values $\pm1$. Determine the set of parameters $\lambda$ for which $\mu_\lambda$ is absolutely continuous.

来源字段：https://doi.org/10.4171/owr/2016/21

15277 | AIM-PROBABILITY-0153

### Absolute-order topology, endpoint conventions, and an exact rank-two family

P / partially_solved | AIM 研讨会问题汇编

**3. Reflection Groups. Problem 3.1 (V. Reiner).** Let \(W\) be a finite Coxeter group and let \(T\) be its set of all reflections. Let \(\ell\) be word length with respect to \(T\), the absolute length. Define the absolute order by \[ a\le_T b \quad\Longleftrightarrow\quad \ell_T(b)=\ell_T(a)+\ell_T(a^{-1}b). \] What is the topology of this poset? In types \(A\) and \(B\), is there an EL-labeling which exhibits a shelling of the order complex? It is known that the absolute length poset is not shellable in type \(D\). Can this be fixed uniformly by retaining only the order ideal of parabolic Coxeter elements?

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15278 | OWR-2487-003

### Action Dependence of Free Indecomposability

O / open | Oberwolfach 报告问题汇编

Does there exist a countable group with one action whose associated probability-measure-preserving equivalence relation is freely indecomposable (FI) and another action whose associated equivalence relation is not FI?

来源字段：https://doi.org/10.4171/owr/2008/49

15279 | OWR-16774-001

### Adaptive Moment Selection Without Reconstruction Overfitting

O / open | Oberwolfach 报告问题汇编

How should the number $R$ of moments be selected in practical reconstruction algorithms so that overestimation does not cause overfitting or inaccurate reconstructions?

来源字段：https://doi.org/10.4171/owr/2019/12

15280 | AIM-PROBABILITY-0044

### Additive reversibilization, nonnormality, and a sharp normal-chain comparison

P / partially_solved | AIM 研讨会问题汇编

**Nonreversible vs. reversible chains.** Let \(P\) be the transition matrix of a general nonreversible chain with, say, bounded degree vertices and transition probabilities uniformly bounded from below. Denote its mixing time by \(t_{\mbox{mix}}^P\) and let \(\widetilde t_{\mbox{mix}}\) be the mixing time of its symmetrization. Is it always true that \(t_{\mbox{mix}}^P\lesssim \widetilde t_{\mbox{mix}}\)? Is the same also true of the cover time?

来源字段：http://aimpl.org/markovmixing/3/

15281 | OWR-15992-001

### Adjusted Profile Confidence Intervals for Large Monte Carlo Models

P / partially_solved | Oberwolfach 报告问题汇编

Do Monte Carlo adjusted profile confidence intervals, built from likelihood estimates obtained by iterated filtering and sequential Monte Carlo, have asymptotically valid coverage in large models such as PanelPOMPs and GenPOMPs?

来源字段：https://doi.org/10.4171/owr/2018/7

15282 | OWR-17138-003

### Airy Limits for Last Paths in Aztec Diamonds

P / partially_solved | Oberwolfach 报告问题汇编

For the lattice paths obtained from the height function of the two-periodic Aztec diamond, prove that an extremal path exists whose suitably centered and scaled limit is the Airy process, and determine the parameter regimes in which this limit holds.

来源字段：https://doi.org/10.4171/owr/2019/42

15283 | AMR-096-0038

### Aldous-Lyons soficity conjecture

P / partially_solved | AMR 问题清单汇编

Does every unimodular random countable locally finite rooted graph arise as a local weak limit of finite graphs?

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15284 | AIM-PROBABILITY-0006

### Algebraic score equations and an independence saddle for RBM learning

P / partially_solved | AIM 研讨会问题汇编

What are the algebraic structures relevant for studying learning algorithms for RBMs?

What are the algebraic structures relevant for studying learning algorithms for RBMs?

来源字段：http://aimpl.org/boltzmann/2/

15285 | OWR-14299518-014

### Algorithmic and Structural Problems in Planted-Tree Detection

O / open | Oberwolfach 报告问题汇编

Determine the detection threshold for planted $d$-ary trees. When $pn<1.1$, can an unknown random tree of size $O(\log n)$ be detected? Can the hairy-path test be implemented in polynomial time?

来源字段：https://doi.org/10.4171/owr/2025/42

15286 | AIM-PROBABILITY-0132

### An analytic block R-transform for unbounded pairs

P / partially_solved | AIM 研讨会问题汇编

**Q:** Can we make sense of the (R)-transform for (x_1,x_2) unbounded (power-series are insufficient to encode all the information)? Easier question is for infinitely divisible unbounded operators.

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15287 | AIM-PROBABILITY-0075

### An exact cumulative-color ASEP array and a forced block-move obstruction

P / partially_solved | AIM 研讨会问题汇编

Associated to q-TASEP there is a triangular array (the q-Whittaker process). This whole array is helpful in computing things about q-TASEP.

Is there such an array for ASEP or for some sort of transformed version of ASEP?

来源字段：http://aimpl.org/kpzuniversality/1/

15288 | AIM-PROBABILITY-0013

### An exact Morse-Bott landscape for leaf-hidden RBMs

P / partially_solved | AIM 研讨会问题汇编

**Characterizing the optimization landscape for RBMs**

How do we characterize local optima on the optimization landscape for RBMs as a function of the architecture and number of empirical samples?

来源字段：http://aimpl.org/boltzmann/5/

15289 | AIM-PROBABILITY-0161

### An exact orthogonal-spike model for infinitesimal type-B cumulants

P / partially_solved | AIM 研讨会问题汇编

**Problem 5.1 (F. Goodman, P. Sniady).** Philippe Biane, Fred Goodman, and Alexandru Nica have defined a type B analogue of free probability [10]. The definition has been motivated by the combinatorics, and there is currently no model of this theory (as the large random matrices are a model for type A free probability). Find a natural model for type B free probability, which motivates the combinatorics. Is there a corresponding notion of free probability in other types?

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15290 | AIM-PROBABILITY-0139

### An exact-flat Boolean model for Coxeter-Catalan numerology

P / partially_solved | AIM 研讨会问题汇编

Explain why the cardinalities of \(NC_W\), \(NN_W\), and the set of facets of \(\Delta_W\) are all \(\operatorname{Cat}(W)\), and why their rank, antichain-cardinality (“height”), and \(h\)-vector distributions are the same Narayana distribution. Find bijections preserving this numerology and an algebraic framework behind it. Explain Chapoton’s \(M\)-, \(F\)-, and \(H\)-triangle identities, give a theoretical proof of the Athanasiadis-Reiner refinement by \(W\)-orbits of intersection flats, relate it to a statistic on \(\Delta_W\), formalize Bessis’s dual Coxeter idea, and interpret \(\operatorname{Cat}(W)\) invariant-theoretically.

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15291 | AMR-096-0002

### Analytic toy model for a percolation-fragmentation congestion transition

O / open | AMR 问题清单汇编

Find a simple network-and-demand toy model in which the marginal satisfiability proportion $r(t)$ can be calculated analytically and exhibits the proposed percolation-fragmentation phase transition under optimization or adaptive capacity growth.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15292 | OWR-2493-004

### Analytical Regimes of Martensitic-Interface Depinning

P / partially_solved | Oberwolfach 报告问题汇编

For the shallow-slope martensitic-interface depinning model, can the different pinning regimes and their associated velocity exponents, produced by the random nature of obstacles, be characterized analytically?

来源字段：https://doi.org/10.4171/owr/2008/55

15293 | AIM-PROBABILITY-0127

## Approximating and pointwise local-growth spaces can differ

S / solved | AIM 研讨会问题汇编

- the parameter \(r\) does not occur on the right-hand side of either displayed definition; - a scalar written as a limsup is followed by a further arrow to \(0\); - the second display uses \(\xi_n\), although no sequence is quantified there.

The primary source that grew out of this circle of questions, Haagerup--Schultz (2009), Definition 3.1(i), uses the coherent approximation condition \[ \xi_n\longrightarrow\xi, \qquad \limsup_{n\to\infty}\|T^n\xi_n\|^{1/n}\le r. \tag{1} \] The natural comparison requested by the workshop question removes the exponent-dependent approximation and uses \(\xi\) itself. To prevent collision with later literature (where the approximating space is denoted \(E(T,r)\)), this report uses descriptive subscripts: \[ \begin{split} \mathcal K_{\mathrm{app}}(T,r) &=\left\{\xi\in H:\exists\xi_n\to\xi, \ \limsup_{n\to\infty}\|T^n\xi_n\|^{1/n}\le r\right\},\\ \mathcal E_{\mathrm{pt}}(T,r) &=\left\{\xi\in H: \limsup_{n\to\infty}\|T^n\xi\|^{1/n}\le r\right\}. \tag{2} \end{split} \] Here \(T\in M\), where \(M\) is a \(\mathrm{II}_1\) factor acting on \(H\), and \(r\ge0\). The recovered question is whether the two spaces in (2) are always equal.

This reconstruction is explicit rather than silent. The literal printed formula has no well-defined pointwise second set and makes \(r\) irrelevant, so it cannot be the intended mathematical question.

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15294 | AMR-094-0007

## Are shy couplings necessarily rigid?

P / partially_solved | AMR 问题清单汇编

Let $D\subset\mathbb{R}^d$, $d\ge2$, be bounded, connected, and open. Suppose there are coupled reflected Brownian motions $X_t,Y_t$ in $D$ and $\varepsilon>0$ such that $\inf_{t\ge0}|X_t-Y_t|\ge\varepsilon$ with positive probability. Must there also exist coupled reflected Brownian motions $X'_t,Y'_t$, a positive $\varepsilon$, and a deterministic function $f$ such that $f(X'_t)=Y'_t$ for every $t\ge0$ almost surely and $\inf_{t\ge0}|X'_t-Y'_t|\ge\varepsilon$ with positive probability?

背景链接：https://sites.math.washington.edu/~burdzy/open_mathjax.php

15295 | OWR-4529-004

## Arrangement Complexity and Random-Complex Homology Thresholds

O / open | Oberwolfach 报告问题汇编

For a complex hyperplane arrangement $\mathcal A$, determine $\operatorname{TC}(M(\mathcal A))$ and test Yuzvinsky's conjecture

$$\operatorname{TC}(M(\mathcal A))=\operatorname{zcl}_{A^*(\mathcal A)}(A^*(\mathcal A))+1.$$

For the Linial-Meshulam random complex $Y(n,n^\alpha)$, determine the threshold for simultaneous vanishing of $H_1(Y,\mathbb Z_p)$ for all primes $p$ and the threshold for having no epimorphism $\pi_1(Y)\twoheadrightarrow\Gamma$ for any finite group $\Gamma$; the latter is conjectured at $\alpha=-1/2$.

来源字段：https://doi.org/10.4171/owr/2010/47

15296 | OWR-15428-001

## Asymptotic Distribution of Maximum Agreement Subtrees

P / partially_solved | Oberwolfach 报告问题汇编

What is the asymptotic distribution of $\operatorname{MAST}(T_1,T_2)$ as $|X|\to\infty$ when $T_1$ and $T_2$ are random binary trees on the leaf set $X$, for distributions such as the uniform and Yule-Harding models?

来源字段：https://doi.org/10.4171/owr/2017/20

15297 | OWR-1960-002

## Asymptotics for Nonstationary GARCH Estimators

P / partially_solved | Oberwolfach 报告问题汇编

Consider two nonstationary GARCH specifications: an additive time-varying-parameter model formed by adding a directly time-dependent component to a GARCH variance equation, and a multiplicative model whose conditional variance is the product of stationary and nonstationary components. Determine the asymptotic properties of their maximum-likelihood estimators.

来源字段：https://doi.org/10.4171/owr/2008/15

15298 | OWR-16931-007

## Asymptotics of Online Random Ramsey Numbers

O / open | Oberwolfach 报告问题汇编

Does the diagonal online random Ramsey number satisfy $$\widetilde r_{\mathrm{rand}}(n,n)=2^{(1+o(1))(2/3)n}$$? For every fixed $m\geq3$, does the off-diagonal number satisfy $$\widetilde r_{\mathrm{rand}}(m,n)=n^{(1+o(1))(2/3)m}$$ as $n\to\infty$?

来源字段：https://doi.org/10.4171/owr/2019/19

15299 | OWR-12591-010

### Atomlessness from Operator-Valued Cauchy Transforms

O / open | Oberwolfach 报告问题汇编

For a selfadjoint polynomial in free variables without atoms, can the absence of atoms in its distribution, and similar qualitative results, be derived from the operator-valued Cauchy-transform and linearization description?

来源字段：https://doi.org/10.4171/owr/2013/43

15300 | OWR-15997-002

### Attainable Limits in Gaussian Projection Pursuit

O / open | Oberwolfach 报告问题汇编

In projection pursuit of structureless Gaussian data with dimension-to-sample ratio $\gamma<1$, is every attainable limiting empirical distribution a mixture of $N(0,q)$, for some $q<(1+\sqrt\gamma)^2$, and a possibly small non-Gaussian component?

来源字段：https://doi.org/10.4171/owr/2018/12

15301 | OWR-13863-002

### Attainable Sum Distributions and Joint Mixability

O / open | Oberwolfach 报告问题汇编

For given marginal distributions $F_1,\ldots,F_n$, characterize $D_n(F_1,\ldots,F_n)=\{\operatorname{cdf}(X_1+\cdots+X_n):X_i\sim F_i\}$ under arbitrary dependence, and find an analytical method to decide when $D_n$ contains a point mass (joint mixability).

来源字段：https://doi.org/10.4171/owr/2015/42

15302 | OWR-1543-004

### Attracting Edges in Strongly Reinforced Random Walks

S / solved | Oberwolfach 报告问题汇编

For strongly edge-reinforced random walk with a reciprocally summable weight function, does the walk on graphs containing odd cycles, in particular a triangle, eventually visit one random edge back and forth as it does on bounded-degree graphs without odd cycles?

来源字段：https://doi.org/10.4171/owr/2007/27

15303 | OWR-15989-008

### Avalanche Tails and Height Correlations in Sandpiles

P / partially_solved | Oberwolfach 报告问题汇编

For the Abelian sandpile model on $\mathbb Z^d$ with infinite-volume measure $\nu$, determine the tail of $\nu(|\operatorname{Av}|>t)$ for $d\geq2$. In dimension $2$, prove or disprove $$ \operatorname{Cov}_\nu(\eta(o)=i,\eta(x)=j)\sim a_{i,j}(\log^2|x|)|x|^{-4} $$ as $|x|\to\infty$, for $i,j\in\{1,2,3\}$.

来源字段：https://doi.org/10.4171/owr/2018/4

15304 | OWR-16633-012

### Average Normal-Cone Width and Random Polytope Diameters

O / open | Oberwolfach 报告问题汇编

Can a notion of normal cones being wide on average yield useful polyhedral-diameter bounds, including a bound for random Gaussian polytopes $\{x:Ax\leq\mathbf1\}$ when the entries of $A$ are independent standard normal variables?

来源字段：https://doi.org/10.4171/owr/2018/50

15305 | OWR-15175-006

### Baik-Rains Fluctuations for General Stationary Exclusion Processes

O / open | Oberwolfach 报告问题汇编

Does the stationary exclusion-process current fluctuation limit with $T^{1/3}$ scaling and the Baik-Rains distribution, established for ASEP, continue to hold for general stationary exclusion processes with non-nearest-neighbor jumps?

来源字段：https://doi.org/10.4171/owr/2016/54

15306 | OWR-13677-003

### Baire Category of Purely Singular Copulas

O / open | Oberwolfach 报告问题汇编

Let $\mathcal C_d$ be the space of $d$-dimensional copulas equipped with the metric

$$D_1(C_1,C_2)=\int_{[0,1]^d}\lvert C_1(u)-C_2(u)\rvert\,du.$$

Is the subset of purely singular copulas nonmeager in $(\mathcal C_d,D_1)$?

来源字段：https://doi.org/10.4171/owr/2015/20

15307 | OWR-16413-003

### Basis Selection for Lagrangian Decision-Rule Approximations

P / partially_solved | Oberwolfach 报告问题汇编

For LDR restrictions in Lagrangian duals of the multistage stochastic mixed-integer program (MSMIP), which basis functions should be used in the LDR approximations, and how can the resulting restricted Lagrangian-dual problems be optimized efficiently?

来源字段：https://doi.org/10.4171/owr/2018/38

15308 | OWR-15221-002

### Bayesian Bias Correction in Support-Boundary Models

P / partially_solved | Oberwolfach 报告问题汇编

In the irregular nonparametric support-boundary model, does a Bayesian procedure automatically produce the finite-sample correction to the maximum-likelihood estimator given by the number of support points on the MLE divided by the sample size?

来源字段：https://doi.org/10.4171/owr/2017/16

15309 | OWR-9790365-014

### Belief-Propagation Uniqueness Boundary for Randomly Colored Ising Models

P / partially_solved | Oberwolfach 报告问题汇编

In the randomly colored stochastic block model with parameters $(q,a,b)$, what is the exact boundary separating the parameter values for which belief propagation has a unique fixed point from those for which it does not?

来源字段：https://doi.org/10.4171/owr/2022/14

15310 | OWR-12861-019

### Biclique Partition Numbers of Random Graphs

S / solved | Oberwolfach 报告问题汇编

Let $\tau(G)$ be the minimum number of pairwise edge-disjoint complete bipartite subgraphs partitioning $E(G)$, and let $\beta(G)$ be the largest order of an induced complete bipartite subgraph. For $G\sim G(n,0.5)$, is $\tau(G)=n-\beta(G)+1$ with high probability?

来源字段：https://doi.org/10.4171/owr/2014/01

15311 | AMR-099-0035

### Binary trees in critical clusters of regular planar triangulations

O / open | AMR 问题清单汇编

Let $H_k$ be a $k$-regular planar triangulation. At the critical percolation parameter for the event that an open cluster contains a full infinite binary tree, prove that the event has positive probability.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=58

15312 | OWR-16413-004

### Bit Complexity of Graver-Basis Augmentation

P / partially_solved | Oberwolfach 报告问题汇编

Given an integer program with equality constraints and a Graver basis $\mathcal G(E)$, is Graver-basis augmentation efficient in bit complexity, even with oracle access to $\mathcal G(E)$?

来源字段：https://doi.org/10.4171/owr/2018/38

15313 | OWR-17138-001

### Boltzmann Limits for Long-Range Lorentz Gases

P / partially_solved | Oberwolfach 报告问题汇编

For Lorentz gases and related particle systems with long-range interaction potentials, determine when the kinetic limit is governed by the Boltzmann equation, when it is governed by a different equation such as the Landau equation, and when no Markovian kinetic description is valid.

来源字段：https://doi.org/10.4171/owr/2019/42

15314 | AIM-PROBABILITY-0024

### Boundary corrections and nonexistence after SURE selection

S / solved | AIM 研讨会问题汇编

Let $X\sim MVN(\theta,I_d)$. Estimate $\theta$ by $X+h(X)$, with $h$ in a class $\mathcal H$. For fixed $h$, \[ \widehat r_h=d+2\nabla\cdot h(X)+\|h(X)\|^2 \] is unbiased for its quadratic risk. Let $h^*$ minimize $\widehat r_h$ over $\mathcal H$, and let $\widehat\theta^*=X+h^*(X)$. Find an unbiased estimator $r^{**}$ of the risk of $\widehat\theta^*$.

The exact mathematical request is:

The hidden issue is that $h^*$ depends on the same $X$ used to evaluate SURE. Write $\widehat h_x$ for the candidate selected when the observed value is $x$. The selected adjustment is the composite map

\[ h_{\mathrm{sel}}(x)=\widehat h_x(x), \]

and the final estimator is

\[ \delta^*(x)=x+h_{\mathrm{sel}}(x). \]

Even if each candidate is smooth, $h_{\mathrm{sel}}$ is usually only piecewise smooth and can jump when the SURE minimizer changes. The source does not specify $\mathcal H$, tie-breaking, differentiability, integrability, or whether generalized functions are permitted. These omissions are mathematically decisive.

来源字段：http://aimpl.org/steinhd/1/

15315 | OWR-2093-001

### Boundary Geometry and Maximal Visibility in Random Media

O / open | Oberwolfach 报告问题汇编

In the Boolean model of visibility from the origin, let $T$ be the set of visible points. Can the boundary of $T$ be described more precisely? More generally, what can be said about maximal visibility in other random media, such as bicolored random tessellations?

来源字段：https://doi.org/10.4171/owr/2008/47

15316 | OWR-4532-003

### Boundary Terms in Dynamical Large Deviations

P / partially_solved | Oberwolfach 报告问题汇编

For the stochastic conservation law with vanishing-viscosity perturbation and dynamic large-deviation functional, can boundary conditions be added so that the corresponding boundary terms for dynamic large deviations are obtained?

来源字段：https://doi.org/10.4171/owr/2010/50

15317 | OWR-11695858-003

### Bourgain-Spencer Regularity Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

If a stationary random conductivity field $A$ satisfies sufficiently strong mixing, prove that the symbol $\overline A$ of the averaged operator is almost $C^{2d}$ near the origin.

来源字段：https://doi.org/10.4171/owr/2022/48

15318 | OWR-15428-013

### Branch Length and Root Identifiability from Split Probabilities

P / partially_solved | Oberwolfach 报告问题汇编

For a species tree $\sigma$ under the multispecies coalescent model, are its branch lengths identifiable from split probabilities? Can an improved NJst algorithm estimate the root, and do higher-degree phylogenetic invariants exist?

来源字段：https://doi.org/10.4171/owr/2017/20

15319 | AIM-PROBABILITY-0120

### Brown measure for unbounded affiliated operators

S / solved | AIM 研讨会问题汇编

For \(T\in\mathcal M^\Delta\), can one define a Brown measure \(\mu_T\) on \(\mathbb C\)?

This is Question 18 from the AIM workshop list *Free analysis*. In particular, it prints the defining integral as fragments such as

and uses the same symbol for a measure on the positive half-line and for the desired Brown measure on the complex plane.

Comparison with Definition 2.1 of Haagerup--Schultz, the work explicitly named by the question, fixes the notation. Let \(\mathcal M\) be a finite von Neumann algebra with faithful normal tracial state \(\tau\), let \(\widetilde{\mathcal M}\) be its closed densely defined affiliated operators, and let \(\mu_{|T|}\) be the spectral distribution of \(|T|\):

\[ \mu_{|T|}(B)=\tau(E_{|T|}(B)),\qquad B\subseteq[0,\infty)\text{ Borel}. \]

The intended class and determinant are

\[ \mathcal M^\Delta =\left\{T\in\widetilde{\mathcal M}: \int_0^\infty \log^+t\,d\mu_{|T|}(t)<\infty\right\}, \qquad \Delta(T)=\exp\!\left(\int_0^\infty\log t\,d\mu_{|T|}(t)\right). \]

Here \(\log^+t=\max(\log t,0)\). The integral in the determinant may equal \(-\infty\), so \(\Delta(T)\in[0,\infty)\); it cannot be \(+\infty\). The recovered question is:

Thus \(\mu_{|T|}\) in the hypothesis is a singular-value distribution on \([0,\infty)\), whereas \(\mu_T\) asked for is a Brown measure on \(\mathbb C\).

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15320 | OWR-16164-012

### Brownian First-Visit Cell Lengths on the Circle

O / open | Oberwolfach 报告问题汇编

Choose starting points $p_1,\ldots,p_k$ on the unit circle and run independent Brownian motions from them. If $B_i$ is the set of points first visited by the motion started at $p_i$, determine the joint distribution of $(|B_1|,\ldots,|B_k|)$.

来源字段：https://doi.org/10.4171/owr/2018/23

15321 | OWR-1970-005

### Bulk Large Deviations in Stationary Point Fields

P / partially_solved | Oberwolfach 报告问题汇编

What large-deviation behavior governs bulk fluctuations of stationary random point fields, as opposed to the boundary-effect fluctuations known for the number of points in a large disk?

来源字段：https://doi.org/10.4171/owr/2008/25

15322 | OWR-15218-007

### Canonical Continuous-Time Martingale Transport

P / partially_solved | Oberwolfach 报告问题汇编

Given probability measures $\mu$ and $\nu$ with $\mu\prec_c\nu$, is there a canonical continuous Markov martingale with initial law $\mu$ and terminal law $\nu$? Is there a canonical continuous-time formulation of the martingale transport problem?

来源字段：https://doi.org/10.4171/owr/2017/13

15323 | OWR-17474-010

### Cardinality of Spin-Glass Ground-State Sets

O / open | Oberwolfach 报告问题汇编

For every graph and every distribution of linear growth, does the set $G(J)$ of spin-glass ground states satisfy $|G(J)|\in\{2,\infty\}$? More generally, is there a connection between random walks and spin-glass ground states on graphs beyond trees?

来源字段：https://doi.org/10.4171/owr/2020/11

15324 | AIM-PROBABILITY-0105

### Cauchy smoothing for free-coordinate tuples

P / partially_solved | AIM 研讨会问题汇编

Q: In the one variable case, if A(t) follows a process dA (t) = dS (t) + kt(A(s)) s≤t then replacing A(t) with A(t) + C[U+000F] (with C having Cauchy distribution and free from A(t)) then kt is replaced by k[U+000F]t = τ (kt|A(t) + C[U+000F]). Thus, k[U+000F]t is smooth. Is there an analog of this smoothing in the several-variable case?

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15325 | OWR-16937-001

### Causal Discovery from Partially Observed Event Processes

P / partially_solved | Oberwolfach 报告问题汇编

When only some coordinates of a multivariate event process are observed, what aspects of the causal structure can be learned from the induced local-independence model?

来源字段：https://doi.org/10.4171/owr/2019/25

15326 | OWR-2488-005

### Càdlàg Solutions of Cylindrical Stable Evolution Equations

O / open | Oberwolfach 报告问题汇编

For $dX_t=AX_t\,dt+dZ_t$ on a Hilbert space, driven by a cylindrical stable process with coefficients $\beta_k$ satisfying $\sum_k\beta_k^\alpha/\gamma_k<\infty$, does the $H$-valued solution have càdlàg trajectories under conditions substantially weaker than $\sum_k\beta_k^\alpha<\infty$?

来源字段：https://doi.org/10.4171/owr/2008/50

15327 | AIM-PROBABILITY-0017

### Certifiably interpretable sparse ferromagnetic RBMs

P / partially_solved | AIM 研讨会问题汇编

Can we design Boltzmann machines that are interpretable?

Can we design Boltzmann machines that are interpretable? In other words, can we construct useful Boltzmann machines that admit a simple explanation in human terms, and behaves in a predictable manner with certainty?

来源字段：http://aimpl.org/boltzmann/7/

15328 | AIM-PROBABILITY-0062

### Certified natural-coordinate continuation for critical trajectories

P / partially_solved | AIM 研讨会问题汇编

**Numerical computation of \(S\)-curves.** A general question about numerical methods for vector equilibrium problems and computation of corresponding \(S\)-curves. More in particular, how to produce critical trajectories in a reliable way, given the spectral curve of the problem?

来源字段：http://aimpl.org/vectorequilib/4/

15329 | OWR-15993-006

### Certified Probability Landscapes for Stochastic Reaction Networks

O / open | Oberwolfach 报告问题汇编

For stochastic chemical reaction networks, determine how to certify that a computed probability landscape is accurate, quantify the best attainable accuracy for laptop-scale and large-scale computation, and control errors such as missed probability peaks and rare-event probabilities.

来源字段：https://doi.org/10.4171/owr/2018/8

15330 | AMR-099-0038

### Cheeger constant and the percolation nonuniqueness phase

P / partially_solved | AMR 问题清单汇编

For every infinite vertex-transitive graph $G$, prove that $p_c(G)<p_u(G)$ if and only if $h(G)>0$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=65

15331 | OWR-14298373-009

### Chemical Distance in Critical Percolation

P / partially_solved | Oberwolfach 报告问题汇编

Obtain stronger results for chemical distance in critical percolation and its relation to nonsimple CLE metrics.

来源字段：https://doi.org/10.4171/owr/2024/25

15332 | OWR-4990373-012

### Chemical Distances in Weighted Random Connection Models

O / open | Oberwolfach 报告问题汇编

For weight-dependent random connection models in the regime without ultrasmallness, determine the asymptotic chemical distance between two vertices; in particular, decide whether the known sublogarithmic lower bound is sharp or whether the distance is linear or polylogarithmic in different parameter regimes.

来源字段：https://doi.org/10.4171/owr/2021/4

15333 | AMR-099-0029

### Circle-packing measure of uniform random triangulations

O / open | AMR 问题清单汇编

Let $T_n$ be a uniform triangulation of the sphere with $n$ faces, normalize its circle packing by its conformal barycenter, and let $\mu_{P T_n}$ be the empirical measure of tangency points. Determine the support dimension and singularity of subsequential limits $\mu_\infty$; prove uniqueness in law; identify any relation to the Gaussian free field and KPZ; and determine how $\mu_\infty$ relates to the limiting random metric $d_\infty$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=51

15334 | OWR-11695859-005

### Classifying Globally Lipschitz Transport Settings

O / open | Oberwolfach 报告问题汇编

In which additional settings is the transport map globally Lipschitz?

来源字段：https://doi.org/10.4171/owr/2022/49

15335 | OWR-14299282-001

### Climate Tipping Probabilities under Parameter Uncertainty

O / open | Oberwolfach 报告问题汇编

For a climate model with uncertain parameters, determine how parameter uncertainty changes the bifurcation structure, identify which parameters dominate the tipping dynamics, and compute the probability that the system tips.

来源字段：https://doi.org/10.4171/owr/2025/21

15336 | OWR-16636-004

### Commutator Models for Nonsymmetric Free Random Variables

O / open | Oberwolfach 报告问题汇编

Extend the available commutator representations for free random variables from symmetric laws $\mu_a,\mu_b$ to arbitrary nonsymmetric laws $\mu_a,\mu_b$.

来源字段：https://doi.org/10.4171/owr/2018/53

15337 | AIM-PROBABILITY-0106

### Commuting multivariable Poisson smoothing and cyclic-gradient closure criteria

P / partially_solved | AIM 研讨会问题汇编

It contains two consecutive questions from the 2006 AIM workshop list *Free Analysis*. In the PDF, the first question ends at the bottom of page 1 (PDF index 0), and the second starts on page 2.

Inspection of the original PDF supports the following reconstruction.

**Question A (multivariable Cauchy/Poisson smoothing).** If \(A\) is an \(n\times n\) Hermitian random matrix and \(f:\mathbb R\to\mathbb R\), scalar Cauchy noise of scale \(\varepsilon\) gives \[ \mathbb E_C f(A+C_\varepsilon)= (P_\varepsilon f)(A),\qquad (P_\varepsilon f)(x)=\int_{\mathbb R}f(y) \frac{\varepsilon}{\pi((y-x)^2+\varepsilon^2)}\,dy. \] Can this be done for several variables?

In the preceding PDF question the noise is said to be free from the original variable; the displayed question itself does not repeat that hypothesis. Thus “several variables” is genuinely underspecified: it could mean a commuting tuple with ordinary joint functional calculus, a tuple of noncommuting matrices with free functional calculus, or a matrix-valued analogue of a multivariate Cauchy law.

**Question B (conjugate variables and cyclic gradients).** Given self-adjoint \(X_1,\ldots,X_m\) in a tracial unital von Neumann algebra \((M,\tau)\), do their conjugate variables belong to the \(L^2\)-closure of the cyclic-gradient space? More precisely, with free difference quotients \[ \partial_i X_j=\delta_{ij}1\otimes1, \qquad J_i=\partial_i^*(1\otimes1), \qquad D_i=m_{\rm flip}\circ\partial_i, \quad m_{\rm flip}(a\otimes b)=ba, \] do there exist \(H_k\in\mathbb C\langle\alpha_1,\ldots,\alpha_m\rangle\) such that \[ (D_1H_k(X),\ldots,D_mH_k(X))\longrightarrow (J_1,\ldots,J_m) \quad\text{in }L^2(M)^m? \] We use the standard convention \(M=W^*(X_1,\ldots,X_m)\) and assume the free difference quotients and conjugates are well-defined. Without this convention, “densely defined on \(L^2(M)\)” and the location of the conjugate variables require extra ambient-algebra data not supplied by the question.

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15338 | OWR-4139-001

### Comparison of Majorization Relations for Probability Vectors

O / open | Oberwolfach 报告问题汇编

For probability vectors $x,y$ with nonzero coordinates, write $x\prec_C y$ if $x\otimes z\prec y\otimes z$ for some probability vector $z$, and write $x\prec_M y$ if $x^{\otimes n}\prec y^{\otimes n}$ for some $n\geq1$. The conditions $N_p(x)\leq N_p(y)$ for $p\geq1$, $N_p(x)\geq N_p(y)$ for $0\leq p\leq1$, and $N_p(x)\leq N_p(y)$ for $p\leq0$, where $N_p(u)=\sum_i u_i^p$, characterize when $x$ is an $\ell^1$-limit of vectors $x_k\prec_C y$. Do the same conditions characterize when $x$ is an $\ell^1$-limit of vectors $x_k\prec_M y$?

来源字段：https://doi.org/10.4171/owr/2009/56

15339 | OWR-15989-001

### Competing First-Passage Percolation in Hostile Environments

P / partially_solved | Oberwolfach 报告问题汇编

For first-passage percolation in a hostile environment on a bounded-degree graph, when does an infinite connected $F_\lambda$ cluster exist? On $\mathbb Z^d$, for $\lambda\in(0,1)$, is there an interval of initial red-particle densities $\mu$ for which infinite $F_1$ and $F_\lambda$ clusters coexist?

来源字段：https://doi.org/10.4171/owr/2018/4

15340 | AIM-PROBABILITY-0123

### Complex Hermitian matrix weights and an exact accretive quadratic sector

P / partially_solved | AIM 研讨会问题汇编

**Q:** Extend the model \[ \exp\!\left[-n\operatorname{Tr}\!\left(P(A_1,\ldots,A_m) +\frac12\sum_{i=1}^m A_i^2\right)\right] dA_1\cdots dA_m \] of Guionnet and Maurel-Segala to non-selfadjoint \(P\) (i.e. polynomials with complex coefficients).

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15341 | AIM-PROBABILITY-0124

### Complex rank-one spherical integrals and a two-atom zero barrier

P / partially_solved | AIM 研讨会问题汇编

Is there a combinatorial interpretation of free cumulants in terms of enumeration of maps and operations on maps? Consider \[ I_n(z,E_n)=\int_{\mathcal U(n)} \exp\{n\operatorname{Tr}(UD_nU^*E_n)\}\,dm_n(U), \qquad D_n=\operatorname{diag}(z,0,\ldots,0),\quad z\in\mathbb C, \] where \(E_n\) is self-adjoint (and may be taken diagonal), its spectrum is uniformly bounded, and its empirical spectral distribution converges to \(\mu_E\). Guionnet and Ma\u00efda showed that \[ f_n(z)=\partial_z\left(\frac1n\log I_n(z,E_n)\right) \] converges to \(R_{\mu_E}(z)\) for \(|z|\) small enough. What is the largest complex domain of convergence? If \(\mu_E\) is \(\boxplus\)-infinitely divisible, does convergence hold on the whole upper half-plane? Can one generalize to noncompactly supported measures? Perhaps this can be approached through normality of \((f_n)\).

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15342 | OWR-11786-021

### Complexity of Random Morse Functions on Simplex Boundaries

P / partially_solved | Oberwolfach 报告问题汇编

What is the expected complexity of a random Morse function on the boundary of a $d$-simplex, and how does it behave under repeated barycentric subdivision?

来源字段：https://doi.org/10.4171/owr/2012/24

15343 | OWR-7155444-004

### Computable Coherent Sets in Stochastic Dynamics

O / open | Oberwolfach 报告问题汇编

Can coherent families of sets for a time-dependent stochastic dynamical system be characterized in a way that permits numerical computation, and can the resulting problem be solved efficiently in higher dimensions?

来源字段：https://doi.org/10.4171/owr/2021/23

15344 | AMR-094-0008

### Concatenated bounded Brownian pieces

O / open | AMR 问题清单汇编

For each $k\in\mathbb{Z}$, let $B^k$ be Brownian motion and $T_k$ a stopping time, with the stopped pieces independent, $0\le T_k<\infty$, and with their total lengths diverging in both time directions. Form a continuous process $X$ by concatenating the pieces $B^k_{[0,T_k)}$. If there is a deterministic $c<\infty$ such that $T_k<c$ almost surely for every $k$, must $X$ be a two-sided Brownian motion?

背景链接：https://sites.math.washington.edu/~burdzy/open_mathjax.php

15345 | OWR-12481-003

### Concentration of Largest Triangular Submatrices in Random Matrices

O / open | Oberwolfach 报告问题汇编

For an $n\times n$ random matrix with independent identically distributed entries, if $T_n$ is the order of its largest lower-triangular submatrix, is $T_n$ concentrated on at most two values?

来源字段：https://doi.org/10.4171/owr/2013/18

15346 | OWR-732-001

### Conditional Correlations in Percolation Models

O / open | Oberwolfach 报告问题汇编

For increasing events $A$, $B$, and $C$ in a percolation model, when does $\mathbb P(A\mid B\cap C)\geq\mathbb P(A\mid C)$ hold: only for product measures or only when the conditioning event is a disconnection event?

来源字段：https://doi.org/10.4171/owr/2004/30

15347 | AIM-PROBABILITY-0115

### Conditional nonmicrostates entropy and a duplicated-generator obstruction

P / partially_solved | AIM 研讨会问题汇编

**Q:** What is the non-microstates analogue of free entropy in the presence, \[ \chi(x_1,\ldots,x_n:y_1,\ldots,y_n)\ ? \]

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15348 | OWR-11101922-004

### Confining Potentials for Generalized Gibbs Ensembles

O / open | Oberwolfach 报告问题汇编

What is the precise class of confining potentials for which the generalized Gibbs ensemble has the desired infinite-volume behavior?

来源字段：https://doi.org/10.4171/owr/2022/41

15349 | OWR-9790363-002

### Consensus Times on Erdős-Rényi Random Graphs

O / open | Oberwolfach 报告问题汇编

For the voter model started from distinct opinions on $G(n,p_n/n)$, determine the asymptotic consensus time as a function of $p_n$, in both discrete and continuous time.

来源字段：https://doi.org/10.4171/owr/2022/12

15350 | OWR-16161-003

### Consistent Noisy Single-Index Regression

O / open | Oberwolfach 报告问题汇编

Can the nonlinear single-index regression estimator based on tangent-field approximations and level-set partitions be modified to give consistent estimation when the noise $\varepsilon\neq0$, despite $|I_k|\not\to0$ even when adjacent level widths are arbitrarily small?

来源字段：https://doi.org/10.4171/owr/2018/20

15351 | AMR-096-0007

### Constant-factor online scheduling of subadditive batches

O / open | AMR 问题清单汇编

Tasks arrive as a rate-one Poisson process and have types in $[0,1]$; batch processing time $S$ is monotone and strictly subadditive and type $a$ incurs waiting cost rate $c(a)$. Is there an explicit online algorithm and universal constant $C$ whose long-run average cost $W$ satisfies $W/W_*<C$ for every $S$ and $c$ whenever the optimal value $W_*$ is finite?

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15352 | AMR-096-0006

### Constrained Ising storage model on a time-varying graph

O / open | AMR 问题清单汇编

Study the constrained Ising storage model described on the page when the underlying graph itself changes in time.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15353 | OWR-9790363-005

### Contact Processes on Exponential-Growth Dynamic Graphs

O / open | Oberwolfach 报告问题汇编

Establish survival and criticality for contact processes on dynamical percolation throughout the untreated parameter regime on exponential-growth graphs.

来源字段：https://doi.org/10.4171/owr/2022/12

15354 | OWR-4990373-010

### Continuity of the Zero-Level Gaussian Percolation Transition

O / open | Oberwolfach 报告问题汇编

For a planar stationary Gaussian field $f$, such as the random plane wave, is the phase transition at level $0$ continuous; equivalently, does the excursion set $\{f\leq 0\}$ have only bounded connected components?

来源字段：https://doi.org/10.4171/owr/2021/4

15355 | OWR-11101922-005

### Continuous-Time Extensions of Attractive Percolation Models

O / open | Oberwolfach 报告问题汇编

For finite-range continuous-time attractive spin systems, establish a critical-parameter dichotomy with positive survival probability above criticality and exponential decay below criticality, extending the corresponding discrete-time result and including the contact process.

来源字段：https://doi.org/10.4171/owr/2022/41

15356 | OWR-14298800-002

### Continuous-Time Limits of Stochastic Bures-Wasserstein Consensus

O / open | Oberwolfach 报告问题汇编

Derive the continuous-time limit as $\Delta t\to0$ of the stochastic multi-agent consensus dynamics on Bures-Wasserstein space, including the limiting stochastic differential equation and conditions for well-posedness and consensus.

来源字段：https://doi.org/10.4171/owr/2024/56

15357 | OWR-14298001-002

### Controlled McKean-Vlasov Problems without Weak Monotonicity

O / open | Oberwolfach 报告问题汇编

Can the weak monotonicity hypotheses in the controlled McKean-Vlasov problem be weakened or removed?

来源字段：https://doi.org/10.4171/owr/2023/44

15358 | AMR-094-0005

### Convergence of synchronous reflected-Brownian couplings

O / open | AMR 问题清单汇编

Let $D\subset\mathbb{R}^2$ be a connected open set with smooth boundary, and let $X,Y$ be synchronously coupled reflected Brownian motions in $D$ driven by the same planar Brownian motion. (i) Does there exist a bounded such domain for which $\limsup_{t\to\infty}|X_t-Y_t|>0$ with positive probability? (ii) If $D$ is the complement of a nondegenerate closed disk, does this inequality hold with positive probability?

背景链接：https://sites.math.washington.edu/~burdzy/open_mathjax.php

15359 | OWR-2654828-004

### Convergence Radii of Autoregressive and Moving Average Persistence Series

O / open | Oberwolfach 报告问题汇编

What is the radius of convergence of the power series in $\rho$ for the persistence exponent of the Gaussian autoregressive process of order one, and is the radius of convergence one for the analogous moving-average process of order one?

来源字段：https://doi.org/10.4171/owr/2020/32

15360 | AMR-096-0032

### Converse implications among SIRSN properties

P / partially_solved | AMR 问题清单汇编

Prove or disprove each of the proposed implications between the SIRSN properties numbered (16), (20), (49), (50), and (51): (16)$\Rightarrow$(20), unique singly-infinite geodesics$\Rightarrow$(49), (49)$\Rightarrow$(50), and (51)$\Rightarrow$(50).

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15361 | OWR-15428-015

## Convexity of Noncompletable Nonnegative Partial Tensors

P / partially_solved | Oberwolfach 报告问题汇编

Is the set of nonnegative partial tensors that cannot be completed to a joint distribution of independent random variables always convex?

来源字段：https://doi.org/10.4171/owr/2017/20

15362 | AIM-PROBABILITY-0052

## Coordinate-start sphere cutoff and a support-activation limit law

P / partially_solved | AIM 研讨会问题汇编

**Kac's random walk.** For a vector \(v\in\mathcal S^n\), choose two coordinates uniformly, choose an angle uniformly on \([0,2\pi)\), and rotate in the circle generated by those coordinates. Does this walk exhibit cutoff? Does the analogous walk on \(SO(n)\) exhibit cutoff?

来源字段：http://aimpl.org/markovmixing/6/

15363 | OWR-13677-001

## Copula Realizations and Centers of Complete Mixes

O / open | Oberwolfach 报告问题汇编

For an $n$-completely mixable distribution $F$, what copula can realize an $n$-complete mix with all margins equal to $F$? Can the uniqueness of the center of a joint mix be proved by a more elegant analytic argument?

来源字段：https://doi.org/10.4171/owr/2015/20

15364 | AIM-PROBABILITY-0041

## Corrected bounds and an exact critical-strip reduction for lattice triangulations

P / partially_solved | AIM 研讨会问题汇编

**Edge-flip chain on triangulations.** Consider the chain over triangulations of \([0,n]^2\), in which, at each step, an edge is randomly chosen and, if it is the diagonal of convex quadrilateral, flipped to the opposite diagonal. Does this chain have polynomial mixing time?

来源字段：http://aimpl.org/markovmixing/2/

15365 | OWR-11581-002

## Corruption-Parameter Choice in Empirical-Bayes Estimation

O / open | Oberwolfach 报告问题汇编

For the monotoneized empirical-Bayes estimator $\Delta_h$, how should the corruption parameter $h$ be chosen, and does inbred cross-validation yield a rate-sharp procedure?

来源字段：https://doi.org/10.4171/owr/2012/14

15366 | AIM-PROBABILITY-0030

## Counterexample to a malformed CCA limit and a corrected scalar Wasserstein bound

S / solved | AIM 研讨会问题汇编

For i.i.d. pairs $(X_i,Y_i)$ with $X_i\in\mathbb R^p$ and $Y_i\in\mathbb R^q$:

1. find $a\in\mathbb R^p$ and $b\in\mathbb R^1$ maximizing $\operatorname{corr}(a^T X,b^T Y)$; 2. choose sample weights $\widehat a,\widehat b$ whose coordinates each sum to one and maximize the sample correlation $\widehat c$; 3. accept the asserted limit, under independence and finite fourth moments, \[ A_n=-\left(n-\frac{p+q+3}{2}\right)\log(1-\widehat c) \ \Rightarrow\ \chi^2_{p+q-2}, \] and bound the distance to that chi-square law.

The record cites Hotelling (1936) and Bartlett (1939).

There are at least four distinct corruptions or conflations.

* Unless $q=1$, $b$ must belong to $\mathbb R^q$, not $\mathbb R^1$. * Correlation is unchanged by nonzero rescaling. Standard population CCA imposes $a^T\Sigma_{XX}a=b^T\Sigma_{YY}b=1$, not coordinate-sum constraints. * Canonical correlations $r_j$ are nonnegative singular values. Wilks' statistic uses $\prod_j(1-r_j^2)$, not $1-\widehat c$ for a largest signed correlation. * The complete-independence Wilks limit has $pq$ degrees of freedom, not $p+q-2$.

Thus there is no finite-sample distance to the displayed target that can tend to zero. Section 7 gives a literal scalar counterexample and then analyzes a standard corrected reading.

来源字段：http://aimpl.org/steinhd/1/

15367 | AIM-PROBABILITY-0021

## Counterexamples and a corrected Cauchy theorem for harmonic means

S / solved | AIM 研讨会问题汇编

Let $\mu$ be a probability density on $\mathbb R$, with positive density near $0$. Let $X_1,X_2,\ldots\overset{\mathrm{i.i.d.}}{\sim}\mu$, and define \[ H_n=\frac{n}{1/X_1+\cdots+1/X_n}. \] Prove, using Stein's method, that $H_n$ converges to a Cauchy distribution as $n\to\infty$, and find its rate.

来源字段：http://aimpl.org/steinhd/1/

15368 | AMR-098-0006

## Coupling characterization of setwise asymptotic stationarity

O / open | AMR 问题清单汇编

If setwise convergence $\mathbb{P}(\theta_nX\in A)\to\mathbb{P}(X'\in A)$ for every measurable path-space set $A$ does not imply total-variation convergence, what is a coupling characterization of this setwise convergence?

背景链接：http://cms.dm.uba.ar/depto/public/Some%20Open%20Problems-preprint.pdf

15369 | OWR-13685-005

### Coupling Markov Shifts to Bernoulli Shifts

O / open | Oberwolfach 报告问题汇编

Is each Markov shift $\rho(x_n)$ representable as a coupling to a Bernoulli shift?

来源字段：https://doi.org/10.4171/owr/2015/28

15370 | OWR-5152-002

### Covariance Estimation Without Logarithmic Oversampling

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a mean-zero random vector in $\mathbb{R}^n$ with covariance matrix $\Sigma$, and let $\Sigma_N$ be the empirical covariance matrix from $N$ independent samples. Characterize the distributions for which $\mathbb{E}\|\Sigma_N-\Sigma\|\leq\varepsilon\|\Sigma\|$ holds with $N=O(n)$. In particular, is linear sample size sufficient for every distribution having $2+\varepsilon$ moments?

来源字段：https://doi.org/10.4171/owr/2011/24

15371 | OWR-1731-002

### Critical Exponents for High-Dimensional Bond Percolation

O / open | Oberwolfach 报告问题汇编

For independent bond percolation on $\mathbb Z^d$ with $d\geq2$, let $\theta(p)$ be the probability that the origin lies in an infinite cluster, let $\chi(p)$ be the susceptibility, and let $p_c=\inf\{p:\theta(p)>0\}=\sup\{p:\chi(p)<\infty\}$. Determine the critical and near-critical behavior around $p_c$, including the existence, values, and universality of the associated critical exponents.

来源字段：https://doi.org/10.4171/owr/2007/48

15372 | AIM-PROBABILITY-0034

### Critical Glauber mixing on random regular graphs: corrected bounds and a one-mode reduction

P / partially_solved | AIM 研讨会问题汇编

**Mixing time for ferromagnetic Ising at critical temperature.** What is the mixing time for the ferromagnetic Ising model at critical temperature $\beta=\beta_c$ on a random $d$-regular graph? Is it $n^c$?

来源字段：http://aimpl.org/markovmixing/1/

15373 | AMR-099-0045

### Critical one-dimensional long-range percolation geometry

P / partially_solved | AMR 问题清单汇编

In one-dimensional long-range percolation with edge probabilities proportional to $\beta|i-j|^{-2}$, study the distance exponent $\theta(\beta)$ defined by typical distances of order $n^{\theta(\beta)}$. Is $\theta$ continuous or monotone in $\beta$? Does the model have a nontrivial metric scaling limit?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=92

15374 | AMR-099-0077

### Critical percolation on distributional planar limits

P / partially_solved | AMR 问题清单汇编

Let $G$ be a distributional local limit of finite planar graphs. Prove that $p_c^{\mathrm{site}}(G)\ge1/2$ almost surely and that there is no infinite cluster at criticality. Determine whether the no-critical-percolation conclusion holds for every unimodular random graph.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/randomplanar2.pdf#page=3

15375 | AMR-099-0052

### Critical probability of polynomial-growth disk triangulations

O / open | AMR 问题清单汇编

Let $G$ be a bounded-degree triangulation of an open disk with polynomial volume growth. Prove that its Bernoulli site-percolation critical probability satisfies $p_c(G)\ge 1/2$. Under natural sparsity hypotheses on vertices of degree greater than six, is $p_c(G)=1/2$?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/resistance_percolation.pdf#page=3

15376 | OWR-12697706-004

### Critical Random Walks on Percolation Clusters

O / open | Oberwolfach 报告问题汇编

What is the anomalous behavior of random walks on percolation clusters at criticality?

来源字段：https://doi.org/10.4171/owr/2023/10

15377 | OWR-12697708-007

### Critical Reinforcement Convergence on the Infinite Line

O / open | Oberwolfach 报告问题汇编

At $\alpha=1$, for $G=\mathbb Z$ with firing rates $p(v)=1$, does $x_v(t)\to1$ almost surely for every vertex $v$ as $t\to\infty$?

来源字段：https://doi.org/10.4171/owr/2023/12

15378 | OWR-11101922-006

### Critical Tails and Cooperative Survival in Attractive Systems

O / open | Oberwolfach 报告问题汇编

Does the infection-time tail above criticality decay exponentially for bootstrap percolation in full generality, and what happens for attractive probabilistic cellular automata in the intermediate regime of cooperative survival but individual death?

来源字段：https://doi.org/10.4171/owr/2022/41

15379 | AIM-PROBABILITY-0072

### Critical two-dimensional stochastic heat flow versus dynamic-GFF chaos

P / partially_solved | AIM 研讨会问题汇编

In 2 spatial dimensions is the measure valued solution of the stochastic heat equation related to the ``exponential'' of the dynamic GFF and hence quantum Louiville gravity?

来源字段：http://aimpl.org/kpzuniversality/1/

15380 | OWR-1731-005

### Crossing Bounds and Shape Functions in Anisotropic Percolation

S / solved | Oberwolfach 报告问题汇编

In critical anisotropic bond percolation on the square lattice $\mathbb{Z}^2$ with horizontal and vertical edge probabilities $p_h, p_v$ satisfying $p_h + p_v = 1$, let $f(n)$ be the characteristic vertical scale (aspect-ratio function) for horizontal crossings of rectangles $[0, n] \times [0, f(n)]$. Can the known rectangular a priori crossing bounds be improved so that $f(n)$ has stronger asymptotics-in particular, so that $n / f(n)$ converges to a nontrivial limit as $n \to \infty$? More broadly, can one find classes of domains with both a uniform a priori crossing bound and a Russo--Seymour--Welsh (RSW) conditional extension theorem?

来源字段：https://doi.org/10.4171/owr/2007/48

15381 | AMR-099-0051

### Crossings in random square tilings

O / open | AMR 问题清单汇编

Tile the unit square by finitely or countably many squares of varying sizes, with at most three squares meeting at a corner, and color the squares independently black or white with equal probabilities. Prove that the probability of a black left-to-right crossing is bounded below by a universal constant $c>0$. Moreover, as the largest tile diameter tends to zero, does this crossing probability tend to $1/2$?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/resistance_percolation.pdf#page=3

15382 | OWR-15988-001

### Curvature Bias in Statistics on Quotient Spaces

P / partially_solved | Oberwolfach 报告问题汇编

For statistical estimators on quotient spaces, determine when curvature-induced bias occurs, whether it affects analyses beyond the mean shape, how large it is in applications, and when and how it should be corrected.

来源字段：https://doi.org/10.4171/owr/2018/3

15383 | OWR-3473-004

### Cutoff for Random Walks on Vertex-Transitive Expanders

O / open | Oberwolfach 报告问题汇编

Does the simple random walk on every family of vertex-transitive expander graphs exhibit total-variation cutoff, and can the comparison between the mixing times of the simple and non-backtracking random walks known for random regular graphs be extended to every such family?

来源字段：https://doi.org/10.4171/owr/2009/23

15384 | OWR-14299904-012

### Cycle Limits for Words in Random Permutations

P / partially_solved | Oberwolfach 报告问题汇编

Let $\sigma$ and $\tau$ be independent uniform random permutations and set $\pi=\sigma\tau\sigma^2\tau^2$. Does the normalized length $L_1/n$ of the cycle containing $1$ converge to the uniform distribution on $[0,1]$, and do the normalized ordered cycle lengths converge to the Poisson-Dirichlet distribution with parameter $1$? More generally, do these limits hold for every primitive word in $\sigma$ and $\tau$, and does the resulting permutation converge to uniformity in total variation?

来源字段：https://doi.org/10.4171/owr/2026/2

15385 | OWR-768-007

### Cycle Statistics of Nonreversible Birational Maps

O / open | Oberwolfach 报告问题汇编

Let $\langle D_0\rangle_p$ denote the average over primes at most $p$ at which $L$ has good reduction. For every nonintegrable birational map $L$ that is not reversible over $\mathbb R$ and has no other symmetry, $\lim_{p\to\infty}\langle D_0\rangle_p(x)=x$.

来源字段：https://doi.org/10.4171/owr/2004/54

15386 | OWR-11570-005

### Decay of Correlations in Random Quadratic Maps

O / open | Oberwolfach 报告问题汇编

For the random quadratic system $T_\lambda(x)=\lambda x(1-x)$ with i.i.d. parameters $\lambda_i\in[1,4]$ distributed by an absolutely continuous probability measure $P$, does the system have random decay of correlations, equivalently, is it relatively exact?

来源字段：https://doi.org/10.4171/owr/2012/03

15387 | OWR-15428-014

### Degree of the Algebraic Boundary in Partial Tensor Completion

P / partially_solved | Oberwolfach 报告问题汇编

In the partial-tensor completion problem for joint distributions of independent random variables, determine the degree of the hypersurface forming the algebraic boundary of the completable set.

来源字段：https://doi.org/10.4171/owr/2017/20

15388 | OWR-15206-011

### Degree-Four Expansion from Random Latin Squares

O / open | Oberwolfach 报告问题汇编

Let $Y(n,d)$ be the random two-dimensional simplicial complex obtained as the union of the complexes associated with $d$ independent uniformly random Latin squares of order $n$, and let $h_1(Y)$ be its first coboundary-expansion constant. Does there exist $\varepsilon>0$ such that $$ \lim_{n\to\infty}\mathbb P\{Y\in Y(n,4):h_1(Y)>\varepsilon\}=1 $$?

来源字段：https://doi.org/10.4171/owr/2017/1

15389 | OWR-13940-005

### Delaunay Simplices in Flag Complexes of Random Triangulations

P / partially_solved | Oberwolfach 报告问题汇编

For a Poisson point process in $\mathbb R^d$ with Delaunay triangulation $T$, form the flag complex of the $1$-skeleton of $T$. How many $d$-simplices, and more generally how many $k$-simplices for $k\leq d$, of this flag complex are Delaunay?

来源字段：https://doi.org/10.4171/owr/2015/45

15390 | OWR-2654836-007

### Density Estimates and Short-Time Asymptotics for Rough Equations

P / partially_solved | Oberwolfach 报告问题汇编

How do sharp uniform density estimates for SDEs driven by fractional Brownian motion relate to existing short-time asymptotic expansions for elliptic and hypoelliptic SDEs?

来源字段：https://doi.org/10.4171/owr/2020/40

15391 | OWR-15218-002

### Dependent Volatility Drivers in Tractable Pricing Models

P / partially_solved | Oberwolfach 报告问题汇编

How far can the assumption that the volatility process $\sigma$ is independent of the Brownian motion $W$ driving the stock price be relaxed while preserving tractable option-pricing models and allowing asymmetric implied-volatility smiles?

来源字段：https://doi.org/10.4171/owr/2017/13

15392 | OWR-12174-001

### Detecting Genuine Modes in Density Deconvolution

P / partially_solved | Oberwolfach 报告问题汇编

In one-dimensional nonparametric density deconvolution, if regularization can introduce an apparent additional mode, can one statistically determine whether the mode is real and significant enough to reject the smooth reconstruction?

来源字段：https://doi.org/10.4171/owr/2012/51

15393 | OWR-14299518-013

### Detection-Threshold Transition for Planted Random Trees

O / open | Oberwolfach 报告问题汇编

As the mean degree $pn$ increases, how does the minimum detectable size of a typical planted tree transition from order $\log^2 n$ to order $\sqrt n$?

来源字段：https://doi.org/10.4171/owr/2025/42

15394 | OWR-2090-009

### Determinant Concentration for Heterogeneous Bernoulli Matrices

O / open | Oberwolfach 报告问题汇编

Let $A$ be a random $n\times n$ matrix with entries $c_{ij}\xi_{ij}$, where $1\leq c_{ij}\leq C=O(1)$ and the $\xi_{ij}$ are independent Bernoulli signs. The known estimate is $|\det A|=\mathbb{E}|\det A|\exp(O(n^{2/3}))$ almost surely. Can the exponential factor be replaced by a polynomial factor $n^{O(1)}$?

来源字段：https://doi.org/10.4171/owr/2008/44

15395 | OWR-12697708-002

### Deterministic Limits of Trace-Reinforced Ant Walks

O / open | Oberwolfach 报告问题汇编

On every finite graph, do the edge-weight sequences in the trace-reinforced ant model, normalized by the number of walks, converge almost surely to deterministic limits? Can the model be extended meaningfully to several food sources?

来源字段：https://doi.org/10.4171/owr/2023/12

15396 | OWR-2654834-001

### Deterministic-to-Stochastic Transition in the Vlasov Approximation

O / open | Oberwolfach 报告问题汇编

Can one rigorously characterize the transition, in the quasilinear approximation of the Vlasov equation, from the short-time deterministic regime to the long-time stochastic regime in which randomness drives the effective diffusion?

来源字段：https://doi.org/10.4171/owr/2020/38

15397 | OWR-14298589-006

### Diagonal Large-Genus Limits of Square-Tiled Surfaces

O / open | Oberwolfach 报告问题汇编

Fix $\alpha>0$. Determine the limiting behavior of random square-tiled surfaces with $N=\lfloor\alpha g\rfloor$ squares as $g\to\infty$, rather than first sending $N$ to infinity at fixed genus.

来源字段：https://doi.org/10.4171/owr/2024/41

15398 | OWR-12861-027

### Diagonal Signings and Real Eigenvalues of Random Matrices

P / partially_solved | Oberwolfach 报告问题汇编

For a random $n\times n$ matrix $M_n$ with entries in $\{\pm1\}$, does the probability tend to $1$ that no choice of signs for its diagonal entries makes it singular? Does $M_n$ have at least two real eigenvalues with probability $1-o(1)$?

来源字段：https://doi.org/10.4171/owr/2014/01

15399 | OWR-16416-001

### Diffusive Limits for the Random Lorentz Gas

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Random Lorentz Gas Invariance Principle [OWR-16772-002]

For the random Lorentz-gas trajectory $X^{r,\rho}(t)$, conditioned on the particle not being trapped, does

$$t\longmapsto\frac{X^{r,\rho}(Tt)}{\sqrt T},\qquad 0\leq t\leq1,$$

converge weakly as $T\to\infty$ to a Wiener process with nondegenerate variance, in either the quenched or averaged-quenched setting?

来源字段：https://doi.org/10.4171/owr/2018/41

15400 | OWR-11695859-006

### Dimension-Free Lipschitz Optimal Transport Maps

O / open | Oberwolfach 报告问题汇编

Is the quadratic optimal-transport map from $\gamma$ onto $\mu$ globally Lipschitz, with Lipschitz norm depending only on $\|f\|_{\mathrm{Lip}}$ and not on the dimension $n$?

来源字段：https://doi.org/10.4171/owr/2022/49

15401 | AIM-PROBABILITY-0073

### Directional limit shapes and exponent changes for a higher-dimensional directed polymer

P / partially_solved | AIM 研讨会问题汇编

Study polymer free energy and growth model fluctuation exponents in higher dimension? Compute limit shapes and fluctuation exponents.

来源字段：http://aimpl.org/kpzuniversality/1/

15402 | OWR-17474-001

### Discontinuities of Two-Sided Specifications

O / open | Oberwolfach 报告问题汇编

Does the set of discontinuity points of the two-sided specification have measure zero, and does the analogous conclusion hold for long-range Ising models?

来源字段：https://doi.org/10.4171/owr/2020/11

15403 | OWR-14298808-013

### Discrete Log-Concavity Under Self-Convolution

S / solved | Oberwolfach 报告问题汇编

Let $p$ be a probability mass function on $\mathbb Z^d$ that is the restriction of a log-concave function on $\mathbb R^d$. Is the self-convolution $p*p$ again the restriction of a log-concave function on $\mathbb R^d$?

来源字段：https://doi.org/10.4171/owr/2024/58

15404 | OWR-2493-003

### Disorder Relevance in Pinning and Renewal Models

P / partially_solved | Oberwolfach 报告问题汇编

In disordered pinning or renewal models, with quenched critical point $h_c(\beta)$ and annealed critical point $h_c^{\mathrm a}(\beta)$, do they differ? Are the quenched and annealed critical exponents different, and how does the answer depend on the renewal-law tail exponent?

来源字段：https://doi.org/10.4171/owr/2008/55

15405 | OWR-12481-004

### Displacement Exponents for Low-Dimensional Self-Avoiding Walks

O / open | Oberwolfach 报告问题汇编

For self-avoiding walk in dimensions $d=2$ and $d=3$, prove that there exists $\varepsilon>0$ such that

$$n^{2/d+\varepsilon}\leq \mathbb E_n[\operatorname{Dist}(\gamma_n,0)^2]\leq n^{2-\varepsilon}$$

for all sufficiently large $n$.

来源字段：https://doi.org/10.4171/owr/2013/18

15406 | OWR-14298373-004

### Distance and Diameter Constants of High-Genus Triangulations

P / partially_solved | Oberwolfach 报告问题汇编

For random triangulations $T_{n,g}$ of high genus, are typical distances asymptotic to $D_\theta\log n$ and the diameter to $D'_\theta\log n$ with $D'_\theta=3D_\theta$?

来源字段：https://doi.org/10.4171/owr/2024/25

15407 | AMR-099-0069

### Distance exponent of random series-parallel graphs

P / partially_solved | AMR 问题清单汇编

Start from one edge and at each stage replace every edge independently by two edges in series with probability $p$ or two edges in parallel with probability $1-p$. For $p>1/2$, write $\mathbb{E}\Delta_n=\exp(n\delta(p)+o(n))$ for the endpoint distance $\Delta_n$. Determine the function $\delta:[1/2,1]\to\mathbb{R}$ and decide whether $\delta(1/2)=0$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=18

15408 | OWR-828-001

### Distinct Higher-Point Motions of Perturbed Arratia Flows

O / open | Oberwolfach 报告问题汇编

Does the stochastic kernel flow obtained from a $\theta$-joining of the Arratia flow have $n$-point motions different from those of the Le Jan-Raimond sticky flow for every $n\geq4$, and are the two flows therefore distinct?

来源字段：https://doi.org/10.4171/owr/2005/23

15409 | OWR-16937-002

### Distinguishing Confounding from Estimation Noise

P / partially_solved | Oberwolfach 报告问题汇编

When an estimated regression vector $\hat a$ differs from the causal vector $a$ because of either finite-sample noise or confounding, why should algorithms distinguish the two causes, and can regularization mitigate confounding as it mitigates overfitting?

来源字段：https://doi.org/10.4171/owr/2019/25

15410 | OWR-4416-006

### Distribution of Profinite Galois Groups

P / partially_solved | Oberwolfach 报告问题汇编

As a finite set $S$ of primes varies, is the probability that $G_S$ is isomorphic to a given $g$-generator, $g$-relator pro-$p$ group $G$ equal to $A(G)/|\operatorname{Aut}(G)|$?

来源字段：https://doi.org/10.4171/owr/2010/35

15411 | OWR-1598-001

### Divergent Conductivity in Low-Dimensional Stochastic Lattices

P / partially_solved | Oberwolfach 报告问题汇编

For an energy-momentum-conserving stochastic lattice model in a nonequilibrium stationary state, what is the divergence order of the conductivity in low-dimensional unpinned systems? Green-Kubo computations predict order $\sqrt{N}$ for $d=1$ and $\log N$ for $d=2$. How do random masses, random interactions, nonlinear interactions, and the noise strength $\gamma$ affect the conductivity and its divergence order?

来源字段：https://doi.org/10.4171/owr/2007/42

15412 | AMR-094-0009

### Do peaks of random labelings repel each other?

O / open | AMR 问题清单汇编

Choose uniformly a bijective labeling of the vertices of the $n\times n$ discrete square by $1,2,\ldots,n^2$, and call a vertex a peak when all adjacent vertices have lower labels. Condition on there being exactly two peaks, and let $\rho_n$ be their graph distance. Does $\rho_n/n$ converge in distribution to $0$ as $n\to\infty$?

背景链接：https://sites.math.washington.edu/~burdzy/open_mathjax.php

15413 | AIM-PROBABILITY-0141

## Domains for reflection-Catalan families and multiplicative convolution

P / partially_solved | AIM 研讨会问题汇编

“A pseudoreflection is a unitary operator ... whose eigenvalues are 0 with multiplicity \(n-1\), and \(-1\) with multiplicity 1.”

The problem asks for the largest natural domains of definition of \(NC_W\), \(NN_W\), and \(\Delta_W\), and of their applications. For an irreducible finite complex reflection group of rank \(n\), with invariant degrees \(d_1,\ldots,d_n\), it proposes \[ \operatorname{Cat}(W):=\prod_{i=1}^{n}\frac{h+d_i}{d_i}, \qquad h=d_n, \tag{1.1} \] and warns that outside the duality/well-generated case the product may not be integral. It then asks about four directions:

- extending \(NC_W\) from finite Coxeter groups to well-generated complex reflection groups; - extending free probability beyond types \(A\) and \(B\), including Speicher's multiplicative functions; - understanding cluster algebras in infinite type; - extending \(NN_W\), the root order, and related objects from Weyl groups to noncrystallographic types, while identifying genuine barriers such as the absence of the usual Lie-algebra and affine-arrangement structures.

The missing continuations are verified as follows. The free-probability bullet ends “See Problem 5.1.” It is followed by “Explain the theory of cluster algebras in infinite types. See Problem 6.5.” The final bullet continues on page 5: it identifies nonnesting partitions as the most glaring crystallographic obstruction, cites Kriloff--Ram, and asks what Chapoton's \(H\)-triangle counts in noncrystallographic types. Problem 1.3 begins only after that bullet.

The PDF itself prints:

The standard corrected definition is that a complex pseudoreflection is a nonidentity finite-order linear map whose fixed subspace is a hyperplane. After choosing a unitary form, its eigenvalues are \[ 1\quad\text{with multiplicity }n-1, \qquad \zeta\ne1\quad\text{with multiplicity }1, \] where \(\zeta\) is a root of unity. Bessis uses exactly the fixed-hyperplane definition.

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15414 | AIM-PROBABILITY-0038

## Dyadic critical mixing solved and an exact general flip-graph reduction

P / partially_solved | AIM 研讨会问题汇编

The preserved input asks about equitable rectangular dissections of an \(n\times n\) lattice square into \(n\) rectangles of area \(n\), with \(n=2^k\), and the edge-flip Glauber chain having stationary weight \(\pi(\sigma)\propto\lambda^{|\sigma|}\). At \(\lambda=1\), it asks for a polynomial mixing-time upper bound in either of two state spaces:

1. dyadic tilings; or 2. all integral rectangular dissections.

The `literature` field is visibly corrupted: it twice contains `$\\lambda1$`, and calls \(O(n\log n)\) a lower bound. The following reconstruction is verified from Theorems 1.2--1.5 of Cannon--Miracle--Randall (CMR):

* for dyadic tilings, the chain is fast mixing in \(O(n^2\log n)\) for every fixed \(\lambda<1\), and has mixing time \(\exp(\Omega(n^2))\) for every fixed \(\lambda>1\); * for general rectangular dissections it is slow mixing both for \(\lambda>1\), with lower bound \(\exp(\Omega(n^2))\), and for \(\lambda<1\), with lower bound \(\exp(\Omega(n\log n))\); * the pre-2017 dyadic lower bound is \(\Omega(n\log n)\), not \(O(n\log n)\). Cannon--Levin--Stauffer (CLS), Theorem 1 and the paragraph preceding Theorem 2, explicitly identify it as an \(\Omega\)-bound coming from the flip-graph diameter.

There is also a normalization issue. CMR works on the \(n\times n\) lattice and, for each legal flip, assigns proposal probability \(1/(4n)\). CLS scales to the unit square, calls the number of rectangles \(n=2^k\), and chooses a rectangle and one of its four sides; each legal flip then has probability \(1/(2n)\). At \(\lambda=1\) the CMR transition matrix is exactly \[ P_{\rm CMR}=\frac{I+P_{\rm CLS}}2. \tag{1} \] Thus the chains differ only by an additional factor-two lazification. All logarithms in the quoted CLS exponents are base two.

CMR assumes \(k\) even in the equitable lattice setup, so that the square ground state exists. The CLS dyadic theorem applies to every \(n=2^k\). Janson--Randall--Spencer use yet another convention: their “\(k\)-tiling” has \(2^k\) rectangles. We use \(n\) for the number of rectangles throughout.

来源字段：http://aimpl.org/markovmixing/2/

15415 | AIM-PROBABILITY-0039

## Dyck random transpositions and a sharp edgewise comparison obstruction

P / partially_solved | AIM 研讨会问题汇编

**Random walks on Dyck's paths.** Consider the Markov chain on the set of Dyck's paths, which moves by choosing uniformly at random two coordinates and exchanging their value ($+$ or $-1$) provided the resulting path remains a Dyck's path. What is the mixing time?

来源字段：http://aimpl.org/markovmixing/2/

15416 | AMR-094-0004

### Efficient couplings in acute triangles

O / open | AMR 问题清单汇编

Let $D$ be a triangle whose angles are all strictly less than $\pi/2$, and let $\mu_2>0$ be the second eigenvalue of the Laplacian on $D$ with Neumann boundary conditions. Can one construct reflected Brownian motions $X_t,Y_t$ in $D$, starting from different points, such that $\tau=\inf\{t\ge0:X_t=Y_t\}<\infty$ almost surely and, for every fixed $\varepsilon>0$, $$\mathbb{P}(\tau>t)\le \exp[-(\mu_2-\varepsilon)t]$$ for all sufficiently large $t$?

背景链接：https://sites.math.washington.edu/~burdzy/open_mathjax.php

15417 | OWR-4532-006

### Efficient Simulation of Metastable Transitions

O / open | Oberwolfach 报告问题汇编

Let $X_t\in\mathbb{R}^{3N}$ follow the overdamped Langevin dynamics $dX_t=-\nabla V(X_t)\,dt+\sqrt{2\beta^{-1}}\,dW_t$, whose invariant density is proportional to $e^{-\beta V}$. How can one efficiently generate equilibrium trajectories that leave one well of the metastable potential $V$ and reach another, and how should the available rare-event methods be classified and improved?

来源字段：https://doi.org/10.4171/owr/2010/50

15418 | OWR-13685-002

### Eigenvalue Counts for Low-Degree Random Regular Graphs

P / partially_solved | Oberwolfach 报告问题汇编

Can the eigenvalue-counting estimate for uniformly random regular graphs be improved when $d\lesssim(\log n)^4$?

来源字段：https://doi.org/10.4171/owr/2015/28

15419 | OWR-15989-011

### Eigenvalue Motion under Ginibre Perturbations

P / partially_solved | Oberwolfach 报告问题汇编

Let $M_N$ and $M_N'$ be independent complex Ginibre matrices and set $$ M_N(\theta)=\cos\theta\,M_N+\sin\theta\,M_N'. $$ How do the eigenvalues $\lambda_i(\theta)$ vary with $\theta$?

来源字段：https://doi.org/10.4171/owr/2018/4

15420 | OWR-15989-010

### Eigenvector Volumes and Angles in Ginibre Matrices

P / partially_solved | Oberwolfach 报告问题汇编

For left and right eigenvectors of an $N\times N$ complex Ginibre matrix, what is the volume of the parallelepiped formed by eigenvectors whose eigenvalues lie near fixed deterministic parameters $\nu_1,\ldots,\nu_\ell$ as $N\to\infty$? How does the minimum angle between eigenvectors depend on $N$?

来源字段：https://doi.org/10.4171/owr/2018/4

15421 | OWR-13863-003

### Elicitability and Backtesting of Expected Shortfall

O / open | Oberwolfach 报告问题汇编

How is elicitability relevant to backtesting risk measures, especially Expected Shortfall, given that Expected Shortfall is not singly elicitable but is jointly elicitable with Value at Risk?

来源字段：https://doi.org/10.4171/owr/2015/42

15422 | OWR-17474-007

### Ends of Critical Long-Range Percolation Clusters

O / open | Oberwolfach 报告问题汇编

In one-dimensional long-range percolation with edge probabilities asymptotic to $\beta|k|^{-s}$, are both of the following true at $s=2$: at criticality the infinite cluster has exactly two ends, and in the supercritical regime it has infinitely many ends?

来源字段：https://doi.org/10.4171/owr/2020/11

15423 | AMR-099-0085

### Ends of infinite clusters in the nonuniqueness phase

S / solved | AMR 问题清单汇编

Let $G$ be a connected quasi-transitive graph and let $p\in(0,1)$. If Bernoulli percolation has more than one infinite cluster almost surely, prove that every infinite cluster has exactly $2^{\aleph_0}$ ends almost surely.

背景链接：https://emis.de/ft/41630#page=6

15424 | AMR-099-0041

### Ends of transient branching random walk

P / partially_solved | AMR 问题清单汇编

Prove that a transient simple branching random walk on any vertex-transitive graph has infinitely many ends.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=70

15425 | OWR-5152-009

### Entry Distributions in Random Contingency Tables

P / partially_solved | Oberwolfach 报告问题汇编

Let $D$ be uniformly random among nonnegative integer matrices with prescribed row margins $R$ and column margins $C$. Does an individual entry of $D$ have an asymptotically geometric distribution as the dimensions grow? In the special case $m=n$ and $R=C=(3n,n,\ldots,n)$, does the expected value of the $(1,1)$ entry grow linearly in $n$?

来源字段：https://doi.org/10.4171/owr/2011/24

15426 | OWR-17474-003

### Equivalence of Gibbs Measures and $g$-Measures

P / partially_solved | Oberwolfach 报告问题汇编

For an equilibrium state $\mu$ of a potential $f$, is the existence of a positive continuous eigenfunction equivalent to $\mu$ being a $g$-measure? More generally, are Gibbs measures and $g$-measures equivalent?

来源字段：https://doi.org/10.4171/owr/2020/11

15427 | OWR-4081-007

### Equivalence of Sznitman Ballisticity Conditions

P / partially_solved | Oberwolfach 报告问题汇编

For a uniformly elliptic i.i.d. random walk in random environment in dimension $d\geq2$, and every $\gamma\in(0,1)$, is Sznitman's condition $(T)_\gamma$ relative to $\ell$ equivalent to condition $(T')$ relative to $\ell$, where $(T')$ means that $(T)_\gamma$ holds for every $\gamma\in(0,1)$?

来源字段：https://doi.org/10.4171/owr/2009/38

15428 | OWR-13111-003

### Estimating Probabilities of Unobserved Species

O / open | Oberwolfach 报告问题汇编

Given a multinomial database sample $X\sim\operatorname{Multinom}(n,p)$ over a very large species set and an unobserved species $s$ with $X_s=0$, develop a statistically sound estimate of $p_s$ (or $-\log_{10}p_s$) and quantify its uncertainty.

来源字段：https://doi.org/10.4171/owr/2014/41

15429 | AMR-096-0017

### Eulerian-circuit continuum limits and SLE

O / open | AMR 问题清单汇编

Is there a relation between space-filling $\operatorname{SLE}_\kappa$ for $\kappa>8$ and the conjectural continuum limit of uniform Eulerian circuits on $\mathbb{Z}_N^2$?

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15430 | AIM-PROBABILITY-0078

### Exact block-update heights for finite-range and speed-change exclusion

P / partially_solved | AIM 研讨会问题汇编

Height functions associated to exclusion processes with longer range jumps, or environment dependent speed-changes.

The exact AIM record is the one-line prompt:

It is Problem 11.6 in the AIM workshop list *The Kardar-Parisi-Zhang equation and universality class*. I interpret it as asking both how to define the microscopic height beyond nearest-neighbor ASEP and how far its hydrodynamic/KPZ fluctuation theory has been established.

“Environment dependent” is ambiguous. In the interacting-particle literature it may mean a translation-invariant rate depending on the nearby particle configuration (a speed-change process), or a rate depending on an external spatial environment. These are separated below. The exact height calculus applies to both, but their scaling theories are different.

来源字段：http://aimpl.org/kpzuniversality/1/

15431 | AMR-098-0004

### Exact coupling of singular non-discrete random walks

S / solved | AMR 问题清单汇编

Let $S$ and $S'$ be random walks on $\mathbb{R}$ with the same i.i.d. step-length distribution, starting at $0$ and $x$. Suppose the step lengths are neither discrete nor spread out. For which initial positions $x$ can the walks be coupled so that $S_n=S'_n$ for every sufficiently large $n$ almost surely?

背景链接：http://cms.dm.uba.ar/depto/public/Some%20Open%20Problems-preprint.pdf

15432 | AIM-PROBABILITY-0005

### Exact determinant-sign characterization of nonnegative rank three for 2x2x2 tensors

S / solved | AIM 研讨会问题汇编

Can we characterize tensors of non-negative rank at most 3 using the description of $RBM_{3,2}$?

Is it possible to use the implicit description of $RBM_{3,2}$ to determine the characterizing properties of non-negative rank $\leq 3$ tensors?

来源字段：http://aimpl.org/boltzmann/2/

15433 | AIM-PROBABILITY-0094

### Exact discrete Cole-Hopf compatibility and the solved Sasamoto-Spohn convergence problem

S / solved | AIM 研讨会问题汇编

Stochastic analysis

Use the well-posedness theory for the KPZ equation to redo Bertini-Giacomin or to prove convergence of simpler discretizations of the KPZ equation such as the one previously studied by Sasamoto and Spohn.

来源字段：http://aimpl.org/kpzuniversality/2/

15434 | AIM-PROBABILITY-0015

### Exact dropout decomposition for RBM likelihoods and an ordinary-RBM no-go example

P / partially_solved | AIM 研讨会问题汇编

Is using dropout during RBM training equivalent to adding a regularizing penalty term on to the training objective function?

The source record also compares the question with statements relating early stopping and an $L_2$ penalty. That comparison must be qualified: early stopping and ridge regression coincide or closely correspond only in particular models/parameterizations, not as an exact identity for arbitrary neural-network training.

There are three materially different readings of “dropout during RBM training”:

1. the expectation, over masks, of the exact conditional negative log-likelihood; 2. the negative log-likelihood of the visible marginal of the dropout mixture model; 3. the stochastic update field produced by the contrastive-divergence (CD) implementation.

They are not the same object. The rigorous result below answers the first reading, compares it with the second, and explains why it does not automatically answer the third.

来源字段：http://aimpl.org/boltzmann/5/

15435 | OWR-14299904-011

### Exact Enumeration and Distribution of Word Trees

P / partially_solved | Oberwolfach 报告问题汇编

Can $w$-trees be enumerated exactly by a formula, recurrence, or functional equation? For two words $W$ and $W^{\prime}$, under what conditions does the total-variation distance between their associated random functions tend to $0$, tend to $1$, or do neither?

来源字段：https://doi.org/10.4171/owr/2026/2

15436 | AIM-PROBABILITY-0009

### Exact fixed-connectivity inference counts for binary RBMs

P / partially_solved | AIM 研讨会问题汇编

For a fixed labeled bipartite graph \(G=(V\sqcup H,E)\) defining a binary RBM, count the generic conditional hidden-MAP maps \[ \pi_\theta:\{0,1\}^{V}\longrightarrow\{0,1\}^{H}, \qquad \pi_\theta(v)=\arg\max_h p_\theta(h\mid v), \] as the biases and the weights on \(E\) vary.

来源字段：http://aimpl.org/boltzmann/3/

15437 | AIM-PROBABILITY-0008

### Exact fixed-graph stability bounds for a sparse semi-quantum RBM

P / partially_solved | AIM 研讨会问题汇编

How do RBMs with quantum effects differ from classical RBMs?

How are the following models different from classical RBMs:

1) RBMs with sparse connectivity.

2) RBMs with sparse connectivity, and with quantum effects.

3) RBMs with sparse connectivity, and with quantum effects and quantum training algorithm.

来源字段：http://aimpl.org/boltzmann/3/

15438 | AIM-PROBABILITY-0016

### Exact graph heat flow on a product-RBM manifold

P / partially_solved | AIM 研讨会问题汇编

How can Wasserstein geometry on RBMs be used to approximate PDE solutions?

The official 2018 AIM workshop report does verify the surrounding context: a working group on Wasserstein distance and optimal transport studied Wasserstein natural gradients for Boltzmann machines [AIM18]. Accordingly, the **plausible but not source-verified reconstruction** used here is:

The reconstructed question is still underspecified: it does not name a PDE, state space, transport metric, RBM type, or approximation criterion. This report makes one rigorous choice suited to binary RBMs. The state space is the hypercube \(\mathcal X=\{0,1\}^n\); the PDE is the coordinate-flip graph heat equation (a finite-state master equation/discrete parabolic PDE); the metric is the logarithmic-mean dynamical transport metric of Maas [Maa11]; and the RBM ansatz is the product-Bernoulli visible family, represented either by a hidden-free RBM or by an RBM in which every hidden unit has degree at most one.

This choice is intentionally not called continuous \(W_2\) geometry. Binary RBM laws are supported on a finite set, and ordinary static \(W_2\) on that discrete ground space has no Riemannian infinitesimal tensor for nontrivial heat curves. A two-state obstruction is proved below.

来源字段：http://aimpl.org/boltzmann/6/

15439 | AIM-PROBABILITY-0169

### Exact matrix recurrence and the irrational-exponent obstruction in noncrystallographic type

P / partially_solved | AIM 研讨会问题汇编

Remarks:

• (N. Reading, D. Speyer) Early calculations suggest that there are “approximate” recurrences, that one returns close to, but bounded away from the original matrix.

Choose an orientation and a real skew-symmetrizable exchange matrix whose Cartan companion encodes a noncrystallographic finite Coxeter diagram. What do its matrix and variable mutations do? Are their orbits finite, periodic, or approximately recurrent?

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15440 | OWR-12873-003

### Exact Minimax Asymptotics in Anisotropic Nonparametric Estimation

O / open | Oberwolfach 报告问题汇编

In the Gaussian white-noise model, determine the exact asymptotics, rather than only the rate order, of the minimax $L^p$-risk over anisotropic Nikol'skii classes.

来源字段：https://doi.org/10.4171/owr/2014/13

15441 | AIM-PROBABILITY-0028

### Exact non-Gaussian SURE bias and transformed shrinkage criteria

P / partially_solved | AIM 研讨会问题汇编

It proposes two non-Gaussian integration-by-parts devices,

\[ \mathbb E\{Yf(Y)\}=\mathbb E\{T(Y)f'(Y)\} \]

\[ \mathbb E\{Yf(Y)\}=\sigma^2\mathbb E f'(Y^*), \]

and asks for the bias of Gaussian SURE for estimators of the form \(\widehat\theta=X+h(X)\), whether the bias divided by dimension tends to zero, and whether this can be applied to wavelet shrinkage.

1. After defining \(\overline Y=(Y_1,\ldots,Y_d)\), the displayed \(X=Y+\theta\) must mean \[ X=\overline Y+\theta\in\mathbb R^d. \] 2. The displayed Gaussian formula \[ \widehat r_h=d+2\nabla\!\cdot h(X)+\|h(X)\|^2 \] presumes \(\operatorname{Var}(Y)=1\). For variance \(\sigma^2\), its correctly scaled version is \[ \widehat r_{h,\sigma} =d\sigma^2+2\sigma^2\nabla\!\cdot h(X)+\|h(X)\|^2. \] 3. The nested expectation in the definition of \(e_d\) is interpreted as the ordinary bias \[ e_d=\mathbb E\widehat r_h- \mathbb E\|\widehat\theta-\theta\|^2. \] The capital \(\Theta\) in the prose is the same unknown vector \(\theta\).

All formulas below state their regularity and integrability assumptions explicitly.

来源字段：http://aimpl.org/steinhd/1/

15442 | AIM-PROBABILITY-0113

### Exact norm of the CAR generator matrix

P / partially_solved | AIM 研讨会问题汇编

To prove the existence of an embedding \(\Gamma_{q,n}\to\mathcal R^\omega\), one uses Speicher's central limit theorem. In this procedure, is it possible to find explicitly uniformly bounded matrices whose mixed moments approach those of \(q\)-Gaussians? More precisely, let \(c_{i,j}\) be unitary generators of the CAR algebra (or \(-1\)-Gaussians); are the matrices \[ \frac1{\sqrt n}[c_{i,j}]_{i,j\leq n} \] uniformly bounded?

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15443 | AIM-PROBABILITY-0002

### Exact parity projection and a certified bracket for RBM(3,1)

P / partially_solved | AIM 研讨会问题汇编

Determine the maximum divergence of $RBM_{3,1}$.

For binary visible and hidden variables, determine \[ \mathcal D_{3,1}^{\rm KL} =\sup_{p\in\Delta(\{0,1\}^3)} \inf_{q\in RBM_{3,1}}D_{\rm KL}(p\Vert q). \]

来源字段：http://aimpl.org/boltzmann/1/

15444 | AIM-PROBABILITY-0035

### Exact Potts censoring on one edge and a monotonicity obstruction

P / partially_solved | AIM 研讨会问题汇编

**Censoring for the Potts model.** Consider the \(q\)-state Potts model (say with \(q=3\) for concreteness) and start from all green. Is it the case that deterministically censoring a sequence of spin flips can only increase the total variation distance to stationarity?

来源字段：http://aimpl.org/markovmixing/1/

15445 | OWR-15218-005

### Exact Smile Calibration in Path-Dependent Volatility Models

P / partially_solved | Oberwolfach 报告问题汇编

For a path-dependent volatility model, can the volatility function $\sigma$ and mean-reversion parameter $\lambda$ be chosen to fit the market volatility smile exactly? Does the calibrated model reproduce features such as large negative short-term forward skews?

来源字段：https://doi.org/10.4171/owr/2017/13

15446 | AIM-PROBABILITY-0107

### Exact special cases and a defect reduction for the chi-star change-of-variables problem

P / partially_solved | AIM 研讨会问题汇编

**Question.** Does the change of variables formula for \(\chi\) also hold for \(\chi^*\)?

The nearby questions fix the intended context. The preceding item asks whether conjugate variables lie in the \(L^2\)-closure of cyclic gradients. The following item asks for a path-process change-of-variables rule for a causal free SDE. Thus the question concerns the multivariable microstates formula for an invertible noncommutative functional calculus, not merely the elementary scalar identity for one classical density.

In the standard analytic version, \(X=(X_1,\ldots,X_n)\) is a self-adjoint tuple, \(F=(F_1,\ldots,F_n)\) and \(G\) are mutually inverse tuples of convergent self-adjoint noncommutative power series, and \[ \mathscr JF(X)=\bigl(\partial_jF_i(X)\bigr)_{i,j} \in M_n\bigl(M\bar\otimes M^{\mathrm{op}}\bigr) \] is the free Jacobian. Voiculescu's microstates formula is \[ \chi(F(X)) =\chi(X)+ (\operatorname{Tr}_n\otimes\tau\otimes\tau^{\mathrm{op}}) \bigl(\log|\mathscr JF(X)|\bigr). \tag{1.1} \] Here \(\operatorname{Tr}_n\) is the **unnormalized** matrix trace. Equivalently, if the Fuglede--Kadison determinant is defined with the normalized trace \(\operatorname{tr}_n\), the last term is \[ n\log\Delta_{\operatorname{tr}_n\otimes\tau\otimes\tau^{\mathrm{op}}} (\mathscr JF(X)). \] This normalization distinction is essential.

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15447 | OWR-15214-002

### Exact Strong Approximation Rates for CIR Processes

P / partially_solved | Oberwolfach 报告问题汇编

Let the Cox-Ingersoll-Ross process satisfy $$ X_t=x+\int_0^t(a-bX_s)\,ds+\int_0^t\sigma\sqrt{X_s}\,dW_s, $$ where $a,\sigma>0$ and $b,x\geq0$. For approximations of $X_T$ based on $N$ equidistant evaluations of $W$, is the optimal absolute-mean convergence order throughout the full parameter range equal to $$ \min\left\{\frac{2a}{\sigma^2},1\right\} $$?

来源字段：https://doi.org/10.4171/owr/2017/9

15448 | AIM-PROBABILITY-0051

### Exact total-variation cutoff profile for random transpositions

S / solved | AIM 研讨会问题汇编

**Cutoff profile of random transpositions.** Can we obtain the cutoff profile for the uniform random transposition walk on \(S_n\)?

来源字段：http://aimpl.org/markovmixing/5/

15449 | AIM-PROBABILITY-0007

### Exact training oracles from sparse RBM topology

P / partially_solved | AIM 研讨会问题汇编

**Efficient methods to train RBMs with fixed, sparse connectivity.** What are the most efficient learning algorithms for training sparsely connected RBMs?

来源字段：http://aimpl.org/boltzmann/3/

15450 | AIM-PROBABILITY-0032

### Exact tree-uniqueness translation and finite-size optimal mixing transfer

S / solved | AIM 研讨会问题汇编

Take a (random) $d$-regular graph, and consider the anti-ferromagnetic Ising model, i.e. $J_{ij}=-1$ for all $i,j$. Can we show fast mixing $O(n\log n)$ up to $$ (d-1)\tanh(\beta)<1\,? $$

来源字段：http://aimpl.org/markovmixing/1/

15451 | AIM-PROBABILITY-0165

### Exceptional polygon models, a counting obstruction, and a folding certificate

P / partially_solved | AIM 研讨会问题汇编

**Problem 6.2 (A. Zelevinsky).** In the classical types (A, B, C, and D), the associahedron \(\Delta_W\) has a visually transparent realization in terms of regular plane polygons and their triangulations. Find a similar interpratation in the exceptional types.

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15452 | OWR-1061-006

### Exceptional Regenerative Composition Structures

O / open | Oberwolfach 报告问题汇编

For sampling distributions induced by coalescents with simultaneous multiple collisions, characterized by measures $\Lambda_j$ on the simplices $\Delta_j$, are all corresponding composition structures nonregenerative except when $\Lambda_j=0$ for every $j\geq2$ and $\Lambda_1$ is concentrated at $0$ or at $1$?

来源字段：https://doi.org/10.4171/owr/2005/40

15453 | AMR-095-0003

### Exchangeability in the mean-zero exclusion process

P / partially_solved | AMR 问题清单汇编

For the exclusion process on $\mathbb{Z}^d$ with translation-invariant kernel $p(x,y)=p(y-x)$ and zero mean $\sum_xxp(x)=0$, prove that every stationary measure is exchangeable.

背景链接：https://web.archive.org/web/20150919235903id_/http://www.math.ucla.edu/~tml/open.pdf

15454 | AMR-096-0015

### Excursion counts in a random Eulerian circuit on a complete graph

O / open | AMR 问题清单汇编

On the bidirected complete $n$-vertex graph, is the expected number of length-$i$ excursions in a uniform Eulerian circuit asymptotic to $e^{-i/n}$?

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15455 | OWR-11695865-009

### Exhaustiveness of Threshold Scenarios for Colored Gaussian Graphical Models

O / open | Oberwolfach 报告问题汇编

Let $G$ be a colored Gaussian graphical model with $I(G,n+1)=0$ and nonzero rank-$n$ elimination ideal $I(G,n)$. Must exactly one of the following mechanisms determine its thresholds: a nonvanishing signed sum of squares in $I(G,n)$, forcing $\operatorname{WMLT}(G)=\operatorname{MLT}(G)=n+1$; or a positive-definite covariance matrix whose sufficient statistics lie in $V(I(G,n))$, forcing $\operatorname{WMLT}(G)=n<\operatorname{MLT}(G)=n+1$?

来源字段：https://doi.org/10.4171/owr/2022/55

15456 | OWR-17474-002

### Existence Criteria for $g$-Measures

O / open | Oberwolfach 报告问题汇编

What conditions on the size or shape of the state set $D$ ensure the existence of a $g$-measure?

来源字段：https://doi.org/10.4171/owr/2020/11

15457 | OWR-2654828-002

### Existence of Gaussian Persistence Exponents

P / partially_solved | Oberwolfach 报告问题汇编

For a Gaussian stationary process with persistence probability $P_f(N)$, what conditions ensure the existence of the exact asymptotic constant $\theta_f=-\lim_{N\to\infty}N^{-1}\log P_f(N)$?

来源字段：https://doi.org/10.4171/owr/2020/32

15458 | OWR-12481-005

### Existence of Random-Graph Coloring Growth Rates

O / open | Oberwolfach 报告问题汇编

For every integer $k\geq3$ and every average degree $d>0$, does the limit

$$\phi(d)=\lim_{n\to\infty}\mathbb E\bigl[\#\{k\text{-colorings of }G(n,m)\}\bigr]^{1/n}$$

exist?

来源字段：https://doi.org/10.4171/owr/2013/18

15459 | AMR-096-0035

### Expected length of a SIRSN spanning subnetwork

O / open | AMR 问题清单汇编

For $k$ uniform random points $Z_1,\ldots,Z_k$ in a square of area $k$, prove $\mathbb{E}\operatorname{len}[\operatorname{span}(Z_1,\ldots,Z_k)]\sim\ell k$.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15460 | OWR-12697684-017

### Expected Real Roots of General Random Polynomials

P / partially_solved | Oberwolfach 报告问题汇编

别名：Unspecified Random Function Root Problem [OWR-12697684-003]

Determine the expected number $\mathbb{E}N_{n,\xi}$ of real roots for general random polynomials, or more generally for random functions, beyond the Gaussian coefficient case.

来源字段：https://doi.org/10.4171/owr/2023/1

15461 | OWR-11132-004

### Explicit Stochastic Averaging Errors in Higher Dimensions

O / open | Oberwolfach 报告问题汇编

Can one obtain explicit error estimates in higher dimensions for stochastic-averaging integrals of the form $\int_0^T X(\tau)Z(\tau\varepsilon^{-2})^q\,d\beta(\tau)$ when $X(t)\in\mathbb{R}^n$ and $n>1$?

来源字段：https://doi.org/10.4171/owr/2011/40

15462 | OWR-9790363-003

### Explosion with Infinite-Mean Offspring

O / open | Oberwolfach 报告问题汇编

Can the contact process with fitness on a Galton-Watson tree explode when the offspring variable $\xi$ satisfies $\mathbb E[\xi]=\infty$?

来源字段：https://doi.org/10.4171/owr/2022/12

15463 | AMR-099-0020

### Exponential intersection tails for loop-erased random walk

O / open | AMR 问题清单汇编

Does the law of loop-erased random walk on $\mathbb{Z}^d$ have the exponential intersection-tail property: for two independent sampled paths $\gamma_1,\gamma_2$, is $\mathbb{P}(|\gamma_1\cap\gamma_2|>n)\le\theta^n$ for some $\theta<1$?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=36

15464 | AMR-099-0022

### Exponential intersection tails in three-dimensional slabs

O / open | AMR 问题清单汇编

For a subset $S=\{(n,f(n),g(n)):n\in\mathbb{N}\}\subset\mathbb{Z}^3$, characterize the conditions on $f$ and $g$ under which $S$ supports a probability measure on infinite paths with exponential intersection tails.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=37

15465 | OWR-15427-014

### Exponential Probability Gaps Between Random Order Types

O / open | Oberwolfach 报告问题汇编

For every probability measure $\mu$ on the plane that assigns measure zero to every line, do there exist order types $\omega_1(\mu)$ and $\omega_2(\mu)$ on $n$ points and an absolute constant $c>1$ such that a random independent $n$-point sample $X$ satisfies $$\Pr[X\text{ realizes }\omega_1(\mu)]>c^n\Pr[X\text{ realizes }\omega_2(\mu)]$$?

来源字段：https://doi.org/10.4171/owr/2017/19

15466 | AMR-099-0083

### Exponential upper bound for linear-time graph covering

S / solved | AMR 问题清单汇编

For every $C<\infty$, prove that there is $c=c(C)<1$ such that, for every simple $n$-vertex graph $G$, the probability that simple random walk covers all of $G$ within $Cn$ steps is at most $c^n$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/randomplanar2.pdf#page=9

15467 | OWR-16416-002

### Extensive Amenability of Wobbling-Group Actions

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Wobbling-Group Nonamenability Conjecture [OWR-16416-003]

Is the action of the wobbling group $W(\mathbb Z^d)$ on $\mathbb Z^d$ extensively amenable? Equivalently, for every symmetric finitely supported probability measure $\mu$ on $W(\mathbb Z^d)$, every $\varepsilon>0$, and all sufficiently large $n$, do the inverted orbits $O_n$ satisfy

$$\mathbb P(\lvert O_n\rvert<\varepsilon n)\geq e^{-\varepsilon n}$$

always hold?

来源字段：https://doi.org/10.4171/owr/2018/41

15468 | AMR-099-0067

### External DLA growth exponent on a hierarchical graph

O / open | AMR 问题清单汇编

On the three-branch hierarchical graph $G_n$ described in Section 9.3, launch external-DLA particles from the sink until a particle settles at the sink, and let $P_n$ be the number launched. Determine $$\lim_{n\to\infty}\frac1n\log\mathbb{E}[P_n].$$

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=16

15469 | AIM-PROBABILITY-0059

### External-source reduction and finite-N response bounds

P / partially_solved | AIM 研讨会问题汇编

The exact AIM record asks to study the Hermitian external-source ensemble

\[ d\mathbb P_{N,A}(M)=\frac{1}{Z_N(A)} \exp\{-N\operatorname{Tr}(V(M)-AM)\}\,dM, \tag{1} \]

where \(N\) is even and

\[ A=aJ,\qquad J=\operatorname{diag}(I_{N/2},-I_{N/2}), \tag{2} \]

so that the eigenvalues \(a\) and \(-a\) have equal multiplicity. It then mentions a source with distinct eigenvalues \(\alpha_1,\ldots,\alpha_j\), multiplicities \(n_1,\ldots,n_j\), and limits \(n_r/N\to c_r\).

The phrase “study” is broad. The official 2012 AIM workshop report makes the intended direction more precise: remove the assumption that \(V\) is even in the Riemann--Hilbert analysis and obtain universal local kernels in the bulk, at edges, and at critical points. This attempt does not silently replace that difficult local question by a global one. It records the current conditional global theory, gives the exact multiple-orthogonal-polynomial reduction, and proves finite-\(N\) identities and bounds which hold without reflection symmetry.

The source also omits a necessary hypothesis: a “general polynomial” does not always define a probability measure. Throughout, \(V\) is real and confining. For a nonconstant real polynomial this means precisely that its degree is even and its leading coefficient is positive.

来源字段：http://aimpl.org/vectorequilib/2/

15470 | OWR-2654828-006

### Extinction Criteria for the Derrida-Retaux Process

O / open | Oberwolfach 报告问题汇编

For the continuous-time Derrida-Retaux process with initial distribution $\mu_0$ on $\mathbb R_+$, find necessary and sufficient conditions on $\mu_0$ for $F_\infty(\mu_0)=0$. Does $F_\infty(\mu_0)=0$ imply that $X_t\to0$ in probability as $t\to\infty$?

来源字段：https://doi.org/10.4171/owr/2020/32

15471 | OWR-14224-004

### Extremal Difficulty of Tree Flow Estimation

P / partially_solved | Oberwolfach 报告问题汇编

For least-squares estimation of noisy tree flows, is the path the hardest flow-estimation problem and the star the easiest?

来源字段：https://doi.org/10.4171/owr/2016/16

15472 | OWR-2093-006

### Face Numbers and Volumes of Poisson Zero Cells

P / partially_solved | Oberwolfach 报告问题汇编

For the zero cell $Z_0$ of a Poisson hyperplane tessellation, determine the mean number $\mathbb E f_\ell(Z_0)$ of $\ell$-dimensional faces for $\ell\geq1$. For the zero cell of a Poisson-Voronoi mosaic, determine its unknown mean volume and face numbers.

来源字段：https://doi.org/10.4171/owr/2008/47

15473 | OWR-3392-002

### Factorizations from Minimal Exchangeable Random Sequences

P / partially_solved | Oberwolfach 报告问题汇编

Does every minimal noncommutatively exchangeable random sequence with trivial tail algebra give rise to a noncommutative factorization over $\mathbb N_0$?

来源字段：https://doi.org/10.4171/owr/2009/09

15474 | OWR-13947-001

### Faithful Simulation of Rare Genetic Variants

O / open | Oberwolfach 报告问题汇编

How should rare-variant data be generated or simulated so that small-sample allele-frequency data faithfully represent rare variants, singletons, and doubletons for association-method comparisons?

来源字段：https://doi.org/10.4171/owr/2015/52

15475 | OWR-14298363-007

### False Discovery Rates in Open-Arrival Multiple Testing

P / partially_solved | Oberwolfach 报告问题汇编

What false-discovery-rate guarantees are possible in multiple-testing models where hypotheses arrive over time, for both p-values and e-values?

来源字段：https://doi.org/10.4171/owr/2024/24

15476 | OWR-4990379-007

### Fast Algorithms for Branching Brownian Unbalanced Transport

O / open | Oberwolfach 报告问题汇编

Can one design a fast algorithm for the regularized unbalanced optimal-transport problem associated with branching Brownian motion?

来源字段：https://doi.org/10.4171/owr/2021/10

15477 | AMR-096-0029

### Feasible statistic triples for SIRSNs

O / open | AMR 问题清单汇编

Determine the set of possible triples $(\Delta=\mathbb{E}D_1,\ell,p(1))$ over all scale-invariant random spatial networks.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15478 | AIM-PROBABILITY-0112

### Filtered q-Hermite word recovery and resolved length cutoff

P / partially_solved | AIM 研讨会问题汇编

The question has four related parts.

Let \(H_{\mathbb R}\) be a real Hilbert space of dimension \(n\), let \(-1<q<1\), and let \[ \Gamma_{q,n}=W^*\!\left(s_g(g)=l(g)+l(g)^*:g\in H_{\mathbb R}\right) \] act on the \(q\)-deformed Fock space.

1. Does the abstract von Neumann algebra \(\Gamma_{q,n}\) depend on \(q\)? 2. In the free case \(q=0\), recover intrinsically the tensor-word basis from the field algebra and seek a \(q\)-analogue with understandable combinatorics. 3. For generalized relations \[ l(e_i)^*l(e_j)=\sum_{s,t}t_{i,j}^{s,t}\,l(e_s)l(e_t)^*+\delta_{ij}I, \tag{1.1} \] determine when the creation-operator \(C^*\)-algebra is an extension of a Cuntz algebra by the compact operators and when the associated fields generate a type \(\mathrm{II}_1\) factor. 4. If \(Q_{\le k}\) denotes the projection onto elements \(x\) for which \(x\Omega\) has Fock length at most \(k\), is \(\|Q_{\le k}\|_{\mathrm{cb}}\) polynomially bounded in \(k\)?

The following repairs are necessary.

- The source prints tensor vectors \[ e_{i_1}^{\otimes\alpha_1}\otimes\cdots\otimes e_{i_r}^{\otimes\alpha_r}. \] The displayed condition \(i_1\ne\cdots\ne i_r\) must mean adjacent inequality \(i_a\ne i_{a+1}\), because these are the maximal constant runs of an arbitrary word. It cannot mean pairwise distinctness: the word \(1,2,1\) must be included. - Every run length is positive, \(\alpha_a\ge1\). The extraction “\(\alpha_1>0\)” is either a typesetting shorthand or a damaged subscript. - The recovered-algebra formula is \[ T_{\alpha_1}(s_0(e_{i_1}))\cdots T_{\alpha_r}(s_0(e_{i_r}))\Omega, \] not the OCR strings \(s_0(e_1)\) and \(s_0(e_q)\). The source calls the polynomials “Chebytchev”; in the monic normalization they are \(U_m(x/2)\), the Chebyshev polynomials of the second kind, rather than the standard first-kind \(T_m\). - The coefficient in (1.1) is \(t_{i,j}^{s,t}\). An arbitrary array does not automatically define a positive Fock inner product, bounded creators, or a tracial vacuum state. Such hypotheses cannot be omitted. - The source projection is a cutoff \(Q_{\le k}\), not the homogeneous projection onto length exactly \(k\). This distinction changes a linear bound into an immediate quadratic bound after summation.

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15479 | OWR-9790359-006

### Finitary-Factor Characterization of Markov Random Fields

O / open | Oberwolfach 报告问题汇编

Characterize the invariant Markov random fields that are finitary factors of independent and identically distributed processes.

来源字段：https://doi.org/10.4171/owr/2022/8

15480 | OWR-12697708-009

### Finite Surviving Components under Supercritical Reinforcement

P / partially_solved | Oberwolfach 报告问题汇编

Suppose the firing rates $p(v)$ are bounded away from zero and infinity and $\alpha>1$. Does $\lim_{t\to\infty}X_e(t)$ exist almost surely for every edge $e$, with the surviving edges forming finite connected components? For $\alpha>2$, are those components trees?

来源字段：https://doi.org/10.4171/owr/2023/12

15481 | AMR-099-0080

### Finite-dimensional distance laws of the Brownian map

P / partially_solved | AMR 问题清单汇编

For every $p\ge4$, determine the joint law of the matrix of pairwise distances among $p$ independent points sampled from the volume measure of the Brownian map.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/randomplanar2.pdf#page=5

15482 | OWR-12010-020

### Finite-Level Selmer Distributions from Isotropic Intersections

O / open | Oberwolfach 报告问题汇编

For fixed $e\geq1$, does the distribution of $\operatorname{Sel}_{p^e}(E)$ over elliptic curves $E$ equal, in the limit $n\to\infty$, the distribution of $Z\cap W$ for independently chosen random maximal isotropic direct summands $Z,W\in\operatorname{OGr}_n(\mathbb Z/p^e\mathbb Z)$?

来源字段：https://doi.org/10.4171/owr/2012/38

15483 | OWR-9790359-005

### Finite-Mean Coding Radius for Finitely Dependent Processes

O / open | Oberwolfach 报告问题汇编

Does there exist a finitely dependent invariant process that is not a finitary factor of any i.i.d. process with finite expected coding radius?

来源字段：https://doi.org/10.4171/owr/2022/8

15484 | AIM-PROBABILITY-0027

### Finite-N boundary and routing-miss Stein comparisons for JSQ approximations

P / partially_solved | AIM 研讨会问题汇编

It asks about \(N\) unit-rate exponential servers, a Poisson arrival stream, and the occupancy vector

\[ Q=(q_1,q_2,\ldots),\qquad q_i=\#\{\text{queues of length at least }i\}. \]

An arrival in the full join-the-shortest-queue (JSQ) system joins a globally shortest queue. The source asks for explicit Stein bounds between its stationary law and two approximations:

1. \(Q_2\): reject an arrival when all queues contain two jobs; 2. \(Q_3\): route an arrival to the shortest among \(k\) randomly chosen queues.

- “Arrive at a constant rate \(\lambda\)” is read as a Poisson process of **total** rate \(\lambda\), while service times are independent exponential random variables of rate one. Thus the load per server is \(\rho=\lambda/N<1\). Papers that write the total arrival rate as \(N\rho\) use the same model with different notation. - Starting from the empty state, the first modification is exactly the capacity-two JSQ chain: its closed state space has no queue longer than two, and an arrival is blocked precisely at \((q_1,q_2)=(N,N)\). This is the interpretation used below. - “\(k\) randomly chosen queues” does not say whether sampling is with or without replacement. Both versions are treated. Only sampling **without replacement** makes \(k=N\) exactly equal to JSQ at finite \(N\).

来源字段：http://aimpl.org/steinhd/1/

15485 | AIM-PROBABILITY-0068

### Finite-time tilt symmetry and the corrected Airy2 scaling

P / partially_solved | AIM 研讨会问题汇编

Compute multi-point (spatial) distribution for KPZ with various initial data. For example, with $\mathcal Z_0(x)=\delta_{x=0}$, let \[ \mathcal Z(t,x)=e^{-x^2/(2t)+t^{1/3}A_t(t^{-2/3}x)-t/24}. \] What is the distribution $F(\xi_1,\xi_2)$ such that \[ \mathbb P\bigl(A_t(x_1)\leq \xi_1,A_t(x_2)\leq \xi_2\bigr)=F(\xi_1,\xi_2)? \] Perhaps easier, show that as $t$ goes to infinity and space is scaled like $t^{2/3}x$, the process $A_t(t^{2/3}x)$ converges to the Airy$_2$ process in $x$.

来源字段：http://aimpl.org/kpzuniversality/1/

15486 | AIM-PROBABILITY-0090

### Finite-torus stationarity and the infinite-bulk gap

P / partially_solved | AIM 研讨会问题汇编

**Interacting particle systems**

Bulk of \(q\)-Whittaker \(2d\) dynamics. Invariant measure.

The AIM page was unavailable (HTTP 502 on 2026-08-12), and nearby records only confirm that this item belongs to the q-TASEP/q-Whittaker interacting-particle-system cluster of the workshop. The fragment omits the state space, update rule, boundary conditions, and meaning of “bulk.” There are at least three plausible readings:

1. **Triangular-array reading.** The continuous-time push-block dynamics on Gelfand--Tsetlin arrays from Macdonald-process theory. Its q-Whittaker measures are evolving fixed-time laws, not automatically invariant laws. 2. **Periodized bulk reading.** Transplant the same local rule away from the triangular boundary to interlacing particles on a two-dimensional torus and find a stationary measure. This is the reading most directly matched by Corwin--Toninelli (2016). 3. **Infinite-plane reading.** Construct a translation-invariant process and invariant Gibbs measure on an infinite interlacing array, or obtain these as a thermodynamic limit of tori.

The finite-torus reading was solved after the workshop. The infinite-plane reading is stronger and is not a formal corollary because a jump may push a vertically aligned string whose length grows with the volume. This report states the finite theorem precisely enough to identify the model and gives a rigorous finite-to-infinite reduction plus a height-gauge obstruction.

来源字段：http://aimpl.org/kpzuniversality/2/

15487 | OWR-14299904-001

### Finiteness and Tail Decay of Infinite-Noodle Components

O / open | Oberwolfach 报告问题汇编

Writing $C(0)$ for the component of the origin in the infinite-noodle model, is $C(0)$ almost surely finite? Does its size satisfy $\mathbb{P}(|C(0)|\geq k)=k^{-(2\sqrt{2}-1)/7+o(1)}$?

来源字段：https://doi.org/10.4171/owr/2026/2

15488 | OWR-15989-006

### Finiteness of Two-Dimensional Sandpile Avalanches

O / open | Oberwolfach 报告问题汇编

For the infinite-volume Abelian sandpile model on $\mathbb{Z}^2$ with stationary measure $\nu$, are avalanches finite almost surely? A sufficient route is to prove that the number of topplings at the addition site is finite $\nu$-almost surely.

来源字段：https://doi.org/10.4171/owr/2018/4

15489 | AIM-PROBABILITY-0093

### Fixed-area excursions select Ferrari-Spohn scales, not the literal KPZ scales

P / partially_solved | AIM 研讨会问题汇编

Consider a Brownian excursion on time interval \([-N,N]\) condition on fixed area \(N\). Take \(1/3, 2/3\) scalings around the edge. Do we get an “Airy-like” limiting fluctuation?

来源字段：http://aimpl.org/kpzuniversality/2/

15490 | AIM-PROBABILITY-0067

### Flat and stationary KPZ statistics with an exact drift audit

P / partially_solved | AIM 研讨会问题汇编

Statistics of KPZ equation in \(1+1\) dimension: Compute statistics for different initial data including \(\mathcal Z_0(x)=1\) (flat) and \(\mathcal Z_0(x)=e^{B(x)}\) for \(B(x)\) a two-sided Brownian motion (equilibrium).

来源字段：http://aimpl.org/kpzuniversality/1/

15491 | OWR-705-002

### Floquet Exponents and Invariant Measures of Schrödinger Operators

O / open | Oberwolfach 报告问题汇编

To what extent does the Floquet exponent of a Schrödinger operator characterize its invariant probability measure?

来源字段：https://doi.org/10.4171/owr/2004/22

15492 | OWR-1598-006

### Fluctuation Limits for Exclusion Processes

P / partially_solved | Oberwolfach 报告问题汇编

For the totally asymmetric simple exclusion process (TASEP), what are the limiting laws of the fluctuations and the limiting fluctuation process as $t\to\infty$, for the various classes of initial conditions?

来源字段：https://doi.org/10.4171/owr/2007/42

15493 | OWR-12584-004

### Fluctuation Order in First-Passage Percolation

O / open | Oberwolfach 报告问题汇编

Determine the order of the fluctuations of the first-passage-percolation metric

$$\rho(x,y)=\inf_{(u_j):x\leadsto y}\sum_j W(u_j)$$

on $\mathbb Z^d$, where the vertex or edge weights $W(u_j)$ are random and nonnegative.

来源字段：https://doi.org/10.4171/owr/2013/36

15494 | AMR-099-0072

### Fluctuations and efficient algorithms in first-passage percolation

P / partially_solved | AMR 问题清单汇编

For i.i.d. first-passage percolation on $\mathbb{Z}^2$, prove or disprove that boundary fluctuations have a Tracy-Widom limit and that the variance of the passage time from $0$ to $(n,0)$ has order $n^{2/3}$. Determine optimal computational bounds for finding a shortest path or estimating its length.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=9

15495 | AMR-099-0066

### Fluctuations in recursive hierarchical first-passage percolation

P / partially_solved | AMR 问题清单汇编

For the hierarchical first-passage distances $D_n$ satisfying $D_n\stackrel d=D_{n-1}+\min(D'_{n-1},D''_{n-1})$, determine concentration around the mean and the lower-order asymptotic terms, analogous to the minimal position in branching random walk.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=15

15496 | OWR-2093-002

### Fluctuations of Poisson-Line Cell Boundary Paths

O / open | Oberwolfach 报告问题汇编

Let $C_n$ be the Poisson-line-process cell containing $(-n/2,0)$ and $(n/2,0)$, and follow $\partial C_n$ clockwise between those points. What are the fluctuations of the path length and its maximum lateral deviation from the Euclidean segment? How far can the true network geodesic differ from the clockwise or counterclockwise boundary paths?

来源字段：https://doi.org/10.4171/owr/2008/47

15497 | AMR-099-0071

### Foliations of Euclidean space by Brownian paths

O / open | AMR 问题清单汇编

For which dimensions $d$ can $\mathbb{R}^d$ be partitioned into pairwise disjoint curves, each of which has the law or geometric regularity of a Brownian path? Give a rigorous construction or prove impossibility.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=5

15498 | OWR-1703875-002

### Force-Flux Gradient Structures for Finite-State Markov Chains

P / partially_solved | Oberwolfach 报告问题汇编

Do analogous force-flux, large-deviation, and gradient-flow structures exist for finite-state Markov chains, including in the irreversible case?

来源字段：https://doi.org/10.4171/owr/2020/29

15499 | OWR-1327-001

### Force-Induced Phase Transitions in Self-Attracting Polymers

O / open | Oberwolfach 报告问题汇编

For the force-extended partially directed self-attracting polymer model on $\mathbb{Z}^2$, whose generating function $H(x,y,z)$ records contacts, span in the force direction, and length, determine the order of its force-induced compact-to-expanded phase transition.

来源字段：https://doi.org/10.4171/owr/2006/41

15500 | AIM-PROBABILITY-0131

### Fourier shadows and the quartic tensor-free defect

P / partially_solved | AIM 研讨会问题汇编

**Q:** Given $x_1,\ldots,x_k$ and $y_1,\ldots,y_k$ in a vNa such that $\{x_1,\ldots,x_k\}$ is tensor-independent of $\{y_1,\ldots,y_k\}$ and such that $\mu_{x_1,\ldots,x_k},\nu_{y_1,\ldots,y_k}$ are freely infinitely divisible, we can apply the Fourier transform to get the power-series of the classical convolution of $\mu_{x_1,\ldots,x_k}$ and $\nu_{y_1,\ldots,y_k}$. How do such power-series relate to the noncommutative power series obtained from free convolution? (In other words how does the set of classically obtainable power-series relate to the set of freely obtainable power-series?)

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15501 | OWR-1458-001

### Free Infinite Divisibility of Semicircle Scale Mixtures

P / partially_solved | Oberwolfach 报告问题汇编

Are the scale mixtures of semicircle laws arising from subordinated matrix Brownian motion freely infinitely divisible? In particular, is the Gaussian distribution freely infinitely divisible?

来源字段：https://doi.org/10.4171/owr/2007/07

15502 | OWR-2654831-001

### Free-Boundary Dimer Convergence to Arc Loop Ensembles

O / open | Oberwolfach 报告问题汇编

For every monomer fugacity $z>0$, does the collection $\mathcal C$ of boundary-touching curves in the free-boundary dimer model on the upper half-plane converge in the scaling limit to $\operatorname{ALE}_{-\lambda,\lambda}$, with loops conditionally independent and distributed as $\operatorname{CLE}_4$ in the connected components of its complement?

来源字段：https://doi.org/10.4171/owr/2020/35

15503 | AMR-099-0044

### From a large percolation component to a giant component on expanders

P / partially_solved | AMR 问题清单汇编

Let $G$ be a bounded-degree expander and suppose some vertex $v$ satisfies $$\mathbb{P}_{1/2}\!\left(\operatorname{diam}(K_v)>\tfrac12\operatorname{diam}(G)\right)>\tfrac12,$$ where $K_v$ is its percolation cluster. Prove that $1/2$-percolation on $G$ contains a giant component with high probability.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=86

15504 | AIM-PROBABILITY-0145

### Fuss-Catalan frameworks and a colored exact-flat Boolean model

P / partially_solved | AIM 研讨会问题汇编

Find the natural Fuss--Catalan generalizations of noncrossing partitions, nonnesting partitions, and the cluster complex; generalize the numerical, bijective, and algebraic questions of Problems 1.1 and 1.2; explain the common Fuss--Catalan and Fuss--Narayana enumerations; extend Chapoton's \(M\)-, \(F\)-, and \(H\)-triangles; and identify algebraic or application-driven frameworks behind the objects.

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15505 | OWR-2053-017

### Følner Growth and Trivial Poisson Boundary

O / open | Oberwolfach 报告问题汇编

If $G$ is a finitely generated group whose Følner function satisfies $\operatorname{Fol}_G(n)\leq\exp(n)$, is the Poisson boundary of its simple random walk trivial?

来源字段：https://doi.org/10.4171/owr/2008/42

15506 | OWR-14298363-003

### Game-Theoretic and Measure-Theoretic Expectations

P / partially_solved | Oberwolfach 报告问题汇编

How do game-theoretic expectations and supermartingales relate to their measure-theoretic counterparts?

来源字段：https://doi.org/10.4171/owr/2024/24

15507 | OWR-13856-002

### Gamma Profiles of Waves Entering a Condensate

O / open | Oberwolfach 报告问题汇编

In the stochastic house-of-cards condensation model, suppose the mutant-fitness distribution satisfies $\mu(1-\varepsilon,1)=\varepsilon^\alpha\ell(\varepsilon)$, where $\alpha>0$ and $\ell$ is slowly varying at zero. If $\Xi_t$ is the empirical fitness distribution and $\gamma(\beta)$ is the condensate mass, is

$$\lim_{t\to\infty}\Xi_t\!\left(1-\frac{x}{t},1\right)=\frac{\gamma(\beta)}{\Gamma(\alpha)}\int_0^x e^{-y}y^{\alpha-1}\,dy$$?

来源字段：https://doi.org/10.4171/owr/2015/35

15508 | OWR-4082-003

### Gaussian Central Limits for Stationary Sequences

P / partially_solved | Oberwolfach 报告问题汇编

Let $(X_i)_{i\geq1}$ be a stationary standard Gaussian sequence with covariance sequence $(r_k)$, and let $g$ be a measurable function with finite variance under the standard Gaussian law. Characterize the conditions on $(r_k)$ and $g$ under which $n^{-1/2}\sum_{i=1}^n g(X_i)$ converges in distribution to a normal law.

来源字段：https://doi.org/10.4171/owr/2009/39

15509 | AIM-PROBABILITY-0058

### Gaussian source-merger reduction for a two-matrix external-source model

P / partially_solved | AIM 研讨会问题汇编

**$2+1/2$ random matrix models**

A general question about Riemann-Hilbert techniques to study the two-matrix plus external source model.

来源字段：http://aimpl.org/vectorequilib/1/

15510 | OWR-10252936-003

### Gaussian-Free-Field Maxima on Percolation Clusters

O / open | Oberwolfach 报告问题汇编

Consider the discrete Gaussian free field on the infinite cluster of two-dimensional Bernoulli bond percolation with parameter $p>1/2$. Does its centered maximum over a box of side length $N$ converge in distribution as $N\to\infty$ for every $p>1/2$, rather than only for $p$ sufficiently close to $1$?

来源字段：https://doi.org/10.4171/owr/2022/27

15511 | AIM-PROBABILITY-0056

### Gaussian-side decoupling and free-convolution density

P / partially_solved | AIM 研讨会问题汇编

Determine eigenvalue information, especially the limiting eigenvalue density of \(M_1\), in the Hermitian two-matrix model

\[ Z_N^{-1}\exp\{-N\operatorname{Tr}[V(M_1)+W(M_2)+tM_1M_2]\}\,dM_1\,dM_2. \tag{1} \]

Its first case is \(V(x)=x^2/2\) and a general, not necessarily even, polynomial \(W\) of even degree. Its broader case takes \(V\) to be an even quartic and \(W\) to have no symmetry. This report uses the standard interpretation that \(M_1,M_2\) are \(N\times N\) Hermitian matrices and \(t\in\mathbb R\). “Even degree (no symmetry)” is understood as allowing odd lower-order terms, not as saying \(W(-x)=W(x)\).

There is an omitted normalization condition. A polynomial of even degree need not be confining: its leading coefficient must be positive. In the quadratic case the interaction imposes a sharper condition described below. Without it, (1) is not a probability density.

来源字段：http://aimpl.org/vectorequilib/1/

15512 | OWR-16937-004

### Generalization across Repeatedly Constructed Test Sets

P / partially_solved | Oberwolfach 报告问题汇编

What does good performance on a held-out test set imply about performance on a new test set created from the same data source and cleaning protocol, and how can such generalization be characterized?

来源字段：https://doi.org/10.4171/owr/2019/25

15513 | AIM-PROBABILITY-0166

### Generalized cluster complexes: solved topology and a simplicial thickening obstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 6.3. (S. Fomin)** Conjecture: The Fomin--Reading generalization of the associahedron \(\Delta_W^{(k)}\) (see Problem 1.3) is Cohen--Macaulay, and is homotopy equivalent to a wedge of \[ \operatorname{Cat}^{(k-1)}(W)=\prod_{i=1}^{n}\frac{(k-1)h+e_i+1}{e_i+1} \] spheres. This has been proved by Eleni Tzanaki in types \(A\) and \(B\) using shelling methods [51]. Moreover, \(\Delta_W^{(k)}\) seems to be the skeleton of a polytopal manifold. Can this be realized geometrically? (This generalizes Problem 6.1 above.)

**Remark (V. Reiner).** Is the Fomin--Reading complex \(k\)-Cohen--Macaulay in the sense of Baclawski?

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15514 | OWR-11695865-006

### Generic Identifiability of Linear Compartment Models

O / open | Oberwolfach 报告问题汇编

For which linear compartment models is the coefficient-map Jacobian generically full rank, equivalently, when is the model generically locally identifiable?

来源字段：https://doi.org/10.4171/owr/2022/55

15515 | OWR-13687-003

### Generic Infinite Index Subgroups of Integral Special Linear Groups

P / partially_solved | Oberwolfach 报告问题汇编

For $G=\mathrm{SL}_n(\mathbb{Z})$ and $B_X=\{g\in G:\lVert g\rVert\leq X\}$, does the probability that $\langle g_1,g_2\rangle$ has infinite index tend to 1 as $X\to\infty$ when $g_1,g_2$ are chosen uniformly from $B_X$?

来源字段：https://doi.org/10.4171/owr/2015/30

15516 | AMR-099-0078

### Geodesics in Gaussian-free-field random metrics

P / partially_solved | AMR 问题清单汇编

On the $n\times n$ grid with a Gaussian free field with no boundary conditions, give every vertex length equal to the exponential of the field. If $\gamma_1(n)$ and $\gamma_2(n)$ are shortest paths joining the two top and the two bottom corners, prove that $\mathbb{P}(\gamma_1(n)\cap\gamma_2(n)\ne\varnothing)>c$ uniformly in $n$. Construct the metric and geodesic scaling limits, determine geodesic dimension and height concentration, and identify any SLE parameter of the limiting paths.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/randomplanar2.pdf#page=3

15517 | OWR-14298589-010

### Geometric Chapuy Bijections for Random Surfaces

O / open | Oberwolfach 报告问题汇编

Construct a geometric adaptation of Chapuy’s bijection between one-face maps and decorated trees that directly produces random surfaces, and explain the associated enumerative identities geometrically.

来源字段：https://doi.org/10.4171/owr/2024/41

15518 | AIM-PROBABILITY-0096

### Geometric-RSK entrance saddle and flag-mirror superpotential

S / solved | AIM 研讨会问题汇编

Exact solvability

Investigate the relationship between the tropical RSK correspondence and mirror symmetry: The entrance law for the Markov process which arises from adding columns under the tropical RSK correspondence arises via a critical point calculation.

来源字段：http://aimpl.org/kpzuniversality/2/

15519 | OWR-1543-002

### Geometry of Directed-Polymer Concentration Points

O / open | Oberwolfach 报告问题汇编

For the directed-polymer endpoint measures in an i.i.d. random environment, characterize geometrically the macroscopic $\varepsilon$-atoms at which the low-temperature polymer measure concentrates.

来源字段：https://doi.org/10.4171/owr/2007/27

15520 | OWR-16931-026

### Giant Components in Nonlinear Degree-Dependent Random Graphs

O / open | Oberwolfach 报告问题汇编

Determine the emergence and size of the giant component in the degree-dependent random-graph processes $G^f_{n,m}$ and $G^{f,*}_{n,m}$ when the attachment function $f$ is genuinely nonlinear.

来源字段：https://doi.org/10.4171/owr/2019/19

15521 | AIM-PROBABILITY-0099

### Gibbs limits depend on the resampling kernel

P / partially_solved | AIM 研讨会问题汇编

**Gibbs line ensembles.** Show tightness and limiting Gibbs property for line ensembles besides the Airy line ensemble (e.g. Bessel process, Peary process, Sine process).

来源字段：http://aimpl.org/kpzuniversality/2/

15522 | OWR-4796-001

### Group Effects in Shape and Deformation Data

O / open | Oberwolfach 报告问题汇编

Given $N$ subjects in two or more groups, each represented by an image or shape, are the shapes group-dependent, and if so, where are they most affected? Given baseline images $B_i$ and follow-up images $F_i$ for each subject, is the deformation from $B_i$ to $F_i$ group-dependent, and if so, where?

来源字段：https://doi.org/10.4171/owr/2011/07

15523 | AMR-096-0024

### Growth exponents in the balanced city-growth regime

O / open | AMR 问题清单汇编

In the balanced regime $0<\alpha<1$ and $\beta>2\alpha$, prove that the upper and lower growth exponents for influence and city population all equal $\beta/(2-2\alpha+\beta)$ and those for nearest-city distance equal $(\alpha-1)/(2-2\alpha+\beta)$.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15524 | AMR-099-0050

### Half-density percolation on transient disk triangulations

O / open | AMR 问题清单汇编

Let $G$ be the one-skeleton of a bounded-degree triangulation of an open disk. If $G$ is transient, prove that Bernoulli site percolation with parameter $1/2$ has an infinite cluster almost surely.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/resistance_percolation.pdf#page=2

15525 | AMR-099-0034

### Half-plane percolation for invariant FKG processes

O / open | AMR 问题清单汇编

Let $X$ be a finite-energy, translation-invariant percolation process on $\mathbb{Z}^2$ satisfying the FKG inequality. If $X$ percolates almost surely, must it also percolate almost surely in each half-plane?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=57

15526 | AIM-PROBABILITY-0102

### Heavy-tail scales for intermediate-disorder polymers

P / partially_solved | AIM 研讨会问题汇编

**Polymer universality.** Determine limiting behavior of polymer without a sixth moment, and with $\beta\rightarrow0$ at the correct rate to have a limit.

来源字段：http://aimpl.org/kpzuniversality/2/

15527 | OWR-4791-032

### High Influence Small Sets in Boolean Functions

O / open | Oberwolfach 报告问题汇编

Let $f:\{0,1\}^n\to\{0,1\}$ be a Boolean function, and call the probability that $f$ remains undetermined after randomly assigning variables outside $S$ the influence of $S$. If the expectation of $f$ is bounded away from 0 and 1, is there a set $S$ with $|S|\leq n/3$ and influence at least $1-c^n$ for some absolute constant $0<c<1$?

来源字段：https://doi.org/10.4171/owr/2011/01

15528 | OWR-11101922-001

### High-Density Aperiodic Lorentz Gases

O / open | Oberwolfach 报告问题汇编

What is the behavior of an aperiodic Lorentz gas in the high-density regime?

来源字段：https://doi.org/10.4171/owr/2022/41

15529 | OWR-10252933-002

### High-Dimensional Limits of Projection-Pursuit Laws

P / partially_solved | Oberwolfach 报告问题汇编

For $n$ independent standard Gaussian points in $\mathbb R^d$ with $n/d\to\alpha\in(0,\infty)$ and fixed $m$, characterize the set $\mathcal F_{m,\alpha}$ of limiting empirical laws of projections onto $m$-dimensional subspaces.

来源字段：https://doi.org/10.4171/owr/2022/25

15530 | OWR-15215-007

### High-Dimensional Volumes of Random Simplices

P / partially_solved | Oberwolfach 报告问题汇编

Determine, as $d\to\infty$, the expected volume of a random $(d+1)$-simplex in a regular simplex, a cube, and other standard convex bodies. Determine the associated normalized asymptotic quantities.

来源字段：https://doi.org/10.4171/owr/2017/10

15531 | AMR-099-0055

### High-intensity hyperbolic Voronoi crossing limits

O / open | AMR 问题清单汇编

In the Poincaré disk, sample a Poisson process of intensity $\lambda$ with respect to hyperbolic area, form its Voronoi tessellation, and color cells black or white independently with probability $1/2$. For four boundary points defining opposite arcs $A$ and $C$, prove that the annealed black-crossing probability from $A$ to $C$ converges as $\lambda\to\infty$, that the limit is Cardy's conformally invariant crossing probability, and that it is bounded away from $0$ and $1$ for nontrivial arcs.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/resistance_percolation.pdf#page=4

15532 | OWR-1463-002

### Higher-Order Correlations at Coincident Energies

O / open | Oberwolfach 报告问题汇编

For the higher-order correlation measures of the random Schrödinger model, determine the behavior of the density near and on the coincident planes $E_j=E_k$ for $j\neq k$.

来源字段：https://doi.org/10.4171/owr/2007/12

15533 | AIM-PROBABILITY-0042

### Hitting a moving trajectory on an Eulerian digraph

P / partially_solved | AIM 研讨会问题汇编

**Hitting time of trajectory.** Consider a nonreversible (say \(2/3\)--\(1/3\)) random walk on an Eulerian digraph \((V,E)\). Prove that the first time, \(\tau_{(X_t^{(y)})}\), the random walk hits a moving target \(X_t^{(y)}\) with \(X_0=y\) satisfies \[ \max_{x,y}\mathbb E_x[\tau_{(X_t^{(y)})}]\leq |E|\,|V|. \]

来源字段：http://aimpl.org/markovmixing/3/

15534 | OWR-15175-003

### Hydrodynamic Limits for Non-Gradient Crystal-Lattice Systems

O / open | Oberwolfach 报告问题汇编

Can the hydrodynamic limit be derived for a non-gradient interacting particle system on a crystal lattice?

来源字段：https://doi.org/10.4171/owr/2016/54

15535 | OWR-757-003

### Hydrodynamics of Ergodic Kob-Andersen Models

P / partially_solved | Oberwolfach 报告问题汇编

For every ergodic Kob-Andersen model at density $\rho<1$, does hydrodynamic behavior emerge on sufficiently large space and time scales?

来源字段：https://doi.org/10.4171/owr/2004/43

15536 | AMR-099-0075

### Hyperbolic local limits of random high-genus quadrangulations

S / solved | AMR 问题清单汇编

Take a uniform quadrangulation with $N$ faces conditioned to have genus $CN$, where $0<C<1/4$. Prove that its rooted local limit is the stochastic hyperbolic infinite quadrangulation constructed from a labeled supercritical Galton-Watson tree, with the predicted mean degree $6/(1-4C)$ and diverging local injectivity radius.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=11

15537 | AMR-100-0012

### Ibragimov's central limit conjecture for $\phi$-mixing sequences

O / open | AMR 问题清单汇编

Let $(X_n)_{n\in\mathbb{Z}}$ be a centered strictly stationary sequence with $\mathbb{E}[X_0^2]<\infty$. For $k\ge1$, define $$\phi_X(k)=\sup_m\sup\bigl\{|\mathbb{P}(B\mid A)-\mathbb{P}(B)|: A\in\sigma(X_j:j\le m),\ \mathbb{P}(A)>0,\ B\in\sigma(X_j:j\ge m+k)\bigr\},$$ and assume $\phi_X(k)\to0$. If $S_n=\sum_{j=1}^n X_j$ and $\operatorname{Var}(S_n)\to\infty$, prove that $$\frac{S_n}{\sqrt{\operatorname{Var}(S_n)}}\xrightarrow{d}\ N(0,1).$$

背景链接：https://mathoverflow.net/a/129732

15538 | OWR-16937-003

### Identifiable Longitudinal Causal Mediation

P / partially_solved | Oberwolfach 报告问题汇编

Develop identifiable causal mediation effects for a longitudinal mediator process and a survival outcome by separating treatment paths with an augmented directed acyclic graph, avoiding the ill-defined nested counterfactual and post-treatment-confounding problems.

来源字段：https://doi.org/10.4171/owr/2019/25

15539 | OWR-2093-005

### Inference for Log-Lévy-Driven Cox Processes

O / open | Oberwolfach 报告问题汇编

Develop statistical inference methods for log-Lévy-driven Cox processes, including inference for their driving intensity and parameters.

来源字段：https://doi.org/10.4171/owr/2008/47

15540 | OWR-2093-004

### Inference from Randomly Oriented Tomographic Marginals

P / partially_solved | Oberwolfach 报告问题汇编

Let $\rho$ be an unknown centered bounded probability density supported on a compact subset of $\mathbb R^3$. Given finitely many two-dimensional marginals of $\rho$ in random unknown orientations and $n$ independent realizations of the resulting stochastic Radon transform, what can be inferred about $\rho$?

来源字段：https://doi.org/10.4171/owr/2008/47

15541 | OWR-15997-001

### Inference in Proportional High-Dimensional Logistic Regression

P / partially_solved | Oberwolfach 报告问题汇编

Develop a rigorous theory of logistic regression when the sample size and number of covariates grow proportionally, including the bias and variance of the maximum-likelihood estimator and the limiting distributions of likelihood-ratio statistics.

来源字段：https://doi.org/10.4171/owr/2018/12

15542 | OWR-4337-001

### Infinite Cycles and Bose-Einstein Condensation

O / open | Oberwolfach 报告问题汇编

For the interacting spatial random-permutation model in dimension $d\geq3$, is the loss of probability mass into infinitely long cycles for an optimizer $P$ of $\chi^{(\leq)}(\beta,\rho)$ equivalent to

$$\chi^{(\leq)}(\beta,\rho)\neq\chi^{(=)}(\beta,\rho)$$

and to the appearance of Bose-Einstein condensation? More specifically, for fixed $\beta$, is this loss absent for sufficiently small density $\rho$ and present for sufficiently large $\rho$?

来源字段：https://doi.org/10.4171/owr/2010/24

15543 | AMR-099-0043

### Infinite-cluster intersections with vertical fibers

O / open | AMR 问题清单汇编

Let $G$ be an infinite graph with $p_c(G)=1$. For Bernoulli percolation on $G\times\mathbb{Z}$, must every infinite cluster intersect each fiber $\{v\}\times\mathbb{Z}$ that it meets in infinitely many vertices almost surely?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=76

15544 | AMR-099-0054

### Infinitely many clusters at half density on transient disk triangulations

O / open | AMR 问题清单汇编

Let $G$ be the one-skeleton of a bounded-degree transient triangulation of an open disk. Prove that Bernoulli site percolation with parameter $1/2$ has infinitely many infinite clusters almost surely.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/resistance_percolation.pdf#page=4

15545 | AIM-PROBABILITY-0097

### Initial-data universality requires the right coupling

P / partially_solved | AIM 研讨会问题汇编

It asks whether two possibly random, mutually independent initial height functions with the same hydrodynamic and fluctuation limits have the same later fluctuation limit, expected to be the KPZ fixed-point semigroup applied to the limiting initial fluctuations.

The record literally writes the hydrodynamic hypothesis as

\[ \epsilon^{-1}h_i(\epsilon^{-1}x;0)\longrightarrow \bar h(x;0), \]

and its fluctuation hypothesis and desired conclusion as

\[ \epsilon^{1/2}\left[h_i(\epsilon^{-1}x;0) -\epsilon^{-3/2}\bar h(\epsilon^{1/2}x;0)\right] \longrightarrow \widetilde h(x;0), \]

\[ \epsilon^{1/2}\left[h_i(\epsilon^{-1}x;\epsilon^{-3/2}t) -\epsilon^{-3/2}\bar h(\epsilon^{1/2}x;t)\right] \longrightarrow \widetilde h(x;t). \]

The first display is almost certainly malformed as written. A microscopic exclusion height has order \(\epsilon^{-1}\) on Euler distance \(\epsilon^{-1}\), so the standard Euler rescaling multiplies it by \(\epsilon\), not \(\epsilon^{-1}\); Euler time is also normally rescaled. This possible typo is not silently repaired here.

The two fluctuation displays are nevertheless dimensionally consistent with KPZ \(1{:}2{:}3\) scaling. Put \(R=\epsilon^{-3/2}\). Then

\[ R^{2/3}=\epsilon^{-1},\qquad R^{-1/3}=\epsilon^{1/2}, \qquad R^{-1/3}x=\epsilon^{1/2}x. \]

\[ h_\epsilon(t,x)=\epsilon^{1/2} \left[h_{2\epsilon^{-3/2}t}(2\epsilon^{-1}x) +\epsilon^{-3/2}t\right]. \]

来源字段：http://aimpl.org/kpzuniversality/2/

15546 | OWR-1458-003

### Injectivity Criteria for Upsilon Transforms

P / partially_solved | Oberwolfach 报告问题汇编

Let $\gamma$ be a $\sigma$-finite Borel measure on $(0,\infty)$, let $\Upsilon_\gamma$ be the Upsilon transform on its Lévy domain, and define $\Upsilon_\gamma(\rho)(A)=\int_0^\infty\rho(Ax^{-1})\,\gamma(dx)$. Find general criteria on $\gamma$ under which $\Upsilon_\gamma$ is one-to-one on its Lévy domain.

来源字段：https://doi.org/10.4171/owr/2007/07

15547 | OWR-15988-003

### Injectivity of Procrustes Means of Covariance Operators

O / open | Oberwolfach 报告问题汇编

Given injective covariance operators $\Sigma_1,\ldots,\Sigma_k$ on a Hilbert space $H$, is their Fréchet mean with respect to the Procrustes metric also injective?

来源字段：https://doi.org/10.4171/owr/2018/3

15548 | AMR-096-0034

### Integrability of all routes to random points in a SIRSN

O / open | AMR 问题清单汇编

Under what additional assumptions, if any, is $\mathbb{E}\sup_{i\ge1}\operatorname{len}[R(0,U_i)]<\infty$ for independent uniform points $U_i$ in the unit disc?

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15549 | OWR-12016-003

### Integrated Super-Brownian Scaling of High-Dimensional Incipient Clusters

O / open | Oberwolfach 报告问题汇编

What is the scaling limit of the entire high-dimensional incipient infinite cluster? The conjecture is that a properly rescaled cluster converges to integrated super-Brownian excursion (ISE).

来源字段：https://doi.org/10.4171/owr/2012/44

15550 | AIM-PROBABILITY-0047

### Interchange mixing on the Boolean hypercube

S / solved | AIM 研讨会问题汇编

**Interchange process on the hypercube.** Obtain good bounds on \(t_{\mbox{mix}}^{IP}\) on the hypercube \(\{0,1\}^n\).

来源字段：http://aimpl.org/markovmixing/4/

15551 | OWR-757-007

### Interchanging Hierarchical Mean-Field and Continuum Limits

P / partially_solved | Oberwolfach 报告问题汇编

In the hierarchical mean-field limit, may the continuum limit and the limit $N\to\infty$ be interchanged?

来源字段：https://doi.org/10.4171/owr/2004/43

15552 | OWR-12342-001

### Intermediate Distributions in Stochastic Numerical Homogenization

P / partially_solved | Oberwolfach 报告问题汇编

For a multiscale numerical approximation $u_\varepsilon^h$ of a stochastic heterogeneous partial differential equation and the corresponding numerical homogenized solution $u_0^h$, determine the limiting distribution of the appropriately normalized fluctuation $u_\varepsilon^h-u_0^h$ as $\varepsilon\to0$ with coarse scale $h$ fixed. Does this intermediate distribution converge, as $h\to0$, to the fluctuation law predicted by continuum corrector theory?

来源字段：https://doi.org/10.4171/owr/2013/14

15553 | OWR-14298589-005

### Intermediate-Area Cylinders on Large-Genus Square-Tiled Surfaces

O / open | Oberwolfach 报告问题汇编

Determine the full cylinder-area spectrum for random square-tiled surfaces in large genus. In particular, estimate the number of cylinders whose normalized area lies between $1/\sqrt{g}$ and $2/\sqrt{g}$, between the known microscopic and macroscopic regimes.

来源字段：https://doi.org/10.4171/owr/2024/41

15554 | OWR-12697706-003

### Invariance Principles for Perturbed Balanced Environments

O / open | Oberwolfach 报告问题汇编

Can one prove an invariance principle for a slightly perturbed balanced environment or for a balanced environment subjected to a percolation perturbation?

来源字段：https://doi.org/10.4171/owr/2023/10

15555 | AMR-099-0036

### Invariant finite-energy percolation with internal threshold one

O / open | AMR 问题清单汇编

Does there exist an automorphism-invariant finite-energy percolation subgraph $X$ of $\mathbb{Z}^d$ that percolates almost surely but whose own Bernoulli critical probability satisfies $p_c(X)=1$?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=61

15556 | OWR-1061-009

### Invariant Laws and Survival of BRACO Processes

O / open | Oberwolfach 报告问题汇编

Is every invariant law of an $(a,b,c,d)$ branching-coalescing process the Poissonization of an invariant law of the corresponding resampling-selection process? Does survival of the $(a,b,c,d)$ branching-coalescing process imply survival of the $(a^\dagger,b,c,d)$ process with reversed jump rates $a^\dagger(i,j)=a(j,i)$?

来源字段：https://doi.org/10.4171/owr/2005/40

15557 | OWR-2042-008

### Inverse Wasserstein Stability of the Geodesic Radon Transform

O / open | Oberwolfach 报告问题汇编

Let $\mathcal R$ be the geodesic Radon transform on probability measures on $S^{n-1}$, sending a measure to its equator averages, and let $d_{W_p}$ be Wasserstein-$p$ distance. Does there exist a uniform reverse inequality $d_{W_p}(\mu,\nu)\leq C'(p,n)d_{W_p}(\mathcal R\mu,\mathcal R\nu)$ for all probability measures $\mu,\nu$?

来源字段：https://doi.org/10.4171/owr/2008/31

15558 | AMR-099-0084

### Isoperimetric bounds for critical probabilities of disk triangulations

P / partially_solved | AMR 问题清单汇编

Let $G$ be a bounded-degree triangulation of a disk. Prove that each of the following conditions implies $p_c(G)\le1/2$: $\operatorname{Dim}(G)\ge2$; $\operatorname{Dim}(G)>1$; or $|\partial A|\ge f(|A|)\log|A|$ for every finite vertex set $A$, where $f(n)\to\infty$. Moreover, prove that $h(G)>0$ implies $p_c(G)<1/2$.

背景链接：https://emis.de/ft/41630#page=4

15559 | AMR-099-0025

### Isoperimetric dimension and connective constants

P / partially_solved | AMR 问题清单汇编

Prove that every graph $G$ with isoperimetric dimension greater than $1$ has self-avoiding-walk connective constant $\mu(G)>1$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=38

15560 | AMR-099-0019

### Isoperimetric dimension and nontrivial percolation threshold

P / partially_solved | AMR 问题清单汇编

Let $G$ be an infinite bounded-degree graph. Prove that $\operatorname{I-dim}(G)>1$ implies $p_c(G)<1$. As a weaker target, prove the conclusion when $\operatorname{I-dim}(G)=\infty$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=34

15561 | AIM-PROBABILITY-0125

### Joint Brown support lies in the Taylor spectrum, with equality for normal tuples

S / solved | AIM 研讨会问题汇编

For a commuting tuple $(T=(T_1,\ldots,T_n)$ in a $(\mathrm{II}_1)$ factor, is $[ \operatorname{supp}\nu_T\subseteq \operatorname{Sp}(T), ]$ where $(\operatorname{Sp}(T))$ is the Taylor joint spectrum?

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15562 | OWR-11579-002

### Joint Event-Trigger and Control Optimization

O / open | Oberwolfach 报告问题汇编

In event-triggered stochastic linear-quadratic control with an average communication budget $\bar m$, what are the optimal control policy $f^*$ and event-triggering policy $\gamma^*$ minimizing $\mathbb E[J_C]$ subject to $\mathbb E[J_E]\leq\bar m$?

来源字段：https://doi.org/10.4171/owr/2012/12

15563 | AMR-098-0003

### Joint limit of total life and relative age

P / partially_solved | AMR 问题清单汇编

For the null-recurrent renewal process of Problems 1.1-1.2, assuming their answers are positive, does $(D_t/\phi(t),U_t)$ converge in distribution to $(D_\phi,U)$ as $t\to\infty$, where $D_\phi$ and the uniform random variable $U$ are independent?

背景链接：http://cms.dm.uba.ar/depto/public/Some%20Open%20Problems-preprint.pdf

15564 | AIM-PROBABILITY-0100

### Karlin-McGregor origin of the Airy2 boundary-value kernel

S / solved | AIM 研讨会问题汇编

Exact solvability

Explain the occurrence of a boundary value problem (BVP) in the kernel for the continuum statistics for the Airy\(_2\) process: \(\mathbb P(A_2(\cdot)\le g(\cdot))=\det(I-M)\), where \(M\) has a kernel given in terms of a BVP involving \(g\). Is this due to the Karlin--McGregor formula for this infinite ensemble of lines? Study the analogous continuum statistics for the top curve of \(N\) non-intersecting Brownian bridges or Brownian motions. This might explain where this BVP comes from.

来源字段：http://aimpl.org/kpzuniversality/2/

15565 | OWR-4803-001

### Kernel Selection and Properties of Kernelized Spatial Depth

O / open | Oberwolfach 报告问题汇编

For kernelized spatial depth $D_\kappa(x,F)$, what properties does it possess, how should the kernel parameter be chosen optimally, and what is its relationship to kernel density estimation?

来源字段：https://doi.org/10.4171/owr/2011/13

15566 | OWR-17138-005

### Kinetic Relaxation and Bootstrap Percolation Infection Times

P / partially_solved | Oberwolfach 报告问题汇编

For a kinetically constrained model with initial law $\mu$, determine whether the relaxation time $T_{\mathrm{rel}}$ and mean infection time $\mathbb E_\mu\tau_0$ have the same asymptotic scale as the typical bootstrap-percolation infection time $\tau_{\mathrm{BP}}$.

来源字段：https://doi.org/10.4171/owr/2019/42

15567 | OWR-12483-002

### KPZ Boundaries in Cutoff Branching Random Walk

O / open | Oberwolfach 报告问题汇编

For a branching random walk on $\mathbb Z^2$ with a cutoff rule limiting the number of particles at each site, is the boundary of the occupied set in the KPZ universality class?

来源字段：https://doi.org/10.4171/owr/2013/20

15568 | AIM-PROBABILITY-0081

### KPZ fixed-point status and a scaling-centering obstruction certificate

P / partially_solved | AIM 研讨会问题汇编

**Universality of the KPZ universality class:** Prove the existence and uniqueness of the KPZ universality class fixed point introduced by Corwin-Quastel in 1 spatial dimension. That is to say, consider any growth process \(h(t,x)\) and show that for an appropriate centering function \(\overline h_\epsilon\), \[ \lim_{\epsilon\rightarrow\infty}\epsilon^{1/2}h(\epsilon^{-3/2}t,\epsilon^{-1}x)-\overline h_\epsilon \] has a limit as \(\epsilon\to0\) as a space-time process? Then show that the properties of this process identify it uniquely (see the conjectured properties in Corwin-Quastel).

来源字段：http://aimpl.org/kpzuniversality/1/

15569 | OPG-60052

### KPZ Universality Conjecture

P / partially_solved | Open Problem Garden

Conjecture Formulate a central limit theorem for the KPZ universality class.

背景链接：http://www.openproblemgarden.org/op/kpz_universality_conjecture

15570 | OWR-9790363-004

### Lamperti Limits of Multitype-Forest Profiles

O / open | Oberwolfach 报告问题汇编

For uniform multitype forests with a prescribed degree sequence, prove convergence of the profile and characterize its limit as a solution of the Lamperti transform.

来源字段：https://doi.org/10.4171/owr/2022/12

15571 | OWR-17469-011

### Large Deviations for Random High-Dimensional Projections

O / open | Oberwolfach 报告问题汇编

Let $\Theta^{(N)}$ be uniform on $S^{N-1}$ and set $\nu_N=\mu_{\Theta^{(N)}}$. Does $(\nu_N)$ satisfy a large-deviation principle with speed $N$ and rate function

$$I(\nu)=-\frac12\log(1-\|\alpha\|_2^2)$$

when $\nu=\kappa_\alpha$ for some $\alpha$ in the open unit ball of $\ell_2$, and $I(\nu)=\infty$ otherwise? As an intermediate step, are the possible Prohorov limits of

$$E_N=\left\{\operatorname{law}\!\left(\sum_{i=1}^N\alpha_iU_i\right):\alpha\in S^{N-1}\right\}$$

exactly $\{\kappa_\alpha:\alpha\in\ell_2,\ \|\alpha\|_2\leq1\}$?

来源字段：https://doi.org/10.4171/owr/2020/6

15572 | OWR-16931-019

### Large-Deviation Asymptotics for Triangle Counts

O / open | Oberwolfach 报告问题汇编

Let $X$ be the number of triangles in $G_{n,p}$. Determine the logarithmic lower-tail asymptotics $$-\log\Pr\!\left(X\leq(1-\delta)\mathbb EX\right)$$ throughout the unresolved sparse regimes, particularly when $p=O(n^{-1/2})$.

来源字段：https://doi.org/10.4171/owr/2019/19

15573 | OWR-13359-001

### Large-Deviation Explanation of Driven TASEP Boundary Behavior

O / open | Oberwolfach 报告问题汇编

Can the different behavior of the boundary conditions, as a function of $\alpha$ in the driven TASEP, be understood through a large-deviation argument?

来源字段：https://doi.org/10.4171/owr/2014/57

15574 | OWR-14298589-004

### Large-Genus Cylinder Decompositions of Square-Tiled Surfaces

P / partially_solved | Oberwolfach 报告问题汇编

Determine the large-genus behavior of the limiting cylinder-area measure $\nu_{k,\mu}$ for every stratum of Abelian, quadratic, and higher-order differentials, extending the known principal-stratum results and the work in progress for quartic differentials.

来源字段：https://doi.org/10.4171/owr/2024/41

15575 | AMR-096-0019

### Largest common subcladogram exponents

O / open | AMR 问题清单汇编

For two independent random $n$-cladograms, under both the uniform and coalescent distributions, prove $\mathbb{E}C_n=n^{\gamma+o(1)}$ for respective constants $\gamma_a,\gamma_b<1/2$, and identify or characterize those exponents.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15576 | AMR-096-0020

### Largest common suborder of two random two-dimensional orders

O / open | AMR 问题清单汇编

For two independent coordinatewise partial orders generated by uniform points in the unit square, prove $\mathbb{E}C_n\sim c n^{1/3}$ and establish the existence or value of $c\in(0,\infty)$.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15577 | OWR-4081-004

### Largest Critical Percolation Clusters with Free Boundaries

O / open | Oberwolfach 报告问题汇编

For critical bond percolation on a high-dimensional box with free boundary conditions, determine the asymptotic order and limiting behavior of the largest cluster.

来源字段：https://doi.org/10.4171/owr/2009/38

15578 | OWR-15219-012

### Largest Roots of Doubly Stochastic Stable Polynomials

P / partially_solved | Oberwolfach 报告问题汇编

Let $D_{m,d}$ consist of the doubly stochastic stable homogeneous polynomials of degree $d$, let $\lambda_1(P)$ be the largest zero of the mixed characteristic polynomial $\chi_P$, and set $P_\infty=(x_1+\cdots+x_m)^d/m^d$. Is $\lambda_1(P)\leq\lambda_1(P_\infty)$ for every $P\in D_{m,d}$?

来源字段：https://doi.org/10.4171/owr/2017/14

15579 | AMR-096-0025

### Largest-city growth at alpha=1

O / open | AMR 问题清单汇编

If $\alpha=1$ and $\beta>2$, prove $N_{(1)}(t)=t(\log t)^{1-2/\beta+o(1)}$ almost surely.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15580 | OWR-11695865-002

### Lattice Supports and Graphical Model Factorization

P / partially_solved | Oberwolfach 报告问题汇编

If the support of a Markov distribution $P\in\mathcal{M}(G)$ is a lattice, must $P$ factorize, that is, must $P\in\mathcal{M}_F(G)$?

来源字段：https://doi.org/10.4171/owr/2022/55

15581 | AIM-PROBABILITY-0158

## LCM balancing, Coxeter-like quotient obstructions, and atomic tameness

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.2 (P. Dehornoy).** (1) Given a cancellative, finitely-generated monoid \(M\) in which lcm's exist, is \(M\) necessarily a Garside monoid? That is, does there exist a Garside element \(\Delta\) in \(M\)?

(2) In the case of Artin groups, the nicest Garside structures come from the Cayley graph of the corresponding Coxeter group. Is there a way to systematize this? Is there some notion of a “Coxeter group” corresponding to each Garside group?

(3) Let \(M\) be a Garside monoid with Garside element \(\Delta\), and let \(\ell\) be a length on \(M\) (\(M\) is atomic). Is it always true that \(\ell(\Delta^k)\leq Ck\,|\Delta|\) for some constant \(C\)?

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15582 | OWR-14741-002

## Learning Related Structural Causal Models from Heterogeneous Data

O / open | Oberwolfach 报告问题汇编

Given observations $(x_1,\ldots,x_n)$ and a similarity measure $k(x_i,x_j)$ indicating whether data points come from the same distribution, how can this information be exploited to learn an overall collection of structural causal models when the data are non-identically distributed across related tasks?

来源字段：https://doi.org/10.4171/owr/2016/33

15583 | OWR-1961-005

## Lifshitz Tails for Sign-Changing Alloy Potentials

O / open | Oberwolfach 报告问题汇编

For a sign-changing alloy-type random Schrödinger operator whose coupling distribution is supported on $[a,b]$, let $E_-(\lambda)$ denote the bottom Neumann ground-state energy at coupling $\lambda$. Do Lifshitz tails occur at the bottom of the spectrum unless the coupling distribution is Bernoulli and $E_-(a)=E_-(b)$?

来源字段：https://doi.org/10.4171/owr/2008/16

15584 | OWR-15215-008

## Limit Laws for Random-Simplex Volume Statistics

P / partially_solved | Oberwolfach 报告问题汇编

Let $P_{d+1}^K$ be the convex hull of $d+1$ independent uniform points in a $d$-dimensional convex body $K$. Can one prove central limit theorems as $d\to\infty$ for the volumes of $P_{d+1}^K$ when $K$ is the Euclidean ball, a regular simplex, a cube, or a cross-polytope? Separately, if $P_n^d$ is the convex hull of $n$ independent uniformly random vertices of the cube $[0,1]^d$, determine the asymptotic behavior of $\mathbb E f_\ell(P_n^d)$ for $2\leq\ell\leq d-2$.

来源字段：https://doi.org/10.4171/owr/2017/10

15585 | AMR-099-0057

## Limit shape in Poisson-Voronoi metrics over $\ell_p$ planes

O / open | AMR 问题清单汇编

Construct the Poisson-Voronoi tessellation of the plane equipped with an $\ell_p$ metric and give the cells their adjacency graph metric. What is the deterministic asymptotic shape of large graph-metric balls?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=4

15586 | AMR-100-0007

## Limit shape of first-passage percolation

O / open | AMR 问题清单汇编

On $\mathbb{Z}^d$, start with the origin black and every other vertex white. Repeatedly choose uniformly an edge having one black and one white endpoint and recolor the white endpoint black. After rescaling the growing black set to have fixed diameter, the shape theorem gives convergence to a deterministic convex limit shape. Determine that limit shape explicitly.

背景链接：https://mathoverflow.net/a/37167

15587 | OWR-11695859-002

## Limit Theorems for Object Data on Metric Spaces

P / partially_solved | Oberwolfach 报告问题汇编

For object-valued data in a metric measure space, define an appropriate notion of mean and determine conditions under which its empirical version satisfies a law of large numbers and a central limit theorem.

来源字段：https://doi.org/10.4171/owr/2022/49

15588 | OWR-1266-001

## Limit Theorems for Random Walks in Random Environments

O / open | Oberwolfach 报告问题汇编

For random walks in random environments in dimension $d>1$, establish laws of large numbers and central limit theorems in full generality, both quenched and annealed.

来源字段：https://doi.org/10.4171/owr/2006/17

15589 | OWR-13856-001

## Limit Theorems in the Condensation Regime

O / open | Oberwolfach 报告问题汇编

In the stochastic house-of-cards model, let $\beta$ be the mutation probability and $\mu$ the mutant-fitness distribution. In the condensation regime

$$\beta\int_0^1\frac{\mu(dx)}{1-x}<1,$$

the population size $N(t)$ has exponential growth rate $\lambda^*=1-\beta$ but is not strictly exponential. Determine the correction to $e^{\lambda^*t}$ needed to obtain a nondegenerate limit theorem for $N(t)$.

来源字段：https://doi.org/10.4171/owr/2015/35

15590 | OWR-14298589-002

### Limiting Area Distribution of Random Self-Overlapping Polygons

O / open | Oberwolfach 报告问题汇编

For a uniformly random $n$-sided self-overlapping polygon with fixed generic side set, prove that its area minus $\frac{1}{2\pi}\log n$ converges in distribution as $n\to\infty$, and identify the limiting law.

来源字段：https://doi.org/10.4171/owr/2024/41

15591 | OWR-13352-007

### Limiting Boolean Functions of Random And-Or Trees

P / partially_solved | Oberwolfach 报告问题汇编

Given random trees $T_n$ with uniformly random AND/OR labels, do the Boolean functions computed by $T_n$ converge in distribution as $n\to\infty$? When is the limiting distribution degenerate, and what can be said when it is nondegenerate?

来源字段：https://doi.org/10.4171/owr/2014/50

15592 | OWR-768-006

### Limiting Cycle Distributions of Birational Maps

P / partially_solved | Oberwolfach 报告问题汇编

Let $D_p(x)$ be the probability that a uniformly chosen point of $\mathbb F_p^2$ lies on a cycle of length at most $px$. Does the limit $D(x)=\lim_{p\to\infty}D_p(x)$ exist for every birational map $L$?

来源字段：https://doi.org/10.4171/owr/2004/54

15593 | OWR-14299577-002

### Limiting Distributions of Mixed Character Sums

O / open | Oberwolfach 报告问题汇编

For each prime $q$, let $\chi_q$ be the prescribed deterministic character and let $S(\chi_q,x,\theta)=\sum_{n\leq x}\chi_q(n)e(n\theta)$. When $\theta$ is uniform on $[0,1]$, determine the limiting distribution of this mixed character sum as $q\to\infty$.

来源字段：https://doi.org/10.4171/owr/2025/51

15594 | OWR-11101920-009

### Limiting Vanishing Probability for Symmetric Group Characters

O / open | Oberwolfach 报告问题汇编

What is the limiting behavior of $\Pr(\chi_\lambda(\mu)=0)$ for random irreducible characters and conjugacy classes of symmetric groups?

来源字段：https://doi.org/10.4171/owr/2022/39

15595 | OWR-1970-004

### Linear Response for Stationary Random Fields

O / open | Oberwolfach 报告问题汇编

Under general hypotheses such as exponential decay of correlations and finite exponential moments, do stationary random fields satisfy a moderate-deviation principle and the linear-response limit $\frac{1}{r^2\varepsilon^2}\log \mathbb{E}\exp(\varepsilon r^2 Z_0(h_r))\to \frac{\sigma^2}{2}\lVert\Delta h\rVert^2$ as $r\to\infty$ and $\varepsilon(\log r)^2\to0$?

来源字段：https://doi.org/10.4171/owr/2008/25

15596 | OWR-12986-001

### Linear Segments in Optimal-Path Shape Functions

P / partially_solved | Oberwolfach 报告问题汇编

Consider lazy paths $S:\{0,\ldots,n\}\to\mathbb Z$ with $\lvert S(i+1)-S(i)\rvert\leq1$ in the product random potential $V(x,i)=F(x)b(i)$, where the fields $\{F(x)\}$ and $\{b(i)\}$ are independent and i.i.d., the potential has a compactly supported density positive on the interior of its support, and $\mathbb E[b(i)]=0$. Does the shape function for the minimum action of point-to-point paths have a nontrivial linear segment near the origin?

来源字段：https://doi.org/10.4171/owr/2014/28

15597 | AMR-099-0081

### Linear support of harmonic measure in recurrent planar triangulations

O / open | AMR 问题清单汇编

Let $G$ be a bounded-degree recurrent planar triangulation with a fixed root. Are there arbitrarily large $r$ and finite domains containing the radius-$r$ ball such that at least $1-o(1)$ of harmonic measure on the domain boundary is supported on only $r^{1+o(1)}$ boundary circles in the circle packing? Does this hold at least under unimodularity, and in particular for the UIPT?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/randomplanar2.pdf#page=8

15598 | OWR-10252936-002

### Liouville Quantum Gravity Above Central Charge One

P / partially_solved | Oberwolfach 报告问题汇编

Can features of Liouville quantum gravity established for $0<\gamma\leq2$, equivalently for matter central charge $c_M\leq1$, be extended to the phase $1<c_M<25$?

来源字段：https://doi.org/10.4171/owr/2022/27

15599 | OWR-14299517-004

### List Sizes Beyond Coordinate-Wise Constraints

O / open | Oberwolfach 报告问题汇编

For a random linear code over $\mathbb F_q$ and a ball defined by constraints more general than coordinate-wise restrictions, is the number of codewords in every such ball at most $q^{\varepsilon n}(1+o(1))$ with high probability?

来源字段：https://doi.org/10.4171/owr/2025/41

15600 | OWR-11695855-001

### Local Complexity of Functional Estimation

O / open | Oberwolfach 报告问题汇编

For which functionals $\rho$ and distribution classes does the local minimax rate for estimating $\rho(P,P_0)$ depend strongly on $P_0$, and how do fuzzy-testing rates transition to functional-estimation rates as the null radius grows?

来源字段：https://doi.org/10.4171/owr/2022/46

15601 | AMR-096-0031

### Local finiteness of SIRSN traffic intensity

O / open | AMR 问题清单汇编

Show, perhaps under regularity hypotheses on a SIRSN, that for $2<\beta<4$ the paper's source-destination measure with displacement density $|z|^{-\beta}$ induces a locally finite traffic-intensity measure on $E(\infty,1)$ and hence on $\bigcup_r E(\infty,r)$.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15602 | OWR-12697708-004

### Local Limits of Preferential Attachment with Deterministic or Random Outdegree

P / partially_solved | Oberwolfach 报告问题汇编

Do the local limits of affine preferential-attachment graphs with fixed outdegree agree with those of the corresponding Bernoulli preferential-attachment model with Poisson outdegree, after attachment is based only on indegree?

来源字段：https://doi.org/10.4171/owr/2023/12

15603 | AIM-PROBABILITY-0001

### Local surjectivity and the boundary-at-infinity gap for RBM(4,3)

P / partially_solved | AIM 研讨会问题汇编

Is \(RBM_{4,3}\) a universal approximator? Does the closure of \(RBM_{4,3}\) fill the simplex \(\Delta_{15}\)?

来源字段：http://aimpl.org/boltzmann/1/

15604 | OWR-11570-006

### Local-Energy Classes of Summable Lattice-Gas Interactions

O / open | Oberwolfach 报告问题汇编

For translation-invariant lattice-gas interactions $\Phi$ on $\mathbb{N}^{\mathbb{Z}^d}$, let $\mathcal{A}_S$ consist of the local energy functions generated by interactions in $S=\{\Phi:\sum_{\Lambda\ni0}\lVert\Phi_\Lambda\rVert_\infty<\infty\}$, and let $\mathcal{R}_d=\{f:\sum_{n\geq1}n^d\delta_n(f)<\infty\}$. Given $\mathcal{R}_d\subseteq\mathcal{A}_S\subseteq\mathcal{R}_{d-1}$, is either inclusion an equality, or are both strict? If both are strict, characterize $\mathcal{A}_S$ by the modulus of continuity of the local energy.

来源字段：https://doi.org/10.4171/owr/2012/03

15605 | AMR-099-0024

### Locality of connective constants

P / partially_solved | AMR 问题清单汇编

Prove that the connective constant $\mu(G)$ is continuous under local convergence of infinite vertex-transitive graphs.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=38

15606 | OWR-7155442-006

### Long Binary Blocks in Uniformly Random Sequences

P / partially_solved | Oberwolfach 报告问题汇编

What can be said about the occurrence of binary blocks of length larger than $\log N$ when a length-$N$ sequence is sampled from the uniform distribution?

来源字段：https://doi.org/10.4171/owr/2021/21

15607 | AMR-096-0005

### Low-density lineage limit of coalescing branching random walk

O / open | AMR 问题清单汇编

For the two stationary branching-coalescing models on $\mathbb{Z}^3$ described by Aldous, prove that as particle intensity tends to zero the suitably time-rescaled lineage process, ignoring spatial positions and short parent-daughter jitter, converges in finite-dimensional distributions to the stated Poisson split-and-merge lineage process.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15608 | OWR-15989-007

### Low-Dimensional Sandpile Scaling Limits

O / open | Oberwolfach 报告问题汇编

For the Abelian sandpile model, does the high-dimensional scaling limit for the toppling probability have an analogue in low dimensions $d<5$?

来源字段：https://doi.org/10.4171/owr/2018/4

15609 | OWR-15175-007

### Low-Vacancy Diffusion in Kinetically Constrained Lattice Gases

O / open | Oberwolfach 报告问题汇编

What is the correct asymptotic order of the diffusion coefficient $D(q)$ as the vacancy density $q$ tends to zero in kinetically constrained lattice-gas models?

来源字段：https://doi.org/10.4171/owr/2016/54

15610 | OWR-12013-003

### Lower Bounds for Fractional Lévy-Area Approximation

O / open | Oberwolfach 报告问题汇编

For fractional Brownian Lévy area with Hurst parameter $H\in(1/4,1)$, sampled at $n$ equally spaced times on $[0,T]$, is the $L^2$ error of the conditional-expectation approximation bounded below by $c_H T^{2H}n^{-2H+1/2}$ for some $c_H>0$?

来源字段：https://doi.org/10.4171/owr/2012/41

15611 | AIM-PROBABILITY-0155

### Lyashko-Looijenga fibers, noncrossing chains, and a dihedral shuffle model

P / partially_solved | AIM 研讨会问题汇编

The Lyashko--Looijenga mapping associates to any complex-valued function on a manifold the polynomial in one variable whose roots are the critical values of the function. The main theorem in [34] states that this mapping is a ramified covering for some families of functions. There is a known relationship between the Lyashko--Looijenga covering of the complex sphere, and the combinatorial cacti of Ian Goulden and David Jackson [27]. Interpret the combinatorics of the type $A$ noncrossing partitions in terms of the Lyashko--Looijenga covering of the sphere. The degree of the covering is $n^{n-2}$, which is also the number of maximal chains in $NC_{A_{n-1}}$. This number is known to count many things, including labelled trees, and cacti. Do these combinatorics generalize to other types?

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15612 | OWR-13352-004

### Marginal Limits for Markov-Source Selection Processes

O / open | Oberwolfach 报告问题汇编

For the bucket-operation process $Y_n$ used for rank selection from strings generated by a Markov source, do the one-dimensional marginals of the appropriately centered and $\sqrt n$-rescaled process converge as $n\to\infty$?

来源字段：https://doi.org/10.4171/owr/2014/50

15613 | OWR-15218-004

### Market Origins of Rough Volatility

O / open | Oberwolfach 报告问题汇编

What mathematical or market-microstructural mechanisms explain the apparently universal roughness of financial volatility, commonly modeled by fractional Brownian motion with a small Hurst parameter $H$?

来源字段：https://doi.org/10.4171/owr/2017/13

15614 | AMR-098-0009

### Markovian-kernel characterization of mass-stationarity

P / partially_solved | AMR 问题清单汇编

Does the invariant-transport characterization of mass-stationarity remain valid if the bounded jointly invariant preserving kernels are restricted to Markovian kernels?

背景链接：http://cms.dm.uba.ar/depto/public/Some%20Open%20Problems-preprint.pdf

15615 | AMR-096-0001

### Martingale for practical purposes

O / open | AMR 问题清单汇编

Give a mathematically useful definition of a process being a 'martingale for practical purposes', so that failure means it is practical to find a stopping time $T$ with $\mathbb{E}X_T\ne\mathbb{E}X_0$. For a discrete process $X_0,\ldots,X_n$, can a natural polynomial-size collection of stopping-time constraints define a tractable such class?

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15616 | AMR-098-0008

### Mass-stationarity of diffuse random measures via allocations

P / partially_solved | AMR 问题清单汇编

Let $(X,\xi)$ consist of a random element and a diffuse random measure on a locally compact second countable Abelian group. Is mass-stationarity of $(X,\xi)$ equivalent to invariance under every measurable, equivariant, $\xi$-preserving allocation, without adjoining an independent stationary random field?

背景链接：http://cms.dm.uba.ar/depto/public/Some%20Open%20Problems-preprint.pdf

15617 | OWR-14298589-008

### Masur-Veech Equidistribution of Individual Twist Tori

P / partially_solved | Oberwolfach 报告问题汇编

别名：Resolved Averaged Twist-Torus Distribution Lead-In [OWR-14298589-007]

Prove that each expanding twist torus, without averaging, converges in the weak-star topology to the Masur-Veech-Smillie measure on the component of the stratum containing it.

来源字段：https://doi.org/10.4171/owr/2024/41

15618 | OWR-820-004

### Matrix Integral Models for Multivariable Free Transforms

O / open | Oberwolfach 报告问题汇编

Find matrix models for free-probability functionals such as the multivariable $R$-transform, particularly for $R$-diagonal elements, and the $S$-transform. Which matrix integrals realize these functionals?

来源字段：https://doi.org/10.4171/owr/2005/15

15619 | OWR-757-006

### Matrix-Algebra Formalisms for Interacting Particle Models

P / partially_solved | Oberwolfach 报告问题汇编

For exponential bricklayers' processes and related locally infinite-state particle models, can one construct matrix-product or quantum matrix-algebra formalisms analogous to those for simple exclusion, relate the stationary and random-walk shock measures, and determine whether the latter is a Markovian integrated law of the second-class particle?

来源字段：https://doi.org/10.4171/owr/2004/43

15620 | OWR-5152-008

### Maximal Submatrix Bounds Without Unconditionality

O / open | Oberwolfach 报告问题汇编

Let $X_1,\ldots,X_n$ be independent isotropic log-concave random vectors in $\mathbb{R}^N$, and let $A$ be the matrix with rows $X_i$. Define $A_{k,m}=\sup\{\|A(J,I):\ell_2^m\to\ell_2^k\|:|J|=k,\ |I|=m\}$. Does the high-probability bound $A_{k,m}\lesssim\sqrt{m\log(3N/m)}+\sqrt{k\log(3n/k)}$ hold without assuming that the vectors are unconditional?

来源字段：https://doi.org/10.4171/owr/2011/24

15621 | OWR-17293-009

### Maximum Additive Multiplicity in Logarithmic Random Sets

O / open | Oberwolfach 报告问题汇编

For a logarithmic random set $A\subseteq\mathbb{N}$, with each $i$ selected independently with probability $1/i$, determine $\max_x r_A(x)$, where $r_A(x)$ counts representations of $x$ as a sum of elements of $A$.

来源字段：https://doi.org/10.4171/owr/2019/50

15622 | OWR-14749-011

### Measurable Local Lemma for Arbitrary Group Actions

O / open | Oberwolfach 报告问题汇编

Does the measurable local lemma hold for every probability-measure-preserving action of every group?

来源字段：https://doi.org/10.4171/owr/2016/41

15623 | OWR-14299914-003

### Metastability and Massive-Arratia Limits for Weakly Interacting Diffusions

O / open | Oberwolfach 报告问题汇编

Prove dynamical metastability for the mean-field partial differential equation governing weakly interacting diffusions. Also prove convergence of the particle system to the massive Arratia flow in the joint regime $N\to\infty$, $\gamma\to\infty$, $\ell\to 0$, and $\gamma\ell/\log N\to\infty$.

来源字段：https://doi.org/10.4171/owr/2026/13

15624 | OWR-13683-001

### Minimax Estimation of Queue Service-Time Distributions

P / partially_solved | Oberwolfach 报告问题汇编

What is the optimal achievable estimation accuracy, in minimax risk, for estimating the service-time distribution in the $M/G/\infty$ queue from queue-length observations?

来源字段：https://doi.org/10.4171/owr/2015/26

15625 | OWR-12873-002

### Minimax Rates for Sparse Varying-Coefficient Models

P / partially_solved | Oberwolfach 报告问题汇编

Determine the minimax-optimal rates of convergence for estimation in sparse, high-dimensional varying-coefficient models.

来源字段：https://doi.org/10.4171/owr/2014/13

15626 | OWR-14741-003

### Minimax Risk of Structured Matrix Estimation

P / partially_solved | Oberwolfach 报告问题汇编

Determine the minimax Frobenius-risk rate for structured matrix estimation from independently sampled entries with independent sub-Gaussian noise over the full structured factor class, beyond the finite-alphabet cases, including how the rate simplifies when either outer factor class consists only of the identity matrix.

来源字段：https://doi.org/10.4171/owr/2016/33

15627 | OWR-1733-001

### Minimaxity and Admissibility Under Ensemble Risks

O / open | Oberwolfach 报告问题汇编

In estimating a $p$-dimensional normal mean vector $\theta$ with squared-error risk $R(\theta,\delta)=\mathbb{E}_\theta[\lVert\delta(X)-\theta\rVert^2]$, define $R_{\gamma^2}(\delta)$ by averaging this risk over a centered $p$-dimensional normal distribution for $\theta$, and define $R_{\mathrm{order}}(\theta_{(\cdot)},\delta)$ by averaging over all permutations of $\theta$'s coordinates. Are standard shrinkage estimators minimax and nearly admissible for both ensemble risks, and is a suitably chosen hierarchical Bayes estimator minimax and admissible?

来源字段：https://doi.org/10.4171/owr/2007/50

15628 | OWR-15206-016

### Minimum Degree after Random Edge Percolation

O / open | Oberwolfach 报告问题汇编

Let $G$ have minimum degree $D$, retain each edge independently with probability $1/2$, and let $d$ be the largest minimum degree of a nonempty subgraph of the resulting random graph. How large must $d$ be in terms of $D$? In particular, is $d\geq D/3-o(D)$ for every $G$?

来源字段：https://doi.org/10.4171/owr/2017/1

15629 | OWR-14299518-012

### Minimum Detectable Planted Trees in Sparse Random Graphs

P / partially_solved | Oberwolfach 报告问题汇编

How small can a planted tree $T$ be while remaining statistically detectable in $G(n,p)\cup T$?

来源字段：https://doi.org/10.4171/owr/2025/42

15630 | OWR-14299518-004

### Minimum Random-Projective-Plane Hitting Sets

O / open | Oberwolfach 报告问题汇编

Let $P$ be the point set of a projective plane of order $q$, and retain each point independently with probability $1/2$ to form $R$. With high probability, must the minimum size of a subset $B\subseteq R$ meeting every nonempty $L\cap R$ grow faster than $q$, perhaps as $\Omega(q\log q)$?

来源字段：https://doi.org/10.4171/owr/2025/42

15631 | AMR-096-0012

### Mixing of branch rotation and triangulation chains

O / open | AMR 问题清单汇编

For both the diagonal-flip chain on triangulations of the regular $n$-gon and the branch-rotation chain on $n$-cladograms, prove that the relaxation time is $O(n^{3/2})$, matching the known lower bound.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15632 | AIM-PROBABILITY-0040

### Mixing of the convex-polygon triangulation flip walk

P / partially_solved | AIM 研讨会问题汇编

Edge-flip on triangulations of a convex polygon. Consider random walk on triangulations driven by flips of randomly chosen diagonal. What mixing time?

来源字段：http://aimpl.org/markovmixing/2/

15633 | OWR-1189-013

### Mixing Rates for Random Edge Flips on Triangulations

O / open | Oberwolfach 报告问题汇编

Determine the mixing rate of the Markov chain on triangulations that repeatedly flips a uniformly random flippable edge.

来源字段：https://doi.org/10.4171/owr/2006/07

15634 | AMR-096-0004

### Mixing times for coagulation-fragmentation processes

O / open | AMR 问题清单汇编

Obtain relaxation- and mixing-time bounds for reversible coagulation-fragmentation Markov chains on finite sets in terms of their model parameters.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15635 | OWR-15428-004

### Mode Counts of Gaussian Mixtures

P / partially_solved | Oberwolfach 报告问题汇编

For mixtures of $k$ Gaussian distributions in dimension $d$, determine the distribution and maximum possible number of modes. In fixed dimension, is the maximum for isotropic mixtures bounded linearly in $k$? Give verifiable conditions ensuring at most $k$ modes, and formulate a Gaussian analogue of strong modes.

来源字段：https://doi.org/10.4171/owr/2017/20

15636 | OWR-1970-002

### Moderate Deviations for Chaotic Analytic Zeros

O / open | Oberwolfach 报告问题汇编

For the centered linear statistic $Z_0(h_r)$ of chaotic analytic zeros, does the Gaussian moderate-deviation asymptotic continue to hold in the intermediate regime $r/(\log r)^2\ll c\ll r$?

来源字段：https://doi.org/10.4171/owr/2008/25

15637 | OWR-14299282-002

### Modes and Maximum A Posteriori Estimators on Metric Spaces

P / partially_solved | Oberwolfach 报告问题汇编

For a probability measure $\mu$ on a metric space $(X,d)$, define and characterize what it means for a point to be a most likely outcome. Determine when competing notions of mode or maximum a posteriori estimator agree and when a unique canonical notion exists.

来源字段：https://doi.org/10.4171/owr/2025/21

15638 | OWR-4081-001

### Monochromatic Arm Exponents in Critical Percolation

P / partially_solved | Oberwolfach 报告问题汇编

For two-dimensional critical percolation, do the probabilities of $j$ monochromatic arms to distance $n$ obey power laws $n^{-\alpha'_j+o(1)}$? If so, how are the monochromatic exponents $\alpha'_j$ related to the polychromatic arm exponents?

来源字段：https://doi.org/10.4171/owr/2009/38

15639 | OWR-15218-003

### Multiple-Maturity Consistency Under Bid-Ask Spreads

P / partially_solved | Oberwolfach 报告问题汇编

Given option prices at more than one maturity and a bound $\varepsilon$ on the discounted bid-ask spread of the underlying asset, characterize $\varepsilon$-consistency and determine the smallest admissible $\varepsilon$.

来源字段：https://doi.org/10.4171/owr/2017/13

15640 | AMR-099-0040

### Multiplicative connection bounds above criticality

O / open | AMR 问题清单汇编

For which $p$ does there exist $C<\infty$ such that, for any vertices $x,y$ and any $z$ on a geodesic from $x$ to $y$, $$\mathbb{P}_p(x\leftrightarrow y)\le C\,\mathbb{P}_p(x\leftrightarrow z)\mathbb{P}_p(z\leftrightarrow y)?$$

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=66

15641 | AMR-099-0074

### Mutually avoiding competing random walks

O / open | AMR 问题清单汇编

Run two walks with a common clock on $\mathbb{Z}^d$, each choosing uniformly among neighbors not previously visited by the other walk. Prove that in $d=2$ one walk is almost surely trapped in a finite region, while in higher dimensions neither walk is trapped almost surely.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=9

15642 | AMR-099-0058

### Near-critical percolation limit shapes

O / open | AMR 问题清单汇编

Delete each edge of the square lattice independently with probability $q<1/2$, condition the origin to lie in the infinite component, and let $K_q$ be the deterministic asymptotic graph-metric ball shape in that component. Prove that, after normalization, $K_q$ converges in the Gromov-Hausdorff sense to a Euclidean disk as $q\uparrow1/2$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=4

15643 | AMR-096-0042

### Near-one asymptotics for oriented-percolation flow

O / open | AMR 问题清单汇编

For the limiting maximum-flow density $v(p)$ in oriented bond percolation on the square lattice, prove $1-v(p)\sim\sqrt{2(1-p)}$ as $p\uparrow1$.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15644 | AMR-095-0004

### Negative association for asymmetric exclusion

P / partially_solved | AMR 问题清单汇编

For nearest-neighbor asymmetric exclusion on $\mathbb{Z}$ with $p(1)=p>q=p(-1)$, start from the deterministic configuration $\cdots11110000\cdots$. Is the distribution at every time negatively associated; that is, does $\int fg\,d\mu\le(\int f\,d\mu)(\int g\,d\mu)$ hold for increasing $f,g$ depending on disjoint coordinate sets?

背景链接：https://web.archive.org/web/20150919235903id_/http://www.math.ucla.edu/~tml/open.pdf

15645 | AIM-PROBABILITY-0057

### Negative-quartic two-matrix model is not a real probability measure

O / open | AIM 研讨会问题汇编

**Coupled random matrix model with negative potentials.** Consider the coupled two-matrix random matrix model on pairs $(M_1, M_2)$ of $n \times n$ Hermitian matrices with formal joint weight $$ \frac{1}{Z_n} \exp\left\{-n \operatorname{Tr}[V(M_1) + W(M_2) + t M_1 M_2]\right\} dM_1\,dM_2 $$ and quartic potentials $$ V(x) = -(x^4 - a x^2), \qquad W(y) = V(y). $$ Is there any new critical behavior for some value of the parameter $a$?

来源字段：http://aimpl.org/vectorequilib/1/

15646 | OWR-15988-006

### Nested Principal Components from Manifold Covariance Fields

O / open | Oberwolfach 报告问题汇编

For PCA and classification on manifolds using covariance tensor fields, formulate a backward nested procedure and characterize when it agrees with the forward procedure. Determine when PCA or classification is appropriate under substantial curvature and whether a multiscale theory analogous to linear PCA exists.

来源字段：https://doi.org/10.4171/owr/2018/3

15647 | AMR-099-0033

### No critical infinite cluster on transitive graphs

P / partially_solved | AMR 问题清单汇编

For every infinite vertex-transitive graph $G$, prove that Bernoulli percolation has no infinite cluster at criticality, i.e. $\theta_G(p_c(G))=0$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=57

15648 | AMR-100-0001

### No percolation at the critical point on $\mathbb{Z}^d$

P / partially_solved | AMR 问题清单汇编

For nearest-neighbor independent bond percolation on $\mathbb{Z}^d$, $d\ge2$, let $p_c(d)$ be the critical edge-retention probability. Prove that at $p=p_c(d)$ there is almost surely no infinite open cluster, for every $d\ge2$.

背景链接：https://mathoverflow.net/a/37160

15649 | OWR-4990373-001

### Noise Sensitivity and Exceptional Times in Spin-System Dynamics

P / partially_solved | Oberwolfach 报告问题汇编

For Ising Glauber dynamics on transitive graphs and heat-bath dynamics in FK random-cluster models, does the weighted-majority characterization of noise sensitivity persist; are the log-Sobolev constants comparable to relaxation times; are critical planar crossing events noise sensitive; and do exceptional times exist?

来源字段：https://doi.org/10.4171/owr/2021/4

15650 | AMR-099-0079

### Noise sensitivity under the Schaeffer bijection

O / open | AMR 问题清单汇编

Generate a quadrangulation from $2n$ bits using the Schaeffer bijection and independently resample each bit with probability $\varepsilon$. Determine the noise sensitivity of natural geometric observables; in particular, estimate the probability that the diameter crosses its median after the perturbation.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/randomplanar2.pdf#page=3

15651 | AIM-PROBABILITY-0045

### Non-backtracking versus simple random-walk mixing

S / solved | AIM 研讨会问题汇编

**Non-backtracking vs. simple random walk.** Is it true that the lazy non-backtracking random walk (NBRW) mixes faster than the lazy simple random walk (SRW)?

来源字段：http://aimpl.org/markovmixing/3/

15652 | AMR-094-0006

### Non-extinction of a Fleming-Viot particle model

P / partially_solved | AMR 问题清单汇编

Let $N$ particles move as independent Brownian motions in a bounded connected open set $D\subset\mathbb{R}^d$. Whenever a particle hits the complement of $D$, replace it by a copy of a uniformly chosen surviving particle. If $\tau_k$ is the time of the $k$th death-and-branching event and $\tau_\infty=\lim_{k\to\infty}\tau_k$, is $\tau_\infty=\infty$ almost surely for every such $D$?

背景链接：https://sites.math.washington.edu/~burdzy/open_mathjax.php

15653 | AIM-PROBABILITY-0137

### Noncommutative Gamma_2 and sharp depolarizing curvature frontiers

P / partially_solved | AIM 研讨会问题汇编

**Q:** What is the analogue of the Bakry-Emery criterion in the noncommutative case? i.e. what is \(\Gamma_2\) for noncommutative Dirichlet forms?

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15654 | AMR-099-0046

### Nonintersecting couplings of random walks in dimensions three and four

P / partially_solved | AMR 问题清单汇编

Can two simple random walks on $\mathbb{Z}^3$ or $\mathbb{Z}^4$, started at vertices at graph distance $10$, be coupled so that their paths are disjoint with positive probability?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=104

15655 | OWR-16940-002

### Nonparametric Bootstrap for High-Dimensional Spectral Statistics

P / partially_solved | Oberwolfach 报告问题汇编

For high-dimensional sample covariance matrices, determine whether a fully nonparametric bootstrap consistently approximates centered linear spectral statistics. The ordinary bootstrap can fail, while an $m$-out-of-$n$ and $q$-out-of-$p$ scheme preserving the dimension-to-sample ratio may succeed under representative-subpopulation assumptions.

来源字段：https://doi.org/10.4171/owr/2019/28

15656 | OWR-12861-026

### Nonsingularity and Limit Laws for Random Matrices

P / partially_solved | Oberwolfach 报告问题汇编

Let $A(n,d)$ be the adjacency matrix of a uniformly random $d$-regular graph, where $d>2$. Is $A(n,d)$ nonsingular with probability tending to $1$ as $n\to\infty$? What are the limiting distributions of the permanent of a random $\{\pm1\}$-matrix and the determinant of a random symmetric $\{\pm1\}$-matrix?

来源字段：https://doi.org/10.4171/owr/2014/01

15657 | OWR-4082-001

### Nonsmoothness of Classifier Error Confidence Intervals

P / partially_solved | Oberwolfach 报告问题汇编

Is the poor coverage of resampling-based confidence intervals for a learned classifier's test error caused in part by the nonsmooth dependence of test error on the learned classifier?

来源字段：https://doi.org/10.4171/owr/2009/39

15658 | AIM-PROBABILITY-0136

### Normal coefficient densities characterize coarse amplification of the gradient correspondence

P / partially_solved | AIM 研讨会问题汇编

**Question.** What conditions on a Dirichlet form \(\delta^*\delta\) guarantee that the bimodule associated to \(\delta\) embeds into \[ \bigoplus L^2N\otimes L^2N? \]

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15659 | OWR-13350-003

### Occupation Times of Manifold Brownian Motion

P / partially_solved | Oberwolfach 报告问题汇编

Let $W$ be Brownian motion on a Riemannian manifold $M$ starting at $o$, and let a hypersurface $N$ locally separate $M$ into $M^+$ and $M^-$. Describe the total time $T$ that $W$ spends in $M^+$ up to time $t$.

来源字段：https://doi.org/10.4171/owr/2014/48

15660 | AMR-096-0039

### Online minimum spanning tree constant

O / open | AMR 问题清单汇编

For the complete graph with i.i.d. uniform edge weights revealed online, prove that the minimum expected cost $\mathbb{E}Y_n$ of an online spanning-tree strategy converges, and determine the limiting constant.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15661 | OWR-12179-002

### Open Problems on Smoothed Laplace Periodograms

P / partially_solved | Oberwolfach 报告问题汇编

For smoothed Laplace periodograms and their rank-based variants, determine their asymptotic distributions, develop their use in signal detection, and extend the methods to locally stationary time series.

来源字段：https://doi.org/10.4171/owr/2012/56

15662 | OWR-9790365-004

### Optimal Distributed Hypothesis Testing beyond Conditional Independence

P / partially_solved | Oberwolfach 报告问题汇编

Determine the optimal distributed hypothesis-testing scheme beyond testing against conditional independence. In particular, is the Shimokawa-Han-Amari scheme optimal when both hypotheses are binary symmetric double sources?

来源字段：https://doi.org/10.4171/owr/2022/14

15663 | OWR-13359-003

### Optimal Geodesic Curvature for Random Transpositions

O / open | Oberwolfach 报告问题汇编

For the random transposition model on the symmetric group $S_n$, determine the correct order in $n$ of the optimal geodesic-convexity constant $\kappa$, known to satisfy $4/(n^2-n) \leq \kappa \leq 2/(n-1)$.

来源字段：https://doi.org/10.4171/owr/2014/57

15664 | OWR-704-003

### Optimal Hermitian Metrics for Random Holomorphic Sections

P / partially_solved | Oberwolfach 报告问题汇编

For a fixed holomorphic line bundle, does a Hermitian metric minimize the expected number of critical points of a random holomorphic section, and is such a metric unique?

来源字段：https://doi.org/10.4171/owr/2004/21

15665 | AMR-096-0030

### Optimal length-route tradeoff for SIRSNs

O / open | AMR 问题清单汇编

Give quantitative estimates improving the known bound on $\ell^*(\Delta)$, the infimum edge intensity among SIRSNs with mean unit-distance route length $\Delta$. Do minimizers exist, and what structure do optimal networks have?

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15666 | OWR-14299517-005

### Optimal List Decoding and Recovery of Random Code Ensembles

O / open | Oberwolfach 报告问题汇编

Which random code ensembles are optimally list-decodable and list-recoverable, and at what radii and list sizes?

来源字段：https://doi.org/10.4171/owr/2025/41

15667 | OWR-15991-002

### Optimal Moment Conditions for Random Conductance Models

O / open | Oberwolfach 报告问题汇编

For random conductance models with stable-like jumps on $\mathbb{Z}^d$, determine the optimal integrability or moment condition replacing the sufficient bounds on positive and negative moments of the conductances that ensure the quenched invariance principle.

来源字段：https://doi.org/10.4171/owr/2018/6

15668 | OWR-11695855-006

### Optimality of Monotonized Asymptotic Risk

O / open | Oberwolfach 报告问题汇编

For an estimator with deterministic asymptotic risk $R_{\mathrm{det}}(\gamma)$, characterize an optimality property of the monotonized risk $\gamma\mapsto\inf_{\zeta\geq\gamma}R_{\mathrm{det}}(\zeta)$.

来源字段：https://doi.org/10.4171/owr/2022/46

15669 | OWR-16630-001

### Orbit Closures and Stationary Measures of Random Walks

O / open | Oberwolfach 报告问题汇编

For a random walk generated by a probability measure $\mu$ acting on $X$ with semigroup closure $\Gamma$, describe all orbit closures $\Gamma x_0$; determine whether the empirical measures $\nu_n$ converge and identify their limits; prove no escape of mass; describe all $\mu$-stationary measures; and describe the topology of the set of $\mu$-ergodic $\mu$-stationary measures.

来源字段：https://doi.org/10.4171/owr/2018/47

15670 | OWR-12016-008

### Parameter Dependence of Directed-Polymer Endpoint Limits

O / open | Oberwolfach 报告问题汇编

In the $(1+1)$-dimensional directed-polymer convergence diagram, does the putative scaling limit at the right-hand end depend on the parameters $\beta$ and $\alpha$?

来源字段：https://doi.org/10.4171/owr/2012/44

15671 | OWR-4200-001

### Parameter Stability in Long-Horizon GARCH Change-Point Models

O / open | Oberwolfach 报告问题汇编

For very long economic time series in a semiparametric change-point model, can the stationary GARCH component still consistently be assumed to have constant parameters?

来源字段：https://doi.org/10.4171/owr/2010/05

15672 | OWR-4803-003

### Penalty Robustness in Tree-Based Level-Set Estimation

O / open | Oberwolfach 报告问题汇编

In smooth tree-based level-set estimation, when relatively simple trees with few branches suffice to obtain a good estimate, is the choice of the tree-complexity penalty parameter $\rho$ inessential?

来源字段：https://doi.org/10.4171/owr/2011/13

15673 | AMR-096-0021

### Percolation criteria for merging planar empires

O / open | AMR 问题清单汇编

For continuous-time processes that merge adjacent polygonal planar regions $A,B$ at a geometry-dependent rate $r(A,B)$, give sufficient conditions on $r$ for percolation and sufficient conditions for non-percolation.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15674 | AMR-094-0003

### Percolation dimension of planar Brownian trace

O / open | AMR 问题清单汇编

For a set $B$, define its percolation dimension as the infimum of the Hausdorff dimensions of Jordan arcs $A\subset B$ containing at least two distinct points. If $X_t$ is two-dimensional Brownian motion, is the percolation dimension of $X[0,1]$ equal to $1$?

背景链接：https://sites.math.washington.edu/~burdzy/open_mathjax.php

15675 | AMR-099-0042

### Percolation nonuniqueness on products with the line

O / open | AMR 问题清单汇编

If $G$ is strongly amenable, can Bernoulli percolation on $G\times\mathbb{Z}$ have infinitely many infinite clusters throughout a nondegenerate interval $[p_1,p_2]$? What if $G$ is also assumed to have polynomial volume growth?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=75

15676 | AMR-096-0022

### Percolation of planar empires at unit merger rate

O / open | AMR 问题清单汇编

When every adjacent pair of planar empires merges at rate $r(A,B)=1$, does percolation occur?

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15677 | AMR-099-0032

### Percolation threshold of fast-growing planar triangulations

O / open | AMR 问题清单汇编

Let $G$ be a planar triangulation with uniform volume growth faster than quadratic. Must $p_c(G)<1$? More strongly, is $p_c(G)=1/2$?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=56

15678 | AMR-099-0027

### Percolation thresholds along expander limits

P / partially_solved | AMR 问题清单汇编

Let $(G_n)$ be a bounded-degree expander family converging locally to an infinite graph $G$. Prove that the finite-graph percolation thresholds $p_c(G_n)$ converge to $p_c(G)$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=40

15679 | OWR-4990373-006

### Percolation Thresholds During Planar Majority Dynamics

P / partially_solved | Oberwolfach 报告问题汇编

For majority dynamics on $\mathbb{Z}^2$ with i.i.d. Bernoulli-$p$ initial opinions, let $p_c(t)$ be the critical probability for an infinite opinion-1 cluster at time $t$. Is $p_c(t)$ strictly decreasing in $t$? Is $t\mapsto p_c(t)$ continuous at $\infty$, and is $p_c(\infty)=1/2$? For the hexagonal lattice, is $p_c(t)=1/2$ for every $t$, and are there exceptional times?

来源字段：https://doi.org/10.4171/owr/2021/4

15680 | OWR-15218-006

### Performance Gains from Robust Mean-Variance Portfolios

P / partially_solved | Oberwolfach 报告问题汇编

How much can robust mean-variance portfolio strategies improve investor performance under uncertainty in volatility, correlation, and model specification?

来源字段：https://doi.org/10.4171/owr/2017/13

15681 | OWR-2654828-001

### Persistence and Exit Times of Integrated Self-Similar Processes

P / partially_solved | Oberwolfach 报告问题汇编

For the integral of a general self-similar Markov process, can one determine persistence and exit-time asymptotics? In particular, can one compute $\mathbb{P}_x(T_a\wedge T_b>t)$ as $t\to\infty$, solve the analogous exit-time problem for the integral of a stable Lévy process, and compute the area under a general stable excursion?

来源字段：https://doi.org/10.4171/owr/2020/32

15682 | OWR-13856-007

### Persistence Exponents of Multiply Integrated Brownian Motion

O / open | Oberwolfach 报告问题汇编

Let $B$ be standard Brownian motion, set $A_t^{(1)}=\int_0^t B_s\,ds$, and recursively define $A_t^{(d)}=\int_0^t A_s^{(d-1)}\,ds$. If $\theta_d$ is the persistence exponent governing the long-time probability that $A^{(d)}$ stays positive, determine $\theta_d$ for every $d\geq2$.

来源字段：https://doi.org/10.4171/owr/2015/35

15683 | OWR-12016-001

### Persistence Times of Dynamical-Percolation Crossings

P / partially_solved | Oberwolfach 报告问题汇编

How long may it take for a crossing event to disappear in dynamical percolation?

来源字段：https://doi.org/10.4171/owr/2012/44

15684 | OWR-12697708-003

### Persistent Hubs under Preferential-Attachment Removal

P / partially_solved | Oberwolfach 报告问题汇编

In preferential-attachment models with vertex removal, is the maximum-degree vertex eventually persistent, or does its identity change infinitely often?

来源字段：https://doi.org/10.4171/owr/2023/12

15685 | OWR-1061-001

### Phase Transitions in Random $L$-Reversal Walks

P / partially_solved | Oberwolfach 报告问题汇编

For the random walk generated by random $L$-reversals, for which values of $L$ does a phase transition occur, and is there a critical value of $L$?

来源字段：https://doi.org/10.4171/owr/2005/40

15686 | OWR-4337-002

### Physical Interpretation of Wasserstein Distance in Particle Systems

O / open | Oberwolfach 报告问题汇编

What is the physical meaning of the 2-Wasserstein distance for particle systems, such as the empirical measures of many independent Brownian motions?

来源字段：https://doi.org/10.4171/owr/2010/24

15687 | OWR-12697706-002

### Planar Bricmont-Kupiainen Invariance Principle

O / open | Oberwolfach 报告问题汇编

Does the full invariance principle hold under the Bricmont-Kupiainen conditions in dimension $d=2$?

来源字段：https://doi.org/10.4171/owr/2023/10

15688 | OWR-11101922-002

### Planar Coexistence in Hostile-Environment First-Passage Percolation

O / open | Oberwolfach 报告问题汇编

Does coexistence occur for hostile-environment first-passage percolation in dimension $d=2$, and can one obtain a coexistence regime in any dimension when $p<p_c^{\mathrm{site}}$?

来源字段：https://doi.org/10.4171/owr/2022/41

15689 | AMR-099-0087

### Planar half-density percolation has no unique infinite cluster

P / partially_solved | AMR 问题清单汇编

Let $G$ be a planar graph and consider Bernoulli percolation at $p=1/2$. If an infinite open cluster exists almost surely, prove that there are almost surely infinitely many infinite open clusters.

背景链接：https://emis.de/ft/41630#page=10

15690 | OWR-2053-006

### Poisson Boundaries of Low-Dimensional Wreath Products

P / partially_solved | Oberwolfach 报告问题汇编

For simple random walks on the wreath products $\mathbb{Z}^d\wr B$, what is the Poisson boundary when $d=3$ or $d=4$?

来源字段：https://doi.org/10.4171/owr/2008/42

15691 | OWR-1463-003

### Poisson Eigenvalue Statistics for Random Band Matrices

O / open | Oberwolfach 报告问题汇编

For the random band matrices $X_{N,W}$, establish Poisson eigenvalue statistics throughout the full regime $W^2/N\to0$, rather than only under a stronger hypothesis $W^\nu/N\to0$.

来源字段：https://doi.org/10.4171/owr/2007/12

15692 | OWR-1061-005

### Poisson Gap Counts in Gapped Local Alignments

O / open | Oberwolfach 报告问题汇编

In the sequence-based model of gapped local alignment, under the asymptotic regime $s,\Delta\to\infty$ with $s e^{-\lambda\Delta}\to\beta>0$, does the number of gaps in the best local alignment converge to a Poisson distribution with finite positive mean, as it does in the heuristic Poisson model?

来源字段：https://doi.org/10.4171/owr/2005/40

15693 | OWR-14298589-011

### Poisson Laws across Boundary Regimes of Random Hyperbolic Surfaces

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Boundary-Regime Poisson-Limit Conjecture [OWR-14298589-012]

Find a unified proof of the known volume identities and determine further volumes for which they hold. Prove a Poisson-limit interpolation for simple closed geodesics on random genus-$g$ hyperbolic surfaces with one boundary of length $L_g$ across the regimes $L_g=o(g)$, $L_g/g\to c$, and $L_g\gg g$, including existence of the required continuous rescaling function.

来源字段：https://doi.org/10.4171/owr/2024/41

15694 | OWR-4990373-011

### Poisson Variance Inequalities for Arbitrary Stopping Sets

O / open | Oberwolfach 报告问题汇编

Let $\eta$ be a Poisson process with intensity measure $\lambda$, let $\eta'$ be an independent copy, and let $Z$ be an arbitrary stopping set. Prove $\mathbb E|f(\eta)-f(\eta')|\leq 2\int\mathbb P(x\in Z)\mathbb E|D_xf(\eta)|\,\lambda(dx)$; equivalently, for $\{0,1\}$-valued $f$, prove $\operatorname{Var}(f(\eta))\leq\int\mathbb P(x\in Z)\mathbb E|D_xf(\eta)|\,\lambda(dx)$.

来源字段：https://doi.org/10.4171/owr/2021/4

15695 | OWR-15214-004

### Polynomial-Rate Approximation of Stochastic Differential Equations

P / partially_solved | Oberwolfach 报告问题汇编

Consider $$ dX(t)=\mu(X(t))\,dt+\sigma(X(t))\,dW(t),\qquad X(0)=x_0, $$ on $[0,1]$. Under what assumptions on the drift $\mu$ and diffusion $\sigma$ does there exist a method using finitely many evaluations of $W$ that approximates $X(1)$ in absolute mean with a polynomial convergence rate?

来源字段：https://doi.org/10.4171/owr/2017/9

15696 | OWR-14740-001

### Polynomial-Rate Surrogates for Noisy Manifold Estimation

O / open | Oberwolfach 报告问题汇编

When estimating a smooth manifold from noisy data in an embedding space with logarithmic minimax rates, what surrogate object can preserve the essential geometric or topological features while being estimable at a polynomial rate of convergence?

来源字段：https://doi.org/10.4171/owr/2016/32

15697 | AIM-PROBABILITY-0164

### Polytopality and an exact dihedral model for noncrystallographic associahedra

S / solved | AIM 研讨会问题汇编

**Problem 6.1.** When \(W\) is a noncrystallographic finite Coxeter group, give a geometric construction that realizes \(\Delta_W\) as a convex polytope. There is a realization of \(\Delta_W\) in all types as a complete simplicial fan, but it is not clear whether this fan is polytopal in the noncrystallographic types.

It is **Problem 6.1**, attributed to H. Thomas and A. Zelevinsky, in the AIM workshop list *Braid groups, clusters and free probability*.

The official PDF gives:

The preceding source paragraph defines \(\Delta_W\) as the flag complex of compatible subsets of the almost-positive roots \[ \Phi_{\ge-1}=\Phi^+\cup(-\Pi). \] It also says that in types \(A,B\) these complexes generalize the **duals** of the classical associahedron and cyclohedron. Thus the precise reading is: construct a simple convex polytope whose polar boundary is \(\Delta_W\), and, more strongly, make the displayed complete fan its normal fan. Confusing the simplicial complex with the face lattice of the simple polytope rather than its dual reverses incidences.

The irreducible noncrystallographic finite types are \(I_2(m)\) other than \(m=3,4,6\), together with \(H_3,H_4\). Reducible cases follow from products/joins after treating irreducible factors.

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15698 | OWR-14299578-005

### Poonen-Rains Selmer Distributions in Twist Families

P / partially_solved | Oberwolfach 报告问题汇编

Let $E/K$ be an elliptic curve and let $E_d$ range through a natural family of twists. Prove that the dimensions of the $p$-Selmer groups $\operatorname{Sel}_p(E_d/K)$ have the Poonen-Rains limiting distribution for the full family, beyond the currently proved primes, fields, and conditional regimes.

来源字段：https://doi.org/10.4171/owr/2025/52

15699 | OWR-1061-008

### Positive Markov Kernels with Dirichlet Stationary Measures

P / partially_solved | Oberwolfach 报告问题汇编

For the Dirichlet or Poisson-Dirichlet transition-density expansion $p_t(x,dy)=d\pi_\theta(y)\{1+\sum_{n\geq1}a_n(\theta)Q_n^{(\theta)}(x,y)\}$, characterize all nonzero coefficient sequences $a_n(\theta)$ that make the kernel nonnegative. In particular, are the standard diffusion coefficients $\rho_n^{(|\theta|)}(t)$ the only time-dependent choices that yield a Markov transition function?

来源字段：https://doi.org/10.4171/owr/2005/40

15700 | OWR-13856-005

### Positive Probability of Zero Total Superfluid Momentum

O / open | Oberwolfach 报告问题汇编

For superfluids, is the total momentum zero with positive asymptotic probability?

来源字段：https://doi.org/10.4171/owr/2015/35

15701 | AIM-PROBABILITY-0074

### Positive-temperature structure and a finite-rank gap criterion

P / partially_solved | AIM 研讨会问题汇编

What is the structure which replaces determinantal point processes and correlation functions and which allows for such computations?

The AIM record asks what replaces determinantal point processes and correlation functions for triangular arrays associated with positive-temperature polymers. These arrays arise as limits of Macdonald processes. Their fixed-level laws resemble GUE/LUE eigenvalue ensembles, but are not determinantal; nevertheless, Laplace transforms of an extremal coordinate can have Fredholm determinant formulas. The question asks for a structure making all analogous eigenvalues, especially joint gap probabilities, accessible.

Its concluding question is:

In modern terminology, geometric or birational RSK is the positive-temperature lift, while tropical RSK is the zero-temperature limit in which a scaled logarithm changes sums into maxima. The positive-temperature level laws are Whittaker measures or related Macdonald degenerations. This report preserves the source wording but calls the finite-temperature variables “Whittaker coordinates.”

来源字段：http://aimpl.org/kpzuniversality/1/

15702 | OWR-14298374-004

### Positivity and Convergence of a Splitting Scheme

O / open | Oberwolfach 报告问题汇编

For the Lie-Trotter splitting scheme applied to $du=\Delta u\,dt+g(u)\,d\beta(t)$ with homogeneous Dirichlet data and the non-globally Lipschitz nonlinearity $g(v)=v^{1.25}$, prove positivity and mean-square convergence of order $1/2$. Also establish a weak convergence rate.

来源字段：https://doi.org/10.4171/owr/2024/26

15703 | OWR-1108-002

### Post-Selection Inference and Growing-Dimension Model Selection

P / partially_solved | Oberwolfach 报告问题汇编

After selecting a model by AIC or FIC, what is the actual coverage probability of a nominal 95 percent confidence interval, and how can the associated over-optimism in confidence intervals and p-values be repaired? What are the limit distributions in Bayesian model-averaging procedures? How does AIC/BIC theory behave when the number of parameters increases with sample size?

来源字段：https://doi.org/10.4171/owr/2005/47

15704 | OWR-14299282-004

### Posterior Sampling for Random-Tree Besov Priors

P / partially_solved | Oberwolfach 报告问题汇编

Develop general, provably convergent algorithms for sampling Bayesian inverse-problem posteriors arising from random-tree Besov priors, with quantitative mixing and complexity guarantees.

来源字段：https://doi.org/10.4171/owr/2025/21

15705 | AIM-PROBABILITY-0026

### Power concavity and mode curvature of symmetric stable densities

P / partially_solved | AIM 研讨会问题汇编

For \(0<\alpha\le2\), let \(Y_\alpha\) have characteristic function \(e^{-|t|^\alpha}\) and density \(f_\alpha\). Prove that \(f_\alpha\) is \(s_\alpha\)-concave, in the standard non-strict sense, for some \(s_\alpha<0\), and prove \(f_\alpha''(0)<0\) at its mode.

Its problem field is preserved exactly:

Taken literally, the definition is impossible: at \(\lambda=0\), or whenever \(x=y\), its two sides are equal. The standard definition uses \(\ge\), as in Han and Wellner (2016). The displayed delimiters and parentheses on the live page are balanced; the missing item is the equality bar in the comparison sign, not a bracket.

Two further hypotheses must be restored.

1. The function \(t\mapsto e^{-|t|^\alpha}\) is a characteristic function precisely in the stable range \(0<\alpha\le2\). That is the range used below. 2. “Curvature at the mode is negative” is read as \(f''(m)<0\). At a stationary point this also says that the signed curvature of the graph is negative, and it is equivalent in sign to \((\log f)''(m)<0\).

The repaired problem is therefore:

来源字段：http://aimpl.org/steinhd/1/

15706 | OWR-4082-004

### Power Variations of Volatility-Modulated Gaussian Processes

P / partially_solved | Oberwolfach 报告问题汇编

Determine the weak limit of normalized power variations of a Gaussian process with centered stationary increments. For the volatility-modulated moving-average process $X_t=\int_{-\infty}^t g(t-s)\sigma_s\,dW_s$, determine the asymptotic behavior of its power variations and establish a central limit theorem.

来源字段：https://doi.org/10.4171/owr/2009/39

15707 | OWR-13683-005

### Practical Confidence Regions for Matrix Estimation

P / partially_solved | Oberwolfach 报告问题汇编

Develop practical trace-norm confidence intervals and trace-norm certificates, characterize confidence intervals in other norms, and analyze alternative construction methods such as Bayesian and bootstrap procedures.

来源字段：https://doi.org/10.4171/owr/2015/26

15708 | AIM-PROBABILITY-0104

### Predictable-step recovery for finite-action free processes

P / partially_solved | AIM 研讨会问题汇编

**Question.** Given a tracial state \(\tau\) corresponding to a free stochastic process, does there exist a sequence of tracial states \(\tau_n\to\tau\) with \(\chi_p^*(\tau_n)\to\chi_p^*(\tau)\), where \(\tau_n\) corresponds to the process \[ dA_i(t)=dS_i(t)+k_t(A_1(s),\ldots,A_m(s))_{s\leq t}\,dt \] with \(k_t\) stepwise constant in \(s\), and \(\chi_p^*\) denotes the quantity \(\chi^*\) defined for processes in the paper of Guionnet and Cabanal-Duvillard?

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15709 | OWR-17474-009

### Probabilistic Equality of Left and Right Tail Fields

O / open | Oberwolfach 报告问题汇编

For a stationary process with finite spin state space, prove using only probability theory, without entropy theory, that the left and right tail sigma-fields coincide modulo null sets.

来源字段：https://doi.org/10.4171/owr/2020/11

15710 | OWR-13352-003

### Probabilistic Interpretation of Waring-Polynomial Series

O / open | Oberwolfach 报告问题汇编

For the Waring-polynomial series $F_n$ arising as approximations to the leading root of the deformed exponential formal power series, is $F_n$ the probability generating function of an explicitly or easily describable random variable?

来源字段：https://doi.org/10.4171/owr/2014/50

15711 | AMR-094-0001

### Probabilistic McMillan theorem in higher dimensions

P / partially_solved | AMR 问题清单汇编

Let $X_t$ be $d$-dimensional Brownian motion starting at the origin, let $D$ be an open subset of $\mathbb{R}^d$ containing the origin, and let $\tau=\inf\{t>0:X_t\notin D\}$. Let $A$ be the set of cluster points of $$(X_t-X_\tau)/|X_t-X_\tau|$$ as $t \uparrow \tau$. For every $d>2$ and every such $D$, is $A$ almost surely either a sphere or a hemisphere?

背景链接：https://sites.math.washington.edu/~burdzy/open_mathjax.php

15712 | OWR-14298592-003

### Probabilities of Properties in Random-Knot Models

O / open | Oberwolfach 报告问题汇编

Fix a knot property $\mathcal P$ and a random-knot model $(K_n)$. Determine $\lim_{n\to\infty}\Pr[K_n\text{ has }\mathcal P]$, when this limit exists.

来源字段：https://doi.org/10.4171/owr/2024/43

15713 | OWR-15176-003

### Processes Associated with Regular Representations of Dirichlet Forms

O / open | Oberwolfach 报告问题汇编

Let $(\mathcal E,W)$ be a Dirichlet form on $L^2(\mathbb R^d,\rho(x)\,dx)$, let $(\mathcal E,H)$ be its closed realization, and suppose a regular representation of $(\mathcal E,W)$ is given. Which stochastic process is associated with the regular representation, and how is it related to the process associated with $(\mathcal E,H)$?

来源字段：https://doi.org/10.4171/owr/2016/55

15714 | OWR-14299911-031

### Property T for Random Free-Product Quotients above One-Third Density

O / open | Oberwolfach 报告问题汇编

For a free product of at least three nontrivial factors, does a random quotient in the free-product density model have Kazhdan's property $(T)$ with high probability at every density $d>1/3$?

来源字段：https://doi.org/10.4171/owr/2026/8

15715 | AIM-PROBABILITY-0148

## q-Fuss-Catalan positivity and a Boolean q-Narayana refinement

S / solved | AIM 研讨会问题汇编

The record comes from Problem 2.1 of Drew Armstrong's outline of the January 2005 AIM workshop *Braid Groups, Clusters, and Free Probability*. I checked the original PDF, including the displayed signs and indices. It asks, for a finite Coxeter group $W$ of rank $n$, Coxeter number $h$, exponents $e_1,\ldots,e_n$, and a positive integer $k$, to define

$$ \operatorname{Cat}^{(k)}_W(q) :=\prod_{i=1}^n\frac{[kh+e_i+1]_q}{[e_i+1]_q}, \qquad [a]_q=q+q^2+\cdots+q^a, \tag{1} $$

and prove that it is a polynomial in $q$ with nonnegative integral coefficients. The source says this was then known in classical types $A,B,D$, and notes that in type $A$, $k=1$, it agrees up to a power of $q$ with the Garsia--Haiman $q,t$-Catalan polynomial specialized at $t=1/q$.

The four follow-up questions in the source are:

1. replace $e_i+1$ by the fundamental degrees $d_i$ for a complex reflection group, and replace $h$ by the highest degree; 2. prove Reiner's cyclic-sieving evaluation at a primitive $d$-th root $\zeta$, for $d\mid h$, counting elements of $NC_W=[1,c]$ fixed by conjugation by $c^{h/d}$; 3. determine positivity of $$ \operatorname{Cat}^{(k),+}_W(q) :=\prod_{i=1}^n\frac{[kh+e_i-1]_q}{[e_i+1]_q}, $$ and find a nonnegative $q$-refinement by Fuss--Narayana numbers; 4. construct $q$-Fuss versions of Chapoton's $M$-, $F$-, and $H$-triangles.

The plus sign in (1), the minus sign in the positive variant, and the conjugating exponent $h/d$ are all verified from the PDF. The source's convention is nonstandard: if

$$ [a]_q^{\mathrm{std}}=1+q+\cdots+q^{a-1}, $$

then $[a]_q=q[a]_q^{\mathrm{std}}$. Every quotient in the problem has one such factor in numerator and denominator, so the rational functions are unchanged by switching conventions. At $q=0$, however, one must first cancel/simplify the rational expression and then evaluate the resulting polynomial; the raw source quotient has the form $0/0$.

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15716 | AIM-PROBABILITY-0109

## Quadratic non-microstates pressure and weighted colored cycles

P / partially_solved | AIM 研讨会问题汇编

The terminal text `#0.3 "Free von Neumann Algebras", Dykema, Ricard` is the next section heading and is not part of the problem.

The page asks whether the Guionnet--Maurel-Segala planar-map interpretation of a microstates variational pressure has an analogue for non-microstates free entropy: $$ F(t)=\sup_{\tau\in TS(A)} \left\{\chi(\tau)-\tau\!\left(\sum_{a=1}^r t_aq_a\right)\right\}, \qquad F^*(t)=\sup_{\tau\in TS(A)} \left\{\chi^*(\tau)-\tau\!\left(\sum_{a=1}^r t_aq_a\right)\right\}. \tag{1.1} $$ Does $F^*$ have a comparable interpretation by enumeration of colored planar maps?

Several symbols in the displayed map series are mangled in the AIM PDF itself. The primary Guionnet--Maurel-Segala paper fixes the intended convention. Relative to $m$ independent GUE matrices, take $$ V_t=\sum_{a=1}^r t_a(q_a+q_a^*) $$ for monomials $q_a$. For small $t$, their normalized free energy is $$ F(t)-F(0)= \sum_{k\in\mathbb N^r\setminus\{0\}} \left(\prod_{a=1}^r\frac{(-t_a)^{k_a}}{k_a!}\right) \mathcal M((q_1,k_1),\ldots,(q_r,k_r)), \tag{1.2} $$ where $\mathcal M$ counts labeled connected colored planar maps with $k_a$ stars of type $q_a$ or $q_a^*$. If the $q_a$ are already self-adjoint or the signs are absorbed into $t_a$, the shorter AIM display is consistent with (1.2).

There is also a normalization ambiguity. Guionnet--Maurel-Segala define the entropy in the relevant theorem relative to product GUE measure; the standard Lebesgue-normalized free entropy differs by the quadratic Gaussian energy and a constant. Consequently the map series is naturally a **pressure difference** with $F(0)=0$. A literal unnormalized standard $\chi$ would have a nonzero semicircular constant at $t=0$.

Finally, `TS(A)` cannot mean only the normal tracial states of a fixed $II_1$ factor: that would usually collapse the optimization to its unique normal trace. Operationally, and consistently with the primary paper, it means an admissible trace-state/noncommutative-law space for the distinguished generators, subject to a norm or moment cutoff. All results below are stated directly on the law space, avoiding this ambiguity.

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15717 | OWR-13350-001

## Quasi-Invariant Random Measures and Associated Diffusions

O / open | Oberwolfach 报告问题汇编

Let a group $G$ act on the space $\mathcal M(X)$ of Radon measures on $X$. Which laws of random measures are quasi-invariant under $G$ and support an associated analysis, such as a Laplace operator or a diffusion on $\mathcal M(X)$?

来源字段：https://doi.org/10.4171/owr/2014/48

15718 | OWR-12726-001

### Quenched Central Limit Theorem for Small-Disorder Random Walks

O / open | Oberwolfach 报告问题汇编

For $d\geq3$, does there exist $\varepsilon(d)>0$ such that, under the i.i.d. small-disorder assumptions,

$$n^{-1/2}(X_n-na(\mu))$$

converges in law, almost surely in the environment, to $N(0,\Sigma(\mu))$, with $\Sigma(\mu)$ positive definite?

来源字段：https://doi.org/10.4171/owr/2013/52

15719 | OWR-12726-003

### Quenched Invariance Principles for Random Conductance Models

P / partially_solved | Oberwolfach 报告问题汇编

For a stationary ergodic random conductance model on $\mathbb Z^d$ with $0<\omega(e)<\infty$ almost surely and finite mean conductance, does finiteness of $\mathbb E[1/\omega(e)]$ for every edge suffice for a quenched functional central limit theorem with deterministic nondegenerate covariance $\Sigma^2$?

来源字段：https://doi.org/10.4171/owr/2013/52

15720 | AIM-PROBABILITY-0050

### Random biased transpositions: two-class cutoff, parity repair, and heterogeneous product reductions

P / partially_solved | AIM 研讨会问题汇编

**Random biased transpositions.** The random uniform transposition walk on $S_n$ is the discrete time Markov chain that at each time step applies a transposition $(i,j)$ uniformly at random.

The random nearly-uniform (biased) transposition walk is the corresponding chain that at each time step applies a transposition $(i,j)$ with probability $p_{ij}$ for $p_{ij}$ comparable to the uniform rate up to uniformly bounded constants.

Does the random nearly-uniform transposition walk on $S_n$ exhibit cutoff?

What about in the simpler case where $p_{ij}=p_i p_j$ for each $1\le i,j\le n$?

来源字段：http://aimpl.org/markovmixing/5/

15721 | AMR-096-0013

### Random Eulerian excursion dichotomy on high-dimensional tori

O / open | AMR 问题清单汇编

On the bidirected torus $\mathbb{Z}_N^d$ with fixed $d\ge3$, let $b^{(N)},t^{(N)},m^{(N)}$ count excursions of a uniform Eulerian circuit longer than $N^d/\omega_N$, shorter than $\omega_N$, and between those scales. For sufficiently slowly growing $\omega_N$, does $(b^{(N)},t^{(N)},m^{(N)})$ converge in distribution to $(S^*,2d-S^*,0)$ for some $S^*$ supported on $\{1,\ldots,2d\}$?

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15722 | OWR-13352-001

### Random Geometric Graphs in the Hyperbolic Plane

P / partially_solved | Oberwolfach 报告问题汇编

How do finite random geometric graphs and continuum percolation behave when the Euclidean plane is replaced by the hyperbolic plane, in particular for the component-size and connectivity questions of the corresponding Poisson or finite-point models?

来源字段：https://doi.org/10.4171/owr/2014/50

15723 | OWR-17469-005

### Random Information for Weighted Approximation Problems

O / open | Oberwolfach 报告问题汇编

For weighted $\ell_p$-balls and other error norms, determine the quality of random information and compare its performance with optimal information.

来源字段：https://doi.org/10.4171/owr/2020/6

15724 | OWR-12010-018

### Random Isotropic Model for Selmer Exact Sequences

O / open | Oberwolfach 报告问题汇编

For $V=\mathbb Z_p^{2n}$, $W=\mathbb Z_p^n\times\{0\}$, and random $Z\in\operatorname{OGr}_n(\mathbb Z_p)$, let $R=(Z\cap W)\otimes\mathbb Q_p/\mathbb Z_p$, $S=(Z\otimes\mathbb Q_p/\mathbb Z_p)\cap(W\otimes\mathbb Q_p/\mathbb Z_p)$, and $T=S/R$. Does the limiting distribution as $n\to\infty$ of $0\to R\to S\to T\to0$ equal the distribution of the corresponding sequences for elliptic curves over $k$?

来源字段：https://doi.org/10.4171/owr/2012/38

15725 | AMR-099-0021

### Random lattice embeddings with exponential intersection tails

O / open | AMR 问题清单汇编

For some $d\ge3$, is there a probability measure on embeddings of $\mathbb{Z}^2$ into $\mathbb{Z}^d$ having an analogue of the exponential intersection-tail property?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=37

15726 | OWR-14299285-004

### Random Moduli of General Brownian Surfaces

O / open | Oberwolfach 报告问题汇编

Derive the random conformal moduli of Brownian surfaces of arbitrary topology, extending the known Brownian-annulus case.

来源字段：https://doi.org/10.4171/owr/2025/26

15727 | OWR-11786-022

### Random Triangulations and Exchangeable Clique Complexes

O / open | Oberwolfach 报告问题汇编

Is there a polynomial-time algorithm that samples uniformly from closed $3$-manifold triangulations with $N$ tetrahedra, including under the restriction to triangulations of the $3$-sphere? Separately, for clique complexes of exchangeable random graphs, determine their homotopy groups and whether this random model is universal in an appropriate sense.

来源字段：https://doi.org/10.4171/owr/2012/24

15728 | OWR-17469-004

### Random Versus Optimal Information for Approximation and Integration

O / open | Oberwolfach 报告问题汇编

For approximation and integration from $n$ measurements, determine the rate of the $n$th minimal error $e_n$, characterize optimal information, and quantify how much random information worsens the error compared with optimal information.

来源字段：https://doi.org/10.4171/owr/2020/6

15729 | AMR-099-0028

### Random-walk displacement exponent on the UIPT

S / solved | AMR 问题清单汇编

For simple random walk $(X_n)$ on the uniform infinite planar triangulation, prove that the graph distance from the starting point has exponent $1/4$, in the sense that $d_{\mathrm{gr}}(X_0,X_n)\asymp n^{1/4}$ up to the intended subpolynomial or constant factors.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=49

15730 | AMR-099-0031

### Random-walk displacement on circle-packed doubling graphs

O / open | AMR 问题清单汇编

In the circle-packed planar-doubling setting of Section 7, prove that the expected distance of simple random walk from its root at time $t$ is at most $t^\alpha$ for some $\alpha<1/2$. Is the optimal exponent $\alpha=d^{-1}$ when volume grows like $r^d$?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=55

15731 | OWR-2093-008

### Randomized-Transport Characterizations of Palm Measures

P / partially_solved | Oberwolfach 报告问题汇编

In defining mass-stationarity for invariant random measures, would allowing external randomization-such as an independent stationary background or transport kernels instead of only allocations-suffice to characterize Palm versions of stationary processes? Can the characterization be narrowed to transport kernels alone?

来源字段：https://doi.org/10.4171/owr/2008/47

15732 | AMR-144-0009

### Randomstrasse101 Open Problems (2024) - Conjecture 11

O / open | AMR 问题清单汇编

Also, to the best of my knowledge, there is no known construction (or proof of existence) of an infinite family achieving a lower bound strictly larger than one. More explicitly, we formulate the following conjecture.

Given the definitions above, we have $$\limsup_{n\to\infty}\sup_{A\in\{\pm1\}^{n\times n}}\frac1{\sqrt{n}}\mathrm{disc}(A) > 1.$$

背景链接：https://arxiv.org/abs/2504.20539

15733 | AMR-144-0010

### Randomstrasse101 Open Problems (2024) - Conjecture 12

O / open | AMR 问题清单汇编

Let $H_k$ be the $2^k\times 2^k$ Hadamard matrix obtained by the Sylvester construction (see above). For odd $k$, we have \[ \mathrm{disc}(H_k) = \sqrt{2}\sqrt{2^k}. \]

背景链接：https://arxiv.org/abs/2504.20539

15734 | AMR-144-0001

### Randomstrasse101 Open Problems (2024) - Conjecture 2 (Globally Synchronizing Regular Graphs)

O / open | AMR 问题清单汇编

Abdalla, Bandeira, Kassabov, Souza, Strogatz, and Townsend [ABKSST-SynchExpander] recently showed that a random Erd\mathbb{H}{o}s-R\'enyi graph is globally synchronizing above the connectivity threshold with high probability (you can also see the Quanta article covering this at https://www.quantamagazine.org/new-proof-shows-that-expander-graphs-synchronize-20230724/). The same paper also shows that a uniform $d$-regular graph is globally synchronizing with high probability for $d\geq 600$, but it leaves open the following question.

A uniform random $3$-regular graph is globally synchronizing with high probability (probability going to $1$ as $n\to\infty$).

背景链接：https://arxiv.org/abs/2504.20539

15735 | AMR-144-0002

### Randomstrasse101 Open Problems (2024) - Conjecture 3 (Global Synchrony with negative edges)

O / open | AMR 问题清单汇编

Given any $\varepsilon>0$, the $n\times n$ random matrix $A$ with zero diagonal and off-diagonal entries given by \[ A_{ij} = \left\{ \begin{array}{ccl} 1 & \text{with probability} & \frac12+\delta\\ -1 & \text{with probability} & \frac12-\delta, \end{array} \right. \] with $\delta \geq (1+\varepsilon)\sqrt{\frac{\log n}{2n}}$, is globally synchronizing with high probability.

背景链接：https://arxiv.org/abs/2504.20539

15736 | AMR-144-0003

### Randomstrasse101 Open Problems (2024) - Conjecture 4 (Density threshold for Global Synchrony)

O / open | AMR 问题清单汇编

Another elusive question about global synchrony is the ``extremal combinatorics'' version of the question.

For any $\varepsilon>0$, there exists $n>0$ and a graph $G$ on $n$ nodes such that the minimum degree of $G$ is at least $\left(\frac34-\varepsilon\right)n$ and $G$ is not globally synchronizing.

背景链接：https://arxiv.org/abs/2504.20539

15737 | AMR-144-0004

### Randomstrasse101 Open Problems (2024) - Conjecture 5 (Ellipsoid fitting)

O / open | AMR 问题清单汇编

Let $n, d \geq 1$ and $x_1, \cdots, x_n \sim \mathcal{N}(0,\mathrm{I}_d/d)$. For any $\varepsilon > 0$: \begin{align}

\limsup_{d \to \infty} \frac{n}{d^2} \leq \frac{1-\varepsilon}{4} \Rightarrow \lim_{d \to \infty} \mathbb{P}[\exists \mathcal{E} \textrm{ an ellipsoid fit to } (x_i)_{i=1}^n] = 1, \\

\liminf_{d \to \infty} \frac{n}{d^2} \geq \frac{1+\varepsilon}{4} \Rightarrow \lim_{d \to \infty} \mathbb{P}[\exists \mathcal{E} \textrm{ an ellipsoid fit to } (x_i)_{i=1}^n] = 0. \end{align}

背景链接：https://arxiv.org/abs/2504.20539

15738 | AMR-144-0006

### Randomstrasse101 Open Problems (2024) - Conjecture 8 (Kikuchi Spectral Threshold)

O / open | AMR 问题清单汇编

Given $r,\ell, n$ positive integers satisfying $n\gg \ell\geq \frac{r}2$ and r even ($r$ and $\ell$ will be fixed and $n\to\infty$). For each $S\subset [n]$ with $|S|=r$ let $Z_S\sim\mathcal{N}(0,1)$, and all independent. Let $\lambda\geq 0$ and let $M$ be the $\binom{n}{\ell} \times \binom{n}{\ell}$ matrix (with rows and columns indexed by subsets $I\subset[n]$ of size $\ell$) given by \[ M(\lambda)_{I,J} = \left\{ \begin{array}{ccl} \lambda + Z_{I \Delta J} & \text{if} & |I \Delta J| = r,\\ 0 &\text{otherwise,}& \end{array} \right. \] where $I\Delta J = (I\cup J) \setminus (I\cap J)$ denotes the symmetric difference.

Let $\lambda^\natural_{r,\ell}$ denote the threshold at which eigenvalues ``pop-out'' of the spectrum of $M(\lambda)$: in other words $\lambda^\natural_{r,\ell}$ is the real number such that, for all $\lambda>\lambda^\natural_{r,\ell}$, there exists $\varepsilon>0$ such that \(\mathbb{E}\lambda_{\max} M(\lambda)>(1+\varepsilon+o(1))\mathbb{E}\lambda_{\max} M(0),\) where $o(1)$ is a term that goes to zero as $n\to\infty$.

For fixed $r$, we have $$n^{\frac{r}4}\lambda^\natural_{r,\ell}\to 0$$ as $\ell\to\infty$ ( note that this is after one has taken $n\to\infty$).

背景链接：https://arxiv.org/abs/2504.20539

15739 | AMR-144-0007

### Randomstrasse101 Open Problems (2024) - Conjecture 9 (K\'{o}mlos Conjecture)

O / open | AMR 问题清单汇编

When the condition of $\pm1$ entries is replaced by an $\ell_2$ condition on the columns, the corresponding question is the famous K\'{o}mlos Conjecture (which coincidentally is the first open problem in my lecture notes from around a decade ago [Afonso10L42P]).

There exists a universal constant $K$ such that, for all square matrices $A$ whose columns have unit $\ell_2$ norm, $\mathrm{disc}(A) \leq K$.

背景链接：https://arxiv.org/abs/2504.20539

15740 | AMR-144-0008

### Randomstrasse101 Open Problems (2024) - Problem 10 (How many deviations are needed afterall?)

O / open | AMR 问题清单汇编

It is easy to see that the statement of this conjecture implies Spencer's Theorem, since the columns of an $n\times n$, $\pm1$ matrix, have $\ell_2$ norm $\sqrt{n}$. The best current lower bound for K\'{o}mlos is a recent construction by Tim Kunisky [Tim-lowerboundKomlos] which shows a lower bound on $K$ of $1+\sqrt{2}$. Unfortunately, the resulting matrix is not a $\pm1$ matrix (even after scaling), and so it does not provide a lower bound for the Spencer setting.

What is the value of the following quantity? $$\sup_{n\in\mathbb{N}}\sup_{A\in\{\pm1\}^{n\times n}}\frac1{\sqrt{n}}\mathrm{disc}(A)$$

背景链接：https://arxiv.org/abs/2504.20539

15741 | AMR-144-0011

### Randomstrasse101 Open Problems (2024) - Problem 14

O / open | AMR 问题清单汇编

Consider the SK-model with an arbitrary external field. Does Glauber Dynamics fast mix (meaning in polynomial time) up to the replica symmetry breaking threshold $\beta_\ast =1$?

背景链接：https://arxiv.org/abs/2504.20539

15742 | AMR-144-0005

### Randomstrasse101 Open Problems (2024) - Problem 6

O / open | AMR 问题清单汇编

Consider synchronization model over $SO(2)$ or synchronization model over a finite group, where the signal $x$ is sampled uniformly over group elements. Find a scaling $L= L_n$ of number of frequencies such that there exists a polynomial-time algorithm that can detect the signal reliably for all $\lambda>\lambda_{\text{comp}, L}$ for some $\lambda_{\text{comp}, L}<1$ as $n\to\infty$.

背景链接：https://arxiv.org/abs/2504.20539

15743 | AMR-145-0002

### Randomstrasse101 Open Problems (2025) - Conjecture 1

O / open | AMR 问题清单汇编

An interesting question is to characterize the Lov\'asz number for random graphs. A well-known family of random graphs is the Erd\mathbb{H}{o}s-R\'enyi random graphs \(G(n, p)\). Here, \(n\) denotes the number of vertices, and \(0 \leq p \leq 1\) is the probability parameter. To sample a graph from \(G(n, p)\), for each pair of vertices \(i \neq j \in [n]\), one adds an edge independently with probability \(p\). The random graphs \(G(n, p)\) are ubiquitous in mathematics; for instance, they are used in combinatorics to lower bound *Ramsey* numbers. There are several results on the Lov\'asz number for Erd\mathbb{H}{o}s-R\'enyi random graphs \(G(n, p)\) (see [juhasz1982asymptotic,coja2005lovasz,aroranote]). However, the precise asymptotics of the Lov\'asz number for \(G(n, 1/2)\) is still open.

Let \(G\) be distributed as \(G(n, 1/2)\). Then, \begin{equation} \mathbb{E} \vartheta(G) = (1 + o(1))\sqrt{n}. \end{equation}

背景链接：https://arxiv.org/abs/2603.29571

15744 | AMR-145-0004

### Randomstrasse101 Open Problems (2025) - Conjecture 1

O / open | AMR 问题清单汇编

In [Bandyopadhyay2002-MUBs] it is shown that $\mathrm{MUB}(d)\leq d+1$ and that equality is achieved when $d$ is a prime power. For $d$ not a prime power, the problem of determining $\mathrm{MUB}(d)$ is still open. The smallest open instance is particularly well known for being a tantalizing open problem that has remained open (see Open Problem 6.2 in [Afonso10L42P]).

Let $\mathrm{MUB}(d)$ denote the largest possible number of (simultaneously) mutually unbiased bases (MUBs) in $\mathbb{C}^d$. We have $\mathrm{MUB}(6)<7$.

背景链接：https://arxiv.org/abs/2603.29571

15745 | AMR-145-0006

### Randomstrasse101 Open Problems (2025) - Conjecture 1

O / open | AMR 问题清单汇编

Let $\omega(G_p)$ denote the clique number of $G_p$ i.e. the size of the largest set of nodes that are all pairwise connected by an edge. Since $G_p$ is isomorphic to its complement this is the same as $\alpha(G_p)$ the independence number of $G_p$ (the size of the largest set of nodes that have no edges between them). We will focus on the clique number of ease of exposition. A random graph of the same degree has logarithmic clique number, which motivates the following well known conjecture (see, e.g., Open Problem 8.4 in [Afonso10L42P])

Let $\omega(G_p)$ denote the clique number of the Paley Graph for $p \equiv 1 \pmod{4}$ prime. \[ \omega(G_p)=O(\mathrm{polylog}(p)}). \]

背景链接：https://arxiv.org/abs/2603.29571

15746 | AMR-145-0011

### Randomstrasse101 Open Problems (2025) - Conjecture 1

O / open | AMR 问题清单汇编

Let $N=4M-5$ and draw $A\in\mathbb{C}^{N\times M}$ at random (here the important thing is for $ \mathrm{im}(A) $ to be drawn uniformly from the Grassmannian of $ M $-dimensional subspaces of $ \mathbb{C}^{4M-5}$, it can e.g. be with iid gaussian entries). Let $p_M$ be the probability that the mapping $ x \bmod \mathbb{T} \mapsto |Ax|^2 $ is injective.

(a) $ p_M < 1 $ for all $ M $.

(b) $ \displaystyle\lim_{M \to \infty} p_M = 0 $.

背景链接：https://arxiv.org/abs/2603.29571

15747 | AMR-145-0001

### Randomstrasse101 Open Problems (2025) - Conjecture 1 (Sharp graph matrix bounds)

O / open | AMR 问题清单汇编

However, there are still examples of shapes $\alpha$ for which iterated linear inequalities fail to produce sharp bounds. Moreover, there are examples of shapes whose norm bounds do have additional non-constant factors. We conjecture that given a fixed shape $\alpha$, the norm of the associated graph matrix grows as a product of a polynomial and a polylogarithmic factor (depending only on $\alpha$).

There are functions $f$ and $g$ depending only on the shape $\alpha$, such that \begin{equation} \mathbb{E}\|M_\alpha\| = \Theta\left(n^{f(\alpha)} (\log n)^{g(\alpha)}\right). \end{equation}

背景链接：https://arxiv.org/abs/2603.29571

15748 | AMR-145-0014

### Randomstrasse101 Open Problems (2025) - Conjecture 1 (Type-$2$ constant of Tensors)

O / open | AMR 问题清单汇编

Let $p \geq 2$, then \begin{equation} \mathbb{E} \left| \left| \sum_{i=1}^n g_i T_i \right| \right|_{ \mathcal{I}_p} \leq \tilde{\mathcal{O}}_{r,p} \left( d^{\frac12 - \frac1p } \sqrt{\sum_{i=1}^n \left| \left| T_i \right| \right|_{ \mathcal{I}_p}^2 }\right). \end{equation} (Here $\tilde{\mathcal{O}}_{r,p}$ hides multiplicative constants depending on $r$ and $p$ as well as polylog factors in $d,n$.)

背景链接：https://arxiv.org/abs/2603.29571

15749 | AMR-145-0003

### Randomstrasse101 Open Problems (2025) - Conjecture 2

O / open | AMR 问题清单汇编

We conjecture, based on numerical observations, that the Lov\'asz number for random dense circulant graphs behaves similarly to that for \(G(n, 1/2)\). This motivates the following question.

Let \(G\) be distributed as a random dense circulant graph. Then, \begin{equation} \mathbb{E} \vartheta(G) = (1 + o(1))\sqrt{n}. \end{equation}

背景链接：https://arxiv.org/abs/2603.29571

15750 | AMR-145-0007

### Randomstrasse101 Open Problems (2025) - Conjecture 2

O / open | AMR 问题清单汇编

We start with the notion of localization [MMP-2019-LPCliques Paley,Kunisky-2023-SpectralPseudorandomnessPaleyClique] : Let $G_{p,1}$ denote the induced subgraph on the vertices adjacent to the node 0, which we call the 1-localization. Because the 1-localization does not depend on the node picked (see [Kunisky-2023-SpectralPseudorandomnessPaleyClique]) we have $\omega(G_p) \leq \omega(G_{p,1}) + 1 \leq \vartheta(\overline{G_p})+1$. It was observed empirically in [MMP-2019-LPCliquesPaley] that $\vartheta(\overline{G_{p,1}}) \sim \sqrt{p/2}$, which would recover the Hanson-Petridis bound (also, note that $G_{p,1}$ is a circulant graph and so the SDP corresponding to $\vartheta(\overline{G_{p,1}})$ reduces to an LP).

$\vartheta(\overline{G_{p,1}}) \sim \sqrt{p/2}$ (for $p \equiv 1 \pmod{4}$ prime) where $G_{p,1}$ is the 1-localization of the Paley Graph described above.

背景链接：https://arxiv.org/abs/2603.29571

15751 | AMR-145-0012

### Randomstrasse101 Open Problems (2025) - Conjecture 2

O / open | AMR 问题清单汇编

\begin{definition} Given $A\in\mathbb{R}^{N\times M}$ let $\omega(A)$ be defined as \begin{equation} \omega(A) = \min_{S\subseteq [N]: \mathrm{rank}(A_{S^c})<n} \sigma_M(A_S), \end{equation} where $A_S$ denotes the submatrix made of the rows indexed by $S$ and $\sigma_M$ is the $M$-th singular value (which is the smallest in non-degenerate situations). \end{definition} The complement property says that $\mathcal{A}$ is injective if and only if $\omega(A)>0$. [balan2013invertibility] shows that stability is tightly connected to the inverse of this quantity. They also make the following intriguing conjecture.

There exist universal constants $C>0$ and $0<\beta<1$ such that, for any $M>1$ and $A\in\mathbb{R}^{2M-1\times M}$ a matrix for which any subset of $M$ rows spans $\mathbb{R}^M$, \[ \omega(A) \leq C \max_{k\in[N]}\|A_k\|\beta^M, \] where $A_k$ is the $k$-th row of $A$.

背景链接：https://arxiv.org/abs/2603.29571

15752 | AMR-145-0008

### Randomstrasse101 Open Problems (2025) - Conjecture 3

O / open | AMR 问题清单汇编

For $p \equiv 1 \pmod{4}$ prime and large enough we have $\vartheta(\overline{G_{p,2}}) \leq \frac23 \sqrt{p}$ where $G_{p,2}$ is the 2-localization of the Paley Graph described above.

背景链接：https://arxiv.org/abs/2603.29571

15753 | AMR-145-0009

### Randomstrasse101 Open Problems (2025) - Conjecture 4

O / open | AMR 问题清单汇编

There is another approach to to use convex relaxation to potentially improve over upper bounds on the $\omega(G_p)$. Recall that the Lovasz $\vartheta$ function can be viewed a as degree-2 Sum-of-Squares relaxation of the clique (or independence) number. A fascinating question is whether the degree-4 Sum-of-Squares relaxation significantly improves on the $\sqrt{p}$ bound, perhaps even improving over the state of the art (see [KY-2022-PaleyDegree4SOS] for a precise construction of this relaxation). There have been several encouraging numerical experiments (for this and other SDPs) [KY-2022-PaleyDegree4SOS,kobzar2023revisiting,wang2024lower]. We particularly note that the numerical experiments in [KY-2022-PaleyDegree4SOS] suggest even a polynomial improvement may be possible, although the authors prove that it cannot improve beyond $\Omega(p^{\frac13})$.

There exists $\varepsilon>0$ such that the Sum-of-Squares relaxation of degree 4 for the clique number of the Paley graph gives a bound of $O(p^{\frac12-\varepsilon})$.

背景链接：https://arxiv.org/abs/2603.29571

15754 | AMR-145-0010

### Randomstrasse101 Open Problems (2025) - Conjecture 5

O / open | AMR 问题清单汇编

The Paley Equiangular Tight Frame construction satisfies the Restricted Isometry Property beyond the square-root bottleneck. In other words, there exists $\varepsilon>0$ such that, for $m \sim p^{1/2+\varepsilon}$, any matrix formed by picking $m$ (distinct) vectors in the Paley ETF has condition number bounded by a constant independent of $p$.

背景链接：https://arxiv.org/abs/2603.29571

15755 | AMR-145-0005

### Randomstrasse101 Open Problems (2025) - Problem 2

O / open | AMR 问题清单汇编

An interesting approach to this problem is to try to build Sum-of-Squares proofs of $\mathrm{MUB}(6)<7$. In [BDK-2022-DualBoundsMUB] it was shown that a certain relaxation related to degree-2 Sum-of-Squares cannot prove that $M(6) < 7$. It is however unclear whether higher degree levels of Sum-of-Squares can provide such a proof.

Is there a Sum-of-Squares degree 4 proof that there are no $7$ Mutually Unbiased Bases in $\mathbb{C}^6$?

背景链接：https://arxiv.org/abs/2603.29571

15756 | AMR-145-0013

### Randomstrasse101 Open Problems (2025) - Problem 3

O / open | AMR 问题清单汇编

This also motivates the following question (for which I do not know the state of the art, but a quick search did not reveal that the answer is already known).

What is the value of $\omega(A)$ for $A$ with iid standard gaussian entries? Does it satisfy the bound in the conjecture? If so, what is $\beta$?

背景链接：https://arxiv.org/abs/2603.29571

15757 | OWR-11695865-003

### Rank Preservation Under Singular Covariance Updates

O / open | Oberwolfach 报告问题汇编

In neighborhood coordinate descent for singular covariance matrices, if $\det\Sigma_{\setminus u,\setminus u}=0$, does the updated matrix still satisfy $\det\widetilde\Sigma=0$?

来源字段：https://doi.org/10.4171/owr/2022/55

15758 | AIM-PROBABILITY-0157

### Rank-two Garside posets and minimal nonuniqueness

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.1 (R. Charney), Questions about classification.** (1) Given an arbitrary poset \(P\), when can it be given a Garside labelling? When such a labelling exists, say that \(P\) is a Garside poset. (2) Given a Garside poset \(P\), what are the relationships between its inequivalent Garside labellings? When does \(P\) have a unique Garside labelling? (3) Given a poset with an edge labelling, when can this be embedded in a Garside structure? When do the corresponding monoids/groups embed? What are the minimal obstructions to doing this?

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15759 | AIM-PROBABILITY-0019

### RBMs as a certified bridge from Ising physics to interaction networks

P / partially_solved | AIM 研讨会问题汇编

How to use RBMs to do statistical inference for a physical system, or properties of a network?

How to use RBMs to do statistical inference for a physical system, or properties of a network?

来源字段：http://aimpl.org/boltzmann/7/

15760 | OWR-12697684-016

### Real Zeros of General Random Functions

P / partially_solved | Oberwolfach 报告问题汇编

别名：Unspecified Random Function Root Problem [OWR-12697684-003]

For broad classes of random functions, determine the mean, variance, and limiting distribution of the number of real zeros.

来源字段：https://doi.org/10.4171/owr/2023/1

15761 | OWR-4795-002

### Real-Time Detection of Asset-Price Bubbles

P / partially_solved | Oberwolfach 报告问题汇编

Can one determine in real time whether a given asset's price process exhibits a bubble?

来源字段：https://doi.org/10.4171/owr/2011/06

15762 | OWR-15436-001

### Recombination, Selection, and Duality in Multilocus Ancestry Models

O / open | Oberwolfach 报告问题汇编

For the multilocus ancestry and diffusion models, can one incorporate recombination and selection, determine the function $m(\cdot)$ in closed form beyond the one-locus reversible case, and extend the strong-recombination diffusion-limit arguments from $L=2$ to $L>2$ with a posterior dual?

来源字段：https://doi.org/10.4171/owr/2017/28

15763 | OWR-4990373-003

### Reconstruction Threshold for Critical Ising Models on the Torus

O / open | Oberwolfach 报告问题汇编

What is the exact reconstruction threshold for the critical Ising model on the torus $\mathbb{Z}_n^2$?

来源字段：https://doi.org/10.4171/owr/2021/4

15764 | OWR-724-001

### Recount Risks under Electoral-College and Popular-Vote Systems

O / open | Oberwolfach 报告问题汇编

Under a probabilistic model of state and national vote margins, is the probability of a recount-triggering Electoral College victory larger than the probability of a recount-triggering national popular-vote margin?

来源字段：https://doi.org/10.4171/owr/2004/14

15765 | OWR-1543-003

### Recurrence and Diffusion in Alternating-Drift Walks

O / open | Oberwolfach 报告问题汇编

For the $H$-walk on $\mathbb Z^d$, whose cookie-induced drift alternates between left and right, determine whether the walk is recurrent in dimension 2 and transient in dimension 3. Determine the analogous behavior for the $Y$-walk, and whether the two-walker Slugs model in dimension 2 is diffusive and transient.

来源字段：https://doi.org/10.4171/owr/2007/27

15766 | OWR-15962-004

### Recurrence of High-Dimensional Scale-Free Percolation

O / open | Oberwolfach 报告问题汇编

Consider scale-free percolation on $\mathbb Z^d$ in a parameter regime with a unique infinite cluster, $\gamma>2$, and $\alpha>2d$. In dimensions $d\geq3$, is the simple random walk on the infinite cluster recurrent or transient? Resolve the analogous open regime for long-range percolation as well.

来源字段：https://doi.org/10.4171/owr/2017/57

15767 | AMR-099-0053

### Recurrence versus half-density percolation in disk triangulations

O / open | AMR 问题清单汇编

Let $G$ be the one-skeleton of a bounded-degree recurrent triangulation of an open disk. Prove that Bernoulli site percolation with parameter $1/2$ has no infinite cluster.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/resistance_percolation.pdf#page=3

15768 | AIM-PROBABILITY-0046

### Reducing interchange comparisons to two label-mixing bounds

P / partially_solved | AIM 研讨会问题汇编

**Comparison of interchange to SRW and Exclusion.** Consider the interchange process with \(n\) particles on \(n\) vertices. Can you say the following? \[ \begin{aligned} t_{\mathrm{mix}}^{IP}&\lesssim t_{\mathrm{mix}}^{SRW}\log n,\\ t_{\mathrm{mix}}^{IP}&\lesssim t_{\mathrm{mix}}^{SRW,n},\\ t_{\mathrm{mix}}^{IP}&\asymp t_{\mathrm{mix}}^{SSEP,n/2}? \end{aligned} \]

来源字段：http://aimpl.org/markovmixing/4/

15769 | OWR-13685-001

### Regularity of Polynomial Laws in Free Probability

P / partially_solved | Oberwolfach 报告问题汇编

What regularity properties does the distribution $\mu_Y$ of $Y=P(X_1,\ldots,X_n)$ have under various assumptions on self-adjoint $X_1,\ldots,X_n$?

来源字段：https://doi.org/10.4171/owr/2015/28

15770 | OWR-12697708-010

### Reinforcement Counterexamples on Integer Lattices

O / open | Oberwolfach 报告问题汇编

Can a supercritical-reinforcement example on a regular tree with infinitely many infinite surviving components be adapted to $\mathbb{Z}^d$ for some $d\geq2$?

来源字段：https://doi.org/10.4171/owr/2023/12

15771 | AMR-098-0001

### Relative age in a null-recurrent renewal process

S / solved | AMR 问题清单汇编

Let $S_n=S_0+X_1+\cdots+X_n$ be a renewal process whose i.i.d. strictly positive recurrence times have infinite mean and a non-lattice distribution. If $A_t$ is the age, $B_t$ the residual life, $D_t=A_t+B_t$, and $U_t=A_t/D_t$, does $U_t$ converge in distribution to a uniform random variable on $[0,1]$ as $t\to\infty$?

背景链接：http://cms.dm.uba.ar/depto/public/Some%20Open%20Problems-preprint.pdf

15772 | AIM-PROBABILITY-0103

### Relative free-difference-quotient constants and spectral gaps

P / partially_solved | AIM 研讨会问题汇编

In a von Neumann algebra \(M\) with a faithful normal trace-state τ let \(X=X^*\in M\) and let \(1\in B\subset M\) be an infinite-dimensional von Neumann subalgebra so that \(B\) and \(X\) are free in the algebraic sense and \(M=W^*(X,B)\). Assume that ∂\(_{X:B}\) is closable in \(L^2(M,τ)\). ... Under what conditions are the \(L^2\) solutions of ∂\(_{X:B}u=0\) in \(L^2(B,τ)\)? A related question ... when does the free Poincare inequality

\[ C\|\partial_{X:B}\xi\|_2\geq \|\xi-E_B\xi\|_2 \]

hold for ξ in \(B\langle X\rangle\)?

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15773 | AMR-096-0008

### Relaxation time of Metropolis chains on Cayley graphs

O / open | AMR 问题清单汇编

For the Metropolis chain on a finite Cayley graph with stationary law $\mu(p)$ obtained by stopping random walk at a geometric time, analyze its relaxation time $\tau(p)$. Is $\tau(p)$ decreasing in $p$? Is it universally bounded by a constant times its endpoint value, and can one give decreasing bounds using $p$, the graph size, and standard graph parameters?

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15774 | OWR-15218-008

### Relaxing Endpoint Conditions in Affine Volatility Large Deviations

P / partially_solved | Oberwolfach 报告问题汇编

In the large-deviation principle for affine stochastic-volatility models, can the endpoint condition $w(u_+)=w(u_-)$ be weakened or removed, and which replacement hypotheses suffice?

来源字段：https://doi.org/10.4171/owr/2017/13

15775 | OWR-1457-001

### Relevant Islands in the Parabolic Anderson Model

O / open | Oberwolfach 报告问题汇编

In the parabolic Anderson model, whose long-time behavior is governed by relevant islands, determine: (1) how large the relevant islands are, and how large the potential and the solution are in them; (2) the shapes of the potential and solution in these islands; and (3) how many relevant islands there are and what can be said about their spatial distribution.

来源字段：https://doi.org/10.4171/owr/2007/06

15776 | OWR-11695859-004

### Removing Contractibility from Metric Measure Reconstruction

O / open | Oberwolfach 报告问题汇编

Can the contractibility assumption in the metric-measure reconstruction theorem be removed?

来源字段：https://doi.org/10.4171/owr/2022/49

15777 | OWR-11695859-003

### Removing Curvature Restrictions from Metric Measure Reconstruction

O / open | Oberwolfach 报告问题汇编

Can the curvature restrictions in the metric-measure reconstruction theorem be removed?

来源字段：https://doi.org/10.4171/owr/2022/49

15778 | OWR-11101922-003

### Removing Monotonicity Assumptions in Interacting Particle Systems

O / open | Oberwolfach 报告问题汇编

Can the monotonicity assumptions used for the Ising and random-cluster interacting-particle-system results be removed?

来源字段：https://doi.org/10.4171/owr/2022/41

15779 | OWR-1186-004

### Representative Volume Scales for Random Microstructures

O / open | Oberwolfach 报告问题汇编

For a specified random microstructure and prescribed accuracy, at what mesoscale is the representative volume element attained?

来源字段：https://doi.org/10.4171/owr/2006/04

15780 | OWR-13861-007

### Resistance Exponents of Critical Branching Random-Walk Traces

O / open | Oberwolfach 报告问题汇编

For the trace of critical branching random walk in dimension $d<6$, what polynomial exponent governs the growth of its effective resistance $R(n)$?

来源字段：https://doi.org/10.4171/owr/2015/40

15781 | AMR-099-0076

### Resistance growth on the UIPT

P / partially_solved | AMR 问题清单汇编

Determine the almost-sure asymptotic growth rate of the effective resistance from the root to graph-distance $r$ in the uniform infinite planar triangulation.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/randomplanar2.pdf#page=3

15782 | OWR-15176-001

### Return Probabilities and Stability of the Liouville Property

O / open | Oberwolfach 报告问题汇编

If a symmetric random walk with step distribution $\mu$ on a group $G$ satisfies $\mu^{(2n)}(e)\geq\exp(-cn^{1/2})$, must $(G,\mu)$ have the Liouville property? For a finitely generated group, is the Liouville property stable under changing the finite generating set?

来源字段：https://doi.org/10.4171/owr/2016/55

15783 | OWR-11570-004

### Rigorous Noise-Induced Phenomena in Random Dynamical Systems

O / open | Oberwolfach 报告问题汇编

Give mathematically complete proofs for various noise-induced phenomena observed in numerical experiments for random dynamical systems, including i.i.d. random dynamics of complex polynomials on the Riemann sphere.

来源字段：https://doi.org/10.4171/owr/2012/03

15784 | AMR-099-0070

### Rotation-, translation-, scale-, and Markov-invariant random tilings

O / open | AMR 问题清单汇编

Does there exist a mixing random tiling of the Euclidean plane whose law is invariant under rotations and translations, is stationary under a local clustering-and-rescaling operation, and has a spatial Markov property in which a tile does not reveal the tiling of its complement along its boundary?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=4

15785 | AMR-099-0039

### Rough-isometry invariance of percolation nonuniqueness

P / partially_solved | AMR 问题清单汇编

For bounded-degree graphs, prove that the property $p_c<p_u$ is invariant under rough isometry.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=65

15786 | OWR-1731-004

### RSW Theory for Anisotropic Square-Lattice Percolation

S / solved | Oberwolfach 报告问题汇编

In critical anisotropic bond percolation on the square lattice, let horizontal and vertical edge probabilities $p_h,p_v$ satisfy $p_h+p_v=1$, and let $H(a,b)$ be the event of an open horizontal crossing of $[0,a]\times[0,b]$. Does an RSW theorem hold for general $p_h,p_v$, namely a uniform positive a priori bound for square crossings and a conditional extension bound for $H(kn,n)$ for every $k$? More generally, is there a class of shapes for which both properties hold?

来源字段：https://doi.org/10.4171/owr/2007/48

15787 | OWR-4081-005

### RWRE Concentration at Transverse Dimension One

O / open | Oberwolfach 报告问题汇编

For an i.i.d. nearest-neighbor elliptic space-time random walk in random environment in dimension $1+1$, is it false that for every sequence $M_N\to\infty$, every $\epsilon>0$, and every partition $\{Q_i^{(N)}\}_{i\geq1}$ of the transverse lattice $\mathbb Z$ into intervals of length $M_N$, the environment probability $$\mathbb P\!\left(\omega:\sum_{i\geq1}\left|P(X_N\in Q_i^{(N)})-P_\omega(X_N\in Q_i^{(N)})\right|>\epsilon\right)$$ decays faster than every power of $N$? Here $P$ and $P_\omega$ are the averaged and quenched laws, respectively.

来源字段：https://doi.org/10.4171/owr/2009/38

15788 | OWR-12729-001

### Scalable Sequential Monte Carlo for Pathogen Sequences

O / open | Oberwolfach 报告问题汇编

Can fundamental algorithmic developments enable sequential Monte Carlo models with about 1000 observed sequences and 20 parameters?

来源字段：https://doi.org/10.4171/owr/2013/55

15789 | AIM-PROBABILITY-0119

### Scalar circular resolvent estimate and the sharp two-thirds obstruction

P / partially_solved | AIM 研讨会问题汇编

Let $x,y$ be two free circular elements, and let $S,T$ be two operators in a $\mathrm{II}_1$ factor which is free from $x,y$. In the Haagerup--Schultz estimate

\[ (\star\star)\qquad \bigl\|(S+xy^{-1})^{-1}-(T+xy^{-1})^{-1}\bigr\|_p \le c(p)\,\|S-T\|_p<\infty, \]

with $0<p<2/3$, can one use $x$ instead of $xy^{-1}$?

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15790 | AMR-099-0013

### Scaling limit of random recursive square subdivision

O / open | AMR 问题清单汇编

Start with a unit square and repeatedly choose a current square uniformly and subdivide it into four squares. Let $D_n$ be the minimum number of current squares in a connected set joining the bottom-left corner to the top-right corner after $n$ subdivisions. Does there exist a deterministic scaling function $a_n$ such that $D_n/a_n$ converges to a nondegenerate random variable? Does a shortest crossing stabilize under further subdivisions?

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=16

15791 | OWR-15989-009

### Scaling Limit of the Sandpile Height-Zero Field

O / open | Oberwolfach 报告问题汇编

Let $D\subset\mathbb C$ be a bounded smooth domain, let $\nu_\varepsilon$ be the Abelian sandpile measure on $D\cap\varepsilon\mathbb Z^2$, and define the centered height-zero field $$ h_\varepsilon(z)=\mathbf1_{\{\eta(z)=0\}}-\nu_\varepsilon(\eta(z)=0). $$ Does a suitably rescaled $h_\varepsilon$ converge, in an appropriate function or distribution space, to the generalized field $:\!|\nabla h|^2\!:$ as $\varepsilon\downarrow0$?

来源字段：https://doi.org/10.4171/owr/2018/4

15792 | OWR-12016-007

### Scaling Limits Across the Directed-Polymer Crossover

O / open | Oberwolfach 报告问题汇编

For $(1+1)$-dimensional directed polymers, does the scaling limit at the $\beta=\infty$ end of the convergence diagram exist; what are the properties of the intermediate-disorder crossover bridge; and, for environment scalings slower than $n^{-1/4}$, does the pair $(\omega_n,S_n)$ have a distributional limit, and if so is it the same as the last-passage-percolation limit?

来源字段：https://doi.org/10.4171/owr/2012/44

15793 | OWR-14298373-008

### Scaling Limits of Random High-Dimensional Dimer Tilings

P / partially_solved | Oberwolfach 报告问题汇编

What are the scaling limits and limit shapes of random dimer tilings of regions in $\mathbb{R}^n$ as $n\to\infty$, especially beyond two dimensions?

来源字段：https://doi.org/10.4171/owr/2024/25

15794 | AMR-099-0068

### Scaling of distances in a random hierarchical graph

O / open | AMR 问题清单汇编

In the random hierarchical graph obtained by repeatedly replacing a uniformly chosen edge by the fixed three-edge pattern of Section 9.4, let $D_n$ be the distance between the two marked endpoints and let $\log\mathbb{E}D_n/\log n\to\gamma$. Determine $\gamma$ and find a normalization under which $D_n$ converges in distribution, equivalently giving a Gromov-Hausdorff scaling limit.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=18

15795 | OWR-1729-002

### Scaling Profiles in Addition-Coagulation Models

O / open | Oberwolfach 报告问题汇编

别名：Duplicate: Addition-Coagulation Similarity Limit with Mismatched Provenance [OWR-1598-004]

Consider the addition model $\dot c_1=J_1(t)-c_1^2-c_1\sum_{j\geq1}j^pc_j$ and $\dot c_j=j^{p-1}c_1c_{j-1}-j^pc_1c_j$ for $j\geq2$, where $p<1$, $J_1(t)=\alpha t^\omega$, $\alpha>0$, and $\omega>-1/2$. Set $$r_p=\frac{1-\omega(1-p)}{(2+\omega)(1-p)}.$$ For initial data satisfying $c_j(0)\leq\rho/j^\mu$ for some $\rho>0$ and $\mu>r_p+p$, define $$\varsigma(t)=\left(\frac{Q_p(\omega)(3-2p)}{2+\omega}\right)^{1/(1-p)}t^{(2+\omega)/(3-2p)},\qquad \widetilde c_j(\varsigma)=c_j(t(\varsigma)).$$ Do constants $A(\omega,p)$ exist such that $A(\omega,p)\varsigma^{r_p}\widetilde c_j(\varsigma)\to\Phi_{1,\omega,p}(\eta)$ as $j,\varsigma\to\infty$ with $\eta=j/\varsigma\neq1$? Here $$\Phi_{1,\omega,p}(\eta)=\begin{cases}\eta^{-p}(1-\eta^{1-p})^{-r_p},&\eta<1,\\0,&\eta>1\end{cases}$$.

来源字段：https://doi.org/10.4171/owr/2007/46

15796 | AMR-098-0002

### Scaling total life in a null-recurrent renewal process

S / solved | AMR 问题清单汇编

For the null-recurrent renewal process of Problem 1.1, is there a non-decreasing function $\phi$ such that $D_t/\phi(t)$ converges in distribution to a non-degenerate random variable $D_\phi$ as $t\to\infty$? In particular, does this hold for $\phi(t)=\mathbb{E}[\min\{X_1,t\}]$?

背景链接：http://cms.dm.uba.ar/depto/public/Some%20Open%20Problems-preprint.pdf

15797 | OWR-14299282-003

### Score-Uncertainty Propagation in Reflected Generative Models

O / open | Oberwolfach 报告问题汇编

Quantify how uncertainty in an estimated score propagates through a reflected score-based generative diffusion. Derive finite-time and distributional error bounds that account for reflection at the domain boundary.

来源字段：https://doi.org/10.4171/owr/2025/21

15798 | AMR-096-0014

### Second-longest Eulerian excursion on the two-dimensional torus

O / open | AMR 问题清单汇编

For a uniform Eulerian circuit on the bidirected two-dimensional torus, does $\log L_2^{(N)}/\log N$ converge in distribution to a random variable with support $[0,2]$?

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15799 | OWR-14740-003

### Second-Order Bias in Kernel Level-Set Estimation

P / partially_solved | Oberwolfach 报告问题汇编

For the kernel estimator $\widehat L_h=\{x\in\mathbb R^d:\widehat f_h(x)\geq c\}$ of $L=\{x:f(x)\geq c\}$, with mean distance-in-measure risk $\operatorname{MDM}(h)=I_1(h)+I_2(h)$, does there exist $B=B(f,K,c)$ such that $h^{-2}I_2(h)\to B$ as $n\to\infty$ without assuming $nh^{d+4}\to0$?

来源字段：https://doi.org/10.4171/owr/2016/32

15800 | AMR-099-0023

### Self-avoiding loops on nonamenable transitive graphs

P / partially_solved | AMR 问题清单汇编

Let $G$ be vertex-transitive with positive Cheeger constant. If $\mu$ is the connective constant of self-avoiding walks and $\mu_{\mathrm{loops}}$ is the exponential growth rate of self-avoiding loops, prove that $\mu_{\mathrm{loops}}<\mu$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=38

15801 | AMR-098-0005

### Setwise convergence versus total-variation convergence of shifted processes

P / partially_solved | AMR 问题清单汇编

Let $X$ and $X'$ be discrete-time stochastic processes on the same state space, and let $\theta_n$ denote the shift. If $\mathbb{P}(\theta_nX\in A)\to\mathbb{P}(X'\in A)$ for every measurable path-space set $A$, must $\mathbb{P}(\theta_nX\in\cdot)$ converge to $\mathbb{P}(X'\in\cdot)$ in total variation?

背景链接：http://cms.dm.uba.ar/depto/public/Some%20Open%20Problems-preprint.pdf

15802 | OWR-10252930-033

### Sharp Concentration of Random-Graph Chromatic Number

P / partially_solved | Oberwolfach 报告问题汇编

For $n^{-1/2+\delta}\ll p\leq1-\delta$, is the $\widetilde O(\sqrt{np}/\log n)$ concentration-window bound for the chromatic number $\chi(G_{n,p})$ close to optimal?

来源字段：https://doi.org/10.4171/owr/2022/22

15803 | OWR-2654836-009

### Sharp Density Estimates for General Gaussian Rough Paths

P / partially_solved | Oberwolfach 报告问题汇编

Can the sharp density-estimate results be extended from fractional Brownian motion to more general Gaussian processes?

来源字段：https://doi.org/10.4171/owr/2020/40

15804 | OWR-2654836-008

### Sharp Fractional Brownian Density Estimates with Drift

P / partially_solved | Oberwolfach 报告问题汇编

Can the sharp density-estimate results for fractional-Brownian rough differential equations be extended to equations with drift, despite the loss of fractional-Brownian scaling and the required sub-Riemannian accessibility conditions?

来源字段：https://doi.org/10.4171/owr/2020/40

15805 | AIM-PROBABILITY-0053

### Sharp Gibbs status and exact proposal-sensitive Metropolis benchmarks

P / partially_solved | AIM 研讨会问题汇编

**Metropolis on $[0,1]^2$.** Consider the stationary distribution on $[0,1]^2$ given by \[ \pi_A(p_1,p_2)=\mathcal Z^{-1}\exp(-A|p_1-p_2|^2), \] parametrized by $A\in[10,1000]$. Obtain a good bound on the mixing time of the Metropolis algorithm for this stationary distribution in terms of $A$. Generalize this to $[0,1]^d$.

来源字段：http://aimpl.org/markovmixing/6/

15806 | OWR-9790365-015

### Sharp Min-Entropy Power Inequalities

O / open | Oberwolfach 报告问题汇编

For sums of $N$ independent random vectors in $\mathbb R^d$, determine the sharp constant and all extremizers in the Rényi entropy-power inequality for orders other than $0$, $1$, and $\infty$.

来源字段：https://doi.org/10.4171/owr/2022/14

15807 | AMR-096-0010

### Sharp phase transition for SIS epidemics on general networks

O / open | AMR 问题清单汇编

For sequences of finite weighted networks with vertex recovery rates and stationary SIS infection counts $X^{(n)}_{\theta,\varepsilon}$ satisfying the talk's subcritical/supercritical assumption (6), prove-perhaps under further weak hypotheses-that there are thresholds $\theta_n$ for which every sufficiently slowly vanishing $\varepsilon_n$ gives vanishing infected proportion below $\theta_n-\delta$ and a nonvanishing proportion above $\theta_n+\delta$, for every $\delta>0$.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15808 | OWR-12697708-008

### Sharp Reinforcement Thresholds for Tree and Whisker Components

P / partially_solved | Oberwolfach 报告问题汇编

For supercritical reinforcement, improve the threshold $\alpha>2$ guaranteeing that every surviving connected component is a tree, and the threshold $\alpha>25$ guaranteeing, under $p(v)=1$, that every component is a whisker. Is $\alpha>4/3$ sharp for both conclusions?

来源字段：https://doi.org/10.4171/owr/2023/12

15809 | OWR-17469-007

### Sharp Singularity Asymptotics for Sparse Bernoulli Matrices

P / partially_solved | Oberwolfach 报告问题汇编

Let $M_n$ be an $n\times n$ matrix with independent Bernoulli entries satisfying $\mathbb P(M_{ij}=1)=p_n$. Let $E_{\mathrm{zero}}$ be the event that $M_n$ has a zero row or column, and let $E_{\mathrm{equal}}$ be the event that it has two equal rows or two equal columns. In the sparse regime, is

$$\mathbb P(M_n\text{ is singular})=(1+o_n(1))\mathbb P(E_{\mathrm{zero}}\cup E_{\mathrm{equal}})?$$

In particular, when $\limsup p_n<1/2$, is this probability

$$2n(1+o_n(1))(1-p_n)^n$$?

来源字段：https://doi.org/10.4171/owr/2020/6

15810 | OWR-17469-006

### Sharp Singularity Probability of Random Sign Matrices

P / partially_solved | Oberwolfach 报告问题汇编

Let $B_n$ be an $n\times n$ random sign matrix with independent entries uniformly distributed in $\{-1,1\}$. Determine the sharp asymptotic probability that $B_n$ is singular. The conjectural scale is

$$\mathbb P(B_n\text{ is singular})=(1/2+o_n(1))^n=(1/2)^{(1+o_n(1))n}$$.

来源字段：https://doi.org/10.4171/owr/2020/6

15811 | OWR-12016-002

### Sharp Tail Decay for Persistent Critical Crossings

O / open | Oberwolfach 报告问题汇编

For critical dynamical percolation and the left-right crossing event $C_Q$ of a square $Q$, is the tail probability bounded by

$$\mathbb P[\omega_s\in C_Q\text{ for all }s\in[0,t]]\leq\exp(-t^{2/3+o(1)})$$

as $t\to\infty$?

来源字段：https://doi.org/10.4171/owr/2012/44

15812 | OWR-1386-002

### Sharp Thresholds in Bootstrap Percolation

O / open | Oberwolfach 报告问题汇编

For bootstrap percolation on a growing sequence of finite graphs $G_n$, let $p_\varepsilon(n)=\inf\{p:\mathbb P(G_n,p)>\varepsilon\}$. Does $p_{1-\varepsilon}(n)-p_\varepsilon(n)\to0$, and in particular does $p_{1-\varepsilon}(n)/p_\varepsilon(n)\to1$ as $n\to\infty$ for every $\varepsilon>0$?

来源字段：https://doi.org/10.4171/owr/2006/48

15813 | AIM-PROBABILITY-0025

### Sharp total-variation asymptotics for Student t versus the standard normal

S / solved | AIM 研讨会问题汇编

It asks, for Student \(t_r\) with \(r\) degrees of freedom and \(Z\sim N(0,1)\):

1. Show that \(d_{\mathrm{TV}}(t_r,Z)\leq c/r\) for some \(c>0\). 2. Find the best such constant. 3. Decide whether \(d_{\mathrm{TV}}(t_r,Z)\geq c'/r\) for some \(c'>0\).

There is one mathematical convention to fix. Throughout this report,

\[ d_{\mathrm{TV}}(P,Q) =\sup_A|P(A)-Q(A)| =\frac12\int_{\mathbb R}|p-q|\,dx. \]

There is also a quantifier ambiguity. Because the prompt begins “as \(r\to\infty\),” “best constant” naturally means the sharp constant for all sufficiently large \(r\), equivalently the sharp first-order asymptotic coefficient. If instead the inequality is required for every positive integer \(r\), the three low degrees must be included separately. Both readings are handled below.

来源字段：http://aimpl.org/steinhd/1/

15814 | OWR-11695855-004

### Sharp Universal Gaussian Mean-Estimation Constant

O / open | Oberwolfach 报告问题汇编

Determine the sharp constant in the universal deviation-minimax rate for Gaussian mean estimation over all pseudonorms.

来源字段：https://doi.org/10.4171/owr/2022/46

15815 | AMR-099-0082

### Sharp vacant-set transition on uniformly transient transitive graphs

P / partially_solved | AMR 问题清单汇编

Let $(G_n)$ be finite transitive graphs with $|G_n|\to\infty$ and uniformly bounded effective resistances between all vertex pairs. Prove that the largest vacant component left by simple random walk drops from order $|G_n|$ to $o(|G_n|)$ before time $C|G_n|$, for a fixed $C$, in a transition window of width $o(|G_n|)$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/randomplanar2.pdf#page=9

15816 | OWR-15962-002

### Sharp Virulence Thresholds for Stationary SIS Infection

O / open | Oberwolfach 报告问题汇编

别名：Epidemic Phase Transitions on General Weighted Networks [OWR-15962-001]

For each $n$, consider a finite connected weighted network with recovery rates $\mu_v$, and let $X^{(n)}_{\theta,\varepsilon}$ be the stationary number of infected vertices in the SIS process whose infection rate at $v$ is $$\varepsilon+\theta\sum_{y\ \mathrm{infected}}w_{vy}.$$ Assume that there are $0<\theta_*<\theta^*<\infty$ such that, for every sufficiently slowly vanishing sequence $\varepsilon_n$, $$X^{(n)}_{\theta_*,\varepsilon_n}=o(n)\quad\text{and}\quad X^{(n)}_{\theta^*,\varepsilon_n}=\Omega(n)$$ in probability. Must there be $\theta_n\in[\theta_*,\theta^*]$ such that, for every fixed $\delta>0$, $$X^{(n)}_{\theta_n-\delta,\varepsilon_n}=o(n)\quad\text{and}\quad X^{(n)}_{\theta_n+\delta,\varepsilon_n}=\Omega(n)$$ in probability?

来源字段：https://doi.org/10.4171/owr/2017/57

15817 | OWR-14299914-004

### Sharper Decay and Algorithms for Non-Gaussian Targets

O / open | Oberwolfach 报告问题汇编

For the projected gradient dynamics with a non-Gaussian log-concave target, prove a sharper decay estimate that does not deteriorate with the convexity parameter $\lambda$, and construct a provably convergent numerical scheme.

来源字段：https://doi.org/10.4171/owr/2026/13

15818 | AMR-096-0016

### Shortest Eulerian excursion on the Hamming cube

O / open | AMR 问题清单汇编

For a uniform Eulerian circuit on the bidirected Hamming cube $\{0,1\}^d$, determine the asymptotic behavior or distribution of the shortest excursion length $L_d^{(d)}$.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15819 | AMR-096-0011

### Shortest routes in random proximity networks

O / open | AMR 问题清单汇编

For random proximity graphs on a planar Poisson point process, determine rigorous orders of magnitude for the transversal deviation $T_r$ of a shortest network route between points at Euclidean distance $r$ and for the variance of its route length $D_r$.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15820 | AIM-PROBABILITY-0061

### Sign normalization and a degree-dependent cut bound for polynomial S-curves

P / partially_solved | AIM 研讨会问题汇编

**$S$-curves for complex potentials in the scalar case**

Analysis of existence and properties of $S$-curves in the scalar case when the potential is $\phi(z)=\textrm{Re}\,z^p$, for $p\geq 3$, and we take $\tau$ as the set of continua that join neighboring valleys of $\phi(z)$ (where $\phi(z)\to-\infty$) in the complex plane. In particular, prove that the support of the equilibrium measure on a curve with the $S$-property is one-cut case, or give bounds on the number of cuts depending on $p$.

来源字段：http://aimpl.org/vectorequilib/4/

15821 | AIM-PROBABILITY-0163

### Signed-charge formulas for finite-order unitary cumulants

S / solved | AIM 研讨会问题汇编

Let \(u\) be a unitary element of a \(*\)-probability space \((A,\varphi)\). Suppose \(u\) has order \(k\) and that \(\varphi(u^i)=0\) for \(1\leq i<k\). Let \(\kappa_n\) denote the multilinear cumulant functionals of \((A,\varphi)\). Give a combinatorial way to compute the cumulants in \(u\) and \(u^*\). Equivalently, give a formula for the \(R\)-transform of \((u,u^*)\).

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15822 | AIM-PROBABILITY-0150

### Signed-cycle traces and the full B2 diagonal coinvariant module

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.2 (C. Kriloff and V. Reiner), recovered.** This is a possible systematic approach to Problem 2.1. As mentioned, in type \(A\) with \(k=1\), the numbers (1) correspond (up to a power of \(q\), and specialized at \(t=1/q\)) with the \(q,t\)-Catalan numbers of Garsia and Haiman, which are given by the \(q,t\)-bigraded Hilbert series for the sign-isotypic component of the ring of diagonal harmonics \[ \mathbb C[V\oplus V^*]/\langle \mathbb C[V\oplus V^*]^W_+\rangle. \] Can this situation be generalized to other \(W\)? When \(W\) is a symmetric group \(A_{n-1}\), it is known that the action of \(W\) on the (ungraded) diagonal harmonics has the same irreducible decomposition as the action of \(W\) on the “finite torus” \(Q/(h+1)Q\), where \(Q\) is the root lattice. Mark Haiman noted this does not hold in type \(B\). However, Iain Gordon has shown that it is possible to take a further quotient which does give the desired combinatorics.

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15823 | OWR-12873-005

### Simultaneous Adaptive Estimation Across Norms and Derivatives

O / open | Oberwolfach 报告问题汇编

For estimation over Hölder balls with unknown smoothness $\beta$ and radius $Q$, can one adapt simultaneously over all $L^p$ losses? Can one simultaneously attain sharp $L^2$ and $L^\infty$ rates for a function and its derivatives? Can adaptation be achieved without coefficientwise wavelet bounds, and can one characterize when simultaneous adaptation is possible and construct an attaining estimator?

来源字段：https://doi.org/10.4171/owr/2014/13

15824 | OWR-9790359-007

### Simultaneous Spatial and Informational Coding Efficiency

O / open | Oberwolfach 报告问题汇编

Let $X$ be a finitary factor of an independent and identically distributed process with exponentially decaying coding-radius tails. Must $X$ also be a finitary factor of a finite-valued independent process having entropy arbitrarily close to that of $X$ and exponentially decaying coding-radius tails?

来源字段：https://doi.org/10.4171/owr/2022/8

15825 | OWR-15988-002

### Singular Curvature and Stickiness of Fréchet Means

O / open | Oberwolfach 报告问题汇编

Are curvature-dependent corrections to Fréchet means and covariances on Riemannian manifolds precursors of stickiness of the mean in singular-curvature limits?

来源字段：https://doi.org/10.4171/owr/2018/3

15826 | AIM-PROBABILITY-0014

### Singular EM fixed manifolds and ML-degree-one determinant strata of RBM(3,2)

P / partially_solved | AIM 研讨会问题汇编

What are the critical points of the EM algorithm for \(RBM_{3,2}\)?

What are the critical points of the EM algorithm for \(RBM_{3,2}\)? Can the MLE degree for the strata of \(RBM_{3,2}\) be determined?

The EM fixed points may be known for \({\cal M}_{3,3}\), but since these points depend on the specific parameterization of the model, the fixed points of \(RBM_{3,2}\) remain unknown.

来源字段：http://aimpl.org/boltzmann/5/

15827 | OWR-11695865-007

### Singular Loci of Compartment Model Jacobians

P / partially_solved | Oberwolfach 报告问题汇编

For which parameter values does the Jacobian matrix of a linear compartment model drop rank?

来源字段：https://doi.org/10.4171/owr/2022/55

15828 | OWR-1183-013

### Singularity Probability of Random Symmetric Sign Matrices

P / partially_solved | Oberwolfach 报告问题汇编

Let $Q_n$ be an $n\times n$ random symmetric matrix whose upper-triangular entries are independent Bernoulli variables taking values $\pm1$. Determine or estimate the probability that $Q_n$ is singular.

来源字段：https://doi.org/10.4171/owr/2006/01

15829 | AMR-096-0036

### SIRSN subnetworks cannot be trees

O / open | AMR 问题清单汇编

Prove that in a scale-invariant random spatial network the subnetwork $S(1)$ cannot be a tree, even allowing Steiner points.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15830 | OWR-1061-007

### Site-Frequency Spectra under Multiple Ancestral Mergers

P / partially_solved | Oberwolfach 报告问题汇编

Determine how multiple mergers of ancestral lineages affect the full site-frequency spectrum, including the number of mutations affecting exactly $k$ lineages for each $k$.

来源字段：https://doi.org/10.4171/owr/2005/40

15831 | OWR-13685-004

### Small Singular Values of Matrix-Valued Free Functions

O / open | Oberwolfach 报告问题汇编

Can the small singular values of general matrix-valued functions $F(X_1,\ldots,X_m)$ be controlled?

来源字段：https://doi.org/10.4171/owr/2015/28

15832 | OWR-4990379-001

### Smooth Optimal Transport Estimation beyond Dimensionality

P / partially_solved | Oberwolfach 报告问题汇编

Can strong smoothness assumptions on the probability distributions be used to overcome the curse of dimensionality in statistical estimation of optimal transport?

来源字段：https://doi.org/10.4171/owr/2021/10

15833 | AIM-PROBABILITY-0128

### Smooth symbols of a circular element: calculi, quantizations, and obstructions

P / partially_solved | AIM 研讨会问题汇编

**Question 26.** Let \(c\) be a circular element, with \(\sigma(c)=\overline{\mathbb D}\), and let \(f\in C^\infty(\mathbb C)\). Can we make sense of \(f(c)\) as an (unbounded) operator affiliated with \(\{c\}''\)?

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15834 | AIM-PROBABILITY-0117

### Smooth-witness and semicircular-attractor obstructions to free strong unimodality

P / partially_solved | AIM 研讨会问题汇编

What are examples or conditions for freely strongly unimodal variables, meaning unimodal random variables whose free additive convolution with every unimodal variable remains unimodal? Here “unimodal” means that the law has a smooth density with a unique maximum; examples include the Gaussian and semicircle laws.

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15835 | AIM-PROBABILITY-0060

### Soft d-bar characterization, defect certificate, and exact radial benchmarks

P / partially_solved | AIM 研讨会问题汇编

**Normal matrix model and d-bar problems.** Is it possible to use the theory of \(\overline{\partial}\)-problems to analyze the normal matrix models?

来源字段：http://aimpl.org/vectorequilib/3/

15836 | AIM-PROBABILITY-0076

### Soft Karlin-McGregor kernels and a Toda determinant obstruction

P / partially_solved | AIM 研讨会问题汇编

The limit of the triangular array is the diffusion of O'Connell based on the quantum Toda lattice, or (after another limit) the KPZ$_T$ line ensemble. Both have Brownian Gibbs properties allowing paths to cross but at exponential cost (a soft analog of the non-intersecting Brownian Gibbs property for the Airy line ensemble.)The Karlin-McGregor formula underlies the solvability of zero temperature system since it writes non-intersecting line ensembles in terms of determinants.

Is there an analog of the Karlin-McGregor formula when the non-intersecting conditioning is replaced by a softer form of conditioning? Does this explain the solvability and occurrence of Fredholm determinants?

来源字段：http://aimpl.org/kpzuniversality/1/

15837 | OWR-9790360-003

### Solution Concepts for Stable SDEs with Distributional Drift

O / open | Oberwolfach 报告问题汇编

For linear and McKean-Vlasov stochastic differential equations driven by a symmetric stable process and having distributional drift, which notion of solution gives a well-defined and unique dynamics, including regimes below the Young threshold?

来源字段：https://doi.org/10.4171/owr/2022/9

15838 | AIM-PROBABILITY-0049

### Solved cutoff and an exact rank-correlation observable

S / solved | AIM 研讨会问题汇编

**Cutoff for random-to-random shuffle.** The random-to-random card shuffle on \(S_n\) has cutoff at \(\frac{3n}{4}\log n\).

The intended chain is the **random-to-random insertion shuffle**. From a deck of \(n\) distinct labeled cards, choose a card uniformly, remove it, choose independently a final position uniformly from \(\{1,\ldots,n\}\), and insert the card there. One step therefore has \(n^2\) equally likely ordered choices. The chain is discrete-time and needs no added laziness: the probability of leaving the deck unchanged is \(1/n\). It is symmetric, irreducible, aperiodic, and has uniform stationary distribution on \(S_n\).

Throughout, \(\log\) is the natural logarithm. For transition kernel \(P_n\), uniform law \(U_n\), and total variation distance, \[ d_n(t)=\max_{\sigma\in S_n}\|P_n^t(\sigma,\cdot)-U_n\|_{\rm TV}. \] Translation invariance makes the maximum independent of the starting permutation. A center \(t_n\) and window \(w_n=o(t_n)\) describe cutoff if \[ \lim_{c\to\infty}\limsup_{n\to\infty}d_n(t_n+cw_n)=0, \qquad \lim_{c\to\infty}\liminf_{n\to\infty}d_n(t_n-cw_n)=1. \]

The later \(1.5324n\log n\) upper bound is joint work of **Chuan Qin and Ben Morris**, not Morris alone, and its coefficient is \(1.5324\), not exactly \(3/2\).

来源字段：http://aimpl.org/markovmixing/5/

15839 | OWR-1108-003

### Sparsity Transfer in Penalized Empirical Risk Minimization

P / partially_solved | Oberwolfach 报告问题汇编

For penalized empirical risk minimization with population minimizer $\lambda_\varepsilon$ and empirical minimizer $\widehat{\lambda}_\varepsilon$, does sparsity of $\lambda_\varepsilon$ imply sparsity of $\widehat{\lambda}_\varepsilon$? How does that sparsity control the excess loss risk and the $\ell^1$ error from the unpenalized population minimizer, and how should the penalty parameter $\varepsilon$ be chosen?

来源字段：https://doi.org/10.4171/owr/2005/47

15840 | OWR-9790360-001

### Spatial Regularity Versus Time Integrability for Fractional SDEs

O / open | Oberwolfach 报告问题汇编

For the stochastic differential equation $$dX_t=b(t,X_t)\,dt+dW_t^H$$ driven by fractional Brownian motion of Hurst parameter $H$, can higher time integrability $b\in L_t^qC_x^\alpha$ lower the spatial regularity threshold below $\alpha>1-1/(2H)$ in accordance with the scaling prediction?

来源字段：https://doi.org/10.4171/owr/2022/9

15841 | OWR-12596-001

### Specification Testing for Higher-Order GARCH Models

P / partially_solved | Oberwolfach 报告问题汇编

Can the specification-testing theory for $\operatorname{GARCH}(1,1)$ models, including its asymptotic and bootstrap results, be extended to higher-order and augmented GARCH processes?

来源字段：https://doi.org/10.4171/owr/2013/48

15842 | OWR-5152-005

### Spectral Bounds for Random DFT Submatrices

O / open | Oberwolfach 报告问题汇编

Let $F_{\Omega_n,T_n}$ be the DFT submatrix formed by independently retaining rows with probability $1-q$ and columns with probability $1-p$. If $p+q>1$, is $\|F_{\Omega_n,T_n}\|<\sqrt{p(1-q)}+\sqrt{q(1-p)}+\varepsilon$ with high probability? If $p\ne q$, is $\|(F_{\Omega_n,T_n})^{-1}\|<\bigl(|\sqrt{p(1-q)}-\sqrt{q(1-p)}|-\varepsilon\bigr)^{-1}$ with high probability?

来源字段：https://doi.org/10.4171/owr/2011/24

15843 | AMR-096-0009

### Spectral gap of a Bayesian graph Laplacian

O / open | AMR 问题清单汇编

For the posterior random weighted graphs $G(t)$ defined from independent Poisson edge counts and flat priors, study the process $\operatorname{gap}(G(t))$. For large graphs is it concentrated near its expectation under weak assumptions, and how large must $t$ be before it is close to $\operatorname{gap}(G(\infty))$?

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15844 | OWR-13683-002

### Spectral Norms of Gaussian Matrices with Variance Profiles

P / partially_solved | Oberwolfach 报告问题汇编

Let $X$ be a symmetric Gaussian random matrix whose entries on and above the diagonal are independent and have prescribed variances $b_{ij}^2$. Determine or sharply bound its operator norm, especially the spectral edge, in terms of the variance profile.

来源字段：https://doi.org/10.4171/owr/2015/26

15845 | OWR-705-003

### Spectral-Edge Spacing in Large Random Band Matrices

O / open | Oberwolfach 报告问题汇编

Strengthen moment-method inequalities for large random band matrices sufficiently to determine the local scale of eigenvalue spacing at the spectral edge.

来源字段：https://doi.org/10.4171/owr/2004/22

15846 | OWR-2654828-007

### Speed Convergence in Convex Hull Avoiding Walks

O / open | Oberwolfach 报告问题汇编

For the finite-memory convex-hull-avoiding process in dimension $d$, is it true that $\lim_{k\to\infty}v_{d,k}=v_{d,\infty}$, where $v_{d,\infty}$ is the conjectural speed of the infinite-memory model?

来源字段：https://doi.org/10.4171/owr/2020/32

15847 | AIM-PROBABILITY-0085

### Spiked polymer formulas and a confluence-safe BBP reduction

S / solved | AIM 研讨会问题汇编

Exact solvability

Use the Macdonald processes formulas to prove the Baik-Ben Arous-P\'{e}ch\'{e} transition for the semi-discrete polymer with a finite number of non-zero drifts for the underlying Brownian motions (tuned critically). In the intermediate disorder scaling, derive the formula for the statistics of the stochastic heat equation started with \(\mathcal{Z}_0(X) = {\bf 1}_{X\geq 0} Z^r(X)\) where \(Z^N(X)\) is the partition function for the semi-discrete polymer with \(r\) levels at time \(X\) (the \(r\) here represents the number of spikes in BBP and the drifts here of the \(r\) Brownian motions can be generally chosen).

来源字段：http://aimpl.org/kpzuniversality/2/

15848 | OWR-14298589-009

### Spines and One-Face Maps of Random Hyperbolic Surfaces

O / open | Oberwolfach 报告问题汇编

Determine the limiting law of the spine of a random hyperbolic surface as the genus tends to infinity, and find other natural geometric constructions of one-face maps on hyperbolic surfaces.

来源字段：https://doi.org/10.4171/owr/2024/41

15849 | OWR-10252930-024

### Spread Bounds for the Random Triangle-Removal Process

O / open | Oberwolfach 报告问题汇编

For the random triangle-removal process on $K_n$, run for $n^2/6-n^{1.99}$ steps with output triangle set $\mathcal{T}_n$, do there exist events $E_n$ of probability $1-o(1)$ and a constant $C$ such that $\Pr(T_1,\ldots,T_k\in\mathcal{T}_n\mid E_n)\leq(C/n)^k$ for every collection of distinct triangles?

来源字段：https://doi.org/10.4171/owr/2022/22

15850 | OWR-4530-006

### Stability Beyond Unstable Population-Process Fluid Limits

O / open | Oberwolfach 报告问题汇编

In the two-chunk deterministic-first-chunk peer-to-peer population process, the large-system-limit dynamical system is unstable while simulations suggest that the stochastic process is stable. Prove or disprove this stability conjecture, and hence determine whether instability of a population process can fail to follow from instability of its large-system limit.

来源字段：https://doi.org/10.4171/owr/2010/48

15851 | AMR-096-0026

### Stability dichotomy for the associated city dynamical system

O / open | AMR 问题清单汇编

For the associated influence-cell dynamical system in general position with positive initial weights, prove that one weight tends to $1$ if $\alpha>1$ or $0<\beta<2\alpha$, while if $0<\alpha<1$ and $\beta>2\alpha$ all weights converge to positive limits independent of the initial values.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15852 | OWR-15428-005

### Stabilization of Hidden Alphabet Distribution Models

O / open | Oberwolfach 报告问题汇编

Let $C_d$ be nested classes of observable distributions, with $C_d\subseteq C_{d'}$ for $d\leq d'$, obtained by increasing a hidden alphabet. Does some finite $d$ satisfy $C_d=C_{d'}$ for every $d'\geq d$? Does stabilization hold under cyclic symmetry or full permutation symmetry of three observed variables without increasing the hidden alphabet?

来源字段：https://doi.org/10.4171/owr/2017/20

15853 | AIM-PROBABILITY-0116

### Stable boundary branches for operator-valued subordination

P / partially_solved | AIM 研讨会问题汇编

What is the boundary behavior of the subordination functions which appear in free convolution of operator-valued random variables?

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15854 | OWR-3399-001

### Stable Low-Rank Matrix Completion from Random Samples

P / partially_solved | Oberwolfach 报告问题汇编

Given a random sample of entries of a sufficiently incoherent low-rank matrix $M$, can nuclear-norm minimization recover every unobserved entry exactly and remain stable under bounded observation noise?

来源字段：https://doi.org/10.4171/owr/2009/16

15855 | OWR-4331-001

### Stable Reconstruction of Poisson Cylinder Directions

O / open | Oberwolfach 报告问题汇编

Given finitely sampled intersection intensities $g$ of a stationary Poisson cylinder process, can one stably estimate its directional distribution by a continuous inversion of the spherical cosine transform?

来源字段：https://doi.org/10.4171/owr/2010/18

15856 | AMR-095-0002

### Stationary distributions in higher dimensions

O / open | AMR 问题清单汇编

On $\mathbb{Z}^2$, take nearest-neighbor jump probabilities $p_1,q_1,p_2,q_2$ in directions $\pm e_1,\pm e_2$, with $p_1>q_1$ and $p_2>q_2$. If the angle between $v$ and the mean vector $(p_1-q_1,p_2-q_2)$ is less than $\pi/2$, prove that there is a stationary distribution invariant under shifts orthogonal to $v$, whose occupation density along $nv$ tends to $0$ as $n\to-\infty$ and to $1$ as $n\to+\infty$.

背景链接：https://web.archive.org/web/20150919235903id_/http://www.math.ucla.edu/~tml/open.pdf

15857 | AMR-095-0001

### Stationary distributions in one dimension

P / partially_solved | AMR 问题清单汇编

For the exclusion process on $\mathbb{Z}$ with $p(x,y)=p(y-x)$, assume $\sum_x|x|p(x)<\infty$, $\sum_xxp(x)>0$, and $\sum_{x<0}x^2p(x)=\infty$. Does there exist a stationary distribution $\mu$ with $\mu(\eta(x)=1)\to0$ as $x\to-\infty$ and $\mu(\eta(x)=1)\to1$ as $x\to+\infty$?

背景链接：https://web.archive.org/web/20150919235903id_/http://www.math.ucla.edu/~tml/open.pdf

15858 | AIM-PROBABILITY-0091

### Stationary flux and a tangent second-class identity for q-TASEP gaps

P / partially_solved | AIM 研讨会问题汇编

Interacting particle systems

Study $q$-TASEP hydrodynamics and other interacting particle systems properties such as second class particles, invariant distributions, and Burkes type theorem.

来源字段：http://aimpl.org/kpzuniversality/2/

15859 | AIM-PROBABILITY-0092

### Stationary gaps and a second-class gap discrepancy for two-sided q-ASEP

P / partially_solved | AIM 研讨会问题汇编

Interacting particle systems

What about $q$-ASEP, where particle can jump to right and left and feel a repulsion from their right or left neighbors: Is there Burkes Theorem? Do second class particle methods work?

来源字段：http://aimpl.org/kpzuniversality/2/

15860 | AMR-096-0040

### Stationary law of a drift-jump particle process

O / open | AMR 问题清单汇编

Give a reasonably explicit description of the unique stationary distribution of the one-dimensional Hammersley-type process whose particles drift right at speed equal to position and jump left at Poisson space-time events.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15861 | OWR-16413-002

### Statistical Optimality and Validation in Learning-and-Optimization Models

P / partially_solved | Oberwolfach 报告问题汇编

For learning-and-optimization models, should statistical optimality be defined through the probability that a solution is $\varepsilon$-optimal rather than through deterministic optimality? How should generalizability be defined, and can hypothesis tests validate such models?

来源字段：https://doi.org/10.4171/owr/2018/38

15862 | OWR-1061-004

### Statistical Significance of Biological Sequence Similarity

O / open | Oberwolfach 报告问题汇编

For two similar DNA or protein sequences, determine whether the similarity is attributable to chance or instead indicates common ancestry or function.

来源字段：https://doi.org/10.4171/owr/2005/40

15863 | OWR-15962-003

### Strategic Starting Vertices in Competing First-Passage Percolation

O / open | Oberwolfach 报告问题汇编

In competing first-passage percolation on a finite-variance configuration model, can a weaker type capture a positive fraction of the vertices when it starts from high-degree vertices and the stronger type starts from a low-degree vertex?

来源字段：https://doi.org/10.4171/owr/2017/57

15864 | AMR-096-0018

### Stretch-length exponent in spatial networks

O / open | AMR 问题清单汇编

Improve the explicit upper and lower bounds for the minimum network length functions $\Psi^{ave}(s)$ and $\Psi^{worst}(s)$, and prove whether there is an exponent $\alpha$ such that each satisfies $\Psi(s)\asymp(s-1)^{-\alpha}$ as $s\downarrow1$.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15865 | OWR-16413-006

### Strong Convergence of Ensemble Kalman Inversion

P / partially_solved | Oberwolfach 报告问题汇编

Does the discrete ensemble Kalman inversion iteration converge strongly to continuous paths of its limiting stochastic system in the general nonlinear setting?

来源字段：https://doi.org/10.4171/owr/2018/38

15866 | OWR-15214-001

### Strong Convergence Rates for Stochastic Cahn-Hilliard Discretizations

O / open | Oberwolfach 报告问题汇编

Determine optimal strong convergence rates for temporal discretizations of the stochastic Cahn-Hilliard-Cook equation.

来源字段：https://doi.org/10.4171/owr/2017/9

15867 | OWR-15214-003

### Strong Convergence without Global Monotonicity

P / partially_solved | Oberwolfach 报告问题汇编

For multidimensional stochastic differential equations arising in applications whose coefficients do not satisfy global monotonicity assumptions, can one prove $L^2$ convergence rates for suitable numerical approximations?

来源字段：https://doi.org/10.4171/owr/2017/9

15868 | OWR-757-004

### Strong Disorder and Polymer Superdiffusivity

O / open | Oberwolfach 报告问题汇编

Does strong disorder imply superdiffusivity?

来源字段：https://doi.org/10.4171/owr/2004/43

15869 | OWR-11134-001

### Strong Prediction for Stationary Ergodic Gaussian Processes

O / open | Oberwolfach 报告问题汇编

For every stationary ergodic Gaussian process $(Y_n)_{n\in\mathbb Z}$, does there exist a prediction rule $g=(g_n)$ such that

$$\left|\mathbb E[Y_n\mid Y_1,\ldots,Y_{n-1}]-g_n(Y_1,\ldots,Y_{n-1})\right|\to0$$

almost surely?

来源字段：https://doi.org/10.4171/owr/2011/42

15870 | OWR-1115-006

### Structural Relations Between Projected Diffusion Generators

O / open | Oberwolfach 报告问题汇编

Let $p:N\to M$ be a smooth surjection between manifolds, and let diffusion generators $A$ on $M$ and $B$ on $N$ satisfy $B(f\circ p)=(Af)\circ p$ for every smooth $f:M\to\mathbb{R}$. Determine structural and spectral relationships between $A$ and $B$, including whether $B$ has a natural decomposition.

来源字段：https://doi.org/10.4171/owr/2005/54

15871 | OWR-1785-002

### Superreplication in Large-Trader Market Models

P / partially_solved | Oberwolfach 报告问题汇编

In a market model with a large trader whose strategies affect asset prices, and under conditions excluding arbitrage for bounded semimartingale strategies, which contingent claims can be superreplicated? Give necessary and sufficient conditions for replication and superreplication.

来源字段：https://doi.org/10.4171/owr/2008/04

15872 | OWR-4990373-008

### Survival Versus Percolation Thresholds in the Contact Process

O / open | Oberwolfach 报告问题汇编

For the classical nearest-neighbor contact process, determine whether the critical infection threshold for survival satisfies $\lambda_c<\lambda_p$, where $\lambda_p$ is the corresponding percolation threshold.

来源字段：https://doi.org/10.4171/owr/2021/4

15873 | AIM-PROBABILITY-0118

### Symmetric unimodality under free additive convolution

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** If \(\mu\) and \(\nu\) are symmetric unimodal probability distributions on \(\mathbb R\), must their free additive convolution \(\mu\boxplus\nu\) be unimodal?

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15874 | OWR-14299285-005

### Symmetries of Open-Boundary Asymmetric Exclusion Processes

O / open | Oberwolfach 报告问题汇编

What role do the non-Abelian symmetries of the intensity matrix play for the asymmetric simple exclusion process with open boundaries?

来源字段：https://doi.org/10.4171/owr/2025/26

15875 | OWR-16409-006

### Symplectic Models for Relative Class Group Distributions

O / open | Oberwolfach 报告问题汇编

After incorporating the contribution of unit groups, is the joint distribution over number fields of relative class-group $\ell$-parts and the two accompanying class-group invariants equal to the distribution predicted by the random symplectic-similitude model?

来源字段：https://doi.org/10.4171/owr/2018/34

15876 | OWR-14298583-001

### Tail-Dependence Estimators Under Long-Range Dependence

P / partially_solved | Oberwolfach 报告问题汇编

Determine the asymptotic distribution of the empirical tail-dependence estimator $\widehat\rho_{B,n}(T)$ for a long-range-dependent max-stable stationary time series, in the regime $u_n\to\infty$ and $n\mathbb P(X_0>u_n)\to\infty$.

来源字段：https://doi.org/10.4171/owr/2024/38

15877 | OWR-2493-007

### Tail-Dependent Ageing in the Parabolic Anderson Model

P / partially_solved | Oberwolfach 报告问题汇编

For the parabolic Anderson model on $\mathbb Z^d$, at what tail threshold does the ageing behavior described by the remaining-lifetime process $R(t)$ change qualitatively as the random potential has a tail lighter than polynomial?

来源字段：https://doi.org/10.4171/owr/2008/55

15878 | AIM-PROBABILITY-0048

### Target-sensitive hitting-time comparison for interchange

P / partially_solved | AIM 研讨会问题汇编

**Hitting time comparison.** Can you compare the hitting times of the interchange process to hitting times for the SRW on a graph \(G\)?

来源字段：http://aimpl.org/markovmixing/4/

15879 | OWR-4532-004

## TASEP Diffusion Coefficients from Weak Asymmetry Limits

O / open | Oberwolfach 报告问题汇编

Does $\lim_{\gamma\downarrow0}D_\gamma(1)$ determine the TASEP diffusion coefficient, which diverges in dimensions $d=1,2$ but is finite for $d\geq3$?

来源字段：https://doi.org/10.4171/owr/2010/50

15880 | AIM-PROBABILITY-0122

## Taylor coefficients and complex HCIZ free energy

P / partially_solved | AIM 研讨会问题汇编

There is another result that states that \(\frac{\partial^n}{\partial t^n}\log IZ(tA_n,B_n)|_{t=0}\) converges. Does this expression match \(\psi\) above? Can we extend Guionnet and Zeitouni's result to complex parameters?

This is Question 20 in the 2006 AIM workshop list *Free analysis*, immediately after the definition of an Itzykson--Zuber integral and a summary of the Guionnet--Zeitouni theorem. The supplied record says

Inspection of the original PDF confirms that the displayed derivative is typeset with order \(n\), the same letter used for matrix dimension. Taken literally, it asks about a derivative whose order grows with dimension. That is not the theorem in the relevant primary source.

The “another result” is Benoît Collins's theorem. Collins uses dimension \(d\), holds the derivative order fixed, and proves convergence of

\[ \left.\frac{\partial^q}{\partial z^q}
\left[d^{-2}\log\int_{U(d)}
e^{zd\operatorname{Tr}(X_dUY_dU^*)}\,dU\right]
\right|_{z=0} \qquad(q\text{ fixed as }d\to\infty). \]

Thus two repairs are required before comparing with the Guionnet--Zeitouni free energy:

1. use a fixed derivative order \(q\), not the diagonal regime \(q=n\); 2. normalize the logarithm by the natural free-energy scale \(n^{-2}\).

The AIM PDF's preceding line also records an asymptotic of the form \(IZ\sim\exp(-n\psi)\), while the standard full-rank HCIZ integral with exponent \(n\operatorname{Tr}\) has logarithm of order \(n^2\). The primary Guionnet--Zeitouni and Collins papers both use the \(n^{-2}\log IZ\) normalization. Accordingly, this report does not silently identify the printed \(\psi\); it defines the normalization explicitly and then states the sign relation.

Let \(A_N,B_N\) be deterministic Hermitian \(N\times N\) matrices and let Haar measure on \(U(N)\) be normalized. Set

\[ I_N(z)=\int_{U(N)}
\exp\!\left\{zN\operatorname{Tr}(A_NUB_NU^*)\right\}\,dU,
\qquad F_N(z)=\frac1{N^2}\Log I_N(z), \]

where near zero \(\Log\) denotes the branch with \(F_N(0)=0\). The source's negative-exponent convention is

\[ IZ_N(tA_N,B_N)=I_N(-t). \]

\[ IZ_N(tA_N,B_N)=\exp\{-N^2\psi(t)+o(N^2)\}, \]

then, wherever the limiting free energy \(F\) is defined,

\[ \psi(t)=-F(-t). \tag{1.1} \]

If the source intended normalized trace instead of ordinary \(\operatorname{Tr}\), the exponent must be rescaled equivalently; the invariant point is that the log partition function is divided by its \(N^2\) full-rank scale.

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15881 | AIM-PROBABILITY-0129

### The Dirac-Brown invariant-subspace problem: a defect-attainment reduction

P / partially_solved | AIM 研讨会问题汇编

**Q:** Let \((\Gamma,\tau)\) be a \(\mathrm{II}_1\) factor, \(T\in\Gamma\), \(\mu_T=\delta_0\). Does \(T\) have a non-trivial invariant subspace affiliated with \(\Gamma\)?

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15882 | AIM-PROBABILITY-0134

### The doubled-Fock copied-letter gradient of the q-Gaussian number operator

S / solved | AIM 研讨会问题汇编

Q: For the q-deformed semicircular, the analogue of \(\partial^*\partial\) exists (it is the number operator). Describe explicitly the associated \(\partial\) (which exists by the work of Sauvageot).

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15883 | OWR-14299081-021

### The Multicolored Geometric Erdős-Hajnal Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For every prime power $q$, all $k,s\in\mathbb{N}$, and every $s$-coloring $c_0$ of $\operatorname{PG}(k-1,q)$, do constants $\delta,C>0$ exist such that, for every $n$ and every $c_0$-free $s$-coloring of $\operatorname{PG}(n-1,q)$, some subspace of dimension at least $Cn^\delta$ uses at most $s-1$ colors?

来源字段：https://doi.org/10.4171/owr/2025/1

15884 | AIM-PROBABILITY-0089

### The q-exponential martingale, its fixed-q bracket obstruction, and the two-stage SHE limit

P / partially_solved | AIM 研讨会问题汇编

Find a microscopic G\"{a}rtner transform (discrete Hopf-Cole transform) which turns the dynamics of \(q\)-TASEP into a discrete stochastic heat equation or better yet a polymer. Then prove this converges to the continuum stochastic heat equation.

来源字段：http://aimpl.org/kpzuniversality/2/

15885 | AIM-PROBABILITY-0142

### Three meanings of infinite type in rank-two cluster algebras

P / partially_solved | AIM 研讨会问题汇编

**Explain the theory of cluster algebras in infinite types. See Problem 6.5.**

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15886 | AIM-PROBABILITY-0156

### Three obstructions to a naive dual BN-pair

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.4. (D. Bessis)** Is there a structure theory of Lie groups and algebraic groups that is analogous to the dual braid monoid [6]? Is there some dual notion of BN-pairs?

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15887 | OWR-14604-009

### Three-Event Correlation Inequalities in Product Spaces

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Three-Event Correlation Question [OWR-14604-010]

Characterize the realizable joint probabilities of three increasing events $A,B,C$ in a finite product probability space. In particular, is $$ 2\mathbb P(A\cap B\cap C)-\mathbb P(A\cap B)\mathbb P(C)-\mathbb P(A\cap C)\mathbb P(B)-\mathbb P(B\cap C)\mathbb P(A)+\mathbb P(A)\mathbb P(B)\mathbb P(C)\geq0 $$ always true?

来源字段：https://doi.org/10.4171/owr/2016/22

15888 | OWR-17464-009

### Thresholds for Universal Random Permutations

P / partially_solved | Oberwolfach 报告问题汇编

For every fixed $\varepsilon>0$, does a uniformly random permutation of length $(1+\varepsilon)k^2/4$ become $k$-universal with high probability as $k\to\infty$?

来源字段：https://doi.org/10.4171/owr/2020/1

15889 | AMR-099-0065

### Time constant in a recursive series-parallel first-passage model

P / partially_solved | AMR 问题清单汇编

Let $D_n$ be the source-to-sink first-passage distance in the recursively substituted hierarchical graph whose distances satisfy $D_n\stackrel{d}{=}D_{n-1}+\min(D'_{n-1},D''_{n-1})$, with i.i.d. edge-weight law $\xi$. Compute $\gamma_{\mathrm{rec}}=\lim 2^{-n}\mathbb{E}D_n$ in terms of $\xi$ and decide whether $\gamma_{\mathrm{rec}}>\gamma_{\mathrm{brw}}$, the corresponding branching-random-walk constant.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/erd100.pdf#page=15

15890 | AIM-PROBABILITY-0055

### Time-convention audit and an exact bottleneck for monotone censored cube walks

P / partially_solved | AIM 研讨会问题汇编

Let \(A\subset\{0,1\}^n\) be monotone increasing. The random walk chooses a coordinate uniformly, flips it, and accepts the move only if the new point remains in \(A\). Can one prove an \(O(n\log n)\) upper bound on its mixing time?

Two omitted conventions materially change the answer.

First, the stated always-flip walk is not the chain in Ding and Mossel's primary 2014 paper. Their chain chooses a coordinate and **rerandomizes** its bit to a fair bit, rejecting only if the proposed configuration is outside \(A\). Equivalently, it is the \(1/2\)-lazy version of the censored flip walk: if a proposed flip is valid, move with probability \(1/2\), and otherwise stay. This is also the convention used explicitly by Fei and Ferreira Pinto Jr. in 2025. It repairs periodicity.

Second, a density-free \(O(n\log n)\) assertion is already false for the lazy chain. The open question in Ding--Mossel assumes \[ \mu(A):=\frac{|A|}{2^n}\geq \alpha>0 \tag{1} \] and asks for \(t_{\rm mix}=O_\alpha(n\log n)\).

1. the literal always-flip chain; 2. the lazy chain for arbitrary nonempty monotone \(A\); 3. the intended lazy, constant-density conjecture (1).

来源字段：http://aimpl.org/markovmixing/6/

15891 | AIM-PROBABILITY-0023

### Time-uniform self-normalization without Cauchy moments

S / solved | AIM 研讨会问题汇编

Let \(X_1,X_2,\ldots\) be i.i.d. symmetric Cauchy random variables, and let \[ S_t=\sum_{i=1}^tX_i,\qquad V_t=\sum_{i=1}^tX_i^2. \] Using Stein's method, prove that for all \(m,x\geq0\), \[ \mathbb P\left(\exists t\in\mathbb N: \frac{S_t}{V_t+m}>x\right)\leq e^{-2mx^2}. \]

来源字段：http://aimpl.org/steinhd/1/

15892 | AMR-096-0041

### Topological realization of compact Markov-chain limits

P / partially_solved | AMR 问题清单汇编

For the measure-theoretic limit transition densities $p_\infty(x,y,t)$ arising from sequences of finite reversible Markov chains, construct a natural topology-such as one induced by an integrated $L^2$ transition-density distance-that makes the state space complete and separable and the limit Markov process Feller.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15893 | AMR-096-0037

### Topology and geometry of a self-similar random planar partition

P / partially_solved | AMR 问题清单汇编

For Aldous's self-similar random partition of the plane, determine its topological properties: in particular, do region boundaries have fractal dimension greater than one? Determine the area law and structural properties of the capital-city adjacency network, including planarity/crossings and whether analogues of the Lewis and Aboav-Weaire laws hold.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15894 | AMR-094-0002

### Topology of planar Brownian trace

O / open | AMR 问题清单汇编

Let $X_t$ be two-dimensional Brownian motion. (i) For every pair $x,y \notin X[0,1]$, is there a Jordan arc $\Gamma$ containing $x$ and $y$ such that $\Gamma\cap X[0,1]$ is finite? (ii) If $\{A_k\}_{k\ge1}$ are the connected components of $\mathbb{R}^2\setminus X[0,1]$ and $K=X[0,1]\setminus\bigcup_{k\ge1}\partial A_k$, is $K$ totally disconnected?

背景链接：https://sites.math.washington.edu/~burdzy/open_mathjax.php

15895 | AIM-PROBABILITY-0010

### Topology overlap and pruning stability for RBMs

P / partially_solved | AIM 研讨会问题汇编

**Given two RBMs with the same number of units but different connectivities, how much do these statistical models overlap?**

How to quantify function approximation given a network topology? In other words, how to measure the proximity between two functions when each are computed by separate networks? Then, can one use this measure to quantify the effects of altering a network's connectivity structure?

来源字段：http://aimpl.org/boltzmann/3/

15896 | AIM-PROBABILITY-0004

### Topology-dependent continuum limits of square binary RBMs

P / partially_solved | AIM 研讨会问题汇编

Characterize the limiting set of distributions for \(\lim_{n\to\infty}RBM_{n,n}\).

Consider the limit as \(n\to\infty\) for \(RBM_{n,n}\). It was suggested that this limiting set of distributions can be seen as a set of measures on the unit interval. Do all possible measures belong to this set? If not, which ones do?

来源字段：http://aimpl.org/boltzmann/1/

15897 | OWR-11695865-001

### Total Positivity and Graphical Model Factorization

P / partially_solved | Oberwolfach 报告问题汇编

For a graph $G$, is $\mathcal{M}_I(G)\cap\mathcal{M}_2(G)$ closed under pointwise limits, and is every multivariate totally positive order-two Markov distribution in $\mathcal{M}_F(G)$?

来源字段：https://doi.org/10.4171/owr/2022/55

15898 | OWR-4081-006

### Transience Implies Ballisticity for Uniformly Elliptic RWRE

O / open | Oberwolfach 报告问题汇编

For a uniformly elliptic i.i.d. random walk in random environment in dimension $d\geq2$, does transience in a direction $\ell$ imply ballisticity in that direction?

来源字段：https://doi.org/10.4171/owr/2009/38

15899 | OWR-2093-003

### Transport Characterizations of Mass-Stationarity

O / open | Oberwolfach 报告问题汇编

For an invariant random measure $\xi$, is invariance of its distribution under invariant mass-preserving transport kernels characteristic of mass-stationarity?

来源字段：https://doi.org/10.4171/owr/2008/47

15900 | AIM-PROBABILITY-0114

### Transported compact resolvent and a Tauberian entropy-dimension reduction

P / partially_solved | AIM 研讨会问题汇编

The original PDF and its raw page-content stream were inspected.

The source asks about a limiting random-matrix/free-Gibbs law whose formal density is \[ \exp\{-N\operatorname{Tr}p(A_1,A_1^*,\ldots,A_m,A_m^*)\}, \] and whose conjugate variables satisfy \(J_i=\mathcal D_iP\). Writing \(\partial_j\) for Voiculescu's free difference quotient at the limiting variables, it asks whether \[ \exp\!\left(-t\sum_j\partial_j^*\partial_j\right) \tag{1.1} \] is compact in the \(N\to\infty\) limit. It proposes as a starting point \[ P=\sum_i A_i^2+\sum_i t_iq_i(A_1,\ldots,A_m), \tag{1.2} \] in the small-perturbation regime studied by Guionnet--Maurel-Segala.

Compactness in (1.1) can only mean compactness as an operator on the limiting Hilbert space \(L^2(M,\tau)\). Compactness is not an intrinsic Banach-space property of a map “on the algebra” unless a topology and norm are specified. The notation \(A_i,A_i^*\) allows genuinely nonnormal models, whereas (1.2) is the self-adjoint starting case. The proved results below treat a finite self-adjoint tuple; the nonnormal case is listed as a gap.

The next source item begins after the heading `0.4 Focus Group on Free Entropy (day 3)`. A font-level audit resolves the damaged title unambiguously: \[ \boxed{\text{Is }\delta^*=\delta^\star\text{?}} \tag{1.3} \] The first superscript is the Computer Modern symbol-font asterisk and the second is character `0x3f` in Computer Modern math italic, namely \(\star\). It is not a question-mark superscript.

The PDF as printed then displays \[ \delta^*=n-\limsup_{t\downarrow0} \frac{\chi^*(x_1+\sqrt t\,s_1,\ldots,x_n+\sqrt t\,s_m)} {\log t^{1/2}}, \tag{1.4} \] and \[ \delta^\star=n-\limsup_{t\to0}\sum_{i=1}^n t\Phi^*(x_1+\sqrt t\,s_1,\ldots,x_m+\sqrt t\,s_m). It also prints `lim sup` in both displays.

The standard definitions in Voiculescu's theory, for an \(n\)-tuple \(X=(x_1,\ldots,x_n)\) and a variance-one semicircular tuple \(S=(s_1,\ldots,s_n)\) free from \(X\), are \[ \delta^*(X) =n-\liminf_{t\downarrow0} \frac{\chi^*(X+\sqrt t\,S)}{\log\sqrt t}, \tag{1.6} \] and \[ \delta^\star(X) =n-\liminf_{t\downarrow0}t\Phi^*(X+\sqrt t\,S). \tag{1.7} \] Equations (1.6)--(1.7), not the malformed displays (1.4)--(1.5), are used below.

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15901 | OWR-14299518-003

### Transversals in Random Subsets of Projective Planes

O / open | Oberwolfach 报告问题汇编

Let $P$ be the point set of a projective plane of order $q$, and retain each point independently with probability $1/2$ to form $R$. With high probability, must every $B\subseteq R$ meeting every nonempty line section $L\cap R$ satisfy $|B|/q>f(q)$ for some $f(q)\to\infty$? Can one take $f(q)=\Omega(\log q)$?

来源字段：https://doi.org/10.4171/owr/2025/42

15902 | OWR-5151-001

### Tree-Cutting Representations of Exchangeable Coalescents

O / open | Oberwolfach 报告问题汇编

Find another exchangeable coalescent that can be represented by cutting down a combinatorial tree.

来源字段：https://doi.org/10.4171/owr/2011/23

15903 | OWR-17474-006

### Triangle Condition in Long-Range Percolation

O / open | Oberwolfach 报告问题汇编

In one-dimensional long-range percolation, is the triangle condition not satisfied for $s>4/3$, and what happens at the critical value $s=4/3$?

来源字段：https://doi.org/10.4171/owr/2020/11

15904 | OWR-3473-009

### Triangle-Count Equidistribution and Local Limits

P / partially_solved | Oberwolfach 报告问题汇编

For fixed $p\in(0,1)$ and triangle count $X$ in $G(n,p)$, how far do asymptotic equidistribution of $X$ modulo $q$ and its local limit theorem extend beyond $q=o(n^2/\log n)$?

来源字段：https://doi.org/10.4171/owr/2009/23

15905 | OWR-14298799-001

### Two-Loop SLE and Loop-Model Measures

O / open | Oberwolfach 报告问题汇编

Relate the two-loop Schramm-Loewner evolution measure to concrete statistical-mechanics measures, such as the $O(n)$ loop model, and to the counting measure on conformal loop ensembles.

来源字段：https://doi.org/10.4171/owr/2024/52

15906 | AMR-098-0007

### Two-process coupling characterization of weak convergence

O / open | AMR 问题清单汇编

Suppose $\theta_nX$ converges in distribution to $X'$ on a separable metric path space. Is there a coupling characterization involving only a joint construction of $X$ and $X'$, rather than a whole family of copies? For example, can they be coupled so that $d(\theta_nX,\theta_nX')\to0$?

背景链接：http://cms.dm.uba.ar/depto/public/Some%20Open%20Problems-preprint.pdf

15907 | AIM-PROBABILITY-0069

### Two-time KPZ laws: fixed-point formulas and a finite-time full-profile obstruction

P / partially_solved | AIM 研讨会问题汇编

Compute multi-time distribution for KPZ. What is the distribution $F'_{t_1,t_2}(\xi_1,\xi_2)$ such that $$ \mathbb P\bigl(A_{t_1}(x)\leq \xi_1, A_{t_2}(x)\leq \xi_2\bigr) =F'_{t_1,t_2}(\xi_1,\xi_2)? $$

The record does not restate the initial condition or define $A_t$. The immediately preceding canonical record, AIM-PROBABILITY-0068, gives the narrow-wedge convention $$ \mathcal Z(0,\cdot)=\delta_0, \qquad \mathcal Z(t,X)= \exp\left\{-\frac{X^2}{2t} +t^{1/3}A_t(t^{-2/3}X)-\frac{t}{24}\right\}. \tag{1.1} $$ I therefore treat narrow-wedge initial data as the intended reading. With $u=t^{-2/3}X$, (1.1) says exactly $$ A_t(u)=t^{-1/3}\log \mathcal Z(t,t^{2/3}u) +\frac{u^2}{2}+\frac{t^{2/3}}{24}. \tag{1.2} $$ There is a small ambiguity in the source: keeping the same argument $x$ in $A_{t_1}(x)$ and $A_{t_2}(x)$ compares the physical locations $X_i=t_i^{2/3}x$, not one fixed physical location. The latter question would instead use $A_{t_i}(t_i^{-2/3}X)$. Both conventions are recorded below.

The two heights must be constructed from the **same space-time white noise**. Using two independent KPZ solutions would change the question and make the joint CDF the product of its marginals.

来源字段：http://aimpl.org/kpzuniversality/1/

15908 | AIM-PROBABILITY-0152

## Type-B extreme diameters and a dihedral meander determinant

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.4 (H. T. Hall).** Suppose that a stream has \(2n\) bridges across it. A classical meander is (the homotopy class of) a closed path which crosses each bridge once without intersecting itself. On each side of the stream, the meander is given by a noncrossing pairing of the set \[ [2n]:=\{1,2,\ldots,2n\}. \] Noncrossing pairings are naturally in bijection with type \(A\) noncrossing partitions of the set \([n]\).

Every ordered pair of noncrossing partitions defines a path (with possibly multiple components) which crosses each bridge exactly once. There is a bijection which says that the meanders (the paths with only one connected component) correspond exactly to pairs of noncrossing partitions that are maximally separated in the Hasse diagram of \(NC_{A_{n-1}}\) (they are diameters in the graph-theoretical sense). Does this bijection suggest a new way to count meanders?

One may also use this bijection to define meanders of type \(W\) (they are the ordered diameters of the Hasse diagram of \(NC_W\)). Is there some combinatorial object that this corresponds to? Is there a type \(B\) meander?

**Remark (A. Nica and J. Scott).** In type \(A\), one may build a “meander determinant” which is known to factor as a product of Chebyshev polynomials. Similarly, one may define a type-\(W\) meander determinant. What factorization properties does it have? Meanders are related to chromatic polynomials of graphs and the Temperley--Lieb algebra. What is the significance of type-\(W\) meanders in this context?

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15909 | OWR-14299586-001

## Typical Cells in Hyperbolic Tessellations with Unbounded Cells

O / open | Oberwolfach 报告问题汇编

Extend the notions of cell intensity and typical cell naturally to hyperbolic tessellations containing unbounded cells.

来源字段：https://doi.org/10.4171/owr/2025/57

15910 | AMR-096-0023

## Unbalanced regimes of the spatial city-growth model

P / partially_solved | AMR 问题清单汇编

For the city-growth model, prove: (a) if $\alpha>1$, the eventual number of cities $M(\infty)$ is finite almost surely; (b) if $\beta<2\alpha$, the largest city satisfies $N_{(1)}(t)/t\to1$ almost surely; and (c) if both hold, the eventual population outside the largest city $N_{(\ge2)}(\infty)$ is finite almost surely.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15911 | AMR-096-0033

## Unbounded component uniqueness in a SIRSN

O / open | AMR 问题清单汇编

Does the major-road subnetwork $E(\infty,1)$ of a SIRSN almost surely have exactly one unbounded connected component?

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15912 | OWR-763-004

## Unconditioned Component Statistics in Branching Processes

O / open | Oberwolfach 报告问题汇编

Determine the statistics of the size and structure of connected components when the age of the root is not conditioned on.

来源字段：https://doi.org/10.4171/owr/2004/49

15913 | OWR-17288-002

## Unification of Free Products in Algebraic Probability

O / open | Oberwolfach 报告问题汇编

Does there exist a reduction or unification of the free product of algebraic probability spaces?

来源字段：https://doi.org/10.4171/owr/2019/45

15914 | OWR-9790365-008

## Unified Gaussian Comparison Framework for Entropic Inequalities

P / partially_solved | Oberwolfach 报告问题汇编

Can a unified Gaussian-comparison framework prove the entropic and geometric inequalities that do not follow from the available Gaussian extremality principle?

来源字段：https://doi.org/10.4171/owr/2022/14

15915 | OWR-13856-006

## Uniform Corrector Moments and Quantitative Stochastic Homogenization

P / partially_solved | Oberwolfach 报告问题汇编

Can one prove bounds on moments of the stochastic corrector that are uniform in space or have a precise divergence rate, and can one prove a corresponding quantitative two-scale expansion for stochastic homogenization?

来源字段：https://doi.org/10.4171/owr/2015/35

15916 | OWR-17474-008

### Uniform Coupling under Gibbs Uniqueness

O / open | Oberwolfach 报告问题汇编

For a specification on $\mathbb Z^d$ with a unique Gibbs state, is it true that for every $\varepsilon>0$ there is an $N$ such that, for all $n\geq N$, every pair of boundary conditions outside the $n$-box induces inside-box measures that can be coupled with expected number of disagreements at most $\varepsilon$?

来源字段：https://doi.org/10.4171/owr/2020/11

15917 | AIM-PROBABILITY-0154

### Uniform finite results and a dihedral infinite-type boundary

P / partially_solved | AIM 研讨会问题汇编

Reading)** under “3. Reflection Groups.” In modern notation the problem fixes a finite Coxeter system \((W,S)\) and a Coxeter element \(c\), and asks:

1. Give a uniform proof that the \(c\)-sortable elements are counted by \[ \operatorname{Cat}(W)=\prod_{i=1}^{r}\frac{h+d_i}{d_i}. \] 2. Give a uniform proof that the map \(\operatorname{nc}_c\) to \(NC(W,c)=[e,c]_T\) is well-defined. 3. Give uniform proofs that \(\operatorname{nc}_c\) and the map \(\operatorname{cl}_c\) to \(c\)-clusters are bijections. 4. The definitions extend to infinite Coxeter groups. What happens there?

The printed PDF writes the maps as “nc” and “cl”; their dependence on the chosen Coxeter element is explicit in the preceding sentence. Reading's standard notation, used here, is \(\operatorname{nc}_c\) and \(\operatorname{cl}_c\). The PDF's item (1) says “Coxeter-sorted elements” once, while the surrounding definition and the cited paper say **Coxeter-sortable**; “sorted” is a source typo, not a second concept. The damaged “\(c.\)There” and “\(NCW\)” are repaired to \(c\), sentence spacing, and \(NC_W\), respectively.

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15918 | AIM-PROBABILITY-0138

### Uniform heat convergence, spectral tails, and affiliated innerness

P / partially_solved | AIM 研讨会问题汇编

Let \(\partial:M\to L^2(M)\bar\otimes L^2(M^o)\) be a closable derivation, let \(\Delta=\partial^*\partial\), and put \(S_t=\exp(-t\Delta)\). If \(S_t\) converges uniformly to the identity in \(\|\cdot\|_2\) on the unit ball, is the derivation inner when considered with values in the algebra of unbounded operators affiliated to \(M\bar\otimes M^o\)?

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15919 | AIM-PROBABILITY-0111

### Uniform spectral tails and compact-resolvent obstruction for the free heat semigroup

P / partially_solved | AIM 研讨会问题汇编

Consider \(\Delta=\sum_{i=1}^m\partial_{x_i}^*\partial_{x_i}\) and the corresponding completely positive map \(\varphi_t=\exp(-t\Delta)\), where \((x_1,\ldots,x_m)\) have finite free Fisher information. Can \(\varphi_t\) converge uniformly to the identity map on the unit ball of \(W^*(x_1,\ldots,x_m)\)? If no, it follows that the von Neumann algebra generated by \((x_1,\ldots,x_m)\) is not weakly rigid if it is non-hyperfinite.

来源字段：https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

15920 | OWR-12697708-006

### Unique Equilibria for Subcritical Reinforcement on Infinite Graphs

P / partially_solved | Oberwolfach 报告问题汇编

For $\alpha<1$ on an infinite bounded-degree graph with bounded firing rates, is there a unique equilibrium $x$, and does $X_e(t)\to x_e$ almost surely for every edge $e$?

来源字段：https://doi.org/10.4171/owr/2023/12

15921 | OWR-828-004

### Uniqueness and Signature Ranges of Rough Paths

P / partially_solved | Oberwolfach 报告问题汇编

Does signature uniqueness extend to rough paths with roughness parameter $p>1$? Can one characterize the range of the signature map?

来源字段：https://doi.org/10.4171/owr/2005/23

15922 | AMR-099-0088

### Uniqueness at the percolation uniqueness threshold

O / open | AMR 问题清单汇编

For a quasi-transitive graph $G$, characterize when Bernoulli percolation has a unique infinite cluster at $p=p_u(G)$. In particular, give necessary and sufficient structural conditions for uniqueness at the threshold.

背景链接：https://emis.de/ft/41630#page=10

15923 | OWR-2488-004

### Uniqueness for Averaged Whitham Equations

O / open | Oberwolfach 报告问题汇编

For the averaged Whitham equation governing the action vector $I$ in the small-noise limit of the randomly perturbed KdV equation, does the equation have a unique solution and a unique stationary measure?

来源字段：https://doi.org/10.4171/owr/2008/50

15924 | OWR-1970-008

### Uniqueness of Dirichlet Forms on the Sierpinski Carpet

O / open | Oberwolfach 报告问题汇编

For the Sierpiński carpet $F$, do the Barlow-Bass and Kusuoka-Zhou Dirichlet forms with the common Besov-Lipschitz domain coincide up to a positive multiplicative constant; equivalently, is there a unique ‘nice’ Dirichlet form on $F$?

来源字段：https://doi.org/10.4171/owr/2008/25

15925 | AMR-099-0037

### Uniqueness of percolation on graphs roughly isometric to lattices

P / partially_solved | AMR 问题清单汇编

Prove that Bernoulli percolation has at most one infinite cluster on every bounded-degree graph roughly isometric to $\mathbb{Z}^d$.

背景链接：https://www.wisdom.weizmann.ac.il/~itai/stflouraug24.pdf#page=64

15926 | AIM-PROBABILITY-0162

### Units, central splitting, and filtration for multivariable boxed convolution

P / partially_solved | AIM 研讨会问题汇编

The official PDF identifies it as **Problem 5.2**, attributed to A. Nica.

The recovered problem asks for the structure of the unit group of \[ \left(\mathbb C_0\langle\!\langle z_1,\ldots,z_s\rangle\!\rangle, \boxtimes_s\right). \] It notes that \(s=1\) is the only commutative case, where Nica--Speicher's free Fourier transform \(F\) identifies the boxed-convolution unit group with invertible ordinary one-variable power series, and asks for a multivariable \(S\)-transform or analogue of \(F\).

来源字段：https://aimath.org/WWN/braidgroups/braidgroups.pdf

15927 | OWR-12873-001

### Universal Almost-Sure Prediction for Stationary Gaussian Processes

O / open | Oberwolfach 报告问题汇编

Let $\{Y_n\}_{n\in\mathbb Z}$ be a stationary, ergodic, mean-zero Gaussian process. Does there exist a sequence of predictors $g_n$ such that

$$\mathbb E(Y_n\mid Y_1,\ldots,Y_{n-1})-g_n(Y_1,\ldots,Y_{n-1})\longrightarrow0$$

almost surely as $n\to\infty$?

来源字段：https://doi.org/10.4171/owr/2014/13

15928 | OWR-14299285-003

### Universal CLE Metrics as Chemical-Distance Limits

P / partially_solved | Oberwolfach 报告问题汇编

For every $\kappa$, does the $\operatorname{CLE}_\kappa$ metric arise as the scaling limit of the intrinsic, or chemical-distance, metric of every discrete model whose interfaces converge to $\operatorname{CLE}_\kappa$?

来源字段：https://doi.org/10.4171/owr/2025/26

15929 | AMR-096-0003

### Universal compression of sparse labeled graphs

O / open | AMR 问题清单汇编

For sparse $n$-vertex graphs of average degree $O(1)$ whose vertices have distinct $O(\log n)$-length labels over a finite alphabet, construct universal codes analogous to Lempel-Ziv that achieve asymptotically optimal compression.

背景链接：https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

15930 | OWR-1327-002

### Universal Fluctuation Limits for Exclusion Processes

P / partially_solved | Oberwolfach 报告问题汇编

For the totally asymmetric simple exclusion process (TASEP), classify the initial conditions that yield a common scaling-limit distribution and common limit process for particle-position fluctuations.

来源字段：https://doi.org/10.4171/owr/2006/41

15931 | OWR-11695855-005

### Universal Gaussian Mean Estimation under Adversarial Corruption

P / partially_solved | Oberwolfach 报告问题汇编

Are there estimators other than the empirical mean with the intended universal estimation property? Is the empirical mean's property explained by its being a sufficient statistic? Can universal estimation be extended beyond Gaussian samples, including to the $\varepsilon$-adversarial corruption model, and what is the largest tolerable corruption proportion $\varepsilon$?

来源字段：https://doi.org/10.4171/owr/2022/46

15932 | OWR-4532-007

### Universal Scaling Limits in Hierarchical Coalescence

O / open | Oberwolfach 报告问题汇编

How can one explain the existence of scaling limits, and why do very different one-dimensional hierarchical coalescence models exhibit the same limiting distributions?

来源字段：https://doi.org/10.4171/owr/2010/50

15933 | OWR-17299-001

### Universality Across the Random-Band-Matrix Localization Transition

P / partially_solved | Oberwolfach 报告问题汇编

What is the universality class of random band matrices as the matrix size $N$ tends to infinity while the bandwidth $W$ varies, including across the localization-delocalization transition?

来源字段：https://doi.org/10.4171/owr/2019/56

15934 | OWR-4532-002

### Universality Classes for Interacting Particle Fluctuations

O / open | Oberwolfach 报告问题汇编

Extend rigorous universality results for one-dimensional interacting particle systems beyond nearest-neighbor simple exclusion: determine conditions under which integrated particle-density fluctuations exhibit the predicted KPZ or Edwards-Wilkinson scaling classes, including for finite-range exclusion.

来源字段：https://doi.org/10.4171/owr/2010/50

15935 | OWR-2040-005

### Universality of Rescaled Grain-Growth Complexes

O / open | Oberwolfach 报告问题汇编

For generic initial conditions, does the cell complex of grains under mean-curvature evolution converge, after rescaling by the increasing average grain size, to a universal random cell complex independent of the material?

来源字段：https://doi.org/10.4171/owr/2008/29

15936 | OWR-2654828-003

### Values and Conditioned Shapes of Gaussian Persistence

P / partially_solved | Oberwolfach 报告问题汇编

For Gaussian stationary processes, can the persistence exponent be computed in specific examples? What is the typical shape of a process conditioned to persist above a level $\ell$ on $[0,T]$, and can the results be extended to non-Gaussian or non-stationary models?

来源字段：https://doi.org/10.4171/owr/2020/32

15937 | OWR-15215-005

### Variance Conjecture for Isotropic Convex Bodies

O / open | Oberwolfach 报告问题汇编

Does there exist an absolute constant $C$ such that every random vector $X$ uniformly distributed on an isotropic convex body in $\mathbb R^n$ satisfies $$ \operatorname{Var}(|X|_2^2)\leq C\sup_{\theta\in S^{n-1}}\mathbb E\langle X,\theta\rangle^2\,\mathbb E|X|_2^2 $$?

来源字段：https://doi.org/10.4171/owr/2017/10

15938 | OWR-12182-002

### Variance Conjecture for Log-Concave Random Vectors

O / open | Oberwolfach 报告问题汇编

Does there exist an absolute constant $C$ such that every centered log-concave random vector $X\in\mathbb R^n$ satisfies $\operatorname{Var}(|X|^2)\leq C\lambda_X^2\mathbb E|X|^2$, where $\lambda_X$ is the largest eigenvalue of its covariance matrix?

来源字段：https://doi.org/10.4171/owr/2012/59

15939 | OWR-12337-002

### Variation Limits for Pure-Jump Semimartingales

O / open | Oberwolfach 报告问题汇编

Let

$$X_t=X_0+\int_{-\infty}^t g(t-s)\,dL_s,$$

where $L$ is a pure-jump Lévy process, $g(x)=x^\alpha f(x)$ with $0<\alpha<1/2$, and $f$ is smooth with exponential decay and $f(0)\neq0$. For

$$V_t^n(X)=\sum_{i=1}^{\lfloor t/\Delta_n\rfloor}|X_{i\Delta_n}-X_{(i-1)\Delta_n}|^2,$$

the compound-Poisson case satisfies

$$\Delta_n^{-2\alpha}V_t^n(X)\xrightarrow{d_{\mathrm{st}}}f(0)^2\sum_{k:T_k\in[0,t]}|\Delta L_{T_k}|^2\sum_{l=1}^{\infty}\left|(l-U_k)_+^\alpha-(l-1-U_k)_+^\alpha\right|^2.$$

Under suitable hypotheses, does the same stable limit hold when the driver is a sufficiently general pure-jump semimartingale?

来源字段：https://doi.org/10.4171/owr/2013/09

15940 | OWR-3386-002

### Variational Principles for Dilute Spin Systems

O / open | Oberwolfach 报告问题汇编

For dilute ferromagnets and sparse spin glasses, does the GREM-associated trial order parameter give the correct variational formula for the pressure? Does the numerator have a direct differential-equation formulation analogous to the Parisi equation? For the dilute ferromagnet, can one prove $\log Z_{N+M}\leq\log Z_N+\log Z_M$ directly? How should the pressure variational principle be formulated to exhibit appropriate trial order parameters, an entropy-minus-internal-energy interpretation, or a Legendre structure?

来源字段：https://doi.org/10.4171/owr/2009/03

15941 | OWR-3473-012

### Vickrey Costs in Random Combinatorial Optimization

O / open | Oberwolfach 报告问题汇编

For random combinatorial optimization problems such as minimum spanning arborescences, non-bipartite perfect matchings, and symmetric or asymmetric traveling-salesman problems with i.i.d. uniform or exponential edge weights, is the expected Vickrey cost asymptotic to twice the expected minimum cost?

来源字段：https://doi.org/10.4171/owr/2009/23

15942 | OWR-15218-001

### Volatility Smile Extrapolation and Latent Stochasticity

O / open | Oberwolfach 报告问题汇编

Can the volatility smile be extrapolated from observable strikes robustly enough to infer accurately the stochasticity of volatility from market data?

来源字段：https://doi.org/10.4171/owr/2017/13

15943 | OWR-14298796-005

### Wasserstein Velocities for Boltzmann Curves

O / open | Oberwolfach 报告问题汇编

Given a family $f_t:\mathbb R^d\to\mathbb R$ and the Boltzmann densities $\rho_t=e^{-f_t}/Z_t$, find general conditions ensuring a globally integrable velocity field $v_t$ satisfying $\partial_t\rho_t+\nabla\cdot(\rho_t v_t)=0$. Equivalently, determine when the Fisher-Rao curve $(\rho_t)$ is absolutely continuous in Wasserstein space.

来源字段：https://doi.org/10.4171/owr/2024/48

15944 | OWR-9790363-001

### Weak Moment Conditions and Branching-Process Limit Laws

O / open | Oberwolfach 报告问题汇编

For the coupled branching process driven by a nondecreasing integer-valued offspring process $X(\lambda)$, determine whether convergence to its martingale limit persists under an $X\log X$ condition, and characterize the law of the limiting process $W$ from its distributional fixed-point equation.

来源字段：https://doi.org/10.4171/owr/2022/12

15945 | OWR-11786-009

### Wedge-of-Spheres Homotopy Type of Random Flag Complexes

P / partially_solved | Oberwolfach 报告问题汇编

Let $X\in X(n,p)$ be a random flag complex. For $d\geq6$ and $$n^{-2/d}\ll p\ll n^{-2/(d+1)},$$ is $X$ asymptotically almost surely homotopy equivalent to a wedge of $\lfloor d/2\rfloor$-dimensional spheres?

来源字段：https://doi.org/10.4171/owr/2012/24

15946 | AIM-PROBABILITY-0083

### Weight universality for LPP and polymers: a finite-volume sensitivity reduction

P / partially_solved | AIM 研讨会问题汇编

For example, prove universality of LPP and polymer (with respect to weight distributions).

This is item 11.85 in the “Big picture questions” section of the AIM workshop *The Kardar-Parisi-Zhang equation and universality class*.

The recovered target must specify at least five choices:

1. **Geometry:** here the primary target is point-to-point up-right paths across rectangles $([1,m]\times[1,n]$, with $m/n\to\gamma\in(0,\infty)$ (full aspect ratio, or curved/droplet geometry). 2. **Temperature:** last-passage percolation (LPP) is zero temperature; a directed polymer at inverse temperature $\beta\in(0,\infty)$ is positive temperature. 3. **Weight class:** a plausible strong-universality class consists of nondegenerate i.i.d. light-tailed weights, with additional regularity when uniqueness or curvature is needed. Heavy-tailed and degenerate laws are not silently included. 4. **Centering and scaling:** the law of the weights changes the limit shape/free-energy density and the scale constant. Only the normalized fluctuations are predicted to be universal. 5. **Universality tier:** equality of exponents, a one-point Tracy-Widom limit, convergence of a spatial profile to an Airy process/KPZ fixed point, and convergence of the full random directed metric and geodesics to the directed landscape are successively stronger statements.

For LPP with i.i.d. law $\nu$, the full-aspect-ratio one-point conjecture has the form

$$ \frac{L_{\lfloor\gamma n\rfloor,n}-n g_\nu(\gamma)} {c_\nu(\gamma)n^{1/3}}\ \Longrightarrow\ \mathrm{TW}_{\mathrm{GUE}}, $$

for a curved characteristic direction and nonuniversal constants $g_\nu,c_\nu$. For a fixed-temperature polymer, writing

$$ F_{m,n}^{(\beta)}=\frac1\beta\log Z_{m,n}^{(\beta)}, $$

the analogous conjecture replaces $g_\nu,c_\nu$ by $f_{\beta,\nu},c_{\beta,\nu}$. Process-level attraction requires simultaneous centering in a $n^{2/3}$ spatial window and convergence under the KPZ $1{:}2{:}3$ scaling, not merely a one-point limit.

来源字段：http://aimpl.org/kpzuniversality/1/

15947 | AIM-PROBABILITY-0064

### Weighted image-charge reduction for Green S-curves

P / partially_solved | AIM 研讨会问题汇编

**Green's potential**

Existence theory for \(S\)-curves in a general sense, if the logarithmic potential is replaced by Green's potential.

Nearby AIM records concern scalar \(S\)-curves and max--min problems. The most faithful precise formulation is therefore: for a planar Greenian domain \(D\), external field \(\phi\), and Hausdorff-topologized class \(\mathcal F\) of admissible compact continua, first minimize weighted Green energy over probability measures on each \(F\), then maximize the equilibrium energy over \(F\), and ask whether a maximizer exists and has the Green \(S\)-property. This report distinguishes that free-boundary problem from the classical fixed-compact problem.

来源字段：http://aimpl.org/vectorequilib/4/

15948 | AIM-PROBABILITY-0066

### When convolution does and does not reduce multiple orthogonality

P / partially_solved | AIM 研讨会问题汇编

Is it possible to relate multiple orthogonal polynomials to standard orthogonal polynomials, via the so-called ``convolution product''?

来源字段：http://aimpl.org/vectorequilib/5/

15949 | AMR-099-0086

### When is the uniqueness threshold below one?

P / partially_solved | AMR 问题清单汇编

Give general conditions implying $p_u(G)<1$. In particular, prove or disprove that every one-ended transitive graph has $p_u(G)<1$.

背景链接：https://emis.de/ft/41630#page=9

15950 | OWR-14298374-003

### Wishart Processes with Noninjective Semigroups

O / open | Oberwolfach 报告问题汇编

For the infinite-dimensional Wishart stochastic differential equation with injective diffusion operator $Q$, does there exist an unbounded semigroup generator $A$ for which a solution exists with $\alpha\notin\mathbb N$?

来源字段：https://doi.org/10.4171/owr/2024/26

15951 | OWR-2488-001

### Zero-Regularity Stochastic Burgers Equations with Lévy Noise

O / open | Oberwolfach 报告问题汇编

For the stochastic Burgers equation driven by a cylindrical Lévy process $W(\beta(t))$ with $\beta\in\operatorname{Sub}(p)$ and $1<p\leq2$, does the theory and well-posedness result extend to spatial regularity $\theta=0$?

来源字段：https://doi.org/10.4171/owr/2008/50

15952 | OWR-12697706-001

### Zero-One Law for Directional Transience

O / open | Oberwolfach 报告问题汇编

For a random walk in a random environment, let $A_+$ be the event of directional transience in a fixed direction. Must $\mathbb P(A_+)\in\{0,1\}$?

来源字段：https://doi.org/10.4171/owr/2023/10

15953 | AIM-PROBABILITY-0012

### Zonotope support functions characterize tropical RBM parameter images

P / partially_solved | AIM 研讨会问题汇编

For fixed numbers \(n\) of visible bits and \(k\) of hidden bits, identify concrete tropical or polyhedral objects whose Boolean evaluation vectors are exactly the max-plus RBM image \(TM_n^k\), and explain its relation to the tropical variety \(TV_n^k\), inference complexes, and regular subdivisions.

来源字段：http://aimpl.org/boltzmann/4/

# 资料汇编 13 | 集合论 / Set Theory

221 条记录。O=开放；P=部分解决；S=已解决。状态与英文/LaTeX 表述均沿用来源。

15954 | OWR-14299082-018

### ABCD Conjecture Under Determinacy

O / open | Oberwolfach 报告问题汇编

Assume $\mathrm{ZF}+\mathrm{AD}$. Let $\alpha,\beta,\gamma,\delta<\Theta$ with $\beta\leq\alpha$ and $\delta\leq\gamma$. Prove that $|\alpha^\beta|\leq|\gamma^\delta|$ if and only if $|\beta|\leq|\delta|$ and $|\alpha|\leq|\gamma|$.

来源字段：https://doi.org/10.4171/owr/2025/2

15955 | OWR-7155442-010

### Admissible Sets and Their Jump Structures

O / open | Oberwolfach 报告问题汇编

For every admissible set $A$, is $A$ $\Sigma$-equivalent to its jump structure $J(A)$ if and only if every structure $B$ that is $\Sigma$-definable in $A$ satisfies $B\leq_\Sigma A$?

来源字段：https://doi.org/10.4171/owr/2021/21

15956 | OWR-15986-002

### Algorithmic Randomness and Uniform Distribution

P / partially_solved | Oberwolfach 报告问题汇编

How is algorithmic randomness related to the classical theory of uniform distribution of sequences of real numbers?

来源字段：https://doi.org/10.4171/owr/2018/1

15957 | OWR-1783-002

### Almost-Disjointness Numbers at Singular Cardinals

O / open | Oberwolfach 报告问题汇编

For infinite $\kappa$, let $\mathfrak a_\kappa$ be the least size of a maximal almost-disjoint family of members of $[\kappa]^\kappa$, subject to $\mathfrak a_\kappa\geq\operatorname{cf}(\kappa)$, and write $\mathfrak a=\mathfrak a_\omega$. Is it consistent that $\mathfrak a_{\aleph_\omega}<\mathfrak a$ while $\mathfrak a>\aleph_\omega$? Is it consistent that $\mathfrak a_{\aleph_{\omega_1}}<\mathfrak a_{\aleph_1}$?

来源字段：https://doi.org/10.4171/owr/2008/02

15958 | OWR-1116-001

### Analytic Preorders from Colored Countable Linear Orders

O / open | Oberwolfach 报告问题汇编

Let $P$ be a well-quasi-order that is not a better-quasi-order. For colored countable total orders $(\sqsubseteq,\varphi),(\sqsubseteq',\varphi')\in LO\times\mathbb{N}^{\mathbb{N}}$, let $(\sqsubseteq,\varphi)\preccurlyeq_P(\sqsubseteq',\varphi')$ if an order embedding $g:\mathbb{N}\to\mathbb{N}$ preserves colors up to $P$. Classify the analytic preorder $\preccurlyeq_P$.

来源字段：https://doi.org/10.4171/owr/2005/55

15959 | OWR-1783-008

### Aronszajn-Line Universality and the Size of the Continuum

O / open | Oberwolfach 报告问题汇编

Assume PFA and let $\eta_C$ be a universal Aronszajn line. Does either the universality of $\eta_C$ or the assertion that Aronszajn lines are well-quasi-ordered by embeddability imply $|\mathbb{R}|\leq\aleph_2$?

来源字段：https://doi.org/10.4171/owr/2008/02

15960 | OWR-1269-005

### Aronszajn-Tree Models from Nonstationary-Ideal Forcing

P / partially_solved | Oberwolfach 报告问题汇编

Can an iteration of the nonstationary-ideal forcing method reprove Abraham's theorem that, if $\kappa<\lambda$, $\kappa$ is $\lambda$-supercompact, and $\lambda$ is weakly compact, then there is a generic extension with no $\aleph_2$- or $\aleph_3$-Aronszajn trees? At minimum, can it construct a model with no special Aronszajn trees on $\aleph_2$ or $\aleph_3$?

来源字段：https://doi.org/10.4171/owr/2006/20

15961 | OWR-15216-011

### Candidate Inner Models for HOD Under Determinacy

P / partially_solved | Oberwolfach 报告问题汇编

Assume $\Delta^1_3$-determinacy. Let $N$ be the iterate arising from the directed system of suitable premice, let $\delta^N$ be its Woodin cardinal, and let $\pi^*$ be the induced cofinal map. For a cone of reals $x$, is $$L[N\mid\delta^N,\pi^*]\equiv\operatorname{HOD}^{L[x]}$$?

来源字段：https://doi.org/10.4171/owr/2017/11

15962 | OWR-14299082-016

### Cardinal-Preserving Namba-Type Forcing

O / open | Oberwolfach 报告问题汇编

Assume $\operatorname{LIP}(\lambda,\mu)$, where $\omega<\kappa<\lambda<\mu$ are regular cardinals. Is there a Namba-type minimal forcing that preserves every cardinal $\nu\leq\lambda$?

来源字段：https://doi.org/10.4171/owr/2025/2

15963 | OWR-15216-013

### Cardinals Carrying Aronszajn Trees

O / open | Oberwolfach 报告问题汇编

Characterize the cardinals $\mu$ for which there exists a tree of height $\mu$, with every level of cardinality less than $\mu$, that has no cofinal branch.

来源字段 : https://doi.org/10.4171/owr/2017/11

15964 | OWR-14299082-017

### Chaining and Weak Mixing for Measure-Class-Preserving Actions

P / partially_solved | Oberwolfach 报告问题汇编

For measure-class-preserving actions, is chaining equivalent to weak mixing? Does there exist a measure-class-preserving action that is chaining but not $1$-chaining?

来源字段 : https://doi.org/10.4171/owr/2025/2

15965 | OWR-1703861-006

### Choice Principles for Discontinuous Polish Group Homomorphisms

P / partially_solved | Oberwolfach 报告问题汇编

How much choice is needed to construct a discontinuous homomorphism between Polish groups?

来源字段 : https://doi.org/10.4171/owr/2020/14

15966 | OWR-14299082-021

### Classification of Symbolic Systems with Specification

O / open | Oberwolfach 报告问题汇编

Classify symbolic dynamical systems with the specification property.

来源字段 : https://doi.org/10.4171/owr/2025/2

15967 | OWR-15957-004

### Complexity of Completely Positive Matrix Recognition and Rank

O / open | Oberwolfach 报告问题汇编

Is membership in the cone $\operatorname{CP}_n$ of completely positive matrices in the complexity class NP? How can the completely positive rank of a given matrix in $\operatorname{CP}_n$ be computed?

来源字段 : https://doi.org/10.4171/owr/2017/52

15968 | OWR-7155442-007

### Computability and Complexity Characterizations of Poisson Genericity

P / partially_solved | Oberwolfach 报告问题汇编

For Poisson-generic reals, does a computable example exist, is every Martin-Löf random real Poisson generic, can Poisson genericity be characterized by Kolmogorov complexity, and is Poisson genericity $\Pi^0_3$-complete?

来源字段 : https://doi.org/10.4171/owr/2021/21

15969 | OWR-7155442-009

### Computable Characterizations of Hausdorff Packing Regularity

P / partially_solved | Oberwolfach 报告问题汇编

For $A\subset 2^\omega$, can computability partially characterize regularity defined by $\dim_H(A)=\dim_P(A)$? Can the point-to-set principles give an alternative proof of the closed-subset result for regular analytic sets, and can the analyticity assumption be removed?

来源字段 : https://doi.org/10.4171/owr/2021/21

15970 | OWR-7155442-004

### Computable Multifunctions Between Effectively Hausdorff Spaces

O / open | Oberwolfach 报告问题汇编

Find and prove a characterization of computable multifunctions when both the domain $X$ and codomain $Y$ are effectively Hausdorff spaces.

来源字段 : https://doi.org/10.4171/owr/2021/21

15971 | OWR-15216-006

### Consistency and Structure of Strongly Surjective Linear Orders

P / partially_solved | Oberwolfach 报告问题汇编

For strongly surjective linear orders, is it consistent that one has cardinality greater than $\aleph_2$, or that one contains no minimal strongly surjective suborder? Does $\mathrm{MA}_{\aleph_1}$ imply the existence of an uncountable strongly surjective order $L\subseteq\mathbb R$? If $L$ is strongly surjective, must $L\setminus\{x\}$ be strongly surjective for every $x\in L$? Do strongly surjective Countryman lines exist under $\mathrm{MA}_{\aleph_1}$ or PFA?

来源字段 : https://doi.org/10.4171/owr/2017/11

15972 | OWR-15216-028

### Consistency Patterns Among Cardinal Characteristics

P / partially_solved | Oberwolfach 报告问题汇编

Are the consistency patterns $\mathfrak b<\mathfrak s<\mathfrak d$ and $\mathfrak b<\mathfrak a<\mathfrak s$ possible? Is $\mathfrak u(\omega_1)<2^{\aleph_1}$ consistent?

来源字段 : https://doi.org/10.4171/owr/2017/11

15973 | OWR-15216-026

### Consistency Strength of Almost Effective Equivalence Reductions

O / open | Oberwolfach 报告问题汇编

What is the consistency strength of $E_{\omega_1}\leq_{a\Delta^1_1}F_{\omega_1}$? Does such an almost-$\Delta^1_1$ reduction imply the existence of a real whose admissibility spectrum is sufficiently sparse that its admissible ordinals are $L$-cardinals and $0^\#$ can be recovered?

来源字段：https://doi.org/10.4171/owr/2017/11

15974 | OWR-1783-007

### Consistency Strength of Forcing-Invariant Elementary Equivalence

S / solved | Oberwolfach 报告问题汇编

Suppose $N\models\mathrm{ZFC}$ and, for every cardinal $\lambda$ of $N$ and every $N$-generic $H\subseteq\operatorname{Col}(\omega,\lambda)$, one has $N[H]\equiv N$. What is the consistency strength of this property, and how does it compare with the large-cardinal hypothesis used to make the club filter on $\omega_1$ an ultrafilter?

来源字段：https://doi.org/10.4171/owr/2008/02

15975 | HIL-001

### Continuum Hypothesis

S / solved | Hilbert 问题

别名：The Continuum Hypothesis [SET-001]

There is no set whose cardinality is strictly between that of the integers and the real numbers.

背景链接：https://doi.org/10.1073/pnas.24.12.556

15976 | OWR-15957-005

### Copositive Hierarchies for the Kissing-Number Problem

O / open | Oberwolfach 报告问题汇编

How do copositive and completely positive approximation hierarchies compare with other known hierarchies for bounding the kissing number?

来源字段：https://doi.org/10.4171/owr/2017/52

15977 | OWR-1116-014

### Decomposition of Well-Quasi-Orders into Better-Quasi-Orders

P / partially_solved | Oberwolfach 报告问题汇编

Is every well-quasi-order a countable union of better-quasi-orders? If $P$ is a well-quasi-order, must $P$ belong to $G_{\mathrm{bqo}}$, or at least $G_{\mathrm{wqo}}$, the classes whose poset algebras are respectively better-quasi-order-generated or well-quasi-order-generated?

来源字段：https://doi.org/10.4171/owr/2005/55

15978 | OWR-1116-003

### Definability Complexity of Subsets of the First Uncountable Ordinal

P / partially_solved | Oberwolfach 报告问题汇编

Given $A\subseteq\omega_1$, what is the least Lévy-hierarchy complexity of a parameter-free formula $\Phi(x)$ such that one can pass to a stationary-set-preserving set-forcing extension in which $A$ is defined by $\Phi$ over $\langle H(\omega_2),\in\rangle$, or over a natural expansion such as $\langle H(\omega_2),\in,\mathrm{NS}_{\omega_1}\rangle$?

来源字段：https://doi.org/10.4171/owr/2005/55

15979 | OWR-1116-004

### Definability Complexity of Well-Orderings of the Reals

P / partially_solved | Oberwolfach 报告问题汇编

What is the least Lévy-hierarchy complexity of a parameter-free formula $\Phi(x,y)$ for which there is a set-forcing extension in which $\Phi$ defines a well-order of the reals over $\langle H(\omega_2),\in\rangle$, or over a natural expansion such as $\langle H(\omega_2),\in,\mathrm{NS}_{\omega_1}\rangle$?

来源字段：https://doi.org/10.4171/owr/2005/55

15980 | OWR-4792-001

### Definable Maximal Almost Disjoint Families

P / partially_solved | Oberwolfach 报告问题汇编

Are $a_{\mathrm{Borel}}=a_{\mathrm{closed}}$ and $a_{\mathrm{Borel}}>\mathfrak b$ consistent? Is it consistent that $\mathfrak b>\aleph_1$ and there is a $\Pi^1_1$ mad family? Is the existence of a $\Pi^1_1$ mad family equivalent to that of a $\Sigma^1_2$ mad family? Does the Ramsey property for $\Sigma^1_2$ sets imply that there is no $\Sigma^1_2$ mad family?

来源字段：https://doi.org/10.4171/owr/2011/02

15981 | OWR-1116-005

### Definable Well-Orders Under Martin's Maximum

S / solved | Oberwolfach 报告问题汇编

Assuming a supercompact cardinal, or another reasonable large-cardinal hypothesis, can one force Martin's Maximum together with a parameter-free definable well-order of $H(\omega_2)$ over $\langle H(\omega_2),\in\rangle$? Does Martin's Maximum itself imply such a well-order definable with at most one real parameter?

来源字段：https://doi.org/10.4171/owr/2005/55

15982 | OWR-1116-002

### Dense-Order-Preserving Preorders on Ordinal-Valued Functions

O / open | Oberwolfach 报告问题汇编

For each countable ordinal $\alpha$, classify the preorder on functions $\varphi,\psi:\mathbb{Q}\to\alpha$ defined by $\varphi\leq_{\alpha,\mathrm{dop}}\psi$ if and only if some dense order-preserving $g:\mathbb{Q}\to\mathbb{Q}$ satisfies $\varphi(q)\leq_\alpha\psi(g(q))$ for every $q\in\mathbb{Q}$.

来源字段：https://doi.org/10.4171/owr/2005/55

15983 | OWR-7155442-002

### Depth Relative to K-Trivial Oracles

O / open | Oberwolfach 报告问题汇编

For a $K$-trivial oracle $A$, is $A$-depth strictly stronger than depth, or does $A$-depth coincide with depth? More generally, which of these two cases occurs for $K$-trivial oracles?

来源字段：https://doi.org/10.4171/owr/2021/21

15984 | OWR-15216-014

### Eliminating Special Aronszajn Trees Along Radin Clubs

O / open | Oberwolfach 报告问题汇编

Can a consistency result asserting the absence of special Aronszajn trees at a prescribed class of regular cardinals be strengthened to assert the absence of all Aronszajn trees at those cardinals?

来源字段：https://doi.org/10.4171/owr/2017/11

15985 | OWR-7155442-008

### Embedding Locally Countable Orders into Turing Degrees

O / open | Oberwolfach 报告问题汇编

Does every locally countable partial order of cardinality continuum embed into the Turing degrees?

来源字段：https://doi.org/10.4171/owr/2021/21

15986 | OWR-15216-010

### Equality of Canonical Models from Universal Projective Sets

O / open | Oberwolfach 报告问题汇编

Let $\widetilde\varphi$ be a $\Delta^1_3$-scale on a good universal $\Pi^1_2$ set, and let $O_{\Sigma^1_3,\widetilde\varphi}$ be the universal $\Sigma^1_3$ set defined relative to that scale. Is $$L_{u_\omega}[O_{\Sigma^1_3}]=L_{u_\omega}[O_{\Sigma^1_3,\widetilde\varphi}]$$?

来源字段：https://doi.org/10.4171/owr/2017/11

15987 | EP-1128

### Erdős Problem #1128: Set Theory, Ramsey Theory

S / solved | Erdős 问题

Let $A,B,C$ be three sets of cardinality $\aleph_1$. Is it true that, in any $2$-colouring of $A\times B\times C$, there must exist $A_1\subset A$, $B_1\subset B$, $C_1\subset C$, all of cardinality $\aleph_0$, such that $A_1\times B_1\times C_1$ is monochromatic?

来源字段：https://www.erdosproblems.com/1128

15988 | EP-1167

### Erdős Problem #1167: Let $\kappa_\alpha$ be cardinals for all $\alpha<\gamma$

O / open | Erdős 问题

Let $r\geq 2$ be finite and $\lambda$ be an infinite cardinal. Let $\kappa_\alpha$ be cardinals for all $\alpha<\gamma$.

Is it true that\[2^\lambda \to (\kappa_\alpha+1)_{\alpha<\gamma}^{r+1}\]implies\[\lambda \to (\kappa_\alpha)_{\alpha<\gamma}^{r}?\]Here $+$ means cardinal addition, so that $\kappa_\alpha+1=\kappa_\alpha$ if $\kappa_\alpha$ is infinite.

来源字段：https://www.erdosproblems.com/1167

15989 | EP-1168

### Erdős Problem #1168: Set Theory, Ramsey Theory

O / open | Erdős 问题

Prove that\[\aleph_{\omega+1}\not\to (\aleph_{\omega+1},3,\ldots,3)_{\aleph_0}^2\]without assuming the generalised continuum hypothesis.

来源字段：https://www.erdosproblems.com/1168

15990 | EP-1169

### Erdős Problem #1169: Is it true that, for all finite $k<\omega$,\[\omega_1^2 \not\to (\omega_1^2, 3)^2?\]

P / partially_solved | Erdős 问题

Is it true that, for all finite $k<\omega$,\[\omega_1^2 \not\to (\omega_1^2, 3)^2?\]

来源字段：https://www.erdosproblems.com/1169

15991 | EP-1170

### Erdős Problem #1170: Is it consistent that\[\omega_2\to (\alpha)_2^2\]for every $\alpha <\omega_2$?

O / open | Erdős 问题

Is it consistent that\[\omega_2\to (\alpha)_2^2\]for every $\alpha <\omega_2$?

来源字段：https://www.erdosproblems.com/1170

15992 | EP-1171

### Erdős Problem #1171: Set Theory, Ramsey Theory

P / partially_solved | Erdős 问题

Is it true that, for all finite $k<\omega$,\[\omega_1^2\to (\omega_1\omega, 3,\ldots,3)_{k+1}^2?\]

来源字段：https://www.erdosproblems.com/1171

15993 | EP-1172

### Erdős Problem #1172: Set Theory, Ramsey Theory

O / open | Erdős 问题

Establish whether the following are true assuming the generalised continuum hypothesis:\[\omega_3 \to (\omega_2,\omega_1+2)^2,\]\[\omega_3\to (\omega_2+\omega_1,\omega_2+\omega)^2,\]\[\omega_2\to (\omega_1^{\omega+2}+2, \omega_1+2)^2.\]Establish whether the following is consistent with the generalised continuum hypothesis:\[\omega_2\to (\omega_1+\omega)_2^2,\]or even $\omega_2 \to (\xi)_2^2$ for all $\xi<\omega_2$.

来源字段：https://www.erdosproblems.com/1172

15994 | EP-1173

### Erdős Problem #1173: Set Theory, Combinatorics

O / open | Erdős 问题

Assume the generalised continuum hypothesis. Let\[f: \omega_{\omega+1}\to [\omega_{\omega+1}]^{\leq \aleph_\omega}\]be a set mapping such that\[\lvert f(\alpha)\cap f(\beta)\rvert <\aleph_\omega\]for all $\alpha\neq \beta$. Does there exist a free set of cardinality $\aleph_{\omega+1}$?

来源字段：https://www.erdosproblems.com/1173

15995 | EP-1174

### Erdős Problem #1174: Set Theory, Ramsey Theory

P / partially_solved | Erdős 问题

Does there exist a graph $G$ with no $K_4$ such that every edge colouring of $G$ with countably many colours contains a monochromatic $K_3$?

Does there exist a graph $G$ with no $K_{\aleph_1}$ such that every edge colouring of $G$ with countably many colours contains a monochromatic $K_{\aleph_0}$?

来源字段：https://www.erdosproblems.com/1174

15996 | EP-1175

### Erdős Problem #1175: Set Theory, Chromatic Number

O / open | Erdős 问题

Let $\kappa$ be an uncountable cardinal. Must there exist a cardinal $\lambda$ such that every graph with chromatic number $\lambda$ contains a triangle-free subgraph with chromatic number $\kappa$?

来源字段：https://www.erdosproblems.com/1175

15997 | EP-1176

### Erdős Problem #1176: Set Theory, Ramsey Theory

P / partially_solved | Erdős 问题

Let $G$ be a graph with chromatic number $\aleph_1$. Is it true that there is a colouring of the edges with $\aleph_1$ many colours such that, in any countable colouring of the vertices, there exists a vertex colour containing all edge colours?

来源字段：https://www.erdosproblems.com/1176

15998 | EP-1177

### Erdős Problem #1177: Set Theory, Chromatic Number

O / open | Erdős 问题

Let $G$ be a finite $3$-uniform hypergraph, and let $F_G(\kappa)$ denote the collection of $3$-uniform hypergraphs with chromatic number $\kappa$ not containing $G$.

If $F_G(\aleph_1)$ is not empty then there exists $X\in F_G(\aleph_1)$ of cardinality at most $2^{2^{\aleph_0}}$.

If both $F_G(\aleph_1)$ and $F_H(\aleph_1)$ are non-empty then $F_G(\aleph_1)\cap F_H(\aleph_1)$ is non-empty.

If $\kappa,\lambda$ are uncountable cardinals and $F_G(\kappa)$ is non-empty then $F_G(\lambda)$ is non-empty.

来源字段：https://www.erdosproblems.com/1177

15999 | EP-118

### Erdős Problem #118: Set Theory, Ramsey Theory

S / solved | Erdős 问题

Let $\alpha$ be a cardinal or ordinal number or an order type such that every two-colouring of $K_\alpha$ contains either a red $K_\alpha$ or a blue $K_3$. For every $n\geq 3$ must every two-colouring of $K_\alpha$ contain either a red $K_\alpha$ or a blue $K_n$?

来源字段：https://www.erdosproblems.com/118

16000 | EP-1218

### Erdős Problem #1218: Ramsey Theory, Set Theory

O / open | Erdős 问题

Assume the Generalised Continuum Hypothesis and let $\beta=\omega_{\omega+1}$. Is it true that\[\aleph_{\beta+1} \not\to (\aleph_\beta,4,\ldots,4)^3\](where there are countably infinite many copies of $4$)?

来源字段：https://www.erdosproblems.com/1218

16001 | EP-1219

### Erdős Problem #1219: Ramsey Theory, Set Theory

S / solved | Erdős 问题

Let $(n_k)$ be an increasing sequence of integers such that $2^{\aleph_{n_k}}$ is strictly increasing, and $2^{\aleph_{n_0}}>\aleph_\omega$. Is it true that\[\sum_{k} 2^{\aleph_{n_k}}\to (\aleph_\omega)^2?\]

来源字段：https://www.erdosproblems.com/1219

16002 | EP-1220

### Erdős Problem #1220: Does\[\lambda \to (\lambda,\aleph_1)^2\]hold?

O / open | Erdős 问题

Let $\lambda$ be a singular cardinal such that both $\lambda$ and $\mathrm{cf}(\lambda)$ are $\aleph_0$-inaccessible. Does\[\lambda \to (\lambda,\aleph_1)^2\]hold?

来源字段：https://www.erdosproblems.com/1220

16003 | EP-474

### Erdős Problem #474: Set Theory, Ramsey Theory

P / partially_solved | Erdős 问题

Under what set theoretic assumptions is it true that $\mathbb{R}^2$ can be $3$-coloured such that, for every uncountable $A\subseteq \mathbb{R}^2$, $A^2$ contains a pair of each colour?

来源字段：https://www.erdosproblems.com/474

16004 | EP-590

### Erdős Problem #590: Set Theory, Ramsey Theory

S / solved | Erdős 问题

Let $\alpha$ be the infinite ordinal $\omega^\omega$. Is it true that in any red/blue colouring of the edges of $K_\alpha$ there is either a red $K_\alpha$ or a blue $K_3$?

来源字段：https://www.erdosproblems.com/590

16005 | EP-70

### Erdős Problem #70

O / open | Erdős 问题

Let $\mathfrak{c}$ be the ordinal of the real numbers, $\beta$ be any countable ordinal, and $2\leq n<\omega$. Is it true that $\mathfrak{c}\to (\beta, n)_2^3$?

来源字段：https://www.erdosproblems.com/70

16006 | OWR-15957-003

### Extremal Completely Positive Plus-Rank

O / open | Oberwolfach 报告问题汇编

For a nonzero symmetric matrix $A\in S^n$, let $\operatorname{cp}(A)$ be its completely positive rank and $\operatorname{cp}^+(A)$ its completely positive plus-rank. What is the maximum finite value of $\operatorname{cp}^+(A)-\operatorname{cp}(A)$, and what is the maximum finite value of $\operatorname{cp}^+(A)$ on $S^n$?

来源字段：https://doi.org/10.4171/owr/2017/52

16007 | OWR-15957-001

### Extremal Copositive Rays and Completely Positive Factorizations

O / open | Oberwolfach 报告问题汇编

For the copositive cone, can the extremal generating rays be completely characterized? For the completely positive cone, can the supporting hyperplanes be completely characterized, and can a general completely positive matrix $A$ be given explicitly as $A=\sum_{i=1}^r b_i b_i^T$ with $b_i\in\mathbb R_+^n$?

来源字段：https://doi.org/10.4171/owr/2017/52

16008 | OWR-15986-004

### Final Segments of Difference-C.E. Degrees

P / partially_solved | Oberwolfach 报告问题汇编

Which finite final segments occur in the degrees of d.c.e. sets?

来源字段：https://doi.org/10.4171/owr/2018/1

16009 | OWR-15986-003

### Finite Families with Two-Element Rogers Semilattices

O / open | Oberwolfach 报告问题汇编

For each $n\geq2$, does there exist a finite family $\mathcal S$ of $\Sigma_n^0$ sets whose Rogers semilattice has exactly two elements $\alpha$ and $\beta$, where $\alpha$ is Friedberg and every numbering $\gamma$ not equivalent to $\alpha$ satisfies $\beta\leq\gamma$?

来源字段：https://doi.org/10.4171/owr/2018/1

16010 | OWR-1269-003

### First Failure of Coherent Realization in a Steel Array

O / open | Oberwolfach 报告问题汇编

Let $\langle N_\xi\mid\xi\leq\Theta\rangle$ be a Steel array in which $e$-robustness is the criterion for adding extenders. What is the first $N_\xi$ that fails to permit a coherent realization for some countable $\sigma_0:P\prec N_\xi$ and countable normal iteration $I$ of $P$? Does this $N_\xi$ have an iteration with an alternating chain?

来源字段：https://doi.org/10.4171/owr/2006/20

16011 | OWR-1783-006

### Forcing-Invariant Elementary Equivalence Models

S / solved | Oberwolfach 报告问题汇编

Does there exist a model $N$ of $\mathrm{ZFC}$ such that, for every cardinal $\lambda$ of $N$ and every $N$-generic $H\subseteq\operatorname{Col}(\omega,\lambda)$, the extension $N[H]$ is elementarily equivalent to $N$? Does such a model exist if the requirement is imposed only when $H$ collapses a regular cardinal of $N$?

来源字段：https://doi.org/10.4171/owr/2008/02

16012 | OWR-14299082-019

### Function-Space Measure Embedding Values

O / open | Oberwolfach 报告问题汇编

Let $\nu$ be the function-space measure on ${}^{\omega_1}\omega_1$. Compute $j_\nu(\omega_2)$, and more generally determine the values of the associated elementary embedding on relevant cardinals.

来源字段：https://doi.org/10.4171/owr/2025/2

16013 | ST-003

### GCH and Diamond Principle

O / open | 独立 / 未分配集合

Does the generalized continuum hypothesis entail the diamond principle $\diamondsuit(E_{\text{cf}(\lambda)}^{\lambda^+})$ for every singular cardinal $\lambda$?

背景链接：https://doi.org/10.1007/3-540-44761-X

16014 | ST-004

### GCH and Suslin Trees

O / open | 独立 / 未分配集合

Does the generalized continuum hypothesis imply the existence of an $\aleph_2$-Suslin tree?

背景链接：https://en.wikipedia.org/wiki/Suslin_tree

16015 | OWR-14299082-003

### Generic Maximal Chains on Exceptional Surfaces

O / open | Oberwolfach 报告问题汇编

Does the action of $\operatorname{Homeo}(X)$ on the hyperspace $\Phi(X)$ of maximal chains of connected closed subsets have a comeager orbit when $X=S^2$? Does it have one when $X=\mathbb{RP}^2$?

来源字段：https://doi.org/10.4171/owr/2025/2

16016 | OWR-14299082-004

### Generic Point Property for Exceptional Surface Homeomorphisms

O / open | Oberwolfach 报告问题汇编

Does $\operatorname{Homeo}_0(S^2)$ have the generic point property? Does $\operatorname{Homeo}_0(\mathbb{RP}^2)$ have the generic point property?

来源字段：https://doi.org/10.4171/owr/2025/2

16017 | OWR-15986-001

### Graph-Cototal Enumeration Degrees and Punctiform Decompositions

O / open | Oberwolfach 报告问题汇编

Is every almost-total enumeration degree graph-cototal? Equivalently, does the Hilbert cube $[0,1]^\omega$ admit a countable decomposition into punctiform spaces?

来源字段：https://doi.org/10.4171/owr/2018/1

16018 | OWR-14299082-014

### HOD Failure at Uncountable Cofinality

O / open | Oberwolfach 报告问题汇编

Is it consistent with ZFC that the HOD hypothesis holds and there are cardinals $\kappa<\lambda$ such that $\kappa$ is $<\lambda$-extendible, $\lambda$ is a strong-limit cardinal with $\operatorname{cf}(\lambda)=\omega_1$, and $(\lambda^+)^{\mathrm{HOD}}<\lambda^+$?

来源字段：https://doi.org/10.4171/owr/2025/2

16019 | OWR-15216-009

### HOD-Pair Capturing Under Determinacy and No Long Extenders

O / open | Oberwolfach 报告问题汇编

别名：Removing Iterability from HOD-Pair Capturing [OWR-15216-008]

Does $(\mathrm{AD}^++\mathrm{NLE})\Rightarrow\mathrm{HPC}$, where NLE asserts that there is no $\omega_1$-iteration strategy for a countable pure extender mouse with a long extender, and HPC denotes HOD-pair capturing?

来源字段：https://doi.org/10.4171/owr/2017/11

16020 | OWR-15216-019

### Hybrid Inner Models at Limits of Woodin Cardinals

O / open | Oberwolfach 报告问题汇编

Assume that there is no inner model with a Woodin cardinal that is a limit of Woodin cardinals, and let $\kappa$ be measurable and a limit of Woodin and strong cardinals. Must there be hybrid structures $M_0\mathrel{\triangleleft}M_1\subseteq H_{\kappa^+}$ such that, for every generic $g\subseteq\operatorname{Coll}(\omega,{<}\kappa)$, one has $H^-\mathrel{\triangleleft}M_0$, $$L(M_0^\omega,\Gamma,\mathbb R)\models\mathrm{AD},$$ $\operatorname{Ord}\cap M_0$ is the largest cardinal of $M_1$, and $M_1$ is close to $V$ at $\kappa$?

来源字段：https://doi.org/10.4171/owr/2017/11

16021 | OWR-14299082-012

### Hyperfiniteness of Generic Wreath-Product Actions

O / open | Oberwolfach 报告问题汇编

Is the generic action of the wreath product $\mathbb Z\wr\mathbb Z$ on Cantor space hyperfinite?

来源字段：https://doi.org/10.4171/owr/2025/2

16022 | OWR-14299082-020

### Hyperfiniteness of Minimal-Subshift Conjugacy

O / open | Oberwolfach 报告问题汇编

Is the topological-conjugacy equivalence relation on minimal symbolic subshifts hyperfinite? In particular, is its restriction to Toeplitz subshifts hyperfinite?

来源字段：https://doi.org/10.4171/owr/2025/2

16023 | OWR-14299082-010

### Idealistic Equivalence Relations and Orbit Relations

O / open | Oberwolfach 报告问题汇编

Is every idealistic equivalence relation on a standard Borel space Borel bireducible with an orbit equivalence relation?

来源字段：https://doi.org/10.4171/owr/2025/2

16024 | OWR-1703861-003

### Indecomposable Ultrafilters Beyond Measurable Cardinals

P / partially_solved | Oberwolfach 报告问题汇编

Can there be an indecomposable ultrafilter on a cardinal that is neither measurable nor the limit of countably many measurable cardinals?

来源字段：https://doi.org/10.4171/owr/2020/14

16025 | ST-009

### Jónsson Algebra on ℵ_ω

O / open | 独立 / 未分配集合

Does there exist a Jónsson algebra on $\aleph_\omega$?

背景链接：https://www.unsolvedmath.com/problems?category=10&status=open

16026 | OWR-1783-003

### Jónsson Cardinals, Trace Filters, and Indecomposable Ultrafilters

O / open | Oberwolfach 报告问题汇编

Let $\kappa$ be a regular Jónsson cardinal. First, do there exist a function $c:[\kappa]^{<\kappa}\to\kappa$ and a uniform filter $\mathcal F$ on $\kappa$ such that, for every $A\in[\kappa]^\kappa$, the range of $c$ on $[A]^{<\omega}$ belongs to $\mathcal F$? Second, can a $C$-system on $\kappa$ have a nontrivial trace filter? Third, if $\lambda=\mu^+$ for a singular strong-limit cardinal $\mu$, is an indecomposable ultrafilter on $\lambda$ compatible with $2^\mu>\mu^+$?

来源字段：https://doi.org/10.4171/owr/2008/02

16027 | OWR-1703861-008

### Large Cardinal Foundations of MAX Universal Baireness

O / open | Oberwolfach 报告问题汇编

Is $\operatorname{MAX}(\mathrm{UB})$ a direct consequence of suitable large-cardinal axioms, where $\operatorname{MAX}(\mathrm{UB})$ asserts class many Woodin cardinals together with the stated universal-Baireness and generic-elementarity conditions for universally Baire sets?

来源字段：https://doi.org/10.4171/owr/2020/14

16028 | OWR-15216-017

### Maximal Almost Disjoint Families and Ramsey Regularity

O / open | Oberwolfach 报告问题汇编

How does the nonexistence of maximal almost-disjoint families of infinite subsets of $\omega$ interact with regularity principles, especially the assertion that every set of reals has the Ramsey property?

来源字段：https://doi.org/10.4171/owr/2017/11

16029 | OWR-15216-023

### Meager Ideal Equalities at Uncountable Regular Cardinals

O / open | Oberwolfach 报告问题汇编

For every uncountable regular cardinal $\kappa$ satisfying $2^{<\kappa}=\kappa$, are $$\operatorname{add}(\mathcal M_\kappa)=\mathfrak b_\kappa\qquad\text{and}\qquad\mathfrak d_\kappa=\operatorname{cof}(\mathcal M_\kappa)$$?

来源字段：https://doi.org/10.4171/owr/2017/11

16030 | OWR-4792-015

### Measurability Assumption in the Prikry Extension Theorem

O / open | Oberwolfach 报告问题汇编

Let $\mathrm{SC}(\kappa)$ assert that $\kappa$ is strongly inaccessible and that $\{x\in[\kappa^+]^{<\kappa}:\operatorname{otp}(x)\text{ is a cardinal}\}$ is stationary. Assuming GCH and $\mathrm{SC}(\kappa)$, can one obtain a Prikry extension satisfying $\neg\square_{\aleph_\omega,<\omega}$ without assuming that $\kappa$ carries a normal measure?

来源字段：https://doi.org/10.4171/owr/2011/02

16031 | OWR-14299082-015

### Measurable-Cardinal Strength of Minimal Namba Collapse

O / open | Oberwolfach 报告问题汇编

Does the consistency of regular cardinals $\nu$ and $\mu=\nu^{++}$ admitting a $|\mu|=\nu^+$-minimal forcing extension that preserves the regularity of $\mu^+$ require the consistency strength of a measurable cardinal?

来源字段：https://doi.org/10.4171/owr/2025/2

16032 | OWR-1269-010

### Minimum $AD_{\mathbb R}$-Like Model at $\lambda$

O / open | Oberwolfach 报告问题汇编

Suppose $M$ is an $AD_{\mathbb R}$-like model at $\lambda$, and let $$M_0=\bigcap\{N:N\text{ is an }AD_{\mathbb R}\text{-like model at }\lambda\text{ with }\Theta_N=\Theta_M\}.$$ Is $M_0$ itself $AD_{\mathbb R}$-like, with $\Theta_{M_0}=\Theta_M$?

来源字段：https://doi.org/10.4171/owr/2006/20

16033 | OWR-14299082-006

### Mouse-Set Property Without a Cone Restriction

O / open | Oberwolfach 报告问题汇编

Assume $\mathrm{ZF}+\mathrm{AD}+V=L(\mathbb R)$. Let $[\alpha,\gamma]$ be a weak $\Sigma_1$ gap, or let $\gamma=\beta+1$ for a strong gap $[\alpha,\beta]$, and let $e$ be least such that $\rho_{e+1}^{J_\gamma(\mathbb R)}=\mathbb R$. Is $\mathrm{OD}^{*\mathbb R}_{\gamma,e+2}$ a mouse set without restricting to a cone of reals; that is, does it equal $\mathbb R\cap M$ for some mouse $M$?

来源字段：https://doi.org/10.4171/owr/2025/2

16034 | OWR-1116-009

### Nonpolishability and Borel Reducibility of Orbit Relations

P / partially_solved | Oberwolfach 报告问题汇编

For a Borel group $H$, is nonpolishability equivalent to the existence of a Polish $H$-space $X$ such that $E_1$ Borel-reduces to $E_H^X$? Here $E_1$ is eventual equality on sequences in $2^{\mathbb{N}}$.

来源字段：https://doi.org/10.4171/owr/2005/55

16035 | ST-010

### Open Coloring Axiom and Continuum Hypothesis

O / open | 独立 / 未分配集合

Is the open coloring axiom (OCA) consistent with $2^{\aleph_0} > \aleph_2$?

背景链接：https://arxiv.org/abs/1904.10516

16036 | OWR-1116-008

### Orbit Equivalence Relations of Nonlocally Compact Borel Groups

O / open | Oberwolfach 报告问题汇编

For a Borel group $H$, is failure of a locally compact second-countable group topology extending its original topology equivalent to the existence of a Polish $H$-space $X$ for which $E_0^{\mathbb{N}}$ Borel-reduces to the orbit relation $E_H^X$? Here $E_0^{\mathbb{N}}$ is coordinatewise eventual equality on $(2^{\mathbb{N}})^{\mathbb{N}}$.

来源字段：https://doi.org/10.4171/owr/2005/55

16037 | ST-001

### Partition Principle Implies Axiom of Choice

O / open | 独立 / 未分配集合

Does the partition principle (PP) imply the axiom of choice (AC)?

背景链接：https://doi.org/10.1007/s00283-026-10531-4

16038 | OWR-7155442-012

### Positive Degrees Within Truth Table and Many One Degrees

P / partially_solved | Oberwolfach 报告问题汇编

Can odd numbers other than $1$ occur as the number of positive degrees inside a truth-table degree? Does every many-one degree that is not recursive either consist of one one-one degree or contain an infinite antichain of one-one degrees?

来源字段：https://doi.org/10.4171/owr/2021/21

16039 | OWR-4792-010

### Precipitous Ideal Separation at Higher Cardinals

O / open | Oberwolfach 报告问题汇编

Can the separation between precipitous and normal precipitous ideals be generalized beyond $\aleph_1$? In particular, is there a model with a precipitous ideal on $\aleph_2$ but no normal precipitous ideal? Is it consistent that a supercompact cardinal exists while $\aleph_1$ carries no precipitous ideal?

来源字段：https://doi.org/10.4171/owr/2011/02

16040 | OWR-4792-009

### Precipitous Ideals Without Normal Precipitous Ideals

P / partially_solved | Oberwolfach 报告问题汇编

Is the assumption $o(\kappa)=\kappa^{++}$ needed for a model with a precipitous ideal on $\aleph_1$ but no normal precipitous ideal? Is it possible to have a GCH model with a precipitous ideal on $\aleph_1$ but no normal one?

来源字段：https://doi.org/10.4171/owr/2011/02

16041 | OWR-12862-007

### Projective Determinacy and Zero Sharp over Third-Order Arithmetic

O / open | Oberwolfach 报告问题汇编

Over third-order arithmetic $Z_3$, does $\Pi^1_1$ determinacy imply the existence of $0^\#$?

来源字段：https://doi.org/10.4171/owr/2014/02

16042 | OWR-15216-015

### Radin Forcing and Simultaneous Tree Properties

O / open | Oberwolfach 报告问题汇编

Is there a Radin forcing that adds a club $C\subseteq\kappa$ such that, for every cardinal $\mu\in C$, there is no special $\mu^+$-Aronszajn tree?

来源字段：https://doi.org/10.4171/owr/2017/11

16043 | OWR-15957-002

### Rational and Integral Completely Positive Factorizations

P / partially_solved | Oberwolfach 报告问题汇编

Does every rational completely positive matrix have a rational completely positive factorization? Does every $2\times2$ integral completely positive matrix have an integral completely positive factorization?

来源字段：https://doi.org/10.4171/owr/2017/52

16044 | OWR-4792-017

### Rectangular Partition Relations on $\omega_3$

O / open | Oberwolfach 报告问题汇编

Is it consistent that there exists a function $f:\omega_3\times\omega_3\to\omega$ that is not constant on any $A\times B\subseteq\omega_3\times\omega_3$ with $\operatorname{otp}(A)=\operatorname{otp}(B)=\omega$?

来源字段：https://doi.org/10.4171/owr/2011/02

16045 | OWR-15216-027

### Reducing Cardinal Invariants from Boolean Ultrapower Constructions

O / open | Oberwolfach 报告问题汇编

Can the Boolean-ultrapower consistency patterns for cardinal invariants be obtained without assuming a supercompact cardinal, or with the affected invariants assigned smaller values below the first ground-model supercompact above $\kappa$?

来源字段：https://doi.org/10.4171/owr/2017/11

16046 | ST-011

### Reinhardt Cardinals without Choice

O / open | 独立 / 未分配集合

Without assuming the axiom of choice, can a nontrivial elementary embedding V→V exist?

背景链接：https://doi.org/10.1016/j.apal.2023.103323

16047 | AMR-133-0001

### Selection Principles in Mathematics - Problem 1 ([coc2])

O / open | AMR 问题清单汇编

- Is $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\Omega)=\mathsf{S}_{\mathrm{fin}}(\Gamma,\Omega)$?

- And if not, does $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\Gamma)$ imply $\mathsf{S}_{\mathrm{fin}}(\Gamma,\Omega)$?

背景链接：https://arxiv.org/abs/math/0609601

16048 | AMR-133-0010

### Selection Principles in Mathematics - Problem 10 ([GN])

O / open | AMR 问题清单汇编

Clearly, $\binom{\Omega}{\Gamma}$ implies the $\delta$-property. $\mathsf{S}_1(\Omega,\Gamma)=\binom{\Omega}{\Gamma}$ [GN].

Is the $\delta$-property equivalent to $\binom{\Omega}{\Gamma}$?

背景链接：https://arxiv.org/abs/math/0609601

16049 | AMR-133-0011

### Selection Principles in Mathematics - Problem 11 ([ideals, MilNonGamma])

O / open | AMR 问题清单汇编

A property of sets of reals is *hereditary* if for each set of reals $X$ satisfying the property, all subsets of $X$ satisfy that property. None of the selective hypotheses involving open covers is provably hereditary [ideals]. However, the property $\mathsf{S}_1(\mathrm{B},\mathrm{B})$ as well as all properties of the form $\Pi(\mathrm{B}_\Gamma,\mathrm{B})$ are hereditary [ideals] (but $\mathsf{S}_1(\mathrm{B}_\Omega,\mathrm{B}_\Gamma)$ is not [MilNonGamma]).

Is $\mathsf{S}_1(\mathrm{B}_\Omega,\mathrm{B}_\Omega)$ or $\mathsf{S}_{\mathrm{fin}}(\mathrm{B}_\Omega,\mathrm{B}_\Omega)$ hereditary?

背景链接：https://arxiv.org/abs/math/0609601

16050 | AMR-133-0012

### Selection Principles in Mathematics - Problem 12 ([CBC])

O / open | AMR 问题清单汇编

$\mathsf{S}_1(\Omega,\Gamma)$, $\mathsf{S}_1(\Omega,\Omega)$, and $\mathsf{S}_{\mathrm{fin}}(\Omega,\Omega)$ are the only properties in the open case which are preserved under taking finite powers [coc2].

Are any of the classes $\mathsf{S}_1(\mathrm{B}_\Omega,\mathrm{B}_\Omega)$ or $\mathsf{S}_{\mathrm{fin}}(\mathrm{B}_\Omega,\mathrm{B}_\Omega)$ preserved by finite powers?

背景链接：https://arxiv.org/abs/math/0609601

16051 | AMR-133-0013

### Selection Principles in Mathematics - Problem 13 ([smzpow])

O / open | AMR 问题清单汇编

Assume that $X,Y\subseteq\mathbb{R}$ satisfy $\mathsf{S}_1(\mathcal{O},\mathcal{O})$, and $X$ also satisfies $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\Gamma)$. Does it follow that $X\times Y$ satisfies $\mathsf{S}_1(\mathcal{O},\mathcal{O})$?

背景链接：https://arxiv.org/abs/math/0609601

16052 | AMR-133-0014

### Selection Principles in Mathematics - Problem 14

O / open | AMR 问题清单汇编

The following problem withstood considerable attacks by several mathematicians. The property in it is equivalent to the Gerlits-Nagy $(*)$ property, and is also equivalent to $\mathsf{S}_1(\Omega,\mathcal{O}^{\gamma\mbox{-*\tiny gp*}})$ [coc7].

Is $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\Gamma)\cap\mathsf{S}_1(\mathcal{O},\mathcal{O})$ preserved by finite products?

背景链接：https://arxiv.org/abs/math/0609601

16053 | AMR-133-0015

### Selection Principles in Mathematics - Problem 15 ([Hdim])

O / open | AMR 问题清单汇编

Assume that $Y$ has Hausdorff dimension zero. The assumption that $X$ satisfies $\mathsf{S}_1(\Omega,\Gamma)$ does not imply that $X\times Y$ has Hausdorff dimension zero. However, if $X$ satisfies $\mathsf{S}_1(\{\mathrm{O}_n\}_{n\in\mathbb{N}},\Gamma)$, ( $\mathsf{S}_1(\{\mathrm{O}_n\}_{n\in\mathbb{N}},\Gamma)$ is the *strong $\gamma$-property* [GM, strongdiags]: For each sequence $\{\mathcal{U}_n\}_{n\in\mathbb{N}}$ where for each $n$, $\mathcal{U}_n$ is an open $n$-cover of $X$ (i.e., each $F\subseteq X$ with $|F|\le n$ is contained in some member of $\mathcal{U}_n$), there are $U_n\in\mathcal{U}_n$, $n\in\mathbb{N}$, such that $\{\mathcal{U}_n : n\in\mathbb{N}\}$ is a $\gamma$-cover of $X$. ) then $X\times Y$ has Hausdorff dimension zero [Hdim].

Assume that $|X|<\mathfrak{p}$. Is it true that for each $Y$ with Hausdorff dimension zero, $X\times Y$ has Hausdorff dimension zero?

背景链接：https://arxiv.org/abs/math/0609601

16054 | AMR-133-0016

### Selection Principles in Mathematics - Problem 16 (Krawczyk)

O / open | AMR 问题清单汇编

Is it consistent (relative to ZFC) that there are uncountable sets of reals satisfying $\mathsf{S}_1(\Omega,\Gamma)$, but for each such set $X$ and each set $Y$ with Hausdorff dimension zero, $X\times Y$ has Hausdorff dimension zero?

背景链接：https://arxiv.org/abs/math/0609601

16055 | AMR-133-0017

### Selection Principles in Mathematics - Problem 17 ([AddQuad])

O / open | AMR 问题清单汇编

The problem is also open for $\mathsf{S}_1(\Gamma,\Omega)$ and $\mathsf{S}_{\mathrm{fin}}(\Gamma,\Omega)$.

Is $\mathsf{S}_{\mathrm{fin}}(\mathrm{B}_\Omega,\mathrm{B}_\Omega)$ consistently additive?

背景链接：https://arxiv.org/abs/math/0609601

16056 | AMR-133-0018

### Selection Principles in Mathematics - Problem 18 ([AddQuad])

O / open | AMR 问题清单汇编

In some cases, there remains the task to determine the *exact* additivity number. The *additivity number* of a nontrivial family $\mathcal{J}$ of sets of reals is $$\operatorname{add}(\mathcal{J})=\min\{|\mathcal{F}| : \mathcal{F}\subseteq\mathcal{I}\mbox{ and }\cup\mathcal{F}\notin\mathcal{J}\}.$$ $\max\{\mathfrak{b},\mathfrak{g}\}\le\operatorname{add}(\mathsf{S}_{\mathrm{fin}}(\mathcal{O},\mathcal{O}))\le\operatorname{cf}(\mathfrak{d})$, $\mathfrak{h}\le\operatorname{add}(\mathsf{S}_1(\Gamma,\Gamma))\le\mathfrak{b}$, and $\operatorname{add}(\mathcal{N})\le\operatorname{add}(\mathsf{S}_1(\mathcal{O},\mathcal{O}))$ [AddQuad].

Is $\operatorname{add}(\mathsf{S}_{\mathrm{fin}}(\mathcal{O},\mathcal{O}))=\max\{\mathfrak{b},\mathfrak{g}\}$?

背景链接：https://arxiv.org/abs/math/0609601

16057 | AMR-133-0002

### Selection Principles in Mathematics - Problem 2

O / open | AMR 问题清单汇编

[{[BZSepar, preprint version]}] Assume that $X\subseteq B\subseteq\mathbb{R}$, $B$ is Borel, and $X$ satisfies $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\Gamma)$. Must there be a $\sigma$-compact $F$ with $X\subset F\subset B$? What if $B$ is $F_{\sigma\delta}$?

背景链接：https://arxiv.org/abs/math/0609601

16058 | AMR-133-0019

### Selection Principles in Mathematics - Problem 20 (Miller, Tsaban)

O / open | AMR 问题清单汇编

Assume that $X$ satisfies $\mathsf{S}_1(\Omega,\Gamma)$ and $|D|<\mathfrak{p}$. Does $X\cup D$ satisfy $\mathsf{S}_1(\Omega,\Gamma)$? Is it true under Martin's Axiom{} when $|D|=\aleph_1$?

背景链接：https://arxiv.org/abs/math/0609601

16059 | AMR-133-0020

### Selection Principles in Mathematics - Problem 21 ([spm])

O / open | AMR 问题清单汇编

Recall that by ``cover of $X$'' we mean one not containing $X$ as an element. $\mathcal{U}$ is a *$\large$-cover* of $X$ if each $x\in X$ is covered by infinitely many members of $\mathcal{U}$. It is a *$\tau$-cover* of $X$ if, in addition, for each $x,y\in X$, either $\{U\in\mathcal{U} : x\in U, y\notin U\}$ is finite, or else $\{U\in\mathcal{U} : y\in U, x\notin U\}$ is finite [tau]. Let $\mathrm{T}$ denote the collection of open $\tau$-covers of $X$. Then $\Gamma \subseteq \mathrm{T} \subseteq \Omega$.

The most important problem concerning $\tau$-covers is the following.

Is $\binom{\Omega}{\Gamma}=\binom{\Omega}{\mathrm{T}}$?

背景链接：https://arxiv.org/abs/math/0609601

16060 | AMR-133-0021

### Selection Principles in Mathematics - Problem 23 ([ideals])

O / open | AMR 问题清单汇编

Are any of the properties $\mathsf{S}_1(\mathrm{B}_\mathrm{T},\mathrm{B}_\Gamma)$, $\mathsf{S}_1(\mathrm{B}_\mathrm{T},\mathrm{B}_\mathrm{T})$, $\mathsf{S}_1(\mathrm{B}_\mathrm{T},\mathrm{B}_\Omega)$, $\mathsf{S}_1(\mathrm{B}_\mathrm{T},\mathrm{B})$, $\mathsf{S}_{\mathrm{fin}}(\mathrm{B}_\mathrm{T},\mathrm{B}_\mathrm{T})$, or $\mathsf{S}_{\mathrm{fin}}(\mathrm{B}_\mathrm{T},\mathrm{B}_\Omega)$, hereditary?

背景链接：https://arxiv.org/abs/math/0609601

16061 | AMR-133-0022

### Selection Principles in Mathematics - Problem 24 ([AddQuad])

O / open | AMR 问题清单汇编

Are any of the properties $\mathsf{S}_1(\mathrm{T},\mathrm{T})$, $\mathsf{S}_{\mathrm{fin}}(\mathrm{T},\mathrm{T})$, $\mathsf{S}_1(\Gamma,\mathrm{T})$, $\mathsf{S}_{\mathrm{fin}}(\Gamma,\mathrm{T})$, and $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\mathrm{T})$ (or any of their Borel versions) additive?

背景链接：https://arxiv.org/abs/math/0609601

16062 | AMR-133-0023

### Selection Principles in Mathematics - Problem 25

O / open | AMR 问题清单汇编

Are any of the properties

- $\mathsf{S}_1(\Omega,\mathrm{T})$, or $\mathsf{S}_{\mathrm{fin}}(\Omega,\mathrm{T})$,

- $\mathsf{S}_1(\mathrm{T},\Gamma)$, $\mathsf{S}_1(\mathrm{T},\mathrm{T})$, $\mathsf{S}_1(\mathrm{T},\Omega)$, $\mathsf{S}_{\mathrm{fin}}(\mathrm{T},\mathrm{T})$, or $\mathsf{S}_{\mathrm{fin}}(\mathrm{T},\Omega)$,

preserved under taking finite powers?

背景链接：https://arxiv.org/abs/math/0609601

16063 | AMR-133-0024

### Selection Principles in Mathematics - Problem 26 ([tautau])

O / open | AMR 问题清单汇编

$Y\subseteq{[\mathbb{N}]^{\aleph_0}}$ is *linearly refinable* if for each $y\in Y$ there exists an infinite subset $\hat y\subseteq y$ such that the family $\hat Y = \{\hat y : y\in Y\}$ is linearly (quasi)ordered by $\subseteq^*$. A cover $\mathcal{U}=\{U_n : n\in\mathbb{N}\}$ of $X$ is a *$\tau^*$-cover* of $X$ if it is large, and the family of all sets $\{n : x\in U_n\}$, $x\in X$, is linearly refinable. $\mathrm{T}^*$ is the collection of all countable open $\tau^*$-covers of $X$.

Every analytic space satisfies $\binom{\mathrm{T}}{\Gamma}$ [tau].

Does $\{0,1\}^{\mathbb{N}}$ satisfy $\binom{\mathrm{T}^*}{\Gamma}$?

背景链接：https://arxiv.org/abs/math/0609601

16064 | AMR-133-0025

### Selection Principles in Mathematics - Problem 27

O / open | AMR 问题清单汇编

$\gimel(\Gamma)\subseteq\gimel(\mathrm{T})\subseteq\gimel(\Omega)$. $\mathsf{S}_{\mathrm{fin}}(\mathcal{O},\mathcal{O})=\mathsf{S}_{\mathrm{fin}}(\Omega,\mathcal{O})$ [coc1], $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\Gamma)=\mathsf{S}_{\mathrm{fin}}(\Omega,\gimel(\Gamma))$ [coc7], and $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\Omega)=\mathsf{S}_{\mathrm{fin}}(\Omega,\gimel(\Omega))$ [coc8]. These results are generalized in [GlCovs]. A positive answer to the following problem is consistent [SF2].

Is $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\mathrm{T})=\mathsf{S}_{\mathrm{fin}}(\Omega,\gimel(\mathrm{T}))$?

背景链接：https://arxiv.org/abs/math/0609601

16065 | AMR-133-0026

### Selection Principles in Mathematics - Problem 28 ([coc8])

O / open | AMR 问题清单汇编

$\mathsf{S}_1(\Omega,\gimel(\Omega))$ is strictly stronger than $\mathsf{S}_1(\mathcal{O},\mathcal{O})$ [strongdiags]. $\mathsf{S}_1(\Omega,\gimel(\Omega))=\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\Omega)\cap\mathsf{S}_1(\mathcal{O},\mathcal{O})$ [strongdiags], so the following problem can also be stated in classical terms.

Is $\mathsf{S}_1(\Omega,\Omega)=\mathsf{S}_1(\Omega,\gimel(\Omega))$?

背景链接：https://arxiv.org/abs/math/0609601

16066 | AMR-133-0027

### Selection Principles in Mathematics - Problem 29

O / open | AMR 问题清单汇编

$\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\Gamma)=\binom{\Lambda}{\gimel(\Gamma)}$ [hureslaloms]. Zdomskyy proved that a positive answer to the following problem follows from NCF.

Is $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\Omega)=\binom{\Lambda}{\gimel(\Omega)}$?

背景链接：https://arxiv.org/abs/math/0609601

16067 | AMR-133-0003

### Selection Principles in Mathematics - Problem 3 ([ideals])

O / open | AMR 问题清单汇编

Dichotomic arguments imply the existence (in ZFC) of a set of reals $X$ satisfying $\mathsf{S}_1(\Gamma,\Gamma)$ such that $|X|=\mathfrak{t}$ [alpha_i], and a set of reals satisfying $\mathsf{S}_{\mathrm{fin}}(\Omega,\Omega)$ such that $|X|=\operatorname{cf}(\mathfrak{d})$ [SFH]. Now, $\operatorname{non}(\mathsf{S}_1(\Gamma,\Gamma))=\mathfrak{b}$, $\operatorname{non}(\mathsf{S}_{\mathrm{fin}}(\Omega,\Omega))=\mathfrak{d}$, and it is consistent that $\mathfrak{b}>\mathfrak{t}$ and $\mathfrak{d}>\operatorname{cf}(\mathfrak{d})$. Thus, these existence results are not satisfactory.

Does there exist (in ZFC) a set of reals $X$ satisfying $\mathsf{S}_1(\Gamma,\Gamma)$ such that $|X|=\mathfrak{b}$?

背景链接：https://arxiv.org/abs/math/0609601

16068 | AMR-133-0028

### Selection Principles in Mathematics - Problem 30 ([split])

O / open | AMR 问题清单汇编

- Does $\mathsf{Split}(\Omega, \mathrm{T})$ imply $\mathsf{Split}(\mathrm{T},\Gamma)$?

- And if not, then does $\mathsf{Split}(\Omega, \mathrm{T})$ imply $\mathsf{Split}(\Lambda, \Lambda)$?

背景链接：https://arxiv.org/abs/math/0609601

16069 | AMR-133-0029

### Selection Principles in Mathematics - Problem 31 (Zdomskyy)

O / open | AMR 问题清单汇编

The product of a $\sigma$-compact $X$ with $Y$ satisfying $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\mathscr{B})$ ($\mathscr{B}\in\{\mathcal{O},\Omega,\Gamma\}$) satisfies $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\mathscr{B})$ [SFT, AddQuad].

Assume that $X$ is compact and $Y$ satisfies $\mathsf{Split}(\Lambda,\Lambda)$. Does $X\times Y$ satisfy $\mathsf{Split}(\Lambda,\Lambda)$?

背景链接：https://arxiv.org/abs/math/0609601

16070 | AMR-133-0030

### Selection Principles in Mathematics - Problem 32 ([AddQuad])

O / open | AMR 问题清单汇编

Can the lower bound $\mathfrak{u}$ on $\operatorname{add}(\mathsf{Split}(\mathrm{T},\mathrm{T}))$ be improved?

背景链接：https://arxiv.org/abs/math/0609601

16071 | AMR-133-0031

### Selection Principles in Mathematics - Problem 33 ([split])

O / open | AMR 问题清单汇编

All problems below are settled for the properties which do not appear in them.

Is $\mathsf{Split}(\Lambda,\Lambda)$ additive?

背景链接：https://arxiv.org/abs/math/0609601

16072 | AMR-133-0032

### Selection Principles in Mathematics - Problem 34 ([split])

O / open | AMR 问题清单汇编

Are any of the properties $\mathsf{Split}(\mathrm{B}_\Omega,\mathrm{B}_\Lambda)$, $\mathsf{Split}(\mathrm{B}_\Omega,\mathrm{B}_\Omega)$, $\mathsf{Split}(\mathrm{B}_\mathrm{T},\mathrm{B}_\mathrm{T})$, and $\mathsf{Split}(\mathrm{B}_\mathrm{T},\mathrm{B}_\Gamma)$ hereditary?

背景链接：https://arxiv.org/abs/math/0609601

16073 | AMR-133-0033

### Selection Principles in Mathematics - Problem 35 ([split])

O / open | AMR 问题清单汇编

Are any of the properties $\mathsf{Split}(\Omega,\Omega)$, $\mathsf{Split}(\Omega,\mathrm{T})$, or $\mathsf{Split}(\mathrm{T},\mathrm{T})$ preserved under taking finite powers?

背景链接：https://arxiv.org/abs/math/0609601

16074 | AMR-133-0034

### Selection Principles in Mathematics - Problem 36 ([FremlinSeq])

O / open | AMR 问题清单汇编

Assume that $\kappa<\mathfrak{b}$, and for each $\alpha<\lambda$, $C_p(X_\alpha)$ satisfies $\mathsf{S}_1(\Gamma_\mathbf{0},\Gamma_\mathbf{0})$. Does $C_p(\bigcup_{\alpha<\kappa}X_\alpha)$ satisfy $\mathsf{S}_1(\Gamma_\mathbf{0},\Gamma_\mathbf{0})$?

背景链接：https://arxiv.org/abs/math/0609601

16075 | AMR-133-0035

### Selection Principles in Mathematics - Problem 37 ([wqn])

O / open | AMR 问题清单汇编

If $X$ satisfies $\mathsf{S}_1(\Gamma,\Gamma)$, then $C_p(X)$ satisfies $\mathsf{S}_1(\Gamma_\mathbf{0},\Gamma_\mathbf{0})$ [wqn].

Is $\mathsf{S}_1(\Gamma_\mathbf{0},\Gamma_\mathbf{0})$ for $C_p(X)$ equivalent to $\mathsf{S}_1(\Gamma,\Gamma)$ for $X$?

背景链接：https://arxiv.org/abs/math/0609601

16076 | AMR-133-0036

### Selection Principles in Mathematics - Problem 38 (Scheepers)

O / open | AMR 问题清单汇编

Is $\mathsf{S}_1(\Gamma_\mathbf{0},\Omega_\mathbf{0})$ for $C_p(X)$ equivalent to $\mathsf{S}_1(\Gamma,\Omega)$ for $X$?

背景链接：https://arxiv.org/abs/math/0609601

16077 | AMR-133-0037

### Selection Principles in Mathematics - Problem 39 ([Sakai06])

O / open | AMR 问题清单汇编

A topological space $Y$ is *$\kappa$-Fr\'echet* if it satisfies $\binom{O(\Omega_x)}{\Gamma_x}$, where $O(\Omega_x)$ is the family of elements of $\Omega_x$ which are open.

What is the minimal cardinality of a set $X\subseteq\mathbb{R}$ such that $C_p(X)$ does not satisfy $\binom{O(\Omega_x)}{\Gamma_x}$?

背景链接：https://arxiv.org/abs/math/0609601

16078 | AMR-133-0004

### Selection Principles in Mathematics - Problem 4 ([SFH])

O / open | AMR 问题清单汇编

\begin{upd}``No'', and the counter-example is Laver's model [BBC]. The answer becomes ``Yes'' if we are allowed to pick, instead of one element from each cover, a union of two elements from each cover [MHP].\end{upd}

Does there exist (in ZFC) a set of reals satisfying $\mathsf{S}_{\mathrm{fin}}(\Omega,\Omega)$ such that $|X|=\mathfrak{d}$?

背景链接：https://arxiv.org/abs/math/0609601

16079 | AMR-133-0038

### Selection Principles in Mathematics - Problem 40 (Tka\v{c}enko)

O / open | AMR 问题清单汇编

Are there, in ZFC, groups $G,H$ satisfying $\mathsf{S}_{\mathrm{fin}}(\mathcal{O}_{\mathrm{nbd}},\mathcal{O})$ such that $G\times H$ does not satisfy $\mathsf{S}_{\mathrm{fin}}(\mathcal{O}_{\mathrm{nbd}},\mathcal{O})$?

背景链接：https://arxiv.org/abs/math/0609601

16080 | AMR-133-0039

### Selection Principles in Mathematics - Problem 41 ([BG])

O / open | AMR 问题清单汇编

Is $\mathsf{S}_1(\Omega_{\mathrm{nbd}},\Omega)=\mathsf{S}_{\mathrm{fin}}(\mathcal{O}_{\mathrm{nbd}},\mathcal{O})$ for separable metrizable groups? Specifically:

- Does the Continuum Hypothesis{} imply the existence of a separable metrizable group $G$ satisfying $\mathsf{S}_{\mathrm{fin}}(\mathcal{O}_{\mathrm{nbd}},\mathcal{O})$ but not $\mathsf{S}_1(\Omega_{\mathrm{nbd}},\allowbreak\Omega)$?

- Is it consistent that $\mathsf{S}_{\mathrm{fin}}(\mathcal{O}_{\mathrm{nbd}},\mathcal{O})=\mathsf{S}_1(\Omega_{\mathrm{nbd}},\Omega)$ for separable metrizable groups?

背景链接：https://arxiv.org/abs/math/0609601

16081 | AMR-133-0040

### Selection Principles in Mathematics - Problem 42 ([BZSPM])

O / open | AMR 问题清单汇编

If $G$ is *analytic* and does not satisfy $\mathsf{S}_1(\Omega_{\mathrm{nbd}},\Gamma)$, then $G^2$ does not satisfy $\mathsf{S}_{\mathrm{fin}}(\mathcal{O}_{\mathrm{nbd}},\mathcal{O})$. Thus, for analytic groups, $\mathsf{S}_1(\Omega_{\mathrm{nbd}},\Gamma)=\mathsf{S}_1(\Omega_{\mathrm{nbd}},\Omega)$ [BZSPM]. Moreover, for analytic *abelian* groups, $\mathsf{S}_{\mathrm{fin}}(\mathcal{O}_{\mathrm{nbd}},\mathcal{O})=\mathsf{S}_1(\Omega_{\mathrm{nbd}},\Gamma)$ [BZSPM]. For general analytic groups this is open.

Is there an analytic group satisfying $\mathsf{S}_{\mathrm{fin}}(\mathcal{O}_{\mathrm{nbd}},\mathcal{O})$ but not $\mathsf{S}_1(\Omega_{\mathrm{nbd}},\Gamma)$?

背景链接：https://arxiv.org/abs/math/0609601

16082 | AMR-133-0041

### Selection Principles in Mathematics - Problem 43 ([BG])

O / open | AMR 问题清单汇编

Say that $G\le{{\mathbb{Z}}^{\mathbb{N}}}$ is *bounded* if $\{|g| : g\in G\}$ is bounded (with respect to $\le^*$), where $|g|$ denotes the absolute value of $g$. For subgroups of ${{\mathbb{Z}}^{\mathbb{N}}}$:

- $G$ satisfies $\mathsf{S}_1(\Omega_{\mathrm{nbd}},\Gamma)$ if, and only if, $G$ is bounded [Bab05].

- $G$ satisfies $\mathsf{S}_1(\mathcal{O}_{\mathrm{nbd}},\mathcal{O}_{\mathrm{nbd}})$ if, and only if, $G$ has strong measure zero [coc11].

Is it consistent that there is $G\le{{\mathbb{Z}}^\mathbb{N}}$ such that $G$ has strong measure zero, is unbounded, and does not satisfy $\mathsf{S}_{\mathrm{fin}}(\mathcal{O},\mathcal{O})$?

背景链接：https://arxiv.org/abs/math/0609601

16083 | AMR-133-0042

### Selection Principles in Mathematics - Problem 44 ([BG])

O / open | AMR 问题清单汇编

Is it consistent that there is $G\le{{\mathbb{Z}}^\mathbb{N}}$ such that $G$ has strong measure zero and satisfies $\mathsf{S}_{\mathrm{fin}}(\mathcal{O},\mathcal{O})$, but is unbounded and does not satisfy $\mathsf{S}_1(\mathcal{O},\mathcal{O})$?

背景链接：https://arxiv.org/abs/math/0609601

16084 | AMR-133-0043

### Selection Principles in Mathematics - Problem 45 ([o-bdd])

O / open | AMR 问题清单汇编

Does the Continuum Hypothesis{} imply the existence of a a group $G\le{{\mathbb{Z}}^{\mathbb{N}}}$ of cardinality $\mathfrak{c}$ satisfying $\mathsf{S}_1(\mathrm{B}_\Omega,\mathrm{B}_\Gamma)$, or at least $\mathsf{S}_1(\Omega,\Gamma)$?

背景链接：https://arxiv.org/abs/math/0609601

16085 | AMR-133-0044

### Selection Principles in Mathematics - Problem 46 ([vD])

O / open | AMR 问题清单汇编

The main open problem in the field is the *Minimal Tower Problem*. This problem has motivated the study of $\tau$-covers.

Is it consistent that $\mathfrak{p}<\mathfrak{t}$?

背景链接：https://arxiv.org/abs/math/0609601

16086 | AMR-133-0045

### Selection Principles in Mathematics - Problem 47 ([tautau, ShTb768])

O / open | AMR 问题清单汇编

The study of $\tau^*$-covers, a variant of $\tau$-covers, led to the following problem. A family $\mathcal{F}\subseteq{[\mathbb{N}]^{\aleph_0}}$ is *linearly refinable* if for each $A\in\mathcal{F}$ there exists an infinite subset $\hat A\subseteq A$ such that the family $\hat\mathcal{F} = \{\hat A : A\in\mathcal{F}\}$ is linearly (quasi)ordered by $\subseteq^*$. $\mathfrak{p}^*$ is the minimal size of a centered family in ${[\mathbb{N}]^{\aleph_0}}$ which is not linearly refinable.

$\mathfrak{p} = \min\{\mathfrak{p}^*,\mathfrak{t}\}$, and $\mathfrak{p}^*\le\mathfrak{d}$ [tautau].

Is $\mathfrak{p}=\mathfrak{p}^*$?

背景链接：https://arxiv.org/abs/math/0609601

16087 | AMR-133-0046

### Selection Principles in Mathematics - Problem 49

O / open | AMR 问题清单汇编

It is easy to see that $\mathfrak{p}=\mathfrak{p}_2=\mathfrak{p}_3=\dots=\mathfrak{p}_{\aleph_0}$, and $\mathfrak{p}_\mathfrak{t}=\mathfrak{t}$. It turns out that $\mathfrak{p} = \mathfrak{p}_{\aleph_1}$ [Pyt]. We get a hierarchy of cardinals between $\mathfrak{p}$ and $\mathfrak{t}$:
$$\mathfrak{p}=\mathfrak{p}_{\aleph_1}\le \mathfrak{p}_{\aleph_2}\le ...\le \mathfrak{p}_\mathfrak{t} = \mathfrak{t}.$$

Is $\mathfrak{p}_{\mathfrak{p}}=\mathfrak{t}$?

背景链接：https://arxiv.org/abs/math/0609601

16088 | AMR-133-0005

### Selection Principles in Mathematics - Problem 5 ([SFH])

O / open | AMR 问题清单汇编

Is there a direct (non-dichotomic) construction of a set of reals $M$ satisfying $\mathsf{S}_{\mathrm{fin}}(\Omega,\Omega)$ but not $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\Gamma)$?

背景链接：https://arxiv.org/abs/math/0609601

16089 | AMR-133-0047

### Selection Principles in Mathematics - Problem 50 ([ShTb768])

O / open | AMR 问题清单汇编

Finally, consider the following Ramsey-theoretic cardinal: For a subset $Y$ of $\mathbb{N}$ and $g\in\mathbb{N}$, we say that $g$ *avoids middles* in $Y$ if:

- for each $f\in Y$, $g\not\le^* f$;

- for all $f,h\in Y$ at least one of the sets $\{n : f(n)<g(n)\le h(n)\}$ and $\{n : h(n)<g(n)\le f(n)\}$ is finite.

$\operatorname{add}(\mathfrak{X},\mathfrak{D})$ is the minimal cardinality $\kappa$ of a dominating $Y\subseteq\mathbb{N}$ such that for each partition of $Y$ into $\kappa$ many pieces, there is a piece such that no $g$ avoids middles in that piece. This cardinal is studied in [ShTb768].

Is $\operatorname{cov}(\mathcal{M})\le\operatorname{add}(\mathfrak{X},\mathfrak{D})$?

背景链接：https://arxiv.org/abs/math/0609601

16090 | AMR-133-0048

### Selection Principles in Mathematics - Problem 51 ([SFT])

O / open | AMR 问题清单汇编

Assume that $X$ satisfies $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\Omega)$. Does it follow that $$G=\{g\in\mathbb{N} : (\exists k)(\exists f_1,\dots,f_k\in Y)\ g\le^* \max\{f_1,\dots,f_k\}\}$$ satisfies $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\Omega)$?

背景链接：https://arxiv.org/abs/math/0609601

16091 | AMR-133-0049

### Selection Principles in Mathematics - Problem 52 ([BrownCox83])

O / open | AMR 问题清单汇编

$X\subseteq\mathbb{R}$ is a *$\nu$-set* if for each $Y\subseteq X$ which is nowhere dense in $X$, $Y$ is countable (i.e., $X$ is Luzin relative to itself). Every continuous image of a $\nu$-set has the property assumed in the following problem.

Assume that $X\subseteq\mathbb{R}$, and for each $Y\subseteq X$, $Y$ is concentrated on a countable subset of $Y$. Does it follow that $X$ is a continuous image of a $\nu$-set?

背景链接：https://arxiv.org/abs/math/0609601

16092 | AMR-133-0050

### Selection Principles in Mathematics - Problem 53 ([BarJu93])

O / open | AMR 问题清单汇编

Is it consistent that $\operatorname{cov}(\mathcal{M})=\aleph_1<\mathfrak{c}=\aleph_{\omega_1}$, and there is a $\mathfrak{c}$-Luzin set (i.e., $L$ with $|L|=\mathfrak{c}$ and $|L\cap M|<\mathfrak{c}$ for all meager $M\subseteq\mathbb{R}$)?

背景链接：https://arxiv.org/abs/math/0609601

16093 | AMR-133-0051

### Selection Principles in Mathematics - Problem 54 ([BarJu93])

O / open | AMR 问题清单汇编

Assume that every strong measure zero set of reals is meager-additive. Does Borel's Conjecture follow?

背景链接：https://arxiv.org/abs/math/0609601

16094 | AMR-133-0052

### Selection Principles in Mathematics - Problem 55 ([prods])

O / open | AMR 问题清单汇编

If $X,Y\subseteq\{0,1\}^{\mathbb{N}}$ are meager-additive, then $X\times Y$ is a meager-additive subset of $\{0,1\}^{\mathbb{N}}\times\{0,1\}^{\mathbb{N}}$. The same is true for null-additive subsets of $\{0,1\}^{\mathbb{N}}$. For the real line this is open.

Assume that $X,Y\subseteq\mathbb{R}$ are meager- (respectively, null-) additive. Does it follow that $X\times Y$ is meager- (respectively, null-) additive?

背景链接：https://arxiv.org/abs/math/0609601

16095 | AMR-133-0053

### Selection Principles in Mathematics - Problem 56 ([prods])

O / open | AMR 问题清单汇编

Assume that $X\subseteq\mathbb{R}$ is meager- (respectively, null-) additive, and $X\subseteq [0,1]$. Does it follow that $X$ is meager- (respectively, null-) additive when viewed as a subset of $\{0,1\}^{\mathbb{N}}$?

背景链接：https://arxiv.org/abs/math/0609601

16096 | AMR-133-0054

### Selection Principles in Mathematics - Problem 57 (Bartoszy\'nski)

O / open | AMR 问题清单汇编

Assume that $X\subseteq\mathbb{N}$ is nonmeager and $Y\subseteq\mathbb{N}$ is dominating. Is there a Borel set $H\subseteq\mathbb{N}\times\mathbb{N}$ such that every meager set is contained in $H_x$ for some $x\in X\cup Y$?

背景链接：https://arxiv.org/abs/math/0609601

16097 | AMR-133-0006

### Selection Principles in Mathematics - Problem 6 ([pawlikowskireclaw])

O / open | AMR 问题清单汇编

Let $\mathcal{J}$ be a property of sets of reals. \emph{Borel's Conjecture for $\mathcal{J}$} is the statement ``All elements of $\mathcal{J}$ are countable''. For all but three of the properties in the Borel case, Borel's Conjecture is consistent.

Is Borel's Conjecture for $\mathsf{S}_{\mathrm{fin}}(\mathrm{B},\mathrm{B})$ consistent?

背景链接：https://arxiv.org/abs/math/0609601

16098 | AMR-133-0007

### Selection Principles in Mathematics - Problem 7 ([CBC])

O / open | AMR 问题清单汇编

Is Borel's Conjecture for $\mathsf{S}_{\mathrm{fin}}(\mathrm{B}_\Omega,\mathrm{B}_\Omega)$ consistent?

背景链接：https://arxiv.org/abs/math/0609601

16099 | AMR-133-0008

### Selection Principles in Mathematics - Problem 8

O / open | AMR 问题清单汇编

Assume the Continuum Hypothesis{}. Settings (c),(h), and (i) were realized in [coc2], Setting (k) was realized in [o-bdd], and Setting (n) was realized in [Rec94, Brendle96, MilNonGamma]. To realize Setting (b), take a set $L$ as in Setting (c) and a set $S$ as in Setting (i), and take $X=L\cup S$. As $\mathsf{S}_{\mathrm{fin}}(\mathrm{B},\mathrm{B})$ is additive, $X$ satisfies this property. But since $\mathsf{S}_1(\mathrm{B}_\Gamma,\mathrm{B}_\Omega)$ and $\mathsf{S}_1(\mathrm{B},\mathrm{B})$ are hereditary for subsets [ideals], $X$ does not satisfy any of these. It seems that using forcing-theoretic arguments similar to those of [Brendle96], we can realize Settings (f) and (m).

Does the Continuum Hypothesis{} imply a realization of the settings (d),(e),(g),(j), and (l)?

背景链接：https://arxiv.org/abs/math/0609601

16100 | AMR-133-0009

### Selection Principles in Mathematics - Problem 9 ([MilNonGamma])

O / open | AMR 问题清单汇编

All constructions mentioned above can be carried out using Martin's Axiom{}. Except perhaps Setting (n).

Does Martin's Axiom{} imply the existence of a set of reals of cardinality continuum, satisfying $\mathsf{S}_1(\mathrm{B}_\Omega,\mathrm{B}_\Gamma)$?

背景链接：https://arxiv.org/abs/math/0609601

16101 | AMR-134-0023

### Selection Principles Milestone Problems - Conjecture 24 ([splittability, Conjecture 8.7])

O / open | AMR 问题清单汇编

None of the classes $\mathsf{Split}(\mathrm{T},\mathrm{T})$ and $\binom{\mathrm{T}}{\Gamma}$ is provably closed under taking finite products.

背景链接：https://arxiv.org/abs/math/0312182

16102 | AMR-134-0034

### Selection Principles Milestone Problems - Conjecture 35 ([strongdiags, Conjecture 1])

O / open | AMR 问题清单汇编

$\mathsf{S}_1(\{\mathcal{O}_n\}_{n\in\mathbb{N}},\mathrm{T})$ is strictly stronger than $\mathsf{S}_1(\Omega,\mathrm{T})$.

背景链接：https://arxiv.org/abs/math/0312182

16103 | AMR-134-0039

### Selection Principles Milestone Problems - Conjecture 40

O / open | AMR 问题清单汇编

In [wqn, FrwQN] it is shown that $s_1$ (for $X$), $\mathsf{wQN}$ (for $X$), and SSP (for $C_p(X)$) are all equivalent. This and other reasons lead to suspecting that all these equivalent properties are equivalent to a standard selection hypothesis. In [wqn], Scheepers shows that $\mathsf{S}_1(\Gamma,\Gamma)$ implies being an $\mathsf{wQN}$-space.

[{Scheepers [wqn, Conjecture 1]}] For sets of reals, $\mathsf{wQN}$ implies $\mathsf{S}_1(\Gamma,\Gamma)$.

背景链接：https://arxiv.org/abs/math/0312182

16104 | AMR-134-0001

### Selection Principles Milestone Problems - Problem 1

O / open | AMR 问题清单汇编

[{[coc2, Problems 1 and 2]}]

- Is $\mathsf{U}_{\mathrm{fin}}(\Gamma,\Omega)$ equivalent to $\mathsf{S}_{\mathrm{fin}}(\Gamma,\Omega)$?

- And if not, does $\mathsf{U}_{\mathrm{fin}}(\Gamma,\Gamma)$ imply $\mathsf{S}_{\mathrm{fin}}(\Gamma,\Omega)$?

背景链接：https://arxiv.org/abs/math/0312182

16105 | AMR-134-0010

### Selection Principles Milestone Problems - Problem 10 ([AddQuad])

O / open | AMR 问题清单汇编

The general problem is to determine the additivity numbers of the properties in question in terms of well known cardinal characteristics of the continuum (like ${\mathfrak{b}}$, ${\mathfrak{d}}$, etc.). See [LecceSurvey] for a list of properties for which the problem is still open. Three of the more interesting ones among these are the following.

Is $\mathsf{add}(\mathsf{U}_{\mathrm{fin}}(\Gamma,\mathcal{O}))={\mathfrak{b}}$?

背景链接：https://arxiv.org/abs/math/0312182

16106 | AMR-134-0011

### Selection Principles Milestone Problems - Problem 11

O / open | AMR 问题清单汇编

It is only known that ${\mathfrak{b}}\le\mathsf{add}(\mathsf{U}_{\mathrm{fin}}(\Gamma,\mathcal{O}))\le\mathsf{cf}({\mathfrak{d}})$, and that the additivity of the corresponding combinatorial notion of smallness is equal to ${\mathfrak{b}}$ [AddQuad].

Is $\mathsf{add}(\mathsf{S}_1(\Gamma,\Gamma))={\mathfrak{b}}$?

背景链接：https://arxiv.org/abs/math/0312182

16107 | AMR-134-0012

### Selection Principles Milestone Problems - Problem 12 (\Cite{\mathbb{S} 4}{spm1})

O / open | AMR 问题清单汇编

The following problem arises in almost every study of $\tau$-covers [tau, tautau, splittability, strongdiags, Hdim, MilNonGamma]. By [GN], $\mathsf{S}_1(\Omega,\Gamma)\Leftrightarrow\binom{\Omega}{\Gamma}$. As $\Gamma\subseteq\mathrm{T}$, this property implies $\binom{\Omega}{\mathrm{T}}$. Thus far, all examples of sets not satisfying $\binom{\Omega}{\Gamma}$ turned out not to satisfy $\binom{\Omega}{\mathrm{T}}$.

Is $\binom{\Omega}{\Gamma}$ equivalent to $\binom{\Omega}{\mathrm{T}}$?

背景链接：https://arxiv.org/abs/math/0312182

16108 | AMR-134-0013

### Selection Principles Milestone Problems - Problem 14 ([ideals, Problem 4])

O / open | AMR 问题清单汇编

Is any of the properties $\mathsf{S}_1(\mathcal{B}_\mathrm{T},\mathcal{B}_\Gamma)$, $\mathsf{S}_1(\mathcal{B}_\mathrm{T},\mathcal{B}_\mathrm{T})$, $\mathsf{S}_1(\mathcal{B}_\mathrm{T},\mathcal{B}_\Omega)$, $\mathsf{S}_1(\mathcal{B}_\mathrm{T},\mathcal{B})$, $\mathsf{S}_{\mathrm{fin}}(\mathcal{B}_\mathrm{T},\mathcal{B}_\mathrm{T})$, or $\mathsf{S}_{\mathrm{fin}}(\mathcal{B}_\mathrm{T},\mathcal{B}_\Omega)$, hereditary?

背景链接：https://arxiv.org/abs/math/0312182

16109 | AMR-134-0014

### Selection Principles Milestone Problems - Problem 15 ([AddQuad])

O / open | AMR 问题清单汇编

Is any of the properties $\mathsf{S}_1(\mathrm{T},\mathrm{T})$, $\mathsf{S}_{\mathrm{fin}}(\mathrm{T},\mathrm{T})$, $\mathsf{S}_1(\Gamma,\mathrm{T})$, $\mathsf{S}_{\mathrm{fin}}(\Gamma,\mathrm{T})$, and $\mathsf{U}_{\mathrm{fin}}(\Gamma,\mathrm{T})$ (or any of their Borel versions) preserved under taking finite unions?

背景链接：https://arxiv.org/abs/math/0312182

16110 | AMR-134-0015

### Selection Principles Milestone Problems - Problem 16

O / open | AMR 问题清单汇编

Is any of the properties

- $\mathsf{S}_1(\Omega,\mathrm{T})$, or $\mathsf{S}_{\mathrm{fin}}(\Omega,\mathrm{T})$,

- $\mathsf{S}_1(\mathrm{T},\Gamma)$, $\mathsf{S}_1(\mathrm{T},\mathrm{T})$, $\mathsf{S}_1(\mathrm{T},\Omega)$, $\mathsf{S}_{\mathrm{fin}}(\mathrm{T},\mathrm{T})$, or $\mathsf{S}_{\mathrm{fin}}(\mathrm{T},\Omega)$,

preserved under taking finite powers?

背景链接：https://arxiv.org/abs/math/0312182

16111 | AMR-134-0016

### Selection Principles Milestone Problems - Problem 17

O / open | AMR 问题清单汇编

Recall that $\mathsf{U}_{\mathrm{fin}}(\Gamma,\mathcal{O})\Leftrightarrow\mathsf{S}_{\mathrm{fin}}(\Omega,\mathcal{O})$. In [coc7] it is proved that $\mathsf{U}_{\mathrm{fin}}(\Gamma,\Gamma)\Leftrightarrow\mathsf{S}_{\mathrm{fin}}(\Omega,\mathcal{O}^{\gamma\mbox{-*\tiny gp*}})$, and in [coc8] it is proved that $\mathsf{U}_{\mathrm{fin}}(\Gamma,\Omega)\Leftrightarrow\mathsf{S}_{\mathrm{fin}}(\Omega,\mathcal{O}^{\omega\mbox{-*\tiny gp*}})$.

Is $\mathsf{U}_{\mathrm{fin}}(\Gamma,\mathrm{T})$ equivalent to $\mathsf{S}_{\mathrm{fin}}(\Omega,\mathcal{O}^{\tau\mbox{-*\tiny gp*}})$?

背景链接：https://arxiv.org/abs/math/0312182

16112 | AMR-134-0017

### Selection Principles Milestone Problems - Problem 18 ([coc8, Problem 4])

O / open | AMR 问题清单汇编

In [strongdiags] it was pointed out that $\mathsf{S}_1(\Omega,\mathcal{O}^{\omega\mbox{-*\tiny gp*}})$ is strictly stronger than $\mathsf{S}_1(\Omega,\mathcal{O})$ (which is the same as $\mathsf{S}_1(\mathcal{O},\mathcal{O})$). The following problem remains open.

Is $\mathsf{S}_1(\Omega,\Omega)$ equivalent to $\mathsf{S}_1(\Omega,\mathcal{O}^{\omega\mbox{-*\tiny gp*}})$?

背景链接：https://arxiv.org/abs/math/0312182

16113 | AMR-134-0018

### Selection Principles Milestone Problems - Problem 19

O / open | AMR 问题清单汇编

- Is $\mathsf{U}_{\mathrm{fin}}(\Gamma,\Omega)$ equivalent to $\binom{\Lambda}{\mathcal{O}^{\omega\mbox{-*\tiny gp*}}}$?

- Is $\mathsf{U}_{\mathrm{fin}}(\Gamma,\mathrm{T})$ equivalent to $\binom{\Lambda}{\mathcal{O}^{\tau\mbox{-*\tiny gp*}}}$?

背景链接：https://arxiv.org/abs/math/0312182

16114 | AMR-134-0002

### Selection Principles Milestone Problems - Problem 2 ([coc2, Problem 3], [BuHa, Problem 1], [LBopenprobs, Problem 1])

O / open | AMR 问题清单汇编

Does there exist (in ZFC) a set of reals $X$ which has the Menger property $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\mathcal{O})$ but not the Hurewicz property $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\Gamma)$?

背景链接：https://arxiv.org/abs/math/0312182

16115 | AMR-134-0019

### Selection Principles Milestone Problems - Problem 20

O / open | AMR 问题清单汇编

It is often the case that properties of the form $\Pi(\Omega,\mathfrak{V})$ where $\Omega\subseteq\mathfrak{V}$ are equivalent to $\Pi(\Lambda,\mathfrak{V})$ [coc1, coc2, coc7, coc8, strongdiags]. But we do not know the answer to the following simple question.

Is $\binom{\Omega}{\mathcal{O}^{\gamma\mbox{-*\tiny gp*}}}$ equivalent to $\binom{\Lambda}{\mathcal{O}^{\gamma\mbox{-*\tiny gp*}}}$?

背景链接：https://arxiv.org/abs/math/0312182

16116 | AMR-134-0020

### Selection Principles Milestone Problems - Problem 21 ([reznicb, Problem 8])

O / open | AMR 问题清单汇编

In [coc7] it is proved that $X$ satisfies $\mathsf{S}_{\mathrm{fin}}(\Omega,\Omega^{gp})$ if, and only if, all finite powers of $X$ satisfy $\mathsf{U}_{\mathrm{fin}}(\Gamma,\Gamma)$, which we now know is the same as $\binom{\Lambda}{\Lambda^{gp}}$.

Is $\mathsf{S}_{\mathrm{fin}}(\Omega,\Omega^{gp})$ equivalent to $\binom{\Omega}{\Omega^{gp}}$?

背景链接：https://arxiv.org/abs/math/0312182

16117 | AMR-134-0021

### Selection Principles Milestone Problems - Problem 22 ([splittability, Problem 6.8])

O / open | AMR 问题清单汇编

We list briefly the only remaining problems concerning preservation of the splittability properties mentioned in the last section under set theoretic operations. All problems below are settled for the properties which do not appear in them.

Is $\mathsf{Split}(\Lambda,\Lambda)$ additive?

背景链接：https://arxiv.org/abs/math/0312182

16118 | AMR-134-0022

### Selection Principles Milestone Problems - Problem 23 ([splittability, Problem 7.5])

O / open | AMR 问题清单汇编

Is any of the properties $\mathsf{Split}(\mathcal{B}_\Omega,\mathcal{B}_\Lambda)$, $\mathsf{Split}(\mathcal{B}_\Omega,\mathcal{B}_\Omega)$, $\mathsf{Split}(\mathcal{B}_\mathrm{T},\mathcal{B}_\mathrm{T})$, and $\mathsf{Split}(\mathcal{B}_\mathrm{T},\mathcal{B}_\Gamma)$ hereditary?

背景链接：https://arxiv.org/abs/math/0312182

16119 | AMR-134-0024

### Selection Principles Milestone Problems - Problem 25 ([splittability, Problem 8.8])

O / open | AMR 问题清单汇编

Is any of the properties $\mathsf{Split}(\Omega,\Omega)$, $\mathsf{Split}(\Omega,\mathrm{T})$, or $\mathsf{Split}(\mathrm{T},\mathrm{T})$ preserved under taking finite powers?

背景链接：https://arxiv.org/abs/math/0312182

16120 | AMR-134-0025

### Selection Principles Milestone Problems - Problem 26 ([GN, p. 160])

O / open | AMR 问题清单汇编

The following problem is classical by now, but it is related to the problems which follow. For a sequence $\{X_n\}_{n\in\mathbb{N}}$ of subsets of $X$, define $\liminf X_n = \bigcup_m\bigcap_{n\ge m} X_n$. For a family $\mathcal{F}$ of subsets of $X$, $L(\mathcal{F})$ denotes its closure under the operation $\liminf$. The following definition appears in the celebrated paper [GN] just after that of the $\gamma$-property: $X$ is a *$\delta$-set* (or: has the *$\delta$-property*) if for each $\omega$-cover $\mathcal{U}$ of $X$, $X\in L(\mathcal{U})$. Observe that if $\{U_n\}_{n\in\mathbb{N}}$ is a $\gamma$-cover of $X$, then $X=\liminf U_n$. Thus, the $\gamma$-property implies the $\delta$-property. Surprisingly, the converse is still open.

Is the $\delta$-property equivalent to the $\gamma$-property?

背景链接：https://arxiv.org/abs/math/0312182

16121 | AMR-134-0026

### Selection Principles Milestone Problems - Problem 27 ([GJ, Question 2.4])

O / open | AMR 问题清单汇编

For an ultrafilter $p$, $\mathcal{O}_p$ denotes the collection of open $\gamma_p$-covers of $X$, that is, open covers $\mathcal{U}$ that can be enumerated as $\{U_n\}_{n\in\mathbb{N}}$ where $\{n : x\in U_n\}\in p$ for all $x\in X$. The property $\binom{\Omega}{\mathcal{O}_p}$ is called *the $\gamma_p$-property* in [GTpseq].

Is the property $\binom{\Omega}{\mathcal{O}_p}$ additive for each ultrafilter $p$?

背景链接：https://arxiv.org/abs/math/0312182

16122 | AMR-134-0027

### Selection Principles Milestone Problems - Problem 28 ([GTpseq, Problem 3.14(2)])

O / open | AMR 问题清单汇编

Assume that $X$ satisfies $\binom{\Omega}{\mathcal{O}_p}$ for each ultrafilter $p$. Must $X$ satisfy $\binom{\Omega}{\Gamma}$?

背景链接：https://arxiv.org/abs/math/0312182

16123 | AMR-134-0028

### Selection Principles Milestone Problems - Problem 29 ([GT1], [GTpseq, Problem 0.10])

O / open | AMR 问题清单汇编

For an ultrafilter $p$, we write $x = {p\mbox{-}\lim}{} x_n$ when for each neighborhood $U$ of $x$, $\{n : x_n\in U\}\in p$. A space $Y$ is *$p$-sequential* if we replace $\lim$ by ${p\mbox{-}\lim}$ in the definition of sequential.

Assume that $C_p(X)$ is $p$-sequential. Must $X$ satisfy $\binom{\Omega}{\mathcal{O}_p}$?

背景链接：https://arxiv.org/abs/math/0312182

16124 | AMR-134-0003

### Selection Principles Milestone Problems - Problem 3 ([ideals, Problem 17], [o-bdd, Problem 2])

O / open | AMR 问题清单汇编

Let $B=\{f_\alpha : \alpha < {\mathfrak{b}}\}\subseteq\mathbb{N}$ be a $\le^*$-unbounded set of strictly increasing elements of $\mathbb{N}$ which forms a \emph{${\mathfrak{b}}$-scale} (that is, for each $\alpha<\beta$, $f_\alpha \leq^* f_\beta$), and set $H = B\cup Q$. In [ideals] it is proved that all finite powers of $H$ satisfy $\mathsf{U}_{\mathrm{fin}}(\mathcal{O},\Gamma)$.

Does $H$ satisfy $\mathsf{S}_1(\Gamma,\Gamma)$?

背景链接：https://arxiv.org/abs/math/0312182

16125 | AMR-134-0029

### Selection Principles Milestone Problems - Problem 30 ([GTpseq, Problem 0.6 (reformulated)])

O / open | AMR 问题清单汇编

Kombarov [Km] introduced the following two generalizations of $p$-sequentiality: Let $M$ be a collection of ultrafilters. $Y$ is *weakly $M$-sequential* if for each non-closed $A\subseteq Y$ there exists $y\in Y\setminus A$ and a sequence $\{a_n\}_{n\in\mathbb{N}}$ in $A$ such that ${p\mbox{-}\lim} a_n = y$ for *some* $p\in M$. $Y$ is *strongly $M$-sequential* if *some* is replaced by *for all* in the last definition.

Assume that $X$ satisfies the $\delta_M$ property. Must $C_p(X)$ be weakly $M$-sequential?

背景链接：https://arxiv.org/abs/math/0312182

16126 | AMR-134-0030

### Selection Principles Milestone Problems - Problem 31 ([BerJu, Question 4.2], [lengthdiags, footnote 1])

O / open | AMR 问题清单汇编

Let $\mathcal{D}$ denote the collection of all families $\mathcal{U}$ of open sets in $X$ such that $\cup\mathcal{U}$ is dense in $X$. In [BerJu], Berner and Juh\'asz introduce the open-point game, which by [coc6] is equivalent to $\mathsf{G}_1(\mathcal{D},\mathcal{D})$ in the sense that a player has a winning strategy in the open-point game on $X$ if, and only if, the *other* player has a winning strategy in $\mathsf{G}_1(\mathcal{D},\mathcal{D})$.

Does there exist in ZFC a space in which $\mathsf{G}_1(\mathcal{D},\mathcal{D})$ is undetermined?

背景链接：https://arxiv.org/abs/math/0312182

16127 | AMR-134-0031

### Selection Principles Milestone Problems - Problem 32 ([coc5])

O / open | AMR 问题清单汇编

Are the properties $\mathsf{S}_1(\mathcal{D}_\Omega,\mathcal{D}_\Omega)$ or $\mathsf{S}_{\mathrm{fin}}(\mathcal{D}_\Omega,\mathcal{D}_\Omega)$ preserved under taking finite powers?

背景链接：https://arxiv.org/abs/math/0312182

16128 | AMR-134-0032

### Selection Principles Milestone Problems - Problem 33 ([coc5, Problem 3])

O / open | AMR 问题清单汇编

Is any of the properties $\mathsf{S}_1(\mathcal{D}_\Omega,\mathcal{D}_\Omega)$ or $\mathsf{S}_{\mathrm{fin}}(\mathcal{D}_\Omega,\mathcal{D}_\Omega)$ equivalent to ONE not having a winning strategy in the corresponding game?

背景链接：https://arxiv.org/abs/math/0312182

16129 | AMR-134-0033

### Selection Principles Milestone Problems - Problem 34 ([coc4, Problem 1])

O / open | AMR 问题清单汇编

Let $\mathcal{K}$ denote the families $\mathcal{U}\in\mathcal{D}$ such that $\{\overline{U} : U\in\mathcal{U}\}$ is a cover of $X$. In [tkachuk] Tkachuk shows that the Continuum Hypothesis{} implies that ONE has a winning strategy in $\mathsf{G}_1(\mathcal{K},\mathcal{D})$ on any space of uncountable cellularity. In [coc4], Scheepers defines $\mathfrak{j}$ as the minimal cardinal $\kappa$ such that ONE has a winning strategy in $\mathsf{G}_1(\mathcal{K},\mathcal{D})$ on each Tychonoff space with cellularity at least $\kappa$, and shows that $\mathsf{cov}(\mathcal{M})\le\mathfrak{j}\le\mathsf{non}(SMZ)$.

Is $\mathfrak{j}$ equal to any standard cardinal characteristic of the continuum?

背景链接：https://arxiv.org/abs/math/0312182

16130 | AMR-134-0035

### Selection Principles Milestone Problems - Problem 36

O / open | AMR 问题清单汇编

Assume that $\mathfrak{V}\in\{\Omega,\Omega^{gp}\}$. Is it true that ONE does not have a winning strategy in $\mathsf{G}_{fin}(\{\mathcal{O}_n\}_{n\in\mathbb{N}},\mathfrak{V})$ if, and only if, $\mathsf{S}_{\mathrm{fin}}(\Omega,\mathfrak{V})$ holds?

背景链接：https://arxiv.org/abs/math/0312182

16131 | AMR-134-0036

### Selection Principles Milestone Problems - Problem 37 ([strongdiags, Problem 5.16])

O / open | AMR 问题清单汇编

Is it true that $X$ is a strong $\gamma$-set (i.e., satisfies $\mathsf{S}_1(\{\mathcal{O}_n\}_{n\in\mathbb{N}},\Gamma)$) if, and only if, ONE has no winning strategy in the game $\mathsf{G}_1(\{\mathcal{O}_n\}_{n\in\mathbb{N}},\Gamma)$?

背景链接：https://arxiv.org/abs/math/0312182

16132 | AMR-134-0037

### Selection Principles Milestone Problems - Problem 38 ([SchLuzin, Problem 4])

O / open | AMR 问题清单汇编

Recall that $\mathcal{K}$ is the collection of families $\mathcal{U}$ of open sets such that $\{\overline{U}:U\in\mathcal{U}\}$ is a cover of $X$. Let $\mathcal{K}_\Omega$ be the collection of all $\mathcal{U}\in\mathcal{K}$ such that no element of $\mathcal{U}$ is dense in $X$, and for each finite $F\subseteq X$, there exists $U\in\mathcal{U}$ such that $F\subseteq\overline{U}$. In [SchLuzin] it is proved that $X$ satisfies $\mathcal{K}_\Omega\rightarrow(\mathcal{K})^2_2$, then $X$ is a Luzin set.

Does the partition relation $\mathcal{K}_\Omega\rightarrow(\mathcal{K})^2_2$ characterize Luzin sets?

背景链接：https://arxiv.org/abs/math/0312182

16133 | AMR-134-0038

### Selection Principles Milestone Problems - Problem 39 ([PolarPar, Problem 1])

O / open | AMR 问题清单汇编

Is {\tiny $\(\begin{matrix}\Omega\\\Omega\end{matrix}\)\to\left[\begin{matrix}\Omega\\\Omega\end{matrix}\right]_{k/<3}$} equivalent to $\mathsf{S}_1(\Omega,\Omega)$? And if not, is its critical cardinality equal to that of $\mathsf{S}_1(\Omega,\Omega)$ (namely, to $\mathsf{cov}(\mathcal{M})$)?

背景链接：https://arxiv.org/abs/math/0312182

16134 | AMR-134-0004

### Selection Principles Milestone Problems - Problem 4 ([pawlikowskireclaw], [CBC, Problem 45], [ideals])

O / open | AMR 问题清单汇编

Borel's Conjecture, which was proved to be consistent by Laver, implies that each set of reals satisfying $\mathsf{S}_1(\mathcal{O},\mathcal{O})$ (and the classes below it) is countable. From our point of view this means that there do not exist ZFC examples of sets satisfying $\mathsf{S}_1(\mathcal{O},\mathcal{O})$. A set of reals $X$ is a *$\sigma$-set* if each $G_\delta$ set in $X$ is also an $F_\sigma$ set in $X$. In [CBC] it is proved that every element of $\mathsf{S}_1(\mathcal{B}_\Gamma,\mathcal{B}_\Gamma)$ is a $\sigma$-set. According to a result of Miller [Mil79Len], it is consistent that every $\sigma$-set of real numbers is countable. Thus, there do not exist uncountable ZFC examples satisfying $\mathsf{S}_1(\mathcal{B}_\Gamma,\mathcal{B}_\Gamma)$. The situation for the other classes, though addressed by top experts, remains open. In particular, we have the following.

Does there exist (in ZFC) an uncountable set of reals satisfying $\mathsf{S}_1(\mathcal{B}_\Gamma,\mathcal{B})$?

背景链接：https://arxiv.org/abs/math/0312182

16135 | AMR-134-0040

### Selection Principles Milestone Problems - Problem 41 ([CpHure, Problem 1])

O / open | AMR 问题清单汇编

The *weak sequence selection property* for $C_p(X)$ [CpHure] is defined as the SSP with the difference that we only require that $\mathbf{0}\in\overline{\{f^n_{k_n} : n\in\mathbb{N}\}}$.

Does countable fan tightness of $C_p(X)$ imply the weak sequence selection property?

背景链接：https://arxiv.org/abs/math/0312182

16136 | AMR-134-0041

### Selection Principles Milestone Problems - Problem 42 ([CpHure, Problem 2])

O / open | AMR 问题清单汇编

The *monotonic sequence selection property* is defined like the SSP with the additional assumption that for each $n$ the sequence $\{f^n_k\}_{k\in\mathbb{N}}$ converges pointwise *monotonically* to $\mathbf{0}$.

Does the monotonic sequence selection property of $C_p(X)$ imply the weak sequence selection property?

背景链接：https://arxiv.org/abs/math/0312182

16137 | AMR-134-0042

### Selection Principles Milestone Problems - Problem 43 ([SakaiRez, Question 3.5], [reznicb])

O / open | AMR 问题清单汇编

In [coc7] it is shown that $C_p(X)$ has countable fan tightness as well as Reznichenko's property if, and only if, each finite power of $X$ has the Hurewicz covering property. Recently, Sakai found an exact dual of the Reznichenko property: An open $\omega$-cover $\mathcal{U}$ of $X$ is *$\omega$-shrinkable* if for each $U \in \mathcal{U}$ there exists a closed subset $C_U\subseteq U$ such that $\{C_U : U \in \mathcal{U}\}$ is an $\omega$-cover of $X$. Then $C_p(X)$ has the Reznichenko property if, and only if, each $\omega$-shrinkable open $\omega$-cover of $X$ is $\omega$-groupable [SakaiRez]. Thus if $X$ satisfies $\binom{\Omega}{\Omega^{gp}}$, then $C_p(X)$ has the Reznichenko property. The other direction is not clear.

Is it true that $C_p(X)$ has the Reznichenko property if, and only if, $X$ satisfies $\binom{\Omega}{\Omega^{\mbox{*\tiny gp*}}}$?

背景链接：https://arxiv.org/abs/math/0312182

16138 | AMR-134-0043

### Selection Principles Milestone Problems - Problem 44 ([SakaiRez, Question 3.6])

O / open | AMR 问题清单汇编

Another simply stated problem is the following.

Does $C_p(\mathbb{N})$ have the Reznicenko property?

背景链接：https://arxiv.org/abs/math/0312182

16139 | AMR-134-0044

### Selection Principles Milestone Problems - Problem 45 ([SakaiRez, Question 2.8])

O / open | AMR 问题清单汇编

A family $\mathcal{P}$ of subsets of of a space $Y$ is a *$\pi$-network* at $y\in Y$ if every neighborhood of $y$ contains some element of $\mathcal{P}$. $Y$ is a *Pytkeev space* if for each $y\in Y$ and $A\subseteq Y$ such that $y\in\overline{A}\setminus A$, there exists a countable $\pi$-network at $y$ which consists of infinite subsets of $A$. In [SakaiRez] it is proved that $C_p(X)$ is a Pytkeev space if, and only if, for each $\omega$-shrinkable open $\omega$-cover $\mathcal{U}$ of $X$ there exist subfamilies $\mathcal{U}_n\subseteq\mathcal{U}$, $n\in\mathbb{N}$, such that $\bigcap_{n\in\mathbb{N}}\mathcal{U}_n$ is an $\omega$-cover of $X$.

Can the term ``$\omega$-shrinkable'' be removed from Sakai's characterization of the Pytkeev property of $C_p(X)$?

背景链接：https://arxiv.org/abs/math/0312182

16140 | AMR-134-0045

### Selection Principles Milestone Problems - Problem 46 ([SakaiRez, Question 2.9])

O / open | AMR 问题清单汇编

If all finite powers of $X$ satisfy $\mathsf{U}_{\mathrm{fin}}(\Gamma,\Gamma)$, then every open $\omega$-cover of $X$ is $\omega$-shrinkable [SakaiRez], thus a positive solution to the following problem would suffice.

Assume that $C_p(X)$ is a Pytkeev space. Is it true that all finite powers of $X$ satisfy $\mathsf{U}_{\mathrm{fin}}(\Gamma,\Gamma)$?

背景链接：https://arxiv.org/abs/math/0312182

16141 | AMR-134-0005

### Selection Principles Milestone Problems - Problem 5 ([ideals, Problem 4], [MilNonGamma, Question 6])

O / open | AMR 问题清单汇编

A property is (provably) *hereditary* if for each space $X$ satisfying the property, all subsets of $X$ satisfy that property. Most of the properties considered in this paper may be considered intuitively as notions of smallness, thus it is somewhat surprising that none of the properties involving open covers is hereditary [ideals]. However, the property $\mathsf{S}_1(\mathcal{B},\mathcal{B})$ as well as all properties of the form $\Pi(\mathcal{B}_\Gamma,\mathcal{B})$ are hereditary [ideals] (but $\mathsf{S}_1(\mathcal{B}_\Omega,\mathcal{B}_\Gamma)$ is not [MilNonGamma]).

Is any of the properties $\mathsf{S}_1(\mathcal{B}_\Omega,\mathcal{B}_\Omega)$ or $\mathsf{S}_{\mathrm{fin}}(\mathcal{B}_\Omega,\mathcal{B}_\Omega)$ hereditary?

背景链接：https://arxiv.org/abs/math/0312182

16142 | AMR-134-0006

### Selection Principles Milestone Problems - Problem 6 ([CBC, Problem 50])

O / open | AMR 问题清单汇编

Is any of the classes $\mathsf{S}_1(\mathcal{B}_\Omega,\mathcal{B}_\Gamma)$, $\mathsf{S}_1(\mathcal{B}_\Omega,\mathcal{B}_\Omega)$, or $\mathsf{S}_{\mathrm{fin}}(\mathcal{B}_\Omega,\mathcal{B}_\Omega)$ closed under taking finite powers?

背景链接：https://arxiv.org/abs/math/0312182

16143 | AMR-134-0007

### Selection Principles Milestone Problems - Problem 7 ([CBC, Problems 19 and 21])

O / open | AMR 问题清单汇编

- Is it true that if $X$ satisfies $\mathsf{S}_1(\mathcal{B}_\Omega,\mathcal{B}_\Omega)$, then all finite powers of $X$ satisfy $\mathsf{S}_1(\mathcal{B},\mathcal{B})$?

- Is it true that if $X$ satisfies $\mathsf{S}_{\mathrm{fin}}(\mathcal{B}_\Omega,\mathcal{B}_\Omega)$, then all finite powers of $X$ satisfy $\mathsf{S}_1(\mathcal{B}_\Gamma,\mathcal{B})$?

背景链接：https://arxiv.org/abs/math/0312182

16144 | AMR-134-0008

### Selection Principles Milestone Problems - Problem 8 ([AddQuad])

O / open | AMR 问题清单汇编

Is any of the classes $\mathsf{S}_{\mathrm{fin}}(\Omega,\Omega)$, $\mathsf{S}_1(\Gamma,\Omega)$, and $\mathsf{S}_{\mathrm{fin}}(\Gamma,\Omega)$ consistently closed under taking finite unions?

背景链接：https://arxiv.org/abs/math/0312182

16145 | AMR-134-0009

### Selection Principles Milestone Problems - Problem 9 ([AddQuad])

O / open | AMR 问题清单汇编

Is $\mathsf{S}_{\mathrm{fin}}(\mathcal{B}_\Omega,\mathcal{B}_\Omega)$ consistently closed under taking finite unions?

背景链接：https://arxiv.org/abs/math/0312182

16146 | OWR-9790353-003

### Set-Theoretic Dichotomy for Borel-Quotient Automorphisms

P / partially_solved | Oberwolfach 报告问题汇编

Does CH imply that $M/E$ has $2^{\mathfrak c}$ automorphisms, while forcing axioms imply that every isomorphism between $M/E$ and $N/F$ is topologically trivial?

来源字段：https://doi.org/10.4171/owr/2022/2

16147 | OWR-1703861-009

### Set-Theoretic Rigidity of Corona-Algebra Automorphisms

P / partially_solved | Oberwolfach 报告问题汇编

For every separable nonunital $C^*$-algebra $A$, does CH imply that $Q(A)$ has $2^{\aleph_1}$ automorphisms that are not topologically trivial, while PFA implies that all automorphisms of $Q(A)$ are topologically trivial?

来源字段：https://doi.org/10.4171/owr/2020/14

16148 | OWR-4792-005

### Simplifying Borel Equivalence Relations on Positive Sets

O / open | Oberwolfach 报告问题汇编

Let $E$ be a Borel equivalence relation on a Polish space $X$ and $I$ a $\sigma$-ideal on $X$. Is there a Borel $I$-positive set $B\subset X$ such that the restricted equivalence relation $E\upharpoonright B$ is significantly simpler than $E$ on $X$?

来源字段：https://doi.org/10.4171/owr/2011/02

16149 | OWR-1703861-002

### Singular Cardinal Arithmetic Without Weak Square

O / open | Oberwolfach 报告问题汇编

For a singular cardinal $\kappa$, is $\operatorname{cof}(\operatorname{pp}(\kappa))>\kappa$? Also, can there be a singular strong-limit $\kappa$ such that weak square at $\kappa$ fails and $2^\kappa>\kappa^+$, or such that the tree property holds at $\kappa^+$ while $2^\kappa>\kappa^+$?

来源字段：https://doi.org/10.4171/owr/2020/14

16150 | SET-002

### Singular Cardinals Hypothesis

S / solved | 独立 / 未分配集合

If $\kappa$ is a singular strong limit cardinal, then $2^\kappa = \kappa^+$.

背景链接：https://doi.org/10.2307/1971065

16151 | OWR-4792-006

### Smoothness on Products of Perfect Sets

O / open | Oberwolfach 报告问题汇编

Let $E$ be a Borel equivalence relation on $\mathbb{R}^\omega$ reducible to an orbit equivalence. Is there a product of perfect sets on which $E$ is smooth?

来源字段：https://doi.org/10.4171/owr/2011/02

16152 | OWR-1703861-005

### Stationary Reflection and Large Power at Aleph Omega

O / open | Oberwolfach 报告问题汇编

Can the consistency result asserting that a strong-limit singular cardinal $\kappa$ of countable cofinality has simultaneous finite stationary-set reflection at $\kappa^+$ together with $2^\kappa=\kappa^{++}$ be obtained with $\kappa=\aleph_\omega$?

来源字段：https://doi.org/10.4171/owr/2020/14

16153 | OWR-12862-006

### Strong Colorings at Singular Cardinals

O / open | Oberwolfach 报告问题汇编

Does $\operatorname{Pr}_1(\lambda,\lambda,\lambda,\operatorname{cf}(\lambda))$ hold for every singular cardinal $\lambda$, or even does $\operatorname{Pr}_1(\lambda,\lambda,\lambda,2)$ hold for every singular cardinal?

来源字段：https://doi.org/10.4171/owr/2014/02

16154 | OWR-1269-002

### Strong Condensation with Large Cardinals

P / partially_solved | Oberwolfach 报告问题汇编

Are large cardinals consistent with stronger forms of Condensation?

来源字段：https://doi.org/10.4171/owr/2006/20

16155 | OWR-12862-005

### Strong Invariant Universality at Singular Cardinals

P / partially_solved | Oberwolfach 报告问题汇编

For generalized trees of size $\kappa$, is $\kappa^\kappa=\kappa$ necessary for embeddability to be strongly invariantly universal among $\kappa$-analytic quasi-orders? For singular $\kappa$, what is the descriptive-set-theoretic complexity of embeddability on $\operatorname{Mod}_\kappa(\varphi)$, and can it consistently be strongly invariantly universal? Can generalized trees be replaced by linear orders or groups while preserving strong invariant universality?

来源字段：https://doi.org/10.4171/owr/2014/02

16156 | OWR-15216-007

### Strong Surjectivity of Countryman Derived Orders Under PFA

O / open | Oberwolfach 报告问题汇编

A *Countryman line* $C$ is an uncountable linear order whose square $C^2$ is a countable union of chains, and a linear order $L$ is *strongly surjective* if for every uncountable suborder $L' \subseteq L$ there is an order-preserving surjection $L' \twoheadrightarrow L$. Under the Proper Forcing Axiom ($\mathrm{PFA}$), is the dense/scattered companion linear order $\eta_C$ constructed from a Countryman line $C$ strongly surjective?

来源字段：https://doi.org/10.4171/owr/2017/11

16157 | ST-008

### Strongly Compact vs Supercompact Cardinals

O / open | 独立 / 未分配集合

Does the consistency of a strongly compact cardinal imply the consistent existence of a supercompact cardinal?

背景链接：https://logic.nankai.edu.cn/_upload/article/files/73/9a/bc6df29a42fe8dd60c911873f3db/07e70874-31c6-4273-a076-0c8f22ca4f07.pdf

16158 | SET-004

### Suslin's Problem

S / solved | 独立 / 未分配集合

If a dense linear order without endpoints is complete and has the countable chain condition, must it be isomorphic to the real numbers?

背景链接：https://eudml.org/doc/16215

16159 | OWR-9790353-002

### Topological Triviality of Borel Quotient Isomorphisms

O / open | Oberwolfach 报告问题汇编

Under what assumptions is every isomorphism between the Borel quotient structures $M/E$ and $N/F$ topologically trivial?

来源字段：https://doi.org/10.4171/owr/2022/2

16160 | OWR-15216-016

### Tree Property at Successors of Countable Limit Alephs

O / open | Oberwolfach 报告问题汇编

Is it consistent that, for every countable limit ordinal $\alpha<\omega_1$, there is no $\aleph_{\alpha+1}$-Aronszajn tree?

来源字段：https://doi.org/10.4171/owr/2017/11

16161 | OWR-4792-016

### Tree Property with Failure of Singular Cardinal Hypothesis

O / open | Oberwolfach 报告问题汇编

Is it consistent that the tree property holds at $\aleph_{\omega+1}$ while the Singular Cardinal Hypothesis fails at $\aleph_\omega$?

来源字段：https://doi.org/10.4171/owr/2011/02

16162 | ST-006

### Ultimate Core Model

O / open | 独立 / 未分配集合

Does there exist an ultimate core model containing all large cardinals?

背景链接：https://math.berkeley.edu/~steel/papers/short.strategycompare.pdf

16163 | OWR-1116-013

### Uniqueness of Bedrock Models

S / solved | Oberwolfach 报告问题汇编

If a universe satisfies the Bedrock Axiom, is its bedrock model unique? A bedrock model is a minimal ground model $W$ such that the universe is a set-forcing extension $V=W[G]$ and $W$ satisfies the Ground Axiom.

来源字段：https://doi.org/10.4171/owr/2005/55

16164 | OWR-12862-001

### Universal Aronszajn Trees at Aleph One

O / open | Oberwolfach 报告问题汇编

Is there a model of set theory in which the universality number $u(T_{\aleph_1})$ of trees of size $\aleph_1$ without an $\aleph_1$-branch equals $1$?

来源字段：https://doi.org/10.4171/owr/2014/02

16165 | OWR-1783-009

### Universality of Quasi-Isometry among Marked Groups

O / open | Oberwolfach 报告问题汇编

Let $\mathcal{G}$ be the Polish space of finitely generated marked groups, and let quasi-isometry be the usual relation defined by a coarsely bilipschitz map with coarsely dense image. Is the quasi-isometry relation on $\mathcal{G}$ a universal $K_\sigma$ Borel equivalence relation; that is, does every $K_\sigma$ Borel equivalence relation Borel-reduce to it?

来源字段：https://doi.org/10.4171/owr/2008/02

16166 | OWR-1703861-007

### Weak Compactness in HOD from Iterated Embeddings

O / open | Oberwolfach 报告问题汇编

Suppose $(V,j)\models\mathrm{ZF}$ and $j:V\to V$ is elementary. Form the iterates $M_0=(V,j)$ and $M_\alpha=(N_\alpha,j_\alpha)$ by setting $M_{\alpha+1}=(N_\alpha,j_\alpha(j_\alpha))$ and taking direct limits at limit ordinals. If $\lambda=\sup_{n<\omega}\sup(\operatorname{crit}(j^n))$, is $\lambda$ weakly compact in $\mathrm{HOD}$?

来源字段：https://doi.org/10.4171/owr/2020/14

16167 | OWR-1269-001

### Weak Condensation with Large Cardinals

P / partially_solved | Oberwolfach 报告问题汇编

Are large cardinals consistent with weaker forms of Condensation?

来源字段：https://doi.org/10.4171/owr/2006/20

16168 | OWR-1269-006

### Weak Uniqueness of Elementary-Embedding Square Roots at $\lambda$

O / open | Oberwolfach 报告问题汇编

别名：Duplicate: Weak Uniqueness of Elementary-Embedding Square Roots [OWR-1269-007]; Duplicate: Weak Uniqueness of Elementary-Embedding Square Roots [OWR-1269-008]; Duplicate: Weak Uniqueness of Elementary-Embedding Square Roots [OWR-1269-009]

Suppose $\lambda$ is a limit of supercompact cardinals, $X\subseteq V_{\lambda+1}$, and $j:L(X,V_{\lambda+1})\to L(X,V_{\lambda+1})$ is a proper elementary embedding. If square roots $k_1,k_2$ of $j$ agree on $V_\lambda$ and both map $L_\omega(X,V_{\lambda+1})$ to itself, must they agree below $\Theta^{L(X,V_{\lambda+1})}$?

来源字段：https://doi.org/10.4171/owr/2006/20

16169 | OWR-7155442-014

### Weihrauch Determinacy and Degree Structure Questions

P / partially_solved | Oberwolfach 报告问题汇编

What are the exact Weihrauch relations among $\Sigma^0_1\text{-DS}$, $\Sigma^0_1\text{-BSLQO}$, $\Sigma^0_1\text{-BSPO}$, $\Sigma^0_1\text{-BS}$, and $\Delta^0_2\text{-DS}$? Does Turing determinacy imply dependent choice for reals, and does Turing determinacy together with dependent choice imply strong Turing determinacy? Given a properly $2$-c.e. degree $D$, does there exist a c.e. set $A$ with $D\not\leq_T A$ such that every c.e. $W\leq_T D$ also satisfies $W\leq_T A$; and does the analogous extension property characterize superlow c.e. degrees?

来源字段：https://doi.org/10.4171/owr/2021/21

16170 | OWR-7155442-013

### Weihrauch Reductions for Paths Through Ill-Founded Trees

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathrm{CNN}$ denote the task of computing a path through an ill-founded subtree of $\mathbb N^{\mathbb N}$. Is $\mathrm{CNN}$ uniformly Weihrauch-reducible to $\Sigma^0_1\text{-RT}$ or to $\mathrm{wFindHS}\text{-}\Pi^0_1$?

来源字段：https://doi.org/10.4171/owr/2021/21

16171 | SET-003

### Whitehead Problem

S / solved | 独立 / 未分配集合

Is every abelian group $A$ such that $\text{Ext}^1(A, \mathbb{Z}) = 0$ a free abelian group?

背景链接：https://doi.org/10.1007/BF02757281

16172 | ST-002

### Woodin's GCH below Strongly Compact Cardinals

P / partially_solved | 独立 / 未分配集合

Does the generalized continuum hypothesis below a strongly compact cardinal imply it everywhere?

背景链接：https://www.dmg.tuwien.ac.at/fb8/2025_slides/Golshani.pdf

16173 | ST-007

### Woodin's Ω-Conjecture

O / open | 独立 / 未分配集合

If there is a proper class of Woodin cardinals, does Ω-logic satisfy an analogue of Gödel's completeness theorem?

背景链接：https://plato.stanford.edu/entries/continuum-hypothesis/

16174 | OWR-7155442-011

### Zigzagging Skips of Arithmetical Degrees

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist an arithmetical degree $\mathbf a$ such that, for every $n$, the $n$th skip of $\mathbf a$ is not below the $(n+1)$st skip, equivalently every $n$th skip is nontotal? Can the HYP degrees be characterized as the degrees bounded by all skip $2$-cycles?

来源字段：https://doi.org/10.4171/owr/2021/21

# 资料汇编 14 | 逻辑 / Logic

374 条记录。O=开放；P=部分解决；S=已解决。状态与英文/LaTeX 表述均沿用来源。

16175 | AIM-LOGIC-0125

## A bounded-class obstruction for quotient Diophantine interpretations

P / partially_solved | AIM 研讨会问题汇编

**Question 12 (Cornelissen).** If \(\mathbb Z\) admits a Diophantine interpretation in \(\mathbb Q\) (that is, using an equivalence relation), does it follow that Mazur's conjecture is wrong? See Cornelissen--Zahidi, Contemp. Math. 270 253--260.

For every variety \(X\) over \(\mathbb Q\), the closure of \(X(\mathbb Q)\) in \(X(\mathbb R)\), with the Euclidean topology, has only finitely many connected components.

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16176 | AIM-LOGIC-0115

## A bounded-existential MRDP fragment and a certificate-growth audit

P / partially_solved | AIM 研讨会问题汇编

Is DPRM a theorem of \(I\Delta_0\)? Given a \(\Sigma_1\) formula \(\psi(\bar x)\), does there exist a polynomial \(P(\bar x,\bar y)\) such that \[ I\Delta_0\vdash (\forall\bar x)\bigl(\psi(\bar x)\leftrightarrow (\exists\bar y)(P(\bar x,\bar y)=0)\bigr)? \]

Poonen and T. Scanlon and recorded by J. Demeyer. On page 1, Question 2 is attributed to P. D'Aquino.

The PDF immediately adds that a positive answer would imply \(\mathsf{NP}=\mathsf{coNP}\), and then says \[ L=\{+,\cdot,0,1,\#,\leq\},\qquad \#(x,y):=x^{\lfloor\log y\rfloor}. Thus this is **not literally** the more common Buss smash notation \(x\#y=2^{|x||y|}\), although the two functions have comparable bit-growth. The PDF does not state the logarithm base or the convention at \(y=0\). Base 2 is the bounded-arithmetic convention; the isolated boundary value can be fixed by an axiom without affecting any result below.

The induction schema printed in the source suppresses parameters. The standard recovered form is \[ \bigl[\varphi(0,\bar a)\wedge \forall x\bigl(\varphi(x,\bar a)\to\varphi(x+1,\bar a)\bigr)\bigr] \to\forall x\,\varphi(x,\bar a), \] for each bounded (\(\Delta_0\)) formula \(\varphi\).

There is a genuine source ambiguity. The question first names ordinary \(I\Delta_0\), whose usual language is \(L_{\rm ar}=\{0,1,+,\cdot,\leq\}\), and only after the displayed MRDP scheme introduces \(\#\). Adding a total function with the stated graph is not automatically a definitional extension of \(I\Delta_0\): its totality is a growth principle of the \(\Omega_1\) type. Accordingly, this report separates:

1. the pure \(I\Delta_0\) question in \(L_{\rm ar}\); and 2. an expanded-language variant in which \(\#(x,y)=x^{\lfloor\log_2 y\rfloor}\) is total and appropriately axiomatized.

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16177 | AIM-LOGIC-0060

## A canonical diamond forcing, its exact omega_3 bottleneck, and small-forcing reflection

P / partially_solved | AIM 研讨会问题汇编

Suppose \(\aleph_\omega\) is a strong limit. Is there a poset \({\mathbb P}\) of size \(<\aleph_\omega\), which adds a club guessing sequence for \(S^{\omega_2}_{\omega_1}\), preserving \(\omega_1\), \(\omega_2\), and \(\omega_3\).

Suppose \(\aleph_\omega\) is a strong limit. Is there a poset \(\mathbb P\) of size \(<\aleph_\omega\) which forces full club guessing on \(S^{\omega_2}_{\omega_1}\) while preserving \(\omega_1,\omega_2,\omega_3\) as regular cardinals?

来源字段：http://aimpl.org/highlowforcing/3/

16178 | AIM-LOGIC-0107

### A central-quotient obstruction to metric ample generics

P / partially_solved | AIM 研讨会问题汇编

**Problem 16.** Study the existence of ample generics in the setting of \(\omega\)-stable separably categorical continuous theory.

**Background.** Generalize the ideas of Hodkinson, Hodges, Lascar and Shelah presented in the paper “The small index property for \(\omega\)-stable, \(\omega\)-categorical structures and the random graph” to the setting of metric structures.

来源字段：https://aimath.org/WWN/continuouslogic/continuouslogic.pdf

16179 | AIM-LOGIC-0044

### A cofinality-stratified normal form for reflection at aleph_(omega+1)

P / partially_solved | AIM 研讨会问题汇编

Is it consistent (relative to large cardinals) that $\mathrm{TP}_{\aleph_{\omega+1}}$ holds along with reflection for all stationary subsets of $\aleph_{\omega+1}$?

来源字段：http://aimpl.org/highlowforcing/2/

16180 | AIM-LOGIC-0147

### A complete Brody-hyperbolicity algorithm for algebraic curves

P / partially_solved | AIM 研讨会问题汇编

**Question 35 (Pheidas).** Let \(X\) be a variety over \(\mathbb Q\). Call \(X\) hyperbolic iff there is no nonconstant holomorphic map \(\mathbb C\to X(\mathbb C)\). Is there an algorithm which can decide whether a variety \(X/\mathbb Q\) over hyperbolic?

**Is there an algorithm which can decide whether a variety \(X/\mathbb Q\) is hyperbolic?**

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16181 | AIM-LOGIC-0141

### A corrected algebraic p-adic field with definable valuation and algebraically closed residue

S / solved | AIM 研讨会问题汇编

**Question 28 (Shlapentokh).** Is there an algebraic extension \(K\) of \(\mathbb Q\) and a nontrivial valuation ring \(R\subset K\), such that the residue field of \(R\) is algebraically closed and \(R\) is definable over \(K\)?

**Answer: YES.** Inside \[ \mathbb Q_p^{\mathrm{alg}} =\overline{\mathbb Q}\cap\mathbb Q_p \subseteq\overline{\mathbb Q_p}, \] the ring \(\mathbb Z_p^{\mathrm{alg}}\) is definable.

**Fact 29 (Pheidas).** \(\mathbb C[[Z]]\) is definable in \(\mathbb C((Z))\): \[ x\in\mathbb C[[Z]]\quad\Longleftrightarrow\quad (\exists y)(1+Zx^2=y^2). \] Proven using Hensel's lemma.

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16182 | AIM-LOGIC-0138

### A dense Diophantine shadow of the Z-adic valuation ring

P / partially_solved | AIM 研讨会问题汇编

**Question 25 (Pheidas).** Is \[ \{f\in\mathbb C(Z)\mid \operatorname{ord}_{Z=0}(f)\ge 0\} \] (existentially) definable in \(\mathbb C(Z)\), where there is a symbol for \(Z\) in the language?

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16183 | AIM-LOGIC-0019

### A false density principle and the finite-component obstruction

S / solved | AIM 研讨会问题汇编

**Problem 5.3.** Let $X/\mathbb Q$ be a variety, and let $\{Y_a\}_{a\in A}$ be a set of uniformly definable subsets of $X$.

1. Is $\overline{Y_a(\mathbb Q)}=Y_a(\mathbb R)$? 2. Does this imply that $\mathbb Z$ is not diophantine in $\mathbb Q$?

来源字段：http://aimpl.org/definedecide/5/

16184 | AIM-LOGIC-0095

### A finite-jump transfer from strongly minimal reducts

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 17.** For any \(\aleph_1\)-categorical \(T\) there is an \(n\) such that if \(T\) has a computable model then every countable model has a presentation computable in \(\mathbf 0^{(n)}\). Note that if \(T\) is strongly minimal then \(n=4\) works.

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16185 | AIM-LOGIC-0089

### A finite-support obstruction to definable isomorphisms with the complex field

P / partially_solved | AIM 研讨会问题汇编

**Question 11.** Let \(\mathcal A=HF(\mathbb C)\), and suppose \(K\cong\mathbb C\) is \(\Sigma\)-definable in \(\mathcal A\). Is there a \(\Sigma\)-definable isomorphism?

The answer is yes if we replace \(\mathbb C\) by \(\mathbb R\) and both \((K,\oplus,\odot)\cong\mathbb R\) and \(K\subseteq\mathbb R\) hold. It is open if merely \(K\subseteq\mathbb R^2\).

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16186 | AIM-LOGIC-0042

## A forcing-axiom sieve and countable-condition reduction at omega two

P / partially_solved | AIM 研讨会问题汇编

$\kappa \rightarrow (\alpha,\beta)^n$ means that for every coloring $F:[\kappa]^n \rightarrow \{0,1\}$, then then there is either a homogeneous set $H_0$ of order-type $\alpha$ such that $F"H_0=\{0\}$, or else a homogeneous set $H_1$ such that $F"H_1=\{1\}$.

Can $\omega_2 \rightarrow (\omega_2,\alpha)^2$ be deduced from one of the forcing axioms at $\aleph_2$.

The record also remarks that PFA implies $\omega_1\rightarrow(\omega_1,\alpha)^2$ for every $\alpha<\omega_1$.

This text has four formulation defects, which are preserved above rather than silently corrected:

1. “then” is duplicated; 2. $F"H_i$ is not the restriction of a map whose domain is $[\kappa]^n$; 3. the order type $\beta$ of $H_1$ is omitted; 4. $\alpha$ is free in the final question, and “one of the forcing axioms at $\aleph_2$” does not name a theory or a class of posets.

The standard repaired definition is \[ \kappa\longrightarrow(\alpha,\beta)^n \] iff every $c:[\kappa]^n\to2$ admits either $H_0\subseteq\kappa$ with $\operatorname{otp}(H_0)=\alpha$ and $c``[H_0]^n=\{0\}$, or $H_1\subseteq\kappa$ with $\operatorname{otp}(H_1)=\beta$ and $c``[H_1]^n=\{1\}$.

Dilip Raghavan's contribution asks whether, relative to large cardinals, it is consistent that \[ \mathsf{CH}+\mathsf S_{\omega_2},\qquad \mathsf S_{\omega_2}:= \forall\alpha<\omega_2\, [\omega_2\longrightarrow(\omega_2,\alpha)^2]. \tag{S} \] Thus (S), with $\alpha<\omega_2$, is the best-verified intended mathematical target. The workshop statement is a consistency question; the problem-list version asks the stronger and less determinate program of deriving (S) from a specified higher forcing axiom. These are kept distinct below.

An existential reading (“for some $\alpha$”) cannot be the intended question: ZFC already gives $\omega_2\to(\omega_2,\omega+1)^2$, hence all smaller second targets. A literal reading with a free variable is not a sentence.

来源字段：http://aimpl.org/highlowforcing/1/

16187 | AIM-LOGIC-0064

## A generic-section obstruction for defining the constants in Q(t)

P / partially_solved | AIM 研讨会问题汇编

**Question 4 (Pheidas).** Is \(\mathbb Q\) existentially definable inside \(\mathbb Q(t)\), in the language of rings expanded by a constant symbol naming \(t\)? It is known to be first-order definable, even without naming \(t\). Using elliptic curves, one can easily existentially define dense subsets of \(\mathbb Q\).

来源字段：https://aimath.org/pastworkshops/definabilityinntproblems.pdf

16188 | AIM-LOGIC-0058

## A higher forcing axiom for proper posets

P / partially_solved | AIM 研讨会问题汇编

**Question 3.2.6.** Does the statement that $\mathcal{P}(\omega_1)/I$ is *strongly generically saturated* (via a countably closed forcing) imply the forcing axiom for $\aleph_2$-many dense sets for every $\sigma$-closed forcing leading to a transitively saturated ultrapower?

来源字段：http://aimpl.org/highlowforcing/3/

16189 | AIM-LOGIC-0028

## A maximal negative answer from overlapping extenders

S / solved | AIM 研讨会问题汇编

**Problem 16.** Suppose there is no inner model with a Woodin cardinal, and \(\kappa\) is a singular strong limit of uncountable cofinality, with \(2^\kappa=\lambda\), some regular \(\lambda>\kappa^+\). Must \(o(\kappa)^K\geq\lambda\)?

A negative answer may have applications in pcf theory.

Known below \(0^{\mathsf P}\) (Gitik--Mitchell).

来源字段：https://aimath.org/pastworkshops/innermodelproblems.pdf

16190 | AIM-LOGIC-0057

## A polarity and status trichotomy for consecutive approachability ideals

S / solved | AIM 研讨会问题汇编

Is there a model where both $I[\aleph_2]$ and $I[\aleph_3]$ are trivial? In other words, is $\text{AP}_{\aleph_2} + \text{AP}_{\aleph_3}$ consistent?

来源字段：http://aimpl.org/highlowforcing/3/

16191 | AIM-LOGIC-0066

## A positive-existential definition of nonzero rationals

P / partially_solved | AIM 研讨会问题汇编

**Question 6 (Koenigsmann).** Consider the language \(\mathcal O_2=\{0,1,+,P_2\}\), where \(P_2\) is the set of squares. Is every \(\exists\)-\(\mathcal O_2\)-definable set in \(\mathbb Q\) already \(\exists^+\)-\(\mathcal O_2\)-definable? Equivalently, are the following sets \(\exists^+\)-\(\mathcal O_2\)-definable? \[ \{x\in\mathbb Q:x\ne0\},\qquad \{x\in\mathbb Q:(\forall y)(y^2\ne x)\}. \]

来源字段：https://aimath.org/pastworkshops/definabilityinntproblems.pdf

16192 | AIM-LOGIC-0131

## A promise-to-total bridge for effective bounds on finite integral point sets

P / partially_solved | AIM 研讨会问题汇编

**Question 18 (Rojas).** Is there a computable bound (in function of \(f\)) on the size of the largest integer solution to \(f(x,y)=0\), when there are finitely many solutions? This is already done for genus 1 curves. There exists an algorithm to decide finiteness of the set of solutions. For rational points, there are papers by Minhyong Kim from Arizona: “Relating decision and search algorithms for rational points on curves of higher genus”, *Arch. Math. Logic* 42 (2003), no. 6, 563-568; “On relative computability for curves”, arXiv:math.NT/0502224.

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16193 | AIM-LOGIC-0135

## A Prym criterion for rank preservation under Denef base change

P / partially_solved | AIM 研讨会问题汇编

**Question 22 (Zahidi).** Look at the Denef curve \[ E:\quad f(t)Y^2=f(X), \] where \(f\) is a cubic. If we choose the curve in a good way, then \(E(k(t))\) has rank \(1\). Define \[ E_u:\quad f(u)Y^2=f(X). \] Try to give conditions on \(u\in k(t)\) such that \(E_u(k(t))\) also has rank \(1\).

The source does not specify the characteristic, separability of \(f\), or what “in a good way” means. The standard Denef setting supplies the missing hypotheses: \(k\) has characteristic zero (more generally, most of the geometry below only needs characteristic different from \(2\)); \(f\in k[x]\) is a separable cubic; and the elliptic curve \[ E_0:\quad y^2=f(x) \] has no \(k\)-endomorphisms beyond the integers. A sufficient standard condition for the last assertion is \(\operatorname{End}_{\bar k}(E_0)=\mathbb Z\), i.e. no complex multiplication.

This report works in characteristic zero and assumes \(f\) separable. It states separately whenever the hypothesis \(\operatorname{End}_k(E_0)=\mathbb Z\) is used.

No finite-generation hypothesis on \(k\) is needed for the rank formula itself. The anti-invariant constant maps are only the finite group \(E_0[2](k)\), while Hom groups of abelian varieties are finitely generated free abelian groups. Number-field or finitely-generated-field hypotheses enter only in the cited Hilbert-subset and effective isogeny tests.

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16194 | AIM-LOGIC-0039

## A reflection criterion for small Hausdorff spaces of spread omega_1

P / partially_solved | AIM 研讨会问题汇编

Do any of the analogs of PFA at \(\aleph_2\) give us \(\lvert X\rvert\leq 2^{\aleph_1}\) for every Hausdorff space with no discrete subspaces of size \(\aleph_2\)?

The stored item has no remarks or literature field. The official workshop report confirms the intended informal context: Neeman lectured on side-condition methods and “higher analogues of PFA,” while Todorčević discussed potential applications of those analogues to topology and combinatorics [AIM16]. The report does not identify a single formal axiom that can be substituted for the plural phrase in the problem.

The immediately preceding item 1.5 defines \[ s(X)=\sup\{\lvert D\rvert:D\subseteq X\text{ is a discrete subspace}\} \] and asks when the ZFC bound \(\lvert X\rvert\leq 2^{2^{s(X)}}\) can be improved to \(2^{s(X)}\). Its literature note contains an internal conflict: it says both that PFA gives a “negative” answer for \(s(X)=\aleph_0\) and that PFA gives the positive bound \(\lvert X\rvert\leq 2^{\aleph_0}\) when there is no uncountable discrete subspace.

The topological convention in item 1.55 is unambiguous. Every subset of a discrete subspace is discrete. Thus a space has no discrete subspace of cardinality \(\omega_2\) if and only if every discrete subspace has size at most \(\omega_1\), equivalently \[ s(X)\leq\omega_1. \] This equivalence does not require the supremum defining spread to be attained.

“Analogs of PFA at \(\aleph_2\)” has at least three plausible readings that must not be conflated:

1. the unnamed Neeman-style two-type side-condition principles contemplated at the 2016 workshop; 2. axioms meeting \(\omega_2\) dense sets for closed or specially restricted forcing classes, including principles discussed in nearby AIM items and [TX26]; 3. the later formal \(\omega_1\)-Strongly Proper Forcing Axiom, the \(\kappa=\omega_1\) instance of [ACKW19], for \(\omega_1\)-lattice, \(\omega_1\)-strongly proper posets and families of \(\omega_2\) dense open sets.

The analysis below assumes no equivalence or implication among these readings.

来源字段：http://aimpl.org/highlowforcing/1/

16195 | AIM-LOGIC-0023

### A smooth everywhere-finite counterexample for linear constraints on symmetric functions

S / solved | AIM 研讨会问题汇编

Consider the system of equations $$ \left\{ \begin{array}{ll} a_1s_1 + \dots + a_4s_4 &= c\\ b_1s_1 + \dots + b_4s_4 &= d \end{array} \right. $$ where $a_i, b_j, c,d\in \mathbb Z$ for all $1\leq i,j\leq 4$. Suppose that this system is solvable with the $s_j$ in $\mathbb Z$ modulo every prime $p$. Fix $k \geq 4$, and let $s_j$ be the $j$-th symmetric function of $x_1, \dots, x_k$. (E.g., $s_1 = x_1 + \dots + x_k$.) Must this system have integer solutions for $x_1, \dots, x_k$? What happens if we add more rows or columns?

来源字段：http://aimpl.org/definedecide/6/

16196 | AIM-LOGIC-0033

### A sparse club at omega_3: a literal construction and preservation obstructions

P / partially_solved | AIM 研讨会问题汇编

Is it consistent that there is a poset which adds a club in \(\omega_3\) which contains no infinite ground model subset over a model of GCH?

来源字段：http://aimpl.org/highlowforcing/1/

16197 | AIM-LOGIC-0048

### A stronger long-interval model and a finite-block strong-limit certificate

P / partially_solved | AIM 研讨会问题汇编

Find a model of $\mathrm{TP}_\kappa$ for all regular $\kappa\in[\aleph_2,\aleph_{\omega^2+2}]$ where $\aleph_{\omega^2}$ is the first strong limit.

来源字段：http://aimpl.org/highlowforcing/2/

16198 | AIM-LOGIC-0034

### A supercompact cardinal does not outright imply the bounded Namba axiom

P / partially_solved | AIM 研讨会问题汇编

Is there a large cardinal hypothesis that proves the bounded forcing axiom for Namba forcing?

来源字段：http://aimpl.org/highlowforcing/1/

16199 | AIM-LOGIC-0053

### A three-gate filter for strong-tree-property counterexamples to SCH

P / partially_solved | AIM 研讨会问题汇编

Suppose $\kappa$ has the strong tree property. Does SCH hold above $\kappa$? (If yes, then the strong tree property at $\aleph_2$ implies SCH.)

来源字段：http://aimpl.org/highlowforcing/2/

16200 | AIM-LOGIC-0020

### A two-endpoint Turing-degree fiber for big subrings of the rationals

P / partially_solved | AIM 研讨会问题汇编

What is the structure of HTP for big rings in $\mathbb Q$ under $\leq_T$? (Here, “big rings” mean rings where infinitely many primes are inverted.)

来源字段：http://aimpl.org/definedecide/5/

16201 | AIM-LOGIC-0117

### A uniform direct Diophantine model via exponent-stride encoding

S / solved | AIM 研讨会问题汇编

Fix a prime \(p\). Is there a Diophantine model of \(\mathbb F_q[Z]\) over \(\mathbb F_p[Z]\), when \(q\) is a power of \(p\), uniformly in \(q\)? In other words, do there exist polynomials \[ f(t,\vec x,\vec x'),\quad g(t,\vec y,\vec y'),\quad h(t,\vec z,\vec z') \] such that, for every power \(q\) of \(p\), \[ S_q:=\{\vec x:(\exists\vec x')f(Z^q,\vec x,\vec x')=0\} \] is in bijection with \(\mathbb F_q[Z]\), and \[ \{\vec y:(\exists\vec y')g(Z^q,\vec y,\vec y')=0\}\subseteq S_q^3 \] and \[ \{\vec z:(\exists\vec z')h(Z^q,\vec z,\vec z')=0\}\subseteq S_q^3 \] correspond respectively to the graphs of addition and multiplication on \(\mathbb F_q[Z]\)? Or can \(Z^q\) be replaced by another reasonable function from the powers of \(p\) to \(\mathbb F_p[Z]\)?

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16202 | AIM-LOGIC-0063

### A uniformity hierarchy and universal-fiber reduction for DPRM over finite-characteristic polynomial rings

P / partially_solved | AIM 研讨会问题汇编

**Question 3 (Demeyer).** A uniform statement of the theorem that r.e. sets are diophantine for \(\mathbb F_p[t]\). How should we even state this precisely?

来源字段：https://aimath.org/pastworkshops/definabilityinntproblems.pdf

16203 | AIM-LOGIC-0052

### A weak-square counterexample and the width-escape mechanism

S / solved | AIM 研讨会问题汇编

Suppose $\forall n<\omega$, $kappa_n$ has the super tree property property. If $\lambda = \sup_{n<\omega}$, does $\lambda^+$ have the tree property?

This text is visibly corrupted in three places: `kappa_n` has lost the backslash of `\kappa_n`; “property” is duplicated; and the expression after the supremum sign has been dropped. These neighboring records support the following high-confidence reconstruction:

It is nevertheless reinforced by Hayut and Unger's paper, whose main ideas were conceived at the same January 2016 AIM workshop and whose main theorem addresses exactly this configuration. The counterexample below also works if “increasing” was not intended, since the exhibited sequence is increasing.

来源字段：http://aimpl.org/highlowforcing/2/

16204 | OWR-14298016-004

### Abstract Formalization of Proof-Theoretic Arguments

O / open | Oberwolfach 报告问题汇编

A much broader open question is how to formalise the underlying proofs in a suitable abstract system. New applications of proof theory: Greedy algorithms, probability, and proof assistants Thomas Powell.

来源字段：https://doi.org/10.4171/owr/2023/53

16205 | OWR-14299587-007

### Abstract-Group Reconstruction of Countably Categorical Structures

P / partially_solved | Oberwolfach 报告问题汇编

When can an $\omega$-categorical structure $M$ be reconstructed from $\operatorname{Aut}(M)$ or $\operatorname{EEmb}(M)$ as an abstract group or monoid, without using the topology of pointwise convergence?

来源字段：https://doi.org/10.4171/owr/2025/58

16206 | AIM-LOGIC-0013

### Addition with directed p-power divisibility

P / partially_solved | AIM 研讨会问题汇编

Let \(p\) be a fixed prime. Define \[ x \mid_p y \Longleftrightarrow \exists s\in \mathbb N,\ y=p^sx. \] Does \((\mathbb Z,+,\mid_p,0,1)\) have a decidable existential theory?

来源字段：http://aimpl.org/definedecide/3/

16207 | OWR-12329-011

### Algebraic Closure and Orderability of Minimal Fields

P / partially_solved | Oberwolfach 报告问题汇编

Is every minimal field algebraically closed, as asserted by Podewski’s conjecture, and can a minimal ordered field of characteristic zero exist?

来源字段：https://doi.org/10.4171/owr/2013/01

16208 | AIM-LOGIC-0030

### Ambiguous nearness and the CH obstruction for higher Countryman lines

O / open | AIM 研讨会问题汇编

Two linear orders $\mathbb{C}_1$ and $\mathbb{C}_2$ are \emph{near} if there is another linear order $\mathbb{C}_0$ that embeds into both of them. $\mathbb{C}_1$ and $\mathbb{C}_2$ are \emph{co-near} if there is a linear order embedding into $\mathbb{C}_1$ and $\mathbb{C}_2^\ast$, where $\mathbb{C}_2^\ast$ is the reverse of $\mathbb{C}_2$.

Is it consistent with the continuum hypothesis that any two $\aleph_2$-Countryman lines are near or co-near?

Two $\kappa$-sized linear orders are $\kappa$-near if some order of size $\kappa$ embeds into both, and are $\kappa$-co-near if one is $\kappa$-near the reverse of the other; take $\kappa=\aleph_2$.

来源字段：http://aimpl.org/highlowforcing/1/

16209 | AIM-LOGIC-0132

### An affine-normalization criterion for infinitude of rational points

P / partially_solved | AIM 研讨会问题汇编

Is there an algorithm to decide whether \(f(x,y)=0\) has infinitely many \(\mathbb Q\)-rational solutions? This seems to be very hard for genus 1 curves. It has been done in other cases. Possible if \(\Sha\) is finite for all elliptic curves over \(\mathbb Q\).

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16210 | AIM-LOGIC-0126

### An inert-prime lifting criterion for the Cornelissen K3 equation

P / partially_solved | AIM 研讨会问题汇编

**Question 13 (Cornelissen).** Solve in integers \(A,B,X,Y\): \[ (A^2+B^2)(A^2+11B^2)=3^2\cdot5^2(X^2-5Y^2)^2. \tag{1} \] This is related to defining the integers in the rational numbers by a \(\Sigma_3^+\)-formula; see Cornelissen--Zahidi, arXiv:math.NT/0412473.

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16211 | AIM-LOGIC-0045

### An unbounded-width obstruction for weak square at aleph_omega

P / partially_solved | AIM 研讨会问题汇编

Is it consistent to have the failure of both $\mathrm{SCH}$ and $\square_{\aleph_\omega}^{*}$?

Is it relatively consistent with ZFC (from suitable large-cardinal hypotheses) that $\kappa=\aleph_\omega$ is a singular strong-limit cardinal, $2^\kappa>\kappa^+$, and Jensen's weak square $\square^*_{\kappa}$ fails?

A positive answer is necessary to get $\mathrm{AP}_{\aleph_{\omega+1}}+\mathrm{AP}_{\aleph_{\omega+2}}$.

来源字段：http://aimpl.org/highlowforcing/2/

16212 | OWR-15430-005

### Arc-Smooth and Whitney-Smooth Functions on Closed Sets

P / partially_solved | Oberwolfach 报告问题汇编

For functions defined on non-open subsets $X\subseteq\mathbb R^d$, when do the arc-generated smoothness classes $A^\infty(X)$ and $A^M(X)$ coincide with the corresponding Whitney-type classes $C^\infty(X)$ and $C^M(X)$?

来源字段：https://doi.org/10.4171/owr/2017/22

16213 | OWR-15430-006

### Arc-Smoothness on Closed Fat Sets

P / partially_solved | Oberwolfach 报告问题汇编

For a closed fat set $X\subseteq\mathbb R^d$, when do $A^\infty(X)=C^\infty(X)$ and $A^M(X)=C^M(X)$ hold?

来源字段：https://doi.org/10.4171/owr/2017/22

16214 | OWR-15181-009

### Arithmetic Integer Parts of Surreal and Real Closed Fields

P / partially_solved | Oberwolfach 报告问题汇编

How do fast-growing functions such as the Ackermann function behave in real closed fields? Does $\mathbf{No}$ have an integer part that is a model of true arithmetic, and does every real closed field have a normal integer part?

来源字段：https://doi.org/10.4171/owr/2016/60

16215 | OWR-14299587-005

### Automatic Action Reconstruction for Countably Categorical Structures

P / partially_solved | Oberwolfach 报告问题汇编

Can the action of $\operatorname{Aut}(A)$ or $\operatorname{EEmb}(A)$ on its underlying set be reconstructed up to conjugation, thereby recovering $A$ up to bi-definability? Does every $\omega$-categorical structure without algebraicity have automatic action reconstruction?

来源字段：https://doi.org/10.4171/owr/2025/58

16216 | AMR-074-0003

### Automorphism Problem for the Turing Degrees

P / partially_solved | AMR 问题清单汇编

Determine the automorphism group of the partial order of Turing degrees.

背景链接：https://mathoverflow.net/questions/227083/what-are-some-important-but-still-unsolved-problems-in-mathematical-logic#227108

16217 | AIM-LOGIC-0022

### Automorphisms of cohesive powers of number fields

P / partially_solved | AIM 研讨会问题汇编

1. What are the automorphisms of cohesive powers $\prod_C F$ for finite algebraic extensions $F$ of $\mathbb Q$? 2. Are there any that do not arise from automorphisms of $F$?

来源字段：http://aimpl.org/definedecide/6/

16218 | OWR-12329-008

### Bad Groups of Finite Morley Rank

O / open | Oberwolfach 报告问题汇编

Does there exist a simple group of finite Morley rank that is not isomorphic to $G(F)$ for any algebraic group $G$ over an algebraically closed field $F$?

来源字段：https://doi.org/10.4171/owr/2013/01

16219 | OWR-1963-001

### Bar-Recursive Interpretation of Second-Order Arithmetic

O / open | Oberwolfach 报告问题汇编

Can a uniform functional interpretation of full second-order arithmetic be obtained by extending the bar-recursive interpretation?

来源字段：https://doi.org/10.4171/owr/2008/18

16220 | OPG-37444

### Blatter-Specker Theorem for ternary relations

P / partially_solved | Open Problem Garden

Let $C$ be a class of finite relational structures. We denote by $f_C(n)$ the number of structures in $C$ over the labeled set $\{0, \dots, n-1 \}$. For any class $C$ definable in monadic second-order logic with unary and binary relation symbols, Specker and Blatter showed that, for every $m \in \mathbb{N}$, the function $f_C(n)$ is ultimately periodic modulo $m$.

Question Does the Blatter-Specker Theorem hold for ternary relations.

背景链接：http://www.openproblemgarden.org/op/blatter_specker_theorem_for_ternary_relations

16221 | OWR-1390-002

## Boffa’s Set-Theoretic Automorphism Conjecture

O / open | Oberwolfach 报告问题汇编

Does there exist a model $M$ of ZF, a cardinal $\kappa\in M$, and a nontrivial $\in$-automorphism $\sigma$ of $M$ such that $2^\kappa=\sigma(\kappa)$?

来源字段：https://doi.org/10.4171/owr/2006/52

16222 | OWR-12329-001

## Borel Cardinalities of Lascar Equivalence Relations

P / partially_solved | Oberwolfach 报告问题汇编

For the level-$\alpha$ Lascar equivalence relation $\equiv_L^\alpha$ in a complete first-order theory, which Borel cardinalities can it have?

来源字段：https://doi.org/10.4171/owr/2013/01

16223 | AIM-LOGIC-0101

## Borel completeness after naming a constant

P / partially_solved | AIM 研讨会问题汇编

Question 24. Suppose ∼=T is Borel complete. Is ∼=T ′ Borel complete?

**Question 23.** Let \(T'\) be an expansion of \(T\) by one constant. Suppose \(\cong_{T'}\) is Borel complete. Is \(\cong_T\) Borel complete?

**Question 24.** Suppose \(\cong_T\) is Borel complete. Is \(\cong_{T'}\) Borel complete?

The original AIM PDF and the preceding record recover the notation and the omitted setup:

Thus “∼=T” is \(\cong_T\), the isomorphism relation on countable models of \(T\), and “T ′” is \(T'\). Question 24 asks whether Borel completeness can be **lost** by naming one constant.

There is a small but important formulation ambiguity. “Expansion by one constant” could mean merely adding an unconstrained constant symbol to the language, but the standard formulation in later specialist sources is the following. Choose \(M\models T\) and \(a\in M\), put

\[ p(x)=\operatorname{tp}^M(a/\varnothing), \]

\[ T_p=\operatorname{Th}(M,a)=T\cup p(c) \]

in \(L\cup\{c\}\). This is a complete theory. The report analyzes this standard naming-a-parameter reading and writes \(T'=T_p\). It does not silently claim that every other incomplete use of “expansion” is equivalent.

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16224 | OWR-11695862-004

## Borovik-Cherlin Generic-Transitivity Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Let a group $G$ of finite Morley rank act definably and generically $k$-transitively on $X$. Prove $k\leq\operatorname{RM}(X)+2$, with equality only when the action is definably isomorphic to $\operatorname{PGL}_n$ acting on $\mathbb P^{n-1}$.

来源字段：https://doi.org/10.4171/owr/2022/52

16225 | AIM-LOGIC-0137

## Bounded square-class width and sign-symmetric factorization in C(Z)

P / partially_solved | AIM 研讨会问题汇编

If \(x\in\mathbb C(Z)\), then \[ \operatorname{ord}_{Z=0}\!\left(\frac{1+Zx^2}{1-Zx^2}\right) =\operatorname{ord}_{Z=\infty}\!\left(\frac{1+Zx^2}{1-Zx^2}\right)=0. \] Can every nonzero \(f\in\mathbb C(Z)\) for which \(\operatorname{ord}_{0}(f)\) and \(\operatorname{ord}_{\infty}(f)\) are even be written \[ f=u^2\prod_{i=1}^{1000}\frac{1+Zx_i^2}{1-Zx_i^2}, \qquad u,x_i\in\mathbb C(Z)? \] Is this at least true for \(f\in\mathbb Q(Z)\), while the witnesses \(u,x_i\) are still allowed to lie in \(\mathbb C(Z)\)?

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16226 | OWR-1390-005

## Bounded-Depth Pigeonhole Proofs and Prime Infinitude

O / open | Oberwolfach 报告问题汇编

(i) Do the propositional pigeonhole-principle tautologies $\mathrm{PHP}_n^{2n}$ and $\mathrm{PHP}_n^{n^2}$ have polynomial-size, constant-depth LK proofs? (ii) Let $\pi(x)$ be the number of primes at most $x$, and let $\mathrm I\Delta_0(\pi)$ be $\mathrm I\Delta_0$ augmented by $\pi$ and its natural recursive definition. Does $\mathrm I\Delta_0(\pi)$ prove that there are arbitrarily large primes?

来源字段：https://doi.org/10.4171/owr/2006/52

16227 | AMR-074-0112

## Cancellation and Schröder-Bernstein for Torsion Abelian Groups

O / open | AMR 问题清单汇编

Are the following statements equivalent to $\Pi^1_1\text{-}\mathsf{CA}_0$? (i) If countable torsion abelian groups $G,H$ satisfy $G\oplus G\cong H\oplus H$, then $G\cong H$. (ii) If $G$ and $H$ are each isomorphic to a direct summand of the other, then $G\cong H$.

背景链接：https://math.berkeley.edu/~antonio/papers/questionsRM.pdf

16228 | OWR-737-008

### Canonical Pseudo-Exponential Fields and Stability

P / partially_solved | Oberwolfach 报告问题汇编

别名：Redundant Extraction - Canonical Pseudo-Exponential Fields [OWR-737-009]

Is there a canonical generic pseudo-exponential field $\widetilde K_{\exp}(\lambda)$ for each uncountable cardinal $\lambda$, and what is the stability status of $\widetilde K_{\exp}$?

来源字段：https://doi.org/10.4171/owr/2004/35

16229 | AIM-LOGIC-0106

### CAT transfer and a metric-presentation criterion for group configuration

P / partially_solved | AIM 研讨会问题汇编

15. Study the group configuration theorem in the setting of continuous logic.

Background:

来源字段：https://aimath.org/WWN/continuouslogic/continuouslogic.pdf

16230 | OWR-11695862-011

### Categoricity of the Complex Exponential Field

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Complex-Exponential Categoricity Formulation [OWR-11695862-012]

Prove that the complex exponential field $\mathbb C_{\exp}$ is isomorphic to Zilber's pseudo-exponential field $\mathbb B_{\exp}$; equivalently, prove Schanuel's conjecture and strong exponential-algebraic closedness for $\mathbb C_{\exp}$.

来源字段：https://doi.org/10.4171/owr/2022/52

16231 | AIM-LOGIC-0136

### Certificate escape for CM j-invariants in the complex rational function field

P / partially_solved | AIM 研讨会问题汇编

**Question 23 (Pheidas).** Consider the elliptic curve \[ E:Y^2=X^3+aX+b. \] The following statement is Diophantine: “\(\operatorname{End}(E)/(2\operatorname{End}(E))\) has more than 2 elements”. Because \(\operatorname{End}(E)\) is a free finitely generated \(\mathbb Z\)-module, this is equivalent with “\(\operatorname{End}(E)\neq\mathbb Z\)”. So, we can existentially define the following set in \(\mathbb C(Z)\): \[ \{j\in\mathbb C\mid j\text{ is the }j\text{-invariant of a CM elliptic curve}\}. \] Can we do anything with this set?

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16232 | OWR-4990369-004

### Classification of Equationally Complete Theories

O / open | Oberwolfach 报告问题汇编

Can equationally complete theories, equivalently minimal varieties, be classified?

来源字段：https://doi.org/10.4171/owr/2021/1

16233 | OWR-4196-001

### Closure Conjectures for Infinite Stable Fields

O / open | Oberwolfach 报告问题汇编

Conjectures: every infinite stable field of finite weight is separably closed, and every infinite strongly stable field is algebraically closed.

来源字段：https://doi.org/10.4171/owr/2010/01

16234 | OWR-4196-002

### Closure Implies Exchange for Generics

O / open | Oberwolfach 报告问题汇编

Let $K$ be an infinite stable saturated field with additional structure. If $K$ is separably closed, must it have the exchange property for generics? If $K$ is algebraically closed, must the following hold: for every small subfield $k$ and elements $g,h$, if $g$ is generic over $k$ and algebraic over $k(h)$, then $h$ is generic over $k$?

来源字段：https://doi.org/10.4171/owr/2010/01

16235 | AIM-LOGIC-0036

### Collapse obstruction and the normal-versus-Knaster frontier for forcing axioms at omega_2

P / partially_solved | AIM 研讨会问题汇编

Assume CH. Is there a strongest forcing axiom of \(\sigma\)-closed posets meeting \(\aleph_2\)-many dense sets? What if we do not assume CH?

来源字段：http://aimpl.org/highlowforcing/1/

16236 | AIM-LOGIC-0047

### Collapsing a successor of a singular to aleph_2

P / partially_solved | AIM 研讨会问题汇编

Is it consistent that there is a poset turning \(\aleph_{\omega+1}\) into \(\aleph_2\)?

Let \(V\models\mathrm{ZFC}\), put \(\mu=(\aleph_\omega)^V\) and \(\lambda=\mu^{+V}=(\aleph_{\omega+1})^V\). Is it consistent that some forcing \(\mathbb P\in V\) has a generic extension \(W=V[G]\) such that \(\lambda=(\aleph_2)^W\)?

来源字段：http://aimpl.org/highlowforcing/2/

16237 | AMR-074-0007

### Compact Interpolation Logic Beyond First-Order Logic

O / open | AMR 问题清单汇编

Does there exist a reasonable logic strictly stronger than first-order logic that has both compactness and Craig's interpolation property?

背景链接：https://mathoverflow.net/questions/227083/what-are-some-important-but-still-unsolved-problems-in-mathematical-logic#227108

16238 | AIM-LOGIC-0112

### Compactness transfer from exact isometry to uniform almost isometry

S / solved | AIM 研讨会问题汇编

**27. Find a mechanism for deducing an “almost isometry theorem” from an isometric theorem using compactness (an ultraproduct).**

来源字段：https://aimath.org/WWN/continuouslogic/continuouslogic.pdf

16239 | OWR-11575-007

### Complete Sets in D-Maximal Orbits

O / open | Oberwolfach 报告问题汇编

For the D-maximal c.e. sets and their infinitely many orbits, do all but finitely many of the remaining orbits contain complete sets?

来源字段：https://doi.org/10.4171/owr/2012/08

16240 | OWR-15958-006

### Completeness of Cut-Free Kozen Modal Calculus

O / open | Oberwolfach 报告问题汇编

Is Kozen's axiomatization of the modal $\mu$-calculus complete after the cut rule is removed?

来源字段：https://doi.org/10.4171/owr/2017/53

16241 | OWR-15181-002

### Composition and Elementary Equivalence in Surreal Differential Fields

O / open | Oberwolfach 报告问题汇编

Can a satisfactory composition operator be defined on all of $\mathbf{No}$? Are the EL-series elementarily equivalent to the omega-series field as differential fields? Does every $\kappa$-computable function on the Cauchy completion of $\mathbf{No}_\kappa$ satisfy the intermediate value theorem? Is there a Kaplansky differential embedding theorem for $\mathbb{R}((G))_\kappa$, and which functions $f$ satisfy $\lambda_{f(x)}=\kappa_x$?

来源字段：https://doi.org/10.4171/owr/2016/60

16242 | AIM-LOGIC-0090

### Computable copies for a finite dimension-vector spectrum

P / partially_solved | AIM 研讨会问题汇编

**Question 12.** (Assume \(V=L\) if it makes things easier.) Suppose that for all \(\lambda\geq\aleph_1\) a theory \(T\) has at most \(\lambda\)-many models of cardinality \(\lambda\). Must each such model have a \(\lambda\)-computable presentation?

Such a \(T\) is necessarily \(\omega\)-stable and non-multidimensional. The question is true if \(T\) is \(\aleph_1\)-categorical.

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16243 | AIM-LOGIC-0079

### Computable structures need not admit low-Scott-rank approximations

S / solved | AIM 研讨会问题汇编

**Is it the case that every computable structure is computably approximable in the preceding sense?**

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16244 | OWR-4801-001

### Computational Interpretation of Univalent Homotopy Type Theory

P / partially_solved | Oberwolfach 报告问题汇编

Do all axioms of homotopy type theory, including univalence, admit a computational interpretation obtained by defining equality on a type recursively on that type, extending Gandy's interpretation for simple type theory?

来源字段：https://doi.org/10.4171/owr/2011/11

16245 | AIM-LOGIC-0099

### Cone spectra for totally categorical theories in infinite unary languages

P / partially_solved | AIM 研讨会问题汇编

**1.12. Spectrum of totally categorical theories (Andrews).** **Definition.** \(\operatorname{Spec}(T)=\{\mathbf d:\text{ there is a model of }T\text{ computable in }\mathbf d\}.\) **Conjecture 21.** If \(T\) is totally categorical then \(\operatorname{Spec}(T)\) is a cone. This is true in a finite language.

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16246 | AIM-LOGIC-0025

### Consistency bounds and a width obstruction for two merged inner-model questions

P / partially_solved | AIM 研讨会问题汇编

**Question 8.** What is the consistency strength of \[ MM(\mathfrak c)? \] The printed annotations give \(AD_{\mathbb R}+\) “\(\Theta\) is regular” as an upper bound (via Woodin's \(\mathbb P_{\max}\) analysis) and \(AD^{L(\mathbb R)}\) as a safe lower bound (Steel--Zoble).

**Question 9.** What is the consistency strength of \[ T_9:=\mathsf{ZFC}+\neg\square_{\omega _2}+\neg\square(\omega _2) +2^{\omega _1}=\omega _2? \] The source then says that the upper bound is weaker than \(AD_{\mathbb R}+\) “\(\Theta\) is Mahlo” and displays the shorthand \[ \{\alpha\mid \operatorname{cof}(\theta_\alpha)\geq\aleph_2 +\theta_\alpha\text{ regular in HOD}\}. \] This displayed set is not by itself a complete theory: in particular, the PDF does not state there whether it is to be stationary or merely unbounded, and the plus sign is being used informally as conjunction. It is therefore preserved as a source annotation rather than silently replaced by a later precise hypothesis. The 2017 theorem stated in Section 2 below is the rigorous published update.

来源字段：https://aimath.org/pastworkshops/innermodelproblems.pdf

16247 | AIM-LOGIC-0037

### Consistency of the lightface Sigma_1 maximality scheme

S / solved | AIM 研讨会问题汇编

The following is inconsistent: “For every \(\Sigma_1\) statement \(\varphi\) with \(\aleph_1\) and \(\aleph_2\) as parameters, if \(\varphi\) can be forced by an \(\aleph_1,\aleph_2\)-preserving forcing, then \(\varphi\) holds.”

The current AIM page agrees verbatim with the repository record and supplies no attribution, definitions, remarks, or status update.

1. \(\Sigma_1\) is read in the full-universe Lévy hierarchy: \(\varphi(a,b)\) has the form \(\exists x\,\theta(x,a,b)\), where every quantifier of \(\theta\) is bounded. 2. The only allowed parameters are the two cardinal constants \(\omega_1\) and \(\omega_2\), not arbitrary elements of \(H_{\omega_3}\). 3. A forcing is \((\omega_1,\omega_2)\)-preserving when it forces the ground-model \(\omega_1\) and \(\omega_2\) to remain, respectively, the first and second uncountable cardinals. 4. The assertion is a first-order **scheme**, one instance for each syntactic \(\Sigma_1\) formula; it is not a single formula quantifying over truth.

Let \(\Gamma_2\) denote this definable class of set forcings. The recovered scheme is \[ \mathsf{LMP}^{\Sigma_1}_2:\qquad \bigl[\Diamond_{\Gamma_2}\varphi(\omega_1,\omega_2)\bigr] \longrightarrow\varphi(\omega_1,\omega_2) \] for every full-universe \(\Sigma_1\) formula \(\varphi\), where \(\Diamond_{\Gamma_2}\) means “true in some \(\Gamma_2\)-forcing extension.”

Under this standard reading, the AIM conjecture is false: the scheme is equiconsistent with ZFC.

来源字段：http://aimpl.org/highlowforcing/1/

16248 | OWR-15958-001

### Constant-Color Proof-Theoretic Finite Ramsey Theorems

O / open | Oberwolfach 报告问题汇编

Can the proof-theoretic finite Ramsey theorem be formulated with a constant number of colors while retaining the optimal induction fragment $I\Sigma_{n-2}$?

来源字段：https://doi.org/10.4171/owr/2017/53

16249 | OWR-14299587-001

### Constraint-Satisfaction Dichotomy for Homogeneous Structures

O / open | Oberwolfach 报告问题汇编

Does the finite-domain constraint-satisfaction dichotomy extend to reducts of finitely bounded homogeneous structures?

来源字段：https://doi.org/10.4171/owr/2025/58

16250 | OWR-2654830-012

## Constructive Definitional Extensions and Morita Equivalence

O / open | Oberwolfach 报告问题汇编

Is the constructive equivalence generated by adding abbreviations, logical constructors, unique-existence terms, and standard sorts such as subsorts, quotient sorts, finite products, and finite disjoint unions the same as constructive Morita equivalence?

来源字段：https://doi.org/10.4171/owr/2020/34

16251 | AIM-LOGIC-0015

## Coordinate dependence and a differential pole-growth criterion

P / partially_solved | AIM 研讨会问题汇编

1. Is $\mathbb C[t_1,\dots,t_n]$ definable in $\mathbb C(t_1,\dots,t_n)$? 2. Is there a diophantine definition?

来源字段：http://aimpl.org/definedecide/4/

16252 | OWR-15170-003

## Decidability of Analytic Solutions to Algebraic Equations

O / open | Oberwolfach 报告问题汇编

Is there an algorithm which, given an algebraic equation in one or more variables with coefficients in $\mathbb Z[z]$, decides whether it has a solution analytic at $0$, on an open superset of the unit disc, or on the whole complex plane?

来源字段：https://doi.org/10.4171/owr/2016/49

16253 | AIM-LOGIC-0010

## Decidability of C(t) with a named coordinate or derivation

P / partially_solved | AIM 研讨会问题汇编

Its earliest available capture, dated 10 December 2019, and a 7 December 2023 capture have the same archived content digest and contain exactly the displayed wording.

The source leaves the signature implicit. Nearby AIM problems explicitly use the language of rings and list its operations together with named coordinates. The standard reconstruction of the first clause is therefore \[ L_t=L_{\mathrm{ring}}\cup\{\tau\}, \qquad L_{\mathrm{ring}}=\{0,1,+,-,\cdot\}, \qquad \tau^{\mathbb C(t)}=t. \] Here “with \(1\) and \(t\)” is shorthand for the ring language with the displayed distinguished constants, not a two-symbol language without ring operations.

There are two coherent readings of the second clause.

1. **One-variable reading:** add the unary function \(D=d/dt\) to \(\mathbb C(t)\), with \(i=1\). 2. **Multivariable reading:** replace the field by \(\mathbb C(t_1,\ldots,t_n)\) and add all standard commuting partial derivations \(D_i=\partial/\partial t_i\).

The official 2019 workshop report supports the second explanation: it discusses defining \(k[x_1,\ldots,x_n]\) in \(k(x_1,\ldots,x_n)\) after all \(\partial/\partial x_i\) are added to the language. It does not repair the wording of Problem 3.2, so both readings are analyzed below.

来源字段：http://aimpl.org/definedecide/3/

16254 | AIM-LOGIC-0011

## Decidable fragments and the Pell obstruction for entire functions

P / partially_solved | AIM 研讨会问题汇编

Is the existential theory of $\{\text{complex holomorphic functions over $\mathbb C$}\}$ in signature $0,1,+,\cdot,z$ decidable?

来源字段：http://aimpl.org/definedecide/3/

16255 | OWR-15170-023

### Definability in Automata-Presentable Ring Expansions

O / open | Oberwolfach 报告问题汇编

Let $R$ be a Euclidean ring with Euclidean function $\delta$, let $\preccurlyeq$ be the preorder induced by $\delta$, and for a noninvertible $r\in R$ let $V_r$ denote the highest power of $r$ dividing an element and $\mathbin{\cdot_r}$ the multiplication relation restricted according to powers of $r$. For multiplicatively independent noninvertible elements $r,s\in R$, characterize the subsets of $R$ definable in both $(R,+,-,0,\preccurlyeq,V_r,\mathbin{\cdot_r})$ and $(R,+,-,0,\preccurlyeq,V_s,\mathbin{\cdot_s})$. Separately, let $$ \mathcal F=(\mathbb F_p[[X]],+,0,V_X,\lambda_X,\pre c,(\mathbin{\cdot_C})_{C\in\mathbb F_q[X]}) $$ and let $\mathcal F_0$ be its substructure of power series algebraic over $\mathbb F_p(X)$. Is the theory of $\mathcal F_0$ decidable?

来源字段：https://doi.org/10.4171/owr/2016/49

16256 | OWR-15430-003

### Definability of Canonical Products with Sparse Zero Sets

O / open | Oberwolfach 报告问题汇编

For $0<s<1$, does the real field define the canonical Weierstrass product with zero sequence $(n^s)$ on $(0,\infty)$? More generally, for $\alpha>1$, what definable sets arise in the expansion of the real field by the canonical product associated with the geometric sequence $a_n=\alpha^n$, beyond the definable set $\alpha^{\mathbb Z}$?

来源字段：https://doi.org/10.4171/owr/2017/22

16257 | OWR-15170-002

### Definability of Evaluation in Rings of Entire Functions

O / open | Oberwolfach 报告问题汇编

For the rings $H_{t_1,t_2}(\mathbb C^2)$ of entire functions and their meromorphic fraction fields $M_{t_1,t_2}(\mathbb C^2)$, is the evaluation predicate positive-existentially definable in the corresponding ring language?

来源字段：https://doi.org/10.4171/owr/2016/49

16258 | OWR-12697685-001

### Definable Endomorphisms of Generic Multiplicative Fields

O / open | Oberwolfach 报告问题汇编

Is every definable endomorphism of a generic multiplicative field $(K,\sigma)$ contained in $\mathbb{Z}[\sigma]$?

来源字段：https://doi.org/10.4171/owr/2023/2

16259 | AIM-LOGIC-0111

### Definable sets reconstruct types in omega-stable theories

P / partially_solved | AIM 研讨会问题汇编

**26. Under which circumstances are there enough definable sets? (i.e. under which assumptions are types determined by the definable sets they belong to?). In particular, do we get a positive answer when \(T\) is \(\omega\)-stable?**

**Background:**

来源字段：https://aimath.org/WWN/continuouslogic/continuouslogic.pdf

16260 | AIM-LOGIC-0139

### Definable valuations on the anisotropic real conic

P / partially_solved | AIM 研讨会问题汇编

**Question 26 (Moret-Bailly).** Is there a nontrivial valuation ring \[ R\subset \operatorname{Frac}\!\left( \frac{\mathbb R[x,y]}{(x^2+y^2+1)} \right) \] which is definable? Same question for “semi-local ring” (finite intersection of valuation rings) instead of “valuation ring”? This is equivalent with the problem for valuation rings.

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16261 | OWR-15170-026

### Degree Definability and Existential Theory of Rational Function Fields

O / open | Oberwolfach 报告问题汇编

Is the relation $E$ on $\mathbb C[t]$, defined by $E(f,g)$ if and only if $\deg(f)=\deg(g)$, Diophantine or definable in the ring language with parameters? For a fixed prime $p$, is the existential theory of $\mathbb F_p(t)$ in the ring language decidable?

来源字段：https://doi.org/10.4171/owr/2016/49

16262 | AIM-LOGIC-0007

### Denominator-support transversal duality for HTP boundary sets

P / partially_solved | AIM 研讨会问题汇编

Given \(f\in \mathbb Z[x_1,\dots,x_n]\), define \[ A(f)=\{\text{subrings of }\mathbb Q\text{ where }f\text{ has a solution}\}, \] \[ \mathcal C(f)=\operatorname{Int}(A(f)^c),\qquad B(f)=\partial(A(f)^c). \] 1. Can the Lebesgue measure of \(B(f)\) be positive? 2. If so, what is the maximal complexity of the measure of \(B(f)\)? 3. Is \(\mathcal C(f)\) always a finite union of basic open sets?

来源字段：http://aimpl.org/definedecide/1/

16263 | AIM-LOGIC-0134

### Density-zero localizations do not force rank-small integral points

P / partially_solved | AIM 研讨会问题汇编

Let \(E\) be an elliptic curve over \(\mathbb Q\) of rank 2. Can we find a subset \(S\) of (infinitely many) primes such that the subgroup generated by \(E(\mathbb Z[S^{-1}])\) has rank one? If \(S\) is finite, the Siegel--Mahler theorem states that \(E(\mathbb Z[S^{-1}])\) is finite. Suppose \(S\) is infinite, but of density 0. Is \(E(\mathbb Z[S^{-1}])\) still “small”?

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16264 | OWR-14299908-007

### Dimension-Independent Nonminimality of Difference Equations

O / open | Oberwolfach 报告问题汇编

Let $K\subset\mathcal U\models\mathrm{ACFA}$ be difference fields with distinguished automorphism $\sigma$, let $V$ be an algebraic variety, and let $\phi:V\dashrightarrow V^\sigma$ be a rational map over $K$. For a generic solution $a$ of $\sigma(a)=\phi(a)$ whose type over $K$ is nonminimal, must $\operatorname{nmdeg}\!\left(\operatorname{tp}(a/K)\right)\leq3$, independently of $\dim V$?

来源字段：https://doi.org/10.4171/owr/2026/3

16265 | AIM-LOGIC-0118

### Diophantine additive polynomials and the failure of finite Artin--Schreier tests

S / solved | AIM 研讨会问题汇编

**Question 5 (Pheidas).** An additive polynomial in \(\mathbb F_p[Z]\) is a polynomial of the form \[ F(Z)=\alpha _0Z+\alpha _1Z^p+\alpha _2Z^{p^2}+\cdots+\alpha _nZ^{p^n}, \qquad \alpha_i\in\mathbb F_p. \] These are the polynomials that satisfy \(f(A+B)=f(A)+f(B)\) for all \(A,B\in\mathbb F_p[Z]\). Can we Diophantinely define the additive polynomials?

I checked page 2 of the official 2005 AIM problem-list PDF. Question 5 is:

Demeyer's comment then rejects the following proposed hierarchy: \[ \begin{aligned} F&=(A^p-A)+\alpha Z,\\ F^2&=(B^p-B)+\beta Z+(C^p-C)Z+\gamma Z^2,\\ F^3&=(L^p-L)+\lambda Z+(M^p-M)Z+\mu Z^2 +(N^p-N)Z^2+\nu Z^3, \end{aligned} \tag{1} \] with witnesses \(A,B,C,L,M,N\in\mathbb F_p[Z]\) and \(\alpha,\beta,\gamma,\lambda,\mu,\nu\in\mathbb F_p\), continued through an ordinary power \(F^n\). The PDF's displayed nonadditive counterexample is \[ H(Z)=\sum_{i=0}^{p-1}\left(Z^{2p}-Z^{p+1}\right)^{p^i}. \]

It concerns the noncommutative ring \(\mathcal A_L\) of additive polynomials over a field \(L\) of characteristic \(p\), with addition and composition, and the centralizer equation \[ f\circ Z^p=Z^p\circ f. \tag{3} \] It is connected to Question 5 because Question 5 seeks the additive-polynomial subset inside an ordinary commutative polynomial ring, while Fact 6 starts with the additive-polynomial composition ring as its ambient structure.

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16266 | AIM-LOGIC-0142

### Diophantine archimedean balls after the all-number-field breakthrough

P / partially_solved | AIM 研讨会问题汇编

**Question 30 (Shlapentokh).** Let \(K\) be a number field and \(\mathcal O_K\) its ring of integers. Fix an embedding \(K\hookrightarrow\mathbb C\), with \(K\nsubseteq\mathbb R\). Is \[ \{\alpha\in\mathcal O_K\mid |\alpha|\leq 1\} \] Diophantine in \(\mathcal O_K\)? If this is true for all \(K\), then Hilbert's Tenth Problem is undecidable for all \(\mathcal O_K\).

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16267 | OWR-15170-027

### Diophantine Definability of Rational Subfields

O / open | Oberwolfach 报告问题汇编

For the relation $S$ on $\mathbb F_p(t)$ defined by $S(f,g)$ if and only if $f\in\mathbb F_p(g)$, is $S$ Diophantine in the parameter-free ring language? In particular, is $\mathbb F_p(t^2)$ Diophantine with parameters in $\mathbb F_p(t)$?

来源字段：https://doi.org/10.4171/owr/2016/49

16268 | AIM-LOGIC-0003

### Directed coding and a forcing bottleneck for the AIM invariant

P / partially_solved | AIM 研讨会问题汇编

Let $\kappa$ be an infinite cardinal and put $\lambda=\kappa^+$. Define \[ u_2(\kappa)=\sup\left\{(\lambda^+)^{L[A]}:A\subseteq\kappa\right\}. \] 1. Is $u_2(\omega_1)=\omega_3$ consistent with (sufficiently) large cardinals? 2. How can one increase $u_2(\omega_1)$ without collapsing $\omega_1$ or $\omega_2$?

来源字段：http://aimpl.org/alephtwo/4/

16269 | AIM-LOGIC-0109

### Displacement classification and a two-level obstruction for a generic Urysohn involution

P / partially_solved | AIM 研讨会问题汇编

19. Study the definable sets in the expansion of the Urysohn space with a predicate for an involution map.

Background:

来源字段：https://aimath.org/WWN/continuouslogic/continuouslogic.pdf

16270 | AIM-LOGIC-0143

### Division-ample sets outside finitely many unit orbits

P / partially_solved | AIM 研讨会问题汇编

**Question 31 (Cornelissen).** Let \(K\) be a number field and \(\mathcal O_K\) its ring of integers. A set \(A\subseteq\mathcal O_K\) is **division-ample** if:

1. \(A\) is Diophantine over \(\mathcal O_K\); 2. every \(x\in\mathcal O_K\) divides some \(a\in A\); 3. there is a positive integer \(\ell\) such that for every \(a\in A\) there is \(\widetilde a\in\mathbb Z\) with \(\widetilde a\mid a\) and \[ N(a)\leq |\widetilde a|^\ell. \]

Give an example of such an \(A\) for which, for every finite \(S\subseteq\mathcal O_K\), \[ A\nsubseteq \mathcal O_K^*\cdot(\mathbb Z\cup S). \]

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16271 | AIM-LOGIC-0021

### Divisorial accumulation for curves and locally dense rational points

P / partially_solved | AIM 研讨会问题汇编

Let \(K\) be a global field and \(v\) a place. Let \(X,Y\) be positive-dimensional irreducible projective varieties over \(K\), let \(f:X\dashrightarrow Y\) be a dominant rational map over \(K\), and let \(U\) be a nonempty \(K\)-defined Zariski open subset of the domain of \(f\). If \(X(K)\) is Zariski dense, then there is a nonzero effective Cartier divisor \(D\) on \(Y\), defined over \(K\), for which the local Weil function \(\lambda_{Y,D,v}\) is unbounded on \[ f(U(K))\setminus \operatorname{Supp}(D). \] Equivalently, there are \(x_n\in U(K)\setminus f^{-1}(\operatorname{Supp}D)\) whose images approach \(\operatorname{Supp}(D)(K_v)\) \(v\)-adically.

来源字段：http://aimpl.org/definedecide/6/

16272 | AIM-LOGIC-0016

### Divisorial valuations and definability obstructions over complex rational function fields

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.2.** Define a non-trivial valuation on $\mathbb C(t_1,\dots,t_n)$.

来源字段：http://aimpl.org/definedecide/4/

16273 | AMR-075-0016

### Does a finitely presented homogeneous structure for a finite relational language have finitely many reducts

O / open | AMR 问题清单汇编

Does a finitely presented homogeneous structure for a finite relational language have finitely many reducts?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Model_theory_and_formal_languages

16274 | AMR-075-0009

### Does every simple first-order theory have stable forking

P / partially_solved | AMR 问题清单汇编

Does every simple first-order theory have stable forking?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Model_theory_and_formal_languages

16275 | AIM-LOGIC-0087

### Downward effective categoricity in a finite-fibre quasiminimal class

P / partially_solved | AIM 研讨会问题汇编

Question 9. Let K be a quasiminimal excellent class, and \(\lambda<\kappa\). Suppose that K is \(\kappa^+\)-computably categorical. Must K be \(\lambda^+\)-computably categorical?

Let \(\kappa\) be a cardinal. Then a set is \(\kappa^+\)-recursively enumerable when it is \(\Sigma_1\) on \(L_{\kappa^+}\).

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16276 | AIM-LOGIC-0088

### Downward reflection of relative computable categoricity

P / partially_solved | AIM 研讨会问题汇编

$K$ is relatively $\kappa^+$-categorical when for any two $\kappa^+$-computable $N,M$ of cardinality $\kappa^+$, they are isomorphic in $L_{\kappa^+}(M,N)$.

**Question 10.** Suppose that $K$ is relatively $\kappa^+$-computably categorical. Must $K$ be relatively $\lambda^+$-computably categorical?

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16277 | OWR-13354-004

## Dynamic Induction for Strictly Positive Definitions

P / partially_solved | Oberwolfach 报告问题汇编

Dynamic induction extends from the natural numbers to inductive definitions of the form $I(x)\leftrightarrow B(x)\lor\exists y<x\,I(y)$. Does it extend to arbitrary strictly positive inductive definitions?

来源字段：https://doi.org/10.4171/owr/2014/52

16278 | AIM-LOGIC-0082

## Effective cocycle criterion for reconstructing a central finite cover

P / partially_solved | AIM 研讨会问题汇编

**Question 4.** How difficult is it to find a presentation of $M$ in terms of $G$?

Let $M$ be a strongly minimal nontrivial locally modular structure. There is a nonorthogonal interpretable strongly minimal group $G$.

Consider a three-to-one map $(\mathbb Q,+)\leftarrow(M,\oplus)$ where $(M,\oplus)$ is strongly minimal.

The immediately preceding source text says:

Inspection of the original PDF resolves the segmentation. The three-to-one paragraph follows Question 4 as a new paragraph, and Question 5 immediately asks whether the operation $\oplus$ is definable in some $\mathbb Q\times F$. Thus the paragraph introduces the notation used by Question 5. It is useful context about finite covers, but it is not part of the sentence constituting Question 4. This report owns and analyzes only Question 4.

The phrase “a presentation of $M$ in terms of $G$” is not formally defined in the source. Three inequivalent readings must be separated:

1. **Existential:** does the isomorphism type of $M$ have a computable copy whenever $G$ does? 2. **Relative to a cover:** from a copy of $G$ and fixed data describing $M\to G$, can one compute a copy of the cover, including the map? 3. **Uniform/functorial:** is there a single computable procedure on copies and isomorphisms of $G$ that constructs copies of $M$? Equivalently, under the usual hypotheses, is $M$ effectively interpretable in $G$?

The results below show that these readings can differ even for elementary finite group covers.

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16279 | AIM-LOGIC-0009

## Effective determination versus streaming enumeration of integral points

P / partially_solved | AIM 研讨会问题汇编

Given an elliptic curve \(E\) given by \(y^2=x^3+Ax+B\): 1. Is there are algorithm that decides if \(E(\mathbb Z)\setminus\{\infty\}\) is empty? 2. Is there an algorithm for enumerating \(E(\mathbb Z)\)? 3. Can these questions be answered if \(E\) is replaced with an arbitrary curve?

来源字段：http://aimpl.org/definedecide/3/

16280 | AIM-LOGIC-0077

## Effective fragments of the maximal abelian and solvable extensions

P / partially_solved | AIM 研讨会问题汇编

Decidability of Qsolv, Qab, Zsolv and Zab.

Determine the decidability of \[ \mathbb Q^{\mathrm{solv}},\qquad \mathbb Q^{\mathrm{ab}},\qquad \mathbb Z^{\mathrm{solv}},\qquad \mathbb Z^{\mathrm{ab}}. \]

来源字段：https://aimath.org/pastworkshops/definabilityinntproblems.pdf

16281 | AIM-LOGIC-0096

## Effective isolation and descent to the prime model

P / partially_solved | AIM 研讨会问题汇编

Question 18 (Millar). Let T be a decidable theory having countably many countable models. When must the prime model have a decidable presentation? (Note that ω-stability suffices.)

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16282 | AIM-LOGIC-0084

## Effective nonforking sections in stable theories

P / partially_solved | AIM 研讨会问题汇编

Consider $T$ stable. Then for all $M\prec N\models T$, the map \[ r:S_1(N)\longrightarrow S_1(M) \] has a continuous section, which sends $p$ to the unique nonforking extension.

**Question 6.** Is there a computable section?

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16283 | AIM-LOGIC-0081

### Effective presentations of interpreted groups and a constant-fiber no-jump criterion

P / partially_solved | AIM 研讨会问题汇编

The original AIM problem list places Question 3 in the following section:

The next two questions ask conversely for a presentation of \(M\) from \(G\), and then discuss a three-to-one map from a strongly minimal group \((M,\oplus)\) to \((\mathbb Q,+)\). This confirms that “presentation” is meant in the computable-model-theoretic sense and that finite covers/quotients are central.

The wording is intentionally qualitative and omits three choices that materially change the answer:

1. whether an interpretation scheme (domain, quotient equality, and group-law formulas) is supplied or must be discovered; 2. whether its finite parameter tuple is supplied or must be selected uniformly from a copy of \(M\); 3. whether “in terms of” asks only for some copy of \(G\), or for a uniform quotient map/functor compatible with copies and isomorphisms.

This report makes those choices explicit rather than silently replacing the question by one formalization.

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16284 | AIM-LOGIC-0121

### Effective semilinearity for polynomial zeros at powers of two

S / solved | AIM 研讨会问题汇编

**Question 8 (Pheidas).** Is the following problem decidable: Given \(P(\vec x)\in\mathbb Z[\vec x]\), do there exist \(n_1,\ldots,n_m\in\mathbb N\) such that \[ P(2^{n_1},\ldots,2^{n_m})=0? \] The answer is YES: this is related to the Mordell--Lang conjecture for tori.

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16285 | AIM-LOGIC-0124

### Effective simple-set transfer and an obstruction from noneffective localizations

P / partially_solved | AIM 研讨会问题汇编

It first recalls that a set \(A\subseteq\mathbb N\) is **simple** when

1. \(A\) is computably enumerable (c.e., called r.e. in the source); 2. \(\mathbb N\setminus A\) is infinite; and 3. every c.e. subset of \(\mathbb N\setminus A\) is finite.

It then proposes the following transfer. Let \(\mathcal O_K\) be a ring of integers, let \(f:\mathcal O_K\hookrightarrow R\) be a ring embedding, and suppose a simple subset is represented by \[ A=\{x\in\mathcal O_K:\exists\mathbf y\in\mathcal O_K^n, P(x,\mathbf y)=0\}. \] Allowing witnesses in \(R\) gives a larger set, asserted to be either simple or cofinite. The stated application is: if \(P\in\mathbb Z[x,\mathbf y]\) defines a simple set using integral witnesses and the set obtained using rational witnesses has infinite complement in \(\mathbb Z\), then Hilbert's tenth problem over \(\mathbb Q\) is undecidable.

* It defines the enlarged set with \(x\in\mathcal O_K\), but then writes \(f(A)\subseteq A'\), comparing a subset of \(R\) with a subset of \(\mathcal O_K\). The type-correct pullback is \[ B_{R,f}=\{x\in\mathcal O_K:\exists\mathbf y\in R^n, P^f(f(x),\mathbf y)=0\}, \tag{1} \] where \(P^f\) is obtained by applying \(f\) to the coefficients. Then \(A\subseteq B_{R,f}\). Equivalently one may work with \(f(B_{R,f})\subseteq f(\mathcal O_K)\). * “Computably enumerable” is meaningful only relative to an effective presentation. An abstract ring embedding does not ensure that the pullback (1) is c.e. This missing hypothesis is essential; Section 7 gives an explicit counterexample to that preservation step.

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16286 | OWR-15181-021

### Elementary Embeddings of Residue Exponential Fields

O / open | Oberwolfach 报告问题汇编

For every o-minimal ordered exponential field $K$ satisfying EXP, does its residue exponential field $K_{\mathrm{res}}$ embed elementarily into $K$?

来源字段：https://doi.org/10.4171/owr/2016/60

16287 | OWR-15181-007

### Elementary Equivalence of Differential Generalized Series Fields

O / open | Oberwolfach 报告问题汇编

If $k\equiv k'$ as fields and the asymptotic couples $(G,\psi)$ and $(G',\psi')$ are elementarily equivalent, are the generalized series fields $k((G))$ and $k'((G'))$, equipped with the derivations induced by their asymptotic couples, elementarily equivalent as differential valued fields?

来源字段：https://doi.org/10.4171/owr/2016/60

16288 | OWR-12697685-007

### Elementary Equivalence of Hyperbolic Groups with Torsion

O / open | Oberwolfach 报告问题汇编

Are two hyperbolic groups $\forall\exists$-equivalent if and only if they are elementarily equivalent, including in the presence of torsion?

来源字段：https://doi.org/10.4171/owr/2023/2

16289 | OWR-1188-015

### Elementary Equivalence of Regular Function Fields

P / partially_solved | Oberwolfach 报告问题汇编

If regular function fields $K/k$ and $L/l$ over reasonable base fields are elementarily equivalent, must $k$ and $l$ be elementarily equivalent? If $k\cong l$, must $K\cong L$ as fields?

来源字段：https://doi.org/10.4171/owr/2006/06

16290 | OWR-5154-010

### Elementary Equivalence of Universal Sofic Groups

O / open | Oberwolfach 报告问题汇编

Are all universal sofic groups $S_U$ elementarily equivalent?

来源字段：https://doi.org/10.4171/owr/2011/26

16291 | AIM-LOGIC-0070

### Elementary equivalence versus isomorphism for finitely generated fields

P / partially_solved | AIM 研讨会问题汇编

For \(K\) and \(L\) finitely generated fields, does \(K\equiv L\) imply \(K\cong L\)?

来源字段：https://aimath.org/pastworkshops/definabilityinntproblems.pdf

16292 | OWR-14299908-008

### Elementary Equivalences for Ordered Exponential Fields

O / open | Oberwolfach 报告问题汇编

Let $R$ be an ordered field equipped with an everywhere continuous and differentiable function $\exp:R\to R$ satisfying $\exp'=\exp$ and $\exp(0)=1$. Are the following conditions equivalent: $R$ has the same first-order theory as the real exponential field $\mathbb R_{\exp}$; $R$ is o-minimal; and $R$ is definably complete?

来源字段：https://doi.org/10.4171/owr/2026/3

16293 | OWR-12697685-006

### Elementary Extensions of Definable Group Dynamics

O / open | Oberwolfach 报告问题汇编

How does the definable topological dynamics of a group $G$ change under elementary extension?

来源字段：https://doi.org/10.4171/owr/2023/2

16294 | OWR-17465-001

### Elementary-Theory Classification of Virtually Free Groups

O / open | Oberwolfach 报告问题汇编

Do the known classifications for finitely generated virtually free groups extend from $\forall\exists$-formulas to arbitrary first-order formulas? More precisely, can elementary equivalence still be characterized by common overgroups obtained through finite sequences of legal large HNN extensions over finite subgroups and legal small replacements of virtually cyclic subgroups, can elementary embeddings still be characterized by iterated legal large extensions, and do the corresponding decision algorithms still exist?

来源字段：https://doi.org/10.4171/owr/2020/2

16295 | AIM-LOGIC-0145

### Endpoint robustness and a tower criterion for Julia Robinson attainment

P / partially_solved | AIM 研讨会问题汇编

**Question 33 (Videla).** Let \(K\subseteq\mathbb Q^{\mathrm{tot.real}}\subseteq\overline{\mathbb Q}\). Define \[ A_K:=\{s\in\mathbb R_{>0}:\text{ there exist infinitely many } \alpha\in\mathcal O_K\text{ such that }\alpha\text{ and its conjugates are all in }[0,s]\}. \] Question of Julia Robinson: Is the infimum of \(A_K\) an element of \(A_K\)? If so, the first-order theory of \(\mathcal O_K\) is undecidable. For \(K=\mathbb Q^{\mathrm{tot.real}}\), \(\inf(A_K)=4\in A_K\).

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16296 | AIM-LOGIC-0075

### Equal degree in polynomial rings: parameters and the place at infinity

P / partially_solved | AIM 研讨会问题汇编

**Question 15 (Demeyer).** For various fields \(K\), consider the relation on \(K[t]\) defined by \[ R(f,g)\Longleftrightarrow \deg(f)=\deg(g). \] (a) Is this relation (existentially) definable for \(K=\mathbb C\)? (b) For \(K=\mathbb F_p\), is there an existential definition, uniform in \(p\)?

来源字段：https://aimath.org/pastworkshops/definabilityinntproblems.pdf

16297 | AIM-LOGIC-0146

### Exact alternation depth and variable bounds for two real rational-function fields

P / partially_solved | AIM 研讨会问题汇编

**Question 34 (Zahidi).** Let \[ \mathbb R^{\mathrm{alg}}:=\overline{\mathbb Q}\cap\mathbb R. \] It is known that \(\mathbb R^{\mathrm{alg}}\equiv\mathbb R\) (elementary equivalence), but that \(\mathbb R^{\mathrm{alg}}(t)\not\equiv\mathbb R(t)\). On the other hand, the existential theories of \(\mathbb R^{\mathrm{alg}}(t)\) and \(\mathbb R(t)\) are the same. What is the minimal quantifier complexity for which they have different theories? What is the minimal number of variables one needs?

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16298 | AIM-LOGIC-0128

### Exact variable accounting for fixed-variable Hilbert's Tenth Problem over the integers

P / partially_solved | AIM 研讨会问题汇编

**Question 15 (Rojas).** What is the smallest \(n\) such that Hilbert's Tenth Problem over \(\mathbb Z\) restricted to one polynomial in \(n\) variables is undecidable?

Minimal \(n\) is known to be \(2\leq n\leq22\) by Matijasevič, and probably \(2\leq n\leq11\) by some Chinese. There is some evidence that \(n=3\).

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16299 | AIM-LOGIC-0041

### Exact-size padding and ZFC bounds for cofinal types at aleph_2

P / partially_solved | AIM 研讨会问题汇编

Under a given analog of PFA for \(\aleph_2\), how many cofinal types of directed sets of cardinality \(\aleph_2\) are there?

Under PFA, \(1,\omega,\omega_1,\omega\times\omega_1\), and \((\omega_1)^{<\omega}\) are the cofinal types of directed sets of cardinality \(\aleph_1\).

For an explicitly named higher forcing axiom \(A\), determine the Tukey-equivalence classes of directed sets of cofinality at most \(\aleph_2\).

来源字段：http://aimpl.org/highlowforcing/1/

16300 | AIM-LOGIC-0040

### Exactification and radial obstructions for the compact cardinality problem

P / partially_solved | AIM 研讨会问题汇编

It is a fact that a compact Hausdorff space \(X\) is countably tight if and only if \(X\) contains no \(\omega_1\)-free sequence.

Suppose \(X\) is a compact Hausdorff space of tightness \(\aleph_1\) (i.e. there are no \(\omega_2\)-free sequences) and density \(\aleph_1\). Is it true that \(|X|\leq 2^{\aleph_1}\)?

来源字段：http://aimpl.org/highlowforcing/1/

16301 | AIM-LOGIC-0072

### Existential analogues of Robinson arithmetic

O / open | AIM 研讨会问题汇编

**Question 12 (Videla). Is there an existential analogue of Robinson's Q?**

来源字段：https://aimath.org/pastworkshops/definabilityinntproblems.pdf

16302 | AIM-LOGIC-0067

### Existential graph coding and empty-graph coordinate rigidity

P / partially_solved | AIM 研讨会问题汇编

**Question 7 (Miller).** Given a countable graph \((V,E)\), find a field \(K\) of characteristic \(0\), polynomials \[ P\in\mathbb Q[x,t],\qquad Q,R\in\mathbb Q[x,y,z], \] and a bijection \[ \alpha:V\longrightarrow\{x\in K:(\exists t\in K)\,P(x,t)=0\} \] such that, for all \(v,w\in V\), \[ (v,w)\in E\Longleftrightarrow(\exists z)\,R(\alpha(v),\alpha(w),z)=0, \] \[ (v,w)\notin E\Longleftrightarrow(\exists z)\,Q(\alpha(v),\alpha(w),z)=0. \]

**Poonen:** let \(P\) define a curve without non-trivial automorphism.

**For the empty graph:** Find \(P\in\mathbb Q[x,t]\) such that \(K\) is generated by \(x_1,x_2,\ldots,t_1,t_2,\ldots\), the elements \(x_1,x_2,\ldots\) are algebraically independent, \(P(x_i,t_i)=0\) for every \(i\), and:

1. for every permutation \(\pi\) of \(\mathbb N\), there is a unique automorphism \(\sigma_\pi\) of \(K\) such that \(\sigma_\pi(x_i)=x_{\pi(i)}\); 2. \(\{x_i:i\in\mathbb N\}=\{y\in K:(\exists t)\,P(y,t)=0\}\).

来源字段：https://aimath.org/pastworkshops/definabilityinntproblems.pdf

16303 | OWR-15170-011

### Existential Theory of Polynomial Rings with Divisibility

P / partially_solved | Oberwolfach 报告问题汇编

Is the existential theory of $\mathbb F_q[t]$ in the language of addition, divisibility, and the predicate for powers of $t$ decidable?

来源字段：https://doi.org/10.4171/owr/2016/49

16304 | OWR-11695862-013

### Exponential-Algebraic Closedness

P / partially_solved | Oberwolfach 报告问题汇编

Let $V\subseteq\mathbb C^n\times(\mathbb C^\times)^n$ be a free rotund algebraic subvariety. Prove that $V$ contains a point $(z,\exp z)$.

来源字段：https://doi.org/10.4171/owr/2022/52

16305 | OWR-12329-004

### Extendable Outer Automorphisms of Subgroup Inclusions

P / partially_solved | Oberwolfach 报告问题汇编

Given a subgroup $H\leq G$, let $\operatorname{Out}(H\hookrightarrow G)$ be the outer automorphisms of $H$ that extend to $G$. What elements can this group contain, and is it finitely generated or finitely presented?

来源字段：https://doi.org/10.4171/owr/2013/01

16306 | AIM-LOGIC-0049

### Extension rigidity and least fresh cardinals for Foreman's maximality principle

P / partially_solved | AIM 研讨会问题汇编

Is it consistent that every poset either adds a real or collapses a cardinal?

来源字段：http://aimpl.org/highlowforcing/2/

16307 | OWR-15170-022

### Extremality Criteria for Positive-Characteristic Valued Fields

O / open | Oberwolfach 报告问题汇编

For a valued field $(K,v)$ of positive residue characteristic, with value group a $\mathbb Z$-group or with divisible value group and large imperfect residue field, is there a simple useful additional assumption ensuring that $(K,v)$ is extremal? More generally, is every maximal valued field with a $\mathbb Z$-group as value group, or with divisible value group and large residue field, extremal?

来源字段：https://doi.org/10.4171/owr/2016/49

16308 | OPG-660

### F_d versus F_{d+1}

P / partially_solved | Open Problem Garden

Problem Find a constant $k$ such that for any $d$ there is a sequence of tautologies of depth $k$ that have polynomial (or quasi-polynomial) size proofs in depth $d+1$ Frege system $F_{d+1}$ but requires exponential size $F_d$ proofs.

背景链接：http://www.openproblemgarden.org/op/f_d_versus_f_d_1

16309 | OWR-14298016-007

### Finite Axiomatizability of TEIP over Open Induction

P / partially_solved | Oberwolfach 报告问题汇编

Is the first-order theory TEIP of exponential integer parts of real-closed exponential fields finitely axiomatizable over open induction?

来源字段：https://doi.org/10.4171/owr/2023/53

16310 | OPG-37863

### Finite entailment of Positive Horn logic

P / partially_solved | Open Problem Garden

Question Positive Horn logic (pH) is the fragment of FO involving exactly $\exists, \forall, \wedge, =$. Does the fragment $pH \wedge \neg pH$ have the finite model property?

背景链接：http://www.openproblemgarden.org/op/finite_satisfiability_of_positive_horn_logic_entailment

16311 | AIM-LOGIC-0148

### Finite homogeneous maps and a sharp fiber bound

S / solved | AIM 研讨会问题汇编

**Question 36 (Jarden).** Given $f_1,\ldots,f_n\in\mathbb C[x_1,\ldots,x_m]$ which are homogeneous of degree $d$, assume that their only common zero is $(0,\ldots,0)$. Prove that \[ V\bigl(f_1(\vec x)=b_1,\ldots,f_n(\vec x)=b_n\bigr) \] is finite for all $b_1,\ldots,b_n\in\mathbb C$.

If the fiber were infinite, the variety in $\mathbb P^m$ defined by the homogenized equations would be positive-dimensional. It would then intersect the hyperplane at infinity, giving a nonzero common zero of the $f_i$.

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16312 | OWR-11575-016

### Finite Lattices in Computably Enumerable Turing Degrees

O / open | Oberwolfach 报告问题汇编

Which finite lattices can be embedded into the computably enumerable Turing degrees while preserving joins and meets?

来源字段：https://doi.org/10.4171/owr/2012/08

16313 | AMR-074-0006

### Finite Spectrum Problem

O / open | AMR 问题清单汇编

Is the complement of the finite spectrum of every first-order sentence also a finite spectrum? Equivalently, is $\mathrm{NE}=\mathrm{coNE}$?

背景链接：https://mathoverflow.net/questions/227083/what-are-some-important-but-still-unsolved-problems-in-mathematical-logic#227108

16314 | AIM-LOGIC-0091

### Finite uncountable spectra and computable presentations

P / partially_solved | AIM 研讨会问题汇编

**Question 12.** (Assume \(V=L\) if it makes things easier.) Suppose that for all \(\lambda\geq\aleph_1\) a theory \(T\) has at most \(\lambda\)-many models of cardinality \(\lambda\). Must each such model have a \(\lambda\)-computable presentation?

Such a \(T\) is necessarily \(\omega\)-stable and non-multidimensional. The question is true if \(T\) is \(\aleph_1\)-categorical.

**Question 13.** What if \(T\) has finitely many models in \(\aleph_1\)? (Maybe look at models in \(\aleph_n\).)

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16315 | AIM-LOGIC-0083

### Finite-cover coordinates for a strongly minimal group

P / partially_solved | AIM 研讨会问题汇编

Consider a three-to-one map \((\mathbb Q,+)\leftarrow(M,\oplus)\) where \((M,\oplus)\) is strongly minimal. **Question 5.** Must \(\oplus\) be definable in some \(\mathbb Q\times F\), where \(F\) is a finite set?

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16316 | AIM-LOGIC-0116

### Finite-fiber Diophantine coding and a degree-at-infinity obstruction

P / partially_solved | AIM 研讨会问题汇编

**Question 3 (Demeyer).** Consider the ring \(\mathbb F_q[W,Z]\). Does there exist a Diophantine predicate \(\alpha(f,\vec g)\), with \(f\in\mathbb F_q[W,Z]\) and \(\vec g\in\mathbb F_q[Z]^n\), such that

1. for every \(f\in\mathbb F_q[W,Z]\), there is \(\vec g\in\mathbb F_q[Z]^n\) for which \(\alpha(f,\vec g)\) holds; and 2. for every \(\vec g\in\mathbb F_q[Z]^n\), the set \[ \{f\in\mathbb F_q[W,Z]:\alpha(f,\vec g)\} \] is finite?

This will imply that r.e. = Diophantine for \(\mathbb F_q[W,Z]\). It is possible to give such a predicate if “\(\alpha(\cdots)\) holds” is replaced by “\(\alpha(\cdots)\) does not hold.”

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16317 | AIM-LOGIC-0122

### Finite-fiber normalization for elliptic Diophantine encodings

P / partially_solved | AIM 研讨会问题汇编

**Question 9 (Pheidas).** Can we redo the proof of Hilbert's Tenth Problem over \(\mathbb Z\), using elliptic curves instead of Pell equations? Hopefully, this would lead to a lower number of variables and/or lower degree. Can this give a finite-fold Diophantine definition of all r.e. sets?

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16318 | AIM-LOGIC-0069

### Finite-fragment and ultraproduct criteria for transcendence-degree parity

P / partially_solved | AIM 研讨会问题汇编

**Question 9 (Flenner).** Is there a sentence \(\varphi\) in the language of rings such that \[ \mathbb Q(t_1,\ldots,t_n)\models\varphi \quad\text{if and only if}\quad n\text{ is even}? \] Using ultrapowers, one might show this is not possible.

来源字段：https://aimath.org/pastworkshops/definabilityinntproblems.pdf

16319 | AIM-LOGIC-0102

### Finite-horizon robustness under automorphism perturbation

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.** Let \(T\) be a complete superstable metric theory, let \(\sigma\) be a new symbol for a unitary function, and let

\[ T_\sigma=T\cup\{\text{“\(\sigma\) is an automorphism”}\}. \]

Assume that \(T_A\), the model companion of \(T_\sigma\), exists. Is \(T_A\) supersimple up to perturbations of the automorphism?

The AIM source PDF was checked directly. The recovered statement owned by this record is only **Problem 1**:

The PDF confirms the notation \(T_\sigma\) and \(T_A\). The hyphenated strings “perturba-tions,” “supersta-ble,” and “automorp-hism” are line-break artifacts. The remainder of the input beginning “2. The Ehrenfeucht constructions…” is a separate problem and is not analyzed here.

来源字段：https://aimath.org/WWN/continuouslogic/continuouslogic.pdf

16320 | OWR-11575-017

### Finite-Lattice Obstructions in Enumerable Reducibility Degrees

O / open | Oberwolfach 报告问题汇编

Does there exist a finite lattice that cannot be embedded into the recursively enumerable cl-degrees or ibT-degrees?

来源字段：https://doi.org/10.4171/owr/2012/08

16321 | OWR-15181-016

### First Root Conjecture for Real Exponential Polynomials

O / open | Oberwolfach 报告问题汇编

For every positive integer $n$ and every $f\in\mathbb{Z}[x_1,\ldots,x_n,e^{x_1},\ldots,e^{x_n}]$, can one effectively find $\eta(n,f)$ such that, whenever $f$ has a zero in $\mathbb{R}^n$, it has a zero $\alpha$ with $\lVert\alpha\rVert<\eta(n,f)$?

来源字段：https://doi.org/10.4171/owr/2016/60

16322 | OWR-15958-005

### Fixed Complexity Separations in Bounded Arithmetic

O / open | Oberwolfach 报告问题汇编

Is there a fixed $k$ such that $T_{2n+1}(\alpha)$ can be separated from $T_{2n}(\alpha)$ by a $\forall\Sigma_k^b(\alpha)$ sentence? In particular, is there a $\forall\Sigma_1^b(\alpha)$ sentence provable in some $T_{2n}(\alpha)$ but not in $\mathrm{APC}_2(\alpha)$?

来源字段：https://doi.org/10.4171/owr/2017/53

16323 | OPG-37424

### Fixed-point logic with counting

S / solved | Open Problem Garden

Question Can either of the following be expressed in fixed-point logic plus counting:

- Given a graph, does it have a perfect matching, i.e., a set $M$ of edges such that every vertex is incident to exactly one edge from $M$? - Given a square matrix over a finite field (regarded as a structure in the natural way, as described in [BGS02]), what is its determinant?

背景链接：http://www.openproblemgarden.org/op/fixed_point_logic_with_counting

16324 | AIM-LOGIC-0100

### Forgetting one constant and Borel completeness

P / partially_solved | AIM 研讨会问题汇编

**Question 23.** Let \(T'\) be an expansion of \(T\) by one constant. Suppose \(\cong_{T'}\) is Borel complete. Is \(\cong_T\) Borel complete?

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16325 | OWR-11695862-005

### Forking Degree and Generic Transitivity

O / open | Oberwolfach 报告问题汇编

Control $\alpha$-forking degree for arbitrary ordinals via Borovik-Cherlin-type bounds; find model-theoretic applications of generalized generic transitivity; extend Borovik-Cherlin theory to supersimple groups and Kolchin polynomials; and relate definable primitivity of binding groups to geometric stability, differential algebraic geometry, and compact complex manifolds.

来源字段：https://doi.org/10.4171/owr/2022/52

16326 | OWR-1453-003

### Formal Group Isomorphisms and Positive-Characteristic CIT

O / open | Oberwolfach 报告问题汇编

Let $(K,D)$ be a Hasse-Schmidt field with field of absolute constants $C$, let $A,B$ be commutative algebraic groups over $C$, and let $\phi:A\to B$ be a nowhere-algebraic formal isomorphism. Suppose $(a,b)\in A(K)\times B(K)$ satisfies $U_\phi(I_D^A(a))=I_D^B(b)$ and $\operatorname{trdeg}_C(a,b)\leq n$, so that $b\in B_0(K)+c$ for some proper algebraic subgroup $B_0<B$ over $C$ and some $c\in B(C)$. If $\phi^{-1}$ is also nowhere algebraic, must $a$ likewise lie in a coset of a proper algebraic subgroup of $A$? Can the special case of a formal map from the multiplicative group to an ordinary elliptic curve prove weak $\mathrm{CIT}_p$? What invariant should replace $\dim_{\mathbb Q}$ in a Schanuel-style formulation?

来源字段：https://doi.org/10.4171/owr/2007/02

16327 | AIM-LOGIC-0152

### Four core-model questions and a modular audit of branch-hypothesis counterexamples

P / partially_solved | AIM 研讨会问题汇编

Some of my goals/questions for the workshop:

A. I would like a better understanding of the relationship between models of the form \(\mathrm{HOD}^M\) and the classical \(L[\vec E]\) models.

B. Very strong axioms are used in the refutations of UBH and CBH (at least in the versions of these results which I have heard about). How exactly are they used? Are they necessary?

C. I would like to know more about Woodin's recent work on promising extender sequences.

D. I would like to get a picture of the status and significance of the \(\Omega\) conjecture.

来源字段：https://aimath.org/WWN/coremodel/coremodel.pdf

16328 | AMR-074-0100

### Friedman-Simpson Interpretability Conjecture

O / open | AMR 问题清单汇编

For any finite sets $X$ and $Y$ of published mathematical theorems expressible in second-order arithmetic, is either $\mathsf{RCA}_0+X$ interpretable in $\mathsf{RCA}_0+Y$, or $\mathsf{RCA}_0+Y$ interpretable in $\mathsf{RCA}_0+X$?

背景链接：https://math.berkeley.edu/~antonio/papers/questionsRM.pdf

16329 | AIM-LOGIC-0073

### Frobenius exceptional families and finite-subfield saturation

P / partially_solved | AIM 研讨会问题汇编

Question 13 (Vidaux). Applications of exceptional sets for B¨ uchi's problem for higher powers in characteristic p?

Question 13 (Vidaux). Applications of exceptional sets for Büchi's problem for higher powers in characteristic p?

来源字段：https://aimath.org/pastworkshops/definabilityinntproblems.pdf

16330 | AIM-LOGIC-0103

### From stable norms to stable theories

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.** Find connections between different notions of stability: quantifier-free stability, stability of structures, stability of theories. What does each of these notions imply?

**Background.** Assume \(M\) is a normed space. If the norm of \(M\) is a stable formula, then there is an isometric copy of \(\ell_p\) inside an ultrapower of \(M\) for some \(p\geq 1\) (Krivine). The group working on “Definable norms and stable Banach spaces” worked on this subject.

来源字段：https://aimath.org/WWN/continuouslogic/continuouslogic.pdf

16331 | OWR-14209-003

### Full-Dimension Intersections in Definable Families

O / open | Oberwolfach 报告问题汇编

Let $(X_a)_{a\in A}$ be a definable family of subsets of a pseudofinite set $Y$, with $\delta(X_a)=\delta(Y)$ for every $a\in A$. Under what hypotheses does generic $a$ make $X_a$ intersect every definable $B\subseteq Y$ of full $\delta$-dimension properly, in the sense that $\delta(X_a\cap B)=\delta(X_a)+\delta(B)-\delta(Y)$?

来源字段：https://doi.org/10.4171/owr/2016/1

16332 | AMR-074-0119

### Furstenberg-Zimmer Structure Theorem

O / open | AMR 问题清单汇编

Over $\mathsf{RCA}_0$, does the Furstenberg-Zimmer structure theorem imply $\Pi^1_1\text{-}\mathsf{CA}_0$?

背景链接：https://math.berkeley.edu/~antonio/papers/questionsRM.pdf

16333 | OWR-14209-001

### Generic Closedness of the Canonical $\Gamma$-Field

O / open | Oberwolfach 报告问题汇编

Let $G$ be either $\mathbb G_m$ or an abelian variety over $k_0$, set $O=\operatorname{End}(G)$, and let $C$ be the canonical $\Gamma$-field with $\Gamma(C)=\{(x,\exp_G(x)):x\in LG(C)\}$, $\Gamma(k_0)=\{0\}$, and $C_0=\Gamma\operatorname{cl}_C(k_0)$. Is $C$ generically $\Gamma$-closed? Explicitly, whenever $k\supseteq C_0$ is a finitely generated $\Gamma$-field and $V\subseteq LG^n\times G^n$ is an irreducible variety over $k$ of dimension $n$ that is not contained in a translate by $\Gamma(k)$ of a proper algebraic subgroup defined over $C_0$, and every connected proper subgroup $H<G^n$ satisfies $\dim(V/(LH\times H))>n-\dim H$ with both quotient projections of positive dimension, must $V(C)\cap\Gamma(C)^n$ contain a point whose coordinates are $O$-linearly independent over $\Gamma(k)$?

来源字段：https://doi.org/10.4171/owr/2016/1

16334 | OWR-14209-007

### Generic Transitivity and Projective Linear Actions

O / open | Oberwolfach 报告问题汇编

Let $(X,G)$ be a transitive, generically $(n+2)$-transitive permutation group of finite Morley rank, with $G$ connected and $\operatorname{rk}(X)=n$. Is $(X,G)$ equivalent to $(\mathbb P^n(K),\operatorname{PGL}_{n+1}(K))$ for some algebraically closed field $K$?

来源字段：https://doi.org/10.4171/owr/2016/1

16335 | OWR-14209-004

### Henselian Valuations on NIP Fields

O / open | Oberwolfach 报告问题汇编

Is every valued NIP field $(K,v)$ henselian? If an NIP field $K$ is neither separably closed nor real closed, does it admit a definable nontrivial henselian valuation?

来源字段：https://doi.org/10.4171/owr/2016/1

16336 | OWR-12329-005

### Higher-Arity VC-Density Bounds in NIP Theories

P / partially_solved | Oberwolfach 报告问题汇编

Assume a theory $T$ has NIP and every formula $\varphi(x;y)$ with $|x|=1$ satisfies $\operatorname{vc}^*(\varphi)\leq k$. For every $n$, does there exist a bound $k(n)$ such that $\operatorname{vc}^*(\varphi(x;y))\leq k(|x|)$ for every formula with $|x|=n$?

来源字段：https://doi.org/10.4171/owr/2013/01

16337 | OWR-15170-009

### Hilbert's Tenth Problem in Generic Rational Subrings

O / open | Oberwolfach 报告问题汇编

Does there exist an HTP-generic subring of $\mathbb Q$ with a Diophantine model of $(\mathbb Z,+,\cdot)$, and does there exist a polynomial $f$ with $\mu(B(f))>0$?

来源字段：https://doi.org/10.4171/owr/2016/49

16338 | OWR-15170-010

### HTP-Complete Polynomials over Rational Subrings

O / open | Oberwolfach 报告问题汇编

Which subrings $R_W\subset\mathbb Q$ have an HTP-complete polynomial, meaning one that uniformly encodes solvability in $R_W$ of every polynomial equation over $\mathbb Q$, and which have an effectively HTP-complete polynomial with a computable encoding map?

来源字段：https://doi.org/10.4171/owr/2016/49

16339 | AMR-075-0018

### If the class of atomic models of a complete first order theory is categorical in the $\aleph_n$, is it categorical in every cardinal

P / partially_solved | AMR 问题清单汇编

If the class of atomic models of a complete first order theory is categorical in the $\aleph_n$, is it categorical in every cardinal?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Model_theory_and_formal_languages

16340 | OWR-2654830-005

### Incompleteness and Self-Consistency Below Robinson Arithmetic

P / partially_solved | Oberwolfach 报告问题汇编

Let $R$ be Robinson’s arithmetic, and let $\mathcal D$ be the set of recursively axiomatizable consistent theories $T\lhd R$ for which $G1$ holds, meaning that every recursively axiomatizable consistent theory interpreting $T$ is incomplete. Such theories are known to exist. Is the interpretability ordering $(\mathcal D,\lhd)$ well founded? Finally, does $G2$ hold for $R$, meaning $R\nvdash\operatorname{Con}(R)$?

来源字段：https://doi.org/10.4171/owr/2020/34

16341 | AMR-074-0108

### Increasing Polarized Ramsey Theorem

P / partially_solved | AMR 问题清单汇编

Over $\mathsf{RCA}_0$, is $\mathsf{IPT}^2_2$ equivalent to $\mathsf{RT}^2_2$?

背景链接：https://math.berkeley.edu/~antonio/papers/questionsRM.pdf

16342 | OWR-15170-024

### Integer Predicates in Models of Totally Real Algebraic Integers

O / open | Oberwolfach 报告问题汇编

Let $(A,Z)$ be a ring of totally real algebraic integers with $Z$ named, and let $(R,S')$ be a model of its theory. If $(R,S)$ is the associated ultrapower model, with $S$ interpreting $Z$, must $S'\subseteq S$?

来源字段：https://doi.org/10.4171/owr/2016/49

16343 | AMR-075-0020

### Is the Borel monadic theory of the real order (BMTO) decidable? Is the monadic theory of well-ordering (MTWO) consistently decidable

P / partially_solved | AMR 问题清单汇编

Is the Borel monadic theory of the real order (BMTO) decidable? Is the monadic theory of well-ordering (MTWO) consistently decidable?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Model_theory_and_formal_languages

16344 | AMR-075-0021

### Is the theory of the field of Laurent series over $\mathbb{Z}_p$ decidable? of the field of polynomials over $\mathbb{C}$

O / open | AMR 问题清单汇编

Is the theory of the field of Laurent series over $\mathbb{Z}_p$ decidable? of the field of polynomials over $\mathbb{C}$?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Model_theory_and_formal_languages

16345 | AMR-075-0022

### Is there a logic L which satisfies both the Beth property and Δ-interpolation, is compact but does not satisfy the interpolation property

O / open | AMR 问题清单汇编

Is there a logic L which satisfies both the Beth property and Δ-interpolation, is compact but does not satisfy the interpolation property?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Model_theory_and_formal_languages

16346 | AIM-LOGIC-0110

### Joint moments as quantifier-free coordinates

P / partially_solved | AIM 研讨会问题汇编

**20. Study non commutative probability spaces.**

**Background:** The working group on Non Commutative Probability Spaces worked in this subject.

来源字段：https://aimath.org/WWN/continuouslogic/continuouslogic.pdf

16347 | AIM-LOGIC-0008

### Jump complexity of term-by-term root taking in Hahn fields

P / partially_solved | AIM 研讨会问题汇编

Let \(K\) be an algebraically closed field of characteristic 0, and let \(G\) be an ordered divisible group. Consider the Hahn field \[ K((G))=\left\{\sum_{g\in S}a_gt^g\mid S\subseteq G\text{ well-ordered},\ a_g\in K\right\}. \] If \(K\) and \(G\) are computable and \(f\in K((G))[X]\), how complex is it to determine a root of \(f\)? At each step, the procedure should produce the next term in a root of \(f\), based on the \(a_g\) in the input coefficients of \(f\). For each step \(\alpha\), how many jumps over these coefficients are needed to do step \(\alpha\)?

来源字段：http://aimpl.org/definedecide/2/

16348 | OWR-14298016-003

### Jump Property of Implication Proof Systems

O / open | Oberwolfach 报告问题汇编

For an implicit propositional proof system $P$, is the system $\operatorname{imp}(P)=[P,P]$ a jump in proof strength?

来源字段：https://doi.org/10.4171/owr/2023/53

16349 | AIM-LOGIC-0098

### Least presentation degrees of differentially closed fields

S / solved | AIM 研讨会问题汇编

**Question 20 (Harizanov), recovered.** Let \(\mathbf d\) be a Turing degree. Is there a countable model \(K\models\mathrm{DCF}_0\) having a copy computable in \(\mathbf d\), while every copy of \(K\) computes \(\mathbf d\)?

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16350 | AMR-074-0117

### Lebesgue Differentiation and Weak Weak König's Lemma

P / partially_solved | AMR 问题清单汇编

Over $\mathsf{RCA}_0$, does the Lebesgue differentiation theorem imply $\mathsf{WWKL}_0$?

背景链接：https://math.berkeley.edu/~antonio/papers/questionsRM.pdf

16351 | AMR-074-0122

### Lengths of Bounded-Rank Linear-Order WQOs

O / open | AMR 问题清单汇编

For an ordinal $\alpha$, determine the length (maximal order type) of the well-quasi-order $L_\alpha$ of countable linear orders of Hausdorff rank below $\alpha$, modulo equimorphism and ordered by embeddability.

背景链接：https://math.berkeley.edu/~antonio/papers/questionsRM.pdf

16352 | AIM-LOGIC-0043

### Linear gaps and row-threshold calibration

P / partially_solved | AIM 研讨会问题汇编

For a specifically named higher forcing axiom \(A\), determine the pairs of regular cardinals \((\kappa,\lambda)\) for which \((\mathcal P(\omega),\subseteq^*)\) has a linear \((\kappa,\lambda^*)\)-gap in models of \(A\).

来源字段：http://aimpl.org/highlowforcing/1/

16353 | AIM-LOGIC-0068

### Listable and height-thin Diophantine subsets of the rationals

P / partially_solved | AIM 研讨会问题汇编

**Question 8 (Pasten).** Do we know an r.e. subset of \(\mathbb Q\) which is not diophantine? There is some evidence that \(\mathbb Z\) is not diophantine, but we have no proof yet.

What is the “thinnest” subset of \(\mathbb Q\) known to be diophantine?

Does Bombieri--Lang imply that the diophantine subsets of \(\mathbb Q\) are either finite or have “fast” growth rates? This would give a different proof, assuming Bombieri--Lang, that \(\mathbb Z\) is not diophantine.

来源字段：https://aimath.org/pastworkshops/definabilityinntproblems.pdf

16354 | OWR-1390-003

### Logical Phase Transitions for Finite Well-Ordering Principles

O / open | Oberwolfach 报告问题汇编

Determine the provability/unprovability phase transition for the additive-norm finite-well-ordering assertions $\mathrm{FWO}_N(F)$ below $\varepsilon_0$, where $\mathrm{FWO}_N(F)$ requires a descending pair in every sufficiently long sequence whose $i$-th ordinal code has additive norm at most $K+F(i)$.

来源字段：https://doi.org/10.4171/owr/2006/52

16355 | OWR-1390-001

### Logical Phase Transitions for Graph-Minor Principles

O / open | Oberwolfach 报告问题汇编

For $f_r(i)=r\sqrt{\log i}$, let $\mathrm{GM}_{f_r}$ assert that every sufficiently long sequence of simple graphs $G_i$ with $|G_i|<K+f_r(i)$ contains $i<j$ such that $G_i$ is a minor of $G_j$. Is $\mathrm{GM}_{f_r}$ provable in $\mathrm I\Delta_0+\exp$ for $r\leq2$ and unprovable in PA for $r>2$?

来源字段：https://doi.org/10.4171/owr/2006/52

16356 | AMR-074-0004

### Martin's Conjecture on Natural Functions of Turing Degrees

P / partially_solved | AMR 问题清单汇编

Classify reasonable increasing functions on the Turing degrees; Martin's conjecture predicts that they are essentially iterates of the Turing jump.

背景链接：https://mathoverflow.net/questions/227083/what-are-some-important-but-still-unsolved-problems-in-mathematical-logic#227108

16357 | OWR-2654830-011

### Maximal Proof-Search Algorithms on Natural Tautology Sets

O / open | Oberwolfach 报告问题汇编

For a natural family of tautology test sets, does there exist a proof-search algorithm $(A,P)$ that is maximal among all proof-search algorithms under the quasi-order comparing running times up to polynomial factors?

来源字段：https://doi.org/10.4171/owr/2020/34

16358 | AIM-LOGIC-0092

### Measure transfer for the one-relator zero-one law

P / partially_solved | AIM 研讨会问题汇编

**1.7. Non-abelian free groups (Knight).** Consider \(n\)-generated groups with a single relator of length at most \(t\). (For each \(t,n\) there are finitely many such groups.) For every sentence \(\varphi\) define \[ h_{n,t,\varphi} = \frac{|\{G\in H_{n,t}:G\models\varphi\}|}{|H_{n,t}|}. \]

**Conjecture 14.** The limit \(\lim_{t\to\infty}h_{n,t,\varphi}\) exists, and always takes the value \(0\) or \(1\), moreover in a way that may depend on \(\varphi\) but not on \(n\).

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16359 | AMR-074-0115

### Metrization of Proper MF Spaces

P / partially_solved | AMR 问题清单汇编

Determine the reverse-mathematical strength of the assertion that a proper MF space is metrizable if and only if it is regular.

背景链接：https://math.berkeley.edu/~antonio/papers/questionsRM.pdf

16360 | OWR-2654830-001

### Minimal Discontinuity Under Continuous Weihrauch Reducibility

P / partially_solved | Oberwolfach 报告问题汇编

Is there a simplest discontinuous multi-valued problem $f:\subseteq\mathbb{N}^{\mathbb{N}}\Rightarrow\mathbb{N}^{\mathbb{N}}$ under continuous Weihrauch reducibility?

来源字段：https://doi.org/10.4171/owr/2020/34

16361 | OWR-12329-012

### Minimal Ordered Structures Beyond Almost Linearity

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a minimal ordered structure, in particular a minimal ordered group, that is not almost linear?

来源字段：https://doi.org/10.4171/owr/2013/01

16362 | OWR-14299908-004

### Model Companions and Geometry of Globally Valued Fields

O / open | Oberwolfach 报告问题汇编

Do existentially closed globally valued fields form a first-order class; equivalently, does the theory $\mathrm{GVF}_e$ have a model companion? Is there a Berkovich-type geometry whose closed points represent global heights on residue rings of algebras? What is the asymptotic first-order theory of number fields viewed as globally valued fields, and of globally valued fields satisfying Northcott's principle?

来源字段：https://doi.org/10.4171/owr/2026/3

16363 | OWR-15181-001

### Model Theory and Analysis of Surreal Exponential Fields

P / partially_solved | Oberwolfach 报告问题汇编

For the logarithmic-exponential series field $\mathbb{T}$ with derivation $\partial$ and exponential map, is $(\mathbb{T},\partial,\exp)$ elementarily equivalent to $(\mathbf{No},\partial,\exp)$, and can every Hardy field closed under exponentiation be embedded in it? Can sine and cosine be defined on $\mathbf{No}$, can exponentiation be extended to $\mathbf{No}[i]$, and does $(\mathbf{No},+,\times,\exp,\widehat f)_{f\in\mathbb{R}\langle\omega\rangle}$ satisfy an intermediate-value property?

来源字段：https://doi.org/10.4171/owr/2016/60

16364 | OWR-11695862-008

### Model Theory and Zilber Dichotomies for SDCF

O / open | Oberwolfach 报告问题汇编

Determine further model-theoretic properties of separably differentially closed fields, describe types of $\operatorname{SDCF}_{\ell,p,\epsilon}$ using separable ideals in differential polynomial rings, and establish an appropriate Zilber dichotomy.

来源字段：https://doi.org/10.4171/owr/2022/52

16365 | OWR-15181-020

### Model Theory of Residue Exponential Fields

O / open | Oberwolfach 报告问题汇编

For every o-minimal ordered exponential field $K$ satisfying EXP, is its residue exponential field $K_{\mathrm{res}}$ elementarily equivalent to $K$?

来源字段：https://doi.org/10.4171/owr/2016/60

16366 | OWR-15430-002

### Model-Theoretic Advantages of Logarithmic-Derivative Transforms

O / open | Oberwolfach 报告问题汇编

For an expansion of the real field by a real entire function $F$, what technical and model-theoretic advantages arise from studying

$$G(z)=\left(z\frac{F'(z)}{F(z)}\right)'$$

instead of $F$ itself?

来源字段：https://doi.org/10.4171/owr/2017/22

16367 | OWR-12329-009

### Model-Theoretic Connected-Component Quotients

O / open | Oberwolfach 报告问题汇编

Let $G$ be a group definable in a first-order structure $M$, let $M^*$ be a saturated elementary extension, and for a small parameter set $A$ let $G_A^{*00}$ and $G_A^{*000}$ denote the model-theoretic connected components of $G(M^*)$ (the smallest $A$-type-definable subgroup of bounded index and the smallest $A$-invariant subgroup of bounded index, respectively). Which compact/quotient groups can occur as the quotient $G_A^{*00} / G_A^{*000}$?

来源字段：https://doi.org/10.4171/owr/2013/01

16368 | OPG-37440

### Monadic second-order logic with cardinality predicates

P / partially_solved | Open Problem Garden

The problem concerns the extension of Monadic Second Order Logic (over a binary relation representing the edge relation) with the following atomic formulas:

- $\text{"}\,\mathrm{Card}(X) = \mathrm{Card}(Y)\,\text{"}$ - $\text{"}\,\mathrm{Card}(X) \text{ belongs to } A\,\text{"}$

where $A$ is a fixed recursive set of integers.

Let us fix $k$ and a closed formula $F$ in this language.

Conjecture Is it true that the validity of $F$ for a graph $G$ of tree-width at most $k$ can be tested in polynomial time in the size of $G$?

背景链接：http://www.openproblemgarden.org/op/monadic_second_order_logic_with_cardinality_predicates

16369 | OWR-15430-011

### Monotone Cylindrical Decompositions of Compact Sets

O / open | Oberwolfach 报告问题汇编

For a compact set $K\subset\mathbb R^n$ and a function $f:K\to\mathbb R$, does there exist a partition of $K$ into monotone cylindrical cells, possibly using different coordinate orders, such that $f$ restricted to every cell is monotone?

来源字段：https://doi.org/10.4171/owr/2017/22

16370 | OWR-1453-011

### Mordell-Lang for Semiabelian Varieties in Positive Characteristic

O / open | Oberwolfach 报告问题汇编

Let $L$ be an algebraically closed field of characteristic $p>0$, let $G$ be a semiabelian variety over $L$, let $X\subset G$ be an irreducible subvariety, and let $\Lambda\leq G(L)$ be a finite-rank subgroup. If $X(L)\cap\Lambda$ is Zariski dense in $X$, must $X$ be special, meaning that $X=g+h^{-1}(X_0)$ for some $g\in G(L)$, a surjective homomorphism $h:G'\to G_0$ from an algebraic subgroup $G'$ of $G$ to a semiabelian variety $G_0$ defined over $\overline{\mathbb F}_p$, and a subvariety $X_0\subset G_0$ defined over $\overline{\mathbb F}_p$?

来源字段：https://doi.org/10.4171/owr/2007/02

16371 | OPG-37448

### MSO alternation hierarchy over pictures

O / open | Open Problem Garden

Question Is the MSO-alternation hierarchy strict for pictures that are balanced, in the sense that the width and the length are polynomially (or linearly) related.

背景链接：http://www.openproblemgarden.org/op/mso_alternation_hierarchy_over_pictures

16372 | AIM-LOGIC-0024

### Mutual stationarity, forcing axioms, and universally Baire absoluteness

P / partially_solved | AIM 研讨会问题汇编

The exact repository `problem` field is:

The exact repository `remarks` entry is:

Thus the printed sentence is ungrammatical. The following is the **recovered statement**, with the single insertion marked by brackets.

Is it consistent that for every sequence \[ \langle S_n\mid n\in\omega\rangle, \qquad S_n\subseteq\aleph_{n+2}\cap\operatorname{cof}(\omega_1), \] with each \(S_n\) **[stationary]**, the sequence is mutually stationary?

The insertion is supported in two independent ways. First, Proposition 5.1 below proves that mutual stationarity forces every component to be stationary, so without the insertion the assertion is immediately false (take an empty component). Second, Ben-Neria restates exactly this question, with indices \(2\leq m<\omega\), and explicitly says “stationary sets.”

The PDF records a Koepke--Welch lower bound: the assertion yields an inner model with infinitely many cardinals \(\kappa_n\) such that, for \(n>m\), the class of measurable \(\lambda<\kappa_n\) of Mitchell order at least \(\kappa_m\) is stationary in \(\kappa_n\) (the workshop shorthand says “stationary in \(V\)”).

Is \(\mathsf{MM}(\mathfrak c^+)\) consistent with Woodin's axiom \((*)\)? Here \(\mathsf{MM}(\kappa)\) is the restriction of Martin's Maximum to stationary-set-preserving partial orders of size at most \(\kappa\). The queried axiom is \(\mathsf{MM}(\mathfrak c^+)\), not \(\mathsf{MM}^{++}\).

Assume a proper class of Woodin cardinals and \(\mathsf{MM}^{++}\). Suppose that for every forcing \(\mathbb P\), \[ \operatorname{Th}\bigl(L(\Gamma_{\mathrm{uB}})\bigr) = \operatorname{Th}\bigl(L(\Gamma_{\mathrm{uB}})\bigr)^{V^{\mathbb P}}, \] in a language with a constant symbol for every ground-model universally Baire set (interpreted canonically in the extension). Does this imply \[ \operatorname{cf}(\theta_{\mathrm{uB}})>\aleph_1? \] Here \(\Gamma_{\mathrm{uB}}\), also denoted \(\Gamma^\infty\), is the collection of universally Baire sets, and \(\theta_{\mathrm{uB}}\) denotes the length/supremum of their Wadge hierarchy in the workshop notation. The PDF reports that, assuming in addition weak UBH, the desired cofinality statement is equivalent to the existence of a semiproper forcing adding a universally Baire set strictly Wadge-above every ground-model universally Baire set.

Remark 1 uses \((*)^+\) for the assertion that every \(A\subseteq\mathbb R\) is captured in a \(\mathbb P_{\max}\)-generic extension \(M[g]\) of an \(\mathsf{AD}^+\)-model \(M\supseteq\mathbb R\). The displayed \((*)^{++}\) strengthens the requirement on this \(M\) to \(M\models\mathsf{AD}_{\mathbb R}+\)“\(\Theta\) is regular.” The workshop notes report \[ \mathsf{MM}^{++}+(*)^{++}\Longrightarrow\theta_{\mathrm{uB}}=\omega_3. \]

来源字段：https://aimath.org/pastworkshops/innermodelproblems.pdf

16373 | AIM-LOGIC-0035

### Namba forcing, side conditions, and the proper-master barrier

P / partially_solved | AIM 研讨会问题汇编

Can Namba forcing be iterated with side conditions? Does this work if we replace Namba forcing with a forcing satisfying Shelah's \(S\)-condition?

来源字段：http://aimpl.org/highlowforcing/1/

16374 | AIM-LOGIC-0123

### Native DPRM over the integers and signed finite-free transfer

P / partially_solved | AIM 研讨会问题汇编

Question 10 (Davis). Find a native proof of DPRM in Z, instead of referring to N.Prove DPRM for some class of rings abstractly, with no reference to N.

**Question 10 (Davis).** Find a native proof of DPRM in \(\mathbb Z\), instead of referring to \(\mathbb N\). Prove DPRM for some class of rings abstractly, with no reference to \(\mathbb N\).

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16375 | OWR-12329-007

### Nearly-Lie Quotients of Automorphism Groups in NIP Theories

P / partially_solved | Oberwolfach 报告问题汇编

Given a saturated model $M$ of an NIP theory $T$ and a formula $\varphi(x,y)$, does there exist a homomorphism $h:\operatorname{Aut}(M)\to G$ into a nearly Lie group $G$ whose kernel fixes every $\varphi$-type over $M$ that does not fork over $\varnothing$?

来源字段：https://doi.org/10.4171/owr/2013/01

16376 | AIM-LOGIC-0054

### Necessary arithmetic and forcing conditions for tree property plus saturation at aleph_2

P / partially_solved | AIM 研讨会问题汇编

**Recovered intended question.** Is there a model in which $TP_{\omega_2}$ holds and there is a proper normal ideal $I$ on $\omega_2$, extending the bounded ideal (hence uniform), such that $\mathcal P(\omega_2)/I$ has the $\omega_3$-chain condition?

来源字段：http://aimpl.org/highlowforcing/2/

16377 | OWR-12697685-010

### NIP Conjecture for k-Dependent Fields

O / open | Oberwolfach 报告问题汇编

Is every $k$-dependent field, whether pure or equipped with valuation, derivation, or similar structure, NIP?

来源字段：https://doi.org/10.4171/owr/2023/2

16378 | OWR-14209-005

### NIP Preservation under Henselian Valuations

S / solved | Oberwolfach 报告问题汇编

If $K$ is an NIP field and $v$ is a henselian valuation on $K$, must the valued field $(K,v)$ be NIP?

来源字段：https://doi.org/10.4171/owr/2016/1

16379 | OWR-11695862-003

### Noiseless Open Cores and Tame Real-Field Expansions

O / open | Oberwolfach 报告问题汇编

Prove that every expansion of the real field that does not define $\mathbb Z$ has a noiseless open core, and classify such expansions under a common notion of tameness.

来源字段：https://doi.org/10.4171/owr/2022/52

16380 | AIM-LOGIC-0120

### One-equation reductions for the rational Hamilton quaternions and Lipschitz order

P / partially_solved | AIM 研讨会问题汇编

\[ H=(-1,-1)_{\mathbb Q} =\mathbb Q\oplus i\mathbb Q\oplus j\mathbb Q\oplus k\mathbb Q, \qquad i^2=j^2=k^2=ijk=-1, \]

be the Hamilton quaternion division algebra over \(\mathbb Q\), and let

\[ O=\mathbb Z+i\mathbb Z+j\mathbb Z+k\mathbb Z \]

be its Lipschitz order. Is there an algorithm deciding whether one noncommutative polynomial equation has a solution in each of the following four coefficient/domain combinations?

1. coefficients in \(\mathbb Q\), solutions in \(H\); 2. coefficients in \(\mathbb Q\), solutions in \(O\); 3. coefficients in \(H\), solutions in \(H\); 4. coefficients in \(H\), solutions in \(O\).

The note then asks whether \(\mathbb Z\) is existentially definable in \(O\) and proposes

\[ x\in\mathbb Z\quad\Longleftrightarrow\quad (\exists I,J,K)\bigl(I^2=-1\wedge J^2=-1\wedge IJ=-JI \wedge xI=Ix\wedge xJ=Jx\bigr). \tag{1.1} \]

It says that this was “very likely done by D. Tunc,” suggests that the analogous definition of \(\mathbb Q\) in \(H\) makes (1) and (3) equivalent to Hilbert's tenth problem over \(\mathbb Q\), and asks the same questions for \(M_n(\mathbb Z)\) and \(M_n(\mathbb Q)\).

The expression after part (2), `PROBLEMS RELATED TO ... 3`, is a page header; the displayed `I2` and `J2` are \(I^2\) and \(J^2\); and `Mn` is \(M_n\). The PDF itself spells the name “D. Tunc.” The variable \(K\) really is quantified but is unused.

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16381 | AIM-LOGIC-0085

### One-jump selection and a pure-equality obstruction for the definition map

P / partially_solved | AIM 研讨会问题汇编

Question 7. How complicated is the map ϕ 7 → dpϕ (possibly with uniformity in p)?

**Question 7.** How complicated is the map \[ \varphi\longmapsto d_p\varphi \] (possibly with uniformity in \(p\))?

Inspection of the original AIM PDF shows that the intended typography is

The question occurs in the section “Continuous sections (Miller).” Immediately before Questions 6-8, the source says that for a stable theory \(T\) and \(M\preccurlyeq N\models T\), restriction \(S_1(N)\to S_1(M)\) has a continuous section sending \(p\) to its unique nonforking extension.

For a formula \(\varphi(x;y)\) and \(p\in S_x(M)\), a \(\varphi\)-definition of \(p\) is an \(M\)-formula \(d_p\varphi(y)\) such that

\[ M\models d_p\varphi(b) \quad\Longleftrightarrow\quad \varphi(x;b)\in p \qquad(b\in M^{|y|}). \]

1. how \(M\) and \(p\) are represented; 2. whether the output is the semantic trace \(b\mapsto\mathbf 1_{\varphi(x;b)\in p}\) or a finite formula code defining that trace; 3. whether a fixed \(p\) is allowed as an oracle, or one functional must work uniformly for all \(p\).

The results below use the standard oracle presentation: formulas with parameters from a countable \(M\) are effectively coded, \(p\) is given by its characteristic function on such formulas, and the elementary diagram \(\operatorname{ElDiag}(M)\) is available as an oracle.

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16382 | AMR-074-0114

### One-Point Compactification for MF Spaces

O / open | AMR 问题清单汇编

Determine the reverse-mathematical strength of Alexandroff's one-point compactification theorem for countably based MF spaces.

背景链接：https://math.berkeley.edu/~antonio/papers/questionsRM.pdf

16383 | AIM-LOGIC-0061

### One-universal definitions of integers in number fields

P / partially_solved | AIM 研讨会问题汇编

Let \(t\) be the free variable and \(u\) the single universal variable. For a number field \(K\): (a) is \(\mathcal O_K\) definable in \(K\) by a formula \(\forall u\,\exists\vec y\,\Theta_K(t,u,\vec y)\)? (b) is \(\mathbb Z\) definable in \(K\) by such a formula? (c) is \(\mathbb Z\) definable in \(K\) by a purely universal formula \(\phi_K(t)\)?

来源字段：https://aimath.org/pastworkshops/definabilityinntproblems.pdf

16384 | AMR-074-0204

### Open Mapping Theorem for Separable Banach Spaces

O / open | AMR 问题清单汇编

Is the open mapping theorem for separable Banach spaces provable in $\mathsf{RCA}_0$, or at least in $\mathsf{WKL}_0$?

背景链接：https://sgslogic.net/t20/talks/cta/problems/

16385 | AMR-150-0001

### Open Problems in Mathematical Logic - Problem 1 (Levin, private communications)

O / open | AMR 问题清单汇编

Are the following true?

(1) Can every chain of random reals in the Turing degrees be combined into a single random real?

(2) Can every Turing ideal generated by random reals be generated by the columns of a single random real?

背景链接：https://arxiv.org/abs/2608.26628

16386 | AMR-150-0010

### Open Problems in Mathematical Logic - Problem 10

O / open | AMR 问题清单汇编

Is there a decision procedure which, given quantifier patterns \[ \overline{P},\overline{Q} \in \{\exists,\forall,\exists^\infty,\forall^\infty\}^{<\omega}, \] decides whether \[ \langle\overline{P}\rangle \leq_m \langle\overline{Q}\rangle? \]

背景链接：https://arxiv.org/abs/2608.26628

16387 | AMR-150-0011

### Open Problems in Mathematical Logic - Problem 11

O / open | AMR 问题清单汇编

A set $A\subseteq\mathbb{R}^2$ is called a *Kakeya set* if, for every $\alpha\in\mathbb{R}$, there exists $C_\alpha\in\mathbb{R}$ such that \[ y=\alpha x+C_\alpha \quad\Longrightarrow\quad (x,y)\in A \] for every $(x,y)\in\mathbb{R}^2$. Equivalently, \[ \bigl\{(x,y)\in\mathbb{R}^2:y=\alpha x+C_\alpha\bigr\} \subseteq A \] for every $\alpha\in\mathbb{R}$.

Is there a Kakeya set $A\subseteq\mathbb{R}^2$ such that \[ x\oplus y\geq_T\emptyset' \] for every $(x,y)\in A$?

背景链接：https://arxiv.org/abs/2608.26628

16388 | AMR-150-0012

### Open Problems in Mathematical Logic - Problem 12

O / open | AMR 问题清单汇编

A Kakeya set $A$ is called *minimal* if every proper subset $B\subsetneq A$ is not a Kakeya set.

Does every Borel Kakeya set $A$ contain a minimal Kakeya set $B\subseteq A$?

背景链接：https://arxiv.org/abs/2608.26628

16389 | AMR-150-0013

### Open Problems in Mathematical Logic - Problem 13

O / open | AMR 问题清单汇编

Since every $F_\sigma$ set is Borel, Larman's theorem rules out the lowest nontrivial possibility in the Borel hierarchy, but it does not settle the following problem.

Is there a Borel two-point set?

背景链接：https://arxiv.org/abs/2608.26628

16390 | AMR-150-0014

### Open Problems in Mathematical Logic - Problem 14

O / open | AMR 问题清单汇编

\begin{definition} Let $F$ be a Borel field. A *Borel algebraically closed extension* of $F$ is a Borel field $K$, together with a Borel injective field homomorphism \[ \iota:F\hookrightarrow K, \] such that $K$ is algebraically closed. \end{definition}

[Montalb\'an--Nies {[MontalbanNies2013, Question 2.8]}] Does every Borel field have a Borel algebraically closed extension?

背景链接：https://arxiv.org/abs/2608.26628

16391 | AMR-150-0015

### Open Problems in Mathematical Logic - Problem 15

O / open | AMR 问题清单汇编

However, for non-well-ordered cardinals, the situation deteriorates quickly. The sequel is the following. For such a theory $T$, we want to show it is very close to a stongly minimal $T'$. It is like a vector space over fixed field or algebraically closed field. And since there is no well order, it is hard to take a basis and match them to one another. So there is the question.

For vector space over fixed $\mathbb{K}$, is choose a basis require the full Axiom of Choice?

背景链接：https://arxiv.org/abs/2608.26628

16392 | AMR-150-0016

### Open Problems in Mathematical Logic - Problem 16

O / open | AMR 问题清单汇编

In 1954, [Mar] shows that for $E\subseteq \mathbb{R}^2$ an analytic set with Hausdorff dimension 1. Then, for almost all $\theta$, the projection of $E$ onto a line with slope $\theta$ has Hausdorff dimension 1. Later in 1969, [Dav] shows that CH could imply there is some $E\subseteq \mathbb{R}^2$ with Hausdorff dimension 1 and for all $\theta$, projection of $E$ onto a line of slope $\theta$ has Hausdorff dimension 0.

Is there a Davies example provably in ZFC?

背景链接：https://arxiv.org/abs/2608.26628

16393 | AMR-150-0017

### Open Problems in Mathematical Logic - Problem 17

O / open | AMR 问题清单汇编

Find more models $M$ and forcing notions $\mathbb{P}\in M$ such that $(*)$ is true.

背景链接：https://arxiv.org/abs/2608.26628

16394 | AMR-150-0018

### Open Problems in Mathematical Logic - Problem 18

O / open | AMR 问题清单汇编

Suppose there are a model $M$ and a forcing notion $\mathbb{P}\in M$ such that $(*)$ is true. Can we deduce that $M\models [\exists\ \text{a large cardinal}]$?

背景链接：https://arxiv.org/abs/2608.26628

16395 | AMR-150-0019

### Open Problems in Mathematical Logic - Problem 19

O / open | AMR 问题清单汇编

\quad If $\mathbf{a}<\mathbf{b}<\mathbf{c}$ are Turing degrees, and $\mathbf{a}$, $\mathbf{c}$ contain different Martin-L\"of randoms, then must $\mathbf{b}$ contain one? The answer is no.

Does this hold with a larger degree notion, for example, LR-degrees?

背景链接：https://arxiv.org/abs/2608.26628

16396 | AMR-150-0002

### Open Problems in Mathematical Logic - Problem 2 (Levin)

O / open | AMR 问题清单汇编

Is it true that, for every $X\subseteq 2^\omega$ with $\mu(X)=1$, there exists $Y\subseteq 2^\omega$ with $\mu(Y)=1$ such that, for every chain $B\subseteq Y$, there is some $\alpha\in X$ satisfying $[B]=[\alpha]$?

背景链接：https://arxiv.org/abs/2608.26628

16397 | AMR-150-0020

### Open Problems in Mathematical Logic - Problem 20

O / open | AMR 问题清单汇编

If $L_\beta\models\mbox{$\mathit{V}=\mathit{HC}$}$, then \begin{align*} \left(\mathcal{P}(\omega)\cap L_\beta\models\mathbf{s}ymbol{\Pi}_{m+1}^1\mathsf{-MI}\right)\iff\beta\mbox{ supports a strong }\Sigma_{m+1}\textrm{-nesting}. \end{align*}

背景链接：https://arxiv.org/abs/2608.26628

16398 | AMR-150-0021

### Open Problems in Mathematical Logic - Problem 21

O / open | AMR 问题清单汇编

Also note the following fact: \begin{fact} (1) There is a complete theory $T$ such that $=$ is Borel bi-reducible to $\cong\restriction_{\mathrm{Mod}(T)}$. For example, $T$ can be chosen as the theory of the equivalence relation on $\omega$ with infinitely many equivalence classes of any finite size.\\ (2) There is a complete theory $T'$ such that $=^+$ is Borel bi-reducible to $\cong\restriction_{\mathrm{Mod}(T')}$. For example, $T'$ can be chosen as the theory of infinitely many independent equivalence relations on $\omega$ and every equivalence relation has infinitely many equivalence classes of any finite size. \end{fact} So we have the following natural question:

Does there exist a complete first-order theory $T$ in a countable language such that $=\ <_B\ \cong\restriction_{\mathrm{Mod}(T)}\ <_B\ =^+$?

背景链接：https://arxiv.org/abs/2608.26628

16399 | AMR-150-0022

### Open Problems in Mathematical Logic - Problem 22

O / open | AMR 问题清单汇编

Maximal cofinitary groups have been studied by Bart Kastermans in [Kas]. It is known that maximal cofinitary groups cannot be $K_\sigma$, abelian, and there is a coanalytic maximal cofinitary group if assuming the axiom of constructibility. So we can ask similar questions for maximal finitely periodic groups:

Is there a countable or $K_\sigma$ or just Borel maximal finitely periodic group?

背景链接：https://arxiv.org/abs/2608.26628

16400 | AMR-150-0023

### Open Problems in Mathematical Logic - Problem 23

O / open | AMR 问题清单汇编

Natural questions about $A \mapsto I(A)$:

- Does the range of $I$ have the same cardinality as $2^{\mathbb{R}}$? (Seems yes under GCH.)

- If we consider the closed set $C$, $I(C)$ is arithmetic ($\mathbf{\Sigma}_2^0$). What's the complexity of the set $$\big\{\mathbf{\Sigma}_2^0 \text{ formula } \varphi : \varphi \text{ defines an } I(C) \text{ for some } C\big\}?$$ Is it $\mathbf{\Sigma}_2^1$-complete? (It is $\mathbf{\Pi}_1^1$-hard.)

- Is the answer to "Borel hierarchy is proper w.r.t. the range of $I$" independent of ZFC?

- Is it consistent with ZFC that for every $f$ and $A\subseteq \mathbb{R}$, if $H^f(A)>0$, then there is a subset $A_0$ such that $H^f(A_0)$ is finite and positive?

背景链接：https://arxiv.org/abs/2608.26628

16401 | AMR-150-0024

### Open Problems in Mathematical Logic - Problem 24

O / open | AMR 问题清单汇编

Let $L_x=\{z : z \leq_K^+ x\}$ and $U_x=\{z : z \geq_K x\}$. Are the following true?

(1) $L_x$ is countable iff $\liminf_{n\to\infty}(K(x\upharpoonright n)-K(n))<\infty$;

(2) $U_x$ is countable iff $\liminf_{n\to\infty}(K(n)+n-K(x\upharpoonright n))<\infty$.

背景链接：https://arxiv.org/abs/2608.26628

16402 | AMR-150-0025

### Open Problems in Mathematical Logic - Problem 25

O / open | AMR 问题清单汇编

Are the following true?

(1) For any c.e. set $A$, there exists a c.e. set $D\subseteq 2\mathbb{N}$, such that $A\equiv_{rK}D$ and $C(A\upharpoonright n|D\upharpoonright n)=C(D\upharpoonright n|A\upharpoonright n)=O(1)$;

(2) For all $x$, there is a random $z$ such that $z \geq_{rK} x$ (or $z \geq_K x$).

背景链接：https://arxiv.org/abs/2608.26628

16403 | AMR-150-0026

### Open Problems in Mathematical Logic - Problem 26

O / open | AMR 问题清单汇编

It was shown in [Ba] that for $k=2,3$, Player 1 has a winning strategy in the game $G_k$. For $k\geq 4$, this is unknown.

For $k\geq 4$, does Player 1 have a winning strategy in the game $G_k$?

背景链接：https://arxiv.org/abs/2608.26628

16404 | AMR-150-0027

### Open Problems in Mathematical Logic - Problem 27

O / open | AMR 问题清单汇编

Using recursion theory, Goh proved that any coanalytic separator of **WF** and **UB** is coanalytic complete. The method also proves Saint Raymond's theorem.

Can this be proved using Descriptive Set Theoretic methods?

背景链接：https://arxiv.org/abs/2608.26628

16405 | AMR-150-0028

### Open Problems in Mathematical Logic - Problem 28 (Downey--Hirschfeldt--Miller--Nies [DHMN])

O / open | AMR 问题清单汇编

Is there an $x$ such that $\Omega^x \geq_T 0^{''}$ where $$\Omega^x = \sum_{\sigma \in 2^{<\omega}} 2^{-K^x(\sigma)}$$ and $$K^x(\sigma) = \min\{|\tau| : U^x(\tau) = \sigma\}?$$

背景链接：https://arxiv.org/abs/2608.26628

16406 | AMR-150-0029

### Open Problems in Mathematical Logic - Problem 29

O / open | AMR 问题清单汇编

- Is there a recursion theoretical proof for the above theorem of Velickovic--Woodin?

- Is the following true: For any $\Sigma_1^1(y)$ set $A$, if there is $x \in A$ such that $\omega_1^{CK(x)} > \omega_1^{CK(y)}$, then the Velickovic--Woodin theorem holds for $A$?

背景链接：https://arxiv.org/abs/2608.26628

16407 | AMR-150-0003

### Open Problems in Mathematical Logic - Problem 3

O / open | AMR 问题清单汇编

The class $\mathrm{ML}(\emptyset')$ of $\emptyset'$-random reals, equivalently the class of $2$-random reals, is a special case.

Is there a chain $B\subseteq\mathrm{ML}(\emptyset')$ such that $[B]\neq[\alpha]$ for every $\alpha\in\mathrm{ML}(\emptyset')$?

背景链接：https://arxiv.org/abs/2608.26628

16408 | AMR-150-0030

### Open Problems in Mathematical Logic - Problem 30

O / open | AMR 问题清单汇编

$P^-$: For every positive tree there are distinct $(X_n : n \in \omega)$ such that $X_n \in [T]$. \end{quote} It is known that $$\text{WKL}_0 \vdash P^+$$ and $$\text{RCA}_0 \vdash P^+ \to P \to P^- \to \text{1-RAN} \leftrightarrow \text{WWKL}_0,$$ where $\text{1-RAN}$ means there exists a 1-random. But we also know that $$P^+ \nvdash \text{WKL}_0,$$ $$\text{RCA}_0 \nvdash P \to P^+,$$ and $$\text{RCA}_0 \nvdash \text{1-RAN} \to P^-.$$

The questions are:

Is it true that $\text{RCA}_0 \vdash P^- \to P$?

背景链接：https://arxiv.org/abs/2608.26628

16409 | AMR-150-0031

### Open Problems in Mathematical Logic - Problem 31

O / open | AMR 问题清单汇编

How about using dimension? (Define the above concepts by replacing measure by dimension.)

背景链接：https://arxiv.org/abs/2608.26628

16410 | AMR-150-0032

### Open Problems in Mathematical Logic - Problem 32

O / open | AMR 问题清单汇编

It is known that $$\text{I}\Sigma_n + \text{WPHP}(\Sigma_{n+1}) + \text{FRT}^e_k(\Sigma_{n+1})(e,k \text{ standard}) \nvdash \text{B}\Sigma_{n+1}$$ and $$\text{FRT}^e_k(\Sigma_{n+1}) \nvdash \text{WPHP}(\Sigma_{n+1}).$$

The questions are:

$\text{I}\Sigma_{n+1} + \text{WPHP}(\Sigma_{n+1}) \vdash \text{FRT}_2^2(\Sigma_{n+1})$? $\text{FRT}_2^2(\Sigma_{n+1}) \vdash \text{FRT}_2^3(\Sigma_{n+1})$?

背景链接：https://arxiv.org/abs/2608.26628

16411 | AMR-150-0033

### Open Problems in Mathematical Logic - Problem 33

O / open | AMR 问题清单汇编

The first-order theory \text{2-RAN} is between $\text{CARD}(\Sigma_2)$ and $\text{WPHP}(\Sigma_2)$, but how about $\text{GPHP}(\Sigma_2)$ and \text{2-RAN}?

背景链接：https://arxiv.org/abs/2608.26628

16412 | AMR-150-0034

### Open Problems in Mathematical Logic - Problem 34

O / open | AMR 问题清单汇编

- Can such a group $L$ have unsolvable word problem?

- Is there a f.g., residually finite $\Pi$-group that is not effectively r.f.?

- Morozov (Higman's question revisited, 2000) constructed a $\Pi$-group that is not isomorphic to a subgroup of $S_{\text{rec}}$. Can we make such a group r.f.? It can't be e.r.f.

背景链接：https://arxiv.org/abs/2608.26628

16413 | AMR-150-0035

### Open Problems in Mathematical Logic - Problem 35

O / open | AMR 问题清单汇编

For each left $\Sigma_2$ real $r \in [0, 1]$ there is a computable profinite group and computable subgroup with Hausdorff dimension $r$.

Which values can occur for $r$ when $G$ is also topologically finitely generated?

背景链接：https://arxiv.org/abs/2608.26628

16414 | AMR-150-0036

### Open Problems in Mathematical Logic - Problem 36

O / open | AMR 问题清单汇编

Nies, Segal, and Tent 2021 [NST] studied expressiveness of f.o. logic for profinite group. Many such groups are axiomatised by single sentence in the language of groups, among this reference class.

For a f.o. sentence $\varphi$, what is the possible complexity of $\{G : G \models \varphi\}$?

背景链接：https://arxiv.org/abs/2608.26628

16415 | AMR-150-0037

### Open Problems in Mathematical Logic - Problem 37

O / open | AMR 问题清单汇编

The following problem was also posed during the workshop, and by the end of the workshop Gao and Nies realized that the answer is yes.

Is there some Borel class of profinite groups with isomorphism relation properly between $S_\infty$-complete and smooth? In particular, how about the Abelian case?

背景链接：https://arxiv.org/abs/2608.26628

16416 | AMR-150-0038

### Open Problems in Mathematical Logic - Problem 38

O / open | AMR 问题清单汇编

Nies, Schlicht, and Tent proved in 2019 [NST2] that the topological isomorphism of oligomorphic groups is $\leq_B E_\infty$. Is it smooth?

背景链接：https://arxiv.org/abs/2608.26628

16417 | AMR-150-0039

### Open Problems in Mathematical Logic - Problem 39

O / open | AMR 问题清单汇编

If $G$ is oligomorphic, is $\text{Out}(G)$ always profinite?

背景链接：https://arxiv.org/abs/2608.26628

16418 | AMR-150-0004

### Open Problems in Mathematical Logic - Problem 4

O / open | AMR 问题清单汇编

Toward a negative answer to Levin's Problem 1.2, are the following true?

(1) Are there $2$-random reals $x<_T z$ such that \[ x'\equiv_T z'? \]

(2) Are there $2$-random reals $x<_T z$ such that \[ \left| K(x\upharpoonright n)-K(z\upharpoonright n) \right| = O(1)? \]

(3) Is there a Turing ideal $B$ generated by a chain of $2$-random reals such that $B$ contains no $x$-random real for any $x\in B$?

(4) What are the jumps of $2$-random reals? See [BDN].

背景链接：https://arxiv.org/abs/2608.26628

16419 | AMR-150-0040

### Open Problems in Mathematical Logic - Problem 40

O / open | AMR 问题清单汇编

Is there a Borel, invariant class $C$ of closed subgroups of $S_\infty$ with isomorphism relation not Borel below graph isomorphism? Analytic complete? How about the class $C$ of pro-countable groups?

背景链接：https://arxiv.org/abs/2608.26628

16420 | AMR-150-0041

### Open Problems in Mathematical Logic - Problem 41

O / open | AMR 问题清单汇编

In [LMNS, Section 3.1] the authors showed the following implications $$\text{1-generic } \Delta^0_2 \Rightarrow \text{ index guessable } \Rightarrow \text{ computes no maximal tower } \Rightarrow \text{ low }$$

In the above diagram, are the first and the second implications reversible?

背景链接：https://arxiv.org/abs/2608.26628

16421 | AMR-150-0042

### Open Problems in Mathematical Logic - Problem 42

O / open | AMR 问题清单汇编

Every degree of decidable categoricity is a degree of categoricity. Is the converse true?

A conjecture for this problem is that it is true for $d \geq 0^{(\alpha)}$, where $\alpha$ is 'sufficiently large' computable ordinal. Say, for $\alpha = \omega$. What happens with d.c.e. degrees $d$?

背景链接：https://arxiv.org/abs/2608.26628

16422 | AMR-150-0043

### Open Problems in Mathematical Logic - Problem 43

O / open | AMR 问题清单汇编

Ceer stands for computably enumerable equivalence relation on $\omega$. A diagonal function (or a fixed-point-free function) for a ceer $E$ is a total function $g(x)$ satisfying $\neg g(x) E x$ for all $x \in \omega$.

The following are three versions of this problem.

- Describe the class **Diag** containing those Turing degrees $d$ such that every ceer $E \neq \text{Id}_1$ admits a $d$-computable diagonal function. Here we have some partial results:

- $\text{**PA**} \subseteq \text{**Diag**} \subseteq \text{**DNR**}$ [Badaev, Bazhenov, Kalmurzayev, Mustafa 2024].

- There exists a Martin-Löf random $x$ such that $x \notin \text{**Diag**}$ [Ng].

- What is the reverse-mathematical strength of the statement “Every $\Sigma_1^0$-definable equivalence relation $E \neq \text{Id}_1$ has a diagonal function”?

- What is the Weihrauch degree of the corresponding problem?

背景链接：https://arxiv.org/abs/2608.26628

16423 | AMR-150-0044

### Open Problems in Mathematical Logic - Problem 44

O / open | AMR 问题清单汇编

Does every (effectively) compact left-c.e. Polish space have a computable (right c.e.) Polish copy? Related to this problem, Melnikov and Ng proved that there is a left c.e. Polish space with no computable Polish presentation. Similarly, Koh, Melnikov, and Ng proved that there is a left c.e. Polish space with no right c.e. Polish presentation. As a positive result, Melnikov and Ng proved that every left c.e. Stone space is homeomorphic to a computable Polish space.

背景链接：https://arxiv.org/abs/2608.26628

16424 | AMR-150-0045

### Open Problems in Mathematical Logic - Problem 45

O / open | AMR 问题清单汇编

A computable Polish space $X$ is $\alpha$-categorical if for every pair of computable metric spaces $M$, $N$ such that $\bar{M} \cong \bar{N} \cong X$, there is an $\alpha$-computable homeomorphism between $\bar{M}$ and $\bar{N}$. The least Turing degree $\alpha$ is the degree of (topological) categoricity of $X$. $X$ is (topologically) computably categorical if it is $0$-categorical.

Is there an infinite computable Polish space that is topologically computably categorical?

背景链接：https://arxiv.org/abs/2608.26628

16425 | AMR-150-0046

### Open Problems in Mathematical Logic - Problem 46

O / open | AMR 问题清单汇编

Is every/any c.e. degree the degree of topological categoricity?

背景链接：https://arxiv.org/abs/2608.26628

16426 | AMR-150-0047

### Open Problems in Mathematical Logic - Problem 47

O / open | AMR 问题清单汇编

\abstractbody{ Let $M$ be a model satisfying $\text{RCA}_0 + \text{B}\Sigma_n + \neg \text{I}\Sigma_n$. Fix a $\Pi_n$ subset $X$ of $M$ and assume $\langle A_s : s \in M \rangle$ is a sequence of pairwise disjoint $M$-finite sets such that $\forall s (A_s \cap X \neq \emptyset)$.

Is there a definable in $M$ choice function $f$ of $\langle A_s \rangle$, i.e., $f(s) \in A_s \cap X$?

背景链接：https://arxiv.org/abs/2608.26628

16427 | AMR-150-0048

### Open Problems in Mathematical Logic - Problem 48

O / open | AMR 问题清单汇编

Consider a tree $T$ which is $\Pi^0_1(\emptyset')$ in $M$. Assume there is a $g \in [T]$ such that $M[g] \models \text{B}\Sigma_2$. Is there a definable path $h \in [T]$ such that $M[h] \models \text{B}\Sigma_2$?

背景链接：https://arxiv.org/abs/2608.26628

16428 | AMR-150-0049

### Open Problems in Mathematical Logic - Problem 49

O / open | AMR 问题清单汇编

In 2021, Matt Foreman asked the following question:

Is the set $\{G \in 2^{\omega \times \omega} : G \text{ is a graph on } \omega \text{ and } G \cong G^c\}$ Borel?

背景链接：https://arxiv.org/abs/2608.26628

16429 | AMR-150-0005

### Open Problems in Mathematical Logic - Problem 5

O / open | AMR 问题清单汇编

Is there an injective one-way function on the reals? More precisely, is there a partial computable real function $f$ such that

(i) $f$ is injective and preserves randomness,

(ii) the domain of $f$ has positive Lebesgue measure, and

(iii) $f(x)<_T x$ for a positive-measure set of reals $x$ in the domain of $f$?

背景链接：https://arxiv.org/abs/2608.26628

16430 | AMR-150-0050

### Open Problems in Mathematical Logic - Problem 50 (The union problem)

O / open | AMR 问题清单汇编

If $E$ is a countable Borel equivalence relation and $E = \cup_n F_n$, where $F_n$ is hyperfinite and $F_n \subseteq F_{n+1}$ for each $n \in \omega$, is it true that $E$ is hyperfinite?

背景链接：https://arxiv.org/abs/2608.26628

16431 | AMR-150-0051

### Open Problems in Mathematical Logic - Problem 51 (The hf/hf problem)

O / open | AMR 问题清单汇编

Analogous to the above, we call a countable Borel equivalence relation *hyperfinite-over-hyperfinite* (or shortly, *hf/hf*) if there is a Borel partial order $\leq$ on $X$ such that

- if $x \leq y$, then $x E y$,

- the order type of $\leq \upharpoonright_{[x]}$ is a suborder of $\mathbb{Z}^2$.

If $E$ is hf/hf, is it hyperfinite?

背景链接：https://arxiv.org/abs/2608.26628

16432 | AMR-150-0052

### Open Problems in Mathematical Logic - Problem 52

O / open | AMR 问题清单汇编

Inspired by the Frisch--Shinko--Vidnyánszky theorem, we ask:

Suppose there is a counterexample to the hf/hf problem. Is the set of all hyperfinite equivalence relations $\mathbf{\Sigma}_2^1$-complete?

背景链接：https://arxiv.org/abs/2608.26628

16433 | AMR-150-0053

### Open Problems in Mathematical Logic - Problem 53 (Weiss' question)

O / open | AMR 问题清单汇编

Is the orbit equivalence relation of Borel action of countable amenable group hyperfinite?

背景链接：https://arxiv.org/abs/2608.26628

16434 | AMR-150-0054

### Open Problems in Mathematical Logic - Problem 54

O / open | AMR 问题清单汇编

Also the connected equivalence relations of the following graphs are proved to be hyperfinite: \begin{quote} Graphs of polynomial growth (Bernshteyn--Yu),

Graph of growth less than $\exp(r^c)$ for some small enough $0<c<1$ (Grebík--Marks--Rozhoň--Shinko). \end{quote}

How about graphs of uniform subexponential growth, i.e., of growth less than $\exp(r^c)$ for some $0<c<1$?

背景链接：https://arxiv.org/abs/2608.26628

16435 | AMR-150-0055

### Open Problems in Mathematical Logic - Problem 55

O / open | AMR 问题清单汇编

For $x\in \text{Act}(F_\infty, ([0,1],\lambda))$, let $\mbox{Sch}(x)$ denote the Schreier graph of the action on $[0,1]$.

Is the set $\{x \in \text{Act}(F_\infty,([0,1],\lambda)) : \text{Sch}(x) \text{ has a measurable perfect matching}\}$ $\mathbf{\Sigma}_1^1$-complete?

背景链接：https://arxiv.org/abs/2608.26628

16436 | AMR-150-0056

### Open Problems in Mathematical Logic - Problem 56 (Halmos, 1956)

O / open | AMR 问题清单汇编

What $T \in \text{Aut}([0,1],\lambda)$ has $S \in \text{Aut}([0,1],\lambda)$ with $T = S^2 = S \circ S$?

Conjecture: $\{T \in \text{Aut}([0,1],\lambda) : \exists S \in \text{Aut}([0,1],\lambda) T = S^2\}$ is $\mathbf{\Sigma}_1^1$-complete.

背景链接：https://arxiv.org/abs/2608.26628

16437 | AMR-150-0057

### Open Problems in Mathematical Logic - Problem 57

O / open | AMR 问题清单汇编

If $\vec{G}$ is a Borel locally finite directed graph whose in-degrees are uniformly bounded and out-neighborhoods have polynomial growth, is the connected equivalence relation of $\vec{G}$ hyperfinite?

背景链接：https://arxiv.org/abs/2608.26628

16438 | AMR-150-0058

### Open Problems in Mathematical Logic - Problem 58

O / open | AMR 问题清单汇编

Let $\alpha$ be a free, probability measure preserving (p.m.p.) and ergodic action of $\mathbb{F}_2$ on measure space $(X,\mu)$.

- If $e \neq x \in \mathbb{F}_2$, is there a free, p.m.p action $\beta$ such that $E_\alpha = E_\beta$ and $x$ acts ergodically?

- (Miller--Tserunyan). Is there a free, p.m.p action $\beta$ such that every nontrivial element of $\mathbb{F}_2$ acts ergodically?

背景链接：https://arxiv.org/abs/2608.26628

16439 | AMR-150-0006

### Open Problems in Mathematical Logic - Problem 6

O / open | AMR 问题清单汇编

Let $M$ be a countably infinite $\omega$-categorical structure in a finite first-order signature, and let \[ G=\operatorname{Aut}(M), \] equipped with the topology of pointwise convergence. Let $\operatorname{Aut}(G)$ denote the group of continuous automorphisms of the topological group $G$, and let \[ \operatorname{Out}(G) := \operatorname{Aut}(G)/\operatorname{Inn}(G) \] be its outer automorphism group.

Let $M$ be an $\omega$-categorical structure in a first order finite language. Is the group $\operatorname{Out}(G)$ finite?

背景链接：https://arxiv.org/abs/2608.26628

16440 | AMR-150-0007

### Open Problems in Mathematical Logic - Problem 7

O / open | AMR 问题清单汇编

\[ J^A(n)\downarrow \Longrightarrow J^A(n)\in T_n. \] Also recall that \[ A\le_{LR} B \iff \mathrm{MLR}^B\subseteq\mathrm{MLR}^A. \]

Does $\le_{SJT}$ imply $\le_{LR}$?

背景链接：https://arxiv.org/abs/2608.26628

16441 | AMR-150-0008

### Open Problems in Mathematical Logic - Problem 8

O / open | AMR 问题清单汇编

\[ A\le_{SJT} B \iff \forall Y \omega\text{-c.a. and }Y\in\mathrm{MLR} \Bigl( A\le_T B\oplus Y \Bigr). \]

Let Rec denote the class of recursive sets.

Is there a $\Pi^0_1$ class $C$ such that $|C\cap \mathrm{Rec}|=\infty$ and $C\cap \mathrm{Rec}$ is maximally almost disjoint among the recursive sets?

背景链接：https://arxiv.org/abs/2608.26628

16442 | AMR-150-0009

### Open Problems in Mathematical Logic - Problem 9

O / open | AMR 问题清单汇编

Determine the equivalence classes of quantifier patterns at the fourth level of the arithmetical hierarchy:

(1) How many $\equiv_m$-equivalence classes of $\Sigma^0_4$ quantifier patterns are there?

(2) How many $\equiv_m$-equivalence classes of $\Pi^0_4$ quantifier patterns are there?

(3) How many $\equiv_{dm}$-equivalence classes of $\Sigma^0_4$ quantifier patterns are there?

背景链接：https://arxiv.org/abs/2608.26628

16443 | OWR-15430-009

### Optimal Complexity of Definable Cylindrical Decompositions

O / open | Oberwolfach 报告问题汇编

Can the quantitative bounds for semi-cylindrical or cylindrical decompositions of definable, even semialgebraic, sets be improved to optimal bounds in higher dimensions?

来源字段：https://doi.org/10.4171/owr/2017/22

16444 | OWR-2654830-010

### Optimality and Practical Efficiency of Proof Search

O / open | Oberwolfach 报告问题汇编

How should proof-search algorithms be compared, and is there an optimal way to search for propositional proofs; why is searching in stronger proof systems apparently unhelpful; and how can practical SAT and automated-theorem-proving algorithms perform well on very long formulas despite exponential lower bounds for associated proof systems?

来源字段：https://doi.org/10.4171/owr/2020/34

16445 | OPG-37429

### Order-invariant queries

P / partially_solved | Open Problem Garden

Question

- Does ${<}\text{-invariant\:FO} = \text{FO}$ hold over graphs of bounded tree-width? - Is ${<}\text{-invariant\:FO}$ included in $\text{MSO}$ over graphs? - Does ${<}\text{-invariant\:FO}$ have a 0-1 law? - Are properties of ${<}\text{-invariant\:FO}$ Hanf-local? - Is there a logic (with an effective syntax) that captures ${<}\text{-invariant\:FO}$?

背景链接：http://www.openproblemgarden.org/op/order_invariant_queries

16446 | OWR-2654830-013

### Partial Proof Objects in Martin-Löf Type Theory

O / open | Oberwolfach 报告问题汇编

In Martin-Löf type theory, which mathematical objects or formal mechanisms can serve as partial proofs, analogously to the partial functions used in the predicative construction of Mahlo universes?

来源字段：https://doi.org/10.4171/owr/2020/34

16447 | OWR-15958-008

### Pathwise Connected Choice in Dimension Two

O / open | Oberwolfach 报告问题汇编

Is pathwise connected choice in dimension $2$ strongly Weihrauch equivalent to Weak König's Lemma, as connected choice is in dimensions at least $2$?

来源字段：https://doi.org/10.4171/owr/2017/53

16448 | AIM-LOGIC-0062

### Pole-support containment at primes above 1 modulo 4

P / partially_solved | AIM 研讨会问题汇编

Let \(K\) be a number field. Does there exist an \(\exists\)-formula \(\delta(x,y)\) such that

\[ \delta(a,b)\Longleftrightarrow (\text{for every }p\text{ above a prime }1\bmod 4) (v_p(a)<0\Longrightarrow v_p(b)<0)? \]

This is known (more or less) for \(K=\mathbb Q\) for primes \(3\bmod 4\). A positive answer would imply existential definability of \(\mathbb Z\) in \(\mathbb Q\), hence a negative answer to HTP over \(\mathbb Q\).

来源字段：https://aimath.org/pastworkshops/definabilityinntproblems.pdf

16449 | OWR-14299908-009

### Polynomial Boundedness of Definably Complete Restricted Exponential Fields

O / open | Oberwolfach 报告问题汇编

Is every definably complete restricted exponential field polynomially bounded?

来源字段：https://doi.org/10.4171/owr/2026/3

16450 | AIM-LOGIC-0051

### Positive solution and parameter descent for the super tree property

S / solved | AIM 研讨会问题汇编

There is a lemma of Magidor and Shelah that if $\lambda=\sup_{n<\omega}\kappa_n$ for supercompact cardinals $\kappa_n$, then $\lambda^+$ has the tree property. Fontanella proved that if the $\kappa_n$'s are strongly compact then $\lambda^+$ has the strong tree property.

Is it possible to obtain the super tree property at a successor of a singular cardinal?

来源字段：http://aimpl.org/highlowforcing/2/

16451 | AIM-LOGIC-0127

### Positive-existential decoder criterion for rational pair storage

P / partially_solved | AIM 研讨会问题汇编

Jeroen Demeyer has observed that the existence of a polynomial bijection \(\mathbb N^2\to\mathbb N\) implies that any first order formula over \(\mathbb N\) in positive prenex form is equivalent to one in which every block of consecutive universal quantifiers is replaced by just one (and the number of existential quantifiers goes up). Such a polynomial bijection can be found in Davis, *Math. Monthly* **80**, 236--237. Does something similar work for \(\mathbb Q\), in other words, can we find a Diophantine injection \(\mathbb Q^2\hookrightarrow\mathbb Q\)? There are some observations related to this in *C.R.A.S. Paris* **328**, 3--8 (1999); for example, this would follow from the generalized \(abc\)-conjecture.

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16452 | AIM-LOGIC-0093

### Presentation-model resolution and relator-normalization robustness

P / partially_solved | AIM 研讨会问题汇编

**1.7. Non-abelian free groups (Knight).** Consider \(n\)-generated groups with a single relator of length at most \(t\). (For each \(t,n\) there are finitely many such groups.) For every sentence \(\varphi\) define \[ h_{n,t,\varphi}= \frac{|\{G\in\mathcal H_{n,t}:G\models\varphi\}|}{|\mathcal H_{n,t}|}. \] **Conjecture 14.** The limit \(\lim_{t\to\infty}h_{n,t,\varphi}\) exists, and always takes the value \(0\) or \(1\), moreover in a way that may depend on \(\varphi\) but not on \(n\).

**Conjecture 15.** Furthermore, the asymptotically almost sure (a.a.s.) theory determined by this \(0-1\) law is that of \(F_2\).

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16453 | OWR-2654830-008

### Proof Mining for Quantitative Tauberian Remainders

P / partially_solved | Oberwolfach 报告问题汇编

Can proof-theoretic methods such as the Dialectica interpretation produce new remainder theorems in Tauberian theory, and are there abstract quantitative metatheorems that unify and generalize related Tauberian theorems?

来源字段：https://doi.org/10.4171/owr/2020/34

16454 | OWR-795-001

### Proof-Theoretic Strength of a Matrix Principle

O / open | Oberwolfach 报告问题汇编

Among the bounded-arithmetic theories corresponding to $\mathrm{AC}^0$, $\mathrm{AC}^0(2)$, $\mathrm{TC}^0$, $\mathrm{NC}^1$, $\mathrm{L}$, and $\mathrm{P}$, what is the weakest theory that proves the matrix principle $AB=I\Rightarrow BA=I$?

来源字段：https://doi.org/10.4171/owr/2005/14

16455 | OWR-2654830-007

### Proof-Theoretic Strength of Minimalist Foundations

O / open | Oberwolfach 报告问题汇编

What is the exact proof-theoretic strength of the Minimalist Foundation MF and of its classical extension MF+EM by the law of excluded middle?

来源字段：https://doi.org/10.4171/owr/2020/34

16456 | AIM-LOGIC-0144

### Proper models transfer exact Mordell-Weil stability

P / partially_solved | AIM 研讨会问题汇编

**Question 32 (Poonen).** Is it true that for all number fields $K$, there exists a variety $X$ (scheme of finite type) over $\mathbb Z$ such that

1. $X(\mathbb Z)$ is infinite. 2. $X(\mathcal O_K)=X(\mathbb Z)$.

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16457 | AIM-LOGIC-0014

### Quantifier-count fragments of Th(Q)

P / partially_solved | AIM 研讨会问题汇编

Which fragments of \(\text{Th}(\mathbb Q)\) are undecidable? Consider the number of quantifiers, not just alternations.

来源字段：http://aimpl.org/definedecide/3/

16458 | AIM-LOGIC-0108

### Quantitative bridge from Vershik genericity to existential closure

S / solved | AIM 研讨会问题汇编

**Recovered statement.** Find connections between different notions of genericity. In particular, connect Vershik's notion of genericity with existential closure. The Urysohn space is generic in both senses.

来源字段：https://aimath.org/WWN/continuouslogic/continuouslogic.pdf

16459 | OWR-12329-010

### Quasiminimal Complex Exponentiation and Abelian Regular Groups

P / partially_solved | Oberwolfach 报告问题汇编

Is the complex exponential field $(\mathbb C,+,\cdot,0,1,\exp)$ quasiminimal, as conjectured by Zilber, and is every regular group abelian?

来源字段：https://doi.org/10.4171/owr/2013/01

16460 | OWR-11695862-010

### Quasiminimal Excellence Above Aleph-Two

O / open | Oberwolfach 报告问题汇编

If a quasiminimal structure $M$ has cardinality at least $\aleph_2$, must $M$ be quasiminimally excellent?

来源字段：https://doi.org/10.4171/owr/2022/52

16461 | AMR-074-0216

### Ramsey's Theorem for Triples over a Weak Base

P / partially_solved | AMR 问题清单汇编

Over $\mathsf{RCA}^*_0$, is Ramsey's theorem for triples equivalent to $\mathsf{ACA}_0$, as it is over $\mathsf{RCA}_0$?

背景链接：https://sgslogic.net/t20/talks/cta/problems/

16462 | OWR-11575-002

### Random Axioms in Computability and Kolmogorov Complexity

O / open | Oberwolfach 报告问题汇编

Are random axioms useful in computability theory and Kolmogorov complexity?

来源字段：https://doi.org/10.4171/owr/2012/08

16463 | OWR-11575-003

### Random Minimal Pairs and Cupping Below the Halting Problem

O / open | Oberwolfach 报告问题汇编

For every noncomputable $A<_T\varnothing'$ that is not $K$-trivial, does there exist a Martin-Löf random $R<_T\varnothing'$ such that $A\oplus R\equiv_T\varnothing'$ and every $B\leq_T A,R$ is $K$-trivial?

来源字段：https://doi.org/10.4171/owr/2012/08

16464 | AIM-LOGIC-0104

### Randomized finite-field vector spaces solve the beautiful-pair problem

S / solved | AIM 研讨会问题汇编

**12.** Find examples of “Non-Modular” stable theories. That is, theories whose beautiful pairs are not essentially separably categorical.

**Background:** Probability algebras and Hilbert spaces are \(\omega\)-stable structures whose beautiful pairs are separably categorical. \(L_p\) spaces have two non-isomorphic beautiful separable pairs, but these two structures are isomorphic up to perturbations of the norm.

来源字段：https://aimath.org/WWN/continuouslogic/continuouslogic.pdf

16465 | AIM-LOGIC-0133

### Rank-one subgroups as Diophantine congruence limits

P / partially_solved | AIM 研讨会问题汇编

Question 20 (Shlapentokh). Let \(E\) be an elliptic curve over \(\mathbb Q\) of rank \(2\). Does there exist an existentially definable rank \(1\) subgroup?

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16466 | OWR-14299587-004

### Reconstructing Structures from Topological Symmetry Monoids

P / partially_solved | Oberwolfach 报告问题汇编

What information about an infinite structure can be reconstructed from its topological automorphism group or from the topological monoid of its elementary self-embeddings?

来源字段：https://doi.org/10.4171/owr/2025/58

16467 | OWR-11575-009

## Recursive Spectra of Strongly Minimal Theories

O / open | Oberwolfach 报告问题汇编

If $T$ is a strongly minimal theory in a finite language satisfying Zilber's conjecture, is the collection of recursively presentable models of $T$ equal to $\varnothing$, all countable models of $T$, or the prime model of $T$?

来源字段：https://doi.org/10.4171/owr/2012/08

16468 | AIM-LOGIC-0065

## Relative algebraic closedness between models of the theory of the rationals

S / solved | AIM 研讨会问题汇编

**Question 5 (Koenigsmann).** Given \(\mathbb Q^\ast\subseteq\mathbb Q^{\ast\ast}\), both elementarily equivalent to \(\mathbb Q\) (in the language of rings), does it imply that \[ (\mathbb Q^\ast)^{\mathrm{alg}}\cap\mathbb Q^{\ast\ast}=\mathbb Q^\ast \] (i.e. is \(\mathbb Q^\ast\) relatively algebraically closed inside \(\mathbb Q^{\ast\ast}\))?

If the answer is “NO” in some case, then we know that \(\mathbb Z\) is not \(\exists\)-definable in \(\mathbb Q\).

We know that \(\mathbb Q^\ast\) is quadratically closed in \(\mathbb Q^{\ast\ast}\).

来源字段：https://aimath.org/pastworkshops/definabilityinntproblems.pdf

16469 | OWR-1453-005

## Relative Galois Reconstruction of One-Variable Function Fields

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Relative Galois Reconstruction Conjecture [OWR-1453-006]

Let $K,K'$ be perfect fields, suppose $K$ has a finite extension of degree greater than $2$ and prime to $\operatorname{char}K$, let $F/K$ be a one-variable function field, and let $F'/K'$ be any function field. Is the canonical map $\operatorname{Isom}(F/K,F'/K')\to\operatorname{Isom}(G_{F/K},G_{F'/K'})/\operatorname{Inn}(G_{F'/K'})$ bijective?

来源字段：https://doi.org/10.4171/owr/2007/02

16470 | AIM-LOGIC-0027

## Repair and resolution of a core-model covering problem

O / open | AIM 研讨会问题汇编

**Recovered Problem 15.** Suppose there is no inner model with a Woodin cardinal, and let \(\kappa\) be a **regular** cardinal in the core model \(K\). Suppose \(\kappa\) is a singular cardinal in \(V\). Must \(\kappa\) be measurable in \(K\)? For \(K\) below \(0^{\P}\), this is known (Cox).

Its exact `problem` field is preserved here:

The original PDF was rendered and inspected visually. It establishes three points.

1. The PDF itself, not merely the OCR, says that \(\kappa\) is “a singular cardinal in \(K\).” This is a source-level mathematical typo: a cardinal cannot be both singular and measurable in the same ZFC model. 2. The symbol in “below \(0\)” is a superscript pilcrow. In standard core-model notation it is \(0^{\P}\), read “zero pistol,” the sharp for an inner model with a strong cardinal. Thus the printed annotation is “For \(K\) below \(0^{\P}\) this is known (Cox).” 3. Footnote 4 belongs to the preceding printed Problem 14. Its marker follows “Must \(N^\#\) exist?” The footnote is physically placed beneath Problem 15 only because both occur at the bottom of the page. It is not a qualification of Problem 15 and is removed from the recovered statement.

The source-supported **recovered statement** is therefore:

The replacement “regular” is not claimed to be the logically unique imaginable edit. It is the unique single-word repair supported by the printed Cox annotation: Cox's main theorem has precisely “regular in \(K\), singular in \(V\)” as its covering configuration. It also removes the direct contradiction between “singular in \(K\)” and “measurable in \(K\).”

来源字段：https://aimath.org/pastworkshops/innermodelproblems.pdf

16471 | AIM-LOGIC-0038

## Repairing the Hausdorff spread bound and a PFA sign error

O / open | AIM 研讨会问题汇编

The answer is negative under $\Diamond$, even for compact Hausdorff spaces. PFA implies that the answer is negative for $X$ with $s(X)=\aleph_0$. PFA also implies that $|X|\leq2^{\aleph_0}$ for every Hausdorff space with no uncountable discrete subspaces.

For which infinite Hausdorff spaces $X$ can the classical bound $|X|\leq 2^{2^{s(X)}}$ be improved to $|X|\leq 2^{s(X)}$? In particular, is it consistent that this improvement holds for every Hausdorff (or every regular) space?

来源字段：http://aimpl.org/highlowforcing/1/

16472 | OWR-15181-015

### Residue Fields of O-Minimal Exponential Fields

O / open | Oberwolfach 报告问题汇编

For every o-minimal ordered exponential field $K$ satisfying EXP, is its residue exponential field o-minimal?

来源字段：https://doi.org/10.4171/owr/2016/60

16473 | OWR-11695862-014

### Residue-Field Domination in Henselian Valued Fields

P / partially_solved | Oberwolfach 报告问题汇编

Extend residue-field domination to broad classes of equicharacteristic-zero henselian valued fields, with the goal of lifting tameness properties of the residue field to the entire valued field.

来源字段：https://doi.org/10.4171/owr/2022/52

16474 | AIM-LOGIC-0029

### Reversal obstructs a globally minimal higher Countryman line

P / partially_solved | AIM 研讨会问题汇编

A *Countryman line* is an uncountable linear ordering whose square is the union of countably many chains. In $\mathrm{ZFC} + \mathrm{CH}$ (and even under weaker hypotheses such as $\diamondsuit$), Countryman lines exist, and under $\mathrm{PFA}$ every two Countryman lines are isomorphic or reverse-isomorphic. Is it consistent with $\mathrm{ZFC} + \mathrm{CH}$ that there is a unique Countryman line up to isomorphism and reverse-isomorphism?

来源字段：http://aimpl.org/highlowforcing/1/

16475 | AMR-074-0121

### Reverse Mathematics of Fraïssé's Conjecture

P / partially_solved | AMR 问题清单汇编

Over $\mathsf{RCA}_0$, is Fraïssé's conjecture for countable linear orders equivalent to $\mathsf{ATR}_0$?

背景链接：https://math.berkeley.edu/~antonio/papers/questionsRM.pdf

16476 | AMR-074-0109

### Reverse-Mathematical Strength of Hindman's Theorem

P / partially_solved | AMR 问题清单汇编

Over $\mathsf{RCA}_0$, is Hindman's theorem equivalent to $\mathsf{ACA}^+_0$, equivalent to $\mathsf{ACA}_0$, or strictly between them?

背景链接：https://math.berkeley.edu/~antonio/papers/questionsRM.pdf

16477 | AIM-LOGIC-0071

### Scalar parameters for divisorial valuation rings

P / partially_solved | AIM 研讨会问题汇编

**Question 11 (Flenner).** Does there exist a formula \(\varphi(x,y)\) such that for any finitely generated field \(K\) and every rank one divisorial valuation \(v\) on \(K\), there exists a parameter \(a\) such that \[ \mathcal O_v=\{b\in K\mid \varphi(b,a)\}? \]

来源字段：https://aimath.org/pastworkshops/definabilityinntproblems.pdf

16478 | OWR-737-007

### Schanuel Conjecture

O / open | Oberwolfach 报告问题汇编

If $x_1,\ldots,x_n\in\mathbb{C}$ are linearly independent over $\mathbb{Q}$, then $\operatorname{trdeg}_{\mathbb{Q}}\mathbb{Q}(x_1,\ldots,x_n,e^{x_1},\ldots,e^{x_n})\geq n$.

来源字段：https://doi.org/10.4171/owr/2004/35

16479 | AIM-LOGIC-0094

### Scott sets and rational cuts in recursively saturated real closed fields

S / solved | AIM 研讨会问题汇编

**1.8. Standard systems of RCF (Marker).**

**Question 16.** What are the possible standard systems of recursively saturated real closed fields? (This is essentially asking: What are the possible sets of cuts of \(\mathbb Q\) that are realized in some models?)

The ideal answer might be “all Scott sets”.

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16480 | AIM-LOGIC-0080

### Scott-rank gaps above infinitary quantifier rank

S / solved | AIM 研讨会问题汇编

Page 2 of the original AIM PDF gives the complete question as follows:

The following Section 1.2 heading begins immediately after the question and is not part of it.

There are two important typographical points.

1. The logic is \(L_{\omega_1,\omega}\): countable conjunctions and disjunctions are allowed, but only finite strings of quantifiers occur along any branch of a formula. 2. The first displayed ordinal is the ordinary first uncountable ordinal \(\omega_1\), not the Church--Kleene ordinal \(\omega_1^{CK}\). This is especially clear from the PDF because the preceding definition in Question 1 explicitly prints a raised “\(CK\)” in \(\omega_1^{CK}\), whereas Question 2 prints only \(\omega_1\).

Scott rank is normally defined here only for countable structures in a countable language. Those are the conventions adopted below. They are also the conventions used in the later paper that resolves the question.

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16481 | OWR-15181-014

### Set-Sized Models of Initial Surreal Substructures

O / open | Oberwolfach 报告问题汇编

If a theory extending the theory of nontrivial densely ordered abelian groups has every model isomorphic to an initial subgroup of $\mathbf{No}$, must it be the theory of divisible ordered abelian groups? Analogously, if every model of an ordered-field theory is isomorphic to an initial subfield of $\mathbf{No}$, must it be the theory of real-closed ordered fields? Can the only-if directions be proved in NBG using set-sized universes?

来源字段：https://doi.org/10.4171/owr/2016/60

16482 | OWR-15958-004

### Shared Conceptual Foundations of Mathematics

O / open | Oberwolfach 报告问题汇编

Can a common core of conceptual foundations for mathematics be developed whose concepts are acceptable across several major philosophical approaches?

来源字段：https://doi.org/10.4171/owr/2017/53

16483 | AIM-LOGIC-0129

### Sharp integer-root bound for binomial-product programs and a tau-convention audit

P / partially_solved | AIM 研讨会问题汇编

Poonen and T. Scanlon and recorded by J. Demeyer. The official PDF, p.

\[ 1,x,g_1,g_2,\ldots, \]

with every \(g_i\) the sum, difference, or product of two earlier terms, defines

\[ \tau(f)=\min\{n:\text{there is such a sequence with }g_n=f\}, \]

and asks for an absolute constant \(c\) such that, for every nonzero \(f\in\mathbb Z[x]\),

\[ \#\{a\in\mathbb Z:f(a)=0\}\le (1+\tau(f))^c. \tag{1} \]

Thus the recovered typography is **\((1+\tau(f))^c\)**. “Number of integer zeros” is interpreted as the number of *distinct* integer zeros. Section 7 proves that counting multiplicity would make (1) false. As is standard, set \(\tau(1)=\tau(x)=0\); the literal minimum in the workshop sentence otherwise omits the two input polynomials.

The ordinary circuit convention permits the same earlier register to be used twice (in particular, squaring is one multiplication). Because “two earlier terms” might be read as requiring two distinct register positions, Section 7 also proves that the two conventions are polynomially equivalent with an explicit constant-factor bound.

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16484 | OWR-15170-008

### Sharp Transcendence Bounds for the Modular J-Function

P / partially_solved | Oberwolfach 报告问题汇编

Let $C$ be $j\operatorname{cl}$-closed, let $\lambda$ be a $j\operatorname{cl}$-independent tuple, and set $F=\mathbb Q(\lambda)$. For a tuple $z$ whose coordinates lie in distinct $G$-orbits and a nonscalar $g\in G_F\setminus G$, is the Ax-Schanuel lower bound $$ \operatorname{td}\bigl(j(z),j'(z),j''(z),j(gz),j'(gz),j''(gz)/F,C\bigr)\geq 3n $$ sharp, or can it be improved without proving a modular Schanuel conjecture? Can the $j$-derivation pregeometry $j\operatorname{cl}$ be described explicitly?

来源字段：https://doi.org/10.4171/owr/2016/49

16485 | AMR-075-0006

### Shelah's categoricity conjecture for $L_{\omega_1,\omega}$

P / partially_solved | AMR 问题清单汇编

Shelah's categoricity conjecture for $L_{\omega_1,\omega}$: If a sentence is categorical above the Hanf number then it is categorical in all cardinals above the Hanf number.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Model_theory_and_formal_languages

16486 | AMR-075-0007

### Shelah's eventual categoricity conjecture

P / partially_solved | AMR 问题清单汇编

Shelah's eventual categoricity conjecture: For every cardinal $\lambda$ there exists a cardinal $\mu(\lambda)$ such that if an AEC K with LS(K)${} \le \lambda$ is categorical in a cardinal above $\mu(\lambda)$ then it is categorical in all cardinals above $\mu(\lambda)$.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Model_theory_and_formal_languages

16487 | AIM-LOGIC-0046

### Silver--Jensen normal form and preparation localization for lower GCH

P / partially_solved | AIM 研讨会问题汇编

Is $\neg\square_\kappa^*+\neg\mathrm{SCH}_\kappa+\mathrm{GCH}_{<\kappa}$ consistent for $\kappa$ singular?

来源字段：http://aimpl.org/highlowforcing/2/

16488 | AIM-LOGIC-0130

### Simplex-supported feasibility over local and global fields

P / partially_solved | AIM 研讨会问题汇编

In particular, the displayed sum was read as a product, vector notation was flattened, and \(\mathbb Q_p^n\) was read as \(\mathbb Q^{np}\).

Let \(c_j\in\mathbb Z\), and let \[ P(x_1,\ldots,x_n)=\sum_{j=0}^{n}c_jx^{a_j}, \qquad a_0,\ldots,a_n\in\mathbb N^n, \] where the exponent vectors are affinely independent. For fixed \(p\), can one decide in polynomial time whether \(P\) has a zero in \(\mathbb Q_p^n\)? The source records a negative complexity answer based on an encoding of \(0/1\)-knapsack, comments on the real analogue, and then asks whether one can decide the existence of a zero in \(\mathbb Q^n\). It notes that the latter contains the rational-solubility problem for diagonal genus-one curves \[ ax^3+by^3=1. \]

The source numbers the terms from \(1\) to \(n+1\); this report shifts the indices to \(0,\ldots,n\). No mathematical content is changed. The complexity phrase “Answer: NO” cannot be an unconditional proof that no polynomial-time algorithm exists unless a complexity-class separation is supplied. The conservative reading is that the stated knapsack encoding gives NP-hardness, so polynomial time would imply \(\mathsf P=\mathsf{NP}\).

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16489 | AIM-LOGIC-0032

### Simultaneous vanishing of the higher limits of the Mardesic-Prasolov system

P / partially_solved | AIM 研讨会问题汇编

Is it consistent with ZFC that \(\lim^p\mathbb A=0\) simultaneously for every positive integer \(p\)?

The supplied source URL is <http://aimpl.org/highlowforcing/1/>.

The record defines

\[ \mathbb A=\langle A_f:f\in\omega^\omega\rangle, \qquad A_f=\bigoplus_{n\in\omega}\bigoplus_{i<f(n)}\mathbb Z =\bigoplus_{n\in\omega}\mathbb Z^{f(n)}, \]

introduces groups denoted by \(G_f\) and an augmented alternating complex beginning

\[ \mathbb Z^{\omega\times\omega}/\mathrm{fin} \longrightarrow \prod_fG_f \longrightarrow \prod_{f_0\leq f_1}G_{f_0} \longrightarrow \prod_{f_0\leq f_1\leq f_2}G_{f_0} \longrightarrow\cdots, \]

The exact source text suppresses several conventions. They can be recovered from the primary literature as follows.

1. The order on \(\omega^\omega\) is the pointwise order: \(f\leq g\) iff \(f(n)\leq g(n)\) for every \(n\). If \(f\leq g\), the bonding map \(p_{gf}:A_g\to A_f\) is coordinate projection. 2. Put \(I_f=\{(n,i)\in\omega^2:i<f(n)\}\), \(B_f=\prod_{I_f}\mathbb Z\), and \(G_f=B_f/A_f\). Thus the ambiguous source expression \(\prod_n\mathbb Z^{f(n)}/\mathrm{fin}\) means functions on \(I_f\) modulo the subgroup of finitely supported functions, or equivalently equality except at finitely many pairs \((n,i)\). The bonding maps on \(B\) and \(G=B/A\) are restrictions. 3. Some papers use \(i\leq f(n)\) instead of \(i<f(n)\). Passing to the cofinal family of everywhere-positive functions, or shifting \(f\) by one, reconciles the conventions and does not change derived limits. 4. If \[ C^{-1}=\mathbb Z^{\omega^2}/\mathrm{fin},\qquad C^q=\prod_{f_0\leq\cdots\leq f_q}G_{f_0}\quad(q\geq0), \] then the augmentation sends a global function to its restrictions modulo finite support. The full differential is \[ (dc)(f_0,\ldots,f_{q+1})= p_{f_1f_0}c(f_1,\ldots,f_{q+1}) +\sum_{i=1}^{q+1}(-1)^i c(f_0,\ldots,\widehat{f_i},\ldots,f_{q+1}). \] The source's formula suppresses the restriction in the zeroth face. Equivalently, all faces are restricted to the common region \(I_{f_0}\).

The recovered question is therefore:

The qualifier “positive” is forced both by the modern theorem statements and by the degree-zero calculation proved below.

来源字段：http://aimpl.org/highlowforcing/1/

16490 | AIM-LOGIC-0006

### Small Martin's Maximum and nonsemiproper stationary-preserving forcing

S / solved | AIM 研讨会问题汇编

- Does Con(ZFC) imply Con(MM for posets of size $\aleph_1$)?

- Is there a stationary set preserving poset of size $\aleph_1$ which is not semiproper?

来源字段：http://aimpl.org/alephtwo/7/

16491 | OWR-14299587-006

### Smooth Classification and Outer Automorphisms of Oligomorphic Groups

P / partially_solved | Oberwolfach 报告问题汇编

Is topological isomorphism of oligomorphic groups smoothly classifiable? In particular, does there exist an oligomorphic group whose outer automorphism group is not profinite?

来源字段：https://doi.org/10.4171/owr/2025/58

16492 | OWR-1453-013

### Solvability of Rank-Two NIP Groups

O / open | Oberwolfach 报告问题汇编

Is every group with NIP, hereditarily FSG, and thorn $U$-rank $2$ solvable-by-finite? If such a group also satisfies $G=G^{00}$, must it be solvable?

来源字段：https://doi.org/10.4171/owr/2007/02

16493 | AIM-LOGIC-0004

### Source repair and the countable chromatic compactness endpoint barrier

O / open | AIM 研讨会问题汇编

Let \(\phi(n)\) be the statement “for every graph of size \(\aleph_{\omega+1}\), if every subgraph of size \(<\aleph_{\omega+1}\) has chromatic number \(\leq\aleph_n\), then the entire graph has chromatic number \(\aleph_n\).” Is \(\phi(0)\) consistent?

It is known that \(\phi(n)\) is consistent whenever \(1\leq n<\omega\). In fact, the sentence \(\forall n\in\omega(n>0\to\phi(n))\) is consistent.

来源字段：http://aimpl.org/alephtwo/5/

16494 | OWR-11575-015

### Splitting Sets Under Differential-Transcendental Extensions

O / open | Oberwolfach 报告问题汇编

If $K$ is a computable differential field and $z$ is differentially transcendental over $K$ in a larger computable differential field, is $T_{K\langle z\rangle}$ Turing-computable from $T_K$?

来源字段：https://doi.org/10.4171/owr/2012/08

16495 | AIM-LOGIC-0113

### Stability, hereditary containment, and the exponent quantifier

P / partially_solved | AIM 研讨会问题汇编

Is there a notion of "explicitly defined norm" such that every Banach space with an explicitly defined norm hereditarily contains an \(\ell_p\) space?

Problem 28 reads:

It is followed by the background sentence saying that the working group on Definable Norms and Stable Banach Spaces studied the subject. The sentence beginning "29. Study asymtotic cones" is the next, independent problem and is not part of Problem 28.

The only corrections made here are typographical: "explictly" is read as "explicitly," and "lp" as the standard sequence space \(\ell_p\). Thus the recovered question is:

The historical working-group report makes the motivation more precise but also exposes two ambiguities. It asks for model-theoretic conditions on a separable infinite-dimensional Banach space \(X\) which ensure that every infinite-dimensional subspace contains one of the classical sequence spaces \(\ell_p\) or \(c_0\). The printed Problem 28 mentions only \(\ell_p\). Neither source says whether the same exponent \(p\) must work in all subspaces, nor whether "contains" means an isomorphic copy, an almost-isometric copy, or finite representability.

来源字段：https://aimath.org/WWN/continuouslogic/continuouslogic.pdf

16496 | OWR-12697685-008

### Stratification by Canonical Definable Partitions

O / open | Oberwolfach 报告问题汇编

Do the canonical definable pieces $S_d$ form a stratification of $X$ in a classical sense?

来源字段：https://doi.org/10.4171/owr/2023/2

16497 | AMR-074-0212

### Strength of Kříž's Labeled-Tree Theorem

O / open | AMR 问题清单汇编

Determine the reverse-mathematical strength of Kříž's labeled-tree generalization of Kruskal's theorem.

背景链接：https://sgslogic.net/t20/talks/cta/problems/

16498 | AMR-074-0208

### Strength of Szemerédi's Theorem

O / open | AMR 问题清单汇编

Is Szemerédi's theorem provable in $\mathsf{ACA}_0$? More generally, determine its reverse-mathematical strength.

背景链接：https://sgslogic.net/t20/talks/cta/problems/

16499 | AMR-074-0118

### Strength of the Auslander-Ellis Theorem

P / partially_solved | AMR 问题清单汇编

Over $\mathsf{RCA}_0$, is the Auslander-Ellis theorem equivalent to $\mathsf{ACA}_0$?

背景链接：https://math.berkeley.edu/~antonio/papers/questionsRM.pdf

16500 | AMR-074-0111

### Strength of the Carlson-Simpson Lemma

P / partially_solved | AMR 问题清单汇编

Determine the reverse-mathematical strength of the Carlson-Simpson infinite-variable-word lemma $\mathsf{CS}$.

背景链接：https://math.berkeley.edu/~antonio/papers/questionsRM.pdf

16501 | AMR-074-0110

### Strength of the Dual Ramsey Theorem

P / partially_solved | AMR 问题清单汇编

Determine the reverse-mathematical strength of the dual Ramsey theorem $\mathsf{DRT}^k$.

背景链接：https://math.berkeley.edu/~antonio/papers/questionsRM.pdf

16502 | AMR-074-0205

### Strength of the Krein-Šmulian Theorem

O / open | AMR 问题清单汇编

Determine the exact reverse-mathematical strength of the Krein-Šmulian theorem for separable Banach spaces.

背景链接：https://sgslogic.net/t20/talks/cta/problems/

16503 | AMR-074-0124

### Strengths of Laver and Nash-Williams BQO Theorems

O / open | AMR 问题清单汇编

Determine the reverse-mathematical strengths of Laver's labeled-linear-order theorem $\mathsf{LAV}$ and the Nash-Williams bqo transfinite-sequence theorem $\mathsf{NWT}$.

背景链接：https://math.berkeley.edu/~antonio/papers/questionsRM.pdf

16504 | AIM-LOGIC-0031

### Strong-homology additivity and one universal earring test

P / partially_solved | AIM 研讨会问题汇编

Its exact `problem` field is:

The old statement about \(\mathfrak d\) is preserved exactly above, but Section 2 treats it as a historical heuristic rather than a theorem.

There are two scope distinctions that are essential to recovering the mathematical question.

1. **Additivity is Milnor's axiom.** For a class \(\mathcal C\), every family \(\{X_\alpha:\alpha\in I\}\) for which all \(X_\alpha\) and the topological coproduct \(\coprod_{\alpha\in I}X_\alpha\) belong to \(\mathcal C\) must induce isomorphisms \[ \bigoplus_{\alpha\in I}\overline H_p(X_\alpha) \longrightarrow \overline H_p\!\left(\coprod_{\alpha\in I}X_\alpha\right) \quad(p\geq 0). \] This is not the group-law assertion suggested by the ordinary-language word “additive.” 2. **The general clause includes nonseparable spaces.** Every closed subspace of a finite-dimensional Euclidean space is locally compact and separable metrizable, but an arbitrary locally compact metric space need not be separable and may have arbitrarily large weight. The modern solution separates these cases.

The literature used below works with the standard integral strong homology groups \(\overline H_p(-;\mathbb Z)\); that coefficient convention is adopted throughout. No assertion is made here that the same consistency statement simultaneously holds for every coefficient group. Strong homology agrees with Steenrod homology on metric compacta but is a different extension beyond compacta; it is not silently replaced by singular, Steenrod--Sitnikov, or Borel--Moore homology.

来源字段：http://aimpl.org/highlowforcing/1/

16505 | OWR-15958-007

### Strong-Induction Admissibility in Cut-Free Modal Calculus

O / open | Oberwolfach 报告问题汇编

Is the strong-induction rule admissible in Kozen's cut-free system for the modal $\mu$-calculus? Equivalently, is the corresponding quantifier-contraction inference admissible?

来源字段：https://doi.org/10.4171/owr/2017/53

16506 | OWR-14299587-003

### Structural Characterization of Exceptional Borel Graphs

P / partially_solved | Oberwolfach 报告问题汇编

What structural property characterizes Borel graphs for which the general results fail? Is there an analogous characterization for Borel graphs whose Borel chromatic number is at most a fixed $k\in\mathbb{N}$?

来源字段：https://doi.org/10.4171/owr/2025/58

16507 | OWR-12697685-011

### Structure of k-Dependent Fields

O / open | Oberwolfach 报告问题汇编

What structural restrictions follow for a $k$-dependent field, and in particular must every such field be NIP?

来源字段：https://doi.org/10.4171/owr/2023/2

16508 | AIM-LOGIC-0056

### Successive tree properties by finite two-type side conditions

S / solved | AIM 研讨会问题汇编

Is there a model for $\mathrm{TP}_{\aleph_2}+\mathrm{TP}_{\aleph_3}$ using side conditions?

来源字段：http://aimpl.org/highlowforcing/3/

16509 | AMR-074-0008

### Superpolynomial Lower Bounds for Frege Proofs

O / open | AMR 问题清单汇编

Prove a superpolynomial lower bound on the size of Frege proofs; in particular, do some tautologies require exponentially large Frege proofs?

背景链接：https://mathoverflow.net/questions/227083/what-are-some-important-but-still-unsolved-problems-in-mathematical-logic#227112

16510 | OWR-14298016-006

### Supremum Computation for Weakly Continuous Functions

P / partially_solved | Oberwolfach 报告问题汇编

For functions satisfying any of the weak continuity notions occurring in decompositions of ordinary continuity, how difficult is it to compute the supremum on a compact interval?

来源字段：https://doi.org/10.4171/owr/2023/53

16511 | AIM-LOGIC-0140

### Symmetry and finite-extension descent for definable valuation rings

P / partially_solved | AIM 研讨会问题汇编

**Question 27 (Shlapentokh).** Can one find an algebraically closed field $K$ and a nontrivial valuation ring $R\subset K(Z)$ (or a finite extension), which is definable in $K(Z)$?

来源字段：https://aimath.org/WWN/hilberts10th/hilberts10th.pdf

16512 | AIM-LOGIC-0074

### Syntax-sensitive reductions for the nonconstant predicate on K(t)

P / partially_solved | AIM 研讨会问题汇编

For a field $K$, is the parameter-free existential theory of $$\mathcal M_K=(K(t);0,1,+,\cdot,T_K),\qquad T_K(f)\Longleftrightarrow f\notin K,$$ undecidable?

来源字段：https://aimath.org/pastworkshops/definabilityinntproblems.pdf

16513 | AMR-075-0005

### The main gap conjecture, e.g. for uncountable first order theories, for AECs, and for $\aleph_1$-saturated models of a countable theory

P / partially_solved | AMR 问题清单汇编

The main gap conjecture, e.g. for uncountable first order theories, for AECs, and for $\aleph_1$-saturated models of a countable theory.

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Model_theory_and_formal_languages

16514 | AIM-LOGIC-0012

### The parameter-free existential theory of F_p(t)

P / partially_solved | AIM 研讨会问题汇编

Is $\exists \operatorname{Th}(\mathbb F_p(t))$, in the language of rings without a name for $t$, decidable?

来源字段：http://aimpl.org/definedecide/3/

16515 | AIM-LOGIC-0017

### The standard integers and a constant-sort barrier

P / partially_solved | AIM 研讨会问题汇编

Is $(\mathbb Z,+,\cdot)$ definable as a subring by first-order formulas in $\mathbb C(t_1,\dots,t_k)$?

来源字段：http://aimpl.org/definedecide/4/

16516 | AIM-LOGIC-0050

### The super tree property at aleph_{omega+1} and the missing assignment quantifier

S / solved | AIM 研讨会问题汇编

Can we obtain the super tree property at $\aleph_{\omega+1}$?

The AIM record (workshop *High and low forcing*, section “Problems in High Forcing,” Problem 2.35) defines a thin $(\kappa,\lambda)$-tree as a set

$$ F\subseteq\{f:x\to2\mid x\in P_\kappa(\lambda)\} $$

that is closed under restriction, has a nonempty level $\operatorname{Lev}_x(F)=\{f\in F:\operatorname{dom}(f)=x\}$ for every $x\in P_\kappa(\lambda)$, and satisfies $|\operatorname{Lev}_x(F)|<\kappa$. It defines a cofinal branch as a function $b:\lambda\to2$ whose restriction to every $x\in P_\kappa(\lambda)$ lies on the $x$-level. Given a level assignment

$$ \vec f=\langle f_x:x\in P_\kappa(\lambda)\rangle, \qquad f_x\in\operatorname{Lev}_x(F), $$

it calls $b$ ineffable when

$$ S(F,\vec f,b)=\{x\in P_\kappa(\lambda):b\mathbin{\upharpoonright}x=f_x\} $$

is stationary. It then asks:

Here $P_\kappa(\lambda)=\{x\subseteq\lambda:|x|<\kappa\}$, and stationarity is with respect to the club filter on this set.

The displayed tree definition is mathematically coherent; `\upharpoonleft` in its first clause is a harmless restriction-symbol variant. The final definition, however, leaves the level assignment free. The standard and literature-confirmed reconstruction is

$$ \boxed{ \forall\lambda\geq\kappa\ \forall F\ \forall\vec f\ \exists b:\lambda\to2\quad S(F,\vec f,b)\text{ is stationary}}, $$

where $F$ ranges over thin $(\kappa,\lambda)$-trees and $\vec f$ ranges over all level assignments for $F$. The universal quantifier over assignments is essential, not stylistic: Section 5 below proves that replacing it by an existential quantifier collapses the purported super tree property exactly to the strong tree property.

来源字段：http://aimpl.org/highlowforcing/2/

16517 | AMR-075-0011

### The universality problem for C-free graphs

P / partially_solved | AMR 问题清单汇编

The universality problem for C-free graphs: For which finite sets C of graphs does the class of C-free countable graphs have a universal member under strong embeddings?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Model_theory_and_formal_languages

16518 | AMR-075-0012

### The universality spectrum problem

O / open | AMR 问题清单汇编

The universality spectrum problem: Is there a first-order theory whose universality spectrum is minimum?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Model_theory_and_formal_languages

16519 | AMR-074-0125

### Three-Element Better-Quasi-Order

P / partially_solved | AMR 问题清单汇编

Is there a subsystem weaker than $\mathsf{ATR}_0$ that proves that the three-element antichain is a better-quasi-order?

背景链接：https://math.berkeley.edu/~antonio/papers/questionsRM.pdf

16520 | OWR-1453-010

### Tits Alternatives for Finite QF-Rank Groups

O / open | Oberwolfach 报告问题汇编

Is there an analogue of the Tits alternative for finitely generated groups of finite quantifier-free rank (QF-rank)? Also, does there exist a CSA $p$-group of finite QF-rank?

来源字段：https://doi.org/10.4171/owr/2007/02

16521 | AIM-LOGIC-0018

### Transcendental endpoints and the curve-witness obstruction

P / partially_solved | AIM 研讨会问题汇编

Is there a diophantine subset \(X\) of \(\mathbb Q\) such that \(\sup\{x\mid x\in X\}\) is transcendental?

来源字段：http://aimpl.org/definedecide/5/

16522 | OWR-15181-017

### Transfer Conjecture for O-Minimal Exponential Fields

O / open | Oberwolfach 报告问题汇编

Is every o-minimal ordered exponential field $K$ satisfying EXP elementarily equivalent to the real exponential field $\mathbb{R}_{\exp}$?

来源字段：https://doi.org/10.4171/owr/2016/60

16523 | OWR-11695862-006

### Transfer Principles for NSOP and NTP Theories

O / open | Oberwolfach 报告问题汇编

Establish transfer principles for NSOP, NTP, and elimination of imaginaries from a model-complete theory of large fields of characteristic zero to its theory of differentially large fields.

来源字段：https://doi.org/10.4171/owr/2022/52

16524 | OWR-11575-013

### Turing Degree of Finite-Set Cardinality Indices

P / partially_solved | Oberwolfach 报告问题汇编

In computability theory and algorithmic randomness, let $(K_e)_{e \in \omega}$ be a standard effective listing of the finite $K$-trivial classes (sets of reals $A$ with prefix-free Kolmogorov complexity $K(A \upharpoonright n) \leq K(n) + e$ for all $n$). Does the cardinality function $e \mapsto |K_e|$ compute the Turing jump $\varnothing'$ or even the double jump $\varnothing''$?

来源字段：https://doi.org/10.4171/owr/2012/08

16525 | OWR-11575-014

### Turing Degrees of Differential-Field Embedding Images

P / partially_solved | Oberwolfach 报告问题汇编

For Harrington's computable differential-field embedding $g$, what is the Turing degree of the image $g(K)$, and what is the appropriate differential analogue of the splitting set?

来源字段：https://doi.org/10.4171/owr/2012/08

16526 | AIM-LOGIC-0076

### Turing degrees of infinite-solution Hilbert's Tenth Problems

P / partially_solved | AIM 研讨会问题汇编

**Question 16 (Miller).** Situate with respect to $\leq_T$ the following sets: \[ \operatorname{HTP}_{\infty}(\mathbb Q),\quad \operatorname{HTP}_{\infty,1}(\mathbb Q),\quad \operatorname{HTP}_{\infty}(\mathbb Z),\quad \operatorname{HTP}_{\infty,1}(\mathbb Z). \] (known: $\mathbf 0'\leq_T\operatorname{HTP}_{\infty}(\mathbb Z)\leq_T\mathbf 0''$)

来源字段：https://aimath.org/pastworkshops/definabilityinntproblems.pdf

16527 | AIM-LOGIC-0055

### Two gates for replacing Mitchell forcing by side conditions at a double successor

P / partially_solved | AIM 研讨会问题汇编

Is there a model for $\mathrm{TP}_{\kappa^{++}}$, with $\kappa$ a singular strong limit of countable cofinality, where we use side conditions in place of the Mitchell poset?

来源字段：http://aimpl.org/highlowforcing/3/

16528 | AIM-LOGIC-0105

### Two-distance graph coding in pure metric spaces

P / partially_solved | AIM 研讨会问题汇编

**Recovered statement.** Understand the **encoding** of graphs in metric structures.

来源字段：https://aimath.org/WWN/continuouslogic/continuouslogic.pdf

16529 | OWR-13354-003

### Two-Valued Realizability Toposes and Computable Dependent Choice

P / partially_solved | Oberwolfach 报告问题汇编

For the classical realizability topos K obtained from the coherence-space domain equation, is K two-valued, and does dependent choice continue to hold when the domain D is restricted to computable elements?

来源字段：https://doi.org/10.4171/owr/2014/52

16530 | AIM-LOGIC-0059

### Type and projection obstructions for a malformed side-condition problem

O / open | AIM 研讨会问题汇编

There is a coloring $F:[\aleph_2]^2 \rightarrow {0,1}$ of pairs from $\aleph_2$ in $2$ colors so that the poset of finite approximations to a $0$- or $1$-homogeneous set has the $\aleph_2$-chain condition. (We mean either the poset $\mathbb P$ of finite functions $f:\omega \rightarrow F^{-1}(0)$ or the poset of $f:\omega \rightarrow F^{-1}(0)$, ordered by inclusion)

Can this forcing be made proper using side conditions? Does this preserve the $\aleph_2$-chain condition?

来源字段：http://aimpl.org/highlowforcing/3/

16531 | OWR-1963-002

### Unavoidable Sequences and Brouwerian Theory $T_2$

O / open | Oberwolfach 报告问题汇编

Determine all sequences that are unavoidable over an arbitrary constructive $L$-theory including bar induction. In particular, is the Brouwerian theory $T_2$ recursively acceptable?

来源字段：https://doi.org/10.4171/owr/2008/18

16532 | AIM-LOGIC-0005

### Undoing precipitousness and the approximation obstruction

S / solved | AIM 研讨会问题汇编

Assume $\mathrm{NS}_{\omega_1}$ is precipitous. Is it possible to force $\mathrm{NS}_{\omega_1}$ to be non-precipitous without adding subsets of $\omega_1$?

来源字段：http://aimpl.org/alephtwo/6/

16533 | AIM-LOGIC-0097

### Uniform principal types and computable prime models

P / partially_solved | AIM 研讨会问题汇编

**1.10. Computable prime models (Andrews).**

**Question 18 (Millar).** Let \(T\) be a decidable theory having countably many countable models. When must the prime model have a decidable presentation? (Note that \(\omega\)-stability suffices.)

**Question 19.** Let \(T\) be a decidable theory having countably many countable models. What do we need to know to build a computable prime model of \(T\)? (Of course \(\omega\)-stability again suffices.)

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16534 | OWR-15958-009

### Uniform Quantitative Rates for Provably Convergent Sequences

P / partially_solved | Oberwolfach 报告问题汇编

For natural classes of provably convergent sequences, which uniform rates, moduli, or other quantitative forms of convergence can be extracted?

来源字段：https://doi.org/10.4171/owr/2017/53

16535 | OWR-4801-003

### Univalent Weakly Universal Kan Fibrations in Toposes

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist a univalent, weakly universal Kan fibration in the category of simplicial objects in every topos?

来源字段：https://doi.org/10.4171/owr/2011/11

16536 | OWR-15181-003

### Valuation and Simplicity Problems for the Surreal Numbers

O / open | Oberwolfach 报告问题汇编

Can the degree map on $\mathbf{No}$ be extended to a field valuation? Can $\mathbb{Q}\subset\mathbf{No}$ be characterized by sign sequences? Can the simplicity order be extended to functions, and is the surreal exponential the simplest homomorphism with the required properties?

来源字段：https://doi.org/10.4171/owr/2016/60

16537 | OPG-38188

### Vertex Cover Integrality Gap

P / partially_solved | Open Problem Garden

Conjecture For every $\varepsilon > 0$ there is $\delta > 0$ such that, for every large $n$, there are $n$-vertex graphs $G$ and $H$ such that $G \equiv_{\delta n}^{\mathrm{C}} H$ and $\mathrm{vc}(G) \ge (2 - \varepsilon) \cdot \mathrm{vc}(H)$.

背景链接：http://www.openproblemgarden.org/op/vertex_cover_integrality_gap

16538 | OWR-13354-001

### Weak Duality Proof Complexity and Strongly Polynomial Linear Programming

O / open | Oberwolfach 报告问题汇编

Does the variable-matrix form of weak linear-programming duality have no polynomial-size proof in the Lovasz-Schrijver system, and do strongly polynomial circuits for linear programming exist?

来源字段：https://doi.org/10.4171/owr/2014/52

16539 | AMR-074-0131

### Weak Infinitary Comprehension versus Weak Choice

O / open | AMR 问题清单汇编

Is weak-$L_{\omega_1,\omega}$-$\mathsf{CA}$ equivalent to weak-$\Sigma^1_1$-$\mathsf{AC}_0$?

背景链接：https://math.berkeley.edu/~antonio/papers/questionsRM.pdf

16540 | AIM-LOGIC-0002

### Weak square after collapsing a successor of a singular

P / partially_solved | AIM 研讨会问题汇编

Suppose \(\kappa\) is singular with cofinality \(\omega_1\). Let \(\lambda=\kappa^+\). Must \(\square_{\omega_1}^*\) hold in a forcing extension where \(\lambda\) is \(\aleph_2\)?

If \(V\subseteq W\), \(\lambda\) is a successor of a singular in \(V\), and \(\lambda=\aleph_2^W\), must weak square hold at \(\lambda\) in \(W\)?

来源字段：http://aimpl.org/alephtwo/3/

16541 | OWR-1390-004

### Weak-Arithmetic Pigeonhole Principles

O / open | Oberwolfach 报告问题汇编

Let $\mathrm{PHP}_n^m(f)$ assert that $f$ does not inject $\{1,\ldots,m\}$ into $\{1,\ldots,n\}$. Does $\mathrm I\Delta_0$ prove $\mathrm{PHP}_n^{n+1}(\Delta_0)$, or even $\mathrm{PHP}_n^{n^2}(\Delta_0)$? Does $\mathrm I\Delta_0(f)$ prove $\mathrm{PHP}_n^{2n}(f)$, or even $\mathrm{PHP}_n^{n^2}(f)$?

来源字段：https://doi.org/10.4171/owr/2006/52

16542 | AIM-LOGIC-0001

### Welch games and the cofinality threshold for terminal extension

P / partially_solved | AIM 研讨会问题汇编

The formal definition in Foreman--Magidor--Zeman (FMZ) is as follows.

A **\(\kappa\)-algebra** is a \(\kappa\)-complete Boolean subalgebra \(\mathcal A\subseteq\mathcal P(\kappa)\) of cardinality \(\kappa\). In the Welch game \(G^W_\gamma(\kappa)\), at every stage \(\alpha<\gamma\), Player I plays \(\mathcal A_\alpha\), extending all earlier algebras, and Player II responds with a uniform \(\kappa\)-complete ultrafilter \(U_\alpha\) on \(\mathcal A_\alpha\), extending all earlier \(U_\beta\). One may require without loss that \(\mathcal A_0\) contain all singletons. Player I moves first at limit stages. Player II wins exactly when legal responses are made at every stage below \(\gamma\).

FMZ also define, for limit \(\gamma\), a stronger terminal payoff \(Q_\gamma\). The corresponding game \(G^*_\gamma\) has the same moves, but Player II wins a full run only when all the \(U_\alpha\)'s have a common uniform \(\kappa\)-complete-ultrafilter extension on the \(\kappa\)-complete Boolean algebra generated by \(\bigcup_{\alpha<\gamma}\mathcal A_\alpha\).

1. the algebras are concrete subalgebras of \(\mathcal P(\kappa)\), have size \(\kappa\), and form an increasing chain; 2. Player II's objects are **uniform ultrafilters**, not arbitrary filters; 3. the ultrafilters must extend one another; and 4. the first literature bullet needs a large-cardinal hypothesis: the length-\(\omega\) winning-strategy statement is the weak-compactness characterization, not an unconditional fact about every regular \(\kappa\).

来源字段：http://aimpl.org/alephtwo/2/

16543 | AMR-074-0120

### Well-Ordered Linearizations

P / partially_solved | AMR 问题清单汇编

Over $\mathsf{RCA}_0$, is $\mathsf{EXT}(\omega^*)$ - the assertion that every well-founded partial order has a well-ordered linearization - equivalent to $\mathsf{ACA}_0$?

背景链接：https://math.berkeley.edu/~antonio/papers/questionsRM.pdf

16544 | AIM-LOGIC-0086

### What a computable nonforking section computes

P / partially_solved | AIM 研讨会问题汇编

Inspection of page 2 of the original AIM problem-session PDF shows that the material beginning “1.4.

The complete relevant source context is Section 1.3, “Continuous sections (Miller)”:

Thus “computable section” in Question 8 refers to a section \[ s=s^{\mathrm{nf}}_{M,N}:S_1(M)\longrightarrow S_1(N) \] of restriction \(r:S_1(N)\to S_1(M)\), where \(s(p)\) is the unique nonforking extension of \(p\) from \(M\) to \(N\). The PDF does not specify representations of models, formula algebras, or points of the Stone spaces. Without such data, “computable” has no invariant formal meaning. The results below therefore state an explicit effective coding convention.

来源字段：https://aimath.org/pastworkshops/computestabproblems.pdf

16545 | AMR-075-0014

### Wikipedia model theory and formal languages item 14: Assume K is the class of models of a countable first order theory omitting countably many types…

P / partially_solved | AMR 问题清单汇编

Assume K is the class of models of a countable first order theory omitting countably many types. If K has a model of cardinality $\aleph_{\omega_1}$ does it have a model of cardinality continuum?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Model_theory_and_formal_languages

16546 | AMR-075-0024

### Wikipedia model theory and formal languages item 24: What is the nature of the proof-theoretic ordinal (the smallest ordinal a theory cannot prove w…

O / open | AMR 问题清单汇编

What is the nature of the proof-theoretic ordinal (the smallest ordinal a theory cannot prove well-founded) for second-order arithmetic, ZFC, or stronger theories?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Model_theory_and_formal_languages

16547 | OWR-15958-003

### Wraith Redundancy for Algebraic Theories

S / solved | Oberwolfach 报告问题汇编

For an algebraic theory $T$, does every $T$-redundant sentence hold in the generic model of $T$?

来源字段：https://doi.org/10.4171/owr/2017/53

16548 | OWR-11695862-009

### Zilber’s Quasiminimality Conjecture for Complex Exponentiation

O / open | Oberwolfach 报告问题汇编

Prove that the complex exponential field $\langle\mathbb C;+ ,\cdot,\exp\rangle$ is quasiminimal: every definable subset is countable or cocountable.

来源字段：https://doi.org/10.4171/owr/2022/52

# 资料汇编 15 | 理论计算机科学 / Computer Science

554 条记录。O=开放；P=部分解决；S=已解决。状态与英文/LaTeX 表述均沿用来源。

16549 | OWR-16634-006

### $d$-to-$1$ Games Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For every integer $d\geq2$ and every $s>0$, does there exist an alphabet size $k$ such that it is NP-hard to distinguish satisfiable $d$-to-$1$ Label-Cover instances from instances of value at most $s$?

来源字段：https://doi.org/10.4171/owr/2018/51

16550 | AMR-027-1001

### 10 Lectures and 42 Open Problems - Angular Synchronization via Projected Power Method

S / solved | AMR 问题清单汇编

In **angular synchronization** (estimating angles $\theta_i\in\mathbb S^1$, $i=1,\dots,n$, from noisy pairwise relative measurements $\tilde Y_{ij}\approx e^{\iota(\theta_i-\theta_j)}$ subject to Gaussian noise), the goal is to recover the phases. **Does the Projected Power Method (PPM - power iteration followed by re-normalization/projection onto the unit circle) converge, with high probability, to the optimal (global) solution of the angular synchronization problem when the noise is Gaussian with sufficiently small variance $\sigma$?**

背景链接：https://afonsobandeira.wordpress.com/2015/12/09/42l10psynchronization/

16551 | AMR-027-0702

### 10 Lectures and 42 Open Problems - Boolean classification and annulus conjecture

P / partially_solved | AMR 问题清单汇编

Prove or disprove: $R_A(\alpha n,\beta n,n)=\alpha+(1-\alpha)R_A(1,\beta n,(1-\alpha)n)+o(1)$.

背景链接：https://afonsobandeira.wordpress.com/2015/11/27/10l42grouptesting/

16552 | AMR-027-0302

### 10 Lectures and 42 Open Problems - Certifying that matrices are PSD

P / partially_solved | AMR 问题清单汇编

Given a symmetric matrix ${M}$ with small condition number, is there a quasi-linear time (on ${n}$ and the number of non-zero entries of ${M}$ ) procedure that certifies that ${M\succeq 0}$ . More specifically, the procedure can be randomized in the sense that it may, with some probably not certify that ${M\succeq 0}$ even if that is the case, what is important is that it never produces erroneous certificates (and that it has a bounded-away-from-zero probably of succeeding, provided that ${M\succeq 0}$ ).

背景链接：https://afonsobandeira.wordpress.com/2015/10/05/10l42pcheeger/

16553 | AMR-027-0502

### 10 Lectures and 42 Open Problems - Certifying the Restricted Isometry Property

S / solved | AMR 问题清单汇编

Let $N = 2M$ . For which $s$ is there a polynomial time algorithm that is guaranteed to, with high probability, certify that a gaussian matrix $A$ is $\left(s,\frac13\right)$ -RIP? In particular, a $\left(s,\frac13\right)$ -RIP matrix has to not have $s$ sparse vectors in its nullspace. This motivates a second question: Let $N = 2M$ . for which $s$ is there a polynomial time algorithm that is guaranteed to, with high probability, certify that a gaussian matrix $A$ does not have $s$ -sparse vectors in its nullspace?

背景链接：https://afonsobandeira.wordpress.com/2015/11/22/10l42gordon/

16554 | AMR-027-1004

### 10 Lectures and 42 Open Problems - Consistency and sample complexity of Multireference Alignment

P / partially_solved | AMR 问题清单汇编

Let $x\in\mathbb R^L$ be the unknown signal in **Multireference Alignment**, observed via $N$ noisy random-phase (cyclic-shift) copies. After **fixing/providing the power spectrum** (to resolve the inherent shift-ambiguity), **is the Maximum Likelihood (ML) estimate consistent** as $N\to\infty$? And **what is the sample complexity** - the minimal number of observations $N$ (as a function of the noise variance $\sigma^2$ and dimension $L$) needed to estimate $x$ to within a fixed accuracy?

背景链接：https://afonsobandeira.wordpress.com/2015/12/09/42l10psynchronization/

16555 | AMR-027-0605

### 10 Lectures and 42 Open Problems - Constructive Kadison-Singer

P / partially_solved | AMR 问题清单汇编

Give a **polynomial-time (constructive)** algorithm producing the tight-frame partition (equivalently, the two-coloring of vectors) guaranteed to exist by the Kadison-Singer problem / Weaver's conjecture $\mathsf{KS}_2$. Marcus, Spielman, and Srivastava (MSS) proved existence **non-constructively** (via interlacing families of polynomials). Is there an efficient algorithm to *find* the partition?

背景链接：https://afonsobandeira.wordpress.com/2015/12/07/10l42pcompressedsensing/

16556 | AMR-027-0501

### 10 Lectures and 42 Open Problems - Deterministic Restricted Isometry Property matrices

P / partially_solved | AMR 问题清单汇编

Construct deterministic matrices $A\in\mathbb{C}^{M\times N}$ (or $A\in\mathbb{R}^{M\times N}$ ) satisfying the $(s,\frac13)$ -RIP for $s\approx\frac{M^{0.6}}{\mathrm{polylog}(N)}$

背景链接：https://afonsobandeira.wordpress.com/2015/11/22/10l42gordon/

16557 | AMR-027-0701

### 10 Lectures and 42 Open Problems - Gilbert-Varshamov bound

P / partially_solved | AMR 问题清单汇编

The Gilbert-Varshamov (GV) bound gives the best known *existential* lower bound on the rate of error-correcting codes of a given relative distance: a random (or random linear) code achieves it with high probability. Two questions: 1. **Explicitness.** Is there an explicit, deterministic polynomial-time construction of binary (or $q$-ary) codes that *achieve* the GV bound (rate-distance tradeoff at least as good as the bound)? 2. **Tightness.** Is the GV bound tight, or can every code be beaten (are there codes strictly better than the GV bound)?

背景链接：https://afonsobandeira.wordpress.com/2015/11/27/10l42grouptesting/

16558 | AMR-027-0301

### 10 Lectures and 42 Open Problems - Hardness at the Cheeger square-root gap

P / partially_solved | AMR 问题清单汇编

Does there exist $c>0$ such that it is NP-hard, given a graph $G$ and $\phi>0$, to distinguish $h_G\le\phi$ from $h_G\ge c\sqrt{\phi}$?

背景链接：https://afonsobandeira.wordpress.com/2015/10/05/10l42pcheeger/

16559 | AMR-027-0805

### 10 Lectures and 42 Open Problems - Maximum and minimum bisections on random regular graphs

P / partially_solved | AMR 问题清单汇编

Given a $d$ -regular graph on $n$ nodes $G$ . Let $MaxBis(G)$ and $MinBis(G)$ denote, respectively, the size of its largest and smallest bisection. Is it true that for every $d$ $MaxBis(G)$ $+ MinBis(G)$ $=$ $\frac{d}{2} + o(1)$ , where $o(1)$ is a term that goes to zero as $n$ grows?

背景链接：https://afonsobandeira.wordpress.com/2015/12/04/10l42pmaxcut/

16560 | AMR-027-0602

### 10 Lectures and 42 Open Problems - Mutually Unbiased Bases

O / open | AMR 问题清单汇编

Let $d=6$. A set of orthonormal bases $\{B_1,\dots,B_m\}$ of $\mathbb C^6$ is **mutually unbiased** if $|\langle u,v\rangle|^2=1/6$ for every pair of vectors $u,v$ taken from two *different* bases. It is a classical fact that at most $d+1=7$ such bases can exist (saturating it gives a complete set, equivalent to a projective plane of order $d$). **How many mutually unbiased bases are there in dimension 6?**

背景链接：https://afonsobandeira.wordpress.com/2015/12/07/10l42pcompressedsensing/

16561 | AMR-027-0303

### 10 Lectures and 42 Open Problems - Open Problem 3.3

P / partially_solved | AMR 问题清单汇编

Let ${G=(V,E,W)}$ be a graph and ${k}$ a positive integer, is the following true? $\rho_G(k) \leq \mathrm{polylog}(k) \sqrt{\lambda_k}. \ \ \ \ (2)$

背景链接：https://afonsobandeira.wordpress.com/2015/10/05/10l42pcheeger/

16562 | AMR-027-0601

### 10 Lectures and 42 Open Problems - Random Partial Discrete Fourier Transform

O / open | AMR 问题清单汇编

Consider a $A\in\mathbb{C}^{M\times N}$ obtained by sampling random rows of a Discrete Fourier Tranform. How large does $M$ need to be in order for, with high probability, $A$ to satisfy the $(s,\frac13)$ -RIP?

背景链接：https://afonsobandeira.wordpress.com/2015/12/07/10l42pcompressedsensing/

16563 | AMR-027-1002

### 10 Lectures and 42 Open Problems - Sharp tightness of the Angular Synchronization SDP

S / solved | AMR 问题清单汇编

Is the SDP for angular synchronization tight (with high probability) for noise levels $\sigma$ essentially until the solution of angular synchronization no longer correlates with the ground truth?

背景链接：https://afonsobandeira.wordpress.com/2015/12/09/42l10psynchronization/

16564 | AMR-027-0802

### 10 Lectures and 42 Open Problems - Sum of Squares approximation ratio for Max-Cut

O / open | AMR 问题清单汇编

Let $\mathrm{SOS}_d$ denote the degree-$d$ Sum-of-Squares (Lasserre) relaxation of the Max-Cut problem, and let $\mathrm{gap}_d$ be its integrality gap (equivalently, the best achievable approximation ratio). The Goemans-Williamson SDP is the degree-2 SOS relaxation with ratio $0.878\ldots$. **What is the approximation ratio / integrality gap of the degree-4 SOS relaxation? What about other constant degrees?**

背景链接：https://afonsobandeira.wordpress.com/2015/12/04/10l42pmaxcut/

16565 | AMR-027-0704

### 10 Lectures and 42 Open Problems - The Deletion Channel

P / partially_solved | AMR 问题清单汇编

What are the asymptotics of $\mathcal{D}\left(n;\frac12\right)$ ? \item An interesting aspect of the Deletion Channel is that different messages may have different difficulties of decoding. This motivates the following question: What are the two (distinct) binary sequences $x^{(1)}$ and $x^{(2)}$ that are more difficult to distinguish (let's say that the receiver knows that either $x^{(1)}$ or $x^{(2)}$ was sent but not which)?

背景链接：https://afonsobandeira.wordpress.com/2015/11/27/10l42grouptesting/

16566 | AMR-027-0803

### 10 Lectures and 42 Open Problems - The Grothendieck Constant

O / open | AMR 问题清单汇编

Determine the exact value of the **real Grothendieck constant** $K_G$, defined as the least $K$ such that, for every real matrix $A=(a_{ij})$ and every choice of signs, one has $$ \Bigl|\sum_{i,j} a_{ij}\langle s_i,t_j\rangle\Bigr| \le K \max_{\varepsilon_i,\delta_j\in\{\pm1\}}\Bigl|\sum_{i,j} a_{ij}\varepsilon_i\delta_j\Bigr| $$ for all unit vectors $s_i,t_j$ in a Hilbert space. The supremum $K_G$ of the ratio over all finite matrices is a universal constant between $1$ and $\approx1.8$; its exact value is unknown.

背景链接：https://afonsobandeira.wordpress.com/2015/12/04/10l42pmaxcut/

16567 | AMR-027-0804

### 10 Lectures and 42 Open Problems - The Paley Clique Problem

O / open | AMR 问题清单汇编

Let $q\equiv 1\pmod4$ be a prime power and let $G=\operatorname{Paley}(q)$ be the **Paley graph** (vertices $\mathbb F_q$; $uv$ adjacent iff $u-v$ is a quadratic residue). Two questions: 1. **What is the clique number $\omega(\operatorname{Paley}(q))$?** (The Paley-Clique Conjecture asserts $\omega(\operatorname{Paley}(q))=O(\sqrt q\,\log_2 q)$, or more precisely that the independence/clique number is $\sim\sqrt q$ up to poly-log factors.) This is tied to the **Hilbert number / Paley graph clique** conjecture of Cohen and to sum-product/pseudorandomness heuristics. 2. **Can the Sum-of-Squares (SOS) degree-4 analogue of the Lovász theta number $\vartheta$ help upper-bound the clique number of the Paley graph** - i.e., does a degree-4 SOS "theta" provably remain near $\sqrt q$ and hence certify a small upper bound on $\omega$?

背景链接：https://afonsobandeira.wordpress.com/2015/12/04/10l42pmaxcut/

16568 | AMR-027-0604

### 10 Lectures and 42 Open Problems - The Paley ETF Conjecture

P / partially_solved | AMR 问题清单汇编

Let $p\equiv 1\pmod 4$ be prime and let $\Phi_p$ be the **Paley equiangular tight frame (ETF)**, a $\tfrac{p+1}{2}\times(p+1)$ matrix built from quadratic residues mod $p$ (a normalized DFT-type matrix whose columns form an ETF). The **square-root bottleneck** says that for most explicit constructions the RIP only holds up to sparsity $K=O(\sqrt M)$ (from mutual coherence / Welch bound). **Does the Paley ETF satisfy the Restricted Isometry Property (RIP) at sparsity $K$ beyond the square-root bottleneck, i.e. $K=\Omega(p^{1/2+\varepsilon})$ for some $\varepsilon>0$ (even by logarithmic factors)?**

背景链接：https://afonsobandeira.wordpress.com/2015/12/07/10l42pcompressedsensing/

16569 | AMR-027-1003

### 10 Lectures and 42 Open Problems - Tightness of the Multireference Alignment SDP

P / partially_solved | AMR 问题清单汇编

In **Multireference Alignment (MRA)** - estimating a signal $x\in\mathbb R^L$ from many noisy observations that are each a **random cyclic shift** of $x$ plus Gaussian noise - a common relaxation is a **Semidefinite Programming (SDP)** approach. **For which levels of noise is the MRA SDP tight** (recovers the underlying signal / its orbit)?

背景链接：https://afonsobandeira.wordpress.com/2015/12/09/42l10psynchronization/

16570 | AMR-108-0018

### 3.10 (Futer, Schleimer) - A practical 3-manifold homeomorphism algorithm

P / partially_solved | AMR 问题清单汇编

Is there a practical algorithm to test whether a pair of $3$-manifolds are homeomorphic?

背景链接：https://arxiv.org/pdf/1512.04620#page=3

16571 | AMR-108-0033

### 5.8 (Schleimer) - Why SnapPy works in practice

O / open | AMR 问题清单汇编

Give a rigorous explanation for why SnapPy works so well in practice.

背景链接：https://arxiv.org/pdf/1512.04620#page=4

16572 | AMR-108-0058

### 8.4 (Schleimer) - Detecting reducible Heegaard splittings

P / partially_solved | AMR 问题清单汇编

Is there an algorithm to detect whether a Heegaard splitting is reducible and, if so, find a reducing curve?

背景链接：https://arxiv.org/pdf/1512.04620#page=6

16573 | AIM-CRYPTOGRAPHY-0018

### A batch-fidelity obstruction in hidden-shift attacks on unique-SVP

P / partially_solved | AIM 研讨会问题汇编

Audit the known Regev reduction from a lattice promise problem to cyclic dihedral coset states, compose its parameter blowup with known hidden-shift solvers when justified, charge state preparation as well as hidden-shift postprocessing, and state precisely what this does and does not imply for approximate SVP.

来源字段：http://aimpl.org/quantumalg/4/

16574 | AIM-COMPUTATION-0022

### A binary-collapse-invisible likelihood face

S / solved | AIM 研讨会问题汇编

Maximum likelihood estimates for an $I\times J\times K$ table under the no 3-way interaction model do not exist if and only if they do not exist for all collapsings of the table to a $2\times2\times2$ table.

The official PDF says:

It immediately adds that Nicholas Eriksson showed later in the week that the conjecture is false already for $3\times3\times3$ tables. It then records Mathias Drton's question about structural zeros and Stephen Roehrig's question about decomposable network flows. The isolated line “Akimichi Takemura” after those questions is the speaker heading for Question 9, not an attribution or a condition attached to Conjecture 8. This boundary was checked directly in the official PDF.

There is a genuine quantifier issue. Read literally, the displayed sentence is

$$ \operatorname{NoMLE}(n) \quad\Longleftrightarrow\quad \text{for every binary collapse }\pi, \operatorname{NoMLE}(\pi n). \tag{1.1} $$

That universal statement is false already for a $2\times2\times3$ table; Proposition 5.4 below gives a two-line example. The contemporaneous paper of Eriksson--Fienberg--Rinaldo--Sullivant (EFRS), together with the workshop's claim that the first failure is at $3^3$, identifies the mathematically substantive intended reading as the **detection formulation**

$$ \operatorname{NoMLE}(n) \quad\Longleftrightarrow\quad \text{there exists at least one binary collapse }\pi \text{ with }\operatorname{NoMLE}(\pi n). \tag{1.2} $$

In other words, one tests the whole family of collapses in order to find a witness; one does not require every member of that family to fail. Formula (1.2) is equivalent to asking whether all relevant facets of the large marginal cone are pulled back from binary marginal cones. This report quotes (1.1) as the literal source and treats (1.2) as a reconstructed formulation, not as a silent rewrite.

EFRS define an **elementary collapsing** as merging a nonempty set of levels of one variable into one level, and a collapsing as a sequence of elementary collapsings. A final binary collapse is therefore a choice of a nontrivial two-block partition of the levels of each variable. Binary labels are irrelevant. Thus a three-level variable has three such partitions, each of singleton-versus-pair form, and a $3^3$ table has $3^3=27$ binary collapses up to binary label swaps.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16575 | AIM-COMPUTATION-0086

### A certificate-first algebraic layer for R

P / partially_solved | AIM 研讨会问题汇编

Question 71 (Sturmfels). What should be the first algebraic functions available in R? Dinwoodie: Toric Markov bases Riccomagno: Ideal of points Pachter: A general R interface with algebra packages Riccomagno: Gr¨ obner bases over fraction fields, with algebraic numbers and parame-ters. Pachter: When do we find out when these things have been implemented? Allman: Improve the documentation!

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16576 | AIM-COMPUTATION-0043

### A cyclic-slice chart for balanced minimal border rank

P / partially_solved | AIM 研讨会问题汇编

Problem 29. Compute the ideal of the secant varieties Sec k(Pk-1 × Pk-1 × Pk-1) for k ≥ 4.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16577 | AIM-CRYPTOGRAPHY-0010

### A fiber--leakage--arity audit for group-theoretic cryptographic building blocks

P / partially_solved | AIM 研讨会问题汇编

**Cryptographic building blocks from problems in group theory**

We would like to use computational problems in group theory to derive fundamental cryptographic objects including one-way functions and multilinear maps.

来源字段：http://aimpl.org/postquantgroup/3/

16578 | AIM-COMPUTATION-0102

### A format-sensitive genus barrier for Schoof-Pila

P / partially_solved | AIM 研讨会问题汇编

**Question 14.** Pila's method for computing roots of unity is exponential in $g$. Is there any reason it can't be made linear? (Elkies)

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16579 | AIM-COMPUTATION-0061

### A four-point counterexample and the border-basis repair

S / solved | AIM 研讨会问题汇编

Let $$ D=\{v_1,\ldots,v_N\}\subseteq\mathbb Q^d $$ be a design of $N$ distinct rational points. Let $$ B=\{x^{\alpha_1},\ldots,x^{\alpha_N}\} $$ be monomials that are linearly independent modulo the design ideal $I(D)$, and suppose $B$ is the set of standard monomials of a zero-dimensional monomial ideal $$ M=\langle m_1,\ldots,m_r\rangle. $$ Because $|B|=N=\dim_{\mathbb Q}\mathbb Q[x]/I(D)$, $B$ is a quotient basis. Interpolate each $m_i$ uniquely in the basis $B$: $$ m_i\equiv h_i\pmod{I(D)},\qquad h_i\in\operatorname{span}_{\mathbb Q}B, $$ and put $f_i=m_i-h_i\in I(D)$. Must $$ V(f_1,\ldots,f_r)=D? $$

The source writes $D\subseteq\mathbb Q^d$ but the polynomial ring as $\mathbb Q[x_1,\ldots,x_n]$. Evaluation requires the number of coordinate variables to agree with the ambient dimension (or an additional coordinate map), so throughout this report the intended reading is $n=d$.

No counterexample is supplied there.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16580 | AIM-COMPUTATION-0051

### A gap-move bound and the exact decomposable-model spectrum

P / partially_solved | AIM 研讨会问题汇编

**Problem 37.** Find the spectra of various models and classes of models.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16581 | AIM-COMPUTATION-0047

### A joint tree and JC69/K80 selector with a finite-sample certificate

P / partially_solved | AIM 研讨会问题汇编

**Question 33.** How can we efficiently decide which trees fit the data best? Which model of mutation is most accurate?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16582 | AIM-COMPUTATION-0098

### A local-to-global screen for number-field ECM torsion

P / partially_solved | AIM 研讨会问题汇编

**Question 10.** To speed up the elliptic curve factorization algorithm, pick $E/\mathbb Q$ with large torsion group so that its reduction modulo $n$ is more likely to be smooth. Mazur's results show that this torsion subgroup can be no larger than $16$. There is a result of Kamienny--Mazur--Merel: there is a bound on $E(K)_{\mathrm{tors}}$ in terms of $[K:\mathbb Q]$. So try to find a number field $K$ where $p$ splits completely and such that $E(K)_{\mathrm{tors}}$ is large. (Pomerance)

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16583 | AIM-COMPUTATION-0100

### A nonresidue-to-witness reduction and exact witness count

P / partially_solved | AIM 研讨会问题汇编

**Question 12 (Kedlaya).** Let \(p\) be prime, write \[ p-1=2^\ell m \] where \(m\) is odd, and assume \(\ell\geq3\). Find in deterministic polynomial time \(x\in\mathbb F_p\) such that \[ x^{2^{\ell-1}}+1 \] is not a \(2^{\ell-1}\)-th power.

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16584 | AIM-COMPUTATION-0089

### A norm obstruction for Agrawal's conjecture

P / partially_solved | AIM 研讨会问题汇编

**Agrawal's conjecture.** Let \(n,r\) be coprime positive integers. If \[ (X-1)^n\equiv X^n-1\pmod{n,\;X^r-1}, \] must either \(n\) be prime or \[ n^2\equiv1\pmod r? \]

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16585 | AIM-CRYPTOGRAPHY-0008

### A one-round action barrier and leave-one-out multiparty key routing

P / partially_solved | AIM 研讨会问题汇编

Generally speaking the field does not seem to offer methods of key establishment. It remains open to construct standard key exchange protocols as well as more complicated multi-party variants.

来源字段：http://aimpl.org/postquantgroup/2/

16586 | AIM-COMPUTATION-0052

### A precise probabilistic repair and algebraic criteria for interval cell spectra

P / partially_solved | AIM 研讨会问题汇编

“Shmuel Onn spoke about the spectra of hierarchical models. The spectra is defined to be the set of all sets of possible cell entries in for a multiway nonnegative integral table with fixed marginal totals.”

**Problem 37.** Find the spectra of various models and classes of models.

**Question 38.** With respect to some distribution is the set of tables whose range of cell entries are intervals dense among all tables?

Under **Shmuel Onn** it says:

The wording of Question 38 does not specify:

1. whether the random objects are tables, margin vectors, or models; 2. a probability law or a sequence of laws; 3. whether “dense” means topologically dense, positive probability, probability one, or asymptotic probability one; 4. whether “range” refers to one distinguished cell or every cell.

Consequently the original sentence is not, by itself, a well-posed yes/no proposition. The mathematical results below first isolate the deterministic property and then give an explicit probabilistic repair.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16587 | AIM-COMPUTATION-0077

### A scale-normalized robustness criterion for algebraic design fans

P / partially_solved | AIM 研讨会问题汇编

**Question 62 (Riccomagno). Is there an algebraic method to decide which Est \(\tau\) factor is best when working with perturbed designs?**

“A design \(D\) is a finite subset of \(\mathbb R^d\). The design ideal \(I(D)\) is the vanishing ideal of polynomial functions vanishing on \(D\). The set of standard monomials modulo \(I(D)\) given a term order \(\tau\) is denoted \(\mathrm{Est}_\tau\). A linear polynomial model \(L\) is a list of monomials.”

**Among the distinct standard-monomial bases \(\mathrm{Est}_\tau\) supplied by term orders, can algebra select one that is best for a perturbed design?**

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16588 | AIM-CRYPTOGRAPHY-0003

### A stabilizer-aware query barrier for vectorization lower bounds

P / partially_solved | AIM 研讨会问题汇编

**Quantum lower bounds for group-theoretic problems in the generic group model**

We would like to solidify the credentials of group-theoretic cryptography as post-quantum by developing lower bounds for the quantum complexity of problems arising in group theory. It would be especially nice to develop such a bound for the vectorisation problem, since this underpins many hardness problems in the area.

来源字段：http://aimpl.org/postquantgroup/1/

16589 | AIM-COMPUTATION-0057

### A support-aware exchange between levels and binary variables

P / partially_solved | AIM 研讨会问题汇编

**Question 43 (Sturmfels).** Which is more important to solve: fixing the model and increasing the number of levels in the model, or fixing binary random variables and letting the number of random variables get large?

**Meek:** It depends on the type of problems you are interested in studying.

**Roehrig:** You can exchange a solution to problems on binary random variables to a problem on nonbinary random variables.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16590 | AIM-COMPUTATION-0037

### A zero-dimensional parity independence variety

P / partially_solved | AIM 研讨会问题汇编

Suppose independence varieties are properly defined, also for the CI constraints involving functional independences, over the field of complex numbers.

**Conjecture 23.** When \(P\) belongs to such a zero dimensional independence variety then \(P\) is the distribution of a random vector with its CI-structure equal to the CI-structure of a matroid.

Donald Richards

Donald Richards spoke on the problem of determining the number of solutions to generic maximum likelihood problems and the use of computational algebra, homotopy continuation, and mixed volume in these calculations.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16591 | OWR-14298371-009

### Abelian Embeddings and Gap-CSP Integrality Gaps

O / open | Oberwolfach 报告问题汇编

Can the complexity of $\operatorname{gap-CSP}\text{-}P[1,s]$ differ from that of $\operatorname{gap-CSP}\text{-}P[1-\varepsilon,s]$ only when $P$ admits a nontrivial Abelian embedding?

来源字段：https://doi.org/10.4171/owr/2024/27

16592 | OWR-13949-003

### Accurate Answers to Adaptively Chosen Statistical Queries

P / partially_solved | Oberwolfach 报告问题汇编

Given a population distribution P on X and an i.i.d. sample S of size n, can a possibly randomized procedure M(S) answer every adaptively chosen sequence of statistical queries accurately with high probability over S and M's randomness?

来源字段：https://doi.org/10.4171/owr/2015/54

16593 | OWR-3394-008

### Achievable External Behaviors Under Controller Interconnection

P / partially_solved | Oberwolfach 报告问题汇编

Given a plant $P$ interconnected with a controller $C$ through control variables $c$, characterize the systems achievable in the external variables $w$, up to bisimulation.

来源字段：https://doi.org/10.4171/owr/2009/11

16594 | OWR-15210-007

### Achievable Relations in Multi-Key Ciphertext Transformations

P / partially_solved | Oberwolfach 报告问题汇编

Given ciphertexts encrypting $x_1,\ldots,x_n$ under independent public keys and transformed ciphertexts decrypting to $y_1,\ldots,y_n$, which relations between $(x_1,\ldots,x_n)$ and $(y_1,\ldots,y_n)$ can an adversary induce?

来源字段：https://doi.org/10.4171/owr/2017/5

16595 | OWR-17129-001

### Adaptive Refinement and Global Optimization of Topological Shapes

O / open | Oberwolfach 报告问题汇编

For adaptive topology optimization using level-set and state-variable discretizations, determine reliable refinement and coarsening strategies, methods for escaping local minima, efficient hierarchical B-spline gradient schemes, couplings with isogeometric shape optimization, and algorithms for extracting surface or volume parametrizations from the level-set field.

来源字段：https://doi.org/10.4171/owr/2019/33

16596 | OWR-13353-017

### Additive Integrality Gap for Ring Loading

P / partially_solved | Oberwolfach 报告问题汇编

For every instance of the ring-loading problem, let $L$ be the optimum unsplittable load, let $L^*$ be the optimum splittable load, and let $D$ be the maximum demand. Is $L\leq L^*+11D/10$?

来源字段：https://doi.org/10.4171/owr/2014/51

16597 | AIM-COMPUTATION-0093

### Adjacent nonisogenous fibers in the Legendre family

P / partially_solved | AIM 研讨会问题汇编

Suppose that you have a nonconstant family of elliptic curves \(E_\lambda\) over \(\mathbb F_p\) (e.g. the Frey curves), with \(\lambda\in\mathbb P^1(\mathbb F_p)\). Can you find deterministically \(\lambda\) such that \(E_\lambda\) and \(E_{\lambda+1}\) are not isogenous (i.e. \[ \#E_\lambda(\mathbb F_p)\ne \#E_{\lambda+1}(\mathbb F_p))? \]

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16598 | OWR-14298796-001

### Adjoint-Free Approximation of Nonsymmetric Low-Rank Matrices

P / partially_solved | Oberwolfach 报告问题汇编

Determine when and with what sample complexity a nonsymmetric low-rank matrix or nonselfadjoint compact operator can be approximated using only forward sketches, without querying its adjoint.

来源字段：https://doi.org/10.4171/owr/2024/48

16599 | OWR-14299525-003

### Agentic and Neuro-Symbolic AI for Mathematical Research

O / open | Oberwolfach 报告问题汇编

Build an agentic AI system that carries out the full workflow of a mathematician, from exploring examples and formulating conjectures to selecting proof strategies, and determine whether it can solve research-level benchmarks such as FATE-X autonomously. Identify an effective architecture that combines neural pattern recognition with rigorous symbolic deduction to produce novel, human-readable, formally checkable proofs.

来源字段：https://doi.org/10.4171/owr/2025/43

16600 | OWR-2489-006

### Alcuin Numbers and Recognition of Small Boat Graphs

O / open | Oberwolfach 报告问题汇编

For a perfect graph, is there a polynomial-time algorithm for computing its Alcuin number? Is recognizing small-boat graphs in NP?

来源字段：https://doi.org/10.4171/owr/2008/51

16601 | OWR-8415359-008

### Algebraic and Graph-Property Tests for Quantum Functions

O / open | Oberwolfach 报告问题汇编

Can one characterize when a test forces a quantum function to be commutative? What happens when graph properties are tested with quantum functions? Beyond dimension testing, can one design tests forcing the measurements of a quantum function to generate a specified algebra? Are there applications to algorithms, complexity theory, or cryptography?

来源字段：https://doi.org/10.4171/owr/2021/54

16602 | AIM-COMPUTATION-0060

### Algebraic certificates for nonparametric mixture likelihood

P / partially_solved | AIM 研讨会问题汇编

**Question 46 (Steele).** What might be the application of these techniques to nonparametric models?

Apply convex and algebraic methods to a genuinely infinite-dimensional nonparametric maximum-likelihood problem in which the unknown is an arbitrary mixing distribution.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16603 | OWR-9790359-008

### Algebraic Complexity of Borel Constraint Problems

O / open | Oberwolfach 报告问题汇编

For a finite relational structure $D$, absence of a homomorphism $f:D^4\to D$ satisfying $f(a,r,e,a)=f(r,a,r,e)$ makes $\operatorname{CSP}_B(D)$ $\boldsymbol{\Sigma}^1_2$-complete. Is the converse true? Can Borel complexity, efficient local solvability for directed graphs, and extension to all locally checkable labelings be characterized algebraically?

来源字段：https://doi.org/10.4171/owr/2022/8

16604 | AIM-COMPUTATION-0033

### Algebraic fibers and sharp sets for missing categorical data

P / partially_solved | AIM 研讨会问题汇编

**Question 19.** Can algebraic techniques be used to better imputing missing categorical variable?

Inspection of the official PDF places it under Russell Steele, immediately after Question 18 about locating modes in a mixture and immediately before Question 20 about collapsing levels of categorical variables for multiple imputation. The source says:

Two natural repairs are “used to better impute a missing categorical variable” and “used to improve imputation of missing categorical variables.” The adjacent multiple-imputation question favors the second, broader reading, which is used below. The original wording is preserved above.

An imputation problem is not defined until the observation pattern, missingness mechanism, substantive model, and output type are specified. This report treats the smallest nontrivial population model in which those distinctions can all be made exact: a fully observed categorical variable \(X\), a partly observed categorical variable \(Y\), and its response indicator \(R\). The output is usually a set of imputation distributions, not a hard category assignment.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16605 | OWR-12697711-014

### Algorithms for Amoeba Dimensions from Tropical Fans

P / partially_solved | Oberwolfach 报告问题汇编

Given a $d$-dimensional rational polyhedral fan with support $F\subseteq\mathbb R^n$, compute $\min_U\{2\dim_{\mathbb R}(U+F)-\dim_{\mathbb R}U\}$ over subspaces $U$ spanned by rational vectors.

来源字段：https://doi.org/10.4171/owr/2023/15

16606 | OWR-8415352-006

### Algorithms for Symmetry-Reduced Semidefinite Programs

O / open | Oberwolfach 报告问题汇编

Can the small number of variables in symmetry-reduced semidefinite programs be exploited to obtain faster algorithms?

来源字段：https://doi.org/10.4171/owr/2021/49

16607 | AIM-COMPUTATION-0025

### An algebraic one-mode certificate for MANOVA

P / partially_solved | AIM 研讨会问题汇编

Can algebraic techniques be used to see the unimodality of the likelihood function in a MANOVA model. These models correspond to the chain graphs as described above but with the extra edges \(X_i\to Y_j\) for each \(i\) and \(j\).

Akimichi Takemura asked, “As the number of samples grows, how quickly does the likelihood converge to a unimodal function in the bivariate SUR model?”

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16608 | AIM-COMPUTATION-0054

### An ancestral-observation certificate for likelihood unimodality

P / partially_solved | AIM 研讨会问题汇编

**Question 40 (Wynn).** Which nodes being hidden imply the multimodality/ unimodality of the likelihood? Given a particular graph, what nodes must be observed to ensure unimodality?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16609 | AIM-CRYPTOGRAPHY-0009

### An epimorphism audit for non-Hopfian cryptography

P / partially_solved | AIM 研讨会问题汇编

**Applications of infinite non-abelian groups to cryptography.** Several cryptanalytic methods against group-based cryptography suggest possible use cases for infinite non-abelian groups; for example, non-Hopfian groups. We would like to develop tools to understand how these infinite objects can fit into a cryptographic context, and whether their use can be used to address some of the cryptanalytic methods deployed against group-based cryptography.

来源字段：http://aimpl.org/postquantgroup/3/

16610 | AIM-COMPUTATION-0074

### An exact MLE fiber for the smallest binary hidden-variable model

P / partially_solved | AIM 研讨会问题汇编

“Question 59 (Ho¸ sten). What can algebra say about MLE's for models with hidden vari-ables?”

**Question 59 (Hoşten). What can algebra say about MLEs for models with hidden variables?**

The preceding text identifies the workshop's MLE project as led in part by Serkan Hoşten. I therefore reconstruct `Ho¸ sten` as **Hoşten**, remove the line-break hyphen from `vari-ables`, and normalize the typography without changing the mathematical content:

This is an open-ended research question, not a theorem with fixed hypotheses. The exact result developed here concerns the smallest common-hidden-cause latent class model: one binary hidden variable and two observed binary variables.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16611 | OWR-730-013

### An FPAS for Constrained Bicriteria Spanning Trees

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Planar Min-Bisection and Bicriteria Extraction [OWR-730-012]

Does the constrained bicriteria minimum spanning-tree problem admit a fully polynomial approximation scheme?

来源字段：https://doi.org/10.4171/owr/2004/28

16612 | AMR-127-0020

### Analysis of Boolean Functions - $\varepsilon$-biased Sets for DNFs

O / open | AMR 问题清单汇编

Is it true for each constant $\delta > 0$ that $s^{-O(1)}$-biased densities $\delta$-fool size-$s$ DNFs? I.e., that if $f \colon \{0,1\}^n \to \{-1,1\}$ is computable by a size-$s$ DNF and $\varphi$ is an $s^{-O(1)}$-biased density on $\{0,1\}$, then $|\mathbb{E}_{\mathbf{x} \sim \{0,1\}^n}[f(\mathbf{x})] - \mathbb{E}_{\mathbf{y} \sim \varphi}[f(\mathbf{y})]| \leq \delta$.

背景链接：https://arxiv.org/abs/1204.6447

16613 | AMR-127-0019

### Analysis of Boolean Functions - $k$-wise Independence for PTFs

O / open | AMR 问题清单汇编

Fix $d \in \mathbb{N}$ and $\varepsilon \in (0,1)$. Determine the least $k = k(d,\varepsilon)$ such that the following holds: If $p \colon \mathbb{R}^n \to \mathbb{R}$ is any degree-$d$ multivariate polynomial, and $\mathbf{X}$ is any $\mathbb{R}^n$-valued random variable with the property that each $\bX_i$ has the standard Gaussian distribution and each collection $\bX_{i_1}, \dots, \bX_{i_k}$ is independent, then $|\operatorname{Pr}[p(\mathbf{X}) \geq 0] - \operatorname{Pr}[p(\mathbb{Z}) \geq 0]| \leq \varepsilon$, where $\mathbb{Z}$ has the standard $n$-dimensional Gaussian distribution.

背景链接：https://arxiv.org/abs/1204.6447

16614 | AMR-127-0015

### Analysis of Boolean Functions - Aaronson--Ambainis Conjecture

O / open | AMR 问题清单汇编

Let $f \colon \{-1,1\}^n \to [-1,1]$ have degree at most $k$. Then there exists $i \in [n]$ with $\Inf_i[f] \geq (\operatorname{Var}[f]/k)^{O(1)}$.

背景链接：https://arxiv.org/abs/1204.6447

16615 | AMR-127-0024

### Analysis of Boolean Functions - Approximate Degree for Approximate Majority

O / open | AMR 问题清单汇编

What is the least possible degree of a function $f \colon \{-1,1\}^n \to [-1, -2/3] \cup [2/3, 1]$ which has $f(x) \in [2/3,1]$ whenever $\sum_{i=1}^n x_i \geq n/2$ and has $f(x) \in [-1,-2/3]$ whenever $\sum_{i=1}^n x_i \leq -n/2$?

背景链接：https://arxiv.org/abs/1204.6447

16616 | AMR-127-0023

### Analysis of Boolean Functions - Average versus Max Sensitivity for Monotone Functions

O / open | AMR 问题清单汇编

Let $f \colon \{-1,1\}^n \to \{-1,1\}$ be monotone. Then $\mathbf{I}[f] < o(\operatorname{sens}[f])$.

背景链接：https://arxiv.org/abs/1204.6447

16617 | AMR-127-0007

### Analysis of Boolean Functions - Bernoulli Conjecture

S / solved | AMR 问题清单汇编

Let $T$ be a finite collection of vectors in $\mathbb{R}^n$. Define $b(T) = \mathbb{E}_{\mathbf{x} \sim \{-1,1\}^n}[\max_{t \in T} \langle t, \mathbf{x} \rangle ]$, and define $g(T)$ to be the same quantity except with $\mathbf{x} \sim \mathbb{R}^n$ Gaussian. Then there exists a finite collection of vectors $T'$ such that $g(T') \leq O(b(T))$ and $\forall t \in T\ \exists t' \in T'\ \|t-t'\|_1 \leq O(b(T))$.

背景链接：https://arxiv.org/abs/1204.6447

16618 | AMR-127-0016

### Analysis of Boolean Functions - Bhattacharyya--Grigorescu--Shapira Conjecture

O / open | AMR 问题清单汇编

Let $M \in \mathbb{F}_2^{m \times k}$ and $\sigma \in \{0,1\}^k$. Say that $f \colon \mathbb{F}_2^n \to \{0,1\}$ is *$(M,\sigma)$-free* if there does not exist $X = (x^{(1)}, \dots, x^{(k)})$ (where each $x^{(j)} \in \mathbb{F}_2^n$ is a row vector) such that $MX = 0$ and $f(x^{(j)}) = \sigma_j$ for all $j \in [k]$. Now fix a (possibly infinite) collection $\{(M^1, \sigma^1), (M^2, \sigma^2), \cdots \}$ and consider the property $\mathcal{P}_n$ of functions $f \colon \mathbb{F}_2^n \to \{0,1\}$ that $f$ is $(M^i,\sigma^i)$-free for all $i$. Then there is a one-sided error, constant-query property-testing algorithm for $\mathcal{P}_n$.

背景链接：https://arxiv.org/abs/1204.6447

16619 | AMR-127-0001

### Analysis of Boolean Functions - Correlation Bounds for Polynomials

O / open | AMR 问题清单汇编

Find an explicit (i.e., in $\mathsf{NP}$) function $f \colon \mathbb{F}_2^n \to \mathbb{F}_2$ such that we have the correlation bound $|\mathbb{E}[(-1)^{\langle f(\mathbf{x}), p(\mathbf{x}) \rangle }]| \leq 1/n$ for every $\mathbb{F}_2$-polynomial $p \colon \mathbb{F}_2^n \to \mathbb{F}_2$ of degree at most $\log_2 n$.

背景链接：https://arxiv.org/abs/1204.6447

16620 | AMR-127-0008

### Analysis of Boolean Functions - Fourier Entropy--Influence Conjecture

O / open | AMR 问题清单汇编

There is a universal constant $C$ such that for any $f \colon \{-1,1\}^n \to \{-1,1\}$ it holds that $\mathbb{H}[\widehat{f}^2] \leq C \cdot \mathbf{I}[f]$, where $\mathbb{H}[\widehat{f}^2] = \sum_{S} \widehat{f}(S)^2 \log_2\frac{1}{\widehat{f}(S)^2}$ is the spectral entropy and $\mathbf{I}[f]$ is the total influence.

背景链接：https://arxiv.org/abs/1204.6447

16621 | AMR-127-0004

### Analysis of Boolean Functions - Gotsman--Linial Conjecture

O / open | AMR 问题清单汇编

Among degree-$k$ polynomial threshold functions $f \colon \{-1,1\}^n \to \{-1,1\}$, the one with maximal total influence is the symmetric one $f(x) = \operatorname{sgn}(p(x_1 + \cdots + x_n))$, where $p$ is a degree-$k$ univariate polynomial which alternates sign on the $k+1$ values of $x_1 + \cdots + x_n$ closest to $0$.

背景链接 : https://arxiv.org/abs/1204.6447

16622 | AMR-127-0009

### Analysis of Boolean Functions - Majority Is Least Stable Conjecture

S / solved | AMR 问题清单汇编

Let $f \colon \{-1,1\}^n \to \{-1,1\}$ be a linear threshold function, $n$ odd. Then for all $\rho \in [0,1]$, $\Stab_\rho[f] \geq \Stab_\rho[\maj_n]$.

背景链接 : https://arxiv.org/abs/1204.6447

16623 | AMR-127-0006

### Analysis of Boolean Functions - Mansour's Conjecture

O / open | AMR 问题清单汇编

Let $f \colon \{-1,1\}^n \to \{-1,1\}$ be computable by a DNF of size $s > 1$ and let $\varepsilon \in (0,1/2]$. Then $f$'s Fourier spectrum is $\varepsilon$-concentrated on a collection $\mathcal{F}$ with $|\mathcal{F}| \leq s^{O(\log(1/\varepsilon))}$.

背景链接 : https://arxiv.org/abs/1204.6447

16624 | AMR-127-0011

### Analysis of Boolean Functions - Noise Sensitivity of Intersections of Halfspaces

O / open | AMR 问题清单汇编

Let $f \colon \{-1,1\}^n \to \{-1,1\}$ be the intersection (AND) of $k$ linear threshold functions. Then $\NS_\delta[f] \leq O(\sqrt{\log k}) \cdot \sqrt{\delta}$.

背景链接 : https://arxiv.org/abs/1204.6447

16625 | AMR-127-0012

### Analysis of Boolean Functions - Non-Interactive Correlation Distillation with Erasures

O / open | AMR 问题清单汇编

Let $f \colon \{-1,1\}^n \to \{-1,1\}$ be an unbiased function. Let $\mathbf{z} \sim \{-1,0,1\}^n$ be a ``random restriction'' in which each coordinate $\bz_i$ is (independently) $\pm 1$ with probability $p/2$ each, and $0$ with probability $1-p$. Assuming $p < 1/2$ and $n$ odd, is it true that $\mathbb{E}_{\mathbf{z}}[|f(\mathbf{z})|]$ is maximized when $f$ is the majority function? (Here we identify $f$ with its multilinear expansion.)

背景链接 : https://arxiv.org/abs/1204.6447

16626 | AMR-127-0010

### Analysis of Boolean Functions - Optimality of Majorities for Non-Interactive Correlation Distillation

O / open | AMR 问题清单汇编

Fix $r \in \mathbb{N}$, $n$ odd, and $0 < \varepsilon < 1/2$. For $f \colon \{-1,1\}^n \to \{-1,1\}$, define $P(f) = \operatorname{Pr}[f(\mathbf{y}^{(1)}) = f(\mathbf{y}^{(2)}) = \cdots f(\mathbf{y}^{(r)})]$, where $\mathbf{x} \sim \{-1,1\}^n$ is chosen uniformly and then each $\mathbf{y}^{(i)}$ is (independently) an $\varepsilon$-noisy copy of $\mathbf{x}$. Is it true that $P(f)$ is maximized among odd functions $f$ by the Majority function $\maj_k$ on *some* odd number of inputs $k$?

背景链接 : https://arxiv.org/abs/1204.6447

16627 | AMR-127-0005

### Analysis of Boolean Functions - Polynomial Freiman--Ruzsa Conjecture (in the $\mathbb{F}_2^n$ setting)

S / solved | AMR 问题清单汇编

Suppose $\emptyset \neq A \subseteq \mathbb{F}_2^n$ satisfies $|A + A| \leq C|A|$. Then $A$ can be covered by the union of $\operatorname{poly}(C)$ affine subspaces, each of cardinality at most $|A|$.

背景链接 : https://arxiv.org/abs/1204.6447

16628 | AMR-127-0021

### Analysis of Boolean Functions - PTF Sparsity for Inner Product Mod 2

O / open | AMR 问题清单汇编

Is it true that any PTF representation of the inner product mod $2$ function on $2n$ bits, $\IP_{2n} \colon \mathbb{F}_2^{2n} \to \{-1,1\}$, requires at least $3^n$ monomials?

背景链接 : https://arxiv.org/abs/1204.6447

16629 | AMR-127-0022

### Analysis of Boolean Functions - Servedio--Tan--Verbin Conjecture (Disproved)

S / solved | AMR 问题清单汇编

Fix any $\varepsilon > 0$. Then every monotone $f \colon \{-1,1\}^n \to \{-1,1\}$ is $\varepsilon$-close to a $\operatorname{poly}(\deg(f))$-junta.

背景链接 : https://arxiv.org/abs/1204.6447

16630 | AMR-127-0018

### Analysis of Boolean Functions - Standard Simplex Conjecture

O / open | AMR 问题清单汇编

Fix $0 \leq \rho \leq 1$. Then among all partitions of $\mathbb{R}^n$ into $3 \leq q \leq n+1$ parts of equal Gaussian measure, the maximal noise stability at $\rho$ occurs for a ``standard simplex partition''. By this it is meant a partition $A_1, \dots, A_q$ satisfying $A_i \supseteq \{x \in \mathbb{R}^n : \langle a_i, x \rangle > \langle a_j, x \rangle \ \forall j \neq i\}$, where $a_1, \dots, a_q \in \mathbb{R}^n$ are unit vectors satisfying $\langle a_i, a_j \rangle = -\frac{1}{q-1}$ for all $i \neq j$. Further, for $-1 \leq \rho \leq 0$ the standard simplex partition minimizes noise stability at $\rho$.

背景链接：https://arxiv.org/abs/1204.6447

16631 | AMR-127-0014

### Analysis of Boolean Functions - Subspaces in Sumsets

O / open | AMR 问题清单汇编

Fix a constant $\alpha > 0$. Let $A \subseteq \mathbb{F}_2^n$ have density at least $\alpha$. Is it true that $A+A$ contains a subspace of codimension $O(\sqrt{n})$?

背景链接：https://arxiv.org/abs/1204.6447

16632 | AMR-127-0017

### Analysis of Boolean Functions - Symmetric Gaussian Problem

O / open | AMR 问题清单汇编

Fix $0 \leq \rho, \mu, \nu \leq 1$. Suppose $A,B \subseteq \mathbb{R}^n$ have Gaussian measure $\mu$, $\nu$ respectively. Further, suppose $A$ is centrally symmetric: $A = -A$. What is the minimal possible value of $\operatorname{Pr}[\mathbf{x} \in A, \mathbf{y} \in B]$, when $(\mathbf{x}, \mathbf{y})$ are $\rho$-correlated $n$-dimensional Gaussians?

背景链接：https://arxiv.org/abs/1204.6447

16633 | AMR-127-0003

### Analysis of Boolean Functions - Talagrand's ``Convolution with a Biased Coin'' Conjecture

O / open | AMR 问题清单汇编

Let $f \colon \{-1,1\}^n \to \mathbb{R}^{\geq 0}$ have $\mathbb{E}[f] = 1$. Fix any $0 < \rho < 1$. Then $\operatorname{Pr}[\mathbb{T}_\rho f \geq t] < o(1/t)$.

背景链接：https://arxiv.org/abs/1204.6447

16634 | AMR-127-0002

### Analysis of Boolean Functions - Tomaszewski's Conjecture

S / solved | AMR 问题清单汇编

Let $a \in \mathbb{R}^n$ have $\|a\|_2 = 1$. Then $\Pr_{\mathbf{x} \sim \{-1,1\}^n}[|\langle a, \mathbf{x} \rangle | \leq 1] \geq 1/2$.

背景链接：https://arxiv.org/abs/1204.6447

16635 | AMR-127-0013

### Analysis of Boolean Functions - Triangle Removal in $\mathbb{F}_2^n$

S / solved | AMR 问题清单汇编

Let $A \subseteq \mathbb{F}_2^n$. Suppose that $\varepsilon 2^n$ elements must be removed from $A$ in order to make it ``triangle-free'' (meaning there does not exist $x, y, x+y \in A$). Is it true that $\Pr_{\mathbf{x}, \mathbf{y}}[\mathbf{x}, \mathbf{y}, \mathbf{x}+\mathbf{y} \in A] \geq \operatorname{poly}(\varepsilon)$, where $\mathbf{x}$ and $\mathbf{y}$ are independent and uniform on $\mathbb{F}_2^n$?

背景链接：https://arxiv.org/abs/1204.6447

16636 | AMR-127-0025

### Analysis of Boolean Functions - Uncertainty Principle for Quadratic Fourier Analysis

O / open | AMR 问题清单汇编

Suppose $q_1, \dots, q_m \colon \mathbb{F}_2^n \to \mathbb{F}_2$ are polynomials of degree at most $2$ and suppose the indicator function of $(1, \dots, 1) \in \mathbb{F}_2^n$, namely $\operatorname{AND} \colon \mathbb{F}_2^n \to \{-1,1\}$, is expressible as $\operatorname{AND}(x) = \sum_{i=1}^m c_i (-1)^{q_i(x)}$ for some real numbers $c_i$. What is a lower bound for $m$?

背景链接：https://arxiv.org/abs/1204.6447

16637 | OWR-8415352-009

### Analytical Performance Theory for QAOA and RQAOA

O / open | Oberwolfach 报告问题汇编

Develop an analytical account of the performance and limitations of QAOA and RQAOA in the resource-limited NISQ regime.

来源字段：https://doi.org/10.4171/owr/2021/49

16638 | OWR-1587-007

### Antisymmetric Association Schemes and Polynomial Factoring

O / open | Oberwolfach 报告问题汇编

Certain antisymmetric association schemes should not exist; if this is true, deterministic polynomial factoring over finite fields would be in $\mathrm{P}$ under the Generalized Riemann Hypothesis.

来源字段：https://doi.org/10.4171/owr/2007/31

16639 | OWR-829-002

### Application-Independent Parallel Algorithms for Directed Grid Dependencies

O / open | Oberwolfach 报告问题汇编

Is it possible to design a uniform underlying parallel algorithm that is independent of applications but suitable for grid-based numerical computations whose data-dependency relation is nonsymmetric and can be represented accurately by a directed graph?

来源字段：https://doi.org/10.4171/owr/2005/24

16640 | OWR-14745-005

### Applications of Noncommutative F5 Gröbner Bases

O / open | Oberwolfach 报告问题汇编

What other mathematical applications are possible for a noncommutative F5 algorithm that computes Gröbner bases?

来源字段：https://doi.org/10.4171/owr/2016/37

16641 | OWR-730-008

### Approximability and Parameterized Complexity of Steiner Forest

P / partially_solved | Oberwolfach 报告问题汇编

Give stronger lower bounds on the approximability of the Steiner Forest problem, and characterize its fixed-parameter complexity when the parameter depends on the number of terminal sets.

来源字段：https://doi.org/10.4171/owr/2004/28

16642 | OWR-14298804-012

### Approximating Fare-Zone Assignment on Trees

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Fare-Zone Approximation Question [OWR-14298804-013]

Given a tree $G=(V,E)$ and weighted commodity paths $(P_i,w_i,u_i)$, choose $y_e\in\{0,1\}$ to maximize $\sum_{e\in E}y_e\sum_{i:e\in P_i}w_i$ subject to $\sum_{e\in P_i}y_e\leq u_i$ for every $i$. What approximation guarantees are achievable on trees, particularly when the maximum number of paths sharing an edge is bounded by a parameter $k$?

来源字段：https://doi.org/10.4171/owr/2024/50

16643 | OWR-8415356-017

### Approximating Maximum Nonsymmetric Principal Subdeterminants

P / partially_solved | Oberwolfach 报告问题汇编

Let $L$ be a matrix such that $L+L^\mathsf T$ is positive semidefinite. Is there a $2^{O(k)}$-approximation algorithm for maximizing $\det L_{S,S}$ over all index sets $S$ of size $k$?

来源字段：https://doi.org/10.4171/owr/2021/53

16644 | OWR-8415356-020

### Approximating Submodular Functions by Matroid Rank Functions

O / open | Oberwolfach 报告问题汇编

Is there a universal constant $\alpha$ such that every normalized nonnegative monotone submodular function $f:2^V\to\mathbb R_{\geq0}$ with $f(\{i\})=1$ for all $i\in V$ admits a matroid rank function $r$ satisfying $r(S)\leq f(S)\leq\alpha r(S)$ for every $S\subseteq V$?

来源字段：https://doi.org/10.4171/owr/2021/53

16645 | AIM-COMPUTATION-0070

### Arbitrary logistic cell gaps with positive margins and ordinary-margin obstructions

P / partially_solved | AIM 研讨会问题汇编

Problem 55 (Karr). Find a set of strictly positive (or strictly \(>5\)) margins for which the set of tables which has these margins has an arbitrarily large gap in a cell entry. Dinwoodie: Are there examples of this phenomenon in logistic regression?

The spectra is defined to be the set of all sets of possible cell entries in for a multiway nonnegative integral table with fixed marginal totals.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16646 | OWR-8415356-007

### Arborescences Induced by Linear-Programming Pivot Rules

P / partially_solved | Oberwolfach 报告问题汇编

For a fixed linear-programming objective, how do the induced arborescences vary with the memoryless pivot rule, and for a fixed memoryless pivot rule, how do they vary with the objective?

来源字段：https://doi.org/10.4171/owr/2021/53

16647 | OWR-831-014

### Arithmetic-Formula Lower Bounds for Permanent and Determinant

O / open | Oberwolfach 报告问题汇编

Prove superpolynomial lower bounds for the arithmetic-formula size of an explicit polynomial, in particular the permanent or determinant.

来源字段：https://doi.org/10.4171/owr/2005/26

16648 | OWR-15210-003

### Assumptions and Function Classes for Function Secret Sharing

P / partially_solved | Oberwolfach 报告问题汇编

Characterize the function classes for which efficient function secret-sharing schemes follow from one-way functions, and determine the necessary and sufficient cryptographic assumptions for useful FSS instances; in particular, improve one-way-function-based FSS for point functions with at least three parties.

来源字段：https://doi.org/10.4171/owr/2017/5

16649 | OWR-10252930-019

### Asymptotic Counts of Growing-Width $k$-SAT Functions

P / partially_solved | Oberwolfach 报告问题汇编

If $k=k(n)\leq(1/2-c)n$ for some constant $c>0$, what is the asymptotic number of $k$-SAT functions on $n$ Boolean variables?

来源字段：https://doi.org/10.4171/owr/2022/22

16650 | OWR-768-005

### Asymptotic Goodness of Function-Field-Tower Codes

O / open | Oberwolfach 报告问题汇编

Are the algebraic-geometric code families obtained by the direct-sum construction from the first levels of recursively defined asymptotically good towers of function fields themselves asymptotically good, with positive asymptotic rate and positive relative minimum distance?

来源字段：https://doi.org/10.4171/owr/2004/54

16651 | OWR-16938-001

### Automatic Recognition of Numerically Unstable Quadratic Programs

O / open | Oberwolfach 报告问题汇编

Consider a quadratically constrained quadratic program with $$ (x_1-1)^2+x_2^2\geq3+s^2,\qquad a_i-a_{i-1}^2=0,\qquad a_0=s/2,$$ where a finite chain of the $a_i$ amplifies numerical error through repeated square roots. How can solvers automatically recognize such severely ill-conditioned instances?

来源字段：https://doi.org/10.4171/owr/2019/26

16652 | OWR-14298166-003

### Average-Case Complexity of Clique on Sparse Graphs

O / open | Oberwolfach 报告问题汇编

For Erdős-Rényi graphs with edge probability near the threshold for containing a $k$-clique, does deciding $k$-Clique require time $n^{\Omega(k)}$? More modestly, is there an $n^{o(k)}$-time algorithm for deciding whether such a graph contains a clique of size $n^\varepsilon$?

来源字段：https://doi.org/10.4171/owr/2024/15

16653 | OWR-15210-006

### Average-Case Hardness of Quasi-Cyclic Code Problems

P / partially_solved | Oberwolfach 报告问题汇编

Let $R_p=\mathbb F_2[x]/(x^p-1)$. Extend average-case decoding-hardness results to quasi-cyclic codes, and prove a polynomial equivalence between the following problems: finding $e_0,e_1\in R_p$ of bounded total Hamming weight with $e_0+e_1h=s$, and deciding whether there exist nonzero $h_0,h_1\in R_p$ of bounded total Hamming weight with $h_0+h_1h=0$.

来源字段：https://doi.org/10.4171/owr/2017/5

16654 | OWR-8415356-021

### Balanced Signings in the Subtour Polytope

O / open | Oberwolfach 报告问题汇编

别名：Balanced Signings in the Subtour Polytope [OWR-8415356-022]

For every point of the subtour-elimination polytope and every even terminal set, does there exist a balanced signing realizable by a capacitated transshipment?

来源字段：https://doi.org/10.4171/owr/2021/53

16655 | AIM-COMPUTATION-0078

### Base-encoded monomial-curve designs

P / partially_solved | AIM 研讨会问题汇编

Given finitely many monomials in \(\mathbb R[x_1,\ldots,x_d]\), construct a finite real design on which their evaluation vectors are linearly independent.

The immediately preceding source text defines a design \(D\) as a finite subset of \(\mathbb R^d\), its design ideal \(I(D)\) as the ideal of polynomials vanishing on \(D\), and a linear polynomial model \(L\) as a list of monomials. Thus the mathematical core is:

“Special statistical properties” is not defined in the source. This report therefore treats two precise and standard properties:

1. **saturation and identifiability:** for \(n=|L|\), use exactly \(n\) runs and obtain a nonsingular model matrix; 2. **restricted exact D-optimality:** after restricting candidate points to a specified compact monomial curve, maximize the determinant of the information matrix among all saturated, equal-replication designs on that curve.

The second property is deliberately qualified. No claim of D-optimality among all designs in \(\mathbb R^d\), nor among all approximate designs, is made.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16656 | AIM-COMPUTATION-0038

### Behrens-Fisher likelihood equations and an equal-scatter bifurcation

S / solved | AIM 研讨会问题汇编

**Problem 24 (Behrens-Fischer Problem).** Solve the maximum likelihood equations for the mixture of two multivariate Gaussians with the same mean and different covariance matrices.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16657 | OWR-14297738-002

### Benign, Tempered, and Catastrophic Overfitting Regimes

P / partially_solved | Oberwolfach 报告问题汇编

Characterize when interpolation and overfitting in neural networks or kernel methods is benign, tempered, or catastrophic.

来源字段：https://doi.org/10.4171/owr/2023/41

16658 | OWR-8415356-024

### Better Approximations for Unrelated Graph Balancing

P / partially_solved | Oberwolfach 报告问题汇编

Does unrelated graph balancing admit an approximation factor below $2$, even for simple graphs in which every orientation-dependent edge weight belongs to $\{1,k\}$ for a fixed $k$?

来源字段：https://doi.org/10.4171/owr/2021/53

16659 | OWR-8415356-018

### Better-Than-Two Approximation for Multiplicative Connectivity Augmentation

P / partially_solved | Oberwolfach 报告问题汇编

别名：Better-Than-Two Approximation for Multiplicative Connectivity Augmentation [OWR-8415356-019]

For every fixed $\alpha>1$, does $\alpha$-multiplicative connectivity augmentation admit an approximation factor strictly below $2$?

来源字段：https://doi.org/10.4171/owr/2021/53

16660 | DARPA-008

### Beyond Convex Optimization

P / partially_solved | DARPA 挑战

Determine whether algebraic geometry can systematically replace linear algebra in optimization.

背景链接：https://web.math.utk.edu/~vasili/refs/darpa07.MathChallenges.html

16661 | AIM-COMPUTATION-0031

### BIC evidence, mixture singularities, and an exact quotient diagnostic

P / partially_solved | AIM 研讨会问题汇编

**Question 17.** Why does the BIC work for model fitting with mixture models? What is being approximated (and why is it good to approximate these things?) This question raised some controversy during the discussion following the talk which was later resolved. The main issue that is interesting in this situation is that the likelihood function for mixture models does not satisfy the necessary regularity conditions to apply the classical results about the BIC.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16662 | AIM-COMPUTATION-0028

### Binary independence and hierarchical models in cumulant coordinates

P / partially_solved | AIM 研讨会问题汇编

Problem 14. Use the transformations from probabilities to moments to cumulants to de-scribe independence models/ hierarchical models in terms of polynomial functions in the cumulants.

**Problem 14.** Use the transformations from probabilities to moments to cumulants to describe independence models/hierarchical models in terms of polynomial functions in the cumulants.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16663 | AIM-COMPUTATION-0095

### Binomial windows for prescribed-degree factors

P / partially_solved | AIM 研讨会问题汇编

Let \(m\geq n\in\mathbb Z\). Prove there is a polynomial \(g\in\mathbb F_q[x]\) of degree \(\leq 2\log n\) so that \(x^m+g(x)\) has an irreducible factor of degree \(n\).

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16664 | OWR-15580-003

### Bit-Width-Independent Verification of Integer Multipliers

P / partially_solved | Oberwolfach 报告问题汇编

Find a general-purpose method for verifying bounded-bit-width integer multiplier properties, including ring identities, without exponential scaling in the bit-width.

来源字段：https://doi.org/10.4171/owr/2017/37

16665 | OWR-1587-001

### Boolean Evaluation and Arithmetic-Circuit Complexity

O / open | Oberwolfach 报告问题汇编

If a multivariate polynomial $P$ can be evaluated on rational inputs by a Boolean polynomial-time algorithm, must $P$ have a polynomial-size arithmetic circuit using only additions, subtractions, and multiplications?

来源字段：https://doi.org/10.4171/owr/2007/31

16666 | OWR-11695855-008

### Cancellation-Sensitive Norms for Two-Layer Networks

O / open | Oberwolfach 报告问题汇编

For a two-layer network with weights $(a,V)$, when can the squared parameter norm be replaced by the tighter product $\lVert a\rVert\lVert V\rVert$, and is there a still tighter cancellation-sensitive expression usable in gradient-descent and test-error bounds?

来源字段：https://doi.org/10.4171/owr/2022/46

16667 | OWR-8415359-009

### Canonical Paths Versus Spectral Independence

O / open | Oberwolfach 报告问题汇编

Establish a general relationship between the canonical-path method and spectral independence, with applications to approximate counting and sampling such as sampling perfect matchings in nonbipartite graphs.

来源字段：https://doi.org/10.4171/owr/2021/54

16668 | OWR-9790365-012

### Capacity and Reliability of Mismatched Decoding

P / partially_solved | Oberwolfach 报告问题汇编

Find a single-letter characterization of mismatched decoding capacity, and determine the exact reliability function for mismatched decoding.

来源字段：https://doi.org/10.4171/owr/2022/14

16669 | OWR-9790365-005

### Capacity Criteria for Arbitrarily Varying Channels with Causal State Information

O / open | Oberwolfach 报告问题汇编

For arbitrarily varying channels with causal state information at the encoder, when do deterministic-code and random-code capacities coincide, and does the criterion necessarily require a multiletter characterization?

来源字段：https://doi.org/10.4171/owr/2022/14

16670 | OWR-2489-003

### Capacity-Preserving Conversion to Unsplittable Flows

O / open | Oberwolfach 报告问题汇编

Given a single-source network with capacities $u$, costs $c$, source $s$, sinks $t_i$, and demands $d_i$ satisfying the no-bottleneck condition $d_{\max}\leq u_e$, does every splittable flow $x$ satisfying the demands admit an unsplittable flow $y$ satisfying the same demands such that $y_e\leq x_e+d_{\max}$ for every edge $e$ and $c(y)\leq c(x)$?

来源字段：https://doi.org/10.4171/owr/2008/51

16671 | OWR-783-004

### Central-Trajectory Integral Bounds

O / open | Oberwolfach 报告问题汇编

For the primal-dual central trajectory $(x(\nu),s(\nu),y(\nu))$ of a linear program with constraint matrix $A$, define $\kappa(\nu)=\lVert\nu\dot x(\nu)\circ\dot s(\nu)\rVert_1^{1/2}$ and $$F(\nu_0,\nu_1)=\int_{\nu_1}^{\nu_0}\frac{\kappa(\nu)}{\nu}\,d\nu.$$ Is $F(\infty,0)$ bounded by a quantity depending only on $A$? More specifically, is $F(\infty,0)=O\!\left(n^{3.5}\log(\bar\chi_A^*+n)\right)$?

来源字段：https://doi.org/10.4171/owr/2005/02

16672 | AIM-CRYPTOGRAPHY-0014

### Certificates for hidden Goppa structure and bounded decoding

P / partially_solved | AIM 研讨会问题汇编

**What witnesses are there of Goppa decodability or non-decodability?**

Given a parity-check matrix \(H\), find the hidden Goppa code in \(H\). How quickly can we decode \(H\), assuming Goppa decoder for \(H\) exists?

**Recovered formulation used here.** We treat “Goppa” as a *classical binary Goppa code in Hamming metric*, as in McEliece/Classic McEliece, not a rank-metric code and not a general algebraic-geometry code. We formulate and prove results for readings (1), (3), and (4). All reconstruction choices are explicit; they are not silently attributed to the AIM text.

来源字段：http://aimpl.org/quantumalg/1/

16673 | OWR-15957-007

### Certificates of Polynomial Nonnegativity in Copositive Optimization

O / open | Oberwolfach 报告问题汇编

For polynomials nonnegative on an unbounded set $S\cap h^{-1}(0)$, under what hypotheses can certificates expressed through polynomials nonnegative on $S$ and the ideal $(h)$ be used to extend copositive-polynomial formulations and polynomial-optimization methods?

来源字段：https://doi.org/10.4171/owr/2017/52

16674 | OWR-1111-002

### Characterization of Ideal $0$-$1$ Matrices

O / open | Oberwolfach 报告问题汇编

Characterize ideal $0$-$1$ matrices $M$, meaning those for which $\{x:Mx\geq\mathbf1,\ x\geq0\}$ is integral; in particular, seek an explicit characterization of minimally nonideal matrices.

来源字段：https://doi.org/10.4171/owr/2005/50

16675 | OWR-14298166-011

### Circuit-Size Hierarchy in Bounded Arithmetic

O / open | Oberwolfach 报告问题汇编

Does the bounded-arithmetic theory $\mathrm{PV}_1$ prove the Circuit-Size Hierarchy theorem? If it does, can one extract from such a proof explicit polynomial-time languages with the corresponding circuit lower bounds?

来源字段：https://doi.org/10.4171/owr/2024/15

16676 | OWR-3394-009

### Circular Assume-Guarantee Reasoning for General Systems

P / partially_solved | Oberwolfach 报告问题汇编

The circular assume-guarantee rule with premises $\Sigma_{P_1}\parallel\Sigma_{Q_2}\preccurlyeq\Sigma_{Q_1}\parallel\Sigma_{Q_2}$ and $\Sigma_{Q_1}\parallel\Sigma_{P_2}\preccurlyeq\Sigma_{Q_1}\parallel\Sigma_{Q_2}$, and conclusion $\Sigma_{P_1}\parallel\Sigma_{P_2}\preccurlyeq\Sigma_{Q_1}\parallel\Sigma_{Q_2}$, is valid for deterministic systems. Is it valid for general systems?

来源字段：https://doi.org/10.4171/owr/2009/11

16677 | OWR-1386-014

### Codegree Thresholds for Hypergraph Matchings

S / solved | Oberwolfach 报告问题汇编

For $n\geq k\geq2$, let $t_k(n)$ be the least $t$ such that every $n$-vertex $k$-uniform hypergraph $H$ with $\delta_{k-1}(H)\geq t$ has a matching of size $\lfloor n/k\rfloor$. If $k\nmid n$, is $t_k(n)=\lfloor n/k\rfloor$?

来源字段：https://doi.org/10.4171/owr/2006/48

16678 | OWR-14298166-002

### Coefficient Compression and Structured Proof-System Lower Bounds

O / open | Oberwolfach 报告问题汇编

Can the coefficient bit length of every $n$-variable linear threshold function be reduced efficiently from $n^{100}$ bits to $n^2$ bits? Can one prove Polynomial Calculus, Sherali-Adams, or Sum-of-Squares degree lower bounds for graph-isomorphism formulas $\operatorname{ISO}(G,H)$ beyond CFI-type graphs? How strong is a deductive proof system whose lines are DNNF formulas?

来源字段：https://doi.org/10.4171/owr/2024/15

16679 | OWR-13353-008

### Combinatorial Decompositions of the All-Ones Vector

P / partially_solved | Oberwolfach 报告问题汇编

Can the all-ones vector be represented as a linear or integer combination of combinatorially interesting vectors, such as incidence vectors of tours?

来源字段：https://doi.org/10.4171/owr/2014/51

16680 | OWR-13353-021

### Compact Constant-Gap Relaxations for Capacitated Facility Location

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Constant-Gap Facility-Location Relaxation [OWR-13353-022]

Does there exist a polynomial-size linear-programming relaxation for the capacitated facility location problem whose integrality gap is bounded by a constant?

来源字段：https://doi.org/10.4171/owr/2014/51

16681 | OWR-8415356-023

### Compact Representations of Four-Thirds Near-Mincuts

O / open | Oberwolfach 报告问题汇编

Can all cuts of weight at most $\tfrac43$ times the minimum-cut value in an $n$-vertex graph be represented compactly, exploiting the fact that there are at most $n^2$ such cuts?

来源字段：https://doi.org/10.4171/owr/2021/53

16682 | OWR-7155444-001

### Comparative Advantages of Physics-Based and Machine-Learned Models

O / open | Oberwolfach 报告问题汇编

Is a physics-based model always advantageous when the underlying system obeys physical laws, and which modelling or controller-design problems can be solved better with machine learning?

来源字段：https://doi.org/10.4171/owr/2021/23

16683 | AIM-COMPUTATION-0020

### Compatible and complete odds-ratio specifications

P / partially_solved | AIM 研讨会问题汇编

**Question 6.** For multi-way tables, what are the complete and partial specifications that come from specifying sets of odds-ratios?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16684 | AMR-054-0038

### Compatible Triangulations

O / open | AMR 问题清单汇编

Is it true that every two sets of $n$ planar points in general position with the same number points on their convex hulls have compatible triangulations? Two triangulations are compatible if they have the same combinatorial structure, i.e., if their face lattices are isomorphic. For compatible triangulations $T_1$ and $T_2$ of point sets $S_1$ and $S_2$, there is a bijection $\phi$ between the points such that $ijk$ is a triangle of $T_1$ empty of points of $S_1$ iff $\phi(i) \phi(j) \phi(k)$ is a triangle of $T_2$ empty of points of $S_2$.

背景链接：https://topp.openproblem.net/p38

16685 | OWR-11576-003

### Competitiveness of High-Order Hybridizable DG Wave Methods

O / open | Oberwolfach 报告问题汇编

Are high-order and hybridizable discontinuous Galerkin methods competitive for demanding wave-propagation computations?

来源字段：https://doi.org/10.4171/owr/2012/09

16686 | AIM-COMPUTATION-0056

### Complex count solved, with an exact real-root cusp slice

P / partially_solved | AIM 研讨会问题汇编

Problem 24 (Behrens-Fischer Problem). Solve the maximum likelihood equations for the mixture of two multivariate Gaussians with the same mean and different covariance matrices.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16687 | OWR-730-011

### Complexity and Approximation of Planar Min-Bisection

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Planar Min-Bisection and Bicriteria Extraction [OWR-730-012]

What is the exact computational complexity of PLANAR MIN-BISECTION, and does PLANAR MIN-BISECTION admit a polynomial-time approximation scheme?

来源字段：https://doi.org/10.4171/owr/2004/28

16688 | OWR-14299916-001

### Complexity and Generalization in Overparameterized Learning

O / open | Oberwolfach 报告问题汇编

Which intrinsic notions of complexity govern learning and generalization in highly and moderately overparameterized models? Why do gradient-based optimization methods consistently select solutions that generalize well? How do useful representations emerge during training, and what are the computational limitations of modern learning systems?

来源字段：https://doi.org/10.4171/owr/2026/11

16689 | OWR-13941-004

### Complexity and Rank of Completely Positive Matrices

O / open | Oberwolfach 报告问题汇编

Let $CP_n=\{BB^T:B\in\mathbb R_{\geq0}^{n\times k}\text{ for some }k\}$ be the cone of completely positive matrices. Is membership in $CP_n$ in NP? How can the cp-rank $\operatorname{cp}(A)=\min\{k:A=BB^T,\ B\in\mathbb R_{\geq0}^{n\times k}\}$ of an arbitrary $A\in CP_n$ be determined?

来源字段：https://doi.org/10.4171/owr/2015/46

16690 | OWR-16931-018

### Complexity of Approximate Counting in Spin Systems

O / open | Oberwolfach 报告问题汇编

Determine the worst-case complexity of approximate counting and sampling for the ferromagnetic Potts model and for the hard-core model on bipartite graphs. Do these problems admit polynomial-time algorithms, or are they NP-hard?

来源字段：https://doi.org/10.4171/owr/2019/19

16691 | OWR-14745-003

### Complexity of Bray’s Involution-Centralizer Algorithm

P / partially_solved | Oberwolfach 报告问题汇编

For a finite classical group of dimension $n$ over a field of odd characteristic, does Bray's algorithm for constructing the centralizer of a strong involution succeed with high probability after examining only $O(\log n)$ random elements?

来源字段：https://doi.org/10.4171/owr/2016/37

16692 | OWR-1386-015

### Complexity of Dense Perfect Hypergraph Matching

S / solved | Oberwolfach 报告问题汇编

For integers $k\geq2$ and $1/k\leq c<1/2$, determine the complexity of deciding whether a $k$-uniform hypergraph $H$ with $|V(H)|\equiv0\pmod k$ and $\delta_{k-1}(H)\geq cn$ has a matching of size $n/k$.

来源字段：https://doi.org/10.4171/owr/2006/48

16693 | OWR-13940-003

### Complexity of Embedding Two-Complexes in Three-Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Determine the decidability and computational complexity of deciding whether a given 2-complex embeds in a 3-manifold, including the case of non-orientable surfaces of even Euler genus and the related thickenability problem.

来源字段：https://doi.org/10.4171/owr/2015/45

16694 | OWR-13353-006

### Complexity of Extremality Testing in Infinite Group Problems

O / open | Oberwolfach 报告问题汇编

Is the pseudo-polynomial running time of the known algorithms for testing extremality of minimal valid functions in the single-row infinite group problem best possible?

来源字段：https://doi.org/10.4171/owr/2014/51

16695 | OWR-13353-012

### Complexity of Maximizing Initial Odd Ears

P / partially_solved | Oberwolfach 报告问题汇编

What is the computational complexity of determining $\beta(G)$ for a $2$-vertex-connected graph $G$, where $\beta(G)$ is the maximum number of initial odd ears in an ear decomposition?

来源字段：https://doi.org/10.4171/owr/2014/51

16696 | OPG-467

### Complexity of square-root sum

P / partially_solved | Open Problem Garden

Question What is the complexity of the following problem?

Given $a_1,\dots,a_n; k$, determine whether or not $\sum_i \sqrt{a_i} \leq k.$

背景链接：http://www.openproblemgarden.org/op/complexity_of_square_root_sum

16697 | OWR-8415356-002

### Complexity of Submodular $k$-Partition

P / partially_solved | Oberwolfach 报告问题汇编

For fixed $k$, what is the computational complexity of minimizing the sum of a submodular function over a partition into $k$ nonempty parts?

来源字段：https://doi.org/10.4171/owr/2021/53

16698 | OWR-14299088-008

### Complexity-Bounded Computation of Maximum Destabilizing Subgroups

O / open | Oberwolfach 报告问题汇编

Construct an algorithm that computes the maximum destabilizing one-parameter subgroup represented by the limiting direction of the relevant gradient flow, and prove explicit complexity bounds for the algorithm.

来源字段：https://doi.org/10.4171/owr/2025/9

16699 | AIM-COMPUTATION-0036

### Componentwise entropy rigidity and a reduced search for a non-multilinear matroid

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 22.** There exists a connected matroid whose CI-structure is p-representable but not multilinear.

Suppose independence vatieties are properly defined, also for the CI constraints involv-ing functional independences, over the field of complex numbers.

Inspection of the official PDF and its HTML transcription shows that this second sentence is not part of Conjecture 22. It is the setup immediately before Conjecture 23. This report therefore treats only the one-sentence Conjecture 22.

The workshop uses František Matúš's augmented CI viewpoint, in which functional dependences are included alongside ordinary conditional independences. This point matters: functional dependence is the equality \(H(X_e\mid X_A)=0\), and is the entropy analogue of \(r(A\cup e)=r(A)\).

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16700 | OWR-8415360-009

### Computable Ground Truths for Learning Problems

O / open | Oberwolfach 报告问题汇编

When is the ground truth of a learning task a computable function, and how does that relate to approximation by computable neural networks?

来源字段：https://doi.org/10.4171/owr/2021/55

16701 | OWR-8415360-008

### Computable Neural-Network Approximation from Noisy Samples

O / open | Oberwolfach 报告问题汇编

For a learning problem specified from point samples, is its ground-truth map computable; are the neural networks supplied by approximation theorems computable; and is there an algorithm that maps noisy samples to such a neural-network approximation?

来源字段：https://doi.org/10.4171/owr/2021/55

16702 | DARPA-020

### Computation at Scale

P / partially_solved | DARPA 挑战

Develop asymptotics for systems with massive degrees of freedom.

背景链接：https://web.math.utk.edu/~vasili/refs/darpa07.MathChallenges.html

16703 | OWR-1455-002

### Computational Complexity of Bruhat-Interval Invariants

O / open | Oberwolfach 报告问题汇编

Given a Coxeter system $(W,S)$ and elements $u\leq v$ with Coxeter lengths $\ell(u)$ and $\ell(v)$, what is the computational complexity of computing the Bruhat-interval quantity $P_{u,v}$, equivalently the coefficients $a_i(u,v)$, or all cd-coefficients $[w]_{u,v}$, as a function of the Coxeter matrix and the input lengths?

来源字段：https://doi.org/10.4171/owr/2007/04

16704 | AMR-092-0002

### Computational Diffie-Hellman problem

O / open | AMR 问题清单汇编

Given a prime modulus $p$, a group generator $g$, and the public values $g^a$ and $g^b$ modulo $p$, can the shared value $g^{ab}\bmod p$ be computed efficiently without knowing $a$ or $b$?

背景链接：https://unsolvedproblems.org/index_files/DiffieHellman.htm

16705 | DARPA-006

### Computational Duality

P / partially_solved | DARPA 挑战

Use mathematical duality and geometry as foundations for developing novel computational algorithms.

背景链接：https://web.math.utk.edu/~vasili/refs/darpa07.MathChallenges.html

16706 | OWR-831-007

### Computational Hardness of Learning Parity with Noise

O / open | Oberwolfach 报告问题汇编

For fixed $0<\varepsilon<1/2$, let $a_i$ be independent uniform vectors in $\mathbb F_2^n$ and let $b_i=\langle s,a_i\rangle+e_i$, where $s\in\mathbb F_2^n$ is hidden and the $e_i$ are independent Bernoulli-$\varepsilon$ errors. Can $s$ be recovered in polynomial time from polynomially many samples $(a_i,b_i)$?

来源字段：https://doi.org/10.4171/owr/2005/26

16707 | OWR-17135-019

### Computational Recognition of Polytope Face Vectors and Cones

O / open | Oberwolfach 报告问题汇编

For fixed $d$, can membership in $f(\mathcal P^d)$ be decided in time polynomial in the binary encoding length of the input vector? Is it decidable whether a hyperplane through $f(\sigma^d)$ supports $\operatorname{Cone}_{f(\sigma^d)}(f(\mathcal P^d))$ or contains an interior ray?

来源字段：https://doi.org/10.4171/owr/2019/39

16708 | OWR-15173-003

### Computing Hecke Operators for Arithmetic-Group Cohomology

P / partially_solved | Oberwolfach 报告问题汇编

Can Hecke operators and their eigenvalues be computed for all arithmetic groups for which the cohomology can be calculated?

来源字段：https://doi.org/10.4171/owr/2016/52

16709 | OWR-10252933-001

### Condition-Number-Robust Low-Rank Matrix Completion

P / partially_solved | Oberwolfach 报告问题汇编

Derive recovery guarantees for factorization-based low-rank matrix-completion algorithms whose dependence on matrix condition numbers matches their empirical performance.

来源字段：https://doi.org/10.4171/owr/2022/25

16710 | OWR-4082-002

### Connectivity Reconstruction in Large Neuronal Networks

O / open | Oberwolfach 报告问题汇编

Reconstruct the connectivity of a large neuronal network from limited, noisy, and incomplete observations.

来源字段：https://doi.org/10.4171/owr/2009/39

16711 | OWR-13949-001

### Consequences of Indistinguishability Obfuscation

O / open | Oberwolfach 报告问题汇编

The indistinguishability obfuscation conjecture asserts the feasibility of obfuscating functionally equivalent programs so their obfuscations cannot be distinguished; what are its precise consequences and relations to one-way functions and $P\neq NP$?

来源字段：https://doi.org/10.4171/owr/2015/54

16712 | OWR-13674-015

### Constructive Selection for Bessel Sequences

O / open | Oberwolfach 报告问题汇编

Let $\{u_j\}_{j\in J}\subset\mathbb{C}^n$ be a finite Bessel sequence with bound $1$ and $\lVert u_j\rVert^2\leq\varepsilon$. Given scalars $t_j\in[0,1]$, design a constructive algorithm that finds $J^\prime\subset J$ satisfying $\left\lVert\sum_{j\in J^\prime}u_ju_j^*-\sum_{j\in J}t_ju_ju_j^*\right\rVert=O(\varepsilon^{1/8})$.

来源字段：https://doi.org/10.4171/owr/2015/17

16713 | OWR-8415360-005

### Convergence and Effective Rates of Unwinding Series

P / partially_solved | Oberwolfach 报告问题汇编

What can be rigorously established about convergence of the unwinding series, and can one obtain effective convergence rates for sufficiently regular functions $F$?

来源字段：https://doi.org/10.4171/owr/2021/55

16714 | OWR-14299914-005

### Convergence of Accelerated Sampling on Probability Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Prove convergence of the proposed accelerated fixed-point and Anderson-mixing methods on suitable regular subsets of general probability measures, including the Stein-geometry sampling algorithm for smooth densities.

来源字段：https://doi.org/10.4171/owr/2026/13

16715 | OWR-16633-006

### Convergence Rates for Convex-Body Bounds

O / open | Oberwolfach 报告问题汇编

Let $p_{\min}$ be the minimum of a polynomial $p$ on a convex body $K$, and let $p(r)$ be the degree-$2r$ sum-of-squares-density upper bound obtained from a reference measure on $K$. What is the sharp asymptotic rate of $p(r)-p_{\min}$ for general convex bodies?

来源字段：https://doi.org/10.4171/owr/2018/50

16716 | OWR-14297738-004

### Convergence to Implicit Geometry under Class Imbalance

P / partially_solved | Oberwolfach 报告问题汇编

Quantify how class imbalance affects the rates at which first-order training methods converge to their asymptotically preferred implicit geometry, and determine how architecture, data, and loss design affect that convergence.

来源字段：https://doi.org/10.4171/owr/2023/41

16717 | OWR-9790365-013

### Convex Relaxations for Planted Graph Matching

P / partially_solved | Oberwolfach 报告问题汇编

Let $\pi$ be a uniformly random permutation, and let $A$ and $B$ be correlated symmetric random matrices under the planted correspondence $\pi$. For which correlations does the minimizer of $$ \min_{X\ \mathrm{doubly\ stochastic}}\|AX-XB\|_F^2 $$ round to $\pi$ with high probability?

来源字段：https://doi.org/10.4171/owr/2022/14

16718 | OWR-2489-004

### Convex-Hull Decomposition for Mixed-Integer Bipartite Systems

O / open | Oberwolfach 报告问题汇编

别名：Subtree Decomposition of Mixed-Integer Convex Hulls [OWR-2489-009]

Let $G$ be bipartite, let $I$ be the set of integral-coordinate vertices, and let $S(G,I)=\{x:x_i+x_j\geq b_{ij}\text{ for }ij\in E,\ x_i\in\mathbb{Z}\text{ for }i\in I\}$. If $T$ ranges over subtrees of $G$ for which $I_T=V(T)\cap I$ is the set of leaves, is $\operatorname{conv}(S(G,I))=\bigcap_T\operatorname{conv}(S(T,I_T))$? The conjecture is known for denominator parameter $k=2$ and open for $k\geq3$.

来源字段：https://doi.org/10.4171/owr/2008/51

16719 | OWR-13941-002

### Convexification and Complexity in Mixed-Integer Optimization

O / open | Oberwolfach 报告问题汇编

Are the intermediate SOS sets $\Sigma^k_{n,d}=\{p:\exists q\in\mathbb{R}[x]^k,\ q\text{ is SOS and }pq\text{ is SOS}\}$ convex? What is the complexity of selecting $s_i\in S_i$ such that $0\in\operatorname{conv}\{s_0,\ldots,s_d\}$? What are the complexities of finding a separating Chvátal-Gomory inequality and of integer quadratic optimization in dimension $3$? What practical rules govern combined SDP and RLT cuts? What practical algorithm solves evaluation-oracle mixed-integer optimization? Is the convex hull of a polyhedron minus finitely many polyhedra polyhedral? Which convex mixed-integer classes admit exact convexification?

来源字段：https://doi.org/10.4171/owr/2015/46

16720 | OWR-16775-004

### Convolutional Codes for Retrieval, Sensing, and Rank Metrics

O / open | Oberwolfach 报告问题汇编

Can convolutional codes exploit sliding-window properties for private information retrieval or compressed sensing, and can they attain strong rank-metric properties?

来源字段：https://doi.org/10.4171/owr/2019/13

16721 | OWR-13941-006

### Core Points of Transitive Permutation Groups

O / open | Oberwolfach 报告问题汇编

For every transitive permutation group that is not $2$-homogeneous, are there infinitely many core points up to integral translations of the fixed space? Equivalently, does the conjecture hold in the remaining case of primitive permutation groups with rational invariant subspaces?

来源字段：https://doi.org/10.4171/owr/2015/46

16722 | AIM-CRYPTOGRAPHY-0020

### Correlation-gap certificates for noisy hidden shift

P / partially_solved | AIM 研讨会问题汇编

How much noise is tolerated in hidden shift algorithms? Consider too many solutions vs. too few.

来源字段：http://aimpl.org/quantumalg/4/

16723 | AIM-COMPUTATION-0076

### Covariance-whitened polynomial invariants for model goodness of fit

P / partially_solved | AIM 研讨会问题汇编

Question 61 (From graphical models notes). Is there any useful statistical measure that can be obtained from evaluating the polynomials that vanish on probabilities coming from a given model at an empirical distribution? What about \[ \sum |f(\widehat p)|? \]

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16724 | OWR-14299916-004

### Cross-Entropy Dynamics Beyond Hadamard Initialization

P / partially_solved | Oberwolfach 报告问题汇编

For the multiclass two-layer linear unconstrained-features model with balanced standard-basis inputs, weights $W_t\in\mathbb R^{k\times d}$ and $H_t\in\mathbb R^{d\times k}$, and cross-entropy loss, determine the gradient-flow dynamics under general initialization. Do the normalized parameters converge to neural-collapse geometry, which quantities are monotone along the trajectory, and how do the answers depend on the initialization and the number of classes $k$?

来源字段：https://doi.org/10.4171/owr/2026/11

16725 | OWR-14299084-010

### Cryptographic Equivalences for High-Energy Phenomena

O / open | Oberwolfach 报告问题汇编

Find further high-energy-physics phenomena whose computational hardness is equivalent to the existence of standard cryptographic primitives, extending the known equivalence for decoding Hawking radiation.

来源字段：https://doi.org/10.4171/owr/2025/3

16726 | OWR-12481-018

### Cryptographic Replacement of Pseudorandom Steganographic Functions

O / open | Oberwolfach 报告问题汇编

Can the pseudorandom functions used in the steganographic constructions be replaced by cryptographic functions?

来源字段：https://doi.org/10.4171/owr/2013/18

16727 | OWR-14299916-003

### Curricula for Algorithmic Reasoning in Sequence Models

O / open | Oberwolfach 报告问题汇编

Beyond currently studied stylized settings, which curricula encourage algorithmic reasoning in transformers? How can disruptive but correctly labeled examples be identified for structured state-space or hybrid architectures? Can these insights be converted into principled data-selection or curriculum-learning methods for training foundation models at scale?

来源字段：https://doi.org/10.4171/owr/2026/11

16728 | OWR-15580-001

### Cutting Planes Lower Bounds for Random CNF Formulas

P / partially_solved | Oberwolfach 报告问题汇编

Can one prove superpolynomial lower bounds on the size of Cutting Planes refutations of random $k$-CNF formulas?

来源字段：https://doi.org/10.4171/owr/2017/37

16729 | AIM-COMPUTATION-0088

### Cyclic-binomial structure in an early Agrawal conjecture

P / partially_solved | AIM 研讨会问题汇编

**Conjecture.** To prove that \(n\) is prime, it is enough to let \(Q(X)\) run over the set \[ \{X-1,X^2-1,\ldots,X^{\log^2 n}-1\}. \]

**Recovered bounded-family conjecture.** For \(n\ge 2\), put \(L(n)=\lfloor(\log_2 n)^2\rfloor\). If \[ (1+X)^n\equiv 1+X^n \pmod{\,n,\ X^r-1\,} \tag{BF} \] for every integer \(1\le r\le L(n)\), then \(n\) is prime.

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16730 | OWR-12975-002

### Decidability and Finiteness of Rational Joint Spectral Radii

O / open | Oberwolfach 报告问题汇编

For rational matrices $A_1,\ldots,A_m$, is there an algorithm that halts in finite time and decides whether their joint spectral radius is strictly less than $1$? If the matrices have rational entries, is the joint spectral radius always attained by the spectral radius of a finite product, i.e., does equality hold for some finite product length $k$?

来源字段：https://doi.org/10.4171/owr/2014/17

16731 | OWR-9790365-009

### Decidability of Linear Information Inequalities

O / open | Oberwolfach 报告问题汇编

Is the validity of linear information inequalities decidable?

来源字段：https://doi.org/10.4171/owr/2022/14

16732 | OWR-17135-024

### Decidability of Piecewise Linear Complex Embeddings

P / partially_solved | Oberwolfach 报告问题汇编

For fixed positive integers $k\leq d$ and a finite $k$-dimensional simplicial complex $K$, decide whether $K$ admits a piecewise-linear embedding into $\mathbb R^d$.

来源字段：https://doi.org/10.4171/owr/2019/39

16733 | OWR-4076-004

### Decidability of Rational Points over Number Fields

O / open | Oberwolfach 报告问题汇编

Given a variety $X$ over a number field $k$, is there an algorithm that decides whether $X$ has a $k$-rational point?

来源字段：https://doi.org/10.4171/owr/2009/33

16734 | OWR-2090-010

### Decidability of Simplicial-Complex Embeddability

P / partially_solved | Oberwolfach 报告问题汇编

For fixed integers $k\leq d$, is there an algorithm that decides whether the polyhedron of a given finite simplicial complex $K$ of dimension at most $k$ admits a piecewise-linear embedding into $\mathbb R^d$?

来源字段：https://doi.org/10.4171/owr/2008/44

16735 | OWR-14299088-005

### Decidability of Symmetric Polynomial Nonnegativity

O / open | Oberwolfach 报告问题汇编

Determine whether nonnegativity of a symmetric polynomial is decidable when there is only one group of variables, and whether the analogous problem is decidable for a single matrix variable.

来源字段：https://doi.org/10.4171/owr/2025/9

16736 | OWR-4425-010

### Decidability of Universal Factor Languages

P / partially_solved | Oberwolfach 报告问题汇编

For a finite set $S$ of words over an alphabet $\Sigma$, where $\operatorname{Fact}(S^*)$ is the set of factors of $S^*$, is it decidable whether $\operatorname{Fact}(S^*)=\Sigma^*$, and what is the computational complexity of testing this equality?

来源字段：https://doi.org/10.4171/owr/2010/37

16737 | AIM-COMPUTATION-0101

### Deformation transport, Fesenko transforms, and a Riemann--Roch size barrier

P / partially_solved | AIM 研讨会问题汇编

**Question 13.** (a) Do the \(p\)-adic point counting methods of Lauder which allow you to go from one curve from another in a family apply to \(p\)-adic methods without the linear factor? (Edixhoven) (b) Do Fesenko's methods for proving good properties of Hasse-Weil \(L\)-functions of curves over number fields provide anything useful for computations? (Edixhoven)

**Remarks.** (i) Does Riemann-Roch provide anything useful? (Apparently not; involves linear algebra over matrices of size the number of points.) (Wan)

Can Lauder's deformation/transport step be combined with a \(p\)-adic point-counting method whose running time no longer has a linear factor in \(p\)?

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16738 | AIM-COMPUTATION-0105

### Deformation, specialization, and plane-curve genus

P / partially_solved | AIM 研讨会问题汇编

(a) Can deformation theory be applied `-adically? (b) Under what circumstances do Betti numbers stay the same under reduction modulo \(p\)? (c) Is finding the genus of a plane curve polynomial time (in the degree)?

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16739 | OWR-14299088-003

### Degree Automatability of Sum-of-Squares Proofs

P / partially_solved | Oberwolfach 报告问题汇编

Characterize meaningful conditions under which low-degree sum-of-squares proofs can be found in polynomial time with polynomially bounded coefficients. In particular, determine when the degree-$d$ sum-of-squares relaxation is algorithmically computable in time polynomial in the natural input size for fixed $d$.

来源字段：https://doi.org/10.4171/owr/2025/9

16740 | AIM-COMPUTATION-0090

## Degree bounds and the dependence on the modulus

P / partially_solved | AIM 研讨会问题汇编

Consider \(\mathbb F_p\) and \(d\in\mathbb Z_{>0}\), and \(S\subset\mathbb F_p\) with \(|S|=d\). Let \(h(X)\in\mathbb F_p[X]\) with \(h(s)\ne0\) for all \(s\in S\). Let \(G\) be the group generated by \(X-s\) for \(s\in S\). In the original AKS paper, we have \(|G|\ge 2^d\). There are techniques for getting \(|G|\ge (5.82)^d\) (using lattice point counting); there seems to be much more room for improvement (perhaps using ABC). Perhaps \[ \frac{\log |G|}{d}\longrightarrow\infty \] as \(d\to\infty\), uniformly over \(h\).

The printed question is:

The main sentence does not specify the ambient group. Remark (v) immediately after the question says “Instead of looking at \(\mathbb F_p[X]/(h)\), look at an elliptic curve...”. I therefore use the following reconstruction: \[ R_h=\mathbb F_p[X]/(h),\qquad G_h=\langle \bar X-s:s\in S\rangle\le R_h^\times . \] The condition \(h(s)\ne0\) is precisely what makes each \(\bar X-s\) a unit: \(\gcd(h,X-s)=1\).

There is a second, consequential omission: the displayed question imposes no degree condition on \(h\). Such a condition is indispensable for the quoted \(2^d\) lower bound. Accordingly, this report distinguishes:

1. the **literal quotient-ring reading**, with arbitrary nonconstant \(h\) satisfying the printed condition; and 2. the **intended AKS reading**, in which \(h\) has additional degree and cyclotomic/Frobenius structure that the question does not state.

The first reading admits an elementary counterexample to the proposed uniform limit. The second remains a meaningful harder question, and this report does not claim to settle it.

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16741 | AIM-INFRASTRUCTURE-0095

## Degree-local Hom computation and certified corner reuse

P / partially_solved | AIM 研讨会问题汇编

There is preliminary code to decompose a module into indecomposables, but it would benefit from being much faster. Right now the bottleneck is the computation of the endomorphisms of module M.

Can we compute all the degree-0 (or degree-≤ e) homomorphisms of a graded module M without computing the entirety of End M? What are other ways to speed up the decomposition of modules into indecomposable summands? If M is the direct sum of \(N_1\) and \(N_2\), how can we extract \(\mathrm{End}(N_1)\) and \(\mathrm{End}(N_2)\) from \(\mathrm{End}(M)\) without recalculating?

来源字段：http://aimpl.org/macaulay2efie/4/

16742 | AIM-COMPUTATION-0016

## Degree-two-and-three Markov basis for the Birkhoff ranking model

S / solved | AIM 研讨会问题汇编

**Problem 1.** Explicitly describe a Markov basis for tables with fixed first-order summary. That is, describe a set of moves which will connect any two surveys that have the same first order summary. In a purely mathematical language: describe a generating set for the toric ideal given by the $p^2\times p!$ matrix whose columns are the $p!$ $p\times p$ permutation matrices.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16743 | OWR-16633-017

## Degree-Utility Optimization in Graphs

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Degree-Utility Graph Optimization Problem [OWR-16633-018]

Given $n,m$ and univariate functions $f_1,\ldots,f_n$, what is the complexity of finding an $m$-edge graph on $[n]$ that maximizes

$$\sum_{i=1}^nf_i(d_i),$$

where $d_i$ is the degree of vertex $i$ and the functions need not be identical?

来源字段：https://doi.org/10.4171/owr/2018/50

16744 | AIM-INFRASTRUCTURE-0089

## Degree-windowed Cech computation and the role of one b-function

P / partially_solved | AIM 研讨会问题汇编

The current method for computing local cohomology requires the iterated computation of Bernstein--Sato polynomials at every step of the Cech complex.

Can computing just the necessary steps of the Cech complex speed up the computation of local cohomology? What about computing just the Budur--Mustata--Saito b-function just once?

来源字段：http://aimpl.org/macaulay2efie/2/

16745 | OWR-11793-003

## Density and Geometry Recovery from Nearest-Neighbor Graphs

P / partially_solved | Oberwolfach 报告问题汇编

Let $X_1,\ldots,X_n$ be i.i.d. samples from a density $p$ on $\mathbb R^d$, and let $A$ be the adjacency matrix of their unweighted $k$-nearest-neighbor graph. Can $p(X_i)$ be estimated up to constant factors from $A$ alone, and can the sample locations be approximately reconstructed up to translation, rotation, and rescaling when $n$ is large?

来源字段：https://doi.org/10.4171/owr/2012/31

16746 | OWR-13105-008

### Depth-Independent Attribute-Based Encryption

P / partially_solved | Oberwolfach 报告问题汇编

Can attribute-based encryption be constructed with parameters independent of circuit depth? What applications does fully key-homomorphic encryption have?

来源字段：https://doi.org/10.4171/owr/2014/35

16747 | OWR-13949-007

### Depth-Three Lower Bounds for an Explicit Trilinear Function

P / partially_solved | Oberwolfach 报告问题汇编

Can one prove a lower bound larger than $\exp(\sqrt n)$ on the size of standard depth-three circuits computing the explicit trilinear function $F_{\mathrm{tet}}(x,y,z)=\sum_{i_1,i_2,i_3:\sum_{j=1}^3|i_j-n/2|\leq n/2}x_{i_1}y_{i_2}z_{i_3}$?

来源字段：https://doi.org/10.4171/owr/2015/54

16748 | OWR-8415359-010

### Derandomizing Space-Bounded Computation

O / open | Oberwolfach 报告问题汇编

Prove that $\mathsf{L}=\mathsf{BPL}$; equivalently, show that randomness is unnecessary for space-efficient computation.

来源字段：https://doi.org/10.4171/owr/2021/54

16749 | OWR-9790359-010

### Determinacy Methods in LOCAL and Borel Coloring

O / open | Oberwolfach 报告问题汇编

Can shift-graph and Galvin-Prikry methods be used in the LOCAL model? Can determinacy arguments construct functions of large Borel chromatic number, and does the Axiom of Determinacy imply that the shift graph $G_S$ has infinite chromatic number?

来源字段：https://doi.org/10.4171/owr/2022/8

16750 | AIM-COMPUTATION-0034

### Determinantal certificates for safe categorical collapse before multiple imputation

P / partially_solved | AIM 研讨会问题汇编

**Question 20.** Can algebraic techniques be used to collapse levels of categorical variables in order to improve the quality of multiple imputation without destroying the quality of inference?

Frantisek Matus Frantisek Matus spoke about the representability of conditional independence (CI) structures.

Direct inspection of the official PDF and its neighboring text shows the following layout:

Those two lines are the heading and introduction of the next speaker section. They are preserved in the exact input record but are not part of Question 20. Questions 17--20 follow the preceding Russell Steele section; Conjecture 21 begins the Frantisek Matus section.

The source does not define “quality,” which variable is missing, or which inferential target must survive. A universal answer is impossible without those choices. This report therefore proves a scoped result:

- \(X\) is a fully observed fine categorical covariate; - \(Z=g(X)\) is a proposed collapse of its levels; - \(W\) is a fully observed finite covariate vector; - \(Y\) is a categorical outcome with missingness indicator \(R\); - the inferential target is the conditional law \(P(Y\mid Z,W)\), and hence any estimand depending only on the joint law of \((Y,Z,W)\); - “imputation quality” means correct asymptotic centering of that target law under a saturated imputation model, while interval validity additionally requires proper and congenial multiple imputation.

This scope directly isolates what is gained and what can be destroyed by collapsing before imputation.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16751 | OWR-15996-001

### Deterministic Components of Stochastic Optimization

O / open | Oberwolfach 报告问题汇编

How can deterministic optimization schemes that work on small test problems be incorporated into stochastic optimization methods, and how should regularization functionals be designed for the learning problem?

来源字段：https://doi.org/10.4171/owr/2018/11

16752 | OWR-8415356-025

### Deterministic Optimization of Top-$k$ Perfect-Matching Edges

P / partially_solved | Oberwolfach 报告问题汇编

别名：Deterministic Optimization of Top-k Perfect-Matching Edges [OWR-8415356-026]

Given a weighted graph and an integer $k$, is there a deterministic polynomial-time algorithm that finds a perfect matching maximizing the total weight of its $k$ heaviest edges?

来源字段：https://doi.org/10.4171/owr/2021/53

16753 | AIM-COMPUTATION-0106

### Deterministic roots of X^3-2: a density-one-eighteenth residual envelope

P / partially_solved | AIM 研讨会问题汇编

**Question 18 (Schoof).** Can you compute roots modulo \(p\) of a fixed polynomial (à la Schoof--Pila) in polynomial time, like \(x^3-2\)?

A problem is realizing a given polynomial as the characteristic polynomial of an endomorphism, if its roots do not lie in a CM field. Does it help to consider modular forms? (Pila)

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16754 | OWR-13353-018

### Dimension Gaps in Lattice Reductions to Learning with Errors

O / open | Oberwolfach 报告问题汇编

Can the discrepancy between the lattice dimensions in the quantum and classical reductions from GapSVP to learning with errors be removed or explained, namely dimension n for the quantum reduction versus order square-root-n for the classical reduction?

来源字段：https://doi.org/10.4171/owr/2014/51

16755 | OWR-14299088-014

### Dimension-Free Sum-of-Squares Certificates for Two Quadratic Constraints

O / open | Oberwolfach 报告问题汇编

Let $Q_1,Q_2,Q_3$ be quadratic forms in $n$ variables, with $Q_1$ nonnegative on the variety $Q_2=Q_3=0$. Find the least degree $2d$ for which there are homogeneous polynomials $P,R_2,R_3$ of degree $2d$, with $P$ a sum of squares, such that $PQ_1+R_2Q_2+R_3Q_3$ is a sum of squares. Can $d$ be bounded independently of $n$?

来源字段：https://doi.org/10.4171/owr/2025/9

16756 | OPG-2150

### Discrete Logarithm Problem

P / partially_solved | Open Problem Garden

If $p$ is prime and $g,h \in {\mathbb Z}_p^*$, we write $\log_g(h) = n$ if $n \in {\mathbb Z}$ satisfies $g^n = h$. The problem of finding such an integer $n$ for a given $g,h \in {\mathbb Z}^*_p$ (with $g \neq 1$ ) is the Discrete Log Problem.

Conjecture There does not exist a polynomial time algorithm to solve the Discrete Log Problem.

背景链接：http://www.openproblemgarden.org/op/discrete_logarithm_problem

16757 | OWR-15173-005

### Discrete Vector Fields for Voronoi-Cohomology Computations

O / open | Oberwolfach 报告问题汇编

Can the discrete vector-field technique be applied effectively to cohomological computations with the Voronoi cell complex?

来源字段：https://doi.org/10.4171/owr/2016/52

16758 | AIM-CRYPTOGRAPHY-0011

### Dominating sets as marked centralizer covers in graph groups

P / partially_solved | AIM 研讨会问题汇编

Computational problems from graph groups

It is known that it is possible to establish correspondence between certain NP-hard graph problems and problems in group theory; we would like to expand our portfolio of such problems.

来源字段：http://aimpl.org/postquantgroup/3/

16759 | OWR-14297745-001

### Duality and Convergence Rates in Lasserre Hierarchies

P / partially_solved | Oberwolfach 报告问题汇编

For Lasserre's hierarchy, when does strong duality hold, do the moment relaxations converge to an optimal measure, and what is their convergence rate?

来源字段：https://doi.org/10.4171/owr/2023/35

16760 | AMR-054-0063

### Dynamic Planar Nearest Neighbors

P / partially_solved | AMR 问题清单汇编

Is there a data structure maintaining a set of $n$ points in the plane subject to insertions, deletions, and nearest-neighbor queries in $O(\log n)$ time? A nearest-neighbor query asks to find a point among the set that is nearest (in Euclidean distance) to a given a point in the plane. This problem reduces to maintaining the convex hull of a set of $n$ points in 3D subject to insertions, deletions, and extreme-point queries.

背景链接：https://topp.openproblem.net/p63

16761 | OWR-14298584-012

### Efficient Computation of Exceptional Galois Representations

P / partially_solved | Oberwolfach 报告问题汇编

Develop an efficient general algorithm for computing exceptional Galois representations, including their projective representations, with substantially better complexity than exhaustive class-field-theoretic or targeted number-field searches.

来源字段：https://doi.org/10.4171/owr/2024/40

16762 | OWR-12975-008

### Efficient Moment Relaxations for Polynomial Optimization

O / open | Oberwolfach 报告问题汇编

Can moment-relaxation methods solve polynomial optimization over closed basic semialgebraic sets efficiently, including computing the infimum and the points where it is attained?

来源字段：https://doi.org/10.4171/owr/2014/17

16763 | OWR-14299286-001

### Efficient Optimization of Neural-Network Weights

O / open | Oberwolfach 报告问题汇编

Given a fixed neural-network architecture, can globally optimal weights be found in reasonable computational time?

来源字段：https://doi.org/10.4171/owr/2025/22

16764 | OWR-10252930-031

### Efficient Recognition of Cyclic Sumsets

O / open | Oberwolfach 报告问题汇编

Given $B\subseteq\mathbb Z/n\mathbb Z$, is there an efficient algorithm that decides whether some $A\subseteq\mathbb Z/n\mathbb Z$ satisfies $B=A+A$?

来源字段：https://doi.org/10.4171/owr/2022/22

16765 | OWR-14299084-003

### Efficient Verification of Distribution Properties

P / partially_solved | Oberwolfach 报告问题汇编

Characterize the properties of an unknown distribution for which an untrusted server can generate, in polynomial time, a short interactive proof of an approximately correct assertion that a verifier can check using fewer samples and less computation than independent analysis would require.

来源字段：https://doi.org/10.4171/owr/2025/3

16766 | OWR-12697684-018

### Enumeration and Sparsity of Growing-$k$ SAT Functions

O / open | Oberwolfach 报告问题汇编

Determine the asymptotic number of distinct $k$-SAT Boolean functions when $k$ grows with the number $n$ of variables. For which growth rates of $k$ are almost all such functions unate, and what are the corresponding sparse-formula thresholds?

来源字段：https://doi.org/10.4171/owr/2023/1

16767 | OWR-10252930-018

### Enumeration and Typical Structure of $k$-SAT Functions

S / solved | Oberwolfach 报告问题汇编

How many $k$-SAT functions on $n$ Boolean variables are there, and what does a typical such function look like? In particular, for fixed $k\geq 2$, is a $1-o(1)$ fraction of all such functions unate?

来源字段：https://doi.org/10.4171/owr/2022/22

16768 | AIM-COMPUTATION-0066

### Equality constraints do not characterize hidden-variable models

P / partially_solved | AIM 研讨会问题汇编

**Question 51 (Meek).** Is there a polynomial time algorithm for describing the constraints on the joint probability distribution on the observed variables in a graphical model with hidden variables?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16769 | OWR-15210-004

### Error-Free Multiparty Homomorphic Secret Sharing

P / partially_solved | Oberwolfach 报告问题汇编

For DDH-based homomorphic secret sharing supporting compact evaluation of branching programs, can the inverse-polynomial evaluation error be eliminated or its running-time dependence improved, and can analogous results be obtained for three or more parties?

来源字段：https://doi.org/10.4171/owr/2017/5

16770 | AIM-COMPUTATION-0080

### Evaluation-rank explanation for the value of knockout time series

P / partially_solved | AIM 研讨会问题汇编

**Question 65 (Laubenbacher).** The model selection problem does not seem to be very effective when only one time series is used but does seem to work well when multiple knockout time series are used in combination. Why?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16771 | AIM-COMPUTATION-0059

### Exact Bayesian evidence and prediction for finite latent discrete models

P / partially_solved | AIM 研讨会问题汇编

**Problem 45 (Fienberg).** Computing the full distributions for observables using Bayesian extensions. Predictive distributions. Dinwoodie: How might you use algebra to pursue Bayesian directions in model fit criteria? Sturmfels: How to you compute this in Macaulay 2? What is Bayesian algebraic geometry? Fienberg: Taking mixtures of things the different models we have discussed at this conference. Meek: Finding the modes of the posterior distributions could be an interesting research problem. Pistone: You could use moment generating functions and cumulant generating functions for these problems. Fienberg: Identifiability is a big issue in the Bayesian context. Wynn: How may Edgeworth corrections play a role in this problem?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16772 | AIM-COMPUTATION-0071

### Exact chain gluing and a sharp cyclic incompatibility certificate

P / partially_solved | AIM 研讨会问题汇编

**Problem 56 (Fienberg).** Consider 3-way table missing data problem. This consists of a 3-way table with holes and 2-way margins which also have holes.. How can you estimate the likelihood and "glue" this information together into a good global picture of the whole table? Can this problem be formulated as an algebraic variety?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16773 | AIM-COMPUTATION-0035

### Exact CI rationalization through rational local charts

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 21.** Every p-representable CI-structure can be p-represented by a random vector living on a finite probability space endowed with the uniform distribution.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16774 | AIM-COMPUTATION-0029

### Exact cumulant ideal of the normalized binary four-cycle

S / solved | AIM 研讨会问题汇编

**Question 15. What is the cumulant ideal of the binary four-cycle model?**

The recovered question is:

In the source PDF they begin the next speaker's topic. Immediately before Question 15, Henry Wynn's section asks for polynomial descriptions obtained by passing from probabilities through moments to cumulants.

I interpret “four-cycle model” as the undirected binary graphical model on

\[ C_4:\qquad E=\{12,23,34,14\}. \]

Its global Markov statements are

\[ X_1\mathbin{\perp\!\!\!\perp}X_3\mid(X_2,X_4), \qquad X_2\mathbin{\perp\!\!\!\perp}X_4\mid(X_1,X_3). \]

This matters because “binary four-cycle” can also refer to models for marginal independence. Pistone and Wynn's four-cycle example uses the isomorphic relabeling whose nonedges are \(12\) and \(34\), and uses \(\{-1,1\}\) coding. The answer below instead fixes the standard edge labeling above and initially uses \(\{0,1\}\) coding. The two labelings are related by a permutation, and affine binary recoding gives an equivalent cumulant ideal.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16775 | AIM-CRYPTOGRAPHY-0025

### Exact DDH leakage and sampler criteria for the CSIDH torsor

P / partially_solved | AIM 研讨会问题汇编

Solve Decisional Diffie--Hellman (DDH) in the context of CSIDH: Let \(E_0\) be a supersingular elliptic curve defined over \(\mathbb F_p\) with endomorphism ring \(\mathbb Z[\pi]\). Distinguish between triples \((\mathfrak a*E_0,\mathfrak b*E_0,\mathfrak c*E_0)\) and triples \((\mathfrak a*E_0,\mathfrak b*E_0,\mathfrak a\mathfrak b*E_0)\), where \(\mathfrak a,\mathfrak b,\mathfrak c\) are ideals in \(\mathbb Z[\pi]\) of odd norm.

来源字段：http://aimpl.org/quantumalg/5/

16776 | AIM-COMPUTATION-0024

### Exact endpoint and stability census for the symmetric triangle

P / partially_solved | AIM 研讨会问题汇编

**Question 10.** Are there always only 0, 1, or 2 statistically interesting solutions to the blocking probability equations in network tomography?

The heading **Ruriko Yoshida** immediately after Question 10 begins the next speaker's topic.

The terminology is made precise by the direct follow-up technical report of Dinwoodie, Gamundi, and Mosteig [DGM04]. Its model is a complete circuit-switched network with dynamic routing: traffic first attempts the direct edge and, if that edge is blocked, may try a two-edge route. An edge variable \(b_{ij}\) is its blocking probability. In the large-capacity Erlang approximation, the fixed-point equations become polynomial after one restricts to positive blocking probabilities. The report calls a real solution statistically/probabilistically interesting precisely when every blocking coordinate is in the open interval \((0,1)\).

This is therefore a dynamic-routing loss-network question, not the unrelated multicast maximum-likelihood model also called “network tomography.”

Source-recovery audit: [DGM04] was recovered from the Internet Archive capture `20060907132830` of its original SAMSI URL. The recovered PDF is 864,647 bytes, has SHA-256 `c2627b2bc2478d90d5a974272f44f54a44d913373b02a0f59ad6fde7143ae150`, and the CDX digest is `UQVCBNMKOPN554HBLYPVOCBKPPJDNMKI`. These details make the equation reconstruction independently checkable.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16777 | OWR-16938-002

### Exact Higher-Order Matrix Decomposability

O / open | Oberwolfach 报告问题汇编

Given a matrix $Q$ and an integer $k>2$, find necessary and sufficient conditions for $Q$ to admit an exact $k\times k$ decomposition into positive-semidefinite principal-block contributions.

来源字段：https://doi.org/10.4171/owr/2019/26

16778 | OWR-13353-020

### Exact Integrality Gap for Capacitated Facility Location

P / partially_solved | Oberwolfach 报告问题汇编

What is the exact integrality gap of the strengthened linear-programming relaxation for the capacitated facility location problem based on partial client assignments?

来源字段：https://doi.org/10.4171/owr/2014/51

16779 | AIM-COMPUTATION-0068

### Exact output-sensitive scaling for decomposable discrete models

P / partially_solved | AIM 研讨会问题汇编

Question 53 (Fienberg). How can we make the algorithms and existing theoretical results apply to problems of sizes that are of practical interest? Scaling up to 5 10 tables is a hopeful goal.

Scaling up to \(5^{10}\) tables is a hopeful goal.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16780 | AIM-CRYPTOGRAPHY-0019

### Exact period-quotient reduction and unary lift cost for hidden shift on Z^d

P / partially_solved | AIM 研讨会问题汇编

Hidden shift on \(\mathbb Z^d\) where \(f_0\) and \(f_1\) are periodic under unary oracle cost.

This is problem 4.5 in the “Hidden Shift” section of the 2019 AIM workshop *Quantum algorithms for analysis of public-key crypto*. The official workshop summary attributes the sentence to Greg Kuperberg and reproduces it verbatim [AIM19]. The adjacent problem asks for hidden shift on \(\mathbb Z\) with a bounded shift “under binary cost of oracle.” That juxtaposition suggests that “unary” may charge in proportion to coordinate magnitude rather than its bit length, but neither the workshop summary nor the workshop page defines the phrase.

Several mathematically inequivalent readings are therefore present in the one-line source:

1. “Periodic” may mean only that there is one nonzero period, that the full period group has rank \(d\), or that a full-rank period lattice and basis are supplied. 2. The requested answer may be an exact vector \(s\in\mathbb Z^d\), a chosen representative, or the identifiable residue class of \(s\) modulo all periods. 3. Standard hidden shift assumes injective quotient functions. Arbitrary periodic functions need not be injective even after quotienting by their full translation stabilizer. 4. “Unary cost” may mean unary address encoding, cost proportional to the largest address in a coherent query, average branch cost, or merely unit cost for each oracle invocation.

The precise reconstruction used below is \[ f_0,f_1:\mathbb Z^d\longrightarrow Y, \qquad f_1(x)=f_0(x+s), \] with exact, coherently queryable oracles. The full period or translation-stabilizer lattice is \[ P=\operatorname{Stab}(f_0) :=\{p\in\mathbb Z^d:f_0(x+p)=f_0(x)\text{ for every }x\in\mathbb Z^d\}. \] The primary recovery target is \([s]\in\mathbb Z^d/P\), since Section 5 proves that this is exactly the information determined by the two functions. Exact \(s\) is not identifiable without a convention or an extra promise.

来源字段：http://aimpl.org/quantumalg/4/

16781 | AIM-CRYPTOGRAPHY-0012

### Exact rank-one decoding and a factorized no-QRAM quantum baseline

P / partially_solved | AIM 研讨会问题汇编

Parameters: \(n \geq 1\), \(r\geq 1\), \(\mathbb F_{q^m}\), \(t \geq 1\). How quickly can we find \(v\in \mathbb F_{q^m}^n\) with \(Hv = s\) and \(\text{wt}(v) = t\), given \(H\in \mathbb F_{q^m}^{r\times n}\) and \(s\in \mathbb F_{q^m}^r\)?

Here, \(\text{wt}(v)\) is defined to be the rank of the \(v\), viewed as a matrix with \(n\) columns.

来源字段：http://aimpl.org/quantumalg/1/

16782 | AIM-COMPUTATION-0083

### Exact sketches versus genuine dynamical reduction

P / partially_solved | AIM 研讨会问题汇编

**Recovered statement.** In many problems for analyzing time series there are very few data points in a high-dimensional space. How can techniques for reducing the dimensionality be used?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16783 | AIM-COMPUTATION-0026

### Exact slack lifts for Markov bases under rounded summaries

P / partially_solved | AIM 研讨会问题汇编

Question 12. What can be done with Markov bases when their are floating point approxi-mations made in marginals or conditionals?

**Question 12.** What can be done with Markov bases when there are floating-point approximations made in marginals or conditionals?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16784 | OWR-3394-006

### Expansion-Point Selection in Time-Domain Model Reduction

P / partially_solved | Oberwolfach 报告问题汇编

In rational Krylov or Laguerre-based time-domain model reduction, how should the expansion point $s_0=\alpha$, equivalently the Laguerre parameter $\alpha$, be chosen to approximate the impulse response accurately?

来源字段：https://doi.org/10.4171/owr/2009/11

16785 | OWR-730-006

### Expected-Optimum Approximation for Random MAX $R$-3SAT

P / partially_solved | Oberwolfach 报告问题汇编

For fixed clause size $r$ and density $m/n$, design a polynomial-time algorithm for random MAX $R$-3SAT instances $F$ that returns an assignment with expected value $\mathbb{E}[A(F)]\geq(1-\varepsilon)\mathbb{E}[\operatorname{OPT}(F)]$ for every $\varepsilon>0$.

来源字段：https://doi.org/10.4171/owr/2004/28

16786 | OWR-4414-001

### Explicit Deterministic Restricted Isometry Matrices

P / partially_solved | Oberwolfach 报告问题汇编

Can one construct deterministic explicit measurement matrices with small restricted isometry constants when the number of measurements is small, or equivalently when the sparsity level is large?

来源字段：https://doi.org/10.4171/owr/2010/33

16787 | AIM-COMPUTATION-0072

### Explicit Gaussian MLE for monotone block missingness

P / partially_solved | AIM 研讨会问题汇编

**Gaussian missing data problem.** Consider a continuous Gaussian population and a random sample of size \(n\) but only some of the characteristics of each individual are observed. Estimate the mean and covariance matrix. Fienberg: First approach the problem using a block structure. Steele: Look at a sequence of regressions as a sequence of polynomials. Can you solve the structure explicitly?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16788 | AIM-CRYPTOGRAPHY-0028

### Explicit near-minimal units from the golden-ratio radical

P / partially_solved | AIM 研讨会问题汇编

Compute short units in the ring of integers of \(\mathbb Q[x]/(x^{312} - x^{156} -1)\). How short can they be, and how fast can you find them?

来源字段：http://aimpl.org/quantumalg/6/

16789 | OWR-1587-005

### Explicit Polynomial Maps Escaping Low-Degree Images

P / partially_solved | Oberwolfach 报告问题汇编

Let $\mathbb{F}$ be a field. First, for functions $s=s(n)$, $m=m(n)$, and $r=r(n)$ satisfying $n^{\omega(1)}\leq s<m=n^r$, with $r$ relatively small (for example, $r=\log\log n$), construct an explicit polynomial map $f:\mathbb{F}^n\to\mathbb{F}^m$ of degree at most $2^n$ whose image is not contained in the image of any polynomial map $\Gamma:\mathbb{F}^s\to\mathbb{F}^m$ of degree at most $r$. Second, for integers $s<m=n^{r+1}$ and $2\leq r\leq O(1)$, given a map $\Gamma:\mathbb{F}^s\to\mathbb{F}^m$ of degree at most $2r-1$, design a polynomial-time algorithm that outputs an explicit map $f:\mathbb{F}^n\to\mathbb{F}^m$ of degree $\operatorname{poly}(n)$ whose image is not contained in the image of $\Gamma$.

来源字段：https://doi.org/10.4171/owr/2007/31

16790 | OWR-16931-009

### Explicit RIP Matrices Beyond Square-Root Sparsity

O / open | Oberwolfach 报告问题汇编

For fixed $\varepsilon>0$, construct explicit real-valued restricted-isometry matrices in the polynomial regime $n=p^{O(1)}$ with sparsity $s\geq n^{1/2+\varepsilon}$.

来源字段：https://doi.org/10.4171/owr/2019/19

16791 | OPG-36311

### Exponential Algorithms for Knapsack

P / partially_solved | Open Problem Garden

Conjecture

The famous 0-1 Knapsack problem is: Given $a_{1},a_{2},\dots,a_{n}$ and $b$ integers, determine whether or not there are $0-1$ values $x_{1},x_{2},\dots,x_{n}$ so that $$\sum_{i=1}^{n} a_{i}x_{i} = b.$$The best known worst-case algorithm runs in time$2^{n/2}$times a polynomial in$n$. Is there an algorithm that runs in time$2^{n/3}$?

背景链接：http://www.openproblemgarden.org/op/exponential_algorithms_for_knapsack

16792 | OWR-831-012

### Exponential Tree Size of Cluster States

O / open | Oberwolfach 报告问题汇编

别名：Rhetorical Quantum-State Vector-Length Question [OWR-831-010]

Do uniform superpositions over random linear codes and the states arising in Shor's algorithm have tree size exponential in $n$; and do cluster states in dimensions two and three likewise have exponential tree size?

来源字段：https://doi.org/10.4171/owr/2005/26

16793 | OWR-14298167-005

### Expressivity and Training Theory for Deep Material Networks

P / partially_solved | Oberwolfach 报告问题汇编

Can Deep Material Networks be made more expressive through improved building blocks or hierarchies; why does elastic training suffice for inelastic applications; and can that theory improve training-data selection?

来源字段：https://doi.org/10.4171/owr/2024/11

16794 | OWR-8415356-006

### Extension Complexity of Inscribed Fixed-Dimensional Polytopes

P / partially_solved | Oberwolfach 报告问题汇编

For every fixed $d\geq2$, does every $d$-dimensional polytope with $n$ vertices on a common sphere have extension complexity $O(\sqrt n)$?

来源字段：https://doi.org/10.4171/owr/2021/53

16795 | OWR-15999-007

### Extrapolating Clusterings from Semidefinite Subsamples

P / partially_solved | Oberwolfach 报告问题汇编

Given a full data set $T$ and a sampled subset $S$, can a near-optimal clustering of $T$ be extrapolated from the Peng-Wei semidefinite program applied to $S$?

来源字段：https://doi.org/10.4171/owr/2018/14

16796 | OWR-14299916-006

### Extremal Top-$m$ Representability of Output Projection Matrices

P / partially_solved | Oberwolfach 报告问题汇编

For a vocabulary of $N$ tokens and embedding dimension $d$, what is the largest $m$ for which there exists an output projection matrix $A\in\mathbb R^{N\times d}$ such that, for every set of $m$ tokens, some embedding $x\in\mathbb R^d$ makes those tokens the $m$ most probable under $\operatorname{softmax}(Ax)$?

来源字段：https://doi.org/10.4171/owr/2026/11

16797 | AIM-COMPUTATION-0085

### Faithful interchange between computational algebra systems

P / partially_solved | AIM 研讨会问题汇编

**Question 70 (Slavkovic).** Are there scripts which convert between input formats for the different computational algebra systems?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16798 | OWR-1592-001

### Fast Instance-Optimal Decoders for Compressed Sensing

O / open | Oberwolfach 报告问题汇编

In compressed sensing, suppose the $\ell^2$ instance-optimal recovery estimate $\lVert x-\Delta(\Phi x)\rVert_2\leq C_0\sigma_k(x)_2$ holds for $k\leq cn\log(N/n)$ with probability at least $1-e^{-bn}$. Can one construct a decoder $\Delta$ with the same guarantee and low algorithmic complexity?

来源字段：https://doi.org/10.4171/owr/2007/36

16799 | OWR-16633-009

### Fast Low-Rank Maximum-Weight Assignment

O / open | Oberwolfach 报告问题汇编

For the maximum-weight assignment problem on a complete bipartite graph with weight matrix $W\in\mathbb R^{n\times n}$ of rank $r$, can the optimum be computed in time $O(nr^2\log n)$?

来源字段：https://doi.org/10.4171/owr/2018/50

16800 | OWR-5160-002

### Fast Solvers for the Electronic Schrödinger Equation

O / open | Oberwolfach 报告问题汇编

Develop a fast computational method for solving the electronic Schrödinger equation, with substantially better complexity than current approaches.

来源字段：https://doi.org/10.4171/owr/2011/32

16801 | OWR-2090-022

### Faster Rank-Two Parametric Assignment

O / open | Oberwolfach 报告问题汇编

For the rank-two parametric assignment problem arising from least-squares matching of planar point sets under rotation, is there an algorithm faster than the standard parametric-assignment approach?

来源字段：https://doi.org/10.4171/owr/2008/44

16802 | AIM-COMPUTATION-0104

### Fields, orbits, and explicit torsion points

P / partially_solved | AIM 研讨会问题汇编

**Question 16 (Edixhoven).** How can you compute with points as in Edixhoven's talk? In that application, you can avoid writing down explicit \(\ell\)-torsion points (only the fields of definition), but can you write them down for other transformation?

**Remarks.**

(i) Can one work out instances of the passage from \(H^1\) to \(H^2\)? E.g., K3 surface of Néron--Severi rank \(19\) (the rank \(20\) case is standard)? (Elkies)

(ii) More comments on \(\ell\)-adic computation of zeta functions? (Pila)

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16803 | AIM-COMPUTATION-0103

### Finite certificate interfaces for finite-field zeta functions

P / partially_solved | AIM 研讨会问题汇编

**Question 15.** Given a variety over a finite field, can you verify that a given function is its zeta function (i.e., is the question in NP)? (Lenstra, Edixhoven)

**Remark.** Yes (for \(g\) fixed), because you can verify the orders of the Jacobian over the first \(g\) extension fields. (Elkies)

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16804 | AIM-COMPUTATION-0041

### Finite Markov templates with several varying level sets

P / partially_solved | AIM 研讨会问题汇编

What types of "finiteness" statements can be made about Markov bases for hierarchical models as more than one set of levels is allowed to vary?

The source places this question immediately after the note that Seth Sullivant spoke about Markov bases for hierarchical models and the structure of minimal Markov-basis elements, and immediately before the problem asking which binary hierarchical models have moves of degree at most four. Thus, “sets of levels” is interpreted here as the state sets of two or more factors of a contingency table. That interpretation is consistent with the later terminology in Hoşten--Sullivant, but it is a reconstruction: the one-sentence question does not specify a category of embeddings or a precise finiteness invariant.

The question is broad rather than a single yes/no conjecture. The results below separate several inequivalent notions, prove an elementary equivalence that is useful for the standard level-permutation system, give a positive two-growing-axis family with a boundary-safe connectivity proof, and contrast it with the known negative three-way universality theorem.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16805 | AIM-CRYPTOGRAPHY-0026

### Finite resource bounds for random spherical covering

P / partially_solved | AIM 研讨会问题汇编

Parameters: \(n \geq 2\) and \(\alpha > 0\). Given independent uniformly random \(x_1, x_2, \dots\in S^{n-1} = \{v \in \mathbb R^n : \|v\|_2 = 1\}\), how quickly can we find a subsequence that covers \(S^{n-1}\)? (Here, ``covers" means that every point of \(S^{n-1}\) is within angle \(\alpha\) of a point in the subsequence.) For example, consider the case when \(n = 1000\) and \(\alpha = 75^\circ\).

来源字段：http://aimpl.org/quantumalg/6/

16806 | OWR-14298804-009

### Finite Strategies in the Distance Game

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Finite-Strategy Question for the Distance Game [OWR-14298804-010]

In the $\delta$-distance game, Peter chooses a leaving vertex of a simplex in the unit ball, and Roland chooses a replacement vertex that preserves the $\delta$-distance property and crosses the opposite face; the game ends when the simplex contains the origin. Does Peter have a strategy that always terminates, and can he force termination within $\operatorname{poly}(d,\delta^{-1})$ rounds?

来源字段：https://doi.org/10.4171/owr/2024/50

16807 | AIM-COMPUTATION-0081

### Finite-field least squares as statistical decoding

P / partially_solved | AIM 研讨会问题汇编

**Question 66 (Laubenbacher).** The model selection problem over finite fields is very sensitive to noise. Is there some sort of “least squares” over finite fields to deal with noisy data?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16808 | AIM-COMPUTATION-0011

### Finite-switch implementation of chattering controls

P / partially_solved | AIM 研讨会问题汇编

How can finite-difference HJB methods be made practical in dimensions 3--5? Given an approximate numerical solution of the PDE for the value function, how should one implement a near-optimal control and bound its suboptimality? If the optimal control switches infinitely many times, how much performance is lost when it is replaced by a control having only finitely many switches?

来源字段：https://aimath.org/WWN/optimalcontrol/optimalcontrol.pdf

16809 | AIM-CRYPTOGRAPHY-0017

### Finite-window reduction and binary-cost accounting for hidden shift on the integers

P / partially_solved | AIM 研讨会问题汇编

What is the cost of hidden shift on \(\mathbb Z\) where the shift \(s\in [a,b]\), under binary cost of oracle?

来源字段：http://aimpl.org/quantumalg/4/

16810 | AIM-INFRASTRUCTURE-0090

### Fixed-ambient extraction of roots of finite Frobenius modules

P / partially_solved | AIM 研讨会问题汇编

If R is a regular ring of characteristic p, then the local cohomology modules have many nice finiteness properties; one can describe many of these properties via the theory of unit $F$-modules. Work on implementing these calculations in M2 was begun at the 2023 Minneapolis workshop.

Given an F-finite F-module, how can one efficiently compute a unit F-module? For example, given the map $\operatorname{Ext}^i(R/I,R)\to \operatorname{Ext}^i(R/I^{[p]},R)$, how can one find a root for the local cohomology module $H^i_I(R)$?

Starting from a finite generating morphism $\beta:M\to F^*M$, compute an injective finite generating morphism (a root) for the same unit $F$-module.

来源字段：http://aimpl.org/macaulay2efie/2/

16811 | OWR-14298804-007

### Fixed-Parameter Tractability of Zonotope Containment

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Dimension-Parameterized Zonotope Containment [OWR-14298804-008]

Given $G\in\mathbb R^{d\times n}$ and $H\in\mathbb R^{d\times m}$, can one decide whether the zonotope $Z(G)$ is contained in $Z(H)$ in time $f(d)\operatorname{poly}(n+m)$ for some computable function $f$?

来源字段：https://doi.org/10.4171/owr/2024/50

16812 | AMR-054-0028

### Flip Graph Connectivity in 3D

O / open | AMR 问题清单汇编

Is the flip graph connected for general-position points in $\mathbb{R}^3$? Given a set of $n$ points in $\mathbb{R}^3$, the flip graph has a node for each tetrahedralization of the set. Two nodes are connected by an arc if there is a 2-to-3 or 3-to-2 ``bistellar flip'' of tetrahedra between the two simplicial complexes. In the plane, the flips correspond to convex quadrilateral diagonal switches; in $\mathbb{R}^3$, a $5$-vertex convex polyhedron is ``flipped'' between two of its tetrahedralizations.

背景链接：https://topp.openproblem.net/p28

16813 | OWR-14299525-001

### Formal Verification of AI-Generated Mathematical Proofs

O / open | Oberwolfach 报告问题汇编

Develop a reliable workflow that translates AI-generated mathematical arguments into Lean, checks every inferential detail, and diagnoses gaps rather than accepting persuasive but invalid prose. Determine how much of this verification and repair process can itself be automated.

来源字段：https://doi.org/10.4171/owr/2025/43

16814 | OWR-14299084-001

### Foundational Assumptions for Obfuscation, FHE, and SNARGs

P / partially_solved | Oberwolfach 报告问题汇编

Determine whether indistinguishability obfuscation can be based on weaker standard assumptions and made post-quantum secure; whether any meaningful virtual-black-box obfuscation can be constructed; whether fully homomorphic encryption can be based on objects other than integer lattices; and whether SNARGs for every NP language can be based on standard cryptographic assumptions.

来源字段：https://doi.org/10.4171/owr/2025/3

16815 | AIM-COMPUTATION-0087

### Four distinct reductions in the American min-put benchmark

P / partially_solved | AIM 研讨会问题汇编

**Recovered statement.** Analyze the independent, identically distributed \(d\)-asset Black-Scholes American min-put benchmark (1), and distinguish exact and approximate ways in which its computational or state dimension can be reduced. In particular, identify what can be reduced without changing the benchmark and what information an exact stopping rule still needs.

It begins with a table-of-contents fragment and then reproduces almost all of “Chapter A: Open problems” from the AIM workshop list *Numerical probabilistic methods for high-dimensional problems in finance* [1]. It includes:

- A.1 on optimal stopping, HJB equations, and reflected forward-backward SDEs; - A.2 on Longstaff-Schwartz regression, Malliavin methods, quantization, dual methods, cubature, and coupled FBSDEs; - A.3, a numerical benchmark; - A.4, several meanings of dimension reduction; and - A.5, the chapter bibliography.

The official eight-page AIM PDF was also inspected directly.

We recover \[ S_t^i=100\exp\!\left((r-\tfrac12\sigma^2)t-\sigma W_t^i\right), \qquad g(x)=\left(K-\min_{1\leq i\leq d}x_i\right)_+,\quad K=100. It is immaterial in law because \(-W\) is again a standard Brownian motion. “Uncorrelated” components of the stated standard Brownian motion are independent.

The recovered research task is:

Section A.4 proposes PCA, low-factor economic models, and maturity randomization. This report develops a precise audit: PCA and low-rank linear factors do not exactly reduce this isotropic benchmark; permutation symmetry removes duplicated labeling but not continuous dimension; the European terminal subproblem has an exact one-dimensional integral; and that integral proves that the present minimum alone is not an exact reduced state.

This is mathematical benchmark analysis, not option-pricing advice or a trading recommendation.

来源字段：https://aimath.org/WWN/highdimfinance/highdimfinance.pdf

16816 | AIM-COMPUTATION-0048

### Fourier edge invariants and a visibility threshold for finite rate-class mixtures

P / partially_solved | AIM 研讨会问题汇编

“Problem 34. Find invariants for other models with rate variation among sites: secant varieties of the phylogenetic variaties.”

Find polynomial phylogenetic invariants for rate-heterogeneous models, in particular for finite mixtures whose complex Zariski closures are secant varieties of fixed-tree phylogenetic varieties.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16817 | OWR-14298166-010

### Fractional Coefficient Savings in Algebraic Proof Systems

P / partially_solved | Oberwolfach 报告问题汇编

Are there natural examples in which fractional coefficients greatly reduce the total coefficient size required for Nullstellensatz or Sherali-Adams proofs?

来源字段：https://doi.org/10.4171/owr/2024/15

16818 | OWR-14299292-016

### Freeness of Two-Generator Integral Matrix Groups

O / open | Oberwolfach 报告问题汇编

Given two integral matrices $A,B\in\mathrm{GL}_n(\mathbb{Z})$, decide whether the subgroup $\langle A,B\rangle$ is freely generated by $A$ and $B$. Is this decision problem solvable uniformly in $n$?

来源字段：https://doi.org/10.4171/owr/2025/27

16819 | OWR-4426-003

### Fundamental-Domain Choice in Symmetric Integer Programming

O / open | Oberwolfach 报告问题汇编

In symmetry-aware linear or integer programming, does the choice among multiple minimal fundamental domains affect the results or efficiency, especially when a variable is a good symmetry-based branching candidate but a poor integer-programming branching candidate?

来源字段：https://doi.org/10.4171/owr/2010/38

16820 | DARPA-013

### Game Theory at Scale

P / partially_solved | DARPA 挑战

Create scalable mathematics for differential games, replacing traditional PDE approaches.

背景链接：https://web.math.utk.edu/~vasili/refs/darpa07.MathChallenges.html

16821 | AIM-COMPUTATION-0065

### Gaussian conditional independence as an algebraic minor

P / partially_solved | AIM 研讨会问题汇编

**Question 50 (Kuhnt).** What about adding continuous random variables into these problems? Might an algebraic framework be useful? Dinwoodie: Can we try to apply algebraic techniques to microarray data? Meek: Gaussian (or similar) assumptions on the random variables might be possible to handle with algebraic techniques.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16822 | OWR-9790365-016

### Gaussian Mean Testing under Information Constraints

P / partially_solved | Oberwolfach 报告问题汇编

Determine optimal tests and matching lower bounds for Gaussian mean testing under communication or information constraints.

来源字段：https://doi.org/10.4171/owr/2022/14

16823 | AIM-COMPUTATION-0039

### Gaussian observed-likelihood equations and a rank-certified monotone solution

P / partially_solved | AIM 研讨会问题汇编

**Problem 25.** Solve the maximum likelihood equations for a multivariate guassian with arbitrary missing data patterns.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16824 | OWR-4426-004

### General Effectiveness of Symmetry-Aware Polyhedral Pivoting

P / partially_solved | Oberwolfach 报告问题汇编

To what extent can pivoting under symmetry be made effective in general for polyhedral computation, beyond the specialized applications where it is currently useful?

来源字段：https://doi.org/10.4171/owr/2010/38

16825 | OWR-14299083-002

### Generalization Guarantees for Self-Supervised Representations

P / partially_solved | Oberwolfach 报告问题汇编

Establish statistical generalization guarantees across downstream prediction tasks for representations obtained by self-supervised pretraining on large unlabeled augmented datasets, beyond the presently understood kernel and infinite-width regimes.

来源字段：https://doi.org/10.4171/owr/2025/4

16826 | OWR-14299525-002

### Generative Sphere Packing across Dimensions and Periodic Geometries

O / open | Oberwolfach 报告问题汇编

Can one conditional generative model trained across different numbers of balls and dimensions extrapolate to larger sphere-packing instances while preserving or improving minimum separation after projection? Extend projection-corrected flow-matching methods from finite boxes to periodic packings and establish when the resulting candidates improve downstream optimization.

来源字段：https://doi.org/10.4171/owr/2025/43

16827 | AIM-COMPUTATION-0046

### Generator-invariant local efficacy for phylogenetic invariants

P / partially_solved | AIM 研讨会问题汇编

**Problem 32. "Finding good invariants": Do some of the invariants have more of a statis-tical/biological significance than others?**

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16828 | OWR-14299088-007

### Generic Point Identification from Polynomial Measurements

O / open | Oberwolfach 报告问题汇编

Extend the known generic-identification result from dimension two to higher dimensions. More generally, characterize polynomial measurement maps for which generic points can be identified from nongeneric measurements.

来源字段：https://doi.org/10.4171/owr/2025/9

16829 | OWR-7155442-005

### Geometric Analogues of Finite Effective Complexity

O / open | Oberwolfach 报告问题汇编

Can one formulate and prove the geometric analogue of the finite effective-complexity statement for points and lines on the projective plane, together with the corresponding sphere-area inequality?

来源字段：https://doi.org/10.4171/owr/2021/21

16830 | OWR-14297738-001

### Geometric Criteria for Benign Overfitting

P / partially_solved | Oberwolfach 报告问题汇编

Identify necessary and sufficient geometric conditions under which, for an arbitrary norm, the associated minimum-norm interpolating estimator exhibits benign overfitting.

来源字段：https://doi.org/10.4171/owr/2023/41

16831 | OWR-16637-004

### Geometric Explanations and Polynomial Pivot Rules for Simplex

O / open | Oberwolfach 报告问题汇编

For the simplex method for linear programming, can one explain its good practical performance by a geometric property of real-world linear programs, and does there exist a polynomial-time pivot rule?

来源字段：https://doi.org/10.4171/owr/2018/54

16832 | AIM-COMPUTATION-0099

### Geometry, residues, and scale thresholds for prescribed-prefix prime products

P / partially_solved | AIM 研讨会问题汇编

Given a (random) number \(n\) of \(10000\) digits, it may be impossible to find \(5000\)-digit primes \(p,q\) such that \(n=pq\). It is much easier to find \(10000\)-digit primes \(p,q\) such that \(n\) is the first \(10000\) digits of \(pq\). Find \(7500\)-digit primes with this property (Coppersmith), and then do this for \(6000\)-digit primes (Bernstein).

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16833 | OWR-14298366-005

### Global Convergence of Low-Rank Multi-Penalty Optimization

O / open | Oberwolfach 报告问题汇编

Under which restricted-isometry or related assumptions on the measurement operator does the iteratively reweighted least-squares algorithm for simultaneously sparse and low-rank matrix recovery converge globally?

来源字段：https://doi.org/10.4171/owr/2024/21

16834 | OWR-11695855-007

### Global Margin Maximization beyond the Neural Tangent Regime

O / open | Oberwolfach 报告问题汇编

Under what conditions does gradient descent globally optimize a maximum-margin objective for deep homogeneous networks, and when does it move beyond the neural-tangent regime to learn stronger representations?

来源字段：https://doi.org/10.4171/owr/2022/46

16835 | OWR-3394-020

### Global Observability from Negative Gramian Curvature

P / partially_solved | Oberwolfach 报告问题汇编

别名：Geometric Criteria for Global Observability [OWR-3394-021]

For the observed system $\dot{x}=f(x)$, $y=h(x)$ on an $n$-dimensional manifold, let $P(x_0)=\int_0^T\Phi(t)^TH(t)^TH(t)\Phi(t)\,dt$ be the local observability Gramian along the trajectory from $x_0$. If $P(x_0)$ is uniformly positive definite and bounded, does uniform negative curvature of the resulting Riemannian metric, together with geodesic completeness, imply global observability on $[0,T]$?

来源字段：https://doi.org/10.4171/owr/2009/11

16836 | OWR-4990379-002

### Global Optimality of Domain-Decomposition Transport Algorithms

O / open | Oberwolfach 报告问题汇编

For the limiting iterates of a domain-decomposition algorithm for optimal transport, what sufficient conditions on the cost function and transport data guarantee convergence to a globally optimal transport plan?

来源字段：https://doi.org/10.4171/owr/2021/10

16837 | OWR-15999-003

### Global Optimization of Overparameterized Shallow Networks

P / partially_solved | Oberwolfach 报告问题汇编

For over-parameterized shallow neural networks, explain and prove when local-search methods such as gradient descent can reach global optimality despite high-dimensional nonconvex training, and characterize the resulting optimization landscape.

来源字段：https://doi.org/10.4171/owr/2018/14

16838 | OWR-8415359-011

### Good Locally Testable Quantum Codes and Linear-Rate PCPs

P / partially_solved | Oberwolfach 报告问题汇编

Construct asymptotically good, locally testable quantum LDPC codes, and construct probabilistically checkable proofs of linear rate.

来源字段：https://doi.org/10.4171/owr/2021/54

16839 | OWR-11451-007

### Greedy Decomposition of the Spanning-Tree Polytope

O / open | Oberwolfach 报告问题汇编

Does the spanning-tree polytope $P$ admit greedy decomposition, meaning that for every positive integer $k$ and integral $w\in kP$, there is an integral $x\in P$ for which $\max\{\lambda:w-\lambda x\in(k-\lambda)P\}$ is an integer?

来源字段：https://doi.org/10.4171/owr/2011/53

16840 | OWR-14299517-003

### Griesmer Bounds and Solomon-Stiffler Constructions in Other Metrics

O / open | Oberwolfach 报告问题汇编

Find analogues of the Griesmer bound, together with Solomon-Stiffler-type constructions attaining them, for coding spaces equipped with metrics other than Hamming distance.

来源字段：https://doi.org/10.4171/owr/2025/41

16841 | AIM-CRYPTOGRAPHY-0015

### Grover optimality and a tail-profile bound for average-value search

P / partially_solved | AIM 研讨会问题汇编

Given a function \(f:\{0,1\}^n \to \{0,1,2\}\), find \(x\in \{0,1\}^n\) to maximize \(f(x)\). Measure average resulting \(f(x)\) if, e.g., \(f\) has one 1, one 2, and all other values 0. Or consider the function \(f:\{0,1\}^n \to \{0,1,2, \dots, 1000\}\). Or consider \(f\) which is i.i.d.

Is it possible to do better than Grover search for \(x\) such that \(f(x) \geq T\) for threshold \(T\)?

来源字段：http://aimpl.org/quantumalg/2/

16842 | OWR-1739-002

### Half-Distance Decoding for Hypergraph Codes

O / open | Oberwolfach 报告问题汇编

For the $t$-partite, $t$-uniform hypergraph codes defined from a local binary code and a regular expanding hypergraph, can one construct an efficient low-complexity decoder that corrects errors up to half the designed minimum distance for each fixed $t$?

来源字段：https://doi.org/10.4171/owr/2007/56

16843 | OWR-14298804-016

### Half-Integral Extreme Points for a Feedback-Vertex-Set Relaxation

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Fiorini Feedback-Vertex-Set Question [OWR-14298804-017]

Does every basic feasible solution of the Chudak-Goemans-Hochbaum-Williamson linear-programming relaxation for a nontrivial feedback vertex set instance have a variable of value at least $1/2$?

来源字段：https://doi.org/10.4171/owr/2024/50

16844 | OWR-8415356-016

### Half-Integral Extreme Points for Feedback Vertex Set

P / partially_solved | Oberwolfach 报告问题汇编

For every extreme point $x^*$ of the standard feedback-vertex-set relaxation, must some vertex $v$ satisfy $x^*(v)\geq\tfrac12$?

来源字段：https://doi.org/10.4171/owr/2021/53

16845 | AMR-054-0029

### Hamiltonian Tetrahedralizations

O / open | AMR 问题清单汇编

Can every convex polytope in $\mathbb{R}^3$ be partitioned into tetrahedra such that the dual graph has a Hamiltonian path?

背景链接：https://topp.openproblem.net/p29

16846 | OWR-13105-002

### Hardness Amplification by Exponent Noise

P / partially_solved | Oberwolfach 报告问题汇编

Does adding noise to the exponent in a group-theoretic learning problem amplify computational intractability, as proposed by Learning with Errors in the Exponent?

来源字段：https://doi.org/10.4171/owr/2014/35

16847 | OWR-831-009

### Hardness of Decoding Random Linear Codes

P / partially_solved | Oberwolfach 报告问题汇编

Determine the computational hardness of decoding from random linear codes.

来源字段：https://doi.org/10.4171/owr/2005/26

16848 | OWR-16633-014

### Hardness of Path-Cost Arborescence Optimization

S / solved | Oberwolfach 报告问题汇编

别名：Duplicate Path-Cost Arborescence Problem [OWR-16633-015]

Given a directed graph $D=(V,A)$, a root $r\in V$, and a cost vector $c^v$ for every $v\neq r$, prove that it is NP-hard to find an arborescence $T$ minimizing

$$\sum_{v\in V\setminus\{r\}}c^v(P^v),$$

where $P^v$ is the path from $r$ to $v$ in $T$.

来源字段：https://doi.org/10.4171/owr/2018/50

16849 | OWR-16633-013

### Hardness of Root-Dependent Spanning-Tree Optimization

S / solved | Oberwolfach 报告问题汇编

Given an undirected graph $G=(V,E)$ and, for each root $r\in V$, an arc-cost vector $c_r$, prove that it is NP-hard to find a spanning tree $T$ minimizing

$$\sum_{r\in V}c_r(T^r),$$

where $T^r$ is the arborescence rooted at $r$ induced by $T$.

来源字段：https://doi.org/10.4171/owr/2018/50

16850 | OWR-730-009

### Hardness of the $\varepsilon-Dense$ Steiner Tree Problem

P / partially_solved | Oberwolfach 报告问题汇编

Prove hardness results for the $\varepsilon$-dense Steiner Tree problem, including whether exact optimization is NP-hard.

来源字段：https://doi.org/10.4171/owr/2004/28

16851 | AMR-093-0101

### Hartmanis-Stearns conjecture

O / open | AMR 问题清单汇编

If the base-$b$ expansion of a real number can be emitted in real time by a multitape Turing machine (bounded time between successive digits), must the number be either rational or transcendental, and therefore never algebraic irrational?

背景链接：https://en.wikipedia.org/w/index.php?title=List_of_unsolved_problems_in_mathematics&oldid=1366636767#Number_theory

16852 | AIM-CRYPTOGRAPHY-0027

### HHL, Macaulay conditioning, and presentation instability

P / partially_solved | AIM 研讨会问题汇编

Does HHL break crypto? (See recent paper.)

Understand condition number over \(\mathbb C\) of matrix of coefficients of \(f_1, f_2, \dots\) (original equations), \(xf_1, xyf_2, \dots\) (only monomial terms).

来源字段：http://aimpl.org/quantumalg/6/

16853 | AIM-CRYPTOGRAPHY-0013

### Hidden Goppa presentations versus public-code decoding

P / partially_solved | AIM 研讨会问题汇编

Given a parity-check matrix \(H\), find the hidden Goppa code in \(H\). How quickly can we decode \(H\), assuming Goppa decoder for \(H\) exists?

来源字段：http://aimpl.org/quantumalg/1/

16854 | OWR-14299916-009

### Hidden-Layer Depth Complexity of Piecewise-Linear Functions

O / open | Oberwolfach 报告问题汇编

What is the exact minimum hidden-layer depth required for a ReLU network to represent the function $(x_1,\ldots,x_d)\mapsto\max\{x_1,\ldots,x_d\}$? More generally, does there exist a continuous piecewise-linear function that cannot be represented by any ReLU network with at most two hidden layers?

来源字段：https://doi.org/10.4171/owr/2026/11

16855 | OWR-7155444-003

### High-Dimensional Regression and PDE Learning Without Dimensional Curse

P / partially_solved | Oberwolfach 报告问题汇编

What error bounds can be obtained for regression of high-dimensional nonsmooth functions, how can PDEs be solved without the curse of dimensionality, and what training and validation data sizes and distributions are efficient and effective for deep learning?

来源字段：https://doi.org/10.4171/owr/2021/23

16856 | OWR-1111-008

### Higher-Row Cuts from Cylindrical Simplices

O / open | Oberwolfach 报告问题汇编

For mixed-integer cutting inequalities derived from cylindrical simplices, determine whether the two-row inequality is generally not finitely generated by mixed-integer Gomory cuts or split cuts; whether useful, easily computed $k$-row inequalities arise from $k$-simplices for $k>2$; and whether these cuts are computationally effective alone or with Gomory mixed-integer cuts.

来源字段：https://doi.org/10.4171/owr/2005/50

16857 | AIM-COMPUTATION-0030

### Holonomic Bayesian normalizers with certified posterior moments

P / partially_solved | AIM 研讨会问题汇编

**Question 16.** Computing normalization constants and posterior distributions for continuous random variables involves the computation of integrals involving rational and transcendental funtions. Can algebraic techniques be used to give exact expressions for these integrals or help give approximations for them?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16858 | OWR-14299916-007

### Human Alignment of Artificial Intelligence Systems

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Human-Alignment Question [OWR-14299916-008]

How can artificial intelligence systems be aligned more closely with human behavior and uncertainty, particularly by representing the full fine-to-coarse hierarchy of human conceptual knowledge?

来源字段：https://doi.org/10.4171/owr/2026/11

16859 | OWR-14299090-001

### Improvements to the Signed Heat Method

O / open | Oberwolfach 报告问题汇编

Improve the Signed Heat Method for computing signed distance, with particular attention to output-sensitive formulations, accuracy guarantees, robustness, and computational complexity across meshes, point clouds, voxels, and nonmanifold geometry.

来源字段：https://doi.org/10.4171/owr/2025/5

16860 | OWR-3473-001

### Independence and Density Thresholds for Cuckoo Hashing

P / partially_solved | Oberwolfach 报告问题汇编

How much universal independence suffices to guarantee that cuckoo hashing works? For the one-table generalization with $d$ hash functions, what is the exact threshold load $n/m$ even when the hash functions are fully random?

来源字段：https://doi.org/10.4171/owr/2009/23

16861 | AIM-COMPUTATION-0017

### Indispensable cubics and a five-move Markov basis on the zero-middle-row face

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.** Find a Markov basis for studying the set $X$ of two-way contingency tables with fixed row sums, column sums, and a fixed quadratic statistic: $$ X = \left\{T : \sum_j T_{ij} = R_i,\quad \sum_i T_{ij} = C_j,\quad \sum_{i,j} (i-2)(j-2) T_{ij} = Q\right\}. $$ For $3 \times 4$ tables (with cells ordered row-major as $(11, 12, 13, 14, 21, 22, 23, 24, 31, 32, 33, 34)$), the design matrix computing the sufficient statistics of the log-linear model is $$ A_0 = \begin{pmatrix} 1&1&1&1&0&0&0&0&0&0&0&0\\ 0&0&0&0&1&1&1&1&0&0&0&0\\ 0&0&0&0&0&0&0&0&1&1&1&1\\ 1&0&0&0&1&0&0&0&1&0&0&0\\ 0&1&0&0&0&1&0&0&0&1&0&0\\ 0&0&1&0&0&0&1&0&0&0&1&0\\ 0&0&0&1&0&0&0&1&0&0&0&1\\ 1&0&-1&-2&0&0&0&0&-1&0&1&2 \end{pmatrix}. $$

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16862 | OWR-1782-004

### Individual Manipulation Power in Neutral Social Choice

O / open | Oberwolfach 报告问题汇编

Let $f$ be a neutral social choice function for $n$ voters and $m$ alternatives, and let $M_i(f)$ be voter $i$'s manipulation power. For $m=3$, if $f$ is $\varepsilon$-far from dictatorship then $\sum_iM_i(f)\geq C\varepsilon^2$. Can the current $\Omega(1/n)$ lower bound on $\max_iM_i(f)$ be improved; can neutrality be replaced by $f$ being far from having range of size at most 2; and does an analogous theorem hold for $m>3$, perhaps with a bound polynomially weaker in $m$?

来源字段：https://doi.org/10.4171/owr/2008/01

16863 | OWR-14299916-005

### Inference and Training Effects of the Softmax Bottleneck

P / partially_solved | Oberwolfach 报告问题汇编

When nucleus sampling is used at inference time, do the probabilities of tokens outside the top one thousand have negligible effects on performance? Does the softmax bottleneck materially limit performance during training even when it does not significantly restrict the probabilities of the most likely tokens at inference time?

来源字段：https://doi.org/10.4171/owr/2026/11

16864 | AIM-COMPUTATION-0096

### Information and conditioning audit for zeta-zero primality and factoring proposals

P / partially_solved | AIM 研讨会问题汇编

Look at \[ \sum n^\rho \] over the zeros \(\rho\) of the Riemann zeta function with the imaginary part of \(\rho\) in the interval \([T,2T]\). This is \[ \approx \frac{T}{2\pi}\Lambda(n). \] Can you make a primality test out of this? (Conrey)

The sum \[ \sum n^\rho\frac{\zeta'(\rho)}{\zeta''(\rho)} \approx \frac{T}{2\pi}\log p\log q \qquad\text{if }n=pq. \] Can you make a factoring algorithm out of this?

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16865 | OWR-14299084-004

### Information-Theoretic Additive Randomized Encodings

P / partially_solved | Oberwolfach 报告问题汇编

Characterize the functions $f(x_1,\ldots,x_n)$ that admit information-theoretically secure additive randomized encodings: local randomized maps $g_i(x_i)$ into a finite abelian group whose sum reveals $f$ and essentially nothing else about the inputs.

来源字段：https://doi.org/10.4171/owr/2025/3

16866 | OWR-15962-007

### Information-Computation Gaps in Sparse Linear Regression

P / partially_solved | Oberwolfach 报告问题汇编

In sparse high-dimensional linear regression, when the sample size $n$ lies between the information-theoretic threshold $n_*$ and the algorithmic threshold $n_{\mathrm{alg}}$, is the signal $\beta_*$ information-theoretically recoverable, and is efficient recovery possible?

来源字段：https://doi.org/10.4171/owr/2017/57

16867 | OWR-11451-006

### Integer Carathéodory Property for Arborescence Polytopes

O / open | Oberwolfach 报告问题汇编

Does the $r$-arborescence polytope-the common-base polytope of the graphic matroid and the appropriate partition matroid-have the Integer Carathéodory Property?

来源字段：https://doi.org/10.4171/owr/2011/53

16868 | OWR-11451-004

### Integer Carathéodory Property for Intersecting Matroid Base Polytopes

O / open | Oberwolfach 报告问题汇编

Does the intersection of two base polytopes of strongly base orderable matroids have the Integer Carathéodory Property?

来源字段：https://doi.org/10.4171/owr/2011/53

16869 | OWR-14298804-004

### Integer Feasibility with Bounded Subdeterminants

O / open | Oberwolfach 报告问题汇编

Let $P=\{x\in\mathbb R^n:Ax\leq b\}$, where $A$ is integral and the absolute values of its full-rank subdeterminants are bounded by $\Delta$. For arbitrary $\Delta\geq3$, can one decide in polynomial time whether $P$ contains an integer vector?

来源字段：https://doi.org/10.4171/owr/2024/50

16870 | OWR-8415356-004

### Integer Programming with Bounded Subdeterminants

P / partially_solved | Oberwolfach 报告问题汇编

For a fixed integer $\Delta\geq3$, can integer programs whose constraint matrices have every square subdeterminant bounded in absolute value by $\Delta$ be solved in polynomial time?

来源字段：https://doi.org/10.4171/owr/2021/53

16871 | OWR-1111-004

### Integral Primal Certificates for Submodular Minimization

O / open | Oberwolfach 报告问题汇编

For submodular function minimization, find an algorithm that directly produces an integral primal optimal point in the base polyhedron together with its representation as a convex combination of greedy vertices.

来源字段：https://doi.org/10.4171/owr/2005/50

16872 | OWR-16633-010

### Integrality Gaps for Directed Linear k-Cut

O / open | Oberwolfach 报告问题汇编

For directed Linear-$k$-Cut, is the integrality gap of the natural linear-programming relaxation $O(1)$ for every $k$, or is it $\omega(1)$?

来源字段：https://doi.org/10.4171/owr/2018/50

16873 | OWR-1111-006

### Integrality-Gap Reduction by Lift-and-Project Closures

O / open | Oberwolfach 报告问题汇编

Determine how much the lift-and-project closure improves a linear-programming relaxation of a $0$-$1$ mixed-integer program in practice, as measured for example by reduction of the integrality gap.

来源字段：https://doi.org/10.4171/owr/2005/50

16874 | OWR-12975-005

### Interior Maximizers of Mixed Characteristic Polynomials

O / open | Oberwolfach 报告问题汇编

In the mixed-characteristic-polynomial optimization problem, maximize the largest zero subject to $u_i$ lying in the hyperbolicity cone, $\operatorname{tr}(u_i)\leq\varepsilon$, and $\sum_i u_i=e$. Is the maximum attained when every $u_i$ is either in the interior of the cone or zero?

来源字段：https://doi.org/10.4171/owr/2014/17

16875 | OWR-13750338-004

### Interoperable Virtual Laboratories for Scientific Simulation

O / open | Oberwolfach 报告问题汇编

Develop interoperable virtual laboratories that combine simulation methods across scientific fields and allow tools to be reused across distinct simulation environments.

来源字段：https://doi.org/10.4171/owr/2023/26

16876 | OWR-792-008

### Interpolation-Point Selection in Rational Krylov Reduction

O / open | Oberwolfach 报告问题汇编

Given a large-scale linear system and a reduced order, how should interpolation points be chosen in projection-based rational Krylov model reduction so that the reduced system has desired properties?

来源字段：https://doi.org/10.4171/owr/2005/11

16877 | OWR-792-007

### Intrinsic Nonlinear Model Reduction

P / partially_solved | Oberwolfach 报告问题汇编

Is there an intrinsic scheme for nonlinear model reduction? Is there a scheme such that both the controllability and observability functions of the reduced-order model are the restrictions of the corresponding functions of the full-order model?

来源字段：https://doi.org/10.4171/owr/2005/11

16878 | OWR-14299088-012

### Inverse-Polynomial Influence Thresholds on the Boolean Cube

O / open | Oberwolfach 报告问题汇编

Let $f$ and $g$ be degree-$d$ multilinear real polynomials on $\{-1,1\}^n$ such that every variable has relative influence at most $\delta$ in both polynomials. Can $\delta$ be chosen inverse-polynomially in $d$ while still guaranteeing a point $x\in\{-1,1\}^n$ for which $f(x)\neq0$ and $g(x)\neq0$?

来源字段：https://doi.org/10.4171/owr/2025/9

16879 | AIM-COMPUTATION-0094

### Irreducible line sections: exact convention-sensitive counts and monodromy status

P / partially_solved | AIM 研讨会问题汇编

**Recovered statement.** Let \(f(X,Y)\in\mathbb F_q[X,Y]\) be irreducible. Let \[ g(Y)=f(aY+b,Y). \] Count (or estimate) the number of pairs \(a,b\) over \(\mathbb F_q\) such that \(g\) is irreducible over \(\mathbb F_q\). (Gao)

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16880 | OWR-14299517-007

### Iterative Quantum LDPC Constructions from Weaker Seeds

O / open | Oberwolfach 报告问题汇编

Is there an iterative construction of asymptotically good quantum low-density parity-check codes that starts from a weaker seed object?

来源字段：https://doi.org/10.4171/owr/2025/41

16881 | OWR-14299095-002

### Joint Design of Bridging Schedules and Transport Maps

O / open | Oberwolfach 报告问题汇编

Combine an optimized schedule of bridging densities, minimizing successive $\alpha$-divergences, with a class of bridging densities and a parametrized transport map $T$ for which the contraction factor $\omega<1$ is guaranteed at every step.

来源字段：https://doi.org/10.4171/owr/2025/10

16882 | OWR-16634-003

### Karchmer-Raz-Wigderson Composition Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

For nonconstant Boolean functions $f$ and $g$, is the circuit-depth complexity of their composition equal, up to lower-order terms, to the sum of their individual depth complexities?

来源字段：https://doi.org/10.4171/owr/2018/51

16883 | OWR-13949-002

### Karchmer-Raz-Wigderson Formula Lower-Bound Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Can the Karchmer-Raz-Wigderson conjecture be proved in full generality, thereby implying superpolynomial formula-size lower bounds?

来源字段：https://doi.org/10.4171/owr/2015/54

16884 | OWR-2041-002

### Learning Capacity of Totally Geodesic Map Families

O / open | Oberwolfach 报告问题汇编

For which pairs of manifolds $M$ and $N$ does the set of totally geodesic mappings from $M$ to $N$ have finite capacity in the learning-theoretic sense?

来源字段：https://doi.org/10.4171/owr/2008/30

16885 | OWR-14298366-007

### Learning Data Manifolds for Augmentation and Nonlinear Components

O / open | Oberwolfach 报告问题汇编

Can the underlying data manifold be learned using local entropy and correlation, can this improve data augmentation, and can nonlinear principal components exploit the manifold hypothesis?

来源字段：https://doi.org/10.4171/owr/2024/21

16886 | AIM-CRYPTOGRAPHY-0001

### Length-faithful algebraic and group-theoretic parameter screens for SL_n(F_p) hashing

P / partially_solved | AIM 研讨会问题汇编

**Parameter determination for post-quantum hash functions**

Ramón Flores presented a Tillich-Zémor-style hash function using the Cayley graph of special linear groups in arbitrary dimension. Questions remain, however, about the optimal choice of parameters for use with this protocol.

In the design of the hash function one has control over two parameters; the dimension of the special linear group considered, and the characteristic of the underlying field. We would like to determine appropriate choices for both via two branches of cryptanalysis: first, by analysis of the multivariate system implied by long matrix products; and second by direct analysis of group-theoretic properties. We should also study the impact of the partial homomorphicity exhibited by the scheme.

来源字段：http://aimpl.org/postquantgroup/1/

16887 | OWR-13949-004

### Limits of False Discovery Control in Interactive Analysis

P / partially_solved | Oberwolfach 报告问题汇编

Can the computational and query-complexity limitations on procedures that prevent false discovery in interactive data analysis be removed?

来源字段：https://doi.org/10.4171/owr/2015/54

16888 | OWR-8415356-014

### Linear Bounds for Minimally Integer-Infeasible Systems

O / open | Oberwolfach 报告问题汇编

别名：Linear Bounds for Minimally Integer-Infeasible Systems [OWR-8415356-015]

Let $P=\{x\in\mathbb R^n:Ax\leq b\}$ be full-dimensional and minimally integer-infeasible, and let $\Delta$ be the maximum absolute subdeterminant of $A$. Do functions $f$ and $g$ exist such that the number $m$ of inequalities satisfies $m\leq nf(\Delta)+g(\Delta)$?

来源字段：https://doi.org/10.4171/owr/2021/53

16889 | AIM-COMPUTATION-0019

### Linear collapse, binomial hypersurfaces, and the image of local-odds fibers

P / partially_solved | AIM 研讨会问题汇编

**Question 5.** What is the algebro-geometric structure of passing from local to global odds ratios? Note that a local odds ratio is obtained by looking at a \(2^k\) subtable of a given table, whereas a global odds ratio is obtained by looking at the odds ratio in a \(2^k\) collapsing of a given table.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16890 | OWR-2489-008

### Linear Optimization over Mixed-Integer Bipartite Systems

O / open | Oberwolfach 报告问题汇编

For a bipartite graph $G=(U\cup V,E)$, an integral-coordinate set $I$, and $S(G,I)=\{x\in\mathbb{R}^{U\cup V}:x_i+x_j\geq b_{ij}\text{ for }ij\in E,\ x_i\in\mathbb{Z}\text{ for }i\in I\}$, determine the complexity of optimizing a linear function over $S(G,I)$, equivalently deciding whether a given $\bar{x}$ belongs to $\operatorname{conv}(S(G,I))$.

来源字段：https://doi.org/10.4171/owr/2008/51

16891 | OPG-474

### Linear-size circuits for stable $0,1 < 2$ sorting?

O / open | Open Problem Garden

Problem Can $O(n)$-size circuits compute the function $f$ on $\{0,1,2\}^*$ defined inductively by $f(\lambda) = \lambda$, $f(0x) = 0f(x)$, $f(1x) = 1f(x)$, and $f(2x) = f(x)2$?

背景链接：http://www.openproblemgarden.org/op/linear_size_circuits_for_stable_0_1_2_sorting

16892 | OWR-16170-002

### Local Algorithms for Semidefinite Programming

O / open | Oberwolfach 报告问题汇编

How accurately can local algorithms solve or approximate semidefinite programs, particularly SDP relaxations for random graph problems?

来源字段：https://doi.org/10.4171/owr/2018/29

16893 | OWR-16170-003

### Local Approximation of Random-Graph Semidefinite Programs

O / open | Oberwolfach 报告问题汇编

For a large class of random graph models, can semidefinite programs be approximately solved by local algorithms, with comparable performance for problems such as minimum bisection and community detection?

来源字段：https://doi.org/10.4171/owr/2018/29

16894 | OWR-9790359-001

### LOCAL Coloring Complexity and Constant-Round Decidability on Trees

P / partially_solved | Oberwolfach 报告问题汇编

Determine the LOCAL complexity of $(\Delta+1)$-coloring. Also decide whether an arbitrary locally checkable labeling problem on trees admits a constant-round algorithm.

来源字段：https://doi.org/10.4171/owr/2022/8

16895 | OWR-9790365-001

### Local-to-Global Tensorization of Information Functionals

O / open | Oberwolfach 报告问题汇编

For which families of information functionals does local tensorization imply global tensorization?

来源字段：https://doi.org/10.4171/owr/2022/14

16896 | AIM-COMPUTATION-0064

### Localization and Fourier coordinates beyond polynomial models

P / partially_solved | AIM 研讨会问题汇编

**Question 49 (Wynn).** Why polynomials? What can be said about algebraic methods when applied to other rings, like rings of rational functions, or rings where Fourier analysis can be used?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16897 | OWR-5160-003

### Low-Rank Subexponential Electronic-Structure Algorithms

O / open | Oberwolfach 报告问题汇编

Can low-rank approximations of wavefunction parameters yield a numerical method for solving the electronic Schrödinger equation with subexponential complexity?

来源字段：https://doi.org/10.4171/owr/2011/32

16898 | OWR-17138-006

### Low-Temperature Sampling by Randomly Initialized Glauber Dynamics

P / partially_solved | Oberwolfach 报告问题汇编

Can Glauber dynamics started from an efficiently samplable random initial state yield polynomial-time counting and sampling algorithms for the hard-core or Potts model at low temperature?

来源字段：https://doi.org/10.4171/owr/2019/42

16899 | OWR-14298166-008

### Lower Bounds for Polynomial Calculus with Inverses

O / open | Oberwolfach 报告问题汇编

Can polynomial calculus with inversion solve Tensor Isomorphism in polynomial time? More generally, prove lower bounds for polynomial calculus with inversion on any system of equations.

来源字段：https://doi.org/10.4171/owr/2024/15

16900 | AIM-COMPUTATION-0063

### Markov-basis complexity depends on representation

P / partially_solved | AIM 研讨会问题汇编

**Question 48 (Meek).** Is there a polynomial time algorithm for describing the Markov basis of an arbitrary hierarchical model? Can you randomly choose elements from the Markov basis in polynomial time?

**Sullivant:** Is the number of symmetry classes a polynomial in the size of the problem?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16901 | OWR-13750338-001

### Mathematical Conjecture Generation with Machine Intelligence

O / open | Oberwolfach 报告问题汇编

Develop machine-guided methods that discover useful bounds between hard matrix quantities and efficiently computable surrogates; characterize invariants of free-group actions on spaces of conjectures; and integrate conjecture-generation pipelines with formal proof assistants and natural-language representations.

来源字段：https://doi.org/10.4171/owr/2023/26

16902 | DARPA-012

### Mathematics of Quantum Computing

P / partially_solved | DARPA 挑战

Develop the mathematics required to control the quantum world for computation.

背景链接：https://web.math.utk.edu/~vasili/refs/darpa07.MathChallenges.html

16903 | OWR-14299084-005

### Minimal Assumptions for Additive Randomized Encodings

P / partially_solved | Oberwolfach 报告问题汇编

Determine the weakest computational assumptions under which efficient, fully secure additive randomized encodings can be constructed.

来源字段：https://doi.org/10.4171/owr/2025/3

16904 | OWR-14299089-005

### Minimal Convex Difference Decompositions of CPWL Functions

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Minimal Convex-Difference Decomposition Problem [OWR-14299089-006]

Given a continuous piecewise-linear function $f$, decompose it as $f=g-h$ with $g$ and $h$ convex continuous piecewise-linear functions having as few linear pieces as possible. Construct a finite procedure that finds a globally minimal decomposition in arbitrary dimension.

来源字段：https://doi.org/10.4171/owr/2025/12

16905 | OWR-1703876-011

### Minimum Crossing Realizations of Spatial Polygon Angle Sequences

O / open | Oberwolfach 报告问题汇编

Given a realizable angle sequence for a polygon in $\mathbb{R}^3$, can an efficient algorithm compute the minimum number of crossings among its generic realizations?

来源字段：https://doi.org/10.4171/owr/2020/30

16906 | OWR-14298371-003

### Mixing of Random Colorings on General Graphs

O / open | Oberwolfach 报告问题汇编

Establish spatial and temporal mixing properties for random proper colorings on general graphs.

来源字段：https://doi.org/10.4171/owr/2024/27

16907 | OWR-14298371-002

### Mixing Times from Spectral Independence

P / partially_solved | Oberwolfach 报告问题汇编

How rapidly do Markov chains mix under spectral independence, and when can the general polynomial bound be improved to the optimal $O(n\log n)$?

来源字段：https://doi.org/10.4171/owr/2024/27

16908 | AIM-COMPUTATION-0055

### ML degree, feasibility, and a Gaussian likelihood-root trichotomy

P / partially_solved | AIM 研讨会问题汇编

**Question 41 (Hosten).** How many complex roots to the likelihood equations can there be? What is the statistical significance of complex roots/ negative real roots or other roots that do not yield a valid statistical model (e.g. covariance matrices that are not positive definite)?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16909 | OWR-14298582-001

### Moment-SOS Convergence for PDE-Constrained Variational Problems

O / open | Oberwolfach 报告问题汇编

Extend the convergence analysis of finite-element moment-sum-of-squares relaxations beyond variational problems with a unique global minimizer and boundary constraints, to variational problems subject to partial differential equation constraints.

来源字段：https://doi.org/10.4171/owr/2024/35

16910 | OWR-12177-010

### Multicolored Sunflower Conjecture over Ternary Vector Spaces

S / solved | Oberwolfach 报告问题汇编

Does there exist $c>0$ such that every $S\subseteq(\mathbb Z_3^n)^3$ with $|S|\geq3^{n(1-c)}$ and $x+y+z=0$ for every $(x,y,z)\in S$ contains three elements $(x_i,y_i,z_i)$ satisfying $x_1+y_2+z_3=0$?

来源字段：https://doi.org/10.4171/owr/2012/54

16911 | AIM-CRYPTOGRAPHY-0016

### Multiple marked inputs: exclusion bounds and tight query regimes

P / partially_solved | AIM 研讨会问题汇编

Find \(x,y\) distinct with \(f(x)=f(y)=1\). Consider the case where there are many solutions at low density. How quickly can this be done? Faster than Grover?

More generally, find distinct \(x_1,\ldots,x_m\) such that \(f(x_1)=\cdots=f(x_m)=1\). Can you do better than \(m\) preimage searches?

来源字段：http://aimpl.org/quantumalg/2/

16912 | OWR-1739-003

### Multiplicity Decoding for Order-Domain Codes

P / partially_solved | Oberwolfach 报告问题汇编

Extend the Geil-Matsumoto Sudan-type list-decoding algorithm for order-domain codes to use multiplicities. In particular, develop an appropriate notion of multiplicity in order-domain theory.

来源字段：https://doi.org/10.4171/owr/2007/56

16913 | OWR-17472-018

### Multiplicity Obstructions in Geometric Complexity Theory

O / open | Oberwolfach 报告问题汇编

For the polynomial orbit closures relevant to geometric complexity theory, can one find a partition $\lambda$ and degree $d$ such that $$ \operatorname{mult}_\lambda\mathbb C[\overline{Gp}]_d>\operatorname{mult}_\lambda\mathbb C[\overline{Gq}]_d, $$ thereby obtaining a multiplicity obstruction separating the orbit closures?

来源字段：https://doi.org/10.4171/owr/2020/9

16914 | OWR-9790365-007

### Mutual-Information Bounds and Dimension-Free Dual Sudakov Minoration

O / open | Oberwolfach 报告问题汇编

Prove the proposed mutual-information inequality for log-concave random vectors and derive a dimension-free dual Sudakov minoration for general log-concave measures.

来源字段：https://doi.org/10.4171/owr/2022/14

16915 | OWR-14298800-003

### Nash Equilibria over Unbounded Measure Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Establish general conditions for existence of Nash equilibria for games whose strategies are probability measures on unbounded sets. Under those conditions, determine whether gradient descent-ascent converges to an equilibrium, beyond the known displacement-convex-concave cases.

来源字段：https://doi.org/10.4171/owr/2024/56

16916 | OWR-14299517-002

### Near-Linear List Decoding of AEL and Ta-Shma Codes

O / open | Oberwolfach 报告问题汇编

Can Alon-Edmonds-Luby codes be list-decoded up to capacity in linear time? Can Ta-Shma codes be list-decoded up to a radius approaching one half, or up to the Johnson bound, in near-linear time?

来源字段：https://doi.org/10.4171/owr/2025/41

16917 | OWR-14298371-001

### Near-Linear Subgraph Isomorphism for Acyclic Patterns

O / open | Oberwolfach 报告问题汇编

Can constant-pattern subgraph isomorphism be solved in near-linear time if and only if the pattern graph is acyclic, and can the surrounding fine-grained conjectures be unified?

来源字段：https://doi.org/10.4171/owr/2024/27

16918 | OWR-8415356-009

### Near-Optimal Algorithms for Mixed-Integer Convex Optimization

O / open | Oberwolfach 报告问题汇编

Can mixed-integer convex optimization be solved with running time matching its information-complexity bound up to lower-order factors?

来源字段：https://doi.org/10.4171/owr/2021/53

16919 | OWR-14298015-001

### Neural Approximation of Strongly Convex Drifts with Linear Error

P / partially_solved | Oberwolfach 报告问题汇编

If $V$ is strongly convex with Lipschitz gradient, does there exist a neural network approximating $-\nabla V$ with error growing only linearly?

来源字段：https://doi.org/10.4171/owr/2023/48

16920 | OWR-11695855-002

### Neural Network Robustness under Adversarial and Distribution Shifts

O / open | Oberwolfach 报告问题汇编

Characterize the robustness of neural-network signal reconstruction under worst-case perturbations and distribution shifts.

来源字段：https://doi.org/10.4171/owr/2022/46

16921 | OWR-14298803-002

### Neural Networks for High-Dimensional Optimal Feedback Control

P / partially_solved | Oberwolfach 报告问题汇编

Determine when neural networks can solve high-dimensional infinite-horizon optimal feedback-control problems with rigorous accuracy, stability, and complexity guarantees that avoid the curse of dimensionality.

来源字段：https://doi.org/10.4171/owr/2024/57

16922 | OWR-14299916-010

### New Theoretical Principles for Foundation Models

O / open | Oberwolfach 报告问题汇编

Do the phenomena observed in contemporary foundation models require fundamentally new mathematical principles, or can they be explained by extensions of existing theories of approximation, optimization, and statistical learning?

来源字段：https://doi.org/10.4171/owr/2026/11

16923 | OWR-792-013

### Nonadaptive Polynomial Stabilization of Minimum-Phase Linear Systems

P / partially_solved | Oberwolfach 报告问题汇编

Consider the single-input, single-output system $\dot{x}(t)=Ax(t)+bu(t)$, $y(t)=cx(t)$, where $A\in\mathbb{R}^{n\times n}$ and $b,c^\top,x^0\in\mathbb{R}^n$ are unknown. Assume that, for some known $r\in\mathbb{N}$, there is a $\rho\in\{1,\ldots,r\}$ such that $cA^ib=0$ for $i=1,\ldots,\rho-2$ and $cA^{\rho-1}b>0$, and that $$\det\begin{bmatrix}sI-A&b\\c&0\end{bmatrix}\ne0\qquad(\operatorname{Re}s\geq0).$$ For an arbitrary performance funnel $$\mathcal{F}_\varphi=\{(t,e)\in\mathbb{R}_{\geq0}\times\mathbb{R}^m:\varphi(t)\lVert e\rVert<1\},$$ where $\varphi\in W^{1,\infty}(\mathbb{R}_{\geq0},\mathbb{R})$, $\varphi(0)=0$, $\varphi(s)>0$ for $s>0$, and $\liminf_{s\to\infty}\varphi(s)>0$, construct a feedback law $$u(t)=-\gamma_r(k(t),y(t),\xi(t)),\qquad k(t)=\bigl[1-(\varphi(t)\lVert y(t)\rVert)^2\bigr]^{-1},\qquad \dot\xi(t)=f(y(t),u(t),\xi(t)),$$ with polynomial $\gamma_r$ and $f$, such that $y$ remains in $\mathcal{F}_\varphi$.

来源字段：https://doi.org/10.4171/owr/2005/11

16924 | AIM-COMPUTATION-0073

### Nondefectivity and low-degree obstructions for the two-component S4 Birkhoff model

P / partially_solved | AIM 研讨会问题汇编

Problem 58 (Sturmfels). Compute a mixture of the S4 model from Diaconis' talk. What are the equations defining this secant variety? Is the likelihood multimodal?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16925 | OWR-11101919-002

### Norm Estimation from Asymmetric Black-Box Linear Operators

P / partially_solved | Oberwolfach 报告问题汇编

Given black-box access to $x\mapsto Ax$ but not to $A^*$, can $\lVert A\rVert$ be estimated by a provably convergent low-memory method? Likewise, can $\lVert A-V\rVert$ be estimated using only black boxes for $A$ and $V^*$?

来源字段：https://doi.org/10.4171/owr/2022/38

16926 | OWR-13948-002

### Numerical Simulation of Dislocation Polygonization

P / partially_solved | Oberwolfach 报告问题汇编

Can numerical methods correctly simulate polygonization in bent single-crystal beams, where dislocations align into polygonization patterns?

来源字段：https://doi.org/10.4171/owr/2015/53

16927 | OWR-792-010

### Observability for Polynomial Systems

P / partially_solved | Oberwolfach 报告问题汇编

What is the right definition of observability for polynomial systems? Moreover, is the more restricted condition of algebraic observability generic?

来源字段：https://doi.org/10.4171/owr/2005/11

16928 | DARPA-007

### Occam's Razor in Many Dimensions

P / partially_solved | DARPA 挑战

Find lower bounds for sensing complexity as data collection grows, addressing entropy maximization.

背景链接：https://web.math.utk.edu/~vasili/refs/darpa07.MathChallenges.html

16929 | AIM-CRYPTOGRAPHY-0029

### Offline Simon, explicit HSP instances, and a classical-transcript obstruction

P / partially_solved | AIM 研讨会问题汇编

Is there a crypto problem that is solved by Simon's algorithm without superposition attackers? Which hidden subgroup problems have crypto applications? Or non-crypto instances?

来源字段：http://aimpl.org/quantumalg/6/

16930 | AIM-COMPUTATION-0053

### One observable mode can lift to infinitely many hidden-parameter maxima

P / partially_solved | AIM 研讨会问题汇编

Question 39 (Sturmfels). In a graphical model with hidden variables how many (nonnegative) real modes might exist in solutions to the likelihood equations?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16931 | OPG-59968

### One-way functions exist

O / open | Open Problem Garden

Conjecture One-way functions exist.

背景链接：http://www.openproblemgarden.org/op/one_way_functions_exist

16932 | OWR-14298363-006

### Online Model Selection for Nonparametric Classes

P / partially_solved | Oberwolfach 报告问题汇编

For which nonparametric model classes is online model selection possible with regret of order $\log M$?

来源字段：https://doi.org/10.4171/owr/2024/24

16933 | OWR-9790359-003

### Open Problems in Local Graph Complexity

O / open | Oberwolfach 报告问题汇编

Classify finitary-factor-of-independent-and-identically-distributed complexity; characterize derandomization on graph classes; determine the limits of Borel and continuous Lovász Local Lemmas; construct finitary-factor-of-independent-and-identically-distributed perfect matchings on trees; decide when Borel unfriendly colorings exist; develop a general complexity theory of local graph problems; and transfer techniques among distributed, finitary, and descriptive settings.

来源字段：https://doi.org/10.4171/owr/2022/8

16934 | OWR-13941-001

### Open Problems in Mixed-Integer Nonlinear Optimization

O / open | Oberwolfach 报告问题汇编

Why is semidefinite programming computationally slow? For $$X(\alpha,\beta,\gamma)=\{(x_1,x_2,t_1,t_2,z_1,z_2)\in\mathbb R^4\times\{0,1\}^2:t_i\geq x_i^2,\ 0\leq\alpha x_1+\beta x_2\leq z_1,\ 0\leq\gamma x_2\leq z_2\},$$ find a scalable compact description of $\operatorname{conv}X(\alpha,\beta,\gamma)$. What is the minimum number of auxiliary variables needed to represent a nonsemialgebraic convex hull semialgebraically? Develop randomized copositivity and complete-positivity tests, and recognize named structures in flat mathematical programs. Finally, if $m\geq12$ points in the shell between the unit sphere and its circumscribed dodecahedron are mutually at least unit distance apart, find concise proofs for $m>16$ of the conjectured Euclidean shell inequality and formulate its $\ell_1$ analogue.

来源字段：https://doi.org/10.4171/owr/2015/46

16935 | OWR-14298166-001

### Open Problems in Proof and Computational Complexity

O / open | Oberwolfach 报告问题汇编

Is there a total NP search class strictly beyond $\mathrm{FP}$ that is easy in the random-oracle model? Can $3$-PIGEON be randomly black-box reduced to $2$-PIGEON? Are Stabbing Planes and Cutting Planes equivalent up to quasipolynomial simulation? Is nonnegativity of binomials of graph densities decidable? Does the two-sorted theory $\mathrm{VPV}$ prove the nondeterministic time-hierarchy theorem?

来源字段：https://doi.org/10.4171/owr/2024/15

16936 | OWR-1587-003

### Optimal Algebraic Immunity of Boolean Functions

O / open | Oberwolfach 报告问题汇编

For every $1\leq m\leq n$, there should exist a Boolean function $f:\{0,1\}^n\to\{0,1\}^m$ whose algebraic immunity equals $d^*(n,m)$, the least $d$ such that $\sum_{i=0}^d\binom{n+m}{i}>2^n$.

来源字段：https://doi.org/10.4171/owr/2007/31

16937 | OWR-14299578-002

### Optimal Bounds for Pairs of Commuting Integer Matrices

P / partially_solved | Oberwolfach 报告问题汇编

For $X\in M_n(\mathbb{Z})$, let $|X|$ be the maximum absolute value of its entries, and define $N(T)=\#\{(X,Y)\in M_n(\mathbb{Z})^2:|X|\leq T,\ |Y|\leq T,\ XY=YX\}$. Prove that, for every $\varepsilon>0$, $N(T)\ll_\varepsilon T^{n^2+1+\varepsilon}$ for arbitrary $n$.

来源字段：https://doi.org/10.4171/owr/2025/52

16938 | OWR-13105-003

### Optimal Fair Multiparty Coin Flipping

O / open | Oberwolfach 报告问题汇编

Does there exist a fair three-party coin-flipping protocol without the $O(\log^2 m)$ factor? More generally, does there exist a fair protocol for any number of parties against any number of corrupted parties?

来源字段：https://doi.org/10.4171/owr/2014/35

16939 | OWR-3394-018

### Optimal Feasible Sets from Quantized Trajectory Observations

O / open | Oberwolfach 报告问题汇编

别名：State Recovery from Quantized Nonlinear Trajectories [OWR-3394-017]

For $x(t+1)=f(x(t))$ and observed quantization regions $P(0),\ldots,P(T)$, define $Q^+(0)=P(0)$, $Q^+(t)=f(Q^+(t-1))\cap P(t)$, $Q^-(T)=P(T)$, $Q^-(t-1)=f^{-1}(Q^-(t))\cap P(t-1)$, and $Q(t)=Q^+(t)\cap Q^-(t)$. Under what conditions on $f$ are some $Q(t)$ small, and is $Q(t)$ the finest information about $x(t)$ obtainable from the quantized observations?

来源字段：https://doi.org/10.4171/owr/2009/11

16940 | OWR-8415360-010

### Optimal Neural-Network Depth for Sparse Recovery

O / open | Oberwolfach 报告问题汇编

For exact recovery of $s$-sparse vectors by ReLU networks, is the known depth bound $\lceil\log_2s+2\rceil$ optimal?

来源字段：https://doi.org/10.4171/owr/2021/55

16941 | OWR-14298377-001

### Optimal Noiseless Rates for Max-Average-Margin Classifiers

O / open | Oberwolfach 报告问题汇编

Why can max-$\ell_p$ margins with $p>1$ improve noiseless recovery, and can the regularized max-average-margin estimator achieve the optimal $O(1/n)$ rate rather than $O(1/\sqrt n)$?

来源字段：https://doi.org/10.4171/owr/2024/30

16942 | OWR-1782-015

### Optimal Rates of Collusion-Secure Fingerprinting Codes

O / open | Oberwolfach 报告问题汇编

For digital fingerprinting codes secure against collusion by $t$ pirates, what is the exact asymptotic rate, and does the known probabilistic construction attain that rate for every $t$?

来源字段：https://doi.org/10.4171/owr/2008/01

16943 | OWR-11695855-003

### Optimal Worst-Case Estimators for Signal Reconstruction

O / open | Oberwolfach 报告问题汇编

Characterize minimax-optimal estimators for broad signal and worst-case noise models. When are robust-optimization formulations equivalent to regularized estimation, and when is in-distribution risk $R_P(f)$ monotonically related to shifted-distribution risk $R_Q(f)$?

来源字段：https://doi.org/10.4171/owr/2022/46

16944 | OWR-730-004

### Optimization with Inexact Real-Valued Inputs

O / open | Oberwolfach 报告问题汇编

How can optimization problems with real-valued input be solved efficiently when the input numbers cannot be represented exactly?

来源字段：https://doi.org/10.4171/owr/2004/28

16945 | OWR-4135-010

### Orbit-Closure Obstructions for Permanent-Determinant Separation

P / partially_solved | Oberwolfach 报告问题汇编

For every positive polynomial $p$ and every $m_0\in\mathbb{N}$, does there exist $m\geq m_0$ such that no equivariant surjection exists from the coordinate ring of the determinant orbit closure to that of the padded permanent orbit closure?

来源字段：https://doi.org/10.4171/owr/2009/52

16946 | OWR-15957-006

### Ordered-Median Facility Location

O / open | Oberwolfach 报告问题汇编

Given $n$ sites, a nonnegative cost matrix $C=(c_{j\ell})$, and $p<n$ facilities, choose $X\subseteq\{1,\ldots,n\}$ with $|X|=p$ to minimize the ordered weighted cost $\sum_{k=1}^n\lambda_k c_{\sigma_X(k)}(X)$, where $c_j(X)=\min_{\ell\in X}c_{j\ell}$ and $\sigma_X$ sorts these costs.

来源字段：https://doi.org/10.4171/owr/2017/52

16947 | OWR-14298371-010

### Output Length and Error of Seedless Extractors

O / open | Oberwolfach 报告问题汇编

Improve the output length and error parameters of seedless extractors.

来源字段：https://doi.org/10.4171/owr/2024/27

16948 | OWR-730-014

### Output-Polynomial Enumeration of Bicriteria Spanning Trees

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Planar Min-Bisection and Bicriteria Extraction [OWR-730-012]; Duplicate Bicriteria Spanning-Tree Enumeration [OWR-730-015]

For bicriteria minimum spanning trees, can all Pareto-optimal solutions be enumerated in time polynomial in the number of solutions?

来源字段：https://doi.org/10.4171/owr/2004/28

16949 | OWR-14604-004

### Overlap Gaps and Algorithmic Hardness of Submatrix Detection

O / open | Oberwolfach 报告问题汇编

In the Gaussian $n\times n$ submatrix problem, suppose near-optimal solutions above a threshold $\alpha^*$ exhibit an overlap gap: their pairwise overlaps avoid a nonempty interval. Does this property imply that no polynomial-time algorithm can find a $k\times k$ submatrix of average at least $(1+o(1))\alpha\sqrt{2\log n/k}$ when $\alpha>\alpha^*$ and $k\to\infty$?

来源字段：https://doi.org/10.4171/owr/2016/22

16950 | MPP-001

### P versus NP Problem

O / open | 千禧年问题

别名：P vs. NP [OPG-661]

Does $P = NP$? More formally: if the solution to a problem can be quickly verified (in polynomial time), can the solution also be quickly found (in polynomial time)?

背景链接：https://www.claymath.org/millennium-problems/

16951 | OPG-51618

### P vs. BPP

O / open | Open Problem Garden

Conjecture Can all problems that can be computed by a probabilistic Turing machine (with error probability < 1/3) in polynomial time be solved by a deterministic Turing machine in polynomial time? That is, does P = BPP?

背景链接：http://www.openproblemgarden.org/op/p_vs_bpp

16952 | OPG-36892

### P vs. PSPACE

O / open | Open Problem Garden

Problem Is there a problem that can be computed by a Turing machine in polynomial space and unbounded time but not in polynomial time? More formally, does P = PSPACE?

背景链接：http://www.openproblemgarden.org/op/p_vs_pspace

16953 | AIM-COMPUTATION-0067

### Parameter tying and exact homogeneous binary-star invariants

P / partially_solved | AIM 研讨会问题汇编

**Problem 52 (Pachter).** Try applying the algebraic techniques to more restrictive families of models. For example, models with homogeneous transition matrices (that is, the same set of parameters is used for more than one edge/ clique in the graph), or genomic models.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16954 | AIM-COMPUTATION-0091

### Perfect-number recognition and simultaneous polynomial divisors

P / partially_solved | AIM 研讨会问题汇编

**Recovered statement.** Find a deterministic polynomial time algorithm for recognizing perfect numbers. (Lenstra)

Given two monic polynomials \(f,g\in\mathbb Z[X]\) with no common irreducible factor, and two integers \(n,m\in\mathbb Z_{>0}\), find in polynomial time all \(x\in\mathbb Z\) such that \[ f(x)\mid m,\qquad g(x)\mid n,\qquad |x|<H(f,g,m,n) \] for some function \(H\).

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16955 | OWR-4135-004

### Permanent-Determinant Separation Through Geometric Complexity Theory

O / open | Oberwolfach 报告问题汇编

Does $\mathrm{VP}\ne\mathrm{VNP}$, equivalently, can the permanent be separated from the determinant through the geometric-complexity-theory orbit-closure program?

来源字段：https://doi.org/10.4171/owr/2009/52

16956 | OWR-8415360-006

### Phase Retrieval from Gabor and Wavelet Magnitudes

P / partially_solved | Oberwolfach 报告问题汇编

What uniqueness and stability guarantees hold for recovering $f\in H$ from magnitude measurements $(|\langle f,\phi_\lambda\rangle|)_{\lambda\in\Lambda}$, particularly for Gabor or wavelet frames?

来源字段：https://doi.org/10.4171/owr/2021/55

16957 | AMR-054-0013

### Point Location in 3D Subdivision

O / open | AMR 问题清单汇编

Is there an $O(n)$-space data structure that supports $O(\log n)$-time point-location queries in a three-dimensional subdivision of $n$ faces?

背景链接：https://topp.openproblem.net/p13

16958 | OWR-4426-006

### Polyhedral Information Derived from Known Symmetries

P / partially_solved | Oberwolfach 报告问题汇编

Given a bounded polyhedron

$$P=\operatorname{conv}\{x_1,\ldots,x_k\}\subset\mathbb R^n$$

and its known symmetry group, what structural or algorithmic information can those symmetries provide about $P$?

来源字段：https://doi.org/10.4171/owr/2010/38

16959 | OWR-4135-007

### Polymorphism Characterization of Constraint Approximation Thresholds

O / open | Oberwolfach 报告问题汇编

Is the approximation threshold of every constraint-satisfaction problem characterized by the existence of approximate non-dictator polymorphisms?

来源字段：https://doi.org/10.4171/owr/2009/52

16960 | OWR-14215-001

### Polynomial Algorithms for Mean-Payoff Games and Linear Programming

O / open | Oberwolfach 报告问题汇编

Does there exist a polynomial-time algorithm for solving mean-payoff games? Does there exist a strongly polynomial algorithm for linear programming?

来源字段：https://doi.org/10.4171/owr/2016/7

16961 | OWR-782-005

### Polynomial Complexity of Greedy Lattice Reduction

P / partially_solved | Oberwolfach 报告问题汇编

Does the greedy lattice-reduction algorithm have polynomial-time complexity in every dimension $d\geq6$?

来源字段：https://doi.org/10.4171/owr/2005/01

16962 | OWR-11451-002

### Polynomial Diameter Bounds for Polyhedra

O / open | Oberwolfach 报告问题汇编

Can the diameter of a polyhedron $P=\{x\in\mathbb{R}^n:Ax\leq b\}$, with $m$ defining inequalities and dimension $n$, be bounded by a polynomial in $m$ and $n$?

来源字段：https://doi.org/10.4171/owr/2011/53

16963 | OWR-12177-008

### Polynomial Freiman-Ruzsa Conjecture over Binary Vector Spaces

S / solved | Oberwolfach 报告问题汇编

If $A\subseteq\mathbb F_2^n$ satisfies $|A+A|\leq K|A|$, is there a subset $A^{\prime}\subseteq A$ with $|A^{\prime}|\geq |A|/\operatorname{poly}(K)$ and $|\operatorname{span}(A^{\prime})|\leq\operatorname{poly}(K)|A^{\prime}|$?

来源字段：https://doi.org/10.4171/owr/2012/54

16964 | OWR-12723-011

### Polynomial Tractability of Permutation-Invariant Integration

O / open | Oberwolfach 报告问题汇编

For integration over the permutation-invariant spaces $S_{I_d}(E_{d,\alpha})$, under what conditions on the coordinate sets $I_d$ is the information complexity $n(\varepsilon,d)$ polynomially bounded in $d$ and $\varepsilon^{-1}$?

来源字段：https://doi.org/10.4171/owr/2013/49

16965 | OWR-12177-007

### Polynomial-Time Algorithms for High-Dimensional Private Queries

O / open | Oberwolfach 报告问题汇编

Can differentially private algorithms for answering many high-dimensional statistical queries achieve polynomial running time?

来源字段：https://doi.org/10.4171/owr/2012/54

16966 | OWR-16633-016

### Polynomial-Time Comparison of Sparse Algebraic Power Sums

O / open | Oberwolfach 报告问题汇编

Given integers $r_1,\ldots,r_n$, is there an algorithm polynomial in

$$\sum_{i=1}^n\log(\lvert r_i\rvert+1)$$

that decides whether $\sum_{i=1}^n2^{r_i/3}>1$? More generally, for fixed $\alpha\in\mathbb Q$ and input coefficients $\beta_i\in\mathbb Q$, decide whether $\sum_i\beta_i\alpha^{r_i}>0$.

来源字段：https://doi.org/10.4171/owr/2018/50

16967 | OWR-741-005

### Polynomial-Time Computation of Ehrhart Periods

O / open | Oberwolfach 报告问题汇编

Can the period of the Ehrhart quasipolynomial $i_P(t)$ of a rational polytope $P$ be determined in polynomial time?

来源字段：https://doi.org/10.4171/owr/2004/39

16968 | OWR-16170-004

### Polynomial-Time Estimation in Binary Latent-Variable Models

O / open | Oberwolfach 报告问题汇编

Under general identifiability conditions for binary latent-variable models, do statistically consistent procedures exist for estimating the parameter matrix $W$ in time polynomial in the number $d$ of hidden variables?

来源字段：https://doi.org/10.4171/owr/2018/29

16969 | OWR-13353-005

### Polynomial-Time Minimization of K-Submodular Functions

O / open | Oberwolfach 报告问题汇编

Can a $k$-submodular function be minimized in polynomial time?

来源字段：https://doi.org/10.4171/owr/2014/51

16970 | OWR-14299084-006

### Power of Shared Randomness in Distributed Computation

P / partially_solved | Oberwolfach 报告问题汇编

Characterize the power of a public random string shared by multiple parties performing a joint task, particularly when the parties' inputs may depend on that string.

来源字段：https://doi.org/10.4171/owr/2025/3

16971 | OWR-1111-005

### Practical Strength of Lift-and-Project Relaxations

O / open | Oberwolfach 报告问题汇编

Determine the practical strength of the Sherali-Adams and Lovász-Schrijver lift-and-project relaxations for mixed-integer programs, including their performance on MIPLIB instances relative to the basic lift-and-project closure.

来源字段：https://doi.org/10.4171/owr/2005/50

16972 | OWR-15580-008

### Practical Sum-of-Squares Tensor Completion

O / open | Oberwolfach 报告问题汇编

Generalize exact sum-of-squares tensor-completion results beyond orthogonal components, and determine whether the sum-of-squares tensor-completion algorithm can be made faster and practically usable.

来源字段：https://doi.org/10.4171/owr/2017/37

16973 | OWR-14298366-008

### Pre-Asymptotic Sampling Numbers of Barron Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Determine the asymptotic nonlinear sampling numbers for variational Barron spaces. For the Fourier-analytic spaces $U_d^\alpha$, determine the pre-asymptotic dependence of $\sigma_m(U_d^\alpha)_{L^p}$ on $d$ and $m$, and determine the optimal reconstruction error when the sample points are uniform random points.

来源字段：https://doi.org/10.4171/owr/2024/21

16974 | OWR-11450-003

### Predicting SAT-Solver Hardness from Formula Structure

P / partially_solved | Oberwolfach 报告问题汇编

What explains the success of SAT solvers, how can one determine whether a particular propositional formula is hard or tractable, and can formula space complexity serve as a useful measure of hardness?

来源字段：https://doi.org/10.4171/owr/2011/52

16975 | OWR-17475-005

### Primal and Dual Geometry of SONC Relaxations

O / open | Oberwolfach 报告问题汇编

For the SONC-cone linear-programming relaxation, what is the relation between the primal polyhedron obtained by fixing the non-auxiliary variables and the SONC cone, and what is the relation between the dual SONC cone and sparse moment cones?

来源字段：https://doi.org/10.4171/owr/2020/12

16976 | AIM-COMPUTATION-0092

### Prime-factor advice in power-residue quotients

P / partially_solved | AIM 研讨会问题汇编

These questions are motivated by the questions posed by AKS concerning finding quadratic nonresidues modulo a prime \(p\).

(a) It is known that finding a single quadratic nonresidue for a given prime is polynomially equivalent to solving all quadratic equations. How far can one do this for finding a single bit of data (or few bits of data) for higher degrees?

(b) We do not know yet that there is a deterministic polynomial time algorithm for finding quadratic nonresidues. Is there a subexponential algorithm?

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

16977 | OWR-8415359-006

### Property Testing for Noncommutative Quantum Functions

O / open | Oberwolfach 报告问题汇编

What kinds of noncommutative properties are there? How does one measure a quantum function’s distance to a noncommutative property? What properties of quantum functions can be tested using few queries?

来源字段：https://doi.org/10.4171/owr/2021/54

16978 | OWR-15999-005

### Provable Advantages of Convolutional Arithmetic Circuits

O / open | Oberwolfach 报告问题汇编

Can the expressive power of convolutional arithmetic circuits be converted through training into a deep-learning architecture with a provable advantage over convolutional rectifier networks?

来源字段：https://doi.org/10.4171/owr/2018/14

16979 | OWR-783-002

### Prox-Regularity of Polynomial Stability Abscissas

O / open | Oberwolfach 报告问题汇编

In the space of monic degree-$k$ polynomials $p(s) = s^k + a_{k-1} s^{k-1} + \cdots + a_0$, let $\alpha(p) = \max\{\operatorname{Re} \lambda : p(\lambda) = 0\}$ be the spectral stability abscissa, and say that $p$ is Hurwitz stable if $\alpha(p) \leq 0$. Does every degree-$k$ polynomial $p(s)$ sufficiently close to $s^k$ have a unique nearest Hurwitz stable polynomial? Equivalently, is the stability abscissa $\alpha$ prox-regular at $s^k$?

来源字段：https://doi.org/10.4171/owr/2005/02

16980 | OWR-16633-003

### Proximity and Sparsity of Optimal Integer Solutions

P / partially_solved | Oberwolfach 报告问题汇编

For integer linear optimization, given an optimal solution of the linear-programming relaxation, how close can an optimal integer solution be in the $\ell_1$ or $\ell_\infty$ norm, and for a system of linear equations with nonnegative variables and a linear objective, how sparse can an optimal integer solution be?

来源字段：https://doi.org/10.4171/owr/2018/50

16981 | OWR-15580-006

### PSD Rank Bounds for Polynomial-Size Selector Matrices

O / open | Oberwolfach 报告问题汇编

Does the positive-semidefinite-rank lower bound for the selector matrix $M_f^{\operatorname{SEL}_m}$ remain valid when $m\leq n^{O(1)}$?

来源字段：https://doi.org/10.4171/owr/2017/37

16982 | OWR-14299906-007

### Quantum Advantage in Threefold Repetition of Feige’s Game

O / open | Oberwolfach 报告问题汇编

Does the quantum value exceed the classical value for the threefold parallel repetition of Feige’s game?

来源字段：https://doi.org/10.4171/owr/2026/4

16983 | OWR-8415352-010

### Quantum Complexity of High-Precision Matrix Scaling

P / partially_solved | Oberwolfach 报告问题汇编

Can a quantum algorithm improve the $\widetilde O(m^{3/2})$ high-precision running time for general matrix scaling while retaining polylogarithmic dependence on $1/\varepsilon$? For a dense nonnegative $n\times n$ matrix whose entries sum to at most $1$, what is the quantum query complexity of approximating its row-sum vector to $\ell_1$ error $\varepsilon$?

来源字段：https://doi.org/10.4171/owr/2021/49

16984 | OWR-831-008

### Quantum Hardness of Approximate Lattice Problems

O / open | Oberwolfach 报告问题汇编

Is it impossible for a quantum polynomial-time algorithm to approximate SVP or SIVP within any polynomial factor?

来源字段：https://doi.org/10.4171/owr/2005/26

16985 | OWR-14298366-006

### Quantum Harmonic Analysis of Data Structure

O / open | Oberwolfach 报告问题汇编

Can quantum harmonic analysis explain structure in data?

来源字段：https://doi.org/10.4171/owr/2024/21

16986 | AIM-COMPUTATION-0042

### Quartic Markov bases and saturated simplicial leaves

P / partially_solved | AIM 研讨会问题汇编

**Problem 28.** Characterize the binary hierarchical models which have Markov bases consisting of moves of degree 4 or less.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16987 | OWR-14298584-009

### Quasilinear Finite-Field Polynomial Multiplication

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Finite-Field Polynomial-Multiplication Conjecture [OWR-14298584-010]

For a prime $p$, let $M_p(d)$ denote the multitape-Turing-machine cost of multiplying polynomials of degree at most $d$ in $\mathbb{F}_p[x]$, and put $n=d\log p$. Prove that $M_p(d)=O(n\log n)$ uniformly in $p$ and $d$, including when $p=2$.

来源字段：https://doi.org/10.4171/owr/2024/40

16988 | OWR-14299089-001

### Quotients and Presentations of L-Convex Objects

O / open | Oberwolfach 报告问题汇编

Describe the correspondence between the two flow-based presentations of $L$-convex sets and functions. Define natural quotient operations for $L$-convex sets and $L$-convex functions, and establish their basic structural properties.

来源字段：https://doi.org/10.4171/owr/2025/12

16989 | OWR-15580-005

### Random CNF Lower Bounds beyond Cutting Planes

O / open | Oberwolfach 报告问题汇编

Can one establish lower bounds in proof systems other than Cutting Planes that are analogous to the exponential Cutting Planes lower bound for random $O(\log n)$-CNF formulas?

来源字段：https://doi.org/10.4171/owr/2017/37

16990 | AIM-INFRASTRUCTURE-0091

### Random-access slices of free resolutions

P / partially_solved | AIM 研讨会问题汇编

Can we compute the n-th step of a resolution quickly, without computing the previous steps?

来源字段：http://aimpl.org/macaulay2efie/3/

16991 | OWR-13949-006

### Randomized Communication Complexity of Distributed Triangle Detection

P / partially_solved | Oberwolfach 报告问题汇编

What is the randomized communication complexity of detecting a triangle when each graph vertex is assigned to a player who receives all incident edges, so each edge is known to its two endpoint players?

来源字段：https://doi.org/10.4171/owr/2015/54

16992 | OWR-8415360-004

### Rank Dependence of Sample Complexity in Matrix Sensing

P / partially_solved | Oberwolfach 报告问题汇编

Is the quadratic dependence of matrix-sensing sample complexity $m$ on the rank $r_\star$ necessary, or is it an artifact of the proof?

来源字段：https://doi.org/10.4171/owr/2021/55

16993 | AIM-COMPUTATION-0082

### Rank-profile compression for finite-field time-series models

P / partially_solved | AIM 研讨会问题汇编

Question 67 (Laubenbacher). What is the complexity of the algorithms used for time series modelling? How can we make the algorithms faster?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

16994 | OWR-12177-002

### Real $\tau$-Conjecture for Sparse Polynomial Sums and Products

P / partially_solved | Oberwolfach 报告问题汇编

Is the number of real roots of a sum of products of sparse univariate polynomials bounded polynomially in the size of that representation?

来源字段：https://doi.org/10.4171/owr/2012/54

16995 | OWR-16634-002

### Real Tau Conjecture for Depth-Four Circuits

P / partially_solved | Oberwolfach 报告问题汇编

Is the number of distinct real zeros of every univariate real polynomial represented by a depth-four arithmetic circuit, consisting of a sum of products of sparse polynomials, bounded by a polynomial in the circuit size?

来源字段：https://doi.org/10.4171/owr/2018/51

16996 | OWR-1189-016

### Recognition and Characterization of Leaf Powers

P / partially_solved | Oberwolfach 报告问题汇编

For each $k\geq5$, characterize $k$-leaf powers and determine the complexity of recognizing them.

来源字段：https://doi.org/10.4171/owr/2006/07

16997 | OWR-782-003

### Recognition and Eutaxy Testing of Bacher Matrices

O / open | Oberwolfach 报告问题汇编

Given an integral matrix $A$, decide whether it is the Bacher matrix of an equivalence class of forms and construct an associated class when one exists. Also give a fast algorithm that decides from a Bacher matrix whether its equivalence class contains a weakly eutactic form.

来源字段：https://doi.org/10.4171/owr/2005/01

16998 | OWR-17471-013

### Recognition of Virtually Special Hyperbolic Three-Manifolds

O / open | Oberwolfach 报告问题汇编

Find an algorithm that, given a triangulated hyperbolic $3$-manifold $M$, decides whether $M$ is homotopy equivalent to a special cube complex; equivalently, decide whether $\pi_1(M)$ embeds in a right-angled Artin group.

来源字段：https://doi.org/10.4171/owr/2020/8

16999 | OWR-15219-002

### Reducing the Exponential Tower in Constructive Degree Bounds

O / open | Oberwolfach 报告问题汇编

Is the tower of five exponentials in the constructive degree bound necessary, or can its height be substantially reduced?

来源字段：https://doi.org/10.4171/owr/2017/14

17000 | OWR-9790365-002

### Reduction of Guessing Problems to Source Compression

O / open | Oberwolfach 报告问题汇编

Can every optimal guessing problem be reduced to its corresponding optimal source-compression problem?

来源字段：https://doi.org/10.4171/owr/2022/14

17001 | OWR-13949-005

### Refutation Lower Bounds for Random k-SAT Instances

O / open | Oberwolfach 报告问题汇编

Does a random $k$-SAT instance with, for example, $m=n^{0.4k}$ clauses lack a polynomial-size refutation with high probability?

来源字段：https://doi.org/10.4171/owr/2015/54

17002 | OPG-36884

### Refuting random 3SAT-instances on $O(n)$ clauses (weak form)

P / partially_solved | Open Problem Garden

Conjecture For every rational $\epsilon > 0$ and every rational $\Delta$, there is no polynomial-time algorithm for the following problem.

Given is a 3SAT (3CNF) formula $I$ on $n$ variables, for some $n$, and $m = \floor{\Delta n}$ clauses drawn uniformly at random from the set of formulas on $n$ variables. Return with probability at least 0.5 (over the instances) that $I$ is typical without returning typical for any instance with at least $(1 - \epsilon)m$ simultaneously satisfiable clauses.

背景链接：http://www.openproblemgarden.org/op/refuting_random_3sat_instances_on_o_n_clauses_weak_form

17003 | AIM-CRYPTOGRAPHY-0002

### Regular-orbit equivalence and affine orbit-size certificates for SDLP

P / partially_solved | AIM 研讨会问题汇编

It asks for progress on four related questions:

1. quantum equivalence of the semidirect discrete logarithm problem (SDLP) and the corresponding computational Diffie--Hellman problem (SCDH); 2. quantum composition-series methods beyond solvable groups; 3. a “Sylow theory” for projected sets arising from the semidirect-product structure; and 4. a classical method to compute the size of the acting object in the semigroup variant of SDLP.

The record does not define “projected sets,” “acting group,” an oracle model, or the success notion for SCDH. Its literature field is empty.

There is also a convention issue. We use the **forward-product convention**

$$ s_0=1,\qquad s_n=g\,\varphi(g)\cdots\varphi^{n-1}(g), $$

for a finite monoid $M$, an endomorphism $\varphi:M\to M$, and $g\in M$. Some recent papers use the reverse product. The two conventions must not be mixed inside a reduction.

来源字段：http://aimpl.org/postquantgroup/1/

17004 | OWR-3394-005

### Reliable Model Reduction for Nonlinear Parameterized Systems

P / partially_solved | Oberwolfach 报告问题汇编

Develop mathematically justified model-reduction methods for nonlinear, parameterized, coupled, and large-scale systems, including error bounds and reliable rules for choosing Krylov/Laguerre design parameters so that the reduced model gives a good approximation to the original system's response for arbitrary input signals while preserving relevant structure such as stability or passivity.

来源字段：https://doi.org/10.4171/owr/2009/11

17005 | AIM-COMPUTATION-0079

### Removing term order without hiding the extrapolation choice

P / partially_solved | AIM 研讨会问题汇编

**Question 64 (Laubenbacher).** Choosing models modulo the design ideal depends on the term order chosen. How can we eliminate this choice of term order from the problem of finding a model that fits the data?

**Richards:** Could you use random search or genetic algorithms to solve this problem?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

17006 | OWR-15580-004

### Resolution Lower Bounds for Clique Certification

P / partially_solved | Oberwolfach 报告问题汇编

Can one prove an unconditional lower bound of the form $n^{\Omega(k)}$ for resolution, or for another restricted proof system, when certifying that an $n$-vertex graph has no $k$-clique?

来源字段：https://doi.org/10.4171/owr/2017/37

17007 | OWR-16931-013

### Restricted Models of Vertex-Distribution-Free Testing

O / open | Oberwolfach 报告问题汇编

How does vertex-distribution-free graph-property testing change if testers are required to work only on sufficiently large graphs, or if the vertex distribution is restricted by lower or upper bounds on its atom weights?

来源字段：https://doi.org/10.4171/owr/2019/19

17008 | OWR-14298166-006

### Restriction Condensation of Deterministic Communication Complexity

O / open | Oberwolfach 报告问题汇编

Can deterministic communication complexity be condensed by restriction, and is some polynomial loss necessary?

来源字段：https://doi.org/10.4171/owr/2024/15

17009 | OWR-13105-007

### Reusable Garbled RAM under Weaker Assumptions

O / open | Oberwolfach 报告问题汇编

Can reusable garbled RAM, including schemes with persistent memory, be constructed under assumptions weaker than indistinguishability obfuscation, functional encryption for circuits, or the stronger obfuscation assumptions used by the known constructions?

来源字段：https://doi.org/10.4171/owr/2014/35

17010 | OWR-783-001

### Robust Radio-Network Planning Under Traffic Misspecification

O / open | Oberwolfach 报告问题汇编

Develop a robust radio-network planning method whose performance remains controlled when the assumed traffic distribution is misspecified by, for example, 10 percent.

来源字段：https://doi.org/10.4171/owr/2005/02

17011 | OWR-12976-001

### Robust Spectral Tensor Methods for Latent-Variable Learning

O / open | Oberwolfach 报告问题汇编

Can spectral tensor-decomposition methods for learning latent Dirichlet allocation and related multi-view latent-variable models be made more robust to noisy data, statistical uncertainty, and incorrect modeling assumptions?

来源字段：https://doi.org/10.4171/owr/2014/18

17012 | OWR-4990384-001

### Robustness-Driven Overparameterization in Two-Layer Neural Networks

P / partially_solved | Oberwolfach 报告问题汇编

Does interpolation by a two-layer neural network with few parameters necessarily force the represented function to have a large Lipschitz constant, thereby making overparameterization necessary for robustness?

来源字段：https://doi.org/10.4171/owr/2021/15

17013 | AIM-COMPUTATION-0050

### Root obstruction and outgroup-oriented algebraic clade certificates

P / partially_solved | AIM 研讨会问题汇编

Question 36. Can we develop algebraic invariant techniques for identifying clades?

Shmuel Onn

Shmuel Onn spoke about the spectra of hierarchical models. The spectra is defined to be the set of all sets of possible cell entries in for a multiway nonnegative integral table with fixed marginal totals.

**Question 36. Can we develop algebraic invariant techniques for identifying clades?**

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

17014 | OWR-2090-020

### Rotational Matching Complexity for Planar Point Sets

O / open | Oberwolfach 报告问题汇编

Let $A_1,\ldots,A_n$ and $B_1,\ldots,B_n$ be $2n$ points in the plane, and define $$f(\theta)=\min_{\pi}\sum_i\|A_i-R(\theta)B_{\pi(i)}\|^2,$$ where $R(\theta)$ is rotation about the origin. How many intervals of $\theta$ can have a constant optimal permutation, and is this number polynomially bounded? How many local minima can $f$ have, and is there a polynomial-time algorithm for finding the best rotation?

来源字段：https://doi.org/10.4171/owr/2008/44

17015 | OWR-15210-005

### Round Complexity of Non-Malleable Commitments

P / partially_solved | Oberwolfach 报告问题汇编

Determine the round complexity of non-malleable commitments and construct round-efficient schemes secure against man-in-the-middle attacks under standard assumptions.

来源字段：https://doi.org/10.4171/owr/2017/5

17016 | OWR-4414-002

### Sample Complexity of Active Coordinate Identification

O / open | Oberwolfach 报告问题汇编

In recovering an approximately low-dimensional function of $N$ variables from samples, how many non-adaptive sample points are required to identify the active coordinates?

来源字段：https://doi.org/10.4171/owr/2010/33

17017 | OWR-16931-012

### Sample Complexity of Vertex-Distribution-Free Graph Testing

O / open | Oberwolfach 报告问题汇编

Does every extendable hereditary graph property admit a vertex-distribution-free tester whose sample complexity is asymptotically comparable to that of a tester in the standard dense-graph model?

来源字段：https://doi.org/10.4171/owr/2019/19

17018 | OWR-15222-001

### Sampling Complexity of Hyperbolic-Cross Approximation

O / open | Oberwolfach 报告问题汇编

Can better sampling measures lower the sample complexity of hyperbolic-cross polynomial approximation, and can sample-complexity estimates be proved for domains that are not unions of rectangles with nonvanishing volume, including triangles, simplices, and spheres?

来源字段：https://doi.org/10.4171/owr/2017/17

17019 | AIM-COMPUTATION-0075

### Saturation-aware exceptional likelihood fibers and the 2x2 independence locus

P / partially_solved | AIM 研讨会问题汇编

**Question 60 (Sullivant).** For a log-linear model, what can be said about the structure of the set of \(b\) with \(Ax=b\) for some \(b\) and the solution space to the maximum likelihood equations is not zero-dimensional?

**Hoşten.** Look at the leading coefficients of polynomials in the elimination ideals (with parameters) and determine when these are zero.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

17020 | OWR-14299525-005

### Scaling Paradigms for Lean Autoformalization

O / open | Oberwolfach 报告问题汇编

Determine which combination of fine-grained tree search, whole-proof generation, iterative debugging, hierarchical decomposition, and reinforcement learning scales most effectively to large research-level theorem-proving and autoformalization tasks in Lean.

来源字段：https://doi.org/10.4171/owr/2025/43

17021 | OWR-15210-002

### Security of Interactive Lattice-Based Group Key Exchange

O / open | Oberwolfach 报告问题汇编

Can provable security be established for the interactive multiparty key exchange protocol of Ding, Xie, and Lin based on LWE and RLWE, viewed as a generalization of their two-party protocol?

来源字段：https://doi.org/10.4171/owr/2017/5

17022 | OWR-768-008

### Semaev's Auxiliary Elliptic-Curve Index-Calculus Algorithm

O / open | Oberwolfach 报告问题汇编

Let $f_{n+1}$ be Semaev's $(n+1)$-st summation polynomial. Given a prime $p$ and $x\in\mathbb{F}_p$, is there an effective general algorithm that finds integers $x_1,\ldots,x_n$, each of size at most $p^{1/n+\delta}$, such that $f_{n+1}(x_1,\ldots,x_n,x)\equiv0\pmod p$ whenever such a tuple is required by the index-calculus construction?

来源字段：https://doi.org/10.4171/owr/2004/54

17023 | AIM-CRYPTOGRAPHY-0004

### Semantic leakage barriers for learned group cryptanalysis

P / partially_solved | AIM 研讨会问题汇编

**Machine learning-based cryptanalysis of group-theoretic protocols.** There have been some results in the application of machine learning to the cryptanalysis of lattice-based schemes; whether these methods translate to the group-theoretic realm remains open.

来源字段：http://aimpl.org/postquantgroup/1/

17024 | OWR-14298377-002

### Semi-Supervised Sparse Gaussian Classification

O / open | Oberwolfach 报告问题汇编

Characterize the statistical and computational regimes in which unlabeled data provably improve sparse Gaussian classification.

来源字段：https://doi.org/10.4171/owr/2024/30

17025 | OWR-1465-013

### Semialgebraic Homeomorphism Types at Fixed Complexity

O / open | Oberwolfach 报告问题汇编

Does the Benedetti-Risler finiteness conjecture hold for every axiomatic complexity measure on semialgebraic sets: for each fixed value of such a complexity measure, are there only finitely many homeomorphism types?

来源字段：https://doi.org/10.4171/owr/2007/14

17026 | AIM-COMPUTATION-0040

### Semialgebraic stratification of maximum likelihood

P / partially_solved | AIM 研讨会问题汇编

Question 26. What are the possible implications of semi-algebraic geometry to maximum likelihood estimation?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

17027 | OWR-16938-005

### Semidefinite Hierarchies for Equiangular Line Bounds

O / open | Oberwolfach 报告问题汇编

For the $k$-point semidefinite-programming bounds $\Delta_k(G_{n,a})^*$ for equiangular lines, explain their stabilization at $$(1/a^2-2)(1/a^2-1)/2,$$ determine whether $\Delta_k(G_{n,a})^*=\Theta(n^2)$ for fixed $k$ and $a$, and compare the corresponding level-$\alpha$ bound with the Lasserre hierarchy.

来源字段：https://doi.org/10.4171/owr/2019/26

17028 | OWR-14299906-008

### Semidefinite Relaxations for Quantum Equilibria

O / open | Oberwolfach 报告问题汇编

Construct a convergent semidefinite-programming hierarchy for maximizing social welfare over quantum correlated equilibria by replacing the explicit tensor-product constraints with suitable commutation constraints.

来源字段：https://doi.org/10.4171/owr/2026/4

17029 | OWR-4135-008

### Sensitivity Growth Under Boolean Conjunctions

O / open | Oberwolfach 报告问题汇编

If Boolean functions $f_1,\ldots,f_n$ have average maximal sensitivity $O((\log n)^d)$, and each $g_i$ is the conjunction of a subset of the $f_j$, must the average maximal sensitivity of the $g_i$ be $O((\log n)^{d+1})$?

来源字段：https://doi.org/10.4171/owr/2009/52

17030 | OWR-14298800-001

### Sensor-Actuator Co-Design and Distributed Controllability Algorithms

O / open | Oberwolfach 报告问题汇编

Given a sensing or actuation budget and a control-cost functional, how should sensors or actuators be selected to minimize cost? Is there a local distributed algorithm for the rank or controllability-volume problem? Characterize when adding or removing links from the system dynamics or communication network improves sensing or controllability rather than amplifying disturbances. Prove convergence of ADMM to the optimal solution in the PDE setting.

来源字段：https://doi.org/10.4171/owr/2024/56

17031 | OWR-1587-002

### Separation of Constant-Free Valiant Classes

O / open | Oberwolfach 报告问题汇编

Determine whether the constant-free Valiant classes satisfy $VP_0\neq VNP_0$.

来源字段：https://doi.org/10.4171/owr/2007/31

17032 | OWR-16775-003

### Sequential Decoding of Nonbinary Convolutional Codes

O / open | Oberwolfach 报告问题汇编

How do nonbinary convolutional codes perform with sequential decoders on channels with errors?

来源字段：https://doi.org/10.4171/owr/2019/13

17033 | OWR-16637-005

### Sharp Smoothed Complexity of Simplex Pivot Rules

O / open | Oberwolfach 报告问题汇编

For smoothed analysis of the simplex method, can the expected number of edges of a projected polyhedron be bounded sharply by $\Theta(d^{3/2}\log n)$ as $\sigma\to\infty$, with a lower bound depending on $\sigma$? What bounds hold if only a random $\varepsilon$-fraction of the entries of $(A,b)$ is perturbed, or if other pivot rules are used?

来源字段：https://doi.org/10.4171/owr/2018/54

17034 | OWR-783-003

### Sharper Complexity Bounds for Full-Newton Interior-Point Methods

O / open | Oberwolfach 报告问题汇编

Can the known iteration-complexity bound for the full-Newton infeasible interior-point method for linear optimization be improved by a factor of $\sqrt n$?

来源字段：https://doi.org/10.4171/owr/2005/02

17035 | AMR-054-0010

### Simple Linear-Time Polygon Triangulation

O / open | AMR 问题清单汇编

Is there a deterministic, linear-time polygon triangulation algorithm significantly simpler than that of Chazelle?

背景链接：https://topp.openproblem.net/p10

17036 | OWR-14741-004

### Simultaneously Adaptive Universal Source Coding

P / partially_solved | Oberwolfach 报告问题汇编

Let $(C(\alpha))_{\alpha\in A}$ be source subclasses with nontrivial minimax redundancy rates. Does there exist a single sequence of coding probabilities $Q_n$, independent of $\alpha$, whose redundancy is asymptotically optimal, or optimal within a fixed factor, simultaneously for every subclass in the collection?

来源字段：https://doi.org/10.4171/owr/2016/33

17037 | AIM-COMPUTATION-0049

### Simultaneously certified invariant quartet scoring and robust assembly

P / partially_solved | AIM 研讨会问题汇编

**Problem 35. Develop techniques to use invariants in combination with quartet methods.**

It occurs in Elizabeth Allman's block immediately after problems on secant varieties, flattening invariants, the stochastic image, “good” invariants, tree/model fit, and among-site rate variation, and immediately before a question about identifying clades. Thus the intended program is to turn local algebraic conditions-especially rank conditions on quartet flattenings-into a scalable full-tree reconstruction method.

Three logically separate tasks are implicit:

1. **Population identifiability:** at the exact site-pattern distribution, does the invariant distinguish the true quartet? 2. **Finite-sample quartet scoring:** from \(N\) sites, how should the three quartet topologies be scored, normalized, tied, or rejected? 3. **Global assembly:** how should possibly dependent, weighted, missing, or conflicting quartet decisions be combined into one tree?

Solving one layer does not solve the other two.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

17038 | OWR-8415359-015

### Size-Efficient Circuits Resilient to Short-Circuit Errors

P / partially_solved | Oberwolfach 报告问题汇编

Given a Boolean circuit $C$, can one construct a circuit $C'$ of comparable size that computes the same function even when an adversary replaces the output of some gates by one of their inputs?

来源字段：https://doi.org/10.4171/owr/2021/54

17039 | SMA-004

### Smale's 4th Problem: Integer Zeros of Polynomials

P / partially_solved | Smale 问题

Find efficient algorithms for deciding whether a polynomial with integer coefficients has an integer root.

背景链接：https://en.wikipedia.org/wiki/Hilbert%27s_tenth_problem

17040 | SMA-009

### Smale's 9th Problem: Linear Programming in Polynomial Time

O / open | Smale 问题

Find a strongly polynomial algorithm for linear programming.

背景链接：https://mathworld.wolfram.com/SmalesProblems.html

17041 | AMR-054-0041

### Sorting $X+Y$ (Pairwise Sums)

O / open | AMR 问题清单汇编

Given two sets of numbers, each of size $n$, how quickly can the set of all pairwise sums be sorted? In symbols, given two sets $X$ and $Y$, our goal is to sort the set $$X+Y = \{x+y \mid x \in X, y \in Y\}.$$

背景链接：https://topp.openproblem.net/p41

17042 | AIM-COMPUTATION-0003

### Source repair and a sharp quadratic specialization theorem

O / open | AIM 研讨会问题汇编

**6.3 Open Question.** Given \(f\in\mathbb Q[x,y,z]\), for which values of \(z\in\mathbb Q\) does \(f\) factor (completely) in \(\mathbb C[x,y]\)? What is the degree of a given \(z\)?

来源字段：https://aimath.org/WWN/polyfactor/polyfactor.pdf

17043 | OWR-831-016

### Space-Efficient Lattice-Basis Reduction

O / open | Oberwolfach 报告问题汇编

Reduce the essentially time-sized space requirement of birthday-method algorithms for lattice-basis reduction while retaining their improved running time.

来源字段：https://doi.org/10.4171/owr/2005/26

17044 | AIM-COMPUTATION-0084

### Specialized algebra kernels and certified interoperability

P / partially_solved | AIM 研讨会问题汇编

**Question 69 (Roehrig).** Why are there so many different systems for computational algebra? Couldn't they be add-ins to other systems? **Stillman:** Not likely to work since those other systems (Mathematica, Maple, etc.) do not give access to their kernels.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

17045 | AIM-CRYPTOGRAPHY-0021

### Spectral laws for Gaussian multiple-shift states

P / partially_solved | AIM 研讨会问题汇编

How does Kuperberg's algorithm behave under multiple hidden shifts (with known relations between shifts, e.g. arithmetic progression) with Gaussians, using the work of Ivanyos, Prakash, and Santha?,

[Elena Kirshanova] How does Kuperberg's algorithm behave under multiple hidden shifts (with known relations between shifts, e.g. arithmetic progression) with Gaussians, using the work of Ivanyos, Prakash, and Santha?

来源字段：http://aimpl.org/quantumalg/4/

17046 | OWR-14298166-005

### Spreadness as a Replacement for Sunflower Structure

O / open | Oberwolfach 报告问题汇编

Can spreadness replace sunflower structure in theoretical-computer-science applications that currently use sunflowers, particularly through robust-sunflower generalizations?

来源字段：https://doi.org/10.4171/owr/2024/15

17047 | OWR-16931-010

### Square-Root Sparsity for Explicit RIP Matrices

O / open | Oberwolfach 报告问题汇编

Does there exist an explicit construction of real-valued restricted-isometry matrices with sparsity $s\geq c\sqrt n$ in the polylogarithmic regime $n=(\log p)^{O(1)}$?

来源字段：https://doi.org/10.4171/owr/2019/19

17048 | OWR-12697684-022

### Stable Algorithms and Overlap Gaps for Friendly Bisections

O / open | Oberwolfach 报告问题汇编

For $h>0$, is there a stable polynomial-time algorithm that finds an $h$-friendly bisection of $G(n,1/2)$? Does the overlap-gap property hold for every $h>0$, perhaps by transfer from random regular graphs to Erdős-Rényi graphs?

来源字段：https://doi.org/10.4171/owr/2023/1

17049 | OWR-792-017

### Stationary and Optimal Linear Differential Behaviors

P / partially_solved | Oberwolfach 报告问题汇编

Characterize the linear time-invariant differential behaviors $\mathcal{B}\in\mathcal{L}^w$ for which there exists a symmetric two-variable polynomial matrix $\Phi\in\mathbb{R}^{w\times w}[\zeta,\eta]$ satisfying $$\mathcal{B}=\ker\Phi\!\left(-\frac{d}{dt},\frac{d}{dt}\right).$$ These are the stationary behaviors. Characterize also the optimal ones, for which one can additionally require $\Phi(-i\omega,i\omega)\succeq0$ for every $\omega\in\mathbb{R}$.

来源字段：https://doi.org/10.4171/owr/2005/11

17050 | OWR-16170-001

### Statistical Recovery of High-Dimensional Network Structure

O / open | Oberwolfach 报告问题汇编

Given observations from a high-dimensional network model, how can the underlying network structure, including its edge-generation rule, be statistically estimated?

来源字段：https://doi.org/10.4171/owr/2018/29

17051 | OWR-8415360-011

### Stochasticity and Sample Complexity in Neural Networks

O / open | Oberwolfach 报告问题汇编

How does stochasticity affect neural-network performance, and how much data is needed to achieve small estimation error with high probability?

来源字段：https://doi.org/10.4171/owr/2021/55

17052 | OWR-14298804-001

### Strongly Polynomial Algorithms for Linear Programming

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Strongly Polynomial Linear-Programming Lead-In [OWR-14298804-002]

Does linear programming admit a strongly polynomial-time algorithm?

来源字段：https://doi.org/10.4171/owr/2024/50

17053 | OWR-15173-004

### Subexponential Completeness Testing for Voronoi Cell Representatives

O / open | Oberwolfach 报告问题汇编

Given a Voronoi cell complex $\operatorname{Vor}_\Gamma$ and a proposed set $C_d$ of representatives of its $d$-cells modulo $\Gamma$, is there a subexponential-time algorithm for deciding whether $C_d$ is complete?

来源字段：https://doi.org/10.4171/owr/2016/52

17054 | OWR-14745-006

### Sublinear Membership Testing in Two-Generator Matrix Groups

P / partially_solved | Oberwolfach 报告问题汇编

For an integer $K\geq2$, let $H(K)\leq\operatorname{SL}(2,\mathbb Q)$ be generated by

$$A(K)=\begin{pmatrix}1&K\\0&1\end{pmatrix},\qquad B(K)=\begin{pmatrix}1&0\\K&1\end{pmatrix}.$$

Is there a sublinear-time algorithm for the subgroup membership problem in $H(K)$?

来源字段：https://doi.org/10.4171/owr/2016/37

17055 | OWR-8415352-003

### Submultiplicativity Gaps for Stabilizer Extent

P / partially_solved | Oberwolfach 报告问题汇编

How large can the submultiplicativity gap between $\xi(\phi\otimes\psi)$ and $\xi(\phi)\xi(\psi)$ be for quantum states $\phi$ and $\psi$?

来源字段：https://doi.org/10.4171/owr/2021/49

17056 | OWR-16634-008

### Subquadratic Approximation for Longest Common Subsequences

P / partially_solved | Oberwolfach 报告问题汇编

Does there exist, for some $\varepsilon>0$, a $1.1$-approximation algorithm for the longest common subsequence of two length-$n$ strings that runs in time $O(n^{2-\varepsilon})$ for every alphabet size?

来源字段：https://doi.org/10.4171/owr/2018/51

17057 | OPG-163

### Subset-sums equality (pigeonhole version)

P / partially_solved | Open Problem Garden

Problem Let $a_1,a_2,\ldots,a_n$ be natural numbers with $\sum_{i=1}^n a_i < 2^n - 1$. It follows from the pigeon-hole principle that there exist distinct subsets $I,J \subseteq \{1,\ldots,n\}$ with $\sum_{i \in I} a_i = \sum_{j \in J} a_j$. Is it possible to find such a pair $I,J$ in polynomial time?

背景链接：http://www.openproblemgarden.org/op/theoretical_computer_science/subset_sums_equality

17058 | AIM-COMPUTATION-0027

### Sufficient releases, conditional fibers, and incompatible Gibbs updates

P / partially_solved | AIM 研讨会问题汇编

**Question 13.** What is the family of distributions that has margins and conditionals as sufficient statistics? What should the stationary distribution be for MCMC? Jon Forster asked, “Is it safe to release full conditionals that have been perturbed?” Aleksandra replied, “Probably not.”

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

17059 | OPG-36887

### Sums of independent random variables with unbounded variance

P / partially_solved | Open Problem Garden

Conjecture If $X_1, \dotsc, X_n \geq 0$ are independent random variables with $\mathbb{E}[X_i] \leq \mu$, then $$\mathrm{Pr} \left( \sum X_i - \mathbb{E} \left[ \sum X_i \right ] < \delta \mu \right) \geq \min \left ( (1 + \delta)^{-1} \delta, e^{-1} \right).$$

背景链接：http://www.openproblemgarden.org/op/sums_of_independent_random_variables_with_unbounded_variance

17060 | OWR-14298166-004

### Superpolynomial Lower Bounds for Dag-Like Resolution over Parities

O / open | Oberwolfach 报告问题汇编

Prove a superpolynomial lower bound on the size of dag-like $\operatorname{Res}(\oplus)$ refutations.

来源字段：https://doi.org/10.4171/owr/2024/15

17061 | OWR-1113-003

### Symmetry-Aware Rank-One Convex Interpolation Algorithms

P / partially_solved | Oberwolfach 报告问题汇编

Develop an efficient algorithm that exploits frame indifference or material symmetry to compute an interpolation with suitable rank-one-convexity properties from values of an energy density on a mesh of deformation gradients.

来源字段：https://doi.org/10.4171/owr/2005/52

17062 | OWR-4426-005

### Symmetry-Preserving Evaluation of Orbitwise Generating Functions

P / partially_solved | Oberwolfach 报告问题汇编

Given an orbitwise rational generating function for the integer points of a symmetric polyhedron, how can its value at the singular point $z=(1,\ldots,1)$ be evaluated in an orbitwise way while preserving the symmetry?

来源字段：https://doi.org/10.4171/owr/2010/38

17063 | OWR-1454-001

### Tangency Handling in Telea Image Inpainting

O / open | Oberwolfach 报告问题汇编

In Telea's image-inpainting algorithm, does changing the order in which pixels are processed suffice to address the singular case of tangency?

来源字段：https://doi.org/10.4171/owr/2007/03

17064 | AIM-CRYPTOGRAPHY-0022

### The cyclic core of Heisenberg hidden shift

P / partially_solved | AIM 研讨会问题汇编

Is there a fast hidden shift algorithm for Heisenberg group over $\mathbb F_p$?

HSP is known.

来源字段：http://aimpl.org/quantumalg/4/

17065 | AIM-CRYPTOGRAPHY-0005

### The inverse-state twist interface, extraction boundary, and exact storage limit

P / partially_solved | AIM 研讨会问题汇编

The connection between SDLP and group actions suggests the availability of generic group-action tools to build more efficient signatures, or signatures with more specialised properties.

We would like to use generic group-action tools to build ring signatures, blind signatures, threshold signatures and more. We would also like to apply the "twist" technique present in similar isogeny schemes to improve the memory performance of group-based digital signatures.

来源字段：http://aimpl.org/postquantgroup/2/

17066 | OWR-14299518-026

### The Log-Rank Conjecture

O / open | Oberwolfach 报告问题汇编

Prove the log-rank conjecture: the deterministic communication complexity of a Boolean function is bounded by a polylogarithmic function of the rank of its communication matrix.

来源字段：https://doi.org/10.4171/owr/2025/42

17067 | AMR-054-0040

### The Number of Pointed Pseudotriangulations

O / open | AMR 问题清单汇编

For a planar point set $S$, is the number of pointed pseudotriangulations always at least the number of triangulations? A pseudotriangle is a planar polygon with exactly three convex vertices. Each pair of convex vertices is connected by a reflex chain, which may be just one segment. (Thus, a triangle is a pseudotriangle.) A pseudotriangulation of a set $S$ of $n$ points in the plane is a partition of the convex hull of $S$ into pseudotriangles using $S$ as a vertex set. A minimum pseudotriangulation, or pointed pseudotriangulation, has the fewest possible number of edges for a given set $S$ of points. See for examples, explanation of the term ``pointed,'' and further details.

背景链接：https://topp.openproblem.net/p40

17068 | CS-002

### The Polynomial Hirsch Conjecture

O / open | 独立 / 未分配集合

别名：The Polynomial Hirsch Conjecture [OWR-16633-011]

The diameter of the graph of a $d$-dimensional polytope with $n$ facets is bounded by a polynomial in $d$ and $n$.

背景链接：https://doi.org/10.1007/s10107-024-02107-x

17069 | AIM-CRYPTOGRAPHY-0024

### The real cost of the CSIDH hidden-shift attack

P / partially_solved | AIM 研讨会问题汇编

Are there CSIDH attacks better than hidden shift? What is the cost of the hidden shift attack?

来源字段：http://aimpl.org/quantumalg/5/

17070 | OPG-445

### The robustness of the tensor product

P / partially_solved | Open Problem Garden

Problem Given two codes $R,C$, their Tensor Product $R \otimes C$ is the code that consists of the matrices whose rows are codewords of $R$ and whose columns are codewords of $C$. The product $R \otimes C$ is said to be robust if whenever a matrix $M$ is far from $R \otimes C$, the rows (columns) of $M$ are far from $R$ ( $C$, respectively).

The problem is to give a characterization of the pairs $R,C$ whose tensor product is robust.

背景链接：http://www.openproblemgarden.org/op/the_robustness_of_the_tensor_product_0

17071 | AIM-COMPUTATION-0045

### The stochastic image of a binary phylogenetic tripod

P / partially_solved | AIM 研讨会问题汇编

Problem 31. Determine the image of the stochastic parameterization (as opposed to the complex parameterization).

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

17072 | CS-001

### The Unique Games Conjecture

O / open | 独立 / 未分配集合

别名：Unique Games Approximation-Threshold Conjecture [OWR-4135-002]; Unique Games Conjecture [OWR-11451-013]; Unique Games Conjecture [OWR-12177-009]; Unique Games Conjecture [OWR-16634-007]

For certain constraint satisfaction problems (unique games), it is NP-hard to approximate the maximum fraction of satisfiable constraints beyond a certain threshold.

背景链接：https://www.theoryofcomputing.org/articles/v021a010/v021a010.pdf

17073 | OWR-16775-009

### Theta Invariants and New Families of MRD Codes

O / open | Oberwolfach 报告问题汇编

Can the theta-dimension and theta-increment invariants of rank-metric codes characterize families of MRD codes other than Gabidulin codes, or yield new code constructions?

来源字段：https://doi.org/10.4171/owr/2019/13

17074 | AIM-COMPUTATION-0097

### Three zeta-like products, degree-count inversion, and the factorization boundary

P / partially_solved | AIM 研讨会问题汇编

**Recovered statement.** Compute the zeta function of \(\operatorname{Spec}\mathbb Z/n\mathbb Z\), \[ Z_n(s)=\prod_{p\mid n}\frac{1}{1-p^{-s}}, \] without knowing the prime factorization of \(n\). The source says that the value at \(s=1\) gives \(\varphi(n)\), which is enough to factor \(n\). It then replaces \(n\) by \(f(x)\in\mathbb F_p[x]\) and literally prints \[ \zeta(s)= \prod_{\substack{g\mid f\\g\ {\rm monic,\ irreducible}}} \frac{1}{1-g}, \tag{1.1} \] asking whether one can recover the factorization of \(f\) in \(\mathbb F_q[x]\) in deterministic polynomial time.

来源字段：https://aimath.org/WWN/primesinp/primesinp.pdf

17075 | OWR-1111-001

### Three-Terminal Distance-Interdiction Complexity

O / open | Oberwolfach 报告问题汇编

Given an undirected graph, with or without edge costs, and three vertices $r,s,t$, determine the complexity of finding a minimum-cost edge set whose deletion increases each of $d(r,s)$, $d(r,t)$, and $d(s,t)$ by at least $1$.

来源字段：https://doi.org/10.4171/owr/2005/50

17076 | OWR-792-014

### Threshold-Avoiding Control for Bank Capital Ratios

O / open | Oberwolfach 报告问题汇编

别名：Redundant Extraction - Bank Capital-Ratio Control [OWR-792-015]

Does there exist a control law for the stochastic model that prevents a bank's capital adequacy ratio from dropping below $0.08$? If not, determine a control law that postpones the first threshold crossing for as long as possible.

来源字段：https://doi.org/10.4171/owr/2005/11

17077 | OWR-9790365-010

### Tight Bounds for Many-User Multiple-Access Coding

O / open | Oberwolfach 报告问题汇编

Determine tight achievable and converse bounds for Gaussian many-user multiple-access coding, including the random-access setting.

来源字段：https://doi.org/10.4171/owr/2022/14

17078 | AIM-CRYPTOGRAPHY-0023

### Torsion-image leakage and rigidity in SIDH

S / solved | AIM 研讨会问题汇编

Can quantum algorithms be used to attack SIDH by using the extra points? Are there SIDH attacks better than claw-finding?

来源字段：http://aimpl.org/quantumalg/5/

17079 | OWR-14297738-003

### Training-Loss Implications for Shallow Neural Networks

O / open | Oberwolfach 报告问题汇编

Determine what guarantees on prediction or generalization follow from small training loss for shallow neural networks.

来源字段：https://doi.org/10.4171/owr/2023/41

17080 | AIM-COMPUTATION-0032

### Transcendence-aware algebraic localization of symmetric Gaussian-mixture modes

P / partially_solved | AIM 研讨会问题汇编

**Question 18. Can algebra be used to better approximate the location of the modes in a mixture?**

The exact AIM question is:

The preceding sentence says that Steele spoke about mixture models, model selection, and the Bayesian Information Criterion. Questions 16 and 17 concern integration and BIC for mixture models; Question 18 contains no additional assumptions. In particular, the source does not say:

- whether a “mode” is a mode of a fitted density, posterior, or likelihood; - which component family is used; - whether component parameters are known or estimated; - the number of components or ambient dimension; or - whether “better” refers to coordinate error, objective-value error, or computational cost.

This report studies the standard fixed-parameter density-mode interpretation for Gaussian mixtures. This is a reconstruction, not extra text from the source. It is also deliberately narrower than the source: arbitrary component densities do not admit one universal algebraic treatment.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

17081 | AIM-COMPUTATION-0018

### Two fixed odds ratios form a line segment, with a complete boundary audit

S / solved | AIM 研讨会问题汇编

For a \(2\times2\) table of probabilities, does it make sense to fix values of two of the odds ratios \[ \alpha=\frac{p_{00}p_{11}}{p_{01}p_{10}},\qquad \alpha^*=\frac{p_{00}p_{01}}{p_{11}p_{10}}, \] and look at the locus of possible values for the third odds ratio \[ \alpha^{**}=\frac{p_{00}p_{10}}{p_{01}p_{11}}? \] Describe the curve (in the probability simplex) obtained by intersecting two of these quadric surfaces.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

17082 | OWR-14298804-014

### Two-Coloring Light Tournaments

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Light-Tournament Two-Coloring Question [OWR-14298804-015]

Is there a polynomial-time algorithm that partitions every two-colorable light tournament into two acyclic vertex sets, or is deciding two-colorability NP-hard even for light tournaments?

来源字段：https://doi.org/10.4171/owr/2024/50

17083 | AIM-COMPUTATION-0069

### Two-sided novelty certificates for sparse binomial output

P / partially_solved | AIM 研讨会问题汇编

**Question 54 (Fienberg).** How can we make it easier to identify "new" interesting details among long complicated computer output? Karr: can we take advantage of sparse representations of problems?

Second Open Problem Session

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

17084 | OWR-9790365-011

### Type-II Error Exponents in Distributed Hypothesis Testing

O / open | Oberwolfach 报告问题汇编

Determine the optimal type-II error exponent for rate-limited distributed hypothesis testing and for its interactive two-sensor variant.

来源字段：https://doi.org/10.4171/owr/2022/14

17085 | OPG-454

### Unconditional derandomization of Arthur-Merlin games

O / open | Open Problem Garden

Problem Prove unconditionally that $\mathcal{AM}$ $\subseteq$ $\Sigma_2$.

背景链接：http://www.openproblemgarden.org/op/unconditional_derandomization_of_mathcal_am

17086 | OWR-14298371-007

### Unconditionally Ruling Out Heuristica

O / open | Oberwolfach 报告问题汇编

Can the complexity-theoretic world Heuristica be ruled out unconditionally?

来源字段：https://doi.org/10.4171/owr/2024/27

17087 | OWR-14299916-002

### Unified Mathematical Theory of Modern Learning

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Unified-Theory Question [OWR-14299916-011]

Can the geometric, probabilistic, computational, representational, and dynamical perspectives on modern learning systems be integrated into a coherent unified mathematical theory?

来源字段：https://doi.org/10.4171/owr/2026/11

17088 | OWR-15999-001

### Unified Simplicity Models for Massive Inverse Problems

O / open | Oberwolfach 报告问题汇编

In phase retrieval and inverse problems for massive complex data, determine how sparsity and minimal information complexity can be used, and whether a unified notion of simplicity can include sparsity, low rank, and low entropy as special cases.

来源字段：https://doi.org/10.4171/owr/2018/14

17089 | OWR-11695865-004

### Unique Genome Reconstruction from Seven Beads

P / partially_solved | Oberwolfach 报告问题汇编

In the noiseless genome-reconstruction model, do the locations of seven unambiguous beads uniquely determine the locations of all remaining beads?

来源字段：https://doi.org/10.4171/owr/2022/55

17090 | AIM-COMPUTATION-0023

### Unique minimal Markov bases, bipartite cycles, and padding monotonicity

P / partially_solved | AIM 研讨会问题汇编

Do \(I\times J\times K\) tables under the no 3-way interaction model always have a unique minimal Markov basis?

The source continues with two related questions: Bernd Sturmfels asks about three-way tables with structural zeros coming from a very regular design, and Henry Wynn asks whether the group structure of circuits plays a role.

After Wynn's sentence, “Emily Gamundi” is set in the larger speaker-heading font; Question 10 then follows under that heading.

The intended statistical model is the hierarchical model with generating class \[ [IJ][IK][JK], \] also called the no-three-factor-interaction model. Conditional inference fixes all three two-dimensional margins \(x_{ij+},x_{i+k},x_{+jk}\). This is not the complete-independence model \([I][J][K]\), which fixes only one-dimensional margins and has different Markov-basis behavior.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

17091 | OWR-16413-005

### Uniqueness of Equilibria in Rank-One Bimatrix Games

O / open | Oberwolfach 报告问题汇编

Is deciding whether a rank-one bimatrix game has a unique equilibrium co-NP-hard?

来源字段：https://doi.org/10.4171/owr/2018/38

17092 | OWR-11101919-003

### Universal Approximation by One-Lipschitz Spline Networks

O / open | Oberwolfach 报告问题汇编

Are weight-constrained neural networks with learnable one-Lipschitz spline activations universal approximators of one-Lipschitz functions? Does universality extend to vector-valued targets, and what depth-width structures are optimal?

来源字段：https://doi.org/10.4171/owr/2022/38

17093 | OWR-9790365-003

### Universal Compression with Large Unknown Alphabets

O / open | Oberwolfach 报告问题汇编

For block size $k>\delta\sqrt{\log n}$, can one construct a universal compressor for the large or unknown alphabets arising from the submatrix decomposition?

来源字段：https://doi.org/10.4171/owr/2022/14

17094 | OWR-1587-008

### Universal Semantic Communication Across Different Meanings

O / open | Oberwolfach 报告问题汇编

Develop a mathematical theory of universal semantic communication: formalize intelligence, understanding, goals, and meaningful interaction when sender and receiver may have different semantics, and characterize when cooperative parties can communicate successfully from their observations and responses.

来源字段：https://doi.org/10.4171/owr/2007/31

17095 | OWR-14299911-023

### Universal-Cover Isomorphism Algorithms for Cube Complexes

O / open | Oberwolfach 报告问题汇编

Is there an algorithm that, given two finite nonpositively curved cube complexes, decides whether their universal covers are isomorphic?

来源字段：https://doi.org/10.4171/owr/2026/8

17096 | AIM-COMPUTATION-0044

### Vertex flattenings generate the phylogenetic ideal

S / solved | AIM 研讨会问题汇编

**Question 30.** Is the ideal obtained by taking the invariants arising from all 3-dimensional flattenings equal to the ideal of all invariants for a phylogenetic tree?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

17097 | OWR-13941-005

### Volume of the Boolean Quadric Polytope

O / open | Oberwolfach 报告问题汇编

For $n\geq2$, let $P_n$ be the Boolean quadric polytope, the convex hull in dimension $d=n(n+1)/2$ of the $0/1$ points satisfying $y_{ij}=x_ix_j$ for $i<j$. Find a formula or nontrivial upper and lower bounds for the $d$-dimensional volume of $P_n$.

来源字段：https://doi.org/10.4171/owr/2015/46

17098 | AIM-COMPUTATION-0058

### When flattening and symmetry-aware collapse are informative

P / partially_solved | AIM 研讨会问题汇编

**Question 44 (Landsberg).** Why is flattening used so frequently in statistics? Might other linear transformations on categorical data be more useful? Why not project onto the irreducible representations of the symmetric group? What about other forms of collapsing data that takes into account group actions?

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

17099 | OWR-13105-006

### Witness Pseudorandom Functions from Witness Encryption

O / open | Oberwolfach 报告问题汇编

Can witness pseudorandom functions be constructed generically from any witness encryption scheme?

来源字段：https://doi.org/10.4171/owr/2014/35

17100 | OWR-12481-013

### Worst-Case Inputs for Smoothed Caching Complexity

O / open | Oberwolfach 报告问题汇编

Characterize the input sequences that have the worst performance under the smoothed complexity of the caching problem, whose competitive-ratio complexity is of order $\log k$ for cache size $k$.

来源字段：https://doi.org/10.4171/owr/2013/18

17101 | AIM-COMPUTATION-0021

### Zero patterns, facial sets, and exact support repair in chordal models

P / partially_solved | AIM 研讨会问题汇编

What conditions on the structures of zeros in a table of counts will guarantee that the maximum likelihood estimates do not exist? In this setting, a maximum likelihood estimate is required to have all strictly positive probabilities.

来源字段：https://aimath.org/WWN/compalgstat/compalgstat.pdf

17102 | OWR-14299518-027

### Zero Submatrices in Sparse Low-Rank Matrices

O / open | Oberwolfach 报告问题汇编

Let $M$ be an $n\times n$ real matrix of rank $r$ with at most $\varepsilon n^2$ nonzero entries, where $0<\varepsilon<1/2$. Must $M$ contain an all-zero square submatrix of size at least $n\exp\!\left(-O(\sqrt{\varepsilon r})\right)$?

来源字段：https://doi.org/10.4171/owr/2025/42

# 资料汇编 16 | 数学物理 / Mathematical Physics

363 条记录。O=开放；P=部分解决；S=已解决。状态与英文/LaTeX 表述均沿用来源。

17103 | AMR-079-0006

## $\beta$-ensembles

S / solved | AMR 问题清单汇编

Random point processes corresponding to $\beta$-ensembles, or, equivalently, log gases at inverse temperature $\beta$, are defined for arbitrary $\beta>0$. The orthogonal, unitary, and symplectic ensembles corresponding to $\beta=1, 2,$ or 4, respectively, are now, of course, well understood, but other values of $\beta$ are also believed to be relevant in applications, for example, in the statistical description of headway in freeway traffic. For certain rational values of $\beta$, $\beta$-ensembles are related to Jack polynomials, but for general $\beta$ much less is known. The analysis of $\beta$-ensembles for general $\beta$ represents an interesting, and increasingly important, challenge. Recently there have been significant developments in the theory of general $\beta$-ensembles. As a result of the work of Edelman and Dumitriu, and also others, we now know that for all $\beta>0$, there exist (tridiagonal) random matrix models whose eigenvalues are distributed according to $\beta$-ensembles. Furthermore, taking an appropriate scaling limit of these tridiagonal matrix ensembles, one arrives at the following remarkable fact. Let $B(x)$, $x\geq0$, denote Brownian motion, and for any $\beta>0$, let $H_\beta$ denote the Schr\"odinger operator $\frac{d^2}{dx^2}-x-\frac{2}{\sqrt \beta} dB(x)$ acting on $L^2((0,\infty),dx)$ with Dirichlet boundary conditions at $x=0$. Then (Edelman-Sutton, Ramirez-Rider-Virag) for almost all realizations $B(x)$, $x\geq0$, $H_\beta$ is self-adjoint with discrete spectrum $\lambda_1(B,\beta)>\lambda_2(B,\beta)>\cdots$ and for each $k$, $\lambda_k(B,\beta)$ has precisely the same distribution as the $k^th$ largest eigenvalues of the corresponding $\beta$-ensemble in the standard edge scaling limit. Part of the challenge in analyzing general $\beta$-ensembles is to use $H_\beta$ to obtain information about these ensembles. Already the variational characterization of the $\lambda_k$'s has been used to give simple proofs of bounds on the asymptotics of the distributions of the $\lambda_k$'s. Question: can one derive the Tracy-Widom formula for $\lambda_1(B;\beta=2)$, say, directly from $H_\beta$? At a more conceptual level, the ($\beta$-ensemble $\leftrightarrow H_\beta$) correspondence brings random matrix theory front and center into the arena and practice of modern day probability theory.

背景链接：https://arxiv.org/abs/0712.0849

17104 | OWR-11787-004

## $L^2$-Cohomology of Strongly Centered Monopole Moduli Spaces

P / partially_solved | Oberwolfach 报告问题汇编

Determine the dimension of the $L^2$-cohomology of the moduli space of strongly centered $\operatorname{SU}(2)$-monopoles on $\mathbb R^3$.

来源字段：https://doi.org/10.4171/owr/2012/25

17105 | OWR-8415362-005

## $u$-Equations for Arbitrary Hyperbolic Surfaces

O / open | Oberwolfach 报告问题汇编

In the positive-geometry and cluster-algebraic formulation of tree and loop string amplitudes (Arkani-Hamed et al.), the $u$-equations are a system of polynomial relations among cross-ratio variables $u_C$ associated to curves on a bordered surface. Do the proposed $u$-equations extend to arbitrary hyperbolic surfaces with boundary and marked cusps?

来源字段：https://doi.org/10.4171/owr/2021/57

17106 | AIM-PHYSICS-0065

## A feasibility sieve for negative-energy binary-single direct scattering

P / partially_solved | AIM 研讨会问题汇编

**Problem 28.** Consider negative energy three-body orbits which are unbound in both directions of time. The two Jacobi vectors of such an orbit then asymptote to Keplerian orbits in both the distant past and the distant future, and so associated to such an orbit we have pair of Kepler elements in the past (one elliptic the other hyperbolic) and another pair in the future. The “direct scattering” problem is: which pair of Kepler elements can be connected to each other in this way? (Richard Montgomery)

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17107 | AIM-PHYSICS-0039

## A gauge-invariant action-gap certificate for barbell transit orbits

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.** Consider the planar restricted three-body problem. For Jacobi constants such that the Hill's region looks like a barbell, do transit orbits always exist? (Rick Moeckel)

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17108 | AIM-PHYSICS-0059

### A Jacobi-Maupertuis and McGehee scale-shape dictionary

P / partially_solved | AIM 研讨会问题汇编

Problem 22. Assume $E=0$ and angular momentum $=0$. Can Jacobi's metric give new insights beyond what McGehee's coordinates give? (Richard Montgomery)

The statement does not specify the number of bodies, the ambient dimension, or the collision regime. The immediately preceding problems discuss three-body syzygies, shape space, and one four-body volume question, while McGehee's original work and most later comparisons concern Newtonian three-body dynamics near triple collision or parabolic infinity. The principal reading adopted here is therefore the Newtonian planar three-body problem after translation and rotation reduction. The scale-shape calculation is first given for an arbitrary translation-reduced Newtonian $N$-body configuration, so the assumption is not hidden.

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17109 | AIM-PHYSICS-0056

### A logarithmic-volume identity and zero-angular-momentum obstruction

P / partially_solved | AIM 研讨会问题汇编

Problem 19. What can be said about the volume of the tetrahedron in the four-body problem? Can it be nonzero forever and stay bounded away from 0 and ∞? (Joseph Gerver)

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17110 | AIM-PHYSICS-0009

### A non-polynomial complex potential with entirely real spectrum

S / solved | AIM 研讨会问题汇编

Is there any non-self-adjoint Schrödinger operator with a non-polynomial potential, either in \(L^2(\mathbb R)\) or in \(L^2(\mathbb R_+)\), which has infinitely many real eigenvalues and at most finitely many non-real eigenvalues?

来源字段：https://aimath.org/pastworkshops/nonselfadjointproblems.pdf

17111 | AIM-PHYSICS-0034

### A quantified Gibbons-Hawking N-body cluster model

P / partially_solved | AIM 研讨会问题汇编

2. (Mazzeo) Combine asymptotics with N -body techniques.

**2. (Mazzeo) Combine asymptotics with N-body techniques.**

- “Wrap-up Session, March 21 2004. R. Mazzeo as moderator.” - “Question (T. Hausel)” and the motto that toric hyperkähler quotients are perhaps the simplest complete hyperkähler metrics. - “Problem: Understand the Hodge cohomology of these spaces from a systematic point of view.” - “Possible first steps:” followed by three numbered items.

Printed item 1 asks for the asymptotics of these spaces and cites Bielawski--Dancer and Gibbons--Rychenkova. Printed item 2 is exactly:

Printed item 3 begins Hausel's discussion of middle-dimensional \(L^2\) Hodge cohomology, self-duality, the \(L^2\)-signature, combinatorics, matroids, and an index theorem. The PDF prints “N-body” without spaces.

The prompt is a research direction rather than a proposition with quantifiers. This report gives it a precise, limited realization for the rank-one toric hyperkähler subclass of multi-center Gibbons--Hawking metrics. It does not duplicate item 1's general asymptotic question and does not claim the Hodge theorem requested by item 3.

来源字段：https://aimath.org/WWN/l2harmonic/l2harmonic.pdf

17112 | AIM-PHYSICS-0036

### A smoothness criterion for projective conic compactifications of ALF twistor models

P / partially_solved | AIM 研讨会问题汇编

**Questions (G. Etesi), 1.** It is known that there exists a good compactification of ALF spaces for \(L^2\)-cohomology. What about its twistor space \(Q\)? (a) Does it compactify? (b) Does the complex structure extend over the compactification?

来源字段：https://aimath.org/WWN/l2harmonic/l2harmonic.pdf

17113 | AIM-PHYSICS-0031

### A synchronized heat-kernel construction on eight-dimensional hyperkähler products

P / partially_solved | AIM 研讨会问题汇编

**Question.** Is there a way to generalize this formalism to 8 dimensional hyperkähler manifolds?

**Reason.** In the non-compact 4-dimensional case or K3 case, the coefficient of the leading order term in the Green's function gave information about the values of the parameters (e.g. sizes of the cycles, metric).

来源字段：https://aimath.org/WWN/l2harmonic/l2harmonic.pdf

17114 | AMR-079-0010

## A Tracy-Widom Central Limit Theorem

O / open | AMR 问题清单汇编

The fact that RMT, and the Tracy-Widom distributions, arise in so many problems in so many different areas leads one to the following question: how can one characterize RMT in purely probabilistic terms? For example, we know that if we take i.i.d.'s $(a_1,a_2,...)$, add them up, and then center and scale appropriately, $$ (a_1,a_2,...)\to(S_1,S_2,...),\qquad S_n=\frac{\sum_{i=1}^n a_i-n\mu}{\sqrt n}, $$ then as $n\to\infty$, $S_n$ converges in distribution to a Gaussian random variable: this is the famous Central Limit Theorem. The analogous situation for RMT is the following: take i.i.d.'s $(a_1,a_2,...)$, perform an operation $X$ on them, $$ (a_1,a_2,...)\to(X_1,X_2,...), $$ and as $n\to\infty$ the $X_n$'s converge to the Gaudin distribution, or the Tracy-Widom distribution. The question is, ``What is $X$?'' Important progress towards answering this question has been made recently, and independently, by Baik-Suidan and Bodineau-Martin, but the full problem remains open and very challenging.

背景链接：https://arxiv.org/abs/0712.0849

17115 | AIM-PHYSICS-0025

## A two-vertex counterexample to color-number monotonicity

S / solved | AIM 研讨会问题汇编

**(5)** Let \(G\) be a finite graph with maximum degree \(\Delta\). This problem is about choosing a random coloring from the uniform distribution. Work with Glauber dynamics: pick a site at random, choose a color at random, and try to change the color at the chosen site. If this results in a proper coloring, make the change. If not, keep the current coloring. For \(q\geq \Delta+2\) it is known that this Markov chain is ergodic, i.e., that it connects the set of proper colorings. Let \(t=t(q)\) be the mixing time: the smallest time such that the total variation distance from the uniform distribution is at most \(1/e\) from any starting configuration. The question is: does \(t(q)\) decrease as \(q\) increases? Even on the square lattice with \(n\) sites there are curious gaps in our knowledge: we know that \(t(3)\) is polynomial in \(n\) (Luby, Randall and Sinclair: Proc. FOCS 1995) and we know that \(t(q)=O(n\log n)\) for \(q\geq6\). But, for \(q=4\) and \(q=5\) we do not even have a polynomial upper bound, even though there is overwhelming numerical evidence that there is a finite correlation length, and therefore that the mixing time is \(O(n\log n)\).

来源字段：https://aimath.org/WWN/phasetransition/phasetransition.pdf

17116 | OWR-13355-001

## Absolutely Continuous Diffraction and Dynamical Spectra

O / open | Oberwolfach 报告问题汇编

别名：Duplicate Diffraction-to-Dynamical-Spectrum Conjecture [OWR-13355-002]

If the diffraction spectrum of a quasicrystalline system has no absolutely continuous component, must the dynamical spectrum also have no absolutely continuous component?

来源字段：https://doi.org/10.4171/owr/2014/53

17117 | AMR-078-0008

## Absolutely Continuous Spectrum Under a Weighted L2 Condition

P / partially_solved | AMR 问题清单汇编

Let $V$ be a function on $\mathbb R^\nu$, $\nu\geq2$, satisfying $\int |x|^{-\nu+1}|V(x)|^2\,d^\nu x<\infty$. Prove that $-\Delta+V$ has absolutely continuous spectrum of infinite multiplicity on $[0,\infty)$.

背景链接：https://en.wikipedia.org/w/index.php?title=Simon_problems&oldid=1335881350#The_2000_list

17118 | OWR-14297742-003

## Accelerated Information Propagation in Bosonic Lattices

P / partially_solved | Oberwolfach 报告问题汇编

When does accelerated information propagation occur in bosonic lattice systems with nearest-neighbor or long-range interactions?

来源字段：https://doi.org/10.4171/owr/2023/39

17119 | OWR-3392-004

## Adapted Cocycles and Tensor Subsystems of Free Filtrations

P / partially_solved | Oberwolfach 报告问题汇编

What is the correct notion of a cocycle adapted to the free filtration, and can it yield adapted unitary cocycles directly from units of the tensor-product system? Does the largest tensor-product subsystem contained in a free product system generate the whole free product system?

来源字段：https://doi.org/10.4171/owr/2009/09

17120 | AMR-080-0007

### additional problems for integrable systems

O / open | AMR 问题清单汇编

In 2002, P. Zhou analyzed the behavior of solutions of the Cauchy problem for perturbations of the defocusing NLS equation i u_t + u_{xx}- 2|u|^2 u - \epsilon |u|^\ell u =0, u(x,t=0) = f(x) \to 0 \qquad as\quad |x|\to \infty. An open problem of great and basic interest is to analyze solutions of the Cauchy problem for perturbations of the focusing NLS equation i\tilde{u}_t+ \tilde{u}_{xx} +2|\tilde{u}|^2 u +\epsilon |\tilde{u}|^{\ell} \tilde{u} =0, \tilde{u}(x,t=0) = \tilde{f}(x) \to 0\qquad as\quad |x|\to \infty. The principal new difficulty in analyzing (eq14) over (eq13), is that for solutions $u(x,t)$ of (eq13) we have decay in the sup norm \sup_x |u(x,t) | \le c/t^{1/2} \to 0 \qquad as\quad t\to\infty. There are two open problems here. The first is to use Riemann--Hilbert/steepest-descent methods to prove the results in Venakides et al.\ rigorously. The second problem concerns the fact that if we replace the exponential forces $e^x$ by a general force $F(x)$, \ddx_k =F(x_{k-1}- x_k) - F(x_k - x_{k+1}), \qquad k\ge 1 then numerical simulations show that the solution of the shock problem for this system behaves in the same way as $t\to\infty$ as the Toda system, provided (eq16) has a 2-periodic solution x_k (t+T) = x_k(t), \quad T>0, \qqquad x_{k+2}(t) = x_k(t) for $k\in \mathbb{Z}$. Second open problem: Analyze the general shock problem (eq16) for suitably small perturbations $F(x)$ of $e^x$. One would like, rather, a definition which is intrinsic to probability

背景链接：https://arxiv.org/abs/1703.04931

17121 | OWR-14297742-002

### Adiabatic Expansions for Gapless Quantum Systems

O / open | Oberwolfach 报告问题汇编

Can the adiabatic-expansion framework, whose convergence follows from imaginary-time correlation estimates for gapped systems, be extended to gapless quantum systems even without interactions?

来源字段：https://doi.org/10.4171/owr/2023/39

17122 | OWR-9790367-005

### Affine-Virasoro Derivation of Nekrasov Blow-Up Identities

O / open | Oberwolfach 报告问题汇编

Derive Nekrasov's blow-up identity for the $N_f=4$, $\operatorname{SU}(2)$ instanton partition function from the affine-Virasoro vertex-operator-algebra identity under $\frac{\epsilon_1-\epsilon_2}{\epsilon_2}=b^2=-\frac{k+3}{k+2}$, including the required normalization of conformal blocks.

来源字段：https://doi.org/10.4171/owr/2022/16

17123 | OWR-16407-007

### All-Order Expansion of the Noncommutative $\Phi^4$ Model

S / solved | Oberwolfach 报告问题汇编

For the function $I_\lambda(a)$ arising from the boundary-value solution of the noncommutative $\Phi^4$ model, does the formal expansion $$ I_\lambda(a)=\sum_{n\geq1}\frac{\lambda^n}{n!}\frac{d^{n-1}}{da^{n-1}}\bigl(-\log(1+a)\bigr)^n -\lambda\sum_{n\geq1}\frac{\lambda^n}{n!}\frac{d^{n-1}}{da^{n-1}}\left[\frac{\bigl(-\log(1+a)\bigr)^n}{a}\right] $$ hold to every order in $\lambda$?

来源字段：https://doi.org/10.4171/owr/2018/32

17124 | OWR-3392-009

### Ambient-Algebra Independence of Exchangeable Independence

O / open | Oberwolfach 报告问题汇编

For the proposed notion of independence of subalgebras $B$ and $C$ contained in an algebra $A$ with state $\phi$, defined through an extremal exchangeable state on the infinite free product, is the dependence on the embedding algebra $A$ necessary?

来源字段：https://doi.org/10.4171/owr/2009/09

17125 | AIM-PHYSICS-0052

### An affine-defect test for Newtonian motions

P / partially_solved | AIM 研讨会问题汇编

Problem 15. Are there solutions to the n-body problem that fall in a certain affine class that are not relative equilibrium or some symmetric solutions? (c.f. Gerver's super-8) (Alain Chenciner) 6

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17126 | AIM-PHYSICS-0027

### An invariant certificate for caustic-forming Kerr rays

P / partially_solved | AIM 研讨会问题汇编

**(14) Plotting the caustics from conjugate points: Which photons (light rays) form the caustics?**

来源字段：https://aimath.org/WWN/lensing/lensing.pdf

17127 | OWR-13496-001

### Analytic Structures for Nonformal Deformation Quantization

P / partially_solved | Oberwolfach 报告问题汇编

What are the essential ingredients of a non-formal deformation quantization, including the associated Hilbert-algebra or von Neumann-algebra structure?

来源字段：https://doi.org/10.4171/owr/2015/11

17128 | OWR-15172-001

### Analytical Foundations of Magnetoelastic Material Models

P / partially_solved | Oberwolfach 报告问题汇编

For mathematical models of magnetoelastic materials, what is the role of the saturation constraint; how can models of physical effects on different length scales be reconciled; what dissipation models are reasonable and derivable microscopically; do energy functionals admit minimizers; can hysteresis and nucleation be read from material parameters; and what are the effective static and dynamical behaviors of composite materials with microscopic magnetic structure?

来源字段：https://doi.org/10.4171/owr/2016/51

17129 | OWR-14299511-010

### AQFT Locality of Unitary Vertex Operator Algebras

O / open | Oberwolfach 报告问题汇编

Prove that every unitary vertex operator algebra is local in the sense required to construct its associated algebraic quantum field theory conformal net.

来源字段：https://doi.org/10.4171/owr/2025/34

17130 | OWR-16411-011

### Area-Angular-Momentum Inequalities with Nonpositive Cosmological Constant

O / open | Oberwolfach 报告问题汇编

Extend the area-angular-momentum inequality $A\geq8\pi|J_S|$ to cosmological constant $\Lambda\leq0$, determine the role of extremal anti-de Sitter geometries in the sharp estimate, and prove the equality rigidity statement for the area-charge inequality.

来源字段：https://doi.org/10.4171/owr/2018/36

17131 | OWR-16407-005

### Arithmetic Nature of Delocalized Eta Invariants

O / open | Oberwolfach 报告问题汇编

What values can delocalized eta invariants of Dirac operators on universal covers take in general; in particular, are they always algebraic numbers?

来源字段：https://doi.org/10.4171/owr/2018/32

17132 | OWR-11787-005

### Asymptotic Metrics for Higher-Rank Monopoles

O / open | Oberwolfach 报告问题汇编

For $\operatorname{SU}(N)$-monopoles with maximal symmetry breaking, establish the quasi-isometric asymptotics of the monopole metric on the analogous compactification and determine whether they agree qualitatively with the $\operatorname{SU}(2)$ case.

来源字段：https://doi.org/10.4171/owr/2012/25

17133 | AMR-078-0010

### Asymptotics of Atomic Ionization Energy

O / open | AMR 问题清单汇编

For the $N$-electron Coulomb ground-state energy $E(N,Z)$, determine the asymptotics of the ionization energy $\delta E(Z)=E(Z,Z-1)-E(Z,Z)$ as $Z\to\infty$.

背景链接：https://en.wikipedia.org/w/index.php?title=Simon_problems&oldid=1335881350#The_2000_list

17134 | OWR-8415343-006

### Asymptotics of Expanding Vacuum Spacetimes Near Singularities

O / open | Oberwolfach 报告问题汇编

For a globally hyperbolic expanding vacuum spacetime with compact Cauchy hypersurfaces and a CMC foliation, what is the asymptotic behavior as the past singularity is approached, and what is the future behavior of the spacetime?

来源字段：https://doi.org/10.4171/owr/2021/40

17135 | OWR-14225-001

### Attainment of Constrained Domain-Wall Energies

O / open | Oberwolfach 报告问题汇编

For the domain-wall minimization problem, is the constrained infimum attained for every $0<\theta\leq\pi/2$ and every $q\in\mathbb{Z}$, when $q$ is the degree or winding number?

来源字段：https://doi.org/10.4171/owr/2016/17

17136 | OWR-14297742-004

### Aubin-Talenti Optimizers for Critical Dirac Potentials

O / open | Oberwolfach 报告问题汇编

In dimensions $d>1$ at the critical parameter $\alpha=\alpha_*(p,d)$, is the optimizing nonnegative potential for the Dirac spectral-gap problem of Aubin-Talenti form $V(x)=A/(1+B|x|^2)$?

来源字段：https://doi.org/10.4171/owr/2023/39

17137 | AIM-PHYSICS-0032

### Audible primitive two-torus areas on a flat three-torus

P / partially_solved | AIM 研讨会问题汇编

On a Riemannian three-manifold, is there a natural class of two-dimensional geometric objects whose data are related to the scalar Laplace spectrum analogously to the relation between closed-geodesic lengths and wave-trace singularities?

**1. When \(M\) is a 3-manifold are there some 2-dimensional objects that are related to \(\operatorname{Spec}(\Delta)\) in a similar way?**

“Closed totally geodesic surfaces are not the correct 2-dimensional objects to consider for #1, since there are too few of them.”

This attempt owns only the first:

The intended question is therefore:

The PDF boundary is unambiguous. Item 1 ends at “way?” Item 2, asking for an “area spectrum” of minimal surfaces, and item 3, asking to count minimal surfaces in \(T^4=T^2\times T^2\), are separate numbered questions and are excluded. The later “Questions (M.

The one source remark explicitly attached to item 1 is Mazzeo's:

The remarks about physical intuition, a Selberg trace formula, and the eta invariant follow the three-question cluster but do not explicitly identify item 1. The Selberg comment is relevant background to the preceding wave-trace discussion, but it is not treated here as a claim that surfaces replace geodesics.

The mathematical analysis below uses that standard convention rather than silently relying on the damaged display.

来源字段：https://aimath.org/WWN/l2harmonic/l2harmonic.pdf

17138 | AIM-PHYSICS-0008

### Axis and low-dimensional cases of Levitin's disk-intersection conjecture

P / partially_solved | AIM 研讨会问题汇编

The spectrum is symmetric with respect to both $\mathbb R$ and $i\mathbb R$. Its non-real eigenvalues are known to lie in \[ D(c,2)\cup D(-c,2),\qquad D(a,2)=\{\lambda\in\mathbb C:|\lambda-a|\leq2\}. \] Prove that every $\lambda\in\sigma(BA)\setminus\mathbb R$ actually lies in the intersection: \[ |\lambda-c|\leq2\quad\text{and}\quad|\lambda+c|\leq2. \]

来源字段：https://aimath.org/pastworkshops/nonselfadjointproblems.pdf

17139 | OWR-16411-001

### Axisymmetry of Spherical Near-Horizon Geometries

P / partially_solved | Oberwolfach 报告问题汇编

Is every solution of the near-horizon-geometry equations on $S^2$ axisymmetric and therefore induced by a degenerate Kerr metric?

来源字段：https://doi.org/10.4171/owr/2018/36

17140 | OWR-14751-005

### Ball-Shaped Minimizers of Competing Nonlocal Energies

S / solved | Oberwolfach 报告问题汇编

For $$ E_\alpha(m)=\inf\left\{E_\alpha[\rho]:0\leq\rho\leq1,\ \int_{\mathbb R^3}\rho=m\right\}, $$ where $$ E_\alpha[\rho]=\frac12\iint_{\mathbb R^3\times\mathbb R^3}\rho(x)\left(|x-y|^{-1}+|x-y|^\alpha\right)\rho(y)\,dx\,dy, $$ does there exist $m'_{c_2}(\alpha)<\infty$ such that, whenever $m>m'_{c_2}(\alpha)$, every minimizer is the characteristic function of a ball of volume $m$?

来源字段：https://doi.org/10.4171/owr/2016/43

17141 | OWR-8415343-005

### Bartnik’s Stationary Vacuum Extension Conjecture

O / open | Oberwolfach 报告问题汇编

Bartnik's stationary vacuum extension conjecture: let $(B^3,g_0,k_0)$ satisfy the dominant energy condition, with boundary $S^2$ whose expansion $\theta=H_{g_0}+\operatorname{tr}_{S^2}k_0$ is not everywhere nonpositive. Does there exist a unique asymptotically flat stationary vacuum initial-data set $(\mathbb{R}^3\setminus B^3,g,k)$ whose Bartnik boundary data $B(g,k)$ agree with $B(g_0,k_0)$ on $S^2$?

来源字段：https://doi.org/10.4171/owr/2021/40

17142 | OWR-2654831-005

### Bilinear Forms behind Resistor and Ising Integrable Equations

O / open | Oberwolfach 报告问题汇编

Why do symmetric bilinear forms lead to the resistor subvariety and the BKP equation, whereas antisymmetric bilinear forms lead to the Ising subvariety and the CKP equation?

来源字段：https://doi.org/10.4171/owr/2020/35

17143 | DARPA-005

### Biological Quantum Field Theory

P / partially_solved | DARPA 挑战

Apply quantum and statistical field theory methods to model and potentially control pathogen evolution.

背景链接：https://web.math.utk.edu/~vasili/refs/darpa07.MathChallenges.html

17144 | OWR-13673-003

### Bisognano-Wichmann Covariance for Standard-Subspace Nets

O / open | Oberwolfach 报告问题汇编

Given a local net of standard subspaces $H(O)$ on double cones $O\subset\mathbb R^4$, covariant under a positive-energy unitary Poincaré representation $U$, does another positive-energy representation $U'$ on the same Hilbert space make $H$ covariant and satisfy the Bisognano-Wichmann property?

来源字段：https://doi.org/10.4171/owr/2015/16

17145 | OWR-4536-006

### Boltzmann-Grad Limits for the Lorentz Gas

S / solved | Oberwolfach 报告问题汇编

Given a Lorentz gas in the Boltzmann-Grad limit with initial distribution $f_0(x,v)$, does $f^\varepsilon(x,v,t)$ converge as $\varepsilon\to0$ to a limit $f(x,v,t)$, and, if so, what equation does $f$ satisfy?

来源字段：https://doi.org/10.4171/owr/2010/54

17146 | AMR-077-0013

### Bose-Einstein Condensation in Continuum Models

O / open | AMR 问题清单汇编

Prove that Bose-Einstein condensation occurs in a continuum model of a weakly interacting Bose gas, or determine whether the long-held assertion fails.

背景链接：https://web.math.princeton.edu/~aizenman/OpenProblems_MathPhys/9801.BEcond.html

17147 | OWR-4536-003

### Bose-Einstein Condensation in Quantum Boltzmann Dynamics

S / solved | Oberwolfach 报告问题汇编

For the quantum Boltzmann equation for Bose-Einstein particles, does there exist a measure solution $F_t$ and $t_0\in(0,T)$ with $F_{t_0}(\{0\})=\mu_{t_0}(\{0\})>0$, or at least with $\limsup_{\varepsilon\to0^+,\ 0\leq t\leq t_0}F_t([0,\varepsilon])>0$?

来源字段：https://doi.org/10.4171/owr/2010/54

17148 | AMR-081-0004

### Bound Entanglement with Negative Partial Transpose

O / open | AMR 问题清单汇编

Determine whether there exist bound entangled bipartite quantum states with negative partial transpose.

背景链接：https://arxiv.org/abs/2002.03233

17149 | AMR-078-0009

### Bounded Excess Electrons

P / partially_solved | AMR 问题清单汇编

For the $N$-electron Coulomb Hamiltonian with nuclear charge $Z$, let $N_0(Z)$ be the least $N$ after which adding electrons no longer lowers the ground-state energy. Prove that $N_0(Z)-Z$ remains bounded as $Z\to\infty$.

背景链接：https://en.wikipedia.org/w/index.php?title=Simon_problems&oldid=1335881350#The_2000_list

17150 | OWR-1109-001

### Bounded Local Observables in Factorizing Quantum Field Theories

P / partially_solved | Oberwolfach 报告问题汇编

For the wedge-local net associated with a factorizing two-dimensional S-matrix, do nontrivial observables localized in bounded double cones exist? Equivalently, for nested wedges $W_1\subset W_2$, is the double-cone algebra $A(W_1)'\cap A(W_2)$ nontrivial?

来源字段：https://doi.org/10.4171/owr/2005/48

17151 | OWR-17292-011

### Canonical Equivalence of Complex and Lagrangian Quantizations

P / partially_solved | Oberwolfach 报告问题汇编

Given a compact symplectic manifold equipped both with a compatible-complex-structure quantization and with a Lagrangian-fibration quantization, construct a canonical isomorphism between the resulting quantum Hilbert spaces.

来源字段：https://doi.org/10.4171/owr/2019/49

17152 | AIM-PHYSICS-0004

### Canonical versus abstract hyper-Kähler structures on cotangent bundles

P / partially_solved | AIM 研讨会问题汇编

**Recovered question.** Let \(M\) be a Kähler manifold. Must \(T^*M\) admit a hyper-Kähler structure?

来源字段：http://aimpl.org/nhquantumsymp/2/

17153 | OWR-4338-002

### Casimir-Connection Monodromy and Quantum Weyl Groups

P / partially_solved | Oberwolfach 报告问题汇编

Is the monodromy of the trigonometric Casimir connection equivalent to the quantum Weyl-group action of the affine braid group on finite-dimensional modules over the quantum loop algebra $U_{\hbar}(L\mathfrak g)$?

来源字段：https://doi.org/10.4171/owr/2010/25

17154 | AIM-PHYSICS-0033

### Centered asymptotics for a Gibbons--Hawking subfamily

P / partially_solved | AIM 研讨会问题汇编

1. Understand the asymptotics of these spaces (for a start see work of Bielawksi-Dancer [2] and Gibbons-Rychnenkova [6]).

Understand the Hodge cohomology of these spaces from a systematic point of view.

来源字段：https://aimath.org/WWN/l2harmonic/l2harmonic.pdf

17155 | AIM-PHYSICS-0048

### Circulant mass compatibility and a six-body temporal Fourier obstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 11. Do choreographies imply equal masses?** (Alain Chenciner)

Here choreography means equal time shifts, same curve.

Alain Chenciner proved it for \(n<6\).

Martin Celli proved it for logarithmic potential for any \(n\).

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17156 | OWR-14299580-002

### Classification and Applications of Yang-Baxter Operators

O / open | Oberwolfach 报告问题汇编

Classify $3\times3$ Yang-Baxter operators up to Drinfeld twists, equivalence of their braid-group representations, and spectral data. What geometry do the irreducible components, dimensions, and singularities of the variety of Yang-Baxter operators have?

来源字段：https://doi.org/10.4171/owr/2025/49

17157 | OWR-11448-002

### Classification of Gapped Ground-State Phases

P / partially_solved | Oberwolfach 报告问题汇编

What is a gapped ground-state phase for an infinite quantum spin system, when phases are defined by differentiable paths of sufficiently short-range interactions with a uniform positive spectral gap?

来源字段：https://doi.org/10.4171/owr/2011/50

17158 | OWR-14743-001

### Classification of Quantizations of Symplectic and Poisson Manifolds

O / open | Oberwolfach 报告问题汇编

Given a symplectic or Poisson manifold, how can all possible quantizations be classified, including deformation quantizations of its algebra of functions?

来源字段：https://doi.org/10.4171/owr/2016/35

17159 | OWR-3392-014

### Classification of Spatial and Nonspatial Markov Semigroups

P / partially_solved | Oberwolfach 报告问题汇编

For Markov semigroups, determine whether nontrivial nonspatial examples exist on $B(H)$; whether spatial Markov semigroups on $B(H)$ can have non-Fock product systems; whether all spatial examples are known or further ones exist; whether such further examples could yield non-type-I Markov semigroups; and whether the theory of form generators can be adapted from $B(H)$ to general von Neumann algebras.

来源字段：https://doi.org/10.4171/owr/2009/09

17160 | OWR-17292-003

### Classification of Subnets of Conformal Nets

P / partially_solved | Oberwolfach 报告问题汇编

Given a conformal net $\mathcal A$, characterize all subnets $\mathcal B\subseteq\mathcal A$ by necessary and sufficient conditions that include information not determined by $\operatorname{Rep}(\mathcal A)$ alone.

来源字段：https://doi.org/10.4171/owr/2019/49

17161 | AIM-PHYSICS-0054

### Closure-area limit sets and geometric phase on the shape sphere

P / partially_solved | AIM 研讨会问题汇编

**Problem 17. Problems on the shape space for the three-body problem (Wu-Yi Hsiang).**

- Are there solutions whose \(\alpha\)- (or \(\omega\)-) limit set is a limit cycle? - The area of the unit shape sphere is \(\pi\). Conjecture: for any \(0<A_0\leq\pi\), there is a solution whose shape curve has its closure with area \(A_0\). - Can two closed shape curves with the same homotopy class be deformed to one another by closed shape curves? Conjecture: No. But then determine connected components.

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17162 | OWR-14298586-004

### Cloud-Organization Limit Cycles

O / open | Oberwolfach 报告问题汇编

Explain the mechanism producing limit cycles in shallow cumulus organization. Determine how external forcings affect their size, frequency, and amplitude; whether they respond quasi-statically to time-dependent forcing; how the diurnal radiative cycle and realistic open boundaries modify them; and how relevant they are to cloud feedback.

来源字段：https://doi.org/10.4171/owr/2024/31

17163 | AIM-PHYSICS-0043

### Coercivity and collision boundaries for the Broucke-Henon class (1,0,1)

P / partially_solved | AIM 研讨会问题汇编

Prove the existence of Broucke--Hénon orbit for the planar 3-body problem. Do Broucke--Hénon orbits exist for small + big masses? Is the Broucke--Hénon solution a minimizer in the class (1,0,1)? (Andrea Venturelli)

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17164 | AIM-PHYSICS-0044

### Coercivity, collision support, and the identity braid for the class (1,0,-1)

P / partially_solved | AIM 研讨会问题汇编

Is \((1,0,-1)\) realized by a collision-free periodic orbit (see figure 3)? If exists, is it minimizing or not? Is trivial braid realized by a collision-free periodic solution? (Andrea Venturelli)

\[ k_{ij}=\operatorname{Deg}(x_i-x_j,0), \]

the number of oriented turns of the side vector \(x_i-x_j\). Problem 4 fixes the displayed coordinate order as \((k_{12},k_{23},k_{31})\). The exact Problem 7 is:

Thus the first clause means

\[ k_{12}=1,\qquad k_{23}=0,\qquad k_{31}=-1. \tag{1} \]

There is no sign reversal hidden in the last coordinate: multiplication of a loop in \(\mathbb C\setminus\{0\}\) by \(-1\) is a fixed rotation and does not change degree, so \(k_{31}=k_{13}\).

Figure 3 is on the following page and is captioned exactly “A possible solution with homology class \((1,0,-1)\).” Inspection of the original vector figure shows a schematic collection of directed tracks. The PDF supplies no masses, period, initial conditions, residual, computer-assisted enclosure, or symmetry group for it. Consequently the drawing is evidence for the intended topology, not a proof or reproducible numerical orbit. The bullet below the figure about a systematic numerical search belongs to Problem 8, not Problem 7.

The source does not specify masses or a braid quotient. Below, the analytic statements use arbitrary fixed positive masses and fixed period. “Full trivial pure braid” means the identity in the labeled inertial configuration space. Reduced/projective conventions are discussed separately.

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17165 | AIM-PHYSICS-0063

### Compatibility and reversible shooting for connections between relative equilibria

P / partially_solved | AIM 研讨会问题汇编

Existence proof for homoclinic and heteroclinic orbits between unstable relative equilibrium solutions. (Rick Moeckel)

In the collision-free planar Newtonian \(n\)-body problem, after fixing nonzero angular momentum and reducing rotations, find homoclinic or heteroclinic trajectories between unstable reduced equilibria; then reconstruct them as trajectories asymptotic to the corresponding rotating relative-equilibrium orbits.

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17166 | OWR-8415352-007

### Completeness of Realignment and SIC-POVM Entanglement Tests

P / partially_solved | Oberwolfach 报告问题汇编

Without preliminary reshuffling of the subsystems, is the family of realignment and SIC-POVM entanglement tests complete? Which multipartite mixed states can these tests detect, and how do the two families compare in detection performance?

来源字段：https://doi.org/10.4171/owr/2021/49

17167 | AIM-PHYSICS-0002

### Complex Bohr-Sommerfeld action and a stability obstruction

P / partially_solved | AIM 研讨会问题汇编

**Bohr-Sommerfeld**

Is there a Bohr-Sommerfeld quantization rule for \(a^w(x,hD)\)? Is there a link to the usual Bohr-Sommerfeld rule for Hermitian operators?

来源字段：http://aimpl.org/nhquantumsymp/1/

17168 | OWR-13673-010

### Conformal Field Theory Realization of the Asaeda-Haagerup Subfactor

O / open | Oberwolfach 报告问题汇编

Does the Asaeda-Haagerup subfactor arise from a conformal field theory?

来源字段：https://doi.org/10.4171/owr/2015/16

17169 | OWR-13673-001

### Conformal Field Theory Realizations of Subfactors

O / open | Oberwolfach 报告问题汇编

Does every subfactor arise in conformal field theory?

来源字段：https://doi.org/10.4171/owr/2015/16

17170 | OWR-17292-001

### Conformal Field Theory Realizations of Subfactors

O / open | Oberwolfach 报告问题汇编

Does every finite-index subfactor admit a counterpart in conformal field theory, and can such a conformal net or vertex operator algebra be constructed directly from the subfactor data?

来源字段：https://doi.org/10.4171/owr/2019/49

17171 | OWR-11101914-001

### Conformal Invariance of Interacting Long-Range Fermionic Models

O / open | Oberwolfach 报告问题汇编

Prove conformal invariance of the non-Gaussian renormalization-group fixed point for long-range fermionic models with symplectic symmetry.

来源字段：https://doi.org/10.4171/owr/2022/33

17172 | OWR-785-005

### Connections Between CGLMP Inequalities and Quantum Field Theory

O / open | Oberwolfach 报告问题汇编

What is the mathematical link between the CGLMP inequality and quantum field theory?

来源字段：https://doi.org/10.4171/owr/2005/04

17173 | AIM-PHYSICS-0055

### Constant normalized area in the three-body problem

P / partially_solved | AIM 研讨会问题汇编

Problem 18. Let Δ be the area of the triangle, I be the moment of inertia. Find periodic solutions to the (planar or spatial) three-body problem such that Δ

I

is a constant. (Wu-Yi Hsiang)

If the angular momentum is zero, then there is none.

What motions have shapes on the meridian passing the Euler configurations?

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17174 | OWR-1736-001

### Constitutive Models for Oscillatory Micellar Sedimentation

O / open | Oberwolfach 报告问题汇编

For the nontransient oscillatory sedimentation of a sphere at low Reynolds number in a wormlike micellar fluid, what constitutive equation or constitutive model gives a mathematical understanding of the observed oscillations?

来源字段：https://doi.org/10.4171/owr/2007/53

17175 | OWR-13495-002

### Construction of K-Matrices for Coideal Subalgebras

P / partially_solved | Oberwolfach 报告问题汇编

Develop a general method for constructing rational and trigonometric $K$-matrices satisfying the matrix reflection equations associated with twisted Yangians and quantum-symmetric-pair coideal subalgebras.

来源字段：https://doi.org/10.4171/owr/2015/10

17176 | OWR-14297737-003

### Continuation of Nordheim Condensation Beyond Dirac-Mass Formation

O / open | Oberwolfach 报告问题汇编

After finite-time formation of a Dirac mass at $\omega=0$ in the Nordheim equation, determine whether the solution can be continued as $f(t,\omega)=n(t)\delta_0/\sqrt\omega+h(t,\omega)$ with $h(t,\cdot)$ a function.

来源字段：https://doi.org/10.4171/owr/2023/38

17177 | AMR-078-0014

### Continuity of the Integrated Density of States

P / partially_solved | AMR 问题清单汇编

Prove that the integrated density of states $k(E)$ is continuous as a function of the energy $E$.

背景链接：https://en.wikipedia.org/w/index.php?title=Simon_problems&oldid=1335881350#The_2000_list

17178 | OWR-1266-004

### Continuous-Spectrum Analysis of Nelson-Type Models

P / partially_solved | Oberwolfach 报告问题汇编

Can the limiting-absorption and local-decay method be extended to $\sigma(H)\setminus\sigma_{\mathrm{pp}}(H)$ for the Nelson model and to $\sigma(H)\setminus(\sigma_{\mathrm{pp}}(H)\cup(-\infty,\Sigma_{\mathrm{el}}))$ for the standard model of nonrelativistic QED, where $\Sigma_{\mathrm{el}}=\inf\sigma(H_{\mathrm{el}})$?

来源字段：https://doi.org/10.4171/owr/2006/17

17179 | OWR-14299085-005

### Counterexamples to Strongly Continuous Many-Fermion Dynamics

O / open | Oberwolfach 报告问题汇编

For the unregularized continuum fermion Hamiltonian, find an interaction $V$ and states for which the Heisenberg evolution of $a(f)$ fails to be strongly continuous.

来源字段：https://doi.org/10.4171/owr/2025/7

17180 | OWR-4415-007

### Coupling and Learning Strategies for Climate Supermodels

P / partially_solved | Oberwolfach 报告问题汇编

For climate supermodels, which connections and variables should be used, how much data and what learning strategies are needed, how should slow oceanic scales and changing climates be handled, and what restrictions follow from model class, balances, conservation laws, or physical insight?

来源字段：https://doi.org/10.4171/owr/2010/34

17181 | OWR-2495-002

### Critical Transition and Pattern Scales in Shear Turbulence

O / open | Oberwolfach 报告问题汇编

In shear flows, why does transition to turbulence occur near the critical Reynolds number $R_{e,c}$ so often, and why is the resulting pattern's length scale so large?

来源字段：https://doi.org/10.4171/owr/2008/57

17182 | OWR-12009-003

### Critical Ultrarelativistic Black-Hole Collision Solutions

O / open | Oberwolfach 报告问题汇编

For equal-mass nonspinning ultrarelativistic black-hole collisions near critical impact parameter, does a unique critical solution exist, and can its high-energy behavior be described by an adiabatic, helical-Killing, or colliding-impulsive-wave approximation?

来源字段：https://doi.org/10.4171/owr/2012/37

17183 | OWR-1544-002

### Current Criteria for Phase Separation in Driven Systems

O / open | Oberwolfach 报告问题汇编

Characterize when phase separation occurs in one-dimensional driven density-conserving systems and how this is related to traffic-jamming phase transitions, including a criterion in terms of the size dependence of steady-state currents.

来源字段：https://doi.org/10.4171/owr/2007/28

17184 | OWR-1114-003

### Deformations and Massless Spectra of Smooth Deligne-Mumford Stacks

P / partially_solved | Oberwolfach 报告问题汇编

For string compactification on a smooth Deligne-Mumford stack, how should deformations and massless spectra be defined so that they agree with the physical theory?

来源字段：https://doi.org/10.4171/owr/2005/53

17185 | OWR-1463-001

### Density of Current-Current Correlation Measures

P / partially_solved | Oberwolfach 报告问题汇编

For the random Schrödinger operator $H_\omega$, let the current-current correlation measures be $$dm_{ij}(E,E')=\mathbb E\!\left[\langle0|\nabla_iH_\omega\,E_{H_\omega}(dE)\,\nabla_jH_\omega\,E_{H_\omega}(dE')|0\rangle\right],$$ and put $dm=\sum_i dm_{ii}$. Does the positive measure $dm$ have a density $m(E,E')$ with respect to Lebesgue measure?

来源字段：https://doi.org/10.4171/owr/2007/12

17186 | OWR-14298167-003

### Dissipation and Interface Misorientation in Phase-Field Plasticity

O / open | Oberwolfach 报告问题汇编

How can energy storage be separated from dissipation in plasticity, crystal misorientation represented at phase-field interfaces, sharp-interface processes calibrated, dissipation in discrete phase transformations quantified, and energetically degenerate microstructures distinguished computationally?

来源字段：https://doi.org/10.4171/owr/2024/11

17187 | OWR-14298591-007

### Dynamics Near Generic Spacetime Singularities

O / open | Oberwolfach 报告问题汇编

Give a rigorous dynamical description of generic solutions of the Einstein equations near spacetime singularities, including the extent to which Belinski-Khalatnikov-Lifshitz dynamics governs their asymptotic behavior.

来源字段：https://doi.org/10.4171/owr/2024/36

17188 | OWR-3392-003

### E0-Semigroups from Free Product Systems

O / open | Oberwolfach 报告问题汇编

Which $E_0$-semigroups (necessarily noises) arise from the free-product-system construction and its associated free left dilation?

来源字段：https://doi.org/10.4171/owr/2009/09

17189 | AIM-PHYSICS-0050

### Effective full-isotropy constraints for time-space-label symmetries

P / partially_solved | AIM 研讨会问题汇编

Problem 13. What subgroups of \(O(2)\times O(d)\times S_n\) can be realized as symmetries of solutions? (Davide Ferrario)

Can symmetry group arise differently? (Alain Chenciner)

Remarks. There is no upper bound to the order of the subgroup.

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17190 | OWR-2493-001

### Efficient Coarse-Grained Free-Energy Computation

O / open | Oberwolfach 报告问题汇编

For a one-dimensional atomistic system with repatom coordinates $X_r$ and eliminated coordinates $X_c$, define $A(X_r)=-\beta^{-1}\log\int\exp(-\beta V(X_r,X_c))\,dX_c$. Can $A(X_r)/N_c$ be computed efficiently as $N_c\to\infty$?

来源字段：https://doi.org/10.4171/owr/2008/55

17191 | OWR-3479-001

### Efficient Reduced Free-Energy Computation for Atomistic Systems

O / open | Oberwolfach 报告问题汇编

For $A(X_r)=-\beta^{-1}\log\int_{\mathbb R^{dN_c}}e^{-\beta V(X_r,X_c)}\,dX_c$, can $A(X_r)/N_c$ be computed efficiently as $N_c\to+\infty$?

来源字段：https://doi.org/10.4171/owr/2009/29

17192 | OWR-8415343-003

### Emergence of Isotropic Big-Bang Singularities

O / open | Oberwolfach 报告问题汇编

Within mathematical models of cosmological singularity formation, what mechanism or initial-data principle explains the emergence of a homogeneous, isotropic Big-Bang singularity with vanishing Weyl curvature from more general initial conditions?

来源字段：https://doi.org/10.4171/owr/2021/40

17193 | OWR-5160-001

### Energy Cost of Density Defects in Fermi Seas

P / partially_solved | Oberwolfach 报告问题汇编

Determine the optimal energy cost, or sharp Lieb-Thirring-type lower bound, for creating a density perturbation including a hole in an infinitely extended free Fermi sea of prescribed positive density $\rho_0$.

来源字段：https://doi.org/10.4171/owr/2011/32

17194 | OWR-1598-003

### Energy Transmission in Strongly Pinned Networks

O / open | Oberwolfach 报告问题汇编

For a heat-conduction network whose pinning potential $V$ is much stronger than its interaction potential $U$, can one prove that there is sufficient energy transmission along the network?

来源字段：https://doi.org/10.4171/owr/2007/42

17195 | OWR-14298586-003

### Entrainment-Mediated Tropical-Cyclone Intensification

O / open | Oberwolfach 报告问题汇编

Using three-dimensional high-resolution simulations, verify the asymptotically derived pathway in which cloud-tower entrainment produces a radial mean flow that transports angular momentum inward and intensifies a tropical cyclone.

来源字段：https://doi.org/10.4171/owr/2024/31

17196 | AMR-077-0008

### Entropy Production in Nonequilibrium Statistical Mechanics

P / partially_solved | AMR 问题清单汇编

Give a fundamental and experimentally accessible definition of the entropy creation rate for general classical systems in stationary nonequilibrium states under external nonconservative forces balanced on average by thermostat forces, without restricting to systems close to equilibrium.

背景链接：https://web.math.princeton.edu/~aizenman/OpenProblems_MathPhys/9806.EntropyProd.html

17197 | OWR-8415343-014

### Equality of Capacity-Volume and ADM Mass

S / solved | Oberwolfach 报告问题汇编

For every asymptotically flat Riemannian three-manifold $(M^3,g)$, does the capacity-volume mass equal the ADM mass, $m_{\mathrm{CV}}(M,g)=m_{\mathrm{ADM}}(M,g)$?

来源字段：https://doi.org/10.4171/owr/2021/40

17198 | AIM-PHYSICS-0026

### Equatorial symmetry and a Jacobian-jet test for local Kerr caustic cusps

P / partially_solved | AIM 研讨会问题汇编

(1) No analytic description of relativistic caustics (except for the intersections on the equatorial plane. (2) No analytic description of gravitational lensing out of the equatorial plane. (3) No idea (analytical or numerical) for what happens to higher order caustics.

**(1) No analytic description of relativistic caustics (except for the intersections on the equatorial plane).**

Describe the critical values of the Kerr lens map analytically, including the local singularity type and the way equatorial caustic intersections extend into off-equatorial caustic arcs or tubes.

来源字段：https://aimath.org/WWN/lensing/lensing.pdf

17199 | OWR-14744-001

### Equivalence of Euclidean AdS/CFT Generating Functionals

P / partially_solved | Oberwolfach 报告问题汇编

For the Euclidean AdS/CFT model on hyperbolic space with exponential interaction and an infrared cutoff, is the boundary-value generating functional equivalent, up to a rescaling $h\mapsto ch$ for some $c>0$, to the ordinary quantum-field-theoretic generating functional with a boundary source term? What happens to this equivalence in the thermodynamic limit?

来源字段：https://doi.org/10.4171/owr/2016/36

17200 | OWR-17292-002

### Equivalent Bicommutant Categories from Nonisomorphic Conformal Nets

O / open | Oberwolfach 报告问题汇编

Do there exist large families of pairwise nonisomorphic conformal nets whose associated bicommutant categories are equivalent as tensor categories?

来源字段：https://doi.org/10.4171/owr/2019/49

17201 | OWR-14298167-004

### Ericksen Universal Deformations of Incompressible Solids

O / open | Oberwolfach 报告问题汇编

Classify all constant-principal-invariant universal deformations of homogeneous incompressible isotropic solids.

来源字段：https://doi.org/10.4171/owr/2024/11

17202 | OWR-11101914-002

### Euclidean Extension of Hierarchical Susceptibility Asymptotics

O / open | Oberwolfach 报告问题汇编

Extend the universal finite-size susceptibility-scaling asymptotics for the four-dimensional hierarchical $|\phi|^4$ model to the Euclidean model with periodic boundary conditions.

来源字段：https://doi.org/10.4171/owr/2022/33

17203 | OWR-15960-001

### Euclidean Reconstruction of Relativistic Quantum Field Theories

P / partially_solved | Oberwolfach 报告问题汇编

Given a Euclidean classical field theory, when does analytic continuation from Euclidean time to real time produce a sensible relativistic quantum field theory?

来源字段：https://doi.org/10.4171/owr/2017/55

17204 | AIM-PHYSICS-0005

### Exact complex-linear spin propagation and a nonlinear coherence-loss benchmark

P / partially_solved | AIM 研讨会问题汇编

**Evolution of Coherent States on SU(2).** Determine the quantum evolution in the semiclassical limit of initial coherent states on SU(2) with a Hamiltonian that is either non-Hermitian or nonlinear in the Lie algebra generators.

来源字段：http://aimpl.org/nhquantumsymp/3/

17205 | AIM-PHYSICS-0064

### Exact reversible shooting criterion for halo-orbit bifurcation

P / partially_solved | AIM 研讨会问题汇编

Prove the existence of spatial, symmetric periodic halo orbits which bifurcate from a planar Lyapunov periodic-orbit family near one of the collinear libration points $L_1,L_2,L_3$, at a vertical critical orbit (a $+1$ vertical Floquet multiplier).

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17206 | AIM-PHYSICS-0003

### Exact zero-Kerr spectrum and high-energy Kerr localization

P / partially_solved | AIM 研讨会问题汇编

The canonical AIMPL record asks for the eigenvalue distribution of $$ H=\omega a^\dagger a+\chi(a^\dagger a)^2-i\gamma aa^\dagger +\beta(a^\dagger+a), \qquad \omega,\chi,\gamma,\beta\in\mathbb R. $$ With the standard bosonic commutation relation $[a,a^\dagger]=I$ and number operator $N=a^\dagger a$, one has $$ aa^\dagger=N+1. $$ Therefore the exact canonical operator is $$ \boxed{H=\chi N^2+(\omega-i\gamma)N-i\gamma I+\beta(a+a^\dagger).} $$

There is a source-convention discrepancy worth recording. The two operators differ by the scalar $-i\gamma I$. Thus their spectra have exactly the same shape and multiplicities, shifted vertically by $-i\gamma$.

The natural setting is the one-mode bosonic Fock space $\mathcal F=\ell^2(\mathbb N_0)$, with orthonormal number states $e_n$ and $$ ae_n=\sqrt n\,e_{n-1},\qquad a^\dagger e_n=\sqrt{n+1}\,e_{n+1},\qquad Ne_n=ne_n. $$ The prompt does not state a domain. For $\chi\ne0$, the closed realization used below has domain $\mathcal D(N^2)$. For $\chi=0$ but $\alpha:=\omega-i\gamma\ne0$, it has domain $\mathcal D(N)$. The degenerate case $\chi=\alpha=0$ is treated separately.

Physically, $\chi N^2$ is a Kerr nonlinearity up to a linear frequency renormalization, $\beta(a+a^\dagger)$ is a coherent one-photon drive, and $-i\gamma N$ is the loss term in an effective no-jump Hamiltonian (up to normalization of $\gamma$). The full open-system dynamics is governed by a Lindblad generator, whose spectrum is not the spectrum asked for here.

来源字段：http://aimpl.org/nhquantumsymp/1/

17207 | AMR-078-0013

### Existence of Quantum Crystals

O / open | AMR 问题清单汇编

Prove that, as the number of nuclei tends to infinity, the ground state of some neutral system of nuclei and electrons approaches a periodic limit, establishing the existence of crystals from quantum principles.

背景链接：https://en.wikipedia.org/w/index.php?title=Simon_problems&oldid=1335881350#The_2000_list

17208 | OWR-13684-002

### Existence of Time-Dependent Bivariational Quantum Dynamics

O / open | Oberwolfach 报告问题汇编

For the time-dependent bivariational initial-value problem used to approximate quantum dynamics, does an approximate trajectory exist, and under what mathematical conditions?

来源字段：https://doi.org/10.4171/owr/2015/27

17209 | AMR-078-0001

### Extended States in the Anderson Model

O / open | AMR 问题清单汇编

Prove that the Anderson model has purely absolutely continuous spectrum in dimension $\nu\geq 3$, for suitable disorder width $b-a$, in some energy range.

背景链接：https://en.wikipedia.org/w/index.php?title=Simon_problems&oldid=1335881350#The_2000_list

17210 | AMR-077-0005

### Extended States with Extensive Disorder

P / partially_solved | AMR 问题清单汇编

Establish, in some energy range, the existence of extended eigenstates or continuous spectrum for linear operators with extensive disorder, such as a discrete Schrödinger operator with an i.i.d. random potential acting on $\ell^2(\mathbb{Z}^d)$. Also clarify the borderline case $d=2$.

背景链接：https://web.math.princeton.edu/~aizenman/OpenProblems_MathPhys/9809.ExtStates.html

17211 | OWR-3392-012

### Extending Dilated E0-Semigroups Beyond GNS Representations

O / open | Oberwolfach 报告问题汇编

Let a Markov semigroup have a dilation to an automorphism or $E_0$-semigroup on an algebra $A$, and suppose $A$ is generated by the image of the original algebra and its time shifts. Must the dilation respect the GNS kernel, and must the induced $E_0$-semigroup on the represented subalgebra extend to all of $\mathcal B^a(E)$? What weaker minimality hypotheses on $A$ suffice?

来源字段：https://doi.org/10.4171/owr/2009/09

17212 | OWR-14299913-005

### Extending Trisection Invariants to Four-Dimensional TQFTs

O / open | Oberwolfach 报告问题汇编

Extend the trisection invariants of smooth four-manifolds to a four-dimensional topological quantum field theory, or identify an obstruction to such an extension.

来源字段：https://doi.org/10.4171/owr/2026/12

17213 | OWR-3392-010

### Extremal Exchangeability Beyond Natural Noncommutative Products

P / partially_solved | Oberwolfach 报告问题汇编

For the exchangeability-based notion of independence, does extremality hold for the free and Boolean products? Are there examples beyond the tensor, free, Boolean, monotone, and antimonotone universal products, and what changes when extremal shift-invariance is replaced by shift-invariance with trivial tail algebra?

来源字段：https://doi.org/10.4171/owr/2009/09

17214 | OWR-785-004

### Facial Structure of Classical and Quantum Correlation Sets

P / partially_solved | Oberwolfach 报告问题汇编

In a multipartite Bell correlation scenario, let $P \subset Q \subset R$ denote respectively the local hidden-variable polytope ($P$), the convex body of quantum-mechanically realizable correlation distributions ($Q$), and the nonsignalling polytope ($R$). Are all facets (faces) of $P$ that are not contained in faces of $R$ nontrivial Bell inequalities? Furthermore, is the boundary of the quantum correlation body $Q$ covered by probability distributions arising from projective measurements on pure states in Hilbert spaces of minimal dimension, with all measurement projectors having rank one?

来源字段：https://doi.org/10.4171/owr/2005/04

17215 | OWR-15998-002

### Factorial Lattice-Regularity Bounds for Equilibrium Displacements

O / open | Oberwolfach 报告问题汇编

Let $\bar u:\mathbb Z^d\to\mathbb R^d$ be an equilibrium displacement field for a crystalline defect. Is there a constant $C>0$ such that, for every $j\geq1$ and lattice site $\ell$ away from the defect,

$$\lvert D^j\bar u(\ell)\rvert\leq Cj!\lvert\ell\rvert^{1-d-j}$$

hold?

来源字段：https://doi.org/10.4171/owr/2018/13

17216 | OWR-15993-005

### Far-From-Threshold Buckling of Indented Spherical Shells

P / partially_solved | Oberwolfach 报告问题汇编

For indentation of pressureless spherical shells and related shells on substrates with vanishing properties, characterize the far-from-threshold buckled or crumpled pattern and explain how indentation-induced localization and buckling arise and evolve.

来源字段：https://doi.org/10.4171/owr/2018/8

17217 | OWR-14299090-002

### Feasible High-Performance Plasma Flows for Stellarators

O / open | Oberwolfach 报告问题汇编

Does there exist a stellarator plasma-flow configuration that is simultaneously highly efficient, dynamically stable, and physically buildable? Develop methods for exploring the configuration space while excluding self-intersections and other nonphysical deformations.

来源字段：https://doi.org/10.4171/owr/2025/5

17218 | OWR-14298014-006

### Fermionic Character of Low-Lying Excitations

O / open | Oberwolfach 报告问题汇编

Do the low-lying excitations of a weakly interacting fermionic system have the character of fermionic quasiparticles?

来源字段：https://doi.org/10.4171/owr/2023/49

17219 | OWR-14299913-006

### Finite-Depth Channel Classification of Mixed-State Topological Order

P / partially_solved | Oberwolfach 报告问题汇编

Classify mixed-state topological orders under finite-depth quantum channels: given two states, determine whether a finite-depth channel maps one to the other, and characterize the obstructions supplied by anyons and nontrivial braiding.

来源字段：https://doi.org/10.4171/owr/2026/12

17220 | OWR-14298167-007

### Finite-Shear Hashin-Shtrikman Bounds and Rod-Hinge Deformations

O / open | Oberwolfach 报告问题汇编

Can the Hashin-Shtrikman bounds be improved, or can one construct microstructures that attain the Hashin-Shtrikman shear-modulus bound without attaining the bulk-modulus bound and remain optimal under finite shear? Separately, can non-macroscopically-affine deformations of periodic rod-hinge structures be obtained as limits of periodic deformations whose unit cells grow relative to the underlying structure?

来源字段：https://doi.org/10.4171/owr/2024/11

17221 | OWR-1966-001

### First-Principles Derivation of Adaptive-Resolution Molecular Dynamics

P / partially_solved | Oberwolfach 报告问题汇编

For the adaptive-resolution interpolation formula in molecular dynamics, why is it not naturally formulated using potentials, and can the force-based interpolation formula be formally derived from first principles of statistical physics?

来源字段：https://doi.org/10.4171/owr/2008/21

17222 | AMR-078-0012

### First-Principles Molecular Configurations

P / partially_solved | AMR 问题清单汇编

Give a mathematically rigorous justification of the techniques used to determine molecular configurations from first principles.

背景链接：https://en.wikipedia.org/w/index.php?title=Simon_problems&oldid=1335881350#The_2000_list

17223 | OWR-9790367-007

### Fivebrane Filtrations and Holomorphic Field Theories

P / partially_solved | Oberwolfach 报告问题汇编

Let $F^\bullet E(5|10)$ be the conjectural filtration of the exceptional Lie superalgebra. Identify its $N$th level with the holomorphic field theory $\chi_{\mathrm{hol}}(\mathfrak{sl}(N))$ for every $N>1$.

来源字段：https://doi.org/10.4171/owr/2022/16

17224 | AIM-PHYSICS-0041

### Fixed winding numbers: exact pair bounds, graph coercivity, and collision loss

P / partially_solved | AIM 研讨会问题汇编

Poincare characterizes a homology class in the planar three-body problem by three integers \(k_{ij}=\operatorname{Deg}(x_i-x_j,0)\); that is, the number of oriented turns of the side \(x_i-x_j\).

**Problem 4 (due to Poincare, 1896).** Can we obtain periodic solutions by minimizing while fixing the \(k_{ij}\)'s? (Alain Chenciner)

The printed text reads “Poincare characterize homology class ...”; in normalized English, its mathematical content is:

The problem is then:

The source remarks are:

- answers are known when all \(k_{ij}\) are nonzero; - for \((k_{12},k_{23},k_{31})=(1,1,1)\) or \((-1,-1,-1)\), minimizers are Lagrange equilateral solutions, while the other all-nonzero cases have collisions; - in choreographies the three \(k_{ij}\)'s are equal; - Montgomery asks whether the answer is the same in the restricted problem.

The PDF does **not** specify whether the loops are labeled, whether translations and rotations have been reduced, which period is fixed, or what precise restricted problem and frame are intended. Since \(x_i-x_j\) uses labeled positions, the natural reading is the labeled collision-free configuration space. Fixed-period minimization and center-of-mass reduction are necessary analytic normalizations and are made explicit below. Choreography is an additional symmetry constraint, not part of the bare homology constraint. The phrase “restricted case” remains ambiguous; the massless limit is analyzed in Section 9 without pretending that this resolves every possible restricted formulation.

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17225 | OWR-14297737-002

### Fluctuation Limits for Moderately Interacting Particle Systems

P / partially_solved | Oberwolfach 报告问题汇编

Prove mean-field convergence for Keller-Segel-type and Dirac-approximating potentials, and obtain the better-than-$N^{-1/2}$ mean-square rate needed to derive intermediate fluctuation limits from smoothed empirical measures.

来源字段：https://doi.org/10.4171/owr/2023/38

17226 | OWR-13673-002

### Four-Dimensional Standard-Subspace Nets with Minimal Length

O / open | Oberwolfach 报告问题汇编

Does there exist a natural local net of standard subspaces on double cones in Minkowski spacetime $\mathbb R^4$ with positive minimal length?

来源字段：https://doi.org/10.4171/owr/2015/16

17227 | AIM-PHYSICS-0049

### Fourier obstructions to unequally phased single-trace motions

P / partially_solved | AIM 研讨会问题汇编

Problem 12. Existence of choreographies with distinct time shifts.

Find a periodic \(N\)-body solution for which \[ q_j(t)=x(t+\tau_j),\qquad j=1,\ldots,N, \] on one common trace, but the phase set \(\{\tau_j\bmod T\}\) is not an equally spaced coset of \((T/N)\mathbb Z/T\mathbb Z\).

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17228 | OWR-14298167-001

### From Dislocation Mechanics to Continuum Plasticity

O / open | Oberwolfach 报告问题汇编

Develop a rigorous scale-transition theory connecting discrete dislocation mechanics to continuum models of crystal plasticity.

来源字段：https://doi.org/10.4171/owr/2024/11

17229 | AIM-PHYSICS-0066

### Full-space mass-splitting criterion for prograde relative periodic orbits

P / partially_solved | AIM 研讨会问题汇编

**Problem 29.** Given three different masses that are comparable in size, prove the existence of prograde orbits without assuming one ratio of mutual distances is nearly zero. (Kuo-Chang Chen)

**Remarks.**

- Poincaré continuation method can be applied to two cases: one mass is nearly zero, or one ratio of mutual distances is nearly zero. - Known if two masses are equal.

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17230 | OWR-15998-003

### Functional Differential Equations for Many-Body Green Functions

P / partially_solved | Oberwolfach 报告问题汇编

How can the interacting Green functions of many-body quantum systems be obtained from a functional differential equation rather than the usual Dyson integral equation?

来源字段：https://doi.org/10.4171/owr/2018/13

17231 | DARPA-023

### Fundamental Laws of Biology

P / partially_solved | DARPA 挑战

Identify governing principles for biological systems, analogous to physical laws.

背景链接：https://web.math.utk.edu/~vasili/refs/darpa07.MathChallenges.html

17232 | OWR-3392-013

### Future Restrictions of Inner Automorphic Dilations

O / open | Oberwolfach 报告问题汇编

Under suitable minimality conditions, can every inner automorphic dilation be restricted to a proper $E_0$-semigroup on the future part $E_+$ of $\mathcal B^a(E)$? Does it extend to all of $\mathcal B^a(E_+)$, and can $\mathcal B^a(E_+)$ be unitally embedded in $\mathcal B^a(E)$ so that the extension is the restriction of the original automorphism semigroup? The two-sided time shift on $q$-Fock space is proposed as a case study.

来源字段：https://doi.org/10.4171/owr/2009/09

17233 | OWR-1114-006

### Gauge Fixing for Quantum Vertex Algebra Deformations

P / partially_solved | Oberwolfach 报告问题汇编

For the deformation theory of quantum vertex algebras, find an analogue of the Tian gauge enabling perturbation theory, and identify the physical degrees of freedom described by the resulting action principle.

来源字段：https://doi.org/10.4171/owr/2005/53

17234 | OWR-3392-005

### Generation of Free Product Systems by Tensor Subsystems

P / partially_solved | Oberwolfach 报告问题汇编

For a free product system $E^{\ast}$, let its largest tensor-product subsystem be the intersection, over all decompositions $t=t_1+\cdots+t_n$, of the subsystems $E_{t_n}\odot\cdots\odot E_{t_1}$. Does this intersection subsystem generate $E^{\ast}$ as a free product system?

来源字段：https://doi.org/10.4171/owr/2009/09

17235 | OWR-4084-009

### Geometric Collapse in the Toroidal Six-Vertex Model

P / partially_solved | Oberwolfach 报告问题汇编

Does the six-vertex model on a torus undergo a phase transition at which the emergent torus geometry collapses, with critical modular parameter $\tau$ satisfying $\operatorname{Im}(\tau)\neq0$?

来源字段：https://doi.org/10.4171/owr/2009/41

17236 | DARPA-017

### Geometric Langlands and Quantum Physics

P / partially_solved | DARPA 挑战

Connect the Langlands program to fundamental physics symmetries.

背景链接：https://web.math.utk.edu/~vasili/refs/darpa07.MathChallenges.html

17237 | OWR-1109-003

### Geometric Spin-Statistics under Weakened Assumptions

P / partially_solved | Oberwolfach 报告问题汇编

Find a geometric proof of the spin-statistics relation. Determine whether the relation remains valid under weaker assumptions and identify the minimal sufficient hypotheses.

来源字段：https://doi.org/10.4171/owr/2005/48

17238 | OWR-16411-014

### Global Attractors for Expanding Torus Symmetric Spacetimes

P / partially_solved | Oberwolfach 报告问题汇编

For expanding $T^2$-symmetric spacetimes, is the local attractor global for $B_0$ solutions? For non-$B_0$ solutions, does there exist a local attractor satisfying $V_\tau\to1/2$ and $Q_\tau=O(e^{-3\tau/2})$, and is it global within the $T^2$-symmetric class?

来源字段：https://doi.org/10.4171/owr/2018/36

17239 | OWR-14298591-002

### Global Dynamics of the Vlasov-Maxwell System

O / open | Oberwolfach 报告问题汇编

Do arbitrary sufficiently regular initial data for the relativistic Vlasov-Maxwell system generate global-in-time classical solutions, beyond the known perturbative and symmetry-restricted regimes?

来源字段：https://doi.org/10.4171/owr/2024/36

17240 | OWR-16411-012

### Global Geometric Inequalities for Material Bodies

O / open | Oberwolfach 报告问题汇编

Establish purely global geometric inequalities for material bodies and determine whether their sharp forms require convexity, roundedness, or a related shape condition.

来源字段：https://doi.org/10.4171/owr/2018/36

17241 | OWR-14298591-012

### Global Stability and Asymptotics of Cosmological Spacetimes

P / partially_solved | Oberwolfach 报告问题汇编

For $\Lambda>0$, do sufficiently small perturbations of de Sitter or Kerr-de Sitter initial data yield global solutions of $\operatorname{Ric}(g)+\Lambda g=0$, and can those solutions be described asymptotically?

来源字段：https://doi.org/10.4171/owr/2024/36

17242 | OWR-1388-002

### Global Well-Posedness and Uniqueness of Gravity-Capillary Waves

O / open | Oberwolfach 报告问题汇编

For the gravity-capillary water-wave problem, is the evolution globally well posed in the energy space used for the variational minimization, and is the resulting minimizer unique up to spatial translations?

来源字段：https://doi.org/10.4171/owr/2006/50

17243 | OWR-3392-016

### Group-Valued Lévy Processes with Type-II Arveson Systems

P / partially_solved | Oberwolfach 报告问题汇编

Do there exist Lévy processes with values in a group, for example a group of homeomorphisms of the circle, whose Arveson system is of type $\mathrm{II}$ rather than type $\mathrm I$?

来源字段：https://doi.org/10.4171/owr/2009/09

17244 | OWR-16411-013

### Hawking Energy Limits and the Null Penrose Inequality

O / open | Oberwolfach 报告问题汇编

For general null cones, characterize convergence of the Hawking energy, compare its limit with the Trautman-Bondi mass, and derive the corresponding null Penrose inequality.

来源字段：https://doi.org/10.4171/owr/2018/36

17245 | OWR-14217-003

### Hemisphere Partition Functions and Stability Across GLSM Phases

O / open | Oberwolfach 报告问题汇编

In the GLSM setting, is the hemisphere partition function $Z_{D^2}:K_0(D)\to\mathbb C$ a central-charge function for a Bridgeland stability condition on $D$, and for the grade-restriction subcategory $W_n$ are the phase quotient categories $D_{\zeta\gg0}$ and $D_{\zeta\ll0}$ related to $W_n$ by the conjectured commuting equivalences?

来源字段：https://doi.org/10.4171/owr/2016/9

17246 | OWR-1114-002

### Heterotic Mirror Symmetry for Bundled Calabi-Yau Manifolds

O / open | Oberwolfach 报告问题汇编

Construct a general formulation and construction procedure for heterotic $(0,2)$ mirror pairs of Calabi-Yau manifolds with holomorphic vector bundles; in particular, determine whether such mirror symmetry can be realized functorially by Fourier-Mukai transforms in the elliptically fibered K3 setting.

来源字段：https://doi.org/10.4171/owr/2005/53

17247 | OWR-14299913-002

### Higher-Categorical Models of Continuous and Spacetime Symmetries

O / open | Oberwolfach 报告问题汇编

Develop precise higher-categorical data and constructions that recover standard continuous internal symmetries and spacetime symmetries of quantum field theories, and establish their functorial and physical properties.

来源字段：https://doi.org/10.4171/owr/2026/12

17248 | OWR-14299913-008

### Higher-Dimensional Quantum Cellular Automata beyond Translations

P / partially_solved | Oberwolfach 报告问题汇编

Classify nontrivial elements in the quotient of higher-dimensional quantum cellular automata by finite-depth circuits and shifts. In particular, determine which elements occur beyond lattice translations and the currently known three-dimensional examples.

来源字段：https://doi.org/10.4171/owr/2026/12

17249 | OWR-14298014-004

### Higher-Dimensional Semiclassical Kinetic-Energy Corrections

O / open | Oberwolfach 报告问题汇编

In dimensions $d\geq2$, can one justify the predicted next-order semiclassical correction to the kinetic energy, proportional to $\int_{\mathbb{R}^d}|\nabla\sqrt{\rho(x)}|^2\,dx$?

来源字段：https://doi.org/10.4171/owr/2023/49

17250 | OWR-1266-005

### Higher-Dimensional Thermodynamic Limits

O / open | Oberwolfach 报告问题汇编

Establish rigorous control of the thermodynamic limit in spatial dimension greater than one for nonlinear sigma models with noncompact target spaces and nonamenable symmetry groups.

来源字段：https://doi.org/10.4171/owr/2006/17

17251 | OWR-16772-001

### Higher-Order Failure of the Quantum Linear Boltzmann Limit

O / open | Oberwolfach 报告问题汇编

For the quantum Lorentz gas with scatterers placed periodically on $\mathbb Z^d$, does the Boltzmann-Grad limit agree with the linear Boltzmann equation through second order in the coupling constant but disagree at every order greater than four?

来源字段：https://doi.org/10.4171/owr/2019/10

17252 | OWR-16163-001

### Higher-Order Tsirelson, Kirchberg, and Connes Problems

P / partially_solved | Oberwolfach 报告问题汇编

Does the bipartite Tsirelson problem imply the tripartite Tsirelson problem? Are the third-order Kirchberg and Tsirelson problems equivalent? Does there exist a version of Connes' embedding problem equivalent to the third-order Connes or Kirchberg problem?

来源字段：https://doi.org/10.4171/owr/2018/22

17253 | HIL-006

### Hilbert's 6th Problem: Axiomatization of Physics

P / partially_solved | Hilbert 问题

Develop a mathematical framework that axiomatizes physics, particularly mechanics, thermodynamics, and probability theory.

背景链接：https://encyclopediaofmath.org/wiki/Hilbert_problems

17254 | OWR-3392-001

### Homogeneous Factorizations Producing Type-II Arveson Systems

O / open | Oberwolfach 报告问题汇编

If $A$ is a type $\mathrm{II}_1$ or type $\mathrm{III}$ factor with state $\phi$, does there exist a homogeneous noncommutative factorization of $(A,\phi)$ over $\mathbb R$ that leads to an Arveson system of type $\mathrm{II}$? Here homogeneity means a $\phi$-preserving translation action on the factorization over measurable subsets of $\mathbb R$.

来源字段：https://doi.org/10.4171/owr/2009/09

17255 | AIM-PHYSICS-0042

### Homology-only escape and a strict zero-winding choreography gap

P / partially_solved | AIM 研讨会问题汇编

**Problem 5.** Is the figure-8 orbit a minimizer with homology class \((0,0,0)\)? (Alain Chenciner)

One can ask, for example, if the Eight is a minimizing choreography in its homology class \((0,0,0)\) (each side of the triangle has zero total rotation).

The source-verified AIM item is exactly:

The preceding text is essential. Immediately before Problem 4, the PDF says that Poincaré describes a homology class in the planar three-body problem by \[ k_{ij}=\operatorname{Deg}(x_i-x_j,0), \] the oriented number of turns made by the side \(x_i-x_j\). Problem 4 asks whether one can obtain periodic solutions by minimizing with the \(k_{ij}\) fixed. Its remarks say that the answer is understood when all three integers are nonzero, that the classes \((1,1,1)\) and \((-1,-1,-1)\) give the Lagrange equilateral solutions, and that in choreographies all three integers are equal. Problem 5 then follows as a separate one-line item.

The one-line AIM version does **not** state whether minimization is among all loops, all choreographies, a fixed symmetry space, a fixed braid class, or periodic solutions only. Nor does it state a period, centre-of-mass condition, or size normalization. These omissions cannot be silently filled.

There is, however, strong source evidence for the intended reading. In the contemporary ICM article from which the question arose, Chenciner writes:

He introduces this as a problem with **mixed constraints: symmetry and homology or homotopy**. Thus this report distinguishes:

1. the **literal homology-only reading** of the AIM sentence; and 2. the **intended mixed reading**, namely fixed-period, centred simple choreographies with common winding number zero.

It is not Gerver's four-body super-eight from AIM Problem 3.

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17256 | OWR-1184-001

### Hoop Conjecture for Black-Hole Formation

O / open | Oberwolfach 报告问题汇编

Do black holes with horizons form if and only if a mass $M$ is compacted into a region whose circumference in every direction is at most $4\pi M$?

来源字段：https://doi.org/10.4171/owr/2006/02

17257 | OWR-16411-003

### Horizon Topology Changes and Stable Higher-Dimensional Black Holes

O / open | Oberwolfach 报告问题汇编

Is the Gregory-Laflamme instability the only mechanism by which general relativity changes horizon topology? In spacetime dimension $D>4$, is every stable black hole a Myers-Perry black hole satisfying $J/M^{D-3}=O(1)$?

来源字段：https://doi.org/10.4171/owr/2018/36

17258 | OWR-17292-006

### Hypergroup Correspondence for Discrete Conformal Subnets

P / partially_solved | Oberwolfach 报告问题汇编

For a strongly additive conformal net $\mathcal A$, establish a bijection between discrete subnets $\mathcal B\subseteq\mathcal A$ and compact hypergroups $K\subseteq\operatorname{QuOp}(\mathcal A)$, including intrinsic definitions that cover the infinite-index case.

来源字段：https://doi.org/10.4171/owr/2019/49

17259 | OWR-15219-003

### Infinite-Dimensional K-Moment Problems in Quantum Field Theory

O / open | Oberwolfach 报告问题汇编

Develop existence, determinacy, and support criteria for representing measures in infinite-dimensional $K$-moment problems arising in quantum field theory and stochastic analysis.

来源字段：https://doi.org/10.4171/owr/2017/14

17260 | OWR-1114-004

### Infrared Equivalence of Stack Presentations

O / open | Oberwolfach 报告问题汇编

Physical theories associated with different physically realizable presentations of the same stack should become the same infrared theory under renormalization-group flow.

来源字段：https://doi.org/10.4171/owr/2005/53

17261 | AMR-080-0005

### initial boundary value problems for integrable systems (IBVP)

P / partially_solved | AMR 问题清单汇编

Initial boundary value problems for integrable systems in $1+1$ dimensions are of great interest. It turns out, however, that in Fokas' approach the general initial value problem is overdetermined. In certain ``integrable'' cases (see below) $u_x(0,t)$ can be determined a priori, but this is not the case in general. As demonstrated by Fokas, his approach leads to a Riemann--Hilbert reformulation of the IBVP, but one can only apply the powerful non-linear steepest descent method to determine the asymptotic behavior of $u(x,t)$ as $t\to \infty$ in the case that $u_x(0,t)$ is known a priori. A particular stand out open problem is to compute the long-time behavior of the solution $u(x,t)$ of the focusing NLS equation, $i u_t= u_{xx} + 2|u|^2 u$, with $u(x,0)=f(x)$, $x\ge 0$, given and $u(0,t) = e^{iwt}$, $t\ge 0$ with $w\in \mathbb{R}$. In other words, the effect of the oscillatory component of the driver does not propagate into the lattice and away from the boundary at $k=0$.; If $\ga_1 > \gamma> \ga_2$, then the asymptotic motion is described by a travelling wave x_k(t) = c_1 k + Y_1 (\beta_1 k + \gamma t), \qquad 1 \ll k \ll t transporting energy away from the driver. Here $c_1$ and $\beta_1$ are certain determined constants and $Y_1(1)$ is $2\pi$-periodic 1-gap solution of the periodic Toda lattice.; More generally, if $\ga_j > \gamma > \ga_{j+1}$, a multi-phase wave emerges of the form x_k = c_j k + Y_j (\beta_1 k + \gamma t, \beta_2 k + \gamma t, \dots, \beta_j k + \gamma t), \qquad 1 \ll k \ll t again transporting energy away from the driver, for certain constants $c_j, \beta_1, \dots, \beta_j$. This is an open problem of the first order, illustrating how the natural modes of an extended physical system ($k$-gap solutions in the case of NLS) can be excited by driving the system externally at one end. This problem arises, in particular, in analyzing the Gross--Pitaevskii equation with a delta function external potential at $x=0$ and even initial data. It turns out that the NLS equation with Robin boundary conditions is an example of an integrable system which is not overdetermined from the point of view of Fokas' method, and indeed for this problem the method can be used to obtain a well-defined and effective Riemann--Hilbert representation for the solution $u(x,t)$ of the IBVP (see). The method of Bikbaev and Tarasov is a non-linear version of the method of images in linear theory, where the solution in $x>0$ is extended to $x<0$, not by reflection as in the linear case, but by a suitable B\"acklund transformation. It is an interesting open problem to use the Bikbaev--Tarasov method to solve the IBVP for defocusing NLS, $iu_t=u_{xx}-2|u|^2 u$, with Robin boundary conditions at $x=0$. In this case the B\"acklund transformation extending the solution from $x>0$ to $x<0$ introduces certain singularities which must be controlled and the long-time behavior of the solution of the IBVP will be very different from the case of focusing NLS. Another open problem, which has broad implications, concerns the smoothness of the solutions $u(x,t)$ of the IBVP for NLS. However in the case of focusing NLS with smooth $f(x)=u(x,0)$ and Robin boundary...

背景链接：https://arxiv.org/abs/1703.04931

17262 | OWR-8415343-011

## Initial Data for Physical Late-Time Schwarzschild Wave Tails

O / open | Oberwolfach 报告问题汇编

For solutions of the scalar wave equation $\Box_g\phi=0$ on a Schwarzschild background arising from physically realistic isolated-system data, how should one choose or characterize the data on an asymptotically hyperboloidal hypersurface $\Sigma$ so as to obtain physically meaningful predictions for the late-time asymptotics along future null infinity $\mathcal{I}^+$?

来源字段：https://doi.org/10.4171/owr/2021/40

17263 | AMR-079-0014

## Initial/boundary value problems for integrable systems

P / partially_solved | AMR 问题清单汇编

In the late 1990s Fokas introduced a new, more flexible approach to inverse scattering theory, and in recent years a number of researchers (Fokas, Its,..., Anne Boutet de Monvel, Shepelsky,...) have applied Fokas' approach to the initial/boundary value problem for various integrable systems such as NLS. Much progress has been made, but there is a basic, and puzzling, obstacle to applying the method, viz. one needs to know certain dependent data in order to proceed. For example, the initial/boundary value problem for NLS is well-posed if one gives the initial data $u(x,0)=u_0(x)$, $x\geq 0$, and the boundary data $u(0,t)=u_1(t)$, $t\geq 0$. However, in implementing the method it turns out that one needs to know the dependent data $u_x(0,t)$, $t\geq 0$, as such information appears explicitly in the solution formulae. Progress has been made in determining $u_x(0,t)$, $t\geq0$, from $u_0(x)$ and $u_1(t)$ intrinsically via the method, but the control one obtains on $u_x(0,t)$ is not sufficient in order to obtain the long-time behavior of the initial/boundary value problem. This is true even in simple cases, such as the following: suppose $u(x,t)$ solves NLS in $(x\geq0,t\geq0)$ with $u(0,t)=\sin(\omega t)$, $\omega\neq0$, and $u(x,0)=u_0(x)$, where $u_0$ is smooth with compact support. How does $u(x,t)$ behave as $t\to\infty$? The solution of this problem for any such $u_0(x)$ would be a very significant development in the theory, and would also be of considerable interest in science and engineering. There is a philosophical point at stake here. The evolution of NLS in $x>0$ represents the interplay of forces which are ``integrable'' at some fundamental, algebraic level, and indeed, if no other forces are present, as in the full-line scattering or periodic cases, the equation can be integrated explicitly. When one considers the initial/boundary value problem, however, new physical forces come into play which describe the interaction of the particles on the boundary with the NLS particles in the interior. There is absolutely no a priori guarantee that the enlarged system, ``NLS particles in $x>0$''+``particles on the boundary'', is integrable. It may be that the long-time behavior of the composite system can only be solved for ``generic, Cantor-set'' like data, as is familiar from KAM theory. In other words, an explicit description of the long-time behavior of the solution of the initial/boundary value problem for NLS for general initial data may not be possible. This is a very intriguing situation.

背景链接：https://arxiv.org/abs/0712.0849

17264 | OWR-13750338-003

## Instance-Independent Drivers in Quantum Annealing

O / open | Oberwolfach 报告问题汇编

Extend the no-quantum-speedup theorem for the transverse-field driver to every instance-independent transverse Hamiltonian.

来源字段：https://doi.org/10.4171/owr/2023/26

17265 | AMR-080-0003

## interacting particle systems and KPZ

P / partially_solved | AMR 问题清单汇编

In 1999, J. All these early examples were ``integrable'' in the sense that certain powerful algebraic tools, e.g., determinantal particle systems, the Robinson--Schensted--Knuth correspondence, Fredholm operator theory, $\dots$, could be applied to reduce the problems at hand to analytically viable forms from which their asymptotic behavior could be deduced using standard asymptotic methods such as the Laplace method/stationary phase, or sometimes, Riemann--Hilbert/nonlinear steepest-descent methods. In a very significant and unanticipated development during 2007--2009, C. Widom analyzed ASEP (the asymmetric simple exclusion process) with step initial data, a system to which the above algebraic tools cannot be applied. Nevertheless, in an algebraic/combinatorial/analytical tour de force, they were able to show that ASEP exhibited the same asymptotic RMT behavior as the integrable models. Spohn were the first to show that $h(x,t)$ converged to the Airy 2 process, but their argument was not rigorous (see for more information on spatial correlations). A basic question is whether KPZ is in the KPZ universality class. In other words, can one show rigorously that solutions of KPZ indeed have the KPZ scaling $3:2:1$? The first difficulty here is to show that the Cauchy problem for the non-linear stochastic KPZ equation indeed has a solution. Quastel, used results from the 2008--2009 work of Tracy and Widom on ASEP to find an explicit formula for the solution of the Cole--Hopf transformed KPZ equation with so-called ``narrow wedge'' initial data. Hammond (see for the strongest and most precise statement of the KPZ universality conjecture for the KPZ equation). Again for the stochastic heat equation with narrow wedge initial data, Corwin and Hammond were able to show tightness of the suitably $3:2:1$ scaled height function $h(x,t)$ in a ``good'' sense which implied convergence for suitable subsequences to limiting processes with ``good'' properties, but they were not able to show that the limits were unique and given by the physically anticipated process. The most challenging open problem regarding the KPZ equation is to prove $3:2:1$ KPZ universality in the strong sense of Corwin, Quastel and Remenik alluded to above, for solutions of the Cauchy problem of the KPZ equation for a wide set of initial data, and to identify the limiting process. Another open problem is to analyze solutions of KPZ at more than one space-time point. Johansson computed correlations for two space-time points for the Brownian LPP problem, from which it easily follows, in particular, that Brownian LPP exhibits $3:2:1$ KPZ scaling. Also, regarding the original longest increasing subsequence problem, KPZ universality for the fluctuations has been proved so far only for the uniform distribution on the permutations: The problem for the fluctuations with more general distributions is very much open. In another direction, it remains an open, and somewhat mysterious, problem to understand how Tracy and Widom's solution of ASEP actually ``works'', and to extend their method of solution to more general initial data (however, see, and more recently,). Developing such a perturbation...

背景链接：https://arxiv.org/abs/1703.04931

17266 | OWR-8415343-012

## Interior Singularity Structure of Dynamically Formed Black Holes

O / open | Oberwolfach 报告问题汇编

What is the global singularity structure inside a black hole formed by gravitational collapse, in particular the relationship between a weakly singular Cauchy horizon, stronger terminal singularities, and the center?

来源字段：https://doi.org/10.4171/owr/2021/40

17267 | OWR-14299088-009

## Irreducible Components of Landau Varieties

P / partially_solved | Oberwolfach 报告问题汇编

Determine whether the proposed elimination-theoretic construction identifies every irreducible component of the Landau variety, and characterize which candidate components genuinely occur.

来源字段：https://doi.org/10.4171/owr/2025/9

17268 | OWR-13861-005

## Joint Boltzmann-Grad and Long-Time Limits for Lorentz Gases

O / open | Oberwolfach 报告问题汇编

For the periodic Lorentz gas with scatterer radius $r$ and time $t$, can one establish a joint limit as $r\to0$ and $t\to\infty$, rather than taking the limits successively?

来源字段：https://doi.org/10.4171/owr/2015/40

17269 | OWR-8415343-008

## Kasner Singularities Beyond the Subcritical Regime

O / open | Oberwolfach 报告问题汇编

For general sufficiently small perturbations, possibly without symmetry, of generalized Kasner solutions of the Einstein equations, what behavior occurs near the singularity when the sub-criticality condition $\max_{I<J,B}(\widetilde q_I+\widetilde q_J-\widetilde q_B)<1$ fails?

来源字段：https://doi.org/10.4171/owr/2021/40

17270 | AMR-080-0001

### KdV with almost periodic initial data

P / partially_solved | AMR 问题清单汇编

Consider the Korteweg--de Vries (KdV) equation u_t + u u_x + u_{xxx} =0 with initial data u(x, t=0) = u_0(x),\qquad x\in \mathbb{R}. In the 1970's, McKean and Trubowitz proved the remarkable result that if the initial data $u_0$ is periodic, $u_0(x+p)= u_0(x)$ for some $p>0$, then the solution $u(x,t)$ of (eq1) is almost periodic in time. This result leads to the following natural conjecture: The same is true if $u_0(x)$ is almost periodic, i.e., if the initial data is almost periodic in space, the solution evolves almost periodically in time. This is a very challenging problem with many new and unique features. For example, for $u_0(x)$ almost periodic, one does not know, by standard PDE methods, whether a solution $u(x,t)$ exists, even for a small time, never mind globally. In spectacular and seminal work over the last few years, D. Damanik and M. Goldstein, joined later by I. Binder and M. Lukic, have partially resolved this conjecture in the affirmative, namely, for small quasi-periodic analytic initial data with Diophantine frequencies, unique global solutions exist and are almost periodic in time. The key insight of Damanik et al.\ was how to utilize the formal integrability of KdV in an effective analytical manner. Not only are the arguments in Damanik et al.\ applicable to other (formally) integrable systems (for example proves a Toda lattice analog of the KdV theorem), but some of their arguments are also applicable to non-integrable systems. It is a natural, open and very challenging conjecture that the result of Damanik et al.\ remains true for (suitable) perturbations of KdV, i.e., for (suitable) perturbations of KdV, the Cauchy problem with (suitable) almost periodic initial data $u_0(x)$, has a unique global solution $u(x,t)$, which evolves almost periodically in time. After all, the finite-dimensional analog of this result is precisely the content of classical KAM theory. We note that such results for perturbations of KdV and the nonlinear Schr\"{o}dinger equation (NLS) with periodic initial data $u_0(x)$, have already been proved by S. Kuksin, T. Kappeler and others. With the discovery of quasi-crystals, one can anticipate, in particular, that interest in PDE problems on quasi-periodic backgrounds, will surely grow.

背景链接：https://arxiv.org/abs/1703.04931

17271 | OWR-11695854-003

### Knots-Quivers Correspondence for All Knots

P / partially_solved | Oberwolfach 报告问题汇编

Does the knots-quivers correspondence hold for every knot, relating its colored HOMFLY-PT generating series $P(x,a,q)$ to the motivic generating series $P_C(x,q)$ of a corresponding quiver?

来源字段：https://doi.org/10.4171/owr/2022/45

17272 | AMR-132-0001

### Lattice Gases and Antiferromagnetic Potts Models - Conjecture 1

O / open | AMR 问题清单汇编

For any countably infinite graph $G$ of maximum degree $\Delta$, the hard-core lattice gas on $G$ has a unique Gibbs measure whenever $0 \le w_x \le (1-\epsilon) \times$\break $(\Delta-1)^{\Delta-1}/(\Delta-2)^\Delta$ $[\sim e/\Delta$ for large $\Delta]$ for all vertices $x$, for some $\epsilon > 0$. [Perhaps this is true even with $\epsilon = 0$.]

背景链接：https://arxiv.org/abs/cond-mat/0004231

17273 | AMR-132-0003

### Lattice Gases and Antiferromagnetic Potts Models - Conjecture 3 (Hamidoune \protect[Hamidoune_90], Stanley \protect[Stanley_98])

O / open | AMR 问题清单汇编

If $G$ is claw-free (i.e.\ has no induced subgraph $K_{1,3}$), then all the zeros of $Z(w)$ are negative real.

背景链接：https://arxiv.org/abs/cond-mat/0004231

17274 | AMR-132-0004

### Lattice Gases and Antiferromagnetic Potts Models - Problem 4

O / open | AMR 问题清单汇编

Salas and Sokal [Salas_97] have some improvements of this result for specific regular lattices (square, triangular, hexagonal, Kagom\'e), based on decimation and a computer-assisted proof. But their results are far from sharp.

Prove uniqueness of the infinite-volume Gibbs measure and exponential decay of correlations for $q=3$ on the hexagonal lattice, $q=4,5,6$ on the square lattice, etc.

背景链接：https://arxiv.org/abs/cond-mat/0004231

17275 | AMR-132-0002

### Lattice Gases and Antiferromagnetic Potts Models - Question 2

O / open | AMR 问题清单汇编

What is the largest complex domain containing $w=0$ on which $Z(w)$ is zero-free for all graphs of maximum degree $\Delta$? [This question has two versions, depending on whether the $w_x$ are assumed equal or unequal. In the latter version, one seeks domains $D$ for which $Z(w) \neq 0$ whenever $w_x \in D$ for all $x$; but there need not exist a unique maximal such domain.]

背景链接：https://arxiv.org/abs/cond-mat/0004231

17276 | AMR-132-0005

### Lattice Gases and Antiferromagnetic Potts Models - Question 5

O / open | AMR 问题清单汇编

\quad (a) What is the sharp bound $C_{opt}(\Delta)$? Might $2\Delta$ work? Is $C_{opt}(3) = 3$? What about regions of the $q$-plane other than discs centered at $q=0$?

(b) Can ``second-largest degree'' be improved to ``max flow''? (See [Sokal_00a, Section 7] for details.)

(c) Can one obtain a *sublinear*\/ bound for suitable subclasses of graphs?

- For theta graphs $\Theta_{s_1,\ldots,s_p}$ [consisting of a pair of endvertices connected by $p$ internally disjoint paths of lengths $s_1,\ldots,s_p$] the answer is *yes*\/: there is a bound of order $p/\log p$ [Brown_et_al].

- For series-parallel graphs, I conjecture that the answer is *yes*\/ (again $p/\log p$) for maximum degree, and *no*\/ for second-largest degree or max flow; I am currently trying to prove these claims.

- For planar graphs, it is an open question.

背景链接：https://arxiv.org/abs/cond-mat/0004231

17277 | OWR-14297742-005

### Lee-Huang-Yang Upper Bound for Hard-Sphere Bosons

S / solved | Oberwolfach 报告问题汇编

For a dilute hard-sphere Bose gas in the thermodynamic limit, prove the second-order upper bound matching the Lee-Huang-Yang expansion $e(\rho)=4\pi a\rho^2\bigl(1+128(\rho a^3)^{1/2}/(15\sqrt\pi)+o((\rho a^3)^{1/2})\bigr)$.

来源字段：https://doi.org/10.4171/owr/2023/39

17278 | AIM-PHYSICS-0058

### Level-orbit potentials: quantifier audit, speed criterion, and transnormal rigidity

S / solved | AIM 研讨会问题汇编

Does there exist potential \(U(x)\) in the plane such that if \(\dot x(0)\) is tangential to the level curve of \(U\), then \(U\) remains constant? (Mark Levi)

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17279 | OWR-14299085-002

### Lieb-Robinson Bounds for Nonrelativistic Fermions

O / open | Oberwolfach 报告问题汇编

Can the local estimate $\lvert\operatorname{tr} O_z(\widehat{x})(\gamma^{(1)}_{N,t}-\omega_{N,t})\rvert\leq C(t)\varepsilon^{-2}$, proved for pseudo-relativistic high-density fermions, be extended to nonrelativistic fermion dynamics, where the one-particle velocity is unbounded?

来源字段：https://doi.org/10.4171/owr/2025/7

17280 | AIM-PHYSICS-0060

### Limits of action minimization: collisions, constraints, stability, and fixed energy

P / partially_solved | AIM 研讨会问题汇编

**Problem 23.** Among the many interesting open questions is the understanding of the natural limits of the minimization method: (Alain Chenciner)

- To what extent is it connected to symmetry constraints?

- May topological constraints be imposed without forcing the minimizers to have collisions?

- To what extent interesting results may be obtained for arbitrary masses?

- What can be said of the hyperbolic and elliptic dimensions of a minimizer?

- Is it interesting to look at minimizers with fixed energy?

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17281 | OWR-1325-004

### Linear Baroclinic Prediction of Ocean Eddies

O / open | Oberwolfach 报告问题汇编

Can observed ocean eddy structure be largely predicted and understood through linear baroclinic-instability theory?

来源字段：https://doi.org/10.4171/owr/2006/39

17282 | OWR-16163-014

### Linear Entropy and Rank Inequalities for Quantum Marginals

P / partially_solved | Oberwolfach 报告问题汇编

For a positive semidefinite quantum state $\rho$ on $A\otimes B\otimes C\otimes D$, determine the linear inequalities constraining the von Neumann entropies and logarithms of ranks of its marginals; in particular, decide the proposed higher-order entropy inequalities and whether $r_{AB}\leq r_{AC}r_{BC}$ for tripartite reductions.

来源字段：https://doi.org/10.4171/owr/2018/22

17283 | OWR-14299913-004

### Liouville Theory from the Virasoro Conformal Net

O / open | Oberwolfach 报告问题汇编

Prove that applying the fully extended two-dimensional conformal-field-theory construction to the Virasoro conformal net produces Liouville conformal field theory.

来源字段：https://doi.org/10.4171/owr/2026/12

17284 | OWR-4080-001

### Local Approximations of Electronic Spectral Projectors

O / open | Oberwolfach 报告问题汇编

Given a discrete electronic Hamiltonian $H$ with occupied eigenvectors $\psi_1,\ldots,\psi_{N_e}$, develop effective approximations of the spectral projector $$P=\sum_{i=1}^{N_e}|\psi_i\rangle\langle\psi_i|$$ onto the $N_e$ lowest eigenstates, particularly approximations exploiting locality and a spectral gap.

来源字段：https://doi.org/10.4171/owr/2009/37

17285 | OWR-13673-008

### Local Conformal Field Theories from Non-Semisimple Tensor Categories

P / partially_solved | Oberwolfach 报告问题汇编

For a nonsemisimple finite tensor category $\mathcal C$, do full local conformal field theories with chiral data $\mathcal C$ exist, and how can their bulk objects and sewing-compatible, mapping-class-group-invariant correlators be constructed?

来源字段：https://doi.org/10.4171/owr/2015/16

17286 | OWR-14298014-003

### Local Density Approximation for the Canonical Functional

O / open | Oberwolfach 报告问题汇编

Can the local-density approximation be rigorously justified for the canonical density functional $F(\rho)$?

来源字段：https://doi.org/10.4171/owr/2023/49

17287 | OWR-8415343-013

### Local Weak Cosmic Censorship in Charged Spherical Collapse

O / open | Oberwolfach 报告问题汇编

Local Weak Cosmic Censorship Conjecture: for generic spacetimes arising from charged spherical collapse, the center endpoint $\widehat{\Gamma}$ is not a locally naked singularity.

来源字段：https://doi.org/10.4171/owr/2021/40

17288 | OWR-13854-006

### Local Well-Posedness of Lovelock Equations

P / partially_solved | Oberwolfach 报告问题汇编

Can one establish local existence and uniqueness of solutions, together with continuous dependence on initial data, for the Lovelock-theory equations?

来源字段：https://doi.org/10.4171/owr/2015/33

17289 | AMR-078-0002

### Localization in Two Dimensions

O / open | AMR 问题清单汇编

Prove that the spectrum of the Anderson model in dimension $\nu=2$ is dense pure point.

背景链接：https://en.wikipedia.org/w/index.php?title=Simon_problems&oldid=1335881350#The_2000_list

17290 | OWR-785-003

### LOCC Dense-Codeability of the Four-Qubit W State

O / open | Oberwolfach 报告问题汇编

Is the so-called four-qubit $W$ state in the LOCC dense-codeable class?

来源字段：https://doi.org/10.4171/owr/2005/04

17291 | OWR-17137-002

### Logarithmic Entanglement Enhancement from Partial Delocalization

P / partially_solved | Oberwolfach 报告问题汇编

For entanglement entropy in disordered free-fermion systems, how much delocalization of the one-particle eigenvalues and eigenfunctions is needed to produce a logarithmic enhancement of the area law, even when the spectrum is not absolutely continuous?

来源字段：https://doi.org/10.4171/owr/2019/41

17292 | AMR-077-0004

### Long-Range Order for the Quantum Heisenberg Model

P / partially_solved | AMR 问题清单汇编

(A) Prove long-range order for the quantum Heisenberg ferromagnet in dimension $D>2$ at temperature $T>0$. (B) Prove long-range order for spin $1/2$ in two dimensions in the ground state. (C) Find a robust proof for these continuous-symmetry cases that does not require exact translation invariance.

背景链接：https://web.math.princeton.edu/~aizenman/OpenProblems_MathPhys/9901.HeisenbergFerr.html

17293 | AMR-079-0008

## Long-time behavior with non-generic initial data

P / partially_solved | AMR 问题清单汇编

The long-time behavior of the solution of the Cauchy problem for a great many integrable systems on the line is now well-understood, using, for example, the Riemann-Hilbert/steepest descent method. The method depends on full knowledge of the nature of the spectrum of the associated Lax operator. For systems such as KdV, MKdV, defocusing NLS and the Toda lattice, for example, one is able to describe the solution asymptotically in complete detail for general initial data. But for focusing NLS, where the associated AKNS Lax operator $T$ is non-self-adjoint (here we make contact with Problem 7), the situation is different. For generic $T$ (i.e. an open dense set of $T$'s in any reasonable topology) the spectrum consists of the real line, where the spectrum is absolutely continuous, together with a finite number of simple eigenvalues (corresponding to solitons) off the real axis. The analysis of the long-time behavior of focusing NLS with such generic initial data proceeds in a straightforward manner similar to KdV, MKdV, etc. For general initial data, however, the situation is more complicated. For example, let $z_0>0$ be any positive number and let $D$ be any arbitrarily small open disk in the complex plane centered at $z_0$, such that $D\subset\{z\,:\,\mathbb{R}e z>0\}$. Let $D^+$ denote the intersection of $D$ with the upper half plane, and let $u(z)$ be an arbitrary function analytic in $D^+$, and continuous in $\overline{D^+}$. Let $B=\{z\in D^+\,:\, u(z)=0\}$. Then there exists (Zhou) an AKNS operator $T$ with infinitely smooth, rapidly decaying coefficients with the property that each point in $B$ is an $L^2(\mathbb{R})$-eigenvalue of $T$. In other words, there exist (non-generic) operators $T$ with Schwartz space coefficients which have $L^2$ spectrum accumulating on the real line at an essentially arbitrary rate. It is a very interesting question to determine what effect such singularities would have on the long-time behavior of the solution of NLS. In particular, recalling that focusing NLS provides a model for data transmission along communication cables, are the effects measurable? The difficulty that we encounter here is not limited to situations where the associated Lax operator is non-self-adjoint. Even in situations when the associated operator is self-adjoint, but of order greater than two, similar difficulties can arise. This is true, in particular, for the Boussinesq equation, where the associated Lax operator is third order. The long-time behavior of the solutions of the Boussinesq equation with general initial data is a very interesting problem with many challenges. Even in the case with generic initial data the situation is only partially understood.

背景链接：https://arxiv.org/abs/0712.0849

17294 | OWR-17137-001

## Long-Time Landau-Pekar Dynamics of Strongly Coupled Polarons

O / open | Oberwolfach 报告问题汇编

For a strongly coupled polaron as $\alpha\to\infty$, can approximation of the quantum dynamics by the Landau-Pekar ansatz be proved on time scales longer than $\lvert t\rvert\lesssim\alpha^2$?

来源字段：https://doi.org/10.4171/owr/2019/41

17295 | OWR-17137-004

## Loop Operators and Microscopic Nonabelian Anyons

O / open | Oberwolfach 报告问题汇编

Can loop operators and the corresponding anyon description be constructed for microscopic models with nonabelian anyons, extending the known abelian construction?

来源字段：https://doi.org/10.4171/owr/2019/41

17296 | OWR-15993-001

## Low-Parameter Hyperelastic Models of Brain Tissue

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Brain-Tissue Constitutive-Model Extraction [OWR-15993-002]

Determine a low-parameter family of isotropic hyperelastic strain-energy functions for brain tissue that captures its observed response under combined shear and axial compression or tension, remains suitable for small-to-moderate in-vivo strains, and can be calibrated and validated on multiaxial data.

来源字段：https://doi.org/10.4171/owr/2018/8

17297 | OWR-14299085-003

## Low-Temperature Order in Quantum Heisenberg Ferromagnets

O / open | Oberwolfach 报告问题汇编

For the short-range quantum Heisenberg ferromagnet in dimension $d\geq 3$, prove that sufficiently low temperature produces long-range order; specifically, prove that $\lim_{\Lambda\nearrow\mathbb{Z}^d}|\Lambda|^{-2}\sum_{x,y\in\Lambda}\langle\widetilde S_x\cdot\widetilde S_y\rangle_{T,\Lambda}>0$ for all sufficiently small $T>0$.

来源字段：https://doi.org/10.4171/owr/2025/7

17298 | OWR-14298170-008

## Lower Bounds for Hamiltonian Mass

O / open | Oberwolfach 报告问题汇编

For every globally well-behaved initial data set whose matter satisfies the dominant-energy-type positivity assumptions, prove that its Hamiltonian mass is bounded below under asymptotic symmetries.

来源字段：https://doi.org/10.4171/owr/2024/9

17299 | OWR-15584-001

### Lower-Bounded Hamiltonians for Many-Body Point Interactions

O / open | Oberwolfach 报告问题汇编

Construct a self-adjoint Hamiltonian, bounded from below, that models pairwise point interactions in a many-body quantum system.

来源字段：https://doi.org/10.4171/owr/2017/41

17300 | OWR-15172-002

### Macroscopic Limits and Phase Diagrams of Ferroelectric Models

O / open | Oberwolfach 报告问题汇编

Do macroscopic polarization models for ferroelectric solids arise as the large-body limit of microscopic electron-density models, including the still-open lower-bound direction? Can one also systematically analyze the proposed random-field-Ising-like model with long-range interactions for the phase diagram of lead titanate-lead zirconate solutions?

来源字段：https://doi.org/10.4171/owr/2016/51

17301 | OWR-11448-001

### Many-Body Localization at Positive Density

O / open | Oberwolfach 报告问题汇编

For systems of infinitely many interacting particles at positive density, what is the precise localization regime and is it stable under interactions?

来源字段：https://doi.org/10.4171/owr/2011/50

17302 | OWR-1789-002

### Massless and Point-Charge Limits of the Pauli-Fierz $g$-Factor

P / partially_solved | Oberwolfach 报告问题汇编

For the Pauli-Fierz Hamiltonian $H(p)$ with photon dispersion $\omega(k)=\sqrt{|k|^2+m_{\mathrm{ph}}^2}$, does its $g$-factor have a finite limit as $m_{\mathrm{ph}}\to0$ and $\widehat\varphi(k)\to(2\pi)^{-3/2}$? If the point-charge limit exists, determine its dependence on the fine-structure constant $\alpha$, including the coefficient of $O(\alpha^2)$ in $$g_\infty=2\left(1+\frac{4\alpha}{3\pi}+O(\alpha^2)\right)$$.

来源字段：https://doi.org/10.4171/owr/2008/08

17303 | OWR-16411-010

### Mass-Angular-Momentum Inequalities with Trapped and Multiple Horizons

O / open | Oberwolfach 报告问题汇编

For axisymmetric initial data with ADM mass $m$ and total angular momentum $J_\infty$, prove $m\geq\sqrt{|J_\infty|}$ for manifolds with a trapped inner boundary. For multiple black holes satisfying $m\geq\mathcal M_{\mathrm{many}}$, determine the value and parameter dependence of $\mathcal M_{\mathrm{many}}$ and identify the geometric meaning of its minimizer.

来源字段：https://doi.org/10.4171/owr/2018/36

17304 | OWR-4083-001

### Mathematical Classification of Atomic Chemical Groups

O / open | Oberwolfach 报告问题汇编

For atoms described by the many-electron Schrödinger equation, determine mathematically meaningful equivalence classes corresponding to chemical groups, determine which atom belongs to which group, and obtain a rigorous electronic-structure description with the simplicity of hydrogen orbital configurations.

来源字段：https://doi.org/10.4171/owr/2009/40

17305 | OWR-11695854-002

### Mathematical Construction of the Negative-Phase Coulomb Category

O / open | Oberwolfach 报告问题汇编

For the $U(1)$ gauged linear sigma model with target $\mathbb P^n$ and stability parameter $\zeta<0$, construct mathematically the Coulomb-branch category $\mathcal C$ predicted by physics and prove the predicted equivalence $D^b(\mathbb P^n)\simeq\mathcal C$.

来源字段：https://doi.org/10.4171/owr/2022/45

17306 | AMR-078-0011

### Mathematical Nuclear Shell Model

O / open | AMR 问题清单汇编

Give a mathematically rigorous formulation and justification of the nuclear shell model.

背景链接：https://en.wikipedia.org/w/index.php?title=Simon_problems&oldid=1335881350#The_2000_list

17307 | AIM-PHYSICS-0051

### Matrix virial decomposition and a collision-free tensor stress test

P / partially_solved | AIM 研讨会问题汇编

Is there a conceptual proof for Saari's conjecture? Why not fix the moment of inertia tensor and ask the same question (maybe in higher dimensions)? (Alain Chenciner)

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17308 | OWR-1703865-001

### Maximum Excess Charge of Atoms

O / open | Oberwolfach 报告问题汇编

What is the maximal excess charge of atoms?

来源字段：https://doi.org/10.4171/owr/2020/18

17309 | AMR-077-0014

### Meaning and Impossibility of Exact Helium Energy Levels

P / partially_solved | AMR 问题清单汇编

Give a mathematically precise meaning to determining the energy levels of the helium atom exactly, in the sense that hydrogen energy levels are exact, and prove or disprove that such exact determination is impossible.

背景链接：https://web.math.princeton.edu/~aizenman/OpenProblems_MathPhys/9808.ImposThms.html

17310 | AMR-077-0006

### Meaning and Nonexistence of an Exact Three-Dimensional Ising Formula

O / open | AMR 问题清单汇编

Give a mathematically precise meaning to an exact formula comparable to Onsager's formula for the two-dimensional Ising model, and prove or disprove that no such formula exists for the three-dimensional Ising model.

背景链接：https://web.math.princeton.edu/~aizenman/OpenProblems_MathPhys/9808.ImposThms.html

17311 | OWR-14298586-002

### Mechanisms and Scales of Tropical-Cyclone Edge Intensification

O / open | Oberwolfach 报告问题汇编

For edge-intensified tropical convection, determine (1) what sets the number and separation distance of convergence lines, including without a sea-surface-temperature gradient; (2) why convection does not occur over the highest sea-surface temperature and whether a predictive model of edge intensification can be derived; and (3) how rotation changes these effects.

来源字段：https://doi.org/10.4171/owr/2024/31

17312 | OWR-1325-001

### Mechanisms Producing Equatorial Superrotation

P / partially_solved | Oberwolfach 报告问题汇编

What features of planetary-scale heating and upscale momentum flux are necessary to create equatorial superrotation?

来源字段：https://doi.org/10.4171/owr/2006/39

17313 | OWR-15176-004

### Metric-Tensor Scaling in Liouville Quantum Gravity

S / solved | Oberwolfach 报告问题汇编

For $\gamma$-Liouville quantum gravity, is the scaling-limit metric tensor associated with the field $h$ given formally by

$$e^{2\gamma h/\beta}(dx^2+dy^2),$$

where $\beta$ is the Hausdorff dimension of the Liouville metric?

来源字段：https://doi.org/10.4171/owr/2016/55

17314 | OWR-14298009-001

### Modular Generators for Massive Double Cones

O / open | Oberwolfach 报告问题汇编

For a massive free scalar field localized in a double cone, determine whether the modular-generator component $M_-$ is independent of the mass and whether $M_-$ is a multiplication operator.

来源字段：https://doi.org/10.4171/owr/2023/50

17315 | OWR-11135-004

### Momentum-Space Scarring on Irrational Point-Scatterer Tori

O / open | Oberwolfach 报告问题汇编

For an irrational flat torus with a point scatterer, does there exist a subsequence of eigenfunctions that scars in momentum space?

来源字段：https://doi.org/10.4171/owr/2011/43

17316 | OWR-17476-002

### Monolithic versus Staggered Magnetomechanical Relaxation

O / open | Oberwolfach 报告问题汇编

For coupled magnetomechanical systems, should relaxation be formulated monolithically or as a staggered sequence of partial relaxations, and under what conditions are the two procedures equivalent?

来源字段：https://doi.org/10.4171/owr/2020/13

17317 | AMR-079-0015

### Multi-matrix models and models with an external field

P / partially_solved | AMR 问题清单汇编

There has been considerable progress (Kuijlaars,...) in understanding basic statistics such as the correlation functions for the 2-matrix random matrix model, and also matrix models with a source. The key element in these developments has been the successful extension by Kuijlaars et al of the Riemann-Hilbert/steepest descent method to $3\times3$ Riemann-Hilbert problems. So far only the simplest situations have been considered. In order to consider the generic situation, one must, in particular, extend the Riemann-Hilbert/steepest descent method to $n\times n$ Riemann-Hilbert problems. This is a challenging problem which would have important implications, not only for random matrix models, but also for problems in other areas, such as Pad\'e-Hermite approximations and irrationality questions for distinguished real numbers, and multi-orthogonal polynomials.

背景链接：https://arxiv.org/abs/0712.0849

17318 | OWR-3392-015

### Multiparameter E0-Semigroups from Free Product Systems

P / partially_solved | Oberwolfach 报告问题汇编

Given $n$ one-parameter free product systems $E^1,\ldots,E^n$, carry out the proposed construction

$$E(t_1,\ldots,t_n)=E^1_{t_1}*\cdots*E^n_{t_n}$$

and its inductive limit over the reference vector to obtain candidate $n$-parameter $E_0$-semigroups, and analyze the resulting systems and their units.

来源字段：https://doi.org/10.4171/owr/2009/09

17319 | OWR-785-002

### Multipartite Dense-Coding Capacity

P / partially_solved | Oberwolfach 报告问题汇编

For distributed dense coding with more than two receivers, determine the capacity, or at least a nontrivial upper bound. Can multipartite bound-entangled states be useful for distributed dense coding?

来源字段：https://doi.org/10.4171/owr/2005/04

17320 | OWR-13111-004

### Multipartite Quantum Data Hiding with Separable States

O / open | Oberwolfach 报告问题汇编

For multipartite quantum systems with more than two parties, can the bipartite data-hiding results for separable states be generalized, including asymptotically optimal information efficiency and the best hiding-quality scaling?

来源字段：https://doi.org/10.4171/owr/2014/41

17321 | OWR-14298167-006

### Multiphysics Frameworks for Computational Homogenization

P / partially_solved | Oberwolfach 报告问题汇编

How should computational homogenization incorporate thermoviscous dissipation and relax strict resonance confinement; which frameworks best balance accuracy, efficiency, and flexibility; and can they extend to electro-magneto-mechanical, thermomechanical, and chemomechanical systems?

来源字段：https://doi.org/10.4171/owr/2024/11

17322 | OWR-14299094-002

### Multitask Hamiltonian Learning across Levels of Theory

O / open | Oberwolfach 报告问题汇编

Develop and analyze a multitask learning framework for Hamiltonians trained on data generated at different levels of electronic-structure theory. Determine how information should be shared across tasks and when transfer improves accuracy and sample efficiency.

来源字段：https://doi.org/10.4171/owr/2025/15

17323 | AMR-079-0005

## Multivariate analysis

S / solved | AMR 问题清单汇编

Random matrix theory was introduced into theoretical physics by Wigner in the 1950s in his study of neutron scattering resonances, but as a subject, RMT goes back to the work of statisticians at the beginning of the 20th century. Recently, advances in RMT have opened the way to the statistical analysis of data sets in cases where the number of variables is comparable to the number of samples, and both are large. One might, for example, be interested in the daily temperature in hundreds of cities around the world, over a 365 day time period. At the technical level, one considers the statistics of the singular values of (appropriately centered and scaled) $p\times n$ matrices $M=(M_{ij})$, where $p\sim n\to\infty$. Here $p$ is the number of variables and $n$ is the sample size. More precisely, one centers the $M_{ij}$'s around their sample averages, $$ M_{ij}\to\hat M_{ij}=M_{ij}-\frac1n\sum_{k=1}^n M_{ik}, $$ and considers the eigenvalues $l_1\geq\cdots\geq l_p\geq0$ and associated eigenvectors $w_1,\dots,w_p$ of the $p\times p$ sample matrix $S=\frac1n \hat M \hat M^T$. The $l_i$'s and $w_i$'s are known as the principal component eigenvalues and eigenvectors, respectively. In Principal Component Analysis (PCA) ``significant'' dimension reduction in the data occurs if the first few principal components $l_1, l_2,...$ account for a ``high'' proportion of the total variance $\operatorname{tr} S=\sum_{j=1}^p l_j$. A common model for the variables $M_{ij}$ is to assume that they follow a (real) $p$-variate Gaussian distribution $N_p(\mu,\Sigma)$ with mean $\mu$ and covariance matrix $\Sigma$. Thus the columns $(M_{1j}, M_{2j},\dots, M_{pj})^T$ provide $n$ independent samples for $N_p(\mu,\Sigma)$. Using recent results from RMT, much has now been proved about the statistics of $l_1, l_2,...$ as $p,n\to\infty$, $p/n\to\gamma\in(0,\infty)$, in the case $\Sigma=I$. In particular, we know that in the limit, $l_1$, appropriately centered and scaled, satisfies the Tracy-Widom distribution for the largest eigenvalues of a GOE matrix. However, most interesting applications involve so-called spiked populations, a terminology introduced by Johnstone, i.e. situations where most of the eigenvalues $\eta_1,...,\eta_p$ of $\Sigma$ are equal to a common value, say 1, but the first few eigenvalues are greater than 1. Thus $$ \eta_1\geq\eta_2\geq\cdots\geq\eta_k>\eta_{k+1}=\cdots=\eta_p=1 $$ for some fixed $k<<p$. It is a major problem in multivariate analysis to analyze the statistics of the eigenvalues $l_1, l_2,...$ as $p,n\to\infty$, $p/n\to\gamma\in(0,\infty)$ for such spiked populations. There are very interesting phase transitions in the theory. For example if $\eta_1>1+\sqrt{\gamma}$, then, as $p\sim n\to\infty$, $l_1$ emerges from the Marchenko-Pastur continuum $\bigl((\sqrt\gamma-1)^2,(\sqrt\gamma+1)^2\bigr)$, where most of the $l_j$'s tend to accumulate, and almost surely $$ l_1\to\eta_1 \cdot \Bigl(1+\frac{\gamma}{\eta_1-1}\Bigr)>(1+\sqrt\gamma)^2\,. $$ In the spiked, complex case, i.e. when the columns $(M_{1j},M_{2j},...,M_{pj})^T$ are sampled from the complex $p$-variate Gaussian distribution, much is known about the asymptotic distribution of the $l_j$'s, as $p,n\to\infty$, $p/n\to\gamma\in(0,\infty)$. By contrast in the real case, apart from a.s. convergence of the $l_i$'s, very little is known about their asymptotic distributions. In the spiked, complex case the analysis is enabled by a particular technical tool, the Harish-Chandra-Itzykson-Zuber formula: unfortunately, no analog of this formula is known in the real case. While there are relatively few applications of complex spiked populations, knowledge of the asymptotic distributions of the $l_i$'s for real spiked populations would have immediate applications to a wide variety of problems in signal processing, genetics and finance.

背景链接：https://arxiv.org/abs/0712.0849

17324 | AMR-081-0002

## Mutually Unbiased Bases in Dimension Six

O / open | AMR 问题清单汇编

Construct a set of at least four mutually unbiased bases in dimension six, or prove that there are no seven mutually unbiased bases in $\mathcal{H}_6$.

背景链接：https://arxiv.org/abs/2002.03233

17325 | OWR-8415352-004

## Mutually Unbiased Bases in Non-Prime-Power Dimensions

P / partially_solved | Oberwolfach 报告问题汇编

For which pairs $(d,k)$ do there exist $k$ mutually unbiased bases in dimension $d$, particularly when $d$ is not a prime power?

来源字段：https://doi.org/10.4171/owr/2021/49

17326 | OWR-14299517-006

## New Quantum LDPC Codes with Good Parameters

O / open | Oberwolfach 报告问题汇编

Construct new families of quantum low-density parity-check codes with asymptotically good parameters.

来源字段：https://doi.org/10.4171/owr/2025/41

17327 | OWR-12009-006

## Non-Disappearance of Nonnegative Scalar-Curvature Limits

O / open | Oberwolfach 报告问题汇编

Do sequences of manifolds with nonnegative scalar curvature and no closed interior minimal surfaces other than their boundaries avoid disappearing in the limit unless their volumes tend to zero?

来源字段：https://doi.org/10.4171/owr/2012/37

17328 | AMR-079-0007

### Non-self adjoint spectral problems

P / partially_solved | AMR 问题清单汇编

Much is now known, theoretically and also numerically, about the spectrum of self-adjoint operators. This is in great contrast to the situation regarding non-self-adjoint operators, where the spectral theory is far more subtle and numerical schemes must overcome significant, inherent instabilities. The heart of the difficulties and instabilities lies in the following fact: in the self-adjoint case $A=A^*$, the resolvent $(A-\lambda)^{-1}$ is bounded by the distance of $\lambda$ to the spectrum of $A$, but in the non-self-adjoint case, this is no longer true, as we see already in the $2\times2$ case, $A=\begin{bmatrix}0 n\\0 0\end{bmatrix}$, $n\to\infty$. In recent years, a number of authors (e.g. Trefethan, Davies,...) have initiated a systematic approach to non-self-adjoint spectral problems, notions such as pseudospectrum have come into prominence, and other authors have conducted in depth studies of particular non-self-adjoint spectrum problems which arise in practice. For example, in analyzing the semi-classical limit of the focusing NLS equation, one must analyze the spectrum of the associated AKNS operator $T(h)$, as Planck's constant $h$ goes to zero. The operator is non-self-adjoint, and as $h$ goes to zero more and more eigenvalues, corresponding to solitons, emerge in the complex plane. It is of critical importance to the analysis of the semi-classical limits for NLS to determine where in the plane, and at what rate, the eigenvalues accumulate. The difficulty in doing this, theoretically and numerically, is illustrated by the following fact: the spectrum of $T(h)$ off the real axis is a discrete set, whereas the numerical range of $T(h)$ is an open subset of the plane. Nevertheless, every point $\lambda$ lying in the numerical range of $T(h)$ is an eigenvalue of $T(h)$ to all orders in $h$, $\|(T(h)-\lambda)u\|=O(|h|^k)$ for any $k\geq 1$, for some $u=u(h)$, $\|u\|=1$. In the language of Kruskal, the computation of the spectrum of $T(h)$ is a problem ``beyond all orders.'' Much has been done (Kamvissis-Miller-McLaughlin, Tovbis-Venakides-Zhou,...) in analyzing the semi-classical limit for NLS in special situations where the spectral problem can be solved explicitly. The spectral problem with general data, both for NLS and also other related non-self-adjoint problems, however, is far from understood, and poses a great challenge whose resolution is still only in the initial stages.

背景链接：https://arxiv.org/abs/0712.0849

17329 | OWR-12009-002

### Nonaxisymmetric Stability of Extremal Kerr Waves

O / open | Oberwolfach 报告问题汇编

For sufficiently regular solutions $\psi$ of the scalar wave equation on extremal Kerr, does the pointwise decay estimate up to and including the event horizon that is known for axisymmetric solutions remain valid without assuming axisymmetry?

来源字段：https://doi.org/10.4171/owr/2012/37

17330 | OWR-2493-005

### Nonconvex Gradient Gibbs States and Free Energy

P / partially_solved | Oberwolfach 报告问题汇编

For gradient fields and Gibbs models with non-convex interactions between microscopic particles, what can be proved about the free energy and the Gibbs states?

来源字段：https://doi.org/10.4171/owr/2008/55

17331 | OWR-16411-004

### Nonlinear Instability of Anti-de Sitter Spacetime

O / open | Oberwolfach 报告问题汇编

Can arbitrarily small perturbations of anti-de Sitter initial data, evolved by the vacuum Einstein equations with reflecting conformal-boundary conditions, form black-hole regions at sufficiently late times?

来源字段：https://doi.org/10.4171/owr/2018/36

17332 | OWR-11788-003

### Nonlinear Stability of Subextremal Kerr Black Holes

P / partially_solved | Oberwolfach 报告问题汇编

Black-hole stability conjecture: do sufficiently small perturbations of sub-extremal Kerr initial data for the vacuum Einstein equations evolve toward a nearby Kerr black hole?

来源字段：https://doi.org/10.4171/owr/2012/26

17333 | OWR-4129-001

### Nonlinear Stability of the Full Kerr Family

P / partially_solved | Oberwolfach 报告问题汇编

Do sufficiently small perturbations of Kerr initial data for the vacuum Einstein equations have maximal developments with complete future null infinity whose exterior asymptotically approaches another Kerr solution?

来源字段：https://doi.org/10.4171/owr/2009/46

17334 | OWR-14298167-002

### Nonperiodic and Nonquadratic FFT Schemes for Materials

O / open | Oberwolfach 报告问题汇编

Can FFT-based material schemes be extended beyond periodic boundary conditions, to nonquadratic energy potentials, and to models with grain-boundary behavior?

来源字段：https://doi.org/10.4171/owr/2024/11

17335 | OWR-1266-002

### Nonperturbative Infrared Behavior of Nonamenable Sigma Models

O / open | Oberwolfach 报告问题汇编

For two-dimensional nonlinear sigma models with noncompact target spaces and nonamenable symmetry groups, does a rigorous nonperturbative analysis confirm the perturbative renormalization-group prediction of infrared asymptotic freedom, or is that prediction an artifact of perturbation theory?

来源字段：https://doi.org/10.4171/owr/2006/17

17336 | AIM-PHYSICS-0062

### Normal Hessian blocks, an exact four-body index jump, and stability obstructions

P / partially_solved | AIM 研讨会问题汇编

**Problem 25.** Give a sharp upper bound for the Morse index of a nonplanar central configuration (as a critical point of the potential on the normalized configuration space). Give a sharp lower bound for the Morse index of a planar central configuration when it is viewed as part of the nonplanar configuration space. Is the Morse index related to the stability of the rigidly rotating periodic orbits? For example, does a linearly stable relative equilibrium necessarily arise from a minimum of the Newtonian potential? (Rick Moeckel)

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17337 | AMR-080-0004

### numerical computation with random data

P / partially_solved | AMR 问题清单汇编

Standard algorithms to compute the eigenvalues of a random matrix $H$ are completely integrable Hamiltonian systems (see ). The question raised in was: What happens if we try to marry these two integrabilities? This question was taken up by P. The authors in then recorded \tau_{\epsilon, n, \mathcal{A}, \mathcal{E}}(H) \equiv \frac{T_{\epsilon, n, \mathcal{A}, \mathcal{E}}(H)- \langle T_{\epsilon, n, \mathcal{A}, \mathcal{E}}\rangle}{\sg_{\epsilon, n, \mathcal{A}, \mathcal{E}}}, where $\langle T_{\epsilon, n \mathcal{A}, \mathcal{E}}\rangle$ and $\sigma^2_{\epsilon, n, \mathcal{A}, \mathcal{E}}$ are respectively the sample average and sample variance for $T_{\epsilon, n,\mathcal{A}, \mathcal{E}}(H)$ computed for a very large sample of matrices $H\in \mathcal{E}$. Trogdon raised the question of whether the universality results in were limited to eigenvalue algorithms, or whether they were present more generally in numerical computations. And indeed in the authors found similar universality results for a wide variety of numerical algorithms, including; more general eigenvalue algorithms such as the Jacobi algorithm, and also algorithms for Hermitian ensembles,; the conjugate gradient and GMRES algorithms to solve linear $n\times n$ systems $Hx=b$, $H$ and $b$ random,; an iterative algorithm to solve the Dirichlet problem $\Delta u=0$ on a random star-shaped region $\Omega \subset \mathbb{R}^2$ with random boundary data $f$ on $\partial \Omega$, and; a genetic algorithm to compute the equilibrium measure for orthogonal polynomials on the line, and was also for a decision making process in recent laboratory experiments of Yu. Thus the Toda algorithm with stopping time $T^{(1)}(H)$ is an algorithm to compute the largest eigenvalue of a given matrix

背景链接：https://arxiv.org/abs/1703.04931

17338 | OWR-17476-003

### Numerical Schemes for Coupled Fracture and Dissipation

O / open | Oberwolfach 报告问题汇编

For rate-independent fracture and damage models coupled with rate-dependent processes, are there efficient feasible numerical schemes that reliably approximate the intended solution class, and what mathematical and mechanical framework should describe the coupling?

来源字段：https://doi.org/10.4171/owr/2020/13

17339 | AMR-080-0006

## numerical solution of integrable systems

P / partially_solved | AMR 问题清单汇编

Solutions of the Cauchy problem for linear dispersive equations can be expressed in terms of Fourier integral operators. This is a very challenging problem in numerical analysis. Again the problem of the solution of the Cauchy problem is in two parts. Concerning the second problem, in a very significant development starting in 2011, S. Olver introduced a methodology to solve RHP's numerically and applied this methodology to compute solutions of the Painlev\'e II equation, which has its own Riemann--Hilbert formulation. Rather than a single, or several, open problems, what we have here is an ``open program'', viz., to apply the Olver/Deconinck--Olver--Trogdon methodology to solve the many different numerical problems for integrable systems as they arise. Particularly difficult are situations where the forward problem is not self-adjoint, as in the case of focusing NLS. It remains an open problem (see ) to solve the forward problem for focusing NLS with general smooth initial data, and then implement the RHP methodology, to obtain the solution of the Cauchy problem in the small dispersion limit $\epsilon=\hbar\downarrow 0$. At the technical level, the difficulty in solving the forward problem is that the relevant physical quantities only appear, in the language of Martin Kruskal, ``beyond all orders''. The forward problem for focusing NLS should, of course, be seen as one more example of the general open problem of computing the spectrum of non-self-adjoint problems (see, e.g., N. Dyson in the 1960's, followed by in the 70's, have contributed to the analysis of $F_s$ as $s\to\infty$ (see P. In a striking development in 2008, F. Bornemann devised a very general and flexible method to evaluate Fredholm determinants, and he then applied the method to compute a variety of determinants arising in modern mathematical physics, including $F_s$. Although Fredholm determinants were introduced more than 100 years, it is surprising that, prior to Bornemann's work, no such general method to evaluate Fredholm determinants numerically and accurately was developed. However, the behavior of the eigenfunctions for finite values of $s$, a problem of great practical interest, presents a serious numerical challenge (a priori, the problem is ill conditioned), which was taken up only very recently by A. A related open problem concerns the Airy operator $A_s$ with kernel $\frac{A_i(\lambda) A'_i(\mu)- A'_i(\lambda) A_i(\mu)}{\lambda-\mu}$ acting on $L^2(s,\infty)$ (here $A_i(\lambda)$ is the standard Airy function). Bothner, confirming an earlier conjecture of C. It is an open problem to analyze the behavior of the eigenfunctions $\tilde{u}_j(s)$ as $s\to\infty$, and to compute their behavior for finite $s$. Finally, in 2005 (see ), the idea was raised of a ``Painlev\'e project'', which would provide a forum to assemble all current information on the algebraic, analytical and numerical properties of the Painlev\'e

背景链接：https://arxiv.org/abs/1703.04931

17340 | AMR-077-0007

## One-Dimensional Fermi Gas with Attractive Interaction

P / partially_solved | AMR 问题清单汇编

Determine the large-distance behavior of the one-particle reduced density matrix for a one-dimensional Fermi gas with spin and attractive interaction.

背景链接：https://web.math.princeton.edu/~aizenman/OpenProblems_MathPhys/9807.FermiGas.html

17341 | AMR-078-0015

## One-Dimensional Lieb-Thirring Constants

P / partially_solved | AMR 问题清单汇编

For spatial dimension $\nu=1$ and $1/2<\gamma<3/2$, determine the optimal constants $L_{\gamma,1}$ in the Lieb-Thirring inequality as predicted by the Lieb-Thirring conjecture.

背景链接：https://en.wikipedia.org/w/index.php?title=Simon_problems&oldid=1335881350#The_2000_list

17342 | OWR-12341-001

## Open Problems in Geometric Nonlinear Quantum Dynamics

O / open | Oberwolfach 报告问题汇编

Develop a theory of densely defined vector fields on Hilbert manifolds that generate flows, generalizing Stone’s theorem. What are the best special-relativistic and general-relativistic analogues of the diffeomorphism group’s role in quantum mechanics? Develop nonlinear gauge transformations that may leave Hilbert space, extend nonlinear quantum time evolutions to composite systems with or without the no-signaling property, and apply differential geometry to generalized, possibly dissipative, quantum hydrodynamics.

来源字段：https://doi.org/10.4171/owr/2013/13

17343 | OWR-14299580-001

## Open Problems on Spectral-Parameter Yang-Baxter Solutions

O / open | Oberwolfach 报告问题汇编

Let $V$ be a possibly infinite-dimensional vector space and let $R:\mathbb R\to\operatorname{GL}(V\otimes V)$ satisfy the Yang-Baxter equation with spectral parameter. Can such families be studied by a unified algebraic or categorical theory, and how do analytic properties of $R(x)$ control the associated braided or quasi-braided categories?

来源字段：https://doi.org/10.4171/owr/2025/49

17344 | OWR-14298373-006

### Open Structures in Compactified Imaginary Liouville Theory

O / open | Oberwolfach 报告问题汇编

Are compactified imaginary Liouville models related to CILT, do they admit a path-integral formulation, are they logarithmic CFTs, and is there a probabilistic theory when $\beta^2\notin\mathbb{Q}$?

来源字段：https://doi.org/10.4171/owr/2024/25

17345 | AMR-077-0012

### Optimal Flux for the Quarter-Filled Band

O / open | AMR 问题清单汇编

For the two-dimensional square-lattice model of independent electrons at density $1/4$, does magnetic flux $\pi/2$ per plaquette minimize the ground-state energy, as predicted by the conjecture that the minimizing flux is $2\pi$ times the electron density?

背景链接：https://web.math.princeton.edu/~aizenman/OpenProblems_MathPhys/9802.OptFlux.html

17346 | OWR-14299913-001

### Orthogonal Symmetry Conjecture for Rigid Higher Categories

O / open | Oberwolfach 报告问题汇编

Let $\operatorname{Cat}^{\mathrm{rigid}}_{(\infty,n)}(\mathcal{V})$ be the $\infty$-category of rigid $(\infty,n)$-categories enriched over a presentably symmetric monoidal $\infty$-category $\mathcal{V}$. Taking opposites in each categorical tier gives an action of $O(1)^n$. Prove that this action extends to an action of the orthogonal group $O(n)$.

来源字段：https://doi.org/10.4171/owr/2026/12

17347 | OWR-1186-001

### Orthotropic Extensions of Anisotropic Creep-Damage Models

O / open | Oberwolfach 报告问题汇编

For continuum-mechanics models of creep damage, how can transversally isotropic material behavior be extended to the orthotropic case; how can secondary anisotropic equations be extended to the tertiary creep regime; and how should the identification procedures be realized?

来源字段：https://doi.org/10.4171/owr/2006/04

17348 | OWR-15960-003

### Osterwalder-Schrader Positivity in Free Probability

P / partially_solved | Oberwolfach 报告问题汇编

Do analogues of Osterwalder-Schrader positivity exist in free probability or for noncommuting random variables?

来源字段：https://doi.org/10.4171/owr/2017/55

17349 | AMR-079-0004

### Painlev\'e equations

O / open | AMR 问题清单汇编

What I have in mind here is not a specific problem, but a project, a very large scale project. The six (nonlinear) Painlev\'e equations form the core of ``modern special function theory.'' The role that the classical special functions, such as the Airy, Bessel, and Legendre functions, started to play in the 19$^th$ century, has now been greatly expanded by the Painlev\'e functions. Increasingly, as nonlinear science develops, people are finding that the solutions to an extraordinarily broad array of scientific problems, from neutron scattering theory, to PDEs, to transportation problems, to combinatorics,..., can be expressed in terms of Painlev\'e functions. What is needed is a project, similar to the Bateman project, or a new volume of Abramowitz and Stegun, devoted to the Painlev\'e equations. Much can be, and has been, proved regarding the algebraic and asymptotic properties of Painlev\'e functions. Here the role of integral representations and the classical steepest descent method in deriving precise asymptotics and connection formulae for the classical special functions is played, and expanded, by a Riemann-Hilbert representation of the Painlev\'e equations, together with the non-commutative steepest descent method introduced in 1993. Very little is known, however, beyond ad hoc calculations, about the numerical solution of the Painlev\'e equations. If $u(x)$ is the solution of the Painlev\'e II equation, say, which is asymptotic to the Airy function $Ai(x)$ as $x\to+\infty$, one would like to know, for example, the location of its poles in the complex $x$-plane. A modern ``Bateman Project: Painlev\'e equations'' would not/should not provide tables for such solutions. Rather, it should provide reliable, easy to use software to compute the solutions. Writing useful software for such nonlinear equations presents many challenges, conceptual, philosophical and technical. Without the help of linearity, it is not at all clear how to select a broad enough class of ``representative problems.'' The software should be in the form of a living document where new numerical problems can be addressed by a pool of experts as they arise. And at the technical level, how does one combine asymptotic information about the solutions obtained from the Riemann-Hilbert problem, together with efficient numerical codes in order to compute the solution $u(x)$ at finite values of $x$? I believe that the importance of the Painlev\'e Project will only grow with time. It should be viewed as creating a national resource and should probably be funded and led at the national level. The NIST Project ``Digital Library of Mathematical Functions'', where Peter Clarkson has a contribution on Painlev\'e functions, is an encouraging first step.

背景链接：https://arxiv.org/abs/0712.0849

17350 | OWR-13355-003

### Periodic Gibbs Phases in Two Dimensions

O / open | Oberwolfach 报告问题汇编

At positive temperature, is every extremal Gibbs measure for a finite-range interaction in two dimensions periodic?

来源字段：https://doi.org/10.4171/owr/2014/53

17351 | OWR-14297742-001

## Periodic Ground States in Gross-Pitaevskii Theory

O / open | Oberwolfach 报告问题汇编

For superstable, sufficiently decaying pair interactions $w$ whose Fourier transform takes negative values, must every infinite-volume ground state of $(-\Delta+w*|u|^2)u=\mu u$ be periodic when $\mu$ exceeds the fluid-solid transition value $\mu_c$?

来源字段：https://doi.org/10.4171/owr/2023/39

17352 | AMR-079-0013

## Perturbation theory for exactly solvable combinatorial problems

S / solved | AMR 问题清单汇编

The asymptotic behavior of a variety of combinatorial problems has now been analyzed in great detail. Here we have in mind Ulam's problem for the length of the longest increasing subsequence, the tiling problem for the Aztec diamond, the hexagon tiling problem, and the last passage percolation problem, amongst many others. In all cases, in an appropriate scaling limit, the statistical fluctuations in the systems at hand are described by random matrix theory. The asymptotic analysis, however, depends in a critical, and rigid way, on the underlying probability measures for the systems. For example, the analysis of the last passage percolation problem requires the waiting time $w_i$ at each site $i$ to be either geometrically or exponentially distributed (Johansson): if the statistics of the waiting times is neither geometric nor exponential, the analysis fails completely. What happens if the geometric distribution, say, is slightly perturbed? The challenge here is to develop an effective perturbation theory for such systems. One expects that the random matrix behavior of the fluctuations should persist. Related work on this problem has been done by Baik-Suidan and Bouchard-Martin (cf. Problem 10).

背景链接：https://arxiv.org/abs/0712.0849

17353 | AMR-079-0012

## Perturbation theory for infinite dimensional integrable systems

P / partially_solved | AMR 问题清单汇编

The bijective mapping properties of the scattering transform for a variety of integrable systems on the line between appropriate weighted Sobolev spaces have now been established (X. This has made it possible, in particular, to analyze (Deift-Zhou) the long-time behavior of the solution of the Cauchy problem for a variety of integrable systems in a fixed space without a ``loss of derivatives''. This in turn has made it possible to analyze perturbations of integrable systems. For example, Deift-Zhou analyzed the perturbed defocusing NLS equation iu_t+u_{xx}-2|u|^2u-\varepsilon V(|u|)u=0 $$ u(x,t=0)=u_0(x)\in H^{1,1}=\{f\in L^2\,|\, f', xf\in L^2(\mathbb{R})\}\,, $$ where $V(|u|)\sim |u|^p$ as $|u|\to0$, for some $p>2$ sufficiently large. In the perturbation theory of the linear Schr\"odinger equation -- when the term $-2|u|^2u$ is absent from (E:7) -- the key role in the analysis is played by the Fourier transform which diagonalizes the linear part of the equation. Such solutions do not decay uniformly in time (i.e. $\sup_{t,x\in\mathbb{R}}|u(x,t)|>0$) and this complicates the analysis of the perturbed equation enormously, as the perturbation term is no longer small with respect to $2|u|^2u$ (see above). The solution of (E:7) in the focusing case in the neighborhood of a $k$-soliton for (cubic) NLS ($k\geq2$), together with a detailed description of the long-time asymptotics, would be regarded as a very significant development in the theory of PDEs/Mathematical Physics. Here KAM methods apply and the authors show that certain finite dimensional tori corresponding to finite gap solutions in the integrable case survive under perturbation. The periodic problem is more complicated than the problem on the line because of the action of

背景链接：https://arxiv.org/abs/0712.0849

17354 | OWR-14608-001

## Phase Diagram of the Interacting Bose Gas

O / open | Oberwolfach 报告问题汇编

Establish the conjectured phase diagram for the three-dimensional interacting Bose gas: below the critical chemical potential the correlations should cluster exponentially, at criticality by a power law, and above it the system should exhibit Bose-Einstein condensation with no clustering.

来源字段：https://doi.org/10.4171/owr/2016/26

17355 | OWR-832-003

## Physical Interpretation of Elliptic Partition Functions

O / open | Oberwolfach 报告问题汇编

Do the elliptic-cohomology partition functions for type II string theory coincide with partition functions of twelve-dimensional F-theory compactified on an elliptic curve? Test this proposed interpretation by computing suitable loop versions of the $E_8$ and Rarita-Schwinger indices.

来源字段：https://doi.org/10.4171/owr/2005/27

17356 | OWR-4135-001

### Physical Realizability of Quantum Computation

P / partially_solved | Oberwolfach 报告问题汇编

Is the standard model of quantum computation feasibly realizable?

来源字段：https://doi.org/10.4171/owr/2009/52

17357 | OWR-1105-001

### Physical Regularity Criteria for Massless Scattering Theory

P / partially_solved | Oberwolfach 报告问题汇编

Can Herbst's technical regularity assumption in the extension of Haag-Ruelle scattering theory to theories with massless particles be replaced by a physically motivated criterion?

来源字段：https://doi.org/10.4171/owr/2005/44

17358 | OWR-785-006

### Positive Maps and Multipartite Bound Entanglement

O / open | Oberwolfach 报告问题汇编

For $i=1,2$, let $$\Lambda_i[X]=\sum_{\mu=0}^{d_i^2-1} \lambda_\mu^{(i)}F_\mu^{(i)}XF_\mu^{(i)},$$ where the $F_\mu^{(i)}$ form Hermitian, Hilbert-Schmidt orthonormal bases of $M_{d_i}(\mathbb{C})$. If exactly one coefficient of $\Lambda_2$ is negative and every positive coefficient is at least its absolute value, then $\Lambda=\Lambda_1\otimes\operatorname{id}_{d_2}+\operatorname{id}_{d_1}\otimes\Lambda_2$ is positive. Can this family be generalized by relaxing the orthonormal-basis hypothesis? For $$\rho_I=\frac1{|I|}\sum_{(\alpha,\beta)\in I}|\psi_{\alpha\beta}\rangle\langle\psi_{\alpha\beta}|,$$ determine the properties of these states and, in particular, whether their entanglement can be detected by other criteria.

来源字段：https://doi.org/10.4171/owr/2005/04

17359 | OWR-4415-003

### Potential Temperature and Potential Vorticity Inversion

O / open | Oberwolfach 报告问题汇编

Is potential-temperature inversion complementary to potential-vorticity inversion?

来源字段：https://doi.org/10.4171/owr/2010/34

17360 | OWR-4415-004

### Predictive Thermostated Dynamics and Subgrid Reservoir Modeling

O / open | Oberwolfach 报告问题汇编

Can thermostated dynamics be predictive on short time scales and can its correlations be computed accurately; how should subgrid-scale reservoir energetics and perturbed vector fields be modeled?

来源字段：https://doi.org/10.4171/owr/2010/34

17361 | OWR-15429-009

### QQ Systems for Spectral Determinants and Integrable Models

P / partially_solved | Oberwolfach 报告问题汇编

What structural mechanism produces the QQ-tilde system for spectral determinants of affine opers, and is there an analogous system on the integrable-model side?

来源字段：https://doi.org/10.4171/owr/2017/21

17362 | AIM-PHYSICS-0061

### Quantitative collinear finiteness and maximal-dimensional simplex rigidity

P / partially_solved | AIM 研讨会问题汇编

For the \(n\)-body problem, \(n\ge 4\), show that the number of central configurations is finite for all choices of masses \(m_i>0\) (or find a counterexample). (Rick Moeckel, Marshall Hampton)

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17363 | AIM-PHYSICS-0047

### Quantitative simplex and dimension bounds for minimum-potential equal-mass configurations

P / partially_solved | AIM 研讨会问题汇编

Is the central configuration for equal masses with minimal potential on the ellipsoid of moment of inertia 1 necessarily symmetric? (Rick Moeckel)

Appears to be false for n = 47. Should require that the orbit is a minimizer of the normalized potential.

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17364 | OWR-14608-002

### Quantized Transport in Interacting Disordered Systems

P / partially_solved | Oberwolfach 报告问题汇编

Develop a theory of quantized charge transport, including sharply quantized Hall plateaux and universal critical conductivities, for interacting disordered systems.

来源字段：https://doi.org/10.4171/owr/2016/26

17365 | OWR-15584-003

### Quantum Backflow Bounds with External Potentials

P / partially_solved | Oberwolfach 报告问题汇编

For a quantum particle scattering from an external potential, is the smeared probability-flux operator on positive incoming states bounded below, analogously to the free-particle quantum-backflow inequality?

来源字段：https://doi.org/10.4171/owr/2017/41

17366 | OWR-14299913-009

### Quantum Cellular Automata Realizing the Full Witt Group

O / open | Oberwolfach 报告问题汇编

Construct higher-dimensional quantum cellular automata corresponding to every element of the Witt group of nondegenerate braided fusion categories, extending the existing constructions for its pointed part.

来源字段：https://doi.org/10.4171/owr/2026/12

17367 | AMR-078-0003

### Quantum Diffusion in the Anderson Model

O / open | AMR 问题清单汇编

For the Anderson model in dimension $\nu\geq3$ and disorder strengths $|b-a|$ admitting absolutely continuous spectrum, prove that $\sum_{n\in\mathbb Z^\nu} n^2|e^{itH}(n,0)|^2$ grows asymptotically like $ct$ as $t\to\infty$.

背景链接：https://en.wikipedia.org/w/index.php?title=Simon_problems&oldid=1335881350#The_2000_list

17368 | OWR-15961-002

### Quantum Optimal Transport for Mean-Field Dynamics

P / partially_solved | Oberwolfach 报告问题汇编

Can the pseudo-metric used for quantum mean-field dynamics be formulated through a quantum analogue of optimal transport, such as a Benamou-Brenier variational formula?

来源字段：https://doi.org/10.4171/owr/2017/56

17369 | OWR-11447-003

### Quantum Unique Ergodicity on Negatively Curved Manifolds

P / partially_solved | Oberwolfach 报告问题汇编

Quantum unique ergodicity conjecture: eigenmodes of manifolds with negative sectional curvature become equidistributed in the high-energy limit.

来源字段：https://doi.org/10.4171/owr/2011/49

17370 | OWR-11447-002

### Quantum Variance Conjectures for Matrix Elements

P / partially_solved | Oberwolfach 报告问题汇编

For an observable $A$, do the off-diagonal matrix elements satisfy $|A_{jk}|^2=S(0)/(2\pi\hbar\rho(E))$, hence scale as $\hbar^{d-1}$? Are the diagonal matrix elements Gaussian with mean $\{A\}$ and variance $gS(0)/(2\pi\hbar\rho(E))$, where $g=2$ in the presence of time-reversal symmetry and $g=1$ otherwise?

来源字段：https://doi.org/10.4171/owr/2011/49

17371 | OWR-1270-002

### Quantum Zeta Functions for Mirror Calabi-Yau Threefolds

P / partially_solved | Oberwolfach 报告问题汇编

For mirror pairs of Calabi-Yau threefolds over $\mathbb Q$, find a satisfactory quantum zeta function and quantum $L$-series that retain the essential weight-motive factor and satisfy the expected $T$-duality.

来源字段：https://doi.org/10.4171/owr/2006/21

17372 | OWR-13495-003

### Quantum-Group and Affine-Hecke Solutions of Boundary qKZ Equations

O / open | Oberwolfach 报告问题汇编

Determine the relation between the special boundary quantum Knizhnik-Zamolodchikov solutions obtained as quantum correlation functions or $q$-integrals and the affine-Hecke-algebra basis characterized by its asymptotics in a Weyl chamber.

来源字段：https://doi.org/10.4171/owr/2015/10

17373 | OWR-14297741-012

### Quantum-Symmetric Classes of Artin-Schelter Regular Algebras

O / open | Oberwolfach 报告问题汇编

Determine the full quantum-symmetric equivalence class of an Artin-Schelter regular algebra.

来源字段：https://doi.org/10.4171/owr/2023/40

17374 | OWR-14299579-002

### Quasi-Bosonic Ground States and Excitation Spectra

O / open | Oberwolfach 报告问题汇编

Can the quasi-bosonic approximation established for the first eigenvalue be extended to the true many-body ground state, possibly after suitable unitary transformations, and does it yield a quasi-bosonic description of the higher excitation spectrum?

来源字段：https://doi.org/10.4171/owr/2025/46

17375 | OWR-13355-004

### Quasicrystalline Gibbs Phases in Higher Dimensions

P / partially_solved | Oberwolfach 报告问题汇编

In every dimension at least $3$, does there exist a translation-invariant Gibbs measure for a short-range interaction whose extremal decomposition contains nonperiodic Gibbs measures?

来源字段：https://doi.org/10.4171/owr/2014/53

17376 | OWR-14298801-003

### Query Complexity and Achievable Functions in Multivariable Quantum Signal Processing

O / open | Oberwolfach 报告问题汇编

For multivariable quantum signal processing, lower-bound the query complexity required to realize a prescribed function class when ancillary space, intermediate measurements, or different oracle types are allowed. Identify a nontrivial, simply described subclass of achievable functions with improved numerical properties.

来源字段：https://doi.org/10.4171/owr/2024/46

17377 | OWR-14751-003

### Reconstruction of Many-Body Dynamics from One-Particle Density

P / partially_solved | Oberwolfach 报告问题汇编

For time-dependent $N$-boson or $N$-fermion systems with known interactions and initial state, can the full quantum dynamics, and in particular the external potential, be reconstructed from the one-particle density?

来源字段：https://doi.org/10.4171/owr/2016/43

17378 | OWR-3392-006

### Recovering Spatial Tensor Systems by Intersection

O / open | Oberwolfach 报告问题汇编

Given a spatial tensor product subsystem $E^{\odot}$ of a free product system, is $E^{\odot}$ recovered by the intersection procedure over all decompositions of each time parameter? More generally, characterize when a free product system can be obtained in this way from a spatial tensor product subsystem.

来源字段：https://doi.org/10.4171/owr/2009/09

17379 | OWR-14299511-001

### Reflection-Positive Two-Dimensional Quantum Field Theories with Boundary

O / open | Oberwolfach 报告问题汇编

Identify the generalized $O(2)$-action on algebras whose fixed points are von Neumann algebras. Classify reflection-positive, remote-detectable, flat-space Lorentz-invariant two-dimensional quantum field theories with boundary: are von Neumann algebras the complete list, or do other examples exist?

来源字段：https://doi.org/10.4171/owr/2025/34

17380 | OWR-13357-003

### Reflection-Positivity Research Questions

O / open | Oberwolfach 报告问题汇编

Resolve the following questions about reflection positivity: describe super versions of massive free fields and reflection-positive Hilbert spaces; determine the role of antiunitary involutions; construct Euclidean realizations of Borchers triples and algebraic quantum field theories; find diffeomorphism-invariant examples and a noncommutative-geometric theory; clarify links with complex measures, stochastic quantization, number theory, string theory, one-dimensional conformal field theory, cluster expansions, modular theory, and the Bessis-Moussa-Villani theorem; remove growth assumptions from the equivalence of the Osterwalder-Schrader and Wightman axioms; and characterize the common structural core of these phenomena.

来源字段：https://doi.org/10.4171/owr/2014/55

17381 | OWR-12009-005

### Regular Conformal Equations for Matter with Nonzero Trace

O / open | Oberwolfach 报告问题汇编

For matter with a nonzero-trace energy-momentum tensor, does there exist a choice of conformal variables yielding a regular conformal field-equation system with matter?

来源字段：https://doi.org/10.4171/owr/2012/37

17382 | OWR-15998-001

### Regularized Density-to-Potential Maps in Density Functional Theory

P / partially_solved | Oberwolfach 报告问题汇编

Establish rigorous conditions for existence of the density-to-potential map in density functional theory, and use functional-analytic Moreau-Yosida regularization to address the nondifferentiability of the constrained-search Levy-Lieb functional.

来源字段：https://doi.org/10.4171/owr/2018/13

17383 | OWR-827-001

### Relationship Between Two-Species Condensates

O / open | Oberwolfach 报告问题汇编

When both particle species have condensates, give a rigorous analysis of the relation between the two condensates.

来源字段：https://doi.org/10.4171/owr/2005/22

17384 | OWR-1109-002

### Renormalization through Langmann-Szabo Duality

P / partially_solved | Oberwolfach 报告问题汇编

Can Langmann-Szabo duality be used to construct renormalizable noncommutative quantum field theories?

来源字段：https://doi.org/10.4171/owr/2005/48

17385 | OWR-12009-007

### Resonant Mode Mixing in Anti-de-Sitter Instability

P / partially_solved | Oberwolfach 报告问题汇编

Is the instability of anti-de-Sitter space caused by resonant mode mixing that transfers energy from low to high frequencies until an apparent horizon forms?

来源字段：https://doi.org/10.4171/owr/2012/37

17386 | OWR-16411-009

### Reverse Burnett Problem for Einstein-Vlasov Limits

P / partially_solved | Oberwolfach 报告问题汇编

Given a solution $g_0$ of the Einstein equations coupled to a massless Vlasov field, can one construct vacuum Einstein metrics $g_\lambda$ converging to $g_0$ in a high-frequency weak limit with $g_\lambda\to g_0$ strongly in $L^2_{\mathrm{loc}}$ and $\partial g_\lambda$ bounded in $L^2_{\mathrm{loc}}$?

来源字段：https://doi.org/10.4171/owr/2018/36

17387 | AMR-079-0003

### Riemann-Hilbert Problem with non-analytic data

P / partially_solved | AMR 问题清单汇编

In many situations one is concerned with the asymptotic behavior of Riemann-Hilbert problems with exponentially varying data of the form $e^{in\phi(z)} r(z)$, $n \to \infty$. The Deift-Zhou nonlinear steepest descent method for such problems requires $\phi(z)$ to be analytic. The analyticity is used in two ways: to control the equilibrium measure associated with the problem, and then to deform the contour for the RHP. By contrast, the method only requires minimal smoothness for $r(z)$ (see eg. Deift-Zhou in the context of Problem 12 below). It is of considerable theoretical and practical interest to extend the nonlinear steepest descent method to situations where $\phi$ is no longer analytic, and has, for example, only a finite number of derivatives. For very interesting work on the analyticity problem, we refer the reader to a recent paper of Miller and McLaughlin. There is also interesting, older work due to Varzugin.

背景链接：https://arxiv.org/abs/0712.0849

17388 | OWR-1544-003

### Rigid Order-Disorder Interfaces in Continuous-Spin Models

O / open | Oberwolfach 报告问题汇编

For the continuous-spin model on $\mathbb Z^d$ with spins $\sigma_i\in S^1=\mathbb R/\mathbb Z$ and Hamiltonian $H(\sigma)=\sum_{\langle i,j\rangle}U(|\sigma_i-\sigma_j|)$, where $U(r)=-((1+\cos r)/2)^p$ for large integer $p$, is the order-disorder interface rigid when $d\geq3$? Equivalently, does the model admit three-dimensional Dobrushin states?

来源字段：https://doi.org/10.4171/owr/2007/28

17389 | OWR-14298373-005

### Rigorous Construction of Conformal Field Theories

O / open | Oberwolfach 报告问题汇编

How can rigorous examples of conformal field theories be constructed, including rigorous path integrals and justification of the bootstrap hypothesis?

来源字段：https://doi.org/10.4171/owr/2024/25

17390 | AMR-080-0008

### Rigorous Diffraction from Two Slits

O / open | AMR 问题清单汇编

Give a rigorous explicit solution of the fixed-frequency scalar-wave diffraction problem for two finite slits in the plane, including asymptotics of the solution.

背景链接：https://arxiv.org/abs/1703.04931

17391 | OWR-14297737-001

### Rigorous Mesoscopic Limits of Many-Particle Systems

O / open | Oberwolfach 报告问题汇编

Rigorously justify mesoscopic evolution equations, including the Boltzmann equation, as limits of many-particle systems.

来源字段：https://doi.org/10.4171/owr/2023/38

17392 | OWR-14298586-001

### Robustness of Climate Models at Unresolved Scales

O / open | Oberwolfach 报告问题汇编

Determine whether reduced geophysical-flow models admit sufficiently rough solution spaces, whether their numerical approximations are robust under grid-scale perturbations, and which stochastic structures correctly represent unresolved processes. In particular, decide whether white-noise and Brownian-motion models suffice or whether rough paths, regularity structures, or paracontrolled distributions are required.

来源字段：https://doi.org/10.4171/owr/2024/31

17393 | OWR-4084-010

### Rooted-Tree Expansions of Renormalized Two-Point Functions

O / open | Oberwolfach 报告问题汇编

At every perturbative order, is the renormalized planar connected two-point function $G_{\alpha,\beta}$ a polynomial with rational coefficients in $\alpha$, $\beta$, $A$, $B$, and iterated integrals indexed by rooted trees?

来源字段：https://doi.org/10.4171/owr/2009/41

17394 | OWR-4431-001

### Rooted-Tree Expansions of Renormalized Two-Point Functions

O / open | Oberwolfach 报告问题汇编

At every order in the perturbative expansion, is the renormalized planar connected two-point function $G_{\alpha\beta}$ a polynomial with rational coefficients in $\alpha$, $\beta$, $A=(1-\alpha)/(1-\alpha\beta)$, $B=(1-\beta)/(1-\alpha\beta)$, and iterated integrals labeled by rooted trees?

来源字段：https://doi.org/10.4171/owr/2010/43

17395 | OWR-1109-006

### Scaling Limits of Tensor Products in AQFT

O / open | Oberwolfach 报告问题汇编

Find general conditions under which scaling-limit and tensor-product operations commute for nets in algebraic quantum field theory.

来源字段：https://doi.org/10.4171/owr/2005/48

17396 | OWR-15998-004

### Selecting Physical Solutions of Many-Body Green-Function Equations

O / open | Oberwolfach 报告问题汇编

For equations governing physical many-body Green functions, determine whether there is a link to phase transitions, whether the equations can be modified to reduce nonphysical solution multiplicity, how the results affect strong-coupling many-body perturbation theory, and how to parametrize the physical solution domain.

来源字段：https://doi.org/10.4171/owr/2018/13

17397 | OWR-8415343-009

### Self-Consistent Motion of Massive Charged Particles in Relativity

O / open | Oberwolfach 报告问题汇编

How can one rigorously model the motion of massive and charged particles in general relativity, including their coupled electromagnetic and gravitational fields and the timelike singularities they generate?

来源字段：https://doi.org/10.4171/owr/2021/40

17398 | OWR-11447-001

### Semiclassical Measures and Exceptional Eigenmodes

O / open | Oberwolfach 报告问题汇编

What are the possible semiclassical measures, and do they include all classically invariant measures? What is the rate of convergence to Liouville measure, and can exceptional eigenmodes exhibit the behavior of exceptional invariant measures?

来源字段：https://doi.org/10.4171/owr/2011/49

17399 | AMR-077-0009

### Separatrix Splitting under Quasiperiodic Forcing

P / partially_solved | AMR 问题清单汇编

Find an asymptotic expression for the splitting of the separatrix of a quasiperiodically forced pendulum in the regime where the perturbation series in the coupling $\varepsilon$ converges but the series of terms leading at each order as the fast-frequency parameter tends to zero does not appear to converge.

背景链接：https://web.math.princeton.edu/~aizenman/OpenProblems_MathPhys/9805.Separatrix.html

17400 | OWR-15213-001

### Sharp Convergence Domains for Cluster Expansions

O / open | Oberwolfach 报告问题汇编

For a polymer-model cluster expansion $$ \log Z_L(w)=\sum_X a(X)w^X, $$ determine its exact domain of convergence and sharp conditions on the contour weights and graph-dependent radii that guarantee convergence.

来源字段：https://doi.org/10.4171/owr/2017/8

17401 | OWR-17137-006

### Sharp Free Energy of the Two-Dimensional Heisenberg Ferromagnet

O / open | Oberwolfach 报告问题汇编

For the two-dimensional quantum Heisenberg ferromagnet, the free energy obeys

$$f^{2d}(\beta,S)\leq-\frac\pi{24}S^{-1}\beta^{-2}\left(1-O\bigl((\beta S)^{-\kappa}\bigr)\right)$$

for every $\kappa<1/3$. Can one prove a matching lower bound?

来源字段：https://doi.org/10.4171/owr/2019/41

17402 | OWR-14299579-001

### Sharp Second-Order Energy of Hard-Sphere Bosons

P / partially_solved | Oberwolfach 报告问题汇编

Prove the upper bound with the Lee-Huang-Yang constant in the second-order ground-state energy expansion for the dilute hard-sphere Bose gas.

来源字段：https://doi.org/10.4171/owr/2025/46

17403 | OWR-14751-002

### Sharp Spectral-Cluster Density Bounds on Manifolds

O / open | Oberwolfach 报告问题汇编

Let $M$ be smooth, compact, and boundaryless, let $E_\lambda=\mathbf{1}_{\{x:\lambda^2\leq x<(\lambda+1)^2\}}(\Delta_g)L^2(M)$, and suppose $1\ll r_\lambda\ll\dim E_\lambda$. For $2\leq p\leq\infty$, do there exist subspaces $Q_\lambda\subset E_\lambda$ with $\dim Q_\lambda\sim r_\lambda$ such that $\lVert\rho_{Q_\lambda}\rVert_{L^{p/2}(M)}\geq c\lambda^{2s(p)}r_\lambda^{1/\alpha(p)}$ for some $c>0$ independent of $\lambda$, where $\rho_Q$ is the density of an orthonormal basis of $Q$?

来源字段：https://doi.org/10.4171/owr/2016/43

17404 | AMR-077-0011

### Short-Range Spin Glasses

O / open | AMR 问题清单汇编

For the Edwards-Anderson Ising spin glass on $\mathbb{Z}^d$ with i.i.d. mean-zero finite-variance nearest-neighbor couplings, prove or disprove the existence of a thermodynamic phase transition above a lower critical dimension $d_c$, determine $d_c$, decide whether a low-temperature phase breaks spin-flip symmetry, and characterize the number and metastate organization of pure-state pairs.

背景链接：https://web.math.princeton.edu/~aizenman/OpenProblems_MathPhys/9803.SpinGlass.html

17405 | OWR-4084-002

### Sigma Models with Noncommutative Target Spaces

P / partially_solved | Oberwolfach 报告问题汇编

When the target space $X$ in the sigma model of maps $g:\Sigma\to X$ is replaced by a noncommutative space such as an irrational rotation algebra, what should replace the maps and the action functional?

来源字段：https://doi.org/10.4171/owr/2009/41

17406 | OWR-17292-005

### Small-Weight Quantum Operations on Lattice Conformal Nets

O / open | Oberwolfach 报告问题汇编

For a lattice conformal net or a strongly local vertex-operator-algebra net $\mathcal A$, find $\phi\in\operatorname{QuOp}(\mathcal A)$ satisfying $$\phi\circ\phi=w^{-1}\operatorname{id}+(1-w^{-1})\phi$$ for arbitrarily small real $w>1$.

来源字段：https://doi.org/10.4171/owr/2019/49

17407 | AMR-077-0001

### Soft Phases in Two-Dimensional O(N) Models

P / partially_solved | AMR 问题清单汇编

Do spin correlations in the classical Heisenberg model, and other $O(N)$ models with $N>2$, decay exponentially at every nonzero temperature, or do these models have soft phases like the XY model? Prove or disprove that the low-temperature perturbative expansion is asymptotically correct, and determine whether there is a nonzero-temperature critical point and, if so, its critical indices.

背景链接：https://web.math.princeton.edu/~aizenman/OpenProblems_MathPhys/9904.O(N

17408 | OWR-1109-005

### Spectral Geometry and Cyclic Homology of $SU_q(2)$

O / open | Oberwolfach 报告问题汇编

For the spectral geometry of $SU_q(2)$, what are the geometric roles of Paschke duality and the ideal $K_q$? Can the Hochschild-cycle axiom be satisfied or suitably modified? Is there a canonical local-index cocycle from the spectral data? What differential calculus and metric arise, and what freedom in the Dirac operator changes that metric? What is the correct equivariant cyclic-homology framework, and how can it be computed?

来源字段：https://doi.org/10.4171/owr/2005/48

17409 | OWR-8415352-008

### Spin-Alignment Entropy Conjecture

P / partially_solved | Oberwolfach 报告问题汇编

Fix a mixed one-spin state $Q$. Consider $n$-spin density operators formed as convex combinations of states in which a chosen subset of spins is fixed to a product of copies of $Q$ and the remaining spins have variable joint states. Is the entropy minimized when every variable joint state is a product of copies of one pure state $|\phi\rangle$, where $|\phi\rangle$ is an eigenvector of $Q$ for its largest eigenvalue?

来源字段：https://doi.org/10.4171/owr/2021/49

17410 | OWR-4083-002

### Stability of Atomic Ground-State Quantum Numbers

O / open | Oberwolfach 报告问题汇编

Prove that the angular-momentum and spin quantum numbers and the ground-state dimensions obtained theoretically for fixed electron number $N$ do not change as the nuclear charge $Z$ decreases to the neutral-atom value $Z=N$.

来源字段：https://doi.org/10.4171/owr/2009/40

17411 | OWR-14298591-011

### Stability of Extremal Critical Collapse

O / open | Oberwolfach 报告问题汇编

Does there exist a teleologically determined hypersurface in the moduli space of initial data that consists of asymptotically extremal black holes, contains the known critical family $D_{\lambda_*}$, and locally separates future-complete data from collapsing data?

来源字段：https://doi.org/10.4171/owr/2024/36

17412 | OWR-1597-005

### Stability of Schwarzschild Black Holes Under Perturbation

P / partially_solved | Oberwolfach 报告问题汇编

For vacuum initial data that are a sufficiently small perturbation of Schwarzschild data on a Cauchy surface $S$, does the maximal development $(M,g)$ still contain a black hole; equivalently, is future null infinity $I^+$ complete and is $M\setminus J^-(I^+)$ nonempty?

来源字段：https://doi.org/10.4171/owr/2007/41

17413 | OWR-4536-001

### Strict Positivity for Fourth-Order Quantum Diffusion

O / open | Oberwolfach 报告问题汇编

Let $n\geq0$ solve $\partial_t n+\operatorname{div}\!\left(n\nabla\left(\frac{\Delta\sqrt n}{\sqrt n}\right)\right)=0$ on the $d$-dimensional torus, with nonnegative initial datum $n_0$. Is $n(\cdot,t)$ strictly positive for every $t>0$ even if $n_0$ vanishes somewhere?

来源字段：https://doi.org/10.4171/owr/2010/54

17414 | OWR-14299085-004

### Strong Continuity Without Ultraviolet Regularization

O / open | Oberwolfach 报告问题汇编

Does strong continuity of the many-body dynamics persist when the ultraviolet regularization is removed, that is, in the limit $\sigma\to 0$?

来源字段：https://doi.org/10.4171/owr/2025/7

17415 | OWR-13854-002

### Strong Cosmic Censorship at Weak Regularity

O / open | Oberwolfach 报告问题汇编

For generic perturbations of Kerr initial data whose maximal globally hyperbolic developments have continuously extendible Cauchy horizons, are those developments nevertheless inextendible by Lorentzian metrics of regularity $C^0\cap W_{\mathrm{loc}}^{1,2}$? More generally, for generic asymptotically flat vacuum initial data, is the maximal globally hyperbolic development inextendible as a weak solution of the Einstein vacuum equations?

来源字段：https://doi.org/10.4171/owr/2015/33

17416 | OWR-4129-004

### Strong Cosmic Censorship for Spherical Einstein-Vlasov Collapse

O / open | Oberwolfach 报告问题汇编

For spherically symmetric Einstein-Vlasov initial data in Eddington-Finkelstein coordinates that evolve from $a>0$ to form a trapped surface, must a curvature singularity arise at the center, and does strong cosmic censorship hold?

来源字段：https://doi.org/10.4171/owr/2009/46

17417 | OWR-15212-002

### Strong-Interaction Limits of the Hohenberg-Kohn Functional

P / partially_solved | Oberwolfach 报告问题汇编

For the Hohenberg-Kohn functional and its multimarginal optimal-transport limit, determine the strong-interaction asymptotics; prove the zero-point-energy correction in one and, if possible, three dimensions; describe corrections due to quantum statistics; establish deterministic Monge solutions in three dimensions with more than two marginals; and construct effective approximate transport maps or functional derivatives for density-functional theory.

来源字段：https://doi.org/10.4171/owr/2017/7

17418 | OWR-14299511-003

### SU(3) Braided Fusion Spin Systems and Haah Nets

O / open | Oberwolfach 报告问题汇编

Prove that the braided fusion spin system associated with $\mathrm{SU}(3)_1$, or with its reverse braiding, is bounded-spread isomorphic to Haah's proposed two-dimensional net of algebras with the specified local clock-and-shift generators.

来源字段：https://doi.org/10.4171/owr/2025/34

17419 | OWR-14299511-002

### Superselection Sectors of Braided Fusion Spin Systems

O / open | Oberwolfach 报告问题汇编

Let $\mathcal{B}$ be a unitary braided fusion category with strong tensor generator $X$, and let its braided fusion spin system be the associated two-dimensional net with local algebras $\operatorname{End}_{\mathcal{B}}(X^{\otimes I})$. Prove that the superselection-sector category of this net is equivalent to the $W^*$-categorical completion of $\mathcal{B}$.

来源字段：https://doi.org/10.4171/owr/2025/34

17420 | DARPA-016

### Symmetries and Action Principles for Biology

P / partially_solved | DARPA 挑战

Extend understanding of symmetries and action principles in biology to include robustness, modularity, evolvability, and variability.

背景链接：https://web.math.utk.edu/~vasili/refs/darpa07.MathChallenges.html

17421 | OWR-15960-002

### Symmetry Identities Beyond Renormalization-Group Methods

P / partially_solved | Oberwolfach 报告问题汇编

Can symmetry-generated identities help study models for which standard renormalization-group techniques do not apply?

来源字段：https://doi.org/10.4171/owr/2017/55

17422 | OWR-790-001

### Systematic Multiscale Design of Hierarchical Materials

O / open | Oberwolfach 报告问题汇编

How can a hierarchical material be designed systematically to achieve a particular set of properties? How can different levels of structural hierarchy be bridged? What criteria should determine the characteristic length scales at all hierarchical levels? What is the theoretical basis for bottom-up materials design?

来源字段：https://doi.org/10.4171/owr/2005/09

17423 | AIM-PHYSICS-0045

### Syzygy-word parity and the angular-momentum holonomy obstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 8.** Can any symbol sequence be realized in the planar three-body problem? Are they minimizing or not? Do they have zero angular momentum? (Richard Montgomery)

“Symbol sequence” is given by the crossing of the equator in the shape space, count each time bodies as collinear with which body is in the middle.

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17424 | OWR-14298591-013

### Tensorial Mode Stability of Kerr-de Sitter Spacetimes

O / open | Oberwolfach 报告问题汇编

Prove ungauged tensorial Einstein mode stability throughout the full subextremal Kerr-de Sitter parameter range.

来源字段：https://doi.org/10.4171/owr/2024/36

17425 | OWR-14298591-010

### Teukolsky Energy Decay on Extremal Kerr Spacetimes

P / partially_solved | Oberwolfach 报告问题汇编

On an extremal Kerr spacetime with $|a|=M$, prove energy boundedness and energy decay for fixed-azimuthal-mode solutions of the Teukolsky equations, with the predicted pointwise decay away from the future event horizon, decay of the outgoing component on the horizon, and generic growth of the ingoing component there.

来源字段：https://doi.org/10.4171/owr/2024/36

17426 | AIM-PHYSICS-0040

### The four-body super-eight and the rotating-square obstruction

S / solved | AIM 研讨会问题汇编

**Problem 3.** Prove the existence of super-eight solution for the 4-body problem with equal masses. (Joseph Gerver)

**Remarks.**

- One difficulty: the square relative equilibrium is absolute minimum in that symmetry class. - There is a computer assisted proof by interval arithmetic.

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17427 | OWR-14299579-005

### The Gell-Mann-Brueckner Formula in the Thermodynamic Limit

O / open | Oberwolfach 报告问题汇编

Establish the Gell-Mann-Brueckner formula in the thermodynamic limit.

来源字段：https://doi.org/10.4171/owr/2025/46

17428 | AIM-PHYSICS-0035

### The L2-signature of toric hyperkahler quotients

P / partially_solved | AIM 研讨会问题汇编

**3. (Hausel)** We know that all the \(L^2\) Hodge cohomology of these spaces must be in the middle dimension and that any \(L^2\)-harmonic form in middle dimension is \(\pm\) self-dual. So it is enough to understand the \(L^2\)-signature. This will have consequences for the sign of \(\sigma_{L^2}(X)\) and is related to combinatorics and matroids.

**More general problem:** Prove a suitable index theorem on such manifolds. Will there be contributions from subsystems?

来源字段：https://aimath.org/WWN/l2harmonic/l2harmonic.pdf

17429 | AMR-079-0009

## The parking problem

P / partially_solved | AMR 问题清单汇编

A number of so-called ``transportation'' problems have now been analyzed in terms of RMT. These include: the ``vicious'' walker problem of M. Fisher, the bus problem in Cuernavaca, Mexico, the headway traffic problem on highways, and the airline boarding problem of Bachmat et al. Recently, researchers in London, Prague, and also Ann Arbor, have noticed an intriguing phenomenon. They found that the fluctuations in the spacings between cars parked on a long street exhibited RMT behavior. Furthermore, \v{S}eba found that there was a difference whether the street is two-way or one-way (On a two-way street, the cars park only on the right, while on a one-way street one of course has the option of also parking on the left.) Quite remarkably, for two-way streets \v{S}eba found GUE statistics, but for left-side parking on one-way streets he found GOE statistics. It is a great challenge to develop a microscopic model for the parking problem, in analogy, perhaps, with the microscopic model introduced by Baik et al. to explain the RMT statistics for the bus problem in Cuernavaca. \v Seba's recent, intriguing calculations on the parking problem can be found posted on the web.

背景链接：https://arxiv.org/abs/0712.0849

17430 | OWR-16636-006

## The Quantum Unique Ergodicity Conjecture

O / open | Oberwolfach 报告问题汇编

On every compact negatively curved manifold, do all high-energy Laplace eigenfunctions equidistribute?

来源字段：https://doi.org/10.4171/owr/2018/53

17431 | AIM-PHYSICS-0007

## The sharp Laptev-Safronov threshold and a Lorentz-stable tube obstruction

P / partially_solved | AIM 研讨会问题汇编

Frank's item 7.1 and then appends the complete items 8.1 (Marcel Hansmann), 9.1 (Michael Hitrik), and 10.1 (David Krejčiřík). The record key is `number: 7`, so this attempt treats only item 7.1.

The exponent in the radius is \((p-d/2)^{-1}\), and the power of \(|V|\) in (1.1) is \(\gamma+d/2\).

Spillover catalog: item 8 asks about removing an \(\varepsilon\)-loss in trace-class perturbation sums; item 9 asks whether an exponential semiclassical resolvent upper bound is sharp; item 10 asks for pseudomodes with discontinuous potentials. None is analyzed here.

来源字段：https://aimath.org/pastworkshops/nonselfadjointproblems.pdf

17432 | AIM-PHYSICS-0038

## The signed Hill threshold for binary exchange

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.** Consider the three-body problem. Let \(E\) be the energy, \(C\) be the angular momentum, and \(L=C^2E\) for circular Euler motion. An exchange orbit is an unbounded solution with the property that the distance between one mass \(m_3\) and the binary \(\{m_1,m_2\}\) goes to infinity as \(t\to-\infty\), \(m_3\) exchanges with \(m_2\), and the distance between \(m_2\) and the binary \(\{m_1,m_3\}\) goes to infinity as \(t\to+\infty\). A solution has Hill type stability if the three bodies cannot undergo an exchange of binary. For exchange orbits, how close to \(1\) can

\[ \frac{C^2E}{L} \]

be? (Christian Marchal)

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17433 | AIM-PHYSICS-0053

## The syzygy-free three-body solution and quantitative crossing certificates

S / solved | AIM 研讨会问题汇编

Problem 16. Is the only solution satisfying (i), (ii) and having no syzygies (i.e. eclipses) the Lagrange homothety solutions? (i) Energy < 0. (ii) Angular momentum = 0. (Richard Montgomery)

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17434 | AMR-079-0011

## The Toda lattice with random initial data

P / partially_solved | AMR 问题清单汇编

Let $J$ be a random tridiagonal matrix drawn from the tridiagonal Gaussian orthogonal ensemble and evolve it by the finite nonperiodic Toda flow, so its off-diagonal entries are $b_1(t),\ldots,b_{n-1}(t)$. Given $\varepsilon>0$, determine the expected time until $\max_{1\leq i\leq n-1}b_i(t)<\varepsilon$; equivalently, determine the average time for Toda particles with random tridiagonal-ensemble initial data to become free.

背景链接：https://arxiv.org/abs/0712.0849

17435 | OWR-1598-002

## Thermal Conductivity Under Asymmetric Momentum Noise

P / partially_solved | Oberwolfach 报告问题汇编

For harmonic oscillators, add to the Hamiltonian dynamics the asymmetric momentum noise $S=\sum_x[(p_{x+1}\partial_{p_x}-p_x\partial_{p_{x+1}})^2+p_xp_{x+1}(p_{x+1}\partial_{p_x}-p_x\partial_{p_{x+1}})]$. Is the resulting thermal conductivity finite?

来源字段：https://doi.org/10.4171/owr/2007/42

17436 | OWR-16632-002

### Thermodynamic Fluid-Solid Phase Transitions

O / open | Oberwolfach 报告问题汇编

Can statistical mechanics prove the existence of a thermodynamic fluid-solid phase transition, for example in the hard-sphere model?

来源字段：https://doi.org/10.4171/owr/2018/49

17437 | OWR-1325-003

### Topographic Rossby Waves and Inertial-Wave Coexistence

O / open | Oberwolfach 报告问题汇编

How is the apparent paradox between topographic Rossby waves and inertial waves resolved: does the series whose zeroth-order term is the topographic Rossby wave diverge locally, or does it coexist with the inertial wave?

来源字段：https://doi.org/10.4171/owr/2006/39

17438 | OWR-16161-002

### Topological Defects in Three-Dimensional Polycrystals

O / open | Oberwolfach 报告问题汇编

For three-dimensional orientation-field models of polycrystalline solidification, are there types of topological defects beyond those already identified in the two-dimensional and quaternionic descriptions?

来源字段：https://doi.org/10.4171/owr/2018/20

17439 | AIM-PHYSICS-0037

### Topology and character sectors in monopole L2-cohomology

P / partially_solved | AIM 研讨会问题汇编

**1.** Is there a topological interpretation for the \(L^2\)-cohomology of higher dimensional monopole moduli spaces?

(i) What are the right tools to use here?

(ii) Can we use Saper's \(\mathcal L\)-module techniques? (See [11, Section 12].)

In the case of the \(SU(2)\)-monopole moduli space over \(\mathbb R^3\), the Segal--Selby paper [12] proves (by pure topology) that there is a harmonic form (coming from the topology of the moduli space) as predicted by Sen's conjecture. What remains is to prove that these are the only \(L^2\)-harmonic forms (surjectivity of the map in the Segal--Selby construction).

来源字段：https://aimath.org/WWN/l2harmonic/l2harmonic.pdf

17440 | OWR-13854-004

### Transferring Resonant Blowup to the Full Einstein System

O / open | Oberwolfach 报告问题汇编

How can the finite-time blowup result for the resonant system be transferred to the full system?

来源字段：https://doi.org/10.4171/owr/2015/33

17441 | OWR-8415354-006

### Trigonometric and Elliptic R-Matrices from Three-Dimensional WZW Models

O / open | Oberwolfach 报告问题汇编

Can three-dimensional Wess-Zumino-Witten models be generalized to produce trigonometric and elliptic R-matrices for broader choices of the defining differential form?

来源字段：https://doi.org/10.4171/owr/2021/51

17442 | OWR-14299913-003

### Two-Categorical Interpretation of a Quantum Lattice Model

O / open | Oberwolfach 报告问题汇编

Give a categorical interpretation of the computed symmetry structure in the two-dimensional quantum lattice model, and determine whether it is naturally governed by a two-category.

来源字段：https://doi.org/10.4171/owr/2026/12

17443 | OWR-14299094-001

### Unique Ground-State Representability under Linear Constraints

P / partially_solved | Oberwolfach 报告问题汇编

For a constrained-search problem with observables $B_1,\ldots,B_m$, characterize when a prescribed expectation vector has a unique representing parameter $\beta\in\mathbb{R}^m$ and a unique ground-state representation, including critical values and degenerate ground spaces.

来源字段：https://doi.org/10.4171/owr/2025/15

17444 | OWR-4805-001

### Uniqueness in the Boltzmann-Nordheim Condensate Model

P / partially_solved | Oberwolfach 报告问题汇编

For the regular part of the isotropic Boltzmann-Nordheim condensate model, consider the mass-conserving evolution equation $\partial_t f_{\mathrm{reg}}(x,t)=C_4[f_{\mathrm{reg}}(\cdot,t)](x)+(m_{\mathrm{tot}}(0)-m[f_{\mathrm{reg}}(\cdot,t)])C_3[f_{\mathrm{reg}}^{(1)}(\cdot,t)](x)$. Does this equation have a unique solution when a condensate is generated? Can the perturbative construction be extended to initial data that are not $\ell^1$-clustering, thereby deriving this equation rather than the standard Boltzmann-Nordheim equation?

来源字段：https://doi.org/10.4171/owr/2011/15

17445 | OWR-14298591-009

### Uniqueness of Spherical Schwarzschild-Anti-de Sitter Spacetimes

O / open | Oberwolfach 报告问题汇编

Prove a uniqueness theorem for spherical Schwarzschild-anti-de Sitter spacetimes among static Einstein spacetimes with negative cosmological constant.

来源字段：https://doi.org/10.4171/owr/2024/36

17446 | OWR-4129-002

### Uniqueness of Stationary Black-Hole Exteriors

O / open | Oberwolfach 报告问题汇编

Are the Kerr solutions the only sufficiently regular stationary black-hole exteriors with a simply connected, non-degenerate bifurcate event horizon?

来源字段：https://doi.org/10.4171/owr/2009/46

17447 | OWR-14751-004

### Universal Bounds on Atomic Excess Charge

O / open | Oberwolfach 报告问题汇编

Does there exist a universal constant $C$ such that a nucleus of charge $Z$ can bind at most $Z+C$ electrons in the full many-body Schrödinger theory?

来源字段：https://doi.org/10.4171/owr/2016/43

17448 | OWR-14298170-020

### Vacuum Failure of the Third Law of Black-Hole Thermodynamics

O / open | Oberwolfach 报告问题汇编

Construct a vacuum spacetime in which gravitational waves collapse to form a subextremal Kerr apparent horizon and the spacetime later forms an exactly extremal Kerr event horizon.

来源字段：https://doi.org/10.4171/owr/2024/9

17449 | OWR-4536-002

### Vanishing and Uniqueness of Quantum Diffusion Solutions

P / partially_solved | Oberwolfach 报告问题汇编

For the sixth-order equation $\partial_t n=\operatorname{div}\!\left[n\nabla\left(\frac12|\nabla^2\log n|^2+\frac1n\nabla^2\!:\!(n\nabla^2\log n)\right)\right]$ on the $d$-dimensional torus, with $n(\cdot,0)=n_0\geq0$, do there exist weak solutions that vanish at some points? Are weak solutions unique?

来源字段：https://doi.org/10.4171/owr/2010/54

17450 | OWR-16769-001

### Variance Feedback in Cloud Boundary-Layer Models

O / open | Oberwolfach 报告问题汇编

For Reynolds-averaged equations modeling the variances of liquid-water potential temperature and total-water mixing ratio in a precipitating cloudy atmospheric boundary layer, can the equations represent explicitly the slow positive feedback that produces the variance without an additional stochastic submodel?

来源字段：https://doi.org/10.4171/owr/2019/7

17451 | OWR-15959-002

### Variational GENERIC Structure and Preserving Integrators

P / partially_solved | Oberwolfach 报告问题汇编

别名：Variational Principle for Complete GENERIC Thermodynamics [OWR-15959-001]

Does there exist a variational principle that embodies the complete GENERIC structure of nonequilibrium thermodynamics, including its reversible Poisson and irreversible dissipative parts? If so, can its discretization yield structure-preserving GENERIC integrators?

来源字段：https://doi.org/10.4171/owr/2017/54

17452 | OWR-14299511-007

### Vertex Operator Algebras with Haagerup Symmetry

O / open | Oberwolfach 报告问题汇编

Construct a vertex operator algebra arising in conformal field theory whose representation category contains the Haagerup fusion category or its symmetry data, or prove that no such vertex operator algebra exists.

来源字段：https://doi.org/10.4171/owr/2025/34

17453 | OWR-1533-001

### Wang-Ostrik Finiteness Conjecture for Fusion Categories

O / open | Oberwolfach 报告问题汇编

For each fixed rank, are there only finitely many fusion categories, and likewise only finitely many ribbon categories, up to tensor equivalence?

来源字段：https://doi.org/10.4171/owr/2007/17

17454 | OWR-8415343-010

### Weak Bianchi Identities Across Timelike Singularities

O / open | Oberwolfach 报告问题汇编

Does the weak contracted second Bianchi identity $\int_M(R_{\mu\nu}-\tfrac12R g_{\mu\nu})\nabla^\mu\psi^\nu\,dV_g=0$ hold for every spacetime whose timelike singularity restricts on spacelike slices to a regular zero-area singularity with negative ZAS mass in the sense of Bray and Jauregui, and for every compactly supported vector field $\psi$?

来源字段：https://doi.org/10.4171/owr/2021/40

17455 | OWR-4129-003

### Weak Cosmic Censorship for Einstein-Vlasov

O / open | Oberwolfach 报告问题汇编

Does weak cosmic censorship hold for the asymptotically flat Einstein-Vlasov system, in particular for generic spherically symmetric initial data?

来源字段：https://doi.org/10.4171/owr/2009/46

17456 | OWR-8415343-007

### Weak Cosmic Censorship for Vacuum Spacetimes

P / partially_solved | Oberwolfach 报告问题汇编

Weak Cosmic Censorship for the Einstein vacuum equations: for generic complete, regular, asymptotically flat Cauchy data, are all singularities in the corresponding maximal development hidden inside a black hole (so that naked singularities do not occur)?

来源字段：https://doi.org/10.4171/owr/2021/40

17457 | OWR-16411-002

### Weak Cosmic Censorship in Asymptotically Flat Spacetimes

O / open | Oberwolfach 报告问题汇编

Does weak cosmic censorship hold for black-hole spacetimes in asymptotically flat settings, and under which geometric and dimensional hypotheses can it be proved or disproved?

来源字段：https://doi.org/10.4171/owr/2018/36

17458 | OWR-13855-007

### Weak Limits of Discrete Wave-Turbulence Spectra

P / partially_solved | Oberwolfach 报告问题汇编

For the damped, randomly forced nonlinear Schrödinger equation on the torus $\mathbb T_L^d$, impose $\nu=\varepsilon^2$, $\rho=L^{1/2}\rho_0$, and $b_k=L^{-d/2}b_k^0$. Under the iterated limit $\lim_{L\to\infty}\lim_{\nu\to0}$, do the averaged actions

$$n_k^{\nu,L}(\tau)=\frac12\mathbb E\lvert v_k^{\nu,L}(\tau)\rvert^2$$

converge weakly to a function $n_k^0(\tau)$ satisfying the corresponding damped-driven wave kinetic equation?

来源字段：https://doi.org/10.4171/owr/2015/34

17459 | OWR-13673-011

### Weak Verlinde Formula beyond Rational Conformal Field Theory

P / partially_solved | Oberwolfach 报告问题汇编

Does the weak Verlinde formula hold beyond rational conformal field theory; specifically, do asymptotic quantum dimensions define a one-dimensional representation of the fusion ring for nonrational or logarithmic conformal field theories?

来源字段：https://doi.org/10.4171/owr/2015/16

17460 | AIM-PHYSICS-0046

### When constrained minimizers gain symmetry

P / partially_solved | AIM 研讨会问题汇编

The printed source reads:

The grammatical slips “minimizers ... has,” “more symmetry that,” and “would ... by” occur in the original PDF and are not silently corrected here. In Chenciner's convention, the displayed \(D_6\) is the order-12 dihedral group; some later papers use \(D_{12}\) for that same abstract group.

The question is qualitative rather than a single yes/no conjecture. The answer developed below is:

1. symmetry enhancement is forced when invariance combines with rigidity of the minimizing set (uniqueness, isolation modulo phase, convexity, or sharp equality conditions); 2. it is not forced merely by minimizing in a symmetric class; and 3. the Italian-symmetry Kepler example admits a complete sharp proof by equality rigidity.

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17461 | AIM-PHYSICS-0001

### When semiclassical spectra follow curves-and when they fill area

P / partially_solved | AIM 研讨会问题汇编

**Eigenvalues and Curves.** Do the eigenvalues of \(a^w(x,hD)\) lie on or cluster around curves in the complex plane \(\mathbb C\)?

来源字段：http://aimpl.org/nhquantumsymp/1/

17462 | AIM-PHYSICS-0030

### Why gravitational instantons arise from monopoles

O / open | AIM 研讨会问题汇编

Questions [Self-dual Gravitational Instantons -- S. Cherkis's lecture]

**Question 1.** Why do you expect self-dual gravitational instantons to arise as monopole moduli spaces?

**Answer (Cherkis).** String theory argument.

Thus the verified reconstruction is:

Those items concern higher-dimensional hyperkähler spaces, Wick rotation, hyperkähler reduction, circle actions at infinity, and Ricci-flat ALE manifolds; they are not silently merged into Question 1 and are not addressed here.

No mathematical symbol in the genuine item needs correction.

来源字段：https://aimath.org/WWN/l2harmonic/l2harmonic.pdf

17463 | MPP-003

### Yang-Mills Existence and Mass Gap

O / open | 千禧年问题

别名：Yang-Mills Existence and Mass Gap [GEOM-009]

Prove that Yang-Mills theory exists and has a mass gap on $\mathbb{R}^4$, meaning the quantum particles have positive masses.

背景链接：https://www.claymath.org/millennium-problems/

17464 | AIM-PHYSICS-0057

### Zero-total-mass reduction and obstruction to the Euler integral

P / partially_solved | AIM 研讨会问题汇编

**Problem 20.** For circular restricted three-body problem, there is a Jacobi integral. For a complete integrability, need two more integrals. Extra integral exists when \(m_1+m_2=0\). Is there one more integral? (Christian Marchal)

来源字段：https://aimath.org/WWN/varcelest/varcelest.pdf

17465 | OWR-14751-001

### Zero-Velocity Lieb-Robinson Bounds for Disordered Spin Systems

O / open | Oberwolfach 报告问题汇编

Can zero-velocity Lieb-Robinson bounds be proved for general disordered finite-range quantum spin models on $\mathbb Z^d$, thereby establishing a universal many-body-localized phase?

来源字段：https://doi.org/10.4171/owr/2016/43

# 资料汇编 17 | 其他 / Miscellaneous

149 条记录。O=开放；P=部分解决；S=已解决。状态与英文/LaTeX 表述均沿用来源。

17466 | AIM-OTHER-0021

## A birational I2(4)-diamond conjugacy and an obstruction for the A-row fiber

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.6.** Relate both sides of the \(L\) identities under promotion, and both sides of the \(J\) identities under (birational) promotion/rowmotion.

8--9. Its exact question is:

Here \(\mathcal L(P)\) is the set of linear extensions and \(\mathcal J(P)\) the set of order ideals. The four rows of Theorem 3.4 immediately preceding the problem are labelled \(A,B,H,I\). To avoid relying on the missing fonts, write

\[ R_n=[n]\times[n],\qquad S_n=J([2]\times[n-1]),\qquad Q_m=J^{m-2}([2]\times[2])\quad(m\geq2). \]

This normalization agrees with the sizes in Theorem 3.4: \(|S_n|=n(n+1)/2=|\Phi^+(A_n)|\), \(|S_5|=15=|\Phi^+(H_3)|\), and \(|Q_m|=2m=|\Phi^+(I_2(2m))|\). The theorem being refined is

\[ \begin{aligned} |\mathcal L(\Phi^+(A_n))| &=2^{n(n-1)/2}|\mathcal L(S_n)|,\\ 2^n|\mathcal J(\Phi^+(A_n)\times[k])| &=|\mathcal J(S_n\times[2k+1])|;\\ |\mathcal L(\Phi^+(B_n))|&=|\mathcal L(R_n)|, &|\mathcal J(\Phi^+(B_n)\times[k])|&=|\mathcal J(R_n\times[k])|;\\ |\mathcal L(\Phi^+(H_3))|&=|\mathcal L(S_5)|, &|\mathcal J(\Phi^+(H_3)\times[k])|&=|\mathcal J(S_5\times[k])|;\\ |\mathcal L(\Phi^+(I_2(2m)))|&=|\mathcal L(Q_m)|, &|\mathcal J(\Phi^+(I_2(2m))\times[k])|&=|\mathcal J(Q_m\times[k])|. \end{aligned} \]

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17467 | AIM-OTHER-0081

## A compact-tail reduction for adelic unipotent measure rigidity

P / partially_solved | AIM 研讨会问题汇编

**Problem 7 (L. Silberman).** Extend the results on measure rigidity of unipotent flows to adelic setting.

It seems natural to expect (and is known in some cases) that the set of finite ergodic invariant measures for other dynamical systems with parabolic behavior has a manageable structure, which is possible to described in algebraic terms.

Suppose that \(H\) is a connected semisimple Lie subgroup of a Lie group \(G\), and let \(P\) be a parabolic subgroup of \(H\). One of manifestations of the measure rigidity of unipotent flows is the fact that every finite \(P\)-invariant measure on \(G/\Gamma\) is \(H\)-invariant (see [146]).

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17468 | AIM-OTHER-0051

## A conjugate-order ambiguity and an above-2 spectral obstruction

P / partially_solved | AIM 研讨会问题汇编

Does there exist a positive real cyclotomic integer \(\alpha\) such that \[ |\sigma(\alpha)|\leq \alpha \quad\text{for every embedding }\sigma:\mathbb Q(\alpha)\hookrightarrow\mathbb C, \] but \(\alpha\neq\operatorname{FPdim}(X)\) for every object \(X\) in every fusion category?

来源字段：http://aimpl.org/fusioncat/4/

17469 | AIM-OTHER-0122

## A countable-ray strengthening of bounded diagonal-ray coexistence

S / solved | AIM 研讨会问题汇编

**Conjecture 33.** Let \(A\) be the group of all diagonal matrices in \(\operatorname{SL}(3,\mathbb R)\), and \(A_1,A_2\subset A\) are rays in \(A\). Then there exists \(x\in \operatorname{SL}(3,\mathbb R)/\operatorname{SL}(3,\mathbb Z)\) such that \(A_1x\) and \(A_2x\) are bounded, but \(Ax\) is not bounded.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17470 | AIM-OTHER-0131

## A critical imbalance parameter for the B_s interpolation

P / partially_solved | AIM 研讨会问题汇编

Write \(\|x\|\) for the distance from \(x\) to the nearest integer. The statement verified against the source PDF is

\[ B_s=\left\{(\alpha,\beta)\in\mathbb R^2: \inf_{n\geq 1}n\bigl(\min\{\|n\alpha\|,\|n\beta\|\}\bigr)^s \bigl(\max\{\|n\alpha\|,\|n\beta\|\}\bigr)^{2-s}>0\right\}. \]

**Question.** Compute \(\dim_H B_s\) for \(0<s<1\).

The source adds that \(B_0\) is the set of badly approximable vectors and has Hausdorff dimension \(2\), whereas \(B_1\) is the exceptional set for Littlewood's conjecture and has Hausdorff dimension \(0\).

Comparison with the PDF shows that the problem ends after the preceding endpoint sentence. The quantum-chaos text is unrelated and is not used here.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17471 | AIM-OTHER-0166

## A cusp obstruction and periodic arithmetic tiling models

P / partially_solved | AIM 研讨会问题汇编

Construct analogs of the geometric coding and the arithmetic codings for the Weyl chamber flow (i.e., the action of the diagonal group) on \(\mathrm{SL}(n,\mathbb R)/\mathrm{SL}(n,\mathbb Z)\).

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17472 | AIM-OTHER-0012

### A cyclotomic audit and the complete type-A2 family

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.8 ([CLS14]).** Prove a cyclic sieving phenomenon for multitriangulations, using the analogue of Cambrian rotation and the polynomial \[ P_{W,k}(q)= \prod_{0\le j<k}\prod_{1\le i\le n} \frac{[d_i+h+2j]_q}{[d_i+2j]_q}. \tag{1} \]

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17473 | AIM-OTHER-0047

### A doubled-Fibonacci Hamiltonian with dense projective braid image

P / partially_solved | AIM 研讨会问题汇编

Is there a physical model which gives infinite image for the braid group?

来源字段：http://aimpl.org/fusioncat/3/

17474 | AIM-OTHER-0059

### A finite four-generator target for a diagrammatic Haagerup-vine obstruction

P / partially_solved | AIM 研讨会问题汇编

Find a non-number theoretic argument to rule out the rest of the Haagerup family vine.

For example, is there a diagram that evaluates in two different ways?

来源字段：http://aimpl.org/fusioncat/8/

17475 | AIM-OTHER-0035

### A finite generalized-toggle shadow of birational rowvacuation

P / partially_solved | AIM 研讨会问题汇编

**Problem 5.4.** Start with a known action in the birational (or piecewise-linear) realm and find the corresponding generalized toggle group action in the combinatorial realm. One such example of a birational map is the pentagram map.

The recovered statement is:

The PDF line-break artifact “combi-natorial” has been repaired to “combinatorial”; no mathematical wording was changed.

This is an open-ended construction problem, not a proposition with a fixed universal quantifier. Below, the known birational action is **birational order-filter rowvacuation** on a graded finite poset. Its finite combinatorial shadow is order-filter rowvacuation, an element of the generalized toggle group on the finite family of order filters. The construction is exact. It does **not** identify a toggle model for the pentagram map.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17476 | AIM-OTHER-0040

### A finite ground-field test for pivotal and spherical structures

P / partially_solved | AIM 研讨会问题汇编

**AIM problem 1.4.** Are all fusion categories pivotal? Are all fusion categories spherical? Does it depend on the ground field (\(k\) v.s. \(\mathbb C\))?

来源字段：http://aimpl.org/fusioncat/1/

17477 | AIM-OTHER-0037

### A finite positivity criterion and determinant obstruction for unitarizability

P / partially_solved | AIM 研讨会问题汇编

Is any integral fusion category unitarizable?

来源字段：http://aimpl.org/fusioncat/1/

17478 | AIM-OTHER-0129

### A finite-superlevel certificate for inhomogeneous minima

P / partially_solved | AIM 研讨会问题汇编

Conjecture 36 (E. Barnes, H. Swinnerton-Dyer [11]). For any quadratic form Q as above, the supremum m(Q) is rational and isolated. Both m(Q) and m2(Q) are attained at points with coordinates in the the splitting field of Q.

\[ Q(X,Y)=aX^2+bXY+cY^2\qquad(a,b,c\in\mathbb Q) \]

be nondegenerate and indefinite, and suppose that it does not represent zero nontrivially over \(\mathbb Q\). Write \(\Delta=b^2-4ac\). Then \(\Delta>0\) and anisotropy over \(\mathbb Q\) is equivalent to \(\Delta\) not being a rational square, so the splitting field is the real quadratic field

\[ K=\mathbb Q(\sqrt\Delta). \]

For \(x\in\mathbb R^2\), with its class also denoted by \(x\) in the torus \(\mathbb T^2=\mathbb R^2/\mathbb Z^2\), define

\[ \mu_Q(x)=\inf_{z\in\mathbb Z^2}|Q(x+z)|, \qquad m(Q)=\sup_{x\in\mathbb R^2}\mu_Q(x). \]

If \(m(Q)\) is isolated from below in the value set of \(\mu_Q\), define

\[ m_2(Q)=\sup\{\mu_Q(x):\mu_Q(x)<m(Q)\}. \]

Thus the recovered conjecture is that \(m(Q)\) is rational and isolated, and that both \(m(Q)\) and \(m_2(Q)\) are realized by torus classes admitting representatives in \(K^2\). The notation \(m_2(Q)\) is meaningful here only after isolation of \(m(Q)\) has been established. The PDF also warns, citing Godwin, that \(m_2(Q)\) itself need not be isolated.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17479 | AIM-OTHER-0026

## A four-element d-complete obstruction to birational periodicity

P / partially_solved | AIM 研讨会问题汇编

Jim Propp, Tom Roby, Jessica Striker, and Nathan Williams, *Problems for 2015 AIM workshop on Dynamical Algebraic Combinatorics*, May 29, 2015.

**Problem 4.3.** Study birational rowmotion on Proctor's d-complete posets.

On PDF page 11 (PDF index 10), the end of Section 4.2 reads:

The very next line is the new subsection heading “4.3 When is birational rowmotion periodic?” Thus the recovered statement for this record is exactly the one sentence above. The duplicate number 4.3 is present in the source: one occurrence numbers the problem and the other numbers the subsection.

None of that material is used to reconstruct Problem 4.3.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17480 | AIM-OTHER-0139

## A Fourier certificate and sharp limits for Gaussian eigenfunction value laws

P / partially_solved | AIM 研讨会问题汇编

**Recovered Conjecture 41 (Marklof).** For real numbers $a<b$,

The record is Conjecture 41, attributed to J. Marklof, in the AIM workshop problem list *Emerging applications of measure rigidity* (dated 1 November 2004). The source PDF gives the surrounding conventions: $X$ is a compact or finite-volume Riemannian manifold (possibly with boundary), $dV$ is **normalized** Riemannian volume, and

$$ 0=\lambda _0<\lambda _1\leq\lambda _2\leq\cdots, \qquad -\Delta\phi_i=\lambda_i\phi_i, \qquad \int_X|\phi_i|^2\,dV=1, $$

with Dirichlet conditions if there is a boundary. Immediately before Conjecture 41, the list says that Berry's random-wave picture should in particular give the following statement for generic negatively curved compact manifolds.

$$ \operatorname{Vol}\{x\in X:a\leq \phi_i(x)\leq b\} \longrightarrow \frac1{\sqrt{2\pi}}\int_a^b e^{-t^2/2}\,dt \qquad(i\to\infty). $$

Here the only normalization consistent with both the nearby definition and the right-hand side is

$$ \operatorname{Vol}(A)=\int_A dV, $$

so $\operatorname{Vol}(X)=1$. The order relation also requires real-valued eigenfunctions; because the Laplacian is real, a real orthonormal eigenbasis can be chosen. The quantifier “for every $a<b$” is the standard distributional reading of the displayed formula, not an additional phrase visible in the source.

Define the value-distribution probability measure

$$ \nu_i=(\phi_i)_*dV, \qquad \nu_i(B)=\int_X\mathbf 1_{\{\phi_i(x)\in B\}}\,dV(x). $$

If $\gamma=N(0,1)$, the conjecture is exactly $\nu_i\Rightarrow\gamma$. Moreover

$$ \int_{\mathbb R}t^2\,d\nu_i(t)=1. $$

On a closed connected manifold, every positive-eigenvalue eigenfunction is also orthogonal to the constants, hence $\int t\,d\nu_i(t)=0$. (That centering conclusion is not asserted here for the more general Dirichlet-boundary convention.)

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17481 | AIM-OTHER-0039

## A gauge-invariant two-channel obstruction to unitarizability

P / partially_solved | AIM 研讨会问题汇编

**Problem.** Does pseudo-unitary imply unitarizable?

来源字段：http://aimpl.org/fusioncat/1/

17482 | AIM-OTHER-0028

### A graded fixed-point cone and the complete chain case

P / partially_solved | AIM 研讨会问题汇编

When \(P\) is minuscule or coincidental, is there a cyclic sieving phenomenon [for] the integer points of \(kO(P)\) under the induced actions of promotion/rowmotion?

This record is Problem 4.5 in the pre-workshop document for the AIM workshop *Dynamical algebraic combinatorics*. The source text, with line-break hyphenation removed and the visibly omitted word “for” supplied in brackets, asks:

Here \(O(P)\) is the order polytope. The paragraph following the question describes, for the type \(A\) positive-root poset, a statistic defined after applying Stanley's transfer map. If \(x_1,\ldots,x_p\) are the resulting chain-polytope coordinates, then the exponent is a linear form \(\lambda(x)\) that weights entries in column \(j\) of the enlarged triangular array by \(j-n-1\), for \(1\le j\le 2n+1\). The source proposes a \(q\)-Ehrhart interpretation of this column-weighted statistic and asks whether one can combine all dilations into one cone. This \(\lambda\) is distinct from the ordinary size statistic used later in this report for the separate minuscule-chain special case.

The preceding source paragraph fixes the intended action. The lattice points of \(kO(P)\) are height-\(k\) \(P\)-partitions, and the action is obtained by extending the usual order-ideal toggles piecewise-linearly. Although the same set can be encoded, up to reversing \(P\), as order ideals of \(P\times[k]\), this induced piecewise-linear action is generally **not** ordinary combinatorial rowmotion on that product.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17483 | AIM-OTHER-0049

### A height-density model for non-cyclotomic candidate fusion graphs

P / partially_solved | AIM 研讨会问题汇编

Is there a sense in which a randomly chosen fusion graph does not have **cyclotomic** dimensions?

来源字段：http://aimpl.org/fusioncat/4/

17484 | AIM-OTHER-0025

### A local geometric certificate for minuscule birational rowmotion

P / partially_solved | AIM 研讨会问题汇编

Problem 4.2. Generalize Grinberg and Roby's proof of periodicity of birational rowmotion on rectangles uniformly to G/P.

Grinberg and Roby's proof corresponds to the case when G = GL (n) and

P is a maximal parabolic subgroup-their coordinates appear to be related to its Plücker embedding, which has generalizations for other quotients (see, for example, [Hil82, page 184] or [FZ00, Section 3.1]. When P is minuscule, this would be a birational generalization of [RS13]. See also [RSW04].

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17485 | AIM-OTHER-0038

### A Morita-stable sieve for minimal integral non-WGT categories

P / partially_solved | AIM 研讨会问题汇编

Is every integral fusion category weakly group theoretical?

来源字段：http://aimpl.org/fusioncat/1/

17486 | AIM-OTHER-0132

### A multiplicity- and cusp-safe QUE certificate

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 38, multiplicity-safe compact form.** Let \(X\) be a closed negatively curved Riemannian manifold and let \(dV\) be normalized volume. For every sequence of \(L^2\)-normalized Laplace eigenfunctions \(\phi_j\), with eigenvalues tending to infinity, \[ |\phi_j|^2dV\Longrightarrow dV. \] Equivalently, for every \(a\in C(X)\), \[ \int_X a|\phi_j|^2\,dV\longrightarrow\int_X a\,dV. \]

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17487 | AIM-OTHER-0096

### A parity-aware discrepancy reduction for quadratic pair correlation

P / partially_solved | AIM 研讨会问题汇编

For irrational $\alpha$ and a fixed interval $[a,b] \subset [0,1]$, define the pair-correlation statistic $$ R_2([a,b], N, \alpha) = \frac{1}{N} \#\left\{1 \leq i \neq j \leq N : \alpha i^2 - \alpha j^2 \in \frac{1}{N}[a,b] \pmod{1}\right\}. $$ Does $R_2([a,b], N, \alpha)$ converge to $b - a$ as $N \to \infty$ for every Diophantine irrational $\alpha$, or can one characterize the irrational $\alpha$ for which Poissonian pair correlation holds?

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17488 | AIM-OTHER-0111

### A quantitative rational-dependence family for Littlewood's conjecture

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 28 (Littlewood).** For any \(\alpha,\beta\in\mathbb R\), \[ \liminf_{q\to\infty}q\langle q\alpha\rangle\langle q\beta\rangle=0. \]

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17489 | AIM-OTHER-0155

### A rank-three positive cone and a strict/closed-cone audit

P / partially_solved | AIM 研讨会问题汇编

**Question 49 (B. Weiss).** Construct examples of cones \(D\) in \(A\) with \(\dim D\geq 3\) and no non-obvious divergent trajectories.

The source record itself has not been modified.

The preceding paragraph in the AIM source says that Weyl chambers give examples having non-obvious divergent trajectories, while Weiss had an \(\mathrm{SL}_3(\mathbb R)\) example of a two-dimensional cone for which all divergent trajectories are obvious. Thus the request is meant to seek higher-dimensional, nonvacuous examples, even though “nonvacuous” is not an explicit logical condition in the one-sentence question.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17490 | AIM-OTHER-0041

### A stabilizer certificate for Frobenius divisibility

P / partially_solved | AIM 研讨会问题汇编

**“Kaplansky's sixth conjecture” for fusion categories.** Is \[ \frac{\operatorname{FPdim}(\mathcal C)} {\operatorname{FPdim}(X)} \] an algebraic integer for every $X\in\operatorname{Irr}(\mathcal C)$?

来源字段：http://aimpl.org/fusioncat/1/

17491 | AIM-OTHER-0053

### A stabilizer trichotomy at Frobenius--Perron dimension two

P / partially_solved | AIM 研讨会问题汇编

What are all the $\mathcal C$ generated by $X$ with $\operatorname{FPdim}(X)\leq 2$, and $X$ not self-dual? Are they group theoretical if $\dim(X)=2$?

来源字段：http://aimpl.org/fusioncat/5/

17492 | AIM-OTHER-0142

### A support obstruction and a Wasserstein repair of Conjecture 42

S / solved | AIM 研讨会问题汇编

Marklof after Feingold--Peres, in the AIM workshop list *Emerging applications of measure rigidity*.

Let \(X\) be a compact negatively curved surface normalized by \(\operatorname{area}(X)=4\pi\). Write

\[ -\Delta\phi_i=\lambda_i\phi_i,\qquad N(\Lambda)=\#\{i:\lambda_i\leq\Lambda\}\sim\Lambda . \]

For real \(a\in C_c^\infty(S^*X,\mathbb R)\) with \(\int_{S^*X}a\,d\lambda=0\), the printed source defines

\[ \boxed{\quad \xi_i(a)=\lambda_i^{1/4} \left|\langle\operatorname{Op}(a)\phi_i,\phi_i\rangle\right|. \quad} \]

The classical autocorrelation quantity is

\[ V(a):=\int_{-\infty}^{\infty}\int_{S^*X} a(xg^t)a(x)\,d\lambda(x)\,dt , \]

where \(g^t\) is the geodesic flow. As \(\Lambda\to\infty\), the source conjectures

\[ \tag{1} \frac1{\Lambda}\sum_{\lambda_i\leq\Lambda}\xi_i(a)^2 \longrightarrow V(a) \]

and, for every interval \(I\subset\mathbb R\),

\[ \tag{2} \frac1{\Lambda} \#\left\{i:\lambda_i\leq\Lambda: V(a)^{-1/2}\xi_i(a)\in I\right\} \longrightarrow \frac1{\sqrt{2\pi}}\int_I e^{-t^2/2}\,dt . \]

Section 6.3 and A. Katok's suggestion about polygonal billiards start after the discussion of Conjecture 42 and are not part of it.

The same paragraph says that Luo and Sarnak proved, for the modular surface, a quantum-variance asymptotic involving a quadratic form \(B\) related to but distinct from \(V\). The workshop bibliography identifies its reference [169] as Sarnak's survey and its proof reference [126] as Luo--Sarnak.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17493 | AIM-OTHER-0088

### A toral finite-orbit-or-Haar theorem and the intermediate-support obstruction

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 14 (A. Furman).** Consider one of the following actions of a group \(\Gamma\):

1. \(\Gamma\) is a “large” subgroup of the group of automorphism of a nilmanifold of finite volume. 2. \(\Gamma\) is a “large” subgroup of a Lie group acting by translations on \(G/\Lambda\), where \(\Lambda\) is a lattice in \(G\).

Then the only ergodic \(\Gamma\)-invariant probability measures are the measures supported on finite \(\Gamma\)-orbits and the Haar measure.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17494 | AIM-OTHER-0091

### A two-gate effective reduction and a Diophantine torus model

P / partially_solved | AIM 研讨会问题汇编

**§4.2.** Let \(V\) be a connected Ad-unipotent subgroup of the Lie group \(G\) such that \(Vx\) is dense in \(G/\Gamma\) for some \(x\in G/\Gamma\).

**Question 16 (G. Margulis [130]).** Show that for a “good” sequence of subsets \(A_n\subset V\) and every \(f\in C_c(G/\Gamma)\), \[ \lim_{n\to\infty}\frac{1}{\operatorname{Vol}(A_n)} \int_{A_n}f(vx)\,dv=\int_{G/\Gamma}f\,d\mu \] with effective error term, where \(dv\) is a Haar measure on \(V\), and \(\mu\) is the probability Haar measure on \(G/\Gamma\).

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17495 | AIM-OTHER-0011

### A universal cyclic quotient and factor obstructions for ASM superpromotion

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.7.** What action of order \(3n-2\), on combinatorial objects of some unspecified kind, plays the role that rotation of link-patterns does in the case of gyration?

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17496 | AIM-OTHER-0055

### Accumulation from above: simple-object and unbounded-rank reductions

P / partially_solved | AIM 研讨会问题汇编

There are accumulation points from below for $\operatorname{FPdim}(X)$ for an object in a fusion category or $[M:N]$ for finite depth subfactors. Are there any accumulation points from above?

It adds: “Are there non-integer accumulation points?” and notes that there are no accumulation points for $\operatorname{FPdim}(\mathcal C)$ itself.

The wording agrees with Problem 9.30 in the AIM workshop document *Classifying Fusion Categories*. The immediately preceding Problem 9.29 asks for a non-number-theoretic way to rule out the remaining Haagerup-family vine, which helps explain the graph-theoretic context.

The following quantifiers are made explicit.

- Let $\mathscr F_{\rm all}$ be the union, over all fusion categories over $\mathbb C$, of the FP dimensions of all nonzero objects. - Let $\mathscr F_{\rm simp}$ be the analogous union restricted to simple objects. - Let $\mathscr I_{\rm fd}$ be the set of Jones indices of finite-depth finite-index subfactors (with repeated realizations identified). - A number $L$ is an accumulation point **from above** of a set $S\subset\mathbb R$ if there are distinct $s_n\in S$ with $s_n>L$ and $s_n\to L$. By passing to a subsequence, one may and will require $s_n\downarrow L$ strictly. “From below” is defined with $s_n<L$ and an increasing subsequence.

Thus infinitely many categories realizing one fixed number do not produce an accumulation point. Also, a finite bipartite graph of norm $\lambda$ supplies only a possible numerical index $\lambda^2$; it is an actual finite-depth subfactor index only when the required principal-graph/connection/flatness data exist.

来源字段：http://aimpl.org/fusioncat/5/

17497 | AIM-OTHER-0148

### Admissibility-certified periodic paths for triangles and the 2026 polygon-existence preprint

P / partially_solved | AIM 研讨会问题汇编

**Question 46 (A. Katok).** Construct periodic orbits for triangular billiards.

Every acute triangle has one obvious periodic orbit, but it is not known whether a general acute triangle has other periodic orbits. It is also not known whether a general obtuse triangle has at least one periodic orbit. Periodic orbits were constructed for some special classes of triangles. A computer-aided proof using McBilliards was reported to show that every triangle with all angles less than 100 degrees has a periodic orbit.

For rational triangles and polygons, unfolding produces a compact flat surface. The number $N(T)$ of periodic-orbit families of length at most $T$ has upper and lower quadratic bounds. Quadratic asymptotics was known for special tables, but not stated for rational polygons in general.

This record is Question 46, attributed to A. Katok, in the June 2004 AIM workshop list *Emerging applications of measure rigidity*. Inspection of page 19 of the original PDF verifies the following statement.

The source then notes nonuniformity near the $30$-$60$-$90$ triangle. The PDF itself misspells “Schwartz” as “Schwatz,” repeats “any” in “for any given any $\varepsilon$),” and contains line-break hyphenation; these are source typographical errors, not mathematical ambiguities.

There is one important implicit convention. A nonsingular periodic path on a rational flat surface lies in a cylinder of parallel closed geodesics. Thus a finite quadratic $N(T)$ cannot count all individual initial conditions; it counts maximal cylinders or periodic families, under a specified convention.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17498 | AIM-OTHER-0005

### All size moments for simultaneous cores with one modulus two

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.2.** Find and prove formulas for higher moments of cores: $\sum_{c\ {\rm an}\ (a,b)\text{-core}} |c|^i$.

Direct inspection of the PDF text around the damaged extraction recovers the display as $$ \boxed{\quad \sum_{\substack{c\ \mathrm{an}\ (a,b)\text{-core}}}|c|^i . \quad} $$ Thus the exact problem is:

The reconstruction makes the following repairs and conventions explicit.

The correct factor is $a+b+1$, as verified in the primary sources: $$ \mathbb E|c|=\frac{(a-1)(b-1)(a+b+1)}{24}. $$ The specialization derived below recovers this plus sign independently.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17499 | AIM-OTHER-0020

### An elementary gap-swap bijection for the dihedral row and a path reduction of AJ

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.5.** Give combinatorial proofs of the equalities above.

We will refer to an equation in Theorem 3.4 by its row (A, B, H, or I) and its column (L or J). Note that AJ is already interesting for $k=1$.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17500 | AIM-OTHER-0098

### An endpoint Fourier-moment and cusp-tail reduction

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 21 (Marklof--Strömbergsson).** Let $f$ be a continuous function on $T^1X$, subject to a specified cusp-growth condition from [MS]. If $\alpha$ is badly approximable and $0<c_1<c_2$, then $$ \frac1M\sum_{m=1}^M f(u_y(m\alpha)) \longrightarrow \int_{T^1X} f\,d\lambda $$ uniformly for $c_1M^{-2}\leq y\leq c_2M^{-2}$, where $\lambda$ is normalized Liouville measure.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17501 | AIM-OTHER-0162

### An ergodic reduction and a finite-place obstruction for CM packets

P / partially_solved | AIM 研讨会问题汇编

**Question 53 (L. Silberman).** Give an ergodic-theoretic proof of the equidistribution of special points on $\mathrm{SL}_2(\mathbb Z)\backslash\mathbb H^2$.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17502 | AIM-OTHER-0056

### An exact center-free Frobenius--Schur exponent procedure

S / solved | AIM 研讨会问题汇编

Is there a way to find the Frobenius-Schur exponent of $\mathcal C$ without computing $Z(\mathcal C)$?

来源字段：http://aimpl.org/fusioncat/6/

17503 | AIM-OTHER-0016

### An orbit-code criterion and the complete type-A2 case

P / partially_solved | AIM 研讨会问题汇编

Let \(W\) be a finite Weyl group of rank \(n\), let \(c=s_1\cdots s_n\) be a Coxeter element, let \(h\) be the Coxeter number, and put \(N=|\Phi^+(W)|=nh/2\). If \(w_0(c)\) is the \(c\)-sorting word for the longest element, set

\[ Q=c\,w_0(c). \]

The set \(\operatorname{Asoc}(W,c)\) consists of the \(N\)-element position subsets \(P\) of \(Q\) for which the selected letters, in order, form a reduced word for \(w_0\).

\[ Q\,\psi(Q)\ \sim\ c^{h+2}, \]

where \(\sim\) allows commuting adjacent commuting generators. Thus \(P\subset Q\) is represented in the completed word by the doubled position set \(P\sqcup\psi(P)\). This repair is forced already in type \(A_2\): \(Q=12121\), so literal \(QQ\) is not the alternating word \((12)^5\), whereas \(Q\psi(Q)=12121\,21212=(12)^5\).

Write \(\rho=\operatorname{Camb}_c\) for the nonnesting \(c\)-Cambrian rotation on order ideals of \(\Phi^+(W)\). In the notation imported from the preceding record,

\[ \rho=\operatorname{Tog}_{\operatorname{inv}(w_0(c))}\operatorname{Tog}^{+}_{\operatorname{inv}(w_0(c))}, \]

where the positive toggle fixes the simple roots and agrees with the usual order-ideal toggle on nonsimple roots.

The recovered conjecture is the following explicit recipe. For \(I\in J(\Phi^+(W))\), form

\[ (I,\rho I,\rho^2I,\ldots,\rho^{h+1}I). \]

In the \(k\)-th copy of \(c\) in \(c^{h+2}\), select \(s\in S\) exactly when its simple root \(\alpha_s\) lies in \(\rho^kI\). Conjecture 3.1 asserts that the resulting position set is the doubled encoding of an element of \(\operatorname{Asoc}(W,c)\), and that the resulting map is a bijection intertwining \(\rho\) with noncrossing Cambrian rotation.

The superscript-looking “1” after “bijaction” is a footnote marker (“a bijection induced by an action”), not part of the formula. Those lines begin the next problem and are not part of Conjecture 3.1.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17504 | AIM-OTHER-0157

### Andre-Oort status and rectangular CM closure classification

S / solved | AIM 研讨会问题汇编

Conjecture 51 was proved in the case when the dimension of the Zariski closure is one assuming generalized Riemannian hypothesis (see [203]).

The phrase “generalized Riemannian hypothesis” is reproduced here because it is what the source prints. The intended term is the **Generalized Riemann Hypothesis (GRH)**; reference [203] is A. Yafaev, *A conjecture of Yves André*, then a 2004 preprint.

Correcting only grammar, the mathematical assertion is:

**André-Oort closure formulation.** Let \(S\) be a (pure) Shimura variety and let \(\Sigma\) be any set of special points of \(S\). Every irreducible component of

\[ \overline{\Sigma}^{\mathrm{Zar}} \]

is a special subvariety. Equivalently, the closure is a finite union of special subvarieties.

The quantifier is “any set,” not merely a sequence, a Galois-stable set, or a dense set. By convention a special subvariety is irreducible; the entire closure need not be.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17505 | AIM-OTHER-0149

### Angular-factor dichotomy for triangular billiards

P / partially_solved | AIM 研讨会问题汇编

**Question 47 (A. Katok).** Prove ergodicity for triangular billiards with irrational angles.

It was proved that the set of ergodic (topologically transitive) triangular billiard table is residual in the sense of Baire category (see [103, 99]). However, it seems unknown whether the set of ergodic billiard tables has positive measure. It is easy to see that rational billiards cannot be ergodic. On the other hand, it was shown in [196] that if an irrational billiard table is very well approximable by a rational one, then it is ergodic. In the case when one of the angles of a triangles is rational, there is a useful unfolding procedure (see [189]) that may lead to a proof of ergodicity.

Also, it seems unknown whether there is a weakly mixing polygonal billiard (see [76] for a positive result in this direction). It was shown in [5] that typical interval exchange transformation which is not defined by a cyclic permutation is weakly mixing (see also [98, 192]).

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17506 | AIM-OTHER-0133

### Arithmetic QUE through Hecke packets

P / partially_solved | AIM 研讨会问题汇编

Conjecture 39 (Arithmetic quantum unique ergodicity). The Riemannian volume is the only arithmetic quantum limit.

Positive results towards this conjecture were obtained for the case \(X=\Gamma\backslash\mathbb H^2\), where \(\Gamma\) is a congruence subgroup in either \(\mathrm{SL}_2(\mathbb Z)\) or in the group of quaternions of norm one. In this case, T. Watson [197] proved

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17507 | AIM-OTHER-0163

### Arithmeticity of groups generated by opposite unipotent lattices

P / partially_solved | AIM 研讨会问题汇编

Conjecture 54 (G. Margulis, A. Selberg). If the group \(\langle\Gamma^+,\Gamma^-\rangle\) is discrete, then it an arithmetic lattice in \(G\). It was observed in [151] that

**Conjecture 54 (Margulis--Selberg).** If \(\langle\Gamma^+,\Gamma^-\rangle\) is discrete, then it is an arithmetic lattice in \(G\).

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17508 | OWR-14298369-001

### Bioinformatic Frontiers in Complex-System Epigenetics

O / open | Oberwolfach 报告问题汇编

Which bioinformatic and mathematical problems are central to understanding epigenetics as a complex dynamical system?

来源字段：https://doi.org/10.4171/owr/2024/18

17509 | AIM-OTHER-0036

### Block-count homomesy under noncrossing-partition toggles

S / solved | AIM 研讨会问题汇编

Let $NC(n)$ denote the lattice of noncrossing partitions of $\{1, 2, \ldots, n\}$, and let $\sigma: NC(n) \to NC(n)$ be the Kreweras complementation (or rowmotion/promotion) map defined by $\sigma(\pi) = K(\pi)$ (or composed with cyclic rotation).

**Conjecture 5.8.** The number of blocks (considered as a statistic on $NC(n)$) is homomesic under $\sigma$, with average value $(n+1)/2$ on each $\sigma$-orbit.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17510 | AIM-OTHER-0060

### Boundary-mass obstructions for rooted principal-graph truncations

P / partially_solved | AIM 研讨会问题汇编

**Is there a polymer theory of principal graphs? What graphs can appear as subgraphs of principal graphs?**

One would need a bound on the rank at each depth.

Penneys: Note that there are examples of forbidden subgraphs, e.g., part of the bad seed.

来源字段：http://aimpl.org/fusioncat/8/

17511 | AIM-OTHER-0002

### Canonical, finite-order, and rowmotion primitives for zero-mesies

P / partially_solved | AIM 研讨会问题汇编

**Problem 1.2.** Find simple and uniform constructions for such functions \(g\) for a wide range of \(0\)-mesies \(f\).

- \(T\) acts invertibly on a set \(X\), with every orbit finite; - \(f:X\to\mathbb R\) is a \(0\)-mesy, so its sum on each orbit is zero; - one seeks \(g:X\to\mathbb R\) such that \(f(x)=g(x)-g(Tx)\).

The exact problem is:

Everything beginning “1.3 Dynamical closures” belongs to the next section. It is used here only as nearby context. The word “action” is read as a permutation, consistently with the cited homomesy literature; noninvertible maps with transient tails would require a different formulation.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17512 | AIM-OTHER-0069

### Centralizer models and root-wall obstructions for split-Cartan restrictions

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 1.** Let \(G\) be a connected semisimple Lie group, \(\Gamma\) an irreducible lattice in \(G\), and \(A\) a closed subgroup of a split Cartan subgroup of \(G\) with \(\dim A>1\). Then any \(C^1\)-small perturbation of the action of \(A\) on \(G/\Gamma\) is \(C^\infty\)-conjugate to the action of \(A\) defined by a continuous homomorphism from \(A\) to the centralizer of \(A\) in \(G\).

We also state one of important partial cases of Conjecture 1:

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17513 | AIM-OTHER-0034

### Chain toggles on an antichain: CSP, homomesy, and an affine lift

P / partially_solved | AIM 研讨会问题汇编

Explore these (and other) generalized toggle groups, look for homomesy and CSP, look at piecewise-linear and birational liftings.

For example, one could look at toggling chains, then the piecewise-linear extension should be to the order complex.

The operative text is:

The source immediately before the problem defines generalized toggles as follows. For a finite set \(E\), a family \(\mathcal L\subseteq 2^E\), and \(e\in E\), the toggle \(t_e\) adds or removes \(e\) when the resulting set remains in \(\mathcal L\), and otherwise fixes the state. It explicitly lists chains of a poset among the examples.

This report answers all three concrete directions-homomesy, cyclic sieving, and a piecewise-linear lift-for the chain family of an antichain. It does not propose a birational lift.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17514 | OWR-12697686-006

### Coincidence of Detected Slopes for Knot Manifolds

O / open | Oberwolfach 报告问题汇编

For every knot manifold, do the sets of left-orderability-detected, cooriented-taut-foliation-detected, and non-L-space-detected boundary slopes coincide?

来源字段：https://doi.org/10.4171/owr/2023/3

17515 | AIM-OTHER-0032

### Common toggle energy and obstructions for locomotion

P / partially_solved | AIM 研讨会问题汇编

Perhaps it is better to approach the birational toggle group from above. Certain algebraic combinations of the variables are invariant under all the birational toggles.

**Problem 4.9.** Can we say what those combinations are? Can we then characterize the birational toggle group as being precisely the group of birational transformations that preserve those algebraic combinations?

Apropos of invariance under toggling, we mention the case of the full toggle group on the poset \([2]\times[2]\). Of the 24 orders in which one can compose all 4 toggles, 8 are not conjugate to rowmotion or promotion or their inverses; indeed, the PL and birational versions of these compositions were shown by Einstein to be of infinite order. Nevertheless, there are things to be proved about the map (called “locomotion” for present purposes). For instance, the pictures at http://jamespropp.org/locomotion.pdf, which shows some projections of an orbit of locomotion strongly suggests that the orbit lies in a 2-dimensional surface in \(\mathbb R^4\); that is, there are two conserved quantities, of which we know only one. Likewise, experimental studies suggest that other quantities are homomesic under locomotion, in an appropriately asymptotic sense of the word “average.” (Similar, polyhedral pictures appear when one replaces birational locomotion by PL locomotion.)

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17516 | AIM-OTHER-0116

### Compact diagonal orbits and product minima

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 30 (G. Margulis).** For every compact set \(K\) of \(\mathrm{SL}(3,\mathbb R)/\mathrm{SL}(3,\mathbb Z)\), there are only finitely many closed \(A\)-orbits contained in \(K\).

Its mathematical statement is:

Checking the official PDF, pp. 12--13, recovers the displayed definitions

\[ F_B(x)=\prod_{i=1}^{3}\left(\sum_{j=1}^{3}a_{ij}x_j\right), \qquad B=(a_{ij})\in M_3(\mathbb R), \] \[ \Delta(F_B)=\det B, \qquad m(F_B)=\inf_{x\in\mathbb Z^3\setminus\{0\}} \left|\frac{F_B(x)}{\Delta(F_B)}\right|. \]

The definition requires \(\det B\ne0\), an assumption implicit in the quotient by \(\Delta(F_B)\). The next sentence in the PDF reads, with the source's apparent numbering slip, "Then Question 30 is equivalent to the following question," followed by Question 31:

\[ \text{For every }\varepsilon>0, \quad [\varepsilon,\infty)\cap\{m(F)\}\text{ is finite.} \tag{Q31} \]

They are quoted here only to recover the boundary and to audit the reformulation; this attempt owns only AIM-OTHER-0116.

We use the positive diagonal identity component

\[ A=\{\operatorname{diag}(a_1,a_2,a_3):a_i>0,\ a_1a_2a_3=1\}. \]

Passing to all real diagonal matrices changes only finite sign components and does not affect compactness or finiteness assertions.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17517 | AIM-OTHER-0118

### Compact thresholds and the two logical gates for the cubic multiplicative spectrum

P / partially_solved | AIM 研讨会问题汇编

**Question 31 (G. Margulis).** Show that, for every \(\varepsilon>0\), \[ [\varepsilon,\infty)\cap \{m(F_B):B\in\mathrm{GL}_3(\mathbb R)\} \quad\text{is finite.} \]

and then continues with the heading “5.4” and a definition of sets \(C_s\).

The notation is supplied by the immediately preceding paragraph of the source. For an invertible real \(3\)-by-\(3\) matrix \(B=(a_{ij})\), put \[ F_B(x)=\prod_{i=1}^3\left(\sum_{j=1}^3a_{ij}x_j\right), \qquad \Delta(F_B)=\det B, \] and \[ m(F_B)=\inf_{x\in\mathbb Z^3\setminus\{0\}} \left|\frac{F_B(x)}{\Delta(F_B)}\right|. \] Thus the exact recovered problem is:

The assumption \(\Delta(F_B)\ne0\) is forced by the displayed quotient and by the preceding description as a product of three independent linear forms. The absolute value makes the sign of \(\det B\) immaterial.

The PDF says just before Question 31 that “Question 30 is equivalent to the following question”; the preceding numbered item is in fact **Conjecture 30**, the finiteness of closed diagonal orbits in compact subsets. Sections 5 and 7 below audit the hypotheses needed to make this advertised reformulation logically exact.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17518 | OWR-12697686-007

### Concordance Versus Isotopy for Positive Knots

O / open | Oberwolfach 报告问题汇编

For positive knots, does concordance imply isotopy?

来源字段：https://doi.org/10.4171/owr/2023/3

17519 | AIM-OTHER-0072

### Conditional fiber collapse and periodic-data tests for the Anosov infranil conjecture

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 3.** Every Anosov diffeomorphism is topologically conjugate to a hyperbolic automorphism on an infranilmanifold.

The AIM PDF was checked directly.

Dekimpe showed that the older, strict definition of an *infra-nilmanifold automorphism* is too narrow: the general algebraic model must allow an affine translation. Accordingly, the safe modern reading is

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17520 | AIM-OTHER-0092

### Conditions for expanding analytic curves and an exact toral model

P / partially_solved | AIM 研讨会问题汇编

**Question 17 (N. Shah).** Under what condition on \(\gamma\) do we have \[ a^n\nu\overset{w^*}{\longrightarrow}\lambda \qquad(n\to\infty)? \]

Let \(L\) be a Lie group, let \(G\) be a closed subgroup of \(L\), and let \(\Lambda\) be a lattice in \(L\). For a semisimple element \(a\in G\), define its expanding horospherical subgroup by

\[ U=\{g\in G:a^{-n}ga^n\longrightarrow e\text{ as }n\to\infty\}. \]

Assume that \(Gx_0\) is dense in \(L/\Lambda\). If a probability measure on \(U\), absolutely continuous with respect to Haar measure, is pushed to \(Ux_0\), then Shah's earlier theorem gives convergence of its \(a^n\)-translates to the probability Haar measure \(\lambda\) on \(L/\Lambda\). The proposed refinement takes an analytic curve

\[ \gamma:[0,1]\longrightarrow U \]

\[ \nu=(t\longmapsto\gamma(t)x_0)_*(dt). \]

The recovered question is therefore:

The PDF's word “Recently” begins the following status sentence; it is not part of the question. That sentence says that Shah had solved the case \(L=\mathrm{SO}(m,1)\), \(G=\mathrm{SO}(n,1)\), \(m>n\), under the sufficient condition that \(\gamma([0,1])\) lies in neither a proper affine subspace nor an \((n-2)\)-dimensional sphere in \(U\cong\mathbb R^{n-1}\). The PDF then states Conjecture 18 for arbitrary ambient Lie groups containing \(G=\mathrm{SO}(n,1)\).

All actions in the setup are left actions on \(L/\Lambda\), and the definition \(a^{-n}ua^n\to e\) is the convention making \(U\) expanding under positive powers of \(a\).

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17521 | AIM-OTHER-0024

### Cyclic itinerary refinements for piecewise-linear rowmotion

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.1.** Explicitly describe the decomposition of \(\mathcal O(P)\) under piecewise-linear promotion/rowmotion. Of particular interest are \([a]\times[b]\) and the type \(B_n\) root poset. The case \(P=[2]\times[n]\) is settled: the polytope is divided into Catalan-many simplices so that every power of the map restricts to a linear map on every simplex. The case \(P=[3]\times[n]\) is more complicated and was unresolved.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17522 | AIM-OTHER-0057

### Cyclic pointed module categories and their Brauer-Picard divisor action

P / partially_solved | AIM 研讨会问题汇编

**Classify module categories and Brauer-Picard groups for known examples.**

来源字段：http://aimpl.org/fusioncat/7/

17523 | AIM-OTHER-0052

### Dimensions between 2 and 3: a quantifier reduction and infinite modular family

P / partially_solved | AIM 研讨会问题汇编

What values can $\dim(X)$ take in $[2,3]$ for $X\in\mathcal C$, a braided fusion category?

来源字段：http://aimpl.org/fusioncat/5/

17524 | AIM-OTHER-0084

### Directional and interval-exchange reductions for polygonal billiards

P / partially_solved | AIM 研讨会问题汇编

**Question 10 (A. Katok).** Classify ergodic invariant probability measure and closed invariant subsets for polygonal billiards.

For rational polygonal billiards, the phase space decomposes into invariant subsets \(P_\theta\) that correspond to directions \(\theta\) of the flow. It was shown in [103] that the billiard flow is uniquely ergodic on \(P_\theta\) for the set of directions \(\theta\) of full measure. There is an estimate of the Hausdorff dimension of this set, which may be positive (see [140, 139, 34]). It is also known that the restriction of the billiard flow on all but countably many of the subsets \(P_\theta\) is minimal.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17525 | AIM-OTHER-0156

### Directional nondivergence, invariant cluster measures, and the closed-orbit case

P / partially_solved | AIM 研讨会问题汇编

Let \(A_T\) denote the ball of radius \(T\) in \(A\) and \(\lambda\) a Haar measure on \(A\).

**Question 50 (B. Weiss).** Suppose that \(\dim A\geq2\) and for some \(x\in G/\Gamma\) and every one-parameter subgroup \(D\) of \(A\), the orbit \(Dx\) is not divergent in \(G/\Gamma\). Is it true that there exists a compact set \(K\subset G/\Gamma\) such that \[ \limsup_{T\to\infty} \frac{1}{\lambda(A_T)} \lambda\bigl(\{a\in A_T:ax\in K\}\bigr)>0? \]

**9. André--Oort Conjecture**

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17526 | OWR-12006-005

### Discrete Green Kernels, Escaping Geodesics, and Random Triangulations

O / open | Oberwolfach 报告问题汇编

Three independent questions are considered. First, under mild regularity assumptions on orthogonal dual planar graphs $\Gamma$ and $\Gamma^*$, can one prove the uniform asymptotics

$$G(u;u_0)=\frac{1}{2\pi}\log|u-u_0|+O(1),\qquad K(v;z_0)=O(|v-z_0|^{-1})?$$

Second, from every point of a metrically complete noncompact Euclidean cone surface with finitely many negatively curved cone singularities, is there a geodesic ray that avoids every singularity and escapes to infinity? Third, what is the typical graph diameter obtained by gluing two independent uniform triangulations of an $n$-gon, conjecturally $n^{1/4+o(1)}$?

来源字段：https://doi.org/10.4171/owr/2012/34

17527 | AIM-OTHER-0023

### Divisor-complete intermediate full-twist actions for Hurwitz cyclic sieving

P / partially_solved | AIM 研讨会问题汇编

**Problem 3.8 (recovered).** The polynomial \(F_W(q)\) appears, when evaluated at roots of unity, to prescribe orbit sizes for other multiples \(ph\) of \(h\) between \(h\) and \(nh\). Can Conjecture 3.7 be generalized by describing corresponding Hurwitz actions? Such elements should be related to solutions of \(\boldsymbol w^p=\boldsymbol c^n\) in the braid group \(B_n\). The endpoint examples are \(\boldsymbol w_0^2=\boldsymbol c^n\) and \(\boldsymbol c^p=\boldsymbol c^n\) for \(p=n\).

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17528 | AIM-OTHER-0108

### Dual approximation on manifolds and an elementary moment-curve bound

P / partially_solved | AIM 研讨会问题汇编

**Question 26 (D. Kleinbock).** Give estimates on Hausdorff dimension of \(v\)-approximable vectors in a nondegenerate submanifold of \(\mathbb R^n\) using dynamics.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17529 | OWR-12977-005

### Dynamic Constraints on Cancer-Cell Passage Through Tissue

P / partially_solved | Oberwolfach 报告问题汇编

How is cancer-cell movement restricted by obstacles, extracellular matrix, and fibers? Can a static squeezing criterion be replaced by a dynamic, time-dependent pore-passage condition such as $F-hv=0$?

来源字段：https://doi.org/10.4171/owr/2014/19

17530 | AIM-OTHER-0043

### Effective Ocneanu rigidity in the total-multiplicity parameter

P / partially_solved | AIM 研讨会问题汇编

Is there an effective version of Ocneanu rigidity? Is there a sub-exponential bound on the number of unitary fusion categories with respect to \(N\), the sum of all the fusion multiplicities \(N_{i,j}^k\)?

来源字段：http://aimpl.org/fusioncat/2/

17531 | AIM-OTHER-0128

### Effective Oppenheim heights and an exact four-square transfer family

S / solved | AIM 研讨会问题汇编

**Question 35 (G. Margulis).** Give an effective estimate \(T=T_Q(\varepsilon)\) such that there is an \(x\in\mathbb Z^d\) with \[ 0<|Q(x)|<\varepsilon,\qquad \|x\|<T. \]

After correcting one duplicated word in the printed PDF, they are:

- \(Q\) is a nondegenerate indefinite quadratic form in \(d\ge 3\) variables; - \(Q\) is not a scalar multiple of a quadratic form with rational coefficients.

The recovered question is therefore:

The subscript on \(T_Q\) is inserted only to expose the dependence that the next sentence of the source explicitly requires: the estimate must depend on the Diophantine properties of the coefficients of \(Q\).

There are three source-recovery issues.

1. The PDF itself prints “nondegenerate indefinite definite quadratic form.” The second “definite” is a typographical duplication. “Indefinite” is forced both by the Oppenheim context and by every cited result. 2. The canonical record continues past “5.6.” into a new discussion of binary rational forms and inhomogeneous minima. That text is not part of Question 35. The present job stops at the end of the sentence about the upcoming work of Margulis and Götze. OCR strings such as `ax 21`, `cx 22`, and stray `>` symbols belong to that unrelated Section 5.6 extraction. 3. The source calls the inequalities \[ |Q(x)|<\varepsilon,\qquad \|x\|<T \] an “easier question,” but does not repeat \(x\in\mathbb Z^d\), does not say \(x\ne0\), and drops the strict lower inequality. Literally this is trivial by \(x=0\). The surrounding sentence can reasonably carry over integrality, but even the reading \(x\in\mathbb Z^d\setminus\{0\}\) can be trivial when \(Q\) has an integral isotropic vector. This defect is not silently repaired below. The later published work formulates its small-value conclusion for a **nonzero integral point** and allows \(|Q(x)|<\varepsilon\); Question 35 itself has the unambiguous strict condition \(0<|Q(x)|\).

The source uses an unspecified norm. The new results below use the sup norm \(\|x\|_\infty\). Any two fixed norms on \(\mathbb R^d\) are equivalent, so this changes constants but not the exponents or conclusions.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17532 | OWR-12181-002

### Efficient Covariance Models for Atmospheric-Chemistry Data Assimilation

O / open | Oberwolfach 报告问题汇编

Can the forecast-error covariance matrices for atmospheric chemistry, especially chemically consistent multivariate and temporal covariances, be treated efficiently in numerical data assimilation?

来源字段：https://doi.org/10.4171/owr/2012/58

17533 | AIM-OTHER-0144

### Equivariant unfolding reduction for polygonal quantum limits

P / partially_solved | AIM 研讨会问题汇编

**Problem 44 (J. Marklof).** Classify all quantum limits of the eigenfunctions of a polygonal billiard.

According to the Berry--Tabor conjecture ...". The PDF confirms that this is the start of the next subsection, about eigenvalue statistics of integrable systems and quadratic forms; it is not part of Problem 44.

The surrounding AIM definitions matter. In §6.1, the eigenfunctions satisfy the **Dirichlet** boundary condition, and a "quantum limit" means a weak-* limit on configuration space of \[ |u_j(x)|^2\,dx, \qquad -\Delta_Pu_j=E_j u_j,\quad \|u_j\|_{L^2(P)}=1,\quad E_j\to\infty. \] Thus this is the primary reading below. Modern papers also call phase-space semiclassical measures quantum limits; §3 distinguishes that stronger notion.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17534 | AIM-OTHER-0145

### Exact A2 symmetry and collision strata in Marklof's raw triple-gap count

P / partially_solved | AIM 研讨会问题汇编

**Question 45 (J. Marklof).** What is the distribution of the set \[ \{(Q(x_1)-Q(x_2),Q(x_2)-Q(x_3)):x_i\in\mathbb Z^2, \ x_i\ne x_j\text{ for }i\ne j\}\subset\mathbb R^2? \] More precisely, determine the asymptotics of \[ N_T((a,b),(c,d)):=\#\left\{(x_1,x_2,x_3)\in(\mathbb Z^2)^3: \begin{array}{l} a<Q(x_1)-Q(x_2)<b,\\ c<Q(x_2)-Q(x_3)<d,\\ x_i\ne x_j\ (i\ne j),\quad \|x_i\|<T\ (i=1,2,3) \end{array}\right\}. \]

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17535 | AIM-OTHER-0004

### Exact rational counterexamples to the middle-runner conjecture

S / solved | AIM 研讨会问题汇编

1. Subscripts and the distinction between instantaneous and averaged positions were flattened: the PDF uses \(p_i(t)\) for the \(i\)-th sorted position and then \(p_i\) for its full-period average. In this report the latter is written \(\bar p_i\) to avoid ambiguity. 2. Everything beginning "2.2 Cores" belongs to the next section and is not part of Conjecture 2.1.

There are \(n\) runners on a unit circular track. Their speeds are nonzero and commensurable, and they need not start at the marked starting line. At time \(t\), sort their representatives in \([0,1)\): \[ p_1(t)\leq p_2(t)\leq\cdots\leq p_n(t). \] For a common return period \(L\), define \[ \bar p_i=\frac1L\int_0^L p_i(t)\,dt. \] The conjecture is \[ \boxed{\bar p_i+\bar p_j=1\qquad\text{whenever }i+j=n+1.} \] For odd \(n\), the central case asserts that the average middle position is \(1/2\).

The phrase "though runners are not required to start at the starting line" naturally permits arbitrary initial phases. The counterexample below also works when all runners start together at the same unmarked point, so it does not depend on the more permissive independent-phase reading.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17536 | AIM-OTHER-0066

### Excluding prime Frobenius--Perron dimension 37

P / partially_solved | AIM 研讨会问题汇编

Suppose \(\mathcal C\) is a finite tensor category over \(\mathbb C\) with prime Frobenius--Perron dimension. Is \(\mathcal C\) fusion?

(Hence, it would be of the form \(\operatorname{Vect}(\mathbb Z/p,\omega)\). This would be an extension of a result in Hopf algebras.)

来源字段：http://aimpl.org/fusioncat/12/

17537 | AIM-OTHER-0031

### Explicit presentations for unions of chains and a diamond kernel reduction

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.8.** When is the birational toggle group finitely presented?

We know that this is the case when \(P\) is just a chain, for then the PL toggles are actual linear maps, and their birational lifts are just monomial maps. But even for as simple a poset as \([2]\times[2]\), we do not know the answer to this question.

Perhaps it is better to approach the birational toggle group from above. Certain algebraic combinations of the variables are invariant under all the birational toggles.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17538 | AIM-OTHER-0107

### Extremality criteria for matrix-valued maps

P / partially_solved | AIM 研讨会问题汇编

**Question 25 (D. Kleinbock, G. Margulis; see [109], Section 6.2).** Find reasonable and checkable conditions for a smooth map \[ f:U\;(\subset\mathbb R^k)\longrightarrow M_{m\times n}(\mathbb R) \] which generalize nondegeneracy of vector-valued maps and imply that almost every point of \(f(U)\) is not VWA. Such conditions were obtained in some cases in [115,116,105].

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17539 | AIM-OTHER-0150

### Factor overload refutes the total-rank trichotomy

S / solved | AIM 研讨会问题汇编

Let \(G\) be a semisimple real algebraic group, \(\Gamma\) a noncocompact arithmetic lattice in \(G\), and \(D\) a closed subgroup of a maximal \(\mathbb R\)-split torus \(A\). An orbit \(Dx\) of \(D\) in \(G/\Gamma\) is called divergent if the map \(d\mapsto dx\), \(d\in D\), is proper.

**Conjecture 48 (B. Weiss).** 1. If \(\dim D>\operatorname{rank}_{\mathbb Q}G\), then there are no divergent orbits of \(D\). 2. If \(\dim D=\operatorname{rank}_{\mathbb Q}G\), then the only divergent trajectories are obvious ones. 3. If \(\dim D<\operatorname{rank}_{\mathbb Q}G\), then there are non-obvious divergent trajectories.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17540 | AIM-OTHER-0101

### Fast-stable invariant measures in the affine toral model

P / partially_solved | AIM 研讨会问题汇编

**Question 22 (F. Ledrappier).** Describe the invariant ergodic measures for the fast stable foliation \(W^s_+\). One may ask the same question for Anosov diffeomorphisms as well.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17541 | AIM-OTHER-0044

### Finiteness versus an exact cyclic pointed count

P / partially_solved | AIM 研讨会问题汇编

**How many fusion categories have the same given fusion rules?**

来源字段：http://aimpl.org/fusioncat/2/

17542 | AIM-OTHER-0007

### Forced freeness and small-order edge homomesy for Aztec diamonds

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.4.** Is there a cyclic action of order \(2^n\) on the set of perfect matchings of the Aztec diamond graph of order \(n\), such that the edge-inclusion indicator functions associated with all the edges of the Aztec diamond graph are all homomesic?

A possible avenue to pursue in solving Problem 2.4 may be an analysis of domino shuffling, since the set of Aztec diamonds is naturally divided into “equivalence classes” of size \(2^n\) by the shuffling algorithm.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17543 | AIM-OTHER-0061

### Fusion-ring extensions: the one-new-simple grading obstruction

P / partially_solved | AIM 研讨会问题汇编

Develop existence, obstruction, and classification machinery for enlarging a fusion category by sectors whose multiplication is multi-valued according to a fusion ring, rather than deterministically indexed by a group. Near-group rules, obtained by adjoining one noninvertible sector to a pointed ring, are the basic test case.

This agrees exactly with Problem 9.2 in the official notes from the 2011 AIM workshop *Classifying Fusion Categories*. The immediately preceding workshop section, “New from old,” lists \(G\)-extensions-categories \(\mathcal D=\bigoplus_{g\in G}\mathcal D_g\) with \(\mathcal D_e=\mathcal C\)-among standard constructions. It also lists short exact sequences, equivariantization, de-equivariantization, Hopf monads, and other genuinely categorical constructions. The next open problem asks for “fusion rings” in place of groups.

That context supports the following reconstruction:

Three notions must not be conflated.

1. A **faithful \(H\)-graded extension** has \[ \mathcal D=\bigoplus_{h\in H}\mathcal D_h,\qquad \mathcal D_h\otimes\mathcal D_k\subseteq\mathcal D_{hk}. \] The product of degrees is a single degree. 2. A **categorification of a fusion ring** asks for a monoidal category whose Grothendieck ring is the prescribed based ring. Ring associativity is only the decategorified first step; an associator satisfying the pentagon is additional data. 3. “A category graded by a fusion ring” is not part of the standard definition of a grading. If \(bc=\sum_dN_{bc}^d d\), multiplying two sectors may feed several sectors. One must specify what the sectors are, what multiplicity \(N_{bc}^d\) means categorically, and which bimodule tensor products and coherence maps realize it.

来源字段：http://aimpl.org/fusioncat/9/

17544 | AIM-OTHER-0086

### Haar squashability on nilpotent regular covers

P / partially_solved | AIM 研讨会问题汇编

**Question 12 (F. Ledrappier, O. Sarig).** Let \(\Gamma\) be a normal conilpotent subgroup of a uniform lattice in \(\mathrm{SL}(2,\mathbb R)\). Is the Haar measure nonsquashable?

For general discrete subgroup \(\Gamma\subset \mathrm{SL}(2,\mathbb R)\), it is not known whether the Haar measure on \(\mathrm{SL}(2,\mathbb R)/\Gamma\) is nonsquashable, or whether there exist other nonsquashable \(u_t\)-ergodic Radon measures.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17545 | OWR-12181-001

### Heterogeneous-Parameter Inversion in Atmospheric Chemistry

O / open | Oberwolfach 报告问题汇编

For atmospheric-chemistry data assimilation, can one formulate and solve the generalized inversion problem of estimating heterogeneous parameters, including emissions and initial values, with realistic multivariate and temporal error covariances?

来源字段：https://doi.org/10.4171/owr/2012/58

17546 | AIM-OTHER-0076

### Higher-rank Anosov actions and the rank-one-factor boundary

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 4.** Every “irreducible” Anosov \(\mathbb Z^k\)- and \(\mathbb R^k\)-action, \(k\geq 2\), is \(C^\infty\)-conjugate to an algebraic action.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17547 | OWR-4131-003

### Historical Data in Modern Epidemic Models

O / open | Oberwolfach 报告问题汇编

What can historical records reveal about the spread and determinants of modern infectious-disease epidemics when combined with epidemic models?

来源字段：https://doi.org/10.4171/owr/2009/48

17548 | AIM-OTHER-0085

### Horospherical Radon measures and a centralizer-character obstruction

P / partially_solved | AIM 研讨会问题汇编

**Question 11 (F. Ledrappier, O. Sarig).** Let \(G\) be a noncompact semisimple Lie group of rank one, \(\Gamma\) a discrete subgroup of \(G\), and \(U\) a horospherical subgroup of \(G\). What are the \(U\)-ergodic Radon measures on \(G/\Gamma\)? In particular, are they either carried by closed \(U\)-orbits or given by the harmonic function construction?

A \(u_t\)-invariant measure \(\mu\) is called squashable if the centralizer of \(u_t\) contains an invertible nonsingular transformation that does not preserve \(\mu\). F. Ledrappier and O. Sarig showed recently that for a normal coabelian subgroup \(\Gamma\) of a uniform lattice, the only nonsquashable \(u_t\)-ergodic measure on \(\mathrm{SL}(2,\mathbb R)/\Gamma\) is Haar. If \(\Gamma\) is conilpotent, then any \(u_t\)-ergodic measure on \(\mathrm{SL}(2,\mathbb R)/\Gamma\) is squashable except possibly the Haar measure.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17549 | OWR-4131-002

### Hospital Infection Control and Colonization Pressure

O / open | Oberwolfach 报告问题汇编

What are the effects of isolation and other control measures, how is ward spread related to colonization pressure, what role do antibiotics play, and how useful are rapid molecular tests compared with culture-based swabs?

来源字段：https://doi.org/10.4171/owr/2009/48

17550 | AIM-OTHER-0015

### Ideal-antichain transfer and boundary-coboundary homomesies for three-chain products

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.11.** What are the homomesies of rowmotion acting [on] order ideals in a product of three chains? Likewise, what are the homomesies of Panyushev complementation acting on antichains in a product of three chains?

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17551 | OWR-705-004

### Intrinsic Characterizations of Regular Model Sets

P / partially_solved | Oberwolfach 报告问题汇编

Find an intrinsic and simple characterization of regular model sets, the Delone sets arising from cut-and-project schemes.

来源字段：https://doi.org/10.4171/owr/2004/22

17552 | AIM-OTHER-0095

### Invariance certificates and obstructions for representation-ball averages

P / partially_solved | AIM 研讨会问题汇编

What is the limiting distribution of \(\mu_T\) as \(T\to\infty\)?

The exact Question 19 occupies the end of Section 4.3; all text beginning with “4.4. For irrational \(\alpha\), the sequence \(\{\alpha n^2\pmod 1\}\) ...” belongs to the next section and is not part of this record.

The preceding paragraph supplies the missing setup. Let \(G\) be a semisimple Lie group acting linearly on a finite-dimensional real vector space \(V\), equipped with a norm. Fix \(p\in V\), and put \[ R_T=\{g\in G:\|gp\|<T\}. \tag{1.1} \] The AIM text assumes that the stabilizer \(G_p\) is finite, says that this makes \(R_T\) compact, fixes a lattice \(\Gamma<G\), and defines \(\mu_T\) to be the pushforward to \(G/\Gamma\) of normalized Haar measure on \(R_T\). Question 19, contributed by N. Shah, is:

The next sentence says that examples are known in references [53, 73] and motivates the problem by distribution of \(\Gamma\)-orbits on \(G/H\), either when \(\Gamma\cap H\) is a lattice in \(H\) or when \(\Gamma H\) is dense in \(G\). The official bibliography identifies [53] as Eskin--Mozes--Shah (1996) and [73] as the 2004 preprint of Gorodnik--Weiss, published in 2007.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17553 | OWR-12977-004

### Leaderless Collective Cell Motion and Macroscopic Limits

P / partially_solved | Oberwolfach 报告问题汇编

Can collective cell movement develop without a leader, and what macroscopic limits result from microscopic or mesoscopic models of such movement?

来源字段：https://doi.org/10.4171/owr/2014/19

17554 | AIM-OTHER-0089

### Literal counterexamples and an exact torus model for sparse unipotent samples

P / partially_solved | AIM 研讨会问题汇编

Let \(G\) be a Lie group, \(\Gamma\) a lattice in \(G\), and \(U=\{u(t)\}\subset G\) a one-parameter Ad-unipotent subgroup. Suppose that, for \(x\in G/\Gamma\), the orbit \(Ux\) is dense in \(G/\Gamma\).

Section 4.1 fixes the following setting:

Question 15 (attributed to G. Margulis) then asks for equidistribution of \[ \{u(t_n)x\}_{n\geq 1} \] for each of \[ \text{(1) }t_n=\lfloor n^\alpha\rfloor\quad(\alpha>1),\qquad \text{(2) }t_n=\lfloor P(n)\rfloor,\qquad \text{(3) }t_n=p_n, \] where \(P\) is a polynomial and \(p_n\) is the \(n\)-th prime.

The continuation on the next page says that A. Venkatesh had suggested a proof of (1) for \(\alpha\) close to 1, recalls almost-everywhere pointwise ergodic results for these sequences, and explicitly warns that \(\lfloor n^\alpha\rfloor\) and \(n^\alpha\) behave differently. Thus the floor in the recovered question is intentional.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17555 | OWR-14299289-009

### Local Geometric Langlands for Hyperspherical Varieties

O / open | Oberwolfach 报告问题汇编

For a hyperspherical $G$-variety $X$, is the category of $G(\mathcal D)$-equivariant sheaves on $X(\mathring{\mathcal D})$ equivalent to the relative tensor product of the corresponding automorphic and spherical sheaf categories?

来源字段：https://doi.org/10.4171/owr/2025/18

17556 | OWR-13944-001

### Long-Range Mechanical Signaling in Tissue Morphogenesis

O / open | Oberwolfach 报告问题汇编

In morphogenesis, how can global tissue-size information be transmitted when cell regulation is local; in particular, can active mechanical stress or its derivative provide a scale-dependent signal beyond the fixed characteristic length of reaction-diffusion mechanisms?

来源字段：https://doi.org/10.4171/owr/2015/49

17557 | OWR-12977-002

### Mathematical Principles for Modelling Living Systems

O / open | Oberwolfach 报告问题汇编

Do the common structures of complex living systems provide a paradigm for deriving life-science models, and can applications reveal hallmarks of a mathematical theory of living systems?

来源字段：https://doi.org/10.4171/owr/2014/19

17558 | AIM-OTHER-0080

### Measure-rigidity obstructions require an entropy and genuine-rank gate

P / partially_solved | AIM 研讨会问题汇编

**Question 6 (D. Fisher).** Do the new results on measure rigidity for actions of higher rank abelian groups give rise to obstructions to smooth or continuous actions of a higher rank abelian group on a compact manifold?

Some basic obstructions for smooth volume preserving actions of higher rank abelian groups are already known, see particularly work of H. Hu and A. Katok. The question is whether one can use results on measure rigidity to obtain more information.

The results on arithmetic quotients for actions of semisimple groups and their lattices have no straightforward analogues here, since the proofs of those results use not only Ratner's theorem but applications of the cocycle superrigidity theorems to cocycles which are necessarily only measurable. Though some cocycle superrigidity theorems are known for particular classes of actions of higher rank abelian groups, none of these apply to measurable cocycles because of the Dye theorem [43] and its generalizations [154, 37].

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17559 | AIM-OTHER-0010

### Measure-sensitive resonance and PL-versus-birational period rigidity

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.6.** Find/define analogues of resonance phenomena for the PL dynamical systems mentioned above and their birational lifts.

For background on piecewise-linear and birational lifts of toggle-group actions, see [EPb] and the more detailed article-in-progress [EPa].

The preceding paragraph is essential context. It says that observed piecewise-linear (PL) lifts can have a spectrum of periods, each supported on a positive-measure part of the ambient polytope. For the PL lift of gyration on order-4 alternating-sign matrices, the list reports many periods 8 and 24, larger periods, and essentially no period 16. The problem asks for a precise analogue that relates such interior behavior to the earlier, discrete notion of resonance.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17560 | OWR-12733-001

### Mechanisms Maintaining Flat Plant Leaves

O / open | Oberwolfach 报告问题汇编

Which growth-coordination and mechanical-feedback mechanisms explain why most plant leaves remain approximately flat despite heterogeneous cellular growth?

来源字段：https://doi.org/10.4171/owr/2013/59

17561 | OWR-12977-001

### Minimal Mechanistic Models of Cancer-Cell Invasion

O / open | Oberwolfach 报告问题汇编

Which subcellular processes are essential for cancer-cell invasive behavior, and what level of mechanistic detail is needed for a model that is realistic yet sufficiently simple?

来源字段：https://doi.org/10.4171/owr/2014/19

17562 | OWR-2050-001

### Mixture versus Single-Component Models of Tissue Growth

O / open | Oberwolfach 报告问题汇编

For modeling growth and remodeling of soft biological tissues, is mixture theory a more effective mathematical framework than the usual one-component continuum theory with volumetric mass sources?

来源字段：https://doi.org/10.4171/owr/2008/39

17563 | AIM-OTHER-0114

### Multiplicative minimum and bounded diagonal orbits

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 29 (G. Margulis).** Let \(A\) be the group of all diagonal matrices in \(\mathrm{SL}(3,\mathbb R)\). Then every bounded \(A\)-orbit in \(\mathrm{SL}(3,\mathbb R)/\mathrm{SL}(3,\mathbb Z)\) is closed.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17564 | OWR-12977-003

### Multiscale Problems in Cancer-Cell Migration

O / open | Oberwolfach 报告问题汇编

Open cancer-migration problems include physical cell passage through obstacles; mechanics and modeling of collective motion; optimization of DTI and Q-ball methods; benchmarks for movement models; the moving-boundary problem for acid-mediated invasion; comparison of amoeboid, mesenchymal, and collective strategies and their invasion speeds; global bounds for chemotaxis-haptotaxis models with tissue remodeling; inverse recovery of chemotaxis sensitivity from observations; and simple microscopic or mesoscopic collective-motion models, including leaderless motion and its macroscopic limits.

来源字段：https://doi.org/10.4171/owr/2014/19

17565 | OWR-13346-005

### Nyström Approximation Bounds for Gaussian Kernels

P / partially_solved | Oberwolfach 报告问题汇编

For the Gaussian kernel $k(x,y)=\exp\bigl(-\lVert x-y\rVert^2/(2\sigma^2)\bigr)$, prove an asymptotic-in-$m$ approximation result, with explicit dependence on $m,X,Z$, of the form $K(X,X)\approx K(X,Z)K(Z,Z)^{-1}K(Z,X)$.

来源字段：https://doi.org/10.4171/owr/2014/44

17566 | AIM-OTHER-0078

### One partially hyperbolic element can hide an entire higher-rank factor

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 5.** Every action of a connected semisimple Lie group of higher rank (i.e., all simple factors have real rank at least 2) or its lattice, which that has an element which acts non-trivially uniformly partially hyperbolicly, is $C^\infty$-conjugate to an algebraic action.

Every action of a connected semisimple Lie group of higher rank (that is, every simple factor has real rank at least $2$), or of a lattice in such a group, that has an element acting nontrivially and uniformly partially hyperbolically, is $C^\infty$-conjugate to an algebraic action.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17567 | AIM-OTHER-0001

### Orbit matrices and explicit zero-mesy bases for unions of chains

P / partially_solved | AIM 研讨会问题汇编

What is the dimension of $V_0$, and what is a natural basis for it?

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17568 | AIM-OTHER-0045

### Orbit-balance constraints for fusion categories with two irreducible Frobenius-Perron degrees

P / partially_solved | AIM 研讨会问题汇编

What can you say about all fusion categories $\mathcal C$ for which $\#\{\dim(X)\mid X\in\mathcal C\}=2$?

来源字段：http://aimpl.org/fusioncat/2/

17569 | OWR-13359-002

### Origins of Synchronization in Interacting Systems

O / open | Oberwolfach 报告问题汇编

What is the origin of synchronization and rhythmic behavior in systems of interacting units?

来源字段：https://doi.org/10.4171/owr/2014/57

17570 | AIM-OTHER-0082

### Parabolic stiffness, the solved moduli example, and finite root/factor diagnostics

P / partially_solved | AIM 研讨会问题汇编

Let $H$ be a connected semisimple Lie group, let $P<H$ be a parabolic subgroup, and suppose $H$ acts on a space $X$ preserving some geometric structure. Under what conditions is every finite $P$-invariant measure on $X$ $H$-invariant? Equivalently, which $H$-actions are stiff? As a motivating case, consider the natural $\mathrm{SL}(2,\mathbb R)$-action on moduli spaces of quadratic differentials on compact Riemann surfaces.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17571 | AIM-OTHER-0087

### Partial-adelic diagonal measures: an entropy dichotomy and a projection obstruction

P / partially_solved | AIM 研讨会问题汇编

**Question (E. Lindenstrauss).** Let $\mathbb A'$ be defined as the ring of adeles, but the product is taken over a subset of places of $\mathbb Q$, and $\Gamma$ an “irreducible” lattice in $\mathrm{SL}(2,\mathbb A')$. What are the finite ergodic $D(\mathbb A')$-invariant measures on $\mathrm{SL}(2,\mathbb A')/\Gamma$?

The sentence beginning “3.6. One may also expect measure rigidity...” is the heading and opening sentence of the next subsection. It is not part of the question.

The preceding paragraph in the PDF defines $\mathbb A$ as the full adele ring, calls $D$ the diagonal subgroup of $\mathrm{SL}_2$, and records Lindenstrauss's full-adelic Haar-uniqueness result. Thus the most natural reconstruction for a set $V$ of places is $$\mathbb A_V=\prod_{v\in V}'\mathbb Q_v, \qquad G_V=\mathrm{SL}_2(\mathbb A_V) =\prod_{v\in V}'\mathrm{SL}_2(\mathbb Q_v),$$ where the finite-place restrictions are with respect to $\mathbb Z_p$ and $\mathrm{SL}_2(\mathbb Z_p)$, respectively. If $V$ is finite, these are ordinary products. An infinite unrestricted Cartesian product would generally not be locally compact, so the usual definitions of Haar measure and lattice would no longer apply.

The source does not define the scare-quoted word “irreducible.” For a finite product, the standard strong reading is that the projection of $\Gamma$ to every nonempty proper subproduct is dense (variants ask only nondiscreteness or impose conditions factor by factor). For an infinite restricted product, even the intended family of projections is unspecified. This ambiguity is mathematically consequential and prevents a literal all-cases classification.

The AIM quotient is the left-action space $G_V/\Gamma$. Most cited papers use $\Gamma\backslash G_V$ with a right action. The inversion homeomorphism $$\iota:G_V/\Gamma\longrightarrow \Gamma\backslash G_V, \qquad \iota(g\Gamma)=\Gamma g^{-1},$$ intertwines left translation by $d$ with right translation by $d^{-1}$. Since $D_V=D_V^{-1}$, the two formulations have exactly the same invariant measures and entropies (with $d$ replaced by $d^{-1}$).

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17572 | AIM-OTHER-0143

### Periodic-orbit scars: a negative full-support result and quantitative transfer obstruction

S / solved | AIM 研讨会问题汇编

**Question 43 (A. Katok).** Do periodic orbits in a triangular billiard correspond to scars? More precisely, are there quantum limits supported on periodic orbits?

Based on the investigation [21], it seems likely that the answer to this question is ‘yes’ for rational billiards.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17573 | AIM-OTHER-0018

### Piecewise-linear Panyushev dynamics and the complete affine A2 case

S / solved | AIM 研讨会问题汇编

Can we give piecewise-linear and birational analogues of Armstrong, Stump, and Thomas' proof (see [AST13]) of Panyushev's Conjecture 2.1 (see [Pan09]) asserting homomesy of antichain cardinality under rowmotion?

For the type-\(A_n\) positive-root poset \(P\), let \(\mathcal O(P)\) and \(\mathcal C(P)\) be the order and chain polytopes, let \(\rho:\mathcal O(P)\to\mathcal O(P)\) be piecewise-linear rowmotion, and let \(\phi:\mathcal O(P)\to\mathcal C(P)\) be Stanley's transfer map. It defines

\[ \rho'=\phi\circ\rho\circ\phi^{-1}:\mathcal C(P)\longrightarrow\mathcal C(P). \]

On vertices, \(\rho'\) is Panyushev complementation. The source then defines

\[ f(y)=\sum_{p\in P}y_p \]

as the piecewise-linear analogue of antichain cardinality.

1. The canonical field `number: "2.1"` is not the number of this AIM item. It was picked up from the citation to **Panyushev's Conjecture 2.1**. The AIM question itself is unnumbered text following AIM Conjecture 3.2. 2. The next sentence in the PDF is **Conjecture 3.3**, which asserts the \(n/2\) average. That sentence begins the next canonical record, AIM-OTHER-0019. It is used here only to identify the intended question and its subsequent status; it is not silently appended to this record.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17574 | OWR-15993-003

### Programming Branching and Growth in Developing Organs

P / partially_solved | Oberwolfach 报告问题汇编

别名：Duplicate Organogenesis Branching Extraction [OWR-15993-004]

For growing animals and branched organs such as lungs and kidneys, what mechanisms program size and shape, determine branch points and epithelial behavior, and terminate growth?

来源字段：https://doi.org/10.4171/owr/2018/8

17575 | AIM-OTHER-0136

### Projective vague convergence and the missing mass scale

S / solved | AIM 研讨会问题汇编

Problem 40 (E. Lindenstrauss). Let X = Γ \H2 where Γ is a noncocompact arithmetic lattice. Show that for all f, g ∈ Cc(X),

∫ > X f dμ i ∫ > X g dμ i → ∫ > X f dV ∫ > X g dV as i → ∞.

The analog of

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17576 | AIM-OTHER-0046

### Property F: current frontier and quantifier obstructions

P / partially_solved | AIM 研讨会问题汇编

**Property F conjecture**

If a fusion category has property F, then can't use braiding alone for a universal quantum computer.

Given a braided, weakly integral fusion category, is the image of the braid group finite?

Is braided and weakly integral fusion equivalent to finite image of the braid group?

来源字段：http://aimpl.org/fusioncat/3/

17577 | OWR-14299512-008

### Quadratic Growth of Periodic Points for Hamiltonian Maps

O / open | Oberwolfach 报告问题汇编

Let $\phi$ be a Hamiltonian diffeomorphism with infinitely many nontrivial periodic points, and let $h(\phi)$ denote their polynomial growth exponent. Must $h(\phi)\geq2$?

来源字段：https://doi.org/10.4171/owr/2025/31

17578 | AIM-OTHER-0064

### Rank-22 Haagerup centers and their two canonical condensations

P / partially_solved | AIM 研讨会问题汇编

Compute the center of the even half of the Asaeda-Haagerup and extended Haagerup subfactors.

来源字段：http://aimpl.org/fusioncat/11/

17579 | AIM-OTHER-0006

### Rational Catalan cyclic sieving under rotation

S / solved | AIM 研讨会问题汇编

**Conjecture 2.3 (recovered).** Let \(a<b\) be coprime positive integers. The set \(\mathrm{NC}(a,b)\) of \((a,b)\)-noncrossing partitions under rotation exhibits the cyclic sieving phenomenon using \[ \operatorname{Cat}_q(a,b) =\frac{1}{[a+b]_q} \begin{bmatrix}a+b\\ b\end{bmatrix}_q. \]

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17580 | OPG-581

### Rendezvous on a line

P / partially_solved | Open Problem Garden

Problem Two players start at a distance of 2 on an (undirected) line (so, neither player knows the direction of the other) and both move at a maximum speed of 1. What is the infimum expected meeting time $R$ (first time when the players occupy the same point) which can be achieved assuming the two players must adopt the same strategy?

背景链接：http://www.openproblemgarden.org/op/rendezvous_on_a_line

17581 | AIM-OTHER-0009

### Resonance as a cyclic factor and an orbit profile

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.5.** What is resonance? And why are there systems that exhibit it?

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17582 | OWR-702-001

### Resource-Capacity Formulations and Relax-and-Fix Heuristics

P / partially_solved | Oberwolfach 报告问题汇编

Find tighter formulations for resource-capacity constraints, and determine how relax-and-fix heuristics can be combined with RINS.

来源字段：https://doi.org/10.4171/owr/2004/19

17583 | OWR-12728-002

### Rigid Elasticity as a Variational Limit

P / partially_solved | Oberwolfach 报告问题汇编

Can rigid-elasticity models be obtained as a variational limit of models whose elastic constants tend to infinity?

来源字段：https://doi.org/10.4171/owr/2013/54

17584 | AIM-OTHER-0030

### Row-sum homomesies for the symmetric action on Gelfand-Tsetlin patterns

P / partially_solved | AIM 研讨会问题汇编

Problem 4.7. What are the homomesies of the symmetric group action on Gelfand-Tsetlin triangles?

One might expect (at least naively) that there are more homomesies for the symmetric group action than for its cyclic subactions: making the group bigger means merging orbits, and this merging permits more averaging to take place. Note that there is significant overlap with the question raised in Section??.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17585 | AIM-OTHER-0120

### Schmidt's weighted intersection conjecture

S / solved | AIM 研讨会问题汇编

**Conjecture 32 (W. Schmidt).** For every $(s,t\in[0,1])$, $$ C_s\cap C_t\ne\varnothing. $$

for $(0\le u\le1)$, $$ C_u=\left\{(\alpha,\beta)\in\mathbb R^2: \inf_{q\ge1}\max\left\{q^u\langle q\alpha\rangle, q^{1-u}\langle q\beta\rangle\right\}>0\right\}. $$ In this Diophantine-approximation context, $(\langle x\rangle)$ denotes distance to the nearest integer. Below it is written in the now-standard notation $$ \|x\|:=\min_{p\in\mathbb Z}|x-p|. $$

The recovered conjecture is therefore:

The source correctly notes that Schmidt's original 1983 question was the specific swapped-weight case $(s=1/3)$, $(t=2/3)$. The assertion for every pair of weights is the generalized form that became known as Schmidt's conjecture.

There is an endpoint subtlety in the displayed formula. Modern papers define the zero-weight coordinate to be omitted, so $$ \mathbf{Bad}(0,1)=\mathbb R\times\mathbf{Bad}, \qquad \mathbf{Bad}(1,0)=\mathbf{Bad}\times\mathbb R, $$ where $$ \mathbf{Bad}=\{x\in\mathbb R:\inf_{q\ge1}q\|qx\|>0\}. $$ The AIM formula literally retains the term $(q^0\|q\alpha\|)$. The endpoint lemma proved below shows that the two readings give exactly the same sets; no convention is silently imposed.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17586 | AIM-OTHER-0054

### Simple tensor powers, packing bounds, and the closed fish route

P / partially_solved | AIM 研讨会问题汇编

**A fusion category version of supertransitivity.** In a fusion category, is there an upper bound on the $(N)$ such that $(X^{\otimes N})$ is a simple object (where $(\dim(X)>1)$)?

来源字段：http://aimpl.org/fusioncat/5/

17587 | AIM-OTHER-0063

### Small algebra objects and a pointed classification

P / partially_solved | AIM 研讨会问题汇编

**Can you classify all algebras in fusion categories with small Frobenius-Perron dimension (e.g., less than $(3+\sqrt3)$)?**

来源字段：http://aimpl.org/fusioncat/11/

17588 | AIM-OTHER-0022

### Solved Hurwitz cyclic sieving conjecture and a dihedral order diagnostic

S / solved | AIM 研讨会问题汇编

**Conjecture 3.7 (recovered).** For \(w=w_0\) and for \(w=\gamma\), \((\operatorname{Red}_T(c_W),F_W(q),T_w)\) exhibits the cyclic sieving phenomenon, using the cyclic orders \(2h\) and \(nh\), respectively.

The surrounding subsection of the AIM pre-workshop document supplies the notation. Let \(W\) be a finite irreducible Coxeter group of rank \(n\), with reflection set \(T\), degrees \(d_1,\ldots,d_n\), Coxeter number \(h\), and a fixed Coxeter element \(c_W\in W\). Write

\[ \operatorname{Red}_T(c_W)= \{(t_1,\ldots,t_n)\in T^n:t_1\cdots t_n=c_W\}. \]

On such words the positive Hurwitz generator is

\[ T_i(\ldots,t_i,t_{i+1},\ldots) = (\ldots,t_{i+1},t_{i+1}t_it_{i+1},\ldots). \]

For \(w=s_{i_1}\cdots s_{i_k}\in\mathfrak S_n\) reduced, set \(T_w=T_{i_1}\cdots T_{i_k}\); braid relations make this independent of the reduced expression. To avoid the source's reuse of \(c\), denote by \(w_0\in\mathfrak S_n\) the longest element and by \(\gamma=s_1s_2\cdots s_{n-1}\in\mathfrak S_n\) the standard Coxeter element. Set

\[ F_W(q)=\prod_{i=1}^n\frac{[ih]_q}{[d_i]_q}, \qquad [a]_q=\frac{1-q^a}{1-q}. \]

The recovered conjecture is:

The repairs are: `wo` is \(w_0\), `Two` is \(T_{w_0}\), `Red T(c)` is \(\operatorname{Red}_T(c_W)\), and the displayed product is \(\prod_i[ih]_q/[d_i]_q\). The superscript `2` sentence about the identities AJ and HJ is a footnote belonging to Problem 3.6, not to this conjecture; `10` is a page number. The subsequent quadrangulation and conjugation comments are context, not part of the displayed conjecture.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17589 | OWR-2040-001

### Stability of Topological Components versus Statistical Clusters

O / open | Oberwolfach 报告问题汇编

Why is the zeroth Betti number, which counts connected components of a topological space, mathematically stable and easy to recover from a space, whereas the number of clusters of a finite sampled statistical space is unstable and difficult to define or recover?

来源字段：https://doi.org/10.4171/owr/2008/29

17590 | AIM-OTHER-0067

### Stable tensor ideals, blocks, supports, and vertices in finite tensor categories

P / partially_solved | AIM 研讨会问题汇编

How much from modular representations of finite groups can be carried to finite tensor categories?

来源字段：http://aimpl.org/fusioncat/12/

17591 | AIM-OTHER-0160

### Strict special-point sequences, Galois component weights, and the correct Haar target

P / partially_solved | AIM 研讨会问题汇编

For some partial results on this conjecture, see [78] and references therein.

In particular, for the above example, Conjecture 52 was established in [42].

**Conjecture 52 (source formulation).** Let \((x_n)\) be special points on a Shimura variety. If, for all sufficiently large \(n\), the point \(x_n\) lies outside any special subvariety, then the normalized counting measures on the finite Galois orbits of \(x_n\) converge, as \(n\to\infty\), to normalized Haar measure.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17592 | AIM-OTHER-0042

### Structural sieves for a fixed small simple object

P / partially_solved | AIM 研讨会问题汇编

Can we find all fusion categories with a given smallest simple object (which is not invertible)?

来源字段：http://aimpl.org/fusioncat/2/

17593 | OWR-4131-001

### Study-Period Estimation Under Imperfect Infection Testing

O / open | Oberwolfach 报告问题汇编

Can one construct a simple method for estimating the ratio of study periods for infectious and non-infectious diseases when admission and discharge data are known but colonization status is observed imperfectly and depends on the testing scheme?

来源字段：https://doi.org/10.4171/owr/2009/48

17594 | AIM-OTHER-0058

### Supertransitivity above index four: formulation correction and complexity bounds

P / partially_solved | AIM 研讨会问题汇编

The supertransitivity with respect to an object $X$ with $\dim(X)>2$ is the largest $N$ such that $\operatorname{Hom}(1,X^{\otimes n})$ is Temperley--Lieb.

Supertransitivity is the analog of transitivity of group actions.

Is there an upper bound on the supertransitivity of a subfactor planar algebra?

Is there a universal upper bound on the finite supertransitivity of non-Temperley--Lieb, finite-depth subfactor standard invariants of index $\delta^2>4$?

来源字段：http://aimpl.org/fusioncat/8/

17595 | AIM-OTHER-0062

### Tensor functors for pointed cyclic categories and the wider group-theoretical program

P / partially_solved | AIM 研讨会问题汇编

Describe functors between group-theoretical categories.

What is known for Verlinde categories? (quantum groups at roots of unity)

来源字段：http://aimpl.org/fusioncat/10/

17596 | AIM-OTHER-0065

### The 4442 even half as a Z3-equivariantization

S / solved | AIM 研讨会问题汇编

Is there a conceptual construction of the even half of 4442?

The even half of 4442 looks like a copy of the even part of affine $E_6$ fusion category (it is also $\operatorname{Rep}(A_4)$) and another copy of affine $E_6$, but as a module category. It looks like $\operatorname{Rep}(A_4)$, graded by the Fibonacci category.

来源字段：http://aimpl.org/fusioncat/11/

17597 | AIM-OTHER-0029

### The A2 cluster pentagon and a local toggle obstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 4.6.** Make the analogy between birational toggles and cluster flips explicit.

Its actual problem ends immediately before the heading “4.6 Gelfand-Tsetlin triangles.” The recovered text is:

The accompanying paragraphs compare cluster flips on a Dynkin diagram with birational toggles on its positive-root poset. They ask, in particular, for an analogue of cluster variables and almost-positive roots along rowmotion orbits, suggest piecewise-linear and birational friezes, and identify the simplest nontrivial frieze shift with the Lyness 5-cycle.

No mathematical claim below uses the spilled Gelfand-Tsetlin text.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17598 | AIM-OTHER-0003

### The affine-hull collapse for a genuine function basis

S / solved | AIM 研讨会问题汇编

Let \(T\) be a finite-order action on a set \(X\), and let \(V\) be a finite-dimensional real vector space of functions \(X\to\mathbb R\). The dynamical closure \(V^T\) is the smallest vector space of functions containing \(V\) and closed under \(f\mapsto f\circ T\). Let \(f_1,\ldots,f_N\) be a basis of \(V^T\), and define \[ V^T(X)=\{(f_1(x),\ldots,f_N(x)):x\in X\}. \] **Problem 1.3.** Characterize the affine closure of \(V^T(X)\) in \(V^T\).

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17599 | AIM-OTHER-0017

### The dihedral root-order product and the meaning of order 2h

P / partially_solved | AIM 研讨会问题汇编

Birational toggles in \(w_0(c)\) root orders appear to continue to have order \(2h\) in type \(A_n\). This fails, for example, in type \(D_4\) (as does birational rowmotion).

**Conjecture 3.2.** Birational toggles in \(w_0(c)\) root orders have order \(2h\) in the coincidental types \(A,B,H_3,I_2(m)\).

None is part of Conjecture 3.2.

The phrase “root order” imports the notation of §3.1 and Williams's thesis. If

\[ w_0(c)=a_1a_2\cdots a_N, \]

then its inversion-root sequence is

\[ \beta_k=a_1a_2\cdots a_{k-1}(\alpha_{a_k}),\qquad 1\leq k\leq N. \]

This is the \(w_0(c)\) root order. With the composition convention used in the source,

\[ \operatorname{Tog}_{\beta_1\beta_2\cdots\beta_N} =T_{\beta_N}\circ\cdots\circ T_{\beta_2}\circ T_{\beta_1}; \tag{2} \]

thus \(T_{\beta_1}\) acts first. Williams calls the combinatorial version of (2) the nonnesting \(c\)-Kreweras complement. Conjecture 3.2 concerns its birational lift, not automatically birational rowmotion: a reflection/root order need not be a linear extension of the root poset.

There is an important convention issue. “Has order \(2h\)” can mean either:

1. the rational map has **minimal** positive order exactly \(2h\); or 2. \(2h\) is a **period/common exponent**, so the minimal order merely divides \(2h\).

These readings differ already in rank one and in even dihedral type. The results below resolve the entire \(I_2(m)\) family and make that distinction explicit.

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17600 | AIM-OTHER-0050

### The exact finiteness threshold for cyclotomic spoke norms

S / solved | AIM 研讨会问题汇编

Look at all spoke graphs with \(N>0\) arms. Are there finitely many \(N\)-tuples \((\ell_1,\dots,\ell_N)\) such that the spoke graph with \(N\) arms of lengths \(\ell_1,\dots,\ell_N\) has cyclotomic norm squared?

来源字段：http://aimpl.org/fusioncat/4/

17601 | AIM-OTHER-0109

### The logarithmic threshold and a malformed epsilon quantifier

O / open | AIM 研讨会问题汇编

For \(\alpha\in\mathbb R\), let \(\langle\alpha\rangle=\operatorname{dist}(\alpha,\mathbb Z)\).

**Question 27 (A. Pollington).** Are there \(\alpha,\beta\in\mathbb R\) such that for every \(\varepsilon>0\), \[ \liminf_{q\to\infty} q(\log q)^{2-\varepsilon} \langle q\alpha\rangle\langle q\beta\rangle>0? \] It follows from [65] that \[ \liminf_{q\to\infty} q(\log q)^2 \langle q\alpha\rangle\langle q\beta\rangle=0 \] for almost all \((\alpha,\beta)\). Thus, the set of \((\alpha,\beta)\) in Question 27 has measure zero.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17602 | AIM-OTHER-0014

### The tableau-rowmotion bridge and its three-chain boundary

P / partially_solved | AIM 研讨会问题汇编

Bloom et al. [BPS13] proved a homomesy result for rectangular semistandard tableaux under promotion. But we also know that promotion of semistandard tableaux is a special case of PL promotion in the order polytope of a product of two chains, and we also know homomesy results for rowmotion and promotion in products of two chains (Propp and Roby).

**Problem 2.10.** How do the results of Bloom, Pechenik, and Saracino relate to the results of Propp and Roby?

Also, there is work to be done regarding rowmotion on products of three chains. The obvious cardinality statistic is not in general homomesic under rowmotion in general, but various other statistics are (or at least appear to be, experimentally).

**Problem 2.11.** What are the homomesies of rowmotion acting [on] order ideals in a product of three chains? ...

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17603 | AIM-OTHER-0083

### The three-measure Katok Finsler flow

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 9 (A. Katok).** The only ergodic probability measures for this example are the smooth measure and the measures supported on periodic orbits.

“This example” is defined by the immediately preceding paragraph: Katok's 1973 non-reversible Finsler metric on \(S^2\), whose geodesic flow is ergodic and has two periodic phase-space orbits. The hyphen in “ex-ample” is only line breaking. The appended sentence beginning “3.3. Since a polygonal billiard ...” starts the next subsection and belongs with AIM-OTHER-0084, not Conjecture 9.

The referent is an approximation-by-conjugation existence construction, not one uniquely specified formula. Katok starts with the round geodesic flow and an \(S^1\)-isometry action, constructs an ergodic energy-level Hamiltonian flow, proves there are no periodic trajectories outside the exceptional set, and identifies its closed geodesics with the round geodesics invariant under the isometry action. For axial rotations of \(S^2\), these are the two oriented phase-space lifts of the equator. Katok explicitly notes that orientation counting matters for a non-symmetric metric.

This is not the elementary integrable irrational-wind/ellipsoid model also often called a Katok or Katok--Ziller example. Those explicit models transparently have two closed geodesics but have extra first integrals and are not the full-bundle ergodic construction selected by AIM.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17604 | AIM-OTHER-0033

### The tropical quotient and equality for unions of chains

P / partially_solved | AIM 研讨会问题汇编

Problem 4.10. Is the PL toggle group isomorphic to the birational toggle group? Or is it a proper quotient group?

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17605 | AIM-OTHER-0102

### Transfer principles for counting compact flats

P / partially_solved | AIM 研讨会问题汇编

**Question 23 (H. Oh).** Determine the asymptotics of the number of compact flats with volume less than \(T\) as \(T\to\infty\).

A flat is a totally geodesic zero-curvature submanifold; maximal flats arise from a Cartan subgroup \(A\). Oh had proved that only finitely many compact maximal flats have bounded volume.

The exact assigned question is:

The PDF then distinguishes a different known/accessible sum in the compact case, \[ \sum_{\substack{F\ \mathrm{regular}\\ \operatorname{systol}(F)<T}} \operatorname{Vol}(F). \] Here “regular” means that a shortest closed geodesic of \(F\) points in a regular direction, and \(\operatorname{systol}(F)\) is its length. This is **not** a volume cutoff and should not be conflated with Question 23.

On \(\Gamma\backslash G\), write \(\mu_F\) for Lebesgue measure on a compact \(A\)-orbit \(F\), \(\bar\mu_F=\mu_F/\operatorname{Vol}(F)\), and \[ \nu_T=\frac{\sum_{\operatorname{Vol}(F)<T}\mu_F} {\sum_{\operatorname{Vol}(F)<T}\operatorname{Vol}(F)},\qquad \bar\nu_T=\frac{\sum_{\operatorname{Vol}(F)<T}\bar\mu_F} {\#\{F:\operatorname{Vol}(F)<T\}}. The question whether these measures converge is Question 24 in the next record; the definitions are retained here only because they legitimately occur in the assigned source record and motivate the comparison below.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17606 | AIM-OTHER-0019

### Type-A antichain rowmotion homomesy and an endpoint certificate

S / solved | AIM 研讨会问题汇编

**Conjecture 3.3.** The statistic \(F\) is homomesic under \(\rho'\), with average \(n/2\).

That later material is not part of this problem.

The definitions immediately preceding the conjecture are essential. Let \[ P=\Phi^+(A_n)=\{[i,j]:1\leq i\leq j\leq n\} \] be the type-\(A_n\) positive-root poset, where \([i,j]\) denotes \(\alpha_i+\cdots+\alpha_j\). Let \(\mathcal O(P)\) and \(\mathcal C(P)\) be its order and chain polytopes. Let \(\rho:\mathcal O(P)\to\mathcal O(P)\) be piecewise-linear (PL) rowmotion and let \(\phi:\mathcal O(P)\to\mathcal C(P)\) be Stanley's transfer map. Set \[ \rho'=\phi\circ\rho\circ\phi^{-1},\qquad F(y)=\sum_{p\in P}y_p. \] The recovered statement is:

The source then proposes a birational analogue. In standard modern normalization, the birational transfer coordinates are positive labels on the same triangular array, the multiplicative statistic is their product, and the stated \(\widehat F\) is its logarithm. With lower and upper boundary labels both equal to \(1\), the conjectured logarithmic average is \(0\). The source reported symbolic checks only for \(n\leq3\).

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17607 | AIM-OTHER-0013

### Type-B middle Narayana and Baxter slices with a Fuss-cardinality obstruction

P / partially_solved | AIM 研讨会问题汇编

**Problem 2.9.** Is there a corresponding multi-noncrossing partition model?

One would expect that this has something to do with the root configuration, and the case $k=1$ is well-known in classical types. Using the correspondence between the type $B_n$ root poset and $[n]\times[n]$, the case $k=2$ for the type $B_n$ root poset corresponds to certain Narayana numbers [SW12]. Similarly, the case $k=3$ corresponds to Baxter numbers [Dil14]. Both of these have noncrossing models: Narayana numbers are well-known to be noncrossing partitions with a specified number of blocks; N. Reading has given a combinatorial description of Baxter permutations as noncrossing diagrams on a horizontal line of points using arcs that stay above or below the points [Rea14].

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17608 | AIM-OTHER-0126

### Uniform gap transfer and an elementary irrational diagonal family

P / partially_solved | AIM 研讨会问题汇编

Let $Q$ be a nondegenerate positive definite quadratic form of dimension $d\geq3$.

Conjecture 34 (Davenport--Lewis). Suppose that $Q$ is not a multiple of a rational form. Then the gaps between consecutive elements of the set $$ \{Q(x):x\in\mathbb Z^d\} $$ go to zero as $Q(x)\to\infty$.

The record has lost the hypotheses printed at the start of Section 5.5.

Thus the recovered AIM statement is: if $Q$ is a nondegenerate positive-definite real quadratic form in $d\geq3$ variables and its coefficient matrix is not a real scalar multiple of a rational matrix, then the gaps between successive **distinct** elements of $Q(\mathbb Z^d)$ tend to zero. Positivity makes the distinct value set locally finite, so it can be ordered

$$ 0=v_0<v_1<v_2<\cdots, $$

and the conclusion is $v_{j+1}-v_j\to0$. Repeated representations of one value are not additional elements.

There is a historical qualification. Later primary papers describe the original Davenport--Lewis conjecture with $d\geq5$, whereas the AIM list places Conjecture 34 in a section beginning with $d\geq3$. This report treats the AIM formulation: its $d\geq5$ part is solved and its $d=3,4$ extension remains open.

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17609 | AIM-OTHER-0048

### Unitary F-gauges are equivalent to categorical unitarizability

S / solved | AIM 研讨会问题汇编

For unitary theories, we can choose a gauge so the $F$ matrices formed by the $6j$ symbols are unitary, and the braiding matrices are diagonal with respect to a certain basis. Can this happen for some non-unitary fusion category?

Is the unitarity of the $F$ matrices equivalent to unitarity?

Does a complex fusion category admit a unitary $F$-gauge if and only if it is unitarizable? If it is braided, what then follows for its $R$-matrices?

来源字段：http://aimpl.org/fusioncat/3/

17610 | OWR-2050-004

### Universal Mechanisms of Biological Pattern Formation

O / open | Oberwolfach 报告问题汇编

What physical or chemical mechanisms produce the recurring periodic patterns in plants, including phyllotaxis and ridges, and in mollusk shells, and why do only a few patterns-often similar to patterns elsewhere in nature and in laboratory experiments-dominate in both?

来源字段：https://doi.org/10.4171/owr/2008/39

17611 | AIM-OTHER-0103

### Volume-weighted versus flat-weighted equidistribution

P / partially_solved | AIM 研讨会问题汇编

Question 24 (H. Oh). Do the measures νT and ˜νT converge to the normalized Haar measure on Γ\G?

According to [113], the normalized Haar measure on Γ \G is the unique ergodic measure of maximal entropy for the geodesic flow. Thus, to resolve

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17612 | AIM-OTHER-0070

### Wall avoidance, root detection, and gaps between local-rigidity hypotheses

P / partially_solved | AIM 研讨会问题汇编

**Conjecture 2.** If in Conjecture 1 the group \(A\) is not contained in a wall of a Weyl chamber of the split Cartan subgroup \(D\), then any \(C^1\)-small perturbation of the action of \(A\) on \(G/\Gamma\) is \(C^\infty\)-conjugate to the action of \(A\) defined by a continuous homomorphism from \(A\) to \(D\).

Its preceding paragraph is Conjecture 1, and the damaged record is:

Thus “Conjecture 1 the” must be replaced by “Conjecture 2. If in Conjecture 1”. Conjecture 1 assumes \(G\) connected semisimple, \(\Gamma<G\) an irreducible lattice, \(D<G\) a split Cartan, and \(A<D\) closed with \(\dim A>1\). The PDF immediately says Katok--Spatzier proved Conjecture 2 when \(A=D\).

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

17613 | AIM-OTHER-0027

### Williams's birational trapezoid conjecture

S / solved | AIM 研讨会问题汇编

**Conjecture 4.4 (Williams, as recorded by AIM).** Let \(p>1\) be an integer and \(s\in\mathbb N\). Then the order of birational rowmotion on \(P_{p,s}\) divides \(p\).

来源字段：http://aimath.org/pastworkshops/dac_preworkshop.pdf

17614 | AIM-OTHER-0094

### Yang's resolution and a finite-stabilizer compactness correction

S / solved | AIM 研讨会问题汇编

Conjecture 18 (N. Shah). The same result holds for all Lie groups L con-taining G = SO( n, 1).

**Recovered Conjecture 18.** Let \(L\) be a Lie group containing \(G=\operatorname{SO}(n,1)\), let \(\Lambda\) be a lattice in \(L\), and let \(a\in G\) have expanding horospherical subgroup \(U\). Suppose \(Gx_0\) is dense in \(L/\Lambda\). If an analytic curve \(\gamma:[0,1]\to U\cong\mathbb R^{n-1}\) is contained in neither a proper affine subspace nor an \((n-2)\)-sphere, then its curve measure \(\nu\) satisfies \[ a^k_*\nu\longrightarrow m_{L/\Lambda}\qquad(k\to\infty). \]

来源字段：https://aimath.org/WWN/measrigid/measrigid.pdf

# 附录 A | 猜想类题名索引 ( 1,117 条 )

原始 title 命中 conjecture(s) 或 hypothesis/hypotheses。按英文题名排序；页码指向主体。此索引不包含只在别名或表述中出现关键词的所有记录。

**"1/3 - 2/3 Conjecture" For every poset that is not a chain, there is some pair of elements x and y so that x appears abo**
AMR-030-0045 | P | 第 1620 页

**"Seymour's Second Neighborhood Conjecture" Any oriented graph has a vertex whose outdegree is at most its second outdegr**
AMR-030-0021 | P | 第 1620 页

**$C^r$ Stability Conjecture**
OPG-725 | P | 第 1265 页

**$d$-to-$1$ Games Conjecture**
OWR-16634-006 | P | 第 2049 页

**(-1)-Curve Conjecture for Plane Blowups**
OWR-16628-008 | O | 第 620 页

**1-Factorization Conjecture**
GRAPH-008 | P | 第 1760 页

**1/3-2/3 Conjecture**
COMB-001 | P | 第 1620 页

**10 Lectures and 42 Open Problems - Boolean classification and annulus conjecture**
AMR-027-0702 | P | 第 2049 页

**10 Lectures and 42 Open Problems - Feige's conjecture**
AMR-027-0406 | P | 第 1877 页

**10 Lectures and 42 Open Problems - The Paley ETF Conjecture**
AMR-027-0604 | P | 第 2051 页

**3-Decomposition Conjecture**
OPG-60027 | P | 第 1760 页

**3-Edge-Coloring Conjecture**
OPG-60046 | P | 第 1760 页

**3-flow conjecture**
OPG-128 | O | 第 1760 页

**4-flow conjecture**
OPG-127 | O | 第 1761 页

**4.2 (Futer) - The surface subgroup conjecture for cubulated hyperbolic groups**
AMR-108-0021 | O | 第 320 页

**5-flow conjecture**
OPG-126 | O | 第 1761 页

**A BMO Approach to the Lindelöf Hypothesis**
OWR-12866-010 | O | 第 15 页

**A coefficient-diagonal volume conjecture for the Gukov-Manolescu invariant**
AIM-TOPOLOGY-0037 | P | 第 1006 页

**A conjecture about direct product of funcoids**
OPG-37540 | P | 第 1007 页

**A conjecture on iterated circumcentres**
OPG-357 | O | 第 761 页

**A convention-sensitive counterexample to Hamidoune's reverse local-connectivity conjecture**
AIM-COMBINATORICS-0191 | S | 第 1622 页

**A cyclic Euler--Fourier obstruction for the volume conjecture**
AIM-TOPOLOGY-0052 | P | 第 1007 页

**A finite height-kernel case of the quasiconvex-subgroup conjecture**
AIM-GEOMETRIC_GROUP_THEORY-0001 | P | 第 323 页

**A finite-order marked-moduli certificate for the fixed-field Poisson conjecture**
AIM-ALGEBRAIC_NUMBER_THEORY-0038 | P | 第 18 页

**A Neumann-point incenter-moment expansion for the discrete Bareket conjecture**
AIM-ANALYSIS-0019 | P | 第 1271 页

**A norm obstruction for Agrawal's conjecture**
AIM-COMPUTATION-0089 | P | 第 2054 页

**A normalization audit of the higher-rank Serre weight conjecture**
AIM-REPRESENTATION_THEORY-0113 | P | 第 203 页

**A pole-safe dual-paired Ewald reduction for the remaining D4 conjecture**
AIM-GEOMETRY-0285 | P | 第 771 页

**A quantitative rational-dependence family for Littlewood's conjecture**
AIM-OTHER-0111 | P | 第 2164 页

**A sextic counterexample to Euler's sum of powers conjecture**
OPG-508 | O | 第 23 页

**A sharp three-edge-color counterexample to the literal conjecture**
AIM-COMBINATORICS-0156 | P | 第 1632 页

**A sharp two-label frontier for Seymour's load conjecture**
AIM-COMBINATORICS-0167 | P | 第 1632 页

**A square-root shifted-divisor hypothesis for the Conrey--Keating one-swap block**
AIM-ANALYTIC_NUMBER_THEORY-0062 | P | 第 23 页

**A support obstruction and a Wasserstein repair of Conjecture 42**
AIM-OTHER-0142 | S | 第 2165 页

**A. Navas: A Conjecture on Delone Sets BL to Lattices (after P. Alestalo, D.A. Trotsenko and J. V\"ais\"al\"a). - Problem**
AMR-065-0014 | S | 第 776 页

**ABC Conjecture**
NT-005 | P | 第 24 页

**ABCD Conjecture Under Determinacy**
OWR-14299082-018 | O | 第 1971 页

**Abelian Covering Conjectures for Curves over Number Fields**
OWR-1053-005 | O | 第 24 页

**Abstract Pierce-Birkhoff Conjecture**
OWR-1465-008 | O | 第 206 页

**Adic Effective-Period Complex Conjecture**
OWR-14299908-003 | O | 第 633 页

**Agoh-Giuga Conjecture**
NT-087 | P | 第 25 页

**Agrawal's conjecture**
AMR-093-0114 | O | 第 25 页

**Aharoni-Berger conjecture**
OPG-382 | O | 第 1635 页

**Albert's Cyclicity Conjecture**
OWR-1393-004 | O | 第 26 页

**Albertson Conjecture**
GRAPH-015 | P | 第 1762 页

**Aldous-Lyons soficity conjecture**
AMR-096-0038 | P | 第 1887 页

**Alexa's Conjecture on Primality**
OPG-37413 | P | 第 26 页

**Algebraic Effective-Period Complex Conjecture**
OWR-14299908-002 | O | 第 636 页

**Algebraic K-Theory and Rigidity Conjectures for Discrete Groups**
OWR-5155-002 | P | 第 329 页

**Alperin Weight Conjecture and Perfect Localities**
OWR-3400-008 | P | 第 210 页

**Alperin-McKay Conjecture**
OWR-13750328-009 | O | 第 329 页

**An odd-subsequence counterexample to the literal ratio-one Volume Conjecture**
AIM-TOPOLOGY-0029 | S | 第 1017 页

**An order-statistic form of Shen's deficit conjecture**
AIM-COMBINATORICS-0201 | P | 第 1638 页

**Analysis of Boolean Functions - Aaronson--Ambainis Conjecture**
AMR-127-0015 | O | 第 2058 页

**Analysis of Boolean Functions - Bernoulli Conjecture**
AMR-127-0007 | S | 第 2058 页

**Analysis of Boolean Functions - Bhattacharyya--Grigorescu--Shapira Conjecture**
AMR-127-0016 | O | 第 2058 页

**Analysis of Boolean Functions - Fourier Entropy--Influence Conjecture**
AMR-127-0008 | O | 第 2058 页

**Analysis of Boolean Functions - Gotsman--Linial Conjecture**
AMR-127-0004 | O | 第 2059 页

**Analysis of Boolean Functions - Majority Is Least Stable Conjecture**
AMR-127-0009 | S | 第 2059 页

**Analysis of Boolean Functions - Mansour's Conjecture**
AMR-127-0006 | O | 第 2059 页

**Analysis of Boolean Functions - Polynomial Freiman--Ruzsa Conjecture (in the $\mathbb{F}_2^n$ setting)**
AMR-127-0005 | S | 第 2059 页

**Analysis of Boolean Functions - Servedio--Tan--Verbin Conjecture (Disproved)**
AMR-127-0022 | S | 第 2059 页

**Analysis of Boolean Functions - Standard Simplex Conjecture**
AMR-127-0018 | O | 第 2060 页

**Analysis of Boolean Functions - Talagrand's ``Convolution with a Biased Coin'' Conjecture**
AMR-127-0003 | O | 第 2060 页

**Analysis of Boolean Functions - Tomaszewski's Conjecture**
AMR-127-0002 | S | 第 2060 页

**Andersen-Masbaum-Ueno Conjecture**
OWR-1967-005 | O | 第 331 页

**Andrews-Curtis Conjecture**
ALG-009 | P | 第 211 页

**Another conjecture about reloids and funcoids**
OPG-59970 | P | 第 1017 页

**Approximation Conjectures for $L^2$-Torsion and Homology**
OWR-5154-004 | P | 第 1019 页

**Arithmetic Volume Conjecture for Kashaev Invariants**
OWR-4416-001 | P | 第 1019 页

**Arnold and Arnold-Givental conjectures**
AMR-048-0001 | P | 第 1514 页

**Around Kouchnirenko's conjecture I**
AMR-046-0028 | P | 第 212 页

**Around Kouchnirenko's conjecture II**
AMR-046-0029 | P | 第 212 页

**Arthur's Conjectures**
ALG-034 | P | 第 212 页

**Artin's Conjecture on Primitive Roots**
NT-029 | P | 第 33 页

**Atiyah Conjecture on $L^2$-Betti Numbers**
OWR-820-003 | P | 第 1020 页

**Atiyah Conjectures and Irrational-Covolume Lattices for TDLC Groups**
OWR-14298004-015 | O | 第 332 页

**Auslander Conjecture for Complete Affine Manifolds**
OWR-12581-002 | O | 第 785 页

**Automorphy Conjecture for Motives**
OWR-1734-009 | P | 第 35 页

**Axiomatic stable homotopy 1 - In our memoir, we give a conjecture for the thick subcategories in a Noetherian stable homotopy catego…**
AMR-110-0033 | P | 第 1020 页

**Axis and low-dimensional cases of Levitin's disk-intersection conjecture**
AIM-PHYSICS-0008 | P | 第 2118 页

**Ádám's Conjecture**
OPG-46432 | O | 第 1764 页

**Babai Polylogarithmic Diameter Conjecture for Finite Simple Groups**
OWR-14299097-009 | O | 第 333 页

**Babai's Diameter Conjecture for Finite Simple Groups**
OWR-1965-006 | O | 第 333 页

**Babai’s Polylogarithmic Diameter Conjecture**
OWR-14298365-007 | O | 第 333 页

**Ball Conjecture for Twisted Bruhat Intervals**
OWR-1455-006 | O | 第 1639 页

**Bang Plank Conjecture for Nonsymmetric Convex Bodies**
OWR-13110-004 | O | 第 788 页

**Barnette's Conjecture**
GRAPH-027 | P | 第 1764 页

**Bartnik’s Stationary Vacuum Extension Conjecture**
OWR-8415343-005 | O | 第 2118 页

**Bass conjecture**
AMR-071-0002 | O | 第 641 页

**Bass-Quillen conjecture**
AMR-071-0003 | P | 第 641 页

**Baum-Connes Conjecture**
TOP-007 | P | 第 1022 页

**Baum-Connes Conjecture for Locally Compact Groups**
OWR-832-005 | O | 第 333 页

**Baum-Connes Conjecture for Torsion-Free Groups**
OWR-1060-009 | O | 第 1022 页

**Bárány-Larman Colored Tverberg Conjecture**
OWR-4798-007 | P | 第 788 页

**Beal's Conjecture**
NT-078 | O | 第 35 页

**Beauville-Voisin Conjecture for Hyperkähler Manifolds**
OWR-13101-006 | P | 第 641 页

**Beck's solution of van den Berg's conjecture and an exact prism refinement**
AIM-ANALYSIS-0024 | S | 第 1284 页

**Beilinson conjectures on special values of motivic L-functions**
AMR-093-0060 | P | 第 35 页

**Beilinson's Regulator Conjecture**
OWR-16171-007 | O | 第 35 页

**Beilinson-Soulé Vanishing Conjecture**
OWR-5155-006 | P | 第 213 页

**Beneš Conjecture**
OPG-37181 | P | 第 1640 页

**Berge Conjecture**
TOP-008 | P | 第 1022 页

**Berge's conjecture on path partitions**
AMR-029-0006 | P | 第 1640 页

**Berry-Tabor conjecture**
AMR-048-0002 | P | 第 1515 页

**Bestvina-Davis Candidates for the Eilenberg-Ganea Conjecture**
OWR-13488-011 | O | 第 333 页

**Birch and Swinnerton-Dyer Conjecture**
MPP-005 | O | 第 35 页

**Birch-Tate Conjecture**
ALG-017 | P | 第 214 页

**Birch-Swinnerton-Dyer Conjecture**
OWR-16171-009 | P | 第 36 页

**Birkhoff Conjecture**
DYN-004 | P | 第 1457 页

**Birkhoff Integrable-Billiards Conjecture**
OWR-1383-006 | O | 第 1515 页

**Birkhoff Rigidity Conjecture for Convex Billiards**
OWR-17127-001 | P | 第 1515 页

**Bloch-Beilinson Torsion Conjecture for Albanese Kernels**
OWR-1318-001 | O | 第 642 页

**Boffa’s Set-Theoretic Automorphism Conjecture**
OWR-1390-002 | O | 第 2005 页

**Bollobás-Erdős-Tuza Independent-Set Hitting Conjecture**
OWR-9790352-031 | P | 第 1765 页

**Bombieri-Lang conjecture**
AMR-093-0081 | P | 第 36 页

**Borel Conjecture**
TOP-005 | P | 第 1023 页

**Borovik-Cherlin Generic-Transitivity Conjecture**
OWR-11695862-004 | P | 第 2005 页

**Borwein Sign Conjectures**
OWR-11695867-008 | P | 第 1641 页

**Bost Assembly Conjecture with $C^*$-Algebra Coefficients**
OWR-2048-002 | P | 第 215 页

**Bouchet's 6-flow conjecture**
OPG-131 | O | 第 1765 页

**Bounded Negativity Conjecture for Complex Projective Surfaces**
OWR-17296-011 | O | 第 643 页

**Bounded Negativity Conjecture for Projective Surfaces**
OWR-14222-011 | O | 第 643 页

**Bourgain-Spencer Regularity Conjecture**
OWR-11695858-003 | P | 第 1891 页

**Branko Grunbaum in Many Dimensions - Conjecture 2**
AMR-142-0002 | O | 第 810 页

**Branko Grunbaum in Many Dimensions - Conjecture 3**
AMR-142-0003 | O | 第 810 页

**Branko Grunbaum in Many Dimensions - Conjecture 6 (Ramos's conjecture)**
AMR-142-0006 | O | 第 811 页

**Branko Grunbaum in Many Dimensions - Conjecture 7**
AMR-142-0007 | O | 第 811 页

**Branko Grunbaum in Many Dimensions - Conjecture 8**
AMR-142-0008 | O | 第 811 页

**Brauer-Manin Conjecture for K3 Surfaces**
OWR-4076-003 | O | 第 37 页

**Broué Abelian Defect-Group Conjecture**
OWR-1265-017 | O | 第 335 页

**Broué's Abelian Defect Group Conjecture**
OWR-11583-002 | O | 第 335 页

**Broué’s Abelian Defect Group Conjecture**
OWR-13675-004 | O | 第 335 页

**Brouwer's Conjecture on Graph Laplacians**
GRAPH-001 | P | 第 1766 页

**Brumer-Stark Class Group Annihilation Conjecture**
OWR-4537-002 | S | 第 37 页

**Brumer-Stark conjecture**
AMR-093-0182 | S | 第 37 页

**Bunyakovsky Conjecture**
NT-084 | O | 第 38 页

**Bøgvad's Normal-Koszul Conjecture**
OWR-1971-008 | O | 第 645 页

**Caccetta-Häggkvist Conjecture**
OPG-46385 | O | 第 1766 页

**Caccetta-Häggkvist Conjecture for Directed Triangles**
OWR-1532-003 | O | 第 1766 页

**Cameron Greedy Base-Size Conjecture**
OWR-14299097-005 | P | 第 336 页

**Campana-Peternell Conjecture for Fano Manifolds**
OWR-13750327-001 | P | 第 645 页

**Cannon's Conjecture**
OWR-717-005 | O | 第 812 页

**Cannon's Conjecture**
OWR-1319-010 | O | 第 812 页

**Cannon’s Hyperbolic Three-Space Conjecture**
OWR-16406-003 | O | 第 336 页

**Carleson-Jones quarter conjecture**
AMR-036-0033 | O | 第 1288 页

**Carmichael's totient function conjecture**
AMR-093-0002 | O | 第 38 页

**Carrell Minimality Conjecture**
OWR-788-003 | S | 第 645 页

**Cartan Determinant Conjecture**
OWR-14218-006 | O | 第 216 页

**Cartan-Hadamard Isoperimetric Conjecture**
OWR-15578-008 | P | 第 1025 页

**Casas-Alvero Conjecture**
ALG-008 | P | 第 216 页

**Catalan-Mersenne Conjecture**
NT-016 | O | 第 38 页

**Catalan-Dickson conjecture on aliquot sequences**
AMR-093-0003 | O | 第 39 页

**Categorification of the Quantum Volume Conjecture**
OWR-12014-008 | O | 第 1025 页

**Cereceda's Conjecture**
GT-008 | P | 第 1767 页

**Charalambous-Reeves Standard-Matching Conjecture**
OWR-734-006 | O | 第 219 页

**Charbit's spectral short-cycle conjecture: the sharp digon case and a Perron-period obstruction**
AIM-COMBINATORICS-0188 | P | 第 1644 页

**Chern's Affine Conjecture**
GEO-016 | P | 第 814 页

**Chern's Conjecture (Affine Geometry)**
GEOM-016 | O | 第 814 页

**Chern's Conjecture for Hypersurfaces in Spheres**
GEO-017 | P | 第 814 页

**Chromatic Lichtenbaum-Quillen and Redshift Conjectures**
OWR-10252932-002 | P | 第 1026 页

**Chvátal's Toughness Conjecture**
GRAPH-028 | P | 第 1768 页

**Closure Conjectures for Infinite Stable Fields**
OWR-4196-001 | O | 第 2006 页

**Cocircuit Graph Diameter Conjecture for Oriented Matroids**
OWR-17135-011 | O | 第 1644 页

**Coincidence Rank Conjecture**
AMR-065-0007 | P | 第 819 页

**Coloring Dense Triangle-Free Graphs and Hajós' Conjecture**
OWR-711-009 | P | 第 1770 页

**Combinatorial Approaches to the Gasca-Maeztu Conjecture**
OWR-9790358-002 | O | 第 224 页

**Combinatorial Explanation of the Clasp Conjecture**
OWR-17135-015 | O | 第 341 页

**Common-Neighbor Conjecture for Generalized Saxl Graphs**
OWR-14299292-003 | P | 第 341 页

**Commuting Locally Nilpotent Derivations Conjecture**
OWR-1452-017 | O | 第 224 页

**Commuting powers conjecture**
AMR-045-0016 | O | 第 1519 页

**Comparable truncators and a radial reduction for the localized Santalo conjecture**
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**Complementarity Form of the Schumaker Strang Conjecture**
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**Complex Extension Conjecture for Polyhedral Kähler Manifolds**
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**Complexified Volume Conjecture for Hyperbolic Knots**
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**Complexity in Enumerative Combinatorics - Conjecture 1 (Wilf)**
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**Complexity in Enumerative Combinatorics - Conjecture 13 (cf. [Ges-sem])**
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**Complexity in Enumerative Combinatorics - Conjecture 2**
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**Complexity in Enumerative Combinatorics - Conjecture 3**
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**Complexity in Enumerative Combinatorics - Conjecture 4**
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**Complexity in Enumerative Combinatorics - Conjecture 6**
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**Complexity in Enumerative Combinatorics - Conjecture 7**
AMR-121-0007 | O | 第 1648 页

**Complexity in Enumerative Combinatorics - Conjecture 8 (cf. [Zei])**
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**Conditional fiber collapse and periodic-data tests for the Anosov infranil conjecture**
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**Conforti-Cornuéjols conjecture on the MFMC property**
AMR-029-0019 | O | 第 1650 页

**Conjecture 0.1 - Is there an infinite expander?**
AMR-099-0001 | O | 第 1772 页

**Conjecture 1.19 - (The volume conjecture, [198, 296]) For any knot K, 2π·lim N →∞ log|JN (K)| N = v3||S3- K||, (2) where||·||denotes th…**
AMR-103-0019 | P | 第 1032 页

**Conjecture 1.21 - (H.**
AMR-103-0021 | P | 第 1032 页

**Conjecture 1.3 - Pillai**
AMR-086-0002 | P | 第 1772 页

**Conjecture 1.4 - Shorey**
AMR-086-0003 | O | 第 1772 页

**Conjecture 1.5**
AMR-086-0004 | O | 第 1772 页

**Conjecture 1.7**
AMR-086-0006 | O | 第 1772 页

**Conjecture 1.8 - Langevin**
AMR-086-0007 | O | 第 1773 页

**Conjecture 1.9**
AMR-086-0008 | P | 第 1773 页

**Conjecture 10.14 - The map (50) is an isomorphism.**
AMR-103-0189 | O | 第 1033 页

**Conjecture 10.15 - {M ~ Y2d S3}/ ~ Y2d+1 is torsion free for each d.**
AMR-103-0190 | O | 第 1033 页

**Conjecture 10.17 - (M.**
AMR-103-0192 | P | 第 1033 页

**Conjecture 10.7 - F as d (ZM)/F as d+1(ZM) (resp.F b d (ZM)/F b d+1(ZM)) is torsion free for each d.**
AMR-103-0182 | O | 第 1033 页

**Conjecture 10.8 - A(∅; Z) is torsion free.**
AMR-103-0183 | O | 第 1033 页

**Conjecture 10.9 - Finite type invariants distinguish integral homology 3-spheres.**
AMR-103-0184 | O | 第 1033 页

**Conjecture 11.2 - The LMO invariant distinguishes integral homology 3-spheres.**
AMR-103-0198 | O | 第 1033 页

**Conjecture 11.3 - 1.**
AMR-109-0324 | O | 第 1033 页

**Conjecture 12.10 - (D.**
AMR-103-0215 | O | 第 1033 页

**Conjecture 12.16 - (X.-S.**
AMR-103-0221 | O | 第 1033 页

**Conjecture 12.2 - (R.**
AMR-103-0208 | O | 第 1034 页

**Conjecture 12.26 - (V.**
AMR-103-0231 | O | 第 1034 页

**Conjecture 12.4 - (3-move conjecture, Y.**
AMR-103-0210 | S | 第 1034 页

**Conjecture 12.5 - (Y.**
AMR-103-0211 | S | 第 1034 页

**Conjecture 12.7 - (D.**
AMR-103-0213 | O | 第 1034 页

**Conjecture 2.1 - ([220, Problem 1.92 (N)]) Fd(ZK)/Fd+1(ZK) is torsion free for each d.**
AMR-103-0023 | O | 第 1034 页

**Conjecture 2.1 - Let Ω∗(Hom(π,G )/G) be the de Rham algebra consisting of all measurable differential forms on Hom(π,G )/G.**
AMR-109-0186 | O | 第 1034 页

**Conjecture 2.1 - The natural homomorphisms ( Λ∗Λ3HQ )Sp →H ∗(Mg,∗; Q), (Λ∗UQ)Sp→H ∗(Mg; Q) induce isomorphisms ( Λ∗Λ3H ∗ Q/ ( [12]tore…**
AMR-109-0303 | O | 第 1035 页

**Conjecture 2.11 - (S.**
AMR-103-0033 | O | 第 1035 页

**Conjecture 2.12**
AMR-086-0016 | O | 第 1773 页

**Conjecture 2.14 - Mahler**
AMR-086-0017 | O | 第 1773 页

**Conjecture 2.15 - Mahler**
AMR-086-0018 | P | 第 1773 页

**Conjecture 2.16 - (M.**
AMR-103-0038 | O | 第 1035 页

**Conjecture 2.17 - [154] Every Vassiliev invariant of classical knots can be extended to a finite type invariant of long virtual knots.**
AMR-103-0039 | S | 第 1035 页

**Conjecture 2.2 - A(S1; Z) is torsion free.**
AMR-103-0024 | O | 第 1035 页

**Conjecture 2.2 - Erdős-Woods**
AMR-086-0010 | P | 第 1773 页

**Conjecture 2.2 - Suppose C∞(Hom(π,G )/G) D-→C∞(Hom(π,G )/G) is a differential operator which commutes with the ModΣ-action on Hom(π,G…**
AMR-109-0187 | O | 第 1035 页

**Conjecture 2.21 - (A.**
AMR-103-0043 | O | 第 1036 页

**Conjecture 2.22 - The map (15) is an isomorphism.**
AMR-103-0044 | O | 第 1036 页

**Conjecture 2.23 - {K~ Cd O}/~ Cd+1 is torsion free for each d.**
AMR-103-0045 | O | 第 1036 页

**Conjecture 2.24 - (K.**
AMR-103-0046 | S | 第 1036 页

**Conjecture 2.3 - (X.-S.**
AMR-103-0025 | P | 第 1036 页

**Conjecture 2.3 - Erdős-Dressler**
AMR-086-0011 | P | 第 1773 页

**Conjecture 2.3 - If r≥ 3, the action of Out(π) on Hom(π,G ) is ergodic.**
AMR-109-0190 | S | 第 1036 页

**Conjecture 2.4 - Philippon**
AMR-086-0012 | O | 第 1773 页

**Conjecture 2.5 - Lang-Waldschmidt**
AMR-086-0013 | O | 第 1773 页

**Conjecture 2.5 - Vassiliev invariants distinguish oriented knots.**
AMR-103-0027 | O | 第 1036 页

**Conjecture 2.6**
AMR-086-0014 | O | 第 1773 页

**Conjecture 2.7 - (see [220, Problem 1.89 (B)]) For any oriented knot K, no Vassiliev invariants distinguish K from -K.**
AMR-103-0029 | O | 第 1036 页

**Conjecture 2.7 - Hall**
AMR-086-0015 | O | 第 1773 页

**Conjecture 3.1 - Suppose that b = 0 (Σ is closed).**
AMR-109-0193 | P | 第 1036 页

**Conjecture 3.10 - Gel’fond-Schneider**
AMR-086-0028 | O | 第 1774 页

**Conjecture 3.11 - $p$-adic analog of Lindemann-Weierstrass's Theorem**
AMR-086-0029 | O | 第 1774 页

**Conjecture 3.12 - $p$-adic analog of an algebraic independence result of Gel’fond**
AMR-086-0030 | O | 第 1774 页

**Conjecture 3.13 - Blum, Cucker, Shub and Smale**
AMR-086-0031 | O | 第 1774 页

**Conjecture 3.14**
AMR-086-0032 | O | 第 1774 页

**Conjecture 3.15 - (Density of pseudo-Anosovs).**
AMR-109-0036 | P | 第 1036 页

**Conjecture 3.15 - Goncharov**
AMR-086-0033 | P | 第 1774 页

**Conjecture 3.16 - d(Ig) = 0.**
AMR-109-0037 | O | 第 1037 页

**Conjecture 3.16 - Zagier**
AMR-086-0034 | P | 第 1774 页

**Conjecture 3.17**
AMR-086-0035 | O | 第 1774 页

**Conjecture 3.18**
AMR-086-0036 | O | 第 1774 页

**Conjecture 3.19 - Rohrlich**
AMR-086-0037 | P | 第 1774 页

**Conjecture 3.2 - If k = 1, then U is onto.**
AMR-109-0199 | P | 第 1037 页

**Conjecture 3.2 - Roy**
AMR-086-0020 | O | 第 1775 页

**Conjecture 3.2 - The Kontsevich invariant distinguishes oriented knots.**
AMR-103-0052 | O | 第 1037 页

**Conjecture 3.20 - Nesterenko**
AMR-086-0038 | P | 第 1775 页

**Conjecture 3.21 - Bertolin**
AMR-086-0039 | O | 第 1775 页

**Conjecture 3.22 - [357, 139] The map (29) is injective.**
AMR-103-0072 | S | 第 1037 页

**Conjecture 3.23**
AMR-086-0041 | O | 第 1775 页

**Conjecture 3.24 - Bertrand**
AMR-086-0042 | O | 第 1775 页

**Conjecture 3.25 - Bertrand**
AMR-086-0043 | O | 第 1775 页

**Conjecture 3.26**
AMR-086-0044 | O | 第 1775 页

**Conjecture 3.3 - Algebraic Independence of Logarithms of Algebraic Numbers**
AMR-086-0021 | O | 第 1775 页

**Conjecture 3.4 - Strong Four Exponentials Conjecture**
AMR-086-0022 | O | 第 1776 页

**Conjecture 3.4 - Z(K) = Z(-K) for any oriented knot K, where -K de- notes K with the opposite orientation.**
AMR-103-0054 | P | 第 1037 页

**Conjecture 3.5 - Strong Five Exponentials Conjecture**
AMR-086-0023 | O | 第 1776 页

**Conjecture 3.6 - Roy**
AMR-086-0024 | O | 第 1776 页

**Conjecture 3.8 - (D.**
AMR-103-0058 | O | 第 1037 页

**Conjecture 3.8 - Gel’fond**
AMR-086-0026 | O | 第 1776 页

**Conjecture 3.9 - Schneider**
AMR-086-0027 | O | 第 1776 页

**Conjecture 4.1 - (Inhomogeneity of all metrics).**
AMR-109-0040 | P | 第 1037 页

**Conjecture 4.1 - Lehmer's Problem**
AMR-086-0046 | P | 第 1776 页

**Conjecture 4.11 - Wirsing and Schmidt**
AMR-086-0053 | P | 第 1776 页

**Conjecture 4.12**
AMR-086-0054 | P | 第 1776 页

**Conjecture 4.13 - Laurent-Roy**
AMR-086-0055 | O | 第 1776 页

**Conjecture 4.14**
AMR-086-0056 | O | 第 1777 页

**Conjecture 4.15**
AMR-086-0057 | O | 第 1777 页

**Conjecture 4.15 - (J.**
AMR-103-0093 | O | 第 1038 页

**Conjecture 4.16 - Quantitative Refinement of Schanuel's Conjecture**
AMR-086-0058 | O | 第 1777 页

**Conjecture 4.19**
AMR-086-0061 | P | 第 1777 页

**Conjecture 4.2 - ( Mg is maximal).**
AMR-109-0041 | O | 第 1038 页

**Conjecture 4.2 - The classes μi are non-trivial for all i = 1, 2,⋯.**
AMR-109-0309 | O | 第 1038 页

**Conjecture 4.20**
AMR-086-0062 | O | 第 1778 页

**Conjecture 4.21**
AMR-086-0063 | O | 第 1778 页

**Conjecture 4.3 - Amoroso-David**
AMR-086-0048 | O | 第 1778 页

**Conjecture 4.3 - If every closed incompressible surface in M is parallel to ∂M, then S2,∞(M ) is torsion free.**
AMR-103-0081 | O | 第 1038 页

**Conjecture 4.4 - (Nonpositive curvature).**
AMR-109-0043 | O | 第 1039 页

**Conjecture 4.4 - Amoroso-David**
AMR-086-0049 | O | 第 1778 页

**Conjecture 4.4 - Let M be a closed hyperbolic 4-manifold.**
AMR-109-0252 | O | 第 1039 页

**Conjecture 4.5 - Amoroso-David**
AMR-086-0050 | P | 第 1778 页

**Conjecture 4.6 - LetS be any surface with d(S)≥ 1.**
AMR-109-0044 | O | 第 1040 页

**Conjecture 4.7 - ( Q-rank of moduli space).**
AMR-109-0045 | S | 第 1040 页

**Conjecture 4.7 - David-Hindry**
AMR-086-0052 | P | 第 1778 页

**Conjecture 4.8 - (Mod g is Kahler).**
AMR-109-0046 | P | 第 1040 页

**Conjecture 4.8 - Each component of Hc g is simply connected.**
AMR-109-0093 | S | 第 1041 页

**Conjecture 4.9 - The stable rational cohomology of OutFn is trivial.**
AMR-109-0315 | S | 第 1041 页

**Conjecture 5.12 - (R.**
AMR-103-0108 | S | 第 1041 页

**Conjecture 5.12 - Ig is finitely presented for g≥ 4.**
AMR-109-0058 | O | 第 1041 页

**Conjecture 5.24 - (Picard number one conjecture for level L structures).**
AMR-109-0068 | P | 第 1041 页

**Conjecture 5.3**
AMR-086-0066 | O | 第 1778 页

**Conjecture 5.3 - Let hX be as above.**
AMR-103-0099 | S | 第 1041 页

**Conjecture 5.4 - Loxton and van der Poorten**
AMR-086-0067 | S | 第 1779 页

**Conjecture 5.5 - For each k≥ 1, the group (Ig)k is not finitely generated.**
AMR-109-0052 | S | 第 1042 页

**Conjecture 6.1 - The classes e1,t 3,t 5,⋯ are all non-trivial.**
AMR-109-0318 | O | 第 1042 页

**Conjecture 7.10 - (The growth rate conjecture) Let d = max{d0,..., dn}.**
AMR-103-0125 | P | 第 1042 页

**Conjecture 7.11 - There is a construct of the right measure, say τM (A)1/2 for A∈M i, from the square root of the Reidemeister torsion…**
AMR-103-0126 | P | 第 1042 页

**Conjecture 7.12 - (H.**
AMR-103-0127 | P | 第 1042 页

**Conjecture 7.25 - (S.**
AMR-103-0140 | O | 第 1042 页

**Conjecture 7.29 - (K.**
AMR-103-0144 | S | 第 1042 页

**Conjecture 7.30 - (K.**
AMR-103-0145 | O | 第 1042 页

**Conjecture 7.5 - For non-vanishing $\tau_r^G(M)$, the absolute value $|\tau_r^G(M)|$ depends only on the fundamental group $\pi_1(M)$.**
AMR-103-0120 | P | 第 1042 页

**Conjecture 7.6 - (The perturbative expansion conjecture) The asymptotic expansion of Z G k (M ) of a closed oriented 3-manifold M is g…**
AMR-103-0121 | P | 第 1043 页

**Conjecture 7.7 - (The asymptotic expansion conjecture, J.E.**
AMR-103-0122 | P | 第 1043 页

**Conjecture 7.9 - (Topological interpretations of the dj ’s) Let Mj be the union of components of the moduli space of flat connections…**
AMR-103-0124 | P | 第 1043 页

**Conjecture I - f f is pseudo-Anosov element of a mapping class group ModS with sufficiently big dilatation coefficient, then the subgrou…**
AMR-109-0103 | S | 第 1043 页

**Conjecture I - f Γ is an irreducible arithmetic group of rank ≥ 2, then every homomorphism Γ→ ModS has finite image.**
AMR-109-0098 | S | 第 1044 页

**Conjecture of multiplicative persistence**
AMR-046-0031 | P | 第 1650 页

**Conjecture - Every subgroup of finite index in ModS contains a congruence subgroup.**
AMR-109-0095 | P | 第 1044 页

**Conjecture - For every finitely generated subgroups G of ModS, the group Φf (G) is nilpotent.**
AMR-109-0107 | O | 第 1044 页

**Conjecture - LetS and R be closed surfaces.**
AMR-109-0106 | O | 第 1044 页

**Conlon-Lee Conjecture for Graph Subdivisions**
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**Connelly’s blooming conjecture**
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**Continous analogue of Hirsch conjecture**
OPG-610 | P | 第 827 页

**Continuum Hypothesis**
HIL-001 | S | 第 1973 页

**Conway's Thrackle Conjecture**
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**Coxeter Groups and Unimodality - Conjecture 13**
AMR-125-0013 | O | 第 1652 页

**Coxeter Groups and Unimodality - Conjecture 14**
AMR-125-0014 | O | 第 1652 页

**Coxeter Groups and Unimodality - Conjecture 15**
AMR-125-0015 | O | 第 1653 页

**Coxeter Groups and Unimodality - Conjecture 2**
AMR-125-0002 | O | 第 1653 页

**Coxeter Groups and Unimodality - Conjecture 3**
AMR-125-0003 | O | 第 1653 页

**Coxeter Groups and Unimodality - Conjecture 4**
AMR-125-0004 | O | 第 1653 页

**Coxeter Groups and Unimodality - Conjecture 5**
AMR-125-0005 | O | 第 1653 页

**Coxeter Groups and Unimodality - Conjecture 8**
AMR-125-0008 | O | 第 1653 页

**Coxeter Groups and Unimodality - Conjecture 9**
AMR-125-0009 | O | 第 1653 页

**Crouzeix's Conjecture**
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OWR-11695867-009 | P | 第 1654 页

**Cusp Covering Conjecture for Hyperbolic Manifolds**
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**Cut covers and small cases of Mader's local-connectivity conjecture**
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**Cyclic-binomial structure in an early Agrawal conjecture**
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**Cyclotomic-Anticyclotomic Iwasawa Main Conjecture**
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**D. Damanik: Quantum Mechanics and Quasicrystals - Conjecture**
AMR-065-0002 | P | 第 833 页

**D. Damanik: Quantum Mechanics and Quasicrystals - Conjecture**
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**Dade's Conjecture**
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**Dar's conjecture: homogeneous repair and Cartesian-product closure**
AIM-FUNCTIONAL_ANALYSIS-0006 | P | 第 1295 页

**De Giorgi Conjecture for Allen-Cahn**
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**De Giorgi Conjecture for the Allen-Cahn Equation**
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**Debarre’s Minimal Cohomology-Class Conjecture**
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**Deligne's Conjecture for Critical Motive Values**
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**Deligne's conjecture on Hochschild cohomology about the operadic structure on Hochschild cochain complex**
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**Determinantal Conjecture**
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**Dickson's Conjecture**
NT-085 | P | 第 53 页

**Ding's tau_r vs. tau conjecture**
OPG-696 | P | 第 1657 页

**Diophantine quintuple conjecture**
OPG-16555 | P | 第 54 页

**Dirac's Conjecture**
OPG-59888 | P | 第 837 页

**Directed Green-Griffiths-Lang Conjecture Beyond Rank One**
OWR-7155446-003 | O | 第 660 页

**Dittert-Hajek conjecture**
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OWR-1452-013 | P | 第 232 页

**Dolnikov’s Colorful Transversal Conjecture**
OWR-14298163-008 | P | 第 839 页

**Donaldson's Almost-Complex Calabi-Yau Conjecture**
OWR-4339-006 | O | 第 839 页

**Donkin’s Good Filtration Conjecture**
OWR-13681-004 | O | 第 232 页

**Donovan’s Finiteness Conjecture for Fixed Defect Groups**
OWR-14218-001 | O | 第 232 页

**Douady Conjecture on Polynomial Linearizability**
OWR-2094-004 | O | 第 1525 页

**Double Cap Conjecture**
OWR-14297745-002 | O | 第 839 页

**Double-critical graph conjecture**
OPG-34839 | P | 第 1784 页

**Dubner's conjecture**
AMR-093-0120 | O | 第 56 页

**Durer's Conjecture**
OPG-37459 | O | 第 839 页

**Dynamical Alekseevskii Conjecture for Homogeneous Ricci Flows**
OWR-10252937-005 | O | 第 1466 页

**Eden's conjecture on local Lyapunov dimension**
AMR-048-0006 | O | 第 1525 页

**Edge list coloring conjecture**
OPG-2110 | P | 第 1784 页

**Edge Reconstruction Conjecture**
OPG-804 | O | 第 1784 页

**Effective Mordell-Vojta Conjecture**
OWR-1104-003 | O | 第 56 页

**Ehrhart's Volume Conjecture**
GEO-028 | P | 第 840 页

**Eilenberg-Ganea Conjecture**
ALG-044 | P | 第 233 页

**Eisenbud's Fiber-Regularity Conjecture**
OWR-1971-004 | O | 第 662 页

**Eisenbud-Mazur Symbolic-Square Conjecture**
OWR-824-009 | O | 第 233 页

**Elliott-Halberstam Conjecture**
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**Erdös-Szekeres conjecture**
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**Erdős Conjecture on Arithmetic Progressions**
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**Erdős-Gyárfás Conjecture**
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**Erdős-Turán Conjecture on Additive Bases**
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**Erdős-Faber-Lovász conjecture**
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**Erdős-Gallai Cycle-Decomposition Conjecture**
OWR-10252930-004 | P | 第 1815 页

**Erdős-Gyárfás Power-of-Two Cycle Conjecture**
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**Erdős-Hajnal Clique-or-Independent-Set Conjecture**
OWR-12697684-011 | O | 第 1815 页

**Erdős-Hajnal Conjecture**
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**Erdős-Hajnal Conjecture for Tournaments**
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**Erdős-Mollin-Walsh conjecture**
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**Failure of Faber’s Conjecture in Genus Five**
OWR-14605-003 | O | 第 671 页

**Falconer Distance Conjecture**
OWR-2090-014 | O | 第 858 页

**Falconer Distance Conjecture in the Plane**
OWR-15952-006 | P | 第 858 页

**Falconer's Conjecture**
GEO-011 | O | 第 858 页

**Fargues’ Cohomological Conjectures for the Curve**
OWR-12697688-016 | P | 第 672 页

**Farrell-Jones Conjecture**
ALG-045 | P | 第 237 页

**Farrell-Jones Conjecture for Torsion-Free Groups**
OWR-4333-004 | P | 第 351 页

**Feit-Thompson conjecture**
AMR-093-0123 | O | 第 121 页

**Fermat-Catalan Conjecture**
NT-081 | P | 第 121 页

**Firoozbakht's Conjecture**
NT-027 | O | 第 123 页

**First Hardy-Littlewood zeta-function conjecture**
AMR-093-0064 | S | 第 123 页

**First Root Conjecture for Real Exponential Polynomials**
OWR-15181-016 | O | 第 2018 页

**Five-Cycle Near-Triangulation Decomposition Conjecture**
OWR-16763-005 | O | 第 1819 页

**Foliations and Diffeomorphism Groups - Question 1 (Virtual Euler class one conjecture)**
AMR-115-0001 | O | 第 1059 页

**Foliations and Diffeomorphism Groups - Question 63 (``Haefliger-Thurston conjecture")**
AMR-115-0063 | O | 第 1067 页

**Fontaine-Mazur Conjecture for Unramified Galois Representations**
OWR-14298588-016 | O | 第 124 页

**Fontaine-Mazur geometric Galois-representation conjecture**
AMR-093-0048 | P | 第 124 页

**Forcing Conjecture for Bipartite Cyclic Graphs**
OWR-4791-019 | O | 第 1820 页

**Fortune's conjecture**
AMR-093-0124 | O | 第 124 页

**Four core-model questions and a modular audit of branch-hypothesis counterexamples**
AIM-LOGIC-0152 | P | 第 2019 页

**Four Exponentials Conjecture**
NT-073 | O | 第 124 页

**Fourier Restriction Conjecture in Higher Dimensions**
OWR-11101912-012 | O | 第 1316 页

**Frankl's Union-Closed Sets Conjecture**
COMB-005 | P | 第 1694 页

**Free-Group Word-Measure Conjecture**
OWR-14299086-010 | P | 第 354 页

**Frey-Jarden Infinite-Rank Conjecture**
OWR-1188-014 | O | 第 125 页

**Friedman-Simpson Interpretability Conjecture**
AMR-074-0100 | O | 第 2019 页

**Fröberg conjecture on the Hilbert functions of a set of forms**
AMR-071-0006 | O | 第 677 页

**Fuglede's conjecture in dimensions one and two**
AMR-023-0002 | P | 第 1317 页

**Fujita's Conjecture for Effective Global Generation**
OWR-17296-005 | O | 第 677 页

**Fujita's Global Generation and Very-Ampleness Conjecture**
OWR-12870-009 | O | 第 677 页

**Gallai’s Path-Decomposition Conjecture**
OWR-9790352-004 | O | 第 1820 页

**Gasca-Maeztu Maximal-Line Conjecture**
OWR-9790358-001 | P | 第 242 页

**Geck's Unitriangularity Conjecture**
OWR-16776-009 | P | 第 355 页

**General Kakeya Dimension Conjecture**
OWR-13750336-006 | P | 第 1317 页

**Generalised Empty Hexagon Conjecture**
OPG-59923 | P | 第 866 页

**Generalized Fried Conjecture for Anosov Flows**
OWR-14299091-002 | P | 第 1549 页

**Generalized Hodge Conjecture**
OWR-1971-001 | O | 第 679 页

**Generalized Ramanujan conjecture for automorphic representations**
AMR-093-0078 | P | 第 127 页

**Generalized Riemann Hypothesis for Automorphic L-Functions**
OWR-15583-003 | O | 第 127 页

**Generalized Riemann hypothesis for Selberg class**
AMR-093-0071 | O | 第 127 页

**Generalized spectral conjecture**
AMR-045-0006 | S | 第 1549 页

**Generating Hypothesis and Nilpotence for Finite Spectra**
OWR-1190-003 | O | 第 1072 页

**Gilbreath Conjecture for Iterated Prime Differences**
OWR-17293-011 | O | 第 128 页

**Gilbreath's Conjecture**
NT-009 | O | 第 128 页

**Gillies' conjecture**
AMR-093-0126 | O | 第 129 页

**Giuga's Conjecture on Primality**
OPG-37411 | O | 第 129 页

**Giuga's Conjecture on Prime Characterization**
GUY-A17 | P | 第 129 页

**Global Gan-Gross-Prasad conjecture**
AMR-093-0178 | P | 第 244 页

**GNRS Conjecture**
GRAPH-017 | P | 第 1821 页

**Goddyn's conjecture on thin spanning trees**
AMR-029-0039 | P | 第 1696 页

**Goldbach's Conjecture**
LAN-001 | P | 第 129 页

**Goldberg Conjecture in Four Dimensions**
OWR-15579-001 | O | 第 875 页

**Goldberg conjecture: corrected statement and counterexample sieve**
AIM-PDES-0036 | P | 第 1478 页

**Goldberg's conjecture**
OPG-2242 | P | 第 1821 页

**Gonek’s Conjecture on Shifted Negative Zeta Moments**
OWR-11695860-006 | P | 第 129 页

**Good finitary conjecture**
AMR-045-0009 | O | 第 1550 页

**Goormaghtigh conjecture**
AMR-093-0108 | O | 第 129 页

**Graded Classification Conjecture for Leavitt Path Algebras**
OWR-14298169-006 | P | 第 244 页

**Graham's conjecture on tree reconstruction**
OPG-164 | P | 第 1821 页

**Graham's Pebbling Conjecture**
GRAPH-003 | O | 第 1821 页

**Graham's reciprocal-mass square conjecture: a dyadic extremal reduction**
AIM-COMBINATORICS-0216 | P | 第 1697 页

**Graham’s Rearrangement Conjecture over Prime Fields**
OWR-14299518-006 | P | 第 356 页

**Grand Riemann Hypothesis**
NT-069 | P | 第 129 页

**Green's conjecture**
AMR-015-0011 | P | 第 245 页

**Green's Syzygy Conjecture for Arbitrary Curves**
OWR-13101-001 | O | 第 683 页

**Greenberg's Iwasawa-invariants conjecture**
AMR-093-0050 | P | 第 130 页

**Greenberg's p-rationality conjecture**
AMR-093-0181 | P | 第 130 页

**Greenberg's pseudo-null conjecture**
AMR-093-0179 | O | 第 130 页

**Greenberg's Pseudo-Nullity Conjecture**
OWR-1274-005 | P | 第 130 页

**Green-Griffiths-Lang Conjecture for General-Type Varieties**
OWR-7155446-002 | P | 第 683 页

**Green-Lazarsfeld Gonality Conjecture**
OWR-4527-004 | S | 第 683 页

**Green’s Canonical Syzygy Conjecture for Arbitrary Curves**
OWR-16408-002 | O | 第 683 页

**Grimm's Conjecture**
NT-033 | P | 第 130 页

**Gromov Macroscopic Dimension Conjecture**
OWR-13106-011 | O | 第 875 页

**Gromov's Scalar Curvature Band-Width Conjecture**
OWR-11101916-009 | P | 第 875 页

**Gromov's Simplicial-Volume Conjecture for $L^2$-Betti Numbers**
OWR-1268-004 | O | 第 1074 页

**Grothendieck Section Conjecture for Hyperbolic Curves**
OWR-12010-003 | O | 第 683 页

**Grothendieck's Section Conjecture over Number Fields**
OWR-3388-009 | O | 第 130 页

**Grothendieck-Katz $p$-Curvature Conjecture**
OWR-1188-010 | O | 第 130 页

**Grothendieck-Katz p-curvature conjecture**
AMR-015-0012 | O | 第 245 页

**Grothendieck-Katz p-Curvature Conjecture**
OWR-13952-005 | O | 第 130 页

**Grothendieck-Serre Conjecture for Regular Local Rings**
OWR-12494-006 | P | 第 245 页

**Grothendieck-Serre Conjecture in Mixed Characteristic**
OWR-14211-011 | P | 第 683 页

**Grothendieck’s Anabelian Conjectures over Finitely Generated Fields**
OWR-16158-006 | P | 第 683 页

**Grothendieck’s Section Conjecture**
OWR-14298795-004 | O | 第 684 页

**Grunbaum's Conjecture**
OPG-177 | S | 第 1822 页

**Guy's Crossing Number Conjecture**
GRAPH-021 | P | 第 1822 页

**Gyárfás Tree-Packing Conjecture**
OWR-10252930-017 | O | 第 1822 页

**Gyárfás-Sumner Conjecture**
GRAPH-011 | P | 第 1822 页

**Gyárfás-Sumner Conjecture for Forest-Free Graphs**
OWR-15206-005 | O | 第 1822 页

**Hadamard Conjecture**
ALG-004 | O | 第 245 页

**Hadamard Matrix Conjecture**
ALG-047 | O | 第 245 页

**Hadwiger Conjecture**
GT-001 | P | 第 1823 页

**Hadwiger’s Coloring Conjecture and Its Odd-Minor Variant**
OWR-16763-010 | O | 第 1823 页

**Hadwiger’s Graph-Coloring Conjecture**
OWR-9790352-018 | P | 第 1823 页

**Hall's Conjecture**
NT-034 | P | 第 130 页

**Halperin conjecture**
AMR-112-0002 | P | 第 1074 页

**Harborth's Conjecture**
GRAPH-018 | P | 第 1824 页

**Hardy-Littlewood Conjecture A (Prime k-tuples)**
HL-A | O | 第 131 页

**Hardy-Littlewood Conjecture B (Second Conjecture)**
HL-B | O | 第 131 页

**Hardy-Littlewood Conjecture F (Primes in Quadratic Polynomials)**
HL-F | P | 第 131 页

**Hardy-Littlewood Prime Tuple Conjecture**
OWR-8415337-001 | O | 第 131 页

**Hartmanis-Stearns conjecture**
AMR-093-0101 | O | 第 2086 页

**Hazrat Conjecture for Leavitt Path Algebras**
OWR-14298169-003 | P | 第 246 页

**Hedetniemi's Conjecture**
OPG-806 | S | 第 1824 页

**Henon Maps: Open Problems - Conjecture 24**
AMR-152-0024 | O | 第 1554 页

**Henon Maps: Open Problems - Conjecture 47**
AMR-152-0047 | O | 第 1554 页

**Henon Maps: Open Problems - Conjecture 48 (see [I])**
AMR-152-0048 | O | 第 1554 页

**Henon Maps: Open Problems - Conjecture 49**
AMR-152-0049 | O | 第 1554 页

**Henon Maps: Open Problems - Conjecture 50**
AMR-152-0050 | O | 第 1554 页

**Herzog-Schönheim Conjecture**
ALG-011 | P | 第 247 页

**Higher-Dimensional Hamilton-Lott Conjecture**
OWR-14298170-009 | P | 第 878 页

**Higher-Dimensional Smale Conjecture for 2-Convex Spheres**
OWR-14610-005 | P | 第 1075 页

**Hilbert-Pólya Conjecture**
NT-068 | O | 第 134 页

**Hilbert-Smith Conjecture**
TOP-009 | P | 第 1076 页

**Hill Conjecture for Complete Graph Crossing Numbers**
OWR-11136-009 | O | 第 1825 页

**Hiraga-Ichino-Ikeda Formal Degree Conjecture**
OWR-8415342-003 | P | 第 134 页

**Hiraga-Ichino-Ikeda-Langlands Plancherel Density Conjecture**
OWR-8415342-002 | P | 第 134 页

**Hoàng-Reed Conjecture**
OPG-47282 | P | 第 1825 页

**Hodge Conjecture**
MPP-006 | O | 第 687 页

**Holomorphic Symplectic Geometry - Conjecture 10**
AMR-118-0010 | O | 第 688 页

**Holomorphic Symplectic Geometry - Conjecture 11**
AMR-118-0011 | O | 第 688 页

**Holomorphic Symplectic Geometry - Conjecture 12**
AMR-118-0012 | O | 第 688 页

**Holomorphic Symplectic Geometry - Conjecture 13**
AMR-118-0013 | O | 第 689 页

**Holomorphic Symplectic Geometry - Conjecture 4**
AMR-118-0004 | O | 第 689 页

**Holomorphic Symplectic Geometry - Conjecture 6**
AMR-118-0006 | O | 第 689 页

**Homeomorphism Conjecture for Moduli of Fundamental Groups**
OWR-14297740-017 | P | 第 134 页

**Homological Hilbert-Scheme Conjecture for Algebraic Links**
OWR-12018-006 | P | 第 690 页

**Homological Pisot Conjecture**
AMR-065-0006 | P | 第 880 页

**Hoop Conjecture for Black-Hole Formation**
OWR-1184-001 | O | 第 2131 页

**Hopf Conjecture**
OWR-11568-006 | O | 第 880 页

**Hopkins Conjecture for Morava-Stabilizer Subgroups**
OWR-1728-004 | O | 第 1079 页

**Hypotheses for analytic invariant curves**
AMR-036-0009 | P | 第 1561 页

**Ibragimov's central limit conjecture for $\phi$-mixing sequences**
AMR-100-0012 | O | 第 1918 页

**Igusa Monodromy Conjecture**
OWR-4086-009 | P | 第 691 页

**Iitaka Subadditivity Conjecture for Kodaira Dimension**
OWR-11101913-002 | P | 第 691 页

**Imbalance Conjecture**
GRAPH-051 | P | 第 1826 页

**Implications Among Conjectures on Principally Polarized Abelian Varieties**
OWR-16408-006 | O | 第 691 页

**Implicit Graph Conjecture**
GRAPH-052 | S | 第 1826 页

**Inductive Proofs of Alperin Conjectures**
OWR-11583-012 | P | 第 358 页

**Inscribed Square Problem (Toeplitz Conjecture)**
GEO-005 | O | 第 883 页

**Irrational Seshadri Constants from the Nagata Conjecture**
OWR-16628-010 | O | 第 693 页

**Is the 1/3-2/3 Conjecture for Pressing Sequences true**
AMR-030-0047 | O | 第 1701 页

**Iwasawa Main Conjecture for Dihedral Ring Class Towers**
OWR-3480-006 | P | 第 139 页

**Iwasawa mu-invariant conjecture for number fields**
AMR-093-0180 | P | 第 140 页

**Iwasawa Reciprocity in Sharifi’s Conjecture**
OWR-1586-002 | P | 第 140 页

**Jacob Palis Conjecture(Finitude of Attractors)(Dynamical Systems)**
OPG-48770 | P | 第 1082 页

**Jacobson's Conjecture**
ALG-006 | P | 第 255 页

**Jaeger's modular orientation conjecture**
OPG-130 | S | 第 1827 页

**Jaeger's Petersen Coloring Conjecture**
GRAPH-012 | P | 第 1827 页

**Jones' conjecture**
OPG-638 | P | 第 1827 页

**Jorgensen's Conjecture**
OPG-154 | P | 第 1828 页

**Kadison-Kastler Conjecture for Separable $C^*$-Algebras**
OWR-12175-005 | P | 第 256 页

**Kahn’s Matroidal Rota Basis Conjecture**
OWR-16763-025 | P | 第 1702 页

**Kakeya Conjecture in Dimensions Four and Higher**
OWR-14299906-010 | P | 第 1327 页

**Kalai's $3^d$ Conjecture**
GEO-025 | O | 第 886 页

**Kalai-Sarkaria Algebraic-Shifting Conjecture**
OWR-1455-009 | O | 第 1702 页

**Kannan-Lovász-Simonovits Isoperimetric Conjecture**
OWR-8415364-006 | O | 第 886 页

**Kaplansky's Conjectures**
ALG-002 | O | 第 256 页

**Kaplansky's Conjectures for Group Rings**
OWR-9790362-007 | P | 第 360 页

**Kaplan-Yorke dimension conjecture**
AMR-048-0009 | P | 第 1563 页

**Karchmer-Raz-Wigderson Composition Conjecture**
OWR-16634-003 | P | 第 2089 页

**Karchmer-Raz-Wigderson Formula Lower-Bound Conjecture**
OWR-13949-002 | P | 第 2089 页

**Kato’s Main Conjecture for Modular Galois Representations**
OWR-5159-009 | P | 第 140 页

**Kawaguchi-Silverman Arithmetic Degree Conjecture**
OWR-14603-004 | P | 第 140 页

**Kawamata Derived-Equivalence Conjecture**
OWR-1114-001 | O | 第 695 页

**Keating-Snaith moment conjecture for the Riemann zeta function**
AMR-093-0065 | P | 第 140 页

**Keating-Snaith Moment Conjecture for the Zeta Function**
OWR-15956-005 | O | 第 140 页

**Kim's Nonabelian Selmer-Dimension Conjecture**
OWR-1542-008 | O | 第 141 页

**Kim’s Nonabelian Chabauty Conjecture**
OWR-14298795-014 | O | 第 695 页

**Kirby’s Whitehead-Double Sliceness Conjecture**
OWR-11790-007 | O | 第 1124 页

**Kisin's framed Breuil-Mezard multiplicity conjecture**
AIM-REPRESENTATION_THEORY-0104 | S | 第 257 页

**Klingenberg's conjecture after the 2025 hyperbolicity advance**
AIM-GEOMETRY-0218 | P | 第 887 页

**Kneser-Poulsen conjecture**
OPG-2089 | P | 第 887 页

**Knot Volume Conjecture**
OWR-13108-014 | P | 第 1124 页

**Komlos Discrepancy Conjecture and Unsatisfiable Formulas**
OWR-14299088-011 | O | 第 1702 页

**Kontsevich's Period Conjecture**
OWR-16171-014 | O | 第 257 页

**Kontsevich-Soibelman Conjecture for Calabi-Yau Degenerations**
OWR-11101913-008 | P | 第 696 页

**KPZ Universality Conjecture**
OPG-60052 | P | 第 1922 页

**Kriesell's Conjecture**
OPG-56233 | P | 第 1828 页

**Kummer-Vandiver conjecture**
AMR-093-0055 | O | 第 141 页

**Kummer-Vandiver Conjecture**
OWR-14211-001 | O | 第 141 页

**Kung-Traub conjecture**
AMR-023-0005 | O | 第 1327 页

**Kühnel Conjecture for Embedded Simplicial Skeletons**
OWR-13110-016 | P | 第 1125 页

**L-Space Conjecture for Rational Homology Spheres**
OWR-14298580-004 | O | 第 1125 页

**Lander-Parkin-Selfridge Conjecture**
NT-058 | O | 第 141 页

**Landis Unique-Continuation Conjecture**
OWR-14299093-002 | P | 第 1481 页

**Lang and Trotter's conjecture**
AMR-093-0056 | P | 第 141 页

**Lang-Vojta and Shafarevich Finiteness Conjectures**
OWR-14298795-019 | O | 第 697 页

**Lattice Gases and Antiferromagnetic Potts Models - Conjecture 1**
AMR-132-0001 | O | 第 2135 页

**Lattice Gases and Antiferromagnetic Potts Models - Conjecture 3 (Hamidoune \protect[Hamidoune_90], Stanley \protect[Stanley_98])**
AMR-132-0003 | O | 第 2135 页

**Lattice Packing Conjecture for Origin-Symmetric Convex Bodies**
OWR-14299583-004 | O | 第 889 页

**Lech Multiplicity Conjecture for Flat Local Extensions**
OWR-15178-003 | O | 第 258 页

**Legendre's Conjecture**
LAN-003 | O | 第 142 页

**Lehmer Conjecture on Mahler Measure**
OWR-4416-009 | O | 第 142 页

**Lemoine's Conjecture**
NT-059 | P | 第 142 页

**Leopoldt's Conjecture**
NT-066 | P | 第 142 页

**Lichnerowicz Conjecture for Conformal Isometry Groups**
OWR-13106-007 | P | 第 890 页

**Lieb-Thirring Refinements and the Ovals Conjecture**
OWR-3389-017 | P | 第 1328 页

**Lindelöf Hypothesis**
NT-067 | P | 第 143 页

**Linear Arboricity Conjecture**
GRAPH-032 | P | 第 1829 页

**List Coloring Conjecture**
GRAPH-013 | P | 第 1830 页

**List Total Colouring Conjecture**
OPG-56449 | P | 第 1830 页

**Little shift equivalence conjecture**
AMR-045-0001 | O | 第 1565 页

**Littlewood Conjecture**
OWR-11784-005 | O | 第 143 页

**Littlewood's Simultaneous Approximation Conjecture**
OWR-15169-001 | O | 第 143 页

**Li-Szegedy Smooth-Retract Conjecture**
OWR-12479-006 | O | 第 1831 页

**Local $p$-Adic Section Conjecture**
OWR-3388-002 | O | 第 143 页

**Local edge budgets for Thomassé's nonneighbor conjecture**
AIM-COMBINATORICS-0180 | P | 第 1705 页

**Local Gan-Gross-Prasad conjecture**
AMR-093-0049 | S | 第 266 页

**Loewner's conjecture**
AMR-046-0025 | P | 第 1571 页

**Looijenga's Dolbeault Dimension Conjecture**
OWR-12875-007 | P | 第 699 页

**Lovász Conjecture**
GRAPH-005 | O | 第 1832 页

**Lovász Path Removal Conjecture**
OPG-46629 | P | 第 1832 页

**Lovász Removable-Path Conjecture for Three Prescribed Vertices**
OWR-1532-014 | O | 第 1832 页

**LP obstructions and cyclic blow-ups for DeVos's distribution conjecture**
AIM-COMBINATORICS-0198 | P | 第 1707 页

**MacEachen Conjecture**
OPG-671 | P | 第 146 页

**Maeda Conjecture for Level-One Cusp Forms**
OWR-12865-004 | O | 第 146 页

**Mahler Volume Product Conjecture**
OWR-4136-014 | P | 第 895 页

**Mahler's Conjecture**
GEO-006 | P | 第 895 页

**Mahler's Symmetric Convex-Body Conjecture**
OWR-1394-007 | O | 第 895 页

**Mahler’s Volume Conjecture**
OWR-16637-003 | O | 第 895 页

**Main Conjecture for Modular Towers**
OWR-1274-025 | P | 第 147 页

**Major problems 2 - The generating hypothesis, which asserts that the stable homotopy functor is faithful on the category…**
AMR-110-0002 | O | 第 1129 页

**Major problems 5 - The chromatic splitting conjecture, which is considerably more complicated to state.**
AMR-110-0005 | P | 第 1130 页

**Makienko conjecture**
AMR-036-0005 | P | 第 1572 页

**Malle Conjecture for Number Field Counts**
OWR-16409-008 | O | 第 147 页

**Manin conjecture**
AMR-093-0083 | P | 第 147 页

**Margulis measure-classification conjecture**
AMR-048-0010 | P | 第 1572 页

**Martin's Conjecture on Natural Functions of Turing Degrees**
AMR-074-0004 | P | 第 2023 页

**Mathematical Conjecture Generation with Machine Intelligence**
OWR-13750338-001 | O | 第 2091 页

**Matrix Spencer Signing Conjecture**
OWR-13674-017 | O | 第 1331 页

**Matveev-Jaco-Rubinstein Lens-Space Complexity Conjecture**
OWR-11786-020 | P | 第 1130 页

**Maulik-Nekrasov-Okounkov-Pandharipande conjecture on an equivalence between Gromov-Witten theory and Donaldson-Thomas theory**
AMR-071-0012 | P | 第 701 页

**Mazur's finite-components conjecture for rational points**
AMR-112-0003 | P | 第 1131 页

**McKay Conjecture**
ALG-014 | S | 第 268 页

**Melzak's Shortest-Edge Polyhedron Conjecture**
AMR-058-0024 | P | 第 898 页

**Meyniel's Conjecture on Cop Number**
GRAPH-006 | P | 第 1833 页

**Minimal-Filling Conjecture for Simple Riemannian Manifolds**
OWR-1383-001 | O | 第 899 页

**Missing unit-disc hypothesis and a universal unbounded counterexample**
AIM-ANALYTIC_NUMBER_THEORY-0041 | P | 第 149 页

**Mixed Littlewood Conjecture for Pseudo-Absolute Values**
OWR-11784-010 | O | 第 149 页

**MLC Conjecture**
DYN-003 | P | 第 1483 页

**Mod-$\ell$ Bogomolov-Pop Reconstruction Conjecture**
OWR-13351-017 | P | 第 150 页

**Modified Cartan conjecture**
AMR-036-0042 | P | 第 1333 页

**Mond's Conjecture for Finite Map Germs**
OWR-14754-010 | O | 第 704 页

**Montgomery's Pair Correlation Conjecture**
NT-070 | P | 第 157 页

**Morrison’s Cone Conjecture for Calabi-Yau Pairs**
OWR-4427-006 | P | 第 704 页

**Multicolored Sunflower Conjecture over Ternary Vector Spaces**
OWR-12177-010 | S | 第 2092 页

**Multicolour Erdős--Hajnal Conjecture**
OPG-60042 | P | 第 1835 页

**Multipoint Seshadri Constants and Nagata-Type Conjectures**
OWR-4527-001 | O | 第 707 页

**Mumford-Tate Conjectures for Gushel-Mukai Varieties**
OWR-11101913-009 | P | 第 707 页

**Nagata Conjecture for Multipoint Seshadri Constants**
OWR-17296-002 | O | 第 707 页

**Nakai conjecture**
AMR-071-0015 | O | 第 707 页

**Nearby Lagrangian conjecture**
AMR-112-0005 | P | 第 1138 页

**Negami's Conjecture**
GRAPH-019 | P | 第 1836 页

**New Mersenne conjecture**
AMR-093-0133 | O | 第 158 页

**Newman's Conjecture**
NT-038 | P | 第 158 页

**NIP Conjecture for k-Dependent Fields**
OWR-12697685-010 | O | 第 2025 页

**Nivat's conjecture**
AMR-045-0054 | P | 第 1578 页

**Non-Integrality Conjecture for Rational Hypersurfaces**
OWR-3388-004 | O | 第 159 页

**Noncompact Casacuberta-Dicks Conjecture**
OWR-14298007-009 | O | 第 1139 页

**Nondegeneracy in the Spectral Edges Conjecture**
OWR-10252925-009 | P | 第 1335 页

**Novikov Conjecture**
TOP-006 | P | 第 1140 页

**Oda and Sturmfels Conjectures for Smooth Toric Varieties**
OWR-1971-007 | O | 第 711 页

**Oda’s Strong Factorization Conjecture for Toric Varieties**
OWR-14299288-011 | O | 第 712 页

**Odd-prime-power periodicity conjecture**
AMR-090-0001 | P | 第 1841 页

**Off-diagonal van der Waerden numbers: refuted quadratic conjecture and a difference-sensitive transversal criterion**
AIM-COMBINATORICS-0023 | P | 第 1712 页

**Olson's Conjecture**
OPG-155 | P | 第 161 页

**Oort’s Hecke-Orbit Conjecture**
OWR-12697688-005 | P | 第 713 页

**Open Coloring Axiom and Continuum Hypothesis**
ST-010 | O | 第 1979 页

**Open Problems and Questions about Geodesics - Conjecture 1 (Long, Bangert, Problem 15 from [Alvarez2006])**
AMR-149-0001 | O | 第 911 页

**Open Problems and Questions about Geodesics - Conjecture 11 (Hingston)**
AMR-149-0011 | O | 第 911 页

**Open Problems and Questions about Geodesics - Conjecture 12 (Klingenberg)**
AMR-149-0012 | O | 第 911 页

**Open Problems and Questions about Geodesics - Conjecture 15 ([Bur-Kat1985])**
AMR-149-0015 | O | 第 911 页

**Open Problems and Questions about Geodesics - Conjecture 18**
AMR-149-0018 | O | 第 911 页

**Open Problems and Questions about Geodesics - Conjecture 19 ([Katok1982])**
AMR-149-0019 | O | 第 911 页

**Open Problems and Questions about Geodesics - Conjecture 2 (Long)**
AMR-149-0002 | O | 第 911 页

**Open Problems and Questions about Geodesics - Conjecture 22**
AMR-149-0022 | O | 第 912 页

**Open Problems and Questions about Geodesics - Conjecture 3 (Long)**
AMR-149-0003 | O | 第 912 页

**Open Problems and Questions about Geodesics - Conjecture 30 (Spatzier)**
AMR-149-0030 | O | 第 912 页

**Open Problems and Questions about Geodesics - Conjecture 31 (Burago-Kleiner)**
AMR-149-0031 | O | 第 912 页

**Open Problems and Questions about Geodesics - Conjecture 51**
AMR-149-0050 | O | 第 912 页

**Open Problems and Questions about Geodesics - Conjecture 60 ([Bo-Ko-Fo1995])**
AMR-149-0059 | O | 第 912 页

**Open Problems and Questions about Geodesics - Conjecture 61**
AMR-149-0060 | O | 第 912 页

**Open Problems in Arithmetic Dynamics - Conjecture 11**
AMR-126-0011 | O | 第 1579 页

**Open Problems in Arithmetic Dynamics - Conjecture 12**
AMR-126-0012 | O | 第 1579 页

**Open Problems in Arithmetic Dynamics - Conjecture 13**
AMR-126-0013 | O | 第 1580 页

**Open Problems in Arithmetic Dynamics - Conjecture 17 (Dynamical Andr{\'e}--Oort Conjecture)**
AMR-126-0017 | O | 第 1580 页

**Open Problems in Arithmetic Dynamics - Conjecture 18 (Dynamical Andr{\'e}--Oort Conjecture for 1-Dimensional Families)**
AMR-126-0018 | O | 第 1580 页

**Open Problems in Arithmetic Dynamics - Conjecture 2**
AMR-126-0002 | O | 第 1580 页

**Open Problems in Arithmetic Dynamics - Conjecture 23 (Orbit Density Conjecture for $\mathbb{A}^N$)**
AMR-126-0023 | O | 第 1580 页

**Open Problems in Arithmetic Dynamics - Conjecture 24 (Orbit Density Conjecture)**
AMR-126-0024 | O | 第 1580 页

**Open Problems in Arithmetic Dynamics - Conjecture 26 (Integrality Conjecture)**
AMR-126-0026 | O | 第 1581 页

**Open Problems in Arithmetic Dynamics - Conjecture 3**
AMR-126-0003 | O | 第 1581 页

**Open Problems in Arithmetic Dynamics - Conjecture 31 (Dynamical Lang Height Conjecture)**
AMR-126-0031 | O | 第 1581 页

**Open Problems in Arithmetic Dynamics - Conjecture 34**
AMR-126-0034 | O | 第 1581 页

**Open Problems in Arithmetic Dynamics - Conjecture 36**
AMR-126-0036 | O | 第 1581 页

**Open Problems in Arithmetic Dynamics - Conjecture 37**
AMR-126-0037 | O | 第 1581 页

**Open Problems in Arithmetic Dynamics - Conjecture 39**
AMR-126-0039 | O | 第 1581 页

**Open Problems in Arithmetic Dynamics - Conjecture 4**
AMR-126-0004 | O | 第 1582 页

**Open Problems in Arithmetic Dynamics - Conjecture 44**
AMR-126-0044 | O | 第 1582 页

**Open Problems in Arithmetic Dynamics - Conjecture 45**
AMR-126-0045 | O | 第 1582 页

**Open Problems in Arithmetic Dynamics - Conjecture 46**
AMR-126-0046 | O | 第 1582 页

**Open Problems in Arithmetic Dynamics - Conjecture 49**
AMR-126-0049 | O | 第 1582 页

**Open Problems in Arithmetic Dynamics - Conjecture 5**
AMR-126-0005 | O | 第 1582 页

**Open Problems in Arithmetic Dynamics - Conjecture 50**
AMR-126-0050 | O | 第 1582 页

**Open Problems in Arithmetic Dynamics - Conjecture 51**
AMR-126-0051 | O | 第 1582 页

**Open Problems in Arithmetic Dynamics - Conjecture 52 (Dynamical Lang--Siegel Conjecture for $\mathbb{P}^N$)**
AMR-126-0052 | O | 第 1583 页

**Open Problems in Arithmetic Dynamics - Conjecture 54 (Dynamical Brauer--Manin Conjecture)**
AMR-126-0054 | O | 第 1583 页

**Open Problems in Arithmetic Dynamics - Conjecture 6**
AMR-126-0006 | O | 第 1583 页

**Open Problems in Arithmetic Dynamics - Conjecture 8 (Odoni, 1985, [MR805714])**
AMR-126-0008 | O | 第 1583 页

**Open Problems in Arithmetic Dynamics - Conjecture 9**
AMR-126-0009 | O | 第 1583 页

**Open Problems in Integrable Systems - Around the Birkhoff conjecture**
AMR-020-0401 | P | 第 1588 页

**Open Problems in Integrable Systems - Around the Birkhoff conjecture**
AMR-020-0402 | O | 第 1588 页

**Open Problems in Integrable Systems - Around the Birkhoff conjecture**
AMR-020-0403 | P | 第 1588 页

**Open Problems in Integrable Systems - Around the Birkhoff conjecture**
AMR-020-0404 | O | 第 1589 页

**Open Problems in the Theory of Ample Fields - Conjecture 2 (Shafarevich)**
AMR-139-0002 | O | 第 272 页

**Open Problems in the Theory of Ample Fields - Conjecture 7 (Podewski)**
AMR-139-0007 | O | 第 272 页

**Open Problems in the Theory of Ample Fields - Conjecture 8 (D\`ebes-Deschamps)**
AMR-139-0008 | O | 第 272 页

**Open Problems in Wavelet Theory - Conjecture 2**
AMR-155-0002 | O | 第 1342 页

**Open Problems on Central Simple Algebras - Conjecture 24 (Tate's Conjecture T)**
AMR-136-0022 | O | 第 273 页

**Open Problems on Central Simple Algebras - Conjecture 27 (BSD Conjecture)**
AMR-136-0023 | O | 第 273 页

**Open Problems on Central Simple Algebras - Conjecture 32 (Amitsur's Conjecture)**
AMR-136-0028 | O | 第 274 页

**Open Problems on Central Simple Algebras - Conjecture 39**
AMR-136-0034 | O | 第 274 页

**Open Problems on Central Simple Algebras - Conjecture 41**
AMR-136-0036 | O | 第 274 页

**Optimal Distributed Hypothesis Testing beyond Conditional Independence**
OWR-9790365-004 | P | 第 1932 页

**Orientation conjecture of Nash-Williams**
AMR-029-0059 | P | 第 1714 页

**Orthogonal Symmetry Conjecture for Rigid Higher Categories**
OWR-14299913-001 | O | 第 2146 页

**Overfull Conjecture**
GRAPH-014 | P | 第 1842 页

**Ozsváth-Szabó Conjecture for Integer Homology Spheres**
OWR-12697686-005 | O | 第 1145 页

**Paramodular Conjecture for Abelian Varieties**
OWR-1734-006 | P | 第 163 页

**Parity Conjectures for Elliptic Curves over Number Fields**
OWR-1590-009 | P | 第 163 页

**Parshin's conjecture**
AMR-071-0016 | P | 第 715 页

**Parshin's Conjecture**
OWR-5155-005 | P | 第 715 页

**Perfect 1-Factorization Conjecture**
GRAPH-009 | P | 第 1843 页

**Period Conjecture for Nori Motives**
OWR-16171-015 | P | 第 715 页

**Period Conjectures for Fano Polygons and Del Pezzo Surfaces**
OWR-13500-014 | P | 第 716 页

**Petersen coloring conjecture**
OPG-143 | P | 第 1843 页

**Petty’s Projection-Body Conjecture**
OWR-13951-015 | O | 第 923 页

**Pierce-Birkhoff Supremum-Infimum Representation Conjecture**
OWR-1465-007 | O | 第 280 页

**Pillai's Conjecture**
NT-023 | P | 第 164 页

**Pisot substitution conjecture**
AMR-045-0053 | P | 第 1601 页

**Pisot Substitution Conjecture for Pure Point Spectrum**
OWR-3396-002 | O | 第 1601 页

**Pointwise Equidistribution Under Toral Endomorphisms with Fewer Hypotheses**
OWR-17298-003 | O | 第 1601 页

**Polignac Conjecture for Consecutive Prime Gaps**
OWR-12725-006 | O | 第 165 页

**Polignac's Conjecture**
NT-022 | P | 第 166 页

**Pollock's octahedral-number conjecture**
AMR-093-0177 | O | 第 166 页

**Pollock's tetrahedral-number conjecture**
AMR-093-0041 | O | 第 166 页

**Polyhedral Frankel Conjecture**
OWR-12581-005 | P | 第 925 页

**Polynomial Freiman-Ruzsa Conjecture over Binary Vector Spaces**
OWR-2043-007 | S | 第 1718 页

**Polynomial Freiman-Ruzsa Conjecture over Binary Vector Spaces**
OWR-12177-008 | S | 第 2097 页

**Polynomial Hirsch Conjecture for Superstrongly Connected Complexes**
OWR-11786-017 | P | 第 1844 页

**Positively Curved Projective Varieties - Conjecture 12**
AMR-140-0012 | O | 第 718 页

**Positively Curved Projective Varieties - Conjecture 2**
AMR-140-0002 | O | 第 718 页

**Power-of-two periodicity conjecture**
AMR-090-0002 | O | 第 1845 页

**Problem 15 - [Ehrenpreis Conjecture] Given two closed Riemann surfaces, there are finite un-branched covers with homeomorphic tot…**
AMR-109-0144 | S | 第 1157 页

**Problem 9 - [Bounded Distortion Conjecture] Given a hyperbolic structure on F, associate its combinatorial invariant, namely, an…**
AMR-109-0139 | O | 第 1204 页

**Problems Around Polynomials - Conjecture 1**
AMR-021-0001 | P | 第 283 页

**Problems Around Polynomials - Conjecture 10**
AMR-021-0015 | O | 第 283 页

**Problems Around Polynomials - Conjecture 11**
AMR-021-0018 | P | 第 283 页

**Problems Around Polynomials - Conjecture 12**
AMR-021-0019 | P | 第 283 页

**Problems Around Polynomials - Conjecture 13**
AMR-021-0020 | O | 第 283 页

**Problems Around Polynomials - Conjecture 2**
AMR-021-0002 | O | 第 283 页

**Problems Around Polynomials - Conjecture 3**
AMR-021-0003 | O | 第 284 页

**Problems Around Polynomials - Conjecture 4**
AMR-021-0007 | O | 第 284 页

**Problems Around Polynomials - Conjecture 5**
AMR-021-0009 | O | 第 284 页

**Problems Around Polynomials - Conjecture 6**
AMR-021-0010 | O | 第 284 页

**Problems Around Polynomials - Conjecture 7**
AMR-021-0012 | S | 第 284 页

**Problems Around Polynomials - Conjecture 8**
AMR-021-0013 | O | 第 284 页

**Problems Around Polynomials - Conjecture 9**
AMR-021-0014 | O | 第 284 页

**Product branching and the missing irreducibility hypothesis**
AIM-SEVERAL_COMPLEX_VARIABLES-0013 | S | 第 1348 页

**Purdy Plane Line Incidence Conjecture**
OWR-11136-004 | O | 第 931 页

**Pure-diagonal deficits in Lev's restricted Scherk conjecture**
AIM-COMBINATORICS-0226 | P | 第 1720 页

**Quantum Variance Conjectures for Matrix Elements**
OWR-11447-002 | P | 第 2149 页

**Question 2.1 - Do mapping class groups or Out(Fn) satisfy the Baum-Connes conjecture?**
AMR-109-0269 | P | 第 1210 页

**Quillen's $p$-Subgroup Conjecture**
OWR-11786-012 | P | 第 603 页

**Rado's Boundedness Conjecture**
GREEN-065 | P | 第 1722 页

**Ramanujan Conjecture for Cuspidal Representations**
OWR-1060-008 | O | 第 170 页

**Ramos Conjecture for Equipartitions by Hyperplanes**
OWR-13490-006 | O | 第 934 页

**Randomstrasse101 Open Problems (2024) - Conjecture 11**
AMR-144-0009 | O | 第 1941 页

**Randomstrasse101 Open Problems (2024) - Conjecture 12**
AMR-144-0010 | O | 第 1941 页

**Randomstrasse101 Open Problems (2024) - Conjecture 2 (Globally Synchronizing Regular Graphs)**
AMR-144-0001 | O | 第 1942 页

**Randomstrasse101 Open Problems (2024) - Conjecture 3 (Global Synchrony with negative edges)**
AMR-144-0002 | O | 第 1942 页

**Randomstrasse101 Open Problems (2024) - Conjecture 4 (Density threshold for Global Synchrony)**
AMR-144-0003 | O | 第 1942 页

**Randomstrasse101 Open Problems (2024) - Conjecture 5 (Ellipsoid fitting)**
AMR-144-0004 | O | 第 1942 页

**Randomstrasse101 Open Problems (2024) - Conjecture 8 (Kikuchi Spectral Threshold)**
AMR-144-0006 | O | 第 1942 页

**Randomstrasse101 Open Problems (2024) - Conjecture 9 (K\'{o}mlos Conjecture)**
AMR-144-0007 | O | 第 1943 页

**Randomstrasse101 Open Problems (2025) - Conjecture 1**
AMR-145-0002 | O | 第 1943 页

**Randomstrasse101 Open Problems (2025) - Conjecture 1**
AMR-145-0004 | O | 第 1943 页

**Randomstrasse101 Open Problems (2025) - Conjecture 1**
AMR-145-0006 | O | 第 1943 页

**Randomstrasse101 Open Problems (2025) - Conjecture 1**
AMR-145-0011 | O | 第 1944 页

**Randomstrasse101 Open Problems (2025) - Conjecture 1 (Sharp graph matrix bounds)**
AMR-145-0001 | O | 第 1944 页

**Randomstrasse101 Open Problems (2025) - Conjecture 1 (Type-$2$ constant of Tensors)**
AMR-145-0014 | O | 第 1944 页

**Randomstrasse101 Open Problems (2025) - Conjecture 2**
AMR-145-0003 | O | 第 1944 页

**Randomstrasse101 Open Problems (2025) - Conjecture 2**
AMR-145-0007 | O | 第 1944 页

**Randomstrasse101 Open Problems (2025) - Conjecture 2**
AMR-145-0012 | O | 第 1945 页

**Randomstrasse101 Open Problems (2025) - Conjecture 3**
AMR-145-0008 | O | 第 1945 页

**Randomstrasse101 Open Problems (2025) - Conjecture 4**
AMR-145-0009 | O | 第 1945 页

**Randomstrasse101 Open Problems (2025) - Conjecture 5**
AMR-145-0010 | O | 第 1945 页

**Rank Conjecture for Free Actions on Products of Spheres**
OWR-3477-017 | P | 第 1229 页

**Rank Conjecture for Free Elementary-Abelian Actions**
OWR-14299915-009 | P | 第 1229 页

**Rank Growth and the Tame Fontaine-Mazur Conjecture**
OWR-1274-021 | O | 第 603 页

**Rank-two verification and peripheral reductions for the relatively hyperbolic LERF conjecture**
AIM-GEOMETRIC_GROUP_THEORY-0007 | P | 第 604 页

**Real $\tau$-Conjecture for Sparse Polynomial Sums and Products**
OWR-12177-002 | P | 第 2100 页

**Real Baum-Connes Isomorphism Conjecture**
OWR-14743-002 | O | 第 604 页

**Real Tau Conjecture for Depth-Four Circuits**
OWR-16634-002 | P | 第 2100 页

**Reconstruction Conjecture**
GT-002 | O | 第 1848 页

**Reed's Conjecture on Graph Coloring**
OWR-1189-001 | O | 第 1848 页

**Reed's omega, delta, and chi conjecture**
OPG-335 | P | 第 1848 页

**Reed’s Chromatic Number Conjecture**
OWR-12330-010 | P | 第 1849 页

**Reed’s Chromatic-Number Conjecture**
OWR-12697684-019 | O | 第 1849 页

**Reid’s Fundamental-Group Conjecture for General-Type Surfaces**
OWR-1730-007 | O | 第 728 页

**Reinhardt's conjecture**
AMR-071-0029 | P | 第 938 页

**Relative Bogomolov Conjecture for Abelian Schemes**
OWR-10252929-007 | P | 第 728 页

**Resolving the average in DeVos's ordered-label conjecture**
AIM-COMBINATORICS-0177 | P | 第 1726 页

**Reverse Mathematics of Fraïssé's Conjecture**
AMR-074-0121 | P | 第 2038 页

**Revised Donkin Tilting-Module Conjectures**
OWR-14299915-004 | O | 第 290 页

**Rigidity and Assembly Conjectures for Discrete Groups**
OWR-4333-003 | P | 第 606 页

**Robertson’s Surface Four-Flow Conjecture**
OWR-14299081-018 | O | 第 1849 页

**Rota's Basis Conjecture**
ALG-048 | P | 第 291 页

**Rota's conjecture on disjoint bases**
AMR-029-0072 | P | 第 1728 页

**Rota's unimodal conjecture**
OPG-361 | P | 第 1728 页

**Rudin's conjecture on squares in progressions**
AMR-031-0008 | P | 第 1728 页

**Ryser's Conjecture**
GRAPH-053 | P | 第 1850 页

**Ryser-Brualdi-Stein Near-Transversal Conjecture**
OWR-9790352-021 | P | 第 1850 页

**Ryser-Brualdi-Stein Transversal Conjecture**
OWR-12697684-020 | P | 第 1729 页

**Sabidussi's compatibility conjecture**
AMR-029-0075 | P | 第 1729 页

**Sarnak's Möbius Disjointness Conjecture**
OWR-12173-005 | O | 第 1608 页

**Schanuel Conjecture**
OWR-737-007 | O | 第 2038 页

**Schanuel's Conjecture**
NUM-016 | O | 第 176 页

**Schinzel's Hypothesis H**
NT-028 | O | 第 176 页

**Schmidt's weighted intersection conjecture**
AIM-OTHER-0120 | S | 第 2181 页

**Schneider Higher-Order Difference-Body Conjecture**
OWR-14298808-007 | P | 第 959 页

**Scholz Conjecture**
NT-039 | P | 第 176 页

**Scrambled Rota conjecture**
AMR-029-0076 | P | 第 1729 页

**Section Conjecture for Hyperbolic Curves Without Boundary Points**
OWR-3480-017 | O | 第 176 页

**Selberg Eigenvalue Conjecture for Congruence Groups**
OWR-14299577-014 | O | 第 177 页

**Selberg's 1/4 conjecture**
AMR-093-0079 | P | 第 177 页

**Selberg's orthogonality conjecture**
AMR-093-0080 | O | 第 177 页

**Selection Principles Milestone Problems - Conjecture 24 ([splittability, Conjecture 8.7])**
AMR-134-0023 | O | 第 1988 页

**Selection Principles Milestone Problems - Conjecture 35 ([strongdiags, Conjecture 1])**
AMR-134-0034 | O | 第 1988 页

**Selection Principles Milestone Problems - Conjecture 40**
AMR-134-0039 | O | 第 1988 页

**Selfridge's conjecture**
AMR-093-0136 | O | 第 177 页

**Semifree-Module Form of the Toral-Rank Conjecture**
OWR-5146-007 | S | 第 1234 页

**Serre Conjecture II for Type $E_7$ Groups**
OWR-14213-003 | O | 第 606 页

**Serre's Conjecture II**
ALG-013 | P | 第 293 页

**Serre's Positivity Conjecture**
ALG-022 | P | 第 293 页

**Seymour One-Flowing Conjecture**
OWR-11451-009 | P | 第 1730 页

**Seymour's r-graph conjecture**
OPG-2226 | S | 第 1851 页

**Seymour's Second Neighbourhood Conjecture**
OPG-646 | P | 第 1851 页

**Seymour's self-minor conjecture**
OPG-333 | O | 第 1851 页

**Sharp Exponents in the Fourier Restriction Conjecture**
OWR-15577-011 | O | 第 1426 页

**Shelah's categoricity conjecture for $L_{\omega_1,\omega}$**
AMR-075-0006 | P | 第 2039 页

**Shelah's eventual categoricity conjecture**
AMR-075-0007 | P | 第 2040 页

**SHGH Conjecture for Linear Systems on Blown-Up Planes**
OWR-3471-007 | O | 第 732 页

**Shokurov Finite-Generation Conjecture**
OWR-788-005 | O | 第 732 页

**Shuffle-Exchange Conjecture**
OPG-37167 | P | 第 1732 页

**Sidorenko's Conjecture**
OPG-60039 | P | 第 1852 页

**Sidorenko's Conjecture for Bipartite Graphons**
OWR-14604-021 | O | 第 1852 页

**Siegel's conjecture**
AMR-093-0160 | O | 第 179 页

**Simonovits-Sós and Sumner Tournament Conjectures**
OWR-4791-001 | P | 第 1853 页

**Simon’s Epimorphism Conjecture for Knot Groups**
OWR-2089-001 | S | 第 1235 页

**Simpson’s Weak-Units Conjecture for Higher Categories**
OWR-14298590-003 | O | 第 295 页

**Singer’s Unobstructedness Conjecture for Anti-Self-Dual Four-Manifolds**
OWR-13750335-006 | P | 第 965 页

**Singular Cardinals Hypothesis**
SET-002 | S | 第 1994 页

**Smooth 4-dimensional Poincare conjecture**
OPG-37125 | P | 第 1236 页

**Smooth 4-Dimensional Poincaré Conjecture**
TOP-001 | O | 第 1236 页

**Snevily's conjecture**
OPG-655 | P | 第 181 页

**Solved Audin conjecture and a partial structural result on Maslov-two areas**
AIM-GEOMETRY-0421 | S | 第 968 页

**Solved Finsler two-or-infinity conjecture and an isometry-orbit certificate**
AIM-GEOMETRY-0207 | S | 第 968 页

**Solved Hurwitz cyclic sieving conjecture and a dihedral order diagnostic**
AIM-OTHER-0022 | S | 第 2182 页

**Soundararajan Conjecture on Zeta Zero Spacing**
OWR-1795-015 | P | 第 182 页

**Spaces with Lower Ricci Curvature - Conjecture 1**
AMR-128-0001 | O | 第 969 页

**Spaces with Lower Ricci Curvature - Conjecture 2**
AMR-128-0002 | O | 第 969 页

**Spaces with Lower Ricci Curvature - Conjecture 3**
AMR-128-0003 | O | 第 969 页

**Spaces with Lower Ricci Curvature - Conjecture 4**
AMR-128-0004 | O | 第 970 页

**Spaces with Lower Ricci Curvature - Conjecture 5**
AMR-128-0005 | O | 第 970 页

**Spaces with Lower Ricci Curvature - Conjecture 6 (noncollapsing Yau's conjecture)**
AMR-128-0006 | O | 第 970 页

**Spaces with Lower Ricci Curvature - Conjecture 7 (energy identity)**
AMR-128-0007 | O | 第 970 页

**Spaces with Lower Ricci Curvature - Conjecture 8**
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**Spaces with Lower Ricci Curvature - Conjecture 9 (local Yau conjecture)**
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**Stable Banach Isometric Subspace Conjecture**
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**Stable Coordinate Conjecture for Complex Polynomials**
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**Stable Field Conjecture**
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**Stable Prasad Multiplicity Conjecture**
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**Standard k-Bubble Conjectures**
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**Stanley's Positivity Problems in Algebraic Combinatorics - Problem 17: Stembridge's monomial immanant positivity conjecture for totally nonnegative matrices**
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**Stanley's Positivity Problems in Algebraic Combinatorics - Problem 18: Stembridge's Schur-positivity conjecture for monomial immanants of Jacobi-Trudi matrices**
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**Stanley's Positivity Problems in Algebraic Combinatorics - Problem 19: Haiman's unimodality and positivity conjecture for Hecke algebra monomial characters on Kazhdan-Lusztig basis elements**
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**Stanley's Positivity Problems in Algebraic Combinatorics - Problem 20: Neggers-Stanley Poset Conjecture on real zeros of the P-Eulerian polynomial W(P,x)**
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**Stanley's Positivity Problems in Algebraic Combinatorics - Problem 21: Stanley-Stembridge e-positivity conjecture for chromatic symmetric functions of (3+1)-free posets**
AMR-156-0022 | O | 第 1738 页

**Stanley's Positivity Problems in Algebraic Combinatorics - Problem 22: Gasharov's Schur-positivity conjecture for chromatic symmetric functions of claw-free graphs**
AMR-156-0023 | O | 第 1738 页

**Stanley's Positivity Problems in Algebraic Combinatorics - Problem 24: Haglund-Ono-Wagner real-rootedness conjecture for per(A + xJ)**
AMR-156-0025 | S | 第 1738 页

**Stanley's Positivity Problems in Algebraic Combinatorics - Problem 25: Unimodality and log-concavity of matroid and geometric lattice sequences (Read-Hoggar-Welsh-Mason conjectures)**
AMR-156-0026 | S | 第 1739 页

**Stanley's Positivity Problems in Algebraic Combinatorics - Problem 4: Charney-Davis conjecture for Gorenstein* flag complexes**
AMR-156-0003 | O | 第 1739 页

**Stanley-Stembridge Chromatic Symmetric Function Conjecture**
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**Stark conjectures on leading terms of Artin L-functions**
AMR-093-0058 | P | 第 184 页

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**Strong 5-cycle double cover conjecture**
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**Strong Atiyah Conjecture for Free-Group Quotients**
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**Strong edge colouring conjecture**
OPG-46475 | P | 第 1858 页

**Strong Intersection Conjecture for Hamiltonian Isotopies**
OWR-2053-005 | O | 第 977 页

**Strong Nakayama Conjecture**
OWR-787-006 | O | 第 299 页

**Strong Onsager Conjecture for Euler Flows**
OWR-11101918-005 | P | 第 1499 页

**Strong Reverse Minkowski Conjecture**
OWR-13951-003 | P | 第 977 页

**Strong Uniform Torsion Conjecture for Abelian Varieties**
OWR-1188-007 | P | 第 184 页

**Structural Explanation of the McKay Conjecture**
OWR-13675-009 | O | 第 610 页

**Structure of Counterexamples to the p-Adic Littlewood Conjecture**
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**Sum of prime and semiprime conjecture**
OPG-37404 | P | 第 185 页

**Sumner's Conjecture**
GRAPH-037 | P | 第 1860 页

**Switching reconstruction conjecture**
OPG-46951 | O | 第 1860 页

**Symbolic Rees Rings and Nagata’s Conjecture**
OWR-3472-012 | O | 第 300 页

**Szpiro's conjecture**
AMR-093-0090 | P | 第 186 页

**Tame Fontaine-Mazur Conjecture**
OWR-1274-020 | O | 第 187 页

**Terao Conjecture for Line Arrangements**
OWR-14222-007 | O | 第 737 页

**Teschner's Bondage Number Conjecture**
GRAPH-055 | P | 第 1861 页

**The "cycle double cover conjecture" states that every bridgeless graph contains a set of cycles which cover each edge of**
AMR-030-0019 | P | 第 1743 页

**The $L_p$ Brunn-Minkowski Conjecture**
OWR-16637-008 | O | 第 981 页

**The 196 conjecture**
AMR-046-0032 | O | 第 1743 页

**The Abelian Rank-One Stark Conjecture**
OWR-11127-017 | P | 第 187 页

**The Abundance Conjecture**
AG-002 | P | 第 738 页

**The additive basis conjecture**
OPG-150 | P | 第 1743 页

**The Alekseev-Torossian Symmetry-Group Conjecture**
OWR-8415345-006 | O | 第 301 页

**The Alexandrov boundary conjecture and a polyhedral Gauss formula**
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**The Alon-Tarsi basis conjecture**
OPG-152 | O | 第 1743 页

**The Alon-Tarsi Conjecture for Latin Squares**
OWR-14299518-016 | O | 第 1743 页

**The Atiyah conjecture on configurations on the invertibility of a certain $n$-by-$n$ matrix depending on $n$ points in $\mathbb{R}^{3}$**
AMR-071-0059 | O | 第 981 页

**The Babai Conjecture on Graph Isomorphism**
ALG-016 | S | 第 301 页

**The Berge Conjecture for Lens-Space Surgeries**
OWR-17471-003 | P | 第 1240 页

**The Berge-Fulkerson conjecture**
OPG-142 | P | 第 1861 页

**The Bermond-Thomassen Conjecture**
OPG-611 | P | 第 1861 页

**The Bismut-Lott Transfer Index Conjecture**
OWR-11137-009 | O | 第 1241 页

**The Bollobás-Eldridge-Catlin Conjecture on graph packing**
OPG-48232 | P | 第 1861 页

**The Bondal-Orlov D-Equivalence Conjecture**
OWR-14299284-001 | P | 第 738 页

**The Borodin-Kostochka Conjecture**
OPG-37670 | P | 第 1861 页

**The Boundary Singer Conjecture**
OWR-17470-002 | P | 第 1241 页

**The Bounded-Height Conjecture for Hyperbolic Dehn Fillings**
OWR-13108-010 | P | 第 1241 页

**The Caccetta-Häggkvist Directed-Cycle Conjecture**
OWR-12330-019 | O | 第 1861 页

**The Carathéodory Conjecture**
GEO-014 | O | 第 982 页

**The Cartan-Hadamard Conjecture**
GEO-015 | P | 第 982 页

**The Cherlin-Zilber Conjecture**
ALG-039 | P | 第 302 页

**The circular embedding conjecture**
OPG-138 | P | 第 1861 页

**The Cycle Double Cover Conjecture**
GT-004 | P | 第 1862 页

**The Demazure Conjecture**
ALG-036 | P | 第 302 页

**The density hypothesis for zeroes of the Riemann zeta function**
AMR-093-0068 | P | 第 187 页

**The Double Cap Conjecture**
OPG-60000 | P | 第 1744 页

**The doubled-label slice is exactly Seymour's second-neighborhood conjecture**
AIM-COMBINATORICS-0162 | P | 第 1744 页

**The Dupont-Monod Comparison-Map Conjecture**
OWR-15438-003 | P | 第 613 页

**The Dynamical Mordell-Lang Conjecture for Endomorphisms**
OWR-14603-019 | P | 第 188 页

**The Erdos-Turan conjecture on additive bases**
OPG-367 | O | 第 188 页

**The Erdös-Hajnal Conjecture**
OPG-162 | P | 第 1862 页

**The Erdős-Oler conjecture**
AMR-071-0026 | O | 第 982 页

**The Erdős-Rado Sunflower Conjecture**
OWR-8415359-002 | P | 第 1745 页

**The Erdős-Rado Sunflower Conjecture**
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**The Euclidean Steinitz conjecture and a sharp ell_1 obstruction**
AIM-COMBINATORICS-0041 | P | 第 1745 页

**The even-color Volume Conjecture needs a logarithm**
AIM-TOPOLOGY-0028 | S | 第 1241 页

**The Filling Area Conjecture**
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**The Generalized Étale-Homotopy Section Conjecture**
OWR-14298795-005 | O | 第 740 页

**The Generalized Hamada p-Rank Conjecture**
OWR-16775-005 | O | 第 1747 页

**The Global Monodromy Conjecture for Calabi-Yau Varieties**
OWR-4427-002 | P | 第 740 页

**The Graceful Tree Conjecture**
GT-003 | O | 第 1862 页

**The Graphic Kahn-Kalai Conjecture**
OWR-14298371-006 | O | 第 1747 页

**The Green-Griffiths-Lang Entire-Curve Conjecture**
OWR-11139-007 | O | 第 740 页

**The Grothendieck-Katz p-Curvature Conjecture**
OWR-14298795-008 | O | 第 741 页

**The Guralnick-Thompson Conjecture**
ALG-025 | S | 第 303 页

**The Hadwiger Covering Conjecture**
GEO-023 | O | 第 983 页

**The Hamiltonicity Conjecture for Four-Connected Line Graphs**
OWR-4212-010 | O | 第 1862 页

**The Hexagonal Conjecture for Spectral Minimal Partitions**
OWR-3389-002 | O | 第 984 页

**The Hofer-Zehnder Periodic-Orbit Conjecture**
OWR-12582-002 | P | 第 1613 页

**The Hopf Conjectures**
GEO-020 | O | 第 984 页

**The Hyperplane Slicing Conjecture**
OWR-12337-001 | S | 第 984 页

**The Ichino-Ikeda Period Conjecture**
OWR-14299289-005 | P | 第 188 页

**The Illumination Conjecture**
GEO-007 | O | 第 984 页

**The Jacobian Conjecture**
SMA-016 | P | 第 303 页

**The Kakeya Conjecture**
GEO-003 | P | 第 984 页

**The Köthe Conjecture**
ALG-003 | P | 第 303 页

**The Kusner conjecture**
AMR-071-0051 | O | 第 985 页

**The L-Space and Left-Orderability Conjecture**
OWR-13108-002 | O | 第 1242 页

**The L-Space and Left-Orderability Conjecture**
OWR-14298592-008 | O | 第 1242 页

**The large sets conjecture**
OPG-373 | O | 第 1747 页

**The Lex-Plus-Powers Betti Number Conjecture**
OWR-3472-002 | O | 第 303 页

**The Local Geometric Langlands Conjecture**
OWR-14299289-008 | O | 第 741 页

**The Log-Rank Conjecture**
OWR-14299518-026 | O | 第 2108 页

**The Lonely Runner Conjecture**
COMB-004 | P | 第 1747 页

**The main gap conjecture, e.g. for uncountable first order theories, for AECs, and for $\aleph_1$-saturated models of a countable theory**
AMR-075-0005 | P | 第 2043 页

**The maximal symmetry-rank conjecture and a sharp product defect formula**
AIM-TOPOLOGY-0243 | P | 第 1242 页

**The missing just-infinite hypothesis and an explicit branch-group counterexample**
AIM-GEOMETRIC_GROUP_THEORY-0111 | S | 第 614 页

**The Mixed Littlewood Conjecture for Arbitrary Pseudo-Absolute Values**
OWR-14603-028 | O | 第 188 页

**The Multicolored Geometric Erdős-Hajnal Conjecture**
OWR-14299081-021 | P | 第 1961 页

**The n-conjecture**
AMR-093-0088 | P | 第 188 页

**The N-or-Infinity Conjecture for Reeb Pseudo-Rotations**
OWR-14299512-006 | O | 第 1242 页

**The Nekovář-Scholl Plectic Cohomology Conjecture**
OWR-14299291-003 | O | 第 741 页

**The Novikov Higher-Signature Conjecture**
OWR-1703864-004 | O | 第 1242 页

**The Numerical Unramified Plancherel Conjecture**
OWR-14299289-007 | P | 第 614 页

**The Odd Perfect Number Conjecture**
NT-026 | P | 第 189 页

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GEO-021 | P | 第 985 页

**The p-Adic Variational Hodge Conjecture**
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DYN-002 | S | 第 1613 页

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OPG-151 | O | 第 1748 页

**The Pierce-Birkhoff Conjecture**
ALG-021 | P | 第 309 页

**The Polynomial Hirsch Conjecture**
CS-002 | O | 第 2108 页

**The Pólya-Szegő buckling conjecture and a radial projection gap on balls**
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**The Q-Form Classification Conjecture**
OWR-8415345-007 | P | 第 1243 页

**The Quantum Unique Ergodicity Conjecture**
OWR-16636-006 | O | 第 2156 页

**The Rational Toral-Rank Conjecture**
OWR-5146-003 | O | 第 1243 页

**The Real-Zero Polynomial Amalgamation Conjecture**
OWR-17475-002 | S | 第 742 页

**The Refined Formal-Degree Conjecture**
OWR-8415342-025 | P | 第 189 页

**The Refined Unstable-Module Realization Conjecture**
OWR-2089-011 | P | 第 1243 页

**The refuted C1 conjecture and radial rigidity**
AIM-PDES-0028 | S | 第 1501 页

**The refuted de Caen clique-covering conjecture and an exact rational recursion**
AIM-COMBINATORICS-0149 | S | 第 1748 页

**The residual all-hyperbolic branch of Long's elliptic geodesic conjecture**
AIM-GEOMETRY-0210 | P | 第 986 页

**The Restricted Geometric Langlands Conjecture**
OWR-8415342-018 | P | 第 743 页

**The Riemann Hypothesis**
MPP-002 | O | 第 189 页

**The Segal Conjecture for Infinite Groups**
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**The sharp symmetric 2n-vertex case of the Euclidean-ball vertex-index conjecture**
AIM-CONVEX_GEOMETRY-0007 | P | 第 986 页

**The Sigma-Rho Poisson Summation Conjecture**
OWR-8415342-012 | O | 第 189 页

**The Sofic Groups Conjecture**
ALG-033 | P | 第 310 页

**The solved Finsler two-or-infinity conjecture and an orbit-covering audit**
AIM-GEOMETRY-0208 | S | 第 987 页

**The solved two-step flag puzzle conjecture and projection-lift support tests**
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**The sparse-removal conjecture fails literally and at the true extremal exponent**
AIM-COMBINATORICS-0157 | S | 第 1749 页

**The Standard Conjectures on Algebraic Cycles**
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**The Strong Slope Conjecture for Colored Jones Polynomials**
OWR-15581-004 | P | 第 1243 页

**The Surjunctivity Conjecture**
ALG-032 | O | 第 311 页

**The Tamed-to-Compatible Symplectic Conjecture**
OWR-2654829-004 | P | 第 988 页

**The Tate Conjecture**
AG-004 | P | 第 744 页

**The Tempered Numerical Period Conjecture**
OWR-14299289-020 | O | 第 189 页

**The three 4-flows conjecture**
OPG-132 | O | 第 1863 页

**The Total Coloring Conjecture**
GT-010 | P | 第 1863 页

**The Twisted Fourier-Jacobi Gan-Gross-Prasad Conjecture**
OWR-8415342-008 | P | 第 190 页

**The Two Color Conjecture**
OPG-169 | P | 第 1863 页

**The Two-Dimensional Time-Bomb Conjecture**
OWR-1183-006 | O | 第 1750 页

**The Uniform Boundedness Conjecture**
ALG-020 | P | 第 311 页

**The Unique Games Conjecture**
CS-001 | O | 第 2109 页

**The Volume Conjecture**
TOP-002 | P | 第 1244 页

**The Weak Gordon-Luecke Conjecture**
OWR-8415345-003 | O | 第 1244 页

**The Weinstein Conjecture**
DYN-001 | P | 第 1613 页

**The Weinstein Periodic-Orbit Conjecture**
OWR-4411-001 | O | 第 1244 页

**The Whitehead Conjecture**
TOP-003 | P | 第 1244 页

**Thiagarajan Grid-Free Decidability Conjecture**
OWR-14299911-005 | S | 第 1863 页

**Thin-Shell Conjecture for Schatten Balls**
OWR-14219-010 | O | 第 988 页

**Thomasse's path-girth conjecture: refutation and exact cyclic blow-ups**
AIM-COMBINATORICS-0204 | S | 第 1750 页

**Thomassen's Hamiltonicity Conjecture for Infinite Line Graphs**
OWR-1532-004 | O | 第 1863 页

**Thom’s Conjecture for the Hyperfinite Factor’s Unitary Group**
OWR-14748-012 | O | 第 1438 页

**Tits’ Center Conjecture for Spherical Buildings**
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**Toms-Winter Conjecture for Nuclear $C^*$-Algebras**
OWR-4214-007 | O | 第 1439 页

**Toms-Winter Regularity Conjecture**
OWR-12175-001 | P | 第 312 页

**Toms-Winter Regularity Conjecture**
OWR-15585-003 | P | 第 312 页

**Topological Period-Index Conjecture**
OWR-13681-012 | S | 第 1245 页

**Torsion-Anomalous Conjecture for Semiabelian Varieties**
OWR-14299578-007 | O | 第 745 页

**Total Colouring Conjecture**
OPG-815 | P | 第 1864 页

**Transfer Conjecture for O-Minimal Exponential Fields**
OWR-15181-017 | O | 第 2044 页

**Transfer Index Conjecture in Differential Algebraic $K$-Theory**
OWR-11787-006 | O | 第 1246 页

**Tree Property with Failure of Singular Cardinal Hypothesis**
OWR-4792-016 | O | 第 1995 页

**Tutte's 4-Flow Conjecture for Petersen-Minor-Free Graphs**
GRAPH-057 | P | 第 1866 页

**Tutte's 5-Flow Conjecture**
GRAPH-056 | O | 第 1866 页

**Tutte’s Four-Flow Conjecture**
OWR-14299081-015 | O | 第 1866 页

**Tuza's Conjecture**
GRAPH-038 | O | 第 1866 页

**Twin Prime Conjecture**
LAN-002 | O | 第 192 页

**Two-Term Weyl Asymptotics and Ivrii's Billiard Conjecture**
OWR-8415336-012 | P | 第 1441 页

**Type-II Error Exponents in Distributed Hypothesis Testing**
OWR-9790365-011 | O | 第 2110 页

**Ulam's Packing Conjecture**
GEO-031 | P | 第 993 页

**Unfriendly Partition Conjecture**
GRAPH-039 | O | 第 1867 页

**Uniform Boundedness Conjecture for Rational Points**
ALG-049 | O | 第 313 页

**Uniform Mordell Conjecture**
OWR-13952-010 | P | 第 194 页

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OWR-12495-006 | P | 第 750 页

**Valuative Form of Grothendieck's Section Conjecture**
OWR-12493-007 | O | 第 195 页

**Vandiver's conjecture**
AMR-088-0006 | O | 第 1868 页

**Vanishing Conjecture for Powers of Homogeneous Polynomials**
OWR-1452-012 | P | 第 750 页

**Vanishing Conjectures for Birational Cohomology of Projective Space**
OWR-13750339-004 | O | 第 751 页

**Variance Conjecture for Isotropic Convex Bodies**
OWR-15215-005 | O | 第 1968 页

**Variance Conjecture for Log-Concave Random Vectors**
OWR-12182-002 | O | 第 1968 页

**Vasconcelos Conjecture for Kähler Differentials**
OWR-12697689-016 | O | 第 315 页

**Vaught Conjecture**
LOGIC-001 | P | 第 315 页

**Virasoro conjecture**
AMR-071-0020 | P | 第 752 页

**Virtual FOG conjecture**
AMR-045-0036 | O | 第 1618 页

**Virtual-knot problem 51 - Prove or disprove the conjecture about the non-uniqueness of minimal representative of a free link, i.e.**
AMR-105-0051 | O | 第 1260 页

**Visibility Graphs of Points, Segments and Polygons - Conjecture 14**
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**Visibility Graphs of Points, Segments and Polygons - Conjecture 15**
AMR-131-0013 | O | 第 1869 页

**Visibility Graphs of Points, Segments and Polygons - Conjecture 16**
AMR-131-0014 | O | 第 1869 页

**Visibility Graphs of Points, Segments and Polygons - Conjecture 4**
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AMR-131-0005 | O | 第 1869 页

**Viterbo's Symplectic Capacity-Volume Conjecture**
OWR-15575-001 | S | 第 997 页

**Vizing's Conjecture**
GRAPH-041 | O | 第 1874 页

**Vizing's Conjecture and Algebraic Sum-of-Squares Certificates**
OWR-16938-003 | O | 第 1874 页

**Vizing’s Cartesian-Product Domination Conjecture**
OWR-10252928-002 | O | 第 1874 页

**Vogan Conjecture for Quasisplit Classical Groups**
OWR-14298805-007 | P | 第 195 页

**Vojta's conjecture**
AMR-093-0087 | P | 第 196 页

**Vojta's Diophantine Approximation Conjecture**
OWR-10252929-011 | O | 第 196 页

**Wahl's Conjecture for Nodal Curves**
OWR-13946-013 | P | 第 752 页

**Wall avoidance, root detection, and gaps between local-rigidity hypotheses**
AIM-OTHER-0070 | P | 第 2187 页

**Wang-Ostrik Finiteness Conjecture for Fusion Categories**
OWR-1533-001 | O | 第 2158 页

**Weak Greenberg Conjecture for CM Fields**
OWR-1274-016 | P | 第 196 页

**Weak Knaster Conjecture**
OWR-4798-016 | P | 第 998 页

**Weak Section Conjecture over Number Fields**
OWR-3388-010 | O | 第 196 页

**Weight Conjectures for Spetsial Reflection Groups**
OWR-11101920-015 | P | 第 618 页

**Wide partition conjecture**
OPG-2063 | P | 第 1759 页

**Wikipedia geometry item 13: Nagata's conjecture on curves, specifically the minimal degree required for a plane algebraic c…**
AMR-071-0013 | O | 第 754 页

**Wikipedia geometry item 14: Nagata-Biran conjecture that if $X$ is a smooth algebraic surface and $L$ is an ample line bund…**
AMR-071-0014 | O | 第 754 页

**Wikipedia geometry item 17: Section conjecture on splittings of group homomorphisms from fundamental groups of complete smo…**
AMR-071-0017 | O | 第 754 页

**Wikipedia geometry item 7: Fujita conjecture regarding the line bundle $K_{M} \otimes L^{\otimes m}$ constructed from a po…**
AMR-071-0007 | O | 第 754 页

**Wikipedia number-theory item 104: Congruent number problem (a corollary to Birch and Swinnerton-Dyer conjecture, per Tunnell's theorem…**
AMR-093-0104 | P | 第 198 页

**Wikipedia number-theory item 109: The uniqueness conjecture for Markov numbers that every Markov number is the largest number in exact…**
AMR-093-0109 | O | 第 198 页

**Williams's birational trapezoid conjecture**
AIM-OTHER-0027 | S | 第 2187 页

**Wolff Cone-Decomposition and Local-Smoothing Conjectures**
OWR-2043-005 | P | 第 1451 页

**Woodall's Conjecture**
GRAPH-058 | P | 第 1876 页

**Woodin's Ω-Conjecture**
ST-007 | O | 第 1997 页

**Wu's Higher-Dimensional Picard Conjecture**
AMR-055-0009 | S | 第 1001 页

**Yau Stability Conjecture for Constant Scalar Curvature Metrics**
OWR-2043-008 | P | 第 755 页

**Yau's Conjecture on First Eigenvalue**
GEOM-018 | P | 第 1001 页

**Yau's Conjecture on First Eigenvalues**
GEO-022 | P | 第 1001 页

**Yau-Tian-Donaldson Conjecture for Polarized Varieties**
OWR-13109-010 | O | 第 755 页

**Zaremba Conjecture for Bounded Continued Fraction Digits**
OWR-8415353-003 | O | 第 198 页

**Zaremba’s Bounded Continued-Fraction Conjecture**
OWR-11135-005 | P | 第 199 页

**Zariski multiplicity conjecture on the topological equisingularity and equimultiplicity of varieties at singular points**
AMR-071-0021 | O | 第 755 页

**Zariski's Multiplicity Conjecture**
OWR-8415350-011 | O | 第 755 页

**Zariski-Lipman conjecture**
AMR-015-0031 | O | 第 318 页

**Zauner's Conjecture**
ALG-007 | P | 第 318 页

**Zeeman Conjecture**
TOP-010 | O | 第 1264 页

**Zilber Conjecture for Torsion-Anomalous Intersections**
OWR-11784-008 | O | 第 199 页

**Zilber-Pink conjecture**
AMR-093-0091 | O | 第 199 页

**Zilber’s Quasiminimality Conjecture for Complex Exponentiation**
OWR-11695862-009 | O | 第 2048 页

# 附录 B | 归档记录索引（1,417 条）

以下记录 published=false，不计入 17,614。索引保留原编号、题名、归档原因和已提供的主条目关系。归档不一定意味着重复，更不意味着已证明。

### AG-003 | The Birch and Swinnerton-Dyer Conjecture

Duplicate of canonical problem MPP-005

MPP-005 / p.35

### AG-005 | The Hodge Conjecture

Duplicate of canonical problem MPP-006

MPP-006 / p.687

### AIM-ALGEBRAIC_GEOMETRY-0144 | Convention split for cycle-Iitaka integrality and a finite-additive-presentation criterion

Truncated or non-self-contained workshop heading/fragment

### AIM-ALGEBRAIC_GEOMETRY-0217 | Brion's degeneration theorem and a birational basis-exchange law

Context-only setup for AIM-ALGEBRAIC_GEOMETRY-0218 (Problem 2 in Degenerations in Algebraic Geometry)

### AIM-ALGEBRAIC_GEOMETRY-0219 | The maximal-contact linear-space arrangement of a plane elliptic cubic

Context-only header for AIM-ALGEBRAIC_GEOMETRY-0220 (Problem 3 in Degenerations in Algebraic Geometry)

### AIM-ALGEBRAIC_GEOMETRY-0223 | Exact gluing and transverse-padding formulas for plane-curve H-constants

Context-only setup for AIM-ALGEBRAIC_GEOMETRY-0224 (Problem 5 in Degenerations in Algebraic Geometry)

### AIM-ALGEBRAIC_GEOMETRY-0228 | Exact convex certificates from square Nagata rays

Duplicate extraction of Problem 7 (already in AIM-ALGEBRAIC_GEOMETRY-0227).

### AIM-ALGEBRAIC_GEOMETRY-0230 | A quadratic-to-cubic maximal-rank wall and its pure resolution

Truncated or non-self-contained workshop heading/fragment

### AIM-ALGEBRAIC_GEOMETRY-0235 | Real smoothness of osculating Schubert curves

Context-only setup and theorem statements for AIM-ALGEBRAIC_GEOMETRY-0236 (Problem 10 in Degenerations in Algebraic Geometry)

### AIM-ALGEBRAIC_GEOMETRY-0247 | Singular free-summand descent and a cuspidal top-Chern counterexample

Multi-problem workshop PDF dump or non-problem placeholder

### AIM-ALGEBRAIC_NUMBER_THEORY-0039 | Exact sparse-incidence dichotomy for tricanonically embedded curves

Multi-problem workshop PDF dump or non-problem placeholder

### AIM-ALGEBRAIC_NUMBER_THEORY-0060 | Quantitative averages of function-field prime-tuple singular series

Multi-problem workshop PDF dump or non-problem placeholder

### AIM-ALGEBRAIC_NUMBER_THEORY-0067 | Zarhin's cubic S3 theorem and propagated CM factors

Literature/reference comment attached to Problem 2 (AIM-ALGEBRAIC_NUMBER_THEORY-0066), not a standalone problem.

### AIM-ALGEBRAIC_NUMBER_THEORY-0096 | Rational points, index transport, and the C_1^0 boundary

Historical remark (iv) attached to Question 11 (AIM-ALGEBRAIC_NUMBER_THEORY-0095), not a standalone problem.

### AIM-ALGEBRAIC_NUMBER_THEORY-0115 | A withdrawn duplicate of the D4 and E6 cubic counting problems

Explicitly withdrawn problem entry ('Question 30. This problem has been withdrawn.') duplicating the cubic counting problems.

### AIM-ANALYSIS-0059 | Explicit projective-space solution to item 9 of the 2016 complex Monge-Ampere list

Multi-problem workshop PDF dump or non-problem placeholder

### AIM-ANALYSIS-0083 | A spectral multiplicity refinement of the Casazza DFT near-counterexample

Unrecoverable AIM statement:

### AIM-ANALYSIS-0101 | A portable tail-certificate principle behind interlacing families

Informal workshop discussion comment about Kadison-Singer polynomial tools, not a mathematical problem statement.

### AIM-ANALYSIS-0157 | A correction-transfer triangle for Polya, Riesz means, and normalized eigenvalue means

Multi-page workshop dump containing 15 distinct problems rather than a single problem statement.

### AIM-ANALYTIC_NUMBER_THEORY-0102 | Conductor collisions in an aggregated workshop record

Workshop table of contents / multi-session dump, not a single mathematical problem.

### AIM-ANALYTIC_NUMBER_THEORY-0105 | A spliced large-sieve/zeta record and a local 1-line zeta certificate

Unrecoverable AIM statement:

### AIM-ARITHMETIC_GEOMETRY-0043 | Uniform degrees for splitting quadratic forms

Truncated or non-self-contained workshop heading/fragment

### AIM-ARITHMETIC_GEOMETRY-0046 | Classification of an AIM test fixture

Website testing fixture ('Name of the problem block / Intro to problem block / This is the problem statement.'), not a mathematical problem.

### AIM-ARITHMETIC_GEOMETRY-0047 | A self-identifying throw-away fixture

Website testing fixture ('Test introduction / the statement of a throw-away problem'), not a mathematical problem.

### AIM-ARITHMETIC_GEOMETRY-0081 | Malformed aggregate and a genus obstruction to Sullivan's cover-density question

Unrecoverable AIM statement:

### AIM-BIOLOGY-0001 | Identifiability of stage-resolved retinal vascular change in diabetes

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0002 | A conditional shunt-steal criterion for retinal oxygenation

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0003 | Repeatability-calibrated sequential warning from retinal microvascular monitoring

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0004 | Separating retinal barrier influx from delivery, volume, and clearance

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0005 | A deployable landmark model for imaging-based diabetic-retinopathy progression warnings

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0006 | A compensation and identifiability framework for aging, IOP, and ocular structure

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0007 | Flow certification for healthy-age retrobulbar Doppler comparisons

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0008 | Supply-demand thresholds for retinal oxygenation in light and darkness

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0009 | A two-number epithelial transport reduction with an aqueous-production identifiability test

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0010 | A rate-based criterion for targeting plastic tumor states

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0011 | A mechanism gate for resistance-driven drug scheduling

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0012 | A closure test from signaling dynamics to tumor growth

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0013 | A blinded finite-portfolio audit for preclinical-to-human prediction

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0014 | A feedback-aware clearance threshold for metabolism and signaling

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0015 | Geometry-specific transport thresholds for 2D and 3D tumor assays

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0016 | A size-phase audit for receptor, signaling, and drug-response variation

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0017 | Landmark selection can reverse metastatic and adjuvant drug rankings

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0018 | Nuisance-adjusted intervention criteria for testing biological model structures

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0019 | A coordinate-aware audit for uncertain model parameters

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0020 | Bounded disclosure delay, participation, and milestone thresholds

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0021 | A robust allocation certificate for student research clinics

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0022 | A dated funding pointer and a staged-funding decision rule

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0023 | What aggregate V1 suppression cannot identify

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0024 | A space-time exclusion certificate for visual-cortical route attribution

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0025 | A rate-delay abstraction certificate for filtered spiking circuits

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0026 | V1 saliency context and a delay margin for symmetry breaking

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0027 | When agreement is a complement, and when extra modalities help

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0028 | A clock-risk audit of the square-root distributed-synchrony code

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0029 | Generic views, Bayesian flexibility, and a falsifiable invariance test

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0030 | Auditing timing and measurement claims in a border-ownership workshop Q&A

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0031 | Peak-window bias and circuit nonidentifiability in push-pull response analysis

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0032 | A robust target-offset test for late 3D pop-out signals

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0033 | A model-discrimination and reporting certificate for Breakout Session I

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0034 | Shared-input model discrimination and recording-overlap identifiability

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0035 | Provenance-safe multidisciplinary priority setting

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0036 | Majority topology and the 0.66-or-1 branch-length question

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0037 | Exact counting with prescribed phylogenetic splits

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0038 | Exact split-residual certificates in BHV tree space

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0039 | Sharp scale-bounded comparison of BHV and Robinson-Foulds distance

Non-mathematical clinical / biological / grant administration workshop item

AIM-BIOLOGY-0040 | When concatenation is score averaging

Non-mathematical clinical / biological / grant administration workshop item

AIM-COMBINATORICS-0134 | Quarantining a test placeholder by an item-identifiability criterion

Multi-problem workshop PDF dump or non-problem placeholder

AIM-COMBINATORICS-0146 | Unique balance and a codegree-variance obstruction for the tetrahedron conjecture

Truncated or non-self-contained workshop heading/fragment

AIM-COMBINATORICS-0148 | An editorial placeholder and a two-axis non-recovery certificate

Multi-problem workshop PDF dump or non-problem placeholder

AIM-COMBINATORICS-0165 | Repairing the safe-completion reduction from rainbow reachability to Sullivan-2

Mid-proof split fragment from the reduction between Conjectures 6.1 and 6.4 (AIM-COMBINATORICS-0164), not an independent problem.

AIM-COMBINATORICS-0166 | The exact fresh-color safety criterion behind the Sullivan-2 reduction

Mid-proof split fragment following Conjecture 6.4 (AIM-COMBINATORICS-0164), not an independent problem.

AIM-COMBINATORICS-0169 | A surplus lattice for Seymour and Sullivan neighborhood inequalities

Truncated transition sentence before Conjecture 6.5, not a standalone problem.

AIM-COMBINATORICS-0170 | The exact interpolation among second-neighborhood inequalities

Second half of a truncated transition sentence before Conjecture 6.5, not a standalone problem.

AIM-COMBINATORICS-0173 | A tight overlap-core reduction for Conjecture 6.8

Truncated or non-self-contained workshop heading/fragment

AIM-COMBINATORICS-0175 | A cut-critical kernel for the vertexwise rainbow-cycle conjecture

Truncated or non-self-contained workshop heading/fragment

AIM-COMBINATORICS-0270 | Malformed perfect-graph workshop mega-record and an NP-certificate bridge

Unrecoverable AIM statement:

AIM-COMBINATORICS-0278 | The chromatic-to-divisibility implication hidden in a source fragment

Truncated or non-self-contained workshop heading/fragment

AIM-COMBINATORICS-0279 | An extraction fragment and an exact triangle-free list-coloring corollary

Mid-sentence PDF extraction splice ('Conjecture 2. Let G be an even-hole-free graph.'); actual Conjecture 2 is AIM-COMBINATORICS-0278

AIM-COMBINATORICS-0283 | Block-local safe edges for the Conforti--Rao conjecture

Mid-sentence PDF extraction fragment ('Conjecture 2 from the section Balanced Graphs.')

AIM-COMPUTATION-0001 | Overmerged factorization questions and a batched splitting criterion

Unrecoverable AIM statement:

### AIM-COMPUTATION-0002 | A malformed composite record and an Artin--Schreier certificate family

Unrecoverable AIM statement:

### AIM-COMPUTATION-0004 | Exact ridge reduction for Lévy PIDEs

Participant research-interest statement (Appendix A.1, Mariko Arisawa), not a standalone open problem.

### AIM-COMPUTATION-0005 | Nuisance-adjusted information for semiparametric MRS design

Participant bio/discussion note (Appendix A.2, Enrico Capobianco), not a standalone mathematical problem.

### AIM-COMPUTATION-0006 | Bellman residuals, policy performance, and coverage

Participant research-interest statement (Appendix A.3, Daniela de Farias), not a standalone open problem.

### AIM-COMPUTATION-0007 | A finite-sample Rao-Blackwell diagnostic between Kalman and particle filtering

Talk outline (Appendix A.4, Simon Haykin), not an open problem.

### AIM-COMPUTATION-0008 | Exact support and trace-controlled ellipsoidal reachability

Truncated participant note (Appendix A.5, Alexander Kurzhanski), not a standalone open problem.

### AIM-COMPUTATION-0009 | From an approximate HJB value to implemented control

Participant research-interest statement (Appendix A.6, Harold Kushner), not a standalone open problem.

### AIM-COMPUTATION-0010 | Certified weak-coupling tubes for structured reachability

Participant research-interest statement (Appendix A.7, Ian Mitchell), not a standalone open problem.

### AIM-COMPUTATION-0012 | An intrinsic-dimension certificate for projected Hamilton--Jacobi equations

Participant research-interest statement (Appendix A.9, Stanley Osher), not a standalone open problem.

### AIM-COMPUTATION-0013 | Coordinate-covariant orientation laws for no-trade regions

Participant research-interest statement (Appendix A.10, Daniel Ostrov), not a standalone open problem.

### AIM-COMPUTATION-0014 | When frequency selectivity can and cannot improve multi-user capacity

Merged open-discussion session notes on wireless communication standards, not a standalone mathematical problem.

### AIM-COMPUTATION-0015 | What is actually exponential in single-carrier equalization?

Merged open-discussion session notes on single-carrier vs. multi-carrier equalization, not a standalone mathematical problem.

### AIM-COMPUTATION-0062 | An exact witness and certificate for a recorded negative answer

Status note ('Question 47 was answered in the negative by Bernd Sturmfels at the conference') attached to AIM-COMPUTATION-0061, not a standalone problem.

### AIM-CONVEX_GEOMETRY-0012 | Four recovered problems and a normalization obstruction

Multi-problem workshop PDF dump or non-problem placeholder

### AIM-CONVEX_GEOMETRY-0042 | Six recovered convex-geometric problems and an inradius lattice-point sandwich

Multi-problem workshop PDF dump or non-problem placeholder

### AIM-CRYPTOGRAPHY-0006 | A reproducible challenge-instance profile

Community benchmark/challenge instance call rather than a mathematical open problem

### AIM-CRYPTOGRAPHY-0007 | A capability-sound kernel for group-based protocol implementations

Software implementation library request rather than a mathematical open problem

### AIM-DYNAMICAL_SYSTEMS-0006 | Letter involutions in no-repeat subshifts

Unrecoverable AIM statement:

### AIM-DYNAMICAL_SYSTEMS-0018 | A centralizer-density invariant for tree automorphisms

Truncated or non-self-contained workshop heading/fragment

### AIM-GEOMETRIC_GROUP_THEORY-0050 | Exact endomorphism monoids for the Artin family Z^k x F_n

Truncated or non-self-contained workshop heading/fragment

### AIM-GEOMETRIC_GROUP_THEORY-0052 | Eliminating the center in the CAT(0) braid-group problem

Duplicate of canonical problem AMR-061-0102

AMR-061-0102 / p.800

### AIM-GEOMETRIC_GROUP_THEORY-0080 | Universal rank-two injectivity for the generalized Tits map

Truncated or non-self-contained workshop heading/fragment

### AIM-GEOMETRIC_GROUP_THEORY-0087 | Exact escape of exponent rectangles in Thompson's group F

Duplicate of canonical problem ALG-009

ALG-009 / p.211

### AIM-GEOMETRIC_GROUP_THEORY-0147 | Editorial bridge to the Out(F_n) curve-complex-analogue problems

Unrecoverable AIM statement:

### AIM-GEOMETRIC_GROUP_THEORY-0150 | A PDF section boundary and a reverse-stretch diagnostic for Outer Space

Mid-sentence PDF extraction fragment ('Problem 1.1?') plus Section 2 introduction; actual question is AIM-GEOMETRIC_GROUP_THEORY-0151

### AIM-GEOMETRY-0085 | Finite exact certification of numerical Cohn--Elkies bounds

Truncated or non-self-contained workshop heading/fragment

### AIM-GEOMETRY-0204 | A Hadamard row-deletion sieve and a 2026 phase-retrieval resolution

Multi-problem workshop PDF dump or non-problem placeholder

### AIM-GEOMETRY-0223 | A spliced rigidity transition and a topological branch-locus reduction

Spliced transition fragment between Sections 5.1 and 5.2 of the geodesics problem list, not a standalone problem.

### AIM-GEOMETRY-0227 | Source-boundary audit and invariant ledger for geodesic-flow conjugacy

Commentary continuation after Conjecture 5.2.1 (AIM-GEOMETRY-0226), not a standalone problem.

### AIM-GEOMETRY-0229 | Entropy-volume context and an exact scaling/cover ledger

Literature status and transition fragment following Conjecture 5.3.1 (AIM-GEOMETRY-0228), not a standalone problem.

### AIM-GEOMETRY-0233 | A transition fragment and bounded-degree simple lifts

Concatenated transition sentence and section header ('Question 5.4.1 above asks whether there are other examples. 5.6. Infinitely many simple closed geodesics.'), not a standalone problem.

### AIM-GEOMETRY-0237 | Flat strips, counting conventions, and a width bound

Discussion continuation following Question 6.2.1 (AIM-GEOMETRY-0236), not an independent question.

### AIM-GEOMETRY-0279 | The Cohn-Elkies implication in dimensions 8 and 24

Single commentary sentence ('Question 1 implies the complete solution to the sphere packing problem in dimensions 8 and 24.') attached to AIM-GEOMETRY-0278, not a standalone problem.

### AIM-GEOMETRY-0288 | Status of the hyperkähler Kirwan comments and a plus-construction bridge

Extracted commentary (Comments 1.3-1.5) following Conjecture 1.1 (AIM-GEOMETRY-0287), not a standalone problem.

### AIM-GEOMETRY-0299 | Contact convexity context and properness on a symplectization

Literature status commentary on Question 5.8 (AIM-GEOMETRY-0298) spliced with the next section header, not a standalone problem.

### AIM-GEOMETRY-0300 | A boundary fragment and the specialization behind it

Incomplete sentence fragment ('Conjecture 1.1, but we state it again here in T. Hausel\'s formulation, for good measure:') preceding AIM-GEOMETRY-0301.

### AIM-GEOMETRY-0344 | A finite recognition certificate for planar tropical fans

Outline bullet list of general research topics ('C.1 Relevant lines of research'), not a mathematical problem.

### AIM-GEOMETRY-0369 | Distinguishing intrinsic and boundary order-three conformal operators

Spliced sentence fragment from Problem 11 and background prose before Problem 12, not a standalone problem.

### AIM-GEOMETRY-0375 | Global descent and flat-torus classification for Lorentzian spinors

Unrecoverable AIM statement:

### AIM-GEOMETRY-0392 | Exact-divergence form of a global conformal invariant

Mid-sentence split fragment from Problem 31 (AIM-GEOMETRY-0391), not a standalone problem.

### AIM-GEOMETRY-0394 | A source and tractor-slot audit of Other routes to Q

Page-split fragment combining the end of Problem 32 with the background paragraph of Problem 33, not a standalone problem.

### AIM-GEOMETRY-0397 | Generalized Q-operators on closed forms

Explanatory transition sentence between Problem 34 and Problem 35, not an open problem.

### AIM-GEOMETRY-0410 | The function in a Weinstein structure: terminology and minimal data

Multi-problem workshop PDF dump or non-problem placeholder

### AIM-GEOMETRY-0428 | A Fourier--Dolbeault model for Kohn--Rossi geometry

One-sentence talk log ('Question 1.32. Akahori discussed some problems involving understanding the geometry of Kohn-Rossi cohomology.') with no problem statement.

### AIM-GEOMETRY-0431 | Source-boundary correction and a twisted-cohomology calculation

Talk log note ('Question 1.35. Eliashberg made some remarks on the above topics...') with no standalone problem statement.

### AIM-GEOMETRY-0436 | A two-scale linking certificate for Reeb flows

One-sentence talk log ('Question 1.40. Mitsumatsu discussed something about linking numbers...') with no problem statement.

### AIM-GEOMETRY-0437 | A structured audit of what HF=HM preserves

One-sentence talk log ('Question 1.41. Yi-Jen Lee gave some further discussion...') with no problem statement.

### AIM-INFRASTRUCTURE-0001 | A rights-portability-cost rule for choosing educational assessment tools

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

### AIM-INFRASTRUCTURE-0002 | A deployment gate for open mathematical assessment

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

### AIM-INFRASTRUCTURE-0003 | A motivation-first deployment gate for educational technology

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

### AIM-INFRASTRUCTURE-0004 | An estimand-anchor-policy contract for educational data

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

### AIM-INFRASTRUCTURE-0005 | A version-aware evidence pipeline for input during educational-technology development

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

### AIM-INFRASTRUCTURE-0006 | Separating feedback content from the opportunity to revise

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0007 | Behavioral transfer and cycle-consistent course crosswalks

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0008 | From interaction traces to defensible collaboration evidence

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0009 | What an authenticity check can and cannot certify

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0010 | A role-specific causal evaluation contract for STACK

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0011 | A version-aware design for longitudinal studies of open assessment systems

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0012 | Blueprint alignment and exam-weight sensitivity

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0013 | A format-replacement-communication audit for short online assessment

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0014 | Relating questions to student goals without pretending to know the learner

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0015 | A distance-without-displacement audit

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0016 | A Coverage--Contract--Cut certificate for educational tool suites

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0017 | Decision-safe STACK reports and a governed user community

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0018 | A versioned closed-loop audit for STACK

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0019 | A trace-validity obstruction and engagement claim certificate

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0020 | A testable training-support-renewal framework for online instructional tools

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0021 | A barrier-lattice audit for lecturer adoption of online assessment

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0022 | Rank-safe credit for collaboration inside a contest

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0023 | Separating exposure from retained mastery under a curriculum time budget

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0024 | An access-validity workflow certificate for online assessment

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0025 | A dependence-robust certificate for informal learning

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0026 | A transport-and-adaptation certificate for structured pedagogy

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0027 | Auditable collaborative reinvention without false authorship claims

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0028 | A versioned research spine for curriculum innovation

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0029 | Construct-preserving portability beyond mathematics

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0030 | A versioned localization and transport certificate

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0031 | Participation-task dual randomization for contributor support and task evaluation

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0032 | Observation sufficiency for non-computational mathematical work

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0033 | A purpose-indexed evidence certificate for mathematical tasks

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0034 | A versioned robustness certificate for positive task impact

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0035 | Calibration-preserving encouragement in assessment feedback

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0036 | Sharp claim bounds for mathematical engagement

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0037 | A causal and decision framework for engaging institutions

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0038 | Certified open-world routing for a database of databases

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0039 | A censored lineage-and-cut audit for open-source sustainability

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0040 | A replicable and inclusive developer on-ramp

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0041 | A certifiable registry profile for mathematical visualizations

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0042 | A typed plural-paper certificate for mathematical communication

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0043 | A semantics-bound certified bridge between Lean and computer algebra

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0044 | A scope-locked robust decision certificate for Space Math

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0045 | Ordered proof traces for sound educational interoperability

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0046 | A claim-limited audit contract for open knowledge tracing

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0047 | A dual-fiber adequacy contract for the LaTeX ecosystem

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0048 | Provenance-closed workbook extraction

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0049 | Version-pinned workspace semantics for activities

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0050 | Context closure for stand-alone PreTeXt worksheets

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0051 | Context-closed fragments and version-pinned overlays

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0052 | A commuting dual-output contract for worksheet exercises

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0053 | A query-complete syllabus model for PreTeXt

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0054 | A role-separated worksheet portability contract

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0055 | Two closure tests for one-file sharing

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0056 | Certified landing-page staging for PreTeXt

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0057 | A semantic contract for journal-style theorem output

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0058 | A loss-accounted PreTeXt-to-JATS contract

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0059 | A contract for a truthful PreTeXt conversion atlas

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0060 | A sound resolver contract for links from extracted documents to a full PreTeXt publication

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0061 | Coherent numbering for extracted PreTeXt content

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0062 | Destination-aligned text for extracted cross-references

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0063 | A certifiable bridge for intermediate PreTeXt users

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0064 | A version-pinned audit of unfinished Guide signals

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0065 | An edge-complete certificate for type-specific quick starts

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0066 | Witnessed discovery for PreTeXt samples

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0067 | Residual-context certificates for PreTeXt snippets

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0068 | Closure-certified copyable PreTeXt starters

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0069 | Certified LaTeX/Markdown import into PreTeXt

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0070 | Context-aware certified selection conversion

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0071 | A dependency-closed LaTeX subset for certified PreTeXt import

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0072 | Profile-scoped evidence for PreTeXt interactivity and accessibility

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0073 | A certified component-scaffolding contract for course syllabi and worksheets

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0074 | Hermetic modules for sharing PreTeXt policy and style

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0075 | A safe publication-first launch contract

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0076 | A release contract for private instructor publishing

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0077 | Collision-safe assembly of legacy worksheets

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0078 | A versioned capability contract for PreTeXt programming languages

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0079 | Task portability for heterogeneous embeddables

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0080 | A semantic degradation contract and state lower bound for EPUB knowl fallbacks

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0081 | Certifying offline PreTeXt slides on removable media

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0082 | Coherent publication of a course slide collection

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0083 | A deterministic contract for parallel Macaulay2 core routines

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0084 | Certified portfolios for a responsive Macaulay2 top level

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0085 | A degree-slice fast path for basis

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0086 | Certified degree windows for truncated algebra commands

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0087 | A versioned contract for reproducible Macaulay2 family benchmarks

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0088 | A localization-safe action of Weyl operators

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0092 | A fixed-basis semibasis certificate for DG-module interoperability

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0093 | Simplicial resolutions in Macaulay2 and a join-complete verification contract

Macaulay2 software feature request rather than a mathematical open problem

AIM-INFRASTRUCTURE-0094 | Truncation representatives, semantic equality certificates, and functorial cohomology maps

Macaulay2 software data-type engineering question rather than a mathematical open problem

AIM-INFRASTRUCTURE-0096 | A consent-bounded two-channel and four-lane community response to harassment

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0097 | A gated equity-action registry for mathematics organizations

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0098 | A provenance-bounded design for the history of women in combinatorics

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0099 | Locally governed translation-before-transfer for non-US departmental equity conversations

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0100 | An event access assurance case for mathematics

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0101 | A mobility-and-care adjusted compensation and incidence framework

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0102 | A co-governed category-and-access charter beyond a single gender binary

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0103 | A choice-point audit for social framing in mathematical practice

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0104 | Breach-centered sampling of IBL classroom culture

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0105 | Identification and repair of an AB/BA proof-learning study

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-INFRASTRUCTURE-0106 | A causal design for long-term effects of Modified Moore Method instruction on baseline-strong students

Non-mathematical workshop administration / infrastructure / UI / pedagogy item

AIM-LOGIC-0026 | Questions 10--14 and a full quantifier collapse in literal dual covering

Multi-problem workshop PDF dump or non-problem placeholder

AIM-LOGIC-0078 | Unrecovered Jarden questions and a Galois-core decidability obstruction

Placeholder heading ('Question 18 (Jarden). Various questions from Jarden\'s talk.') with no mathematical problem statement.

AIM-LOGIC-0114 | Fermat's little theorem and a bounded modular-factorial code

Multi-problem workshop PDF dump or non-problem placeholder

AIM-LOGIC-0119 | What the Drinfeld commutator actually defines

Technical remark attached to Question 5 of Pheidas (AIM-LOGIC-0118), not a standalone problem.

AIM-LOGIC-0149 | A branch-profile audit of the CBH participant note

Participant bio/interest statement (Appendix A.1, David Aspero), not an open problem.

AIM-LOGIC-0150 | A forcing diagnostic for CH and Omega-logic

Participant bio/interest statement (Appendix A.2, Andrew Brooke-Taylor), not an open problem.

AIM-LOGIC-0151 | A parameter-exact sharp-to-perfect-set bridge

Participant bio/interest statement (Appendix A.3, Elizabeth Brown), not an open problem.

AIM-LOGIC-0153 | Games, distributivity, and stationarity preservation

Multi-problem workshop PDF dump or non-problem placeholder

AIM-LOGIC-0154 | From Jensen covering to a singular-cardinal obstruction

Participant bio/interest statement (Appendix A.6, Mirna Dzamonja), not an open problem.

AIM-LOGIC-0155 | A limit-gap obstruction for partial iteration strategies

Participant bio/interest statement (Appendix A.7, Gunter Fuchs), not an open problem.

AIM-LOGIC-0156 | The relativized Church-Kleene boundary as a fine-structure bridge

Participant bio/interest statement (Appendix A.8, Noam Greenberg), not an open problem.

AIM-LOGIC-0157 | An internal coding criterion for HOD to satisfy V = HOD

Participant bio/interest statement (Appendix A.9, Peter Koellner), not an open problem.

AIM-LOGIC-0158 | Local branch hypotheses as uniqueness and gluing principles for strategies

Participant bio/interest statement (Appendix A.10, Grigor Sargsyan), not an open problem.

AIM-LOGIC-0159 | A delegated problem list and a ZFC envelope for Theta of L(R)

Broken URL pointer in participant appendix (Appendix A.11, Ernest Schimmerling), not a self-contained problem.

AIM-LOGIC-0160 | A necessary HOD diagnostic inside determinacy models

Participant bio/interest statement (Appendix A.12, Yasuo Yoshinobu), not an open problem.

AIM-OTHER-0008 | Source-boundary repair and a quantitative resonance profile

Spliced discussion fragment following Problem 2.4 and section header 2.4, not a standalone problem.

AIM-OTHER-0068 | A spurious contents-page match and an entropy obstruction to strict local rigidity

Multi-problem workshop PDF dump or non-problem placeholder

AIM-OTHER-0071 | A section-boundary fragment and a periodic-data obstruction

Literature status fragment following Conjecture 2 (AIM-OTHER-0070), not a standalone problem.

AIM-OTHER-0073 | A truncated transitivity status sentence and a Fourier mixing criterion

Mid-sentence status fragment following Conjecture 3 (AIM-OTHER-0072), not a standalone problem.

AIM-OTHER-0074 | Context around Anosov classification: transitivity and smooth-conjugacy obstructions

Continuation of the status commentary on Conjecture 3 (AIM-OTHER-0072), not a standalone problem.

AIM-OTHER-0075 | Higher-rank conjugacy regularity: scope, obstruction, and a toral affine criterion

Mid-sentence continuation of the commentary on Conjecture 3 (AIM-OTHER-0072), not a standalone problem.

AIM-OTHER-0077 | Higher-rank rigidity context and a finite-phase circle normal form

Mid-sentence status fragment following Conjecture 4 (AIM-OTHER-0076), not a standalone problem.

AIM-OTHER-0079 | Open-dense classification and the equivariant blow-up obstruction

Mid-sentence status fragment following Conjecture 5 (AIM-OTHER-0078), not a standalone problem.

AIM-OTHER-0090 | Boundary fragment and rounding-sensitive Kronecker sampling

Mid-sentence status commentary following Question 15 (AIM-OTHER-0089), not a standalone problem.

AIM-OTHER-0093 | Exact certificates for Shah's rank-one curve condition

Explanatory paragraph describing N. Shah's partial result on Question 17 (AIM-OTHER-0092), not a standalone problem.

AIM-OTHER-0097 | A quantitative fixed-horocycle bridge to the critical simultaneous limit

Introductory setup paragraph between Conjecture 20 and Conjecture 21 (AIM-OTHER-0098), not a standalone problem.

AIM-OTHER-0099 | The critical exponent is a quantifier boundary, not a new conjecture

Mid-sentence status fragment between Conjecture 21 and Conjecture 22, not a standalone problem.

AIM-OTHER-0100 | Source-boundary recovery and a fast/slow suspension model

Spliced status fragment between Conjecture 21 and Section 5.1, not a standalone problem.

AIM-OTHER-0104 | Entropy and no escape as a convergence criterion

Mid-sentence commentary fragment following Question 24 (AIM-OTHER-0103), not a standalone problem.

AIM-OTHER-0105 | Status fragment for equidistribution of compact flats

Mid-sentence status fragment following Question 24 (AIM-OTHER-0103), not a standalone problem.

AIM-OTHER-0106 | A quantitative Borel-Cantelli estimate for VWA matrices

Mid-sentence status fragment following Question 24 (AIM-OTHER-0103), not a standalone problem.

AIM-OTHER-0110 | Bridge from Question 27 to Littlewood's conjecture

Single transition sentence ('Question 27 is related to the well-known conjecture of Littlewood:') before Conjecture 28, not a standalone problem.

AIM-OTHER-0112 | The exact logical consequence of the EKL Littlewood exceptional-set theorem

Mid-sentence status fragment on Conjecture 28 (AIM-OTHER-0111), not a standalone problem.

AIM-OTHER-0113 | The Cassels-Swinnerton-Dyer bridge from Margulis to Littlewood

Single transition sentence ('Conjecture 28 is implied by the following conjecture:') before Conjecture 29, not a standalone problem.

### AIM-OTHER-0115 | Logical separation of the neighboring Margulis conjectures

Background/transition paragraph between Conjecture 29 and Question 30, not a standalone problem.

### AIM-OTHER-0117 | Auditing the compact-orbit/Markov-tail bridge

Single transition sentence ('Question 30 is equivalent to the following question:') before Question 31, not a standalone problem.

### AIM-OTHER-0119 | The all-weights strategy for Littlewood's conjecture

Transition fragment between Conjecture 31 and Conjecture 32, not a standalone problem.

### AIM-OTHER-0121 | A transition from weighted approximation to diagonal flows

Single transition sentence ('Conjecture 32 is related to the following conjecture: OPEN PROBLEMS 14') before Conjecture 33, not a standalone problem.

### AIM-OTHER-0123 | The weighted-cone and opposite-Weyl-chamber cases of Conjecture 33

Incomplete sentence fragment following Conjecture 33 (AIM-OTHER-0122), not a standalone problem.

### AIM-OTHER-0124 | A continuation fragment about opposite Weyl chambers and full dimension

Mid-sentence fragment following Conjecture 33 (AIM-OTHER-0122), not a standalone problem.

### AIM-OTHER-0125 | Full dimension for the lattice solutions of Conjecture 33

Spliced sentence ending and section setup before Conjecture 34 (AIM-OTHER-0126), not a standalone problem.

### AIM-OTHER-0127 | Progression lifting gives a four-dimensional shrinking-gap family

Status commentary fragment following Conjecture 34 (AIM-OTHER-0126), not a standalone problem.

### AIM-OTHER-0130 | Non-isolation of the second inhomogeneous minimum

Status commentary and transition sentence between Conjecture 36 and Question 37, not a standalone problem.

### AIM-OTHER-0134 | Watson's conditional rate in arithmetic quantum unique ergodicity

Mid-sentence status fragment on Conjecture 39 (AIM-OTHER-0133), not a standalone problem.

### AIM-OTHER-0135 | A height-moment criterion for recovering escaped arithmetic quantum mass

Mid-sentence status fragment on Conjecture 39 (AIM-OTHER-0133), not a standalone problem.

### AIM-OTHER-0137 | Projective normalization for continuous-spectrum quantum measures

Mid-sentence status fragment on Problem 40 (AIM-OTHER-0136), not a standalone problem.

### AIM-OTHER-0138 | Higher-rank arithmetic QUE and a product-joining obstruction

Spliced status sentence on Conjecture 39 and setup paragraph for Conjecture 41, not a standalone problem.

### AIM-OTHER-0140 | Evidence sentence and a correlation obstruction to rate CLTs

Mid-sentence transition fragment between Conjecture 41 and Conjecture 42, not a standalone problem.

### AIM-OTHER-0141 | Semiclassical setup and commutator-coboundary invariance

Setup definitions fragment preceding Conjecture 42 (AIM-OTHER-0142), not a standalone problem.

### AIM-OTHER-0146 | Coarea normalization for the Question 45 continuation

Mid-sentence continuation fragment between Question 45 and Conjecture 46, not a standalone problem.

### AIM-OTHER-0147 | Shared-anchor obstructions for a two-gap quadratic-form fragment

Five-word boundary fragment ('Conjecture 45 is dense in R2.7. Polygonal billiards'), not a standalone problem.

### AIM-OTHER-0151 | A split status sentence for Weiss's divergent-orbit conjecture

Incomplete status sentence fragment following Conjecture 48 (AIM-OTHER-0150), not a standalone problem.

### AIM-OTHER-0152 | The one-parameter and full-torus status cases of Conjecture 48

Mid-sentence status fragment following Conjecture 48 (AIM-OTHER-0150), not a standalone problem.

### AIM-OTHER-0153 | The general rational-rank obstruction and a finite-isogeny audit

Mid-sentence status fragment following Conjecture 48 (AIM-OTHER-0150), not a standalone problem.

### AIM-OTHER-0154 | The former exceptional diagonal subgroup in SL_4

Status commentary and transition paragraph between Conjecture 48 and Question 49, not a standalone problem.

### AIM-OTHER-0158 | The GRH curve theorem and a Hecke-correlated CM conductor ray

Single status sentence on Conjecture 51 (AIM-OTHER-0157), not a standalone problem.

### AIM-OTHER-0159 | Equidistribution as a route to special closure

Explanatory motivation fragment following Conjecture 51 (AIM-OTHER-0157), not a standalone problem.

### AIM-OTHER-0161 | Duke packets, conductors, and stable discriminant windows

Truncated or non-self-contained workshop heading/fragment

### AIM-OTHER-0164 | Dimension correction and unconditional arithmeticity bridge

Truncated or non-self-contained workshop heading/fragment

### AIM-OTHER-0165 | Dimension and normalization of eligible horospherical lattices

Mid-sentence setup fragment preceding Question 54 (AIM-OTHER-0166), not a standalone problem.

AIM-PDES-0005 | Exact SQG sector and sharp interior-PV boundary drift

Truncated or non-self-contained workshop heading/fragment

AIM-PDES-0022 | Landau equation: context recovery and a structure-preserving kernel audit

Truncated or non-self-contained workshop heading/fragment

AIM-PDES-0023 | A virial clock for homogeneous aggregation-diffusion

Truncated or non-self-contained workshop heading/fragment

AIM-PDES-0024 | Applied Models heading recovery and a gang-model spectral benchmark

Truncated or non-self-contained workshop heading/fragment

AIM-PDES-0066 | Branch-aware well balancing and a resonance obstruction for oscillatory forcing

Participant research-interest statement (Appendix A.1, Debora Amadori), not a standalone open problem.

AIM-PDES-0067 | A viscous duality benchmark for particle methods with point sources

Participant research-interest statement (Appendix A.2, Alina Chertock), not a standalone open problem.

AIM-PDES-0068 | Uniform-time matrix-viscosity estimates at a fully damped equilibrium

Participant research-interest statement (Appendix A.3, Cleopatra Christoforou), not a standalone open problem.

AIM-PDES-0069 | Energy/AP audit and a kinetic-cone counterexample for a stiff two-moment model

Participant research-interest statement (Appendix A.4, Bruno Després), not a standalone open problem.

AIM-PDES-0070 | Well-balanced reconstruction for hyperbolic chemotaxis

Contributed-talk abstract (Appendix A.5, Francis Filbet), not an open problem.

AIM-PDES-0071 | A smooth moment-realizability obstruction for non-Gaussian kinetic closures

Participant research statement (A.6 Irene Gamba) in Stiff Sources and Numerical Methods for Conservation Laws, not a standalone open problem

AIM-PDES-0072 | A mass-drift certificate for spherical shallow-water patch coupling

Participant research-interest statement (Appendix A.7, Anne Gelb), not a standalone open problem.

AIM-PDES-0073 | A commutator and equilibrium-projection obstruction for stiff splitting

Participant research-interest statement (Appendix A.8, Margot Gerritsen and Rami Younis), not a standalone open problem.

AIM-PDES-0074 | Exact interface balance and a raw-density current diagnostic for semiconductor drift-diffusion

Participant research-interest statement (Appendix A.9, Cory Hauck), not a standalone open problem.

AIM-PDES-0075 | Flux-conservative Liouville scattering at a potential step

Participant research-interest statement (Appendix A.10, Shi Jin), not a standalone open problem.

AIM-PDES-0076 | A boundary-work concentration diagnostic for shear-band blow-up

Participant research-interest statement (Appendix A.11, Theodoros Katsaounis), not a standalone open problem.

AIM-PHYSICS-0006 | Completeness status and the singular small-exponent limit

Multi-page workshop dump spanning contributions from five participants rather than a single problem statement.

AIM-PHYSICS-0010 | A one-coupling J-Schur certificate for projected spectral enclosures

Exact duplicate of AIM-ANALYTIC_NUMBER_THEORY-0077 (same AIM workshop list extracted under two subjects)

AIM-ANALYTIC_NUMBER_THEORY-0077 / p.145

AIM-PHYSICS-0011 | When holomorphic anomaly equations survive fiber integration

Exact duplicate of AIM-ANALYTIC_NUMBER_THEORY-0078 (same AIM workshop list extracted under two subjects)

AIM-ANALYTIC_NUMBER_THEORY-0078 / p.137

AIM-PHYSICS-0012 | Differential fingerprints of eta-products

Exact duplicate of AIM-ANALYTIC_NUMBER_THEORY-0079 (same AIM workshop list extracted under two subjects)

AIM-ANALYTIC_NUMBER_THEORY-0079 / p.54

AIM-PHYSICS-0013 | Holomorphic anomaly, fibrations, and the BPS multiple-cover transform

Exact duplicate of AIM-ANALYTIC_NUMBER_THEORY-0080 (same AIM workshop list extracted under two subjects)

AIM-ANALYTIC_NUMBER_THEORY-0080 / p.120

AIM-PHYSICS-0014 | Compact-fixed-locus definitions and pole obstructions for noncompact elliptic genera

Exact duplicate of AIM-ANALYTIC_NUMBER_THEORY-0081 (same AIM workshop list extracted under two subjects)

AIM-ANALYTIC_NUMBER_THEORY-0081 / p.19

AIM-PHYSICS-0015 | A normal-symbol criterion for lower-dimensional modular reduction

Exact duplicate of AIM-ANALYTIC_NUMBER_THEORY-0082 (same AIM workshop list extracted under two subjects)

AIM-ANALYTIC_NUMBER_THEORY-0082 / p.17

AIM-PHYSICS-0016 | Coefficientwise indices and a level-two Calabi--Yau/CFT test

Exact duplicate of AIM-ANALYTIC_NUMBER_THEORY-0083 (same AIM workshop list extracted under two subjects)

AIM-ANALYTIC_NUMBER_THEORY-0083 / p.40

AIM-PHYSICS-0017 | Paramodular consistency tests for Calabi-Yau counting series

Exact duplicate of AIM-ANALYTIC_NUMBER_THEORY-0084 (same AIM workshop list extracted under two subjects)

AIM-ANALYTIC_NUMBER_THEORY-0084 / p.173

AIM-PHYSICS-0018 | Normalization-aware leading quintic amplitudes at the Gepner point

Exact duplicate of AIM-ANALYTIC_NUMBER_THEORY-0085 (same AIM workshop list extracted under two subjects)

AIM-ANALYTIC_NUMBER_THEORY-0085 / p.162

### AIM-PHYSICS-0019 | Hurwitz class numbers, CM enumeration, and local conductor layers

Exact duplicate of AIM-ANALYTIC_NUMBER_THEORY-0086 (same AIM workshop list extracted under two subjects)

AIM-ANALYTIC_NUMBER_THEORY-0086 / p.134

### AIM-PHYSICS-0020 | Formal, Jacobi, and wave-function transformations of the string coupling

Exact duplicate of AIM-ANALYTIC_NUMBER_THEORY-0087 (same AIM workshop list extracted under two subjects)

AIM-ANALYTIC_NUMBER_THEORY-0087 / p.50

### AIM-PHYSICS-0021 | Mirror-quintic periods, modular-type structures, and a conifold obstruction

Exact duplicate of AIM-ANALYTIC_NUMBER_THEORY-0088 (same AIM workshop list extracted under two subjects)

AIM-ANALYTIC_NUMBER_THEORY-0088 / p.149

### AIM-PHYSICS-0022 | Functional equations and an Euler-product obstruction for mixed mock L-series

Exact duplicate of AIM-ANALYTIC_NUMBER_THEORY-0089 (same AIM workshop list extracted under two subjects)

AIM-ANALYTIC_NUMBER_THEORY-0089 / p.178

### AIM-PHYSICS-0023 | Hauptmodul differential equations for an even-weight mixed-mock class

Exact duplicate of AIM-ANALYTIC_NUMBER_THEORY-0090 (same AIM workshop list extracted under two subjects)

AIM-ANALYTIC_NUMBER_THEORY-0090 / p.22

### AIM-PHYSICS-0024 | Mock modularity, geometric invariants, and finite pole tomography

Exact duplicate of AIM-ANALYTIC_NUMBER_THEORY-0091 (same AIM workshop list extracted under two subjects)

AIM-ANALYTIC_NUMBER_THEORY-0091 / p.150

### AIM-PHYSICS-0028 | Rellich compactness and spectrum on open manifolds

Recorded answer/commentary fragment attached to the preceding L^2-harmonic forms question, not a standalone problem.

### AIM-PHYSICS-0029 | Melrose's answer, compact Sobolev embedding, and escape of mass

Corrupted answer/boundary fragment ('3. (Melrose) There is no good general answer.'), not a standalone problem.

### AIM-PROBABILITY-0080 | Eden conventions, exponential FPP clocks, and an exact finite-cluster comparison

Truncated or non-self-contained workshop heading/fragment

### AIM-PROBABILITY-0140 | A recovered Coxeter-Catalan context fragment and a parabolic face bridge

Mid-sentence literature commentary fragment split from Problem 3.2 (AIM-PROBABILITY-0139), not a standalone problem.

### AIM-PROBABILITY-0144 | A split source fragment and the dihedral noncrystallographic H-triangle

Mid-sentence continuation fragment split from Problem 1.2 (AIM-PROBABILITY-0143), not a standalone problem.

### AIM-PROBABILITY-0146 | Fuss-nonnesting orders, a rank-one obstruction, and a Boolean q-model

Mid-sentence commentary fragment split from Problem 1.3 (AIM-PROBABILITY-0145), not a standalone problem.

### AIM-PROBABILITY-0147 | An extraction boundary and stable edge coefficients

Four-word section boundary artifact ('Problem 2.1 below. 2. Enumerative Combinatorics'), not a problem.

### AIM-PROBABILITY-0149 | Diagonal coinvariants beyond type A and an explicit Boolean model

Truncated ten-word prefix of Problem 2.2 (whose body is in AIM-PROBABILITY-0150), not a standalone problem.

### AIM-PROBABILITY-0167 | Flag and homology obstructions to a polytopal skeleton realization

Mid-sentence status fragment split from Problem 6.3 (AIM-PROBABILITY-0166), not a standalone problem.

### AIM-PROBABILITY-0168 | Higher Cohen-Macaulay connectivity of generalized cluster complexes

Truncated or non-self-contained workshop heading/fragment

### AIM-REPRESENTATION_THEORY-0071 | Lurie's twisted Whittaker conjecture: status and a torus formulation test

Unrecoverable AIM statement:

### AIM-TOPOLOGY-0174 | Royden motivation for Thurston-norm rigidity: source recovery and a globalization reduction

Introductory motivation fragment split from Problem 1.1 (AIM-TOPOLOGY-0173), not a standalone problem.

### AIM-TOPOLOGY-0204 | Critical-exponent exhaustions and the ergodic endpoint

Commentary remark attached to Problem 4.3 (AIM-TOPOLOGY-0203), not a standalone problem.

### AIM-TOPOLOGY-0211 | Monday section aggregate and the overtwisted absorber principle

Multi-problem workshop PDF dump or non-problem placeholder

### AIM-TOPOLOGY-0212 | Tuesday problem-session bundle and primitive height in a symplectization

Multi-problem workshop PDF dump or non-problem placeholder

### AIM-TOPOLOGY-0213 | Friday contact-topology session and Hofer-large commutators

Multi-problem workshop PDF dump or non-problem placeholder

### ALG-018 | Hilbert's Fifteenth Problem

Duplicate of canonical problem HIL-015

HIL-015 / p.686

### ALG-019 | Hilbert's Sixteenth Problem

Duplicate of canonical problem HIL-016

HIL-016 / p.879

### ALG-023 | The Andrews-Curtis Conjecture

Internal Seed duplicate of ALG-008 (id=1108, 'Andrews-Curtis Conjecture')

ALG-009 / p.211

### ALG-024 | The Bounded Burnside Problem

Internal Seed duplicate of ALG-009 (id=1109, 'Bounded Burnside Problem')

ALG-010 / p.215

### ALG-026 | The Herzog-Schönheim Conjecture

Internal Seed duplicate of ALG-010 (id=1110, 'Herzog-Schönheim Conjecture')

ALG-011 / p.247

### ALG-027 | The Inverse Galois Problem

Internal Seed duplicate of ALG-001 (id=20, 'Inverse Galois Problem')

ALG-001 / p.252

### ALG-050 | Köthe Conjecture

Internal Seed duplicate of ALG-003 (id=38, 'The Köthe Conjecture')

ALG-003 / p.303

### ALG-051 | Birch-Tate Conjecture

Internal Seed duplicate of ALG-017 (id=1160, 'Birch-Tate Conjecture')

ALG-017 / p.214

### ALG-052 | Zauner's Conjecture (SIC-POVM)

Internal Seed duplicate of ALG-006 (id=1106, 'Zauner's Conjecture')

ALG-007 / p.318

### ALG-053 | Andrews-Curtis Conjecture

Internal Seed duplicate of ALG-008 (id=1108, 'Andrews-Curtis Conjecture')

ALG-009 / p.211

### ALG-054 | Herzog-Schönheim Conjecture

Internal Seed duplicate of ALG-010 (id=1110, 'Herzog-Schönheim Conjecture')

ALG-011 / p.247

### ALG-055 | Birch-Tate Conjecture

Internal Seed duplicate of ALG-017 (id=1160, 'Birch-Tate Conjecture')

ALG-017 / p.214

### ALG-056 | Casas-Alvero Conjecture

Internal Seed duplicate of ALG-007 (id=1107, 'Casas-Alvero Conjecture')

ALG-008 / p.216

### ALG-057 | Connes Embedding Problem

Internal Seed duplicate of ALG-004 (id=1104, 'Connes Embedding Problem')

ALG-005 / p.226

### ALG-058 | Crouzeix's Conjecture

Internal Seed duplicate of ALG-004 (id=1300, 'Crouzeix's Conjecture')

ALG-041 / p.228

### ALG-059 | Köthe Conjecture

Internal Seed duplicate of ALG-003 (id=38, 'The Köthe Conjecture')

ALG-003 / p.303

### ALG-060 | Perfect Cuboid

Internal Seed duplicate of ALG-006 (id=1302, 'Perfect Cuboid')

ALG-042 / p.279

### ALG-061 | Serre's Positivity Conjecture

Internal Seed duplicate of ALG-022 (id=1182, 'Serre's Positivity Conjecture')

ALG-022 / p.293

### ALGGEOM-001 | Abundance Conjecture

Internal Seed duplicate of AG-002 (id=37, 'The Abundance Conjecture')

AG-002 / p.738

### AMR-022-1004 | Research Problems in Function Theory - Problem 1.4

Answered in the source or not an open problem (solved (no; answered in the source))

### AMR-022-1006 | Research Problems in Function Theory - Problem 1.6

Answered in the source or not an open problem (solved (first question yes, conjecture false; answered in the source))

### AMR-022-1011 | Research Problems in Function Theory - Problem 1.11

Answered in the source or not an open problem (solved (no; answered in the source))

### AMR-022-1012 | Research Problems in Function Theory - Problem 1.12

Answered in the source or not an open problem (solved (no; answered in the source))

### AMR-022-1013 | Research Problems in Function Theory - Problem 1.13

Answered in the source or not an open problem (solved (answered in the source))

### AMR-022-1017 | Research Problems in Function Theory - Problem 1.17

Answered in the source or not an open problem (solved (yes; answered in the source))

### AMR-022-1021 | Research Problems in Function Theory - Problem 1.21

Answered in the source or not an open problem (solved (yes, it can exceed 1; it is never infinite; answered in the source))

### AMR-022-1022 | Research Problems in Function Theory - Problem 1.22

Answered in the source or not an open problem (solved (no; answered in the source))

### AMR-022-1024 | Research Problems in Function Theory - Problem 1.24

Answered in the source or not an open problem (solved (yes; answered in the source))

### AMR-022-1025 | Research Problems in Function Theory - Problem 1.25

Answered in the source or not an open problem (solved (yes; answered in the source))

### AMR-022-1028 | Research Problems in Function Theory - Problem 1.28

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-1033 | Research Problems in Function Theory - Problem 1.33

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-1034 | Research Problems in Function Theory - Problem 1.34

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-1042 | Research Problems in Function Theory - Problem 1.42

Answered in the source or not an open problem (solved (answered in the source))

AMR-022-2002 | Research Problems in Function Theory - Problem 2.2

Answered in the source or not an open problem (solved (answered in the source))

AMR-022-2004 | Research Problems in Function Theory - Problem 2.4

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-2005 | Research Problems in Function Theory - Problem 2.5

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-2007 | Research Problems in Function Theory - Problem 2.7

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-2018 | Research Problems in Function Theory - Problem 2.18

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-2020 | Research Problems in Function Theory - Problem 2.20

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-2021 | Research Problems in Function Theory - Problem 2.21

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-2022 | Research Problems in Function Theory - Problem 2.22

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-2023 | Research Problems in Function Theory - Problem 2.23

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-2024 | Research Problems in Function Theory - Problem 2.24

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-2025 | Research Problems in Function Theory - Problem 2.25

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-2028 | Research Problems in Function Theory - Problem 2.28

Answered in the source or not an open problem (solved ((a) yes, (b) no; answered in the source))

AMR-022-2029 | Research Problems in Function Theory - Problem 2.29

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-2030 | Research Problems in Function Theory - Problem 2.30

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-2031 | Research Problems in Function Theory - Problem 2.31

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-2032 | Research Problems in Function Theory - Problem 2.32

Answered in the source or not an open problem (solved (no, only ce^z; answered in the source))

AMR-022-2035 | Research Problems in Function Theory - Problem 2.35

Answered in the source or not an open problem (solved (yes, A = 1; answered in the source))

AMR-022-2037 | Research Problems in Function Theory - Problem 2.37

Answered in the source or not an open problem (solved (answered in the source; yes for Polya peaks of N(r,0), no for those of log M or T))

AMR-022-2038 | Research Problems in Function Theory - Problem 2.38

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-2040 | Research Problems in Function Theory - Problem 2.40

Answered in the source or not an open problem (solved (answered in the source: Hayman's conjecture true and sharp; second question no))

AMR-022-2042 | Research Problems in Function Theory - Problem 2.42

Answered in the source or not an open problem (solved (answered in the source; b_k = 1 impossible))

AMR-022-2044 | Research Problems in Function Theory - Problem 2.44

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-2052 | Research Problems in Function Theory - Problem 2.52

Answered in the source or not an open problem (solved (answered in the source))

AMR-022-2057 | Research Problems in Function Theory - Problem 2.57

Answered in the source or not an open problem (solved (smoothness of growth suffices; answered in the source))

AMR-022-2058 | Research Problems in Function Theory - Problem 2.58

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-2065 | Research Problems in Function Theory - Problem 2.65

Answered in the source or not an open problem (solved (yes, k = 6; answered in the source))

AMR-022-2066 | Research Problems in Function Theory - Problem 2.66

Answered in the source or not an open problem (solved (independent of ZFC; answered in the source, unpublished attribution))

AMR-022-2070 | Research Problems in Function Theory - Problem 2.70

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-2072 | Research Problems in Function Theory - Problem 2.72

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-2076 | Research Problems in Function Theory - Problem 2.76

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-2086 | Research Problems in Function Theory - Problem 2.86

Answered in the source or not an open problem (solved (answered in the source; (a) via a thesis))

AMR-022-3001 | Research Problems in Function Theory - Problem 3.1

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-3002 | Research Problems in Function Theory - Problem 3.2

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-3003 | Research Problems in Function Theory - Problem 3.3

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-3006 | Research Problems in Function Theory - Problem 3.6

Answered in the source or not an open problem (solved (answered in the source))

AMR-022-3007 | Research Problems in Function Theory - Problem 3.7

Answered in the source or not an open problem (solved (answered in the source))

AMR-022-3011 | Research Problems in Function Theory - Problem 3.11

Answered in the source or not an open problem (solved (answered in the source))

AMR-022-3017 | Research Problems in Function Theory - Problem 3.17

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-3019 | Research Problems in Function Theory - Problem 3.19

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-3021 | Research Problems in Function Theory - Problem 3.21

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-3023 | Research Problems in Function Theory - Problem 3.23

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-3024 | Research Problems in Function Theory - Problem 3.24

Answered in the source or not an open problem (solved (answered in the source))

AMR-022-3025 | Research Problems in Function Theory - Problem 3.25

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-3026 | Research Problems in Function Theory - Problem 3.26

Answered in the source or not an open problem (solved (answered in the source; n >= 4 via a remark of Ancona reported there))

AMR-022-3027 | Research Problems in Function Theory - Problem 3.27

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-3032 | Research Problems in Function Theory - Problem 3.32

Answered in the source or not an open problem (solved (answered in the source))

AMR-022-4001 | Research Problems in Function Theory - Problem 4.1

Answered in the source or not an open problem (solved (answered in the source; growth rate by Beck))

AMR-022-4002 | Research Problems in Function Theory - Problem 4.2

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-4008 | Research Problems in Function Theory - Problem 4.8

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-4009 | Research Problems in Function Theory - Problem 4.9

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-4019 | Research Problems in Function Theory - Problem 4.19

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-4028 | Research Problems in Function Theory - Problem 4.28

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-4029 | Research Problems in Function Theory - Problem 4.29

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-5001 | Research Problems in Function Theory - Problem 5.1

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-5003 | Research Problems in Function Theory - Problem 5.3

Answered in the source or not an open problem (solved (answered in the source))

AMR-022-5005 | Research Problems in Function Theory - Problem 5.5

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-5007 | Research Problems in Function Theory - Problem 5.7

Answered in the source or not an open problem (solved (answered in the source))

AMR-022-5011 | Research Problems in Function Theory - Problem 5.11

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-5013 | Research Problems in Function Theory - Problem 5.13

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-5014 | Research Problems in Function Theory - Problem 5.14

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-5015 | Research Problems in Function Theory - Problem 5.15

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-5024 | Research Problems in Function Theory - Problem 5.24

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-5028 | Research Problems in Function Theory - Problem 5.28

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-5029 | Research Problems in Function Theory - Problem 5.29

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-5033 | Research Problems in Function Theory - Problem 5.33

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-5040 | Research Problems in Function Theory - Problem 5.40

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-5050 | Research Problems in Function Theory - Problem 5.50

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-5061 | Research Problems in Function Theory - Problem 5.61

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-5065 | Research Problems in Function Theory - Problem 5.65

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-6008 | Research Problems in Function Theory - Problem 6.8

Answered in the source or not an open problem (partially_solved (class S, means of f, answered in the source))

AMR-022-6011 | Research Problems in Function Theory - Problem 6.11

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-6016 | Research Problems in Function Theory - Problem 6.16

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-6036 | Research Problems in Function Theory - Problem 6.36

Answered in the source or not an open problem (partially_solved (answered in the source for p >= 1))

AMR-022-6038 | Research Problems in Function Theory - Problem 6.38

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-6041 | Research Problems in Function Theory - Problem 6.41

Answered in the source or not an open problem (solved (answered in the source))

AMR-022-6044 | Research Problems in Function Theory - Problem 6.44

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-6045 | Research Problems in Function Theory - Problem 6.45

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-6046 | Research Problems in Function Theory - Problem 6.46

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-6053 | Research Problems in Function Theory - Problem 6.53

Answered in the source or not an open problem (partially_solved (two of four questions answered in the source))

AMR-022-6059 | Research Problems in Function Theory - Problem 6.59

Answered in the source or not an open problem (partially_solved (finite-range case answered in the source))

AMR-022-6066 | Research Problems in Function Theory - Problem 6.66

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-6067 | Research Problems in Function Theory - Problem 6.67

Answered in the source or not an open problem (partially_solved (first question answered in the source))

AMR-022-6069 | Research Problems in Function Theory - Problem 6.69

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-6070 | Research Problems in Function Theory - Problem 6.70

Answered in the source or not an open problem (partially_solved (one of two questions answered in the source))

AMR-022-6073 | Research Problems in Function Theory - Problem 6.73

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-6079 | Research Problems in Function Theory - Problem 6.79

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-6085 | Research Problems in Function Theory - Problem 6.85

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-7006 | Research Problems in Function Theory - Problem 7.6

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-7009 | Research Problems in Function Theory - Problem 7.9

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-7011 | Research Problems in Function Theory - Problem 7.11

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-7012 | Research Problems in Function Theory - Problem 7.12

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-7014 | Research Problems in Function Theory - Problem 7.14

Answered in the source or not an open problem (partially_solved (first question answered in the source))

AMR-022-7017 | Research Problems in Function Theory - Problem 7.17

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-7022 | Research Problems in Function Theory - Problem 7.22

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-7030 | Research Problems in Function Theory - Problem 7.30

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-7032 | Research Problems in Function Theory - Problem 7.32

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-7040 | Research Problems in Function Theory - Problem 7.40

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-7041 | Research Problems in Function Theory - Problem 7.41

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-7043 | Research Problems in Function Theory - Problem 7.43

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-7044 | Research Problems in Function Theory - Problem 7.44

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-7045 | Research Problems in Function Theory - Problem 7.45

Answered in the source or not an open problem (solved (answered in the source))

AMR-022-7046 | Research Problems in Function Theory - Problem 7.46

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-7056 | Research Problems in Function Theory - Problem 7.56

Answered in the source or not an open problem (partially_solved (second question answered in the source))

AMR-022-7059 | Research Problems in Function Theory - Problem 7.59

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-7060 | Research Problems in Function Theory - Problem 7.60

Answered in the source or not an open problem (solved (yes to both; answered in the source))

AMR-022-7063 | Research Problems in Function Theory - Problem 7.63

Answered in the source or not an open problem (solved (yes, for the intended exponent 2n/(n-1); answered in the source))

AMR-022-7078 | Research Problems in Function Theory - Problem 7.78

Answered in the source or not an open problem (solved (no, in the plane; answered in the source))

AMR-022-7082 | Research Problems in Function Theory - Problem 7.82

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-8002 | Research Problems in Function Theory - Problem 8.2

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-022-8006 | Research Problems in Function Theory - Problem 8.6

Answered in the source or not an open problem (partially_solved (the converse is answered in the source))

AMR-022-8016 | Research Problems in Function Theory - Problem 8.16

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-8020 | Research Problems in Function Theory - Problem 8.20

Answered in the source or not an open problem (solved (no; answered in the source under Update 8.2))

AMR-022-8022 | Research Problems in Function Theory - Problem 8.22

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-022-9003 | Research Problems in Function Theory - Problem 9.3

Answered in the source or not an open problem (solved (no to both; answered in the source))

AMR-027-0103 | 10 Lectures and 42 Open Problems - Open Problem 1.3

Answered in the source or not an open problem ()

AMR-029-0080 | Smooth well-balanced orientations with prescribed in-degrees

Answered in the source or not an open problem (solved (NP-complete))

AMR-029-0089 | Well-balanced orientations of hypergraphs

Answered in the source or not an open problem (solved (NP-complete))

AMR-038-0001 | Antipodes of symmetric convex bodies

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-041-0011 | Mañé's last theorem

Answered in the source or not an open problem (solved (yes, on compact surfaces; answered in the source))

AMR-045-0022 | Stable cellular-automaton limit sets

Answered in the source or not an open problem (partially_solved (partly answered in the source))

AMR-054-0074 | Slicing Axes-Parallel Rectangles

Answered in the source or not an open problem (partially solved)

AMR-061-0035 | Boundaries of Groups and Kleinian Groups - Problem 35

Answered in the source or not an open problem (partially_solved (answered in the source except n = 4))

AMR-061-0097 | Boundaries of Groups and Kleinian Groups - Problem 97

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-067-0028 | Closed geodesics

Truncated or non-self-contained workshop heading/fragment

AMR-086-0059 | Question 4.17 - Mazur

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-086-0060 | Conjecture 4.18

Answered in the source or not an open problem (Statement fix: define V and Z.)

AMR-093-0084 | Generalized Sato-Tate conjecture

Duplicate of canonical problem AG-004 ('The Tate Conjecture')

AG-004 / p.744

AMR-093-0110 | Pillai's conjecture

Duplicate of canonical problem NT-023 ('Pillai's Conjecture')

NT-023 / p.164

AMR-093-0149 | Are there infinitely many Fibonacci primes

Duplicate of canonical problem NT-065

NT-065 [not in active corpus]

AMR-093-0164 | Are there infinitely many Wieferich primes

Duplicate of canonical problem NT-067

NT-067 / p.143

AMR-093-0165 | Are there infinitely many Wilson primes

Duplicate of canonical problem NT-066

NT-066 / p.142

AMR-093-0183 | Second Hardy-Littlewood zeta-function conjecture

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-103-0020 | Problem 1.20 - Justify the above arguments rigorously.

Truncated or non-self-contained workshop heading/fragment

AMR-103-0090 | Problem 4.12 - Construct invariants of 3-manifolds via a linear skein theo ry based on the Kauffman skein module.

Answered in the source or not an open problem (solved (answered in the source))

AMR-103-0110 | Problem 6.2 - Is the Burau representation of B4 faithful?

Duplicate of canonical problem ALG-006

ALG-006 / p.255

AMR-103-0117 | Problem 7.2 - (S.K.

Answered in the source or not an open problem (solved (answered in the source))

AMR-103-0219 | Problem 12.14 - (N.

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-109-0109 | Problem 2.4 - Let Γ be a subgroup of finite index in the mapping class group Mod1,2 and let φ: Γ → Γ be an automorphism.

Answered in the source or not an open problem (solved (no; answered in the source))

AMR-109-0146 | Question - Do there exist any other examples of non commuting homeomorphisms g andh which are not both Dehn twists and satisfy a...

Answered in the source or not an open problem (solved (yes; answered in the source))

AMR-109-0219 | Problem 12 - (Analog of Ratner theorem).

Truncated or non-self-contained workshop heading/fragment

AMR-112-0001 | Bing-Borsuk conjecture

Duplicate of canonical problem EP-505 ('Erdős Problem #505: Geometry')

EP-505 / p.846

ANA-006 | Navier-Stokes Regularity

Duplicate of canonical problem MPP-004

MPP-004 / p.1484

COMB-002 | Lonely Runner Conjecture

Internal Seed duplicate of COMB-004 (id=30, 'The Lonely Runner Conjecture')

COMB-004 / p.1747

COMB-006 | Sunflower Conjecture

Duplicate of canonical problem EP-20

EP-20 / p.1670

COMB-012 | The Hadwiger-Nelson Problem

Duplicate of canonical problem EP-508

EP-508 / p.1794

COMB-013 | The Union-Closed Sets Conjecture

Internal Seed duplicate of COMB-005 (id=55, 'Frankl's Union-Closed Sets Conjecture')

COMB-005 / p.1694

COMB-014 | Singmaster's Conjecture

Duplicate of canonical problem EP-849

EP-849 / p.108

COMB-015 | The Sunflower Conjecture

Duplicate of canonical problem EP-20

EP-20 / p.1670

COMB-016 | Ramsey Number $R(5,5)$

Internal Seed duplicate of COMB-003 (id=29, 'Ramsey Number R(5,5)')

COMB-003 / p.1722

COMB-017 | Union-Closed Sets Conjecture

Internal Seed duplicate of COMB-005 (id=55, 'Frankl's Union-Closed Sets Conjecture')

COMB-005 / p.1694

COMB-018 | 1/3-2/3 Conjecture

Internal Seed duplicate of COMB-001 (id=1319, '1/3-2/3 Conjecture')

COMB-001 / p.1620

COMB-019 | Lonely Runner Conjecture

Internal Seed duplicate of COMB-004 (id=30, 'The Lonely Runner Conjecture')

COMB-004 / p.1747

COMB-020 | Sunflower Conjecture

Duplicate of canonical problem EP-20

EP-20 / p.1670

COMB-021 | Union-Closed Sets Conjecture

Internal Seed duplicate of COMB-005 (id=55, 'Frankl's Union-Closed Sets Conjecture')

COMB-005 / p.1694

COMB-022 | Ramsey Number R(5,5)

Internal Seed duplicate of COMB-003 (id=29, 'Ramsey Number R(5,5)')

COMB-003 / p.1722

DYN-005 | Weinstein Conjecture

Internal Seed duplicate of DYN-001 (id=1141, 'The Weinstein Conjecture')

DYN-001 / p.1613

GAME-007 | Cap Set Problem

Internal Seed duplicate of COMB-010 (id=1231, 'The Cap Set Problem')

COMB-010 / p.1744

GEO-001 | Sphere Packing Problem in Higher Dimensions

Internal Seed duplicate of GEO-002 (id=19, 'Sphere Packing in Higher Dimensions')

GEO-002 / p.972

GEO-024 | The Heilbronn Triangle Problem

Duplicate of canonical problem EP-507

EP-507 / p.846

GEO-026 | The Unit Distance Problem

Duplicate of canonical problem EP-90

EP-90 / p.1812

GEO-029 | Borsuk's Conjecture

Duplicate of canonical problem EP-505

EP-505 / p.846

GEO-034 | The Inscribed Square Problem

Internal Seed duplicate of GEO-005 (id=56, 'Inscribed Square Problem (Toeplitz Conjecture)')

GEO-005 / p.883

GEO-035 | The Happy Ending Problem

Duplicate of canonical problem EP-107

EP-107 / p.844

GEOM-001 | Sphere Packing Problem Higher Dimensions

Internal Seed duplicate of GEO-002 (id=19, 'Sphere Packing in Higher Dimensions')

GEO-002 / p.972

GEOM-007 | Kakeya Conjecture

Internal Seed duplicate of GEO-003 (id=32, 'The Kakeya Conjecture')

GEO-003 / p.984

GEOM-008 | Illumination Problem

Internal Seed duplicate of GEO-003 (id=1123, 'The Illumination Conjecture')

GEO-007 / p.984

GEOM-009 | Yang-Mills Existence and Mass Gap

Duplicate of canonical problem MPP-003

MPP-003 / p.2159

GEOM-010 | Kissing Number Problem

Internal Seed duplicate of GEO-030 (id=1225, 'The Kissing Number Problem')

GEO-030 / p.984

GEOM-014 | Carathéodory Conjecture

Internal Seed duplicate of GEO-013 (id=1204, 'The Carathéodory Conjecture')

GEO-014 / p.982

GEOM-015 | Cartan-Hadamard Conjecture

Internal Seed duplicate of GEO-014 (id=1205, 'The Cartan-Hadamard Conjecture')

GEO-015 / p.982

GEOM-017 | Hopf Conjectures

Internal Seed duplicate of GEO-019 (id=1210, 'The Hopf Conjectures')

GEO-020 / p.984

GEOM-019 | Hadwiger Conjecture (Covering)

Internal Seed duplicate of GEO-022 (id=1213, 'The Hadwiger Covering Conjecture')

GEO-023 / p.983

GEOM-020 | Happy Ending Problem

Duplicate of canonical problem EP-107

EP-107 / p.844

GEOM-021 | Heilbronn Triangle Problem

Duplicate of canonical problem EP-507

EP-507 / p.846

GEOM-022 | Kalai's 3^d Conjecture

Internal Seed duplicate of GEO-025 (id=1216, 'Kalai's $3^d$ Conjecture')

GEO-025 / p.886

GEOM-024 | Unit Distance Problem

Duplicate of canonical problem EP-90

EP-90 / p.1812

GEOM-025 | Bellman's Lost-in-a-Forest Problem

Internal Seed duplicate of GEO-005 (id=1125, 'Bellman's Lost in a Forest Problem')

GEO-009 / p.788

GRAPH-029 | Cycle Double Cover Conjecture

Internal Seed duplicate of GT-004 (id=59, 'The Cycle Double Cover Conjecture')

GT-004 / p.1862

GRAPH-031 | Erdős-Hajnal Conjecture

Internal Seed duplicate of GRAPH-004 (id=1328, 'Erdős-Hajnal Conjecture')

GRAPH-004 / p.1815

GRAPH-033 | Lovász Conjecture

Internal Seed duplicate of GRAPH-005 (id=1329, 'Lovász Conjecture')

GRAPH-005 / p.1832

GRAPH-059 | Cycle Double Cover Conjecture

Internal Seed duplicate of GT-004 (id=59, 'The Cycle Double Cover Conjecture')

GT-004 / p.1862

GRAPH-060 | Hadwiger-Nelson Problem

Duplicate of canonical problem EP-508

EP-508 / p.1794

GRAPH-061 | Cereceda's Conjecture

Internal Seed duplicate of GT-008 (id=1144, 'Cereceda's Conjecture')

GT-008 / p.1767

GRAPH-062 | Total Coloring Conjecture

Internal Seed duplicate of GT-010 (id=1165, 'The Total Coloring Conjecture')

GT-010 / p.1863

GREEN-097 | N-Queens Problem Asymptotics

Exact duplicate of GREEN-061 in Ben Green's Open Problems

GREEN-061 / p.1711

### GREEN-098 | Bounds for Homogeneous Polynomial Zeros

Exact duplicate of GREEN-062 in Ben Green's Open Problems

GREEN-062 / p.37

### GREEN-099 | Polynomial Solutions in Dense Sets

Exact duplicate of GREEN-063 in Ben Green's Open Problems

GREEN-063 / p.181

### GREEN-100 | Sofic Groups

Exact duplicate of GREEN-064 in Ben Green's Open Problems

GREEN-064 / p.290

### GUY-A10 | Gilbreath's Conjecture

Duplicate of canonical problem NT-009 ('Gilbreath's Conjecture')

NT-009 / p.128

### GUY-A11 | Erdős $100 Problem on Increasing and Decreasing Gaps

Duplicate of canonical problem EP-6

EP-6 / p.103

### GUY-A13 | Erdős Conjecture on Carmichael Numbers

Duplicate of canonical problem EP-1057

EP-1057 / p.60

### GUY-A14a | Pomerance's Questions on Good Primes

Duplicate of canonical problem EP-453

EP-453 / p.92

### GUY-A15 | Congruent Products of Consecutive Numbers

Duplicate of canonical problem EP-677

EP-677 / p.100

### GUY-A16 | Walking to Infinity on Gaussian Primes

Duplicate of canonical problem EP-952

EP-952 / p.114

### GUY-A19a | Erdős Conjecture on $n - 2^k$ Prime

Duplicate of canonical problem EP-1142

EP-1142 / p.64

### GUY-A5b | Erdős $3000 Conjecture on Arithmetic Progressions

Duplicate of canonical problem EP-3

EP-3 / p.78

### GUY-A6 | Consecutive Primes in Arithmetic Progression

Duplicate of canonical problem EP-141

EP-141 / p.69

### GUY-A8a | Erdős $5000 Problem on Prime Gaps

Duplicate of canonical problem EP-4

EP-4 / p.96

### GUY-A8b | Twin Prime Conjecture

Duplicate of canonical problem LAN-002

LAN-002 / p.192

### GUY-A9 | General Patterns of Consecutive Primes

Duplicate of canonical problem HL-A

HL-A / p.131

### LOGIC-002 | Cherlin-Zilber Conjecture

Internal Seed duplicate of ALG-039 (id=1220, 'The Cherlin-Zilber Conjecture')

ALG-039 / p.302

### MOD-001 | Cherlin-Zilber Conjecture

Internal Seed duplicate of ALG-039 (id=1220, 'The Cherlin-Zilber Conjecture')

ALG-039 / p.302

### MOD-002 | Generalized Star Height Problem

Internal Seed duplicate of ALG-040 (id=1221, 'The Generalized Star Height Problem')

ALG-040 / p.303

### MOD-003 | Hilbert's Tenth Problem for Number Fields

Duplicate of canonical problem HIL-010

HIL-010 / p.134

### MOD-004 | Vaught Conjecture

Internal Seed duplicate of LOGIC-001 (id=1343, 'Vaught Conjecture')

LOGIC-001 / p.315

### NT-002 | Collatz Conjecture

Duplicate of canonical problem EP-1135

EP-1135 / p.63

### NT-010 | Brocard's Problem

Duplicate of canonical problem EP-398

EP-398 / p.87

### NT-012 | The Erdős-Straus Conjecture

Duplicate of canonical problem EP-242

EP-242 / p.72

### NT-017 | Are There Infinitely Many Mersenne Primes?

Internal Seed duplicate of NT-007 (id=26, 'Are there infinitely many Mersenne primes?')

NT-007 / p.31

### NT-024 | Erdős-Straus Conjecture

Duplicate of canonical problem EP-242

EP-242 / p.72

### NT-030 | The abc Conjecture

Internal Seed duplicate of NT-005 (id=13, 'ABC Conjecture')

NT-005 / p.24

### NT-032 | Gauss Circle Problem

Internal Seed duplicate of NT-025 (id=1159, 'The Gauss Circle Problem')

NT-025 / p.188

### NT-065 | Hilbert's Twelfth Problem

Duplicate of canonical problem HIL-012

HIL-012 / p.133

NT-076 | Littlewood Conjecture

Duplicate of canonical problem EP-495

EP-495 / p.95

NT-086 | Brocard's Conjecture (Prime Gaps)

Duplicate of canonical problem EP-398

EP-398 / p.87

NUM-001 | Singmaster's Conjecture

Duplicate of canonical problem EP-849

EP-849 / p.108

NUM-002 | Odd Perfect Numbers

Internal Seed duplicate of NT-001 (id=8, 'Odd Perfect Numbers')

NT-001 / p.161

NUM-004 | Quasiperfect Numbers

Internal Seed duplicate of NT-043 (id=1250, 'Quasiperfect Numbers')

NT-043 / p.169

NUM-006 | Odd Weird Numbers

Internal Seed duplicate of NT-050 (id=1257, 'Odd Weird Numbers')

NT-050 / p.161

NUM-010 | Gilbreath's Conjecture

Internal Seed duplicate of NT-009 (id=28, 'Gilbreath's Conjecture')

NT-009 / p.128

NUM-011 | Lander-Parkin-Selfridge Conjecture

Internal Seed duplicate of NT-058 (id=1265, 'Lander-Parkin-Selfridge Conjecture')

NT-058 / p.141

NUM-012 | Class Number Problem

Internal Seed duplicate of NT-064 (id=1271, 'Class Number Problem')

NT-064 / p.40

NUM-013 | Hilbert's 12th Problem

Duplicate of canonical problem HIL-012

HIL-012 / p.133

NUM-014 | Leopoldt's Conjecture

Internal Seed duplicate of NT-066 (id=1273, 'Leopoldt's Conjecture')

NT-066 / p.142

NUM-018 | Littlewood Conjecture

Duplicate of canonical problem EP-495

EP-495 / p.95

NUM-019 | Four Exponentials Conjecture

Internal Seed duplicate of NT-073 (id=1281, 'Four Exponentials Conjecture')

NT-073 / p.124

NUM-020 | Integer Factorization Polynomial Time

Internal Seed duplicate of NT-077 (id=1285, 'Integer Factorization in Polynomial Time')

NT-077 / p.136

OPG-137 | Cycle double cover conjecture

Duplicate of canonical problem GT-004 ('The Cycle Double Cover Conjecture')

GT-004 / p.1862

OPG-165 | Ryser's conjecture

Duplicate of canonical problem GRAPH-053 ('Ryser's Conjecture')

GRAPH-053 / p.1850

OPG-16570 | Magic square of squares

Duplicate of canonical problem NT-036 ('Magic Square of Squares')

NT-036 / p.147

OPG-1768 | Jacobian Conjecture

Duplicate of canonical problem SMA-016

SMA-016 / p.303

OPG-1788 | Schanuel's Conjecture

Duplicate of canonical problem NUM-016 ('Schanuel's Conjecture')

NUM-016 / p.176

OPG-179 | Woodall's Conjecture

Duplicate of canonical problem GRAPH-058 ('Woodall's Conjecture')

GRAPH-058 / p.1876

OPG-1790 | Tarski's exponential function problem

Duplicate of canonical problem MOD-005 ('Tarski's Exponential Function Problem')

MOD-005 / p.301

OPG-1803 | The Hodge Conjecture

Duplicate of canonical problem MPP-006 ('Hodge Conjecture')

MPP-006 / p.687

OPG-2108 | Frankl's union-closed sets conjecture

Duplicate of canonical problem COMB-005 ('Frankl's Union-Closed Sets Conjecture')

COMB-005 / p.1694

OPG-2147 | Odd perfect numbers

Duplicate of canonical problem NT-001 ('Odd Perfect Numbers')

NT-001 / p.161

OPG-2379 | Termination of the sixth Goodstein Sequence

Historical solved problem (Mersenne's Laws, 1636), not an open problem

OPG-3572 | Inverse Galois Problem

Duplicate of canonical problem ALG-001 ('Inverse Galois Problem')

ALG-001 / p.252

OPG-36952 | Twin prime conjecture

Duplicate of canonical problem LAN-002 ('Twin Prime Conjecture')

LAN-002 / p.192

OPG-37079 | A gold-grabbing game

Retired: the finite-game existence question follows from backward induction, and no algorithmic efficiency target is specified.

OPG-37089 | Shuffle-Exchange Conjecture (graph-theoretic form)

Duplicate of canonical problem OPG-37167

OPG-37167 / p.1732

OPG-37210 | Beneš Conjecture (graph-theoretic form)

Duplicate of canonical problem OPG-37181

OPG-37181 / p.1640

OPG-37221 | Perfect cuboid

Duplicate of canonical problem ALG-042 ('Perfect Cuboid')

ALG-042 / p.279

OPG-37255 | Lindelöf hypothesis

Duplicate of canonical problem NT-067 ('Lindelöf Hypothesis')

NT-067 / p.143

OPG-37283 | Finite Lattice Representation Problem

Duplicate of canonical problem ALG-046 ('Finite Lattice Representation Problem')

ALG-046 / p.238

OPG-37289 | Polignac's Conjecture

Duplicate of canonical problem NT-022 ('Polignac's Conjecture')

NT-022 / p.166

OPG-37297 | Hilbert-Smith conjecture

Duplicate of canonical problem TOP-009 ('Hilbert-Smith Conjecture')

TOP-009 / p.1076

OPG-37378 | Funcoidal products inside an inward reloid

Trivial/degenerate statement without open research content

OPG-37397 | Erdős-Straus conjecture

Duplicate of canonical problem EP-242 ('Erdős Problem #242: Number Theory, Unit Fractions')

EP-242 / p.72

OPG-37423 | Birch & Swinnerton-Dyer conjecture

Duplicate of canonical problem MPP-005 ('Birch and Swinnerton-Dyer Conjecture')

MPP-005 / p.35

OPG-385 | Barnette's Conjecture

Duplicate of canonical problem GRAPH-027 ('Barnette's Conjecture')

GRAPH-027 / p.1764

OPG-416 | Lonely runner conjecture

Duplicate of canonical problem COMB-004 ('The Lonely Runner Conjecture')

COMB-004 / p.1747

OPG-432 | The 3n+1 conjecture

Duplicate of canonical problem EP-1135 ('Erdős Problem #1135')

EP-1135 / p.63

OPG-439 | Graceful Tree Conjecture

Duplicate of canonical problem GT-003 ('The Graceful Tree Conjecture')

GT-003 / p.1862

OPG-46900 | Earth-Moon Problem

Duplicate of canonical problem GRAPH-010 ('Earth-Moon Problem')

GRAPH-010 / p.1784

OPG-50149 | inverse of an integer matrix

Non-research homework/calculus submission on Open Problem Garden

OPG-573 | The Riemann Hypothesis

Duplicate of canonical problem MPP-002

MPP-002 / p.189

OPG-57613 | Imbalance conjecture

Duplicate of canonical problem GRAPH-051 ('Imbalance Conjecture')

GRAPH-051 / p.1826

OPG-57824 | Graphs of exact colorings

Duplicate of canonical problem OPG-37229

OPG-37229 / p.1816

OPG-59908 | Fractional Hadwiger

Duplicate of canonical problem GEO-023 ('The Hadwiger Covering Conjecture')

GEO-023 / p.983

OPG-59943 | List Hadwiger Conjecture

Duplicate of canonical problem GT-001 ('Hadwiger Conjecture')

GT-001 / p.1823

OPG-60026 | Which outer reloids are equal to inner ones

Forum meta-post / non-problem entry

OPG-60034 | Singmaster's conjecture

Duplicate of canonical problem EP-849 ('Erdős Problem #849')

EP-849 / p.108

OPG-60037 | Edge-Unfolding Convex Polyhedra

Duplicate of canonical problem OPG-37459

OPG-37459 / p.839

OPG-658 | Reconstruction conjecture

Duplicate of canonical problem GT-002 ('Reconstruction Conjecture')

GT-002 / p.1848

OPG-661 | P vs. NP

Duplicate of canonical problem MPP-001 ('P versus NP Problem')

MPP-001 / p.2096

OPG-706 | Goldbach conjecture

Duplicate of canonical problem LAN-001 ('Goldbach's Conjecture')

LAN-001 / p.129

OPG-739 | Chowla's cosine problem

Duplicate of canonical problem EP-510 ('Erdős Problem #510')

EP-510 / p.1674

OPG-757 | Inscribed Square Problem

Duplicate of canonical problem GEO-005 ('Inscribed Square Problem (Toeplitz Conjecture)')

GEO-005 / p.883

OPG-778 | Cube-Simplex conjecture

Duplicate of canonical problem EP-90

EP-90 / p.1812

OPG-788 | Partial List Coloring

Duplicate of canonical problem OPG-771

OPG-771 / p.1842

### OWR-10252925-006 | Białynicki-Birula Cell Construction

Unrecoverable OWR statement (The official report passage contains definitions and a proved construction but no open mathematical claim.)

### OWR-10252926-001 | Unspecified Geometric Representation Theory Conjecture

Unrecoverable OWR statement (The official report shows that this is a table-of-contents fragment without a recoverable standalone conjecture. No separate clean statement)

### OWR-10252928-001 | Unspecified Conic Optimization Research Problem

Unrecoverable OWR statement (The official report shows that this is workshop-level introductory prose rather than a standalone conic-optimization problem. No separate cl)

### OWR-10252928-003 | Unspecified Flag Algebra Extremal Problem

Unrecoverable OWR statement (The official report gives a descriptive account of flag-algebra methods but does not formulate a standalone open extremal problem. No separa)

### OWR-10252929-001 | Zilber-Pink Unlikely Intersection Conjecture

Duplicate/merged OWR extraction (The official report contains only workshop-level context mentioning the Zilber-Pink conjecture; the complete modular formulation is retained)

OWR-10252929-008 / p.756

### OWR-10252930-002 | Talagrand's Conjecture on Selector Processes

Answered in the source or not an open problem (solved (a title line, not a question))

### OWR-10252930-010 | Ryser-Brualdi-Stein Transversal Conjecture

Duplicate/merged OWR extraction (The official report states the Ryser-Brualdi-Stein conjecture, but the identical conjecture is already represented by the earlier canonical )

OWR-16763-027 / p.1729

### OWR-10252931-002 | Partial Fermionization of Cohomological Hall Algebras

Duplicate/merged OWR extraction (The official report confirms that this is a duplicate extraction of the quiver-with-potential formulation already retained in OWR-10252931-0)

OWR-10252931-001 / p.238

### OWR-10252932-001 | Open Problems in K-Theory of Ring Pushouts

Unrecoverable OWR statement (The official source passage is a table-of-contents fragment ending in the word conjecture and contains no recoverable claim. No separate cle)

### OWR-10252932-004 | Comparison Maps in p-Adic Cohomology

Unrecoverable OWR statement (The official source passage is a damaged splice between a conjectural comparison-map fragment and a proved theorem, so it does not identify )

### OWR-10252932-008 | Hopkins-Morel Input to Motivic Stable Homotopy

Unrecoverable OWR statement (The official source passage explains that a proved main theorem follows from the Hopkins-Morel conjecture; it is not a separate unresolved p)

### OWR-10252935-003 | Foundations of Stochastic Quantization

Unrecoverable OWR statement (The phrase is the title of an expository contribution that compares established constructions of stochastic quantization, not a standalone o)

### OWR-10252936-001 | Definition and Scope of Stochastic Quantization

Unrecoverable OWR statement (The official report uses this phrase as the rhetorical title of an expository discussion and does not formulate a single mathematical questi)

### OWR-1052-001 | Weinstein Conjecture for Contact-Type Energy Hypersurfaces

Duplicate of canonical problem DYN-001

DYN-001 / p.1613

### OWR-1052-003 | Harmonic One-Forms in Perturbed Pseudoholomorphic-Curve Equations

Answered in the source or not an open problem (solved (answered in the source))

### OWR-1053-008 | Birch-Swinnerton-Dyer Conjecture

Duplicate of canonical problem MPP-005

MPP-005 / p.35

### OWR-1054-002 | Hausdorff Dimension of Singular Sets for Variational Minimizers

Answered in the source or not an open problem (solved (yes; answered in the source))

### OWR-1062-010 | Lower Semicontinuity of Free Entropy Dimension

Answered in the source or not an open problem (solved (no; answered in the source))

### OWR-1063-003 | Rhetorical Primitive-Central Cohomology Question

Unrecoverable OWR statement (The extracted record converts a rhetorical transition into an open problem, but the official abstract immediately answers it with structural)

### OWR-1106-006 | Duplicate: Finiteness of Degree-One Three-Manifold Targets

Duplicate/merged OWR extraction (The official Oberwolfach report was checked directly. This extraction repeats the same source problem as OWR-1106-005 and contains no indepe)

OWR-1106-005 / p.1057

### OWR-1106-007 | Duplicate: Finiteness of Degree-One Three-Manifold Targets

Duplicate/merged OWR extraction (The official Oberwolfach report was checked directly. This extraction repeats the same source problem as OWR-1106-005 and contains no indepe)

OWR-1106-005 / p.1057

### OWR-1108-001 | Statistical Model Selection and Prediction Targets

Terminological discussion prompt on the meaning of the phrase 'model selection' rather than a mathematical research problem

### OWR-11101913-003 | Known Cases of Iitaka Subadditivity

Duplicate/merged OWR extraction (The official report confirms that this passage lists established cases of Iitaka subadditivity and does not pose a separate open problem; th)

OWR-11101913-002 / p.691

### OWR-11101913-004 | Methods for Kodaira-Dimension Superadditivity

Unrecoverable OWR statement (The official report confirms that this extraction is commentary on methods and known cases rather than a self-contained open problem. No sep)

### OWR-11101916-003 | Unspecified Assembly Map Comparison Conjecture

Unrecoverable OWR statement (The extraction combines introductory assembly-map discussion and definitions without stating a standalone unresolved target. No separate cle)

### OWR-11101916-006 | Field-Theoretic Description of the Full Spectral Action

Unrecoverable OWR statement (The contribution asks this question rhetorically and then answers it with its main spectral-action expansion theorem. No separate clean stat)

### OWR-11101916-008 | Surjectivity in Algebraic K-Theory of Varieties

Answered in the source or not an open problem (partially_solved (no, for finite fields of odd characteristic; answered in the source))

### OWR-11101916-010 | Generalized Group Actions for Crossed-Product Operator Algebras

Unrecoverable OWR statement (The contribution immediately answers its motivating question by identifying Fell bundles as the appropriate generalized actions. No separate)

### OWR-11101917-001 | Connes-Embedding Table-of-Contents Fragment

Unrecoverable OWR statement (The official report confirms that the extraction is a table-of-contents fragment, not a standalone open mathematical problem.)

### OWR-11101917-013 | Resolved Coarseness Conjecture for Free Group Factors

Unrecoverable OWR statement (The official report states that the coarseness conjecture under discussion was resolved positively; the extracted passage is not an open pro)

### OWR-11101918-001 | Workshop Summary on Geometric Measure Theory and Flows

Unrecoverable OWR statement (The official report confirms that the extraction is workshop-summary prose and a list of topics, not a standalone mathematical problem. No s)

### OWR-11101918-002 | Workshop Contents Listing for Geometric Variational Analysis

Unrecoverable OWR statement (The official report confirms that the extraction is a table-of-contents fragment, not a mathematical problem. No separate clean statement is)

### OWR-11101918-003 | Classical Limit of Nonlocal Minimal Surfaces

Unrecoverable OWR statement (The official report raises this question only as a lead-in and immediately states that it is answered by the reported work, so it is not pre)

### OWR-11101919-001 | Unspecified Three-Dimensional Shape Analysis Problem

Unrecoverable OWR statement (The official report gives only broad workshop context about three-dimensional shape analysis and states no standalone mathematical target. N)

### OWR-11101924-001 | High-Frequency Helmholtz Green-Function Separability

Unrecoverable OWR statement (The official report presents separability bounds answering this rhetorical title and does not isolate an unresolved mathematical claim.)

### OWR-11101924-002 | Phase-Space Quantization for Variable-Coefficient PDEs

Unrecoverable OWR statement (The official report immediately answers this question by a semiclassical Fourier-integral-operator construction, so it is not an open proble)

### OWR-11101924-005 | Resolved Helmholtz Degrees-of-Freedom Question

Unrecoverable OWR statement (The official report answers the degrees-of-freedom question and develops its extension; the extraction does not retain an open problem.)

### OWR-11101924-007 | Diffractive High-Frequency Wave Propagation

Unrecoverable OWR statement (The official report treats this heading through sharp results and counterexamples rather than stating an unresolved mathematical problem.)

### OWR-11126-004 | Invariant Measures for Thermostated Mechanical Systems

Unrecoverable OWR statement (The official report states affirmative results for the invariant-measure, uniqueness, convergence, and rate questions in the thermostated sy)

### OWR-11126-006 | Frequency Maps of Central N-Body Equilibria

Duplicate/merged OWR extraction (This is a less precise repeated extraction of the frequency-image conjecture preserved in OWR-11126-005. No separate clean statement is prom)

OWR-11126-005 / p.1548

### OWR-11126-007 | Reeb Orbits Bounding Global Disk Sections

Answered in the source or not an open problem (solved (for non-degenerate contact forms; answered in the source))

### OWR-11126-010 | Rigidity of Maximal Rank Abelian Actions

Unrecoverable OWR statement (The official report introduces these rigidity questions and then immediately states a theorem resolving them in the setting considered. No s)

### OWR-11127-022 | Duplicate Schur $\sigma$-Group Frequency Heuristic

Duplicate/merged OWR extraction (This record exactly repeats the same-report conjecture represented canonically by OWR-11127-021, with no distinct hypothesis or conclusion. )

OWR-11127-021 / p.176

### OWR-11128-010 | Higher-Dimensional Riesz Transforms and Uniform Rectifiability

Duplicate/merged OWR extraction (This is a narrower repeated extraction of the David-Semmes Riesz-transform problem preserved in OWR-11128-002. No separate clean statement i)

OWR-11128-002 / p.1447

### OWR-11132-003 | Stochastic Averaging with Topologically Complex Fast Orbits

Unrecoverable OWR statement (The wording is an expository transition to examples whose behavior is then analyzed in the contribution, not a standalone open problem. No s)

### OWR-11133-004 | Realizability of Divisor Volumes

Answered in the source or not an open problem (solved (no; answered in the source))

### OWR-11134-003 | Answered Characterization of Lookback Adjusters

Unrecoverable OWR statement (The source asks which trading guarantees are possible and immediately gives necessary-and-sufficient characterizations in Theorems 4 and 5. )

### OWR-11136-011 | Lower Bounds for Higher-Dimensional Permutation Counts

Duplicate/merged OWR extraction (This repeats the asymptotic enumeration conjecture preserved in OWR-11136-010 and also corrupts its exponential factor. No separate clean st)

OWR-11136-010 / p.1639

### OWR-11138-003 | Koszul Decompositions in Homotopy Theory

Unrecoverable OWR statement (The official report presents the Koszul property as an established result and then explains its meaning rather than posing an unresolved pro)

### OWR-1115-003 | Redundant Extraction - Active Robin Boundaries

Duplicate/merged OWR extraction (The primary Oberwolfach report was checked directly and confirms that this extraction repeats OWR-1115-002 with the same domain, harmonic Ro)

OWR-1115-002 / p.1318

### OWR-1116-010 | Polychromatic Subsets for Bounded Colorings of the First Uncountable Ordinal

Answered in the source or not an open problem (solved (independent of ZFC; answered in the source))

### OWR-1116-011 | Ground Axiom Consequences for Cardinal Arithmetic

Answered in the source or not an open problem (solved (no; answered in the source))

### OWR-11446-001 | Simultaneous Exact Tests for Contingency Tables

Unrecoverable OWR statement (The phrase was extracted from a contribution title and abstract that present methods and results, not a standalone open conjecture. No separ)

### OWR-11451-005 | Composite Repetition of Three Matroid Questions

Unrecoverable OWR statement (The official report contains three separate questions, but this extraction conflates all three and repeats the first two records in this chu)

### OWR-11451-013 | Unique Games Conjecture

Duplicate of canonical problem CS-001

CS-001 / p.2109

### OWR-11571-001 | Recovering Tacit Mathematical Skills from Historical Sources

Unrecoverable OWR statement (The official history-of-mathematics workshop report treats this as a historical, pedagogical, or methodological discussion prompt rather tha)

### OWR-11571-002 | Tacit and Explicit Knowledge in Mathematical Practice

Unrecoverable OWR statement (The official history-of-mathematics workshop report treats this as a historical, pedagogical, or methodological discussion prompt rather tha)

### OWR-11571-003 | Tacit Knowledge in Mathematical History

Unrecoverable OWR statement (The official history-of-mathematics workshop report treats this as a historical, pedagogical, or methodological discussion prompt rather tha)

### OWR-11571-004 | Explicit and Tacit Knowledge in Mathematical Instruments

Unrecoverable OWR statement (The official history-of-mathematics workshop report treats this as a historical, pedagogical, or methodological discussion prompt rather tha)

### OWR-11571-005 | Bicycle Mechanics and Tacit Mathematical Knowledge

Unrecoverable OWR statement (The official history-of-mathematics workshop report treats this as a historical, pedagogical, or methodological discussion prompt rather tha)

### OWR-11571-006 | Modeling and Solving Bicycle Motion

Unrecoverable OWR statement (The official history-of-mathematics workshop report treats this as a historical, pedagogical, or methodological discussion prompt rather tha)

### OWR-11571-007 | Interpreting Obscure Medieval Mathematical Algorithms

Unrecoverable OWR statement (The official history-of-mathematics workshop report treats this as a historical, pedagogical, or methodological discussion prompt rather tha)

### OWR-11571-008 | Tacit Steps and Textual Corruption in Medieval Algorithms

Unrecoverable OWR statement (The official history-of-mathematics workshop report treats this as a historical, pedagogical, or methodological discussion prompt rather tha)

### OWR-11571-009 | Omitted Steps in Historical Mathematical Algorithms

Unrecoverable OWR statement (The official history-of-mathematics workshop report treats this as a historical, pedagogical, or methodological discussion prompt rather tha)

### OWR-11571-010 | Mathematical Signs and Model-Theoretic Semantics

Unrecoverable OWR statement (The official history-of-mathematics workshop report treats this as a historical, pedagogical, or methodological discussion prompt rather tha)

### OWR-11571-011 | Practices of Nineteenth-Century Mathematical Education

Unrecoverable OWR statement (The official history-of-mathematics workshop report treats this as a historical, pedagogical, or methodological discussion prompt rather tha)

### OWR-11571-012 | Mathematical Learning as Evidence of Tacit Knowledge

Unrecoverable OWR statement (The official history-of-mathematics workshop report treats this as a historical, pedagogical, or methodological discussion prompt rather tha)

### OWR-11571-013 | Polanyi's Tacit Knowledge in Mathematical History

Unrecoverable OWR statement (The official history-of-mathematics workshop report treats this as a historical, pedagogical, or methodological discussion prompt rather tha)

### OWR-11571-014 | Rigor and Antiformalism in Mathematics

Unrecoverable OWR statement (The official history-of-mathematics workshop report treats this as a historical, pedagogical, or methodological discussion prompt rather tha)

### OWR-11571-015 | Tree Metaphors for Mathematical Development

Unrecoverable OWR statement (The official history-of-mathematics workshop report treats this as a historical, pedagogical, or methodological discussion prompt rather tha)

### OWR-11571-017 | Logical Strength and Usefulness in Number Theory

Unrecoverable OWR statement (The official history-of-mathematics workshop report treats this as a historical, pedagogical, or methodological discussion prompt rather tha)

### OWR-11573-001 | Equivalent-Dynamics Construction Report

Unrecoverable OWR statement (The official contribution introduces the desire for faster equivalent dynamics and then supplies the construction and results; the extracted)

### OWR-11573-002 | Ballistic-Annihilation Scaling-Limit Report

Unrecoverable OWR statement (The official contribution presents the continuum-limit question as motivation and then derives the announced Boltzmann scaling result; the e)

### OWR-11575-006 | Complexity of Maximal Chains in Computable Well-Partial-Orders

Answered in the source or not an open problem (solved (answered in the source))

### OWR-11581-001 | Poisson Empirical-Bayes Estimator Report

Unrecoverable OWR statement (The official contribution states this estimator-construction goal and immediately develops the estimator and its asymptotic risk theorem, so)

### OWR-11582-001 | Dense-Granular-Flow Results Report

Answered in the source or not an open problem (solved (answered in the source))

### OWR-11695854-001 | Workshop Contents Listing for Quantum Geometry and Character Varieties

Unrecoverable OWR statement (The official report confirms that the extraction is a table-of-contents fragment, not a mathematical problem. No separate clean statement is)

### OWR-11695858-001 | Invariance Principle for Reversible Random Conductance Walks

Unrecoverable OWR statement (The official report answers the extracted question with an invariance-principle theorem and does not leave it as an open problem.)

### OWR-11695862-001 | O-Minimality Workshop Table-of-Contents Fragment

Unrecoverable OWR statement (The official report confirms that the extraction is a table-of-contents fragment, not a standalone open mathematical problem.)

### OWR-11695862-012 | Duplicate Complex-Exponential Categoricity Formulation

Duplicate/merged OWR extraction (The official Oberwolfach report confirms that this extraction repeats the equivalent Schanuel and strong exponential-algebraic-closedness fo)

OWR-11695862-011 / p.2006

### OWR-11695863-003 | Geometric Spectral Bounds for Infinite Quantum Graphs

Unrecoverable OWR statement (The official report poses a motivating question about geometric spectral estimates and then supplies the relevant theorem and corollary in t)

### OWR-11695864-006 | Nonexistence of Geproci Sets in Linear General Position

Duplicate/merged OWR extraction (The official report confirms that this is a narrower duplicate of the geproci classification questions retained in OWR-11695864-001. No sepa)

OWR-11695864-001 / p.648

### OWR-11695864-007 | Ten-Point Blowup Interpolation Consequences

Unrecoverable OWR statement (The extracted Pell-equation and vector-bundle passage reports established constructions and consequences but states no standalone open probl)

### OWR-11695866-001 | Handwriting and Chronology in Veblen’s Notebooks

Unrecoverable OWR statement (The official report passage poses a historical and archival research inquiry about Veblen, not an open mathematical problem.)

### OWR-11695866-002 | Authorship and Composition of Private Mathematical Manuscripts

Unrecoverable OWR statement (The official report passage concerns historical attribution and manuscript analysis, not an open mathematical problem.)

### OWR-11695866-003 | Veblen’s Private Life and Archival Self-Presentation

Unrecoverable OWR statement (The official report passage concerns biographical and archival history, not an open mathematical problem.)

### OWR-11695866-004 | Veblen’s Role in Princeton’s Institutional Rise

Unrecoverable OWR statement (The official report passage concerns institutional history, not an open mathematical problem.)

### OWR-11695866-005 | Moore’s Laboratory Method in Veblen’s Geometry

Unrecoverable OWR statement (The official report passage concerns the history of mathematical practice and pedagogy, not an open mathematical problem.)

### OWR-11780-004 | Nonspherical Orbit-Parameterization Report

Unrecoverable OWR statement (The official contribution introduces the nonspherical orbit question and then states its orbit-parameterization theorem, so the extraction t)

### OWR-11784-009 | Conditional Sparse-Decomposition Theorem

Answered in the source or not an open problem (solved (a conditional theorem, not a problem))

### OWR-11788-005 | Duplicate Quintic Resonant-System Scattering Problem

Duplicate/merged OWR extraction (This record exactly repeats the same-report resonant-system conjecture represented canonically by OWR-11788-004. No separate clean statement)

OWR-11788-004 / p.1492

### OWR-11788-009 | Duplicate Three-Dimensional Navier-Stokes Problem

Duplicate/merged OWR extraction (This record restates the same-report three-dimensional Navier-Stokes global regularity and uniqueness problem represented canonically by OWR)

OWR-11788-002 / p.1477

### OWR-11789-008 | Shrinking-Soliton Degeneration Report

Unrecoverable OWR statement (The official contribution uses a broad question about Gromov-Hausdorff limits as motivation and then states its compactness and singular-loc)

### OWR-1183-007 | Duplicate Two-Dimensional Time-Bomb Conjecture

Duplicate/merged OWR extraction (This is a redundant extraction of OWR-1183-006 from the same statement of the Time-Bomb Conjecture in the official report. No separate clean)

OWR-1183-006 / p.1750

### OWR-1183-012 | Duplicate Point-to-Hyperplane Matching Conjecture

Duplicate/merged OWR extraction (This is a redundant extraction of the matching-points-to-hyperplanes conjecture recorded canonically as OWR-1183-011. No separate clean stat)

OWR-1183-011 / p.1700

### OWR-1186-002 | Answered Canonical-Momentum Balance Question

Answered in the source or not an open problem (solved (yes, with a qualification; answered in the source))

### OWR-1186-003 | Answered Material-Momentum Terminology Question

Answered in the source or not an open problem (solved (yes; answered in the source))

### OWR-1187-002 | Tilting Bundles of Line Bundles on Smooth Toric Varieties

Answered in the source or not an open problem (solved (no; answered in the source))

### OWR-1188-013 | Duplicate: Bounded Local Solutions of Global Differential Modules

Duplicate/merged OWR extraction (The official Oberwolfach report was checked directly. This extraction repeats the same source problem as OWR-1188-012 and contains no indepe)

OWR-1188-012 / p.36

### OWR-1190-001 | Realization of Algebraic Cones over Complete Intersections

Answered in the source or not an open problem (solved (yes; answered in the source))

### OWR-1190-005 | Repeated Compact Lie-Group Algebraic-Model Conjecture

Duplicate/merged OWR extraction (The official report contains a single conjecture followed by its consequences; this extraction repeats its core equivalence and adds no inde)

OWR-1190-004 / p.1014

### OWR-12005-010 | Duplicate Round-Sphere Eigenvalue Question

Unrecoverable OWR statement (The official report contains one round-sphere extremal question, and this extraction is textually identical in substance to the preceding re)

### OWR-12006-003 | Derived Schläfli Formula for Spherical Cone Metrics

Answered in the source or not an open problem (solved (a completed calculation, not an open problem))

### OWR-12007-013 | Answered Survey of Zero-Entropy Disk Dynamics

Duplicate/merged OWR extraction (The source uses this broad prompt to survey several notions of complexity, then records positive constructions for dense orbits, ergodicity,)

OWR-12007-014 / p.1573

### OWR-12007-015 | Duplicate Mixing Zero-Entropy Disk Question

Duplicate/merged OWR extraction (This record exactly repeats the same-report mixing question represented canonically by OWR-12007-014. No separate clean statement is promote)

OWR-12007-014 / p.1573

### OWR-12010-014 | Serre-Weight Modularity Theorem

Unrecoverable OWR statement (The official contribution states and proves the claimed modularity criterion under its hypotheses; the extraction strips those hypotheses an)

### OWR-12014-005 | L-Space Conjecture

Duplicate of canonical problem KP-3.48

KP-3.48 / p.1104

### OWR-12018-005 | Duplicate: HOMFLY Polynomials from Nested Hilbert Schemes

Duplicate/merged OWR extraction (Comparison with the official report and OWR-12018-003 confirms that this is a shortened repeat of the same ordinary HOMFLY conjecture. No se)

OWR-12018-003 / p.690

### OWR-12018-007 | Duplicate: Homological Hilbert-Scheme Conjecture

Duplicate/merged OWR extraction (The official report presents this passage as verification context for the conjecture already recorded in OWR-12018-006, not as a separate pr)

OWR-12018-006 / p.690

### OWR-12172-002 | Toric Orbits Meeting Subvariety Closures

Answered in the source or not an open problem (solved (answered in the source))

### OWR-12173-007 | Duplicate: Sarnak's Möbius Disjointness Conjecture

Duplicate/merged OWR extraction (Comparison with the official report and OWR-12173-005 confirms that this is a second extraction of the same conjecture plus evidence about s)

OWR-12173-005 / p.1608

### OWR-12175-002 | Duplicate: Toms-Winter Regularity Conjecture

Duplicate/merged OWR extraction (Comparison with the official report and OWR-12175-001 confirms that this is a restatement of the same three-way equivalence. No separate cle)

OWR-12175-001 / p.312

### OWR-12175-006 | Duplicate: Kadison-Kastler Conjecture for $C^*$-Algebras

Duplicate/merged OWR extraction (Comparison with the official report and OWR-12175-005 confirms that this is a shortened repeat of the same conjecture. No separate clean sta)

OWR-12175-005 / p.256

### OWR-12177-001 | Differential-Privacy Workshop Framing

Unrecoverable OWR statement (The official report uses this broad complexity prompt as workshop framing and supplies no determinate claim distinct from the concrete high-)

### OWR-12177-005 | Useful Differential-Privacy Framing

Unrecoverable OWR statement (The official report uses this broad sentence as expository framing and does not pose it as a determinate problem separate from the explicit )

### OWR-12177-006 | Private-Data-Analysis Framing

Unrecoverable OWR statement (The official contribution uses this broad prompt to motivate differential privacy and then narrows it to the concrete high-dimensional runni)

### OWR-12177-009 | Unique Games Conjecture

Duplicate of canonical problem CS-001

CS-001 / p.2109

### OWR-12177-011 | Machine-Learning Algorithms Results Report

Unrecoverable OWR statement (The official contribution asks rhetorically whether provable algorithms are possible and immediately answers yes by presenting three results)

### OWR-12182-007 | Generalized Wills-Polynomial Results Report

Unrecoverable OWR statement (The official contribution introduces an interest in measure-independent root properties and then states the properties it proves; the extrac)

### OWR-12182-008 | Geometric-Comparison Stability Framework

Unrecoverable OWR statement (The official contribution presents this generic inequality as a framework before applying it to specified geometric characteristics; without)

### OWR-12182-009 | Geometric-Comparison Separation Framework

Unrecoverable OWR statement (The official contribution presents this generic separation inequality as the companion framework to its proved applications; the extraction )

### OWR-12329-013 | Duplicate Minimal Ordered-Structure Question

Unrecoverable OWR statement (The official report contains one existence question beyond almost linearity, and this extraction duplicates the preceding canonical record f)

### OWR-12330-013 | Counterexamples to Averaged Local Fractional-Coloring Bounds

Already answered in the original Oberwolfach workshop report (not an open problem)

### OWR-12330-015 | Duplicate Weight-Modified Reed Coloring Question

Duplicate/merged OWR extraction (This record exactly repeats the same-report fractional-coloring question represented canonically by OWR-12330-014. No separate clean stateme)

OWR-12330-014 / p.1770

### OWR-12331-003 | Sign-Definite Variational Formulations of Helmholtz Impedance Problems

Unrecoverable OWR statement (The official report asks for other sign-definite formulations and immediately supplies examples and a general transformation, so the extract)

### OWR-12338-003 | Duplicate Einstein-Type Fluid-Dynamics Question

Duplicate/merged OWR extraction (The official report poses one question about deriving an Einstein-type law for localized fluid-mechanical objects and then cites concentrati)

OWR-12338-004 / p.1466

### OWR-12340-001 | Historical Foundations of Mathematical Classification

Early-Modern Mixed Mathematics workshop report (16th-century historiography, not a mathematical research problem)

### OWR-12340-002 | From Pure and Mixed to Pure and Applied Mathematics

Early-Modern Mixed Mathematics workshop report (16th-century historiography, not a mathematical research problem)

### OWR-12340-003 | Mathematical Modelling as Scientific Knowledge Production

Early-Modern Mixed Mathematics workshop report (16th-century historiography, not a mathematical research problem)

### OWR-12340-005 | Mathematics in Early-Modern Landscape Design

Early-Modern Mixed Mathematics workshop report (16th-century historiography, not a mathematical research problem)

### OWR-12340-006 | Mixed Mathematics in Sixteenth-Century Practice

Early-Modern Mixed Mathematics workshop report (16th-century historiography, not a mathematical research problem)

### OWR-12340-007 | Practice-Based Histories of Applied Mathematics

Early-Modern Mixed Mathematics workshop report (16th-century historiography, not a mathematical research problem)

### OWR-12342-002 | Homogenization of Nonlinear Plasticity

Unrecoverable OWR statement (The official report immediately answers this explanatory question by identifying the nonlinear equations, differential inclusion, and insuff)

### OWR-12487-001 | Expository Automorphic Spectral Decomposition

Unrecoverable OWR statement (The official report presents this as expository background and immediately gives the discrete-continuous decomposition, finite-multiplicity )

### OWR-12487-002 | Expository Transition to Adelic Hecke Operators

Unrecoverable OWR statement (The source’s “What does one know?” is a rhetorical transition in an introductory account; the following paragraphs answer it by passing to t)

### OWR-12493-006 | Duplicate: Valuative Form of Grothendieck's Section Conjecture

Duplicate/merged OWR extraction (Comparison with the official report and OWR-12493-007 confirms that this is another extraction of the same valuative section conjecture. No )

OWR-12493-007 / p.195

### OWR-12495-002 | Secondary K-Theory Comparison Map

Unrecoverable OWR statement (Checked against the official Oberwolfach report. The extracted sentence states a construction and result proved in the source, rather than a)

### OWR-12495-003 | Lefschetz Classes and the Tate and Hodge Conjectures

Unrecoverable OWR statement (Checked against the official Oberwolfach report. The extracted sentence is an explanatory sufficient-condition statement, not an open proble)

### OWR-12495-005 | Tate and Hodge Classes on CM Abelian Varieties

Unrecoverable OWR statement (Checked against the official Oberwolfach report. The extraction paraphrases background already represented by the adjacent source questions )

### OWR-12588-003 | Existence of Solutions to the Inverse $\sigma_k$ Equation

Answered in the source or not an open problem (solved (answered in the source))

### OWR-12592-005 | Duplicate Linear $\sigma$-Model Phase Equivalence

Duplicate/merged OWR extraction (The official report states one conjecture that the D-brane categories associated with the respective phases of the constructed nonabelian li)

OWR-12592-004 / p.656

### OWR-12596-002 | Stationarity of a GARCH Recurrence

Unrecoverable OWR statement (Checked against the official Oberwolfach report. The source presents the implication as a standard explanatory step establishing stationarit)

### OWR-1265-001 | Workshop-Scope List of Representation Conjectures

Unrecoverable OWR statement (The official report introduction only lists several conjectures as workshop themes; it does not state a distinct combined problem, and indiv)

### OWR-1265-004 | McKay Conjecture for Finite Groups

Duplicate of canonical problem ALG-014

ALG-014 / p.268

### OWR-1265-014 | Repeated Local Grothendieck-Group Conjecture

Duplicate/merged OWR extraction (The official report states one integral equivariant isomorphism conjecture; this extraction paraphrases the same conjecture preserved in OWR)

OWR-1265-015 / p.359

### OWR-1265-016 | Repeated Local Grothendieck-Group Conjecture

Duplicate/merged OWR extraction (The official report states one integral equivariant isomorphism conjecture; this shortened label restates the same claim preserved in OWR-12)

OWR-1265-015 / p.359

### OWR-1265-024 | Answered Shortest-Route Enumeration

Already answered in the original Oberwolfach workshop report (not an open problem)

### OWR-1269-007 | Duplicate: Weak Uniqueness of Elementary-Embedding Square Roots

Duplicate/merged OWR extraction (The official Oberwolfach report was checked directly. This extraction repeats the same source problem as OWR-1269-006 and restates only part)

OWR-1269-006 / p.1996

### OWR-1269-008 | Duplicate: Weak Uniqueness of Elementary-Embedding Square Roots

Duplicate/merged OWR extraction (The official Oberwolfach report was checked directly. This extraction repeats the same source problem as OWR-1269-006 and restates only part)

OWR-1269-006 / p.1996

### OWR-1269-009 | Duplicate: Weak Uniqueness of Elementary-Embedding Square Roots

Duplicate/merged OWR extraction (The official Oberwolfach report was checked directly. This extraction repeats the same source problem as OWR-1269-006 and restates the same )

OWR-1269-006 / p.1996

### OWR-12697684-001 | Ryser-Brualdi-Stein Latin Square Transversal Conjecture

Duplicate/merged OWR extraction (The extraction is workshop context for the Ryser-Brualdi-Stein conjecture, whose complete formulation is retained in OWR-12697684-020. No se)

OWR-12697684-020 / p.1729

### OWR-12697684-002 | Unspecified Extremal Combinatorics Conjecture

Unrecoverable OWR statement (The official report shows that this is an incomplete table-of-contents fragment without a mathematical statement. No separate clean statemen)

### OWR-12697684-003 | Unspecified Random Function Root Problem

Duplicate/merged OWR extraction (The extraction is only the title of a contribution; its specific random-root questions are retained in OWR-12697684-016 and OWR-12697684-017)

OWR-12697684-016 / p.1946 OWR-12697684-017 / p.1909

### OWR-12697684-012 | Repeated Erdős-Hajnal Conjecture Context

Duplicate/merged OWR extraction (The official report confirms that this is contextual repetition of the Erdős-Hajnal conjecture retained in OWR-12697684-011. No separate cle)

OWR-12697684-011 / p.1815

### OWR-12697686-001 | Workshop Contents Listing for Low-Dimensional Topology

Unrecoverable OWR statement (The official report confirms that the extraction is a table-of-contents fragment, not a mathematical problem. No separate clean statement is)

### OWR-12697686-004 | Background on Circle Actions and the L-Space Conjecture

Unrecoverable OWR statement (The official report confirms that the extraction is introductory exposition rather than a standalone problem; the adjacent records represent)

### OWR-12697686-013 | Slice-Ribbon Conjecture

Duplicate of canonical problem TOP-004

TOP-004 / p.1248

### OWR-12697686-014 | Background on Cable Sliceness and Genus Bounds

Unrecoverable OWR statement (The official report confirms that the extraction is explanatory prose surrounding a proved argument, not a separately stated open problem. N)

### OWR-12697686-017 | Incomplete Slice-Missing-Slice-Disc Lead-In

Unrecoverable OWR statement (The official report confirms that this extraction is the first half of a figure-dependent problem and is not independently self-contained. N)

### OWR-12697686-018 | Figure-Dependent Slice Curves in Disc Complements

Unrecoverable OWR statement (The official report confirms that the knot and the curves in this question are defined only by a figure absent from the dataset, so a self-c)

### OWR-12697687-007 | Homeomorphism Classification of Compact Oriented Four-Manifolds

Unrecoverable OWR statement (The extracted sentence is an expository heading that is immediately followed by the modified-surgery strategy; it is not posed as a standalo)

### OWR-12697688-007 | Central Leaves Under Prime-to-p Hecke Correspondences

Unrecoverable OWR statement (The official report passage states a proved theorem on the stability of central leaves under Hecke correspondences, not an open problem.)

### OWR-12697688-009 | Local Nonvanishing in Arithmetic Gan-Gross-Prasad Theory

Unrecoverable OWR statement (The official report gives a local nonvanishing fact and then states proved cases; the passage does not formulate an unresolved problem.)

### OWR-12697688-011 | Langlands-Rapoport Conjecture for Parahoric Level

Unrecoverable OWR statement (The official report explicitly says that the conjecture discussed in this extraction has been proved, so it is not an open problem.)

### OWR-12697688-019 | Fargues Eigensheaf Construction Context

Unrecoverable OWR statement (The official report confirms that this extraction is contextual setup duplicated by the following complete open problem.)

### OWR-12697688-021 | Naive Local-Global Compatibility Formulation

Unrecoverable OWR statement (The official report presents this naive formulation only as exposition before a conditional theorem; the extraction does not preserve a stan)

### OWR-12697688-022 | Local-Model Description of Shimura Varieties

Unrecoverable OWR statement (The official report passage is a proof and local-model exposition, not an open mathematical problem.)

### OWR-12697689-001 | Conjectures on Resolutions and Singular Rings

Unrecoverable OWR statement (The official report confirms that this record is a table-of-contents fragment rather than a mathematical question. No separate clean stateme)

### OWR-12697689-009 | Introduction to the Weak-Implies-Strong Conjecture

Duplicate/merged OWR extraction (The official report confirms that this record is an introductory fragment for the weak-implies-strong conjecture, whose complete statement a)

OWR-12697689-010 / p.316

### OWR-12697692-001 | Behavioral and Seasonal Effects on COVID-19 Transmission

Unrecoverable OWR statement (The question is a presentation heading answered empirically in the following discussion of mobility, seasonality, and viral variants, not a )

### OWR-12697692-002 | Exchangeability Tests for Branching in Timed Phylogenies

Unrecoverable OWR statement (The question is a talk heading introducing a developed exchangeability test for branching times, not an unresolved mathematical problem. No )

### OWR-12697708-001 | Unspecified Reinforced Random Process Problem

Unrecoverable OWR statement (The official report passage only describes the workshop's discussion sessions and lists contribution titles without stating a mathematical p)

### OWR-12697708-005 | Mathematical Characterization of Condensation in Reinforced Systems

Unrecoverable OWR statement (The extracted phrase is a section heading that the ensuing exposition answers by defining and illustrating condensation; it is not an open p)

### OWR-12697709-001 | Unspecified Weak Space-Time PDE Optimization Problem

Unrecoverable OWR statement (The official report passage presents optimization models and proved first-order conditions but does not state an unresolved mathematical que)

### OWR-12697710-010 | Context-Free Open-Problem Label

Unrecoverable OWR statement (The official report contains only the words “Open Problem” at this extraction boundary, with no recoverable mathematical content. No separat)

### OWR-12697711-001 | Real Algebraic Geometry Workshop Overview

Unrecoverable OWR statement (The official report confirms that this extraction is a workshop-summary paragraph and contains no standalone open problem.)

### OWR-12697711-007 | Orphaned Discotope Conjecture Label

Unrecoverable OWR statement (The official report confirms that the extraction is only an orphaned conjecture label; the complete conjecture is preserved in the adjacent )

### OWR-12697712-001 | Complexity of Counting versus Recovering Planted Communities

Unrecoverable OWR statement (The contribution asks whether community statistics can be easier to infer than communities themselves and then states that its low-degree lo)

### OWR-12697712-003 | Chromatic Numbers of Exchangeable Graphon Random Graphs

Duplicate/merged OWR extraction (The passage summarizes proved results for block and approximable graphons and does not itself state the complete general conjecture; the con)

OWR-12697712-004 / p.1779

### OWR-12723-003 | Duplicate Halton-Kronecker Discrepancy Question

Duplicate/merged OWR extraction (This record is the same Halton-Kronecker discrepancy problem represented canonically and with the missing definition of $\phi_2$ by OWR-1272)

OWR-12723-004 / p.1426

### OWR-12724-005 | Monotonicity of Sofic Model Counts

Answered in the source or not an open problem (solved (answered in the source; not a research problem))

### OWR-12725-008 | Independent-Kernel Heuristic for Refined abc Bounds

Unrecoverable OWR statement (Checked against the official Oberwolfach report. The source explicitly labels independence of the three kernels as a heuristic assumption us)

### OWR-12727-002 | Answered Reconstruction of Contact Structures from Contact Lines

Answered in the source or not an open problem (solved (yes, if X is not projective space; answered in the source))

### OWR-12733-002 | Cellular Response to Mechanical Stress

Unrecoverable OWR statement (Checked against the official Oberwolfach report. The source poses this as a rhetorical question and immediately reports an experimental and )

### OWR-1274-017 | Duplicate: Weak Greenberg Conjecture for CM Fields

Duplicate/merged OWR extraction (The official Oberwolfach report was checked directly. This extraction repeats the same source problem as OWR-1274-016 and contains no indepe)

OWR-1274-016 / p.196

### OWR-12861-017 | Unit Distances in Rational Directions

Answered in the source or not an open problem (solved (a theorem, answered in the source))

### OWR-12861-024 | Determinant of a Longest-Increasing-Subsequence Matrix

Unrecoverable OWR statement (The official problem session itself inconsistently names the matrix $M$ and $A$, indexes its columns by an apparent proper subset of $S_n$ w)

### OWR-12866-003 | Duplicate Composite: Nyman Criterion and Autocorrelation Maximum

Duplicate/merged OWR extraction (This extraction merely combines the two source questions represented canonically and separately by OWR-12866-002 and OWR-12866-004. No separ)

OWR-12866-002 / p.173 OWR-12866-004 / p.138

### OWR-12870-007 | Finiteness of Negative Extremal Rays for KLT Pairs

Answered in the source or not an open problem (solved (no; answered in the source))

### OWR-12870-012 | Probability Convergence of Metrics on Linear Series

Answered in the source or not an open problem (solved (yes, for ample type; answered in the source))

### OWR-12872-010 | Enumeration of Pairwise-Compatible Triple Families

Statement proved later in the same workshop report (marked unrecoverable/not an open problem in owr_statement_audit.json)

### OWR-12979-011 | Answered Drury-Arveson Kernel-Norm Question

Answered in the source or not an open problem (solved (no, for n ≥ 2; answered in the source))

### OWR-12988-001 | Peridynamics as a Multiscale Mechanics Model

Unrecoverable OWR statement (Checked against the official Oberwolfach report. The phrase is a rhetorical heading followed by an explanation of why peridynamics is useful)

### OWR-12988-002 | Adaptive Galerkin Discretization of Peridynamics

Unrecoverable OWR statement (Checked against the official Oberwolfach report. The source presents the listed finite-element choices as a proposed computational construct)

### OWR-13101-002 | Duplicate Summary: Canonical Bases for Log Calabi-Yau Varieties

Duplicate/merged OWR extraction (The source sentence is an introductory summary of the detailed Gross-Hacking-Keel conjecture represented by OWR-13101-003, not a separate ma)

OWR-13101-003 / p.747

### OWR-13103-002 | Mass-Preserving Decomposition of Normal Currents

Answered in the source or not an open problem (solved (no; answered in the source))

### OWR-13108-011 | Duplicate Hyperbolic Dehn-Filling Bounded-Height Question

Duplicate/merged OWR extraction (The official report contains one bounded-height conjecture for Dehn fillings, represented canonically by OWR-13108-010. This record repeats )

OWR-13108-010 / p.1241

### OWR-13109-001 | Abundance Conjecture for Projective Minimal Models

Duplicate of canonical problem AG-002

AG-002 / p.738

### OWR-13110-015 | Embedding Triangular 2-Complexes in 3-Space

Answered in the source or not an open problem (solved (decidable; answered in the source))

### OWR-1323-002 | Paths of Hypercyclic Operators with Common Vectors

Answered in the source or not an open problem (solved (yes; answered in the source))

### OWR-13351-014 | Birational Anabelian Reconstruction over Finitely Generated Fields

Answered in the source or not an open problem (solved (yes; answered in the source))

### OWR-13351-016 | Duplicate Statement of the Mod-$\ell$ Bogomolov-Pop Conjecture

Duplicate/merged OWR extraction (The official report shows that this declarative extraction and OWR-13351-017 state the same mod-$\ell$ Bogomolov-Pop conjecture; OWR-13351-0)

OWR-13351-017 / p.150

### OWR-13353-016 | Duplicate Triangle-Cone Characterization

Duplicate/merged OWR extraction (This extraction repeats verbatim the graph-triangle-cone problem represented canonically by OWR-13353-015. No separate clean statement is pr)

OWR-13353-015 / p.1865

### OWR-13353-022 | Duplicate Constant-Gap Facility-Location Relaxation

Duplicate/merged OWR extraction (This extraction repeats the capacitated-facility-location relaxation question represented canonically by OWR-13353-021. No separate clean st)

OWR-13353-021 / p.2065

### OWR-13355-002 | Duplicate Diffraction-to-Dynamical-Spectrum Conjecture

Duplicate/merged OWR extraction (This declarative extraction is the same conjecture as the question represented canonically by OWR-13355-001. No separate clean statement is )

OWR-13355-001 / p.2115

### OWR-13358-003 | Individual Conceptions and Classroom Mathematical Norms

Mathematics Education and Classroom Pedagogy workshop report (didactics, not a mathematical research problem)

### OWR-13358-004 | Classroom Participation and Mathematical Growth Trajectories

Mathematics Education and Classroom Pedagogy workshop report (didactics, not a mathematical research problem)

### OWR-13358-005 | Learning Trajectories in History-Based Mathematics Curricula

Mathematics Education and Classroom Pedagogy workshop report (didactics, not a mathematical research problem)

### OWR-13358-006 | Perceived Difficulty in Mathematical Activity

Mathematics Education and Classroom Pedagogy workshop report (didactics, not a mathematical research problem)

### OWR-13358-007 | Proof Scripts and Proof Comprehension

Mathematics Education and Classroom Pedagogy workshop report (didactics, not a mathematical research problem)

### OWR-13358-008 | Subject Matter Effects on Mathematical Structures

Mathematics Education and Classroom Pedagogy workshop report (didactics, not a mathematical research problem)

### OWR-13488-012 | Composite of Unrelated Hyperbolic-Group Questions

Unrecoverable OWR statement (The official report shows that the extraction combines Steenbock's Kaplansky zero-divisor question with Rüping's separate Farrell-Jones ques)

### OWR-13489-005 | Minimal Ambient Dimension for Complete Bounded Complex Submanifolds

Answered in the source or not an open problem (solved (no; answered in the source))

### OWR-13490-010 | Duplicate Lefschetz-Injectivity Extraction

Duplicate/merged OWR extraction (The passage duplicates the same Lefschetz-injectivity question recorded in OWR-13490-011 and adds no independent problem. No separate clean )

OWR-13490-011 / p.258

### OWR-13497-001 | Duplicate Four-Agent Formation-Control Heading

Duplicate/merged OWR extraction (The official report poses a specific saddle-point question for incorrect equilibria, represented by OWR-13497-006. This broad heading-level )

OWR-13497-006 / p.1608

### OWR-13497-002 | Expository Packet-Scheduling and Traffic-Control Results

Unrecoverable OWR statement (The official report presents two illustrative cyber-physical-systems results, not unresolved problems: it says a sharp characterization of f)

### OWR-13497-004 | Inferred Stronger Four-Agent Formation Claim

Duplicate/merged OWR extraction (The official report asks whether incorrect equilibria are saddle points; it does not pose this stronger global-convergence assertion as a se)

OWR-13497-006 / p.1608

### OWR-13497-005 | Duplicate General Four-Agent Formation Question

Duplicate/merged OWR extraction (This is a broad restatement of the same-report question represented precisely by OWR-13497-006, rather than a separate mathematical problem.)

OWR-13497-006 / p.1608

### OWR-13499-001 | Duplicate Chromatic-Splitting Extraction

Duplicate/merged OWR extraction (The passage duplicates the chromatic splitting conjecture recorded in OWR-13499-002 and does not state a distinct problem. No separate clean)

OWR-13499-002 / p.1026

### OWR-13674-011 | Marcus-Spielman-Srivastava Theorem Extraction

Answered in the source or not an open problem (solved (a proven theorem, not a question))

### OWR-13674-012 | Kadison-Singer Paving Theorem Extraction

Answered in the source or not an open problem (Suggest: solved for both, or flag both as not a problem.)

### OWR-13678-007 | Hermite-Projector Theorem Extraction

Answered in the source or not an open problem (solved (a theorem, not a problem))

### OWR-13678-009 | Duplicate Complete-Intersection Extraction

Duplicate/merged OWR extraction (The passage is a duplicate extraction of the commuting-matrix problem recorded in OWR-13678-008. No separate clean statement is promoted; th)

OWR-13678-008 / p.267

### OWR-13686-010 | Duplicate Reduced-Structure-Group Lower Bound

Duplicate/merged OWR extraction (This extraction repeats the reduced-structure-group rank conjecture represented canonically and with definitions by OWR-13686-009. No separa)

OWR-13686-009 / p.1128

### OWR-13750328-001 | Representation-Theory Workshop Overview

Unrecoverable OWR statement (The official report confirms that this extraction is a workshop overview and contains no standalone open problem.)

### OWR-13750328-002 | Representation-Theory Table-of-Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a table-of-contents fragment, not a standalone open problem.)

### OWR-13750328-016 | Character-Field Conjecture Introduction

Unrecoverable OWR statement (The official report confirms that this extraction is only an incomplete setup; the following record preserves the full conjecture.)

### OWR-13750331-001 | Energy Costs of Unspecified Molecular Mechanisms

Unrecoverable OWR statement (The extracted question introduces a talk's already-developed model of energetic cost in molecular-clock synchronization and is followed by i)

### OWR-13750331-002 | Reconstruction of Chromosome Structure from Contact Maps

Unrecoverable OWR statement (The extracted phrase is a figure caption answered by the pictured contact map; the surrounding contribution describes established three-dime)

### OWR-13750334-001 | Galkin-Usnich Floer-Potential Conjecture Fragment

Unrecoverable OWR statement (The official report confirms that this workshop-introduction fragment refers broadly to the Galkin-Usnich conjecture but does not state its )

### OWR-13750336-001 | Fourier Restriction and Kakeya Incidence Conjectures

Unrecoverable OWR statement (The official report confirms that this is workshop-level motivation about incidence problems rather than a separately formulated conjecture.)

### OWR-13750336-002 | Unspecified Harmonic Analysis Incidence Problem

Unrecoverable OWR statement (The extraction is a table-of-contents fragment and contains no mathematical problem. No separate clean statement is promoted, and the origin)

### OWR-13750336-009 | Equidistribution Rates for Semisimple Torus Random Walks

Unrecoverable OWR statement (The motivating equidistribution question is answered immediately in the contribution by the two stated semisimple random-walk theorems. No s)

### OWR-13750336-013 | Endpoint Sparse Bounds for Planar Averaging Operators

Duplicate/merged OWR extraction (The extraction is a sentence fragment from the same Bochner-Riesz contribution whose complete open boundary-segment question is retained in )

OWR-13750336-014 / p.1286

### OWR-13750341-001 | Unspecified P-Adic Euler System Conjecture

Unrecoverable OWR statement (The extraction begins with the trailing word of a Bloch-Kato reference and then states proved Euler-system and regulator theorems; it contai)

### OWR-13750341-002 | Unspecified Abelian Surface BSD Conjecture

Unrecoverable OWR statement (The extraction joins the end of a proved Gross-Zagier converse discussion to an abelian-surface BSD application theorem, without stating a s)

### OWR-13852-002 | Duplicate Deformed Hermitian-Yang-Mills Stability Criterion

Duplicate/merged OWR extraction (The official report contains one numerical subvariety criterion for solvability of the deformed Hermitian-Yang-Mills equation, represented c)

OWR-13852-001 / p.711

### OWR-13852-005 | Cross-Report Duplicate of the Normal Mod-Two Current Question

Duplicate/merged OWR extraction (This report repeats exactly the normal mod-two current question already represented by the earlier OWR-13103-005 record from Oberwolfach Rep)

OWR-13103-005 / p.1325

### OWR-13856-003 | Expository Metastability Heading for a Nonlocal Dirichlet Problem

Duplicate/merged OWR extraction (The official report uses this broad question as the title and motivation for an account that proves existence of stationary metastable profi)

OWR-13856-004 / p.1504

### OWR-13856-008 | Motivating Fixed-Point Question for Chinese Restaurant Permutations

Unrecoverable OWR statement (The official report introduces this pathwise question as motivation and then gives the relevant time-change limit results rather than presen)

### OWR-13857-004 | Duplicate Weighted-Curvature Extraction

Duplicate/merged OWR extraction (The passage duplicates the weighted-curvature flow alternatives recorded in OWR-13857-005 and does not state an independent problem. No sepa)

OWR-13857-005 / p.863

### OWR-13859-001 | Pointwise Subsequence Convergence of Approximation Operators

Answered in the source or not an open problem (solved (no; answered in the source))

### OWR-1386-003 | Sampling Algorithms for Maximum-Cut Density

Answered in the source or not an open problem (solved (max-cut density is testable; answered in the source))

### OWR-1386-013 | Codegree Thresholds for Hypergraph Hamilton Cycles

Duplicate extraction; the canonical question is OWR-1782-009.

OWR-1782-009 / p.1644

### OWR-1386-020 | Runtime Thresholds for Ant-Colony Optimization

Answered in the source or not an open problem (solved (answered in the source))

### OWR-1392-001 | Displacement Convexity and Ricci Curvature

Answered in the source or not an open problem (solved (a theorem recalled in the source))

### OWR-1393-003 | Reconstruction of Extended Affine Lie Algebras from Lie Tori

Answered in the source or not an open problem (solved (answered in the source))

### OWR-1394-002 | Alexandrov Uniqueness Conjecture for Convex Surfaces

Answered in the source or not an open problem (solved (no; answered in the source))

### OWR-1394-008 | Thin-Shell Concentration for Isotropic Convex Bodies

Answered in the source or not an open problem (solved (yes; answered in the source))

### OWR-13941-003 | Unspecified Convex-Hull Conjecture for a Pooling Relaxation

Unrecoverable OWR statement (The official report only says that a structured nonconvex pooling relaxation has a conjectured full convex-hull description in the remaining)

### OWR-13948-003 | Duplicate Quasiconvexity Extraction

Duplicate/merged OWR extraction (The passage duplicates the multiple-valued quasiconvexity question recorded in OWR-13948-004. No separate clean statement is promoted; the o)

OWR-13948-004 / p.1349

### OWR-13951-014 | Duplicate Petty-Conjecture Extraction

Duplicate/merged OWR extraction (The passage duplicates Petty’s projection-body conjecture recorded in OWR-13951-015. No separate clean statement is promoted; the original r)

OWR-13951-015 / p.923

### OWR-14222-008 | Combinatorial Multiplicity Indices and Terao Conjecture

Answered in the source or not an open problem (solved (the implication is proved; answered in the source))

### OWR-14224-001 | Detection Gaps Between Parametric and Kernel Tests

Answered in the source or not an open problem (solved (answered in the source))

### OWR-14224-002 | Adaptive Honest Confidence Intervals Under Moderate Sparsity

Answered in the source or not an open problem (solved (answered in the source))

### OWR-14224-003 | Oracle Quality Confidence Intervals with Unknown Sparsity

Answered in the source or not an open problem (solved (no in general; answered in the source))

### OWR-14297732-001 | Geometric Analysis Table-of-Contents Fragment

Unrecoverable OWR statement (The official report confirms that this record is a table-of-contents fragment rather than a mathematical question. No separate clean stateme)

### OWR-14297732-002 | Einstein Metrics Table-of-Contents Fragment

Unrecoverable OWR statement (The official report confirms that this record is a continuation of the table of contents rather than a mathematical question. No separate cl)

### OWR-14297733-004 | Uniform Long-Time Sobolev Bounds for Nonlinear Beam Equations

Unrecoverable OWR statement (The long-time Sobolev-bound question is followed immediately by the contribution's stability theorem, which supplies the requested constant )

### OWR-14297736-001 | Spectral-Geometry Workshop Overview

Unrecoverable OWR statement (The official report confirms that this extraction is a workshop summary and open-session introduction, not a standalone mathematical problem)

### OWR-14297736-012 | Spherical Neumann-Eigenvalue Context

Unrecoverable OWR statement (The official report confirms that this extraction is explanatory material duplicated by the adjacent complete open problems.)

### OWR-14297736-016 | Duplicate Spherical Robin-Eigenvalue Question

Unrecoverable OWR statement (The official report confirms that this extraction duplicates the complete open problem preserved in the preceding record.)

### OWR-14297736-019 | Genus-Free Steklov-Eigenvalue Bound

Unrecoverable OWR statement (The official report gives a counterexample and explicitly explains why the genus dependence cannot be removed; the extracted question is res)

### OWR-14297736-020 | Universal Steklov Isoperimetric Questions

Unrecoverable OWR statement (The official report answers both extracted questions negatively by giving unbounded constructions, so they are not open problems.)

### OWR-14297740-001 | Unspecified Anabelian Galois Representation Conjecture

Unrecoverable OWR statement (The extraction is workshop-introduction prose ending in a reference to the Coleman conjecture; the report states no separate conjecture at t)

### OWR-14297740-002 | Unspecified Anabelian Geometry Conjecture

Unrecoverable OWR statement (The extraction is a composite workshop overview of several research programs rather than a formulated mathematical problem. No separate clea)

### OWR-14297740-008 | Anderson-Ihara Maximal Pro-p Extension Conjecture

Unrecoverable OWR statement (The report explicitly says that this historical Anderson-Ihara question has an affirmative answer under the former Deligne-Ihara conjecture,)

### OWR-14297740-010 | Jannsen Finiteness for Galois Cohomology

Unrecoverable OWR statement (The extraction only notes that one finiteness hypothesis in a proved theorem is a special case of Jannsen's broader conjecture and does not )

### OWR-14297741-004 | Kasparov-Theory Heading Fragment

Unrecoverable OWR statement (The official report confirms that this extraction combines the end of preceding exposition with the heading of the Kasparov-theory contribut)

### OWR-14297744-001 | Equivariant Homotopy-Theory Workshop Abstract

Unrecoverable OWR statement (The official report confirms that this extraction is a workshop abstract and contains no standalone open problem.)

### OWR-14297744-002 | Equivariant Homotopy-Theory Table-of-Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a table-of-contents fragment, not a standalone open problem.)

### OWR-14297744-004 | Orphaned Equivariant h-Cobordism Conjecture Label

Unrecoverable OWR statement (The official report confirms that this extraction is only an orphaned heading and conjecture label; the complete problems are preserved in t)

### OWR-14297746-005 | Random Perturbations of Aperiodic Schrödinger Potentials

Unrecoverable OWR statement (The official report asks this introductory question and then answers it with localization and spectral-structure theorems; the remaining Fib)

### OWR-14297746-007 | Duplicate Random-Perturbation Question

Unrecoverable OWR statement (The official report confirms that this extraction repeats Question 4, which is preserved in the preceding record.)

### OWR-14298007-011 | Simplicial-Polytope Reconstruction Overview

Unrecoverable OWR statement (The official report uses this broad question to introduce known results and two precise conjectures; the surviving open case is preserved in)

### OWR-14298008-001 | Multiplication of Low-Regularity Functions

Unrecoverable OWR statement (The official report asks this expository question and immediately answers it with Young's theorem, including the exact regularity criterion.)

### OWR-14298008-002 | Equilibration from Log-Sobolev Inequalities

Unrecoverable OWR statement (The official report uses this question to introduce log-Sobolev inequalities and then gives quantitative exponential equilibration bounds; i)

### OWR-14298009-002 | Information Loss and an Intrinsic Arrow of Time

Answered in the source or not an open problem (solved (answered in the source))

### OWR-14298009-003 | Quantum Operations from Temporal Asymmetry

Answered in the source or not an open problem (solved (yes; answered in the source))

### OWR-14298009-005 | Channel-Divergence Discussion

Unrecoverable OWR statement (The official report only says that open questions were discussed; it does not state any question that can be reconstructed as a standalone p)

### OWR-14298013-006 | Poisson Ideals of the Virasoro Algebra

Unrecoverable OWR statement (The official report lists these questions as motivation and then states that the authors answer all of them; the extraction is not an unreso)

### OWR-14298014-001 | Reduced-Density-Matrix and RPA Discussion Fragment

Unrecoverable OWR statement (The official report mentions possible links between reduced-density-matrix conditions and random-phase approximation as a broad research dir)

### OWR-14298158-001 | Permutation-Patterns Table-of-Contents Fragment

Unrecoverable OWR statement (The official report confirms that this record is a table-of-contents and heading fragment rather than a mathematical question. No separate c)

### OWR-14298158-010 | Separable-Permutation Enumeration Introduction

Duplicate/merged OWR extraction (The official report confirms that this extraction repeats the rationality question already represented by OWR-14298158-002 and otherwise con)

OWR-14298158-002 / p.1723

### OWR-14298158-016 | Point-Probability Criterion Commentary

Unrecoverable OWR statement (The official report confirms that this fragment comments on a proved point-probability criterion and does not state an additional conjecture)

OWR-14298158-015 / p.1657

### OWR-14298159-006 | Balmer-Spectrum Motivation

Unrecoverable OWR statement (The official report uses this broad question to motivate the three precise tensor-triangular questions preserved in the following record.)

### OWR-14298160-001 | Artin-Group Workshop Introduction

Unrecoverable OWR statement (The official report confirms that this is introductory workshop context duplicating the concrete Artin-group problems preserved below.)

### OWR-14298160-002 | Homological Motivation from Stability Conditions

Unrecoverable OWR statement (The official report states a broad research motivation rather than a mathematically delimited open problem.)

### OWR-14298160-006 | Definition of the Coxeter Galaxy

Unrecoverable OWR statement (The official report uses this phrase as a section heading and then defines the Coxeter galaxy; it is not an open problem.)

### OWR-14298160-007 | Coxeter-Galaxy Structure Overview

Unrecoverable OWR statement (The official report uses this phrase as a section heading and immediately states structural theorems; the remaining questions are preserved )

### OWR-14298160-010 | Profinite Rigidity in the Coxeter Galaxy

Unrecoverable OWR statement (The official report uses this phrase as a section heading and follows it with a proved profinite-rigidity theorem for ranks at most three, n)

### OWR-14298162-001 | K-Stability Table-of-Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a table-of-contents fragment ending in the word conjecture, not a mathematical problem.)

### OWR-14298162-003 | Duplicate Negative-Part Valuation Question

Unrecoverable OWR statement (The official report confirms that this extraction repeats the same negative-part valuation question preserved in the preceding record.)

### OWR-14298163-001 | Discrete-Geometry Problems Table-of-Contents Entry

Unrecoverable OWR statement (The official report confirms that this extraction is a table-of-contents fragment, not a mathematical problem.)

### OWR-14298163-013 | No-Three-in-Line Problem

Duplicate of canonical problem COMB-011

COMB-011 / p.1711

### OWR-14298169-002 | Workshop Contents Listing for Graph and Groupoid Algebras

Unrecoverable OWR statement (The official report confirms that this extraction is a contents-listing fragment, not a mathematical problem. No separate clean statement is)

### OWR-14298169-004 | Move Classification of Countable Vertex-Finite Graphs

Unrecoverable OWR statement (The official report was checked and confirms a genuine eight-part move-classification conjecture, but the extracted text omits the definitio)

### OWR-14298170-006 | Positive-Scalar-Curvature Gluing Overview

Unrecoverable OWR statement (The official report confirms that this extraction is only the heading and introduction to three open questions, all preserved in the followi)

### OWR-14298170-014 | Scalar-Curvature Rigidity Context

Unrecoverable OWR statement (The official report uses this broad question to introduce a proved rigidity theorem; the remaining precise smoothing question is preserved i)

### OWR-14298170-019 | Rotating-Black-Hole Motivation

Unrecoverable OWR statement (The official report confirms that this extraction is contextual motivation duplicated by the complete vacuum Kerr conjecture in the followin)

### OWR-14298363-001 | Higher-Dimensional Generalization of a Strassen-Type Result

Unrecoverable OWR statement (The official report says only that a recent one-dimensional generalization of Strassen’s theorem should be extended to higher dimensions; it)

### OWR-14298363-009 | Hedge-Supermartingale Exposition

Unrecoverable OWR statement (The official report confirms that this extraction is an expository derivation of the Hedge supermartingale and does not pose a standalone op)

### OWR-14298364-008 | Informal Question about a Rescaled Einstein Limit

Unrecoverable OWR statement (The phrase asks only whether a newly obtained limiting metric is 'new or interesting' and does not specify a falsifiable mathematical target)

### OWR-14298365-001 | Workshop Summary on Growth and Expansion in Groups

Unrecoverable OWR statement (The official report confirms that the extraction is workshop-summary prose and a list of lecture topics, not a standalone conjecture. No sep)

### OWR-14298365-016 | Duplicate Alternating-Group Expander-Power Question

Duplicate/merged OWR extraction (The official report confirms that this extraction repeats the power-of-alternating-groups question already represented canonically by OWR-14)

OWR-14298365-015 / p.603

### OWR-14298367-005 | Weighted-Shift Supercyclicity Converse

Unrecoverable OWR statement (The official report asks whether supercyclicity of the associated weighted shift implies supercyclicity of the composition operator and imme)

### OWR-14298372-002 | Quantum-Channel Interpolation Characterization

Unrecoverable OWR statement (The official report poses this interpolation question and then gives a necessary-and-sufficient characterization as Theorem 5.)

### OWR-14298372-004 | Setup for Noncommutative Stability Conjectures

Unrecoverable OWR statement (The official report confirms that this extraction is introductory setup for Conjectures 10 and 12, both preserved in the following records.)

### OWR-14298372-007 | Free-Conjugate-Variable Regularity Overview

Unrecoverable OWR statement (The official report uses this broad question to organize a survey of several established regularity results; it does not isolate an unresolv)

### OWR-14298374-001 | Random-Surface Modeling Motivation

Unrecoverable OWR statement (The official report uses this question to introduce the stochastic surface models developed in the remainder of the abstract; it is not a de)

### OWR-14298374-002 | Infinite-Resolution Random-Surface Question

Unrecoverable OWR statement (The official report asks whether the limit surface is well defined and immediately answers it through continuity and differentiability crite)

### OWR-14298374-006 | Autoencoder Optimization Perspectives

Unrecoverable OWR statement (The official report describes three equivalent optimization viewpoints and their modeling consequences; it does not state an unresolved math)

### OWR-14298375-001 | Unspecified Hyperplane Arrangement Conjecture

Answered in the source or not an open problem (solved (not a problem; retire))

### OWR-14298376-001 | Unspecified Real Enumerative Geometry Conjecture

Unrecoverable OWR statement (The extraction is a table-of-contents fragment ending in a contribution title, not a mathematical statement. No separate clean statement is )

### OWR-14298580-002 | Unspecified Low-Dimensional Topology Conjecture

Unrecoverable OWR statement (The extraction consists only of the trailing word 'conjecture' from the workshop introduction and contains no recoverable target at that loc)

### OWR-14298580-005 | Taut Foliations on Rational Homology Spheres

Duplicate/merged OWR extraction (The extraction repeats the taut-foliation and geography context for the L-space conjecture retained in OWR-14298580-004. No separate clean s)

OWR-14298580-004 / p.1125

### OWR-14298580-006 | Strong Geography Restrictions in Heegaard Floer Homology

Duplicate/merged OWR extraction (The extraction supplies context and a theorem before introducing the strong-geography question, which is retained completely in OWR-14298580)

OWR-14298580-008 / p.1239

### OWR-14298580-010 | Motivic Generalization of Milnor Product Formulas

Answered in the source or not an open problem (solved (answered in the source; the general case is a preprint))

### OWR-14298581-001 | Calabi-Yau Torelli Table-of-Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a table-of-contents fragment and contains no mathematical conjecture.)

### OWR-14298581-004 | Isolated Complex-Geometry Question Label

Unrecoverable OWR statement (The extraction contains only the label “Question 1.” The official report has no additional standalone problem at this position beyond the co)

### OWR-14298581-009 | Strict Calabi-Yau Automorphism Preamble

Unrecoverable OWR statement (The official report confirms that this extraction stops immediately before the actual questions, which are preserved in the following record)

### OWR-14298581-012 | Equivariant Fukaya-Cohomology Discussion

Unrecoverable OWR statement (The official report poses these questions to motivate the talk, immediately rejects the naive first two formulations, and develops a more el)

### OWR-14298581-013 | Isolated Hyperkähler Conjecture Label

Unrecoverable OWR statement (The official report confirms that this extraction is an isolated table-of-contents fragment containing only the word “conjecture.”)

### OWR-14298583-002 | Observed Extreme-Rainfall Trends

Unrecoverable OWR statement (The official report poses this empirical question and then gives the study’s answer: events became more frequent, while their individual-siz)

### OWR-14298584-010 | Duplicate Finite-Field Polynomial-Multiplication Conjecture

Duplicate/merged OWR extraction (The official report confirms that this overlapping extraction repeats the finite-field polynomial-multiplication conjecture represented cano)

OWR-14298584-009 / p.2100

### OWR-14298587-001 | Stable Phase-Transition Table-of-Contents Entry

Unrecoverable OWR statement (The official report confirms that this extraction is only a table-of-contents entry, not a mathematical problem.)

### OWR-14298588-006 | Categorical Local Langlands Program Summary

Duplicate/merged OWR extraction (The official report confirms that this extraction is introductory programmatic exposition; the categorical local Langlands conjecture itself)

OWR-14298588-008 / p.858

### OWR-14298588-007 | Equivalent Categorical Local Langlands Formulation

Duplicate/merged OWR extraction (The official report confirms that this extraction gives an equivalent functorial formulation of the categorical local Langlands conjecture r)

OWR-14298588-008 / p.858

### OWR-14298588-012 | Arthur-Eigensheaf Spectral-Side Continuation

Duplicate/merged OWR extraction (The official report confirms that this extraction is discussion and evidence following the Arthur-parameter eigensheaf conjecture represente)

OWR-14298588-010 / p.876

### OWR-14298588-015 | Reported Height Theorem for Abelian Isogeny Classes

Unrecoverable OWR statement (The official report states the bounded-height finiteness result as a theorem conditional on the Mumford-Tate conjecture and notes the cases )

### OWR-14298589-003 | Mixed Square-Tiled-Surface Talk Transition

Unrecoverable OWR statement (The official report confirms that this extraction joins the end of one contribution to the title and opening sentence of the next and contai)

### OWR-14298589-007 | Resolved Averaged Twist-Torus Distribution Lead-In

Duplicate/merged OWR extraction (The official report poses this broad question as a lead-in and immediately gives the authors’ averaged equidistribution theorem; the remaini)

OWR-14298589-008 / p.1927

### OWR-14298589-012 | Duplicate Boundary-Regime Poisson-Limit Conjecture

Duplicate/merged OWR extraction (The official report confirms that this overlapping extraction repeats the volume-identity question and boundary-regime Poisson-limit conject)

OWR-14298589-011 / p.1935

### OWR-14298591-001 | Workshop Summary on Mathematical General Relativity

Unrecoverable OWR statement (The official report confirms that this extraction is workshop-summary prose covering several talks and contains no standalone problem. No se)

### OWR-14298591-003 | Reported Asymptotics for Silent Wave Equations

Unrecoverable OWR statement (The official report presents asymptotic estimates of all orders and a Maxwell-field application as results of the reported work; it does not)

### OWR-14298591-004 | Incomplete Lorentzian-Geometry Result Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a damaged fragment of a result and surrounding technical hypotheses, not a standalone o)

### OWR-14298592-001 | Workshop Contents Listing for Low-Dimensional Topology

Unrecoverable OWR statement (The official report confirms that this extraction is a contents-listing fragment, not a mathematical problem. No separate clean statement is)

### OWR-14298592-005 | Ribbon-Clasp Obstruction Methods

Unrecoverable OWR statement (The official report describes attempted applications of proved propositions and says that no examples had yet been obtained; it does not for)

### OWR-14298795-001 | Workshop Summary on Anabelian Geometry

Unrecoverable OWR statement (The official report confirms that this extraction is workshop-summary prose covering many talks and contains no standalone conjecture. No se)

### OWR-14298795-002 | Workshop Contents Listing for Anabelian Geometry

Unrecoverable OWR statement (The official report confirms that this extraction is a contents-listing fragment, not a mathematical problem. No separate clean statement is)

### OWR-14298795-009 | Incomplete Nonlinear Eisenstein-Conjecture Continuation

Unrecoverable OWR statement (The official report confirms that this extraction is only a dangling sentence continuing a preceding nonlinear differential-equation conject)

### OWR-14298798-001 | Resolved Equilibration-Time Lead-In for Open-Boundary ASEP

Unrecoverable OWR statement (The official report raises the equilibration-time question only as a lead-in and immediately states that the reported joint work obtains the)

### OWR-14298802-003 | Workshop Contents Listing for Algebraic K-Theory

Unrecoverable OWR statement (The official report confirms that this extraction is a contents-listing fragment, not a mathematical problem. No separate clean statement is)

### OWR-14298802-004 | Expository Telescope-Conjecture Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is background exposition on the telescope conjecture, whose higher-height form is discusse)

### OWR-14298802-007 | Reported Structure of Circle-Cochain THH

Unrecoverable OWR statement (The official report discusses structural results for the topological Hochschild homology of spherical cochains on the circle and does not po)

### OWR-14298804-002 | Duplicate Strongly Polynomial Linear-Programming Lead-In

Duplicate/merged OWR extraction (The official report confirms that this extraction is contextual commentary about progress toward the strongly polynomial linear-programming )

OWR-14298804-001 / p.2107

### OWR-14298804-003 | Workshop Contents Listing for Combinatorial Optimization

Unrecoverable OWR statement (The official report confirms that this extraction is a contents listing of six open-problem contributions, not a standalone problem. No sepa)

### OWR-14298804-008 | Duplicate Dimension-Parameterized Zonotope Containment

Duplicate/merged OWR extraction (The official report confirms that this extraction repeats the fixed-parameter tractability question represented canonically by OWR-14298804-)

OWR-14298804-007 / p.2081

### OWR-14298804-010 | Duplicate Finite-Strategy Question for the Distance Game

Duplicate/merged OWR extraction (The official report confirms that this extraction repeats the finite-strategy question represented canonically by OWR-14298804-009. No separ)

OWR-14298804-009 / p.2080

### OWR-14298804-013 | Duplicate Fare-Zone Approximation Question

Duplicate/merged OWR extraction (The official report confirms that this extraction repeats the tree fare-zone approximation question represented canonically by OWR-14298804-)

OWR-14298804-012 / p.2061

### OWR-14298804-015 | Duplicate Light-Tournament Two-Coloring Question

Duplicate/merged OWR extraction (The official report confirms that this extraction repeats the complexity question represented canonically by OWR-14298804-014. No separate c)

OWR-14298804-014 / p.2110

### OWR-14298804-017 | Duplicate Fiorini Feedback-Vertex-Set Question

Duplicate/merged OWR extraction (The official report confirms that this extraction is commentary on the still-open Fiorini question represented canonically by OWR-14298804-0)

OWR-14298804-016 / p.2085

### OWR-14298805-001 | Survey of Smooth Modular Representations

Unrecoverable OWR statement (The official report presents “what is known and what is not known?” as the heading of a broad survey, not as a standalone mathematical probl)

### OWR-14298805-008 | Supercuspidal Character Survey Heading

Unrecoverable OWR statement (The official report uses “What do characters look like?” as an expository section heading and historical organizing question, not as a preci)

### OWR-14298808-004 | Classical Busemann-Petty Problem

Answered in the source or not an open problem (solved (answered in the source))

### OWR-14298808-005 | Busemann-Petty Problem for Arbitrary Measures

Answered in the source or not an open problem (solved (answered in the source))

### OWR-14298808-012 | Duplicate Growing-Parameter Limit Question

Unrecoverable OWR statement (The official report confirms that this extraction is the preamble to, and repeats, the precise asymptotic-distribution question preserved in)

### OWR-14298808-014 | Centroidal-Section Introductory Question

Unrecoverable OWR statement (The official report uses this broad sentence to introduce the precise centroidal-section problem preserved in the following record.)

### OWR-14299081-001 | Workshop Summary Mention of the Erdős-Hajnal Conjecture

Unrecoverable OWR statement (The official report confirms that this extraction is workshop-summary prose mentioning the Erdős-Hajnal conjecture, which is stated separate)

### OWR-14299081-002 | Corrupted Graph-Theory Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a damaged contents-listing fragment, not a mathematical problem. No separate clean stat)

### OWR-14299081-009 | Duplicate Erdős-Hajnal Conjecture Passage

Duplicate/merged OWR extraction (The official report confirms that this extraction repeats the Erdős-Hajnal conjecture represented canonically by OWR-14299081-008. No separa)

OWR-14299081-008 / p.1844

### OWR-14299081-010 | Duplicate Erdős-Hajnal Conjecture Passage

Duplicate/merged OWR extraction (The official report confirms that this extraction repeats the Erdős-Hajnal conjecture represented canonically by OWR-14299081-008. No separa)

OWR-14299081-008 / p.1844

### OWR-14299081-011 | Truncated Extremal-Graph-Theory Conjecture

Unrecoverable OWR statement (The official report confirms that this extraction contains only an isolated conjecture label and no independently recoverable mathematical s)

### OWR-14299081-020 | Duplicate Erdős-Hajnal Conjecture Lead-In

Duplicate/merged OWR extraction (The official report confirms that this extraction is another occurrence of the Erdős-Hajnal conjecture represented canonically by OWR-142990)

OWR-14299081-008 / p.1844

### OWR-14299081-023 | Truncated Graph-Theory Conjecture

Unrecoverable OWR statement (The official report confirms that this extraction is a truncated conjecture fragment with no independently recoverable mathematical statemen)

### OWR-14299081-024 | Solved Design-Conjecture Prose Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is prose about a new proof of the design Existence Conjecture, not a new open problem. No )

### OWR-14299082-001 | Descriptive-Set-Theory Table-of-Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is table-of-contents debris ending in the word “conjecture,” not a recoverable mathematica)

### OWR-14299082-005 | Duplicate Generic Point Property Question

Unrecoverable OWR statement (The official report confirms that this extraction repeats Question 5, which is preserved in the preceding record.)

### OWR-14299083-001 | Similarity Metrics for Stochastic Processes

Unrecoverable OWR statement (The official report poses this broad question and then presents an optimal-transport construction, proving that the relevant bisimulation me)

### OWR-14299083-003 | Adam Optimization Performance

Unrecoverable OWR statement (The official report uses this question as the title of a talk and then supplies its proposed mechanism: heavy-tailed class imbalance creates)

### OWR-14299083-004 | Definition of a Causal Representation

Unrecoverable OWR statement (The official report presents this as a broad epistemological framing question rather than a precise mathematical problem with stated hypothe)

### OWR-14299083-005 | Criteria for a Good Representation

Unrecoverable OWR statement (The official report poses this as a philosophical organizing question in an expository discussion, not as a standalone mathematical problem.)

### OWR-14299084-007 | Duplicate Distribution-Verification Question

Unrecoverable OWR statement (The official report repeats the two motivating questions already preserved in the earlier verifiable-data-science record and then gives the )

### OWR-14299084-008 | Evasive-LWE Research-Direction Discussion

Unrecoverable OWR statement (The official report records an open-ended discussion about which research direction to pursue after counterexamples, not a mathematically te)

### OWR-14299086-009 | Duplicate Free-Group Word-Measure Conjecture

Unrecoverable OWR statement (The official report confirms that this extraction contains contextual material followed by the same Conjecture 4 preserved in the next recor)

### OWR-14299088-001 | Optimization Workshop Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a damaged table-of-contents fragment, not a mathematical conjecture. No separate clean )

### OWR-14299088-015 | Duplicate Dimension-Free Separation-Exponent Problem

Duplicate/merged OWR extraction (The official report confirms that this open-problem-session extraction repeats the dimension-free separation-exponent problem represented ca)

OWR-14299088-010 / p.660

### OWR-14299089-006 | Duplicate Minimal Convex-Difference Decomposition Problem

Duplicate/merged OWR extraction (The official report confirms that this overlapping extraction repeats the minimal convex-difference decomposition problem represented canoni)

OWR-14299089-005 / p.2091

### OWR-14299093-001 | Landis-Conjecture Table-of-Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is table-of-contents debris, not the statement of a conjecture.)

### OWR-14299095-001 | Ensemble Necessity in Data Assimilation

Unrecoverable OWR statement (The official report poses this question as a talk title and then gives a negative experimental answer in its studied setting: a learned sing)

### OWR-14299097-001 | Finite-Group-Theory Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a table-of-contents fragment, not a mathematical conjecture. No separate clean statemen)

### OWR-14299097-002 | Finite-Group-Theory Contents Continuation

Unrecoverable OWR statement (The official report confirms that this extraction is a continuation of the table of contents, not a mathematical problem. No separate clean )

### OWR-14299281-002 | Algebraic-Groups Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a table-of-contents fragment spanning unrelated talks and does not state a mathematical)

### OWR-14299281-003 | Truncated Algebraic-Group Conjecture

Unrecoverable OWR statement (The official report confirms that the extracted sentence omits both the conjecture and the name to which it refers, so a mathematical statem)

### OWR-14299283-006 | Duplicate Thin-Affine-Grassmannian Question Continuation

Duplicate/merged OWR extraction (The official report confirms that this extraction is a concluding sentence pointing back to the question represented canonically by OWR-1429)

OWR-14299283-002 / p.251

### OWR-14299283-011 | Reported Quasi-BPS Construction

Unrecoverable OWR statement (The official report presents the quasi-BPS category as the authors' proposed and partially established answer to a preceding geometric-reali)

### OWR-14299284-003 | Truncated Derived-Category Conjecture

Unrecoverable OWR statement (The official report confirms that the extraction contains only the phrase introducing a conjecture and omits the conjecture itself. No separ)

### OWR-14299284-011 | Duplicate Atomic-Decomposition Extraction

Duplicate/merged OWR extraction (The official report confirms that this extraction is an overlapping lead-in to the same atomic semiorthogonal-decomposition conjecture repre)

OWR-14299284-012 / p.640

### OWR-14299285-002 | Resolved Directed-Landscape Converse Lead-In

Unrecoverable OWR statement (The official report asks this converse question only as a lead-in and immediately answers it with the stated theorem, so the extraction is n)

### OWR-14299288-001 | Toric-Geometry Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a damaged table-of-contents fragment, not a mathematical conjecture. No separate clean )

### OWR-14299288-007 | Duplicate Zero-Mutable Log-Resolution Conjecture

Duplicate/merged OWR extraction (The official report confirms that this introductory extraction states the same zero-mutable log-resolution conjecture represented canonicall)

OWR-14299288-008 / p.728

### OWR-14299289-001 | Relative-Langlands Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a table-of-contents fragment and does not contain the conjecture whose final word prece)

### OWR-14299289-004 | Period-Sheaf Expository Fragment

Unrecoverable OWR statement (The official report confirms that this extraction consists of a definition of the period sheaf followed by the setup for a different period )

### OWR-14299289-006 | Solved-Conjecture Tail for Spherical Varieties

Unrecoverable OWR statement (The official report confirms that the extraction is the concluding sentence of an earlier conjecture, immediately saying that it is known in)

### OWR-14299289-011 | Duplicate Local Geometric Duality Formulation

Duplicate/merged OWR extraction (The official report confirms that this extraction repeats the local geometric-duality equivalence represented canonically by OWR-14299289-01)

OWR-14299289-010 / p.347

### OWR-14299289-016 | Duplicate Global Geometric Duality Conjecture

Duplicate/merged OWR extraction (The official report confirms that this extraction repeats the global geometric-duality conjecture represented canonically by OWR-14299289-01)

OWR-14299289-015 / p.733

### OWR-14299289-018 | Geometric Numerical Conjecture Lead-In

Duplicate/merged OWR extraction (The official report confirms that this extraction is an expository overview and setup; the actual geometric numerical conjecture is represen)

OWR-14299289-019 / p.32

### OWR-14299291-001 | Corrupted Non-Archimedean-Geometry Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a damaged contents listing, not a mathematical problem. No separate clean statement is )

### OWR-14299292-001 | Computational Group Theory Workshop Narrative

Unrecoverable OWR statement (The official report confirms that this extraction is workshop-summary prose ending at a dangling occurrence of "conjecture", not a standalon)

### OWR-14299292-002 | Unnamed Candidate Monster-Subgroup Extensions

Unrecoverable OWR statement (The official report says only that conjugacy-class information produced two candidate groups for the prospective Monster maximal subgroup of)

### OWR-14299292-008 | Reported Tensor Null-Pattern Algorithm

Unrecoverable OWR statement (The official report presents an algorithm that either finds a change of basis revealing the prescribed parameterized null pattern or determi)

### OWR-14299511-009 | Reported Quantum Search for Binary Fusion Data

Unrecoverable OWR statement (The official report asks how to find binary fusion data efficiently and immediately presents the authors' quantum algorithm as a solution; i)

### OWR-14299513-003 | Truncated Harmonic-Analysis Conjecture

Unrecoverable OWR statement (The official report confirms that this extraction contains only an isolated conjecture label and no independently recoverable mathematical s)

### OWR-14299516-001 | Truncated Algebraic-K-Theory Abstract Fragment

Unrecoverable OWR statement (The official report confirms that this extraction begins with the tail of a sentence from the workshop abstract and does not contain a stand)

### OWR-14299516-004 | Rhetorical Condensed-Cohomology Talk Heading

Unrecoverable OWR statement (The official report confirms that this extraction is a rhetorical talk heading followed by its title and author, not a standalone open probl)

### OWR-14299518-002 | Two Erdős Problems on Partial Block Designs

Already answered in the original Oberwolfach workshop report (not an open problem)

### OWR-14299518-020 | Resolved Root-Distribution Lead-In for Littlewood Polynomials

Unrecoverable OWR statement (The official report uses the broad root-distribution question as a lead-in to theorems proved in the reported work rather than as a separate)

### OWR-14299520-001 | Interdisciplinary Renormalization Discussion Prompt

Unrecoverable OWR statement (The official report confirms that this is a broad workshop-motivation prompt about interaction between physicists and mathematicians, not a )

### OWR-14299521-001 | Corrupted Differential-Geometry Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a damaged contents-listing fragment, not a mathematical problem. No separate clean stat)

### OWR-14299574-009 | Proved Symplectic-Cohomology Factorization Fragment

Unrecoverable OWR statement (The official report confirms that this extraction states a refined conjecture only to report that it is proved for all quasihomogeneous comp)

### OWR-14299577-008 | Reported Uniform Stability Bound for Twisted Braid-Group Homology

Unrecoverable OWR statement (The official report explains that the required uniform stability bound is proved in the cited work; this extraction is proof exposition surr)

### OWR-14299577-009 | Duplicate Reported Twisted Braid-Group Stability Result

Unrecoverable OWR statement (The official report confirms that this overlapping extraction repeats the stable-homology calculation and uniform stability bound already de)

### OWR-14299577-010 | Reported Function-Field Moment Result

Unrecoverable OWR statement (The official report states that the cited work proves the fixed-function-field moment formula for sufficiently large odd prime powers and de)

### OWR-14299577-022 | Primitive-Root Orders of Elliptic-Curve Frobenius Traces

Answered in the source or not an open problem (partially_solved)

### OWR-14299579-003 | Huang-Yang Derivation Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a formula and exposition from the derivation of a proved Huang-Yang energy expansion, n)

### OWR-14299579-004 | Huang-Yang Completion-of-Squares Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a technical proof fragment describing a completion-of-squares argument, not an open pro)

### OWR-14299580-004 | Truncated Yang-Baxter Conjecture

Unrecoverable OWR statement (The official report confirms that the extracted fragment contains only a reference to a conjecture and no recoverable formulation. No separa)

### OWR-14299580-011 | Unidentified Hecke-Type Equivalence Class

Unrecoverable OWR statement (The official report asks whether the last item in an unpublished classification list is realizable, but it does not identify that equivalenc)

### OWR-14299583-001 | Functional Inequalities Workshop Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a damaged table-of-contents fragment, not a mathematical conjecture. No separate clean )

### OWR-14299583-006 | Reported Mean-Norm Bound for Nonsymmetric Convex Bodies

Unrecoverable OWR statement (The official report identifies the nonsymmetric case as a formerly open problem and then states the authors' theorem $M(K)\leq C\log(n)/\sqr)

### OWR-14299583-007 | Reported Nonexistence of Critical Loop-Soup Double Points

Unrecoverable OWR statement (The official report first motivates the critical double-point question and then states the authors' result proving that such double points d)

### OWR-14299904-003 | Upper Elements in Grothendieck Schubert Supports

Duplicate extraction; the canonical question is OWR-14299904-002.

OWR-14299904-002 / p.1757

### OWR-14299904-006 | Triangle-Counting Figure Caption

Unrecoverable OWR statement (The official report confirms that this extraction is a figure caption whose missing diagram and parameters prevent recovery of a standalone )

### OWR-14299904-007 | Duplicate Homotopy Question in Permutahedron-Triangulation Exposition

Duplicate/merged OWR extraction (The official report confirms that this extraction is mostly expository discussion and repeats the explicit homotopy problem represented cano)

OWR-14299904-008 / p.357

### OWR-14299905-001 | Workshop-Highlights Fragment

Unrecoverable OWR statement (The official report confirms that this extraction splices conclusions about permanent anticoncentration and Rota’s basis conjecture without )

### OWR-14299905-002 | Spectral-Graph-Theory Highlight Fragment

Unrecoverable OWR statement (The official report confirms that this extraction summarizes proved workshop highlights and begins with a stray word from an earlier conject)

### OWR-14299905-003 | Combinatorics Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a table-of-contents fragment spanning unrelated talks and does not state a problem. No )

### OWR-14299905-004 | Extremal-Combinatorics Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is another table-of-contents fragment and does not state a problem. No separate clean stat)

### OWR-14299905-011 | Solved Sparse-Graph Conjecture Fragment

Unrecoverable OWR statement (The official report confirms that the extraction attaches a stray word to two theorems proving the formerly conjectured sparse-graph stateme)

### OWR-14299906-006 | Existence of Nonsofic Groups

Duplicate of canonical problem ALG-008

ALG-008 / p.216

### OWR-14299906-011 | Duplicate Higher-Dimensional Kakeya Status

Unrecoverable OWR statement (The official report confirms that this extraction only repeats the higher-dimensional status of the Kakeya conjecture preserved in the prece)

### OWR-14299908-001 | Workshop-Summary Extraction Artifact

Unrecoverable OWR statement (The extraction starts with the final word of a preceding reference to Schanuel's conjecture and then continues with workshop-summary prose; )

### OWR-14299908-006 | Expository h-Minimality Passage

Unrecoverable OWR statement (The official contribution is expository text introducing h-minimality and summarizing applications; the extracted passage contains no separa)

### OWR-14299909-003 | Duplicate Multiple-Eisenstein Relations Preamble

Unrecoverable OWR statement (The official report confirms that this extraction is an informal lead-in to the precise kernel conjecture preserved in the following record.)

### OWR-14299909-005 | Critical-Value Conjecture Preamble

Unrecoverable OWR statement (The official report confirms that this extraction is background on period polynomials immediately preceding the complete critical-value conj)

### OWR-14299909-007 | Duplicate Critical-Value and Maeda Discussion

Unrecoverable OWR statement (The official report confirms that this extraction is the explanatory continuation of the critical-value conjecture preserved in the precedin)

### OWR-14299911-001 | Median Geometry Workshop Contents Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a table-of-contents fragment, not a mathematical conjecture. No separate clean statemen)

### OWR-14299911-009 | Special Cube Complexes Talk Transition

Unrecoverable OWR statement (The official report confirms that this extraction is expository narrative ending with an unspecific reference to questions and then crossing)

### OWR-14299911-011 | Median Geometry Problem-List Introduction

Unrecoverable OWR statement (The official report confirms that this extraction is only an introductory sentence for a problem list; the individual mathematical questions)

### OWR-14299912-001 | Nonlinear-Waves Workshop Introduction Fragment

Unrecoverable OWR statement (The official report confirms that this extraction is a truncated workshop-summary fragment containing no identifiable conjecture. No separat)

### OWR-14299914-002 | Causal-Attention Retrieval Motivation

Unrecoverable OWR statement (The official report uses this question to define its retrieval observable and then derives an asymptotic formula and a rigorous U-shaped ret)

### OWR-14299915-001 | Cohomological Finite-Generation Survey Summary

Unrecoverable OWR statement (The official report confirms that this extraction is a programmatic summary of a survey and does not state a standalone open problem. No sep)

### OWR-14299915-015 | Synthetic-Building Classification Preamble

Unrecoverable OWR statement (The official report confirms that these broad questions introduce the precise minimal-synthetic-building conjectures preserved in the follow)

### OWR-14299915-017 | Duplicate Noncompact Casacuberta-Dicks Question

Duplicate/merged OWR extraction (The official report states that the compact contractible two-dimensional case is solved and leaves exactly the noncompact Casacuberta-Dicks )

OWR-14298007-009 / p.1139

### OWR-14299915-025 | Exotic-Four-Sphere Special-Case Discussion

Unrecoverable OWR statement (The official report identifies this fragment as a special case of the preceding doubles question rather than a separate extracted problem. N)

### OWR-14299916-008 | Duplicate Human-Alignment Question

Duplicate/merged OWR extraction (This extraction repeats the same human-alignment question retained in OWR-14299916-007 and appends an author name and page footer; it is not)

OWR-14299916-007 / p.2086

### OWR-14299916-011 | Duplicate Unified-Theory Question

Duplicate/merged OWR extraction (This concluding rhetorical question repeats the unified-theory problem retained in OWR-14299916-002 and does not introduce a distinct mathem)

OWR-14299916-002 / p.2111

### OWR-1452-011 | Jacobian Conjecture for Polynomial Maps

Duplicate of canonical problem SMA-016

SMA-016 / p.303

### OWR-1452-018 | Rational Periodic Points from Finite-Field Periodicity

Answered in the source or not an open problem ()

### OWR-1453-004 | Reconstructing Complete Local Rings from Finite Quotients

Answered in the source or not an open problem (solved (no in general; answered in the source itself))

### OWR-1453-006 | Duplicate Relative Galois Reconstruction Conjecture

Duplicate/merged OWR extraction (This is a redundant, assumption-incomplete paraphrase of the relative Galois reconstruction conjecture recorded precisely as OWR-1453-005 fr)

OWR-1453-005 / p.2037

### OWR-14602-009 | Refined Local Langlands Parametrization of Tempered Packets

Duplicate/merged OWR extraction (The official report shows that this record is a second extraction of the refined local Langlands conjecture retained as OWR-14602-010. No se)

OWR-14602-010 / p.605

### OWR-14603-003 | Algebraic Values and Relations of Mahler Functions

Answered in the source or not an open problem (solved (answered in the source; made effective by Adamczewski-Faverjon))

### OWR-14603-011 | Duplicate Small-Value Sparsity Question for Littlewood Polynomials

Duplicate/merged OWR extraction (The official report contains this small-value counting question once; OWR-14603-013 gives its complete formula and is the canonical clean re)

OWR-14603-013 / p.121

### OWR-14604-003 | Discrepancy versus Spectral Expansion in Sparse Regular Graphs

Answered in the source or not an open problem (solved (answered in the source: no in general; yes for Cayley and vertex-transitive graphs))

### OWR-14604-010 | Duplicate Three-Event Correlation Question

Duplicate/merged OWR extraction (The official report shows that this extraction duplicates the three-increasing-event problem represented canonically by OWR-14604-009. No se)

OWR-14604-009 / p.1961

### OWR-14604-015 | Duplicate Progression-Free-Set Enumeration Question

Duplicate/merged OWR extraction (This extraction duplicates the arithmetic-progression-free enumeration problem represented more completely by OWR-14604-016. No separate cle)

OWR-14604-016 / p.1639

### OWR-14605-006 | Expository Automorphic-Correction Description of Lorentzian Kac-Moody Algebras

Duplicate/merged OWR extraction (The official report states that Lorentzian Kac-Moody Lie and superalgebras are automorphic corrections of hyperbolic Kac-Moody algebras as e)

OWR-14605-007 / p.192

### OWR-1461-003 | Cramér-Wold Uniqueness for Infinite Signed Measures

Answered in the source or not an open problem (solved (answered in the source))

### OWR-14611-006 | Deformation Invariance of Rationality

Answered in the source or not an open problem (solved (no; answered in the source))

### OWR-14746-009 | Duplicate Dense Frobenius-Eigenvector Conjecture

Duplicate/merged OWR extraction (The official report states one dense Frobenius-eigenvector conjecture for the convergent completed group algebra, represented canonically by)

OWR-14746-008 / p.52

### OWR-14748-011 | Duplicate Thom Conjecture for the Hyperfinite Factor

Duplicate/merged OWR extraction (The official report states Andreas Thom's conjecture once; OWR-14748-012 is the canonical clean record. This entry asks the same question wi)

OWR-14748-012 / p.1438

### OWR-15170-005 | Duplicate Northcott-Property Question

Duplicate/merged OWR extraction (This extraction duplicates the bounded-degree-composita Northcott question represented canonically by OWR-15170-004. No separate clean state)

OWR-15170-004 / p.160

### OWR-15170-025 | Composite of Unrelated Field-Theory Questions

Unrecoverable OWR statement (The official problem-session list shows that this extraction concatenates several unrelated questions about definable valuations, finitely a)

### OWR-15170-028 | Composite of Unrelated Decidability Questions

Unrecoverable OWR statement (The official problem-session list shows that this extraction joins the decidability problem for rigid analytic function rings to a separate )

### OWR-15173-002 | Cuspidal Homology for Gaussian Bianchi Congruence Groups

Answered in the source or not an open problem ()

### OWR-15181-012 | Irreducible Factorization in Generalized Power Series Rings

Unrecoverable OWR statement (The official report presents the asserted factorization as a proved theorem, with its extension to omnific integers, rather than as an open )

### OWR-15207-001 | Gendered Descriptions and the Leaky Pipeline in Mathematics

Unrecoverable OWR statement (The official report presents this as a historical, biographical, or sociological research prompt rather than an open mathematical problem. N)

### OWR-15207-003 | Emmy Noether’s Formation of a Mathematical School

Unrecoverable OWR statement (The official report presents this as a historical, biographical, or sociological research prompt rather than an open mathematical problem. N)

### OWR-15207-004 | Barriers to Women in Danish Research Mathematics

Unrecoverable OWR statement (The official report presents this as a historical, biographical, or sociological research prompt rather than an open mathematical problem. N)

### OWR-15207-005 | Careers of Danish Women Mathematicians

Unrecoverable OWR statement (The official report presents this as a historical, biographical, or sociological research prompt rather than an open mathematical problem. N)

### OWR-15207-006 | Historical Scientific Personae and Women Mathematicians

Unrecoverable OWR statement (The official report presents this as a historical, biographical, or sociological research prompt rather than an open mathematical problem. N)

### OWR-15207-007 | Practice-Based Personae of Women in Science

Unrecoverable OWR statement (The official report presents this as a historical, biographical, or sociological research prompt rather than an open mathematical problem. N)

### OWR-15207-008 | The Woytinskys’ Path to Statistical Prominence

Unrecoverable OWR statement (The official report presents this as a historical, biographical, or sociological research prompt rather than an open mathematical problem. N)

### OWR-15207-010 | Gender, Politics, and Rózsa Péter’s Academic Career

Unrecoverable OWR statement (The official report presents this as a historical, biographical, or sociological research prompt rather than an open mathematical problem. N)

### OWR-15208-001 | Higher Homotopy of Positive Ricci Observer Moduli Spaces

Answered in the source or not an open problem (solved (yes, for large odd n; answered in the source))

### OWR-15208-024 | Infinite Components of Positive Ricci Metric Moduli

Duplicate/merged OWR extraction (The official report shows that this discussion-summary record duplicates the positive-Ricci moduli question retained as OWR-15208-006. No se)

OWR-15208-006 / p.1081

### OWR-15208-025 | Components of Nonnegative Curvature Metrics on Homotopy Spheres

Duplicate/merged OWR extraction (The official report shows that this discussion-summary record duplicates the homotopy-sphere moduli question retained as OWR-15208-007. No s)

OWR-15208-007 / p.1031

### OWR-15208-028 | Curvature Detection of Exotic Complex Projective Space Bundles

Duplicate/merged OWR extraction (The official report shows that this discussion-summary record restates the projective 3-space bundle question retained as OWR-15208-010. No )

OWR-15208-010 / p.832

### OWR-15216-008 | Removing Iterability from HOD-Pair Capturing

Duplicate/merged OWR extraction (The official report shows that this record is a second extraction of the HOD-pair-capturing conjecture retained as OWR-15216-009. No separat)

OWR-15216-009 / p.1977

### OWR-15219-018 | Sum of Squares Symmetric Quartics in Higher Dimensions

Unrecoverable OWR statement (The official report answers the higher-dimensional sum-of-squares question immediately by displaying an identity valid for every dimension, )

### OWR-1536-008 | Duplicate Game-Chromatic-Index Upper-Bound Question

Duplicate/merged OWR extraction (This is a redundant extraction of OWR-1536-007 from the same official source. No separate clean statement is promoted; the original remains )

OWR-1536-007 / p.1859

### OWR-1536-010 | Duplicate Game-Chromatic-Index Lower-Bound Question

Duplicate/merged OWR extraction (This is a redundant extraction of OWR-1536-009 from the same official source. No separate clean statement is promoted; the original remains )

OWR-1536-009 / p.1834

### OWR-1537-001 | Linear Independence of Tschakaloff Function Values

Duplicate/merged OWR extraction (The official report confirms that this is the same broad open problem stated more completely, with the defining series, in OWR-1537-011.)

OWR-1537-011 / p.143

### OWR-1542-010 | Duplicate: Integral-Point Non-Density on Plane-Divisor Complements

Duplicate/merged OWR extraction (The official Oberwolfach report was checked directly. This extraction repeats the same source problem as OWR-1542-009; no separate clean sta)

OWR-1542-009 / p.137

### OWR-15426-002 | Frame Bundles as Stack Fiber Products

Unrecoverable OWR statement (The official report immediately identifies the displayed fiber product with the frame bundle of the vector bundle; this is an answered clari)

### OWR-15426-003 | Shtuka Strata as Automorphism Torsors

Unrecoverable OWR statement (The official report immediately explains the stratum as an automorphism-group torsor and a twisted form; this is an answered clarification q)

### OWR-15426-004 | Multiplicity One in Cuspidal Automorphic Decompositions

Unrecoverable OWR statement (The official report states and proves the multiplicity-one assertion in the surrounding theorem; this extraction is an answered expository q)

### OWR-15427-007 | Flag Sphere Gamma Vectors and Polytope Separation

Duplicate/merged OWR extraction (The official report shows that this record duplicates the flag-sphere gamma-vector problem retained as OWR-15427-004. No separate clean stat)

OWR-15427-004 / p.1058

### OWR-15428-011 | Membership Testing in Likelihood Correspondences

Unrecoverable OWR statement (The official report presents the trace test and witness-set construction as the method for deciding this membership question; the extraction)

### OWR-15433-008 | Stable Prasad Multiplicity Conjecture

Duplicate/merged OWR extraction (The official report shows that this record is a second extraction of the stable Prasad multiplicity conjecture retained as OWR-15433-007. No)

OWR-15433-007 / p.610

### OWR-15433-010 | Geometric Description of Pure Local Arthur Packets

Duplicate/merged OWR extraction (The official report shows that this record duplicates the pure local Arthur-packet conjecture retained as OWR-15433-009. No separate clean s)

OWR-15433-009 / p.355

### OWR-15579-006 | Duplicate Edge-Cone Yamabe Nonexistence Conjecture

Duplicate/merged OWR extraction (The official report restates the same edge-cone Yamabe nonexistence conjecture represented canonically and with its standard metric notation)

OWR-15579-002 / p.908

### OWR-15581-014 | Duplicate Complexified Volume Conjecture

Duplicate/merged OWR extraction (The official report contains one complexified Volume Conjecture for the colored Jones invariant, represented canonically by OWR-15581-013. T)

OWR-15581-013 / p.1031

### OWR-15582-003 | Theorem on Orbicurve Factorization of Holonomy Representations

Duplicate/merged OWR extraction (The official report labels this factorization assertion as Theorem 2 and explains that it supplies evidence for the distinct foliation conje)

OWR-15582-002 / p.1518

### OWR-15585-009 | Composite of Unrelated Operator-Algebra Questions

Unrecoverable OWR statement (The official problem-session list shows that this extraction combines the UCT question for finite cyclic crossed products of AF algebras wit)

### OWR-1595-008 | Smoothness and Very Ampleness in the Ohsugi-Hibi Example

Answered in the source or not an open problem (solved (no to both; (1) answered in the source, (2) ours))

### OWR-15955-001 | Heegaard-Floer Homology and Lattice Cohomology Conjecture

Duplicate/merged OWR extraction (The official report shows that this summary extraction duplicates the precise parity-refined Heegaard-Floer and lattice-cohomology conjectur)

OWR-15955-003 / p.1145

### OWR-15956-012 | Odd-Density Limits for Multiplicative Functions

Answered in the source or not an open problem ()

### OWR-15956-013 | Contractivity Abscissas of Dirichlet Partial Sums

Answered in the source or not an open problem ()

### OWR-15959-001 | Variational Principle for Complete GENERIC Thermodynamics

Duplicate/merged OWR extraction (The official report shows that this record repeats the first half of the two-part GENERIC variational-principle problem retained in full as )

OWR-15959-002 / p.2158

### OWR-1596-014 | Duplicate Novikov-Invertibility Polyhedrality Question

Duplicate/merged OWR extraction (The official report and extracted provenance identify this as a redundant copy of the Novikov-invertibility question retained canonically as)

OWR-1596-003 / p.589

### OWR-15962-001 | Epidemic Phase Transitions on General Weighted Networks

Duplicate/merged OWR extraction (The official report shows that this is an overview-level summary of the explicit SIS phase-transition conjecture retained as OWR-15962-002. )

OWR-15962-002 / p.1953

### OWR-15963-001 | Historical Research Questions on Embodied Mathematics

Unrecoverable OWR statement (The official report presents these as framing questions for historical, philosophical, and educational inquiry rather than as a standalone m)

### OWR-15963-003 | Historical Proposal for Concrete Infinitesimal Analysis

Unrecoverable OWR statement (The source is a historical and foundational research proposal about machine-based calculus, not a sufficiently specified mathematical conjec)

### OWR-15963-004 | Historical Study of Instruments in Medieval Monasteries

Unrecoverable OWR statement (The source poses historical questions about evidence, motivations, and practice in medieval monasteries rather than a mathematical problem. )

### OWR-15963-005 | Historical Uses of Portable Astronomical Instruments

Unrecoverable OWR statement (The source asks an archaeological and historical question about the use and portability of medieval instruments, not a mathematical problem.)

### OWR-15963-006 | Historical Classification of Optical Diagrams as Instruments

Unrecoverable OWR statement (The source develops a historiographical classification question about optical diagrams rather than a mathematical problem. No separate clean)

### OWR-1598-004 | Duplicate: Addition-Coagulation Similarity Limit with Mismatched Provenance

Duplicate/merged OWR extraction (The extracted text duplicates the addition-coagulation similarity-limit problem represented canonically by OWR-1729-002, but this record poi)

OWR-1729-002 / p.1950

### OWR-15986-005 | Composite of Randomness and Hindman-Theorem Questions

Unrecoverable OWR statement (The official open-problem section shows that this extraction concatenates independent questions on low discrepancy of a Martin-Löf random re)

### OWR-15986-006 | Composite of Randomness and Scott-Sentence Questions

Unrecoverable OWR statement (The official open-problem section shows that this extraction concatenates the low-discrepancy question with a separate computable-group Scot)

### OWR-15986-007 | Composite of Three Independent Computability Questions

Unrecoverable OWR statement (The official open-problem section shows that this extraction combines independent questions about Weihrauch degrees, computable regularity, )

### OWR-15987-005 | Duplicate Universal Smooth-Cubic Multisection Problem

Duplicate/merged OWR extraction (The official report contains one classification problem for multisections of the universal smooth cubic family, represented canonically by O)

OWR-15987-004 / p.1045

### OWR-15987-018 | Duplicate FI-Type Stability Problem for Braid Groups

Duplicate/merged OWR extraction (The official report states one conjectural FI-type classification of slowly growing braid-group representations, represented canonically by )

OWR-15987-017 / p.352

### OWR-15993-002 | Duplicate Brain-Tissue Constitutive-Model Extraction

Duplicate/merged OWR extraction (The official report presents this as the same brain-tissue constitutive-model program represented canonically by OWR-15993-001. No separate )

OWR-15993-001 / p.2138

### OWR-15993-004 | Duplicate Organogenesis Branching Extraction

Duplicate/merged OWR extraction (The official report presents this as the same organogenesis problem represented canonically and more completely by OWR-15993-003. No separat)

OWR-15993-003 / p.2180

### OWR-15993-007 | Expository Questions About Mathematical Biology

Unrecoverable OWR statement (The official report presents these broad prompts as the framing of an expository discussion about living systems, not as a well-posed open m)

### OWR-15995-001 | Optimal Control with Multiplicative Measurement Noise

Duplicate/merged OWR extraction (The official report shows that this is a summary extraction of the scalar multiplicative-observation-noise problem stated precisely in OWR-1)

OWR-15995-002 / p.1611

### OWR-16158-009 | Conditional Homogeneous-Space Density Theorem

Unrecoverable OWR statement (The official report states that the asserted Brauer-Manin density conclusion follows from its external Conjecture 9.1; the extracted conditi)

### OWR-16158-010 | Established Stabilizer-Induction Step

Unrecoverable OWR statement (The official report proves this inductive step under its external Conjecture 9.1. The extracted conditional assertion is not a separate open)

### OWR-16160-014 | Duplicate Weak Warped-Cone Equivalence Question

Duplicate/merged OWR extraction (This record repeats the weak-equivalence question represented canonically by OWR-16160-013 in the same problem session. No separate clean st)

OWR-16160-013 / p.1525

### OWR-16162-012 | Finiteness of Genera of Simply Connected Groups

Duplicate/merged OWR extraction (The official report shows that this record is a duplicate extraction of the particular conjectural case already retained in OWR-16162-013. N)

OWR-16162-013 / p.355

### OWR-16166-002 | Split Values of Linear and Quadratic Polynomials

Answered in the source or not an open problem (solved (a theorem, not a problem))

### OWR-16166-007 | Decidability of Finite Field Extensions

Answered in the source or not an open problem (solved (no; answered in the source))

### OWR-16167-018 | Biembeddability and the Golden-Ratio Thompson Group

Unrecoverable OWR statement (The official report shows that this extraction merges two independent numbered questions, one about the biembeddability poset and another ab)

### OWR-16167-021 | Homology and Hopficity of Braided Thompson Groups

Unrecoverable OWR statement (The official report shows that this extraction merges three independent numbered questions concerning Hopficity, acyclicity, and a Stein-Far)

### OWR-16167-022 | Nonamenable Groups without Free Subgroups

Unrecoverable OWR statement (The official report shows that this extraction merges two independent numbered questions concerning nonamenable groups and Wajnryb-Witowicz )

### OWR-16167-024 | Finitely Presented Piecewise-Projective Groups

Unrecoverable OWR statement (The official report shows that this extraction merges two independent numbered questions concerning fast one-bump groups and an embedding of)

### OWR-16167-026 | Finiteness Properties of Lodha-Moore Kernels

Unrecoverable OWR statement (The official report shows that this extraction merges two independent numbered questions concerning a Lodha-Moore kernel and a non-finitely-)

### OWR-16171-002 | Canonical Comparison of Geometric Local-System Realizations

Duplicate/merged OWR extraction (The official report confirms that this is a duplicate extraction of the same conjecture retained in OWR-16171-001. No separate clean stateme)

OWR-16171-001 / p.620

### OWR-16406-004 | Duplicate Cannon-Conjecture Extraction

Duplicate/merged OWR extraction (This record restates Cannon’s conjecture, represented canonically by OWR-16406-003 in the same report. No separate clean statement is promot)

OWR-16406-003 / p.336

### OWR-16408-008 | Expository Motivation for Nonreductive GIT

Unrecoverable OWR statement (The official report immediately answers this motivational question by explaining why nonreductive geometric invariant theory enters the modu)

### OWR-16409-004 | Geometric Reduction Types of Plane Quartics

Unrecoverable OWR statement (The official report states that the authors answer these reduction-type questions and characterize the answer using Dixmier-Ohno invariants,)

### OWR-16411-008 | Well Posed Perturbative Einstein Field Equations

Duplicate/merged OWR extraction (The official report contains one overarching open question about handling perturbatively corrected equations; this generic restatement adds )

OWR-16411-007 / p.1490

### OWR-16413-007 | Source-Resolved Random-Domain Modeling Agenda

Unrecoverable OWR statement (The official report poses these questions as the agenda of a contribution and then supplies its well-posedness framework, expectation and de)

### OWR-16415-002 | Duplicate Good-Curve-Graph Conjecture

Duplicate/merged OWR extraction (This record is the conjectural answer to the same good-curve-graph question represented canonically and completely by OWR-16415-001. No sepa)

OWR-16415-001 / p.1073

### OWR-16416-003 | Duplicate Wobbling-Group Nonamenability Conjecture

Duplicate/merged OWR extraction (The official report presents this as the conjectured negative answer to the extensive-amenability question represented canonically by OWR-16)

OWR-16416-002 / p.1910

### OWR-16629-002 | Decomposition of Equiangular Tight Frames

Duplicate/merged OWR extraction (The official report shows that this overview extraction duplicates the precise equiangular-tight-frame decomposition question retained as OW)

OWR-16629-009 / p.834

### OWR-16633-011 | The Polynomial Hirsch Conjecture

Duplicate of canonical problem CS-002

CS-002 / p.2108

### OWR-16633-015 | Duplicate Path-Cost Arborescence Problem

Duplicate/merged OWR extraction (The official problem session contains one path-cost arborescence hardness problem, represented canonically by OWR-16633-014. This extraction)

OWR-16633-014 / p.2085

### OWR-16633-018 | Duplicate Degree-Utility Graph Optimization Problem

Duplicate/merged OWR extraction (This record repeats the graph-complexity question represented canonically and with its objective by OWR-16633-017. No separate clean stateme)

OWR-16633-017 / p.2072

### OWR-16633-022 | Duplicate Clique-Sequence Submodularity Conjecture

Duplicate/merged OWR extraction (This extraction repeats the submodularity clause represented canonically and more completely by OWR-16633-021. No separate clean statement i)

OWR-16633-021 / p.1644

### OWR-16633-024 | Duplicate Represented-Matroid Minimization Problem

Duplicate/merged OWR extraction (This record repeats the represented-matroid submodular-minimization problem represented canonically by OWR-16633-023. No separate clean stat)

OWR-16633-023 / p.1691

### OWR-16634-004 | Defining High-Dimensional Expanders Through Local Spectra

Unrecoverable OWR statement (The official report uses this sentence as an expository heading, immediately supplies a local-spectral definition, and lists different concr)

### OWR-16634-005 | Explicit Equations Separating Matrix Multiplication from Low-Rank Tensors

Unrecoverable OWR statement (The official report poses this as a rhetorical transition and immediately answers it by introducing representation-theoretic methods for fin)

### OWR-16634-007 | Unique Games Conjecture

Duplicate of canonical problem CS-001

CS-001 / p.2109

### OWR-16763-004 | Duplicate Five-Cycle Near-Triangulation Conjecture

Duplicate/merged OWR extraction (This extraction repeats the near-triangulation decomposition conjecture represented canonically and with its computational evidence by OWR-1)

OWR-16763-005 / p.1819

### OWR-16763-012 | Duplicate Clustered Hadwiger-Type Conjecture

Duplicate/merged OWR extraction (This record restates the clustered-coloring conjecture represented canonically by OWR-16763-011. No separate clean statement is promoted; th)

OWR-16763-011 / p.1769

### OWR-16763-026 | Duplicate Latin-Square Transversal Conjecture

Duplicate/merged OWR extraction (This record repeats the Latin-square transversal conjecture represented canonically and with its matching formulation by OWR-16763-027. No s)

OWR-16763-027 / p.1729

### OWR-16763-028 | Composite Duplicate of Two Rainbow-Matching Conjectures

Duplicate/merged OWR extraction (The first clause is represented canonically by OWR-16763-027 and the second by OWR-16763-030. This composite extraction adds no distinct pro)

OWR-16763-027 / p.1729 OWR-16763-030 / p.1695

### OWR-16765-002 | Convergence of Finite-Element Mean-Curvature Flow

Answered in the source or not an open problem (solved (yes, as posed; answered in the source))

### OWR-16766-002 | Exact Recovery by Lagrangian-Dual Binary Tomography

Duplicate/merged OWR extraction (The official report shows that this is a partial duplicate of the binary-tomography recovery conjecture retained in OWR-16766-003. No separa)

OWR-16766-003 / p.1298

### OWR-16766-004 | Binary Tomography Recovery Conjecture

Duplicate/merged OWR extraction (The official report confirms that this is a second complete duplicate of the binary-tomography recovery conjecture retained in OWR-16766-003)

OWR-16766-003 / p.1298

### OWR-16767-005 | Duplicate Finite-Element Mean-Curvature-Flow Problem

Answered in the source or not an open problem (solved (yes, as posed; answered in the source))

OWR-16765-002 [not in active corpus]

### OWR-16768-010 | Duplicate Localization Conjecture for Finitistic Test Ideals

Duplicate/merged OWR extraction (This record repeats the localization-compatibility problem represented canonically by OWR-16768-003. No separate clean statement is promoted)

OWR-16768-003 / p.266

### OWR-16770-003 | Admissibility and Finite Length on the Drinfeld Tower

Duplicate/merged OWR extraction (The official report contains a single two-part conjecture on admissibility and finite length; this extraction duplicates the complete statem)

OWR-16770-002 / p.328

### OWR-16772-002 | Duplicate Random Lorentz Gas Invariance Principle

Duplicate/merged OWR extraction (This record repeats the quenched and averaged-quenched random-Lorentz-gas invariance-principle question represented canonically by OWR-16416)

OWR-16416-001 / p.1901

### OWR-16776-008 | Robinson's Defect Conjecture and Commuting Sylow Subgroups

Unrecoverable OWR statement (The official report shows that this extraction concatenates two independent conjectures: Robinson's defect conjecture and a criterion for co)

### OWR-16931-001 | Oberwolfach Seating Problem

Unrecoverable OWR statement (The official report explicitly answers the sufficiently-large-order Oberwolfach seating problem affirmatively, so this extraction is a prove)

### OWR-16931-017 | Decomposing Complete Graphs into Bounded-Degree Regular Factors

Duplicate/merged OWR extraction (The official report confirms the bounded-degree regular-factor conjecture, but the identical conjecture is already represented by the earlie)

OWR-16763-018 / p.1867

### OWR-16940-001 | Relative Perturbation Bounds for Empirical Covariance Operators

Unrecoverable OWR statement (The official report presents an approach to relative perturbation bounds for empirical covariance operators, not a standalone unresolved pro)

### OWR-16941-011 | Sheaf Models for Modular Category O

Unrecoverable OWR statement (The official report immediately answers this rhetorical question by giving the motivic-sheaf model and its categorical equivalence. No separ)

### OWR-1703864-001 | The Riemann Hypothesis

Duplicate of canonical problem MPP-002

MPP-002 / p.189

### OWR-1703868-002 | Imprecise Kashiwara-Vergne Symmetry Relation

Unrecoverable OWR statement (The official report says only that the approach illuminates a close relationship conjectured elsewhere between the Kashiwara-Vergne symmetry)

### OWR-1703871-003 | Galois Arithmetic of Special Subvarieties in Hodge Variations

Duplicate/merged OWR extraction (Cross-report comparison confirms that this is a second extraction of the special-subvariety descent conjecture retained in OWR-1703867-008. )

OWR-1703867-008 / p.678

### OWR-1703872-002 | Large Values of Arithmetic Hyperbolic Three-Manifold Eigenfunctions

Answered in the source or not an open problem (solved (yes, for infinitely many f; answered in the source))

### OWR-1703876-017 | Duplicate Discrete-Geometry Problem Collection

Duplicate/merged OWR extraction (The official report confirms that this extraction repeats the hypergraph-coloring, partial-triangulation, polytope, spherical-cap, and pseud)

OWR-1703876-016 / p.1690

### OWR-17135-032 | Nonnegative Square Decompositions of Bipartite Determinant Differences

Duplicate/merged OWR extraction (The official report contains one nonnegative square-decomposition question; this extraction duplicates the source-verified formulation retai)

OWR-17135-031 / p.1712

### OWR-1728-003 | Stable Homotopy Type of Open-Closed Surface Categories

Answered in the source or not an open problem (solved (a theorem, not a problem))

### OWR-17293-007 | Duplicate Erdős-Hooley Typical-Size Extraction

Duplicate/merged OWR extraction (The official report confirms that this extraction repeats the typical-size question and optimal-exponent conjecture represented canonically )

OWR-17293-006 / p.193

### OWR-17293-008 | Duplicate Erdős-Hooley Optimal-Exponent Extraction

Duplicate/merged OWR extraction (The official report confirms that this extraction repeats the optimal-exponent part of the Erdős-Hooley question represented canonically by )

OWR-17293-006 / p.193

### OWR-17293-016 | Coefficient Normalization of Positive Definite Homogeneous Forms

Answered in the source or not an open problem ()

### OWR-17293-017 | Absolutely Convergent Multiplicative Expansions of the von Mangoldt Function

Answered in the source or not an open problem ()

### OWR-17296-018 | Duplicate Bounded Negativity Conjecture

Duplicate/merged OWR extraction (The official report confirms that this is a second formulation of the Bounded Negativity Conjecture represented canonically by OWR-17296-011)

OWR-17296-011 / p.643

### OWR-17300-007 | Conditional Harder-Congruence Theorem

Duplicate/merged OWR extraction (The official report identifies this extraction as a proved theorem under four explicit hypotheses, not as a separate open problem; the under)

OWR-17300-006 / p.131

### OWR-17300-008 | Conditional Modularity Theorem for Chow Cycles

Unrecoverable OWR statement (The official report identifies this extraction as a theorem proved assuming the Bloch-Beilinson conjecture, not as a separately posed open p)

### OWR-17300-017 | Cuspidal Spectral-Multiplier Theorem

Unrecoverable OWR statement (The official report identifies this extraction as Theorem 1, a proved spectral-isolation result for non-S-CAP cuspidal representations, not )

### OWR-1735-006 | Evaluating Teachers' Responses to Mathematics Test Items

Mathematics Teacher Education workshop report (teacher evaluation and pedagogy policy questions, not mathematical research problems)

### OWR-1735-007 | Societal Responsibilities of Mathematics Teacher Educators

Mathematics Teacher Education workshop report (teacher evaluation and pedagogy policy questions, not mathematical research problems)

### OWR-1735-008 | Frameworks for Mathematical Knowledge in Teaching

Mathematics Teacher Education workshop report (teacher evaluation and pedagogy policy questions, not mathematical research problems)

### OWR-1735-009 | Developing Teachers' Usable Mathematical Knowledge

Mathematics Teacher Education workshop report (teacher evaluation and pedagogy policy questions, not mathematical research problems)

### OWR-1738-001 | Structural Problems for Units of Integral Group Rings

Unrecoverable OWR statement (The official report confirms that this extraction is the workshop's broad topic overview rather than a discrete mathematical problem; the de)

### OWR-17466-008 | Geometry and Mutation of Quiver Grassmannians

Unrecoverable OWR statement (The official report shows that this extraction concatenates seven independent numbered conjectures and problems rather than one publishable )

### OWR-17466-016 | Weighted-Surface Classification of Quaternion-Type Algebras

Duplicate/merged OWR extraction (The official report confirms that this is a duplicate extraction of the Galactic Conjecture retained in OWR-17466-011. No separate clean sta)

OWR-17466-011 / p.220

### OWR-17466-017 | Galactic Conjecture for Quaternion-Type Algebras

Duplicate/merged OWR extraction (The official report confirms that this is a duplicate extraction of the Galactic Conjecture retained in OWR-17466-011. No separate clean sta)

OWR-17466-011 / p.220

### OWR-17470-009 | Duplicate Ordering-Ratio Dichotomy

Duplicate/merged OWR extraction (This extraction repeats the second question represented canonically, together with its motivating travelling-salesman question, by OWR-17470)

OWR-17470-008 / p.991

### OWR-17470-012 | Duplicate Integral-Approximation Question

Duplicate/merged OWR extraction (This record repeats the second question represented canonically by OWR-17470-011. No separate clean statement is promoted; the original reco)

OWR-17470-011 / p.1053

### OWR-17470-014 | Duplicate Bestvina-Complex Retraction Question

Duplicate/merged OWR extraction (This record repeats the second question represented canonically by OWR-17470-013. No separate clean statement is promoted; the original reco)

OWR-17470-013 / p.1025

### OWR-17475-007 | Duplicate Moment-Matrix Question

Duplicate/merged OWR extraction (This record repeats the first question represented canonically, together with its localization-matrix extension, by OWR-17475-006. No separa)

OWR-17475-006 / p.703

### OWR-1782-003 | Quantitative Manipulability of Social Choice Functions

Answered in the source or not an open problem (solved (no, for functions far from non-manipulable ones; answered in the source for three alternatives))

### OWR-1795-011 | Twin Prime Conjecture

Duplicate of canonical problem LAN-002

LAN-002 / p.192

### OWR-1967-007 | Duplicate: Quantum Invariants and Boundary Character Varieties

Duplicate/merged OWR extraction (The official Oberwolfach report was checked directly. This extraction repeats the same source problem as OWR-1967-006; no separate clean sta)

OWR-1967-006 / p.1206

### OWR-1967-009 | Duplicate: Quantum Knot Invariants and Boundary Character Varieties

Duplicate/merged OWR extraction (The official Oberwolfach report was checked directly. This extraction repeats the same source problem as OWR-1967-008; no separate clean sta)

OWR-1967-008 / p.1207

### OWR-1970-001 | Characterization of Chaotic Analytic Zero Processes

Unrecoverable OWR statement (The official report uses this as a rhetorical opening to its exposition, not as a discrete research problem. No separate clean statement is )

### OWR-2049-005 | Vanishing Pattern for Voronoi Compactification Intersections

Duplicate/merged OWR extraction (The official source confirms that this is a duplicate extraction of OWR-2049-004, not a separate problem. No separate clean statement is pro)

OWR-2049-004 / p.693

### OWR-2090-003 | Realizability of Allowable Double-Permutation Sequences

Answered in the source or not an open problem (solved (yes, as intended; answered in the source))

### OWR-2489-009 | Subtree Decomposition of Mixed-Integer Convex Hulls

Duplicate/merged OWR extraction (The official report confirms that this repeats the same Conjecture 2 extracted as OWR-2489-004. No separate clean statement is promoted; the)

OWR-2489-004 / p.2069

### OWR-2492-003 | Smooth Decomposition of Local Group Action Deformations

Duplicate/merged OWR extraction (The official report confirms that this overview sentence and OWR-2492-004 are two extractions of one source question; OWR-2492-004 preserves)

OWR-2492-004 / p.722

### OWR-2654827-008 | Computed Exceptional Hilbert Series

Answered in the source or not an open problem (solved (a theorem, not an open problem))

### OWR-2654828-005 | Resolved Weighted-Gaussian-Persistence Lead-In

Unrecoverable OWR statement (The official report presents this as a lead-in to results on weighted sums of correlated Gaussian stationary processes and then states how t)

### OWR-2654835-002 | Single-Point Recovery of Time-Dependent Heat Sources

Answered in the source or not an open problem (solved (answered qualitatively in the source))

### OWR-2654837-001 | Cultural and Social Variation in Numeracy

History of Everyday Numeracy workshop report (19th-century/medieval historiography, not a mathematical research problem)

### OWR-2654837-002 | Objects and Scope of Ancient Mathematics

History of Everyday Numeracy workshop report (19th-century/medieval historiography, not a mathematical research problem)

### OWR-2654837-003 | Reach of Axiomatic Deduction and Mathematical Explanation

History of Everyday Numeracy workshop report (19th-century/medieval historiography, not a mathematical research problem)

### OWR-2654837-004 | Demographics of Medieval Mathematical Practitioners

History of Everyday Numeracy workshop report (19th-century/medieval historiography, not a mathematical research problem)

### OWR-2654837-005 | Professional Status of Medieval Mathematical Practitioners

History of Everyday Numeracy workshop report (19th-century/medieval historiography, not a mathematical research problem)

### OWR-2654837-006 | Acquisition of Everyday Numeracy in the Nineteenth Century

History of Everyday Numeracy workshop report (19th-century/medieval historiography, not a mathematical research problem)

### OWR-2654837-007 | Arithmetic Education and Numerical Resources in the Nineteenth Century

History of Everyday Numeracy workshop report (19th-century/medieval historiography, not a mathematical research problem)

### OWR-2654837-008 | Worldviews as Historical Categories in Mathematics

History of Everyday Numeracy workshop report (19th-century/medieval historiography, not a mathematical research problem)

### OWR-3384-005 | Duplicate: Smooth Schubert Cycles from Gelfand-Zetlin Faces

Duplicate/merged OWR extraction (Comparison of the two same-report source-derived extractions confirms that this record repeats OWR-3384-004; no separate clean statement is )

OWR-3384-004 / p.634

### OWR-3391-001 | Mathematical Uses of Category Theory

History and Prospects of Category Theory workshop report (sociological and philosophical discussion prompts, not mathematical problems)

### OWR-3391-002 | Natural Limits of Categorical Methods

History and Prospects of Category Theory workshop report (sociological and philosophical discussion prompts, not mathematical problems)

### OWR-3392-007 | Non-Specific Time-Reflection Research Direction

Unrecoverable OWR statement (The official report records only a suspected deep relationship between time reflection and commutants of von Neumann bimodules, without a ma)

### OWR-3393-003 | Duplicate Non-$p\times p$ Specht-Complexity Conjecture

Duplicate/merged OWR extraction (After distinguishing the proved $p\times p$ direction, this record repeats the unresolved maximal-complexity question represented canonicall)

OWR-3393-002 / p.267

### OWR-3394-017 | State Recovery from Quantized Nonlinear Trajectories

Duplicate/merged OWR extraction (The official report confirms that this is the broad introductory formulation of the single observability problem stated fully in OWR-3394-01)

OWR-3394-018 / p.2095

### OWR-3394-021 | Geometric Criteria for Global Observability

Duplicate/merged OWR extraction (The official report confirms that this restates the same global-observability problem extracted fully as OWR-3394-020. No separate clean sta)

OWR-3394-020 / p.2084

### OWR-3396-003 | Model Set Formulations of the Pisot Conjecture

Unrecoverable OWR statement (The official report passage is explanatory prose saying that model sets are natural in constant-length examples and that equivalent formulat)

### OWR-3400-005 | Uglov Bipartitions and Hecke Decomposition Numbers

Unrecoverable OWR statement (The official report states these claims as a theorem under Lusztig's Conjecture 14.2 and its parameter hypotheses, not as an open problem. N)

### OWR-3477-014 | Duplicate: Cohomology Classes of Undistorted Central Extensions

Duplicate/merged OWR extraction (Comparison of the two same-report source-derived extractions confirms that this record repeats OWR-3477-013; no separate clean statement is )

OWR-3477-013 / p.340

### OWR-3478-002 | Positive Ricci Curvature and String Orientation Classes

Answered in the source or not an open problem (solved (no; answered in the source))

### OWR-3479-002 | Descriptive Cellular-Membrane Research Theme

Unrecoverable OWR statement (The official report passage describes broad biological motivations for studying cell and organelle shapes but supplies no mathematical varia)

### OWR-3480-013 | $K_1$ Groups and Theta Images of Iwasawa Algebras

Unrecoverable OWR statement (The official report presents the computation and theta-image characterization as Theorem 1.1 and its localized consequence, not as an open p)

### OWR-3480-018 | Section Conjecture for Hyperbolic Curves

Duplicate/merged OWR extraction (The official report confirms that this is a second extraction of the same section conjecture preserved fully in OWR-3480-017. No separate cl)

OWR-3480-017 / p.176

### OWR-3481-001 | Descriptive Singular-Complex Observation

Answered in the source or not an open problem (solved (a reported theorem, not a problem))

### OWR-4085-002 | Uniform Stability of Linearized Quasicontinuum Operators

Answered in the source or not an open problem (solved (no when $\varphi''(2) \neq 0$; answered in the source))

### OWR-4127-006 | Incomplete Fourier-Mukai Limit-Cycle Extraction

Unrecoverable OWR statement (The official-source extraction names a degeneration, a Fourier-Mukai transform, and cycles $\phi_\eta,\phi_0$ without defining any of them. )

### OWR-4130-001 | Literal Symbolism in the Development of Algebra

History of Mathematics workshop report (historiographical question, not a mathematical research problem)

### OWR-4130-002 | Material Supports for Mathematical Symbolism

History of Mathematics workshop report (historiographical question, not a mathematical research problem)

### OWR-4135-002 | Unique Games Approximation-Threshold Conjecture

Duplicate of canonical problem CS-001

CS-001 / p.2109

### OWR-4138-001 | Incomplete Strong-Coupling Spectral Formula

Unrecoverable OWR statement (The extracted formula uses the configurations $\nu^*$ and $\omega^1$ without defining them, so the official-source fragment is not self-cont)

### OWR-4139-002 | Bell Inequality Criteria for Special Unitary Dynamics

Answered in the source or not an open problem (solved (yes; answered in the source))

### OWR-4196-005 | Cardinality Characterization of NIP Theories

Unrecoverable OWR statement (The official OWR report proves the displayed gross-model cardinality property for NIP theories, observes that it does not characterize NIP, )

### OWR-4211-009 | Incomplete Self-Similar-Tree DtN Uniqueness Claim

Unrecoverable OWR statement (The extracted functional equation asks for “homeomorphic solutions” without defining that solution class, and the phrase is not mathematical)

### OWR-4211-010 | Duplicate Acoustic Boundary-Operator Coercivity Question

Duplicate/merged OWR extraction (This is a less explicit duplicate of the acoustic combined-boundary-operator coercivity problem represented canonically by OWR-4211-008 from)

OWR-4211-008 / p.1289

### OWR-4213-001 | Source-Mismatched Extremal-Graph Questions

Duplicate/merged OWR extraction (The cited History of Numerical and Graphical Tables report does not contain these extremal-graph questions. Closely related wording occurs i)

OWR-4212-011 / p.1819

### OWR-4213-002 | The Odd-Order Simple-Group Problem

History of Mathematics lecture note mentioning the 1963 Feit-Thompson Odd Order Theorem, not an open problem

### OWR-4214-003 | Repeated Extraction of Naimark's Problem

Duplicate/merged OWR extraction (The official report presents this Naimark question twice in separate talks; OWR-4214-010 retains the more precise nonseparable formulation. )

OWR-4214-010 / p.1334

### OWR-4214-005 | Repeated Extraction of the Toms-Winter Conjecture

Duplicate/merged OWR extraction (The official report contains one statement of this conjecture, duplicated by the extraction as OWR-4214-005 and OWR-4214-007. OWR-4214-007 i)

OWR-4214-007 / p.1439

### OWR-4330-002 | Birkhoff Conjecture for Integrable Convex Billiards

Duplicate of canonical problem DYN-004

DYN-004 / p.1457

### OWR-4339-007 | Duplicate: $\varepsilon$-Regularity for Donaldson's Equation

Duplicate/merged OWR extraction (Comparison of same-report source-derived extractions confirms that this record repeats OWR-4339-008; no separate clean statement is promoted)

OWR-4339-008 / p.1452

### OWR-4341-001 | Novikov Homotopy Invariance Conjecture

Duplicate of canonical problem TOP-006

TOP-006 / p.1140

### OWR-4415-001 | Large-Scale Mechanisms Driving Atmospheric Monsoons

Empirical atmospheric-science question ('What drives large-scale monsoons in the Earth's atmosphere?') without a mathematical formulation

### OWR-4415-002 | Oceanic Mid-Depth Stratification and Overturning Circulation

Empirical physical-oceanography question ('What sets the oceanic mid-depth stratification and overturning circulation?') without a mathematical formulation

### OWR-4415-005 | Nonlinear Coupling of Buoyancy Reversal and Radiation

Descriptive atmospheric-physics question ('Can buoyancy reversal remain nonlinearly coupled to external forcings such as radiation?') without a mathematical model specification

### OWR-4415-006 | Errors in Long Equatorial Wave Models

Empirical weather-model error question ('How can the significant errors in the longest equatorial waves be explained or reduced?') without a mathematical formulation

### OWR-4425-019 | Repeated Extraction of the Equal-Sum-Factor Problem

Duplicate/merged OWR extraction (The official report contains one occurrence of Problem 47, duplicated by the extraction as OWR-4425-018 and OWR-4425-019. OWR-4425-018 is re)

OWR-4425-018 / p.1700

### OWR-4425-021 | Repeated Extraction of the PVHH-Cube Conjecture

Duplicate/merged OWR extraction (The official report contains one occurrence of Conjecture 48, duplicated by the extraction as OWR-4425-020 and OWR-4425-021. OWR-4425-020 is)

OWR-4425-020 / p.1720

### OWR-4427-007 | Answered Definition of Burniat Surfaces

Unrecoverable OWR statement (The source heading asks what Burniat surfaces are and the following text immediately gives their construction as bidouble covers from nine-l)

### OWR-4528-001 | Defining Higher-Dimensional Elliptic Fibrations

Unrecoverable OWR statement (The official OWR report labels this section “What is an elliptic fibration?”, gives the workshop's common definition, and lists optional ref)

### OWR-4531-018 | Duplicate: Boundary Behavior in the Dirichlet Weak Product

Duplicate/merged OWR extraction (Comparison of same-report source-derived extractions confirms that this record repeats OWR-4531-017; no separate clean statement is promoted)

OWR-4531-017 / p.1311

### OWR-4532-005 | Answered Discussion of Adaptive Biasing Dynamics

Unrecoverable OWR statement (In the official OWR report this is a rhetorical transition immediately followed by the answer: the adaptive dynamics $Y_t$ converges essenti)

### OWR-4535-008 | Duplicate Takhtajan-Zograf Representation Question

Duplicate/merged OWR extraction (This is the first half of the same-report two-part problem represented canonically by OWR-4535-007, with no distinct hypothesis or target. N)

OWR-4535-007 / p.874

### OWR-4537-003 | Brumer-Stark Conjecture for Ray Class Groups

Unrecoverable OWR statement (The official OWR report states the $T$-class-group annihilation conjecture only “modulo a minor technical condition on $T$” and does not spe)

### OWR-4791-006 | Erdős-Shelah Bounds for Union-Free Families

Already answered in the original Oberwolfach workshop report (not an open problem)

### OWR-4791-011 | Optimal Constructions for Cycle Saturation

Unrecoverable OWR statement (The claim refers to two diagram-dependent constructions that the report does not specify in recoverable standalone text. No separate clean s)

### OWR-4791-016 | Functional Inequality on the Boolean Cube

Duplicate/merged OWR extraction (This is a second extraction of the same Boolean-cube functional inequality preserved in OWR-4791-015. No separate clean statement is promote)

OWR-4791-015 / p.1695

### OWR-4791-021 | Maximum Simplicial Depth of Point Configurations

Answered in the source or not an open problem (solved (answered in the source))

### OWR-4791-024 | Rosenfeld Conjecture for Oriented Paths

Answered in the source or not an open problem (solved (yes for all n except 3, 5 and 7; answered in the source))

### OWR-4798-022 | Duplicate: Polynomial Bounds for Polytope Diameters

Duplicate/merged OWR extraction (Comparison of same-report source-derived extractions confirms that this record repeats OWR-4798-005; no separate clean statement is promoted)

OWR-4798-005 / p.1717

### OWR-4800-010 | Theorem: Mutation Compatibility of Generic-Basis Maps

Unrecoverable OWR statement (The official Oberwolfach report states this extraction as a theorem rather than an open problem; no separate clean statement is promoted, an)

### OWR-4802-001 | Mathematical Definition of Numerical Tables

History of Numerical Tables workshop report (historiography of ancient tables, not a mathematical research problem)

### OWR-4802-002 | Boundaries Between Formal Tables and Material Arrays

History of Numerical Tables workshop report (historiography of ancient tables, not a mathematical research problem)

### OWR-4802-003 | Mathematical Tables in the Digital Era

History of Numerical Tables workshop report (historiography of ancient tables, not a mathematical research problem)

### OWR-4802-004 | Scope of Digital Mathematical Tables

History of Numerical Tables workshop report (historiography of ancient tables, not a mathematical research problem)

### OWR-4802-005 | Interpolation in Medieval Astronomical Tables

History of Numerical Tables workshop report (historiography of ancient tables, not a mathematical research problem)

### OWR-4802-006 | Mathematical Classification of Plimpton 322

History of Numerical Tables workshop report (historiography of ancient tables, not a mathematical research problem)

### OWR-4802-007 | Generation Algorithms for Plimpton 322

History of Numerical Tables workshop report (historiography of ancient tables, not a mathematical research problem)

### OWR-4802-008 | Tabular Meaning of Plimpton 322

History of Numerical Tables workshop report (historiography of ancient tables, not a mathematical research problem)

### OWR-4807-004 | Frame-Norm-Preserving Operators

Unrecoverable OWR statement (The official report poses the classification and then immediately gives its answer as a theorem, so the extracted question is not left open.)

### OWR-4807-005 | Signal Approximation by Gabor Multipliers

Unrecoverable OWR statement (The official report presents this as an approximation setup followed by variational solutions and inverse algorithms rather than as a standa)

### OWR-4807-007 | Stable Coefficient Recovery from Bandlimited Samples

Unrecoverable OWR statement (The official report asks these recovery questions and then answers both affirmatively under explicit hypotheses, so the extraction is not an)

### OWR-4990375-008 | Paramodular Correspondence for Abelian Surfaces and Fourfolds

Duplicate/merged OWR extraction (The official report confirms that this is a duplicate extraction of the detailed paramodular correspondence retained in OWR-4990375-007. No )

OWR-4990375-007 / p.163

### OWR-4990375-011 | Kodaira Dimension of Odd Minimal Strata

Duplicate/merged OWR extraction (The official report confirms that this is a duplicate extraction of the Kodaira-dimension conjecture retained in OWR-4990375-010. No separat)

OWR-4990375-010 / p.696

### OWR-5146-001 | Answered Sullivan-Model Existence Question

Answered in the source or not an open problem (solved (yes; answered in the source))

### OWR-5146-004 | Duplicate Rational Toral-Rank Conjecture

Duplicate/merged OWR extraction (This record restates the rational toral-rank conjecture represented canonically by OWR-5146-003 from the same official report, changing only)

OWR-5146-003 / p.1243

### OWR-5154-008 | Existence of Nonsofic Groups

Duplicate of canonical problem ALG-008

ALG-008 / p.216

### OWR-5155-003 | Homological Computability of Restricted Singularity Classes

Unrecoverable OWR statement (The official report poses this as a rhetorical transition and immediately explains the homological computability of the class of maps. No se)

### OWR-5157-002 | Expository Random-Matrix Framing Question

Unrecoverable OWR statement (The sentence is a rhetorical framing question at the start of a survey of established spectral and eigenvector universality results; the rep)

### OWR-5158-004 | Duplicate Teichmüller-Metric Question

Duplicate/merged OWR extraction (This is the third subquestion of the same-report three-part problem represented canonically by OWR-5158-003, with only examples of metrics a)

OWR-5158-003 / p.993

### OWR-701-003 | Entropy-Stable Paths for Approximate Riemann States

Unrecoverable OWR statement (The official report asks this as an expository design question and then answers it in the same abstract by choosing subpaths along approxima)

### OWR-7155447-001 | Reading Practices and Arguments in Mathematical Classics

Duplicate/merged OWR extraction (The official report shows that this workshop-level overview merges availability, reading-practice, and mathematical-argument themes that are)

OWR-7155447-003 [not in active corpus] OWR-7155447-004 [not in active corpus]

### OWR-7155447-002 | The Persistence of Ancient Mathematical Classics

History of Mathematical Classics and Marginalia workshop report (historiography, not a mathematical research problem)

### OWR-7155447-003 | Mathematical Classics across Scholarly Traditions

History of Mathematical Classics and Marginalia workshop report (historiography, not a mathematical research problem)

### OWR-7155447-004 | Classical Authority in Mathematical Innovation and Argument

History of Mathematical Classics and Marginalia workshop report (historiography, not a mathematical research problem)

### OWR-7155447-005 | Yang Hui's Reclassification of Problems from The Nine Chapters

History of Mathematical Classics and Marginalia workshop report (historiography, not a mathematical research problem)

### OWR-7155447-006 | Mathematical Work in Commentarial Readings of Classics

History of Mathematical Classics and Marginalia workshop report (historiography, not a mathematical research problem)

### OWR-7155447-007 | Meanings and Purposes in Mathematical Commentaries

History of Mathematical Classics and Marginalia workshop report (historiography, not a mathematical research problem)

### OWR-7155447-008 | Reading Mathematical Classics from Lost Traditions

History of Mathematical Classics and Marginalia workshop report (historiography, not a mathematical research problem)

### OWR-7155447-009 | Chasles's Reconstruction of Brahmagupta and Euclid

History of Mathematical Classics and Marginalia workshop report (historiography, not a mathematical research problem)

### OWR-7155447-010 | Mathematical Knowledge Transferred to Practitioners and Craftsmen

History of Mathematical Classics and Marginalia workshop report (historiography, not a mathematical research problem)

### OWR-7155447-011 | Material Forms and Reading of Mathematical Classics

History of Mathematical Classics and Marginalia workshop report (historiography, not a mathematical research problem)

### OWR-7155449-006 | Duplicate K3 Hilbert-Scheme Conjecture Suite

Duplicate/merged OWR extraction (The official Oberwolfach report confirms that Conjectures 2-4 form one contiguous suite on quasi-Jacobi structure, multiple covers, and the )

OWR-7155449-005 / p.723

### OWR-723-001 | Circle Discrepancy in Balanced Lattice Colorings

Already answered in the original Oberwolfach workshop report (not an open problem)

### OWR-730-012 | Duplicate Planar Min-Bisection and Bicriteria Extraction

Duplicate/merged OWR extraction (The official Oberwolfach report was checked directly. This extraction repeats the same source problem as OWR-730-011 and OWR-730-013/OWR-730)

OWR-730-011 / p.2066 OWR-730-013 / p.2057 OWR-730-014 / p.2096

### OWR-730-015 | Duplicate Bicriteria Spanning-Tree Enumeration

Duplicate/merged OWR extraction (The official Oberwolfach report was checked directly. This extraction repeats the same source problem as OWR-730-014 and is only a trailing )

OWR-730-014 / p.2096

### OWR-734-010 | Duplicate Lefschetz Property Conjecture

Duplicate/merged OWR extraction (This is a redundant prose extraction of the Lefschetz conjecture stated precisely in OWR-734-011. No separate clean statement is promoted; t)

OWR-734-011 / p.1703

### OWR-734-017 | Single-Wave Uniqueness for Impenetrable Scatterers

Unrecoverable OWR statement (The cited official report on combinatorial commutative algebra contains no inverse-scattering problem; this statement appears to have been s)

### OWR-735-002 | Fourier Reconstruction of Discrete Objects

Unrecoverable OWR statement (The cited official report on fast methods for non-local operators contains no discrete Fourier-reconstruction problem; the statement belongs)

### OWR-736-002 | Duplicate Warped Wavelet-Basis Problem

Duplicate/merged OWR extraction (The official Oberwolfach report was checked directly. This extraction repeats the same source problem as OWR-736-001 and paraphrases the sam)

OWR-736-001 / p.1428

### OWR-736-003 | Duplicate Warped Wavelet-Basis Extraction

Duplicate/merged OWR extraction (The official Oberwolfach report was checked directly. This extraction repeats the same source problem as OWR-736-001 and is byte-identical a)

OWR-736-001 / p.1428 OWR-736-002 [not in active corpus]

### OWR-737-009 | Redundant Extraction - Canonical Pseudo-Exponential Fields

Duplicate/merged OWR extraction (The primary Oberwolfach report was checked directly. The extracted text duplicates OWR-737-008 verbatim and does not contain the distinct pr)

OWR-737-008 / p.2006

### OWR-742-001 | Duplicate Complex-Interpolation Compactness Question

Duplicate/merged OWR extraction (This is a redundant broad extraction of the complex-interpolation compactness question stated precisely as OWR-742-002. No separate clean st)

OWR-742-002 / p.1299

### OWR-782-002 | Duplicate Extremal Modular-Form Congruence

Duplicate/merged OWR extraction (The official Oberwolfach report was checked directly. This extraction repeats the same source problem as OWR-782-001 and contains no indepen)

OWR-782-001 / p.48

### OWR-782-006 | Low-Moment Detection of Reductive Subgroups

Answered in the source or not an open problem (solved (yes for $d \geq 5$, with one exception; answered in the source))

### OWR-782-008 | Duplicate Saito Positivity Conjecture

Duplicate/merged OWR extraction (The official Oberwolfach report was checked directly. This extraction repeats the same source problem as OWR-782-007 and restates the same S)

OWR-782-007 / p.166

### OWR-787-007 | Minimal Families of Nonisomorphic Subgroup Embeddings

Unrecoverable OWR statement (The official report explicitly says that its final theorem answers this minimal-family question positively. No separate clean open statement)

### OWR-792-015 | Redundant Extraction - Bank Capital-Ratio Control

Duplicate/merged OWR extraction (The primary Oberwolfach report was checked directly. This record repeats the core text already captured, with fuller report context, by OWR-)

OWR-792-014 / p.2109

### OWR-831-004 | Finite-Field Erdős-Szemerédi Sum-Product Bounds

Answered in the source or not an open problem (solved (answered in the source))

### OWR-831-010 | Rhetorical Quantum-State Vector-Length Question

Duplicate/merged OWR extraction (This is the title-level rhetorical framing of the abstract, not a standalone mathematical statement; the actual tree-size conjectures are re)

OWR-831-012 / p.2079

### OWR-831-013 | Resolved Triple-Product Group Challenge

Unrecoverable OWR statement (The official report poses this challenge and then constructs such groups in the same abstract, obtaining nontrivial bounds on the matrix-mul)

### OWR-832-008 | Volume Conjecture for Knot Polynomials

Duplicate of canonical problem TOP-002

TOP-002 / p.1244

### OWR-8415336-014 | Density of Finite-Order Absolutely Periodic Billiard Orbits

Unrecoverable OWR statement (The official report states this density construction as Theorem 1, not as an open problem. No separate clean statement is promoted, and the )

### OWR-8415342-004 | Duplicate Hiraga-Ichino-Ikeda-Langlands Conjecture

Duplicate/merged OWR extraction (The official Oberwolfach report confirms that this extraction repeats the Hiraga-Ichino-Ikeda-Langlands Plancherel-density conjecture alread)

OWR-8415342-002 / p.134

### OWR-8415342-005 | Formal Degrees for Symplectic and Even Orthogonal Groups

Unrecoverable OWR statement (The official report labels the extracted assertion Theorem 1.3 and states it as a proved special case, not an open problem.)

### OWR-8415342-006 | Spectral Expansion of Twisted Orbital Integrals

Unrecoverable OWR statement (The official report confirms that the extraction is a theorem-level spectral-expansion result used in the proof of Theorem 1.3, not an open )

### OWR-8415342-019 | Beilinson-Drinfeld Equivalence and Restricted Geometric Langlands

Unrecoverable OWR statement (The official report confirms that this extraction is the evidence paragraph following the restricted geometric Langlands conjecture, which i)

### OWR-8415345-005 | Bridge Position for Surfaces in Trisected Four-Manifolds

Answered in the source or not an open problem (solved (a theorem, not a problem))

### OWR-8415347-001 | Unspecified Quantum-Groups Workshop Problem

Unrecoverable OWR statement (The official report confirms that the extraction is a table-of-contents fragment, not a standalone open problem.)

### OWR-8415347-002 | Unspecified Quantum-Groups Open Problem

Unrecoverable OWR statement (The official report confirms that the extraction is a table-of-contents fragment, not a standalone open problem.)

### OWR-8415347-012 | Resolved Witt-Group Question on Anisotropic Modular Categories

Answered in the source or not an open problem (solved (yes; answered in the source))

### OWR-8415350-001 | Unspecified Surface Singularity Conjecture

Unrecoverable OWR statement (The official report confirms that this record is only a table-of-contents fragment and contains no identifiable conjecture. No separate clea)

### OWR-8415350-002 | Non-Archimedean Wilkie-Type Point-Counting Bounds

Unrecoverable OWR statement (The official contribution reports results in each of the listed non-Archimedean settings, including a proved Pfaffian analogue of Wilkie's c)

### OWR-8415350-003 | Floer Cohomology and Contact Loci of Jets

Unrecoverable OWR statement (The official contribution states immediately after the displayed conjecture that the asserted isomorphism is true in general as a theorem in)

### OWR-8415350-008 | Unspecified Rank-One Del Pezzo Surface Conjecture

Duplicate/merged OWR extraction (The phrase 'the latter conjecture remains open' refers back to the Negativity Conjecture retained in OWR-8415350-007; the following rank-one)

OWR-8415350-007 / p.708

### OWR-8415352-005 | Symmetry Structure of Mutually Unbiased Bases

Unrecoverable OWR statement (The official report uses this question as an expository heading and immediately identifies the wreath-product symmetry acting on the mutuall)

### OWR-8415353-001 | Open Problems in Thin Groups and Super-Approximation

Unrecoverable OWR statement (The official source passage is a table-of-contents fragment and contains no mathematical claim. No separate clean statement is promoted; the)

### OWR-8415353-004 | Zaremba Conjecture for Bounded Continued Fraction Digits

Duplicate/merged OWR extraction (The official report states the same Zaremba conjecture already represented by the source-verified record OWR-8415353-003 in this report. No )

OWR-8415353-003 / p.198

### OWR-8415355-001 | Open Problems in Geometric Representation Theory

Unrecoverable OWR statement (The official source passage is a table-of-contents fragment and contains no mathematical claim. No separate clean statement is promoted; the)

### OWR-8415355-005 | Langlands Dual Parametrization from Baxter Polynomiality

Duplicate/merged OWR extraction (The official passage only gives evidence for the Langlands-dual parametrization conjecture already represented by OWR-8415355-004 in this re)

OWR-8415355-004 / p.294

### OWR-8415356-001 | Unspecified Combinatorial Optimization Problem

Unrecoverable OWR statement (The official report shows that the extract is a table-of-contents fragment listing several later contributions, not a standalone mathematica)

### OWR-8415356-015 | Linear Bounds for Minimally Integer-Infeasible Systems

Duplicate/merged OWR extraction (The official report confirms that this is a duplicate extraction of the minimally integer-infeasible systems problem retained in OWR-8415356)

OWR-8415356-014 / p.2090

### OWR-8415356-019 | Better-Than-Two Approximation for Multiplicative Connectivity Augmentation

Duplicate/merged OWR extraction (The official report confirms that this is a duplicate extraction of the multiplicative connectivity augmentation problem retained in OWR-841)

OWR-8415356-018 / p.2063

### OWR-8415356-022 | Balanced Signings in the Subtour Polytope

Duplicate/merged OWR extraction (The official report confirms that this is a duplicate extraction of the balanced-signing question retained in OWR-8415356-021. No separate c)

OWR-8415356-021 / p.2062

### OWR-8415356-026 | Deterministic Optimization of Top-k Perfect-Matching Edges

Duplicate/merged OWR extraction (The official report confirms that this is a duplicate extraction of the top-k perfect-matching question retained in OWR-8415356-025. No sepa)

OWR-8415356-025 / p.2073

### OWR-8415359-005 | Probabilistic Box Models

Unrecoverable OWR statement (The linked report was inspected: this is a rhetorical transition immediately followed by a definition of quantum functions, not a freestandi)

### OWR-8415359-007 | Dimension-Complexity Tradeoffs in Quantum-Function Testing

Unrecoverable OWR statement (The linked report was inspected: the apparent question is immediately answered by the stated corollary of $\mathrm{MIP}^*=\mathrm{RE}$, so t)

### OWR-8415359-014 | Average-Case Metacomplexity Characterizations of One-Way Functions

Unrecoverable OWR statement (The linked report was inspected: this passage surveys proved characterizations of one-way functions and contains no explicit unresolved prob)

### OWR-8415361-001 | Unspecified Higher-Rank Dynamics Conjecture

Duplicate/merged OWR extraction (The official report shows that this is introductory context pointing to the higher-rank rigidity conjecture retained in OWR-8415361-002. No )

OWR-8415361-002 / p.1513

### OWR-8415362-001 | Cluster Algebras Scattering Amplitudes and Positive Geometries

Unrecoverable OWR statement (The official source passage is the workshop abstract describing its themes, not a mathematical problem or conjecture. No separate clean stat)

### OWR-8415364-001 | Kannan-Lovasz-Simonovits Spectral Gap Conjecture

Duplicate/merged OWR extraction (The official report shows that this workshop-level introduction merely mentions the Kannan-Lovász-Simonovits conjecture, which is retained i)

OWR-8415364-006 / p.886

### OWR-8415364-002 | Unspecified Convex Geometry Conjecture

Unrecoverable OWR statement (The official report shows that this extract is an incomplete contents fragment and does not state a standalone mathematical problem. No sepa)

### OWR-8415364-003 | Unspecified Convex Geometry Conjecture

Unrecoverable OWR statement (The official report shows that this extract contains only a conjecture label and no recoverable mathematical statement. No separate clean st)

### OWR-8415364-004 | Foundations of Convex Algebraic Geometry

Unrecoverable OWR statement (The official report immediately answers this expository heading by defining convex algebraic geometry; it is not posed as an open mathematic)

### OWR-8415364-005 | KLS Isoperimetric Coefficient Bound for Log-Concave Measures

Duplicate/merged OWR extraction (The official report confirms that this is a duplicate extraction of the Kannan-Lovász-Simonovits conjecture retained in OWR-8415364-006. No )

OWR-8415364-006 / p.886

### OWR-8415364-007 | KLS Isoperimetric Coefficient Bound for Log-Concave Measures

Duplicate/merged OWR extraction (The official report confirms that this is a duplicate extraction of the Kannan-Lovász-Simonovits conjecture retained in OWR-8415364-006. No )

OWR-8415364-006 / p.886

### OWR-8415364-008 | Unspecified Log-Concave Measure Conjecture

Unrecoverable OWR statement (The official report places this fragment inside proved results on log-concave measures and does not state an open problem. No separate clean)

### OWR-9790352-001 | Unspecified Structural Graph Theory Problem

Unrecoverable OWR statement (The official report shows that this is workshop-level introductory prose rather than a standalone mathematical problem. No separate clean st)

### OWR-9790352-002 | Unspecified Structural Graph Theory Conjecture

Unrecoverable OWR statement (The official report shows that this extract is a table-of-contents fragment without a recoverable standalone statement. No separate clean st)

### OWR-9790352-003 | Unspecified Structural Graph Theory Conjecture

Unrecoverable OWR statement (The official report shows that this extract is a second table-of-contents fragment without a recoverable standalone statement. No separate c)

### OWR-9790352-005 | Local Separators in Large Networks

Unrecoverable OWR statement (The official report immediately answers the rhetorical heading by defining local separators, so it is not an open problem. No separate clean)

### OWR-9790352-006 | Unspecified Graph Structure Problem

Unrecoverable OWR statement (The official report answers this figure-specific question in the following sentence; the extract is contextual exposition rather than an ope)

### OWR-9790352-008 | Report-Proved Logarithmic Treewidth Bound

Unrecoverable OWR statement (The official report states this logarithmic-treewidth conjecture and then proves it as a theorem in the same contribution, so it must not be)

### OWR-9790352-015 | Unspecified Colin de Verdiere Graph Parameter Problem

Unrecoverable OWR statement (The official report shows that the extract states a theorem and corollary about the Colin de Verdière parameter, not an open problem. No sep)

### OWR-9790352-016 | Unspecified Nonrepetitive Graph Coloring Problem

Unrecoverable OWR statement (The official report uses the extracted formula to record a proved nonrepetitive-coloring bound rather than a standalone open problem. No sep)

### OWR-9790352-020 | Chi-Bounds and Stable Sets in Burling Graphs

Duplicate/merged OWR extraction (The official report confirms that this is a partial duplicate of the Burling-graph questions retained in OWR-9790352-019. No separate clean )

OWR-9790352-019 / p.1859

### OWR-9790352-022 | Unspecified Alspach Rainbow Matching Conjecture

Unrecoverable OWR statement (The official report shows that this fragment reports the asymptotic resolution of Alspach's conjecture and does not state a complete open pr)

### OWR-9790352-023 | Aharoni-Berger Rainbow Matching Conjecture

Duplicate/merged OWR extraction (This is a later cross-report duplicate of the Aharoni-Berger rainbow-matching conjecture canonically retained in OWR-16763-030. No separate )

OWR-16763-030 / p.1695

### OWR-9790352-024 | Unspecified Alspach Rainbow Matching Conjecture

Unrecoverable OWR statement (The official report shows that this is another fragment reporting the asymptotic resolution of Alspach's conjecture, not a standalone open p)

### OWR-9790352-028 | Density Threshold for Rainbow Clique Subdivisions

Duplicate/merged OWR extraction (The official report confirms that this is a corrupted duplicate of the rainbow clique-subdivision question retained in OWR-9790352-027. No s)

OWR-9790352-027 / p.1782

### OWR-9790354-002 | Type-I Conjecture Section Heading

Unrecoverable OWR statement (The official report confirms that the extraction is only a section heading and an incomplete setup duplicated by the following complete conj)

### OWR-9790355-001 | Unspecified Multiscale Complex Media Modeling Problem

Unrecoverable OWR statement (The official report shows that this is broad workshop context mentioning many questions without stating a recoverable standalone target. No )

### OWR-9790357-001 | Convolutional Reconstruction in Many Interface Field Models

Unrecoverable OWR statement (The official source says only that several questions remain and then describes a reconstruction procedure; it does not state an independentl)

### OWR-9790358-003 | Unspecified Polynomial Interpolation Conjecture

Unrecoverable OWR statement (The official report shows that this extract is a contents fragment and does not state a recoverable standalone problem. No separate clean st)

### OWR-9790358-005 | Higher-Dimensional GC Set Counterexamples

Unrecoverable OWR statement (The official report recounts known higher-dimensional counterexamples and unsuccessful approaches rather than posing a new standalone open p)

### OWR-9790358-006 | Gasca-Maeztu Maximal Line Conjecture

Duplicate/merged OWR extraction (The official report confirms that this is a duplicate extraction of the Gasca-Maeztu conjecture retained in OWR-9790358-001. No separate cle)

OWR-9790358-001 / p.242

### OWR-9790358-007 | Structure of n-Independent and GCn Interpolation Sets

Unrecoverable OWR statement (The official report presents descriptive background about interpolation sets but does not formulate a standalone open mathematical target. N)

OWR-9790358-001 / p.242

### OWR-9790359-004 | Constraint-Satisfaction Table Caption

Unrecoverable OWR statement (The official report confirms that the extraction is a table caption summarizing a taxonomy, not a standalone open mathematical problem.)

### OWR-9790364-001 | Unspecified Hypergeometric Positivity Problem

Unrecoverable OWR statement (The official report shows that this is a contents fragment without a recoverable standalone hypergeometric-positivity problem. No separate c)

### OWR-9790364-003 | Unspecified Laguerre-Polya Density Criterion Problem

Unrecoverable OWR statement (The official report presents survey context about a density criterion but does not state a complete standalone open problem. No separate cle)

### OWR-9790367-004 | Strength of Semisimple TQFT Invariants

Answered in the source or not an open problem (solved (answered in the source))

### SET-001 | The Continuum Hypothesis

Duplicate of canonical problem HIL-001

HIL-001 / p.1973

### TOP-011 | Volume Conjecture

Internal Seed duplicate of TOP-002 (id=34, 'The Volume Conjecture')

TOP-002 / p.1244

### TOP-012 | Borel Conjecture

Internal Seed duplicate of TOP-002 (id=1332, 'Borel Conjecture')

TOP-005 / p.1023

### TOP-013 | Novikov Conjecture

Internal Seed duplicate of TOP-004 (id=1334, 'Novikov Conjecture')

TOP-006 / p.1140

### TOP-014 | Unknotting Problem

Internal Seed duplicate of TOP-001 (id=1331, 'Unknotting Problem')

TOP-004 / p.1248

### TOP-015 | Volume Conjecture

Internal Seed duplicate of TOP-002 (id=34, 'The Volume Conjecture')

TOP-002 / p.1244

### TOP-016 | Whitehead Conjecture

Internal Seed duplicate of TOP-003 (id=1146, 'The Whitehead Conjecture')

TOP-003 / p.1244